

STAR RX72N - rx_hal Library
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Chapter 5

Topic Documentation

5.1 RX72N S12AD Register Definitions

12-bit A/D Converter hardware register interface

Data Structures

- struct [rx_s12ad_regs_t](#)
S12ADFa Register Map for 12-bit A/D Converter.

Enumerations

- enum [rx_s12ad_addresses_t](#) : [uintptr_t](#) { [k_s12ad0_base_addr](#) = 0x00089000 , [k_s12ad1_base_addr](#) = 0x00089100 }
S12ADFa peripheral base addresses.
- enum [s12ad_offsets_t](#) : [uint8_t](#) {
[k_s12ad_offset_adcsr](#) = 0x00 , [k_s12ad_offset_adansa0](#) = 0x04 , [k_s12ad_offset_adads0](#) = 0x08 ,
[k_s12ad_offset_adadc](#) = 0x0C ,
[k_s12ad_offset_adcer](#) = 0x0E , [k_s12ad_offset_adstrgr](#) = 0x10 , [k_s12ad_offset_adansb0](#) = 0x14 ,
[k_s12ad_offset_adbldr](#) ,
[k_s12ad_offset_addrd](#) = 0x1E , [k_s12ad_offset_addr0](#) = 0x20 , [k_s12ad_offset_addr1](#) = 0x22 ,
[k_s12ad_offset_addr2](#) = 0x24 ,
[k_s12ad_offset_addr3](#) = 0x26 , [k_s12ad_offset_addr4](#) = 0x28 , [k_s12ad_offset_addr5](#) = 0x2A ,
[k_s12ad_offset_addr6](#) = 0x2C ,
[k_s12ad_offset_addr7](#) = 0x2E }
S12ADFa register offsets from base address.
- enum [s12ad_reserved_sizes_t](#) : [uint8_t](#) { [k_s12ad_reserved_02_03](#) = 2 , [k_s12ad_reserved_0d](#) = 1 , [k_s12ad_reserved_12_13](#) = 2 , [k_s12ad_reserved_16_1d](#) = 8 }
S12ADFa register reserved field sizes for structure padding.

Functions

- static volatile [rx_s12ad_regs_t](#) * [s12ad0](#) (void)
Get pointer to S12AD0 (Unit 0) registers.
- static volatile [rx_s12ad_regs_t](#) * [s12ad1](#) (void)
Get pointer to S12AD1 (Unit 1) registers.

5.1.1 Detailed Description

12-bit A/D Converter hardware register interface

Provides memory-mapped register access to the RX72N S12AD peripheral for motor current sensing applications. All register addresses and offsets are compile-time verified via static assertions.

Quick Start Example:

```
// Read motor 0 current from AN007 (P47)
volatile rx_s12ad_regs_t* adc = s12ad0();

// Enable channel 7 for single scan
adc->adansa0 = (1U « 7); // AN007 only

// Configure: single scan, software trigger
adc->adcsr = 0x0000;

// Start conversion
adc->adcsr |= 0x8000; // ADST bit

// Wait for completion
while (adc->adcsr & 0x8000) {}

// Read 12-bit result (right-aligned)
uint16_t raw_value = adc->addr7;

// Convert to voltage (3.3V reference)
static const float s_vref_volts = 3.3F;
static const float s_adc_full_scale_12bit = 4095.0F;
float voltage_v = (float)raw_value * s_vref_volts / s_adc_full_scale_12bit;
```

5.1.2 Data Structure Documentation

5.1.2.1 struct rx's12ad'regs't

S12ADFa Register Map for 12-bit A/D Converter.

Memory-mapped register structure for the S12AD peripheral. This structure provides direct hardware access to ADC control and data registers for motor current sensing and analog input measurement.

Memory Layout (Total: 48 bytes = 0x30):

Offset	Size	Field	Description
0x00	2	adcsr	Control/Status (mode, start, IRQ)
0x02	2	reserved	Reserved padding
0x04	2	adansa0	Channel select A (ch 0-15)
0x06	2	adansa1	Channel select A (ch 16-20, Unit1)
0x08	2	adads0	Addition mode (ch 0-15)
0x0A	2	adads1	Addition mode (ch 16-20, Unit1)
0x0C	1	adadc	Addition count (1, 2, 4, 16)
0x0D	1	reserved	Reserved padding
0x0E	2	adcer	Control extended (alignment)

Offset	Size	Field	Description
0x10	2	adstrgr	Trigger source select
0x12	2	reserved	Reserved padding
0x14	2	adansb0	Channel select B (group B scan)
0x16	8	reserved	Reserved (ADDBLDR at 0x84, not in struct)
0x1E	2	adrd	Self-diagnosis data
0x20	2	addr0	Result register AN0/AN100
0x22	2	addr1	Result register AN1/AN101
0x24	2	addr2	Result register AN2/AN102
0x26	2	addr3	Result register AN3/AN103
0x28	2	addr4	Result register AN4/AN104
0x2A	2	addr5	Result register AN5/AN105
0x2C	2	addr6	Result register AN6/AN106
0x2E	2	addr7	Result register AN7/AN107

Usage Example - Single Channel Conversion:

```
// Initialize ADC for motor 0 current sensing (AN007, P47)
volatile rx_s12ad_regs_t* adc = s12ad0();

// 1. Select channel AN007 for group A scan
adc->adansa0 = (1U << 7); // Bit 7 = AN007

// 2. Configure single scan mode, software trigger
adc->adcsr = 0x0000; // ADCS[1:0]=00 (single), TRGE=0 (software)

// 3. Set right-alignment (bits 11:0 contain result)
adc->adcr = 0x0000; // ADRFMT=0 (right-align)

// 4. Start conversion (set ADST bit)
adc->adcsr |= 0x8000;

// 5. Wait for conversion complete (ADST clears)
while (adc->adcsr & 0x8000) {
    // ~1 us per conversion
}

// 6. Read 12-bit result
uint16_t raw_count = adc->addr7 & 0xFFFF;

// 7. Convert to current (using DRV8263H-Q1 IPROPI current sensing)
// IPROPI mirror: 202 uA/A, sense resistor: 5100 ohm
// V_IPROPI = I_motor * 202e-6 * 5100 = I_motor * 1.0302
// so I_motor = V_IPROPI / 1.0302
static const float s_avcc_volts = 3.3F;
static const float s_adc_full_scale_divisor = 4095.0F; // 2^12 - 1 (12-bit full scale)
static const float s_ipropi_mirror = 202e-6F; // 202 uA/A current mirror ratio
static const float s_ipropi_sense_ohm = 5100.0F; // 5.1k sense resistor
static const float s_ipropi_divisor = 1.0302F; // = 202e-6 * 5100
float voltage = (float)raw_count * s_avcc_volts / s_adc_full_scale_divisor;
float current_amps = voltage / s_ipropi_divisor;
```

Usage Example - Continuous 4-Channel Scan:

```
volatile rx_s12ad_regs_t* adc = s12ad0();

// Enable channels AN004-AN007 (all 4 motor currents)
adc->adansa0 = 0x00F0; // Bits [7:4] = channels 4-7

// Configure continuous scan mode
adc->adcsr = 0x4000; // ADCS[1:0]=10 (continuous scan), bits[14:13]=10

// Start continuous conversion
adc->adcsr |= 0x8000;

// Read results any time (updated continuously)
```

```
uint16_t motor0_current = adc->addr7; // AN007 = M0
uint16_t motor1_current = adc->addr6; // AN006 = M1
uint16_t motor2_current = adc->addr5; // AN005 = M2
uint16_t motor3_current = adc->addr4; // AN004 = M3

// Stop conversion when done
adc->adcsr &= ~0x8000;
```

Note

This struct covers registers common to both S12AD0 and S12AD1. S12AD1 has additional registers (ADDR8-20, temp sensor, etc.) beyond offset 0x30 that are not included in this structure.

Warning

All registers are memory-mapped and must be accessed as volatile. Do not cache register values in local variables for repeated use.

Reserved fields must never be written. Reading returns undefined values.

Invariant

`sizeof(rx_s12ad_regs_t) == 0x30` (48 bytes)

All register offsets match RX72N Manual Ch56 Table 56.1

Data registers ADDR0-ADDR7 have 2-byte spacing

See also

[s12ad0\(\)](#) Accessor for S12AD Unit 0

[s12ad1\(\)](#) Accessor for S12AD Unit 1

RX72N Manual Ch56 Section 56.2 for register bit definitions

Definition at line 397 of file [rx72n_adc_regs.h](#).

Collaboration diagram for `rx_s12ad_regs_t`:



Data Fields

volatile uint8_t	adadc	A/D Addition Count Select Register (ADADC) @ offset 0x0C. Sets the number of conversions to add/↵ average for oversampling. Valid Values: <table><tr><th>Value</th><th>Additio</th><th>Effect on Data</th><th>Time Multiplier</th></tr><tr><td>0x00</td><td>1x</td><td>No addition</td><td>1x</td></tr><tr><td>0x01</td><td>2x</td><td>Sum or average</td><td>2x</td></tr><tr><td>0x02</td><td>3x</td><td>Sum or average</td><td>3x</td></tr><tr><td>0x03</td><td>4x</td><td>Sum or average</td><td>4x</td></tr><tr><td>0x05</td><td>16x</td><td>Sum or average</td><td>16x</td></tr></table>	Value	Additio	Effect on Data	Time Multiplier	0x00	1x	No addition	1x	0x01	2x	Sum or average	2x	0x02	3x	Sum or average	3x	0x03	4x	Sum or average	4x	0x05	16x	Sum or average	16x
Value	Additio	Effect on Data	Time Multiplier																							
0x00	1x	No addition	1x																							
0x01	2x	Sum or average	2x																							
0x02	3x	Sum or average	3x																							
0x03	4x	Sum or average	4x																							
0x05	16x	Sum or average	16x																							
volatile uint16_t	adads0	A/D Addition Mode Select Register 0 (ADADS0) @ offset 0x08. Selects channels 0-15 for addition/↵ averaging mode. When enabled, multiple conversions are accumulated or averaged per the ADADC setting. Oversampling Benefit: • 4x averaging: Reduces noise by ~6 dB (factor of 2) • 16x averaging: Reduces noise by ~12 dB (factor of 4)																								
volatile uint16_t	adads1	A/D Addition Mode Select Register 1 (ADADS1) @ offset 0x0A. Selects channels 16-20 for addition mode (S12AD1 only).																								

volatile uint16_t	adansa0	A/D Channel Select Register A0 (ADANSA0) @ offset 0x04. Selects channels 0-15 for Group A scan. Each bit enables the corresponding analog input channel. <table><tr><th>Bit</th><th>Channel</th><th>Pin</th><th>Usage</th></tr><tr><td>0-3</td><td>AN000-3</td><td>P40-43</td><td>Available</td></tr><tr><td>4</td><td>AN004</td><td>P44</td><td>M3 current</td></tr><tr><td>5</td><td>AN005</td><td>P45</td><td>M2 current</td></tr><tr><td>6</td><td>AN006</td><td>P46</td><td>M1 current</td></tr><tr><td>7</td><td>AN007</td><td>P47</td><td>M0 current</td></tr></table> Channel Mapping (S12AD0, 144-pin LFP): Example: Select motor 0 and 1 current channels <pre>adc->adansa0 = 0x00C0; // Enable AN007 (M0), AN006 (M1)</pre>	Bit	Channel	Pin	Usage	0-3	AN000-3	P40-43	Available	4	AN004	P44	M3 current	5	AN005	P45	M2 current	6	AN006	P46	M1 current	7	AN007	P47	M0 current
Bit	Channel	Pin	Usage																							
0-3	AN000-3	P40-43	Available																							
4	AN004	P44	M3 current																							
5	AN005	P45	M2 current																							
6	AN006	P46	M1 current																							
7	AN007	P47	M0 current																							
volatile uint16_t	adansa1	A/D Channel Select Register A1 (ADANSA1) @ offset 0x06. Selects channels 16-20 for Group A scan (S12AD1 only). For S12AD0, this register exists but bits [7:0] are reserved. Note Only valid for S12AD1. S12AD0 has only 8 channels (0-7).																								
volatile uint16_t	adansb0	A/D Channel Select Register B0 (ADANSB0) @ offset 0x14. Selects channels for Group B scan (dual-group scanning mode). Group B runs in parallel with Group A for interleaved sampling.																								

volatile uint16_t	adcer	A/D Control Extended Register (AD-CER) @ offset 0x0E. Configures data alignment and self-diagnosis mode. Key Bits: • Bit 15 (ADRFMT): 0=right-align (bits [11:0]), 1=left-align (bits [15:4]) • Bit 5 (ACE): Auto-clear enable for data registers Alignment Example: • Right-align: Result in bits ↵ [11:0], bits [15:12] = 0 • Left-align: Result in bits [15:↵ 4], bits [3:0] = 0
-------------------	-------	--

volatile uint16_t	adcsr	A/D Control/Status Register (ADCSR) @ offset 0x00. Controls conversion mode, trigger source, and interrupt enable. <table><tr><th>Bits</th><th>Name</th><th>R/W</th><th>Description</th></tr><tr><td>[15]</td><td>ADST</td><td>R/W</td><td>Conversion start (1=start/↔running)</td></tr><tr><td>[14:↔13]</td><td>ADCS</td><td>R/W</td><td>Scan mode (00=single, 01=group, 10=continuous)</td></tr><tr><td>[12]</td><td>ADIE</td><td>R/W</td><td>Scan end interrupt enable</td></tr><tr><td>[9]</td><td>TRGE</td><td>R/W</td><td>Trigger start enable (0=software only, 1=sync/↔async trigger)</td></tr><tr><td>[8]</td><td>EXTRG</td><td>R/W</td><td>Trigger select (0=synchronous, 1=asynchronous)</td></tr><tr><td>[7]</td><td>DBLE</td><td>R/W</td><td>Double trigger mode select</td></tr><tr><td>[6]</td><td>GBADIER</td><td>R/W</td><td>Group B scan end interrupt enable</td></tr></table> Bit Field Description: Common Configurations:	Bits	Name	R/W	Description	[15]	ADST	R/W	Conversion start (1=start/↔running)	[14:↔13]	ADCS	R/W	Scan mode (00=single, 01=group, 10=continuous)	[12]	ADIE	R/W	Scan end interrupt enable	[9]	TRGE	R/W	Trigger start enable (0=software only, 1=sync/↔async trigger)	[8]	EXTRG	R/W	Trigger select (0=synchronous, 1=asynchronous)	[7]	DBLE	R/W	Double trigger mode select	[6]	GBADIER	R/W	Group B scan end interrupt enable
Bits	Name	R/W	Description																															
[15]	ADST	R/W	Conversion start (1=start/↔running)																															
[14:↔13]	ADCS	R/W	Scan mode (00=single, 01=group, 10=continuous)																															
[12]	ADIE	R/W	Scan end interrupt enable																															
[9]	TRGE	R/W	Trigger start enable (0=software only, 1=sync/↔async trigger)																															
[8]	EXTRG	R/W	Trigger select (0=synchronous, 1=asynchronous)																															
[7]	DBLE	R/W	Double trigger mode select																															
[6]	GBADIER	R/W	Group B scan end interrupt enable																															
		Generated by Doxygen 0x0000: Single scan, software trigger, no interrupt 0x4000: Continuous scan, software trigger																																

volatile uint16_t	addr0	A/D Data Register 0 (ADDR0) @ offset 0x20. 12-bit conversion result for channel AN000 (Unit 0) or AN100 (Unit 1). Data Format (ADCER.ADRFMT=0, right-aligned): • Bits [11:0]: 12-bit result (0-4095) • Bits [15:12]: Always 0 Conversion Formula: $V_{in} = \frac{ADC_{result} \times V_{ref}}{4095}$ where Vref = 3.3V (AVCC0)
volatile uint16_t	addr1	A/D Data Register 1 @ 0x22: AN001/↔ AN101
volatile uint16_t	addr2	A/D Data Register 2 @ 0x24: AN002/↔ AN102
volatile uint16_t	addr3	A/D Data Register 3 @ 0x26: AN003/↔ AN103
volatile uint16_t	addr4	A/D Data Register 4 @ 0x28: AN004/↔ AN104 (Motor 3 current)
volatile uint16_t	addr5	A/D Data Register 5 @ 0x2A: AN005/↔ AN105 (Motor 2 current)
volatile uint16_t	addr6	A/D Data Register 6 @ 0x2C: AN006/↔ AN106 (Motor 1 current)
volatile uint16_t	addr7	A/D Data Register 7 @ 0x2E: AN007/↔ AN107 (Motor 0 current)
volatile uint16_t	adrd	A/D Self-Diagnosis Data Register (ADRD) @ offset 0x1E. Contains result of self-diagnosis conversion when ADCER.DIAGM = 1. Used for ADC calibration and health monitoring.

volatile uint16_t	adstrgr	A/D Start Trigger Select Register (ADSTRGR) @ offset 0x10. Selects the trigger source for starting A/D conversion when hardware triggering is enabled (ADCSR.TRGE = 1). Trigger Sources (TRSA[5:0] at bits [13:8]; TRSB[5:0] at bits [5:0]): <table> <tr> <th>Value</th><th>Source</th><th>Use Case</th></tr> <tr> <td>0x00</td><td>Software (ADST)</td><td>Polled conversion</td></tr> <tr> <td>0x01</td><td>MTU0 TGIA</td><td>Motor PWM sync</td></tr> <tr> <td>0x02</td><td>MTU0 TGIB</td><td>Motor PWM sync</td></tr> <tr> <td>0x09</td><td>MTU4 TAD-CORA</td><td>MTU4 PWM sync</td></tr> </table>	Value	Source	Use Case	0x00	Software (ADST)	Polled conversion	0x01	MTU0 TGIA	Motor PWM sync	0x02	MTU0 TGIB	Motor PWM sync	0x09	MTU4 TAD-CORA	MTU4 PWM sync
Value	Source	Use Case															
0x00	Software (ADST)	Polled conversion															
0x01	MTU0 TGIA	Motor PWM sync															
0x02	MTU0 TGIB	Motor PWM sync															
0x09	MTU4 TAD-CORA	MTU4 PWM sync															
uint8_t	reserved0[k_s12ad_reserved_02_03]	Reserved @ 0x02-0x03 - do not access															
uint8_t	reserved1[k_s12ad_reserved_0d]	Reserved @ 0x0D - do not access															
uint8_t	reserved2[k_s12ad_reserved_12_13]	Reserved @ 0x12-0x13 - do not access															
uint8_t	reserved3[k_s12ad_reserved_16_1d]	Reserved @ 0x16-0x1D (ADDBLDR at 0x84, not in struct)															

5.1.3 Enumeration Type Documentation

5.1.3.1 rx's12ad'addresses't

```
enum rx_s12ad_addresses_t : uintptr_t
```

S12ADFa peripheral base addresses.

Hardware base addresses for the two S12AD units on RX72N. Each unit occupies 256 bytes (0x100) of address space in the peripheral area.

Per RX72N Manual Ch56 and Ch04 Memory Map:

- S12AD0: 0x00089000 (Unit 0, 8 high-speed channels)
- S12AD1: 0x00089100 (Unit 1, 21 channels + temperature sensor)

Address Space Layout:

Address Range	Description
0x00089000 - 0x000890FF	S12AD0 registers (Unit 0)
0x00089100 - 0x000891FF	S12AD1 registers (Unit 1)

Unit Selection Guidelines:

- Use S12AD0 for motor current sensing (lower latency, 8 channels)
- Use S12AD1 for temperature monitoring (has internal sensor)

Invariant

Base addresses are 256-byte aligned (bits [7:0] = 0)

Unit spacing is exactly 0x100 bytes

See also

[s12ad0\(\)](#) Accessor function for Unit 0

[s12ad1\(\)](#) Accessor function for Unit 1

Enumerator

k_s12ad0_base_addr	S12AD Unit 0 base address. 8-channel high-speed ADC optimized for motor current sensing. Channels AN000-AN007 mapped to ports P40-P47. Usage: Motor current sensing (all 4 motors) Throughput: Up to 1 MSPS in continuous scan mode
k_s12ad1_base_addr	S12AD Unit 1 base address. 21-channel ADC with additional features including temperature sensor and internal voltage reference channels. Channels AN100-AN120 mapped to various ports. Usage: Temperature monitoring, analog sensing Special Channels: Temperature sensor (AN127), Ref voltage (AN128)

Definition at line 186 of file [rx72n_adc_regs.h](#).

```
00186 : uintptr_t {
00187 /**
00188  * @brief S12AD Unit 0 base address
```

```

00189  * @details
00190  * 8-channel high-speed ADC optimized for motor current sensing.
00191  * Channels AN000-AN007 mapped to ports P40-P47.
00192  *
00193  * @par Usage: Motor current sensing (all 4 motors)
00194  * @par Throughput: Up to 1 MSPS in continuous scan mode
00195  */
00196 k_s12ad0_base_addr = 0x00089000,
00197
00198 /**
00199  * @brief S12AD Unit 1 base address
00200  * @details
00201  * 21-channel ADC with additional features including temperature
00202  * sensor and internal voltage reference channels.
00203  * Channels AN100-AN120 mapped to various ports.
00204  *
00205  * @par Usage: Temperature monitoring, analog sensing
00206  * @par Special Channels: Temperature sensor (AN127), Ref voltage (AN128)
00207  */
00208 k_s12ad1_base_addr = 0x00089100,
00209 } rx_s12ad_addresses_t;

```

5.1.3.2 s12ad_offsets_t

```
enum s12ad_offsets_t : uint8_t
```

S12ADFa register offsets from base address.

Register byte offsets for accessing S12AD peripheral registers. All offsets verified against RX72N Manual Ch56, section 56.2.

Register Map Overview:

Offset	Register	Size	Description
0x00	ADCSR	16	Control/Status (start, mode)
0x04	ADANSA0	16	Channel select Group A (0-15)
0x08	ADADS0	16	Addition mode select
0x0C	ADADC	8	Addition count (1, 2, 4, 16)
0x0E	ADCER	16	Control extended (alignment)
0x10	ADSTRGR	16	Trigger source select
0x14	ADANSB0	16	Channel select Group B
0x84	ADDBLDR	16	Double trigger data (not in struct; accessed directly)
0x1E	ADRD	16	Self-diagnosis data
0x20	ADDR0	16	Conversion result channel 0
...	...	...	(2-byte spacing per channel)
0x2E	ADDR7	16	Conversion result channel 7

Note

Offsets are 8-bit values (max 255), sufficient for S12AD register space

Invariant

Data registers ADDR0-ADDR7 are contiguous with 2-byte spacing

All 16-bit registers are halfword-aligned (even addresses)

Enumerator

k_s12ad_offset_adcsr	A/D Control Status: start, mode, interrupt control
k_s12ad_offset_adansa0	Channel Select A0: channels 0-15 for group A scan
k_s12ad_offset_adads0	Addition Select 0: channels 0-15 for averaging
k_s12ad_offset_adadc	Addition Count: 1x, 2x, 4x, or 16x oversampling
k_s12ad_offset_adcer	Control Extended: data alignment, self-diag mode
k_s12ad_offset_adstrgr	Trigger Select: software, MTU, GPT, external
k_s12ad_offset_adansb0	Channel Select B0: channels for group B scan
k_s12ad_offset_addblldr	Double Trigger Data: double-trigger mode result (not in struct)
k_s12ad_offset_adrd	Self-Diagnosis Data: internal reference result
k_s12ad_offset_addr0	Data Register 0: AN000/AN100 conversion result
k_s12ad_offset_addr1	Data Register 1: AN001/AN101 conversion result
k_s12ad_offset_addr2	Data Register 2: AN002/AN102 conversion result
k_s12ad_offset_addr3	Data Register 3: AN003/AN103 conversion result
k_s12ad_offset_addr4	Data Register 4: AN004/AN104 conversion result
k_s12ad_offset_addr5	Data Register 5: AN005/AN105 conversion result
k_s12ad_offset_addr6	Data Register 6: AN006/AN106 conversion result
k_s12ad_offset_addr7	Data Register 7: AN007/AN107 conversion result

Definition at line 241 of file `rx72n_adc_regs.h`.

```

00241      : uint8_t {
00242  k_s12ad_offset_adcsr = 0x00, /**< A/D Control Status: start, mode, interrupt control */
00243  k_s12ad_offset_adansa0 = 0x04, /**< Channel Select A0: channels 0-15 for group A scan */
00244  k_s12ad_offset_adads0 = 0x08, /**< Addition Select 0: channels 0-15 for averaging */
00245  k_s12ad_offset_adadc = 0x0C, /**< Addition Count: 1x, 2x, 4x, or 16x oversampling */
00246  k_s12ad_offset_adcer = 0x0E, /**< Control Extended: data alignment, self-diag mode */
00247  k_s12ad_offset_adstrgr = 0x10, /**< Trigger Select: software, MTU, GPT, external */
00248  k_s12ad_offset_adansb0 = 0x14, /**< Channel Select B0: channels for group B scan */
00249  k_s12ad_offset_addblldr =
00250    0x84, /**< Double Trigger Data: double-trigger mode result (not in struct) */
00251  k_s12ad_offset_adrd = 0x1E, /**< Self-Diagnosis Data: internal reference result */
00252  k_s12ad_offset_addr0 = 0x20, /**< Data Register 0: AN000/AN100 conversion result */
00253  k_s12ad_offset_addr1 = 0x22, /**< Data Register 1: AN001/AN101 conversion result */
00254  k_s12ad_offset_addr2 = 0x24, /**< Data Register 2: AN002/AN102 conversion result */
00255  k_s12ad_offset_addr3 = 0x26, /**< Data Register 3: AN003/AN103 conversion result */
00256  k_s12ad_offset_addr4 = 0x28, /**< Data Register 4: AN004/AN104 conversion result */
00257  k_s12ad_offset_addr5 = 0x2A, /**< Data Register 5: AN005/AN105 conversion result */
00258  k_s12ad_offset_addr6 = 0x2C, /**< Data Register 6: AN006/AN106 conversion result */
00259  k_s12ad_offset_addr7 = 0x2E, /**< Data Register 7: AN007/AN107 conversion result */
00260 } s12ad_offsets_t;

```

5.1.3.3 s12ad_reserved_sizes_t

```
enum s12ad_reserved_sizes_t : uint8_t
```

S12ADFa register reserved field sizes for structure padding.

Defines byte sizes for reserved regions in the S12AD register map. These ensure proper alignment and offset matching between the C structure and actual hardware register locations.

Warning

Do not modify these values - they are derived from the RX72N hardware register layout specification.

Note

Reserved bytes must be declared as `uint8_t` arrays, not accessed

Enumerator

<code>k_s12ad_reserved_02_03</code>	Padding after ADCSR (16-bit) to ADANSA0 (32-bit aligned)
<code>k_s12ad_reserved_0d</code>	Padding after ADADC (8-bit) to ADCER (16-bit aligned)
<code>k_s12ad_reserved_12_13</code>	Padding after ADSTRGR to ADANSB0
<code>k_s12ad_reserved_16_1d</code>	Padding after ADANSB0 to ADRD (ADDBLDR at 0x84, not in struct)

Definition at line 276 of file `rx72n_adc_regs.h`.

```

00276     : uint8_t {
00277   k_s12ad_reserved_02_03 = 2, /**< Padding after ADCSR (16-bit) to ADANSA0 (32-bit aligned) */
00278   k_s12ad_reserved_0d   = 1, /**< Padding after ADADC (8-bit) to ADCER (16-bit aligned) */
00279   k_s12ad_reserved_12_13 = 2, /**< Padding after ADSTRGR to ADANSB0 */
00280   k_s12ad_reserved_16_1d = 8, /**< Padding after ADANSB0 to ADRD (ADDBLDR at 0x84, not in struct) */
00281 } s12ad_reserved_sizes_t;

```

5.1.4 Function Documentation

5.1.4.1 `s12ad0()`

```

volatile rx_s12ad_regs_t * s12ad0 (
    void ) [inline], [static]

```

Get pointer to S12AD0 (Unit 0) registers.

Returns a volatile pointer to the S12AD Unit 0 register structure, enabling direct hardware register access for ADC operations. Unit 0 is optimized for motor current sensing with 8 high-speed channels.

Returns

Volatile pointer to S12AD0 register structure at 0x00089000

Precondition

Module stop state must be released (`MSTPCRA.MSTPA17 = 0`)
 AVCC0 and AVSS0 power supplies must be stable

Postcondition

Pointer is immediately usable for register access
 No hardware state is modified by this function

Note

Thread Safety: Function is reentrant and thread-safe (returns constant address)

Performance: Zero overhead - compiles to single load instruction

Example - Motor Current Sensing (144-pin LFQFP):

```
// Read all 4 motor currents (AN004-AN007 on P44-P47)
volatile rx_s12ad_regs_t* adc = s12ad0();

// Configure for 4-channel scan
adc->adansa0 = 0x00F0; // AN004-AN007
adc->adcsr = 0x0000; // Single scan, software trigger
adc->adcer = 0x0000; // Right-aligned data

// Perform conversion
adc->adcsr |= 0x8000; // Start
while (adc->adcsr & 0x8000) {} // Wait (~4 us for 4 channels)

// Read currents (M0=AN007, M1=AN006, M2=AN005, M3=AN004)
uint16_t motor_currents[4] = {
    adc->addr7, adc->addr6, adc->addr5, adc->addr4
};
```

See also

[s12ad1\(\)](#) For temperature sensing and additional channels

[rx_s12ad_regs_t](#) Register structure definition

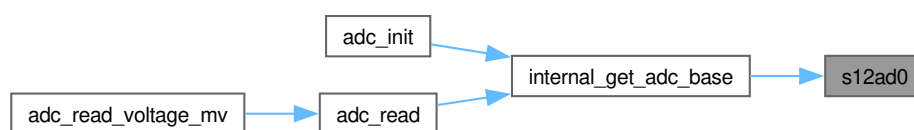
Definition at line 622 of file [rx72n_adc_regs.h](#).

```
00623 {
00624     return (volatile rx_s12ad_regs_t*)k_s12ad0_base_addr;
00625 }
```

References [k_s12ad0_base_addr](#).

Referenced by [internal_get_adc_base\(\)](#).

Here is the caller graph for this function:



5.1.4.2 s12ad1()

```
volatile rx_s12ad_regs_t * s12ad1 (
    void ) [inline], [static]
```

Get pointer to S12AD1 (Unit 1) registers.

Returns a volatile pointer to the S12AD Unit 1 register structure. Unit 1 provides 21 analog channels plus temperature sensor and internal voltage reference for calibration and monitoring.

Returns

Volatile pointer to S12AD1 register structure at 0x00089100

Precondition

Module stop state must be released (MSTPCRA.MSTPA16 = 0)
 AVCC0 and AVSS0 power supplies must be stable

Postcondition

Pointer is immediately usable for register access
 No hardware state is modified by this function

Note

Thread Safety: Function is reentrant and thread-safe
 Performance: Zero overhead - compiles to single load instruction

Warning

The [rx_s12ad_regs_t](#) structure only covers registers up to ADDR7. S12AD1 has additional registers (ADDR8-20, temperature sensor, internal reference) at higher offsets that require direct access.

Example - Temperature Sensor Reading:

```
volatile rx_s12ad_regs_t* adc = s12ad1();

// Enable internal temperature sensor (AN127)
// Note: Requires accessing registers beyond rx_s12ad_regs_t
// See RX72N Manual Ch56 for temperature sensor configuration
```

See also

[s12ad0\(\)](#) For motor current sensing (Unit 0)
[rx_s12ad_regs_t](#) Register structure definition

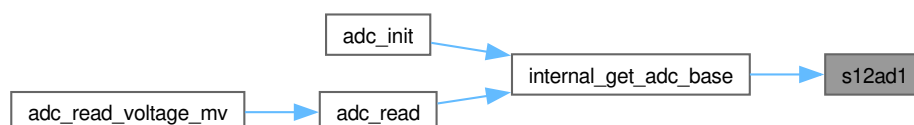
Definition at line 662 of file [rx72n_adc_regs.h](#).

```
00663 {
00664     return (volatile rx_s12ad_regs_t*)k_s12ad1_base_addr;
00665 }
```

References [k_s12ad1_base_addr](#).

Referenced by [internal_get_adc_base\(\)](#).

Here is the caller graph for this function:



5.2 CMT Register Definitions

Compare Match Timer (CMT) hardware register interface.

Collaboration diagram for CMT Register Definitions:



Topics

- [CMT Static Assertions](#)
Compile-time verification of CMT register layout.

Data Structures

- struct [rx_cmt_channel_regs_t](#)
CMT Channel Register Map (6 bytes per channel).
- struct [rx_cmt_control_regs_t](#)
CMT Start/Stop Control Register Map.

Enumerations

- enum [rx_cmt_addresses_t](#) : `uintptr_t` {
`k_cmt_ctrl_base_addr = 0x00088000` , `k_cmt1_ctrl_base_addr = 0x00088010` , `k_cmt0_base_addr = 0x00088002` , `k_cmt1_base_addr = 0x00088008` ,
`k_cmt2_base_addr = 0x00088012` , `k_cmt3_base_addr = 0x00088018` }
CMT base addresses (verified against RX72N Hardware Manual Ch31).
- enum [rx_cmt_layout_t](#) : `uint8_t` {
`k_cmt_channel_regs_size_bytes = 6` , `k_cmt_control_regs_size_bytes = 4` , `k_cmt_cmcr_offset = 0x00` , `k_cmt_cmcr_offset = 0x02` ,
`k_cmt_cmcr_offset = 0x04` , `k_cmt_cmstr_reg_size_bytes = 2` }
CMT register layout constants used for compile-time verification.
- enum [rx_cmt_cmstr0_bits_t](#) : `uint8_t` { `k_rx72n_cmstr0_cmt0_enable = 0x01` , `k_rx72n_cmstr0_cmt1_enable = 0x02` }
CMSTR0 register bit definitions (CMT Unit 0 start control).
- enum [rx_cmt_cmstr1_bits_t](#) : `uint8_t` { `k_rx72n_cmstr1_cmt2_enable = 0x01` , `k_rx72n_cmstr1_cmt3_enable = 0x02` }
CMSTR1 register bit definitions (CMT Unit 1 start control).

Functions

- static volatile [rx_cmt_channel_regs_t](#) * `cmt0` (void)
Get pointer to CMT0 registers.
- static volatile [rx_cmt_channel_regs_t](#) * `cmt1` (void)
Get pointer to CMT1 registers.
- static volatile [rx_cmt_channel_regs_t](#) * `cmt2` (void)
Get pointer to CMT2 registers.
- static volatile [rx_cmt_channel_regs_t](#) * `cmt3` (void)
Get pointer to CMT3 registers.
- static volatile [rx_cmt_control_regs_t](#) * `cmt_ctrl` (void)
Get pointer to CMT control registers (CMSTR0).
- static volatile [rx_cmt_control_regs_t](#) * `cmt_ctrl1` (void)
Get pointer to CMT Unit 1 control register (CMSTR1).

5.2.1 Detailed Description

Compare Match Timer (CMT) hardware register interface.

5.2.2 Data Structure Documentation

5.2.2.1 struct rx_cmt_channel_regs_t

CMT Channel Register Map (6 bytes per channel).

Compare Match Timer (CMT) channel registers for periodic interrupt generation. Each channel is a 16-bit up-counter that generates an interrupt when CMCNT matches CMCOR, then resets to 0 and continues counting.

Register Layout (6 bytes)

* Offset	* Size	* Register	* Description
* -----+-----+-----+-----			
* 0x00	* 2	* CMCR	* Control: clock div, IRQ enable
* 0x02	* 2	* CMCNT	* Counter: current count value
* 0x04	* 2	* CMCOR	* Compare: match value (period)
*			

CMCR Register Bit Layout (RX72N Manual Ch31, p.1583)

* Bit	* Name	* Description
* -----+-----+-----		
* 15-8	* -	* Reserved
* 7	* -	* Reserved (write value should be 1)
* 6	* CMIE	* Compare Match Interrupt Enable
* 5-2	* -	* Reserved
* 1-0	* CKS	* Clock Select (00=PCLK/8, 01=/32, 10=/128, 11=/512)
*		

Timer Operation

1. Configure CMCR with clock divider and interrupt enable
2. Set CMCOR to desired period value
3. Clear CMCNT to 0
4. Enable timer via CMSTRn register
5. Counter increments at PCLKB/DIV rate
6. When CMCNT == CMCOR, interrupt fires and CMCNT resets to 0

Invariant

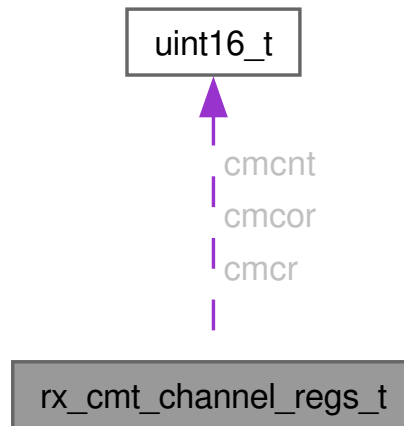
CMCNT ≤ CMCOR during normal operation

See also

[rx_cmt_control_regs_t](#) For start/stop control (CMSTR0/1)

Definition at line 260 of file [rx72n_cmt_regs.h](#).

Collaboration diagram for [rx_cmt_channel_regs_t](#):



Data Fields

volatile uint16_t	cmcnt	Compare Match Timer Counter Register. 16-bit up-counter incremented by selected clock. Cleared to 0 on compare match and when timer is stopped. - Read: Current counter value - Write: Set counter value (typically write 0 before starting)																
volatile uint16_t	cmcor	Compare Match Timer Constant Register. Compare match value. When CMCNT equals this value, an interrupt is generated (if CMIE=1) and CMCNT resets to 0. Period calculation: $T = \frac{(CMCOR+1) \times DIV}{f_{PCLKB}}$ <table><tr><th>CMCOR Value</th><th>Divider</th><th>Period @ 60MHz PCLKB</th></tr><tr><td>18749</td><td>/32</td><td>10 ms (100 Hz)</td></tr><tr><td>1874</td><td>/32</td><td>1 ms (1000 Hz)</td></tr><tr><td>7499</td><td>$\frac{PCLKB}{8}$</td><td>1 ms (1000 Hz)</td></tr></table> Note: PCLKB/8 cannot reach 100 Hz (requires CMCOR=74999 > uint16_t max 65535).	CMCOR Value	Divider	Period @ 60MHz PCLKB	18749	/32	10 ms (100 Hz)	1874	/32	1 ms (1000 Hz)	7499	$\frac{PCLKB}{8}$	1 ms (1000 Hz)				
CMCOR Value	Divider	Period @ 60MHz PCLKB																
18749	/32	10 ms (100 Hz)																
1874	/32	1 ms (1000 Hz)																
7499	$\frac{PCLKB}{8}$	1 ms (1000 Hz)																
volatile uint16_t	cmcr	Compare Match Timer Control Register. Controls clock source divider and interrupt enable. <table><tr><th>Bits</th><th>Name</th><th>R/W</th><th>Description</th></tr><tr><td>7</td><td>-</td><td>R/W</td><td>Reserved; write value should be 1 (manual p.1583)</td></tr><tr><td>6</td><td>CMIE</td><td>R/W</td><td>1 = Enable compare match interrupt</td></tr><tr><td>1-0</td><td>CKS</td><td>R/W</td><td>Clock select: 00=/8, 01=/32, 10=/128, 11=/512</td></tr></table>	Bits	Name	R/W	Description	7	-	R/W	Reserved; write value should be 1 (manual p.1583)	6	CMIE	R/W	1 = Enable compare match interrupt	1-0	CKS	R/W	Clock select: 00=/8, 01=/32, 10=/128, 11=/512
Bits	Name	R/W	Description															
7	-	R/W	Reserved; write value should be 1 (manual p.1583)															
6	CMIE	R/W	1 = Enable compare match interrupt															
1-0	CKS	R/W	Clock select: 00=/8, 01=/32, 10=/128, 11=/512															

5.2.2.2 struct rx_cmt_control_regs_t

CMT Start/Stop Control Register Map.

Controls starting and stopping of CMT channels. Each unit has a separate start register controlling two channels.

Memory Layout

```
*  Address   | Register | Controls
*  -----+-----+-----
*  0x00088000 | CMSTR0   | CMT0, CMT1
*  0x00088010 | CMSTR1   | CMT2, CMT3
*
```

Note

CMSTR0 and CMSTR1 are NOT contiguous! CMSTR1 is at +0x10 from CMSTR0. This struct is only for documentation; use [cmt_ctrl\(\)](#) for CMSTR0.

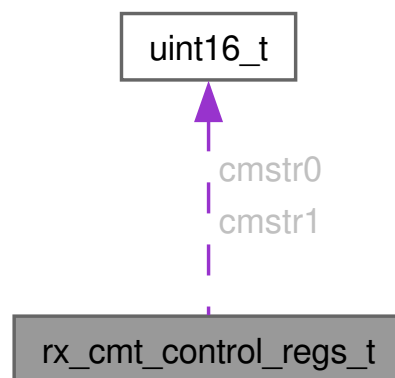
Usage Example

```
// Start CMT0
cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;

// Stop CMT0
cmt_ctrl()->cmstr0 &= ~k_rx72n_cmstr0_cmt0_enable;
```

Definition at line 335 of file [rx72n_cmt_regs.h](#).

Collaboration diagram for rx_cmt_control_regs_t:



Data Fields

volatile uint16_t	cmstr0	Compare Match Timer Start Register 0. Controls CMT0 and CMT1 operation. • Bit 0 (STR0): 1 = Start CMT0, 0 = Stop CMT0 • Bit 1 (STR1): 1 = Start CMT1, 0 = Stop CMT1
-------------------	--------	---

volatile uint16_t	cmstr1	Compare Match Timer Start Register 1. Controls CMT2 and CMT3 operation. • Bit 0 (STR2): 1 = Start CMT2, 0 = Stop CMT2 • Bit 1 (STR3): 1 = Start CMT3, 0 = Stop CMT3 Warning This field is NOT at the expected offset! CMSTR1 is at 0x00088010, not 0x00088002. Access via separate function.
-------------------	--------	--

5.2.3 Enumeration Type Documentation

5.2.3.1 rx'cmt'addresses't

enum `rx_cmt_addresses_t` : `uintptr_t`

CMT base addresses (verified against RX72N Hardware Manual Ch31).

The RX72N has 4 CMT channels organized into 2 units:

- Unit 0: CMT0 and CMT1 (controlled by CMSTR0)
- Unit 1: CMT2 and CMT3 (controlled by CMSTR1)

Memory Layout

* Address	Content
* -----+	
* 0x00088000	CMSTR0 (Unit 0 start)
* 0x00088002	CMT0 registers (6 bytes)
* 0x00088008	CMT1 registers (6 bytes)
* 0x00088010	CMSTR1 (Unit 1 start)
* 0x00088012	CMT2 registers (6 bytes)
* 0x00088018	CMT3 registers (6 bytes)
*	

Note

CMSTR registers are at base, channel registers follow at +2 offset.

Enumerator

k_cmt_ctrl_base_addr	CMSTR0 base: Unit 0 start register (CMT0/CMT1)
k_cmt1_ctrl_base_addr	CMSTR1 base: Unit 1 start register (CMT2/CMT3)
k_cmt0_base_addr	CMT0 registers (CMCR, CMCNT, CMCOR)
k_cmt1_base_addr	CMT1 registers (CMCR, CMCNT, CMCOR)
k_cmt2_base_addr	CMT2 registers (CMCR, CMCNT, CMCOR)
k_cmt3_base_addr	CMT3 registers (CMCR, CMCNT, CMCOR)

Definition at line 187 of file `rx72n_cmt_regs.h`.

```

00187         : uintptr_t {
00188     k_cmt_ctrl_base_addr = 0x00088000, /**< CMSTR0 base: Unit 0 start register (CMT0/CMT1) */
00189     k_cmt1_ctrl_base_addr = 0x00088010, /**< CMSTR1 base: Unit 1 start register (CMT2/CMT3) */
00190     k_cmt0_base_addr      = 0x00088002, /**< CMT0 registers (CMCR, CMCNT, CMCOR) */
00191     k_cmt1_base_addr      = 0x00088008, /**< CMT1 registers (CMCR, CMCNT, CMCOR) */
00192     k_cmt2_base_addr      = 0x00088012, /**< CMT2 registers (CMCR, CMCNT, CMCOR) */
00193     k_cmt3_base_addr      = 0x00088018, /**< CMT3 registers (CMCR, CMCNT, CMCOR) */
00194 } rx_cmt_addresses_t;

```

5.2.3.2 rx_cmt_cmstr0_bits_t

```
enum rx_cmt_cmstr0_bits_t : uint8_t
```

CMSTR0 register bit definitions (CMT Unit 0 start control).

Bit field definitions for Compare Match Timer Start Register 0. Controls starting and stopping of CMT0 and CMT1 channels.

Register Layout (16-bit, only bits 0-1 used)

```
* Bit | Name | Description
* ----+-----+-----
* 0 | STR0 | 1 = Start CMT0, 0 = Stop CMT0
* 1 | STR1 | 1 = Start CMT1, 0 = Stop CMT1
* 2-15| - | Reserved (write 0)
*
```

Enumerator

k_rx72n_cmstr0_cmt0_enable	Bit 0: Start/stop CMT0 counter
k_rx72n_cmstr0_cmt1_enable	Bit 1: Start/stop CMT1 counter

Definition at line 373 of file [rx72n_cmt_regs.h](#).

```
00373 : uint8_t {
00374 k_rx72n_cmstr0_cmt0_enable = 0x01, /**< Bit 0: Start/stop CMT0 counter */
00375 k_rx72n_cmstr0_cmt1_enable = 0x02, /**< Bit 1: Start/stop CMT1 counter */
00376 } rx_cmt_cmstr0_bits_t;
```

5.2.3.3 rx_cmt_cmstr1_bits_t

```
enum rx_cmt_cmstr1_bits_t : uint8_t
```

CMSTR1 register bit definitions (CMT Unit 1 start control).

Bit field definitions for Compare Match Timer Start Register 1. Controls starting and stopping of CMT2 and CMT3 channels.

Note

CMSTR1 is at address 0x00088010, which is 16 bytes after CMSTR0.

Register Layout (16-bit, only bits 0-1 used)

```
* Bit | Name | Description
* ----+-----+-----
* 0 | STR2 | 1 = Start CMT2, 0 = Stop CMT2
* 1 | STR3 | 1 = Start CMT3, 0 = Stop CMT3
* 2-15| - | Reserved (write 0)
*
```

Enumerator

k_rx72n_cmstr1_cmt2_enable	Bit 0: Start/stop CMT2 counter
k_rx72n_cmstr1_cmt3_enable	Bit 1: Start/stop CMT3 counter

Definition at line 397 of file [rx72n_cmt_regs.h](#).

```
00397      : uint8_t {
00398  k_rx72n_cmstr1_cmt2_enable = 0x01, /**< Bit 0: Start/stop CMT2 counter */
00399  k_rx72n_cmstr1_cmt3_enable = 0x02, /**< Bit 1: Start/stop CMT3 counter */
00400 } rx_cmt_cmstr1_bits_t;
```

5.2.3.4 rx_cmt_layout_t

```
enum rx_cmt_layout_t : uint8_t
```

CMT register layout constants used for compile-time verification.

Physical byte sizes and register offsets for CMT structures. Used exclusively in static_assert expressions so no magic numbers appear in code.

See also

[rx_cmt_channel_regs_t](#)

[rx_cmt_control_regs_t](#)

Since

Version 1.0.0

Enumerator

k_cmt_channel_regs_size_bytes	Channel struct: 3 x 16-bit registers
k_cmt_control_regs_size_bytes	Control struct: 2 x 16-bit registers
k_cmt_cmcr_offset	CMCR byte offset from channel base
k_cmt_cmcnt_offset	CMCNT byte offset from channel base
k_cmt_cmcpr_offset	CMCOR byte offset from channel base
k_cmt_cmstr_reg_size_bytes	CMSTR register width (one 16-bit word)

Definition at line 210 of file [rx72n_cmt_regs.h](#).

```
00210      : uint8_t {
00211  k_cmt_channel_regs_size_bytes = 6, /**< Channel struct: 3 x 16-bit registers */
00212  k_cmt_control_regs_size_bytes = 4, /**< Control struct: 2 x 16-bit registers */
00213  k_cmt_cmcr_offset             = 0x00, /**< CMCR byte offset from channel base */
00214  k_cmt_cmcnt_offset            = 0x02, /**< CMCNT byte offset from channel base */
00215  k_cmt_cmcpr_offset            = 0x04, /**< CMCOR byte offset from channel base */
00216  k_cmt_cmstr_reg_size_bytes    = 2,  /**< CMSTR register width (one 16-bit word) */
00217 } rx_cmt_layout_t;
```

5.2.4 Function Documentation

5.2.4.1 cmt0()

```
volatile rx_cmt_channel_regs_t * cmt0 (
    void )    [inline], [static]
```

Get pointer to CMT0 registers.

Returns a volatile pointer to CMT channel 0 registers at address 0x00088002. CMT0 is typically used for the ThreadX system tick (100 Hz).

Returns

Volatile pointer to CMT0 register structure

Example

```
// Configure CMT0 for 100 Hz interrupt with PCLKB/32 divider
cmt0()->cmcr = 0x00C1; // CKS=01 (PCLKB/32), CMIE=1, bit7=1(rsvd)
cmt0()->cmcor = 18749; // 60MHz/32/100Hz - 1
cmt0()->cmcnt = 0;    // Clear counter
```

See also

[cmt_ctrl\(\)](#) For start/stop control

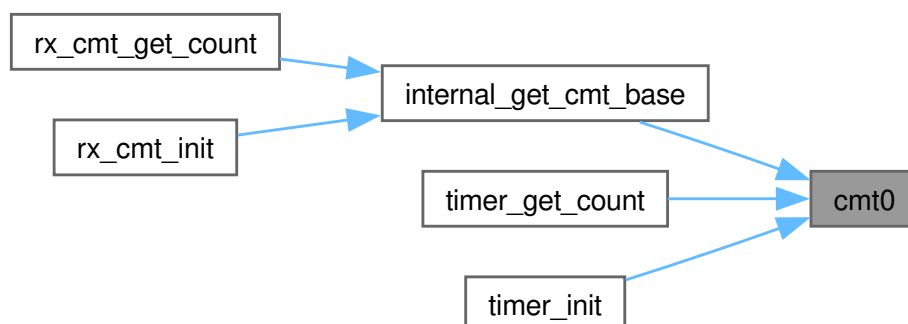
Definition at line 431 of file [rx72n_cmt_regs.h](#).

```
00432 {
00433     return (volatile rx_cmt_channel_regs_t*)k_cmt0_base_addr;
00434 }
```

References [k_cmt0_base_addr](#).

Referenced by [internal_get_cmt_base\(\)](#), [timer_get_count\(\)](#), and [timer_init\(\)](#).

Here is the caller graph for this function:



5.2.4.2 `cmt1()`

```
volatile rx_cmt_channel_regs_t * cmt1 (
    void ) [inline], [static]
```

Get pointer to CMT1 registers.

Returns a volatile pointer to CMT channel 1 registers at address 0x00088008. CMT1 is available for general-purpose timing.

Returns

Volatile pointer to CMT1 register structure

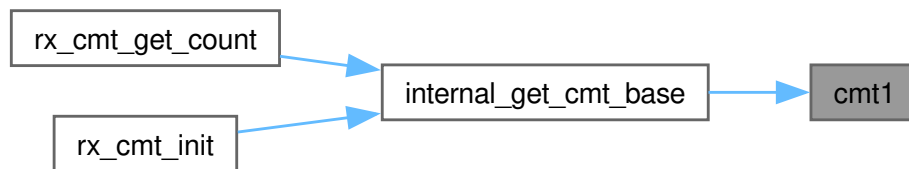
Definition at line 445 of file `rx72n_cmt_regs.h`.

```
00446 {
00447     return (volatile rx_cmt_channel_regs_t*)k_cmt1_base_addr;
00448 }
```

References `k_cmt1_base_addr`.

Referenced by `internal_get_cmt_base()`.

Here is the caller graph for this function:

5.2.4.3 `cmt2()`

```
volatile rx_cmt_channel_regs_t * cmt2 (
    void ) [inline], [static]
```

Get pointer to CMT2 registers.

Returns a volatile pointer to CMT channel 2 registers at address 0x00088012. CMT2 is part of Unit 1 (controlled by CMSTR1).

Returns

Volatile pointer to CMT2 register structure

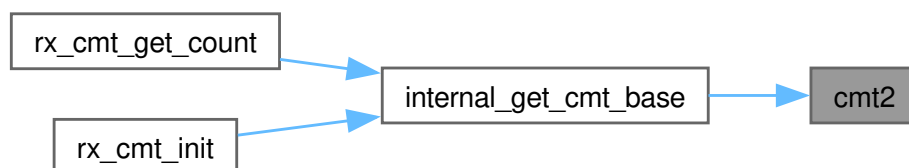
Definition at line 459 of file `rx72n_cmt_regs.h`.

```
00460 {
00461     return (volatile rx_cmt_channel_regs_t*)k_cmt2_base_addr;
00462 }
```

References `k_cmt2_base_addr`.

Referenced by `internal_get_cmt_base()`.

Here is the caller graph for this function:



5.2.4.4 cmt3()

```
volatile rx_cmt_channel_regs_t * cmt3 (
    void ) [inline], [static]
```

Get pointer to CMT3 registers.

Returns a volatile pointer to CMT channel 3 registers at address 0x00088018. CMT3 is part of Unit 1 (controlled by CMSTR1).

Returns

Volatile pointer to CMT3 register structure

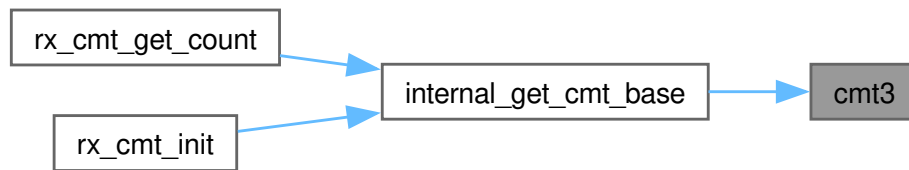
Definition at line 473 of file rx72n_cmt_regs.h.

```
00474 {
00475     return (volatile rx_cmt_channel_regs_t*)k_cmt3_base_addr;
00476 }
```

References [k_cmt3_base_addr](#).

Referenced by [internal_get_cmt_base\(\)](#).

Here is the caller graph for this function:



5.2.4.5 cmt_ctrl()

```
volatile rx_cmt_control_regs_t * cmt_ctrl (
    void ) [inline], [static]
```

Get pointer to CMT control registers (CMSTR0).

Returns a volatile pointer to CMT control register structure at address 0x00088000. Provides access to CMSTR0 for starting/stopping CMT0 and CMT1.

Warning

CMSTR1 (for CMT2/3) is at 0x00088010, not adjacent to CMSTR0. The [rx_cmt_control_regs_t](#) struct has incorrect cmstr1 offset. For CMSTR1 access, use: `*(volatile uint16_t*)0x00088010`

Returns

Volatile pointer to CMT control register structure

Example

```
// Start CMT0
cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;

// Stop CMT0 and CMT1
cmt_ctrl()->cmstr0 = 0;
```

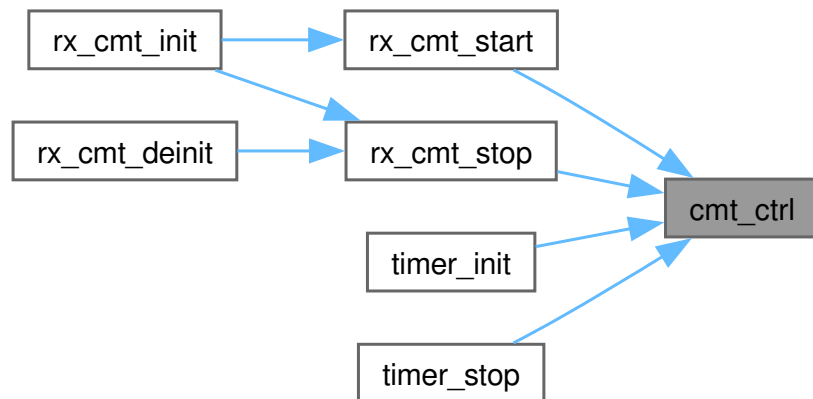
Definition at line 500 of file [rx72n_cmt_regs.h](#).

```
00501 {
00502     return (volatile rx_cmt_control_regs_t*)k_cmt_ctrl_base_addr;
00503 }
```

References [k_cmt_ctrl_base_addr](#).

Referenced by [rx_cmt_start\(\)](#), [rx_cmt_stop\(\)](#), [timer_init\(\)](#), and [timer_stop\(\)](#).

Here is the caller graph for this function:

5.2.4.6 `cmt_ctrl()`

```
volatile rx_cmt_control_regs_t * cmt_ctrl (
    void ) [inline], [static]
```

Get pointer to CMT Unit 1 control register (CMSTR1).

Returns a volatile pointer to CMSTR1 at address 0x00088010. CMSTR1 is NOT contiguous with CMSTR0 – it is 0x10 bytes after CMSTR0. This accessor provides correct access to CMSTR1 for starting/↵ stopping CMT2 and CMT3.

Warning

Do NOT use `cmt_ctrl()->cmstr1` to access CMSTR1 – the struct field `cmstr1` sits at offset +2 from CMSTR0 (address 0x00088002), which is CMT0.CMCR, not CMSTR1.

Returns

Volatile pointer to CMT Unit 1 control register structure

Example

```
// Start CMT2
cmt_ctrl1()->cmstr0 |= k_rx72n_cmstr1_cmt2_enable;

// Stop CMT3
cmt_ctrl1()->cmstr0 &= ~k_rx72n_cmstr1_cmt3_enable;
```

See also

[cmt_ctrl\(\)](#) For CMSTR0 (CMT0/CMT1 control)

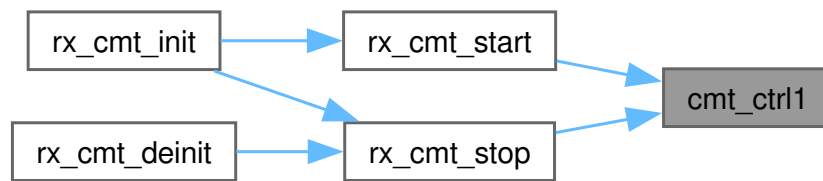
Definition at line 530 of file [rx72n_cmt_regs.h](#).

```
00531 {
00532     return (volatile rx_cmt_control_regs_t*)k_cmt1_ctrl_base_addr;
00533 }
```

References [k_cmt1_ctrl_base_addr](#).

Referenced by [rx_cmt_start\(\)](#), and [rx_cmt_stop\(\)](#).

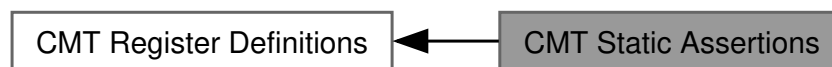
Here is the caller graph for this function:



5.2.5 CMT Static Assertions

Compile-time verification of CMT register layout.

Collaboration diagram for CMT Static Assertions:



Compile-time verification of CMT register layout.

These static assertions are verified at compile time to ensure the C structure definitions match the actual hardware register layout. A failed assertion indicates a bug in the register definitions that must be fixed before the code can compile.

5.3 RX72N GPTW Register Definitions

General PWM Timer hardware register interface.

Data Structures

- struct `rx_gptw_channel_regs_t`
GPTW Channel Register Map for 32-bit PWM generation.
- struct `rx_gptw_common_regs_t`
GPTW Common Register Map.

Enumerations

- enum `gptw_padding_sizes_t` : `uint8_t` { `k_gptw_ch_reserved_count` = 6U }
- enum `gptw_addresses_t` : `uintptr_t` {
`k_gptw_channel_offset` = 0x100 , `k_gptw0_base_addr` = 0x000C2000 , `k_gptw1_base_addr` = 0x000C2100 , `k_gptw2_base_addr` = 0x000C2200 ,
`k_gptw3_base_addr` = 0x000C2300 , `k_gptw_common_base_addr` }
GPTW hardware addresses (verified against RX72N Hardware Manual).
- enum `gptw_gtwp_bits_t` : `uint16_t` {
`k_gptw_gtwp_wp0` = (1 << 0) , `k_gptw_gtwp_wp1` = (1 << 1) , `k_gptw_gtwp_wp2` = (1 << 2) , `k_gptw_gtwp_wp3` = (1 << 3) ,
`k_gptw_gtwp_unlock` = 0xA500 , `k_gptw_gtwp_lock` = 0xA501 }
- enum `gptw_gtcr_bits_t` : `uint32_t` {
`k_gptw_gtcr_cst` = (1 << 0) , `k_gptw_gtcr_icds` = (1 << 8) , `k_gptw_gtcr_md_shift` = 16 ,
`k_gptw_gtcr_md_mask` = (0x07 << `k_gptw_gtcr_md_shift`) ,
`k_gptw_gtcr_md_saw_pwm` = (0x00 << `k_gptw_gtcr_md_shift`) , `k_gptw_gtcr_md_saw_oneshot` = (0x01 << `k_gptw_gtcr_md_shift`) , `k_gptw_gtcr_md_tri_pwm1` = (0x04 << `k_gptw_gtcr_md_shift`) ,
`k_gptw_gtcr_md_tri_pwm2` = (0x05 << `k_gptw_gtcr_md_shift`) ,
`k_gptw_gtcr_md_tri_pwm3` = (0x06 << `k_gptw_gtcr_md_shift`) , `k_gptw_gtcr_tpcs_shift` = 23 , `k_gptw_gtcr_tpcs_mask` = (0x0F << `k_gptw_gtcr_tpcs_shift`) , `k_gptw_gtcr_tpcs_1` = (0x00 << `k_gptw_gtcr_tpcs_shift`) ,
`k_gptw_gtcr_tpcs_2` = (0x01 << `k_gptw_gtcr_tpcs_shift`) , `k_gptw_gtcr_tpcs_4` = (0x02 << `k_gptw_gtcr_tpcs_shift`) , `k_gptw_gtcr_tpcs_8` = (0x03 << `k_gptw_gtcr_tpcs_shift`) ,
`k_gptw_gtcr_tpcs_16` = (0x04 << `k_gptw_gtcr_tpcs_shift`) ,
`k_gptw_gtcr_tpcs_32` = (0x05 << `k_gptw_gtcr_tpcs_shift`) , `k_gptw_gtcr_tpcs_64` = (0x06 << `k_gptw_gtcr_tpcs_shift`) , `k_gptw_gtcr_tpcs_256` = (0x08 << `k_gptw_gtcr_tpcs_shift`) ,
`k_gptw_gtcr_tpcs_1024` = (0x0A << `k_gptw_gtcr_tpcs_shift`) }
- enum `gptw_gtior_bits_t` : `uint32_t` {
`k_gptw_gtior_oa_shift` = 0 , `k_gptw_gtior_oa_mask` = 0x1F , `k_gptw_gtior_oa_disabled` = 0x00 , `k_gptw_gtior_oa_low_cmpup` = 0x01 ,
`k_gptw_gtior_oa_high_cmpup` = 0x02 , `k_gptw_gtior_oa_toggle` = 0x03 , `k_gptw_gtior_oa_init_low` ,
`k_gptw_gtior_oa_init_high` ,
`k_gptw_gtior_oae` = (1 << 8) , `k_gptw_gtior_ob_shift` = 16 , `k_gptw_gtior_ob_mask` = (0x1F << 16) , `k_gptw_gtior_ob_disabled` = (0x00 << 16) ,
`k_gptw_gtior_ob_low_cmpup` = (0x01 << 16) , `k_gptw_gtior_ob_high_cmpup` = (0x02 << 16) , `k_gptw_gtior_ob_toggle` = (0x03 << 16) , `k_gptw_gtior_ob_init_low` ,
`k_gptw_gtior_ob_init_high` , `k_gptw_gtior_obe` = (1 << 24) }
- enum `gptw_gtintad_bits_t` : `uint32_t` {
`k_gptw_gtintad_grp_shift` = 24 , `k_gptw_gtintad_grp_mask` = (0x03U << 24) , `k_gptw_gtintad_grp_a` = (0x00U << 24) , `k_gptw_gtintad_grp_b` = (0x01U << 24) ,
`k_gptw_gtintad_grp_c` = (0x02U << 24) , `k_gptw_gtintad_grp_d` = (0x03U << 24) ,
`k_gptw_gtintad_grpdte` = (1U << 28) , `k_gptw_gtintad_grpabh` = (1U << 29) ,
`k_gptw_gtintad_grpabl` = (1U << 30) }
GTINTAD register bit definitions for POEG group linkage.
- enum `gptw_gtber_bits_t` : `uint32_t` {
`k_gptw_gtber_bd0` = (1U << 0) , `k_gptw_gtber_bd1` = (1U << 1) , `k_gptw_gtber_bd2` = (1U << 2) , `k_gptw_gtber_bd3` = (1U << 3) ,
`k_gptw_gtber_ccra_single` = (0x01U << 16) , `k_gptw_gtber_ccra_double` = (0x02U << 16) ,
`k_gptw_gtber_ccrb_single` = (0x01U << 18) , `k_gptw_gtber_ccrb_double` = (0x02U << 18) ,
`k_gptw_gtber_pr_single` = (0x01U << 20) , `k_gptw_gtber_pr_double` = (0x02U << 20) }

- enum `gptw_gtdtcr_bits_t` : `uint8_t` { `k_gptw_gtdtcr_tde` = (1 << 0) , `k_gptw_gtdtcr_tdfc` = (1 << 4) }
- enum `gptw_gtst_bits_t` : `uint16_t` {
`k_gptw_gtst_tcfa` = (1 << 0) , `k_gptw_gtst_tcfb` = (1 << 1) , `k_gptw_gtst_tcfc` = (1 << 2) ,
`k_gptw_gtst_tcfb` = (1 << 3) ,
`k_gptw_gtst_tcfb` = (1 << 4) , `k_gptw_gtst_tcff` = (1 << 5) , `k_gptw_gtst_tcfpo` = (1 << 6) ,
`k_gptw_gtst_tcfpu` = (1 << 7) ,
`k_gptw_gtst_tucf` = (1 << 15) }
- enum `gptw_gtstr_bits_t` : `uint8_t` { `k_gptw_gtstr_cst0` = (1 << 0) , `k_gptw_gtstr_cst1` = (1 << 1) , `k_gptw_gtstr_cst2` = (1 << 2) , `k_gptw_gtstr_cst3` = (1 << 3) }
- enum `gptw_gtuddtyc_bits_t` : `uint32_t` { `k_gptw_gtuddtyc_ud` = (1 << 0) , `k_gptw_gtuddtyc_udf` = (1 << 1) }

Up/Down Count Duty Setting Register (GTUDDTYC) Bits.

- enum `gptw_gtadsmr_bits_t` : `uint32_t` { `k_gptw_gtadsmr_adsm0` = (1 << 0) , `k_gptw_gtadsmr_adsm1` = (1 << 1) }
- enum `gptw_gteitc_bits_t` : `uint32_t` {
`k_gptw_gteitc_eivtc_mask` = 0x0000FFFF , `k_gptw_gteitc_eivtt_shift` = 16 , `k_gptw_gteitc_eivtt_mask` = (0x03 << 16) , `k_gptw_gteitc_eivtt_none` = (0x00 << 16) ,
`k_gptw_gteitc_eivtt_trough` = (0x01 << 16) , `k_gptw_gteitc_eivtt_crest` = (0x02 << 16) ,
`k_gptw_gteitc_eivtt_both` = (0x03 << 16) , `k_gptw_gteitc_eivten_shift` = 24 ,
`k_gptw_gteitc_eivten_mask` = (0xFFU << 24) }
- enum `gptw_gteitli1_bits_t` : `uint32_t` { `k_gptw_gteitli1_eadt1` = (1 << 0) , `k_gptw_gteitli1_eivtciv` = (1 << 8) , `k_gptw_gteitli1_eivtcu` = (1 << 9) }
- enum `gptw_gteitli2_bits_t` : `uint32_t` {
`k_gptw_gteitli2_eivtciv_shift` = 0 , `k_gptw_gteitli2_eivtciv_mask` = 0x000000FF , `k_gptw_gteitli2_eivtcia` = (1 << 0) , `k_gptw_gteitli2_eivtcib` = (1 << 1) ,
`k_gptw_gteitli2_eivtcic` = (1 << 2) , `k_gptw_gteitli2_eivtcid` = (1 << 3) , `k_gptw_gteitli2_eivtcie` = (1 << 4) , `k_gptw_gteitli2_eivtcif` = (1 << 5) }
- enum `gptw_gteitlb_bits_t` : `uint32_t` { `k_gptw_gteitlb_eivtcpbl` = (1 << 0) , `k_gptw_gteitlb_eivtcabl` = (1 << 1) , `k_gptw_gteitlb_eivtcibl` = (1 << 2) }
- enum `gptw_gtsecsr_bits_t` : `uint32_t` { `k_gptw_gtsecsr_secsl0` = (1 << 0) , `k_gptw_gtsecsr_secsl1` = (1 << 1) , `k_gptw_gtsecsr_secsl2` = (1 << 2) , `k_gptw_gtsecsr_secsl3` = (1 << 3) }
- enum `gptw_gtsecl_bits_t` : `uint32_t` { `k_gptw_gtsecl_sbdce` = (1 << 0) , `k_gptw_gtsecl_sbdcd` = (1 << 8) , `k_gptw_gtsecl_sbdpe` = (1 << 9) }
- enum `gptw_module_stop_bits_t` : `uint8_t` { `k_mstpa_gptw` = 7 }
- enum `gptw_mpc_psel_t` : `uint8_t` { `k_pfs_psel_gptw` = 0x1E }
- enum `gptw_ch_reg_offsets_t` : `uint8_t` {
`k_gptw_ch_off_gtwp` = 0x00U , `k_gptw_ch_off_gtstr` = 0x04U , `k_gptw_ch_off_gtstp` = 0x08U , `k_gptw_ch_off_gtclr` = 0x0CU ,
`k_gptw_ch_off_gtssr` = 0x10U , `k_gptw_ch_off_gtpsr` = 0x14U , `k_gptw_ch_off_gtcsr` = 0x18U , `k_gptw_ch_off_gtupsr` = 0x1CU ,
`k_gptw_ch_off_gtdnsr` = 0x20U , `k_gptw_ch_off_gticasr` = 0x24U , `k_gptw_ch_off_gticbsr` = 0x28U , `k_gptw_ch_off_gtcr` = 0x2CU ,
`k_gptw_ch_off_gtuddtyc` = 0x30U , `k_gptw_ch_off_gtior` = 0x34U , `k_gptw_ch_off_gtintad` = 0x38U , `k_gptw_ch_off_gtst` = 0x3CU ,
`k_gptw_ch_off_gtber` = 0x40U , `k_gptw_ch_off_gtite` = 0x44U , `k_gptw_ch_off_gtent` = 0x48U , `k_gptw_ch_off_gtcera` = 0x4CU ,
`k_gptw_ch_off_gtcerb` = 0x50U , `k_gptw_ch_off_gtcerc` = 0x54U , `k_gptw_ch_off_gtcere` = 0x58U , `k_gptw_ch_off_gtcerd` = 0x5CU ,
`k_gptw_ch_off_gtcerf` = 0x60U , `k_gptw_ch_off_gtpr` = 0x64U , `k_gptw_ch_off_gtpbr` = 0x68U , `k_gptw_ch_off_gtpdbr` = 0x6CU ,
`k_gptw_ch_off_gtadtra` = 0x70U , `k_gptw_ch_off_gtadtbra` = 0x74U , `k_gptw_ch_off_gtadtbra` , `k_gptw_ch_off_gtadtrb` = 0x7CU ,
`k_gptw_ch_off_gtadtbrb` = 0x80U , `k_gptw_ch_off_gtadtbrb` , `k_gptw_ch_off_gtdtcr` = 0x88U , `k_gptw_ch_off_gtdvu` = 0x8CU ,
`k_gptw_ch_off_gtdvd` = 0x90U , `k_gptw_ch_off_gtdbu` = 0x94U , `k_gptw_ch_off_gtdbd` = 0x98U , `k_gptw_ch_off_gtsos` = 0x9CU ,

- ```

k_gptw_ch_off_gtsotr = 0xA0U , k_gptw_ch_off_gtadsmr = 0xA4U , k_gptw_ch_off_gteitc
= 0xA8U , k_gptw_ch_off_gteitli1 = 0xACU ,
k_gptw_ch_off_gteitli2 = 0xB0U , k_gptw_ch_off_gteitlb = 0xB4U , k_gptw_ch_off_reserved
= 0xB8U , k_gptw_ch_off_gtsecsr = 0xD0U ,
k_gptw_ch_off_gtsecr = 0xD4U , k_gptw_ch_struct_size = 0xD8U }

```
- enum `gptw_common_reg_offsets_t` : `uint8_t` { `k_gptw_common_off_gtstpa` = 0x04U , `k_gptw_common_off_gtclra` = 0x08U , `k_gptw_common_struct_size` = 0x0CU }
  - enum `gptw_periph_space_t` : `uintptr_t` { `k_gptw_periph_space_mask` = 0xFFFF0000U , `k_gptw_periph_space_base` = 0x000C0000U }

## Functions

- static volatile `rx_gptw_channel_regs_t` \* `gptw0` (void)  
GPTW register inline accessor functions.
- static volatile `rx_gptw_channel_regs_t` \* `gptw1` (void)
- static volatile `rx_gptw_channel_regs_t` \* `gptw2` (void)
- static volatile `rx_gptw_channel_regs_t` \* `gptw3` (void)
- static volatile `rx_gptw_common_regs_t` \* `gptw_common` (void)  
GPTW common register accessor.

### 5.3.1 Detailed Description

General PWM Timer hardware register interface.

Provides memory-mapped register access to the RX72N GPTW peripheral for 32-bit PWM generation in motor control applications.

Quick Start Example - Configure 20 kHz PWM:

```

// Get GPTW channel 0 for motor 0
volatile rx_gptw_channel_regs_t* gptw = gptw0();

// 1. Unlock write protection
gptw->gtwp = k_gptw_gtwp_unlock;

// 2. Stop counter during configuration
gptw->gter &= ~k_gptw_gter_cst;

// 3. Set mode: Sawtooth PWM, no prescaler (PCLKA/1 = 120 MHz)
gptw->gter = k_gptw_gter_md_saw_pwm | k_gptw_gter_tpcs_1;

// 4. Set period for 20 kHz: 120 MHz / 20 kHz - 1 = 5999
gptw->gtpr = 5999;

// 5. Set initial duty to 0%
gptw->gtccra = 0;

// 6. Configure GTIOCA output: initial low, high at end-of-cycle, low at compare match
gptw->gtior = k_gptw_gtior_oa_init_low | k_gptw_gtior_oae;

// 7. Clear counter
gptw->gtcnt = 0;

// 8. Start counting
gptw->gter |= k_gptw_gter_cst;

// 9. Lock write protection
gptw->gtwp = k_gptw_gtwp_lock;

// Update duty cycle to 50%
gptw->gtwp = k_gptw_gtwp_unlock;
gptw->gtccra = 3000; // 50% of 5999
gptw->gtwp = k_gptw_gtwp_lock;

```

### 5.3.2 Data Structure Documentation

#### 5.3.2.1 struct rx'gptw'channel'regs't

GPTW Channel Register Map for 32-bit PWM generation.

Memory-mapped register structure for a single GPTW channel. Each of the 4 channels (GPTW0↔GPTW3) has an identical register layout at different base addresses spaced 0x100 bytes apart.

Channel Base Addresses:

| Channel | Base Address | Motor | Output Pins      |
|---------|--------------|-------|------------------|
| GPTW0   | 0x000C2000   | M0    | PE5 (A), PE2 (B) |
| GPTW1   | 0x000C2100   | M1    | PE4 (A), PE1 (B) |
| GPTW2   | 0x000C2200   | M2    | PE3 (A), PE0 (B) |
| GPTW3   | 0x000C2300   | M3    | PE7 (A), PE6 (B) |

Key Registers for Motor PWM:

| Register | Offset | Description                           |
|----------|--------|---------------------------------------|
| GTWP     | 0x00   | Write protection (unlock: 0xA500)     |
| GTCR     | 0x2C   | Control (mode, prescaler, start/stop) |
| GTIOR    | 0x34   | I/O control (output polarity/enable)  |
| GTCNT    | 0x48   | 32-bit counter value                  |
| GTCCRA   | 0x4C   | Compare A (duty cycle for GTIOCA)     |
| GTCCRB   | 0x50   | Compare B (duty cycle for GTIOCB)     |
| GTPR     | 0x64   | Period (PWM frequency)                |
| GTDTCR   | 0x88   | Dead-time control                     |
| GTDVU    | 0x8C   | Dead-time value (rising edge)         |
| GTDVD    | 0x90   | Dead-time value (falling edge)        |

PWM Frequency Calculation:

$$f_{PWM} = \frac{f_{PCLKA}}{(GTPR+1) \times Prescaler}$$

Duty Cycle Calculation:

$$Duty\% = \frac{GTCCRA}{GTPR+1} \times 100\%$$

Usage Example - Dead-Time Configuration:

```
volatile rx_gptw_channel_regs_t* gptw = gptw0();

// Enable dead-time generation
gptw->gtdtcr = k_gptw_gtdtcr_tde;

// Set dead-time: 500 ns @ 120 MHz = 60 counts
gptw->gtdvu = 60; // Rising edge dead-time
gptw->gtdvd = 60; // Falling edge dead-time
```

Invariant

`sizeof(rx_gptw_channel_regs_t) == 0xD8` (216 bytes)

Channel spacing is exactly 0x100 bytes

See also

[gptw0\(\)](#), [gptw1\(\)](#), [gptw2\(\)](#), [gptw3\(\)](#) Accessor functions

[rx\\_gptw\\_common\\_regs\\_t](#) Common registers for synchronized control

Definition at line 266 of file [rx72n\\_gptw\\_regs.h](#).

Collaboration diagram for `rx_gptw_channel_regs_t`:



## Data Fields

|                   |           |                                                      |
|-------------------|-----------|------------------------------------------------------|
| volatile uint32_t | gtadsmr   | 0xA4: A/D Conversion Start Request Signal Monitoring |
| volatile uint32_t | gtadtbra  | 0x74: A/D Trigger Buffer Register A                  |
| volatile uint32_t | gtadtbrb  | 0x80: A/D Trigger Buffer Register B                  |
| volatile uint32_t | gtadtdbra | 0x78: A/D Trigger Double Buffer A                    |
| volatile uint32_t | gtadtdbrb | 0x84: A/D Trigger Double Buffer B                    |
| volatile uint32_t | gtadtra   | 0x70: A/D Trigger Register A                         |
| volatile uint32_t | gtadtrb   | 0x7C: A/D Trigger Register B                         |
| volatile uint32_t | gtber     | 0x40: Buffer Enable Register                         |
| volatile uint32_t | gtccra    | 0x4C: Compare Capture Register A                     |
| volatile uint32_t | gtccrb    | 0x50: Compare Capture Register B                     |
| volatile uint32_t | gtccrc    | 0x54: Compare Capture Register C                     |
| volatile uint32_t | gtccrd    | 0x5C: Compare Capture Register D                     |
| volatile uint32_t | gtccre    | 0x58: Compare Capture Register E                     |
| volatile uint32_t | gtccrf    | 0x60: Compare Capture Register F                     |
| volatile uint32_t | gtclr     | 0x0C: Software Clear Register                        |
| volatile uint32_t | gtcnt     | 0x48: Counter                                        |
| volatile uint32_t | gtcr      | 0x2C: Control Register                               |
| volatile uint32_t | gtcsr     | 0x18: Clear Source Select Register                   |
| volatile uint32_t | gtdbd     | 0x98: Dead Time Buffer Down                          |
| volatile uint32_t | gtdbu     | 0x94: Dead Time Buffer Upper                         |
| volatile uint32_t | gtdnsr    | 0x20: Count-Down Source Select Register              |
| volatile uint32_t | gdtcr     | 0x88: Dead Time Control Register                     |
| volatile uint32_t | gtdvd     | 0x90: Dead Time Value Down                           |
| volatile uint32_t | gtdvu     | 0x8C: Dead Time Value Upper                          |
| volatile uint32_t | gteitc    | 0xA8: Extended Interrupt Skipping Counter Control    |
| volatile uint32_t | gteitlb   | 0xB4: Extended Buffer Transfer Skipping Setting      |
| volatile uint32_t | gteitli1  | 0xAC: Extended Interrupt Skipping Setting Register 1 |
| volatile uint32_t | gteitli2  | 0xB0: Extended Interrupt Skipping Setting Register 2 |
| volatile uint32_t | gticasr   | 0x24: Input Capture Source Select A                  |
| volatile uint32_t | gticbsr   | 0x28: Input Capture Source Select B                  |
| volatile uint32_t | gtintad   | 0x38: Interrupt Output Setting                       |
| volatile uint32_t | gtior     | 0x34: I/O Control Register                           |
| volatile uint32_t | gtitc     | 0x44: Interrupt/ADC Trigger Control                  |
| volatile uint32_t | gtpbr     | 0x68: Cycle Setting Buffer Register                  |
| volatile uint32_t | gtpdbr    | 0x6C: Cycle Setting Double Buffer                    |
| volatile uint32_t | gtpcr     | 0x64: Cycle Setting Register (Period)                |
| volatile uint32_t | gtpsr     | 0x14: Stop Source Select Register                    |



|                   |                                                         |                                                                |
|-------------------|---------------------------------------------------------|----------------------------------------------------------------|
| volatile uint32_t | gtsecr                                                  | 0xD4: Operation Enable Bit Simultaneous Control                |
| volatile uint32_t | gtsecsr                                                 | 0xD0: Operation Enable Bit Simultaneous Control Channel Select |
| volatile uint32_t | gtsos                                                   | 0x9C: Output Protection Status                                 |
| volatile uint32_t | gtsotr                                                  | 0xA0: Output Protection Trigger                                |
| volatile uint32_t | gtssr                                                   | 0x10: Start Source Select Register                             |
| volatile uint32_t | gtst                                                    | 0x3C: Status Register                                          |
| volatile uint32_t | gtstp                                                   | 0x08: Software Stop Register                                   |
| volatile uint32_t | gtstr                                                   | 0x04: Software Start Register                                  |
| volatile uint32_t | gtuddtyc                                                | 0x30: Up/Down Count Duty Setting                               |
| volatile uint32_t | gtupsr                                                  | 0x1C: Count-Up Source Select Register                          |
| volatile uint32_t | gtwp                                                    | 0x00: Write Protection Register                                |
| uint32_t          | reserved↵<br><a href="#">[k_gptw_ch_reserved_count]</a> | 0xB8-0xCC: Reserved                                            |

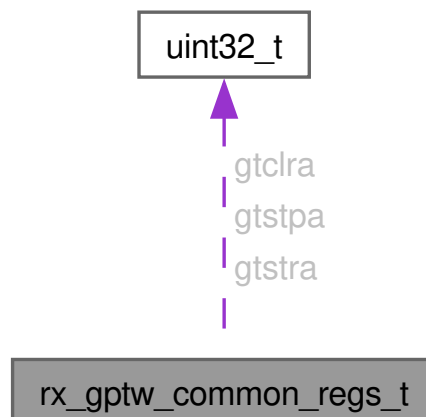
### 5.3.2.2 struct rx\_gptw\_common\_regs\_t

GPTW Common Register Map.

Overlays GPTW0.GTSTR/GTSTP/GTCLR to provide a named handle for multi-channel atomic start/↵ stop/clear. Writing bits [3:0] in any of these registers affects channels 0-3 simultaneously (per RX72N HW manual section 26.2.2-26.2.4). Base address: [k\\_gptw\\_common\\_base\\_addr](#) (0x000C2004) == GPTW0.↵ GTSTR. The RX72N address map has no separate "common" GPTW block; the per-channel GTSTR/↵ GTSTP/GTCLR registers are cross-channel by design.

Definition at line 328 of file [rx72n\\_gptw\\_regs.h](#).

Collaboration diagram for [rx\\_gptw\\_common\\_regs\\_t](#):



#### Data Fields

|                   |        |                                                           |
|-------------------|--------|-----------------------------------------------------------|
| volatile uint32_t | gtclra | 0x08: Software Clear (GPTW0.GTCLR, bits [3:0] = Ch3..Ch0) |
|-------------------|--------|-----------------------------------------------------------|

|                   |        |                                                           |
|-------------------|--------|-----------------------------------------------------------|
| volatile uint32_t | gtstpa | 0x04: Software Stop (GPTW0.GTSTP, bits [3:0] = Ch3..Ch0)  |
| volatile uint32_t | gtstra | 0x00: Software Start (GPTW0.GTSTR, bits [3:0] = Ch3..Ch0) |

### 5.3.3 Enumeration Type Documentation

#### 5.3.3.1 gptw\_addresses\_t

enum [gptw\\_addresses\\_t](#) : uintptr\_t

GPTW hardware addresses (verified against RX72N Hardware Manual).

Enumerator

|                         |                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_gptw_channel_offset   | Channel spacing between GPTW registers                                                                                                                                                                                                                                                                             |
| k_gptw0_base_addr       | GPTW channel 0 base address                                                                                                                                                                                                                                                                                        |
| k_gptw1_base_addr       | GPTW channel 1 base address                                                                                                                                                                                                                                                                                        |
| k_gptw2_base_addr       | GPTW channel 2 base address                                                                                                                                                                                                                                                                                        |
| k_gptw3_base_addr       | GPTW channel 3 base address                                                                                                                                                                                                                                                                                        |
| k_gptw_common_base_addr | GPTW0.GTSTR – used as multi-channel start/stop/clear base. RX72N has NO separate "common" GPTW block (Chapter 5 Table 5.1 shows GPTW space ending at 0x000C23D4; nothing at 0x000C2B00). GTSTR/GTSTP/GTCLR in any channel affect all channels whose corresponding bit [3:0] is set (manual section 26.2.2-26.2.4). |

Definition at line 342 of file [rx72n\\_gptw\\_regs.h](#).

```

00342 : uintptr_t {
00343 k_gptw_channel_offset = 0x100, /**< Channel spacing between GPTW registers */
00344 k_gptw0_base_addr = 0x000C2000, /**< GPTW channel 0 base address */
00345 k_gptw1_base_addr = 0x000C2100, /**< GPTW channel 1 base address */
00346 k_gptw2_base_addr = 0x000C2200, /**< GPTW channel 2 base address */
00347 k_gptw3_base_addr = 0x000C2300, /**< GPTW channel 3 base address */
00348 k_gptw_common_base_addr =
00349 0x000C2004, /**< GPTW0.GTSTR -- used as multi-channel start/stop/clear base.
00350 * RX72N has NO separate "common" GPTW block (Chapter 5 Table 5.1
00351 * shows GPTW space ending at 0x000C23D4; nothing at 0x000C2B00).
00352 * GTSTR/GTSTP/GTCLR in any channel affect all channels whose
00353 * corresponding bit [3:0] is set (manual section 26.2.2-26.2.4). */
00354 } gptw_addresses_t;
```

#### 5.3.3.2 gptw\_ch\_reg\_offsets\_t

enum [gptw\\_ch\\_reg\\_offsets\\_t](#) : uint8\_t

Enumerator

|                     |                           |
|---------------------|---------------------------|
| k_gptw_ch_off_gtwp  | Write Protection Register |
| k_gptw_ch_off_gtstr | Software Start Register   |
| k_gptw_ch_off_gtstp | Software Stop Register    |
| k_gptw_ch_off_gtclr | Software Clear Register   |

|                         |                                                              |
|-------------------------|--------------------------------------------------------------|
| k_gptw_ch_off_gtssr     | Start Source Select Register                                 |
| k_gptw_ch_off_gtpsr     | Stop Source Select Register                                  |
| k_gptw_ch_off_gtcsr     | Clear Source Select Register                                 |
| k_gptw_ch_off_gtupsr    | Count-Up Source Select Register                              |
| k_gptw_ch_off_gtdnsr    | Count-Down Source Select Register                            |
| k_gptw_ch_off_gticasr   | Input Capture Source Select A                                |
| k_gptw_ch_off_gticbsr   | Input Capture Source Select B                                |
| k_gptw_ch_off_gtcr      | Control Register                                             |
| k_gptw_ch_off_gtuddtyc  | Count Direction and Duty Setting Register                    |
| k_gptw_ch_off_gtior     | I/O Control Register                                         |
| k_gptw_ch_off_gtintad   | Interrupt Output Setting Register                            |
| k_gptw_ch_off_gtst      | Status Register                                              |
| k_gptw_ch_off_gtber     | Buffer Enable Register                                       |
| k_gptw_ch_off_gtitc     | Interrupt Skipping Setting Register                          |
| k_gptw_ch_off_gtcnt     | Counter                                                      |
| k_gptw_ch_off_gtcra     | Compare Capture Register A                                   |
| k_gptw_ch_off_gtccrb    | Compare Capture Register B                                   |
| k_gptw_ch_off_gtccrc    | Compare Capture Register C                                   |
| k_gptw_ch_off_gtccre    | Compare Capture Register E                                   |
| k_gptw_ch_off_gtccrd    | Compare Capture Register D                                   |
| k_gptw_ch_off_gtccrf    | Compare Capture Register F                                   |
| k_gptw_ch_off_gtpr      | Cycle Setting Register                                       |
| k_gptw_ch_off_gtpbr     | Cycle Setting Buffer Register                                |
| k_gptw_ch_off_gtpdbr    | Cycle Setting Double-Buffer Register                         |
| k_gptw_ch_off_gtadtra   | A/D Conversion Start Request Timing Register A               |
| k_gptw_ch_off_gtadtbra  | A/D Conversion Start Request Timing Buffer Register A        |
| k_gptw_ch_off_gtadtdbra | A/D Conversion Start Request Timing Double-Buffer Register A |
| k_gptw_ch_off_gtadtrb   | A/D Conversion Start Request Timing Register B               |
| k_gptw_ch_off_gtadtbrb  | A/D Conversion Start Request Timing Buffer Register B        |
| k_gptw_ch_off_gtadtdbrb | A/D Conversion Start Request Timing Double-Buffer Register B |
| k_gptw_ch_off_gtdtcr    | Dead-Time Control Register                                   |
| k_gptw_ch_off_gtdvu     | Dead-Time Value Register U                                   |
| k_gptw_ch_off_gtdvd     | Dead-Time Value Register D                                   |
| k_gptw_ch_off_gtdbu     | Dead-Time Buffer Register U                                  |
| k_gptw_ch_off_gtdbd     | Dead-Time Buffer Register D                                  |
| k_gptw_ch_off_gtsos     | Output Protection Function Status Register                   |
| k_gptw_ch_off_gtsotr    | Output Protection Function Temporary Release Register        |
| k_gptw_ch_off_gtadsmr   | A/D Conversion Start Request Signal Monitoring Register      |
| k_gptw_ch_off_gteitc    | Extended Interrupt Skipping Counter Control Register         |
| k_gptw_ch_off_gteitli1  | Extended Interrupt Skipping Setting Register 1               |
| k_gptw_ch_off_gteitli2  | Extended Interrupt Skipping Setting Register 2               |
| k_gptw_ch_off_gteitlb   | Extended Buffer Transfer Skipping Setting Register           |

|                        |                                                          |
|------------------------|----------------------------------------------------------|
| k_gptw_ch_off_reserved | Reserved region start                                    |
| k_gptw_ch_off_gtsecsr  | Operation Enable Bit Simultaneous Control Channel Select |
| k_gptw_ch_off_gtsecl   | Operation Enable Bit Simultaneous Control Register       |
| k_gptw_ch_struct_size  | Total channel structure size in bytes                    |

Definition at line 735 of file rx72n\_gptw\_regs.h.

```

00735 : uint8_t {
00736 k_gptw_ch_off_gtwp = 0x00U, /**< Write Protection Register */
00737 k_gptw_ch_off_gtstr = 0x04U, /**< Software Start Register */
00738 k_gptw_ch_off_gtstp = 0x08U, /**< Software Stop Register */
00739 k_gptw_ch_off_gtclr = 0x0CU, /**< Software Clear Register */
00740 k_gptw_ch_off_gtssr = 0x10U, /**< Start Source Select Register */
00741 k_gptw_ch_off_gtpsr = 0x14U, /**< Stop Source Select Register */
00742 k_gptw_ch_off_gtcsr = 0x18U, /**< Clear Source Select Register */
00743 k_gptw_ch_off_gtuupsr = 0x1CU, /**< Count-Up Source Select Register */
00744 k_gptw_ch_off_gtdnsr = 0x20U, /**< Count-Down Source Select Register */
00745 k_gptw_ch_off_gticasr = 0x24U, /**< Input Capture Source Select A */
00746 k_gptw_ch_off_gticbsr = 0x28U, /**< Input Capture Source Select B */
00747 k_gptw_ch_off_gtcr = 0x2CU, /**< Control Register */
00748 k_gptw_ch_off_gtudtctc = 0x30U, /**< Count Direction and Duty Setting Register */
00749 k_gptw_ch_off_gtior = 0x34U, /**< I/O Control Register */
00750 k_gptw_ch_off_gtintad = 0x38U, /**< Interrupt Output Setting Register */
00751 k_gptw_ch_off_gtst = 0x3CU, /**< Status Register */
00752 k_gptw_ch_off_gtber = 0x40U, /**< Buffer Enable Register */
00753 k_gptw_ch_off_gtitc = 0x44U, /**< Interrupt Skipping Setting Register */
00754 k_gptw_ch_off_gtcnt = 0x48U, /**< Counter */
00755 k_gptw_ch_off_gtcra = 0x4CU, /**< Compare Capture Register A */
00756 k_gptw_ch_off_gtcrcb = 0x50U, /**< Compare Capture Register B */
00757 k_gptw_ch_off_gtcrc = 0x54U, /**< Compare Capture Register C */
00758 k_gptw_ch_off_gtcrc = 0x58U, /**< Compare Capture Register E */
00759 k_gptw_ch_off_gtcrcd = 0x5CU, /**< Compare Capture Register D */
00760 k_gptw_ch_off_gtccrf = 0x60U, /**< Compare Capture Register F */
00761 k_gptw_ch_off_gtpr = 0x64U, /**< Cycle Setting Register */
00762 k_gptw_ch_off_gtpbr = 0x68U, /**< Cycle Setting Buffer Register */
00763 k_gptw_ch_off_gtpdbr = 0x6CU, /**< Cycle Setting Double-Buffer Register */
00764 k_gptw_ch_off_gtadtra = 0x70U, /**< A/D Conversion Start Request Timing Register A */
00765 k_gptw_ch_off_gtadtbra = 0x74U, /**< A/D Conversion Start Request Timing Buffer Register A */
00766 k_gptw_ch_off_gtadtbra =
00767 0x78U, /**< A/D Conversion Start Request Timing Double-Buffer Register A */
00768 k_gptw_ch_off_gtadtrb = 0x7CU, /**< A/D Conversion Start Request Timing Register B */
00769 k_gptw_ch_off_gtadtbrb = 0x80U, /**< A/D Conversion Start Request Timing Buffer Register B */
00770 k_gptw_ch_off_gtadtbrb =
00771 0x84U, /**< A/D Conversion Start Request Timing Double-Buffer Register B */
00772 k_gptw_ch_off_gtdtcr = 0x88U, /**< Dead-Time Control Register */
00773 k_gptw_ch_off_gtdvu = 0x8CU, /**< Dead-Time Value Register U */
00774 k_gptw_ch_off_gtdvd = 0x90U, /**< Dead-Time Value Register D */
00775 k_gptw_ch_off_gtdbu = 0x94U, /**< Dead-Time Buffer Register U */
00776 k_gptw_ch_off_gtdbd = 0x98U, /**< Dead-Time Buffer Register D */
00777 k_gptw_ch_off_gtsos = 0x9CU, /**< Output Protection Function Status Register */
00778 k_gptw_ch_off_gtsotr = 0xA0U, /**< Output Protection Function Temporary Release Register */
00779 k_gptw_ch_off_gtadsmr = 0xA4U, /**< A/D Conversion Start Request Signal Monitoring Register */
00780 k_gptw_ch_off_gteitc = 0xA8U, /**< Extended Interrupt Skipping Counter Control Register */
00781 k_gptw_ch_off_gteitl1 = 0xACU, /**< Extended Interrupt Skipping Setting Register 1 */
00782 k_gptw_ch_off_gteitl2 = 0xB0U, /**< Extended Interrupt Skipping Setting Register 2 */
00783 k_gptw_ch_off_gteitlb = 0xB4U, /**< Extended Buffer Transfer Skipping Setting Register */
00784 k_gptw_ch_off_reserved = 0xB8U, /**< Reserved region start */
00785 k_gptw_ch_off_gtsecsr = 0xD0U, /**< Operation Enable Bit Simultaneous Control Channel Select */
00786 k_gptw_ch_off_gtsecl = 0xD4U, /**< Operation Enable Bit Simultaneous Control Register */
00787 k_gptw_ch_struct_size = 0xD8U, /**< Total channel structure size in bytes */
00788 } gptw_ch_reg_offsets_t;

```

### 5.3.3.3 gptw'common'reg'offsets't

```
enum gptw_common_reg_offsets_t : uint8_t
```

Enumerator

|                           |                                                     |
|---------------------------|-----------------------------------------------------|
| k_gptw_common_off_gtstpa  | Software Stop Register (GPTW0.GTSTP)                |
| k_gptw_common_off_gtclra  | Software Clear Register (GPTW0.GTCLR)               |
| k_gptw_common_struct_size | Total common structure size in bytes (3 x uint32_t) |

Definition at line 790 of file [rx72n\\_gptw\\_regs.h](#).

```
00790 : uint8_t {
00791 k_gptw_common_off_gtstpa = 0x04U, /**< Software Stop Register (GPTW0.GTSTP) */
00792 k_gptw_common_off_gtclra = 0x08U, /**< Software Clear Register (GPTW0.GTCLR) */
00793 k_gptw_common_struct_size = 0x0CU, /**< Total common structure size in bytes (3 x uint32_t) */
00794 } gptw_common_reg_offsets_t;
```

#### 5.3.3.4 gptw\_gtadsmr\_bits\_t

enum [gptw\\_gtadsmr\\_bits\\_t](#) : uint32\_t

Enumerator

|                      |                               |
|----------------------|-------------------------------|
| k_gptw_gtadsmr_adsm0 | GTADSM0 output status monitor |
| k_gptw_gtadsmr_adsm1 | GTADSM1 output status monitor |

Definition at line 605 of file [rx72n\\_gptw\\_regs.h](#).

```
00605 : uint32_t {
00606 k_gptw_gtadsmr_adsm0 = (1 << 0), /**< GTADSM0 output status monitor */
00607 k_gptw_gtadsmr_adsm1 = (1 << 1), /**< GTADSM1 output status monitor */
00608 } gptw_gtadsmr_bits_t;
```

#### 5.3.3.5 gptw\_gtber\_bits\_t

enum [gptw\\_gtber\\_bits\\_t](#) : uint32\_t

Enumerator

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| k_gptw_gtber_bd0         | Disable buffer op for GTCCRA/GTCCRB (1=disabled)   |
| k_gptw_gtber_bd1         | Disable buffer op for GTPR (1=disabled)            |
| k_gptw_gtber_bd2         | Disable buffer op for GTADTRA/GTADTRB (1=disabled) |
| k_gptw_gtber_bd3         | Disable buffer op for GTDVU/GTDVD (1=disabled)     |
| k_gptw_gtber_ccra_single | GTCCRA single-buffer: GTCCRC -> GTCCRA             |
| k_gptw_gtber_ccra_double | GTCCRA double-buffer: GTCCRD -> GTCCRC -> GTCCRA   |
| k_gptw_gtber_ccrb_single | GTCCRB single-buffer: GTCCRE -> GTCCRB             |
| k_gptw_gtber_ccrb_double | GTCCRB double-buffer: GTCCRF -> GTCCRE -> GTCCRB   |
| k_gptw_gtber_pr_single   | GTPR single-buffer: GTPBR -> GTPR                  |
| k_gptw_gtber_pr_double   | GTPR double-buffer: GTPDBR -> GTPBR -> GTPR        |

Definition at line 537 of file [rx72n\\_gptw\\_regs.h](#).

```
00537 : uint32_t {
00538 /* BD[3:0] - Buffer Operation Disable (b3:b0).
00539 * Set 1 to DISABLE buffer operation for the register group. Reset = 0 (buffer enabled). */
00540 k_gptw_gtber_bd0 = (1U << 0), /**< Disable buffer op for GTCCRA/GTCCRB (1=disabled) */
00541 k_gptw_gtber_bd1 = (1U << 1), /**< Disable buffer op for GTPR (1=disabled) */
00542 k_gptw_gtber_bd2 = (1U << 2), /**< Disable buffer op for GTADTRA/GTADTRB (1=disabled) */
00543 k_gptw_gtber_bd3 = (1U << 3), /**< Disable buffer op for GTDVU/GTDVD (1=disabled) */
00544 /* CCRA[1:0] - GTCCRA Buffer Operation (b17:b16). BD[0]=0 required for buffer to be active. */
00545 k_gptw_gtber_ccra_single = (0x01U << 16), /**< GTCCRA single-buffer: GTCCRC -> GTCCRA */
00546 k_gptw_gtber_ccra_double = (0x02U << 16), /**< GTCCRA double-buffer: GTCCRD -> GTCCRC -> GTCCRA */
00547 /* CCRB[1:0] - GTCCRB Buffer Operation (b19:b18). BD[0]=0 required for buffer to be active. */
00548 k_gptw_gtber_ccrb_single = (0x01U << 18), /**< GTCCRB single-buffer: GTCCRE -> GTCCRB */
00549 k_gptw_gtber_ccrb_double = (0x02U << 18), /**< GTCCRB double-buffer: GTCCRF -> GTCCRE -> GTCCRB */
00550 /* PR[1:0] - GTPR Buffer Operation (b21:b20). BD[1]=0 required for buffer to be active. */
00551 k_gptw_gtber_pr_single = (0x01U << 20), /**< GTPR single-buffer: GTPBR -> GTPR */
00552 k_gptw_gtber_pr_double = (0x02U << 20), /**< GTPR double-buffer: GTPDBR -> GTPBR -> GTPR */
00553 } gptw_gtber_bits_t;
```

## 5.3.3.6 gptw\_gtcr\_bits\_t

```
enum gptw_gtcr_bits_t : uint32_t
```

## Enumerator

|                            |                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------|
| k_gptw_gtcr_cst            | Counter Start (0=stop, 1=count)                                                      |
| k_gptw_gtcr_icds           | 0=capture at stop, 1=no capture at stop                                              |
| k_gptw_gtcr_md_shift       |                                                                                      |
| k_gptw_gtcr_md_mask        |                                                                                      |
| k_gptw_gtcr_md_saw_pwm     | Sawtooth-wave PWM mode (single buffer or double buffer possible)                     |
| k_gptw_gtcr_md_saw_oneshot | Sawtooth-wave one-shot pulse mode (fixed buffer operation)                           |
| k_gptw_gtcr_md_tri_pwm1    | Triangle-wave PWM mode 1: 32-bit transfer at trough (single/double buffer)           |
| k_gptw_gtcr_md_tri_pwm2    | Triangle-wave PWM mode 2: 32-bit transfer at crest and trough (single/double buffer) |
| k_gptw_gtcr_md_tri_pwm3    | Triangle-wave PWM mode 3: 64-bit transfer at trough (fixed buffer operation)         |
| k_gptw_gtcr_tpcs_shift     |                                                                                      |
| k_gptw_gtcr_tpcs_mask      |                                                                                      |
| k_gptw_gtcr_tpcs_1         | PCLKA/1 (120 MHz)                                                                    |
| k_gptw_gtcr_tpcs_2         | PCLKA/2                                                                              |
| k_gptw_gtcr_tpcs_4         | PCLKA/4                                                                              |
| k_gptw_gtcr_tpcs_8         | PCLKA/8                                                                              |
| k_gptw_gtcr_tpcs_16        | PCLKA/16                                                                             |
| k_gptw_gtcr_tpcs_32        | PCLKA/32                                                                             |
| k_gptw_gtcr_tpcs_64        | PCLKA/64                                                                             |
| k_gptw_gtcr_tpcs_256       | PCLKA/256                                                                            |
| k_gptw_gtcr_tpcs_1024      | PCLKA/1024                                                                           |

Definition at line 403 of file [rx72n\\_gptw\\_regs.h](#).

```
00403 : uint32_t {
00404 /* b0: CST - Count Start */
00405 k_gptw_gtcr_cst = (1 « 0), /**< Counter Start (0=stop, 1=count) */
00406
00407 /* b8: ICDS - Input Capture Operation Select at Count Stop */
00408 k_gptw_gtcr_icds = (1 « 8), /**< 0=capture at stop, 1=no capture at stop */
00409
00410 /* b18-b16: MD[2:0] - Mode Select */
00411 k_gptw_gtcr_md_shift = 16,
00412 k_gptw_gtcr_md_mask = (0x07 « k_gptw_gtcr_md_shift),
00413
00414 /** Sawtooth-wave PWM mode (single buffer or double buffer possible) */
00415 k_gptw_gtcr_md_saw_pwm = (0x00 « k_gptw_gtcr_md_shift),
00416 /** Sawtooth-wave one-shot pulse mode (fixed buffer operation) */
00417 k_gptw_gtcr_md_saw_oneshot = (0x01 « k_gptw_gtcr_md_shift),
00418 /* 0x02 (010): Setting prohibited */
00419 /* 0x03 (011): Setting prohibited */
00420 /** Triangle-wave PWM mode 1: 32-bit transfer at trough (single/double buffer) */
00421 k_gptw_gtcr_md_tri_pwm1 = (0x04 « k_gptw_gtcr_md_shift),
00422 /** Triangle-wave PWM mode 2: 32-bit transfer at crest and trough (single/double buffer) */
```

```

00423 k_gptw_gtr_md_tri_pwm2 = (0x05 « k_gptw_gtr_md_shift),
00424 /** Triangle-wave PWM mode 3: 64-bit transfer at trough (fixed buffer operation) */
00425 k_gptw_gtr_md_tri_pwm3 = (0x06 « k_gptw_gtr_md_shift),
00426 /* 0x07 (111): Setting prohibited */
00427
00428 /* b26-b23: TPCS[3:0] - Timer Prescaler Select */
00429 k_gptw_gtr_tpcs_shift = 23,
00430 k_gptw_gtr_tpcs_mask = (0x0F « k_gptw_gtr_tpcs_shift),
00431
00432 k_gptw_gtr_tpcs_1 = (0x00 « k_gptw_gtr_tpcs_shift), /** < PCLKA/1 (120 MHz) */
00433 k_gptw_gtr_tpcs_2 = (0x01 « k_gptw_gtr_tpcs_shift), /** < PCLKA/2 */
00434 k_gptw_gtr_tpcs_4 = (0x02 « k_gptw_gtr_tpcs_shift), /** < PCLKA/4 */
00435 k_gptw_gtr_tpcs_8 = (0x03 « k_gptw_gtr_tpcs_shift), /** < PCLKA/8 */
00436 k_gptw_gtr_tpcs_16 = (0x04 « k_gptw_gtr_tpcs_shift), /** < PCLKA/16 */
00437 k_gptw_gtr_tpcs_32 = (0x05 « k_gptw_gtr_tpcs_shift), /** < PCLKA/32 */
00438 k_gptw_gtr_tpcs_64 = (0x06 « k_gptw_gtr_tpcs_shift), /** < PCLKA/64 */
00439 /* 0x07 (0111): Setting prohibited */
00440 k_gptw_gtr_tpcs_256 = (0x08 « k_gptw_gtr_tpcs_shift), /** < PCLKA/256 */
00441 /* 0x09 (1001): Setting prohibited */
00442 k_gptw_gtr_tpcs_1024 = (0x0A « k_gptw_gtr_tpcs_shift), /** < PCLKA/1024 */
00443 /* 0x0B (1011): Setting prohibited */
00444 /* 0x0C-0x0F: GTETRGA/B/C/D via POEG (external clock) */
00445 } gptw_gtr_bits_t;

```

### 5.3.3.7 gptw\_gtdtc`bits`

```
enum gptw_gtdtc_bits_t : uint8_t
```

Enumerator

|                   |                         |
|-------------------|-------------------------|
| k_gptw_gtdtc_tde  | Dead time enable        |
| k_gptw_gtdtc_tdfe | Dead time buffer enable |

Definition at line 560 of file rx72n\_gptw\_regs.h.

```

00560 : uint8_t {
00561 k_gptw_gtdtc_tde = (1 « 0), /** < Dead time enable */
00562 k_gptw_gtdtc_tdfe = (1 « 4), /** < Dead time buffer enable */
00563 } gptw_gtdtc_bits_t;

```

### 5.3.3.8 gptw\_gteitc`bits`

```
enum gptw_gteitc_bits_t : uint32_t
```

Enumerator

|                            |                                         |
|----------------------------|-----------------------------------------|
| k_gptw_gteitc_eivtc_mask   | Extended interval counter (0-65535)     |
| k_gptw_gteitc_eivtt_shift  | Extended interval transfer timing shift |
| k_gptw_gteitc_eivtt_mask   | Extended interval transfer timing mask  |
| k_gptw_gteitc_eivtt_none   | No transfer                             |
| k_gptw_gteitc_eivtt_trough | Transfer at trough                      |
| k_gptw_gteitc_eivtt_crest  | Transfer at crest                       |
| k_gptw_gteitc_eivtt_both   | Transfer at both crest and trough       |
| k_gptw_gteitc_eivtcn_shift | Extended interval counter shift         |
| k_gptw_gteitc_eivtcn_mask  | Extended interval counter mask          |

Definition at line 618 of file rx72n\_gptw\_regs.h.

```

00618 : uint32_t {
00619 k_gptw_gteitc_eivtc_mask = 0x0000FFFF, /** < Extended interval counter (0-65535) */

```

```

00620 k_gptw_gteitc_eivtt_shift = 16, /**< Extended interval transfer timing shift */
00621 k_gptw_gteitc_eivtt_mask = (0x03 « 16), /**< Extended interval transfer timing mask */
00622 k_gptw_gteitc_eivtt_none = (0x00 « 16), /**< No transfer */
00623 k_gptw_gteitc_eivtt_trough = (0x01 « 16), /**< Transfer at trough */
00624 k_gptw_gteitc_eivtt_crest = (0x02 « 16), /**< Transfer at crest */
00625 k_gptw_gteitc_eivtt_both = (0x03 « 16), /**< Transfer at both crest and trough */
00626 k_gptw_gteitc_eivtcn_shift = 24, /**< Extended interval counter shift */
00627 k_gptw_gteitc_eivtcn_mask = (0xFFU « 24), /**< Extended interval counter mask */
00628 } gptw_gteitc_bits_t;

```

### 5.3.3.9 gptw`gteitlb`bits`

```
enum gptw_gteitlb_bits_t : uint32_t
```

Enumerator

|                         |                                 |
|-------------------------|---------------------------------|
| k_gptw_gteitlb_eivtcpbl | Period buffer transfer skipping |
| k_gptw_gteitlb_eivtcabl | GTCCRA buffer transfer skipping |
| k_gptw_gteitlb_eivtcdbl | GTCCRB buffer transfer skipping |

Definition at line 662 of file [rx72n\\_gptw\\_regs.h](#).

```

00662 : uint32_t {
00663 k_gptw_gteitlb_eivtcpbl = (1 « 0), /**< Period buffer transfer skipping */
00664 k_gptw_gteitlb_eivtcabl = (1 « 1), /**< GTCCRA buffer transfer skipping */
00665 k_gptw_gteitlb_eivtcdbl = (1 « 2), /**< GTCCRB buffer transfer skipping */
00666 } gptw_gteitlb_bits_t;

```

### 5.3.3.10 gptw`gteitli1`bits`

```
enum gptw_gteitli1_bits_t : uint32_t
```

Enumerator

|                         |                                             |
|-------------------------|---------------------------------------------|
| k_gptw_gteitli1_eadtl   | A/D conversion start request link enable    |
| k_gptw_gteitli1_eivtciv | GTCIV (overflow) interrupt skipping enable  |
| k_gptw_gteitli1_eivtcui | GTCIU (underflow) interrupt skipping enable |

Definition at line 635 of file [rx72n\\_gptw\\_regs.h](#).

```

00635 : uint32_t {
00636 k_gptw_gteitli1_eadtl = (1 « 0), /**< A/D conversion start request link enable */
00637 k_gptw_gteitli1_eivtciv = (1 « 8), /**< GTCIV (overflow) interrupt skipping enable */
00638 k_gptw_gteitli1_eivtcui = (1 « 9), /**< GTCIU (underflow) interrupt skipping enable */
00639 } gptw_gteitli1_bits_t;

```

### 5.3.3.11 gptw`gteitli2`bits`

```
enum gptw_gteitli2_bits_t : uint32_t
```

Enumerator

|                               |                                 |
|-------------------------------|---------------------------------|
| k_gptw_gteitli2_eivtciv_shift | Extended interval counter shift |
| k_gptw_gteitli2_eivtciv_mask  | Extended interval counter mask  |



|                         |                                        |
|-------------------------|----------------------------------------|
| k_gptw_gteitli2_eivtcia | GTCIA compare match interrupt skipping |
| k_gptw_gteitli2_eivtcib | GTCIB compare match interrupt skipping |
| k_gptw_gteitli2_eivtcic | GTCIC compare match interrupt skipping |
| k_gptw_gteitli2_eivtcid | GTCID compare match interrupt skipping |
| k_gptw_gteitli2_eivtcie | GTCIE compare match interrupt skipping |
| k_gptw_gteitli2_eivtcif | GTCIF compare match interrupt skipping |

Definition at line 646 of file [rx72n\\_gptw\\_regs.h](#).

```

00646 : uint32_t {
00647 k_gptw_gteitli2_eivtciv_shift = 0, /**< Extended interval counter shift */
00648 k_gptw_gteitli2_eivtciv_mask = 0x000000FF, /**< Extended interval counter mask */
00649 k_gptw_gteitli2_eivtcia = (1 < 0), /**< GTCIA compare match interrupt skipping */
00650 k_gptw_gteitli2_eivtcib = (1 < 1), /**< GTCIB compare match interrupt skipping */
00651 k_gptw_gteitli2_eivtcic = (1 < 2), /**< GTCIC compare match interrupt skipping */
00652 k_gptw_gteitli2_eivtcid = (1 < 3), /**< GTCID compare match interrupt skipping */
00653 k_gptw_gteitli2_eivtcie = (1 < 4), /**< GTCIE compare match interrupt skipping */
00654 k_gptw_gteitli2_eivtcif = (1 < 5), /**< GTCIF compare match interrupt skipping */
00655 } gptw_gteitli2_bits_t;

```

### 5.3.3.12 gptw\_gtintad\_bits\_t

```
enum gptw_gtintad_bits_t : uint32_t
```

GTINTAD register bit definitions for POEG group linkage.

The GTINTAD register (offset 0x38) configures the link between a GPTW channel and a POEG group for emergency output stop. When a fault is detected (dead-time error, simultaneous high/low output), the GPTW signals the selected POEG group to disable PWM outputs.

STAR Project Usage:

| GPTW Channel | POEG Group | GRP[1:↔0] | Motor |
|--------------|------------|-----------|-------|
| GPTW0        | A          | 00        | 0     |
| GPTW1        | B          | 01        | 1     |
| GPTW2        | C          | 10        | 2     |
| GPTW3        | D          | 11        | 3     |

Warning

GRP[1:0] must be set while OAE and OBE in GTIOR are 0

See also

[rx72n\\_poeg\\_regs.h](#) POEG register definitions

RX72N Manual Section 26.2.15 - GTINTAD register

Since

Version 1.0.0

Enumerator

|                          |                                    |
|--------------------------|------------------------------------|
| k_gptw_gtintad_grp_shift | GRP field shift                    |
| k_gptw_gtintad_grp_mask  | GRP field mask                     |
| k_gptw_gtintad_grp_a     | Select POEG Group A                |
| k_gptw_gtintad_grp_b     | Select POEG Group B                |
| k_gptw_gtintad_grp_c     | Select POEG Group C                |
| k_gptw_gtintad_grp_d     | Select POEG Group D                |
| k_gptw_gtintad_grpdte    | Enable dead-time error detection   |
| k_gptw_gtintad_grpabh    | Enable simultaneous high detection |
| k_gptw_gtintad_grpabl    | Enable simultaneous low detection  |

Definition at line 513 of file [rx72n\\_gptw\\_regs.h](#).

```

00513 : uint32_t {
00514 /* GRP[1:0] - Output Stop Group Select (bits 25:24) */
00515 k_gptw_gtintad_grp_shift = 24, /*< GRP field shift */
00516 k_gptw_gtintad_grp_mask = (0x03U « 24), /*< GRP field mask */
00517 k_gptw_gtintad_grp_a = (0x00U « 24), /*< Select POEG Group A */
00518 k_gptw_gtintad_grp_b = (0x01U « 24), /*< Select POEG Group B */
00519 k_gptw_gtintad_grp_c = (0x02U « 24), /*< Select POEG Group C */
00520 k_gptw_gtintad_grp_d = (0x03U « 24), /*< Select POEG Group D */
00521
00522 /* GRPDTE - Dead Time Error Output Stop Detection Enable (bit 28) */
00523 k_gptw_gtintad_grpdte = (1U « 28), /*< Enable dead-time error detection */
00524
00525 /* GRPABH - Simultaneous High Output Stop Detection Enable (bit 29) */
00526 k_gptw_gtintad_grpabh = (1U « 29), /*< Enable simultaneous high detection */
00527
00528 /* GRPABL - Simultaneous Low Output Stop Detection Enable (bit 30) */
00529 k_gptw_gtintad_grpabl = (1U « 30), /*< Enable simultaneous low detection */
00530 } gptw_gtintad_bits_t;

```

### 5.3.3.13 gptw\_gtior\_bits\_t

```
enum gptw_gtior_bits_t : uint32_t
```

Enumerator

|                            |                                                                                    |
|----------------------------|------------------------------------------------------------------------------------|
| k_gptw_gtior_oa_shift      |                                                                                    |
| k_gptw_gtior_oa_mask       |                                                                                    |
| k_gptw_gtior_oa_disabled   | Output disabled                                                                    |
| k_gptw_gtior_oa_low_cmpup  | Low on compare match (up)                                                          |
| k_gptw_gtior_oa_high_cmpup | High on compare match (up)                                                         |
| k_gptw_gtior_oa_toggle     | Toggle on compare match                                                            |
| k_gptw_gtior_oa_init_low   | Initial LOW; high at end-of-cycle; low at compare match (active-high sawtooth PWM) |

|                            |                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------|
| k_gptw_gtior_oa_init_high  | Initial HIGH; low at end-of-cycle; high at compare match (complement of init_low) |
| k_gptw_gtior_oae           | GTIOCA Output Enable                                                              |
| k_gptw_gtior_ob_shift      |                                                                                   |
| k_gptw_gtior_ob_mask       |                                                                                   |
| k_gptw_gtior_ob_disabled   | Output disabled                                                                   |
| k_gptw_gtior_ob_low_cmpup  | Low on compare match (up)                                                         |
| k_gptw_gtior_ob_high_cmpup | High on compare match (up)                                                        |
| k_gptw_gtior_ob_toggle     | Toggle on compare match                                                           |
| k_gptw_gtior_ob_init_low   | Initial LOW; high at end-of-cycle; low at compare match                           |
| k_gptw_gtior_ob_init_high  | Initial HIGH; low at end-of-cycle; high at compare match                          |
| k_gptw_gtior_obe           | GTIOCB Output Enable                                                              |

Definition at line 452 of file rx72n\_gptw\_regs.h.

```

00452 : uint32_t {
00453 /* GTIOCA Output Control (bits 0-4) */
00454 k_gptw_gtior_oa_shift = 0,
00455 k_gptw_gtior_oa_mask = 0x1F,
00456 k_gptw_gtior_oa_disabled = 0x00, /**< Output disabled */
00457 k_gptw_gtior_oa_low_cmpup = 0x01, /**< Low on compare match (up) */
00458 k_gptw_gtior_oa_high_cmpup = 0x02, /**< High on compare match (up) */
00459 k_gptw_gtior_oa_toggle = 0x03, /**< Toggle on compare match */
00460 k_gptw_gtior_oa_init_low =
00461 0x09, /**< Initial LOW; high at end-of-cycle; low at compare match (active-high sawtooth PWM) */
00462 k_gptw_gtior_oa_init_high =
00463 0x16, /**< Initial HIGH; low at end-of-cycle; high at compare match (complement of init_low) */
00464
00465 /* GTIOCA Output Enable (bit 8) */
00466 k_gptw_gtior_oae = (1 « 8), /**< GTIOCA Output Enable */
00467
00468 /* GTIOCB Output Control (bits 16-20) */
00469 k_gptw_gtior_ob_shift = 16,
00470 k_gptw_gtior_ob_mask = (0x1F « 16),
00471 k_gptw_gtior_ob_disabled = (0x00 « 16), /**< Output disabled */
00472 k_gptw_gtior_ob_low_cmpup = (0x01 « 16), /**< Low on compare match (up) */
00473 k_gptw_gtior_ob_high_cmpup = (0x02 « 16), /**< High on compare match (up) */
00474 k_gptw_gtior_ob_toggle = (0x03 « 16), /**< Toggle on compare match */
00475 k_gptw_gtior_ob_init_low =
00476 (0x09 « 16), /**< Initial LOW; high at end-of-cycle; low at compare match */
00477 k_gptw_gtior_ob_init_high =
00478 (0x16 « 16), /**< Initial HIGH; low at end-of-cycle; high at compare match */
00479
00480 /* GTIOCB Output Enable (bit 24) */
00481 k_gptw_gtior_obe = (1 « 24), /**< GTIOCB Output Enable */
00482 } gptw_gtior_bits_t;

```

#### 5.3.3.14 gptw\_gtsecl\_bits\_t

```
enum gptw_gtsecl_bits_t : uint32_t
```

Enumerator

|                      |                                      |
|----------------------|--------------------------------------|
| k_gptw_gtsecl_sbdcce | Simultaneous buffer transfer disable |
| k_gptw_gtsecl_sbdccl | Buffer transfer disable clear        |
| k_gptw_gtsecl_sbdcpe | Simultaneous buffer transfer enable  |

Definition at line 685 of file rx72n\_gptw\_regs.h.

```

00685 : uint32_t {
00686 k_gptw_gtsecl_sbdcce = (1 « 0), /**< Simultaneous buffer transfer disable */
00687 k_gptw_gtsecl_sbdccl = (1 « 8), /**< Buffer transfer disable clear */
00688 k_gptw_gtsecl_sbdcpe = (1 « 9), /**< Simultaneous buffer transfer enable */
00689 } gptw_gtsecl_bits_t;

```

## 5.3.3.15 gptw\_gtsecsr\_bits\_t

```
enum gptw_gtsecsr_bits_t : uint32_t
```

Enumerator

|                        |                                       |
|------------------------|---------------------------------------|
| k_gptw_gtsecsr_secsel0 | Channel 0 simultaneous control enable |
| k_gptw_gtsecsr_secsel1 | Channel 1 simultaneous control enable |
| k_gptw_gtsecsr_secsel2 | Channel 2 simultaneous control enable |
| k_gptw_gtsecsr_secsel3 | Channel 3 simultaneous control enable |

Definition at line 673 of file rx72n\_gptw\_regs.h.

```
00673 : uint32_t {
00674 k_gptw_gtsecsr_secsel0 = (1 « 0), /**< Channel 0 simultaneous control enable */
00675 k_gptw_gtsecsr_secsel1 = (1 « 1), /**< Channel 1 simultaneous control enable */
00676 k_gptw_gtsecsr_secsel2 = (1 « 2), /**< Channel 2 simultaneous control enable */
00677 k_gptw_gtsecsr_secsel3 = (1 « 3), /**< Channel 3 simultaneous control enable */
00678 } gptw_gtsecsr_bits_t;
```

## 5.3.3.16 gptw\_gtst\_bits\_t

```
enum gptw_gtst_bits_t : uint16_t
```

Enumerator

|                   |                             |
|-------------------|-----------------------------|
| k_gptw_gtst_tcfa  | Compare match flag A        |
| k_gptw_gtst_tcfb  | Compare match flag B        |
| k_gptw_gtst_tcfc  | Compare match flag C        |
| k_gptw_gtst_tcfD  | Compare match flag D        |
| k_gptw_gtst_tcfE  | Compare match flag E        |
| k_gptw_gtst_tcff  | Compare match flag F        |
| k_gptw_gtst_tcfpo | Overflow flag (at period)   |
| k_gptw_gtst_tcfpu | Underflow flag (at 0)       |
| k_gptw_gtst_tucf  | Count direction flag (1=up) |

Definition at line 570 of file rx72n\_gptw\_regs.h.

```
00570 : uint16_t {
00571 k_gptw_gtst_tcfa = (1 « 0), /**< Compare match flag A */
00572 k_gptw_gtst_tcfb = (1 « 1), /**< Compare match flag B */
00573 k_gptw_gtst_tcfc = (1 « 2), /**< Compare match flag C */
00574 k_gptw_gtst_tcfD = (1 « 3), /**< Compare match flag D */
00575 k_gptw_gtst_tcfE = (1 « 4), /**< Compare match flag E */
00576 k_gptw_gtst_tcff = (1 « 5), /**< Compare match flag F */
00577 k_gptw_gtst_tcfpo = (1 « 6), /**< Overflow flag (at period) */
00578 k_gptw_gtst_tcfpu = (1 « 7), /**< Underflow flag (at 0) */
00579 k_gptw_gtst_tucf = (1 « 15), /**< Count direction flag (1=up) */
00580 } gptw_gtst_bits_t;
```

## 5.3.3.17 gptw\_gtstr\_bits\_t

```
enum gptw_gtstr_bits_t : uint8_t
```

## Enumerator

|                   |                       |
|-------------------|-----------------------|
| k_gptw_gtstr_cst0 | Channel 0 count start |
| k_gptw_gtstr_cst1 | Channel 1 count start |
| k_gptw_gtstr_cst2 | Channel 2 count start |
| k_gptw_gtstr_cst3 | Channel 3 count start |

Definition at line 587 of file [rx72n\\_gptw\\_regs.h](#).

```
00587 : uint8_t {
00588 k_gptw_gtstr_cst0 = (1 << 0), /**< Channel 0 count start */
00589 k_gptw_gtstr_cst1 = (1 << 1), /**< Channel 1 count start */
00590 k_gptw_gtstr_cst2 = (1 << 2), /**< Channel 2 count start */
00591 k_gptw_gtstr_cst3 = (1 << 3), /**< Channel 3 count start */
00592 } gptw_gtstr_bits_t;
```

## 5.3.3.18 gptw'gtuddtyc'bits't

```
enum gptw_gtuddtyc_bits_t : uint32_t
```

Up/Down Count Duty Setting Register (GTUDDTYC) Bits.

## Enumerator

|                     |                               |
|---------------------|-------------------------------|
| k_gptw_gtuddtyc_ud  | Count direction: 0=down, 1=up |
| k_gptw_gtuddtyc_udf | Forcibly set count direction  |

Definition at line 595 of file [rx72n\\_gptw\\_regs.h](#).

```
00595 : uint32_t {
00596 k_gptw_gtuddtyc_ud = (1 << 0), /**< Count direction: 0=down, 1=up */
00597 k_gptw_gtuddtyc_udf = (1 << 1), /**< Forcibly set count direction */
00598 } gptw_gtuddtyc_bits_t;
```

## 5.3.3.19 gptw'gtpw'bits't

```
enum gptw_gtpw_bits_t : uint16_t
```

## Enumerator

|                    |                              |
|--------------------|------------------------------|
| k_gptw_gtpw_wp0    | Register write protect bit 0 |
| k_gptw_gtpw_wp1    | Register write protect bit 1 |
| k_gptw_gtpw_wp2    | Register write protect bit 2 |
| k_gptw_gtpw_wp3    | Register write protect bit 3 |
| k_gptw_gtpw_unlock | Write protection unlock key  |
| k_gptw_gtpw_lock   | Write protection lock key    |

Definition at line 388 of file [rx72n\\_gptw\\_regs.h](#).

```
00388 : uint16_t {
00389 k_gptw_gtpw_wp0 = (1 << 0), /**< Register write protect bit 0 */
00390 k_gptw_gtpw_wp1 = (1 << 1), /**< Register write protect bit 1 */
00391 k_gptw_gtpw_wp2 = (1 << 2), /**< Register write protect bit 2 */
00392 k_gptw_gtpw_wp3 = (1 << 3), /**< Register write protect bit 3 */
00393 k_gptw_gtpw_unlock = 0xA500, /**< Write protection unlock key */
00394 k_gptw_gtpw_lock = 0xA501, /**< Write protection lock key */
00395 } gptw_gtpw_bits_t;
```

## 5.3.3.20 gptw`module`stop`bits`

```
enum gptw_module_stop_bits_t : uint8_t
```

## Enumerator

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_mstpa_gptw | GPTW module stop bit in MSTPCRA. Per RX72N Hardware Manual Ch.11 (Low Power Consumption) and Renesas iodefne.h for RX72N, the GPTW peripheral is controlled by MSTPCRA bit 7 (MSTPA7). Previous HAL code erroneously targeted MSTPCRC.MSTPC6 which is a different, unrelated peripheral on this chip.<br>MSTPCRA lives at 0x00080010 and requires PRC1 unlock (0xA502) before write, lock (0xA500) after. GPTW module stop bit in MSTPCRA (MSTPA7) |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 696 of file [rx72n\\_gptw\\_regs.h](#).

```
00696 : uint8_t {
00697 /**
00698 * @brief GPTW module stop bit in MSTPCRA
00699 *
00700 * @details
00701 * Per RX72N Hardware Manual Ch.11 (Low Power Consumption) and Renesas
00702 * iodefne.h for RX72N, the GPTW peripheral is controlled by MSTPCRA
00703 * bit 7 (MSTPA7). Previous HAL code erroneously targeted MSTPCRC.MSTPC6
00704 * which is a different, unrelated peripheral on this chip.
00705 *
00706 * MSTPCRA lives at 0x00080010 and requires PRC1 unlock (0xA502) before
00707 * write, lock (0xA500) after.
00708 */
00709 k_mstpa_gptw = 7, /**< GPTW module stop bit in MSTPCRA (MSTPA7) */
00710 } gptw_module_stop_bits_t;
```

## 5.3.3.21 gptw`mpc`psel`

```
enum gptw_mpc_psel_t : uint8_t
```

## Enumerator

|                 |                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_pfs_psel_gptw | Peripheral select value for GPTW outputs on Port E.<br><br>Warning<br><br>Verify this value against RX72N Hardware Manual. PSEL values are chip-specific. PSEL value for GPTW (GTIOC*) alternate function. Per RX72N HW manual Tables 23.4, 23.6, 23.10, 23.14, 23.16: 0b011110 selects GTIOC on every port that exposes it. Old 0x14 selected an unrelated function and silently left the pin un-muxed. |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 717 of file [rx72n\\_gptw\\_regs.h](#).

```
00717 : uint8_t {
00718 /**
00719 * @brief Peripheral select value for GPTW outputs on Port E
00720 * @warning Verify this value against RX72N Hardware Manual.
00721 * PSEL values are chip-specific.
00722 */
00723 k_pfs_psel_gptw = 0x1E, /**< PSEL value for GPTW (GTIOC*) alternate function.
00724 * Per RX72N HW manual Tables 23.4, 23.6, 23.10, 23.14,
00725 * 23.16: 0b011110 selects GTIOC on every port that
00726 * exposes it. Old 0x14 selected an unrelated
00727 * function and silently left the pin un-muxed. */
00728 } gptw_mpc_psel_t;
```

## 5.3.3.22 gptw\_padding\_sizes\_t

```
enum gptw_padding_sizes_t : uint8_t
```

Enumerator

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| k_gptw_ch_reserved_count | Reserved uint32_t count in channel struct (0xB8-0xCC) |
|--------------------------|-------------------------------------------------------|

Definition at line 205 of file [rx72n\\_gptw\\_regs.h](#).

```
00205 : uint8_t {
00206 k_gptw_ch_reserved_count = 6U, /**< Reserved uint32_t count in channel struct (0xB8-0xCC) */
00207 } gptw_padding_sizes_t;
```

## 5.3.3.23 gptw\_periph\_space\_t

```
enum gptw_periph_space_t : uintptr_t
```

Enumerator

|                          |                                               |
|--------------------------|-----------------------------------------------|
| k_gptw_periph_space_mask | Mask to extract peripheral space from address |
| k_gptw_periph_space_base | GPTW peripheral space base (0x000Cxxxx)       |

Definition at line 796 of file [rx72n\\_gptw\\_regs.h](#).

```
00796 : uintptr_t {
00797 k_gptw_periph_space_mask = 0xFFFF0000U, /**< Mask to extract peripheral space from address */
00798 k_gptw_periph_space_base = 0x000C0000U, /**< GPTW peripheral space base (0x000Cxxxx) */
00799 } gptw_periph_space_t;
```

## 5.3.4 Function Documentation

## 5.3.4.1 gptw0()

```
volatile rx_gptw_channel_regs_t * gptw0 (
 void) [inline], [static]
```

GPTW register inline accessor functions.

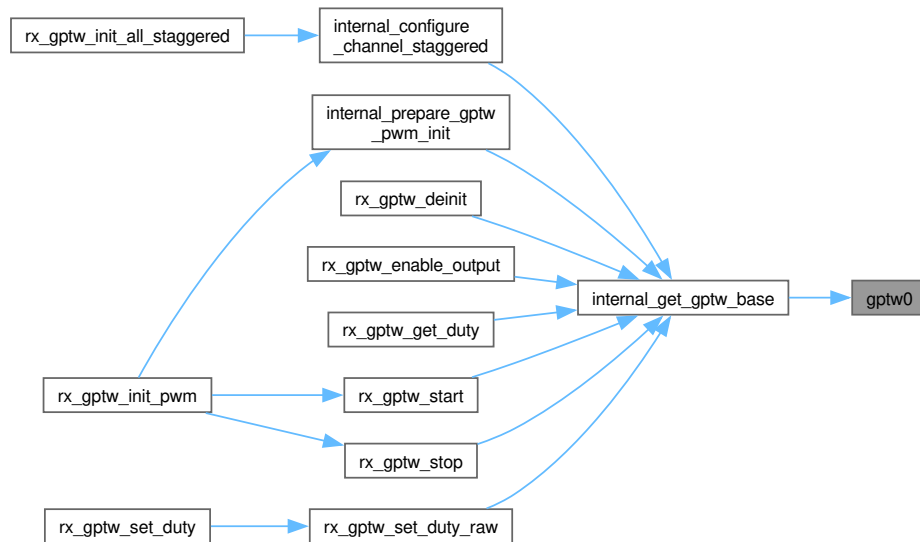
Definition at line 357 of file [rx72n\\_gptw\\_regs.h](#).

```
00358 {
00359 return (volatile rx_gptw_channel_regs_t*)k_gptw0_base_addr;
00360 }
```

References [k\\_gptw0\\_base\\_addr](#).

Referenced by [internal\\_get\\_gptw\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.3.4.2 gptw1()

```
volatile rx_gptw_channel_regs_t * gptw1 (
 void) [inline], [static]
```

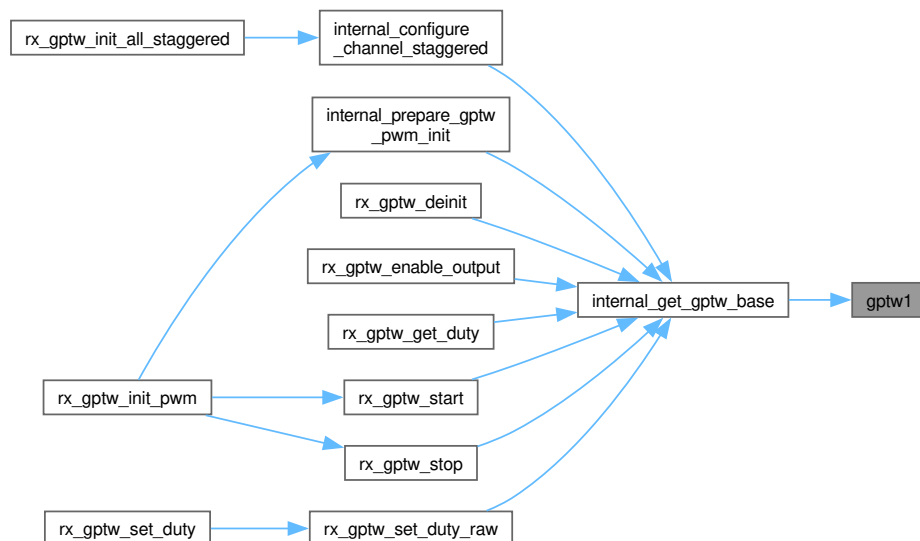
Definition at line 362 of file `rx72n_gptw_regs.h`.

```
00363 {
00364 return (volatile rx_gptw_channel_regs_t*)k_gptw1_base_addr;
00365 }
```

References `k_gptw1_base_addr`.

Referenced by `internal_get_gptw_base()`.

Here is the caller graph for this function:





## 5.3.4.3 gptw2()

```
volatile rx_gptw_channel_regs_t * gptw2 (
 void) [inline], [static]
```

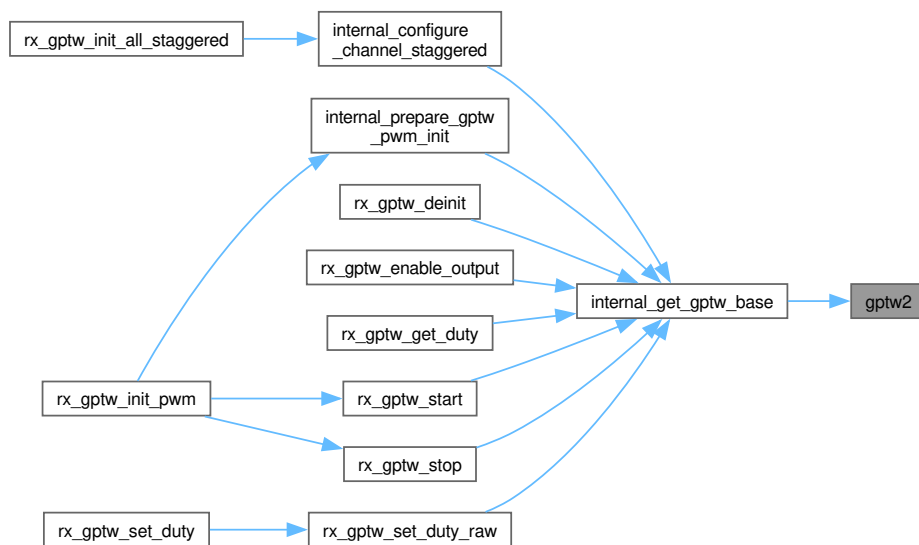
Definition at line 367 of file [rx72n\\_gptw\\_regs.h](#).

```
00368 {
00369 return (volatile rx_gptw_channel_regs_t*)k_gptw2_base_addr;
00370 }
```

References [k\\_gptw2\\_base\\_addr](#).

Referenced by [internal\\_get\\_gptw\\_base\(\)](#).

Here is the caller graph for this function:



## 5.3.4.4 gptw3()

```
volatile rx_gptw_channel_regs_t * gptw3 (
 void) [inline], [static]
```

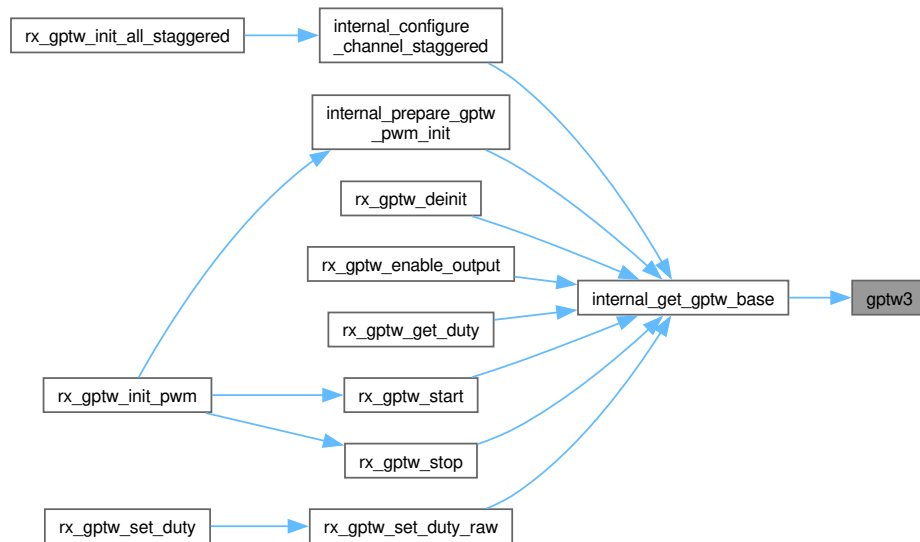
Definition at line 372 of file [rx72n\\_gptw\\_regs.h](#).

```
00373 {
00374 return (volatile rx_gptw_channel_regs_t*)k_gptw3_base_addr;
00375 }
```

References [k\\_gptw3\\_base\\_addr](#).

Referenced by [internal\\_get\\_gptw\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.3.4.5 gptw\_common()

```
volatile rx_gptw_common_regs_t * gptw_common (
 void) [inline], [static]
```

GPTW common register accessor.

Definition at line 378 of file rx72n\_gptw\_regs.h.

```
00379 {
00380 return (volatile rx_gptw_common_regs_t*)k_gptw_common_base_addr;
00381 }
```

References [k\\_gptw\\_common\\_base\\_addr](#).

Referenced by [rx\\_gptw\\_init\\_all\\_staggered\(\)](#).

Here is the caller graph for this function:



## 5.4 RX72N ICU Register Definitions

Interrupt Controller Unit hardware register interface.

Data Structures

- struct [rx\\_dmrsr\\_entry\\_t](#)  
DMAC Module Start Register entry with 4-byte alignment.
- struct [rx\\_icu\\_regs\\_t](#)  
ICU Register Map for Interrupt Controller Unit.

## Enumerations

- enum `rx_icu_addresses_t` : `uintptr_t` { `k_icu_base_addr` = 0x00087000 , `k_icu_slibr_base_addr` = 0x00087700 , `k_icu_sliar_base_addr` = 0x00087900 }  
ICU peripheral base address.
- enum `icu_reserved_sizes_t` : `uint16_t` {  
`k_icu_reserved_after_ier_bytes` = 192 , `k_icu_reserved_after_swint2r_bytes` = 14 ,  
`k_icu_reserved_after_fir_bytes` = 14 , `k_icu_reserved_after_dmrsrc_bytes` = 224 ,  
`k_icu_reserved_after_irqcr_bytes` = 16 , `k_icu_reserved_after_irqfite_bytes` = 6 , `k_icu_reserved_after_irqfite`  
= 84 , `k_icu_reserved_after_exnmiclr_bytes` = 9 ,  
`k_icu_reserved_after_nmifite_bytes` = 3 , `k_icu_reserved_after_nmifite_bytes` = 491 ,  
`k_icu_reserved_after_slibr_bytes` = 512 , `k_icu_dmrsrc_entry_padding` = 3 }  
ICU register reserved field sizes for structure padding.
- enum `icu_array_sizes_t` : `uint16_t` {  
`k_icu_ir_count` = 256 , `k_icu_dtcer_count` = 256 , `k_icu_ier_count` = 32 , `k_icu_ier_bits_per_reg`  
= 8 ,  
`k_icu_ipr_count` = 256 , `k_icu_dmrsrc_count` = 8 , `k_icu_irqcr_count` = 16 , `k_icu_irqfite_count`  
= 2 ,  
`k_icu_irqfite_count` = 2 , `k_icu_slibxr_count` = 16 , `k_icu_slibr_count` = 64 , `k_icu_sliar_count`  
= 48 }  
ICU register array dimensions.
- enum `rx_cmt_interrupt_vector_t` : `uint8_t` { `k_vect_cmt0_cmi0` = 28 , `k_vect_cmt1_cmi1` = 29 }  
CMT (Compare Match Timer) interrupt vector numbers.
- enum `rx_sci_interrupt_vector_t` : `uint8_t` { `k_vect_sci9_rx9` = 102 }  
SCI (Serial Communications Interface) interrupt vector numbers.
- enum `rx_interrupt_priority_t` : `uint8_t` { `k_ipr_level_disable` = 0 , `k_ipr_level_min` = 1 ,  
`k_ipr_level_max` = 15 }  
Interrupt priority levels for IPR registers.

## Functions

- static volatile `rx_icu_regs_t` \* `icu` (void)  
Get pointer to ICU registers.
- static volatile `uint8_t` \* `icu_slibr` (`uint8_t` vec)  
Get pointer to ICU SLIBR source-select register for a SELECTB vector.

## 5.4.1 Detailed Description

Interrupt Controller Unit hardware register interface.

Provides memory-mapped register access to the RX72N ICU peripheral for interrupt configuration, enabling, and priority management.

Quick Start Example - Enable CMT0 Interrupt:

```
volatile rx_icu_regs_t* icu_regs = icu();

// 1. Clear any pending interrupt request
icu_regs->ir[k_vect_cmt0_cmi0] = 0x00;

// 2. Set priority (1=lowest, 15=highest, 0=disabled)
icu_regs->ipr[k_vect_cmt0_cmi0] = 10; // Priority 10

// 3. Enable interrupt in IER register
// IER[3] bit 4 = vector 28 (CMT0_CMI0)
// IER index = vector / 8, bit = vector % 8
uint8_t ier_index = k_vect_cmt0_cmi0 / 8; // = 3
uint8_t ier_bit = k_vect_cmt0_cmi0 % 8; // = 4
icu_regs->ier[ier_index] |= (1U << ier_bit);

// 4. Enable global interrupts (in PSW register, done separately)
```

## 5.4.2 Data Structure Documentation

### 5.4.2.1 struct rx\_dmr\_entry\_t

DMAC Module Start Register entry with 4-byte alignment.

The DMRSR registers are 8-bit registers but located at 4-byte aligned addresses (0x400, 0x404, 0x408, 0x40C, 0x410, 0x414, 0x418, 0x41C). This structure provides proper padding to match the hardware layout.

Manual Reference:

Ch15 Section 15.2.8 - DMRSR0-7 addresses:

- DMRSR0: 0x00087400
- DMRSR1: 0x00087404
- DMRSR2: 0x00087408
- DMRSR3: 0x0008740C
- DMRSR4: 0x00087410
- DMRSR5: 0x00087414
- DMRSR6: 0x00087418
- DMRSR7: 0x0008741C

Usage:

```
// Set DMAC channel 0 trigger to vector 28 (CMT0)
icu()->dmrsr[0].reg = 28;

// Set DMAC channel 3 trigger to USB vector
icu()->dmrsr[3].reg = 102;
```

Note

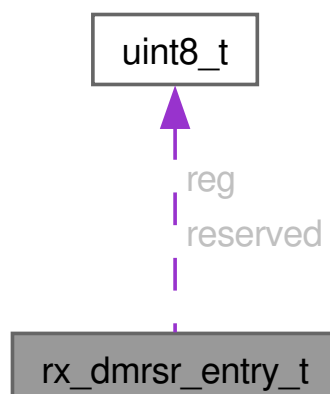
Each DMRSR register selects the interrupt vector number that triggers the corresponding DMAC channel transfer.

Warning

Do not set the same vector number for multiple DMRSR registers.  
Do not set the same interrupt as both DTC and DMAC trigger.

Definition at line 288 of file [rx72n\\_icu\\_regs.h](#).

Collaboration diagram for rx\_dmr\_entry\_t:



## Data Fields

|                  |                                          |                              |
|------------------|------------------------------------------|------------------------------|
| volatile uint8_t | reg                                      | 8-bit vector number register |
| uint8_t          | reserved↵<br>[k_icu_dmrsr_entry_padding] | Padding to 4-byte boundary   |

## 5.4.2.2 struct rx\_icu\_regs\_t

ICU Register Map for Interrupt Controller Unit.

Memory-mapped register structure for the ICU peripheral. This structure provides direct hardware access to interrupt control registers including request flags, enable control, priorities, and external IRQ configuration.

Interrupt Handling Workflow:

1. Peripheral generates interrupt request -> IR[vector] = 1
2. ICU checks IER[vector/8] bit (vector%8) for enable
3. If enabled, ICU checks IPR[vector] for priority (1-15)
4. Higher priority interrupts preempt lower priority
5. CPU vectors to interrupt handler
6. Handler clears IR[vector] = 0 before returning

Usage Example - Configure Multiple Interrupts:

```
volatile rx_icu_regs_t* icu_regs = icu();

// Disable all interrupts during configuration
// (typically done with PSW.I bit, not shown here)

// Configure CMT0 (vector 28) - System tick at priority 10
icu_regs->ir[28] = 0; // Clear pending
icu_regs->ipr[28] = 10; // Priority 10
icu_regs->ier[3] |= 0x10; // Enable (IER3 bit 4)

// Configure CMT1 (vector 29) - Motor control at priority 12
icu_regs->ir[29] = 0; // Clear pending
icu_regs->ipr[29] = 12; // Priority 12 (higher than tick)
icu_regs->ier[3] |= 0x20; // Enable (IER3 bit 5)

// Configure USB (vector 102) - Communication at priority 8
icu_regs->ir[102] = 0; // Clear pending
icu_regs->ipr[102] = 8; // Priority 8
icu_regs->ier[12] |= 0x40; // Enable (IER12 bit 6)

// Re-enable global interrupts (PSW.I = 1)
```

Memory Layout (Total: ~2562 bytes):

| Offset | Size | Field         | Description             |
|--------|------|---------------|-------------------------|
| 0x000  | 256  | ir↵<br>[256]↵ | Interrupt Request flags |

| Offset | Size | Field           | Description                           |
|--------|------|-----------------|---------------------------------------|
| 0x100  | 256  | dtcer↔<br>[256] | DTC Enable registers                  |
| 0x200  | 32   | ier↔<br>[32]    | Interrupt Enable registers            |
| 0x220  | 192  | reserved        | Reserved gap                          |
| 0x2E0  | 1    | swintr          | Software Interrupt register           |
| 0x2E1  | 1    | swint2r         | Software Interrupt 2 register         |
| 0x2F0  | 2    | fir             | Fast Interrupt register               |
| 0x300  | 256  | ipr↔<br>[256]   | Interrupt Priority registers          |
| 0x400  | 32   | dmrsr↔<br>[8]   | DMAC Module Start registers           |
| 0x500  | 16   | irqcr↔<br>[16]  | External IRQ Control                  |
| 0x520  | 2    | irqfte↔<br>[2]  | IRQ Filter Enable                     |
| 0x528  | 4    | irqftc↔<br>[2]  | IRQ Filter Clock Select               |
| 0x580  | 4    | nmisr..nmicr    | NMI Status/Enable/Clear/Ctrl          |
| 0x584  | 3    | exnmisr..clr    | Extended NMI registers                |
| 0x590  | 1    | nmifte          | NMI Filter Enable                     |
| 0x594  | 1    | nmiftc          | NMI Filter Clock                      |
| 0x595  | 491  | reserved        | Reserved gap                          |
| 0x780  | 16   | slibxr↔<br>[16] | Sel. Int B source sel. (vect 128-143) |
| 0x790  | 64   | slibr↔<br>[64]  | Sel. Int B source sel. (vect 144-207) |
| 0x7D0  | 512  | reserved        | Reserved gap                          |
| 0x9D0  | 48   | sliar↔<br>[48]  | Sel. Int A source sel. (vect 208-255) |
| 0xA00  | 1    | sliprcr         | SLIBXR/SLIBR/SLIAR write-protect      |
| 0xA01  | 1    | selexdr         | Selectable ext int direction          |

#### Note

Register layout verified against RX72N Manual Ch15, Rev.1.20

#### Warning

All registers are memory-mapped and must be accessed as volatile.  
Some IR/IPR indices (0-15) are reserved and must not be accessed.

## Invariant

IR flags are read-write: write 0 to clear, read to check status

IPR values must be 0-15 (0 disables, 1-15 sets priority)

IER bits: 1=enabled, 0=disabled

## See also

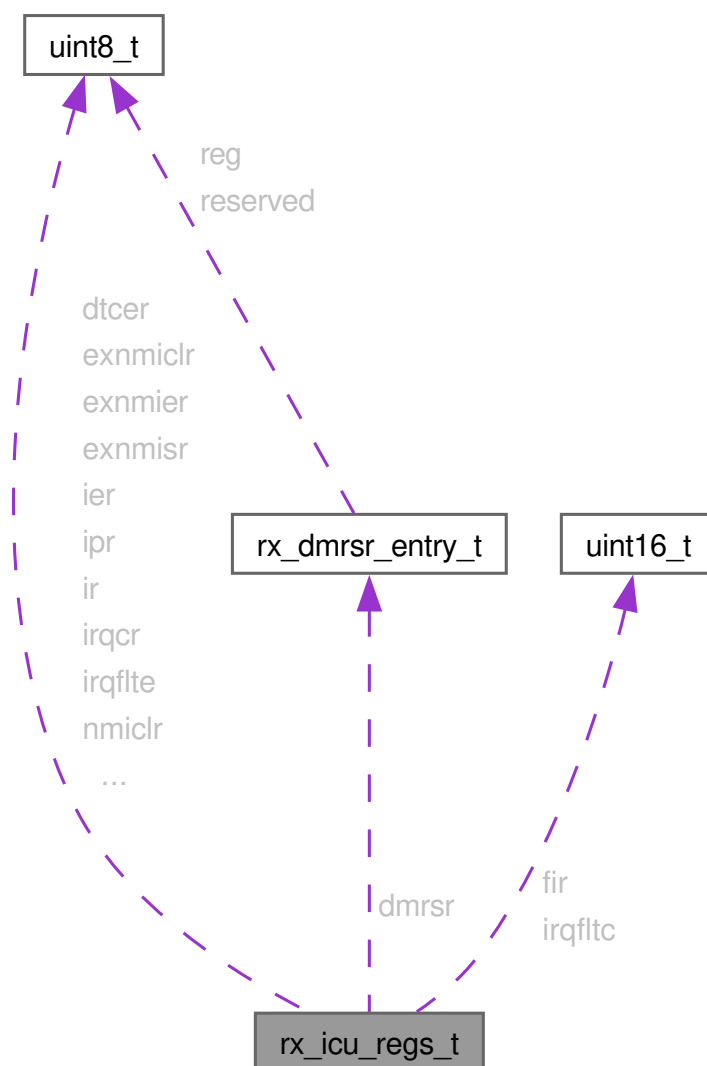
[icu\(\)](#) Accessor function

[rx\\_cmt\\_interrupt\\_vector\\_t](#) Common interrupt vector numbers

[rx\\_interrupt\\_priority\\_t](#) Priority level definitions

Definition at line 376 of file [rx72n\\_icu\\_regs.h](#).

Collaboration diagram for `rx_icu_regs_t`:



## Data Fields

|                                  |                                                                  |                                                                       |
|----------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| <a href="#">rx_dmrsr_entry_t</a> | dmrsr[ <a href="#">k_icu_dmrsr_count</a> ]                       | @0x400 DMRSR0-↵<br>DMRSR7 (4-byte spacing!)                           |
| volatile uint8_t                 | dtcer[ <a href="#">k_icu_dtcer_count</a> ]                       | @0x100 DTCER026-↵<br>DTCER255                                         |
| volatile uint8_t                 | exnmiclr                                                         | @0x586 Extended NMI<br>Clear Register                                 |
| volatile uint8_t                 | exnmier                                                          | @0x585 Extended NMI<br>Enable Register                                |
| volatile uint8_t                 | exnmisr                                                          | @0x584 Extended NMI<br>Status Register                                |
| volatile uint16_t                | fir                                                              | @0x2F0 Fast Interrupt<br>Register                                     |
| volatile uint8_t                 | ier[ <a href="#">k_icu_ier_count</a> ]                           | @0x200 IER02-IER1F                                                    |
| volatile uint8_t                 | ipr[ <a href="#">k_icu_ipr_count</a> ]                           | @0x300 IPR000-IPR255                                                  |
| volatile uint8_t                 | ir[ <a href="#">k_icu_ir_count</a> ]                             | @0x000 IR016-IR255                                                    |
| volatile uint8_t                 | irqcr[ <a href="#">k_icu_irqcr_count</a> ]                       | @0x500 IRQCR0-↵<br>IRQCR15                                            |
| volatile uint16_t                | irqfltcr[ <a href="#">k_icu_irqfltcr_count</a> ]                 | @0x528 IRQFLTC0-↵<br>IRQFLTC1                                         |
| volatile uint8_t                 | irqflte[ <a href="#">k_icu_irqflte_count</a> ]                   | @0x520 IRQFLTE0-↵<br>IRQFLTE1                                         |
| volatile uint8_t                 | nmiclr                                                           | @0x582 NMI Clear Register                                             |
| volatile uint8_t                 | nmier                                                            | @0x583 NMI Control<br>Register (FIXED: was<br>uint32_t)               |
| volatile uint8_t                 | nmier                                                            | @0x581 NMI Enable Register                                            |
| volatile uint8_t                 | nmiflter                                                         | @0x594 NMI Filter Clock<br>Select Register (RE-<br>NAMED from nmiflt) |
| volatile uint8_t                 | nmiflte                                                          | @0x590 NMI Filter Enable<br>Register (ADDED -<br>was missing)         |
| volatile uint8_t                 | nmisr                                                            | @0x580 NMI Status Register                                            |
| uint8_t                          | reserved0[ <a href="#">k_icu_reserved_after_ier_bytes</a> ]      | @0x220 Reserved                                                       |
| uint8_t                          | reserved1↵<br><a href="#">k_icu_reserved_after_swint2r_bytes</a> | @0x2E2 Reserved                                                       |
| uint8_t                          | reserved10[ <a href="#">k_icu_reserved_after_slibr_bytes</a> ]   | @0x7D0 Reserved                                                       |
| uint8_t                          | reserved2[ <a href="#">k_icu_reserved_after_fir_bytes</a> ]      | @0x2F2 Reserved                                                       |



|                  |                                                     |                                                                          |
|------------------|-----------------------------------------------------|--------------------------------------------------------------------------|
| uint8_t          | reserved3[k_icu_reserved_after_dmrsrc_bytes]        | @0x420 Reserved<br>(FIXED from 0x408)                                    |
| uint8_t          | reserved4[k_icu_reserved_after_irqcr_bytes]         | @0x510 Reserved                                                          |
| uint8_t          | reserved5[k_icu_reserved_after_irqflte_bytes]       | @0x522 Reserved                                                          |
| uint8_t          | reserved6[k_icu_reserved_after_irqfltc_bytes]       | @0x52C Reserved                                                          |
| uint8_t          | reserved7↵<br>[k_icu_reserved_after_exnmiclr_bytes] | @0x587 Reserved                                                          |
| uint8_t          | reserved8↵<br>[k_icu_reserved_after_nmiflte_bytes]  | @0x591 Reserved                                                          |
| uint8_t          | reserved9↵<br>[k_icu_reserved_after_nmifltc_bytes]  | @0x595 Reserved                                                          |
| volatile uint8_t | selexdr                                             | @0xA01 SELEXDR: selectable external interrupt direction                  |
| volatile uint8_t | sliar[k_icu_sliar_count]                            | @0x9D0 SLIARn: source select for INTA208-255                             |
| volatile uint8_t | slibr[k_icu_slibr_count]                            | @0x790 SLIBRn: source select for INTB144-207                             |
| volatile uint8_t | slibxr[k_icu_slibxr_count]                          | @0x780 SLIBXRn:↵<br>source select for<br>INTB128-143                     |
| volatile uint8_t | sliprcr                                             | @0xA00 SLIPRCR:↵<br>write-protect for<br>SLIBXR/SLIBR/↵<br>SLIAR/SELEXDR |
| volatile uint8_t | swint2r                                             | @0x2E1 Software Interrupt 2 Register                                     |
| volatile uint8_t | swintr                                              | @0x2E0 Software Interrupt Register                                       |

### 5.4.3 Enumeration Type Documentation

#### 5.4.3.1 icu\_array\_sizes\_t

```
enum icu_array_sizes_t : uint16_t
```

ICU register array dimensions.

Defines the sizes of register arrays in the ICU. These values are used to dimension arrays in the `rx_icu_regs_t` structure.

Register Array Purpose:

- IR[256]: One flag per interrupt vector (vectors 0-255)
- DTCER[256]: One enable bit per DTC-capable vector
- IER[32]: 8 bits per register, covering 256 vectors (32 x 8 = 256)
- IPR[256]: One priority level per interrupt vector
- DMRSR[8]: DMAC channel 0-7 start source selection (4-byte aligned)
- IRQCR[16]: External IRQ0-15 edge/level configuration

Enumerator

|                        |                                                |
|------------------------|------------------------------------------------|
| k_icu_ir_count         | 256 interrupt request flags (one per vector)   |
| k_icu_dtcer_count      | 256 DTC enable flags                           |
| k_icu_ier_count        | 32 enable registers x 8 bits = 256 vectors     |
| k_icu_ier_bits_per_reg | Each IER controls 8 interrupt enables          |
| k_icu_ipr_count        | 256 priority registers (0-15 each)             |
| k_icu_dmrsr_count      | 8 DMAC channels                                |
| k_icu_irqcr_count      | 16 external IRQ pins (IRQ0-15)                 |
| k_icu_irqfte_count     | 2 filter enable registers (16 bits total)      |
| k_icu_irqftc_count     | 2 filter clock registers                       |
| k_icu_slibxr_count     | SLIBXRn entries: vectors 128-143 (INTB128-143) |
| k_icu_slibr_count      | SLIBRn entries: vectors 144-207 (INTB144-207)  |
| k_icu_sliar_count      | SLIARn entries: vectors 208-255 (INTA208-255)  |

Definition at line 238 of file `rx72n_icu_regs.h`.

```

00238 : uint16_t {
00239 k_icu_ir_count = 256, /**< 256 interrupt request flags (one per vector) */
00240 k_icu_dtcer_count = 256, /**< 256 DTC enable flags */
00241 k_icu_ier_count = 32, /**< 32 enable registers x 8 bits = 256 vectors */
00242 k_icu_ier_bits_per_reg = 8, /**< Each IER controls 8 interrupt enables */
00243 k_icu_ipr_count = 256, /**< 256 priority registers (0-15 each) */
00244 k_icu_dmrsr_count = 8, /**< 8 DMAC channels */
00245 k_icu_irqcr_count = 16, /**< 16 external IRQ pins (IRQ0-15) */
00246 k_icu_irqfte_count = 2, /**< 2 filter enable registers (16 bits total) */
00247 k_icu_irqftc_count = 2, /**< 2 filter clock registers */
00248 k_icu_slibxr_count = 16, /**< SLIBXRn entries: vectors 128-143 (INTB128-143) */
00249 k_icu_slibr_count = 64, /**< SLIBRn entries: vectors 144-207 (INTB144-207) */
00250 k_icu_sliar_count = 48, /**< SLIARn entries: vectors 208-255 (INTA208-255) */
00251 } icu_array_sizes_t;

```

#### 5.4.3.2 icu\_reserved\_sizes\_t

```
enum icu_reserved_sizes_t : uint16_t
```

ICU register reserved field sizes for structure padding.

Defines byte sizes for reserved regions in the ICU register map. These ensure proper alignment and offset matching between the C structure and actual hardware register locations.

## Warning

Do not modify these values - they are derived from the RX72N hardware register layout specification (Ch15).

## Enumerator

|                                     |                                            |
|-------------------------------------|--------------------------------------------|
| k_icu_reserved_after_ier_bytes      | 0x220-0x2DF: Gap to SWINTR                 |
| k_icu_reserved_after_swint2r_bytes  | 0x2E2-0x2EF: Gap to FIR                    |
| k_icu_reserved_after_fir_bytes      | 0x2F2-0x2FF: Gap to IPR                    |
| k_icu_reserved_after_dmrsr_bytes    | 0x420-0x4FF: Gap to IRQCR (was 248, FIXED) |
| k_icu_reserved_after_irqcr_bytes    | 0x510-0x51F: Gap to IRQFLTE                |
| k_icu_reserved_after_irqfte_bytes   | 0x522-0x527: Gap to IRQFLTC                |
| k_icu_reserved_after_irqftc_bytes   | 0x52C-0x57F: Gap to NMISR                  |
| k_icu_reserved_after_exnmiclr_bytes | 0x587-0x58F: Gap to NMIFLTE                |
| k_icu_reserved_after_nmiflte_bytes  | 0x591-0x593: Gap to NMIFLTC                |
| k_icu_reserved_after_nmifltc_bytes  | 0x595-0x77F: Gap to SLIBXRn                |
| k_icu_reserved_after_slibr_bytes    | 0x7D0-0x9CF: Gap to SLIARn                 |
| k_icu_dmrsr_entry_padding           | DMRSR 8-bit regs at 4-byte boundaries      |

Definition at line 207 of file rx72n\_icu\_regs.h.

```

00207 : uint16_t {
00208 k_icu_reserved_after_ier_bytes = 192, /**< 0x220-0x2DF: Gap to SWINTR */
00209 k_icu_reserved_after_swint2r_bytes = 14, /**< 0x2E2-0x2EF: Gap to FIR */
00210 k_icu_reserved_after_fir_bytes = 14, /**< 0x2F2-0x2FF: Gap to IPR */
00211 k_icu_reserved_after_dmrsr_bytes = 224, /**< 0x420-0x4FF: Gap to IRQCR (was 248, FIXED) */
00212 k_icu_reserved_after_irqcr_bytes = 16, /**< 0x510-0x51F: Gap to IRQFLTE */
00213 k_icu_reserved_after_irqfte_bytes = 6, /**< 0x522-0x527: Gap to IRQFLTC */
00214 k_icu_reserved_after_irqftc_bytes = 84, /**< 0x52C-0x57F: Gap to NMISR */
00215 k_icu_reserved_after_exnmiclr_bytes = 9, /**< 0x587-0x58F: Gap to NMIFLTE */
00216 k_icu_reserved_after_nmiflte_bytes = 3, /**< 0x591-0x593: Gap to NMIFLTC */
00217 k_icu_reserved_after_nmifltc_bytes = 491, /**< 0x595-0x77F: Gap to SLIBXRn */
00218 k_icu_reserved_after_slibr_bytes = 512, /**< 0x7D0-0x9CF: Gap to SLIARn */
00219 k_icu_dmrsr_entry_padding = 3, /**< DMRSR 8-bit regs at 4-byte boundaries */
00220 } icu_reserved_sizes_t;

```

## 5.4.3.3 rx\_cmt\_interrupt\_vector\_t

```
enum rx_cmt_interrupt_vector_t : uint8_t
```

CMT (Compare Match Timer) interrupt vector numbers.

Vector numbers for CMT peripheral interrupts. These are used to index into IR[], IER[], and IPR[] arrays.

IER/Bit Calculation:

- IER index = vector / 8
- IER bit = vector % 8

Example: CMT0 (vector 28)

- IER[3] (28/8=3), bit 4 (28%8=4)
- IPR[28] for priority

See also

RX72N Manual Ch15 Table 15.4 for complete vector list

Enumerator

|                  |                                                          |
|------------------|----------------------------------------------------------|
| k_vect_cmt0_cmi0 | CMT0 compare match: IER[3] bit 4, used for 1ms tick      |
| k_vect_cmt1_cmi1 | CMT1 compare match: IER[3] bit 5, used for motor control |

Definition at line 531 of file rx72n\_icu\_regs.h.

```
00531 : uint8_t {
00532 k_vect_cmt0_cmi0 = 28, /**< CMT0 compare match: IER[3] bit 4, used for 1ms tick */
00533 k_vect_cmt1_cmi1 = 29, /**< CMT1 compare match: IER[3] bit 5, used for motor control */
00534 } rx_cmt_interrupt_vector_t;
```

#### 5.4.3.4 rx\_icu\_addresses\_t

```
enum rx_icu_addresses_t : uintptr_t
```

ICU peripheral base address.

The ICU occupies a contiguous block of memory starting at 0x00087000. All ICU registers are accessed relative to this base address.

Address Space:

| Offset Range | Registers                         | Description                            |
|--------------|-----------------------------------|----------------------------------------|
| 0x000-0x0FF  | IR $\leftrightarrow$<br>[256]     | Interrupt Request Flags                |
| 0x100-0x1FF  | DTCER $\leftrightarrow$<br>[256]  | DTC Enable Registers                   |
| 0x200-0x21F  | IER $\leftrightarrow$<br>[32]     | Interrupt Enable Regs                  |
| 0x2E0-0x2E1  | SWINTR, SWINT2R                   | Software Interrupt                     |
| 0x2F0        | FIR                               | Fast Interrupt Register                |
| 0x300-0x3FF  | IPR $\leftrightarrow$<br>[256]    | Interrupt Priority Regs                |
| 0x400-0x41F  | DMRSR $\leftrightarrow$<br>[8]    | DMAC Module Start Regs                 |
| 0x500-0x50F  | IRQCR $\leftrightarrow$<br>[16]   | External IRQ Control                   |
| 0x520-0x52D  | IRQFLTE, IRQFLTC                  | IRQ Digital Filter                     |
| 0x580-0x594  | NMI Registers                     | Non-Maskable Interrupt                 |
| 0x780-0x78F  | SLIBXRn $\leftrightarrow$<br>[16] | Selectable Int B source (vect 128-143) |
| 0x790-0x7CF  | SLIBRn $\leftrightarrow$<br>[64]  | Selectable Int B source (vect 144-207) |

| Offset Range | Registers                    | Description                            |
|--------------|------------------------------|----------------------------------------|
| 0x9D0-0x9FF  | SLIAR <sub>n</sub> ↔<br>[48] | Selectable Int A source (vect 208-255) |
| 0xA00        | SLIPRCR                      | SLIBXR/SLIBR/SLIAR write-protect       |
| 0xA01        | SELEXDR                      | Selectable ext int direction           |

## Invariant

Base address is fixed and cannot be relocated

## Enumerator

|                       |                                          |
|-----------------------|------------------------------------------|
| k_icu_base_addr       | ICU register base address (fixed)        |
| k_icu_slibr_base_addr | SELECTB source-select table (n=144..207) |
| k_icu_sliar_base_addr | SELECTA source-select table (n=208..255) |

Definition at line 189 of file `rx72n_icu_regs.h`.

```

00189 : uintptr_t {
00190 k_icu_base_addr = 0x00087000, /**< ICU register base address (fixed) */
00191 k_icu_slibr_base_addr = 0x00087700, /**< SELECTB source-select table (n=144..207) */
00192 k_icu_sliar_base_addr = 0x00087900, /**< SELECTA source-select table (n=208..255) */
00193 } rx_icu_addresses_t;

```

## 5.4.3.5 rx\_interrupt\_priority\_t

```
enum rx_interrupt_priority_t : uint8_t
```

Interrupt priority levels for IPR registers.

Priority values for the IPR[n] registers. Higher values mean higher priority. A running interrupt can only be preempted by an interrupt with strictly higher priority.

Priority Guidelines for STAR Project:

| Priority | Usage                   | Example           |
|----------|-------------------------|-------------------|
| 15       | Reserved for NMI/faults | Hardware fault    |
| 12-14    | Time-critical motor     | Motor PWM sync    |
| 9-11     | Real-time control       | CMT tick, encoder |
| 6-8      | Communication           | USB, UART RX      |
| 3-5      | Background processing   | UART TX, DMA      |
| 1-2      | Low priority            | Diagnostics       |
| 0        | Disabled                | Unused interrupts |

## Warning

Setting two interrupts to the same priority is allowed but undefined which will be serviced first if both are pending

## Enumerator

|                     |                                          |
|---------------------|------------------------------------------|
| k_ipr_level_disable | Interrupt disabled (will not trigger)    |
| k_ipr_level_min     | Minimum enabled priority (lowest)        |
| k_ipr_level_max     | Maximum priority (highest, preempts all) |

Definition at line 582 of file [rx72n\\_icu\\_regs.h](#).

```
00582 : uint8_t {
00583 k_ipr_level_disable = 0, /**< Interrupt disabled (will not trigger) */
00584 k_ipr_level_min = 1, /**< Minimum enabled priority (lowest) */
00585 k_ipr_level_max = 15, /**< Maximum priority (highest, preempts all) */
00586 } rx_interrupt_priority_t;
```

## 5.4.3.6 rx\_sci\_interrupt\_vector\_t

```
enum rx_sci_interrupt_vector_t : uint8_t
```

SCI (Serial Communications Interface) interrupt vector numbers.

Vector numbers for SCI peripheral receive/transmit interrupts. Used to index into IR[], IER[], and IPR[] arrays per RX72N HUM Table 14.3.

IER/Bit Calculation:

- IER index = vector / 8
- IER bit = vector % 8

Example: SCI9 RXI9 (vector 102)

- IER[12] (102/8=12), bit 6 (102%8=6)
- IPR[102] for priority

See also

RX72N HUM Ch14 Table 14.3 for complete vector list

## Enumerator

|                  |                                                           |
|------------------|-----------------------------------------------------------|
| k_vect_sci9_rxi9 | SCI9 receive data full: IER[12] bit 6, STAR debug UART RX |
|------------------|-----------------------------------------------------------|

Definition at line 554 of file [rx72n\\_icu\\_regs.h](#).

```
00554 : uint8_t {
00555 k_vect_sci9_rxi9 = 102, /**< SCI9 receive data full: IER[12] bit 6, STAR debug UART RX */
00556 } rx_sci_interrupt_vector_t;
```

## 5.4.4 Function Documentation

### 5.4.4.1 icu()

```
volatile rx_icu_regs_t * icu (
 void) [inline], [static]
```

Get pointer to ICU registers.

Returns a volatile pointer to the ICU register structure, enabling direct hardware register access for interrupt configuration.

Returns

Volatile pointer to ICU register structure at 0x00087000

Precondition

No prerequisites - ICU is always accessible

Postcondition

Pointer is immediately usable for register access

No hardware state is modified by this function

Note

Thread Safety: Function is reentrant (returns constant address)

Performance: Zero overhead - compiles to single load instruction

Warning

Interrupt configuration should be done with global interrupts disabled to prevent race conditions

Example - Enable/Disable Specific Interrupt:

```
volatile rx_icu_regs_t* icu_regs = icu();

// Calculate IER index and bit for vector 28
uint8_t vector = 28;
uint8_t ier_idx = vector / 8; // = 3
uint8_t ier_bit = vector % 8; // = 4

// Enable interrupt
icu_regs->ier[ier_idx] |= (1U « ier_bit);

// Disable interrupt
icu_regs->ier[ier_idx] &= ~(1U « ier_bit);

// Check if interrupt is pending
if (icu_regs->ir[vector]) {
 // Interrupt is pending
 icu_regs->ir[vector] = 0; // Clear it
}
```

See also

[rx\\_icu\\_regs\\_t](#) Register structure definition

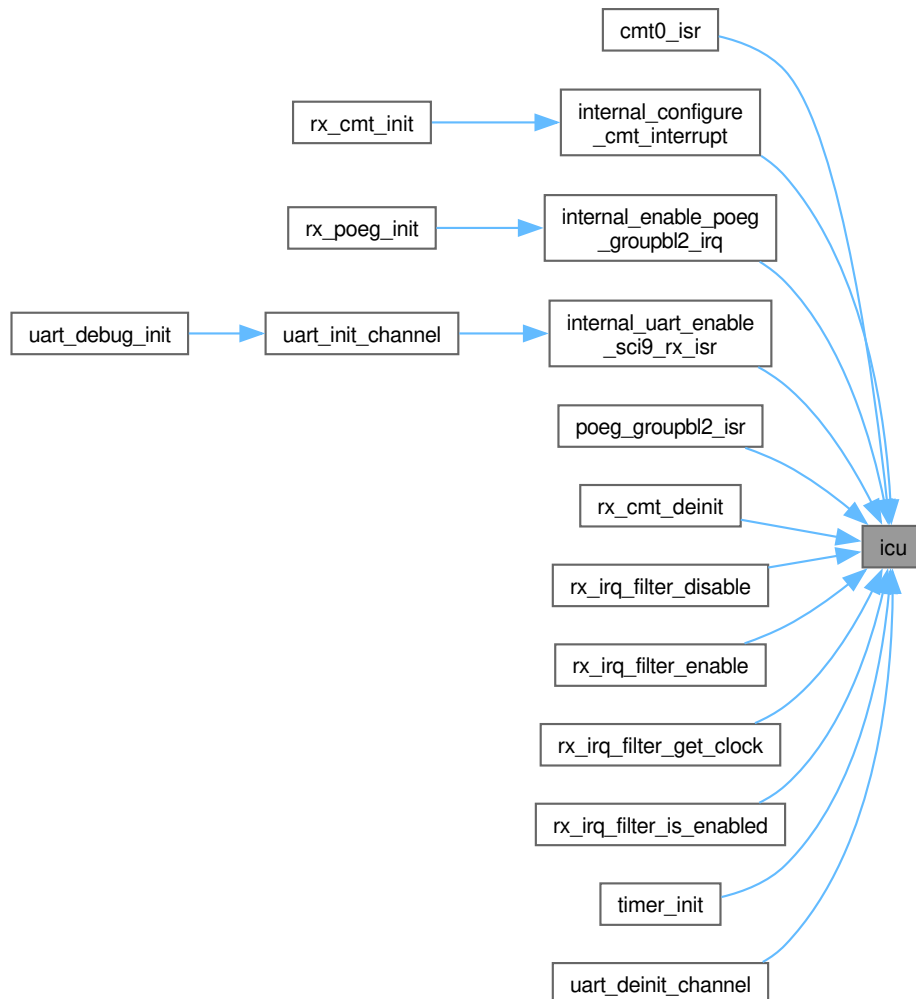
Definition at line 490 of file [rx72n\\_icu\\_regs.h](#).

```
00491 {
00492 return (volatile rx_icu_regs_t*)k_icu_base_addr;
00493 }
```

References [k\\_icu\\_base\\_addr](#).

Referenced by [cmt0\\_isr\(\)](#), [internal\\_configure\\_cmt\\_interrupt\(\)](#), [internal\\_enable\\_poeg\\_groupbl2\\_irq\(\)](#), [internal\\_uart\\_enable\\_sci9\\_rx\\_isr\(\)](#), [poeg\\_groupbl2\\_isr\(\)](#), [rx\\_cmt\\_deinit\(\)](#), [rx\\_irq\\_filter\\_disable\(\)](#), [rx\\_irq\\_filter\\_enable\(\)](#), [rx\\_irq\\_filter\\_get\\_clock\(\)](#), [rx\\_irq\\_filter\\_is\\_enabled\(\)](#), [timer\\_init\(\)](#), and [uart\\_deinit\\_channel\(\)](#).

Here is the caller graph for this function:



#### 5.4.4.2 icu\_slibr()

```
volatile uint8_t * icu_slibr (
 uint8_t vec) [inline], [static]
```



Get pointer to ICU SLIBR source-select register for a SELECTB vector.

SLIBR is a byte-wide table at 0x00087700 indexed by absolute vector number, covering SELECTB slots 144..207 (RX72N HW manual Ch15 Section 15.2 and hirakuni45/RX RX72N/icu.hpp line 824). Writing a SELECTB source code (e.g. `k_usb0_usbi_sli_src = 62`) into `SLIBR[vec]` routes that peripheral interrupt onto vec. Vectors outside 144..207 have no backing register – writes are ignored and reads return 0.

Parameters

|    |     |                                                        |
|----|-----|--------------------------------------------------------|
| in | vec | Absolute vector number in the SELECTB range (144..207) |
|----|-----|--------------------------------------------------------|

Returns

`volatile uint8_t*` Pointer to `SLIBR[vec]`

Definition at line 508 of file [rx72n\\_icu\\_regs.h](#).

```
00509 {
00510 return (volatile uint8_t*)(k_icu_slibr_base_addr + (uintptr_t)vec);
00511 }
```

References [k\\_icu\\_slibr\\_base\\_addr](#).

## 5.5 MPC Register Definitions

Multi-Function Pin Controller (MPC) hardware register interface.

Collaboration diagram for MPC Register Definitions:



Topics

- [MPC Static Assertions](#)  
Compile-time verification of MPC register layout.

Data Structures

- struct [rx\\_pfs\\_regs\\_t](#)  
Pin Function Select Register (C bitfield).
- struct [rx\\_mpc\\_regs\\_t](#)  
MPC Register Map.

## Enumerations

- enum `rx_mpc_addresses_t` : `uintptr_t` { `k_mpc_base_addr` = 0x0008C100 }  
MPC base address (verified against RX72N Hardware Manual Ch23).
- enum `mpc_register_offsets_t` : `uint8_t` {  
`k_mpc_offset_pwpr` = 0x1F , `k_mpc_offset_pfs` = 0x40 , `k_mpc_offset_port5_pfs` = 0x28 ,  
`k_mpc_offset_porta_pfs` = 0x50 ,  
`k_mpc_offset_porte_pfs` = 0x70 , `k_mpc_offset_portj_pfs` = 0x90 , `k_mpc_pfs_total_bytes` }  
Byte offsets of key MPC registers from the MPC base address (0x0008C100).
- enum `mpc_reserved_sizes_t` : `uint8_t` {  
`k_mpc_reserved1_bytes` = 1 , `k_mpc_reserved2_bytes` = 4 , `k_mpc_reserved3_bytes` = 16 ,  
`k_mpc_reserved4_bytes` = 8 ,  
`k_mpc_dscr2_count` = 24 , `k_port3_reserved_bytes` = 3 , `k_port5_reserved_bytes` = 1 ,  
`k_port6_reserved_bytes` = 1 ,  
`k_port7_reserved_bytes` = 1 , `k_portf_reserved1_bytes` = 2 , `k_portf_reserved2_bytes` = 2 ,  
`k_portj_reserved1_bytes` = 1 ,  
`k_portj_reserved2_bytes` = 2 }  
MPC register reserved field sizes.
- enum `mpc_pwpr_bits_t` : `uint8_t` { `k_mpc_pwpr_pfswe` = (1 << 6) , `k_mpc_pwpr_b0wi` = (1 << 7) }  
MPC Write Protect Register (PWPR) bit definitions.
- enum `pfs_bits_t` : `uint8_t` { `k_pfs_psel_mask` = 0x1F , `k_pfs_isel` = (1 << 6) , `k_pfs_asel` = (1 << 7) }  
Pin Function Select (PFS) register bit definitions.

## Functions

- static volatile `rx_mpc_regs_t` \* `mpc` (void)  
Get pointer to MPC registers.

### 5.5.1 Detailed Description

Multi-Function Pin Controller (MPC) hardware register interface.

### 5.5.2 Data Structure Documentation

#### 5.5.2.1 struct rx`pfs`regs`t

Pin Function Select Register (C bitfield).

Each pin has a PFS register controlling peripheral function selection.

This struct uses C bitfields (the : N syntax) to pack multiple fields into a single byte, matching the hardware register layout exactly:

Bit layout in hardware register (1 byte):

```

7 6 5 4 3 2 1 0
+-----+
| ASEL | ISEL | Reserved | PSEL[4:0] |
+-----+
```

The syntax `volatile uint8_t` name : bits defines a bitfield where:

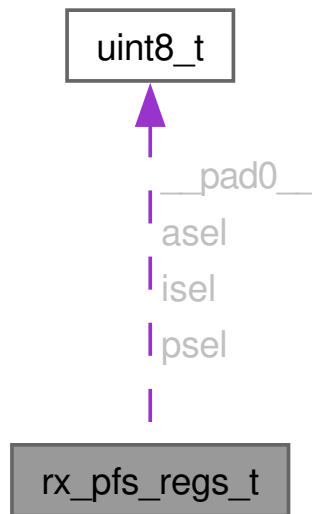
- name is the field name (or omitted for padding/reserved bits)
- bits is the number of bits the field occupies

Example: `psel : 5` means `psel` occupies 5 bits (bits 0-4)

This is standard C (not C++) and commonly used in embedded systems for hardware register definitions where bits are packed together.

Definition at line 234 of file `rx72n_mpc_regs.h`.

Collaboration diagram for `rx_pfs_regs_t`:



#### Data Fields

|                               |                            |                                                     |
|-------------------------------|----------------------------|-----------------------------------------------------|
| <code>uint8_t</code>          | <code>___pad0___: 1</code> | Reserved bit (bit 5) - unnamed bitfield for padding |
| <code>volatile uint8_t</code> | <code>asel: 1</code>       | Analog Input Select (bit 7)                         |
| <code>volatile uint8_t</code> | <code>isel: 1</code>       | Interrupt Input Select (bit 6)                      |
| <code>volatile uint8_t</code> | <code>psel: 5</code>       | Peripheral Select (bits 0-4)                        |

#### 5.5.2.2 struct `rx_mpc_regs_t`

MPC Register Map.

Multi-Function Pin Controller (MPC) registers for pin function selection. Controls which peripheral function each GPIO pin is assigned to. Base address: 0x0008C100

CRITICAL: This struct matches the exact hardware memory layout!

- 0x0008C100: Bus control registers (PFCSE, PFCSS, PFAOE, PFBCR, PFENET)
- 0x0008C11F: PWPR (Write Protect Register)
- 0x0008C140: PFS registers start (P00PFS)

Package Support (144-pin LFQFP R5F572NNHxFB):

- Available ports: 0-9 (partial), A-E (full), F (partial), J (partial: PJ3, PJ5)
- Not available on 144-pin: G, H
- Note: All ports are defined for compatibility with larger packages

Complete Memory Layout:

- 0x0008C100: PFCSE (CS Output Enable Register)
- 0x0008C101: Reserved (1 byte)
- 0x0008C102-0x0008C103: PFCSS0, PFCSS1 (CS Output Pin Select)
- 0x0008C104-0x0008C105: PFAOE0, PFAOE1 (Address Output Enable)
- 0x0008C106-0x0008C109: PFBCR0-3 (External Bus Control)
- 0x0008C10A-0x0008C10D: Reserved (4 bytes)
- 0x0008C10E: PFENET (Ethernet Control)
- 0x0008C10F-0x0008C11E: Reserved (16 bytes)
- 0x0008C11F: PWPR (Write Protect Register)
- 0x0008C120-0x0008C127: Reserved (8 bytes)
- 0x0008C128-0x0008C13F: DSCR2 (Drive Capacity Control, 24 bytes)
- 0x0008C140+: PFS registers (P00PFS through PJPFS, 152 bytes)

Definition at line [272](#) of file [rx72n\\_mpc\\_regs.h](#).

Collaboration diagram for `rx_mpc_regs_t`:



## Data Fields

|                  |                          |                                      |
|------------------|--------------------------|--------------------------------------|
| uint8_t          | dscr2[k_mpc_dscr2_count] | +0x28-0x3F: Drive Capacity Control 2 |
| volatile uint8_t | p00pfs                   | Port 0 Pin 0 Function Select         |
| volatile uint8_t | p01pfs                   | Port 0 Pin 1 Function Select         |
| volatile uint8_t | p02pfs                   | Port 0 Pin 2 Function Select         |
| volatile uint8_t | p03pfs                   | Port 0 Pin 3 Function Select         |
| volatile uint8_t | p04pfs                   | Port 0 Pin 4 Function Select         |
| volatile uint8_t | p05pfs                   | Port 0 Pin 5 Function Select         |
| volatile uint8_t | p06pfs                   | Port 0 Pin 6 Function Select         |
| volatile uint8_t | p07pfs                   | Port 0 Pin 7 Function Select         |
| volatile uint8_t | p10pfs                   | Port 1 Pin 0 Function Select         |
| volatile uint8_t | p11pfs                   | Port 1 Pin 1 Function Select         |
| volatile uint8_t | p12pfs                   | Port 1 Pin 2 Function Select         |
| volatile uint8_t | p13pfs                   | Port 1 Pin 3 Function Select         |
| volatile uint8_t | p14pfs                   | Port 1 Pin 4 Function Select         |
| volatile uint8_t | p15pfs                   | Port 1 Pin 5 Function Select         |
| volatile uint8_t | p16pfs                   | Port 1 Pin 6 Function Select         |
| volatile uint8_t | p17pfs                   | Port 1 Pin 7 Function Select         |

|                  |        |                              |
|------------------|--------|------------------------------|
| volatile uint8_t | p20pfs | Port 2 Pin 0 Function Select |
| volatile uint8_t | p21pfs | Port 2 Pin 1 Function Select |
| volatile uint8_t | p22pfs | Port 2 Pin 2 Function Select |
| volatile uint8_t | p23pfs | Port 2 Pin 3 Function Select |
| volatile uint8_t | p24pfs | Port 2 Pin 4 Function Select |
| volatile uint8_t | p25pfs | Port 2 Pin 5 Function Select |
| volatile uint8_t | p26pfs | Port 2 Pin 6 Function Select |
| volatile uint8_t | p27pfs | Port 2 Pin 7 Function Select |
| volatile uint8_t | p30pfs | Port 3 Pin 0 Function Select |
| volatile uint8_t | p31pfs | Port 3 Pin 1 Function Select |
| volatile uint8_t | p32pfs | Port 3 Pin 2 Function Select |
| volatile uint8_t | p33pfs | Port 3 Pin 3 Function Select |
| volatile uint8_t | p34pfs | Port 3 Pin 4 Function Select |
| volatile uint8_t | p40pfs | Port 4 Pin 0 Function Select |
| volatile uint8_t | p41pfs | Port 4 Pin 1 Function Select |
| volatile uint8_t | p42pfs | Port 4 Pin 2 Function Select |
| volatile uint8_t | p43pfs | Port 4 Pin 3 Function Select |
| volatile uint8_t | p44pfs | Port 4 Pin 4 Function Select |
| volatile uint8_t | p45pfs | Port 4 Pin 5 Function Select |
| volatile uint8_t | p46pfs | Port 4 Pin 6 Function Select |
| volatile uint8_t | p47pfs | Port 4 Pin 7 Function Select |
| volatile uint8_t | p50pfs | Port 5 Pin 0 Function Select |
| volatile uint8_t | p51pfs | Port 5 Pin 1 Function Select |
| volatile uint8_t | p52pfs | Port 5 Pin 2 Function Select |
| volatile uint8_t | p54pfs | Port 5 Pin 4 Function Select |
| volatile uint8_t | p55pfs | Port 5 Pin 5 Function Select |
| volatile uint8_t | p56pfs | Port 5 Pin 6 Function Select |
| volatile uint8_t | p57pfs | Port 5 Pin 7 Function Select |
| volatile uint8_t | p60pfs | Port 6 Pin 0 Function Select |
| volatile uint8_t | p61pfs | Port 6 Pin 1 Function Select |
| volatile uint8_t | p62pfs | Port 6 Pin 2 Function Select |
| volatile uint8_t | p63pfs | Port 6 Pin 3 Function Select |
| volatile uint8_t | p64pfs | Port 6 Pin 4 Function Select |
| volatile uint8_t | p66pfs | Port 6 Pin 6 Function Select |
| volatile uint8_t | p67pfs | Port 6 Pin 7 Function Select |
| volatile uint8_t | p71pfs | Port 7 Pin 1 Function Select |
| volatile uint8_t | p72pfs | Port 7 Pin 2 Function Select |
| volatile uint8_t | p73pfs | Port 7 Pin 3 Function Select |
| volatile uint8_t | p74pfs | Port 7 Pin 4 Function Select |
| volatile uint8_t | p75pfs | Port 7 Pin 5 Function Select |
| volatile uint8_t | p76pfs | Port 7 Pin 6 Function Select |
| volatile uint8_t | p77pfs | Port 7 Pin 7 Function Select |

|                  |        |                              |
|------------------|--------|------------------------------|
| volatile uint8_t | p80pfs | Port 8 Pin 0 Function Select |
| volatile uint8_t | p81pfs | Port 8 Pin 1 Function Select |
| volatile uint8_t | p82pfs | Port 8 Pin 2 Function Select |
| volatile uint8_t | p83pfs | Port 8 Pin 3 Function Select |
| volatile uint8_t | p84pfs | Port 8 Pin 4 Function Select |
| volatile uint8_t | p85pfs | Port 8 Pin 5 Function Select |
| volatile uint8_t | p86pfs | Port 8 Pin 6 Function Select |
| volatile uint8_t | p87pfs | Port 8 Pin 7 Function Select |
| volatile uint8_t | p90pfs | Port 9 Pin 0 Function Select |
| volatile uint8_t | p91pfs | Port 9 Pin 1 Function Select |
| volatile uint8_t | p92pfs | Port 9 Pin 2 Function Select |
| volatile uint8_t | p93pfs | Port 9 Pin 3 Function Select |
| volatile uint8_t | p94pfs | Port 9 Pin 4 Function Select |
| volatile uint8_t | p95pfs | Port 9 Pin 5 Function Select |
| volatile uint8_t | p96pfs | Port 9 Pin 6 Function Select |
| volatile uint8_t | p97pfs | Port 9 Pin 7 Function Select |
| volatile uint8_t | pa0pfs | Port A Pin 0 Function Select |
| volatile uint8_t | pa1pfs | Port A Pin 1 Function Select |
| volatile uint8_t | pa2pfs | Port A Pin 2 Function Select |
| volatile uint8_t | pa3pfs | Port A Pin 3 Function Select |
| volatile uint8_t | pa4pfs | Port A Pin 4 Function Select |
| volatile uint8_t | pa5pfs | Port A Pin 5 Function Select |
| volatile uint8_t | pa6pfs | Port A Pin 6 Function Select |
| volatile uint8_t | pa7pfs | Port A Pin 7 Function Select |
| volatile uint8_t | pb0pfs | Port B Pin 0 Function Select |
| volatile uint8_t | pb1pfs | Port B Pin 1 Function Select |
| volatile uint8_t | pb2pfs | Port B Pin 2 Function Select |
| volatile uint8_t | pb3pfs | Port B Pin 3 Function Select |
| volatile uint8_t | pb4pfs | Port B Pin 4 Function Select |
| volatile uint8_t | pb5pfs | Port B Pin 5 Function Select |
| volatile uint8_t | pb6pfs | Port B Pin 6 Function Select |
| volatile uint8_t | pb7pfs | Port B Pin 7 Function Select |
| volatile uint8_t | pc0pfs | Port C Pin 0 Function Select |
| volatile uint8_t | pc1pfs | Port C Pin 1 Function Select |
| volatile uint8_t | pc2pfs | Port C Pin 2 Function Select |
| volatile uint8_t | pc3pfs | Port C Pin 3 Function Select |
| volatile uint8_t | pc4pfs | Port C Pin 4 Function Select |
| volatile uint8_t | pc5pfs | Port C Pin 5 Function Select |
| volatile uint8_t | pc6pfs | Port C Pin 6 Function Select |
| volatile uint8_t | pc7pfs | Port C Pin 7 Function Select |
| volatile uint8_t | pd0pfs | Port D Pin 0 Function Select |
| volatile uint8_t | pd1pfs | Port D Pin 1 Function Select |

|                  |        |                                  |
|------------------|--------|----------------------------------|
| volatile uint8_t | pd2pfs | Port D Pin 2 Function Select     |
| volatile uint8_t | pd3pfs | Port D Pin 3 Function Select     |
| volatile uint8_t | pd4pfs | Port D Pin 4 Function Select     |
| volatile uint8_t | pd5pfs | Port D Pin 5 Function Select     |
| volatile uint8_t | pd6pfs | Port D Pin 6 Function Select     |
| volatile uint8_t | pd7pfs | Port D Pin 7 Function Select     |
| volatile uint8_t | pe0pfs | Port E Pin 0 Function Select     |
| volatile uint8_t | pe1pfs | Port E Pin 1 Function Select     |
| volatile uint8_t | pe2pfs | Port E Pin 2 Function Select     |
| volatile uint8_t | pe3pfs | Port E Pin 3 Function Select     |
| volatile uint8_t | pe4pfs | Port E Pin 4 Function Select     |
| volatile uint8_t | pe5pfs | Port E Pin 5 Function Select     |
| volatile uint8_t | pe6pfs | Port E Pin 6 Function Select     |
| volatile uint8_t | pe7pfs | Port E Pin 7 Function Select     |
| volatile uint8_t | pf0pfs | Port F Pin 0 Function Select     |
| volatile uint8_t | pf1pfs | Port F Pin 1 Function Select     |
| volatile uint8_t | pf2pfs | Port F Pin 2 Function Select     |
| volatile uint8_t | pf5pfs | Port F Pin 5 Function Select     |
| volatile uint8_t | pfaoe0 | +0x04: Address Output Enable 0   |
| volatile uint8_t | pfaoe1 | +0x05: Address Output Enable 1   |
| volatile uint8_t | pfbcr0 | +0x06: External Bus Control 0    |
| volatile uint8_t | pfbcr1 | +0x07: External Bus Control 1    |
| volatile uint8_t | pfbcr2 | +0x08: External Bus Control 2    |
| volatile uint8_t | pfbcr3 | +0x09: External Bus Control 3    |
| volatile uint8_t | pfcese | +0x00: CS Output Enable Register |
| volatile uint8_t | pfcss0 | +0x02: CS Output Pin Select 0    |
| volatile uint8_t | pfcss1 | +0x03: CS Output Pin Select 1    |
| volatile uint8_t | pfenet | +0x0E: Ethernet Control Register |
| volatile uint8_t | pg0pfs | Port G Pin 0 Function Select     |
| volatile uint8_t | pg1pfs | Port G Pin 1 Function Select     |
| volatile uint8_t | pg2pfs | Port G Pin 2 Function Select     |
| volatile uint8_t | pg3pfs | Port G Pin 3 Function Select     |
| volatile uint8_t | pg4pfs | Port G Pin 4 Function Select     |
| volatile uint8_t | pg5pfs | Port G Pin 5 Function Select     |
| volatile uint8_t | pg6pfs | Port G Pin 6 Function Select     |
| volatile uint8_t | pg7pfs | Port G Pin 7 Function Select     |
| volatile uint8_t | ph0pfs | Port H Pin 0 Function Select     |
| volatile uint8_t | ph1pfs | Port H Pin 1 Function Select     |
| volatile uint8_t | ph2pfs | Port H Pin 2 Function Select     |
| volatile uint8_t | ph3pfs | Port H Pin 3 Function Select     |
| volatile uint8_t | ph4pfs | Port H Pin 4 Function Select     |
| volatile uint8_t | ph5pfs | Port H Pin 5 Function Select     |



|                  |                                               |                                                        |
|------------------|-----------------------------------------------|--------------------------------------------------------|
| volatile uint8_t | ph6pfs                                        | Port H Pin 6 Function Select                           |
| volatile uint8_t | ph7pfs                                        | Port H Pin 7 Function Select                           |
| volatile uint8_t | pj0pfs                                        | Port J Pin 0 Function Select                           |
| volatile uint8_t | pj1pfs                                        | Port J Pin 1 Function Select                           |
| volatile uint8_t | pj2pfs                                        | Port J Pin 2 Function Select                           |
| volatile uint8_t | pj3pfs                                        | Port J Pin 3 Function Select                           |
| volatile uint8_t | pj5pfs                                        | Port J Pin 5 Function Select                           |
| uint8_t          | port3_reserved↔<br>[k_port3_reserved_bytes]   | Port 3 reserved (P35=NMI, P36-↔<br>P37 N/A)            |
| uint8_t          | port5_reserved↔<br>[k_port5_reserved_bytes]   | Port 5 reserved (P53=BCLK dedi-<br>cated)              |
| uint8_t          | port6_reserved↔<br>[k_port6_reserved_bytes]   | Port 6 reserved (P65=CKE dedi-<br>cated)               |
| uint8_t          | port7_reserved↔<br>[k_port7_reserved_bytes]   | Port 7 reserved (P70 N/A)                              |
| uint8_t          | portf_reserved1↔<br>[k_portf_reserved1_bytes] | Port F reserved (PF3-PF4 N/A)                          |
| uint8_t          | portf_reserved2↔<br>[k_portf_reserved2_bytes] | Port F reserved (PF6-PF7 N/A)                          |
| uint8_t          | portj_reserved1↔<br>[k_portj_reserved1_bytes] | Port J reserved (PJ4 N/A)                              |
| uint8_t          | portj_reserved2↔<br>[k_portj_reserved2_bytes] | Port J reserved (PJ6-PJ7 N/A)                          |
| volatile uint8_t | pwpr                                          | +0x1F: Write Protect Register (en-<br>able PFS writes) |
| uint8_t          | reserved1[k_mpc_reserved1_bytes]              | +0x01: Reserved                                        |
| uint8_t          | reserved2[k_mpc_reserved2_bytes]              | +0x0A-0x0D: Reserved                                   |
| uint8_t          | reserved3[k_mpc_reserved3_bytes]              | +0x0F-0x1E: Reserved                                   |
| uint8_t          | reserved4[k_mpc_reserved4_bytes]              | +0x20-0x27: Reserved                                   |

### 5.5.3 Enumeration Type Documentation

#### 5.5.3.1 mpc\_pwpr\_bits\_t

```
enum mpc_pwpr_bits_t : uint8_t
```

MPC Write Protect Register (PWPR) bit definitions.

The PWPR register protects PFS registers from accidental writes. A specific unlock sequence is required to modify any PFS register.

## PWPR Register Layout

| * Bit | Name  | Description                                 |
|-------|-------|---------------------------------------------|
| * 7   | B0WI  | PFSWE Bit Write Disable (1=PFSWE protected) |
| * 6   | PFSWE | PFS Write Enable (1=PFS writes allowed)     |
| * 5-0 | -     | Reserved                                    |
| *     |       |                                             |

## Unlock Sequence

1. Write 0x00 to PWPR (clear B0WI)
2. Write 0x40 to PWPR (set PFSWE)
3. Write to desired PFS register(s)
4. Write 0x00 to PWPR (clear PFSWE)
5. Write 0x80 to PWPR (set B0WI)

## Warning

Failing to re-lock after configuration leaves PFS vulnerable!

## Enumerator

|                  |                                              |
|------------------|----------------------------------------------|
| k_mpc_pwpr_pfswe | Bit 6: PFS Write Enable (1=allow writes)     |
| k_mpc_pwpr_b0wi  | Bit 7: PFSWE Write Disable (1=protect PFSWE) |

Definition at line 495 of file [rx72n\\_mpc\\_regs.h](#).

```

00495 : uint8_t {
00496 k_mpc_pwpr_pfswe = (1 « 6), /**< Bit 6: PFS Write Enable (1=allow writes) */
00497 k_mpc_pwpr_b0wi = (1 « 7), /**< Bit 7: PFSWE Write Disable (1=protect PFSWE) */
00498 } mpc_pwpr_bits_t;

```

## 5.5.3.2 mpc`register`offsets`t

```
enum mpc_register_offsets_t : uint8_t
```

Byte offsets of key MPC registers from the MPC base address (0x0008C100).

These constants are used in `static_assert` calls to verify the struct layout matches the hardware register map. Each constant encodes the exact byte offset from either the MPC base or the PFS block base (P00PFS), as noted.

## Note

All values verified against RX72N Manual Chapter 23, Section 23.2

Since

Version 1.0.0

Enumerator

|                        |                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------|
| k_mpc_offset_pwpr      | PWPR offset from MPC base (0x0008C11F)                                                      |
| k_mpc_offset_pfs       | PFS block start offset from MPC base (P00PFS = 0x0008C140)                                  |
| k_mpc_offset_port5_pfs | Port 5 PFS block offset from P00PFS                                                         |
| k_mpc_offset_porta_pfs | Port A PFS block offset from P00PFS                                                         |
| k_mpc_offset_porte_pfs | Port E PFS block offset from P00PFS                                                         |
| k_mpc_offset_portj_pfs | Port J PFS block offset from P00PFS                                                         |
| k_mpc_pfs_total_bytes  | Total size in bytes of the PFS register block (P00PFS through PJ5PFS with reserved padding) |

Definition at line 179 of file `rx72n_mpc_regs.h`.

```

00179 : uint8_t {
00180 k_mpc_offset_pwpr = 0x1F, /**< PWPR offset from MPC base (0x0008C11F) */
00181 k_mpc_offset_pfs = 0x40, /**< PFS block start offset from MPC base (P00PFS = 0x0008C140) */
00182 k_mpc_offset_port5_pfs = 0x28, /**< Port 5 PFS block offset from P00PFS */
00183 k_mpc_offset_porta_pfs = 0x50, /**< Port A PFS block offset from P00PFS */
00184 k_mpc_offset_porte_pfs = 0x70, /**< Port E PFS block offset from P00PFS */
00185 k_mpc_offset_portj_pfs = 0x90, /**< Port J PFS block offset from P00PFS */
00186 k_mpc_pfs_total_bytes =
00187 152, /**< Total size in bytes of the PFS register block (P00PFS through PJ5PFS with reserved padding) */
00188 } mpc_register_offsets_t;

```

### 5.5.3.3 mpc reserved sizes

```
enum mpc_reserved_sizes_t : uint8_t
```

MPC register reserved field sizes.

Enumerator

|                         |                                               |
|-------------------------|-----------------------------------------------|
| k_mpc_reserved1_bytes   | Reserved after PFCSE (0x0008C101)             |
| k_mpc_reserved2_bytes   | Reserved after PFBCR3 (0x0008C10A-0x0008C10D) |
| k_mpc_reserved3_bytes   | Reserved after PFENET (0x0008C10F-0x0008C11E) |
| k_mpc_reserved4_bytes   | Reserved after PWPR (0x0008C120-0x0008C127)   |
| k_mpc_dscr2_count       | DSCR2 registers count (0x0008C128-0x0008C13F) |
| k_port3_reserved_bytes  | Port 3: P35-P37 (P35 is NMI-only)             |
| k_port5_reserved_bytes  | Port 5: P53 (dedicated BCLK)                  |
| k_port6_reserved_bytes  | Port 6: P65 (dedicated CKE)                   |
| k_port7_reserved_bytes  | Port 7: P70 (does not exist)                  |
| k_portf_reserved1_bytes | Port F: PF3-PF4                               |
| k_portf_reserved2_bytes | Port F: PF6-PF7                               |
| k_portj_reserved1_bytes | Port J: PJ4                                   |
| k_portj_reserved2_bytes | Port J: PJ6-PJ7                               |

Definition at line 191 of file `rx72n_mpc_regs.h`.

```
00191 : uint8_t {
```

```

00192 k_mpc_reserved1_bytes = 1, /**< Reserved after PFCSE (0x0008C101) */
00193 k_mpc_reserved2_bytes = 4, /**< Reserved after PFBCR3 (0x0008C10A-0x0008C10D) */
00194 k_mpc_reserved3_bytes = 16, /**< Reserved after PFENET (0x0008C10F-0x0008C11E) */
00195 k_mpc_reserved4_bytes = 8, /**< Reserved after PWPR (0x0008C120-0x0008C127) */
00196 k_mpc_dscr2_count = 24, /**< DSCR2 registers count (0x0008C128-0x0008C13F) */
00197
00198 /* Port reserved byte counts (for ports with <8 pins) */
00199 k_port3_reserved_bytes = 3, /**< Port 3: P35-P37 (P35 is NMI-only) */
00200 k_port5_reserved_bytes = 1, /**< Port 5: P53 (dedicated BCLK) */
00201 k_port6_reserved_bytes = 1, /**< Port 6: P65 (dedicated CKE) */
00202 k_port7_reserved_bytes = 1, /**< Port 7: P70 (does not exist) */
00203 k_portf_reserved1_bytes = 2, /**< Port F: PF3-PF4 */
00204 k_portf_reserved2_bytes = 2, /**< Port F: PF6-PF7 */
00205 k_portj_reserved1_bytes = 1, /**< Port J: PJ4 */
00206 k_portj_reserved2_bytes = 2, /**< Port J: PJ6-PJ7 */
00207 } mpc_reserved_sizes_t;

```

#### 5.5.3.4 pfs\_bits\_t

```
enum pfs_bits_t : uint8_t
```

Pin Function Select (PFS) register bit definitions.

Each GPIO pin has a PFS register controlling its function assignment. The PSEL field selects which peripheral function the pin is connected to.

#### PFS Register Layout

```

* Bit | Name | Description
* ----+-----+-----
* 7 | ASEL | Analog Input Select (1=analog function)
* 6 | ISEL | IRQ Input Select (1=IRQ input enabled)
* 5 | Reserved | Always write 0
* 4-0 | PSEL[4:0] | Peripheral function select (0x00-0x1F)
*

```

#### Note

PSEL values are pin-specific. Check RX72N Manual Appendix for the function assignment table for each pin.

#### Enumerator

|                 |                                  |
|-----------------|----------------------------------|
| k_pfs_psel_mask | Bits 0-4: Peripheral Select mask |
| k_pfs_isel      | Bit 6: Interrupt Input Select    |
| k_pfs_asel      | Bit 7: Analog Input Select       |

Definition at line 521 of file [rx72n\\_mpc\\_regs.h](#).

```

00521 : uint8_t {
00522 k_pfs_psel_mask = 0x1F, /**< Bits 0-4: Peripheral Select mask */
00523 k_pfs_isel = (1 « 6), /**< Bit 6: Interrupt Input Select */
00524 k_pfs_asel = (1 « 7), /**< Bit 7: Analog Input Select */
00525 } pfs_bits_t;

```

## 5.5.3.5 rx\_mpc\_addresses\_t

enum `rx_mpc_addresses_t` : `uintptr_t`

MPC base address (verified against RX72N Hardware Manual Ch23).

The MPC module is located at a single base address. All registers are accessed relative to this base.

Enumerator

|                              |                           |
|------------------------------|---------------------------|
| <code>k_mpc_base_addr</code> | MPC register base address |
|------------------------------|---------------------------|

Definition at line 163 of file `rx72n_mpc_regs.h`.

```
00163 : uintptr_t {
00164 k_mpc_base_addr = 0x0008C100, /**< MPC register base address */
00165 } rx_mpc_addresses_t;
```

## 5.5.4 Function Documentation

## 5.5.4.1 mpc()

volatile `rx_mpc_regs_t` \* `mpc` (  
void ) [inline], [static]

Get pointer to MPC registers.

Returns a volatile pointer to the MPC register structure at address 0x0008C100. Provides access to all MPC registers including PWPR and PFS registers.

Returns

Volatile pointer to MPC register structure

Example

```
// Unlock, configure pin, re-lock
mpc()->pwpr = 0x00; // Clear B0WI
mpc()->pwpr = k_mpc_pwpr_pfswe; // Enable PFS write
mpc()->pb7pfs = 0x0A; // SC19 TXD
mpc()->pwpr = 0x00; // Clear PFSWE
mpc()->pwpr = k_mpc_pwpr_b0wi; // Protect PFSWE
```

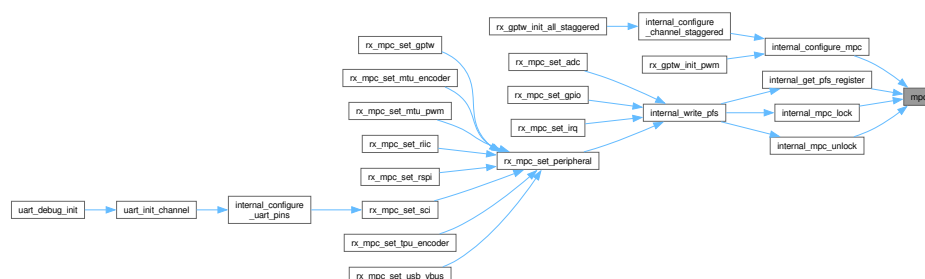
Definition at line 464 of file `rx72n_mpc_regs.h`.

```
00465 {
00466 return (volatile rx_mpc_regs_t*)k_mpc_base_addr;
00467 }
```

References `k_mpc_base_addr`.

Referenced by `internal_configure_mpc()`, `internal_get_pfs_register()`, `internal_mpc_lock()`, and `internal_mpc_unlock()`.

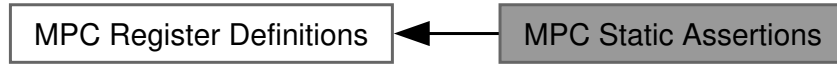
Here is the caller graph for this function:



### 5.5.5 MPC Static Assertions

Compile-time verification of MPC register layout.

Collaboration diagram for MPC Static Assertions:



Compile-time verification of MPC register layout.

These static assertions are verified at compile time to ensure the C structure definitions match the actual hardware register layout. A failed assertion indicates a bug in the register definitions that must be fixed before the code can compile.

## 5.6 MTU Register Definitions

### Data Structures

- struct [rx\\_mtu\\_channel\\_regs\\_t](#)  
Standard MTU channel register map (MTU0-MTU2).
- struct [rx\\_mtu3\\_channel\\_regs\\_t](#)  
MTU3 register map accessed via base address 0x000C1200.
- struct [rx\\_mtu12\\_phase\\_regs\\_t](#)  
MTU1/MTU2 extended register map for 32-bit phase counting mode.
- struct [rx\\_mtu\\_tstr\\_regs\\_t](#)  
MTU Start Register Map.

### Enumerations

- enum [rx\\_mtu\\_addresses\\_t](#) : [uintptr\\_t](#) {  
[k\\_mtu0\\_base\\_addr](#) = 0x000C1300 , [k\\_mtu1\\_base\\_addr](#) = 0x000C1380 , [k\\_mtu2\\_base\\_addr](#) = 0x000C1400 , [k\\_mtu3\\_base\\_addr](#) = 0x000C1200 ,  
[k\\_mtu4\\_base\\_addr](#) = ([k\\_mtu3\\_base\\_addr](#) + 1U) , [k\\_mtu5u\\_base\\_addr](#) = 0x000C1C80 ,  
[k\\_mtu5v\\_base\\_addr](#) = 0x000C1C90 , [k\\_mtu5w\\_base\\_addr](#) = 0x000C1CA0 ,  
[k\\_mtu6\\_base\\_addr](#) = 0x000C1A00 , [k\\_mtu7\\_base\\_addr](#) = ([k\\_mtu6\\_base\\_addr](#) + 1U) ,  
[k\\_mtu8\\_base\\_addr](#) = 0x000C1600 , [k\\_mtu\\_tstra\\_base\\_addr](#) = 0x000C1280 ,  
[k\\_mtu\\_tstrb\\_base\\_addr](#) = 0x000C1A80 }  
 MTU3a peripheral base addresses (RX72N memory map Chapter 4).
- enum [mtu\\_bit\\_values\\_t](#) : [uint8\\_t](#) { [k\\_mtu\\_bit\\_value\\_set](#) = 1 }  
 Bit value constants for shift operations.
- enum [mtu3\\_padding\\_sizes\\_t](#) : [uint8\\_t](#) {  
[k\\_mtu3\\_pad\\_after\\_tiorl](#) = 2U , [k\\_mtu3\\_pad\\_after\\_tier](#) = 7U , [k\\_mtu3\\_pad\\_after\\_tcnt](#) = 6U ,  
[k\\_mtu3\\_pad\\_after\\_tgrb](#) = 8U ,  
[k\\_mtu3\\_pad\\_after\\_tgrd](#) = 4U , [k\\_mtu3\\_pad\\_after\\_tsr](#) = 11U , [k\\_mtu3\\_pad\\_after\\_tbtm](#) = 19U , [k\\_mtu3\\_pad\\_after\\_tcr2](#) = 37U }  
 Named padding array sizes for [rx\\_mtu3\\_channel\\_regs\\_t](#) reserved fields.
- enum [mtu12\\_phase\\_padding\\_sizes\\_t](#) : [uint8\\_t](#) { [k\\_mtu12p\\_pad\\_before\\_tcntlw](#) = 14U }  
 Named padding array sizes for [rx\\_mtu12\\_phase\\_regs\\_t](#) reserved fields.

- enum `mtu_tcr_bits_t` : `uint8_t` {  
`k_mtu_tcr_cclr_shift` = 5 , `k_mtu_tcr_tpssc_mask` = 0x07 , `k_mtu_tcr_ckege_mask` = 0x18 ,  
`k_mtu_tcr_cclr_mask` = 0xE0 ,  
`k_mtu_tcr_tpssc_1` = 0x00 , `k_mtu_tcr_tpssc_4` = 0x01 , `k_mtu_tcr_tpssc_16` = 0x02 ,  
`k_mtu_tcr_tpssc_64` = 0x03 ,  
`k_mtu_tcr_cclr_tgra` = (`k_mtu_bit_value_set` << `k_mtu_tcr_cclr_shift`) }

Timer Control Register (TCR) bit field definitions.

- enum `mtu_tmdr_bits_t` : `uint8_t` {  
`k_mtu_tmdr_bfa_shift` = 4 , `k_mtu_tmdr_bfb_shift` = 5 , `k_mtu_tmdr_md_mask` = 0x0F ,  
`k_mtu_tmdr_md_normal` = 0x00 ,  
`k_mtu_tmdr_md_pwm1` = 0x02 , `k_mtu_tmdr_md_pwm2` = 0x03 , `k_mtu_tmdr_bfa` = (`k_mtu_bit_value_set` << `k_mtu_tmdr_bfa_shift`) , `k_mtu_tmdr_bfb` = (`k_mtu_bit_value_set` << `k_mtu_tmdr_bfb_shift`) }

Timer Mode Register (TMDR) bit field definitions.

- enum `mtu_tmdr3_bits_t` : `uint8_t` { `k_mtu_tmdr3_lwa_shift` = 0 , `k_mtu_tmdr3_phcksel_shift` = 1 , `k_mtu_tmdr3_lwa` , `k_mtu_tmdr3_phcksel` }

Timer Mode Register 3 (TMDR3) bits for MTU1/MTU2 32-bit cascaded mode.

- enum `mtu_tior_bits_t` : `uint8_t` {  
`k_mtu_tior_ioa_mask` = 0x0F , `k_mtu_tior_iob_mask` = 0xF0 , `k_mtu_tior_init_low` = 0x02 ,  
`k_mtu_tior_init_high` = 0x05 ,  
`k_mtu_tior_toggle` = 0x03 }

Timer I/O Control Register (TIOR) bit field definitions.

- enum `mtu_tstr_bits_t` : `uint8_t` {  
`k_mtu_tstr_cst0_shift` = 0 , `k_mtu_tstr_cst1_shift` = 1 , `k_mtu_tstr_cst2_shift` = 2 ,  
`k_mtu_tstr_cst8_shift` = 3 ,  
`k_mtu_tstr_cst3_shift` = 6 , `k_mtu_tstr_cst4_shift` = 7 , `k_mtu_tstr_cst0` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst0_shift`) , `k_mtu_tstr_cst1` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst1_shift`) ,  
`k_mtu_tstr_cst2` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst2_shift`) , `k_mtu_tstr_cst3` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst3_shift`) , `k_mtu_tstr_cst4` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst4_shift`) , `k_mtu_tstr_cst8` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst8_shift`) }

Timer Start Register A (TSTRA) bit definitions - controls MTU0-4, MTU8.

- enum `mtu_tstrb_bits_t` : `uint8_t` { `k_mtu_tstr_cst6_shift` = 6 , `k_mtu_tstr_cst7_shift` = 7 ,  
`k_mtu_tstr_cst6` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst6_shift`) , `k_mtu_tstr_cst7` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst7_shift`) }

Timer Start Register B (TSTRB) bit definitions - controls MTU6-7.

- enum `mtu_channel_reg_offsets_t` : `uint8_t` { `k_mtu_ch_off_tcr` = 0x00U , `k_mtu_ch_off_tcncnt` = 0x06U , `k_mtu_ch_off_tgra` = 0x08U , `k_mtu_ch_off_tgrb` = 0x0AU }

Byte offsets of registers within `rx_mtu_channel_regs_t`.

- enum `mtu3_reg_offsets_t` : `uint8_t` {  
`k_mtu3_off_tcr` = 0x00U , `k_mtu3_off_tmdr1` = 0x02U , `k_mtu3_off_tiorh` = 0x04U ,  
`k_mtu3_off_tiorl` = 0x05U ,  
`k_mtu3_off_tier` = 0x08U , `k_mtu3_off_tcncnt` = 0x10U , `k_mtu3_off_tgra` = 0x18U ,  
`k_mtu3_off_tgrb` = 0x1AU ,  
`k_mtu3_off_tgrc` = 0x24U , `k_mtu3_off_tgrd` = 0x26U , `k_mtu3_off_tsr` = 0x2CU ,  
`k_mtu3_off_tbtm` = 0x38U ,  
`k_mtu3_off_tcr2` = 0x4CU , `k_mtu3_off_tgre` = 0x72U , `k_mtu3_struct_size` = 0x74U }

Byte offsets of MTU3 registers from base 0x000C1200.

- enum `mtu12_phase_reg_offsets_t` : `uint8_t` {  
`k_mtu12p_off_tcr` = 0x00U , `k_mtu12p_off_tmdr` = 0x01U , `k_mtu12p_off_tcncnt` = 0x06U ,  
`k_mtu12p_off_tgra` = 0x08U ,  
`k_mtu12p_off_tgrb` = 0x0AU , `k_mtu12p_off_tmdr3` = 0x11U , `k_mtu12p_off_tcncntlw` = 0x20U ,  
`k_mtu12p_off_tgralw` = 0x24U ,  
`k_mtu12p_off_tgrblw` = 0x28U , `k_mtu12p_struct_size` = 0x2CU }

Byte offsets of registers within `rx_mtu12_phase_regs_t`.

- enum `mtu_periph_space_t` : `uintptr_t` { `k_mtu_periph_space_mask` = `0xFFFF0000U` , `k_mtu_periph_space_base` = `0x000C0000U` }

Address space mask and expected base for MTU peripheral verification.

## Functions

- static volatile `rx_mtu_channel_regs_t` \* `mtu0` (void)  
Get pointer to MTU0 registers.
- static volatile `rx_mtu_channel_regs_t` \* `mtu1` (void)  
Get pointer to MTU1 registers (16-bit mode).
- static volatile `rx_mtu_channel_regs_t` \* `mtu2` (void)  
Get pointer to MTU2 registers (16-bit mode).
- static volatile `rx_mtu12_phase_regs_t` \* `mtu1_phase` (void)  
Get pointer to MTU1 extended registers for 32-bit phase counting mode.
- static volatile `rx_mtu12_phase_regs_t` \* `mtu2_phase` (void)  
Get pointer to MTU2 extended registers for 32-bit phase counting mode.
- static volatile `rx_mtu3_channel_regs_t` \* `mtu3` (void)  
Get pointer to MTU3 registers via `rx_mtu3_channel_regs_t` struct.
- static volatile void \* `mtu4` (void)  
Get raw base address pointer for MTU4 register access.
- static volatile `rx_mtu3_channel_regs_t` \* `mtu6` (void)  
Get pointer to MTU6 registers (interleaved layout, identical to MTU3).
- static volatile void \* `mtu7` (void)  
Get raw base address pointer for MTU7 register access.
- static volatile `rx_mtu_tstr_regs_t` \* `mtu_tstra` (void)  
Get pointer to MTU TSTRA registers (MTU0-4, MTU8).
- static volatile `rx_mtu_tstr_regs_t` \* `mtu_tstrb` (void)  
Get pointer to MTU TSTRB registers (MTU6-7).

### 5.6.1 Detailed Description

### 5.6.2 Data Structure Documentation

#### 5.6.2.1 struct rx\_mtu\_channel\_regs\_t

Standard MTU channel register map (MTU0-MTU2).

Memory-mapped register structure for consecutive-layout 16-bit MTU channels. This structure applies to MTU0, MTU1, and MTU2 only. MTU6 and MTU7 have an interleaved register layout identical to MTU3/MTU4 and must be accessed via `mtu6()` which returns `rx_mtu3_channel_regs_t*`.

Register Memory Layout (18 bytes total)

| * Offset | Size | Register | Description                                   |
|----------|------|----------|-----------------------------------------------|
| * 0x00   | 1    | TCR      | Timer Control (prescaler, edge, clear source) |
| * 0x01   | 1    | TMDR     | Timer Mode (normal, PWM1, PWM2, phase count)  |
| * 0x02   | 1    | TIORH    | Timer I/O Control H (TIOCA/TIOCB output mode) |
| * 0x03   | 1    | TIORL    | Timer I/O Control L (TIOCC/TIOCD output mode) |
| * 0x04   | 1    | TIER     | Timer Interrupt Enable (TGIA-D, overflow)     |
| * 0x05   | 1    | TSR      | Timer Status (interrupt flags, read to clear) |
| * 0x06   | 2    | TCNT     | Timer Counter (16-bit, counts up/down)        |
| * 0x08   | 2    | TGRA     | Timer General Register A (compare/capture)    |
| * 0x0A   | 2    | TGRB     | Timer General Register B (compare/capture)    |
| * 0x0C   | 2    | TGRC     | Timer General Register C (compare/capture)    |
| * 0x0E   | 2    | TGRD     | Timer General Register D (compare/capture)    |
| *        |      |          |                                               |



### Timer Operating Modes

The TMDR register MD bits select the operating mode:

- Normal (0x00): Free-running counter with compare match
- PWM Mode 1 (0x02): TGRA=period, TGRB=duty cycle
- PWM Mode 2 (0x03): TGRC/TGRD as buffer registers
- Phase Counting (0x04-0x07): Quadrature encoder input

### PWM Generation Example

```
// Configure MTU0 for 20 kHz PWM, 50% duty cycle
volatile rx_mtu_channel_regs_t* mtu = mtu0();

// Stop timer during configuration
mtu->tstra()->tstr &= ~k_mtu_tstr_cst0;

// Configure prescaler and clear source
mtu->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra;

// Set PWM mode 1
mtu->tmdr = k_mtu_tmdr_md_pwm1;

// Configure output: low initially, high on compare match
mtu->tiorh = k_mtu_tior_init_low;

// Set period and duty cycle (120 MHz / 6000 = 20 kHz)
mtu->tgra = 5999; // Period
mtu->tgrb = 3000; // 50% duty (high for 3000 counts)

// Clear counter and start timer
mtu->tcnt = 0;
mtu_tstra()->tstr |= k_mtu_tstr_cst0;
```

### Invariant

Structure size must be exactly 18 bytes (0x10 offset)

All registers are volatile for hardware access

See also

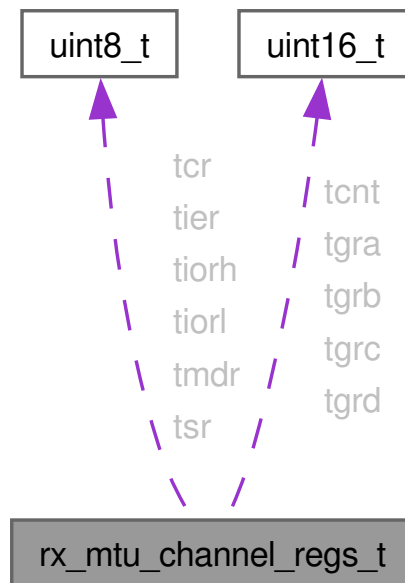
[rx\\_mtu34\\_channel\\_regs\\_t](#) for extended MTU3/MTU4 with TGR E/F

[rx\\_mtu12\\_phase\\_regs\\_t](#) for 32-bit phase counting mode

RX72N Hardware Manual Section 24.2 (Register Descriptions)

Definition at line 328 of file [rx72n\\_mtu\\_regs.h](#).

Collaboration diagram for [rx\\_mtu\\_channel\\_regs\\_t](#):



## Data Fields

|                   |       |                                                                                                                                                                                                            |
|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | tcnt  | Timer Counter (TCNT) - 16-bit counter value.<br>Current counter value. Cleared by compare match or overflow depending on TCR.CCLR setting.                                                                 |
| volatile uint8_t  | tcr   | Timer Control Register (TCR) - prescaler, edge, clear source.<br>Controls clock source selection and counter clear conditions.<br><br>See also<br><a href="#">mtu_tcr_bits_t</a> for bit field definitions |
| volatile uint16_t | tgra  | Timer General Register A (TGRA) - compare/capture.<br>In PWM mode: defines period. In capture mode: stores captured value.                                                                                 |
| volatile uint16_t | tgrb  | Timer General Register B (TGRB) - compare/capture.<br>In PWM mode: defines duty cycle (when to toggle output).                                                                                             |
| volatile uint16_t | tgrc  | Timer General Register C (TGRC) - compare/capture.<br>Additional compare register; can buffer TGRA in some modes.                                                                                          |
| volatile uint16_t | tgrd  | Timer General Register D (TGRD) - compare/capture.<br>Additional compare register; can buffer TGRB in some modes.                                                                                          |
| volatile uint8_t  | tier  | Timer Interrupt Enable Register (TIER).<br>Enables interrupts for compare match and overflow events.                                                                                                       |
| volatile uint8_t  | tiorh | Timer I/O Control Register H - TIOCA/TIOCB output control.<br>Controls output pin behavior on compare match A/B.<br><br>See also<br><a href="#">mtu_tior_bits_t</a> for output mode values                 |
| volatile uint8_t  | tiorl | Timer I/O Control Register L - TIOCC/TIOCD output control.<br>Controls output pin behavior on compare match C/D.<br><br>See also<br><a href="#">mtu_tior_bits_t</a> for output mode values                 |

|                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | tmdr | <p>Timer Mode Register (TMDR) - operating mode selection. Selects normal, PWM, or phase counting mode.</p> <p>See also</p> <p><a href="#">mtu_tmdr_bits_t</a> for mode values</p>                                                                                                                                                                                                                                                                                                                                          |
| volatile uint8_t | tsr  | <p>Timer Status Register (TSR) - interrupt flags. Contains interrupt flags; read to check, write 0 to clear.</p> <p>Warning</p> <p>Per RX72N HW manual R01UH0824EJ0111 Section 21.2 (MTU register tables), MTU0 has NO TSR at this offset – accessing this field via the MTU0 base (0x000C1300) targets a reserved byte and the result is undefined. This field is only valid on MTU channels that define a TSR (e.g., MTU1, MTU2, MTU3/4 TSR copies); callers must gate access by channel before reading/writing tsr.</p> |

### 5.6.2.2 struct rx\_mtu3\_channel\_regs\_t

MTU3 register map accessed via base address 0x000C1200.

Memory-mapped register structure for MTU3 channel, which supports complementary PWM generation for H-bridge motor driver control along with standard timer/counter and phase-counting encoder modes.

The MTU3/MTU4 peripheral block places MTU3 and MTU4 registers at INTERLEAVED addresses starting at 0x000C1200. MTU3 registers are NOT at consecutive byte offsets – there are large gaps between them where MTU4-specific registers and shared MTU3/4 control registers reside.

#### Warning

This struct is ONLY valid for MTU3 ([mtu3\(\)](#) base 0x000C1200). MTU4 registers have DIFFERENT offsets relative to the MTU4 base address (0x000C1201) and CANNOT use this struct. For MTU4, compute each register address individually using the Ch 5 I/O address table (e.g. MTU4.TIORH = 0x000C1206, not base+0x04). See RX72N Hardware Manual Ch 5 I/O Register Address Table.

Register Memory Layout (Ch 5 I/O Register Address Table, verified)

| * Offset    | * Size | * Register      | * Address    | * Description                                 |
|-------------|--------|-----------------|--------------|-----------------------------------------------|
| * 0x00      | * 1    | * tcr           | * 0x000C1200 | * Timer Control (prescaler, clear source)     |
| * 0x01      | * -    | * [MTU4.TCR]    |              | * (reserved -- do not access via this struct) |
| * 0x02      | * 1    | * tmdr1         | * 0x000C1202 | * Timer Mode Register 1 (normal/PWM/phase)    |
| * 0x03      | * -    | * [MTU4.TMDR1]  |              | * (reserved)                                  |
| * 0x04      | * 1    | * tiorh         | * 0x000C1204 | * Timer I/O Control H (TIOCA/TIOCB output)    |
| * 0x05      | * 1    | * tiorl         | * 0x000C1205 | * Timer I/O Control L (TIOCC/TIOCD output)    |
| * 0x06      | * -    | * [MTU4.TIORH]  |              | * (reserved)                                  |
| * 0x07      | * -    | * [MTU4.TIORL]  |              | * (reserved)                                  |
| * 0x08      | * 1    | * tier          | * 0x000C1208 | * Timer Interrupt Enable                      |
| * 0x09-0x0F | * -    | * [shared/MTU4] |              | * (reserved -- TOERA, TGCRA, TOCR1A, TOCR2A)  |
| * 0x10      | * 2    | * tcnt          | * 0x000C1210 | * Timer Counter (16-bit)                      |
| * 0x12-0x17 | * -    | * [shared/MTU4] |              | * (reserved -- MTU4.TCNT, TCDRA, TDDRA)       |
| * 0x18      | * 2    | * tgra          | * 0x000C1218 | * Timer General Register A (period/capture)   |
| * 0x1A      | * 2    | * tgrb          | * 0x000C121A | * Timer General Register B (duty/capture)     |
| * 0x1C-0x23 | * -    | * [shared/MTU4] |              | * (reserved -- MTU4.TGRA/B, TCNTSA, TCBRA)    |
| * 0x24      | * 2    | * tgrc          | * 0x000C1224 | * Timer General Register C (buffer/capture)   |
| * 0x26      | * 2    | * tgrd          | * 0x000C1226 | * Timer General Register D (buffer/capture)   |

```

* 0x28-0x2B - [MTU4.TGRC/D] (reserved)
* 0x2C 1 tsr 0x000C122C Timer Status Register (interrupt flags)
* 0x2D-0x37 - [shared/MTU4] (reserved -- MTU4.TSR, TITCR1A, TBTERA, etc.)
* 0x38 1 tbtm 0x000C1238 Timer Buffer Transfer Mode
* 0x39-0x4B - [shared/MTU4] (reserved -- MTU4.TBTM, TITMRA, MTU4.TADCR etc.)
* 0x4C 1 tcr2 0x000C124C Timer Control Register 2 (phase count edge)
* 0x4D-0x71 - [shared/MTU4] (reserved -- MTU4.TCR2, TWCRA, TMDR2A, etc.)
* 0x72 2 tgre 0x000C1272 Timer General Register E (dead-time/extended)
*

```

#### Note

MTU3 has NO TGRF register. Only MTU4 has TGRF (at 0x000C1276).

#### Dead-Time Insertion via TGRE

TGRE provides dead-time values to prevent H-bridge shoot-through. MTU4.TGRE and MTU4.TGRF must be set separately using direct address access.

$$t_{dead} = \frac{TGRE}{f_{PCLKA}} = \frac{TGRE}{120 \text{ MHz}}$$

Example: 500ns dead-time requires TGRE = 60 (at 120 MHz PCLKA)

#### Complementary PWM Example

```

// Configure MTU3/4 for complementary PWM with 500ns dead-time
volatile rx_mtu3_channel_regs_t* mtu = mtu3();

// Stop both timers
mtu->tstra()->tstr &= ~(k_mtu_tstr_cst3 | k_mtu_tstr_cst4);

// Configure for complementary PWM mode (via TMDR1)
mtu->tmdr1 = 0x0D; // Complementary PWM mode

// Set period (TGRA) and duty cycle (TGRB)
mtu->tgra = 5999; // 20 kHz period at 120 MHz PCLKA
mtu->tgrb = 3000; // 50% duty cycle

// Set dead-time for MTU3 (TGRE at 0x000C1272)
mtu->tgre = 60; // 500ns at 120 MHz
// MTU4.TGRF must be set via direct address (MTU4 uses different struct offsets):
(volatile uint16_t)0x000C1276U = 60; // MTU4.TGRF = 500ns dead-time

// Enable buffer transfer on counter clear
mtu->tbtm = 0x03;

// Start both timers synchronously
mtu_tstra()->tstr |= (k_mtu_tstr_cst3 | k_mtu_tstr_cst4);

```

#### Invariant

Struct size is 0x74 bytes (116) covering MTU3 registers up to TGRE

Only valid when cast from `mtu3()` base address (0x000C1200)

MTU4 registers CANNOT use this struct (different relative offsets)

See also

[rx\\_mtu\\_channel\\_regs\\_t](#) for MTU0/1/2 standard (consecutive) channel struct  
[mtu6\(\)](#) for MTU6 access using this interleaved struct  
 RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a registers)  
 RX72N Hardware Manual Section 24.2 (MTU3/MTU4 Register Descriptions)

Definition at line 552 of file [rx72n\\_mtu\\_regs.h](#).

Collaboration diagram for `rx_mtu3_channel_regs_t`:



#### Data Fields

|                  |                                                          |                                                                                                                       |
|------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | reserved_01                                              | Reserved – MTU4.TCR at this address (+0x01 = 0x000C1201).                                                             |
| volatile uint8_t | reserved_03                                              | Reserved – MTU4.TMDR1 at this address (+0x03 = 0x000C1203).                                                           |
| volatile uint8_t | reserved_06↔<br><a href="#">[k_mtu3_pad_after_tiorl]</a> | Reserved – MTU4.TIORH and MTU4.TIORL occupy +0x06 and +0x07.                                                          |
| volatile uint8_t | reserved_09↔<br><a href="#">[k_mtu3_pad_after_tier]</a>  | Reserved (+0x09-0x0F) – MTU4.↔TIER, MTU.TOERA, reserved x2, MTU.TGCRA, MTU.TOCR1A, MTU.TOCR2A occupy these addresses. |

|                   |                                         |                                                                                                                                                                                                                      |
|-------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | reserved_12↔<br>[k_mtu3_pad_after_tcnc] | Reserved (+0x12-0x17) – MTU4.↔ TCNT(2), MTU.TCDRA(2), MTU.↔ TDDRA(2) occupy these addresses.                                                                                                                         |
| volatile uint8_t  | reserved_1c↔<br>[k_mtu3_pad_after_tgrb] | Reserved (+0x1C-0x23) – MTU4.↔ TGRA(2), MTU4.TGRB(2), MTU.↔ TCNTSA(2), MTU.TCBRA(2) occupy these addresses.                                                                                                          |
| volatile uint8_t  | reserved_28↔<br>[k_mtu3_pad_after_tgrd] | Reserved (+0x28-0x2B) – MTU4.↔ TGRC(2), MTU4.TGRD(2).                                                                                                                                                                |
| volatile uint8_t  | reserved_2d[k_mtu3_pad_after_tsr]       | Reserved (+0x2D-0x37) – MTU4.↔ TSR, reserved x2, MTU.TITCR1A, MTU.TITCNT1A, MTU.TBTERA, reserved, MTU.TDERA, reserved, MTU.TOLBRA, reserved occupy these addresses.                                                  |
| volatile uint8_t  | reserved_39↔<br>[k_mtu3_pad_after_tbtm] | Reserved (+0x39-0x4B) – MTU4.↔ TBTM, MTU.TITMRA, MTU.↔ TITCR2A, MTU.TITCNT2A, reserved, MTU4.TADCR(2), reserved(2), MTU4.TADCORA(2), MTU4.TADCORB(2), MTU4.↔ TADCOBRA(2), MTU4.↔ TADCOBRB(2) occupy these addresses. |
| volatile uint8_t  | reserved_4d↔<br>[k_mtu3_pad_after_tcr2] | Reserved (+0x4D-0x71) – MTU4.↔ TCR2, reserved x18, MTU.TWCRA, reserved x15, MTU.TMDR2A, reserved occupy these addresses.                                                                                             |
| volatile uint8_t  | tbtm                                    | TBTM – Timer Buffer Transfer Mode Register (+0x38 = 0x000C1238).                                                                                                                                                     |
| volatile uint16_t | tcnc                                    | TCNT – Timer Counter 16-bit (+0x10 = 0x000C1210).                                                                                                                                                                    |
| volatile uint8_t  | tcr                                     | TCR – Timer Control Register (+0x00 = 0x000C1200).                                                                                                                                                                   |
| volatile uint8_t  | tcr2                                    | TCR2 – Timer Control Register 2:↔ phase count clock edge (+0x4C = 0x000C124C).                                                                                                                                       |
| volatile uint16_t | tgra                                    | TGRA – Timer General Register A: period in PWM mode (+0x18 = 0x000C1218).                                                                                                                                            |
| volatile uint16_t | tgrb                                    | TGRB – Timer General Register B:↔ duty cycle in PWM mode (+0x1A = 0x000C121A).                                                                                                                                       |
| volatile uint16_t | tgrc                                    | TGRC – Timer General Register C: buffer for TGRA (+0x24 = 0x000↔ C1224).                                                                                                                                             |
| volatile uint16_t | tgrd                                    | TGRD – Timer General Register D: buffer for TGRB (+0x26 = 0x000↔ C1226).                                                                                                                                             |

|                   |       |                                                                                                                                                                                                    |
|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | tgreh | TGRE – Timer General Register E:↔ dead-time / extended TGR (+0x72 = 0x000C1272).<br>MTU3 has TGRE only. MTU4 has both TGRE (0x000C1274) and TGRF (0x000C1276). MTU3 does NOT have a TGRF register. |
| volatile uint8_t  | tier  | TIER – Timer Interrupt Enable Register (+0x08 = 0x000C1208).                                                                                                                                       |
| volatile uint8_t  | tiorh | TIORH – Timer I/O Control H: TIOCA/TIOCB output modes (+0x04 = 0x000C1204).                                                                                                                        |
| volatile uint8_t  | tiorl | TIORL – Timer I/O Control L: TIOCC/TIOCD output modes (+0x05 = 0x000C1205).                                                                                                                        |
| volatile uint8_t  | tmdr1 | TMDR1 – Timer Mode Register 1 (+0x02 = 0x000C1202).                                                                                                                                                |
| volatile uint8_t  | tsr   | TSR – Timer Status Register: interrupt flags, write 0 to clear (+0x2C = 0x000C122C).                                                                                                               |

### 5.6.2.3 struct rx'mtu12'phase'regs't

MTU1/MTU2 extended register map for 32-bit phase counting mode.

Extended register structure for MTU1 and MTU2 when operating in 32-bit cascaded phase counting mode. This mode is ideal for quadrature encoder applications requiring high-resolution position tracking over extended ranges without software overflow handling.

### STAR Robot Encoder Configuration

The STAR robot uses 4x brushed DC gearmotors with Hall effect quadrature encoders. MTU1 and MTU2 are cascaded to provide 32-bit position tracking:

```

* Encoder Signals MTU1/MTU2 32-bit Cascaded Mode
* -----
*
* Motor 1 (FL) +-----+
* +-----+ | 32-bit TCNTLW |
* | Hall A --->+> MTCLKA --->| [MTU2.TCNT | MTU1.TCNT] |
* | Hall B --->+> MTCLKB --->| Upper 16 Lower 16 |
* +-----+ |
* 341 PPR x 4 edges | Position = TCNTLW / 1364 |
* = 1364 counts/rev | Range: +/-3.1M revolutions |
* +-----+
*

```

### 32-bit Cascaded Mode Benefits

- Extended Range:  $2^{32}$  counts vs  $2^{16}$  counts (65536x more range)
- Atomic Access: TCNTLW, TGRALW, TGRBLW are 32-bit atomic operations
- No Race Conditions: No need to disable interrupts during read
- Hardware Implementation: Zero software overhead for overflow tracking

## Position Calculation

For STAR robot Hall encoders (341 PPR, 4x quadrature = 1364 counts/rev):

$$\text{Position (revolutions)} = \frac{\text{TCNTLW}}{1364}$$

$$\text{Position (degrees)} = \frac{\text{TCNTLW} \times 360deg}{1364}$$

## Overflow Analysis

At maximum motor speed (210 RPM = 3.5 rev/sec):

$$t_{\text{overflow}} = \frac{2^{31}}{1364 \times 3.5} = 449,686 \text{ seconds} = 10.4 \text{ days}$$

## Register Memory Layout (44 bytes total)

| * Offset | Size | Register | Description                        |
|----------|------|----------|------------------------------------|
| * 0x00   | 1    | TCR      | Timer Control (prescaler, edge)    |
| * 0x01   | 1    | TMDR     | Timer Mode (phase counting mode)   |
| * 0x02   | 1    | TIORH    | Timer I/O Control H                |
| * 0x03   | 1    | TIORL    | Timer I/O Control L                |
| * 0x04   | 1    | TIER     | Timer Interrupt Enable             |
| * 0x05   | 1    | TSR      | Timer Status (flags)               |
| * 0x06   | 2    | TCNT     | Timer Counter (16-bit, lower word) |
| * 0x08   | 2    | TGRA     | Timer General Register A (16-bit)  |
| * 0x0A   | 2    | TGRB     | Timer General Register B (16-bit)  |
| * 0x0C   | 2    | TGRC     | Timer General Register C           |
| * 0x0E   | 2    | TGRD     | Timer General Register D           |
| * 0x10   | 1    | Reserved | -                                  |
| * 0x11   | 1    | TMDR3    | 32-bit mode control (LWA, PHCKSEL) |
| * 0x12   | 14   | Reserved | -                                  |
| * 0x20   | 4    | TCNTLW   | 32-bit cascaded counter (atomic)   |
| * 0x24   | 4    | TGRALW   | 32-bit general register A (atomic) |
| * 0x28   | 4    | TGRBLW   | 32-bit general register B (atomic) |
| *        |      |          |                                    |

## 32-bit Mode Configuration Example

```
// Configure MTU1+MTU2 for 32-bit quadrature encoder (Hall sensors)
volatile rx_mtu12_phase_regs_t* mtu = mtu1_phase();

// Stop timers during configuration
mtu->tstra()->tstr &= ~(k_mtu_tstr_cst1 | k_mtu_tstr_cst2);

// Configure for phase counting mode 4 (count on both edges)
mtu->tmdr = 0x04; // Phase counting mode 4

// Enable 32-bit cascaded mode (LWA bit) with both-edge counting
mtu->tmdr3 = k_mtu_tmdr3_lwa | k_mtu_tmdr3_phcksel;

// Clear 32-bit counter
mtu->tcntlw = 0;

// Start both timers
mtu->tstra()->tstr |= (k_mtu_tstr_cst1 | k_mtu_tstr_cst2);

// Read 32-bit position (atomic, no race condition)
int32_t position = (int32_t)mtu->tcntlw;
```



## Invariant

Structure size must be exactly 44 bytes (0x2C)

TMDR3 at offset 0x11 controls 32-bit cascaded mode

TCNTLW/TGRALW/TGRBLW are 32-bit aligned for atomic access

## See also

[mtu\\_tmdr3\\_bits\\_t](#) for TMDR3 bit definitions

[rx\\_mtu\\_channel\\_regs\\_t](#) for standard 16-bit mode

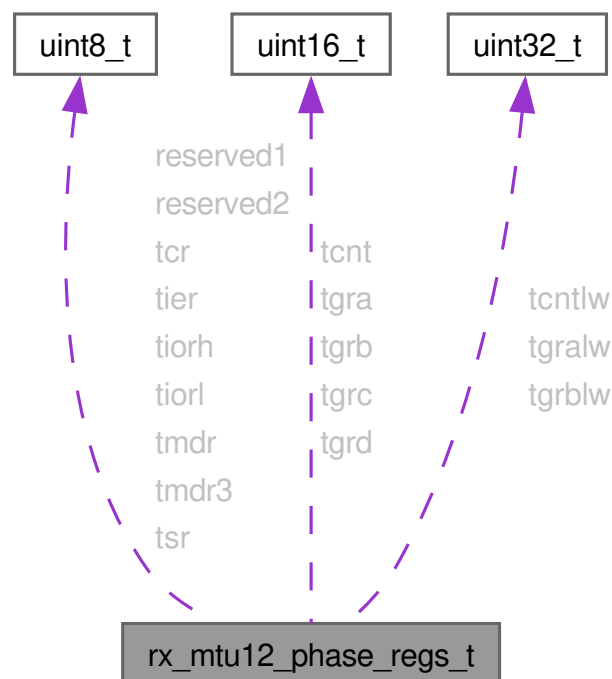
RX72N Hardware Manual Section 24.2.25 (TMDR3 Register)

RX72N Hardware Manual Section 24.2.26 (32-bit Registers)

RX72N Hardware Manual Section 24.3.5 (Phase Counting Mode)

Definition at line 773 of file [rx72n\\_mtu\\_regs.h](#).

Collaboration diagram for `rx_mtu12_phase_regs_t`:



## Data Fields

|                                |                                                                          |                                                                                                                       |
|--------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>uint8_t</code>           | <code>reserved1</code>                                                   | Reserved byte at offset 0x10.                                                                                         |
| <code>uint8_t</code>           | <code>reserved2</code> ↔<br><a href="#">[k_mtu12p_pad_before_tcntlw]</a> | Reserved bytes at offset 0x12-0x1F (14 bytes, before 32-bit long-word regs).                                          |
| <code>volatile uint16_t</code> | <code>tcnt</code>                                                        | Timer Counter (TCNT) - lower 16 bits of cascaded counter.<br>Use TCNTLW for atomic 32-bit access when LWA bit is set. |

|                   |        |                                                                                                                                                                                                                                                                                                                                           |
|-------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint32_t | tcntlw | <p>32-bit Cascaded Counter (TCNTLW) - atomic position value<br/>Combined MTU1.TCNT (lower 16 bits) and MTU2.TCNT (upper 16 bits). Provides atomic 32-bit read/write without race conditions. Sign-extended for bidirectional counting (use int32_t cast).</p> <p>Note</p> <p>Only valid when TMDR3.<br/>LWA = 1 (32-bit mode enabled)</p> |
| volatile uint8_t  | tcr    | <p>Timer Control Register (TCR) - prescaler and edge selection.<br/>For phase counting, TCR.CKEG selects which edge of MTCLKA/B to count.</p>                                                                                                                                                                                             |
| volatile uint16_t | tgra   | <p>Timer General Register A - 16-bit compare (use TGRALW for 32-bit).</p>                                                                                                                                                                                                                                                                 |
| volatile uint32_t | tgralw | <p>32-bit General Register A (TGRALW) - atomic compare value<br/>32-bit compare register for position threshold interrupts.</p> <p>Note</p> <p>Only valid when TMDR3.<br/>LWA = 1</p>                                                                                                                                                     |
| volatile uint16_t | tgrb   | <p>Timer General Register B - 16-bit compare (use TGRBLW for 32-bit).</p>                                                                                                                                                                                                                                                                 |
| volatile uint32_t | tgrblw | <p>32-bit General Register B (TGRBLW) - atomic compare value<br/>32-bit compare register for position threshold interrupts.</p> <p>Note</p> <p>Only valid when TMDR3.<br/>LWA = 1</p>                                                                                                                                                     |
| volatile uint16_t | tgrc   | <p>Timer General Register C - 16-bit compare.</p>                                                                                                                                                                                                                                                                                         |
| volatile uint16_t | tgrd   | <p>Timer General Register D - 16-bit compare.</p>                                                                                                                                                                                                                                                                                         |
| volatile uint8_t  | tier   | <p>Timer Interrupt Enable Register (TIER).<br/>Enable overflow/underflow interrupts for position limit detection.</p>                                                                                                                                                                                                                     |
| volatile uint8_t  | tiorh  | <p>Timer I/O Control Register H (typically unused in phase counting).</p>                                                                                                                                                                                                                                                                 |
| volatile uint8_t  | tiorl  | <p>Timer I/O Control Register L (typically unused in phase counting).</p>                                                                                                                                                                                                                                                                 |

|                  |       |                                                                                                                                                                                                                                                                                                                   |
|------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | tmdr  | Timer Mode Register (TMDR) - phase counting mode select.<br>Mode 4-7 select different phase counting configurations. Mode 4:↔ Count up/down on MTCLKA, direction from MTCLKB.                                                                                                                                     |
| volatile uint8_t | tmdr3 | Timer Mode Register 3 (TMDR3) - 32-bit mode control.<br><br><ul style="list-style-type: none"> <li>• Bit 0 (LWA): Long Word Access enable (32-bit cascaded mode)</li> <li>• Bit 1 (PHCKSEL): Phase counting clock select (both edges)</li> </ul> See also<br><a href="#">mtu_tmdr3_bits_t</a> for bit definitions |
| volatile uint8_t | tsr   | Timer Status Register (TSR) - interrupt flags.<br>Contains overflow/underflow flags for position monitoring.                                                                                                                                                                                                      |

#### 5.6.2.4 struct rx\_mtu\_tstr\_regs\_t

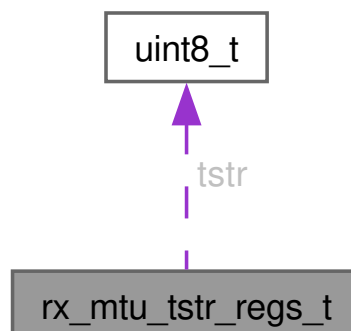
MTU Start Register Map.

Timer start/stop control registers. Base addresses:

- TSTRA: 0x000C1280 (MTU0-4, MTU8)
- TSTRB: 0x000C1A80 (MTU6-7)

Definition at line 876 of file [rx72n\\_mtu\\_regs.h](#).

Collaboration diagram for rx\_mtu\_tstr\_regs\_t:



## Data Fields

|                  |      |                                            |
|------------------|------|--------------------------------------------|
| volatile uint8_t | tstr | Timer Start Register (channel enable bits) |
|------------------|------|--------------------------------------------|

## 5.6.3 Enumeration Type Documentation

## 5.6.3.1 mtu12`phase`padding`sizes` t

```
enum mtu12_phase_padding_sizes_t : uint8_t
```

Named padding array sizes for [rx\\_mtu12\\_phase\\_regs\\_t](#) reserved fields.

Each constant names the number of reserved bytes between two adjacent registers in the 32-bit phase counting register layout. Verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.

Since

Version 1.0.0

## Enumerator

|                            |                                                                          |
|----------------------------|--------------------------------------------------------------------------|
| k_mtu12p_pad_before_tcntlw | 14 reserved bytes at offset 0x12-0x1F (gap before 32-bit long-word regs) |
|----------------------------|--------------------------------------------------------------------------|

Definition at line 662 of file [rx72n\\_mtu\\_regs.h](#).

```
00662 : uint8_t {
00663 /** @brief 14 reserved bytes at offset 0x12-0x1F (gap before 32-bit long-word regs) */
00664 k_mtu12p_pad_before_tcntlw = 14U,
00665 } mtu12_phase_padding_sizes_t;
```

## 5.6.3.2 mtu12`phase`reg`offsets` t

```
enum mtu12_phase_reg_offsets_t : uint8_t
```

Byte offsets of registers within [rx\\_mtu12\\_phase\\_regs\\_t](#).

Named constants for compile-time layout verification of the 32-bit phase counting register struct. Verified against RX72N Hardware Manual Ch 5 I/O Register Address Table and Ch 24 Table 24.11.

Since

Version 1.0.0

## Enumerator

|                   |                      |
|-------------------|----------------------|
| k_mtu12p_off_tcr  | TCR at offset 0x00.  |
| k_mtu12p_off_tmdr | TMDR at offset 0x01. |
| k_mtu12p_off_tcnt | TCNT at offset 0x06. |

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| k_mtu12p_off_tgra    | TGRA at offset 0x08.                                                |
| k_mtu12p_off_tgrb    | TGRB at offset 0x0A.                                                |
| k_mtu12p_off_tmdr3   | TMDR3 at offset 0x11 (32-bit mode enable register).                 |
| k_mtu12p_off_tcntlw  | TCNTLW at offset 0x20 (32-bit counter in long-word mode).           |
| k_mtu12p_off_tgralw  | TGRALW at offset 0x24 (32-bit compare A in long-word mode).         |
| k_mtu12p_off_tgrblw  | TGRBLW at offset 0x28 (32-bit compare B in long-word mode).         |
| k_mtu12p_struct_size | Total size of <code>rx_mtu12_phase_regs_t</code> = 0x2C bytes (44). |

Definition at line 1507 of file `rx72n_mtu_regs.h`.

```

01507 : uint8_t {
01508 /** @brief TCR at offset 0x00 */
01509 k_mtu12p_off_tcr = 0x00U,
01510 /** @brief TMDR at offset 0x01 */
01511 k_mtu12p_off_tmdr = 0x01U,
01512 /** @brief TCNT at offset 0x06 */
01513 k_mtu12p_off_tcnt = 0x06U,
01514 /** @brief TGRA at offset 0x08 */
01515 k_mtu12p_off_tgra = 0x08U,
01516 /** @brief TGRB at offset 0x0A */
01517 k_mtu12p_off_tgrb = 0x0AU,
01518 /** @brief TMDR3 at offset 0x11 (32-bit mode enable register) */
01519 k_mtu12p_off_tmdr3 = 0x11U,
01520 /** @brief TCNTLW at offset 0x20 (32-bit counter in long-word mode) */
01521 k_mtu12p_off_tcntlw = 0x20U,
01522 /** @brief TGRALW at offset 0x24 (32-bit compare A in long-word mode) */
01523 k_mtu12p_off_tgralw = 0x24U,
01524 /** @brief TGRBLW at offset 0x28 (32-bit compare B in long-word mode) */
01525 k_mtu12p_off_tgrblw = 0x28U,
01526 /** @brief Total size of rx_mtu12_phase_regs_t = 0x2C bytes (44) */
01527 k_mtu12p_struct_size = 0x2CU,
01528 } mtu12_phase_reg_offsets_t;

```

### 5.6.3.3 mtu3'padding'sizes't

```
enum mtu3_padding_sizes_t : uint8_t
```

Named padding array sizes for `rx_mtu3_channel_regs_t` reserved fields.

Each constant names the number of reserved bytes between two adjacent MTU3 registers in the hardware layout. The comments show which hardware registers occupy those byte ranges (verified against RX72N Ch 5 I/O Register Address Table).

Since

Version 1.0.0

Enumerator

|                        |                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------|
| k_mtu3_pad_after_tiorl | 2 reserved bytes at +0x06-0x07: MTU4.TIORH, MTU4.TIORL                                               |
| k_mtu3_pad_after_tier  | 7 reserved bytes at +0x09-0x0F: MTU4.TIER, MTU.TOERA, reserved x2, MTU.TGCRA, MTU.TOCR1A, MTU.TOCR2A |
| k_mtu3_pad_after_tcnt  | 6 reserved bytes at +0x12-0x17: MTU4.TCNT(2), MTU.TCDRA(2), MTU.TDDRA(2)                             |
| k_mtu3_pad_after_tgrb  | 8 reserved bytes at +0x1C-0x23: MTU4.TGRA(2), MTU4.TGRB(2), MTU.TCNTSA(2), MTU.TCBRA(2)              |
| k_mtu3_pad_after_tgrd  | 4 reserved bytes at +0x28-0x2B: MTU4.TGRC(2), MTU4.TGRD(2)                                           |

|                       |                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_mtu3_pad_after_tsr  | 11 reserved bytes at +0x2D-0x37: MTU4.TSR, reserved x2, MTU4.TITCR1A, MTU4.TITCNT1A, MTU4.TBTERA, reserved, MTU4.TDERA, reserved, MTU4.TOLBRA, reserved                                          |
| k_mtu3_pad_after_tbtm | 19 reserved bytes at +0x39-0x4B: MTU4.TBTM, MTU4.TITMRA, MTU4.TITCR2A, MTU4.TITCNT2A, reserved, MTU4.TADCR(2), reserved(2), MTU4.TADCORA(2), MTU4.TADCORB(2), MTU4.TADCOBRA(2), MTU4.TADCOBRB(2) |
| k_mtu3_pad_after_tcr2 | 37 reserved bytes at +0x4D-0x71: MTU4.TCR2, reserved x18, MTU4.TWCRA, reserved x15, MTU4.TMDR2A, reserved                                                                                        |

Definition at line 420 of file [rx72n\\_mtu\\_regs.h](#).

```

00420 : uint8_t {
00421 /** @brief 2 reserved bytes at +0x06-0x07: MTU4.TIORH, MTU4.TIORL */
00422 k_mtu3_pad_after_tiorl = 2U,
00423
00424 /** @brief 7 reserved bytes at +0x09-0x0F: MTU4.TIER, MTU4.TOERA, reserved x2,
00425 * MTU4.TGCRA, MTU4.TOCR1A, MTU4.TOCR2A */
00426 k_mtu3_pad_after_tier = 7U,
00427
00428 /** @brief 6 reserved bytes at +0x12-0x17: MTU4.TCNT(2), MTU4.TCDRA(2), MTU4.TDDRA(2) */
00429 k_mtu3_pad_after_tcnc = 6U,
00430
00431 /** @brief 8 reserved bytes at +0x1C-0x23: MTU4.TGRA(2), MTU4.TGRB(2),
00432 * MTU4.TCNTSA(2), MTU4.TCBRA(2) */
00433 k_mtu3_pad_after_tgrb = 8U,
00434
00435 /** @brief 4 reserved bytes at +0x28-0x2B: MTU4.TGRC(2), MTU4.TGRD(2) */
00436 k_mtu3_pad_after_tgrd = 4U,
00437
00438 /** @brief 11 reserved bytes at +0x2D-0x37: MTU4.TSR, reserved x2, MTU4.TITCR1A,
00439 * MTU4.TITCNT1A, MTU4.TBTERA, reserved, MTU4.TDERA, reserved, MTU4.TOLBRA,
00440 * reserved */
00441 k_mtu3_pad_after_tsr = 11U,
00442
00443 /** @brief 19 reserved bytes at +0x39-0x4B: MTU4.TBTM, MTU4.TITMRA, MTU4.TITCR2A,
00444 * MTU4.TITCNT2A, reserved, MTU4.TADCR(2), reserved(2), MTU4.TADCORA(2),
00445 * MTU4.TADCORB(2), MTU4.TADCOBRA(2), MTU4.TADCOBRB(2) */
00446 k_mtu3_pad_after_tbtm = 19U,
00447
00448 /** @brief 37 reserved bytes at +0x4D-0x71: MTU4.TCR2, reserved x18, MTU4.TWCRA,
00449 * reserved x15, MTU4.TMDR2A, reserved */
00450 k_mtu3_pad_after_tcr2 = 37U,
00451 } mtu3_padding_sizes_t;

```

#### 5.6.3.4 mtu3`reg`offsets`t

```
enum mtu3_reg_offsets_t : uint8_t
```

Byte offsets of MTU3 registers from base 0x000C1200.

Named constants for compile-time layout verification of [rx\\_mtu3\\_channel\\_regs\\_t](#). The MTU3/MTU4 interleaved layout means offsets are sparse – see the struct definition for the full picture of which bytes belong to MTU4 vs. shared regs. All values verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.

Since

Version 1.0.0

Enumerator

|                |                                  |
|----------------|----------------------------------|
| k_mtu3_off_tcr | TCR at offset 0x00 = 0x000C1200. |
|----------------|----------------------------------|

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| k_mtu3_off_tmdr1   | TMDR1 at offset 0x02 = 0x000C1202.                       |
| k_mtu3_off_tiorh   | TIORH at offset 0x04 = 0x000C1204.                       |
| k_mtu3_off_tiorl   | TIORL at offset 0x05 = 0x000C1205.                       |
| k_mtu3_off_tier    | TIER at offset 0x08 = 0x000C1208.                        |
| k_mtu3_off_tcnc    | TCNT at offset 0x10 = 0x000C1210.                        |
| k_mtu3_off_tgra    | TGRA at offset 0x18 = 0x000C1218.                        |
| k_mtu3_off_tgrb    | TGRB at offset 0x1A = 0x000C121A.                        |
| k_mtu3_off_tgrc    | TGRC at offset 0x24 = 0x000C1224.                        |
| k_mtu3_off_tgrd    | TGRD at offset 0x26 = 0x000C1226.                        |
| k_mtu3_off_tsr     | TSR at offset 0x2C = 0x000C122C.                         |
| k_mtu3_off_tbtm    | TBTM at offset 0x38 = 0x000C1238.                        |
| k_mtu3_off_tcr2    | TCR2 at offset 0x4C = 0x000C124C.                        |
| k_mtu3_off_tgre    | TGRE at offset 0x72 = 0x000C1272.                        |
| k_mtu3_struct_size | Total size of rx_mtu3_channel_regs_t = 0x74 bytes (116). |

Definition at line 1463 of file rx72n\_mtu\_regs.h.

```

01463 : uint8_t {
01464 /** @brief TCR at offset 0x00 = 0x000C1200 */
01465 k_mtu3_off_tcr = 0x00U,
01466 /** @brief TMDR1 at offset 0x02 = 0x000C1202 */
01467 k_mtu3_off_tmdr1 = 0x02U,
01468 /** @brief TIORH at offset 0x04 = 0x000C1204 */
01469 k_mtu3_off_tiorh = 0x04U,
01470 /** @brief TIORL at offset 0x05 = 0x000C1205 */
01471 k_mtu3_off_tiorl = 0x05U,
01472 /** @brief TIER at offset 0x08 = 0x000C1208 */
01473 k_mtu3_off_tier = 0x08U,
01474 /** @brief TCNT at offset 0x10 = 0x000C1210 */
01475 k_mtu3_off_tcnc = 0x10U,
01476 /** @brief TGRA at offset 0x18 = 0x000C1218 */
01477 k_mtu3_off_tgra = 0x18U,
01478 /** @brief TGRB at offset 0x1A = 0x000C121A */
01479 k_mtu3_off_tgrb = 0x1AU,
01480 /** @brief TGRC at offset 0x24 = 0x000C1224 */
01481 k_mtu3_off_tgrc = 0x24U,
01482 /** @brief TGRD at offset 0x26 = 0x000C1226 */
01483 k_mtu3_off_tgrd = 0x26U,
01484 /** @brief TSR at offset 0x2C = 0x000C122C */
01485 k_mtu3_off_tsr = 0x2CU,
01486 /** @brief TBTM at offset 0x38 = 0x000C1238 */
01487 k_mtu3_off_tbtm = 0x38U,
01488 /** @brief TCR2 at offset 0x4C = 0x000C124C */
01489 k_mtu3_off_tcr2 = 0x4CU,
01490 /** @brief TGRE at offset 0x72 = 0x000C1272 */
01491 k_mtu3_off_tgre = 0x72U,
01492 /** @brief Total size of rx_mtu3_channel_regs_t = 0x74 bytes (116) */
01493 k_mtu3_struct_size = 0x74U,
01494 } mtu3_reg_offsets_t;

```

### 5.6.3.5 mtu\_bit\_values\_t

```
enum mtu_bit_values_t : uint8_t
```

Bit value constants for shift operations.

Named constants to eliminate magic numbers in bit shift expressions.

Enumerator

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| k_mtu_bit_value_set | Bit value to shift (replaces 1, 1U, 0x01 in shifts) |
|---------------------|-----------------------------------------------------|

Definition at line 257 of file [rx72n\\_mtu\\_regs.h](#).

```
00257 : uint8_t {
00258 k_mtu_bit_value_set = 1, /**< Bit value to shift (replaces 1, 1U, 0x01 in shifts) */
00259 } mtu_bit_values_t;
```

### 5.6.3.6 mtu\_channel\_reg\_offsets\_t

```
enum mtu_channel_reg_offsets_t : uint8_t
```

Byte offsets of registers within [rx\\_mtu\\_channel\\_regs\\_t](#).

Named constants used in compile-time layout verification. Applies to standard MTU channels (MTU0, MTU1, MTU2, MTU6, MTU7). Verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.

Since

Version 1.0.0

Enumerator

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| k_mtu_ch_off_tcr    | TCR at offset 0x00 (first register in channel block). |
| k_mtu_ch_off_tcncnt | TCNT at offset 0x06.                                  |
| k_mtu_ch_off_tgra   | TGRA at offset 0x08.                                  |
| k_mtu_ch_off_tgrb   | TGRB at offset 0x0A.                                  |

Definition at line 1440 of file [rx72n\\_mtu\\_regs.h](#).

```
01440 : uint8_t {
01441 /** @brief TCR at offset 0x00 (first register in channel block) */
01442 k_mtu_ch_off_tcr = 0x00U,
01443 /** @brief TCNT at offset 0x06 */
01444 k_mtu_ch_off_tcncnt = 0x06U,
01445 /** @brief TGRA at offset 0x08 */
01446 k_mtu_ch_off_tgra = 0x08U,
01447 /** @brief TGRB at offset 0x0A */
01448 k_mtu_ch_off_tgrb = 0x0AU,
01449 } mtu_channel_reg_offsets_t;
```

### 5.6.3.7 mtu\_periph\_space\_t

```
enum mtu_periph_space_t : uintptr_t
```

Address space mask and expected base for MTU peripheral verification.

Used in `static_assert` to verify MTU base addresses fall in the correct peripheral address space (0x000↔Cxxxx), as required by the RX72N memory map.



Since

Version 1.0.0

Enumerator

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| k_mtu_periph_space_mask | Mask selecting the upper 16 bits of a 32-bit address. |
| k_mtu_periph_space_base | Expected upper 16 bits of all MTU base addresses.     |

Definition at line 1540 of file rx72n\_mtu\_regs.h.

```
01540 : uintptr_t {
01541 /** @brief Mask selecting the upper 16 bits of a 32-bit address */
01542 k_mtu_periph_space_mask = 0xFFFF0000U,
01543 /** @brief Expected upper 16 bits of all MTU base addresses */
01544 k_mtu_periph_space_base = 0x000C0000U,
01545 } mtu_periph_space_t;
```

### 5.6.3.8 mtu\_tcr\_bits\_t

```
enum mtu_tcr_bits_t : uint8_t
```

Timer Control Register (TCR) bit field definitions.

The TCR register controls timer clock source selection, counting edge, and counter clear source. Each MTU channel has its own TCR register.

Register Bit Layout

```
* Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
* -----
* | CCLR[2:0] | CKEG[1:0] | TPSC[2:0] |
* | Counter Clear | Clock Edge | Timer Prescaler |
* -----
*
```

Prescaler Values (TPSC)

| Value | Prescaler           | Frequency (120 MHz PCLKA) |
|-------|---------------------|---------------------------|
| 0x00  | $\text{PCLKA} / 1$  | 120 MHz                   |
| 0x01  | $\text{PCLKA} / 4$  | 30 MHz                    |
| 0x02  | $\text{PCLKA} / 16$ | 7.5 MHz                   |
| 0x03  | $\text{PCLKA} / 64$ | 1.875 MHz                 |

## Counter Clear Source (CCLR)

| Value | Source | Description                 |
|-------|--------|-----------------------------|
| 0x01  | TGRA   | Clear on TGRA compare match |
| 0x02  | TGRB   | Clear on TGRB compare match |
| 0x03  | Sync   | Synchronous clear (MTU3/4)  |

See also

[rx\\_mtu\\_channel\\_regs\\_t::tcr](#)

RX72N Hardware Manual Section 24.2.1 (TCR Register)

## Enumerator

|                      |                                                 |
|----------------------|-------------------------------------------------|
| k_mtu_tcr_cclr_shift | Counter Clear Source bit position (bits 5-7).   |
| k_mtu_tcr_tpsc_mask  | Timer Prescaler mask - bits 0-2.                |
| k_mtu_tcr_ckege_mask | Clock Edge mask - bits 3-4.                     |
| k_mtu_tcr_cclr_mask  | Counter Clear Source mask - bits 5-7.           |
| k_mtu_tcr_tpsc_1     | Prescaler PCLKA/1 (120 MHz @ 120 MHz PCLKA).    |
| k_mtu_tcr_tpsc_4     | Prescaler PCLKA/4 (30 MHz @ 120 MHz PCLKA).     |
| k_mtu_tcr_tpsc_16    | Prescaler PCLKA/16 (7.5 MHz @ 120 MHz PCLKA).   |
| k_mtu_tcr_tpsc_64    | Prescaler PCLKA/64 (1.875 MHz @ 120 MHz PCLKA). |
| k_mtu_tcr_cclr_tgra  | Clear counter on TGRA compare match.            |

Definition at line 1161 of file [rx72n\\_mtu\\_regs.h](#).

```

01161 : uint8_t {
01162 /** @brief Counter Clear Source bit position (bits 5-7) */
01163 k_mtu_tcr_cclr_shift = 5,
01164
01165 /** @brief Timer Prescaler mask - bits 0-2 */
01166 k_mtu_tcr_tpsc_mask = 0x07,
01167
01168 /** @brief Clock Edge mask - bits 3-4 */
01169 k_mtu_tcr_ckege_mask = 0x18,
01170
01171 /** @brief Counter Clear Source mask - bits 5-7 */
01172 k_mtu_tcr_cclr_mask = 0xE0,
01173
01174 /** @brief Prescaler PCLKA/1 (120 MHz @ 120 MHz PCLKA) */
01175 k_mtu_tcr_tpsc_1 = 0x00,
01176
01177 /** @brief Prescaler PCLKA/4 (30 MHz @ 120 MHz PCLKA) */
01178 k_mtu_tcr_tpsc_4 = 0x01,
01179
01180 /** @brief Prescaler PCLKA/16 (7.5 MHz @ 120 MHz PCLKA) */
01181 k_mtu_tcr_tpsc_16 = 0x02,
01182
01183 /** @brief Prescaler PCLKA/64 (1.875 MHz @ 120 MHz PCLKA) */
01184 k_mtu_tcr_tpsc_64 = 0x03,
01185
01186 /** @brief Clear counter on TGRA compare match */
01187 k_mtu_tcr_cclr_tgra = (k_mtu_bit_value_set « k_mtu_tcr_cclr_shift),
01188 } mtu_tcr_bits_t;

```

## 5.6.3.9 mtu\_tior\_bits\_t

```
enum mtu_tior_bits_t : uint8_t
```

Timer I/O Control Register (TIOR) bit field definitions.

The TIOR register controls the behavior of timer output pins on compare match events. TIORH controls TIOCA/TIOCB outputs, TIORL controls TIOCC/TIOCD.

Register Bit Layout (TIORH)

```

* Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
* -----
* | IOB[3:0] | IOA[3:0] |
* | TIOCB Output Mode | TIOCA Output Mode |
* -----
*

```

Output Modes

| Value | Initial | On Match | PWM Application |
|-------|---------|----------|-----------------|
| 0x02  | Low     | High     | Active-high PWM |
| 0x05  | High    | Low      | Active-low PWM  |
| 0x03  | -       | Toggle   | 50% duty square |

PWM Configuration Example

```

// Configure TIOCA0 for active-high PWM output
mtu0()->tiorh = k_mtu_tior_init_low; // Low initially, high on match

```

See also

[rx\\_mtu\\_channel\\_regs\\_t::tiorh](#)

[rx\\_mtu\\_channel\\_regs\\_t::tiorl](#)

RX72N Hardware Manual Section 24.2.4 (TIOR Register)

Enumerator

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| k_mtu_tior_ioa_mask  | I/O Control A mask - bits 0-3 (TIOCA output mode).               |
| k_mtu_tior_iob_mask  | I/O Control B mask - bits 4-7 (TIOCB output mode).               |
| k_mtu_tior_init_low  | Initial output low, set high on compare match (active-high PWM). |
| k_mtu_tior_init_high | Initial output high, set low on compare match (active-low PWM).  |
| k_mtu_tior_toggle    | Toggle output on compare match (50% duty square wave).           |

Definition at line 1309 of file [rx72n\\_mtu\\_regs.h](#).

```

01309 : uint8_t {
01310 /** @brief I/O Control A mask - bits 0-3 (TIOCA output mode) */
01311 k_mtu_tior_ioa_mask = 0x0F,
01312
01313 /** @brief I/O Control B mask - bits 4-7 (TIOCB output mode) */
01314 k_mtu_tior_iob_mask = 0xF0,
01315
01316 /** @brief Initial output low, set high on compare match (active-high PWM) */
01317 k_mtu_tior_init_low = 0x02,
01318
01319 /** @brief Initial output high, set low on compare match (active-low PWM) */
01320 k_mtu_tior_init_high = 0x05,
01321
01322 /** @brief Toggle output on compare match (50% duty square wave) */
01323 k_mtu_tior_toggle = 0x03,
01324 } mtu_tior_bits_t;

```

## 5.6.3.10 mtu\_tmdr3\_bits\_t

```
enum mtu_tmdr3_bits_t : uint8_t
```

Timer Mode Register 3 (TMDR3) bits for MTU1/MTU2 32-bit cascaded mode.

TMDR3 register controls 32-bit cascaded counter operation for MTU1/MTU2 in phase counting mode. Available at offset 0x11 in [rx\\_mtu12\\_phase\\_regs\\_t](#).

Key features:

- LWA: Enables 32-bit cascaded counter (TCNTLW, TGRALW, TGRBLW)
- PHCKSEL: Selects phase counting clock edge (both edges vs single edge)

See also

RX72N Hardware Manual Section 24.2.25 (TMDR3 Register)

Enumerator

|                           |                                                   |
|---------------------------|---------------------------------------------------|
| k_mtu_tmdr3_lwa_shift     | Long Word Access bit position                     |
| k_mtu_tmdr3_phcksel_shift | Phase Counting Clock Select bit position          |
| k_mtu_tmdr3_lwa           | Enable 32-bit cascaded counter (LWA)              |
| k_mtu_tmdr3_phcksel       | Phase counting clock select (both edges when set) |

Definition at line 1265 of file [rx72n\\_mtu\\_regs.h](#).

```
01265 : uint8_t {
01266 k_mtu_tmdr3_lwa_shift = 0, /**< Long Word Access bit position */
01267 k_mtu_tmdr3_phcksel_shift = 1, /**< Phase Counting Clock Select bit position */
01268 k_mtu_tmdr3_lwa =
01269 (k_mtu_bit_value_set « k_mtu_tmdr3_lwa_shift), /**< Enable 32-bit cascaded counter (LWA) */
01270 k_mtu_tmdr3_phcksel =
01271 (k_mtu_bit_value_set
01272 « k_mtu_tmdr3_phcksel_shift), /**< Phase counting clock select (both edges when set) */
01273 } mtu_tmdr3_bits_t;
```

## 5.6.3.11 mtu\_tmdr\_bits\_t

```
enum mtu_tmdr_bits_t : uint8_t
```

Timer Mode Register (TMDR) bit field definitions.

The TMDR register selects the timer operating mode: normal operation, PWM generation, phase counting, or complementary PWM.

Register Bit Layout

```
* Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
* -----
* | - | - | BFB | BFA | MD[3:0] |
* | Rsrvd | Rsrvd | BufferB | BufferA | Mode Select |
* -----
*
```

Operating Modes (MD[3:0

)]

| Value | Mode      | Description                        |
|-------|-----------|------------------------------------|
| 0x00  | Normal    | Free-running with compare match    |
| 0x02  | PWM 1     | TGRA=period, TGRB=duty             |
| 0x03  | PWM 2     | TGRC/D as buffer registers         |
| 0x04  | Phase 1   | Up/down on MTCLKA, dir from MTCLKB |
| 0x05  | Phase 2   | Up on MTCLKA, down on MTCLKB       |
| 0x06  | Phase 3   | Count on MTCLKA edges only         |
| 0x07  | Phase 4   | Count on MTCLKA and MTCLKB edges   |
| 0x0D  | Compl PWM | Complementary PWM (MTU3/4, 6/7)    |

Buffer Transfer Mode

When BFA or BFB is set, buffer registers (TGRC->TGRA, TGRD->TGRB) are automatically transferred on counter clear, enabling glitch-free duty cycle updates during PWM operation.

See also

[rx\\_mtu\\_channel\\_regs\\_t::tmdr](#)

RX72N Hardware Manual Section 24.2.2 (TMDR Register)

Enumerator

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| k_mtu_tmdr_bfa_shift | Buffer mode A bit position.                                         |
| k_mtu_tmdr_bfb_shift | Buffer mode B bit position.                                         |
| k_mtu_tmdr_md_mask   | Mode select mask (bits 0-3).                                        |
| k_mtu_tmdr_md_normal | Normal mode - free-running counter with compare match.              |
| k_mtu_tmdr_md_pwm1   | PWM mode 1 - TGRA defines period, TGRB defines duty cycle.          |
| k_mtu_tmdr_md_pwm2   | PWM mode 2 - TGRC/TGRD as buffer registers for glitch-free updates. |
| k_mtu_tmdr_bfa       | Enable TGRC->TGRA buffer transfer on counter clear.                 |
| k_mtu_tmdr_bfb       | Enable TGRD->TGRB buffer transfer on counter clear.                 |

Definition at line 1227 of file [rx72n\\_mtu\\_regs.h](#).

```

01227 : uint8_t {
01228 /** @brief Buffer mode A bit position */
01229 k_mtu_tmdr_bfa_shift = 4,
01230
01231 /** @brief Buffer mode B bit position */
01232 k_mtu_tmdr_bfb_shift = 5,
01233
01234 /** @brief Mode select mask (bits 0-3) */
01235 k_mtu_tmdr_md_mask = 0x0F,
01236
01237 /** @brief Normal mode - free-running counter with compare match */
01238 k_mtu_tmdr_md_normal = 0x00,
01239
01240 /** @brief PWM mode 1 - TGRA defines period, TGRB defines duty cycle */
01241 k_mtu_tmdr_md_pwm1 = 0x02,
01242
01243 /** @brief PWM mode 2 - TGRC/TGRD as buffer registers for glitch-free updates */
01244 k_mtu_tmdr_md_pwm2 = 0x03,

```

```

01245
01246 /** @brief Enable TGRC->TGRA buffer transfer on counter clear */
01247 k_mtu_tmdr_bfa = (k_mtu_bit_value_set « k_mtu_tmdr_bfa_shift),
01248
01249 /** @brief Enable TGRD->TGRB buffer transfer on counter clear */
01250 k_mtu_tmdr_bfb = (k_mtu_bit_value_set « k_mtu_tmdr_bfb_shift),
01251 } mtu_tmdr_bits_t;

```

#### 5.6.3.12 mtu\_tstr\_bits\_t

```
enum mtu_tstr_bits_t : uint8_t
```

Timer Start Register A (TSTRA) bit definitions - controls MTU0-4, MTU8.

TSTRA (at 0x000C1280) controls the start/stop state of MTU channels 0-4 and 8. Setting a CSTn bit starts the corresponding timer; clearing it stops the timer.

Register Bit Layout (TSTRA)

```

* Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
* -----
* | CST4 | CST3 | - | - | CST8 | CST2 | CST1 | CST0 |
* | MTU4 | MTU3 | Rsvd | Rsvd | MTU8 | MTU2 | MTU1 | MTU0 |
* -----
*

```

Synchronized Start Example

```

// Start MTU1 and MTU2 simultaneously for 32-bit cascaded mode
mtu_tstra()->tstr |= (k_mtu_tstr_cst1 | k_mtu_tstr_cst2);

// Stop all timers for configuration
mtu_tstra()->tstr = 0;

```

See also

[rx\\_mtu\\_tstr\\_regs\\_t](#)

[mtu\\_tstra\(\)](#) for register accessor

RX72N Hardware Manual Section 24.2.15 (TSTRA Register)

Enumerator

|                       |                                      |
|-----------------------|--------------------------------------|
| k_mtu_tstr_cst0_shift | Counter Start 0 bit position (MTU0). |
| k_mtu_tstr_cst1_shift | Counter Start 1 bit position (MTU1). |
| k_mtu_tstr_cst2_shift | Counter Start 2 bit position (MTU2). |
| k_mtu_tstr_cst8_shift | Counter Start 8 bit position (MTU8). |
| k_mtu_tstr_cst3_shift | Counter Start 3 bit position (MTU3). |
| k_mtu_tstr_cst4_shift | Counter Start 4 bit position (MTU4). |
| k_mtu_tstr_cst0       | Start MTU0 counter (TSTRA bit 0).    |
| k_mtu_tstr_cst1       | Start MTU1 counter (TSTRA bit 1).    |
| k_mtu_tstr_cst2       | Start MTU2 counter (TSTRA bit 2).    |
| k_mtu_tstr_cst3       | Start MTU3 counter (TSTRA bit 6).    |

|                 |                                   |
|-----------------|-----------------------------------|
| k_mtu_tstr_cst4 | Start MTU4 counter (TSTRA bit 7). |
| k_mtu_tstr_cst8 | Start MTU8 counter (TSTRA bit 3). |

Definition at line 1356 of file rx72n\_mtu\_regs.h.

```

01356 : uint8_t {
01357 /** @brief Counter Start 0 bit position (MTU0) */
01358 k_mtu_tstr_cst0_shift = 0,
01359
01360 /** @brief Counter Start 1 bit position (MTU1) */
01361 k_mtu_tstr_cst1_shift = 1,
01362
01363 /** @brief Counter Start 2 bit position (MTU2) */
01364 k_mtu_tstr_cst2_shift = 2,
01365
01366 /** @brief Counter Start 8 bit position (MTU8) */
01367 k_mtu_tstr_cst8_shift = 3,
01368
01369 /** @brief Counter Start 3 bit position (MTU3) */
01370 k_mtu_tstr_cst3_shift = 6,
01371
01372 /** @brief Counter Start 4 bit position (MTU4) */
01373 k_mtu_tstr_cst4_shift = 7,
01374
01375 /** @brief Start MTU0 counter (TSTRA bit 0) */
01376 k_mtu_tstr_cst0 = (k_mtu_bit_value_set « k_mtu_tstr_cst0_shift),
01377
01378 /** @brief Start MTU1 counter (TSTRA bit 1) */
01379 k_mtu_tstr_cst1 = (k_mtu_bit_value_set « k_mtu_tstr_cst1_shift),
01380
01381 /** @brief Start MTU2 counter (TSTRA bit 2) */
01382 k_mtu_tstr_cst2 = (k_mtu_bit_value_set « k_mtu_tstr_cst2_shift),
01383
01384 /** @brief Start MTU3 counter (TSTRA bit 6) */
01385 k_mtu_tstr_cst3 = (k_mtu_bit_value_set « k_mtu_tstr_cst3_shift),
01386
01387 /** @brief Start MTU4 counter (TSTRA bit 7) */
01388 k_mtu_tstr_cst4 = (k_mtu_bit_value_set « k_mtu_tstr_cst4_shift),
01389
01390 /** @brief Start MTU8 counter (TSTRA bit 3) */
01391 k_mtu_tstr_cst8 = (k_mtu_bit_value_set « k_mtu_tstr_cst8_shift),
01392 } mtu_tstr_bits_t;

```

### 5.6.3.13 mtu\_tstrb\_bits\_t

```
enum mtu_tstrb_bits_t : uint8_t
```

Timer Start Register B (TSTRB) bit definitions - controls MTU6-7.

TSTRB (at 0x000C1A80) controls the start/stop state of MTU channels 6 and 7. These channels are typically used together for complementary PWM generation.

Register Bit Layout (TSTRB)

```

* Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
* -----
* | CST7 | CST6 | - | - | - | - | - | - |
* | MTU7 | MTU6 | Reserved |
* -----
*

```

See also

[rx\\_mtu\\_tstr\\_regs\\_t](#)

[mtu\\_tstrb\(\)](#) for register accessor

RX72N Hardware Manual Section 24.2.16 (TSTRB Register)

Enumerator

|                       |                                      |
|-----------------------|--------------------------------------|
| k_mtu_tstr_cst6_shift | Counter Start 6 bit position (MTU6). |
| k_mtu_tstr_cst7_shift | Counter Start 7 bit position (MTU7). |
| k_mtu_tstr_cst6       | Start MTU6 counter (TSTRB bit 6).    |
| k_mtu_tstr_cst7       | Start MTU7 counter (TSTRB bit 7).    |

Definition at line 1415 of file [rx72n\\_mtu\\_regs.h](#).

```

01415 : uint8_t {
01416 /** @brief Counter Start 6 bit position (MTU6) */
01417 k_mtu_tstr_cst6_shift = 6,
01418
01419 /** @brief Counter Start 7 bit position (MTU7) */
01420 k_mtu_tstr_cst7_shift = 7,
01421
01422 /** @brief Start MTU6 counter (TSTRB bit 6) */
01423 k_mtu_tstr_cst6 = (k_mtu_bit_value_set « k_mtu_tstr_cst6_shift),
01424
01425 /** @brief Start MTU7 counter (TSTRB bit 7) */
01426 k_mtu_tstr_cst7 = (k_mtu_bit_value_set « k_mtu_tstr_cst7_shift),
01427 } mtu_tstrb_bits_t;

```

#### 5.6.3.14 rx\_mtu\_addresses\_t

enum [rx\\_mtu\\_addresses\\_t](#) : uintptr\_t

MTU3a peripheral base addresses (RX72N memory map Chapter 4).

Base addresses for all MTU3a timer channels. The MTU3a module occupies memory region 0x000C1200-0x000C1CFF in the RX72N peripheral address space.

Memory Map Overview

| * Address Range         | Channel | Notes                                |
|-------------------------|---------|--------------------------------------|
| * 0x000C1200-0x000C127F | MTU3/4  | Interleaved (complementary PWM pair) |
| * 0x000C1280-0x000C12FF | TSTRA   | Timer Start Register A (MTU0-4, 8)   |
| * 0x000C1300-0x000C137F | MTU0    | General-purpose timer                |
| * 0x000C1380-0x000C13FF | MTU1    | Phase counting capable               |
| * 0x000C1400-0x000C147F | MTU2    | Phase counting capable               |
| * 0x000C1600-0x000C16FF | MTU8    | 32-bit native counter                |
| * 0x000C1A00-0x000C1A7F | MTU6/7  | Interleaved (complementary PWM pair) |
| * 0x000C1A80-0x000C1AFF | TSTRB   | Timer Start Register B (MTU6-7)      |
| * 0x000C1C80-0x000C1CFF | MTU5    | Waveform generator (U/V/W phases)    |
| *                       |         |                                      |

Hardware Design Notes

- MTU3/MTU4 share base address 0x000C1200 (interleaved registers)
- MTU6/MTU7 share base address 0x000C1A00 (interleaved registers)
- Interleaving enables synchronized complementary PWM generation
- MTU1/MTU2 can cascade to form 32-bit counter (TCNTLW)



## Code Example

```
// Access MTU0 timer counter
volatile rx_mtu_channel_regs_t* mtu = mtu0();
uint16_t count = mtu->tcnt;

// Configure MTU0 for 20 kHz PWM
mtu->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra; // PCLKA/1, clear on TGRA
mtu->tgra = 5999; // Period: 120 MHz / 6000 = 20 kHz
mtu->tgrb = 3000; // 50% duty cycle
```

See also

RX72N Hardware Manual Chapter 4 (Address Map)

RX72N Hardware Manual Section 24.2 (MTU Register Descriptions)

## Enumerator

|                       |                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_mtu0_base_addr      | MTU0 base address (0x000C1300) - general purpose 16-bit timer.                                                                                                                                                                                |
| k_mtu1_base_addr      | MTU1 base address (0x000C1380) - 16-bit with phase counting, cascable to 32-bit.                                                                                                                                                              |
| k_mtu2_base_addr      | MTU2 base address (0x000C1400) - 16-bit with phase counting, cascable to 32-bit.                                                                                                                                                              |
| k_mtu3_base_addr      | MTU3 base address (0x000C1200) - complementary PWM pair with MTU4.                                                                                                                                                                            |
| k_mtu4_base_addr      | MTU4 base address (MTU3 + 1) - interleaved with MTU3. MTU4 registers are at odd offsets relative to MTU3 at even offsets. This interleaving enables synchronized complementary PWM operation.                                                 |
| k_mtu5u_base_addr     | MTU5U base address (0x000C1C80) - waveform generator U-phase.                                                                                                                                                                                 |
| k_mtu5v_base_addr     | MTU5V base address (0x000C1C90) - waveform generator V-phase.                                                                                                                                                                                 |
| k_mtu5w_base_addr     | MTU5W base address (0x000C1CA0) - waveform generator W-phase.                                                                                                                                                                                 |
| k_mtu6_base_addr      | MTU6 base address (0x000C1A00) - complementary PWM pair with MTU7. MTU6 and MTU7 share the same base address with interleaved registers.                                                                                                      |
| k_mtu7_base_addr      | MTU7 base address (MTU6 + 1) - interleaved with MTU6. MTU7 registers are at odd offsets relative to MTU6 at even offsets. This interleaving enables synchronized complementary PWM operation. (Manual: MTU7.TCR = 0x000C1A01 = MTU6 base + 1) |
| k_mtu8_base_addr      | MTU8 base address (0x000C1600) - native 32-bit counter.                                                                                                                                                                                       |
| k_mtu_tstra_base_addr | TSTRA base address (0x000C1280) - timer start control for MTU0-4, MTU8.                                                                                                                                                                       |
| k_mtu_tstrb_base_addr | TSTRB base address (0x000C1A80) - timer start control for MTU6-7.                                                                                                                                                                             |

Definition at line 199 of file [rx72n\\_mtu\\_regs.h](#).

```
00199 : uintptr_t {
00200 /** @brief MTU0 base address (0x000C1300) - general purpose 16-bit timer */
00201 k_mtu0_base_addr = 0x000C1300,
00202
00203 /** @brief MTU1 base address (0x000C1380) - 16-bit with phase counting, cascable to 32-bit */
00204 k_mtu1_base_addr = 0x000C1380,
00205
00206 /** @brief MTU2 base address (0x000C1400) - 16-bit with phase counting, cascable to 32-bit */
00207 k_mtu2_base_addr = 0x000C1400,
00208
00209 /** @brief MTU3 base address (0x000C1200) - complementary PWM pair with MTU4 */
00210 k_mtu3_base_addr = 0x000C1200,
00211
00212 /**
```

```

00213 * @brief MTU4 base address (MTU3 + 1) - interleaved with MTU3
00214 * @details MTU4 registers are at odd offsets relative to MTU3 at even offsets.
00215 * This interleaving enables synchronized complementary PWM operation.
00216 */
00217 k_mtu4_base_addr = (k_mtu3_base_addr + 1U),
00218
00219 /** @brief MTU5U base address (0x000C1C80) - waveform generator U-phase */
00220 k_mtu5u_base_addr = 0x000C1C80,
00221
00222 /** @brief MTU5V base address (0x000C1C90) - waveform generator V-phase */
00223 k_mtu5v_base_addr = 0x000C1C90,
00224
00225 /** @brief MTU5W base address (0x000C1CA0) - waveform generator W-phase */
00226 k_mtu5w_base_addr = 0x000C1CA0,
00227
00228 /**
00229 * @brief MTU6 base address (0x000C1A00) - complementary PWM pair with MTU7
00230 * @details MTU6 and MTU7 share the same base address with interleaved registers.
00231 */
00232 k_mtu6_base_addr = 0x000C1A00,
00233
00234 /**
00235 * @brief MTU7 base address (MTU6 + 1) - interleaved with MTU6
00236 * @details MTU7 registers are at odd offsets relative to MTU6 at even offsets.
00237 * This interleaving enables synchronized complementary PWM operation.
00238 * (Manual: MTU7.TCR = 0x000C1A01 = MTU6 base + 1)
00239 */
00240 k_mtu7_base_addr = (k_mtu6_base_addr + 1U),
00241
00242 /** @brief MTU8 base address (0x000C1600) - native 32-bit counter */
00243 k_mtu8_base_addr = 0x000C1600,
00244
00245 /** @brief TSTRA base address (0x000C1280) - timer start control for MTU0-4, MTU8 */
00246 k_mtu_tstra_base_addr = 0x000C1280,
00247
00248 /** @brief TSTRB base address (0x000C1A80) - timer start control for MTU6-7 */
00249 k_mtu_tstrb_base_addr = 0x000C1A80,
00250 } rx_mtu_addresses_t;

```

## 5.6.4 Function Documentation

### 5.6.4.1 mtu0()

```
volatile rx_mtu_channel_regs_t * mtu0 (
 void) [inline], [static]
```

Get pointer to MTU0 registers.

Returns a volatile pointer to MTU0's register structure at base address 0x000C1300. MTU0 is a general-purpose 16-bit timer commonly used for PWM generation or interval timing.

Returns

Volatile pointer to MTU0 register structure

### Example

```

// Read MTU0 counter value
uint16_t count = mtu0()->tcnt;

// Configure MTU0 for 20 kHz PWM
mtu0()->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra;
mtu0()->tgra = 5999; // Period: 120 MHz / 6000 = 20 kHz

```

Note

Thread-safe: returns constant hardware address

See also

[rx\\_mtu\\_channel\\_regs\\_t](#) for register layout

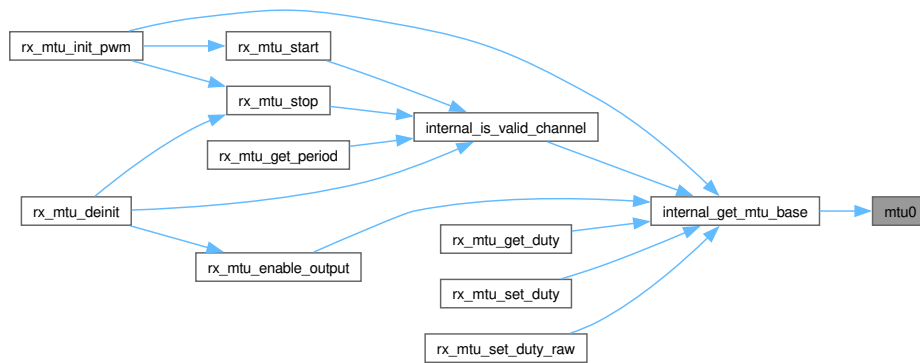
Definition at line 908 of file [rx72n\\_mtu\\_regs.h](#).

```
00909 {
00910 return (volatile rx_mtu_channel_regs_t*)k_mtu0_base_addr;
00911 }
```

References [k\\_mtu0\\_base\\_addr](#).

Referenced by [internal\\_get\\_mtu\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.6.4.2 mtu1()

```
volatile rx_mtu_channel_regs_t * mtu1 (
 void) [inline], [static]
```

Get pointer to MTU1 registers (16-bit mode).

Returns a volatile pointer to MTU1's standard 16-bit register structure. For 32-bit cascaded phase counting mode, use [mtu1\\_phase\(\)](#) instead.

Returns

Volatile pointer to MTU1 register structure

See also

[mtu1\\_phase\(\)](#) for 32-bit cascaded phase counting

[rx\\_mtu\\_channel\\_regs\\_t](#) for register layout

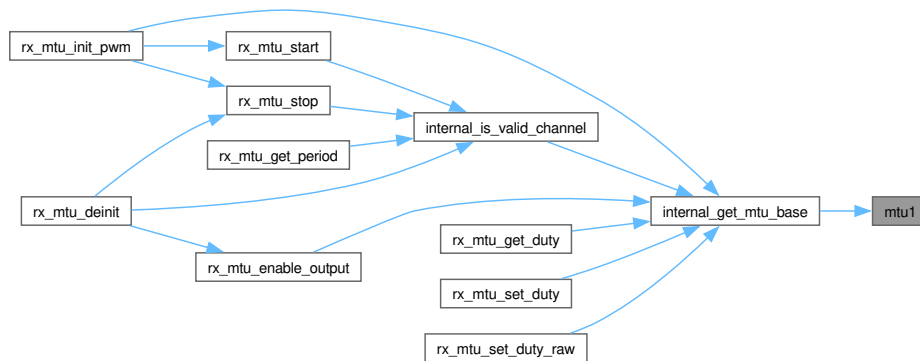
Definition at line 928 of file [rx72n\\_mtu\\_regs.h](#).

```
00929 {
00930 return (volatile rx_mtu_channel_regs_t*)k_mtu1_base_addr;
00931 }
```

References [k\\_mtu1\\_base\\_addr](#).

Referenced by [internal\\_get\\_mtu\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.6.4.3 mtu1`phase()

```
volatile rx_mtu12_phase_regs_t * mtu1_phase (
 void) [inline], [static]
```

Get pointer to MTU1 extended registers for 32-bit phase counting mode.

Provides access to MTU1's extended register set including TMDR3 and 32-bit cascaded counter registers (TCNTLW, TGRALW, TGRBLW). Use this accessor when configuring or reading 32-bit cascaded phase counting mode.

For standard 16-bit phase counting, use [mtu1\(\)](#) instead.

Returns

Volatile pointer to MTU1 extended register structure

See also

- [rx\\_mtu12\\_phase\\_regs\\_t](#) for complete register map
- RX72N Hardware Manual Section 24.2.25 for TMDR3 register
- RX72N Hardware Manual Section 24.2.26 for 32-bit registers

Definition at line 970 of file [rx72n\\_mtu\\_regs.h](#).

```
00971 {
00972 return (volatile rx_mtu12_phase_regs_t*)k_mtu1_base_addr;
00973 }
```

References [k\\_mtu1\\_base\\_addr](#).

#### 5.6.4.4 mtu2()

```
volatile rx_mtu_channel_regs_t * mtu2 (
 void) [inline], [static]
```

Get pointer to MTU2 registers (16-bit mode).

Returns a volatile pointer to MTU2's standard 16-bit register structure. For 32-bit cascaded phase counting mode, use [mtu2\\_phase\(\)](#) instead.

## Returns

Volatile pointer to MTU2 register structure

## See also

[mtu2\\_phase\(\)](#) for 32-bit cascaded phase counting

[rx\\_mtu\\_channel\\_regs\\_t](#) for register layout

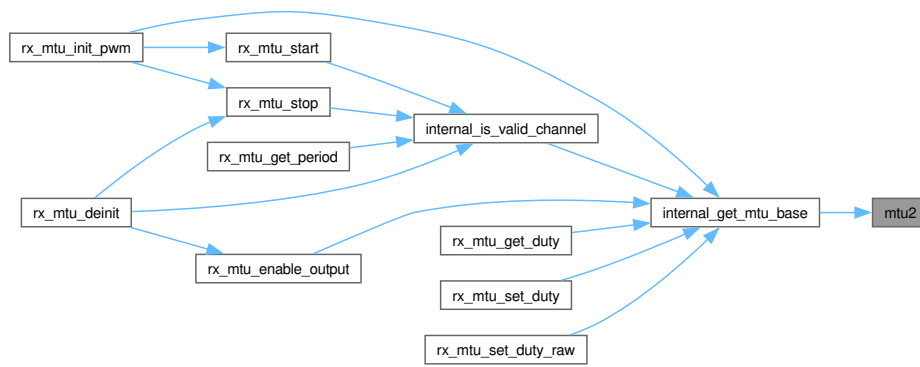
Definition at line 948 of file [rx72n\\_mtu\\_regs.h](#).

```
00949 {
00950 return (volatile rx_mtu_channel_regs_t*)k_mtu2_base_addr;
00951 }
```

References [k\\_mtu2\\_base\\_addr](#).

Referenced by [internal\\_get\\_mtu\\_base\(\)](#).

Here is the caller graph for this function:



## 5.6.4.5 mtu2\_phase()

```
volatile rx_mtu12_phase_regs_t * mtu2_phase (
 void) [inline], [static]
```

Get pointer to MTU2 extended registers for 32-bit phase counting mode.

Provides access to MTU2's extended register set including TMDR3 and 32-bit cascaded counter registers (TCNTLW, TGRALW, TGRBLW). Use this accessor when configuring or reading 32-bit cascaded phase counting mode.

For standard 16-bit phase counting, use [mtu2\(\)](#) instead.

## Returns

Volatile pointer to MTU2 extended register structure

## See also

[rx\\_mtu12\\_phase\\_regs\\_t](#) for complete register map

RX72N Hardware Manual Section 24.2.25 for TMDR3 register

RX72N Hardware Manual Section 24.2.26 for 32-bit registers

Definition at line 991 of file [rx72n\\_mtu\\_regs.h](#).

```
00992 {
00993 return (volatile rx_mtu12_phase_regs_t*)k_mtu2_base_addr;
00994 }
```

References [k\\_mtu2\\_base\\_addr](#).

#### 5.6.4.6 mtu3()

```
volatile rx_mtu3_channel_regs_t * mtu3 (
 void) [inline], [static]
```

Get pointer to MTU3 registers via [rx\\_mtu3\\_channel\\_regs\\_t](#) struct.

Returns a pointer to MTU3's register space cast to [rx\\_mtu3\\_channel\\_regs\\_t](#). The struct has been carefully padded to match the actual hardware layout verified against the RX72N Ch 5 I/O Register Address Table.

MTU3 and MTU4 registers are interleaved at 0x000C1200: MTU3 at 0x000C1200, MTU4 at 0x000C1201, with shared MTU3/4 registers interspersed. The struct fields (tcr, tmdr1, tiorh, tiorl, tier, tcnt, tgra, tgrb, tgrc, tgrd, tsr, tbtm, tcr2, tgre) are placed at the correct byte offsets for MTU3. The gap bytes in the struct are reserved/MTU4/shared-register space.

#### Warning

Do NOT use this pointer for MTU4 register access. MTU4 registers are at different offsets relative to the MTU4 base (0x000C1201). Access MTU4 registers using the individual address constants or by computing addresses manually from `k_mtu4_base_addr`.

#### Returns

Volatile pointer to MTU3 register structure (base 0x000C1200)

#### See also

[rx\\_mtu3\\_channel\\_regs\\_t](#) for complete register layout with addresses  
 RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)

Definition at line 1020 of file [rx72n\\_mtu\\_regs.h](#).

```
01021 {
01022 return (volatile rx_mtu3_channel_regs_t*)k_mtu3_base_addr;
01023 }
```

References [k\\_mtu3\\_base\\_addr](#).

#### 5.6.4.7 mtu4()

```
volatile void * mtu4 (
 void) [inline], [static]
```

Get raw base address pointer for MTU4 register access.

Returns the MTU4 base address (0x000C1201) as a void pointer. MTU4 registers are at DIFFERENT relative offsets than MTU3 registers, so the [rx\\_mtu3\\_channel\\_regs\\_t](#) struct CANNOT be used for MTU4.

MTU4 register addresses from the RX72N Ch 5 I/O Address Table:

- MTU4.TCR = 0x000C1201 (`k_mtu4_base_addr + 0x00`)
- MTU4.TMDR1 = 0x000C1203 (`k_mtu4_base_addr + 0x02`)

- MTU4.TIORH = 0x000C1206 (k\_mtu4\_base\_addr + 0x05) – NOTE: +0x05, not +0x04
- MTU4.TIORL = 0x000C1207 (k\_mtu4\_base\_addr + 0x06)
- MTU4.TIER = 0x000C1209 (k\_mtu4\_base\_addr + 0x08)
- MTU4.TCNT = 0x000C1212 (k\_mtu4\_base\_addr + 0x11) – NOTE: odd address
- MTU4.TGRA = 0x000C121C (k\_mtu4\_base\_addr + 0x1B)
- MTU4.TGRB = 0x000C121E (k\_mtu4\_base\_addr + 0x1D)
- MTU4.TGRC = 0x000C1228 (k\_mtu4\_base\_addr + 0x27)
- MTU4.TGRD = 0x000C122A (k\_mtu4\_base\_addr + 0x29)
- MTU4.TSR = 0x000C122D (k\_mtu4\_base\_addr + 0x2C)
- MTU4.TBTM = 0x000C1239 (k\_mtu4\_base\_addr + 0x38)
- MTU4.TCR2 = 0x000C124D (k\_mtu4\_base\_addr + 0x4C)
- MTU4.TGRE = 0x000C1274 (k\_mtu4\_base\_addr + 0x73)
- MTU4.TGRF = 0x000C1276 (k\_mtu4\_base\_addr + 0x75) – MTU3 has no TGRF

Access MTU4 registers by casting the returned pointer to the appropriate volatile integer pointer and applying the correct offset, e.g.: (volatile uint8\_t)((uintptr\_t)mtu4() + 0x05U) = value; // MTU4.TIORH

Returns

Volatile void pointer to MTU4 base address (0x000C1201)

See also

RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)

Definition at line 1058 of file rx72n\_mtu\_regs.h.

```
01059 {
01060 return (volatile void*)k_mtu4_base_addr;
01061 }
```

References [k\\_mtu4\\_base\\_addr](#).

#### 5.6.4.8 mtu6()

```
volatile rx_mtu3_channel_regs_t * mtu6 (
 void) [inline], [static]
```

Get pointer to MTU6 registers (interleaved layout, identical to MTU3).

MTU6/MTU7 registers are interleaved at 0x000C1A00 in the same pattern as MTU3/MTU4 at 0x000C1200. MTU6 registers are at even offsets; MTU7 registers are at odd offsets. Use [rx\\_mtu3\\_channel\\_regs\\_t](#) to access MTU6 fields at their correct byte offsets (TCR+=+0x00, TMDR1+=+0x02, TIORH+=+0x04, etc.).

## Warning

Do NOT cast this pointer to `rx_mtu_channel_regs_t` – that struct assumes consecutive layout (TMDR at +0x01) which is wrong for MTU6.

## Returns

Volatile pointer to MTU6 register structure (interleaved layout)

## See also

[rx\\_mtu3\\_channel\\_regs\\_t](#) for register layout and field offsets  
[mtu7\(\)](#) for raw MTU7 base pointer (void\*)

Definition at line 1080 of file `rx72n_mtu_regs.h`.

```
01081 {
01082 return (volatile rx_mtu3_channel_regs_t*)k_mtu6_base_addr;
01083 }
```

References [k\\_mtu6\\_base\\_addr](#).

## 5.6.4.9 mtu7()

```
volatile void * mtu7 (
 void) [inline], [static]
```

Get raw base address pointer for MTU7 register access.

Returns the MTU7 base address (`k_mtu6_base_addr + 1 = 0x000C1A01`) as a void pointer. MTU7 registers are at DIFFERENT relative offsets than MTU6 registers (same situation as MTU4 relative to MTU3), so no struct type can be directly applied. Access MTU7 registers by computing addresses manually from `k_mtu7_base_addr` using the Ch 5 I/O Address Table.

## Returns

Volatile void pointer to MTU7 base address (0x000C1A01)

## See also

[mtu6\(\)](#) for MTU6 access via [rx\\_mtu3\\_channel\\_regs\\_t](#)  
 RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)

Definition at line 1100 of file `rx72n_mtu_regs.h`.

```
01101 {
01102 return (volatile void*)k_mtu7_base_addr;
01103 }
```

References [k\\_mtu7\\_base\\_addr](#).



5.6.4.10 `mtu_tstra()`

```
volatile rx_mtu_tstr_regs_t * mtu_tstra (
 void) [inline], [static]
```

Get pointer to MTU TSTRA registers (MTU0-4, MTU8).

Returns

Volatile pointer to MTU TSTRA register structure

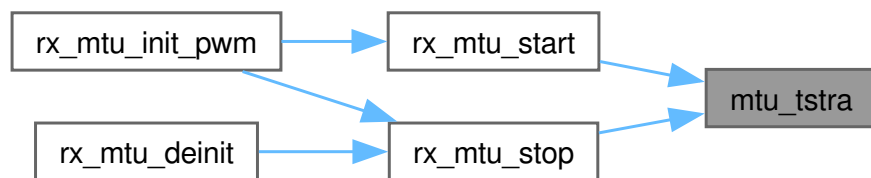
Definition at line 1111 of file `rx72n_mtu_regs.h`.

```
01112 {
01113 return (volatile rx_mtu_tstr_regs_t*)k_mtu_tstra_base_addr;
01114 }
```

References `k_mtu_tstra_base_addr`.

Referenced by `rx_mtu_start()`, and `rx_mtu_stop()`.

Here is the caller graph for this function:

5.6.4.11 `mtu_tstrb()`

```
volatile rx_mtu_tstr_regs_t * mtu_tstrb (
 void) [inline], [static]
```

Get pointer to MTU TSTRB registers (MTU6-7).

Returns

Volatile pointer to MTU TSTRB register structure

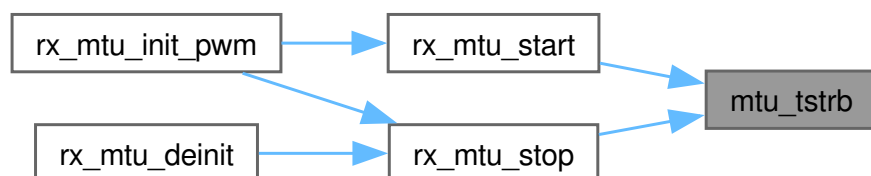
Definition at line 1121 of file `rx72n_mtu_regs.h`.

```
01122 {
01123 return (volatile rx_mtu_tstr_regs_t*)k_mtu_tstrb_base_addr;
01124 }
```

References `k_mtu_tstrb_base_addr`.

Referenced by `rx_mtu_start()`, and `rx_mtu_stop()`.

Here is the caller graph for this function:



## 5.7 PORT Register Definitions

### Data Structures

- struct `rx_port_regs_t`  
Port Register Map providing logical per-port view of RX72N GPIO.

### Enumerations

- enum `rx_port_reg_bases_t` : `uintptr_t` {  
`k_port_pdr_base` = 0x0008C000 , `k_port_podr_base` = 0x0008C020 , `k_port_pidr_base` = 0x0008C040 , `k_port_pmr_base` = 0x0008C060 ,  
`k_port_odr0_base` = 0x0008C080 , `k_port_pcr_base` = 0x0008C0C0 , `k_port_dscr_base` = 0x0008C0E0 , `k_port_dscr2_base` = 0x0008C128 }  
PORT register block base addresses (organized by register type).
- enum `rx_port_offsets_t` : `uint8_t` {  
`k_port_offset_0` = 0x00 , `k_port_offset_1` = 0x01 , `k_port_offset_2` = 0x02 , `k_port_offset_3` = 0x03 ,  
`k_port_offset_4` = 0x04 , `k_port_offset_5` = 0x05 , `k_port_offset_6` = 0x06 , `k_port_offset_7` = 0x07 ,  
`k_port_offset_8` = 0x08 , `k_port_offset_9` = 0x09 , `k_port_offset_a` = 0x0A , `k_port_offset_b` = 0x0B ,  
`k_port_offset_c` = 0x0C , `k_port_offset_d` = 0x0D , `k_port_offset_e` = 0x0E , `k_port_offset_f` = 0x0F ,  
`k_port_offset_j` = 0x12 }  
Port number offsets for register access (144-pin LQFP).

### Functions

- static volatile `rx_port_regs_t * port0` (void)  
Get pointer to PORT0 registers.
- static volatile `rx_port_regs_t * port1` (void)  
Get pointer to PORT1 registers.
- static volatile `rx_port_regs_t * port2` (void)  
Get pointer to PORT2 registers.
- static volatile `rx_port_regs_t * port3` (void)  
Get pointer to PORT3 registers.
- static volatile `rx_port_regs_t * port4` (void)  
Get pointer to PORT4 registers.
- static volatile `rx_port_regs_t * port5` (void)  
Get pointer to PORT5 registers.
- static volatile `rx_port_regs_t * port6` (void)  
Get pointer to PORT6 registers.
- static volatile `rx_port_regs_t * port7` (void)  
Get pointer to PORT7 registers.
- static volatile `rx_port_regs_t * port8` (void)  
Get pointer to PORT8 registers.
- static volatile `rx_port_regs_t * port9` (void)  
Get pointer to PORT9 registers.
- static volatile `rx_port_regs_t * porta` (void)  
Get pointer to PORTA registers.

- static volatile `rx_port_regs_t * portb` (void)  
Get pointer to PORTB registers.
- static volatile `rx_port_regs_t * portc` (void)  
Get pointer to PORTC registers.
- static volatile `rx_port_regs_t * portd` (void)  
Get pointer to PORTD registers.
- static volatile `rx_port_regs_t * porte` (void)  
Get pointer to PORTE registers.
- static volatile `rx_port_regs_t * portf` (void)  
Get pointer to PORTF registers.
- static volatile `rx_port_regs_t * portj` (void)  
Get pointer to PORTJ registers.
- static volatile `uint16_t * port0_odr` (void)  
Get pointer to PORT0 ODR register.
- static volatile `uint16_t * port1_odr` (void)  
Get pointer to PORT1 ODR register.
- static volatile `uint16_t * port2_odr` (void)  
Get pointer to PORT2 ODR register.
- static volatile `uint16_t * port3_odr` (void)  
Get pointer to PORT3 ODR register.
- static volatile `uint16_t * port4_odr` (void)  
Get pointer to PORT4 ODR register.
- static volatile `uint16_t * port5_odr` (void)  
Get pointer to PORT5 ODR register.
- static volatile `uint16_t * port6_odr` (void)  
Get pointer to PORT6 ODR register.
- static volatile `uint16_t * port7_odr` (void)  
Get pointer to PORT7 ODR register.
- static volatile `uint16_t * port8_odr` (void)  
Get pointer to PORT8 ODR register.
- static volatile `uint16_t * port9_odr` (void)  
Get pointer to PORT9 ODR register.
- static volatile `uint16_t * porta_odr` (void)  
Get pointer to PORTA ODR register.
- static volatile `uint16_t * portb_odr` (void)  
Get pointer to PORTB ODR register.
- static volatile `uint16_t * portc_odr` (void)  
Get pointer to PORTC ODR register.
- static volatile `uint16_t * portd_odr` (void)  
Get pointer to PORTD ODR register.
- static volatile `uint16_t * porte_odr` (void)  
Get pointer to PORTE ODR register.
- static volatile `uint16_t * portf_odr` (void)  
Get pointer to PORTF ODR register.
- static volatile `uint16_t * portj_odr` (void)  
Get pointer to PORTJ ODR register.

### 5.7.1 Detailed Description

### 5.7.2 Data Structure Documentation

#### 5.7.2.1 struct rx'port'regs't

Port Register Map providing logical per-port view of RX72N GPIO.

This structure provides a convenient PORT-centric view of the GPIO registers, abstracting away the hardware's type-grouped layout. Padding bytes ensure correct alignment between register types.

Register Memory Layout (297 bytes total per port view)

| * Offset | Size | Register | Description                                    |
|----------|------|----------|------------------------------------------------|
| * 0x00   | 1    | PDR      | Port Direction (0=input, 1=output)             |
| * 0x01   | 31   | [pad]    | Reserved padding to PODR                       |
| * 0x20   | 1    | PODR     | Port Output Data (0=low, 1=high)               |
| * 0x21   | 31   | [pad]    | Reserved padding to PIDR                       |
| * 0x40   | 1    | PIDR     | Port Input Data (read-only, current pin level) |
| * 0x41   | 31   | [pad]    | Reserved padding to PMR                        |
| * 0x60   | 1    | PMR      | Port Mode (0=GPIO, 1=peripheral function)      |
| * 0x61   | 95   | [pad]    | Reserved (includes ODR space, see note below)  |
| * 0xC0   | 1    | PCR      | Pull-up Control (0=disabled, 1=enabled)        |
| * 0xC1   | 31   | [pad]    | Reserved padding to DSCR                       |
| * 0xE0   | 1    | DSCR     | Drive Capacity (0=normal, 1=high drive)        |
| * 0xE1   | 71   | [pad]    | Reserved padding to DSCR2                      |
| * 0x128  | 1    | DSCR2    | Drive Capacity 2 (enhanced high drive)         |
| *        |      |          |                                                |

Note

ODR (Open Drain Control) registers are NOT part of this structure. They use word addressing and are accessed via separate port\*\_odr() functions at hardware addresses: 0x0008C080 + (port\_offset \* 2)

#### GPIO Configuration Sequence

Typical GPIO output configuration sequence:

1. Set PMR = 0 (GPIO mode, not peripheral)
2. Set PDR = 1 (output direction)
3. Set PODR = desired output level
4. Optionally set DSCR = 1 for high drive current

Typical GPIO input configuration sequence:

1. Set PMR = 0 (GPIO mode)
2. Set PDR = 0 (input direction)
3. Optionally set PCR = 1 for internal pull-up
4. Read PIDR for current pin state

## LED Blink Example

```
// Configure PD0 as output and blink LED
volatile rx_port_regs_t* pd = portd();

pd->pmr &= ~(1 << 0); // GPIO mode
pd->pdr |= (1 << 0); // Output direction

while (1) {
 pd->podr ^= (1 << 0); // Toggle LED
 delay_ms(500);
}
```

## Button Read Example

```
// Configure PA0 as input with pull-up and read button
volatile rx_port_regs_t* pa = porta();

pa->pmr &= ~(1 << 0); // GPIO mode
pa->pdr &= ~(1 << 0); // Input direction
pa->pcr |= (1 << 0); // Enable pull-up (button connects to GND)

bool pressed = ((pa->pidr & (1 << 0)) == 0); // Active-low
```

## Invariant

Structure size must be exactly 297 bytes (0x129)  
 All register offsets must match hardware layout exactly  
 Structure must be packed to ensure correct padding sizes

## Note

Uses attribute((packed)) to prevent compiler padding optimizations

## Warning

This structure assumes PORT0 padding layout (offset=0x00). The type-grouped hardware layout places ODR at varying struct offsets for different ports (0x80 + port\_offset\*2). ODR is accessed via separate port\*\_odr() functions to ensure correct word addressing.

Do not use sizeof() for port address calculations; use offset enum

## Note

PORT3 lacks DSCR (has only DSCR2).  
 PORT4 lacks both DSCR and DSCR2 (hardware limitation).  
 All other ports (0-2, 5, A-E, J) have both DSCR and DSCR2.

## See also

[rx\\_port\\_reg\\_bases\\_t](#) for register base addresses  
[rx\\_port\\_offsets\\_t](#) for port offset values  
 RX72N Hardware Manual Section 22.3 (Register Descriptions)

Definition at line 463 of file [rx72n\\_port\\_regs.h](#).

Collaboration diagram for rx\_port\_regs\_t:



#### Data Fields

|                  |                                 |                                                                                                                                                                                                                                          |
|------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | <code>__pad1</code> ↔<br>[0x1F] | Reserved padding (0x01-0x1F) to align PODR at +0x20.                                                                                                                                                                                     |
| volatile uint8_t | <code>__pad2</code> ↔<br>[0x1F] | Reserved padding (0x21-0x3F) to align PIDR at +0x40.                                                                                                                                                                                     |
| volatile uint8_t | <code>__pad3</code> ↔<br>[0x1F] | Reserved padding (0x41-0x5F) to align PMR at +0x60.                                                                                                                                                                                      |
| volatile uint8_t | <code>__pad4</code> ↔<br>[0x5F] | Reserved padding (0x61-0xBF) to align PCR at +0xC0.                                                                                                                                                                                      |
| volatile uint8_t | <code>__pad5</code> ↔<br>[0x1F] | Reserved padding (0xC1-0xDF) to align DSCR at +0xE0.                                                                                                                                                                                     |
| volatile uint8_t | <code>__pad6</code> ↔<br>[0x47] | Reserved padding (0xE1-0x127) to align DSCR2 at +0x128.                                                                                                                                                                                  |
| volatile uint8_t | dscr                            | Drive Capacity Control Register (DSCR) - +0xE0.<br>Configures output drive strength. DSCR bit = 0: Normal drive capacity. DSCR bit = 1: High drive capacity.<br><br>Note<br>High drive useful for long traces or high capacitance loads. |

|                  |       |                                                                                                                                                                                                                                                                                                                                       |
|------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | dscr2 | <p>Drive Capacity Control Register 2 (DSCR2) - +0x128.<br/>Secondary drive capacity control for high-current applications. DSCR2 bit = 0: Normal drive capacity (same as DSCR). DSCR2 bit = 1: Enhanced high drive capacity.</p> <p>Note</p> <p>Not available on PORT4 (hardware limitation).<br/>PORT3 lacks DSCR but has DSCR2.</p> |
| volatile uint8_t | pcr   | <p>Pull-up Control Register (PCR) - +0xC0.<br/>Enables internal pull-up resistor (approx 50kOhm typical). PCR bit = 0: Pull-up disabled. PCR bit = 1: Pull-up enabled.</p> <p>Note</p> <p>Only effective when PDR bit = 0 (input direction).</p>                                                                                      |
| volatile uint8_t | pdr   | <p>Port Direction Register (PDR) - +0x00.<br/>Configures each pin as input (0) or output (1).</p> <p>Note</p> <p>Before writing PDR, ensure PMR = 0 for GPIO mode.</p>                                                                                                                                                                |
| volatile uint8_t | pidr  | <p>Port Input Data Register (PIDR) - +0x40.<br/>Read-only register returning current pin logic level.</p> <p>Note</p> <p>Always readable regardless of PDR setting.</p>                                                                                                                                                               |
| volatile uint8_t | pmr   | <p>Port Mode Register (PMR) - +0x60.<br/>Selects GPIO (0) or peripheral function (1) mode.</p> <p>Note</p> <p>Set PMR = 0 before configuring GPIO direction/output.</p> <p>See also</p> <p>MPC (Multi-function Pin Controller) for peripheral selection.</p>                                                                          |
| volatile uint8_t | podr  | <p>Port Output Data Register (PODR) - +0x20.<br/>Sets output level: 0 = low, 1 = high.</p> <p>Note</p> <p>Only effective when PDR bit = 1 (output direction).</p>                                                                                                                                                                     |

### 5.7.3 Enumeration Type Documentation

#### 5.7.3.1 rx\_port\_offsets\_t

```
enum rx_port_offsets_t : uint8_t
```

Port number offsets for register access (144-pin LFQFP).

Offset values to add to register base addresses for accessing specific ports. The 144-pin LFQFP package has broader port availability compared to the 144-pin package. Ports G, H, K, L, M, N, Q are not bonded out.

#### Available Pins by Port (144-pin LFQFP)

| Port | Offset | Available Pins    | Notes                       |
|------|--------|-------------------|-----------------------------|
| 0    | 0x00   | P00-P03, P05, P07 | 6 pins (no P04, P06)        |
| 1    | 0x01   | P12-P17           | 6 pins (no P10, P11)        |
| 2    | 0x02   | P20-P27           | Full 8 pins                 |
| 3    | 0x03   | P30-P37           | Full 8 pins, P35 input-only |
| 4    | 0x04   | P40-P47           | Full 8 pins                 |
| 5    | 0x05   | P50-P56           | 7 pins (no P57)             |
| 6    | 0x06   | P60-P67           | Full 8 pins                 |
| 7    | 0x07   | P70-P77           | Full 8 pins                 |
| 8    | 0x08   | P80-P83, P86-P87  | 6 pins (no P84, P85)        |
| 9    | 0x09   | P90-P93           | 4 pins (no P94-P97)         |
| A    | 0x0A   | PA0-PA7           | Full 8 pins                 |
| B    | 0x0B   | PB0-PB7           | Full 8 pins                 |
| C    | 0x0C   | PC0-PC7           | Full 8 pins                 |
| D    | 0x0D   | PD0-PD7           | Full 8 pins                 |
| E    | 0x0E   | PE0-PE7           | Full 8 pins                 |
| F    | 0x0F   | PF5               | 1 pin only                  |
| J    | 0x12   | PJ3, PJ5          | 2 pins only                 |

#### Usage Example

```
// Calculate PORTB PDR address (uintptr_t preserves full pointer width on all hosts)
uintptr_t portb_pdr_addr = k_port_pdr_base + k_port_offset_b; // 0x0008C00B

// Calculate PORTD PODR address
uintptr_t portd_podr_addr = k_port_podr_base + k_port_offset_d; // 0x0008C02D
```

See also

[rx\\_port\\_reg\\_bases\\_t](#) for register base addresses

RX72N Hardware Manual Chapter 22.1.1 (Package Pin Availability)

#### Enumerator

|                 |                                             |
|-----------------|---------------------------------------------|
| k_port_offset_0 | Port 0 offset - P00-P03, P05, P07 (6 pins). |
| k_port_offset_1 | Port 1 offset - P12-P17 (6 pins).           |
| k_port_offset_2 | Port 2 offset - P20-P27 (8 pins).           |



|                 |                                                   |
|-----------------|---------------------------------------------------|
| k_port_offset_3 | Port 3 offset - P30-P37 (8 pins, P35 input-only). |
| k_port_offset_4 | Port 4 offset - P40-P47 (8 pins).                 |
| k_port_offset_5 | Port 5 offset - P50-P56 (7 pins).                 |
| k_port_offset_6 | Port 6 offset - P60-P67 (8 pins).                 |
| k_port_offset_7 | Port 7 offset - P70-P77 (8 pins).                 |
| k_port_offset_8 | Port 8 offset - P80-P83, P86-P87 (6 pins).        |
| k_port_offset_9 | Port 9 offset - P90-P93 (4 pins).                 |
| k_port_offset_a | Port A offset - PA0-PA7 (8 pins).                 |
| k_port_offset_b | Port B offset - Full: PB0-PB7 (8 pins).           |
| k_port_offset_c | Port C offset - Full: PC0-PC7 (8 pins).           |
| k_port_offset_d | Port D offset - Full: PD0-PD7 (8 pins).           |
| k_port_offset_e | Port E offset - PE0-PE7 (8 pins).                 |
| k_port_offset_f | Port F offset - PF5 (1 pin).                      |
| k_port_offset_j | Port J offset - PJ3, PJ5 (2 pins).                |

Definition at line 313 of file [rx72n\\_port\\_regs.h](#).

```

00313 : uint8_t {
00314 /** @brief Port 0 offset - P00-P03, P05, P07 (6 pins) */
00315 k_port_offset_0 = 0x00,
00316
00317 /** @brief Port 1 offset - P12-P17 (6 pins) */
00318 k_port_offset_1 = 0x01,
00319
00320 /** @brief Port 2 offset - P20-P27 (8 pins) */
00321 k_port_offset_2 = 0x02,
00322
00323 /** @brief Port 3 offset - P30-P37 (8 pins, P35 input-only) */
00324 k_port_offset_3 = 0x03,
00325
00326 /** @brief Port 4 offset - P40-P47 (8 pins) */
00327 k_port_offset_4 = 0x04,
00328
00329 /** @brief Port 5 offset - P50-P56 (7 pins) */
00330 k_port_offset_5 = 0x05,
00331
00332 /** @brief Port 6 offset - P60-P67 (8 pins) */
00333 k_port_offset_6 = 0x06,
00334
00335 /** @brief Port 7 offset - P70-P77 (8 pins) */
00336 k_port_offset_7 = 0x07,
00337
00338 /** @brief Port 8 offset - P80-P83, P86-P87 (6 pins) */
00339 k_port_offset_8 = 0x08,
00340
00341 /** @brief Port 9 offset - P90-P93 (4 pins) */
00342 k_port_offset_9 = 0x09,
00343
00344 /** @brief Port A offset - PA0-PA7 (8 pins) */
00345 k_port_offset_a = 0x0A,

```

```

00346
00347 /** @brief Port B offset - Full: PB0-PB7 (8 pins) */
00348 k_port_offset_b = 0x0B,
00349
00350 /** @brief Port C offset - Full: PC0-PC7 (8 pins) */
00351 k_port_offset_c = 0x0C,
00352
00353 /** @brief Port D offset - Full: PD0-PD7 (8 pins) */
00354 k_port_offset_d = 0x0D,
00355
00356 /** @brief Port E offset - PE0-PE7 (8 pins) */
00357 k_port_offset_e = 0x0E,
00358
00359 /** @brief Port F offset - PF5 (1 pin) */
00360 k_port_offset_f = 0x0F,
00361
00362 /** @brief Port J offset - PJ3, PJ5 (2 pins) */
00363 k_port_offset_j = 0x12,
00364 } rx_port_offsets_t;

```

### 5.7.3.2 rx\_port\_reg\_bases\_t

```
enum rx_port_reg_bases_t : uintptr_t
```

PORT register block base addresses (organized by register type).

The RX72N organizes port registers by type rather than by port number. All PDR registers are contiguous, all PODR registers are contiguous, etc. This enum provides base addresses for each register type block.

### Memory Map Overview

| * Address Range         | Register Type | Description                          |
|-------------------------|---------------|--------------------------------------|
| * -----                 |               |                                      |
| * 0x0008C000-0x0008C017 | PDR[]         | Port Direction Registers             |
| * 0x0008C020-0x0008C037 | PODR[]        | Port Output Data Registers           |
| * 0x0008C040-0x0008C057 | PIDR[]        | Port Input Data Registers            |
| * 0x0008C060-0x0008C077 | PMR[]         | Port Mode Registers                  |
| * 0x0008C080-0x0008C0B7 | ODR0/ODR1[]   | Open Drain Control (16-bit per port) |
| * 0x0008C0C0-0x0008C0D7 | PCR[]         | Pull-up Control Registers            |
| * 0x0008C0E0-0x0008C0F7 | DSCR[]        | Drive Capacity Registers             |
| * 0x0008C128-0x0008C13F | DSCR2[]       | Drive Capacity 2 Registers           |
| *                       |               |                                      |

### Register Access Pattern

To access a specific port's register, add the port offset to the base:

- PORT0.PDR = k\_port\_pdr\_base + k\_port\_offset\_0 = 0x0008C000
- PORTB.PDR = k\_port\_pdr\_base + k\_port\_offset\_b = 0x0008C00B
- PORTD.PODR = k\_port\_podr\_base + k\_port\_offset\_d = 0x0008C02D

See also

[rx\\_port\\_offsets\\_t](#) for port offset values

RX72N Hardware Manual Section 22.3 (Register Descriptions)

## Enumerator

|                   |                                                                                                                                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_port_pdr_base   | PDR (Port Direction Register) base address. All PDRn registers at 0x0008C000 + port_offset. PDR bit = 0: Input, PDR bit = 1: Output.                                                                                                                                          |
| k_port_podr_base  | PODR (Port Output Data Register) base address. All PODRn registers at 0x0008C020 + port_offset. PODR bit = 0: Output low, PODR bit = 1: Output high.                                                                                                                          |
| k_port_pidr_base  | PIDR (Port Input Data Register) base address. All PIDRn registers at 0x0008C040 + port_offset. Read-only. Returns current pin logic level.                                                                                                                                    |
| k_port_pmr_base   | PMR (Port Mode Register) base address. All PMRn registers at 0x0008C060 + port_offset. PMR bit = 0: GPIO mode, PMR bit = 1: Peripheral function.                                                                                                                              |
| k_port_odr0_base  | <p>ODR0 (Open Drain Control 0) base address. ODR0/ODR1 form a 16-bit register per port at 0x0008C080 + 2*port_offset. ODR bit = 0: CMOS output, ODR bit = 1: Open-drain output.</p> <p>Note</p> <p>ODR1 immediately follows ODR0; accessed as 16-bit odr field in struct.</p> |
| k_port_pcr_base   | PCR (Pull-up Control Register) base address. All PCRN registers at 0x0008C0C0 + port_offset. PCR bit = 0: Pull-up disabled, PCR bit = 1: Pull-up enabled.                                                                                                                     |
| k_port_dscr_base  | DSCR (Drive Capacity Control Register) base address. All DSCRn registers at 0x0008C0E0 + port_offset. DSCR bit = 0: Normal drive, DSCR bit = 1: High drive.                                                                                                                   |
| k_port_dscr2_base | <p>DSCR2 (Drive Capacity Control Register 2) base address. All DSCR2n registers at 0x0008C128 + port_offset. DSCR2 bit = 0: Normal drive (same as DSCR). DSCR2 bit = 1: Enhanced high drive capacity.</p> <p>Note</p> <p>Not available on PORT4.</p>                          |

Definition at line 210 of file rx72n\_port\_regs.h.

```

00210 : uintptr_t {
00211 /**
00212 * @brief PDR (Port Direction Register) base address
00213 * @details All PDRn registers at 0x0008C000 + port_offset.
00214 * PDR bit = 0: Input, PDR bit = 1: Output.
00215 */
00216 k_port_pdr_base = 0x0008C000,
00217
00218 /**
00219 * @brief PODR (Port Output Data Register) base address
00220 * @details All PODRn registers at 0x0008C020 + port_offset.
00221 * PODR bit = 0: Output low, PODR bit = 1: Output high.
00222 */
00223 k_port_podr_base = 0x0008C020,
00224
00225 /**
00226 * @brief PIDR (Port Input Data Register) base address
00227 * @details All PIDRn registers at 0x0008C040 + port_offset.
00228 * Read-only. Returns current pin logic level.
00229 */
00230 k_port_pidr_base = 0x0008C040,
00231
00232 /**
00233 * @brief PMR (Port Mode Register) base address
00234 * @details All PMRn registers at 0x0008C060 + port_offset.
00235 * PMR bit = 0: GPIO mode, PMR bit = 1: Peripheral function.
00236 */
00237 k_port_pmr_base = 0x0008C060,
00238
00239 /**
00240 * @brief ODR0 (Open Drain Control 0) base address
00241 * @details ODR0/ODR1 form a 16-bit register per port at 0x0008C080 + 2*port_offset.

```

```

00242 * ODR bit = 0: CMOS output, ODR bit = 1: Open-drain output.
00243 * @note ODR1 immediately follows ODR0; accessed as 16-bit odr field in struct.
00244 */
00245 k_port_odr_base = 0x0008C080,
00246
00247 /**
00248 * @brief PCR (Pull-up Control Register) base address
00249 * @details All PCRn registers at 0x0008C0C0 + port_offset.
00250 * PCR bit = 0: Pull-up disabled, PCR bit = 1: Pull-up enabled.
00251 */
00252 k_port_pcr_base = 0x0008C0C0,
00253
00254 /**
00255 * @brief DSCR (Drive Capacity Control Register) base address
00256 * @details All DSCRn registers at 0x0008C0E0 + port_offset.
00257 * DSCR bit = 0: Normal drive, DSCR bit = 1: High drive.
00258 */
00259 k_port_dscr_base = 0x0008C0E0,
00260
00261 /**
00262 * @brief DSCR2 (Drive Capacity Control Register 2) base address
00263 * @details All DSCR2n registers at 0x0008C128 + port_offset.
00264 * DSCR2 bit = 0: Normal drive (same as DSCR).
00265 * DSCR2 bit = 1: Enhanced high drive capacity.
00266 * @note Not available on PORT4.
00267 */
00268 k_port_dscr2_base = 0x0008C128,
00269 } rx_port_reg_bases_t;

```

## 5.7.4 Function Documentation

### 5.7.4.1 port0()

```
volatile rx_port_regs_t * port0 (
 void) [inline], [static]
```

Get pointer to PORT0 registers.

Returns a volatile pointer to PORT0's register structure. PORT0 has P00-P03, P05, P07 available on the 144-pin LFQFP package (6 pins).

Returns

Volatile pointer to PORT0 register structure

Register Addresses

- PDR: 0x0008C000
- PODR: 0x0008C020
- PIDR: 0x0008C040
- PMR: 0x0008C060

Available Pins (144-pin)

P00-P03, P05, P07 (6 pins)

Note

Thread-safe: returns constant hardware address

See also

[rx\\_port\\_regs\\_t](#) for register layout

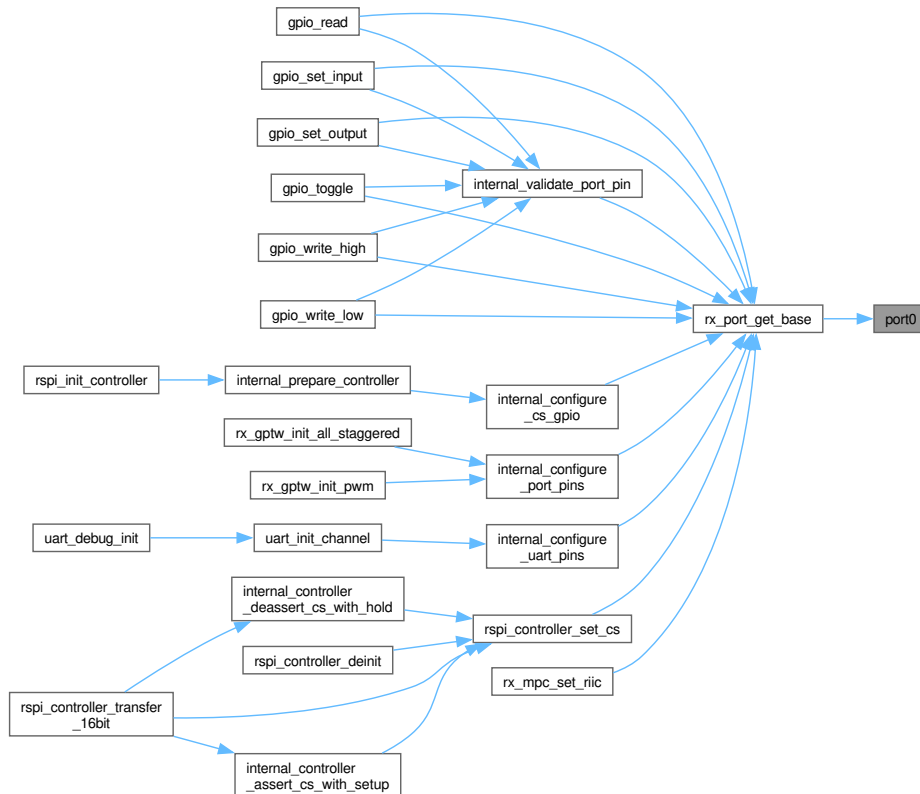
Definition at line 608 of file [rx72n\\_port\\_regs.h](#).

```
00609 {
00610 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_0);
00611 }
```

References [k\\_port\\_offset\\_0](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.2 port0`odr()

```
volatile uint16_t * port0_odr (
 void) [inline], [static]
```

Get pointer to PORT0 ODR register.

Returns

Volatile pointer to 16-bit ODR register (combines ODR0+ODR1)

Note

PORT0.ODR = 0x0008C080

Definition at line 826 of file [rx72n\\_port\\_regs.h](#).

```
00827 {
00828 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_0 * 2));
00829 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_0](#).

### 5.7.4.3 port1()

```
volatile rx_port_regs_t * port1 (
 void) [inline], [static]
```

Get pointer to PORT1 registers.

Returns

Volatile pointer to PORT1 register structure

Note

PORT1 PDR = 0x0008C001, PODR = 0x0008C021, PIDR = 0x0008C041

144-pin: Only P12-P17 available

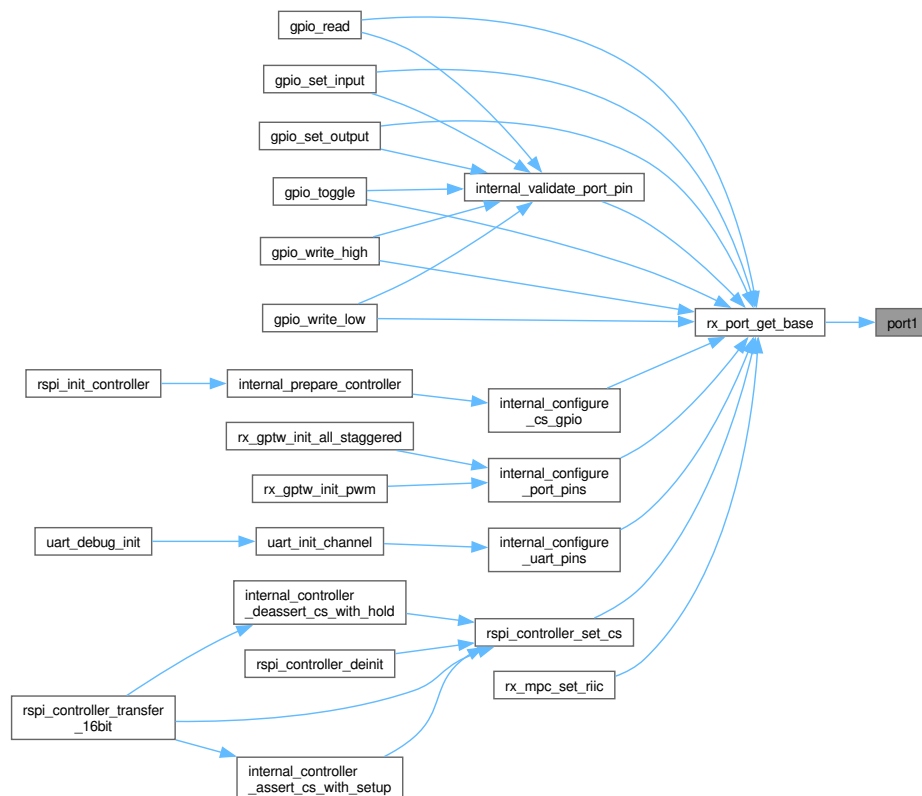
Definition at line 619 of file [rx72n\\_port\\_regs.h](#).

```
00620 {
00621 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_1);
00622 }
```

References [k\\_port\\_offset\\_1](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



## 5.7.4.4 port1\_odr()

```
volatile uint16_t * port1_odr (
 void) [inline], [static]
```

Get pointer to PORT1 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT1.ODR = 0x0008C082 (word addressing: base + offset\*2)

Definition at line 836 of file [rx72n\\_port\\_regs.h](#).

```
00837 {
00838 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_1 * 2));
00839 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_1](#).

## 5.7.4.5 port2()

```
volatile rx_port_regs_t * port2 (
 void) [inline], [static]
```

Get pointer to PORT2 registers.

Returns

Volatile pointer to PORT2 register structure

Note

PORT2 PDR = 0x0008C002, PODR = 0x0008C022, PIDR = 0x0008C042  
144-pin: P20-P27 available

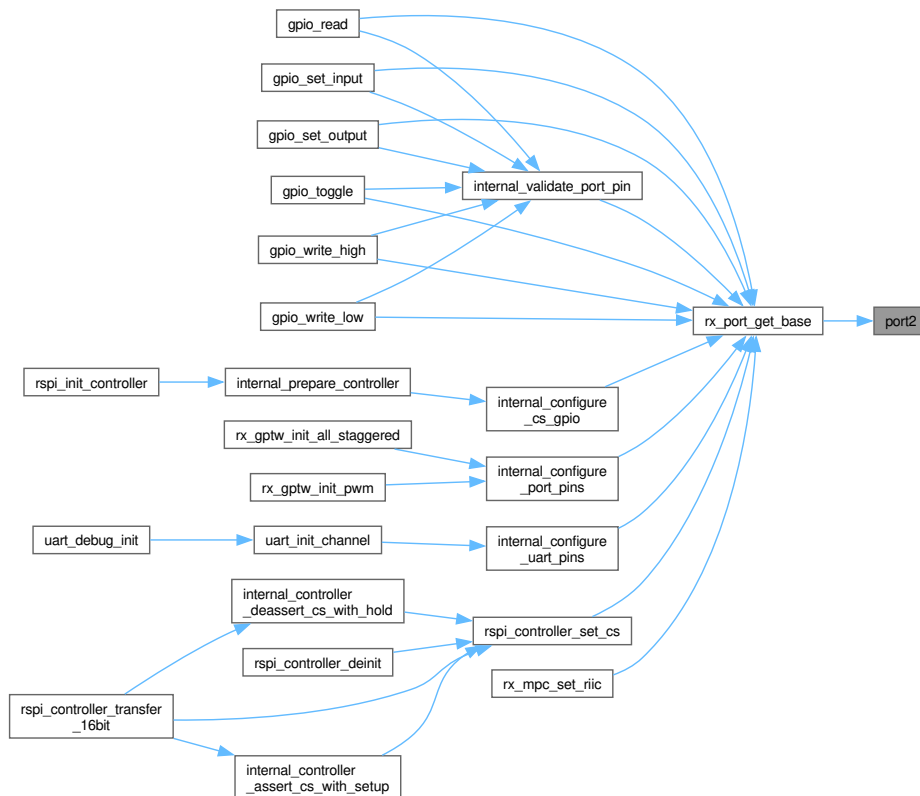
Definition at line 630 of file [rx72n\\_port\\_regs.h](#).

```
00631 {
00632 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_2);
00633 }
```

References [k\\_port\\_offset\\_2](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.6 port2odr()

```
volatile uint16_t * port2_odr (
 void) [inline], [static]
```

Get pointer to PORT2 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT2.ODR = 0x0008C084

Definition at line 846 of file [rx72n\\_port\\_regs.h](#).

```
00847 {
00848 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_2 * 2));
00849 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_2](#).



## 5.7.4.7 port3()

```
volatile rx_port_regs_t * port3 (
 void) [inline], [static]
```

Get pointer to PORT3 registers.

Returns

Volatile pointer to PORT3 register structure

Note

PORT3 PDR = 0x0008C003, PODR = 0x0008C023, PIDR = 0x0008C043  
 144-pin: P30-P37 available (P35 is input-only)

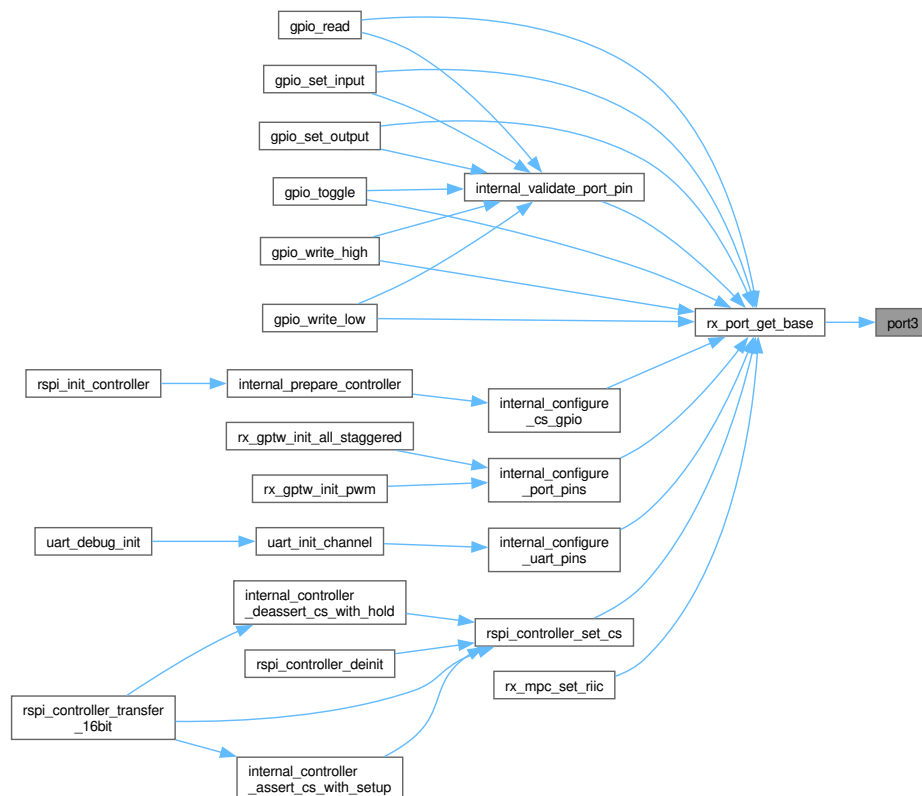
Definition at line 642 of file rx72n\_port\_regs.h.

```
00643 {
00644 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_3);
00645 }
```

References [k\\_port\\_offset\\_3](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.8 port3odr()

```
volatile uint16_t * port3odr (
 void) [inline], [static]
```

Get pointer to PORT3 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT3.ODR = 0x0008C086

Definition at line 856 of file [rx72n\\_port\\_regs.h](#).

```
00857 {
00858 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_3 * 2));
00859 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_3](#).

#### 5.7.4.9 port4()

```
volatile rx_port_regs_t * port4 (
 void) [inline], [static]
```

Get pointer to PORT4 registers.

Returns

Volatile pointer to PORT4 register structure

Note

PORT4 PDR = 0x0008C004, PODR = 0x0008C024, PIDR = 0x0008C044  
144-pin: P40-P47 available

Definition at line 655 of file [rx72n\\_port\\_regs.h](#).

```
00656 {
00657 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_4);
00658 }
```

References [k\\_port\\_offset\\_4](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



## 5.7.4.11 port5()

```
volatile rx_port_regs_t * port5 (
 void) [inline], [static]
```

Get pointer to PORT5 registers.

Returns

Volatile pointer to PORT5 register structure

Note

PORT5 PDR = 0x0008C005, PODR = 0x0008C025, PIDR = 0x0008C045  
144-pin: P50-P56 available (7 pins)

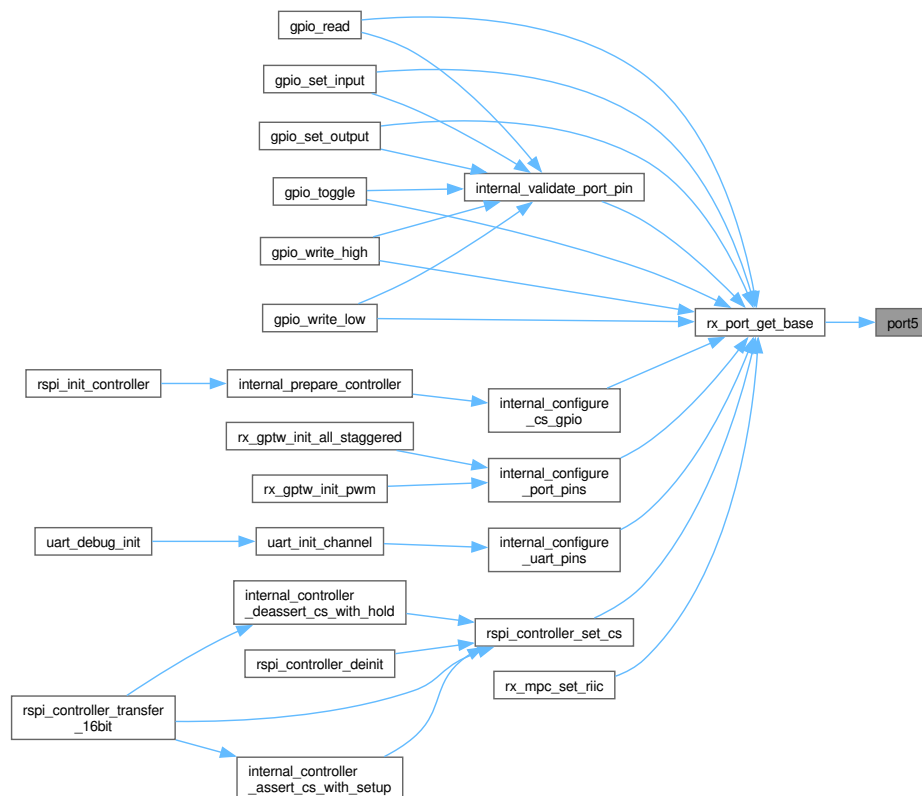
Definition at line 668 of file rx72n\_port\_regs.h.

```
00669 {
00670 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_5);
00671 }
```

References [k\\_port\\_offset\\_5](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



5.7.4.12 `port5_odr()`

```
volatile uint16_t * port5_odr (
 void) [inline], [static]
```

Get pointer to PORT5 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT5.ODR = 0x0008C08A

Definition at line 876 of file [rx72n\\_port\\_regs.h](#).

```
00877 {
00878 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_5 * 2));
00879 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_5](#).

5.7.4.13 `port6()`

```
volatile rx_port_regs_t * port6 (
 void) [inline], [static]
```

Get pointer to PORT6 registers.

Returns

Volatile pointer to PORT6 register structure

Note

PORT6 PDR = 0x0008C006, PODR = 0x0008C026, PIDR = 0x0008C046  
144-pin: P60-P67 available (8 pins)

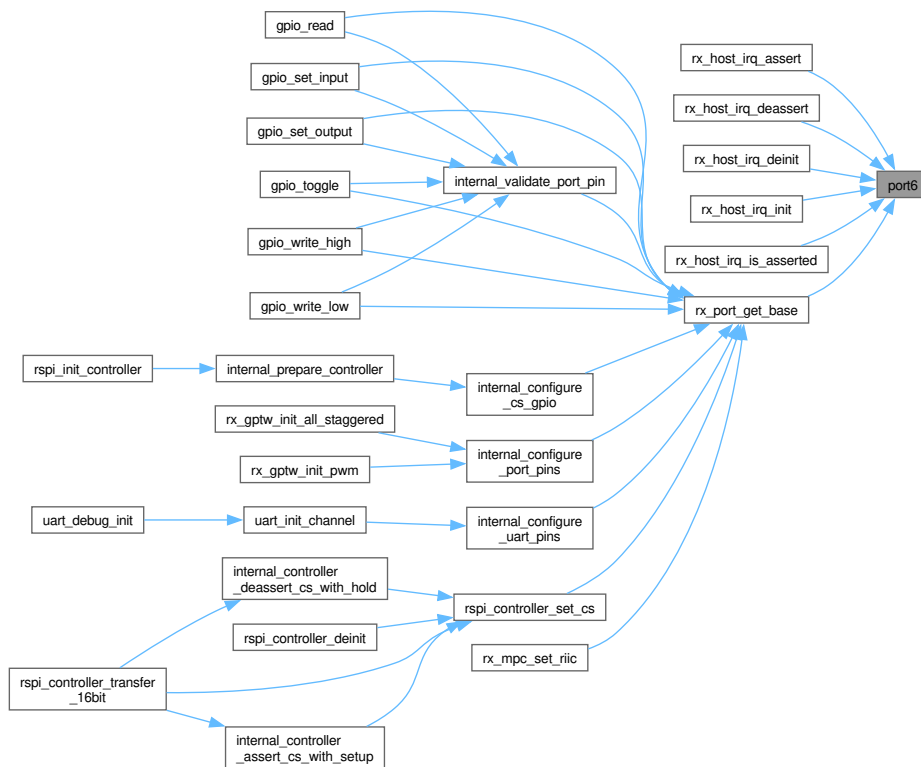
Definition at line 681 of file [rx72n\\_port\\_regs.h](#).

```
00682 {
00683 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_6);
00684 }
```

References [k\\_port\\_offset\\_6](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_host\\_irq\\_assert\(\)](#), [rx\\_host\\_irq\\_deassert\(\)](#), [rx\\_host\\_irq\\_deinit\(\)](#), [rx\\_host\\_irq\\_init\(\)](#), [rx\\_host\\_irq\\_is\\_asserted\(\)](#), and [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.14 port6 odr()

```
volatile uint16_t * port6_odr (
 void) [inline], [static]
```

Get pointer to PORT6 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT6.ODR = 0x0008C08C (offset 0x06 \* 2)

Definition at line 886 of file [rx72n\\_port\\_regs.h](#).

```
00887 {
00888 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_6 * 2));
00889 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_6](#).

## 5.7.4.15 port7()

```
volatile rx_port_regs_t * port7 (
 void) [inline], [static]
```

Get pointer to PORT7 registers.

Returns

Volatile pointer to PORT7 register structure

Note

PORT7 PDR = 0x0008C007, PODR = 0x0008C027, PIDR = 0x0008C047  
144-pin: P70-P77 available (8 pins)

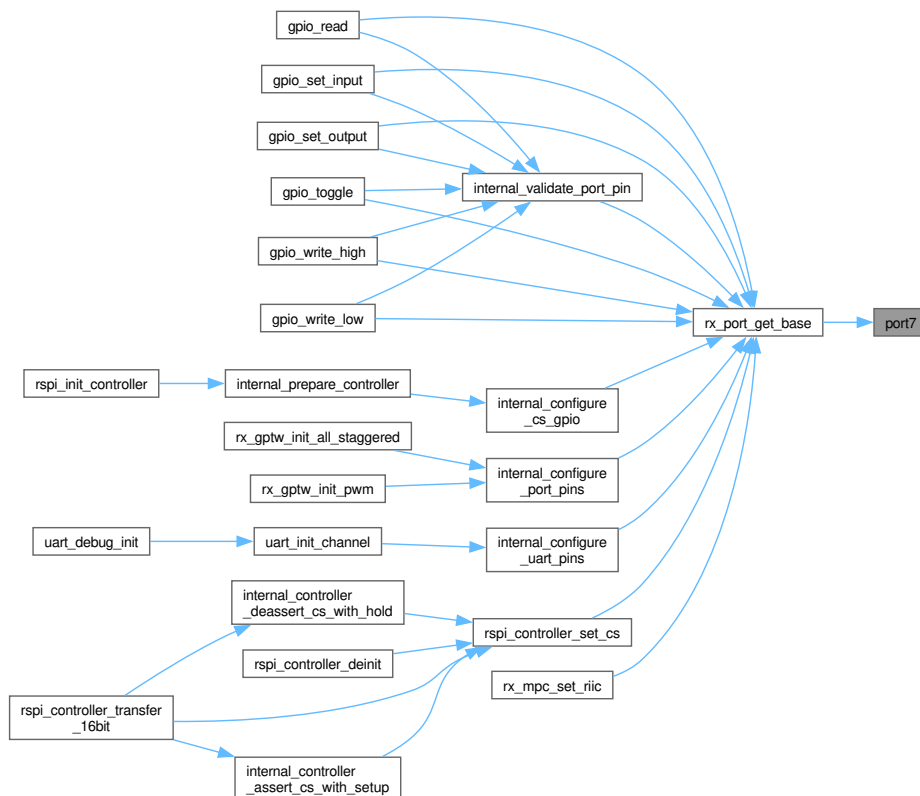
Definition at line 694 of file rx72n\_port\_regs.h.

```
00695 {
00696 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_7);
00697 }
```

References [k\\_port\\_offset\\_7](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.16 port7\_odr()

```
volatile uint16_t * port7_odr (
 void) [inline], [static]
```

Get pointer to PORT7 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT7.ODR = 0x0008C08E (offset 0x07 \* 2)

Definition at line 896 of file [rx72n\\_port\\_regs.h](#).

```
00897 {
00898 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_7 * 2));
00899 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_7](#).

#### 5.7.4.17 port8()

```
volatile rx_port_regs_t * port8 (
 void) [inline], [static]
```

Get pointer to PORT8 registers.

Returns

Volatile pointer to PORT8 register structure

Note

PORT8 PDR = 0x0008C008, PODR = 0x0008C028, PIDR = 0x0008C048  
144-pin: P80-P83, P86-P87 available (6 pins)

Definition at line 707 of file [rx72n\\_port\\_regs.h](#).

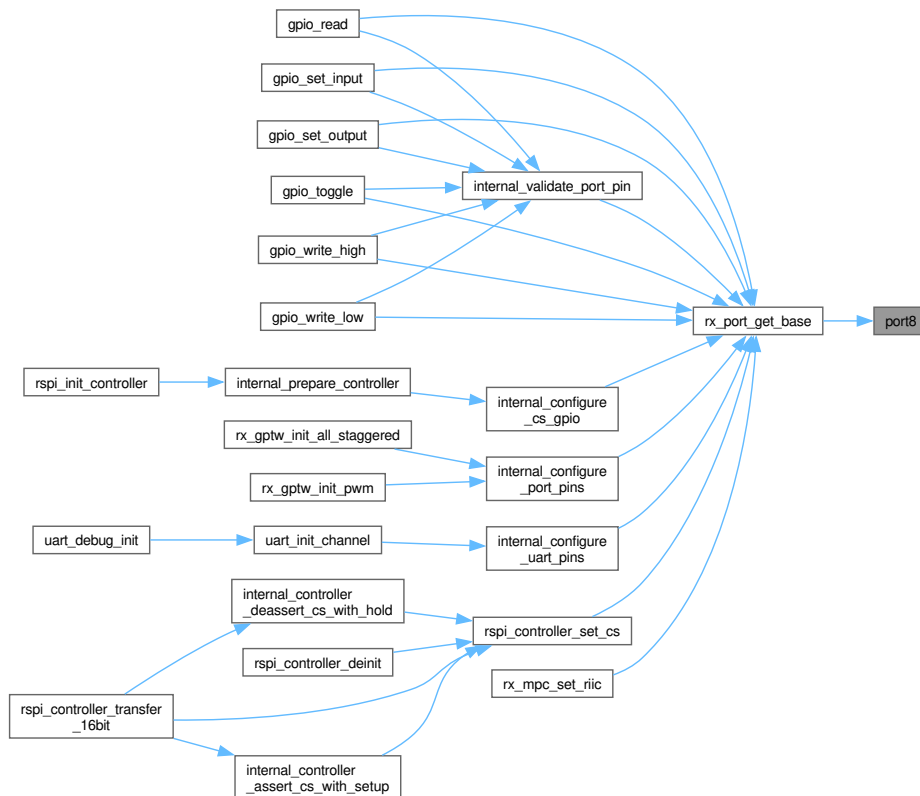
```
00708 {
00709 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_8);
00710 }
```

References [k\\_port\\_offset\\_8](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:





#### 5.7.4.18 port8 odr()

```
volatile uint16_t * port8_odr (
 void) [inline], [static]
```

Get pointer to PORT8 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT8.ODR = 0x0008C090 (offset 0x08 \* 2)

Definition at line 906 of file [rx72n\\_port\\_regs.h](#).

```
00907 {
00908 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_8 * 2));
00909 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_8](#).

### 5.7.4.19 port9()

```
volatile rx_port_regs_t * port9 (
 void) [inline], [static]
```

Get pointer to PORT9 registers.

Returns

Volatile pointer to PORT9 register structure

Note

PORT9 PDR = 0x0008C009, PODR = 0x0008C029, PIDR = 0x0008C049

144-pin: P90-P93 available (4 pins)

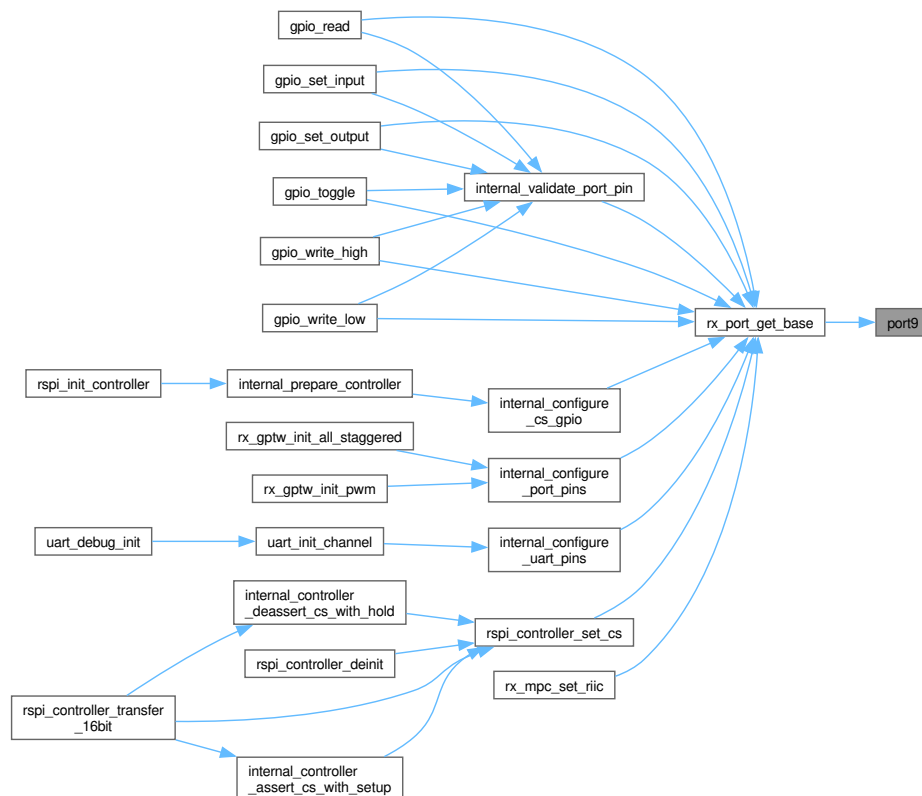
Definition at line 720 of file [rx72n\\_port\\_regs.h](#).

```
00721 {
00722 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_9);
00723 }
```

References [k\\_port\\_offset\\_9](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



## 5.7.4.20 port9odr()

```
volatile uint16_t * port9odr (
 void) [inline], [static]
```

Get pointer to PORT9 ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORT9.ODR = 0x0008C092 (offset 0x09 \* 2)

Definition at line 916 of file [rx72n\\_port\\_regs.h](#).

```
00917 {
00918 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_9 * 2));
00919 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_9](#).

## 5.7.4.21 porta()

```
volatile rx_port_regs_t * porta (
 void) [inline], [static]
```

Get pointer to PORTA registers.

Returns

Volatile pointer to PORTA register structure

Note

PORTA PDR = 0x0008C00A, PODR = 0x0008C02A, PIDR = 0x0008C04A  
144-pin: PA0-PA7 available

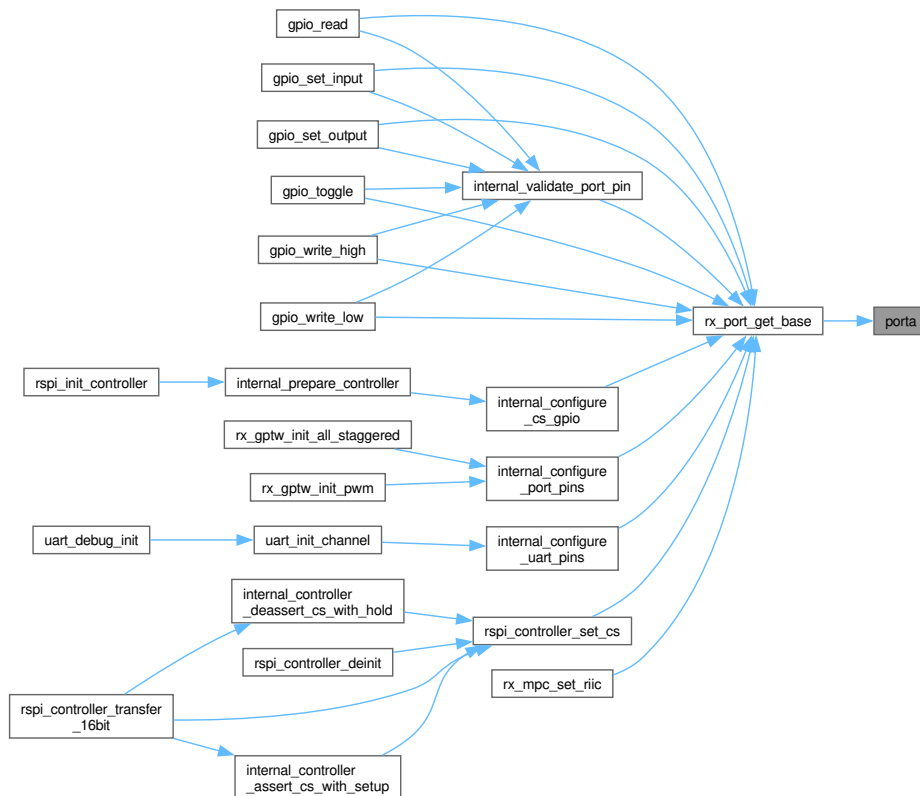
Definition at line 732 of file [rx72n\\_port\\_regs.h](#).

```
00733 {
00734 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_a);
00735 }
```

References [k\\_port\\_offset\\_a](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.22 `portaodr()`

```
volatile uint16_t * portaodr (
 void) [inline], [static]
```

Get pointer to PORTA ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

$\text{PORTA.ODR} = 0x0008C094$  (offset  $0x0A * 2$ )

Definition at line 926 of file `rx72n_port_regs.h`.

```
00927 {
00928 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_a * 2));
00929 }
```

References `k_port_odr0_base`, and `k_port_offset_a`.

## 5.7.4.23 portb()

```
volatile rx_port_regs_t * portb (
 void) [inline], [static]
```

Get pointer to PORTB registers.

Returns

Volatile pointer to PORTB register structure

Note

PORTB PDR = 0x0008C00B, PODR = 0x0008C02B, PIDR = 0x0008C04B

144-pin: PB0-PB7 available

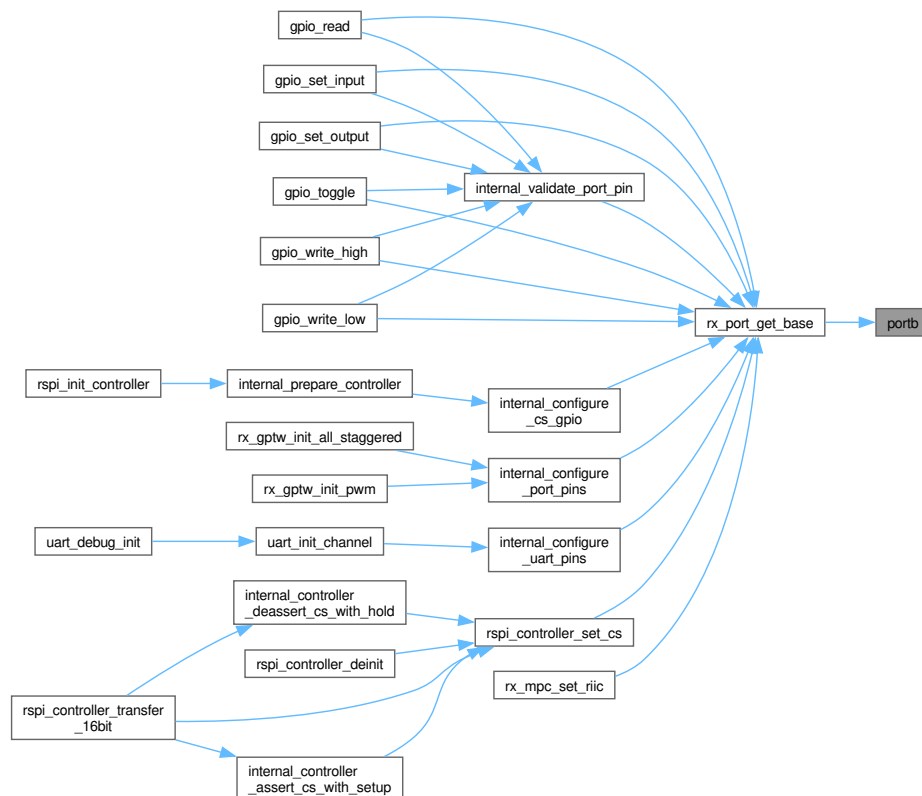
Definition at line 743 of file rx72n\_port\_regs.h.

```
00744 {
00745 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_b);
00746 }
```

References [k\\_port\\_offset\\_b](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.24 portb\_odr()

```
volatile uint16_t * portb_odr (
 void) [inline], [static]
```

Get pointer to PORTB ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORTB.ODR = 0x0008C096

Definition at line 936 of file [rx72n\\_port\\_regs.h](#).

```
00937 {
00938 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_b * 2));
00939 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_b](#).

#### 5.7.4.25 portc()

```
volatile rx_port_regs_t * portc (
 void) [inline], [static]
```

Get pointer to PORTC registers.

Returns

Volatile pointer to PORTC register structure

Note

PORTC PDR = 0x0008C00C, PODR = 0x0008C02C, PIDR = 0x0008C04C  
144-pin: PC0-PC7 available

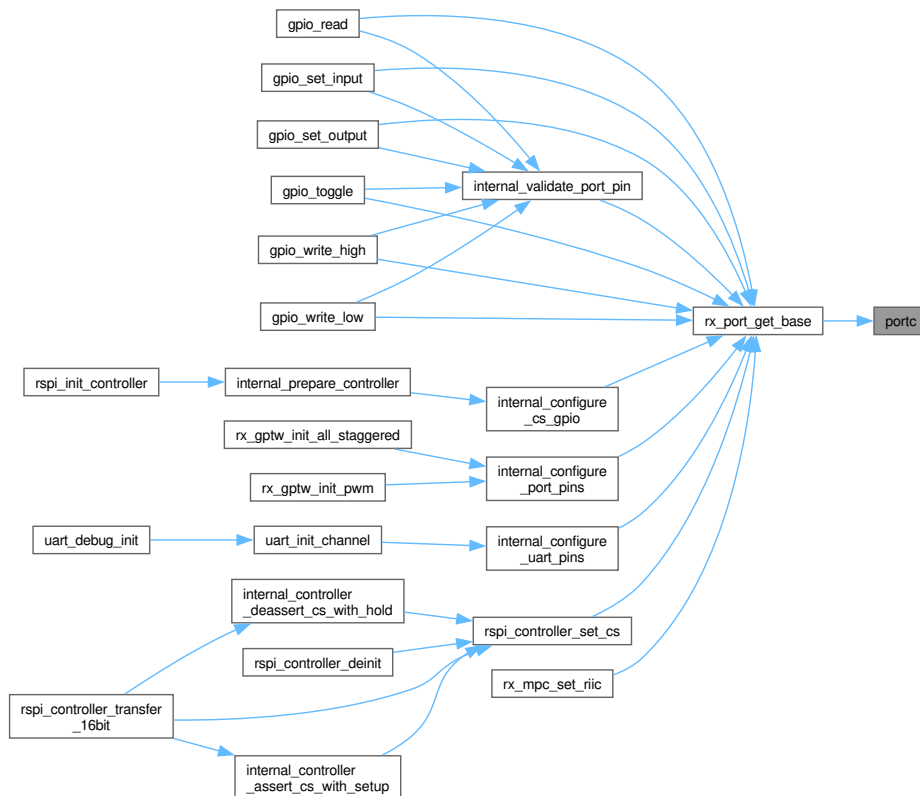
Definition at line 754 of file [rx72n\\_port\\_regs.h](#).

```
00755 {
00756 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_c);
00757 }
```

References [k\\_port\\_offset\\_c](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.26 portc odr()

```
volatile uint16_t * portc_odr (
 void) [inline], [static]
```

Get pointer to PORTC ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORTC.ODR = 0x0008C098

Definition at line 946 of file [rx72n\\_port\\_regs.h](#).

```
00947 {
00948 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_c * 2));
00949 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_c](#).

### 5.7.4.27 portd()

```
volatile rx_port_regs_t * portd (
 void) [inline], [static]
```

Get pointer to PORTD registers.

Returns

Volatile pointer to PORTD register structure

Note

PORTD PDR = 0x0008C00D, PODR = 0x0008C02D, PIDR = 0x0008C04D

144-pin: PD0-PD7 available

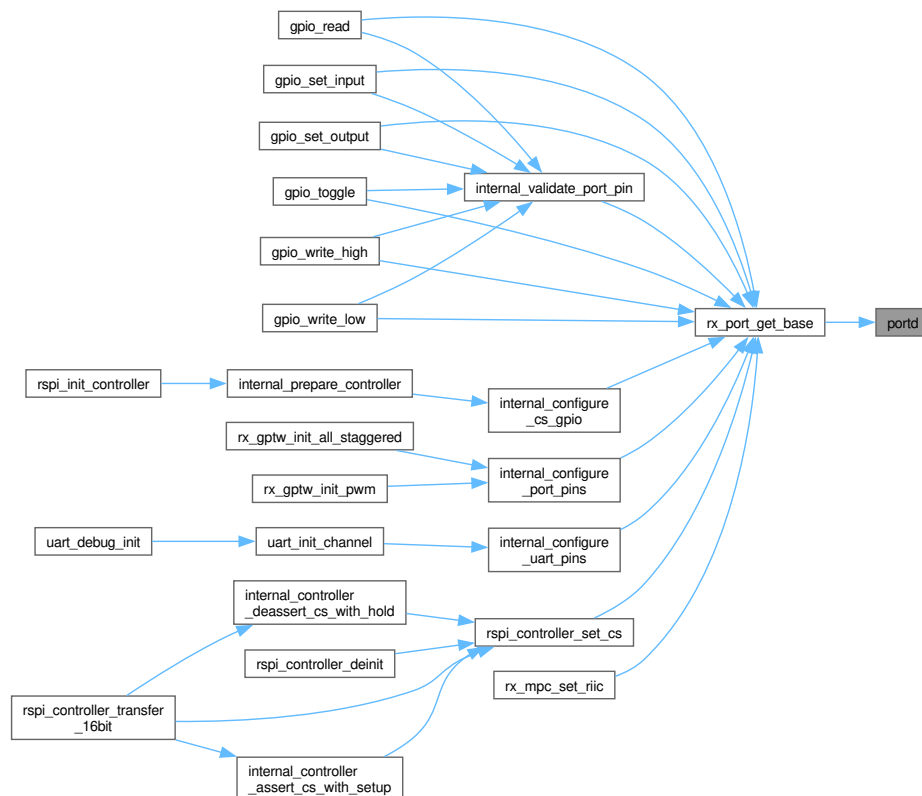
Definition at line 765 of file [rx72n\\_port\\_regs.h](#).

```
00766 {
00767 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_d);
00768 }
```

References [k\\_port\\_offset\\_d](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:





## 5.7.4.28 portd odr()

```
volatile uint16_t * portd_odr (
 void) [inline], [static]
```

Get pointer to PORTD ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORTD.ODR = 0x0008C09A

Definition at line 956 of file [rx72n\\_port\\_regs.h](#).

```
00957 {
00958 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_d * 2));
00959 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_d](#).

## 5.7.4.29 porte()

```
volatile rx_port_regs_t * porte (
 void) [inline], [static]
```

Get pointer to PORTE registers.

Returns

Volatile pointer to PORTE register structure

Note

PORTE PDR = 0x0008C00E, PODR = 0x0008C02E, PIDR = 0x0008C04E  
144-pin: PE0-PE7 available

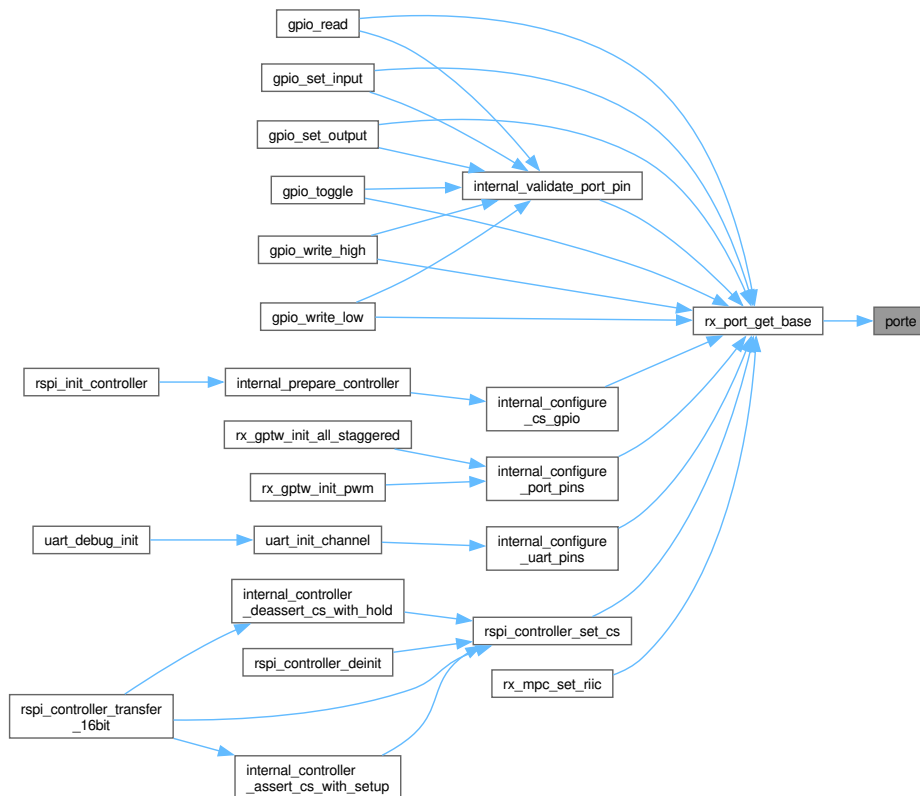
Definition at line 776 of file [rx72n\\_port\\_regs.h](#).

```
00777 {
00778 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_e);
00779 }
```

References [k\\_port\\_offset\\_e](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



#### 5.7.4.30 porteodr()

```
volatile uint16_t * porteodr (
 void) [inline], [static]
```

Get pointer to PORTE ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

`PORTE.ODR = 0x0008C09C`

Definition at line 966 of file `rx72n_port_regs.h`.

```
00967 {
00968 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_e * 2));
00969 }
```

References `k_port_odr0_base`, and `k_port_offset_e`.

## 5.7.4.31 portf()

```
volatile rx_port_regs_t * portf (
 void) [inline], [static]
```

Get pointer to PORTF registers.

Returns

Volatile pointer to PORTF register structure

Note

PORTF PDR = 0x0008C00F, PODR = 0x0008C02F, PIDR = 0x0008C04F  
 144-pin: PF5 available (1 pin)

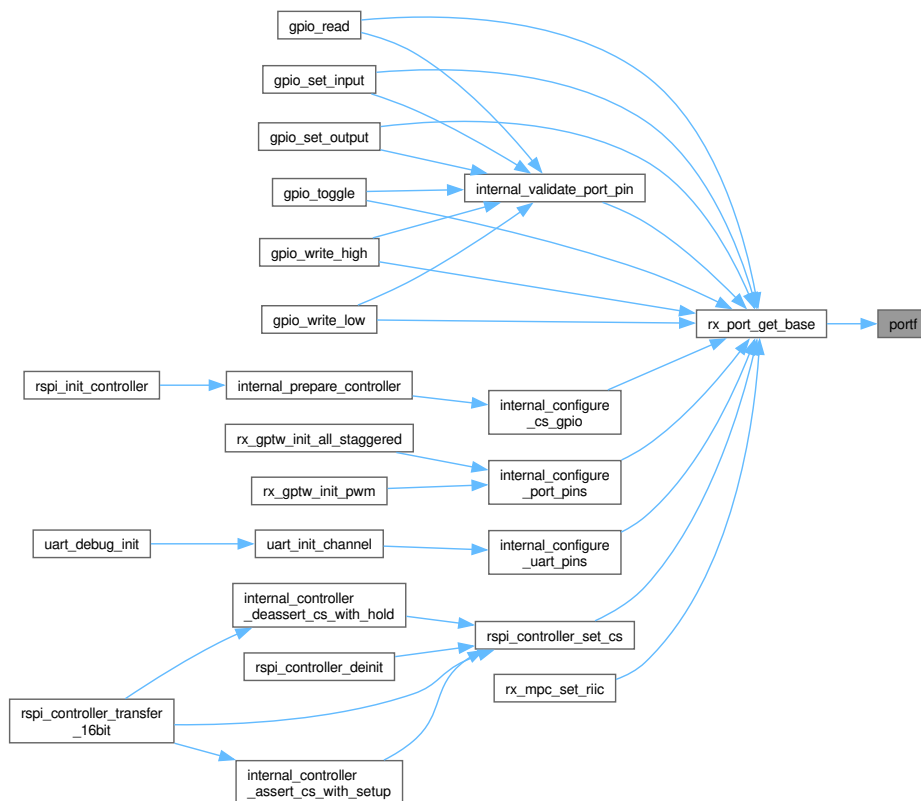
Definition at line 788 of file rx72n\_port\_regs.h.

```
00789 {
00790 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_f);
00791 }
```

References [k\\_port\\_offset\\_f](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



### 5.7.4.32 portfodr()

```
volatile uint16_t * portfodr (
 void) [inline], [static]
```

Get pointer to PORTF ODR register.

Returns

Volatile pointer to 16-bit ODR register

Note

PORTF.ODR = 0x0008C09E (offset 0x0F \* 2)

Definition at line 976 of file [rx72n\\_port\\_regs.h](#).

```
00977 {
00978 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_f * 2));
00979 }
```

References [k\\_port\\_odr0\\_base](#), and [k\\_port\\_offset\\_f](#).

### 5.7.4.33 portj()

```
volatile rx_port_regs_t * portj (
 void) [inline], [static]
```

Get pointer to PORTJ registers.

Returns

Volatile pointer to PORTJ register structure

Note

PORTJ PDR = 0x0008C012, PODR = 0x0008C032, PIDR = 0x0008C052  
144-pin: PJ3, PJ5 available (2 pins)

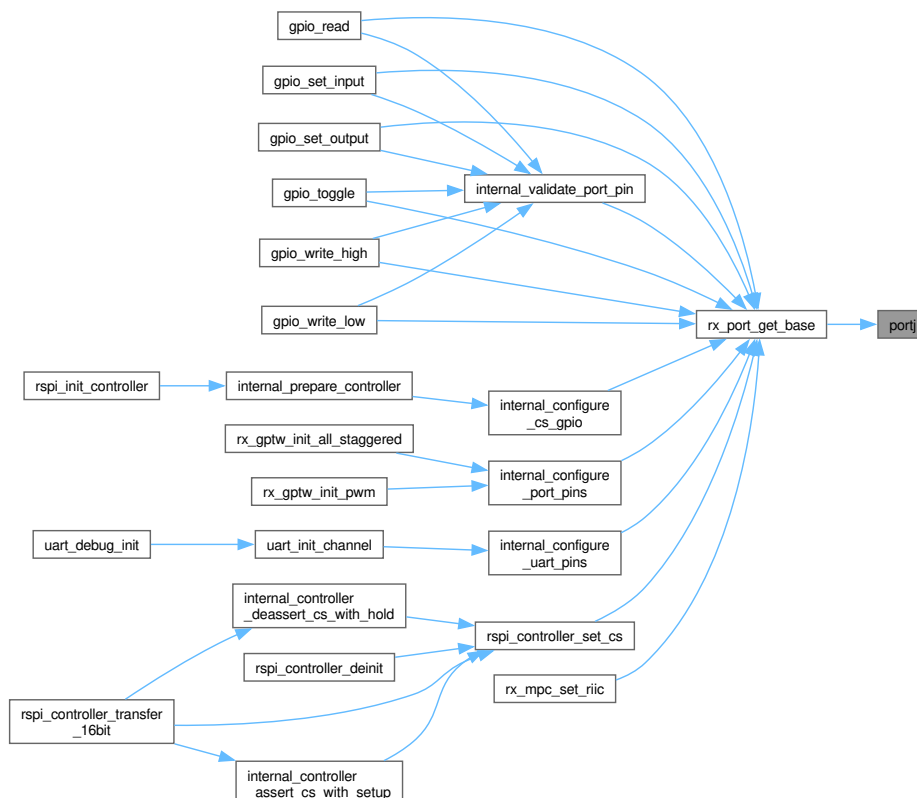
Definition at line 800 of file [rx72n\\_port\\_regs.h](#).

```
00801 {
00802 return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_j);
00803 }
```

References [k\\_port\\_offset\\_j](#), and [k\\_port\\_pdr\\_base](#).

Referenced by [rx\\_port\\_get\\_base\(\)](#).

Here is the caller graph for this function:



5.7.4.34 `portj odr()`

```
volatile uint16_t * portj_odr (
 void) [inline], [static]
```

Get pointer to PORTJ ODR register.

## Returns

Volatile pointer to 16-bit ODR register

Note

PORTJ.ODR = 0x0008C0A4 (offset 0x12 \* 2)

Definition at line 986 of file rx72n\_port\_regs.h.

```
00987 {
00988 return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_j * 2));
00989 }
```

References `k_port_odr0_base`, and `k_port_offset_j`.

## 5.8 RIIC Register Definitions

I2C Bus Interface (RIIC) hardware register interface.

Collaboration diagram for RIIC Register Definitions:



## Topics

- [RIIC Static Assertions](#)  
Compile-time verification of RIIC register layout.

## Data Structures

- struct [rx\\_riic\\_regs\\_t](#)  
RIIC Channel Register Map (20 bytes per channel).

## Enumerations

- enum [rx\\_riic\\_addresses\\_t](#) : [uintptr\\_t](#) { [k\\_riic0\\_base\\_addr](#) = 0x00088300 , [k\\_riic1\\_base\\_addr](#) = 0x00088320 , [k\\_riic2\\_base\\_addr](#) = 0x00088340 }
- RIIC channel base addresses (verified against RX72N Hardware Manual Ch42).
- enum [riic\\_iccr1\\_bits\\_t](#) : [uint8\\_t](#) {  
[k\\_riic\\_iccr1\\_ice](#) = (1 << 7) , [k\\_riic\\_iccr1\\_iicrst](#) = (1 << 6) , [k\\_riic\\_iccr1\\_clo](#) = (1 << 5) ,  
[k\\_riic\\_iccr1\\_sowp](#) = (1 << 4) ,  
[k\\_riic\\_iccr1\\_sclo](#) = (1 << 3) , [k\\_riic\\_iccr1\\_sdao](#) = (1 << 2) , [k\\_riic\\_iccr1\\_scli](#) = (1 << 1) ,  
[k\\_riic\\_iccr1\\_sdai](#) = (1 << 0) }
- ICCR1 (I2C Control Register 1) bit definitions.
- enum [riic\\_iccr2\\_bits\\_t](#) : [uint8\\_t](#) {  
[k\\_riic\\_iccr2\\_bbsy](#) = (1 << 7) , [k\\_riic\\_iccr2\\_mst](#) = (1 << 6) , [k\\_riic\\_iccr2\\_trs](#) = (1 << 5) ,  
[k\\_riic\\_iccr2\\_sp](#) = (1 << 3) ,  
[k\\_riic\\_iccr2\\_rs](#) = (1 << 2) , [k\\_riic\\_iccr2\\_st](#) = (1 << 1) }
- ICCR2 (I2C Control Register 2) bit definitions.
- enum [riic\\_icsr1\\_bits\\_t](#) : [uint8\\_t](#) {  
[k\\_riic\\_icsr1\\_hoa](#) = (1 << 7) , [k\\_riic\\_icsr1\\_did](#) = (1 << 5) , [k\\_riic\\_icsr1\\_gca](#) = (1 << 3) ,  
[k\\_riic\\_icsr1\\_aas2](#) = (1 << 2) ,  
[k\\_riic\\_icsr1\\_aas1](#) = (1 << 1) , [k\\_riic\\_icsr1\\_aas0](#) = (1 << 0) }
- ICSR1 (I2C Status Register 1) bit definitions.
- enum [riic\\_icsr2\\_bits\\_t](#) : [uint8\\_t](#) {  
[k\\_riic\\_icsr2\\_tdre](#) = (1 << 7) , [k\\_riic\\_icsr2\\_tend](#) = (1 << 6) , [k\\_riic\\_icsr2\\_rdrf](#) = (1 << 5) ,  
[k\\_riic\\_icsr2\\_nackf](#) = (1 << 4) ,  
[k\\_riic\\_icsr2\\_stop](#) = (1 << 3) , [k\\_riic\\_icsr2\\_start](#) = (1 << 2) , [k\\_riic\\_icsr2\\_al](#) = (1 << 1) ,  
[k\\_riic\\_icsr2\\_tmof](#) = (1 << 0) }
- ICSR2 (I2C Status Register 2) bit definitions.

## Functions

- static volatile [rx\\_riic\\_regs\\_t](#) \* [riic0](#) (void)  
Get pointer to RIIC0 registers.
- static volatile [rx\\_riic\\_regs\\_t](#) \* [riic1](#) (void)  
Get pointer to RIIC1 registers.
- static volatile [rx\\_riic\\_regs\\_t](#) \* [riic2](#) (void)  
Get pointer to RIIC2 registers.

## 5.8.1 Detailed Description

I2C Bus Interface (RIIC) hardware register interface.

## 5.8.2 Data Structure Documentation

### 5.8.2.1 struct rx\_riic\_regs\_t

RIIC Channel Register Map (20 bytes per channel).

I2C Bus Interface (RIIC) registers for I2C controller/peripheral communication. All registers are 8-bit and byte-accessible.

Register Layout (20 bytes)

| * Offset | Size | Register | Description                      |
|----------|------|----------|----------------------------------|
| * 0x00   | 1    | ICCR1    | Control 1 (enable, reset)        |
| * 0x01   | 1    | ICCR2    | Control 2 (start, stop, restart) |
| * 0x02   | 1    | ICMR1    | Mode 1 (clock divider)           |
| * 0x03   | 1    | ICMR2    | Mode 2 (timeout, SDA delay)      |
| * 0x04   | 1    | ICMR3    | Mode 3 (ACK/NACK control)        |
| * 0x05   | 1    | ICFER    | Function Enable                  |
| * 0x06   | 1    | ICSER    | Status Enable                    |
| * 0x07   | 1    | ICIER    | Interrupt Enable                 |
| * 0x08   | 1    | ICSR1    | Status 1 (ACK status)            |
| * 0x09   | 1    | ICSR2    | Status 2 (TX/RX flags)           |
| * 0x0A   | 1    | SARL0    | Peripheral Address 0 Low         |
| * 0x0B   | 1    | SARU0    | Peripheral Address 0 Upper       |
| * 0x0C   | 1    | SARL1    | Peripheral Address 1 Low         |
| * 0x0D   | 1    | SARU1    | Peripheral Address 1 Upper       |
| * 0x0E   | 1    | SARL2    | Peripheral Address 2 Low         |
| * 0x0F   | 1    | SARU2    | Peripheral Address 2 Upper       |
| * 0x10   | 1    | ICBRL    | Bit Rate Low period              |
| * 0x11   | 1    | ICBRH    | Bit Rate High period             |
| * 0x12   | 1    | ICDRT    | Transmit Data                    |
| * 0x13   | 1    | ICDRR    | Receive Data                     |
| *        |      |          |                                  |

Initialization Sequence

1. Set IICRST bit in ICCR1 (internal reset)
2. Configure ICMR1/2/3 (clock, timeouts)
3. Configure ICBRL/ICBRH (bit rate)
4. Configure ICSER (status detection)
5. Configure ICIER (interrupts)
6. Clear IICRST and set ICE in ICCR1 (enable)

Invariant

```
sizeof(rx_riic_regs_t) == 20
```

See also

[riic0\(\)](#), [riic1\(\)](#), [riic2\(\)](#) Accessor functions for each channel

Definition at line 264 of file [rx72n\\_riic\\_regs.h](#).

Collaboration diagram for rx\_riic\_regs\_t:



## Data Fields

|                  |       |                                                                                               |        |                                            |
|------------------|-------|-----------------------------------------------------------------------------------------------|--------|--------------------------------------------|
| volatile uint8_t | icbrh | I2C Bus Bit Rate High-Level Width Setting Register.<br>Sets the high period of SCL clock.     |        |                                            |
|                  |       | Bits                                                                                          | Name   | Description                                |
|                  |       | 4-0                                                                                           | BRH    | High period count (affects SCL high time)  |
| volatile uint8_t | icbrl | I2C Bus Bit Rate Low-Level Width Setting Register.<br>Sets the low period of SCL clock.       |        |                                            |
|                  |       | Bits                                                                                          | Name   | Description                                |
|                  |       | 4-0                                                                                           | BRL    | Low period count (affects SCL low time)    |
| volatile uint8_t | iccr1 | I2C Bus Control Register 1.<br>Enables RIIC module and controls internal reset and pin state. |        |                                            |
|                  |       | Bit                                                                                           | Name   | Description                                |
|                  |       | 7                                                                                             | ICE    | RIIC Enable (1=enabled)                    |
|                  |       | 6                                                                                             | IICRST | Internal Reset (1=reset state)             |
|                  |       | 5                                                                                             | CLO    | Additional SCL Output (bus recovery clock) |
|                  |       | 4                                                                                             | SOWP   | SCLO and SDAO Write Protect                |
|                  |       | 3                                                                                             | SCLO   | SCL Output Control/Monitor                 |
|                  |       | 2                                                                                             | SDAO   | SDA Output Control/Monitor                 |
|                  |       | 1                                                                                             | SCLI   | SCL Line Input Monitor                     |
|                  |       | 0                                                                                             | SDAI   | SDA Line Input Monitor                     |



| volatile uint8_t | iccr2 | <p>I2C Bus Control Register 2.<br/>Controls start/stop/restart conditions and bus state.</p> <table> <tr> <th>Bit</th><th>Name</th><th>Description</th></tr> <tr> <td>7</td><td>BBSY</td><td>Bus Busy (read-only)</td></tr> <tr> <td>6</td><td>MST</td><td>Controller Mode (1=controller)</td></tr> <tr> <td>5</td><td>TRS</td><td>Transmit/Receive (1=transmit)</td></tr> <tr> <td>3</td><td>SP</td><td>Stop Condition Request</td></tr> <tr> <td>2</td><td>RS</td><td>Restart Condition Request</td></tr> <tr> <td>1</td><td>ST</td><td>Start Condition Request</td></tr> </table> | Bit  | Name | Description | 7   | BBSY | Bus Busy (read-only)                                   | 6 | MST   | Controller Mode (1=controller)            | 5   | TRS   | Transmit/Receive (1=transmit) | 3 | SP    | Stop Condition Request              | 2 | RS | Restart Condition Request | 1 | ST | Start Condition Request |
|------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|-------------|-----|------|--------------------------------------------------------|---|-------|-------------------------------------------|-----|-------|-------------------------------|---|-------|-------------------------------------|---|----|---------------------------|---|----|-------------------------|
| Bit              | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 7                | BBSY  | Bus Busy (read-only)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 6                | MST   | Controller Mode (1=controller)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 5                | TRS   | Transmit/Receive (1=transmit)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 3                | SP    | Stop Condition Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 2                | RS    | Restart Condition Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 1                | ST    | Start Condition Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icdrr | <p>I2C Bus Receive Data Register.<br/>Read received data from here. Read clears RDRF flag.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icdrt | <p>I2C Bus Transmit Data Register.<br/>Write data here to transmit on I2C bus.</p> <ul style="list-style-type: none"> <li>• First byte after START should be peripheral address + R/W bit</li> <li>• Subsequent bytes are data</li> </ul>                                                                                                                                                                                                                                                                                                                                            |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icfer | <p>I2C Bus Function Enable Register.<br/>Enables various RIIC functions (timeout, noise filter, etc.).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icier | <p>I2C Bus Interrupt Enable Register.<br/>Enables interrupt generation for various conditions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icmr1 | <p>I2C Bus Mode Register 1.<br/>Configures internal reference clock divider.</p> <table> <tr> <th>Bits</th><th>Name</th><th>Description</th></tr> <tr> <td>6-4</td><td>CKS</td><td>Clock Select (0-7, divides PCLKB by 2<sup>CKS</sup>)</td></tr> <tr> <td>3</td><td>BCWP</td><td>ICCR2 Write Protect (1=BC bits read-only)</td></tr> <tr> <td>2-0</td><td>BC</td><td>Bit Counter</td></tr> </table>                                                                                                                                                                                 | Bits | Name | Description | 6-4 | CKS  | Clock Select (0-7, divides PCLKB by 2 <sup>CKS</sup> ) | 3 | BCWP  | ICCR2 Write Protect (1=BC bits read-only) | 2-0 | BC    | Bit Counter                   |   |       |                                     |   |    |                           |   |    |                         |
| Bits             | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 6-4              | CKS   | Clock Select (0-7, divides PCLKB by 2 <sup>CKS</sup> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 3                | BCWP  | ICCR2 Write Protect (1=BC bits read-only)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 2-0              | BC    | Bit Counter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icmr2 | <p>I2C Bus Mode Register 2.<br/>Configures timeout and SDA output delay.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icmr3 | <p>I2C Bus Mode Register 3.<br/>Controls ACK/NACK transmission and wait state.</p> <table> <tr> <th>Bit</th><th>Name</th><th>Description</th></tr> <tr> <td>6</td><td>WAIT</td><td>Wait insertion (1=insert wait)</td></tr> <tr> <td>5</td><td>RDRFS</td><td>RDRF Flag Set Timing</td></tr> <tr> <td>4</td><td>ACKWP</td><td>ACKBT Write Protect</td></tr> <tr> <td>3</td><td>ACKBT</td><td>ACK Bit to Transmit (0=ACK, 1=NACK)</td></tr> </table>                                                                                                                                   | Bit  | Name | Description | 6   | WAIT | Wait insertion (1=insert wait)                         | 5 | RDRFS | RDRF Flag Set Timing                      | 4   | ACKWP | ACKBT Write Protect           | 3 | ACKBT | ACK Bit to Transmit (0=ACK, 1=NACK) |   |    |                           |   |    |                         |
| Bit              | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 6                | WAIT  | Wait insertion (1=insert wait)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 5                | RDRFS | RDRF Flag Set Timing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 4                | ACKWP | ACKBT Write Protect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| 3                | ACKBT | ACK Bit to Transmit (0=ACK, 1=NACK)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |
| volatile uint8_t | icser | <p>I2C Bus Status Enable Register.<br/>Selects which conditions set status flags.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |      |             |     |      |                                                        |   |       |                                           |     |       |                               |   |       |                                     |   |    |                           |   |    |                         |

| volatile uint8_t | icr1  | I2C Bus Status Register 1.<br>Contains peripheral address detection and host/device-ID detection flags. <table border="1"> <thead> <tr> <th>Bit</th><th>Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>7</td><td>HOA</td><td>Host Address Detection Flag</td></tr> <tr> <td>5</td><td>DID</td><td>Device-ID Address Detection Flag</td></tr> <tr> <td>3</td><td>GCA</td><td>General Call Address Detection Flag</td></tr> <tr> <td>2</td><td>AAS2</td><td>Peripheral Address 2 Detection Flag</td></tr> <tr> <td>1</td><td>AAS1</td><td>Peripheral Address 1 Detection Flag</td></tr> <tr> <td>0</td><td>AAS0</td><td>Peripheral Address 0 Detection Flag</td></tr> </tbody> </table>                                        | Bit | Name | Description | 7 | HOA  | Host Address Detection Flag | 5 | DID  | Device-ID Address Detection Flag | 3 | GCA  | General Call Address Detection Flag | 2 | AAS2  | Peripheral Address 2 Detection Flag | 1 | AAS1 | Peripheral Address 1 Detection Flag | 0 | AAS0  | Peripheral Address 0 Detection Flag |   |    |                       |   |      |                        |
|------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|-------------|---|------|-----------------------------|---|------|----------------------------------|---|------|-------------------------------------|---|-------|-------------------------------------|---|------|-------------------------------------|---|-------|-------------------------------------|---|----|-----------------------|---|------|------------------------|
| Bit              | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 7                | HOA   | Host Address Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 5                | DID   | Device-ID Address Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 3                | GCA   | General Call Address Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 2                | AAS2  | Peripheral Address 2 Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 1                | AAS1  | Peripheral Address 1 Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 0                | AAS0  | Peripheral Address 0 Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | icr2  | I2C Bus Status Register 2.<br>Contains transmit/receive status flags and condition detection. <table border="1"> <thead> <tr> <th>Bit</th><th>Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>7</td><td>TDRE</td><td>Transmit Data Empty</td></tr> <tr> <td>6</td><td>TEND</td><td>Transmit End</td></tr> <tr> <td>5</td><td>RDRF</td><td>Receive Data Full</td></tr> <tr> <td>4</td><td>NACKF</td><td>NACK Detection Flag</td></tr> <tr> <td>3</td><td>STOP</td><td>Stop Condition Detection</td></tr> <tr> <td>2</td><td>START</td><td>Start Condition Detection</td></tr> <tr> <td>1</td><td>AL</td><td>Arbitration-Lost Flag</td></tr> <tr> <td>0</td><td>TMOF</td><td>Timeout Detection Flag</td></tr> </tbody> </table> | Bit | Name | Description | 7 | TDRE | Transmit Data Empty         | 6 | TEND | Transmit End                     | 5 | RDRF | Receive Data Full                   | 4 | NACKF | NACK Detection Flag                 | 3 | STOP | Stop Condition Detection            | 2 | START | Start Condition Detection           | 1 | AL | Arbitration-Lost Flag | 0 | TMOF | Timeout Detection Flag |
| Bit              | Name  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 7                | TDRE  | Transmit Data Empty                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 6                | TEND  | Transmit End                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 5                | RDRF  | Receive Data Full                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 4                | NACKF | NACK Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 3                | STOP  | Stop Condition Detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 2                | START | Start Condition Detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 1                | AL    | Arbitration-Lost Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| 0                | TMOF  | Timeout Detection Flag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar10 | Peripheral Address Register L0 (7-bit: bits 7-1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar11 | Peripheral Address Register L1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar12 | Peripheral Address Register L2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar0  | Peripheral Address Register U0 (10-bit: bits 9-8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar1  | Peripheral Address Register U1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |
| volatile uint8_t | sar2  | Peripheral Address Register U2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |      |             |   |      |                             |   |      |                                  |   |      |                                     |   |       |                                     |   |      |                                     |   |       |                                     |   |    |                       |   |      |                        |

### 5.8.3 Enumeration Type Documentation

#### 5.8.3.1 riic'icr1'bits't

enum `riic_icr1_bits_t` : `uint8_t`

ICCR1 (I2C Control Register 1) bit definitions.

Controls RIIC module enable/disable, internal reset, bus recovery clock, and provides pin output control and input monitoring.

Register Layout

```

* Bit | Name | Description
* ----+-----+-----
* 7 | ICE | RIIC Enable (1=enabled, 0=disabled)
* 6 | IICRST | Internal Reset (1=reset state)
* 5 | CLO | Additional SCL Output (bus recovery clock)
* 4 | SOWP | SCLO and SDAO Write Protect
* 3 | SCLO | SCL Output Control/Monitor
* 2 | SDAO | SDA Output Control/Monitor
* 1 | SCLI | SCL Line Input Monitor (read-only)
* 0 | SDAI | SDA Line Input Monitor (read-only)
*

```

## Enumerator

|                     |                                             |
|---------------------|---------------------------------------------|
| k_riic_iccr1_ice    | Bit 7: I2C Bus Interface Enable             |
| k_riic_iccr1_iicrst | Bit 6: Internal Reset (hold in reset)       |
| k_riic_iccr1_clo    | Bit 5: Additional SCL Output (bus recovery) |
| k_riic_iccr1_sowp   | Bit 4: SCLO and SDAO Write Protect          |
| k_riic_iccr1_sclo   | Bit 3: SCL Output Control/Monitor           |
| k_riic_iccr1_sdao   | Bit 2: SDA Output Control/Monitor           |
| k_riic_iccr1_scli   | Bit 1: SCL Line Input Monitor (read-only)   |
| k_riic_iccr1_sdai   | Bit 0: SDA Line Input Monitor (read-only)   |

Definition at line 495 of file `rx72n_riic_regs.h`.

```

00495 : uint8_t {
00496 k_riic_iccr1_ice = (1 << 7), /**< Bit 7: I2C Bus Interface Enable */
00497 k_riic_iccr1_iicrst = (1 << 6), /**< Bit 6: Internal Reset (hold in reset) */
00498 k_riic_iccr1_clo = (1 << 5), /**< Bit 5: Additional SCL Output (bus recovery) */
00499 k_riic_iccr1_sowp = (1 << 4), /**< Bit 4: SCLO and SDAO Write Protect */
00500 k_riic_iccr1_sclo = (1 << 3), /**< Bit 3: SCL Output Control/Monitor */
00501 k_riic_iccr1_sdao = (1 << 2), /**< Bit 2: SDA Output Control/Monitor */
00502 k_riic_iccr1_scli = (1 << 1), /**< Bit 1: SCL Line Input Monitor (read-only) */
00503 k_riic_iccr1_sdai = (1 << 0), /**< Bit 0: SDA Line Input Monitor (read-only) */
00504 } riic_iccr1_bits_t;

```

## 5.8.3.2 riic'iccr2'bits't

```
enum riic_iccr2_bits_t : uint8_t
```

ICCR2 (I2C Control Register 2) bit definitions.

Controls bus conditions (start, stop, restart) and provides bus status.

## Register Layout

```

* Bit | Name | R/W | Description
* ----+-----+-----+-----
* 7 | BBSY | R | Bus Busy (1=bus busy, 0=bus free)
* 6 | MST | R | Controller Mode (1=controller, 0=peripheral)
* 5 | TRS | R | Transmit Mode (1=transmit, 0=receive)
* 4 | - | - | Reserved
* 3 | SP | W | Stop Condition Request (write 1 to issue)
* 2 | RS | W | Restart Condition Request
* 1 | ST | W | Start Condition Request
* 0 | - | - | Reserved
*

```

## Note

ST, RS, SP bits are write-only and always read as 0.

## Enumerator

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| k_riic_iccr2_bbsy | Bit 7: Bus Busy (read-only)                            |
| k_riic_iccr2_mst  | Bit 6: Controller Mode (read-only)                     |
| k_riic_iccr2_trs  | Bit 5: Transmit/Receive Select (1=transmit, read-only) |

|                 |                                  |
|-----------------|----------------------------------|
| k_riic_iccr2_sp | Bit 3: Stop Condition Request    |
| k_riic_iccr2_rs | Bit 2: Restart Condition Request |
| k_riic_iccr2_st | Bit 1: Start Condition Request   |

Definition at line 529 of file [rx72n\\_riic\\_regs.h](#).

```

00529 : uint8_t {
00530 k_riic_iccr2_bbsy = (1 « 7), /**< Bit 7: Bus Busy (read-only) */
00531 k_riic_iccr2_mst = (1 « 6), /**< Bit 6: Controller Mode (read-only) */
00532 k_riic_iccr2_trs = (1 « 5), /**< Bit 5: Transmit/Receive Select (1=transmit, read-only) */
00533 k_riic_iccr2_sp = (1 « 3), /**< Bit 3: Stop Condition Request */
00534 k_riic_iccr2_rs = (1 « 2), /**< Bit 2: Restart Condition Request */
00535 k_riic_iccr2_st = (1 « 1), /**< Bit 1: Start Condition Request */
00536 } riic_iccr2_bits_t;

```

### 5.8.3.3 riic\_icsr1\_bits\_t

enum [riic\\_icsr1\\_bits\\_t](#) : uint8\_t

ICSR1 (I2C Status Register 1) bit definitions.

Contains peripheral address detection flags, general call detection, device-ID detection, and host address detection (peripheral mode).

Register Layout

```

* Bit | Name | Description
* ----+-----+-----
* 7 | HOA | Host Address Detection Flag
* 6 | - | Reserved
* 5 | DID | Device-ID Address Detection Flag
* 4 | - | Reserved
* 3 | GCA | General Call Address Detection Flag
* 2 | AAS2 | Peripheral Address 2 Detection Flag
* 1 | AAS1 | Peripheral Address 1 Detection Flag
* 0 | AAS0 | Peripheral Address 0 Detection Flag
*

```

Usage

```

if (riic0()->icsr1 & k_riic_icsr1_aas0) {
 // Peripheral address 0 detected - we are being addressed
}

```

Enumerator

|                   |                                            |
|-------------------|--------------------------------------------|
| k_riic_icsr1_hoa  | Bit 7: Host Address Detection Flag         |
| k_riic_icsr1_did  | Bit 5: Device-ID Address Detection Flag    |
| k_riic_icsr1_gca  | Bit 3: General Call Address Detection Flag |
| k_riic_icsr1_aas2 | Bit 2: Peripheral Address 2 Detection Flag |
| k_riic_icsr1_aas1 | Bit 1: Peripheral Address 1 Detection Flag |
| k_riic_icsr1_aas0 | Bit 0: Peripheral Address 0 Detection Flag |

Definition at line 567 of file [rx72n\\_riic\\_regs.h](#).

```

00567 : uint8_t {
00568 k_riic_icsr1_hoa = (1 « 7), /**< Bit 7: Host Address Detection Flag */
00569 k_riic_icsr1_did = (1 « 5), /**< Bit 5: Device-ID Address Detection Flag */
00570 k_riic_icsr1_gca = (1 « 3), /**< Bit 3: General Call Address Detection Flag */
00571 k_riic_icsr1_aas2 = (1 « 2), /**< Bit 2: Peripheral Address 2 Detection Flag */
00572 k_riic_icsr1_aas1 = (1 « 1), /**< Bit 1: Peripheral Address 1 Detection Flag */
00573 k_riic_icsr1_aas0 = (1 « 0), /**< Bit 0: Peripheral Address 0 Detection Flag */
00574 } riic_icsr1_bits_t;

```

## 5.8.3.4 riic\_icsr2\_bits\_t

```
enum riic_icsr2_bits_t : uint8_t
```

ICSR2 (I2C Status Register 2) bit definitions.

Contains transmit/receive status flags and condition detection.

Register Layout

```
* Bit | Name | Description
* ----+-----+-----
* 7 | TDRE | Transmit Data Empty (1=can write ICDRT)
* 6 | TEND | Transmit End (1=all data transmitted)
* 5 | RDRF | Receive Data Full (1=data in ICDRR)
* 4 | NACKF | NACK Detection Flag
* 3 | STOP | Stop Condition Detected
* 2 | START | Start Condition Detected
* 1 | AL | Arbitration-Lost Flag
* 0 | TMOF | Timeout Detection Flag
*
```

Typical Polling Sequence

```
// Wait for transmit ready
while (!(riic0()->icsr2 & k_riic_icsr2_tdre)) {}

// Wait for receive complete
while (!(riic0()->icsr2 & k_riic_icsr2_rdrf)) {}
```

Enumerator

|                    |                                  |
|--------------------|----------------------------------|
| k_riic_icsr2_tdre  | Bit 7: Transmit Data Empty       |
| k_riic_icsr2_tend  | Bit 6: Transmit End              |
| k_riic_icsr2_rdrf  | Bit 5: Receive Data Full         |
| k_riic_icsr2_nackf | Bit 4: NACK Detection Flag       |
| k_riic_icsr2_stop  | Bit 3: Stop Condition Detection  |
| k_riic_icsr2_start | Bit 2: Start Condition Detection |
| k_riic_icsr2_al    | Bit 1: Arbitration-Lost Flag     |
| k_riic_icsr2_tmof  | Bit 0: Timeout Detection Flag    |

Definition at line 606 of file rx72n\_riic\_regs.h.

```
00606 : uint8_t {
00607 k_riic_icsr2_tdre = (1 << 7), /**< Bit 7: Transmit Data Empty */
00608 k_riic_icsr2_tend = (1 << 6), /**< Bit 6: Transmit End */
00609 k_riic_icsr2_rdrf = (1 << 5), /**< Bit 5: Receive Data Full */
00610 k_riic_icsr2_nackf = (1 << 4), /**< Bit 4: NACK Detection Flag */
00611 k_riic_icsr2_stop = (1 << 3), /**< Bit 3: Stop Condition Detection */
00612 k_riic_icsr2_start = (1 << 2), /**< Bit 2: Start Condition Detection */
00613 k_riic_icsr2_al = (1 << 1), /**< Bit 1: Arbitration-Lost Flag */
00614 k_riic_icsr2_tmof = (1 << 0), /**< Bit 0: Timeout Detection Flag */
00615 } riic_icsr2_bits_t;
```

### 5.8.3.5 rx\_riic\_addresses\_t

```
enum rx_riic_addresses_t : uintptr_t
```

RIIC channel base addresses (verified against RX72N Hardware Manual Ch42).

The RX72N has 3 independent RIIC channels. Each channel has a 32-byte register block (though only 20 bytes are used).

#### Memory Layout

```
* Address | Channel | Pins
* -----+-----+-----
* 0x00088300 | RIIC0 | P16/SCL0, P17/SDA0
* 0x00088320 | RIIC1 | P21/SCL1, P20/SDA1
* 0x00088340 | RIIC2 | P66/SCL2, P67/SDA2
*
```

#### Note

Channel spacing is 0x20 (32 bytes) between adjacent channels.

#### Enumerator

|                   |                             |
|-------------------|-----------------------------|
| k_riic0_base_addr | RIIC0 register base address |
| k_riic1_base_addr | RIIC1 register base address |
| k_riic2_base_addr | RIIC2 register base address |

Definition at line 212 of file [rx72n\\_riic\\_regs.h](#).

```
00212 : uintptr_t {
00213 k_riic0_base_addr = 0x00088300, /**< RIIC0 register base address */
00214 k_riic1_base_addr = 0x00088320, /**< RIIC1 register base address */
00215 k_riic2_base_addr = 0x00088340, /**< RIIC2 register base address */
00216 } rx_riic_addresses_t;
```

## 5.8.4 Function Documentation

### 5.8.4.1 riic0()

```
volatile rx_riic_regs_t * riic0 (
 void) [inline], [static]
```

Get pointer to RIIC0 registers.

Returns a volatile pointer to RIIC channel 0 registers at address 0x00088300. RIIC0 pins are P16 (SCL0) and P17 (SDA0).

Returns

Volatile pointer to RIIC0 register structure

## Example

```
// Enable RIIC0
riic0()->iccr1 = k_riic_iccr1_ice;

// Start condition
riic0()->iccr2 = k_riic_iccr2_st;
```

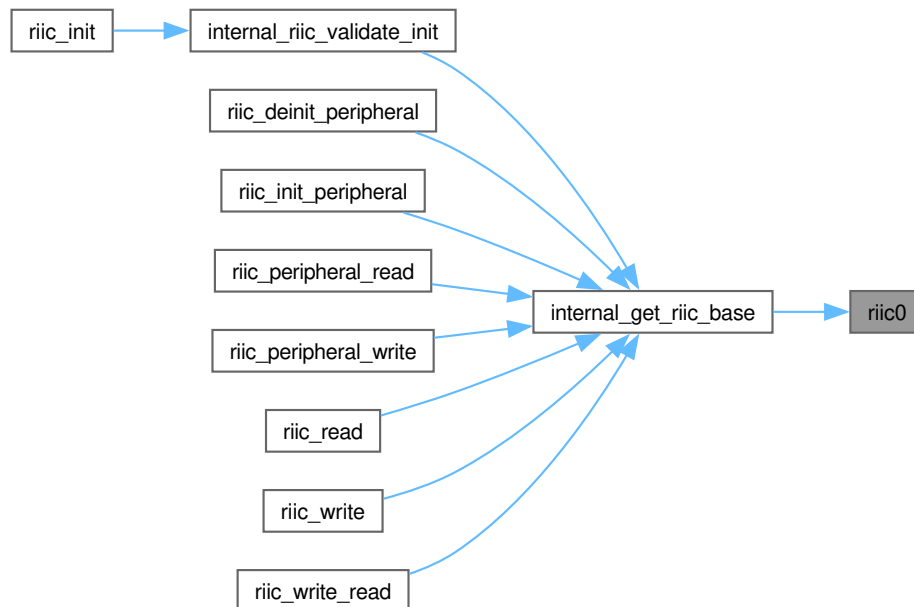
Definition at line 440 of file `rx72n_riic_regs.h`.

```
00441 {
00442 return (volatile rx_riic_regs_t*)k_riic0_base_addr;
00443 }
```

References `k_riic0_base_addr`.

Referenced by `internal_get_riic_base()`.

Here is the caller graph for this function:

5.8.4.2 `riic1()`

```
volatile rx_riic_regs_t * riic1 (
 void) [inline], [static]
```

Get pointer to RIIC1 registers.

Returns a volatile pointer to RIIC channel 1 registers at address 0x00088320. RIIC1 pins are P21 (SCL1) and P20 (SDA1).

## Returns

Volatile pointer to RIIC1 register structure

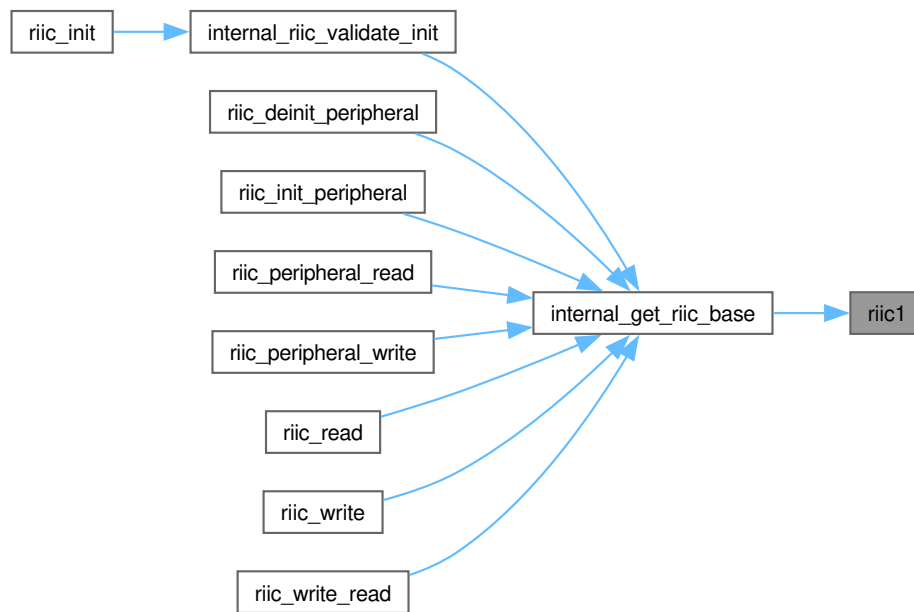
Definition at line 454 of file [rx72n\\_riic\\_regs.h](#).

```
00455 {
00456 return (volatile rx_riic_regs_t*)k_riic1_base_addr;
00457 }
```

References [k\\_riic1\\_base\\_addr](#).

Referenced by [internal\\_get\\_riic\\_base\(\)](#).

Here is the caller graph for this function:



### 5.8.4.3 riic2()

```
volatile rx_riic_regs_t * riic2 (
 void) [inline], [static]
```

Get pointer to RIIC2 registers.

Returns a volatile pointer to RIIC channel 2 registers at address 0x00088340. RIIC2 pins are P66 (SCL2) and P67 (SDA2).

## Returns

Volatile pointer to RIIC2 register structure

Definition at line 468 of file [rx72n\\_riic\\_regs.h](#).

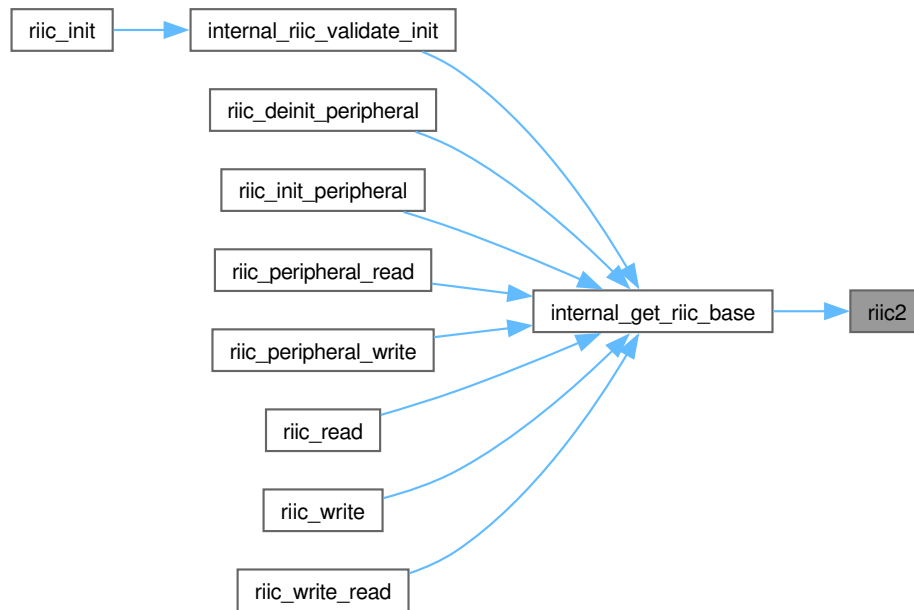
```
00469 {
00470 return (volatile rx_riic_regs_t*)k_riic2_base_addr;
00471 }
```

References [k\\_riic2\\_base\\_addr](#).

Referenced by [internal\\_get\\_riic\\_base\(\)](#).

Here is the caller graph for this function:

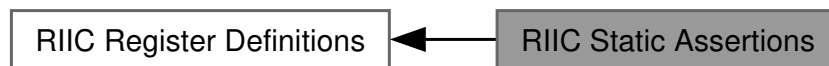




### 5.8.5 RIIC Static Assertions

Compile-time verification of RIIC register layout.

Collaboration diagram for RIIC Static Assertions:



Compile-time verification of RIIC register layout.

These static assertions are verified at compile time to ensure the C structure definitions match the actual hardware register layout. A failed assertion indicates a bug in the register definitions that must be fixed before the code can compile.

## 5.9 SCI Register Definitions

### Data Structures

- struct `rx_sci_regs_t`  
SCI register map covering UART and SPI mode registers (14 bytes).

## Enumerations

- enum `rx_sci_addresses_t` : `uintptr_t` {  
`k_sci0_base_addr` = 0x0008A000 , `k_sci1_base_addr` = 0x0008A020 , `k_sci2_base_addr` = 0x0008A040 , `k_sci3_base_addr` = 0x0008A060 ,  
`k_sci4_base_addr` = 0x0008A080 , `k_sci5_base_addr` = 0x0008A0A0 , `k_sci6_base_addr` = 0x0008A0C0 , `k_sci7_base_addr` = 0x000D00E0 ,  
`k_sci8_base_addr` = 0x000D0000 , `k_sci9_base_addr` = 0x000D0020 , `k_sci10_base_addr` = 0x000D0040 , `k_sci11_base_addr` = 0x000D0060 ,  
`k_sci12_base_addr` = 0x0008B300 }

SCI channel base addresses (verified against RX72N Hardware Manual).

- enum `rx_sci_ssr_flags_t` : `uint8_t` {  
`k_sci_ssr_tdre_flag` = 0x80U , `k_sci_ssr_rdrf_flag` = 0x40U , `k_sci_ssr_orer_flag` = 0x20U ,  
`k_sci_ssr_fer_flag` = 0x10U ,  
`k_sci_ssr_per_flag` = 0x08U , `k_sci_ssr_tend_flag` = 0x04U , `k_sci_ssr_error_mask` = 0x38U  
}

SCI Serial Status Register (SSR) flag masks.

- enum `sci_channel_limits_t` : `uint8_t` { `k_sci_channel_max` = 13 }

SCI channel count.

## Functions

- static volatile `rx_sci_regs_t * sci0` (void)  
Get pointer to SCI0 registers.
- static volatile `rx_sci_regs_t * sci1` (void)  
Get pointer to SCI1 registers.
- static volatile `rx_sci_regs_t * sci2` (void)  
Get pointer to SCI2 registers.
- static volatile `rx_sci_regs_t * sci3` (void)  
Get pointer to SCI3 registers.
- static volatile `rx_sci_regs_t * sci4` (void)  
Get pointer to SCI4 registers.
- static volatile `rx_sci_regs_t * sci5` (void)  
Get pointer to SCI5 registers.
- static volatile `rx_sci_regs_t * sci6` (void)  
Get pointer to SCI6 registers.
- static volatile `rx_sci_regs_t * sci7` (void)  
Get pointer to SCI7 registers.
- static volatile `rx_sci_regs_t * sci8` (void)  
Get pointer to SCI8 registers.
- static volatile `rx_sci_regs_t * sci9` (void)  
Get pointer to SCI9 registers (Debug UART).
- static volatile `rx_sci_regs_t * sci10` (void)  
Get pointer to SCI10 registers.
- static volatile `rx_sci_regs_t * sci11` (void)  
Get pointer to SCI11 registers.
- static volatile `rx_sci_regs_t * sci12` (void)  
Get pointer to SCI12 registers.
- static volatile `rx_sci_regs_t * sci_get_channel` (uint8\_t channel)  
Get SCI register base for a channel.

## 5.9.1 Detailed Description

## 5.9.2 Data Structure Documentation

### 5.9.2.1 struct rx\_sci\_regs\_t

SCI register map covering UART and SPI mode registers (14 bytes).

Memory-mapped register structure for the RX72N SCI peripheral. Covers registers from SMR (0x00) through SPMR (0x0D) to support both asynchronous UART and clock-synchronous SPI modes.

Register Memory Layout (14 bytes total)

| * Offset | Size | Register | Description                                     |
|----------|------|----------|-------------------------------------------------|
| * 0x00   | 1    | SMR      | Serial Mode (data format, clock source)         |
| * 0x01   | 1    | BRR      | Bit Rate (baud rate divisor)                    |
| * 0x02   | 1    | SCR      | Serial Control (TX/RX enable, interrupts)       |
| * 0x03   | 1    | TDR      | Transmit Data (write-only data buffer)          |
| * 0x04   | 1    | SSR      | Serial Status (flags, error conditions)         |
| * 0x05   | 1    | RDR      | Receive Data (read-only data buffer)            |
| * 0x06   | 1    | SCMR     | Smart Card Mode (SDIR for bit order)            |
| * 0x07   | 1    | SEMR     | Serial Extended Mode (noise filter, modulation) |
| * 0x08   | 1    | SNFR     | Noise Filter Setting                            |
| * 0x09   | 1    | SIMR1    | I2C Mode Register 1                             |
| * 0x0A   | 1    | SIMR2    | I2C Mode Register 2                             |
| * 0x0B   | 1    | SIMR3    | I2C Mode Register 3                             |
| * 0x0C   | 1    | SISR     | I2C Status Register                             |
| * 0x0D   | 1    | SPMR     | SPI Mode Register (CKPH, CKPOL)                 |

### UART Transmission Sequence

- \* 1. Wait for SSR.TDRE = 1 (transmit data register empty)
- \* 2. Write data byte to TDR
- \* 3. SSR.TDRE automatically clears to 0
- \* 4. Data shifts out serially via TXD pin
- \* 5. SSR.TEND = 1 when transmission complete
- \*

### UART Reception Sequence

- \* 1. Data arrives via RXD pin
- \* 2. SSR.RDRF = 1 when byte received
- \* 3. Read data byte from RDR
- \* 4. SSR.RDRF automatically clears to 0
- \* 5. Check SSR.ORER/FER/PER for errors
- \*

### Transmit Character Example

```
void uart_putc(volatile rx_sci_regs_t* uart, char c)
{
 // Wait for transmit buffer empty
 while ((uart->ssr & 0x80) == 0) {
 // SSR.TDRE bit 7
 }
 // Write character to transmit
 uart->tdr = (uint8_t)c;
}
```

## Receive Character Example

```
char uart_getc(volatile rx_sci_regs_t* uart)
{
 // Wait for receive data ready
 while ((uart->ssr & 0x40) == 0) {
 // SSR.RDRF bit 6
 }
 // Read received character
 return (char)uart->rdr;
}
```

## Invariant

Structure size must be exactly 14 bytes  
All registers are 8-bit (byte-accessible)

## Note

Offsets 0x00-0x07 are unchanged from the original 8-byte structure. Existing UART code works unmodified. Offsets 0x08-0x0D add I2C mode and SPI mode register access needed for clock-synchronous SPI.

## See also

[rx\\_sci\\_addresses\\_t](#) for channel base addresses  
RX72N Hardware Manual Section 41.2 (Register Descriptions)

Definition at line 319 of file [rx72n\\_sci\\_regs.h](#).

Collaboration diagram for rx\_sci\_regs\_t:



## Data Fields

|                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | brr  | <p>Bit Rate Register (BRR) - baud rate divisor.</p> <p>Async mode: Baud rate = <math>PCLK / (64 * 2^{(2n-1)} * (BRR + 1))</math> Sync mode: Bit rate = <math>PCLK / (4 * (BRR + 1))</math> for CKS=0 PCLK = PCLKB (60 MHz) for SCI0-6/12; PCLKA (120 MHz) for SCI7-11.</p> <p>Common async values at 60 MHz PCLKB (SCI0-6/12), CKS=0:</p> <ul style="list-style-type: none"> <li>• 9600 baud: BRR = 194</li> <li>• 38400 baud: BRR = 47</li> <li>• 115200 baud: BRR = 15</li> </ul> <p>Common sync values at 60 MHz PCLKB (SCI0-6/12), CKS=0:</p> <ul style="list-style-type: none"> <li>• BRR=1: 7.5 MHz SPI clock</li> <li>• BRR=3: 3.75 MHz SPI clock</li> </ul>              |
| volatile uint8_t | rdr  | <p>Receive Data Register (RDR) - read-only data buffer.</p> <p>Read received data here. Wait for SSR.RDRF = 1 first.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| volatile uint8_t | scmr | <p>Smart Card Mode Register (SCMR) - smart card and bit order settings.</p> <ul style="list-style-type: none"> <li>• Bit 7: BCP2 - Base clock pulse 2 (smart card clock cycles per ETU)</li> <li>• Bits 6-5: Reserved (write 1)</li> <li>• Bit 4: CHR1 - Character length 1 (0=9/8 bit, 1=8/7 bit; use with SMR.CHR)</li> <li>• Bit 3: SDIR - Data transfer direction (0=LSB first, 1=MSB first)</li> <li>• Bit 2: SINV - Smart card data inversion (0=no inversion, 1=inverted)</li> <li>• Bit 1: Reserved (write 1)</li> <li>• Bit 0: SMIF - Smart card interface mode (0=disabled, 1=enabled)</li> </ul> <p>Note</p> <p>For SPI mode, set SDIR=1 for MSB-first transfers.</p> |
| volatile uint8_t | scr  | <p>Serial Control Register (SCR) - TX/RX enable and interrupts.</p> <ul style="list-style-type: none"> <li>• Bit 7: TIE - Transmit interrupt enable</li> <li>• Bit 6: RIE - Receive interrupt enable</li> <li>• Bit 5: TE - Transmit enable</li> <li>• Bit 4: RE - Receive enable</li> <li>• Bit 3: MPIE - Multi-processor interrupt enable</li> <li>• Bit 2: TEIE - Transmit end interrupt enable</li> <li>• Bits 1-0: CKE - Clock enable (external clock source)</li> </ul>                                                                                                                                                                                                    |

|                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | semr  | <p>Serial Extended Mode Register (SEMR) - advanced settings.</p> <ul style="list-style-type: none"> <li>• Bit 7: RXDESEL - RXD extended input control</li> <li>• Bit 6: BGDM - Baud rate generator double-speed mode</li> <li>• Bit 5: NFEN - Digital noise filter enable</li> <li>• Bit 4: ABCS - Asynchronous mode base clock select</li> <li>• Bit 3: ABCSE - Asynchronous mode extended base clock select</li> <li>• Bit 2: BRME - Bit rate modulation enable</li> <li>• Bit 1: Reserved</li> <li>• Bit 0: ACS0 - Asynchronous mode clock source select 0</li> </ul> |
| volatile uint8_t | simr1 | <p>I2C Mode Register 1 (SIMR1).<br/>Used in I2C mode. Not used for UART or SPI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| volatile uint8_t | simr2 | <p>I2C Mode Register 2 (SIMR2).<br/>Used in I2C mode. Not used for UART or SPI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| volatile uint8_t | simr3 | <p>I2C Mode Register 3 (SIMR3).<br/>Used in I2C mode. Not used for UART or SPI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| volatile uint8_t | sir   | <p>I2C Status Register (SISR).<br/>Used in I2C mode. Not used for UART or SPI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| volatile uint8_t | smr   | <p>Serial Mode Register (SMR) - data format configuration.</p> <ul style="list-style-type: none"> <li>• Bits 7: CM - Clock mode (0=async, 1=sync)</li> <li>• Bits 6: CHR - Character length (0=8-bit, 1=7-bit)</li> <li>• Bits 5: PE - Parity enable (0=disable, 1=enable)</li> <li>• Bits 4: PM - Parity mode (0=even, 1=odd)</li> <li>• Bits 3: STOP - Stop bits (0=1 bit, 1=2 bits)</li> <li>• Bits 2: MP - Multi-processor mode</li> <li>• Bits 1-0: CKS - Clock select (00=PCLK/1, 01=PCLK/4, 10=PCLK/16, 11=PCLK/64)</li> </ul>                                    |
| volatile uint8_t | snfr  | <p>Noise Filter Setting Register (SNFR).</p> <ul style="list-style-type: none"> <li>• Bits 2-0: NFCS - Noise filter clock select</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t | spmr | <p>SPI Mode Register (SPMR) - SPI clock polarity and phase.</p> <ul style="list-style-type: none"> <li>• Bit 7: CKPH - Clock phase (inverted from standard CPHA in some Renesas docs)</li> <li>• Bit 6: CKPOL - Clock polarity (0=idle LOW, 1=idle HIGH)</li> <li>• Bit 4: MFF - Mode fault flag</li> <li>• Bit 3: Reserved</li> <li>• Bit 2: MSS - Controller/peripheral select (0=controller, 1=peripheral)</li> <li>• Bit 1: CTSE - CTS enable (0=disabled, 1=CTS output enabled)</li> <li>• Bit 0: SSE - SS pin enable</li> </ul>                                                                                                                                         |
| volatile uint8_t | ssr  | <p>Serial Status Register (SSR) - status flags and errors.</p> <ul style="list-style-type: none"> <li>• Bit 7: TDRE - Transmit data register empty (1=ready to write)</li> <li>• Bit 6: RDRF - Receive data register full (1=data available)</li> <li>• Bit 5: ORER - Overrun error (1=error occurred)</li> <li>• Bit 4: FER - Framing error (1=error occurred)</li> <li>• Bit 3: PER - Parity error (1=error occurred)</li> <li>• Bit 2: TEND - Transmit end (1=transmission complete)</li> <li>• Bit 1: MPB - Multi-processor bit</li> <li>• Bit 0: MPBT - Multi-processor bit transfer</li> </ul> <p>Note</p> <p>Write 0 to error bits to clear them (do not write 1).</p> |
| volatile uint8_t | tdr  | <p>Transmit Data Register (TDR) - write-only data buffer.<br/>Write data here to transmit. Wait for SSR.TDRE = 1 first.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### 5.9.3 Enumeration Type Documentation

#### 5.9.3.1 rx'sci'addresses't

```
enum rx_sci_addresses_t : uintptr_t
```

SCI channel base addresses (verified against RX72N Hardware Manual).

Base addresses for all 13 SCI channels. The RX72N organizes SCI channels into three modules in different memory regions:

## Memory Map Overview

| * Address Range         | Module | Channels | Notes                            |
|-------------------------|--------|----------|----------------------------------|
| * -----                 |        |          |                                  |
| * 0x0008A000-0x0008A0DF | SCIj   | SCI0-6   | Standard region, 0x20 spacing    |
| * 0x000D0000-0x000D00FF | SCIi   | SCI7-11  | Extended region, varying spacing |
| * 0x0008B300-0x0008B31F | SCIh   | SCI12    | Standard region, separate        |
| *                       |        |          |                                  |

## Channel Spacing

- SCIj (SCI0-6): 0x20 bytes between channels
- SCIi (SCI7-11): Non-uniform spacing (see individual addresses)
- SCIh (SCI12): Standalone at 0x0008B300

## STAR Robot Usage

SCI9 is designated as the debug UART, connected to the CY7C65213 USB-UART bridge IC for host PC communication at 115200 baud.

See also

RX72N Hardware Manual Section 41.2 (Register Addresses)

## Enumerator

|                   |                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| k_sci0_base_addr  | SCI0 base address - standard region (SCIj module).                                                                         |
| k_sci1_base_addr  | SCI1 base address - standard region (SCIj module).                                                                         |
| k_sci2_base_addr  | SCI2 base address - standard region (SCIj module).                                                                         |
| k_sci3_base_addr  | SCI3 base address - standard region (SCIj module).                                                                         |
| k_sci4_base_addr  | SCI4 base address - standard region (SCIj module).                                                                         |
| k_sci5_base_addr  | SCI5 base address - standard region (SCIj module).                                                                         |
| k_sci6_base_addr  | SCI6 base address - standard region (SCIj module).                                                                         |
| k_sci7_base_addr  | SCI7 base address - extended region (SCIi module).                                                                         |
| k_sci8_base_addr  | SCI8 base address - extended region (SCIi module).                                                                         |
| k_sci9_base_addr  | SCI9 base address - extended region (SCIi module).<br><br>Note<br>STAR Debug UART - Connected to CY7C65213 USB-UART bridge |
| k_sci10_base_addr | SCI10 base address - extended region (SCIi module).                                                                        |
| k_sci11_base_addr | SCI11 base address - extended region (SCIi module).                                                                        |
| k_sci12_base_addr | SCI12 base address - standard region (SCIh module).                                                                        |

Definition at line 192 of file [rx72n\\_sci\\_regs.h](#).

```

00192 : uintptr_t {
00193 /** @brief SCI0 base address - standard region (SCIj module) */
00194 k_sci0_base_addr = 0x0008A000,
00195
00196 /** @brief SCI1 base address - standard region (SCIj module) */
00197 k_sci1_base_addr = 0x0008A020,
00198
```



```

00199 /** @brief SCI2 base address - standard region (SCIj module) */
00200 k_sci2_base_addr = 0x0008A040,
00201
00202 /** @brief SCI3 base address - standard region (SCIj module) */
00203 k_sci3_base_addr = 0x0008A060,
00204
00205 /** @brief SCI4 base address - standard region (SCIj module) */
00206 k_sci4_base_addr = 0x0008A080,
00207
00208 /** @brief SCI5 base address - standard region (SCIj module) */
00209 k_sci5_base_addr = 0x0008A0A0,
00210
00211 /** @brief SCI6 base address - standard region (SCIj module) */
00212 k_sci6_base_addr = 0x0008A0C0,
00213
00214 /** @brief SCI7 base address - extended region (SCIi module) */
00215 k_sci7_base_addr = 0x000D00E0,
00216
00217 /** @brief SCI8 base address - extended region (SCIi module) */
00218 k_sci8_base_addr = 0x000D0000,
00219
00220 /**
00221 * @brief SCI9 base address - extended region (SCIi module)
00222 * @note **STAR Debug UART** - Connected to CY7C65213 USB-UART bridge
00223 */
00224 k_sci9_base_addr = 0x000D0020,
00225
00226 /** @brief SCI10 base address - extended region (SCIi module) */
00227 k_sci10_base_addr = 0x000D0040,
00228
00229 /** @brief SCI11 base address - extended region (SCIi module) */
00230 k_sci11_base_addr = 0x000D0060,
00231
00232 /** @brief SCI12 base address - standard region (SCIh module) */
00233 k_sci12_base_addr = 0x0008B300,
00234 } rx_sci_addresses_t;

```

### 5.9.3.2 rx\_sci\_ssr\_flags\_t

```
enum rx_sci_ssr_flags_t : uint8_t
```

SCI Serial Status Register (SSR) flag masks.

Bit masks for the SSR register per RX72N HUM 34.2.7. Error flags (ORER, FER, PER) are sticky and must be cleared by read-modify-write with a 0 written to the bit position (writing 1 has no effect).

Note

Bit positions match both the SCIIa/b/c/d/e/f (standard) and SCIIg/h/i (extended) SSR layouts – the same mask works for all SCI channels on the RX72N.

Enumerator

|                      |                                                 |
|----------------------|-------------------------------------------------|
| k_sci_ssr_tdre_flag  | Transmit data register empty (HUM 34.2.7 bit 7) |
| k_sci_ssr_rdrf_flag  | Receive data register full (bit 6)              |
| k_sci_ssr_orer_flag  | Overrun error (bit 5)                           |
| k_sci_ssr_fer_flag   | Framing error (bit 4)                           |
| k_sci_ssr_per_flag   | Parity error (bit 3)                            |
| k_sci_ssr_tend_flag  | Transmit end (bit 2)                            |
| k_sci_ssr_error_mask | ORER FER PER – receive error flags              |

Definition at line 479 of file [rx72n\\_sci\\_regs.h](#).

```

00479 : uint8_t {
00480 k_sci_ssr_tdre_flag = 0x80U, /**< Transmit data register empty (HUM 34.2.7 bit 7) */
00481 k_sci_ssr_rdrf_flag = 0x40U, /**< Receive data register full (bit 6) */
00482 k_sci_ssr_orer_flag = 0x20U, /**< Overrun error (bit 5) */
00483 k_sci_ssr_fer_flag = 0x10U, /**< Framing error (bit 4) */
00484 k_sci_ssr_per_flag = 0x08U, /**< Parity error (bit 3) */
00485 k_sci_ssr_tend_flag = 0x04U, /**< Transmit end (bit 2) */
00486 k_sci_ssr_error_mask = 0x38U, /**< ORER|FER|PER -- receive error flags */
00487 } rx_sci_ssr_flags_t;

```

### 5.9.3.3 sci\_channel\_limits\_t

enum [sci\\_channel\\_limits\\_t](#) : uint8\_t

SCI channel count.

Enumerator

|                   |                           |
|-------------------|---------------------------|
| k_sci_channel_max | Total SCI channels (0-12) |
|-------------------|---------------------------|

Definition at line 614 of file [rx72n\\_sci\\_regs.h](#).

```
00614 : uint8_t {
00615 k_sci_channel_max = 13, /**< Total SCI channels (0-12) */
00616 } sci_channel_limits_t;
```

## 5.9.4 Function Documentation

### 5.9.4.1 sci0()

volatile [rx\\_sci\\_regs\\_t](#) \* sci0 (  
void ) [inline], [static]

Get pointer to SCI0 registers.

Returns

Volatile pointer to SCI0 register structure

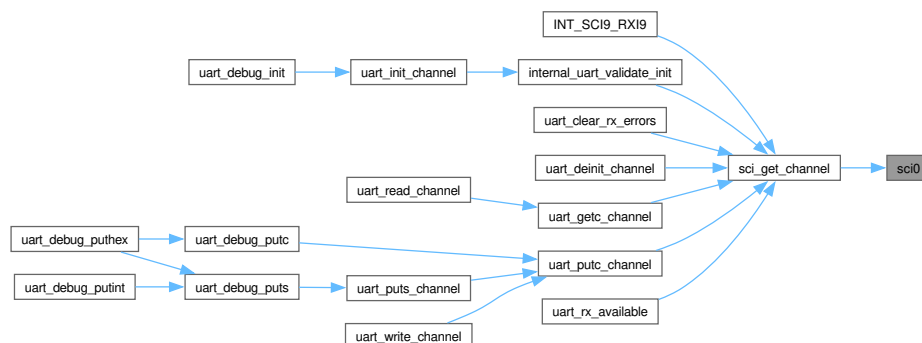
Definition at line 493 of file [rx72n\\_sci\\_regs.h](#).

```
00494 {
00495 return (volatile rx_sci_regs_t*)k_sci0_base_addr;
00496 }
```

References [k\\_sci0\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.2 sci1()

```
volatile rx_sci_regs_t * sci1 (
 void) [inline], [static]
```

Get pointer to SCI1 registers.

Returns

Volatile pointer to SCI1 register structure

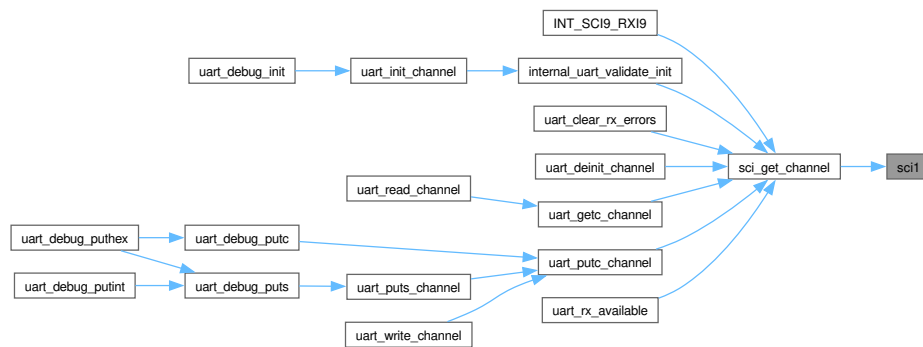
Definition at line 502 of file rx72n\_sci\_regs.h.

```
00503 {
00504 return (volatile rx_sci_regs_t*)k_sci1_base_addr;
00505 }
```

References [k\\_sci1\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.3 sci10()

```
volatile rx_sci_regs_t * sci10 (
 void) [inline], [static]
```

Get pointer to SCI10 registers.

Returns

Volatile pointer to SCI10 register structure

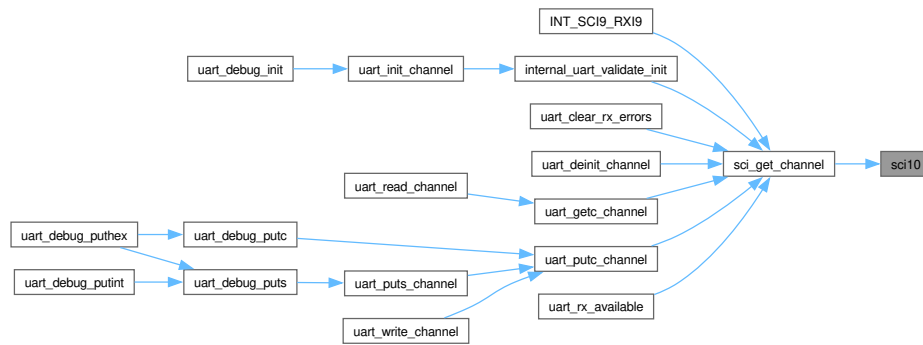
Definition at line 583 of file rx72n\_sci\_regs.h.

```
00584 {
00585 return (volatile rx_sci_regs_t*)k_sci10_base_addr;
00586 }
```

References [k\\_sci10\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



#### 5.9.4.4 sci11()

```
volatile rx_sci_regs_t * sci11 (
 void) [inline], [static]
```

Get pointer to SCI11 registers.

Returns

Volatile pointer to SCI11 register structure

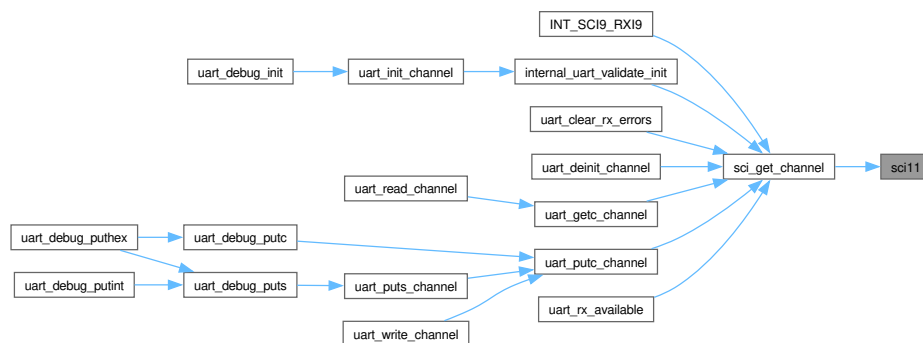
Definition at line 592 of file rx72n\_sci\_regs.h.

```
00593 {
00594 return (volatile rx_sci_regs_t*)k_sci11_base_addr;
00595 }
```

References [k\\_sci11\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.5 sci12()

```
volatile rx_sci_regs_t * sci12 (
 void) [inline], [static]
```

Get pointer to SCI12 registers.

Returns

Volatile pointer to SCI12 register structure

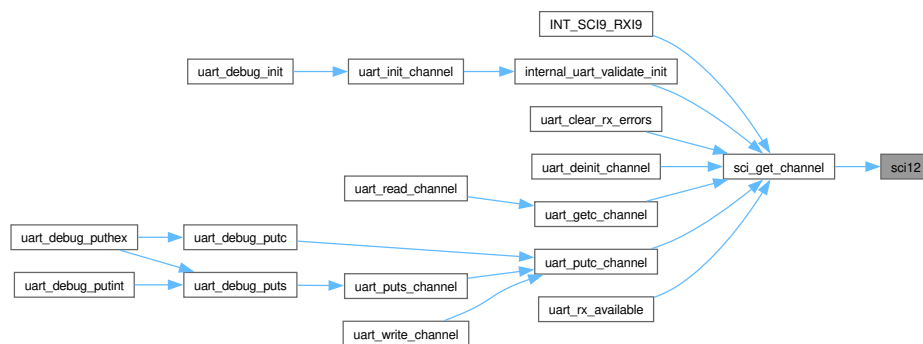
Definition at line 601 of file rx72n\_sci\_regs.h.

```
00602 {
00603 return (volatile rx_sci_regs_t*)k_sci12_base_addr;
00604 }
```

References [k\\_sci12\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.6 sci2()

```
volatile rx_sci_regs_t * sci2 (
 void) [inline], [static]
```

Get pointer to SCI2 registers.

Returns

Volatile pointer to SCI2 register structure

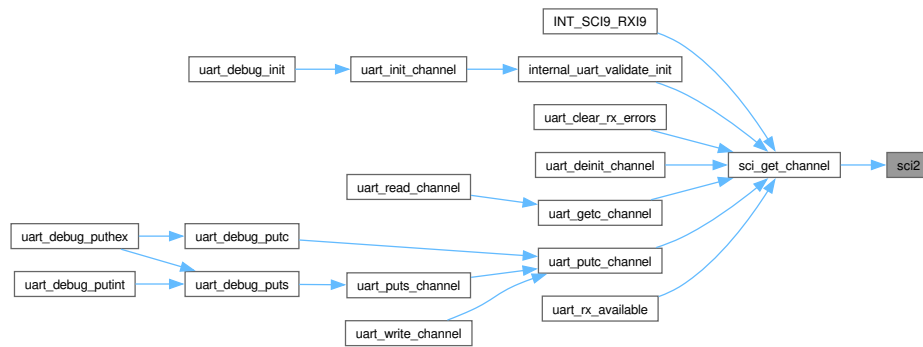
Definition at line 511 of file rx72n\_sci\_regs.h.

```
00512 {
00513 return (volatile rx_sci_regs_t*)k_sci2_base_addr;
00514 }
```

References [k\\_sci2\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



#### 5.9.4.7 sci3()

```
volatile rx_sci_regs_t * sci3 (
 void) [inline], [static]
```

Get pointer to SCI3 registers.

Returns

Volatile pointer to SCI3 register structure

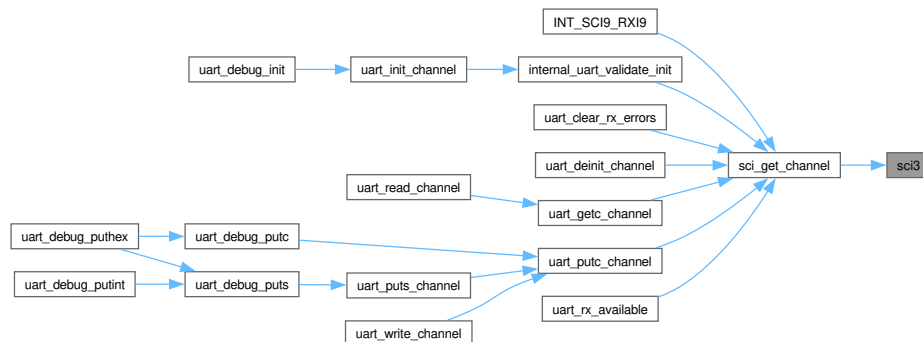
Definition at line 520 of file rx72n\_sci\_regs.h.

```
00521 {
00522 return (volatile rx_sci_regs_t*)k_sci3_base_addr;
00523 }
```

References [k\\_sci3\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.8 sci4()

```
volatile rx_sci_regs_t * sci4 (
 void) [inline], [static]
```

Get pointer to SCI4 registers.

Returns

Volatile pointer to SCI4 register structure

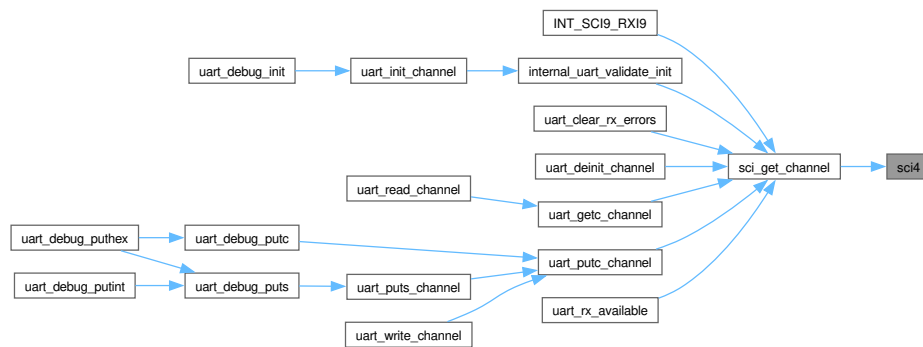
Definition at line 529 of file rx72n\_sci\_regs.h.

```
00530 {
00531 return (volatile rx_sci_regs_t*)k_sci4_base_addr;
00532 }
```

References [k\\_sci4\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.9 sci5()

```
volatile rx_sci_regs_t * sci5 (
 void) [inline], [static]
```

Get pointer to SCI5 registers.

Returns

Volatile pointer to SCI5 register structure

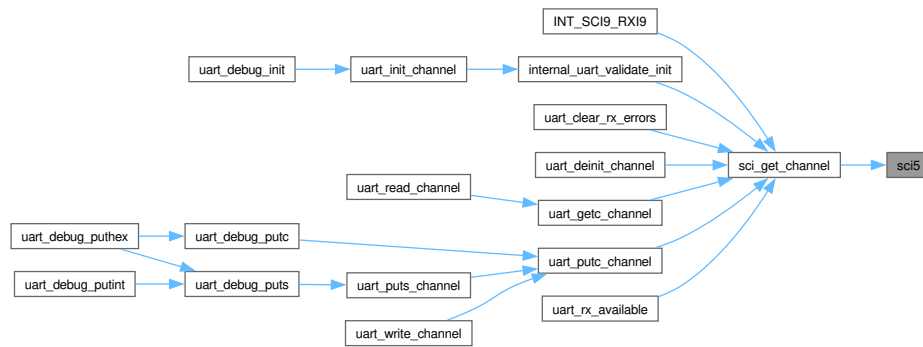
Definition at line 538 of file rx72n\_sci\_regs.h.

```
00539 {
00540 return (volatile rx_sci_regs_t*)k_sci5_base_addr;
00541 }
```

References [k\\_sci5\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



#### 5.9.4.10 sci6()

```
volatile rx_sci_regs_t * sci6 (
 void) [inline], [static]
```

Get pointer to SCI6 registers.

Returns

Volatile pointer to SCI6 register structure

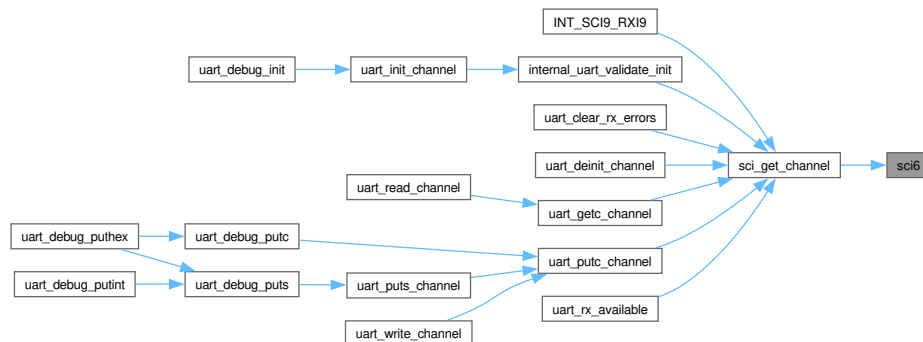
Definition at line 547 of file rx72n\_sci\_regs.h.

```
00548 {
00549 return (volatile rx_sci_regs_t*)k_sci6_base_addr;
00550 }
```

References [k\\_sci6\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:





## 5.9.4.11 sci7()

```
volatile rx_sci_regs_t * sci7 (
 void) [inline], [static]
```

Get pointer to SCI7 registers.

Returns

Volatile pointer to SCI7 register structure

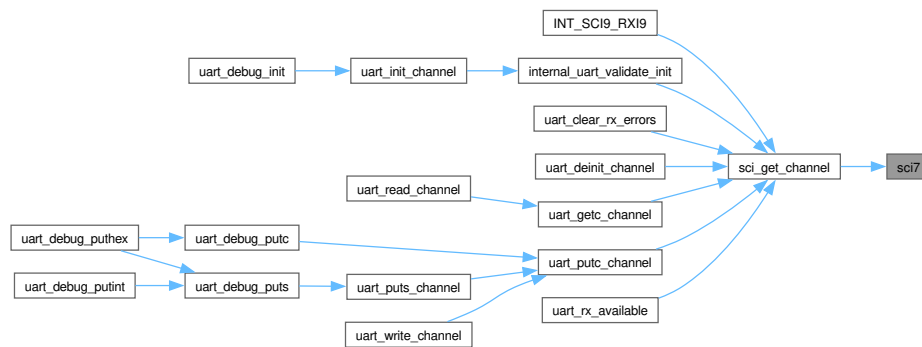
Definition at line 556 of file rx72n\_sci\_regs.h.

```
00557 {
00558 return (volatile rx_sci_regs_t*)k_sci7_base_addr;
00559 }
```

References [k\\_sci7\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.12 sci8()

```
volatile rx_sci_regs_t * sci8 (
 void) [inline], [static]
```

Get pointer to SCI8 registers.

Returns

Volatile pointer to SCI8 register structure

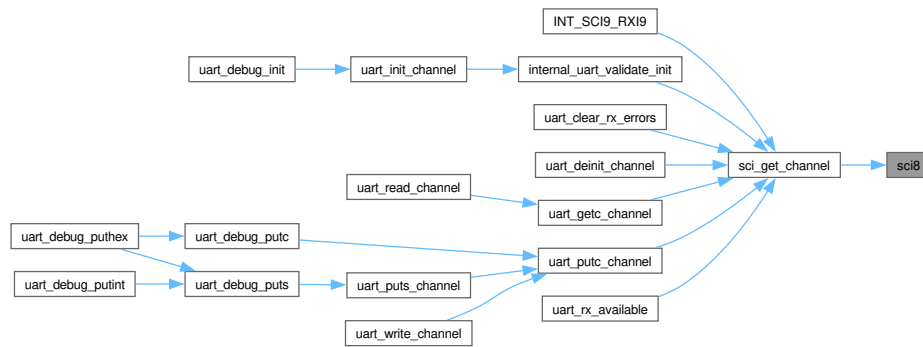
Definition at line 565 of file rx72n\_sci\_regs.h.

```
00566 {
00567 return (volatile rx_sci_regs_t*)k_sci8_base_addr;
00568 }
```

References [k\\_sci8\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



#### 5.9.4.13 sci9()

```
volatile rx_sci_regs_t * sci9 (
 void) [inline], [static]
```

Get pointer to SCI9 registers (Debug UART).

Returns

Volatile pointer to SCI9 register structure

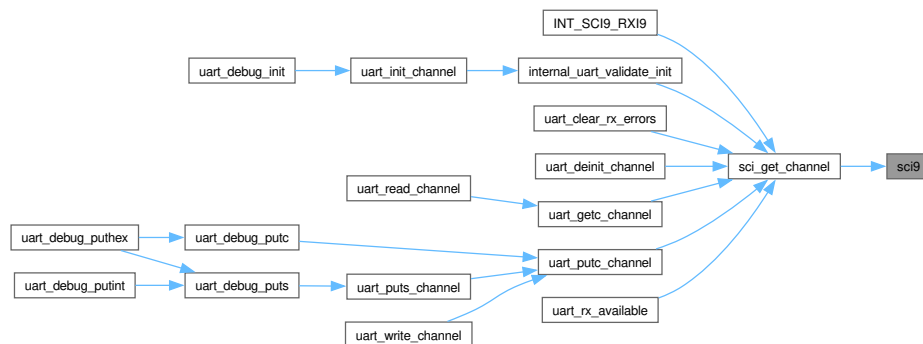
Definition at line 574 of file rx72n\_sci\_regs.h.

```
00575 {
00576 return (volatile rx_sci_regs_t*)k_sci9_base_addr;
00577 }
```

References [k\\_sci9\\_base\\_addr](#).

Referenced by [sci\\_get\\_channel\(\)](#).

Here is the caller graph for this function:



## 5.9.4.14 sci\_get\_channel()

```
volatile rx_sci_regs_t * sci_get_channel (
 uint8_t channel) [inline], [static]
```

Get SCI register base for a channel.

Parameters

|    |         |                           |
|----|---------|---------------------------|
| in | channel | SCI channel number (0-12) |
|----|---------|---------------------------|

Returns

Pointer to SCI registers, or nullptr if invalid channel

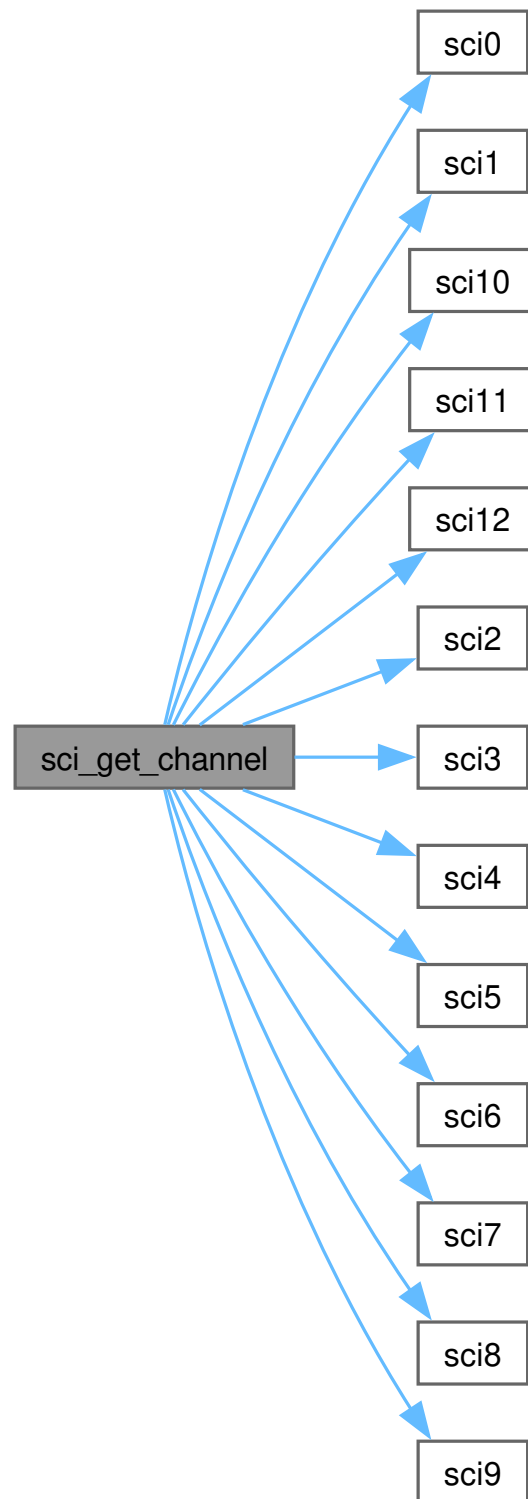
Definition at line 626 of file rx72n\_sci\_regs.h.

```
00627 {
00628 switch (channel) {
00629 case 0:
00630 return sci0();
00631 case 1:
00632 return sci1();
00633 case 2:
00634 return sci2();
00635 case 3:
00636 return sci3();
00637 case 4:
00638 return sci4();
00639 case 5:
00640 return sci5();
00641 case 6:
00642 return sci6();
00643 case 7:
00644 return sci7();
00645 case 8:
00646 return sci8();
00647 case 9:
00648 return sci9();
00649 case 10:
00650 return sci10();
00651 case 11:
00652 return sci11();
00653 case 12:
00654 return sci12();
00655 default:
00656 return nullptr; /* Invalid channel */
00657 }
00658 }
```

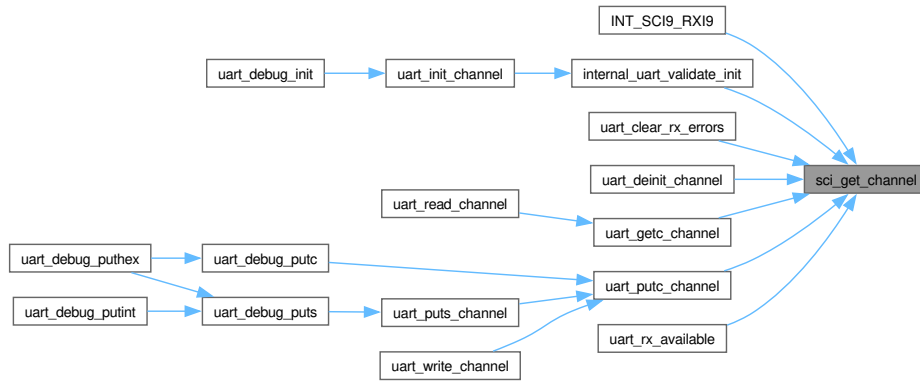
References [sci0\(\)](#), [sci1\(\)](#), [sci10\(\)](#), [sci11\(\)](#), [sci12\(\)](#), [sci2\(\)](#), [sci3\(\)](#), [sci4\(\)](#), [sci5\(\)](#), [sci6\(\)](#), [sci7\(\)](#), [sci8\(\)](#), and [sci9\(\)](#).

Referenced by [INT\\_SCI9\\_RXI9\(\)](#), [internal\\_uart\\_validate\\_init\(\)](#), [uart\\_clear\\_rx\\_errors\(\)](#), [uart\\_deinit\\_channel\(\)](#), [uart\\_getc\\_channel\(\)](#), [uart\\_putc\\_channel\(\)](#), and [uart\\_rx\\_available\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



## 5.10 Task-Level Watchdog Monitoring

Software layer for per-task heartbeat tracking.

### Data Structures

- struct `task_monitor_t`  
Task monitoring state for a single registered task.
- struct `task_monitor_state_t`  
Complete task monitoring module state.

### Enumerations

- enum `task_registration_state_t` : `uint8_t` { `k_task_not_registered` = 0 , `k_task_registered` = 1 }
  - enum `task_search_constants_t` : `int32_t` { `k_task_not_found` = -1 , `k_task_idx_min` = 0 }
- Task registration state tracking.
- Constants for task search operations.

### Functions

- static `int32_t` `internal_find_task` (`const char *task_name`)  
Find task index by name in the registered tasks array.
- `rx_err_t` `rx_hal_iwdt_register_task` (`const char *task_name`, `uint32_t timeout_ms`)  
Register a task for watchdog monitoring.
- void `rx_hal_iwdt_task_heartbeat` (`const char *task_name`)  
Record a task heartbeat to indicate liveness.
- `rx_err_t` `rx_hal_iwdt_check_tasks` (void)  
Check all monitored tasks for timeout violations.
- `const char *` `rx_hal_iwdt_get_failed_task` (void)  
Get name of last failed task for post-mortem analysis.

### Variables

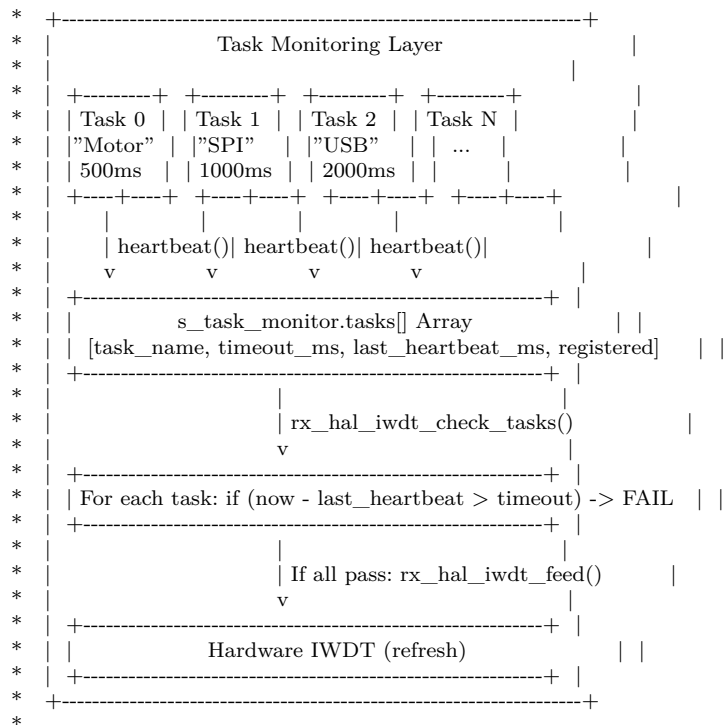
- static `task_monitor_state_t` `s_task_monitor` = {}  
Module state for task monitoring (static allocation).

### 5.10.1 Detailed Description

Software layer for per-task heartbeat tracking.

Provides fine-grained monitoring of individual RTOS tasks beyond hardware watchdog. Tasks register with unique names and timeout periods, then must send periodic heartbeats. A monitoring task checks for timeouts and identifies which specific task has hung.

#### Architecture



### 5.10.2 Data Structure Documentation

#### 5.10.2.1 struct task\_monitor\_t

Task monitoring state for a single registered task.

Stores all information needed to track one task's heartbeat status. Each registered task gets one slot in the `s_task_monitor.tasks[]` array.

Memory Layout (~32 bytes per task)

```

* Offset 0: task_name[16] [16 bytes] - Null-terminated name
* Offset 16: timeout_ms [4 bytes] - Heartbeat timeout
* Offset 20: last_heartbeat_ms [4 bytes] - Last heartbeat timestamp
* Offset 24: registered [1 byte] - Registration state
* Offset 25: padding [7 bytes] - Alignment padding
* Total: ~32 bytes (depends on alignment)
*

```

## Timeout Check Algorithm

```

* elapsed = current_time - last_heartbeat_ms
* if (elapsed > timeout_ms) -> Task has timed out
*

```

## Invariant

```

task_name is null-terminated
timeout_ms >= k_iwdt_timeout_min_ms
If registered == k_task_registered, task_name[0] != '\0'

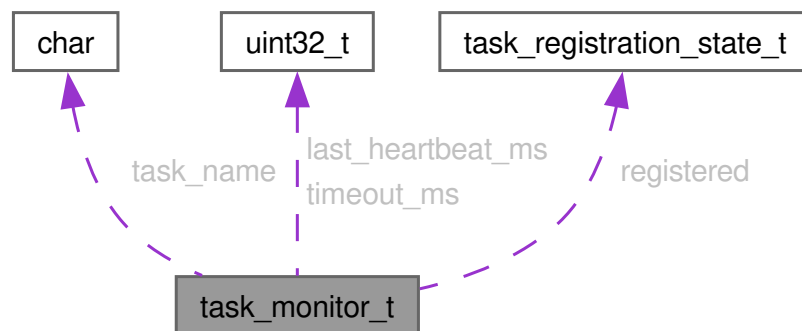
```

Since

Version 1.0.0

Definition at line 1169 of file [rx\\_iwdt.c](#).

Collaboration diagram for task\_monitor\_t:



## Data Fields

|                                           |                                                     |                                                          |
|-------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|
| uint32_t                                  | last_heartbeat_ms                                   | Timestamp (tx_time_get) of last heartbeat                |
| <a href="#">task_registration_state_t</a> | registered                                          | Whether this slot is in use                              |
| char                                      | task_name↔<br><a href="#">[k_task_name_max_len]</a> | Task identifier (null-terminated, truncated to 15 chars) |
| uint32_t                                  | timeout_ms                                          | Maximum time between heartbeats in milliseconds          |

## 5.10.2.2 struct task\_monitor\_state\_t

Complete task monitoring module state.

Contains all state for the task monitoring subsystem. Statically allocated to avoid dynamic memory (NASA Power of 10 Rule 3). One instance exists as `s_task_monitor`.

Memory Layout (~520 bytes total)

```
* Offset 0: tasks[8] [~256 bytes] - Task array (8 x 32 bytes)
* Offset 256: task_count [1 byte] - Number of registered tasks
* Offset 257: failed_task[16] [16 bytes] - Failed task name buffer
* Offset 273: padding [7 bytes] - Alignment
* Total: ~280 bytes (actual layout depends on alignment)
*
```

Invariant

`task_count <= k_iwdt_max_tasks`

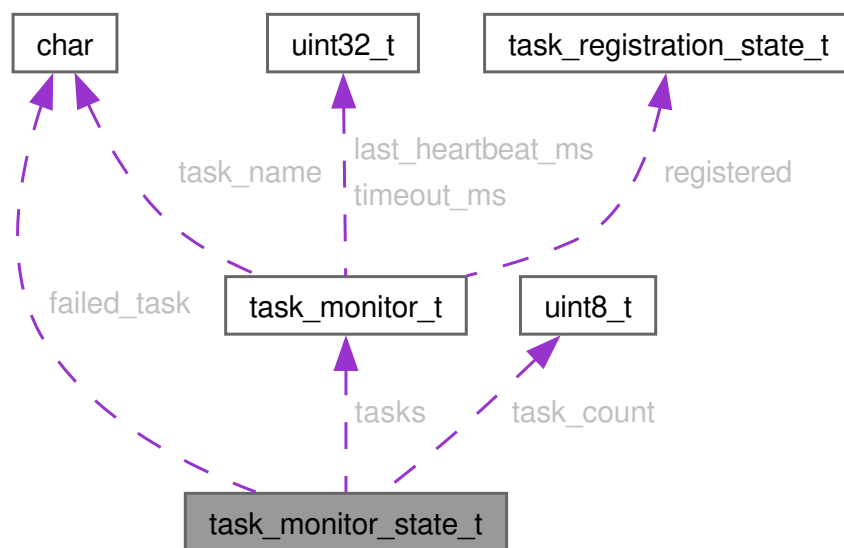
`failed_task` is null-terminated (or all zeros if no failure)

Since

Version 1.0.0

Definition at line 1200 of file `rx_iwdt.c`.

Collaboration diagram for `task_monitor_state_t`:



Data Fields

|                             |                                                                 |                                  |
|-----------------------------|-----------------------------------------------------------------|----------------------------------|
| <code>char</code>           | <code>failed_task↔</code><br><code>[k_task_name_max_len]</code> | Name of most recent failed task  |
| <code>uint8_t</code>        | <code>task_count</code>                                         | Number of registered tasks (0-8) |
| <code>task_monitor_t</code> | <code>tasks[k_iwdt_max_tasks]</code>                            | Array of task monitoring slots   |



### 5.10.3 Enumeration Type Documentation

#### 5.10.3.1 task`registration`state`t

enum [task\\_registration\\_state\\_t](#) : uint8\_t

Task registration state tracking.

Indicates whether a [task\\_monitor\\_t](#) slot is in use. Allows sparse array usage and efficient task lookup.

Since

Version 1.0.0

Enumerator

|                       |                                          |
|-----------------------|------------------------------------------|
| k_task_not_registered | Slot available for new task registration |
| k_task_registered     | Slot in use by registered task           |

Definition at line [1119](#) of file [rx\\_iwtdt.c](#).

```
01119 : uint8_t {
01120 k_task_not_registered = 0, /**< Slot available for new task registration */
01121 k_task_registered = 1, /**< Slot in use by registered task */
01122 } task_registration_state_t;
```

#### 5.10.3.2 task`search`constants`t

enum [task\\_search\\_constants\\_t](#) : int32\_t

Constants for task search operations.

Defines sentinel values and bounds for task array operations. Using signed int32\_t allows -1 sentinel for "not found" result.

Since

Version 1.0.0

Enumerator

|                  |                                          |
|------------------|------------------------------------------|
| k_task_not_found | Return value when task name not in array |
| k_task_idx_min   | Minimum valid task index (array start)   |

Definition at line [1134](#) of file [rx\\_iwtdt.c](#).

```
01134 : int32_t {
01135 k_task_not_found = -1, /**< Return value when task name not in array */
01136 k_task_idx_min = 0, /**< Minimum valid task index (array start) */
01137 } task_search_constants_t;
```

## 5.10.4 Function Documentation

### 5.10.4.1 internal\_find\_task()

```
int32_t internal_find_task (
 const char * task_name) [static]
```

Find task index by name in the registered tasks array.

Performs linear search through `s_task_monitor.tasks[]` to find a task matching the given name. Uses `strncmp` for safe string comparison.

#### Algorithm

- \* 1. Validate task\_name is not nullptr
- \* 2. Validate task\_name is not empty string
- \* 3. Linear scan through registered tasks (0 to task\_count-1):
- \*   - Compare names using `strncmp(task_name, k_task_name_cmp_len)`
- \*   - If match found: return index
- \* 4. If no match: return `k_task_not_found (-1)`
- \*

#### Complexity

- Time:  $O(n \times m)$  where  $n = \text{task\_count}$ ,  $m = \text{k\_task\_name\_cmp\_len}$
- Space:  $O(1)$  - no additional memory

#### Precondition

```
task_name != nullptr
task_name[0] != '\0'
```

#### Postcondition

```
Return value < s_task_monitor.task_count (if found)
Return value == k_task_not_found (if not found or invalid input)
```

#### Note

Uses `k_task_name_cmp_len (15)` for comparison, not full buffer  
 Empty string returns `k_task_not_found` (invalid task name)

#### See also

[rx\\_hal\\_iwdt\\_task\\_heartbeat\(\)](#) Primary caller  
[rx\\_hal\\_iwdt\\_register\\_task\(\)](#) Uses for duplicate detection

Since

Version 1.0.0

Definition at line 1269 of file `rx_iwdt.c`.

```

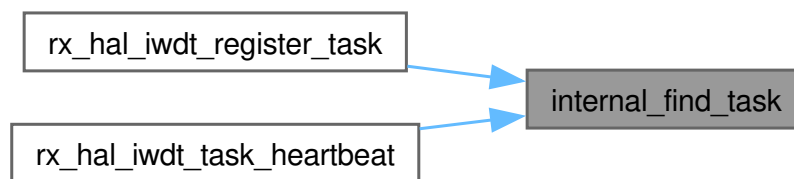
01270 {
01271 if (task_name == nullptr) {
01272 return k_task_not_found;
01273 }
01274
01275 if (task_name[0] == '\0') {
01276 return k_task_not_found; /* Empty task name invalid */
01277 }
01278
01279 for (uint32_t i = (uint32_t)k_task_idx_min; i < s_task_monitor.task_count; i++) {
01280 if (strcmp(s_task_monitor.tasks[i].task_name, task_name, k_task_name_cmp_len) == 0) {
01281 return (int32_t)i;
01282 }
01283 }
01284
01285 return k_task_not_found;
01286 }

```

References `k_task_idx_min`, `k_task_name_cmp_len`, `k_task_not_found`, and `s_task_monitor`.

Referenced by `rx_hal_iwdt_register_task()`, and `rx_hal_iwdt_task_heartbeat()`.

Here is the caller graph for this function:



#### 5.10.4.2 `rx_hal_iwdt_check_tasks()`

```

rx_err_t rx_hal_iwdt_check_tasks (
 void) [nodiscard]

```

Check all monitored tasks for timeout violations.

Check for task deadlocks.

Iterates through all registered tasks and compares elapsed time since last heartbeat against each task's configured timeout. If any task has exceeded its timeout, logs the failure and records the task name.

#### Check Algorithm

- \* 1. Validate IWDT is initialized
- \* 2. Validate task\_count is within bounds (detect corruption)
- \* 3. Get current time from ThreadX
- \* 4. For each registered task:
  - \* a. Calculate elapsed = current\_time - last\_heartbeat\_ms
  - \* b. If elapsed > timeout\_ms:
    - \* - Log error with task name and timeout
    - \* - Record failed task name
    - \* - Set result = k\_rx\_err\_timeout
- \* 5. Return result (k\_rx\_ok if all pass, k\_rx\_err\_timeout if any fail)
- \*

## Typical Usage Pattern

```
* Monitoring Task Loop:
* +-----+
* | 1. rx_hal_iwdt_check_tasks() |
* | 2. If all tasks OK: rx_hal_iwdt_feed() |
* | 3. If timeout: DON'T feed (let WDT reset) |
* | 4. tx_thread_sleep(check_interval) |
* | 5. Repeat |
* +-----+
*
```

## Precondition

IWDT must be initialized via [rx\\_hal\\_iwdt\\_init\(\)](#)

Called periodically from monitoring task

## Postcondition

If timeout detected, failed task name recorded in `s_task_monitor.failed_task`

Error logged for each timed-out task

Continues checking all tasks even if one fails (logs all failures)

## Note

Does NOT automatically feed hardware watchdog

Checks all tasks even after finding failure (comprehensive logging)

Uses `tx_time_get()` which returns `uint32_t` tick count

## Warning

Do NOT feed watchdog if this function returns `k_rx_err_timeout`

## Example:

```
static void watchdog_monitor_task(ULONG input) {
 while (1) {
 rx_err_t result = rx_hal_iwdt_check_tasks();

 if (result == k_rx_ok) {
 // All tasks healthy - feed hardware watchdog
 rx_hal_iwdt_feed();
 } else if (result == k_rx_err_timeout) {
 // Task deadlock detected!
 // DON'T feed - let hardware watchdog reset system
 const char* failed = rx_hal_iwdt_get_failed_task();
 uart_debug_printf("DEADLOCK: %s\\r\\n", failed);
 }

 tx_thread_sleep(100); // Check every 100ms
 }
}
```

## See also

[rx\\_hal\\_iwdt\\_register\\_task\(\)](#) Register tasks before checking

[rx\\_hal\\_iwdt\\_task\\_heartbeat\(\)](#) Tasks report liveness

[rx\\_hal\\_iwdt\\_get\\_failed\\_task\(\)](#) Get name of failed task

[rx\\_hal\\_iwdt\\_feed\(\)](#) Feed hardware watchdog if all tasks OK

Since

Version 1.0.0

Definition at line 1541 of file `rx_iwdt.c`.

```

01542 {
01543 uint32_t current_time_ms;
01544 rx_err_t result;
01545
01546 const task_monitor_t* task = nullptr;
01547 uint32_t elapsed_ms;
01548 char log_msg[k_log_msg_buffer_size];
01549 int32_t snprintf_result;
01550
01551 if (s_iwdt_initialized != k_iwdt_initialized) {
01552 return k_rx_err_invalid_state;
01553 }
01554
01555 if (s_task_monitor.task_count > k_iwdt_max_tasks) {
01556 return k_rx_err_invalid_state; /* Corrupted state */
01557 }
01558
01559 current_time_ms = tx_time_get();
01560 result = k_rx_ok;
01561
01562 for (uint32_t i = (uint32_t)k_task_idx_min; i < s_task_monitor.task_count; i++) {
01563 task = &s_task_monitor.tasks[i];
01564 elapsed_ms = current_time_ms - task->last_heartbeat_ms;
01565
01566 if (elapsed_ms > task->timeout_ms) {
01567 /* Task has exceeded heartbeat timeout - deadlock detected */
01568 snprintf_result = snprintf(log_msg,
01569 sizeof(log_msg),
01570 "Task deadlock: %s (timeout %" PRIu32 " ms)",
01571 task->task_name,
01572 task->timeout_ms);
01573
01574 if (snprintf_result > 0 && (uint32_t)snprintf_result < sizeof(log_msg)) {
01575 rx_log_error(s_tag, log_msg);
01576 }
01577
01578 /* Record failed task name for post-mortem */
01579 strncpy(s_task_monitor.failed_task, task->task_name, k_task_name_cmp_len);
01580 s_task_monitor.failed_task[k_task_name_cmp_len] = '\0';
01581
01582 result = k_rx_err_timeout;
01583 /* Continue checking other tasks to log all failures */
01584 }
01585 }
01586
01587 return result;
01588 }

```

References `k_iwdt_initialized`, `k_iwdt_max_tasks`, `k_log_msg_buffer_size`, `k_task_idx_min`, `k_task_name_cmp_len`, `task_monitor_t::last_heartbeat_ms`, `s_iwdt_initialized`, `s_tag`, `s_task_monitor`, `task_monitor_t::task_name`, and `task_monitor_t::timeout_ms`.

#### 5.10.4.3 `rx_hal_iwdt_get_failed_task()`

```

const char * rx_hal_iwdt_get_failed_task (
 void)

```

Get name of last failed task for post-mortem analysis.

Get the name of the last failed task (if any).

Returns the name of the task that most recently exceeded its heartbeat timeout. This information is recorded by `rx_hal_iwdt_check_tasks()` when it detects a timeout. Useful for diagnosing which task caused a watchdog reset.

## Use Cases

- \* 1. Post-reset diagnostics:
- \* - After watchdog reset, call `rx_hal_iwdt_get_failed_task()`
- \* - Log failed task name for debugging
- \* - Helps identify deadlock/hang location
- \* 2. Pre-reset logging:
- \* - When `rx_hal_iwdt_check_tasks()` returns `k_rx_err_timeout`
- \* - Get failed task name and log before reset occurs
- \*

## Precondition

IWDT must be initialized via `rx_hal_iwdt_init()`  
`rx_hal_iwdt_check_tasks()` has been called and detected a timeout

## Postcondition

Returns pointer to `s_task_monitor.failed_task` (do not modify)  
 Returned pointer valid until next timeout detection

## Note

Returns pointer to internal static buffer - do not free  
 Buffer contents may change on next `rx_hal_iwdt_check_tasks()` failure  
 Returns NULL if `failed_task[0] == '\0'` (no failure recorded)

## Warning

Do not modify returned buffer (const pointer)  
 Buffer may be overwritten by next timeout detection

## Example:

```
// After detecting timeout
rx_err_t result = rx_hal_iwdt_check_tasks();
if (result == k_rx_err_timeout) {
 const char* failed = rx_hal_iwdt_get_failed_task();
 if (failed != nullptr) {
 uart_debug_printf("FATAL: Task '%s' deadlocked\r\n", failed);
 }
 // Let watchdog reset the system...
}

// After system restart
if (rx_hal_iwdt_was_reset()) {
 const char* failed = rx_hal_iwdt_get_failed_task();
 // Note: May be NULL if buffer not preserved across reset
}
```

## See also

`rx_hal_iwdt_check_tasks()` Records failed task name  
`rx_hal_iwdt_was_reset()` Check if watchdog caused reset

Since

Version 1.0.0

Definition at line 1648 of file `rx_iwdt.c`.

```
01649 {
01650 if (s_iwdt_initialized != k_iwdt_initialized) {
01651 return nullptr;
01652 }
01653
01654 if (s_task_monitor.failed_task[0] != '\0') {
01655 return s_task_monitor.failed_task;
01656 }
01657
01658 return nullptr;
01659 }
```

References `k_iwdt_initialized`, `s_iwdt_initialized`, and `s_task_monitor`.

#### 5.10.4.4 `rx_hal_iwdt_register_task()`

```
rx_err_t rx_hal_iwdt_register_task (
 const char * task_name,
 uint32_t timeout_ms) [nodiscard]
```

Register a task for watchdog monitoring.

Register a task for heartbeat monitoring.

Adds a task to the monitoring array with specified timeout period. The task must subsequently call `rx_hal_iwdt_task_heartbeat()` within each timeout period to indicate liveness. Failed tasks are detected by `rx_hal_iwdt_check_tasks()`.

#### Registration Process

- \* 1. Validate task\_name (not nullptr, not empty)
- \* 2. Validate timeout\_ms (within allowed range)
- \* 3. Check capacity (task\_count < k\_iwdt\_max\_tasks)
- \* 4. Check for duplicate registration (same name)
- \* 5. Copy task name to slot (truncate if > 15 chars)
- \* 6. Initialize timeout and heartbeat timestamp
- \* 7. Mark slot as registered
- \* 8. Increment task\_count
- \*

#### Precondition

IWDT module initialized (but not strictly required for registration)

task\_name points to valid null-terminated string

timeout\_ms appropriate for task's execution frequency

#### Postcondition

Task added to `s_task_monitor.tasks[]`

`s_task_monitor.task_count` incremented

Initial heartbeat timestamp set to current time

## Invariant

`s_task_monitor.task_count`  $\leq$  `k_iwdt_max_tasks` at all times

## Note

Task name truncated to 15 characters if longer

First heartbeat auto-recorded at registration time

## Warning

Not thread-safe - call during initialization only

## Example:

```
// Register motor control task (must heartbeat every 500ms)
rx_err_t err = rx_hal_iwdt_register_task("MotorCtrl", 500);
if (err != k_rx_ok) {
 // Handle registration failure
 return err;
}

// In motor task loop
while (1) {
 motor_control_update();
 rx_hal_iwdt_task_heartbeat("MotorCtrl");
 tx_thread_sleep(10); // 100Hz loop
}
```

## See also

[rx\\_hal\\_iwdt\\_task\\_heartbeat\(\)](#) Send heartbeat from registered task

[rx\\_hal\\_iwdt\\_check\\_tasks\(\)](#) Check all tasks for timeout

## Since

Version 1.0.0

Definition at line 1347 of file [rx\\_iwdt.c](#).

```
01348 {
01349 /* Validate parameters */
01350 if (task_name == nullptr) {
01351 return k_rx_err_invalid_arg;
01352 }
01353
01354 if (task_name[0] == '\0') {
01355 return k_rx_err_invalid_arg;
01356 }
01357
01358 if (timeout_ms < k_iwdt_timeout_min_ms || timeout_ms > k_iwdt_timeout_max_ms) {
01359 return k_rx_err_invalid_arg;
01360 }
01361
01362 if (s_task_monitor.task_count >= k_iwdt_max_tasks) {
01363 return k_rx_err_no_mem;
01364 }
01365
01366 /* Check for duplicate registration */
01367 if (internal_find_task(task_name) != k_task_not_found) {
01368 return k_rx_err_exists;
01369 }
01370
01371 /* Register new task */
01372 task_monitor_t* task = &s_task_monitor.tasks[s_task_monitor.task_count];
01373
01374 strncpy(task->task_name, task_name, k_task_name_cmp_len);
```



```

01375 task->task_name[k_task_name_cmp_len] = '\0'; /* Ensure null termination */
01376
01377 task->timeout_ms = timeout_ms;
01378 task->last_heartbeat_ms = tx_time_get();
01379 task->registered = k_task_registered;
01380
01381 s_task_monitor.task_count++;
01382
01383 rx_log_info(s_tag, "Registered task for monitoring");
01384
01385 return k_rx_ok;
01386 }

```

References [internal\\_find\\_task\(\)](#), [k\\_iwdt\\_max\\_tasks](#), [k\\_iwdt\\_timeout\\_max\\_ms](#), [k\\_task\\_name\\_cmp\\_len](#), [k\\_task\\_not\\_found](#), [k\\_task\\_registered](#), [task\\_monitor\\_t::last\\_heartbeat\\_ms](#), [task\\_monitor\\_t::registered](#), [s\\_tag](#), [s\\_task\\_monitor](#), [task\\_monitor\\_t::task\\_name](#), and [task\\_monitor\\_t::timeout\\_ms](#).

Here is the call graph for this function:



#### 5.10.4.5 rx\_hal\_iwdt\_task\_heartbeat()

```

void rx_hal_iwdt_task_heartbeat (
 const char * task_name)

```

Record a task heartbeat to indicate liveness.

Record a task heartbeat.

Updates the `last_heartbeat_ms` timestamp for the specified task. Must be called by each monitored task at a rate faster than its configured timeout to avoid being flagged as failed by [rx\\_hal\\_iwdt\\_check\\_tasks\(\)](#).

#### Heartbeat Algorithm

- \* 1. Validate `task_name` (not nullptr)
- \* 2. Find task in registered array ([internal\\_find\\_task](#))
- \* 3. If found: Update `last_heartbeat_ms = tx_time_get()`
- \* 4. If not found: Log error (graceful degradation)
- \*

#### Timing Requirements

- \* - Must be called faster than task's `timeout_ms`
- \* - Typical pattern: Call at end of task's main loop iteration
- \* - Example: 500ms timeout -> call at least every 400ms
- \*

#### Precondition

`task_name` must be non-NULL

Task must be registered via [rx\\_hal\\_iwdt\\_register\\_task\(\)](#)

Called faster than task's configured timeout

## Postcondition

Task's last\_heartbeat\_ms updated to current time  
 No state change if task not found (logs error only)

## Note

Void return - errors logged but not returned  
 Unregistered task heartbeats are logged and ignored  
 Uses tx\_time\_get() for timestamp (ThreadX tick count)

## Warning

Missing heartbeats cause [rx\\_hal\\_iwdt\\_check\\_tasks\(\)](#) to detect timeout

## Example:

```
static void communication_task(ULONG input) {
 while (1) {
 // Process incoming messages
 process_messages();

 // Indicate task is alive
 rx_hal_iwdt_task_heartbeat("CommTask");

 // Wait for next iteration
 tx_thread_sleep(50); // 50ms loop
 }
}
```

## See also

[rx\\_hal\\_iwdt\\_register\\_task\(\)](#) Register task before heartbeating  
[rx\\_hal\\_iwdt\\_check\\_tasks\(\)](#) Verifies heartbeats are received

## Since

Version 1.0.0

Definition at line 1446 of file [rx\\_iwdt.c](#).

```
01447 {
01448 if (task_name == nullptr) {
01449 rx_log_error(s_tag, "Heartbeat called with nullptr task_name");
01450 return;
01451 }
01452 int32_t idx = internal_find_task(task_name);
01453
01454 if (idx == k_task_not_found) {
01455 /* Task not registered - log warning but don't fail */
01456 rx_log_error(s_tag, "Heartbeat from unregistered task");
01457 return;
01458 }
01459
01460 /* Update heartbeat timestamp */
01461 s_task_monitor.tasks[idx].last_heartbeat_ms = tx_time_get();
01462 }
01463 }
```

References [internal\\_find\\_task\(\)](#), [k\\_task\\_not\\_found](#), [s\\_tag](#), and [s\\_task\\_monitor](#).

Here is the call graph for this function:



### 5.10.5 Variable Documentation

#### 5.10.5.1 s'task'monitor

```
task_monitor_state_t s_task_monitor = {} [static]
```

Module state for task monitoring (static allocation).

Single instance of task monitoring state. Zero-initialized at startup, ensuring all tasks start unregistered and no failed task recorded.

##### Initial State

```
* - task_count = 0
* - All tasks[].registered = k_task_not_registered
* - failed_task[0] = '\0'
*
```

##### Thread Safety

- Registration: Call from single thread during initialization
- Heartbeats: Not thread-safe, caller must synchronize
- Check: Not thread-safe, typically called from single monitoring task

##### Warning

Do not access directly - use API functions

##### Since

Version 1.0.0

Definition at line 1230 of file rx\_iwdt.c.

```
01230 {};
```

Referenced by [internal\\_find\\_task\(\)](#), [rx\\_hal\\_iwdt\\_check\\_tasks\(\)](#), [rx\\_hal\\_iwdt\\_get\\_failed\\_task\(\)](#), [rx\\_hal\\_iwdt\\_register\\_task\(\)](#), and [rx\\_hal\\_iwdt\\_task\\_heartbeat\(\)](#).

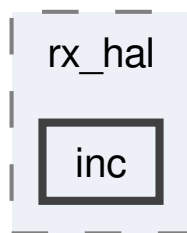


# Chapter 6

## Directory Documentation

### 6.1 libs/rx\_hal/inc Directory Reference

Directory dependency graph for inc:



#### Files

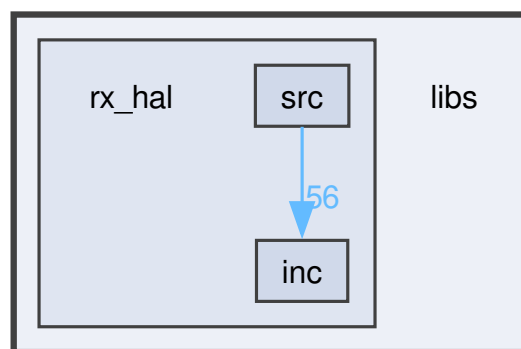
- file [hardware.h](#)  
Hardware Abstraction Layer (HAL) - Unified Interface for RX72N Peripherals.
- file [rx72n\\_adc\\_regs.h](#)  
RX72N S12AD 12-bit A/D Converter Register Definitions.
- file [rx72n\\_cac\\_regs.h](#)  
RX72N CAC (Clock Frequency Accuracy Measurement Circuit) Register Definitions.
- file [rx72n\\_clock.h](#)  
RX72N Clock Frequency Definitions and Configuration.
- file [rx72n\\_cmt\\_regs.h](#)  
RX72N Compare Match Timer (CMT) Register Definitions.
- file [rx72n\\_crc\\_regs.h](#)  
RX72N CRC Calculator Register Definitions.
- file [rx72n\\_dmac\\_regs.h](#)  
RX72N DMA Controller (DMAC) Register Definitions.
- file [rx72n\\_doc\\_regs.h](#)  
RX72N DOC (Data Operation Circuit) Register Definitions.
- file [rx72n\\_dtc\\_regs.h](#)  
RX72N Data Transfer Controller (DTCb) Register Definitions.

- file [rx72n\\_eccram\\_regs.h](#)  
RX72N ECCRAM (ECC-protected RAM) Register Definitions.
- file [rx72n\\_elc\\_regs.h](#)  
RX72N Event Link Controller (ELC) Register Definitions.
- file [rx72n\\_flash\\_regs.h](#)  
RX72N Flash Memory Register Definitions.
- file [rx72n\\_gptw\\_regs.h](#)  
RX72N GPTW (General PWM Timer) Register Definitions.
- file [rx72n\\_icu\\_regs.h](#)  
RX72N ICU Interrupt Controller Unit Register Definitions.
- file [rx72n\\_iwdt\\_regs.h](#)  
RX72N IWDT Independent Watchdog Timer Register Definitions.
- file [rx72n\\_lpc\\_regs.h](#)  
RX72N Low Power Consumption Register Definitions.
- file [rx72n\\_lvda\\_regs.h](#)  
RX72N Voltage Detection Circuit (LVDA) Register Definitions.
- file [rx72n\\_mpc\\_regs.h](#)  
RX72N Multi-Function Pin Controller (MPC) Register Definitions.
- file [rx72n\\_mpu\\_regs.h](#)  
RX72N Memory Protection Unit (MPU) Register Definitions.
- file [rx72n\\_mtu\\_regs.h](#)  
RX72N Multi-Function Timer Unit (MTU3a) register definitions.
- file [rx72n\\_poe3\\_regs.h](#)  
RX72N Port Output Enable 3 (POE3) Register Definitions.
- file [rx72n\\_poeg\\_regs.h](#)  
RX72N GPTW Port Output Enable (POEG) Register Definitions.
- file [rx72n\\_port\\_regs.h](#)  
RX72N PORT (GPIO) register definitions for digital pin control.
- file [rx72n\\_ram\\_regs.h](#)  
RX72N RAM and Expansion RAM register definitions.
- file [rx72n\\_regs.h](#)  
RX72N Hardware Register Definitions - Main Aggregator Header.
- file [rx72n\\_riic\\_regs.h](#)  
RX72N I2C Bus Interface (RIIC) Register Definitions.
- file [rx72n\\_rspi\\_regs.h](#)  
RX72N RSPI Serial Peripheral Interface Register Definitions.
- file [rx72n\\_rtc\\_regs.h](#)  
RX72N Real-Time Clock (RTC) Register Definitions.
- file [rx72n\\_sci\\_regs.h](#)  
RX72N Serial Communication Interface (SCI) register definitions.
- file [rx72n\\_system\\_regs.h](#)  
RX72N System Control Register Definitions.
- file [rx72n\\_temps\\_regs.h](#)  
RX72N Temperature Sensor (TEMPS) Register Definitions.
- file [rx72n\\_tmr\\_regs.h](#)  
RX72N 8-Bit Timer (TMRb) Register Definitions.
- file [rx72n\\_tpu\\_regs.h](#)  
RX72N 16-Bit Timer Pulse Unit (TPU) Register Definitions.
- file [rx72n\\_usb\\_regs.h](#)  
RX72N USB 2.0 Full-Speed Host/Function Module Register Definitions.
- file [rx72n\\_wdt\\_regs.h](#)

- RX72N WDT Watchdog Timer Register Definitions.
- file [rx\\_cac.h](#)  
Clock Frequency Accuracy Measurement Circuit (CAC) Driver API for RX72N.
- file [rx\\_cmt.h](#)  
CMT (Compare Match Timer) Driver API for RX72N - Periodic Interrupt Generation.
- file [rx\\_doc.h](#)  
RX72N DOC (Data Operation Circuit) HAL Public API.
- file [rx\\_eccram.h](#)  
ECC-Protected RAM (ECCRAM) Driver API.
- file [rx\\_gptw.h](#)  
GPTW PWM Driver for RX72N Motor Control.
- file [rx\\_hal\\_iwdt.h](#)  
Independent Watchdog Timer (IWDt) HAL Driver for RX72N.
- file [rx\\_host\\_irq.h](#)  
HOST\_IRQ GPIO Output Driver for RPi5 Data-Ready Signaling.
- file [rx\\_irq\\_filter.h](#)  
IRQ Digital Filter Driver API for RX72N.
- file [rx\\_lpc.h](#)  
Low Power Consumption (LPC) HAL Driver for RX72N.
- file [rx\\_mpc.h](#)  
Multi-Function Pin Controller (MPC) Driver API for RX72N.
- file [rx\\_mtu.h](#)  
MTU PWM Driver for RX72N Multi-Function Timer Unit.
- file [rx\\_poeg.h](#)  
POEG Motor Fault Protection Driver API.
- file [rx\\_port\\_utils.h](#)  
PORT utility functions for RX72N GPIO register access.
- file [rx\\_tmr.h](#)  
TMR HAL Driver for RX72N 8-Bit Timer (TMRb).
- file [rx\\_tpu.h](#)  
TPU HAL Driver for RX72N 16-Bit Timer Pulse Unit (Phase Counting).

## 6.2 libs Directory Reference

Directory dependency graph for libs:

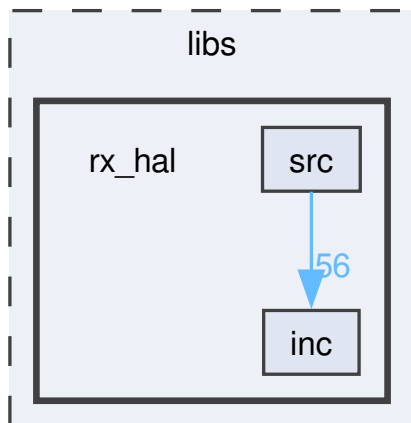


Directories

- directory [rx\\_hal](#)

## 6.3 libs/rx\_hal Directory Reference

Directory dependency graph for rx\_hal:

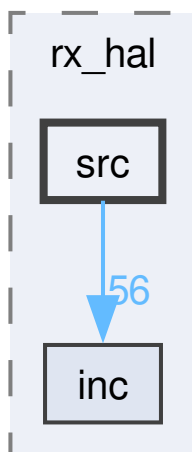


Directories

- directory [inc](#)
- directory [src](#)

## 6.4 libs/rx\_hal/src Directory Reference

Directory dependency graph for src:





## Files

- file [adc.c](#)  
ADC Driver for RX72N S12ADFa - 12-bit Analog-to-Digital Conversion.
- file [gpio.c](#)  
GPIO Driver for RX72N - General Purpose Input/Output Control.
- file [riic.c](#)  
RIIC (I2C) Driver Implementation for RX72N.
- file [rspi.c](#)  
RSPI (Renesas Serial Peripheral Interface) Driver for RX72N.
- file [rx\\_cac.c](#)  
Clock Frequency Accuracy Measurement Circuit (CAC) Driver Implementation.
- file [rx\\_cmt.c](#)  
CMT (Compare Match Timer) Driver Implementation - Periodic Interrupt Generation.
- file [rx\\_doc.c](#)  
RX72N DOC (Data Operation Circuit) Driver Implementation.
- file [rx\\_eccram.c](#)  
ECC-Protected RAM (ECCRAM) Driver Implementation.
- file [rx\\_gptw.c](#)  
GPTW PWM Driver Implementation for Motor Control.
- file [rx\\_host\\_irq.c](#)  
HOST\_IRQ GPIO Output Driver Implementation.
- file [rx\\_irq\\_filter.c](#)  
IRQ Digital Filter Driver Implementation for RX72N.
- file [rx\\_iwdt.c](#)  
Independent Watchdog Timer (IWDT) Driver Implementation for RX72N.
- file [rx\\_lpc.c](#)  
Low Power Consumption (LPC) HAL Driver Implementation for RX72N.
- file [rx\\_mpc.c](#)  
Multi-Function Pin Controller (MPC) Driver Implementation.
- file [rx\\_mtu.c](#)  
MTU PWM Driver Implementation for RX72N.
- file [rx\\_poeg.c](#)  
POEG Motor Fault Protection Driver Implementation.
- file [rx\\_tmr.c](#)  
TMR HAL Driver Implementation (8-bit + 16-bit cascade).
- file [rx\\_tpu.c](#)  
TPU HAL Driver Implementation for Phase Counting Mode.
- file [timer.c](#)  
ThreadX system tick timer implementation using RX72N CMT0.
- file [uart.c](#)  
Multi-Channel UART Driver for RX72N SCI Peripherals.



## Chapter 7

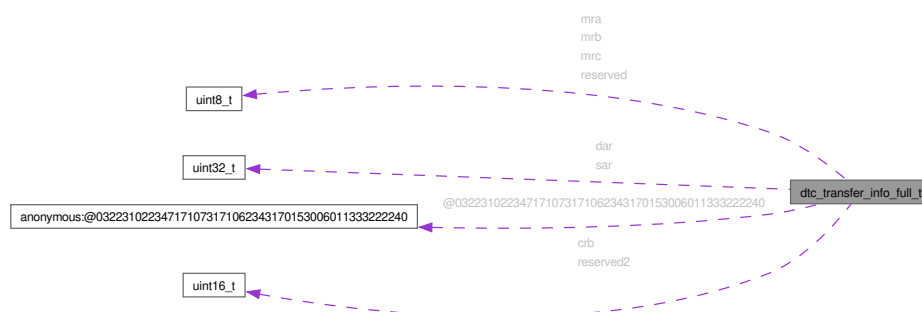
# Data Structure Documentation

### 7.1 dtc\_transfer\_info\_full\_t Struct Reference

DTC transfer information structure for full-address mode.

```
#include <rx72n_dtc_regs.h>
```

Collaboration diagram for `dtc_transfer_info_full_t`:



#### Data Fields

- `uint8_t` `mra`
- `uint8_t` `mrb`
- `uint8_t` `mrc`
- `uint8_t` `reserved`
- `uint32_t` `sar`
- `uint32_t` `dar`
- `union` {
  - `struct` {
    - `uint16_t` `cra_low`
    - `uint16_t` `cra_high`
  - `uint32_t` `cra`
- `uint16_t` `crb`
- `uint16_t` `reserved2`

### 7.1.1 Detailed Description

DTC transfer information structure for full-address mode.

This structure is placed in RAM and read by the DTC hardware. The address is calculated from DTCVBR + (vector\_number x 4).

Warning

Must be aligned to 4-byte boundary

Note

For short-address mode, use [dte\\_transfer\\_info\\_short\\_t](#)

Since

Version 1.0.0

Definition at line [443](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.1.2 Field Documentation

#### 7.1.2.1 [union]

union { ... } [dte\\_transfer\\_info\\_full\\_t](#)

#### 7.1.2.2 crb

uint16\_t dte\_transfer\_info\_full\_t::crb

Transfer Count Register B (block count)

Definition at line [457](#) of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.3 dar

uint32\_t dte\_transfer\_info\_full\_t::dar

Destination Address Register

Definition at line [449](#) of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.4 mra

uint8\_t dte\_transfer\_info\_full\_t::mra

Mode Register A (transfer mode, size)

Definition at line [444](#) of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.5 mrb

uint8\_t dtc\_transfer\_info\_full\_t::mrb

Mode Register B (chain, interrupt)

Definition at line 445 of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.6 mrc

uint8\_t dtc\_transfer\_info\_full\_t::mrc

Mode Register C (sequence control)

Definition at line 446 of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.7 reserved

uint8\_t dtc\_transfer\_info\_full\_t::reserved

Reserved (set to 0)

Definition at line 447 of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.8 reserved2

uint16\_t dtc\_transfer\_info\_full\_t::reserved2

Reserved

Definition at line 458 of file [rx72n\\_dtc\\_regs.h](#).

#### 7.1.2.9 sar

uint32\_t dtc\_transfer\_info\_full\_t::sar

Source Address Register

Definition at line 448 of file [rx72n\\_dtc\\_regs.h](#).

The documentation for this struct was generated from the following file:

- [libs/rx\\_hal/inc/rx72n\\_dtc\\_regs.h](#)

## 7.2 dtc\_transfer\_info\_full\_t ``unnamed0`` Union Reference

### Data Fields

- struct {  
  uint16\_t [cra\\_low](#)  
  uint16\_t [cra\\_high](#)  
}
- uint32\_t [cra](#)

### 7.2.1 Detailed Description

Definition at line [450](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.2.2 Field Documentation

7.2.2.1 "006051261071317356052225056071032263073360152353

7.2.2.2 [cra](#)

Transfer Count Register A (full)

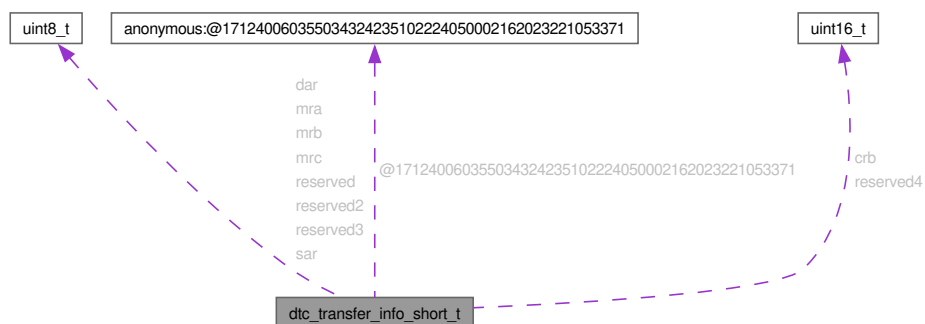
The documentation for this union was generated from the following files:

## 7.3 dtc\_transfer\_info\_short\_t Struct Reference

DTC transfer information structure for short-address mode.

```
#include <rx72n_dtc_regs.h>
```

Collaboration diagram for `dtc_transfer_info_short_t`:



## Data Fields

- `uint8_t mra`
- `uint8_t mrb`
- `uint8_t mrc`
- `uint8_t reserved`
- `uint8_t sar [3]`
- `uint8_t reserved2`
- `uint8_t dar [3]`
- `uint8_t reserved3`
- `union {`  
     `struct {`  
         `uint16_t cra_low`  
         `uint16_t cra_high`  
     }  
     `uint32_t cra`  
 };
- `uint16_t crb`
- `uint16_t reserved4`

## 7.3.1 Detailed Description

DTC transfer information structure for short-address mode.

Compact format for 16 MB address space transfers. Uses 3-byte addresses to save memory.

## Warning

Must be aligned to 4-byte boundary

## Note

For full-address mode, use `dtc_transfer_info_full_t`

## Since

Version 1.0.0

Definition at line 473 of file `rx72n_dtc_regs.h`.

## 7.3.2 Field Documentation

## 7.3.2.1 [union]

`union { ... } dtc_transfer_info_short_t`

## 7.3.2.2 crb

`uint16_t dtc_transfer_info_short_t::crb`

Definition at line 489 of file `rx72n_dtc_regs.h`.

### 7.3.2.3 dar

`uint8_t dtc_transfer_info_short_t::dar[3]`

Destination Address (24-bit)

Definition at line [480](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.3.2.4 mra

`uint8_t dtc_transfer_info_short_t::mra`

Mode Register A

Definition at line [474](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.3.2.5 mrb

`uint8_t dtc_transfer_info_short_t::mrb`

Mode Register B

Definition at line [475](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.3.2.6 mrc

`uint8_t dtc_transfer_info_short_t::mrc`

Mode Register C

Definition at line [476](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.3.2.7 reserved

`uint8_t dtc_transfer_info_short_t::reserved`

Reserved

Definition at line [477](#) of file [rx72n\\_dtc\\_regs.h](#).

### 7.3.2.8 reserved2

`uint8_t dtc_transfer_info_short_t::reserved2`

Reserved

Definition at line [479](#) of file [rx72n\\_dtc\\_regs.h](#).



## 7.3.2.9 reserved3

```
uint8_t dtc_transfer_info_short_t::reserved3
```

Reserved

Definition at line 481 of file [rx72n\\_dtc\\_regs.h](#).

## 7.3.2.10 reserved4

```
uint16_t dtc_transfer_info_short_t::reserved4
```

Definition at line 490 of file [rx72n\\_dtc\\_regs.h](#).

## 7.3.2.11 sar

```
uint8_t dtc_transfer_info_short_t::sar[3]
```

Source Address (24-bit)

Definition at line 478 of file [rx72n\\_dtc\\_regs.h](#).

The documentation for this struct was generated from the following file:

- [libs/rx\\_hal/inc/rx72n\\_dtc\\_regs.h](#)

## 7.4 dtc\_transfer\_info\_short\_t ``unnamed0`` Union Reference

Data Fields

- struct {  
  uint16\_t [cra\\_low](#)  
  uint16\_t [cra\\_high](#)  
}
- uint32\_t [cra](#)

## 7.4.1 Detailed Description

Definition at line 482 of file [rx72n\\_dtc\\_regs.h](#).

## 7.4.2 Field Documentation

## 7.4.2.1 ”@077227312060121117242220021160226234243054272306

## 7.4.2.2 cra

The documentation for this union was generated from the following files:



## Chapter 8

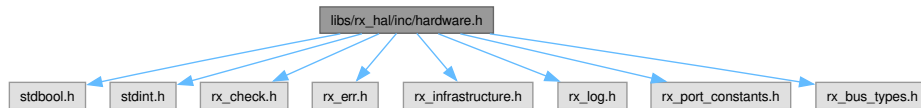
# File Documentation

### 8.1 libs/rx\_hal/inc/hardware.h File Reference

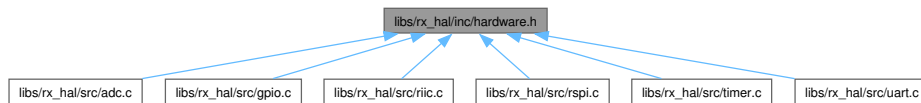
Hardware Abstraction Layer (HAL) - Unified Interface for RX72N Peripherals.

```
#include <stdbool.h>
#include <stdint.h>
#include "rx_check.h"
#include "rx_err.h"
#include "rx_infrastructure.h"
#include "rx_log.h"
#include "rx_port_constants.h"
#include "rx_bus_types.h"
```

Include dependency graph for hardware.h:



This graph shows which files directly or indirectly include this file:



#### Data Structures

- struct [riic\\_channel\\_t](#)  
RIIC (I2C) channel wrapper providing strong type safety.
- struct [i2c\\_device\\_addr\\_t](#)  
I2C device address wrapper providing strong type safety.
- struct [rspi\\_config\\_t](#)  
RSPI peripheral mode configuration structure.
- struct [rspi\\_controller\\_config\\_t](#)  
RSPI controller mode configuration structure.
- struct [uart\\_channel\\_config\\_t](#)  
UART channel configuration structure.

## Typedefs

- typedef rx\_adc\_resolution\_t [adc\\_resolution\\_t](#)

## Enumerations

- enum [adc\\_unit\\_t](#) : uint8\_t { [k\\_adc\\_unit\\_0](#) , [k\\_adc\\_unit\\_1](#) }  
ADC unit enumeration with type safety for 12-bit SAR ADC modules.
- enum [adc\\_channel\\_t](#) : uint8\_t {  
[k\\_adc\\_channel\\_0](#) = 0 , [k\\_adc\\_channel\\_1](#) = 1 , [k\\_adc\\_channel\\_2](#) = 2 , [k\\_adc\\_channel\\_3](#) = 3 ,  
[k\\_adc\\_channel\\_4](#) = 4 , [k\\_adc\\_channel\\_5](#) = 5 , [k\\_adc\\_channel\\_6](#) = 6 , [k\\_adc\\_channel\\_7](#) = 7  
}  
ADC channel enumeration with type safety for analog input selection.
- enum [riic\\_peripheral\\_limits\\_t](#) : uint16\_t { [k\\_riic\\_peripheral\\_transfer\\_limit](#) = 256 }  
Public size limits for RIIC peripheral mode callers.
- enum [rspi\\_mode\\_t](#) : uint8\_t { [k\\_rspi\\_mode\\_0](#) , [k\\_rspi\\_mode\\_1](#) , [k\\_rspi\\_mode\\_2](#) = 2 ,  
[k\\_rspi\\_mode\\_3](#) }  
SPI mode enumeration defining clock polarity (CPOL) and phase (CPHA) combinations.
- enum [rspi\\_channel\\_t](#) : uint8\_t { [k\\_rspi\\_channel\\_0](#) = 0 , [k\\_rspi\\_channel\\_1](#) = 1 ,  
[k\\_rspi\\_channel\\_2](#) = 2 }  
RSPI channel enumeration with type safety for SPI peripheral selection.
- enum [rspi\\_freq\\_constants\\_t](#) : uint32\_t { [k\\_rspi\\_freq\\_1mhz](#) = 1000000 , [k\\_rspi\\_freq\\_10mhz](#) = 10000000 }  
Common SPI clock frequency constants for controller mode.
- enum [uart\\_defaults\\_t](#) : uint8\_t { [k\\_uart\\_debug\\_channel](#) = 9 }  
Default debug UART channel (SCI9 - CY7C65213 USB bridge).
- enum [uart\\_channel\\_t](#) : uint8\_t {  
[k\\_uart\\_channel\\_0](#) = 0 , [k\\_uart\\_channel\\_1](#) = 1 , [k\\_uart\\_channel\\_2](#) = 2 , [k\\_uart\\_channel\\_3](#) = 3 ,  
[k\\_uart\\_channel\\_4](#) = 4 , [k\\_uart\\_channel\\_5](#) = 5 , [k\\_uart\\_channel\\_6](#) = 6 , [k\\_uart\\_channel\\_7](#) = 7 ,  
[k\\_uart\\_channel\\_8](#) = 8 , [k\\_uart\\_channel\\_9](#) = 9 , [k\\_uart\\_channel\\_10](#) = 10 , [k\\_uart\\_channel\\_11](#) = 11 ,  
[k\\_uart\\_channel\\_12](#) = 12 }  
UART channel enumeration with type safety.

## Functions

- rx\_err\_t [gpio\\_set\\_output](#) (rx\_port\_pin\_t pin)  
Configure GPIO pin as digital output.
- rx\_err\_t [gpio\\_set\\_input](#) (rx\_port\_pin\_t pin)  
Configure GPIO pin as digital input.
- rx\_err\_t [gpio\\_write\\_high](#) (rx\_port\_pin\_t pin)  
Set GPIO output pin to logic high (VDD).
- rx\_err\_t [gpio\\_write\\_low](#) (rx\_port\_pin\_t pin)  
Set GPIO output pin to logic low (GND).
- rx\_err\_t [gpio\\_toggle](#) (rx\_port\_pin\_t pin)  
Toggle GPIO output pin state (high to low, or low to high).
- rx\_err\_t [gpio\\_read](#) (rx\_port\_pin\_t pin, bool \*value)  
Read current logic level of GPIO pin.
- rx\_err\_t [timer\\_init](#) (void)  
Initialize CMT0 for ThreadX system tick (100 Hz).

- rx\_err\_t [timer\\_stop](#) (void)  
Stop CMT0 timer and disable compare match interrupts.
- rx\_err\_t [timer\\_get\\_count](#) (uint16\_t \*count)  
Get current CMT0 counter value.
- rx\_err\_t [adc\\_init](#) (adc\_unit\_t unit, adc\_channel\_t channel, adc\_resolution\_t bits)  
Initialize ADC unit and channel.
- rx\_err\_t [adc\\_read](#) (adc\_unit\_t unit, adc\_channel\_t channel, uint16\_t \*value)  
Read ADC value.
- rx\_err\_t [adc\\_read\\_voltage\\_mv](#) (adc\_unit\_t unit, adc\_channel\_t channel, adc\_resolution\_t bits, uint32\_t \*voltage\_mv)  
Read ADC value and convert to millivolts.
- rx\_err\_t [riic\\_init](#) (riic\_channel\_t channel, uint32\_t frequency\_hz)  
Initialize RIIC channel for I2C controller communication.
- rx\_err\_t [riic\\_write](#) (riic\_channel\_t channel, i2c\_device\_addr\_t device\_addr, const uint8\_t \*data, uint16\_t length)  
Write data to I2C device on specified RIIC channel.
- rx\_err\_t [riic\\_read](#) (riic\_channel\_t channel, i2c\_device\_addr\_t device\_addr, uint8\_t \*data, uint16\_t length)  
Read data from I2C device on specified RIIC channel.
- rx\_err\_t [riic\\_write\\_read](#) (riic\_channel\_t channel, i2c\_device\_addr\_t device\_addr, const uint8\_t \*write\_data, uint16\_t write\_length, uint8\_t \*read\_data, uint16\_t read\_length)  
Write then read from I2C device using repeated START (combined transaction).
- rx\_err\_t [riic\\_init\\_peripheral](#) (riic\_channel\_t channel, i2c\_device\_addr\_t device\_addr)  
Initialize RIIC channel in peripheral (device) mode.
- rx\_err\_t [riic\\_deinit\\_peripheral](#) (riic\_channel\_t channel)  
Deinitialize an RIIC channel previously initialized in peripheral mode.
- rx\_err\_t [riic\\_peripheral\\_read](#) (riic\_channel\_t channel, uint8\_t \*data, uint16\_t max\_length, uint16\_t \*bytes\_read)  
Read bytes written by the I2C controller to this peripheral.
- rx\_err\_t [riic\\_peripheral\\_write](#) (riic\_channel\_t channel, const uint8\_t \*data, uint16\_t length)  
Write bytes for the I2C controller to read from this peripheral.
- rx\_err\_t [rspi\\_init\\_peripheral](#) (rspi\_channel\_t channel, const rspi\_config\_t \*config)  
Initialize RSPI channel in peripheral mode for RPi5 communication.
- rx\_err\_t [rspi\\_peripheral\\_transfer](#) (rspi\_channel\_t channel, const uint8\_t \*tx\_data, uint8\_t \*rx\_data, uint16\_t length)  
Full-duplex SPI transfer in peripheral mode.
- rx\_err\_t [rspi\\_peripheral\\_read\\_available](#) (rspi\_channel\_t channel, bool \*available)  
Check if receive data is available in the RSPI receive buffer.
- rx\_err\_t [rspi\\_peripheral\\_write\\_ready](#) (rspi\_channel\_t channel, bool \*ready)  
Check if the RSPI transmit buffer is ready for new data.
- rx\_err\_t [rspi\\_deinit](#) (rspi\_channel\_t channel)  
Deinitialize and disable an RSPI peripheral-mode channel.
- rx\_err\_t [rspi\\_init\\_controller](#) (rspi\_channel\_t channel, const rspi\_controller\_config\_t \*config)  
Initialize RSPI channel in controller mode for external device communication.
- rx\_err\_t [rspi\\_controller\\_set\\_cs](#) (rspi\_channel\_t channel, bool active)  
Set chip select line state for an RSPI controller-mode channel.
- rx\_err\_t [rspi\\_controller\\_transfer\\_16bit](#) (rspi\_channel\_t channel, uint16\_t tx\_data, uint16\_t \*rx\_data)  
Perform a 16-bit full-duplex SPI transfer in controller mode with automatic CS.
- rx\_err\_t [rspi\\_controller\\_deinit](#) (rspi\_channel\_t channel)  
Deinitialize and disable an RSPI controller-mode channel.
- rx\_err\_t [uart\\_init\\_channel](#) (const uart\_channel\_config\_t \*config)

- Initialize SCI channel for UART communication.
- rx\_err\_t [uart\\_deinit\\_channel](#) ([uart\\_channel\\_t](#) channel)
- Deinitialize SCI channel.
- rx\_err\_t [uart\\_putc\\_channel](#) ([uart\\_channel\\_t](#) channel, char data)
- Transmit a single character on specified channel.
- rx\_err\_t [uart\\_puts\\_channel](#) ([uart\\_channel\\_t](#) channel, const char \*str)
- Transmit a null-terminated string on specified channel.
- rx\_err\_t [uart\\_write\\_channel](#) ([uart\\_channel\\_t](#) channel, const uint8\_t \*data, uint16\_t length)
- Write buffer to specified channel.
- rx\_err\_t [uart\\_getc\\_channel](#) ([uart\\_channel\\_t](#) channel, char \*data)
- Receive a single character from specified channel (non-blocking).
- rx\_err\_t [uart\\_read\\_channel](#) ([uart\\_channel\\_t](#) channel, uint8\_t \*data, uint16\_t length, uint16\_t \*bytes\_read)
- Read available data from specified channel (non-blocking).
- rx\_err\_t [uart\\_rx\\_available](#) ([uart\\_channel\\_t](#) channel, bool \*available)
- Check if receive data is available on channel.
- rx\_err\_t [uart\\_clear\\_rx\\_errors](#) ([uart\\_channel\\_t](#) channel)
- Clear sticky receive error flags (ORER/FER/PER) on a UART channel.
- rx\_err\_t [uart\\_debug\\_init](#) (void)
- Initialize debug UART (SCI9, 115200 bps, 8N1).
- void [uart\\_debug\\_putc](#) (char data)
- Transmit a single character on debug UART (SCI9).
- void [uart\\_debug\\_puts](#) (const char \*str)
- Transmit a null-terminated string on debug UART (SCI9).
- void [uart\\_debug\\_putint](#) (int32\_t value)
- Print a signed integer on debug UART (SCI9).
- void [uart\\_debug\\_puthex](#) (uint32\_t value, uint8\_t digits)
- Print a hexadecimal value on debug UART (SCI9).

### 8.1.1 Detailed Description

Hardware Abstraction Layer (HAL) - Unified Interface for RX72N Peripherals.

#### 8.1.1.1 Overview

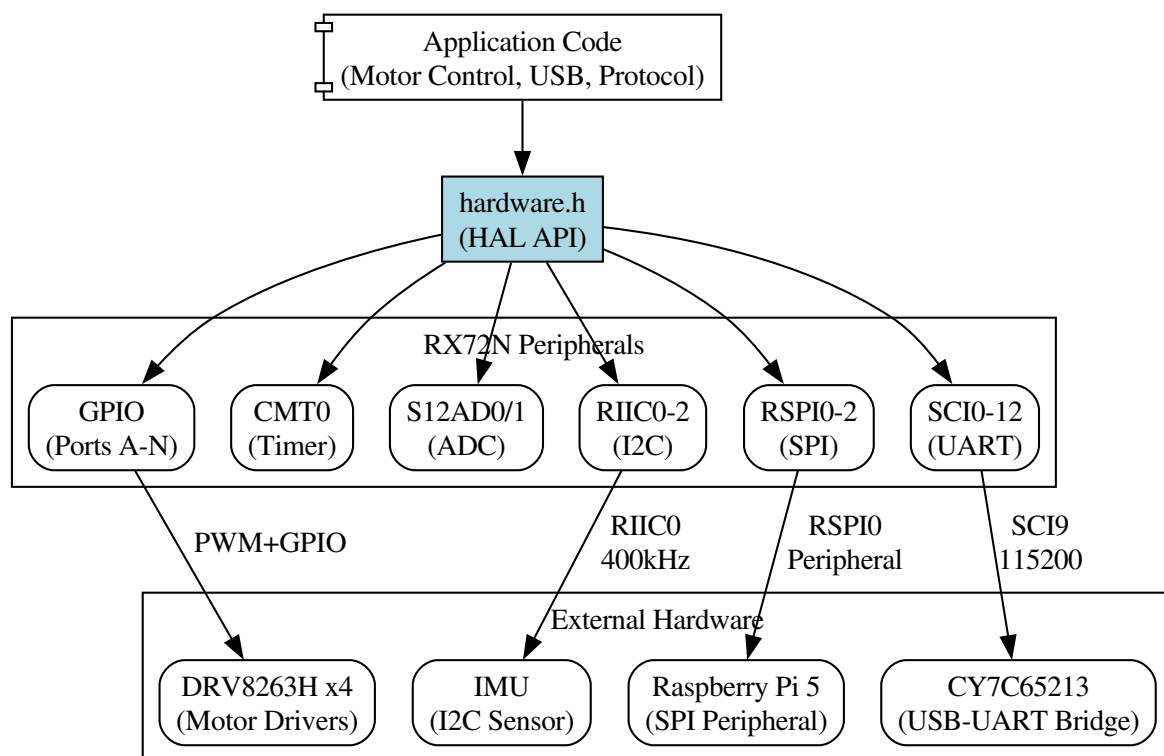
Comprehensive HAL providing type-safe, unified API for all RX72N microcontroller peripherals used in the STAR motor controller. Abstracts hardware register manipulation and provides safe, validated access to GPIO, timers, ADC, I2C, SPI, and UART peripherals.

#### 8.1.1.2 Supported Peripherals

| Peripheral | Channels       | Purpose                            | Configuration                      |
|------------|----------------|------------------------------------|------------------------------------|
| GPIO       | 14 ports (A-N) | Digital I/O, motor control signals | Output/Input, Read/Write/↔ Toggle  |
| CMT        | 4 channels     | ThreadX system tick (100 Hz)       | CMT0 configured for 10ms intervals |

| Peripheral | Channels           | Purpose                                           | Configuration           |
|------------|--------------------|---------------------------------------------------|-------------------------|
| S12AD      | 2 units (0-1)      | Motor current sensing, analog inputs              | 8/10/12-bit resolution  |
| RIIC       | 3 channels (0-2)   | I2C sensors (IMU); temperature sensor uses 1-Wire | 100kHz, 400kHz, 1MHz    |
| RSPI       | 3 channels (0-2)   | SPI: RPi5 comm (peripheral), sensors (controller) | Mode 0-3, 100kHz-10MHz  |
| SCI        | 13 channels (0-12) | UART: Debug (SCI9 @ 115200), expansion            | Configurable baud rates |

### 8.1.1.3 Hardware Architecture



### 8.1.1.4 Key Design Principles

#### 8.1.1.4.1 1. Type Safety (C23 Typed Enums + Struct Wrappers)

- Typed enums: All constants use C23 : `uint8_t` syntax for guaranteed size
- Struct wrappers: I2C channel/address use struct wrappers to prevent argument swaps
- Compile-time validation: Invalid types caught at compile time, not runtime

Example - Type Safety Prevents Bugs:

```

// WRONG - Swapped arguments with plain uint8_t
riic_write(0x48, 0, data, len); // Accidentally swapped channel and address

// CORRECT - Struct wrappers catch this at compile time
riic_channel_t ch = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};
riic_write(ch, addr, data, len); // Type-checked, no accidental swaps possible

```

#### 8.1.1.4.2 2. Unified Error Handling

- Consistent returns: All functions return `rx_err_t` (except debug helpers)
- Comprehensive error codes: Covers NULL pointers, invalid arguments, timeouts, conflicts
- Validation: All functions validate inputs before hardware access

#### 8.1.1.4.3 3. Configuration Structs Over Positional Parameters

- Maintainability: Adding new config options doesn't break API
- Clarity: Named fields > positional parameters
- Safety: Reduces parameter order mistakes

Example:

```
// OLD STYLE (error-prone):
uart_init(9, 115200, k_port_b_pin_7, k_port_b_pin_6); // Which pin is TX vs RX?

// NEW STYLE (clear and safe):
uart_channel_config_t config = {
 .channel = k_uart_channel_9,
 .baudrate = 115200,
 .tx_gpio = k_port_b_pin_7,
 .rx_gpio = k_port_b_pin_6
};
uart_init_channel(&config); // Self-documenting
```

#### 8.1.1.4.4 4. Zero Dynamic Allocation (NASA Rule 3)

- All functions use stack-based parameters
- No malloc/free anywhere in HAL
- Predictable memory behavior for safety-critical systems

### 8.1.1.5 Performance Characteristics

#### 8.1.1.5.1 Function Call Overhead (RX72N @ 240 MHz)

| Function Category     | Cycles | Time    | Notes                          |
|-----------------------|--------|---------|--------------------------------|
| GPIO write            | ~20    | 83 ns   | Direct register access         |
| GPIO read             | ~25    | 104 ns  | Includes pin validation        |
| UART putc             | ~150   | 625 ns  | Polling TDRE flag              |
| UART puts (10 chars)  | ~1500  | 6.25 us | 150 cycles/char                |
| I2C write (1 byte)    | ~24000 | 100 us  | 400kHz I2C clock               |
| SPI transfer (1 byte) | ~480   | 2 us    | 10 MHz SPI clock               |
| ADC read              | ~7200  | 30 us   | 12-bit conversion + validation |



## 8.1.1.5.2 Memory Footprint

| Component    | Flash  | RAM                 | Stack (max) |
|--------------|--------|---------------------|-------------|
| GPIO drivers | ~2 KB  | 16 bytes (state)    | 32 bytes    |
| UART drivers | ~3 KB  | 128 bytes (buffers) | 48 bytes    |
| I2C drivers  | ~4 KB  | 64 bytes (state)    | 64 bytes    |
| SPI drivers  | ~5 KB  | 96 bytes (state)    | 64 bytes    |
| ADC drivers  | ~2 KB  | 32 bytes (state)    | 32 bytes    |
| Total HAL    | ~16 KB | ~336 bytes          | ~240 bytes  |

## 8.1.1.6 Thread Safety

- NOT thread-safe: Hardware registers are not protected by mutexes
- Single-threaded access: Each peripheral should be accessed from one thread only
- RTOS integration: Use ThreadX mutexes if multi-threaded access needed
- Interrupt safety: All HAL functions disable interrupts during critical sections

## 8.1.1.7 Initialization Sequence

Recommended initialization order for STAR motor controller:

```
void hardware_init(void) {
 // 1. Debug UART (for early logging)
 uart_debug_init();
 rx_log_info("INIT", "Starting hardware initialization");

 // 2. System timer (for ThreadX tick)
 timer_init();

 // 3. GPIO (for motor control signals)
 gpio_set_output(k_port_c_pin_0); // Motor enable
 gpio_set_output(k_port_c_pin_1); // Direction

 // 4. SPI controller (for sensors; DRV8263H uses PWM+GPIO, not SPI)
 rspi_controller_config_t spi_cfg = {
 .spi_mode = k_rspi_mode_0,
 .freq_hz = k_rspi_freq_1mhz,
 .cs = k_port_c_pin_4
 };
 rspi_init_controller(k_rspi_channel_1, &spi_cfg);

 // 5. SPI peripheral (for RPi5 communication)
 rspi_config_t rpi_cfg = {
 .spi_mode = k_rspi_mode_0,
 .use_16bit = false
 };
 rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);

 // 6. ADC (for motor current sensing on AN007/P47)
 adc_init(k_adc_unit_0, k_adc_channel_7, k_adc_resolution_12bit);

 // 7. I2C (for IMU sensor)
 riic_init((riic_channel_t){.value = 0}, 400000);

 rx_log_info("INIT", "Hardware initialization complete");
}
```

### 8.1.1.8 Pin Assignments (From Project Schematic)

See docs/sections/03\_hardware\_pinout.tex for complete pinout documentation.

Critical Signals:

| Signal    | Pin | Peripheral | Purpose                          |
|-----------|-----|------------|----------------------------------|
| SPI0_CIPO | P30 | RSPI0      | RPi5 <- RX72N (peripheral mode)  |
| SPI0_COPI | P31 | RSPI0      | RPi5 -> RX72N                    |
| SPI0_CLK  | P32 | RSPI0      | RPi5 clock                       |
| SPI0_CS   | P33 | GPIO       | RPi5 chip select                 |
| SPI1_CIPO | PC5 | RSPI1      | Not used (RSPI1 controller mode) |
| SPI1_COPI | PC6 | RSPI1      | Not used (RSPI1 controller mode) |
| SPI1_CLK  | PC7 | RSPI1      | Not used (RSPI1 clock)           |
| UART_TX   | PB7 | SCI9       | Debug output (CY7C65213)         |
| UART_RX   | PB6 | SCI9       | Debug input                      |

### 8.1.1.9 Dependencies

- rx\_err.h - Error code definitions
- rx\_check.h - Validation macros
- rx\_log.h - Logging infrastructure
- rx\_port\_constants.h - Port/pin definitions
- rx\_bus\_types.h - Bus enumeration types
- [rx72n\\_regs.h](#) - Hardware register definitions (internal)

NASA Power of 10 Compliance:

- Rule 1 [YES]: No goto, setjmp, recursion (pure validation + register access)
- Rule 2 [YES]: All loops bounded (timeout loops use compile-time max iterations)
- Rule 3 [YES]: No dynamic allocation (all stack-based or static configuration)
- Rule 4 [YES]: All functions <60 lines (HAL functions are typically 20-40 lines)
- \*\*Rule 5 [YES]\*\*: Minimum 2 validations per function (NULL checks + range validation)
- \*\*Rule 6 [YES]\*\*: Data at smallest scope (loop counters, temporary variables)
- \*\*Rule 7 [YES]\*\*: All returns validated (rx\_err\_t checked at every call site)
- \*\*Rule 8 [YES]\*\*: C23 typed enums for constants, minimal preprocessor use
- \*\*Rule 9 [YES]\*\*: Single-level dereferencing (pointer parameters for output values)
- \*\*Rule 10 [YES]\*\*: Compiles with -Wall -Wextra -Werror

SOLID Principles:

- **\*\*S (Single Responsibility):\*\*** Each peripheral has dedicated functions (GPIO, UART, SPI, I2C, ADC)
- **\*\*O (Open/Closed):\*\*** Config structs allow extension without modifying function signatures
- **\*\*L (Liskov Substitution):\*\*** All HAL functions return `rx_err_t` with consistent semantics
- **\*\*I (Interface Segregation):\*\*** Separate controller/peripheral SPI APIs, debug UART wrappers
- **\*\*D (Dependency Inversion):\*\*** Depends on abstract `rx_err_t`, `rx_log`, not concrete implementations

Module Dependencies:

```
hardware.h
+--> rx_err.h (error codes)
+--> rx_check.h (validation macros)
+--> rx_log.h (logging)
+--> rx_port_constants.h (port/pin enums)
+--> rx_bus_types.h (ADC resolution enums)
+--> [internal] rx72n_regs.h (register definitions)
```

Used by:

```
+--> Motor control (rx_motor_control.c)
+--> USB communication (rx_usb.c)
+--> Protocol handling (rx_comm_manager.c)
+--> Application tasks (comm, motor, sensors)
```

Author

Locked, Inc.

Date

2026-01-27

Version

1.0.0

Copyright

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See also

`rx_err.h` for error code definitions  
`rx_port_constants.h` for complete port/pin enumerations  
`docs/sections/03_hardware_pinout.tex` for hardware pinout documentation

Definition in file [hardware.h](#).

## 8.1.2 Data Structure Documentation

### 8.1.2.1 struct `riic_channel_t`

RIIC (I2C) channel wrapper providing strong type safety.

Wraps RIIC channel number in a struct to prevent accidental argument swaps with I2C device addresses. This is a stronger type safety mechanism than simple typedefs because structs cannot be implicitly converted.

### 8.1.2.2 Design Rationale: Struct Wrapper vs. Typedef

Why struct wrapper instead of simple typedef?

I2C functions have multiple `uint8_t` parameters that are easily confused:

- Channel number (0-2)
- Device address (0x00-0x7F)
- Register address (0x00-0xFF)
- Data bytes (0x00-0xFF)

#### 8.1.2.2.1 Problem with Simple Typedefs

```
typedef uint8_t riic_channel_t; // Simple typedef
typedef uint8_t i2c_device_addr_t; // Simple typedef

riic_channel_t ch = 0;
i2c_device_addr_t addr = 0x48;

// PROBLEM: Arguments can be swapped without compiler error!
riic_write(addr, ch, data, len); // Compiles! Wrong order, but types are compatible
```

#### 8.1.2.2.2 Solution with Struct Wrappers

```
typedef struct { uint8_t value; } riic_channel_t; // Struct wrapper
typedef struct { uint8_t value; } i2c_device_addr_t; // Struct wrapper

riic_channel_t ch = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};

// CORRECT: Compiler catches swapped arguments!
riic_write(addr, ch, data, len); // Compile error: type mismatch!
riic_write(ch, addr, data, len); // Correct order, compiles successfully
```

### 8.1.2.3 Comparison: ADC vs. I2C Type Safety

| Peripheral | Type Safety Mechanism | Rationale                                                    |
|------------|-----------------------|--------------------------------------------------------------|
| ADC        | Simple typed enums    | Parameters are distinct (unit vs channel vs resolution bits) |
| I2C        | Struct wrappers       | Multiple <code>uint8_t</code> parameters easily confused     |
| SPI        | Simple typed enums    | Mode, channel, frequency are distinct types                  |
| UART       | Simple typed enums    | Channel and baudrate are distinct types                      |

ADC: `adc_read(k_adc_unit_0, k_adc_channel_1, &value)` -> Different types prevent swaps I2C:↵  
`riic_write({.value = 0}, {.value = 0x48}, data, len)` -> Struct wrappers prevent swaps

#### 8.1.2.4 RX72N RIIC Hardware

- Channels Available: 3 (RIIC0, RIIC1, RIIC2)
- Speeds: 100 kHz (standard), 400 kHz (fast), 1 MHz (fast-plus)
- Addressing: 7-bit (0x00-0x7F) or 10-bit (not used in STAR project)

## 8.1.2.5 STAR Project Usage

| Channel | SCL Pin | SDA Pin | Device        | Address | Purpose             |
|---------|---------|---------|---------------|---------|---------------------|
| 0       | P16     | P17     | IMU (MPU6050) | 0x68    | Orientation sensing |
| 1       | P54     | P55     | Reserved      | -       | Future expansion    |
| 2       | P52     | P53     | Reserved      | -       | Future expansion    |

## 8.1.2.6 Usage Example

```
#include "hardware.h"

void imu_init(void) {
 // Initialize I2C channel 0 at 400 kHz (fast mode)
 riic_channel_t ch = {.value = 0};
 riic_init(ch, 400000);

 // MPU6050 IMU device address
 i2c_device_addr_t imu_addr = {.value = 0x68};

 // Write to WHO_AM_I register (0x75) to verify device
 uint8_t reg_addr = 0x75;
 uint8_t who_am_i;
 riic_write_read(ch, imu_addr, ®_addr, 1, &who_am_i, 1);

 if (who_am_i == 0x68) {
 rx_log_info("IMU", "MPU6050 detected");
 }
}
```

Example: Compiler Catches Mistakes

```
riic_channel_t ch = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};
uint8_t data[] = {0x01, 0x02};

// WRONG - Arguments swapped (compiler error!)
riic_write(addr, ch, data, 2);
// Error: cannot convert 'i2c_device_addr_t' to 'riic_channel_t'

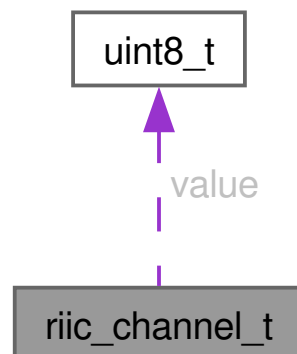
// CORRECT - Proper order
riic_write(ch, addr, data, 2); // Compiles successfully
```

See also

[i2c\\_device\\_addr\\_t](#) I2C device address wrapper  
[riic\\_init\(\)](#) Initialize I2C channel  
[riic\\_write\(\)](#) Write data to I2C device  
[riic\\_read\(\)](#) Read data from I2C device  
[riic\\_write\\_read\(\)](#) Combined write-then-read transaction

Definition at line 1131 of file [hardware.h](#).

Collaboration diagram for `riic_channel_t`:



## Data Fields

|         |       |                                                             |
|---------|-------|-------------------------------------------------------------|
| uint8_t | value | RIIC channel number (0-2). Use {.value = N} initialization. |
|---------|-------|-------------------------------------------------------------|

## 8.1.2.7 struct i2c\_device\_addr\_t

I2C device address wrapper providing strong type safety.

Wraps 7-bit I2C device address in a struct to prevent accidental swaps with channel numbers or register addresses. Part of the type safety system that uses struct wrappers to catch parameter ordering mistakes at compile time.

## 8.1.2.8 I2C Addressing

- 7-bit addressing: Standard mode used in STAR project
- Valid range: 0x08-0x77 (0x00-0x07 and 0x78-0x7F reserved)
- Reserved addresses:

0x00: General call

0x01-0x07: CBUS, different bus formats

0x78-0x7F: 10-bit addressing, reserved

## 8.1.2.9 Common Device Addresses

| Device              | Address   | Alt Address | Configurable   |
|---------------------|-----------|-------------|----------------|
| MPU6050 (IMU)       | 0x68      | 0x69        | Via AD0 pin    |
| BMP280 (Pressure)   | 0x76      | 0x77        | Via SDO pin    |
| ADS1115 (ADC)       | 0x48-0x4B | -           | Via ADDR pin   |
| PCA9685 (PWM)       | 0x40-0x7F | -           | Via A0-A5 pins |
| EEPROM (24C series) | 0x50-0x57 | -           | Via A0-A2 pins |

## 8.1.2.10 Address Format on Bus

The 7-bit address is shifted left by 1 bit and combined with R/W bit:

[A6 A5 A4 A3 A2 A1 A0 R/W]  
 \\_\_\_\_ 7-bit address \_\_\_\_/ \\_ 0=Write, 1=Read

Note: HAL automatically handles this shifting. Always provide 7-bit address.

## 8.1.2.11 STAR Project Devices

| Device          | Address | Channel | Purpose                |
|-----------------|---------|---------|------------------------|
| MPU6050 IMU     | 0x68    | RIIC0   | 6-axis motion tracking |
| Future sensor 1 | TBD     | RIIC1   | Expansion              |
| Future sensor 2 | TBD     | RIIC2   | Expansion              |

## 8.1.2.12 Usage Example

```
// MPU6050 IMU on RIIC0
riic_channel_t ch = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

// Read power management register (0x6B)
uint8_t pwr_mgmt_reg = 0x6B;
uint8_t pwr_mgmt_val;

riic_write_read(ch, imu_addr, &pwr_mgmt_reg, 1, &pwr_mgmt_val, 1);
rx_log_info_hex("IMU", "Power management", pwr_mgmt_val, 2);
```

## Example: Multi-Device Bus

```
// Multiple devices on same I2C bus
riic_channel_t ch = {.value = 0};

i2c_device_addr_t imu_addr = {.value = 0x68}; // MPU6050
i2c_device_addr_t baro_addr = {.value = 0x76}; // BMP280
i2c_device_addr_t adc_addr = {.value = 0x48}; // ADS1115

// Read from different devices on same bus
uint8_t imu_data[6];
riic_read(ch, imu_addr, imu_data, 6);

uint8_t baro_data[3];
riic_read(ch, baro_addr, baro_data, 3);

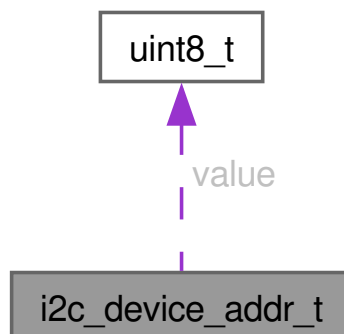
uint8_t adc_data[2];
riic_read(ch, adc_addr, adc_data, 2);
```

See also

[riic\\_channel\\_t](#) RIIC channel wrapper  
[riic\\_write\(\)](#) Write data to I2C device  
[riic\\_read\(\)](#) Read data from I2C device  
[riic\\_write\\_read\(\)](#) Register read pattern (write reg addr, then read)

Definition at line 1215 of file [hardware.h](#).

Collaboration diagram for `i2c_device_addr_t`:



## Data Fields

|                      |       |                                                                                        |
|----------------------|-------|----------------------------------------------------------------------------------------|
| <code>uint8_t</code> | value | 7-bit I2C device address (0x08-0x77). Use <code>{.value = 0xNN}</code> initialization. |
|----------------------|-------|----------------------------------------------------------------------------------------|

8.1.2.13 `struct rspi_config_t`

RSPI peripheral mode configuration structure.

Configuration parameters for initializing an RSPI channel in peripheral mode. Used when the RX72N acts as an SPI peripheral device, receiving clock from an external controller (e.g., Raspberry Pi 5). The struct encapsulates SPI mode and frame size settings to avoid positional parameter errors.

```
rspi_config_t rpi_cfg = {
 .spi_mode = k_rspi_mode_0, // CPOL=0, CPHA=0
 .use_16bit = false // 8-bit frame size
};
```

## Invariant

`spi_mode` must be a valid [rspi\\_mode\\_t](#) value (0-3, CPOL/CPHA combinations)

`use_16bit` is boolean: false for 8-bit frames, true for 16-bit frames

## See also

[rspi\\_mode\\_t](#) SPI mode enumeration (CPOL/CPHA combinations)

[rspi\\_init\\_peripheral\(\)](#) Initialize channel with this configuration

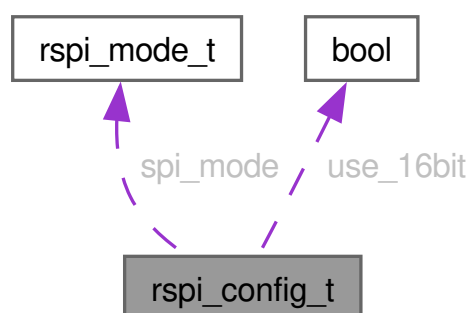
[rspi\\_controller\\_config\\_t](#) Controller mode configuration (separate struct)

## Since

Version 1.0.0

Definition at line 1838 of file [hardware.h](#).

Collaboration diagram for `rspi_config_t`:



## Data Fields

|                             |                       |                                             |
|-----------------------------|-----------------------|---------------------------------------------|
| <a href="#">rspi_mode_t</a> | <code>spi_mode</code> | SPI mode (0-3): CPOL and CPHA configuration |
|-----------------------------|-----------------------|---------------------------------------------|



|      |           |                                                |
|------|-----------|------------------------------------------------|
| bool | use_16bit | True for 16-bit frames, false for 8-bit frames |
|------|-----------|------------------------------------------------|

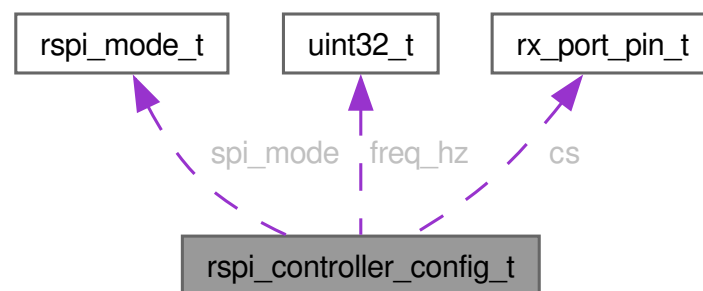
#### 8.1.2.14 struct rspi\_controller\_config\_t

RSPI controller mode configuration structure.

Configuration for initializing RSPI in controller mode. Used for communicating with SPI peripheral devices (sensors, etc.).

Definition at line 2098 of file [hardware.h](#).

Collaboration diagram for rspi\_controller\_config\_t:



#### Data Fields

|                            |                       |                                               |
|----------------------------|-----------------------|-----------------------------------------------|
| <code>rx_port_pin_t</code> | <code>cs</code>       | GPIO pin for chip select (type-safe port+pin) |
| <code>uint32_t</code>      | <code>freq_hz</code>  | SPI clock frequency in Hz (100kHz - 10MHz)    |
| <code>rspi_mode_t</code>   | <code>spi_mode</code> | SPI mode (0-3): CPOL and CPHA configuration   |

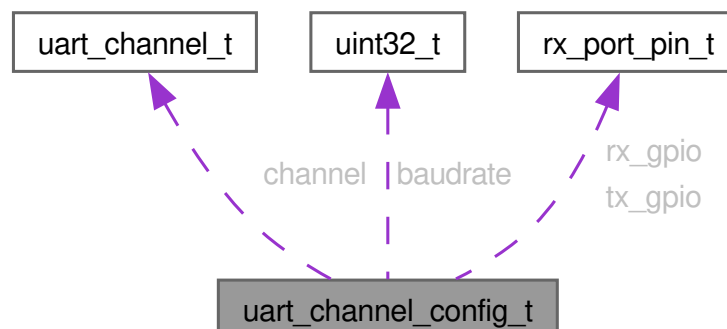
#### 8.1.2.15 struct uart\_channel\_config\_t

UART channel configuration structure.

Encapsulates all parameters needed to initialize a UART channel. Using a struct instead of positional parameters prevents accidental parameter swapping and makes the API more maintainable.

Definition at line 2332 of file [hardware.h](#).

Collaboration diagram for uart\_channel\_config\_t:



## Data Fields

|                |          |                                |
|----------------|----------|--------------------------------|
| uint32_t       | baudrate | Baud rate (e.g., 9600, 115200) |
| uart_channel_t | channel  | SCI channel (0-12)             |
| rx_port_pin_t  | rx_gpio  | RX pin (rx_port_pin_t)         |
| rx_port_pin_t  | tx_gpio  | TX pin (rx_port_pin_t)         |

## 8.1.3 Typedef Documentation

## 8.1.3.1 adc\_resolution\_t

```
typedef rx_adc_resolution_t adc_resolution_t
```

Definition at line 966 of file [hardware.h](#).

## 8.1.4 Enumeration Type Documentation

## 8.1.4.1 adc\_channel\_t

```
enum adc_channel_t : uint8_t
```

ADC channel enumeration with type safety for analog input selection.

Identifies which analog input channel (0-7) to use within an ADC unit. Each unit (S12AD0, S12AD1) has 8 independent channels with dedicated analog input pins. Using a typed enum prevents accidental parameter swaps with unit or resolution.

## 8.1.4.2 Pin Mapping (RX72N 144-pin LQFP, R5F572NNHxFB)

## 8.1.4.2.1 S12AD0 Channels (use with k\_adc\_unit\_0)

| Channel         | Pin | Signal Name | STAR Usage            |
|-----------------|-----|-------------|-----------------------|
| k_adc_channel_0 | P40 | AN000       | Available (expansion) |
| k_adc_channel_1 | P41 | AN001       | Available (expansion) |
| k_adc_channel_2 | P42 | AN002       | Available (expansion) |
| k_adc_channel_3 | P43 | AN003       | Available (expansion) |
| k_adc_channel_4 | P44 | AN004       | Motor 3 current sense |
| k_adc_channel_5 | P45 | AN005       | Motor 2 current sense |

| Channel         | Pin | Signal Name | STAR Usage            |
|-----------------|-----|-------------|-----------------------|
| k_adc_channel_6 | P46 | AN006       | Motor 1 current sense |
| k_adc_channel_7 | P47 | AN007       | Motor 0 current sense |

#### 8.1.4.3 Input Specifications

- Voltage Range: 0V to AVCC (3.3V)
- Input Impedance: 1 MOhm typical
- Max Input Current: +/-10 uA
- Protection: Internal clamping diodes to AVCC/AVSS

#### 8.1.4.4 Usage Example

```
// Motor current sensing
adc_init(k_adc_unit_0, k_adc_channel_0, k_adc_resolution_12bit);

uint32_t voltage_mv;
adc_read_voltage_mv(k_adc_unit_0, k_adc_channel_0, k_adc_resolution_12bit, &voltage_mv);
rx_log_info_val("ADC", "Current sense (mV)", voltage_mv);
```

Underlying Type: uint8\_t (C23 typed enum)

- Range: 0-7 (8 channels per unit)
- Size: 1 byte
- Rationale: Matches hardware channel select bits (ADANSA register)

See also

[adc\\_unit\\_t](#) ADC unit selection  
[adc\\_resolution\\_t](#) Resolution configuration  
[adc\\_read\(\)](#) Read raw ADC value  
[adc\\_read\\_voltage\\_mv\(\)](#) Read voltage in millivolts

#### Enumerator

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| k_adc_channel_0 | Channel 0 - AN000 (unit 0) or AN100 (unit 1). Available.          |
| k_adc_channel_1 | Channel 1 - AN001 (unit 0) or AN101 (unit 1). Available.          |
| k_adc_channel_2 | Channel 2 - AN002 (unit 0) or AN102 (unit 1). Available.          |
| k_adc_channel_3 | Channel 3 - AN003 (unit 0) or AN103 (unit 1). Available.          |
| k_adc_channel_4 | Channel 4 - AN004/P44 (unit 0, Motor 3 current) or AN104 (unit 1) |

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| k_adc_channel_5 | Channel 5 - AN005/P45 (unit 0, Motor 2 current) or AN105 (unit 1) |
| k_adc_channel_6 | Channel 6 - AN006/P46 (unit 0, Motor 1 current) or AN106 (unit 1) |
| k_adc_channel_7 | Channel 7 - AN007/P47 (unit 0, Motor 0 current) or AN107 (unit 1) |

Definition at line 954 of file [hardware.h](#).

```

00954 : uint8_t {
00955 k_adc_channel_0 = 0, /**< Channel 0 - AN000 (unit 0) or AN100 (unit 1). Available. */
00956 k_adc_channel_1 = 1, /**< Channel 1 - AN001 (unit 0) or AN101 (unit 1). Available. */
00957 k_adc_channel_2 = 2, /**< Channel 2 - AN002 (unit 0) or AN102 (unit 1). Available. */
00958 k_adc_channel_3 = 3, /**< Channel 3 - AN003 (unit 0) or AN103 (unit 1). Available. */
00959 k_adc_channel_4 = 4, /**< Channel 4 - AN004/P44 (unit 0, Motor 3 current) or AN104 (unit 1) */
00960 k_adc_channel_5 = 5, /**< Channel 5 - AN005/P45 (unit 0, Motor 2 current) or AN105 (unit 1) */
00961 k_adc_channel_6 = 6, /**< Channel 6 - AN006/P46 (unit 0, Motor 1 current) or AN106 (unit 1) */
00962 k_adc_channel_7 = 7, /**< Channel 7 - AN007/P47 (unit 0, Motor 0 current) or AN107 (unit 1) */
00963 } adc_channel_t;

```

#### 8.1.4.5 adc\_unit\_t

```
enum adc_unit_t : uint8_t
```

ADC unit enumeration with type safety for 12-bit SAR ADC modules.

Identifies which of the two independent 12-bit successive approximation ADC units (S12AD) on the RX72N to use. Each unit has separate channels, registers, and conversion control. Using a typed enum prevents accidental swapping of unit and channel parameters.

#### 8.1.4.6 Hardware Specifications

| Unit   | Base Address | Channels Available | Max Sample Rate |
|--------|--------------|--------------------|-----------------|
| S12AD0 | 0x00089000   | 8 (AN000-AN007)    | 1 Msps          |
| S12AD1 | 0x00089100   | 21 (AN100-AN120)   | 1 Msps          |

#### 8.1.4.7 STAR Project Usage (144-pin LFQFP)

| Unit              | Channel              | Pin | Signal   | Purpose                                 |
|-------------------|----------------------|-----|----------|-----------------------------------------|
| k_adc_unit↔<br>_0 | k_adc_channel↔<br>_7 | P47 | MOTOR_I0 | Motor 0 current sense (DRV8263H IPROPI) |
| k_adc_unit↔<br>_0 | k_adc_channel↔<br>_6 | P46 | MOTOR_I1 | Motor 1 current sense (DRV8263H IPROPI) |
| k_adc_unit↔<br>_0 | k_adc_channel↔<br>_5 | P45 | MOTOR_I2 | Motor 2 current sense                   |
| k_adc_unit↔<br>_0 | k_adc_channel↔<br>_4 | P44 | MOTOR_I3 | Motor 3 current sense                   |

## 8.1.4.8 Type Safety Benefit

```
// WRONG - Easy to swap unit and channel with plain uint8_t
adc_read(0, 1, &value); // Is 0 the unit or channel?

// CORRECT - Typed enums make intent clear
adc_read(k_adc_unit_0, k_adc_channel_1, &value); // Unambiguous
```

Underlying Type: `uint8_t` (C23 typed enum)

- Range: 0-1 (only 2 units on RX72N)
- Size: 1 byte
- Rationale: Smallest type sufficient, explicit size guarantees ABI stability

See also

[adc\\_channel\\_t](#) Channel enumeration  
[adc\\_resolution\\_t](#) Resolution configuration  
[adc\\_init\(\)](#) ADC initialization function

Enumerator

|                           |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| <code>k_adc_unit_0</code> | S12AD0 - ADC unit 0 (channels AN000-AN007). Used for motor current sensing. |
| <code>k_adc_unit_1</code> | S12AD1 - ADC unit 1 (channels AN100-AN120). Reserved for expansion sensors. |

Definition at line 899 of file [hardware.h](#).

```
00899 : uint8_t {
00900 k_adc_unit_0 =
00901 0, /**< S12AD0 - ADC unit 0 (channels AN000-AN007). Used for motor current sensing. */
00902 k_adc_unit_1 =
00903 1 /**< S12AD1 - ADC unit 1 (channels AN100-AN120). Reserved for expansion sensors. */
00904 } adc_unit_t;
```

8.1.4.9 `riic_peripheral_limits_t`

enum [riic\\_peripheral\\_limits\\_t](#) : `uint16_t`

Public size limits for RIIC peripheral mode callers.

Exposes the I/O size ceiling enforced by [riic\\_peripheral\\_read\(\)](#) and [riic\\_peripheral\\_write\(\)](#) so that callers can statically size their buffers and validate arguments without needing to inspect the driver implementation.

See also

[riic\\_peripheral\\_read\(\)](#) Enforces this limit on `max_length`  
[riic\\_peripheral\\_write\(\)](#) Enforces this limit on `length`

Since

Version 1.0.0

Enumerator

|                                               |                                                 |
|-----------------------------------------------|-------------------------------------------------|
| <code>k_riic_peripheral_transfer_limit</code> | Maximum bytes per peripheral read or write call |
|-----------------------------------------------|-------------------------------------------------|

Definition at line 1511 of file [hardware.h](#).

```
01511 : uint16_t {
01512 k_riic_peripheral_transfer_limit = 256, /**< Maximum bytes per peripheral read or write call */
01513 } riic_peripheral_limits_t;
```

#### 8.1.4.10 `rspi_channel_t`

```
enum rspi_channel_t : uint8_t
```

RSPI channel enumeration with type safety for SPI peripheral selection.

Identifies which of the three RSPI channels (RSPI0, RSPI1, RSPI2) on the RX72N to use. Using a typed enum instead of plain `uint8_t` prevents accidental argument swaps and provides compile-time validation of channel numbers.

#### 8.1.4.11 RX72N RSPI Channel Assignments (STAR Project)

| Channel | Role       | Connected Device   | Pins                                        |
|---------|------------|--------------------|---------------------------------------------|
| RSPI0   | Peripheral | Raspberry Pi 5     | P30 (CIPO), P31 (COPI), P32 (CLK), P33 (CS) |
| RSPI1   | Controller | Reserved (sensors) | PC5 (CIPO), PC6 (COPI), PC7 (CLK)           |
| RSPI2   | Unused     | -                  | -                                           |

Underlying Type: `uint8_t` (C23 typed enum)

- Range: 0-2 (3 channels on RX72N)
- Size: 1 byte

Invariant

Values must be in range `[0, 2]` corresponding to RSPI0, RSPI1, RSPI2

```
// Initialize RSPI channel as peripheral
rspi_config_t cfg = { .spi_mode = k_rspi_mode_0, .use_16bit = false };
rx_err_t err = rspi_init_peripheral(k_rspi_channel_0, &cfg);
```

See also

[rspi\\_init\\_peripheral\(\)](#) Initialize channel in peripheral mode

[rspi\\_init\\_controller\(\)](#) Initialize channel in controller mode

Since

Version 1.0.0

Enumerator

|                  |                                                              |
|------------------|--------------------------------------------------------------|
| k_rspi_channel_0 | RSPI0 - RPi5 communication (peripheral mode in STAR project) |
| k_rspi_channel_1 | RSPI1 - External sensor SPI (controller mode, reserved)      |
| k_rspi_channel_2 | RSPI2 - Unused (available for expansion)                     |

Definition at line 1806 of file [hardware.h](#).

```
01806 : uint8_t {
01807 k_rspi_channel_0 = 0, /**< RSPI0 - RPi5 communication (peripheral mode in STAR project) */
01808 k_rspi_channel_1 = 1, /**< RSPI1 - External sensor SPI (controller mode, reserved) */
01809 k_rspi_channel_2 = 2 /**< RSPI2 - Unused (available for expansion) */
01810 } rspi_channel_t;
```

#### 8.1.4.12 rspi`freq`constants`t

```
enum rspi_freq_constants_t : uint32_t
```

Common SPI clock frequency constants for controller mode.

Named constants for commonly used SPI clock frequencies. Use these instead of magic literals when configuring controller-mode SPI clock rate. The RSPI bit rate register (SPBR) is calculated from these values and the peripheral clock (PCLKB).

Invariant

All values must be positive and expressible by the SPBR register

See also

[rspi\\_init\\_controller\(\)](#) Uses freq\_hz to calculate SPBR

Since

Version 1.0.0

Enumerator

|                   |                                        |
|-------------------|----------------------------------------|
| k_rspi_freq_1mhz  | 1 MHz SPI clock frequency              |
| k_rspi_freq_10mhz | 10 MHz SPI clock frequency (RPi5 link) |

Definition at line 2087 of file [hardware.h](#).

```
02087 : uint32_t {
02088 k_rspi_freq_1mhz = 1000000, /**< 1 MHz SPI clock frequency */
02089 k_rspi_freq_10mhz = 10000000, /**< 10 MHz SPI clock frequency (RPi5 link) */
02090 } rspi_freq_constants_t;
```

#### 8.1.4.13 `rspi_mode_t`

enum `rspi_mode_t` : `uint8_t`

SPI mode enumeration defining clock polarity (CPOL) and phase (CPHA) combinations.

Defines the four standard SPI modes based on clock polarity and phase settings. These modes determine when data is sampled relative to the clock edge and what the idle state of the clock line is.

#### 8.1.4.14 SPI Mode Definitions

| Mode | CPOL | CPHA | Clock Idle | Data Sampled | Data Shifted | Common Devices     |
|------|------|------|------------|--------------|--------------|--------------------|
| 0    | 0    | 0    | Low        | Rising edge  | Falling edge | RPi5, most sensors |
| 1    | 0    | 1    | Low        | Falling edge | Rising edge  | SD cards           |
| 2    | 1    | 0    | High       | Falling edge | Rising edge  | Rare               |
| 3    | 1    | 1    | High       | Rising edge  | Falling edge | Rare               |

#### 8.1.4.15 STAR Project Usage

| Peripheral                  | Mode   | Rationale                                   |
|-----------------------------|--------|---------------------------------------------|
| RPi5 (RSPI0 peripheral)     | Mode 0 | Linux spidev default, maximum compatibility |
| RSPI1 controller (not used) | TBD    | RSPI1 not used in current design            |

#### 8.1.4.16 Clock Polarity (CPOL)

- CPOL=0: Clock idle state is low (GND). Clock starts low, pulses high.
- CPOL=1: Clock idle state is high (VDD). Clock starts high, pulses low.

#### 8.1.4.17 Clock Phase (CPHA)

- CPHA=0: Data sampled on leading clock edge, shifted on trailing edge.
- CPHA=1: Data shifted on leading clock edge, sampled on trailing edge.

#### 8.1.4.18 Timing Diagrams

##### 8.1.4.18.1 Mode 0 (CPOL=0, CPHA=0) - Most Common

```

CLK: ____/---____/---____
COP: ===[D0]===[D1]===
CIP: ===[D0]===[D1]===
 ^ Sample ^ Sample

```

##### 8.1.4.18.2 Mode 3 (CPOL=1, CPHA=1)

```

CLK: ---____/---____/---
COP: ===[D0]===[D1]===
CIP: ===[D0]===[D1]===
 ^ Sample ^ Sample

```



## 8.1.4.19 Selection Guidelines

1. Check device datasheet: Most devices specify preferred mode
2. Mode 0 default: Use Mode 0 if device supports multiple modes
3. Controller/Peripheral match: Both sides must use same mode
4. Avoid Modes 1 and 2: Less common, higher chance of incompatibility

## 8.1.4.20 Hardware Configuration

- SPCR.CPOL bit: Set for CPOL=1, clear for CPOL=0
- SPCR.CPHA bit: Set for CPHA=1, clear for CPHA=0
- Both configurable: HAL automatically sets registers based on mode

## 8.1.4.21 Usage Example

```
// RPi5 communication (peripheral mode, Mode 0)
rspi_config_t rpi_cfg = {
 .spi_mode = k_rspi_mode_0, // CPOL=0, CPHA=0
 .use_16bit = false
};
rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);

// External sensor communication (controller mode, Mode 0)
rspi_controller_config_t sensor_cfg = {
 .spi_mode = k_rspi_mode_0, // Standard SPI mode 0
 .freq_hz = k_rspi_freq_1mhz, // 1 MHz
 .cs = k_port_c_pin_4
};
rspi_init_controller(k_rspi_channel_1, &sensor_cfg); // RSPI1
```

Underlying Type: uint8\_t (C23 typed enum)

- Range: 0-3 (4 SPI modes)
- Size: 1 byte
- Rationale: Matches SPCR register bit pattern (CPOL | (CPHA << 1))

See also

[rspi\\_init\\_peripheral\(\)](#) Initialize SPI in peripheral mode

[rspi\\_init\\_controller\(\)](#) Initialize SPI in controller mode

[rspi\\_config\\_t](#) Configuration structure for peripheral mode

[rspi\\_controller\\_config\\_t](#) Configuration structure for controller mode

## Enumerator

|               |                                                                            |
|---------------|----------------------------------------------------------------------------|
| k_rspi_mode_0 | CPOL=0, CPHA=0 - Data sampled on rising edge, idle low. Most common mode.  |
| k_rspi_mode_1 | CPOL=0, CPHA=1 - Data sampled on falling edge, idle low. Used by SD cards. |
| k_rspi_mode_2 | CPOL=1, CPHA=0 - Data sampled on falling edge, idle high. Rarely used.     |

|               |                                                                                        |
|---------------|----------------------------------------------------------------------------------------|
| k_rspi_mode_3 | CPOL=1, CPHA=1 - Data sampled on rising edge, idle high. Alternative for some devices. |
|---------------|----------------------------------------------------------------------------------------|

Definition at line 1763 of file [hardware.h](#).

```

01763 : uint8_t {
01764 k_rspi_mode_0 =
01765 0, /**< CPOL=0, CPHA=0 - Data sampled on rising edge, idle low. Most common mode. */
01766 k_rspi_mode_1 =
01767 1, /**< CPOL=0, CPHA=1 - Data sampled on falling edge, idle low. Used by SD cards. */
01768 k_rspi_mode_2 = 2, /**< CPOL=1, CPHA=0 - Data sampled on falling edge, idle high. Rarely used. */
01769 k_rspi_mode_3 =
01770 3 /**< CPOL=1, CPHA=1 - Data sampled on rising edge, idle high. Alternative for some devices. */
01771 } rspi_mode_t;

```

#### 8.1.4.22 uart\_channel\_t

```
enum uart_channel_t : uint8_t
```

UART channel enumeration with type safety.

Prevents accidental argument swaps and provides compile-time type checking. RX72N has SCI channels 0-12 (13 channels total).

Enumerator

|                   |  |
|-------------------|--|
| k_uart_channel_0  |  |
| k_uart_channel_1  |  |
| k_uart_channel_2  |  |
| k_uart_channel_3  |  |
| k_uart_channel_4  |  |
| k_uart_channel_5  |  |
| k_uart_channel_6  |  |
| k_uart_channel_7  |  |
| k_uart_channel_8  |  |
| k_uart_channel_9  |  |
| k_uart_channel_10 |  |
| k_uart_channel_11 |  |
| k_uart_channel_12 |  |

Definition at line 2309 of file [hardware.h](#).

```

02309 : uint8_t {
02310 k_uart_channel_0 = 0,
02311 k_uart_channel_1 = 1,

```

```

02312 k_uart_channel_2 = 2,
02313 k_uart_channel_3 = 3,
02314 k_uart_channel_4 = 4,
02315 k_uart_channel_5 = 5,
02316 k_uart_channel_6 = 6,
02317 k_uart_channel_7 = 7,
02318 k_uart_channel_8 = 8,
02319 k_uart_channel_9 = 9,
02320 k_uart_channel_10 = 10,
02321 k_uart_channel_11 = 11,
02322 k_uart_channel_12 = 12
02323 } uart_channel_t;

```

#### 8.1.4.23 uart\_defaults\_t

```
enum uart_defaults_t : uint8_t
```

Default debug UART channel (SCI9 - CY7C65213 USB bridge).

Enumerator

|                      |                                         |
|----------------------|-----------------------------------------|
| k_uart_debug_channel | Debug UART on SCI9 (PB7/TXD9, PB6/RXD9) |
|----------------------|-----------------------------------------|

Definition at line [2299](#) of file [hardware.h](#).

```

02299 : uint8_t {
02300 k_uart_debug_channel = 9 /**< Debug UART on SCI9 (PB7/TXD9, PB6/RXD9) */
02301 } uart_defaults_t;

```

### 8.1.5 Function Documentation

#### 8.1.5.1 adc\_init()

```

rx_err_t adc_init (
 const adc_unit_t unit,
 const adc_channel_t channel,
 const adc_resolution_t bits) [nodiscard]

```

Initialize ADC unit and channel.

Parameters

|    |         |                                |
|----|---------|--------------------------------|
| in | unit    | ADC unit (0 or 1)              |
| in | channel | ADC channel (0-7)              |
| in | bits    | Resolution (8, 10, or 12 bits) |

Returns

k\_rx\_ok on success, k\_rx\_err\_invalid\_arg if unit, channel, or bits is invalid, k\_rx\_err\_gpio\_conflict if pin already reserved

Initialize ADC unit and channel.

Configures the S12ADFa peripheral for A/D conversion. On first call for each unit, enables the module clock, configures resolution, and initializes the hardware. Subsequent calls to the same unit enable additional channels.

## 8.1.5.2 Initialization Behavior

| Scenario                        | Action              | Resolution    | Result               |
|---------------------------------|---------------------|---------------|----------------------|
| First init for unit             | Full init           | Set to 'bits' | k_rx_ok              |
| Subsequent init, same bits      | Enable channel only | Unchanged     | k_rx_ok              |
| Subsequent init, different bits | Error               | Unchanged     | k_rx_err_invalid_arg |

## Precondition

None (can be first call for unit)

Resolution must match if unit already initialized

## Postcondition

Module clock enabled (if first init)

Channel enabled in ADANSA register

Unit marked as initialized

## Note

NOT thread-safe: Call from initialization code only

Performance: ~1-2 us for first init, ~0.5 us for subsequent

## Example:

```
// Initialize ADC unit 0, channel 2 with 12-bit resolution
rx_err_t err = adc_init(k_adc_unit_0, k_adc_channel_2, k_adc_resolution_12bit);
if (err != k_rx_ok) {
 rx_log_error("SENSOR", "ADC init failed: %d", err);
 return err;
}
```

## See also

[adc\\_read\(\)](#) Read after initialization

[adc\\_read\\_voltage\\_mv\(\)](#) Read as voltage

## Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 3 postconditions

Definition at line 708 of file [adc.c](#).

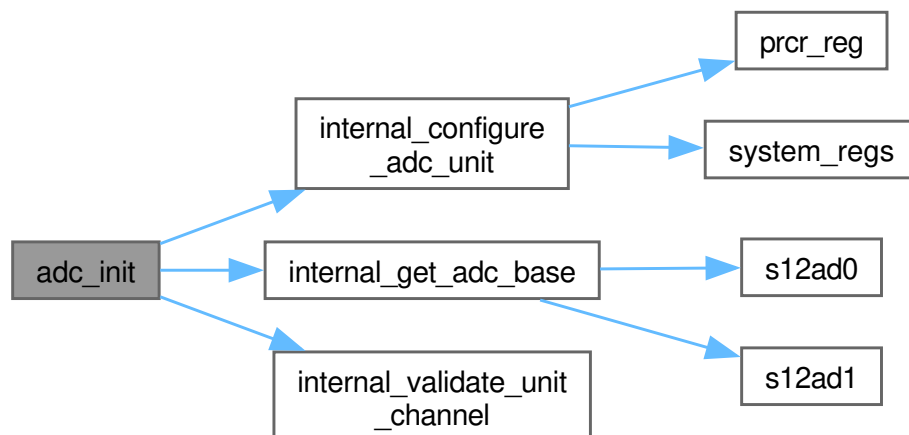
```

00709 {
00710 /* Validate parameters */
00711 rx_err_t err = internal_validate_unit_channel(unit, channel);
00712 RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00713
00714 /* Validate resolution */
00715 if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00716 bits != k_adc_resolution_12bit) {
00717 rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00718 return k_rx_err_invalid_arg;
00719 }
00720
00721 /* Get ADC base */
00722 volatile rx_s12ad_regs_t* const adc = internal_get_adc_base(unit);
00723 if (adc == nullptr) {
00724 return k_rx_err_invalid_arg;
00725 }
00726
00727 /* Initialize ADC unit if not already initialized */
00728 if (!s_adc_unit_initialized[unit]) {
00729 /* Store resolution for validation on subsequent calls */
00730 s_adc_unit_resolution[unit] = bits;
00731
00732 err = internal_configure_adc_unit(unit, adc, bits);
00733 if (err != k_rx_ok) {
00734 return err;
00735 }
00736 } else {
00737 /* Unit already initialized - verify resolution matches */
00738 if (bits != s_adc_unit_resolution[unit]) {
00739 rx_log_error(s_tag, "ADC resolution mismatch on subsequent init");
00740 return k_rx_err_invalid_arg;
00741 }
00742 }
00743
00744 /* Enable the specified channel */
00745 if ((uint8_t)channel <= k_adc_channel_adansa0_max) {
00746 adc->adansa0 |= ((uint16_t)k_adc_bit_one << channel);
00747 } else {
00748 adc->adansa1 |= ((uint16_t)k_adc_bit_one << ((uint8_t)channel - k_adc_channel_adansa1_base));
00749 }
00750
00751 rx_log_debug(s_tag, "ADC channel enabled");
00752
00753 return k_rx_ok;
00754 }

```

References [rx\\_s12ad\\_regs\\_t::adansa0](#), [rx\\_s12ad\\_regs\\_t::adansa1](#), [internal\\_configure\\_adc\\_unit\(\)](#), [internal\\_get\\_adc\\_base\(\)](#), [internal\\_validate\\_unit\\_channel\(\)](#), [k\\_adc\\_bit\\_one](#), [k\\_adc\\_channel\\_adansa0\\_max](#), [k\\_adc\\_channel\\_adansa1\\_base](#), [s\\_adc\\_unit\\_initialized](#), [s\\_adc\\_unit\\_resolution](#), and [s\\_tag](#).

Here is the call graph for this function:



### 8.1.5.3 `adc_read()`

```
rx_err_t adc_read (
 const adc_unit_t unit,
 const adc_channel_t channel,
 uint16_t * value) [nodiscard]
```

Read ADC value.

Parameters

|     |         |                            |
|-----|---------|----------------------------|
| in  | unit    | ADC unit (0 or 1)          |
| in  | channel | ADC channel (0-7)          |
| out | value   | Pointer to store ADC value |

Returns

`k_rx_ok` on success, `k_rx_err_null_ptr` if value is nullptr, `k_rx_err_invalid_arg` if unit or channel is invalid, `k_rx_err_invalid_state` if ADC unit not initialized, `k_rx_err_timeout` if conversion times out

Read ADC value.

Performs a single-scan A/D conversion and returns the raw digital value. This is a blocking operation that polls the ADST bit until conversion completes or timeout occurs.

### 8.1.5.4 Conversion Sequence

1. Validate parameters and initialization state
2. Start conversion by setting ADCSR.ADST bit
3. Poll ADCSR.ADST until cleared (conversion complete)
4. Read result from appropriate ADDRn register

### 8.1.5.5 Output Value Ranges

| Resolution | Range  | Full-Scale |
|------------|--------|------------|
| 12-bit     | 0-4095 | 3.3V       |
| 10-bit     | 0-1023 | 3.3V       |
| 8-bit      | 0-255  | 3.3V       |

Precondition

ADC unit must be initialized via `adc_init()`  
 value must point to valid `uint16_t` storage

## Postcondition

\*value contains raw ADC reading (if k\_rx\_ok)  
 ADC ready for next conversion

## Note

NOT thread-safe: Caller must synchronize if shared  
 Blocking: Waits up to ~100ms for conversion  
 Performance: ~1-10 us typical (depends on conversion time)

## Warning

Do not call from ISR (blocking wait)

## Example:

```
uint16_t raw_value;
rx_err_t err = adc_read(k_adc_unit_0, k_adc_channel_2, &raw_value);
if (err == k_rx_ok) {
 // Example: 12-bit resolution (0-4095 raw -> 0-3.3 V)
 static const float s_vref_volts = 3.3F;
 static const float s_adc_full_scale_12bit = 4095.0F;
 float voltage_v = (raw_value * s_vref_volts) / s_adc_full_scale_12bit;
}
```

## See also

[adc\\_init\(\)](#) Must be called first  
[adc\\_read\\_voltage\\_mv\(\)](#) Convenience function for voltage

## Since

Version 1.0.0

## NASA Power of 10 Compliance:

- Rule 2: [OK] Timeout loop has fixed upper bound
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 814 of file [adc.c](#).

```
00815 {
00816 /* Validate parameters */
00817 RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00818
00819 const rx_err_t err = internal_validate_unit_channel(unit, channel);
00820 RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00821
00822 /* Check if unit is initialized */
00823 if (!s_adc_unit_initialized[unit]) {
00824 rx_log_error(s_tag, "ADC unit not initialized");
00825 return k_rx_err_invalid_state;
00826 }
00827
00828 /* Get ADC base */
00829 volatile rx_s12ad_regs_t* const adc = internal_get_adc_base(unit);
00830 if (adc == nullptr) {
00831 return k_rx_err_invalid_arg;
00832 }
00833 }
```

```

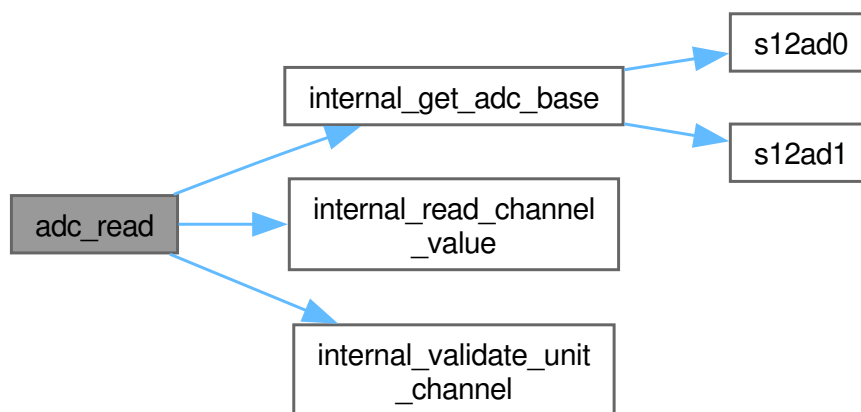
00834 /* Start single-scan conversion */
00835 adc->adcsr = k_adc_adcsr_adst;
00836
00837 /* Wait for conversion to complete (poll ADCSR.ADST bit) */
00838 uint32_t timeout = k_adc_timeout_ms * k_adc_timeout_multiplier;
00839 while ((adc->adcsr & k_adc_adcsr_adst) != k_adc_adcsr_idle && timeout > k_adc_timeout_expired) {
00840 timeout--;
00841 }
00842
00843 if (timeout == k_adc_timeout_expired) {
00844 rx_log_error(s_tag, "ADC conversion timeout");
00845 return k_rx_err_timeout;
00846 }
00847
00848 /* Read conversion result from appropriate data register */
00849 return internal_read_channel_value(adc, channel, value);
00850 }

```

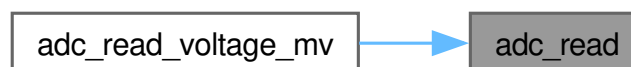
References [rx\\_s12ad\\_regs\\_t::adcsr](#), [internal\\_get\\_adc\\_base\(\)](#), [internal\\_read\\_channel\\_value\(\)](#), [internal\\_validate\\_unit\\_channel\(\)](#), [k\\_adc\\_adcsr\\_adst](#), [k\\_adc\\_adcsr\\_idle](#), [k\\_adc\\_timeout\\_expired](#), [k\\_adc\\_timeout\\_ms](#), [k\\_adc\\_timeout\\_multiplier](#), [s\\_adc\\_unit\\_initialized](#), and [s\\_tag](#).

Referenced by [adc\\_read\\_voltage\\_mv\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.6 `adc_read_voltage_mv()`

```

rx_err_t adc_read_voltage_mv (
 const adc_unit_t unit,
 adc_channel_t channel,
 adc_resolution_t bits,
 uint32_t * voltage_mv) [nodiscard]

```

Read ADC value and convert to millivolts.



## Parameters

|     |            |                                            |
|-----|------------|--------------------------------------------|
| in  | unit       | ADC unit (0 or 1)                          |
| in  | channel    | ADC channel (0-7)                          |
| in  | bits       | Resolution used during init (8, 10, or 12) |
| out | voltage_mv | Pointer to store voltage in millivolts     |

## Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if voltage\_mv is nullptr, k\_rx\_err\_invalid\_arg if unit or channel is invalid, k\_rx\_err\_invalid\_state if ADC unit not initialized, k\_rx\_err\_timeout if conversion times out

Performs an A/D conversion and scales the result to millivolts. This is a convenience function that calls [adc\\_read\(\)](#) and applies the voltage calculation formula.

## 8.1.5.7 Voltage Calculation Formula

$$V_{mV} = \frac{V_{raw} \times 3300}{2^{bits} - 1}$$

## 8.1.5.8 Example Calculations

| Resolution | Raw Value | Voltage (mV) | Calculation        |
|------------|-----------|--------------|--------------------|
| 12-bit     | 2048      | 1650         | 2048 x 3300 / 4095 |
| 10-bit     | 512       | 1650         | 512 x 3300 / 1023  |
| 8-bit      | 128       | 1655         | 128 x 3300 / 255   |

## Precondition

ADC unit must be initialized via [adc\\_init\(\)](#)

bits must match the resolution used in [adc\\_init\(\)](#)

## Postcondition

\*voltage\_mv contains voltage in millivolts (0-3300)

ADC ready for next conversion

## Note

NOT thread-safe: Caller must synchronize if shared

Blocking: Waits for conversion (see [adc\\_read](#))

Reference: Uses 3.3V reference (k\_adc\_reference\_voltage\_mv = 3300)

Example:

```
// Read motor current sense (DRV8263H IPROPI output)
uint32_t voltage_mv;
rx_err_t err = adc_read_voltage_mv(k_adc_unit_0, k_adc_channel_0,
 k_adc_resolution_12bit, &voltage_mv);

if (err == k_rx_ok) {
 // voltage_mv: 0-3300 mV
 // For DRV8263H-Q1 IPROPI (202 uA/A mirror, 5100 Ohm sense):
 // I_motor = V_IPROPI / k_ipropi_divisor
 // (where k_ipropi_divisor = 1.0302 = 202e-6 * 5100)
 // Fixed-point: scale = 10000, divisor = 10302 (1.0302 * 10000)
 enum : uint32_t { k_scale = 10000, k_divisor = 10302 };
 uint32_t current_ma = (voltage_mv * k_scale) / k_divisor;
 rx_log_info("ADC", "Current: %lu mA", current_ma);
}
```

See also

[adc\\_read\(\)](#) Lower-level raw value read

[adc\\_init\(\)](#) Initialize before reading

Since

Version 1.0.0

NASA Power of 10 Compliance:

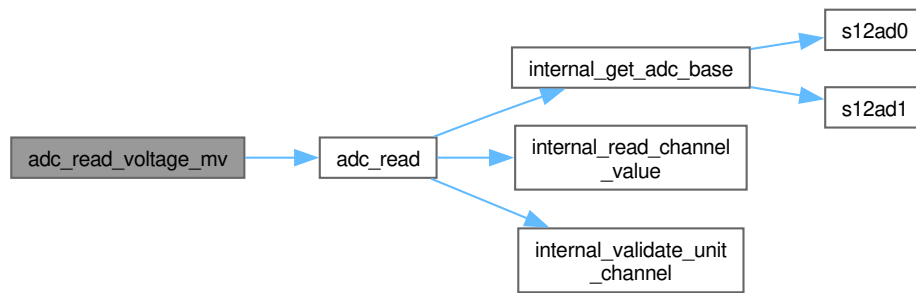
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 912 of file [adc.c](#).

```
00916 {
00917 /* Parameter order: unit, channel, resolution bits. */
00918 RX_CHECK_NULL_PTR(voltage_mv, s_tag, "Voltage pointer is nullptr");
00919
00920 if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00921 bits != k_adc_resolution_12bit) {
00922 rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00923 return k_rx_err_invalid_arg;
00924 }
00925
00926 /* Validate that passed resolution matches unit's configured resolution */
00927 if (unit > k_adc_unit_1) {
00928 rx_log_error(s_tag, "Invalid ADC unit");
00929 return k_rx_err_invalid_arg;
00930 }
00931
00932 if (bits != s_adc_unit_resolution[unit]) {
00933 rx_log_error(s_tag, "ADC resolution mismatch in voltage read");
00934 return k_rx_err_invalid_arg;
00935 }
00936
00937 /* Read raw ADC value */
00938 uint16_t raw_value = k_adc_raw_value_reset;
00939 const rx_err_t err = adc_read(unit, channel, &raw_value);
00940 RX_RETURN_ON_ERROR(err, s_tag, "ADC read failed");
00941
00942 /* Calculate voltage (using ADC reference voltage) */
00943 const uint32_t max_value = ((uint32_t)k_adc_bit_one << bits) - k_adc_bit_one;
00944 *voltage_mv = ((uint32_t)raw_value * k_adc_reference_voltage_mv) / max_value;
00945
00946 return k_rx_ok;
00947 }
```

References [adc\\_read\(\)](#), [k\\_adc\\_bit\\_one](#), [k\\_adc\\_raw\\_value\\_reset](#), [k\\_adc\\_reference\\_voltage\\_mv](#), [k\\_adc\\_unit\\_1](#), [s\\_adc\\_unit\\_resolution](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.9 gpio\_read()

```

rx_err_t gpio_read (
 const rx_port_pin_t pin,
 bool * value) [nodiscard]

```

Read current logic level of GPIO pin.

Reads the current logic level from the specified GPIO pin's Port Input Data Register (PIDR). Works for both input and output pins. For output pins, reads back the actual pin state (not the output register), which may differ if external circuitry pulls the pin.

#### 8.1.5.10 Algorithm

1. Validate pin parameter (port exists, pin number valid)
2. Validate value pointer (NULL check)
3. Extract port letter and pin number from rx\_port\_pin\_t
4. Read PIDR register for the specified port
5. Extract bit corresponding to pin number
6. Store result (true if bit=1, false if bit=0) in value pointer

#### 8.1.5.11 Performance

- Cycles: ~25 @ 240 MHz
- Time: ~104 ns
- Stack: 8 bytes (pin extraction, register read)

## Parameters

|     |       |                                                                                                                                                                                                                                                                                         |
|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in  | pin   | GPIO pin identifier from rx_port_constants.h <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Can be input or output pin</li> </ul>                                                        |
| out | value | Pointer to boolean for storing pin state <ul style="list-style-type: none"> <li>• Must point to valid bool variable</li> <li>• Set to true if pin reads logic high (VDD)</li> <li>• Set to false if pin reads logic low (GND)</li> <li>• Unchanged if function returns error</li> </ul> |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| k_rx_ok                    | Pin level successfully read and stored in *value |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N)               |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)                |
| k_rx_err_null_ptr          | value pointer is nullptr                         |

## Precondition

value pointer must be valid (not nullptr)

## Postcondition

\*value contains current pin state if function succeeds

\*value unchanged if function returns error

## Note

Thread Safety: NOT thread-safe. Reading PIDR register is atomic but value pointer access is not protected.

Output Pins: For output pins, this reads the actual pin level (PIDR), not the output register (PODR). This detects external pull-up/pull-down or short circuits.

Input Pins: Ensure pin is configured as input via [gpio\\_set\\_input\(\)](#) before reading for sensor/button inputs.

## Example: Reading Button State

```
#include "hardware.h"

void check_button(void) {
 // Button connected to Port E Pin 5 (active low with pull-up)
 bool button_state;
 rx_err_t err = gpio_read(k_port_e_pin_5, &button_state);

 if (err != k_rx_ok) {
 rx_log_error_val("BUTTON", "Failed to read button", err);
 return;
 }

 if (!button_state) { // Active low (button pressed)
 rx_log_info("BUTTON", "Button pressed");
 handle_button_press();
 }
}
```

## Example: Verifying Output State

```
// Set motor enable high
gpio_write_high(k_port_c_pin_0);

// Verify pin actually went high (detects short circuit)
bool actual_state;
gpio_read(k_port_c_pin_0, &actual_state);
if (!actual_state) {
 rx_log_error("MOTOR", "Enable pin stuck low - possible short circuit");
 emergency_shutdown();
}
```

## Example: Polling Multiple Inputs

```
typedef struct {
 rx_port_pin_t pin;
 const char* name;
} sensor_input_t;

void poll_sensors(void) {
 const sensor_input_t sensors[] = {
 {k_port_a_pin_0, "LIMIT_SWITCH_1"},
 {k_port_a_pin_1, "LIMIT_SWITCH_2"},
 {k_port_a_pin_2, "ESTOP"}
 };

 for (uint8_t i = 0; i < 3; i++) {
 bool state;
 if (gpio_read(sensors[i].pin, &state) == k_rx_ok) {
 rx_log_debug_val(sensors[i].name, "State", state ? 1 : 0);
 }
 }
}
```

See also

[gpio\\_set\\_input\(\)](#) Configure pin as digital input  
[gpio\\_set\\_output\(\)](#) Configure pin as digital output  
[gpio\\_write\\_high\(\)](#) Set output pin to logic high  
[gpio\\_write\\_low\(\)](#) Set output pin to logic low

Read current logic level of GPIO pin.

Reads the current logic level of a GPIO pin. Uses the PIDR (Port Input Data Register) which reflects the actual voltage level on the pin, regardless of whether the pin is configured as input or output.

## 8.1.5.12 Register Operation

```
value = (PIDR & (1 « pin)) != 0
```

### 8.1.5.13 Input vs Output Reading

- Input pin: PIDR reflects external voltage on pin
- Output pin: PIDR reflects actual pin state (may differ from PODR if pin is shorted or overdriven)

### 8.1.5.14 Voltage Thresholds (RX72N)

| Level     | Voltage Range | Returned Value |
|-----------|---------------|----------------|
| Low       | 0V to ~0.8V   | false          |
| High      | ~2.0V to VCC  | true           |
| Undefined | 0.8V to 2.0V  | Unpredictable  |

#### Precondition

value must be non-nullptr

Pin should be configured as input for external signals

#### Postcondition

\*value contains current pin level (true=high, false=low)

No hardware state changes

#### Note

Thread-safe: Read-only operation

Performance: ~100 ns @ 240 MHz

ISR-safe: Can be called from interrupt handlers

#### Warning

Floating inputs may return unpredictable values

Use pull-up/pull-down for stable readings on open pins

#### Example:

```
// Read button state (active low with pull-up)
bool button_pressed;
rx_err_t err = gpio_read(RX_PORT_PIN(2, 0), &button_pressed);

if (err == k_rx_ok) {
 // Invert because button is active low
 if (!button_pressed) {
 rx_log_info("BTN", "Button pressed");
 handle_button_press();
 }
}
```

Example: Debounced button read

```
bool debounced_read(rx_port_pin_t pin) {
 bool first, second;
 gpio_read(pin, &first);
 tx_thread_sleep(1); // 1 tick delay
 gpio_read(pin, &second);
 return (first == second) ? first : false; // Only return if stable
}
```

See also

[gpio\\_set\\_input\(\)](#) Configure pin as input first  
[gpio\\_write\\_high\(\)](#) Set output high  
[gpio\\_write\\_low\(\)](#) Set output low

Since

Version 1.0.0

NASA Power of 10 Compliance:

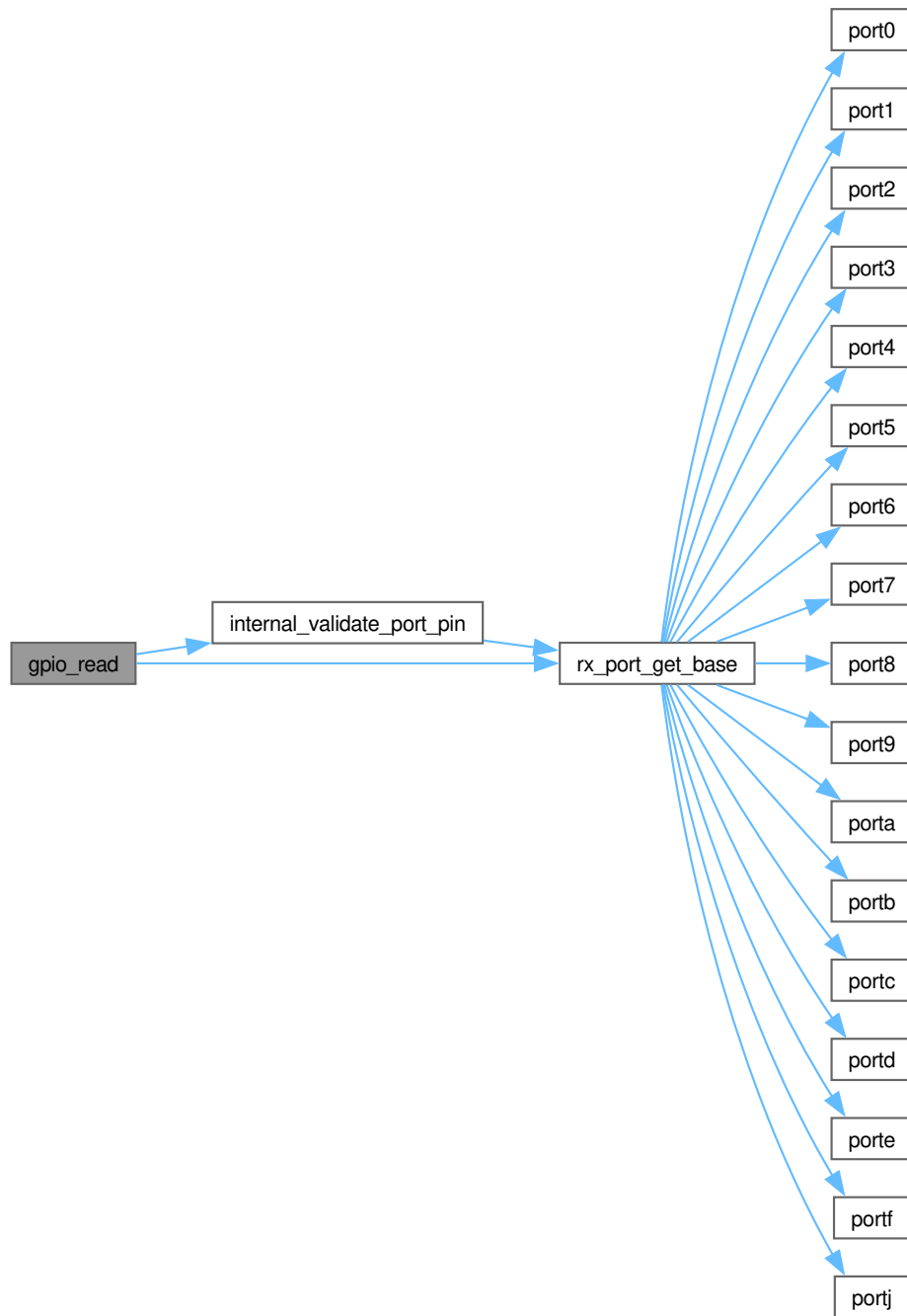
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 803 of file [gpio.c](#).

```
00804 {
00805 /* Check null pointer */
00806 RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00807
00808 /* Extract port and pin number from rx_port_pin_t */
00809 const uint8_t port = rx_port_from_pin(pin);
00810 const uint8_t pin_num = rx_pin_from_pin(pin);
00811
00812 /* Validate parameters */
00813 const rx_err_t err = internal_validate_port_pin(port, pin_num);
00814 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00815
00816 /* Get port base address */
00817 const volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00818
00819 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00820 *value = (port_base->pidr & pin_mask) != k_gpio_bit_clear;
00821
00822 return k_rx_ok;
00823 }
```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_clear](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::pidr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.15 gpio\_set\_input()

```
rx_err_t gpio_set_input (
 const rx_port_pin_t pin) [nodiscard]
```

Configure GPIO pin as digital input.

Configures the specified GPIO pin as a digital input with high-impedance state. Performs full hardware initialization including port direction register (PDR), port mode register (PMR), and input buffer control. Validates pin assignment and checks for peripheral conflicts.



## 8.1.5.16 Algorithm

1. Validate pin parameter (port exists, pin number valid)
2. Check pin is not already reserved by another peripheral
3. Unlock port write protection (PWPR register)
4. Set PMR bit to 0 (GPIO mode, not peripheral function)
5. Set PDR bit to 0 (input direction)
6. Enable input buffer (ICR register)
7. Lock port write protection
8. Mark pin as reserved in allocation tracker

## Parameters

| in | pin | GPIO pin identifier from rx_port_constants.h                                                                                                                                                                                                                             |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     | <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Example: k_port_e_pin_5 (Port E, Pin 5)</li> <li>• Range: All valid combinations defined in rx_port_constants.h</li> </ul> |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| k_rx_ok                    | Pin successfully configured as input                        |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N)                          |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)                           |
| k_rx_err_gpio_conflict     | Pin already reserved by UART, SPI, I2C, or other peripheral |

## Precondition

- Pin must not be currently in use by another peripheral
- Pin must correspond to a valid physical pin on the RX72N package

## Postcondition

- Pin configured as high-impedance input with input buffer enabled
- Pin marked as reserved in internal allocation tracker

Note

Thread Safety: NOT thread-safe. Do not call concurrently from multiple threads.

Example: Button Input

```
#include "hardware.h"

void button_init(void) {
 rx_err_t err = gpio_set_input(k_port_e_pin_5);
 if (err != k_rx_ok) {
 rx_log_error_val("BTN", "Failed to configure button pin", err);
 return;
 }
 rx_log_info("BTN", "Button input configured");
}
```

See also

[gpio\\_set\\_output\(\)](#) Configure pin as digital output

[gpio\\_read\(\)](#) Read pin logic level

Since

Version 1.0.0

Configures a GPIO pin for digital input mode. This function performs:

1. Parameter validation (port and pin number)
2. Pin reservation via the global pin validator (conflict detection)
3. PMR register clear (select GPIO mode instead of peripheral function)
4. PDR register clear (configure as input direction)

#### 8.1.5.17 Register Operations

```
PMR &= ~(1 « pin) // Clear bit: Select GPIO mode (not peripheral)
PDR &= ~(1 « pin) // Clear bit: Configure as input
```

#### 8.1.5.18 Pull-up Configuration

This function does NOT configure internal pull-up resistors. To enable pull-ups, additional configuration of the PCR register is required.

### 8.1.5.19 Reading Input State

After configuration, use `gpio_read()` to read the current pin state. The PIDR register is read for input state, not PODR.

#### Precondition

- Pin must be a valid GPIO-capable pin on RX72N
- System infrastructure should be initialized (for pin validation)

#### Postcondition

- Pin is configured as GPIO input
- PMR bit for pin is cleared (GPIO mode)
- PDR bit for pin is cleared (input direction)
- Pin is reserved in validator (if available)

#### Note

- Thread-safe: Pin reservation uses mutex
- Performance: ~1-2 us @ 240 MHz (includes reservation)
- Pull-ups: Not enabled by this function (use PCR register)

#### Warning

- Floating inputs may read unpredictable values - use pull-up/pull-down
- Do not call from ISR (pin reservation may block)

#### Example:

```
// Configure button pin as input
rx_err_t err = gpio_set_input(RX_PORT_PIN(2, 0)); // Port 2, Pin 0
if (err != k_rx_ok) {
 rx_log_error("BTN", "Failed to configure button pin");
 return err;
}

// Read button state
bool pressed;
err = gpio_read(RX_PORT_PIN(2, 0), &pressed);
if (err == k_rx_ok && pressed) {
 rx_log_info("BTN", "Button pressed");
}
```

#### See also

- `gpio_set_output()` Configure pin as output
- `gpio_read()` Read input pin state
- `RX_PORT_PIN()` Macro to create rx\_port\_pin\_t

#### Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 4 postconditions

Definition at line 481 of file [gpio.c](#).

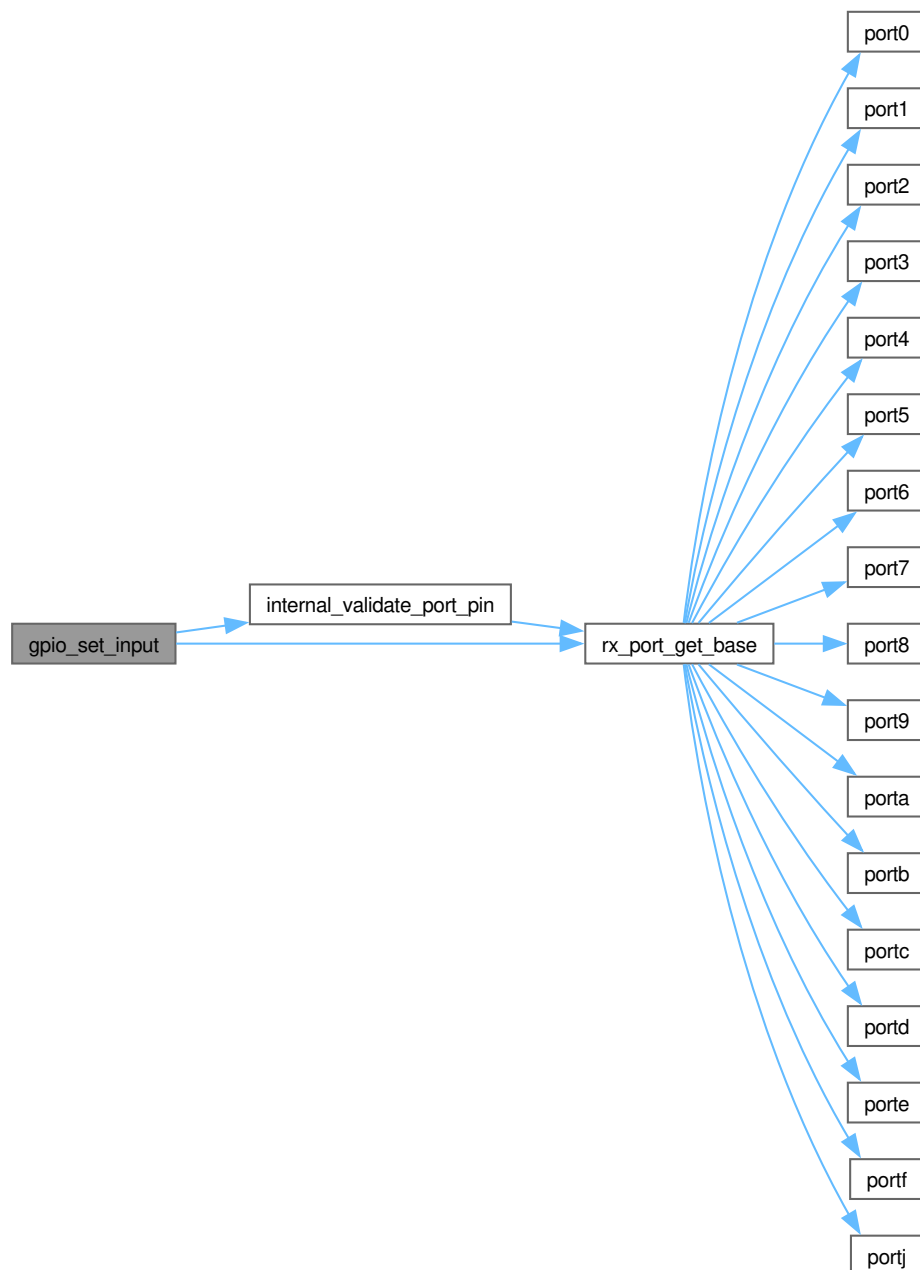
```

00482 {
00483 /* Extract port and pin number from rx_port_pin_t */
00484 const uint8_t port = rx_port_from_pin(pin);
00485 const uint8_t pin_num = rx_pin_from_pin(pin);
00486
00487 /* Validate parameters */
00488 rx_err_t err = internal_validate_port_pin(port, pin_num);
00489 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00490
00491 /* Reserve pin through global pin validator */
00492 const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00493 if (pin_iface != nullptr) {
00494 err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_IN");
00495 if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00496 /* Allow conflict (pin already reserved), but fail on other errors */
00497 RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00498 }
00499 }
00500
00501 /* Get port base address */
00502 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00503
00504 /* Set as GPIO mode */
00505 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00506 port_base->pmr &= (uint8_t)~pin_mask;
00507
00508 /* Set as input */
00509 port_base->pdr &= (uint8_t)~pin_mask;
00510
00511 rx_log_debug(s_tag, "Pin configured as input");
00512
00513 return k_rx_ok;
00514 }

```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::pdr](#), [rx\\_port\\_regs\\_t::pmr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.20 gpio\_set\_output()

```
rx_err_t gpio_set_output (
 const rx_port_pin_t pin) [nodiscard]
```

Configure GPIO pin as digital output.

Configures the specified GPIO pin as a digital output with push-pull drive. Performs full hardware initialization including port direction register (PDR), port mode register (PMR), and open-drain control (ODR). Validates pin assignment and checks for peripheral conflicts.

## 8.1.5.21 Algorithm

1. Validate pin parameter (port exists, pin number valid)
2. Check pin is not already reserved by another peripheral
3. Unlock port write protection (PWPR register)
4. Set PMR bit to 0 (GPIO mode, not peripheral function)
5. Set PDR bit to 1 (output direction)
6. Ensure ODR is 0 (push-pull, not open-drain)
7. Lock port write protection
8. Mark pin as reserved in allocation tracker

## 8.1.5.22 Performance

- Cycles: ~120 @ 240 MHz
- Time: ~500 ns
- Stack: 8 bytes (pin validation, temporary registers)

## Parameters

|    |     |                                                                                                                                                                                                                                                                                                                       |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in | pin | GPIO pin identifier from rx_port_constants.h <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Example: k_port_c_pin_0 (Port C, Pin 0)</li> <li>• Range: All valid combinations defined in rx_port_constants.h</li> </ul> |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| k_rx_ok                    | Pin successfully configured as output                       |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N)                          |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)                           |
| k_rx_err_gpio_conflict     | Pin already reserved by UART, SPI, I2C, or other peripheral |

## Precondition

Pin must not be currently in use by another peripheral

## Postcondition

Pin configured as push-pull output, initial state undefined (use `gpio_write_*` to set level)  
 Pin marked as reserved in internal allocation tracker

## Note

Thread Safety: NOT thread-safe. Do not call concurrently from multiple threads.  
 Initial State: Output level is undefined after configuration. Call [gpio\\_write\\_high\(\)](#) or [gpio\\_write\\_low\(\)](#) to set initial state.  
 Peripheral Conflicts: Automatically detects conflicts with UART, SPI, I2C pin assignments.

## Warning

Changing pin direction while peripheral is active may cause undefined behavior.

## Example: Motor Enable Pin

```
#include "hardware.h"

void motor_init(void) {
 // Configure motor enable pin as output
 rx_err_t err = gpio_set_output(k_port_c_pin_0);
 if (err != k_rx_ok) {
 rx_log_error_val("MOTOR", "Failed to configure enable pin", err);
 return;
 }

 // Set initial state (motors disabled)
 gpio_write_low(k_port_c_pin_0);
 rx_log_info("MOTOR", "Enable pin configured");
}
```

## Example: Multiple GPIO Outputs

```
void configure_leds(void) {
 const rx_port_pin_t led_pins[] = {
 k_port_e_pin_5, // LED1
 k_port_e_pin_6, // LED2
 k_port_e_pin_7, // LED3
 };

 for (uint8_t i = 0; i < 3; i++) {
 rx_err_t err = gpio_set_output(led_pins[i]);
 if (err != k_rx_ok) {
 rx_log_error_hex("GPIO", "LED init failed", i, 2);
 } else {
 gpio_write_low(led_pins[i]); // LEDs off initially
 }
 }
}
```

## Example: Error Handling

```
rx_err_t setup_motor_control_pins(void) {
 // Try to configure direction pin
 rx_err_t err = gpio_set_output(k_port_c_pin_1);
 if (err == k_rx_err_gpio_conflict) {
 rx_log_error("MOTOR", "Pin conflict - already used by SPI?");
 return err;
 } else if (err != k_rx_ok) {
 rx_log_error_val("MOTOR", "Unexpected error", err);
 return err;
 }

 return k_rx_ok;
}
```

See also

[gpio\\_set\\_input\(\)](#) Configure pin as digital input  
[gpio\\_write\\_high\(\)](#) Set output pin to logic high  
[gpio\\_write\\_low\(\)](#) Set output pin to logic low  
[gpio\\_toggle\(\)](#) Toggle output pin state  
[rx\\_port\\_constants.h](#) for complete pin definitions

Configures a GPIO pin for digital output mode. This function performs:

1. Parameter validation (port and pin number)
2. Pin reservation via the global pin validator (conflict detection)
3. PMR register clear (select GPIO mode instead of peripheral function)
4. PDR register set (configure as output direction)

#### 8.1.5.23 Register Operations

```
PMR &= ~(1 « pin) // Clear bit: Select GPIO mode (not peripheral)
PDR |= (1 « pin) // Set bit: Configure as output
```

#### 8.1.5.24 Pin Reservation

The function attempts to reserve the pin through `rx_infrastructure_get_pin_interface()`. If the pin is already reserved (`k_rx_err_gpio_conflict`), the function continues without error - this allows reconfiguration of already-owned pins.

#### 8.1.5.25 Initial Output State

This function does NOT set the initial output level. After calling [gpio\\_set\\_output\(\)](#), call [gpio\\_write\\_high\(\)](#) or [gpio\\_write\\_low\(\)](#) to set the desired level.

Precondition

Pin must be a valid GPIO-capable pin on RX72N  
 System infrastructure should be initialized (for pin validation)

Postcondition

Pin is configured as GPIO output  
 PMR bit for pin is cleared (GPIO mode)  
 PDR bit for pin is set (output direction)  
 Pin is reserved in validator (if available)

Note

Thread-safe: Pin reservation uses mutex, register writes are not atomic  
 Performance: ~1-2 us @ 240 MHz (includes reservation)  
 Idempotent: Safe to call multiple times on same pin



## Warning

Output level is undefined after this call - set with `gpio_write_*`  
 Do not call from ISR (pin reservation may block)

## Example:

```
// Configure LED pin as output
rx_err_t err = gpio_set_output(RX_PORT_PIN(0, 3)); // Port 0, Pin 3
if (err != k_rx_ok) {
 rx_log_error("LED", "Failed to configure LED pin");
 return err;
}

// Set initial state to low (LED off)
gpio_write_low(RX_PORT_PIN(0, 3));

// Toggle LED
gpio_toggle(RX_PORT_PIN(0, 3));
```

## See also

[gpio\\_set\\_input\(\)](#) Configure pin as input  
[gpio\\_write\\_high\(\)](#) Set output high after configuration  
[gpio\\_write\\_low\(\)](#) Set output low after configuration  
[RX\\_PORT\\_PIN\(\)](#) Macro to create `rx_port_pin_t`

## Since

Version 1.0.0

## NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 4 postconditions

Definition at line 375 of file [gpio.c](#).

```
00376 {
00377 /* Extract port and pin number from rx_port_pin_t */
00378 const uint8_t port = rx_port_from_pin(pin);
00379 const uint8_t pin_num = rx_pin_from_pin(pin);
00380
00381 /* Validate parameters */
00382 rx_err_t err = internal_validate_port_pin(port, pin_num);
00383 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00384
00385 /* Reserve pin through global pin validator */
00386 const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00387 if (pin_iface != nullptr) {
00388 err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_OUT");
00389 if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00390 /* Allow conflict (pin already reserved), but fail on other errors */
00391 RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00392 }
00393 }
00394
00395 /* Get port base address */
00396 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00397
00398 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00399
00400 /* Set as GPIO mode (not peripheral) */
00401 port_base->pmr &= (uint8_t)~pin_mask;
00402
00403 /* Set as output */
00404 port_base->pdr |= pin_mask;
00405
00406 rx_log_debug(s_tag, "Pin configured as output");
```

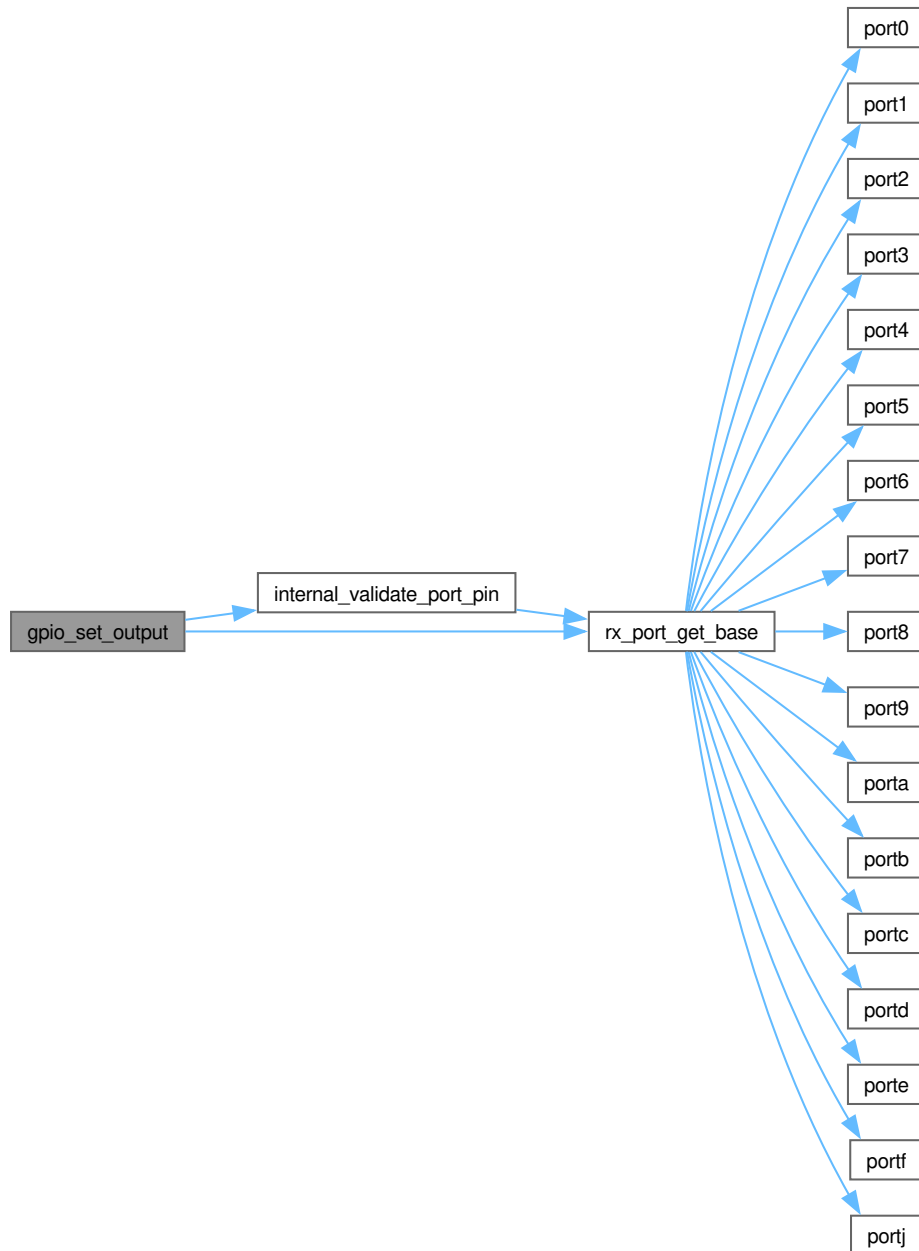
```

00407
00408 return k_rx_ok;
00409 }

```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::pdr](#), [rx\\_port\\_regs\\_t::pmr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.26 gpio\_toggle()

```

rx_err_t gpio_toggle (
 const rx_port_pin_t pin) [nodiscard]

```

Toggle GPIO output pin state (high to low, or low to high).

Inverts the current output level of the specified GPIO pin by XOR-ing the corresponding bit in the Port Output Data Register (PODR). If the pin is currently high it becomes low, and vice versa. The pin must have been previously configured as output via [gpio\\_set\\_output\(\)](#).

## 8.1.5.27 Algorithm

1. Validate pin parameter (port exists, pin number valid)
2. Extract port letter and pin number from rx\_port\_pin\_t
3. XOR corresponding bit in PODR register (read-modify-write)

## Parameters

| in | pin | GPIO pin identifier from rx_port_constants.h                                                                                                                                                                                |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     | <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Must be configured as output via <a href="#">gpio_set_output()</a></li> </ul> |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                    |
|----------------------------|------------------------------------|
| k_rx_ok                    | Pin successfully toggled           |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N) |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)  |

## Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)  
 Pin port and number must be within valid hardware range

## Postcondition

Pin output level is inverted from its previous state  
 PODR register bit for the pin is toggled

## Note

Thread Safety: NOT thread-safe. Concurrent toggles or writes to pins on the same port may cause read-modify-write races on the PODR register.

## Example: LED Heartbeat

```
// Toggle heartbeat LED every 500 ms
void heartbeat_task(void) {
 gpio_set_output(k_port_e_pin_5);
 while (1) {
 gpio_toggle(k_port_e_pin_5);
 tx_thread_sleep(50); // 500 ms at 100 Hz tick
 }
}
```

See also

[gpio\\_write\\_high\(\)](#) Set output pin to logic high  
[gpio\\_write\\_low\(\)](#) Set output pin to logic low  
[gpio\\_set\\_output\(\)](#) Configure pin as output first

Since

Version 1.0.0

Toggle GPIO output pin state (high to low, or low to high).

Inverts the current output state of a GPIO pin. If the pin is currently high, it becomes low. If currently low, it becomes high. Uses XOR operation for efficient toggling.

#### 8.1.5.28 Register Operation

```
PODR ^= (1 << pin) // XOR bit: Toggle output
```

#### 8.1.5.29 Atomicity Warning

This operation is a read-modify-write sequence and is NOT atomic. If an interrupt modifies the same port register between the read and write, the interrupt's change may be lost.

For thread-safe toggling, use explicit [write\\_high/write\\_low](#) with external synchronization, or disable interrupts around the toggle.

Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)  
Port and pin numbers must be valid

Postcondition

PODR bit for pin is inverted  
Pin output level is opposite of previous state

Note

NOT thread-safe: Read-modify-write operation  
NOT atomic: Can race with ISRs on same port  
Performance: ~150 ns @ 240 MHz

Warning

Race condition possible with ISRs modifying same port  
Use [gpio\\_write\\_high/low](#) with sync for critical applications

Example:

```
// Blink LED in loop
while (running) {
 gpio_toggle(RX_PORT_PIN(0, 3)); // Toggle LED
 tx_thread_sleep(50); // 50ms delay (20 Hz blink)
}
```

See also

[gpio\\_write\\_high\(\)](#) Set output high explicitly  
[gpio\\_write\\_low\(\)](#) Set output low explicitly  
[gpio\\_read\(\)](#) Read current output state

Since

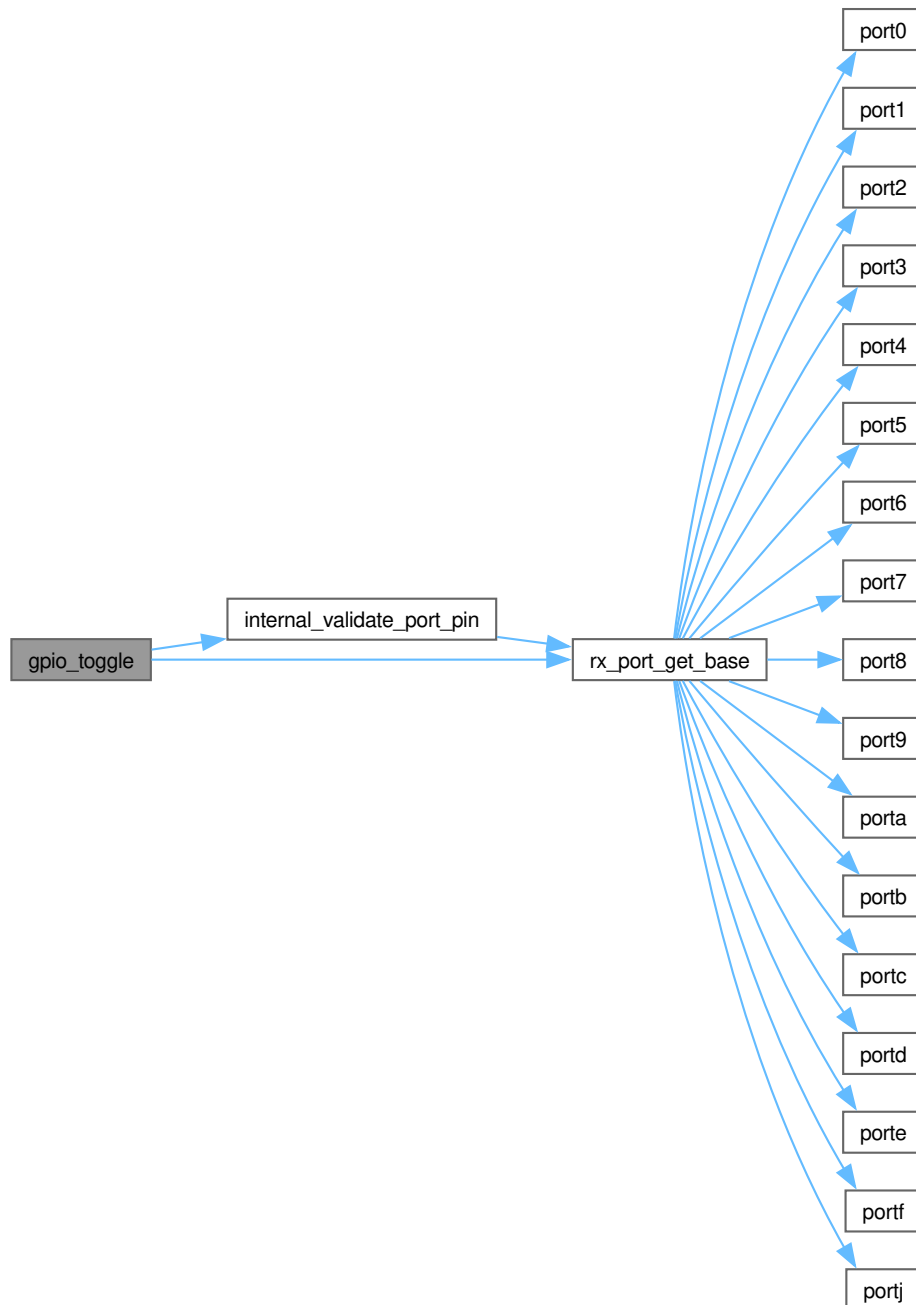
Version 1.0.0

Definition at line 706 of file [gpio.c](#).

```
00707 {
00708 /* Extract port and pin number from rx_port_pin_t */
00709 const uint8_t port = rx_port_from_pin(pin);
00710 const uint8_t pin_num = rx_pin_from_pin(pin);
00711
00712 /* Validate parameters */
00713 const rx_err_t err = internal_validate_port_pin(port, pin_num);
00714 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00715
00716 /* Get port base address */
00717 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00718
00719 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00720 port_base->podr ^= pin_mask;
00721
00722 return k_rx_ok;
00723 }
```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::podr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.30 gpio\_write\_high()

```
rx_err_t gpio_write_high (
 const rx_port_pin_t pin) [nodiscard]
```

Set GPIO output pin to logic high (VDD).

Drives the specified GPIO pin to logic high by setting the corresponding bit in the Port Output Data Register (PODR). The pin must have been previously configured as output via [gpio\\_set\\_output\(\)](#). Writing to an input pin has no effect on the physical pin state.

## 8.1.5.31 Algorithm

1. Validate pin parameter (port exists, pin number valid)
2. Extract port letter and pin number from rx\_port\_pin\_t
3. Set corresponding bit in PODR register (atomic read-modify-write)

## Parameters

| in | pin | GPIO pin identifier from rx_port_constants.h                                                                                                                                                                                |
|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     | <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Must be configured as output via <a href="#">gpio_set_output()</a></li> </ul> |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                    |
|----------------------------|------------------------------------|
| k_rx_ok                    | Pin successfully driven high       |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N) |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)  |

## Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)  
 Pin port and number must be within valid hardware range

## Postcondition

Pin output level is logic high (VDD, typically 3.3V)  
 PODR register bit for the pin is set to 1

## Note

Thread Safety: NOT thread-safe. Concurrent writes to pins on the same port may cause read-modify-write races on the PODR register.

## Example: Enable Motor

```
// Enable motor driver (active high)
rx_err_t err = gpio_write_high(k_port_c_pin_0);
if (err != k_rx_ok) {
 rx_log_error_val("MOTOR", "Failed to enable motor", err);
}
```

See also

[gpio\\_write\\_low\(\)](#) Set output pin to logic low  
[gpio\\_toggle\(\)](#) Toggle output pin state  
[gpio\\_set\\_output\(\)](#) Configure pin as output first

Since

Version 1.0.0

Set GPIO output pin to logic high (VDD).

Sets a GPIO output pin to logic high level. On the RX72N, this means the pin outputs VCC (typically 3.3V). The pin must be configured as an output before calling this function.

#### 8.1.5.32 Register Operation

```
PODR |= (1 « pin) // Set bit: Output high
```

#### 8.1.5.33 Timing

The output level changes on the next peripheral clock cycle after the register write completes. GPIO toggle rate is limited by bus speed.

Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)  
Port and pin numbers must be valid

Postcondition

PODR bit for pin is set (output high)  
Pin outputs VCC level (typically 3.3V)

Note

NOT thread-safe: Direct register write without synchronization  
Performance: ~100 ns @ 240 MHz  
ISR-safe: Can be called from interrupt handlers

Warning

No check that pin is configured as output  
Calling on input pin has no visible effect



Example:

```
// Turn on LED (active high)
rx_err_t err = gpio_write_high(RX_PORT_PIN(0, 3));
if (err != k_rx_ok) {
 // Handle error
}
```

See also

[gpio\\_write\\_low\(\)](#) Set output low  
[gpio\\_toggle\(\)](#) Toggle output state  
[gpio\\_set\\_output\(\)](#) Configure pin as output first

Since

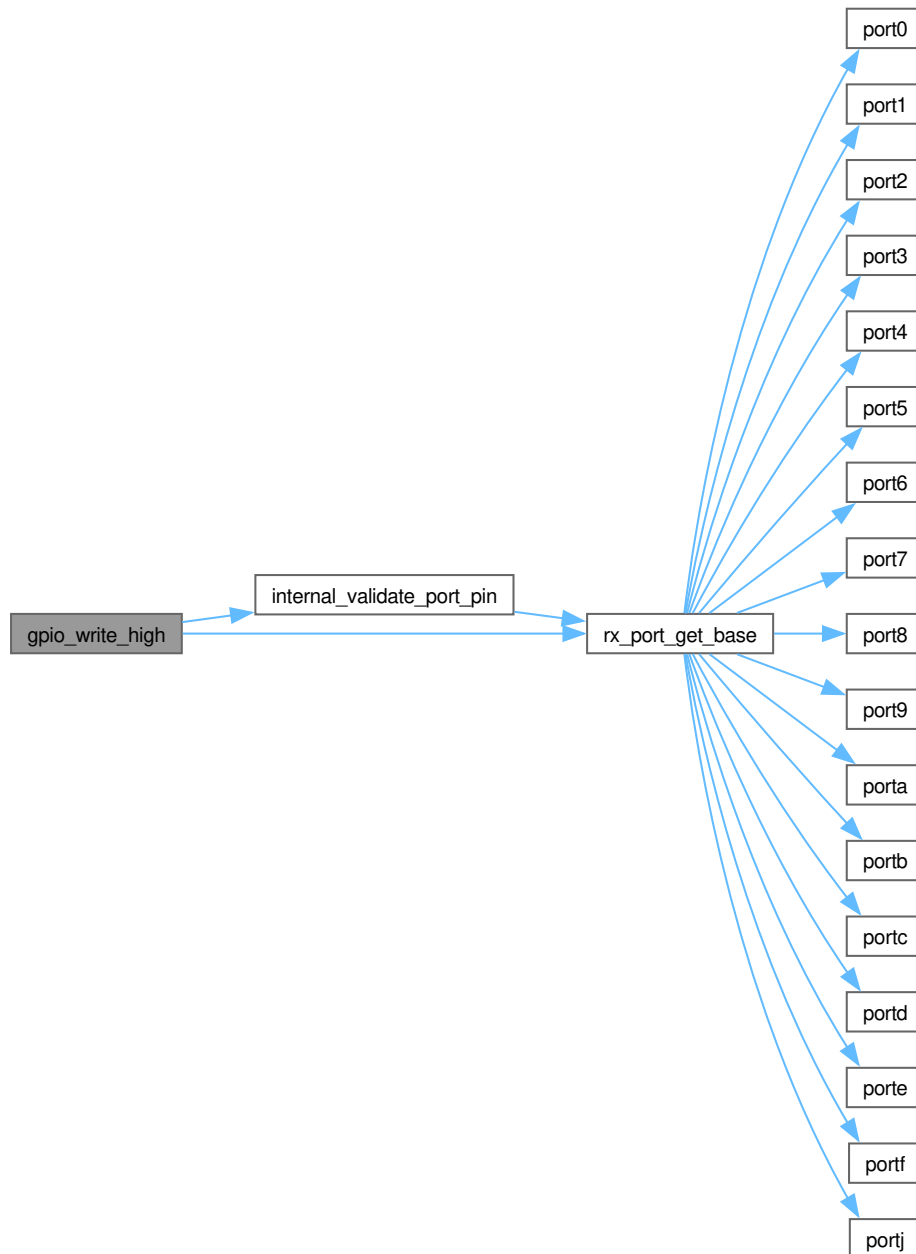
Version 1.0.0

Definition at line 565 of file [gpio.c](#).

```
00566 {
00567 /* Extract port and pin number from rx_port_pin_t */
00568 const uint8_t port = rx_port_from_pin(pin);
00569 const uint8_t pin_num = rx_pin_from_pin(pin);
00570
00571 /* Validate parameters */
00572 const rx_err_t err = internal_validate_port_pin(port, pin_num);
00573 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00574
00575 /* Get port base address */
00576 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00577
00578 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00579 port_base->podr |= pin_mask;
00580
00581 return k_rx_ok;
00582 }
```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::podr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.34 gpio\_write\_low()

```
rx_err_t gpio_write_low (
 const rx_port_pin_t pin) [nodiscard]
```

Set GPIO output pin to logic low (GND).

Drives the specified GPIO pin to logic low by clearing the corresponding bit in the Port Output Data Register (PODR). The pin must have been previously configured as output via [gpio\\_set\\_output\(\)](#). Writing to an input pin has no effect on the physical pin state.

#### 8.1.5.35 Algorithm

1. Validate pin parameter (port exists, pin number valid)

2. Extract port letter and pin number from rx\_port\_pin\_t
3. Clear corresponding bit in PODR register (atomic read-modify-write)

## Parameters

|    |     |                                                                                                                                                                                                                                                                          |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in | pin | GPIO pin identifier from rx_port_constants.h <ul style="list-style-type: none"> <li>• Type: rx_port_pin_t (C23 typed enum)</li> <li>• Format: k_port_X_pin_Y (X = A-N, Y = 0-7)</li> <li>• Must be configured as output via <a href="#">gpio_set_output()</a></li> </ul> |
|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                            |                                    |
|----------------------------|------------------------------------|
| k_rx_ok                    | Pin successfully driven low        |
| k_rx_err_gpio_invalid_port | Port letter out of range (not A-N) |
| k_rx_err_gpio_invalid_pin  | Pin number out of range (not 0-7)  |

## Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)  
Pin port and number must be within valid hardware range

## Postcondition

Pin output level is logic low (GND, 0V)  
PODR register bit for the pin is cleared to 0

## Note

Thread Safety: NOT thread-safe. Concurrent writes to pins on the same port may cause read-modify-write races on the PODR register.

## Example: Disable Motor

```
// Disable motor driver (active high enable)
rx_err_t err = gpio_write_low(k_port_c_pin_0);
if (err != k_rx_ok) {
 rx_log_error_val("MOTOR", "Failed to disable motor", err);
}
```

## See also

[gpio\\_write\\_high\(\)](#) Set output pin to logic high  
[gpio\\_toggle\(\)](#) Toggle output pin state  
[gpio\\_set\\_output\(\)](#) Configure pin as output first

Since

Version 1.0.0

Set GPIO output pin to logic low (GND).

Sets a GPIO output pin to logic low level. On the RX72N, this means the pin outputs GND (0V). The pin must be configured as an output before calling this function.

#### 8.1.5.36 Register Operation

```
PODR &= ~(1 « pin) // Clear bit: Output low
```

#### 8.1.5.37 Open-Drain Mode

If the pin is configured for open-drain mode (ODR register), setting low actively pulls the pin to GND. Setting high makes the pin float (high-impedance), requiring an external pull-up.

Precondition

Pin must be configured as output via [gpio\\_set\\_output\(\)](#)

Port and pin numbers must be valid

Postcondition

PODR bit for pin is cleared (output low)

Pin outputs GND level (0V)

Note

NOT thread-safe: Direct register write without synchronization

Performance: ~100 ns @ 240 MHz

ISR-safe: Can be called from interrupt handlers

Warning

No check that pin is configured as output

Calling on input pin has no visible effect

Example:

```
// Turn off LED (active high)
rx_err_t err = gpio_write_low(RX_PORT_PIN(0, 3));
if (err != k_rx_ok) {
 // Handle error
}
```

See also

[gpio\\_write\\_high\(\)](#) Set output high  
[gpio\\_toggle\(\)](#) Toggle output state  
[gpio\\_set\\_output\(\)](#) Configure pin as output first

Since

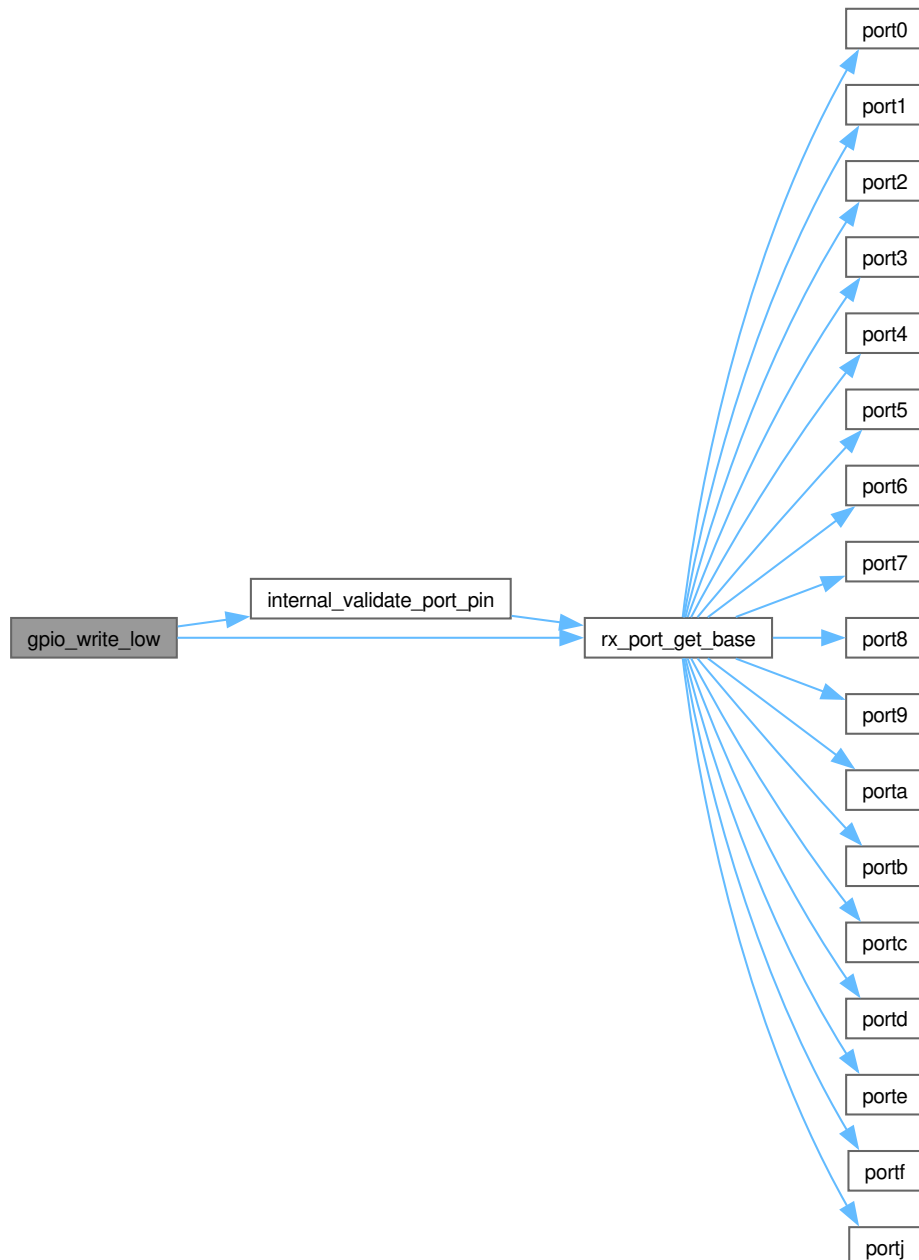
Version 1.0.0

Definition at line 634 of file [gpio.c](#).

```
00635 {
00636 /* Extract port and pin number from rx_port_pin_t */
00637 const uint8_t port = rx_port_from_pin(pin);
00638 const uint8_t pin_num = rx_pin_from_pin(pin);
00639
00640 /* Validate parameters */
00641 const rx_err_t err = internal_validate_port_pin(port, pin_num);
00642 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00643
00644 /* Get port base address */
00645 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00646
00647 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00648 port_base->podr &= (uint8_t)~pin_mask;
00649
00650 return k_rx_ok;
00651 }
```

References [internal\\_validate\\_port\\_pin\(\)](#), [k\\_gpio\\_bit\\_set](#), [rx\\_port\\_regs\\_t::podr](#), [rx\\_port\\_get\\_base\(\)](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.38 riic\_deinit\_peripheral()

```

rx_err_t riic_deinit_peripheral (
 const riic_channel_t channel) [nodiscard]

```

Deinitialize an RIIC channel previously initialized in peripheral mode.

Disables peripheral address match (ICSER.SAR0E), clears SARL0/SARU0 address registers, and resets the RIIC module. After this call, the channel may be reinitialized via [riic\\_init\(\)](#) or [riic\\_init\\_peripheral\(\)](#).

##### Parameters

|    |         |                                                         |
|----|---------|---------------------------------------------------------|
| in | channel | RIIC channel to deinitialize (channel.value range: 0-2) |
|----|---------|---------------------------------------------------------|

## Returns

rx\_err\_t Error code

## Return values

|                        |                                                |
|------------------------|------------------------------------------------|
| k_rx_ok                | Channel deinitialized successfully             |
| k_rx_err_invalid_arg   | channel.value >= k_riic_max_channels           |
| k_rx_err_invalid_state | Channel was not initialized in peripheral mode |

## Precondition

channel.value must be in range [0, k\_riic\_max\_channels - 1]

Channel must have been initialized via [riic\\_init\\_peripheral\(\)](#)

## Postcondition

Channel mode reset to uninitialized; safe to call [riic\\_init\\_peripheral\(\)](#) again

Peripheral address match disabled on the hardware

## Note

Only available on RX72N target (guarded by RX preprocessor)

Not thread-safe; provide external synchronization if needed

## Example:

```
riic_channel_t ch = { .value = k_i2c_comm_default_channel };
rx_err_t err = riic_deinit_peripheral(ch);
if (err != k_rx_ok) {
 // Channel was not in peripheral mode or index out of range
 return err;
}
// Channel is now free; may be reinitialized for controller or peripheral use
```

## See also

[riic\\_init\\_peripheral\(\)](#) Complementary initialization function

## Since

Version 1.0.0

Deinitialize an RIIC channel previously initialized in peripheral mode.

Disables the peripheral address match (ICSER.SAR0E), clears the SARL0/SARU0 address registers, resets the RIIC module, and marks the channel as uninitialized. After this call, [riic\\_init\(\)](#) or [riic\\_init\\_peripheral\(\)](#) may be called again on the same channel.

## Precondition

channel.value must be in range [0, k\_riic\_max\_channels - 1]  
 Channel must have been initialized via [riic\\_init\\_peripheral\(\)](#)

## Postcondition

s\_riic\_channel\_mode[channel.value] == k\_riic\_mode\_uninitialized on success  
 RIIC peripheral address match disabled; hardware in reset state

## Note

Only available on RX72N target (guarded by RX preprocessor)  
 Not thread-safe; provide external synchronization if needed

## See also

[riic\\_init\\_peripheral\(\)](#) Complementary initialization function

## Since

Version 1.0.0

Definition at line 3100 of file [riic.c](#).

```

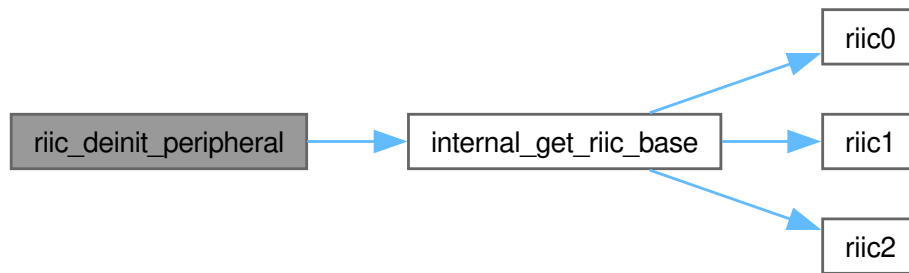
03101 {
03102 if (channel.value >= k_riic_max_channels) {
03103 return k_rx_err_invalid_arg;
03104 }
03105
03106 if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03107 return k_rx_err_invalid_state;
03108 }
03109
03110 volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03111 if (riic == nullptr) {
03112 return k_rx_err_invalid_arg;
03113 }
03114
03115 /* Disable peripheral address match (clear SAR0E in IC SER) */
03116 riic->icser = (uint8_t)(riic->icser & ~(uint8_t)k_riic_icser_sar0e);
03117
03118 /* Clear peripheral address registers */
03119 riic->sar0 = k_riic_register_clear;
03120 riic->saru0 = k_riic_register_clear;
03121
03122 /* Reset the RIIC module */
03123 riic->iccr1 = k_riic_iccr1_iicrst;
03124 riic->iccr1 = k_riic_register_clear;
03125
03126 /* Mark channel as uninitialized */
03127 s_riic_channel_mode[channel.value] = k_riic_mode_uninitialized;
03128
03129 rx_log_debug(s_tag, "RIIC peripheral mode deinitialized");
03130
03131 return k_rx_ok;
03132 }

```

References [rx\\_riic\\_regs\\_t::iccr1](#), [rx\\_riic\\_regs\\_t::icser](#), [internal\\_get\\_riic\\_base\(\)](#), [k\\_riic\\_iccr1\\_iicrst](#), [k\\_riic\\_icser\\_sar0e](#), [k\\_riic\\_max\\_channels](#), [k\\_riic\\_mode\\_peripheral](#), [k\\_riic\\_mode\\_uninitialized](#), [k\\_riic\\_register\\_clear](#), [s\\_riic\\_channel\\_mode](#), [s\\_tag](#), [rx\\_riic\\_regs\\_t::sar0](#), [rx\\_riic\\_regs\\_t::saru0](#), and [riic\\_channel\\_t::value](#).

Here is the call graph for this function:





#### 8.1.5.39 riic\_init()

```

rx_err_t riic_init (
 riic_channel_t channel,
 uint32_t frequency_hz) [nodiscard]

```

Initialize RIIC channel for I2C controller communication.

Configures the specified RIIC peripheral channel for I2C controller mode at the requested clock frequency. Performs full hardware initialization including module stop release, pin function selection (SCL/SDA via MPC), internal reset sequence, bit rate register (BRH/BRL) calculation from PCLKB, and interrupt configuration.

#### 8.1.5.40 Algorithm

1. Validate channel number (0-2) and frequency
2. Release RIIC module from module stop (MSTPCRB)
3. Assert internal reset (ICCR1.IICRST = 1)
4. Configure SCL/SDA pins via MPC register
5. Set bit rate registers (BRH/BRL) from PCLKB and frequency\_hz
6. Configure noise filter, timeout detection
7. Release internal reset (ICCR1.IICRST = 0)
8. Enable I2C bus (ICCR1.ICE = 1)

#### 8.1.5.41 Supported Frequencies

| Frequency  | Mode      | BRL/BRH         | Max Bus Length |
|------------|-----------|-----------------|----------------|
| 100000 Hz  | Standard  | Auto-calculated | 1 m            |
| 400000 Hz  | Fast      | Auto-calculated | 0.3 m          |
| 1000000 Hz | Fast-plus | Auto-calculated | 0.1 m          |

## Parameters

|    |              |                                                                                                                                                                  |
|----|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in | channel      | RIIC channel wrapper <ul style="list-style-type: none"><li>channel.value range: 0-2</li><li>Use <code>{.value = N}</code> designated initializer</li></ul>       |
| in | frequency_hz | I2C clock frequency in Hz <ul style="list-style-type: none"><li>Valid values: 100000, 400000, 1000000</li><li>Must match connected device capabilities</li></ul> |

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| <code>k_rx_ok</code>                | Channel successfully initialized     |
| <code>k_rx_err_invalid_arg</code>   | Channel > 2 or unsupported frequency |
| <code>k_rx_err_invalid_state</code> | Channel already initialized          |

## Precondition

Channel must not be already initialized (call `riic deinit` first to reinitialize)  
System clocks must be configured (PCLKB running at expected frequency)

## Postcondition

RIIC channel configured and ready for I2C transactions  
SCL/SDA pins configured with open-drain output and internal pull-ups

## Note

Thread Safety: NOT thread-safe. Initialize each channel from a single thread during system startup.

## Example: Initialize IMU I2C Bus

```
riic_channel_t ch = {.value = 0};
rx_err_t err = riic_init(ch, 400000); // 400 kHz fast mode
if (err != k_rx_ok) {
 rx_log_error_val("I2C", "RIIC0 init failed", err);
 return;
}
rx_log_info("I2C", "RIIC0 initialized at 400 kHz");
```

See also

[riic\\_write\(\)](#) Write data to I2C device  
[riic\\_read\(\)](#) Read data from I2C device  
[riic\\_write\\_read\(\)](#) Combined write-then-read transaction

Since

Version 1.0.0

Definition at line 2231 of file [riic.c](#).

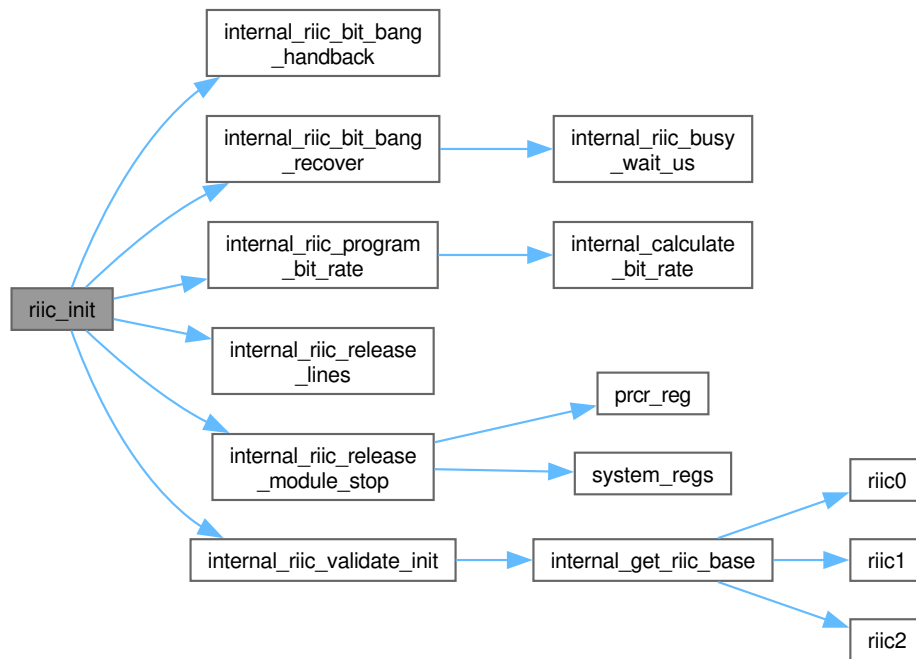
```

02232 {
02233 volatile rx_riic_regs_t* riic = nullptr;
02234 const rx_err_t check = internal_riic_validate_init(channel, frequency_hz, &riic);
02235 if (check != k_rx_ok) {
02236 return check;
02237 }
02238
02239 internal_riic_release_module_stop(channel.value);
02240
02241 /* Unconditional bit-bang bus recovery BEFORE the peripheral observes the
02242 * bus. Per RX72N HW manual section 42.2.1, the RIIC comes out of
02243 * module-stop + IICRST with SCLO=SDAO=0 internally -- it drives the bus
02244 * low until the SOWP dance below writes SCLO=SDAO=1. hardware_init has
02245 * already raised PMR for SCL/SDA by the time riic_init runs, so without
02246 * this step the low-drive glitches the external bus and can latch
02247 * BBSY=1 as soon as ICE=1. Channels without an entry in
02248 * k_riic_recovery_pins[] (RIIC0, RIIC2) are no-ops. */
02249 internal_riic_bit_bang_recover(channel.value); /* leaves PMR=0 */
02250
02251 /* Reset RIIC */
02252 riic->iccr1 = k_riic_iccr1_iicrst;
02253 riic->iccr1 = k_riic_register_clear;
02254
02255 RX_RETURN_ON_ERROR(internal_riic_program_bit_rate(riic, frequency_hz),
02256 s_tag,
02257 "RIIC bit-rate program failed");
02258
02259 /* Enable I2C bus interface */
02260 riic->iccr1 = k_riic_iccr1_ice;
02261
02262 /* Post-enable bus release: per RX72N HW manual section 42.2.1 the RIIC
02263 * leaves SCLO/SDAO=0 after ICE=1 -- which drives both lines LOW -- until
02264 * it detects its first STOP on the wire. */
02265 internal_riic_release_lines(riic);
02266
02267 /* Pair to the unconditional bit-bang above: now that SCLO=SDAO=1 in the
02268 * peripheral (set by the SOWP dance), it is safe to route the pads back
02269 * to RIIC. */
02270 internal_riic_bit_bang_handback(channel.value);
02271
02272 s_riic_channel_mode[channel.value] = k_riic_mode_controller;
02273 rx_log_debug(s_tag, "RIIC channel initialized");
02274
02275 return k_rx_ok;
02276 }

```

References [rx\\_riic\\_regs\\_t::iccr1](#), [internal\\_riic\\_bit\\_bang\\_handback\(\)](#), [internal\\_riic\\_bit\\_bang\\_recover\(\)](#), [internal\\_riic\\_program\\_bit\\_rate\(\)](#), [internal\\_riic\\_release\\_lines\(\)](#), [internal\\_riic\\_release\\_module\\_stop\(\)](#), [internal\\_riic\\_validate\\_init\(\)](#), [k\\_riic\\_iccr1\\_ice](#), [k\\_riic\\_iccr1\\_iicrst](#), [k\\_riic\\_mode\\_controller](#), [k\\_riic\\_register\\_clear](#), [s\\_riic\\_channel\\_mode](#), [s\\_tag](#), and [riic\\_channel\\_t::value](#).

Here is the call graph for this function:



#### 8.1.5.42 riic\_init\_peripheral()

```

rx_err_t riic_init_peripheral (
 const riic_channel_t channel,
 const i2c_device_addr_t device_addr) [nodiscard]

```

Initialize RIIC channel in peripheral (device) mode.

Configures the RIIC peripheral to act as an I2C peripheral device, responding to a fixed 7-bit I2C address issued by a controller (RPi5). Sets up the peripheral address register (SARL0/SARU0), enables address match detection (ICSER.SAR0E), and leaves the module in peripheral mode (MS=0 in ICMR1).

Peripheral Mode Architecture:

- RPi5 acts as I2C controller (generates SCL, START, STOP)
- RX72N acts as I2C peripheral (responds to address, provides/accepts data)
- Address matching handled in hardware (ICSER.SAR0E)
- Data transfer polled via ICSR2.RDRF and ICSR2.TDRE

#### Parameters

|    |             |                                                 |
|----|-------------|-------------------------------------------------|
| in | channel     | RIIC channel wrapper (channel.value range: 0-2) |
| in | device_addr | 7-bit peripheral address for this device        |

## Returns

`rx_err_t` Error code

## Return values

|                                     |                                                                 |
|-------------------------------------|-----------------------------------------------------------------|
| <code>k_rx_ok</code>                | Channel initialized in peripheral mode                          |
| <code>k_rx_err_invalid_arg</code>   | Channel value out of range                                      |
| <code>k_rx_err_invalid_state</code> | Channel already initialized; deinitialize before reinitializing |

## Precondition

`channel.value` must be in range `[0, 2]` (i.e., `< k_riic_max_channels`)

`device_addr.value` must be in range `[0x08, 0x77]` (valid non-reserved 7-bit I2C address)

## Postcondition

RIIC peripheral responds to `device_addr` on the I2C bus

`s_riic_channel_initialized[channel.value] == true` on success

## Note

Only available on RX72N target (guarded by RX preprocessor)

Not thread-safe during initialization

## See also

[riic\\_peripheral\\_read\(\)](#) Read data written by I2C controller

[riic\\_peripheral\\_write\(\)](#) Provide data to I2C controller read

## Since

Version 1.0.0

Configures the RIIC peripheral to act as an I2C peripheral device, responding to a fixed 7-bit I2C address issued by a controller (RPI5). Sets up the peripheral address register (SARL0/SARU0), enables address match detection (ICSER.SAR0E), and leaves the module in peripheral mode (MS=0 in ICMR1).

## 8.1.5.43 Initialization Sequence

1. Validate channel number and device address
2. Enable RIIC module clock (clear MSTPCRB bit)
3. Reset RIIC internal state (IICRST)
4. Configure peripheral address in SARL0/SARU0
5. Enable SAR0 address match detection (ICSER.SAR0E)
6. Configure peripheral mode (MS=0), 7-bit addressing
7. Enable I2C bus interface (ICE)
8. Mark channel as initialized

## Precondition

channel.value must be in range [0, k\_riic\_max\_channels - 1]  
 device\_addr.value must be <= k\_riic\_addr\_max\_7bit (0x7F)

## Postcondition

RIIC peripheral responds to device\_addr on the I2C bus  
 s\_riic\_channel\_mode[channel.value] == k\_riic\_mode\_peripheral

## Note

Only available on RX72N target (guarded by RX preprocessor)  
 Not thread-safe during initialization

## See also

[riic\\_peripheral\\_read\(\)](#) Read data written by I2C controller  
[riic\\_peripheral\\_write\(\)](#) Provide data to I2C controller read

## Since

Version 1.0.0

Definition at line 2860 of file [riic.c](#).

```

02861 {
02862 if (channel.value >= k_riic_max_channels) {
02863 return k_rx_err_invalid_arg;
02864 }
02865
02866 if (device_addr.value > k_riic_addr_max_7bit) {
02867 rx_log_error(s_tag, "Invalid peripheral device address");
02868 return k_rx_err_invalid_arg;
02869 }
02870
02871 volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02872 if (riic == nullptr) {
02873 return k_rx_err_invalid_arg;
02874 }
02875
02876 /* Enable RIIC module (clear module stop bit) */
02877 *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
02878
02879 if (channel.value == k_riic_channel_0) {
02880 system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic0;
02881 } else if (channel.value == k_riic_channel_1) {
02882 system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic1;
02883 } else {
02884 system_regs()->mstpcrc &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpc_riic2;
02885 }
02886
02887 *prcr_reg() = k_rx_prcr_lock;
02888
02889 /* Reset RIIC */
02890 riic->iccr1 = k_riic_iccr1_iicrst;
02891 riic->iccr1 = k_riic_register_clear;
02892
02893 /* Set peripheral (device) address in SARL0/SARU0
02894 * SARL0[7:1] = device_addr.value (7-bit address)
02895 * SARU0 = 0 for 7-bit mode */
02896 riic->sarl0 = (uint8_t)(device_addr.value « k_riic_sarl_addr_shift);
02897 riic->saru0 = k_riic_saru0_7bit;
02898
02899 /* Enable SAR0 address match detection */
02900 riic->icser = k_riic_icser_sar0e;
02901
02902 /* Configure for peripheral mode (MS=0), 7-bit addressing, CKS=0 */
02903 riic->icmr1 = k_riic_icmr1_peripheral_7bit;

```

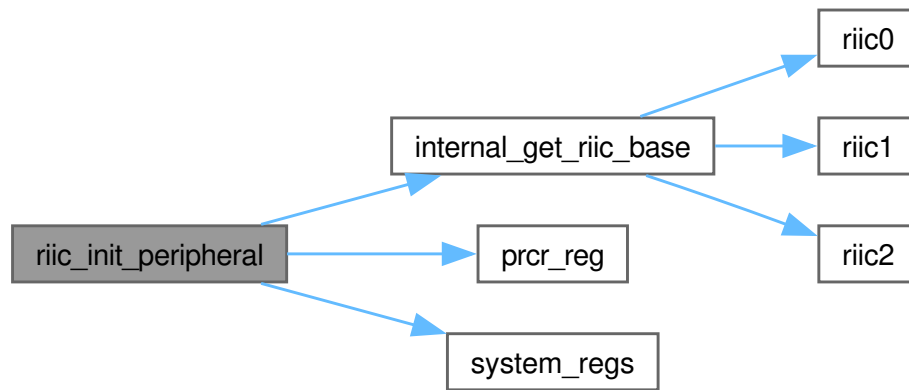
```

02904 riic->icmr2 = k_riic_icmr2_default;
02905 riic->icmr3 = k_riic_icmr2_default;
02906
02907 /* Enable I2C bus interface */
02908 riic->iccr1 = k_riic_iccr1_ice;
02909
02910 /* Mark channel as initialized in peripheral mode */
02911 s_riic_channel_mode[channel.value] = k_riic_mode_peripheral;
02912
02913 rx_log_debug(s_tag, "RIIC peripheral mode initialized");
02914
02915 return k_rx_ok;
02916 }

```

References `rx_riic_regs_t::iccr1`, `rx_riic_regs_t::icmr1`, `rx_riic_regs_t::icmr2`, `rx_riic_regs_t::icmr3`, `rx_riic_regs_t::icser`, `internal_get_riic_base()`, `k_riic_addr_max_7bit`, `k_riic_channel_0`, `k_riic_channel_1`, `k_riic_iccr1_ice`, `k_riic_iccr1_icrst`, `k_riic_icmr1_peripheral_7bit`, `k_riic_icmr2_default`, `k_riic_icser_sar0e`, `k_riic_max_channels`, `k_riic_mode_peripheral`, `k_riic_mstpb_bit_value`, `k_riic_mstpb_riic0`, `k_riic_mstpb_riic1`, `k_riic_mstpc_riic2`, `k_riic_register_clear`, `k_riic_sarl_addr_shift`, `k_riic_saru0_7bit`, `rx_system_regs_t::mstpcrb`, `rx_system_regs_t::mstpcrc`, `prcr_reg()`, `s_riic_channel_mode`, `s_tag`, `rx_riic_regs_t::sarl0`, `rx_riic_regs_t::saru0`, `system_regs()`, `i2c_device_addr_t::value`, and `riic_channel_t::value`.

Here is the call graph for this function:



#### 8.1.5.44 riic\_peripheral\_read()

```

rx_err_t riic_peripheral_read (
 const riic_channel_t channel,
 uint8_t * data,
 const uint16_t max_length,
 uint16_t * bytes_read) [nodiscard]

```

Read bytes written by the I2C controller to this peripheral.

Polls ICSR2.RDRF (Receive Data Register Full) to detect incoming bytes from the I2C controller (RPI5). Reads up to `max_length` bytes. Returns immediately with `bytes_read=0` if no data is currently available.

##### Parameters

|     |            |                                                 |
|-----|------------|-------------------------------------------------|
| in  | channel    | RIIC channel wrapper (channel.value range: 0-2) |
| out | data       | Buffer to store received bytes                  |
| in  | max_length | Maximum number of bytes to read                 |
| out | bytes_read | Actual number of bytes read (may be 0)          |

## Returns

`rx_err_t` Error code

## Return values

|                                     |                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Read completed (check <code>bytes_read</code> for count)                                                                    |
| <code>k_rx_err_null_ptr</code>      | <code>data</code> or <code>bytes_read</code> is <code>nullptr</code>                                                        |
| <code>k_rx_err_invalid_arg</code>   | Channel value out of range or <code>max_length == 0</code> or <code>max_length &gt; k_riic_peripheral_transfer_limit</code> |
| <code>k_rx_err_invalid_state</code> | Channel not initialized via <a href="#">riic_init_peripheral()</a>                                                          |

## Precondition

channel must be initialized via [riic\\_init\\_peripheral\(\)](#)

`data != nullptr` with capacity for at least `max_length` bytes

`max_length >= 1 && max_length <= k_riic_peripheral_transfer_limit`

## Postcondition

`*bytes_read` contains number of bytes placed in `data` (0 to `max_length`)

`data[0..*bytes_read-1]` contain valid received bytes on `k_rx_ok`

## Note

Non-blocking: returns immediately with available data

Only available on RX72N target (guarded by RX preprocessor)

## See also

[riic\\_init\\_peripheral\(\)](#) Initialize channel first

[riic\\_peripheral\\_write\(\)](#) Provide read data to controller

## Since

Version 1.0.0

Polls ICSR2.RDRF (Receive Data Register Full) to detect incoming bytes from the I2C controller (RPi5). Reads up to `max_length` bytes into the caller-supplied buffer. Returns immediately with `bytes_read == 0` if no data is currently available (non-blocking). Each byte read clears the RDRF flag automatically on the RX72N.



## 8.1.5.45 Read Algorithm

1. Validate pointers, channel, and max\_length
2. Clear \*bytes\_read to 0
3. Poll ICSR2.RDRF; if set, read ICDRR into data[\*bytes\_read], increment count
4. Repeat until RDRF clear or max\_length reached
5. Return k\_rx\_ok with actual count in \*bytes\_read

## Precondition

channel must be initialized via [riic\\_init\\_peripheral\(\)](#)  
 data != nullptr with at least max\_length bytes of capacity  
 max\_length >= 1 && max\_length <= k\_riic\_peripheral\_transfer\_limit

## Postcondition

\*bytes\_read contains the number of bytes placed in data (0 to max\_length)  
 data[0..(\*bytes\_read)-1] contain valid received bytes on k\_rx\_ok

## Note

Non-blocking: returns immediately with available data (bytes\_read may be 0)  
 Only available on RX72N target (guarded by RX preprocessor)  
 Not thread-safe, caller must provide external synchronization

## See also

[riic\\_init\\_peripheral\(\)](#) Initialize channel first  
[riic\\_peripheral\\_write\(\)](#) Provide read data to controller

## Since

Version 1.0.0

Definition at line 2952 of file [riic.c](#).

```

02956 {
02957 if (data == nullptr || bytes_read == nullptr) {
02958 return k_rx_err_null_ptr;
02959 }
02960
02961 if (channel.value >= k_riic_max_channels) {
02962 return k_rx_err_invalid_arg;
02963 }
02964
02965 if (max_length == 0) {
02966 return k_rx_err_invalid_arg;
02967 }
02968
02969 if (max_length > k_riic_peripheral_transfer_limit) {
02970 rx_log_error(s_tag, "Peripheral read max_length exceeds limit");
02971 return k_rx_err_invalid_arg;
02972 }
02973
02974 if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
02975 return k_rx_err_invalid_state;

```

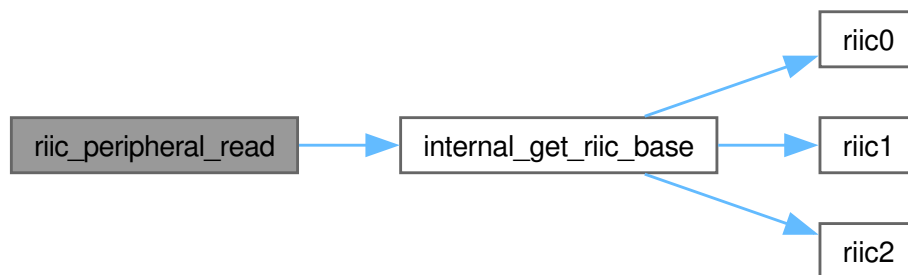
```

02976 }
02977
02978 volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02979 if (riic == nullptr) {
02980 return k_rx_err_invalid_arg;
02981 }
02982
02983 *bytes_read = 0;
02984
02985 /* Poll ICSR2.RDRF (bit 1) for each available byte - non-blocking */
02986 while (*bytes_read < max_length) {
02987 if (!(riic->icsr2 & k_riic_icsr2_rdrf)) {
02988 break; /* No more data available right now */
02989 }
02990 data[*bytes_read] = riic->icdrr;
02991 (*bytes_read)++;
02992 }
02993
02994 return k_rx_ok;
02995 }

```

References [rx\\_riic\\_regs\\_t::icdrr](#), [rx\\_riic\\_regs\\_t::icsr2](#), [internal\\_get\\_riic\\_base\(\)](#), [k\\_riic\\_icsr2\\_rdrf](#), [k\\_riic\\_max\\_channels](#), [k\\_riic\\_mode\\_peripheral](#), [k\\_riic\\_peripheral\\_transfer\\_limit](#), [s\\_riic\\_channel\\_mode](#), [s\\_tag](#), and [riic\\_channel\\_t::value](#).

Here is the call graph for this function:



#### 8.1.5.46 riic\_peripheral\_write()

```

rx_err_t riic_peripheral_write (
 const riic_channel_t channel,
 const uint8_t * data,
 const uint16_t length) [nodiscard]

```

Write bytes for the I2C controller to read from this peripheral.

Polls ICSR2.TDRE (Transmit Data Register Empty) to detect when the I2C controller is reading from this peripheral. Writes up to length bytes into the ICDRT transmit data register. Returns immediately if the transmit register is not yet empty (controller not currently reading).

#### Parameters

|    |         |                                                                             |
|----|---------|-----------------------------------------------------------------------------|
| in | channel | RIIC channel wrapper (channel.value range: 0-2)                             |
| in | data    | Pointer to data buffer to write                                             |
| in | length  | Number of bytes to write (valid range: 1..k_riic_peripheral_transfer_limit) |

## Returns

`rx_err_t` Error code

## Return values

|                                     |                                                                                                    |
|-------------------------------------|----------------------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Write completed                                                                                    |
| <code>k_rx_err_null_ptr</code>      | data is nullptr                                                                                    |
| <code>k_rx_err_invalid_arg</code>   | Channel value out of range, length == 0, or length > <code>k_riic_peripheral_transfer_limit</code> |
| <code>k_rx_err_invalid_state</code> | Channel not initialized via <a href="#">riic_init_peripheral()</a>                                 |
| <code>k_rx_err_timeout</code>       | Transmit data register not empty within timeout                                                    |

## Precondition

channel must be initialized via [riic\\_init\\_peripheral\(\)](#)

data != nullptr with at least length valid bytes

length >= 1 && length <= `k_riic_peripheral_transfer_limit`

## Postcondition

On `k_rx_ok`: all length bytes have been written to ICDRT

On `k_rx_err_timeout`: partial write may have occurred

## Note

Only available on RX72N target (guarded by RX preprocessor)

Not thread-safe, caller must provide external synchronization

## See also

[riic\\_init\\_peripheral\(\)](#) Initialize channel first

[riic\\_peripheral\\_read\(\)](#) Receive data from controller

## Since

Version 1.0.0

Loads data bytes into the RIIC transmit data register (ICDRT) one at a time. For each byte, polls ICSR2.TDRE (Transmit Data Register Empty) until the register is available or a 1 ms timeout expires, then writes the byte. The controller clock-shifts bytes out automatically after each write.

## 8.1.5.47 Write Algorithm

1. Validate pointer, channel, and length
2. For each byte in data[0..length-1]: a. Poll ICSR2.TDRE with k\_riic\_periph\_timeout\_us count-down b. If timeout, return k\_rx\_err\_timeout c. Write data[i] to ICDRT
3. Return k\_rx\_ok

## Precondition

channel must be initialized via [riic\\_init\\_peripheral\(\)](#)  
 data != nullptr with at least length valid bytes  
 length >= 1

## Postcondition

On k\_rx\_ok: all length bytes have been written to ICDRT  
 On k\_rx\_err\_timeout: partial write may have occurred (i bytes written)

## Note

Only available on RX72N target (guarded by RX preprocessor)  
 Not thread-safe, caller must provide external synchronization

## See also

[riic\\_init\\_peripheral\(\)](#) Initialize channel first  
[riic\\_peripheral\\_read\(\)](#) Receive data from controller

## Since

Version 1.0.0

Definition at line 3032 of file [riic.c](#).

```

03033 {
03034 if (data == nullptr) {
03035 return k_rx_err_null_ptr;
03036 }
03037
03038 if (channel.value >= k_riic_max_channels) {
03039 return k_rx_err_invalid_arg;
03040 }
03041
03042 if (length == 0 || length > k_riic_peripheral_transfer_limit) {
03043 rx_log_error(
03044 s_tag,
03045 "riic_peripheral_write: length out of range (1..k_riic_peripheral_transfer_limit)");
03046 return k_rx_err_invalid_arg;
03047 }
03048
03049 if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03050 return k_rx_err_invalid_state;
03051 }
03052
03053 volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03054 if (riic == nullptr) {
03055 return k_rx_err_invalid_arg;
03056 }
03057

```

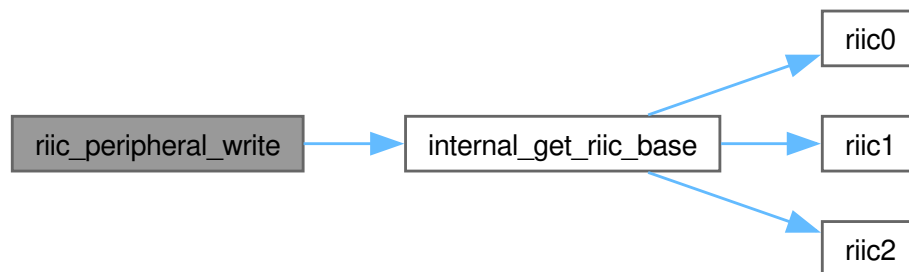
```

03058 for (uint16_t i = 0; i < length; i++) {
03059 /* Wait for ICSR2.TDRE (transmit data register empty) */
03060 uint32_t timeout = k_riic_periph_timeout_us;
03061 while (!(riic->icsr2 & k_riic_icsr2_tdre) && timeout > k_riic_timeout_zero) {
03062 timeout--;
03063 }
03064 if (timeout == k_riic_timeout_zero) {
03065 return k_rx_err_timeout;
03066 }
03067 riic->icdrt = data[i];
03070 }
03071 rx_log_debug(s_tag, "RIIC peripheral write complete");
03073 return k_rx_ok;
03075 }

```

References [rx\\_riic\\_regs\\_t::icdrt](#), [rx\\_riic\\_regs\\_t::icsr2](#), [internal\\_get\\_riic\\_base\(\)](#), [k\\_riic\\_icsr2\\_tdre](#), [k\\_riic\\_max\\_channels](#), [k\\_riic\\_mode\\_peripheral](#), [k\\_riic\\_periph\\_timeout\\_us](#), [k\\_riic\\_peripheral\\_transfer\\_limit](#), [k\\_riic\\_timeout\\_zero](#), [s\\_riic\\_channel\\_mode](#), [s\\_tag](#), and [riic\\_channel\\_t::value](#).

Here is the call graph for this function:



#### 8.1.5.48 riic\_read()

```

rx_err_t riic_read (
 const riic_channel_t channel,
 const i2c_device_addr_t device_addr,
 uint8_t * data,
 const uint16_t length) [nodiscard]

```

Read data from I2C device on specified RIIC channel.

Performs a complete I2C read transaction: generates START condition, transmits device address with read bit (R/W=1), receives data bytes with ACK (NACK on last byte), and generates STOP condition. The function blocks until all bytes are received or an error occurs.

#### 8.1.5.49 Transaction Sequence

1. Generate START condition
2. Send device address byte (7-bit address << 1 | 1)
3. Wait for ACK from peripheral
4. Receive data bytes, sending ACK after each (NACK after last byte)
5. Generate STOP condition

6. Wait for bus idle (BBSY = 0)

#### Parameters

|     |                  |                                                                                                                                                                                                           |
|-----|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in  | channel          | RIIC channel wrapper <ul style="list-style-type: none"> <li>channel.value range: 0-2</li> </ul>                                                                                                           |
| in  | device_↔<br>addr | 7-bit I2C device address wrapper <ul style="list-style-type: none"> <li>device_addr.value range: 0x08-0x77</li> </ul>                                                                                     |
| out | data             | Pointer to buffer for received data <ul style="list-style-type: none"> <li>Must not be nullptr</li> <li>Must have space for at least length bytes</li> <li>Unchanged if function returns error</li> </ul> |
| in  | length           | Number of bytes to read <ul style="list-style-type: none"> <li>Range: 1-65535</li> </ul>                                                                                                                  |

#### Returns

rx\_err\_t Error code indicating success or failure

#### Return values

|                        |                                                            |
|------------------------|------------------------------------------------------------|
| k_rx_ok                | All bytes successfully received                            |
| k_rx_err_null_ptr      | data pointer is nullptr                                    |
| k_rx_err_invalid_state | Channel not initialized via <a href="#">riic_init()</a>    |
| k_rx_err_timeout       | Bus timeout waiting for data or bus idle                   |
| k_rx_err_nack          | Device sent NACK on address (wrong address or device busy) |

#### Precondition

Channel must be initialized via [riic\\_init\(\)](#)

data pointer must be valid with space for at least length bytes

#### Postcondition

data buffer contains received bytes if function succeeds

I2C bus returned to idle state (STOP condition generated)

#### Note

Thread Safety: NOT thread-safe. Do not perform concurrent transactions on the same RIIC channel. Different channels may be used from different threads.

## Example: Read IMU Accelerometer Data

```

riic_channel_t ch = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

uint8_t accel_data[6];
rx_err_t err = riic_read(ch, imu_addr, accel_data, 6);
if (err != k_rx_ok) {
 rx_log_error_val("IMU", "Failed to read accel data", err);
}

```

See also

[riic\\_write\(\)](#) Write data to I2C device  
[riic\\_write\\_read\(\)](#) Combined write-then-read (register read pattern)  
[riic\\_init\(\)](#) Initialize RIIC channel first

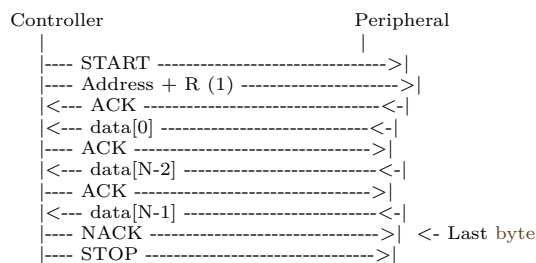
Since

Version 1.0.0

Read data from I2C device on specified RIIC channel.

Performs a complete I2C read transaction: START, address+R, receive data bytes (ACK all except last), NACK last byte, STOP. This function is suitable for reading device status, sensor data, or sequential memory reads where the device auto-increments its internal pointer.

## 8.1.5.50 Transaction Sequence

8.1.5.51 When to Use `riic_read()` vs `riic_write_read()`

| Function                          | Use When                                          |
|-----------------------------------|---------------------------------------------------|
| <a href="#">riic_read()</a>       | Device auto-increments address, or reading status |
| <a href="#">riic_write_read()</a> | Need to specify register address first            |

## 8.1.5.52 Common Use Cases

- Read device status register (if device supports direct read)
- Continue reading after previous write set address pointer
- Read from devices with auto-increment address mode

## Precondition

[riic\\_init\(\)](#) called for this channel  
 data != nullptr  
 Buffer size >= length  
 length >= 1 && length <= 256  
 device\_addr.value <= 127

## Postcondition

On success: data contains received bytes, bus released  
 On failure: Bus released via STOP

## Note

This is a blocking function - returns only after transfer complete.  
 NACK is automatically sent before reading last byte.

## Warning

Most peripherals require [riic\\_write\\_read\(\)](#) to specify which register to read from. Use this function only when the device supports direct reads or address auto-increment.

## Example: Read Temperature Sensor (Direct Read)

```

// Read 2 bytes from temperature sensor at address 0x48
// (sensor auto-reports temperature on read)
uint8_t temp_data[2];
riic_channel_t channel = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};

rx_err_t err = riic_read(channel, addr, temp_data, 2);
if (err == k_rx_ok) {
 int16_t raw = (temp_data[0] « 8) | temp_data[1];
 float celsius = raw * 0.0625f;
}

```

## See also

[riic\\_init\(\)](#) Initialize channel before use  
[riic\\_write\(\)](#) Write-only transaction  
[riic\\_write\\_read\(\)](#) Combined write-then-read (register read)

Definition at line 2507 of file [riic.c](#).

```

02511 {
02512 RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02513 RX_CHECK_RANGE_TAG(channel.value,
02514 k_riic_channel_0,
02515 (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02516 k_rx_err_invalid_arg,
02517 s_tag);
02518
02519 if (device_addr.value > k_riic_addr_max_7bit) {
02520 rx_log_error(s_tag, "Invalid device address");
02521 return k_rx_err_invalid_arg;
02522 }
02523
02524 if (length == k_riic_length_zero) {
02525 rx_log_error(s_tag, "Read length cannot be zero");
02526 return k_rx_err_invalid_arg;

```



```

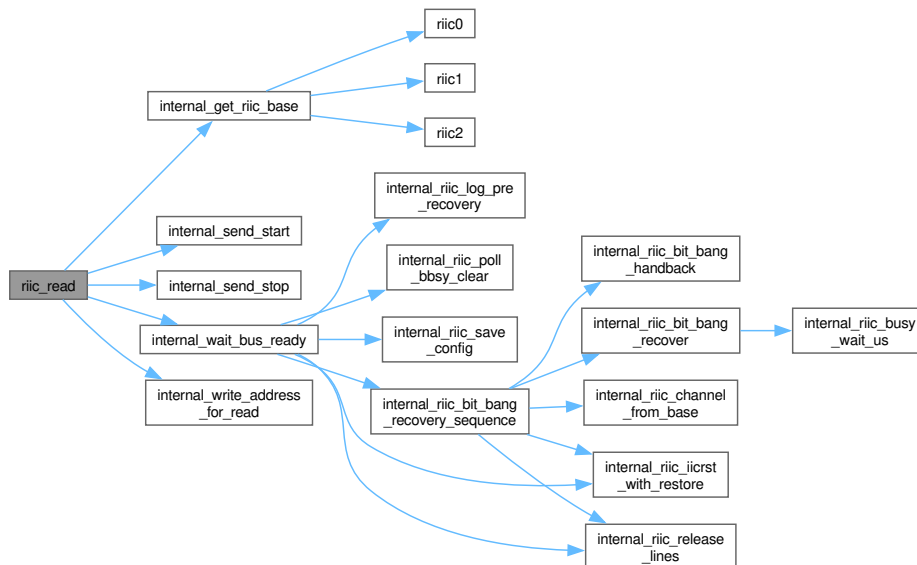
02527 }
02528
02529 if (length > k_riic_max_transfer_length) {
02530 rx_log_error(s_tag, "Read length exceeds maximum");
02531 return k_rx_err_invalid_arg;
02532 }
02533
02534 /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02535 if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02536 rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02537 return k_rx_err_invalid_state;
02538 }
02539
02540 /* Get RIIC base */
02541 volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02542
02543 /* Wait for bus ready */
02544 rx_err_t err = internal_wait_bus_ready(riic);
02545 RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02546
02547 /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02548 riic->icsr2 = k_riic_register_clear;
02549
02550 /* Set controller receive mode */
02551 riic->iccr2 = k_riic_iccr2_mst;
02552
02553 /* Send start condition */
02554 err = internal_send_start(riic);
02555 RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02556
02557 /* Send device address (read) -- use the receive-mode helper, NOT
02558 * internal_write_byte: TEND/TDRE-reassertion don't fire after addr[R in
02559 * controller-receive mode; RDRF does. */
02560 err = internal_write_address_for_read(
02561 riic,
02562 (uint8_t)((device_addr.value « k_riic_addr_shift) | k_riic_addr_read_bit));
02563 if (err != k_rx_ok) {
02564 rx_err_t stop_err = internal_send_stop(riic);
02565 (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02566 return err;
02567 }
02568
02569 /* 1-byte read: pre-set ACKBT=1 before the dummy read so the sole byte's
02570 * 9th-clock ACK slot carries NACK (HUM Fig 42.10). */
02571 if (length == 1U) {
02572 riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02573 }
02574
02575 (void)riic->icdrr; /* dummy read kicks reception */
02576
02577 /* Mirror of the read loop in internal_riic_read_phase -- see that
02578 * function for the why-ACKBT-on-N-2 rationale. */
02579 for (uint16_t i = 0; i < length; i++) {
02580 uint32_t timeout = k_riic_timeout_us;
02581 while (!(riic->icsr2 & k_riic_icsr2_rdrf) && timeout > k_riic_timeout_zero) {
02582 timeout--;
02583 }
02584 if (timeout == k_riic_timeout_zero) {
02585 rx_log_error_val(s_tag, "Read byte RDRF timeout i=", (uint8_t)i);
02586 rx_err_t stop_err = internal_send_stop(riic);
02587 (void)stop_err;
02588 return k_rx_err_timeout;
02589 }
02590
02591 if (length >= 2U && i == (uint16_t)(length - 2U)) {
02592 riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02593 }
02594
02595 if (i == (uint16_t)(length - k_riic_last_index_offset)) {
02596 riic->icsr2 &= (uint8_t) ~(uint8_t)k_riic_icsr2_stop;
02597 riic->iccr2 |= k_riic_iccr2_sp;
02598 }
02599
02600 data[i] = riic->icdrr;
02601 }
02602
02603 uint32_t stop_timeout = k_riic_timeout_us;
02604 while (!(riic->icsr2 & k_riic_icsr2_stop) && stop_timeout > k_riic_timeout_zero) {
02605 stop_timeout--;
02606 }
02607 if (stop_timeout == k_riic_timeout_zero) {
02608 rx_log_error(s_tag, "STOP after final ICDRR timeout");
02609 return k_rx_err_timeout;
02610 }
02611 riic->icsr2 &= (uint8_t) ~(uint8_t)(k_riic_icsr2_stop | k_riic_icsr2_nackf);
02612
02613 return k_rx_ok;

```

02614 }

References `rx_riic_regs_t::iccr2`, `rx_riic_regs_t::icdrr`, `rx_riic_regs_t::icmr3`, `rx_riic_regs_t::icsr2`, `internal_get_riic_base()`, `internal_send_start()`, `internal_send_stop()`, `internal_wait_bus_ready()`, `internal_write_address_for_read()`, `k_riic_addr_max_7bit`, `k_riic_addr_read_bit`, `k_riic_addr_shift`, `k_riic_channel_0`, `k_riic_iccr2_mst`, `k_riic_iccr2_sp`, `k_riic_icmr3_ackbt_mask`, `k_riic_icsr2_nackf`, `k_riic_icsr2_rdrf`, `k_riic_icsr2_stop`, `k_riic_last_index_offset`, `k_riic_length_zero`, `k_riic_max_channels`, `k_riic_max_transfer_length`, `k_riic_mode_controller`, `k_riic_register_clear`, `k_riic_timeout_us`, `k_riic_timeout_zero`, `s_riic_channel_mode`, `s_tag`, `i2c_device_addr_t::value`, and `riic_channel_t::value`.

Here is the call graph for this function:



### 8.1.5.53 riic\_write()

```

rx_err_t riic_write (
 const riic_channel_t channel,
 const i2c_device_addr_t device_addr,
 const uint8_t * data,
 const uint16_t length) [nodiscard]

```

Write data to I2C device on specified RIIC channel.

Performs a complete I2C write transaction: generates START condition, transmits device address with write bit (R/W=0), sends data bytes, and generates STOP condition. Each byte is acknowledged by the peripheral device. The function blocks until all bytes are transmitted or an error occurs.

### 8.1.5.54 Transaction Sequence

1. Generate START condition
2. Send device address byte (7-bit address << 1 | 0)
3. Wait for ACK from peripheral
4. Send data bytes one at a time, waiting for ACK after each

5. Generate STOP condition
6. Wait for bus idle (BBSY = 0)

#### Parameters

|    |                  |                                                                                                                                                |
|----|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| in | channel          | RIIC channel wrapper <ul style="list-style-type: none"><li>channel.value range: 0-2</li></ul>                                                  |
| in | device_↵<br>addr | 7-bit I2C device address wrapper <ul style="list-style-type: none"><li>device_addr.value range: 0x08-0x77</li></ul>                            |
| in | data             | Pointer to data buffer to write <ul style="list-style-type: none"><li>Must not be nullptr</li><li>Must contain at least length bytes</li></ul> |
| in | length           | Number of bytes to write <ul style="list-style-type: none"><li>Range: 1-65535</li></ul>                                                        |

#### Returns

rx\_err\_t Error code indicating success or failure

#### Return values

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| k_rx_ok                | All bytes successfully written and ACKed                |
| k_rx_err_null_ptr      | data pointer is nullptr                                 |
| k_rx_err_invalid_state | Channel not initialized via <a href="#">riic_init()</a> |
| k_rx_err_timeout       | Bus timeout waiting for ACK or bus idle                 |
| k_rx_err_nack          | Device sent NACK (wrong address or device busy)         |

#### Precondition

Channel must be initialized via [riic\\_init\(\)](#)

data pointer must be valid and contain at least length bytes

#### Postcondition

All data bytes transmitted to the I2C device if successful

I2C bus returned to idle state (STOP condition generated)

#### Note

Thread Safety: NOT thread-safe. Do not perform concurrent transactions on the same RIIC channel. Different channels may be used from different threads.

## Example: Write IMU Configuration Register

```

riic_channel_t ch = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

// Write power management register (0x6B) to wake up MPU6050
uint8_t wake_cmd[] = {0x6B, 0x00};
rx_err_t err = riic_write(ch, imu_addr, wake_cmd, 2);
if (err != k_rx_ok) {
 rx_log_error_val("IMU", "Failed to wake IMU", err);
}

```

See also

[riic\\_read\(\)](#) Read data from I2C device  
[riic\\_write\\_read\(\)](#) Combined write-then-read (register read pattern)  
[riic\\_init\(\)](#) Initialize RIIC channel first

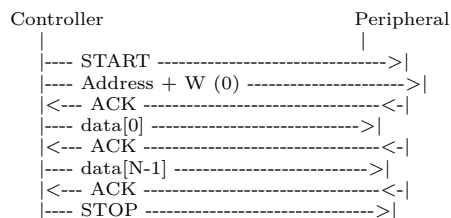
Since

Version 1.0.0

Write data to I2C device on specified RIIC channel.

Performs a complete I2C write transaction: START, address+W, data bytes, STOP. This function is suitable for writing configuration registers, sending commands, or storing data in EEPROM.

## 8.1.5.55 Transaction Sequence



## 8.1.5.56 Common Use Cases

| Use Case              | Data Format                 |
|-----------------------|-----------------------------|
| Config register write | [reg_addr, value]           |
| Multi-byte register   | [reg_addr, val0, val1, ...] |
| EEPROM page write     | [addr_hi, addr_lo, data...] |
| Sensor command        | [command_↵ byte]            |

## 8.1.5.57 Error Handling

On any error (timeout or NACK), the function issues a STOP condition to release the bus before returning. The original error code is preserved.

## Precondition

`riic_init()` called for this channel  
`data != nullptr`  
`length >= 1 && length <= 256`  
`device__addr.value <= 127`

## Postcondition

On success: All bytes transferred, bus released  
 On failure: Bus released via STOP

## Note

This is a blocking function - returns only after transfer complete.  
 STOP is always issued (success or failure) to release the bus.

## Warning

NACK on address usually means device not present or wrong address.  
 NACK on data may indicate device buffer full or write-protected.

## Example: Write Configuration Register

```
// Write value 0x42 to register 0x05 of device at address 0x48
uint8_t config_data[2] = {0x05, 0x42}; // [reg_addr, value]
riic_channel_t channel = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};

rx_err_t err = riic_write(channel, addr, config_data, 2);
if (err != k_rx_ok) {
 handle_i2c_error(err);
}
```

## See also

`riic_init()` Initialize channel before use  
`riic_read()` Read-only transaction  
`riic_write_read()` Combined write-then-read transaction

Definition at line 2349 of file `riic.c`.

```
02353 {
02354 RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02355 RX_CHECK_RANGE_TAG(channel.value,
02356 k_riic_channel_0,
02357 (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02358 k_rx_err_invalid_arg,
02359 s_tag);
02360
02361 if (device__addr.value > k_riic_addr_max_7bit) {
02362 rx_log_error(s_tag, "Invalid device address");
02363 }
02364 }
```

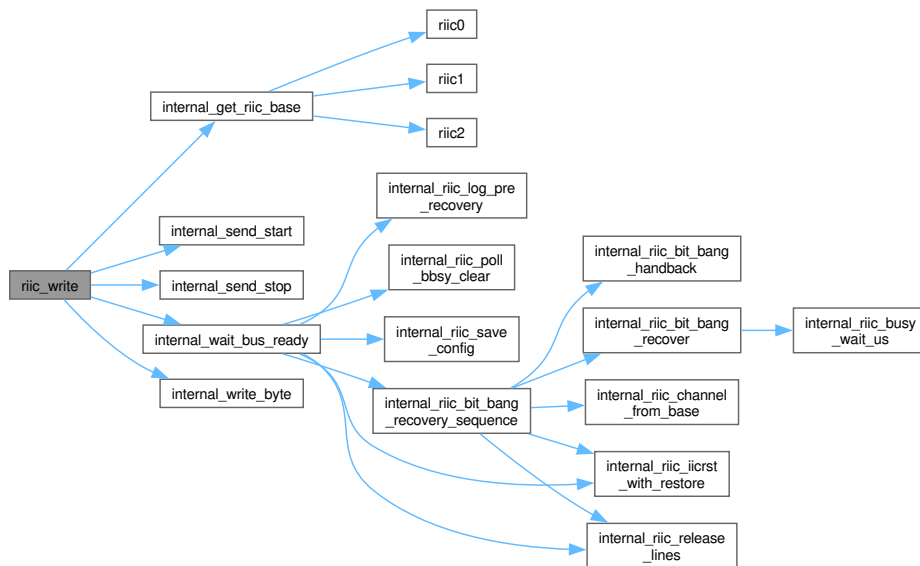
```

02363 return k_rx_err_invalid_arg;
02364 }
02365
02366 if (length == k_riic_length_zero) {
02367 rx_log_error(s_tag, "Write length cannot be zero");
02368 return k_rx_err_invalid_arg;
02369 }
02370
02371 if (length > k_riic_max_transfer_length) {
02372 rx_log_error(s_tag, "Write length exceeds maximum");
02373 return k_rx_err_invalid_arg;
02374 }
02375
02376 /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02377 if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02378 rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02379 return k_rx_err_invalid_state;
02380 }
02381
02382 /* Get RIIC base */
02383 volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02384
02385 /* Wait for bus ready */
02386 rx_err_t err = internal_wait_bus_ready(riic);
02387 RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02388
02389 /* Pre-clear any status flags left over from a previous transaction (e.g.
02390 * NACKF, STOP, AL) that normal bus activity does not clear. Per RX72N
02391 * HW manual section 42.2.7, these are sticky software-clear flags that
02392 * IICRST wipes but normal transactions do not. Without this pre-clear a
02393 * stale NACKF survives into the next write and internal_write_byte
02394 * returns k_rx_err_nack on a byte that was actually ACKed; a stale AL
02395 * silently clears MST on the next START. */
02396 riic->iccr2 = k_riic_register_clear;
02397
02398 /* Set controller transmit mode */
02399 riic->iccr2 = k_riic_iccr2_mst | k_riic_iccr2_trs;
02400
02401 /* Send start condition */
02402 err = internal_send_start(riic);
02403 RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02404
02405 /* Send device address (write) */
02406 err = internal_write_byte(riic, (device_addr.value « k_riic_addr_shift) | k_riic_addr_write_bit);
02407 if (err != k_rx_ok) {
02408 rx_err_t stop_err = internal_send_stop(riic);
02409 (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02410 return err;
02411 }
02412
02413 /* Send data bytes */
02414 for (uint16_t i = 0; i < length; i++) {
02415 err = internal_write_byte(riic, data[i]);
02416 if (err != k_rx_ok) {
02417 rx_err_t stop_err = internal_send_stop(riic);
02418 (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02419 return err;
02420 }
02421 }
02422
02423 /* Send stop condition */
02424 err = internal_send_stop(riic);
02425 RX_RETURN_ON_ERROR(err, s_tag, "Stop condition failed");
02426
02427 return k_rx_ok;
02428 }

```

References [rx\\_riic\\_regs\\_t::iccr2](#), [rx\\_riic\\_regs\\_t::icsr2](#), [internal\\_get\\_riic\\_base\(\)](#), [internal\\_send\\_start\(\)](#), [internal\\_send\\_stop\(\)](#), [internal\\_wait\\_bus\\_ready\(\)](#), [internal\\_write\\_byte\(\)](#), [k\\_riic\\_addr\\_max\\_7bit](#), [k\\_riic\\_addr\\_shift](#), [k\\_riic\\_addr\\_write\\_bit](#), [k\\_riic\\_channel\\_0](#), [k\\_riic\\_iccr2\\_mst](#), [k\\_riic\\_iccr2\\_trs](#), [k\\_riic\\_last\\_index\\_offset](#), [k\\_riic\\_length\\_zero](#), [k\\_riic\\_max\\_channels](#), [k\\_riic\\_max\\_transfer\\_length](#), [k\\_riic\\_mode\\_controller](#), [k\\_riic\\_register\\_clear](#), [s\\_riic\\_channel\\_mode](#), [s\\_tag](#), [i2c\\_device\\_addr\\_t::value](#), and [riic\\_channel\\_t::value](#).

Here is the call graph for this function:



#### 8.1.5.58 riic\_write\_read()

```

rx_err_t riic_write_read (
 const riic_channel_t channel,
 const i2c_device_addr_t device_addr,
 const uint8_t * write_data,
 const uint16_t write_length,
 uint8_t * read_data,
 const uint16_t read_length) [nodiscard]

```

Write then read from I2C device using repeated START (combined transaction).

Performs an I2C combined write-then-read transaction using a repeated START condition. This is the standard pattern for reading device registers: write the register address, then issue a repeated START and read back the register data without releasing the bus between operations.

#### 8.1.5.59 Transaction Sequence

1. Generate START condition
2. Send device address byte (7-bit address << 1 | 0) – write mode
3. Wait for ACK from peripheral
4. Send write\_data bytes (typically register address), ACK after each
5. Generate repeated START condition (no STOP)
6. Send device address byte (7-bit address << 1 | 1) – read mode
7. Wait for ACK from peripheral
8. Receive read\_data bytes, ACK each (NACK on last byte)
9. Generate STOP condition
10. Wait for bus idle (BBSY = 0)

## Parameters

|     |              |                                                                                                                                                                                                                |
|-----|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in  | channel      | RIIC channel wrapper <ul style="list-style-type: none"> <li>channel.value range: 0-2</li> </ul>                                                                                                                |
| in  | device_addr  | 7-bit I2C device address wrapper <ul style="list-style-type: none"> <li>device_addr.value range: 0x08-0x77</li> </ul>                                                                                          |
| in  | write_data   | Pointer to data to write (typically register address) <ul style="list-style-type: none"> <li>Must not be nullptr</li> <li>Must contain at least write_length bytes</li> </ul>                                  |
| in  | write_length | Number of bytes to write <ul style="list-style-type: none"> <li>Typically 1 (single register address byte)</li> </ul>                                                                                          |
| out | read_data    | Pointer to buffer for received data <ul style="list-style-type: none"> <li>Must not be nullptr</li> <li>Must have space for at least read_length bytes</li> <li>Unchanged if function returns error</li> </ul> |
| in  | read_length  | Number of bytes to read <ul style="list-style-type: none"> <li>Range: 1-65535</li> </ul>                                                                                                                       |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| k_rx_ok                | Write and read completed successfully                   |
| k_rx_err_null_ptr      | write_data or read_data pointer is nullptr              |
| k_rx_err_invalid_state | Channel not initialized via <a href="#">riic_init()</a> |
| k_rx_err_timeout       | Bus timeout during write or read phase                  |
| k_rx_err_nack          | Device sent NACK (wrong address or device busy)         |

## Precondition

Channel must be initialized via [riic\\_init\(\)](#)

write\_data and read\_data pointers must be valid with sufficient size

## Postcondition

read\_data buffer contains received register data if function succeeds

I2C bus returned to idle state (STOP condition generated)



## Note

Thread Safety: NOT thread-safe. Do not perform concurrent transactions on the same RIIC channel. Different channels may be used from different threads.

Example: Read IMU WHO\_AM\_I Register

```
riic_channel_t ch = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

uint8_t reg_addr = 0x75; // WHO_AM_I register
uint8_t who_am_i;
rx_err_t err = riic_write_read(ch, imu_addr, ®_addr, 1, &who_am_i, 1);
if (err == k_rx_ok && who_am_i == 0x68) {
 rx_log_info("IMU", "MPU6050 detected");
}
```

See also

[riic\\_write\(\)](#) Write-only transaction  
[riic\\_read\(\)](#) Read-only transaction  
[riic\\_init\(\)](#) Initialize RIIC channel first

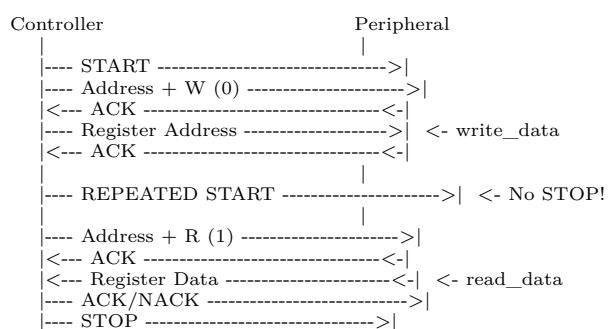
Since

Version 1.0.0

Write then read from I2C device using repeated START (combined transaction).

Performs a complete I2C write-read transaction using a repeated START condition. This is the most common pattern for reading registers from I2C peripherals: write the register address, then read the data.

#### 8.1.5.60 Transaction Sequence (Register Read)



#### 8.1.5.61 Why Repeated START?

The repeated START (Sr) is critical for atomic register access:

| Approach     | Bus Release | Risk                               |
|--------------|-------------|------------------------------------|
| STOP + START | Yes         | Another controller could interrupt |

| Approach       | Bus Release | Risk                  |
|----------------|-------------|-----------------------|
| Repeated START | No          | Transaction is atomic |

Many I2C peripherals require repeated START and may not work correctly with separate write-then-read transactions.

#### 8.1.5.62 Common Use Cases

| Peripheral Type | write_data            | read_length | Purpose            |
|-----------------|-----------------------|-------------|--------------------|
| Temperature     | [reg_↵<br>addr]       | 2           | Read temp register |
| EEPROM          | [addr_hi,<br>addr_lo] | N           | Read N bytes       |
| Accelerometer   | [reg_↵<br>addr]       | 6           | Read XYZ axes      |
| RTC             | [seconds_↵<br>reg]    | 7           | Read time          |

#### Precondition

```

riic_init() called for this channel
write_data != nullptr
read_data != nullptr
write_length >= 1 && write_length <= 256
read_length >= 1 && read_length <= 256
device_addr.value <= 127

```

#### Postcondition

On success: read\_data contains register contents, bus released  
On failure: Bus released via STOP

#### Note

This is a blocking function - returns only after transfer complete.  
Uses repeated START (no bus release between write and read phases).  
NACK is automatically sent before reading last byte.

#### Example: Read WHO\_AM\_I Register from IMU

```

// Read WHO_AM_I register (0x0F) from MPU6050 at address 0x68
riic_channel_t channel = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

uint8_t reg_addr = 0x0F; // WHO_AM_I register
uint8_t who_am_i;

rx_err_t err = riic_write_read(channel, imu_addr,
 ®_addr, 1, // Write: register address
 &who_am_i, 1); // Read: 1 byte

if (err == k_rx_ok && who_am_i == 0x68) {
 // IMU detected and responding correctly
}

```

Example: Read 6 Bytes of Accelerometer Data

```
// Read ACCEL_XOUT_H through ACCEL_ZOUT_L (6 bytes starting at 0x3B)
uint8_t reg_addr = 0x3B;
uint8_t accel_data[6];

rx_err_t err = riic_write_read(channel, imu_addr,
 ®_addr, 1,
 accel_data, 6);

if (err == k_rx_ok) {
 int16_t accel_x = (accel_data[0] « 8) | accel_data[1];
 int16_t accel_y = (accel_data[2] « 8) | accel_data[3];
 int16_t accel_z = (accel_data[4] « 8) | accel_data[5];
}
```

See also

[riic\\_init\(\)](#) Initialize channel before use

[riic\\_write\(\)](#) Write-only transaction

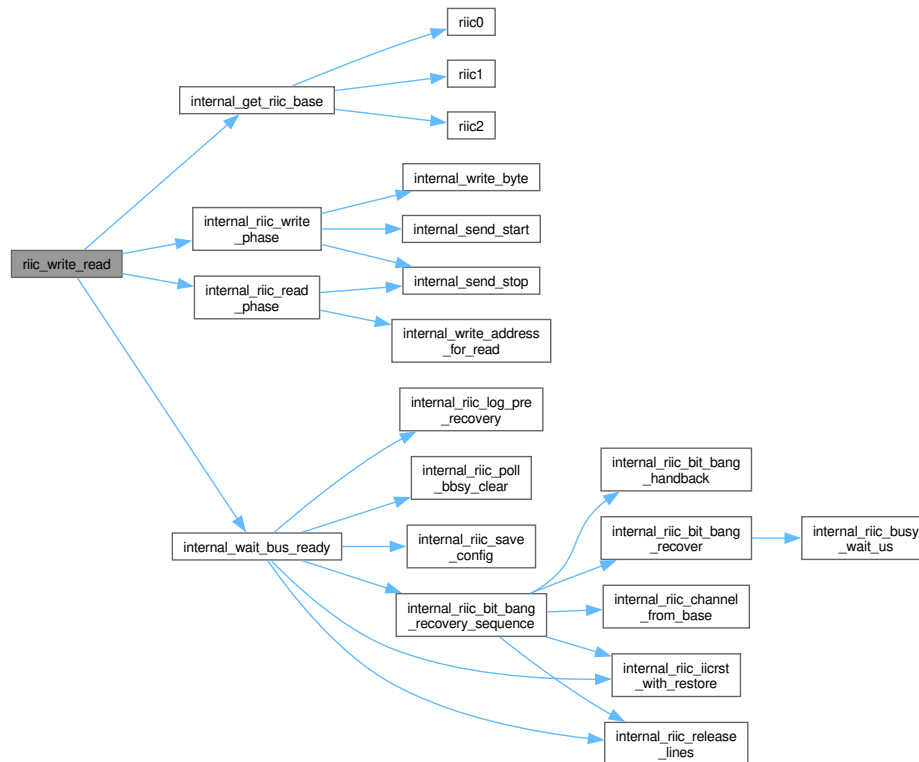
[riic\\_read\(\)](#) Read-only transaction (no address write)

Definition at line 2721 of file [riic.c](#).

```
02727 {
02728 RX_CHECK_NULL_PTR(write_data, s_tag, "Write data pointer is nullptr");
02729 RX_CHECK_NULL_PTR(read_data, s_tag, "Read data pointer is nullptr");
02730 RX_CHECK_RANGE_TAG(channel.value,
02731 k_riic_channel_0,
02732 (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02733 k_rx_err_invalid_arg,
02734 s_tag);
02735
02736 if (device_addr.value > k_riic_addr_max_7bit) {
02737 rx_log_error(s_tag, "Invalid device address");
02738 return k_rx_err_invalid_arg;
02739 }
02740
02741 if (write_length == k_riic_length_zero || read_length == k_riic_length_zero) {
02742 rx_log_error(s_tag, "Read/write length cannot be zero");
02743 return k_rx_err_invalid_arg;
02744 }
02745
02746 if (write_length > k_riic_max_transfer_length || read_length > k_riic_max_transfer_length) {
02747 rx_log_error(s_tag, "Read/write length exceeds maximum");
02748 return k_rx_err_invalid_arg;
02749 }
02750
02751 /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02752 if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02753 rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02754 return k_rx_err_invalid_state;
02755 }
02756
02757 /* Get RIIC base */
02758 volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02759
02760 /* Wait for bus ready */
02761 rx_err_t err = internal_wait_bus_ready(riic);
02762 RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02763
02764 /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02765 riic->icr2 = k_riic_register_clear;
02766
02767 /* Perform write phase */
02768 err = internal_riic_write_phase(riic, device_addr, write_data, write_length);
02769 if (err != k_rx_ok) {
02770 return err;
02771 }
02772
02773 /* Perform read phase. On success the final byte's NACK + STOP is
02774 * issued atomically inside internal_read_byte(), so no separate
02775 * internal_send_stop() call is needed here. */
02776 err = internal_riic_read_phase(riic, device_addr, read_data, read_length);
02777 if (err != k_rx_ok) {
02778 return err;
02779 }
02780
02781 return k_rx_ok;
02782 }
```

References `rx_riic_regs_t::icr2`, `internal_get_riic_base()`, `internal_riic_read_phase()`, `internal_riic_write_phase()`, `internal_wait_bus_ready()`, `k_riic_addr_max_7bit`, `k_riic_channel_0`, `k_riic_last_index_offset`, `k_riic_length_zero`, `k_riic_max_channels`, `k_riic_max_transfer_length`, `k_riic_mode_controller`, `k_riic_register_clear`, `s_riic_channel_mode`, `s_tag`, `i2c_device_addr_t::value`, and `riic_channel_t::value`.

Here is the call graph for this function:



#### 8.1.5.63 `rspi_controller_deinit()`

```

rx_err_t rspi_controller_deinit (
 const rspi_channel_t channel) [nodiscard]

```

Deinitialize and disable an RSPI controller-mode channel.

Performs a safe shutdown of the specified RSPI channel previously initialized in controller mode. The deinitialization sequence is ordered to prevent bus glitches: first deasserts CS to release any connected peripheral, then disables SPI (`SPCR.SPE = 0`) to stop clock generation, then gates the module clock via `MSTPCRB` for power savings, and finally clears internal state (CS pin config and initialization flag). Register protection (`PRCR`) is temporarily unlocked for module stop control.

This function is safe to call on channels that were never initialized in controller mode; it will still disable the hardware and clear the state.

#### Parameters

|    |         |                                                                                                                                        |
|----|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| in | channel | RSPI channel number (valid: <code>k_rspi_channel_0</code> , <code>k_rspi_channel_1</code> , <code>k_rspi_channel_2</code> ; range 0-2) |
|----|---------|----------------------------------------------------------------------------------------------------------------------------------------|

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                   |                                                                     |
|-----------------------------------|---------------------------------------------------------------------|
| <code>k_rx_ok</code>              | Channel successfully deinitialized and module stopped               |
| <code>k_rx_err_invalid_arg</code> | channel $\geq 3$ or RSPI base address lookup failed for the channel |

## Precondition

channel must be in range [0, 2] (`k_rspi_channel_0` through `k_rspi_channel_2`)

No active SPI transfer should be in progress on this channel

## Postcondition

CS GPIO deasserted (driven high) if channel was initialized

RSPI peripheral disabled (`SPCR.SPE = 0`)

Module clock gated (`MSTPCRB` module stop bit set)

`s_rspi_cs_config[channel]` cleared to defaults (port=0, pin=0)

`s_rspi_controller_initialized[channel]` set to false

## Note

Not thread-safe. Caller must ensure no concurrent transfers or accesses to the same channel during deinitialization.

## See also

[rspi\\_init\\_controller\(\)](#) Reinitialize channel after deinit

[rspi\\_deinit\(\)](#) Deinit for peripheral-mode channels

## Since

Version 1.0.0

Deinitialize and disable an RSPI controller-mode channel.

Disables the RSPI controller, deasserts chip select, clears configuration, and stops the module to reduce power consumption.

## Precondition

Should be called after [rspi\\_init\\_controller\(\)](#) to properly clean up

## Postcondition

If successful:

- CS is deasserted via `rspi_controller_set_cs(channel, false)`
- RSPI is disabled (`rspi->spr = k_rspi_spr_disabled`)
- Module clock is stopped via `internal_set_mstpcrb_for_channel()`
- `s_rspi_cs_config[channel]` is cleared (`port=0, pin=0`)
- `s_rspi_controller_initialized[channel] = false`
- Register protection is unlocked/locked via `*prcr_reg()`

## Note

Thread-safe only if caller ensures no concurrent access to the same channel

Definition at line 1656 of file `rspi.c`.

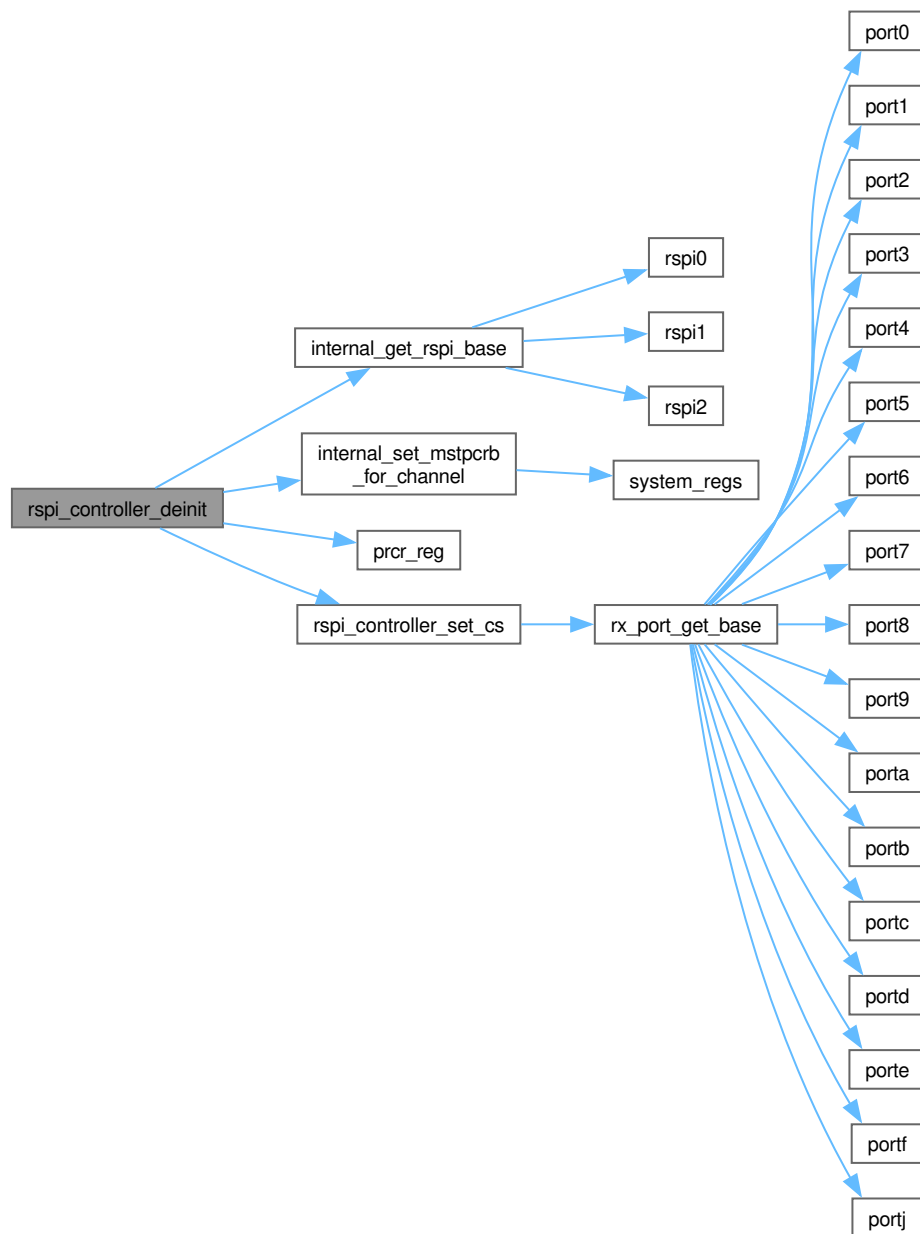
```

01657 {
01658 /* Validate channel */
01659 if ((uint8_t)channel >= k_rspi_max_channels) {
01660 rx_log_error(s_tag, "Invalid RSPI channel");
01661 return k_rx_err_invalid_arg;
01662 }
01663
01664 /* Get RSPI base */
01665 volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01666 if (rspi == nullptr) {
01667 return k_rx_err_invalid_arg;
01668 }
01669
01670 /*
01671 * RSPI Controller Deinitialization Sequence
01672 * =====
01673 * The deinit order ensures safe shutdown without bus glitches:
01674 * 1. Deassert CS (drive high) - release peripheral before disabling SPI
01675 * 2. Disable SPI (SPCR.SPE=0) - stop controller clock generation
01676 * 3. Disable module clock (MSTPCRB) - gate clock only after SPI is idle
01677 * 4. Clear software state - CS config and init flag after hardware is off
01678 */
01679
01680 /* Step 1: Deassert CS to release any connected peripheral before shutdown */
01681 if (s_rspi_controller_initialized[channel]) {
01682 (void)rspi_controller_set_cs(channel, false);
01683 }
01684
01685 /* Step 2: Disable SPI to stop controller clock generation */
01686 rspi->spr = k_rspi_spr_disabled;
01687
01688 /* Step 3: Disable RSPI module clock (set module stop bit) - must be done
01689 * after SPI is disabled to avoid undefined behavior from clock gating
01690 * during an active transfer */
01691 *prcr_reg() = k_rx_pcr_unlock_prc1_prc3;
01692 internal_set_mstpcrb_for_channel(channel, false);
01693 *prcr_reg() = k_rx_pcr_lock;
01694
01695 /* Step 4: Clear software state after hardware is safely off */
01696 s_rspi_cs_config[channel].port = k_rspi_cs_default_port;
01697 s_rspi_cs_config[channel].pin = k_rspi_cs_default_pin;
01698
01699 s_rspi_controller_initialized[channel] = false;
01700
01701 rx_log_debug(s_tag, "RSPI controller deinitialized");
01702
01703 return k_rx_ok;
01704 }

```

References `internal_get_rspi_base()`, `internal_set_mstpcrb_for_channel()`, `k_rspi_cs_default_pin`, `k_rspi_cs_default_port`, `k_rspi_max_channels`, `k_rspi_spr_disabled`, `prcr_reg()`, `rspi_controller_set_cs()`, `s_rspi_controller_initialized`, `s_rspi_cs_config`, `s_tag`, and `rx_rspi_regs_t::spr`.

Here is the call graph for this function:



#### 8.1.5.64 rspi\_controller\_set\_cs()

```

rx_err_t rspi_controller_set_cs (
 const rspi_channel_t channel,
 const bool active) \[nodiscard\]

```

Set chip select line state for an RSPI controller-mode channel.

Controls the GPIO chip select pin associated with the specified RSPI channel operating in controller mode. The CS signal is active-low: when active is true, the GPIO pin is driven low (CS asserted, peripheral selected); when active is false, the pin is driven high (CS deasserted, peripheral released). The port and pin are read from the internal `s_rspi_cs_config[]` array populated during [rspi\\_init\\_controller\(\)](#).

This function provides direct CS control for protocols that require multi-transfer CS framing. For single 16-bit transfers with automatic CS handling, use [rspi\\_controller\\_transfer\\_16bit\(\)](#) instead.

## Parameters

|    |         |                                                                                                                                        |
|----|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| in | channel | RSPI channel number (valid: <code>k_rspi_channel_0</code> , <code>k_rspi_channel_1</code> , <code>k_rspi_channel_2</code> ; range 0-2) |
| in | active  | true to assert CS (drive GPIO low), false to deassert CS (drive GPIO high)                                                             |

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                     |                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | CS line successfully driven to requested state                                                            |
| <code>k_rx_err_invalid_state</code> | channel $\geq 3$ or channel not initialized in controller mode via <a href="#">rspi_init_controller()</a> |
| <code>k_rx_err_invalid_arg</code>   | GPIO port base address lookup failed for the stored CS port configuration                                 |

## Precondition

channel must be initialized via [rspi\\_init\\_controller\(\)](#)

`s_rspi_cs_config[channel]` must contain valid port/pin from init

## Postcondition

GPIO PODR bit for the CS pin reflects the requested state

No other GPIO pins on the same port are modified (read-modify-write)

## Note

Not thread-safe. Caller must provide external synchronization when accessing the same channel or GPIO port from multiple threads.

## See also

[rspi\\_init\\_controller\(\)](#) Configures CS pin during initialization

[rspi\\_controller\\_transfer\\_16bit\(\)](#) Automatic CS handling for transfers

[rspi\\_controller\\_deinit\(\)](#) Deasserts CS during cleanup

## Since

Version 1.0.0

Set chip select line state for an RSPI controller-mode channel.

Controls the GPIO chip select pin for the specified RSPI channel operating in controller mode. CS is active-low (asserted low, deasserted high).



## Precondition

Channel must be initialized via [rspi\\_init\\_controller\(\)](#)  
`channel < k_rspi_max_channels`  
`s_rspi_controller_initialized[channel] == true`

## Postcondition

GPIO `port_regs->podr` is modified to assert/deassert CS pin

## Note

Reads `s_rspi_cs_config[channel]` for port and pin configuration

Definition at line 1382 of file [rspi.c](#).

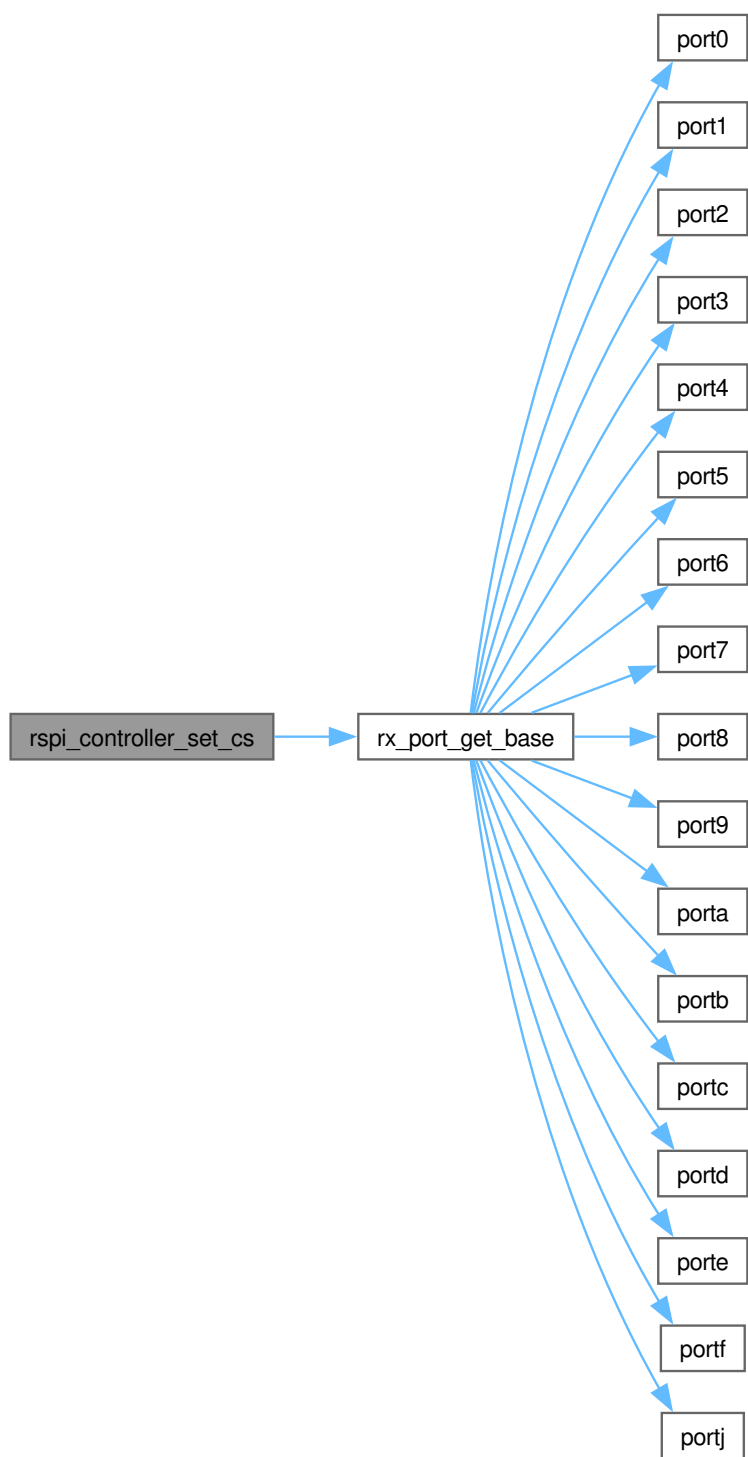
```

01383 {
01384 /* Validate channel */
01385 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01386 rx_log_error(s_tag, "RSPI controller channel not initialized");
01387 return k_rx_err_invalid_state;
01388 }
01389
01390 const uint8_t port = s_rspi_cs_config[channel].port;
01391 const uint8_t pin = s_rspi_cs_config[channel].pin;
01392 volatile rx_port_regs_t* const port_regs = rx_port_get_base(port);
01393
01394 if (port_regs == nullptr) {
01395 return k_rx_err_invalid_arg;
01396 }
01397
01398 /* Compute pin mask once (identical for both assert and deassert) */
01399 const uint8_t pin_mask = (uint8_t)((uint32_t)k_rspi_bit_set « pin);
01400
01401 if (active) {
01402 /* CS active (low) */
01403 port_regs->podr &= (uint8_t)~pin_mask;
01404 } else {
01405 /* CS inactive (high) */
01406 port_regs->podr |= pin_mask;
01407 }
01408
01409 return k_rx_ok;
01410 }
```

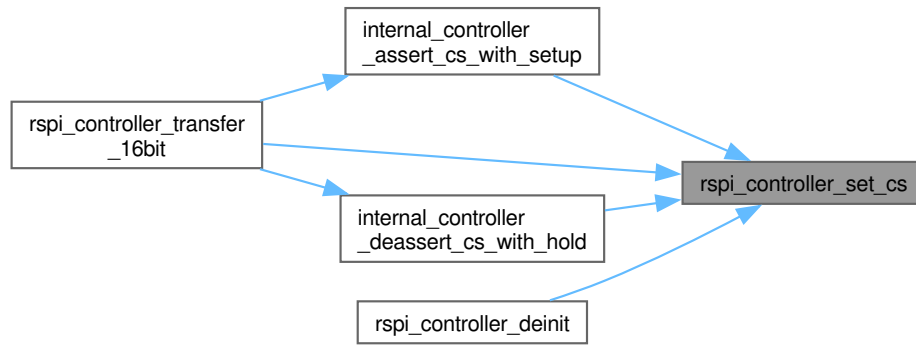
References [k\\_rspi\\_bit\\_set](#), [k\\_rspi\\_max\\_channels](#), [rx\\_port\\_regs\\_t::podr](#), [rx\\_port\\_get\\_base\(\)](#), [s\\_rspi\\_controller\\_initialized](#), [s\\_rspi\\_cs\\_config](#), and [s\\_tag](#).

Referenced by [internal\\_controller\\_assert\\_cs\\_with\\_setup\(\)](#), [internal\\_controller\\_deassert\\_cs\\_with\\_hold\(\)](#), [rspi\\_controller\\_deinit\(\)](#), and [rspi\\_controller\\_transfer\\_16bit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.65 rspi\_controller\_transfer\_16bit()

```

rx_err_t rspi_controller_transfer_16bit (
 const rspi_channel_t channel,
 const uint16_t tx_data,
 uint16_t *const rx_data) [nodiscard]

```

Perform a 16-bit full-duplex SPI transfer in controller mode with automatic CS.

Executes a single 16-bit SPI transfer with full chip select lifecycle management. The transfer sequence is:

1. Assert CS (drive low) with ~300 ns setup delay
2. Wait for TX buffer empty, write 16-bit tx\_data to SPDR
3. Wait for RX buffer full, read 16-bit response from SPDR
4. Clear SPSR status flags (SPRF, OVRF)
5. Apply ~300 ns hold delay, then deassert CS (drive high)

On transfer error (timeout), CS is forcibly deasserted before returning to avoid leaving the peripheral in a selected state. The CS setup and hold delays use cycle-accurate NOP loops calibrated for 240 MHz core clock.

#### Parameters

|     |         |                                                                                              |
|-----|---------|----------------------------------------------------------------------------------------------|
| in  | channel | RSPI channel number (valid: k_rspi_channel_0, k_rspi_channel_1, k_rspi_channel_2; range 0-2) |
| in  | tx_data | 16-bit data word to transmit                                                                 |
| out | rx_data | Pointer to store the received 16-bit response (must not be nullptr)                          |

#### Returns

rx\_err\_t Error code indicating success or failure

#### Return values

|         |                                                               |
|---------|---------------------------------------------------------------|
| k_rx_ok | Transfer completed, *rx_data contains response, CS deasserted |
|---------|---------------------------------------------------------------|

|                                     |                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>k_rx_err_null_ptr</code>      | rx_data pointer is nullptr                                                                                  |
| <code>k_rx_err_invalid_state</code> | channel $\geq 3$ or channel not initialized via <a href="#">rspi_init_controller()</a>                      |
| <code>k_rx_err_invalid_arg</code>   | RSPI base address lookup failed, or GPIO port lookup failed during CS assertion/deassertion                 |
| <code>k_rx_err_timeout</code>       | TX buffer did not empty or RX buffer did not fill within the 10 ms timeout (CS is deasserted on this error) |

#### Precondition

channel must be initialized via [rspi\\_init\\_controller\(\)](#)

rx\_data must be a valid non-null pointer

#### Postcondition

\*rx\_data contains received 16-bit value on success

CS is deasserted (GPIO driven high) on both success and error

SPSR status flags SPRF and OVRF are cleared on success

#### Note

Not thread-safe. Caller must provide external synchronization when accessing the same channel from multiple threads.

#### See also

[rspi\\_init\\_controller\(\)](#) Must be called before this function

[rspi\\_controller\\_set\\_cs\(\)](#) Manual CS control for multi-transfer framing

[rspi\\_controller\\_deinit\(\)](#) Deinitialize controller-mode channel

#### Since

Version 1.0.0

Perform a 16-bit full-duplex SPI transfer in controller mode with automatic CS.

Executes a full-duplex 16-bit SPI transfer with automatic chip select assertion/deassertion and timing delays.

#### Sequence:

1. Assert CS (active low) with setup delay ( $\sim 300\text{ns}$ )
2. Wait for TX buffer ready, then transmit 16-bit data
3. Wait for RX buffer full, then read 16-bit response
4. Hold CS for minimum hold time ( $\sim 300\text{ns}$ )
5. Deassert CS (inactive high)
6. Clear status flags

## Precondition

Channel must be initialized via [rspi\\_init\\_controller\(\)](#)  
 rx\_data must be a valid non-nullptr

## Postcondition

If successful:

- rx\_data is filled with received 16-bit value
- Status flags (SPRF, OVRF) are cleared
- CS is deasserted (high)

On error:

- CS is deasserted if possible
- rx\_data contents are undefined

## Note

CS setup delay: k\_rspi\_cs\_setup\_delay NOP loops (~300ns)

CS hold delay: k\_rspi\_cs\_hold\_delay NOP loops (~300ns)

Uses inline assembly NOP for precise timing control

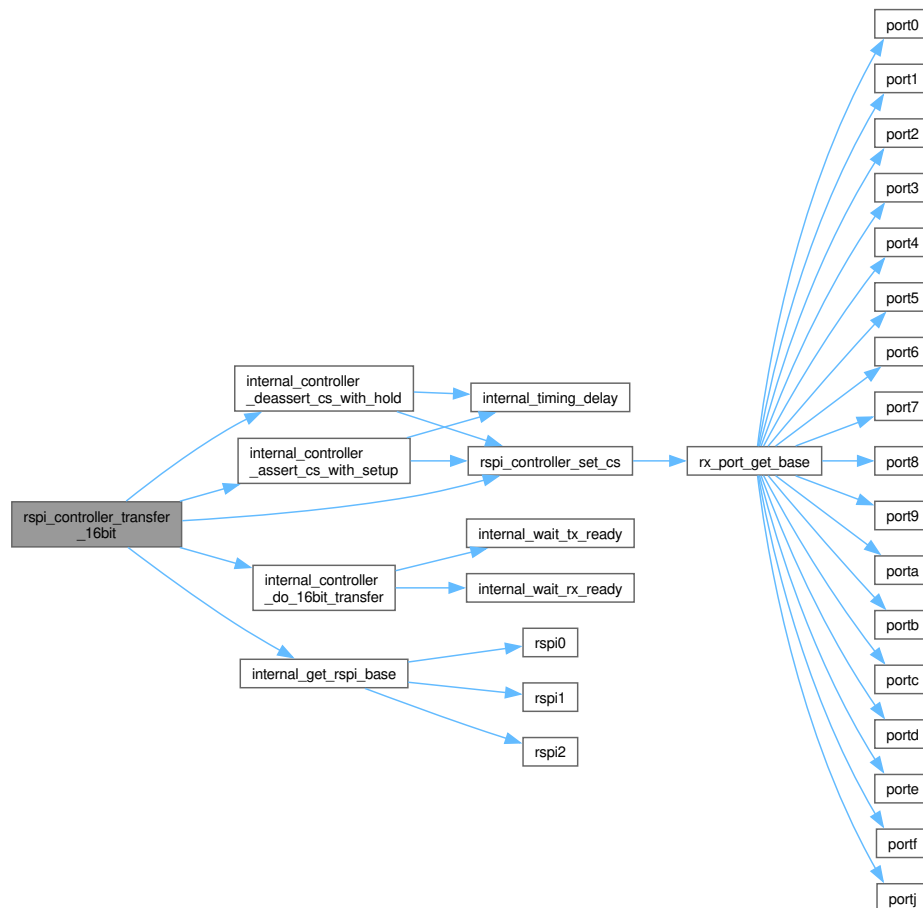
Definition at line 1600 of file [rspi.c](#).

```

01603 {
01604 RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
01605
01606 /* Validate channel */
01607 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01608 rx_log_error(s_tag, "RSPI controller channel not initialized");
01609 return k_rx_err_invalid_state;
01610 }
01611
01612 /* Get RSPI base */
01613 volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01614 if (rspi == nullptr) {
01615 return k_rx_err_invalid_arg;
01616 }
01617
01618 /* Assert CS with setup delay */
01619 rx_err_t err = internal_controller_assert_cs_with_setup(channel);
01620 if (err != k_rx_ok) {
01621 return err;
01622 }
01623
01624 /* Perform 16-bit TX/RX transfer */
01625 err = internal_controller_do_16bit_transfer(rspi, tx_data, rx_data);
01626 if (err != k_rx_ok) {
01627 /* Deassert CS on error (ignore CS deassert errors) */
01628 (void)rspi_controller_set_cs(channel, false);
01629 return err;
01630 }
01631
01632 /* Deassert CS with hold delay */
01633 return internal_controller_deassert_cs_with_hold(channel);
01634 }
```

References [internal\\_controller\\_assert\\_cs\\_with\\_setup\(\)](#), [internal\\_controller\\_deassert\\_cs\\_with\\_hold\(\)](#), [internal\\_controller\\_do\\_16bit\\_transfer\(\)](#), [internal\\_get\\_rspi\\_base\(\)](#), [k\\_rspi\\_max\\_channels](#), [rspi\\_controller\\_set\\_cs\(\)](#), [s\\_rspi\\_controller\\_initialized](#), and [s\\_tag](#).

Here is the call graph for this function:



### 8.1.5.66 rspi\_deinit()

```
rx_err_t rspi_deinit (
 const rspi_channel_t channel) [nodiscard]
```

Deinitialize and disable an RSPI peripheral-mode channel.

Performs a safe shutdown of the specified RSPI channel previously initialized in peripheral mode. The deinitialization sequence disables SPI bus activity (SPCR.SPE = 0), gates the module clock via MST-PCRB to reduce power consumption, and clears the internal initialization flag. Register protection (PRCR) is temporarily unlocked for module stop control.

This function is safe to call on channels that were never initialized; it will still disable the hardware and clear the init flag. To reinitialize a channel after deinit, call [rspi\\_init\\_peripheral\(\)](#) again.

#### Parameters

|    |         |                                                                                              |
|----|---------|----------------------------------------------------------------------------------------------|
| in | channel | RSPI channel number (valid: k_rspi_channel_0, k_rspi_channel_1, k_rspi_channel_2; range 0-2) |
|----|---------|----------------------------------------------------------------------------------------------|

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                   |                                                                     |
|-----------------------------------|---------------------------------------------------------------------|
| <code>k_rx_ok</code>              | Channel successfully deinitialized and module stopped               |
| <code>k_rx_err_invalid_arg</code> | channel $\geq 3$ or RSPI base address lookup failed for the channel |

## Precondition

channel must be in range  $[0, 2]$  (`k_rspi_channel_0` through `k_rspi_channel_2`)

No active SPI transfer should be in progress on this channel

## Postcondition

RSPI peripheral disabled (`SPCR.SPE = 0`)

Module clock gated (`MSTPCRB` module stop bit set)

`s_rspi_channel_initialized[channel]` set to false

## Note

Not thread-safe. Caller must ensure no concurrent transfers or accesses to the same channel during deinitialization.

## See also

[rspi\\_init\\_peripheral\(\)](#) Reinitialize channel after deinit

[rspi\\_controller\\_deinit\(\)](#) Deinit for controller-mode channels

## Since

Version 1.0.0

Deinitialize and disable an RSPI peripheral-mode channel.

Disables the specified RSPI channel, clears the initialization flag, and optionally disables the module (via module stop bit) to reduce power consumption.

## Precondition

Channel should have been initialized via [rspi\\_init\\_peripheral\(\)](#) previously

## Postcondition

If successful:

- RSPI peripheral is disabled (SPCR.SPE = 0)
- `s_rspi_channel_initialized[channel]` is set to false
- Channel is no longer available for SPI operations

Definition at line 884 of file [rspi.c](#).

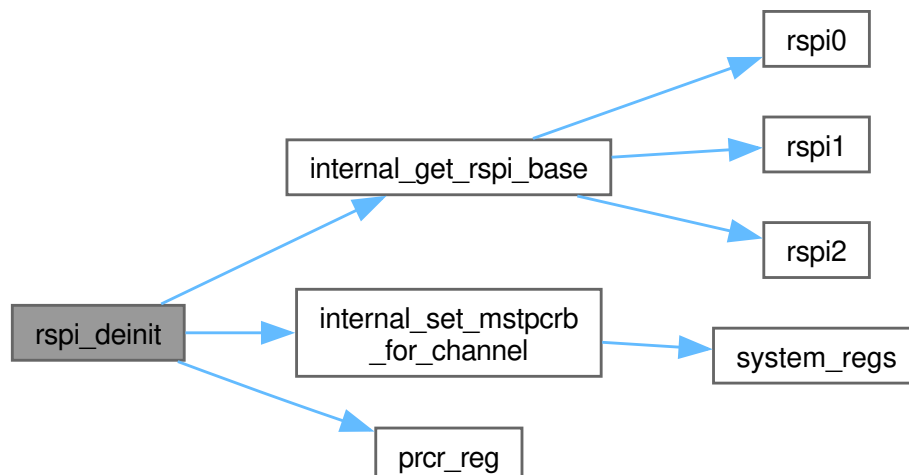
```

00885 {
00886 /* Validate channel */
00887 if ((uint8_t)channel >= k_rspi_max_channels) {
00888 rx_log_error(s_tag, "Invalid RSPI channel");
00889 return k_rx_err_invalid_arg;
00890 }
00891
00892 /* Get RSPI base */
00893 volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00894 if (rspi == nullptr) {
00895 return k_rx_err_invalid_arg;
00896 }
00897
00898 /*
00899 * RSPI Deinitialization Register Write Sequence
00900 * =====
00901 * The deinit order is the reverse of initialization:
00902 * 1. Disable SPI (SPCR.SPE=0) - stop all bus activity before removing clock
00903 * 2. Disable module clock (MSTPCRB) - safe to gate clock only after SPI is idle
00904 * 3. Clear software state - mark channel uninitialized after hardware is off
00905 */
00906
00907 /* Step 1: Disable SPI to stop all bus activity */
00908 rspi->spcr = k_rspi_spcr_disabled;
00909
00910 /* Step 2: Disable RSPI module clock (set module stop bit) - must be done
00911 * after SPI is disabled to avoid undefined behavior from clock gating
00912 * during an active transfer */
00913 *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00914 internal_set_mstpcrb_for_channel(channel, false);
00915
00916 *prcr_reg() = k_rx_prcr_lock;
00917
00918 /* Step 3: Mark channel as uninitialized after hardware is safely off */
00919 s_rspi_channel_initialized[channel] = false;
00920
00921 rx_log_debug(s_tag, "RSPI deinitialized");
00922
00923 return k_rx_ok;
00924 }

```

References [internal\\_get\\_rspi\\_base\(\)](#), [internal\\_set\\_mstpcrb\\_for\\_channel\(\)](#), [k\\_rspi\\_max\\_channels](#), [k\\_rspi\\_spcr\\_disabled](#), [prcr\\_reg\(\)](#), [s\\_rspi\\_channel\\_initialized](#), [s\\_tag](#), and [rx\\_rspi\\_regs\\_t::spcr](#).

Here is the call graph for this function:





8.1.5.67 `rspi_init_controller()`

```

rx_err_t rspi_init_controller (
 const rspi_channel_t channel,
 const rspi_controller_config_t * config) [nodiscard]

```

Initialize RSPI channel in controller mode for external device communication.

Configures the specified RSPI channel as an SPI controller for communicating with external SPI peripheral devices such as sensors. Performs argument validation, SPBR bit rate calculation, CS GPIO pin configuration, and full hardware register setup following the RX72N Hardware Manual Section 38.3.6 initialization sequence. Always configures 16-bit data frames with word access mode.

The initialization is split into three internal phases: argument validation, hardware resource preparation (SPBR calculation, base address lookup, GPIO setup), and register configuration (module clock enable, SPI mode, bit rate, SPI enable). On any failure, no partial hardware state is left behind.

## Parameters

|    |         |                                                                                                                                                                                                                                                                                                                                                        |
|----|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in | channel | RSPI channel number (valid: <code>k_rspi_channel_0</code> , <code>k_rspi_channel_1</code> , <code>k_rspi_channel_2</code> ; range 0-2)                                                                                                                                                                                                                 |
| in | config  | Pointer to <code>rspi_controller_config_t</code> containing: <ul style="list-style-type: none"> <li><code>spi_mode</code>: SPI mode (0-3) for CPOL/CPHA configuration</li> <li><code>freq_hz</code>: SPI clock frequency in Hz (100 kHz to 10 MHz)</li> <li><code>cs</code>: GPIO pin for chip select (<code>rx_port_pin_t</code> encoding)</li> </ul> |

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                     |                                                                                                                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Channel successfully initialized in controller mode                                                                                                                   |
| <code>k_rx_err_null_ptr</code>      | config pointer is nullptr                                                                                                                                             |
| <code>k_rx_err_invalid_arg</code>   | channel $\geq 3$ , <code>spi_mode</code> $> 3$ , <code>freq_hz</code> outside 100 kHz - 10 MHz range, RSPI base address lookup failed, or CS GPIO port/pin is invalid |
| <code>k_rx_err_invalid_state</code> | channel already initialized in controller mode (call <code>rspi_controller_deinit()</code> first)                                                                     |

## Precondition

config must be a valid non-null pointer

channel must not already be initialized in controller mode

System clocks must be configured (PCLKB running at 60 MHz)

## Postcondition

RSPI module clock enabled, registers configured, SPI enabled in controller mode (SPCR.MSTR = 1, SPCR.SPE = 1)

CS GPIO configured as output, driven high (inactive / deasserted)

s\_rspi\_controller\_initialized[channel] set to true

## Note

Not thread-safe. Initialize each channel from a single thread during system startup.

## See also

[rspi\\_controller\\_transfer\\_16bit\(\)](#) Perform 16-bit transfers after init

[rspi\\_controller\\_set\\_cs\(\)](#) Manual chip select control

[rspi\\_controller\\_deinit\(\)](#) Deinitialize controller-mode channel

[rspi\\_controller\\_config\\_t](#) Configuration structure

## Since

Version 1.0.0

Initialize RSPI channel in controller mode for external device communication.

Configures the specified RSPI channel as a SPI controller to communicate with SPI peripheral devices. Enables the module, configures clock frequency, SPI mode, and chip select GPIO pin.

## Precondition

config must be non-NULL

channel < k\_rspi\_max\_channels

channel must not be already initialized

CS GPIO port and pin must be valid and available

## Postcondition

If successful:

- RSPI module is enabled via [internal\\_set\\_mstpcrb\\_for\\_channel\(\)](#)
- \*prcr\_reg() is unlocked/locked for register protection
- RSPI registers configured: spbr, spcmd0, spcr, spdcr, sppcr
- CS GPIO configured as output, driven high (inactive)
- s\_rspi\_cs\_config[channel] stores CS port/pin
- s\_rspi\_controller\_initialized[channel] = true

On error:

- Hardware state is unchanged
- Channel remains uninitialized

## Note

Bit rate calculation:  $\text{freq} = \text{PCLKB} / (2 * (\text{SPBR} + 1))$

Always configures 16-bit data frames with word access mode

CS is active-low (asserted=0, deasserted=1)

Definition at line 1339 of file [rspi.c](#).

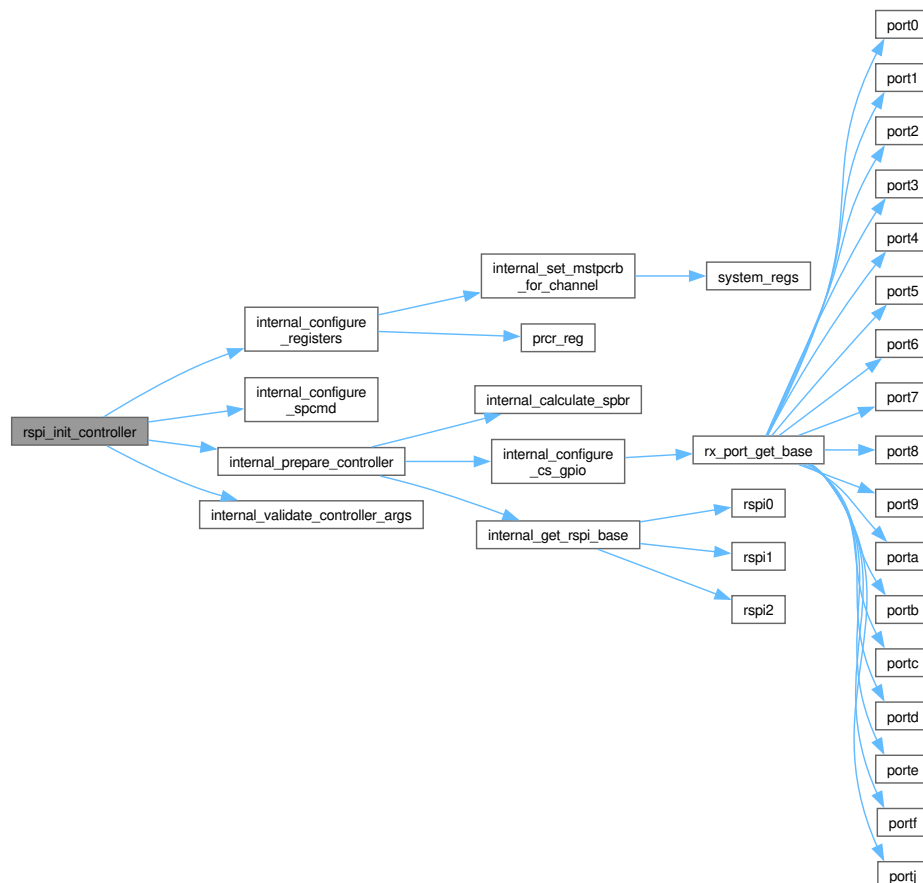
```

01340 {
01341 /* Validate all arguments and preconditions */
01342 rx_err_t err = internal_validate_controller_args(channel, config);
01343 if (err != k_rx_ok) {
01344 return err;
01345 }
01346 /* Prepare hardware resources (SPBR, RSPI base, CS GPIO) */
01347 volatile rx_rspi_regs_t* rspi = nullptr;
01348 uint8_t spbr = k_rspi_spbr_init;
01349 err = internal_prepare_controller(channel, config, &rspi, &spbr);
01350 if (err != k_rx_ok) {
01351 return err;
01352 }
01353 }
01354
01355 /* Configure SPI mode (CPOL and CPHA) */
01356 uint16_t spcmd = k_rspi_spcmd_init;
01357 internal_configure_spcmd(&spcmd, config->spi_mode);
01358
01359 /* Configure 16-bit data length */
01360 spcmd |= (k_rspi_spcmd_16bit << k_rspi_spcmd_spl_shift);
01361
01362 /* Configure and enable RSPI hardware registers */
01363 return internal_configure_registers(channel, rspi, spbr, spcmd, config);
01364 }

```

References [internal\\_configure\\_registers\(\)](#), [internal\\_configure\\_spcmd\(\)](#), [internal\\_prepare\\_controller\(\)](#), [internal\\_validate\\_controller\\_args\(\)](#), [k\\_rspi\\_spbr\\_init](#), [k\\_rspi\\_spcmd\\_16bit](#), [k\\_rspi\\_spcmd\\_init](#), [k\\_rspi\\_spcmd\\_spl\\_shift](#), and [rspi\\_controller\\_config\\_t::spi\\_mode](#).

Here is the call graph for this function:



8.1.5.68 `rspi_init_peripheral()`

```
rx_err_t rspi_init_peripheral (
 rspi_channel_t channel,
 const rspi_config_t * config) [nodiscard]
```

Initialize RSPI channel in peripheral mode for RPi5 communication.

Configures the specified RSPI channel as an SPI peripheral device. In peripheral mode the RX72N receives the clock signal from an external SPI controller (Raspberry Pi 5). Performs full hardware initialization including module stop release, pin function selection (COPI/CIPO/CLK/CS via MPC), SPI mode register configuration, and frame size setup.

## 8.1.5.69 Algorithm

1. Validate channel (0-2) and config pointer
2. Release RSPI module from module stop (MSTPCRB)
3. Disable RSPI (SPCR.SPE = 0)
4. Configure pin functions via MPC (COPI, CIPO, CLK, CS)
5. Set SPI mode (CPOL/CPHA in SPCMD0)
6. Set frame length (8-bit or 16-bit in SPCMD0)
7. Configure as peripheral (SPCR.MSTR = 0)
8. Enable RSPI (SPCR.SPE = 1)

## Parameters

|    |         |                                                                                                                                                                     |
|----|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| in | channel | RSPI channel number (0-2)                                                                                                                                           |
| in | config  | Pointer to peripheral mode configuration <ul style="list-style-type: none"> <li>• Must not be nullptr</li> <li>• config-&gt;spi_mode must be valid (0-3)</li> </ul> |

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                     |                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Channel successfully initialized in peripheral mode                           |
| <code>k_rx_err_invalid_arg</code>   | Channel > 2 or invalid SPI mode                                               |
| <code>k_rx_err_null_ptr</code>      | config pointer is nullptr                                                     |
| <code>k_rx_err_invalid_state</code> | Channel already initialized (call <code>rspi_deinit()</code> to reinitialize) |

## Precondition

- Channel must not be already initialized (call [rspi\\_deinit\(\)](#) first to reinitialize)
- System clocks must be configured (PCLKB running at expected frequency)

## Postcondition

- RSPI channel configured as SPI peripheral, ready for transfers
- COPI/CIPO/CLK/CS pins configured for SPI peripheral function

## Note

- Thread Safety: NOT thread-safe. Initialize each channel from a single thread during system startup.

## Example: RPi5 Communication Setup

```
rspi_config_t rpi_cfg = {
 .spi_mode = k_rspi_mode_0,
 .use_16bit = false
};
rx_err_t err = rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);
if (err != k_rx_ok) {
 rx_log_error_val("SPI", "RSPI0 peripheral init failed", err);
}
```

## See also

- [rspi\\_peripheral\\_transfer\(\)](#) Full-duplex SPI transfer
- [rspi\\_deinit\(\)](#) Deinitialize RSPI channel
- [rspi\\_config\\_t](#) Configuration structure

## Since

Version 1.0.0

Definition at line 601 of file [rspi.c](#).

```
00602 {
00603 volatile rx_rspi_regs_t* rspi = nullptr;
00604 const rx_err_t check = internal_rspi_validate_init(channel, config, &rspi);
00605 if (check != k_rx_ok) {
00606 return check;
00607 }
00608 }
00609 /*
00610 * RSPI Peripheral Mode Register Write Sequence (RX72N HW Manual 38.3.6):
00611 * 1. Enable module clock (MSTPCRB)
00612 * 2. Disable SPI (SPCR.SPE=0)
00613 * 3. Configure SPCMD0 (CPOL/CPHA/data length)
00614 * 4. Configure SPDCR (byte vs longword access)
00615 * 5. Configure SPPCR (no loopback) BEFORE enabling SPI
00616 * 6. Enable SPI (SPCR.SPE=1) -- LAST write so partial config is never active
00617 */
00618 internal_rspi_release_module_stop(channel);
00619 rspi->spcr = k_rspi_spcr_disabled;
00620
00621 rspi->spcmd0 = internal_rspi_build_spcmd(rspi, config);
00622
00623 /* SPPCR MUST be written before SPCR.SPE per HW manual to avoid glitches
00624 * on COPI/CIPO lines during enable */
00625 rspi->sppcr = k_rspi_sppcr_no_loopback;
00626
00627 /* SPCR.SPE=1 MUST be the last register write so all configuration is
00628 * stable before the peripheral becomes active. MSTR=0 -> peripheral mode. */
00629 rspi->spcr = k_rspi_spcr_spe;
```

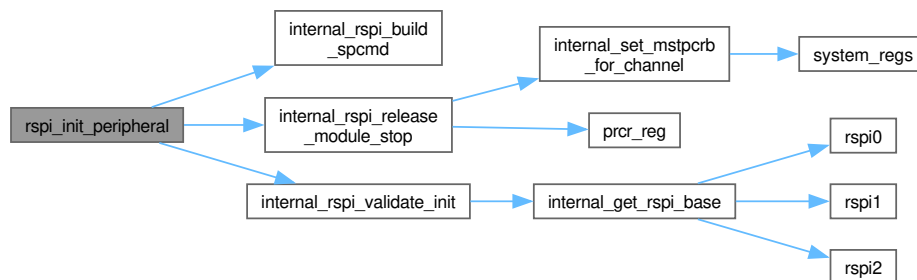
```

00630
00631 s_rspi_channel_initialized[channel] = true;
00632 rx_log_debug(s_tag, "RSPi peripheral mode initialized");
00633 return k_rx_ok;
00634 }

```

References [internal\\_rspi\\_build\\_spcmd\(\)](#), [internal\\_rspi\\_release\\_module\\_stop\(\)](#), [internal\\_rspi\\_validate\\_init\(\)](#), [k\\_rspi\\_spcr\\_disabled](#), [k\\_rspi\\_spcr\\_spe](#), [k\\_rspi\\_sppcr\\_no\\_loopback](#), [s\\_rspi\\_channel\\_initialized](#), [s\\_tag](#), [rx\\_rspi\\_regs\\_t::spcmd0](#), [rx\\_rspi\\_regs\\_t::spcr](#), and [rx\\_rspi\\_regs\\_t::sppcr](#).

Here is the call graph for this function:



#### 8.1.5.70 rspi\_peripheral\_read\_available()

```

rx_err_t rspi_peripheral_read_available (
 const rspi_channel_t channel,
 bool * available) [nodiscard]

```

Check if receive data is available in the RSPi receive buffer.

Polls the RSPi Status Register (SPSR) SPRF flag to determine whether the receive buffer contains data ready for reading. This is a non-blocking status check that returns immediately without waiting for data. Use this to implement polled receive patterns without incurring the overhead of a full transfer call.

##### Parameters

|     |           |                                                                                                             |
|-----|-----------|-------------------------------------------------------------------------------------------------------------|
| in  | channel   | RSPi channel number (valid: k_rspi_channel_0, k_rspi_channel_1, k_rspi_channel_2; range 0-2)                |
| out | available | Pointer to bool set to true if SPRF flag indicates data is available, false otherwise (must not be nullptr) |

##### Returns

rx\_err\_t Error code indicating success or failure

##### Return values

|                        |                                                                                    |
|------------------------|------------------------------------------------------------------------------------|
| k_rx_ok                | Status successfully read into *available                                           |
| k_rx_err_null_ptr      | available pointer is nullptr                                                       |
| k_rx_err_invalid_state | channel >= 3 or channel not initialized via <a href="#">rspi_init_peripheral()</a> |
| k_rx_err_invalid_arg   | RSPi base address lookup failed for the channel                                    |

## Precondition

channel must be initialized via [rspi\\_init\\_peripheral\(\)](#)  
 available must be a valid non-null pointer

## Postcondition

\*available reflects current SPRF flag state on success  
 \*available set to false on RSPI base address lookup failure

## Note

Not thread-safe. Caller must provide external synchronization when accessing the same channel from multiple threads.

## See also

[rspi\\_peripheral\\_transfer\(\)](#) Full-duplex transfer in peripheral mode  
[rspi\\_peripheral\\_write\\_ready\(\)](#) Non-blocking transmit buffer check  
[rspi\\_init\\_peripheral\(\)](#) Must be called before this function

## Since

Version 1.0.0

Check if receive data is available in the RSPI receive buffer.

Polls the RSPI status register to determine if the receive buffer contains data ready for reading. Returns the status without blocking.

## Precondition

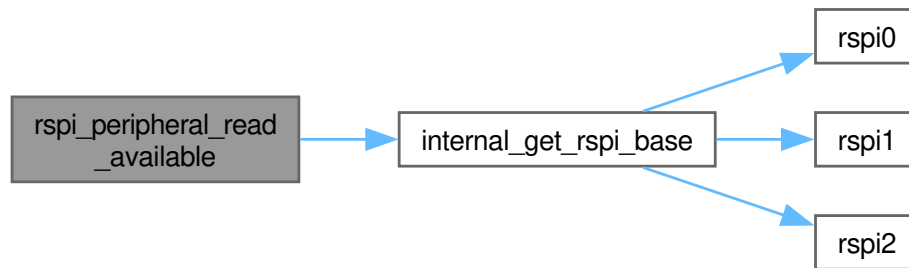
Channel must be initialized via [rspi\\_init\\_peripheral\(\)](#) before calling this function  
 available must be a valid non-nullptr

Definition at line 812 of file [rspi.c](#).

```
00813 {
00814 RX_CHECK_NULL_PTR(available, s_tag, "Available pointer is nullptr");
00815
00816 /* Validate channel */
00817 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00818 rx_log_error(s_tag, "RSPI channel not initialized");
00819 return k_rx_err_invalid_state;
00820 }
00821
00822 /* Get RSPI base */
00823 const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00824 if (rspi == nullptr) {
00825 *available = false;
00826 return k_rx_err_invalid_arg;
00827 }
00828
00829 /* Check if receive buffer has data */
00830 *available = (rspi->spsr & k_rspi_spsr_sprf) != 0;
00831
00832 return k_rx_ok;
00833 }
```

References [internal\\_get\\_rspi\\_base\(\)](#), [k\\_rspi\\_max\\_channels](#), [k\\_rspi\\_spsr\\_sprf](#), [s\\_rspi\\_channel\\_initialized](#), [s\\_tag](#), and [rx\\_rspi\\_regs\\_t::spsr](#).

Here is the call graph for this function:



#### 8.1.5.71 rspi\_peripheral\_transfer()

```

rx_err_t rspi_peripheral_transfer (
 const rspi_channel_t channel,
 const uint8_t * tx_data,
 uint8_t * rx_data,
 const uint16_t length) [nodiscard]

```

Full-duplex SPI transfer in peripheral mode.

Executes a byte-by-byte full-duplex SPI transfer on the specified RSPI channel operating in peripheral mode. For each byte, the function waits for the transmit buffer to empty, writes the TX byte, waits for the receive buffer to fill, then reads the RX byte. Both TX and RX wait operations are bounded by a 10 ms timeout to prevent infinite loops.

The transfer loop is bounded by `k_rspi_transfer_len_max` (65535) per NASA Rule 2 to ensure statically provable loop termination.

#### Parameters

|     |         |                                                                                                                                        |
|-----|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| in  | channel | RSPI channel number (valid: <code>k_rspi_channel_0</code> , <code>k_rspi_channel_1</code> , <code>k_rspi_channel_2</code> ; range 0-2) |
| in  | tx_data | Pointer to transmit data buffer (must not be nullptr)                                                                                  |
| out | rx_data | Pointer to receive data buffer (must not be nullptr, must have capacity for at least length bytes)                                     |
| in  | length  | Number of bytes to transfer (valid: 1 to 65535)                                                                                        |

#### Returns

`rx_err_t` Error code indicating success or failure

#### Return values

|                                     |                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Transfer completed successfully, <code>rx_data</code> filled                          |
| <code>k_rx_err_null_ptr</code>      | <code>tx_data</code> or <code>rx_data</code> pointer is nullptr                       |
| <code>k_rx_err_invalid_arg</code>   | length is zero, or RSPI base address lookup failed for the channel                    |
| <code>k_rx_err_invalid_state</code> | channel $\geq 3$ or channel not initialized via <code>rspi_init_peripheral()</code>   |
| <code>k_rx_err_timeout</code>       | Transmit buffer did not empty or receive buffer did not fill within the 10 ms timeout |



#### Precondition

channel must be initialized via [rspi\\_init\\_peripheral\(\)](#)  
tx\_data and rx\_data must be valid non-null pointers  
length must be in range [1, 65535]

#### Postcondition

rx\_data buffer contains received bytes corresponding to each transmitted byte offset on success  
SPSR status flags (SPRF, OVRF) cleared after each byte transfer

#### Note

Not thread-safe. Caller must provide external synchronization when accessing the same channel from multiple threads.

#### See also

[rspi\\_init\\_peripheral\(\)](#) Must be called before this function  
[rspi\\_peripheral\\_read\\_available\(\)](#) Non-blocking receive check  
[rspi\\_peripheral\\_write\\_ready\(\)](#) Non-blocking transmit check  
[rspi\\_deinit\(\)](#) Deinitialize RSPI channel

#### Since

Version 1.0.0

#### Full-duplex SPI transfer in peripheral mode.

Executes a full-duplex (simultaneous transmit and receive) SPI transfer on the specified RSPI channel operating in peripheral mode. Transfers data byte-by-byte with timeout protection on both transmit buffer ready and receive buffer full operations.

#### Precondition

channel must be initialized via [rspi\\_init\\_peripheral\(\)](#) before calling  
tx\_data pointer must be non-NULL  
rx\_data pointer must be non-NULL  
length must be in range [1, 65535]

## Postcondition

If successful, rx\_data buffer is filled with received bytes corresponding to transmitted data at each byte offset

Definition at line 742 of file [rspi.c](#).

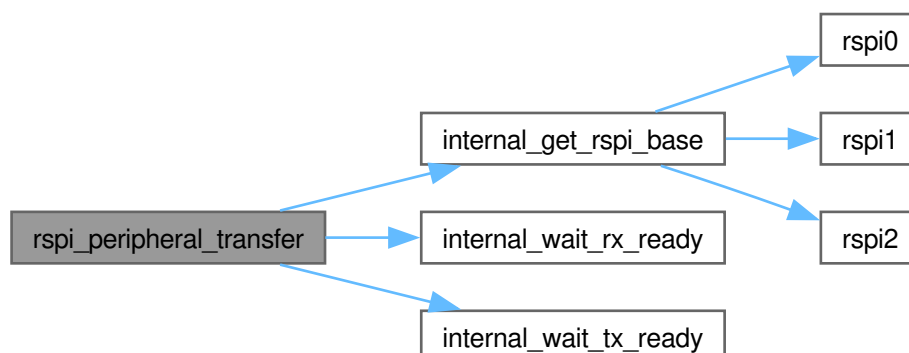
```

00746 {
00747 RX_CHECK_NULL_PTR(tx_data, s_tag, "TX data pointer is nullptr");
00748 RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
00749
00750 if (length == k_rspi_len_zero) {
00751 rx_log_error(s_tag, "Invalid transfer length");
00752 return k_rx_err_invalid_arg;
00753 }
00754
00755 /* Validate channel */
00756 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00757 rx_log_error(s_tag, "RSPI channel not initialized");
00758 return k_rx_err_invalid_state;
00759 }
00760
00761 /* Get RSPI base */
00762 volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
00763 if (rspi == nullptr) {
00764 rx_log_error(s_tag, "Failed to get RSPI base address");
00765 return k_rx_err_invalid_arg;
00766 }
00767
00768 /* NASA Rule 2: Statically bounded loop */
00769 for (uint16_t i = k_rspi_loop_start; i < k_rspi_transfer_len_max; i++) {
00770 if (i >= length) {
00771 break;
00772 }
00773
00774 /* Wait for transmit buffer empty */
00775 rx_err_t err = internal_wait_tx_ready(rspi);
00776 if (err != k_rx_ok) {
00777 rx_log_error(s_tag, "SPI transmit timeout");
00778 return err;
00779 }
00780
00781 /* Write transmit data */
00782 rspi->spdr = tx_data[i];
00783
00784 /* Wait for receive buffer full */
00785 err = internal_wait_rx_ready(rspi);
00786 if (err != k_rx_ok) {
00787 rx_log_error(s_tag, "SPI receive timeout");
00788 return err;
00789 }
00790
00791 /* Read receive data */
00792 rx_data[i] = (uint8_t)rspi->spdr;
00793
00794 /* Clear status flags */
00795 rspi->spsr &= (uint8_t) ~(uint8_t)(k_rspi_spsr_sprf | k_rspi_spsr_ovrf);
00796 }
00797 return k_rx_ok;
00798 }
00799

```

References [internal\\_get\\_rspi\\_base\(\)](#), [internal\\_wait\\_rx\\_ready\(\)](#), [internal\\_wait\\_tx\\_ready\(\)](#), [k\\_rspi\\_len\\_zero](#), [k\\_rspi\\_loop\\_start](#), [k\\_rspi\\_max\\_channels](#), [k\\_rspi\\_spsr\\_ovrf](#), [k\\_rspi\\_spsr\\_sprf](#), [k\\_rspi\\_transfer\\_len\\_max](#), [s\\_rspi\\_channel\\_initialized](#), [s\\_tag](#), [rx\\_rspi\\_regs\\_t::spdr](#), and [rx\\_rspi\\_regs\\_t::spsr](#).

Here is the call graph for this function:



8.1.5.72 `rspi_peripheral_write_ready()`

```
rx_err_t rspi_peripheral_write_ready (
 const rspi_channel_t channel,
 bool * ready) [nodiscard]
```

Check if the RSPi transmit buffer is ready for new data.

Polls the RSPi Status Register (SPSR) SPTEF flag to determine whether the transmit buffer is empty and ready to accept new data. This is a non-blocking status check that returns immediately without waiting for the buffer to drain. Use this to implement polled transmit patterns or to verify readiness before writing to the SPI data register.

## Parameters

|     |         |                                                                                                                                        |
|-----|---------|----------------------------------------------------------------------------------------------------------------------------------------|
| in  | channel | RSPi channel number (valid: <code>k_rspi_channel_0</code> , <code>k_rspi_channel_1</code> , <code>k_rspi_channel_2</code> ; range 0-2) |
| out | ready   | Pointer to bool set to true if SPTEF flag indicates the transmit buffer is empty, false otherwise (must not be nullptr)                |

## Returns

`rx_err_t` Error code indicating success or failure

## Return values

|                                     |                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------|
| <code>k_rx_ok</code>                | Status successfully read into *ready                                                   |
| <code>k_rx_err_null_ptr</code>      | ready pointer is nullptr                                                               |
| <code>k_rx_err_invalid_state</code> | channel $\geq 3$ or channel not initialized via <a href="#">rspi_init_peripheral()</a> |
| <code>k_rx_err_invalid_arg</code>   | RSPi base address lookup failed for the channel                                        |

## Precondition

channel must be initialized via [rspi\\_init\\_peripheral\(\)](#)

ready must be a valid non-null pointer

## Postcondition

\*ready reflects current SPTEF flag state on success

\*ready set to false on RSPi base address lookup failure

## Note

Not thread-safe. Caller must provide external synchronization when accessing the same channel from multiple threads.

See also

[rspi\\_peripheral\\_transfer\(\)](#) Full-duplex transfer in peripheral mode  
[rspi\\_peripheral\\_read\\_available\(\)](#) Non-blocking receive buffer check  
[rspi\\_init\\_peripheral\(\)](#) Must be called before this function

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Check if the RSPI transmit buffer is ready for new data.

Polls the RSPI status register to determine if the transmit buffer is empty and ready to accept new data. Returns the status without blocking.

Precondition

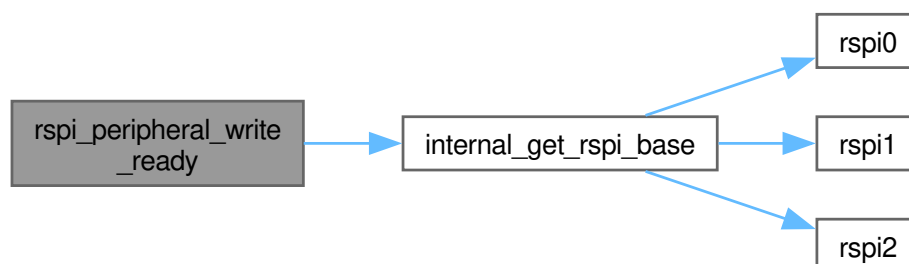
Channel must be initialized via [rspi\\_init\\_peripheral\(\)](#)  
 ready must be a valid non-nullptr

Definition at line 846 of file [rspi.c](#).

```
00847 {
00848 RX_CHECK_NULL_PTR(ready, s_tag, "Ready pointer is nullptr");
00849
00850 /* Validate channel */
00851 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00852 rx_log_error(s_tag, "RSPI channel not initialized");
00853 return k_rx_err_invalid_state;
00854 }
00855
00856 /* Get RSPI base */
00857 const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00858 if (rspi == nullptr) {
00859 *ready = false;
00860 return k_rx_err_invalid_arg;
00861 }
00862
00863 /* Check if transmit buffer is empty */
00864 *ready = (rspi->spsr & k_rspi_spsr_sptef) != 0;
00865
00866 return k_rx_ok;
00867 }
```

References [internal\\_get\\_rspi\\_base\(\)](#), [k\\_rspi\\_max\\_channels](#), [k\\_rspi\\_spsr\\_sptef](#), [s\\_rspi\\_channel\\_initialized](#), [s\\_tag](#), and [rx\\_rspi\\_regs\\_t::spsr](#).

Here is the call graph for this function:



## 8.1.5.73 timer`get`count()

```
rx_err_t timer_get_count (
 uint16_t * count) [nodiscard]
```

Get current CMT0 counter value.

Reads the 16-bit free-running counter register (CMCNT) from CMT0. The counter increments at PCLKB/8 rate and resets to 0 on compare match with CMCOR. This can be used for sub-tick timing measurements or to check elapsed time within a 10 ms system tick period. Works whether the timer is running or stopped.

## Parameters

| out | count | Pointer to store 16-bit counter value                                                                                                                                                                |
|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |       | <ul style="list-style-type: none"> <li>• Must point to valid uint16_t variable</li> <li>• Value range: 0 to CMCOR (compare match constant)</li> <li>• Unchanged if function returns error</li> </ul> |

## Returns

rx\_err\_t Error code indicating success or failure

## Return values

|                   |                                                      |
|-------------------|------------------------------------------------------|
| k_rx_ok           | Counter value successfully read and stored in *count |
| k_rx_err_null_ptr | count pointer is nullptr                             |

## Precondition

count pointer must be valid (not nullptr)

CMT0 must have been initialized via [timer\\_init\(\)](#) for meaningful values

## Postcondition

\*count contains current CMCNT register value if function succeeds

\*count unchanged if function returns error

## Note

Thread Safety: NOT thread-safe. The 16-bit register read is atomic on RX72N but the pointer write is not protected.

## Example: Sub-Tick Timing

```
uint16_t start, end;
timer_get_count(&start);
// ... perform operation ...
timer_get_count(&end);
uint16_t elapsed_ticks = (end >= start) ? (end - start) : (0xFFFF - start + end);
```

See also

[timer\\_init\(\)](#) Initialize CMT0 timer

[timer\\_stop\(\)](#) Stop CMT0 timer

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Get current CMT0 counter value.

Reads the current value of CMT0.CMCNT register, which increments from 0 to CMCOR (4687) at 468.75 kHz. This provides sub-millisecond timing resolution for diagnostics, performance measurement, and jitter analysis.

Algorithm steps:

1. Validate count pointer is not nullptr
2. Validate CMT0 hardware accessibility
3. Read CMCNT register (atomic 16-bit read)
4. Validate count is within expected range (0-4687)
5. Store count value to output parameter

Counter behavior:

- Increments every  $1/(1.875 \text{ MHz}) = 0.533 \text{ us}$
- Resets to 0 when reaching CMCOR (18749)
- Wraps every 10 ms (one system tick)
- Can be read at any time without affecting timer operation

Timing resolution:

$$\text{Resolution} = \frac{1}{f_{CMT}} = \frac{1}{1875000 \text{ Hz}} = 0.533 \text{ us}$$

Maximum readable value:

$$\text{Max count} = CMCOR = 18749 \approx 10 \text{ ms}$$

Return Value Details:

| Return Value         | Condition                            | *count Value |
|----------------------|--------------------------------------|--------------|
| k_rx_ok              | Normal operation                     | 0 to 18749   |
| k_rx_err_null_ptr    | count == nullptr                     | Unchanged    |
| k_rx_err_hw_unmapped | <a href="#">cmt0()</a> returned NULL | Unchanged    |

| Return Value      | Condition     | *count Value  |
|-------------------|---------------|---------------|
| k_rx_err_hw_error | *count > 4687 | Invalid value |

#### Precondition

`timer_init()` must have been called successfully  
count pointer must be valid and non-NULL

#### Postcondition

If `k_rx_ok`: \*count contains current CMCNT value (0-4687)  
If error: \*count unchanged (nullptr) or contains invalid value (hw\_error)  
Timer operation not affected (read-only access)

#### Invariant

Return value `k_rx_ok` guarantees \*count in [0, 18749]

#### Note

Thread Safety: Thread-safe for reading. Multiple threads can call simultaneously. CMCNT is atomic 16-bit register on RX72N. No need for mutual exclusion.

Re-entrancy: Fully reentrant. Safe to call from ISRs and tasks concurrently. No shared state modified.

Performance: Executes in ~500 ns @ 240 MHz (single register read). Suitable for high-frequency polling.

Memory: No heap allocation. Stack usage: ~4 bytes (local variable).

#### Warning

Counter wraps: Counter resets to 0 every 10 ms. For measuring intervals longer than 10 ms, use ThreadX system time (`tx_time_get`) instead.

Race condition: Counter value is snapshot at time of read. By the time caller processes value, hardware counter has advanced. Don't assume static value.

#### Example - Basic Usage:

```
uint16_t count;
rx_err_t err = timer_get_count(&count);
if (err == k_rx_ok) {
 // Convert to microseconds: count * 0.533 us
 float us = count * 0.533f;
 rx_log_info("TIMER", "Current time in tick: %.1f us", us);
}
```

## Example - Performance Measurement:

```
// Measure execution time of function (sub-tick resolution)
uint16_t start, end;
timer_get_count(&start);

// Execute function to measure
motor_update();

timer_get_count(&end);

// Calculate elapsed time (handle wrap)
uint16_t elapsed;
if (end >= start) {
 elapsed = end - start;
} else {
 elapsed = (18749 - start) + end; // Wrapped
}

float us = elapsed * 0.533f;
rx_log_info("PERF", "motor_update took %.1f us", us);
```

## Example - Jitter Analysis:

```
// Measure interrupt jitter
static uint16_t last_count = 0;

void analyze_jitter(void)
{
 uint16_t current;
 timer_get_count(¤t);

 int16_t jitter = current - last_count;
 if (abs(jitter) > 10) { // >21 us jitter
 rx_log_warn("TIMER", "High jitter detected: %d counts", jitter);
 }

 last_count = current;
}
```

## Example - Error Handling:

```
uint16_t count;
rx_err_t err = timer_get_count(&count);
switch (err) {
 case k_rx_ok:
 // Normal operation
 break;
 case k_rx_err_null_ptr:
 rx_log_error("TIMER", "Invalid pointer");
 return err;
 case k_rx_err_hw_unmapped:
 rx_log_error("TIMER", "CMT0 not initialized");
 return err;
 case k_rx_err_hw_error:
 rx_log_error("TIMER", "Counter overflow: %u", count);
 return err;
}
```

See also

- [timer\\_init\(\)](#) Initialize timer before reading
- [tx\\_time\\_get\(\)](#) For millisecond-resolution system time
- [RX72N User's Manual Section 23.2.5 - CMCNT Register](#)

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## NASA Power of 10 Compliance

- Rule 1 [OK] No goto, setjmp, recursion
- Rule 2 [OK] No loops
- Rule 3 [OK] No dynamic allocation
- Rule 4 [OK] Function is 13 lines (under 60 line limit)
- Rule 5 [OK] 2 preconditions, 3 postconditions
- Rule 6 [OK] Minimal scope (single output parameter)
- Rule 7 [OK] All return values validated (RX\_CHECK\_NULL\_PTR)
- Rule 8 [OK] Uses C23 typed enums (k\_cmt0\_compare\_match)
- Rule 9 [OK] Single level of pointer dereferencing
- Rule 10 [OK] Compiles with -Wall -Wextra -Werror

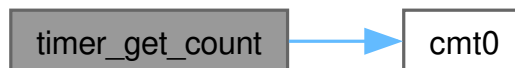
Definition at line 1024 of file [timer.c](#).

```

01025 {
01026 RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
01027
01028 if (cmt0() == nullptr) {
01029 rx_log_error(s_tag, "CMT0 register block is nullptr");
01030 return k_rx_err_hw_unmapped;
01031 }
01032
01033 *count = cmt0()->cmcnt;
01034
01035 if (*count > k_cmt0_compare_match) {
01036 return k_rx_err_hw_error;
01037 }
01038
01039 return k_rx_ok;
01040 }
```

References [rx\\_cmt\\_channel\\_regs\\_t::cmcnt](#), [cmt0\(\)](#), [k\\_cmt0\\_compare\\_match](#), and [s\\_tag](#).

Here is the call graph for this function:



## 8.1.5.74 timer\_init()

```

rx_err_t timer_init (
 void) [nodiscard]
```

Initialize CMT0 for ThreadX system tick (100 Hz).

Configures Compare Match Timer channel 0 (CMT0) to generate periodic interrupts at 100 Hz (10 ms interval) for the ThreadX RTOS system tick. Performs full hardware initialization including module stop release, clock source selection (PCLKB/8), compare match constant calculation, and interrupt priority setup.

### 8.1.5.75 Algorithm

1. Release CMT0 from module stop (MSTPCRA.MSTPA15 = 0)
2. Stop counter (CMSTR0.STR0 = 0)
3. Configure CMCR: PCLKB/8 clock source, compare match interrupt enable
4. Calculate CMCOR from PCLKB frequency and desired 100 Hz rate
5. Clear counter (CMCNT = 0)
6. Set compare match value (CMCOR)
7. Configure interrupt priority (IPR = 5)
8. Start counter (CMSTR0.STR0 = 1)

### Returns

rx\_err\_t Error code indicating success or failure

### Return values

|                        |                                                     |
|------------------------|-----------------------------------------------------|
| k_rx_ok                | CMT0 successfully initialized and running at 100 Hz |
| k_rx_err_invalid_state | CMT0 already initialized                            |

### Precondition

System clocks must be configured (PCLKB running at expected frequency)  
 CMT0 must not be already initialized (call [timer\\_stop\(\)](#) first to reinitialize)

### Postcondition

CMT0 generating 100 Hz interrupts for ThreadX system tick  
 CMT0 interrupt enabled at priority 5

### Note

Thread Safety: NOT thread-safe. Must be called once during system initialization before ThreadX is started.

### Example: System Initialization

```
void hardware_init(void) {
 rx_err_t err = timer_init();
 if (err != k_rx_ok) {
 rx_log_error_val("TIMER", "CMT0 init failed", err);
 return;
 }
 rx_log_info("TIMER", "System tick running at 100 Hz");
}
```

See also

[timer\\_stop\(\)](#) Stop the CMT0 timer  
[timer\\_get\\_count\(\)](#) Read current counter value

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Initialize CMT0 for ThreadX system tick (100 Hz).

Configures the RX72N CMT0 (Compare Match Timer 0) peripheral to generate precise periodic interrupts at 100.00 Hz for ThreadX RTOS scheduling. This function must be called during system initialization, before starting ThreadX kernel.

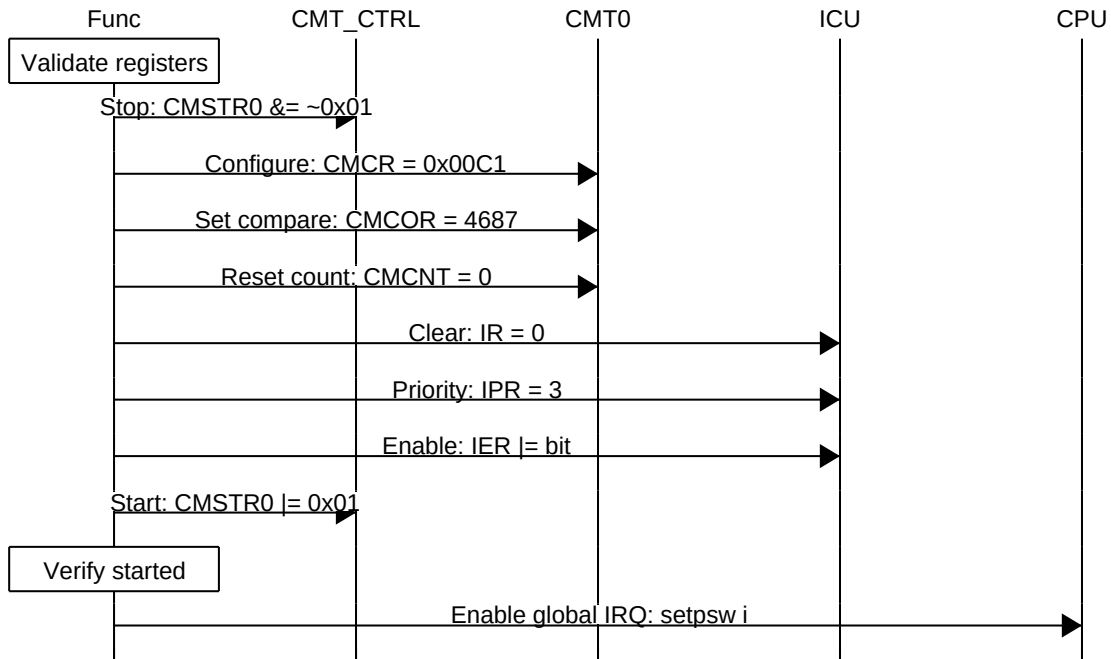
Algorithm steps:

1. Validate CMT0 and CMT\_CTRL hardware accessibility
2. Stop CMT0 if currently running (safe reconfiguration)
3. Configure CMCR register: PCLKB/128 clock, interrupt enabled
4. Set CMCOR compare match value: 4687 (for 100 Hz)
5. Reset CMCNT counter to 0 (known initial state)
6. Configure ICU: clear pending interrupt, set priority, enable interrupt
7. Start CMT0 timer
8. Enable global interrupts (PSW I-flag)
9. Verify timer started successfully

Configuration summary:

- Clock source: PCLKB = 60 MHz
- Divider: /32 -> 1.875 MHz
- Compare value: 18749 -> 100.00 Hz
- Interrupt priority: 3/15 (higher than app tasks, lower than motor control)
- Tick period: 10.000 ms (exact, zero drift)

Hardware configuration sequence:



Return Value Details:

| Return Value           | Condition                                                          |
|------------------------|--------------------------------------------------------------------|
| k_rx_ok                | Timer initialized successfully                                     |
| k_rx_err_invalid_state | <a href="#">cmt_ctrl()</a> or <a href="#">cmt0()</a> returned NULL |
| k_rx_err_hw_error      | CMSTR0 start bit failed to set                                     |

#### Precondition

- System clocks must be configured (PCLKB = 60 MHz)
- CMT0 must not be in use by another module
- Must be called before [tx\\_kernel\\_enter\(\)](#)
- Interrupts can be disabled (function enables at end)

#### Postcondition

- CMT0 running at 100 Hz, generating periodic interrupts
- [cmt0\\_isr\(\)](#) will be called every 10 ms
- ThreadX system tick operational
- Global interrupts enabled (PSW I-flag set)

#### Invariant

- CMT0 compare value = 4687 (100.00 Hz)
- CMT0 interrupt priority = 3/15

## Note

Thread Safety: Not thread-safe. Must be called during single-threaded initialization before ThreadX kernel starts. Do not call from running RTOS tasks.

Re-entrancy: Not reentrant. Calling multiple times without `timer_stop()` is undefined behavior (timer will be reconfigured).

Performance: Initialization takes ~10 us @ 240 MHz (multiple register writes). This is one-time startup cost, not runtime overhead.

Memory: Zero heap allocation. All configuration is register I/O.

## Warning

Call order critical: Must call `timer_init()` BEFORE `tx_kernel_enter()`. ThreadX depends on system tick being operational. Calling after kernel start causes undefined behavior.

Clock dependency: Assumes PCLKB = 60 MHz. If system clock configuration changes, `k_cmt0←_compare_match` must be recalculated. Incorrect clock frequency causes incorrect tick rate.

## Attention

Global interrupt enable: This function enables global interrupts via "setpsw i". If interrupts were intentionally disabled, they will be enabled after this call.

## Example - System Initialization:

```
#include "timer.h"
#include "tx_api.h"

int main(void)
{
 // 1. Initialize hardware (clocks, GPIO, etc.)
 hardware_init();

 // 2. Initialize system tick BEFORE ThreadX
 rx_err_t err = timer_init();
 if (err != k_rx_ok) {
 rx_log_error("MAIN", "System tick init failed: %d", err);
 while (1) {} // Halt on critical error
 }

 // 3. Define ThreadX application
 tx_application_define(NULLptr);

 // 4. Start ThreadX kernel (never returns)
 tx_kernel_enter();
}
```

## Example - Error Handling:

```
// Robust initialization with error checking
rx_err_t err = timer_init();
switch (err) {
 case k_rx_ok:
 rx_log_info("MAIN", "System tick running at 100 Hz");
 break;
 case k_rx_err_invalid_state:
 rx_log_error("MAIN", "Clock not configured - check hardware_init()");
 return err;
 case k_rx_err_hw_error:
 rx_log_error("MAIN", "CMT0 hardware fault");
 return err;
 default:
 rx_log_error("MAIN", "Unexpected error: %d", err);
 return err;
}
```

Example - Verification:

```
// After initialization, verify tick is running
timer_init();

uint16_t count1, count2;
timer_get_count(&count1);
tx_thread_sleep(50); // Sleep 500ms (50 ticks)
timer_get_count(&count2);

// Counter should have rolled over ~50 times
rx_log_info("MAIN", "Tick verified: count changed from %u to %u", count1, count2);
```

See also

[timer\\_stop\(\)](#) Stop system tick timer  
[timer\\_get\\_count\(\)](#) Read current counter value  
[cmt0\\_isr\(\)](#) Interrupt handler called every 10 ms  
[tx\\_kernel\\_enter\(\)](#) ThreadX kernel entry point  
 RX72N User's Manual Chapter 23 - Compare Match Timer

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 1 [OK] No goto, setjmp, recursion (sequential register writes)
- Rule 2 [OK] No loops (straight-line initialization code)
- Rule 3 [OK] No dynamic allocation (all register I/O)
- Rule 4 [OK] Function is 47 lines (under 60 line limit)
- Rule 5 [OK] 4 preconditions, 4 postconditions documented
- Rule 6 [OK] Minimal scope (no local variables)
- Rule 7 [OK] All register accessor return values validated
- Rule 8 [OK] C23 typed enums for ALL constants (k\_cmt0\_\*)
- Rule 9 [OK] Single level of dereferencing for registers
- Rule 10 [OK] Compiles clean with -Wall -Wextra -Werror

Definition at line 726 of file [timer.c](#).

```
00727 {
00728 if (cmt_ctrl() == nullptr || cmt0() == nullptr) {
00729 return k_rx_err_invalid_state;
00730 }
00731
00732 rx_log_info(s_tag, "Initializing CMT0 for ThreadX tick");
00733
00734 /* Stop CMT0 if running */
00735 cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t) k_cmt0_cmstr_start_bit;
00736
00737 /* Configure CMT0 */
00738 /* CMCOR = 0x00C1: bit7=1 (reserved, write-1 per manual), bit6=1 (CMIE), bits1-0=01 (PCLKB/32) */
00739 cmt0()->cmcr = k_cmt0_cmcr_config;
00740
00741 /* Set compare match value for 100 Hz tick
00742 * CMCOR = (60,000,000 / 32 / 100) - 1 = 18749 */
00743 cmt0()->cmcor = k_cmt0_compare_match;
00744
00745 /* Reset counter */
00746 cmt0()->cmcnt = k_cmt0_counter_init;
00747
00748 /* Configure interrupt controller (ICU) */
00749 /* Clear any pending interrupt */
```

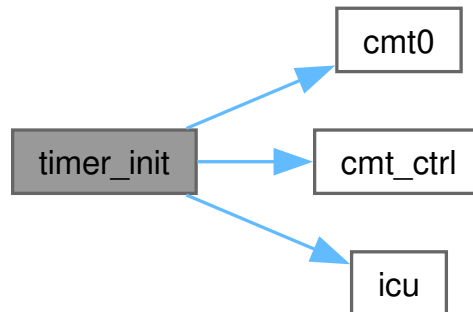
```

00750 icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear;
00751
00752 /* Set interrupt priority (3 out of 15). RX72N IPR array is SPARSE --
00753 * ipr[vector] is wrong; CMT0_CMI0 actually lives at ipr[4]. See
00754 * k_cmt0_ipr_index declaration above for the full rationale. */
00755 icu()->ipr[k_cmt0_ipr_index] = k_cmt0_irq_priority;
00756
00757 /* Enable CMT0 interrupt in ICU */
00758 const uint8_t ier_bit = (uint8_t)(k_vect_cmt0_cmi0 % k_icu_ier_bits_per_reg);
00759 const uint8_t ier_mask = (uint8_t)(k_ier_bit_enable_base « ier_bit);
00760 icu()->ier[k_vect_cmt0_cmi0 / k_icu_ier_bits_per_reg] |= ier_mask;
00761
00762 /* Start CMT0 */
00763 cmt_ctrl()->cmstr0 |= (uint16_t)k_cmt0_cmstr_start_bit;
00764
00765 if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) == 0) {
00766 return k_rx_err_hw_error;
00767 }
00768
00769 /* Enable interrupts globally (set I flag in PSW) */
00770 /* Enable interrupts globally via PSW I-flag
00771 * Note: Direct assembly required as RX GCC has no built-in for PSW manipulation */
00772 __asm__ volatile("setpsw i");
00773
00774 rx_log_info(s_tag, "CMT0 initialized successfully");
00775
00776 return k_rx_ok;
00777 }

```

References [rx\\_cmt\\_channel\\_regs\\_t::cment](#), [rx\\_cmt\\_channel\\_regs\\_t::cmcor](#), [rx\\_cmt\\_channel\\_regs\\_t::cmcr](#), [rx\\_cmt\\_control\\_regs\\_t::cmstr0](#), [cmt0\(\)](#), [cmt\\_ctrl\(\)](#), [icu\(\)](#), [rx\\_icu\\_regs\\_t::ier](#), [rx\\_icu\\_regs\\_t::ipr](#), [rx\\_icu\\_regs\\_t::ir](#), [k\\_cmt0\\_cmcr\\_config](#), [k\\_cmt0\\_cmstr\\_start\\_bit](#), [k\\_cmt0\\_compare\\_match](#), [k\\_cmt0\\_counter\\_init](#), [k\\_cmt0\\_ipr\\_index](#), [k\\_cmt0\\_irq\\_priority](#), [k\\_icu\\_ier\\_bits\\_per\\_reg](#), [k\\_icu\\_ir\\_clear](#), [k\\_ier\\_bit\\_enable\\_base](#), [k\\_vect\\_cmt0\\_cmi0](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.76 timer\_stop()

```

rx_err_t timer_stop (
 void) [nodiscard]

```

Stop CMT0 timer and disable compare match interrupts.

Halts the CMT0 counter by clearing the start bit (CMSTR0.STR0 = 0) and disables the compare match interrupt. The counter value is preserved and can be read via [timer\\_get\\_count\(\)](#). After stopping, [timer\\_init\(\)](#) may be called to restart the timer.

### 8.1.5.77 Algorithm

1. Disable compare match interrupt (CMCR.CMIE = 0)
2. Stop counter (CMSTR0.STR0 = 0)
3. Mark timer as stopped in internal state

#### Returns

`rx_err_t` Error code indicating success or failure

#### Return values

|                                     |                           |
|-------------------------------------|---------------------------|
| <code>k_rx_ok</code>                | CMT0 successfully stopped |
| <code>k_rx_err_invalid_state</code> | CMT0 was not initialized  |

#### Precondition

CMT0 must have been initialized via [timer\\_init\(\)](#)

Caller must ensure stopping the system tick will not break RTOS scheduling

#### Postcondition

CMT0 counter halted, counter value preserved

Compare match interrupts disabled

#### Note

Thread Safety: NOT thread-safe. Stopping the system tick from a ThreadX thread may cause scheduling to freeze.

#### Warning

Stopping CMT0 halts the ThreadX system tick. Only stop during shutdown or low-power entry.

#### Example: Entering Low Power Mode

```
void enter_low_power(void) {
 rx_err_t err = timer_stop();
 if (err != k_rx_ok) {
 rx_log_error_val("TIMER", "Failed to stop timer", err);
 return;
 }
 // Enter low-power mode
}
```

#### See also

[timer\\_init\(\)](#) Reinitialize CMT0 after stopping

[timer\\_get\\_count\(\)](#) Read counter value (works while stopped)



Since

Version 1.0.0

Stop CMT0 timer and disable compare match interrupts.

Halts the CMT0 timer by clearing the start bit in CMSTR0 register. This stops the system tick interrupts, effectively freezing ThreadX scheduling and time tracking. Use only during system shutdown or for testing.

Algorithm steps:

1. Validate CMT\_CTRL hardware accessibility
2. Check if timer is currently running
3. Clear CMSTR0 start bit to stop counter
4. Verify timer stopped successfully

Use cases:

- System shutdown sequence
- Power management (entering low-power mode)
- Testing and diagnostics
- Emergency stop during fault condition

Warning

Stopping system tick will freeze ThreadX scheduler. All time-based operations (tx\_thread\_sleep, timeouts) will hang. Only call during controlled shutdown or with ThreadX suspended.

Precondition

[timer\\_init\(\)](#) must have been called successfully

CMT0 should be running (otherwise returns k\_rx\_err\_invalid\_state)

Postcondition

CMT0 timer stopped, CMCNT frozen

No more [cmt0\\_isr\(\)](#) interrupts will occur

ThreadX time tracking frozen (system time no longer increments)

Note

Thread Safety: Not thread-safe. Should only be called from single-threaded shutdown sequence or with ThreadX suspended.

Performance: Executes in ~1 us @ 240 MHz (register read/write).

Example - Controlled Shutdown:

```
// Shutdown sequence
tx_thread_suspend(&all_app_threads); // Suspend all app threads

rx_err_t err = timer_stop();
if (err != k_rx_ok) {
 rx_log_error("SHUTDOWN", "Failed to stop timer: %d", err);
}

// System is now halted, safe to power down
```

See also

[timer\\_init\(\)](#) Initialize and start timer  
[timer\\_get\\_count\(\)](#) Read current counter value

Since

Version 1.0.0

Definition at line 837 of file [timer.c](#).

```
00838 {
00839 if (cmt_ctrl() == nullptr) {
00840 return k_rx_err_hw_unmapped;
00841 }
00842
00843 if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) == 0) {
00844 return k_rx_err_invalid_state;
00845 }
00846
00847 rx_log_info(s_tag, "Stopping CMT0");
00848
00849 /* Stop CMT0 */
00850 cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t)k_cmt0_cmstr_start_bit;
00851
00852 if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) != 0) {
00853 return k_rx_err_hw_error;
00854 }
00855
00856 return k_rx_ok;
00857 }
```

References [rx\\_cmt\\_control\\_regs\\_t::cmstr0](#), [cmt\\_ctrl\(\)](#), [k\\_cmt0\\_cmstr\\_start\\_bit](#), and [s\\_tag](#).

Here is the call graph for this function:



#### 8.1.5.78 `uart_clear_rx_errors()`

```
rx_err_t uart_clear_rx_errors (
 uart_channel_t channel) [nodiscard]
```

Clear sticky receive error flags (ORER/FER/PER) on a UART channel.

Reads SSR and writes back with the receive error bits masked to 0. Per HUM 34.2.7, those flags must be cleared before the receiver will latch further bytes. The RXI ISR clears them on every byte; this helper is the periodic-fallback path when an error latches without RDRF firing (e.g. line glitch on an idle bus).

## Parameters

|    |         |                     |
|----|---------|---------------------|
| in | channel | SCI channel (0-12). |
|----|---------|---------------------|

## Returns

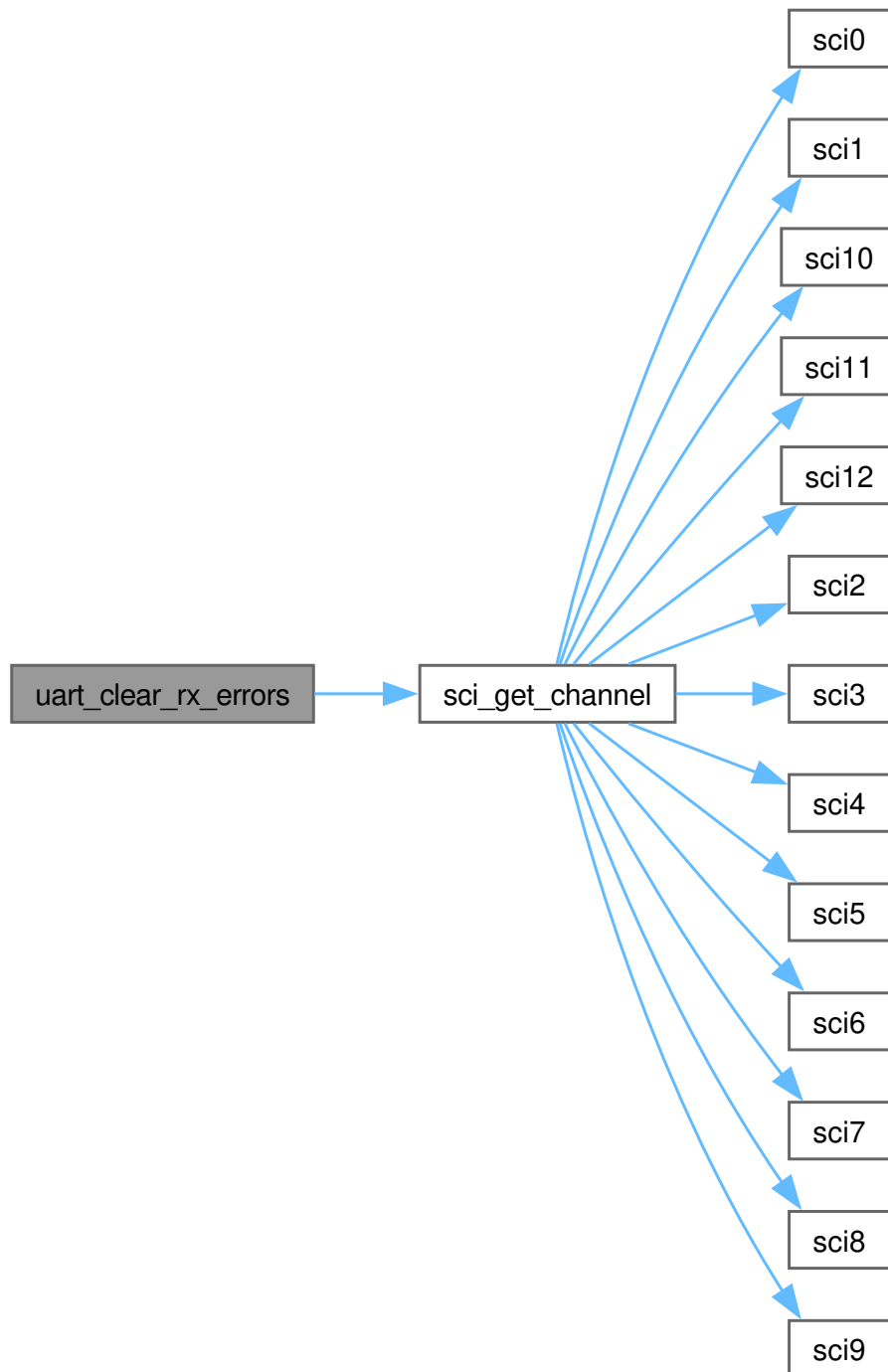
k\_rx\_ok on success, k\_rx\_err\_invalid\_arg if channel is invalid.

Definition at line 1834 of file [uart.c](#).

```
01835 {
01836 if ((uint8_t)channel >= k_uart_max_channels) {
01837 return k_rx_err_invalid_arg;
01838 }
01839 volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01840 if (sci == nullptr) {
01841 return k_rx_err_invalid_arg;
01842 }
01843 const uint8_t ssr = sci->ssr;
01844 if ((ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01845 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_error_mask);
01846 }
01847 return k_rx_ok;
01848 }
```

References [k\\_sci\\_ssr\\_error\\_mask](#), [k\\_uart\\_flag\\_clear](#), [k\\_uart\\_max\\_channels](#), [sci\\_get\\_channel\(\)](#), and [rx\\_sci\\_regs\\_t::ssr](#).

Here is the call graph for this function:



#### 8.1.5.79 `uart_debug_init()`

```
rx_err_t uart_debug_init (
 void) [nodiscard]
```

Initialize debug UART (SCI9, 115200 bps, 8N1).

Convenience wrapper for `uart_init_channel(k_uart_debug_channel, 115200)`. Used by `rx_log` for early boot logging before ThreadX is initialized.

### Returns

k\_rx\_ok on success, error code on failure

Initialize debug UART (SCI9, 115200 bps, 8N1).

Convenience function to initialize SCI9 with default debug settings:

- Channel: SCI9
- Baud rate: 115200 bps
- TX pin: PB7 (TXD9)
- RX pin: PB6 (RXD9)
- Format: 8N1 (8 data bits, no parity, 1 stop bit)

SCI9 is connected to the CY7C65213 USB-UART bridge on the STAR PCB, providing a virtual COM port when connected via USB.

### Algorithm steps:

1. Create `uart_channel_config_t` with default values
2. Call `uart_init_channel()` with the configuration
3. Return result

### Precondition

System clocks must be initialized (PCLKB = 60 MHz)

SCI9 must not be already initialized

### Postcondition

SCI9 configured and ready for TX/RX at 115200 baud

`uart_debug_puts()`, `uart_debug_putc()` etc. are functional

### Note

Call once during early system initialization

Call before any other debug output functions

### Warning

Must be called before ThreadX starts if debug output is needed during init

### Thread Safety:

Not thread-safe. Call once during startup before ThreadX.

Performance:

- Execution time: ~50 us
- Stack usage: 64 bytes

Example:

```
void system__init(void)
{
 // Initialize clocks first
 rx_clock_init();

 // Initialize debug UART for early output
 rx_err_t err = uart_debug_init();
 if (err != k_rx_ok) {
 // Cannot print error - no UART!
 // Could blink LED or set error flag
 return;
 }

 // Now we can print debug messages
 uart_debug_puts("[INFO] System starting...\n");
}
```

See also

[uart\\_init\\_channel\(\)](#) Generic channel initialization

[uart\\_debug\\_puts\(\)](#) Debug string output

[uart\\_debug\\_putc\(\)](#) Debug character output

Since

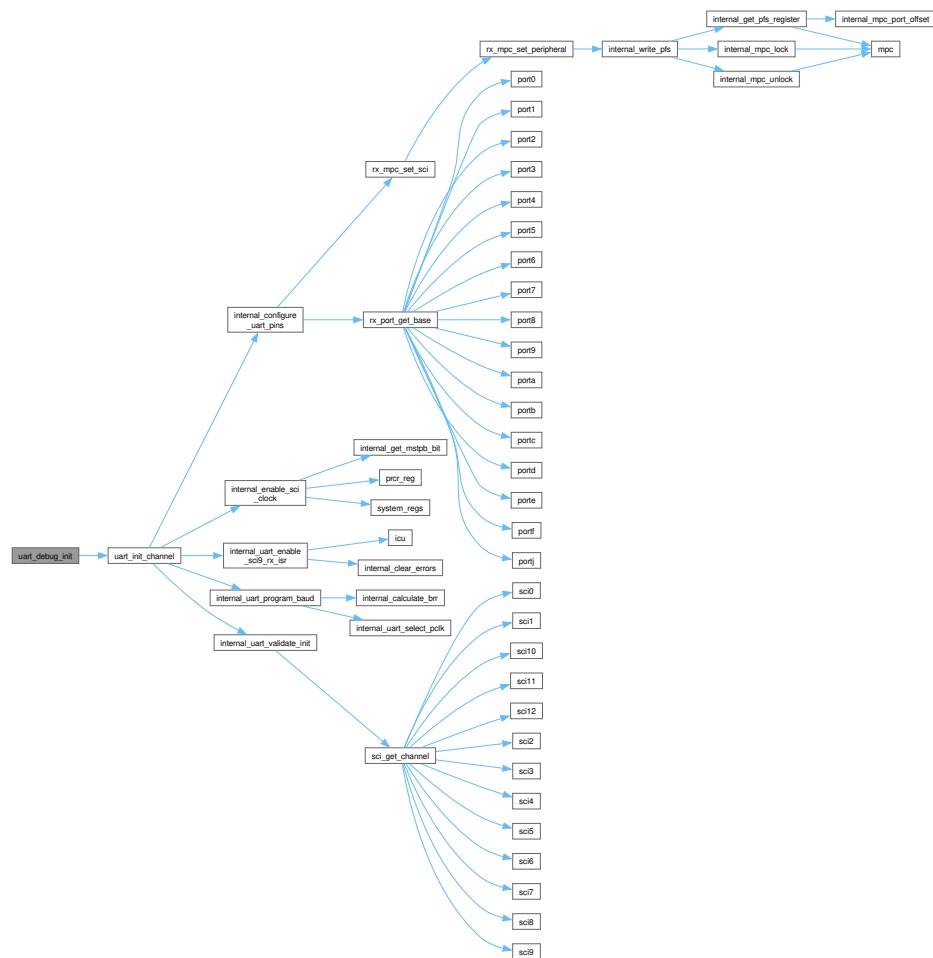
Version 1.0.0

Definition at line 1977 of file [uart.c](#).

```
01978 {
01979 const uart_channel_config_t config = {
01980 .channel = (uart_channel_t)k_uart_debug_channel,
01981 .baudrate = k_uart_default_baudrate,
01982 .tx_gpio = (rx_port_pin_t)k_uart_debug_tx_gpio,
01983 .rx_gpio = (rx_port_pin_t)k_uart_debug_rx_gpio,
01984 };
01985 return uart_init_channel(&config);
01986 }
```

References [k\\_uart\\_debug\\_channel](#), [k\\_uart\\_debug\\_rx\\_gpio](#), [k\\_uart\\_debug\\_tx\\_gpio](#), [k\\_uart\\_default\\_baudrate](#), and [uart\\_init\\_channel\(\)](#).

Here is the call graph for this function:



### 8.1.5.80 uart\_debug\_putc()

```
void uart_debug_putc (
 const char data)
```

Transmit a single character on debug UART (SCI9).

#### Parameters

|    |      |                       |
|----|------|-----------------------|
| in | data | Character to transmit |
|----|------|-----------------------|

Transmit a single character on debug UART (SCI9).

Convenience wrapper around [uart\\_putc\\_channel\(\)](#) for the fixed debug channel (SCI9). Errors are silently ignored to allow use in early initialization contexts where error propagation is not yet possible.

Algorithm steps:

1. Call `uart_putc_channel(k_uart_debug_channel, data)`
2. Cast return value to (void) - errors discarded

#### Precondition

[uart\\_debug\\_init\(\)](#) must have been called successfully  
SCI9 must be initialized and TX enabled

#### Postcondition

Character written to SCI9 TDR and transmission started  
Any errors are silently discarded

#### Note

Error return from [uart\\_putc\\_channel\(\)](#) is intentionally discarded  
For error-checked output use [uart\\_putc\\_channel\(\)](#) directly

#### Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

#### Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

#### Performance:

- Execution time: ~90 us @ 115200 baud
- Stack usage: 16 bytes

#### Example:

```
uart_debug_putc('A'); // Send 'A'
uart_debug_putc('\r'); // Carriage return
uart_debug_putc('\n'); // Line feed
```

#### See also

[uart\\_debug\\_puts\(\)](#) Transmit string with newline conversion  
[uart\\_putc\\_channel\(\)](#) Error-checked single character transmit

#### Since

Version 1.0.0

Definition at line 2040 of file [uart.c](#).

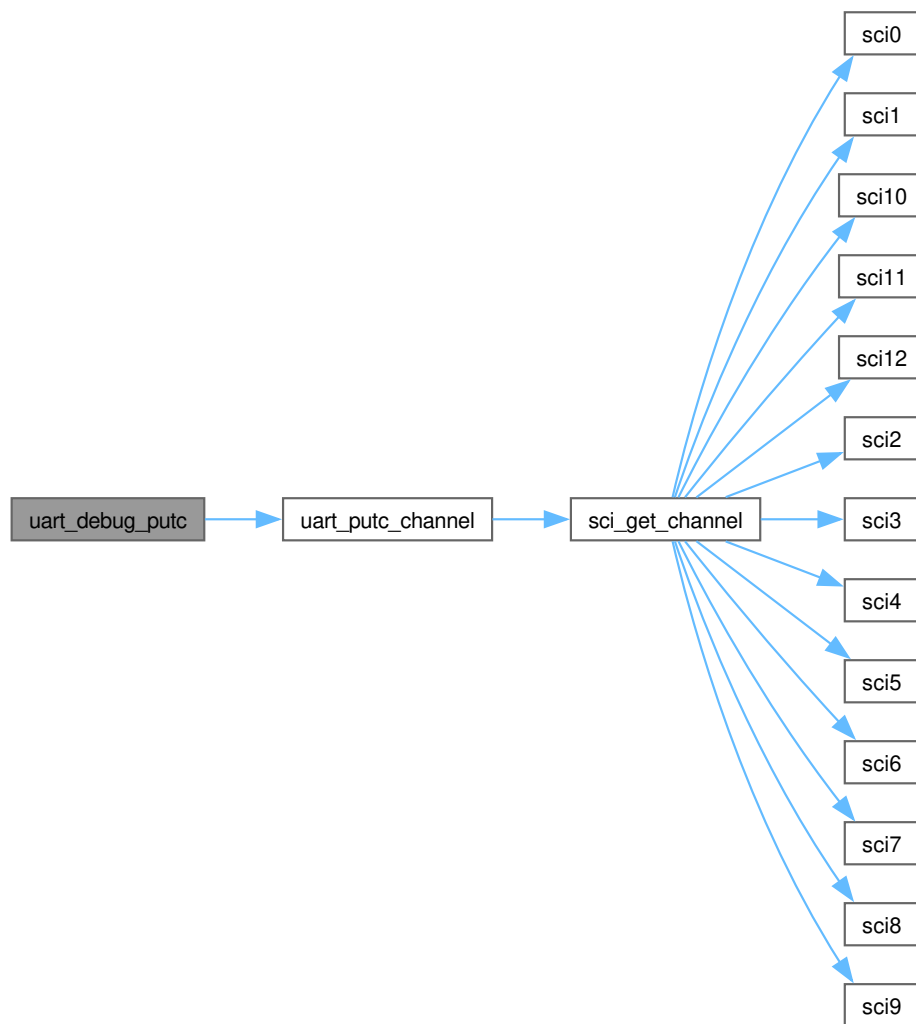
```
02041 {
02042 /* Ignore errors for debug output (used in early init before error handling) */
02043 (void)uart_putc_channel((uart_channel_t)k_uart_debug_channel, data);
02044 }
```

References [k\\_uart\\_debug\\_channel](#), and [uart\\_putc\\_channel\(\)](#).

Referenced by [uart\\_debug\\_puthex\(\)](#).

Here is the call graph for this function:





Here is the caller graph for this function:



#### 8.1.5.81 uart\_debug\_puthex()

```
void uart_debug_puthex (
 const uint32_t value,
 uint8_t digits)
```

Print a hexadecimal value on debug UART (SCI9).

Parameters

|    |        |                            |
|----|--------|----------------------------|
| in | value  | Value to print             |
| in | digits | Number of hex digits (1-8) |

Print a hexadecimal value on debug UART (SCI9).

Transmits "0x" prefix followed by the specified number of uppercase hex digits of value on SCI9. Digits are always printed most-significant first. The digits parameter is clamped to [k\_uart\_hex\_min\_digits, k\_uart\_hex\_max\_digits] (1-8) before use.

Algorithm steps:

1. Send "0x" prefix via `uart_debug_puts()`
2. Clamp digits to [1, 8]
3. Iterate nibbles from MSN to LSN (statically bounded by k\_uart\_hex\_max\_digits)
4. For each nibble index < digits, extract nibble and print uppercase hex char

Precondition

`uart_debug_init()` must have been called successfully  
`uart_debug_puts()` and `uart_debug_putc()` must be functional

Postcondition

"0x" followed by digits uppercase hex characters transmitted on SCI9  
digits clamped to [1, 8] if out of range

Note

Uses static lookup table `s_hex[]` for digit-to-character conversion  
Always prefixes output with "0x"

Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes

Example:

```
uart_debug_puts("Addr: ");
uart_debug_puthex(0xDEADBEEF, 8); // Sends "0xDEADBEEF"
uart_debug_puthex(0x42, 2); // Sends "0x42"
uart_debug_putc('\n');
```

See also

[uart\\_debug\\_putint\(\)](#) Print signed decimal value  
[uart\\_debug\\_puts\(\)](#) Underlying string transmit

Since

Version 1.0.0

Definition at line 2230 of file [uart.c](#).

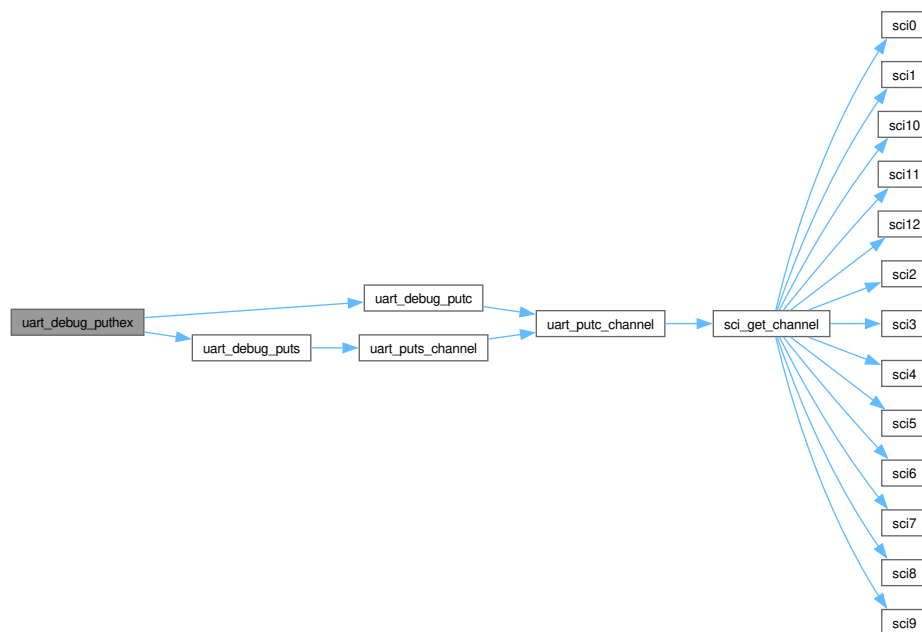
```

02231 {
02232 static const char s_hex[] = "0123456789ABCDEF";
02233
02234 uart_debug_puts("0x");
02235
02236 /* Clamp digits to valid range */
02237 if (digits > k_uart_hex_max_digits) {
02238 digits = k_uart_hex_max_digits;
02239 }
02240 if (digits == k_uart_hex_zero_digits) {
02241 digits = k_uart_hex_min_digits;
02242 }
02243
02244 /* Print hex digits from most significant (statically bounded, unsigned counter) */
02245 for (uint8_t i = k_uart_hex_zero_digits; i < k_uart_hex_max_digits; ++i) {
02246 const uint8_t digit_idx = (k_uart_hex_max_digits - k_uart_hex_min_digits) - i;
02247 if (digit_idx < digits) {
02248 const uint8_t nibble =
02249 (uint8_t)((value » (digit_idx * k_uart_hex_nibble_bits)) & k_uart_hex_nibble_mask);
02250 uart_debug_putc(s_hex[nibble]);
02251 }
02252 }
02253 }

```

References [k\\_uart\\_hex\\_max\\_digits](#), [k\\_uart\\_hex\\_min\\_digits](#), [k\\_uart\\_hex\\_nibble\\_bits](#), [k\\_uart\\_hex\\_nibble\\_mask](#), [k\\_uart\\_hex\\_zero\\_digits](#), [uart\\_debug\\_putc\(\)](#), and [uart\\_debug\\_puts\(\)](#).

Here is the call graph for this function:



8.1.5.82 `uart_debug_putint()`

```
void uart_debug_putint (
 const int32_t value)
```

Print a signed integer on debug UART (SCI9).

Parameters

|    |       |                        |
|----|-------|------------------------|
| in | value | Integer value to print |
|----|-------|------------------------|

Print a signed integer on debug UART (SCI9).

Converts a signed 32-bit integer to a decimal ASCII string and transmits it on SCI9. Handles INT32\_MIN correctly by using int64\_t for the negation. Uses a fixed-size stack buffer (`k_uart_int_buffer_size` = 12) built in reverse then passed to [uart\\_debug\\_puts\(\)](#).

Algorithm steps:

1. Determine sign; compute `abs_value` as `uint32_t`
2. Null-terminate the buffer end
3. Build digits right-to-left (statically bounded by `k_uart_int_buffer_size`)
4. Prepend '-' if negative
5. Call [uart\\_debug\\_puts\(\)](#) with the resulting substring pointer

Precondition

[uart\\_debug\\_init\(\)](#) must have been called successfully  
[uart\\_debug\\_puts\(\)](#) must be functional

Postcondition

Decimal representation of value transmitted on SCI9  
 Leading zeros suppressed; negative values prefixed with '-'

Note

No return value - output errors are silently discarded  
 Buffer is stack-allocated; safe for re-entrant calls on different tasks

Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us per digit @ 115200 baud + digit-loop overhead
- Stack usage: 32 bytes (buffer + locals)

Example:

```
uart_debug_puts("Value: ");
uart_debug_putint(42); // Sends "42"
uart_debug_putint(-2147483648); // Sends "-2147483648"
uart_debug_putc('\n');
```

See also

[uart\\_debug\\_puthex\(\)](#) Print value in hexadecimal  
[uart\\_debug\\_puts\(\)](#) Underlying string transmit

Since

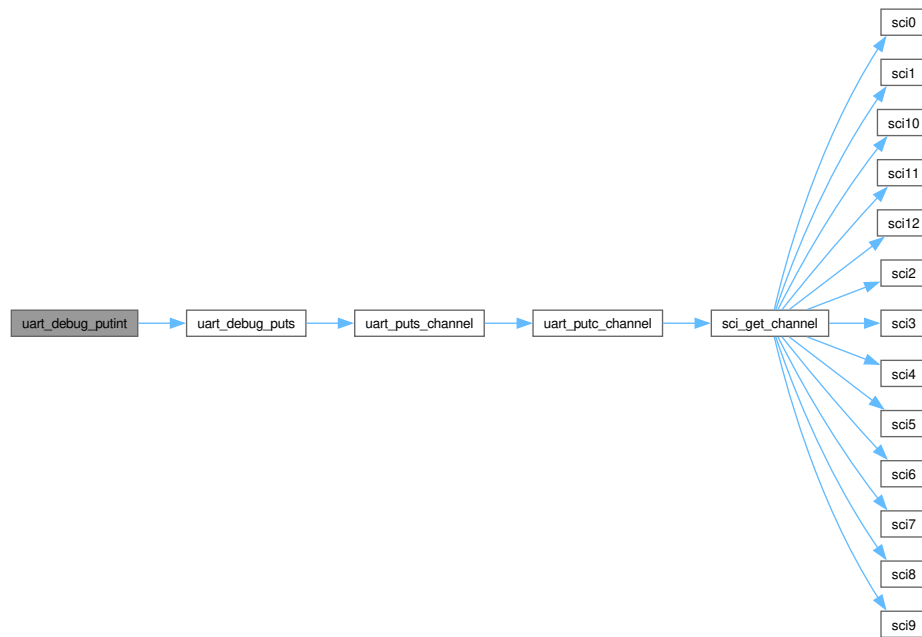
Version 1.0.0

Definition at line 2150 of file [uart.c](#).

```
02151 {
02152 char buffer[k_uart_int_buffer_size]; /* Enough for -2147483648 */
02153 char* p = buffer + sizeof(buffer) - 1;
02154
02155 /* Handle negative numbers */
02156 const bool is_negative = (value < 0);
02157 uint32_t abs_value = is_negative ? (uint32_t)(-(int64_t)value) : (uint32_t)value;
02158
02159 /* Null terminate */
02160 *p = '\0';
02161
02162 /* Convert to string (reverse order, statically bounded) */
02163 for (uint8_t i = 0; i < k_uart_int_buffer_size - 1; ++i) {
02164 *(&p) = (char)('0' + (abs_value % k_uart_base_10));
02165 abs_value /= k_uart_base_10;
02166 if (abs_value == 0) {
02167 break; /* All digits processed */
02168 }
02169 if (p <= buffer) {
02170 break; /* Buffer limit reached (should not happen with correct sizing) */
02171 }
02172 }
02173
02174 /* Add minus sign if negative */
02175 if (is_negative && p > buffer) {
02176 *(&p) = '-';
02177 }
02178
02179 /* Transmit the string */
02180 uart_debug_puts(p);
02181 }
```

References [k\\_uart\\_base\\_10](#), [k\\_uart\\_int\\_buffer\\_size](#), and [uart\\_debug\\_puts\(\)](#).

Here is the call graph for this function:



#### 8.1.5.83 `uart_debug_puts()`

```
void uart_debug_puts (
 const char * str)
```

Transmit a null-terminated string on debug UART (SCI9).

##### Parameters

|    |     |                   |
|----|-----|-------------------|
| in | str | Pointer to string |
|----|-----|-------------------|

Transmit a null-terminated string on debug UART (SCI9).

Transmits a null-terminated string to SCI9 with automatic LF-to-`\r`+LF conversion for terminal compatibility. Enforces a maximum length of `k_uart_max_str_len` (256) characters per NASA Power of 10 Rule 2. Silently returns on NULL pointer to allow safe use in early init.

Algorithm steps:

1. Return immediately if `str` is `nullptr` (defensive, no error return)
2. For each character up to `k_uart_max_str_len`:
  - a. Stop at null terminator
  - b. If LF, send `\r` first
  - c. Send character via [uart\\_debug\\_putc\(\)](#)

##### Precondition

[uart\\_debug\\_init\(\)](#) must have been called successfully  
`str` should point to a null-terminated string in valid memory

## Postcondition

All characters up to null terminator transmitted to SCI9  
Each LF replaced by `\r\n` in the transmitted output

## Note

No return value - errors from `uart_debug_putc()` are silently discarded  
String truncated at `k_uart_max_str_len` (256) characters

## Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

## Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

## Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes

## Example:

```
uart_debug_puts("Hello, World!\n"); // Sends "Hello, World!\r\n"
uart_debug_puts("[INFO] Boot complete\n");
```

## See also

[uart\\_debug\\_putc\(\)](#) Transmit single character  
[uart\\_debug\\_putint\(\)](#) Transmit decimal integer  
[uart\\_puts\\_channel\(\)](#) Error-checked string transmit

## Since

Version 1.0.0

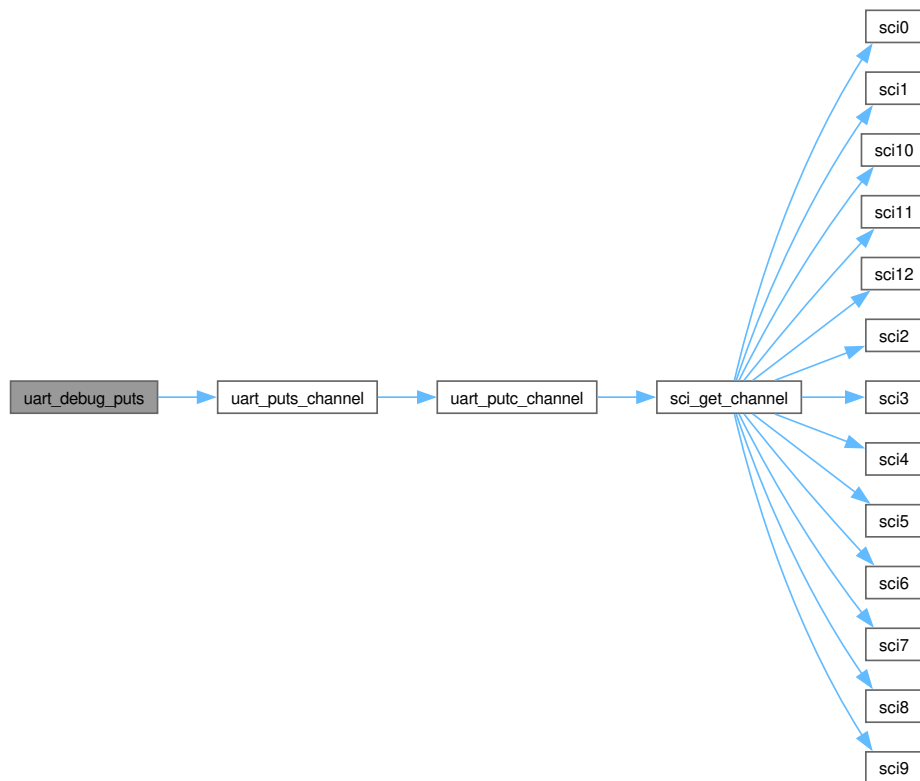
Definition at line 2092 of file [uart.c](#).

```
02093 {
02094 if (str == nullptr) {
02095 return;
02096 }
02097
02098 /* Delegate to uart_puts_channel which already handles LF->CRLF conversion,
02099 * bounded iteration (k_uart_max_str_len), and per-character error checking. */
02100 (void)uart_puts_channel((uart_channel_t)k_uart_debug_channel, str);
02101 }
```

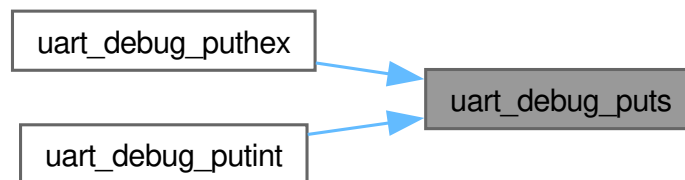
References [k\\_uart\\_debug\\_channel](#), and [uart\\_puts\\_channel\(\)](#).

Referenced by [uart\\_debug\\_puthex\(\)](#), and [uart\\_debug\\_putint\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.84 uart\_deinit\_channel()

```
rx_err_t uart_deinit_channel (
 const uart_channel_t channel) [nodiscard]
```

Deinitialize SCI channel.

Parameters

|    |         |                    |
|----|---------|--------------------|
| in | channel | SCI channel (0-12) |
|----|---------|--------------------|



### Returns

k\_rx\_ok on success, k\_rx\_err\_invalid\_arg if channel is invalid

### Deinitialize SCI channel.

Disables transmit and receive on the specified SCI channel and marks it as uninitialized. The module clock is left running (safe to re-initialize without re-enabling). After this call the channel may be re-initialized with `uart_init_channel()`.

### Algorithm steps:

1. Validate channel number (0-12)
2. Get SCI register base address
3. Write S\r = 0 to disable TX and RX
4. Clear s\_channel\_initialized[channel]

### Precondition

channel must be in range [0, 12]

Channel should be initialized before deinitializing (safe to call on uninit channel)

### Postcondition

TX and RX disabled (S\r = 0)

s\_channel\_initialized[channel] set to false

### Note

Not thread-safe - call during shutdown, not during normal operation

Module stop clock is NOT re-asserted (peripheral clock left enabled)

### Thread Safety:

Not thread-safe. Do not call while another thread is using the channel.

### Performance:

- Execution time: ~1 us
- Stack usage: 16 bytes

### Example:

```
rx_err_t err = uart_deinit_channel(k_uart_channel_9);
if (err != k_rx_ok) {
 // Handle error (invalid channel)
}
// Channel is now disabled and can be re-initialized
```

See also

[uart\\_init\\_channel\(\)](#) Initialize channel  
[uart\\_debug\\_init\(\)](#) Initialize debug channel (SCI9)

Since

Version 1.0.0

Definition at line 1215 of file [uart.c](#).

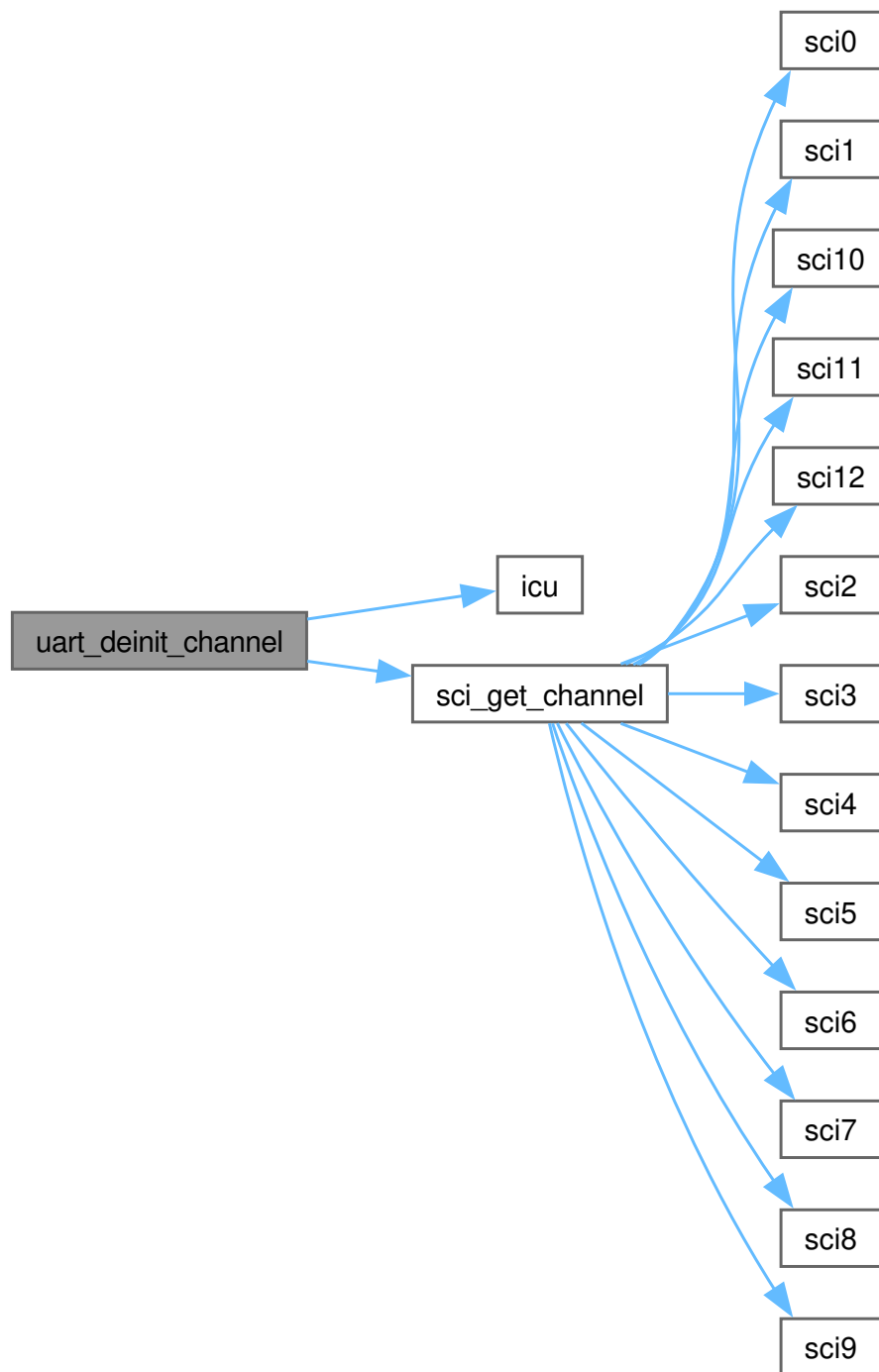
```

01216 {
01217 /* Validate channel */
01218 if ((uint8_t)channel >= k_uart_max_channels) {
01219 return k_rx_err_invalid_arg;
01220 }
01221
01222 /* Get SCI register base */
01223 volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01224 if (sci == nullptr) {
01225 return k_rx_err_invalid_arg;
01226 }
01227
01228 /* Disable TX/RX (this also drops SCR.RIE for SCI9). */
01229 sci->scr = k_sci_scr_disabled;
01230
01231 /* For SCI9, mask the RXI9 interrupt in the ICU and flush the ring so a
01232 * subsequent re-init starts clean. */
01233 if (channel == k_uart_channel_9) {
01234 const uint16_t vec_rxi9 = (uint16_t)k_vect_sci9_rxi9;
01235 icu()->ier[vec_rxi9 / k_icu_ier_bits_per_reg] &=
01236 (uint8_t) ~(1U « (vec_rxi9 % k_icu_ier_bits_per_reg));
01237 icu()->ir[vec_rxi9] = 0U;
01238 s_sci9_rx_head = 0U;
01239 s_sci9_rx_tail = 0U;
01240 }
01241
01242 /* Mark channel as not initialized */
01243 s_channel_initialized[channel] = false;
01244
01245 return k_rx_ok;
01246 }

```

References [icu\(\)](#), [rx\\_icu\\_regs\\_t::ier](#), [rx\\_icu\\_regs\\_t::ir](#), [k\\_icu\\_ier\\_bits\\_per\\_reg](#), [k\\_sci\\_scr\\_disabled](#), [k\\_uart\\_channel\\_9](#), [k\\_uart\\_max\\_channels](#), [k\\_vect\\_sci9\\_rxi9](#), [s\\_channel\\_initialized](#), [s\\_sci9\\_rx\\_head](#), [s\\_sci9\\_rx\\_tail](#), [sci\\_get\\_channel\(\)](#), and [rx\\_sci\\_regs\\_t::scr](#).

Here is the call graph for this function:



#### 8.1.5.85 `uart_getc_channel()`

```
rx_err_t uart_getc_channel (
 const uart_channel_t channel,
 char * data) [nodiscard]
```

Receive a single character from specified channel (non-blocking).

## Parameters

|     |         |                                     |
|-----|---------|-------------------------------------|
| in  | channel | SCI channel (0-12)                  |
| out | data    | Pointer to store received character |

## Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if data is nullptr, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized, k\_rx\_err\_empty if no data available

Receive a single character from specified channel (non-blocking).

Attempts to read a single byte from the specified SCI channel's receive data register (RDR). This is a non-blocking operation - returns immediately with k\_rx\_err\_empty if no data is available.

## Algorithm steps:

1. Validate data pointer and channel number
2. Check channel initialization state
3. Get SCI register base address
4. Clear any error flags (ORER, FER, PER)
5. Check RDRF flag (Receive Data Register Full)
  - If not set, return k\_rx\_err\_empty immediately
6. Read data byte from RDR register
7. Clear RDRF flag (read SSR, write back with RDRF=0)

## Error flag handling:

- ORER (Overflow Error): Previous data overwritten before read
- FER (Framing Error): Stop bit not detected
- PER (Parity Error): Parity mismatch (not used in 8N1 mode)

## Precondition

Channel must be initialized via [uart\\_init\\_channel\(\)](#)  
data must point to valid memory for single char

## Postcondition

On success, \*data contains received byte  
RDRF flag cleared (ready for next byte)  
Error flags cleared if any were set

#### Note

- Non-blocking - returns immediately if no data
- Clears error flags automatically before read
- Single byte buffer - call frequently to avoid overrun

#### Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

#### Performance:

- Execution time: ~5 us (no wait)
- Stack usage: 16 bytes

#### Example (Polling Loop):

```
char c;
rx_err_t err;

// Poll for incoming data
do {
 err = uart_getc_channel(k_uart_channel_9, &c);
 if (err == k_rx_ok) {
 process_character(c);
 }
} while (err == k_rx_ok); // Continue while data available
```

#### Example (Timeout Loop):

```
char c;
uint32_t timeout = 100000;

while (timeout > 0) {
 rx_err_t err = uart_getc_channel(k_uart_channel_9, &c);
 if (err == k_rx_ok) {
 // Got data
 break;
 }
 timeout--;
}

if (timeout == 0) {
 // No data received within timeout
}
```

#### See also

- [uart\\_read\\_channel\(\)](#) Read multiple bytes
- [uart\\_rx\\_available\(\)](#) Check if data is available

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Definition at line 1606 of file `uart.c`.

```

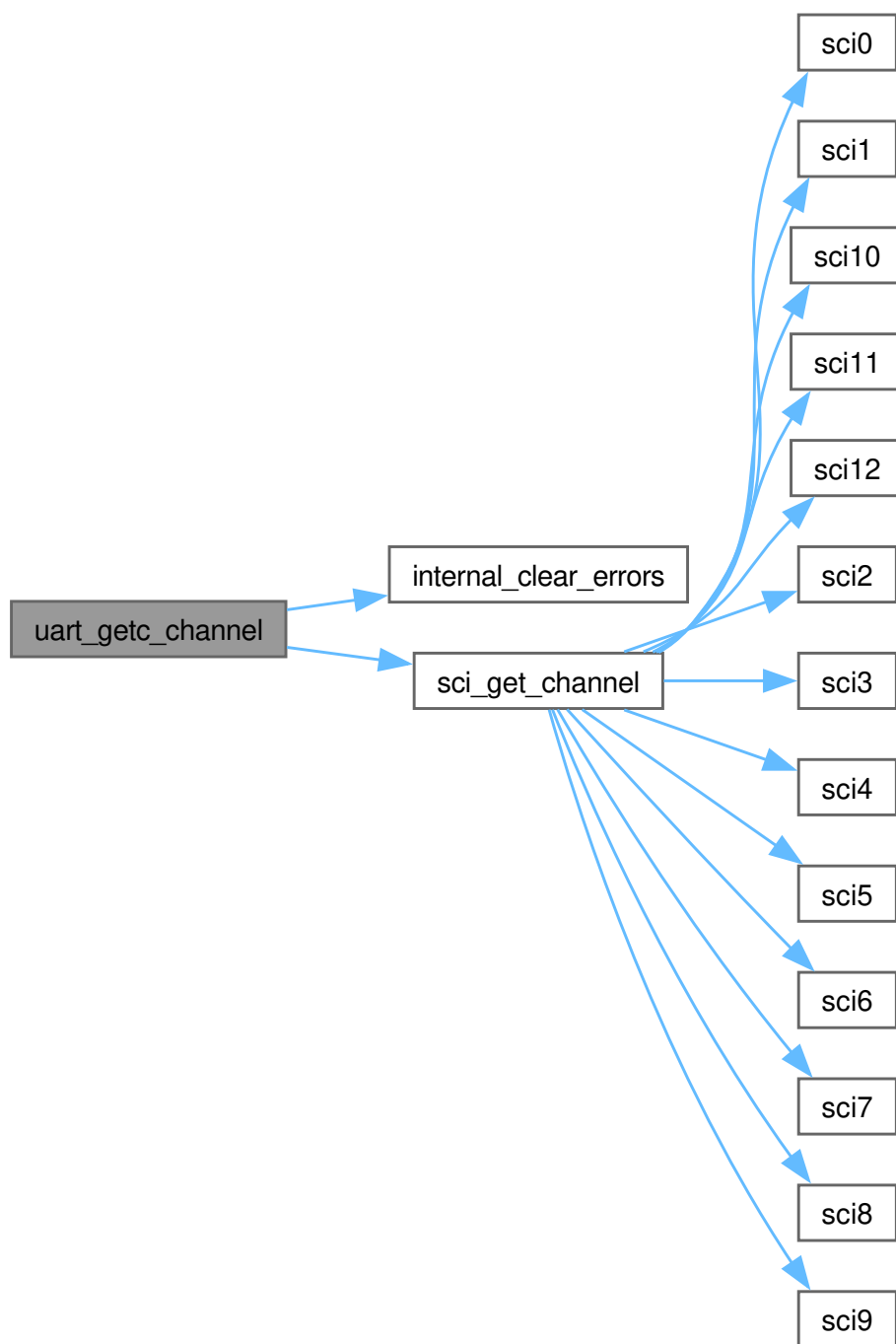
01607 {
01608 /* Validate parameters */
01609 if (data == nullptr) {
01610 return k_rx_err_null_ptr;
01611 }
01612
01613 if ((uint8_t)channel >= k_uart_max_channels) {
01614 return k_rx_err_invalid_arg;
01615 }
01616
01617 if (!s_channel_initialized[channel]) {
01618 return k_rx_err_invalid_state;
01619 }
01620
01621 /* SCI9 reads pull from the RXI9-fed ring buffer so polled callers cannot
01622 * lose bytes that arrive between polls. Head/tail word accesses are atomic
01623 * on RX72N (aligned 16-bit loads/stores are single-cycle), so lock-free
01624 * SPSC is safe between the ISR producer and the task consumer. */
01625 if (channel == k_uart_channel_9) {
01626 if (s_sci9_rx_tail == s_sci9_rx_head) {
01627 return k_rx_err_empty;
01628 }
01629 *data = (char)s_sci9_rx_ring[s_sci9_rx_tail];
01630 s_sci9_rx_tail = (uint16_t)((s_sci9_rx_tail + 1U) & k_sci9_rx_ring_mask);
01631 return k_rx_ok;
01632 }
01633
01634 /* Get SCI register base */
01635 volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01636 if (sci == nullptr) {
01637 return k_rx_err_invalid_arg;
01638 }
01639
01640 /* Clear any error flags first */
01641 if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01642 internal_clear_errors(sci);
01643 }
01644
01645 /* Check if receive data is available (RDRF flag) */
01646 if ((sci->ssr & k_sci_ssr_rdrf_flag) == k_uart_flag_clear) {
01647 return k_rx_err_empty;
01648 }
01649
01650 /* Read received data */
01651 *data = (char)sci->rdr;
01652
01653 /* Clear RDRF flag by reading SSR then writing 0 */
01654 /* Some RX MCUs require explicit clear after reading RDR */
01655 const uint8_t ssr = sci->ssr;
01656 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01657
01658 return k_rx_ok;
01659 }

```

References `internal_clear_errors()`, `k_sci9_rx_ring_mask`, `k_sci_ssr_error_mask`, `k_sci_ssr_rdrf_flag`, `k_uart_channel_9`, `k_uart_flag_clear`, `k_uart_max_channels`, `rx_sci_regs_t::rdr`, `s_channel_initialized`, `s_sci9_rx_head`, `s_sci9_rx_ring`, `s_sci9_rx_tail`, `sci_get_channel()`, and `rx_sci_regs_t::ssr`.

Referenced by `uart_read_channel()`.

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.86 uart\_init\_channel()

```
rx_err_t uart_init_channel (
 const uart_channel_config_t * config) [nodiscard]
```

Initialize SCI channel for UART communication.

Performs full hardware initialization including:

- Module stop control (MSTPCRB)
- Pin function configuration (MPC)
- GPIO direction setup (PDR/PMR)
- SCI register configuration

#### Parameters

|    |        |                                               |
|----|--------|-----------------------------------------------|
| in | config | UART channel configuration (must not be NULL) |
|----|--------|-----------------------------------------------|

#### Returns

k\_rx\_ok on success, k\_rx\_err\_invalid\_arg if config is nullptr or contains invalid values, k\_rx\_err\_invalid\_state if channel already initialized

Definition at line 1135 of file [uart.c](#).

```

01136 {
01137 volatile rx_sci_regs_t* sci = nullptr;
01138 const rx_err_t check = internal_uart_validate_init(config, &sci);
01139 if (check != k_rx_ok) {
01140 return check;
01141 }
01142
01143 rx_err_t err = internal_enable_sci_clock(config->channel);
01144 if (err != k_rx_ok) {
01145 return err;
01146 }
01147 err = internal_configure_uart_pins(config->tx_gpio, config->rx_gpio);
01148 if (err != k_rx_ok) {
01149 return err;
01150 }
01151
01152 /* Disable TX/RX while reprogramming registers */
01153 sci->scr = k_sci_scr_disabled;
01154
01155 internal_uart_program_baud(sci, config->channel, config->baudrate);
01156
01157 /* Configure serial control: Enable TX and RX */
01158 sci->scr = k_sci_scr_txrx_enabled;
01159
01160 if (config->channel == k_uart_channel_9) {
01161 internal_uart_enable_sci9_rx_isr(sci);
01162 }
01163
01164 s_channel_initialized[config->channel] = true;
01165 return k_rx_ok;
01166 }

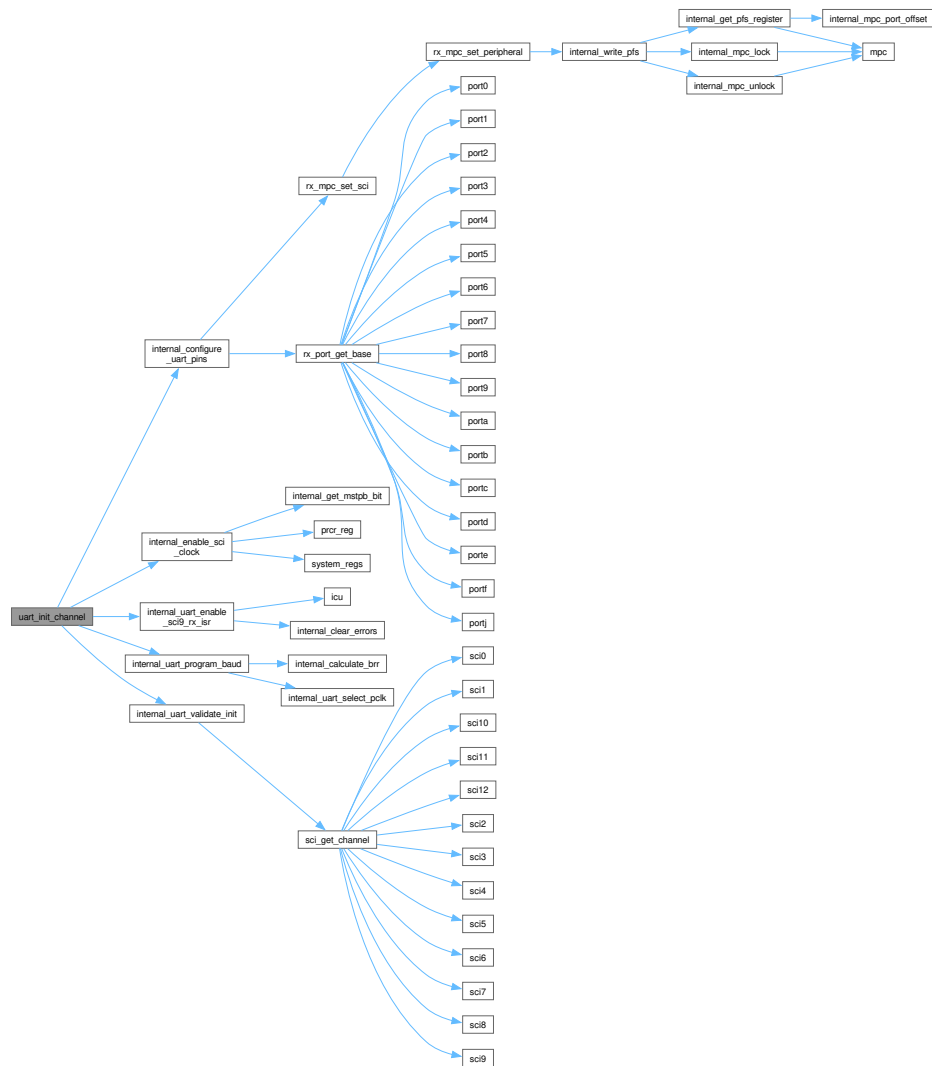
```

References [uart\\_channel\\_config\\_t::baudrate](#), [uart\\_channel\\_config\\_t::channel](#), [internal\\_configure\\_uart\\_pins\(\)](#), [internal\\_enable\\_sci\\_clock\(\)](#), [internal\\_uart\\_enable\\_sci9\\_rx\\_isr\(\)](#), [internal\\_uart\\_program\\_baud\(\)](#), [internal\\_uart\\_validate\\_init\(\)](#), [k\\_sci\\_scr\\_disabled](#), [k\\_sci\\_scr\\_txrx\\_enabled](#), [k\\_uart\\_channel\\_9](#), [uart\\_channel\\_config\\_t::rx\\_gpio](#), [s\\_channel\\_initialized](#), [rx\\_sci\\_regs\\_t::scr](#), and [uart\\_channel\\_config\\_t::tx\\_gpio](#).

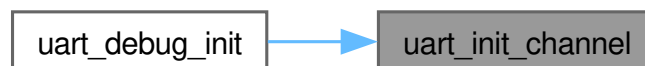
Referenced by [uart\\_debug\\_init\(\)](#).

Here is the call graph for this function:





Here is the caller graph for this function:



#### 8.1.5.87 uart\_putc\_channel()

```

rx_err_t uart_putc_channel (
 const uart_channel_t channel,
 const char data) [nodiscard]

```

Transmit a single character on specified channel.

Parameters

|    |         |                       |
|----|---------|-----------------------|
| in | channel | SCI channel (0-12)    |
| in | data    | Character to transmit |

### Returns

k\_rx\_ok on success, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized

Transmit a single character on specified channel.

Transmits a single byte on the specified SCI channel. Waits for the transmit data register to become empty (TDRE flag) with timeout protection, then writes the data to TDR. Uses polling mode (no interrupts).

Algorithm steps:

1. Validate channel number (0-12)
2. Check channel is initialized
3. Get SCI register base address
4. Wait for TDRE flag with timeout (k\_uart\_tx\_timeout iterations)
5. Write data byte to TDR register
6. Clear TDRE flag (read SSR, write back with TDRE=0)

Character transmission timing at 115200 baud:

- Start bit: 8.68 us
- 8 data bits: 69.44 us
- Stop bit: 8.68 us
- Total: ~86.8 us per character

### Precondition

Channel must be initialized via [uart\\_init\\_channel\(\)](#)  
UART TX must be enabled (happens during init)

### Postcondition

Character written to TDR and transmission started  
TDRE flag cleared (TDR occupied)

### Note

Blocking call - waits for TDRE flag  
Timeout prevents infinite wait if hardware fails

### Warning

May block for up to ~100ms if TX line is held low

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~90 us @ 115200 baud (one character time)
- Stack usage: 16 bytes

Example:

```
// Send single character
rx_err_t err = uart_putc_channel(k_uart_channel_9, 'A');
if (err != k_rx_ok) {
 // Handle error
}

// Send carriage return + line feed
uart_putc_channel(k_uart_channel_9, '\\r');
uart_putc_channel(k_uart_channel_9, '\\n');
```

See also

[uart\\_puts\\_channel\(\)](#) Transmit string with newline conversion

[uart\\_write\\_channel\(\)](#) Transmit binary data

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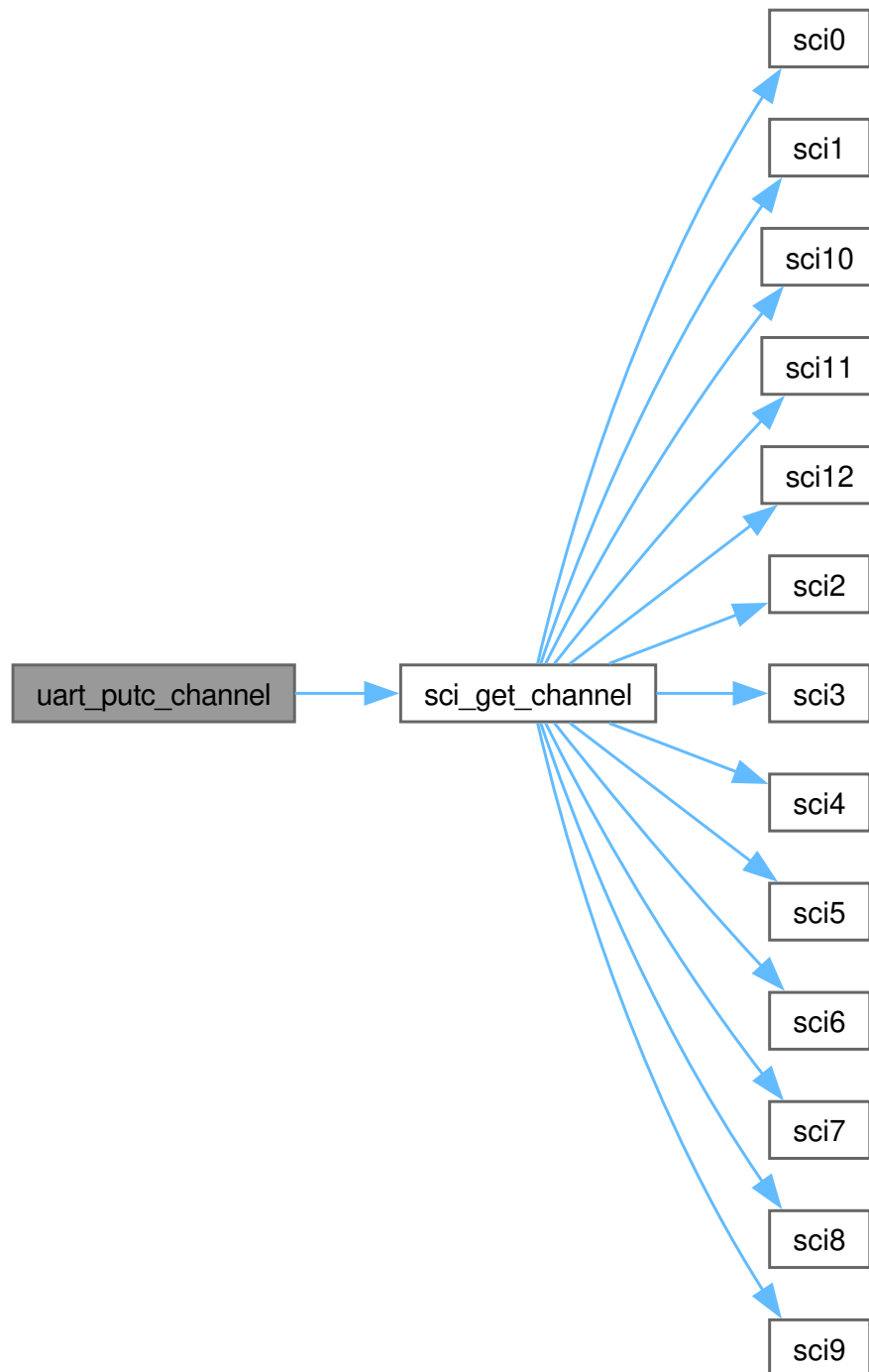
Definition at line 1308 of file [uart.c](#).

```
01309 {
01310 /* Validate channel */
01311 if ((uint8_t)channel >= k_uart_max_channels) {
01312 return k_rx_err_invalid_arg;
01313 }
01314
01315 /* Check initialization */
01316 if (!s_channel_initialized[channel]) {
01317 return k_rx_err_invalid_state;
01318 }
01319
01320 /* Get SCI register base */
01321 volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01322 if (sci == nullptr) {
01323 return k_rx_err_invalid_arg;
01324 }
01325
01326 /* Wait for transmit buffer to be empty (TDRE flag) with timeout */
01327 uint32_t timeout = k_uart_tx_timeout;
01328 while ((sci->ssr & k_sci_ssr_tdre_flag) == k_uart_flag_clear &&
01329 timeout > k_uart_timeout_expired) {
01330 timeout -= k_uart_timeout_decrement;
01331 }
01332
01333 /* Check if timeout occurred */
01334 if (timeout == k_uart_timeout_expired) {
01335 return k_rx_err_timeout;
01336 }
01337
01338 /* Write data to transmit register */
01339 sci->tdr = (uint8_t)data;
01340
01341 /* Clear TDRE flag by reading SSR then writing 0 */
01342 const uint8_t ssr = sci->ssr;
01343 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_tdre_flag);
01344
01345 return k_rx_ok;
01346 }
```

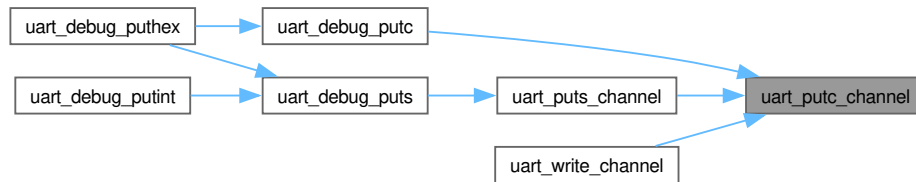
References [k\\_sci\\_ssr\\_tdre\\_flag](#), [k\\_uart\\_flag\\_clear](#), [k\\_uart\\_max\\_channels](#), [k\\_uart\\_timeout\\_decrement](#), [k\\_uart\\_timeout\\_expired](#), [k\\_uart\\_tx\\_timeout](#), [s\\_channel\\_initialized](#), [sci\\_get\\_channel\(\)](#), [rx\\_sci\\_regs\\_t::ssr](#), and [rx\\_sci\\_regs\\_t::tdr](#).

Referenced by [uart\\_debug\\_putc\(\)](#), [uart\\_puts\\_channel\(\)](#), and [uart\\_write\\_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.88 uart\_puts\_channel()

```

rx_err_t uart_puts_channel (
 const uart_channel_t channel,
 const char * str) [nodiscard]

```

Transmit a null-terminated string on specified channel.

Parameters

|    |         |                    |
|----|---------|--------------------|
| in | channel | SCI channel (0-12) |
| in | str     | Pointer to string  |

Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if str is nullptr, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized

Transmit a null-terminated string on specified channel.

Transmits a null-terminated string with automatic LF to \r+LF conversion for terminal compatibility. Enforces maximum string length (k\_uart\_max\_str\_len = 256) to comply with NASA Power of 10 Rule 2 (bounded loops).

Algorithm steps:

1. Validate string pointer (NULL check)
2. Validate channel number and initialization state
3. For each character up to k\_uart\_max\_str\_len: a. Check for null terminator (success exit) b. If ' ', send '\r' first (\r+LF conversion) c. Transmit character via [uart\\_putc\\_channel\(\)](#) d. Propagate any errors immediately
4. If limit reached without terminator, return k\_rx\_err\_invalid\_size

Newline conversion:

- Input ' ' (LF) -> Output '\r\n' (\r+LF)
- Required for Windows terminals (e.g., PuTTY, TeraTerm)
- Ensures cursor returns to column 0 before line feed

#### Precondition

Channel must be initialized via `uart_init_channel()`  
str must point to null-terminated string in valid memory

#### Postcondition

All characters transmitted including converted newlines  
On error, partial string may have been transmitted

#### Note

Blocking call - waits for each character to transmit  
String length limited to 256 characters for safety

#### Warning

Long strings may take significant time (256 chars ~ 22ms @ 115200)

#### Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

#### Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes
- Example: 100-char string ~ 9 ms

#### Example:

```
// Simple string
uart_puts_channel(k_uart_channel_9, "Hello, World!\n");
// Output: "Hello, World!\r\n"

// Multiple lines
uart_puts_channel(k_uart_channel_9, "Line 1\nLine 2\nLine 3\n");
// Output: "Line 1\r\nLine 2\r\nLine 3\r\n"

// Error handling
rx_err_t err = uart_puts_channel(k_uart_channel_9, msg);
if (err != k_rx_ok) {
 // Partial transmission may have occurred
}
```

#### See also

`uart_putc_channel()` Transmit single character  
`uart_write_channel()` Transmit binary data (no conversion)  
`uart_debug_puts()` Simplified debug wrapper

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Definition at line 1415 of file [uart.c](#).

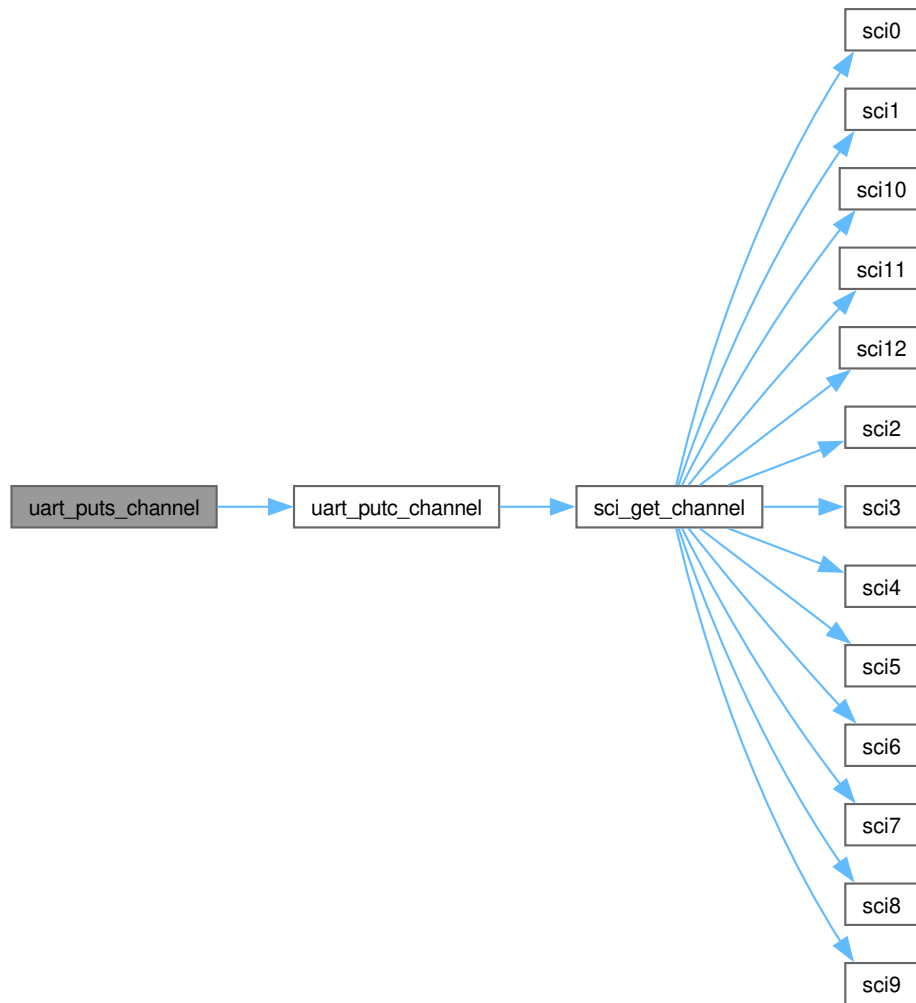
```

01416 {
01417 /* Validate parameters */
01418 if (str == nullptr) {
01419 return k_rx_err_null_ptr;
01420 }
01421
01422 if ((uint8_t)channel >= k_uart_max_channels) {
01423 return k_rx_err_invalid_arg;
01424 }
01425
01426 if (!s_channel_initialized[channel]) {
01427 return k_rx_err_invalid_state;
01428 }
01429
01430 /* Transmit string with \n to \r\n conversion (statically bounded) */
01431 for (uint32_t i = 0; i < k_uart_max_str_len; ++i) {
01432 if (str[i] == '\0') {
01433 return k_rx_ok; /* Terminator found, success */
01434 }
01435 if (str[i] == '\n') {
01436 const rx_err_t err = uart_putc_channel(channel, '\r');
01437 if (err != k_rx_ok) {
01438 return err;
01439 }
01440 }
01441 const rx_err_t err = uart_putc_channel(channel, str[i]);
01442 if (err != k_rx_ok) {
01443 return err;
01444 }
01445 }
01446
01447 /* Reached limit without finding terminator */
01448 return k_rx_err_invalid_size;
01449 }
```

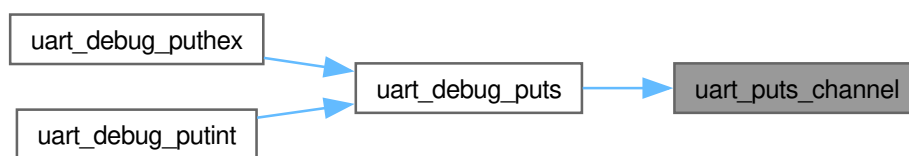
References [k\\_uart\\_max\\_channels](#), [k\\_uart\\_max\\_str\\_len](#), [s\\_channel\\_initialized](#), and [uart\\_putc\\_channel\(\)](#).

Referenced by [uart\\_debug\\_puts\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



#### 8.1.5.89 uart\_read\_channel()

```

rx_err_t uart_read_channel (
 const uart_channel_t channel,
 uint8_t * data,
 uint16_t length,
 uint16_t * bytes_read) [nodiscard]

```

Read available data from specified channel (non-blocking).

Reads up to length bytes that are currently available.



## Parameters

|     |            |                                     |
|-----|------------|-------------------------------------|
| in  | channel    | SCI channel (0-12)                  |
| out | data       | Pointer to buffer for received data |
| in  | length     | Maximum number of bytes to read     |
| out | bytes_read | Pointer to store actual bytes read  |

## Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if data or bytes\_read is nullptr, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized

Read available data from specified channel (non-blocking).

Reads up to length bytes from the specified SCI channel using [uart\\_getc\\_channel\(\)](#). Stops immediately when no more data is available (RDRE flag not set) rather than waiting. Actual bytes received is reported via bytes\_read.

Algorithm steps:

1. Validate data and bytes\_read pointers (NULL check)
2. Validate channel number and initialization state
3. Set \*bytes\_read = 0
4. For each slot in [0, length): a. Call [uart\\_getc\\_channel\(\)](#); if k\_rx\_err\_empty, break (done) b. On other error, propagate immediately c. Store byte and increment \*bytes\_read
5. Return k\_rx\_ok (even if zero bytes were read)

## Precondition

Channel must be initialized via [uart\\_init\\_channel\(\)](#)  
data must point to at least length bytes of writable memory

## Postcondition

\*bytes\_read set to actual number of bytes received  
data[0..\*bytes\_read-1] contain the received bytes

## Note

Non-blocking - returns immediately with whatever data is available  
k\_rx\_ok with \*bytes\_read == 0 means no data was available

## Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time:  $\sim 5$  us per byte received +  $\sim 5$  us when no data
- Stack usage: 24 bytes

Example:

```
uint8_t buf[64];
uint16_t count = 0;
rx_err_t err = uart_read_channel(k_uart_channel_9, buf, sizeof(buf), &count);
if (err == k_rx_ok && count > 0) {
 process_bytes(buf, count);
}
```

See also

[uart\\_getc\\_channel\(\)](#) Read single character  
[uart\\_rx\\_available\(\)](#) Check if data is available  
[uart\\_write\\_channel\(\)](#) Write binary data

Since

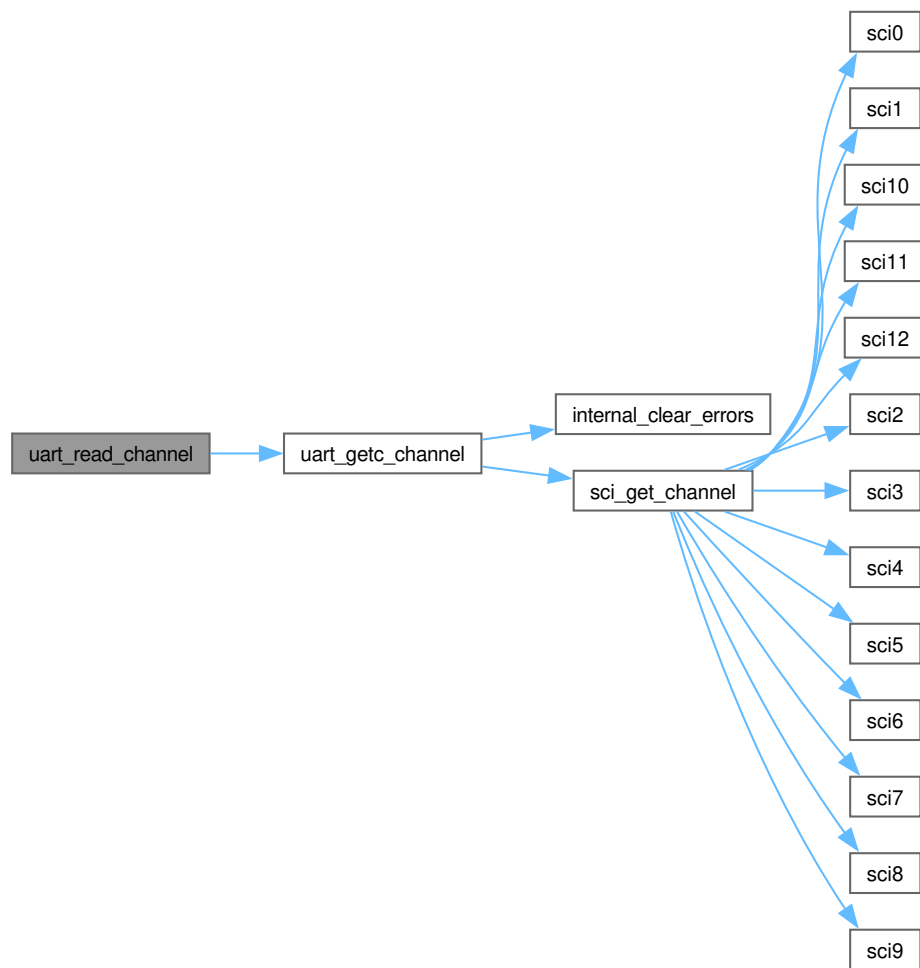
Version 1.0.0

Definition at line 1717 of file [uart.c](#).

```
01721 {
01722 /* Validate parameters */
01723 if (data == nullptr || bytes_read == nullptr) {
01724 return k_rx_err_null_ptr;
01725 }
01726
01727 if ((uint8_t)channel >= k_uart_max_channels) {
01728 return k_rx_err_invalid_arg;
01729 }
01730
01731 if (!s_channel_initialized[channel]) {
01732 return k_rx_err_invalid_state;
01733 }
01734
01735 /* Read available bytes */
01736 *bytes_read = 0;
01737 for (uint16_t i = 0; i < length; i++) {
01738 char c;
01739 const rx_err_t err = uart_getc_channel(channel, &c);
01740 if (err == k_rx_err_empty) {
01741 /* No more data available */
01742 break;
01743 }
01744 if (err != k_rx_ok) {
01745 return err;
01746 }
01747 data[i] = (uint8_t)c;
01748 (*bytes_read)++;
01749 }
01750
01751 return k_rx_ok;
01752 }
```

References [k\\_uart\\_max\\_channels](#), [s\\_channel\\_initialized](#), and [uart\\_getc\\_channel\(\)](#).

Here is the call graph for this function:



#### 8.1.5.90 uart`rx`available()

```

rx_err_t uart_rx_available (
 const uart_channel_t channel,
 bool * available) [nodiscard]

```

Check if receive data is available on channel.

##### Parameters

|     |           |                                      |
|-----|-----------|--------------------------------------|
| in  | channel   | SCI channel (0-12)                   |
| out | available | Pointer to store availability status |

##### Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if available is nullptr, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized

Check if receive data is available on channel.

Checks the RDRF (Receive Data Register Full) flag in the SSR register of the specified SCI channel. Provides a non-destructive peek at receive availability without consuming any data from the buffer.

Algorithm steps:

1. Validate available pointer (NULL check)
2. Validate channel number and initialization state
3. Get SCI register base address
4. Read SSR and test RDRF bit; store boolean result in \*available

#### Precondition

Channel must be initialized via [uart\\_init\\_channel\(\)](#)  
available must point to valid bool storage

#### Postcondition

\*available reflects current RDRF state  
No data is consumed from the receive buffer

#### Note

Non-destructive - does not read RDR or clear any flags  
RDRF can be set again immediately after reading if UART is receiving

#### Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

#### Performance:

- Execution time: ~2 us
- Stack usage: 16 bytes

#### Example:

```
bool ready = false;
if (uart_rx_available(k_uart_channel_9, &ready) == k_rx_ok && ready) {
 char c;
 uart_getc_channel(k_uart_channel_9, &c);
}
```

#### See also

[uart\\_getc\\_channel\(\)](#) Read single character  
[uart\\_read\\_channel\(\)](#) Read multiple bytes

Since

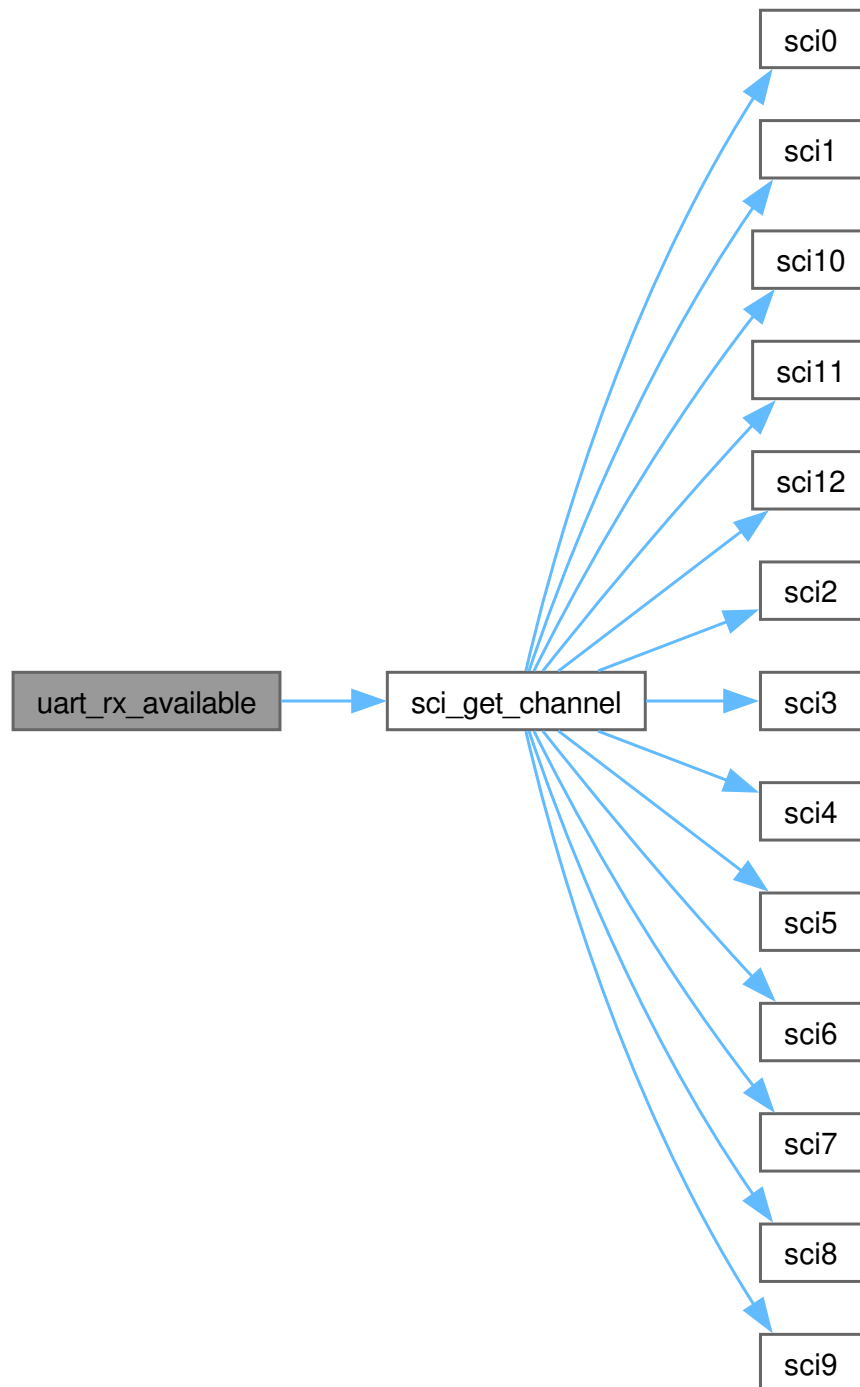
Version 1.0.0

Definition at line 1801 of file [uart.c](#).

```
01802 {
01803 /* Validate parameters */
01804 if (available == nullptr) {
01805 return k_rx_err_null_ptr;
01806 }
01807
01808 if ((uint8_t)channel >= k_uart_max_channels) {
01809 return k_rx_err_invalid_arg;
01810 }
01811
01812 if (!s_channel_initialized[channel]) {
01813 return k_rx_err_invalid_state;
01814 }
01815
01816 /* SCI9 uses the RXI9 ring buffer; non-empty = bytes available. */
01817 if (channel == k_uart_channel_9) {
01818 *available = (s_sci9_rx_head != s_sci9_rx_tail);
01819 return k_rx_ok;
01820 }
01821
01822 /* Get SCI register base */
01823 const volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01824 if (sci == nullptr) {
01825 return k_rx_err_invalid_arg;
01826 }
01827
01828 /* Check RDRF flag */
01829 *available = ((sci->ssr & k_sci_ssr_rdrf_flag) != k_uart_flag_clear);
01830
01831 return k_rx_ok;
01832 }
```

References [k\\_sci\\_ssr\\_rdrf\\_flag](#), [k\\_uart\\_channel\\_9](#), [k\\_uart\\_flag\\_clear](#), [k\\_uart\\_max\\_channels](#), [s\\_channel\\_initialized](#), [s\\_sci9\\_rx\\_head](#), [s\\_sci9\\_rx\\_tail](#), [sci\\_get\\_channel\(\)](#), and [rx\\_sci\\_regs\\_t::ssr](#).

Here is the call graph for this function:



#### 8.1.5.91 `uart_write_channel()`

```
rx_err_t uart_write_channel (
 const uart_channel_t channel,
 const uint8_t * data,
 uint16_t length) [nodiscard]
```

Write buffer to specified channel.

## Parameters

|    |         |                          |
|----|---------|--------------------------|
| in | channel | SCI channel (0-12)       |
| in | data    | Pointer to data buffer   |
| in | length  | Number of bytes to write |

## Returns

k\_rx\_ok on success, k\_rx\_err\_null\_ptr if data is nullptr, k\_rx\_err\_invalid\_arg if channel is invalid, k\_rx\_err\_invalid\_state if channel not initialized

Write buffer to specified channel.

Transmits a block of raw bytes on the specified SCI channel using [uart\\_putc\\_channel\(\)](#) for each byte. No LF-to-\r+LF conversion is performed, making this function suitable for binary protocol data.

Algorithm steps:

1. Validate data pointer (NULL check)
2. Validate channel number and initialization state
3. For each byte in [0, length): call [uart\\_putc\\_channel\(\)](#) and propagate errors

## Precondition

Channel must be initialized via [uart\\_init\\_channel\(\)](#)  
data must point to at least length bytes of valid memory

## Postcondition

All length bytes transmitted in order  
On error, partial data may have been transmitted

## Note

Blocking call - waits for each byte to be accepted by TDR  
No newline conversion - use [uart\\_puts\\_channel\(\)](#) for text output

## Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

## Performance:

- Execution time: ~90 us per byte @ 115200 baud
- Stack usage: 24 bytes

Example:

```
const uint8_t frame[] = {0xAA, 0x55, 0x01, 0x02, 0x03};
rx_err_t err = uart_write_channel(k_uart_channel_9, frame, sizeof(frame));
if (err != k_rx_ok) {
 // Handle partial write
}
```

See also

[uart\\_puts\\_channel\(\)](#) Transmit text string with newline conversion  
[uart\\_read\\_channel\(\)](#) Read binary data

Since

Version 1.0.0

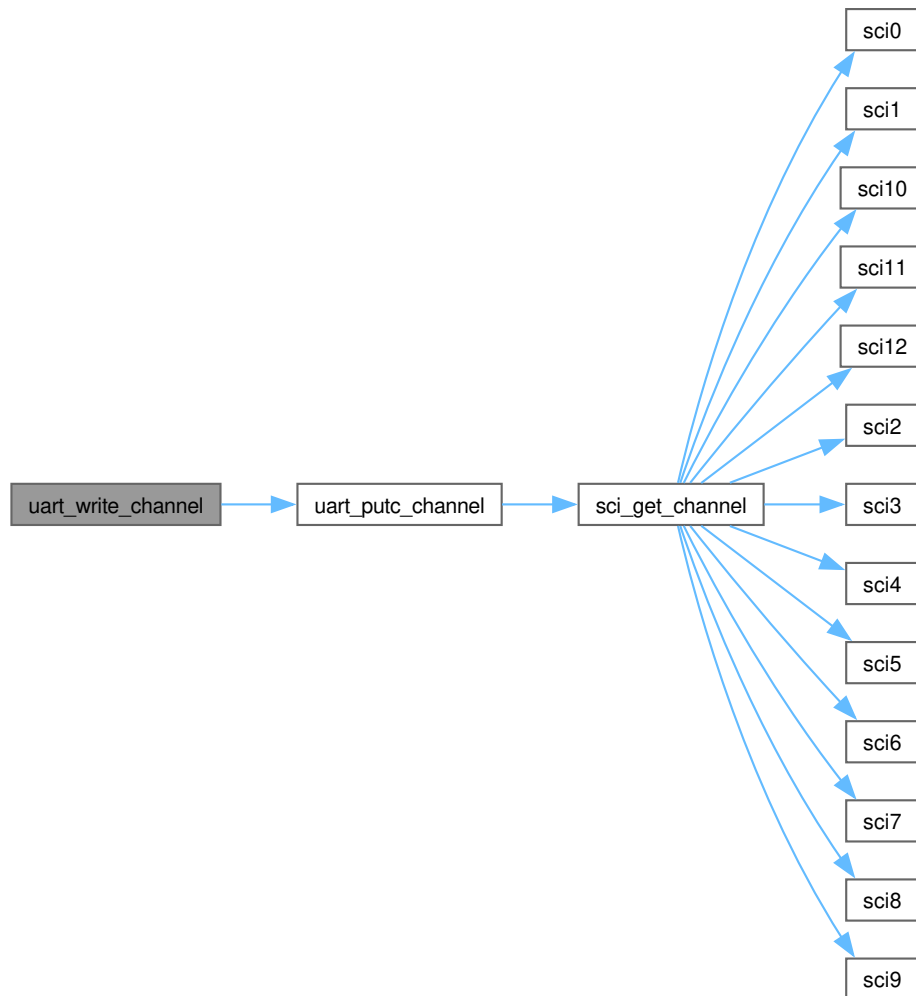
Definition at line 1498 of file [uart.c](#).

```
01499 {
01500 /* Validate parameters */
01501 if (data == nullptr) {
01502 return k_rx_err_null_ptr;
01503 }
01504
01505 if ((uint8_t)channel >= k_uart_max_channels) {
01506 return k_rx_err_invalid_arg;
01507 }
01508
01509 if (!s_channel_initialized[channel]) {
01510 return k_rx_err_invalid_state;
01511 }
01512
01513 /* Write each byte */
01514 for (uint16_t i = 0; i < length; i++) {
01515 const rx_err_t err = uart_putc_channel(channel, (char)data[i]);
01516 if (err != k_rx_ok) {
01517 return err;
01518 }
01519 }
01520
01521 return k_rx_ok;
01522 }
```

References [k\\_uart\\_max\\_channels](#), [s\\_channel\\_initialized](#), and [uart\\_putc\\_channel\(\)](#).

Here is the call graph for this function:





## 8.2 hardware.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file hardware.h
00003 * @brief Hardware Abstraction Layer (HAL) - Unified Interface for RX72N Peripherals
00004 *
00005 * @details
00006 * ## Overview
00007 * Comprehensive HAL providing type-safe, unified API for all RX72N microcontroller peripherals
00008 * used in the STAR motor controller. Abstracts hardware register manipulation and provides
00009 * safe, validated access to GPIO, timers, ADC, I2C, SPI, and UART peripherals.
00010 *
00011 * ## Supported Peripherals
00012 *
00013 * | Peripheral | Channels | Purpose | Configuration |
00014 * |-----|-----|-----|-----|
00015 * | **GPIO** | 14 ports (A-N) | Digital I/O, motor control signals | Output/Input, Read/Write/Toggle |
00016 * | **CMT** | 4 channels | ThreadX system tick (100 Hz) | CMT0 configured for 10ms intervals |
00017 * | **S12AD** | 2 units (0-1) | Motor current sensing, analog inputs | 8/10/12-bit resolution |
00018 * | **RIIC** | 3 channels (0-2) | I2C sensors (IMU); temperature sensor uses 1-Wire | 100kHz, 400kHz, 1MHz |
00019 * | **RSPI** | 3 channels (0-2) | SPI: RPi5 comm (peripheral), sensors (controller) | Mode 0-3, 100kHz-10MHz |
00020 * | **SCI** | 13 channels (0-12) | UART: Debug (SCI9 @ 115200), expansion | Configurable baud rates |
00021 *
00022 * ## Hardware Architecture
00023 *
00024 * @dot
00025 * digraph hal_architecture {
00026 * rankdir=TB;
00027 * node [shape=box, style=rounded];
00028 *
00029 * app [label="Application Code\n(Motor Control, USB, Protocol)", shape=component];
00030 * hal [label="hardware.h\n(HAL API)", style=filled, fillcolor=lightblue];

```

```

00031 *
00032 * subgraph cluster_peripherals {
00033 * label="RX72N Peripherals";
00034 * gpio [label="GPIO\n(Ports A-N)"];
00035 * cmt [label="CMT0\n(Timer)"];
00036 * adc [label="S12AD0/1\n(ADC)"];
00037 * riic [label="RIIC0-2\n(I2C)"];
00038 * rspi [label="RSPi0-2\n(SPI)"];
00039 * sci [label="SCI0-12\n(UART)"];
00040 * }
00041 *
00042 * subgraph cluster_external {
00043 * label="External Hardware";
00044 * rpi [label="Raspberry Pi 5\n(SPI Peripheral)"];
00045 * drv [label="DRV8263H x4\n(Motor Drivers)"];
00046 * imu [label="IMU\n(I2C Sensor)"];
00047 * uart_bridge [label="CY7C65213\n(USB-UART Bridge)"];
00048 * }
00049 *
00050 * app -> hal;
00051 * hal -> gpio;
00052 * hal -> cmt;
00053 * hal -> adc;
00054 * hal -> riic;
00055 * hal -> rspi;
00056 * hal -> sci;
00057 *
00058 * rspi -> rpi [label="RSPi0\nPeripheral"];
00059 * gpio -> drv [label="PWM+GPIO"];
00060 * riic -> imu [label="RIIC0\n400kHz"];
00061 * sci -> uart_bridge [label="SCI9\n115200"];
00062 * }
00063 * @enddot
00064 *
00065 * ## Key Design Principles
00066 *
00067 * ### 1. Type Safety (C23 Typed Enums + Struct Wrappers)
00068 * - **Typed enums**: All constants use C23 `uint8_t` syntax for guaranteed size
00069 * - **Struct wrappers**: I2C channel/address use struct wrappers to prevent argument swaps
00070 * - **Compile-time validation**: Invalid types caught at compile time, not runtime
00071 *
00072 * **Example - Type Safety Prevents Bugs**:
00073 * @code{.c}
00074 * // WRONG - Swapped arguments with plain uint8_t
00075 * riic_write(0x48, 0, data, len); // Accidentally swapped channel and address
00076 *
00077 * // CORRECT - Struct wrappers catch this at compile time
00078 * riic_channel_t ch = {.value = 0};
00079 * i2c_device_addr_t addr = {.value = 0x48};
00080 * riic_write(ch, addr, data, len); // Type-checked, no accidental swaps possible
00081 * @endcode
00082 *
00083 * ### 2. Unified Error Handling
00084 * - **Consistent returns**: All functions return `rx_err_t` (except debug helpers)
00085 * - **Comprehensive error codes**: Covers NULL pointers, invalid arguments, timeouts, conflicts
00086 * - **Validation**: All functions validate inputs before hardware access
00087 *
00088 * ### 3. Configuration Structs Over Positional Parameters
00089 * - **Maintainability**: Adding new config options doesn't break API
00090 * - **Clarity**: Named fields > positional parameters
00091 * - **Safety**: Reduces parameter order mistakes
00092 *
00093 * **Example**:
00094 * @code{.c}
00095 * // OLD STYLE (error-prone):
00096 * uart_init(9, 115200, k_port_b_pin_7, k_port_b_pin_6); // Which pin is TX vs RX?
00097 *
00098 * // NEW STYLE (clear and safe):
00099 * uart_channel_config_t config = {
00100 * .channel = k_uart_channel_9,
00101 * .baudrate = 115200,
00102 * .tx_gpio = k_port_b_pin_7,
00103 * .rx_gpio = k_port_b_pin_6
00104 * };
00105 * uart_init_channel(&config); // Self-documenting
00106 * @endcode
00107 *
00108 * ### 4. Zero Dynamic Allocation (NASA Rule 3)
00109 * - All functions use stack-based parameters
00110 * - No malloc/free anywhere in HAL
00111 * - Predictable memory behavior for safety-critical systems
00112 *
00113 * ## Performance Characteristics
00114 *
00115 * ### Function Call Overhead (RX72N @ 240 MHz)
00116 * | Function Category | Cycles | Time | Notes |
00117 * |-----|-----|-----|-----|

```

```

00118 * | GPIO write | ~20 | 83 ns | Direct register access |
00119 * | GPIO read | ~25 | 104 ns | Includes pin validation |
00120 * | UART putc | ~150 | 625 ns | Polling TDRE flag |
00121 * | UART puts (10 chars) | ~1500 | 6.25 us | 150 cycles/char |
00122 * | I2C write (1 byte) | ~24000 | 100 us | 400kHz I2C clock |
00123 * | SPI transfer (1 byte) | ~480 | 2 us | 10 MHz SPI clock |
00124 * | ADC read | ~7200 | 30 us | 12-bit conversion + validation |
00125
00126 * ### Memory Footprint
00127 * | Component | Flash | RAM | Stack (max) |
00128 * |-----|-----|-----|-----|
00129 * | GPIO drivers | ~2 KB | 16 bytes (state) | 32 bytes |
00130 * | UART drivers | ~3 KB | 128 bytes (buffers) | 48 bytes |
00131 * | I2C drivers | ~4 KB | 64 bytes (state) | 64 bytes |
00132 * | SPI drivers | ~5 KB | 96 bytes (state) | 64 bytes |
00133 * | ADC drivers | ~2 KB | 32 bytes (state) | 32 bytes |
00134 * | **Total HAL** | ~16 KB | ~336 bytes | ~240 bytes |
00135
00136 * ## Thread Safety
00137 * - **NOT thread-safe**: Hardware registers are not protected by mutexes
00138 * - **Single-threaded access**: Each peripheral should be accessed from one thread only
00139 * - **RTOS integration**: Use ThreadX mutexes if multi-threaded access needed
00140 * - **Interrupt safety**: All HAL functions disable interrupts during critical sections
00141
00142 * ## Initialization Sequence
00143
00144 * Recommended initialization order for STAR motor controller:
00145 * @code{.c}
00146 void hardware_init(void) {
00147 // 1. Debug UART (for early logging)
00148 uart_debug_init();
00149 rx_log_info("INIT", "Starting hardware initialization");
00150
00151 // 2. System timer (for ThreadX tick)
00152 timer_init();
00153
00154 // 3. GPIO (for motor control signals)
00155 gpio_set_output(k_port_c_pin_0); // Motor enable
00156 gpio_set_output(k_port_c_pin_1); // Direction
00157
00158 // 4. SPI controller (for sensors; DRV8263H uses PWM+GPIO, not SPI)
00159 rspi_controller_config_t spi_cfg = {
00160 .spi_mode = k_rspi_mode_0,
00161 .freq_hz = k_rspi_freq_1mhz,
00162 .cs = k_port_c_pin_4
00163 };
00164 rspi_init_controller(k_rspi_channel_1, &spi_cfg);
00165
00166 // 5. SPI peripheral (for RPi5 communication)
00167 rspi_config_t rpi_cfg = {
00168 .spi_mode = k_rspi_mode_0,
00169 .use_16bit = false
00170 };
00171 rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);
00172
00173 // 6. ADC (for motor current sensing on AN007/P47)
00174 adc_init(k_adc_unit_0, k_adc_channel_7, k_adc_resolution_12bit);
00175
00176 // 7. I2C (for IMU sensor)
00177 riic_init((riic_channel_t){.value = 0}, 400000);
00178
00179 rx_log_info("INIT", "Hardware initialization complete");
00180 }
00181 * @endcode
00182
00183 * ## Pin Assignments (From Project Schematic)
00184
00185 * See `docs/sections/03_hardware_pinout.tex` for complete pinout documentation.
00186
00187 * **Critical Signals**:
00188 * | Signal | Pin | Peripheral | Purpose |
00189 * |-----|-----|-----|-----|
00190 * | SPI0_CIPO | P30 | RSPI0 | RPi5 <- RX72N (peripheral mode) |
00191 * | SPI0_COPI | P31 | RSPI0 | RPi5 -> RX72N |
00192 * | SPI0_CLK | P32 | RSPI0 | RPi5 clock |
00193 * | SPI0_CS | P33 | GPIO | RPi5 chip select |
00194 * | SPI1_CIPO | PC5 | RSPI1 | Not used (RSPI1 controller mode) |
00195 * | SPI1_COPI | PC6 | RSPI1 | Not used (RSPI1 controller mode) |
00196 * | SPI1_CLK | PC7 | RSPI1 | Not used (RSPI1 clock) |
00197 * | UART_TX | PB7 | SCI9 | Debug output (CY7C65213) |
00198 * | UART_RX | PB6 | SCI9 | Debug input |
00199
00200 * ## Dependencies
00201 * - `rx_err.h` - Error code definitions
00202 * - `rx_check.h` - Validation macros
00203 * - `rx_log.h` - Logging infrastructure
00204 * - `rx_port_constants.h` - Port/pin definitions

```

```

00205 * - `rx_bus_types.h` - Bus enumeration types
00206 * - `rx72n_regs.h` - Hardware register definitions (internal)
00207 *
00208 * @par NASA Power of 10 Compliance:
00209 * - **Rule 1 [YES]** No goto, setjmp, recursion (pure validation + register access)
00210 * - **Rule 2 [YES]** All loops bounded (timeout loops use compile-time max iterations)
00211 * - **Rule 3 [YES]** No dynamic allocation (all stack-based or static configuration)
00212 * - **Rule 4 [YES]** All functions <60 lines (HAL functions are typically 20-40 lines)
00213 * - **Rule 5 [YES]** Minimum 2 validations per function (NULL checks + range validation)
00214 * - **Rule 6 [YES]** Data at smallest scope (loop counters, temporary variables)
00215 * - **Rule 7 [YES]** All returns validated (rx_err_t checked at every call site)
00216 * - **Rule 8 [YES]** C23 typed enums for constants, minimal preprocessor use
00217 * - **Rule 9 [YES]** Single-level dereferencing (pointer parameters for output values)
00218 * - **Rule 10 [YES]** Compiles with -Wall -Wextra -Werror
00219 *
00220 * @par SOLID Principles:
00221 * - **S (Single Responsibility)** Each peripheral has dedicated functions (GPIO, UART, SPI, I2C, ADC)
00222 * - **O (Open/Closed)** Config structs allow extension without modifying function signatures
00223 * - **L (Liskov Substitution)** All HAL functions return rx_err_t with consistent semantics
00224 * - **I (Interface Segregation)** Separate controller/peripheral SPI APIs, debug UART wrappers
00225 * - **D (Dependency Inversion)** Depends on abstract rx_err_t, rx_log, not concrete implementations
00226 *
00227 * @par Module Dependencies:
00228 * ```
00229 * hardware.h
00230 * +--> rx_err.h (error codes)
00231 * +--> rx_check.h (validation macros)
00232 * +--> rx_log.h (logging)
00233 * +--> rx_port_constants.h (port/pin enums)
00234 * +--> rx_bus_types.h (ADC resolution enums)
00235 * +--> [internal] rx72n_regs.h (register definitions)
00236 *
00237 * Used by:
00238 * +--> Motor control (rx_motor_control.c)
00239 * +--> USB communication (rx_usb.c)
00240 * +--> Protocol handling (rx_comm_manager.c)
00241 * +--> Application tasks (comm, motor, sensors)
00242 * ```
00243 *
00244 * @author Locked, Inc.
00245 * @date 2026-01-27
00246 * @version 1.0.0
00247 * @copyright Copyright (c) 2026 Locked Inc.
00248 * SPDX-License-Identifier: MIT
00249 *
00250 * @see rx_err.h for error code definitions
00251 * @see rx_port_constants.h for complete port/pin enumerations
00252 * @see docs/sections/03_hardware_pinout.tex for hardware pinout documentation
00253 */
00254
00255 #pragma once
00256
00257 #include <stdbool.h>
00258 #include <stdint.h>
00259
00260 /* Core infrastructure */
00261 #include "rx_check.h"
00262 #include "rx_err.h"
00263 #include "rx_infrastructure.h"
00264 #include "rx_log.h"
00265
00266 /* Port/pin constants and types */
00267 #include "rx_port_constants.h"
00268
00269 /* Bus types (includes ADC enums) */
00270 #include "rx_bus_types.h"
00271
00272 #ifdef __cplusplus
00273 extern "C" {
00274 #endif
00275
00276 /*
=====
00277 * GPIO Functions
00278 *
=====
00279 */
00280 /**
00281 * @brief Configure GPIO pin as digital output
00282 *
00283 * @details
00284 * Configures the specified GPIO pin as a digital output with push-pull drive.
00285 * Performs full hardware initialization including port direction register (PDR),
00286 * port mode register (PMR), and open-drain control (ODR). Validates pin assignment
00287 * and checks for peripheral conflicts.
00288 *
00289 */

```

```

00290 * ## Algorithm
00291 * 1. Validate pin parameter (port exists, pin number valid)
00292 * 2. Check pin is not already reserved by another peripheral
00293 * 3. Unlock port write protection (PWPR register)
00294 * 4. Set PMR bit to 0 (GPIO mode, not peripheral function)
00295 * 5. Set PDR bit to 1 (output direction)
00296 * 6. Ensure ODR is 0 (push-pull, not open-drain)
00297 * 7. Lock port write protection
00298 * 8. Mark pin as reserved in allocation tracker
00299 *
00300 * ## Performance
00301 * - **Cycles**: ~120 @ 240 MHz
00302 * - **Time**: ~500 ns
00303 * - **Stack**: 8 bytes (pin validation, temporary registers)
00304 *
00305 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00306 * - Type: rx_port_pin_t (C23 typed enum)
00307 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00308 * - Example: k_port_c_pin_0 (Port C, Pin 0)
00309 * - Range: All valid combinations defined in rx_port_constants.h
00310 *
00311 * @return rx_err_t Error code indicating success or failure
00312 * @retval k_rx_ok Pin successfully configured as output
00313 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00314 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00315 * @retval k_rx_err_gpio_conflict Pin already reserved by UART, SPI, I2C, or other peripheral
00316 *
00317 * @pre Pin must not be currently in use by another peripheral
00318 * @post Pin configured as push-pull output, initial state undefined (use gpio_write_* to set level)
00319 * @post Pin marked as reserved in internal allocation tracker
00320 *
00321 * @note **Thread Safety**: NOT thread-safe. Do not call concurrently from multiple threads.
00322 * @note **Initial State**: Output level is undefined after configuration. Call gpio_write_high() or gpio_write_low() to
00323 * set initial state.
00324 * @note **Peripheral Conflicts**: Automatically detects conflicts with UART, SPI, I2C pin assignments.
00325 *
00326 * @warning Changing pin direction while peripheral is active may cause undefined behavior.
00327 *
00328 * @par Example: Motor Enable Pin
00329 * @code{.c}
00330 * #include "hardware.h"
00331 *
00332 * void motor_init(void) {
00333 * // Configure motor enable pin as output
00334 * rx_err_t err = gpio_set_output(k_port_c_pin_0);
00335 * if (err != k_rx_ok) {
00336 * rx_log_error_val("MOTOR", "Failed to configure enable pin", err);
00337 * return;
00338 * }
00339 *
00340 * // Set initial state (motors disabled)
00341 * gpio_write_low(k_port_c_pin_0);
00342 * rx_log_info("MOTOR", "Enable pin configured");
00343 * }
00344 * @endcode
00345 *
00346 * @par Example: Multiple GPIO Outputs
00347 * @code{.c}
00348 * void configure_leds(void) {
00349 * const rx_port_pin_t led_pins[] = {
00350 * k_port_e_pin_5, // LED1
00351 * k_port_e_pin_6, // LED2
00352 * k_port_e_pin_7 // LED3
00353 * };
00354 *
00355 * for (uint8_t i = 0; i < 3; i++) {
00356 * rx_err_t err = gpio_set_output(led_pins[i]);
00357 * if (err != k_rx_ok) {
00358 * rx_log_error_hex("GPIO", "LED init failed", i, 2);
00359 * } else {
00360 * gpio_write_low(led_pins[i]); // LEDs off initially
00361 * }
00362 * }
00363 * }
00364 * @endcode
00365 *
00366 * @par Example: Error Handling
00367 * @code{.c}
00368 * rx_err_t setup_motor_control_pins(void) {
00369 * // Try to configure direction pin
00370 * rx_err_t err = gpio_set_output(k_port_c_pin_1);
00371 * if (err == k_rx_err_gpio_conflict) {
00372 * rx_log_error("MOTOR", "Pin conflict - already used by SPI?");
00373 * return err;
00374 * } else if (err != k_rx_ok) {
00375 * rx_log_error_val("MOTOR", "Unexpected error", err);
00376 * return err;
00377 * }
00378 * }

```

```

00376 * }
00377 *
00378 * return k_rx_ok;
00379 * }
00380 * @endcode
00381 *
00382 * @see gpio_set_input() Configure pin as digital input
00383 * @see gpio_write_high() Set output pin to logic high
00384 * @see gpio_write_low() Set output pin to logic low
00385 * @see gpio_toggle() Toggle output pin state
00386 * @see rx_port_constants.h for complete pin definitions
00387 */
00388 [[nodiscard]] rx_err_t gpio_set_output(rx_port_pin_t pin);
00389
00390 /**
00391 * @brief Configure GPIO pin as digital input
00392 *
00393 * @details
00394 * Configures the specified GPIO pin as a digital input with high-impedance state.
00395 * Performs full hardware initialization including port direction register (PDR),
00396 * port mode register (PMR), and input buffer control. Validates pin assignment
00397 * and checks for peripheral conflicts.
00398 *
00399 * ## Algorithm
00400 * 1. Validate pin parameter (port exists, pin number valid)
00401 * 2. Check pin is not already reserved by another peripheral
00402 * 3. Unlock port write protection (PWPR register)
00403 * 4. Set PMR bit to 0 (GPIO mode, not peripheral function)
00404 * 5. Set PDR bit to 0 (input direction)
00405 * 6. Enable input buffer (ICR register)
00406 * 7. Lock port write protection
00407 * 8. Mark pin as reserved in allocation tracker
00408 *
00409 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00410 * - Type: rx_port_pin_t (C23 typed enum)
00411 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00412 * - Example: k_port_e_pin_5 (Port E, Pin 5)
00413 * - Range: All valid combinations defined in rx_port_constants.h
00414 *
00415 * @return rx_err_t Error code indicating success or failure
00416 * @retval k_rx_ok Pin successfully configured as input
00417 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00418 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00419 * @retval k_rx_err_gpio_conflict Pin already reserved by UART, SPI, I2C, or other peripheral
00420 *
00421 * @pre Pin must not be currently in use by another peripheral
00422 * @pre Pin must correspond to a valid physical pin on the RX72N package
00423 * @post Pin configured as high-impedance input with input buffer enabled
00424 * @post Pin marked as reserved in internal allocation tracker
00425 *
00426 * @note **Thread Safety**: NOT thread-safe. Do not call concurrently from multiple threads.
00427 *
00428 * @par Example: Button Input
00429 * @code{.c}
00430 * #include "hardware.h"
00431 *
00432 * void button_init(void) {
00433 * rx_err_t err = gpio_set_input(k_port_e_pin_5);
00434 * if (err != k_rx_ok) {
00435 * rx_log_error_val("BTN", "Failed to configure button pin", err);
00436 * return;
00437 * }
00438 * rx_log_info("BTN", "Button input configured");
00439 * }
00440 * @endcode
00441 *
00442 * @see gpio_set_output() Configure pin as digital output
00443 * @see gpio_read() Read pin logic level
00444 *
00445 * @since Version 1.0.0
00446 */
00447 [[nodiscard]] rx_err_t gpio_set_input(rx_port_pin_t pin);
00448
00449 /**
00450 * @brief Set GPIO output pin to logic high (VDD)
00451 *
00452 * @details
00453 * Drives the specified GPIO pin to logic high by setting the corresponding bit
00454 * in the Port Output Data Register (PODR). The pin must have been previously
00455 * configured as output via gpio_set_output(). Writing to an input pin has no
00456 * effect on the physical pin state.
00457 *
00458 * ## Algorithm
00459 * 1. Validate pin parameter (port exists, pin number valid)
00460 * 2. Extract port letter and pin number from rx_port_pin_t
00461 * 3. Set corresponding bit in PODR register (atomic read-modify-write)
00462 *

```

```

00463 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00464 * - Type: rx_port_pin_t (C23 typed enum)
00465 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00466 * - Must be configured as output via gpio_set_output()
00467 *
00468 * @return rx_err_t Error code indicating success or failure
00469 * @retval k_rx_ok Pin successfully driven high
00470 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00471 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00472 *
00473 * @pre Pin must be configured as output via gpio_set_output()
00474 * @pre Pin port and number must be within valid hardware range
00475 * @post Pin output level is logic high (VDD, typically 3.3V)
00476 * @post PODR register bit for the pin is set to 1
00477 *
00478 * @note **Thread Safety**: NOT thread-safe. Concurrent writes to pins on the same port may
00479 * cause read-modify-write races on the PODR register.
00480 *
00481 * @par Example: Enable Motor
00482 * @code{.c}
00483 * // Enable motor driver (active high)
00484 * rx_err_t err = gpio_write_high(k_port_c_pin_0);
00485 * if (err != k_rx_ok) {
00486 * rx_log_error_val("MOTOR", "Failed to enable motor", err);
00487 * }
00488 * @endcode
00489 *
00490 * @see gpio_write_low() Set output pin to logic low
00491 * @see gpio_toggle() Toggle output pin state
00492 * @see gpio_set_output() Configure pin as output first
00493 *
00494 * @since Version 1.0.0
00495 */
00496 [[nodiscard]] rx_err_t gpio_write_high(rx_port_pin_t pin);
00497
00498 /**
00499 * @brief Set GPIO output pin to logic low (GND)
00500 *
00501 * @details
00502 * Drives the specified GPIO pin to logic low by clearing the corresponding bit
00503 * in the Port Output Data Register (PODR). The pin must have been previously
00504 * configured as output via gpio_set_output(). Writing to an input pin has no
00505 * effect on the physical pin state.
00506 *
00507 * ## Algorithm
00508 * 1. Validate pin parameter (port exists, pin number valid)
00509 * 2. Extract port letter and pin number from rx_port_pin_t
00510 * 3. Clear corresponding bit in PODR register (atomic read-modify-write)
00511 *
00512 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00513 * - Type: rx_port_pin_t (C23 typed enum)
00514 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00515 * - Must be configured as output via gpio_set_output()
00516 *
00517 * @return rx_err_t Error code indicating success or failure
00518 * @retval k_rx_ok Pin successfully driven low
00519 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00520 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00521 *
00522 * @pre Pin must be configured as output via gpio_set_output()
00523 * @pre Pin port and number must be within valid hardware range
00524 * @post Pin output level is logic low (GND, 0V)
00525 * @post PODR register bit for the pin is cleared to 0
00526 *
00527 * @note **Thread Safety**: NOT thread-safe. Concurrent writes to pins on the same port may
00528 * cause read-modify-write races on the PODR register.
00529 *
00530 * @par Example: Disable Motor
00531 * @code{.c}
00532 * // Disable motor driver (active high enable)
00533 * rx_err_t err = gpio_write_low(k_port_c_pin_0);
00534 * if (err != k_rx_ok) {
00535 * rx_log_error_val("MOTOR", "Failed to disable motor", err);
00536 * }
00537 * @endcode
00538 *
00539 * @see gpio_write_high() Set output pin to logic high
00540 * @see gpio_toggle() Toggle output pin state
00541 * @see gpio_set_output() Configure pin as output first
00542 *
00543 * @since Version 1.0.0
00544 */
00545 [[nodiscard]] rx_err_t gpio_write_low(rx_port_pin_t pin);
00546
00547 /**
00548 * @brief Toggle GPIO output pin state (high to low, or low to high)
00549 *

```



```

00550 * @details
00551 * Inverts the current output level of the specified GPIO pin by XOR-ing the
00552 * corresponding bit in the Port Output Data Register (PODR). If the pin is
00553 * currently high it becomes low, and vice versa. The pin must have been
00554 * previously configured as output via gpio_set_output().
00555 *
00556 * ## Algorithm
00557 * 1. Validate pin parameter (port exists, pin number valid)
00558 * 2. Extract port letter and pin number from rx_port_pin_t
00559 * 3. XOR corresponding bit in PODR register (read-modify-write)
00560 *
00561 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00562 * - Type: rx_port_pin_t (C23 typed enum)
00563 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00564 * - Must be configured as output via gpio_set_output()
00565 *
00566 * @return rx_err_t Error code indicating success or failure
00567 * @retval k_rx_ok Pin successfully toggled
00568 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00569 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00570 *
00571 * @pre Pin must be configured as output via gpio_set_output()
00572 * @pre Pin port and number must be within valid hardware range
00573 * @post Pin output level is inverted from its previous state
00574 * @post PODR register bit for the pin is toggled
00575 *
00576 * @note **Thread Safety**: NOT thread-safe. Concurrent toggles or writes to pins on the same
00577 * port may cause read-modify-write races on the PODR register.
00578 *
00579 * @par Example: LED Heartbeat
00580 * @code{.c}
00581 * // Toggle heartbeat LED every 500 ms
00582 * void heartbeat_task(void) {
00583 * gpio_set_output(k_port_e_pin_5);
00584 * while (1) {
00585 * gpio_toggle(k_port_e_pin_5);
00586 * tx_thread_sleep(50); // 500 ms at 100 Hz tick
00587 * }
00588 * }
00589 * @endcode
00590 *
00591 * @see gpio_write_high() Set output pin to logic high
00592 * @see gpio_write_low() Set output pin to logic low
00593 * @see gpio_set_output() Configure pin as output first
00594 *
00595 * @since Version 1.0.0
00596 */
00597 [[nodiscard]] rx_err_t gpio_toggle(rx_port_pin_t pin);
00598
00599 /**
00600 * @brief Read current logic level of GPIO pin
00601 *
00602 * @details
00603 * Reads the current logic level from the specified GPIO pin's Port Input Data Register (PIDR).
00604 * Works for both input and output pins. For output pins, reads back the actual pin state
00605 * (not the output register), which may differ if external circuitry pulls the pin.
00606 *
00607 * ## Algorithm
00608 * 1. Validate pin parameter (port exists, pin number valid)
00609 * 2. Validate value pointer (NULL check)
00610 * 3. Extract port letter and pin number from rx_port_pin_t
00611 * 4. Read PIDR register for the specified port
00612 * 5. Extract bit corresponding to pin number
00613 * 6. Store result (true if bit=1, false if bit=0) in value pointer
00614 *
00615 * ## Performance
00616 * - **Cycles**: ~25 @ 240 MHz
00617 * - **Time**: ~104 ns
00618 * - **Stack**: 8 bytes (pin extraction, register read)
00619 *
00620 * @param[in] pin GPIO pin identifier from rx_port_constants.h
00621 * - Type: rx_port_pin_t (C23 typed enum)
00622 * - Format: k_port_X_pin_Y (X = A-N, Y = 0-7)
00623 * - Can be input or output pin
00624 * @param[out] value Pointer to boolean for storing pin state
00625 * - Must point to valid bool variable
00626 * - Set to true if pin reads logic high (VDD)
00627 * - Set to false if pin reads logic low (GND)
00628 * - Unchanged if function returns error
00629 *
00630 * @return rx_err_t Error code indicating success or failure
00631 * @retval k_rx_ok Pin level successfully read and stored in *value
00632 * @retval k_rx_err_gpio_invalid_port Port letter out of range (not A-N)
00633 * @retval k_rx_err_gpio_invalid_pin Pin number out of range (not 0-7)
00634 * @retval k_rx_err_null_ptr value pointer is nullptr
00635 *
00636 * @pre value pointer must be valid (not nullptr)

```



```

00637 * @post *value contains current pin state if function succeeds
00638 * @post *value unchanged if function returns error
00639 *
00640 * @note **Thread Safety**: NOT thread-safe. Reading PIDR register is atomic but value pointer access is not protected.
00641 * @note **Output Pins**: For output pins, this reads the actual pin level (PIDR), not the output register (PODR). This
 detects external pull-up/pull-down or short circuits.
00642 * @note **Input Pins**: Ensure pin is configured as input via gpio_set_input() before reading for sensor/button inputs.
00643 *
00644 * @par Example: Reading Button State
00645 * @code{.c}
00646 * #include "hardware.h"
00647 *
00648 * void check_button(void) {
00649 * // Button connected to Port E Pin 5 (active low with pull-up)
00650 * bool button_state;
00651 * rx_err_t err = gpio_read(k_port_e_pin_5, &button_state);
00652 *
00653 * if (err != k_rx_ok) {
00654 * rx_log_error_val("BUTTON", "Failed to read button", err);
00655 * return;
00656 * }
00657 *
00658 * if (!button_state) { // Active low (button pressed)
00659 * rx_log_info("BUTTON", "Button pressed");
00660 * handle_button_press();
00661 * }
00662 * }
00663 * @endcode
00664 *
00665 * @par Example: Verifying Output State
00666 * @code{.c}
00667 * // Set motor enable high
00668 * gpio_write_high(k_port_c_pin_0);
00669 *
00670 * // Verify pin actually went high (detects short circuit)
00671 * bool actual_state;
00672 * gpio_read(k_port_c_pin_0, &actual_state);
00673 * if (!actual_state) {
00674 * rx_log_error("MOTOR", "Enable pin stuck low - possible short circuit");
00675 * emergency_shutdown();
00676 * }
00677 * @endcode
00678 *
00679 * @par Example: Polling Multiple Inputs
00680 * @code{.c}
00681 * typedef struct {
00682 * rx_port_pin_t pin;
00683 * const char* name;
00684 * } sensor_input_t;
00685 *
00686 * void poll_sensors(void) {
00687 * const sensor_input_t sensors[] = {
00688 * {k_port_a_pin_0, "LIMIT_SWITCH_1"},
00689 * {k_port_a_pin_1, "LIMIT_SWITCH_2"},
00690 * {k_port_a_pin_2, "ESTOP"}
00691 * };
00692 *
00693 * for (uint8_t i = 0; i < 3; i++) {
00694 * bool state;
00695 * if (gpio_read(sensors[i].pin, &state) == k_rx_ok) {
00696 * rx_log_debug_val(sensors[i].name, "State", state ? 1 : 0);
00697 * }
00698 * }
00699 * }
00700 * @endcode
00701 *
00702 * @see gpio_set_input() Configure pin as digital input
00703 * @see gpio_set_output() Configure pin as digital output
00704 * @see gpio_write_high() Set output pin to logic high
00705 * @see gpio_write_low() Set output pin to logic low
00706 */
00707 [[nodiscard]] rx_err_t gpio_read(rx_port_pin_t pin, bool* value);
00708
00709 /*
=====
00710 * Timer Functions
00711 *
=====
00712 */
00713
00714 /**
00715 * @brief Initialize CMT0 for ThreadX system tick (100 Hz)
00716 *
00717 * @details
00718 * Configures Compare Match Timer channel 0 (CMT0) to generate periodic interrupts
00719 * at 100 Hz (10 ms interval) for the ThreadX RTOS system tick. Performs full
00720 * hardware initialization including module stop release, clock source selection

```

```

00721 * (PCLKB/8), compare match constant calculation, and interrupt priority setup.
00722 *
00723 * ## Algorithm
00724 * 1. Release CMT0 from module stop (MSTPCRA.MSTPA15 = 0)
00725 * 2. Stop counter (CMSTR0.STR0 = 0)
00726 * 3. Configure CMCR: PCLKB/8 clock source, compare match interrupt enable
00727 * 4. Calculate CMCOR from PCLKB frequency and desired 100 Hz rate
00728 * 5. Clear counter (CMCNT = 0)
00729 * 6. Set compare match value (CMCOR)
00730 * 7. Configure interrupt priority (IPR = 5)
00731 * 8. Start counter (CMSTR0.STR0 = 1)
00732 *
00733 * @return rx_err_t Error code indicating success or failure
00734 * @retval k_rx_ok CMT0 successfully initialized and running at 100 Hz
00735 * @retval k_rx_err_invalid_state CMT0 already initialized
00736 *
00737 * @pre System clocks must be configured (PCLKB running at expected frequency)
00738 * @pre CMT0 must not be already initialized (call timer_stop() first to reinitialize)
00739 * @post CMT0 generating 100 Hz interrupts for ThreadX system tick
00740 * @post CMT0 interrupt enabled at priority 5
00741 *
00742 * @note **Thread Safety**: NOT thread-safe. Must be called once during system initialization
00743 * before ThreadX is started.
00744 *
00745 * @par Example: System Initialization
00746 * @code{.c}
00747 * void hardware_init(void) {
00748 * rx_err_t err = timer_init();
00749 * if (err != k_rx_ok) {
00750 * rx_log_error_val("TIMER", "CMT0 init failed", err);
00751 * return;
00752 * }
00753 * rx_log_info("TIMER", "System tick running at 100 Hz");
00754 * }
00755 * @endcode
00756 *
00757 * @see timer_stop() Stop the CMT0 timer
00758 * @see timer_get_count() Read current counter value
00759 *
00760 * @since Version 1.0.0
00761 */
00762 [[nodiscard]] rx_err_t timer_init(void);
00763
00764 /**
00765 * @brief Stop CMT0 timer and disable compare match interrupts
00766 *
00767 * @details
00768 * Halts the CMT0 counter by clearing the start bit (CMSTR0.STR0 = 0) and
00769 * disables the compare match interrupt. The counter value is preserved and can
00770 * be read via timer_get_count(). After stopping, timer_init() may be called
00771 * to restart the timer.
00772 *
00773 * ## Algorithm
00774 * 1. Disable compare match interrupt (CMCR.CMIE = 0)
00775 * 2. Stop counter (CMSTR0.STR0 = 0)
00776 * 3. Mark timer as stopped in internal state
00777 *
00778 * @return rx_err_t Error code indicating success or failure
00779 * @retval k_rx_ok CMT0 successfully stopped
00780 * @retval k_rx_err_invalid_state CMT0 was not initialized
00781 *
00782 * @pre CMT0 must have been initialized via timer_init()
00783 * @pre Caller must ensure stopping the system tick will not break RTOS scheduling
00784 * @post CMT0 counter halted, counter value preserved
00785 * @post Compare match interrupts disabled
00786 *
00787 * @note **Thread Safety**: NOT thread-safe. Stopping the system tick from a ThreadX thread
00788 * may cause scheduling to freeze.
00789 * @warning Stopping CMT0 halts the ThreadX system tick. Only stop during shutdown or
00790 * low-power entry.
00791 *
00792 * @par Example: Entering Low Power Mode
00793 * @code{.c}
00794 * void enter_low_power(void) {
00795 * rx_err_t err = timer_stop();
00796 * if (err != k_rx_ok) {
00797 * rx_log_error_val("TIMER", "Failed to stop timer", err);
00798 * return;
00799 * }
00800 * // Enter low-power mode
00801 * }
00802 * @endcode
00803 *
00804 * @see timer_init() Reinitialize CMT0 after stopping
00805 * @see timer_get_count() Read counter value (works while stopped)
00806 *
00807 * @since Version 1.0.0

```

```

00808 */
00809 [[nodiscard]] rx_err_t timer_stop(void);
00810
00811 /**
00812 * @brief Get current CMT0 counter value
00813 *
00814 * @details
00815 * Reads the 16-bit free-running counter register (CMCNT) from CMT0. The counter
00816 * increments at PCLKB/8 rate and resets to 0 on compare match with CMCOR. This
00817 * can be used for sub-tick timing measurements or to check elapsed time within
00818 * a 10 ms system tick period. Works whether the timer is running or stopped.
00819 *
00820 * @param[out] count Pointer to store 16-bit counter value
00821 * - Must point to valid uint16_t variable
00822 * - Value range: 0 to CMCOR (compare match constant)
00823 * - Unchanged if function returns error
00824 *
00825 * @return rx_err_t Error code indicating success or failure
00826 * @retval k_rx_ok Counter value successfully read and stored in *count
00827 * @retval k_rx_err_null_ptr count pointer is nullptr
00828 *
00829 * @pre count pointer must be valid (not nullptr)
00830 * @pre CMT0 must have been initialized via timer_init() for meaningful values
00831 * @post *count contains current CMCNT register value if function succeeds
00832 * @post *count unchanged if function returns error
00833 *
00834 * @note **Thread Safety**: NOT thread-safe. The 16-bit register read is atomic on RX72N
00835 * but the pointer write is not protected.
00836 *
00837 * @par Example: Sub-Tick Timing
00838 * @code{.c}
00839 * uint16_t start, end;
00840 * timer_get_count(&start);
00841 * // ... perform operation ...
00842 * timer_get_count(&end);
00843 * uint16_t elapsed_ticks = (end >= start) ? (end - start) : (0xFFFF - start + end);
00844 * @endcode
00845 *
00846 * @see timer_init() Initialize CMT0 timer
00847 * @see timer_stop() Stop CMT0 timer
00848 *
00849 * @since Version 1.0.0
00850 */
00851 [[nodiscard]] rx_err_t timer_get_count(uint16_t* count);
00852
00853 /*
=====
00854 * ADC Functions
00855 *
=====
00856 */
00857
00858 /**
00859 * @enum adc_unit_t
00860 * @brief ADC unit enumeration with type safety for 12-bit SAR ADC modules
00861 *
00862 * @details
00863 * Identifies which of the two independent 12-bit successive approximation ADC units (S12AD)
00864 * on the RX72N to use. Each unit has separate channels, registers, and conversion control.
00865 * Using a typed enum prevents accidental swapping of unit and channel parameters.
00866 *
00867 * ## Hardware Specifications
00868 * | Unit | Base Address | Channels Available | Max Sample Rate |
00869 * |-----|-----|-----|-----|
00870 * | S12AD0 | 0x00089000 | 8 (AN000-AN007) | 1 Msps |
00871 * | S12AD1 | 0x00089100 | 21 (AN100-AN120) | 1 Msps |
00872 *
00873 * ## STAR Project Usage (144-pin LQFP)
00874 * | Unit | Channel | Pin | Signal | Purpose |
00875 * |-----|-----|-----|-----|-----|
00876 * | k_adc_unit_0 | k_adc_channel_7 | P47 | MOTOR_I0 | Motor 0 current sense (DRV8263H IPROPI) |
00877 * | k_adc_unit_0 | k_adc_channel_6 | P46 | MOTOR_I1 | Motor 1 current sense (DRV8263H IPROPI) |
00878 * | k_adc_unit_0 | k_adc_channel_5 | P45 | MOTOR_I2 | Motor 2 current sense |
00879 * | k_adc_unit_0 | k_adc_channel_4 | P44 | MOTOR_I3 | Motor 3 current sense |
00880 *
00881 * ## Type Safety Benefit
00882 * @code{.c}
00883 * // WRONG - Easy to swap unit and channel with plain uint8_t
00884 * adc_read(0, 1, &value); // Is 0 the unit or channel?
00885 *
00886 * // CORRECT - Typed enums make intent clear
00887 * adc_read(k_adc_unit_0, k_adc_channel_1, &value); // Unambiguous
00888 * @endcode
00889 *
00890 * @par Underlying Type: uint8_t (C23 typed enum)
00891 * - **Range**: 0-1 (only 2 units on RX72N)
00892 * - **Size**: 1 byte

```

```

00893 * - **Rationale**: Smallest type sufficient, explicit size guarantees ABI stability
00894 *
00895 * @see adc_channel_t Channel enumeration
00896 * @see adc_resolution_t Resolution configuration
00897 * @see adc_init() ADC initialization function
00898 */
00899 typedef enum : uint8_t {
00900 k_adc_unit_0 =
00901 0, /**< S12AD0 - ADC unit 0 (channels AN000-AN007). Used for motor current sensing. */
00902 k_adc_unit_1 =
00903 1 /**< S12AD1 - ADC unit 1 (channels AN100-AN120). Reserved for expansion sensors. */
00904 } adc_unit_t;
00905 /**
00906 * @enum adc_channel_t
00907 * @brief ADC channel enumeration with type safety for analog input selection
00908 *
00909 * @details
00910 * Identifies which analog input channel (0-7) to use within an ADC unit. Each unit
00911 * (S12AD0, S12AD1) has 8 independent channels with dedicated analog input pins.
00912 * Using a typed enum prevents accidental parameter swaps with unit or resolution.
00913 *
00914 * ## Pin Mapping (RX72N 144-pin LQFP, R5F572NNHxFB)
00915 * ### S12AD0 Channels (use with k_adc_unit_0)
00916 * | Channel | Pin | Signal Name | STAR Usage |
00917 * |-----|----|-----|-----|
00918 * | k_adc_channel_0 | P40 | AN000 | Available (expansion) |
00919 * | k_adc_channel_1 | P41 | AN001 | Available (expansion) |
00920 * | k_adc_channel_2 | P42 | AN002 | Available (expansion) |
00921 * | k_adc_channel_3 | P43 | AN003 | Available (expansion) |
00922 * | k_adc_channel_4 | P44 | AN004 | Motor 3 current sense |
00923 * | k_adc_channel_5 | P45 | AN005 | Motor 2 current sense |
00924 * | k_adc_channel_6 | P46 | AN006 | Motor 1 current sense |
00925 * | k_adc_channel_7 | P47 | AN007 | Motor 0 current sense |
00926 *
00927 * ## Input Specifications
00928 * - **Voltage Range**: 0V to AVCC (3.3V)
00929 * - **Input Impedance**: 1 MOhm typical
00930 * - **Max Input Current**: +/-10 uA
00931 * - **Protection**: Internal clamping diodes to AVCC/AVSS
00932 *
00933 * ## Usage Example
00934 * @code{.c}
00935 * // Motor current sensing
00936 * adc_init(k_adc_unit_0, k_adc_channel_0, k_adc_resolution_12bit);
00937 *
00938 * uint32_t voltage_mv;
00939 * adc_read_voltage_mv(k_adc_unit_0, k_adc_channel_0, k_adc_resolution_12bit, &voltage_mv);
00940 * rx_log_info_val("ADC", "Current sense (mV)", voltage_mv);
00941 * @endcode
00942 *
00943 * @par Underlying Type: uint8_t (C23 typed enum)
00944 * - **Range**: 0-7 (8 channels per unit)
00945 * - **Size**: 1 byte
00946 * - **Rationale**: Matches hardware channel select bits (ADANSA register)
00947 *
00948 * @see adc_unit_t ADC unit selection
00949 * @see adc_resolution_t Resolution configuration
00950 * @see adc_read() Read raw ADC value
00951 * @see adc_read_voltage_mv() Read voltage in millivolts
00952 */
00953
00954 typedef enum : uint8_t {
00955 k_adc_channel_0 = 0, /**< Channel 0 - AN000 (unit 0) or AN100 (unit 1). Available. */
00956 k_adc_channel_1 = 1, /**< Channel 1 - AN001 (unit 0) or AN101 (unit 1). Available. */
00957 k_adc_channel_2 = 2, /**< Channel 2 - AN002 (unit 0) or AN102 (unit 1). Available. */
00958 k_adc_channel_3 = 3, /**< Channel 3 - AN003 (unit 0) or AN103 (unit 1). Available. */
00959 k_adc_channel_4 = 4, /**< Channel 4 - AN004/P44 (unit 0, Motor 3 current) or AN104 (unit 1) */
00960 k_adc_channel_5 = 5, /**< Channel 5 - AN005/P45 (unit 0, Motor 2 current) or AN105 (unit 1) */
00961 k_adc_channel_6 = 6, /**< Channel 6 - AN006/P46 (unit 0, Motor 1 current) or AN106 (unit 1) */
00962 k_adc_channel_7 = 7 /**< Channel 7 - AN007/P47 (unit 0, Motor 0 current) or AN107 (unit 1) */
00963 } adc_channel_t;
00964
00965 /* ADC resolution type alias (enum defined in rx_bus_types.h) */
00966 typedef rx_adc_resolution_t adc_resolution_t;
00967 /**
00968 * @brief Initialize ADC unit and channel
00969 *
00970 * @param[in] unit ADC unit (0 or 1)
00971 * @param[in] channel ADC channel (0-7)
00972 * @param[in] bits Resolution (8, 10, or 12 bits)
00973 *
00974 * @return k_rx_ok on success,
00975 * k_rx_err_invalid_arg if unit, channel, or bits is invalid,
00976 * k_rx_err_gpio_conflict if pin already reserved
00977 */
00978
00979 [[nodiscard]] rx_err_t adc_init(adc_unit_t unit, adc_channel_t channel, adc_resolution_t bits);

```

```

00980
00981 /**
00982 * @brief Read ADC value
00983 *
00984 * @param[in] unit ADC unit (0 or 1)
00985 * @param[in] channel ADC channel (0-7)
00986 * @param[out] value Pointer to store ADC value
00987 *
00988 * @return k_rx_ok on success,
00989 * k_rx_err_null_ptr if value is nullptr,
00990 * k_rx_err_invalid_arg if unit or channel is invalid,
00991 * k_rx_err_invalid_state if ADC unit not initialized,
00992 * k_rx_err_timeout if conversion times out
00993 */
00994 [[nodiscard]] rx_err_t adc_read(adc_unit_t unit, adc_channel_t channel, uint16_t* value);
00995
00996 /**
00997 * @brief Read ADC value and convert to millivolts
00998 *
00999 * @param[in] unit ADC unit (0 or 1)
01000 * @param[in] channel ADC channel (0-7)
01001 * @param[in] bits Resolution used during init (8, 10, or 12)
01002 * @param[out] voltage_mv Pointer to store voltage in millivolts
01003 *
01004 * @return k_rx_ok on success,
01005 * k_rx_err_null_ptr if voltage_mv is nullptr,
01006 * k_rx_err_invalid_arg if unit or channel is invalid,
01007 * k_rx_err_invalid_state if ADC unit not initialized,
01008 * k_rx_err_timeout if conversion times out
01009 */
01010 [[nodiscard]] rx_err_t adc_read_voltage_mv(adc_unit_t unit,
01011 adc_channel_t channel,
01012 adc_resolution_t bits,
01013 uint32_t* voltage_mv);
01014
01015 /*
=====
01016 * RIIC (I2C) Functions
01017 *
=====
01018 */
01019
01020 /**
01021 * @struct riic_channel_t
01022 * @brief RIIC (I2C) channel wrapper providing strong type safety
01023 *
01024 * @details
01025 * Wraps RIIC channel number in a struct to prevent accidental argument swaps with
01026 * I2C device addresses. This is a stronger type safety mechanism than simple
01027 * typedefs because structs cannot be implicitly converted.
01028 *
01029 * ### Design Rationale: Struct Wrapper vs. Typedef
01030 *
01031 * **Why struct wrapper instead of simple typedef?**
01032 *
01033 * I2C functions have multiple `uint8_t` parameters that are easily confused:
01034 * - Channel number (0-2)
01035 * - Device address (0x00-0x7F)
01036 * - Register address (0x00-0xFF)
01037 * - Data bytes (0x00-0xFF)
01038 *
01039 * ### Problem with Simple Typedefs
01040 * @code{.c}
01041 * typedef uint8_t riic_channel_t; // Simple typedef
01042 * typedef uint8_t i2c_device_addr_t; // Simple typedef
01043 *
01044 * riic_channel_t ch = 0;
01045 * i2c_device_addr_t addr = 0x48;
01046 *
01047 * // PROBLEM: Arguments can be swapped without compiler error!
01048 * riic_write(addr, ch, data, len); // Compiles! Wrong order, but types are compatible
01049 * @endcode
01050 *
01051 * ### Solution with Struct Wrappers
01052 * @code{.c}
01053 * typedef struct { uint8_t value; } riic_channel_t; // Struct wrapper
01054 * typedef struct { uint8_t value; } i2c_device_addr_t; // Struct wrapper
01055 *
01056 * riic_channel_t ch = {.value = 0};
01057 * i2c_device_addr_t addr = {.value = 0x48};
01058 *
01059 * // CORRECT: Compiler catches swapped arguments!
01060 * riic_write(addr, ch, data, len); // Compile error: type mismatch!
01061 * riic_write(ch, addr, data, len); // Correct order, compiles successfully
01062 * @endcode
01063 *
01064 * ## Comparison: ADC vs. I2C Type Safety

```

```

01065 *
01066 * | Peripheral | Type Safety Mechanism | Rationale |
01067 * |-----|-----|-----|
01068 * | **ADC** | Simple typed enums | Parameters are distinct (unit vs channel vs resolution bits) |
01069 * | **I2C** | Struct wrappers | Multiple uint8_t parameters easily confused |
01070 * | **SPI** | Simple typed enums | Mode, channel, frequency are distinct types |
01071 * | **UART** | Simple typed enums | Channel and baudrate are distinct types |
01072 *
01073 * ADC: `adc_read(k_adc_unit_0, k_adc_channel_1, &value)` -> Different types prevent swaps
01074 * I2C: `riic_write({.value = 0}, {.value = 0x48}, data, len)` -> Struct wrappers prevent swaps
01075 *
01076 * ## RX72N RIIC Hardware
01077 * - **Channels Available**: 3 (RIIC0, RIIC1, RIIC2)
01078 * - **Speeds**: 100 kHz (standard), 400 kHz (fast), 1 MHz (fast-plus)
01079 * - **Addressing**: 7-bit (0x00-0x7F) or 10-bit (not used in STAR project)
01080 *
01081 * ## STAR Project Usage
01082 * | Channel | SCL Pin | SDA Pin | Device | Address | Purpose |
01083 * |-----|-----|-----|-----|-----|-----|
01084 * | 0 | P16 | P17 | IMU (MPU6050) | 0x68 | Orientation sensing |
01085 * | 1 | P54 | P55 | Reserved | - | Future expansion |
01086 * | 2 | P52 | P53 | Reserved | - | Future expansion |
01087 *
01088 * ## Usage Example
01089 * @code{.c}
01090 * #include "hardware.h"
01091 *
01092 * void imu_init(void) {
01093 * // Initialize I2C channel 0 at 400 kHz (fast mode)
01094 * riic_channel_t ch = {.value = 0};
01095 * riic_init(ch, 400000);
01096 *
01097 * // MPU6050 IMU device address
01098 * i2c_device_addr_t imu_addr = {.value = 0x68};
01099 *
01100 * // Write to WHO_AM_I register (0x75) to verify device
01101 * uint8_t reg_addr = 0x75;
01102 * uint8_t who_am_i;
01103 * riic_write_read(ch, imu_addr, ®_addr, 1, &who_am_i, 1);
01104 *
01105 * if (who_am_i == 0x68) {
01106 * rx_log_info("IMU", "MPU6050 detected");
01107 * }
01108 * }
01109 * @endcode
01110 *
01111 * @par Example: Compiler Catches Mistakes
01112 * @code{.c}
01113 * riic_channel_t ch = {.value = 0};
01114 * i2c_device_addr_t addr = {.value = 0x48};
01115 * uint8_t data[] = {0x01, 0x02};
01116 *
01117 * // WRONG - Arguments swapped (compiler error!)
01118 * riic_write(addr, ch, data, 2);
01119 * // Error: cannot convert 'i2c_device_addr_t' to 'riic_channel_t'
01120 *
01121 * // CORRECT - Proper order
01122 * riic_write(ch, addr, data, 2); // Compiles successfully
01123 * @endcode
01124 *
01125 * @see i2c_device_addr_t I2C device address wrapper
01126 * @see riic_init() Initialize I2C channel
01127 * @see riic_write() Write data to I2C device
01128 * @see riic_read() Read data from I2C device
01129 * @see riic_write_read() Combined write-then-read transaction
01130 */
01131 typedef struct {
01132 uint8_t value; /**< RIIC channel number (0-2). Use {.value = N} initialization. */
01133 } riic_channel_t;
01134
01135 /**
01136 * @struct i2c_device_addr_t
01137 * @brief I2C device address wrapper providing strong type safety
01138 *
01139 * @details
01140 * Wraps 7-bit I2C device address in a struct to prevent accidental swaps with
01141 * channel numbers or register addresses. Part of the type safety system that
01142 * uses struct wrappers to catch parameter ordering mistakes at compile time.
01143 *
01144 * ## I2C Addressing
01145 * - **7-bit addressing**: Standard mode used in STAR project
01146 * - **Valid range**: 0x08-0x77 (0x00-0x07 and 0x78-0x7F reserved)
01147 * - **Reserved addresses**:
01148 * - 0x00: General call
01149 * - 0x01-0x07: CBUS, different bus formats
01150 * - 0x78-0x7F: 10-bit addressing, reserved
01151 *

```

```

01152 * ## Common Device Addresses
01153 * | Device | Address | Alt Address | Configurable |
01154 * |-----|-----|-----|-----|
01155 * | MPU6050 (IMU) | 0x68 | 0x69 | Via AD0 pin |
01156 * | BMP280 (Pressure) | 0x76 | 0x77 | Via SDO pin |
01157 * | ADS1115 (ADC) | 0x48-0x4B | - | Via ADDR pin |
01158 * | PCA9685 (PWM) | 0x40-0x7F | - | Via A0-A5 pins |
01159 * | EEPROM (24C series) | 0x50-0x57 | - | Via A0-A2 pins |
01160 *
01161 * ## Address Format on Bus
01162 * The 7-bit address is shifted left by 1 bit and combined with R/W bit:
01163 * ...
01164 * [A6 A5 A4 A3 A2 A1 A0 R/W]
01165 * ____7-bit address____/ _ 0=Write, 1=Read
01166 * ...
01167 * **Note**: HAL automatically handles this shifting. Always provide 7-bit address.
01168 *
01169 * ## STAR Project Devices
01170 * | Device | Address | Channel | Purpose |
01171 * |-----|-----|-----|-----|
01172 * | MPU6050 IMU | 0x68 | RIIC0 | 6-axis motion tracking |
01173 * | Future sensor 1 | TBD | RIIC1 | Expansion |
01174 * | Future sensor 2 | TBD | RIIC2 | Expansion |
01175 *
01176 * ## Usage Example
01177 * @code{.c}
01178 * // MPU6050 IMU on RIIC0
01179 * riic_channel_t ch = {.value = 0};
01180 * i2c_device_addr_t imu_addr = {.value = 0x68};
01181 *
01182 * // Read power management register (0x6B)
01183 * uint8_t pwr_mgmt_reg = 0x6B;
01184 * uint8_t pwr_mgmt_val;
01185 *
01186 * riic_write_read(ch, imu_addr, &pwr_mgmt_reg, 1, &pwr_mgmt_val, 1);
01187 * rx_log_info_hex("IMU", "Power management", pwr_mgmt_val, 2);
01188 * @endcode
01189 *
01190 * @par Example: Multi-Device Bus
01191 * @code{.c}
01192 * // Multiple devices on same I2C bus
01193 * riic_channel_t ch = {.value = 0};
01194 *
01195 * i2c_device_addr_t imu_addr = {.value = 0x68}; // MPU6050
01196 * i2c_device_addr_t baro_addr = {.value = 0x76}; // BMP280
01197 * i2c_device_addr_t adc_addr = {.value = 0x48}; // ADS1115
01198 *
01199 * // Read from different devices on same bus
01200 * uint8_t imu_data[6];
01201 * riic_read(ch, imu_addr, imu_data, 6);
01202 *
01203 * uint8_t baro_data[3];
01204 * riic_read(ch, baro_addr, baro_data, 3);
01205 *
01206 * uint8_t adc_data[2];
01207 * riic_read(ch, adc_addr, adc_data, 2);
01208 * @endcode
01209 *
01210 * @see riic_channel_t RIIC channel wrapper
01211 * @see riic_write() Write data to I2C device
01212 * @see riic_read() Read data from I2C device
01213 * @see riic_write_read() Register read pattern (write reg addr, then read)
01214 */
01215 typedef struct {
01216 uint8_t value; /** < 7-bit I2C device address (0x08-0x77). Use {.value = 0xNN} initialization. */
01217 } i2c_device_addr_t;
01218
01219 /**
01220 * @brief Initialize RIIC channel for I2C controller communication
01221 *
01222 * @details
01223 * Configures the specified RIIC peripheral channel for I2C controller mode at the
01224 * requested clock frequency. Performs full hardware initialization including module
01225 * stop release, pin function selection (SCL/SDA via MPC), internal reset sequence,
01226 * bit rate register (BRH/BRL) calculation from PCLKB, and interrupt configuration.
01227 *
01228 * ## Algorithm
01229 * 1. Validate channel number (0-2) and frequency
01230 * 2. Release RIIC module from module stop (MSTPCRB)
01231 * 3. Assert internal reset (ICCR1.IICRST = 1)
01232 * 4. Configure SCL/SDA pins via MPC register
01233 * 5. Set bit rate registers (BRH/BRL) from PCLKB and frequency_hz
01234 * 6. Configure noise filter, timeout detection
01235 * 7. Release internal reset (ICCR1.IICRST = 0)
01236 * 8. Enable I2C bus (ICCR1.ICE = 1)
01237 *
01238 * ## Supported Frequencies

```



```

01239 * | Frequency | Mode | BRL/BRH | Max Bus Length |
01240 * |-----|-----|-----|-----|
01241 * | 100000 Hz | Standard | Auto-calculated | 1 m |
01242 * | 400000 Hz | Fast | Auto-calculated | 0.3 m |
01243 * | 1000000 Hz | Fast-plus | Auto-calculated | 0.1 m |
01244 *
01245 * @param[in] channel RIIC channel wrapper
01246 * - channel.value range: 0-2
01247 * - Use {.value = N} designated initializer
01248 * @param[in] frequency_hz I2C clock frequency in Hz
01249 * - Valid values: 100000, 400000, 1000000
01250 * - Must match connected device capabilities
01251 *
01252 * @return rx_err_t Error code indicating success or failure
01253 * @retval k_rx_ok Channel successfully initialized
01254 * @retval k_rx_err_invalid_arg Channel > 2 or unsupported frequency
01255 * @retval k_rx_err_invalid_state Channel already initialized
01256 *
01257 * @pre Channel must not be already initialized (call riic_deinit first to reinitialize)
01258 * @pre System clocks must be configured (PCLKB running at expected frequency)
01259 * @post RIIC channel configured and ready for I2C transactions
01260 * @post SCL/SDA pins configured with open-drain output and internal pull-ups
01261 *
01262 * @note **Thread Safety**: NOT thread-safe. Initialize each channel from a single thread
01263 * during system startup.
01264 *
01265 * @par Example: Initialize IMU I2C Bus
01266 * @code{.c}
01267 * riic_channel_t ch = {.value = 0};
01268 * rx_err_t err = riic_init(ch, 400000); // 400 kHz fast mode
01269 * if (err != k_rx_ok) {
01270 * rx_log_error_val("I2C", "RIIC0 init failed", err);
01271 * return;
01272 * }
01273 * rx_log_info("I2C", "RIIC0 initialized at 400 kHz");
01274 * @endcode
01275 *
01276 * @see riic_write() Write data to I2C device
01277 * @see riic_read() Read data from I2C device
01278 * @see riic_write_read() Combined write-then-read transaction
01279 *
01280 * @since Version 1.0.0
01281 */
01282 [[nodiscard]] rx_err_t riic_init(riic_channel_t channel, uint32_t frequency_hz);
01283
01284 /**
01285 * @brief Write data to I2C device on specified RIIC channel
01286 *
01287 * @details
01288 * Performs a complete I2C write transaction: generates START condition, transmits
01289 * device address with write bit (R/W=0), sends data bytes, and generates STOP
01290 * condition. Each byte is acknowledged by the peripheral device. The function
01291 * blocks until all bytes are transmitted or an error occurs.
01292 *
01293 * ## Transaction Sequence
01294 * 1. Generate START condition
01295 * 2. Send device address byte (7-bit address « 1 | 0)
01296 * 3. Wait for ACK from peripheral
01297 * 4. Send data bytes one at a time, waiting for ACK after each
01298 * 5. Generate STOP condition
01299 * 6. Wait for bus idle (BBSY = 0)
01300 *
01301 * @param[in] channel RIIC channel wrapper
01302 * - channel.value range: 0-2
01303 * @param[in] device_addr 7-bit I2C device address wrapper
01304 * - device_addr.value range: 0x08-0x77
01305 * @param[in] data Pointer to data buffer to write
01306 * - Must not be nullptr
01307 * - Must contain at least length bytes
01308 * @param[in] length Number of bytes to write
01309 * - Range: 1-65535
01310 *
01311 * @return rx_err_t Error code indicating success or failure
01312 * @retval k_rx_ok All bytes successfully written and ACKed
01313 * @retval k_rx_err_null_ptr data pointer is nullptr
01314 * @retval k_rx_err_invalid_state Channel not initialized via riic_init()
01315 * @retval k_rx_err_timeout Bus timeout waiting for ACK or bus idle
01316 * @retval k_rx_err_nack Device sent NACK (wrong address or device busy)
01317 *
01318 * @pre Channel must be initialized via riic_init()
01319 * @pre data pointer must be valid and contain at least length bytes
01320 * @post All data bytes transmitted to the I2C device if successful
01321 * @post I2C bus returned to idle state (STOP condition generated)
01322 *
01323 * @note **Thread Safety**: NOT thread-safe. Do not perform concurrent transactions on the
01324 * same RIIC channel. Different channels may be used from different threads.
01325 */

```



```

01326 * @par Example: Write IMU Configuration Register
01327 * @code{.c}
01328 * riic_channel_t ch = {.value = 0};
01329 * i2c_device_addr_t imu_addr = {.value = 0x68};
01330 *
01331 * // Write power management register (0x6B) to wake up MPU6050
01332 * uint8_t wake_cmd[] = {0x6B, 0x00};
01333 * rx_err_t err = riic_write(ch, imu_addr, wake_cmd, 2);
01334 * if (err != k_rx_ok) {
01335 * rx_log_error_val("IMU", "Failed to wake IMU", err);
01336 * }
01337 * @endcode
01338 *
01339 * @see riic_read() Read data from I2C device
01340 * @see riic_write_read() Combined write-then-read (register read pattern)
01341 * @see riic_init() Initialize RIIC channel first
01342 *
01343 * @since Version 1.0.0
01344 */
01345 [[nodiscard]] rx_err_t riic_write(riic_channel_t channel,
01346 i2c_device_addr_t device_addr,
01347 const uint8_t* data,
01348 uint16_t length);
01349
01350 /**
01351 * @brief Read data from I2C device on specified RIIC channel
01352 *
01353 * @details
01354 * Performs a complete I2C read transaction: generates START condition, transmits
01355 * device address with read bit (R/W=1), receives data bytes with ACK (NACK on
01356 * last byte), and generates STOP condition. The function blocks until all bytes
01357 * are received or an error occurs.
01358 *
01359 * ## Transaction Sequence
01360 * 1. Generate START condition
01361 * 2. Send device address byte (7-bit address « 1 | 1)
01362 * 3. Wait for ACK from peripheral
01363 * 4. Receive data bytes, sending ACK after each (NACK after last byte)
01364 * 5. Generate STOP condition
01365 * 6. Wait for bus idle (BBSY = 0)
01366 *
01367 * @param[in] channel RIIC channel wrapper
01368 * - channel.value range: 0-2
01369 * @param[in] device_addr 7-bit I2C device address wrapper
01370 * - device_addr.value range: 0x08-0x77
01371 * @param[out] data Pointer to buffer for received data
01372 * - Must not be nullptr
01373 * - Must have space for at least length bytes
01374 * - Unchanged if function returns error
01375 * @param[in] length Number of bytes to read
01376 * - Range: 1-65535
01377 *
01378 * @return rx_err_t Error code indicating success or failure
01379 * @retval k_rx_ok All bytes successfully received
01380 * @retval k_rx_err_null_ptr data pointer is nullptr
01381 * @retval k_rx_err_invalid_state Channel not initialized via riic_init()
01382 * @retval k_rx_err_timeout Bus timeout waiting for data or bus idle
01383 * @retval k_rx_err_nack Device sent NACK on address (wrong address or device busy)
01384 *
01385 * @pre Channel must be initialized via riic_init()
01386 * @pre data pointer must be valid with space for at least length bytes
01387 * @post data buffer contains received bytes if function succeeds
01388 * @post I2C bus returned to idle state (STOP condition generated)
01389 *
01390 * @note **Thread Safety**: NOT thread-safe. Do not perform concurrent transactions on the
01391 * same RIIC channel. Different channels may be used from different threads.
01392 *
01393 * @par Example: Read IMU Accelerometer Data
01394 * @code{.c}
01395 * riic_channel_t ch = {.value = 0};
01396 * i2c_device_addr_t imu_addr = {.value = 0x68};
01397 *
01398 * uint8_t accel_data[6];
01399 * rx_err_t err = riic_read(ch, imu_addr, accel_data, 6);
01400 * if (err != k_rx_ok) {
01401 * rx_log_error_val("IMU", "Failed to read accel data", err);
01402 * }
01403 * @endcode
01404 *
01405 * @see riic_write() Write data to I2C device
01406 * @see riic_write_read() Combined write-then-read (register read pattern)
01407 * @see riic_init() Initialize RIIC channel first
01408 *
01409 * @since Version 1.0.0
01410 */
01411 [[nodiscard]] rx_err_t
01412 riic_read(riic_channel_t channel, i2c_device_addr_t device_addr, uint8_t* data, uint16_t length);

```

```

01413
01414 /**
01415 * @brief Write then read from I2C device using repeated START (combined transaction)
01416 *
01417 * @details
01418 * Performs an I2C combined write-then-read transaction using a repeated START
01419 * condition. This is the standard pattern for reading device registers: write the
01420 * register address, then issue a repeated START and read back the register data
01421 * without releasing the bus between operations.
01422 *
01423 * ## Transaction Sequence
01424 * 1. Generate START condition
01425 * 2. Send device address byte (7-bit address « 1 | 0) -- write mode
01426 * 3. Wait for ACK from peripheral
01427 * 4. Send write_data bytes (typically register address), ACK after each
01428 * 5. Generate repeated START condition (no STOP)
01429 * 6. Send device address byte (7-bit address « 1 | 1) -- read mode
01430 * 7. Wait for ACK from peripheral
01431 * 8. Receive read_data bytes, ACK each (NACK on last byte)
01432 * 9. Generate STOP condition
01433 * 10. Wait for bus idle (BBSY = 0)
01434 *
01435 * @param[in] channel RIIC channel wrapper
01436 * - channel.value range: 0-2
01437 * @param[in] device_addr 7-bit I2C device address wrapper
01438 * - device_addr.value range: 0x08-0x77
01439 * @param[in] write_data Pointer to data to write (typically register address)
01440 * - Must not be nullptr
01441 * - Must contain at least write_length bytes
01442 * @param[in] write_length Number of bytes to write
01443 * - Typically 1 (single register address byte)
01444 * @param[out] read_data Pointer to buffer for received data
01445 * - Must not be nullptr
01446 * - Must have space for at least read_length bytes
01447 * - Unchanged if function returns error
01448 * @param[in] read_length Number of bytes to read
01449 * - Range: 1-65535
01450 *
01451 * @return rx_err_t Error code indicating success or failure
01452 * @retval k_rx_ok Write and read completed successfully
01453 * @retval k_rx_err_null_ptr write_data or read_data pointer is nullptr
01454 * @retval k_rx_err_invalid_state Channel not initialized via riic_init()
01455 * @retval k_rx_err_timeout Bus timeout during write or read phase
01456 * @retval k_rx_err_nack Device sent NACK (wrong address or device busy)
01457 *
01458 * @pre Channel must be initialized via riic_init()
01459 * @pre write_data and read_data pointers must be valid with sufficient size
01460 * @post read_data buffer contains received register data if function succeeds
01461 * @post I2C bus returned to idle state (STOP condition generated)
01462 *
01463 * @note **Thread Safety**: NOT thread-safe. Do not perform concurrent transactions on the
01464 * same RIIC channel. Different channels may be used from different threads.
01465 *
01466 * @par Example: Read IMU WHO_AM_I Register
01467 * @code{.c}
01468 * riic_channel_t ch = {.value = 0};
01469 * i2c_device_addr_t imu_addr = {.value = 0x68};
01470 *
01471 * uint8_t reg_addr = 0x75; // WHO_AM_I register
01472 * uint8_t who_am_i;
01473 * rx_err_t err = riic_write_read(ch, imu_addr, ®_addr, 1, &who_am_i, 1);
01474 * if (err == k_rx_ok && who_am_i == 0x68) {
01475 * rx_log_info("IMU", "MPU6050 detected");
01476 * }
01477 * @endcode
01478 *
01479 * @see riic_write() Write-only transaction
01480 * @see riic_read() Read-only transaction
01481 * @see riic_init() Initialize RIIC channel first
01482 *
01483 * @since Version 1.0.0
01484 */
01485 [[nodiscard]] rx_err_t riic_write_read(riic_channel_t channel,
01486 i2c_device_addr_t device_addr,
01487 const uint8_t* write_data,
01488 uint16_t write_length,
01489 uint8_t* read_data,
01490 uint16_t read_length);
01491
01492 /**
=====
01493 * RIIC (I2C) Functions - Peripheral Mode
01494 *
=====
01495 */
01496
01497 /**

```

```

01498 * @enum riic_peripheral_limits_t
01499 * @brief Public size limits for RIIC peripheral mode callers
01500 *
01501 * @details
01502 * Exposes the I/O size ceiling enforced by riic_peripheral_read() and
01503 * riic_peripheral_write() so that callers can statically size their buffers
01504 * and validate arguments without needing to inspect the driver implementation.
01505 *
01506 * @see riic_peripheral_read() Enforces this limit on max_length
01507 * @see riic_peripheral_write() Enforces this limit on length
01508 *
01509 * @since Version 1.0.0
01510 */
01511 typedef enum : uint16_t {
01512 k_riic_peripheral_transfer_limit = 256, /**< Maximum bytes per peripheral read or write call */
01513 } riic_peripheral_limits_t;
01514
01515 /**
01516 * @brief Initialize RIIC channel in peripheral (device) mode
01517 *
01518 * @details
01519 * Configures the RIIC peripheral to act as an I2C peripheral device, responding
01520 * to a fixed 7-bit I2C address issued by a controller (RPi5). Sets up the
01521 * peripheral address register (SARL0/SARU0), enables address match detection
01522 * (ICSER.SAR0E), and leaves the module in peripheral mode (MS=0 in ICMR1).
01523 *
01524 * @par Peripheral Mode Architecture:
01525 * - RPi5 acts as I2C controller (generates SCL, START, STOP)
01526 * - RX72N acts as I2C peripheral (responds to address, provides/accepts data)
01527 * - Address matching handled in hardware (ICSER.SAR0E)
01528 * - Data transfer polled via ICSR2.RDRF and ICSR2.TDRE
01529 *
01530 * @param[in] channel RIIC channel wrapper (channel.value range: 0-2)
01531 * @param[in] device_addr 7-bit peripheral address for this device
01532 *
01533 * @return rx_err_t Error code
01534 * @retval k_rx_ok Channel initialized in peripheral mode
01535 * @retval k_rx_err_invalid_arg Channel value out of range
01536 * @retval k_rx_err_invalid_state Channel already initialized; deinitialize before reinitializing
01537 *
01538 * @pre channel.value must be in range [0, 2] (i.e., < k_riic_max_channels)
01539 * @pre device_addr.value must be in range [0x08, 0x77] (valid non-reserved 7-bit I2C address)
01540 * @post RIIC peripheral responds to device_addr on the I2C bus
01541 * @post s_riic_channel_initialized[channel.value] == true on success
01542 *
01543 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
01544 * @note Not thread-safe during initialization
01545 *
01546 * @see riic_peripheral_read() Read data written by I2C controller
01547 * @see riic_peripheral_write() Provide data to I2C controller read
01548 *
01549 * @since Version 1.0.0
01550 */
01551 [[nodiscard]] rx_err_t riic_init_peripheral(riic_channel_t channel, i2c_device_addr_t device_addr);
01552
01553 /**
01554 * @brief Deinitialize an RIIC channel previously initialized in peripheral mode
01555 *
01556 * @details
01557 * Disables peripheral address match (ICSER.SAR0E), clears SARL0/SARU0
01558 * address registers, and resets the RIIC module. After this call, the channel
01559 * may be reinitialized via riic_init() or riic_init_peripheral().
01560 *
01561 * @param[in] channel RIIC channel to deinitialize (channel.value range: 0-2)
01562 *
01563 * @return rx_err_t Error code
01564 * @retval k_rx_ok Channel deinitialized successfully
01565 * @retval k_rx_err_invalid_arg channel.value >= k_riic_max_channels
01566 * @retval k_rx_err_invalid_state Channel was not initialized in peripheral mode
01567 *
01568 * @pre channel.value must be in range [0, k_riic_max_channels - 1]
01569 * @pre Channel must have been initialized via riic_init_peripheral()
01570 * @post Channel mode reset to uninitialized; safe to call riic_init_peripheral() again
01571 * @post Peripheral address match disabled on the hardware
01572 *
01573 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
01574 * @note Not thread-safe; provide external synchronization if needed
01575 *
01576 * @par Example:
01577 * @code{.c}
01578 * riic_channel_t ch = { .value = k_i2c_comm_default_channel };
01579 * rx_err_t err = riic_deinit_peripheral(ch);
01580 * if (err != k_rx_ok) {
01581 * // Channel was not in peripheral mode or index out of range
01582 * return err;
01583 * }
01584 * // Channel is now free; may be reinitialized for controller or peripheral use

```

```

01585 * @endcode
01586 *
01587 * @see riic_init_peripheral() Complementary initialization function
01588 *
01589 * @since Version 1.0.0
01590 */
01591 [[nodiscard]] rx_err_t riic_deinit_peripheral(riic_channel_t channel);
01592
01593 /**
01594 * @brief Read bytes written by the I2C controller to this peripheral
01595 *
01596 * @details
01597 * Polls ICSR2.RDRF (Receive Data Register Full) to detect incoming bytes from
01598 * the I2C controller (RPi5). Reads up to max_length bytes. Returns immediately
01599 * with bytes_read=0 if no data is currently available.
01600 *
01601 * @param[in] channel RIIC channel wrapper (channel.value range: 0-2)
01602 * @param[out] data Buffer to store received bytes
01603 * @param[in] max_length Maximum number of bytes to read
01604 * @param[out] bytes_read Actual number of bytes read (may be 0)
01605 *
01606 * @return rx_err_t Error code
01607 * @retval k_rx_ok Read completed (check bytes_read for count)
01608 * @retval k_rx_err_null_ptr data or bytes_read is nullptr
01609 * @retval k_rx_err_invalid_arg Channel value out of range or max_length == 0
01610 * or max_length > k_riic_peripheral_transfer_limit
01611 * @retval k_rx_err_invalid_state Channel not initialized via riic_init_peripheral()
01612 *
01613 * @pre channel must be initialized via riic_init_peripheral()
01614 * @pre data != nullptr with capacity for at least max_length bytes
01615 * @pre max_length >= 1 && max_length <= k_riic_peripheral_transfer_limit
01616 * @post *bytes_read contains number of bytes placed in data (0 to max_length)
01617 * @post data[0..*bytes_read-1] contain valid received bytes on k_rx_ok
01618 *
01619 * @note Non-blocking: returns immediately with available data
01620 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
01621 *
01622 * @see riic_init_peripheral() Initialize channel first
01623 * @see riic_peripheral_write() Provide read data to controller
01624 *
01625 * @since Version 1.0.0
01626 */
01627 [[nodiscard]] rx_err_t riic_peripheral_read(riic_channel_t channel,
01628 uint8_t* data,
01629 uint16_t max_length,
01630 uint16_t* bytes_read);
01631
01632 /**
01633 * @brief Write bytes for the I2C controller to read from this peripheral
01634 *
01635 * @details
01636 * Polls ICSR2.TDRE (Transmit Data Register Empty) to detect when the I2C
01637 * controller is reading from this peripheral. Writes up to length bytes into
01638 * the ICDRT transmit data register. Returns immediately if the transmit
01639 * register is not yet empty (controller not currently reading).
01640 *
01641 * @param[in] channel RIIC channel wrapper (channel.value range: 0-2)
01642 * @param[in] data Pointer to data buffer to write
01643 * @param[in] length Number of bytes to write (valid range: 1..k_riic_peripheral_transfer_limit)
01644 *
01645 * @return rx_err_t Error code
01646 * @retval k_rx_ok Write completed
01647 * @retval k_rx_err_null_ptr data is nullptr
01648 * @retval k_rx_err_invalid_arg Channel value out of range, length == 0,
01649 * or length > k_riic_peripheral_transfer_limit
01650 * @retval k_rx_err_invalid_state Channel not initialized via riic_init_peripheral()
01651 * @retval k_rx_err_timeout Transmit data register not empty within timeout
01652 *
01653 * @pre channel must be initialized via riic_init_peripheral()
01654 * @pre data != nullptr with at least length valid bytes
01655 * @pre length >= 1 && length <= k_riic_peripheral_transfer_limit
01656 * @post On k_rx_ok: all length bytes have been written to ICDRT
01657 * @post On k_rx_err_timeout: partial write may have occurred
01658 *
01659 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
01660 * @note Not thread-safe, caller must provide external synchronization
01661 *
01662 * @see riic_init_peripheral() Initialize channel first
01663 * @see riic_peripheral_read() Receive data from controller
01664 *
01665 * @since Version 1.0.0
01666 */
01667 [[nodiscard]] rx_err_t
01668 riic_peripheral_write(riic_channel_t channel, const uint8_t* data, uint16_t length);
01669
01670 */
=====

```

```

01671 * RSPI (SPI) Functions - Peripheral Mode
01672 *
=====
01673 */
01674
01675 /**
01676 * @enum rspi_mode_t
01677 * @brief SPI mode enumeration defining clock polarity (CPOL) and phase (CPHA) combinations
01678 *
01679 * @details
01680 * Defines the four standard SPI modes based on clock polarity and phase settings.
01681 * These modes determine when data is sampled relative to the clock edge and what
01682 * the idle state of the clock line is.
01683 *
01684 * ## SPI Mode Definitions
01685 * | Mode | CPOL | CPHA | Clock Idle | Data Sampled | Data Shifted | Common Devices |
01686 * |-----|-----|-----|-----|-----|-----|-----|
01687 * | 0 | 0 | 0 | Low | Rising edge | Falling edge | RPi5, most sensors |
01688 * | 1 | 0 | 1 | Low | Falling edge | Rising edge | SD cards |
01689 * | 2 | 1 | 0 | High | Falling edge | Rising edge | Rare |
01690 * | 3 | 1 | 1 | High | Rising edge | Falling edge | Rare |
01691 *
01692 * ## STAR Project Usage
01693 * | Peripheral | Mode | Rationale |
01694 * |-----|-----|-----|
01695 * | RPi5 (RSPI0 peripheral) | Mode 0 | Linux spidev default, maximum compatibility |
01696 * | RSPI1 controller (not used) | TBD | RSPI1 not used in current design |
01697 *
01698 * ## Clock Polarity (CPOL)
01699 * - **CPOL=0**: Clock idle state is low (GND). Clock starts low, pulses high.
01700 * - **CPOL=1**: Clock idle state is high (VDD). Clock starts high, pulses low.
01701 *
01702 * ## Clock Phase (CPHA)
01703 * - **CPHA=0**: Data sampled on leading clock edge, shifted on trailing edge.
01704 * - **CPHA=1**: Data shifted on leading clock edge, sampled on trailing edge.
01705 *
01706 * ## Timing Diagrams
01707 *
01708 * ### Mode 0 (CPOL=0, CPHA=0) - Most Common
01709 * ```
01710 * CLK: ____/---____/---____
01711 * COPI: ===[D0]===[D1]===
01712 * CIPO: ===[D0]===[D1]===
01713 * ^ Sample ^ Sample
01714 * ```
01715 *
01716 * ### Mode 3 (CPOL=1, CPHA=1)
01717 * ```
01718 * CLK: ---____/---____/---
01719 * COPI: ===[D0]===[D1]===
01720 * CIPO: ===[D0]===[D1]===
01721 * ^ Sample ^ Sample
01722 * ```
01723 *
01724 * ## Selection Guidelines
01725 * 1. **Check device datasheet**: Most devices specify preferred mode
01726 * 2. **Mode 0 default**: Use Mode 0 if device supports multiple modes
01727 * 3. **Controller/Peripheral match**: Both sides must use same mode
01728 * 4. **Avoid Modes 1 and 2**: Less common, higher chance of incompatibility
01729 *
01730 * ## Hardware Configuration
01731 * - **SPCR.CPOL bit**: Set for CPOL=1, clear for CPOL=0
01732 * - **SPCR.CPHA bit**: Set for CPHA=1, clear for CPHA=0
01733 * - **Both configurable**: HAL automatically sets registers based on mode
01734 *
01735 * ## Usage Example
01736 * @code{.c}
01737 * // RPi5 communication (peripheral mode, Mode 0)
01738 * rspi_config_t rpi_cfg = {
01739 * .spi_mode = k_rspi_mode_0, // CPOL=0, CPHA=0
01740 * .use_16bit = false
01741 * };
01742 * rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);
01743 *
01744 * // External sensor communication (controller mode, Mode 0)
01745 * rspi_controller_config_t sensor_cfg = {
01746 * .spi_mode = k_rspi_mode_0, // Standard SPI mode 0
01747 * .freq_hz = k_rspi_freq_1mhz, // 1 MHz
01748 * .cs = k_port_c_pin_4
01749 * };
01750 * rspi_init_controller(k_rspi_channel_1, &sensor_cfg); // RSPI1
01751 * @endcode
01752 *
01753 * @par Underlying Type: uint8_t (C23 typed enum)
01754 * - **Range**: 0-3 (4 SPI modes)
01755 * - **Size**: 1 byte
01756 * - **Rationale**: Matches SPCR register bit pattern (CPOL | (CPHA « 1))

```

```

01757 *
01758 * @see rspi_init_peripheral() Initialize SPI in peripheral mode
01759 * @see rspi_init_controller() Initialize SPI in controller mode
01760 * @see rspi_config_t Configuration structure for peripheral mode
01761 * @see rspi_controller_config_t Configuration structure for controller mode
01762 */
01763 typedef enum : uint8_t {
01764 k_rspi_mode_0 =
01765 0, /**< CPOL=0, CPHA=0 - Data sampled on rising edge, idle low. Most common mode. */
01766 k_rspi_mode_1 =
01767 1, /**< CPOL=0, CPHA=1 - Data sampled on falling edge, idle low. Used by SD cards. */
01768 k_rspi_mode_2 = 2, /**< CPOL=1, CPHA=0 - Data sampled on falling edge, idle high. Rarely used. */
01769 k_rspi_mode_3 =
01770 3 /**< CPOL=1, CPHA=1 - Data sampled on rising edge, idle high. Alternative for some devices. */
01771 } rspi_mode_t;
01772
01773 /**
01774 * @enum rspi_channel_t
01775 * @brief RSPI channel enumeration with type safety for SPI peripheral selection
01776 *
01777 * @details
01778 * Identifies which of the three RSPI channels (RSPI0, RSPI1, RSPI2) on the RX72N
01779 * to use. Using a typed enum instead of plain uint8_t prevents accidental argument
01780 * swaps and provides compile-time validation of channel numbers.
01781 *
01782 * ## RX72N RSPI Channel Assignments (STAR Project)
01783 * | Channel | Role | Connected Device | Pins |
01784 * |-----|-----|-----|-----|
01785 * | RSPI0 | Peripheral | Raspberry Pi 5 | P30 (CIPO), P31 (COPI), P32 (CLK), P33 (CS) |
01786 * | RSPI1 | Controller | Reserved (sensors) | PC5 (CIPO), PC6 (COPI), PC7 (CLK) |
01787 * | RSPI2 | Unused | - | - |
01788 *
01789 * @par Underlying Type: uint8_t (C23 typed enum)
01790 * - **Range**: 0-2 (3 channels on RX72N)
01791 * - **Size**: 1 byte
01792 *
01793 * @invariant Values must be in range [0, 2] corresponding to RSPI0, RSPI1, RSPI2
01794 *
01795 * @code{.c}
01796 * // Initialize RSPI channel as peripheral
01797 * rspi_config_t cfg = { .spi_mode = k_rspi_mode_0, .use_16bit = false };
01798 * rx_err_t err = rspi_init_peripheral(k_rspi_channel_0, &cfg);
01799 * @endcode
01800 *
01801 * @see rspi_init_peripheral() Initialize channel in peripheral mode
01802 * @see rspi_init_controller() Initialize channel in controller mode
01803 *
01804 * @since Version 1.0.0
01805 */
01806 typedef enum : uint8_t {
01807 k_rspi_channel_0 = 0, /**< RSPI0 - RPi5 communication (peripheral mode in STAR project) */
01808 k_rspi_channel_1 = 1, /**< RSPI1 - External sensor SPI (controller mode, reserved) */
01809 k_rspi_channel_2 = 2 /**< RSPI2 - Unused (available for expansion) */
01810 } rspi_channel_t;
01811
01812 /**
01813 * @struct rspi_config_t
01814 * @brief RSPI peripheral mode configuration structure
01815 *
01816 * @details
01817 * Configuration parameters for initializing an RSPI channel in peripheral mode.
01818 * Used when the RX72N acts as an SPI peripheral device, receiving clock from an external
01819 * controller (e.g., Raspberry Pi 5). The struct encapsulates SPI mode and frame size
01820 * settings to avoid positional parameter errors.
01821 *
01822 * @code{.c}
01823 * rspi_config_t rpi_cfg = {
01824 * .spi_mode = k_rspi_mode_0, // CPOL=0, CPHA=0
01825 * .use_16bit = false // 8-bit frame size
01826 * };
01827 * @endcode
01828 *
01829 * @invariant spi_mode must be a valid rspi_mode_t value (0-3, CPOL/CPHA combinations)
01830 * @invariant use_16bit is boolean: false for 8-bit frames, true for 16-bit frames
01831 *
01832 * @see rspi_mode_t SPI mode enumeration (CPOL/CPHA combinations)
01833 * @see rspi_init_peripheral() Initialize channel with this configuration
01834 * @see rspi_controller_config_t Controller mode configuration (separate struct)
01835 *
01836 * @since Version 1.0.0
01837 */
01838 typedef struct {
01839 rspi_mode_t spi_mode; /**< SPI mode (0-3): CPOL and CPHA configuration */
01840 bool use_16bit; /**< True for 16-bit frames, false for 8-bit frames */
01841 } rspi_config_t;
01842
01843 /**

```



```

01844 * @brief Initialize RSPI channel in peripheral mode for RPi5 communication
01845 *
01846 * @details
01847 * Configures the specified RSPI channel as an SPI peripheral device.
01848 * In peripheral mode the RX72N receives the clock signal from an external SPI
01849 * controller (Raspberry Pi 5). Performs full hardware initialization including
01850 * module stop release, pin function selection (COPI/CIPO/CLK/CS via MPC),
01851 * SPI mode register configuration, and frame size setup.
01852 *
01853 * ## Algorithm
01854 * 1. Validate channel (0-2) and config pointer
01855 * 2. Release RSPI module from module stop (MSTPCRB)
01856 * 3. Disable RSPI (SPCR.SPE = 0)
01857 * 4. Configure pin functions via MPC (COPI, CIPO, CLK, CS)
01858 * 5. Set SPI mode (CPOL/CPHA in SPCMD0)
01859 * 6. Set frame length (8-bit or 16-bit in SPCMD0)
01860 * 7. Configure as peripheral (SPCR.MSTR = 0)
01861 * 8. Enable RSPI (SPCR.SPE = 1)
01862 *
01863 * @param[in] channel RSPI channel number (0-2)
01864 * @param[in] config Pointer to peripheral mode configuration
01865 * - Must not be nullptr
01866 * - config->spi_mode must be valid (0-3)
01867 *
01868 * @return rx_err_t Error code indicating success or failure
01869 * @retval k_rx_ok Channel successfully initialized in peripheral mode
01870 * @retval k_rx_err_invalid_arg Channel > 2 or invalid SPI mode
01871 * @retval k_rx_err_null_ptr config pointer is nullptr
01872 * @retval k_rx_err_invalid_state Channel already initialized (call rspi_deinit() to reinitialize)
01873 *
01874 * @pre Channel must not be already initialized (call rspi_deinit() first to reinitialize)
01875 * @pre System clocks must be configured (PCLKB running at expected frequency)
01876 * @post RSPI channel configured as SPI peripheral, ready for transfers
01877 * @post COPI/CIPO/CLK/CS pins configured for SPI peripheral function
01878 *
01879 * @note **Thread Safety**: NOT thread-safe. Initialize each channel from a single thread
01880 * during system startup.
01881 *
01882 * @par Example: RPi5 Communication Setup
01883 * @code{.c}
01884 * rspi_config_t rpi_cfg = {
01885 * .spi_mode = k_rspi_mode_0,
01886 * .use_16bit = false
01887 * };
01888 * rx_err_t err = rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);
01889 * if (err != k_rx_ok) {
01890 * rx_log_error_val("SPI", "RSPIO peripheral init failed", err);
01891 * }
01892 * @endcode
01893 *
01894 * @see rspi_peripheral_transfer() Full-duplex SPI transfer
01895 * @see rspi_deinit() Deinitialize RSPI channel
01896 * @see rspi_config_t Configuration structure
01897 *
01898 * @since Version 1.0.0
01899 */
01900 [[nodiscard]] rx_err_t rspi_init_peripheral(rspi_channel_t channel, const rspi_config_t* config);
01901
01902 /**
01903 * @brief Full-duplex SPI transfer in peripheral mode
01904 *
01905 * @details
01906 * Executes a byte-by-byte full-duplex SPI transfer on the specified RSPI
01907 * channel operating in peripheral mode. For each byte, the function waits
01908 * for the transmit buffer to empty, writes the TX byte, waits for the
01909 * receive buffer to fill, then reads the RX byte. Both TX and RX wait
01910 * operations are bounded by a 10 ms timeout to prevent infinite loops.
01911 *
01912 * The transfer loop is bounded by k_rspi_transfer_len_max (65535) per
01913 * NASA Rule 2 to ensure statically provable loop termination.
01914 *
01915 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
01916 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
01917 * @param[in] tx_data Pointer to transmit data buffer (must not be nullptr)
01918 * @param[out] rx_data Pointer to receive data buffer (must not be nullptr,
01919 * must have capacity for at least length bytes)
01920 * @param[in] length Number of bytes to transfer (valid: 1 to 65535)
01921 *
01922 * @return rx_err_t Error code indicating success or failure
01923 * @retval k_rx_ok Transfer completed successfully, rx_data filled
01924 * @retval k_rx_err_null_ptr tx_data or rx_data pointer is nullptr
01925 * @retval k_rx_err_invalid_arg length is zero, or RSPI base address
01926 * lookup failed for the channel
01927 * @retval k_rx_err_invalid_state channel >= 3 or channel not initialized
01928 * via rspi_init_peripheral()
01929 * @retval k_rx_err_timeout Transmit buffer did not empty or receive buffer
01930 * did not fill within the 10 ms timeout

```

```

01931 *
01932 * @pre channel must be initialized via rspi_init_peripheral()
01933 * @pre tx_data and rx_data must be valid non-null pointers
01934 * @pre length must be in range [1, 65535]
01935 * @post rx_data buffer contains received bytes corresponding to each
01936 * transmitted byte offset on success
01937 * @post SPSR status flags (SPRF, OVRF) cleared after each byte transfer
01938 *
01939 * @note Not thread-safe. Caller must provide external synchronization when
01940 * accessing the same channel from multiple threads.
01941 *
01942 * @see rspi_init_peripheral() Must be called before this function
01943 * @see rspi_peripheral_read_available() Non-blocking receive check
01944 * @see rspi_peripheral_write_ready() Non-blocking transmit check
01945 * @see rspi_deinit() Deinitialize RSPI channel
01946 *
01947 * @since Version 1.0.0
01948 */
01949 [[nodiscard]] rx_err_t rspi_peripheral_transfer(rspi_channel_t channel,
01950 const uint8_t* tx_data,
01951 uint8_t* rx_data,
01952 uint16_t length);
01953
01954 /**
01955 * @brief Check if receive data is available in the RSPI receive buffer
01956 *
01957 * @details
01958 * Polls the RSPI Status Register (SPSR) SPRF flag to determine whether the
01959 * receive buffer contains data ready for reading. This is a non-blocking
01960 * status check that returns immediately without waiting for data. Use this
01961 * to implement polled receive patterns without incurring the overhead of a
01962 * full transfer call.
01963 *
01964 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
01965 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
01966 * @param[out] available Pointer to bool set to true if SPRF flag indicates
01967 * data is available, false otherwise (must not be nullptr)
01968 *
01969 * @return rx_err_t Error code indicating success or failure
01970 * @retval k_rx_ok Status successfully read into *available
01971 * @retval k_rx_err_null_ptr available pointer is nullptr
01972 * @retval k_rx_err_invalid_state channel >= 3 or channel not initialized
01973 * via rspi_init_peripheral()
01974 * @retval k_rx_err_invalid_arg RSPI base address lookup failed for the channel
01975 *
01976 * @pre channel must be initialized via rspi_init_peripheral()
01977 * @pre available must be a valid non-null pointer
01978 * @post *available reflects current SPRF flag state on success
01979 * @post *available set to false on RSPI base address lookup failure
01980 *
01981 * @note Not thread-safe. Caller must provide external synchronization when
01982 * accessing the same channel from multiple threads.
01983 *
01984 * @see rspi_peripheral_transfer() Full-duplex transfer in peripheral mode
01985 * @see rspi_peripheral_write_ready() Non-blocking transmit buffer check
01986 * @see rspi_init_peripheral() Must be called before this function
01987 *
01988 * @since Version 1.0.0
01989 */
01990 [[nodiscard]] rx_err_t rspi_peripheral_read_available(rspi_channel_t channel, bool* available);
01991
01992 /**
01993 * @brief Check if the RSPI transmit buffer is ready for new data
01994 *
01995 * @details
01996 * Polls the RSPI Status Register (SPSR) SPTEF flag to determine whether the
01997 * transmit buffer is empty and ready to accept new data. This is a non-blocking
01998 * status check that returns immediately without waiting for the buffer to drain.
01999 * Use this to implement polled transmit patterns or to verify readiness before
02000 * writing to the SPI data register.
02001 *
02002 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02003 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02004 * @param[out] ready Pointer to bool set to true if SPTEF flag indicates the
02005 * transmit buffer is empty, false otherwise (must not be nullptr)
02006 *
02007 * @return rx_err_t Error code indicating success or failure
02008 * @retval k_rx_ok Status successfully read into *ready
02009 * @retval k_rx_err_null_ptr ready pointer is nullptr
02010 * @retval k_rx_err_invalid_state channel >= 3 or channel not initialized
02011 * via rspi_init_peripheral()
02012 * @retval k_rx_err_invalid_arg RSPI base address lookup failed for the channel
02013 *
02014 * @pre channel must be initialized via rspi_init_peripheral()
02015 * @pre ready must be a valid non-null pointer
02016 * @post *ready reflects current SPTEF flag state on success
02017 * @post *ready set to false on RSPI base address lookup failure

```



```

02018 *
02019 * @note Not thread-safe. Caller must provide external synchronization when
02020 * accessing the same channel from multiple threads.
02021 *
02022 * @see rspi_peripheral_transfer() Full-duplex transfer in peripheral mode
02023 * @see rspi_peripheral_read_available() Non-blocking receive buffer check
02024 * @see rspi_init_peripheral() Must be called before this function
02025 *
02026 * @since Version 1.0.0
02027 */
02028 [[nodiscard]] rx_err_t rspi_peripheral_write_ready(rspi_channel_t channel, bool* ready);
02029
02030 /**
02031 * @brief Deinitialize and disable an RSPI peripheral-mode channel
02032 *
02033 * @details
02034 * Performs a safe shutdown of the specified RSPI channel previously initialized
02035 * in peripheral mode. The deinitialization sequence disables SPI bus activity
02036 (SPCR.SPE = 0), gates the module clock via MSTPCRB to reduce power
02037 consumption, and clears the internal initialization flag. Register
02038 protection (PRCR) is temporarily unlocked for module stop control.
02039 *
02040 * This function is safe to call on channels that were never initialized; it
02041 * will still disable the hardware and clear the init flag. To reinitialize
02042 * a channel after deinit, call rspi_init_peripheral() again.
02043 *
02044 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02045 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02046 *
02047 * @return rx_err_t Error code indicating success or failure
02048 * @retval k_rx_ok Channel successfully deinitialized and module stopped
02049 * @retval k_rx_err_invalid_arg channel >= 3 or RSPI base address lookup
02050 * failed for the channel
02051 *
02052 * @pre channel must be in range [0, 2] (k_rspi_channel_0 through k_rspi_channel_2)
02053 * @pre No active SPI transfer should be in progress on this channel
02054 * @post RSPI peripheral disabled (SPCR.SPE = 0)
02055 * @post Module clock gated (MSTPCRB module stop bit set)
02056 * @post s_rspi_channel_initialized[channel] set to false
02057 *
02058 * @note Not thread-safe. Caller must ensure no concurrent transfers or
02059 * accesses to the same channel during deinitialization.
02060 *
02061 * @see rspi_init_peripheral() Reinitialize channel after deinit
02062 * @see rspi_controller_deinit() Deinit for controller-mode channels
02063 *
02064 * @since Version 1.0.0
02065 */
02066 [[nodiscard]] rx_err_t rspi_deinit(rspi_channel_t channel);
02067
02068 /*
=====
02069 * RSPI (SPI) Functions - Controller Mode
02070 *
=====
02071 */
02072
02073 /**
02074 * @enum rspi_freq_constants_t
02075 * @brief Common SPI clock frequency constants for controller mode
02076 *
02077 * @details
02078 * Named constants for commonly used SPI clock frequencies. Use these
02079 * instead of magic literals when configuring controller-mode SPI clock rate.
02080 * The RSPI bit rate register (SPBR) is calculated from these values and
02081 * the peripheral clock (PCLKB).
02082 *
02083 * @invariant All values must be positive and expressible by the SPBR register
02084 * @see rspi_init_controller() Uses freq_hz to calculate SPBR
02085 * @since Version 1.0.0
02086 */
02087 typedef enum : uint32_t {
02088 k_rspi_freq_1mhz = 1000000, /**< 1 MHz SPI clock frequency */
02089 k_rspi_freq_10mhz = 10000000, /**< 10 MHz SPI clock frequency (RPi5 link) */
02090 } rspi_freq_constants_t;
02091
02092 /**
02093 * @brief RSPI controller mode configuration structure
02094 *
02095 * Configuration for initializing RSPI in controller mode.
02096 * Used for communicating with SPI peripheral devices (sensors, etc.).
02097 */
02098 typedef struct {
02099 rspi_mode_t spi_mode; /**< SPI mode (0-3): CPOL and CPHA configuration */
02100 uint32_t freq_hz; /**< SPI clock frequency in Hz (100kHz - 10MHz) */
02101 rx_port_pin_t cs; /**< GPIO pin for chip select (type-safe port+pin) */
02102 } rspi_controller_config_t;

```

```

02103
02104 /**
02105 * @brief Initialize RSPI channel in controller mode for external device communication
02106 *
02107 * @details
02108 * Configures the specified RSPI channel as an SPI controller for
02109 * communicating with external SPI peripheral devices such as sensors. Performs
02110 * argument validation, SPBR bit rate calculation, CS GPIO pin configuration,
02111 * and full hardware register setup following the RX72N Hardware Manual Section
02112 * 38.3.6 initialization sequence. Always configures 16-bit data frames with
02113 * word access mode.
02114 *
02115 * The initialization is split into three internal phases: argument validation,
02116 * hardware resource preparation (SPBR calculation, base address lookup, GPIO
02117 * setup), and register configuration (module clock enable, SPI mode, bit rate,
02118 * SPI enable). On any failure, no partial hardware state is left behind.
02119 *
02120 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02121 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02122 * @param[in] config Pointer to rspi_controller_config_t containing:
02123 * - spi_mode: SPI mode (0-3) for CPOL/CPHA configuration
02124 * - freq_hz: SPI clock frequency in Hz (100 kHz to 10 MHz)
02125 * - cs: GPIO pin for chip select (rx_port_pin_t encoding)
02126 *
02127 * @return rx_err_t Error code indicating success or failure
02128 * @retval k_rx_ok Channel successfully initialized in controller mode
02129 * @retval k_rx_err_null_ptr config pointer is nullptr
02130 * @retval k_rx_err_invalid_arg channel >= 3, spi_mode > 3, freq_hz outside
02131 * 100 kHz - 10 MHz range, RSPI base address lookup failed, or CS
02132 * GPIO port/pin is invalid
02133 * @retval k_rx_err_invalid_state channel already initialized in controller mode
02134 * (call rspi_controller_deinit() first)
02135 *
02136 * @pre config must be a valid non-null pointer
02137 * @pre channel must not already be initialized in controller mode
02138 * @pre System clocks must be configured (PCLKB running at 60 MHz)
02139 * @post RSPI module clock enabled, registers configured, SPI enabled in
02140 * controller mode (SPCR.MSTR = 1, SPCR.SPE = 1)
02141 * @post CS GPIO configured as output, driven high (inactive / deasserted)
02142 * @post s_rspi_controller_initialized[channel] set to true
02143 *
02144 * @note Not thread-safe. Initialize each channel from a single thread during
02145 * system startup.
02146 *
02147 * @see rspi_controller_transfer_16bit() Perform 16-bit transfers after init
02148 * @see rspi_controller_set_cs() Manual chip select control
02149 * @see rspi_controller_deinit() Deinitialize controller-mode channel
02150 * @see rspi_controller_config_t Configuration structure
02151 *
02152 * @since Version 1.0.0
02153 */
02154 [[nodiscard]] rx_err_t rspi_init_controller(rspi_channel_t channel,
02155 const rspi_controller_config_t* config);
02156
02157 /**
02158 * @brief Set chip select line state for an RSPI controller-mode channel
02159 *
02160 * @details
02161 * Controls the GPIO chip select pin associated with the specified RSPI
02162 * channel operating in controller mode. The CS signal is active-low: when
02163 * active is true, the GPIO pin is driven low (CS asserted, peripheral
02164 * selected); when active is false, the pin is driven high (CS deasserted,
02165 * peripheral released). The port and pin are read from the internal
02166 * s_rspi_cs_config[] array populated during rspi_init_controller().
02167 *
02168 * This function provides direct CS control for protocols that require
02169 * multi-transfer CS framing. For single 16-bit transfers with automatic
02170 * CS handling, use rspi_controller_transfer_16bit() instead.
02171 *
02172 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02173 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02174 * @param[in] active true to assert CS (drive GPIO low),
02175 * false to deassert CS (drive GPIO high)
02176 *
02177 * @return rx_err_t Error code indicating success or failure
02178 * @retval k_rx_ok CS line successfully driven to requested state
02179 * @retval k_rx_err_invalid_state channel >= 3 or channel not initialized
02180 * in controller mode via rspi_init_controller()
02181 * @retval k_rx_err_invalid_arg GPIO port base address lookup failed for
02182 * the stored CS port configuration
02183 *
02184 * @pre channel must be initialized via rspi_init_controller()
02185 * @pre s_rspi_cs_config[channel] must contain valid port/pin from init
02186 * @post GPIO P0DR bit for the CS pin reflects the requested state
02187 * @post No other GPIO pins on the same port are modified (read-modify-write)
02188 *
02189 * @note Not thread-safe. Caller must provide external synchronization when

```

```

02190 * accessing the same channel or GPIO port from multiple threads.
02191 *
02192 * @see rspi_init_controller() Configures CS pin during initialization
02193 * @see rspi_controller_transfer_16bit() Automatic CS handling for transfers
02194 * @see rspi_controller_deinit() Deasserts CS during cleanup
02195 *
02196 * @since Version 1.0.0
02197 */
02198 [[nodiscard]] rx_err_t rspi_controller_set_cs(rspi_channel_t channel, bool active);
02199
02200 /**
02201 * @brief Perform a 16-bit full-duplex SPI transfer in controller mode with automatic CS
02202 *
02203 * @details
02204 * Executes a single 16-bit SPI transfer with full chip select lifecycle
02205 * management. The transfer sequence is:
02206 * 1. Assert CS (drive low) with ~300 ns setup delay
02207 * 2. Wait for TX buffer empty, write 16-bit tx_data to SPDR
02208 * 3. Wait for RX buffer full, read 16-bit response from SPDR
02209 * 4. Clear SPSR status flags (SPRF, OVRF)
02210 * 5. Apply ~300 ns hold delay, then deassert CS (drive high)
02211 *
02212 * On transfer error (timeout), CS is forcibly deasserted before returning
02213 * to avoid leaving the peripheral in a selected state. The CS setup and
02214 * hold delays use cycle-accurate NOP loops calibrated for 240 MHz core clock.
02215 *
02216 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02217 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02218 * @param[in] tx_data 16-bit data word to transmit
02219 * @param[out] rx_data Pointer to store the received 16-bit response
02220 * (must not be nullptr)
02221 *
02222 * @return rx_err_t Error code indicating success or failure
02223 * @retval k_rx_ok Transfer completed, *rx_data contains response, CS deasserted
02224 * @retval k_rx_err_null_ptr rx_data pointer is nullptr
02225 * @retval k_rx_err_invalid_state channel >= 3 or channel not initialized
02226 * via rspi_init_controller()
02227 * @retval k_rx_err_invalid_arg RSPI base address lookup failed, or GPIO port
02228 * lookup failed during CS assertion/deassertion
02229 * @retval k_rx_err_timeout TX buffer did not empty or RX buffer did not fill
02230 * within the 10 ms timeout (CS is deasserted on this error)
02231 *
02232 * @pre channel must be initialized via rspi_init_controller()
02233 * @pre rx_data must be a valid non-null pointer
02234 * @post *rx_data contains received 16-bit value on success
02235 * @post CS is deasserted (GPIO driven high) on both success and error
02236 * @post SPSR status flags SPRF and OVRF are cleared on success
02237 *
02238 * @note Not thread-safe. Caller must provide external synchronization when
02239 * accessing the same channel from multiple threads.
02240 *
02241 * @see rspi_init_controller() Must be called before this function
02242 * @see rspi_controller_set_cs() Manual CS control for multi-transfer framing
02243 * @see rspi_controller_deinit() Deinitialize controller-mode channel
02244 *
02245 * @since Version 1.0.0
02246 */
02247 [[nodiscard]] rx_err_t
02248 rspi_controller_transfer_16bit(rspi_channel_t channel, uint16_t tx_data, uint16_t* rx_data);
02249
02250 /**
02251 * @brief Deinitialize and disable an RSPI controller-mode channel
02252 *
02253 * @details
02254 * Performs a safe shutdown of the specified RSPI channel previously initialized
02255 * in controller mode. The deinitialization sequence is ordered to prevent bus
02256 * glitches: first deasserts CS to release any connected peripheral, then
02257 * disables SPI (SPCR.SPE = 0) to stop clock generation, then gates the
02258 * module clock via MSTPCRB for power savings, and finally clears internal
02259 * state (CS pin config and initialization flag). Register protection (PRCR)
02260 * is temporarily unlocked for module stop control.
02261 *
02262 * This function is safe to call on channels that were never initialized in
02263 * controller mode; it will still disable the hardware and clear the state.
02264 *
02265 * @param[in] channel RSPI channel number (valid: k_rspi_channel_0,
02266 * k_rspi_channel_1, k_rspi_channel_2; range 0-2)
02267 *
02268 * @return rx_err_t Error code indicating success or failure
02269 * @retval k_rx_ok Channel successfully deinitialized and module stopped
02270 * @retval k_rx_err_invalid_arg channel >= 3 or RSPI base address lookup
02271 * failed for the channel
02272 *
02273 * @pre channel must be in range [0, 2] (k_rspi_channel_0 through k_rspi_channel_2)
02274 * @pre No active SPI transfer should be in progress on this channel
02275 * @post CS GPIO deasserted (driven high) if channel was initialized
02276 * @post RSPI peripheral disabled (SPCR.SPE = 0)

```

```

02277 * @post Module clock gated (MSTPCRB module stop bit set)
02278 * @post s_rspi_cs_config[channel] cleared to defaults (port=0, pin=0)
02279 * @post s_rspi_controller_initialized[channel] set to false
02280 *
02281 * @note Not thread-safe. Caller must ensure no concurrent transfers or
02282 * accesses to the same channel during deinitialization.
02283 *
02284 * @see rspi_init_controller() Reinitialize channel after deinit
02285 * @see rspi_deinit() Deinit for peripheral-mode channels
02286 *
02287 * @since Version 1.0.0
02288 */
02289 [[nodiscard]] rx_err_t rspi_controller_deinit(rspi_channel_t channel);
02290
02291 /*
=====
02292 * UART Configuration Types
02293 *
=====
02294 */
02295
02296 /**
02297 * @brief Default debug UART channel (SCI9 - CY7C65213 USB bridge)
02298 */
02299 typedef enum : uint8_t {
02300 k_uart_debug_channel = 9 /**< Debug UART on SCI9 (PB7/TXD9, PB6/RXD9) */
02301 } uart_defaults_t;
02302
02303 /**
02304 * @brief UART channel enumeration with type safety
02305 *
02306 * Prevents accidental argument swaps and provides compile-time type checking.
02307 * RX72N has SCI channels 0-12 (13 channels total).
02308 */
02309 typedef enum : uint8_t {
02310 k_uart_channel_0 = 0,
02311 k_uart_channel_1 = 1,
02312 k_uart_channel_2 = 2,
02313 k_uart_channel_3 = 3,
02314 k_uart_channel_4 = 4,
02315 k_uart_channel_5 = 5,
02316 k_uart_channel_6 = 6,
02317 k_uart_channel_7 = 7,
02318 k_uart_channel_8 = 8,
02319 k_uart_channel_9 = 9,
02320 k_uart_channel_10 = 10,
02321 k_uart_channel_11 = 11,
02322 k_uart_channel_12 = 12
02323 } uart_channel_t;
02324
02325 /**
02326 * @brief UART channel configuration structure
02327 *
02328 * Encapsulates all parameters needed to initialize a UART channel.
02329 * Using a struct instead of positional parameters prevents accidental
02330 * parameter swapping and makes the API more maintainable.
02331 */
02332 typedef struct {
02333 uart_channel_t channel; /**< SCI channel (0-12) */
02334 uint32_t baudrate; /**< Baud rate (e.g., 9600, 115200) */
02335 rx_port_pin_t tx_gpio; /**< TX pin (rx_port_pin_t) */
02336 rx_port_pin_t rx_gpio; /**< RX pin (rx_port_pin_t) */
02337 } uart_channel_config_t;
02338
02339 /*
=====
02340 * UART HAL Functions
02341 *
=====
02342 */
02343
02344 /**
02345 * @brief Initialize SCI channel for UART communication
02346 *
02347 * Performs full hardware initialization including:
02348 * - Module stop control (MSTPCRB)
02349 * - Pin function configuration (MPC)
02350 * - GPIO direction setup (PDR/PMR)
02351 * - SCI register configuration
02352 *
02353 * @param[in] config UART channel configuration (must not be NULL)
02354 *
02355 * @return k_rx_ok on success,
02356 * k_rx_err_invalid_arg if config is nullptr or contains invalid values,
02357 * k_rx_err_invalid_state if channel already initialized
02358 */
02359 [[nodiscard]] rx_err_t uart_init_channel(const uart_channel_config_t* config);

```

```

02360
02361 /**
02362 * @brief Deinitialize SCI channel
02363 *
02364 * @param[in] channel SCI channel (0-12)
02365 *
02366 * @return k_rx_ok on success,
02367 * k_rx_err_invalid_arg if channel is invalid
02368 */
02369 [[nodiscard]] rx_err_t uart_deinit_channel(uart_channel_t channel);
02370
02371 /**
02372 * @brief Transmit a single character on specified channel
02373 *
02374 * @param[in] channel SCI channel (0-12)
02375 * @param[in] data Character to transmit
02376 *
02377 * @return k_rx_ok on success,
02378 * k_rx_err_invalid_arg if channel is invalid,
02379 * k_rx_err_invalid_state if channel not initialized
02380 */
02381 [[nodiscard]] rx_err_t uart_putc_channel(uart_channel_t channel, char data);
02382
02383 /**
02384 * @brief Transmit a null-terminated string on specified channel
02385 *
02386 * @param[in] channel SCI channel (0-12)
02387 * @param[in] str Pointer to string
02388 *
02389 * @return k_rx_ok on success,
02390 * k_rx_err_null_ptr if str is nullptr,
02391 * k_rx_err_invalid_arg if channel is invalid,
02392 * k_rx_err_invalid_state if channel not initialized
02393 */
02394 [[nodiscard]] rx_err_t uart_puts_channel(uart_channel_t channel, const char* str);
02395
02396 /**
02397 * @brief Write buffer to specified channel
02398 *
02399 * @param[in] channel SCI channel (0-12)
02400 * @param[in] data Pointer to data buffer
02401 * @param[in] length Number of bytes to write
02402 *
02403 * @return k_rx_ok on success,
02404 * k_rx_err_null_ptr if data is nullptr,
02405 * k_rx_err_invalid_arg if channel is invalid,
02406 * k_rx_err_invalid_state if channel not initialized
02407 */
02408 [[nodiscard]] rx_err_t
02409 uart_write_channel(uart_channel_t channel, const uint8_t* data, uint16_t length);
02410
02411 /**
02412 * @brief Receive a single character from specified channel (non-blocking)
02413 *
02414 * @param[in] channel SCI channel (0-12)
02415 * @param[out] data Pointer to store received character
02416 *
02417 * @return k_rx_ok on success,
02418 * k_rx_err_null_ptr if data is nullptr,
02419 * k_rx_err_invalid_arg if channel is invalid,
02420 * k_rx_err_invalid_state if channel not initialized,
02421 * k_rx_err_empty if no data available
02422 */
02423 [[nodiscard]] rx_err_t uart_getc_channel(uart_channel_t channel, char* data);
02424
02425 /**
02426 * @brief Read available data from specified channel (non-blocking)
02427 *
02428 * Reads up to length bytes that are currently available.
02429 *
02430 * @param[in] channel SCI channel (0-12)
02431 * @param[out] data Pointer to buffer for received data
02432 * @param[in] length Maximum number of bytes to read
02433 * @param[out] bytes_read Pointer to store actual bytes read
02434 *
02435 * @return k_rx_ok on success,
02436 * k_rx_err_null_ptr if data or bytes_read is nullptr,
02437 * k_rx_err_invalid_arg if channel is invalid,
02438 * k_rx_err_invalid_state if channel not initialized
02439 */
02440 [[nodiscard]] rx_err_t
02441 uart_read_channel(uart_channel_t channel, uint8_t* data, uint16_t length, uint16_t* bytes_read);
02442
02443 /**
02444 * @brief Check if receive data is available on channel
02445 *
02446 * @param[in] channel SCI channel (0-12)

```

```

02447 * @param[out] available Pointer to store availability status
02448 *
02449 * @return k_rx_ok on success,
02450 * k_rx_err_null_ptr if available is nullptr,
02451 * k_rx_err_invalid_arg if channel is invalid,
02452 * k_rx_err_invalid_state if channel not initialized
02453 */
02454 [[nodiscard]] rx_err_t uart_rx_available(uart_channel_t channel, bool* available);
02455
02456 /**
02457 * @brief Clear sticky receive error flags (ORER/FER/PER) on a UART channel
02458 *
02459 * @details
02460 * Reads SSR and writes back with the receive error bits masked to 0.
02461 * Per HUM 34.2.7, those flags must be cleared before the receiver will
02462 * latch further bytes. The RXI ISR clears them on every byte; this
02463 * helper is the periodic-fallback path when an error latches without
02464 * RDRF firing (e.g. line glitch on an idle bus).
02465 *
02466 * @param[in] channel SCI channel (0-12).
02467 * @return k_rx_ok on success, k_rx_err_invalid_arg if channel is invalid.
02468 */
02469 [[nodiscard]] rx_err_t uart_clear_rx_errors(uart_channel_t channel);
02470
02471 /*
=====
02472 * UART Debug Functions - Convenience Wrappers for SCI9
02473 *
=====
02474 */
02475
02476 /**
02477 * @brief Initialize debug UART (SCI9, 115200 bps, 8N1)
02478 *
02479 * Convenience wrapper for uart_init_channel(k_uart_debug_channel, 115200).
02480 * Used by rx_log for early boot logging before ThreadX is initialized.
02481 *
02482 * @return k_rx_ok on success, error code on failure
02483 */
02484 [[nodiscard]] rx_err_t uart_debug_init(void);
02485
02486 /**
02487 * @brief Transmit a single character on debug UART (SCI9)
02488 *
02489 * @param[in] data Character to transmit
02490 */
02491 void uart_debug_putc(char data);
02492
02493 /**
02494 * @brief Transmit a null-terminated string on debug UART (SCI9)
02495 *
02496 * @param[in] str Pointer to string
02497 */
02498 void uart_debug_puts(const char* str);
02499
02500 /**
02501 * @brief Print a signed integer on debug UART (SCI9)
02502 *
02503 * @param[in] value Integer value to print
02504 */
02505 void uart_debug_putint(int32_t value);
02506
02507 /**
02508 * @brief Print a hexadecimal value on debug UART (SCI9)
02509 *
02510 * @param[in] value Value to print
02511 * @param[in] digits Number of hex digits (1-8)
02512 */
02513 void uart_debug_puthex(uint32_t value, uint8_t digits);
02514
02515 #ifdef __cplusplus
02516 }
02517 #endif

```

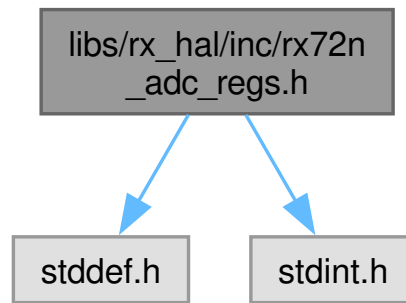
### 8.3 libs/rx`hal/inc/rx72n`adc`regs.h File Reference

RX72N S12AD 12-bit A/D Converter Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_adc\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct [rx\\_s12ad\\_regs\\_t](#)  
S12ADFa Register Map for 12-bit A/D Converter.

## Enumerations

- enum [rx\\_s12ad\\_addresses\\_t](#): [uintptr\\_t](#) { [k\\_s12ad0\\_base\\_addr](#) = 0x00089000 , [k\\_s12ad1\\_base\\_addr](#) = 0x00089100 }  
S12ADFa peripheral base addresses.
- enum [s12ad\\_offsets\\_t](#): [uint8\\_t](#) {  
[k\\_s12ad\\_offset\\_adcsr](#) = 0x00 , [k\\_s12ad\\_offset\\_adansa0](#) = 0x04 , [k\\_s12ad\\_offset\\_adads0](#) = 0x08 ,  
[k\\_s12ad\\_offset\\_adadc](#) = 0x0C ,  
[k\\_s12ad\\_offset\\_adcerc](#) = 0x0E , [k\\_s12ad\\_offset\\_adstrgr](#) = 0x10 , [k\\_s12ad\\_offset\\_adansb0](#) = 0x14 , [k\\_s12ad\\_offset\\_adtbldr](#) ,  
[k\\_s12ad\\_offset\\_addrd](#) = 0x1E , [k\\_s12ad\\_offset\\_addr0](#) = 0x20 , [k\\_s12ad\\_offset\\_addr1](#) = 0x22 ,  
[k\\_s12ad\\_offset\\_addr2](#) = 0x24 ,  
[k\\_s12ad\\_offset\\_addr3](#) = 0x26 , [k\\_s12ad\\_offset\\_addr4](#) = 0x28 , [k\\_s12ad\\_offset\\_addr5](#) = 0x2A ,  
[k\\_s12ad\\_offset\\_addr6](#) = 0x2C ,  
[k\\_s12ad\\_offset\\_addr7](#) = 0x2E }  
S12ADFa register offsets from base address.
- enum [s12ad\\_reserved\\_sizes\\_t](#): [uint8\\_t](#) { [k\\_s12ad\\_reserved\\_02\\_03](#) = 2 , [k\\_s12ad\\_reserved\\_0d](#) = 1 , [k\\_s12ad\\_reserved\\_12\\_13](#) = 2 , [k\\_s12ad\\_reserved\\_16\\_1d](#) = 8 }  
S12ADFa register reserved field sizes for structure padding.

## Functions

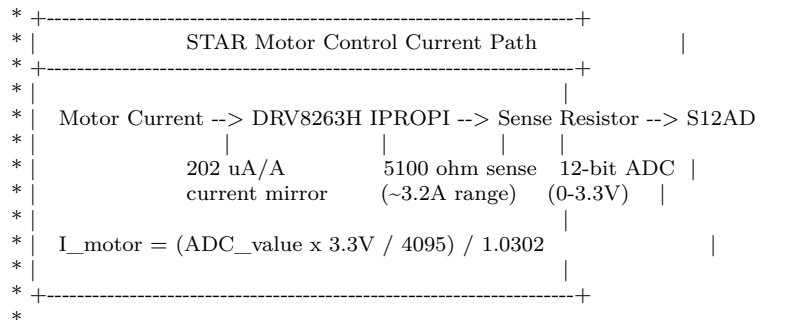
- static volatile [rx\\_s12ad\\_regs\\_t](#) \* [s12ad0](#) (void)  
Get pointer to S12AD0 (Unit 0) registers.
- static volatile [rx\\_s12ad\\_regs\\_t](#) \* [s12ad1](#) (void)  
Get pointer to S12AD1 (Unit 1) registers.

### 8.3.1 Detailed Description

#### RX72N S12AD 12-bit A/D Converter Register Definitions.

This header provides comprehensive register definitions for the RX72N's 12-bit Successive Approximation A/D Converter (S12ADFa). The S12AD is essential for motor current sensing in the STAR project's DRV8263H H-bridge motor control system.

#### System Architecture Context:



#### S12AD Unit Specifications:

| Unit   | Base Address | Channels    | Features                        |
|--------|--------------|-------------|---------------------------------|
| S12AD0 | 0x00089000   | AN000-AN007 | 8ch, high-speed, motor sensing  |
| S12AD1 | 0x00089100   | AN100-AN120 | 21ch, temp sensor, internal ref |

#### Conversion Timing (@ 60 MHz PCLKB):

| Mode        | Conversion Time | Throughput Rate |
|-------------|-----------------|-----------------|
| Single scan | ~1.0 us/ch      | ~1 MSPS         |
| Group scan  | ~0.9 us/ch      | ~1.1 MSPS       |
| Continuous  | ~0.85 us/ch     | ~1.2 MSPS       |

#### Motor Current Sensing Pin Mapping (144-pin LFQFP):

| Motor | DRV8263H Pin | ADC Channel | RX72N Pin |
|-------|--------------|-------------|-----------|
| M0    | IPROPI       | AN007       | P47       |



| Motor | DRV8263H Pin | ADC Channel | RX72N Pin |
|-------|--------------|-------------|-----------|
| M1    | IPROPI       | AN006       | P46       |
| M2    | IPROPI       | AN005       | P45       |
| M3    | IPROPI       | AN004       | P44       |

#### Hardware Requirements:

- CPU: Renesas RX72N @ 240 MHz (ICLK)
- Peripheral clock: PCLKB = 60 MHz (ADC clock source)
- ADC reference: AVCC0 = 3.3V, AVSS0 = GND
- Sample hold: Internal S&H capacitor charged during sampling phase

#### Memory Usage:

- Register structure size: 48 bytes (0x30)
- No dynamic allocation required
- Direct memory-mapped register access

#### See also

docs/sections/03\_hardware\_pinout.tex GPIO pin assignments

lib/rx\_hal/src/adc.c ADC implementation

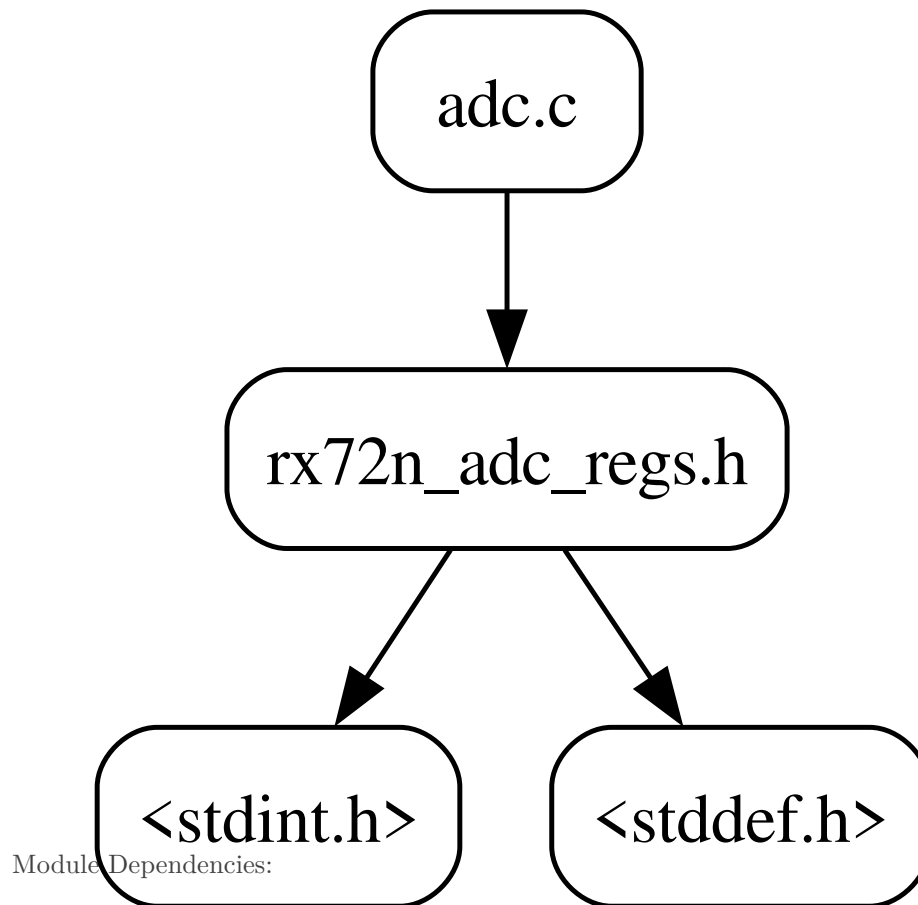
RX72N Group User's Manual: Hardware, Chapter 56 (12-Bit A/D Converter)

#### NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, or recursion in inline functions
- Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
- Rule 8: [OK] All constants as C23 typed enums (base addresses, offsets)
- Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings

#### SOLID Principles:

- S: Single responsibility - only ADC register definitions
- O: Open for extension via accessor functions, closed for modification
- I: Interface segregation - minimal required registers exposed
- D: Dependency inversion - hardware accessed via typed pointers



## Verification Status:

[PASS] VERIFIED (2026-04-20) - All base addresses and register offsets verified against RX72N Manual R01UH0824EJ0111 Table 5.1. Critical bug fixed: spurious reserved5[2] field removed; ADDR0-7 corrected from 0x22-0x30 to 0x20-0x2E.

## Author

Locked, Inc.

## Date

2026-01-28

## Version

1.0.0

## Copyright

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Definition in file [rx72n\\_adc\\_regs.h](#).

## 8.4 rx72n\_adc\_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_adc_regs.h
00003 * @brief RX72N S12AD 12-bit A/D Converter Register Definitions
00004 *
00005 * @details
00006 * This header provides comprehensive register definitions for the RX72N's
00007 * 12-bit Successive Approximation A/D Converter (S12ADFa). The S12AD is
00008 * essential for motor current sensing in the STAR project's DRV8263H H-bridge
00009 * motor control system.
00010 *
00011 * @par System Architecture Context:
00012 * @verbatim
00013 * +-----+
00014 * | STAR Motor Control Current Path |
00015 * +-----+
00016 * | | | |
00017 * | Motor Current --> DRV8263H IPROPI --> Sense Resistor --> S12AD |
00018 * | | | |
00019 * | 202 uA/A | 5100 ohm sense | 12-bit ADC |
00020 * | current mirror | (~3.2A range) | (0-3.3V) |
00021 * | | | |
00022 * | I_motor = (ADC_value x 3.3V / 4095) / 1.0302 |
00023 * | | | |
00024 * +-----+
00025 * @endverbatim
00026 *
00027 * @par S12AD Unit Specifications:
00028 *
00029 * | Unit | Base Address | Channels | Features |
00030 * |-----|-----|-----|-----|
00031 * | S12AD0 | 0x00089000 | AN000-AN007 | 8ch, high-speed, motor sensing |
00032 * | S12AD1 | 0x00089100 | AN100-AN120 | 21ch, temp sensor, internal ref |
00033 *
00034 * @par Conversion Timing (@ 60 MHz PCLKB):
00035 *
00036 * | Mode | Conversion Time | Throughput Rate |
00037 * |-----|-----|-----|
00038 * | Single scan | ~1.0 us/ch | ~1 MSPS |
00039 * | Group scan | ~0.9 us/ch | ~1.1 MSPS |
00040 * | Continuous | ~0.85 us/ch | ~1.2 MSPS |
00041 *
00042 * @par Motor Current Sensing Pin Mapping (144-pin LFQFP):
00043 *
00044 * | Motor | DRV8263H Pin | ADC Channel | RX72N Pin |
00045 * |-----|-----|-----|-----|
00046 * | M0 | IPROPI | AN007 | P47 |
00047 * | M1 | IPROPI | AN006 | P46 |
00048 * | M2 | IPROPI | AN005 | P45 |
00049 * | M3 | IPROPI | AN004 | P44 |
00050 *
00051 * @par Hardware Requirements:
00052 * - CPU: Renesas RX72N @ 240 MHz (ICLK)
00053 * - Peripheral clock: PCLKB = 60 MHz (ADC clock source)
00054 * - ADC reference: AVCC0 = 3.3V, AVSS0 = GND
00055 * - Sample hold: Internal S&H capacitor charged during sampling phase
00056 *
00057 * @par Memory Usage:
00058 * - Register structure size: 48 bytes (0x30)
00059 * - No dynamic allocation required
00060 * - Direct memory-mapped register access
00061 *
00062 * @see docs/sections/03_hardware_pinout.tex GPIO pin assignments
00063 * @see lib/rx_hal/src/adc.c ADC implementation
00064 * @see RX72N Group User's Manual: Hardware, Chapter 56 (12-Bit A/D Converter)
00065 *
00066 * @par NASA Power of 10 Compliance:
00067 * - Rule 1: [OK] No goto, setjmp, or recursion in inline functions
00068 * - Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
00069 * - Rule 8: [OK] All constants as C23 typed enums (base addresses, offsets)
00070 * - Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings
00071 *
00072 * @par SOLID Principles:
00073 * - S: Single responsibility - only ADC register definitions
00074 * - O: Open for extension via accessor functions, closed for modification
00075 * - I: Interface segregation - minimal required registers exposed
00076 * - D: Dependency inversion - hardware accessed via typed pointers
00077 *
00078 * @par Module Dependencies:
00079 * @dot
00080 * digraph adc_deps {
00081 * rankdir=TB;
00082 * node [shape=box, style=rounded];

```

```

00083 * adc_regs [label="rx72n_adc_regs.h"];
00084 * stdint [label="<stdint.h>"];
00085 * stddef [label="<stddef.h>"];
00086 * adc_impl [label="adc.c"];
00087 * adc_regs -> stdint;
00088 * adc_regs -> stddef;
00089 * adc_impl -> adc_regs;
00090 * }
00091 * @enddot
00092 *
00093 * @par Verification Status:
00094 * [PASS] VERIFIED (2026-04-20) - All base addresses and register offsets verified
00095 * against RX72N Manual R01UH0824EJ0111 Table 5.1. Critical bug fixed: spurious
00096 * reserved5[2] field removed; ADDR0-7 corrected from 0x22-0x30 to 0x20-0x2E.
00097 *
00098 * @author Locked, Inc.
00099 * @date 2026-01-28
00100 * @version 1.0.0
00101 * @copyright Copyright (c) 2026 Locked Inc.
00102 * SPDX-License-Identifier: MIT
00103 */
00104
00105 #pragma once
00106
00107 #include <stddef.h>
00108 #include <stdint.h>
00109
00110 #ifdef __cplusplus
00111 extern "C" {
00112 #endif
00113
00114 /**
00115 * @defgroup rx_adc_regs RX72N S12AD Register Definitions
00116 * @brief 12-bit A/D Converter hardware register interface
00117 *
00118 * @details
00119 * Provides memory-mapped register access to the RX72N S12AD peripheral
00120 * for motor current sensing applications. All register addresses and
00121 * offsets are compile-time verified via static assertions.
00122 *
00123 * @par Quick Start Example:
00124 * @code{.c}
00125 * // Read motor 0 current from AN007 (P47)
00126 * volatile rx_s12ad_regs_t* adc = s12ad0();
00127 *
00128 * // Enable channel 7 for single scan
00129 * adc->adansa0 = (1U « 7); // AN007 only
00130 *
00131 * // Configure: single scan, software trigger
00132 * adc->adcsr = 0x0000;
00133 *
00134 * // Start conversion
00135 * adc->adcsr |= 0x8000; // ADST bit
00136 *
00137 * // Wait for completion
00138 * while (adc->adcsr & 0x8000) {}
00139 *
00140 * // Read 12-bit result (right-aligned)
00141 * uint16_t raw_value = adc->addr7;
00142 *
00143 * // Convert to voltage (3.3V reference)
00144 * static const float s_vref_volts = 3.3F;
00145 * static const float s_adc_full_scale_12bit = 4095.0F;
00146 * float voltage_v = (float)raw_value * s_vref_volts / s_adc_full_scale_12bit;
00147 * @endcode
00148 *
00149 * @{
00150 */
00151
00152 /**
=====
00153 * 12-bit A/D Converter (S12ADFa) - For Current Sensing
00154 *
=====
00155 */
00156
00157 /**
00158 * @enum rx_s12ad_addresses_t
00159 * @brief S12ADFa peripheral base addresses
00160 *
00161 * @details
00162 * Hardware base addresses for the two S12AD units on RX72N. Each unit
00163 * occupies 256 bytes (0x100) of address space in the peripheral area.
00164 *
00165 * Per RX72N Manual Ch56 and Ch04 Memory Map:
00166 * - S12AD0: 0x00089000 (Unit 0, 8 high-speed channels)
00167 * - S12AD1: 0x00089100 (Unit 1, 21 channels + temperature sensor)

```

```

00168 *
00169 * @par Address Space Layout:
00170 *
00171 * | Address Range | Description |
00172 * |-----|-----|
00173 * | 0x00089000 - 0x000890FF | S12AD0 registers (Unit 0) |
00174 * | 0x00089100 - 0x000891FF | S12AD1 registers (Unit 1) |
00175 *
00176 * @par Unit Selection Guidelines:
00177 * - Use S12AD0 for motor current sensing (lower latency, 8 channels)
00178 * - Use S12AD1 for temperature monitoring (has internal sensor)
00179 *
00180 * @invariant Base addresses are 256-byte aligned (bits [7:0] = 0)
00181 * @invariant Unit spacing is exactly 0x100 bytes
00182 *
00183 * @see s12ad0() Accessor function for Unit 0
00184 * @see s12ad1() Accessor function for Unit 1
00185 */
00186 typedef enum : uintptr_t {
00187 /**
00188 * @brief S12AD Unit 0 base address
00189 * @details
00190 * 8-channel high-speed ADC optimized for motor current sensing.
00191 * Channels AN000-AN007 mapped to ports P40-P47.
00192 *
00193 * @par Usage: Motor current sensing (all 4 motors)
00194 * @par Throughput: Up to 1 MSPS in continuous scan mode
00195 */
00196 k_s12ad0_base_addr = 0x00089000,
00197
00198 /**
00199 * @brief S12AD Unit 1 base address
00200 * @details
00201 * 21-channel ADC with additional features including temperature
00202 * sensor and internal voltage reference channels.
00203 * Channels AN100-AN120 mapped to various ports.
00204 *
00205 * @par Usage: Temperature monitoring, analog sensing
00206 * @par Special Channels: Temperature sensor (AN127), Ref voltage (AN128)
00207 */
00208 k_s12ad1_base_addr = 0x00089100,
00209 } rx_s12ad_addresses_t;
00210
00211 /**
00212 * @enum s12ad_offsets_t
00213 * @brief S12ADFa register offsets from base address
00214 *
00215 * @details
00216 * Register byte offsets for accessing S12AD peripheral registers.
00217 * All offsets verified against RX72N Manual Ch56, section 56.2.
00218 *
00219 * @par Register Map Overview:
00220 *
00221 * | Offset | Register | Size | Description |
00222 * |-----|-----|-----|-----|
00223 * | 0x00 | ADCSR | 16 | Control/Status (start, mode) |
00224 * | 0x04 | ADANSA0 | 16 | Channel select Group A (0-15) |
00225 * | 0x08 | ADADS0 | 16 | Addition mode select |
00226 * | 0x0C | ADADC | 8 | Addition count (1, 2, 4, 16) |
00227 * | 0x0E | ADCER | 16 | Control extended (alignment) |
00228 * | 0x10 | ADSTRGR | 16 | Trigger source select |
00229 * | 0x14 | ADANSB0 | 16 | Channel select Group B |
00230 * | 0x84 | ADDBLDR | 16 | Double trigger data (not in struct; accessed directly) |
00231 * | 0x1E | ADDR | 16 | Self-diagnosis data |
00232 * | 0x20 | ADDR0 | 16 | Conversion result channel 0 |
00233 * | ... | ... | ... | (2-byte spacing per channel) |
00234 * | 0x2E | ADDR7 | 16 | Conversion result channel 7 |
00235 *
00236 * @note Offsets are 8-bit values (max 255), sufficient for S12AD register space
00237 *
00238 * @invariant Data registers ADDR0-ADDR7 are contiguous with 2-byte spacing
00239 * @invariant All 16-bit registers are halfword-aligned (even addresses)
00240 */
00241 typedef enum : uint8_t {
00242 k_s12ad_offset_adcsr = 0x00, /**< A/D Control Status: start, mode, interrupt control */
00243 k_s12ad_offset_adansa0 = 0x04, /**< Channel Select A0: channels 0-15 for group A scan */
00244 k_s12ad_offset_adads0 = 0x08, /**< Addition Select 0: channels 0-15 for averaging */
00245 k_s12ad_offset_adadc = 0x0C, /**< Addition Count: 1x, 2x, 4x, or 16x oversampling */
00246 k_s12ad_offset_adcerc = 0x0E, /**< Control Extended: data alignment, self-diag mode */
00247 k_s12ad_offset_adstrgr = 0x10, /**< Trigger Select: software, MTU, GPT, external */
00248 k_s12ad_offset_adansb0 = 0x14, /**< Channel Select B0: channels for group B scan */
00249 k_s12ad_offset_addblldr =
00250 0x84, /**< Double Trigger Data: double-trigger mode result (not in struct) */
00251 k_s12ad_offset_addrd = 0x1E, /**< Self-Diagnosis Data: internal reference result */
00252 k_s12ad_offset_addr0 = 0x20, /**< Data Register 0: AN000/AN100 conversion result */
00253 k_s12ad_offset_addr1 = 0x22, /**< Data Register 1: AN001/AN101 conversion result */
00254 k_s12ad_offset_addr2 = 0x24, /**< Data Register 2: AN002/AN102 conversion result */

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```

00255 k_s12ad_offset_addr3 = 0x26, /**< Data Register 3: AN003/AN103 conversion result */
00256 k_s12ad_offset_addr4 = 0x28, /**< Data Register 4: AN004/AN104 conversion result */
00257 k_s12ad_offset_addr5 = 0x2A, /**< Data Register 5: AN005/AN105 conversion result */
00258 k_s12ad_offset_addr6 = 0x2C, /**< Data Register 6: AN006/AN106 conversion result */
00259 k_s12ad_offset_addr7 = 0x2E, /**< Data Register 7: AN007/AN107 conversion result */
00260 } s12ad_offsets_t;
00261
00262 /**
00263 * @enum s12ad_reserved_sizes_t
00264 * @brief S12ADFa register reserved field sizes for structure padding
00265 *
00266 * @details
00267 * Defines byte sizes for reserved regions in the S12AD register map.
00268 * These ensure proper alignment and offset matching between the C
00269 * structure and actual hardware register locations.
00270 *
00271 * @warning Do not modify these values - they are derived from the
00272 * RX72N hardware register layout specification.
00273 *
00274 * @note Reserved bytes must be declared as uint8_t arrays, not accessed
00275 */
00276 typedef enum : uint8_t {
00277 k_s12ad_reserved_02_03 = 2, /**< Padding after ADCSR (16-bit) to ADANSA0 (32-bit aligned) */
00278 k_s12ad_reserved_0d = 1, /**< Padding after ADADC (8-bit) to ADCER (16-bit aligned) */
00279 k_s12ad_reserved_12_13 = 2, /**< Padding after ADSTRGR to ADANSB0 */
00280 k_s12ad_reserved_16_1d = 8, /**< Padding after ADANSB0 to ADRD (ADDBLDR at 0x84, not in struct) */
00281 } s12ad_reserved_sizes_t;
00282
00283 /**
00284 * @struct rx_s12ad_regs_t
00285 * @brief S12ADFa Register Map for 12-bit A/D Converter
00286 *
00287 * @details
00288 * Memory-mapped register structure for the S12AD peripheral. This structure
00289 * provides direct hardware access to ADC control and data registers for
00290 * motor current sensing and analog input measurement.
00291 *
00292 * @par Memory Layout (Total: 48 bytes = 0x30):
00293 *
00294 * | Offset | Size | Field | Description |
00295 * |-----|-----|-----|-----|
00296 * | 0x00 | 2 | adcsr | Control/Status (mode, start, IRQ) |
00297 * | 0x02 | 2 | reserved | Reserved padding |
00298 * | 0x04 | 2 | adansa0 | Channel select A (ch 0-15) |
00299 * | 0x06 | 2 | adansa1 | Channel select A (ch 16-20, Unit1) |
00300 * | 0x08 | 2 | adads0 | Addition mode (ch 0-15) |
00301 * | 0x0A | 2 | adads1 | Addition mode (ch 16-20, Unit1) |
00302 * | 0x0C | 1 | adadc | Addition count (1, 2, 4, 16) |
00303 * | 0x0D | 1 | reserved | Reserved padding |
00304 * | 0x0E | 2 | adcer | Control extended (alignment) |
00305 * | 0x10 | 2 | adstrgr | Trigger source select |
00306 * | 0x12 | 2 | reserved | Reserved padding |
00307 * | 0x14 | 2 | adansb0 | Channel select B (group B scan) |
00308 * | 0x16 | 8 | reserved | Reserved (ADDBLDR at 0x84, not in struct) |
00309 * | 0x1E | 2 | adrd | Self-diagnosis data |
00310 * | 0x20 | 2 | addr0 | Result register AN0/AN100 |
00311 * | 0x22 | 2 | addr1 | Result register AN1/AN101 |
00312 * | 0x24 | 2 | addr2 | Result register AN2/AN102 |
00313 * | 0x26 | 2 | addr3 | Result register AN3/AN103 |
00314 * | 0x28 | 2 | addr4 | Result register AN4/AN104 |
00315 * | 0x2A | 2 | addr5 | Result register AN5/AN105 |
00316 * | 0x2C | 2 | addr6 | Result register AN6/AN106 |
00317 * | 0x2E | 2 | addr7 | Result register AN7/AN107 |
00318 *
00319 * @par Usage Example - Single Channel Conversion:
00320 * @code{.c}
00321 * // Initialize ADC for motor 0 current sensing (AN007, P47)
00322 * volatile rx_s12ad_regs_t* adc = s12ad0();
00323 *
00324 * // 1. Select channel AN007 for group A scan
00325 * adc->adansa0 = (1U << 7); // Bit 7 = AN007
00326 *
00327 * // 2. Configure single scan mode, software trigger
00328 * adc->adcsr = 0x0000; // ADCS[1:0]=00 (single), TRGE=0 (software)
00329 *
00330 * // 3. Set right-alignment (bits 11:0 contain result)
00331 * adc->adcer = 0x0000; // ADRFMT=0 (right-align)
00332 *
00333 * // 4. Start conversion (set ADST bit)
00334 * adc->adcsr |= 0x8000;
00335 *
00336 * // 5. Wait for conversion complete (ADST clears)
00337 * while (adc->adcsr & 0x8000) {
00338 * // ~1 us per conversion
00339 * }
00340 *
00341 * // 6. Read 12-bit result

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00342 * uint16_t raw_count = adc->addr7 & 0x0FFF;
00343 *
00344 * // 7. Convert to current (using DRV8263H-Q1 IPROPI current sensing)
00345 * // IPROPI mirror: 202 uA/A, sense resistor: 5100 ohm
00346 * // V_IPROPI = I_motor * 202e-6 * 5100 = I_motor * 1.0302
00347 * // so I_motor = V_IPROPI / 1.0302
00348 * static const float s_avcc_volts = 3.3F;
00349 * static const float s_adc_full_scale_divisor = 4095.0F; // 2^12 - 1 (12-bit full scale)
00350 * static const float s_ipropi_mirror = 202e-6F; // 202 uA/A current mirror ratio
00351 * static const float s_ipropi_sense_ohm = 5100.0F; // 5.1k sense resistor
00352 * static const float s_ipropi_divisor = 1.0302F; // = 202e-6 * 5100
00353 * float voltage = (float)raw_count * s_avcc_volts / s_adc_full_scale_divisor;
00354 * float current_amps = voltage / s_ipropi_divisor;
00355 * @endcode
00356 *
00357 * @par Usage Example - Continuous 4-Channel Scan:
00358 * @code{.c}
00359 * volatile rx_s12ad_regs_t* adc = s12ad0();
00360 *
00361 * // Enable channels AN004-AN007 (all 4 motor currents)
00362 * adc->adansa0 = 0x00F0; // Bits [7:4] = channels 4-7
00363 *
00364 * // Configure continuous scan mode
00365 * adc->adcsr = 0x4000; // ADCS[1:0]=10 (continuous scan), bits[14:13]=10
00366 *
00367 * // Start continuous conversion
00368 * adc->adcsr |= 0x8000;
00369 *
00370 * // Read results any time (updated continuously)
00371 * uint16_t motor0_current = adc->addr7; // AN007 = M0
00372 * uint16_t motor1_current = adc->addr6; // AN006 = M1
00373 * uint16_t motor2_current = adc->addr5; // AN005 = M2
00374 * uint16_t motor3_current = adc->addr4; // AN004 = M3
00375 *
00376 * // Stop conversion when done
00377 * adc->adcsr &= ~0x8000;
00378 * @endcode
00379 *
00380 * @note This struct covers registers common to both S12AD0 and S12AD1.
00381 * S12AD1 has additional registers (ADDR8-20, temp sensor, etc.)
00382 * beyond offset 0x30 that are not included in this structure.
00383 *
00384 * @warning All registers are memory-mapped and must be accessed as volatile.
00385 * Do not cache register values in local variables for repeated use.
00386 *
00387 * @warning Reserved fields must never be written. Reading returns undefined values.
00388 *
00389 * @invariant sizeof(rx_s12ad_regs_t) == 0x30 (48 bytes)
00390 * @invariant All register offsets match RX72N Manual Ch56 Table 56.1
00391 * @invariant Data registers ADDR0-ADDR7 have 2-byte spacing
00392 *
00393 * @see s12ad0() Accessor for S12AD Unit 0
00394 * @see s12ad1() Accessor for S12AD Unit 1
00395 * @see RX72N Manual Ch56 Section 56.2 for register bit definitions
00396 */
00397 typedef struct {
00398 /**
00399 * @brief A/D Control/Status Register (ADCSR) @ offset 0x00
00400 *
00401 * @details
00402 * Controls conversion mode, trigger source, and interrupt enable.
00403 *
00404 * @par Bit Field Description:
00405 * | Bits | Name | R/W | Description |
00406 * |-----|-----|-----|-----|
00407 * | [15] | ADST | R/W | Conversion start (1=start/running) |
00408 * | [14:13] | ADCS | R/W | Scan mode (00=single, 01=group, 10=continuous) |
00409 * | [12] | ADIE | R/W | Scan end interrupt enable |
00410 * | [9] | TRGE | R/W | Trigger start enable (0=software only, 1=sync/async trigger) |
00411 * | [8] | EXTRG | R/W | Trigger select (0=synchronous, 1=asynchronous) |
00412 * | [7] | DBLE | R/W | Double trigger mode select |
00413 * | [6] | GBADIE | R/W | Group B scan end interrupt enable |
00414 *
00415 * @par Common Configurations:
00416 * - 0x0000: Single scan, software trigger, no interrupt
00417 * - 0x4000: Continuous scan, software trigger
00418 * - 0x8000: Start conversion (write 1 to ADST)
00419 */
00420 volatile uint16_t adcsr;
00421
00422 uint8_t reserved0[k_s12ad_reserved_02_03]; /**< Reserved @ 0x02-0x03 - do not access */
00423
00424 /**
00425 * @brief A/D Channel Select Register A0 (ADANSA0) @ offset 0x04
00426 *
00427 * @details
00428 * Selects channels 0-15 for Group A scan. Each bit enables the

```

```

00429 * corresponding analog input channel.
00430 *
00431 * @par Channel Mapping (S12AD0, 144-pin LFQFP):
00432 * | Bit | Channel | Pin | Usage |
00433 * |-----|-----|-----|-----|
00434 * | 0-3 | AN000-3 | P40-43 | Available |
00435 * | 4 | AN004 | P44 | M3 current |
00436 * | 5 | AN005 | P45 | M2 current |
00437 * | 6 | AN006 | P46 | M1 current |
00438 * | 7 | AN007 | P47 | M0 current |
00439 *
00440 * @par Example: Select motor 0 and 1 current channels
00441 * @code
00442 * adc->adansa0 = 0x00C0; // Enable AN007 (M0), AN006 (M1)
00443 * @endcode
00444 */
00445 volatile uint16_t adansa0;
00446
00447 /**
00448 * @brief A/D Channel Select Register A1 (ADANSA1) @ offset 0x06
00449 *
00450 * @details
00451 * Selects channels 16-20 for Group A scan (S12AD1 only).
00452 * For S12AD0, this register exists but bits [7:0] are reserved.
00453 *
00454 * @note Only valid for S12AD1. S12AD0 has only 8 channels (0-7).
00455 */
00456 volatile uint16_t adansa1;
00457
00458 /**
00459 * @brief A/D Addition Mode Select Register 0 (ADADS0) @ offset 0x08
00460 *
00461 * @details
00462 * Selects channels 0-15 for addition/averaging mode. When enabled,
00463 * multiple conversions are accumulated or averaged per the ADADC setting.
00464 *
00465 * @par Oversampling Benefit:
00466 * - 4x averaging: Reduces noise by ~6 dB (factor of 2)
00467 * - 16x averaging: Reduces noise by ~12 dB (factor of 4)
00468 */
00469 volatile uint16_t adads0;
00470
00471 /**
00472 * @brief A/D Addition Mode Select Register 1 (ADADS1) @ offset 0x0A
00473 *
00474 * @details
00475 * Selects channels 16-20 for addition mode (S12AD1 only).
00476 */
00477 volatile uint16_t adads1;
00478
00479 /**
00480 * @brief A/D Addition Count Select Register (ADADC) @ offset 0x0C
00481 *
00482 * @details
00483 * Sets the number of conversions to add/average for oversampling.
00484 *
00485 * @par Valid Values:
00486 * | Value | Additions | Effect on Data | Time Multiplier |
00487 * |-----|-----|-----|-----|
00488 * | 0x00 | 1x | No addition | 1x |
00489 * | 0x01 | 2x | Sum or average | 2x |
00490 * | 0x02 | 3x | Sum or average | 3x |
00491 * | 0x03 | 4x | Sum or average | 4x |
00492 * | 0x05 | 16x | Sum or average | 16x |
00493 */
00494 volatile uint8_t adadc;
00495
00496 uint8_t reserved1[k_s12ad_reserved_0d]; /**< Reserved @ 0x0D - do not access */
00497
00498 /**
00499 * @brief A/D Control Extended Register (ADCER) @ offset 0x0E
00500 *
00501 * @details
00502 * Configures data alignment and self-diagnosis mode.
00503 *
00504 * @par Key Bits:
00505 * - Bit 15 (ADRFMT): 0=right-align (bits [11:0]), 1=left-align (bits [15:4])
00506 * - Bit 5 (ACE): Auto-clear enable for data registers
00507 *
00508 * @par Alignment Example:
00509 * - Right-align: Result in bits [11:0], bits [15:12] = 0
00510 * - Left-align: Result in bits [15:4], bits [3:0] = 0
00511 */
00512 volatile uint16_t adcer;
00513
00514 /**
00515 * @brief A/D Start Trigger Select Register (ADSTRGR) @ offset 0x10

```



```

00516 *
00517 * @details
00518 * Selects the trigger source for starting A/D conversion when
00519 * hardware triggering is enabled (ADCSR.TRGE = 1).
00520 *
00521 * @par Trigger Sources (TRSA[5:0] at bits [13:8]; TRSB[5:0] at bits [5:0]):
00522 * | Value | Source | Use Case |
00523 * |-----|-----|-----|
00524 * | 0x00 | Software (ADST) | Polled conversion |
00525 * | 0x01 | MTU0 TGIA | Motor PWM sync |
00526 * | 0x02 | MTU0 TGIB | Motor PWM sync |
00527 * | 0x09 | MTU4 TADCORA | MTU4 PWM sync |
00528 */
00529 volatile uint16_t adstrgr;
00530
00531 uint8_t reserved2[k_s12ad_reserved_12_13]; /**< Reserved @ 0x12-0x13 - do not access */
00532
00533 /**
00534 * @brief A/D Channel Select Register B0 (ADANSB0) @ offset 0x14
00535 *
00536 * @details
00537 * Selects channels for Group B scan (dual-group scanning mode).
00538 * Group B runs in parallel with Group A for interleaved sampling.
00539 */
00540 volatile uint16_t adansb0;
00541
00542 uint8_t
00543 reserved3[k_s12ad_reserved_16_1d]; /**< Reserved @ 0x16-0x1D (ADDBLDR at 0x84, not in struct) */
00544
00545 /**
00546 * @brief A/D Self-Diagnosis Data Register (ADDRD) @ offset 0x1E
00547 *
00548 * @details
00549 * Contains result of self-diagnosis conversion when ADCER.DIAGM = 1.
00550 * Used for ADC calibration and health monitoring.
00551 */
00552 volatile uint16_t addrd;
00553
00554 /**
00555 * @brief A/D Data Register 0 (ADDR0) @ offset 0x20
00556 *
00557 * @details
00558 * 12-bit conversion result for channel AN000 (Unit 0) or AN100 (Unit 1).
00559 *
00560 * @par Data Format (ADCER.ADRFMT=0, right-aligned):
00561 * - Bits [11:0]: 12-bit result (0-4095)
00562 * - Bits [15:12]: Always 0
00563 *
00564 * @par Conversion Formula:
00565 * @f$ V_{in} = \frac{ADC_{result}}{4095} \times V_{ref} $ @f$
00566 *
00567 * where Vref = 3.3V (AVCC0)
00568 */
00569 volatile uint16_t addr0;
00570
00571 volatile uint16_t addr1; /**< A/D Data Register 1 @ 0x22: AN001/AN101 */
00572 volatile uint16_t addr2; /**< A/D Data Register 2 @ 0x24: AN002/AN102 */
00573 volatile uint16_t addr3; /**< A/D Data Register 3 @ 0x26: AN003/AN103 */
00574 volatile uint16_t addr4; /**< A/D Data Register 4 @ 0x28: AN004/AN104 (Motor 3 current) */
00575 volatile uint16_t addr5; /**< A/D Data Register 5 @ 0x2A: AN005/AN105 (Motor 2 current) */
00576 volatile uint16_t addr6; /**< A/D Data Register 6 @ 0x2C: AN006/AN106 (Motor 1 current) */
00577 volatile uint16_t addr7; /**< A/D Data Register 7 @ 0x2E: AN007/AN107 (Motor 0 current) */
00578 } rx_s12ad_regs_t;
00579
00580 /**
00581 * @brief Get pointer to S12AD0 (Unit 0) registers
00582 *
00583 * @details
00584 * Returns a volatile pointer to the S12AD Unit 0 register structure,
00585 * enabling direct hardware register access for ADC operations.
00586 * Unit 0 is optimized for motor current sensing with 8 high-speed channels.
00587 *
00588 * @return Volatile pointer to S12AD0 register structure at 0x00089000
00589 *
00590 * @pre Module stop state must be released (MSTPCRA.MSTPA17 = 0)
00591 * @pre AVCC0 and AVSS0 power supplies must be stable
00592 *
00593 * @post Pointer is immediately usable for register access
00594 * @post No hardware state is modified by this function
00595 *
00596 * @note Thread Safety: Function is reentrant and thread-safe (returns constant address)
00597 * @note Performance: Zero overhead - compiles to single load instruction
00598 *
00599 * @par Example - Motor Current Sensing (144-pin LQFP):
00600 * @code{.c}
00601 * // Read all 4 motor currents (AN004-AN007 on P44-P47)
00602 * volatile rx_s12ad_regs_t* adc = s12ad0();

```

```

00603 *
00604 * // Configure for 4-channel scan
00605 * adc->adansa0 = 0x00F0; // AN004-AN007
00606 * adc->adcsr = 0x0000; // Single scan, software trigger
00607 * adc->adcer = 0x0000; // Right-aligned data
00608 *
00609 * // Perform conversion
00610 * adc->adcsr |= 0x8000; // Start
00611 * while (adc->adcsr & 0x8000) {} // Wait (~4 us for 4 channels)
00612 *
00613 * // Read currents (M0=AN007, M1=AN006, M2=AN005, M3=AN004)
00614 * uint16_t motor_currents[4] = {
00615 * adc->addr7, adc->addr6, adc->addr5, adc->addr4
00616 * };
00617 * @endcode
00618 *
00619 * @see s12ad1() For temperature sensing and additional channels
00620 * @see rx_s12ad_regs_t Register structure definition
00621 */
00622 static inline volatile rx_s12ad_regs_t* s12ad0(void)
00623 {
00624 return (volatile rx_s12ad_regs_t*)k_s12ad0_base_addr;
00625 }
00626
00627 /**
00628 * @brief Get pointer to S12AD1 (Unit 1) registers
00629 *
00630 * @details
00631 * Returns a volatile pointer to the S12AD Unit 1 register structure.
00632 * Unit 1 provides 21 analog channels plus temperature sensor and
00633 * internal voltage reference for calibration and monitoring.
00634 *
00635 * @return Volatile pointer to S12AD1 register structure at 0x00089100
00636 *
00637 * @pre Module stop state must be released (MSTPCRA.MSTPA16 = 0)
00638 * @pre AVCC0 and AVSS0 power supplies must be stable
00639 *
00640 * @post Pointer is immediately usable for register access
00641 * @post No hardware state is modified by this function
00642 *
00643 * @note Thread Safety: Function is reentrant and thread-safe
00644 * @note Performance: Zero overhead - compiles to single load instruction
00645 *
00646 * @warning The rx_s12ad_regs_t structure only covers registers up to ADDR7.
00647 * S12AD1 has additional registers (ADDR8-20, temperature sensor,
00648 * internal reference) at higher offsets that require direct access.
00649 *
00650 * @par Example - Temperature Sensor Reading:
00651 * @code{.c}
00652 * volatile rx_s12ad_regs_t* adc = s12ad1();
00653 *
00654 * // Enable internal temperature sensor (AN127)
00655 * // Note: Requires accessing registers beyond rx_s12ad_regs_t
00656 * // See RX72N Manual Ch56 for temperature sensor configuration
00657 * @endcode
00658 *
00659 * @see s12ad0() For motor current sensing (Unit 0)
00660 * @see rx_s12ad_regs_t Register structure definition
00661 */
00662 static inline volatile rx_s12ad_regs_t* s12ad1(void)
00663 {
00664 return (volatile rx_s12ad_regs_t*)k_s12ad1_base_addr;
00665 }
00666
00667 /**
=====
00668 * Static Assertions - Verify Register Layout at Compile Time
00669 *
=====
00670 */
00671
00672 /* Verify base addresses match Hardware Manual Ch56 */
00673 static_assert(k_s12ad0_base_addr == 0x00089000, "S12AD0 base address incorrect");
00674 static_assert(k_s12ad1_base_addr == 0x00089100, "S12AD1 base address incorrect");
00675
00676 /* Verify channel spacing (0x100 bytes between units) */
00677 static_assert((k_s12ad1_base_addr - k_s12ad0_base_addr) == 0x100,
00678 "S12AD0 to S12AD1 spacing incorrect");
00679
00680 /* Verify register structure layout - offsets per RX72N Manual Ch56 */
00681 static_assert(sizeof(rx_s12ad_regs_t) == 0x30,
00682 "S12AD register structure size incorrect (expected 0x30)");
00683 static_assert(offsetof(rx_s12ad_regs_t, adcsr) == 0x00, "ADCSR offset incorrect (expected 0x00)");
00684 static_assert(offsetof(rx_s12ad_regs_t, adansa0) == 0x04,
00685 "ADANSA0 offset incorrect (expected 0x04)");
00686 static_assert(offsetof(rx_s12ad_regs_t, adansa1) == 0x06,
00687 "ADANSA1 offset incorrect (expected 0x06)");

```

```

00688 static_assert(offsetof(rx_s12ad_regs_t, adads0) == 0x08, "ADADS0 offset incorrect (expected 0x08)");
00689 static_assert(offsetof(rx_s12ad_regs_t, adads1) == 0x0A, "ADADS1 offset incorrect (expected 0x0A)");
00690 static_assert(offsetof(rx_s12ad_regs_t, adadc) == 0x0C, "ADADC offset incorrect (expected 0x0C)");
00691 static_assert(offsetof(rx_s12ad_regs_t, adcdr) == 0x0E, "ADCER offset incorrect (expected 0x0E)");
00692 static_assert(offsetof(rx_s12ad_regs_t, adstrgr) == 0x10,
00693 "ADSTRGR offset incorrect (expected 0x10)");
00694 static_assert(offsetof(rx_s12ad_regs_t, adansb0) == 0x14,
00695 "ADANSB0 offset incorrect (expected 0x14)");
00696 static_assert(offsetof(rx_s12ad_regs_t, adrd) == 0x1E, "ADRD offset incorrect (expected 0x1E)");
00697 static_assert(offsetof(rx_s12ad_regs_t, addr0) == k_s12ad_offset_addr0, "ADDR0 offset incorrect");
00698 static_assert(offsetof(rx_s12ad_regs_t, addr1) == k_s12ad_offset_addr1, "ADDR1 offset incorrect");
00699 static_assert(offsetof(rx_s12ad_regs_t, addr2) == k_s12ad_offset_addr2, "ADDR2 offset incorrect");
00700 static_assert(offsetof(rx_s12ad_regs_t, addr3) == k_s12ad_offset_addr3, "ADDR3 offset incorrect");
00701 static_assert(offsetof(rx_s12ad_regs_t, addr4) == k_s12ad_offset_addr4, "ADDR4 offset incorrect");
00702 static_assert(offsetof(rx_s12ad_regs_t, addr5) == k_s12ad_offset_addr5, "ADDR5 offset incorrect");
00703 static_assert(offsetof(rx_s12ad_regs_t, addr6) == k_s12ad_offset_addr6, "ADDR6 offset incorrect");
00704 static_assert(offsetof(rx_s12ad_regs_t, addr7) == k_s12ad_offset_addr7, "ADDR7 offset incorrect");
00705
00706 /** @} */ /* End of rx_adc_regs group */
00707
00708 #ifdef __cplusplus
00709 }
00710 #endif

```

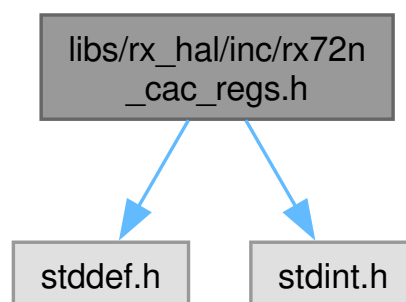
## 8.5 libs/rx72n/inc/rx72n\_cac\_regs.h File Reference

RX72N CAC (Clock Frequency Accuracy Measurement Circuit) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_cac\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct rx\_cac\_regs\_t  
CAC Register Map Structure.

## Enumerations

- enum `rx_cac_addresses_t` : `uintptr_t` { `k_cac_base_addr` = 0x0008B000 }  
CAC base address constant.
- enum `cac_reserved_sizes_t` : `uint8_t` { `k_cac_reserved_after_castr_bytes` = 1 }  
CAC register reserved field sizes.
- enum `cac_cacr0_bits_t` : `uint8_t` { `k_cac_cacr0_cfme_pos` = 0 , `k_cac_cacr0_cfme_mask` = (1U << 0) , `k_cac_cacr0_cfme_stop` = 0x00 , `k_cac_cacr0_cfme_start` = (1U << 0) }  
CAC Control Register 0 (CACR0) bit values.
- enum `cac_cacr1_shifts_t` : `uint8_t` { `k_cac_cacr1_caie_pos` = 0 , `k_cac_cacr1_fmcs_shift` = 1 , `k_cac_cacr1_tcsc_shift` = 4 , `k_cac_cacr1_edges_shift` = 6 }  
CAC Control Register 1 (CACR1) field shift positions.
- enum `cac_cacr1_bits_t` : `uint8_t` {  
`k_cac_cacr1_caie_mask` = (1U << 0) , `k_cac_cacr1_caie_disable` = 0x00 , `k_cac_cacr1_caie_enable` = (1U << 0) , `k_cac_cacr1_fmcs_mask` = (0x07U << 1) ,  
`k_cac_cacr1_fmcs_main` = (0x00U << 1) , `k_cac_cacr1_fmcs_sub` = (0x01U << 1) ,  
`k_cac_cacr1_fmcs_hoco` = (0x02U << 1) , `k_cac_cacr1_fmcs_loco` = (0x03U << 1) ,  
`k_cac_cacr1_fmcs_pclkb` = (0x04U << 1) , `k_cac_cacr1_fmcs_iwdtclk` = (0x05U << 1) ,  
`k_cac_cacr1_fmcs_uclk` = (0x06U << 1) , `k_cac_cacr1_fmcs_clkout25` = (0x07U << 1) ,  
`k_cac_cacr1_tcsc_mask` = (0x03U << 4) , `k_cac_cacr1_tcsc_div_1` = (0x00U << 4) ,  
`k_cac_cacr1_tcsc_div_4` = (0x01U << 4) , `k_cac_cacr1_tcsc_div_8` = (0x02U << 4) ,  
`k_cac_cacr1_tcsc_div_32` = (0x03U << 4) , `k_cac_cacr1_edges_mask` = (0x03U << 6) ,  
`k_cac_cacr1_edges_rising` = (0x00U << 6) , `k_cac_cacr1_edges_falling` = (0x01U << 6) ,  
`k_cac_cacr1_edges_both` = (0x02U << 6) }  
CAC Control Register 1 (CACR1) bit values.
- enum `cac_cacr2_shifts_t` : `uint8_t` { `k_cac_cacr2_rps_pos` = 0 , `k_cac_cacr2_rscs_shift` = 1 , `k_cac_cacr2_rcds_shift` = 4 , `k_cac_cacr2_dfs_shift` = 6 }  
CAC Control Register 2 (CACR2) field shift positions.
- enum `cac_cacr2_bits_t` : `uint8_t` {  
`k_cac_cacr2_rps_mask` = (1U << 0) , `k_cac_cacr2_rps_internal` = 0x00 , `k_cac_cacr2_rps_cacref` = (1U << 0) , `k_cac_cacr2_rscs_mask` = (0x07U << 1) ,  
`k_cac_cacr2_rscs_main` = (0x00U << 1) , `k_cac_cacr2_rscs_sub` = (0x01U << 1) ,  
`k_cac_cacr2_rscs_hoco` = (0x02U << 1) , `k_cac_cacr2_rscs_loco` = (0x03U << 1) ,  
`k_cac_cacr2_rscs_pclkb` = (0x04U << 1) , `k_cac_cacr2_rscs_iwdtclk` = (0x05U << 1) ,  
`k_cac_cacr2_rcds_mask` = (0x03U << 4) , `k_cac_cacr2_rcds_div_32` = (0x00U << 4) ,  
`k_cac_cacr2_rcds_div_128` = (0x01U << 4) , `k_cac_cacr2_rcds_div_1024` = (0x02U << 4) ,  
`k_cac_cacr2_rcds_div_8192` = (0x03U << 4) , `k_cac_cacr2_dfs_mask` = (0x03U << 6) ,  
`k_cac_cacr2_dfs_disabled` = (0x00U << 6) , `k_cac_cacr2_dfs_div_1` = (0x01U << 6) ,  
`k_cac_cacr2_dfs_div_4` = (0x02U << 6) , `k_cac_cacr2_dfs_div_16` = (0x03U << 6) }  
CAC Control Register 2 (CACR2) bit values.
- enum `cac_caicr_bits_t` : `uint8_t` {  
`k_cac_caicr_ferrie_mask` = (1U << 0) , `k_cac_caicr_mendie_mask` = (1U << 1) ,  
`k_cac_caicr_ovfie_mask` = (1U << 2) , `k_cac_caicr_ferrfcl_mask` = (1U << 4) ,  
`k_cac_caicr_mendfcl_mask` = (1U << 5) , `k_cac_caicr_ovffcl_mask` = (1U << 6) ,  
`k_cac_caicr_clear_all_flags` = (1U << 4) | (1U << 5) | (1U << 6) }  
CAC Interrupt Control Register (CAICR) bit values.
- enum `cac_castr_bits_t` : `uint8_t` { `k_cac_castr_ferrf_mask` = (1U << 0) , `k_cac_castr_mendf_mask` = (1U << 1) , `k_cac_castr_ovff_mask` = (1U << 2) }  
CAC Status Register (CASTR) bit values.
- enum `cac_mstp_info_t` : `uint8_t` { `k_cac_mstpc_bit` = 19 }  
Location of CAC module-stop bit in MSTPCRC (RX72N HW Manual pp.410-411).
- enum `cac_groupbl0_bits_t` : `uint8_t` { `k_cac_groupbl0_vector` = 110 , `k_cac_groupbl0_ferrf` = 26 , `k_cac_groupbl0_mendf` = 27 , `k_cac_groupbl0_ovff` = 28 }  
CAC interrupt vector bits inside GROUPBL0 (RX72N HW Manual pp.509-519).

## Functions

- static volatile `rx_cac_regs_t * cac` (void)  
Get pointer to CAC registers.

### 8.5.1 Detailed Description

#### RX72N CAC (Clock Frequency Accuracy Measurement Circuit) Register Definitions.

Register definitions for the Clock Frequency Accuracy Measurement Circuit (CAC) module on the RX72N microcontroller. CAC measures the frequency of a target clock source against a reference clock source and raises an interrupt (FERRF / MENDF / OVFF) when the ratio drifts outside configured bounds.

#### Feature Summary

- 16-bit up-counter clocked by the measurement target clock
- Counter is sampled on the edge of the reference clock
- Upper-limit (CAULVR) / lower-limit (CALLVR) comparators
- FERRF flag + interrupt on frequency-error
- MENDF flag + interrupt on measurement-end
- OVFF flag + interrupt on counter overflow
- Optional digital filter on the external reference clock input (CACREF)

#### Use Case in STAR

MOSC (24 MHz crystal) is measured against HOCO (16 MHz internal RC). A drifted crystal (aging, mechanical damage, PCB leakage) is detected before it corrupts motor control timing; the FERRF ISR drives the motor e-stop path.

#### Hardware Requirements

| Parameter       | Value              | Notes                             |
|-----------------|--------------------|-----------------------------------|
| Base Address    | 0x0008B000         | 11-byte register block            |
| Module Stop Bit | MSTPCRC.MSTPC19    | 1 = stopped, 0 = active           |
| Group Interrupt | GROUPBL0 (vec 110) | Bits: FERRF=26, MENDF=27, OVFF=28 |
| Reset           | PORF               | All registers cleared on reset    |

#### Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0111)

- Chapter 10: Clock Frequency Accuracy Measurement Circuit (CAC)
- Section 10.2: Register Descriptions, pp.392-397
- Address summary: p.236

## NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops in register definitions)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessor functions are single-statement
- Rule 5: [OK] N/A (hardware layer)
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A (no return values to check)
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

See also

[rx\\_cac.h](#) Higher-level CAC driver API

[rx72n\\_system\\_regs.h](#) MSTPCRC and PRCR accessors

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx72n\\_cac\\_regs.h](#).

## 8.5.2 Data Structure Documentation

### 8.5.2.1 struct rx\_cac\_regs\_t

CAC Register Map Structure.

Clock Frequency Accuracy Measurement Circuit registers, laid out to match the hardware memory map described in the RX72N HW Manual Ch10.2 (pp.392-397). Packed to guarantee byte offsets; all fields volatile because hardware can update them (CASTR, CACNTBR) outside of program control.

Memory Layout Table

| Offset | Size | Field     | Type          | Description                            |
|--------|------|-----------|---------------|----------------------------------------|
| 0x00   | 1    | cacr0     | uint8↵<br>_t  | CAC Control Register 0 (p.393)         |
| 0x01   | 1    | cacr1     | uint8↵<br>_t  | CAC Control Register 1 (p.393)         |
| 0x02   | 1    | cacr2     | uint8↵<br>_t  | CAC Control Register 2 (p.394)         |
| 0x03   | 1    | caicr     | uint8↵<br>_t  | CAC Interrupt Control Register (p.395) |
| 0x04   | 1    | castr     | uint8↵<br>_t  | CAC Status Register (p.395)            |
| 0x05   | 1    | reserved0 | -             | Reserved                               |
| 0x06   | 2    | caulvr    | uint16↵<br>_t | CAC Upper-Limit Value Register (p.396) |
| 0x08   | 2    | callvr    | uint16↵<br>_t | CAC Lower-Limit Value Register (p.397) |
| 0x0A   | 2    | cacntbr   | uint16↵<br>_t | CAC Counter Buffer Register (p.397)    |
| Total  | 12   |           |               |                                        |

## Invariant

`sizeof(rx_cac_regs_t) == 12 bytes`

Offsets match RX72N HW Manual R01UH0824EJ0111 Ch10.2

## See also

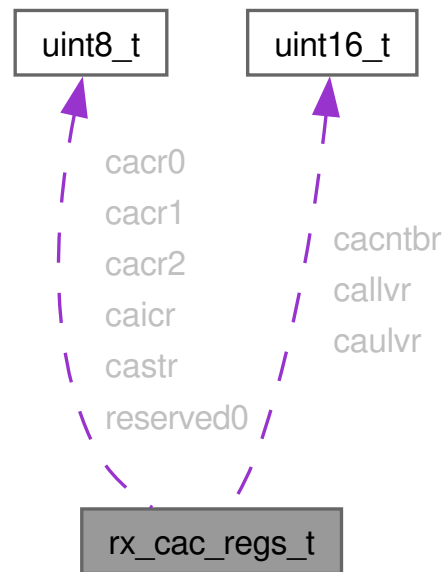
[cac\(\)](#) Accessor function

## Since

Version 1.0.0

Definition at line 151 of file [rx72n\\_cac\\_regs.h](#).

Collaboration diagram for `rx_cac_regs_t`:



## Data Fields

|                   |         |                                                                                                                                                                                                             |
|-------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | cacntbr | CAC Counter Buffer Register (CACNTBR) @ offset 0x0A. 16-bit read-only snapshot of the last completed measurement. RX72N HW Manual p.397.                                                                    |
| volatile uint8_t  | cacr0   | CAC Control Register 0 (CACR0) @ offset 0x00. CFME bit (0) enables/disables frequency measurement (p.393).<br><br>See also<br><a href="#">cac_cacr0_bits_t</a> Bit definitions                              |
| volatile uint8_t  | cacr1   | CAC Control Register 1 (CACR1) @ offset 0x01. Selects measured-clock source, edge, and enables limit comparison. RX72N HW Manual p.393.<br><br>See also<br><a href="#">cac_cacr1_bits_t</a> Bit definitions |



|                   |        |                                                                                                                                                                                                                                                                      |
|-------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | cacr2  | <p>CAC Control Register 2 (CACR2) @ offset 0x02. Reference-clock source, reference-clock divider, and CACREF digital-filter select. RX72N HW Manual p.394.</p> <p>See also</p> <p><a href="#">cac_cacr2_bits_t</a> Bit definitions</p>                               |
| volatile uint8_t  | caicr  | <p>CAC Interrupt Control Register (CAICR) @ offset 0x03. Individual interrupt enables (FERRIE/MENDIE/OVFIE) plus status-flag clear bits (FERRFCL/MENDFCL/OVFFCL). RX72N HW Manual p.395.</p> <p>See also</p> <p><a href="#">cac_caicr_bits_t</a> Bit definitions</p> |
| volatile uint16_t | callvr | <p>CAC Lower-Limit Value Register (CALLVR) @ offset 0x08. 16-bit lower comparison threshold. When the sampled counter is below this value, FERRF is set. RX72N HW Manual p.397.</p> <p>Warning</p> <p>Write only while CACR0.CFME = 0.</p>                           |
| volatile uint8_t  | castr  | <p>CAC Status Register (CASTR) @ offset 0x04. Read-only; hardware sets FERRF (bit 0), MENDF (bit 1), OVFF (bit 2). Software clears via CAICR. RX72N HW Manual p.395.</p> <p>See also</p> <p><a href="#">cac_castr_bits_t</a> Bit definitions</p>                     |

|                   |                                                  |                                                                                                                                                                                                                                           |
|-------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | caulvr                                           | <p>CAC Upper-Limit Value Register (CAULVR) @ offset 0x06. 16-bit upper comparison threshold. When the sampled counter exceeds this value, FERRF is set. RX72N HW Manual p.396.</p> <p>Warning</p> <p>Write only while CACR0.CFME = 0.</p> |
| uint8_t           | reserved0↔<br>[k_cac_reserved_after_castr_bytes] | Reserved @ 0x05                                                                                                                                                                                                                           |

### 8.5.3 Enumeration Type Documentation

#### 8.5.3.1 cac\_cacr0\_bits\_t

enum [cac\\_cacr0\\_bits\\_t](#) : uint8\_t

CAC Control Register 0 (CACR0) bit values.

CACR0 has a single active bit - CFME (Clock Frequency Measurement Enable). All other bits are reserved and must be written 0.

Register Layout (8-bit)

| Bits     | Field    | R/W | Description                           |
|----------|----------|-----|---------------------------------------|
| 0        | CFME     | R/W | 0 = measurement disabled, 1 = enabled |
| 7:↔<br>1 | Reserved | R   | Read as 0, write 0                    |

See also

[rx\\_cac\\_regs\\_t::cacr0](#)

Since

Version 1.0.0

Enumerator

|                       |                               |
|-----------------------|-------------------------------|
| k_cac_cacr0_cfme_pos  | CFME bit position             |
| k_cac_cacr0_cfme_mask | CFME bit mask                 |
| k_cac_cacr0_cfme_stop | Measurement disabled (CFME=0) |

|                        |                              |
|------------------------|------------------------------|
| k_cac_cacr0_cfme_start | Measurement enabled (CFME=1) |
|------------------------|------------------------------|

Definition at line 261 of file rx72n\_cac\_regs.h.

```
00261 : uint8_t {
00262 k_cac_cacr0_cfme_pos = 0, /**< CFME bit position */
00263 k_cac_cacr0_cfme_mask = (1U « 0), /**< CFME bit mask */
00264 k_cac_cacr0_cfme_stop = 0x00, /**< Measurement disabled (CFME=0) */
00265 k_cac_cacr0_cfme_start = (1U « 0), /**< Measurement enabled (CFME=1) */
00266 } cac_cacr0_bits_t;
```

### 8.5.3.2 cac`cacr1`bits`t

enum [cac\\_cacr1\\_bits\\_t](#) : uint8\_t

CAC Control Register 1 (CACR1) bit values.

Selects the measurement target clock (FMCS), its division (TCSS), the sampling edge of CACREF (EDGES), and the CAC-interrupt enable (CAIE).

Register Layout (8-bit) - RX72N HW Manual p.393

| Bits     | Field | R/W | Description                                       |
|----------|-------|-----|---------------------------------------------------|
| 0        | CAIE  | R/W | 0 = IRQ disabled, 1 = enabled                     |
| 3:↔<br>1 | FMCS  | R/W | Measurement target clock source select            |
| 5:↔<br>4 | TCSS  | R/W | Target-clock division (div 1/4/8/32)              |
| 7:↔<br>6 | EDGES | R/W | CACREF edge: rising / falling / both / (reserved) |

FMCS[2:0

target-clock options (RX72N HW Manual p.393)]

| FMCS | Clock source       |
|------|--------------------|
| 000  | Main clock         |
| 001  | Sub-clock          |
| 010  | HOCO               |
| 011  | LOCO               |
| 100  | Peripheral B       |
| 101  | IWDTCLK            |
| 110  | UCLK (USB)         |
| 111  | 25 MHz (CLKOUT25M) |

TCSS[1:0]

target-clock divider (RX72N HW Manual p.393)]

| TCSS | Divider        |
|------|----------------|
| 00   | $\frac{1}{1}$  |
| 01   | $\frac{1}{4}$  |
| 10   | $\frac{1}{8}$  |
| 11   | $\frac{1}{32}$ |

EDGES[1:0]

CACREF valid edge (RX72N HW Manual p.393)]

| EDGES | Edge         |
|-------|--------------|
| 00    | Rising edge  |
| 01    | Falling edge |
| 10    | Both edges   |
| 11    | Reserved     |

See also

[rx\\_cac\\_regs\\_t::cacr1](#)

Since

Version 1.0.0

Enumerator

|                           |                      |
|---------------------------|----------------------|
| k_cac_cacr1_caie_mask     |                      |
| k_cac_cacr1_caie_disable  |                      |
| k_cac_cacr1_caie_enable   |                      |
| k_cac_cacr1_fmcs_mask     |                      |
| k_cac_cacr1_fmcs_main     | Main clock (MOSC)    |
| k_cac_cacr1_fmcs_sub      | Sub-clock oscillator |
| k_cac_cacr1_fmcs_hoco     | HOCO                 |
| k_cac_cacr1_fmcs_loco     | LOCO                 |
| k_cac_cacr1_fmcs_pclkb    | Peripheral clock B   |
| k_cac_cacr1_fmcs_iwdtclk  | IWDT-dedicated clock |
| k_cac_cacr1_fmcs_uclk     | USB clock            |
| k_cac_cacr1_fmcs_clkout25 | 25 MHz CLKOUT        |
| k_cac_cacr1_tcsc_mask     |                      |

|                           |  |
|---------------------------|--|
| k_cac_cacr1_tcsc_div_1    |  |
| k_cac_cacr1_tcsc_div_4    |  |
| k_cac_cacr1_tcsc_div_8    |  |
| k_cac_cacr1_tcsc_div_32   |  |
| k_cac_cacr1_edges_mask    |  |
| k_cac_cacr1_edges_rising  |  |
| k_cac_cacr1_edges_falling |  |
| k_cac_cacr1_edges_both    |  |

Definition at line 332 of file rx72n\_cac\_regs.h.

```

00332 : uint8_t {
00333 /* CAIE - CAC Interrupt Enable */
00334 k_cac_cacr1_caie_mask = (1U « 0),
00335 k_cac_cacr1_caie_disable = 0x00,
00336 k_cac_cacr1_caie_enable = (1U « 0),
00337
00338 /* FMCS[2:0] - Measurement target clock select */
00339 k_cac_cacr1_fmcs_mask = (0x07U « 1),
00340 k_cac_cacr1_fmcs_main = (0x00U « 1), /**< Main clock (MOSC) */
00341 k_cac_cacr1_fmcs_sub = (0x01U « 1), /**< Sub-clock oscillator */
00342 k_cac_cacr1_fmcs_hoco = (0x02U « 1), /**< HOCO */
00343 k_cac_cacr1_fmcs_loco = (0x03U « 1), /**< LOCO */
00344 k_cac_cacr1_fmcs_pclkb = (0x04U « 1), /**< Peripheral clock B */
00345 k_cac_cacr1_fmcs_iwdtclk = (0x05U « 1), /**< IWDt-dedicated clock */
00346 k_cac_cacr1_fmcs_uclk = (0x06U « 1), /**< USB clock */
00347 k_cac_cacr1_fmcs_clkout25 = (0x07U « 1), /**< 25 MHz CLKOUT */
00348
00349 /* TCSS[1:0] - Target-clock division select */
00350 k_cac_cacr1_tcsc_mask = (0x03U « 4),
00351 k_cac_cacr1_tcsc_div_1 = (0x00U « 4),
00352 k_cac_cacr1_tcsc_div_4 = (0x01U « 4),
00353 k_cac_cacr1_tcsc_div_8 = (0x02U « 4),
00354 k_cac_cacr1_tcsc_div_32 = (0x03U « 4),
00355
00356 /* EDGES[1:0] - CACREF valid-edge select */
00357 k_cac_cacr1_edges_mask = (0x03U « 6),
00358 k_cac_cacr1_edges_rising = (0x00U « 6),
00359 k_cac_cacr1_edges_falling = (0x01U « 6),
00360 k_cac_cacr1_edges_both = (0x02U « 6),
00361 } cac_cacr1_bits_t;

```

### 8.5.3.3 cac\_cacr1\_shifts\_t

```
enum cac_cacr1_shifts_t : uint8_t
```

CAC Control Register 1 (CACR1) field shift positions.

Since

Version 1.0.0

Enumerator

|                        |                                             |
|------------------------|---------------------------------------------|
| k_cac_cacr1_caie_pos   | CAIE - CAC interrupt enable bit             |
| k_cac_cacr1_fmcs_shift | FMCS[2:0] - Measurement target clock select |
| k_cac_cacr1_tcsc_shift | TCSS[1:0] - Target-clock division select    |

|                         |                                   |
|-------------------------|-----------------------------------|
| k_cac_cacr1_edges_shift | EDGES[1:0] - Valid edge of CACREF |
|-------------------------|-----------------------------------|

Definition at line 278 of file rx72n\_cac\_regs.h.

```
00278 : uint8_t {
00279 k_cac_cacr1_caie_pos = 0, /**< CAIE - CAC interrupt enable bit */
00280 k_cac_cacr1_fmcs_shift = 1, /**< FMCS[2:0] - Measurement target clock select */
00281 k_cac_cacr1_tcsc_shift = 4, /**< TCSS[1:0] - Target-clock division select */
00282 k_cac_cacr1_edges_shift = 6, /**< EDGES[1:0] - Valid edge of CACREF */
00283 } cac_cacr1_shifts_t;
```

#### 8.5.3.4 cac\_cacr2\_bits\_t

```
enum cac_cacr2_bits_t : uint8_t
```

CAC Control Register 2 (CACR2) bit values.

Selects the reference clock source (RSCS), its divider (RCDS), the digital-filter tap on the external CACREF pin (DFS), and whether the reference clock is taken from the external CACREF pin or the internal source selected by RSCS (RPS).

Register Layout (8-bit) - RX72N HW Manual p.394

| Bits     | Field | R/W | Description                              |
|----------|-------|-----|------------------------------------------|
| 0        | RPS   | R/W | 0 = internal RSCS source, 1 = CACREF pin |
| 3:↔<br>1 | RSCS  | R/W | Internal reference source select         |
| 5:↔<br>4 | RCDS  | R/W | Reference-clock divider                  |
| 7:↔<br>6 | DFS   | R/W | Digital-filter enable for CACREF pin     |

RSCS[2:0]

internal reference source (RX72N HW Manual p.394)]

| RSCS | Source       |
|------|--------------|
| 000  | Main clock   |
| 001  | Sub-clock    |
| 010  | HOCO         |
| 011  | LOCO         |
| 100  | Peripheral B |
| 101  | IWDTCLK      |

## RCDS[1:0]

reference-clock divider (RX72N HW Manual p.394)]

| RCDS | Divider |
|------|---------|
| 00   | /32     |
| 01   | /128    |
| 10   | /1024   |
| 11   | /8192   |

## DFS[1:0]

digital-filter sampling (RX72N HW Manual p.394)]

| DFS | Filter                      |
|-----|-----------------------------|
| 00  | Filter disabled             |
| 01  | Sample at target-clock rate |
| 10  | Sample at target-clock / 4  |
| 11  | Sample at target-clock / 16 |

See also

[rx\\_cac\\_regs\\_t::cacr2](#)

Since

Version 1.0.0

## Enumerator

|                           |                          |
|---------------------------|--------------------------|
| k_cac_cacr2_rps_mask      |                          |
| k_cac_cacr2_rps_internal  | Use internal RSCS source |
| k_cac_cacr2_rps_cacref    | Use external CACREF pin  |
| k_cac_cacr2_rscs_mask     |                          |
| k_cac_cacr2_rscs_main     |                          |
| k_cac_cacr2_rscs_sub      |                          |
| k_cac_cacr2_rscs_hoco     |                          |
| k_cac_cacr2_rscs_loco     |                          |
| k_cac_cacr2_rscs_pclkb    |                          |
| k_cac_cacr2_rscs_iwdtclk  |                          |
| k_cac_cacr2_rcds_mask     |                          |
| k_cac_cacr2_rcds_div_32   |                          |
| k_cac_cacr2_rcds_div_128  |                          |
| k_cac_cacr2_rcds_div_1024 |                          |
| k_cac_cacr2_rcds_div_8192 |                          |

|                          |  |
|--------------------------|--|
| k_cac_cacr2_dfs_mask     |  |
| k_cac_cacr2_dfs_disabled |  |
| k_cac_cacr2_dfs_div_1    |  |
| k_cac_cacr2_dfs_div_4    |  |
| k_cac_cacr2_dfs_div_16   |  |

Definition at line 427 of file [rx72n\\_cac\\_regs.h](#).

```

00427 : uint8_t {
00428 /* RPS - Reference Clock Pin Select */
00429 k_cac_cacr2_rps_mask = (1U « 0),
00430 k_cac_cacr2_rps_internal = 0x00, /*< Use internal RSCS source */
00431 k_cac_cacr2_rps_cacref = (1U « 0), /*< Use external CACREF pin */
00432
00433 /* RSCS[2:0] - Reference-clock source select (ignored when RPS=1) */
00434 k_cac_cacr2_rscs_mask = (0x07U « 1),
00435 k_cac_cacr2_rscs_main = (0x00U « 1),
00436 k_cac_cacr2_rscs_sub = (0x01U « 1),
00437 k_cac_cacr2_rscs_hoco = (0x02U « 1),
00438 k_cac_cacr2_rscs_loco = (0x03U « 1),
00439 k_cac_cacr2_rscs_pclkb = (0x04U « 1),
00440 k_cac_cacr2_rscs_iwdtclk = (0x05U « 1),
00441
00442 /* RCDS[1:0] - Reference-clock divider select (RX72N HW Manual p.394) */
00443 k_cac_cacr2_rcds_mask = (0x03U « 4),
00444 k_cac_cacr2_rcds_div_32 = (0x00U « 4),
00445 k_cac_cacr2_rcds_div_128 = (0x01U « 4),
00446 k_cac_cacr2_rcds_div_1024 = (0x02U « 4),
00447 k_cac_cacr2_rcds_div_8192 = (0x03U « 4),
00448
00449 /* DFS[1:0] - CACREF digital-filter sampling (RX72N HW Manual p.394) */
00450 k_cac_cacr2_dfs_mask = (0x03U « 6),
00451 k_cac_cacr2_dfs_disabled = (0x00U « 6),
00452 k_cac_cacr2_dfs_div_1 = (0x01U « 6),
00453 k_cac_cacr2_dfs_div_4 = (0x02U « 6),
00454 k_cac_cacr2_dfs_div_16 = (0x03U « 6),
00455 } cac_cacr2_bits_t;

```

### 8.5.3.5 cac\_cacr2\_shifts\_t

```
enum cac_cacr2_shifts_t : uint8_t
```

CAC Control Register 2 (CACR2) field shift positions.

Since

Version 1.0.0

Enumerator

|                        |                                            |
|------------------------|--------------------------------------------|
| k_cac_cacr2_rps_pos    | RPS - Reference clock pin select           |
| k_cac_cacr2_rscs_shift | RSCS[2:0] - Reference clock source select  |
| k_cac_cacr2_rcds_shift | RCDS[1:0] - Reference-clock divider select |
| k_cac_cacr2_dfs_shift  | DFS[1:0] - Digital filter select (CACREF)  |

Definition at line 373 of file [rx72n\\_cac\\_regs.h](#).

```

00373 : uint8_t {
00374 k_cac_cacr2_rps_pos = 0, /*< RPS - Reference clock pin select */
00375 k_cac_cacr2_rscs_shift = 1, /*< RSCS[2:0] - Reference clock source select */
00376 k_cac_cacr2_rcds_shift = 4, /*< RCDS[1:0] - Reference-clock divider select */
00377 k_cac_cacr2_dfs_shift = 6, /*< DFS[1:0] - Digital filter select (CACREF) */
00378 } cac_cacr2_shifts_t;

```



## 8.5.3.6 cac\_caicr\_bits\_t

```
enum cac_caicr_bits_t : uint8_t
```

CAC Interrupt Control Register (CAICR) bit values.

Enables the individual interrupt sources (FERRIE/MENDIE/OVFIE) and provides write-1-to-clear bits for the status flags in CASTR (FERRFCL/MENDFCL/OVFFCL).

Register Layout (8-bit) - RX72N HW Manual p.395

| Bit | Field   | R/W | Description                       |
|-----|---------|-----|-----------------------------------|
| 0   | FERRIE  | R/W | Frequency-error interrupt enable  |
| 1   | MENDIE  | R/W | Measurement-end interrupt enable  |
| 2   | OVFIE   | R/W | Counter-overflow interrupt enable |
| 3   | -       | R   | Reserved                          |
| 4   | FERRFCL | W1C | Write 1 to clear CASTR.FERRF      |
| 5   | MENDFCL | W1C | Write 1 to clear CASTR.MENDF      |
| 6   | OVFFCL  | W1C | Write 1 to clear CASTR.OVFF       |
| 7   | -       | R   | Reserved                          |

See also

[rx\\_cac\\_regs\\_t::caicr](#)

Since

Version 1.0.0

Enumerator

|                                             |                                                                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">k_cac_caicr_ferrie_mask</a>     | FERRIE - frequency-error IRQ enable                                                                                                                          |
| <a href="#">k_cac_caicr_mendie_mask</a>     | MENDIE - measurement-end IRQ enable                                                                                                                          |
| <a href="#">k_cac_caicr_ovfie_mask</a>      | OVFIE - overflow IRQ enable                                                                                                                                  |
| <a href="#">k_cac_caicr_ferrfcl_mask</a>    | FERRFCL - write-1-clear FERRF                                                                                                                                |
| <a href="#">k_cac_caicr_mendfcl_mask</a>    | MENDFCL - write-1-clear MENDF                                                                                                                                |
| <a href="#">k_cac_caicr_ovffcl_mask</a>     | OVFFCL - write-1-clear OVFF                                                                                                                                  |
| <a href="#">k_cac_caicr_clear_all_flags</a> | Mask covering all three write-1-clear bits. Write this value to CAICR to acknowledge all pending frequency-error / measurement-end / overflow flags at once. |

Definition at line 485 of file [rx72n\\_cac\\_regs.h](#).

```
00485 : uint8_t {
00486 k_cac_caicr_ferrie_mask = (1U « 0), /**< FERRIE - frequency-error IRQ enable */
00487 k_cac_caicr_mendie_mask = (1U « 1), /**< MENDIE - measurement-end IRQ enable */
00488 k_cac_caicr_ovfie_mask = (1U « 2), /**< OVFIE - overflow IRQ enable */
00489 k_cac_caicr_ferrfcl_mask = (1U « 4), /**< FERRFCL - write-1-clear FERRF */
00490 k_cac_caicr_mendfcl_mask = (1U « 5), /**< MENDFCL - write-1-clear MENDF */
00491 k_cac_caicr_ovffcl_mask = (1U « 6), /**< OVFFCL - write-1-clear OVFF */
00492
00493 /**
00494 * @brief Mask covering all three write-1-clear bits
00495 * @details Write this value to CAICR to acknowledge all pending
00496 * frequency-error / measurement-end / overflow flags at once.
00497 */
00498 k_cac_caicr_clear_all_flags = (1U « 4) | (1U « 5) | (1U « 6),
00499 } cac_caicr_bits_t;
```

### 8.5.3.7 `cac_castr_bits_t`

```
enum cac_castr_bits_t : uint8_t
```

CAC Status Register (CASTR) bit values.

Read-only status flags. Hardware sets each flag on its respective event; software clears them by writing 1 to the corresponding bit in CAICR.

Register Layout (8-bit) - RX72N HW Manual p.395

| Bit      | Field | R | Description                              |
|----------|-------|---|------------------------------------------|
| 0        | FERRF | R | 1 = counter outside [CALLVR, CAULVR]     |
| 1        | MENDF | R | 1 = measurement completed                |
| 2        | OVFF  | R | 1 = counter overflowed (exceeded 0xFFFF) |
| 7:↔<br>3 | -     | R | Reserved, read as 0                      |

See also

[`rx\_cac\_regs\_t::castr`](#)

Since

Version 1.0.0

Enumerator

|                                     |                              |
|-------------------------------------|------------------------------|
| <code>k_cac_castr_ferrf_mask</code> | FERRF - frequency-error flag |
| <code>k_cac_castr_mendf_mask</code> | MENDF - measurement-end flag |
| <code>k_cac_castr_ovff_mask</code>  | OVFF - counter overflow flag |

Definition at line 525 of file [rx72n\\_cac\\_regs.h](#).

```
00525 : uint8_t {
00526 k_cac_castr_ferrf_mask = (1U « 0), /**< FERRF - frequency-error flag */
00527 k_cac_castr_mendf_mask = (1U « 1), /**< MENDF - measurement-end flag */
00528 k_cac_castr_ovff_mask = (1U « 2), /**< OVFF - counter overflow flag */
00529 } cac_castr_bits_t;
```

### 8.5.3.8 `cac_groupbl0_bits_t`

```
enum cac_groupbl0_bits_t : uint8_t
```

CAC interrupt vector bits inside GROUPBL0 (RX72N HW Manual pp.509-519).

CAC shares GROUPBL0 (interrupt vector 110) with several other peripherals. The CAC-specific bits inside GRPBL0 / GCRBL0 / GENBL0 are:

| Bit | Source |
|-----|--------|
| 26  | FERRF  |
| 27  | MENDF  |
| 28  | OVFF   |

Since

Version 1.0.0

Enumerator

|                       |                           |
|-----------------------|---------------------------|
| k_cac_groupbl0_vector | GROUPBL0 vector number    |
| k_cac_groupbl0_ferrf  | FERRF bit inside GROUPBL0 |
| k_cac_groupbl0_mendf  | MENDF bit inside GROUPBL0 |
| k_cac_groupbl0_ovff   | OVFF bit inside GROUPBL0  |

Definition at line 570 of file [rx72n\\_cac\\_regs.h](#).

```

00570 : uint8_t {
00571 k_cac_groupbl0_vector = 110, /**< GROUPBL0 vector number */
00572 k_cac_groupbl0_ferrf = 26, /**< FERRF bit inside GROUPBL0 */
00573 k_cac_groupbl0_mendf = 27, /**< MENDF bit inside GROUPBL0 */
00574 k_cac_groupbl0_ovff = 28, /**< OVFF bit inside GROUPBL0 */
00575 } cac_groupbl0_bits_t;

```

### 8.5.3.9 cac\_mstp\_info\_t

```
enum cac_mstp_info_t : uint8_t
```

Location of CAC module-stop bit in MSTPCRC (RX72N HW Manual pp.410-411).

The CAC module is gated by MSTPCRC.MSTPC19 (bit 19 of the MSTPCRC register at SYSTEM base + 0x18). Writing 1 stops the module, 0 enables it. Access requires prior PRCR unlock (PRC1).

Warning

This is informational only - the driver uses the mstpcrc field of [rx\\_system\\_regs\\_t](#) directly rather than a dedicated accessor.

Since

Version 1.0.0

Enumerator

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| k_cac_mstpc_bit | CAC is MSTPCRC.MSTPC19 (RX72N HW Manual pp.410-411) |
|-----------------|-----------------------------------------------------|

Definition at line 550 of file [rx72n\\_cac\\_regs.h](#).

```

00550 : uint8_t {
00551 k_cac_mstpc_bit = 19, /**< CAC is MSTPCRC.MSTPC19 (RX72N HW Manual pp.410-411) */
00552 } cac_mstp_info_t;

```

### 8.5.3.10 `cac_reserved_sizes_t`

```
enum cac_reserved_sizes_t : uint8_t
```

CAC register reserved field sizes.

Since

Version 1.0.0

Enumerator

|                                               |                                                |
|-----------------------------------------------|------------------------------------------------|
| <code>k_cac_reserved_after_castr_bytes</code> | Reserved byte @ 0x05, between CASTR and CAULVR |
|-----------------------------------------------|------------------------------------------------|

Definition at line 112 of file [rx72n\\_cac\\_regs.h](#).

```
00112 : uint8_t {
00113 k_cac_reserved_after_castr_bytes = 1, /**< Reserved byte @ 0x05, between CASTR and CAULVR */
00114 } cac_reserved_sizes_t;
```

### 8.5.3.11 `rx_cac_addresses_t`

```
enum rx_cac_addresses_t : uintptr_t
```

CAC base address constant.

Base address for the Clock Frequency Accuracy Measurement Circuit register block at 0x0008B000 (RX72N HW Manual R01UH0824EJ0111, p.236 address summary).

Memory Map (RX72N HW Manual Ch10.2, pp.392-397)

| Address    | Register | Size | Description                      |
|------------|----------|------|----------------------------------|
| 0x0008B000 | CACR0    | 1    | CAC Control Register 0           |
| 0x0008B001 | CACR1    | 1    | CAC Control Register 1           |
| 0x0008B002 | CACR2    | 1    | CAC Control Register 2           |
| 0x0008B003 | CAICR    | 1    | CAC Interrupt Control Register   |
| 0x0008B004 | CASTR    | 1    | CAC Status Register              |
| 0x0008B005 | -        | 1    | Reserved                         |
| 0x0008B006 | CAULVR   | 2    | CAC Upper-Limit Value Register   |
| 0x0008B008 | CALLVR   | 2    | CAC Lower-Limit Value Register   |
| 0x0008B00A | CACNTBR  | 2    | CAC Counter Buffer Register (RO) |

Since

Version 1.0.0

Enumerator

|                              |                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------|
| <code>k_cac_base_addr</code> | CAC register base address (0x0008B000). Verified against RX72N HW Manual R01UH0824EJ0111 p.236. |
|------------------------------|-------------------------------------------------------------------------------------------------|

Definition at line 99 of file [rx72n\\_cac\\_regs.h](#).

```

00099 : uintptr_t {
00100 /**
00101 * @brief CAC register base address (0x0008B000)
00102 * @details Verified against RX72N HW Manual R01UH0824EJ0111 p.236.
00103 */
00104 k_cac_base_addr = 0x0008B000,
00105 } rx_cac_addresses_t;

```

## 8.5.4 Function Documentation

### 8.5.4.1 `cac()`

```
volatile rx_cac_regs_t * cac (
 void) [inline], [static]
```

Get pointer to CAC registers.

Returns a volatile pointer to the CAC register structure at `k_cac_base_addr` (0x0008B000).

Returns

Volatile pointer to CAC register structure

Return values

|          |                                                 |
|----------|-------------------------------------------------|
| Non-NULL | Always returns valid pointer (hardware address) |
|----------|-------------------------------------------------|

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

See also

[k\\_cac\\_base\\_addr](#) Base address constant

[rx\\_cac\\_init\(\)](#) Higher-level initialization

Since

Version 1.0.0

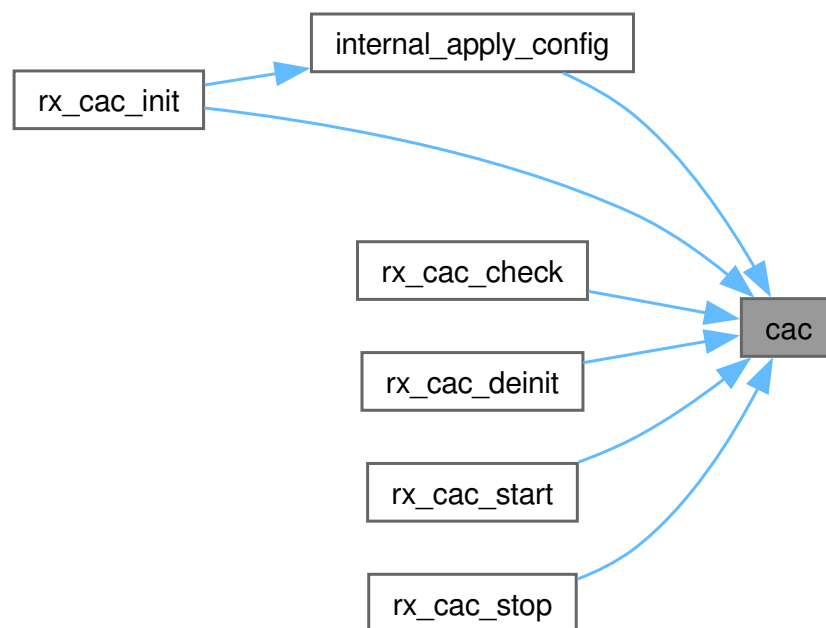
Definition at line 234 of file [rx72n\\_cac\\_regs.h](#).

```
00235 {
00236 return (volatile rx_cac_regs_t*)k_cac_base_addr;
00237 }
```

References [k\\_cac\\_base\\_addr](#).

Referenced by [internal\\_apply\\_config\(\)](#), [rx\\_cac\\_check\(\)](#), [rx\\_cac\\_deinit\(\)](#), [rx\\_cac\\_init\(\)](#), [rx\\_cac\\_start\(\)](#), and [rx\\_cac\\_stop\(\)](#).

Here is the caller graph for this function:



## 8.6 rx72n\_cac\_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002 * @file rx72n_cac_regs.h
00003 * @brief RX72N CAC (Clock Frequency Accuracy Measurement Circuit) Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the Clock Frequency Accuracy Measurement Circuit
00007 * (CAC) module on the RX72N microcontroller. CAC measures the frequency of a
00008 * target clock source against a reference clock source and raises an interrupt
00009 * (FERRF / MENDF / OVFF) when the ratio drifts outside configured bounds.
00010 *
00011 * @par Feature Summary
00012 * - 16-bit up-counter clocked by the measurement target clock
00013 * - Counter is sampled on the edge of the reference clock
00014 * - Upper-limit (CAULVR) / lower-limit (CALLVR) comparators
00015 * - FERRF flag + interrupt on frequency-error
00016 * - MENDF flag + interrupt on measurement-end
00017 * - OVFF flag + interrupt on counter overflow
00018 * - Optional digital filter on the external reference clock input (CACREF)
00019 *
00020 * @par Use Case in STAR
00021 * MOSC (24 MHz crystal) is measured against HOCO (16 MHz internal RC). A
00022 * drifted crystal (aging, mechanical damage, PCB leakage) is detected before
00023 * it corrupts motor control timing; the FERRF ISR drives the motor e-stop
```

```

00024 * path.
00025 *
00026 * @par Hardware Requirements
00027 * | Parameter | Value | Notes |
00028 * |-----|-----|-----|
00029 * | Base Address | 0x0008B000 | 11-byte register block |
00030 * | Module Stop Bit | MSTPCRC.MSTPC19 | 1 = stopped, 0 = active |
00031 * | Group Interrupt | GROUPBL0 (vec 110) | Bits: FERRF=26, MENDF=27, OVFF=28 |
00032 * | Reset | PORF | All registers cleared on reset |
00033 *
00034 * @par Manual Reference
00035 * RX72N Group User's Manual: Hardware (R01UH0824EJ0111)
00036 * - Chapter 10: Clock Frequency Accuracy Measurement Circuit (CAC)
00037 * - Section 10.2: Register Descriptions, pp.392-397
00038 * - Address summary: p.236
00039 *
00040 * @par NASA Power of 10 Compliance
00041 * - Rule 1: [OK] No goto, setjmp, or recursion
00042 * - Rule 2: [OK] N/A (no loops in register definitions)
00043 * - Rule 3: [OK] No dynamic memory allocation
00044 * - Rule 4: [OK] All accessor functions are single-statement
00045 * - Rule 5: [OK] N/A (hardware layer)
00046 * - Rule 6: [OK] Minimal scope
00047 * - Rule 7: [OK] N/A (no return values to check)
00048 * - Rule 8: [OK] All constants use C23 typed enums
00049 * - Rule 9: [OK] No function pointers
00050 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00051 *
00052 * @see rx_cac.h Higher-level CAC driver API
00053 * @see rx72n_system_regs.h MSTPCRC and PRCR accessors
00054 *
00055 * @author Locked, Inc.
00056 * @date 2026-04-21
00057 * @copyright Copyright (c) 2026 Locked Inc.
00058 * SPDX-License-Identifier: MIT
00059 * @since Version 1.0.0
00060 */
00061
00062 #pragma once
00063
00064 #include <stddef.h>
00065 #include <stdint.h>
00066
00067 #ifdef __cplusplus
00068 extern "C" {
00069 #endif
00070
00071 /*
=====
00072 * CAC Base Address (RX72N HW Manual p.236)
00073 *
=====
00074 */
00075
00076 /**
00077 * @enum rx_cac_addresses_t
00078 * @brief CAC base address constant
00079 *
00080 * @details
00081 * Base address for the Clock Frequency Accuracy Measurement Circuit register
00082 * block at 0x0008B000 (RX72N HW Manual R01UH0824EJ0111, p.236 address summary).
00083 *
00084 * @par Memory Map (RX72N HW Manual Ch10.2, pp.392-397)
00085 * | Address | Register | Size | Description |
00086 * |-----|-----|-----|-----|
00087 * | 0x0008B000 | CACR0 | 1 | CAC Control Register 0 |
00088 * | 0x0008B001 | CACR1 | 1 | CAC Control Register 1 |
00089 * | 0x0008B002 | CACR2 | 1 | CAC Control Register 2 |
00090 * | 0x0008B003 | CAICR | 1 | CAC Interrupt Control Register |
00091 * | 0x0008B004 | CASTR | 1 | CAC Status Register |
00092 * | 0x0008B005 | - | 1 | Reserved |
00093 * | 0x0008B006 | CAULVR | 2 | CAC Upper-Limit Value Register |
00094 * | 0x0008B008 | CALLVR | 2 | CAC Lower-Limit Value Register |
00095 * | 0x0008B00A | CACNTBR | 2 | CAC Counter Buffer Register (RO) |
00096 *
00097 * @since Version 1.0.0
00098 */
00099 typedef enum : uintptr_t {
00100 /**
00101 * @brief CAC register base address (0x0008B000)
00102 * @details Verified against RX72N HW Manual R01UH0824EJ0111 p.236.
00103 */
00104 k_cac_base_addr = 0x0008B000,
00105 } rx_cac_addresses_t;
00106
00107 /**
00108 * @enum cac_reserved_sizes_t

```

```

00109 * @brief CAC register reserved field sizes
00110 * @since Version 1.0.0
00111 */
00112 typedef enum : uint8_t {
00113 k_cac_reserved_after_castr_bytes = 1, /**< Reserved byte @ 0x05, between CASTR and CAULVR */
00114 } cac_reserved_sizes_t;
00115
00116 /*
=====
00117 * CAC Register Map Structure
00118 *
=====
00119 */
00120
00121 /**
00122 * @struct rx_cac_regs_t
00123 * @brief CAC Register Map Structure
00124 *
00125 * @details
00126 * Clock Frequency Accuracy Measurement Circuit registers, laid out to match
00127 * the hardware memory map described in the RX72N HW Manual Ch10.2 (pp.392-397).
00128 * Packed to guarantee byte offsets; all fields volatile because hardware can
00129 * update them (CASTR, CACNTBR) outside of program control.
00130 *
00131 * @par Memory Layout Table
00132 * | Offset | Size | Field | Type | Description |
00133 * |-----|-----|-----|-----|-----|
00134 * | 0x00 | 1 | cacr0 | uint8_t | CAC Control Register 0 (p.393) |
00135 * | 0x01 | 1 | cacr1 | uint8_t | CAC Control Register 1 (p.393) |
00136 * | 0x02 | 1 | cacr2 | uint8_t | CAC Control Register 2 (p.394) |
00137 * | 0x03 | 1 | caicr | uint8_t | CAC Interrupt Control Register (p.395)|
00138 * | 0x04 | 1 | castr | uint8_t | CAC Status Register (p.395) |
00139 * | 0x05 | 1 | reserved0 | - | Reserved |
00140 * | 0x06 | 2 | caulvr | uint16_t | CAC Upper-Limit Value Register (p.396)|
00141 * | 0x08 | 2 | callvr | uint16_t | CAC Lower-Limit Value Register (p.397)|
00142 * | 0x0A | 2 | cacntbr | uint16_t | CAC Counter Buffer Register (p.397) |
00143 * | **Total** | **12** | | | |
00144 *
00145 * @invariant sizeof(rx_cac_regs_t) == 12 bytes
00146 * @invariant Offsets match RX72N HW Manual R01UH0824EJ0111 Ch10.2
00147 *
00148 * @see cac() Accessor function
00149 * @since Version 1.0.0
00150 */
00151 typedef struct __attribute__((packed)) {
00152 /**
00153 * @brief CAC Control Register 0 (CACR0) @ offset 0x00
00154 * @details CFME bit (0) enables/disables frequency measurement (p.393).
00155 * @see cac_cacr0_bits_t Bit definitions
00156 */
00157 volatile uint8_t cacr0;
00158
00159 /**
00160 * @brief CAC Control Register 1 (CACR1) @ offset 0x01
00161 * @details Selects measured-clock source, edge, and enables limit comparison.
00162 * RX72N HW Manual p.393.
00163 * @see cac_cacr1_bits_t Bit definitions
00164 */
00165 volatile uint8_t cacr1;
00166
00167 /**
00168 * @brief CAC Control Register 2 (CACR2) @ offset 0x02
00169 * @details Reference-clock source, reference-clock divider, and CACREF
00170 * digital-filter select. RX72N HW Manual p.394.
00171 * @see cac_cacr2_bits_t Bit definitions
00172 */
00173 volatile uint8_t cacr2;
00174
00175 /**
00176 * @brief CAC Interrupt Control Register (CAICR) @ offset 0x03
00177 * @details Individual interrupt enables (FERRIE/MENDIE/OVFIE) plus status-
00178 * flag clear bits (FERRFCL/MENDFCL/OVFFCL). RX72N HW Manual p.395.
00179 * @see cac_caicr_bits_t Bit definitions
00180 */
00181 volatile uint8_t caicr;
00182
00183 /**
00184 * @brief CAC Status Register (CASTR) @ offset 0x04
00185 * @details Read-only; hardware sets FERRF (bit 0), MENDF (bit 1), OVFF
00186 * (bit 2). Software clears via CAICR. RX72N HW Manual p.395.
00187 * @see cac_castr_bits_t Bit definitions
00188 */
00189 volatile uint8_t castr;
00190
00191 uint8_t reserved0[k_cac_reserved_after_castr_bytes]; /**< Reserved @ 0x05 */
00192
00193 /**

```



```

00194 * @brief CAC Upper-Limit Value Register (CAULVR) @ offset 0x06
00195 * @details 16-bit upper comparison threshold. When the sampled counter
00196 * exceeds this value, FERRF is set. RX72N HW Manual p.396.
00197 * @warning Write only while CACR0.CFME = 0.
00198 */
00199 volatile uint16_t caulvr;
00200
00201 /**
00202 * @brief CAC Lower-Limit Value Register (CALLVR) @ offset 0x08
00203 * @details 16-bit lower comparison threshold. When the sampled counter
00204 * is below this value, FERRF is set. RX72N HW Manual p.397.
00205 * @warning Write only while CACR0.CFME = 0.
00206 */
00207 volatile uint16_t callvr;
00208
00209 /**
00210 * @brief CAC Counter Buffer Register (CACNTBR) @ offset 0x0A
00211 * @details 16-bit read-only snapshot of the last completed measurement.
00212 * RX72N HW Manual p.397.
00213 */
00214 volatile uint16_t cacntbr;
00215 } rx_cac_regs_t;
00216
00217 /**
00218 * @brief Get pointer to CAC registers
00219 *
00220 * @details
00221 * Returns a volatile pointer to the CAC register structure at
00222 * k_cac_base_addr (0x0008B000).
00223 *
00224 * @return Volatile pointer to CAC register structure
00225 * @retval Non-NULL Always returns valid pointer (hardware address)
00226 *
00227 * @note Thread Safety: Safe - returns constant hardware address
00228 * @note This function is always inlined for zero overhead
00229 *
00230 * @see k_cac_base_addr Base address constant
00231 * @see rx_cac_init() Higher-level initialization
00232 * @since Version 1.0.0
00233 */
00234 static inline volatile rx_cac_regs_t* cac(void)
00235 {
00236 return (volatile rx_cac_regs_t*)k_cac_base_addr;
00237 }
00238
00239 /*
=====
00240 * CACR0 Bit Definitions (RX72N HW Manual p.393)
00241 *
=====
00242 */
00243
00244 /**
00245 * @enum cac_cacr0_bits_t
00246 * @brief CAC Control Register 0 (CACR0) bit values
00247 *
00248 * @details
00249 * CACR0 has a single active bit - CFME (Clock Frequency Measurement Enable).
00250 * All other bits are reserved and must be written 0.
00251 *
00252 * @par Register Layout (8-bit)
00253 * | Bits | Field | R/W | Description |
00254 * |-----|-----|-----|-----|
00255 * | 0 | CFME | R/W | 0 = measurement disabled, 1 = enabled |
00256 * | 7:1 | Reserved | R | Read as 0, write 0 |
00257 *
00258 * @see rx_cac_regs_t::cacr0
00259 * @since Version 1.0.0
00260 */
00261 typedef enum : uint8_t {
00262 k_cac_cacr0_cfme_pos = 0, /**< CFME bit position */
00263 k_cac_cacr0_cfme_mask = (1U « 0), /**< CFME bit mask */
00264 k_cac_cacr0_cfme_stop = 0x00, /**< Measurement disabled (CFME=0) */
00265 k_cac_cacr0_cfme_start = (1U « 0), /**< Measurement enabled (CFME=1) */
00266 } cac_cacr0_bits_t;
00267
00268 /*
=====
00269 * CACR1 Bit Definitions (RX72N HW Manual p.393)
00270 *
=====
00271 */
00272
00273 /**
00274 * @enum cac_cacr1_shifts_t
00275 * @brief CAC Control Register 1 (CACR1) field shift positions
00276 * @since Version 1.0.0

```

```

00277 */
00278 typedef enum : uint8_t {
00279 k_cac_cacr1_caie_pos = 0, /**< CAIE - CAC interrupt enable bit */
00280 k_cac_cacr1_fmcs_shift = 1, /**< FMCS[2:0] - Measurement target clock select */
00281 k_cac_cacr1_tcss_shift = 4, /**< TCSS[1:0] - Target-clock division select */
00282 k_cac_cacr1_edges_shift = 6, /**< EDGES[1:0] - Valid edge of CACREF */
00283 } cac_cacr1_shifts_t;
00284
00285 /**
00286 * @enum cac_cacr1_bits_t
00287 * @brief CAC Control Register 1 (CACR1) bit values
00288 *
00289 * @details
00290 * Selects the measurement target clock (FMCS), its division (TCSS), the
00291 * sampling edge of CACREF (EDGES), and the CAC-interrupt enable (CAIE).
00292 *
00293 * @par Register Layout (8-bit) - RX72N HW Manual p.393
00294 * | Bits | Field | R/W | Description |
00295 * |-----|-----|-----|-----|
00296 * | 0 | CAIE | R/W | 0 = IRQ disabled, 1 = enabled |
00297 * | 3:1 | FMCS | R/W | Measurement target clock source select |
00298 * | 5:4 | TCSS | R/W | Target-clock division (div 1/4/8/32) |
00299 * | 7:6 | EDGES | R/W | CACREF edge: rising / falling / both / (reserved) |
00300 *
00301 * @par FMCS[2:0] target-clock options (RX72N HW Manual p.393)
00302 * | FMCS | Clock source |
00303 * |-----|-----|
00304 * | 000 | Main clock |
00305 * | 001 | Sub-clock |
00306 * | 010 | HOCO |
00307 * | 011 | LOCO |
00308 * | 100 | Peripheral B |
00309 * | 101 | IWDTCCLK |
00310 * | 110 | UCLK (USB) |
00311 * | 111 | 25 MHz (CLKOUT25M) |
00312 *
00313 * @par TCSS[1:0] target-clock divider (RX72N HW Manual p.393)
00314 * | TCSS | Divider |
00315 * |-----|-----|
00316 * | 00 | /1 |
00317 * | 01 | /4 |
00318 * | 10 | /8 |
00319 * | 11 | /32 |
00320 *
00321 * @par EDGES[1:0] CACREF valid edge (RX72N HW Manual p.393)
00322 * | EDGES | Edge |
00323 * |-----|-----|
00324 * | 00 | Rising edge |
00325 * | 01 | Falling edge |
00326 * | 10 | Both edges |
00327 * | 11 | Reserved |
00328 *
00329 * @see rx_cac_regs_t::cacr1
00330 * @since Version 1.0.0
00331 */
00332 typedef enum : uint8_t {
00333 /** CAIE - CAC Interrupt Enable */
00334 k_cac_cacr1_caie_mask = (1U « 0),
00335 k_cac_cacr1_caie_disable = 0x00,
00336 k_cac_cacr1_caie_enable = (1U « 0),
00337
00338 /** FMCS[2:0] - Measurement target clock select */
00339 k_cac_cacr1_fmcs_mask = (0x07U « 1),
00340 k_cac_cacr1_fmcs_main = (0x00U « 1), /**< Main clock (MOSC) */
00341 k_cac_cacr1_fmcs_sub = (0x01U « 1), /**< Sub-clock oscillator */
00342 k_cac_cacr1_fmcs_hoco = (0x02U « 1), /**< HOCO */
00343 k_cac_cacr1_fmcs_loco = (0x03U « 1), /**< LOCO */
00344 k_cac_cacr1_fmcs_pclb = (0x04U « 1), /**< Peripheral clock B */
00345 k_cac_cacr1_fmcs_iwdtclk = (0x05U « 1), /**< IWDTC-dedicated clock */
00346 k_cac_cacr1_fmcs_uclk = (0x06U « 1), /**< USB clock */
00347 k_cac_cacr1_fmcs_clkout25 = (0x07U « 1), /**< 25 MHz CLKOUT */
00348
00349 /** TCSS[1:0] - Target-clock division select */
00350 k_cac_cacr1_tcss_mask = (0x03U « 4),
00351 k_cac_cacr1_tcss_div_1 = (0x00U « 4),
00352 k_cac_cacr1_tcss_div_4 = (0x01U « 4),
00353 k_cac_cacr1_tcss_div_8 = (0x02U « 4),
00354 k_cac_cacr1_tcss_div_32 = (0x03U « 4),
00355
00356 /** EDGES[1:0] - CACREF valid-edge select */
00357 k_cac_cacr1_edges_mask = (0x03U « 6),
00358 k_cac_cacr1_edges_rising = (0x00U « 6),
00359 k_cac_cacr1_edges_falling = (0x01U « 6),
00360 k_cac_cacr1_edges_both = (0x02U « 6),
00361 } cac_cacr1_bits_t;
00362
00363 */

```

```

=====
00364 * CACR2 Bit Definitions (RX72N HW Manual p.394)
00365 *
=====
00366 */
00367
00368 /**
00369 * @enum cac_cacr2_shifts_t
00370 * @brief CAC Control Register 2 (CACR2) field shift positions
00371 * @since Version 1.0.0
00372 */
00373 typedef enum : uint8_t {
00374 k_cac_cacr2_rps_pos = 0, /**< RPS - Reference clock pin select */
00375 k_cac_cacr2_rscs_shift = 1, /**< RSCS[2:0] - Reference clock source select */
00376 k_cac_cacr2_rcds_shift = 4, /**< RCDS[1:0] - Reference-clock divider select */
00377 k_cac_cacr2_dfs_shift = 6, /**< DFS[1:0] - Digital filter select (CACREF) */
00378 } cac_cacr2_shifts_t;
00379
00380 /**
00381 * @enum cac_cacr2_bits_t
00382 * @brief CAC Control Register 2 (CACR2) bit values
00383 *
00384 * @details
00385 * Selects the reference clock source (RSCS), its divider (RCDS), the
00386 * digital-filter tap on the external CACREF pin (DFS), and whether the
00387 * reference clock is taken from the external CACREF pin or the internal
00388 * source selected by RSCS (RPS).
00389 *
00390 * @par Register Layout (8-bit) - RX72N HW Manual p.394
00391 * | Bits | Field | R/W | Description |
00392 * |-----|-----|-----|-----|
00393 * | 0 | RPS | R/W | 0 = internal RSCS source, 1 = CACREF pin |
00394 * | 3:1 | RSCS | R/W | Internal reference source select |
00395 * | 5:4 | RCDS | R/W | Reference-clock divider |
00396 * | 7:6 | DFS | R/W | Digital-filter enable for CACREF pin |
00397 *
00398 * @par RSCS[2:0] internal reference source (RX72N HW Manual p.394)
00399 * | RSCS | Source |
00400 * |-----|-----|
00401 * | 000 | Main clock |
00402 * | 001 | Sub-clock |
00403 * | 010 | HOCO |
00404 * | 011 | LOCO |
00405 * | 100 | Peripheral B |
00406 * | 101 | IWDTCCLK |
00407 *
00408 * @par RCDS[1:0] reference-clock divider (RX72N HW Manual p.394)
00409 * | RCDS | Divider |
00410 * |-----|-----|
00411 * | 00 | /32 |
00412 * | 01 | /128 |
00413 * | 10 | /1024 |
00414 * | 11 | /8192 |
00415 *
00416 * @par DFS[1:0] digital-filter sampling (RX72N HW Manual p.394)
00417 * | DFS | Filter |
00418 * |-----|-----|
00419 * | 00 | Filter disabled |
00420 * | 01 | Sample at target-clock rate |
00421 * | 10 | Sample at target-clock / 4 |
00422 * | 11 | Sample at target-clock / 16 |
00423 *
00424 * @see rx_cac_regs_t::cacr2
00425 * @since Version 1.0.0
00426 */
00427 typedef enum : uint8_t {
00428 /* RPS - Reference Clock Pin Select */
00429 k_cac_cacr2_rps_mask = (1U « 0),
00430 k_cac_cacr2_rps_internal = 0x00, /**< Use internal RSCS source */
00431 k_cac_cacr2_rps_cacref = (1U « 0), /**< Use external CACREF pin */
00432
00433 /* RSCS[2:0] - Reference-clock source select (ignored when RPS=1) */
00434 k_cac_cacr2_rscs_mask = (0x07U « 1),
00435 k_cac_cacr2_rscs_main = (0x00U « 1),
00436 k_cac_cacr2_rscs_sub = (0x01U « 1),
00437 k_cac_cacr2_rscs_hoco = (0x02U « 1),
00438 k_cac_cacr2_rscs_loco = (0x03U « 1),
00439 k_cac_cacr2_rscs_pclkb = (0x04U « 1),
00440 k_cac_cacr2_rscs_iwdtclk = (0x05U « 1),
00441
00442 /* RCDS[1:0] - Reference-clock divider select (RX72N HW Manual p.394) */
00443 k_cac_cacr2_rcds_mask = (0x03U « 4),
00444 k_cac_cacr2_rcds_div_32 = (0x00U « 4),
00445 k_cac_cacr2_rcds_div_128 = (0x01U « 4),
00446 k_cac_cacr2_rcds_div_1024 = (0x02U « 4),
00447 k_cac_cacr2_rcds_div_8192 = (0x03U « 4),
00448

```

```

00449 /* DFS[1:0] - CACREF digital-filter sampling (RX72N HW Manual p.394) */
00450 k_cac_cacr2_dfs_mask = (0x03U « 6),
00451 k_cac_cacr2_dfs_disabled = (0x00U « 6),
00452 k_cac_cacr2_dfs_div_1 = (0x01U « 6),
00453 k_cac_cacr2_dfs_div_4 = (0x02U « 6),
00454 k_cac_cacr2_dfs_div_16 = (0x03U « 6),
00455 } cac_cacr2_bits_t;
00456
00457 /*
=====
00458 * CAICR Bit Definitions (RX72N HW Manual p.395)
00459 *
=====
00460 */
00461
00462 /**
00463 * @enum cac_caicr_bits_t
00464 * @brief CAC Interrupt Control Register (CAICR) bit values
00465 *
00466 * @details
00467 * Enables the individual interrupt sources (FERRIE/MENDIE/OVFIE) and provides
00468 * write-1-to-clear bits for the status flags in CASTR (FERRFCL/MENDFCL/OVFFCL).
00469 *
00470 * @par Register Layout (8-bit) - RX72N HW Manual p.395
00471 * | Bit | Field | R/W | Description |
00472 * |-----|-----|-----|-----|
00473 * | 0 | FERRIE | R/W | Frequency-error interrupt enable |
00474 * | 1 | MENDIE | R/W | Measurement-end interrupt enable |
00475 * | 2 | OVFIE | R/W | Counter-overflow interrupt enable |
00476 * | 3 | - | R | Reserved |
00477 * | 4 | FERRFCL | W1C | Write 1 to clear CASTR.FERRF |
00478 * | 5 | MENDFCL | W1C | Write 1 to clear CASTR.MENDF |
00479 * | 6 | OVFFCL | W1C | Write 1 to clear CASTR.OVFF |
00480 * | 7 | - | R | Reserved |
00481 *
00482 * @see rx_cac_regs_t::caicr
00483 * @since Version 1.0.0
00484 */
00485 typedef enum : uint8_t {
00486 k_cac_caicr_ferrie_mask = (1U « 0), /**< FERRIE - frequency-error IRQ enable */
00487 k_cac_caicr_mendie_mask = (1U « 1), /**< MENDIE - measurement-end IRQ enable */
00488 k_cac_caicr_ovfie_mask = (1U « 2), /**< OVFIE - overflow IRQ enable */
00489 k_cac_caicr_ferrfcl_mask = (1U « 4), /**< FERRFCL - write-1-clear FERRF */
00490 k_cac_caicr_mendfcl_mask = (1U « 5), /**< MENDFCL - write-1-clear MENDF */
00491 k_cac_caicr_ovffcl_mask = (1U « 6), /**< OVFFCL - write-1-clear OVFF */
00492
00493 /**
00494 * @brief Mask covering all three write-1-clear bits
00495 * @details Write this value to CAICR to acknowledge all pending
00496 * frequency-error / measurement-end / overflow flags at once.
00497 */
00498 k_cac_caicr_clear_all_flags = (1U « 4) | (1U « 5) | (1U « 6),
00499 } cac_caicr_bits_t;
00500
00501 /*
=====
00502 * CASTR Bit Definitions (RX72N HW Manual p.395)
00503 *
=====
00504 */
00505
00506 /**
00507 * @enum cac_castr_bits_t
00508 * @brief CAC Status Register (CASTR) bit values
00509 *
00510 * @details
00511 * Read-only status flags. Hardware sets each flag on its respective event;
00512 * software clears them by writing 1 to the corresponding bit in CAICR.
00513 *
00514 * @par Register Layout (8-bit) - RX72N HW Manual p.395
00515 * | Bit | Field | R | Description |
00516 * |-----|-----|-----|-----|
00517 * | 0 | FERRF | R | 1 = counter outside [CALLVR, CAULVR] |
00518 * | 1 | MENDF | R | 1 = measurement completed |
00519 * | 2 | OVFF | R | 1 = counter overflowed (exceeded 0xFFFF) |
00520 * | 7:3 | - | R | Reserved, read as 0 |
00521 *
00522 * @see rx_cac_regs_t::castr
00523 * @since Version 1.0.0
00524 */
00525 typedef enum : uint8_t {
00526 k_cac_castr_ferrf_mask = (1U « 0), /**< FERRF - frequency-error flag */
00527 k_cac_castr_mendf_mask = (1U « 1), /**< MENDF - measurement-end flag */
00528 k_cac_castr_ovff_mask = (1U « 2), /**< OVFF - counter overflow flag */
00529 } cac_castr_bits_t;
00530
00531 /*

```

```

=====
00532 * Module Stop + Group Interrupt References (cross-reference only)
00533 *
=====
00534 */
00535
00536 /**
00537 * @enum cac_mstp_info_t
00538 * @brief Location of CAC module-stop bit in MSTPCRC (RX72N HW Manual pp.410-411)
00539 *
00540 * @details
00541 * The CAC module is gated by MSTPCRC.MSTPC19 (bit 19 of the MSTPCRC register
00542 * at SYSTEM base + 0x18). Writing 1 stops the module, 0 enables it. Access
00543 * requires prior PRCR unlock (PRC1).
00544 *
00545 * @warning This is informational only - the driver uses the mstpcrc field
00546 * of rx_system_regs_t directly rather than a dedicated accessor.
00547 *
00548 * @since Version 1.0.0
00549 */
00550 typedef enum : uint8_t {
00551 k_cac_mstp_bit = 19, /**< CAC is MSTPCRC.MSTPC19 (RX72N HW Manual pp.410-411) */
00552 } cac_mstp_info_t;
00553
00554 /**
00555 * @enum cac_groupbl0_bits_t
00556 * @brief CAC interrupt vector bits inside GROUPBL0 (RX72N HW Manual pp.509-519)
00557 *
00558 * @details
00559 * CAC shares GROUPBL0 (interrupt vector 110) with several other peripherals.
00560 * The CAC-specific bits inside GRPBL0 / GCRBL0 / GENBL0 are:
00561 *
00562 * | Bit | Source |
00563 * |-----|-----|
00564 * | 26 | FERRF |
00565 * | 27 | MENDF |
00566 * | 28 | OVFF |
00567 *
00568 * @since Version 1.0.0
00569 */
00570 typedef enum : uint8_t {
00571 k_cac_groupbl0_vector = 110, /**< GROUPBL0 vector number */
00572 k_cac_groupbl0_ferrf = 26, /**< FERRF bit inside GROUPBL0 */
00573 k_cac_groupbl0_mendf = 27, /**< MENDF bit inside GROUPBL0 */
00574 k_cac_groupbl0_ovff = 28, /**< OVFF bit inside GROUPBL0 */
00575 } cac_groupbl0_bits_t;
00576
00577 /*
=====
00578 * Static Assertions - Verify register layout at compile time
00579 *
=====
00580 */
00581
00582 /* The k_cac_base_addr enum declaration above is itself the contract for the
00583 * CAC base address (per Hardware Manual). A static_assert that re-states the
00584 * same literal here would duplicate the magic number without adding a real
00585 * cross-check, so we rely on the enum declaration alone. */
00586
00587 static_assert(sizeof(rx_cac_regs_t) == 12, "CAC register structure size incorrect");
00588 static_assert(offsetof(rx_cac_regs_t, cacr0) == 0x00, "CACR0 offset incorrect");
00589 static_assert(offsetof(rx_cac_regs_t, cacr1) == 0x01, "CACR1 offset incorrect");
00590 static_assert(offsetof(rx_cac_regs_t, cacr2) == 0x02, "CACR2 offset incorrect");
00591 static_assert(offsetof(rx_cac_regs_t, caicr) == 0x03, "CAICR offset incorrect");
00592 static_assert(offsetof(rx_cac_regs_t, castr) == 0x04, "CASTR offset incorrect");
00593 static_assert(offsetof(rx_cac_regs_t, caulvr) == 0x06, "CAULVR offset incorrect");
00594 static_assert(offsetof(rx_cac_regs_t, callvr) == 0x08, "CALLVR offset incorrect");
00595 static_assert(offsetof(rx_cac_regs_t, cacntbr) == 0x0A, "CACNTBR offset incorrect");
00596
00597 #ifdef __cplusplus
00598 }
00599 #endif

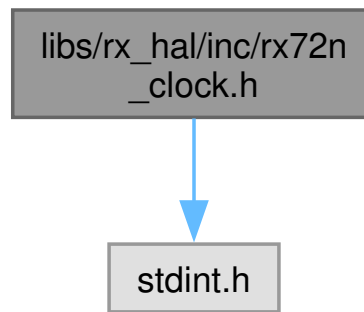
```

## 8.7 libs/rx72n/clock.h File Reference

RX72N Clock Frequency Definitions and Configuration.

```
#include <stdint.h>
```

Include dependency graph for rx72n\_clock.h:



This graph shows which files directly or indirectly include this file:



## Enumerations

- enum `rx_clock_frequencies_t` : `uint32_t` {  
`k_ick_hz` = 24000000UL , `k_pclka_hz` = 120000000UL , `k_pclkb_hz` = 60000000UL ,  
`k_pclkc_hz` = 60000000UL ,  
`k_pclkd_hz` = 60000000UL , `k_bclk_hz` = 120000000UL , `k_fclk_hz` = 60000000UL }

System clock frequencies for RX72N @ 240 MHz operation.

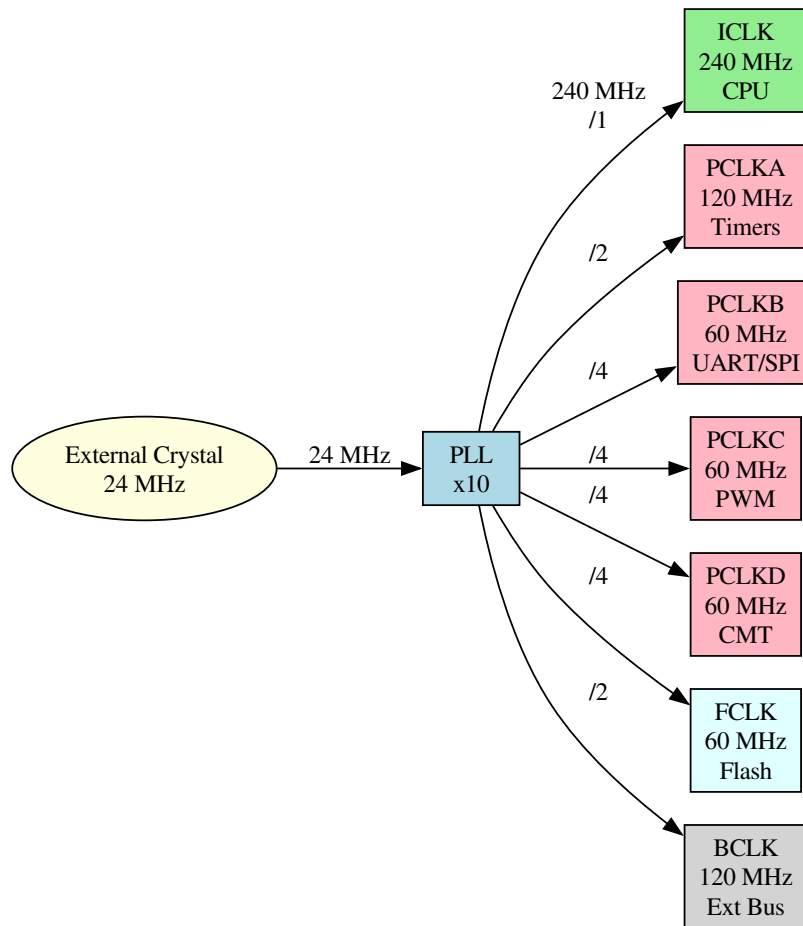
## 8.7.1 Detailed Description

### RX72N Clock Frequency Definitions and Configuration.

Defines all system clock frequencies for the RX72N microcontroller configured with external 24 MHz crystal and PLL x10 for maximum 240 MHz CPU operation. These constants are used throughout the firmware for timing calculations, baud rate generation, PWM frequency configuration, and peripheral initialization.

#### 8.7.1.1 Clock Generation Architecture (STAR Project)

The RX72N has a sophisticated clock generation system with multiple clock domains optimized for different purposes:



### 8.7.1.2 Clock Domain Assignments

| Clock | Frequency | Peripherals            | Purpose                        |
|-------|-----------|------------------------|--------------------------------|
| ICLK  | 240 MHz   | CPU core, DMA          | Maximum instruction throughput |
| PCLKA | 120 MHz   | CMT, TMR, MTU, GPT     | High-resolution timers for PWM |
| PCLKB | 60 MHz    | SCI (UART), RSPI, RIIC | Communication interfaces       |
| PCLKC | 60 MHz    | (Reserved)             | User peripherals               |
| PCLKD | 60 MHz    | (Reserved)             | User peripherals               |
| FCLK  | 60 MHz    | Flash controller       | Flash access (max 60 MHz)      |
| BCLK  | 120 MHz   | External bus           | External memory interface      |

### 8.7.1.3 Clock Configuration (SYSTEM Registers)

Clock frequencies are set during boot by `rx_clock_power_init()`:

1. Main Clock Enable: MOSCWTCR, MOSCCR, MOFCR registers
2. PLL Configuration: PLLCR (x10 multiplier, STC=0x13), PLLCR2 (enable)
3. Clock Dividers: SCKCR register sets ICK, PCKA, PCKB, PCKC, PCKD, FCK, BCK dividers
4. Clock Source Selection: SCKCR3 selects PLL as system clock

Register Configuration (from rx\_clock\_power\_init.c):

```
SYSTEM.SCKCR.LONG = 0x20011222; // ICK=0(/1)->240MHz, BCK=1(/2)->120MHz, PCKA=1(/2)->120MHz,
 PCKB/C/D=2(/4)->60MHz, FCK=2(/4)->60MHz
SYSTEM.PLLCR.WORD = 0x1300; // STC=19 -> x10.0 (manual p345); 24 MHz x 10 = 240 MHz
```

#### 8.7.1.4 Performance Implications

| Operation        | Clock           | Impact of Frequency                             |
|------------------|-----------------|-------------------------------------------------|
| CPU instructions | ICLK (240 MHz)  | ~4.2 ns per cycle, critical for control loops   |
| Flash access     | FCLK (60 MHz)   | ~17 ns per access, affects code execution speed |
| PWM resolution   | PCLKA (120 MHz) | 8.3 ns resolution, 20 kHz PWM = 6000 steps      |
| UART baud rate   | PCLKB (60 MHz)  | Max 3.75 Mbaud (60M/16), typical 115200 baud    |
| Timer tick       | PCLKD (60 MHz)  | ThreadX 10ms tick = 600,000 cycles per tick     |

#### 8.7.1.5 Hardware Requirements

| Requirement         | Value              | Notes                             |
|---------------------|--------------------|-----------------------------------|
| External Crystal    | 24 MHz             | +/-50 ppm accuracy required       |
| PLL Input Range     | 8-24 MHz           | 24 MHz nominal for STAR project   |
| PLL Output Range    | 200-520 MHz        | 240 MHz output (/1 for CPU)       |
| CPU Max Frequency   | 240 MHz            | Absolute max per datasheet        |
| Flash Max Frequency | 60 MHz             | Must not exceed! (hardware limit) |
| Operating Voltage   | 3.3V +/-10%        | Required for 240 MHz operation    |
| Temperature Range   | -40degC to +85degC | Industrial grade                  |

#### 8.7.1.6 Timing Calculation Examples

PWM Frequency:

$\text{PWM\_freq} = \text{PCLKA} / (\text{prescaler} \times \text{period})$   
 20 kHz = 120 MHz / (1 x 6000)

UART Baud Rate:

$\text{Baud\_rate} = \text{PCLKB} / (32 \times (N + 1))$   
 115200 = 60 MHz / (32 x (16 + 1)) -> N=16 (BRR register value)

Timer Period:

$\text{Period\_ms} = (\text{counter\_max} \times \text{prescaler}) / \text{PCLKD\_MHz}$   
 1 ms = (60000 x 1) / 60 MHz

Module Dependencies:

- NO dependencies (pure constant definitions)
- Used by: ALL timing-related modules (CMT, PWM, UART, SPI, ADC, etc.)



NASA Power of 10 Compliance:

- Rule 1: [OK] No code (only data definitions)
- Rule 2: [OK] No loops
- Rule 3: [OK] No dynamic allocation (compile-time constants)
- Rule 4: [OK] No functions
- Rule 5: [OK] N/A (no functions)
- Rule 6: [OK] File scope only (header file)
- Rule 7: [OK] No functions
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [OK] No pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles:

- Single Responsibility: Only defines clock frequencies, no logic
- Open/Closed: Extensible (add new clocks) without modification
- Liskov Substitution: N/A (no inheritance)
- Interface Segregation: Minimal interface (single enum)
- Dependency Inversion: N/A (no dependencies)

Usage Example - Baud Rate Calculation:

```
#include "rx72n_clock.h"

// Calculate UART BRR register value for desired baud rate
uint8_t uart_calculate_brr(uint32_t baud_rate) {
 // Formula: BRR = (PCLKB / (32 x baud)) - 1
 uint32_t brr = (k_pclkb_hz / (32 * baud_rate)) - 1;
 return (uint8_t)brr;
}

// Example: 115200 baud
uint8_t brr = uart_calculate_brr(115200); // Returns 16
```

Usage Example - PWM Configuration:

```
// Configure PWM for 20 kHz at 50% duty cycle
void pwm_configure_20khz(void) {
 typedef enum : uint32_t {
 k_pwm_freq_hz = 20000, // Desired PWM frequency
 k_prescaler = 1, // Timer prescaler
 } pwm_config_t;

 // Calculate timer period: Period = PCLKA / (prescaler x freq)
 uint32_t period = k_pclka_hz / (k_prescaler * k_pwm_freq_hz);
 // period = 120,000,000 / (1 x 20,000) = 6000 counts

 uint32_t duty_50pct = period / 2; // 3000 counts = 50%

 gpt_set_period(GPT_CHANNEL_0, period);
 gpt_set_duty(GPT_CHANNEL_0, duty_50pct);
}
```

## Usage Example - Delay Timing:

```
// Calculate cycles for precise delay
void delay_us(uint32_t us) {
 // Cycles per microsecond at ICLK frequency
 typedef enum : uint32_t {
 k_cycles_per_us = k_ickl_hz / 1000000, // 240 cycles/us
 } delay_config_t;

 uint32_t cycles = us * k_cycles_per_us;

 // Busy-wait loop (NOP instructions)
 for (uint32_t i = 0; i < cycles / 4; i++) {
 __asm volatile("nop\nnop\nnop\nnop");
 }
}
```

## Usage Example - Timer Configuration:

```
// Configure CMT for 1ms tick (1 kHz interrupt)
void cmt_configure_1ms(void) {
 typedef enum : uint32_t {
 k_desired_freq_hz = 1000, // 1 kHz = 1ms period
 k_cmt_prescaler = 8, // CMT prescaler (PCLKD/8)
 } cmt_config_t;

 // Calculate compare match value
 // CMCOR = (PCLKD / prescaler / freq) - 1
 uint32_t cmcor = (k_pclkd_hz / k_cmt_prescaler / k_desired_freq_hz) - 1;
 // cmcor = (60,000,000 / 8 / 1000) - 1 = 7499

 CMT0.CMCOR = (uint16_t)cmcor;
}
```

See also

[RX72N User's Manual](#) Chapter 9: Clock Generation Circuit

[RX72N Group Hardware Manual](#) Section 59: Electrical Characteristics

[rx\\_clock\\_power\\_init.c](#) Clock initialization code

[hardware\\_init.c](#) Calls clock initialization

Author

Locked, Inc.

Date

2026-01-27

Version

1.0.0

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Since

Version 1.0.0

Definition in file [rx72n\\_clock.h](#).

## 8.7.2 Enumeration Type Documentation

### 8.7.2.1 rx\_clock\_frequencies\_t

```
enum rx_clock_frequencies_t : uint32_t
```

System clock frequencies for RX72N @ 240 MHz operation.

Defines all clock domain frequencies for the STAR project's RX72N configuration. These values are determined by the PLL and clock divider settings in rx\_clock\_power\_init() and are guaranteed to be accurate for timing calculations.

Clock Source: 24 MHz external crystal -> PLL (x10) -> 240 MHz -> divided per domain

Frequency Derivation:

- PLL output: 24 MHz x 10 = 240 MHz
- ICLK: 240 MHz / 1 = 240 MHz (CPU)
- PCLKA: 240 MHz / 2 = 120 MHz (high-speed peripherals)
- PCLKB/C/D: 240 MHz / 4 = 60 MHz (standard peripherals)
- FCLK: 240 MHz / 4 = 60 MHz (flash - max allowed)
- BCLK: 240 MHz / 2 = 120 MHz (external bus - not used in STAR)

### 8.7.2.2 Clock Domain Usage Table

| Clock | Freq    | Used By            | STAR Project Usage                          |
|-------|---------|--------------------|---------------------------------------------|
| ICLK  | 240 MHz | CPU core           | PID control loop, communication protocol    |
| PCLKA | 120 MHz | MTU, GPT, CMT, TMR | Encoder capture (MTU), PWM generation (GPT) |
| PCLKB | 60 MHz  | SCI, RSPI, RIIC    | UART debug, SPI to RPi5, I2C sensors        |
| PCLKC | 60 MHz  | (Reserved)         | Not used in STAR project                    |
| PCLKD | 60 MHz  | CMT                | ThreadX system tick (10ms)                  |
| FCLK  | 60 MHz  | Flash              | Code execution, const data access           |
| BCLK  | 120 MHz | External bus       | Not used (no external memory)               |

Field Constraints:

| Constant   | Min   | Max     | Notes                                     |
|------------|-------|---------|-------------------------------------------|
| k_ick_hz   | 1 MHz | 240 MHz | 240 MHz is absolute max per datasheet     |
| k_pclka_hz | 1 MHz | 240 MHz | Limited by ICLK (cannot exceed CPU clock) |
| k_pclkb_hz | 1 MHz | 60 MHz  | Communication peripherals rated to 60 MHz |
| k_fclk_hz  | 1 MHz | 60 MHz  | MUST NOT EXCEED 60 MHz (flash limit)      |

Usage Example - Calculate Timer Period:

```
// Calculate CMT compare value for 100 Hz interrupt (10ms period)
typedef enum : uint32_t {
 k_desired_freq_hz = 100, // 100 Hz = 10ms period
 k_cmt_prescaler = 8, // PCLKD/8
} timer_config_t;

uint32_t cmcor = (k_pclkd_hz / k_cmt_prescaler / k_desired_freq_hz) - 1;
// cmcor = (60,000,000 / 8 / 100) - 1 = 74,999
```

See also

`rx_clock_power_init()` Function that configures these frequencies

Since

Version 1.0.0

## Enumerator

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_ick_hz | <p>CPU core clock frequency (ICK) - 240 MHz. The instruction clock (ICK) drives the RX72N CPU core, DMA controller, and internal buses. This is the primary determinant of computational throughput and interrupt latency.</p> <p>Value: 240,000,000 Hz (240 MHz) Source: PLL output (240 MHz) divided by 1 Cycle time: 4.17 nanoseconds per cycle Register: SYSTEM.SCKCR.ICK (divider = b'0000' = /1 from PLL)</p> <p>Performance Metrics @ 240 MHz:</p> <ul style="list-style-type: none"> <li>• Single-cycle instructions: 4.17 ns (e.g., MOV, ADD)</li> <li>• Flash access: ~20 ns (5 cycles with wait states)</li> <li>• Branch misprediction penalty: ~8 ns (2 cycles)</li> <li>• Interrupt latency: ~200 ns (minimum, no context save)</li> </ul> <p>Use Cases in STAR Project:</p> <ul style="list-style-type: none"> <li>• PID control loop execution (~2 us per motor update @ -O2)</li> <li>• Protocol parsing (nanopb decode ~50 us for 256-byte frame)</li> <li>• Floating-point math (division ~20 cycles = 83 ns)</li> <li>• Critical timing operations (encoder pulse measurement)</li> </ul> <p>Rationale: Maximum frequency selected for:</p> <ul style="list-style-type: none"> <li>• Minimize control loop latency (100 Hz loop = 10ms budget)</li> <li>• Fast protocol processing (SPI @ 10 Mbps = 25.6 us per 32-byte frame)</li> <li>• Headroom for future features (current usage ~30% of CPU)</li> </ul> <p>Impact of Changing:</p> <ul style="list-style-type: none"> <li>• Increase: Not possible (240 MHz is absolute max)</li> <li>• Decrease: Proportional reduction in throughput, may cause control loop overruns</li> </ul> <p>Warning</p> <p>Flash wait states increase CPU power consumption. Use -O2 optimization to minimize instruction count and reduce power.</p> <p>See also</p> <p><a href="#">k_fclk_hz</a> Flash clock must stay &lt;=60 MHz regardless of ICLK</p> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_pclka_hz | <p>Peripheral Clock A (PCLKA) - 120 MHz. High-speed peripheral clock for time-critical functions: timers for PWM generation (GPT), encoder capture (MTU), and high-resolution timing (CMT).</p> <p>Value: 120,000,000 Hz (120 MHz) Source: 24 MHz main oscillator -&gt; PLL x10 = 240 MHz -&gt; PCKA divider /2 Cycle time: ~8.33 nanoseconds per cycle Register: SYSTEM.SCKCR = 0x20011222 (PCKA[15:12] = 1 = divide-by-2 from 240 MHz PLL)</p> <p>Connected Peripherals:</p> <ul style="list-style-type: none"> <li>• MTU (Multi-Function Timer): Encoder quadrature capture (4 motors)</li> <li>• GPT (General PWM Timer): Motor PWM generation (4 H-bridges)</li> <li>• CMT (Compare Match Timer): System tick, delay timers</li> <li>• TMR (8/16-bit Timer): Miscellaneous timing functions</li> </ul> <p>Performance Analysis:</p> <p>PWM Resolution @ 20 kHz:</p> <p>Period = <math>PCLKA / PWM\_freq = 120,000,000 / 20,000 = 6000</math> counts<br/> Resolution = <math>100\% / 6000 = 0.0167\%</math> per count<br/> Voltage resolution (12V supply) = <math>12V \times 0.0167\% = 2</math> mV per step</p> <p>Encoder Capture Precision:</p> <p>Max velocity = 2.5 m/s (STAR robot spec)<br/> Wheel diameter = 0.1 m -&gt; circumference = 0.314 m<br/> Max RPM = <math>(2.5 \text{ m/s} / 0.314 \text{ m}) \times 60 = 477</math> RPM<br/> Encoder PPR = <math>341 \times 4</math> (quadrature) = 1364 edges/rev<br/> Max edge rate = <math>477 \text{ RPM} \times 1364 \text{ edges} / 60 = 10,836</math> edges/sec<br/> Min edge period = <math>1 / 10,836 = 92.3</math> us<br/> Timer counts per edge @ 120 MHz = <math>92.3 \text{ us} \times 120 \text{ MHz} = 11,076</math> counts<br/> Velocity resolution = <math>1 \text{ count} / 11,076 = 0.009\%</math> per timer count [OK]</p> <p>Rationale: 120 MHz chosen for:</p> <ul style="list-style-type: none"> <li>• PWM resolution: 6000 steps @ 20 kHz (excellent granularity)</li> <li>• Encoder precision: 0.009% velocity measurement resolution</li> <li>• Timer headroom: Max edge rate only uses 10% of timer bandwidth</li> </ul> <p>Impact of Changing:</p> <ul style="list-style-type: none"> <li>• Increase to 240 MHz: Double PWM resolution (12000 steps), but wastes power</li> <li>• Decrease to 60 MHz: Half PWM resolution (3000 steps), still acceptable</li> </ul> <p>Warning</p> <p>GPT, MTU, CMT baud rate calculations assume 120 MHz. Changing PCLKA requires recalculating all timer prescalers and compare values.</p> |
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| k_pclkb_hz | <p>Peripheral Clock B (PCLKB) - 60 MHz. Standard-speed peripheral clock for communication interfaces: UART serial console, SPI to Raspberry Pi 5, and I2C sensor bus.</p> <p>Value: 60,000,000 Hz (60 MHz) Source: ICLK (240 MHz) divided by 4 Cycle time: 16.67 nanoseconds per cycle Register: SYSTEM.SCKCR.PCKB (divider = b'0010' = /4 from ICLK)</p> <p>Connected Peripherals:</p> <ul style="list-style-type: none"> <li>• SCI (Serial Communication Interface): UART for USB CDC debug console</li> <li>• RSPI (Renesas SPI): SPI controller for RPi5 communication</li> <li>• RIIC (Renesas I2C): I2C controller for sensor bus (IMU, lidar interface)</li> </ul> <p>Baud Rate Capabilities:</p> <p>UART (SCI) @ 115200 baud:</p> $\text{BRR} = (\text{PCLKB} / (32 \times \text{baud\_rate})) - 1$ $\text{BRR} = (60,000,000 / (32 \times 115200)) - 1 = 16.276... \sim 16$ <p>Actual baud = 60,000,000 / (32 x 17) = 110,294 baud</p> <p>Error = (110,294 - 115200) / 115200 = -4.3% (within +/-5% tolerance) [OK]</p> <p>SPI (RSPI) @ 10 Mbps (STAR project: communication with RPi5):</p> $\text{SPI\_bitrate} = \text{PCLKB} / (2 \times (\text{SPBR} + 1))$ $10 \text{ Mbps} = 60,000,000 / (2 \times (\text{SPBR} + 1))$ $\text{SPBR} = (60,000,000 / (2 \times 10,000,000)) - 1 = 2$ <p>Actual rate = 60,000,000 / (2 x 3) = 10 Mbps exact [OK]</p> <p>I2C (RIIC) @ 400 kHz (Fast-mode I2C for IMU):</p> $\text{I2C\_freq} = \text{PCLKB} / (\text{CKS} \times (\text{BRL} + \text{BRH} + 4))$ <p>400 kHz typical with CKS=0 (prescaler /1), BRL=BRH=73</p> <p>Actual = 60,000,000 / (1 x (73 + 73 + 4)) = 400 kHz exact [OK]</p> <p>Rationale: 60 MHz chosen for:</p> <ul style="list-style-type: none"> <li>• UART: Standard baud rates (9600, 19200, 115200) achievable with &lt;5% error</li> <li>• SPI: 10 Mbps target rate (STAR requirement) achievable exactly</li> <li>• I2C: 100 kHz (standard) and 400 kHz (fast-mode) both supported</li> <li>• Power: Lower clock = lower power for communication peripherals</li> </ul> <p>Impact of Changing:</p> <ul style="list-style-type: none"> <li>• Increase to 120 MHz: Would allow 20 Mbps SPI, but RPi5 limited to 10 Mbps</li> <li>• Decrease to 30 MHz: Would limit SPI to 5 Mbps, still acceptable</li> </ul> <p>Warning</p> <p>All UART/SPI/I2C drivers assume 60 MHz PCLKB. Changing requires recalculating all BRR/SPBR register values.</p> |
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| k_pclkc_hz | <p>Peripheral Clock C (PCLKC) - 60 MHz (unused in STAR project). General-purpose peripheral clock, currently unused in STAR project. Available for future expansion or custom peripherals.</p> <p>Value: 60,000,000 Hz (60 MHz) Source: ICLK (240 MHz) divided by 4 Register: SYSTEM.SCKCR.PCKC (divider = b'0010' = /4 from ICLK)</p> <p>Rationale: Configured to 60 MHz (matching PCLKB/D) for:</p> <ul style="list-style-type: none"> <li>• Consistency: All general-purpose clocks run at same frequency</li> <li>• Power: Lower clock reduces power consumption of unused peripherals</li> <li>• Future-proofing: Standard communication peripheral clock if needed</li> </ul> <p>Warning</p> <p>Not currently used. Changing has no effect on STAR functionality.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| k_pclkd_hz | <p>Peripheral Clock D (PCLKD) - 60 MHz. General-purpose peripheral clock, used in STAR project for CMT system tick timer that drives ThreadX RTOS scheduling.</p> <p>Value: 60,000,000 Hz (60 MHz) Source: ICLK (240 MHz) divided by 4 Cycle time: 16.67 nanoseconds per cycle Register: SYSTEM.SCKCR.PCKD (divider = b'0010' = /4 from ICLK)</p> <p>Connected Peripherals:</p> <ul style="list-style-type: none"> <li>• CMT (Compare Match Timer): ThreadX 10ms system tick</li> </ul> <p>ThreadX Tick Configuration:</p> <p>Tick frequency = 100 Hz (10ms period)<br/> CMT prescaler = <math>PCLKD / 8 = 60 \text{ MHz} / 8 = 7.5 \text{ MHz}</math><br/> Compare value = <math>(7.5 \text{ MHz} / 100 \text{ Hz}) - 1 = 75,000 - 1 = 74,999</math><br/> Actual period = <math>75,000 / 7.5 \text{ MHz} = 10.000 \text{ ms exact [OK]}</math></p> <p>Rationale: 60 MHz chosen for:</p> <ul style="list-style-type: none"> <li>• ThreadX tick: Integer divider achieves exact 10ms period</li> <li>• Power: Lower clock reduces CMT power consumption</li> <li>• Flexibility: High enough for sub-millisecond timing if needed</li> </ul> <p>Impact of Changing:</p> <ul style="list-style-type: none"> <li>• Increase to 120 MHz: Would double timer resolution, but overkill for 10ms tick</li> <li>• Decrease to 30 MHz: Would halve resolution, still sufficient for 10ms tick</li> </ul> <p>Warning</p> <p>ThreadX configuration assumes 100 Hz tick (10ms). Changing PCLKD requires reconfiguring CMT compare value in <a href="#">rx_cmt.c</a>.</p> |



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| k_bclk_hz | <p>External Bus Clock (BCLK) - 120 MHz (unused in STAR project). Clock for external memory interface (SDRAM, SRAM, NOR flash). Not used in STAR project as all code/data fits in internal 4MB flash + 512KB SRAM.</p> <p>Value: 120,000,000 Hz (120 MHz) Source: ICLK (240 MHz) divided by 2 Register: SYSTEM.SCKCR.BCK (divider = b'0001' = /2 from ICLK)</p> <p>Rationale: Configured to 120 MHz despite being unused because:</p> <ul style="list-style-type: none"><li>• Default safe value (does not affect internal operation)</li><li>• Matches PCLKA (high-speed peripheral clock)</li><li>• Future-proofing if external memory added later</li></ul> <p>Warning</p> <p>Not used in STAR (no external bus devices). Changing has no effect.</p> |
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| k_fclk_hz | <p>Flash Memory Clock (FCLK) - 60 MHz (hardware limit). Clock for internal flash controller (code fetch, const data access). This frequency MUST NOT EXCEED 60 MHz per RX72N datasheet, regardless of CPU clock speed.</p> <p>Value: 60,000,000 Hz (60 MHz) Source: ICLK (240 MHz) divided by 4 Cycle time: 16.67 nanoseconds per cycle Register: SYSTEM.SCKCR.FCK (divider = b'0010' = /4 from ICLK)</p> <p>Flash Access Characteristics:</p> <ul style="list-style-type: none"> <li>• 1-wait access: 16.67 ns + wait state ~ 20 ns per access</li> <li>• CPU cycles per access: 20 ns / 4.17 ns = ~5 CPU cycles</li> <li>• Code throughput: ~48 million instructions/sec (if 1 fetch per instruction)</li> <li>• Actual throughput: Higher due to caching and instruction pipelining</li> </ul> <p>Wait State Configuration (FACI.FWEPROR register):</p> <p>ICLK = 240 MHz, FCLK = 60 MHz<br/> Wait states = 1 (minimum for 60 MHz FCLK @ 240 MHz ICLK)<br/> Access time = 1 FCLK cycle + 1 wait = 33.3 ns</p> <p>Performance Impact:</p> <ul style="list-style-type: none"> <li>• Flash access is slower than SRAM (5 cycles vs 1 cycle)</li> <li>• Critical loops should use SRAM (<code>__attribute__((section(".data")))</code>)</li> <li>• Const data in flash adds latency vs SRAM variables</li> </ul> <p>Rationale: 60 MHz is MAXIMUM allowed by hardware:</p> <ul style="list-style-type: none"> <li>• Exceeding 60 MHz causes flash corruption and data loss</li> <li>• Lower frequencies (30 MHz) would reduce throughput unnecessarily</li> <li>• 60 MHz balances speed and hardware safety limit</li> </ul> <p>Impact of Changing:</p> <ul style="list-style-type: none"> <li>• Increase &gt;60 MHz: FORBIDDEN - Violates datasheet, causes corruption</li> <li>• Decrease to 30 MHz: Halves flash throughput, increases code execution time</li> </ul> <p>Warning</p> <p>CRITICAL: FCLK MUST NOT EXCEED 60 MHz under any circumstances. Flash controller will malfunction if overclocked, leading to data corruption, crashes, and potential permanent flash damage.</p> <p>Attention</p> <p>If increasing ICLK in future, MUST adjust FCK divider to keep FCLK &lt;=60 MHz. Example: ICLK=480 MHz requires FCK divider /8.</p> |
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Definition at line 286 of file [rx72n\\_clock.h](#).

```

00286 : uint32_t {
00287 /**
00288 * @brief CPU core clock frequency (ICLK) - 240 MHz
00289 *
00290 * @details
00291 * The instruction clock (ICLK) drives the RX72N CPU core, DMA controller,
00292 * and internal buses. This is the primary determinant of computational
00293 * throughput and interrupt latency.
00294 */

```

```

00295 * **Value**: 240,000,000 Hz (240 MHz)
00296 * **Source**: PLL output (240 MHz) divided by 1
00297 * **Cycle time**: 4.17 nanoseconds per cycle
00298 * **Register**: SYSTEM.SCKCR.ICK (divider = b'0000' = /1 from PLL)
00299 *
00300 * **Performance Metrics** @ 240 MHz:
00301 * - Single-cycle instructions: 4.17 ns (e.g., MOV, ADD)
00302 * - Flash access: ~20 ns (5 cycles with wait states)
00303 * - Branch misprediction penalty: ~8 ns (2 cycles)
00304 * - Interrupt latency: ~200 ns (minimum, no context save)
00305 *
00306 * **Use Cases in STAR Project**:
00307 * - PID control loop execution (~2 us per motor update @ -O2)
00308 * - Protocol parsing (nanopb decode ~50 us for 256-byte frame)
00309 * - Floating-point math (division ~20 cycles = 83 ns)
00310 * - Critical timing operations (encoder pulse measurement)
00311 *
00312 * **Rationale**: Maximum frequency selected for:
00313 * - Minimize control loop latency (100 Hz loop = 10ms budget)
00314 * - Fast protocol processing (SPI @ 10 Mbps = 25.6 us per 32-byte frame)
00315 * - Headroom for future features (current usage ~30% of CPU)
00316 *
00317 * @par Impact of Changing:
00318 * - **Increase**: Not possible (240 MHz is absolute max)
00319 * - **Decrease**: Proportional reduction in throughput, may cause control loop overruns
00320 *
00321 * @warning Flash wait states increase CPU power consumption. Use '-O2' optimization
00322 * to minimize instruction count and reduce power.
00323 *
00324 * @see k_fclk_hz Flash clock must stay <=60 MHz regardless of ICLK
00325 */
00326 k_ickl_hz = 240000000UL,
00327
00328 /**
00329 * @brief Peripheral Clock A (PCLKA) - 120 MHz
00330 *
00331 * @details
00332 * High-speed peripheral clock for time-critical functions: timers for PWM
00333 * generation (GPT), encoder capture (MTU), and high-resolution timing (CMT).
00334 *
00335 * **Value**: 120,000,000 Hz (120 MHz)
00336 * **Source**: 24 MHz main oscillator -> PLL x10 = 240 MHz -> PCKA divider /2
00337 * **Cycle time**: ~8.33 nanoseconds per cycle
00338 * **Register**: SYSTEM.SCKCR = 0x20011222 (PCKA[15:12] = 1 = divide-by-2 from 240 MHz PLL)
00339 *
00340 * **Connected Peripherals**:
00341 * - **MTU (Multi-Function Timer)**: Encoder quadrature capture (4 motors)
00342 * - **GPT (General PWM Timer)**: Motor PWM generation (4 H-bridges)
00343 * - **CMT (Compare Match Timer)**: System tick, delay timers
00344 * - **TMR (8/16-bit Timer)**: Miscellaneous timing functions
00345 *
00346 * **Performance Analysis**:
00347 *
00348 * **PWM Resolution @ 20 kHz**:
00349 * ```
00350 * Period = PCLKA / PWM_freq = 120,000,000 / 20,000 = 6000 counts
00351 * Resolution = 100% / 6000 = 0.0167% per count
00352 * Voltage resolution (12V supply) = 12V x 0.0167% = 2 mV per step
00353 * ```
00354 *
00355 * **Encoder Capture Precision**:
00356 * ```
00357 * Max velocity = 2.5 m/s (STAR robot spec)
00358 * Wheel diameter = 0.1 m -> circumference = 0.314 m
00359 * Max RPM = (2.5 m/s / 0.314 m) x 60 = 477 RPM
00360 * Encoder PPR = 341 x 4 (quadrature) = 1364 edges/rev
00361 * Max edge rate = 477 RPM x 1364 edges / 60 = 10,836 edges/sec
00362 * Min edge period = 1 / 10,836 = 92.3 us
00363 * Timer counts per edge @ 120 MHz = 92.3 us x 120 MHz = 11,076 counts
00364 * Velocity resolution = 1 count / 11,076 = 0.009% per timer count [OK]
00365 * ```
00366 *
00367 * **Rationale**: 120 MHz chosen for:
00368 * - PWM resolution: 6000 steps @ 20 kHz (excellent granularity)
00369 * - Encoder precision: 0.009% velocity measurement resolution
00370 * - Timer headroom: Max edge rate only uses 10% of timer bandwidth
00371 *
00372 * @par Impact of Changing:
00373 * - **Increase to 240 MHz**: Double PWM resolution (12000 steps), but wastes power
00374 * - **Decrease to 60 MHz**: Half PWM resolution (3000 steps), still acceptable
00375 *
00376 * @warning GPT, MTU, CMT baud rate calculations assume 120 MHz. Changing PCLKA
00377 * requires recalculating all timer prescalers and compare values.
00378 */
00379 k_pclka_hz = 120000000UL,
00380
00381 /**

```

```

00382 * @brief Peripheral Clock B (PCLKB) - 60 MHz
00383 *
00384 * @details
00385 * Standard-speed peripheral clock for communication interfaces: UART serial
00386 * console, SPI to Raspberry Pi 5, and I2C sensor bus.
00387 *
00388 * **Value**: 60,000,000 Hz (60 MHz)
00389 * **Source**: ICLK (240 MHz) divided by 4
00390 * **Cycle time**: 16.67 nanoseconds per cycle
00391 * **Register**: SYSTEM.SCKCR.PCKB (divider = b'0010' = /4 from ICLK)
00392 *
00393 * **Connected Peripherals**:
00394 * - **SCI (Serial Communication Interface)**: UART for USB CDC debug console
00395 * - **RSPi (Renesas SPI)**: SPI controller for RPi5 communication
00396 * - **RIIC (Renesas I2C)**: I2C controller for sensor bus (IMU, lidar interface)
00397 *
00398 * **Baud Rate Capabilities**:
00399 *
00400 * **UART (SCI) @ 115200 baud**:
00401 * ```
00402 * BRR = (PCLKB / (32 x baud_rate)) - 1
00403 * BRR = (60,000,000 / (32 x 115200)) - 1 = 16.276... ~ 16
00404 * Actual baud = 60,000,000 / (32 x 17) = 110,294 baud
00405 * Error = (110,294 - 115200) / 115200 = -4.3% (within +/-5% tolerance) [OK]
00406 * ```
00407 *
00408 * **SPI (RSPi) @ 10 Mbps** (STAR project: communication with RPi5):
00409 * ```
00410 * SPI_bitrate = PCLKB / (2 x (SPBR + 1))
00411 * 10 Mbps = 60,000,000 / (2 x (SPBR + 1))
00412 * SPBR = (60,000,000 / (2 x 10,000,000)) - 1 = 2
00413 * Actual rate = 60,000,000 / (2 x 3) = 10 Mbps exact [OK]
00414 * ```
00415 *
00416 * **I2C (RIIC) @ 400 kHz** (Fast-mode I2C for IMU):
00417 * ```
00418 * I2C_freq = PCLKB / (CKS x (BRL + BRH + 4))
00419 * 400 kHz typical with CKS=0 (prescaler /1), BRL=BRH=73
00420 * Actual = 60,000,000 / (1 x (73 + 73 + 4)) = 400 kHz exact [OK]
00421 * ```
00422 *
00423 * **Rationale**: 60 MHz chosen for:
00424 * - UART: Standard baud rates (9600, 19200, 115200) achievable with <5% error
00425 * - SPI: 10 Mbps target rate (STAR requirement) achievable exactly
00426 * - I2C: 100 kHz (standard) and 400 kHz (fast-mode) both supported
00427 * - Power: Lower clock = lower power for communication peripherals
00428 *
00429 * @par Impact of Changing:
00430 * - **Increase to 120 MHz**: Would allow 20 Mbps SPI, but RPi5 limited to 10 Mbps
00431 * - **Decrease to 30 MHz**: Would limit SPI to 5 Mbps, still acceptable
00432 *
00433 * @warning All UART/SPI/I2C drivers assume 60 MHz PCLKB. Changing requires
00434 * recalculating all BRR/SPBR register values.
00435 */
00436 k_pclkb_hz = 60000000UL,
00437
00438 /**
00439 * @brief Peripheral Clock C (PCLKC) - 60 MHz (unused in STAR project)
00440 *
00441 * @details
00442 * General-purpose peripheral clock, currently unused in STAR project.
00443 * Available for future expansion or custom peripherals.
00444 *
00445 * **Value**: 60,000,000 Hz (60 MHz)
00446 * **Source**: ICLK (240 MHz) divided by 4
00447 * **Register**: SYSTEM.SCKCR.PCKC (divider = b'0010' = /4 from ICLK)
00448 *
00449 * **Rationale**: Configured to 60 MHz (matching PCLKB/D) for:
00450 * - Consistency: All general-purpose clocks run at same frequency
00451 * - Power: Lower clock reduces power consumption of unused peripherals
00452 * - Future-proofing: Standard communication peripheral clock if needed
00453 *
00454 * @warning Not currently used. Changing has no effect on STAR functionality.
00455 */
00456 k_pclkc_hz = 60000000UL,
00457
00458 /**
00459 * @brief Peripheral Clock D (PCLKD) - 60 MHz
00460 *
00461 * @details
00462 * General-purpose peripheral clock, used in STAR project for CMT system tick
00463 * timer that drives ThreadX RTOS scheduling.
00464 *
00465 * **Value**: 60,000,000 Hz (60 MHz)
00466 * **Source**: ICLK (240 MHz) divided by 4
00467 * **Cycle time**: 16.67 nanoseconds per cycle
00468 * **Register**: SYSTEM.SCKCR.PCKD (divider = b'0010' = /4 from ICLK)

```

```

00469 *
00470 * **Connected Peripherals**:
00471 * - **CMT (Compare Match Timer)**: ThreadX 10ms system tick
00472 *
00473 * **ThreadX Tick Configuration**:
00474 * ```
00475 * Tick frequency = 100 Hz (10ms period)
00476 * CMT prescaler = PCLKD / 8 = 60 MHz / 8 = 7.5 MHz
00477 * Compare value = (7.5 MHz / 100 Hz) - 1 = 75,000 - 1 = 74,999
00478 * Actual period = 75,000 / 7.5 MHz = 10.000 ms exact [OK]
00479 * ```
00480 *
00481 * **Rationale**: 60 MHz chosen for:
00482 * - ThreadX tick: Integer divider achieves exact 10ms period
00483 * - Power: Lower clock reduces CMT power consumption
00484 * - Flexibility: High enough for sub-millisecond timing if needed
00485 *
00486 * @par Impact of Changing:
00487 * - **Increase to 120 MHz**: Would double timer resolution, but overkill for 10ms tick
00488 * - **Decrease to 30 MHz**: Would halve resolution, still sufficient for 10ms tick
00489 *
00490 * @warning ThreadX configuration assumes 100 Hz tick (10ms). Changing PCLKD
00491 * requires reconfiguring CMT compare value in rx_cmt.c.
00492 */
00493 k_pclkd_hz = 60000000UL,
00494
00495 /**
00496 * @brief External Bus Clock (BCLK) - 120 MHz (unused in STAR project)
00497 *
00498 * @details
00499 * Clock for external memory interface (SDRAM, SRAM, NOR flash). Not used in
00500 * STAR project as all code/data fits in internal 4MB flash + 512KB SRAM.
00501 *
00502 * **Value**: 120,000,000 Hz (120 MHz)
00503 * **Source**: ICLK (240 MHz) divided by 2
00504 * **Register**: SYSTEM.SCKCR.BCK (divider = b'0001' = /2 from ICLK)
00505 *
00506 * **Rationale**: Configured to 120 MHz despite being unused because:
00507 * - Default safe value (does not affect internal operation)
00508 * - Matches PCLKA (high-speed peripheral clock)
00509 * - Future-proofing if external memory added later
00510 *
00511 * @warning Not used in STAR (no external bus devices). Changing has no effect.
00512 */
00513 k_bclk_hz = 120000000UL,
00514
00515 /**
00516 * @brief Flash Memory Clock (FCLK) - 60 MHz (hardware limit)
00517 *
00518 * @details
00519 * Clock for internal flash controller (code fetch, const data access). This
00520 * frequency MUST NOT EXCEED 60 MHz per RX72N datasheet, regardless of CPU
00521 * clock speed.
00522 *
00523 * **Value**: 60,000,000 Hz (60 MHz)
00524 * **Source**: ICLK (240 MHz) divided by 4
00525 * **Cycle time**: 16.67 nanoseconds per cycle
00526 * **Register**: SYSTEM.SCKCR.FCK (divider = b'0010' = /4 from ICLK)
00527 *
00528 * **Flash Access Characteristics**:
00529 * - **1-wait access**: 16.67 ns + wait state ~ 20 ns per access
00530 * - **CPU cycles per access**: 20 ns / 4.17 ns = ~5 CPU cycles
00531 * - **Code throughput**: ~48 million instructions/sec (if 1 fetch per instruction)
00532 * - **Actual throughput**: Higher due to caching and instruction pipelining
00533 *
00534 * **Wait State Configuration** (FACI.FWEPROR register):
00535 * ```
00536 * ICLK = 240 MHz, FCLK = 60 MHz
00537 * Wait states = 1 (minimum for 60 MHz FCLK @ 240 MHz ICLK)
00538 * Access time = 1 FCLK cycle + 1 wait = 33.3 ns
00539 * ```
00540 *
00541 * **Performance Impact**:
00542 * - Flash access is slower than SRAM (5 cycles vs 1 cycle)
00543 * - Critical loops should use SRAM ('__attribute__((section(".data")))')
00544 * - Const data in flash adds latency vs SRAM variables
00545 *
00546 * **Rationale**: 60 MHz is MAXIMUM allowed by hardware:
00547 * - Exceeding 60 MHz causes flash corruption and data loss
00548 * - Lower frequencies (30 MHz) would reduce throughput unnecessarily
00549 * - 60 MHz balances speed and hardware safety limit
00550 *
00551 * @par Impact of Changing:
00552 * - **Increase >60 MHz**: **FORBIDDEN** - Violates datasheet, causes corruption
00553 * - **Decrease to 30 MHz**: Halves flash throughput, increases code execution time
00554 *
00555 * @warning **CRITICAL**: FCLK MUST NOT EXCEED 60 MHz under any circumstances.

```

```

00556 * Flash controller will malfunction if overclocked, leading to data
00557 * corruption, crashes, and potential permanent flash damage.
00558 *
00559 * @attention If increasing ICLK in future, MUST adjust FCK divider to keep
00560 * FCLK <=60 MHz. Example: ICLK=480 MHz requires FCK divider /8.
00561 */
00562 k_fclk_hz = 60000000UL,
00563
00564 } rx_clock_frequencies_t;

```

## 8.8 rx72n\_clock.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_clock.h
00003 * @brief RX72N Clock Frequency Definitions and Configuration
00004 *
00005 * @details
00006 * Defines all system clock frequencies for the RX72N microcontroller configured
00007 * with external 24 MHz crystal and PLL x10 for maximum 240 MHz CPU operation.
00008 * These constants are used throughout the firmware for timing calculations,
00009 * baud rate generation, PWM frequency configuration, and peripheral initialization.
00010 *
00011 * ## Clock Generation Architecture (STAR Project)
00012 *
00013 * The RX72N has a sophisticated clock generation system with multiple clock
00014 * domains optimized for different purposes:
00015 *
00016 * @dot
00017 * digraph clock_tree {
00018 * rankdir=LR;
00019 * node [shape=box, style=rounded];
00020 *
00021 * EXTAL [label="External Crystal\n24 MHz", shape=ellipse, fillcolor=lightyellow, style=filled];
00022 * PLL [label="PLL\nx10", fillcolor=lightblue, style=filled];
00023 * ICLK [label="ICLK\n240 MHz\nCPU", fillcolor=lightgreen, style=filled];
00024 * PCLKA [label="PCLKA\n120 MHz\nTimers", fillcolor=lightpink, style=filled];
00025 * PCLKB [label="PCLKB\n60 MHz\nUART/SPI", fillcolor=lightpink, style=filled];
00026 * PCLKC [label="PCLKC\n60 MHz\nPWM", fillcolor=lightpink, style=filled];
00027 * PCLKD [label="PCLKD\n60 MHz\nCMT", fillcolor=lightpink, style=filled];
00028 * FCLK [label="FCLK\n60 MHz\nFlash", fillcolor=lightcyan, style=filled];
00029 * BCLK [label="BCLK\n120 MHz\nExt Bus", fillcolor=lightgray, style=filled];
00030 *
00031 * EXTAL -> PLL [label="24 MHz"];
00032 * PLL -> ICLK [label="240 MHz\n/1"];
00033 * PLL -> PCLKA [label="/2"];
00034 * PLL -> PCLKB [label="/4"];
00035 * PLL -> PCLKC [label="/4"];
00036 * PLL -> PCLKD [label="/4"];
00037 * PLL -> FCLK [label="/4"];
00038 * PLL -> BCLK [label="/2"];
00039 * }
00040 * @enddot
00041 *
00042 * ## Clock Domain Assignments
00043 *
00044 * | Clock | Frequency | Peripherals | Purpose |
00045 * |-----|-----|-----|-----|
00046 * | ICLK | 240 MHz | CPU core, DMA | Maximum instruction throughput |
00047 * | PCLKA | 120 MHz | CMT, TMR, MTU, GPT | High-resolution timers for PWM |
00048 * | PCLKB | 60 MHz | SCI (UART), RSPI, RIIC | Communication interfaces |
00049 * | PCLKC | 60 MHz | (Reserved) | User peripherals |
00050 * | PCLKD | 60 MHz | (Reserved) | User peripherals |
00051 * | FCLK | 60 MHz | Flash controller | Flash access (max 60 MHz) |
00052 * | BCLK | 120 MHz | External bus | External memory interface |
00053 *
00054 * ## Clock Configuration (SYSTEM Registers)
00055 *
00056 * Clock frequencies are set during boot by `rx_clock_power_init()`:
00057 *
00058 * 1. **Main Clock Enable**: MOSCWTCR, MOSCCR, MOFCR registers
00059 * 2. **PLL Configuration**: PLLCR (x10 multiplier, STC=0x13), PLLCR2 (enable)
00060 * 3. **Clock Dividers**: SCKCR register sets ICK, PCKA, PCKB, PCKC, PCKD, FCK, BCK dividers
00061 * 4. **Clock Source Selection**: SCKCR3 selects PLL as system clock
00062 *
00063 * **Register Configuration** (from rx_clock_power_init.c):
00064 * ```
00065 * SYSTEM.SCKCR.LONG = 0x20011222; // ICK=0/(1)->240MHz, BCK=1/(2)->120MHz, PCKA=1/(2)->120MHz,
 PCKB/C/D=2/(4)->60MHz, FCK=2/(4)->60MHz
00066 * SYSTEM.PLLCR.WORD = 0x1300; // STC=19 -> x10.0 (manual p345); 24 MHz x 10 = 240 MHz
00067 * ```

```

```

00068 *
00069 * ## Performance Implications
00070 *
00071 * | Operation | Clock | Impact of Frequency |
00072 * |-----|-----|-----|
00073 * | CPU instructions | ICLK (240 MHz) | ~4.2 ns per cycle, critical for control loops |
00074 * | Flash access | FCLK (60 MHz) | ~17 ns per access, affects code execution speed |
00075 * | PWM resolution | PCLKA (120 MHz) | 8.3 ns resolution, 20 kHz PWM = 6000 steps |
00076 * | UART baud rate | PCLKB (60 MHz) | Max 3.75 Mbaud (60M/16), typical 115200 baud |
00077 * | Timer tick | PCLKD (60 MHz) | ThreadX 10ms tick = 600,000 cycles per tick |
00078 *
00079 * ## Hardware Requirements
00080 *
00081 * | Requirement | Value | Notes |
00082 * |-----|-----|-----|
00083 * | External Crystal | 24 MHz | +/-50 ppm accuracy required |
00084 * | PLL Input Range | 8-24 MHz | 24 MHz nominal for STAR project |
00085 * | PLL Output Range | 200-520 MHz | 240 MHz output (/1 for CPU) |
00086 * | CPU Max Frequency | 240 MHz | Absolute max per datasheet |
00087 * | Flash Max Frequency | 60 MHz | Must not exceed! (hardware limit) |
00088 * | Operating Voltage | 3.3V +/-10% | Required for 240 MHz operation |
00089 * | Temperature Range | -40degC to +85degC | Industrial grade |
00090 *
00091 * ## Timing Calculation Examples
00092 *
00093 * **PWM Frequency**:
00094 * ```
00095 * PWM_freq = PCLKA / (prescaler x period)
00096 * 20 kHz = 120 MHz / (1 x 6000)
00097 * ```
00098 *
00099 * **UART Baud Rate**:
00100 * ```
00101 * Baud_rate = PCLKB / (32 x (N + 1))
00102 * 115200 = 60 MHz / (32 x (16 + 1)) -> N=16 (BRR register value)
00103 * ```
00104 *
00105 * **Timer Period**:
00106 * ```
00107 * Period_ms = (counter_max x prescaler) / PCLKD_MHz
00108 * 1 ms = (60000 x 1) / 60 MHz
00109 * ```
00110 *
00111 * @par Module Dependencies:
00112 * - **NO dependencies** (pure constant definitions)
00113 * - Used by: ALL timing-related modules (CMT, PWM, UART, SPI, ADC, etc.)
00114 *
00115 * @par NASA Power of 10 Compliance:
00116 * - **Rule 1**: [OK] No code (only data definitions)
00117 * - **Rule 2**: [OK] No loops
00118 * - **Rule 3**: [OK] No dynamic allocation (compile-time constants)
00119 * - **Rule 4**: [OK] No functions
00120 * - **Rule 5**: [OK] N/A (no functions)
00121 * - **Rule 6**: [OK] File scope only (header file)
00122 * - **Rule 7**: [OK] No functions
00123 * - **Rule 8**: [OK] C23 typed enums for all constants
00124 * - **Rule 9**: [OK] No pointers
00125 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00126 *
00127 * @par SOLID Principles:
00128 * - **Single Responsibility**: Only defines clock frequencies, no logic
00129 * - **Open/Closed**: Extensible (add new clocks) without modification
00130 * - **Liskov Substitution**: N/A (no inheritance)
00131 * - **Interface Segregation**: Minimal interface (single enum)
00132 * - **Dependency Inversion**: N/A (no dependencies)
00133 *
00134 * @par Usage Example - Baud Rate Calculation:
00135 * @code{.c}
00136 * #include "rx72n_clock.h"
00137 *
00138 * // Calculate UART BRR register value for desired baud rate
00139 * uint8_t uart_calculate_brr(uint32_t baud_rate) {
00140 * // Formula: BRR = (PCLKB / (32 x baud)) - 1
00141 * uint32_t brr = (k_pclkb_hz / (32 * baud_rate)) - 1;
00142 * return (uint8_t)brr;
00143 * }
00144 *
00145 * // Example: 115200 baud
00146 * uint8_t brr = uart_calculate_brr(115200); // Returns 16
00147 * @endcode
00148 *
00149 * @par Usage Example - PWM Configuration:
00150 * @code{.c}
00151 * // Configure PWM for 20 kHz at 50% duty cycle
00152 * void pwm_configure_20khz(void) {
00153 * typedef enum : uint32_t {
00154 * k_pwm_freq_hz = 20000, // Desired PWM frequency

```



```

00155 * k_prescaler = 1, // Timer prescaler
00156 * } pwm_config_t;
00157 *
00158 * // Calculate timer period: Period = PCLKA / (prescaler x freq)
00159 * uint32_t period = k_pclka_hz / (k_prescaler * k_pwm_freq_hz);
00160 * // period = 120,000,000 / (1 x 20,000) = 6000 counts
00161 *
00162 * uint32_t duty_50pct = period / 2; // 3000 counts = 50%
00163 *
00164 * gpt_set_period(GPT_CHANNEL_0, period);
00165 * gpt_set_duty(GPT_CHANNEL_0, duty_50pct);
00166 * }
00167 * @endcode
00168 *
00169 * @par Usage Example - Delay Timing:
00170 * @code{.c}
00171 * // Calculate cycles for precise delay
00172 * void delay_us(uint32_t us) {
00173 * // Cycles per microsecond at ICLK frequency
00174 * typedef enum : uint32_t {
00175 * k_cycles_per_us = k_ickl_hz / 1000000, // 240 cycles/us
00176 * } delay_config_t;
00177 *
00178 * uint32_t cycles = us * k_cycles_per_us;
00179 *
00180 * // Busy-wait loop (NOP instructions)
00181 * for (uint32_t i = 0; i < cycles / 4; i++) {
00182 * __asm volatile("nop\nnop\nnop\nnop");
00183 * }
00184 * }
00185 * @endcode
00186 *
00187 * @par Usage Example - Timer Configuration:
00188 * @code{.c}
00189 * // Configure CMT for 1ms tick (1 kHz interrupt)
00190 * void cmt_configure_1ms(void) {
00191 * typedef enum : uint32_t {
00192 * k_desired_freq_hz = 1000, // 1 kHz = 1ms period
00193 * k_cmt_prescaler = 8, // CMT prescaler (PCLKD/8)
00194 * } cmt_config_t;
00195 *
00196 * // Calculate compare match value
00197 * // CMCOR = (PCLKD / prescaler / freq) - 1
00198 * uint32_t cmcor = (k_pclkd_hz / k_cmt_prescaler / k_desired_freq_hz) - 1;
00199 * // cmcor = (60,000,000 / 8 / 1000) - 1 = 7499
00200 *
00201 * CMT0.CMCOR = (uint16_t)cmcor;
00202 * }
00203 * @endcode
00204 *
00205 * @see [RX72N User's Manual](../docs/rx72n_manual.pdf) Chapter 9: Clock Generation Circuit
00206 * @see [RX72N Group Hardware Manual](../docs/rx72n_hardware.pdf) Section 59: Electrical Characteristics
00207 * @see [rx_clock_power_init.c](../src/rx_clock_power_init.c) Clock initialization code
00208 * @see [hardware_init.c](../src/hardware_init.c) Calls clock initialization
00209 *
00210 * @author Locked, Inc.
00211 * @date 2026-01-27
00212 * @version 1.0.0
00213 * @copyright Copyright (c) 2026 Locked Inc.
00214 * SPDX-License-Identifier: MIT
00215 *
00216 * @since Version 1.0.0
00217 */
00218
00219 #pragma once
00220
00221 #include <stdint.h>
00222
00223 #ifdef __cplusplus
00224 extern "C" {
00225 #endif
00226
00227 /*
=====
00228 * Clock Frequencies (assume PLL configured for 240 MHz)
00229 *
=====
00230 */
00231
00232 /**
00233 * @enum rx_clock_frequencies_t
00234 * @brief System clock frequencies for RX72N @ 240 MHz operation
00235 *
00236 * @details
00237 * Defines all clock domain frequencies for the STAR project's RX72N configuration.
00238 * These values are determined by the PLL and clock divider settings in
00239 * rx_clock_power_init() and are guaranteed to be accurate for timing calculations.

```



```

00240 *
00241 * **Clock Source**: 24 MHz external crystal -> PLL (x10) -> 240 MHz -> divided per domain
00242 *
00243 * **Frequency Derivation**:
00244 * - PLL output: 24 MHz x 10 = 240 MHz
00245 * - ICLK: 240 MHz / 1 = 240 MHz (CPU)
00246 * - PCLKA: 240 MHz / 2 = 120 MHz (high-speed peripherals)
00247 * - PCLKB/C/D: 240 MHz / 4 = 60 MHz (standard peripherals)
00248 * - FCLK: 240 MHz / 4 = 60 MHz (flash - max allowed)
00249 * - BCLK: 240 MHz / 2 = 120 MHz (external bus - not used in STAR)
00250 *
00251 * ## Clock Domain Usage Table
00252 *
00253 * | Clock | Freq | Used By | STAR Project Usage |
00254 * |-----|-----|-----|-----|
00255 * | ICLK | 240 MHz | CPU core | PID control loop, communication protocol |
00256 * | PCLKA | 120 MHz | MTU, GPT, CMT, TMR | Encoder capture (MTU), PWM generation (GPT) |
00257 * | PCLKB | 60 MHz | SCI, RSPI, RIIC | UART debug, SPI to RPi5, I2C sensors |
00258 * | PCLKC | 60 MHz | (Reserved) | Not used in STAR project |
00259 * | PCLKD | 60 MHz | CMT | ThreadX system tick (10ms) |
00260 * | FCLK | 60 MHz | Flash | Code execution, const data access |
00261 * | BCLK | 120 MHz | External bus | Not used (no external memory) |
00262 *
00263 * @par Field Constraints:
00264 * | Constant | Min | Max | Notes |
00265 * |-----|-----|-----|-----|
00266 * | k_ickl_hz | 1 MHz | 240 MHz | 240 MHz is absolute max per datasheet |
00267 * | k_pclka_hz | 1 MHz | 240 MHz | Limited by ICLK (cannot exceed CPU clock) |
00268 * | k_pclkb_hz | 1 MHz | 60 MHz | Communication peripherals rated to 60 MHz |
00269 * | k_fclk_hz | 1 MHz | 60 MHz | **MUST NOT EXCEED 60 MHz** (flash limit) |
00270 *
00271 * @par Usage Example - Calculate Timer Period:
00272 * @code{.c}
00273 * // Calculate CMT compare value for 100 Hz interrupt (10ms period)
00274 * typedef enum : uint32_t {
00275 * k_desired_freq_hz = 100, // 100 Hz = 10ms period
00276 * k_cmt_prescaler = 8, // PCLKD/8
00277 * } timer_config_t;
00278 *
00279 * uint32_t cmcwr = (k_pclkd_hz / k_cmt_prescaler / k_desired_freq_hz) - 1;
00280 * // cmcwr = (60,000,000 / 8 / 100) - 1 = 74,999
00281 * @endcode
00282 *
00283 * @see rx_clock_power_init() Function that configures these frequencies
00284 * @since Version 1.0.0
00285 */
00286 typedef enum : uint32_t {
00287 /**
00288 * @brief CPU core clock frequency (ICLK) - 240 MHz
00289 *
00290 * @details
00291 * The instruction clock (ICLK) drives the RX72N CPU core, DMA controller,
00292 * and internal buses. This is the primary determinant of computational
00293 * throughput and interrupt latency.
00294 *
00295 * **Value**: 240,000,000 Hz (240 MHz)
00296 * **Source**: PLL output (240 MHz) divided by 1
00297 * **Cycle time**: 4.17 nanoseconds per cycle
00298 * **Register**: SYSTEM.SCKCR.ICK (divider = b'0000' = /1 from PLL)
00299 *
00300 * **Performance Metrics** @ 240 MHz:
00301 * - Single-cycle instructions: 4.17 ns (e.g., MOV, ADD)
00302 * - Flash access: ~20 ns (5 cycles with wait states)
00303 * - Branch misprediction penalty: ~8 ns (2 cycles)
00304 * - Interrupt latency: ~200 ns (minimum, no context save)
00305 *
00306 * **Use Cases in STAR Project**:
00307 * - PID control loop execution (~2 us per motor update @ -O2)
00308 * - Protocol parsing (nanopb decode ~50 us for 256-byte frame)
00309 * - Floating-point math (division ~20 cycles = 83 ns)
00310 * - Critical timing operations (encoder pulse measurement)
00311 *
00312 * **Rationale**: Maximum frequency selected for:
00313 * - Minimize control loop latency (100 Hz loop = 10ms budget)
00314 * - Fast protocol processing (SPI @ 10 Mbps = 25.6 us per 32-byte frame)
00315 * - Headroom for future features (current usage ~30% of CPU)
00316 *
00317 * @par Impact of Changing:
00318 * - **Increase**: Not possible (240 MHz is absolute max)
00319 * - **Decrease**: Proportional reduction in throughput, may cause control loop overruns
00320 *
00321 * @warning Flash wait states increase CPU power consumption. Use -O2 optimization
00322 * to minimize instruction count and reduce power.
00323 *
00324 * @see k_fclk_hz Flash clock must stay <=60 MHz regardless of ICLK
00325 */
00326 k_ickl_hz = 240000000UL,

```

```

00327
00328 /**
00329 * @brief Peripheral Clock A (PCLKA) - 120 MHz
00330 *
00331 * @details
00332 * High-speed peripheral clock for time-critical functions: timers for PWM
00333 * generation (GPT), encoder capture (MTU), and high-resolution timing (CMT).
00334 *
00335 * **Value**: 120,000,000 Hz (120 MHz)
00336 * **Source**: 24 MHz main oscillator -> PLL x10 = 240 MHz -> PCKA divider /2
00337 * **Cycle time**: ~8.33 nanoseconds per cycle
00338 * **Register**: SYSTEM.SCKCR = 0x20011222 (PCKA[15:12] = 1 = divide-by-2 from 240 MHz PLL)
00339 *
00340 * **Connected Peripherals**:
00341 * - **MTU (Multi-Function Timer)**: Encoder quadrature capture (4 motors)
00342 * - **GPT (General PWM Timer)**: Motor PWM generation (4 H-bridges)
00343 * - **CMT (Compare Match Timer)**: System tick, delay timers
00344 * - **TMR (8/16-bit Timer)**: Miscellaneous timing functions
00345 *
00346 * **Performance Analysis**:
00347 *
00348 * **PWM Resolution @ 20 kHz**:
00349 * ```
00350 * Period = PCLKA / PWM_freq = 120,000,000 / 20,000 = 6000 counts
00351 * Resolution = 100% / 6000 = 0.0167% per count
00352 * Voltage resolution (12V supply) = 12V x 0.0167% = 2 mV per step
00353 * ```
00354 *
00355 * **Encoder Capture Precision**:
00356 * ```
00357 * Max velocity = 2.5 m/s (STAR robot spec)
00358 * Wheel diameter = 0.1 m -> circumference = 0.314 m
00359 * Max RPM = (2.5 m/s / 0.314 m) x 60 = 477 RPM
00360 * Encoder PPR = 341 x 4 (quadrature) = 1364 edges/rev
00361 * Max edge rate = 477 RPM x 1364 edges / 60 = 10,836 edges/sec
00362 * Min edge period = 1 / 10,836 = 92.3 us
00363 * Timer counts per edge @ 120 MHz = 92.3 us x 120 MHz = 11,076 counts
00364 * Velocity resolution = 1 count / 11,076 = 0.009% per timer count [OK]
00365 * ```
00366 *
00367 * **Rationale**: 120 MHz chosen for:
00368 * - PWM resolution: 6000 steps @ 20 kHz (excellent granularity)
00369 * - Encoder precision: 0.009% velocity measurement resolution
00370 * - Timer headroom: Max edge rate only uses 10% of timer bandwidth
00371 *
00372 * @par Impact of Changing:
00373 * - **Increase to 240 MHz**: Double PWM resolution (12000 steps), but wastes power
00374 * - **Decrease to 60 MHz**: Half PWM resolution (3000 steps), still acceptable
00375 *
00376 * @warning GPT, MTU, CMT baud rate calculations assume 120 MHz. Changing PCLKA
00377 * requires recalculating all timer prescalers and compare values.
00378 */
00379 k_pclka_hz = 120000000UL,
00380
00381 /**
00382 * @brief Peripheral Clock B (PCLKB) - 60 MHz
00383 *
00384 * @details
00385 * Standard-speed peripheral clock for communication interfaces: UART serial
00386 * console, SPI to Raspberry Pi 5, and I2C sensor bus.
00387 *
00388 * **Value**: 60,000,000 Hz (60 MHz)
00389 * **Source**: ICLK (240 MHz) divided by 4
00390 * **Cycle time**: 16.67 nanoseconds per cycle
00391 * **Register**: SYSTEM.SCKCR.PCKB (divider = b'0010' = /4 from ICLK)
00392 *
00393 * **Connected Peripherals**:
00394 * - **SCI (Serial Communication Interface)**: UART for USB CDC debug console
00395 * - **RSPI (Renesas SPI)**: SPI controller for RPi5 communication
00396 * - **RIIC (Renesas I2C)**: I2C controller for sensor bus (IMU, lidar interface)
00397 *
00398 * **Baud Rate Capabilities**:
00399 *
00400 * **UART (SCI) @ 115200 baud**:
00401 * ```
00402 * BRR = (PCLKB / (32 x baud_rate)) - 1
00403 * BRR = (60,000,000 / (32 x 115200)) - 1 = 16.276... ~ 16
00404 * Actual baud = 60,000,000 / (32 x 17) = 110,294 baud
00405 * Error = (110,294 - 115200) / 115200 = -4.3% (within +/-5% tolerance) [OK]
00406 * ```
00407 *
00408 * **SPI (RSPI) @ 10 Mbps** (STAR project: communication with RPi5):
00409 * ```
00410 * SPI_bitrate = PCLKB / (2 x (SPBR + 1))
00411 * 10 Mbps = 60,000,000 / (2 x (SPBR + 1))
00412 * SPBR = (60,000,000 / (2 x 10,000,000)) - 1 = 2
00413 * Actual rate = 60,000,000 / (2 x 3) = 10 Mbps exact [OK]

```

```

00414 * ````
00415 *
00416 * **I2C (RIIC) @ 400 kHz** (Fast-mode I2C for IMU):
00417 * ````
00418 * I2C_freq = PCLKB / (CKS x (BRL + BRH + 4))
00419 * 400 kHz typical with CKS=0 (prescaler /1), BRL=BRH=73
00420 * Actual = 60,000,000 / (1 x (73 + 73 + 4)) = 400 kHz exact [OK]
00421 * ````
00422 *
00423 * **Rationale**: 60 MHz chosen for:
00424 * - UART: Standard baud rates (9600, 19200, 115200) achievable with <5% error
00425 * - SPI: 10 Mbps target rate (STAR requirement) achievable exactly
00426 * - I2C: 100 kHz (standard) and 400 kHz (fast-mode) both supported
00427 * - Power: Lower clock = lower power for communication peripherals
00428 *
00429 * @par Impact of Changing:
00430 * - **Increase to 120 MHz**: Would allow 20 Mbps SPI, but RPi5 limited to 10 Mbps
00431 * - **Decrease to 30 MHz**: Would limit SPI to 5 Mbps, still acceptable
00432 *
00433 * @warning All UART/SPI/I2C drivers assume 60 MHz PCLKB. Changing requires
00434 * recalculating all BRR/SPBR register values.
00435 */
00436 k_pclkb_hz = 60000000UL,
00437
00438 /**
00439 * @brief Peripheral Clock C (PCLKC) - 60 MHz (unused in STAR project)
00440 *
00441 * @details
00442 * General-purpose peripheral clock, currently unused in STAR project.
00443 * Available for future expansion or custom peripherals.
00444 *
00445 * **Value**: 60,000,000 Hz (60 MHz)
00446 * **Source**: ICLK (240 MHz) divided by 4
00447 * **Register**: SYSTEM.SCKCR.PCKC (divider = b'0010' = /4 from ICLK)
00448 *
00449 * **Rationale**: Configured to 60 MHz (matching PCLKB/D) for:
00450 * - Consistency: All general-purpose clocks run at same frequency
00451 * - Power: Lower clock reduces power consumption of unused peripherals
00452 * - Future-proofing: Standard communication peripheral clock if needed
00453 *
00454 * @warning Not currently used. Changing has no effect on STAR functionality.
00455 */
00456 k_pclkc_hz = 60000000UL,
00457
00458 /**
00459 * @brief Peripheral Clock D (PCLKD) - 60 MHz
00460 *
00461 * @details
00462 * General-purpose peripheral clock, used in STAR project for CMT system tick
00463 * timer that drives ThreadX RTOS scheduling.
00464 *
00465 * **Value**: 60,000,000 Hz (60 MHz)
00466 * **Source**: ICLK (240 MHz) divided by 4
00467 * **Cycle time**: 16.67 nanoseconds per cycle
00468 * **Register**: SYSTEM.SCKCR.PCKD (divider = b'0010' = /4 from ICLK)
00469 *
00470 * **Connected Peripherals**:
00471 * - **CMT (Compare Match Timer)**: ThreadX 10ms system tick
00472 *
00473 * **ThreadX Tick Configuration**:
00474 * ````
00475 * Tick frequency = 100 Hz (10ms period)
00476 * CMT prescaler = PCLKD / 8 = 60 MHz / 8 = 7.5 MHz
00477 * Compare value = (7.5 MHz / 100 Hz) - 1 = 75,000 - 1 = 74,999
00478 * Actual period = 75,000 / 7.5 MHz = 10.000 ms exact [OK]
00479 * ````
00480 *
00481 * **Rationale**: 60 MHz chosen for:
00482 * - ThreadX tick: Integer divider achieves exact 10ms period
00483 * - Power: Lower clock reduces CMT power consumption
00484 * - Flexibility: High enough for sub-millisecond timing if needed
00485 *
00486 * @par Impact of Changing:
00487 * - **Increase to 120 MHz**: Would double timer resolution, but overkill for 10ms tick
00488 * - **Decrease to 30 MHz**: Would halve resolution, still sufficient for 10ms tick
00489 *
00490 * @warning ThreadX configuration assumes 100 Hz tick (10ms). Changing PCLKD
00491 * requires reconfiguring CMT compare value in rx_cmt.c.
00492 */
00493 k_pclkd_hz = 60000000UL,
00494
00495 /**
00496 * @brief External Bus Clock (BCLK) - 120 MHz (unused in STAR project)
00497 *
00498 * @details
00499 * Clock for external memory interface (SDRAM, SRAM, NOR flash). Not used in
00500 * STAR project as all code/data fits in internal 4MB flash + 512KB SRAM.

```

```

00501 *
00502 * **Value**: 120,000,000 Hz (120 MHz)
00503 * **Source**: ICLK (240 MHz) divided by 2
00504 * **Register**: SYSTEM.SCKCR.BCK (divider = b'0001' = /2 from ICLK)
00505 *
00506 * **Rationale**: Configured to 120 MHz despite being unused because:
00507 * - Default safe value (does not affect internal operation)
00508 * - Matches PCLKA (high-speed peripheral clock)
00509 * - Future-proofing if external memory added later
00510 *
00511 * @warning Not used in STAR (no external bus devices). Changing has no effect.
00512 */
00513 k_bclk_hz = 120000000UL,
00514
00515 /**
00516 * @brief Flash Memory Clock (FCLK) - 60 MHz (hardware limit)
00517 *
00518 * @details
00519 * Clock for internal flash controller (code fetch, const data access). This
00520 * frequency MUST NOT EXCEED 60 MHz per RX72N datasheet, regardless of CPU
00521 * clock speed.
00522 *
00523 * **Value**: 60,000,000 Hz (60 MHz)
00524 * **Source**: ICLK (240 MHz) divided by 4
00525 * **Cycle time**: 16.67 nanoseconds per cycle
00526 * **Register**: SYSTEM.SCKCR.FCK (divider = b'0010' = /4 from ICLK)
00527 *
00528 * **Flash Access Characteristics**:
00529 * - **1-wait access**: 16.67 ns + wait state ~ 20 ns per access
00530 * - **CPU cycles per access**: 20 ns / 4.17 ns = ~5 CPU cycles
00531 * - **Code throughput**: ~48 million instructions/sec (if 1 fetch per instruction)
00532 * - **Actual throughput**: Higher due to caching and instruction pipelining
00533 *
00534 * **Wait State Configuration** (FACI.FWEPROR register):
00535 * ```
00536 * ICLK = 240 MHz, FCLK = 60 MHz
00537 * Wait states = 1 (minimum for 60 MHz FCLK @ 240 MHz ICLK)
00538 * Access time = 1 FCLK cycle + 1 wait = 33.3 ns
00539 * ```
00540 *
00541 * **Performance Impact**:
00542 * - Flash access is slower than SRAM (5 cycles vs 1 cycle)
00543 * - Critical loops should use SRAM ('__attribute__((section(".data"))')
00544 * - Const data in flash adds latency vs SRAM variables
00545 *
00546 * **Rationale**: 60 MHz is MAXIMUM allowed by hardware:
00547 * - Exceeding 60 MHz causes flash corruption and data loss
00548 * - Lower frequencies (30 MHz) would reduce throughput unnecessarily
00549 * - 60 MHz balances speed and hardware safety limit
00550 *
00551 * @par Impact of Changing:
00552 * - **Increase >60 MHz**: **FORBIDDEN** - Violates datasheet, causes corruption
00553 * - **Decrease to 30 MHz**: Halves flash throughput, increases code execution time
00554 *
00555 * @warning **CRITICAL**: FCLK MUST NOT EXCEED 60 MHz under any circumstances.
00556 * Flash controller will malfunction if overclocked, leading to data
00557 * corruption, crashes, and potential permanent flash damage.
00558 *
00559 * @attention If increasing ICLK in future, MUST adjust FCK divider to keep
00560 * FCLK <=60 MHz. Example: ICLK=480 MHz requires FCK divider /8.
00561 */
00562 k_fclk_hz = 60000000UL,
00563
00564 } rx_clock_frequencies_t;
00565
00566 #ifdef __cplusplus
00567 }
00568 #endif

```

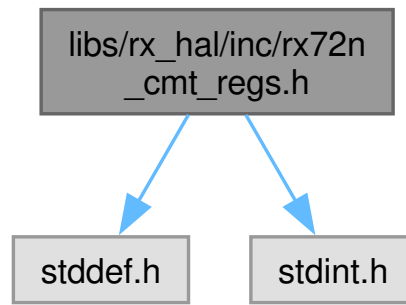
## 8.9 libs/rx-hal/inc/rx72n-cmt-regs.h File Reference

RX72N Compare Match Timer (CMT) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_cmt\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct [rx\\_cmt\\_channel\\_regs\\_t](#)  
CMT Channel Register Map (6 bytes per channel).
- struct [rx\\_cmt\\_control\\_regs\\_t](#)  
CMT Start/Stop Control Register Map.

## Enumerations

- enum [rx\\_cmt\\_addresses\\_t](#) : `uintptr_t` {  
`k_cmt_ctrl_base_addr = 0x00088000` , `k_cmt1_ctrl_base_addr = 0x00088010` , `k_cmt0_base_addr = 0x00088002` , `k_cmt1_base_addr = 0x00088008` ,  
`k_cmt2_base_addr = 0x00088012` , `k_cmt3_base_addr = 0x00088018` }  
 CMT base addresses (verified against RX72N Hardware Manual Ch31).
- enum [rx\\_cmt\\_layout\\_t](#) : `uint8_t` {  
`k_cmt_channel_regs_size_bytes = 6` , `k_cmt_control_regs_size_bytes = 4` , `k_cmt_cmcr_offset = 0x00` , `k_cmt_cmcr_offset = 0x02` ,  
`k_cmt_cmcr_offset = 0x04` , `k_cmt_cmstr_reg_size_bytes = 2` }  
 CMT register layout constants used for compile-time verification.
- enum [rx\\_cmt\\_cmstr0\\_bits\\_t](#) : `uint8_t` { `k_rx72n_cmstr0_cmt0_enable = 0x01` , `k_rx72n_cmstr0_cmt1_enable = 0x02` }  
 CMSTR0 register bit definitions (CMT Unit 0 start control).
- enum [rx\\_cmt\\_cmstr1\\_bits\\_t](#) : `uint8_t` { `k_rx72n_cmstr1_cmt2_enable = 0x01` , `k_rx72n_cmstr1_cmt3_enable = 0x02` }  
 CMSTR1 register bit definitions (CMT Unit 1 start control).

## Functions

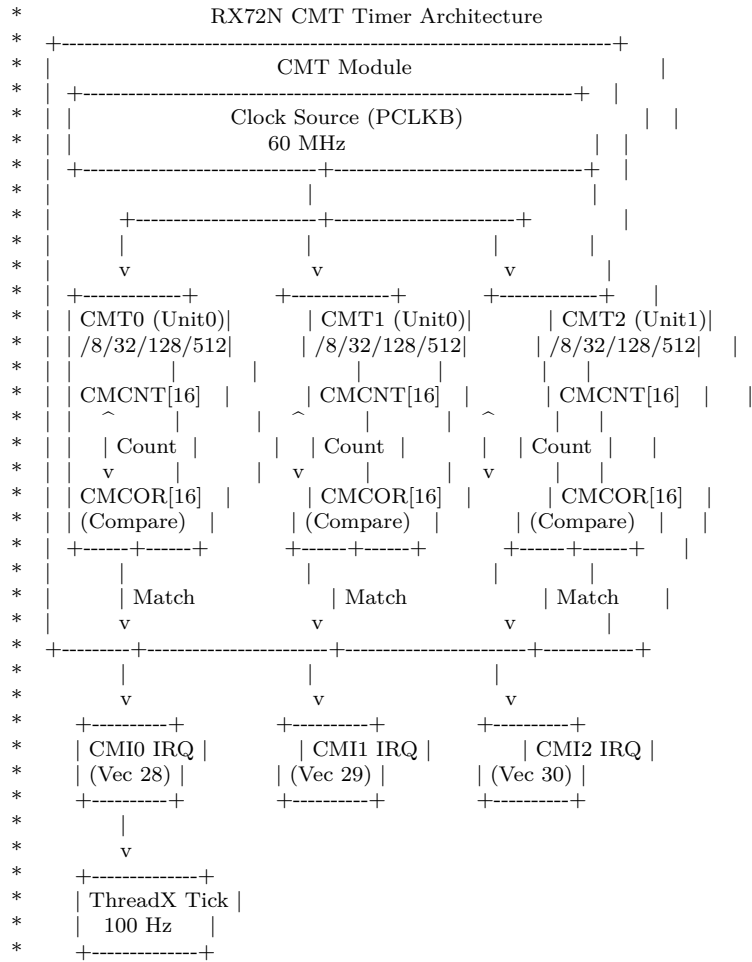
- static volatile [rx\\_cmt\\_channel\\_regs\\_t](#) \* `cmt0` (void)  
Get pointer to CMT0 registers.
- static volatile [rx\\_cmt\\_channel\\_regs\\_t](#) \* `cmt1` (void)  
Get pointer to CMT1 registers.
- static volatile [rx\\_cmt\\_channel\\_regs\\_t](#) \* `cmt2` (void)  
Get pointer to CMT2 registers.
- static volatile [rx\\_cmt\\_channel\\_regs\\_t](#) \* `cmt3` (void)  
Get pointer to CMT3 registers.
- static volatile [rx\\_cmt\\_control\\_regs\\_t](#) \* `cmt_ctrl` (void)  
Get pointer to CMT control registers (CMSTR0).
- static volatile [rx\\_cmt\\_control\\_regs\\_t](#) \* `cmt_ctrl1` (void)  
Get pointer to CMT Unit 1 control register (CMSTR1).

### 8.9.1 Detailed Description

#### RX72N Compare Match Timer (CMT) Register Definitions.

Register definitions for the Compare Match Timer (CMT) peripheral on the RX72N microcontroller. CMT provides simple periodic interrupt generation used for RTOS system tick and general timing purposes.

#### CMT System Architecture



Memory Map (Verified against RX72N Manual Ch31):

| Address    | Register                               |
|------------|----------------------------------------|
| 0x00088000 | CMSTR0 (Start Register Unit 0: CMT0/1) |
| 0x00088002 | CMT0.CMCR (Control Register)           |
| 0x00088004 | CMT0.CMCNT (Counter)                   |
| 0x00088006 | CMT0.CMCOR (Compare Match Register)    |
| 0x00088008 | CMT1.CMCR                              |
| 0x0008800A | CMT1.CMCNT                             |
| 0x0008800C | CMT1.CMCOR                             |
| 0x00088010 | CMSTR1 (Start Register Unit 1: CMT2/3) |
| 0x00088012 | CMT2.CMCR                              |
| 0x00088014 | CMT2.CMCNT                             |
| 0x00088016 | CMT2.CMCOR                             |
| 0x00088018 | CMT3.CMCR                              |
| 0x0008801A | CMT3.CMCNT                             |
| 0x0008801C | CMT3.CMCOR                             |

### Timer Period Calculation

The CMT generates an interrupt when CMCNT matches CMCOR. The timer period is calculated as:

$$T_{period} = \frac{(CMCOR + 1) \times DIV}{f_{PCLKB}}$$

Where:

- $T_{period}$  = Timer period in seconds
- $CMCOR$  = Compare match register value (0-65535)
- $DIV$  = Clock divider (8, 32, 128, or 512)
- $f_{PCLKB}$  = Peripheral clock B frequency (60 MHz)

### STAR Project ThreadX Configuration (100 Hz tick)

$$CMCOR = \frac{f_{PCLKB}}{DIV \times f_{tick}} - 1 = \frac{60,000,000}{32 \times 100} - 1 = 18,749$$

Note: PCLKB/8 cannot reach 100 Hz – it requires CMCOR=74999 which exceeds the 16-bit register maximum of 65535. Use PCLKB/32 (CKS=01) for exact 100 Hz.

Example: CMT0 Initialization for 100 Hz ThreadX Tick

```
#include "rx72n_cmt_regs.h"

// Stop CMT0 before configuration
cmt_ctrl()->cmstr0 &= ~k_rx72n_cmstr0_cmt0_enable;

// Configure CMT0: PCLKB/32 = 1.875 MHz, interrupt enable, reserved bit7=1
cmt0()->cmcr = 0x00C1; // CKS=01 (PCLKB/32), CMIE=1, bit7=1(rsvd)

// Set compare match value for 100 Hz: 1.875 MHz / 100 - 1 = 18749
cmt0()->cmcor = 18749;

// Clear counter
cmt0()->cmcnt = 0;

// Configure interrupt in ICU (priority 5, vector 28)
// ... (see rx72n_icu_regs.h for ICU configuration)

// Start CMT0
cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;
```

### NASA Power of 10 Compliance

- Rule 3: No dynamic memory - all register access via static inline functions
- Rule 5: Compile-time assertions verify all register layout assumptions
- Rule 8: Uses typed enums (C23) for all constants, no magic numbers
- Rule 10: Clean build with -Wall -Wextra -Werror

### SOLID Principles

- Single Responsibility: CMT register definitions only, no driver logic
- Open/Closed: Accessors can be used with any CMT configuration
- Interface Segregation: Minimal API - just register access functions

## Manual Reference

RX72N Group User's Manual: Hardware, Chapter 31 (Compare Match Timer)

## Verification Status

[PASS] VERIFIED (2026-01-28) - All register addresses and offsets verified against RX72N Manual Chapter 31.

## See also

[rx\\_cmt.h](#) Higher-level CMT driver API

[rx72n\\_icu\\_regs.h](#) ICU interrupt configuration for CMI interrupts

## Date

2026-01-28

## Copyright

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Definition in file [rx72n\\_cmt\\_regs.h](#).

## 8.10 rx72n\_cmt\_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_cmt_regs.h
00003 * @brief RX72N Compare Match Timer (CMT) Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the Compare Match Timer (CMT) peripheral on the
00007 * RX72N microcontroller. CMT provides simple periodic interrupt generation
00008 * used for RTOS system tick and general timing purposes.
00009 *
00010 * @par CMT System Architecture
00011 * @verbatim
00012 *
00013 * RX72N CMT Timer Architecture
00014 * +-----+
00015 * | CMT Module |
00016 * +-----+
00017 * | Clock Source (PCLKB) |
00018 * | 60 MHz |
00019 * +-----+
00020 * | | | |
00021 * | v | v | v |
00022 * +-----+ +-----+ +-----+
00023 * | CMT0 (Unit0) | | CMT1 (Unit0) | | CMT2 (Unit1) | |
00024 * | /8/32/128/512 | | /8/32/128/512 | | /8/32/128/512 | |
00025 * | CMCNT[16] | | CMCNT[16] | | CMCNT[16] | |
00026 * | ^ | | | ^ | | | ^ | | |
00027 * | Count | | | Count | | | Count | | |
00028 * | v | | | v | | | v | | |
00029 * | CMCOR[16] | | CMCOR[16] | | CMCOR[16] | |
00030 * | (Compare) | | (Compare) | | (Compare) | |
00031 * +-----+ +-----+ +-----+
00032 * | Match | | | Match | | | Match | | |
00033 * | v | | | v | | | v | | |
00034 * +-----+ +-----+ +-----+
00035 *
00036 * +-----+
00037 *

```



```

00038 * | | |
00039 * v v v
00040 * +-----+ +-----+ +-----+
00041 * | CMI0 IRQ | | CMI1 IRQ | | CMI2 IRQ |
00042 * | (Vec 28) | | (Vec 29) | | (Vec 30) |
00043 * +-----+ +-----+ +-----+
00044 * | |
00045 * v
00046 * +-----+
00047 * | ThreadX Tick |
00048 * | 100 Hz |
00049 * +-----+
00050 *
00051 * Memory Map (Verified against RX72N Manual Ch31):
00052 * +-----+
00053 * | Address | Register |
00054 * +-----+
00055 * | 0x00088000 | CMSTR0 (Start Register Unit 0: CMT0/1) |
00056 * | 0x00088002 | CMT0.CMCR (Control Register) |
00057 * | 0x00088004 | CMT0.CMCNT (Counter) |
00058 * | 0x00088006 | CMT0.CMCOR (Compare Match Register) |
00059 * | 0x00088008 | CMT1.CMCR |
00060 * | 0x0008800A | CMT1.CMCNT |
00061 * | 0x0008800C | CMT1.CMCOR |
00062 * | 0x00088010 | CMSTR1 (Start Register Unit 1: CMT2/3) |
00063 * | 0x00088012 | CMT2.CMCR |
00064 * | 0x00088014 | CMT2.CMCNT |
00065 * | 0x00088016 | CMT2.CMCOR |
00066 * | 0x00088018 | CMT3.CMCR |
00067 * | 0x0008801A | CMT3.CMCNT |
00068 * | 0x0008801C | CMT3.CMCOR |
00069 * +-----+
00070 * @endverbatim
00071 *
00072 * @par Timer Period Calculation
00073 * The CMT generates an interrupt when CMCNT matches CMCOR. The timer period
00074 * is calculated as:
00075 *
00076 * @f[
00077 * T_{period} = \frac{(CMCOR + 1) \times DIV}{f_{PCLKB}}
00078 * @f]
00079 *
00080 * Where:
00081 * - @f$ T_{period} @f$ = Timer period in seconds
00082 * - @f$ CMCOR @f$ = Compare match register value (0-65535)
00083 * - @f$ DIV @f$ = Clock divider (8, 32, 128, or 512)
00084 * - @f$ f_{PCLKB} @f$ = Peripheral clock B frequency (60 MHz)
00085 *
00086 * @par STAR Project ThreadX Configuration (100 Hz tick)
00087 * @f[
00088 * CMCOR = \frac{f_{PCLKB}}{DIV \times f_{tick}} - 1
00089 * = \frac{60,000,000}{32 \times 100} - 1 = 18,749
00090 * @f]
00091 *
00092 * Note: PCLKB/8 cannot reach 100 Hz -- it requires CMCOR=74999 which exceeds the
00093 * 16-bit register maximum of 65535. Use PCLKB/32 (CKS=01) for exact 100 Hz.
00094 *
00095 * @par Example: CMT0 Initialization for 100 Hz ThreadX Tick
00096 * @code
00097 * #include "rx72n_cmt_regs.h"
00098 *
00099 * // Stop CMT0 before configuration
00100 * cmt_ctrl()->cmstr0 &= ~k_rx72n_cmstr0_cmt0_enable;
00101 *
00102 * // Configure CMT0: PCLKB/32 = 1.875 MHz, interrupt enable, reserved bit7=1
00103 * cmt0()->cmcr = 0x00C1; // CKS=01 (PCLKB/32), CMIE=1, bit7=1(rsvd)
00104 *
00105 * // Set compare match value for 100 Hz: 1.875 MHz / 100 - 1 = 18749
00106 * cmt0()->cmcor = 18749;
00107 *
00108 * // Clear counter
00109 * cmt0()->cmcnt = 0;
00110 *
00111 * // Configure interrupt in ICU (priority 5, vector 28)
00112 * // ... (see rx72n_icu_regs.h for ICU configuration)
00113 *
00114 * // Start CMT0
00115 * cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;
00116 * @endcode
00117 *
00118 * @par NASA Power of 10 Compliance
00119 * - Rule 3: No dynamic memory - all register access via static inline functions
00120 * - Rule 5: Compile-time assertions verify all register layout assumptions
00121 * - Rule 8: Uses typed enums (C23) for all constants, no magic numbers
00122 * - Rule 10: Clean build with -Wall -Wextra -Werror
00123 *
00124 * @par SOLID Principles

```

```

00125 * - Single Responsibility: CMT register definitions only, no driver logic
00126 * - Open/Closed: Accessors can be used with any CMT configuration
00127 * - Interface Segregation: Minimal API - just register access functions
00128 *
00129 * @par Manual Reference
00130 * RX72N Group User's Manual: Hardware, Chapter 31 (Compare Match Timer)
00131 *
00132 * @par Verification Status
00133 * [PASS] VERIFIED (2026-01-28) - All register addresses and offsets verified
00134 * against RX72N Manual Chapter 31.
00135 *
00136 * @see rx_cmt.h Higher-level CMT driver API
00137 * @see rx72n_icu_regs.h ICU interrupt configuration for CMI interrupts
00138 *
00139 * @date 2026-01-28
00140 * @copyright Copyright (c) 2026 Locked Inc.
00141 * SPDX-License-Identifier: MIT
00142 */
00143
00144 #pragma once
00145
00146 #include <stddef.h>
00147 #include <stdint.h>
00148
00149 #ifdef __cplusplus
00150 extern "C" {
00151 #endif
00152
00153 /**
00154 * @defgroup cmt_regs CMT Register Definitions
00155 * @brief Compare Match Timer (CMT) hardware register interface
00156 * @{
00157 */
00158
00159 /*
=====
00160 * Compare Match Timer (CMT) - For ThreadX System Tick
00161 *
=====
00162 */
00163
00164 /**
00165 * @enum rx_cmt_addresses_t
00166 * @brief CMT base addresses (verified against RX72N Hardware Manual Ch31)
00167 *
00168 * @details
00169 * The RX72N has 4 CMT channels organized into 2 units:
00170 * - Unit 0: CMT0 and CMT1 (controlled by CMSTR0)
00171 * - Unit 1: CMT2 and CMT3 (controlled by CMSTR1)
00172 *
00173 * @par Memory Layout
00174 * @verbatim
00175 * Address | Content
00176 * -----+-----
00177 * 0x00088000 | CMSTR0 (Unit 0 start)
00178 * 0x00088002 | CMT0 registers (6 bytes)
00179 * 0x00088008 | CMT1 registers (6 bytes)
00180 * 0x00088010 | CMSTR1 (Unit 1 start)
00181 * 0x00088012 | CMT2 registers (6 bytes)
00182 * 0x00088018 | CMT3 registers (6 bytes)
00183 * @endverbatim
00184 *
00185 * @note CMSTR registers are at base, channel registers follow at +2 offset.
00186 */
00187 typedef enum : uintptr_t {
00188 k_cmt_ctrl_base_addr = 0x00088000, /**< CMSTR0 base: Unit 0 start register (CMT0/CMT1) */
00189 k_cmt1_ctrl_base_addr = 0x00088010, /**< CMSTR1 base: Unit 1 start register (CMT2/CMT3) */
00190 k_cmt0_base_addr = 0x00088002, /**< CMT0 registers (CMCR, CMCNT, CMCOR) */
00191 k_cmt1_base_addr = 0x00088008, /**< CMT1 registers (CMCR, CMCNT, CMCOR) */
00192 k_cmt2_base_addr = 0x00088012, /**< CMT2 registers (CMCR, CMCNT, CMCOR) */
00193 k_cmt3_base_addr = 0x00088018, /**< CMT3 registers (CMCR, CMCNT, CMCOR) */
00194 } rx_cmt_addresses_t;
00195
00196 /**
00197 * @enum rx_cmt_layout_t
00198 * @brief CMT register layout constants used for compile-time verification
00199 *
00200 * @details
00201 * Physical byte sizes and register offsets for CMT structures.
00202 * Used exclusively in static_assert expressions so no magic numbers
00203 * appear in code.
00204 *
00205 * @see rx_cmt_channel_regs_t
00206 * @see rx_cmt_control_regs_t
00207 *
00208 * @since Version 1.0.0
00209 */

```

```

00210 typedef enum : uint8_t {
00211 k_cmt_channel_regs_size_bytes = 6, /**< Channel struct: 3 x 16-bit registers */
00212 k_cmt_control_regs_size_bytes = 4, /**< Control struct: 2 x 16-bit registers */
00213 k_cmt_cmcr_offset = 0x00, /**< CMCR byte offset from channel base */
00214 k_cmt_cmcnt_offset = 0x02, /**< CMCNT byte offset from channel base */
00215 k_cmt_cmcpr_offset = 0x04, /**< CMCOR byte offset from channel base */
00216 k_cmt_cmstr_reg_size_bytes = 2, /**< CMSTR register width (one 16-bit word) */
00217 } rx_cmt_layout_t;
00218
00219 /**
00220 * @struct rx_cmt_channel_regs_t
00221 * @brief CMT Channel Register Map (6 bytes per channel)
00222 *
00223 * @details
00224 * Compare Match Timer (CMT) channel registers for periodic interrupt generation.
00225 * Each channel is a 16-bit up-counter that generates an interrupt when CMCNT
00226 * matches CMCOR, then resets to 0 and continues counting.
00227 *
00228 * @par Register Layout (6 bytes)
00229 * @verbatim
00230 * Offset | Size | Register | Description
00231 * -----+-----+-----+-----
00232 * 0x00 | 2 | CMCR | Control: clock div, IRQ enable
00233 * 0x02 | 2 | CMCNT | Counter: current count value
00234 * 0x04 | 2 | CMCOR | Compare: match value (period)
00235 * @endverbatim
00236 *
00237 * @par CMCR Register Bit Layout (RX72N Manual Ch31, p.1583)
00238 * @verbatim
00239 * Bit | Name | Description
00240 * ----+-----+-----
00241 * 15-8 | - | Reserved
00242 * 7 | - | Reserved (write value should be 1)
00243 * 6 | CMIE | Compare Match Interrupt Enable
00244 * 5-2 | - | Reserved
00245 * 1-0 | CKS | Clock Select (00=PCLK/8, 01=/32, 10=/128, 11=/512)
00246 * @endverbatim
00247 *
00248 * @par Timer Operation
00249 * 1. Configure CMCR with clock divider and interrupt enable
00250 * 2. Set CMCOR to desired period value
00251 * 3. Clear CMCNT to 0
00252 * 4. Enable timer via CMSTRn register
00253 * 5. Counter increments at PCLKB/DIV rate
00254 * 6. When CMCNT == CMCOR, interrupt fires and CMCNT resets to 0
00255 *
00256 * @invariant CMCNT <= CMCOR during normal operation
00257 *
00258 * @see rx_cmt_control_regs_t For start/stop control (CMSTR0/1)
00259 */
00260 typedef struct {
00261 /**
00262 * @brief Compare Match Timer Control Register
00263 *
00264 * @details
00265 * Controls clock source divider and interrupt enable.
00266 *
00267 * | Bits | Name | R/W | Description |
00268 * |-----+-----+-----+-----|
00269 * | 7 | - | R/W | Reserved; write value should be 1 (manual p.1583) |
00270 * | 6 | CMIE | R/W | 1 = Enable compare match interrupt |
00271 * | 1-0 | CKS | R/W | Clock select: 00=/8, 01=/32, 10=/128, 11=/512 |
00272 */
00273 volatile uint16_t cmcr;
00274
00275 /**
00276 * @brief Compare Match Timer Counter Register
00277 *
00278 * @details
00279 * 16-bit up-counter incremented by selected clock. Cleared to 0
00280 * on compare match and when timer is stopped.
00281 *
00282 * - Read: Current counter value
00283 * - Write: Set counter value (typically write 0 before starting)
00284 */
00285 volatile uint16_t cmcnt;
00286
00287 /**
00288 * @brief Compare Match Timer Constant Register
00289 *
00290 * @details
00291 * Compare match value. When CMCNT equals this value, an interrupt
00292 * is generated (if CMIE=1) and CMCNT resets to 0.
00293 *
00294 * Period calculation: @f$ T = \frac{(CMCOR + 1)}{\text{times DIV}} \{f_{PCLKB}\} @f$
00295 *
00296 * | CMCOR Value | Divider | Period @ 60MHz PCLKB |

```

```

00297 * |-----|-----|-----|
00298 * | 18749 | /32 | 10 ms (100 Hz) |
00299 * | 1874 | /32 | 1 ms (1000 Hz) |
00300 * | 7499 | /8 | 1 ms (1000 Hz) |
00301 *
00302 * Note: PCLKB/8 cannot reach 100 Hz (requires CMCOR=74999 > uint16_t max 65535).
00303 */
00304 volatile uint16_t cmcor;
00305 } rx_cmt_channel_regs_t;
00306
00307 /**
00308 * @struct rx_cmt_control_regs_t
00309 * @brief CMT Start/Stop Control Register Map
00310 *
00311 * @details
00312 * Controls starting and stopping of CMT channels. Each unit has a
00313 * separate start register controlling two channels.
00314 *
00315 * @par Memory Layout
00316 * @verbatim
00317 * Address | Register | Controls
00318 * +-----+ +-----+
00319 * 0x00088000 | CMSTR0 | CMT0, CMT1
00320 * 0x00088010 | CMSTR1 | CMT2, CMT3
00321 * @endverbatim
00322 *
00323 * @note CMSTR0 and CMSTR1 are NOT contiguous! CMSTR1 is at +0x10 from CMSTR0.
00324 * This struct is only for documentation; use cmt_ctrl() for CMSTR0.
00325 *
00326 * @par Usage Example
00327 * @code
00328 * // Start CMT0
00329 * cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;
00330 *
00331 * // Stop CMT0
00332 * cmt_ctrl()->cmstr0 &= ~k_rx72n_cmstr0_cmt0_enable;
00333 * @endcode
00334 */
00335 typedef struct {
00336 /**
00337 * @brief Compare Match Timer Start Register 0
00338 * @details Controls CMT0 and CMT1 operation.
00339 * - Bit 0 (STR0): 1 = Start CMT0, 0 = Stop CMT0
00340 * - Bit 1 (STR1): 1 = Start CMT1, 0 = Stop CMT1
00341 */
00342 volatile uint16_t cmstr0;
00343
00344 /**
00345 * @brief Compare Match Timer Start Register 1
00346 * @details Controls CMT2 and CMT3 operation.
00347 * - Bit 0 (STR2): 1 = Start CMT2, 0 = Stop CMT2
00348 * - Bit 1 (STR3): 1 = Start CMT3, 0 = Stop CMT3
00349 *
00350 * @warning This field is NOT at the expected offset! CMSTR1 is at
00351 * 0x00088010, not 0x00088002. Access via separate function.
00352 */
00353 volatile uint16_t cmstr1;
00354 } rx_cmt_control_regs_t;
00355
00356 /**
00357 * @enum rx_cmt_cmstr0_bits_t
00358 * @brief CMSTR0 register bit definitions (CMT Unit 0 start control)
00359 *
00360 * @details
00361 * Bit field definitions for Compare Match Timer Start Register 0.
00362 * Controls starting and stopping of CMT0 and CMT1 channels.
00363 *
00364 * @par Register Layout (16-bit, only bits 0-1 used)
00365 * @verbatim
00366 * Bit | Name | Description
00367 * +---+ +-----+
00368 * 0 | STR0 | 1 = Start CMT0, 0 = Stop CMT0
00369 * 1 | STR1 | 1 = Start CMT1, 0 = Stop CMT1
00370 * 2-15| - | Reserved (write 0)
00371 * @endverbatim
00372 */
00373 typedef enum : uint8_t {
00374 k_rx72n_cmstr0_cmt0_enable = 0x01, /**< Bit 0: Start/stop CMT0 counter */
00375 k_rx72n_cmstr0_cmt1_enable = 0x02, /**< Bit 1: Start/stop CMT1 counter */
00376 } rx_cmt_cmstr0_bits_t;
00377
00378 /**
00379 * @enum rx_cmt_cmstr1_bits_t
00380 * @brief CMSTR1 register bit definitions (CMT Unit 1 start control)
00381 *
00382 * @details
00383 * Bit field definitions for Compare Match Timer Start Register 1.

```

```

00384 * Controls starting and stopping of CMT2 and CMT3 channels.
00385 *
00386 * @note CMSTR1 is at address 0x00088010, which is 16 bytes after CMSTR0.
00387 *
00388 * @par Register Layout (16-bit, only bits 0-1 used)
00389 * @verbatim
00390 * Bit | Name | Description
00391 * ----+-----+-----
00392 * 0 | STR2 | 1 = Start CMT2, 0 = Stop CMT2
00393 * 1 | STR3 | 1 = Start CMT3, 0 = Stop CMT3
00394 * 2-15| - | Reserved (write 0)
00395 * @endverbatim
00396 */
00397 typedef enum : uint8_t {
00398 k_rx72n_cmstr1_cmt2_enable = 0x01, /**< Bit 0: Start/stop CMT2 counter */
00399 k_rx72n_cmstr1_cmt3_enable = 0x02, /**< Bit 1: Start/stop CMT3 counter */
00400 } rx_cmt_cmstr1_bits_t;
00401
00402 /*
=====
00403 * Inline Register Accessor Functions
00404 *
=====
00405 * These functions provide type-safe access to CMT hardware registers.
00406 * Using functions instead of macros enables:
00407 * - Type checking at compile time
00408 * - Debugger-friendly register access
00409 * - Consistent volatile semantics
00410 */
00411
00412 /**
00413 * @brief Get pointer to CMT0 registers
00414 *
00415 * @details
00416 * Returns a volatile pointer to CMT channel 0 registers at address 0x00088002.
00417 * CMT0 is typically used for the ThreadX system tick (100 Hz).
00418 *
00419 * @return Volatile pointer to CMT0 register structure
00420 *
00421 * @par Example
00422 * @code
00423 * // Configure CMT0 for 100 Hz interrupt with PCLKB/32 divider
00424 * cmt0()->cmcr = 0x00C1; // CKS=01 (PCLKB/32), CMIE=1, bit7=1(rsvd)
00425 * cmt0()->cmcor = 18749; // 60MHz/32/100Hz - 1
00426 * cmt0()->cmcnt = 0; // Clear counter
00427 * @endcode
00428 *
00429 * @see cmt_ctrl() For start/stop control
00430 */
00431 static inline volatile rx_cmt_channel_regs_t* cmt0(void)
00432 {
00433 return (volatile rx_cmt_channel_regs_t*)k_cmt0_base_addr;
00434 }
00435
00436 /**
00437 * @brief Get pointer to CMT1 registers
00438 *
00439 * @details
00440 * Returns a volatile pointer to CMT channel 1 registers at address 0x00088008.
00441 * CMT1 is available for general-purpose timing.
00442 *
00443 * @return Volatile pointer to CMT1 register structure
00444 */
00445 static inline volatile rx_cmt_channel_regs_t* cmt1(void)
00446 {
00447 return (volatile rx_cmt_channel_regs_t*)k_cmt1_base_addr;
00448 }
00449
00450 /**
00451 * @brief Get pointer to CMT2 registers
00452 *
00453 * @details
00454 * Returns a volatile pointer to CMT channel 2 registers at address 0x00088012.
00455 * CMT2 is part of Unit 1 (controlled by CMSTR1).
00456 *
00457 * @return Volatile pointer to CMT2 register structure
00458 */
00459 static inline volatile rx_cmt_channel_regs_t* cmt2(void)
00460 {
00461 return (volatile rx_cmt_channel_regs_t*)k_cmt2_base_addr;
00462 }
00463
00464 /**
00465 * @brief Get pointer to CMT3 registers
00466 *
00467 * @details
00468 * Returns a volatile pointer to CMT channel 3 registers at address 0x00088018.

```

```

00469 * CMT3 is part of Unit 1 (controlled by CMSTR1).
00470 *
00471 * @return Volatile pointer to CMT3 register structure
00472 */
00473 static inline volatile rx_cmt_channel_regs_t* cmt3(void)
00474 {
00475 return (volatile rx_cmt_channel_regs_t*)k_cmt3_base_addr;
00476 }
00477
00478 /**
00479 * @brief Get pointer to CMT control registers (CMSTR0)
00480 *
00481 * @details
00482 * Returns a volatile pointer to CMT control register structure at address
00483 * 0x00088000. Provides access to CMSTR0 for starting/stopping CMT0 and CMT1.
00484 *
00485 * @warning CMSTR1 (for CMT2/3) is at 0x00088010, not adjacent to CMSTR0.
00486 * The rx_cmt_control_regs_t struct has incorrect cmstr1 offset.
00487 * For CMSTR1 access, use: `(volatile uint16_t*)0x00088010`
00488 *
00489 * @return Volatile pointer to CMT control register structure
00490 *
00491 * @par Example
00492 * @code
00493 * // Start CMT0
00494 * cmt_ctrl()->cmstr0 |= k_rx72n_cmstr0_cmt0_enable;
00495 *
00496 * // Stop CMT0 and CMT1
00497 * cmt_ctrl()->cmstr0 = 0;
00498 * @endcode
00499 */
00500 static inline volatile rx_cmt_control_regs_t* cmt_ctrl(void)
00501 {
00502 return (volatile rx_cmt_control_regs_t*)k_cmt_ctrl_base_addr;
00503 }
00504
00505 /**
00506 * @brief Get pointer to CMT Unit 1 control register (CMSTR1)
00507 *
00508 * @details
00509 * Returns a volatile pointer to CMSTR1 at address 0x00088010. CMSTR1 is NOT
00510 * contiguous with CMSTR0 -- it is 0x10 bytes after CMSTR0. This accessor
00511 * provides correct access to CMSTR1 for starting/stopping CMT2 and CMT3.
00512 *
00513 * @warning Do NOT use cmt_ctrl()->cmstr1 to access CMSTR1 -- the struct field
00514 * cmstr1 sits at offset +2 from CMSTR0 (address 0x00088002), which is
00515 * CMT0.CMCR, not CMSTR1.
00516 *
00517 * @return Volatile pointer to CMT Unit 1 control register structure
00518 *
00519 * @par Example
00520 * @code
00521 * // Start CMT2
00522 * cmt_ctrl1()->cmstr0 |= k_rx72n_cmstr1_cmt2_enable;
00523 *
00524 * // Stop CMT3
00525 * cmt_ctrl1()->cmstr0 &= ~k_rx72n_cmstr1_cmt3_enable;
00526 * @endcode
00527 *
00528 * @see cmt_ctrl() For CMSTR0 (CMT0/CMT1 control)
00529 */
00530 static inline volatile rx_cmt_control_regs_t* cmt_ctrl1(void)
00531 {
00532 return (volatile rx_cmt_control_regs_t*)k_cmt1_ctrl_base_addr;
00533 }
00534
00535 /**
=====
00536 * Static Assertions - Compile-time verification of register layout
00537 *
=====
00538 * These assertions verify our register structures match the RX72N hardware
00539 * exactly. If any assertion fails, the struct definition must be corrected.
00540 *
00541 * Reference: RX72N Group User's Manual: Hardware, Chapter 31, Section 21.2
00542 *
=====
00543 */
00544
00545 /**
00546 * @defgroup cmt_static_assert CMT Static Assertions
00547 * @brief Compile-time verification of CMT register layout
00548 * @details
00549 * These static assertions are verified at compile time to ensure the
00550 * C structure definitions match the actual hardware register layout.
00551 * A failed assertion indicates a bug in the register definitions that
00552 * must be fixed before the code can compile.

```

```

00553 * @{
00554 */
00555
00556 /* Verify CMT channel register structure size (3 x 16-bit = 6 bytes) */
00557 static_assert(sizeof(rx_cmt_channel_regs_t) == k_cmt_channel_regs_size_bytes,
00558 "CMT channel register structure size mismatch");
00559
00560 /* Verify CMT channel register offsets */
00561 static_assert(offsetof(rx_cmt_channel_regs_t, cmcr) == k_cmt_cmcr_offset,
00562 "CMT CMCR register offset incorrect");
00563 static_assert(offsetof(rx_cmt_channel_regs_t, cmcnt) == k_cmt_cmcnt_offset,
00564 "CMT CMCNT register offset incorrect");
00565 static_assert(offsetof(rx_cmt_channel_regs_t, cmcor) == k_cmt_cmcor_offset,
00566 "CMT CMCOR register offset incorrect");
00567
00568 /* Verify CMT control register structure size (2 x 16-bit = 4 bytes) */
00569 static_assert(sizeof(rx_cmt_control_regs_t) == k_cmt_control_regs_size_bytes,
00570 "CMT control register structure size mismatch");
00571
00572 /* Verify address relationships: each channel immediately follows prior struct */
00573 static_assert(k_cmt0_base_addr - k_cmt_ctrl_base_addr == k_cmt_cmstr_reg_size_bytes,
00574 "CMT0 must start one 16-bit CMSTR0 word after unit 0 base");
00575 static_assert(k_cmt1_base_addr - k_cmt0_base_addr == k_cmt_channel_regs_size_bytes,
00576 "CMT0 to CMT1 spacing must equal one channel struct");
00577 static_assert(k_cmt2_base_addr - k_cmt1_ctrl_base_addr == k_cmt_cmstr_reg_size_bytes,
00578 "CMT2 must start one 16-bit CMSTR1 word after unit 1 base");
00579 static_assert(k_cmt3_base_addr - k_cmt2_base_addr == k_cmt_channel_regs_size_bytes,
00580 "CMT2 to CMT3 spacing must equal one channel struct");
00581
00582 /** @} */ /* End of cmt_static_assert group */
00583
00584 /** @} */ /* End of cmt_regs group */
00585
00586 #ifdef __cplusplus
00587 }
00588 #endif

```

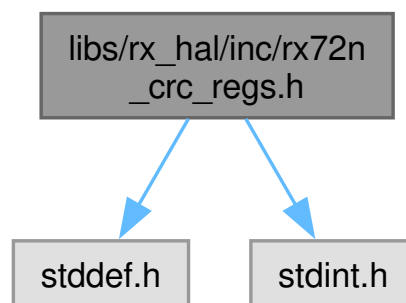
## 8.11 libs/rx\_hal/inc/rx72n\_crc\_regs.h File Reference

RX72N CRC Calculator Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_crc\_regs.h:



This graph shows which files directly or indirectly include this file:



### Data Structures

- struct `rx_crc_regs_t`  
CRC Calculator Register Map Structure.

## Enumerations

- enum `rx_crc_addresses_t` : `uintptr_t` { `k_crc_base_addr` = 0x00088280 }  
CRC module base address.
- enum `rx_crc_reserved_sizes_t` : `uint8_t` { `k_crc_reserved_after_crccr_bytes` = 3 }  
CRC register reserved field sizes.
- enum `rx_crc_crccr_shifts_t` : `uint8_t` { `k_crc_bit_mask` = 1 , `k_crc_crccr_lms_shift` = 6 , `k_crc_crccr_dorclr_shift` = 7 }  
CRC Control Register (CRCCR) Bit Field Positions.
- enum `rx_crc_crccr_masks_t` : `uint8_t` { `k_crc_crccr_lms_mask` , `k_crc_crccr_dorclr_mask` }  
CRC Control Register (CRCCR) Bit Masks.
- enum `rx_crc_crccr_bits_t` : `uint8_t` { `k_crc_crccr_gps_crc32` = 0x03 , `k_crc_crccr_lms` = `k_crc_crccr_lms_mask` , `k_crc_crccr_dorclr` = `k_crc_crccr_dorclr_mask` }  
CRC Control Register (CRCCR) Bit Definitions.

## Functions

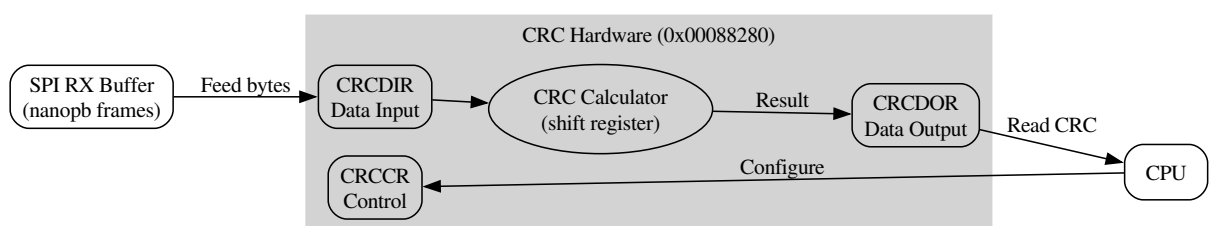
- static volatile `rx_crc_regs_t` \* `crc_regs` (void)  
Get pointer to CRC registers.

## 8.11.1 Detailed Description

## RX72N CRC Calculator Register Definitions.

Register definitions for the hardware CRC Calculator used for CRC-32 acceleration. The CRC module provides hardware-accelerated cyclic redundancy check computation, significantly faster than software implementations.

## System Architecture



## STAR Project CRC Usage

| Use Case                 | Polynomial  | Direction | Notes                   |
|--------------------------|-------------|-----------|-------------------------|
| SPI Protocol (RPI5)      | CRC-32 IEEE | LSB-first | nanopb frame validation |
| Flash Integrity (future) | CRC-32 IEEE | MSB-first | Firmware verification   |



### Key Features

- Hardware-accelerated CRC calculation (single cycle per byte)
- CRC-32 IEEE 802.3 polynomial (0x04C11DB7) for compatibility with Go
- Configurable bit order (MSB-first or LSB-first/reflected)
- Direct memory-mapped register access
- Supports CRC-8, CRC-16, CRC-CCITT, CRC-32, CRC-32C (only CRC-32 used)

### CRC-32 IEEE 802.3 Specification

- Polynomial: 0x04C11DB7 (normal form)
- Initial Value: 0xFFFFFFFF
- Reflect Input: Yes (LSB-first in STAR project)
- Reflect Output: Yes
- Final XOR: 0xFFFFFFFF

### Performance

| Method       | Time for 64 bytes | Notes                       |
|--------------|-------------------|-----------------------------|
| Hardware CRC | ~64 cycles        | 1 cycle per byte @ 240 MHz  |
| Software CRC | ~2000+ cycles     | Table lookup method         |
| Speedup      | ~31x              | Critical for SPI throughput |

### Hardware Requirements

| Parameter    | Value           | Notes                      |
|--------------|-----------------|----------------------------|
| Base Address | 0x00088280      | 12-byte register block     |
| Module Stop  | MSTPCRB.MSTPB23 | Clear to enable module     |
| Clock        | PCLKB (60 MHz)  | No additional clock needed |

### NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops in register definitions)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessor functions are single-statement
- Rule 5: [OK] N/A (hardware layer)
- Rule 6: [OK] Minimal scope - inline accessors only
- Rule 7: [OK] N/A (no return values to check)
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

## SOLID Principles

- S: Single responsibility - only CRC register definitions
- O: Open for extension via additional polynomial support
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessor
- D: N/A (hardware register layer)

## Module Dependencies

- `<stdint.h>` - Fixed-width integer types
- `<stddef.h>` - `offsetof` macro for static assertions

## Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 46: CRC Calculator, pages 2411-2420

## Compatibility

The CRC-32 IEEE 802.3 output is compatible with:

- Go: `crc32.ChecksumIEEE()`
- Python: `binascii.crc32()`
- Linux: `cksum -o 3`

## See also

[rx\\_crc.h](#) Higher-level CRC API

[rx72n\\_regs.h](#) Main register include file

[docs/sections/01\\_nanopb\\_protocol.tex](#) Protocol specification

## Author

Locked, Inc.

## Date

2026-01-28

## Copyright

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## Since

Version 1.0.0

Definition in file [rx72n\\_crc\\_regs.h](#).

## 8.11.2 Data Structure Documentation

### 8.11.2.1 struct rx\_crc\_regs\_t

CRC Calculator Register Map Structure.

Hardware CRC Calculator registers for accelerated CRC computation. The CRC module computes CRC checksums in hardware, significantly faster than software implementations.

Memory Layout Table

| Offset | Size | Field    | Type          | Description          |
|--------|------|----------|---------------|----------------------|
| 0x00   | 1    | crccr    | uint8↔<br>_t  | Control Register     |
| 0x01   | 3    | reserved | -             | Reserved             |
| 0x04   | 4    | crkdir   | uint32↔<br>_t | Data Input Register  |
| 0x08   | 4    | crcdor   | uint32↔<br>_t | Data Output Register |
| Total  | 12   |          |               |                      |

### CRC Calculation Process

1. Configure CRCCR with polynomial and bit order
2. Clear the output register by writing DORCLR=1
3. Feed data bytes/words to CRCDIR
4. Read final CRC from CRCDOR

### Initialization Example (CRC-32 IEEE)

```
// Configure CRC-32 IEEE 802.3 with LSB-first (reflected) mode
volatile rx_crc_regs_t* crc = crc_regs();

// 1. Set polynomial to CRC-32 IEEE, LSB-first
crc->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms;

// 2. Clear output register (initialize to 0xFFFFFFFF)
crc->crccr |= k_crc_crccr_dorclr;

// 3. Feed data (example: 4 bytes)
crc->crkdir = 0x12345678; // Feed 4 bytes at once

// 4. Read result
uint32_t crc_result = crc->crcdor;
```

### Byte-by-Byte Example

```
// Calculate CRC of a buffer
uint32_t calculate_crc32(const uint8_t* data, uint32_t len) {
 volatile rx_crc_regs_t* crc = crc_regs();

 // Configure for CRC-32 IEEE, LSB-first
 crc->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms | k_crc_crccr_dorclr;

 // Feed data byte by byte
 for (uint32_t i = 0; i < len; i++) {
 ((volatile uint8_t)&crc->crkdir) = data[i];
 }

 // Return CRC (XOR with 0xFFFFFFFF for IEEE compliance)
 return crc->crcdor ^ 0xFFFFFFFF;
}
```

## Invariant

`sizeof(rx_crc_regs_t) == 12 bytes`  
All offsets match RX72N Hardware Manual Ch46

## Note

Hardware automatically processes each byte written to CRCDIR

## Warning

Must clear CRCDOR (DORCLR) before starting new calculation

## Attention

CRC module must be enabled in MSTPCRB before use

## See also

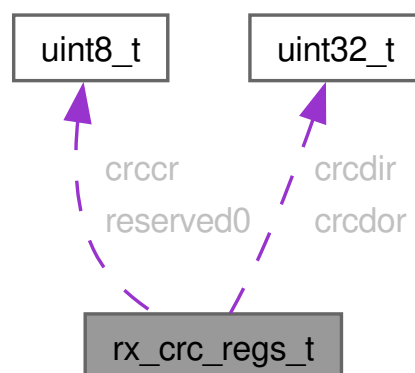
[crc\\_regs\(\)](#) Accessor function  
[rx\\_crc\\_crccr\\_bits\\_t](#) Control register bit definitions

## Since

Version 1.0.0

Definition at line 242 of file [rx72n\\_crc\\_regs.h](#).

Collaboration diagram for `rx_crc_regs_t`:



## Data Fields

|                   |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | cccr   | <p>CRC Control Register (CR-CCR) @ offset 0x00.<br/>Configures polynomial selection and bit order.</p> <p>Bit Fields:</p> <ul style="list-style-type: none"> <li>• Bits 2:0 (GPS): Generator Polynomial Select</li> <li>• Bit 6 (LMS): LSB/↔ MSB First (0=MSB, 1=LSB/reflected)</li> <li>• Bit 7 (DORCLR):↔ Data Output Register Clear (write 1 to clear)</li> </ul> <p>See also</p> <p><a href="#">rx_crc_cccr_bits_t</a> Bit definitions</p> |
| volatile uint32_t | crkdir | <p>CRC Data Input Register (CR-CDIR) @ offset 0x04.<br/>Write data here to feed into CRC calculator.</p> <p>Access:</p> <ul style="list-style-type: none"> <li>• Write: Feed data bytes/words into calculator</li> <li>• Read: Returns last written value (not useful)</li> </ul> <p>Note</p> <p>Can write 8, 16, or 32 bits - all processed by hardware</p>                                                                                   |

|                   |                                                  |                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint32_t | crcdor                                           | <p>CRC Data Output Register (CRCDOR) @ offset 0x08. Read computed CRC result from here.</p> <p>Access:</p> <ul style="list-style-type: none"> <li>• Read: Current CRC value</li> <li>• Write: Clears to initial value (same as DORCLR)</li> </ul> <p>Note</p> <p>For IEEE compliance, XOR result with 0x←FFFFFFF</p> |
| uint8_t           | reserved0←<br>[k_crc_reserved_after_crccr_bytes] | Reserved @ 0x01-0x03                                                                                                                                                                                                                                                                                                 |

### 8.11.3 Enumeration Type Documentation

#### 8.11.3.1 rx\_crc\_addresses\_t

enum [rx\\_crc\\_addresses\\_t](#) : uintptr\_t

CRC module base address.

Base address for the CRC Calculator register block. The CRC module provides hardware-accelerated CRC computation with configurable polynomials.

#### Memory Map

| Address    | Register | Size | Description          |
|------------|----------|------|----------------------|
| 0x00088280 | CRCCR    | 1    | Control Register     |
| 0x00088281 | Reserved | 3    | -                    |
| 0x00088284 | CRCDIR   | 4    | Data Input Register  |
| 0x00088288 | CRCDOR   | 4    | Data Output Register |

See also

[crc\\_regs\(\)](#) Accessor function

Since

Version 1.0.0

Enumerator

|                              |                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------|
| <code>k_crc_base_addr</code> | CRC register base address (0x00088280). Verified against RX72N Hardware Manual Ch46 |
|------------------------------|-------------------------------------------------------------------------------------|

Definition at line 149 of file `rx72n_crc_regs.h`.

```

00149 : uintptr_t {
00150 /**
00151 * @brief CRC register base address (0x00088280)
00152 * @details Verified against RX72N Hardware Manual Ch46
00153 */
00154 k_crc_base_addr = 0x00088280,
00155 } rx_crc_addresses_t;

```

### 8.11.3.2 rx\_crc\_crccr\_bits\_t

```
enum rx_crc_crccr_bits_t : uint8_t
```

CRC Control Register (CRCCR) Bit Definitions.

Configuration values for CRC polynomial selection and bit order. Only CRC-32 IEEE 802.3 is used in the STAR project.

Generator Polynomial Select (GPS) Values

| GPS  | Polynomial       | Name        | Used in STAR |
|------|------------------|-------------|--------------|
| 0x00 | $X^8 + \dots$    | CRC-8       | No           |
| 0x01 | $X^{16} + \dots$ | CRC-16      | No           |
| 0x02 | $X^{16} + \dots$ | CRC-CCITT   | No           |
| 0x03 | 0x04C11DB7       | CRC-32 IEEE | Yes          |
| 0x04 | 0x1EDC6F41       | CRC-32C     | No           |

Bit Order Selection (LMS)

- MSB-first (LMS=0): Standard CRC, data processed from bit 7 to bit 0
- LSB-first (LMS=1): Reflected CRC, data processed from bit 0 to bit 7

For IEEE 802.3 compatibility (Go `crc32.ChecksumIEEE`), use LSB-first.

Configuration Example

```

// CRC-32 IEEE 802.3, LSB-first (reflected), clear output
crc_regs()->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms | k_crc_crccr_dorclr;

```

See also

[rx\\_crc\\_regs\\_t::crccr](#) Control register

Since

Version 1.0.0

#### Enumerator

|                       |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_crc_crccr_gps_crc32 | <p>CRC-32 IEEE 802.3 polynomial (GPS=0x03). Polynomial: <math>0x04C11DB7 (x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1)</math></p> <p>Compatibility:</p> <ul style="list-style-type: none"> <li>• Go: <code>crc32.ChecksumIEEE()</code></li> <li>• Python: <code>binascii.crc32()</code></li> <li>• Ethernet, ZIP, PNG</li> </ul> |
| k_crc_crccr_lms       | <p>LSB-first (reflected) bit order (LMS=1). Process data LSB first. Required for IEEE 802.3 compatibility. Set this bit for standard Ethernet/ZIP CRC-32 behavior.</p>                                                                                                                                                                                                                           |
| k_crc_crccr_dorclr    | <p>Clear Data Output Register (DORCLR=1). Writing 1 clears CRCDOR to initial value (0xFFFFFFFF for CRC-32). Must be done before starting a new CRC calculation.</p> <p>Note</p> <p>This bit is write-only and reads as 0</p>                                                                                                                                                                     |

Definition at line 384 of file [rx72n\\_crc\\_regs.h](#).

```

00384 : uint8_t {
00385 /**
00386 * @brief CRC-32 IEEE 802.3 polynomial (GPS=0x03)
00387 * @details Polynomial: 0x04C11DB7 ($x^{32} + x^{26} + x^{23} + x^{22} + x^{16} +$
00388 * $x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$)
00389 * @par Compatibility:
00390 * - Go: crc32.ChecksumIEEE()
00391 * - Python: binascii.crc32()
00392 * - Ethernet, ZIP, PNG
00393 */
00394 k_crc_crccr_gps_crc32 = 0x03,
00395
00396 /**
00397 * @brief LSB-first (reflected) bit order (LMS=1)
00398 * @details Process data LSB first. Required for IEEE 802.3 compatibility.
00399 * Set this bit for standard Ethernet/ZIP CRC-32 behavior.
00400 */
00401 k_crc_crccr_lms = k_crc_crccr_lms_mask,
00402
00403 /**
00404 * @brief Clear Data Output Register (DORCLR=1)
00405 * @details Writing 1 clears CRCDOR to initial value (0xFFFFFFFF for CRC-32).
00406 * Must be done before starting a new CRC calculation.
00407 * @note This bit is write-only and reads as 0
00408 */
00409 k_crc_crccr_dorclr = k_crc_crccr_dorclr_mask,
00410 } rx_crc_crccr_bits_t;

```

#### 8.11.3.3 rx\_crc\_crccr\_masks\_t

```
enum rx_crc_crccr_masks_t : uint8_t
```

CRC Control Register (CRCCR) Bit Masks.

Bit masks for isolating specific fields in the CRCCR register.



Since

Version 1.0.0

Enumerator

|                         |                            |
|-------------------------|----------------------------|
| k_crc_crccr_lms_mask    | LSB/MSB First mask (bit 6) |
| k_crc_crccr_dorclr_mask | DORCLR mask (bit 7)        |

Definition at line 345 of file [rx72n\\_crc\\_regs.h](#).

```
00345 : uint8_t {
00346 k_crc_crccr_lms_mask =
00347 (k_crc_bit_mask « k_crc_crccr_lms_shift), /**< LSB/MSB First mask (bit 6) */
00348 k_crc_crccr_dorclr_mask =
00349 (k_crc_bit_mask « k_crc_crccr_dorclr_shift), /**< DORCLR mask (bit 7) */
00350 } rx_crc_crccr_masks_t;
```

#### 8.11.3.4 rx\_crc\_crccr\_shifts\_t

enum [rx\\_crc\\_crccr\\_shifts\\_t](#) : uint8\_t

CRC Control Register (CRCCR) Bit Field Positions.

Bit positions for CRCCR register fields. Used to construct field values.

Register Layout (8-bit @ offset 0x00)

| Bits    | Field  | Description                 |
|---------|--------|-----------------------------|
| 2↔<br>0 | GPS    | Generator Polynomial Select |
| 5↔<br>3 | -      | Reserved                    |
| 6       | LMS    | LSB/MSB First selection     |
| 7       | DORCLR | Data Output Register Clear  |

Since

Version 1.0.0

Enumerator

|                          |                               |
|--------------------------|-------------------------------|
| k_crc_bit_mask           | Single-bit mask               |
| k_crc_crccr_lms_shift    | LMS field position (bit 6)    |
| k_crc_crccr_dorclr_shift | DORCLR field position (bit 7) |

Definition at line 330 of file [rx72n\\_crc\\_regs.h](#).

```
00330 : uint8_t {
00331 k_crc_bit_mask = 1, /**< Single-bit mask */
00332 k_crc_crccr_lms_shift = 6, /**< LMS field position (bit 6) */
00333 k_crc_crccr_dorclr_shift = 7, /**< DORCLR field position (bit 7) */
00334 } rx_crc_crccr_shifts_t;
```

### 8.11.3.5 rx\_crc\_reserved\_sizes\_t

enum [rx\\_crc\\_reserved\\_sizes\\_t](#) : uint8\_t

CRC register reserved field sizes.

Defines reserved byte counts for structure padding to match hardware layout.

Since

Version 1.0.0

Enumerator

|                                  |                            |
|----------------------------------|----------------------------|
| k_crc_reserved_after_crccr_bytes | Reserved bytes @ 0x01-0x03 |
|----------------------------------|----------------------------|

Definition at line 166 of file [rx72n\\_crc\\_regs.h](#).

```
00166 : uint8_t {
00167 k_crc_reserved_after_crccr_bytes = 3, /**< Reserved bytes @ 0x01-0x03 */
00168 } rx_crc_reserved_sizes_t;
```

## 8.11.4 Function Documentation

### 8.11.4.1 crc\_regs()

volatile [rx\\_crc\\_regs\\_t](#) \* [crc\\_regs](#) (  
void ) [inline], [static]

Get pointer to CRC registers.

Returns a volatile pointer to the CRC Calculator register structure at address 0x00088280. The CRC module provides hardware-accelerated CRC computation for protocol integrity checking.

Returns

Volatile pointer to CRC register structure

Return values

|          |                                                 |
|----------|-------------------------------------------------|
| Non-NULL | Always returns valid pointer (hardware address) |
|----------|-------------------------------------------------|

Precondition

CRC module must be enabled (MSTPCRB.MSTPB23 = 0)

Postcondition

Pointer is valid for the lifetime of program execution

## Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

## Warning

Ensure module stop bit is cleared before accessing registers

## Example:

```
// Quick CRC calculation
volatile rx_crc_regs_t* crc = crc_regs();
crc->cccr = k_crc_cccr_gps_crc32 | k_crc_cccr_lms | k_crc_cccr_dorclr;
crc->credir = data_word;
uint32_t result = crc->credor ^ 0xFFFFFFFF;
```

## See also

[k\\_crc\\_base\\_addr](#) Base address constant

[rx\\_crc32\\_init\(\)](#) Higher-level initialization

## Since

Version 1.0.0

Definition at line 308 of file [rx72n\\_crc\\_regs.h](#).

```
00309 {
00310 return (volatile rx_crc_regs_t*)k_crc_base_addr;
00311 }
```

References [k\\_crc\\_base\\_addr](#).

## 8.12 rx72n`crc`regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002 * @file rx72n_crc_regs.h
00003 * @brief RX72N CRC Calculator Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the hardware CRC Calculator used for CRC-32
00007 * acceleration. The CRC module provides hardware-accelerated cyclic redundancy
00008 * check computation, significantly faster than software implementations.
00009 *
00010 * @par System Architecture
00011 * @dot
00012 * digraph crc_architecture {
00013 * rankdir=LR;
00014 * node [shape=box, style=rounded];
00015 *
00016 * subgraph cluster_crc {
00017 * label="CRC Hardware (0x00088280)";
00018 * style=filled;
00019 * color=lightgrey;
00020 *
00021 * ctrl [label="CRCCR\nControl"];
00022 * din [label="CRCDIR\nData Input"];
00023 * dout [label="CRCDOR\nData Output"];
00024 * calc [label="CRC Calculator\n(shift register)" shape=ellipse];
00025 * }
00026 * }
```

```

00027 * cpu [label="CPU"];
00028 * spi [label="SPI RX Buffer\n(nanopb frames)"];
00029 *
00030 * cpu -> ctrl [label="Configure"];
00031 * spi -> din [label="Feed bytes"];
00032 * din -> calc;
00033 * calc -> dout [label="Result"];
00034 * dout -> cpu [label="Read CRC"];
00035 * }
00036 * @enddot
00037
00038 * @par STAR Project CRC Usage
00039 * | Use Case | Polynomial | Direction | Notes |
00040 * |-----|-----|-----|-----|
00041 * | SPI Protocol (RPi5) | CRC-32 IEEE | LSB-first | nanopb frame validation |
00042 * | Flash Integrity (future) | CRC-32 IEEE | MSB-first | Firmware verification |
00043 *
00044 * @par Key Features
00045 * - Hardware-accelerated CRC calculation (single cycle per byte)
00046 * - CRC-32 IEEE 802.3 polynomial (0x04C11DB7) for compatibility with Go
00047 * - Configurable bit order (MSB-first or LSB-first/reflected)
00048 * - Direct memory-mapped register access
00049 * - Supports CRC-8, CRC-16, CRC-CCITT, CRC-32, CRC-32C (only CRC-32 used)
00050 *
00051 * @par CRC-32 IEEE 802.3 Specification
00052 * - Polynomial: 0x04C11DB7 (normal form)
00053 * - Initial Value: 0xFFFFFFFF
00054 * - Reflect Input: Yes (LSB-first in STAR project)
00055 * - Reflect Output: Yes
00056 * - Final XOR: 0xFFFFFFFF
00057 *
00058 * @par Performance
00059 * | Method | Time for 64 bytes | Notes |
00060 * |-----|-----|-----|
00061 * | Hardware CRC | ~64 cycles | 1 cycle per byte @ 240 MHz |
00062 * | Software CRC | ~2000+ cycles | Table lookup method |
00063 * | Speedup | ~31x | Critical for SPI throughput |
00064 *
00065 * @par Hardware Requirements
00066 * | Parameter | Value | Notes |
00067 * |-----|-----|-----|
00068 * | Base Address | 0x00088280 | 12-byte register block |
00069 * | Module Stop | MSTPCRB.MSTPB23 | Clear to enable module |
00070 * | Clock | PCLKB (60 MHz) | No additional clock needed |
00071 *
00072 * @par NASA Power of 10 Compliance
00073 * - Rule 1: [OK] No goto, setjmp, or recursion
00074 * - Rule 2: [OK] N/A (no loops in register definitions)
00075 * - Rule 3: [OK] No dynamic memory allocation
00076 * - Rule 4: [OK] All accessor functions are single-statement
00077 * - Rule 5: [OK] N/A (hardware layer)
00078 * - Rule 6: [OK] Minimal scope - inline accessors only
00079 * - Rule 7: [OK] N/A (no return values to check)
00080 * - Rule 8: [OK] All constants use C23 typed enums
00081 * - Rule 9: [OK] No function pointers
00082 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00083 *
00084 * @par SOLID Principles
00085 * - **S**: Single responsibility - only CRC register definitions
00086 * - **O**: Open for extension via additional polynomial support
00087 * - **I**: N/A (no inheritance hierarchy)
00088 * - **I**: Minimal interface - only necessary accessor
00089 * - **D**: N/A (hardware register layer)
00090 *
00091 * @par Module Dependencies
00092 * - `<stdint.h>` - Fixed-width integer types
00093 * - `<stddef.h>` - offsetof macro for static assertions
00094 *
00095 * @par Manual References
00096 * - RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00097 * - Chapter 46: CRC Calculator, pages 2411-2420
00098 *
00099 * @par Compatibility
00100 * - The CRC-32 IEEE 802.3 output is compatible with:
00101 * - Go: `crc32.ChecksumIEEE()`
00102 * - Python: `binascii.crc32()`
00103 * - Linux: `cksum -o 3`
00104 *
00105 * @see rx_crc.h Higher-level CRC API
00106 * @see rx72n_regs.h Main register include file
00107 * @see docs/sections/01_nanopb_protocol.tex Protocol specification
00108 *
00109 * @author Locked, Inc.
00110 * @date 2026-01-28
00111 * @copyright Copyright (c) 2026 Locked Inc.
00112 * SPDX-License-Identifier: MIT
00113 * @since Version 1.0.0

```

```

00114 */
00115
00116 #pragma once
00117
00118 #include <stddef.h>
00119 #include <stdint.h>
00120
00121 #ifndef __cplusplus
00122 extern "C" {
00123 #endif
00124
00125 /*
=====
00126 * CRC Calculator - Hardware CRC-32 Acceleration
00127 *
=====
00128 */
00129
00130 /**
00131 * @enum rx_crc_addresses_t
00132 * @brief CRC module base address
00133 *
00134 * @details
00135 * Base address for the CRC Calculator register block. The CRC module
00136 * provides hardware-accelerated CRC computation with configurable polynomials.
00137 *
00138 * @par Memory Map
00139 * | Address | Register | Size | Description |
00140 * |-----|-----|-----|-----|
00141 * | 0x00088280 | CRCCR | 1 | Control Register |
00142 * | 0x00088281 | Reserved | 3 | - |
00143 * | 0x00088284 | CRCDIR | 4 | Data Input Register |
00144 * | 0x00088288 | CRCDOR | 4 | Data Output Register |
00145 *
00146 * @see crc_regs() Accessor function
00147 * @since Version 1.0.0
00148 */
00149 typedef enum : uintptr_t {
00150 /**
00151 * @brief CRC register base address (0x00088280)
00152 * @details Verified against RX72N Hardware Manual Ch46
00153 */
00154 k_crc_base_addr = 0x00088280,
00155 } rx_crc_addresses_t;
00156
00157 /**
00158 * @enum rx_crc_reserved_sizes_t
00159 * @brief CRC register reserved field sizes
00160 *
00161 * @details
00162 * Defines reserved byte counts for structure padding to match hardware layout.
00163 *
00164 * @since Version 1.0.0
00165 */
00166 typedef enum : uint8_t {
00167 k_crc_reserved_after_crccr_bytes = 3, /**< Reserved bytes @ 0x01-0x03 */
00168 } rx_crc_reserved_sizes_t;
00169
00170 /**
00171 * @struct rx_crc_regs_t
00172 * @brief CRC Calculator Register Map Structure
00173 *
00174 * @details
00175 * Hardware CRC Calculator registers for accelerated CRC computation.
00176 * The CRC module computes CRC checksums in hardware, significantly faster
00177 * than software implementations.
00178 *
00179 * @par Memory Layout Table
00180 * | Offset | Size | Field | Type | Description |
00181 * |-----|-----|-----|-----|-----|
00182 * | 0x00 | 1 | crccr | uint8_t | Control Register |
00183 * | 0x01 | 3 | reserved | - | Reserved |
00184 * | 0x04 | 4 | crcdir | uint32_t | Data Input Register |
00185 * | 0x08 | 4 | crcdor | uint32_t | Data Output Register |
00186 * | **Total** | **12** | | | |
00187 *
00188 * @par CRC Calculation Process
00189 * 1. Configure CRCCR with polynomial and bit order
00190 * 2. Clear the output register by writing DORCLR=1
00191 * 3. Feed data bytes/words to CRCDIR
00192 * 4. Read final CRC from CRCDOR
00193 *
00194 * @par Initialization Example (CRC-32 IEEE)
00195 * @code{.c}
00196 * // Configure CRC-32 IEEE 802.3 with LSB-first (reflected) mode
00197 * volatile rx_crc_regs_t* crc = crc_regs();
00198 */

```

```

00199 * // 1. Set polynomial to CRC-32 IEEE, LSB-first
00200 * crc->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms;
00201 *
00202 * // 2. Clear output register (initialize to 0xFFFFFFFF)
00203 * crc->crccr |= k_crc_crccr_dorclr;
00204 *
00205 * // 3. Feed data (example: 4 bytes)
00206 * crc->crkdir = 0x12345678; // Feed 4 bytes at once
00207 *
00208 * // 4. Read result
00209 * uint32_t crc_result = crc->crcdor;
00210 * @endcode
00211 *
00212 * @par Byte-by-Byte Example
00213 * @code{.c}
00214 * // Calculate CRC of a buffer
00215 * uint32_t calculate_crc32(const uint8_t* data, uint32_t len) {
00216 * volatile rx_crc_regs_t* crc = crc_regs();
00217 *
00218 * // Configure for CRC-32 IEEE, LSB-first
00219 * crc->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms | k_crc_crccr_dorclr;
00220 *
00221 * // Feed data byte by byte
00222 * for (uint32_t i = 0; i < len; i++) {
00223 * *((volatile uint8_t*)&crc->crkdir) = data[i];
00224 * }
00225 *
00226 * // Return CRC (XOR with 0xFFFFFFFF for IEEE compliance)
00227 * return crc->crcdor ^ 0xFFFFFFFF;
00228 * }
00229 * @endcode
00230 *
00231 * @invariant sizeof(rx_crc_regs_t) == 12 bytes
00232 * @invariant All offsets match RX72N Hardware Manual Ch46
00233 *
00234 * @note Hardware automatically processes each byte written to CRCDIR
00235 * @warning Must clear CRCDOR (DORCLR) before starting new calculation
00236 * @attention CRC module must be enabled in MSTPCRB before use
00237 *
00238 * @see crc_regs() Accessor function
00239 * @see rx_crc_crccr_bits_t Control register bit definitions
00240 * @since Version 1.0.0
00241 */
00242 typedef struct {
00243 /**
00244 * @brief CRC Control Register (CRCCR) @ offset 0x00
00245 * @details Configures polynomial selection and bit order.
00246 * @par Bit Fields:
00247 * - Bits 2:0 (GPS): Generator Polynomial Select
00248 * - Bit 6 (LMS): LSB/MSB First (0=MSB, 1=LSB/reflected)
00249 * - Bit 7 (DORCLR): Data Output Register Clear (write 1 to clear)
00250 * @see rx_crc_crccr_bits_t Bit definitions
00251 */
00252 volatile uint8_t crccr;
00253
00254 uint8_t reserved0[k_crc_reserved_after_crccr_bytes]; /**< Reserved @ 0x01-0x03 */
00255
00256 /**
00257 * @brief CRC Data Input Register (CRCDIR) @ offset 0x04
00258 * @details Write data here to feed into CRC calculator.
00259 * @par Access:
00260 * - Write: Feed data bytes/words into calculator
00261 * - Read: Returns last written value (not useful)
00262 * @note Can write 8, 16, or 32 bits - all processed by hardware
00263 */
00264 volatile uint32_t crkdir;
00265
00266 /**
00267 * @brief CRC Data Output Register (CRCDOR) @ offset 0x08
00268 * @details Read computed CRC result from here.
00269 * @par Access:
00270 * - Read: Current CRC value
00271 * - Write: Clears to initial value (same as DORCLR)
00272 * @note For IEEE compliance, XOR result with 0xFFFFFFFF
00273 */
00274 volatile uint32_t crcdor;
00275 } rx_crc_regs_t;
00276
00277 /**
00278 * @brief Get pointer to CRC registers
00279 *
00280 * @details
00281 * Returns a volatile pointer to the CRC Calculator register structure at
00282 * address 0x00088280. The CRC module provides hardware-accelerated CRC
00283 * computation for protocol integrity checking.
00284 *
00285 * @return Volatile pointer to CRC register structure

```

```

00286 * @retval Non-NULL Always returns valid pointer (hardware address)
00287 *
00288 * @pre CRC module must be enabled (MSTPCRB.MSTPB23 = 0)
00289 * @post Pointer is valid for the lifetime of program execution
00290 *
00291 * @note Thread Safety: Safe - returns constant hardware address
00292 * @note This function is always inlined for zero overhead
00293 * @warning Ensure module stop bit is cleared before accessing registers
00294 *
00295 * @par Example:
00296 * @code{.c}
00297 * // Quick CRC calculation
00298 * volatile rx_crc_regs_t* crc = crc_regs();
00299 * crc->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms | k_crc_crccr_dorclr;
00300 * crc->crkdir = data_word;
00301 * uint32_t result = crc->crclor ^ 0xFFFFFFFF;
00302 * @endcode
00303 *
00304 * @see k_crc_base_addr Base address constant
00305 * @see rx_crc32_init() Higher-level initialization
00306 * @since Version 1.0.0
00307 */
00308 static inline volatile rx_crc_regs_t* crc_regs(void)
00309 {
00310 return (volatile rx_crc_regs_t*)k_crc_base_addr;
00311 }
00312 /**
00313 * @enum rx_crc_crccr_shifts_t
00314 * @brief CRC Control Register (CRCCR) Bit Field Positions
00315 *
00316 * @details
00317 * Bit positions for CRCCR register fields. Used to construct field values.
00318 *
00319 * @par Register Layout (8-bit @ offset 0x00)
00320 * | Bits | Field | Description |
00321 * |-----|-----|-----|
00322 * | 2:0 | GPS | Generator Polynomial Select |
00323 * | 5:3 | - | Reserved |
00324 * | 6 | LMS | LSB/MSB First selection |
00325 * | 7 | DORCLR | Data Output Register Clear |
00326 *
00327 * @since Version 1.0.0
00328 */
00329 typedef enum : uint8_t {
00330 k_crc_bit_mask = 1, /**< Single-bit mask */
00331 k_crc_crccr_lms_shift = 6, /**< LMS field position (bit 6) */
00332 k_crc_crccr_dorclr_shift = 7, /**< DORCLR field position (bit 7) */
00333 } rx_crc_crccr_shifts_t;
00334 /**
00335 * @enum rx_crc_crccr_masks_t
00336 * @brief CRC Control Register (CRCCR) Bit Masks
00337 *
00338 * @details
00339 * Bit masks for isolating specific fields in the CRCCR register.
00340 *
00341 * @since Version 1.0.0
00342 */
00343 typedef enum : uint8_t {
00344 k_crc_crccr_lms_mask =
00345 (k_crc_bit_mask << k_crc_crccr_lms_shift), /**< LSB/MSB First mask (bit 6) */
00346 k_crc_crccr_dorclr_mask =
00347 (k_crc_bit_mask << k_crc_crccr_dorclr_shift), /**< DORCLR mask (bit 7) */
00348 } rx_crc_crccr_masks_t;
00349 /**
00350 * @enum rx_crc_crccr_bits_t
00351 * @brief CRC Control Register (CRCCR) Bit Definitions
00352 *
00353 * @details
00354 * Configuration values for CRC polynomial selection and bit order.
00355 * Only CRC-32 IEEE 802.3 is used in the STAR project.
00356 *
00357 * @par Generator Polynomial Select (GPS) Values
00358 * | GPS | Polynomial | Name | Used in STAR |
00359 * |-----|-----|-----|-----|
00360 * | 0x00 | X^8+... | CRC-8 | No |
00361 * | 0x01 | X^16+... | CRC-16 | No |
00362 * | 0x02 | X^16+... | CRC-CCITT | No |
00363 * | 0x03 | 0x04C11DB7 | CRC-32 IEEE | **Yes** |
00364 * | 0x04 | 0x1EDC6F41 | CRC-32C | No |
00365 *
00366 * @par Bit Order Selection (LMS)
00367 * - **MSB-first (LMS=0)**: Standard CRC, data processed from bit 7 to bit 0
00368 * - **LSB-first (LMS=1)**: Reflected CRC, data processed from bit 0 to bit 7
00369 */

```

```

00373 * For IEEE 802.3 compatibility (Go crc32.ChecksumIEEE), use LSB-first.
00374 *
00375 * @par Configuration Example
00376 * @code{.c}
00377 * // CRC-32 IEEE 802.3, LSB-first (reflected), clear output
00378 * crc_regs()->crccr = k_crc_crccr_gps_crc32 | k_crc_crccr_lms | k_crc_crccr_dorclr;
00379 * @endcode
00380 *
00381 * @see rx_crc_regs_t::crccr Control register
00382 * @since Version 1.0.0
00383 */
00384 typedef enum : uint8_t {
00385 /**
00386 * @brief CRC-32 IEEE 802.3 polynomial (GPS=0x03)
00387 * @details Polynomial: 0x04C11DB7 ($x^{32} + x^{26} + x^{23} + x^{16} +$
00388 * $x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$)
00389 * @par Compatibility:
00390 * - Go: crc32.ChecksumIEEE()
00391 * - Python: binascii.crc32()
00392 * - Ethernet, ZIP, PNG
00393 */
00394 k_crc_crccr_gps_crc32 = 0x03,
00395
00396 /**
00397 * @brief LSB-first (reflected) bit order (LMS=1)
00398 * @details Process data LSB first. Required for IEEE 802.3 compatibility.
00399 * Set this bit for standard Ethernet/ZIP CRC-32 behavior.
00400 */
00401 k_crc_crccr_lms = k_crc_crccr_lms_mask,
00402
00403 /**
00404 * @brief Clear Data Output Register (DORCLR=1)
00405 * @details Writing 1 clears CRCDOR to initial value (0xFFFFFFFF for CRC-32).
00406 * Must be done before starting a new CRC calculation.
00407 * @note This bit is write-only and reads as 0
00408 */
00409 k_crc_crccr_dorclr = k_crc_crccr_dorclr_mask,
00410 } rx_crc_crccr_bits_t;
00411
00412 /*
00413 =====
00413 * Static Assertions - Verify Register Layout at Compile Time
00414 *
00415 =====
00415 */
00416
00417 /* The k_crc_base_addr enum declaration above is itself the contract for the
00418 * CRC base address (per Hardware Manual). A static_assert that re-states the
00419 * same literal here would duplicate the magic number without adding a real
00420 * cross-check, so we rely on the enum declaration alone. */
00421
00422 /* Verify register structure layout */
00423 static_assert(sizeof(rx_crc_regs_t) == 12, "CRC register structure size incorrect");
00424 static_assert(offsetof(rx_crc_regs_t, crccr) == 0x00, "CRCCR offset incorrect");
00425 static_assert(offsetof(rx_crc_regs_t, crcdir) == 0x04, "CRCDIR offset incorrect");
00426 static_assert(offsetof(rx_crc_regs_t, crcdor) == 0x08, "CRCDOR offset incorrect");
00427
00428 #ifdef __cplusplus
00429 }
00430 #endif

```

## 8.13 libs/rx72n/inc/rx72n\_dmac\_regs.h File Reference

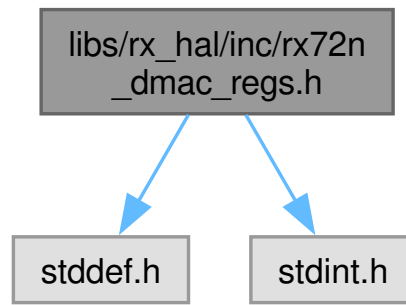
RX72N DMA Controller (DMAC) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_dmac\_regs.h:





This graph shows which files directly or indirectly include this file:



## Data Structures

- struct `rx_dmac_channel_regs_t`  
DMAC per-channel register structure.

## Enumerations

- enum `dmac_addresses_t` : `uintptr_t` {  
`k_dmac_base_addr` = 0x00082000U , `k_dmac_channel_size` = 0x40U , `k_dmac0_base_addr` = 0x00082000U , `k_dmac1_base_addr` = 0x00082040U ,  
`k_dmac2_base_addr` = 0x00082080U , `k_dmac3_base_addr` = 0x000820C0U , `k_dmac4_base_addr` = 0x00082100U , `k_dmac5_base_addr` = 0x00082140U ,  
`k_dmac6_base_addr` = 0x00082180U , `k_dmac7_base_addr` = 0x000821C0U , `k_dmac_dmast_addr` = 0x00082200U , `k_dmac_dmist_addr` = 0x00082204U }  
DMAC module addresses.
- enum `dmac_offsets_t` : `uint8_t` {  
`k_dmac_offset_dmsar` = 0x00U , `k_dmac_offset_dmdar` = 0x04U , `k_dmac_offset_dmcra` = 0x08U , `k_dmac_offset_dmcra` = 0x08U , `k_dmac_offset_dmcra` = 0x08U ,  
`k_dmac_offset_dmtmd` = 0x10U , `k_dmac_offset_dmint` = 0x13U , `k_dmac_offset_dmamd` = 0x14U , `k_dmac_offset_dmo` = 0x18U ,  
`k_dmac_offset_dmcnt` = 0x1CU , `k_dmac_offset_dmreq` = 0x1DU , `k_dmac_offset_dmsts` = 0x1EU , `k_dmac_offset_dmcsl` = 0x1FU }  
Per-channel register offsets.
- enum `dmac_channel_t` : `uint8_t` {  
`k_dmac_channel_0` = 0U , `k_dmac_channel_1` = 1U , `k_dmac_channel_2` = 2U ,  
`k_dmac_channel_3` = 3U ,  
`k_dmac_channel_4` = 4U , `k_dmac_channel_5` = 5U , `k_dmac_channel_6` = 6U ,  
`k_dmac_channel_7` = 7U ,  
`k_dmac_channel_count` = 8U }  
DMAC channel numbers.
- enum `dmtmd_bits_t` : `uint16_t` {  
`k_dmtmd_dctg_software` = 0x0000U , `k_dmtmd_dctg_interrupt` = 0x0001U , `k_dmtmd_dctg_mask` = 0x0003U , `k_dmtmd_sz_8bit` = 0x0000U ,  
`k_dmtmd_sz_16bit` = 0x0100U , `k_dmtmd_sz_32bit` = 0x0200U , `k_dmtmd_sz_mask` = 0x0300U , `k_dmtmd_dts_dest` = 0x0000U ,  
`k_dmtmd_dts_source` = 0x1000U , `k_dmtmd_dts_none` = 0x2000U , `k_dmtmd_dts_mask` = 0x3000U , `k_dmtmd_md_normal` = 0x0000U ,  
`k_dmtmd_md_repeat` = 0x4000U , `k_dmtmd_md_block` = 0x8000U , `k_dmtmd_md_mask` = 0xC000U }

DMA Transfer Mode Register (DMTMD) bit definitions.

- enum [dmint\\_bits\\_t](#) : [uint8\\_t](#) {  
[k\\_dmint\\_disabled](#) = 0x00U , [k\\_dmint\\_darie](#) = 0x01U , [k\\_dmint\\_sarie](#) = 0x02U , [k\\_dmint\\_rptie](#) = 0x04U ,  
[k\\_dmint\\_esie](#) = 0x08U , [k\\_dmint\\_dtie](#) = 0x10U , [k\\_dmint\\_mask](#) = 0x1FU }

DMA Interrupt Setting Register (DMINT) bit definitions.

- enum [dmamd\\_bits\\_t](#) : [uint16\\_t](#) {  
[k\\_dmamd\\_dara\\_mask](#) = 0x0003U , [k\\_dmamd\\_dm\\_fixed](#) = 0x0000U , [k\\_dmamd\\_dm\\_offset](#) = 0x0040U , [k\\_dmamd\\_dm\\_increment](#) = 0x0080U ,  
[k\\_dmamd\\_dm\\_decrement](#) = 0x00C0U , [k\\_dmamd\\_dm\\_mask](#) = 0x00C0U , [k\\_dmamd\\_sara\\_mask](#) = 0x0300U , [k\\_dmamd\\_sm\\_fixed](#) = 0x0000U ,  
[k\\_dmamd\\_sm\\_offset](#) = 0x4000U , [k\\_dmamd\\_sm\\_increment](#) = 0x8000U , [k\\_dmamd\\_sm\\_decrement](#) = 0xC000U , [k\\_dmamd\\_sm\\_mask](#) = 0xC000U }

DMA Address Mode Register (DMAMD) bit definitions.

- enum [dmcnt\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmcnt\\_dte](#) = 0x01U }

DMA Transfer Enable Register (DMCNT) bit definitions.

- enum [dmreq\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmreq\\_swreq](#) = 0x01U , [k\\_dmreq\\_clrs](#) = 0x10U }

DMA Software Start Register (DMREQ) bit definitions.

- enum [dmsts\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmsts\\_esif](#) = 0x01U , [k\\_dmsts\\_dtif](#) = 0x10U , [k\\_dmsts\\_act](#) = 0x80U }

DMA Status Register (DMSTS) bit definitions.

- enum [dmcs1\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmcs1\\_disel](#) = 0x01U }

DMA Channel Select Register (DMCSL) bit definitions.

- enum [dmast\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmast\\_dmst](#) = 0x01U }

DMAC Module Start Register (DMAST) bit definitions.

- enum [dmist\\_bits\\_t](#) : [uint8\\_t](#) { [k\\_dmist\\_dmis4](#) = 0x10U , [k\\_dmist\\_dmis5](#) = 0x20U ,  
[k\\_dmist\\_dmis6](#) = 0x40U , [k\\_dmist\\_dmis7](#) = 0x80U }

DMAC74 Interrupt Status Monitor Register (DMIST) bit definitions.

- enum [dmac\\_mstpcr\\_t](#) : [uint32\\_t](#) { [k\\_dmac\\_mstpcra\\_bit](#) = (1U << 28U) }

DMAC module stop control bits.

## Functions

- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac\\_channel](#) ([uint8\\_t](#) channel)  
Get pointer to DMAC channel registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac0](#) (void)  
Get pointer to DMAC channel 0 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac1](#) (void)  
Get pointer to DMAC channel 1 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac2](#) (void)  
Get pointer to DMAC channel 2 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac3](#) (void)  
Get pointer to DMAC channel 3 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac4](#) (void)  
Get pointer to DMAC channel 4 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac5](#) (void)  
Get pointer to DMAC channel 5 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac6](#) (void)  
Get pointer to DMAC channel 6 registers.
- static volatile [rx\\_dmac\\_channel\\_regs\\_t](#) \* [dmac7](#) (void)  
Get pointer to DMAC channel 7 registers.
- static volatile [uint8\\_t](#) \* [dmac\\_dmast\\_reg](#) (void)  
Get pointer to DMAC Module Start Register (DMAST).
- static volatile [uint8\\_t](#) \* [dmac\\_dmist\\_reg](#) (void)  
Get pointer to DMAC74 Interrupt Status Monitor Register (DMIST).

### 8.13.1 Detailed Description

#### RX72N DMA Controller (DMAC) Register Definitions.

Complete register definitions for the RX72N 8-channel DMA Controller. The DMAC transfers data without CPU intervention, supporting normal, repeat, and block transfer modes.

#### DMAC Overview

- 8 independent channels (DMAC0-DMAC7)
- 512 MB transfer space (0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF)
- Maximum 64M data transfers (1024 data x 65536 blocks)
- 8/16/32-bit transfer sizes
- Software or peripheral interrupt triggers
- Channel priority: Ch0 > Ch1 > Ch2 > ... > Ch7

#### Register Organization

| Address    | Size | Description                      |
|------------|------|----------------------------------|
| 0x00082000 | -    | DMAC channel 0 base              |
| 0x00082040 | -    | DMAC channel 1 base              |
| ...        | ...  | (channels spaced 0x40 apart)     |
| 0x000821C0 | -    | DMAC channel 7 base              |
| 0x00082200 | 8    | DMAST - Module Start Register    |
| 0x00082204 | 8    | DMIST - Interrupt Status Monitor |

#### Per-Channel Register Offsets

| Offset | Size | Register | Description                 |
|--------|------|----------|-----------------------------|
| 0x00   | 32   | DMSAR    | Source Address              |
| 0x04   | 32   | DMDAR    | Destination Address         |
| 0x08   | 32   | DMCRA    | Transfer Count A            |
| 0x0C   | 16   | DMCRB    | Transfer Count B            |
| 0x10   | 16   | DMTMD    | Transfer Mode               |
| 0x13   | 8    | DMINT    | Interrupt Setting           |
| 0x14   | 16   | DMAMD    | Address Mode                |
| 0x18   | 32   | DMOFR    | Offset Register (ch0 only!) |
| 0x1C   | 8    | DMCNT    | DMA Transfer Enable         |
| 0x1D   | 8    | DMREQ    | DMA Software Start          |
| 0x1E   | 8    | DMSTS    | DMA Status                  |
| 0x1F   | 8    | DMCSL    | DMA Channel Select          |

## NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops)
- Rule 3: No dynamic memory allocation
- Rule 4: Short accessor functions
- Rule 5: Static assertions verify addresses
- Rule 6: Minimal scope
- Rule 7: N/A
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

## Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 18: DMA Controller (DMAC<sub>Aa</sub>), pages 677-746
- Section 18.2: Register Descriptions

## Author

Locked, Inc.

## Date

2026-01-29

## Copyright

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## Since

Version 1.0.0

Definition in file [rx72n\\_dmac\\_regs.h](#).

## 8.13.2 Data Structure Documentation

### 8.13.2.1 struct rx\_dmac\_channel\_regs\_t

DMAC per-channel register structure.

Memory-mapped register layout for one DMAC channel. Each channel has 64 bytes (0x40) of register space.

## Warning

DMOFR register at offset 0x18 is ONLY valid for channel 0! For channels 1-7, this location is reserved.

## Note

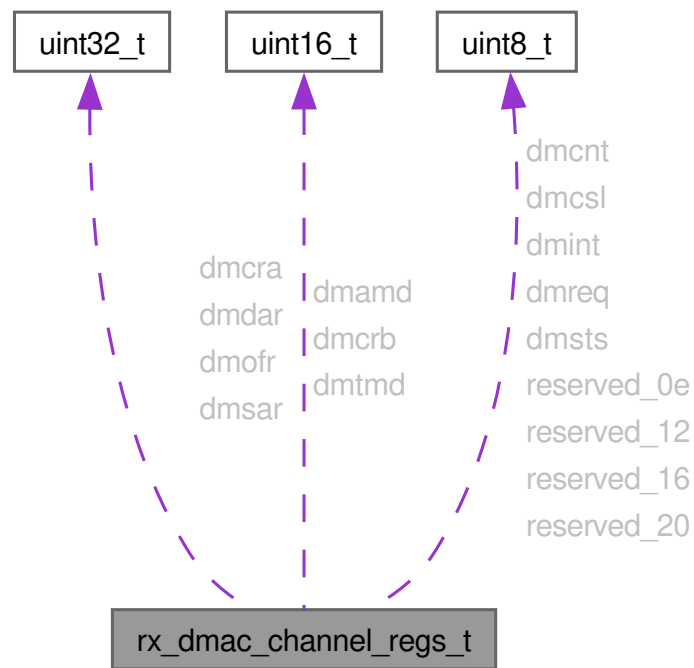
Structure size verified as 0x20 (32 bytes of active registers) with reserved padding to fill 0x40 channel spacing.

## Since

Version 1.0.0

Definition at line 196 of file [rx72n\\_dmac\\_regs.h](#).

Collaboration diagram for rx\_dmac\_channel\_regs\_t:



## Data Fields

|                   |       |                                                                                                                                                                                        |
|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | dmamd | <p>DMA Address Mode Register (DMAMD). Controls source/destination address update mode and extended repeat.</p> <p>See also</p> <p><a href="#">dmamd_bits_t</a> for bit definitions</p> |
|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                   |       |                                                                                                                                                                                                                                            |
|-------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | dmcnt | <p>DMA Transfer Enable Register (DMCNT).<br/>Enables/disables DMA transfer for this channel.</p> <p>See also</p> <p><a href="#">dmcnt_bits_t</a> for bit definitions</p>                                                                   |
| volatile uint32_t | dmcra | <p>DMA Transfer Count Register A (DMCRA).</p> <ul style="list-style-type: none"> <li>• Normal mode: bits [15:0] = transfer count, bits [25:16] = block size</li> <li>• Repeat/Block mode: bits [9:0] = repeat/block size</li> </ul>        |
| volatile uint16_t | dmcrb | <p>DMA Transfer Count Register B (DMCRB).</p> <ul style="list-style-type: none"> <li>• Normal mode (free-running): Not used</li> <li>• Repeat mode: bits [15:0] = repeat count</li> <li>• Block mode: bits [15:0] = block count</li> </ul> |
| volatile uint8_t  | dmcs1 | <p>DMA Channel Select Register (DMCSL).<br/>Selects interrupt source channel (used with grouped interrupts).</p> <p>See also</p> <p><a href="#">dmcs1_bits_t</a> for bit definitions</p>                                                   |
| volatile uint32_t | dmdar | <p>DMA Destination Address Register (DMDAR).<br/>32-bit destination start address. Bits 31:29 extend bit 28. Valid range: 0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF</p>                                                                 |
| volatile uint8_t  | dmint | <p>DMA Interrupt Setting Register (DMINT).<br/>Enables/disables various interrupt sources.</p> <p>See also</p> <p><a href="#">dmint_bits_t</a> for bit definitions</p>                                                                     |
| volatile uint32_t | dmofr | <p>DMA Offset Register (DMOFR).<br/>32-bit signed offset value for offset addition mode.</p> <p>Warning</p> <p>ONLY VALID FOR CHANNEL 0! Reserved for channels 1-7.</p>                                                                    |
| volatile uint8_t  | dmreq | <p>DMA Software Start Register (DMREQ).<br/>Software trigger control and clear settings.</p> <p>See also</p> <p><a href="#">dmreq_bits_t</a> for bit definitions</p>                                                                       |
| volatile uint32_t | dmsar | <p>DMA Source Address Register (DMSAR).<br/>32-bit source start address. Bits 31:29 extend bit 28. Valid range: 0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF</p>                                                                           |

|                   |                      |                                                                                                                                                                                |
|-------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | dmsts                | DMA Status Register (DMSTS).<br>Transfer status flags (active, end, escape end, etc.).<br><br>See also<br><a href="#">dmsts_bits_t</a> for bit definitions                     |
| volatile uint16_t | dmtmd                | DMA Transfer Mode Register (DMTMD).<br>Controls transfer mode, data size, repeat area, and trigger source.<br><br>See also<br><a href="#">dmtmd_bits_t</a> for bit definitions |
| volatile uint8_t  | reserved_0e↔<br>[2]  | Reserved (2 bytes padding between DMCRB and DMTMD).                                                                                                                            |
| volatile uint8_t  | reserved_12          | Reserved (1 byte padding between DMTMD and DMINT).                                                                                                                             |
| volatile uint8_t  | reserved_16↔<br>[2]  | Reserved (2 bytes padding between DMAMD and DMOFR).                                                                                                                            |
| volatile uint8_t  | reserved_20↔<br>[32] | Reserved padding to 0x40 (64 bytes) channel spacing.                                                                                                                           |

### 8.13.3 Enumeration Type Documentation

#### 8.13.3.1 dmac\_addresses\_t

enum [dmac\\_addresses\\_t](#) : uintptr\_t

DMAC module addresses.

Base addresses for DMAC channels and shared registers. All addresses verified against RX72N Hardware Manual Ch18.

Note

Addresses verified 2026-01-29 against manual section 18.2

Since

Version 1.0.0

Enumerator

|                     |                                                            |
|---------------------|------------------------------------------------------------|
| k_dmac_base_addr    | DMAC module base address (channel 0).                      |
| k_dmac_channel_size | Channel register block size (0x40 = 64 bytes per channel). |
| k_dmac0_base_addr   | DMAC channel 0 base address.                               |
| k_dmac1_base_addr   | DMAC channel 1 base address.                               |
| k_dmac2_base_addr   | DMAC channel 2 base address.                               |
| k_dmac3_base_addr   | DMAC channel 3 base address.                               |

|                   |                                                           |
|-------------------|-----------------------------------------------------------|
| k_dmac4_base_addr | DMAC channel 4 base address.                              |
| k_dmac5_base_addr | DMAC channel 5 base address.                              |
| k_dmac6_base_addr | DMAC channel 6 base address.                              |
| k_dmac7_base_addr | DMAC channel 7 base address.                              |
| k_dmac_dmast_addr | DMAC Module Start Register (DMAST) address.               |
| k_dmac_dmist_addr | DMAC74 Interrupt Status Monitor Register (DMIST) address. |

Definition at line 93 of file [rx72n\\_dmac\\_regs.h](#).

```

00093 : uintptr_t {
00094 /** @brief DMAC module base address (channel 0) */
00095 k_dmac_base_addr = 0x00082000U,
00096
00097 /** @brief Channel register block size (0x40 = 64 bytes per channel) */
00098 k_dmac_channel_size = 0x40U,
00099
00100 /** @brief DMAC channel 0 base address */
00101 k_dmac0_base_addr = 0x00082000U,
00102
00103 /** @brief DMAC channel 1 base address */
00104 k_dmac1_base_addr = 0x00082040U,
00105
00106 /** @brief DMAC channel 2 base address */
00107 k_dmac2_base_addr = 0x00082080U,
00108
00109 /** @brief DMAC channel 3 base address */
00110 k_dmac3_base_addr = 0x000820C0U,
00111
00112 /** @brief DMAC channel 4 base address */
00113 k_dmac4_base_addr = 0x00082100U,
00114
00115 /** @brief DMAC channel 5 base address */
00116 k_dmac5_base_addr = 0x00082140U,
00117
00118 /** @brief DMAC channel 6 base address */
00119 k_dmac6_base_addr = 0x00082180U,
00120
00121 /** @brief DMAC channel 7 base address */
00122 k_dmac7_base_addr = 0x000821C0U,
00123
00124 /** @brief DMAC Module Start Register (DMAST) address */
00125 k_dmac_dmast_addr = 0x00082200U,
00126
00127 /** @brief DMAC74 Interrupt Status Monitor Register (DMIST) address */
00128 k_dmac_dmist_addr = 0x00082204U,
00129 } dmac_addresses_t;

```

### 8.13.3.2 dmac\_channel\_t

enum [dmac\\_channel\\_t](#) : uint8\_t

DMAC channel numbers.

Channel indices for use with accessor functions.

Since

Version 1.0.0

Enumerator

|                  |                                   |
|------------------|-----------------------------------|
| k_dmac_channel_0 | DMAC channel 0 (highest priority) |
|------------------|-----------------------------------|



|                      |                                  |
|----------------------|----------------------------------|
| k_dmac_channel_1     | DMAC channel 1                   |
| k_dmac_channel_2     | DMAC channel 2                   |
| k_dmac_channel_3     | DMAC channel 3                   |
| k_dmac_channel_4     | DMAC channel 4                   |
| k_dmac_channel_5     | DMAC channel 5                   |
| k_dmac_channel_6     | DMAC channel 6                   |
| k_dmac_channel_7     | DMAC channel 7 (lowest priority) |
| k_dmac_channel_count | Total number of DMAC channels    |

Definition at line 164 of file rx72n\_dmac\_regs.h.

```

00164 : uint8_t {
00165 k_dmac_channel_0 = 0U, /**< DMAC channel 0 (highest priority) */
00166 k_dmac_channel_1 = 1U, /**< DMAC channel 1 */
00167 k_dmac_channel_2 = 2U, /**< DMAC channel 2 */
00168 k_dmac_channel_3 = 3U, /**< DMAC channel 3 */
00169 k_dmac_channel_4 = 4U, /**< DMAC channel 4 */
00170 k_dmac_channel_5 = 5U, /**< DMAC channel 5 */
00171 k_dmac_channel_6 = 6U, /**< DMAC channel 6 */
00172 k_dmac_channel_7 = 7U, /**< DMAC channel 7 (lowest priority) */
00173 k_dmac_channel_count = 8U, /**< Total number of DMAC channels */
00174 } dmac_channel_t;

```

### 8.13.3.3 dmac\_mstpcr\_t

enum [dmac\\_mstpcr\\_t](#) : uint32\_t

DMAC module stop control bits.

The DMAC is controlled by MSTPCRA.MSTPA28. Must be cleared (0) before accessing DMAC registers.

Note

Manual ref: Ch11 (Low Power Consumption) module stop tables

Since

Version 1.0.0

Enumerator

|                    |                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dmac_mstpcra_bit | MSTPCRA bit for DMAC (bit 28). <ul style="list-style-type: none"> <li>0: DMAC module is running</li> <li>1: DMAC module is stopped (default)</li> </ul> |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 599 of file [rx72n\\_dmac\\_regs.h](#).

```
00599 : uint32_t {
00600 /**
00601 * @brief MSTPCRA bit for DMAC (bit 28)
00602 * @details
00603 * - 0: DMAC module is running
00604 * - 1: DMAC module is stopped (default)
00605 */
00606 k_dmac_mstpcra_bit = (1U « 28U),
00607 } dmac_mstpcr_t;
```

#### 8.13.3.4 dmac`offsets`

```
enum dmac_offsets_t : uint8_t
```

Per-channel register offsets.

Offsets from channel base address for each register. All offsets verified against RX72N Hardware Manual Ch18 section 18.2.

Note

Offsets verified 2026-01-29 against manual

Since

Version 1.0.0

Enumerator

|                      |                                       |
|----------------------|---------------------------------------|
| k_dmac_offset_dmsar  | Source Address Register (32-bit)      |
| k_dmac_offset_dmdar  | Destination Address Register (32-bit) |
| k_dmac_offset_dmcra  | Transfer Count Register A (32-bit)    |
| k_dmac_offset_dmcra  | Transfer Count Register B (16-bit)    |
| k_dmac_offset_dmtmd  | Transfer Mode Register (16-bit)       |
| k_dmac_offset_dmint  | Interrupt Setting Register (8-bit)    |
| k_dmac_offset_dmamd  | Address Mode Register (16-bit)        |
| k_dmac_offset_dmoifr | Offset Register (32-bit, ch0 only!)   |
| k_dmac_offset_dmcnt  | DMA Transfer Enable Register (8-bit)  |
| k_dmac_offset_dmreq  | DMA Software Start Register (8-bit)   |
| k_dmac_offset_dmsts  | DMA Status Register (8-bit)           |
| k_dmac_offset_dmcsl  | DMA Channel Select Register (8-bit)   |

Definition at line 142 of file [rx72n\\_dmac\\_regs.h](#).

```
00142 : uint8_t {
00143 k_dmac_offset_dmsar = 0x00U, /**< Source Address Register (32-bit) */
00144 k_dmac_offset_dmdar = 0x04U, /**< Destination Address Register (32-bit) */
00145 k_dmac_offset_dmcra = 0x08U, /**< Transfer Count Register A (32-bit) */
00146 k_dmac_offset_dmcra = 0x0CU, /**< Transfer Count Register B (16-bit) */
00147 k_dmac_offset_dmtmd = 0x10U, /**< Transfer Mode Register (16-bit) */
00148 k_dmac_offset_dmint = 0x13U, /**< Interrupt Setting Register (8-bit) */
00149 k_dmac_offset_dmamd = 0x14U, /**< Address Mode Register (16-bit) */
00150 k_dmac_offset_dmoifr = 0x18U, /**< Offset Register (32-bit, ch0 only!) */
00151 k_dmac_offset_dmcnt = 0x1CU, /**< DMA Transfer Enable Register (8-bit) */
00152 k_dmac_offset_dmreq = 0x1DU, /**< DMA Software Start Register (8-bit) */
00153 k_dmac_offset_dmsts = 0x1EU, /**< DMA Status Register (8-bit) */
00154 k_dmac_offset_dmcsl = 0x1FU, /**< DMA Channel Select Register (8-bit) */
00155 } dmac_offsets_t;
```

## 8.13.3.5 dmamd\_bits\_t

enum [dmamd\\_bits\\_t](#) : uint16\_t

DMA Address Mode Register (DMAMD) bit definitions.

Controls source/destination address update mode and extended repeat area.

Register Layout (16-bit @ offset 0x14)

| Bits    | Field | Description                              |
|---------|-------|------------------------------------------|
| 1:<br>0 | DARA  | Destination Address Extended Repeat Area |
| 5:<br>2 | -     | Reserved                                 |
| 7:<br>6 | DM    | Destination Address Update Mode          |
| 9:<br>8 | SARA  | Source Address Extended Repeat Area      |
| 13:10   | -     | Reserved                                 |
| 15:14   | SM    | Source Address Update Mode               |

Note

Manual ref: Ch18 section 18.2.7

Since

Version 1.0.0

Enumerator

|                      |                                      |
|----------------------|--------------------------------------|
| k_dmamd_dara_mask    | DARA field mask                      |
| k_dmamd_dm_fixed     | Destination address fixed            |
| k_dmamd_dm_offset    | Add offset after transfer (ch0 only) |
| k_dmamd_dm_increment | Increment after transfer             |
| k_dmamd_dm_decrement | Decrement after transfer             |
| k_dmamd_dm_mask      | DM field mask                        |
| k_dmamd_sara_mask    | SARA field mask                      |
| k_dmamd_sm_fixed     | Source address fixed                 |
| k_dmamd_sm_offset    | Add offset after transfer (ch0 only) |
| k_dmamd_sm_increment | Increment after transfer             |
| k_dmamd_sm_decrement | Decrement after transfer             |
| k_dmamd_sm_mask      | SM field mask                        |

Definition at line [415](#) of file [rx72n\\_dmac\\_regs.h](#).

```

00415 : uint16_t {
00416 /* DARA - Destination Address Extended Repeat Area (bits 1:0) */
00417 k_dmamd_dara_mask = 0x0003U, /**< DARA field mask */
00418
00419 /* DM - Destination Address Update Mode (bits 7:6) */
00420 k_dmamd_dm_fixed = 0x0000U, /**< Destination address fixed */
00421 k_dmamd_dm_offset = 0x0040U, /**< Add offset after transfer (ch0 only) */
00422 k_dmamd_dm_increment = 0x0080U, /**< Increment after transfer */
00423 k_dmamd_dm_decrement = 0x00C0U, /**< Decrement after transfer */
00424 k_dmamd_dm_mask = 0x00C0U, /**< DM field mask */
00425
00426 /* SARA - Source Address Extended Repeat Area (bits 9:8) */
00427 k_dmamd_sara_mask = 0x0300U, /**< SARA field mask */
00428
00429 /* SM - Source Address Update Mode (bits 15:14) */
00430 k_dmamd_sm_fixed = 0x0000U, /**< Source address fixed */
00431 k_dmamd_sm_offset = 0x4000U, /**< Add offset after transfer (ch0 only) */
00432 k_dmamd_sm_increment = 0x8000U, /**< Increment after transfer */
00433 k_dmamd_sm_decrement = 0xC000U, /**< Decrement after transfer */
00434 k_dmamd_sm_mask = 0xC000U, /**< SM field mask */
00435 } dmamd_bits_t;

```

### 8.13.3.6 dmast`bits`

```
enum dmast_bits_t : uint8_t
```

DMAC Module Start Register (DMAST) bit definitions.

Register Layout (8-bit @ 0x00082200)

| Bit      | Field | Description       |
|----------|-------|-------------------|
| 0        | DMST  | DMAC Module Start |
| 7:↔<br>1 | -     | Reserved          |

Note

Manual ref: Ch18 section 18.2.13

Since

Version 1.0.0

Enumerator

|              |                                              |
|--------------|----------------------------------------------|
| k_dmast_dmst | DMAC module start (0=suspended, 1=operating) |
|--------------|----------------------------------------------|

Definition at line 551 of file rx72n\_dmac\_regs.h.

```

00551 : uint8_t {
00552 k_dmast_dmst = 0x01U, /**< DMAC module start (0=suspended, 1=operating) */
00553 } dmast_bits_t;

```

8.13.3.7 dmcnt`bits``t`

```
enum dmcnt_bits_t : uint8_t
```

DMA Transfer Enable Register (DMCNT) bit definitions.

Register Layout (8-bit @ offset 0x1C)

| Bit     | Field | Description         |
|---------|-------|---------------------|
| 0       | DTE   | DMA Transfer Enable |
| 7:<br>1 | -     | Reserved            |

Note

Manual ref: Ch18 section 18.2.9

Since

Version 1.0.0

Enumerator

|                          |                                             |
|--------------------------|---------------------------------------------|
| <code>k_dmcnt_dte</code> | DMA Transfer Enable (0=disabled, 1=enabled) |
|--------------------------|---------------------------------------------|

Definition at line 455 of file [rx72n\\_dmac\\_regs.h](#).

```
00455 : uint8_t {
00456 k_dmcnt_dte = 0x01U, /**< DMA Transfer Enable (0=disabled, 1=enabled) */
00457 } dmcnt_bits_t;
```

8.13.3.8 dmcsl`bits``t`

```
enum dmcsl_bits_t : uint8_t
```

DMA Channel Select Register (DMCSL) bit definitions.

Register Layout (8-bit @ offset 0x1F)

| Bit     | Field | Description      |
|---------|-------|------------------|
| 0       | DISEL | Interrupt Select |
| 7:<br>1 | -     | Reserved         |

Note

Manual ref: Ch18 section 18.2.12

Since

Version 1.0.0

Enumerator

|                |                                           |
|----------------|-------------------------------------------|
| k_dmcs_l_disel | Interrupt select (0=transfer end, 1=DTIF) |
|----------------|-------------------------------------------|

Definition at line 529 of file rx72n\_dmac\_regs.h.

```
00529 : uint8_t {
00530 k_dmcs_l_disel = 0x01U, /**< Interrupt select (0=transfer end, 1=DTIF) */
00531 } dmcs_l_bits_t;
```

### 8.13.3.9 dmint`bits`

```
enum dmint_bits_t : uint8_t
```

DMA Interrupt Setting Register (DMINT) bit definitions.

Controls interrupt enable for various DMA events.

Register Layout (8-bit @ offset 0x13)

| Bit     | Field | Description                                |
|---------|-------|--------------------------------------------|
| 0       | DARIE | Dest Address Extended Repeat Overflow IE   |
| 1       | SARIE | Source Address Extended Repeat Overflow IE |
| 2       | RPTIE | Repeat Size End Interrupt Enable           |
| 3       | ESIE  | Transfer Escape End Interrupt Enable       |
| 4       | DTIE  | Transfer End Interrupt Enable              |
| 7↔<br>5 | -     | Reserved                                   |

Note

Manual ref: Ch18 section 18.2.6

Since

Version 1.0.0

Enumerator

|                  |                                                 |
|------------------|-------------------------------------------------|
| k_dmint_disabled | All interrupts disabled (polling mode)          |
| k_dmint_darie    | Dest addr extended repeat overflow int enable   |
| k_dmint_sarie    | Source addr extended repeat overflow int enable |
| k_dmint_rptie    | Repeat size end interrupt enable                |
| k_dmint_esie     | Transfer escape end interrupt enable            |

|              |                                              |
|--------------|----------------------------------------------|
| k_dmint_dtie | Transfer end interrupt enable                |
| k_dmint_mask | Mask of all interrupt enable bits (bits 4:0) |

Definition at line 380 of file [rx72n\\_dmac\\_regs.h](#).

```

00380 : uint8_t {
00381 k_dmint_disabled = 0x00U, /**< All interrupts disabled (polling mode) */
00382 k_dmint_darie = 0x01U, /**< Dest addr extended repeat overflow int enable */
00383 k_dmint_sarie = 0x02U, /**< Source addr extended repeat overflow int enable */
00384 k_dmint_rptie = 0x04U, /**< Repeat size end interrupt enable */
00385 k_dmint_esie = 0x08U, /**< Transfer escape end interrupt enable */
00386 k_dmint_dtie = 0x10U, /**< Transfer end interrupt enable */
00387 k_dmint_mask = 0x1FU, /**< Mask of all interrupt enable bits (bits 4:0) */
00388 } dmint_bits_t;

```

#### 8.13.3.10 dmist\_bits\_t

enum [dmist\\_bits\\_t](#) : uint8\_t

DMAC74 Interrupt Status Monitor Register (DMIST) bit definitions.

Register Layout (8-bit @ 0x00082204)

| Bit     | Field | Description                |
|---------|-------|----------------------------|
| 3↔<br>0 | -     | Reserved                   |
| 4       | DMIS4 | Channel 4 interrupt status |
| 5       | DMIS5 | Channel 5 interrupt status |
| 6       | DMIS6 | Channel 6 interrupt status |
| 7       | DMIS7 | Channel 7 interrupt status |

Note

Manual ref: Ch18 section 18.2.14

Since

Version 1.0.0

Enumerator

|               |                                    |
|---------------|------------------------------------|
| k_dmist_dmis4 | Channel 4 interrupt status (bit 4) |
| k_dmist_dmis5 | Channel 5 interrupt status (bit 5) |
| k_dmist_dmis6 | Channel 6 interrupt status (bit 6) |
| k_dmist_dmis7 | Channel 7 interrupt status (bit 7) |

Definition at line 576 of file [rx72n\\_dmac\\_regs.h](#).

```

00576 : uint8_t {
00577 k_dmist_dmis4 = 0x10U, /**< Channel 4 interrupt status (bit 4) */
00578 k_dmist_dmis5 = 0x20U, /**< Channel 5 interrupt status (bit 5) */
00579 k_dmist_dmis6 = 0x40U, /**< Channel 6 interrupt status (bit 6) */
00580 k_dmist_dmis7 = 0x80U, /**< Channel 7 interrupt status (bit 7) */
00581 } dmist_bits_t;

```

### 8.13.3.11 dmreq\_bits\_t

```
enum dmreq_bits_t : uint8_t
```

DMA Software Start Register (DMREQ) bit definitions.

Register Layout (8-bit @ offset 0x1D)

| Bit      | Field | Description                  |
|----------|-------|------------------------------|
| 0        | SWREQ | DMA Software Start           |
| 3:↵<br>1 | -     | Reserved                     |
| 4        | CLRS  | DMA Software Start Bit Clear |
| 7:↵<br>5 | -     | Reserved                     |

Note

Manual ref: Ch18 section 18.2.10

Since

Version 1.0.0

Enumerator

|               |                                       |
|---------------|---------------------------------------|
| k_dmreq_swreq | Software start (write 1 to trigger)   |
| k_dmreq_clrs  | Auto-clear SWREQ after transfer start |

Definition at line 479 of file rx72n\_dmac\_regs.h.

```
00479 : uint8_t {
00480 k_dmreq_swreq = 0x01U, /**< Software start (write 1 to trigger) */
00481 k_dmreq_clrs = 0x10U, /**< Auto-clear SWREQ after transfer start */
00482 } dmreq_bits_t;
```

### 8.13.3.12 dmsts\_bits\_t

```
enum dmsts_bits_t : uint8_t
```

DMA Status Register (DMSTS) bit definitions.



Register Layout (8-bit @ offset 0x1E)

| Bit      | Field | Description              |
|----------|-------|--------------------------|
| 0        | ESIF  | Transfer Escape End Flag |
| 3:↵<br>1 | -     | Reserved                 |
| 4        | DTIF  | Transfer End Flag        |
| 6:↵<br>5 | -     | Reserved                 |
| 7        | ACT   | DMA Active Flag          |

Note

Manual ref: Ch18 section 18.2.11

Since

Version 1.0.0

Enumerator

|              |                                          |
|--------------|------------------------------------------|
| k_dmsts_esif | Transfer escape end flag                 |
| k_dmsts_dtif | Transfer end flag                        |
| k_dmsts_act  | DMA active flag (1=transfer in progress) |

Definition at line 505 of file [rx72n\\_dmac\\_regs.h](#).

```

00505 : uint8_t {
00506 k_dmsts_esif = 0x01U, /**< Transfer escape end flag */
00507 k_dmsts_dtif = 0x10U, /**< Transfer end flag */
00508 k_dmsts_act = 0x80U, /**< DMA active flag (1=transfer in progress) */
00509 } dmsts_bits_t;

```

### 8.13.3.13 dmtmd`bits``t`

enum [dmtmd\\_bits\\_t](#) : uint16\_t

DMA Transfer Mode Register (DMTMD) bit definitions.

Controls transfer mode, data size, repeat area selection, and trigger source.

Register Layout (16-bit @ offset 0x10)

| Bits     | Field | Description                    |
|----------|-------|--------------------------------|
| 1:↵<br>0 | DCTG  | Transfer Request Source Select |
| 7:↵<br>2 | -     | Reserved                       |

| Bits  | Field | Description               |
|-------|-------|---------------------------|
| 9:8   | SZ    | Transfer Data Size Select |
| 11:10 | -     | Reserved                  |
| 13:12 | DTS   | Repeat Area Select        |
| 15:14 | MD    | Transfer Mode Select      |

Note

Manual ref: Ch18 section 18.2.5

Since

Version 1.0.0

Enumerator

|                        |                                  |
|------------------------|----------------------------------|
| k_dmtmd_dctg_software  | Software trigger                 |
| k_dmtmd_dctg_interrupt | Peripheral/external interrupt    |
| k_dmtmd_dctg_mask      | DCTG field mask                  |
| k_dmtmd_sz_8bit        | 8-bit transfer                   |
| k_dmtmd_sz_16bit       | 16-bit transfer                  |
| k_dmtmd_sz_32bit       | 32-bit transfer                  |
| k_dmtmd_sz_mask        | SZ field mask                    |
| k_dmtmd_dts_dest       | Destination is repeat/block area |
| k_dmtmd_dts_source     | Source is repeat/block area      |
| k_dmtmd_dts_none       | No repeat/block area             |
| k_dmtmd_dts_mask       | DTS field mask                   |
| k_dmtmd_md_normal      | Normal transfer mode             |
| k_dmtmd_md_repeat      | Repeat transfer mode             |
| k_dmtmd_md_block       | Block transfer mode              |
| k_dmtmd_md_mask        | MD field mask                    |

Definition at line 330 of file [rx72n\\_dmac\\_regs.h](#).

```

00330 : uint16_t {
00331 /* DCTG - Transfer Request Source Select (bits 1:0) */
00332 k_dmtmd_dctg_software = 0x0000U, /**< Software trigger */
00333 k_dmtmd_dctg_interrupt = 0x0001U, /**< Peripheral/external interrupt */
00334 k_dmtmd_dctg_mask = 0x0003U, /**< DCTG field mask */
00335
00336 /* SZ - Transfer Data Size Select (bits 9:8) */
00337 k_dmtmd_sz_8bit = 0x0000U, /**< 8-bit transfer */
00338 k_dmtmd_sz_16bit = 0x0100U, /**< 16-bit transfer */
00339 k_dmtmd_sz_32bit = 0x0200U, /**< 32-bit transfer */
00340 k_dmtmd_sz_mask = 0x0300U, /**< SZ field mask */
00341
00342 /* DTS - Repeat Area Select (bits 13:12) */
00343 k_dmtmd_dts_dest = 0x0000U, /**< Destination is repeat/block area */
00344 k_dmtmd_dts_source = 0x1000U, /**< Source is repeat/block area */
00345 k_dmtmd_dts_none = 0x2000U, /**< No repeat/block area */
00346 k_dmtmd_dts_mask = 0x3000U, /**< DTS field mask */
00347
00348 /* MD - Transfer Mode Select (bits 15:14) */
00349 k_dmtmd_md_normal = 0x0000U, /**< Normal transfer mode */
00350 k_dmtmd_md_repeat = 0x4000U, /**< Repeat transfer mode */
00351 k_dmtmd_md_block = 0x8000U, /**< Block transfer mode */
00352 k_dmtmd_md_mask = 0xC000U, /**< MD field mask */
00353 } dmtmd_bits_t;

```

## 8.13.4 Function Documentation

### 8.13.4.1 dmac0()

```
volatile rx_dmac_channel_regs_t * dmac0 (
 void) [inline], [static]
```

Get pointer to DMAC channel 0 registers.

Returns

Volatile pointer to DMAC0 register structure

Since

Version 1.0.0

Definition at line 649 of file [rx72n\\_dmac\\_regs.h](#).

```
00650 {
00651 return (volatile rx_dmac_channel_regs_t*)k_dmac0_base_addr;
00652 }
```

References [k\\_dmac0\\_base\\_addr](#).

### 8.13.4.2 dmac1()

```
volatile rx_dmac_channel_regs_t * dmac1 (
 void) [inline], [static]
```

Get pointer to DMAC channel 1 registers.

Returns

Volatile pointer to DMAC1 register structure

Since

Version 1.0.0

Definition at line 659 of file [rx72n\\_dmac\\_regs.h](#).

```
00660 {
00661 return (volatile rx_dmac_channel_regs_t*)k_dmac1_base_addr;
00662 }
```

References [k\\_dmac1\\_base\\_addr](#).

#### 8.13.4.3 dmac2()

```
volatile rx_dmac_channel_regs_t * dmac2 (
 void) [inline], [static]
```

Get pointer to DMAC channel 2 registers.

Returns

Volatile pointer to DMAC2 register structure

Since

Version 1.0.0

Definition at line 669 of file [rx72n\\_dmac\\_regs.h](#).

```
00670 {
00671 return (volatile rx_dmac_channel_regs_t*)k_dmac2_base_addr;
00672 }
```

References [k\\_dmac2\\_base\\_addr](#).

#### 8.13.4.4 dmac3()

```
volatile rx_dmac_channel_regs_t * dmac3 (
 void) [inline], [static]
```

Get pointer to DMAC channel 3 registers.

Returns

Volatile pointer to DMAC3 register structure

Since

Version 1.0.0

Definition at line 679 of file [rx72n\\_dmac\\_regs.h](#).

```
00680 {
00681 return (volatile rx_dmac_channel_regs_t*)k_dmac3_base_addr;
00682 }
```

References [k\\_dmac3\\_base\\_addr](#).

#### 8.13.4.5 dmac4()

```
volatile rx_dmac_channel_regs_t * dmac4 (
 void) [inline], [static]
```

Get pointer to DMAC channel 4 registers.

Returns

Volatile pointer to DMAC4 register structure

Since

Version 1.0.0

Definition at line 689 of file [rx72n\\_dmac\\_regs.h](#).

```
00690 {
00691 return (volatile rx_dmac_channel_regs_t*)k_dmac4_base_addr;
00692 }
```

References [k\\_dmac4\\_base\\_addr](#).

#### 8.13.4.6 dmac5()

```
volatile rx_dmac_channel_regs_t * dmac5 (
 void) [inline], [static]
```

Get pointer to DMAC channel 5 registers.

Returns

Volatile pointer to DMAC5 register structure

Since

Version 1.0.0

Definition at line 699 of file [rx72n\\_dmac\\_regs.h](#).

```
00700 {
00701 return (volatile rx_dmac_channel_regs_t*)k_dmac5_base_addr;
00702 }
```

References [k\\_dmac5\\_base\\_addr](#).

#### 8.13.4.7 dmac6()

```
volatile rx_dmac_channel_regs_t * dmac6 (
 void) [inline], [static]
```

Get pointer to DMAC channel 6 registers.

Returns

Volatile pointer to DMAC6 register structure

Since

Version 1.0.0

Definition at line 709 of file [rx72n\\_dmac\\_regs.h](#).

```
00710 {
00711 return (volatile rx_dmac_channel_regs_t*)k_dmac6_base_addr;
00712 }
```

References [k\\_dmac6\\_base\\_addr](#).

#### 8.13.4.8 dmac7()

```
volatile rx_dmac_channel_regs_t * dmac7 (
 void) [inline], [static]
```

Get pointer to DMAC channel 7 registers.

Returns

Volatile pointer to DMAC7 register structure

Since

Version 1.0.0

Definition at line 719 of file [rx72n\\_dmac\\_regs.h](#).

```
00720 {
00721 return (volatile rx_dmac_channel_regs_t*)k_dmac7_base_addr;
00722 }
```

References [k\\_dmac7\\_base\\_addr](#).

### 8.13.4.9 `dmac_channel()`

```
volatile rx_dmac_channel_regs_t * dmac_channel (
 uint8_t channel) [inline], [static]
```

Get pointer to DMAC channel registers.

#### Parameters

|         |                      |
|---------|----------------------|
| channel | Channel number (0-7) |
|---------|----------------------|

#### Returns

Volatile pointer to channel register structure

#### Precondition

channel < `k_dmac_channel_count` (8)

DMAC module stop bit (MSTPCRA.MSTPA28) must be cleared

#### Note

Thread Safety: Safe - returns constant hardware address

#### Example:

```
// Configure channel 0 for memory-to-memory transfer
volatile rx_dmac_channel_regs_t* ch0 = dmac_channel(k_dmac_channel_0);
ch0->dmsar = (uint32_t)src_buffer;
ch0->dmdar = (uint32_t)dst_buffer;
ch0->dmcr = transfer_count;
ch0->dmtmd = k_dmtmd_md_normal | k_dmtmd_sz_32bit | k_dmtmd_dctg_software;
ch0->dmamd = k_dmamd_sm_increment | k_dmamd_dm_increment;
ch0->dmcnt = k_dmcnt_dte; // Enable transfer
```

#### Since

Version 1.0.0

Definition at line 639 of file `rx72n_dmac_regs.h`.

```
00640 {
00641 return (volatile rx_dmac_channel_regs_t*)(k_dmac_base_addr + (channel * k_dmac_channel_size));
00642 }
```

References `k_dmac_base_addr`, and `k_dmac_channel_size`.

#### 8.13.4.10 dmac\_dmast\_reg()

```
volatile uint8_t * dmac_dmast_reg (
 void) [inline], [static]
```

Get pointer to DMAC Module Start Register (DMAST).

Returns

Volatile pointer to DMAST register (8-bit)

Since

Version 1.0.0

Definition at line 729 of file [rx72n\\_dmac\\_regs.h](#).

```
00730 {
00731 return (volatile uint8_t*)k_dmac_dmast_addr;
00732 }
```

References [k\\_dmac\\_dmast\\_addr](#).

#### 8.13.4.11 dmac\_dmist\_reg()

```
volatile uint8_t * dmac_dmist_reg (
 void) [inline], [static]
```

Get pointer to DMAC74 Interrupt Status Monitor Register (DMIST).

Returns

Volatile pointer to DMIST register (8-bit)

Since

Version 1.0.0

Definition at line 739 of file [rx72n\\_dmac\\_regs.h](#).

```
00740 {
00741 return (volatile uint8_t*)k_dmac_dmist_addr;
00742 }
```

References [k\\_dmac\\_dmist\\_addr](#).

## 8.14 rx72n\_dmac\_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_dmac_regs.h
00003 * @brief RX72N DMA Controller (DMAC) Register Definitions
00004 *
00005 * @details
00006 * Complete register definitions for the RX72N 8-channel DMA Controller.
00007 * The DMAC transfers data without CPU intervention, supporting normal,
00008 * repeat, and block transfer modes.
00009 *
00010 * @par DMAC Overview
00011 * - 8 independent channels (DMAC0-DMAC7)
00012 * - 512 MB transfer space (0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF)
00013 * - Maximum 64M data transfers (1024 data x 65536 blocks)
00014 * - 8/16/32-bit transfer sizes
00015 * - Software or peripheral interrupt triggers
00016 * - Channel priority: Ch0 > Ch1 > Ch2 > ... > Ch7
00017 *
00018 * @par Register Organization
00019 * | Address | Size | Description |
00020 * |-----|-----|-----|
00021 * | 0x00082000 | - | DMAC channel 0 base |
00022 * | 0x00082040 | - | DMAC channel 1 base |
00023 * | ... | ... | (channels spaced 0x40 apart) |
00024 * | 0x000821C0 | - | DMAC channel 7 base |
00025 * | 0x00082200 | 8 | DMAST - Module Start Register |
00026 * | 0x00082204 | 8 | DMIST - Interrupt Status Monitor |
00027 *
00028 * @par Per-Channel Register Offsets
00029 * | Offset | Size | Register | Description |
00030 * |-----|-----|-----|-----|
00031 * | 0x00 | 32 | DMSAR | Source Address |
00032 * | 0x04 | 32 | DMDAR | Destination Address |
00033 * | 0x08 | 32 | DMCRA | Transfer Count A |
00034 * | 0x0C | 16 | DMCRB | Transfer Count B |
00035 * | 0x10 | 16 | DMTMD | Transfer Mode |
00036 * | 0x13 | 8 | DMINT | Interrupt Setting |
00037 * | 0x14 | 16 | DMAMD | Address Mode |
00038 * | 0x18 | 32 | DMOFR | Offset Register (ch0 only!) |
00039 * | 0x1C | 8 | DMCNT | DMA Transfer Enable |
00040 * | 0x1D | 8 | DMREQ | DMA Software Start |
00041 * | 0x1E | 8 | DMSTS | DMA Status |
00042 * | 0x1F | 8 | DMCSL | DMA Channel Select |
00043 *
00044 * @par NASA Power of 10 Compliance
00045 * - Rule 1: No goto, setjmp, or recursion
00046 * - Rule 2: N/A (no loops)
00047 * - Rule 3: No dynamic memory allocation
00048 * - Rule 4: Short accessor functions
00049 * - Rule 5: Static assertions verify addresses
00050 * - Rule 6: Minimal scope
00051 * - Rule 7: N/A
00052 * - Rule 8: All constants use C23 typed enums
00053 * - Rule 9: No function pointers
00054 * - Rule 10: Compiles with -Wall -Wextra -Werror
00055 *
00056 * @par Manual References
00057 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00058 * - Chapter 18: DMA Controller (DMACaA), pages 677-746
00059 * - Section 18.2: Register Descriptions
00060 *
00061 * @author Locked, Inc.
00062 * @date 2026-01-29
00063 * @copyright Copyright (c) 2026 Locked Inc.
00064 * SPDX-License-Identifier: MIT
00065 * @since Version 1.0.0
00066 */
00067
00068 #pragma once
00069
00070 #include <stdint.h>
00071 #include <stdint.h>
00072
00073 #ifdef __cplusplus
00074 extern "C" {
00075 #endif
00076
00077 /*
=====
00078 * DMAC Base Addresses and Channel Configuration
00079 *
=====
00080 */

```



```

00081
00082 /**
00083 * @enum dmac_addresses_t
00084 * @brief DMAC module addresses
00085 *
00086 * @details
00087 * Base addresses for DMAC channels and shared registers.
00088 * All addresses verified against RX72N Hardware Manual Ch18.
00089 *
00090 * @note Addresses verified 2026-01-29 against manual section 18.2
00091 * @since Version 1.0.0
00092 */
00093 typedef enum : uintptr_t {
00094 /** @brief DMAC module base address (channel 0) */
00095 k_dmac_base_addr = 0x00082000U,
00096
00097 /** @brief Channel register block size (0x40 = 64 bytes per channel) */
00098 k_dmac_channel_size = 0x40U,
00099
00100 /** @brief DMAC channel 0 base address */
00101 k_dmac0_base_addr = 0x00082000U,
00102
00103 /** @brief DMAC channel 1 base address */
00104 k_dmac1_base_addr = 0x00082040U,
00105
00106 /** @brief DMAC channel 2 base address */
00107 k_dmac2_base_addr = 0x00082080U,
00108
00109 /** @brief DMAC channel 3 base address */
00110 k_dmac3_base_addr = 0x000820C0U,
00111
00112 /** @brief DMAC channel 4 base address */
00113 k_dmac4_base_addr = 0x00082100U,
00114
00115 /** @brief DMAC channel 5 base address */
00116 k_dmac5_base_addr = 0x00082140U,
00117
00118 /** @brief DMAC channel 6 base address */
00119 k_dmac6_base_addr = 0x00082180U,
00120
00121 /** @brief DMAC channel 7 base address */
00122 k_dmac7_base_addr = 0x000821C0U,
00123
00124 /** @brief DMAC Module Start Register (DMAST) address */
00125 k_dmac_dmast_addr = 0x00082200U,
00126
00127 /** @brief DMAC74 Interrupt Status Monitor Register (DMIST) address */
00128 k_dmac_dmist_addr = 0x00082204U,
00129 } dmac_addresses_t;
00130
00131 /**
00132 * @enum dmac_offsets_t
00133 * @brief Per-channel register offsets
00134 *
00135 * @details
00136 * Offsets from channel base address for each register.
00137 * All offsets verified against RX72N Hardware Manual Ch18 section 18.2.
00138 *
00139 * @note Offsets verified 2026-01-29 against manual
00140 * @since Version 1.0.0
00141 */
00142 typedef enum : uint8_t {
00143 k_dmac_offset_dmsar = 0x00U, /**< Source Address Register (32-bit) */
00144 k_dmac_offset_dmdar = 0x04U, /**< Destination Address Register (32-bit) */
00145 k_dmac_offset_dmcra = 0x08U, /**< Transfer Count Register A (32-bit) */
00146 k_dmac_offset_dmcra = 0x0CU, /**< Transfer Count Register B (16-bit) */
00147 k_dmac_offset_dmtmd = 0x10U, /**< Transfer Mode Register (16-bit) */
00148 k_dmac_offset_dmint = 0x13U, /**< Interrupt Setting Register (8-bit) */
00149 k_dmac_offset_dmadm = 0x14U, /**< Address Mode Register (16-bit) */
00150 k_dmac_offset_dmoft = 0x18U, /**< Offset Register (32-bit, ch0 only!) */
00151 k_dmac_offset_dmcfr = 0x1CU, /**< DMA Transfer Enable Register (8-bit) */
00152 k_dmac_offset_dmreq = 0x1DU, /**< DMA Software Start Register (8-bit) */
00153 k_dmac_offset_dmsts = 0x1EU, /**< DMA Status Register (8-bit) */
00154 k_dmac_offset_dmcsl = 0x1FU, /**< DMA Channel Select Register (8-bit) */
00155 } dmac_offsets_t;
00156
00157 /**
00158 * @enum dmac_channel_t
00159 * @brief DMAC channel numbers
00160 *
00161 * @details Channel indices for use with accessor functions.
00162 * @since Version 1.0.0
00163 */
00164 typedef enum : uint8_t {
00165 k_dmac_channel_0 = 0U, /**< DMAC channel 0 (highest priority) */
00166 k_dmac_channel_1 = 1U, /**< DMAC channel 1 */
00167 k_dmac_channel_2 = 2U, /**< DMAC channel 2 */

```

```

00168 k_dmac_channel_3 = 3U, /**< DMAC channel 3 */
00169 k_dmac_channel_4 = 4U, /**< DMAC channel 4 */
00170 k_dmac_channel_5 = 5U, /**< DMAC channel 5 */
00171 k_dmac_channel_6 = 6U, /**< DMAC channel 6 */
00172 k_dmac_channel_7 = 7U, /**< DMAC channel 7 (lowest priority) */
00173 k_dmac_channel_count = 8U, /**< Total number of DMAC channels */
00174 } dmac_channel_t;
00175
00176 /*
=====
00177 * DMAC Register Structures
00178 *
=====
00179 */
00180
00181 /**
00182 * @struct rx_dmac_channel_regs_t
00183 * @brief DMAC per-channel register structure
00184 *
00185 * @details
00186 * Memory-mapped register layout for one DMAC channel.
00187 * Each channel has 64 bytes (0x40) of register space.
00188 *
00189 * @warning DMOFR register at offset 0x18 is ONLY valid for channel 0!
00190 * For channels 1-7, this location is reserved.
00191 *
00192 * @note Structure size verified as 0x20 (32 bytes of active registers)
00193 * with reserved padding to fill 0x40 channel spacing.
00194 * @since Version 1.0.0
00195 */
00196 typedef struct __attribute__((packed)) {
00197 /**
00198 * @brief DMA Source Address Register (DMSAR)
00199 * @details 32-bit source start address. Bits 31:29 extend bit 28.
00200 * Valid range: 0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF
00201 */
00202 volatile uint32_t dmsar;
00203
00204 /**
00205 * @brief DMA Destination Address Register (DMDAR)
00206 * @details 32-bit destination start address. Bits 31:29 extend bit 28.
00207 * Valid range: 0x00000000-0xFFFFFFFF, 0xF0000000-0xFFFFFFFF
00208 */
00209 volatile uint32_t dmdar;
00210
00211 /**
00212 * @brief DMA Transfer Count Register A (DMCRA)
00213 * @details
00214 * - Normal mode: bits [15:0] = transfer count, bits [25:16] = block size
00215 * - Repeat/Block mode: bits [9:0] = repeat/block size
00216 */
00217 volatile uint32_t dmcra;
00218
00219 /**
00220 * @brief DMA Transfer Count Register B (DMCRB)
00221 * @details
00222 * - Normal mode (free-running): Not used
00223 * - Repeat mode: bits [15:0] = repeat count
00224 * - Block mode: bits [15:0] = block count
00225 */
00226 volatile uint16_t dmcrb;
00227
00228 /** @brief Reserved (2 bytes padding between DMCRB and DMTMD) */
00229 volatile uint8_t reserved_0e[2];
00230
00231 /**
00232 * @brief DMA Transfer Mode Register (DMTMD)
00233 * @details
00234 * Controls transfer mode, data size, repeat area, and trigger source.
00235 * @see dmtmd_bits_t for bit definitions
00236 */
00237 volatile uint16_t dmtmd;
00238
00239 /** @brief Reserved (1 byte padding between DMTMD and DMINT) */
00240 volatile uint8_t reserved_12;
00241
00242 /**
00243 * @brief DMA Interrupt Setting Register (DMINT)
00244 * @details
00245 * Enables/disables various interrupt sources.
00246 * @see dmint_bits_t for bit definitions
00247 */
00248 volatile uint8_t dmint;
00249
00250 /**
00251 * @brief DMA Address Mode Register (DMAMD)
00252 * @details

```

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00253 * Controls source/destination address update mode and extended repeat.
00254 * @see dmamd_bits_t for bit definitions
00255 */
00256 volatile uint16_t dmamd;
00257
00258 /** @brief Reserved (2 bytes padding between DMAMD and DMOFR) */
00259 volatile uint8_t reserved_16[2];
00260
00261 /**
00262 * @brief DMA Offset Register (DMOFR)
00263 * @details
00264 * 32-bit signed offset value for offset addition mode.
00265 * @warning ONLY VALID FOR CHANNEL 0! Reserved for channels 1-7.
00266 */
00267 volatile uint32_t dmofr;
00268
00269 /**
00270 * @brief DMA Transfer Enable Register (DMCNT)
00271 * @details
00272 * Enables/disables DMA transfer for this channel.
00273 * @see dmcnt_bits_t for bit definitions
00274 */
00275 volatile uint8_t dmcnt;
00276
00277 /**
00278 * @brief DMA Software Start Register (DMREQ)
00279 * @details
00280 * Software trigger control and clear settings.
00281 * @see dmreq_bits_t for bit definitions
00282 */
00283 volatile uint8_t dmreq;
00284
00285 /**
00286 * @brief DMA Status Register (DMSTS)
00287 * @details
00288 * Transfer status flags (active, end, escape end, etc.).
00289 * @see dmsts_bits_t for bit definitions
00290 */
00291 volatile uint8_t dmsts;
00292
00293 /**
00294 * @brief DMA Channel Select Register (DMCSL)
00295 * @details
00296 * Selects interrupt source channel (used with grouped interrupts).
00297 * @see dmcsl_bits_t for bit definitions
00298 */
00299 volatile uint8_t dmcs;
00300
00301 /** @brief Reserved padding to 0x40 (64 bytes) channel spacing */
00302 volatile uint8_t reserved_20[32];
00303 } rx_dmac_channel_regs_t;
00304
00305 /*
=====
00306 * DMTMD Register Bits (Transfer Mode)
00307 *
=====
00308 */
00309
00310 /**
00311 * @enum dmtmd_bits_t
00312 * @brief DMA Transfer Mode Register (DMTMD) bit definitions
00313 *
00314 * @details
00315 * Controls transfer mode, data size, repeat area selection, and trigger source.
00316 *
00317 * @par Register Layout (16-bit @ offset 0x10)
00318 * | Bits | Field | Description |
00319 * |-----|-----|-----|
00320 * | 1:0 | DCTG | Transfer Request Source Select |
00321 * | 7:2 | - | Reserved |
00322 * | 9:8 | SZ | Transfer Data Size Select |
00323 * | 11:10 | - | Reserved |
00324 * | 13:12 | DTS | Repeat Area Select |
00325 * | 15:14 | MD | Transfer Mode Select |
00326 *
00327 * @note Manual ref: Ch18 section 18.2.5
00328 * @since Version 1.0.0
00329 */
00330 typedef enum : uint16_t {
00331 /* DCTG - Transfer Request Source Select (bits 1:0) */
00332 k_dmtmd_dctg_software = 0x0000U, /**< Software trigger */
00333 k_dmtmd_dctg_interrupt = 0x0001U, /**< Peripheral/external interrupt */
00334 k_dmtmd_dctg_mask = 0x0003U, /**< DCTG field mask */
00335
00336 /* SZ - Transfer Data Size Select (bits 9:8) */
00337 k_dmtmd_sz_8bit = 0x0000U, /**< 8-bit transfer */

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00338 k_dmtmd_sz_16bit = 0x0100U, /**< 16-bit transfer */
00339 k_dmtmd_sz_32bit = 0x0200U, /**< 32-bit transfer */
00340 k_dmtmd_sz_mask = 0x0300U, /**< SZ field mask */
00341
00342 /* DTS - Repeat Area Select (bits 13:12) */
00343 k_dmtmd_dts_dest = 0x0000U, /**< Destination is repeat/block area */
00344 k_dmtmd_dts_source = 0x1000U, /**< Source is repeat/block area */
00345 k_dmtmd_dts_none = 0x2000U, /**< No repeat/block area */
00346 k_dmtmd_dts_mask = 0x3000U, /**< DTS field mask */
00347
00348 /* MD - Transfer Mode Select (bits 15:14) */
00349 k_dmtmd_md_normal = 0x0000U, /**< Normal transfer mode */
00350 k_dmtmd_md_repeat = 0x4000U, /**< Repeat transfer mode */
00351 k_dmtmd_md_block = 0x8000U, /**< Block transfer mode */
00352 k_dmtmd_md_mask = 0xC000U, /**< MD field mask */
00353 } dmtmd_bits_t;
00354
00355 /*
=====
00356 * DMINT Register Bits (Interrupt Setting)
00357 *
=====
00358 */
00359
00360 /**
00361 * @enum dmint_bits_t
00362 * @brief DMA Interrupt Setting Register (DMINT) bit definitions
00363 *
00364 * @details
00365 * Controls interrupt enable for various DMA events.
00366 *
00367 * @par Register Layout (8-bit @ offset 0x13)
00368 * | Bit | Field | Description |
00369 * |-----|-----|-----|
00370 * | 0 | DARIE | Dest Address Extended Repeat Overflow IE |
00371 * | 1 | SARIE | Source Address Extended Repeat Overflow IE |
00372 * | 2 | RPTIE | Repeat Size End Interrupt Enable |
00373 * | 3 | ESIE | Transfer Escape End Interrupt Enable |
00374 * | 4 | DTIE | Transfer End Interrupt Enable |
00375 * | 7:5 | - | Reserved |
00376 *
00377 * @note Manual ref: Ch18 section 18.2.6
00378 * @since Version 1.0.0
00379 */
00380 typedef enum : uint8_t {
00381 k_dmint_disabled = 0x00U, /**< All interrupts disabled (polling mode) */
00382 k_dmint_darie = 0x01U, /**< Dest addr extended repeat overflow int enable */
00383 k_dmint_sarie = 0x02U, /**< Source addr extended repeat overflow int enable */
00384 k_dmint_rptie = 0x04U, /**< Repeat size end interrupt enable */
00385 k_dmint_esie = 0x08U, /**< Transfer escape end interrupt enable */
00386 k_dmint_dtie = 0x10U, /**< Transfer end interrupt enable */
00387 k_dmint_mask = 0x1FU, /**< Mask of all interrupt enable bits (bits 4:0) */
00388 } dmint_bits_t;
00389
00390 /*
=====
00391 * DMAMD Register Bits (Address Mode)
00392 *
=====
00393 */
00394
00395 /**
00396 * @enum dmamd_bits_t
00397 * @brief DMA Address Mode Register (DMAMD) bit definitions
00398 *
00399 * @details
00400 * Controls source/destination address update mode and extended repeat area.
00401 *
00402 * @par Register Layout (16-bit @ offset 0x14)
00403 * | Bits | Field | Description |
00404 * |-----|-----|-----|
00405 * | 1:0 | DARA | Destination Address Extended Repeat Area |
00406 * | 5:2 | - | Reserved |
00407 * | 7:6 | DM | Destination Address Update Mode |
00408 * | 9:8 | SARA | Source Address Extended Repeat Area |
00409 * | 13:10 | - | Reserved |
00410 * | 15:14 | SM | Source Address Update Mode |
00411 *
00412 * @note Manual ref: Ch18 section 18.2.7
00413 * @since Version 1.0.0
00414 */
00415 typedef enum : uint16_t {
00416 /* DARA - Destination Address Extended Repeat Area (bits 1:0) */
00417 k_dmamd_dara_mask = 0x0003U, /**< DARA field mask */
00418
00419 /* DM - Destination Address Update Mode (bits 7:6) */
00420 k_dmamd_dm_fixed = 0x0000U, /**< Destination address fixed */

```

```

00421 k_dmamd_dm_offset = 0x0040U, /**< Add offset after transfer (ch0 only) */
00422 k_dmamd_dm_increment = 0x0080U, /**< Increment after transfer */
00423 k_dmamd_dm_decrement = 0x00C0U, /**< Decrement after transfer */
00424 k_dmamd_dm_mask = 0x00C0U, /**< DM field mask */
00425
00426 /* SARA - Source Address Extended Repeat Area (bits 9:8) */
00427 k_dmamd_sara_mask = 0x0300U, /**< SARA field mask */
00428
00429 /* SM - Source Address Update Mode (bits 15:14) */
00430 k_dmamd_sm_fixed = 0x0000U, /**< Source address fixed */
00431 k_dmamd_sm_offset = 0x4000U, /**< Add offset after transfer (ch0 only) */
00432 k_dmamd_sm_increment = 0x8000U, /**< Increment after transfer */
00433 k_dmamd_sm_decrement = 0xC000U, /**< Decrement after transfer */
00434 k_dmamd_sm_mask = 0xC000U, /**< SM field mask */
00435 } dmamd_bits_t;
00436
00437 /*
=====
00438 * DMCNT Register Bits (Transfer Enable)
00439 *
=====
00440 */
00441
00442 /**
00443 * @enum dmcnt_bits_t
00444 * @brief DMA Transfer Enable Register (DMCNT) bit definitions
00445 *
00446 * @par Register Layout (8-bit @ offset 0x1C)
00447 * | Bit | Field | Description |
00448 * |-----|-----|-----|
00449 * | 0 | DTE | DMA Transfer Enable |
00450 * | 7:1 | - | Reserved |
00451 *
00452 * @note Manual ref: Ch18 section 18.2.9
00453 * @since Version 1.0.0
00454 */
00455 typedef enum : uint8_t {
00456 k_dmcnt_dte = 0x01U, /**< DMA Transfer Enable (0=disabled, 1=enabled) */
00457 } dmcnt_bits_t;
00458
00459 /*
=====
00460 * DMREQ Register Bits (Software Start)
00461 *
=====
00462 */
00463
00464 /**
00465 * @enum dmreq_bits_t
00466 * @brief DMA Software Start Register (DMREQ) bit definitions
00467 *
00468 * @par Register Layout (8-bit @ offset 0x1D)
00469 * | Bit | Field | Description |
00470 * |-----|-----|-----|
00471 * | 0 | SWREQ | DMA Software Start |
00472 * | 3:1 | - | Reserved |
00473 * | 4 | CLRS | DMA Software Start Bit Clear |
00474 * | 7:5 | - | Reserved |
00475 *
00476 * @note Manual ref: Ch18 section 18.2.10
00477 * @since Version 1.0.0
00478 */
00479 typedef enum : uint8_t {
00480 k_dmreq_swreq = 0x01U, /**< Software start (write 1 to trigger) */
00481 k_dmreq_chrs = 0x10U, /**< Auto-clear SWREQ after transfer start */
00482 } dmreq_bits_t;
00483
00484 /*
=====
00485 * DMSTS Register Bits (Status)
00486 *
=====
00487 */
00488
00489 /**
00490 * @enum dmsts_bits_t
00491 * @brief DMA Status Register (DMSTS) bit definitions
00492 *
00493 * @par Register Layout (8-bit @ offset 0x1E)
00494 * | Bit | Field | Description |
00495 * |-----|-----|-----|
00496 * | 0 | ESIF | Transfer Escape End Flag |
00497 * | 3:1 | - | Reserved |
00498 * | 4 | DTIF | Transfer End Flag |
00499 * | 6:5 | - | Reserved |
00500 * | 7 | ACT | DMA Active Flag |
00501 *

```

```

00502 * @note Manual ref: Ch18 section 18.2.11
00503 * @since Version 1.0.0
00504 */
00505 typedef enum : uint8_t {
00506 k_dmsts_esif = 0x01U, /**< Transfer escape end flag */
00507 k_dmsts_dtif = 0x10U, /**< Transfer end flag */
00508 k_dmsts_act = 0x80U, /**< DMA active flag (1=transfer in progress) */
00509 } dmsts_bits_t;
00510
00511 /*
=====
00512 * DMCSL Register Bits (Channel Select)
00513 *
=====
00514 */
00515
00516 /**
00517 * @enum dmcsl_bits_t
00518 * @brief DMA Channel Select Register (DMCSL) bit definitions
00519 *
00520 * @par Register Layout (8-bit @ offset 0x1F)
00521 * | Bit | Field | Description |
00522 * |-----|-----|-----|
00523 * | 0 | DISEL | Interrupt Select |
00524 * | 7:1 | - | Reserved |
00525 *
00526 * @note Manual ref: Ch18 section 18.2.12
00527 * @since Version 1.0.0
00528 */
00529 typedef enum : uint8_t {
00530 k_dmcsl_disel = 0x01U, /**< Interrupt select (0=transfer end, 1=DTIF) */
00531 } dmcsl_bits_t;
00532
00533 /*
=====
00534 * DMAST Register Bits (Module Start)
00535 *
=====
00536 */
00537
00538 /**
00539 * @enum dmast_bits_t
00540 * @brief DMAC Module Start Register (DMAST) bit definitions
00541 *
00542 * @par Register Layout (8-bit @ 0x00082200)
00543 * | Bit | Field | Description |
00544 * |-----|-----|-----|
00545 * | 0 | DMST | DMAC Module Start |
00546 * | 7:1 | - | Reserved |
00547 *
00548 * @note Manual ref: Ch18 section 18.2.13
00549 * @since Version 1.0.0
00550 */
00551 typedef enum : uint8_t {
00552 k_dmast_dmst = 0x01U, /**< DMAC module start (0=suspended, 1=operating) */
00553 } dmast_bits_t;
00554
00555 /*
=====
00556 * DMIST Register Bits (Interrupt Status Monitor)
00557 *
=====
00558 */
00559
00560 /**
00561 * @enum dmist_bits_t
00562 * @brief DMAC74 Interrupt Status Monitor Register (DMIST) bit definitions
00563 *
00564 * @par Register Layout (8-bit @ 0x00082204)
00565 * | Bit | Field | Description |
00566 * |-----|-----|-----|
00567 * | 3:0 | - | Reserved |
00568 * | 4 | DMIS4 | Channel 4 interrupt status |
00569 * | 5 | DMIS5 | Channel 5 interrupt status |
00570 * | 6 | DMIS6 | Channel 6 interrupt status |
00571 * | 7 | DMIS7 | Channel 7 interrupt status |
00572 *
00573 * @note Manual ref: Ch18 section 18.2.14
00574 * @since Version 1.0.0
00575 */
00576 typedef enum : uint8_t {
00577 k_dmist_dmis4 = 0x10U, /**< Channel 4 interrupt status (bit 4) */
00578 k_dmist_dmis5 = 0x20U, /**< Channel 5 interrupt status (bit 5) */
00579 k_dmist_dmis6 = 0x40U, /**< Channel 6 interrupt status (bit 6) */
00580 k_dmist_dmis7 = 0x80U, /**< Channel 7 interrupt status (bit 7) */
00581 } dmist_bits_t;
00582

```

```

00583 /*
=====
00584 * Module Stop Control
00585 *
=====
00586 */
00587
00588 /**
00589 * @enum dmac_mstpcr_t
00590 * @brief DMAC module stop control bits
00591 *
00592 * @details
00593 * The DMAC is controlled by MSTPCRA.MSTPA28.
00594 * Must be cleared (0) before accessing DMAC registers.
00595 *
00596 * @note Manual ref: Ch11 (Low Power Consumption) module stop tables
00597 * @since Version 1.0.0
00598 */
00599 typedef enum : uint32_t {
00600 /**
00601 * @brief MSTPCRA bit for DMAC (bit 28)
00602 * @details
00603 * - 0: DMAC module is running
00604 * - 1: DMAC module is stopped (default)
00605 */
00606 k_dmac_mstpcra_bit = (1U « 28U),
00607 } dmac_mstpcr_t;
00608
00609 /*
=====
00610 * Register Accessor Functions
00611 *
=====
00612 */
00613
00614 /**
00615 * @brief Get pointer to DMAC channel registers
00616 *
00617 * @param channel Channel number (0-7)
00618 * @return Volatile pointer to channel register structure
00619 *
00620 * @pre channel < k_dmac_channel_count (8)
00621 * @pre DMAC module stop bit (MSTPCRA.MSTPA28) must be cleared
00622 *
00623 * @note Thread Safety: Safe - returns constant hardware address
00624 *
00625 * @par Example:
00626 * @code{.c}
00627 * // Configure channel 0 for memory-to-memory transfer
00628 * volatile rx_dmac_channel_regs_t* ch0 = dmac_channel(k_dmac_channel_0);
00629 * ch0->dmsar = (uint32_t)src_buffer;
00630 * ch0->dmdar = (uint32_t)dst_buffer;
00631 * ch0->dmcra = transfer_count;
00632 * ch0->dmtmd = k_dmtmd_md_normal | k_dmtmd_sz_32bit | k_dmtmd_dctg_software;
00633 * ch0->dmamd = k_dmamd_sm_increment | k_dmamd_dm_increment;
00634 * ch0->dmcnt = k_dmcnt_dte; // Enable transfer
00635 * @endcode
00636 *
00637 * @since Version 1.0.0
00638 */
00639 static inline volatile rx_dmac_channel_regs_t* dmac_channel(uint8_t channel)
00640 {
00641 return (volatile rx_dmac_channel_regs_t*)(k_dmac_base_addr + (channel * k_dmac_channel_size));
00642 }
00643
00644 /**
00645 * @brief Get pointer to DMAC channel 0 registers
00646 * @return Volatile pointer to DMAC0 register structure
00647 * @since Version 1.0.0
00648 */
00649 static inline volatile rx_dmac_channel_regs_t* dmac0(void)
00650 {
00651 return (volatile rx_dmac_channel_regs_t*)k_dmac0_base_addr;
00652 }
00653
00654 /**
00655 * @brief Get pointer to DMAC channel 1 registers
00656 * @return Volatile pointer to DMAC1 register structure
00657 * @since Version 1.0.0
00658 */
00659 static inline volatile rx_dmac_channel_regs_t* dmac1(void)
00660 {
00661 return (volatile rx_dmac_channel_regs_t*)k_dmac1_base_addr;
00662 }
00663
00664 /**
00665 * @brief Get pointer to DMAC channel 2 registers

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00666 * @return Volatile pointer to DMAC2 register structure
00667 * @since Version 1.0.0
00668 */
00669 static inline volatile rx_dmac_channel_regs_t* dmac2(void)
00670 {
00671 return (volatile rx_dmac_channel_regs_t*)k_dmac2_base_addr;
00672 }
00673
00674 /**
00675 * @brief Get pointer to DMAC channel 3 registers
00676 * @return Volatile pointer to DMAC3 register structure
00677 * @since Version 1.0.0
00678 */
00679 static inline volatile rx_dmac_channel_regs_t* dmac3(void)
00680 {
00681 return (volatile rx_dmac_channel_regs_t*)k_dmac3_base_addr;
00682 }
00683
00684 /**
00685 * @brief Get pointer to DMAC channel 4 registers
00686 * @return Volatile pointer to DMAC4 register structure
00687 * @since Version 1.0.0
00688 */
00689 static inline volatile rx_dmac_channel_regs_t* dmac4(void)
00690 {
00691 return (volatile rx_dmac_channel_regs_t*)k_dmac4_base_addr;
00692 }
00693
00694 /**
00695 * @brief Get pointer to DMAC channel 5 registers
00696 * @return Volatile pointer to DMAC5 register structure
00697 * @since Version 1.0.0
00698 */
00699 static inline volatile rx_dmac_channel_regs_t* dmac5(void)
00700 {
00701 return (volatile rx_dmac_channel_regs_t*)k_dmac5_base_addr;
00702 }
00703
00704 /**
00705 * @brief Get pointer to DMAC channel 6 registers
00706 * @return Volatile pointer to DMAC6 register structure
00707 * @since Version 1.0.0
00708 */
00709 static inline volatile rx_dmac_channel_regs_t* dmac6(void)
00710 {
00711 return (volatile rx_dmac_channel_regs_t*)k_dmac6_base_addr;
00712 }
00713
00714 /**
00715 * @brief Get pointer to DMAC channel 7 registers
00716 * @return Volatile pointer to DMAC7 register structure
00717 * @since Version 1.0.0
00718 */
00719 static inline volatile rx_dmac_channel_regs_t* dmac7(void)
00720 {
00721 return (volatile rx_dmac_channel_regs_t*)k_dmac7_base_addr;
00722 }
00723
00724 /**
00725 * @brief Get pointer to DMAC Module Start Register (DMAST)
00726 * @return Volatile pointer to DMAST register (8-bit)
00727 * @since Version 1.0.0
00728 */
00729 static inline volatile uint8_t* dmac_dmast_reg(void)
00730 {
00731 return (volatile uint8_t*)k_dmac_dmast_addr;
00732 }
00733
00734 /**
00735 * @brief Get pointer to DMAC74 Interrupt Status Monitor Register (DMIST)
00736 * @return Volatile pointer to DMIST register (8-bit)
00737 * @since Version 1.0.0
00738 */
00739 static inline volatile uint8_t* dmac_dmist_reg(void)
00740 {
00741 return (volatile uint8_t*)k_dmac_dmist_addr;
00742 }
00743
00744 /*
=====
00745 * Static Assertions - Address Verification
00746 *
=====
00747 */
00748
00749 /**
00750 * @name Address Verification

```



```

00751 * @brief Compile-time verification of register addresses and offsets
00752 * @{
00753 */
00754
00755 /* Verify base addresses */
00756 static_assert(k_dmac_base_addr == 0x00082000U,
00757 "DMAC base address must be 0x00082000 per Ch18 manual");
00758
00759 static_assert(k_dmac0_base_addr == 0x00082000U,
00760 "DMAC0 base address must be 0x00082000 per Ch18 manual");
00761
00762 static_assert(k_dmac1_base_addr == 0x00082040U,
00763 "DMAC1 base address must be 0x00082040 per Ch18 manual");
00764
00765 static_assert(k_dmac7_base_addr == 0x000821C0U,
00766 "DMAC7 base address must be 0x000821C0 per Ch18 manual");
00767
00768 /* Verify channel spacing */
00769 static_assert(k_dmac_channel_size == 0x40U, "DMAC channel spacing must be 0x40 (64 bytes)");
00770
00771 static_assert(k_dmac1_base_addr - k_dmac0_base_addr == k_dmac_channel_size,
00772 "Channel 0 to 1 spacing must be 0x40");
00773
00774 static_assert(k_dmac7_base_addr - k_dmac0_base_addr == 7 * k_dmac_channel_size,
00775 "Channel 0 to 7 spacing must be 7 * 0x40");
00776
00777 /* Verify shared register addresses */
00778 static_assert(k_dmac_dmast_addr == 0x00082200U, "DMAST address must be 0x00082200 per Ch18 manual");
00779
00780 static_assert(k_dmac_dmist_addr == 0x00082204U, "DMIST address must be 0x00082204 per Ch18 manual");
00781
00782 /* Verify register offsets */
00783 static_assert(k_dmac_offset_dmsar == 0x00U, "DMSAR offset must be 0x00 per Ch18 manual");
00784
00785 static_assert(k_dmac_offset_dmdar == 0x04U, "DMDAR offset must be 0x04 per Ch18 manual");
00786
00787 static_assert(k_dmac_offset_dmcra == 0x08U, "DMCRA offset must be 0x08 per Ch18 manual");
00788
00789 static_assert(k_dmac_offset_dmcra == 0x08U, "DMCRA offset must be 0x08 per Ch18 manual");
00790
00791 static_assert(k_dmac_offset_dmtmd == 0x10U, "DMTMD offset must be 0x10 per Ch18 manual");
00792
00793 static_assert(k_dmac_offset_dmint == 0x13U, "DMINT offset must be 0x13 per Ch18 manual");
00794
00795 static_assert(k_dmac_offset_dmamd == 0x14U, "DMAMD offset must be 0x14 per Ch18 manual");
00796
00797 static_assert(k_dmac_offset_dmofo == 0x18U, "DMOFR offset must be 0x18 per Ch18 manual");
00798
00799 static_assert(k_dmac_offset_dmcnt == 0x1CU, "DMCNT offset must be 0x1C per Ch18 manual");
00800
00801 static_assert(k_dmac_offset_dmreq == 0x1DU, "DMREQ offset must be 0x1D per Ch18 manual");
00802
00803 static_assert(k_dmac_offset_dmsts == 0x1EU, "DMSTS offset must be 0x1E per Ch18 manual");
00804
00805 static_assert(k_dmac_offset_dmcsl == 0x1FU, "DMCSL offset must be 0x1F per Ch18 manual");
00806
00807 /* Verify structure layout matches manual offsets */
00808 static_assert(sizeof(rx_dmac_channel_regs_t) == k_dmac_channel_size,
00809 "DMAC channel struct size must be 0x40 (64 bytes)");
00810
00811 static_assert(offsetof(rx_dmac_channel_regs_t, dmsar) == k_dmac_offset_dmsar,
00812 "DMSAR struct offset must match manual");
00813
00814 static_assert(offsetof(rx_dmac_channel_regs_t, dmdar) == k_dmac_offset_dmdar,
00815 "DMDAR struct offset must match manual");
00816
00817 static_assert(offsetof(rx_dmac_channel_regs_t, dmcra) == k_dmac_offset_dmcra,
00818 "DMCRA struct offset must match manual");
00819
00820 static_assert(offsetof(rx_dmac_channel_regs_t, dmcra) == k_dmac_offset_dmcra,
00821 "DMCRA struct offset must match manual");
00822
00823 static_assert(offsetof(rx_dmac_channel_regs_t, dmtmd) == k_dmac_offset_dmtmd,
00824 "DMTMD struct offset must match manual");
00825
00826 static_assert(offsetof(rx_dmac_channel_regs_t, dmint) == k_dmac_offset_dmint,
00827 "DMINT struct offset must match manual");
00828
00829 static_assert(offsetof(rx_dmac_channel_regs_t, dmamd) == k_dmac_offset_dmamd,
00830 "DMAMD struct offset must match manual");
00831
00832 static_assert(offsetof(rx_dmac_channel_regs_t, dmofo) == k_dmac_offset_dmofo,
00833 "DMOFR struct offset must match manual");
00834
00835 static_assert(offsetof(rx_dmac_channel_regs_t, dmcnt) == k_dmac_offset_dmcnt,
00836 "DMCNT struct offset must match manual");
00837

```

```

00838 static_assert(offsetof(rx_dmac_channel_regs_t, dmreq) == k_dmac_offset_dmreq,
00839 "DMREQ struct offset must match manual");
00840
00841 static_assert(offsetof(rx_dmac_channel_regs_t, dmsts) == k_dmac_offset_dmsts,
00842 "DMSTS struct offset must match manual");
00843
00844 static_assert(offsetof(rx_dmac_channel_regs_t, dmcs1) == k_dmac_offset_dmcs1,
00845 "DMCSL struct offset must match manual");
00846
00847 /** @} */
00848
00849 #ifdef __cplusplus
00850 }
00851 #endif

```

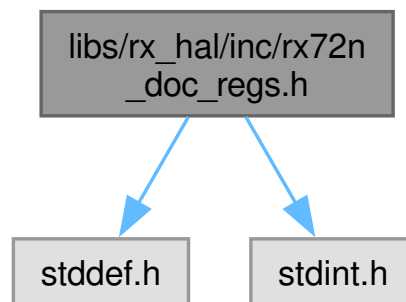
## 8.15 libs/rx\_hal/inc/rx72n\_doc\_regs.h File Reference

RX72N DOC (Data Operation Circuit) Register Definitions.

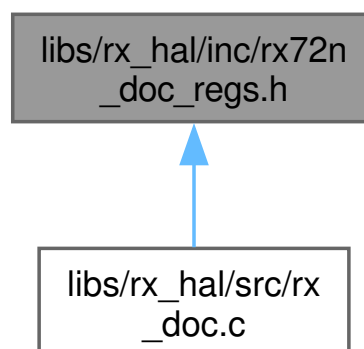
```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_doc\_regs.h:



This graph shows which files directly or indirectly include this file:



### Data Structures

- struct [rx\\_doc\\_regs\\_t](#)  
DOC register map structure.

## Enumerations

- enum `rx_doc_addresses_t` : `uintptr_t` { `k_doc_base_addr` = 0x0008B080 }  
DOC register block base address.
- enum `rx_doc_mstpb_bits_t` : `uint8_t` { `k_doc_mstpb_bit` = 6 }  
DOC module stop bit position in MSTPCRB.
- enum `rx_doc_reserved_sizes_t` : `uint8_t` { `k_doc_reserved_after_docr_bytes` = 1 }  
DOC register block reserved field sizes.
- enum `rx_doc_docr_shifts_t` : `uint8_t` {  
`k_doc_docr_oms_shift` = 0 , `k_doc_docr_dtsel_shift` = 2 , `k_doc_docr_dopcie_shift` = 4 ,  
`k_doc_docr_dopcf_shift` = 5 ,  
`k_doc_docr_dopcfcl_shift` = 6 }  
DOCR bit-field positions.
- enum `rx_doc_docr_bits_t` : `uint8_t` {  
`k_doc_oms_compare` = (0x00 << `k_doc_docr_oms_shift`) , `k_doc_oms_add` = (0x01 << `k_doc_docr_oms_shift`) , `k_doc_oms_subtract` = (0x02 << `k_doc_docr_oms_shift`) ,  
`k_doc_dtsel_mismatch` = (0x00 << `k_doc_docr_dtsel_shift`) ,  
`k_doc_dtsel_match` = (0x01 << `k_doc_docr_dtsel_shift`) , `k_doc_dopcie_disabled` = (0x00 << `k_doc_docr_dopcie_shift`) , `k_doc_dopcie_enabled` = (0x01 << `k_doc_docr_dopcie_shift`) ,  
`k_doc_docr_dopcf_mask` = (0x01 << `k_doc_docr_dopcf_shift`) ,  
`k_doc_dopcfcl_clear` = (0x01 << `k_doc_docr_dopcfcl_shift`) , `k_doc_docr_oms_mask` = (0x03 << `k_doc_docr_oms_shift`) , `k_doc_docr_dtsel_mask` = (0x01 << `k_doc_docr_dtsel_shift`) ,  
`k_doc_docr_dopcie_mask` = (0x01 << `k_doc_docr_dopcie_shift`) }  
DOCR bit-field values.

## Functions

- static volatile `rx_doc_regs_t` \* `doc` (void)  
Get pointer to DOC register block.

## 8.15.1 Detailed Description

## RX72N DOC (Data Operation Circuit) Register Definitions.

Register definitions for the Data Operation Circuit (DOC) peripheral on the RX72N microcontroller. The DOC is a small fixed-function accelerator that performs one of four 16-bit operations on two operands:

| Mode<br>(OMS[1:0]) | Operation                                  | Flag trigger             |
|--------------------|--------------------------------------------|--------------------------|
| 00 (DCSEL=0)       | 16-bit compare (mismatch detection)        | DODIR != DODSR           |
| 00 (DCSEL=1)       | 16-bit compare (match detection)           | DODIR == DODSR           |
| 01                 | 16-bit addition (DODSR = DODSR + DODIR)    | Carry (result > 0xFFFF)  |
| 10                 | 16-bit subtraction (DODSR = DODSR - DODIR) | Borrow (result < 0x0000) |
| 11                 | Setting prohibited                         | N/A                      |

The DOC sets DOPCF whenever the configured condition occurs; DOPCF is read-only and must be cleared by writing 1 to DOPCFCL. An interrupt (GROUPBL0::DOPCI) can optionally be generated via DOPCIE.

## 8.15.1.1 Memory Map

| Offset | Size | Register | Access | Description                    |
|--------|------|----------|--------|--------------------------------|
| 0x00   | 1    | DOCR     | R/W    | DOC Control Register           |
| 0x01   | 1    | -        | -      | Reserved                       |
| 0x02   | 2    | DODIR    | R/W    | Data Input Register            |
| 0x04   | 2    | DODSR    | R/W    | Data Setting Register / Result |

## 8.15.1.2 DOCR Bit Layout (8-bit)

| Bit      | Field         | Access | Description                                    |
|----------|---------------|--------|------------------------------------------------|
| 1:↵<br>0 | OMS[1:↵<br>0] | R/W    | Operating Mode Select                          |
| 2        | DCSEL         | R/W    | Detection Condition Select (compare mode only) |
| 3        | -             | R      | Reserved (read as 0)                           |
| 4        | DOPCIE        | R/W    | DOPCI Interrupt Enable (1=enabled)             |
| 5        | DOPCF         | R      | DOPC Flag (1=condition occurred)               |
| 6        | DOPCFCL       | W1C    | DOPCF Clear (write 1 to clear DOPCF)           |
| 7        | -             | R      | Reserved (read as 0)                           |

## 8.15.1.3 Module Stop Bit

DOC is clock-gated via MSTPCRB.MSTPB6. The DOC module stop bit must be cleared (under PRCR.↵  
PRC1 unlock) before accessing any DOC register, or writes will be silently ignored.

## 8.15.1.4 Interrupt Source

DOC raises one interrupt request: GROUPBL0.DOPCI (vector 110, group bit 29). When DOPCIE=1, setting DOPCF triggers GROUPBL0.DOPCI.

## Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0111)

- Chapter 59: Data Operation Circuit (DOC), pages 2969-2978
- Section 11.2.2 MSTPCRB: page 409 (DOC = MSTPB6)
- Section 14.4 Group Interrupt Request Sources: GROUPBL0.DOPCI at bit 29

## NASA Power of 10 Compliance

- Rule 1: No goto / setjmp / recursion (header-only data)
- Rule 2: N/A (no loops)
- Rule 3: No dynamic allocation
- Rule 4: Accessor is a single-statement inline function
- Rule 5: N/A (hardware layer; enforced by higher-level API)

- Rule 6: Minimal scope (no file-scope variables)
- Rule 7: N/A (no return values to check)
- Rule 8: All constants use C23 typed enums (uintptr\_t for addresses)
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

See also

[rx\\_doc.h](#) Higher-level DOC API

[rx72n\\_system\\_regs.h](#) PRCR and MSTPCRB accessors

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx72n\\_doc\\_regs.h](#).

## 8.15.2 Data Structure Documentation

### 8.15.2.1 struct rx`doc`regs`t

DOC register map structure.

Packed volatile structure mirroring the DOC memory-mapped register layout. Access via the [doc\(\)](#) accessor returns a pointer to this structure at k\_doc\_base\_addr.

Memory Layout (6 bytes total)

| Offset | Size | Field     | Type         | Description          |
|--------|------|-----------|--------------|----------------------|
| 0x00   | 1    | docr      | uint8↔<br>_t | DOC Control Register |
| 0x01   | 1    | reserved0 | -            | Reserved             |

| Offset | Size | Field | Type     | Description                    |
|--------|------|-------|----------|--------------------------------|
| 0x02   | 2    | dodir | uint16_t | Data Input Register            |
| 0x04   | 2    | dodsr | uint16_t | Data Setting / Result Register |

Invariant

```
sizeof(rx_doc_regs_t) == 6
```

All offsets match RX72N HW Manual Chapter 59 register table

See also

[doc\(\)](#) Accessor function

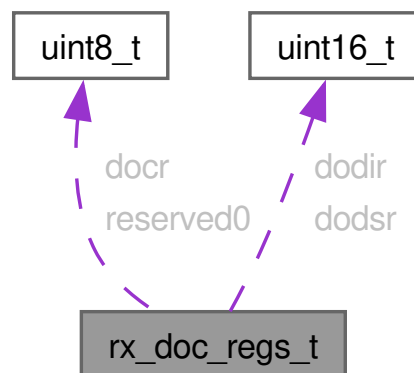
[rx\\_doc\\_docr\\_bits\\_t](#) DOCR bit field values

Since

Version 1.0.0

Definition at line 188 of file [rx72n\\_doc\\_regs.h](#).

Collaboration diagram for rx\_doc\_regs\_t:



## Data Fields

|                   |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | docr                                                            | <p>DOC Control Register (DOCR) @ offset 0x00.<br/>Selects operating mode, detection condition, enables interrupt, reports and clears the operation completion flag.</p> <p>See also</p> <p><a href="#">rx_doc_docr_bits_t</a> Bit-field values for this register</p>                                                                                                                                                                                                |
| volatile uint16_t | dodir                                                           | <p>Data Input Register (DODIR) @ offset 0x02.<br/>16-bit operand register. Holds the compare reference, or the second addend (for add mode), or the subtrahend (for subtract mode).</p> <p>Note</p> <p>Writes to DODIR itself do not trigger the DOC operation.</p>                                                                                                                                                                                                 |
| volatile uint16_t | dodsr                                                           | <p>Data Setting / Result Register (DODSR) @ offset 0x04.<br/>16-bit operand register and result register:</p> <ul style="list-style-type: none"> <li>• Compare mode: holds the comparison value</li> <li>• Add mode: before write = initial accumulator; after write = accumulator + DODIR</li> <li>• Subtract mode: before write = minuend; after write = minuend - DODIR</li> </ul> <p>Note</p> <p>Writing DODSR is what triggers a DOC operation to execute.</p> |
| uint8_t           | reserved0↵<br><a href="#">[k_doc_reserved_after_docr_bytes]</a> | Reserved byte @ offset 0x01.                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### 8.15.3 Enumeration Type Documentation

#### 8.15.3.1 rx`doc`addresses`t

enum [rx\\_doc\\_addresses\\_t](#) : uintptr\_t

DOC register block base address.

Base address for the Data Operation Circuit register block at 0x0008B080. The DOC occupies a 6-byte register window (DOCR, reserved byte, DODIR, DODSR).

#### Manual Reference

RX72N HW Manual, Chapter 59 (Data Operation Circuit), Register address table, page 2969.

See also

[doc\(\)](#) Accessor function

Since

Version 1.0.0

#### Enumerator

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| k_doc_base_addr | DOC register base address (0x0008B080). Verified against RX72N HW Manual, Chapter 59, page 2969. |
|-----------------|--------------------------------------------------------------------------------------------------|

Definition at line 111 of file [rx72n\\_doc\\_regs.h](#).

```
00111 : uintptr_t {
00112 /**
00113 * @brief DOC register base address (0x0008B080)
00114 * @details Verified against RX72N HW Manual, Chapter 59, page 2969.
00115 */
00116 k_doc_base_addr = 0x0008B080,
00117 } rx_doc_addresses_t;
```

#### 8.15.3.2 rx`doc`docr`bits`t

enum [rx\\_doc\\_docr\\_bits\\_t](#) : uint8\_t

DOCR bit-field values.

Pre-shifted values for writing DOCR fields. Values may be bitwise-OR'd together except that OMS fields are mutually exclusive.



## DOCR Layout (8-bit)

| Bit     | Field   | Notes                                            |
|---------|---------|--------------------------------------------------|
| 1:<br>0 | OMS     | 00=compare, 01=add, 10=subtract, 11=prohibited   |
| 2       | DCSEL   | 0=mismatch detect, 1=match detect (compare only) |
| 4       | DOPCIE  | 0=interrupt disabled, 1=enabled                  |
| 5       | DOPCF   | read-only completion flag                        |
| 6       | DOPCFCL | write 1 to clear DOPCF                           |

Since

Version 1.0.0

## Enumerator

|                        |                      |
|------------------------|----------------------|
| k_doc_oms_compare      | 16-bit compare       |
| k_doc_oms_add          | 16-bit addition      |
| k_doc_oms_subtract     | 16-bit subtraction   |
| k_doc_dtsel_mismatch   | flag on mismatch     |
| k_doc_dtsel_match      | flag on match        |
| k_doc_dopcie_disabled  | IRQ disabled         |
| k_doc_dopcie_enabled   | IRQ enabled          |
| k_doc_docr_dopcf_mask  | DOPCF mask           |
| k_doc_dopcfcl_clear    | write to clear DOPCF |
| k_doc_docr_oms_mask    |                      |
| k_doc_docr_dtsel_mask  |                      |
| k_doc_docr_dopcie_mask |                      |

Definition at line 297 of file [rx72n\\_doc\\_regs.h](#).

```

00297 : uint8_t {
00298 /* OMS[1:0] - Operating Mode Select (bits 1:0) */
00299 k_doc_oms_compare = (0x00 « k_doc_docr_oms_shift), /**< 16-bit compare */
00300 k_doc_oms_add = (0x01 « k_doc_docr_oms_shift), /**< 16-bit addition */
00301 k_doc_oms_subtract = (0x02 « k_doc_docr_oms_shift), /**< 16-bit subtraction */
00302
00303 /* DCSEL - Detection Condition Select (bit 2, compare mode only) */
00304 k_doc_dtsel_mismatch = (0x00 « k_doc_docr_dtsel_shift), /**< flag on mismatch */
00305 k_doc_dtsel_match = (0x01 « k_doc_docr_dtsel_shift), /**< flag on match */
00306
00307 /* DOPCIE - Interrupt Enable (bit 4) */
00308 k_doc_dopcie_disabled = (0x00 « k_doc_docr_dopcie_shift), /**< IRQ disabled */
00309 k_doc_dopcie_enabled = (0x01 « k_doc_docr_dopcie_shift), /**< IRQ enabled */
00310
00311 /* DOPCF - Completion Flag (bit 5, read-only) */
00312 k_doc_docr_dopcf_mask = (0x01 « k_doc_docr_dopcf_shift), /**< DOPCF mask */
00313
00314 /* DOPCFCL - DOPCF Clear (bit 6, write-1-to-clear) */
00315 k_doc_dopcfcl_clear = (0x01 « k_doc_docr_dopcfcl_shift), /**< write to clear DOPCF */
00316
00317 /* Field masks for read-back validation */
00318 k_doc_docr_oms_mask = (0x03 « k_doc_docr_oms_shift),
00319 k_doc_docr_dtsel_mask = (0x01 « k_doc_docr_dtsel_shift),
00320 k_doc_docr_dopcie_mask = (0x01 « k_doc_docr_dopcie_shift),
00321 } rx_doc_docr_bits_t;

```

8.15.3.3 rx`doc`docr`shifts``t`

```
enum rx_doc_docr_shifts_t : uint8_t
```

DOCR bit-field positions.

Since

Version 1.0.0

Enumerator

|                          |                              |
|--------------------------|------------------------------|
| k_doc_docr_oms_shift     | OMS[1:0] position (bits 1:0) |
| k_doc_docr_dcsel_shift   | DCSEL position (bit 2)       |
| k_doc_docr_dopcie_shift  | DOPCIE position (bit 4)      |
| k_doc_docr_dopcf_shift   | DOPCF position (bit 5)       |
| k_doc_docr_dopcfcl_shift | DOPCFCL position (bit 6)     |

Definition at line 265 of file [rx72n\\_doc\\_regs.h](#).

```
00265 : uint8_t {
00266 k_doc_docr_oms_shift = 0, /**< OMS[1:0] position (bits 1:0) */
00267 k_doc_docr_dcsel_shift = 2, /**< DCSEL position (bit 2) */
00268 k_doc_docr_dopcie_shift = 4, /**< DOPCIE position (bit 4) */
00269 k_doc_docr_dopcf_shift = 5, /**< DOPCF position (bit 5) */
00270 k_doc_docr_dopcfcl_shift = 6, /**< DOPCFCL position (bit 6) */
00271 } rx_doc_docr_shifts_t;
```

8.15.3.4 rx`doc`mstpb`bits``t`

```
enum rx_doc_mstpb_bits_t : uint8_t
```

DOC module stop bit position in MSTPCRB.

The DOC peripheral is clock-gated by bit 6 of MSTPCRB (MSTPB6). Writing 0 to this bit (under PRCR.PRC1 unlock) enables DOC; writing 1 stops the module.

Manual Reference

RX72N HW Manual, Section 11.2.2 MSTPCRB, page 409: MSTPB6 = DOC.

Since

Version 1.0.0

Enumerator

|                 |                                                  |
|-----------------|--------------------------------------------------|
| k_doc_mstpb_bit | DOC module stop bit position in MSTPCRB (bit 6). |
|-----------------|--------------------------------------------------|

Definition at line 137 of file [rx72n\\_doc\\_regs.h](#).

```
00137 : uint8_t {
00138 /** @brief DOC module stop bit position in MSTPCRB (bit 6) */
00139 k_doc_mstpb_bit = 6,
00140 } rx_doc_mstpb_bits_t;
```

### 8.15.3.5 rx\_doc\_reserved\_sizes\_t

```
enum rx_doc_reserved_sizes_t : uint8_t
```

DOC register block reserved field sizes.

Number of reserved bytes used to pad the DOC register layout so the compiler-generated struct matches the hardware memory map.

Since

Version 1.0.0

Enumerator

|                                 |                                                     |
|---------------------------------|-----------------------------------------------------|
| k_doc_reserved_after_docr_bytes | Reserved byte between DOCR (0x00) and DODIR (0x02). |
|---------------------------------|-----------------------------------------------------|

Definition at line 154 of file [rx72n\\_doc\\_regs.h](#).

```
00154 : uint8_t {
00155 /** @brief Reserved byte between DOCR (0x00) and DODIR (0x02) */
00156 k_doc_reserved_after_docr_bytes = 1,
00157 } rx_doc_reserved_sizes_t;
```

## 8.15.4 Function Documentation

### 8.15.4.1 doc()

```
volatile rx_doc_regs_t * doc (
 void) [inline], [static]
```

Get pointer to DOC register block.

Returns a volatile pointer to the DOC register structure mapped at k\_doc\_base\_addr (0x0008B080).

Returns

Volatile pointer to DOC register structure (never nullptr)

Return values

|          |                                       |
|----------|---------------------------------------|
| Non-NULL | Always returns valid hardware pointer |
|----------|---------------------------------------|

Precondition

Caller must have cleared MSTPCRB.MSTPB6 before accessing any register

Postcondition

Pointer is valid for program lifetime (hardware address is constant)

## Note

Thread Safety: Safe (returns a constant hardware address)  
 Always inlined; zero overhead

## Example

```
doc()->docr = k_doc_oms_compare | k_doc_dtsel_match;
doc()->dodir = 0xCAFE;
doc()->dodsr = 0xCAFE; // triggers DOC operation
```

## See also

[k\\_doc\\_base\\_addr](#) Base address constant

## Since

Version 1.0.0

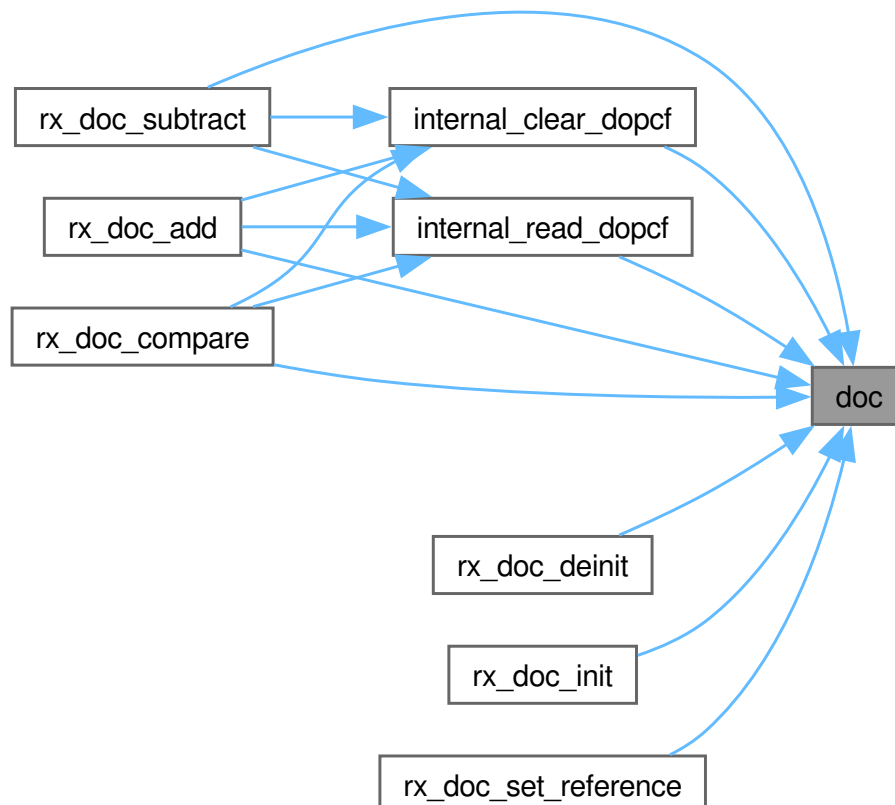
Definition at line 250 of file [rx72n\\_doc\\_regs.h](#).

```
00251 {
00252 return (volatile rx_doc_regs_t*)k_doc_base_addr;
00253 }
```

References [k\\_doc\\_base\\_addr](#).

Referenced by [internal\\_clear\\_dopcf\(\)](#), [internal\\_read\\_dopcf\(\)](#), [rx\\_doc\\_add\(\)](#), [rx\\_doc\\_compare\(\)](#), [rx\\_doc\\_deinit\(\)](#), [rx\\_doc\\_init\(\)](#), [rx\\_doc\\_set\\_reference\(\)](#), and [rx\\_doc\\_subtract\(\)](#).

Here is the caller graph for this function:



## 8.16 rx72n`doc`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n`doc`regs.h
00003 * @brief RX72N DOC (Data Operation Circuit) Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the Data Operation Circuit (DOC) peripheral on the
00007 * RX72N microcontroller. The DOC is a small fixed-function accelerator that
00008 * performs one of four 16-bit operations on two operands:
00009 *
00010 * | Mode (OMS[1:0]) | Operation | Flag trigger |
00011 * |-----|-----|-----|
00012 * | 00 (DCSEL=0) | 16-bit compare (mismatch detection) | DODIR != DODSR |
00013 * | 00 (DCSEL=1) | 16-bit compare (match detection) | DODIR == DODSR |
00014 * | 01 | 16-bit addition (DODSR = DODSR + DODIR) | Carry (result > 0xFFFF) |
00015 * | 10 | 16-bit subtraction (DODSR = DODSR - DODIR) | Borrow (result < 0x0000) |
00016 * | 11 | Setting prohibited | N/A |
00017 *
00018 * The DOC sets DOPCF whenever the configured condition occurs; DOPCF is read-only
00019 * and must be cleared by writing 1 to DOPCFCL. An interrupt (GROUPBL0::DOPCI) can
00020 * optionally be generated via DOPCIE.
00021 *
00022 * ## Memory Map
00023 *
00024 * | Offset | Size | Register | Access | Description |
00025 * |-----|-----|-----|-----|-----|
00026 * | 0x00 | 1 | DOCR | R/W | DOC Control Register |
00027 * | 0x01 | 1 | - | - | Reserved |
00028 * | 0x02 | 2 | DODIR | R/W | Data Input Register |
00029 * | 0x04 | 2 | DODSR | R/W | Data Setting Register / Result |
00030 *
00031 * ## DOCR Bit Layout (8-bit)
00032 *
00033 * | Bit | Field | Access | Description |
00034 * |-----|-----|-----|-----|
00035 * | 1:0 | OMS[1:0] | R/W | Operating Mode Select |
00036 * | 2 | DCSEL | R/W | Detection Condition Select (compare mode only) |
00037 * | 3 | - | R | Reserved (read as 0) |
00038 * | 4 | DOPCIE | R/W | DOPCI Interrupt Enable (1=enabled) |
00039 * | 5 | DOPCF | R | DOPC Flag (1=condition occurred) |
00040 * | 6 | DOPCFCL | W1C | DOPCF Clear (write 1 to clear DOPCF) |
00041 * | 7 | - | R | Reserved (read as 0) |
00042 *
00043 * ## Module Stop Bit
00044 *
00045 * DOC is clock-gated via MSTPCRB.MSTPB6. The DOC module stop bit must be
00046 * cleared (under PRCR.PRC1 unlock) before accessing any DOC register, or writes
00047 * will be silently ignored.
00048 *
00049 * ## Interrupt Source
00050 *
00051 * DOC raises one interrupt request: GROUPBL0.DOPCI (vector 110, group bit 29).
00052 * When DOPCIE=1, setting DOPCF triggers GROUPBL0.DOPCI.
00053 *
00054 * @par Manual Reference
00055 * RX72N Group User's Manual: Hardware (R01UH0824EJ0111)
00056 * - Chapter 59: Data Operation Circuit (DOC), pages 2969-2978
00057 * - Section 11.2.2 MSTPCRB: page 409 (DOC = MSTPB6)
00058 * - Section 14.4 Group Interrupt Request Sources: GROUPBL0.DOPCI at bit 29
00059 *
00060 * @par NASA Power of 10 Compliance
00061 * - Rule 1: No goto / setjmp / recursion (header-only data)
00062 * - Rule 2: N/A (no loops)
00063 * - Rule 3: No dynamic allocation
00064 * - Rule 4: Accessor is a single-statement inline function
00065 * - Rule 5: N/A (hardware layer; enforced by higher-level API)
00066 * - Rule 6: Minimal scope (no file-scope variables)
00067 * - Rule 7: N/A (no return values to check)
00068 * - Rule 8: All constants use C23 typed enums (uintptr_t for addresses)
00069 * - Rule 9: No function pointers
00070 * - Rule 10: Compiles with -Wall -Wextra -Werror
00071 *
00072 * @see rx`doc`h Higher-level DOC API
00073 * @see rx72n`system`regs.h PRCR and MSTPCRB accessors
00074 *
00075 * @author Locked, Inc.
00076 * @date 2026-04-21
00077 * @copyright Copyright (c) 2026 Locked Inc.
00078 * SPDX-License-Identifier: MIT
00079 * @since Version 1.0.0
00080 */
00081
00082 #pragma once

```

```

00083
00084 #include <stddef.h>
00085 #include <stdint.h>
00086
00087 #ifdef __cplusplus
00088 extern "C" {
00089 #endif
00090
00091 /*
=====
00092 * DOC Base Address
00093 *
=====
00094 */
00095
00096 /**
00097 * @enum rx_doc_addresses_t
00098 * @brief DOC register block base address
00099 *
00100 * @details
00101 * Base address for the Data Operation Circuit register block at 0x0008B080.
00102 * The DOC occupies a 6-byte register window (DOCR, reserved byte, DODIR, DODSR).
00103 *
00104 * @par Manual Reference
00105 * RX72N HW Manual, Chapter 59 (Data Operation Circuit), Register address table,
00106 * page 2969.
00107 *
00108 * @see doc() Accessor function
00109 * @since Version 1.0.0
00110 */
00111 typedef enum : uintptr_t {
00112 /**
00113 * @brief DOC register base address (0x0008B080)
00114 * @details Verified against RX72N HW Manual, Chapter 59, page 2969.
00115 */
00116 k_doc_base_addr = 0x0008B080,
00117 } rx_doc_addresses_t;
00118
00119 /*
=====
00120 * DOC Module Stop Bit (MSTPCRB)
00121 *
=====
00122 */
00123
00124 /**
00125 * @enum rx_doc_mstpb_bits_t
00126 * @brief DOC module stop bit position in MSTPCRB
00127 *
00128 * @details
00129 * The DOC peripheral is clock-gated by bit 6 of MSTPCRB (MSTPB6). Writing 0 to
00130 * this bit (under PRCR.PRC1 unlock) enables DOC; writing 1 stops the module.
00131 *
00132 * @par Manual Reference
00133 * RX72N HW Manual, Section 11.2.2 MSTPCRB, page 409: MSTPB6 = DOC.
00134 *
00135 * @since Version 1.0.0
00136 */
00137 typedef enum : uint8_t {
00138 /** @brief DOC module stop bit position in MSTPCRB (bit 6) */
00139 k_doc_mstpb_bit = 6,
00140 } rx_doc_mstpb_bits_t;
00141
00142 /*
=====
00143 * DOC Reserved Field Sizes
00144 *
=====
00145 */
00146
00147 /**
00148 * @enum rx_doc_reserved_sizes_t
00149 * @brief DOC register block reserved field sizes
00150 * @details Number of reserved bytes used to pad the DOC register layout so the
00151 * compiler-generated struct matches the hardware memory map.
00152 * @since Version 1.0.0
00153 */
00154 typedef enum : uint8_t {
00155 /** @brief Reserved byte between DOCR (0x00) and DODIR (0x02) */
00156 k_doc_reserved_after_docr_bytes = 1,
00157 } rx_doc_reserved_sizes_t;
00158
00159 /*
=====
00160 * DOC Register Structure
00161 *
=====

```

```

00162 */
00163
00164 /**
00165 * @struct rx_doc_regs_t
00166 * @brief DOC register map structure
00167 *
00168 * @details
00169 * Packed volatile structure mirroring the DOC memory-mapped register layout.
00170 * Access via the doc() accessor returns a pointer to this structure at
00171 * k_doc_base_addr.
00172 *
00173 * @par Memory Layout (6 bytes total)
00174 * | Offset | Size | Field | Type | Description |
00175 * |-----|-----|-----|-----|-----|
00176 * | 0x00 | 1 | docr | uint8_t | DOC Control Register |
00177 * | 0x01 | 1 | reserved0 | - | Reserved |
00178 * | 0x02 | 2 | dodir | uint16_t | Data Input Register |
00179 * | 0x04 | 2 | dodsr | uint16_t | Data Setting / Result Register |
00180 *
00181 * @invariant sizeof(rx_doc_regs_t) == 6
00182 * @invariant All offsets match RX72N HW Manual Chapter 59 register table
00183 *
00184 * @see doc() Accessor function
00185 * @see rx_doc_docr_bits_t DOCR bit field values
00186 * @since Version 1.0.0
00187 */
00188 typedef struct __attribute__((packed)) {
00189 /**
00190 * @brief DOC Control Register (DOCR) @ offset 0x00
00191 * @details Selects operating mode, detection condition, enables interrupt,
00192 * reports and clears the operation completion flag.
00193 * @see rx_doc_docr_bits_t Bit-field values for this register
00194 */
00195 volatile uint8_t docr;
00196
00197 /** @brief Reserved byte @ offset 0x01 */
00198 uint8_t reserved0[k_doc_reserved_after_docr_bytes];
00199
00200 /**
00201 * @brief Data Input Register (DODIR) @ offset 0x02
00202 * @details 16-bit operand register. Holds the compare reference, or the
00203 * second addend (for add mode), or the subtrahend (for subtract mode).
00204 * @note Writes to DODIR itself do not trigger the DOC operation.
00205 */
00206 volatile uint16_t dodir;
00207
00208 /**
00209 * @brief Data Setting / Result Register (DODSR) @ offset 0x04
00210 * @details 16-bit operand register and result register:
00211 * - Compare mode: holds the comparison value
00212 * - Add mode: before write = initial accumulator; after write = accumulator + DODIR
00213 * - Subtract mode: before write = minuend; after write = minuend - DODIR
00214 * @note Writing DODSR is what triggers a DOC operation to execute.
00215 */
00216 volatile uint16_t dodsr;
00217 } rx_doc_regs_t;
00218
00219 /*
=====
00220 * DOC Register Accessor
00221 *
=====
00222 */
00223
00224 /**
00225 * @brief Get pointer to DOC register block
00226 *
00227 * @details
00228 * Returns a volatile pointer to the DOC register structure mapped at
00229 * k_doc_base_addr (0x0008B080).
00230 *
00231 * @return Volatile pointer to DOC register structure (never nullptr)
00232 * @retval Non-NULL Always returns valid hardware pointer
00233 *
00234 * @pre Caller must have cleared MSTPCRB.MSTPB6 before accessing any register
00235 * @post Pointer is valid for program lifetime (hardware address is constant)
00236 *
00237 * @note Thread Safety: Safe (returns a constant hardware address)
00238 * @note Always inlined; zero overhead
00239 *
00240 * @par Example
00241 * @code{.c}
00242 * doc()->docr = k_doc_oms_compare | k_doc_dcsel_match;
00243 * doc()->dodir = 0xCAFE;
00244 * doc()->dodsr = 0xCAFE; // triggers DOC operation
00245 * @endcode
00246 */

```

```

00247 * @see k_doc_base_addr Base address constant
00248 * @since Version 1.0.0
00249 */
00250 static inline volatile rx_doc_regs_t* doc(void)
00251 {
00252 return (volatile rx_doc_regs_t*)k_doc_base_addr;
00253 }
00254
00255 /*
=====
00256 * DOCR Bit Positions
00257 *
=====
00258 */
00259 /**
00260 **
00261 * @enum rx_doc_docr_shifts_t
00262 * @brief DOCR bit-field positions
00263 * @since Version 1.0.0
00264 */
00265 typedef enum : uint8_t {
00266 k_doc_docr_oms_shift = 0, /**< OMS[1:0] position (bits 1:0) */
00267 k_doc_docr_dcsel_shift = 2, /**< DCSEL position (bit 2) */
00268 k_doc_docr_dopcie_shift = 4, /**< DOPCIE position (bit 4) */
00269 k_doc_docr_dopcf_shift = 5, /**< DOPCF position (bit 5) */
00270 k_doc_docr_dopcfcl_shift = 6, /**< DOPCFCL position (bit 6) */
00271 } rx_doc_docr_shifts_t;
00272
00273 /*
=====
00274 * DOCR Bit Field Values
00275 *
=====
00276 */
00277 /**
00278 **
00279 * @enum rx_doc_docr_bits_t
00280 * @brief DOCR bit-field values
00281 *
00282 * @details
00283 * Pre-shifted values for writing DOCR fields. Values may be bitwise-OR'd
00284 * together except that OMS fields are mutually exclusive.
00285 *
00286 * @par DOCR Layout (8-bit)
00287 * | Bit | Field | Notes |
00288 * |-----|-----|-----|
00289 * | 1:0 | OMS | 00=compare, 01=add, 10=subtract, 11=prohibited |
00290 * | 2 | DCSEL | 0=mismatch detect, 1=match detect (compare only) |
00291 * | 4 | DOPCIE | 0=interrupt disabled, 1=enabled |
00292 * | 5 | DOPCF | read-only completion flag |
00293 * | 6 | DOPCFCL | write 1 to clear DOPCF |
00294 *
00295 * @since Version 1.0.0
00296 */
00297 typedef enum : uint8_t {
00298 /* OMS[1:0] - Operating Mode Select (bits 1:0) */
00299 k_doc_oms_compare = (0x00 « k_doc_docr_oms_shift), /**< 16-bit compare */
00300 k_doc_oms_add = (0x01 « k_doc_docr_oms_shift), /**< 16-bit addition */
00301 k_doc_oms_subtract = (0x02 « k_doc_docr_oms_shift), /**< 16-bit subtraction */
00302
00303 /* DCSEL - Detection Condition Select (bit 2, compare mode only) */
00304 k_doc_dcsel_mismatch = (0x00 « k_doc_docr_dcsel_shift), /**< flag on mismatch */
00305 k_doc_dcsel_match = (0x01 « k_doc_docr_dcsel_shift), /**< flag on match */
00306
00307 /* DOPCIE - Interrupt Enable (bit 4) */
00308 k_doc_dopcie_disabled = (0x00 « k_doc_docr_dopcie_shift), /**< IRQ disabled */
00309 k_doc_dopcie_enabled = (0x01 « k_doc_docr_dopcie_shift), /**< IRQ enabled */
00310
00311 /* DOPCF - Completion Flag (bit 5, read-only) */
00312 k_doc_docr_dopcf_mask = (0x01 « k_doc_docr_dopcf_shift), /**< DOPCF mask */
00313
00314 /* DOPCFCL - DOPCF Clear (bit 6, write-1-to-clear) */
00315 k_doc_dopcfcl_clear = (0x01 « k_doc_docr_dopcfcl_shift), /**< write to clear DOPCF */
00316
00317 /* Field masks for read-back validation */
00318 k_doc_docr_oms_mask = (0x03 « k_doc_docr_oms_shift),
00319 k_doc_docr_dcsel_mask = (0x01 « k_doc_docr_dcsel_shift),
00320 k_doc_docr_dopcie_mask = (0x01 « k_doc_docr_dopcie_shift),
00321 } rx_doc_docr_bits_t;
00322
00323 /*
=====
00324 * Static Assertions - Verify Register Layout at Compile Time
00325 *
=====
00326 */
00327

```



```

00328 /* The k_doc_base_addr and k_doc_mstpb_bit enum declarations above are
00329 * themselves the contract (per Hardware Manual p.2969 and p.409). Re-stating
00330 * the same literals in static_asserts would duplicate the magic numbers
00331 * without adding a real cross-check, so we rely on the enum declarations
00332 * alone. */
00333 static_assert(sizeof(rx_doc_regs_t) == 6, "DOC register structure size incorrect");
00334 static_assert(offsetof(rx_doc_regs_t, docr) == 0x00, "DOCR offset incorrect");
00335 static_assert(offsetof(rx_doc_regs_t, dodir) == 0x02, "DODIR offset incorrect");
00336 static_assert(offsetof(rx_doc_regs_t, dodsr) == 0x04, "DODSR offset incorrect");
00337
00338 #ifdef __cplusplus
00339 }
00340 #endif

```

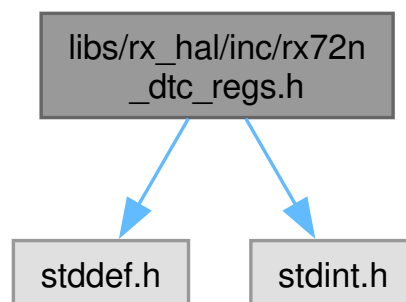
### 8.17 libs/rx/hal/inc/rx72n\_dtc\_regs.h File Reference

RX72N Data Transfer Controller (DTCb) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_dtc\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct `dtc_transfer_info_full_t`  
DTC transfer information structure for full-address mode.
- struct `dtc_transfer_info_short_t`  
DTC transfer information structure for short-address mode.
- union `dtc_transfer_info_full_t.__unnamed0__`
- struct `dtc_transfer_info_full_t.__unnamed0__.__unnamed0__`
- union `dtc_transfer_info_short_t.__unnamed0__`
- struct `dtc_transfer_info_short_t.__unnamed0__.__unnamed0__`

## Enumerations

- enum `dtc_addresses_t` : `uintptr_t` {  
`k_dtc_base_addr` = 0x00082400U , `k_dtc_dtccr_addr` = 0x00082400U , `k_dtc_dtcvbr_addr` = 0x00082404U , `k_dtc_dtcadmod_addr` = 0x00082408U ,  
`k_dtc_dtcst_addr` = 0x0008240CU , `k_dtc_dtcsts_addr` = 0x0008240EU , `k_dtc_dtcibr_addr` = 0x00082410U , `k_dtc_dtcdr_addr` = 0x00082414U ,  
`k_dtc_dtcsqe_addr` = 0x00082416U , `k_dtc_dtcdisp_addr` = 0x00082418U }  
DTC module addresses.
- enum `dtc_offsets_t` : `uint8_t` {  
`k_dtc_offset_dtccr` = 0x00U , `k_dtc_offset_dtcvbr` = 0x04U , `k_dtc_offset_dtcadmod` = 0x08U ,  
`k_dtc_offset_dtcst` = 0x0CU ,  
`k_dtc_offset_dtcsts` = 0x0EU , `k_dtc_offset_dtcibr` = 0x10U , `k_dtc_offset_dtcdr` = 0x14U ,  
`k_dtc_offset_dtcsqe` = 0x16U ,  
`k_dtc_offset_dtcdisp` = 0x18U }  
Register offsets from DTC base address.
- enum `dtccr_bits_t` : `uint8_t` { `k_dtccr_reserved_bit3` = 0x08U , `k_dtccr_rrs` = 0x10U }  
DTC Control Register (DTCCR) bit definitions.
- enum `dtcadmod_bits_t` : `uint8_t` { `k_dtcadmod_short` = 0x01U }  
DTC Address Mode Register (DTCADMOD) bit definitions.
- enum `dtcst_bits_t` : `uint8_t` { `k_dtcst_dtcst` = 0x01U }  
DTC Module Start Register (DTCST) bit definitions.
- enum `dtcsts_bits_t` : `uint16_t` { `k_dtcsts_vecn_mask` = 0x00FFU , `k_dtcsts_act` = 0x8000U }  
DTC Status Register (DTCSTS) bit definitions.
- enum `dtcor_bits_t` : `uint8_t` { `k_dtcdr_sqtfrrl` = 0x01U }  
DTC Operation Register (DTCOR) bit definitions.
- enum `dtcsqe_bits_t` : `uint16_t` { `k_dtcsqe_vecn_mask` = 0x00FFU , `k_dtcsqe_espssel` = 0x8000U }  
DTC Sequence Transfer Enable Register (DTCSQE) bit definitions.
- enum `dtc_mra_bits_t` : `uint8_t` {  
`k_dtc_mra_wbdis` = 0x01U , `k_dtc_mra_sm_fixed` = 0x00U , `k_dtc_mra_sm_increment` = 0x08U , `k_dtc_mra_sm_decrement` = 0x0CU ,  
`k_dtc_mra_sm_mask` = 0x0CU , `k_dtc_mra_sz_byte` = 0x00U , `k_dtc_mra_sz_word` = 0x10U ,  
`k_dtc_mra_sz_long` = 0x20U ,  
`k_dtc_mra_sz_mask` = 0x30U , `k_dtc_mra_md_normal` = 0x00U , `k_dtc_mra_md_repeat` = 0x40U , `k_dtc_mra_md_block` = 0x80U ,  
`k_dtc_mra_md_mask` = 0xC0U }  
DTC Mode Register A (MRA) bit definitions for transfer info.
- enum `dtc_mrb_bits_t` : `uint8_t` {  
`k_dtc_mrb_sqend` = 0x01U , `k_dtc_mrb_indx` = 0x02U , `k_dtc_mrb_dm_fixed` = 0x00U ,  
`k_dtc_mrb_dm_increment` = 0x08U ,  
`k_dtc_mrb_dm_decrement` = 0x0CU , `k_dtc_mrb_dm_mask` = 0x0CU , `k_dtc_mrb_dts` = 0x10U , `k_dtc_mrb_disel` = 0x20U ,  
`k_dtc_mrb_chns` = 0x40U , `k_dtc_mrb_chne` = 0x80U }  
DTC Mode Register B (MRB) bit definitions for transfer info.
- enum `dtc_mstpcr_t` : `uint32_t` { `k_dtc_mstpcra_bit` = (1U << 28U) }  
DTC module stop control bits.

## Functions

- static volatile `uint8_t` \* `dtc_dtccr_reg` (void)  
Get pointer to DTC Control Register (DTCCR).
- static volatile `uint32_t` \* `dtc_dtcvbr_reg` (void)

- Get pointer to DTC Vector Base Register (DTCVBR).
- static volatile uint8\_t \* [dtc\\_dtcadmod\\_reg](#) (void)
  - Get pointer to DTC Address Mode Register (DTCADMOD).
- static volatile uint8\_t \* [dtc\\_dtcst\\_reg](#) (void)
  - Get pointer to DTC Module Start Register (DTCST).
- static volatile uint16\_t \* [dtc\\_dtcsts\\_reg](#) (void)
  - Get pointer to DTC Status Register (DTCSTS).
- static volatile uint32\_t \* [dtc\\_dtcibr\\_reg](#) (void)
  - Get pointer to DTC Index Table Base Register (DTCIBR).
- static volatile uint8\_t \* [dtc\\_dtc\\_cor\\_reg](#) (void)
  - Get pointer to DTC Operation Register (DTCOR).
- static volatile uint16\_t \* [dtc\\_dtc\\_sqe\\_reg](#) (void)
  - Get pointer to DTC Sequence Transfer Enable Register (DTCSQE).
- static volatile uint32\_t \* [dtc\\_dtc\\_disp\\_reg](#) (void)
  - Get pointer to DTC Address Displacement Register (DTCDISP).

### 8.17.1 Detailed Description

RX72N Data Transfer Controller (DTCb) Register Definitions.

Complete register definitions for the RX72N Data Transfer Controller. The DTC performs interrupt-driven data transfers without CPU intervention, supporting normal, repeat, block, and sequence transfer modes.

#### DTC Overview

- Triggered by interrupt requests
- Number of channels = number of interrupt sources that can start DTC
- Transfer modes: normal, repeat, block, chain, sequence
- Transfer space: 16 MB (short) or 4 GB (full address mode)
- Single data sizes: 8/16/32-bit
- Block sizes: 1-256 data units
- Up to 256 sequences per trigger source

#### Register Organization

| Address    | Size | Register | Description                      |
|------------|------|----------|----------------------------------|
| 0x00082400 | 8    | DTCCR    | DTC Control Register             |
| 0x00082404 | 32   | DTCVBR   | DTC Vector Base Register         |
| 0x00082408 | 8    | DTCADMOD | DTC Address Mode Register        |
| 0x0008240C | 8    | DTCST    | DTC Module Start Register        |
| 0x0008240E | 16   | DTCSTS   | DTC Status Register              |
| 0x00082410 | 32   | DTCIBR   | DTC Index Table Base Register    |
| 0x00082414 | 8    | DTCOR    | DTC Operation Register           |
| 0x00082416 | 16   | DTCSQE   | DTC Sequence Transfer Enable Reg |

| Address    | Size | Register | Description                       |
|------------|------|----------|-----------------------------------|
| 0x00082418 | 32   | DTCDISP  | DTC Address Displacement Register |

#### DTC vs DMAC

- DTC: Low overhead, triggered by any interrupt, lightweight transfers
- DMAC: High throughput, dedicated 8 channels, advanced features
- DTC is ideal for small periodic transfers (ADC, UART, SPI)
- DMAC is better for bulk memory-to-memory transfers

#### NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops)
- Rule 3: No dynamic memory allocation
- Rule 4: Short accessor functions
- Rule 5: Static assertions verify addresses
- Rule 6: Minimal scope
- Rule 7: N/A
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

#### Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 20: Data Transfer Controller (DTCb), pages 786-867
- Section 20.2: Register Descriptions

#### Author

Locked, Inc.

#### Date

2026-01-29

#### Copyright

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#### Since

Version 1.0.0

Definition in file [rx72n\\_dtc\\_regs.h](#).

## 8.17.2 Data Structure Documentation

### 8.17.2.1 struct dtc\_transfer\_info\_full\_t ``unnamed0`` ``unnamed0``

Definition at line [451](#) of file [rx72n\\_dtc\\_regs.h](#).

Data Fields

|          |          |                                |
|----------|----------|--------------------------------|
| uint16_t | cra_high | Block size (block mode)        |
| uint16_t | cra_low  | Transfer count (normal/repeat) |

### 8.17.2.2 struct dtc\_transfer\_info\_short\_t ``unnamed0`` ``unnamed0``

Definition at line [483](#) of file [rx72n\\_dtc\\_regs.h](#).

Data Fields

|          |          |  |
|----------|----------|--|
| uint16_t | cra_high |  |
| uint16_t | cra_low  |  |

## 8.17.3 Enumeration Type Documentation

### 8.17.3.1 dtc\_addresses\_t

```
enum dtc_addresses_t : uintptr_t
```

DTC module addresses.

Base address and individual register addresses for DTC control registers. All addresses verified against RX72N Hardware Manual Ch20.

Note

Addresses verified 2026-01-29 against manual section 20.2

Since

Version 1.0.0

Enumerator

|                     |                                    |
|---------------------|------------------------------------|
| k_dtc_base_addr     | DTC base address.                  |
| k_dtc_dtccr_addr    | DTC Control Register address.      |
| k_dtc_dtcvbr_addr   | DTC Vector Base Register address.  |
| k_dtc_dtcadmod_addr | DTC Address Mode Register address. |
| k_dtc_dtcst_addr    | DTC Module Start Register address. |

|                    |                                                |
|--------------------|------------------------------------------------|
| k_dtc_dtcsts_addr  | DTC Status Register address.                   |
| k_dtc_dtcibr_addr  | DTC Index Table Base Register address.         |
| k_dtc_dtcop_addr   | DTC Operation Register address.                |
| k_dtc_dtcseqe_addr | DTC Sequence Transfer Enable Register address. |
| k_dtc_dtcdisp_addr | DTC Address Displacement Register address.     |

Definition at line 87 of file `rx72n_dtc_regs.h`.

```

00087 : uintptr_t {
00088 /** @brief DTC base address */
00089 k_dtc_base_addr = 0x00082400U,
00090
00091 /** @brief DTC Control Register address */
00092 k_dtc_dtccr_addr = 0x00082400U,
00093
00094 /** @brief DTC Vector Base Register address */
00095 k_dtc_dtcvbr_addr = 0x00082404U,
00096
00097 /** @brief DTC Address Mode Register address */
00098 k_dtc_dtcadmod_addr = 0x00082408U,
00099
00100 /** @brief DTC Module Start Register address */
00101 k_dtc_dtcst_addr = 0x0008240CU,
00102
00103 /** @brief DTC Status Register address */
00104 k_dtc_dtcsts_addr = 0x0008240EU,
00105
00106 /** @brief DTC Index Table Base Register address */
00107 k_dtc_dtcibr_addr = 0x00082410U,
00108
00109 /** @brief DTC Operation Register address */
00110 k_dtc_dtcop_addr = 0x00082414U,
00111
00112 /** @brief DTC Sequence Transfer Enable Register address */
00113 k_dtc_dtcseqe_addr = 0x00082416U,
00114
00115 /** @brief DTC Address Displacement Register address */
00116 k_dtc_dtcdisp_addr = 0x00082418U,
00117 } dtc_addresses_t;

```

### 8.17.3.2 dtc`mra`bits`

```
enum dtc_mra_bits_t : uint8_t
```

DTC Mode Register A (MRA) bit definitions for transfer info.

MRA is part of DTC transfer information stored in RAM, not directly accessible as a memory-mapped register.

Register Layout (8-bit in transfer info)

| Bits    | Symbol      | Description                    |
|---------|-------------|--------------------------------|
| 0       | WBDIS       | Write-back Disable             |
| 1       | -           | Reserved (set to 0)            |
| 3↔<br>2 | SM[1↔<br>0] | Source Address Addressing Mode |
| 5↔<br>4 | SZ[1↔<br>0] | Transfer Data Size             |

| Bits     | Symbol       | Description          |
|----------|--------------|----------------------|
| 7:↔<br>6 | MD[1:↔<br>0] | Transfer Mode Select |

Note

Manual ref: Ch20 section 20.2.1

Since

Version 1.0.0

Enumerator

|                        |                                |
|------------------------|--------------------------------|
| k_dtc_mra_wbdis        | Don't write back transfer info |
| k_dtc_mra_sm_fixed     | Source address fixed           |
| k_dtc_mra_sm_increment | Source address increment       |
| k_dtc_mra_sm_decrement | Source address decrement       |
| k_dtc_mra_sm_mask      | SM field mask                  |
| k_dtc_mra_sz_byte      | 8-bit transfer                 |
| k_dtc_mra_sz_word      | 16-bit transfer                |
| k_dtc_mra_sz_long      | 32-bit transfer                |
| k_dtc_mra_sz_mask      | SZ field mask                  |
| k_dtc_mra_md_normal    | Normal transfer mode           |
| k_dtc_mra_md_repeat    | Repeat transfer mode           |
| k_dtc_mra_md_block     | Block transfer mode            |
| k_dtc_mra_md_mask      | MD field mask                  |

Definition at line 370 of file rx72n\_dtc\_regs.h.

```

00370 : uint8_t {
00371 /* WBDIS - Write-back Disable (bit 0) */
00372 k_dtc_mra_wbdis = 0x01U, /**< Don't write back transfer info */
00373
00374 /* SM - Source Address Addressing Mode (bits 3:2) */
00375 k_dtc_mra_sm_fixed = 0x00U, /**< Source address fixed */
00376 k_dtc_mra_sm_increment = 0x08U, /**< Source address increment */
00377 k_dtc_mra_sm_decrement = 0x0CU, /**< Source address decrement */
00378 k_dtc_mra_sm_mask = 0x0CU, /**< SM field mask */
00379
00380 /* SZ - Transfer Data Size (bits 5:4) */
00381 k_dtc_mra_sz_byte = 0x00U, /**< 8-bit transfer */
00382 k_dtc_mra_sz_word = 0x10U, /**< 16-bit transfer */
00383 k_dtc_mra_sz_long = 0x20U, /**< 32-bit transfer */
00384 k_dtc_mra_sz_mask = 0x30U, /**< SZ field mask */
00385
00386 /* MD - Transfer Mode Select (bits 7:6) */
00387 k_dtc_mra_md_normal = 0x00U, /**< Normal transfer mode */
00388 k_dtc_mra_md_repeat = 0x40U, /**< Repeat transfer mode */
00389 k_dtc_mra_md_block = 0x80U, /**< Block transfer mode */
00390 k_dtc_mra_md_mask = 0xC0U, /**< MD field mask */
00391 } dtc_mra_bits_t;

```

### 8.17.3.3 dtc`mrb`bits`*t*

```
enum dtc_mrb_bits_t : uint8_t
```

DTC Mode Register B (MRB) bit definitions for transfer info.

MRB is part of DTC transfer information stored in RAM, not directly accessible as a memory-mapped register.

Register Layout (8-bit in transfer info)

| Bit      | Symbol | Description                         |
|----------|--------|-------------------------------------|
| 0        | SQEND  | Sequence Transfer End               |
| 1        | INDX   | Index Table Reference               |
| 3:↔<br>2 | DM     | Destination Address Addressing Mode |
| 4        | DTS    | Repeat/Block Area Select            |
| 5        | DISEL  | DTC Interrupt Select                |
| 6        | CHNS   | Chain Transfer Select               |
| 7        | CHNE   | Chain Transfer Enable               |

Note

Manual ref: Ch20 section 20.2.2

Since

Version 1.0.0

Enumerator

|                        |                                              |
|------------------------|----------------------------------------------|
| k_dtc_mrb_sqend        | End sequence transfer                        |
| k_dtc_mrb_indx         | Refer to index table                         |
| k_dtc_mrb_dm_fixed     | Destination address fixed                    |
| k_dtc_mrb_dm_increment | Destination address increment                |
| k_dtc_mrb_dm_decrement | Destination address decrement                |
| k_dtc_mrb_dm_mask      | DM field mask                                |
| k_dtc_mrb_dts          | Source is repeat/block area (1=src, 0=dest)  |
| k_dtc_mrb_disel        | Interrupt after each transfer (not just end) |
| k_dtc_mrb_chns         | Chain only when counter becomes 0            |
| k_dtc_mrb_chne         | Chain transfer enable                        |

Definition at line 415 of file [rx72n\\_dtc\\_regs.h](#).

```

00415 : uint8_t {
00416 k_dtc_mrb_sqend = 0x01U, /**< End sequence transfer */
00417 k_dtc_mrb_indx = 0x02U, /**< Refer to index table */
00418
00419 /* DM - Destination Address Addressing Mode (bits 3:2) */
00420 k_dtc_mrb_dm_fixed = 0x00U, /**< Destination address fixed */
00421 k_dtc_mrb_dm_increment = 0x08U, /**< Destination address increment */
00422 k_dtc_mrb_dm_decrement = 0x0CU, /**< Destination address decrement */
00423 k_dtc_mrb_dm_mask = 0x0CU, /**< DM field mask */
00424
00425 k_dtc_mrb_dts = 0x10U, /**< Source is repeat/block area (1=src, 0=dest) */
00426 k_dtc_mrb_disel = 0x20U, /**< Interrupt after each transfer (not just end) */
00427 k_dtc_mrb_chns = 0x40U, /**< Chain only when counter becomes 0 */
00428 k_dtc_mrb_chne = 0x80U, /**< Chain transfer enable */
00429 } dtc_mrb_bits_t;

```



#### 8.17.3.4 dtc\_mstpcr\_t

enum [dtc\\_mstpcr\\_t](#) : uint32\_t

DTC module stop control bits.

The DTC is controlled by MSTPCRA.MSTPA28. Must be cleared (0) before accessing DTC registers.

Note

Manual ref: Ch11 (Low Power Consumption) module stop tables

DTC and DMAC share the same module stop bit (MSTPA28)

Since

Version 1.0.0

Enumerator

|                                   |                                                                                                                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">k_dtc_mstpcra_bit</a> | <div>MSTPCRA bit for DTC (bit 28).</div> <ul style="list-style-type: none"><li>• 0: DTC module is running</li><li>• 1: DTC module is stopped (default)</li></ul> <div>Note</div> <div>Same bit as DMAC (<a href="#">k_dmac_mstpcra_bit</a>)</div> |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 510 of file [rx72n\\_dtc\\_regs.h](#).

```
00510 : uint32_t {
00511 /**
00512 * @brief MSTPCRA bit for DTC (bit 28)
00513 * @details
00514 * - 0: DTC module is running
00515 * - 1: DTC module is stopped (default)
00516 * @note Same bit as DMAC (k_dmac_mstpcra_bit)
00517 */
00518 k_dtc_mstpcra_bit = (1U << 28U),
00519 } dtc_mstpcr_t;
```

#### 8.17.3.5 dtc\_offsets\_t

enum [dtc\\_offsets\\_t](#) : uint8\_t

Register offsets from DTC base address.

Offsets for each register relative to [k\\_dtc\\_base\\_addr](#).

Note

Offsets verified 2026-01-29 against manual

Since

Version 1.0.0

Enumerator

|                       |                         |
|-----------------------|-------------------------|
| k_dtc_offset_dtccr    | DTCCR offset (8-bit)    |
| k_dtc_offset_dtcvbr   | DTCVBR offset (32-bit)  |
| k_dtc_offset_dtcadmod | DTCADMOD offset (8-bit) |
| k_dtc_offset_dtcst    | DTCST offset (8-bit)    |
| k_dtc_offset_dtcsts   | DTCSTS offset (16-bit)  |
| k_dtc_offset_dtcibr   | DTCIBR offset (32-bit)  |
| k_dtc_offset_dtcior   | DTCOR offset (8-bit)    |
| k_dtc_offset_dtcscqe  | DTCSCQE offset (16-bit) |
| k_dtc_offset_dtcdisp  | DTCDISP offset (32-bit) |

Definition at line 129 of file `rx72n_dtc_regs.h`.

```

00129 : uint8_t {
00130 k_dtc_offset_dtccr = 0x00U, /**< DTCCR offset (8-bit) */
00131 k_dtc_offset_dtcvbr = 0x04U, /**< DTCVBR offset (32-bit) */
00132 k_dtc_offset_dtcadmod = 0x08U, /**< DTCADMOD offset (8-bit) */
00133 k_dtc_offset_dtcst = 0x0CU, /**< DTCST offset (8-bit) */
00134 k_dtc_offset_dtcsts = 0x0EU, /**< DTCSTS offset (16-bit) */
00135 k_dtc_offset_dtcibr = 0x10U, /**< DTCIBR offset (32-bit) */
00136 k_dtc_offset_dtcior = 0x14U, /**< DTCOR offset (8-bit) */
00137 k_dtc_offset_dtcscqe = 0x16U, /**< DTCSCQE offset (16-bit) */
00138 k_dtc_offset_dtcdisp = 0x18U, /**< DTCDISP offset (32-bit) */
00139 } dtc_offsets_t;

```

### 8.17.3.6 dtcadmod`bits`

enum `dtcadmod_bits_t` : `uint8_t`

DTC Address Mode Register (DTCADMOD) bit definitions.

Selects short-address or full-address mode for DTC transfers.

Register Layout (8-bit @ 0x00082408)

| Bit      | Symbol | Description                   |
|----------|--------|-------------------------------|
| 0        | SHORT  | Short-Address Mode Set        |
| 7:↵<br>1 | -      | Reserved (read as 0, write 0) |

Note

Manual ref: Ch20 section 20.2.10

Since

Version 1.0.0

Enumerator

|                  |                                                                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dtcadmod_short | <p>Short-Address Mode Set (bit 0).</p> <ul style="list-style-type: none"> <li>0: Full-address mode (4 GB space)</li> <li>1: Short-address mode (16 MB space)</li> </ul> <p>Warning</p> <p>Set to 0 (full-address mode) when using sequence transfer</p> |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 199 of file rx7n\_dtc\_regs.h.

```

00199 : uint8_t {
00200 /**
00201 * @brief Short-Address Mode Set (bit 0)
00202 * @details
00203 * - 0: Full-address mode (4 GB space)
00204 * - 1: Short-address mode (16 MB space)
00205 * @warning Set to 0 (full-address mode) when using sequence transfer
00206 */
00207 k_dtcadmod_short = 0x01U,
00208 } dtcadmod_bits_t;

```

### 8.17.3.7 dtccr\_bits\_t

enum dtccr\_bits\_t : uint8\_t

DTC Control Register (DTCCR) bit definitions.

Controls DTC read skip functionality.

Register Layout (8-bit @ 0x00082400)

| Bit     | Symbol | Description                           |
|---------|--------|---------------------------------------|
| 2↔<br>0 | -      | Reserved (read as 0, write 0)         |
| 3       | -      | Reserved (read as 1, write 1)         |
| 4       | RRS    | Transfer Information Read Skip Enable |
| 7↔<br>5 | -      | Reserved (read as 0, write 0)         |

Note

Manual ref: Ch20 section 20.2.8

Since

Version 1.0.0

Enumerator

|                       |                                                                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dtccr_reserved_bit3 | Reserved bit 3 (must write 1).                                                                                                                                                                                                                                               |
| k_dtccr_rrs           | Transfer Information Read Skip Enable (bit 4). <ul style="list-style-type: none"> <li>• 0: Transfer information read is not skipped</li> <li>• 1: Transfer information read is skipped when vector numbers match</li> </ul> Warning<br>Set to 0 when using sequence transfer |

Definition at line 164 of file [rx72n\\_dtc\\_regs.h](#).

```

00164 : uint8_t {
00165 /** @brief Reserved bit 3 (must write 1) */
00166 k_dtccr_reserved_bit3 = 0x08U,
00167
00168 /**
00169 * @brief Transfer Information Read Skip Enable (bit 4)
00170 * @details
00171 * - 0: Transfer information read is not skipped
00172 * - 1: Transfer information read is skipped when vector numbers match
00173 * @warning Set to 0 when using sequence transfer
00174 */
00175 k_dtccr_rrs = 0x10U,
00176 } dtccr_bits_t;

```

### 8.17.3.8 dtcor`bits`

```
enum dtcor_bits_t : uint8_t
```

DTC Operation Register (DTCOR) bit definitions.

Controls DTC operation, specifically sequence transfer termination.

Register Layout (8-bit @ 0x00082414)

| Bit      | Symbol | Description                   |
|----------|--------|-------------------------------|
| 0        | SQTFRL | Sequence Transfer Terminate   |
| 7:↔<br>1 | -      | Reserved (read as 0, write 0) |

Note

Manual ref: Ch20 section 20.2.14

Since

Version 1.0.0

Enumerator

|                |                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| k_dtcor_sqtfrl | Sequence Transfer Terminate (bit 0, write-only). Writing 1 terminates the sequence transfer in progress. This bit is always read as 0. |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 299 of file [rx72n\\_dtc\\_regs.h](#).

```
00299 : uint8_t {
00300 /**
00301 * @brief Sequence Transfer Terminate (bit 0, write-only)
00302 * @details
00303 * Writing 1 terminates the sequence transfer in progress.
00304 * This bit is always read as 0.
00305 */
00306 k_dtcor_sqtfrl = 0x01U,
00307 } dtcor_bits_t;
```

### 8.17.3.9 dtcsqe\_bits\_t

```
enum dtcsqe_bits_t : uint16_t
```

DTC Sequence Transfer Enable Register (DTCSQE) bit definitions.

Enables sequence transfer and specifies the trigger vector number.

Register Layout (16-bit @ 0x00082416)

| Bits      | Symbol         | Description                             |
|-----------|----------------|-----------------------------------------|
| 7:↵<br>0  | VECN[7:↵<br>0] | Sequence Transfer Vector Number Setting |
| 14:↵<br>8 | -              | Reserved (read as 0, write 0)           |
| 15        | ESPSEL         | Sequence Transfer Enable                |

Note

Manual ref: Ch20 section 20.2.15

Since

Version 1.0.0

Enumerator

|                    |                                      |
|--------------------|--------------------------------------|
| k_dtcsqe_vecn_mask | Vector number field mask (bits 7:0). |
|--------------------|--------------------------------------|

|                 |                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dtcsqe_espsel | Sequence Transfer Enable (bit 15). <ul style="list-style-type: none"> <li>0: Sequence transfer is disabled</li> <li>1: Sequence transfer is enabled</li> </ul> Note<br>Set DTCADMOD.SHORT to 0 when enabling sequence transfer |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 331 of file rx72n\_dtc\_regs.h.

```

00331 : uint16_t {
00332 /** @brief Vector number field mask (bits 7:0) */
00333 k_dtcsqe_vecn_mask = 0x00FFU,
00334
00335 /**
00336 * @brief Sequence Transfer Enable (bit 15)
00337 * @details
00338 * - 0: Sequence transfer is disabled
00339 * - 1: Sequence transfer is enabled
00340 * @note Set DTCADMOD.SHORT to 0 when enabling sequence transfer
00341 */
00342 k_dtcsqe_espsel = 0x8000U,
00343 } dtcsqe_bits_t;

```

#### 8.17.3.10 dtcst`bits`

enum dtcst\_bits\_t : uint8\_t

DTC Module Start Register (DTCST) bit definitions.

Enables or disables the DTC module.

Register Layout (8-bit @ 0x0008240C)

| Bit      | Symbol | Description                   |
|----------|--------|-------------------------------|
| 0        | DTCST  | DTC Module Start              |
| 7:↵<br>1 | -      | Reserved (read as 0, write 0) |

Note

Manual ref: Ch20 section 20.2.11

Since

Version 1.0.0

Enumerator

|               |                                                                                                                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dtcst_dtcst | <p>DTC Module Start (bit 0).</p> <ul style="list-style-type: none"> <li>0: DTC module stop (no transfer requests accepted)</li> <li>1: DTC module start (transfer requests accepted)</li> </ul> <p>Note</p> <p>Set to 0 before entering module stop or standby modes</p> |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 231 of file [rx72n\\_dtc\\_regs.h](#).

```

00231 : uint8_t {
00232 /**
00233 * @brief DTC Module Start (bit 0)
00234 * @details
00235 * - 0: DTC module stop (no transfer requests accepted)
00236 * - 1: DTC module start (transfer requests accepted)
00237 * @note Set to 0 before entering module stop or standby modes
00238 */
00239 k_dtcst_dtcst = 0x01U,
00240 } dtcst_bits_t;

```

#### 8.17.3.11 dtcsts\_bits\_t

enum [dtcsts\\_bits\\_t](#) : uint16\_t

DTC Status Register (DTCSTS) bit definitions.

Read-only register providing DTC transfer status.

Register Layout (16-bit @ 0x0008240E)

| Bits      | Symbol         | Description                      |
|-----------|----------------|----------------------------------|
| 7:↔<br>0  | VECN[7:↔<br>0] | Active Vector Number (read-only) |
| 14:↔<br>8 | -              | Reserved (read as 0)             |
| 15        | ACT            | DTC Active Flag (read-only)      |

Note

Manual ref: Ch20 section 20.2.12

Since

Version 1.0.0

Enumerator

|                    |                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_dtcsts_vecn_mask | Vector number field mask (bits 7:0).                                                                                                                                                                  |
| k_dtcsts_act       | DTC Active Flag (bit 15). <ul style="list-style-type: none"> <li>0: Data transfer is not in progress</li> <li>1: Data transfer is in progress</li> </ul> Note<br>VECN[7:0] is only valid when ACT = 1 |

Definition at line 264 of file [rx72n\\_dtc\\_regs.h](#).

```

00264 : uint16_t {
00265 /** @brief Vector number field mask (bits 7:0) */
00266 k_dtcsts_vecn_mask = 0x00FFU,
00267
00268 /**
00269 * @brief DTC Active Flag (bit 15)
00270 * @details
00271 * - 0: Data transfer is not in progress
00272 * - 1: Data transfer is in progress
00273 * @note VECN[7:0] is only valid when ACT = 1
00274 */
00275 k_dtcsts_act = 0x8000U,
00276 } dtcsts_bits_t;

```

## 8.17.4 Function Documentation

### 8.17.4.1 dtc\_dtcadmod\_reg()

```
volatile uint8_t * dtc_dtcadmod_reg (
 void) [inline], [static]
```

Get pointer to DTC Address Mode Register (DTCADMOD).

Returns

Volatile pointer to DTCADMOD register (8-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 554 of file [rx72n\\_dtc\\_regs.h](#).

```

00555 {
00556 return (volatile uint8_t*)k_dtc_dtcadmod_addr;
00557 }

```

References [k\\_dtc\\_dtcadmod\\_addr](#).



#### 8.17.4.2 dtc\_dtccr\_reg()

```
volatile uint8_t * dtc_dtccr_reg (
 void) [inline], [static]
```

Get pointer to DTC Control Register (DTCCR).

Returns

Volatile pointer to DTCCR register (8-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 532 of file [rx72n\\_dtc\\_regs.h](#).

```
00533 {
00534 return (volatile uint8_t*)k_dtc_dtccr_addr;
00535 }
```

References [k\\_dtc\\_dtccr\\_addr](#).

#### 8.17.4.3 dtc\_dtcdisp\_reg()

```
volatile uint32_t * dtc_dtcdisp_reg (
 void) [inline], [static]
```

Get pointer to DTC Address Displacement Register (DTCDISP).

Returns

Volatile pointer to DTCDISP register (32-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 620 of file [rx72n\\_dtc\\_regs.h](#).

```
00621 {
00622 return (volatile uint32_t*)k_dtc_dtcdisp_addr;
00623 }
```

References [k\\_dtc\\_dtcdisp\\_addr](#).

#### 8.17.4.4 `dtc`dtcibr`reg()`

```
volatile uint32_t * dtc_dtcibr_reg (
 void) [inline], [static]
```

Get pointer to DTC Index Table Base Register (DTCIBR).

Returns

Volatile pointer to DTCIBR register (32-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 587 of file [rx72n\\_dtc\\_regs.h](#).

```
00588 {
00589 return (volatile uint32_t*)k_dtc_dtcibr_addr;
00590 }
```

References [k\\_dtc\\_dtcibr\\_addr](#).

#### 8.17.4.5 `dtc`dtcor`reg()`

```
volatile uint8_t * dtc_dtcor_reg (
 void) [inline], [static]
```

Get pointer to DTC Operation Register (DTCOR).

Returns

Volatile pointer to DTCOR register (8-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 598 of file [rx72n\\_dtc\\_regs.h](#).

```
00599 {
00600 return (volatile uint8_t*)k_dtc_dtcor_addr;
00601 }
```

References [k\\_dtc\\_dtcor\\_addr](#).

#### 8.17.4.6 dtc\_dtcseq\_reg()

```
volatile uint16_t * dtc_dtcseq_reg (
 void) [inline], [static]
```

Get pointer to DTC Sequence Transfer Enable Register (DTCSQE).

Returns

Volatile pointer to DTCSQE register (16-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 609 of file [rx72n\\_dtc\\_regs.h](#).

```
00610 {
00611 return (volatile uint16_t*)k_dtc_dtcseq_addr;
00612 }
```

References [k\\_dtc\\_dtcseq\\_addr](#).

#### 8.17.4.7 dtc\_dtcst\_reg()

```
volatile uint8_t * dtc_dtcst_reg (
 void) [inline], [static]
```

Get pointer to DTC Module Start Register (DTCST).

Returns

Volatile pointer to DTCST register (8-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 565 of file [rx72n\\_dtc\\_regs.h](#).

```
00566 {
00567 return (volatile uint8_t*)k_dtc_dtcst_addr;
00568 }
```

References [k\\_dtc\\_dtcst\\_addr](#).

#### 8.17.4.8 `dtc`dtcsts`reg()`

```
volatile uint16_t * dtc_dtcsts_reg (
 void) [inline], [static]
```

Get pointer to DTC Status Register (DTCSTS).

Returns

Volatile pointer to DTCSTS register (16-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 576 of file [rx72n\\_dtc\\_regs.h](#).

```
00577 {
00578 return (volatile uint16_t*)k_dtc_dtcsts_addr;
00579 }
```

References [k\\_dtc\\_dtcsts\\_addr](#).

#### 8.17.4.9 `dtc`dtcvbr`reg()`

```
volatile uint32_t * dtc_dtcvbr_reg (
 void) [inline], [static]
```

Get pointer to DTC Vector Base Register (DTCVBR).

Returns

Volatile pointer to DTCVBR register (32-bit)

Precondition

DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

Since

Version 1.0.0

Definition at line 543 of file [rx72n\\_dtc\\_regs.h](#).

```
00544 {
00545 return (volatile uint32_t*)k_dtc_dtcvbr_addr;
00546 }
```

References [k\\_dtc\\_dtcvbr\\_addr](#).

## 8.18 rx72n`dtc`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_dtc_regs.h
00003 * @brief RX72N Data Transfer Controller (DTCb) Register Definitions
00004 *
00005 * @details
00006 * Complete register definitions for the RX72N Data Transfer Controller.
00007 * The DTC performs interrupt-driven data transfers without CPU intervention,
00008 * supporting normal, repeat, block, and sequence transfer modes.
00009 *
00010 * @par DTC Overview
00011 * - Triggered by interrupt requests
00012 * - Number of channels = number of interrupt sources that can start DTC
00013 * - Transfer modes: normal, repeat, block, chain, sequence
00014 * - Transfer space: 16 MB (short) or 4 GB (full address mode)
00015 * - Single data sizes: 8/16/32-bit
00016 * - Block sizes: 1-256 data units
00017 * - Up to 256 sequences per trigger source
00018 *
00019 * @par Register Organization
00020 * | Address | Size | Register | Description |
00021 * |-----|-----|-----|-----|
00022 * | 0x00082400 | 8 | DTCCR | DTC Control Register |
00023 * | 0x00082404 | 32 | DTCVBR | DTC Vector Base Register |
00024 * | 0x00082408 | 8 | DTCADMOD | DTC Address Mode Register |
00025 * | 0x0008240C | 8 | DTCST | DTC Module Start Register |
00026 * | 0x0008240E | 16 | DTCSTS | DTC Status Register |
00027 * | 0x00082410 | 32 | DTCIBR | DTC Index Table Base Register |
00028 * | 0x00082414 | 8 | DTCOR | DTC Operation Register |
00029 * | 0x00082416 | 16 | DTCSQE | DTC Sequence Transfer Enable Reg |
00030 * | 0x00082418 | 32 | DTCDISP | DTC Address Displacement Register |
00031 *
00032 * @par DTC vs DMAC
00033 * - DTC: Low overhead, triggered by any interrupt, lightweight transfers
00034 * - DMAC: High throughput, dedicated 8 channels, advanced features
00035 * - DTC is ideal for small periodic transfers (ADC, UART, SPI)
00036 * - DMAC is better for bulk memory-to-memory transfers
00037 *
00038 * @par NASA Power of 10 Compliance
00039 * - Rule 1: No goto, setjmp, or recursion
00040 * - Rule 2: N/A (no loops)
00041 * - Rule 3: No dynamic memory allocation
00042 * - Rule 4: Short accessor functions
00043 * - Rule 5: Static assertions verify addresses
00044 * - Rule 6: Minimal scope
00045 * - Rule 7: N/A
00046 * - Rule 8: All constants use C23 typed enums
00047 * - Rule 9: No function pointers
00048 * - Rule 10: Compiles with -Wall -Wextra -Werror
00049 *
00050 * @par Manual References
00051 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00052 * - Chapter 20: Data Transfer Controller (DTCb), pages 786-867
00053 * - Section 20.2: Register Descriptions
00054 *
00055 * @author Locked, Inc.
00056 * @date 2026-01-29
00057 * @copyright Copyright (c) 2026 Locked Inc.
00058 * SPDX-License-Identifier: MIT
00059 * @since Version 1.0.0
00060 */
00061
00062 #pragma once
00063
00064 #include <stdint.h>
00065 #include <stdint.h>
00066
00067 #ifdef __cplusplus
00068 extern "C" {
00069 #endif
00070
00071 /*
=====
00072 * DTC Base Address and Register Addresses
00073 *
=====
00074 */
00075
00076 /**
00077 * @enum dtc_addresses_t
00078 * @brief DTC module addresses
00079 *
00080 * @details

```

```

00081 * Base address and individual register addresses for DTC control registers.
00082 * All addresses verified against RX72N Hardware Manual Ch20.
00083 *
00084 * @note Addresses verified 2026-01-29 against manual section 20.2
00085 * @since Version 1.0.0
00086 */
00087 typedef enum : uintptr_t {
00088 /** @brief DTC base address */
00089 k_dtc_base_addr = 0x00082400U,
00090
00091 /** @brief DTC Control Register address */
00092 k_dtc_dtccr_addr = 0x00082400U,
00093
00094 /** @brief DTC Vector Base Register address */
00095 k_dtc_dtcvbr_addr = 0x00082404U,
00096
00097 /** @brief DTC Address Mode Register address */
00098 k_dtc_dtcadmod_addr = 0x00082408U,
00099
00100 /** @brief DTC Module Start Register address */
00101 k_dtc_dtctst_addr = 0x0008240CU,
00102
00103 /** @brief DTC Status Register address */
00104 k_dtc_dtcts_addr = 0x0008240EU,
00105
00106 /** @brief DTC Index Table Base Register address */
00107 k_dtc_dtcibr_addr = 0x00082410U,
00108
00109 /** @brief DTC Operation Register address */
00110 k_dtc_dtcor_addr = 0x00082414U,
00111
00112 /** @brief DTC Sequence Transfer Enable Register address */
00113 k_dtc_dtcseq_addr = 0x00082416U,
00114
00115 /** @brief DTC Address Displacement Register address */
00116 k_dtc_dtcdisp_addr = 0x00082418U,
00117 } dtc_addresses_t;
00118
00119 /**
00120 * @enum dtc_offsets_t
00121 * @brief Register offsets from DTC base address
00122 *
00123 * @details
00124 * Offsets for each register relative to k_dtc_base_addr.
00125 *
00126 * @note Offsets verified 2026-01-29 against manual
00127 * @since Version 1.0.0
00128 */
00129 typedef enum : uint8_t {
00130 k_dtc_offset_dtccr = 0x00U, /**< DTCCR offset (8-bit) */
00131 k_dtc_offset_dtcvbr = 0x04U, /**< DTCVBR offset (32-bit) */
00132 k_dtc_offset_dtcadmod = 0x08U, /**< DTCADMOD offset (8-bit) */
00133 k_dtc_offset_dtctst = 0x0CU, /**< DTCTST offset (8-bit) */
00134 k_dtc_offset_dtcts = 0x0EU, /**< DTCTS offset (16-bit) */
00135 k_dtc_offset_dtcibr = 0x10U, /**< DTCIBR offset (32-bit) */
00136 k_dtc_offset_dtcor = 0x14U, /**< DTCOR offset (8-bit) */
00137 k_dtc_offset_dtcseq = 0x16U, /**< DTCSQE offset (16-bit) */
00138 k_dtc_offset_dtcdisp = 0x18U, /**< DTCDISP offset (32-bit) */
00139 } dtc_offsets_t;
00140
00141 /*
=====
00142 * DTCCR Register Bits (Control Register)
00143 *
=====
00144 */
00145
00146 /**
00147 * @enum dtccr_bits_t
00148 * @brief DTC Control Register (DTCCR) bit definitions
00149 *
00150 * @details
00151 * Controls DTC read skip functionality.
00152 *
00153 * @par Register Layout (8-bit @ 0x00082400)
00154 * | Bit | Symbol | Description |
00155 * |-----|-----|-----|
00156 * | 2:0 | - | Reserved (read as 0, write 0) |
00157 * | 3 | - | Reserved (read as 1, write 1) |
00158 * | 4 | RRS | Transfer Information Read Skip Enable |
00159 * | 7:5 | - | Reserved (read as 0, write 0) |
00160 *
00161 * @note Manual ref: Ch20 section 20.2.8
00162 * @since Version 1.0.0
00163 */
00164 typedef enum : uint8_t {
00165 /** @brief Reserved bit 3 (must write 1) */

```

```

00166 k_dtccr_reserved_bit3 = 0x08U,
00167
00168 /**
00169 * @brief Transfer Information Read Skip Enable (bit 4)
00170 * @details
00171 * - 0: Transfer information read is not skipped
00172 * - 1: Transfer information read is skipped when vector numbers match
00173 * @warning Set to 0 when using sequence transfer
00174 */
00175 k_dtccr_rrs = 0x10U,
00176 } dtccr_bits_t;
00177
00178 /*
=====
00179 * DTCADMOD Register Bits (Address Mode)
00180 *
=====
00181 */
00182
00183 /**
00184 * @enum dtcadmod_bits_t
00185 * @brief DTC Address Mode Register (DTCADMOD) bit definitions
00186 * @details
00187 * Selects short-address or full-address mode for DTC transfers.
00188
00189 * @par Register Layout (8-bit @ 0x00082408)
00190 * | Bit | Symbol | Description |
00191 * |-----|-----|-----|
00192 * | 0 | SHORT | Short-Address Mode Set |
00193 * | 7:1 | - | Reserved (read as 0, write 0) |
00194 *
00195 * @note Manual ref: Ch20 section 20.2.10
00196 * @since Version 1.0.0
00197 */
00198
00199 typedef enum : uint8_t {
00200 /**
00201 * @brief Short-Address Mode Set (bit 0)
00202 * @details
00203 * - 0: Full-address mode (4 GB space)
00204 * - 1: Short-address mode (16 MB space)
00205 * @warning Set to 0 (full-address mode) when using sequence transfer
00206 */
00207 k_dtcadmod_short = 0x01U,
00208 } dtcadmod_bits_t;
00209
00210 /*
=====
00211 * DTCST Register Bits (Module Start)
00212 *
=====
00213 */
00214
00215 /**
00216 * @enum dtcst_bits_t
00217 * @brief DTC Module Start Register (DTCST) bit definitions
00218 * @details
00219 * Enables or disables the DTC module.
00220
00221 * @par Register Layout (8-bit @ 0x0008240C)
00222 * | Bit | Symbol | Description |
00223 * |-----|-----|-----|
00224 * | 0 | DTCST | DTC Module Start |
00225 * | 7:1 | - | Reserved (read as 0, write 0) |
00226 *
00227 * @note Manual ref: Ch20 section 20.2.11
00228 * @since Version 1.0.0
00229 */
00230
00231 typedef enum : uint8_t {
00232 /**
00233 * @brief DTC Module Start (bit 0)
00234 * @details
00235 * - 0: DTC module stop (no transfer requests accepted)
00236 * - 1: DTC module start (transfer requests accepted)
00237 * @note Set to 0 before entering module stop or standby modes
00238 */
00239 k_dtcst_dtcst = 0x01U,
00240 } dtcst_bits_t;
00241
00242 /*
=====
00243 * DTCSTS Register Bits (Status)
00244 *
=====
00245 */
00246

```

```

00247 /**
00248 * @enum dtcsts_bits_t
00249 * @brief DTC Status Register (DTCSTS) bit definitions
00250 *
00251 * @details
00252 * Read-only register providing DTC transfer status.
00253 *
00254 * @par Register Layout (16-bit @ 0x0008240E)
00255 * | Bits | Symbol | Description |
00256 * |-----|-----|-----|
00257 * | 7:0 | VECN[7:0] | Active Vector Number (read-only) |
00258 * | 14:8 | - | Reserved (read as 0) |
00259 * | 15 | ACT | DTC Active Flag (read-only) |
00260 *
00261 * @note Manual ref: Ch20 section 20.2.12
00262 * @since Version 1.0.0
00263 */
00264 typedef enum : uint16_t {
00265 /** @brief Vector number field mask (bits 7:0) */
00266 k_dtcsts_vecn_mask = 0x00FFU,
00267
00268 /**
00269 * @brief DTC Active Flag (bit 15)
00270 * @details
00271 * - 0: Data transfer is not in progress
00272 * - 1: Data transfer is in progress
00273 * @note VECN[7:0] is only valid when ACT = 1
00274 */
00275 k_dtcsts_act = 0x8000U,
00276 } dtcsts_bits_t;
00277
00278 /*
=====
00279 * DTCOR Register Bits (Operation)
00280 *
=====
00281 */
00282
00283 /**
00284 * @enum dtcor_bits_t
00285 * @brief DTC Operation Register (DTCOR) bit definitions
00286 *
00287 * @details
00288 * Controls DTC operation, specifically sequence transfer termination.
00289 *
00290 * @par Register Layout (8-bit @ 0x00082414)
00291 * | Bit | Symbol | Description |
00292 * |-----|-----|-----|
00293 * | 0 | SQTFR1 | Sequence Transfer Terminate |
00294 * | 7:1 | - | Reserved (read as 0, write 0) |
00295 *
00296 * @note Manual ref: Ch20 section 20.2.14
00297 * @since Version 1.0.0
00298 */
00299 typedef enum : uint8_t {
00300 /**
00301 * @brief Sequence Transfer Terminate (bit 0, write-only)
00302 * @details
00303 * Writing 1 terminates the sequence transfer in progress.
00304 * This bit is always read as 0.
00305 */
00306 k_dtcor_sqtf1 = 0x01U,
00307 } dtcor_bits_t;
00308
00309 /*
=====
00310 * DTCSQE Register Bits (Sequence Transfer Enable)
00311 *
=====
00312 */
00313
00314 /**
00315 * @enum dtcsqe_bits_t
00316 * @brief DTC Sequence Transfer Enable Register (DTCSQE) bit definitions
00317 *
00318 * @details
00319 * Enables sequence transfer and specifies the trigger vector number.
00320 *
00321 * @par Register Layout (16-bit @ 0x00082416)
00322 * | Bits | Symbol | Description |
00323 * |-----|-----|-----|
00324 * | 7:0 | VECN[7:0] | Sequence Transfer Vector Number Setting |
00325 * | 14:8 | - | Reserved (read as 0, write 0) |
00326 * | 15 | ESPSEL | Sequence Transfer Enable |
00327 *
00328 * @note Manual ref: Ch20 section 20.2.15
00329 * @since Version 1.0.0

```



```

00330 */
00331 typedef enum : uint16_t {
00332 /** @brief Vector number field mask (bits 7:0) */
00333 k_dtcsqe_vecn_mask = 0x00FFU,
00334
00335 /**
00336 * @brief Sequence Transfer Enable (bit 15)
00337 * @details
00338 * - 0: Sequence transfer is disabled
00339 * - 1: Sequence transfer is enabled
00340 * @note Set DTCADMOD.SHORT to 0 when enabling sequence transfer
00341 */
00342 k_dtcsqe_espsel = 0x8000U,
00343 } dtcsqe_bits_t;
00344
00345 /*
=====
00346 * DTC Transfer Information Structure (stored in RAM)
00347 *
=====
00348 */
00349
00350 /**
00351 * @enum dtc_mra_bits_t
00352 * @brief DTC Mode Register A (MRA) bit definitions for transfer info
00353 *
00354 * @details
00355 * MRA is part of DTC transfer information stored in RAM, not directly
00356 * accessible as a memory-mapped register.
00357 *
00358 * @par Register Layout (8-bit in transfer info)
00359 * | Bits | Symbol | Description |
00360 * |-----|-----|-----|
00361 * | 0 | WBDIS | Write-back Disable |
00362 * | 1 | - | Reserved (set to 0) |
00363 * | 3:2 | SM[1:0] | Source Address Addressing Mode |
00364 * | 5:4 | SZ[1:0] | Transfer Data Size |
00365 * | 7:6 | MD[1:0] | Transfer Mode Select |
00366 *
00367 * @note Manual ref: Ch20 section 20.2.1
00368 * @since Version 1.0.0
00369 */
00370 typedef enum : uint8_t {
00371 /* WBDIS - Write-back Disable (bit 0) */
00372 k_dtc_mra_wbdis = 0x01U, /**< Don't write back transfer info */
00373
00374 /* SM - Source Address Addressing Mode (bits 3:2) */
00375 k_dtc_mra_sm_fixed = 0x00U, /**< Source address fixed */
00376 k_dtc_mra_sm_increment = 0x08U, /**< Source address increment */
00377 k_dtc_mra_sm_decrement = 0x0CU, /**< Source address decrement */
00378 k_dtc_mra_sm_mask = 0x0CU, /**< SM field mask */
00379
00380 /* SZ - Transfer Data Size (bits 5:4) */
00381 k_dtc_mra_sz_byte = 0x00U, /**< 8-bit transfer */
00382 k_dtc_mra_sz_word = 0x10U, /**< 16-bit transfer */
00383 k_dtc_mra_sz_long = 0x20U, /**< 32-bit transfer */
00384 k_dtc_mra_sz_mask = 0x30U, /**< SZ field mask */
00385
00386 /* MD - Transfer Mode Select (bits 7:6) */
00387 k_dtc_mra_md_normal = 0x00U, /**< Normal transfer mode */
00388 k_dtc_mra_md_repeat = 0x40U, /**< Repeat transfer mode */
00389 k_dtc_mra_md_block = 0x80U, /**< Block transfer mode */
00390 k_dtc_mra_md_mask = 0xC0U, /**< MD field mask */
00391 } dtc_mra_bits_t;
00392
00393 /**
00394 * @enum dtc_mrb_bits_t
00395 * @brief DTC Mode Register B (MRB) bit definitions for transfer info
00396 *
00397 * @details
00398 * MRB is part of DTC transfer information stored in RAM, not directly
00399 * accessible as a memory-mapped register.
00400 *
00401 * @par Register Layout (8-bit in transfer info)
00402 * | Bit | Symbol | Description |
00403 * |-----|-----|-----|
00404 * | 0 | SQEND | Sequence Transfer End |
00405 * | 1 | INDX | Index Table Reference |
00406 * | 3:2 | DM | Destination Address Addressing Mode |
00407 * | 4 | DTS | Repeat/Block Area Select |
00408 * | 5 | DISEL | DTC Interrupt Select |
00409 * | 6 | CHNS | Chain Transfer Select |
00410 * | 7 | CHNE | Chain Transfer Enable |
00411 *
00412 * @note Manual ref: Ch20 section 20.2.2
00413 * @since Version 1.0.0
00414 */

```

```

00415 typedef enum : uint8_t {
00416 k_dtc_mrb_sqend = 0x01U, /**< End sequence transfer */
00417 k_dtc_mrb_indx = 0x02U, /**< Refer to index table */
00418
00419 /* DM - Destination Address Addressing Mode (bits 3:2) */
00420 k_dtc_mrb_dm_fixed = 0x00U, /**< Destination address fixed */
00421 k_dtc_mrb_dm_increment = 0x08U, /**< Destination address increment */
00422 k_dtc_mrb_dm_decrement = 0x0CU, /**< Destination address decrement */
00423 k_dtc_mrb_dm_mask = 0x0CU, /**< DM field mask */
00424
00425 k_dtc_mrb_dts = 0x10U, /**< Source is repeat/block area (1=src, 0=dest) */
00426 k_dtc_mrb_disel = 0x20U, /**< Interrupt after each transfer (not just end) */
00427 k_dtc_mrb_chns = 0x40U, /**< Chain only when counter becomes 0 */
00428 k_dtc_mrb_chne = 0x80U, /**< Chain transfer enable */
00429 } dtc_mrb_bits_t;
00430
00431 /**
00432 * @struct dtc_transfer_info_full_t
00433 * @brief DTC transfer information structure for full-address mode
00434 *
00435 * @details
00436 * This structure is placed in RAM and read by the DTC hardware.
00437 * The address is calculated from DTCVBR + (vector_number x 4).
00438 *
00439 * @warning Must be aligned to 4-byte boundary
00440 * @note For short-address mode, use dtc_transfer_info_short_t
00441 * @since Version 1.0.0
00442 */
00443 typedef struct __attribute__((packed, aligned(4))) {
00444 uint8_t mra; /**< Mode Register A (transfer mode, size) */
00445 uint8_t mrb; /**< Mode Register B (chain, interrupt) */
00446 uint8_t mrc; /**< Mode Register C (sequence control) */
00447 uint8_t reserved; /**< Reserved (set to 0) */
00448 uint32_t sar; /**< Source Address Register */
00449 uint32_t dar; /**< Destination Address Register */
00450 union {
00451 struct {
00452 uint16_t cra_low; /**< Transfer count (normal/repeat) */
00453 uint16_t cra_high; /**< Block size (block mode) */
00454 };
00455 uint32_t cra; /**< Transfer Count Register A (full) */
00456 };
00457 uint16_t crb; /**< Transfer Count Register B (block count) */
00458 uint16_t reserved2; /**< Reserved */
00459 } dtc_transfer_info_full_t;
00460
00461 /**
00462 * @struct dtc_transfer_info_short_t
00463 * @brief DTC transfer information structure for short-address mode
00464 *
00465 * @details
00466 * Compact format for 16 MB address space transfers.
00467 * Uses 3-byte addresses to save memory.
00468 *
00469 * @warning Must be aligned to 4-byte boundary
00470 * @note For full-address mode, use dtc_transfer_info_full_t
00471 * @since Version 1.0.0
00472 */
00473 typedef struct __attribute__((packed, aligned(4))) {
00474 uint8_t mra; /**< Mode Register A */
00475 uint8_t mrb; /**< Mode Register B */
00476 uint8_t mrc; /**< Mode Register C */
00477 uint8_t reserved; /**< Reserved */
00478 uint8_t sar[3]; /**< Source Address (24-bit) */
00479 uint8_t reserved2; /**< Reserved */
00480 uint8_t dar[3]; /**< Destination Address (24-bit) */
00481 uint8_t reserved3; /**< Reserved */
00482 union {
00483 struct {
00484 uint16_t cra_low;
00485 uint16_t cra_high;
00486 };
00487 uint32_t cra;
00488 };
00489 uint16_t crb;
00490 uint16_t reserved4;
00491 } dtc_transfer_info_short_t;
00492
00493 /**
=====
00494 * Module Stop Control
00495 *
=====
00496 */
00497
00498 /**
00499 * @enum dtc_mstpcr_t

```

```

00500 * @brief DTC module stop control bits
00501 *
00502 * @details
00503 * The DTC is controlled by MSTPCRA.MSTPA28.
00504 * Must be cleared (0) before accessing DTC registers.
00505 *
00506 * @note Manual ref: Ch11 (Low Power Consumption) module stop tables
00507 * @note DTC and DMAC share the same module stop bit (MSTPA28)
00508 * @since Version 1.0.0
00509 */
00510 typedef enum : uint32_t {
00511 /**
00512 * @brief MSTPCRA bit for DTC (bit 28)
00513 * @details
00514 * - 0: DTC module is running
00515 * - 1: DTC module is stopped (default)
00516 * @note Same bit as DMAC (k_dmac_mstpcra_bit)
00517 */
00518 k_dtc_mstpcra_bit = (1U « 28U),
00519 } dtc_mstpcr_t;
00520
00521 /*
=====
00522 * Register Accessor Functions
00523 *
=====
00524 */
00525
00526 /**
00527 * @brief Get pointer to DTC Control Register (DTCCR)
00528 * @return Volatile pointer to DTCCR register (8-bit)
00529 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00530 * @since Version 1.0.0
00531 */
00532 static inline volatile uint8_t* dtc_dtccr_reg(void)
00533 {
00534 return (volatile uint8_t*)k_dtc_dtccr_addr;
00535 }
00536
00537 /**
00538 * @brief Get pointer to DTC Vector Base Register (DTCVBR)
00539 * @return Volatile pointer to DTCVBR register (32-bit)
00540 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00541 * @since Version 1.0.0
00542 */
00543 static inline volatile uint32_t* dtc_dtcvbr_reg(void)
00544 {
00545 return (volatile uint32_t*)k_dtc_dtcvbr_addr;
00546 }
00547
00548 /**
00549 * @brief Get pointer to DTC Address Mode Register (DTCADMOD)
00550 * @return Volatile pointer to DTCADMOD register (8-bit)
00551 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00552 * @since Version 1.0.0
00553 */
00554 static inline volatile uint8_t* dtc_dtcadmod_reg(void)
00555 {
00556 return (volatile uint8_t*)k_dtc_dtcadmod_addr;
00557 }
00558
00559 /**
00560 * @brief Get pointer to DTC Module Start Register (DTCST)
00561 * @return Volatile pointer to DTCST register (8-bit)
00562 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00563 * @since Version 1.0.0
00564 */
00565 static inline volatile uint8_t* dtc_dtcst_reg(void)
00566 {
00567 return (volatile uint8_t*)k_dtc_dtcst_addr;
00568 }
00569
00570 /**
00571 * @brief Get pointer to DTC Status Register (DTCSTS)
00572 * @return Volatile pointer to DTCSTS register (16-bit)
00573 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00574 * @since Version 1.0.0
00575 */
00576 static inline volatile uint16_t* dtc_dtcsts_reg(void)
00577 {
00578 return (volatile uint16_t*)k_dtc_dtcsts_addr;
00579 }
00580
00581 /**
00582 * @brief Get pointer to DTC Index Table Base Register (DTCIBR)
00583 * @return Volatile pointer to DTCIBR register (32-bit)
00584 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared

```

```

00585 * @since Version 1.0.0
00586 */
00587 static inline volatile uint32_t* dtc_dtcibr_reg(void)
00588 {
00589 return (volatile uint32_t*)k_dtc_dtcibr_addr;
00590 }
00591
00592 /**
00593 * @brief Get pointer to DTC Operation Register (DTCOR)
00594 * @return Volatile pointer to DTCOR register (8-bit)
00595 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00596 * @since Version 1.0.0
00597 */
00598 static inline volatile uint8_t* dtc_dtcor_reg(void)
00599 {
00600 return (volatile uint8_t*)k_dtc_dtcor_addr;
00601 }
00602
00603 /**
00604 * @brief Get pointer to DTC Sequence Transfer Enable Register (DTCSQE)
00605 * @return Volatile pointer to DTCSQE register (16-bit)
00606 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00607 * @since Version 1.0.0
00608 */
00609 static inline volatile uint16_t* dtc_dtcsqe_reg(void)
00610 {
00611 return (volatile uint16_t*)k_dtc_dtcsqe_addr;
00612 }
00613
00614 /**
00615 * @brief Get pointer to DTC Address Displacement Register (DTCDISP)
00616 * @return Volatile pointer to DTCDISP register (32-bit)
00617 * @pre DTC module stop bit (MSTPCRA.MSTPA28) must be cleared
00618 * @since Version 1.0.0
00619 */
00620 static inline volatile uint32_t* dtc_dtcdisp_reg(void)
00621 {
00622 return (volatile uint32_t*)k_dtc_dtcdisp_addr;
00623 }
00624
00625 /*
=====
00626 * Static Assertions - Address Verification
00627 *
=====
00628 */
00629
00630 /**
00631 * @name Address Verification
00632 * @brief Compile-time verification of register addresses and offsets
00633 * @{
00634 */
00635
00636 /* Verify base address */
00637 static_assert(k_dtc_base_addr == 0x00082400U,
00638 "DTC base address must be 0x00082400 per Ch20 manual");
00639
00640 /* Verify register addresses */
00641 static_assert(k_dtc_dtccr_addr == 0x00082400U, "DTCCR address must be 0x00082400 per Ch20 manual");
00642
00643 static_assert(k_dtc_dtcvbr_addr == 0x00082404U,
00644 "DTCVBR address must be 0x00082404 per Ch20 manual");
00645
00646 static_assert(k_dtc_dtcadmod_addr == 0x00082408U,
00647 "DTCADMOD address must be 0x00082408 per Ch20 manual");
00648
00649 static_assert(k_dtc_dtcst_addr == 0x0008240CU, "DTCST address must be 0x0008240C per Ch20 manual");
00650
00651 static_assert(k_dtc_dtcsts_addr == 0x0008240EU,
00652 "DTCSTS address must be 0x0008240E per Ch20 manual");
00653
00654 static_assert(k_dtc_dtcibr_addr == 0x00082410U,
00655 "DTCIBR address must be 0x00082410 per Ch20 manual");
00656
00657 static_assert(k_dtc_dtcor_addr == 0x00082414U, "DTCOR address must be 0x00082414 per Ch20 manual");
00658
00659 static_assert(k_dtc_dtcsqe_addr == 0x00082416U,
00660 "DTCSQE address must be 0x00082416 per Ch20 manual");
00661
00662 static_assert(k_dtc_dtcdisp_addr == 0x00082418U,
00663 "DTCDISP address must be 0x00082418 per Ch20 manual");
00664
00665 /* Verify register offsets */
00666 static_assert(k_dtc_offset_dtccr == 0x00U, "DTCCR offset must be 0x00");
00667
00668 static_assert(k_dtc_offset_dtcvbr == 0x04U, "DTCVBR offset must be 0x04");
00669

```

```

00670 static_assert(k_dtc_offset_dtcadmod == 0x08U, "DTCADMOD offset must be 0x08");
00671
00672 static_assert(k_dtc_offset_dtcst == 0x0CU, "DTCST offset must be 0x0C");
00673
00674 static_assert(k_dtc_offset_dtcsts == 0x0EU, "DTCSTS offset must be 0x0E");
00675
00676 static_assert(k_dtc_offset_dtcibr == 0x10U, "DTCIBR offset must be 0x10");
00677
00678 static_assert(k_dtc_offset_dtcdr == 0x14U, "DTCOR offset must be 0x14");
00679
00680 static_assert(k_dtc_offset_dtcscqe == 0x16U, "DTCSCQE offset must be 0x16");
00681
00682 static_assert(k_dtc_offset_dtcdisp == 0x18U, "DTCDISP offset must be 0x18");
00683
00684 /* Verify offset + base = address */
00685 static_assert(k_dtc_base_addr + k_dtc_offset_dtcdr == k_dtc_dtcdr_addr,
00686 "DTCOR: base + offset must equal address");
00687
00688 static_assert(k_dtc_base_addr + k_dtc_offset_dtcvbr == k_dtc_dtcvbr_addr,
00689 "DTCVBR: base + offset must equal address");
00690
00691 static_assert(k_dtc_base_addr + k_dtc_offset_dtcadmod == k_dtc_dtcadmod_addr,
00692 "DTCADMOD: base + offset must equal address");
00693
00694 static_assert(k_dtc_base_addr + k_dtc_offset_dtcst == k_dtc_dtcst_addr,
00695 "DTCST: base + offset must equal address");
00696
00697 static_assert(k_dtc_base_addr + k_dtc_offset_dtcsts == k_dtc_dtcsts_addr,
00698 "DTCSTS: base + offset must equal address");
00699
00700 static_assert(k_dtc_base_addr + k_dtc_offset_dtcibr == k_dtc_dtcibr_addr,
00701 "DTCIBR: base + offset must equal address");
00702
00703 static_assert(k_dtc_base_addr + k_dtc_offset_dtcdr == k_dtc_dtcdr_addr,
00704 "DTCOR: base + offset must equal address");
00705
00706 static_assert(k_dtc_base_addr + k_dtc_offset_dtcscqe == k_dtc_dtcscqe_addr,
00707 "DTCSCQE: base + offset must equal address");
00708
00709 static_assert(k_dtc_base_addr + k_dtc_offset_dtcdisp == k_dtc_dtcdisp_addr,
00710 "DTCDISP: base + offset must equal address");
00711
00712 /* Verify transfer info structure sizes */
00713 static_assert(sizeof(dtc_transfer_info_full_t) == 20,
00714 "Full transfer info structure must be 20 bytes");
00715
00716 /** @} */
00717
00718 #ifdef __cplusplus
00719 }
00720 #endif

```

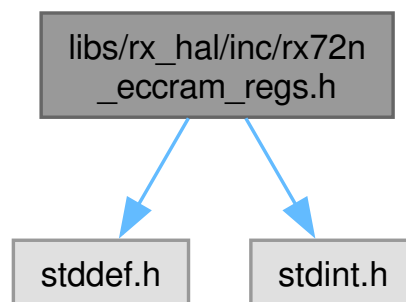
## 8.19 libs/rx\_hal/inc/rx72n\_eccram\_regs.h File Reference

RX72N ECCRAM (ECC-protected RAM) Register Definitions.

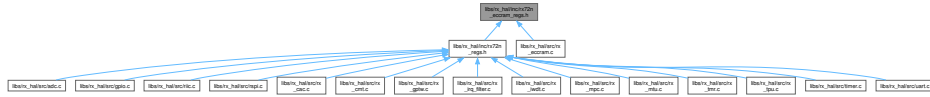
```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_eccram\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct `rx_eccram_regs_t`  
ECCRAM control register block.

## Enumerations

- enum `rx_eccram_region_addr_t` : `uintptr_t` { `k_rx_eccram_region_base_addr` = 0x00FF8000 , `k_rx_eccram_region_end_addr` = 0x00FFFFFF }
- enum `rx_eccram_region_size_t` : `uint32_t` { `k_rx_eccram_region_size_bytes` = 0x00008000 }
- enum `rx_eccram_reg_base_t` : `uintptr_t` { `k_rx_eccram_reg_base_addr` = 0x000812C0 }
- enum `rx_eccram_reg_offset_t` : `uint8_t` {  
`k_rx_eccram_offset_eccrammode` = 0x00 , `k_rx_eccram_offset_eccram2sts` = 0x01 ,  
`k_rx_eccram_offset_eccram1sts` = 0x02 , `k_rx_eccram_offset_eccram1sts` = 0x03 ,  
`k_rx_eccram_offset_eccramprcr` = 0x04 , `k_rx_eccram_offset_eccram2ecad` = 0x08 ,  
`k_rx_eccram_offset_eccram1ecad` = 0x0C , `k_rx_eccram_offset_eccramprcr2` = 0x10 ,  
`k_rx_eccram_offset_eccrametst` = 0x14 }  
 Offsets of ECCRAM control registers from `k_rx_eccram_reg_base_addr`.
- enum `rx_eccrammode_bits_t` : `uint8_t` {  
`k_rx_eccrammode_ecc_disabled` = 0x00 , `k_rx_eccrammode_prohibited` = 0x01 , `k_rx_eccrammode_ecc_no_ch`  
 = 0x02 , `k_rx_eccrammode_ecc_with_check` = 0x03 ,  
`k_rx_eccrammode_rammod_mask` = 0x03 }  
 ECCRAMMODE operating mode select values (RAMMOD[1:0]).
- enum `rx_eccram2sts_bits_t` : `uint8_t` { `k_rx_eccram2sts_ecc2err_shift` = 0 , `k_rx_eccram2sts_ecc2err_mask`  
 = 0x01 , `k_rx_eccram2sts_clear` = 0x00 }  
 ECCRAM2STS 2-bit error flag (ECC2ERR).
- enum `rx_eccram1sts_bits_t` : `uint8_t` { `k_rx_eccram1sts_ecc1sts` = 0 ,  
`k_rx_eccram1sts_ecc1sts_mask` = 0x01 , `k_rx_eccram1sts_disable` = 0x00 , `k_rx_eccram1sts_enable`  
 = 0x01 }  
 ECCRAM1STSEN 1-bit error info update enable (ECC1STSEN).
- enum `rx_eccram1sts_bits_t` : `uint8_t` { `k_rx_eccram1sts_ecc1err_shift` = 0 , `k_rx_eccram1sts_ecc1err_mask`  
 = 0x01 , `k_rx_eccram1sts_clear` = 0x00 }  
 ECCRAM1STS 1-bit error flag (ECC1ERR).
- enum `rx_eccramprcr_bits_t` : `uint8_t` {  
`k_rx_eccramprcr_prcr_shift` = 0 , `k_rx_eccramprcr_prcr_mask` = 0x01 , `k_rx_eccramprcr_key`  
 = 0xF0 , `k_rx_eccramprcr_unlock` = 0xF1 ,  
`k_rx_eccramprcr_lock` = 0xF0 }  
 ECCRAMPRCR protection register values.
- enum `rx_eccramprcr2_bits_t` : `uint8_t` {  
`k_rx_eccramprcr2_prcr2_shift` = 0 , `k_rx_eccramprcr2_prcr2_mask` = 0x01 , `k_rx_eccramprcr2_key`  
 = 0xF0 , `k_rx_eccramprcr2_unlock` = 0xF1 ,  
`k_rx_eccramprcr2_lock` = 0xF0 }  
 ECCRAMPRCR2 protection register 2 values (guards ECCRAMETST).
- enum `rx_eccrametst_bits_t` : `uint8_t` { `k_rx_eccrametst_tstbyp_shift` = 0 , `k_rx_eccrametst_tstbyp_mask`  
 = 0x01 , `k_rx_eccrametst_normal` = 0x00 , `k_rx_eccrametst_bypass` = 0x01 }  
 ECCRAMETST ECC decoder bypass select (TSTBYP) for test mode.
- enum `rx_eccram_mstpcrc_bits_t` : `uint32_t` { `k_rx_eccram_mstpcrc_bit_shift` = 6 ,  
`k_rx_eccram_mstpcrc_bit_mask` = (1U << 6) }  
 MSTPCRC.MSTPC6 bit position for ECCRAM module-stop control.

## Functions

- static volatile [rx\\_eccram\\_regs\\_t](#) \* [eccram\\_regs](#) (void)  
Get pointer to the ECCRAM control register block.

### 8.19.1 Detailed Description

#### RX72N ECCRAM (ECC-protected RAM) Register Definitions.

Register definitions for the 32 KB ECC-protected RAM region on the RX72N. The ECCRAM supports Single-Error Correction / Double-Error Detection (SEC-DED) Hamming-style ECC over every aligned 32-bit word: one detected single-bit error is corrected on read, while any detected double-bit error raises an interrupt so that software can refuse to act on corrupt state (typically by e-stopping the system).

Memory Map (RX72N HW Manual Chapter 60, Table 60.1)

| Region | Start      | End        | Size  | Bus          |
|--------|------------|------------|-------|--------------|
| ECCRAM | 0x00FF8000 | 0x00FFFFFF | 32 KB | Memory Bus 3 |

Control Register Block (Manual Chapter 60, sections 60.2.9-60.2.17)

| Offset | Register     | Size | Description                                                              |
|--------|--------------|------|--------------------------------------------------------------------------|
| 0x00   | ECCRAMMODE   | 8    | ECC mode select<br>(disabled/correct/<br>correct-and-detect/detect-only) |
| 0x01   | ECCRAM2STS   | 8    | 2-bit error status flag                                                  |
| 0x02   | ECCRAM1STSEN | 8    | 1-bit error info update enable                                           |
| 0x03   | ECCRAM1STS   | 8    | 1-bit error status flag                                                  |
| 0x04   | ECCRAMPRCR   | 8    | Protection register (MODE/1STSEN)                                        |
| 0x08   | ECCRAM2ECAD  | 32   | 2-bit error address capture                                              |
| 0x0C   | ECCRAM1ECAD  | 32   | 1-bit error address capture                                              |
| 0x10   | ECCRAMPRCR2  | 8    | Protection register 2 (ETST)                                             |
| 0x14   | ECCRAMETST   | 8    | Test control (ECC decoder bypass)                                        |

Initialization Sequence (Manual section 60.3.3)

Before enabling error checking, every 32-bit word of ECCRAM must be written so that the ECC syndrome matches the data. Uninitialized ECC bits are random and would falsely flag errors on first read. The driver zeroes the entire region using 32-bit stores while the mode is set to `k_rx_eccrammode_ecc_no_check`, then switches to the final mode.

## NASA Power of 10 Compliance

- Rule 5: Static assertions verify all register offsets and base addresses
- Rule 8: All constants use C23 typed enums, no macros for numeric values
- Rule 9: Inline accessor (single-level pointer return), no function pointers

## Terminology

ECC = Error Correcting Code SEC-DED = Single Error Correction, Double Error Detection PRCR = Protection Register

## Author

Locked, Inc.

## Date

2026-04-21

## Copyright

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## Since

Version 1.0.0

Definition in file [rx72n\\_eccram\\_regs.h](#).

## 8.19.2 Data Structure Documentation

### 8.19.2.1 struct rx\_eccram\_regs\_t

ECCRAM control register block.

Memory-mapped layout of the ECCRAM control registers starting at `k_rx_eccram_reg_base_addr` (0x000812C0). Reserved gaps preserve the hardware layout exactly as described in the manual.

## Invariant

`sizeof(rx_eccram_regs_t) == 24` bytes (21 documented + 3 tail padding)

All field offsets match RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17

## See also

[eccram\\_regs\(\)](#) Accessor that returns a pointer to this block at its hardware address

RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17, pages 2981-2990

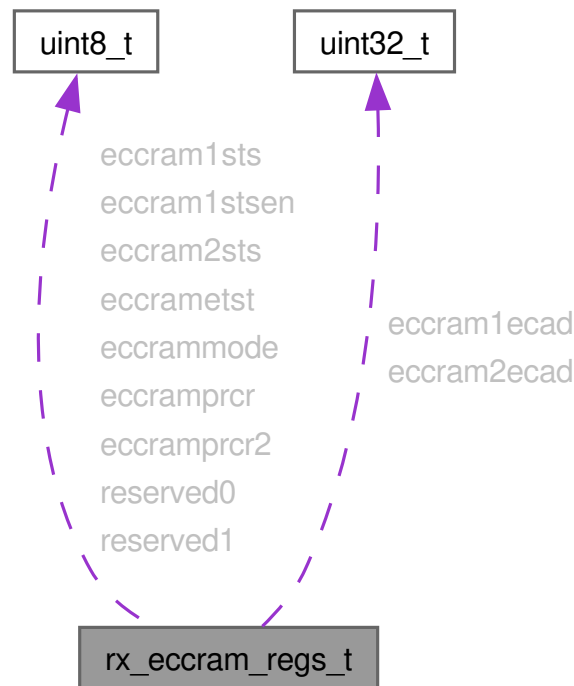


Since

Version 1.0.0

Definition at line 389 of file [rx72n\\_eccram\\_regs.h](#).

Collaboration diagram for rx\_eccram\_regs\_t:



#### Data Fields

|                   |              |                                                                                    |
|-------------------|--------------|------------------------------------------------------------------------------------|
| volatile uint32_t | eccram1ecad  | ECCRAM1ECAD @ 0x0C - 1-bit error address (Ch60, section 60.2.15, page 2987).       |
| volatile uint8_t  | eccram1sts   | ECCRAM1STS @ 0x03 - 1-bit error status flag (Ch60, section 60.2.12, page 2984).    |
| volatile uint8_t  | eccram1stsen | ECCRAM1STSEN @ 0x02 - 1-bit info update enable (Ch60, section 60.2.11, page 2983). |
| volatile uint32_t | eccram2ecad  | ECCRAM2ECAD @ 0x08 - 2-bit error address (Ch60, section 60.2.14, page 2986).       |
| volatile uint8_t  | eccram2sts   | ECCRAM2STS @ 0x01 - 2-bit error status flag (Ch60, section 60.2.10, page 2982).    |
| volatile uint8_t  | eccrametst   | ECCRAMETST @ 0x14 - Test control (Ch60, section 60.2.17, page 2990).               |
| volatile uint8_t  | eccrammode   | ECCRAMMODE @ 0x00 - ECC operating mode select (Ch60, section 60.2.9, page 2981).   |
| volatile uint8_t  | eccramprcr   | ECCRAMPRCR @ 0x04 - Protection register (Ch60, section 60.2.13, page 2985).        |
| volatile uint8_t  | eccramprcr2  | ECCRAMPRCR2 @ 0x10 - Protection register 2 (Ch60, section 60.2.16, page 2989).     |

|         |                   |                       |
|---------|-------------------|-----------------------|
| uint8_t | reserved0↵<br>[3] | Reserved @ 0x05-0x07. |
| uint8_t | reserved1↵<br>[3] | Reserved @ 0x11-0x13. |

### 8.19.3 Enumeration Type Documentation

#### 8.19.3.1 rx'eccram1sts'bits't

```
enum rx_eccram1sts_bits_t : uint8_t
```

ECCRAM1STS 1-bit error flag (ECC1ERR).

Set to 1 by hardware when a correctable 1-bit error is detected. Cleared by writing 0 to bit 0. While this flag is set, ECCRAM1ECAD holds the failing address.

See also

RX72N HW Manual Chapter 60, section 60.2.12, page 2984

Since

Version 1.0.0

Enumerator

|                               |                                            |
|-------------------------------|--------------------------------------------|
| k_rx_eccram1sts_ecc1err_shift | ECC1ERR bit position.                      |
| k_rx_eccram1sts_ecc1err_mask  | ECC1ERR mask.                              |
| k_rx_eccram1sts_clear         | Value to clear ECC1ERR (write 0 to bit 0). |

Definition at line 253 of file [rx72n\\_eccram\\_regs.h](#).

```
00253 : uint8_t {
00254 /** @brief ECC1ERR bit position */
00255 k_rx_eccram1sts_ecc1err_shift = 0,
00256 /** @brief ECC1ERR mask */
00257 k_rx_eccram1sts_ecc1err_mask = 0x01,
00258 /** @brief Value to clear ECC1ERR (write 0 to bit 0) */
00259 k_rx_eccram1sts_clear = 0x00,
00260 } rx_eccram1sts_bits_t;
```

#### 8.19.3.2 rx'eccram1stsen'bits't

```
enum rx_eccram1stsen_bits_t : uint8_t
```

ECCRAM1STSEN 1-bit error info update enable (ECC1STSEN).

See also

RX72N HW Manual Chapter 60, section 60.2.11, page 2983

Since

Version 1.0.0

Enumerator

|                                   |                                     |
|-----------------------------------|-------------------------------------|
| k_rx_eccram1stsen_ecc1stsen_shift | ECC1STSEN bit position.             |
| k_rx_eccram1stsen_ecc1stsen_mask  | ECC1STSEN mask.                     |
| k_rx_eccram1stsen_disable         | 1-bit error status updates disabled |
| k_rx_eccram1stsen_enable          | 1-bit error status updates enabled  |

Definition at line 225 of file [rx72n\\_eccram\\_regs.h](#).

```

00225 : uint8_t {
00226 /** @brief ECC1STSEN bit position */
00227 k_rx_eccram1stsen_ecc1stsen_shift = 0,
00228 /** @brief ECC1STSEN mask */
00229 k_rx_eccram1stsen_ecc1stsen_mask = 0x01,
00230 /** @brief 1-bit error status updates disabled */
00231 k_rx_eccram1stsen_disable = 0x00,
00232 /** @brief 1-bit error status updates enabled */
00233 k_rx_eccram1stsen_enable = 0x01,
00234 } rx_eccram1stsen_bits_t;

```

### 8.19.3.3 rx\_eccram2sts\_bits\_t

```
enum rx_eccram2sts_bits_t : uint8_t
```

ECCRAM2STS 2-bit error flag (ECC2ERR).

Set to 1 by hardware when a 2-bit error is detected on an ECCRAM read. Cleared by writing 0 to bit 0. While this flag is set, ECCRAM2ECAD holds the failing address.

See also

RX72N HW Manual Chapter 60, section 60.2.10, page 2982

Since

Version 1.0.0

Enumerator

|                               |                                            |
|-------------------------------|--------------------------------------------|
| k_rx_eccram2sts_ecc2err_shift | ECC2ERR bit position.                      |
| k_rx_eccram2sts_ecc2err_mask  | ECC2ERR mask.                              |
| k_rx_eccram2sts_clear         | Value to clear ECC2ERR (write 0 to bit 0). |

Definition at line 204 of file [rx72n\\_eccram\\_regs.h](#).

```

00204 : uint8_t {
00205 /** @brief ECC2ERR bit position */
00206 k_rx_eccram2sts_ecc2err_shift = 0,
00207 /** @brief ECC2ERR mask */
00208 k_rx_eccram2sts_ecc2err_mask = 0x01,
00209 /** @brief Value to clear ECC2ERR (write 0 to bit 0) */
00210 k_rx_eccram2sts_clear = 0x00,
00211 } rx_eccram2sts_bits_t;

```

#### 8.19.3.4 rx'eccram'mstpcrc'bits't

```
enum rx_eccram_mstpcrc_bits_t : uint32_t
```

MSTPCRC.MSTPC6 bit position for ECCRAM module-stop control.

Setting MSTPC6 = 1 stops the clock supply to the ECCRAM control logic and the region becomes inaccessible. It must be cleared (bit = 0) before any ECCRAM access – including the zero-initialization pass. The write requires the main system PRCR to be in the "PRC1 unlocked" state (see k\_rx\_prcr←\_unlock\_prc1 in rx\_register\_protection.h).

See also

RX72N HW Manual Chapter 60, section 60.4.1 (MSTPCRC), page 2994

RX72N HW Manual Chapter 11 (Low Power Consumption)

Since

Version 1.0.0

Enumerator

|                               |                                                |
|-------------------------------|------------------------------------------------|
| k_rx_eccram_mstpcrc_bit_shift | Bit position of ECCRAM module-stop in MSTPCRC. |
| k_rx_eccram_mstpcrc_bit_mask  | Bit mask for MSTPCRC.MSTPC6.                   |

Definition at line 361 of file rx72n\_eccram\_regs.h.

```
00361 : uint32_t {
00362 /** @brief Bit position of ECCRAM module-stop in MSTPCRC */
00363 k_rx_eccram_mstpcrc_bit_shift = 6,
00364 /** @brief Bit mask for MSTPCRC.MSTPC6 */
00365 k_rx_eccram_mstpcrc_bit_mask = (1U « 6),
00366 } rx_eccram_mstpcrc_bits_t;
```

#### 8.19.3.5 rx'eccram'reg'base't

```
enum rx_eccram_reg_base_t : uintptr_t
```

ECCRAM control register block base address.

See also

RX72N HW Manual Chapter 60, section 60.2 (register list), page 2979

Since

Version 1.0.0

Enumerator

|                           |                                     |
|---------------------------|-------------------------------------|
| k_rx_eccram_reg_base_addr | ECCRAM control register block base. |
|---------------------------|-------------------------------------|

Definition at line 119 of file rx72n\_eccram\_regs.h.

```
00119 : uintptr_t {
00120 /** @brief ECCRAM control register block base */
00121 k_rx_eccram_reg_base_addr = 0x000812C0,
00122 } rx_eccram_reg_base_t;
```

## 8.19.3.6 rx\_eccram\_reg\_offset\_t

```
enum rx_eccram_reg_offset_t : uint8_t
```

Offsets of ECCRAM control registers from k\_rx\_eccram\_reg\_base\_addr.

See also

RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17, pages 2981-2990

Since

Version 1.0.0

Enumerator

|                                 |                                                               |
|---------------------------------|---------------------------------------------------------------|
| k_rx_eccram_offset_eccrammode   | ECCRAMMODE @ 0x000812C0 (Ch60, section 60.2.9, page 2981).    |
| k_rx_eccram_offset_eccram2sts   | ECCRAM2STS @ 0x000812C1 (Ch60, section 60.2.10, page 2982).   |
| k_rx_eccram_offset_eccram1stsen | ECCRAM1STSEN @ 0x000812C2 (Ch60, section 60.2.11, page 2983). |
| k_rx_eccram_offset_eccram1sts   | ECCRAM1STS @ 0x000812C3 (Ch60, section 60.2.12, page 2984).   |
| k_rx_eccram_offset_eccramprcr   | ECCRAMPRCR @ 0x000812C4 (Ch60, section 60.2.13, page 2985).   |
| k_rx_eccram_offset_eccram2ecad  | ECCRAM2ECAD @ 0x000812C8 (Ch60, section 60.2.14, page 2986).  |
| k_rx_eccram_offset_eccram1ecad  | ECCRAM1ECAD @ 0x000812CC (Ch60, section 60.2.15, page 2987).  |
| k_rx_eccram_offset_eccramprcr2  | ECCRAMPRCR2 @ 0x000812D0 (Ch60, section 60.2.16, page 2989).  |
| k_rx_eccram_offset_eccrametst   | ECCRAMETST @ 0x000812D4 (Ch60, section 60.2.17, page 2990).   |

Definition at line 136 of file rx72n\_eccram\_regs.h.

```
00136 : uint8_t {
00137 /** @brief ECCRAMMODE @ 0x000812C0 (Ch60, section 60.2.9, page 2981) */
00138 k_rx_eccram_offset_eccrammode = 0x00,
00139 /** @brief ECCRAM2STS @ 0x000812C1 (Ch60, section 60.2.10, page 2982) */
00140 k_rx_eccram_offset_eccram2sts = 0x01,
00141 /** @brief ECCRAM1STSEN @ 0x000812C2 (Ch60, section 60.2.11, page 2983) */
00142 k_rx_eccram_offset_eccram1stsen = 0x02,
00143 /** @brief ECCRAM1STS @ 0x000812C3 (Ch60, section 60.2.12, page 2984) */
00144 k_rx_eccram_offset_eccram1sts = 0x03,
00145 /** @brief ECCRAMPRCR @ 0x000812C4 (Ch60, section 60.2.13, page 2985) */
00146 k_rx_eccram_offset_eccramprcr = 0x04,
00147 /** @brief ECCRAM2ECAD @ 0x000812C8 (Ch60, section 60.2.14, page 2986) */
00148 k_rx_eccram_offset_eccram2ecad = 0x08,
00149 /** @brief ECCRAM1ECAD @ 0x000812CC (Ch60, section 60.2.15, page 2987) */
00150 k_rx_eccram_offset_eccram1ecad = 0x0C,
00151 /** @brief ECCRAMPRCR2 @ 0x000812D0 (Ch60, section 60.2.16, page 2989) */
00152 k_rx_eccram_offset_eccramprcr2 = 0x10,
00153 /** @brief ECCRAMETST @ 0x000812D4 (Ch60, section 60.2.17, page 2990) */
00154 k_rx_eccram_offset_eccrametst = 0x14,
00155 } rx_eccram_reg_offset_t;
```

### 8.19.3.7 rx'eccram'region'addr't

enum `rx_eccram_region_addr_t` : `uintptr_t`

ECCRAM data region start, end, and size.

Physical address range of the ECC-protected RAM region. These are the addresses that application code reads and writes – the control registers live at a separate base (see `rx_eccram_reg_addr_t`).

See also

RX72N HW Manual Chapter 60, Table 60.1 (RAM memory map), page 2977

Since

Version 1.0.0

Enumerator

|                                           |                                          |
|-------------------------------------------|------------------------------------------|
| <code>k_rx_eccram_region_base_addr</code> | ECCRAM region start address (inclusive). |
| <code>k_rx_eccram_region_end_addr</code>  | ECCRAM region end address (inclusive).   |

Definition at line 82 of file `rx72n_eccram_regs.h`.

```
00082 : uintptr_t {
00083 /** @brief ECCRAM region start address (inclusive) */
00084 k_rx_eccram_region_base_addr = 0x00FF8000,
00085 /** @brief ECCRAM region end address (inclusive) */
00086 k_rx_eccram_region_end_addr = 0x00FFFFFF,
00087 } rx_eccram_region_addr_t;
```

### 8.19.3.8 rx'eccram'region'size't

enum `rx_eccram_region_size_t` : `uint32_t`

ECCRAM region size in bytes.

Total usable bytes in the ECC-protected region. The driver's zero-initialization loop iterates over this many bytes in 32-bit steps so that every ECC syndrome matches the data before error checking is enabled.

See also

RX72N HW Manual Chapter 60, Table 60.1, page 2977

Since

Version 1.0.0

Enumerator

|                                            |                               |
|--------------------------------------------|-------------------------------|
| <code>k_rx_eccram_region_size_bytes</code> | ECCRAM size in bytes (32 KB). |
|--------------------------------------------|-------------------------------|

Definition at line 102 of file `rx72n_eccram_regs.h`.

```
00102 : uint32_t {
00103 /** @brief ECCRAM size in bytes (32 KB) */
00104 k_rx_eccram_region_size_bytes = 0x00008000,
00105 } rx_eccram_region_size_t;
```

## 8.19.3.9 rx'eccrametst'bits't

```
enum rx_eccrametst_bits_t : uint8_t
```

ECCRAMETST ECC decoder bypass select (TSTBYP) for test mode.

See also

RX72N HW Manual Chapter 60, section 60.2.17, page 2990

Since

Version 1.0.0

Enumerator

|                              |                                             |
|------------------------------|---------------------------------------------|
| k_rx_eccrametst_tstbyp_shift | TSTBYP bit position.                        |
| k_rx_eccrametst_tstbyp_mask  | TSTBYP mask.                                |
| k_rx_eccrametst_normal       | Normal operation (ECC decoder active).      |
| k_rx_eccrametst_bypass       | Bypass ECC decoder (for HW self-test only). |

Definition at line 330 of file [rx72n\\_eccram\\_regs.h](#).

```
00330 : uint8_t {
00331 /** @brief TSTBYP bit position */
00332 k_rx_eccrametst_tstbyp_shift = 0,
00333 /** @brief TSTBYP mask */
00334 k_rx_eccrametst_tstbyp_mask = 0x01,
00335 /** @brief Normal operation (ECC decoder active) */
00336 k_rx_eccrametst_normal = 0x00,
00337 /** @brief Bypass ECC decoder (for HW self-test only) */
00338 k_rx_eccrametst_bypass = 0x01,
00339 } rx_eccrametst_bits_t;
```

## 8.19.3.10 rx'eccrammode'bits't

```
enum rx_eccrammode_bits_t : uint8_t
```

ECCRAMMODE operating mode select values (RAMMOD[1:0]).

Selects whether ECC generation and checking are active on ECCRAM accesses. The recommended mode for production is k\_rx\_eccrammode\_ecc\_with\_check (correct-and-detect).

See also

RX72N HW Manual Chapter 60, section 60.2.9, page 2981

Since

Version 1.0.0

Enumerator

|                                |                                                                              |
|--------------------------------|------------------------------------------------------------------------------|
| k_rx_eccrammode_ecc_disabled   | ECC disabled (RAMMOD[1:0] = 00b).                                            |
| k_rx_eccrammode_prohibited     | Setting prohibited (RAMMOD[1:0] = 01b) – do not use.                         |
| k_rx_eccrammode_ecc_no_check   | ECC generated, no error checking (RAMMOD[1:0] = 10b).                        |
| k_rx_eccrammode_ecc_with_check | ECC generated with 1-bit correction and 2-bit detection (RAMMOD[1:0] = 11b). |
| k_rx_eccrammode_rammod_mask    | RAMMOD[1:0] field mask.                                                      |

Definition at line 174 of file [rx72n\\_eccram\\_regs.h](#).

```

00174 : uint8_t {
00175 /** @brief ECC disabled (RAMMOD[1:0] = 00b) */
00176 k_rx_eccrammode_ecc_disabled = 0x00,
00177 /** @brief Setting prohibited (RAMMOD[1:0] = 01b) -- do not use */
00178 k_rx_eccrammode_prohibited = 0x01,
00179 /** @brief ECC generated, no error checking (RAMMOD[1:0] = 10b) */
00180 k_rx_eccrammode_ecc_no_check = 0x02,
00181 /** @brief ECC generated with 1-bit correction and 2-bit detection (RAMMOD[1:0] = 11b) */
00182 k_rx_eccrammode_ecc_with_check = 0x03,
00183 /** @brief RAMMOD[1:0] field mask */
00184 k_rx_eccrammode_rammod_mask = 0x03,
00185 } rx_eccrammode_bits_t;

```

#### 8.19.3.11 rx\_eccramprcr2\_bits\_t

enum [rx\\_eccramprcr2\\_bits\\_t](#) : uint8\_t

ECCRAMPRCR2 protection register 2 values (guards ECCRAMETST).

See also

RX72N HW Manual Chapter 60, section 60.2.16, page 2989

Since

Version 1.0.0

Enumerator

|                              |                                            |
|------------------------------|--------------------------------------------|
| k_rx_eccramprcr2_prcr2_shift | PRCR2 bit position.                        |
| k_rx_eccramprcr2_prcr2_mask  | PRCR2 mask.                                |
| k_rx_eccramprcr2_key         | KW2[6:0] = 1111000b in bits [7:1].         |
| k_rx_eccramprcr2_unlock      | Full value to unlock writes to ECCRAMETST. |
| k_rx_eccramprcr2_lock        | Full value to re-lock.                     |

Definition at line 305 of file [rx72n\\_eccram\\_regs.h](#).

```

00305 : uint8_t {
00306 /** @brief PRCR2 bit position */
00307 k_rx_eccramprcr2_prcr2_shift = 0,
00308 /** @brief PRCR2 mask */
00309 k_rx_eccramprcr2_prcr2_mask = 0x01,
00310 /** @brief KW2[6:0] = 1111000b in bits [7:1] */
00311 k_rx_eccramprcr2_key = 0xF0,
00312 /** @brief Full value to unlock writes to ECCRAMETST */
00313 k_rx_eccramprcr2_unlock = 0xF1,
00314 /** @brief Full value to re-lock */
00315 k_rx_eccramprcr2_lock = 0xF0,
00316 } rx_eccramprcr2_bits_t;

```



8.19.3.12 rx`eccramprcr`bits``t`

```
enum rx_eccramprcr_bits_t : uint8_t
```

ECCRAMPRCR protection register values.

Write-enables ECCRAMMODE and ECCRAM1STSEN. Bits [7:1] are the key KW[6:0] = 1111000b; bit [0] is PRCR. Write 0xF1 to unlock (key + PRCR=1), 0xF0 to lock (key + PRCR=0). Writes with any other key pattern are ignored.

See also

RX72N HW Manual Chapter 60, section 60.2.13, page 2985

Since

Version 1.0.0

Enumerator

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| k_rx_eccramprcr_prcr_shift | PRCR bit position.                                        |
| k_rx_eccramprcr_prcr_mask  | PRCR mask.                                                |
| k_rx_eccramprcr_key        | KW[6:0] = 1111000b in bits [7:1] (mandatory key).         |
| k_rx_eccramprcr_unlock     | Full value to unlock writes to ECCRAMMODE / ECCRAM1STSEN. |
| k_rx_eccramprcr_lock       | Full value to re-lock writes.                             |

Definition at line 280 of file rx72n\_eccram\_regs.h.

```
00280 : uint8_t {
00281 /** @brief PRCR bit position */
00282 k_rx_eccramprcr_prcr_shift = 0,
00283 /** @brief PRCR mask */
00284 k_rx_eccramprcr_prcr_mask = 0x01,
00285 /** @brief KW[6:0] = 1111000b in bits [7:1] (mandatory key) */
00286 k_rx_eccramprcr_key = 0xF0,
00287 /** @brief Full value to unlock writes to ECCRAMMODE / ECCRAM1STSEN */
00288 k_rx_eccramprcr_unlock = 0xF1,
00289 /** @brief Full value to re-lock writes */
00290 k_rx_eccramprcr_lock = 0xF0,
00291 } rx_eccramprcr_bits_t;
```

## 8.19.4 Function Documentation

## 8.19.4.1 eccram`regs`()

```
volatile rx_eccram_regs_t * eccram_regs (
 void) [inline], [static]
```

Get pointer to the ECCRAM control register block.

Returns a volatile pointer to the ECCRAM control registers at their fixed hardware address. This is the sole, NASA-Rule-8-compliant way to reach the registers – no macros are used for addresses.

## Returns

Volatile pointer to [rx\\_eccram\\_regs\\_t](#)

## Return values

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| Non-NULL | Always returns a valid pointer to <a href="#">k_rx_eccram_reg_base_addr</a> |
|----------|-----------------------------------------------------------------------------|

## Precondition

None (hardware register address is always mapped)

## Postcondition

Pointer is valid for the lifetime of program execution

## Note

Thread Safety: Safe. Returns a constant hardware address with no shared state. Concurrent register reads/writes through the pointer are the caller's concern.

## Example:

```
eccram_regs()->eccramprcr = k_rx_eccramprcr_unlock;
eccram_regs()->eccrammode = k_rx_eccrammode_ecc_with_check;
eccram_regs()->eccramprcr = k_rx_eccramprcr_lock;
```

## See also

[k\\_rx\\_eccram\\_reg\\_base\\_addr](#)

## Since

Version 1.0.0

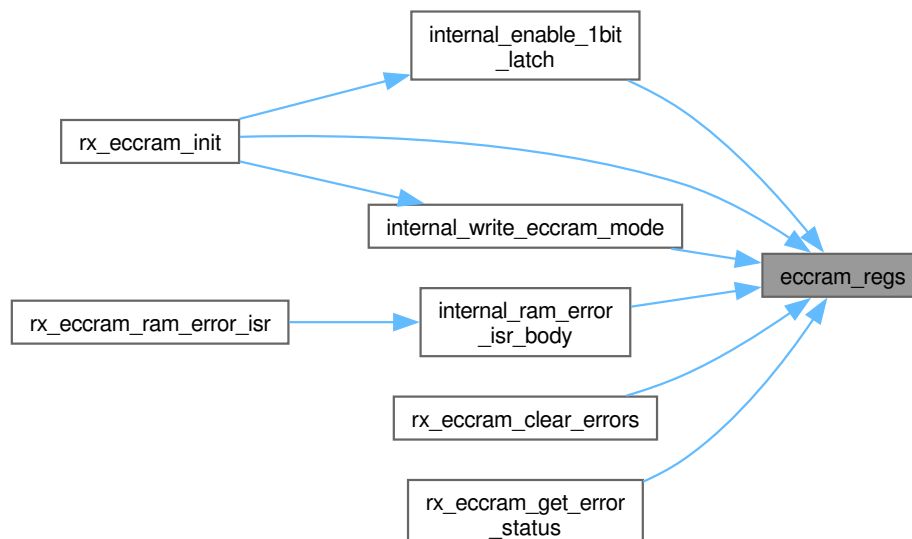
Definition at line 447 of file [rx72n\\_eccram\\_regs.h](#).

```
00448 {
00449 return (volatile rx_eccram_regs_t*)k_rx_eccram_reg_base_addr;
00450 }
```

References [k\\_rx\\_eccram\\_reg\\_base\\_addr](#).

Referenced by [internal\\_enable\\_1bit\\_latch\(\)](#), [internal\\_ram\\_error\\_isr\\_body\(\)](#), [internal\\_write\\_eccram\\_mode\(\)](#), [rx\\_eccram\\_clear\\_errors\(\)](#), [rx\\_eccram\\_get\\_error\\_status\(\)](#), and [rx\\_eccram\\_init\(\)](#).

Here is the caller graph for this function:



## 8.20 rx72n\_eccram\_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_eccram_regs.h
00003 * @brief RX72N ECCRAM (ECC-protected RAM) Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the 32 KB ECC-protected RAM region on the RX72N.
00007 * The ECCRAM supports Single-Error Correction / Double-Error Detection
00008 * (SEC-DED) Hamming-style ECC over every aligned 32-bit word: one detected
00009 * single-bit error is corrected on read, while any detected double-bit
00010 * error raises an interrupt so that software can refuse to act on corrupt
00011 * state (typically by e-stopping the system).
00012 *
00013 * @par Memory Map (RX72N HW Manual Chapter 60, Table 60.1)
00014 * | Region | Start | End | Size | Bus |
00015 * |-----|-----|-----|-----|-----|
00016 * | ECCRAM | 0x00FF8000 | 0x00FFFFFF | 32 KB | Memory Bus 3 |
00017 *
00018 * @par Control Register Block (Manual Chapter 60, sections 60.2.9-60.2.17)
00019 * | Offset | Register | Size | Description |
00020 * |-----|-----|-----|-----|
00021 * | 0x00 | ECCRAMMODE | 8 | ECC mode select (disabled/correct/ |
00022 * | | | | correct-and-detect/detect-only) |
00023 * | 0x01 | ECCRAM2STS | 8 | 2-bit error status flag |
00024 * | 0x02 | ECCRAM1STSEN | 8 | 1-bit error info update enable |
00025 * | 0x03 | ECCRAM1STS | 8 | 1-bit error status flag |
00026 * | 0x04 | ECCRAMPRCR | 8 | Protection register (MODE/1STSEN) |
00027 * | 0x08 | ECCRAM2ECAD | 32 | 2-bit error address capture |
00028 * | 0x0C | ECCRAM1ECAD | 32 | 1-bit error address capture |
00029 * | 0x10 | ECCRAMPRCR2 | 8 | Protection register 2 (ETST) |
00030 * | 0x14 | ECCRAMETST | 8 | Test control (ECC decoder bypass) |
00031 *
00032 * @par Initialization Sequence (Manual section 60.3.3)
00033 * Before enabling error checking, every 32-bit word of ECCRAM must be
00034 * written so that the ECC syndrome matches the data. Uninitialized ECC
00035 * bits are random and would falsely flag errors on first read. The driver
00036 * zeroes the entire region using 32-bit stores while the mode is set to
00037 * k_rx_eccrammode_ecc_no_check, then switches to the final mode.
00038 *
00039 * @par NASA Power of 10 Compliance
00040 * - Rule 5: Static assertions verify all register offsets and base addresses
00041 * - Rule 8: All constants use C23 typed enums, no macros for numeric values
00042 * - Rule 9: Inline accessor (single-level pointer return), no function pointers
00043 *
00044 * @par Terminology
00045 * ECC = Error Correcting Code
00046 * SEC-DED = Single Error Correction, Double Error Detection
00047 * PRCR = Protection Register
00048 *
00049 * @author Locked, Inc.
00050 * @date 2026-04-21
00051 * @copyright Copyright (c) 2026 Locked Inc.
00052 * SPDX-License-Identifier: MIT
00053 * @since Version 1.0.0
00054 */
00055
00056 #pragma once
00057
00058 #include <stddef.h>
00059 #include <stdint.h>
00060
00061 #ifdef __cplusplus
00062 extern "C" {
00063 #endif
00064
00065 /*
=====
00066 * Memory Region Addresses (Manual Ch60, Table 60.1, page 2977)
00067 *
=====
*/
00068
00069
00070 /**
00071 * @enum rx_eccram_region_addr_t
00072 * @brief ECCRAM data region start, end, and size
00073 *
00074 * @details
00075 * Physical address range of the ECC-protected RAM region. These are the
00076 * addresses that application code reads and writes -- the control
00077 * registers live at a separate base (see rx_eccram_reg_addr_t).
00078 *
00079 * @see RX72N HW Manual Chapter 60, Table 60.1 (RAM memory map), page 2977
00080 * @since Version 1.0.0

```

```

00081 */
00082 typedef enum : uintptr_t {
00083 /** @brief ECCRAM region start address (inclusive) */
00084 k_rx_eccram_region_base_addr = 0x00FF8000,
00085 /** @brief ECCRAM region end address (inclusive) */
00086 k_rx_eccram_region_end_addr = 0x0FFFFFFF,
00087 } rx_eccram_region_addr_t;
00088
00089 /**
00090 * @enum rx_eccram_region_size_t
00091 * @brief ECCRAM region size in bytes
00092 *
00093 * @details
00094 * Total usable bytes in the ECC-protected region. The driver's
00095 * zero-initialization loop iterates over this many bytes in 32-bit
00096 * steps so that every ECC syndrome matches the data before error
00097 * checking is enabled.
00098 *
00099 * @see RX72N HW Manual Chapter 60, Table 60.1, page 2977
00100 * @since Version 1.0.0
00101 */
00102 typedef enum : uint32_t {
00103 /** @brief ECCRAM size in bytes (32 KB) */
00104 k_rx_eccram_region_size_bytes = 0x00008000,
00105 } rx_eccram_region_size_t;
00106
00107 /*
=====
00108 * Control Register Base Address (Manual Ch60, section 60.2, page 2979)
00109 *
=====
00110 */
00111
00112 /**
00113 * @enum rx_eccram_reg_base_t
00114 * @brief ECCRAM control register block base address
00115 *
00116 * @see RX72N HW Manual Chapter 60, section 60.2 (register list), page 2979
00117 * @since Version 1.0.0
00118 */
00119 typedef enum : uintptr_t {
00120 /** @brief ECCRAM control register block base */
00121 k_rx_eccram_reg_base_addr = 0x000812C0,
00122 } rx_eccram_reg_base_t;
00123
00124 /*
=====
00125 * Register Offsets (Manual Ch60, sections 60.2.9-60.2.17, pages 2981-2990)
00126 *
=====
00127 */
00128
00129 /**
00130 * @enum rx_eccram_reg_offset_t
00131 * @brief Offsets of ECCRAM control registers from k_rx_eccram_reg_base_addr
00132 *
00133 * @see RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17, pages 2981-2990
00134 * @since Version 1.0.0
00135 */
00136 typedef enum : uint8_t {
00137 /** @brief ECCRAMMODE @ 0x000812C0 (Ch60, section 60.2.9, page 2981) */
00138 k_rx_eccram_offset_eccrammode = 0x00,
00139 /** @brief ECCRAM2STS @ 0x000812C1 (Ch60, section 60.2.10, page 2982) */
00140 k_rx_eccram_offset_eccram2sts = 0x01,
00141 /** @brief ECCRAM1STSEN @ 0x000812C2 (Ch60, section 60.2.11, page 2983) */
00142 k_rx_eccram_offset_eccram1stsen = 0x02,
00143 /** @brief ECCRAM1STS @ 0x000812C3 (Ch60, section 60.2.12, page 2984) */
00144 k_rx_eccram_offset_eccram1sts = 0x03,
00145 /** @brief ECCRAMPRCR @ 0x000812C4 (Ch60, section 60.2.13, page 2985) */
00146 k_rx_eccram_offset_eccramprcr = 0x04,
00147 /** @brief ECCRAM2ECAD @ 0x000812C8 (Ch60, section 60.2.14, page 2986) */
00148 k_rx_eccram_offset_eccram2ecad = 0x08,
00149 /** @brief ECCRAM1ECAD @ 0x000812CC (Ch60, section 60.2.15, page 2987) */
00150 k_rx_eccram_offset_eccram1ecad = 0x0C,
00151 /** @brief ECCRAMPRCR2 @ 0x000812D0 (Ch60, section 60.2.16, page 2989) */
00152 k_rx_eccram_offset_eccramprcr2 = 0x10,
00153 /** @brief ECCRAMETST @ 0x000812D4 (Ch60, section 60.2.17, page 2990) */
00154 k_rx_eccram_offset_eccrametst = 0x14,
00155 } rx_eccram_reg_offset_t;
00156
00157 /*
=====
00158 * ECCRAMMODE bit definitions (Manual Ch60, section 60.2.9, page 2981)
00159 *
=====
00160 */
00161

```

```

00162 /**
00163 * @enum rx_eccrammode_bits_t
00164 * @brief ECCRAMMODE operating mode select values (RAMMOD[1:0])
00165 *
00166 * @details
00167 * Selects whether ECC generation and checking are active on ECCRAM
00168 * accesses. The recommended mode for production is
00169 * k_rx_eccrammode_ecc_with_check (correct-and-detect).
00170 *
00171 * @see RX72N HW Manual Chapter 60, section 60.2.9, page 2981
00172 * @since Version 1.0.0
00173 */
00174 typedef enum : uint8_t {
00175 /** @brief ECC disabled (RAMMOD[1:0] = 00b) */
00176 k_rx_eccrammode_ecc_disabled = 0x00,
00177 /** @brief Setting prohibited (RAMMOD[1:0] = 01b) -- do not use */
00178 k_rx_eccrammode_prohibited = 0x01,
00179 /** @brief ECC generated, no error checking (RAMMOD[1:0] = 10b) */
00180 k_rx_eccrammode_ecc_no_check = 0x02,
00181 /** @brief ECC generated with 1-bit correction and 2-bit detection (RAMMOD[1:0] = 11b) */
00182 k_rx_eccrammode_ecc_with_check = 0x03,
00183 /** @brief RAMMOD[1:0] field mask */
00184 k_rx_eccrammode_rammod_mask = 0x03,
00185 } rx_eccrammode_bits_t;
00186
00187 /*
=====
00188 * ECCRAM2STS bit definitions (Manual Ch60, section 60.2.10, page 2982)
00189 *
=====
00190 */
00191
00192 /**
00193 * @enum rx_eccram2sts_bits_t
00194 * @brief ECCRAM2STS 2-bit error flag (ECC2ERR)
00195 *
00196 * @details
00197 * Set to 1 by hardware when a 2-bit error is detected on an ECCRAM read.
00198 * Cleared by writing 0 to bit 0. While this flag is set, ECCRAM2ECAD
00199 * holds the failing address.
00200 *
00201 * @see RX72N HW Manual Chapter 60, section 60.2.10, page 2982
00202 * @since Version 1.0.0
00203 */
00204 typedef enum : uint8_t {
00205 /** @brief ECC2ERR bit position */
00206 k_rx_eccram2sts_ecc2err_shift = 0,
00207 /** @brief ECC2ERR mask */
00208 k_rx_eccram2sts_ecc2err_mask = 0x01,
00209 /** @brief Value to clear ECC2ERR (write 0 to bit 0) */
00210 k_rx_eccram2sts_clear = 0x00,
00211 } rx_eccram2sts_bits_t;
00212
00213 /*
=====
00214 * ECCRAM1STSEN bit definitions (Manual Ch60, section 60.2.11, page 2983)
00215 *
=====
00216 */
00217
00218 /**
00219 * @enum rx_eccram1stsen_bits_t
00220 * @brief ECCRAM1STSEN 1-bit error info update enable (ECC1STSEN)
00221 *
00222 * @see RX72N HW Manual Chapter 60, section 60.2.11, page 2983
00223 * @since Version 1.0.0
00224 */
00225 typedef enum : uint8_t {
00226 /** @brief ECC1STSEN bit position */
00227 k_rx_eccram1stsen_ecc1stsen_shift = 0,
00228 /** @brief ECC1STSEN mask */
00229 k_rx_eccram1stsen_ecc1stsen_mask = 0x01,
00230 /** @brief 1-bit error status updates disabled */
00231 k_rx_eccram1stsen_disable = 0x00,
00232 /** @brief 1-bit error status updates enabled */
00233 k_rx_eccram1stsen_enable = 0x01,
00234 } rx_eccram1stsen_bits_t;
00235
00236 /*
=====
00237 * ECCRAM1STS bit definitions (Manual Ch60, section 60.2.12, page 2984)
00238 *
=====
00239 */
00240
00241 /**
00242 * @enum rx_eccram1sts_bits_t

```

```

00243 * @brief ECCRAM1STS 1-bit error flag (ECC1ERR)
00244 *
00245 * @details
00246 * Set to 1 by hardware when a correctable 1-bit error is detected.
00247 * Cleared by writing 0 to bit 0. While this flag is set, ECCRAM1ECAD
00248 * holds the failing address.
00249 *
00250 * @see RX72N HW Manual Chapter 60, section 60.2.12, page 2984
00251 * @since Version 1.0.0
00252 */
00253 typedef enum : uint8_t {
00254 /** @brief ECC1ERR bit position */
00255 k_rx_eccram1sts_ecc1err_shift = 0,
00256 /** @brief ECC1ERR mask */
00257 k_rx_eccram1sts_ecc1err_mask = 0x01,
00258 /** @brief Value to clear ECC1ERR (write 0 to bit 0) */
00259 k_rx_eccram1sts_clear = 0x00,
00260 } rx_eccram1sts_bits_t;
00261
00262 /*
=====
00263 * ECCRAMPRCR bit definitions (Manual Ch60, section 60.2.13, page 2985)
00264 *
=====
00265 */
00266
00267 /**
00268 * @enum rx_eccramprcr_bits_t
00269 * @brief ECCRAMPRCR protection register values
00270 *
00271 * @details
00272 * Write-enables ECCRAMMODE and ECCRAM1STSEN. Bits [7:1] are the key
00273 * KW[6:0] = 1111000b; bit [0] is PRCR. Write 0xF1 to unlock (key + PRCR=1),
00274 * 0xF0 to lock (key + PRCR=0). Writes with any other key pattern are
00275 * ignored.
00276 *
00277 * @see RX72N HW Manual Chapter 60, section 60.2.13, page 2985
00278 * @since Version 1.0.0
00279 */
00280 typedef enum : uint8_t {
00281 /** @brief PRCR bit position */
00282 k_rx_eccramprcr_prcr_shift = 0,
00283 /** @brief PRCR mask */
00284 k_rx_eccramprcr_prcr_mask = 0x01,
00285 /** @brief KW[6:0] = 1111000b in bits [7:1] (mandatory key) */
00286 k_rx_eccramprcr_key = 0xF0,
00287 /** @brief Full value to unlock writes to ECCRAMMODE / ECCRAM1STSEN */
00288 k_rx_eccramprcr_unlock = 0xF1,
00289 /** @brief Full value to re-lock writes */
00290 k_rx_eccramprcr_lock = 0xF0,
00291 } rx_eccramprcr_bits_t;
00292
00293 /*
=====
00294 * ECCRAMPRCR2 bit definitions (Manual Ch60, section 60.2.16, page 2989)
00295 *
=====
00296 */
00297
00298 /**
00299 * @enum rx_eccramprcr2_bits_t
00300 * @brief ECCRAMPRCR2 protection register 2 values (guards ECCRAMETST)
00301 *
00302 * @see RX72N HW Manual Chapter 60, section 60.2.16, page 2989
00303 * @since Version 1.0.0
00304 */
00305 typedef enum : uint8_t {
00306 /** @brief PRCR2 bit position */
00307 k_rx_eccramprcr2_prcr2_shift = 0,
00308 /** @brief PRCR2 mask */
00309 k_rx_eccramprcr2_prcr2_mask = 0x01,
00310 /** @brief KW2[6:0] = 1111000b in bits [7:1] */
00311 k_rx_eccramprcr2_key = 0xF0,
00312 /** @brief Full value to unlock writes to ECCRAMETST */
00313 k_rx_eccramprcr2_unlock = 0xF1,
00314 /** @brief Full value to re-lock */
00315 k_rx_eccramprcr2_lock = 0xF0,
00316 } rx_eccramprcr2_bits_t;
00317
00318 /*
=====
00319 * ECCRAMETST bit definitions (Manual Ch60, section 60.2.17, page 2990)
00320 *
=====
00321 */
00322
00323 /**

```

```

00324 * @enum rx_eccrametst_bits_t
00325 * @brief ECCRAMETST ECC decoder bypass select (TSTBYP) for test mode
00326 *
00327 * @see RX72N HW Manual Chapter 60, section 60.2.17, page 2990
00328 * @since Version 1.0.0
00329 */
00330 typedef enum : uint8_t {
00331 /** @brief TSTBYP bit position */
00332 k_rx_eccrametst_tstbyp_shift = 0,
00333 /** @brief TSTBYP mask */
00334 k_rx_eccrametst_tstbyp_mask = 0x01,
00335 /** @brief Normal operation (ECC decoder active) */
00336 k_rx_eccrametst_normal = 0x00,
00337 /** @brief Bypass ECC decoder (for HW self-test only) */
00338 k_rx_eccrametst_bypass = 0x01,
00339 } rx_eccrametst_bits_t;
00340
00341 /*
=====
00342 * MSTPCRC bit for ECCRAM module stop (Manual Ch60, section 60.4.1, page 2994)
00343 *
=====
00344 */
00345
00346 /**
00347 * @enum rx_eccram_mstpcrc_bits_t
00348 * @brief MSTPCRC.MSTPC6 bit position for ECCRAM module-stop control
00349 *
00350 * @details
00351 * Setting MSTPC6 = 1 stops the clock supply to the ECCRAM control logic
00352 * and the region becomes inaccessible. It must be cleared (bit = 0)
00353 * before any ECCRAM access -- including the zero-initialization pass.
00354 * The write requires the main system PRCR to be in the "PRC1 unlocked"
00355 * state (see k_rx_prcr_unlock_prc1 in rx_register_protection.h).
00356 *
00357 * @see RX72N HW Manual Chapter 60, section 60.4.1 (MSTPCRC), page 2994
00358 * @see RX72N HW Manual Chapter 11 (Low Power Consumption)
00359 * @since Version 1.0.0
00360 */
00361 typedef enum : uint32_t {
00362 /** @brief Bit position of ECCRAM module-stop in MSTPCRC */
00363 k_rx_eccram_mstpcrc_bit_shift = 6,
00364 /** @brief Bit mask for MSTPCRC.MSTPC6 */
00365 k_rx_eccram_mstpcrc_bit_mask = (1U << 6),
00366 } rx_eccram_mstpcrc_bits_t;
00367
00368 /*
=====
00369 * Register Structure
00370 *
=====
00371 */
00372
00373 /**
00374 * @struct rx_eccram_regs_t
00375 * @brief ECCRAM control register block
00376 *
00377 * @details
00378 * Memory-mapped layout of the ECCRAM control registers starting at
00379 * k_rx_eccram_reg_base_addr (0x000812C0). Reserved gaps preserve the
00380 * hardware layout exactly as described in the manual.
00381 *
00382 * @invariant sizeof(rx_eccram_regs_t) == 24 bytes (21 documented + 3 tail padding)
00383 * @invariant All field offsets match RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17
00384 *
00385 * @see eccram_regs() Accessor that returns a pointer to this block at its hardware address
00386 * @see RX72N HW Manual Chapter 60, sections 60.2.9-60.2.17, pages 2981-2990
00387 * @since Version 1.0.0
00388 */
00389 typedef struct {
00390 /** @brief ECCRAMMODE @ 0x00 - ECC operating mode select (Ch60, section 60.2.9, page 2981) */
00391 volatile uint8_t eccrammode;
00392 /** @brief ECCRAM2STS @ 0x01 - 2-bit error status flag (Ch60, section 60.2.10, page 2982) */
00393 volatile uint8_t eccram2sts;
00394 /** @brief ECCRAM1STSEN @ 0x02 - 1-bit info update enable (Ch60, section 60.2.11, page 2983) */
00395 volatile uint8_t eccram1stsen;
00396 /** @brief ECCRAM1STS @ 0x03 - 1-bit error status flag (Ch60, section 60.2.12, page 2984) */
00397 volatile uint8_t eccram1sts;
00398 /** @brief ECCRAMPRCR @ 0x04 - Protection register (Ch60, section 60.2.13, page 2985) */
00399 volatile uint8_t eccramprcr;
00400 /** @brief Reserved @ 0x05-0x07 */
00401 uint8_t reserved0[3];
00402 /** @brief ECCRAM2ECAD @ 0x08 - 2-bit error address (Ch60, section 60.2.14, page 2986) */
00403 volatile uint32_t eccram2ecad;
00404 /** @brief ECCRAM1ECAD @ 0x0C - 1-bit error address (Ch60, section 60.2.15, page 2987) */
00405 volatile uint32_t eccram1ecad;
00406 /** @brief ECCRAMPRCR2 @ 0x10 - Protection register 2 (Ch60, section 60.2.16, page 2989) */

```

```

00407 volatile uint8_t eccramprcr2;
00408 /** @brief Reserved @ 0x11-0x13 */
00409 uint8_t reserved1[3];
00410 /** @brief ECCRAMETST @ 0x14 - Test control (Ch60, section 60.2.17, page 2990) */
00411 volatile uint8_t eccrametst;
00412 } rx_eccram_regs_t;
00413
00414 /*
=====
00415 * Inline Register Accessor
00416 *
=====
00417 */
00418
00419 /**
00420 * @brief Get pointer to the ECCRAM control register block
00421 *
00422 * @details
00423 * Returns a volatile pointer to the ECCRAM control registers at their
00424 * fixed hardware address. This is the sole, NASA-Rule-8-compliant way
00425 * to reach the registers -- no macros are used for addresses.
00426 *
00427 * @return Volatile pointer to rx_eccram_regs_t
00428 * @retval Non-NULL Always returns a valid pointer to k_rx_eccram_reg_base_addr
00429 *
00430 * @pre None (hardware register address is always mapped)
00431 * @post Pointer is valid for the lifetime of program execution
00432 *
00433 * @note Thread Safety: Safe. Returns a constant hardware address with no
00434 * shared state. Concurrent register reads/writes through the pointer
00435 * are the caller's concern.
00436 *
00437 * @par Example:
00438 * @code{.c}
00439 * eccram_regs()->eccramprcr = k_rx_eccramprcr_unlock;
00440 * eccram_regs()->eccrammode = k_rx_eccrammode_ecc_with_check;
00441 * eccram_regs()->eccramprcr = k_rx_eccramprcr_lock;
00442 * @endcode
00443 *
00444 * @see k_rx_eccram_reg_base_addr
00445 * @since Version 1.0.0
00446 */
00447 static inline volatile rx_eccram_regs_t* eccram_regs(void)
00448 {
00449 return (volatile rx_eccram_regs_t*)k_rx_eccram_reg_base_addr;
00450 }
00451
00452 /*
=====
00453 * Static Assertions -- Layout Verification
00454 *
=====
00455 */
00456
00457 static_assert(sizeof(rx_eccram_regs_t) == 24,
00458 "rx_eccram_regs_t must be 24 bytes (21 documented + 3 tail padding)");
00459
00460 static_assert(offsetof(rx_eccram_regs_t, eccrammode) == k_rx_eccram_offset_eccrammode,
00461 "ECCRAMMODE offset must be 0x00");
00462 static_assert(offsetof(rx_eccram_regs_t, eccram2sts) == k_rx_eccram_offset_eccram2sts,
00463 "ECCRAM2STS offset must be 0x01");
00464 static_assert(offsetof(rx_eccram_regs_t, eccram1stsen) == k_rx_eccram_offset_eccram1stsen,
00465 "ECCRAM1STSEN offset must be 0x02");
00466 static_assert(offsetof(rx_eccram_regs_t, eccram1lsts) == k_rx_eccram_offset_eccram1lsts,
00467 "ECCRAM1STS offset must be 0x03");
00468 static_assert(offsetof(rx_eccram_regs_t, eccramprcr) == k_rx_eccram_offset_eccramprcr,
00469 "ECCRAMPRCR offset must be 0x04");
00470 static_assert(offsetof(rx_eccram_regs_t, eccram2ecad) == k_rx_eccram_offset_eccram2ecad,
00471 "ECCRAM2ECAD offset must be 0x08");
00472 static_assert(offsetof(rx_eccram_regs_t, eccram1ecad) == k_rx_eccram_offset_eccram1ecad,
00473 "ECCRAM1ECAD offset must be 0x0C");
00474 static_assert(offsetof(rx_eccram_regs_t, eccramprcr2) == k_rx_eccram_offset_eccramprcr2,
00475 "ECCRAMPRCR2 offset must be 0x10");
00476 static_assert(offsetof(rx_eccram_regs_t, eccrametst) == k_rx_eccram_offset_eccrametst,
00477 "ECCRAMETST offset must be 0x14");
00478
00479 static_assert(k_rx_eccram_reg_base_addr == 0x000812C0,
00480 "ECCRAM control register base must be 0x000812C0");
00481 static_assert(k_rx_eccram_region_base_addr == 0x00FF8000, "ECCRAM region base must be 0x00FF8000");
00482 static_assert(k_rx_eccram_region_end_addr == 0x0FFFFFFF, "ECCRAM region end must be 0x0FFFFFFF");
00483 static_assert(k_rx_eccram_region_size_bytes == 0x00008000,
00484 "ECCRAM region size must be 32 KB (0x8000)");
00485 static_assert(k_rx_eccramprcr_unlock == 0xF1, "ECCRAMPRCR unlock must be 0xF1");
00486 static_assert(k_rx_eccramprcr2_unlock == 0xF1, "ECCRAMPRCR2 unlock must be 0xF1");
00487 static_assert(k_rx_eccram_mstpcrc_bit_shift == 6, "MSTPCRC.MSTPC6 must be bit 6");
00488
00489 #ifdef __cplusplus

```



```
00490 }
00491 #endif
```

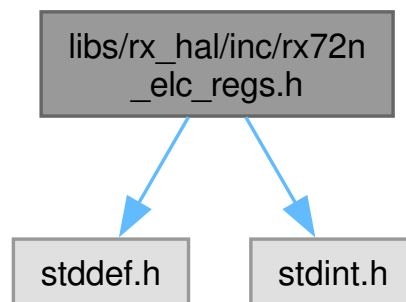
## 8.21 libs/rx\_hal/inc/rx72n\_elc\_regs.h File Reference

RX72N Event Link Controller (ELC) Register Definitions.

```
#include <stdint.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_elc\_regs.h:



This graph shows which files directly or indirectly include this file:



### Enumerations

- enum `elc_addresses_t` : `uintptr_t` {
  - `k_elc_base_addr` = 0x0008B100U , `k_elc_elcr_addr` = 0x0008B100U , `k_elc_elsr0_addr` = 0x0008B101U , `k_elc_elsr3_addr` = 0x0008B104U ,
  - `k_elc_elsr4_addr` = 0x0008B105U , `k_elc_elsr7_addr` = 0x0008B108U , `k_elc_elsr10_addr` = 0x0008B10BU , `k_elc_elsr11_addr` = 0x0008B10CU ,
  - `k_elc_elsr12_addr` = 0x0008B10DU , `k_elc_elsr13_addr` = 0x0008B10EU , `k_elc_elsr15_addr` = 0x0008B110U , `k_elc_elsr16_addr` = 0x0008B111U ,
  - `k_elc_elsr18_addr` = 0x0008B113U , `k_elc_elsr19_addr` = 0x0008B114U , `k_elc_elsr20_addr` = 0x0008B115U , `k_elc_elsr21_addr` = 0x0008B116U ,
  - `k_elc_elsr22_addr` = 0x0008B117U , `k_elc_elsr23_addr` = 0x0008B118U , `k_elc_elsr24_addr` = 0x0008B119U , `k_elc_elsr25_addr` = 0x0008B11AU ,
  - `k_elc_elsr26_addr` = 0x0008B11BU , `k_elc_elsr27_addr` = 0x0008B11CU , `k_elc_elsr28_addr` = 0x0008B11DU , `k_elc_elsr33_addr` = 0x0008B131U ,
  - `k_elc_elsr35_addr` = 0x0008B133U , `k_elc_elsr36_addr` = 0x0008B134U , `k_elc_elsr37_addr` = 0x0008B135U , `k_elc_elsr38_addr` = 0x0008B136U ,
  - `k_elc_elsr45_addr` = 0x0008B13DU , `k_elc_elsr48_addr` = 0x0008B146U , `k_elc_elsr49_addr` = 0x0008B147U , `k_elc_elsr50_addr` = 0x0008B148U ,
  - `k_elc_elsr51_addr` = 0x0008B149U , `k_elc_elsr52_addr` = 0x0008B14AU , `k_elc_elsr53_addr` = 0x0008B14BU , `k_elc_elsr54_addr` = 0x0008B14CU ,
  - `k_elc_elsr55_addr` = 0x0008B14DU , `k_elc_elsr56_addr` = 0x0008B14EU , `k_elc_elsr57_addr` = 0x0008B14FU , `k_elc_elopa_addr` = 0x0008B11FU ,
  - `k_elc_elopb_addr` = 0x0008B120U , `k_elc_elopc_addr` = 0x0008B121U , `k_elc_elopd_addr` =

```
0x0008B122U , k_elc_elpf_addr = 0x0008B13FU ,
k_elc_eloph_addr = 0x0008B141U , k_elc_pgr1_addr = 0x0008B123U , k_elc_pgr2_addr =
0x0008B124U , k_elc_pgc1_addr = 0x0008B125U ,
k_elc_pgc2_addr = 0x0008B126U , k_elc_pdbf1_addr = 0x0008B127U , k_elc_pdbf2_addr =
0x0008B128U , k_elc_pel0_addr = 0x0008B129U ,
k_elc_pel1_addr = 0x0008B12AU , k_elc_pel2_addr = 0x0008B12BU , k_elc_pel3_addr =
0x0008B12CU , k_elc_elsegr_addr = 0x0008B12DU }
```

ELC register addresses.

- enum `elcr_bits_t` : `uint8_t` { `k_elcr_reserved` = 0x7FU , `k_elcr_elcon` = 0x80U }

Event Link Control Register (ELCR) bit definitions.

- enum `elsr_event_t` : `uint8_t` {  
`k_elsr_event_disabled` = 0x00U , `k_elsr_mtu0_cmpa` = 0x01U , `k_elsr_mtu0_cmpb` = 0x02U ,  
`k_elsr_mtu0_cmpc` = 0x03U ,  
`k_elsr_mtu0_cmpd` = 0x04U , `k_elsr_mtu0_cmpe` = 0x05U , `k_elsr_mtu0_cmpf` = 0x06U ,  
`k_elsr_mtu0_ovf` = 0x07U ,  
`k_elsr_mtu3_cmpa` = 0x10U , `k_elsr_mtu3_cmpb` = 0x11U , `k_elsr_mtu3_cmpc` = 0x12U ,  
`k_elsr_mtu3_cmpd` = 0x13U ,  
`k_elsr_mtu3_ovf` = 0x14U , `k_elsr_mtu4_cmpa` = 0x15U , `k_elsr_mtu4_cmpb` = 0x16U ,  
`k_elsr_mtu4_cmpc` = 0x17U ,  
`k_elsr_mtu4_cmpd` = 0x18U , `k_elsr_mtu4_ovf` = 0x19U , `k_elsr_mtu4_udf` = 0x1AU ,  
`k_elsr_cmt1_cmp` = 0x1FU ,  
`k_elsr_tmr0_cmpa` = 0x22U , `k_elsr_tmr0_cmpb` = 0x23U , `k_elsr_tmr0_ovf` = 0x24U ,  
`k_elsr_s12ad0_end` = 0x58U ,  
`k_elsr_s12ad1_end` = 0x6CU , `k_elsr_dmac0_end` = 0x5DU , `k_elsr_dmac1_end` = 0x5EU ,  
`k_elsr_dmac2_end` = 0x5FU ,  
`k_elsr_dmac3_end` = 0x60U , `k_elsr_dtc_end` = 0x61U , `k_elsr_port_grp1` = 0x63U ,  
`k_elsr_port_grp2` = 0x64U ,  
`k_elsr_port_single0` = 0x65U , `k_elsr_port_single1` = 0x66U , `k_elsr_port_single2` = 0x67U ,  
`k_elsr_port_single3` = 0x68U ,  
`k_elsr_software` = 0x69U , `k_elsr_gptw0_cmpa` = 0x80U , `k_elsr_gptw0_cmpb` = 0x81U ,  
`k_elsr_gptw0_cmpc` = 0x82U ,  
`k_elsr_gptw0_cmpd` = 0x83U , `k_elsr_gptw0_cmpe` = 0x84U , `k_elsr_gptw0_cmpf` = 0x85U ,  
`k_elsr_gptw0_ovf` = 0x86U ,  
`k_elsr_gptw0_udf` = 0x87U , `k_elsr_gptw1_cmpa` = 0x88U , `k_elsr_gptw1_cmpb` = 0x89U ,  
`k_elsr_gptw1_cmpc` = 0x8AU ,  
`k_elsr_gptw1_cmpd` = 0x8BU , `k_elsr_gptw1_cmpe` = 0x8CU , `k_elsr_gptw1_cmpf` = 0x8DU ,  
`k_elsr_gptw1_ovf` = 0x8EU ,  
`k_elsr_gptw1_udf` = 0x8FU , `k_elsr_gptw2_cmpa` = 0x90U , `k_elsr_gptw2_cmpb` = 0x91U ,  
`k_elsr_gptw2_cmpc` = 0x92U ,  
`k_elsr_gptw2_cmpd` = 0x93U , `k_elsr_gptw2_cmpe` = 0x94U , `k_elsr_gptw2_cmpf` = 0x95U ,  
`k_elsr_gptw2_ovf` = 0x96U ,  
`k_elsr_gptw2_udf` = 0x97U , `k_elsr_gptw3_cmpa` = 0x98U , `k_elsr_gptw3_cmpb` = 0x99U ,  
`k_elsr_gptw3_cmpc` = 0x9AU ,  
`k_elsr_gptw3_cmpd` = 0x9BU , `k_elsr_gptw3_cmpe` = 0x9CU , `k_elsr_gptw3_cmpf` = 0x9DU ,  
`k_elsr_gptw3_ovf` = 0x9EU ,  
`k_elsr_gptw3_udf` = 0x9FU }

Event signal values for ELSRn registers.

- enum `elop_timer_op_t` : `uint8_t` { `k_elop_timer_count_start` = 0x00U , `k_elop_timer_count_restart` = 0x01U , `k_elop_timer_input_capture` , `k_elop_timer_disabled` = 0x03U }

Timer operation selection bits for ELOPA/B/C/D/F/H registers.

- enum `elopa_bits_t` : `uint8_t` { `k_elopa_mtu0op_shift` = 0U , `k_elopa_mtu0op_mask` = 0x03U ,  
`k_elopa_mtu3op_shift` = 6U , `k_elopa_mtu3op_mask` = 0xC0U }

ELOPA register bit positions (MTU0/MTU3).

- enum `elopb_bits_t` : `uint8_t` { `k_elopb_mtu4op_shift` = 0U , `k_elopb_mtu4op_mask` = 0x03U }

ELOPB register bit positions (MTU4).

- enum `elopc_bits_t` : `uint8_t` { `k_elopc_cmt1op_shift` = 2U , `k_elopc_cmt1op_mask` = 0x0CU }  
ELOPC register bit positions (CMT1).
- enum `elopd_bits_t` : `uint8_t` {  
`k_elopd_tmr0op_shift` = 0U , `k_elopd_tmr0op_mask` = 0x03U , `k_elopd_tmr1op_shift` = 2U ,  
`k_elopd_tmr1op_mask` = 0x0CU ,  
`k_elopd_tmr2op_shift` = 4U , `k_elopd_tmr2op_mask` = 0x30U , `k_elopd_tmr3op_shift` = 6U ,  
`k_elopd_tmr3op_mask` = 0xC0U }  
ELOPD register bit positions (TMR0-3).
- enum `pgc_bits_t` : `uint8_t` {  
`k_pgc_pgci_shift` = 0U , `k_pgc_pgci_mask` = 0x03U , `k_pgc_pgci_rising` = 0x00U ,  
`k_pgc_pgci_falling` = 0x01U ,  
`k_pgc_pgci_both` = 0x02U , `k_pgc_pgcove` = 0x04U , `k_pgc_pgco_shift` = 4U , `k_pgc_pgco_mask` = 0x70U ,  
`k_pgc_pgco_low` = 0x00U , `k_pgc_pgco_high` = 0x10U , `k_pgc_pgco_toggle` = 0x20U ,  
`k_pgc_pgco_buffer` = 0x30U ,  
`k_pgc_pgco_rotate` = 0x40U }  
Port Group Control (PGC1/PGC2) register bit definitions.
- enum `pel_bits_t` : `uint8_t` {  
`k_pel_psb_mask` = 0x07U , `k_pel_psp_shift` = 3U , `k_pel_psp_mask` = 0x18U ,  
`k_pel_psp_disabled` = 0x00U ,  
`k_pel_psp_portb` = 0x08U , `k_pel_psp_porte` = 0x10U , `k_pel_psm_shift` = 5U ,  
`k_pel_psm_mask` = 0x60U ,  
`k_pel_psm_rising` = 0x00U , `k_pel_psm_falling` = 0x20U , `k_pel_psm_both` = 0x40U }  
Port Event Link (PEL0-3) register bit definitions.
- enum `elsegr_bits_t` : `uint8_t` { `k_elsegr_seg` = 0x01U , `k_elsegr_we` = 0x40U , `k_elsegr_wi` = 0x80U }  
Software Event Generation Register (ELSEGR) bit definitions.
- enum `elc_mstpcr_t` : `uint32_t` { `k_elc_mstpcrb_bit` = (1U << 9U) }  
ELC module stop control bits.
- enum `elsr_range_adj_t` : `uint8_t` {  
`k_elsr_range_a_adj` = 0x01U , `k_elsr_range_b_adj` = 0x10U , `k_elsr_range_c_adj` = 0x16U ,  
`k_elsr_range_a_max` = 28U ,  
`k_elsr_range_b_max` = 45U , `k_elsr_range_c_min` = 48U , `k_elsr_range_c_max` = 57U }  
Address computation adjustments for the three ELSRn register ranges.

## Functions

- static volatile `uint8_t` \* `elc_elcr_reg` (void)  
Get pointer to ELCR register (Event Link Control).
- static volatile `uint8_t` \* `elc_elsr_reg` (`uint8_t` n)  
Get pointer to ELSRn register.
- static volatile `uint8_t` \* `elc_elsr0_reg` (void)  
Get pointer to ELSR0 register (MTU0).
- static volatile `uint8_t` \* `elc_elsr15_reg` (void)  
Get pointer to ELSR15 register (S12AD ELCTRG00N).
- static volatile `uint8_t` \* `elc_elsr45_reg` (void)  
Get pointer to ELSR45 register (S12AD1 ELCTRG10N).
- static volatile `uint8_t` \* `elc_elsr48_reg` (void)  
Get pointer to ELSR48 register (GPTW event source A).
- static volatile `uint8_t` \* `elc_elopa_reg` (void)  
Get pointer to ELOPA register (MTU0/MTU3 option).
- static volatile `uint8_t` \* `elc_elopb_reg` (void)

- Get pointer to ELOPB register (MTU4 option).
- static volatile uint8\_t \* [elc\\_pgr1\\_reg](#) (void)
- Get pointer to PGR1 register (Port Group 1).
- static volatile uint8\_t \* [elc\\_pgr2\\_reg](#) (void)
- Get pointer to PGR2 register (Port Group 2).
- static volatile uint8\_t \* [elc\\_pgc1\\_reg](#) (void)
- Get pointer to PGC1 register (Port Group Control 1).
- static volatile uint8\_t \* [elc\\_pgc2\\_reg](#) (void)
- Get pointer to PGC2 register (Port Group Control 2).
- static volatile uint8\_t \* [elc\\_pdbf1\\_reg](#) (void)
- Get pointer to PDBF1 register (Port Data Buffer 1).
- static volatile uint8\_t \* [elc\\_pdbf2\\_reg](#) (void)
- Get pointer to PDBF2 register (Port Data Buffer 2).
- static volatile uint8\_t \* [elc\\_pel\\_reg](#) (uint8\_t n)
- Get pointer to PELn register (Port Event Link).
- static volatile uint8\_t \* [elc\\_elsegr\\_reg](#) (void)
- Get pointer to ELSEGR register (Software Event Generation).

### 8.21.1 Detailed Description

RX72N Event Link Controller (ELC) Register Definitions.

Complete register definitions for the RX72N Event Link Controller. The ELC uses interrupt requests from peripheral modules as event signals to interconnect peripherals without CPU intervention.

#### ELC Overview

- 135 types of event signals can be directly interconnected
- Timer event input control for MTU, CMT, TMR, TPU, CMTW, GPTW
- Port event I/O control for Port B and Port E
- Software event generation capability

#### Register Organization

| Address      | Size | Description                   |
|--------------|------|-------------------------------|
| 0x0008B100   | 8    | ELCR - Event Link Control     |
| 0x0008B101   | 8    | ELSR0 - MTU0 event link       |
| 0x0008B104   | 8    | ELSR3 - MTU3 event link       |
| ...          | ...  | (various ELSRn registers)     |
| 0x0008B11F   | 8    | ELOPA - MTU0/3 option setting |
| 0x0008B120   | 8    | ELOPB - MTU4 option setting   |
| 0x0008B121   | 8    | ELOPC - CMT1 option setting   |
| 0x0008B122   | 8    | ELOPD - TMR0-3 option setting |
| 0x0008B123-4 | 8    | PGR1/2 - Port group registers |
| 0x0008B125-6 | 8    | PGC1/2 - Port group control   |

| Address       | Size | Description                        |
|---------------|------|------------------------------------|
| 0x0008B127-8  | 8    | PDBF1/2 - Port data buffers        |
| 0x0008B129-C  | 8    | PEL0-3 - Port event link           |
| 0x0008B12D    | 8    | ELSEGR - Software event generation |
| 0x0008B13F    | 8    | ELOPF - TPU0-3 option setting      |
| 0x0008B141    | 8    | ELOPH - CMTW0 option setting       |
| 0x0008B146-4F | 8    | ELSR48-57 - GPTW event links       |

#### NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops)
- Rule 3: No dynamic memory allocation
- Rule 4: Short accessor functions
- Rule 5: Static assertions verify addresses
- Rule 6: Minimal scope
- Rule 7: N/A
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

#### Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 21: Event Link Controller (ELC), pages 835-860
- Section 21.2: Register Descriptions

#### Author

Locked, Inc.

#### Date

2026-01-29

#### Copyright

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#### Since

Version 1.0.0

Definition in file [rx72n\\_elc\\_regs.h](#).

## 8.21.2 Enumeration Type Documentation

### 8.21.2.1 `elc_addresses_t`

enum `elc_addresses_t` : `uintptr_t`

ELC register addresses.

All addresses verified against RX72N Hardware Manual Ch21.

Note

Addresses verified 2026-01-29 against manual section 21.2

Since

Version 1.0.0

Enumerator

|                                |                                     |
|--------------------------------|-------------------------------------|
| <code>k_elc_base_addr</code>   | ELC module base address.            |
| <code>k_elc_elcr_addr</code>   | ELCR - Event Link Control Register. |
| <code>k_elc_elsr0_addr</code>  | MTU0                                |
| <code>k_elc_elsr3_addr</code>  | MTU3                                |
| <code>k_elc_elsr4_addr</code>  | MTU4                                |
| <code>k_elc_elsr7_addr</code>  | CMT1                                |
| <code>k_elc_elsr10_addr</code> | TMR0                                |
| <code>k_elc_elsr11_addr</code> | TMR1                                |
| <code>k_elc_elsr12_addr</code> | TMR2                                |
| <code>k_elc_elsr13_addr</code> | TMR3                                |
| <code>k_elc_elsr15_addr</code> | S12AD (ELCTRG00N)                   |
| <code>k_elc_elsr16_addr</code> | DA0                                 |
| <code>k_elc_elsr18_addr</code> | ICU Interrupt 1                     |
| <code>k_elc_elsr19_addr</code> | ICU Interrupt 2                     |
| <code>k_elc_elsr20_addr</code> | Output port group 1                 |
| <code>k_elc_elsr21_addr</code> | Output port group 2                 |
| <code>k_elc_elsr22_addr</code> | Input port group 1                  |
| <code>k_elc_elsr23_addr</code> | Input port group 2                  |
| <code>k_elc_elsr24_addr</code> | Single port 0                       |
| <code>k_elc_elsr25_addr</code> | Single port 1                       |
| <code>k_elc_elsr26_addr</code> | Single port 2                       |
| <code>k_elc_elsr27_addr</code> | Single port 3                       |
| <code>k_elc_elsr28_addr</code> | Clock source switch to LOCO         |
| <code>k_elc_elsr33_addr</code> | CMTW0                               |
| <code>k_elc_elsr35_addr</code> | TPU0                                |

|                   |                           |
|-------------------|---------------------------|
| k_elc_elsr36_addr | TPU1                      |
| k_elc_elsr37_addr | TPU2                      |
| k_elc_elsr38_addr | TPU3                      |
| k_elc_elsr45_addr | S12AD1 (ELCTRG10N)        |
| k_elc_elsr48_addr | GPTW event source A       |
| k_elc_elsr49_addr | GPTW event source B       |
| k_elc_elsr50_addr | GPTW event source C       |
| k_elc_elsr51_addr | GPTW event source D       |
| k_elc_elsr52_addr | GPTW event source E       |
| k_elc_elsr53_addr | GPTW event source F       |
| k_elc_elsr54_addr | GPTW event source G       |
| k_elc_elsr55_addr | GPTW event source H       |
| k_elc_elsr56_addr | S12AD (ELCTRG01N)         |
| k_elc_elsr57_addr | S12AD1 (ELCTRG11N)        |
| k_elc_elopa_addr  | MTU0/MTU3 option setting  |
| k_elc_elopb_addr  | MTU4 option setting       |
| k_elc_elopc_addr  | CMT1 option setting       |
| k_elc_elopd_addr  | TMR0-3 option setting     |
| k_elc_elopf_addr  | TPU0-3 option setting     |
| k_elc_eloph_addr  | CMTW0 option setting      |
| k_elc_pgr1_addr   | Port Group Register 1     |
| k_elc_pgr2_addr   | Port Group Register 2     |
| k_elc_pgc1_addr   | Port Group Control 1      |
| k_elc_pgc2_addr   | Port Group Control 2      |
| k_elc_pdbf1_addr  | Port Data Buffer 1        |
| k_elc_pdbf2_addr  | Port Data Buffer 2        |
| k_elc_pel0_addr   | Port Event Link 0         |
| k_elc_pel1_addr   | Port Event Link 1         |
| k_elc_pel2_addr   | Port Event Link 2         |
| k_elc_pel3_addr   | Port Event Link 3         |
| k_elc_elsegr_addr | Software Event Generation |

Definition at line 84 of file [rx72n\\_elc\\_regs.h](#).

```

00084 : uintptr_t {
00085 /** @brief ELC module base address */
00086 k_elc_base_addr = 0x0008B100U,
00087
00088 /** @brief ELCR - Event Link Control Register */
00089 k_elc_elcr_addr = 0x0008B100U,
00090
00091 /* ELSRn - Event Link Setting Registers (n = peripheral specific) */
00092 k_elc_elsr0_addr = 0x0008B101U, /**< MTU0 */
00093 k_elc_elsr3_addr = 0x0008B104U, /**< MTU3 */
00094 k_elc_elsr4_addr = 0x0008B105U, /**< MTU4 */
00095 k_elc_elsr7_addr = 0x0008B108U, /**< CMT1 */
00096 k_elc_elsr10_addr = 0x0008B10BU, /**< TMR0 */
00097 k_elc_elsr11_addr = 0x0008B10CU, /**< TMR1 */
00098 k_elc_elsr12_addr = 0x0008B10DU, /**< TMR2 */
00099 k_elc_elsr13_addr = 0x0008B10EU, /**< TMR3 */
00100 k_elc_elsr15_addr = 0x0008B110U, /**< S12AD (ELCTRG00N) */
00101 k_elc_elsr16_addr = 0x0008B111U, /**< DA0 */
00102 k_elc_elsr18_addr = 0x0008B113U, /**< ICU Interrupt 1 */

```

```

00103 k_elc_elsr19_addr = 0x0008B114U, /**< ICU Interrupt 2 */
00104 k_elc_elsr20_addr = 0x0008B115U, /**< Output port group 1 */
00105 k_elc_elsr21_addr = 0x0008B116U, /**< Output port group 2 */
00106 k_elc_elsr22_addr = 0x0008B117U, /**< Input port group 1 */
00107 k_elc_elsr23_addr = 0x0008B118U, /**< Input port group 2 */
00108 k_elc_elsr24_addr = 0x0008B119U, /**< Single port 0 */
00109 k_elc_elsr25_addr = 0x0008B11AU, /**< Single port 1 */
00110 k_elc_elsr26_addr = 0x0008B11BU, /**< Single port 2 */
00111 k_elc_elsr27_addr = 0x0008B11CU, /**< Single port 3 */
00112 k_elc_elsr28_addr = 0x0008B11DU, /**< Clock source switch to LOCO */
00113 k_elc_elsr33_addr = 0x0008B131U, /**< CMTW0 */
00114 k_elc_elsr35_addr = 0x0008B133U, /**< TPU0 */
00115 k_elc_elsr36_addr = 0x0008B134U, /**< TPU1 */
00116 k_elc_elsr37_addr = 0x0008B135U, /**< TPU2 */
00117 k_elc_elsr38_addr = 0x0008B136U, /**< TPU3 */
00118 k_elc_elsr45_addr = 0x0008B13DU, /**< S12AD1 (ELCTRG10N) */
00119 k_elc_elsr48_addr = 0x0008B146U, /**< GPTW event source A */
00120 k_elc_elsr49_addr = 0x0008B147U, /**< GPTW event source B */
00121 k_elc_elsr50_addr = 0x0008B148U, /**< GPTW event source C */
00122 k_elc_elsr51_addr = 0x0008B149U, /**< GPTW event source D */
00123 k_elc_elsr52_addr = 0x0008B14AU, /**< GPTW event source E */
00124 k_elc_elsr53_addr = 0x0008B14BU, /**< GPTW event source F */
00125 k_elc_elsr54_addr = 0x0008B14CU, /**< GPTW event source G */
00126 k_elc_elsr55_addr = 0x0008B14DU, /**< GPTW event source H */
00127 k_elc_elsr56_addr = 0x0008B14EU, /**< S12AD (ELCTRG01N) */
00128 k_elc_elsr57_addr = 0x0008B14FU, /**< S12AD1 (ELCTRG11N) */
00129
00130 /* Option Setting Registers */
00131 k_elc_elopa_addr = 0x0008B11FU, /**< MTU0/MTU3 option setting */
00132 k_elc_elopb_addr = 0x0008B120U, /**< MTU4 option setting */
00133 k_elc_elopc_addr = 0x0008B121U, /**< CMT1 option setting */
00134 k_elc_elopd_addr = 0x0008B122U, /**< TMR0-3 option setting */
00135 k_elc_elopf_addr = 0x0008B13FU, /**< TPU0-3 option setting */
00136 k_elc_eloph_addr = 0x0008B141U, /**< CMTW0 option setting */
00137
00138 /* Port Event Registers */
00139 k_elc_pgr1_addr = 0x0008B123U, /**< Port Group Register 1 */
00140 k_elc_pgr2_addr = 0x0008B124U, /**< Port Group Register 2 */
00141 k_elc_pgc1_addr = 0x0008B125U, /**< Port Group Control 1 */
00142 k_elc_pgc2_addr = 0x0008B126U, /**< Port Group Control 2 */
00143 k_elc_pdbf1_addr = 0x0008B127U, /**< Port Data Buffer 1 */
00144 k_elc_pdbf2_addr = 0x0008B128U, /**< Port Data Buffer 2 */
00145 k_elc_pel0_addr = 0x0008B129U, /**< Port Event Link 0 */
00146 k_elc_pel1_addr = 0x0008B12AU, /**< Port Event Link 1 */
00147 k_elc_pel2_addr = 0x0008B12BU, /**< Port Event Link 2 */
00148 k_elc_pel3_addr = 0x0008B12CU, /**< Port Event Link 3 */
00149
00150 /* Software Event Generation */
00151 k_elc_elsegr_addr = 0x0008B12DU, /**< Software Event Generation */
00152 } elc_addresses_t;

```

### 8.21.2.2 elc\_mstpcr\_t

```
enum elc_mstpcr_t : uint32_t
```

ELC module stop control bits.

The ELC is controlled by MSTPCRB.MSTPB9. Must be cleared (0) before accessing ELC registers.

Note

Manual ref: Ch11 (Low Power Consumption) module stop tables



Since

Version 1.0.0

Enumerator

|                   |                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| k_elc_mstpcrb_bit | MSTPCRB bit for ELC (bit 9). <ul style="list-style-type: none"> <li>0: ELC module is running</li> <li>1: ELC module is stopped (default)</li> </ul> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

Definition at line 513 of file rx72n\_elc\_regs.h.

```

00513 : uint32_t {
00514 /**
00515 * @brief MSTPCRB bit for ELC (bit 9)
00516 * @details
00517 * - 0: ELC module is running
00518 * - 1: ELC module is stopped (default)
00519 */
00520 k_elc_mstpcrb_bit = (1U « 9U),
00521 } elc_mstpcr_t;

```

### 8.21.2.3 elcr\_bits\_t

enum [elcr\\_bits\\_t](#) : uint8\_t

Event Link Control Register (ELCR) bit definitions.

Register Layout (8-bit @ 0x0008B100)

| Bit     | Field | Description           |
|---------|-------|-----------------------|
| 6:<br>0 | -     | Reserved (write 1)    |
| 7       | ELCON | All Event Link Enable |

Note

Manual ref: Ch21 section 21.2.1

Since

Version 1.0.0

Enumerator

|                 |                                                |
|-----------------|------------------------------------------------|
| k_elcr_reserved | Reserved bits mask (write 1).                  |
| k_elcr_elcon    | All Event Link Enable (0=disabled, 1=enabled). |

Definition at line 172 of file rx72n\_elc\_regs.h.

```

00172 : uint8_t {
00173 /** @brief Reserved bits mask (write 1) */
00174 k_elcr_reserved = 0x7FU,
00175
00176 /** @brief All Event Link Enable (0=disabled, 1=enabled) */
00177 k_elcr_elcon = 0x80U,
00178 } elcr_bits_t;

```

### 8.21.2.4 elop\_timer\_op\_t

```
enum elop_timer_op_t : uint8_t
```

Timer operation selection bits for ELOPA/B/C/D/F/H registers.

Each timer can be configured to perform different operations when an event signal is received. These 2-bit fields are used in the ELOPx registers.

For MTU/TPU timers (ELOPA, ELOPF): 0x02 = Input capture (TCNT -> TGRA). For CMT/TMR/CMTW timers (ELOPC, ELOPD, ELOPH): 0x02 = Event counter.

Note

Manual ref: Ch21 sections 21.2.3-21.2.8, Table 21.5

Since

Version 1.0.0

Enumerator

|                            |                                                         |
|----------------------------|---------------------------------------------------------|
| k_elop_timer_count_start   | Counting is started                                     |
| k_elop_timer_count_restart | Counting is restarted                                   |
| k_elop_timer_input_capture | Input capture (MTU/TPU) or event counter (CMT/TMR/CMTW) |
| k_elop_timer_disabled      | Event output is disabled                                |

Definition at line 316 of file rx72n\_elc\_regs.h.

```
00316 : uint8_t {
00317 k_elop_timer_count_start = 0x00U, /**< Counting is started */
00318 k_elop_timer_count_restart = 0x01U, /**< Counting is restarted */
00319 k_elop_timer_input_capture =
00320 0x02U, /**< Input capture (MTU/TPU) or event counter (CMT/TMR/CMTW) */
00321 k_elop_timer_disabled = 0x03U, /**< Event output is disabled */
00322 } elop_timer_op_t;
```

### 8.21.2.5 elopa\_bits\_t

```
enum elopa_bits_t : uint8_t
```

ELOPA register bit positions (MTU0/MTU3).

Register Layout (8-bit @ 0x0008B11F)

| Bits    | Field  | Description            |
|---------|--------|------------------------|
| 1:<br>0 | MTU0MD | MTU0 operation setting |

| Bits     | Field  | Description            |
|----------|--------|------------------------|
| 5:↔<br>2 | -      | Reserved               |
| 7:↔<br>6 | MTU3MD | MTU3 operation setting |

Since

Version 1.0.0

Enumerator

|                      |                                       |
|----------------------|---------------------------------------|
| k_elopa_mtu0op_shift | MTU0 operation field shift (bits 1:0) |
| k_elopa_mtu0op_mask  | MTU0 operation field mask             |
| k_elopa_mtu3op_shift | MTU3 operation field shift (bits 7:6) |
| k_elopa_mtu3op_mask  | MTU3 operation field mask             |

Definition at line 337 of file [rx72n\\_elc\\_regs.h](#).

```

00337 : uint8_t {
00338 k_elopa_mtu0op_shift = 0U, /**< MTU0 operation field shift (bits 1:0) */
00339 k_elopa_mtu0op_mask = 0x03U, /**< MTU0 operation field mask */
00340 k_elopa_mtu3op_shift = 6U, /**< MTU3 operation field shift (bits 7:6) */
00341 k_elopa_mtu3op_mask = 0xC0U, /**< MTU3 operation field mask */
00342 } elopa_bits_t;

```

#### 8.21.2.6 elopb\_bits\_t

```
enum elopb_bits_t : uint8_t
```

ELOPB register bit positions (MTU4).

Register Layout (8-bit @ 0x0008B120)

| Bits     | Field  | Description            |
|----------|--------|------------------------|
| 1:↔<br>0 | MTU4OP | MTU4 operation setting |
| 7:↔<br>2 | -      | Reserved               |

Since

Version 1.0.0

Enumerator

|                      |                            |
|----------------------|----------------------------|
| k_elopb_mtu4op_shift | MTU4 operation field shift |
|----------------------|----------------------------|

|                     |                           |
|---------------------|---------------------------|
| k_elopb_mtu4op_mask | MTU4 operation field mask |
|---------------------|---------------------------|

Definition at line 356 of file [rx72n\\_elc\\_regs.h](#).

```
00356 : uint8_t {
00357 k_elopb_mtu4op_shift = 0U, /**< MTU4 operation field shift */
00358 k_elopb_mtu4op_mask = 0x03U, /**< MTU4 operation field mask */
00359 } elopb_bits_t;
```

#### 8.21.2.7 elopc`bits`<sup>t</sup>

enum [elopc\\_bits\\_t](#) : uint8\_t

ELOPC register bit positions (CMT1).

Register Layout (8-bit @ 0x0008B121)

| Bits     | Field  | Description            |
|----------|--------|------------------------|
| 1:↔<br>0 | -      | Reserved               |
| 3:↔<br>2 | CMT1MD | CMT1 operation setting |
| 7:↔<br>4 | -      | Reserved               |

Since

Version 1.0.0

Enumerator

|                      |                                       |
|----------------------|---------------------------------------|
| k_elopc_cmt1op_shift | CMT1 operation field shift (bits 3:2) |
| k_elopc_cmt1op_mask  | CMT1 operation field mask             |

Definition at line 374 of file [rx72n\\_elc\\_regs.h](#).

```
00374 : uint8_t {
00375 k_elopc_cmt1op_shift = 2U, /**< CMT1 operation field shift (bits 3:2) */
00376 k_elopc_cmt1op_mask = 0x0CU, /**< CMT1 operation field mask */
00377 } elopc_bits_t;
```

#### 8.21.2.8 elopd`bits`<sup>t</sup>

enum [elopd\\_bits\\_t](#) : uint8\_t

ELOPD register bit positions (TMR0-3).

Register Layout (8-bit @ 0x0008B122)

| Bits | Field  | Description            |
|------|--------|------------------------|
| 1↔0  | TMR0OP | TMR0 operation setting |
| 3↔2  | TMR1OP | TMR1 operation setting |
| 5↔4  | TMR2OP | TMR2 operation setting |
| 7↔6  | TMR3OP | TMR3 operation setting |

Since

Version 1.0.0

Enumerator

|                     |                            |
|---------------------|----------------------------|
| k_elpd_tmr0op_shift | TMR0 operation field shift |
| k_elpd_tmr0op_mask  | TMR0 operation field mask  |
| k_elpd_tmr1op_shift | TMR1 operation field shift |
| k_elpd_tmr1op_mask  | TMR1 operation field mask  |
| k_elpd_tmr2op_shift | TMR2 operation field shift |
| k_elpd_tmr2op_mask  | TMR2 operation field mask  |
| k_elpd_tmr3op_shift | TMR3 operation field shift |
| k_elpd_tmr3op_mask  | TMR3 operation field mask  |

Definition at line 393 of file [rx72n\\_elc\\_regs.h](#).

```

00393 : uint8_t {
00394 k_elpd_tmr0op_shift = 0U, /**< TMR0 operation field shift */
00395 k_elpd_tmr0op_mask = 0x03U, /**< TMR0 operation field mask */
00396 k_elpd_tmr1op_shift = 2U, /**< TMR1 operation field shift */
00397 k_elpd_tmr1op_mask = 0x0CU, /**< TMR1 operation field mask */
00398 k_elpd_tmr2op_shift = 4U, /**< TMR2 operation field shift */
00399 k_elpd_tmr2op_mask = 0x30U, /**< TMR2 operation field mask */
00400 k_elpd_tmr3op_shift = 6U, /**< TMR3 operation field shift */
00401 k_elpd_tmr3op_mask = 0xC0U, /**< TMR3 operation field mask */
00402 } elpd_bits_t;

```

#### 8.21.2.9 elsegr\_bits\_t

```
enum elsegr_bits_t : uint8_t
```

Software Event Generation Register (ELSEGR) bit definitions.

Register Layout (8-bit @ 0x0008B12D)

| Bit | Field | Description               |
|-----|-------|---------------------------|
| 0   | SEG   | Software Event Generation |

| Bit | Field | Description   |
|-----|-------|---------------|
| 5:1 | -     | Reserved      |
| 6   | WE    | Write Enable  |
| 7   | WI    | Write Disable |

Note

To write: Set WE=1, WI=0, then set SEG

Since

Version 1.0.0

Enumerator

|              |                                  |
|--------------|----------------------------------|
| k_elsegr_seg | Software event generation        |
| k_elsegr_we  | Write enable                     |
| k_elsegr_wi  | Write disable (set 1 to disable) |

Definition at line 491 of file [rx72n\\_elc\\_regs.h](#).

```

00491 : uint8_t {
00492 k_elsegr_seg = 0x01U, /**< Software event generation */
00493 k_elsegr_we = 0x40U, /**< Write enable */
00494 k_elsegr_wi = 0x80U, /**< Write disable (set 1 to disable) */
00495 } elsegr_bits_t;

```

#### 8.21.2.10 elsr\_event\_t

```
enum elsr_event_t : uint8_t
```

Event signal values for ELSRn registers.

Values to set in ELSRn.ELS[7:0] to select the event signal source. Only a subset of commonly-used events are defined here.

Note

Manual ref: Ch21 Table 21.3

Since

Version 1.0.0

Enumerator

|                       |                       |
|-----------------------|-----------------------|
| k_elsr_event_disabled | Event output disabled |
| k_elsr_mtu0_cmpa      | MTU0 compare match A  |

|                     |                           |
|---------------------|---------------------------|
| k_elSr_mtu0_cmpb    | MTU0 compare match B      |
| k_elSr_mtu0_cmpc    | MTU0 compare match C      |
| k_elSr_mtu0_cmpd    | MTU0 compare match D      |
| k_elSr_mtu0_cmpe    | MTU0 compare match E      |
| k_elSr_mtu0_cmpf    | MTU0 compare match F      |
| k_elSr_mtu0_ovf     | MTU0 overflow             |
| k_elSr_mtu3_compa   | MTU3 compare match A      |
| k_elSr_mtu3_cmpb    | MTU3 compare match B      |
| k_elSr_mtu3_cmpc    | MTU3 compare match C      |
| k_elSr_mtu3_cmpd    | MTU3 compare match D      |
| k_elSr_mtu3_ovf     | MTU3 overflow             |
| k_elSr_mtu4_compa   | MTU4 compare match A      |
| k_elSr_mtu4_cmpb    | MTU4 compare match B      |
| k_elSr_mtu4_cmpc    | MTU4 compare match C      |
| k_elSr_mtu4_cmpd    | MTU4 compare match D      |
| k_elSr_mtu4_ovf     | MTU4 overflow             |
| k_elSr_mtu4_udf     | MTU4 underflow            |
| k_elSr_cmt1_cmp     | CMT1 compare match        |
| k_elSr_tmr0_compa   | TMR0 compare match A      |
| k_elSr_tmr0_cmpb    | TMR0 compare match B      |
| k_elSr_tmr0_ovf     | TMR0 overflow             |
| k_elSr_s12ad0_end   | S12AD A/D conversion end  |
| k_elSr_s12ad1_end   | S12AD1 A/D conversion end |
| k_elSr_dmac0_end    | DMAC0 transfer end        |
| k_elSr_dmac1_end    | DMAC1 transfer end        |
| k_elSr_dmac2_end    | DMAC2 transfer end        |
| k_elSr_dmac3_end    | DMAC3 transfer end        |
| k_elSr_dtc_end      | DTC transfer end          |
| k_elSr_port_grp1    | Input port group 1 edge   |
| k_elSr_port_grp2    | Input port group 2 edge   |
| k_elSr_port_single0 | Single port 0 edge        |
| k_elSr_port_single1 | Single port 1 edge        |
| k_elSr_port_single2 | Single port 2 edge        |
| k_elSr_port_single3 | Single port 3 edge        |
| k_elSr_software     | Software event            |
| k_elSr_gptw0_compa  | GPTW0 compare match A     |
| k_elSr_gptw0_cmpb   | GPTW0 compare match B     |
| k_elSr_gptw0_cmpc   | GPTW0 compare match C     |
| k_elSr_gptw0_cmpd   | GPTW0 compare match D     |
| k_elSr_gptw0_cmpe   | GPTW0 compare match E     |
| k_elSr_gptw0_cmpf   | GPTW0 compare match F     |
| k_elSr_gptw0_ovf    | GPTW0 overflow            |

|                   |                       |
|-------------------|-----------------------|
| k_elsr_gptw0_udf  | GPTW0 underflow       |
| k_elsr_gptw1_cmpa | GPTW1 compare match A |
| k_elsr_gptw1_cmpb | GPTW1 compare match B |
| k_elsr_gptw1_cmpc | GPTW1 compare match C |
| k_elsr_gptw1_cmpd | GPTW1 compare match D |
| k_elsr_gptw1_cmpe | GPTW1 compare match E |
| k_elsr_gptw1_cmpf | GPTW1 compare match F |
| k_elsr_gptw1_ovf  | GPTW1 overflow        |
| k_elsr_gptw1_udf  | GPTW1 underflow       |
| k_elsr_gptw2_cmpa | GPTW2 compare match A |
| k_elsr_gptw2_cmpb | GPTW2 compare match B |
| k_elsr_gptw2_cmpc | GPTW2 compare match C |
| k_elsr_gptw2_cmpd | GPTW2 compare match D |
| k_elsr_gptw2_cmpe | GPTW2 compare match E |
| k_elsr_gptw2_cmpf | GPTW2 compare match F |
| k_elsr_gptw2_ovf  | GPTW2 overflow        |
| k_elsr_gptw2_udf  | GPTW2 underflow       |
| k_elsr_gptw3_cmpa | GPTW3 compare match A |
| k_elsr_gptw3_cmpb | GPTW3 compare match B |
| k_elsr_gptw3_cmpc | GPTW3 compare match C |
| k_elsr_gptw3_cmpd | GPTW3 compare match D |
| k_elsr_gptw3_cmpe | GPTW3 compare match E |
| k_elsr_gptw3_cmpf | GPTW3 compare match F |
| k_elsr_gptw3_ovf  | GPTW3 overflow        |
| k_elsr_gptw3_udf  | GPTW3 underflow       |

Definition at line 196 of file [rx72n\\_elc\\_regs.h](#).

```

00196 : uint8_t {
00197 k_elsr_event_disabled = 0x00U, /**< Event output disabled */
00198
00199 /** MTU0 events (0x01-0x07) */
00200 k_elsr_mtu0_cmpa = 0x01U, /**< MTU0 compare match A */
00201 k_elsr_mtu0_cmpb = 0x02U, /**< MTU0 compare match B */
00202 k_elsr_mtu0_cmpc = 0x03U, /**< MTU0 compare match C */
00203 k_elsr_mtu0_cmpd = 0x04U, /**< MTU0 compare match D */
00204 k_elsr_mtu0_cmpe = 0x05U, /**< MTU0 compare match E */
00205 k_elsr_mtu0_cmpf = 0x06U, /**< MTU0 compare match F */
00206 k_elsr_mtu0_ovf = 0x07U, /**< MTU0 overflow */
00207
00208 /** MTU3 events (0x10-0x14) */
00209 k_elsr_mtu3_cmpa = 0x10U, /**< MTU3 compare match A */
00210 k_elsr_mtu3_cmpb = 0x11U, /**< MTU3 compare match B */
00211 k_elsr_mtu3_cmpc = 0x12U, /**< MTU3 compare match C */
00212 k_elsr_mtu3_cmpd = 0x13U, /**< MTU3 compare match D */
00213 k_elsr_mtu3_ovf = 0x14U, /**< MTU3 overflow */
00214
00215 /** MTU4 events (0x15-0x1A) */
00216 k_elsr_mtu4_cmpa = 0x15U, /**< MTU4 compare match A */
00217 k_elsr_mtu4_cmpb = 0x16U, /**< MTU4 compare match B */
00218 k_elsr_mtu4_cmpc = 0x17U, /**< MTU4 compare match C */
00219 k_elsr_mtu4_cmpd = 0x18U, /**< MTU4 compare match D */
00220 k_elsr_mtu4_ovf = 0x19U, /**< MTU4 overflow */
00221 k_elsr_mtu4_udf = 0x1AU, /**< MTU4 underflow */
00222
00223 /** CMT1 event (0x1F) */
00224 k_elsr_cmt1_cmp = 0x1FU, /**< CMT1 compare match */
00225
00226 /** TMR events (0x22-0x2D) */

```



```

00227 k_elsr_tmr0_cmpa = 0x22U, /**< TMR0 compare match A */
00228 k_elsr_tmr0_cmpb = 0x23U, /**< TMR0 compare match B */
00229 k_elsr_tmr0_ovf = 0x24U, /**< TMR0 overflow */
00230
00231 /* S12AD events (0x58, 0x6C) */
00232 k_elsr_s12ad0_end = 0x58U, /**< S12AD A/D conversion end */
00233 k_elsr_s12ad1_end = 0x6CU, /**< S12AD1 A/D conversion end */
00234
00235 /* DMAC transfer end events (0x5D-0x60) */
00236 k_elsr_dmac0_end = 0x5DU, /**< DMAC0 transfer end */
00237 k_elsr_dmac1_end = 0x5EU, /**< DMAC1 transfer end */
00238 k_elsr_dmac2_end = 0x5FU, /**< DMAC2 transfer end */
00239 k_elsr_dmac3_end = 0x60U, /**< DMAC3 transfer end */
00240
00241 /* DTC transfer end (0x61) */
00242 k_elsr_dtc_end = 0x61U, /**< DTC transfer end */
00243
00244 /* Port events (0x63-0x68) */
00245 k_elsr_port_grp1 = 0x63U, /**< Input port group 1 edge */
00246 k_elsr_port_grp2 = 0x64U, /**< Input port group 2 edge */
00247 k_elsr_port_single0 = 0x65U, /**< Single port 0 edge */
00248 k_elsr_port_single1 = 0x66U, /**< Single port 1 edge */
00249 k_elsr_port_single2 = 0x67U, /**< Single port 2 edge */
00250 k_elsr_port_single3 = 0x68U, /**< Single port 3 edge */
00251
00252 /* Software event (0x69) */
00253 k_elsr_software = 0x69U, /**< Software event */
00254
00255 /* GPTW0 events (0x80-0x87) */
00256 k_elsr_gptw0_cmpa = 0x80U, /**< GPTW0 compare match A */
00257 k_elsr_gptw0_cmpb = 0x81U, /**< GPTW0 compare match B */
00258 k_elsr_gptw0_cmpc = 0x82U, /**< GPTW0 compare match C */
00259 k_elsr_gptw0_cmpd = 0x83U, /**< GPTW0 compare match D */
00260 k_elsr_gptw0_cmpe = 0x84U, /**< GPTW0 compare match E */
00261 k_elsr_gptw0_cmpf = 0x85U, /**< GPTW0 compare match F */
00262 k_elsr_gptw0_ovf = 0x86U, /**< GPTW0 overflow */
00263 k_elsr_gptw0_udf = 0x87U, /**< GPTW0 underflow */
00264
00265 /* GPTW1 events (0x88-0x8F) */
00266 k_elsr_gptw1_cmpa = 0x88U, /**< GPTW1 compare match A */
00267 k_elsr_gptw1_cmpb = 0x89U, /**< GPTW1 compare match B */
00268 k_elsr_gptw1_cmpc = 0x8AU, /**< GPTW1 compare match C */
00269 k_elsr_gptw1_cmpd = 0x8BU, /**< GPTW1 compare match D */
00270 k_elsr_gptw1_cmpe = 0x8CU, /**< GPTW1 compare match E */
00271 k_elsr_gptw1_cmpf = 0x8DU, /**< GPTW1 compare match F */
00272 k_elsr_gptw1_ovf = 0x8EU, /**< GPTW1 overflow */
00273 k_elsr_gptw1_udf = 0x8FU, /**< GPTW1 underflow */
00274
00275 /* GPTW2 events (0x90-0x97) */
00276 k_elsr_gptw2_cmpa = 0x90U, /**< GPTW2 compare match A */
00277 k_elsr_gptw2_cmpb = 0x91U, /**< GPTW2 compare match B */
00278 k_elsr_gptw2_cmpc = 0x92U, /**< GPTW2 compare match C */
00279 k_elsr_gptw2_cmpd = 0x93U, /**< GPTW2 compare match D */
00280 k_elsr_gptw2_cmpe = 0x94U, /**< GPTW2 compare match E */
00281 k_elsr_gptw2_cmpf = 0x95U, /**< GPTW2 compare match F */
00282 k_elsr_gptw2_ovf = 0x96U, /**< GPTW2 overflow */
00283 k_elsr_gptw2_udf = 0x97U, /**< GPTW2 underflow */
00284
00285 /* GPTW3 events (0x98-0x9F) */
00286 k_elsr_gptw3_cmpa = 0x98U, /**< GPTW3 compare match A */
00287 k_elsr_gptw3_cmpb = 0x99U, /**< GPTW3 compare match B */
00288 k_elsr_gptw3_cmpc = 0x9AU, /**< GPTW3 compare match C */
00289 k_elsr_gptw3_cmpd = 0x9BU, /**< GPTW3 compare match D */
00290 k_elsr_gptw3_cmpe = 0x9CU, /**< GPTW3 compare match E */
00291 k_elsr_gptw3_cmpf = 0x9DU, /**< GPTW3 compare match F */
00292 k_elsr_gptw3_ovf = 0x9EU, /**< GPTW3 overflow */
00293 k_elsr_gptw3_udf = 0x9FU, /**< GPTW3 underflow */
00294 } elsr_event_t;

```

### 8.21.2.11 elsr\_range\_adj\_t

```
enum elsr_range_adj_t : uint8_t
```

Address computation adjustments for the three ELSRn register ranges.

ELSRn registers are NOT contiguous – option and port registers occupy the gaps. There are three separate address ranges:

Range A n=0..28 : address = base + k\_elsr\_range\_a\_adj + n (0x0008B101..0x0008B11D) Range B n=33..45 : address = base + k\_elsr\_range\_b\_adj + n (0x0008B131..0x0008B13D) Range C n=48..57 : address = base + k\_elsr\_range\_c\_adj + n (0x0008B146..0x0008B14F)

Derivation: ELSR0 = 0x0008B101 = base + 1 + 0 -> adj\_a = 1 ELSR33 = 0x0008B131 = base + 0x10 + 33 -> adj\_b = 0x10 ELSR48 = 0x0008B146 = base + 0x16 + 48 -> adj\_c = 0x16

Note

Manual ref: Ch21 section 21.2.2, address list

Since

Version 1.0.0

Enumerator

|                    |                          |
|--------------------|--------------------------|
| k_elsr_range_a_adj | Adjustment for n=0..28   |
| k_elsr_range_b_adj | Adjustment for n=33..45  |
| k_elsr_range_c_adj | Adjustment for n=48..57  |
| k_elsr_range_a_max | Last valid n in range A  |
| k_elsr_range_b_max | Last valid n in range B  |
| k_elsr_range_c_min | First valid n in range C |
| k_elsr_range_c_max | Last valid n in range C  |

Definition at line 548 of file [rx72n\\_elc\\_regs.h](#).

```

00548 : uint8_t {
00549 k_elsr_range_a_adj = 0x01U, /**< Adjustment for n=0..28 */
00550 k_elsr_range_b_adj = 0x10U, /**< Adjustment for n=33..45 */
00551 k_elsr_range_c_adj = 0x16U, /**< Adjustment for n=48..57 */
00552 k_elsr_range_a_max = 28U, /**< Last valid n in range A */
00553 k_elsr_range_b_max = 45U, /**< Last valid n in range B */
00554 k_elsr_range_c_min = 48U, /**< First valid n in range C */
00555 k_elsr_range_c_max = 57U, /**< Last valid n in range C */
00556 } elsr_range_adj_t;

```

### 8.21.2.12 pel\_bits\_t

```
enum pel_bits_t : uint8_t
```

Port Event Link (PEL0-3) register bit definitions.

Register Layout (8-bit @ 0x0008B129-0x0008B12C)

| Bits     | Field         | Description                                     |
|----------|---------------|-------------------------------------------------|
| 2:↵<br>0 | PSB[2:↵<br>0] | Bit number for single port (pin 0-7)            |
| 4:↵<br>3 | PSP[1:↵<br>0] | Port select (00=disabled, 01=Port B, 10=Port E) |
| 6:↵<br>5 | PSM[1:↵<br>0] | Event link spec (edge or output mode)           |
| 7        | -             | Reserved                                        |

Note

Manual ref: Ch21 section 21.2.12

Since

Version 1.0.0

Enumerator

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| k_pel_psb_mask     | 3-bit pin select mask (pins 0-7)                            |
| k_pel_psp_shift    | PSP field shift                                             |
| k_pel_psp_mask     | PSP field mask                                              |
| k_pel_psp_disabled | Setting disabled                                            |
| k_pel_psp_portb    | Port B (corresponding to PGR1)                              |
| k_pel_psp_porte    | Port E (corresponding to PGR2)                              |
| k_pel_psm_shift    | PSM field shift                                             |
| k_pel_psm_mask     | PSM field mask                                              |
| k_pel_psm_rising   | Rising edge (input port) or output low (output port)        |
| k_pel_psm_falling  | Falling edge (input port) or output high (output port)      |
| k_pel_psm_both     | Both edges (input port) or toggle output (output port) (1x) |

Definition at line 459 of file rx72n\_elc\_regs.h.

```

00459 : uint8_t {
00460 /* PSB[2:0] - Bit Number Specification (bits 2:0) */
00461 k_pel_psb_mask = 0x07U, /**< 3-bit pin select mask (pins 0-7) */
00462 /* PSP[1:0] - Port Number Specification (bits 4:3) */
00463 k_pel_psp_shift = 3U, /**< PSP field shift */
00464 k_pel_psp_mask = 0x18U, /**< PSP field mask */
00465 k_pel_psp_disabled = 0x00U, /**< Setting disabled */
00466 k_pel_psp_portb = 0x08U, /**< Port B (corresponding to PGR1) */
00467 k_pel_psp_porte = 0x10U, /**< Port E (corresponding to PGR2) */
00468 /* PSM[1:0] - Event Link Specification (bits 6:5) */
00469 k_pel_psm_shift = 5U, /**< PSM field shift */
00470 k_pel_psm_mask = 0x60U, /**< PSM field mask */
00471 k_pel_psm_rising = 0x00U, /**< Rising edge (input port) or output low (output port) */
00472 k_pel_psm_falling = 0x20U, /**< Falling edge (input port) or output high (output port) */
00473 k_pel_psm_both = 0x40U, /**< Both edges (input port) or toggle output (output port) (1x) */
00474 } pel_bits_t;

```

### 8.21.2.13 pgc\_bits\_t

```
enum pgc_bits_t : uint8_t
```

Port Group Control (PGC1/PGC2) register bit definitions.

Register Layout (8-bit @ 0x0008B125/0x0008B126)

| Bits    | Field         | Description                            |
|---------|---------------|----------------------------------------|
| 1:<br>0 | PGCI[1:<br>0] | Event output edge select (input ports) |
| 2       | PGCOVE        | PDBF overwrite enable                  |

| Bits | Field     | Description                 |
|------|-----------|-----------------------------|
| 3    | -         | Reserved                    |
| 6:4  | PGCO[2:0] | Port group operation select |
| 7    | -         | Reserved                    |

Note

Manual ref: Ch21 section 21.2.10

Since

Version 1.0.0

Enumerator

|                    |                                                           |
|--------------------|-----------------------------------------------------------|
| k_pgc_pgci_shift   | PGCI field shift                                          |
| k_pgc_pgci_mask    | PGCI field mask                                           |
| k_pgc_pgci_rising  | Event on rising edge of input signal                      |
| k_pgc_pgci_falling | Event on falling edge of input signal                     |
| k_pgc_pgci_both    | Event on both edges of input signal (1x)                  |
| k_pgc_pgcove       | PDBF overwrite enable (0=disabled, 1=enabled)             |
| k_pgc_pgco_shift   | PGCO field shift                                          |
| k_pgc_pgco_mask    | PGCO field mask                                           |
| k_pgc_pgco_low     | Output low when event signal is input                     |
| k_pgc_pgco_high    | Output high when event signal is input                    |
| k_pgc_pgco_toggle  | Toggle output when event signal is input                  |
| k_pgc_pgco_buffer  | Output buffer (PDBF) value when event signal is input     |
| k_pgc_pgco_rotate  | Rotate output MSB-to-LSB when event signal is input (1xx) |

Definition at line 425 of file [rx72n\\_elc\\_regs.h](#).

```

00425 : uint8_t {
00426 /* PGCI[1:0] - Event Output Edge Select for input port groups (bits 1:0) */
00427 k_pgc_pgci_shift = 0U, /**< PGCI field shift */
00428 k_pgc_pgci_mask = 0x03U, /**< PGCI field mask */
00429 k_pgc_pgci_rising = 0x00U, /**< Event on rising edge of input signal */
00430 k_pgc_pgci_falling = 0x01U, /**< Event on falling edge of input signal */
00431 k_pgc_pgci_both = 0x02U, /**< Event on both edges of input signal (1x) */
00432 /* PGCOVE - PDBF Overwrite enable (bit 2) */
00433 k_pgc_pgcove = 0x04U, /**< PDBF overwrite enable (0=disabled, 1=enabled) */
00434 /* PGCO[2:0] - Port Group Operation Select for output port groups (bits 6:4) */
00435 k_pgc_pgco_shift = 4U, /**< PGCO field shift */
00436 k_pgc_pgco_mask = 0x70U, /**< PGCO field mask */
00437 k_pgc_pgco_low = 0x00U, /**< Output low when event signal is input */
00438 k_pgc_pgco_high = 0x10U, /**< Output high when event signal is input */
00439 k_pgc_pgco_toggle = 0x20U, /**< Toggle output when event signal is input */
00440 k_pgc_pgco_buffer = 0x30U, /**< Output buffer (PDBF) value when event signal is input */
00441 k_pgc_pgco_rotate = 0x40U, /**< Rotate output MSB-to-LSB when event signal is input (1xx) */
00442 } pgc_bits_t;

```

## 8.21.3 Function Documentation

### 8.21.3.1 elc\_elcr\_reg()

```
volatile uint8_t * elc_elcr_reg (
 void) [inline], [static]
```

Get pointer to ELCR register (Event Link Control).

## Returns

Volatile pointer to ELCR register (8-bit)

## Since

Version 1.0.0

Definition at line 568 of file [rx72n\\_elc\\_regs.h](#).

```
00569 {
00570 return (volatile uint8_t*)k_elc_elcr_addr;
00571 }
```

References [k\\_elc\\_elcr\\_addr](#).

## 8.21.3.2 elc\_elopa\_reg()

```
volatile uint8_t * elc_elopa_reg (
 void) [inline], [static]
```

Get pointer to ELOPA register (MTU0/MTU3 option).

## Returns

Volatile pointer to ELOPA register (8-bit)

## Since

Version 1.0.0

Definition at line 642 of file [rx72n\\_elc\\_regs.h](#).

```
00643 {
00644 return (volatile uint8_t*)k_elc_elopa_addr;
00645 }
```

References [k\\_elc\\_elopa\\_addr](#).

## 8.21.3.3 elc\_elopb\_reg()

```
volatile uint8_t * elc_elopb_reg (
 void) [inline], [static]
```

Get pointer to ELOPB register (MTU4 option).

## Returns

Volatile pointer to ELOPB register (8-bit)

## Since

Version 1.0.0

Definition at line 652 of file [rx72n\\_elc\\_regs.h](#).

```
00653 {
00654 return (volatile uint8_t*)k_elc_elopb_addr;
00655 }
```

References [k\\_elc\\_elopb\\_addr](#).

#### 8.21.3.4 `elc_elsegr_reg()`

```
volatile uint8_t * elc_elsegr_reg (
 void) [inline], [static]
```

Get pointer to ELSEGR register (Software Event Generation).

Returns

Volatile pointer to ELSEGR register (8-bit)

Since

Version 1.0.0

Definition at line 733 of file [rx72n\\_elc\\_regs.h](#).

```
00734 {
00735 return (volatile uint8_t*)k_elc_elsegr_addr;
00736 }
```

References [k\\_elc\\_elsegr\\_addr](#).

#### 8.21.3.5 `elc_elsr0_reg()`

```
volatile uint8_t * elc_elsr0_reg (
 void) [inline], [static]
```

Get pointer to ELSR0 register (MTU0).

Returns

Volatile pointer to ELSR0 register (8-bit)

Since

Version 1.0.0

Definition at line 602 of file [rx72n\\_elc\\_regs.h](#).

```
00603 {
00604 return (volatile uint8_t*)k_elc_elsr0_addr;
00605 }
```

References [k\\_elc\\_elsr0\\_addr](#).

### 8.21.3.6 elc\_elsr15\_reg()

```
volatile uint8_t * elc_elsr15_reg (
 void) [inline], [static]
```

Get pointer to ELSR15 register (S12AD ELCTRG00N).

Returns

Volatile pointer to ELSR15 register (8-bit)

Since

Version 1.0.0

Definition at line 612 of file [rx72n\\_elc\\_regs.h](#).

```
00613 {
00614 return (volatile uint8_t*)k_elc_elsr15_addr;
00615 }
```

References [k\\_elc\\_elsr15\\_addr](#).

### 8.21.3.7 elc\_elsr45\_reg()

```
volatile uint8_t * elc_elsr45_reg (
 void) [inline], [static]
```

Get pointer to ELSR45 register (S12AD1 ELCTRG10N).

Returns

Volatile pointer to ELSR45 register (8-bit)

Since

Version 1.0.0

Definition at line 622 of file [rx72n\\_elc\\_regs.h](#).

```
00623 {
00624 return (volatile uint8_t*)k_elc_elsr45_addr;
00625 }
```

References [k\\_elc\\_elsr45\\_addr](#).

### 8.21.3.8 `elc_elsr48_reg()`

```
volatile uint8_t * elc_elsr48_reg (
 void) [inline], [static]
```

Get pointer to ELSR48 register (GPTW event source A).

Returns

Volatile pointer to ELSR48 register (8-bit)

Since

Version 1.0.0

Definition at line 632 of file [rx72n\\_elc\\_regs.h](#).

```
00633 {
00634 return (volatile uint8_t*)k_elc_elsr48_addr;
00635 }
```

References [k\\_elc\\_elsr48\\_addr](#).

### 8.21.3.9 `elc_elsr_reg()`

```
volatile uint8_t * elc_elsr_reg (
 uint8_t n) [inline], [static]
```

Get pointer to ELSRn register.

Parameters

|   |                                                                         |
|---|-------------------------------------------------------------------------|
| n | Register index (0, 3, 4, 7, 10-13, 15, 16, 18-28, 33, 35-38, 45, 48-57) |
|---|-------------------------------------------------------------------------|

Returns

Volatile pointer to ELSRn register (8-bit)

Warning

Not all indices in [0..57] are valid – consult Table 21.2 for the valid set. Passing an invalid index returns a pointer to a reserved or wrong address.

Note

ELSRn registers occupy three non-contiguous ranges; see [elsr\\_range\\_adj\\_t](#).

Since

Version 1.0.0

Definition at line 583 of file [rx72n\\_elc\\_regs.h](#).

```
00584 {
00585 if (n <= k_elsr_range_a_max) {
00586 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_a_adj + n);
00587 }
00588 if ((n >= k_elsr_range_a_max + 1U) && (n <= k_elsr_range_b_max)) {
00589 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_b_adj + n);
00590 }
00591 if ((n >= k_elsr_range_c_min) && (n <= k_elsr_range_c_max)) {
00592 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_c_adj + n);
00593 }
00594 return nullptr;
00595 }
```

References [k\\_elc\\_base\\_addr](#), [k\\_elsr\\_range\\_a\\_adj](#), [k\\_elsr\\_range\\_a\\_max](#), [k\\_elsr\\_range\\_b\\_adj](#), [k\\_elsr\\_range\\_b\\_max](#), [k\\_elsr\\_range\\_c\\_adj](#), [k\\_elsr\\_range\\_c\\_max](#), and [k\\_elsr\\_range\\_c\\_min](#).



### 8.21.3.10 elc\_pdbf1\_reg()

```
volatile uint8_t * elc_pdbf1_reg (
 void) [inline], [static]
```

Get pointer to PDBF1 register (Port Data Buffer 1).

Returns

Volatile pointer to PDBF1 register (8-bit)

Since

Version 1.0.0

Definition at line 702 of file [rx72n\\_elc\\_regs.h](#).

```
00703 {
00704 return (volatile uint8_t*)k_elc_pdbf1_addr;
00705 }
```

References [k\\_elc\\_pdbf1\\_addr](#).

### 8.21.3.11 elc\_pdbf2\_reg()

```
volatile uint8_t * elc_pdbf2_reg (
 void) [inline], [static]
```

Get pointer to PDBF2 register (Port Data Buffer 2).

Returns

Volatile pointer to PDBF2 register (8-bit)

Since

Version 1.0.0

Definition at line 712 of file [rx72n\\_elc\\_regs.h](#).

```
00713 {
00714 return (volatile uint8_t*)k_elc_pdbf2_addr;
00715 }
```

References [k\\_elc\\_pdbf2\\_addr](#).

### 8.21.3.12 `elc_pel_reg()`

```
volatile uint8_t * elc_pel_reg (
 uint8_t n) [inline], [static]
```

Get pointer to PELn register (Port Event Link).

Parameters

|   |                      |
|---|----------------------|
| n | Register index (0-3) |
|---|----------------------|

Returns

Volatile pointer to PELn register (8-bit)

Since

Version 1.0.0

Definition at line 723 of file [rx72n\\_elc\\_regs.h](#).

```
00724 {
00725 return (volatile uint8_t*)(uintptr_t)(k_elc_pel0_addr + n);
00726 }
```

References [k\\_elc\\_pel0\\_addr](#).

### 8.21.3.13 `elc_pgc1_reg()`

```
volatile uint8_t * elc_pgc1_reg (
 void) [inline], [static]
```

Get pointer to PGC1 register (Port Group Control 1).

Returns

Volatile pointer to PGC1 register (8-bit)

Since

Version 1.0.0

Definition at line 682 of file [rx72n\\_elc\\_regs.h](#).

```
00683 {
00684 return (volatile uint8_t*)k_elc_pgc1_addr;
00685 }
```

References [k\\_elc\\_pgc1\\_addr](#).

#### 8.21.3.14 elc\_pgc2\_reg()

```
volatile uint8_t * elc_pgc2_reg (
 void) [inline], [static]
```

Get pointer to PGC2 register (Port Group Control 2).

Returns

Volatile pointer to PGC2 register (8-bit)

Since

Version 1.0.0

Definition at line 692 of file [rx72n\\_elc\\_regs.h](#).

```
00693 {
00694 return (volatile uint8_t*)k_elc_pgc2_addr;
00695 }
```

References [k\\_elc\\_pgc2\\_addr](#).

#### 8.21.3.15 elc\_pgr1\_reg()

```
volatile uint8_t * elc_pgr1_reg (
 void) [inline], [static]
```

Get pointer to PGR1 register (Port Group 1).

Returns

Volatile pointer to PGR1 register (8-bit)

Since

Version 1.0.0

Definition at line 662 of file [rx72n\\_elc\\_regs.h](#).

```
00663 {
00664 return (volatile uint8_t*)k_elc_pgr1_addr;
00665 }
```

References [k\\_elc\\_pgr1\\_addr](#).

### 8.21.3.16 `elc_pgr2_reg()`

```
volatile uint8_t * elc_pgr2_reg (
 void) [inline], [static]
```

Get pointer to PGR2 register (Port Group 2).

Returns

Volatile pointer to PGR2 register (8-bit)

Since

Version 1.0.0

Definition at line 672 of file `rx72n_elc_regs.h`.

```
00673 {
00674 return (volatile uint8_t*)k_elc_pgr2_addr;
00675 }
```

References `k_elc_pgr2_addr`.

## 8.22 `rx72n_elc_regs.h`

[Go to the documentation of this file.](#)

```
00001 /**
00002 * @file rx72n_elc_regs.h
00003 * @brief RX72N Event Link Controller (ELC) Register Definitions
00004 *
00005 * @details
00006 * Complete register definitions for the RX72N Event Link Controller.
00007 * The ELC uses interrupt requests from peripheral modules as event signals
00008 * to interconnect peripherals without CPU intervention.
00009 *
00010 * @par ELC Overview
00011 * - 135 types of event signals can be directly interconnected
00012 * - Timer event input control for MTU, CMT, TMR, TPU, CMTW, GPTW
00013 * - Port event I/O control for Port B and Port E
00014 * - Software event generation capability
00015 *
00016 * @par Register Organization
00017 * | Address | Size | Description |
00018 * |-----|-----|-----|
00019 * | 0x0008B100 | 8 | ELCR - Event Link Control |
00020 * | 0x0008B101 | 8 | ELSR0 - MTU0 event link |
00021 * | 0x0008B104 | 8 | ELSR3 - MTU3 event link |
00022 * | ... | ... | (various ELSRn registers) |
00023 * | 0x0008B11F | 8 | ELOPA - MTU0/3 option setting |
00024 * | 0x0008B120 | 8 | ELOPB - MTU4 option setting |
00025 * | 0x0008B121 | 8 | ELOPC - CMT1 option setting |
00026 * | 0x0008B122 | 8 | ELOPD - TMR0-3 option setting |
00027 * | 0x0008B123-4 | 8 | PGR1/2 - Port group registers |
00028 * | 0x0008B125-6 | 8 | PGC1/2 - Port group control |
00029 * | 0x0008B127-8 | 8 | PDBF1/2 - Port data buffers |
00030 * | 0x0008B129-C | 8 | PEL0-3 - Port event link |
00031 * | 0x0008B12D | 8 | ELSEGR - Software event generation |
00032 * | 0x0008B13F | 8 | ELOPF - TPU0-3 option setting |
00033 * | 0x0008B141 | 8 | ELOPH - CMTW0 option setting |
00034 * | 0x0008B146-4F | 8 | ELSR48-57 - GPTW event links |
00035 *
00036 * @par NASA Power of 10 Compliance
00037 * - Rule 1: No goto, setjmp, or recursion
00038 * - Rule 2: N/A (no loops)
00039 * - Rule 3: No dynamic memory allocation
00040 * - Rule 4: Short accessor functions
00041 * - Rule 5: Static assertions verify addresses
00042 * - Rule 6: Minimal scope
00043 * - Rule 7: N/A
00044 * - Rule 8: All constants use C23 typed enums
```

```

00045 * - Rule 9: No function pointers
00046 * - Rule 10: Compiles with -Wall -Wextra -Werror
00047 *
00048 * @par Manual References
00049 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00050 * - Chapter 21: Event Link Controller (ELC), pages 835-860
00051 * - Section 21.2: Register Descriptions
00052 *
00053 * @author Locked, Inc.
00054 * @date 2026-01-29
00055 * @copyright Copyright (c) 2026 Locked Inc.
00056 * SPDX-License-Identifier: MIT
00057 * @since Version 1.0.0
00058 */
00059
00060 #pragma once
00061
00062 #include <stddef.h>
00063 #include <stdint.h>
00064
00065 #ifdef __cplusplus
00066 extern "C" {
00067 #endif
00068
00069 /*
=====
00070 * ELC Base Address and Register Addresses
00071 *
=====
*/
00072
00073 /**
00074 * @enum elc_addresses_t
00075 * @brief ELC register addresses
00076 *
00077 * @details
00078 * All addresses verified against RX72N Hardware Manual Ch21.
00079 *
00080 * @note Addresses verified 2026-01-29 against manual section 21.2
00081 * @since Version 1.0.0
00082 */
00083
00084 typedef enum : uintptr_t {
00085 /** @brief ELC module base address */
00086 k_elc_base_addr = 0x0008B100U,
00087
00088 /** @brief ELCR - Event Link Control Register */
00089 k_elc_elcr_addr = 0x0008B100U,
00090
00091 /** ELSRn - Event Link Setting Registers (n = peripheral specific) */
00092 k_elc_elsr0_addr = 0x0008B101U, /**< MTU0 */
00093 k_elc_elsr3_addr = 0x0008B104U, /**< MTU3 */
00094 k_elc_elsr4_addr = 0x0008B105U, /**< MTU4 */
00095 k_elc_elsr7_addr = 0x0008B108U, /**< CMT1 */
00096 k_elc_elsr10_addr = 0x0008B10BU, /**< TMR0 */
00097 k_elc_elsr11_addr = 0x0008B10CU, /**< TMR1 */
00098 k_elc_elsr12_addr = 0x0008B10DU, /**< TMR2 */
00099 k_elc_elsr13_addr = 0x0008B10EU, /**< TMR3 */
00100 k_elc_elsr15_addr = 0x0008B110U, /**< S12AD (ELCTRG00N) */
00101 k_elc_elsr16_addr = 0x0008B111U, /**< DA0 */
00102 k_elc_elsr18_addr = 0x0008B113U, /**< ICU Interrupt 1 */
00103 k_elc_elsr19_addr = 0x0008B114U, /**< ICU Interrupt 2 */
00104 k_elc_elsr20_addr = 0x0008B115U, /**< Output port group 1 */
00105 k_elc_elsr21_addr = 0x0008B116U, /**< Output port group 2 */
00106 k_elc_elsr22_addr = 0x0008B117U, /**< Input port group 1 */
00107 k_elc_elsr23_addr = 0x0008B118U, /**< Input port group 2 */
00108 k_elc_elsr24_addr = 0x0008B119U, /**< Single port 0 */
00109 k_elc_elsr25_addr = 0x0008B11AU, /**< Single port 1 */
00110 k_elc_elsr26_addr = 0x0008B11BU, /**< Single port 2 */
00111 k_elc_elsr27_addr = 0x0008B11CU, /**< Single port 3 */
00112 k_elc_elsr28_addr = 0x0008B11DU, /**< Clock source switch to LOCO */
00113 k_elc_elsr33_addr = 0x0008B131U, /**< CMTW0 */
00114 k_elc_elsr35_addr = 0x0008B133U, /**< TPU0 */
00115 k_elc_elsr36_addr = 0x0008B134U, /**< TPU1 */
00116 k_elc_elsr37_addr = 0x0008B135U, /**< TPU2 */
00117 k_elc_elsr38_addr = 0x0008B136U, /**< TPU3 */
00118 k_elc_elsr45_addr = 0x0008B13DU, /**< S12AD1 (ELCTRG10N) */
00119 k_elc_elsr48_addr = 0x0008B146U, /**< GPTW event source A */
00120 k_elc_elsr49_addr = 0x0008B147U, /**< GPTW event source B */
00121 k_elc_elsr50_addr = 0x0008B148U, /**< GPTW event source C */
00122 k_elc_elsr51_addr = 0x0008B149U, /**< GPTW event source D */
00123 k_elc_elsr52_addr = 0x0008B14AU, /**< GPTW event source E */
00124 k_elc_elsr53_addr = 0x0008B14BU, /**< GPTW event source F */
00125 k_elc_elsr54_addr = 0x0008B14CU, /**< GPTW event source G */
00126 k_elc_elsr55_addr = 0x0008B14DU, /**< GPTW event source H */
00127 k_elc_elsr56_addr = 0x0008B14EU, /**< S12AD (ELCTRG01N) */
00128 k_elc_elsr57_addr = 0x0008B14FU, /**< S12AD1 (ELCTRG11N) */
00129

```

```

00130 /* Option Setting Registers */
00131 k_elc_elopa_addr = 0x0008B11FU, /**< MTU0/MTU3 option setting */
00132 k_elc_elopb_addr = 0x0008B120U, /**< MTU4 option setting */
00133 k_elc_elopc_addr = 0x0008B121U, /**< CMT1 option setting */
00134 k_elc_elopd_addr = 0x0008B122U, /**< TMR0-3 option setting */
00135 k_elc_elopf_addr = 0x0008B13FU, /**< TPU0-3 option setting */
00136 k_elc_eloph_addr = 0x0008B141U, /**< CMTW0 option setting */
00137
00138 /* Port Event Registers */
00139 k_elc_pgr1_addr = 0x0008B123U, /**< Port Group Register 1 */
00140 k_elc_pgr2_addr = 0x0008B124U, /**< Port Group Register 2 */
00141 k_elc_pgc1_addr = 0x0008B125U, /**< Port Group Control 1 */
00142 k_elc_pgc2_addr = 0x0008B126U, /**< Port Group Control 2 */
00143 k_elc_pdbf1_addr = 0x0008B127U, /**< Port Data Buffer 1 */
00144 k_elc_pdbf2_addr = 0x0008B128U, /**< Port Data Buffer 2 */
00145 k_elc_pel0_addr = 0x0008B129U, /**< Port Event Link 0 */
00146 k_elc_pel1_addr = 0x0008B12AU, /**< Port Event Link 1 */
00147 k_elc_pel2_addr = 0x0008B12BU, /**< Port Event Link 2 */
00148 k_elc_pel3_addr = 0x0008B12CU, /**< Port Event Link 3 */
00149
00150 /* Software Event Generation */
00151 k_elc_elsegr_addr = 0x0008B12DU, /**< Software Event Generation */
00152 } elc_addresses_t;
00153
00154 /*
=====
00155 * ELCR Register Bits (Event Link Control)
00156 *
=====
00157 */
00158
00159 /**
00160 * @enum elcr_bits_t
00161 * @brief Event Link Control Register (ELCR) bit definitions
00162 *
00163 * @par Register Layout (8-bit @ 0x0008B100)
00164 * | Bit | Field | Description |
00165 * |-----|-----|-----|
00166 * | 6:0 | - | Reserved (write 1) |
00167 * | 7 | ELCON | All Event Link Enable |
00168 *
00169 * @note Manual ref: Ch21 section 21.2.1
00170 * @since Version 1.0.0
00171 */
00172 typedef enum : uint8_t {
00173 /** @brief Reserved bits mask (write 1) */
00174 k_elcr_reserved = 0x7FU,
00175
00176 /** @brief All Event Link Enable (0=disabled, 1=enabled) */
00177 k_elcr_elcon = 0x80U,
00178 } elcr_bits_t;
00179
00180 /*
=====
00181 * ELSRn Event Signal Values
00182 *
=====
00183 */
00184
00185 /**
00186 * @enum elsr_event_t
00187 * @brief Event signal values for ELSRn registers
00188 *
00189 * @details
00190 * Values to set in ELSRn.ELSR[7:0] to select the event signal source.
00191 * Only a subset of commonly-used events are defined here.
00192 *
00193 * @note Manual ref: Ch21 Table 21.3
00194 * @since Version 1.0.0
00195 */
00196 typedef enum : uint8_t {
00197 k_elsr_event_disabled = 0x00U, /**< Event output disabled */
00198
00199 /* MTU0 events (0x01-0x07) */
00200 k_elsr_mtu0_cmpa = 0x01U, /**< MTU0 compare match A */
00201 k_elsr_mtu0_cmpb = 0x02U, /**< MTU0 compare match B */
00202 k_elsr_mtu0_cmpc = 0x03U, /**< MTU0 compare match C */
00203 k_elsr_mtu0_cmpd = 0x04U, /**< MTU0 compare match D */
00204 k_elsr_mtu0_cmpe = 0x05U, /**< MTU0 compare match E */
00205 k_elsr_mtu0_cmpf = 0x06U, /**< MTU0 compare match F */
00206 k_elsr_mtu0_ovf = 0x07U, /**< MTU0 overflow */
00207
00208 /* MTU3 events (0x10-0x14) */
00209 k_elsr_mtu3_cmpa = 0x10U, /**< MTU3 compare match A */
00210 k_elsr_mtu3_cmpb = 0x11U, /**< MTU3 compare match B */
00211 k_elsr_mtu3_cmpc = 0x12U, /**< MTU3 compare match C */
00212 k_elsr_mtu3_cmpd = 0x13U, /**< MTU3 compare match D */

```

```

00213 k_elsr_mtu3_ovf = 0x14U, /**< MTU3 overflow */
00214
00215 /* MTU4 events (0x15-0x1A) */
00216 k_elsr_mtu4_cmpa = 0x15U, /**< MTU4 compare match A */
00217 k_elsr_mtu4_cmpb = 0x16U, /**< MTU4 compare match B */
00218 k_elsr_mtu4_cmpc = 0x17U, /**< MTU4 compare match C */
00219 k_elsr_mtu4_cmpd = 0x18U, /**< MTU4 compare match D */
00220 k_elsr_mtu4_ovf = 0x19U, /**< MTU4 overflow */
00221 k_elsr_mtu4_udf = 0x1AU, /**< MTU4 underflow */
00222
00223 /* CMT1 event (0x1F) */
00224 k_elsr_cmt1_cmp = 0x1FU, /**< CMT1 compare match */
00225
00226 /* TMR events (0x22-0x2D) */
00227 k_elsr_tmr0_cmpa = 0x22U, /**< TMR0 compare match A */
00228 k_elsr_tmr0_cmpb = 0x23U, /**< TMR0 compare match B */
00229 k_elsr_tmr0_ovf = 0x24U, /**< TMR0 overflow */
00230
00231 /* S12AD events (0x58, 0x6C) */
00232 k_elsr_s12ad0_end = 0x58U, /**< S12AD A/D conversion end */
00233 k_elsr_s12ad1_end = 0x6CU, /**< S12AD1 A/D conversion end */
00234
00235 /* DMAC transfer end events (0x5D-0x60) */
00236 k_elsr_dmac0_end = 0x5DU, /**< DMAC0 transfer end */
00237 k_elsr_dmac1_end = 0x5EU, /**< DMAC1 transfer end */
00238 k_elsr_dmac2_end = 0x5FU, /**< DMAC2 transfer end */
00239 k_elsr_dmac3_end = 0x60U, /**< DMAC3 transfer end */
00240
00241 /* DTC transfer end (0x61) */
00242 k_elsr_dtc_end = 0x61U, /**< DTC transfer end */
00243
00244 /* Port events (0x63-0x68) */
00245 k_elsr_port_grp1 = 0x63U, /**< Input port group 1 edge */
00246 k_elsr_port_grp2 = 0x64U, /**< Input port group 2 edge */
00247 k_elsr_port_single0 = 0x65U, /**< Single port 0 edge */
00248 k_elsr_port_single1 = 0x66U, /**< Single port 1 edge */
00249 k_elsr_port_single2 = 0x67U, /**< Single port 2 edge */
00250 k_elsr_port_single3 = 0x68U, /**< Single port 3 edge */
00251
00252 /* Software event (0x69) */
00253 k_elsr_software = 0x69U, /**< Software event */
00254
00255 /* GPTW0 events (0x80-0x87) */
00256 k_elsr_gptw0_cmpa = 0x80U, /**< GPTW0 compare match A */
00257 k_elsr_gptw0_cmpb = 0x81U, /**< GPTW0 compare match B */
00258 k_elsr_gptw0_cmpc = 0x82U, /**< GPTW0 compare match C */
00259 k_elsr_gptw0_cmpd = 0x83U, /**< GPTW0 compare match D */
00260 k_elsr_gptw0_cmpe = 0x84U, /**< GPTW0 compare match E */
00261 k_elsr_gptw0_cmpf = 0x85U, /**< GPTW0 compare match F */
00262 k_elsr_gptw0_ovf = 0x86U, /**< GPTW0 overflow */
00263 k_elsr_gptw0_udf = 0x87U, /**< GPTW0 underflow */
00264
00265 /* GPTW1 events (0x88-0x8F) */
00266 k_elsr_gptw1_cmpa = 0x88U, /**< GPTW1 compare match A */
00267 k_elsr_gptw1_cmpb = 0x89U, /**< GPTW1 compare match B */
00268 k_elsr_gptw1_cmpc = 0x8AU, /**< GPTW1 compare match C */
00269 k_elsr_gptw1_cmpd = 0x8BU, /**< GPTW1 compare match D */
00270 k_elsr_gptw1_cmpe = 0x8CU, /**< GPTW1 compare match E */
00271 k_elsr_gptw1_cmpf = 0x8DU, /**< GPTW1 compare match F */
00272 k_elsr_gptw1_ovf = 0x8EU, /**< GPTW1 overflow */
00273 k_elsr_gptw1_udf = 0x8FU, /**< GPTW1 underflow */
00274
00275 /* GPTW2 events (0x90-0x97) */
00276 k_elsr_gptw2_cmpa = 0x90U, /**< GPTW2 compare match A */
00277 k_elsr_gptw2_cmpb = 0x91U, /**< GPTW2 compare match B */
00278 k_elsr_gptw2_cmpc = 0x92U, /**< GPTW2 compare match C */
00279 k_elsr_gptw2_cmpd = 0x93U, /**< GPTW2 compare match D */
00280 k_elsr_gptw2_cmpe = 0x94U, /**< GPTW2 compare match E */
00281 k_elsr_gptw2_cmpf = 0x95U, /**< GPTW2 compare match F */
00282 k_elsr_gptw2_ovf = 0x96U, /**< GPTW2 overflow */
00283 k_elsr_gptw2_udf = 0x97U, /**< GPTW2 underflow */
00284
00285 /* GPTW3 events (0x98-0x9F) */
00286 k_elsr_gptw3_cmpa = 0x98U, /**< GPTW3 compare match A */
00287 k_elsr_gptw3_cmpb = 0x99U, /**< GPTW3 compare match B */
00288 k_elsr_gptw3_cmpc = 0x9AU, /**< GPTW3 compare match C */
00289 k_elsr_gptw3_cmpd = 0x9BU, /**< GPTW3 compare match D */
00290 k_elsr_gptw3_cmpe = 0x9CU, /**< GPTW3 compare match E */
00291 k_elsr_gptw3_cmpf = 0x9DU, /**< GPTW3 compare match F */
00292 k_elsr_gptw3_ovf = 0x9EU, /**< GPTW3 overflow */
00293 k_elsr_gptw3_udf = 0x9FU, /**< GPTW3 underflow */
00294 } elsr_event_t;
00295
00296 /*
=====
00297 * Timer Option Setting Bits
00298 *

```

```

=====
00299 */
00300
00301 /**
00302 * @enum elop_timer_op_t
00303 * @brief Timer operation selection bits for ELOPA/B/C/D/F/H registers
00304 *
00305 * @details
00306 * Each timer can be configured to perform different operations
00307 * when an event signal is received. These 2-bit fields are used
00308 * in the ELOPx registers.
00309 *
00310 * For MTU/TPU timers (ELOPA, ELOPF): 0x02 = Input capture (TCNT -> TGRA).
00311 * For CMT/TMR/CMTW timers (ELOPC, ELOPD, ELOPH): 0x02 = Event counter.
00312 *
00313 * @note Manual ref: Ch21 sections 21.2.3-21.2.8, Table 21.5
00314 * @since Version 1.0.0
00315 */
00316 typedef enum : uint8_t {
00317 k_elop_timer_count_start = 0x00U, /**< Counting is started */
00318 k_elop_timer_count_restart = 0x01U, /**< Counting is restarted */
00319 k_elop_timer_input_capture =
00320 0x02U, /**< Input capture (MTU/TPU) or event counter (CMT/TMR/CMTW) */
00321 k_elop_timer_disabled = 0x03U, /**< Event output is disabled */
00322 } elop_timer_op_t;
00323
00324 /**
00325 * @enum elopa_bits_t
00326 * @brief ELOPA register bit positions (MTU0/MTU3)
00327 *
00328 * @par Register Layout (8-bit @ 0x0008B11F)
00329 * | Bits | Field | Description |
00330 * |-----|-----|-----|
00331 * | 1:0 | MTU0MD | MTU0 operation setting |
00332 * | 5:2 | - | Reserved |
00333 * | 7:6 | MTU3MD | MTU3 operation setting |
00334 *
00335 * @since Version 1.0.0
00336 */
00337 typedef enum : uint8_t {
00338 k_elopa_mtu0op_shift = 0U, /**< MTU0 operation field shift (bits 1:0) */
00339 k_elopa_mtu0op_mask = 0x03U, /**< MTU0 operation field mask */
00340 k_elopa_mtu3op_shift = 6U, /**< MTU3 operation field shift (bits 7:6) */
00341 k_elopa_mtu3op_mask = 0xC0U, /**< MTU3 operation field mask */
00342 } elopa_bits_t;
00343
00344 /**
00345 * @enum elopb_bits_t
00346 * @brief ELOPB register bit positions (MTU4)
00347 *
00348 * @par Register Layout (8-bit @ 0x0008B120)
00349 * | Bits | Field | Description |
00350 * |-----|-----|-----|
00351 * | 1:0 | MTU4OP | MTU4 operation setting |
00352 * | 7:2 | - | Reserved |
00353 *
00354 * @since Version 1.0.0
00355 */
00356 typedef enum : uint8_t {
00357 k_elopb_mtu4op_shift = 0U, /**< MTU4 operation field shift */
00358 k_elopb_mtu4op_mask = 0x03U, /**< MTU4 operation field mask */
00359 } elopb_bits_t;
00360
00361 /**
00362 * @enum elopc_bits_t
00363 * @brief ELOPC register bit positions (CMT1)
00364 *
00365 * @par Register Layout (8-bit @ 0x0008B121)
00366 * | Bits | Field | Description |
00367 * |-----|-----|-----|
00368 * | 1:0 | - | Reserved |
00369 * | 3:2 | CMT1MD | CMT1 operation setting |
00370 * | 7:4 | - | Reserved |
00371 *
00372 * @since Version 1.0.0
00373 */
00374 typedef enum : uint8_t {
00375 k_elopc_cmt1op_shift = 2U, /**< CMT1 operation field shift (bits 3:2) */
00376 k_elopc_cmt1op_mask = 0x0CU, /**< CMT1 operation field mask */
00377 } elopc_bits_t;
00378
00379 /**
00380 * @enum elopd_bits_t
00381 * @brief ELOPD register bit positions (TMR0-3)
00382 *
00383 * @par Register Layout (8-bit @ 0x0008B122)
00384 * | Bits | Field | Description |

```



```

00385 * |-----|-----|-----|
00386 * | 1:0 | TMR0OP | TMR0 operation setting |
00387 * | 3:2 | TMR1OP | TMR1 operation setting |
00388 * | 5:4 | TMR2OP | TMR2 operation setting |
00389 * | 7:6 | TMR3OP | TMR3 operation setting |
00390 *
00391 * @since Version 1.0.0
00392 */
00393 typedef enum : uint8_t {
00394 k_elopd_tmr0op_shift = 0U, /**< TMR0 operation field shift */
00395 k_elopd_tmr0op_mask = 0x03U, /**< TMR0 operation field mask */
00396 k_elopd_tmr1op_shift = 2U, /**< TMR1 operation field shift */
00397 k_elopd_tmr1op_mask = 0x0CU, /**< TMR1 operation field mask */
00398 k_elopd_tmr2op_shift = 4U, /**< TMR2 operation field shift */
00399 k_elopd_tmr2op_mask = 0x30U, /**< TMR2 operation field mask */
00400 k_elopd_tmr3op_shift = 6U, /**< TMR3 operation field shift */
00401 k_elopd_tmr3op_mask = 0xC0U, /**< TMR3 operation field mask */
00402 } elopd_bits_t;
00403
00404 /*
=====
00405 * Port Event Registers
00406 *
=====
00407 */
00408
00409 /**
00410 * @enum pgc_bits_t
00411 * @brief Port Group Control (PGC1/PGC2) register bit definitions
00412 *
00413 * @par Register Layout (8-bit @ 0x0008B125/0x0008B126)
00414 * | Bits | Field | Description |
00415 * |-----|-----|-----|
00416 * | 1:0 | PGCI[1:0] | Event output edge select (input ports) |
00417 * | 2 | PGCOVE | PDBF overwrite enable |
00418 * | 3 | - | Reserved |
00419 * | 6:4 | PGCO[2:0] | Port group operation select |
00420 * | 7 | - | Reserved |
00421 *
00422 * @note Manual ref: Ch21 section 21.2.10
00423 * @since Version 1.0.0
00424 */
00425 typedef enum : uint8_t {
00426 /* PGCI[1:0] - Event Output Edge Select for input port groups (bits 1:0) */
00427 k_pgc_pgci_shift = 0U, /**< PGCI field shift */
00428 k_pgc_pgci_mask = 0x03U, /**< PGCI field mask */
00429 k_pgc_pgci_rising = 0x00U, /**< Event on rising edge of input signal */
00430 k_pgc_pgci_falling = 0x01U, /**< Event on falling edge of input signal */
00431 k_pgc_pgci_both = 0x02U, /**< Event on both edges of input signal (1x) */
00432 /* PGCOVE - PDBF Overwrite enable (bit 2) */
00433 k_pgc_pgcove = 0x04U, /**< PDBF overwrite enable (0=disabled, 1=enabled) */
00434 /* PGCO[2:0] - Port Group Operation Select for output port groups (bits 6:4) */
00435 k_pgc_pgco_shift = 4U, /**< PGCO field shift */
00436 k_pgc_pgco_mask = 0x70U, /**< PGCO field mask */
00437 k_pgc_pgco_low = 0x00U, /**< Output low when event signal is input */
00438 k_pgc_pgco_high = 0x10U, /**< Output high when event signal is input */
00439 k_pgc_pgco_toggle = 0x20U, /**< Toggle output when event signal is input */
00440 k_pgc_pgco_buffer = 0x30U, /**< Output buffer (PDBF) value when event signal is input */
00441 k_pgc_pgco_rotate = 0x40U, /**< Rotate output MSB-to-LSB when event signal is input (1xx) */
00442 } pgc_bits_t;
00443
00444 /**
00445 * @enum pel_bits_t
00446 * @brief Port Event Link (PEL0-3) register bit definitions
00447 *
00448 * @par Register Layout (8-bit @ 0x0008B129-0x0008B12C)
00449 * | Bits | Field | Description |
00450 * |-----|-----|-----|
00451 * | 2:0 | PSB[2:0] | Bit number for single port (pin 0-7) |
00452 * | 4:3 | PSP[1:0] | Port select (00=disabled, 01=Port B, 10=Port E) |
00453 * | 6:5 | PSM[1:0] | Event link spec (edge or output mode) |
00454 * | 7 | - | Reserved |
00455 *
00456 * @note Manual ref: Ch21 section 21.2.12
00457 * @since Version 1.0.0
00458 */
00459 typedef enum : uint8_t {
00460 /* PSB[2:0] - Bit Number Specification (bits 2:0) */
00461 k_pel_psb_mask = 0x07U, /**< 3-bit pin select mask (pins 0-7) */
00462 /* PSP[1:0] - Port Number Specification (bits 4:3) */
00463 k_pel_psp_shift = 3U, /**< PSP field shift */
00464 k_pel_psp_mask = 0x18U, /**< PSP field mask */
00465 k_pel_psp_disabled = 0x00U, /**< Setting disabled */
00466 k_pel_psp_portb = 0x08U, /**< Port B (corresponding to PGR1) */
00467 k_pel_psp_porte = 0x10U, /**< Port E (corresponding to PGR2) */
00468 /* PSM[1:0] - Event Link Specification (bits 6:5) */
00469 k_pel_psm_shift = 5U, /**< PSM field shift */

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00470 k_pel_psm_mask = 0x60U, /**< PSM field mask */
00471 k_pel_psm_rising = 0x00U, /**< Rising edge (input port) or output low (output port) */
00472 k_pel_psm_falling = 0x20U, /**< Falling edge (input port) or output high (output port) */
00473 k_pel_psm_both = 0x40U, /**< Both edges (input port) or toggle output (output port) (1x) */
00474 } pel_bits_t;
00475
00476 /**
00477 * @enum elsegr_bits_t
00478 * @brief Software Event Generation Register (ELSEGR) bit definitions
00479 *
00480 * @par Register Layout (8-bit @ 0x0008B12D)
00481 * | Bit | Field | Description |
00482 * |-----|-----|-----|
00483 * | 0 | SEG | Software Event Generation |
00484 * | 5:1 | - | Reserved |
00485 * | 6 | WE | Write Enable |
00486 * | 7 | WI | Write Disable |
00487 *
00488 * @note To write: Set WE=1, WI=0, then set SEG
00489 * @since Version 1.0.0
00490 */
00491 typedef enum : uint8_t {
00492 k_elsegr_seg = 0x01U, /**< Software event generation */
00493 k_elsegr_we = 0x40U, /**< Write enable */
00494 k_elsegr_wi = 0x80U, /**< Write disable (set 1 to disable) */
00495 } elsegr_bits_t;
00496
00497 /**
=====
00498 * Module Stop Control
00499 *
=====
00500 */
00501
00502 /**
00503 * @enum elc_mstpcr_t
00504 * @brief ELC module stop control bits
00505 *
00506 * @details
00507 * The ELC is controlled by MSTPCRB.MSTPB9.
00508 * Must be cleared (0) before accessing ELC registers.
00509 *
00510 * @note Manual ref: Ch11 (Low Power Consumption) module stop tables
00511 * @since Version 1.0.0
00512 */
00513 typedef enum : uint32_t {
00514 /**
00515 * @brief MSTPCRB bit for ELC (bit 9)
00516 * @details
00517 * - 0: ELC module is running
00518 * - 1: ELC module is stopped (default)
00519 */
00520 k_elc_mstpcrb_bit = (1U « 9U),
00521 } elc_mstpcr_t;
00522
00523 /**
=====
00524 * ELSRn Address Range Adjustments
00525 *
=====
00526 */
00527
00528 /**
00529 * @enum elsr_range_adj_t
00530 * @brief Address computation adjustments for the three ELSRn register ranges
00531 *
00532 * @details
00533 * ELSRn registers are NOT contiguous -- option and port registers occupy the
00534 * gaps. There are three separate address ranges:
00535 *
00536 * Range A n=0..28 : address = base + k_elsr_range_a_adj + n (0x0008B101..0x0008B11D)
00537 * Range B n=33..45 : address = base + k_elsr_range_b_adj + n (0x0008B131..0x0008B13D)
00538 * Range C n=48..57 : address = base + k_elsr_range_c_adj + n (0x0008B146..0x0008B14F)
00539 *
00540 * Derivation:
00541 * ELSR0 = 0x0008B101 = base + 1 + 0 -> adj_a = 1
00542 * ELSR33 = 0x0008B131 = base + 0x10 + 33 -> adj_b = 0x10
00543 * ELSR48 = 0x0008B146 = base + 0x16 + 48 -> adj_c = 0x16
00544 *
00545 * @note Manual ref: Ch21 section 21.2.2, address list
00546 * @since Version 1.0.0
00547 */
00548 typedef enum : uint8_t {
00549 k_elsr_range_a_adj = 0x01U, /**< Adjustment for n=0..28 */
00550 k_elsr_range_b_adj = 0x10U, /**< Adjustment for n=33..45 */
00551 k_elsr_range_c_adj = 0x16U, /**< Adjustment for n=48..57 */
00552 k_elsr_range_a_max = 28U, /**< Last valid n in range A */

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00553 k_elsr_range_b_max = 45U, /**< Last valid n in range B */
00554 k_elsr_range_c_min = 48U, /**< First valid n in range C */
00555 k_elsr_range_c_max = 57U, /**< Last valid n in range C */
00556 } elsr_range_adj_t;
00557
00558 /*
=====
00559 * Register Accessor Functions
00560 *
=====
00561 */
00562
00563 /**
00564 * @brief Get pointer to ELCR register (Event Link Control)
00565 * @return Volatile pointer to ELCR register (8-bit)
00566 * @since Version 1.0.0
00567 */
00568 static inline volatile uint8_t* elc_elcr_reg(void)
00569 {
00570 return (volatile uint8_t*)k_elc_elcr_addr;
00571 }
00572
00573 /**
00574 * @brief Get pointer to ELSRn register
00575 * @param n Register index (0, 3, 4, 7, 10-13, 15, 16, 18-28, 33, 35-38, 45, 48-57)
00576 * @return Volatile pointer to ELSRn register (8-bit)
00577 * @warning Not all indices in [0..57] are valid -- consult Table 21.2 for the
00578 * valid set. Passing an invalid index returns a pointer to a reserved
00579 * or wrong address.
00580 * @note ELSRn registers occupy three non-contiguous ranges; see elsr_range_adj_t.
00581 * @since Version 1.0.0
00582 */
00583 static inline volatile uint8_t* elc_elsr_reg(uint8_t n)
00584 {
00585 if (n <= k_elsr_range_a_max) {
00586 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_a_adj + n);
00587 }
00588 if ((n >= k_elsr_range_a_max + 1U) && (n <= k_elsr_range_b_max)) {
00589 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_b_adj + n);
00590 }
00591 if ((n >= k_elsr_range_c_min) && (n <= k_elsr_range_c_max)) {
00592 return (volatile uint8_t*)(uintptr_t)(k_elc_base_addr + k_elsr_range_c_adj + n);
00593 }
00594 return nullptr;
00595 }
00596
00597 /**
00598 * @brief Get pointer to ELSR0 register (MTU0)
00599 * @return Volatile pointer to ELSR0 register (8-bit)
00600 * @since Version 1.0.0
00601 */
00602 static inline volatile uint8_t* elc_elsr0_reg(void)
00603 {
00604 return (volatile uint8_t*)k_elc_elsr0_addr;
00605 }
00606
00607 /**
00608 * @brief Get pointer to ELSR15 register (S12AD ELCTRG00N)
00609 * @return Volatile pointer to ELSR15 register (8-bit)
00610 * @since Version 1.0.0
00611 */
00612 static inline volatile uint8_t* elc_elsr15_reg(void)
00613 {
00614 return (volatile uint8_t*)k_elc_elsr15_addr;
00615 }
00616
00617 /**
00618 * @brief Get pointer to ELSR45 register (S12AD1 ELCTRG10N)
00619 * @return Volatile pointer to ELSR45 register (8-bit)
00620 * @since Version 1.0.0
00621 */
00622 static inline volatile uint8_t* elc_elsr45_reg(void)
00623 {
00624 return (volatile uint8_t*)k_elc_elsr45_addr;
00625 }
00626
00627 /**
00628 * @brief Get pointer to ELSR48 register (GPTW event source A)
00629 * @return Volatile pointer to ELSR48 register (8-bit)
00630 * @since Version 1.0.0
00631 */
00632 static inline volatile uint8_t* elc_elsr48_reg(void)
00633 {
00634 return (volatile uint8_t*)k_elc_elsr48_addr;
00635 }
00636
00637 /**

```

```

00638 * @brief Get pointer to ELOPA register (MTU0/MTU3 option)
00639 * @return Volatile pointer to ELOPA register (8-bit)
00640 * @since Version 1.0.0
00641 */
00642 static inline volatile uint8_t* elc_elopa_reg(void)
00643 {
00644 return (volatile uint8_t*)k_elc_elopa_addr;
00645 }
00646
00647 /**
00648 * @brief Get pointer to ELOPB register (MTU4 option)
00649 * @return Volatile pointer to ELOPB register (8-bit)
00650 * @since Version 1.0.0
00651 */
00652 static inline volatile uint8_t* elc_elopb_reg(void)
00653 {
00654 return (volatile uint8_t*)k_elc_elopb_addr;
00655 }
00656
00657 /**
00658 * @brief Get pointer to PGR1 register (Port Group 1)
00659 * @return Volatile pointer to PGR1 register (8-bit)
00660 * @since Version 1.0.0
00661 */
00662 static inline volatile uint8_t* elc_pgr1_reg(void)
00663 {
00664 return (volatile uint8_t*)k_elc_pgr1_addr;
00665 }
00666
00667 /**
00668 * @brief Get pointer to PGR2 register (Port Group 2)
00669 * @return Volatile pointer to PGR2 register (8-bit)
00670 * @since Version 1.0.0
00671 */
00672 static inline volatile uint8_t* elc_pgr2_reg(void)
00673 {
00674 return (volatile uint8_t*)k_elc_pgr2_addr;
00675 }
00676
00677 /**
00678 * @brief Get pointer to PGC1 register (Port Group Control 1)
00679 * @return Volatile pointer to PGC1 register (8-bit)
00680 * @since Version 1.0.0
00681 */
00682 static inline volatile uint8_t* elc_pgc1_reg(void)
00683 {
00684 return (volatile uint8_t*)k_elc_pgc1_addr;
00685 }
00686
00687 /**
00688 * @brief Get pointer to PGC2 register (Port Group Control 2)
00689 * @return Volatile pointer to PGC2 register (8-bit)
00690 * @since Version 1.0.0
00691 */
00692 static inline volatile uint8_t* elc_pgc2_reg(void)
00693 {
00694 return (volatile uint8_t*)k_elc_pgc2_addr;
00695 }
00696
00697 /**
00698 * @brief Get pointer to PDBF1 register (Port Data Buffer 1)
00699 * @return Volatile pointer to PDBF1 register (8-bit)
00700 * @since Version 1.0.0
00701 */
00702 static inline volatile uint8_t* elc_pdbf1_reg(void)
00703 {
00704 return (volatile uint8_t*)k_elc_pdbf1_addr;
00705 }
00706
00707 /**
00708 * @brief Get pointer to PDBF2 register (Port Data Buffer 2)
00709 * @return Volatile pointer to PDBF2 register (8-bit)
00710 * @since Version 1.0.0
00711 */
00712 static inline volatile uint8_t* elc_pdbf2_reg(void)
00713 {
00714 return (volatile uint8_t*)k_elc_pdbf2_addr;
00715 }
00716
00717 /**
00718 * @brief Get pointer to PELn register (Port Event Link)
00719 * @param n Register index (0-3)
00720 * @return Volatile pointer to PELn register (8-bit)
00721 * @since Version 1.0.0
00722 */
00723 static inline volatile uint8_t* elc_pel_reg(uint8_t n)
00724 {

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```

00725 return (volatile uint8_t*)(uintptr_t)(k_elc_pel0_addr + n);
00726 }
00727
00728 /**
00729 * @brief Get pointer to ELSEGR register (Software Event Generation)
00730 * @return Volatile pointer to ELSEGR register (8-bit)
00731 * @since Version 1.0.0
00732 */
00733 static inline volatile uint8_t* elc_elsegr_reg(void)
00734 {
00735 return (volatile uint8_t*)k_elc_elsegr_addr;
00736 }
00737
00738 /*
=====
00739 * Static Assertions - Address Verification
00740 *
=====
00741 */
00742
00743 /**
00744 * @name Address Verification
00745 * @brief Compile-time verification of register addresses
00746 * @{
00747 */
00748
00749 /* Verify base address */
00750 static_assert(k_elc_base_addr == 0x0008B100U,
00751 "ELC base address must be 0x0008B100 per Ch21 manual");
00752
00753 /* Verify ELCR address */
00754 static_assert(k_elc_elcr_addr == 0x0008B100U, "ELCR address must be 0x0008B100 per Ch21 manual");
00755
00756 /* Verify ELSRn addresses */
00757 static_assert(k_elc_elsr0_addr == 0x0008B101U, "ELSR0 address must be 0x0008B101 per Ch21 manual");
00758
00759 static_assert(k_elc_elsr3_addr == 0x0008B104U, "ELSR3 address must be 0x0008B104 per Ch21 manual");
00760
00761 static_assert(k_elc_elsr4_addr == 0x0008B105U, "ELSR4 address must be 0x0008B105 per Ch21 manual");
00762
00763 static_assert(k_elc_elsr15_addr == 0x0008B110U,
00764 "ELSR15 address must be 0x0008B110 per Ch21 manual");
00765
00766 static_assert(k_elc_elsr45_addr == 0x0008B13DU,
00767 "ELSR45 address must be 0x0008B13D per Ch21 manual");
00768
00769 static_assert(k_elc_elsr48_addr == 0x0008B146U,
00770 "ELSR48 address must be 0x0008B146 per Ch21 manual");
00771
00772 static_assert(k_elc_elsr57_addr == 0x0008B14FU,
00773 "ELSR57 address must be 0x0008B14F per Ch21 manual");
00774
00775 /* Verify option setting register addresses */
00776 static_assert(k_elc_elopa_addr == 0x0008B11FU, "ELOPA address must be 0x0008B11F per Ch21 manual");
00777
00778 static_assert(k_elc_elopb_addr == 0x0008B120U, "ELOPB address must be 0x0008B120 per Ch21 manual");
00779
00780 static_assert(k_elc_elopc_addr == 0x0008B121U, "ELOPC address must be 0x0008B121 per Ch21 manual");
00781
00782 static_assert(k_elc_elopd_addr == 0x0008B122U, "ELOPD address must be 0x0008B122 per Ch21 manual");
00783
00784 static_assert(k_elc_elopf_addr == 0x0008B13FU, "ELOPF address must be 0x0008B13F per Ch21 manual");
00785
00786 static_assert(k_elc_eloph_addr == 0x0008B141U, "ELOPH address must be 0x0008B141 per Ch21 manual");
00787
00788 /* Verify port event register addresses */
00789 static_assert(k_elc_pgr1_addr == 0x0008B123U, "PGR1 address must be 0x0008B123 per Ch21 manual");
00790
00791 static_assert(k_elc_pgr2_addr == 0x0008B124U, "PGR2 address must be 0x0008B124 per Ch21 manual");
00792
00793 static_assert(k_elc_pgc1_addr == 0x0008B125U, "PGC1 address must be 0x0008B125 per Ch21 manual");
00794
00795 static_assert(k_elc_pgc2_addr == 0x0008B126U, "PGC2 address must be 0x0008B126 per Ch21 manual");
00796
00797 static_assert(k_elc_pdbf1_addr == 0x0008B127U, "PDBF1 address must be 0x0008B127 per Ch21 manual");
00798
00799 static_assert(k_elc_pdbf2_addr == 0x0008B128U, "PDBF2 address must be 0x0008B128 per Ch21 manual");
00800
00801 static_assert(k_elc_pel0_addr == 0x0008B129U, "PEL0 address must be 0x0008B129 per Ch21 manual");
00802
00803 static_assert(k_elc_pel1_addr == 0x0008B12AU, "PEL1 address must be 0x0008B12A per Ch21 manual");
00804
00805 static_assert(k_elc_pel2_addr == 0x0008B12BU, "PEL2 address must be 0x0008B12B per Ch21 manual");
00806
00807 static_assert(k_elc_pel3_addr == 0x0008B12CU, "PEL3 address must be 0x0008B12C per Ch21 manual");
00808
00809 static_assert(k_elc_elsegr_addr == 0x0008B12DU,

```

```

00810 "ELSEGR address must be 0x0008B12D per Ch21 manual");
00811
00812 /* Verify bit definitions */
00813 static_assert(k_elcr_elcon == 0x80U, "ELCR.ELCON must be bit 7");
00814
00815 static_assert(k_elc_mstpcrb_bit == (1U << 9U), "ELC module stop is MSTPCRB bit 9");
00816
00817 /** @} */
00818
00819 #ifdef __cplusplus
00820 }
00821 #endif

```

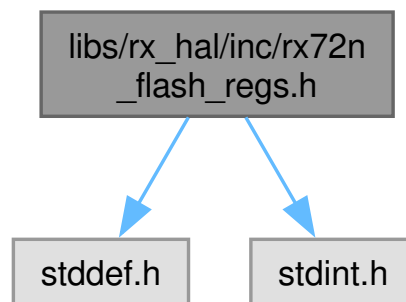
## 8.23 libs/rx\_hal/inc/rx72n\_flash\_regs.h File Reference

RX72N Flash Memory Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_flash\_regs.h:



This graph shows which files directly or indirectly include this file:



### Enumerations

- enum `rx_rom_cache_addresses_t` : `uintptr_t` {  
`k_rom_cache_base_addr` = 0x00081000 , `k_romce_addr` = 0x00081000 , `k_romciv_addr` = 0x00081004 , `k_ncrg0_addr` = 0x00081040 ,  
`k_ncrc0_addr` = 0x00081044 , `k_ncrg1_addr` = 0x00081048 , `k_ncrc1_addr` = 0x0008104C }  
ROM cache register addresses.
- enum `rx_romce_bits_t` : `uint16_t` { `k_romce_romcen` = (1U << 0) }  
ROM Cache Enable Register (ROMCE) bit definitions.
- enum `rx_romciv_bits_t` : `uint16_t` { `k_romciv_romciv` = (1U << 0) }  
ROM Cache Invalidate Register (ROMCIV) bit definitions.
- enum `rx_fwepror_addresses_t` : `uintptr_t` { `k_fwepror_addr` = 0x0008C296 }  
Flash Write/Erase Protect Register addresses.
- enum `rx_fwepror_bits_t` : `uint8_t` { `k_fwepror_flwe_mask` = 0x03 , `k_fwepror_flwe_disable` = 0x00 , `k_fwepror_flwe_enable` = 0x01 , `k_fwepror_reset_value` = 0x02 }  
Flash P/E Protect Register (FWEPROR) bit definitions.

- enum `rx_faci_addresses_t` : `uintptr_t` {  
`k_faci_base_addr` = 0x007FE000 , `k_fastat_addr` = 0x007FE010 , `k_faeint_addr` = 0x007FE014 ,  
`k_frдыie_addr` = 0x007FE018 ,  
`k_fsaddr_addr` = 0x007FE030 , `k_feaddr_addr` = 0x007FE034 , `k_fstattr_addr` = 0x007FE080 ,  
`k_fentryr_addr` = 0x007FE084 ,  
`k_fsuint_r_addr` = 0x007FE08C , `k_fcmdir_addr` = 0x007FE0A0 , `k_fbcnt_addr` = 0x007FE0D0 ,  
`k_fbcstat_addr` = 0x007FE0D4 ,  
`k_fpsaddr_addr` = 0x007FE0D8 , `k_fawmon_addr` = 0x007FE0DC , `k_fcpsr_addr` = 0x007FE0E0 ,  
`k_fpckar_addr` = 0x007FE0E4 ,  
`k_fsuacr_addr` = 0x007FE0E8 , `k_faci_cmd_area` = 0x007E0000 }  
 FACI (Flash Application Command Interface) register addresses.
- enum `rx_eepfclk_addresses_t` : `uintptr_t` { `k_eepfclk_addr` = 0x007FC040 }  
 Data Flash Memory Access Frequency register address.
- enum `rx_fastat_bits_t` : `uint8_t` {  
`k_fastat_dfae_shift` = 3 , `k_fastat_cmdlk_shift` = 4 , `k_fastat_cfae_shift` = 7 , `k_fastat_dfae` = (1U << 3) ,  
`k_fastat_cmdlk` = (1U << 4) , `k_fastat_cfae` = (1U << 7) }  
 Flash Access Status Register (FASTAT) bit definitions.
- enum `rx_faeint_bits_t` : `uint8_t` {  
`k_faeint_dfaeie_shift` = 3 , `k_faeint_cmdlkie_shift` = 4 , `k_faeint_cfaeie_shift` = 7 ,  
`k_faeint_dfaeie` = (1U << 3) ,  
`k_faeint_cmdlkie` = (1U << 4) , `k_faeint_cfaeie` = (1U << 7) , `k_faeint_reset_value` = 0x98 }  
 Flash Access Error Interrupt Enable Register (FAEINT) bit definitions.
- enum `rx_frдыie_bits_t` : `uint8_t` { `k_frдыie_frдыie` = (1U << 0) }  
 Flash Ready Interrupt Enable Register (FRDYIE) bit definitions.
- enum `rx_fstattr_bits_t` : `uint32_t` {  
`k_fstattr_flweerr_shift` = 6 , `k_fstattr_prgspd_shift` = 8 , `k_fstattr_ersspd_shift` = 9 ,  
`k_fstattr_dbfull_shift` = 10 ,  
`k_fstattr_susrdy_shift` = 11 , `k_fstattr_prgerr_shift` = 12 , `k_fstattr_erserr_shift` = 13 ,  
`k_fstattr_ilglerr_shift` = 14 ,  
`k_fstattr_frды_shift` = 15 , `k_fstattr_oterr_shift` = 20 , `k_fstattr_secerr_shift` = 21 ,  
`k_fstattr_feseterr_shift` = 22 ,  
`k_fstattr_ilgcomerr_shift` = 23 , `k_fstattr_flweerr` = (1UL << 6) , `k_fstattr_prgspd` = (1UL << 8) ,  
`k_fstattr_ersspd` = (1UL << 9) ,  
`k_fstattr_dbfull` = (1UL << 10) , `k_fstattr_susrdy` = (1UL << 11) , `k_fstattr_prgerr` = (1UL << 12) ,  
`k_fstattr_erserr` = (1UL << 13) ,  
`k_fstattr_ilglerr` = (1UL << 14) , `k_fstattr_frды` = (1UL << 15) , `k_fstattr_oterr` = (1UL << 20) ,  
`k_fstattr_secerr` = (1UL << 21) ,  
`k_fstattr_feseterr` = (1UL << 22) , `k_fstattr_ilgcomerr` = (1UL << 23) , `k_fstattr_error_mask` ,  
`k_fstattr_reset_value` = 0x00008000 }  
 Flash Status Register (FSTATTR) bit definitions.
- enum `rx_fentryr_bits_t` : `uint16_t` {  
`k_fentryr_fentryc` = (1U << 0) , `k_fentryr_fentryd` = (1U << 7) , `k_fentryr_key_shift` = 8 ,  
`k_fentryr_key` = 0xAA00 ,  
`k_fentryr_code_pe` = 0xAA01 , `k_fentryr_data_pe` = 0xAA80 , `k_fentryr_read_mode` = 0xAA00 }  
 Flash P/E Mode Entry Register (FENTRYR) bit definitions.
- enum `rx_fsuint_r_bits_t` : `uint16_t` { `k_fsuint_r_suint` = (1U << 0) , `k_fsuint_r_key` = 0x2D00 ,  
`k_fsuint_r_init_cmd` = 0x2D01 }  
 Flash Sequencer Setup Initialization Register (FSUINITR) bit defs.
- enum `rx_fcmdir_bits_t` : `uint16_t` { `k_fcmdir_pcmdr_mask` = 0x00FF , `k_fcmdir_pcmdr_shift` = 0 ,  
`k_fcmdir_cmdr_mask` = 0xFF00 , `k_fcmdir_cmdr_shift` = 8 }  
 FACI Command Register (FCMDR) bit definitions.
- enum `rx_faci_commands_t` : `uint8_t` {  
`k_faci_cmd_program` = 0xE8 , `k_faci_cmd_block_erase` = 0x20 , `k_faci_cmd_pe_suspend` = 0xB0 ,  
`k_faci_cmd_pe_resume` = 0xD0 ,



```
k_faci_cmd_status_clear = 0x50 , k_faci_cmd_forced_stop = 0xB3 , k_faci_cmd_blank_check
= 0x71 , k_faci_cmd_config_set = 0x40 ,
k_faci_cmd_d0 = 0xD0 }
```

FACI command codes.

- enum `rx_fpckar_bits_t` : `uint16_t` { `k_fpckar_pcka_mask` = 0x00FF , `k_fpckar_key_shift` = 8 , `k_fpckar_key` = 0x1E00 }

Flash Sequencer Clock Notification Register (FPCKAR) bit definitions.

- enum `rx_fbcent_bits_t` : `uint8_t` { `k_fbcent_bcdir` = (1U << 0) }

Blank Check Control Register (FBCCNT) bit definitions.

- enum `rx_fbcstat_bits_t` : `uint8_t` { `k_fbcstat_bcst` = (1U << 0) }

Blank Check Status Register (FBCSTAT) bit definitions.

- enum `rx_uidr_addresses_t` : `uintptr_t` { `k_uidr0_addr` = 0xFE7F7D90 , `k_uidr1_addr` = 0xFE7F7D94 , `k_uidr2_addr` = 0xFE7F7D98 , `k_uidr3_addr` = 0xFE7F7D9C }

Unique ID Register addresses (Ch62 Section 62.4.23).

- enum `rx_flash_memory_addresses_t` : `uintptr_t` { `k_code_flash_start` = 0xFFC00000 , `k_code_flash_end` = 0xFFFFFFFF , `k_data_flash_start` = 0x00100000 , `k_data_flash_end` = 0x00107FFF }

Flash memory region base and end addresses for the RX72N.

- enum `rx_flash_memory_layout_t` : `uint32_t` { `k_code_flash_size` = 0x00400000 , `k_data_flash_size` = 0x00008000 , `k_code_flash_block_8k` = 0x00002000 , `k_code_flash_block_32k` = 0x00008000 , `k_data_flash_block_64` = 0x00000040 , `k_code_flash_program_unit` , `k_data_flash_program_unit` }

Flash memory region sizes, erase block granularities, and program units.

## Functions

- static volatile `uint16_t` \* `romce_reg` (void)  
Get pointer to ROM Cache Enable Register (ROMCE).
- static volatile `uint16_t` \* `romciv_reg` (void)  
Get pointer to ROM Cache Invalidate Register (ROMCIV).
- static volatile `uint8_t` \* `fweprror_reg` (void)  
Get pointer to Flash P/E Protect Register (FWEPROR).
- static volatile `uint8_t` \* `fastat_reg` (void)  
Get pointer to Flash Access Status Register (FASTAT).
- static volatile `uint8_t` \* `faeint_reg` (void)  
Get pointer to Flash Access Error Interrupt Enable (FAEINT).
- static volatile `uint8_t` \* `frdyie_reg` (void)  
Get pointer to Flash Ready Interrupt Enable (FRDYIE).
- static volatile `uint32_t` \* `fstatr_reg` (void)  
Get pointer to Flash Status Register (FSTATR).
- static volatile `uint16_t` \* `fentryr_reg` (void)  
Get pointer to Flash P/E Mode Entry Register (FENTRYR).
- static volatile `uint16_t` \* `fsuinitr_reg` (void)  
Get pointer to Flash Sequencer Setup Init Register (FSUINITR).
- static volatile `uint32_t` \* `fsaddr_reg` (void)  
Get pointer to FACI Start Address Register (FSADDR).
- static volatile `uint32_t` \* `feaddr_reg` (void)  
Get pointer to FACI End Address Register (FEADDR).
- static volatile `uint16_t` \* `fcmdr_reg` (void)  
Get pointer to FACI Command Register (FCMDR).

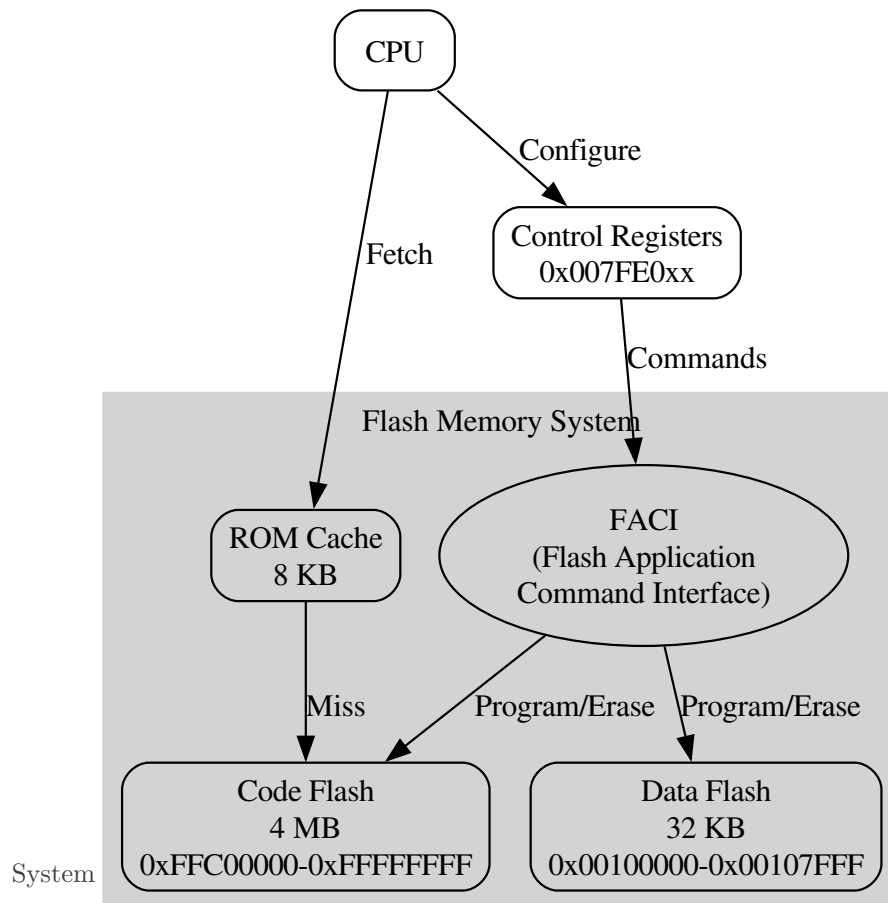


- static volatile uint8\_t \* [faci\\_cmd\\_area](#) (void)  
Get pointer to FACI Command Issuing Area.
- static volatile uint16\_t \* [fpckar\\_reg](#) (void)  
Get pointer to Flash Sequencer Clock Register (FPCKAR).
- static volatile uint8\_t \* [eepfclk\\_reg](#) (void)  
Get pointer to Data Flash Access Frequency Register (EPPFCLK).
- static volatile uint8\_t \* [fbccnt\\_reg](#) (void)  
Get pointer to Blank Check Control Register (FBCCNT).
- static volatile uint8\_t \* [fbcstat\\_reg](#) (void)  
Get pointer to Blank Check Status Register (FBCSTAT).
- static volatile uint32\_t \* [fpsaddr\\_reg](#) (void)  
Get pointer to Data Flash Programming Start Address (FPSADDR).
- static volatile uint32\_t \* [fawmon\\_reg](#) (void)  
Get pointer to Flash Access Window Monitor (FAWMON).
- static volatile uint16\_t \* [fcpsr\\_reg](#) (void)  
Get pointer to Flash Sequencer Processing Switch (FCPSR).
- static volatile uint16\_t \* [fsuacr\\_reg](#) (void)  
Get pointer to Flash Startup Area Select (FSUACR).
- static volatile const uint32\_t \* [uidr0\\_reg](#) (void)  
Get pointer to Unique ID Register 0 (UIDR0).
- static volatile const uint32\_t \* [uidr1\\_reg](#) (void)  
Get pointer to Unique ID Register 1 (UIDR1).
- static volatile const uint32\_t \* [uidr2\\_reg](#) (void)  
Get pointer to Unique ID Register 2 (UIDR2).
- static volatile const uint32\_t \* [uidr3\\_reg](#) (void)  
Get pointer to Unique ID Register 3 (UIDR3).

### 8.23.1 Detailed Description

#### RX72N Flash Memory Register Definitions.

Register definitions for the Flash Memory module including ROM cache control, flash programming/erase operations (FACI), and access protection. The RX72N has 4MB code flash and 32KB data flash with hardware-accelerated operations.



### Flash Memory Specifications

| Parameter     | Code Flash         | Data Flash          |
|---------------|--------------------|---------------------|
| Capacity      | 4 MB               | 32 KB               |
| Address Range | FFC00000-FFFF FFFF | 0010 0000-0010 7FFF |
| Program Unit  | 128 bytes          | 4 bytes             |
| Erase Unit    | 8/32 KB block      | 64 bytes            |
| Read Latency  | 1-3 cycles         | FCLK based          |

### FACI Commands (via 0x007E0000)

| Command      | Code | Description             |
|--------------|------|-------------------------|
| Program      | 0xE8 | Write data to flash     |
| Block Erase  | 0x20 | Erase a single block    |
| Blank Check  | 0x71 | Verify area is erased   |
| Status Clear | 0x50 | Clear error status      |
| Forced Stop  | 0xB3 | Abort current operation |

| Command    | Code | Description                |
|------------|------|----------------------------|
| PE Suspend | 0xB0 | Suspend program/erase      |
| PE Resume  | 0xD0 | Resume suspended operation |

#### Module Dependencies

- Module stop: MSTPCRB.MSTPB31 (FCU) - already running at boot

#### Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 62: Flash Memory (FLASH), pages 2992-3065

#### Note

The flash sequencer (FACI) must be in P/E mode before issuing commands

#### Warning

Programming/erasing code flash while executing from it is restricted

#### Attention

Enable FWEPROR before any programming operations

#### NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 3: No dynamic memory allocation
- Rule 8: All constants use C23 typed enums
- Rule 10: Compiles with -Wall -Wextra -Werror

#### See also

[rx\\_flash.h](#) Higher-level flash API (when implemented)

[rx72n\\_regs.h](#) Main register include file

#### Author

Locked, Inc.

#### Date

2026-01-28

#### Copyright

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#### Since

Version 1.0.0

Definition in file [rx72n\\_flash\\_regs.h](#).

## 8.23.2 Enumeration Type Documentation

### 8.23.2.1 rx\_eepfclk\_addresses\_t

```
enum rx_eepfclk_addresses_t : uintptr_t
```

Data Flash Memory Access Frequency register address.

Since

Version 1.0.0

Enumerator

|                |                                     |
|----------------|-------------------------------------|
| k_eepfclk_addr | Data Flash Access Frequency Setting |
|----------------|-------------------------------------|

Definition at line 287 of file rx72n\_flash\_regs.h.

```
00287 : uintptr_t {
00288 k_eepfclk_addr = 0x007FC040, /**< Data Flash Access Frequency Setting */
00289 } rx_eepfclk_addresses_t;
```

### 8.23.2.2 rx\_faci\_addresses\_t

```
enum rx_faci_addresses_t : uintptr_t
```

FACI (Flash Application Command Interface) register addresses.

FACI registers control flash programming and erase operations. All registers are in the 0x007FE0xx range.

Memory Map

| Address    | Register | Size | Description                          |
|------------|----------|------|--------------------------------------|
| 0x007FE010 | FASTAT   | 8    | Flash Access Status                  |
| 0x007FE014 | FAEINT   | 8    | Flash Access Error Int Enable        |
| 0x007FE018 | FRDYIE   | 8    | Flash Ready Interrupt Enable         |
| 0x007FE030 | FSADDR   | 32   | FACI Start Address                   |
| 0x007FE034 | FEADDR   | 32   | FACI End Address                     |
| 0x007FE080 | FSTATR   | 32   | Flash Status Register                |
| 0x007FE084 | FENTRYR  | 16   | Flash P/E Mode Entry                 |
| 0x007FE08C | FSUINTR  | 16   | Flash Sequencer Setup Init           |
| 0x007FE0A0 | FCMDR    | 16   | FACI Command Register                |
| 0x007FE0D0 | FBCCNT   | 8    | Blank Check Control                  |
| 0x007FE0D4 | FBCSTAT  | 8    | Blank Check Status                   |
| 0x007FE0D8 | FPSADDR  | 32   | Data Flash Programming Start Address |

| Address    | Register | Size | Description                       |
|------------|----------|------|-----------------------------------|
| 0x007FE0DC | FAWMON   | 32   | Flash Access Window Monitor       |
| 0x007FE0E0 | FCPSR    | 16   | Flash Sequencer Processing Switch |
| 0x007FE0E4 | FPCKAR   | 16   | Flash Sequencer Clock Notify      |
| 0x007FE0E8 | FSUACR   | 16   | Flash Startup Area Select         |

Since

Version 1.0.0

Enumerator

|                  |                                      |
|------------------|--------------------------------------|
| k_faci_base_addr | FACI register base (conceptual)      |
| k_fastat_addr    | Flash Access Status Register         |
| k_faieint_addr   | Flash Access Error Interrupt Enable  |
| k_frdyie_addr    | Flash Ready Interrupt Enable         |
| k_fsaddr_addr    | FACI Start Address Register          |
| k_feaddr_addr    | FACI End Address Register            |
| k_fstatr_addr    | Flash Status Register                |
| k_fentryr_addr   | Flash P/E Mode Entry Register        |
| k_fsuinitr_addr  | Flash Sequencer Setup Init Register  |
| k_fcmdr_addr     | FACI Command Register                |
| k_fbcnt_addr     | Blank Check Control Register         |
| k_fbcstat_addr   | Blank Check Status Register          |
| k_fpsaddr_addr   | Data Flash Programming Start Address |
| k_fawmon_addr    | Flash Access Window Monitor          |
| k_fcpsr_addr     | Flash Sequencer Processing Switch    |
| k_fpckar_addr    | Flash Sequencer Clock Frequency      |
| k_fsuacr_addr    | Flash Startup Area Select            |
| k_faci_cmd_area  | FACI Command Issuing Area            |

Definition at line 260 of file rx72n\_flash\_regs.h.

```

00260 : uintptr_t {
00261 k_faci_base_addr = 0x007FE000, /**< FACI register base (conceptual) */
00262 k_fastat_addr = 0x007FE010, /**< Flash Access Status Register */
00263 k_faieint_addr = 0x007FE014, /**< Flash Access Error Interrupt Enable */
00264 k_frdyie_addr = 0x007FE018, /**< Flash Ready Interrupt Enable */
00265 k_fsaddr_addr = 0x007FE030, /**< FACI Start Address Register */
00266 k_feaddr_addr = 0x007FE034, /**< FACI End Address Register */
00267 k_fstatr_addr = 0x007FE080, /**< Flash Status Register */
00268 k_fentryr_addr = 0x007FE084, /**< Flash P/E Mode Entry Register */
00269 k_fsuinitr_addr = 0x007FE08C, /**< Flash Sequencer Setup Init Register */
00270 k_fcmdr_addr = 0x007FE0A0, /**< FACI Command Register */
00271 k_fbcnt_addr = 0x007FE0D0, /**< Blank Check Control Register */
00272 k_fbcstat_addr = 0x007FE0D4, /**< Blank Check Status Register */
00273 k_fpsaddr_addr = 0x007FE0D8, /**< Data Flash Programming Start Address */
00274 k_fawmon_addr = 0x007FE0DC, /**< Flash Access Window Monitor */
00275 k_fcpsr_addr = 0x007FE0E0, /**< Flash Sequencer Processing Switch */
00276 k_fpckar_addr = 0x007FE0E4, /**< Flash Sequencer Clock Frequency */
00277 k_fsuacr_addr = 0x007FE0E8, /**< Flash Startup Area Select */
00278 k_faci_cmd_area = 0x007FE0000, /**< FACI Command Issuing Area */
00279 } rx_faci_addresses_t;

```

### 8.23.2.3 rx'faci'commands't

```
enum rx_faci_commands_t : uint8_t
```

FACI command codes.

Commands are written to the FACI command issuing area @ 0x007E0000. Command sequences vary; see manual Ch62 Section 62.8.1 for details.

Since

Version 1.0.0

Enumerator

|                         |                           |
|-------------------------|---------------------------|
| k_faci_cmd_program      | Program command           |
| k_faci_cmd_block_erase  | Block erase command       |
| k_faci_cmd_pe_suspend   | P/E suspend command       |
| k_faci_cmd_pe_resume    | P/E resume command        |
| k_faci_cmd_status_clear | Status clear command      |
| k_faci_cmd_forced_stop  | Forced stop command       |
| k_faci_cmd_blank_check  | Blank check command       |
| k_faci_cmd_config_set   | Configuration set command |
| k_faci_cmd_d0           | Command terminator        |

Definition at line 625 of file rx72n\_flash\_regs.h.

```
00625 : uint8_t {
00626 k_faci_cmd_program = 0xE8, /**< Program command */
00627 k_faci_cmd_block_erase = 0x20, /**< Block erase command */
00628 k_faci_cmd_pe_suspend = 0xB0, /**< P/E suspend command */
00629 k_faci_cmd_pe_resume = 0xD0, /**< P/E resume command */
00630 k_faci_cmd_status_clear = 0x50, /**< Status clear command */
00631 k_faci_cmd_forced_stop = 0xB3, /**< Forced stop command */
00632 k_faci_cmd_blank_check = 0x71, /**< Blank check command */
00633 k_faci_cmd_config_set = 0x40, /**< Configuration set command */
00634 k_faci_cmd_d0 = 0xD0, /**< Command terminator */
00635 } rx_faci_commands_t;
```

### 8.23.2.4 rx'faeint'bits't

```
enum rx_faeint_bits_t : uint8_t
```

Flash Access Error Interrupt Enable Register (FAEINT) bit definitions.

FAEINT @ 0x007FE014 enables FIFERR interrupt generation. Reset value: 0x98 (all interrupts enabled)

Since

Version 1.0.0

Enumerator

|                        |                                |
|------------------------|--------------------------------|
| k_faeint_dfaeie_shift  | DFAEIE bit position            |
| k_faeint_cmdlkie_shift | CMDLKIE bit position           |
| k_faeint_cfaeie_shift  | CFAEIE bit position            |
| k_faeint_dfaeie        | Data Flash Access Violation IE |
| k_faeint_cmdlkie       | Command Lock Interrupt Enable  |
| k_faeint_cfaeie        | Code Flash Access Violation IE |
| k_faeint_reset_value   | Reset value (all enabled)      |

Definition at line 351 of file [rx72n\\_flash\\_regs.h](#).

```

00351 : uint8_t {
00352 k_faeint_dfaeie_shift = 3, /**< DFAEIE bit position */
00353 k_faeint_cmdlkie_shift = 4, /**< CMDLKIE bit position */
00354 k_faeint_cfaeie_shift = 7, /**< CFAEIE bit position */
00355
00356 k_faeint_dfaeie = (1U « 3), /**< Data Flash Access Violation IE */
00357 k_faeint_cmdlkie = (1U « 4), /**< Command Lock Interrupt Enable */
00358 k_faeint_cfaeie = (1U « 7), /**< Code Flash Access Violation IE */
00359
00360 k_faeint_reset_value = 0x98, /**< Reset value (all enabled) */
00361 } rx_faeint_bits_t;

```

#### 8.23.2.5 rx`fastat`bits`t

```
enum rx_fastat_bits_t : uint8_t
```

Flash Access Status Register (FASTAT) bit definitions.

FASTAT @ 0x007FE010 indicates flash access violations.

Bit Layout (8-bit)

| Bit     | Name  | R/W | Description                     |
|---------|-------|-----|---------------------------------|
| 7       | CFAE  | R/W | Code Flash Access Violation     |
| 6↔<br>5 | -     | R   | Reserved                        |
| 4       | CMDLK | R   | Command Lock (sequencer locked) |
| 3       | DFAE  | R/W | Data Flash Access Violation     |
| 2↔<br>0 | -     | R   | Reserved                        |

Note

Write 0 to clear CFAE/DFAE after reading 1

Since

Version 1.0.0

Enumerator

|                      |                             |
|----------------------|-----------------------------|
| k_fastat_dfae_shift  | DFAE bit position           |
| k_fastat_cmdlk_shift | CMDLK bit position          |
| k_fastat_cfae_shift  | CFAE bit position           |
| k_fastat_dfae        | Data Flash Access Violation |
| k_fastat_cmdlk       | Command Lock Flag           |
| k_fastat_cfae        | Code Flash Access Violation |

Definition at line 316 of file [rx72n\\_flash\\_regs.h](#).

```

00316 : uint8_t {
00317 k_fastat_dfae_shift = 3, /**< DFAE bit position */
00318 k_fastat_cmdlk_shift = 4, /**< CMDLK bit position */
00319 k_fastat_cfae_shift = 7, /**< CFAE bit position */
00320
00321 k_fastat_dfae = (1U « 3), /**< Data Flash Access Violation */
00322 k_fastat_cmdlk = (1U « 4), /**< Command Lock Flag */
00323 k_fastat_cfae = (1U « 7), /**< Code Flash Access Violation */
00324 } rx_fastat_bits_t;

```

#### 8.23.2.6 rx\_fbcnt\_bits\_t

```
enum rx_fbcnt_bits_t : uint8_t
```

Blank Check Control Register (FBCCNT) bit definitions.

Since

Version 1.0.0

Enumerator

|               |                                              |
|---------------|----------------------------------------------|
| k_fbcnt_bcdir | Blank Check Direction (0=forward, 1=reverse) |
|---------------|----------------------------------------------|

Definition at line 714 of file [rx72n\\_flash\\_regs.h](#).

```

00714 : uint8_t {
00715 k_fbcnt_bcdir = (1U « 0), /**< Blank Check Direction (0=forward, 1=reverse) */
00716 } rx_fbcnt_bits_t;

```

#### 8.23.2.7 rx\_fbcstat\_bits\_t

```
enum rx_fbcstat_bits_t : uint8_t
```

Blank Check Status Register (FBCSTAT) bit definitions.



Since

Version 1.0.0

Enumerator

|               |                                           |
|---------------|-------------------------------------------|
| k_fbcstat_bcs | Blank Check Status (0=blank, 1=not blank) |
|---------------|-------------------------------------------|

Definition at line 739 of file [rx72n\\_flash\\_regs.h](#).

```
00739 : uint8_t {
00740 k_fbcstat_bcs = (1U << 0), /**< Blank Check Status (0=blank, 1=not blank) */
00741 } rx_fbcstat_bits_t;
```

#### 8.23.2.8 rx`fcmr`bits`

```
enum rx_fcmr_bits_t : uint16_t
```

FACI Command Register (FCMDR) bit definitions.

FCMDR @ 0x007FE0A0 contains the two most recent commands received. Read-only register for debugging.

Since

Version 1.0.0

Enumerator

|                   |                                  |
|-------------------|----------------------------------|
| k_fcmr_pcmr_mask  | Previous command mask (bits 7:0) |
| k_fcmr_pcmr_shift | Previous command shift           |
| k_fcmr_cmdr_mask  | Current command mask (bits 15:8) |
| k_fcmr_cmdr_shift | Current command shift            |

Definition at line 593 of file [rx72n\\_flash\\_regs.h](#).

```
00593 : uint16_t {
00594 k_fcmr_pcmr_mask = 0x00FF, /**< Previous command mask (bits 7:0) */
00595 k_fcmr_pcmr_shift = 0, /**< Previous command shift */
00596 k_fcmr_cmdr_mask = 0xFF00, /**< Current command mask (bits 15:8) */
00597 k_fcmr_cmdr_shift = 8, /**< Current command shift */
00598 } rx_fcmr_bits_t;
```

#### 8.23.2.9 rx`fentryr`bits`

```
enum rx_fentryr_bits_t : uint16_t
```

Flash P/E Mode Entry Register (FENTRYR) bit definitions.

FENTRYR @ 0x007FE084 controls entry into P/E (Program/Erase) mode. Write with KEY=0xAA to modify FENTRYC/FENTRYD bits.

## Entry Values

- 0xAA01: Enter code flash P/E mode
- 0xAA80: Enter data flash P/E mode
- 0xAA00: Exit P/E mode (return to read mode)

## Warning

Writing 0xAA81 causes ILGLERR error

## Since

Version 1.0.0

## Enumerator

|                     |                           |
|---------------------|---------------------------|
| k_fentryr_fentryc   | Code Flash P/E Mode Entry |
| k_fentryr_fentryd   | Data Flash P/E Mode Entry |
| k_fentryr_key_shift | Key code position         |
| k_fentryr_key       | Key code (write only)     |
| k_fentryr_code_pe   | Enter code flash P/E mode |
| k_fentryr_data_pe   | Enter data flash P/E mode |
| k_fentryr_read_mode | Exit P/E mode             |

Definition at line 500 of file [rx72n\\_flash\\_regs.h](#).

```

00500 : uint16_t {
00501 k_fentryr_fentryc = (1U « 0), /**< Code Flash P/E Mode Entry */
00502 k_fentryr_fentryd = (1U « 7), /**< Data Flash P/E Mode Entry */
00503 k_fentryr_key_shift = 8, /**< Key code position */
00504 k_fentryr_key = 0xAA00, /**< Key code (write only) */
00505
00506 k_fentryr_code_pe = 0xAA01, /**< Enter code flash P/E mode */
00507 k_fentryr_data_pe = 0xAA80, /**< Enter data flash P/E mode */
00508 k_fentryr_read_mode = 0xAA00, /**< Exit P/E mode */
00509 } rx_fentryr_bits_t;

```

## 8.23.2.10 rx'flash'memory'addresses't

```
enum rx_flash_memory_addresses_t : uintptr_t
```

Flash memory region base and end addresses for the RX72N.

Defines the inclusive address range of each flash region on the RX72N (4 MB code flash, 32 KB data flash). Addresses are physical bus addresses as documented in the RX72N Hardware Manual, Section 5 (Memory Map).

Code flash occupies the top of the 32-bit address space (0xFFC00000 - 0xFFFFFFFF). Data flash is a separate EEPROM-emulation region beginning at 0x00100000. Neither region overlaps RAM or peripheral space.

Use these constants as bounds for range checks before issuing erase or program commands via the Flash Control Unit (FCU). Never use raw literals for flash address comparisons; always reference these enum values so that any future memory-map revision is caught at a single point.

## Note

All addresses must be naturally aligned to the program unit defined in [rx\\_flash\\_memory\\_layout\\_t](#) (128 bytes for code flash, 4 bytes for data flash). Unaligned addresses are rejected by the FCU hardware.

Code flash erase granularity differs by block index: blocks 0-7 are 8 KB each, blocks 8 and above are 32 KB each (see [rx\\_flash\\_memory\\_layout\\_t](#)).

## Invariant

`k_code_flash_start < k_code_flash_end`

`k_data_flash_start < k_data_flash_end`

Code flash and data flash address ranges do not overlap

## Usage Example:

```
bool is_code_flash(uintptr_t addr) {
 return addr >= (uintptr_t)k_code_flash_start &&
 addr <= (uintptr_t)k_code_flash_end;
}
```

## See also

[rx\\_flash\\_memory\\_layout\\_t](#) Block sizes and program units for each region

[fentryr\\_reg\(\)](#) FCU entry register that enables erase/program mode

[fsaddr\\_reg\(\)](#) FCU start-address register (set before erase/program)

[feaddr\\_reg\(\)](#) FCU end-address register (set before block erase)

[rx\\_flash.h](#) Higher-level flash API (when implemented)

## Since

Version 1.0.0

## Enumerator

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| <code>k_code_flash_start</code> | Code flash start address (bottom of 4 MB region)   |
| <code>k_code_flash_end</code>   | Code flash end address (top of 32-bit space)       |
| <code>k_data_flash_start</code> | Data flash start address (EEPROM-emulation region) |
| <code>k_data_flash_end</code>   | Data flash end address (32 KB above start)         |

Definition at line 929 of file [rx72n\\_flash\\_regs.h](#).

```
00929 : uintptr_t {
00930 k_code_flash_start = 0xFFC00000, /**< Code flash start address (bottom of 4 MB region) */
00931 k_code_flash_end = 0xFFFFFFFF, /**< Code flash end address (top of 32-bit space) */
00932 k_data_flash_start = 0x00100000, /**< Data flash start address (EEPROM-emulation region) */
00933 k_data_flash_end = 0x00107FFF, /**< Data flash end address (32 KB above start) */
00934 } rx_flash_memory_addresses_t;
```

### 8.23.2.11 rx\_flash\_memory\_layout\_t

```
enum rx_flash_memory_layout_t : uint32_t
```

Flash memory region sizes, erase block granularities, and program units.

Captures the capacity and write/erase geometry of the RX72N flash subsystem as documented in the RX72N Hardware Manual, Sections 7 and 59.

Code flash (4 MB) has a split-block architecture:

- Blocks 0-7: 8 KB each (small blocks at the top of the address space, typically used for bootloader or parameter storage)
- Blocks 8 and above: 32 KB each (large blocks for application code)

Data flash (32 KB) is organised in uniform 64-byte erase blocks and is intended for EEPROM-emulation of configuration data and run-time state.

Program (write) operations must be issued in multiples of the program unit:

- Code flash: 128-byte aligned writes
- Data flash: 4-byte aligned writes

Use these constants when computing erase/program address ranges, validating buffer sizes, or determining which block type covers a given address before issuing FCU commands.

Note

All values are byte counts or byte sizes, not word or block counts.

The program unit constraints are enforced in hardware; the FCU will set FSTATR.PRGERR if an unaligned or undersized write is attempted.

Invariant

```
k_code_flash_size == (k_code_flash_block_8k * 8) + (k_code_flash_block_32k * 126)
k_data_flash_size == k_data_flash_block_64 * 512
```

Usage Example:

```
bool is_aligned_code_flash_write(uintptr_t addr, uint32_t len) {
 return (addr % k_code_flash_program_unit == 0) &&
 (len % k_code_flash_program_unit == 0);
}
```

See also

[rx\\_flash\\_memory\\_addresses\\_t](#) Physical start/end addresses for each region  
[fstatr\\_reg\(\)](#) FCU status register (check PRGERR after program)  
[fsaddr\\_reg\(\)](#) FCU start-address register (set before erase/program)  
[feaddr\\_reg\(\)](#) FCU end-address register (set before block erase)  
[rx\\_flash.h](#) Higher-level flash API (when implemented)

Since

Version 1.0.0

Enumerator

|                           |                                                              |
|---------------------------|--------------------------------------------------------------|
| k_code_flash_size         | Total code flash size in bytes (4 MB)                        |
| k_data_flash_size         | Total data flash size in bytes (32 KB)                       |
| k_code_flash_block_8k     | Erase block size for blocks 0-7 (8 KB each)                  |
| k_code_flash_block_32k    | Erase block size for blocks 8+ (32 KB each)                  |
| k_data_flash_block_64     | Erase block size for data flash (64 bytes each)              |
| k_code_flash_program_unit | Minimum program unit for code flash (bytes, must be aligned) |
| k_data_flash_program_unit | Minimum program unit for data flash (bytes, must be aligned) |

Definition at line 983 of file [rx72n\\_flash\\_regs.h](#).

```

00983 : uint32_t {
00984 k_code_flash_size = 0x00400000, /**< Total code flash size in bytes (4 MB) */
00985 k_data_flash_size = 0x00008000, /**< Total data flash size in bytes (32 KB) */
00986
00987 k_code_flash_block_8k = 0x00002000, /**< Erase block size for blocks 0-7 (8 KB each) */
00988 k_code_flash_block_32k = 0x00008000, /**< Erase block size for blocks 8+ (32 KB each) */
00989 k_data_flash_block_64 = 0x00000040, /**< Erase block size for data flash (64 bytes each) */
00990
00991 k_code_flash_program_unit =
00992 128U, /**< Minimum program unit for code flash (bytes, must be aligned) */
00993 k_data_flash_program_unit =
00994 4U, /**< Minimum program unit for data flash (bytes, must be aligned) */
00995 } rx_flash_memory_layout_t;
```

#### 8.23.2.12 rx7n/flash/regs.h

enum [rx7n/flash/regs.h](#) : uint16\_t

Flash Sequencer Clock Notification Register (FPCKAR) bit definitions.

FPCKAR @ 0x007FE0E4 notifies the flash sequencer of FCLK frequency. Must be set before issuing FACL commands for correct timing. Write with KEY=0x1E to modify PCKA field.

PCKA Values

Set PCKA = FCLK frequency in MHz (e.g., 60 for 60 MHz FCLK) Valid range: 4 to 60 MHz

Since

Version 1.0.0

Enumerator

|                    |                           |
|--------------------|---------------------------|
| k_fpckar_pcka_mask | PCKA frequency field mask |
| k_fpckar_key_shift | Key code position         |
| k_fpckar_key       | Key code for write enable |

Definition at line 672 of file [rx72n\\_flash\\_regs.h](#).

```

00672 : uint16_t {
00673 k_fpckar_pcka_mask = 0x00FF, /**< PCKA frequency field mask */
00674 k_fpckar_key_shift = 8, /**< Key code position */
00675 k_fpckar_key = 0x1E00, /**< Key code for write enable */
00676 } rx_fpckar_bits_t;
```

## 8.23.2.13 rx`frdyie`bits`t

```
enum rx_frdyie_bits_t : uint8_t
```

Flash Ready Interrupt Enable Register (FRDYIE) bit definitions.

Since

Version 1.0.0

Enumerator

|                 |                              |
|-----------------|------------------------------|
| k_frdyie_frdyie | Flash Ready Interrupt Enable |
|-----------------|------------------------------|

Definition at line 384 of file rx72n\_flash\_regs.h.

```
00384 : uint8_t {
00385 k_frdyie_frdyie = (1U « 0), /**< Flash Ready Interrupt Enable */
00386 } rx_frdyie_bits_t;
```

## 8.23.2.14 rx`fstatr`bits`t

```
enum rx_fstatr_bits_t : uint32_t
```

Flash Status Register (FSTATR) bit definitions.

FSTATR @ 0x007FE080 indicates flash sequencer state. This is the primary register for monitoring flash operations.

Bit Layout (32-bit)

| Bit | Name      | Description                   |
|-----|-----------|-------------------------------|
| 6   | FLWEERR   | Flash P/E Protect Error       |
| 8   | PRGSPD    | Program Suspend Status        |
| 9   | ERSSPD    | Erase Suspend Status          |
| 10  | DBFULL    | Data Buffer Full              |
| 11  | SUSRDY    | Suspend Ready                 |
| 12  | PRGERR    | Program Error                 |
| 13  | ERSERR    | Erase Error                   |
| 14  | ILGLERR   | Illegal Command/Access Error  |
| 15  | FRDY      | Flash Ready (1=ready, 0=busy) |
| 20  | OTERR     | Other Error                   |
| 21  | SECERR    | Security Error                |
| 22  | FESETERR  | FENTRY Setting Error          |
| 23  | ILGCOMERR | Illegal Command Error         |

Since

Version 1.0.0

Enumerator

|                          |                         |
|--------------------------|-------------------------|
| k_fstatr_flweerr_shift   |                         |
| k_fstatr_prghpd_shift    |                         |
| k_fstatr_erspd_shift     |                         |
| k_fstatr_dbfull_shift    |                         |
| k_fstatr_susrdy_shift    |                         |
| k_fstatr_prgerr_shift    |                         |
| k_fstatr_erserr_shift    |                         |
| k_fstatr_ilglerr_shift   |                         |
| k_fstatr_frdy_shift      |                         |
| k_fstatr_oterr_shift     |                         |
| k_fstatr_secerr_shift    |                         |
| k_fstatr_feseterr_shift  |                         |
| k_fstatr_ilgcomerr_shift |                         |
| k_fstatr_flweerr         | Flash P/E Protect Error |
| k_fstatr_prgspd          | Program Suspend Status  |
| k_fstatr_erspd           | Erase Suspend Status    |
| k_fstatr_dbfull          | Data Buffer Full        |
| k_fstatr_susrdy          | Suspend Ready           |
| k_fstatr_prgerr          | Program Error           |
| k_fstatr_erserr          | Erase Error             |
| k_fstatr_ilglerr         | Illegal Error           |
| k_fstatr_frdy            | Flash Ready             |
| k_fstatr_oterr           | Other Error             |
| k_fstatr_secerr          | Security Error          |
| k_fstatr_feseterr        | FENTRY Setting Error    |
| k_fstatr_ilgcomerr       | Illegal Command Error   |
| k_fstatr_error_mask      |                         |
| k_fstatr_reset_value     | Reset value (FRDY=1)    |

Definition at line 430 of file rx72n\_flash\_regs.h.

```

00430 : uint32_t {
00431 /* Bit positions */
00432 k_fstatr_flweerr_shift = 6,
00433 k_fstatr_prgspd_shift = 8,
00434 k_fstatr_erspd_shift = 9,
00435 k_fstatr_dbfull_shift = 10,
00436 k_fstatr_susrdy_shift = 11,
00437 k_fstatr_prgerr_shift = 12,
00438 k_fstatr_erserr_shift = 13,
00439 k_fstatr_ilglerr_shift = 14,
00440 k_fstatr_frdy_shift = 15,
00441 k_fstatr_oterr_shift = 20,
00442 k_fstatr_secerr_shift = 21,
00443 k_fstatr_feseterr_shift = 22,
00444 k_fstatr_ilgcomerr_shift = 23,
00445
00446 /* Bit masks */
00447 k_fstatr_flweerr = (1UL « 6), /**< Flash P/E Protect Error */

```

```

00448 k_fstatr_prghpd = (1UL « 8), /**< Program Suspend Status */
00449 k_fstatr_erspd = (1UL « 9), /**< Erase Suspend Status */
00450 k_fstatr_dbfull = (1UL « 10), /**< Data Buffer Full */
00451 k_fstatr_susrdy = (1UL « 11), /**< Suspend Ready */
00452 k_fstatr_prgherr = (1UL « 12), /**< Program Error */
00453 k_fstatr_erserr = (1UL « 13), /**< Erase Error */
00454 k_fstatr_ilgherr = (1UL « 14), /**< Illegal Error */
00455 k_fstatr_frdy = (1UL « 15), /**< Flash Ready */
00456 k_fstatr_oterr = (1UL « 20), /**< Other Error */
00457 k_fstatr_secerr = (1UL « 21), /**< Security Error */
00458 k_fstatr_feseterr = (1UL « 22), /**< FENTRY Setting Error */
00459 k_fstatr_ilghcomerr = (1UL « 23), /**< Illegal Command Error */
00460
00461 /* Combined error mask */
00462 k_fstatr_error_mask = (k_fstatr_flweerr | k_fstatr_prgherr | k_fstatr_erserr | k_fstatr_ilgherr |
00463 k_fstatr_oterr | k_fstatr_secerr | k_fstatr_feseterr | k_fstatr_ilghcomerr),
00464
00465 k_fstatr_reset_value = 0x00008000, /**< Reset value (FRDY=1) */
00466 } rx_fstatr_bits_t;

```

### 8.23.2.15 rx\_fsuinitr\_bits\_t

```
enum rx_fsuinitr_bits_t : uint16_t
```

Flash Sequencer Setup Initialization Register (FSUINITR) bit defs.

FSUINITR @ 0x007FE08C initializes flash sequencer setup registers. Write with KEY=0x2D to enable SUINIT write.

Since

Version 1.0.0

Enumerator

|                     |                                  |
|---------------------|----------------------------------|
| k_fsuinitr_suinit   | Setup Initialization             |
| k_fsuinitr_key      | Key code for write enable        |
| k_fsuinitr_init_cmd | Initialize flash sequencer setup |

Definition at line 536 of file rx72n\_flash\_regs.h.

```

00536 : uint16_t {
00537 k_fsuinitr_suinit = (1U « 0), /**< Setup Initialization */
00538 k_fsuinitr_key = 0x2D00, /**< Key code for write enable */
00539
00540 k_fsuinitr_init_cmd = 0x2D01, /**< Initialize flash sequencer setup */
00541 } rx_fsuinitr_bits_t;

```

### 8.23.2.16 rx\_fweprror\_addresses\_t

```
enum rx_fweprror_addresses_t : uintptr_t
```

Flash Write/Erase Protect Register addresses.



Since

Version 1.0.0

Enumerator

|               |                            |
|---------------|----------------------------|
| k_fwepor_addr | Flash P/E Protect Register |
|---------------|----------------------------|

Definition at line 189 of file rx72n\_flash\_regs.h.

```
00189 : uintptr_t {
00190 k_fwepor_addr = 0x0008C296, /**< Flash P/E Protect Register */
00191 } rx_fwepor_addresses_t;
```

#### 8.23.2.17 rx`fwepor`bits`

```
enum rx_fwepor_bits_t : uint8_t
```

Flash P/E Protect Register (FWEPROR) bit definitions.

FWEPROR @ 0x0008C296 enables/disables flash program and erase operations.

- FLWE = 00b: Program, block erase, blank check disabled
- FLWE = 01b: Program, block erase, blank check enabled
- FLWE = 10b: Disabled
- FLWE = 11b: Disabled

Warning

Register is reset on deep software standby mode entry

Since

Version 1.0.0

Enumerator

|                       |                        |
|-----------------------|------------------------|
| k_fwepor_flwe_mask    | FLWE[1:0] mask         |
| k_fwepor_flwe_disable | Program/erase disabled |
| k_fwepor_flwe_enable  | Program/erase enabled  |
| k_fwepor_reset_value  | Value after reset      |

Definition at line 208 of file rx72n\_flash\_regs.h.

```
00208 : uint8_t {
00209 k_fwepor_flwe_mask = 0x03, /**< FLWE[1:0] mask */
00210 k_fwepor_flwe_disable = 0x00, /**< Program/erase disabled */
00211 k_fwepor_flwe_enable = 0x01, /**< Program/erase enabled */
00212 k_fwepor_reset_value = 0x02, /**< Value after reset */
00213 } rx_fwepor_bits_t;
```

## 8.23.2.18 rx\_rom\_cache\_addresses\_t

```
enum rx_rom_cache_addresses_t : uintptr_t
```

ROM cache register addresses.

Base addresses for ROM cache control registers. The ROM cache improves code fetch performance with 8KB direct-mapped cache.

## Memory Map

| Address    | Register | Description                  |
|------------|----------|------------------------------|
| 0x00081000 | ROMCE    | ROM Cache Enable             |
| 0x00081004 | ROMCIV   | ROM Cache Invalidate         |
| 0x00081040 | NCRG0    | Non-Cacheable Area 0 Address |
| 0x00081044 | NCRC0    | Non-Cacheable Area 0 Control |
| 0x00081048 | NCRG1    | Non-Cacheable Area 1 Address |
| 0x0008104C | NCRC1    | Non-Cacheable Area 1 Control |

Since

Version 1.0.0

## Enumerator

|                       |                               |
|-----------------------|-------------------------------|
| k_rom_cache_base_addr | ROM cache register base       |
| k_romce_addr          | ROM Cache Enable Register     |
| k_romciv_addr         | ROM Cache Invalidate Register |
| k_ncrg0_addr          | Non-Cacheable Area 0 Address  |
| k_ncrc0_addr          | Non-Cacheable Area 0 Control  |
| k_ncrg1_addr          | Non-Cacheable Area 1 Address  |
| k_ncrc1_addr          | Non-Cacheable Area 1 Control  |

Definition at line 120 of file [rx72n\\_flash\\_regs.h](#).

```
00120 : uintptr_t {
00121 k_rom_cache_base_addr = 0x00081000, /**< ROM cache register base */
00122 k_romce_addr = 0x00081000, /**< ROM Cache Enable Register */
00123 k_romciv_addr = 0x00081004, /**< ROM Cache Invalidate Register */
00124 k_ncrg0_addr = 0x00081040, /**< Non-Cacheable Area 0 Address */
00125 k_ncrc0_addr = 0x00081044, /**< Non-Cacheable Area 0 Control */
00126 k_ncrg1_addr = 0x00081048, /**< Non-Cacheable Area 1 Address */
00127 k_ncrc1_addr = 0x0008104C, /**< Non-Cacheable Area 1 Control */
00128 } rx_rom_cache_addresses_t;
```

## 8.23.2.19 rx\_romce\_bits\_t

```
enum rx_romce_bits_t : uint16_t
```

ROM Cache Enable Register (ROMCE) bit definitions.

ROMCE @ 0x00081000 controls ROM cache enable/disable.

Since

Version 1.0.0

Enumerator

|                |                                          |
|----------------|------------------------------------------|
| k_romce_romcen | ROM Cache Enable (1=enabled, 0=disabled) |
|----------------|------------------------------------------|

Definition at line 139 of file rx72n\_flash\_regs.h.

```
00139 : uint16_t {
00140 k_romce_romcen = (1U « 0), /**< ROM Cache Enable (1=enabled, 0=disabled) */
00141 } rx_romce_bits_t;
```

#### 8.23.2.20 rx`romciv`bits` t

```
enum rx_romciv_bits_t : uint16_t
```

ROM Cache Invalidate Register (ROMCIV) bit definitions.

ROMCIV @ 0x00081004 invalidates all cache lines. Read: 0=complete/not started, 1=in progress Write: 1=start invalidation, 0=no effect

Since

Version 1.0.0

Enumerator

|                 |                      |
|-----------------|----------------------|
| k_romciv_romciv | ROM Cache Invalidate |
|-----------------|----------------------|

Definition at line 154 of file rx72n\_flash\_regs.h.

```
00154 : uint16_t {
00155 k_romciv_romciv = (1U « 0), /**< ROM Cache Invalidate */
00156 } rx_romciv_bits_t;
```

#### 8.23.2.21 rx`uidr`addresses` t

```
enum rx_uidr_addresses_t : uintptr_t
```

Unique ID Register addresses (Ch62 Section 62.4.23).

UIDR0-3 contain a 128-bit unique identifier for each chip. These are read-only registers in the FLASH-CONST area (on-chip ROM).

Memory Map (FLASHCONST area)

| Address    | Register | Description         |
|------------|----------|---------------------|
| 0xFE7F7D90 | UIDR0    | Unique ID bits 31-0 |

| Address    | Register | Description           |
|------------|----------|-----------------------|
| 0xFE7F7D94 | UIDR1    | Unique ID bits 63-32  |
| 0xFE7F7D98 | UIDR2    | Unique ID bits 95-64  |
| 0xFE7F7D9C | UIDR3    | Unique ID bits 127-96 |

#### Access Requirements

- Must read in 32-bit units (as per manual specification)
- SYSCR0.ROME must be 1 (on-chip ROM enabled) - this is the default in single-chip mode, so no special setup needed for normal firmware
- ROMCODE/SPCC security settings do NOT block CPU self-reads (only external programmer access)

#### Note

Values are factory-programmed and unique to each chip

These addresses are in FLASHCONST area (FE7F xxxx), not the typical FACL register space (007F xxxx) - this is expected per manual

#### Since

Version 1.0.0

#### Enumerator

|              |                                    |
|--------------|------------------------------------|
| k_uidr0_addr | Unique ID Register 0 (bits 31-0)   |
| k_uidr1_addr | Unique ID Register 1 (bits 63-32)  |
| k_uidr2_addr | Unique ID Register 2 (bits 95-64)  |
| k_uidr3_addr | Unique ID Register 3 (bits 127-96) |

Definition at line 832 of file [rx72n\\_flash\\_regs.h](#).

```

00832 : uintptr_t {
00833 k_uidr0_addr = 0xFE7F7D90, /**< Unique ID Register 0 (bits 31-0) */
00834 k_uidr1_addr = 0xFE7F7D94, /**< Unique ID Register 1 (bits 63-32) */
00835 k_uidr2_addr = 0xFE7F7D98, /**< Unique ID Register 2 (bits 95-64) */
00836 k_uidr3_addr = 0xFE7F7D9C, /**< Unique ID Register 3 (bits 127-96) */
00837 } rx_uidr_addresses_t;

```

## 8.23.3 Function Documentation

### 8.23.3.1 eepfclk\_reg()

```
volatile uint8_t * eepfclk_reg (
 void) [inline], [static]
```

Get pointer to Data Flash Access Frequency Register (EPPFCLK).

Returns

Volatile pointer to EEPFCLK register

Since

Version 1.0.0

Definition at line 698 of file [rx72n\\_flash\\_regs.h](#).

```
00699 {
00700 return (volatile uint8_t*)k_eepfclk_addr;
00701 }
```

References [k\\_eepfclk\\_addr](#).

### 8.23.3.2 faci\_cmd\_area()

```
volatile uint8_t * faci_cmd_area (
 void) [inline], [static]
```

Get pointer to FACI Command Issuing Area.

Commands are issued by writing to this 4-byte area. The write sequence depends on the command type.

Returns

Volatile pointer to FACI command area

Since

Version 1.0.0

Definition at line 647 of file [rx72n\\_flash\\_regs.h](#).

```
00648 {
00649 return (volatile uint8_t*)k_faci_cmd_area;
00650 }
```

References [k\\_faci\\_cmd\\_area](#).

### 8.23.3.3 faeint\_reg()

```
volatile uint8_t * faeint_reg (
 void) [inline], [static]
```

Get pointer to Flash Access Error Interrupt Enable (FAEINT).

Returns

Volatile pointer to FAEINT register

Since

Version 1.0.0

Definition at line 368 of file [rx72n\\_flash\\_regs.h](#).

```
00369 {
00370 return (volatile uint8_t*)k_faeint_addr;
00371 }
```

References [k\\_faeint\\_addr](#).

#### 8.23.3.4 fastat\_reg()

```
volatile uint8_t * fastat_reg (
 void) [inline], [static]
```

Get pointer to Flash Access Status Register (FASTAT).

Returns

Volatile pointer to FASTAT register

Since

Version 1.0.0

Definition at line 331 of file [rx72n\\_flash\\_regs.h](#).

```
00332 {
00333 return (volatile uint8_t*)k_fastat_addr;
00334 }
```

References [k\\_fastat\\_addr](#).

#### 8.23.3.5 fawmon\_reg()

```
volatile uint32_t * fawmon_reg (
 void) [inline], [static]
```

Get pointer to Flash Access Window Monitor (FAWMON).

Returns

Volatile pointer to FAWMON register

Since

Version 1.0.0

Definition at line 773 of file [rx72n\\_flash\\_regs.h](#).

```
00774 {
00775 return (volatile uint32_t*)k_fawmon_addr;
00776 }
```

References [k\\_fawmon\\_addr](#).

### 8.23.3.6 fbcnt\_reg()

```
volatile uint8_t * fbcnt_reg (
 void) [inline], [static]
```

Get pointer to Blank Check Control Register (FBCCNT).

Returns

Volatile pointer to FBCCNT register

Since

Version 1.0.0

Definition at line 723 of file [rx72n\\_flash\\_regs.h](#).

```
00724 {
00725 return (volatile uint8_t*)k_fbcnt_addr;
00726 }
```

References [k\\_fbcnt\\_addr](#).

### 8.23.3.7 fbcstat\_reg()

```
volatile uint8_t * fbcstat_reg (
 void) [inline], [static]
```

Get pointer to Blank Check Status Register (FBCSTAT).

Returns

Volatile pointer to FBCSTAT register

Since

Version 1.0.0

Definition at line 748 of file [rx72n\\_flash\\_regs.h](#).

```
00749 {
00750 return (volatile uint8_t*)k_fbcstat_addr;
00751 }
```

References [k\\_fbcstat\\_addr](#).

### 8.23.3.8 fcmdr`reg()

```
volatile uint16_t * fcmdr_reg (
 void) [inline], [static]
```

Get pointer to FSCI Command Register (FCMDR).

Returns

Volatile pointer to FCMDR register

Since

Version 1.0.0

Definition at line 605 of file [rx72n\\_flash\\_regs.h](#).

```
00606 {
00607 return (volatile uint16_t*)k_fcmdr_addr;
00608 }
```

References [k\\_fcmdr\\_addr](#).

### 8.23.3.9 fcpsr`reg()

```
volatile uint16_t * fcpsr_reg (
 void) [inline], [static]
```

Get pointer to Flash Sequencer Processing Switch (FCPSR).

Returns

Volatile pointer to FCPSR register

Since

Version 1.0.0

Definition at line 783 of file [rx72n\\_flash\\_regs.h](#).

```
00784 {
00785 return (volatile uint16_t*)k_fcpsr_addr;
00786 }
```

References [k\\_fcpsr\\_addr](#).



### 8.23.3.10 feaddr\_reg()

```
volatile uint32_t * feaddr_reg (
 void) [inline], [static]
```

Get pointer to FSCI End Address Register (FEADDR).

Returns

Volatile pointer to FEADDR register

Since

Version 1.0.0

Definition at line 573 of file [rx72n\\_flash\\_regs.h](#).

```
00574 {
00575 return (volatile uint32_t*)k_feaddr_addr;
00576 }
```

References [k\\_feaddr\\_addr](#).

### 8.23.3.11 fentryr\_reg()

```
volatile uint16_t * fentryr_reg (
 void) [inline], [static]
```

Get pointer to Flash P/E Mode Entry Register (FENTRYR).

Returns

Volatile pointer to FENTRYR register

Since

Version 1.0.0

Definition at line 516 of file [rx72n\\_flash\\_regs.h](#).

```
00517 {
00518 return (volatile uint16_t*)k_fentryr_addr;
00519 }
```

References [k\\_fentryr\\_addr](#).

### 8.23.3.12 fpckar\_reg()

```
volatile uint16_t * fpckar_reg (
 void) [inline], [static]
```

Get pointer to Flash Sequencer Clock Register (FPCKAR).

Returns

Volatile pointer to FPCKAR register

Since

Version 1.0.0

Definition at line 683 of file [rx72n\\_flash\\_regs.h](#).

```
00684 {
00685 return (volatile uint16_t*)k_fpckar_addr;
00686 }
```

References [k\\_fpckar\\_addr](#).

### 8.23.3.13 fpsaddr\_reg()

```
volatile uint32_t * fpsaddr_reg (
 void) [inline], [static]
```

Get pointer to Data Flash Programming Start Address (FPSADDR).

Returns

Volatile pointer to FPSADDR register

Since

Version 1.0.0

Definition at line 763 of file [rx72n\\_flash\\_regs.h](#).

```
00764 {
00765 return (volatile uint32_t*)k_fpsaddr_addr;
00766 }
```

References [k\\_fpsaddr\\_addr](#).

#### 8.23.3.14 frdyie\_reg()

```
volatile uint8_t * frdyie_reg (
 void) [inline], [static]
```

Get pointer to Flash Ready Interrupt Enable (FRDYIE).

Returns

Volatile pointer to FRDYIE register

Since

Version 1.0.0

Definition at line 393 of file [rx72n\\_flash\\_regs.h](#).

```
00394 {
00395 return (volatile uint8_t*)k_frdyie_addr;
00396 }
```

References [k\\_frdyie\\_addr](#).

#### 8.23.3.15 fsaddr\_reg()

```
volatile uint32_t * fsaddr_reg (
 void) [inline], [static]
```

Get pointer to FSCI Start Address Register (FSADDR).

Returns

Volatile pointer to FSADDR register

Since

Version 1.0.0

Definition at line 563 of file [rx72n\\_flash\\_regs.h](#).

```
00564 {
00565 return (volatile uint32_t*)k_fsaddr_addr;
00566 }
```

References [k\\_fsaddr\\_addr](#).

### 8.23.3.16 fstatr\_reg()

```
volatile uint32_t * fstatr_reg (
 void) [inline], [static]
```

Get pointer to Flash Status Register (FSTATR).

Returns

Volatile pointer to FSTATR register

Since

Version 1.0.0

Definition at line 473 of file [rx72n\\_flash\\_regs.h](#).

```
00474 {
00475 return (volatile uint32_t*)k_fstatr_addr;
00476 }
```

References [k\\_fstatr\\_addr](#).

### 8.23.3.17 fsuacr\_reg()

```
volatile uint16_t * fsuacr_reg (
 void) [inline], [static]
```

Get pointer to Flash Startup Area Select (FSUACR).

Returns

Volatile pointer to FSUACR register

Since

Version 1.0.0

Definition at line 793 of file [rx72n\\_flash\\_regs.h](#).

```
00794 {
00795 return (volatile uint16_t*)k_fsuacr_addr;
00796 }
```

References [k\\_fsuacr\\_addr](#).

### 8.23.3.18 fsuinitr\_reg()

```
volatile uint16_t * fsuinitr_reg (
 void) [inline], [static]
```

Get pointer to Flash Sequencer Setup Init Register (FSUINITR).

Returns

Volatile pointer to FSUINITR register

Since

Version 1.0.0

Definition at line 548 of file [rx72n\\_flash\\_regs.h](#).

```
00549 {
00550 return (volatile uint16_t*)k_fsuinitr_addr;
00551 }
```

References [k\\_fsuinitr\\_addr](#).

### 8.23.3.19 fwepror\_reg()

```
volatile uint8_t * fwepror_reg (
 void) [inline], [static]
```

Get pointer to Flash P/E Protect Register (FWEPROR).

Returns

Volatile pointer to FWEPROR register

Since

Version 1.0.0

Definition at line 220 of file [rx72n\\_flash\\_regs.h](#).

```
00221 {
00222 return (volatile uint8_t*)k_fwepror_addr;
00223 }
```

References [k\\_fwepror\\_addr](#).

### 8.23.3.20 romce\_reg()

```
volatile uint16_t * romce_reg (
 void) [inline], [static]
```

Get pointer to ROM Cache Enable Register (ROMCE).

Returns

Volatile pointer to ROMCE register

Since

Version 1.0.0

Definition at line 163 of file [rx72n\\_flash\\_regs.h](#).

```
00164 {
00165 return (volatile uint16_t*)k_romce_addr;
00166 }
```

References [k\\_romce\\_addr](#).

### 8.23.3.21 romciv\_reg()

```
volatile uint16_t * romciv_reg (
 void) [inline], [static]
```

Get pointer to ROM Cache Invalidate Register (ROMCIV).

Returns

Volatile pointer to ROMCIV register

Since

Version 1.0.0

Definition at line 173 of file [rx72n\\_flash\\_regs.h](#).

```
00174 {
00175 return (volatile uint16_t*)k_romciv_addr;
00176 }
```

References [k\\_romciv\\_addr](#).

### 8.23.3.22 uidr0\_reg()

```
volatile const uint32_t * uidr0_reg (
 void) [inline], [static]
```

Get pointer to Unique ID Register 0 (UIDR0).

Returns

Volatile pointer to UIDR0 register (read-only)

Note

Access may be restricted depending on security settings

Since

Version 1.0.0

Definition at line 845 of file [rx72n\\_flash\\_regs.h](#).

```
00846 {
00847 return (volatile const uint32_t*)k_uidr0_addr;
00848 }
```

References [k\\_uidr0\\_addr](#).

### 8.23.3.23 uidr1\_reg()

```
volatile const uint32_t * uidr1_reg (
 void) [inline], [static]
```

Get pointer to Unique ID Register 1 (UIDR1).

Returns

Volatile pointer to UIDR1 register (read-only)

Since

Version 1.0.0

Definition at line 855 of file [rx72n\\_flash\\_regs.h](#).

```
00856 {
00857 return (volatile const uint32_t*)k_uidr1_addr;
00858 }
```

References [k\\_uidr1\\_addr](#).

#### 8.23.3.24 uidr2\_reg()

```
volatile const uint32_t * uidr2_reg (
 void) [inline], [static]
```

Get pointer to Unique ID Register 2 (UIDR2).

Returns

Volatile pointer to UIDR2 register (read-only)

Since

Version 1.0.0

Definition at line 865 of file [rx72n\\_flash\\_regs.h](#).

```
00866 {
00867 return (volatile const uint32_t*)k_uidr2_addr;
00868 }
```

References [k\\_uidr2\\_addr](#).

#### 8.23.3.25 uidr3\_reg()

```
volatile const uint32_t * uidr3_reg (
 void) [inline], [static]
```

Get pointer to Unique ID Register 3 (UIDR3).

Returns

Volatile pointer to UIDR3 register (read-only)

Since

Version 1.0.0

Definition at line 875 of file [rx72n\\_flash\\_regs.h](#).

```
00876 {
00877 return (volatile const uint32_t*)k_uidr3_addr;
00878 }
```

References [k\\_uidr3\\_addr](#).



## 8.24 rx72n`flash`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_flash_regs.h
00003 * @brief RX72N Flash Memory Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the Flash Memory module including ROM cache control,
00007 * flash programming/erase operations (FACI), and access protection. The RX72N
00008 * has 4MB code flash and 32KB data flash with hardware-accelerated operations.
00009 *
00010 * @par System Architecture
00011 * @dot
00012 * digraph flash_architecture {
00013 * rankdir=TB;
00014 * node [shape=box, style=rounded];
00015 *
00016 * subgraph cluster_flash {
00017 * label="Flash Memory System";
00018 * style=filled;
00019 * color=lightgrey;
00020 *
00021 * code [label="Code Flash\n4 MB\n0xFFC00000-0xFFFFFFFF"];
00022 * data [label="Data Flash\n32 KB\n0x00100000-0x00107FFF"];
00023 * faci [label="FACI\n(Flash Application\nCommand Interface)" shape=ellipse];
00024 * cache [label="ROM Cache\n8 KB"];
00025 * }
00026 *
00027 * cpu [label="CPU"];
00028 * regs [label="Control Registers\n0x007FE0xx"];
00029 *
00030 * cpu -> cache [label="Fetch"];
00031 * cache -> code [label="Miss"];
00032 * cpu -> regs [label="Configure"];
00033 * regs -> faci [label="Commands"];
00034 * faci -> code [label="Program/Erase"];
00035 * faci -> data [label="Program/Erase"];
00036 * }
00037 * @enddot
00038 *
00039 * @par Flash Memory Specifications
00040 * | Parameter | Code Flash | Data Flash |
00041 * |-----|-----|-----|
00042 * | Capacity | 4 MB | 32 KB |
00043 * | Address Range | FFC00000-FFFF FFFF | 0010 0000-0010 7FFF |
00044 * | Program Unit | 128 bytes | 4 bytes |
00045 * | Erase Unit | 8/32 KB block | 64 bytes |
00046 * | Read Latency | 1-3 cycles | FCLK based |
00047 *
00048 * @par FACI Commands (via 0x007E0000)
00049 * | Command | Code | Description |
00050 * |-----|-----|-----|
00051 * | Program | 0xE8 | Write data to flash |
00052 * | Block Erase | 0x20 | Erase a single block |
00053 * | Blank Check | 0x71 | Verify area is erased |
00054 * | Status Clear | 0x50 | Clear error status |
00055 * | Forced Stop | 0xB3 | Abort current operation |
00056 * | PE Suspend | 0xB0 | Suspend program/erase |
00057 * | PE Resume | 0xD0 | Resume suspended operation |
00058 *
00059 * @par Module Dependencies
00060 * - Module stop: MSTPCRB.MSTPB31 (FCU) - already running at boot
00061 *
00062 * @par Manual References
00063 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00064 * - Chapter 62: Flash Memory (FLASH), pages 2992-3065
00065 *
00066 * @note The flash sequencer (FACI) must be in P/E mode before issuing commands
00067 * @warning Programming/erasing code flash while executing from it is restricted
00068 * @attention Enable FWEPOR before any programming operations
00069 *
00070 * @par NASA Power of 10 Compliance
00071 * - Rule 1: No goto, setjmp, or recursion
00072 * - Rule 3: No dynamic memory allocation
00073 * - Rule 8: All constants use C23 typed enums
00074 * - Rule 10: Compiles with -Wall -Wextra -Werror
00075 *
00076 * @see rx_flash.h Higher-level flash API (when implemented)
00077 * @see rx72n_regs.h Main register include file
00078 *
00079 * @author Locked, Inc.
00080 * @date 2026-01-28
00081 * @copyright Copyright (c) 2026 Locked Inc.
00082 * SPDX-License-Identifier: MIT

```

```

00083 * @since Version 1.0.0
00084 */
00085
00086 #pragma once
00087
00088 #include <stddef.h>
00089 #include <stdint.h>
00090
00091 #ifndef __cplusplus
00092 extern "C" {
00093 #endif
00094
00095 /*
=====
00096 * ROM Cache Control Registers - Base Address 0x00081000
00097 *
=====
00098 */
00099
00100 /**
00101 * @enum rx_rom_cache_addresses_t
00102 * @brief ROM cache register addresses
00103 *
00104 * @details
00105 * Base addresses for ROM cache control registers. The ROM cache improves
00106 * code fetch performance with 8KB direct-mapped cache.
00107 *
00108 * @par Memory Map
00109 * | Address | Register | Description |
00110 * |-----|-----|-----|
00111 * | 0x00081000 | ROMCE | ROM Cache Enable |
00112 * | 0x00081004 | ROMCIV | ROM Cache Invalidate |
00113 * | 0x00081040 | NCRG0 | Non-Cacheable Area 0 Address |
00114 * | 0x00081044 | NCRC0 | Non-Cacheable Area 0 Control |
00115 * | 0x00081048 | NCRG1 | Non-Cacheable Area 1 Address |
00116 * | 0x0008104C | NCRC1 | Non-Cacheable Area 1 Control |
00117 *
00118 * @since Version 1.0.0
00119 */
00120 typedef enum : uintptr_t {
00121 k_rom_cache_base_addr = 0x00081000, /**< ROM cache register base */
00122 k_romce_addr = 0x00081000, /**< ROM Cache Enable Register */
00123 k_romciv_addr = 0x00081004, /**< ROM Cache Invalidate Register */
00124 k_ncrg0_addr = 0x00081040, /**< Non-Cacheable Area 0 Address */
00125 k_ncrc0_addr = 0x00081044, /**< Non-Cacheable Area 0 Control */
00126 k_ncrg1_addr = 0x00081048, /**< Non-Cacheable Area 1 Address */
00127 k_ncrc1_addr = 0x0008104C, /**< Non-Cacheable Area 1 Control */
00128 } rx_rom_cache_addresses_t;
00129
00130 /**
00131 * @enum rx_romce_bits_t
00132 * @brief ROM Cache Enable Register (ROMCE) bit definitions
00133 *
00134 * @details
00135 * ROMCE @ 0x00081000 controls ROM cache enable/disable.
00136 *
00137 * @since Version 1.0.0
00138 */
00139 typedef enum : uint16_t {
00140 k_romce_romcen = (1U « 0), /**< ROM Cache Enable (1=enabled, 0=disabled) */
00141 } rx_romce_bits_t;
00142
00143 /**
00144 * @enum rx_romciv_bits_t
00145 * @brief ROM Cache Invalidate Register (ROMCIV) bit definitions
00146 *
00147 * @details
00148 * ROMCIV @ 0x00081004 invalidates all cache lines.
00149 * Read: 0=complete/not started, 1=in progress
00150 * Write: 1=start invalidation, 0=no effect
00151 *
00152 * @since Version 1.0.0
00153 */
00154 typedef enum : uint16_t {
00155 k_romciv_romciv = (1U « 0), /**< ROM Cache Invalidate */
00156 } rx_romciv_bits_t;
00157
00158 /**
00159 * @brief Get pointer to ROM Cache Enable Register (ROMCE)
00160 * @return Volatile pointer to ROMCE register
00161 * @since Version 1.0.0
00162 */
00163 static inline volatile uint16_t* romce_reg(void)
00164 {
00165 return (volatile uint16_t*)k_romce_addr;
00166 }
00167

```

```

00168 /**
00169 * @brief Get pointer to ROM Cache Invalidate Register (ROMCIV)
00170 * @return Volatile pointer to ROMCIV register
00171 * @since Version 1.0.0
00172 */
00173 static inline volatile uint16_t* romciv_reg(void)
00174 {
00175 return (volatile uint16_t*)k_romciv_addr;
00176 }
00177
00178 /*
=====
00179 * Flash P/E Protect Register - Address 0x0008C296
00180 *
=====
00181 */
00182
00183 /**
00184 * @enum rx_fwepror_addresses_t
00185 * @brief Flash Write/Erase Protect Register addresses
00186 *
00187 * @since Version 1.0.0
00188 */
00189 typedef enum : uintptr_t {
00190 k_fwepror_addr = 0x0008C296, /**< Flash P/E Protect Register */
00191 } rx_fwepror_addresses_t;
00192
00193 /**
00194 * @enum rx_fwepror_bits_t
00195 * @brief Flash P/E Protect Register (FWEPROR) bit definitions
00196 *
00197 * @details
00198 * FWEPROR @ 0x0008C296 enables/disables flash program and erase operations.
00199 * - FLWE = 00b: Program, block erase, blank check disabled
00200 * - FLWE = 01b: Program, block erase, blank check enabled
00201 * - FLWE = 10b: Disabled
00202 * - FLWE = 11b: Disabled
00203 *
00204 * @warning Register is reset on deep software standby mode entry
00205 *
00206 * @since Version 1.0.0
00207 */
00208 typedef enum : uint8_t {
00209 k_fwepror_flwe_mask = 0x03, /**< FLWE[1:0] mask */
00210 k_fwepror_flwe_disable = 0x00, /**< Program/erase disabled */
00211 k_fwepror_flwe_enable = 0x01, /**< Program/erase enabled */
00212 k_fwepror_reset_value = 0x02, /**< Value after reset */
00213 } rx_fwepror_bits_t;
00214
00215 /**
00216 * @brief Get pointer to Flash P/E Protect Register (FWEPROR)
00217 * @return Volatile pointer to FWEPROR register
00218 * @since Version 1.0.0
00219 */
00220 static inline volatile uint8_t* fwepror_reg(void)
00221 {
00222 return (volatile uint8_t*)k_fwepror_addr;
00223 }
00224
00225 /*
=====
00226 * FACI Registers - Base Address 0x007FE000
00227 *
=====
00228 */
00229
00230 /**
00231 * @enum rx_faci_addresses_t
00232 * @brief FACI (Flash Application Command Interface) register addresses
00233 *
00234 * @details
00235 * FACI registers control flash programming and erase operations.
00236 * All registers are in the 0x007FE0xx range.
00237 *
00238 * @par Memory Map
00239 * | Address | Register | Size | Description |
00240 * |-----|-----|-----|-----|
00241 * | 0x007FE010 | FASTAT | 8 | Flash Access Status |
00242 * | 0x007FE014 | FAEINT | 8 | Flash Access Error Int Enable |
00243 * | 0x007FE018 | FRDYIE | 8 | Flash Ready Interrupt Enable |
00244 * | 0x007FE030 | FSADDR | 32 | FACI Start Address |
00245 * | 0x007FE034 | FEADDR | 32 | FACI End Address |
00246 * | 0x007FE080 | FSTATR | 32 | Flash Status Register |
00247 * | 0x007FE084 | FENTRYR | 16 | Flash P/E Mode Entry |
00248 * | 0x007FE08C | FSUINTR | 16 | Flash Sequencer Setup Init |
00249 * | 0x007FE0A0 | FCMDR | 16 | FACI Command Register |
00250 * | 0x007FE0D0 | FBCCNT | 8 | Blank Check Control |

```

```

00251 * | 0x007FE0D4 | FBCSTAT | 8 | Blank Check Status |
00252 * | 0x007FE0D8 | FPSADDR | 32 | Data Flash Programming Start Address |
00253 * | 0x007FE0DC | FAWMON | 32 | Flash Access Window Monitor |
00254 * | 0x007FE0E0 | FCPSR | 16 | Flash Sequencer Processing Switch |
00255 * | 0x007FE0E4 | FPCKAR | 16 | Flash Sequencer Clock Notify |
00256 * | 0x007FE0E8 | FSUACR | 16 | Flash Startup Area Select |
00257 *
00258 * @since Version 1.0.0
00259 */
00260 typedef enum : uintptr_t {
00261 k_faci_base_addr = 0x007FE000, /**< FACI register base (conceptual) */
00262 k_fastat_addr = 0x007FE010, /**< Flash Access Status Register */
00263 k_faeint_addr = 0x007FE014, /**< Flash Access Error Interrupt Enable */
00264 k_frdyie_addr = 0x007FE018, /**< Flash Ready Interrupt Enable */
00265 k_fsaddr_addr = 0x007FE030, /**< FACI Start Address Register */
00266 k_feaddr_addr = 0x007FE034, /**< FACI End Address Register */
00267 k_fstatr_addr = 0x007FE080, /**< Flash Status Register */
00268 k_fentryr_addr = 0x007FE084, /**< Flash P/E Mode Entry Register */
00269 k_fsuinitr_addr = 0x007FE08C, /**< Flash Sequencer Setup Init Register */
00270 k_fcmdr_addr = 0x007FE0A0, /**< FACI Command Register */
00271 k_fbcnt_addr = 0x007FE0D0, /**< Blank Check Control Register */
00272 k_fbcstat_addr = 0x007FE0D4, /**< Blank Check Status Register */
00273 k_fpsaddr_addr = 0x007FE0D8, /**< Data Flash Programming Start Address */
00274 k_fawmon_addr = 0x007FE0DC, /**< Flash Access Window Monitor */
00275 k_fcpsr_addr = 0x007FE0E0, /**< Flash Sequencer Processing Switch */
00276 k_fpckar_addr = 0x007FE0E4, /**< Flash Sequencer Clock Frequency */
00277 k_fsuacr_addr = 0x007FE0E8, /**< Flash Startup Area Select */
00278 k_faci_cmd_area = 0x007E0000, /**< FACI Command Issuing Area */
00279 } rx_faci_addresses_t;
00280
00281 /**
00282 * @enum rx_eepfclk_addresses_t
00283 * @brief Data Flash Memory Access Frequency register address
00284 *
00285 * @since Version 1.0.0
00286 */
00287 typedef enum : uintptr_t {
00288 k_eepfclk_addr = 0x007FC040, /**< Data Flash Access Frequency Setting */
00289 } rx_eepfclk_addresses_t;
00290
00291 /**
=====
00292 * FASTAT - Flash Access Status Register @ 0x007FE010
00293 *
=====
00294 */
00295
00296 /**
00297 * @enum rx_fastat_bits_t
00298 * @brief Flash Access Status Register (FASTAT) bit definitions
00299 *
00300 * @details
00301 * FASTAT @ 0x007FE010 indicates flash access violations.
00302 *
00303 * @par Bit Layout (8-bit)
00304 * | Bit | Name | R/W | Description |
00305 * |-----|-----|-----|-----|
00306 * | 7 | CFAE | R/W | Code Flash Access Violation |
00307 * | 6:5 | - | R | Reserved |
00308 * | 4 | CMDLK | R | Command Lock (sequencer locked) |
00309 * | 3 | DFAE | R/W | Data Flash Access Violation |
00310 * | 2:0 | - | R | Reserved |
00311 *
00312 * @note Write 0 to clear CFAE/DFAE after reading 1
00313 *
00314 * @since Version 1.0.0
00315 */
00316 typedef enum : uint8_t {
00317 k_fastat_dfae_shift = 3, /**< DFAE bit position */
00318 k_fastat_cmdlk_shift = 4, /**< CMDLK bit position */
00319 k_fastat_cfae_shift = 7, /**< CFAE bit position */
00320
00321 k_fastat_dfae = (1U « 3), /**< Data Flash Access Violation */
00322 k_fastat_cmdlk = (1U « 4), /**< Command Lock Flag */
00323 k_fastat_cfae = (1U « 7), /**< Code Flash Access Violation */
00324 } rx_fastat_bits_t;
00325
00326 /**
00327 * @brief Get pointer to Flash Access Status Register (FASTAT)
00328 * @return Volatile pointer to FASTAT register
00329 * @since Version 1.0.0
00330 */
00331 static inline volatile uint8_t* fastat_reg(void)
00332 {
00333 return (volatile uint8_t*)k_fastat_addr;
00334 }
00335

```

```

00336 /*
=====
00337 * FAEINT - Flash Access Error Interrupt Enable @ 0x007FE014
00338 *
=====
00339 */
00340 /**
00341 **
00342 * @enum rx_faeint_bits_t
00343 * @brief Flash Access Error Interrupt Enable Register (FAEINT) bit definitions
00344 *
00345 * @details
00346 * FAEINT @ 0x007FE014 enables FIFERR interrupt generation.
00347 * Reset value: 0x98 (all interrupts enabled)
00348 *
00349 * @since Version 1.0.0
00350 */
00351 typedef enum : uint8_t {
00352 k_faeint_dfaeie_shift = 3, /**< DFAEIE bit position */
00353 k_faeint_cmdlkie_shift = 4, /**< CMDLKIE bit position */
00354 k_faeint_cfaeie_shift = 7, /**< CFAEIE bit position */
00355
00356 k_faeint_dfaeie = (1U « 3), /**< Data Flash Access Violation IE */
00357 k_faeint_cmdlkie = (1U « 4), /**< Command Lock Interrupt Enable */
00358 k_faeint_cfaeie = (1U « 7), /**< Code Flash Access Violation IE */
00359
00360 k_faeint_reset_value = 0x98, /**< Reset value (all enabled) */
00361 } rx_faeint_bits_t;
00362
00363 /**
00364 * @brief Get pointer to Flash Access Error Interrupt Enable (FAEINT)
00365 * @return Volatile pointer to FAEINT register
00366 * @since Version 1.0.0
00367 */
00368 static inline volatile uint8_t* faeint_reg(void)
00369 {
00370 return (volatile uint8_t*)k_faeint_addr;
00371 }
00372
00373 /*
=====
00374 * FRDYIE - Flash Ready Interrupt Enable @ 0x007FE018
00375 *
=====
00376 */
00377 /**
00378 **
00379 * @enum rx_frdyie_bits_t
00380 * @brief Flash Ready Interrupt Enable Register (FRDYIE) bit definitions
00381 *
00382 * @since Version 1.0.0
00383 */
00384 typedef enum : uint8_t {
00385 k_frdyie_frdyie = (1U « 0), /**< Flash Ready Interrupt Enable */
00386 } rx_frdyie_bits_t;
00387
00388 /**
00389 * @brief Get pointer to Flash Ready Interrupt Enable (FRDYIE)
00390 * @return Volatile pointer to FRDYIE register
00391 * @since Version 1.0.0
00392 */
00393 static inline volatile uint8_t* frdyie_reg(void)
00394 {
00395 return (volatile uint8_t*)k_frdyie_addr;
00396 }
00397
00398 /*
=====
00399 * FSTATR - Flash Status Register @ 0x007FE080
00400 *
=====
00401 */
00402 /**
00403 **
00404 * @enum rx_fstatr_bits_t
00405 * @brief Flash Status Register (FSTATR) bit definitions
00406 *
00407 * @details
00408 * FSTATR @ 0x007FE080 indicates flash sequencer state.
00409 * This is the primary register for monitoring flash operations.
00410 *
00411 * @par Bit Layout (32-bit)
00412 * | Bit | Name | Description |
00413 * |-----|-----|-----|
00414 * | 6 | FLWEERR | Flash P/E Protect Error |
00415 * | 8 | PRGSPD | Program Suspend Status |
00416 * | 9 | ERSSPD | Erase Suspend Status |

```

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00417 * | 10 | DBFULL | Data Buffer Full |
00418 * | 11 | SUSRDY | Suspend Ready |
00419 * | 12 | PRGERR | Program Error |
00420 * | 13 | ERSERR | Erase Error |
00421 * | 14 | ILGLERR | Illegal Command/Access Error |
00422 * | 15 | FRDY | Flash Ready (1=ready, 0=busy) |
00423 * | 20 | OTERR | Other Error |
00424 * | 21 | SECERR | Security Error |
00425 * | 22 | FESETERR | FENTRY Setting Error |
00426 * | 23 | ILGCOMERR | Illegal Command Error |
00427
00428 * @since Version 1.0.0
00429 */
00430 typedef enum : uint32_t {
00431 /* Bit positions */
00432 k_fstatr_flweerr_shift = 6,
00433 k_fstatr_prghspd_shift = 8,
00434 k_fstatr_erghspd_shift = 9,
00435 k_fstatr_dbfull_shift = 10,
00436 k_fstatr_susrdy_shift = 11,
00437 k_fstatr_prgerr_shift = 12,
00438 k_fstatr_erserr_shift = 13,
00439 k_fstatr_ilglerr_shift = 14,
00440 k_fstatr_frdy_shift = 15,
00441 k_fstatr_oterr_shift = 20,
00442 k_fstatr_secerr_shift = 21,
00443 k_fstatr_feseterr_shift = 22,
00444 k_fstatr_ilgcomerr_shift = 23,
00445
00446 /* Bit masks */
00447 k_fstatr_flweerr = (1UL « 6), /**< Flash P/E Protect Error */
00448 k_fstatr_prgspd = (1UL « 8), /**< Program Suspend Status */
00449 k_fstatr_erghspd = (1UL « 9), /**< Erase Suspend Status */
00450 k_fstatr_dbfull = (1UL « 10), /**< Data Buffer Full */
00451 k_fstatr_susrdy = (1UL « 11), /**< Suspend Ready */
00452 k_fstatr_prgerr = (1UL « 12), /**< Program Error */
00453 k_fstatr_erserr = (1UL « 13), /**< Erase Error */
00454 k_fstatr_ilglerr = (1UL « 14), /**< Illegal Error */
00455 k_fstatr_frdy = (1UL « 15), /**< Flash Ready */
00456 k_fstatr_oterr = (1UL « 20), /**< Other Error */
00457 k_fstatr_secerr = (1UL « 21), /**< Security Error */
00458 k_fstatr_feseterr = (1UL « 22), /**< FENTRY Setting Error */
00459 k_fstatr_ilgcomerr = (1UL « 23), /**< Illegal Command Error */
00460
00461 /* Combined error mask */
00462 k_fstatr_error_mask = (k_fstatr_flweerr | k_fstatr_prgerr | k_fstatr_erserr | k_fstatr_ilglerr |
00463 k_fstatr_oterr | k_fstatr_secerr | k_fstatr_feseterr | k_fstatr_ilgcomerr),
00464
00465 k_fstatr_reset_value = 0x00008000, /**< Reset value (FRDY=1) */
00466 } rx_fstatr_bits_t;
00467
00468 /**
00469 * @brief Get pointer to Flash Status Register (FSTATR)
00470 * @return Volatile pointer to FSTATR register
00471 * @since Version 1.0.0
00472 */
00473 static inline volatile uint32_t* fstatr_reg(void)
00474 {
00475 return (volatile uint32_t*)k_fstatr_addr;
00476 }
00477
00478 /*
=====
00479 * FENTRYR - Flash P/E Mode Entry Register @ 0x007FE084
00480 *
=====
00481 */
00482
00483 /**
00484 * @enum rx_fentryr_bits_t
00485 * @brief Flash P/E Mode Entry Register (FENTRYR) bit definitions
00486 *
00487 * @details
00488 * FENTRYR @ 0x007FE084 controls entry into P/E (Program/Erase) mode.
00489 * Write with KEY=0xAA to modify FENTRYC/FENTRYD bits.
00490 *
00491 * @par Entry Values
00492 * - 0xAA01: Enter code flash P/E mode
00493 * - 0xAA80: Enter data flash P/E mode
00494 * - 0xAA00: Exit P/E mode (return to read mode)
00495 *
00496 * @warning Writing 0xAA81 causes ILGLERR error
00497 *
00498 * @since Version 1.0.0
00499 */
00500 typedef enum : uint16_t {
00501 k_fentryr_fentryc = (1U « 0), /**< Code Flash P/E Mode Entry */

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00502 k_fentryr_fentryd = (1U « 7), /**< Data Flash P/E Mode Entry */
00503 k_fentryr_key_shift = 8, /**< Key code position */
00504 k_fentryr_key = 0xAA00, /**< Key code (write only) */
00505
00506 k_fentryr_code_pe = 0xAA01, /**< Enter code flash P/E mode */
00507 k_fentryr_data_pe = 0xAA80, /**< Enter data flash P/E mode */
00508 k_fentryr_read_mode = 0xAA00, /**< Exit P/E mode */
00509 } rx_fentryr_bits_t;
00510
00511 /**
00512 * @brief Get pointer to Flash P/E Mode Entry Register (FENTRYR)
00513 * @return Volatile pointer to FENTRYR register
00514 * @since Version 1.0.0
00515 */
00516 static inline volatile uint16_t* fentryr_reg(void)
00517 {
00518 return (volatile uint16_t*)k_fentryr_addr;
00519 }
00520
00521 /**
=====
00522 * FSUINITR - Flash Sequencer Setup Initialization @ 0x007FE08C
00523 *
=====
00524 */
00525
00526 /**
00527 * @enum rx_fsuinitr_bits_t
00528 * @brief Flash Sequencer Setup Initialization Register (FSUINITR) bit defs
00529 *
00530 * @details
00531 * FSUINITR @ 0x007FE08C initializes flash sequencer setup registers.
00532 * Write with KEY=0x2D to enable SUINIT write.
00533 *
00534 * @since Version 1.0.0
00535 */
00536 typedef enum : uint16_t {
00537 k_fsuinitr_suinit = (1U « 0), /**< Setup Initialization */
00538 k_fsuinitr_key = 0x2D00, /**< Key code for write enable */
00539
00540 k_fsuinitr_init_cmd = 0x2D01, /**< Initialize flash sequencer setup */
00541 } rx_fsuinitr_bits_t;
00542
00543 /**
00544 * @brief Get pointer to Flash Sequencer Setup Init Register (FSUINITR)
00545 * @return Volatile pointer to FSUINITR register
00546 * @since Version 1.0.0
00547 */
00548 static inline volatile uint16_t* fsuinitr_reg(void)
00549 {
00550 return (volatile uint16_t*)k_fsuinitr_addr;
00551 }
00552
00553 /**
=====
00554 * FSADDR / FEADDR - FACI Address Registers @ 0x007FE030 / 0x007FE034
00555 *
=====
00556 */
00557
00558 /**
00559 * @brief Get pointer to FACI Start Address Register (FSADDR)
00560 * @return Volatile pointer to FSADDR register
00561 * @since Version 1.0.0
00562 */
00563 static inline volatile uint32_t* fsaddr_reg(void)
00564 {
00565 return (volatile uint32_t*)k_fsaddr_addr;
00566 }
00567
00568 /**
00569 * @brief Get pointer to FACI End Address Register (FEADDR)
00570 * @return Volatile pointer to FEADDR register
00571 * @since Version 1.0.0
00572 */
00573 static inline volatile uint32_t* feaddr_reg(void)
00574 {
00575 return (volatile uint32_t*)k_feaddr_addr;
00576 }
00577
00578 /**
=====
00579 * FCMR - FACI Command Register @ 0x007FE0A0
00580 *
=====
00581 */
00582

```



```

00583 /**
00584 * @enum rx_fcmdr_bits_t
00585 * @brief FACI Command Register (FCMDR) bit definitions
00586 *
00587 * @details
00588 * FCMDR @ 0x007FE0A0 contains the two most recent commands received.
00589 * Read-only register for debugging.
00590 *
00591 * @since Version 1.0.0
00592 */
00593 typedef enum : uint16_t {
00594 k_fcmdr_pcmdr_mask = 0x00FF, /**< Previous command mask (bits 7:0) */
00595 k_fcmdr_pcmdr_shift = 0, /**< Previous command shift */
00596 k_fcmdr_cmdr_mask = 0xFF00, /**< Current command mask (bits 15:8) */
00597 k_fcmdr_cmdr_shift = 8, /**< Current command shift */
00598 } rx_fcmdr_bits_t;
00599
00600 /**
00601 * @brief Get pointer to FACI Command Register (FCMDR)
00602 * @return Volatile pointer to FCMDR register
00603 * @since Version 1.0.0
00604 */
00605 static inline volatile uint16_t* fcmdr_reg(void)
00606 {
00607 return (volatile uint16_t*)k_fcmdr_addr;
00608 }
00609
00610 /*
=====
00611 * FACI Commands - Written to 0x007E0000
00612 *
=====
00613 */
00614
00615 /**
00616 * @enum rx_faci_commands_t
00617 * @brief FACI command codes
00618 *
00619 * @details
00620 * Commands are written to the FACI command issuing area @ 0x007E0000.
00621 * Command sequences vary; see manual Ch62 Section 62.8.1 for details.
00622 *
00623 * @since Version 1.0.0
00624 */
00625 typedef enum : uint8_t {
00626 k_faci_cmd_program = 0xE8, /**< Program command */
00627 k_faci_cmd_block_erase = 0x20, /**< Block erase command */
00628 k_faci_cmd_pe_suspend = 0xB0, /**< P/E suspend command */
00629 k_faci_cmd_pe_resume = 0xD0, /**< P/E resume command */
00630 k_faci_cmd_status_clear = 0x50, /**< Status clear command */
00631 k_faci_cmd_forced_stop = 0xB3, /**< Forced stop command */
00632 k_faci_cmd_blank_check = 0x71, /**< Blank check command */
00633 k_faci_cmd_config_set = 0x40, /**< Configuration set command */
00634 k_faci_cmd_d0 = 0xD0, /**< Command terminator */
00635 } rx_faci_commands_t;
00636
00637 /**
00638 * @brief Get pointer to FACI Command Issuing Area
00639 *
00640 * @details
00641 * Commands are issued by writing to this 4-byte area.
00642 * The write sequence depends on the command type.
00643 *
00644 * @return Volatile pointer to FACI command area
00645 * @since Version 1.0.0
00646 */
00647 static inline volatile uint8_t* faci_cmd_area(void)
00648 {
00649 return (volatile uint8_t*)k_faci_cmd_area;
00650 }
00651
00652 /*
=====
00653 * FPCKAR - Flash Sequencer Clock Frequency Notification @ 0x007FE0E4
00654 *
=====
00655 */
00656
00657 /**
00658 * @enum rx_fpckar_bits_t
00659 * @brief Flash Sequencer Clock Notification Register (FPCKAR) bit definitions
00660 *
00661 * @details
00662 * FPCKAR @ 0x007FE0E4 notifies the flash sequencer of FCLK frequency.
00663 * Must be set before issuing FACI commands for correct timing.
00664 * Write with KEY=0x1E to modify PCKA field.
00665 */

```



```

00666 * @par PCKA Values
00667 * Set PCKA = FCLK frequency in MHz (e.g., 60 for 60 MHz FCLK)
00668 * Valid range: 4 to 60 MHz
00669 *
00670 * @since Version 1.0.0
00671 */
00672 typedef enum : uint16_t {
00673 k_fpckar_pcka_mask = 0x00FF, /**< PCKA frequency field mask */
00674 k_fpckar_key_shift = 8, /**< Key code position */
00675 k_fpckar_key = 0x1E00, /**< Key code for write enable */
00676 } rx_fpckar_bits_t;
00677
00678 /**
00679 * @brief Get pointer to Flash Sequencer Clock Register (FPCKAR)
00680 * @return Volatile pointer to FPCKAR register
00681 * @since Version 1.0.0
00682 */
00683 static inline volatile uint16_t* fpckar_reg(void)
00684 {
00685 return (volatile uint16_t*)k_fpckar_addr;
00686 }
00687
00688 /*
=====
00689 * EEPFCLK - Data Flash Memory Access Frequency @ 0x007FC040
00690 *
=====
00691 */
00692
00693 /**
00694 * @brief Get pointer to Data Flash Access Frequency Register (EEPFCLK)
00695 * @return Volatile pointer to EEPFCLK register
00696 * @since Version 1.0.0
00697 */
00698 static inline volatile uint8_t* eepfelk_reg(void)
00699 {
00700 return (volatile uint8_t*)k_eepfclk_addr;
00701 }
00702
00703 /*
=====
00704 * FBCCNT - Blank Check Control @ 0x007FE0D0
00705 *
=====
00706 */
00707
00708 /**
00709 * @enum rx_fbcnt_bits_t
00710 * @brief Blank Check Control Register (FBCCNT) bit definitions
00711 *
00712 * @since Version 1.0.0
00713 */
00714 typedef enum : uint8_t {
00715 k_fbcnt_bcdir = (1U << 0), /**< Blank Check Direction (0=forward, 1=reverse) */
00716 } rx_fbcnt_bits_t;
00717
00718 /**
00719 * @brief Get pointer to Blank Check Control Register (FBCCNT)
00720 * @return Volatile pointer to FBCCNT register
00721 * @since Version 1.0.0
00722 */
00723 static inline volatile uint8_t* fbcnt_reg(void)
00724 {
00725 return (volatile uint8_t*)k_fbcnt_addr;
00726 }
00727
00728 /*
=====
00729 * FBCSTAT - Blank Check Status @ 0x007FE0D4
00730 *
=====
00731 */
00732
00733 /**
00734 * @enum rx_fbcstat_bits_t
00735 * @brief Blank Check Status Register (FBCSTAT) bit definitions
00736 *
00737 * @since Version 1.0.0
00738 */
00739 typedef enum : uint8_t {
00740 k_fbcstat_bcst = (1U << 0), /**< Blank Check Status (0=blank, 1=not blank) */
00741 } rx_fbcstat_bits_t;
00742
00743 /**
00744 * @brief Get pointer to Blank Check Status Register (FBCSTAT)
00745 * @return Volatile pointer to FBCSTAT register
00746 * @since Version 1.0.0

```

```

00747 */
00748 static inline volatile uint8_t* fbcstat_reg(void)
00749 {
00750 return (volatile uint8_t*)k_fbcstat_addr;
00751 }
00752
00753 /*
=====
00754 * Additional Register Accessors
00755 *
=====
00756 */
00757 /**
00758 * @brief Get pointer to Data Flash Programming Start Address (FPSADDR)
00759 * @return Volatile pointer to FPSADDR register
00760 * @since Version 1.0.0
00761 */
00762 static inline volatile uint32_t* fpsaddr_reg(void)
00763 {
00764 return (volatile uint32_t*)k_fpsaddr_addr;
00765 }
00766 /**
00767 * @brief Get pointer to Flash Access Window Monitor (FAWMON)
00768 * @return Volatile pointer to FAWMON register
00769 * @since Version 1.0.0
00770 */
00771 static inline volatile uint32_t* fawmon_reg(void)
00772 {
00773 return (volatile uint32_t*)k_fawmon_addr;
00774 }
00775 /**
00776 * @brief Get pointer to Flash Sequencer Processing Switch (FCPSR)
00777 * @return Volatile pointer to FCPSR register
00778 * @since Version 1.0.0
00779 */
00780 static inline volatile uint16_t* fcpsr_reg(void)
00781 {
00782 return (volatile uint16_t*)k_fcpsr_addr;
00783 }
00784 /**
00785 * @brief Get pointer to Flash Startup Area Select (FSUACR)
00786 * @return Volatile pointer to FSUACR register
00787 * @since Version 1.0.0
00788 */
00789 static inline volatile uint16_t* fsuacr_reg(void)
00790 {
00791 return (volatile uint16_t*)k_fsuacr_addr;
00792 }
00793
00794 /*
=====
00795 * Unique ID Registers - Read-only chip identification @ FE7F 7D90h-7D9Ch
00796 *
=====
00797 */
00798 /**
00799 * @enum rx_uidr_addresses_t
00800 * @brief Unique ID Register addresses (Ch62 Section 62.4.23)
00801 *
00802 * @details
00803 * UIDR0-3 contain a 128-bit unique identifier for each chip.
00804 * These are read-only registers in the FLASHCONST area (on-chip ROM).
00805 *
00806 * @par Memory Map (FLASHCONST area)
00807 * | Address | Register | Description |
00808 * |-----|-----|-----|
00809 * | 0xFE7F7D90 | UIDR0 | Unique ID bits 31-0 |
00810 * | 0xFE7F7D94 | UIDR1 | Unique ID bits 63-32 |
00811 * | 0xFE7F7D98 | UIDR2 | Unique ID bits 95-64 |
00812 * | 0xFE7F7D9C | UIDR3 | Unique ID bits 127-96 |
00813 *
00814 * @par Access Requirements
00815 * - **Must read in 32-bit units** (as per manual specification)
00816 * - **SYSCR0.ROME must be 1** (on-chip ROM enabled) - this is the default
00817 * in single-chip mode, so no special setup needed for normal firmware
00818 * - ROMCODE/SPCC security settings do NOT block CPU self-reads (only
00819 * external programmer access)
00820 *
00821 * @note Values are factory-programmed and unique to each chip
00822 * @note These addresses are in FLASHCONST area (FE7F xxxx), not the typical
00823 * FACI register space (007F xxxx) - this is expected per manual
00824 */

```

```

00830 * @since Version 1.0.0
00831 */
00832 typedef enum : uintptr_t {
00833 k_uidr0_addr = 0xFE7F7D90, /**< Unique ID Register 0 (bits 31-0) */
00834 k_uidr1_addr = 0xFE7F7D94, /**< Unique ID Register 1 (bits 63-32) */
00835 k_uidr2_addr = 0xFE7F7D98, /**< Unique ID Register 2 (bits 95-64) */
00836 k_uidr3_addr = 0xFE7F7D9C, /**< Unique ID Register 3 (bits 127-96) */
00837 } rx_uidr_addresses_t;
00838
00839 /**
00840 * @brief Get pointer to Unique ID Register 0 (UIDR0)
00841 * @return Volatile pointer to UIDR0 register (read-only)
00842 * @note Access may be restricted depending on security settings
00843 * @since Version 1.0.0
00844 */
00845 static inline volatile const uint32_t* uidr0_reg(void)
00846 {
00847 return (volatile const uint32_t*)k_uidr0_addr;
00848 }
00849
00850 /**
00851 * @brief Get pointer to Unique ID Register 1 (UIDR1)
00852 * @return Volatile pointer to UIDR1 register (read-only)
00853 * @since Version 1.0.0
00854 */
00855 static inline volatile const uint32_t* uidr1_reg(void)
00856 {
00857 return (volatile const uint32_t*)k_uidr1_addr;
00858 }
00859
00860 /**
00861 * @brief Get pointer to Unique ID Register 2 (UIDR2)
00862 * @return Volatile pointer to UIDR2 register (read-only)
00863 * @since Version 1.0.0
00864 */
00865 static inline volatile const uint32_t* uidr2_reg(void)
00866 {
00867 return (volatile const uint32_t*)k_uidr2_addr;
00868 }
00869
00870 /**
00871 * @brief Get pointer to Unique ID Register 3 (UIDR3)
00872 * @return Volatile pointer to UIDR3 register (read-only)
00873 * @since Version 1.0.0
00874 */
00875 static inline volatile const uint32_t* uidr3_reg(void)
00876 {
00877 return (volatile const uint32_t*)k_uidr3_addr;
00878 }
00879
00880 /*
=====
00881 * Flash Memory Address Constants
00882 *
=====
00883 */
00884
00885 /**
00886 * @enum rx_flash_memory_addresses_t
00887 * @brief Flash memory region base and end addresses for the RX72N
00888 *
00889 * @details
00890 * Defines the inclusive address range of each flash region on the RX72N
00891 * (4 MB code flash, 32 KB data flash). Addresses are physical bus addresses
00892 * as documented in the RX72N Hardware Manual, Section 5 (Memory Map).
00893 *
00894 * Code flash occupies the top of the 32-bit address space (0xFFC00000 -
00895 * 0xFFFFFFFF). Data flash is a separate EEPROM-emulation region beginning at
00896 * 0x00100000. Neither region overlaps RAM or peripheral space.
00897 *
00898 * Use these constants as bounds for range checks before issuing erase or
00899 * program commands via the Flash Control Unit (FCU). Never use raw literals
00900 * for flash address comparisons; always reference these enum values so that
00901 * any future memory-map revision is caught at a single point.
00902 *
00903 * @note All addresses must be naturally aligned to the program unit defined
00904 * in rx_flash_memory_layout_t (128 bytes for code flash, 4 bytes for
00905 * data flash). Unaligned addresses are rejected by the FCU hardware.
00906 * @note Code flash erase granularity differs by block index: blocks 0-7 are
00907 * 8 KB each, blocks 8 and above are 32 KB each (see rx_flash_memory_layout_t).
00908 *
00909 * @invariant k_code_flash_start < k_code_flash_end
00910 * @invariant k_data_flash_start < k_data_flash_end
00911 * @invariant Code flash and data flash address ranges do not overlap
00912 *
00913 * @par Usage Example:
00914 * @code

```

```

00915 * bool is_code_flash(uintptr_t addr) {
00916 * return addr >= (uintptr_t)k_code_flash_start &&
00917 * addr <= (uintptr_t)k_code_flash_end;
00918 * }
00919 * @endcode
00920 *
00921 * @see rx_flash_memory_layout_t Block sizes and program units for each region
00922 * @see fentryr_reg() FCU entry register that enables erase/program mode
00923 * @see fsaddr_reg() FCU start-address register (set before erase/program)
00924 * @see feaddr_reg() FCU end-address register (set before block erase)
00925 * @see rx_flash.h Higher-level flash API (when implemented)
00926 *
00927 * @since Version 1.0.0
00928 */
00929 typedef enum : uintptr_t {
00930 k_code_flash_start = 0xFFC00000, /**< Code flash start address (bottom of 4 MB region) */
00931 k_code_flash_end = 0xFFFFFFFF, /**< Code flash end address (top of 32-bit space) */
00932 k_data_flash_start = 0x00100000, /**< Data flash start address (EEPROM-emulation region) */
00933 k_data_flash_end = 0x00107FFF, /**< Data flash end address (32 KB above start) */
00934 } rx_flash_memory_addresses_t;
00935
00936 /**
00937 * @enum rx_flash_memory_layout_t
00938 * @brief Flash memory region sizes, erase block granularities, and program units
00939 *
00940 * @details
00941 * Captures the capacity and write/erase geometry of the RX72N flash subsystem
00942 * as documented in the RX72N Hardware Manual, Sections 7 and 59.
00943 *
00944 * Code flash (4 MB) has a split-block architecture:
00945 * - Blocks 0-7: 8 KB each (small blocks at the top of the address space,
00946 * typically used for bootloader or parameter storage)
00947 * - Blocks 8 and above: 32 KB each (large blocks for application code)
00948 *
00949 * Data flash (32 KB) is organised in uniform 64-byte erase blocks and is
00950 * intended for EEPROM-emulation of configuration data and run-time state.
00951 *
00952 * Program (write) operations must be issued in multiples of the program unit:
00953 * - Code flash: 128-byte aligned writes
00954 * - Data flash: 4-byte aligned writes
00955 *
00956 * Use these constants when computing erase/program address ranges, validating
00957 * buffer sizes, or determining which block type covers a given address before
00958 * issuing FCU commands.
00959 *
00960 * @note All values are byte counts or byte sizes, not word or block counts.
00961 * @note The program unit constraints are enforced in hardware; the FCU will
00962 * set FSTATR.PRGERR if an unaligned or undersized write is attempted.
00963 *
00964 * @invariant k_code_flash_size == (k_code_flash_block_8k * 8) + (k_code_flash_block_32k * 126)
00965 * @invariant k_data_flash_size == k_data_flash_block_64 * 512
00966 *
00967 * @par Usage Example:
00968 * @code
00969 bool is_aligned_code_flash_write(uintptr_t addr, uint32_t len) {
00970 return (addr % k_code_flash_program_unit == 0) &&
00971 (len % k_code_flash_program_unit == 0);
00972 }
00973 * @endcode
00974 *
00975 * @see rx_flash_memory_addresses_t Physical start/end addresses for each region
00976 * @see fstatr_reg() FCU status register (check PRGERR after program)
00977 * @see fsaddr_reg() FCU start-address register (set before erase/program)
00978 * @see feaddr_reg() FCU end-address register (set before block erase)
00979 * @see rx_flash.h Higher-level flash API (when implemented)
00980 *
00981 * @since Version 1.0.0
00982 */
00983 typedef enum : uint32_t {
00984 k_code_flash_size = 0x00400000, /**< Total code flash size in bytes (4 MB) */
00985 k_data_flash_size = 0x00008000, /**< Total data flash size in bytes (32 KB) */
00986
00987 k_code_flash_block_8k = 0x00002000, /**< Erase block size for blocks 0-7 (8 KB each) */
00988 k_code_flash_block_32k = 0x00008000, /**< Erase block size for blocks 8+ (32 KB each) */
00989 k_data_flash_block_64 = 0x00000040, /**< Erase block size for data flash (64 bytes each) */
00990
00991 k_code_flash_program_unit =
00992 128U, /**< Minimum program unit for code flash (bytes, must be aligned) */
00993 k_data_flash_program_unit =
00994 4U, /**< Minimum program unit for data flash (bytes, must be aligned) */
00995 } rx_flash_memory_layout_t;
00996
00997 /**
00998 * Static Assertions - Verify Register Addresses at Compile Time
00999 *
=====

```

```

01000 */
01001
01002 /* ROM Cache register addresses (Ch62 Section 62.4.1-62.4.4) */
01003 static_assert(k_romce_addr == 0x00081000, "ROMCE address mismatch");
01004 static_assert(k_romciv_addr == 0x00081004, "ROMCIV address mismatch");
01005 static_assert(k_ncrg0_addr == 0x00081040, "NCRG0 address mismatch");
01006 static_assert(k_ncrc0_addr == 0x00081044, "NCRC0 address mismatch");
01007 static_assert(k_ncrg1_addr == 0x00081048, "NCRG1 address mismatch");
01008 static_assert(k_ncrc1_addr == 0x0008104C, "NCRC1 address mismatch");
01009
01010 /* Flash P/E Protect register (Ch62 Section 62.4.5) */
01011 static_assert(k_fweprror_addr == 0x0008C296, "FWEPOR address mismatch");
01012
01013 /* FACI registers (Ch62 Section 62.4.6-62.4.22) */
01014 static_assert(k_fastat_addr == 0x007FE010, "FASTAT address mismatch");
01015 static_assert(k_faeint_addr == 0x007FE014, "FAEINT address mismatch");
01016 static_assert(k_frdyie_addr == 0x007FE018, "FRDYIE address mismatch");
01017 static_assert(k_fsaddr_addr == 0x007FE030, "FSADDR address mismatch");
01018 static_assert(k_feaddr_addr == 0x007FE034, "FEADDR address mismatch");
01019 static_assert(k_fstatr_addr == 0x007FE080, "FSTATR address mismatch");
01020 static_assert(k_fentryr_addr == 0x007FE084, "FENTRYR address mismatch");
01021 static_assert(k_fsuinitr_addr == 0x007FE08C, "FSUINTR address mismatch");
01022 static_assert(k_fcmdr_addr == 0x007FE0A0, "FCMDR address mismatch");
01023 static_assert(k_fbcnt_addr == 0x007FE0D0, "FBCCNT address mismatch");
01024 static_assert(k_fbcstat_addr == 0x007FE0D4, "FBCSTAT address mismatch");
01025 static_assert(k_fpsaddr_addr == 0x007FE0D8, "FPSADDR address mismatch");
01026 static_assert(k_fawmon_addr == 0x007FE0DC, "FAWMON address mismatch");
01027 static_assert(k_fcpsr_addr == 0x007FE0E0, "FCPSR address mismatch");
01028 static_assert(k_fpckar_addr == 0x007FE0E4, "FPCKAR address mismatch");
01029 static_assert(k_fsuacr_addr == 0x007FE0E8, "FSUACR address mismatch");
01030
01031 /* FACI command area (Ch62 Section 62.2) */
01032 static_assert(k_faci_cmd_addr == 0x007FE0000, "FACI command area address mismatch");
01033
01034 /* Data flash frequency register (Ch62 Section 62.4.22) */
01035 static_assert(k_eepfclk_addr == 0x007FC040, "EEPFCCLK address mismatch");
01036
01037 /* Unique ID registers (Ch62 Section 62.4.23) - FLASHCONST area
01038 * These are in on-chip ROM area (FE7F xxxx). Readable when SYSCR0.ROME=1 (default).
01039 * Must be read in 32-bit units. ROMCODE/SPCC do NOT block CPU self-reads. */
01040 static_assert(k_uidr0_addr == 0xFE7F7D90, "UIDR0 address mismatch");
01041 static_assert(k_uidr1_addr == 0xFE7F7D94, "UIDR1 address mismatch");
01042 static_assert(k_uidr2_addr == 0xFE7F7D98, "UIDR2 address mismatch");
01043 static_assert(k_uidr3_addr == 0xFE7F7D9C, "UIDR3 address mismatch");
01044
01045 /* Flash memory regions */
01046 static_assert(k_code_flash_start == 0xFFC00000, "Code flash start address mismatch");
01047 static_assert(k_data_flash_start == 0x00100000, "Data flash start address mismatch");
01048
01049 #ifdef __cplusplus
01050 }
01051 #endif

```

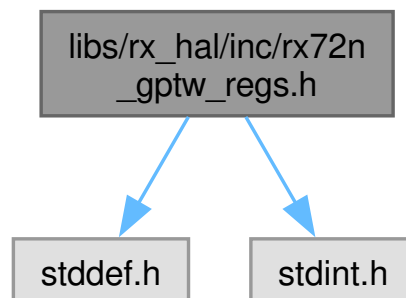
## 8.25 libs/rx\_hal/inc/rx72n\_gptw\_regs.h File Reference

RX72N GPTW (General PWM Timer) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_gptw\_regs.h:



This graph shows which files directly or indirectly include this file:



## Data Structures

- struct [rx\\_gptw\\_channel\\_regs\\_t](#)  
GPTW Channel Register Map for 32-bit PWM generation.
- struct [rx\\_gptw\\_common\\_regs\\_t](#)  
GPTW Common Register Map.

## Enumerations

- enum [gptw\\_padding\\_sizes\\_t](#) : uint8\_t { [k\\_gptw\\_ch\\_reserved\\_count](#) = 6U }
- enum [gptw\\_addresses\\_t](#) : uintptr\_t {  
[k\\_gptw\\_channel\\_offset](#) = 0x100 , [k\\_gptw0\\_base\\_addr](#) = 0x000C2000 , [k\\_gptw1\\_base\\_addr](#) = 0x000C2100 , [k\\_gptw2\\_base\\_addr](#) = 0x000C2200 ,  
[k\\_gptw3\\_base\\_addr](#) = 0x000C2300 , [k\\_gptw\\_common\\_base\\_addr](#) }  
GPTW hardware addresses (verified against RX72N Hardware Manual).
- enum [gptw\\_gtwp\\_bits\\_t](#) : uint16\_t {  
[k\\_gptw\\_gtwp\\_wp0](#) = (1 << 0) , [k\\_gptw\\_gtwp\\_wp1](#) = (1 << 1) , [k\\_gptw\\_gtwp\\_wp2](#) = (1 << 2) , [k\\_gptw\\_gtwp\\_wp3](#) = (1 << 3) ,  
[k\\_gptw\\_gtwp\\_unlock](#) = 0xA500 , [k\\_gptw\\_gtwp\\_lock](#) = 0xA501 }
- enum [gptw\\_gtcr\\_bits\\_t](#) : uint32\_t {  
[k\\_gptw\\_gtcr\\_cst](#) = (1 << 0) , [k\\_gptw\\_gtcr\\_icds](#) = (1 << 8) , [k\\_gptw\\_gtcr\\_md\\_shift](#) = 16 ,  
[k\\_gptw\\_gtcr\\_md\\_mask](#) = (0x07 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_md\\_saw\\_pwm](#) = (0x00 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) , [k\\_gptw\\_gtcr\\_md\\_saw\\_oneshot](#) = (0x01 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) , [k\\_gptw\\_gtcr\\_md\\_tri\\_pwm1](#) = (0x04 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_md\\_tri\\_pwm2](#) = (0x05 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_md\\_tri\\_pwm3](#) = (0x06 << [k\\_gptw\\_gtcr\\_md\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_shift](#) = 23 , [k\\_gptw\\_gtcr\\_tpcs\\_mask](#) = (0x0F << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_1](#) = (0x00 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_tpcs\\_2](#) = (0x01 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_4](#) = (0x02 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_8](#) = (0x03 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_tpcs\\_16](#) = (0x04 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_tpcs\\_32](#) = (0x05 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_64](#) = (0x06 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) , [k\\_gptw\\_gtcr\\_tpcs\\_256](#) = (0x08 << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) ,  
[k\\_gptw\\_gtcr\\_tpcs\\_1024](#) = (0x0A << [k\\_gptw\\_gtcr\\_tpcs\\_shift](#)) }
- enum [gptw\\_gtior\\_bits\\_t](#) : uint32\_t {  
[k\\_gptw\\_gtior\\_oa\\_shift](#) = 0 , [k\\_gptw\\_gtior\\_oa\\_mask](#) = 0x1F , [k\\_gptw\\_gtior\\_oa\\_disabled](#) = 0x00 , [k\\_gptw\\_gtior\\_oa\\_low\\_cmpup](#) = 0x01 ,  
[k\\_gptw\\_gtior\\_oa\\_high\\_cmpup](#) = 0x02 , [k\\_gptw\\_gtior\\_oa\\_toggle](#) = 0x03 , [k\\_gptw\\_gtior\\_oa\\_init\\_low](#) ,  
[k\\_gptw\\_gtior\\_oa\\_init\\_high](#) ,  
[k\\_gptw\\_gtior\\_oae](#) = (1 << 8) , [k\\_gptw\\_gtior\\_ob\\_shift](#) = 16 , [k\\_gptw\\_gtior\\_ob\\_mask](#) = (0x1F << 16) , [k\\_gptw\\_gtior\\_ob\\_disabled](#) = (0x00 << 16) ,  
[k\\_gptw\\_gtior\\_ob\\_low\\_cmpup](#) = (0x01 << 16) , [k\\_gptw\\_gtior\\_ob\\_high\\_cmpup](#) = (0x02 << 16) , [k\\_gptw\\_gtior\\_ob\\_toggle](#) = (0x03 << 16) , [k\\_gptw\\_gtior\\_ob\\_init\\_low](#) ,  
[k\\_gptw\\_gtior\\_ob\\_init\\_high](#) , [k\\_gptw\\_gtior\\_obe](#) = (1 << 24) }
- enum [gptw\\_gtintad\\_bits\\_t](#) : uint32\_t {  
[k\\_gptw\\_gtintad\\_grp\\_shift](#) = 24 , [k\\_gptw\\_gtintad\\_grp\\_mask](#) = (0x03U << 24) , [k\\_gptw\\_gtintad\\_grp\\_a](#) = (0x00U << 24) , [k\\_gptw\\_gtintad\\_grp\\_b](#) = (0x01U << 24) ,  
[k\\_gptw\\_gtintad\\_grp\\_c](#) = (0x02U << 24) , [k\\_gptw\\_gtintad\\_grp\\_d](#) = (0x03U << 24) ,  
[k\\_gptw\\_gtintad\\_grpdte](#) = (1U << 28) , [k\\_gptw\\_gtintad\\_grpabh](#) = (1U << 29) ,  
[k\\_gptw\\_gtintad\\_grpabl](#) = (1U << 30) }

GTINTAD register bit definitions for POEG group linkage.

- enum `gptw_gtber_bits_t` : `uint32_t` {  
`k_gptw_gtber_bd0` = (1U << 0) , `k_gptw_gtber_bd1` = (1U << 1) , `k_gptw_gtber_bd2` = (1U << 2) , `k_gptw_gtber_bd3` = (1U << 3) ,  
`k_gptw_gtber_ccra_single` = (0x01U << 16) , `k_gptw_gtber_ccra_double` = (0x02U << 16) ,  
`k_gptw_gtber_ccrb_single` = (0x01U << 18) , `k_gptw_gtber_ccrb_double` = (0x02U << 18) ,  
`k_gptw_gtber_pr_single` = (0x01U << 20) , `k_gptw_gtber_pr_double` = (0x02U << 20) }
- enum `gptw_gtdtcr_bits_t` : `uint8_t` { `k_gptw_gtdtcr_tde` = (1 << 0) , `k_gptw_gtdtcr_tdf` = (1 << 4) }
- enum `gptw_gtst_bits_t` : `uint16_t` {  
`k_gptw_gtst_tcfa` = (1 << 0) , `k_gptw_gtst_tcfb` = (1 << 1) , `k_gptw_gtst_tcf` = (1 << 2) ,  
`k_gptw_gtst_tdfd` = (1 << 3) ,  
`k_gptw_gtst_tcf` = (1 << 4) , `k_gptw_gtst_tcff` = (1 << 5) , `k_gptw_gtst_tcfpo` = (1 << 6) ,  
`k_gptw_gtst_tcfpu` = (1 << 7) ,  
`k_gptw_gtst_tucf` = (1 << 15) }
- enum `gptw_gtstr_bits_t` : `uint8_t` { `k_gptw_gtstr_cst0` = (1 << 0) , `k_gptw_gtstr_cst1` = (1 << 1) , `k_gptw_gtstr_cst2` = (1 << 2) , `k_gptw_gtstr_cst3` = (1 << 3) }
- enum `gptw_gtuddtyc_bits_t` : `uint32_t` { `k_gptw_gtuddtyc_ud` = (1 << 0) , `k_gptw_gtuddtyc_udf` = (1 << 1) }

Up/Down Count Duty Setting Register (GTUDDTYC) Bits.

- enum `gptw_gtadsmr_bits_t` : `uint32_t` { `k_gptw_gtadsmr_adsm0` = (1 << 0) , `k_gptw_gtadsmr_adsm1` = (1 << 1) }
- enum `gptw_gteitc_bits_t` : `uint32_t` {  
`k_gptw_gteitc_eivtc_mask` = 0x0000FFFF , `k_gptw_gteitc_eivtt_shift` = 16 , `k_gptw_gteitc_eivtt_mask` = (0x03 << 16) , `k_gptw_gteitc_eivtt_none` = (0x00 << 16) ,  
`k_gptw_gteitc_eivtt_trough` = (0x01 << 16) , `k_gptw_gteitc_eivtt_crest` = (0x02 << 16) ,  
`k_gptw_gteitc_eivtt_both` = (0x03 << 16) , `k_gptw_gteitc_eivtcn_shift` = 24 ,  
`k_gptw_gteitc_eivtcn_mask` = (0xFFU << 24) }
- enum `gptw_gteitli1_bits_t` : `uint32_t` { `k_gptw_gteitli1_eadt1` = (1 << 0) , `k_gptw_gteitli1_eivtciv` = (1 << 8) , `k_gptw_gteitli1_eivtciv` = (1 << 9) }
- enum `gptw_gteitli2_bits_t` : `uint32_t` {  
`k_gptw_gteitli2_eivtciv_shift` = 0 , `k_gptw_gteitli2_eivtciv_mask` = 0x000000FF , `k_gptw_gteitli2_eivtcia` = (1 << 0) , `k_gptw_gteitli2_eivtcib` = (1 << 1) ,  
`k_gptw_gteitli2_eivtcic` = (1 << 2) , `k_gptw_gteitli2_eivtcid` = (1 << 3) , `k_gptw_gteitli2_eivtcie` = (1 << 4) , `k_gptw_gteitli2_eivtcif` = (1 << 5) }
- enum `gptw_gteitlb_bits_t` : `uint32_t` { `k_gptw_gteitlb_eivtcibl` = (1 << 0) , `k_gptw_gteitlb_eivtcabl` = (1 << 1) , `k_gptw_gteitlb_eivtcibl` = (1 << 2) }
- enum `gptw_gtsecsr_bits_t` : `uint32_t` { `k_gptw_gtsecsr_secsl0` = (1 << 0) , `k_gptw_gtsecsr_secsl1` = (1 << 1) , `k_gptw_gtsecsr_secsl2` = (1 << 2) , `k_gptw_gtsecsr_secsl3` = (1 << 3) }
- enum `gptw_gtsecr_bits_t` : `uint32_t` { `k_gptw_gtsecr_sbdce` = (1 << 0) , `k_gptw_gtsecr_sbdcd` = (1 << 8) , `k_gptw_gtsecr_sbdpe` = (1 << 9) }
- enum `gptw_module_stop_bits_t` : `uint8_t` { `k_mstop_gptw` = 7 }
- enum `gptw_mpc_psel_t` : `uint8_t` { `k_pfs_psel_gptw` = 0x1E }
- enum `gptw_ch_reg_offsets_t` : `uint8_t` {  
`k_gptw_ch_off_gtwp` = 0x00U , `k_gptw_ch_off_gtstr` = 0x04U , `k_gptw_ch_off_gtstp` = 0x08U , `k_gptw_ch_off_gtclr` = 0x0CU ,  
`k_gptw_ch_off_gtssr` = 0x10U , `k_gptw_ch_off_gtcsr` = 0x14U , `k_gptw_ch_off_gtcsr` = 0x18U , `k_gptw_ch_off_gtupsr` = 0x1CU ,  
`k_gptw_ch_off_gtdnsr` = 0x20U , `k_gptw_ch_off_gticasr` = 0x24U , `k_gptw_ch_off_gticbsr` = 0x28U , `k_gptw_ch_off_gtcr` = 0x2CU ,  
`k_gptw_ch_off_gtuddtyc` = 0x30U , `k_gptw_ch_off_gtior` = 0x34U , `k_gptw_ch_off_gtintad` = 0x38U , `k_gptw_ch_off_gtst` = 0x3CU ,  
`k_gptw_ch_off_gtber` = 0x40U , `k_gptw_ch_off_gtite` = 0x44U , `k_gptw_ch_off_gtent` = 0x48U , `k_gptw_ch_off_gtcra` = 0x4CU ,  
`k_gptw_ch_off_gtcrb` = 0x50U , `k_gptw_ch_off_gtcrc` = 0x54U , `k_gptw_ch_off_gtcrc` = 0x58U , `k_gptw_ch_off_gtcrcd` = 0x5CU ,  
`k_gptw_ch_off_gtcrcf` = 0x60U , `k_gptw_ch_off_gtpr` = 0x64U , `k_gptw_ch_off_gtpbr` =



```

0x68U , k_gptw_ch_off_gtpdbr = 0x6CU ,
k_gptw_ch_off_gtadtra = 0x70U , k_gptw_ch_off_gtadtbra = 0x74U , k_gptw_ch_off_gtadtbra
, k_gptw_ch_off_gtdtrb = 0x7CU ,
k_gptw_ch_off_gtadtbrb = 0x80U , k_gptw_ch_off_gtadtbrb , k_gptw_ch_off_gtdter =
0x88U , k_gptw_ch_off_gtdvu = 0x8CU ,
k_gptw_ch_off_gtdvd = 0x90U , k_gptw_ch_off_gtdbu = 0x94U , k_gptw_ch_off_gtdbd =
0x98U , k_gptw_ch_off_gtsos = 0x9CU ,
k_gptw_ch_off_gtsotr = 0xA0U , k_gptw_ch_off_gtadsmr = 0xA4U , k_gptw_ch_off_gteitc
= 0xA8U , k_gptw_ch_off_gteitli1 = 0xACU ,
k_gptw_ch_off_gteitli2 = 0xB0U , k_gptw_ch_off_gteitlb = 0xB4U , k_gptw_ch_off_reserved
= 0xB8U , k_gptw_ch_off_gtsecsr = 0xD0U ,
k_gptw_ch_off_gtsecr = 0xD4U , k_gptw_ch_struct_size = 0xD8U }
• enum gptw_common_reg_offsets_t : uint8_t { k_gptw_common_off_gtstpa = 0x04U ,
k_gptw_common_off_gtelra = 0x08U , k_gptw_common_struct_size = 0x0CU }
• enum gptw_periph_space_t : uintptr_t { k_gptw_periph_space_mask = 0xFFFF0000U ,
k_gptw_periph_space_base = 0x000C0000U }

```

## Functions

- static volatile `rx_gptw_channel_regs_t` \* `gptw0` (void)  
GPTW register inline accessor functions.
- static volatile `rx_gptw_channel_regs_t` \* `gptw1` (void)
- static volatile `rx_gptw_channel_regs_t` \* `gptw2` (void)
- static volatile `rx_gptw_channel_regs_t` \* `gptw3` (void)
- static volatile `rx_gptw_common_regs_t` \* `gptw_common` (void)  
GPTW common register accessor.

### 8.25.1 Detailed Description

#### RX72N GPTW (General PWM Timer) Register Definitions.

This header provides comprehensive register definitions for the RX72N's General PWM Timer (GPTW). The GPTW is a 32-bit timer optimized for motor PWM generation with dead-time control, complementary outputs, and hardware-synchronized ADC triggering.

#### System Architecture Context:

```

* +-----+
* | STAR Motor PWM Generation (GPTW) |
* +-----+
* |
* | PID Output --> GPTW Compare --> GTIOCA/B --> DRV8263H --> Motor |
* | (0-100%) (32-bit) (PWM pins) (H-bridge) (DC motor) |
* |
* | +-----+ |
* | | GPTW Channel Configuration (per motor) | |
* | |
* | | GTPR (Period) = PCLKA / (Prescaler x PWM_Freq) - 1 | |
* | | GTCCRA (Duty) = GTPR x (duty_percent / 100) | |
* | |
* | | @ PCLKA=120MHz, Prescaler=1, PWM_Freq=20kHz: | |
* | | GTPR = 120MHz / 20kHz - 1 = 5999 | |
* | | 50% duty: GTCCRA = 2999 | |
* | +-----+ |
* |
* +-----+
*

```



## GPTW Key Features:

- 32-bit counter resolution (vs MTU's 16-bit)
- 4 independent channels (GPTW0-GPTW3) for 4 motors
- Each channel has 2 outputs (GTIOCA, GTIOCB) for H-bridge control
- Hardware dead-time insertion for shoot-through prevention
- Synchronized start/stop across multiple channels
- ADC trigger generation for current sensing synchronization
- Double/triple buffering for glitch-free duty updates

## Channel to Motor Mapping (STAR Project):

| Channel | Motor | GTIOCA Pin | GTIOCB Pin | DRV8263H |
|---------|-------|------------|------------|----------|
| GPTW0   | M0    | PE5        | PE2        | U1       |
| GPTW1   | M1    | PE4        | PE1        | U2       |
| GPTW2   | M2    | PE3        | PE0        | U3       |
| GPTW3   | M3    | PE7        | PE6        | U4       |

## PWM Timing Specifications:

| Parameter       | Value          | Notes                    |
|-----------------|----------------|--------------------------|
| Clock source    | PCLKA (120MHz) | From clock generator     |
| PWM frequency   | 20 kHz         | Above audible range      |
| Resolution      | 6000 counts    | GTPR = 5999              |
| Duty resolution | 0.0167%        | $1/6000 = 0.0167\%$      |
| Dead time       | 500 ns         | Shoot-through protection |
| Update rate     | 100 Hz         | PID control loop rate    |

## Hardware Requirements:

- CPU: Renesas RX72N @ 240 MHz (ICLK)
- Peripheral clock: PCLKA = 120 MHz (GPTW clock source)
- GPIO: Port E pins configured for GPTW alternate function
- Module stop: MSTPCRA.MSTPA7 = 0 to enable GPTW (RX72N)

## Memory Usage:

- Channel register structure: 216 bytes (0xD8) per channel
- Common register structure: 44 bytes (0x2C)
- No dynamic allocation required

See also

docs/sections/05\_motor\_control.tex Motor control system design

lib/rx\_motor/inc/rx\_motor.h Motor control API

lib/rx\_drv8263/inc/rx\_drv8263.h H-bridge driver interface

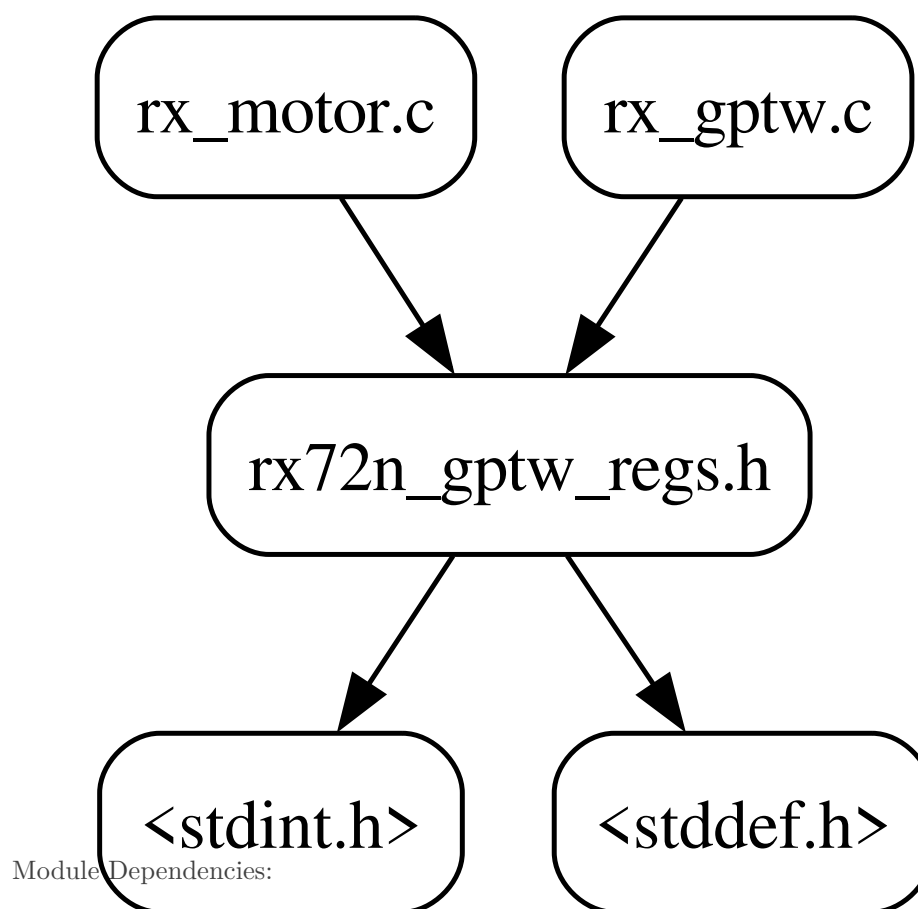
RX72N Group User's Manual: Hardware, Chapter 26 (GPTW)

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
- Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
- Rule 8: [OK] All constants as C23 typed enums (addresses, bit fields)
- Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings

SOLID Principles:

- S: Single responsibility - only GPTW register definitions
- O: Open for extension via accessor functions
- I: Interface segregation - channel and common registers separate
- D: Dependency inversion - hardware accessed via typed pointers



Verification Status:

[PASS] VERIFIED (2026-02-03) - Complete verification against Smart Configurator iodefine.h

- All 55 channel registers present and offset-verified
- All 6 common registers present and offset-verified
- 61/61 static assertions (100% coverage)
- Base addresses verified: 0x000C2000-0x000C2300 (channels), 0x000C2B00 (common)
- Structure sizes verified: 216 bytes (channel), 44 bytes (common)
- Reference: iodefine.h lines 10427-11461 (struct st\_gptw)

Author

Locked, Inc.

Date

2026-01-28

Version

1.0.0

Copyright

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Definition in file [rx72n\\_gptw\\_regs.h](#).

## 8.26 rx72n`gptw`regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002 * @file rx72n_gptw_regs.h
00003 * @brief RX72N GPTW (General PWM Timer) Register Definitions
00004 *
00005 * @details
00006 * This header provides comprehensive register definitions for the RX72N's
00007 * General PWM Timer (GPTW). The GPTW is a 32-bit timer optimized for motor
00008 * PWM generation with dead-time control, complementary outputs, and
00009 * hardware-synchronized ADC triggering.
00010 *
00011 * @par System Architecture Context:
00012 * @verbatim
00013 * +-----+
00014 * | STAR Motor PWM Generation (GPTW) |
00015 * +-----+
00016 * |
00017 * | PID Output --> GPTW Compare --> GTIOCA/B --> DRV8263H --> Motor |
00018 * | (0-100%) (32-bit) (PWM pins) (H-bridge) (DC motor) |
00019 * |
00020 * | +-----+
00021 * | | GPTW Channel Configuration (per motor) | |
00022 * | |
00023 * | | GTPR (Period) = PCLKA / (Prescaler x PWM_Freq) - 1 | |
00024 * | | GTCCRA (Duty) = GTPR x (duty_percent / 100) | |
00025 * | |
00026 * | | @ PCLKA=120MHz, Prescaler=1, PWM_Freq=20kHz: | |
00027 * | | GTPR = 120MHz / 20kHz - 1 = 5999 | |
00028 * | | 50% duty: GTCCRA = 2999 | |
00029 * | +-----+
00030 * |
```

```

00030 * |
00031 * +-----+
00032 * @endverbatim
00033 *
00034 * @par GPTW Key Features:
00035 * - 32-bit counter resolution (vs MTU's 16-bit)
00036 * - 4 independent channels (GPTW0-GPTW3) for 4 motors
00037 * - Each channel has 2 outputs (GTIOCA, GTIOCB) for H-bridge control
00038 * - Hardware dead-time insertion for shoot-through prevention
00039 * - Synchronized start/stop across multiple channels
00040 * - ADC trigger generation for current sensing synchronization
00041 * - Double/triple buffering for glitch-free duty updates
00042 *
00043 * @par Channel to Motor Mapping (STAR Project):
00044 *
00045 * | Channel | Motor | GTIOCA Pin | GTIOCB Pin | DRV8263H |
00046 * |-----|-----|-----|-----|-----|
00047 * | GPTW0 | M0 | PE5 | PE2 | U1 |
00048 * | GPTW1 | M1 | PE4 | PE1 | U2 |
00049 * | GPTW2 | M2 | PE3 | PE0 | U3 |
00050 * | GPTW3 | M3 | PE7 | PE6 | U4 |
00051 *
00052 * @par PWM Timing Specifications:
00053 *
00054 * | Parameter | Value | Notes |
00055 * |-----|-----|-----|
00056 * | Clock source | PCLKA (120MHz) | From clock generator |
00057 * | PWM frequency | 20 kHz | Above audible range |
00058 * | Resolution | 6000 counts | GTPR = 5999 |
00059 * | Duty resolution | 0.0167% | 1/6000 = 0.0167% |
00060 * | Dead time | 500 ns | Shoot-through protection |
00061 * | Update rate | 100 Hz | PID control loop rate |
00062 *
00063 * @par Hardware Requirements:
00064 * - CPU: Renesas RX72N @ 240 MHz (ICLK)
00065 * - Peripheral clock: PCLKA = 120 MHz (GPTW clock source)
00066 * - GPIO: Port E pins configured for GPTW alternate function
00067 * - Module stop: MSTPCRA.MSTPA7 = 0 to enable GPTW (RX72N)
00068 *
00069 * @par Memory Usage:
00070 * - Channel register structure: 216 bytes (0xD8) per channel
00071 * - Common register structure: 44 bytes (0x2C)
00072 * - No dynamic allocation required
00073 *
00074 * @see docs/sections/05_motor_control.tex Motor control system design
00075 * @see lib/rx_motor/inc/rx_motor.h Motor control API
00076 * @see lib/rx_drv8263/inc/rx_drv8263.h H-bridge driver interface
00077 * @see RX72N Group User's Manual: Hardware, Chapter 26 (GPTW)
00078 *
00079 * @par NASA Power of 10 Compliance:
00080 * - Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
00081 * - Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
00082 * - Rule 8: [OK] All constants as C23 typed enums (addresses, bit fields)
00083 * - Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings
00084 *
00085 * @par SOLID Principles:
00086 * - S: Single responsibility - only GPTW register definitions
00087 * - O: Open for extension via accessor functions
00088 * - I: Interface segregation - channel and common registers separate
00089 * - D: Dependency inversion - hardware accessed via typed pointers
00090 *
00091 * @par Module Dependencies:
00092 * @dot
00093 * digraph gptw_deps {
00094 * rankdir=TB;
00095 * node [shape=box, style=rounded];
00096 * gptw_regs [label="rx72n_gptw_regs.h"];
00097 * stdint [label="<stdint.h>"];
00098 * stddef [label="<stddef.h>"];
00099 * motor [label="rx_motor.c"];
00100 * gptw_impl [label="rx_gptw.c"];
00101 * gptw_regs -> stdint;
00102 * gptw_regs -> stddef;
00103 * motor -> gptw_regs;
00104 * gptw_impl -> gptw_regs;
00105 * }
00106 * @enddot
00107 *
00108 * @par Verification Status:
00109 * [PASS] VERIFIED (2026-02-03) - Complete verification against Smart Configurator iodefne.h
00110 * - All 55 channel registers present and offset-verified
00111 * - All 6 common registers present and offset-verified
00112 * - 61/61 static assertions (100% coverage)
00113 * - Base addresses verified: 0x000C2000-0x000C2300 (channels), 0x000C2B00 (common)
00114 * - Structure sizes verified: 216 bytes (channel), 44 bytes (common)
00115 * - Reference: iodefne.h lines 10427-11461 (struct st_gptw)
00116 *

```

```

00117 * @author Locked, Inc.
00118 * @date 2026-01-28
00119 * @version 1.0.0
00120 * @copyright Copyright (c) 2026 Locked Inc.
00121 * SPDX-License-Identifier: MIT
00122 */
00123
00124 #pragma once
00125
00126 #include <stdint.h>
00127 #include <stdint.h>
00128
00129 #ifdef __cplusplus
00130 extern "C" {
00131 #endif
00132
00133 /**
00134 * @defgroup rx_gptw_regs RX72N GPTW Register Definitions
00135 * @brief General PWM Timer hardware register interface
00136 *
00137 * @details
00138 * Provides memory-mapped register access to the RX72N GPTW peripheral
00139 * for 32-bit PWM generation in motor control applications.
00140 *
00141 * @par Quick Start Example - Configure 20 kHz PWM:
00142 * @code{.c}
00143 * // Get GPTW channel 0 for motor 0
00144 * volatile rx_gptw_channel_regs_t* gptw = gptw0();
00145 *
00146 * // 1. Unlock write protection
00147 * gptw->gtwp = k_gptw_gtwp_unlock;
00148 *
00149 * // 2. Stop counter during configuration
00150 * gptw->gtr = ~k_gptw_gtr_cst;
00151 *
00152 * // 3. Set mode: Sawtooth PWM, no prescaler (PCLKA/1 = 120 MHz)
00153 * gptw->gtr = k_gptw_gtr_md_saw_pwm | k_gptw_gtr_tpcs_1;
00154 *
00155 * // 4. Set period for 20 kHz: 120 MHz / 20 kHz - 1 = 5999
00156 * gptw->gtr = 5999;
00157 *
00158 * // 5. Set initial duty to 0%
00159 * gptw->gtcra = 0;
00160 *
00161 * // 6. Configure GTIOCA output: initial low, high at end-of-cycle, low at compare match
00162 * gptw->gtior = k_gptw_gtior_oa_init_low | k_gptw_gtior_oae;
00163 *
00164 * // 7. Clear counter
00165 * gptw->gtcnt = 0;
00166 *
00167 * // 8. Start counting
00168 * gptw->gtr |= k_gptw_gtr_cst;
00169 *
00170 * // 9. Lock write protection
00171 * gptw->gtwp = k_gptw_gtwp_lock;
00172 *
00173 * // Update duty cycle to 50%
00174 * gptw->gtcra = k_gptw_gtwp_unlock;
00175 * gptw->gtcra = 3000; // 50% of 5999
00176 * gptw->gtwp = k_gptw_gtwp_lock;
00177 * @endcode
00178 *
00179 * @{
00180 */
00181
00182 /*
=====
00183 * General PWM Timer (GPTW) - 32-bit PWM for Motor Control
00184 *
=====
00185 *
00186 * GPTW Features:
00187 * - 32-bit counter resolution (vs MTU's 16-bit)
00188 * - 4 independent channels (GPTW0-GPTW3)
00189 * - Each channel has 2 outputs (GTIOCA, GTIOCB)
00190 * - Optimized for motor PWM generation
00191 * - Dead-time control for H-bridge safety
00192 *
00193 * Pin Mapping (Port E):
00194 * - GPTW0: PE5/GTIOC0A, PE2/GTIOC0B
00195 * - GPTW1: PE4/GTIOC1A, PE1/GTIOC1B
00196 * - GPTW2: PE3/GTIOC2A, PE0/GTIOC2B
00197 * - GPTW3: PE7/GTIOC3A, PE6/GTIOC3B
00198 */
00199
00200 /*
=====

```

```

00201 * Struct Padding Sizes - Named array counts for reserved fields
00202 *
=====
00203 */
00204
00205 typedef enum : uint8_t {
00206 k_gptw_ch_reserved_count = 6U, /**< Reserved uint32_t count in channel struct (0xB8-0xCC) */
00207 } gptw_padding_sizes_t;
00208
00209 /**
00210 * @struct rx_gptw_channel_regs_t
00211 * @brief GPTW Channel Register Map for 32-bit PWM generation
00212 *
00213 * @details
00214 * Memory-mapped register structure for a single GPTW channel. Each of
00215 * the 4 channels (GPTW0-GPTW3) has an identical register layout at
00216 * different base addresses spaced 0x100 bytes apart.
00217 *
00218 * @par Channel Base Addresses:
00219 *
00220 * | Channel | Base Address | Motor | Output Pins |
00221 * |-----|-----|-----|-----|
00222 * | GPTW0 | 0x000C2000 | M0 | PE5 (A), PE2 (B) |
00223 * | GPTW1 | 0x000C2100 | M1 | PE4 (A), PE1 (B) |
00224 * | GPTW2 | 0x000C2200 | M2 | PE3 (A), PE0 (B) |
00225 * | GPTW3 | 0x000C2300 | M3 | PE7 (A), PE6 (B) |
00226 *
00227 * @par Key Registers for Motor PWM:
00228 *
00229 * | Register | Offset | Description |
00230 * |-----|-----|-----|
00231 * | GTWP | 0x00 | Write protection (unlock: 0xA500) |
00232 * | GTCR | 0x2C | Control (mode, prescaler, start/stop) |
00233 * | GTIOR | 0x34 | I/O control (output polarity/enable) |
00234 * | GTCNT | 0x48 | 32-bit counter value |
00235 * | GTCCRA | 0x4C | Compare A (duty cycle for GTIOCA) |
00236 * | GTCCRB | 0x50 | Compare B (duty cycle for GTIOCB) |
00237 * | GTPR | 0x64 | Period (PWM frequency) |
00238 * | GTDTCR | 0x88 | Dead-time control |
00239 * | GTDVU | 0x8C | Dead-time value (rising edge) |
00240 * | GTDVD | 0x90 | Dead-time value (falling edge) |
00241 *
00242 * @par PWM Frequency Calculation:
00243 * @f$ f_{\text{PWM}} = \frac{f_{\text{PCLKA}}}{(GTPR + 1) \times \text{Prescaler}} @f$
00244 *
00245 * @par Duty Cycle Calculation:
00246 * @f$ \text{Duty\%} = \frac{GTCCRA}{GTPR + 1} \times 100\% @f$
00247 *
00248 * @par Usage Example - Dead-Time Configuration:
00249 * @code{.c}
00250 * volatile rx_gptw_channel_regs_t* gptw = gptw0();
00251 *
00252 * // Enable dead-time generation
00253 * gptw->gtdtcr = k_gptw_gtdtcr_tde;
00254 *
00255 * // Set dead-time: 500 ns @ 120 MHz = 60 counts
00256 * gptw->gtdvu = 60; // Rising edge dead-time
00257 * gptw->gtdvd = 60; // Falling edge dead-time
00258 * @endcode
00259 *
00260 * @invariant sizeof(rx_gptw_channel_regs_t) == 0xD8 (216 bytes)
00261 * @invariant Channel spacing is exactly 0x100 bytes
00262 *
00263 * @see gptw0(), gptw1(), gptw2(), gptw3() Accessor functions
00264 * @see rx_gptw_common_regs_t Common registers for synchronized control
00265 */
00266 typedef struct {
00267 volatile uint32_t gtwp; /**< 0x00: Write Protection Register */
00268 volatile uint32_t gtstr; /**< 0x04: Software Start Register */
00269 volatile uint32_t gtstp; /**< 0x08: Software Stop Register */
00270 volatile uint32_t gtclr; /**< 0x0C: Software Clear Register */
00271 volatile uint32_t gtssr; /**< 0x10: Start Source Select Register */
00272 volatile uint32_t gtpsr; /**< 0x14: Stop Source Select Register */
00273 volatile uint32_t gtcsr; /**< 0x18: Clear Source Select Register */
00274 volatile uint32_t gtupsr; /**< 0x1C: Count-Up Source Select Register */
00275 volatile uint32_t gtdnsr; /**< 0x20: Count-Down Source Select Register */
00276 volatile uint32_t gticasr; /**< 0x24: Input Capture Source Select A */
00277 volatile uint32_t gticbsr; /**< 0x28: Input Capture Source Select B */
00278 volatile uint32_t gter; /**< 0x2C: Control Register */
00279 volatile uint32_t gtuddtvc; /**< 0x30: Up/Down Count Duty Setting */
00280 volatile uint32_t gtior; /**< 0x34: I/O Control Register */
00281 volatile uint32_t gtintad; /**< 0x38: Interrupt Output Setting */
00282 volatile uint32_t gtst; /**< 0x3C: Status Register */
00283 volatile uint32_t gtber; /**< 0x40: Buffer Enable Register */
00284 volatile uint32_t gtitc; /**< 0x44: Interrupt/ADC Trigger Control */
00285 volatile uint32_t gtcnt; /**< 0x48: Counter */
00286 volatile uint32_t gtccra; /**< 0x4C: Compare Capture Register A */

```

```

00287 volatile uint32_t gtcrcb; /**< 0x50: Compare Capture Register B */
00288 volatile uint32_t gtcrc; /**< 0x54: Compare Capture Register C */
00289 volatile uint32_t gtcrcr; /**< 0x58: Compare Capture Register E */
00290 volatile uint32_t gtcrcd; /**< 0x5C: Compare Capture Register D */
00291 volatile uint32_t gtcrcf; /**< 0x60: Compare Capture Register F */
00292 volatile uint32_t gtpr; /**< 0x64: Cycle Setting Register (Period) */
00293 volatile uint32_t gtpbr; /**< 0x68: Cycle Setting Buffer Register */
00294 volatile uint32_t gtpdbr; /**< 0x6C: Cycle Setting Double Buffer */
00295 volatile uint32_t gtadtra; /**< 0x70: A/D Trigger Register A */
00296 volatile uint32_t gtadtbra; /**< 0x74: A/D Trigger Buffer Register A */
00297 volatile uint32_t gtadtbra; /**< 0x78: A/D Trigger Double Buffer A */
00298 volatile uint32_t gtadtrb; /**< 0x7C: A/D Trigger Register B */
00299 volatile uint32_t gtadtbrb; /**< 0x80: A/D Trigger Buffer Register B */
00300 volatile uint32_t gtadtbrb; /**< 0x84: A/D Trigger Double Buffer B */
00301 volatile uint32_t gtdtcr; /**< 0x88: Dead Time Control Register */
00302 volatile uint32_t gtdvu; /**< 0x8C: Dead Time Value Upper */
00303 volatile uint32_t gtdvd; /**< 0x90: Dead Time Value Down */
00304 volatile uint32_t gtdbu; /**< 0x94: Dead Time Buffer Upper */
00305 volatile uint32_t gtdbd; /**< 0x98: Dead Time Buffer Down */
00306 volatile uint32_t gtsos; /**< 0x9C: Output Protection Status */
00307 volatile uint32_t gtsotr; /**< 0xA0: Output Protection Trigger */
00308 volatile uint32_t gtadsmr; /**< 0xA4: A/D Conversion Start Request Signal Monitoring */
00309 volatile uint32_t gteitc; /**< 0xA8: Extended Interrupt Skipping Counter Control */
00310 volatile uint32_t gteitl1; /**< 0xAC: Extended Interrupt Skipping Setting Register 1 */
00311 volatile uint32_t gteitl2; /**< 0xB0: Extended Interrupt Skipping Setting Register 2 */
00312 volatile uint32_t gteitlb; /**< 0xB4: Extended Buffer Transfer Skipping Setting */
00313 uint32_t reserved[k_gptw_ch_reserved_count]; /**< 0xB8-0xCC: Reserved */
00314 volatile uint32_t gtsecsr; /**< 0xD0: Operation Enable Bit Simultaneous Control Channel Select */
00315 volatile uint32_t gtsecl; /**< 0xD4: Operation Enable Bit Simultaneous Control */
00316 } rx_gptw_channel_regs_t;
00317
00318 /**
00319 * @brief GPTW Common Register Map
00320 * @details
00321 * Overlays GPTW0.GTSTR/GTSTP/GTCLR to provide a named handle for multi-channel
00322 * atomic start/stop/clear. Writing bits [3:0] in any of these registers affects
00323 * channels 0-3 simultaneously (per RX72N HW manual section 26.2.2-26.2.4).
00324 * Base address: k_gptw_common_base_addr (0x000C2004) == GPTW0.GTSTR.
00325 * The RX72N address map has no separate "common" GPTW block; the per-channel
00326 * GTSTR/GTSTP/GTCLR registers are cross-channel by design.
00327 */
00328 typedef struct {
00329 volatile uint32_t gtstra; /**< 0x00: Software Start (GPTW0.GTSTR, bits [3:0] = Ch3..Ch0) */
00330 volatile uint32_t gtstpa; /**< 0x04: Software Stop (GPTW0.GTSTP, bits [3:0] = Ch3..Ch0) */
00331 volatile uint32_t gtclra; /**< 0x08: Software Clear (GPTW0.GTCLR, bits [3:0] = Ch3..Ch0) */
00332 } rx_gptw_common_regs_t;
00333
00334 /**
=====
00335 * Base Address Definitions
00336 *
=====
00337 *
00338 * Addresses verified against RX72N Hardware Manual (R01UH0824EJ0120 Rev.1.20).
00339 */
00340
00341 /** @brief GPTW hardware addresses (verified against RX72N Hardware Manual) */
00342 typedef enum : uintptr_t {
00343 k_gptw_channel_offset = 0x100, /**< Channel spacing between GPTW registers */
00344 k_gptw0_base_addr = 0x000C2000, /**< GPTW channel 0 base address */
00345 k_gptw1_base_addr = 0x000C2100, /**< GPTW channel 1 base address */
00346 k_gptw2_base_addr = 0x000C2200, /**< GPTW channel 2 base address */
00347 k_gptw3_base_addr = 0x000C2300, /**< GPTW channel 3 base address */
00348 k_gptw_common_base_addr =
00349 0x000C2004, /**< GPTW0.GTSTR -- used as multi-channel start/stop/clear base.
00350 * RX72N has NO separate "common" GPTW block (Chapter 5 Table 5.1
00351 * shows GPTW space ending at 0x000C23D4; nothing at 0x000C2B00).
00352 * GTSTR/GTSTP/GTCLR in any channel affect all channels whose
00353 * corresponding bit [3:0] is set (manual section 26.2.2-26.2.4). */
00354 } gptw_addresses_t;
00355
00356 /** @brief GPTW register inline accessor functions */
00357 static inline volatile rx_gptw_channel_regs_t* gptw0(void)
00358 {
00359 return (volatile rx_gptw_channel_regs_t*)k_gptw0_base_addr;
00360 }
00361
00362 static inline volatile rx_gptw_channel_regs_t* gptw1(void)
00363 {
00364 return (volatile rx_gptw_channel_regs_t*)k_gptw1_base_addr;
00365 }
00366
00367 static inline volatile rx_gptw_channel_regs_t* gptw2(void)
00368 {
00369 return (volatile rx_gptw_channel_regs_t*)k_gptw2_base_addr;
00370 }
00371
00372

```

```

00372 static inline volatile rx_gptw_channel_regs_t* gptw3(void)
00373 {
00374 return (volatile rx_gptw_channel_regs_t*)k_gptw3_base_addr;
00375 }
00376
00377 /** @brief GPTW common register accessor */
00378 static inline volatile rx_gptw_common_regs_t* gptw_common(void)
00379 {
00380 return (volatile rx_gptw_common_regs_t*)k_gptw_common_base_addr;
00381 }
00382
00383 /*
=====
00384 * Write Protection Register (GTPW) Bits
00385 *
=====
00386 */
00387
00388 typedef enum : uint16_t {
00389 k_gptw_gtwp_wp0 = (1 « 0), /**< Register write protect bit 0 */
00390 k_gptw_gtwp_wp1 = (1 « 1), /**< Register write protect bit 1 */
00391 k_gptw_gtwp_wp2 = (1 « 2), /**< Register write protect bit 2 */
00392 k_gptw_gtwp_wp3 = (1 « 3), /**< Register write protect bit 3 */
00393 k_gptw_gtwp_unlock = 0xA500, /**< Write protection unlock key */
00394 k_gptw_gtwp_lock = 0xA501, /**< Write protection lock key */
00395 } gptw_gtwp_bits_t;
00396
00397 /*
=====
00398 * Control Register (GTCR) Bits
00399 *
=====
00400 * Reference: RX72N Hardware Manual R01UH0824EJ0120 Rev.1.20, Section 26.2.12
00401 */
00402
00403 typedef enum : uint32_t {
00404 /* b0: CST - Count Start */
00405 k_gptw_gtcr_cst = (1 « 0), /**< Counter Start (0=stop, 1=count) */
00406
00407 /* b8: ICDS - Input Capture Operation Select at Count Stop */
00408 k_gptw_gtcr_icds = (1 « 8), /**< 0=capture at stop, 1=no capture at stop */
00409
00410 /* b18-b16: MD[2:0] - Mode Select */
00411 k_gptw_gtcr_md_shift = 16,
00412 k_gptw_gtcr_md_mask = (0x07 « k_gptw_gtcr_md_shift),
00413
00414 /** Sawtooth-wave PWM mode (single buffer or double buffer possible) */
00415 k_gptw_gtcr_md_saw_pwm = (0x00 « k_gptw_gtcr_md_shift),
00416 /** Sawtooth-wave one-shot pulse mode (fixed buffer operation) */
00417 k_gptw_gtcr_md_saw_oneshot = (0x01 « k_gptw_gtcr_md_shift),
00418 /* 0x02 (010): Setting prohibited */
00419 /* 0x03 (011): Setting prohibited */
00420 /** Triangle-wave PWM mode 1: 32-bit transfer at trough (single/double buffer) */
00421 k_gptw_gtcr_md_tri_pwm1 = (0x04 « k_gptw_gtcr_md_shift),
00422 /** Triangle-wave PWM mode 2: 32-bit transfer at crest and trough (single/double buffer) */
00423 k_gptw_gtcr_md_tri_pwm2 = (0x05 « k_gptw_gtcr_md_shift),
00424 /** Triangle-wave PWM mode 3: 64-bit transfer at trough (fixed buffer operation) */
00425 k_gptw_gtcr_md_tri_pwm3 = (0x06 « k_gptw_gtcr_md_shift),
00426 /* 0x07 (111): Setting prohibited */
00427
00428 /* b26-b23: TPCS[3:0] - Timer Prescaler Select */
00429 k_gptw_gtcr_tpcs_shift = 23,
00430 k_gptw_gtcr_tpcs_mask = (0x0F « k_gptw_gtcr_tpcs_shift),
00431
00432 k_gptw_gtcr_tpcs_1 = (0x00 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/1 (120 MHz) */
00433 k_gptw_gtcr_tpcs_2 = (0x01 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/2 */
00434 k_gptw_gtcr_tpcs_4 = (0x02 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/4 */
00435 k_gptw_gtcr_tpcs_8 = (0x03 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/8 */
00436 k_gptw_gtcr_tpcs_16 = (0x04 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/16 */
00437 k_gptw_gtcr_tpcs_32 = (0x05 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/32 */
00438 k_gptw_gtcr_tpcs_64 = (0x06 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/64 */
00439 /* 0x07 (0111): Setting prohibited */
00440 k_gptw_gtcr_tpcs_256 = (0x08 « k_gptw_gtcr_tpcs_shift), /**< PCLKA/256 */
00441 /* 0x09 (1001): Setting prohibited */
00442 k_gptw_gtcr_tpcs_1024 = (0x0A « k_gptw_gtcr_tpcs_shift), /**< PCLKA/1024 */
00443 /* 0x0B (1011): Setting prohibited */
00444 /* 0x0C-0x0F: GTETRGA/B/C/D via POEG (external clock) */
00445 } gptw_gtcr_bits_t;
00446
00447 /*
=====
00448 * I/O Control Register (GTIOR) Bits
00449 *
=====
00450 */
00451
00452 typedef enum : uint32_t {

```



```

00453 /* GTIOCA Output Control (bits 0-4) */
00454 k_gptw_gtior_oa_shift = 0,
00455 k_gptw_gtior_oa_mask = 0x1F,
00456 k_gptw_gtior_oa_disabled = 0x00, /**< Output disabled */
00457 k_gptw_gtior_oa_low_cmpup = 0x01, /**< Low on compare match (up) */
00458 k_gptw_gtior_oa_high_cmpup = 0x02, /**< High on compare match (up) */
00459 k_gptw_gtior_oa_toggle = 0x03, /**< Toggle on compare match */
00460 k_gptw_gtior_oa_init_low =
00461 0x09, /**< Initial LOW; high at end-of-cycle; low at compare match (active-high sawtooth PWM) */
00462 k_gptw_gtior_oa_init_high =
00463 0x16, /**< Initial HIGH; low at end-of-cycle; high at compare match (complement of init_low) */
00464
00465 /* GTIOCA Output Enable (bit 8) */
00466 k_gptw_gtior_oae = (1 « 8), /**< GTIOCA Output Enable */
00467
00468 /* GTIOCB Output Control (bits 16-20) */
00469 k_gptw_gtior_ob_shift = 16,
00470 k_gptw_gtior_ob_mask = (0x1F « 16),
00471 k_gptw_gtior_ob_disabled = (0x00 « 16), /**< Output disabled */
00472 k_gptw_gtior_ob_low_cmpup = (0x01 « 16), /**< Low on compare match (up) */
00473 k_gptw_gtior_ob_high_cmpup = (0x02 « 16), /**< High on compare match (up) */
00474 k_gptw_gtior_ob_toggle = (0x03 « 16), /**< Toggle on compare match */
00475 k_gptw_gtior_ob_init_low =
00476 (0x09 « 16), /**< Initial LOW; high at end-of-cycle; low at compare match */
00477 k_gptw_gtior_ob_init_high =
00478 (0x16 « 16), /**< Initial HIGH; low at end-of-cycle; high at compare match */
00479
00480 /* GTIOCB Output Enable (bit 24) */
00481 k_gptw_gtior_obe = (1 « 24), /**< GTIOCB Output Enable */
00482 } gptw_gtior_bits_t;
00483
00484 /*
=====
00485 * Interrupt Output Setting Register (GTINTAD) Bits
00486 *
=====
00487 */
00488
00489 /**
00490 * @enum gptw_gtintad_bits_t
00491 * @brief GTINTAD register bit definitions for POEG group linkage
00492 *
00493 * @details
00494 * The GTINTAD register (offset 0x38) configures the link between a GPTW
00495 * channel and a POEG group for emergency output stop. When a fault is
00496 * detected (dead-time error, simultaneous high/low output), the GPTW
00497 * signals the selected POEG group to disable PWM outputs.
00498 *
00499 * @par STAR Project Usage:
00500 * | GPTW Channel | POEG Group | GRP[1:0] | Motor |
00501 * |-----|-----|-----|-----|
00502 * | GPTW0 | A | 00 | 0 |
00503 * | GPTW1 | B | 01 | 1 |
00504 * | GPTW2 | C | 10 | 2 |
00505 * | GPTW3 | D | 11 | 3 |
00506 *
00507 * @warning GRP[1:0] must be set while OAE and OBE in GTIOR are 0
00508 *
00509 * @see rx72n_poeg_regs.h POEG register definitions
00510 * @see RX72N Manual Section 26.2.15 - GTINTAD register
00511 * @since Version 1.0.0
00512 */
00513 typedef enum : uint32_t {
00514 /* GRP[1:0] - Output Stop Group Select (bits 25:24) */
00515 k_gptw_gtintad_grp_shift = 24, /**< GRP field shift */
00516 k_gptw_gtintad_grp_mask = (0x03U « 24), /**< GRP field mask */
00517 k_gptw_gtintad_grp_a = (0x00U « 24), /**< Select POEG Group A */
00518 k_gptw_gtintad_grp_b = (0x01U « 24), /**< Select POEG Group B */
00519 k_gptw_gtintad_grp_c = (0x02U « 24), /**< Select POEG Group C */
00520 k_gptw_gtintad_grp_d = (0x03U « 24), /**< Select POEG Group D */
00521
00522 /* GRPDTE - Dead Time Error Output Stop Detection Enable (bit 28) */
00523 k_gptw_gtintad_grpdte = (1U « 28), /**< Enable dead-time error detection */
00524
00525 /* GRPABH - Simultaneous High Output Stop Detection Enable (bit 29) */
00526 k_gptw_gtintad_grpabh = (1U « 29), /**< Enable simultaneous high detection */
00527
00528 /* GRPABL - Simultaneous Low Output Stop Detection Enable (bit 30) */
00529 k_gptw_gtintad_grpabl = (1U « 30), /**< Enable simultaneous low detection */
00530 } gptw_gtintad_bits_t;
00531
00532 /*
=====
00533 * Buffer Enable Register (GTBER) Bits
00534 *
=====
00535 */

```

```

00536
00537 typedef enum : uint32_t {
00538 /* BD[3:0] - Buffer Operation Disable (b3:b0).
00539 * Set 1 to DISABLE buffer operation for the register group. Reset = 0 (buffer enabled). */
00540 k_gptw_gtber_bd0 = (1U « 0), /**< Disable buffer op for GTCCRA/GTCCRB (1=disabled) */
00541 k_gptw_gtber_bd1 = (1U « 1), /**< Disable buffer op for GTPR (1=disabled) */
00542 k_gptw_gtber_bd2 = (1U « 2), /**< Disable buffer op for GTADTRA/GTADTRB (1=disabled) */
00543 k_gptw_gtber_bd3 = (1U « 3), /**< Disable buffer op for GTDVU/GTDVD (1=disabled) */
00544 /* CCRA[1:0] - GTCCRA Buffer Operation (b17:b16). BD[0]=0 required for buffer to be active. */
00545 k_gptw_gtber_ccra_single = (0x01U « 16), /**< GTCCRA single-buffer: GTCCRC -> GTCCRA */
00546 k_gptw_gtber_ccra_double = (0x02U « 16), /**< GTCCRA double-buffer: GTCCRD -> GTCCRC -> GTCCRA */
00547 /* CCRB[1:0] - GTCCRB Buffer Operation (b19:b18). BD[0]=0 required for buffer to be active. */
00548 k_gptw_gtber_ccrb_single = (0x01U « 18), /**< GTCCRB single-buffer: GTCCRE -> GTCCRB */
00549 k_gptw_gtber_ccrb_double = (0x02U « 18), /**< GTCCRB double-buffer: GTCCRF -> GTCCRE -> GTCCRB */
00550 /* PR[1:0] - GTPR Buffer Operation (b21:b20). BD[1]=0 required for buffer to be active. */
00551 k_gptw_gtber_pr_single = (0x01U « 20), /**< GTPR single-buffer: GTPBR -> GTPR */
00552 k_gptw_gtber_pr_double = (0x02U « 20), /**< GTPR double-buffer: GTPDBR -> GTPBR -> GTPR */
00553 } gptw_gtber_bits_t;
00554
00555 /*
=====
00556 * Dead Time Control Register (GTDTCR) Bits
00557 *
=====
00558 */
00559
00560 typedef enum : uint8_t {
00561 k_gptw_gtdtcr_tde = (1 « 0), /**< Dead time enable */
00562 k_gptw_gtdtcr_tdfe = (1 « 4), /**< Dead time buffer enable */
00563 } gptw_gtdtcr_bits_t;
00564
00565 /*
=====
00566 * Status Register (GTST) Bits
00567 *
=====
00568 */
00569
00570 typedef enum : uint16_t {
00571 k_gptw_gtst_tcfa = (1 « 0), /**< Compare match flag A */
00572 k_gptw_gtst_tcfb = (1 « 1), /**< Compare match flag B */
00573 k_gptw_gtst_tcfc = (1 « 2), /**< Compare match flag C */
00574 k_gptw_gtst_tcfD = (1 « 3), /**< Compare match flag D */
00575 k_gptw_gtst_tcfE = (1 « 4), /**< Compare match flag E */
00576 k_gptw_gtst_tcff = (1 « 5), /**< Compare match flag F */
00577 k_gptw_gtst_tcfpo = (1 « 6), /**< Overflow flag (at period) */
00578 k_gptw_gtst_tcfpu = (1 « 7), /**< Underflow flag (at 0) */
00579 k_gptw_gtst_tucf = (1 « 15), /**< Count direction flag (1=up) */
00580 } gptw_gtst_bits_t;
00581
00582 /*
=====
00583 * Start/Stop Register Bits (GTSTRA)
00584 *
=====
00585 */
00586
00587 typedef enum : uint8_t {
00588 k_gptw_gtstr_cst0 = (1 « 0), /**< Channel 0 count start */
00589 k_gptw_gtstr_cst1 = (1 « 1), /**< Channel 1 count start */
00590 k_gptw_gtstr_cst2 = (1 « 2), /**< Channel 2 count start */
00591 k_gptw_gtstr_cst3 = (1 « 3), /**< Channel 3 count start */
00592 } gptw_gtstr_bits_t;
00593
00594 /** @brief Up/Down Count Duty Setting Register (GTUDDTYC) Bits */
00595 typedef enum : uint32_t {
00596 k_gptw_gtuddtyc_ud = (1 « 0), /**< Count direction: 0=down, 1=up */
00597 k_gptw_gtuddtyc_udf = (1 « 1), /**< Forcibly set count direction */
00598 } gptw_gtuddtyc_bits_t;
00599
00600 /*
=====
00601 * A/D Conversion Start Request Signal Monitoring Register (GTADSMR) Bits
00602 *
=====
00603 */
00604
00605 typedef enum : uint32_t {
00606 k_gptw_gtadsmr_adsm0 = (1 « 0), /**< GTADSM0 output status monitor */
00607 k_gptw_gtadsmr_adsm1 = (1 « 1), /**< GTADSM1 output status monitor */
00608 } gptw_gtadsmr_bits_t;
00609
00610 /*
=====
00611 * Extended Interrupt Skipping Counter Control Register (GTEITC) Bits
00612 *
=====

```

```

00613 * WARNING: This is DIFFERENT from GTITC @ 0x44!
00614 * - GTITC @ 0x44: Interrupt and A/D Conversion Start Request Skipping Setting
00615 * - GTEITC @ 0xA8: Extended Interrupt Skipping Counter Control
00616 */
00617
00618 typedef enum : uint32_t {
00619 k_gptw_gteitc_eivtc_mask = 0x0000FFFF, /**< Extended interval counter (0-65535) */
00620 k_gptw_gteitc_eivtt_shift = 16, /**< Extended interval transfer timing shift */
00621 k_gptw_gteitc_eivtt_mask = (0x03 « 16), /**< Extended interval transfer timing mask */
00622 k_gptw_gteitc_eivtt_none = (0x00 « 16), /**< No transfer */
00623 k_gptw_gteitc_eivtt_trough = (0x01 « 16), /**< Transfer at trough */
00624 k_gptw_gteitc_eivtt_crest = (0x02 « 16), /**< Transfer at crest */
00625 k_gptw_gteitc_eivtt_both = (0x03 « 16), /**< Transfer at both crest and trough */
00626 k_gptw_gteitc_eivtcn_shift = 24, /**< Extended interval counter shift */
00627 k_gptw_gteitc_eivtcn_mask = (0xFFU « 24), /**< Extended interval counter mask */
00628 } gptw_gteitc_bits_t;
00629
00630 /*
=====
00631 * Extended Interrupt Skipping Setting Register 1 (GTEITLI1) Bits
00632 *
=====
00633 */
00634
00635 typedef enum : uint32_t {
00636 k_gptw_gteitli1_eadt1 = (1 « 0), /**< A/D conversion start request link enable */
00637 k_gptw_gteitli1_eivtciv = (1 « 8), /**< GTCIV (overflow) interrupt skipping enable */
00638 k_gptw_gteitli1_eivtcui = (1 « 9), /**< GTCIU (underflow) interrupt skipping enable */
00639 } gptw_gteitli1_bits_t;
00640
00641 /*
=====
00642 * Extended Interrupt Skipping Setting Register 2 (GTEITLI2) Bits
00643 *
=====
00644 */
00645
00646 typedef enum : uint32_t {
00647 k_gptw_gteitli2_eivtciv_shift = 0, /**< Extended interval counter shift */
00648 k_gptw_gteitli2_eivtciv_mask = 0x000000FF, /**< Extended interval counter mask */
00649 k_gptw_gteitli2_eivtcia = (1 « 0), /**< GTCIA compare match interrupt skipping */
00650 k_gptw_gteitli2_eivtcib = (1 « 1), /**< GTCIB compare match interrupt skipping */
00651 k_gptw_gteitli2_eivtcic = (1 « 2), /**< GTCIC compare match interrupt skipping */
00652 k_gptw_gteitli2_eivtcid = (1 « 3), /**< GTCID compare match interrupt skipping */
00653 k_gptw_gteitli2_eivtcie = (1 « 4), /**< GTCIE compare match interrupt skipping */
00654 k_gptw_gteitli2_eivtcif = (1 « 5), /**< GTCIF compare match interrupt skipping */
00655 } gptw_gteitli2_bits_t;
00656
00657 /*
=====
00658 * Extended Buffer Transfer Skipping Setting Register (GTEITLB) Bits
00659 *
=====
00660 */
00661
00662 typedef enum : uint32_t {
00663 k_gptw_gteitlb_eivtcpbl = (1 « 0), /**< Period buffer transfer skipping */
00664 k_gptw_gteitlb_eivtcabl = (1 « 1), /**< GTCCRA buffer transfer skipping */
00665 k_gptw_gteitlb_eivtcbb1 = (1 « 2), /**< GTCCRB buffer transfer skipping */
00666 } gptw_gteitlb_bits_t;
00667
00668 /*
=====
00669 * Operation Enable Bit Simultaneous Control Channel Select Register (GTSECSR) Bits
00670 *
=====
00671 */
00672
00673 typedef enum : uint32_t {
00674 k_gptw_gtsecsr_secse0 = (1 « 0), /**< Channel 0 simultaneous control enable */
00675 k_gptw_gtsecsr_secse1 = (1 « 1), /**< Channel 1 simultaneous control enable */
00676 k_gptw_gtsecsr_secse2 = (1 « 2), /**< Channel 2 simultaneous control enable */
00677 k_gptw_gtsecsr_secse3 = (1 « 3), /**< Channel 3 simultaneous control enable */
00678 } gptw_gtsecsr_bits_t;
00679
00680 /*
=====
00681 * Operation Enable Bit Simultaneous Control Register (GTSECR) Bits
00682 *
=====
00683 */
00684
00685 typedef enum : uint32_t {
00686 k_gptw_gtsecr_sbdc = (1 « 0), /**< Simultaneous buffer transfer disable */
00687 k_gptw_gtsecr_sbdccl = (1 « 8), /**< Buffer transfer disable clear */
00688 k_gptw_gtsecr_sbdpe = (1 « 9), /**< Simultaneous buffer transfer enable */
00689 } gptw_gtsecr_bits_t;

```

```

00690
00691 /*
=====
00692 * Module Stop Control (MSTPCRA)
00693 *
=====
00694 */
00695
00696 typedef enum : uint8_t {
00697 /**
00698 * @brief GPTW module stop bit in MSTPCRA
00699 *
00700 * @details
00701 * Per RX72N Hardware Manual Ch.11 (Low Power Consumption) and Renesas
00702 * iodef.h for RX72N, the GPTW peripheral is controlled by MSTPCRA
00703 * bit 7 (MSTPA7). Previous HAL code erroneously targeted MSTPCRC.MSTPC6
00704 * which is a different, unrelated peripheral on this chip.
00705 *
00706 * MSTPCRA lives at 0x00080010 and requires PRC1 unlock (0xA502) before
00707 * write, lock (0xA500) after.
00708 */
00709 k_mstpa_gptw = 7, /**< GPTW module stop bit in MSTPCRA (MSTPA7) */
00710 } gptw_module_stop_bits_t;
00711
00712 /*
=====
00713 * MPC PFS Select Values for GPTW Pins
00714 *
=====
00715 */
00716
00717 typedef enum : uint8_t {
00718 /**
00719 * @brief Peripheral select value for GPTW outputs on Port E
00720 * @warning Verify this value against RX72N Hardware Manual.
00721 * PSEL values are chip-specific.
00722 */
00723 k_pfs_psel_gptw = 0x1E, /**< PSEL value for GPTW (GTIOC*) alternate function.
00724 * Per RX72N HW manual Tables 23.4, 23.6, 23.10, 23.14,
00725 * 23.16: 0b011110 selects GTIOC on every port that
00726 * exposes it. Old 0x14 selected an unrelated
00727 * function and silently left the pin un-muxed. */
00728 } gptw_mpc_psel_t;
00729
00730 /*
=====
00731 * Register Offset Constants - Named constants for compile-time assertions
00732 *
=====
00733 */
00734
00735 typedef enum : uint8_t {
00736 k_gptw_ch_off_gtwp = 0x00U, /**< Write Protection Register */
00737 k_gptw_ch_off_gtstr = 0x04U, /**< Software Start Register */
00738 k_gptw_ch_off_gtstp = 0x08U, /**< Software Stop Register */
00739 k_gptw_ch_off_gtclr = 0x0CU, /**< Software Clear Register */
00740 k_gptw_ch_off_gtssr = 0x10U, /**< Start Source Select Register */
00741 k_gptw_ch_off_gtpsr = 0x14U, /**< Stop Source Select Register */
00742 k_gptw_ch_off_gtcsr = 0x18U, /**< Clear Source Select Register */
00743 k_gptw_ch_off_gtupsr = 0x1CU, /**< Count-Up Source Select Register */
00744 k_gptw_ch_off_gtdnsr = 0x20U, /**< Count-Down Source Select Register */
00745 k_gptw_ch_off_gticasr = 0x24U, /**< Input Capture Source Select A */
00746 k_gptw_ch_off_gticbsr = 0x28U, /**< Input Capture Source Select B */
00747 k_gptw_ch_off_gtcr = 0x2CU, /**< Control Register */
00748 k_gptw_ch_off_gtudtvc = 0x30U, /**< Count Direction and Duty Setting Register */
00749 k_gptw_ch_off_gtior = 0x34U, /**< I/O Control Register */
00750 k_gptw_ch_off_gtintad = 0x38U, /**< Interrupt Output Setting Register */
00751 k_gptw_ch_off_gtst = 0x3CU, /**< Status Register */
00752 k_gptw_ch_off_gtber = 0x40U, /**< Buffer Enable Register */
00753 k_gptw_ch_off_gtitc = 0x44U, /**< Interrupt Skipping Setting Register */
00754 k_gptw_ch_off_gtcnt = 0x48U, /**< Counter */
00755 k_gptw_ch_off_gtcra = 0x4CU, /**< Compare Capture Register A */
00756 k_gptw_ch_off_gtcrcb = 0x50U, /**< Compare Capture Register B */
00757 k_gptw_ch_off_gtcrc = 0x54U, /**< Compare Capture Register C */
00758 k_gptw_ch_off_gtcrcd = 0x58U, /**< Compare Capture Register D */
00759 k_gptw_ch_off_gtcrcd = 0x5CU, /**< Compare Capture Register D */
00760 k_gptw_ch_off_gtcrf = 0x60U, /**< Compare Capture Register F */
00761 k_gptw_ch_off_gtpr = 0x64U, /**< Cycle Setting Register */
00762 k_gptw_ch_off_gtpbr = 0x68U, /**< Cycle Setting Buffer Register */
00763 k_gptw_ch_off_gtpdbr = 0x6CU, /**< Cycle Setting Double-Buffer Register */
00764 k_gptw_ch_off_gtadtra = 0x70U, /**< A/D Conversion Start Request Timing Register A */
00765 k_gptw_ch_off_gtadtbra = 0x74U, /**< A/D Conversion Start Request Timing Buffer Register A */
00766 k_gptw_ch_off_gtadtdbra =
00767 0x78U, /**< A/D Conversion Start Request Timing Double-Buffer Register A */
00768 k_gptw_ch_off_gtadtrb = 0x7CU, /**< A/D Conversion Start Request Timing Register B */
00769 k_gptw_ch_off_gtadtbrb = 0x80U, /**< A/D Conversion Start Request Timing Buffer Register B */
00770 k_gptw_ch_off_gtadtdbrb =

```

```

00771 0x84U, /**< A/D Conversion Start Request Timing Double-Buffer Register B */
00772 k_gptw_ch_off_gtdtcr = 0x88U, /**< Dead-Time Control Register */
00773 k_gptw_ch_off_gtdvu = 0x8CU, /**< Dead-Time Value Register U */
00774 k_gptw_ch_off_gtdvd = 0x90U, /**< Dead-Time Value Register D */
00775 k_gptw_ch_off_gtdbu = 0x94U, /**< Dead-Time Buffer Register U */
00776 k_gptw_ch_off_gtdbd = 0x98U, /**< Dead-Time Buffer Register D */
00777 k_gptw_ch_off_gtsos = 0x9CU, /**< Output Protection Function Status Register */
00778 k_gptw_ch_off_gtsotr = 0xA0U, /**< Output Protection Function Temporary Release Register */
00779 k_gptw_ch_off_gtadsmr = 0xA4U, /**< A/D Conversion Start Request Signal Monitoring Register */
00780 k_gptw_ch_off_gteitc = 0xA8U, /**< Extended Interrupt Skipping Counter Control Register */
00781 k_gptw_ch_off_gteitli1 = 0xACU, /**< Extended Interrupt Skipping Setting Register 1 */
00782 k_gptw_ch_off_gteitli2 = 0xB0U, /**< Extended Interrupt Skipping Setting Register 2 */
00783 k_gptw_ch_off_gteitlb = 0xB4U, /**< Extended Buffer Transfer Skipping Setting Register */
00784 k_gptw_ch_off_reserved = 0xB8U, /**< Reserved region start */
00785 k_gptw_ch_off_gtsecsr = 0xD0U, /**< Operation Enable Bit Simultaneous Control Channel Select */
00786 k_gptw_ch_off_gtsecr = 0xD4U, /**< Operation Enable Bit Simultaneous Control Register */
00787 k_gptw_ch_struct_size = 0xD8U, /**< Total channel structure size in bytes */
00788 } gptw_ch_reg_offsets_t;
00789
00790 typedef enum : uint8_t {
00791 k_gptw_common_off_gtstpa = 0x04U, /**< Software Stop Register (GPTW0.GTSTP) */
00792 k_gptw_common_off_gtclra = 0x08U, /**< Software Clear Register (GPTW0.GTCLR) */
00793 k_gptw_common_struct_size = 0x0CU, /**< Total common structure size in bytes (3 x uint32_t) */
00794 } gptw_common_reg_offsets_t;
00795
00796 typedef enum : uintptr_t {
00797 k_gptw_periph_space_mask = 0xFFFF0000U, /**< Mask to extract peripheral space from address */
00798 k_gptw_periph_space_base = 0x000C0000U, /**< GPTW peripheral space base (0x000Cxxxx) */
00799 } gptw_periph_space_t;
00800
00801 /*
=====
00802 * Static Assertions - Compile-time verification of register layout
00803 *
=====
00804 */
00805
00806 /* Verify GPTW channel register structure size and critical offsets */
00807 static_assert(sizeof(rx_gptw_channel_regs_t) == k_gptw_ch_struct_size,
00808 "GPTW channel register structure size mismatch");
00809 static_assert(offsetof(rx_gptw_channel_regs_t, gtwp) == k_gptw_ch_off_gtwp,
00810 "GPTW GTWP register offset incorrect");
00811 static_assert(offsetof(rx_gptw_channel_regs_t, gtcra) == k_gptw_ch_off_gtcra,
00812 "GPTW GTCRA register offset incorrect");
00813 static_assert(offsetof(rx_gptw_channel_regs_t, gtcnt) == k_gptw_ch_off_gtcnt,
00814 "GPTW GTCNT register offset incorrect");
00815 static_assert(offsetof(rx_gptw_channel_regs_t, gtcra) == k_gptw_ch_off_gtcra,
00816 "GPTW GTCRA register offset incorrect");
00817 static_assert(offsetof(rx_gptw_channel_regs_t, gtpcr) == k_gptw_ch_off_gtpcr,
00818 "GPTW GTPCR register offset incorrect");
00819 static_assert(offsetof(rx_gptw_channel_regs_t, gtdtcr) == k_gptw_ch_off_gtdtcr,
00820 "GPTW GTDTCR register offset incorrect");
00821 static_assert(offsetof(rx_gptw_channel_regs_t, gtsotr) == k_gptw_ch_off_gtsotr,
00822 "GPTW GTSOTR register offset incorrect");
00823 static_assert(offsetof(rx_gptw_channel_regs_t, gtadsmr) == k_gptw_ch_off_gtadsmr,
00824 "GPTW GTADSMR register offset incorrect");
00825 static_assert(offsetof(rx_gptw_channel_regs_t, gteitc) == k_gptw_ch_off_gteitc,
00826 "GPTW GTEITC register offset incorrect");
00827 static_assert(offsetof(rx_gptw_channel_regs_t, gteitli1) == k_gptw_ch_off_gteitli1,
00828 "GPTW GTEITLI1 register offset incorrect");
00829 static_assert(offsetof(rx_gptw_channel_regs_t, gteitli2) == k_gptw_ch_off_gteitli2,
00830 "GPTW GTEITLI2 register offset incorrect");
00831 static_assert(offsetof(rx_gptw_channel_regs_t, gteitlb) == k_gptw_ch_off_gteitlb,
00832 "GPTW GTEITLB register offset incorrect");
00833 static_assert(offsetof(rx_gptw_channel_regs_t, gtsecsr) == k_gptw_ch_off_gtsecsr,
00834 "GPTW GTSECSR register offset incorrect");
00835 static_assert(offsetof(rx_gptw_channel_regs_t, gtsecr) == k_gptw_ch_off_gtsecr,
00836 "GPTW GTSECR register offset incorrect");
00837
00838 /* Additional channel register offset verification (100% coverage) */
00839 static_assert(offsetof(rx_gptw_channel_regs_t, gtstr) == k_gptw_ch_off_gtstr,
00840 "GPTW GTSTR register offset incorrect");
00841 static_assert(offsetof(rx_gptw_channel_regs_t, gtstp) == k_gptw_ch_off_gtstp,
00842 "GPTW GTSTP register offset incorrect");
00843 static_assert(offsetof(rx_gptw_channel_regs_t, gtclr) == k_gptw_ch_off_gtclr,
00844 "GPTW GTCLR register offset incorrect");
00845 static_assert(offsetof(rx_gptw_channel_regs_t, gtssr) == k_gptw_ch_off_gtssr,
00846 "GPTW GTSSR register offset incorrect");
00847 static_assert(offsetof(rx_gptw_channel_regs_t, gtupsr) == k_gptw_ch_off_gtupsr,
00848 "GPTW GTUPSR register offset incorrect");
00849 static_assert(offsetof(rx_gptw_channel_regs_t, gtcsr) == k_gptw_ch_off_gtcsr,
00850 "GPTW GTCSR register offset incorrect");
00851 static_assert(offsetof(rx_gptw_channel_regs_t, gtupsr) == k_gptw_ch_off_gtupsr,
00852 "GPTW GTUPSR register offset incorrect");
00853 static_assert(offsetof(rx_gptw_channel_regs_t, gtdnsr) == k_gptw_ch_off_gtdnsr,
00854 "GPTW GTDNSR register offset incorrect");
00855 static_assert(offsetof(rx_gptw_channel_regs_t, gticasr) == k_gptw_ch_off_gticasr,

```



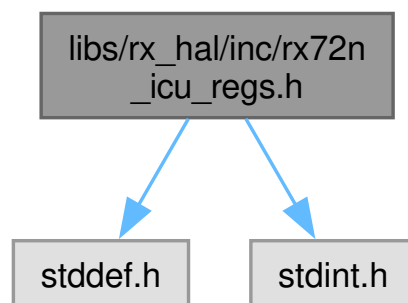
```

00856 "GPTW GTICASR register offset incorrect");
00857 static_assert(offsetof(rx_gptw_channel_regs_t, gticbsr) == k_gptw_ch_off_gticbsr,
00858 "GPTW GTICBSR register offset incorrect");
00859 static_assert(offsetof(rx_gptw_channel_regs_t, gtuddtyc) == k_gptw_ch_off_gtuddtyc,
00860 "GPTW GTUDDTYC register offset incorrect");
00861 static_assert(offsetof(rx_gptw_channel_regs_t, gtior) == k_gptw_ch_off_gtior,
00862 "GPTW GTIOR register offset incorrect");
00863 static_assert(offsetof(rx_gptw_channel_regs_t, gtintad) == k_gptw_ch_off_gtintad,
00864 "GPTW GTINTAD register offset incorrect");
00865 static_assert(offsetof(rx_gptw_channel_regs_t, gtst) == k_gptw_ch_off_gtst,
00866 "GPTW GTST register offset incorrect");
00867 static_assert(offsetof(rx_gptw_channel_regs_t, gtber) == k_gptw_ch_off_gtber,
00868 "GPTW GTBER register offset incorrect");
00869 static_assert(offsetof(rx_gptw_channel_regs_t, gtite) == k_gptw_ch_off_gtite,
00870 "GPTW GTITC register offset incorrect");
00871 static_assert(offsetof(rx_gptw_channel_regs_t, gtccrb) == k_gptw_ch_off_gtccrb,
00872 "GPTW GTCCRB register offset incorrect");
00873 static_assert(offsetof(rx_gptw_channel_regs_t, gtccrc) == k_gptw_ch_off_gtccrc,
00874 "GPTW GTCCRC register offset incorrect");
00875 static_assert(offsetof(rx_gptw_channel_regs_t, gtccre) == k_gptw_ch_off_gtccre,
00876 "GPTW GTCCRE register offset incorrect");
00877 static_assert(offsetof(rx_gptw_channel_regs_t, gtccrd) == k_gptw_ch_off_gtccrd,
00878 "GPTW GTCCRD register offset incorrect");
00879 static_assert(offsetof(rx_gptw_channel_regs_t, gtccrf) == k_gptw_ch_off_gtccrf,
00880 "GPTW GTCCRF register offset incorrect");
00881 static_assert(offsetof(rx_gptw_channel_regs_t, gtpbr) == k_gptw_ch_off_gtpbr,
00882 "GPTW GTPBR register offset incorrect");
00883 static_assert(offsetof(rx_gptw_channel_regs_t, gtpdbr) == k_gptw_ch_off_gtpdbr,
00884 "GPTW GTPDBR register offset incorrect");
00885 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtra) == k_gptw_ch_off_gtadtra,
00886 "GPTW GTADTRA register offset incorrect");
00887 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtbra) == k_gptw_ch_off_gtadtbra,
00888 "GPTW GTADTBRA register offset incorrect");
00889 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtdbra) == k_gptw_ch_off_gtadtdbra,
00890 "GPTW GTADTDBRA register offset incorrect");
00891 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtrb) == k_gptw_ch_off_gtadtrb,
00892 "GPTW GTADTRB register offset incorrect");
00893 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtbrb) == k_gptw_ch_off_gtadtbrb,
00894 "GPTW GTADTBRB register offset incorrect");
00895 static_assert(offsetof(rx_gptw_channel_regs_t, gtadtdbrb) == k_gptw_ch_off_gtadtdbrb,
00896 "GPTW GTADTDBRB register offset incorrect");
00897 static_assert(offsetof(rx_gptw_channel_regs_t, gtdvu) == k_gptw_ch_off_gtdvu,
00898 "GPTW GTDVU register offset incorrect");
00899 static_assert(offsetof(rx_gptw_channel_regs_t, gtdvd) == k_gptw_ch_off_gtdvd,
00900 "GPTW GTDVD register offset incorrect");
00901 static_assert(offsetof(rx_gptw_channel_regs_t, gtdbu) == k_gptw_ch_off_gtdbu,
00902 "GPTW GTDBU register offset incorrect");
00903 static_assert(offsetof(rx_gptw_channel_regs_t, gtdbd) == k_gptw_ch_off_gtdbd,
00904 "GPTW GTDBD register offset incorrect");
00905 static_assert(offsetof(rx_gptw_channel_regs_t, gtsos) == k_gptw_ch_off_gtsos,
00906 "GPTW GTSOS register offset incorrect");
00907 static_assert(offsetof(rx_gptw_channel_regs_t, reserved) == k_gptw_ch_off_reserved,
00908 "GPTW reserved register offset incorrect");
00909
00910 /* Verify GPTW common register structure size */
00911 static_assert(sizeof(rx_gptw_common_regs_t) == k_gptw_common_struct_size,
00912 "GPTW common register structure size mismatch");
00913
00914 /* Verify GPTW common register offsets (100% coverage) */
00915 static_assert(offsetof(rx_gptw_common_regs_t, gtstra) == 0x00,
00916 "GPTW common GTSTRA register offset incorrect");
00917 static_assert(offsetof(rx_gptw_common_regs_t, gtstpa) == k_gptw_common_off_gtstpa,
00918 "GPTW common GTSTPA register offset incorrect");
00919 static_assert(offsetof(rx_gptw_common_regs_t, gtclra) == k_gptw_common_off_gtclra,
00920 "GPTW common GTCLRA register offset incorrect");
00921
00922 /* Verify base addresses are in correct peripheral space (0x000C2xxx) */
00923 static_assert((k_gptw0_base_addr & k_gptw_periph_space_mask) == k_gptw_periph_space_base,
00924 "GPTW0 base address not in GPTW peripheral space");
00925 static_assert((k_gptw1_base_addr & k_gptw_periph_space_mask) == k_gptw_periph_space_base,
00926 "GPTW1 base address not in GPTW peripheral space");
00927 static_assert((k_gptw2_base_addr & k_gptw_periph_space_mask) == k_gptw_periph_space_base,
00928 "GPTW2 base address not in GPTW peripheral space");
00929 static_assert((k_gptw3_base_addr & k_gptw_periph_space_mask) == k_gptw_periph_space_base,
00930 "GPTW3 base address not in GPTW peripheral space");
00931
00932 /* Verify channel spacing is correct */
00933 static_assert(k_gptw1_base_addr - k_gptw0_base_addr == k_gptw_channel_offset,
00934 "GPTW channel spacing incorrect");
00935 static_assert(k_gptw2_base_addr - k_gptw1_base_addr == k_gptw_channel_offset,
00936 "GPTW channel spacing incorrect");
00937 static_assert(k_gptw3_base_addr - k_gptw2_base_addr == k_gptw_channel_offset,
00938 "GPTW channel spacing incorrect");
00939
00940 /** @} */ /* End of rx_gptw_regs group */
00941
00942 #ifdef __cplusplus

```

## 8.27 libs/rx/hal/inc/rx72n-icu-regs.h File Reference

```
#include <stddef.h>
#include <stdint.h>
Include dependency graph for rx72n_icu_regs.h:
```



- enum `icu_array_sizes_t` : `uint16_t` {  
`k_icu_ir_count` = 256 , `k_icu_dtcer_count` = 256 , `k_icu_ier_count` = 32 , `k_icu_ier_bits_per_reg` = 8 ,  
`k_icu_ipr_count` = 256 , `k_icu_dmrsr_count` = 8 , `k_icu_irqcr_count` = 16 , `k_icu_irqfte_count` = 2 ,  
`k_icu_irqfte_count` = 2 , `k_icu_slibr_count` = 16 , `k_icu_slibr_count` = 64 , `k_icu_sliar_count` = 48 }  
ICU register array dimensions.
- enum `rx_cmt_interrupt_vector_t` : `uint8_t` { `k_vect_cmt0_cmi0` = 28 , `k_vect_cmt1_cmi1` = 29 }  
CMT (Compare Match Timer) interrupt vector numbers.
- enum `rx_sci_interrupt_vector_t` : `uint8_t` { `k_vect_sci9_rxi9` = 102 }  
SCI (Serial Communications Interface) interrupt vector numbers.
- enum `rx_interrupt_priority_t` : `uint8_t` { `k_ipr_level_disable` = 0 , `k_ipr_level_min` = 1 , `k_ipr_level_max` = 15 }  
Interrupt priority levels for IPR registers.

## Functions

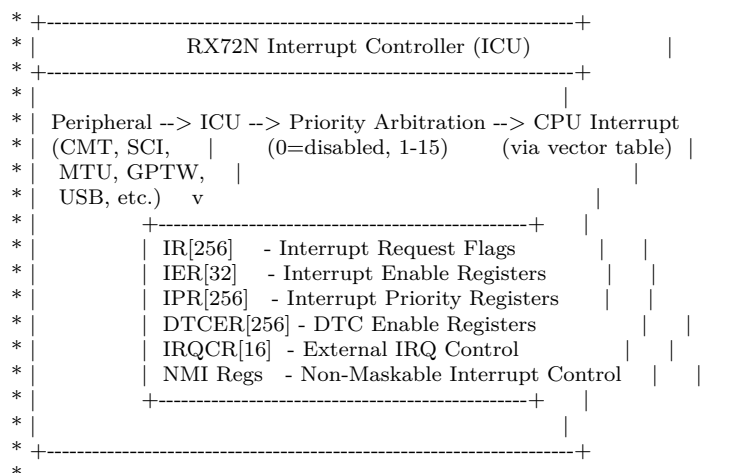
- static volatile `rx_icu_regs_t` \* `icu` (void)  
Get pointer to ICU registers.
- static volatile `uint8_t` \* `icu_slibr` (`uint8_t` vec)  
Get pointer to ICU SLIBR source-select register for a SELECTB vector.

### 8.27.1 Detailed Description

#### RX72N ICU Interrupt Controller Unit Register Definitions.

This header provides comprehensive register definitions for the RX72N's Interrupt Controller Unit (ICU). The ICU manages all 256 interrupt sources with configurable priorities, enabling/disabling, and status monitoring.

#### System Architecture Context:





## ICU Key Features:

- 256 interrupt vectors (16-255 usable, 0-15 reserved)
- 16 priority levels (0=disabled, 1=lowest, 15=highest)
- Per-interrupt enable/disable control via IER registers
- Fast interrupt mode (FIR) for single-cycle response
- DTC (Data Transfer Controller) integration
- 16 external IRQ pins with edge/level detection and filtering
- NMI (Non-Maskable Interrupt) with dedicated registers

## STAR Project Interrupt Usage:

| Vector  | Source          | Priority | Description                          |
|---------|-----------------|----------|--------------------------------------|
| 28      | CMT0_CMI0       | 10       | System tick (1 ms, ThreadX)          |
| 29      | CMT1_CMI1       | 8        | Motor control timer                  |
| 72-75   | IRQ8-IRQ11      | varies   | HC-SR04 echo (4 sensors, both edges) |
| 188-191 | INTB188-INTB191 | 14       | POEG fault protection (4 motors)     |

Note on vectors 188-191 (POEG): these are SLIBR software-configurable

interrupt B vectors. The SLIBXR/SLIBR registers (offsets 0x780-0x7CF) must be programmed to route POEGGAI/BI/CI/DI to INTB188-191 respectively. SCI9 (debug UART) is polled, not interrupt-driven; its fixed vectors are 102 (RXI9) and 103 (TXI9). SCI1 fixed vectors are 60 (RXI1) and 61 (TXI1).

## Hardware Requirements:

- CPU: Renesas RX72N with 256-vector interrupt controller
- ICU base address: 0x00087000 (fixed, non-relocatable)
- Register access: 8-bit for IR/IER/IPR, 16-bit for FIR

## Memory Usage:

- Register structure size: 0xA02 bytes (2562 bytes)
- No dynamic allocation required
- Direct memory-mapped register access

See also

docs/sections/03\_hardware\_pinout.tex External IRQ pin assignments  
 lib/rx\_core/inc/rx\_error\_handler.h Error and fault interrupt handling  
 RX72N Group User's Manual: Hardware, Chapter 15 (Interrupt Controller)

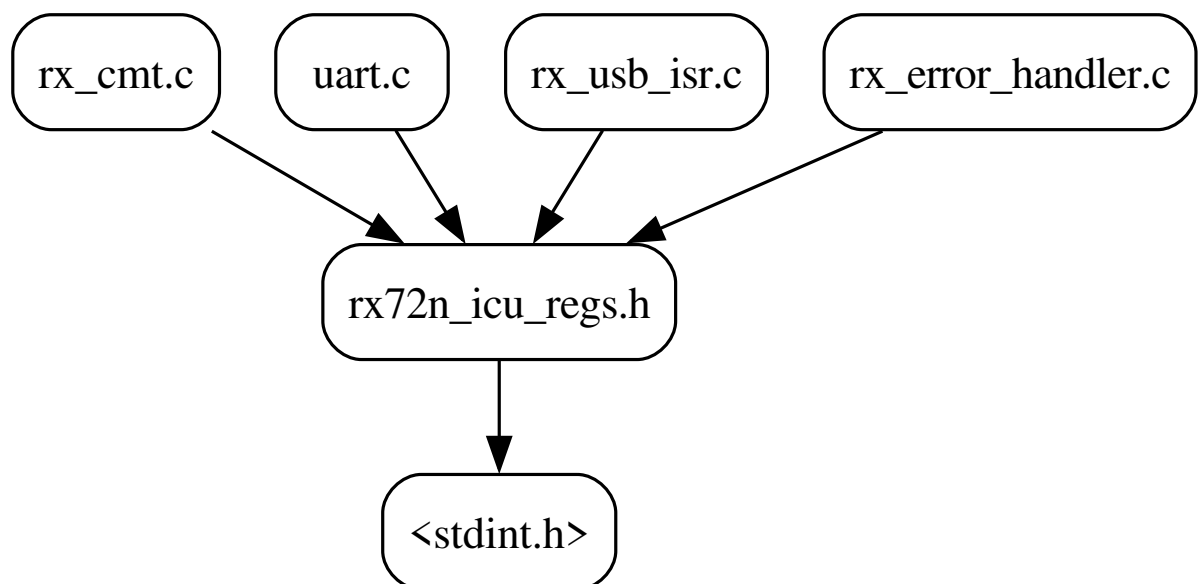
## NASA Power of 10 Compliance:

- Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
- Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
- Rule 8: [OK] All constants as C23 typed enums (vectors, priorities)
- Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings

## SOLID Principles:

- S: Single responsibility - only ICU register definitions
- O: Open for extension via accessor functions
- I: Interface segregation - minimal required registers exposed
- D: Dependency inversion - hardware accessed via typed pointers

## Module Dependencies:



## Verification Status:

[PASS] VERIFIED (2026-01-28) - All register offsets verified against RX72N Manual Ch15 section 15.2. Corrections applied:

- NMI register layout corrected (was wrong order)
- DMRSR registers fixed: 8-bit regs at 4-byte aligned addresses (0x400/404/408/...) NOT consecutive bytes as originally implemented

## Author

Locked, Inc.

## Date

2026-01-28

Version

1.0.0

Copyright

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Definition in file [rx72n`icu`regs.h](#).

## 8.28 rx72n`icu`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002 * @file rx72n_icu_regs.h
00003 * @brief RX72N ICU Interrupt Controller Unit Register Definitions
00004 *
00005 * @details
00006 * This header provides comprehensive register definitions for the RX72N's
00007 * Interrupt Controller Unit (ICU). The ICU manages all 256 interrupt sources
00008 * with configurable priorities, enabling/disabling, and status monitoring.
00009 *
00010 * @par System Architecture Context:
00011 * @verbatim
00012 * +-----+
00013 * | RX72N Interrupt Controller (ICU) |
00014 * +-----+
00015 * | |
00016 * | Peripheral --> ICU --> Priority Arbitration --> CPU Interrupt |
00017 * | (CMT, SCI, | (0=disabled, 1-15) (via vector table) |
00018 * | MTU, GPTW, |
00019 * | USB, etc.) v
00020 * |
00021 * | +-----+ |
00022 * | | IR[256] - Interrupt Request Flags | |
00023 * | | IER[32] - Interrupt Enable Registers | |
00024 * | | IPR[256] - Interrupt Priority Registers | |
00025 * | | DTCCR[256] - DTC Enable Registers | |
00026 * | | IRQCR[16] - External IRQ Control | |
00027 * | | NMI Regs - Non-Maskable Interrupt Control | |
00028 * | +-----+ |
00029 * +-----+
00030 * @endverbatim
00031 *
00032 * @par ICU Key Features:
00033 * - 256 interrupt vectors (16-255 usable, 0-15 reserved)
00034 * - 16 priority levels (0=disabled, 1=lowest, 15=highest)
00035 * - Per-interrupt enable/disable control via IER registers
00036 * - Fast interrupt mode (FIR) for single-cycle response
00037 * - DTC (Data Transfer Controller) integration
00038 * - 16 external IRQ pins with edge/level detection and filtering
00039 * - NMI (Non-Maskable Interrupt) with dedicated registers
00040 *
00041 * @par STAR Project Interrupt Usage:
00042 *
00043 * | Vector | Source | Priority | Description |
00044 * |-----|-----|-----|-----|
00045 * | 28 | CMT0_CMI0 | 10 | System tick (1 ms, ThreadX) |
00046 * | 29 | CMT1_CMI1 | 8 | Motor control timer |
00047 * | 72-75 | IRQ8-IRQ11 | varies | HC-SR04 echo (4 sensors, both edges) |
00048 * | 188-191 | INTB188-INTB191 | 14 | POEG fault protection (4 motors) |
00049 *
00050 * @par Note on vectors 188-191 (POEG): these are SLIBR software-configurable
00051 * interrupt B vectors. The SLIBXR/SLIBR registers (offsets 0x780-0x7CF) must
00052 * be programmed to route POEGGAI/BI/CI/DI to INTB188-191 respectively.
00053 * SCI9 (debug UART) is polled, not interrupt-driven; its fixed vectors are
00054 * 102 (RXI9) and 103 (TXI9). SCI1 fixed vectors are 60 (RXI1) and 61 (TXI1).
00055 *
00056 * @par Hardware Requirements:
00057 * - CPU: Renesas RX72N with 256-vector interrupt controller
00058 * - ICU base address: 0x00087000 (fixed, non-relocatable)
00059 * - Register access: 8-bit for IR/IER/IPR, 16-bit for FIR
00060 *
00061 * @par Memory Usage:
00062 * - Register structure size: 0xA02 bytes (2562 bytes)
00063 * - No dynamic allocation required

```

```

00064 * - Direct memory-mapped register access
00065 *
00066 * @see docs/sections/03_hardware_pinout.tex External IRQ pin assignments
00067 * @see lib/rx_core/inc/rx_error_handler.h Error and fault interrupt handling
00068 * @see RX72N Group User's Manual: Hardware, Chapter 15 (Interrupt Controller)
00069 *
00070 * @par NASA Power of 10 Compliance:
00071 * - Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
00072 * - Rule 3: [OK] Zero dynamic allocation - all registers statically mapped
00073 * - Rule 8: [OK] All constants as C23 typed enums (vectors, priorities)
00074 * - Rule 10: [OK] Compile with -Wall -Wextra -Werror, zero warnings
00075 *
00076 * @par SOLID Principles:
00077 * - S: Single responsibility - only ICU register definitions
00078 * - O: Open for extension via accessor functions
00079 * - I: Interface segregation - minimal required registers exposed
00080 * - D: Dependency inversion - hardware accessed via typed pointers
00081 *
00082 * @par Module Dependencies:
00083 * @dot
00084 * digraph icu_deps {
00085 * rankdir=TB;
00086 * node [shape=box, style=rounded];
00087 * icu_regs [label="rx72n_icu_regs.h"];
00088 * stdint [label="<stdint.h>"];
00089 * cmt [label="rx_cmt.c"];
00090 * uart [label="uart.c"];
00091 * usb_isr [label="rx_usb_isr.c"];
00092 * error [label="rx_error_handler.c"];
00093 * icu_regs -> stdint;
00094 * cmt -> icu_regs;
00095 * uart -> icu_regs;
00096 * usb_isr -> icu_regs;
00097 * error -> icu_regs;
00098 * }
00099 * @enddot
00100 *
00101 * @par Verification Status:
00102 * [PASS] VERIFIED (2026-01-28) - All register offsets verified against
00103 * RX72N Manual Ch15 section 15.2. Corrections applied:
00104 * - NMI register layout corrected (was wrong order)
00105 * - DMRSR registers fixed: 8-bit regs at 4-byte aligned addresses (0x400/404/408/...)
00106 * NOT consecutive bytes as originally implemented
00107 *
00108 * @author Locked, Inc.
00109 * @date 2026-01-28
00110 * @version 1.0.0
00111 * @copyright Copyright (c) 2026 Locked Inc.
00112 * SPDX-License-Identifier: MIT
00113 */
00114
00115 #pragma once
00116
00117 #include <stddef.h>
00118 #include <stdint.h>
00119
00120 #ifdef __cplusplus
00121 extern "C" {
00122 #endif
00123
00124 /**
00125 * @defgroup rx_icu_regs RX72N ICU Register Definitions
00126 * @brief Interrupt Controller Unit hardware register interface
00127 *
00128 * @details
00129 * Provides memory-mapped register access to the RX72N ICU peripheral
00130 * for interrupt configuration, enabling, and priority management.
00131 *
00132 * @par Quick Start Example - Enable CMT0 Interrupt:
00133 * @code{.c}
00134 * volatile rx_icu_regs_t* icu_regs = icu();
00135 *
00136 * // 1. Clear any pending interrupt request
00137 * icu_regs->ir[k_vect_cmt0_cmi0] = 0x00;
00138 *
00139 * // 2. Set priority (1=lowest, 15=highest, 0=disabled)
00140 * icu_regs->ipr[k_vect_cmt0_cmi0] = 10; // Priority 10
00141 *
00142 * // 3. Enable interrupt in IER register
00143 * // IER[3] bit 4 = vector 28 (CMT0_CMI0)
00144 * // IER index = vector / 8, bit = vector % 8
00145 * uint8_t ier_index = k_vect_cmt0_cmi0 / 8; // = 3
00146 * uint8_t ier_bit = k_vect_cmt0_cmi0 % 8; // = 4
00147 * icu_regs->ier[ier_index] |= (1U « ier_bit);
00148 *
00149 * // 4. Enable global interrupts (in PSW register, done separately)
00150 * @endcode

```

```

00151 *
00152 * @{
00153 */
00154
00155 /*
=====
00156 * Interrupt Controller (ICU)
00157 *
=====
00158 */
00159
00160 /**
00161 * @enum rx_icu_addresses_t
00162 * @brief ICU peripheral base address
00163 *
00164 * @details
00165 * The ICU occupies a contiguous block of memory starting at 0x00087000.
00166 * All ICU registers are accessed relative to this base address.
00167 *
00168 * @par Address Space:
00169 * | Offset Range | Registers | Description |
00170 * |-----|-----|-----|
00171 * | 0x000-0x0FF | IR[256] | Interrupt Request Flags |
00172 * | 0x100-0x1FF | DTCER[256] | DTC Enable Registers |
00173 * | 0x200-0x21F | IER[32] | Interrupt Enable Regs |
00174 * | 0x2E0-0x2E1 | SWINTR, SWINT2R | Software Interrupt |
00175 * | 0x2F0 | FIR | Fast Interrupt Register |
00176 * | 0x300-0x3FF | IPR[256] | Interrupt Priority Regs |
00177 * | 0x400-0x41F | DMRSR[8] | DMAC Module Start Regs |
00178 * | 0x500-0x50F | IRQCR[16] | External IRQ Control |
00179 * | 0x520-0x52D | IRQFLTE, IRQFLTC | IRQ Digital Filter |
00180 * | 0x580-0x594 | NMI Registers | Non-Maskable Interrupt |
00181 * | 0x780-0x78F | SLIBXRn[16] | Selectable Int B source (vect 128-143) |
00182 * | 0x790-0x7CF | SLIBRn[64] | Selectable Int B source (vect 144-207) |
00183 * | 0x9D0-0x9FF | SLIARn[48] | Selectable Int A source (vect 208-255) |
00184 * | 0xA00 | SLIPRCR | SLIBXR/SLIBR/SLIAR write-protect |
00185 * | 0xA01 | SELEXDR | Selectable ext int direction |
00186 *
00187 * @invariant Base address is fixed and cannot be relocated
00188 */
00189 typedef enum : uintptr_t {
00190 k_icu_base_addr = 0x00087000, /**< ICU register base address (fixed) */
00191 k_icu_slibr_base_addr = 0x00087700, /**< SELECTB source-select table (n=144..207) */
00192 k_icu_sliar_base_addr = 0x00087900, /**< SELECTA source-select table (n=208..255) */
00193 } rx_icu_addresses_t;
00194
00195 /**
00196 * @enum icu_reserved_sizes_t
00197 * @brief ICU register reserved field sizes for structure padding
00198 *
00199 * @details
00200 * Defines byte sizes for reserved regions in the ICU register map.
00201 * These ensure proper alignment and offset matching between the C
00202 * structure and actual hardware register locations.
00203 *
00204 * @warning Do not modify these values - they are derived from the
00205 * RX72N hardware register layout specification (Ch15).
00206 */
00207 typedef enum : uint16_t {
00208 k_icu_reserved_after_ier_bytes = 192, /**< 0x220-0x2DF: Gap to SWINTR */
00209 k_icu_reserved_after_swint2r_bytes = 14, /**< 0x2E2-0x2EF: Gap to FIR */
00210 k_icu_reserved_after_fir_bytes = 14, /**< 0x2F2-0x2FF: Gap to IPR */
00211 k_icu_reserved_after_dmsr_bytes = 224, /**< 0x420-0x4FF: Gap to IRQCR (was 248, FIXED) */
00212 k_icu_reserved_after_irqcr_bytes = 16, /**< 0x510-0x51F: Gap to IRQFLTE */
00213 k_icu_reserved_after_irqfte_bytes = 6, /**< 0x522-0x527: Gap to IRQFLTC */
00214 k_icu_reserved_after_irqfltc_bytes = 84, /**< 0x52C-0x57F: Gap to NMISR */
00215 k_icu_reserved_after_exnmiclr_bytes = 9, /**< 0x587-0x58F: Gap to NMIFLTE */
00216 k_icu_reserved_after_nmiflte_bytes = 3, /**< 0x591-0x593: Gap to NMIFLTC */
00217 k_icu_reserved_after_nmifltc_bytes = 491, /**< 0x595-0x77F: Gap to SLIBXRn */
00218 k_icu_reserved_after_slibr_bytes = 512, /**< 0x7D0-0x9CF: Gap to SLIARn */
00219 k_icu_dmsr_entry_padding = 3, /**< DMRSR 8-bit regs at 4-byte boundaries */
00220 } icu_reserved_sizes_t;
00221
00222 /**
00223 * @enum icu_array_sizes_t
00224 * @brief ICU register array dimensions
00225 *
00226 * @details
00227 * Defines the sizes of register arrays in the ICU. These values are
00228 * used to dimension arrays in the rx_icu_regs_t structure.
00229 *
00230 * @par Register Array Purpose:
00231 * - IR[256]: One flag per interrupt vector (vectors 0-255)
00232 * - DTCER[256]: One enable bit per DTC-capable vector
00233 * - IER[32]: 8 bits per register, covering 256 vectors (32 x 8 = 256)
00234 * - IPR[256]: One priority level per interrupt vector
00235 * - DMRSR[8]: DMAC channel 0-7 start source selection (4-byte aligned)

```

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00236 * - IRQCR[16]: External IRQ0-15 edge/level configuration
00237 */
00238 typedef enum : uint16_t {
00239 k_icu_ir_count = 256, /**< 256 interrupt request flags (one per vector) */
00240 k_icu_dtcr_count = 256, /**< 256 DTC enable flags */
00241 k_icu_ier_count = 32, /**< 32 enable registers x 8 bits = 256 vectors */
00242 k_icu_ier_bits_per_reg = 8, /**< Each IER controls 8 interrupt enables */
00243 k_icu_ipr_count = 256, /**< 256 priority registers (0-15 each) */
00244 k_icu_dmrsr_count = 8, /**< 8 DMAC channels */
00245 k_icu_irqcr_count = 16, /**< 16 external IRQ pins (IRQ0-15) */
00246 k_icu_irqfte_count = 2, /**< 2 filter enable registers (16 bits total) */
00247 k_icu_irqftc_count = 2, /**< 2 filter clock registers */
00248 k_icu_sliboxr_count = 16, /**< SLIBXRn entries: vectors 128-143 (INTB128-143) */
00249 k_icu_slibr_count = 64, /**< SLIBRn entries: vectors 144-207 (INTB144-207) */
00250 k_icu_sliar_count = 48, /**< SLIARn entries: vectors 208-255 (INTA208-255) */
00251 } icu_array_sizes_t;
00252
00253 /**
00254 * @struct rx_dmrsr_entry_t
00255 * @brief DMAC Module Start Register entry with 4-byte alignment
00256 *
00257 * @details
00258 * The DMRSR registers are 8-bit registers but located at 4-byte aligned
00259 * addresses (0x400, 0x404, 0x408, 0x40C, 0x410, 0x414, 0x418, 0x41C).
00260 * This structure provides proper padding to match the hardware layout.
00261 *
00262 * @par Manual Reference:
00263 * Ch15 Section 15.2.8 - DMRSR0-7 addresses:
00264 * - DMRSR0: 0x00087400
00265 * - DMRSR1: 0x00087404
00266 * - DMRSR2: 0x00087408
00267 * - DMRSR3: 0x0008740C
00268 * - DMRSR4: 0x00087410
00269 * - DMRSR5: 0x00087414
00270 * - DMRSR6: 0x00087418
00271 * - DMRSR7: 0x0008741C
00272 *
00273 * @par Usage:
00274 * @code{.c}
00275 * // Set DMAC channel 0 trigger to vector 28 (CMT0)
00276 * icu()->dmrsr[0].reg = 28;
00277 *
00278 * // Set DMAC channel 3 trigger to USB vector
00279 * icu()->dmrsr[3].reg = 102;
00280 * @endcode
00281 *
00282 * @note Each DMRSR register selects the interrupt vector number that
00283 * triggers the corresponding DMAC channel transfer.
00284 *
00285 * @warning Do not set the same vector number for multiple DMRSR registers.
00286 * @warning Do not set the same interrupt as both DTC and DMAC trigger.
00287 */
00288 typedef struct {
00289 volatile uint8_t reg; /**< 8-bit vector number register */
00290 uint8_t reserved[k_icu_dmrsr_entry_padding]; /**< Padding to 4-byte boundary */
00291 } rx_dmrsr_entry_t;
00292
00293 /**
00294 * @struct rx_icu_regs_t
00295 * @brief ICU Register Map for Interrupt Controller Unit
00296 *
00297 * @details
00298 * Memory-mapped register structure for the ICU peripheral. This structure
00299 * provides direct hardware access to interrupt control registers including
00300 * request flags, enable control, priorities, and external IRQ configuration.
00301 *
00302 * @par Interrupt Handling Workflow:
00303 * 1. Peripheral generates interrupt request -> IR[vector] = 1
00304 * 2. ICU checks IER[vector/8] bit (vector%8) for enable
00305 * 3. If enabled, ICU checks IPR[vector] for priority (1-15)
00306 * 4. Higher priority interrupts preempt lower priority
00307 * 5. CPU vectors to interrupt handler
00308 * 6. Handler clears IR[vector] = 0 before returning
00309 *
00310 * @par Usage Example - Configure Multiple Interrupts:
00311 * @code{.c}
00312 * volatile rx_icu_regs_t* icu_regs = icu();
00313 *
00314 * // Disable all interrupts during configuration
00315 * // (typically done with PSW.I bit, not shown here)
00316 *
00317 * // Configure CMT0 (vector 28) - System tick at priority 10
00318 * icu_regs->ir[28] = 0; // Clear pending
00319 * icu_regs->ipr[28] = 10; // Priority 10
00320 * icu_regs->ier[3] |= 0x10; // Enable (IER3 bit 4)
00321 *
00322 * // Configure CMT1 (vector 29) - Motor control at priority 12

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```

00323 * icu_regs->ir[29] = 0; // Clear pending
00324 * icu_regs->ipr[29] = 12; // Priority 12 (higher than tick)
00325 * icu_regs->ier[3] |= 0x20; // Enable (IER3 bit 5)
00326 *
00327 * // Configure USB (vector 102) - Communication at priority 8
00328 * icu_regs->ir[102] = 0; // Clear pending
00329 * icu_regs->ipr[102] = 8; // Priority 8
00330 * icu_regs->ier[12] |= 0x40; // Enable (IER12 bit 6)
00331 *
00332 * // Re-enable global interrupts (PSW.I = 1)
00333 * @endcode
00334 *
00335 * @par Memory Layout (Total: ~2562 bytes):
00336 *
00337 * | Offset | Size | Field | Description |
00338 * |-----|-----|-----|-----|
00339 * | 0x000 | 256 | ir[256] | Interrupt Request flags |
00340 * | 0x100 | 256 | dtcer[256] | DTC Enable registers |
00341 * | 0x200 | 32 | ier[32] | Interrupt Enable registers |
00342 * | 0x220 | 192 | reserved | Reserved gap |
00343 * | 0x2E0 | 1 | swintr | Software Interrupt register |
00344 * | 0x2E1 | 1 | swint2r | Software Interrupt 2 register |
00345 * | 0x2F0 | 2 | fir | Fast Interrupt register |
00346 * | 0x300 | 256 | ipr[256] | Interrupt Priority registers |
00347 * | 0x400 | 32 | dmrsr[8] | DMAC Module Start registers |
00348 * | 0x500 | 16 | irqcr[16] | External IRQ Control |
00349 * | 0x520 | 2 | irqfte[2] | IRQ Filter Enable |
00350 * | 0x528 | 4 | irqftc[2] | IRQ Filter Clock Select |
00351 * | 0x580 | 4 | nmisr..nmicr | NMI Status/Enable/Clear/Ctrl |
00352 * | 0x584 | 3 | exnmisr..clr | Extended NMI registers |
00353 * | 0x590 | 1 | nmifte | NMI Filter Enable |
00354 * | 0x594 | 1 | nmiftc | NMI Filter Clock |
00355 * | 0x595 | 491 | reserved | Reserved gap |
00356 * | 0x780 | 16 | slibr[16] | Sel. Int B source sel. (vect 128-143)|
00357 * | 0x790 | 64 | slibr[64] | Sel. Int B source sel. (vect 144-207)|
00358 * | 0x7D0 | 512 | reserved | Reserved gap |
00359 * | 0x9D0 | 48 | sliar[48] | Sel. Int A source sel. (vect 208-255)|
00360 * | 0xA00 | 1 | slipcr | SLIBXR/SLIBR/SLIAR write-protect |
00361 * | 0xA01 | 1 | seleldr | Selectable ext int direction |
00362 *
00363 * @note Register layout verified against RX72N Manual Ch15, Rev.1.20
00364 *
00365 * @warning All registers are memory-mapped and must be accessed as volatile.
00366 * @warning Some IR/IPR indices (0-15) are reserved and must not be accessed.
00367 *
00368 * @invariant IR flags are read-write: write 0 to clear, read to check status
00369 * @invariant IPR values must be 0-15 (0 disables, 1-15 sets priority)
00370 * @invariant IER bits: 1=enabled, 0=disabled
00371 *
00372 * @see icu() Accessor function
00373 * @see rx_cmt_interrupt_vector_t Common interrupt vector numbers
00374 * @see rx_interrupt_priority_t Priority level definitions
00375 */
00376 typedef struct {
00377 /* Offset 0x000-0x0FF: Interrupt Request Registers */
00378 volatile uint8_t ir[k_icu_ir_count]; /**< @0x000 IR016-IR255 */
00379
00380 /* Offset 0x100-0x1FF: DTC Enable Registers */
00381 volatile uint8_t dtcer[k_icu_dtcer_count]; /**< @0x100 DTCER026-DTCER255 */
00382
00383 /* Offset 0x200-0x21F: Interrupt Enable Registers */
00384 volatile uint8_t ier[k_icu_ier_count]; /**< @0x200 IER02-IER1F */
00385 uint8_t reserved0[k_icu_reserved_after_ier_bytes]; /**< @0x220 Reserved */
00386
00387 /* Offset 0x2E0-0x2E1: Software Interrupt Registers */
00388 volatile uint8_t swintr; /**< @0x2E0 Software Interrupt Register */
00389 volatile uint8_t swint2r; /**< @0x2E1 Software Interrupt 2 Register */
00390 uint8_t reserved1[k_icu_reserved_after_swint2r_bytes]; /**< @0x2E2 Reserved */
00391
00392 /* Offset 0x2F0: Fast Interrupt Register */
00393 volatile uint16_t fir; /**< @0x2F0 Fast Interrupt Register */
00394 uint8_t reserved2[k_icu_reserved_after_fir_bytes]; /**< @0x2F2 Reserved */
00395
00396 /* Offset 0x300-0x3FF: Interrupt Priority Registers */
00397 volatile uint8_t ipr[k_icu_ipr_count]; /**< @0x300 IPR000-IPR255 */
00398
00399 /* Offset 0x400-0x41F: DMAC Module Start Registers (4-byte aligned 8-bit regs) */
00400 rx_dmrsr_entry_t dmrsr[k_icu_dmrsr_count]; /**< @0x400 DMRSR0-DMRSR7 (4-byte spacing!) */
00401 uint8_t reserved3[k_icu_reserved_after_dmrsr_bytes]; /**< @0x420 Reserved (FIXED from 0x408) */
00402
00403 /* Offset 0x500-0x50F: IRQ Control Registers */
00404 volatile uint8_t irqcr[k_icu_irqcr_count]; /**< @0x500 IRQCR0-IRQCR15 */
00405 uint8_t reserved4[k_icu_reserved_after_irqcr_bytes]; /**< @0x510 Reserved */
00406
00407 /* Offset 0x520-0x52D: IRQ Filter Registers */
00408 volatile uint8_t irqfte[k_icu_irqfte_count]; /**< @0x520 IRQFLTE0-IRQFLTE1 */
00409 uint8_t reserved5[k_icu_reserved_after_irqfte_bytes]; /**< @0x522 Reserved */

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00410 volatile uint16_t irqfltc[k_icu_irqfltc_count]; /**< @0x528 IRQFLTC0-IRQFLTC1 */
00411 uint8_t reserved6[k_icu_reserved_after_irqfltc_bytes]; /**< @0x52C Reserved */
00412
00413 /* Offset 0x580-0x597: NMI Registers (CORRECTED ORDER - Manual Ch15 Section 15.2.13-15.2.22) */
00414 volatile uint8_t nmisr; /**< @0x580 NMI Status Register */
00415 volatile uint8_t nmier; /**< @0x581 NMI Enable Register */
00416 volatile uint8_t nmiclr; /**< @0x582 NMI Clear Register */
00417 volatile uint8_t nmicr; /**< @0x583 NMI Control Register (FIXED: was uint32_t) */
00418
00419 /* Offset 0x584-0x58F: Extended NMI Registers (ADDED - were missing) */
00420 volatile uint8_t exnmisr; /**< @0x584 Extended NMI Status Register */
00421 volatile uint8_t exnmier; /**< @0x585 Extended NMI Enable Register */
00422 volatile uint8_t exnmiclr; /**< @0x586 Extended NMI Clear Register */
00423 uint8_t reserved7[k_icu_reserved_after_exnmiclr_bytes]; /**< @0x587 Reserved */
00424
00425 /* Offset 0x590-0x597: NMI Filter Registers (CORRECTED) */
00426 volatile uint8_t nmiflte; /**< @0x590 NMI Filter Enable Register (ADDED - was missing) */
00427 uint8_t reserved8[k_icu_reserved_after_nmiflte_bytes]; /**< @0x591 Reserved */
00428 volatile uint8_t nmiflfc; /**< @0x594 NMI Filter Clock Select Register (RENAMED from nmiflt) */
00429 uint8_t reserved9[k_icu_reserved_after_nmiflfc_bytes]; /**< @0x595 Reserved */
00430
00431 /* Offset 0x780-0x78F: Selectable Interrupt B Source Registers (vectors 128-143) */
00432 volatile uint8_t slibr[k_icu_slibr_count]; /**< @0x780 SLIBXRn: source select for INTB128-143 */
00433
00434 /* Offset 0x790-0x7CF: Selectable Interrupt B Source Registers (vectors 144-207) */
00435 volatile uint8_t slibr[k_icu_slibr_count]; /**< @0x790 SLIBRn: source select for INTB144-207 */
00436 uint8_t reserved10[k_icu_reserved_after_slibr_bytes]; /**< @0x7D0 Reserved */
00437
00438 /* Offset 0x9D0-0x9FF: Selectable Interrupt A Source Registers (vectors 208-255) */
00439 volatile uint8_t sliar[k_icu_sliar_count]; /**< @0x9D0 SLIARn: source select for INTA208-255 */
00440
00441 /* Offset 0xA00-0xA01: Selectable Interrupt Write-Protect / Direction */
00442 volatile uint8_t sliprcr; /**< @0xA00 SLIPRCR: write-protect for SLIBXR/SLIBR/SLIAR/SELEXDR */
00443 volatile uint8_t seleldr; /**< @0xA01 SELEXDR: selectable external interrupt direction */
00444 } rx_icu_regs_t;
00445
00446 /**
00447 * @brief Get pointer to ICU registers
00448 *
00449 * @details
00450 * Returns a volatile pointer to the ICU register structure,
00451 * enabling direct hardware register access for interrupt configuration.
00452 *
00453 * @return Volatile pointer to ICU register structure at 0x00087000
00454 *
00455 * @pre No prerequisites - ICU is always accessible
00456 *
00457 * @post Pointer is immediately usable for register access
00458 * @post No hardware state is modified by this function
00459 *
00460 * @note Thread Safety: Function is reentrant (returns constant address)
00461 * @note Performance: Zero overhead - compiles to single load instruction
00462 *
00463 * @warning Interrupt configuration should be done with global interrupts
00464 * disabled to prevent race conditions
00465 *
00466 * @par Example - Enable/Disable Specific Interrupt:
00467 * @code{.c}
00468 * volatile rx_icu_regs_t* icu_regs = icu();
00469 *
00470 * // Calculate IER index and bit for vector 28
00471 * uint8_t vector = 28;
00472 * uint8_t ier_idx = vector / 8; // = 3
00473 * uint8_t ier_bit = vector % 8; // = 4
00474 *
00475 * // Enable interrupt
00476 * icu_regs->ier[ier_idx] |= (1U « ier_bit);
00477 *
00478 * // Disable interrupt
00479 * icu_regs->ier[ier_idx] &= ~(1U « ier_bit);
00480 *
00481 * // Check if interrupt is pending
00482 * if (icu_regs->ir[vector]) {
00483 * // Interrupt is pending
00484 * icu_regs->ir[vector] = 0; // Clear it
00485 * }
00486 * @endcode
00487 *
00488 * @see rx_icu_regs_t Register structure definition
00489 */
00490 static inline volatile rx_icu_regs_t* icu(void)
00491 {
00492 return (volatile rx_icu_regs_t*)k_icu_base_addr;
00493 }
00494
00495 /**
00496 * @brief Get pointer to ICU SLIBR source-select register for a SELECTB vector.

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00497 *
00498 * SLIBR is a byte-wide table at 0x00087700 indexed by absolute vector
00499 * number, covering SELECTB slots 144..207 (RX72N HW manual Ch15 Section
00500 * 15.2 and hirakuni45/RX RX72N/icu.hpp line 824). Writing a SELECTB
00501 * source code (e.g. k_usb0_usbi_sli_src = 62) into SLIBR[vec] routes
00502 * that peripheral interrupt onto vec. Vectors outside 144..207 have
00503 * no backing register -- writes are ignored and reads return 0.
00504 *
00505 * @param[in] vec Absolute vector number in the SELECTB range (144..207)
00506 * @return volatile uint8_t* Pointer to SLIBR[vec]
00507 */
00508 static inline volatile uint8_t* icu_slibr(uint8_t vec)
00509 {
00510 return (volatile uint8_t*)(k_icu_slibr_base_addr + (uintptr_t)vec);
00511 }
00512
00513 /**
00514 * @enum rx_cmt_interrupt_vector_t
00515 * @brief CMT (Compare Match Timer) interrupt vector numbers
00516 *
00517 * @details
00518 * Vector numbers for CMT peripheral interrupts. These are used to
00519 * index into IR[], IER[], and IPR[] arrays.
00520 *
00521 * @par IER/Bit Calculation:
00522 * - IER index = vector / 8
00523 * - IER bit = vector % 8
00524 *
00525 * @par Example: CMT0 (vector 28)
00526 * - IER[3] (28/8=3), bit 4 (28%8=4)
00527 * - IPR[28] for priority
00528 *
00529 * @see RX72N Manual Ch15 Table 15.4 for complete vector list
00530 */
00531 typedef enum : uint8_t {
00532 k_vect_cmt0_cmi0 = 28, /**< CMT0 compare match: IER[3] bit 4, used for 1ms tick */
00533 k_vect_cmt1_cmi1 = 29, /**< CMT1 compare match: IER[3] bit 5, used for motor control */
00534 } rx_cmt_interrupt_vector_t;
00535
00536 /**
00537 * @enum rx_sci_interrupt_vector_t
00538 * @brief SCI (Serial Communications Interface) interrupt vector numbers
00539 *
00540 * @details
00541 * Vector numbers for SCI peripheral receive/transmit interrupts. Used to
00542 * index into IR[], IER[], and IPR[] arrays per RX72N HUM Table 14.3.
00543 *
00544 * @par IER/Bit Calculation:
00545 * - IER index = vector / 8
00546 * - IER bit = vector % 8
00547 *
00548 * @par Example: SCI9 RXI9 (vector 102)
00549 * - IER[12] (102/8=12), bit 6 (102%8=6)
00550 * - IPR[102] for priority
00551 *
00552 * @see RX72N HUM Ch14 Table 14.3 for complete vector list
00553 */
00554 typedef enum : uint8_t {
00555 k_vect_sci9_rx9 = 102, /**< SCI9 receive data full: IER[12] bit 6, STAR debug UART RX */
00556 } rx_sci_interrupt_vector_t;
00557
00558 /**
00559 * @enum rx_interrupt_priority_t
00560 * @brief Interrupt priority levels for IPR registers
00561 *
00562 * @details
00563 * Priority values for the IPR[n] registers. Higher values mean higher
00564 * priority. A running interrupt can only be preempted by an interrupt
00565 * with strictly higher priority.
00566 *
00567 * @par Priority Guidelines for STAR Project:
00568 *
00569 * | Priority | Usage | Example |
00570 * |-----|-----|-----|
00571 * | 15 | Reserved for NMI/faults | Hardware fault |
00572 * | 12-14 | Time-critical motor | Motor PWM sync |
00573 * | 9-11 | Real-time control | CMT tick, encoder |
00574 * | 6-8 | Communication | USB, UART RX |
00575 * | 3-5 | Background processing | UART TX, DMA |
00576 * | 1-2 | Low priority | Diagnostics |
00577 * | 0 | Disabled | Unused interrupts |
00578 *
00579 * @warning Setting two interrupts to the same priority is allowed but
00580 * undefined which will be serviced first if both are pending
00581 */
00582 typedef enum : uint8_t {
00583 k_ipr_level_disable = 0, /**< Interrupt disabled (will not trigger) */

```

```

00584 k_ipr_level_min = 1, /**< Minimum enabled priority (lowest) */
00585 k_ipr_level_max = 15, /**< Maximum priority (highest, preempts all) */
00586 } rx_interrupt_priority_t;
00587
00588 /*
=====
00589 * Static Assertions - Verify Register Layout at Compile Time
00590 *
=====
00591 */
00592
00593 /* The k_icu_base_addr enum declaration above is itself the contract for the
00594 * ICU base address (per Hardware Manual Ch15). A static_assert that re-states
00595 * the same literal here would duplicate the magic number without adding a
00596 * cross-check, so we rely on the enum declaration alone. */
00597
00598 /* Verify register structure layout and key offsets (Ch15 Section 15.2) */
00599 static_assert(offsetof(rx_icu_regs_t, ir) == 0x000, "IR offset must be 0x000 (@ 0x00087000)");
00600 static_assert(offsetof(rx_icu_regs_t, dtcer) == 0x100, "DT CER offset must be 0x100 (@ 0x00087100)");
00601 static_assert(offsetof(rx_icu_regs_t, ier) == 0x200, "IER offset must be 0x200 (@ 0x00087200)");
00602 static_assert(offsetof(rx_icu_regs_t, swintr) == 0x2E0,
00603 "SWINTR offset must be 0x2E0 (@ 0x000872E0)");
00604 static_assert(offsetof(rx_icu_regs_t, swint2r) == 0x2E1,
00605 "SWINT2R offset must be 0x2E1 (@ 0x000872E1)");
00606 static_assert(offsetof(rx_icu_regs_t, fir) == 0x2F0, "FIR offset must be 0x2F0 (@ 0x000872F0)");
00607 static_assert(offsetof(rx_icu_regs_t, ipr) == 0x300, "IPR offset must be 0x300 (@ 0x00087300)");
00608 static_assert(offsetof(rx_icu_regs_t, dmrsr) == 0x400, "DMRSR offset must be 0x400 (@ 0x00087400)");
00609 static_assert(offsetof(rx_icu_regs_t, irqcr) == 0x500, "IRQCR offset must be 0x500 (@ 0x00087500)");
00610 static_assert(offsetof(rx_icu_regs_t, irqflte) == 0x520,
00611 "IRQFLTE offset must be 0x520 (@ 0x00087520)");
00612 static_assert(offsetof(rx_icu_regs_t, irqfltc) == 0x528,
00613 "IRQFLTC offset must be 0x528 (@ 0x00087528)");
00614
00615 /* Verify NMI register offsets (Ch15 Section 15.2.13-15.2.22) - CRITICAL */
00616 static_assert(offsetof(rx_icu_regs_t, nmisr) == 0x580, "NMISR offset must be 0x580 (@ 0x00087580)");
00617 static_assert(offsetof(rx_icu_regs_t, nmier) == 0x581, "NMIER offset must be 0x581 (@ 0x00087581)");
00618 static_assert(offsetof(rx_icu_regs_t, nmiclr) == 0x582,
00619 "NMICLR offset must be 0x582 (@ 0x00087582)");
00620 static_assert(offsetof(rx_icu_regs_t, nmicr) == 0x583, "NMICR offset must be 0x583 (@ 0x00087583)");
00621 static_assert(offsetof(rx_icu_regs_t, exnmisr) == 0x584,
00622 "EXNMISR offset must be 0x584 (@ 0x00087584)");
00623 static_assert(offsetof(rx_icu_regs_t, exnmier) == 0x585,
00624 "EXNMIER offset must be 0x585 (@ 0x00087585)");
00625 static_assert(offsetof(rx_icu_regs_t, exnmiclr) == 0x586,
00626 "EXNMICLR offset must be 0x586 (@ 0x00087586)");
00627 static_assert(offsetof(rx_icu_regs_t, nmiflte) == 0x590,
00628 "NMIFLTE offset must be 0x590 (@ 0x00087590)");
00629 static_assert(offsetof(rx_icu_regs_t, nmifltc) == 0x594,
00630 "NMIFLTC offset must be 0x594 (@ 0x00087594)");
00631
00632 /* Verify register sizes */
00633 static_assert(sizeof((rx_icu_regs_t*)0)->nmicr) == 1, "NMICR must be 8-bit (1 byte), not 32-bit");
00634
00635 /* Verify selectable interrupt source register offsets (Ch15 Section 15.2.17-15.2.21) */
00636 static_assert(offsetof(rx_icu_regs_t, slibxr) == 0x780,
00637 "SLIBXRn offset must be 0x780 (@ 0x00087780)");
00638 static_assert(offsetof(rx_icu_regs_t, slibr) == 0x790,
00639 "SLIBRn offset must be 0x790 (@ 0x00087790)");
00640 static_assert(offsetof(rx_icu_regs_t, sliar) == 0x9D0,
00641 "SLIARn offset must be 0x9D0 (@ 0x000879D0)");
00642 static_assert(offsetof(rx_icu_regs_t, sliprcr) == 0xA00,
00643 "SLIPRCR offset must be 0xA00 (@ 0x00087A00)");
00644 static_assert(offsetof(rx_icu_regs_t, selexdr) == 0xA01,
00645 "SELEXDR offset must be 0xA01 (@ 0x00087A01)");
00646
00647 /* Verify DMRSR entry structure - CRITICAL FIX for 4-byte aligned 8-bit registers */
00648 static_assert(sizeof(rx_dmrsr_entry_t) == 4, "DMRSR entry must be 4 bytes (1 reg + 3 pad)");
00649 static_assert(offsetof(rx_dmrsr_entry_t, reg) == 0, "DMRSR reg offset must be 0");
00650
00651 /* Verify individual DMRSR register addresses (Ch15 Section 15.2.8) */
00652 static_assert(offsetof(rx_icu_regs_t, dmrsr[0]) == 0x400, "DMRSR0 must be at 0x400 (@ 0x00087400)");
00653 static_assert(offsetof(rx_icu_regs_t, dmrsr[1]) == 0x404, "DMRSR1 must be at 0x404 (@ 0x00087404)");
00654 static_assert(offsetof(rx_icu_regs_t, dmrsr[2]) == 0x408, "DMRSR2 must be at 0x408 (@ 0x00087408)");
00655 static_assert(offsetof(rx_icu_regs_t, dmrsr[3]) == 0x40C, "DMRSR3 must be at 0x40C (@ 0x0008740C)");
00656 static_assert(offsetof(rx_icu_regs_t, dmrsr[4]) == 0x410, "DMRSR4 must be at 0x410 (@ 0x00087410)");
00657 static_assert(offsetof(rx_icu_regs_t, dmrsr[5]) == 0x414, "DMRSR5 must be at 0x414 (@ 0x00087414)");
00658 static_assert(offsetof(rx_icu_regs_t, dmrsr[6]) == 0x418, "DMRSR6 must be at 0x418 (@ 0x00087418)");
00659 static_assert(offsetof(rx_icu_regs_t, dmrsr[7]) == 0x41C, "DMRSR7 must be at 0x41C (@ 0x0008741C)");
00660
00661 /** @} */ /* End of rx_icu_regs group */
00662
00663 #ifdef __cplusplus
00664 }
00665 #endif

```

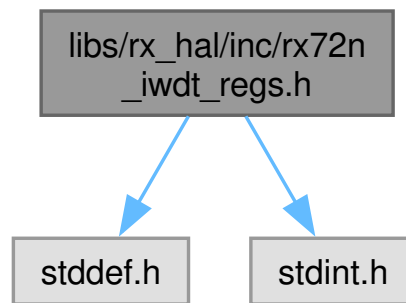
## 8.29 libs/rx\_hal/inc/rx72n\_iwdt\_regs.h File Reference

RX72N IWDT Independent Watchdog Timer Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_iwdt\_regs.h:



This graph shows which files directly or indirectly include this file:



### Data Structures

- struct [rx\\_iwdt\\_regs\\_t](#)  
IWDT Register Map Structure.

### Enumerations

- enum [rx\\_iwdt\\_addresses\\_t](#) : [uintptr\\_t](#) { [k\\_iwdt\\_base\\_addr](#) = 0x00088030 }
- enum [iwdt\\_reserved\\_sizes\\_t](#) : [uint8\\_t](#) { [k\\_iwdt\\_reserved\\_after\\_iwdtrr\\_bytes](#) = 1 , [k\\_iwdt\\_reserved\\_after\\_iwdtrcr\\_bytes](#) = 1 }
- enum [iwdt\\_refresh\\_sequence\\_t](#) : [uint8\\_t](#) { [k\\_iwdt\\_refresh\\_start](#) = 0x00 , [k\\_iwdt\\_refresh\\_end](#) = 0xFF }
- enum [iwdt\\_iwdtcr\\_shifts\\_t](#) : [uint8\\_t](#) { [k\\_iwdt\\_iwdtcr\\_cks\\_shift](#) = 4 , [k\\_iwdt\\_iwdtcr\\_rpes\\_shift](#) = 8 , [k\\_iwdt\\_iwdtcr\\_rpss\\_shift](#) = 12 }

- enum `iwdt_iwdtcr_bits_t` : `uint16_t` {  
`k_iwdt_tops_1024` = 0x0000 , `k_iwdt_tops_4096` = 0x0001 , `k_iwdt_tops_8192` = 0x0002 ,  
`k_iwdt_tops_16384` = 0x0003 ,  
`k_iwdt_cks_div_1` = (0x00 << `k_iwdt_iwdtcr_cks_shift`) , `k_iwdt_cks_div_16` = (0x02 << `k_iwdt_iwdtcr_cks_shift`) ,  
`k_iwdt_cks_div_32` = (0x03 << `k_iwdt_iwdtcr_cks_shift`) ,  
`k_iwdt_cks_div_64` = (0x04 << `k_iwdt_iwdtcr_cks_shift`) ,  
`k_iwdt_cks_div_128` = (0x0F << `k_iwdt_iwdtcr_cks_shift`) , `k_iwdt_cks_div_256` = (0x05 << `k_iwdt_iwdtcr_cks_shift`) ,  
`k_iwdt_rpes_75` = (0x00 << `k_iwdt_iwdtcr_rpes_shift`) ,  
`k_iwdt_rpes_50` = (0x01 << `k_iwdt_iwdtcr_rpes_shift`) ,  
`k_iwdt_rpes_25` = (0x02 << `k_iwdt_iwdtcr_rpes_shift`) , `k_iwdt_rpes_0` = (0x03 << `k_iwdt_iwdtcr_rpes_shift`) ,  
`k_iwdt_rpss_25` = (0x00 << `k_iwdt_iwdtcr_rpss_shift`) ,  
`k_iwdt_rpss_50` = (0x01 << `k_iwdt_iwdtcr_rpss_shift`) ,  
`k_iwdt_rpss_75` = (0x02 << `k_iwdt_iwdtcr_rpss_shift`) , `k_iwdt_rpss_100` = (0x03 << `k_iwdt_iwdtcr_rpss_shift`) }

IWDT Control Register (IWDTCR) Bit Field Values.

- enum `iwdt_iwdtsr_bits_t` : `uint16_t` {  
`k_iwdt_sr_cntval_mask` = 0x3FFF , `k_iwdt_sr_undff_shift` = 14 , `k_iwdt_sr_refef_shift` = 15 ,  
`k_iwdt_sr_undff` = (1U << 14) ,  
`k_iwdt_sr_refef` = (1U << 15) }

IWDT Status Register (IWDTSR) Bit Field Values.

- enum `iwdt_iwdtrcr_bits_t` : `uint8_t` { `k_iwdt_rstirqs_pos` = 7 , `k_iwdt_rcr_rstirqs_mask` = (1U << 7) , `k_iwdt_rstirqs_reset` = (1U << 7) , `k_iwdt_rstirqs_nmi` = 0x00 }

IWDT Reset Control Register (IWDTRCR) Bit Field Values.

- enum `iwdt_iwdtcstpr_shifts_t` : `uint8_t` { `k_iwdt_cstpr_slcstp_pos` = 7 }

IWDTCSTPR bit field positions.

- enum `iwdt_iwdtcstpr_bits_t` : `uint8_t` { `k_iwdt_cstpr_slcstp_mask` = (1U << 7) , `k_iwdt_slcstp_stop` = (1U << 7) , `k_iwdt_slcstp_continue` = 0x00 }

IWDT Count Stop Control Register (IWDTCSTPR) Bit Field Values.

## Functions

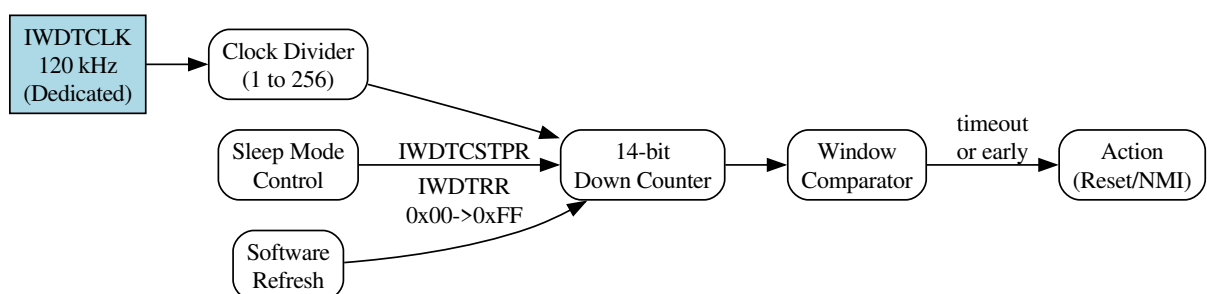
- static volatile `rx_iwdt_regs_t * iwdt` (void)  
Get pointer to IWDT registers.

### 8.29.1 Detailed Description

RX72N IWDT Independent Watchdog Timer Register Definitions.

Register definitions for the Independent Watchdog Timer (IWDT) module on the RX72N microcontroller. The IWDT provides critical system monitoring with a dedicated clock source independent of the main system clock.

#### System Architecture



## IWDT vs WDT Comparison

| Feature           | IWDT (This Module)        | WDT                   |
|-------------------|---------------------------|-----------------------|
| Clock Source      | Dedicated 120 kHz         | PCLKB (60 MHz)        |
| Clock Independent | Yes (survives clock fail) | No                    |
| Stop in Software  | No (once started)         | Yes                   |
| Sleep Mode Ctrl   | Yes (IWDTCSTPR)           | No                    |
| Best For          | Production safety         | Development/debugging |
| Timeout Range     | 8.5 ms to 17.5 s          | 4 us to 2.2 s         |

## Key Features

- 14-bit down counter with dedicated 120 kHz oscillator
- Clock-independent operation - resets system even if main clock fails
- Configurable timeout period: 1024 to 16384 cycles
- Configurable clock division: 1 to 256
- Optional window protection to detect too-early refresh
- Sleep mode count stop control (IWDTCSTPR)
- Generates reset or NMI on timeout (configurable)
- Cannot be stopped in software once started (safety feature)

## Timeout Calculation

$$T_{timeout} = \frac{TOPS \times CKS}{f_{IWDTCCLK}}$$

Example with IWDTCCLK = 120 kHz, TOPS = 16384, CKS = 128:

$$T_{timeout} = \frac{16384 \times 128}{120,000} = 17.5 \text{ seconds}$$

## Timeout Period Table (IWDTCCLK = 120 kHz)

| TOPS  | CKS/ $\leftrightarrow$<br>1 | CKS/16 | CKS/64 | CKS/128 | CKS/256 |
|-------|-----------------------------|--------|--------|---------|---------|
| 1024  | 8.5 ms                      | 137 ms | 546 ms | 1.09 s  | 2.18 s  |
| 4096  | 34 ms                       | 546 ms | 2.18 s | 4.37 s  | 8.74 s  |
| 8192  | 68 ms                       | 1.09 s | 4.37 s | 8.74 s  | 17.5 s  |
| 16384 | 137 ms                      | 2.18 s | 8.74 s | 17.5 s  | 35 s*   |

\*Note: 35s exceeds practical limits; use shorter timeouts for safety.

## Hardware Requirements

| Parameter        | Value      | Notes                        |
|------------------|------------|------------------------------|
| Clock Source     | IWDTCLK    | 120 kHz dedicated oscillator |
| Base Address     | 0x00088030 | 9-byte register block        |
| Interrupt Vector | IWDT_WUNI  | IWDT underflow (NMI mode)    |
| Register Count   | 5          | IWDTRR, IWDTCR, IWDTSR, etc. |

## Production Safety Recommendation

The IWDT is recommended for production systems because:

1. Cannot be disabled by software (protects against runaway code)
2. Survives main clock failure
3. Configurable sleep mode behavior

## NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops in register definitions)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessor functions are single-statement
- Rule 5: [OK] N/A (hardware layer)
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A (no return values to check)
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

## SOLID Principles

- S: Single responsibility - only IWDT register definitions
- O: Open for extension via additional bit field enums
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessor
- D: N/A (hardware register layer)

## Module Dependencies

- <stdint.h> - Fixed-width integer types

## Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 35: Independent Watchdog Timer (IWDT), pages 1559-1572

See also

[rx72n\\_wdt\\_regs.h](#) Software-controllable Watchdog Timer

[rx\\_iwdt.h](#) Higher-level IWDT API

[rx72n\\_regs.h](#) Main register include file

Author

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Date

2026-01-28

Copyright

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Since

Version 1.0.0

Definition in file [rx72n\\_iwdt\\_regs.h](#).

## 8.29.2 Data Structure Documentation

### 8.29.2.1 struct rx`iwdt`regs`t

IWDT Register Map Structure.

Independent Watchdog Timer (IWDT) registers for safety-critical system monitoring. Uses dedicated 120 kHz oscillator that operates independently of the main system clock.

Memory Layout Table

| Offset | Size | Field     | Type          | Description            |
|--------|------|-----------|---------------|------------------------|
| 0x00   | 1    | iwdtrr    | uint8↔<br>_t  | Refresh Register       |
| 0x01   | 1    | reserved0 | -             | Reserved               |
| 0x02   | 2    | iwdtcr    | uint16↔<br>_t | Control Register       |
| 0x04   | 2    | iwdtsr    | uint16↔<br>_t | Status Register        |
| 0x06   | 1    | iwdtrcr   | uint8↔<br>_t  | Reset Control Register |

| Offset | Size | Field     | Type         | Description                     |
|--------|------|-----------|--------------|---------------------------------|
| 0x07   | 1    | reserved1 | -            | Reserved                        |
| 0x08   | 1    | iwdtcstpr | uint8↔<br>_t | Count Stop Control Register     |
| Total  | 9    |           |              | (packed, no additional padding) |

### IWDT Characteristics

- Cannot be stopped once started - This is a safety feature
- Clock-independent - Continues operating if main clock fails
- Sleep mode control - Can stop counting during sleep (IWDTCSSTPR)

### Initialization Example

```
// Configure IWDT for 1-second timeout (production configuration)
volatile rx_iwdt_regs_t* iwdt_regs = iwdt();

// 1. Configure control register BEFORE first refresh
// TOPS=16384 cycles, CKS=div128, RPSS=100%, RPES=0% (no window)
iwdt_regs->iwdtcr = k_iwdt_tops_16384 | k_iwdt_cks_div_128 |
 k_iwdt_rpss_100 | k_iwdt_rpes_0;

// 2. Configure to generate reset on timeout (production)
iwdt_regs->iwdtcr = k_iwdt_rstirqs_reset;

// 3. Configure sleep mode behavior
iwdt_regs->iwdtcstpr = k_iwdt_slcstp_continue; // Keep counting in sleep

// 4. Start IWDT by performing first refresh
// WARNING: IWDT cannot be stopped after this!
iwdt_regs->iwdtrr = k_iwdt_refresh_start; // Write 0x00
iwdt_regs->iwdtrr = k_iwdt_refresh_end; // Write 0xFF - IWDT NOW RUNNING
```

### Refresh Example (in main loop)

```
void app_main_loop(void) {
 while (1) {
 // Perform application tasks
 process_commands();
 update_motors();
 check_safety();

 // Refresh IWDT (must complete both writes atomically)
 iwdt()->iwdtrr = k_iwdt_refresh_start; // 0x00
 iwdt()->iwdtrr = k_iwdt_refresh_end; // 0xFF

 // Wait for next cycle
 tx_thread_sleep(100); // 100ms loop
 }
}
```

### Note

Unlike WDT, IWDT has IWDTCSSTPR for sleep mode control

### Invariant

sizeof(rx\_iwdt\_regs\_t) == 9 bytes

All offsets match RX72N Hardware Manual Ch35



## Warning

IWDT cannot be stopped once started - ensure configuration is correct  
Configuration must be complete before first refresh

## Attention

Recommended for production systems due to clock independence

## See also

[iwdt\(\)](#) Accessor function

[iwdt\\_refresh\\_sequence\\_t](#) Refresh values

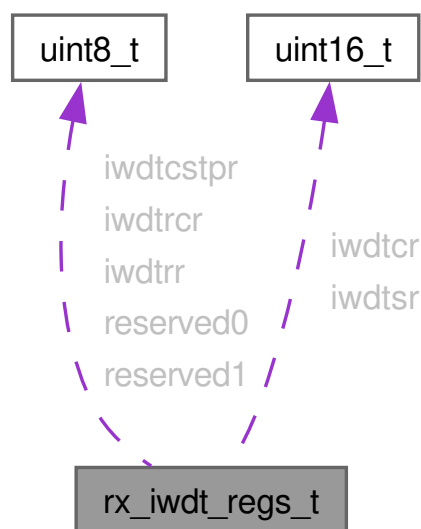
[iwdt\\_iwdtcr\\_bits\\_t](#) Control register bits

## Since

Version 1.0.0

Definition at line 261 of file [rx72n\\_iwdt\\_regs.h](#).

Collaboration diagram for rx\_iwdt\_regs\_t:



## Data Fields

|                   |           |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint16_t | iwdtcr    | <p>IWDT Control Register (IWDTCR) @ offset 0x02. Configures timeout period, clock division, and window parameters. Must be configured before first refresh (before starting IWDT).</p> <p>See also</p> <p><a href="#">iwdt_iwdtcr_bits_t</a><br/>Bit definitions</p>                                                                                                            |
| volatile uint8_t  | iwdtcstpr | <p>IWDT Count Stop Control Register (IWDTCSTPR) @ offset 0x08. Controls whether IWDT counter continues during sleep modes.</p> <p>Bit 7 (SLCSTP):</p> <ul style="list-style-type: none"> <li>• 0: Continue counting during sleep (recommended)</li> <li>• 1: Stop counting during sleep</li> </ul> <p>See also</p> <p><a href="#">iwdt_iwdtcstpr_bits_t</a> Bit definitions</p> |
| volatile uint8_t  | iwdtrcr   | <p>IWDT Reset Control Register (IWDTRCR) @ offset 0x06. Selects action on timeout:↵ reset or NMI.</p> <p>See also</p> <p><a href="#">iwdt_iwdtrcr_bits_t</a><br/>Bit definitions</p>                                                                                                                                                                                            |

|                   |                                                                     |                                                                                                                                                                                                                                                                                                                                |
|-------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volatile uint8_t  | iwdtrr                                                              | <p>IWDT Refresh Register (IWDTRR) @ offset 0x00. Write 0x00 then 0xFF to refresh the watchdog counter. First refresh also starts the IWDT (cannot be stopped after).</p> <p>See also</p> <p><a href="#">iwdt_refresh_sequence_t</a> Refresh values</p> <p>Warning</p> <p>First refresh starts IWDT permanently until reset</p> |
| volatile uint16_t | iwdtsr                                                              | <p>IWDT Status Register (IWDTSR) @ offset 0x04. Contains current counter value and error flags.</p> <p>See also</p> <p><a href="#">iwdt_iwdtsr_bits_t</a> Bit definitions</p>                                                                                                                                                  |
| uint8_t           | reserved0↵<br><a href="#">[k_iwdt_reserved_after_iwdtrr_bytes]</a>  | Reserved @ 0x01                                                                                                                                                                                                                                                                                                                |
| uint8_t           | reserved1↵<br><a href="#">[k_iwdt_reserved_after_iwdtrcr_bytes]</a> | Reserved @ 0x07                                                                                                                                                                                                                                                                                                                |

### 8.29.3 Enumeration Type Documentation

#### 8.29.3.1 iwdt`iwdtcr`bits` t

```
enum iwdt_iwdtcr_bits_t : uint16_t
```

IWDT Control Register (IWDTCR) Bit Field Values.

Configuration values for IWDT timeout period, clock division, and window protection. These values can be OR'd together to configure IWDTCR.

Timeout Calculation

$$T_{timeout} = \frac{TOPS \times CKS}{120,000}$$

Timeout Examples (IWDTCCLK = 120 kHz)

| Configuration            | Timeout |
|--------------------------|---------|
| TOPS_1024 + CKS_DIV_1    | 8.5 ms  |
| TOPS_16384 + CKS_DIV_128 | 17.5 s  |
| TOPS_16384 + CKS_DIV_64  | 8.7 s   |
| TOPS_4096 + CKS_DIV_32   | ~1.1 s  |

Production Configuration Example

```
// 1-second timeout, no window protection
iwdt()->iwdtcr = k_iwdt_tops_16384 | k_iwdt_cks_div_128 |
 k_iwdt_rpss_100 | k_iwdt_rpes_0;
```

See also

[rx\\_iwdt\\_regs\\_t::iwdtcr](#) Control register

Since

Version 1.0.0

Enumerator

|                    |                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| k_iwdt_tops_1024   | 1024 cycles timeout (~8.5ms at 120kHz, no division) Counter loads 0x03FF (1023). Shortest timeout period.                              |
| k_iwdt_tops_4096   | 4096 cycles timeout (~34ms at 120kHz, no division) Counter loads 0x0FFF (4095).                                                        |
| k_iwdt_tops_8192   | 8192 cycles timeout (~68ms at 120kHz, no division) Counter loads 0x1FFF (8191).                                                        |
| k_iwdt_tops_16384  | 16384 cycles timeout (~137ms at 120kHz, no division) Counter loads 0x3FFF (16383). Longest timeout period.                             |
| k_iwdt_cks_div_1   | IWDTCCLK divided by 1 (fastest, shortest timeout). CKS[3:0] = 0000. 120 kHz counter clock.                                             |
| k_iwdt_cks_div_16  | IWDTCCLK divided by 16. CKS[3:0] = 0010. 7.5 kHz counter clock.                                                                        |
| k_iwdt_cks_div_32  | IWDTCCLK divided by 32. CKS[3:0] = 0011. 3.75 kHz counter clock.                                                                       |
| k_iwdt_cks_div_64  | IWDTCCLK divided by 64. CKS[3:0] = 0100. 1.875 kHz counter clock.                                                                      |
| k_iwdt_cks_div_128 | IWDTCCLK divided by 128 (recommended for ~1s timeout). CKS[3:0] = 1111. 937.5 Hz counter clock. With TOPS_16384: ~17.5 second timeout. |
| k_iwdt_cks_div_256 | IWDTCCLK divided by 256 (slowest, longest timeout). CKS[3:0] = 0101. 468.75 Hz counter clock.                                          |
| k_iwdt_rpes_75     | Window closes at 75% counter remaining. Refresh prohibited when counter < 75% of initial value.                                        |
| k_iwdt_rpes_50     | Window closes at 50% counter remaining.                                                                                                |
| k_iwdt_rpes_25     | Window closes at 25% counter remaining.                                                                                                |
| k_iwdt_rpes_0      | No window end (refresh allowed until timeout). Recommended for most applications.                                                      |
| k_iwdt_rpss_25     | Window opens at 25% counter remaining. Strictest early-refresh detection.                                                              |

|                 |                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| k_iwdt_rpss_50  | Window opens at 50% counter remaining.                                                                         |
| k_iwdt_rpss_75  | Window opens at 75% counter remaining.                                                                         |
| k_iwdt_rpss_100 | No window start (refresh allowed immediately). Recommended for most applications (no early-refresh detection). |

Definition at line 439 of file [rx72n\\_iwdt\\_regs.h](#).

```

00439 : uint16_t {
00440 /*
=====
00441 * Timeout Period Select (TOPS) - bits 1:0
00442 * Counter cycles before timeout (14-bit counter max values)
00443 *
=====
00444 */
00445 /**
00446 * @brief 1024 cycles timeout (~8.5ms at 120kHz, no division)
00447 * @details Counter loads 0x03FF (1023). Shortest timeout period.
00448 */
00449 k_iwdt_tops_1024 = 0x0000,
00450
00451 /**
00452 * @brief 4096 cycles timeout (~34ms at 120kHz, no division)
00453 * @details Counter loads 0x0FFF (4095).
00454 */
00455 k_iwdt_tops_4096 = 0x0001,
00456
00457 /**
00458 * @brief 8192 cycles timeout (~68ms at 120kHz, no division)
00459 * @details Counter loads 0x1FFF (8191).
00460 */
00461 k_iwdt_tops_8192 = 0x0002,
00462
00463 /**
00464 * @brief 16384 cycles timeout (~137ms at 120kHz, no division)
00465 * @details Counter loads 0x3FFF (16383). Longest timeout period.
00466 */
00467 k_iwdt_tops_16384 = 0x0003,
00468
00469 /*
=====
00470 * Clock Division Ratio (CKS) - bits 7:4
00471 * IWDTCLK (120 kHz) divided by this value feeds the counter
00472 *
=====
00473 */
00474 /**
00475 * @brief IWDTCLK divided by 1 (fastest, shortest timeout)
00476 * @details CKS[3:0] = 0000. 120 kHz counter clock.
00477 */
00478 k_iwdt_cks_div_1 = (0x00 « k_iwdt_iwdtcr_cks_shift),
00479
00480 /**
00481 * @brief IWDTCLK divided by 16
00482 * @details CKS[3:0] = 0010. 7.5 kHz counter clock.
00483 */
00484 k_iwdt_cks_div_16 = (0x02 « k_iwdt_iwdtcr_cks_shift),
00485
00486 /**
00487 * @brief IWDTCLK divided by 32
00488 * @details CKS[3:0] = 0011. 3.75 kHz counter clock.
00489 */
00490 k_iwdt_cks_div_32 = (0x03 « k_iwdt_iwdtcr_cks_shift),
00491
00492 /**
00493 * @brief IWDTCLK divided by 64
00494 * @details CKS[3:0] = 0100. 1.875 kHz counter clock.
00495 */
00496 k_iwdt_cks_div_64 = (0x04 « k_iwdt_iwdtcr_cks_shift),
00497
00498 /**
00499 * @brief IWDTCLK divided by 128 (recommended for ~1s timeout)
00500 * @details CKS[3:0] = 1111. 937.5 Hz counter clock.
00501 * With TOPS_16384: ~17.5 second timeout.
00502 */
00503 k_iwdt_cks_div_128 = (0x0F « k_iwdt_iwdtcr_cks_shift),
00504
00505 /**
00506 * @brief IWDTCLK divided by 256 (slowest, longest timeout)
00507 * @details CKS[3:0] = 0101. 468.75 Hz counter clock.

```

```

00508 */
00509 k_iwdt_cks_div_256 = (0x05 « k_iwdt_iwdtcr_cks_shift),
00510
00511 /*
=====
00512 * Window End Position (RPES) - bits 9:8
00513 * Specifies when refresh window CLOSES (refresh becomes prohibited)
00514 *
=====
00515 */
00516 /**
00517 * @brief Window closes at 75% counter remaining
00518 * @details Refresh prohibited when counter < 75% of initial value.
00519 */
00520 k_iwdt_rpes_75 = (0x00 « k_iwdt_iwdtcr_rpes_shift),
00521
00522 /**
00523 * @brief Window closes at 50% counter remaining
00524 */
00525 k_iwdt_rpes_50 = (0x01 « k_iwdt_iwdtcr_rpes_shift),
00526
00527 /**
00528 * @brief Window closes at 25% counter remaining
00529 */
00530 k_iwdt_rpes_25 = (0x02 « k_iwdt_iwdtcr_rpes_shift),
00531
00532 /**
00533 * @brief No window end (refresh allowed until timeout)
00534 * @details Recommended for most applications.
00535 */
00536 k_iwdt_rpes_0 = (0x03 « k_iwdt_iwdtcr_rpes_shift),
00537
00538 /*
=====
00539 * Window Start Position (RPSS) - bits 13:12
00540 * Specifies when refresh window OPENS (refresh becomes allowed)
00541 *
=====
00542 */
00543 /**
00544 * @brief Window opens at 25% counter remaining
00545 * @details Strictest early-refresh detection.
00546 */
00547 k_iwdt_rpss_25 = (0x00 « k_iwdt_iwdtcr_rpss_shift),
00548
00549 /**
00550 * @brief Window opens at 50% counter remaining
00551 */
00552 k_iwdt_rpss_50 = (0x01 « k_iwdt_iwdtcr_rpss_shift),
00553
00554 /**
00555 * @brief Window opens at 75% counter remaining
00556 */
00557 k_iwdt_rpss_75 = (0x02 « k_iwdt_iwdtcr_rpss_shift),
00558
00559 /**
00560 * @brief No window start (refresh allowed immediately)
00561 * @details Recommended for most applications (no early-refresh detection).
00562 */
00563 k_iwdt_rpss_100 = (0x03 « k_iwdt_iwdtcr_rpss_shift),
00564 } iwdt_iwdtcr_bits_t;

```

### 8.29.3.2 iwdt\_iwdtcr\_shifts\_t

```
enum iwdt_iwdtcr_shifts_t : uint8_t
```

IWDT Control Register (IWDTCR) Bit Field Positions.

Bit positions for IWDTCR register fields. Used to construct field values.

## Register Layout (16-bit)

| Bits    | Field | Description              |
|---------|-------|--------------------------|
| 1:<br>0 | TOPS  | Timeout period selection |
| 3:<br>2 | -     | Reserved                 |
| 7:<br>4 | CKS   | Clock division ratio     |
| 9:<br>8 | RPES  | Window end position      |
| 11:10   | -     | Reserved                 |
| 13:12   | RPSS  | Window start position    |
| 15:14   | -     | Reserved                 |

Since

Version 1.0.0

## Enumerator

|                          |                                  |
|--------------------------|----------------------------------|
| k_iwdt_iwdtcr_cks_shift  | CKS field position (bits 7:4)    |
| k_iwdt_iwdtcr_rpes_shift | RPES field position (bits 9:8)   |
| k_iwdt_iwdtcr_rpss_shift | RPSS field position (bits 13:12) |

Definition at line 402 of file rx72n\_iwdt\_regs.h.

```

00402 : uint8_t {
00403 k_iwdt_iwdtcr_cks_shift = 4, /**< CKS field position (bits 7:4) */
00404 k_iwdt_iwdtcr_rpes_shift = 8, /**< RPES field position (bits 9:8) */
00405 k_iwdt_iwdtcr_rpss_shift = 12, /**< RPSS field position (bits 13:12) */
00406 } iwdt_iwdtcr_shifts_t;

```

## 8.29.3.3 iwdt\_iwdtcr\_bits\_t

```
enum iwdt_iwdtcr_bits_t : uint8_t
```

IWDT Count Stop Control Register (IWDTCSR) Bit Field Values.

Controls whether the IWDT counter continues during sleep modes. This register is unique to IWDT (WDT does not have this capability).

## Register Layout (8-bit)

| Bit     | Field  | R/W | Description                          |
|---------|--------|-----|--------------------------------------|
| 6:<br>0 | -      | -   | Reserved                             |
| 7       | SLCSTP | R/W | 0=Continue counting, 1=Stop in sleep |

## Sleep Mode Behavior

- Continue (k\_iwdt\_slcstp\_continue): IWDT keeps running in sleep. Ensures system wakes up if sleep takes too long. Recommended.
- Stop (k\_iwdt\_slcstp\_stop): IWDT pauses during sleep. Use only if sleep duration is unpredictable and legitimate.

## Configuration Example

```
// Recommended: Keep counting during sleep
iwdt()->iwdtctpr = k_iwdt_slcstp_continue;

// Alternative: Stop counting during sleep
iwdt()->iwdtctpr = k_iwdt_slcstp_stop;
```

See also

[rx\\_iwdt\\_regs\\_t::iwdtctpr](#) Count stop control register

Since

Version 1.0.0

## Enumerator

|                          |                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k_iwdt_cstpr_slcstp_mask | SLCSTP bit mask.                                                                                                                                                          |
| k_iwdt_slcstp_stop       | <p>Stop counting during sleep (SLCSTP=1). IWDT counter pauses when MCU enters sleep mode.</p> <p>Warning</p> <p>Use with caution - may mask stuck-in-sleep conditions</p> |
| k_iwdt_slcstp_continue   | Continue counting during sleep (SLCSTP=0). IWDT counter continues in sleep mode. Recommended. Ensures system recovery if sleep is not properly exited.                    |

Definition at line 708 of file [rx72n\\_iwdt\\_regs.h](#).

```
00708 : uint8_t {
00709 /**
00710 * @brief SLCSTP bit mask
00711 */
00712 k_iwdt_cstpr_slcstp_mask = (1U « 7),
00713
00714 /**
00715 * @brief Stop counting during sleep (SLCSTP=1)
00716 * @details IWDT counter pauses when MCU enters sleep mode.
00717 * @warning Use with caution - may mask stuck-in-sleep conditions
00718 */
00719 k_iwdt_slcstp_stop = (1U « 7),
00720
00721 /**
00722 * @brief Continue counting during sleep (SLCSTP=0)
00723 * @details IWDT counter continues in sleep mode. **Recommended.**
00724 * Ensures system recovery if sleep is not properly exited.
00725 */
00726 k_iwdt_slcstp_continue = 0x00,
00727 } iwdt_iwdtctpr_bits_t;
```



## 8.29.3.4 iwdt\_iwdtcstpr\_shifts\_t

```
enum iwdt_iwdtcstpr_shifts_t : uint8_t
```

IWDTCSTPR bit field positions.

Since

Version 1.0.0

Enumerator

|                         |                                    |
|-------------------------|------------------------------------|
| k_iwdt_cstpr_slcstp_pos | Sleep Mode Count Stop bit position |
|-------------------------|------------------------------------|

Definition at line 672 of file [rx72n\\_iwdt\\_regs.h](#).

```
00672 : uint8_t {
00673 k_iwdt_cstpr_slcstp_pos = 7, /**< Sleep Mode Count Stop bit position */
00674 } iwdt_iwdtcstpr_shifts_t;
```

## 8.29.3.5 iwdt\_iwdtrcr\_bits\_t

```
enum iwdt_iwdtrcr_bits_t : uint8_t
```

IWDT Reset Control Register (IWDTRCR) Bit Field Values.

Controls the action taken on IWDT timeout.

Register Layout (8-bit)

| Bit      | Field   | R/W | Description               |
|----------|---------|-----|---------------------------|
| 6:↔<br>0 | -       | -   | Reserved                  |
| 7        | RSTIRQS | R/W | 0=NMI, 1=Reset on timeout |

Configuration Example

```
/** Production: Generate reset on timeout (recommended)
iwdt()->iwdtrcr = k_iwdt_rstirqs_reset;
```

See also

[rx\\_iwdt\\_regs\\_t::iwdtrcr](#) Reset control register

Since

Version 1.0.0

Enumerator

|                         |                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------|
| k_iwdt_rstirqs_pos      | RSTIRQS bit position.                                                               |
| k_iwdt_rcr_rstirqs_mask | RSTIRQS bit mask.                                                                   |
| k_iwdt_rstirqs_reset    | Generate internal reset on timeout (RSTIRQS=1). Recommended for production systems. |
| k_iwdt_rstirqs_nmi      | Generate NMI on timeout (RSTIRQS=0). Useful for debugging or logging before reset.  |

Definition at line 643 of file [rx72n\\_iwdt\\_regs.h](#).

```

00643 : uint8_t {
00644 /**
00645 * @brief RSTIRQS bit position
00646 */
00647 k_iwdt_rstirqs_pos = 7,
00648
00649 /**
00650 * @brief RSTIRQS bit mask
00651 */
00652 k_iwdt_rcr_rstirqs_mask = (1U « 7),
00653
00654 /**
00655 * @brief Generate internal reset on timeout (RSTIRQS=1)
00656 * @details Recommended for production systems.
00657 */
00658 k_iwdt_rstirqs_reset = (1U « 7),
00659
00660 /**
00661 * @brief Generate NMI on timeout (RSTIRQS=0)
00662 * @details Useful for debugging or logging before reset.
00663 */
00664 k_iwdt_rstirqs_nmi = 0x00,
00665 } iwdt_iwdtrcr_bits_t;

```

### 8.29.3.6 iwdt\_iwdtsr\_bits\_t

```
enum iwdt_iwdtsr_bits_t : uint16_t
```

IWDT Status Register (IWDTSR) Bit Field Values.

Contains the current down-counter value and error flags.

Register Layout (16-bit)

| Bits      | Field  | R/W | Description                           |
|-----------|--------|-----|---------------------------------------|
| 13:↔<br>0 | CNTVAL | R   | Current counter value                 |
| 14        | UNDFE  | R/W | Underflow flag (write 0 to clear)     |
| 15        | REFEF  | R/W | Refresh error flag (write 0 to clear) |

Error Checking Example

```

uint16_t status = iwdt()->iwdtsr;
if (status & k_iwdt_sr_undff) {
 rx_log_warn("IWDT", "Reset was caused by IWDT timeout");
}

```

See also

[rx\\_iwdt\\_regs\\_t::iwdt\\_sr](#) Status register

Since

Version 1.0.0

Enumerator

|                                    |                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------|
| <code>k_iwdt_sr_cntval_mask</code> | Counter value mask (bits 13:0). Extracts 14-bit down-counter value.                          |
| <code>k_iwdt_sr_undff_shift</code> | Underflow flag bit position.                                                                 |
| <code>k_iwdt_sr_refef_shift</code> | Refresh error flag bit position.                                                             |
| <code>k_iwdt_sr_undff</code>       | Underflow flag - Bit 14. Set when counter reaches 0 (timeout). Write 0 to clear.             |
| <code>k_iwdt_sr_refef</code>       | Refresh error flag - Bit 15. Set when refresh occurs outside valid window. Write 0 to clear. |

Definition at line 591 of file [rx72n\\_iwdt\\_regs.h](#).

```

00591 : uint16_t {
00592 /**
00593 * @brief Counter value mask (bits 13:0)
00594 * @details Extracts 14-bit down-counter value.
00595 */
00596 k_iwdt_sr_cntval_mask = 0x3FFF,
00597
00598 /**
00599 * @brief Underflow flag bit position
00600 */
00601 k_iwdt_sr_undff_shift = 14,
00602
00603 /**
00604 * @brief Refresh error flag bit position
00605 */
00606 k_iwdt_sr_refef_shift = 15,
00607
00608 /**
00609 * @brief Underflow flag - Bit 14
00610 * @details Set when counter reaches 0 (timeout). Write 0 to clear.
00611 */
00612 k_iwdt_sr_undff = (1U « 14),
00613
00614 /**
00615 * @brief Refresh error flag - Bit 15
00616 * @details Set when refresh occurs outside valid window. Write 0 to clear.
00617 */
00618 k_iwdt_sr_refef = (1U « 15),
00619 } iwdt_iwdt_sr_bits_t;

```

### 8.29.3.7 iwdt`refresh`sequence`

enum [iwdt\\_refresh\\_sequence\\_t](#) : uint8\_t

IWDT Refresh Register (IWDTRR) Write Sequence Values.

The IWDT counter is refreshed by writing a specific two-byte sequence:

1. Write 0x00 (`k_iwdt_refresh_start`)
2. Write 0xFF (`k_iwdt_refresh_end`)

The first refresh also starts the IWDT, which cannot be stopped until reset.

## Refresh Example

```
// Refresh IWDT (atomic - disable interrupts if needed)
iwdt()->iwdtrr = k_iwdt_refresh_start; // Step 1: Write 0x00
iwdt()->iwdtrr = k_iwdt_refresh_end; // Step 2: Write 0xFF
```

## Warning

First refresh starts IWDT permanently

Both writes must complete in sequence - partial refresh is invalid

See also

[rx\\_iwdt\\_regs\\_t::iwdtrr](#) Refresh register

Since

Version 1.0.0

## Enumerator

|                                   |                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>k_iwdt_refresh_start</code> | First write value for refresh sequence (0x00). Must be written first, followed immediately by <code>k_iwdt_refresh_end</code> .                                       |
| <code>k_iwdt_refresh_end</code>   | Second write value for refresh sequence (0xFF). Must be written immediately after <code>k_iwdt_refresh_start</code> . Counter reloads to initial value on completion. |

Definition at line 367 of file [rx72n\\_iwdt\\_regs.h](#).

```
00367 : uint8_t {
00368 /**
00369 * @brief First write value for refresh sequence (0x00)
00370 * @details Must be written first, followed immediately by k_iwdt_refresh_end.
00371 */
00372 k_iwdt_refresh_start = 0x00,
00373
00374 /**
00375 * @brief Second write value for refresh sequence (0xFF)
00376 * @details Must be written immediately after k_iwdt_refresh_start.
00377 * Counter reloads to initial value on completion.
00378 */
00379 k_iwdt_refresh_end = 0xFF,
00380 } iwdt_refresh_sequence_t;
```

### 8.29.3.8 iwdt`reserved`sizes`t

```
enum iwdt_reserved_sizes_t : uint8_t
```

IWDT register reserved field sizes.

Defines reserved byte counts for structure padding to match hardware layout.

Since

Version 1.0.0

Enumerator

|                                     |                                    |
|-------------------------------------|------------------------------------|
| k_iwdt_reserved_after_iwdtrr_bytes  | Reserved byte after IWDTRR @ 0x01  |
| k_iwdt_reserved_after_iwdtrcr_bytes | Reserved byte after IWDTRCR @ 0x07 |

Definition at line 175 of file rx72n\_iwdt\_regs.h.

```
00175 : uint8_t {
00176 k_iwdt_reserved_after_iwdtrr_bytes = 1, /**< Reserved byte after IWDTRR @ 0x01 */
00177 k_iwdt_reserved_after_iwdtrcr_bytes = 1, /**< Reserved byte after IWDTRCR @ 0x07 */
00178 } iwdt_reserved_sizes_t;
```

### 8.29.3.9 rx\_iwdt\_addresses\_t

enum rx\_iwdt\_addresses\_t : uintptr\_t

IWDT base address constant.

Base address for the Independent Watchdog Timer register block at 0x00088030. The IWDT has a 9-byte register space with 5 registers and 2 reserved bytes.

Memory Map

| Address    | Register  | Size | Description                 |
|------------|-----------|------|-----------------------------|
| 0x00088030 | IWDTRR    | 1    | Refresh Register            |
| 0x00088031 | Reserved  | 1    | -                           |
| 0x00088032 | IWDTCR    | 2    | Control Register            |
| 0x00088034 | IWDTSR    | 2    | Status Register             |
| 0x00088036 | IWDTRCR   | 1    | Reset Control Register      |
| 0x00088037 | Reserved  | 1    | -                           |
| 0x00088038 | IWDTCSTPR | 1    | Count Stop Control Register |

See also

[iwdt\(\)](#) Accessor function

Since

Version 1.0.0

Enumerator

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| k_iwdt_base_addr | IWDT register base address (0x00088030). Verified against RX72N Hardware Manual Ch35 |
|------------------|--------------------------------------------------------------------------------------|

Definition at line 161 of file rx72n\_iwdt\_regs.h.

```
00161 : uintptr_t {
00162 /**
00163 * @brief IWDT register base address (0x00088030)
00164 * @details Verified against RX72N Hardware Manual Ch35
00165 */
00166 k_iwdt_base_addr = 0x00088030,
00167 } rx_iwdt_addresses_t;
```

## 8.29.4 Function Documentation

### 8.29.4.1 iwdt()

```
volatile rx_iwdt_regs_t * iwdt (
 void) [inline], [static]
```

Get pointer to IWDT registers.

Returns a volatile pointer to the IWDT register structure at address 0x00088030. The IWDT is recommended for production systems due to its clock-independent operation.

Returns

Volatile pointer to IWDT register structure

Return values

|          |                                                 |
|----------|-------------------------------------------------|
| Non-NULL | Always returns valid pointer (hardware address) |
|----------|-------------------------------------------------|

Precondition

None (hardware register always accessible)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example:

```
// Refresh IWDT
iwdt()->iwdtrr = k_iwdt_refresh_start;
iwdt()->iwdtrr = k_iwdt_refresh_end;

// Check counter value
uint16_t counter = iwdt()->iwdtsr & k_iwdt_sr_cntval_mask;
```

See also

[k\\_iwdt\\_base\\_addr](#) Base address constant

[rx\\_iwdt\\_init\(\)](#) Higher-level initialization

Since

Version 1.0.0

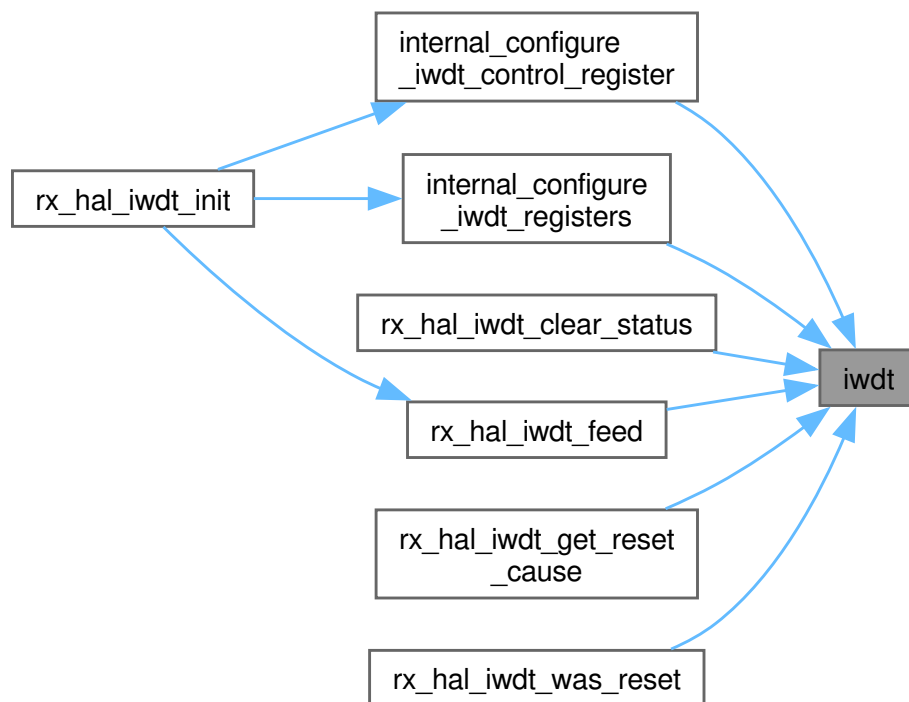
Definition at line 339 of file [rx72n\\_iwdt\\_regs.h](#).

```
00340 {
00341 return (volatile rx_iwdt_regs_t*)k_iwdt_base_addr;
00342 }
```

References [k\\_iwdt\\_base\\_addr](#).

Referenced by [internal\\_configure\\_iwdt\\_control\\_register\(\)](#), [internal\\_configure\\_iwdt\\_registers\(\)](#), [rx\\_hal\\_iwdt\\_clear\\_status\(\)](#), [rx\\_hal\\_iwdt\\_feed\(\)](#), [rx\\_hal\\_iwdt\\_get\\_reset\\_cause\(\)](#), and [rx\\_hal\\_iwdt\\_was\\_reset\(\)](#).

Here is the caller graph for this function:



## 8.30 rx72n\_iwdt\_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002 * @file rx72n_iwdt_regs.h
00003 * @brief RX72N IWDT Independent Watchdog Timer Register Definitions
00004 *
00005 * @details
00006 * Register definitions for the Independent Watchdog Timer (IWDT) module on
00007 * the RX72N microcontroller. The IWDT provides critical system monitoring
00008 * with a dedicated clock source independent of the main system clock.
00009 *
00010 * @par System Architecture
00011 * @dot
00012 * digraph iwdt_architecture {
00013 * rankdir=LR;
00014 * node [shape=box, style=rounded];
00015 *
00016 * osc [label="IWDTCLK\n120 kHz\n(Dedicated)" style="filled" fillcolor="lightblue"];
00017 * divider [label="Clock Divider\n(1 to 256)"];
00018 * counter [label="14-bit\nDown Counter"];
00019 * window [label="Window\nComparator"];
```

```

00020 * action [label="Action\n(Reset/NMI)"];
00021 * sleep [label="Sleep Mode\nControl"];
00022 *
00023 * sw [label="Software\nRefresh"];
00024 *
00025 * osc -> divider;
00026 * divider -> counter;
00027 * counter -> window;
00028 * window -> action [label="timeout\nor early"];
00029 * sw -> counter [label="IWDTRR\n0x00->0xFF"];
00030 * sleep -> counter [label="IWDTCSTPR"];
00031 * }
00032 * @enddot
00033 *
00034 * @par IWDT vs WDT Comparison
00035 * | Feature | IWDT (This Module) | WDT |
00036 * |-----|-----|-----|
00037 * | Clock Source | Dedicated 120 kHz | PCLKB (60 MHz) |
00038 * | Clock Independent | Yes (survives clock fail) | No |
00039 * | Stop in Software | No (once started) | Yes |
00040 * | Sleep Mode Ctrl | Yes (IWDTCSTPR) | No |
00041 * | Best For | **Production safety** | Development/debugging |
00042 * | Timeout Range | 8.5 ms to 17.5 s | 4 us to 2.2 s |
00043 *
00044 * @par Key Features
00045 * - 14-bit down counter with dedicated 120 kHz oscillator
00046 * - **Clock-independent operation** - resets system even if main clock fails
00047 * - Configurable timeout period: 1024 to 16384 cycles
00048 * - Configurable clock division: 1 to 256
00049 * - Optional window protection to detect too-early refresh
00050 * - Sleep mode count stop control (IWDTCSTPR)
00051 * - Generates reset or NMI on timeout (configurable)
00052 * - **Cannot be stopped in software once started** (safety feature)
00053 *
00054 * @par Timeout Calculation
00055 * @f[
00056 * T_{timeout} = \frac{TOPS \times CKS}{f_{IWDTCCLK}}
00057 * @f]
00058 *
00059 * Example with IWDTCCLK = 120 kHz, TOPS = 16384, CKS = 128:
00060 * @f[
00061 * T_{timeout} = \frac{16384 \times 128}{120,000} = 17.5 \text{ seconds}
00062 * @f]
00063 *
00064 * @par Timeout Period Table (IWDTCCLK = 120 kHz)
00065 * | TOPS | CKS/1 | CKS/16 | CKS/64 | CKS/128 | CKS/256 |
00066 * |-----|-----|-----|-----|-----|-----|
00067 * | 1024 | 8.5 ms | 137 ms | 546 ms | 1.09 s | 2.18 s |
00068 * | 4096 | 34 ms | 546 ms | 2.18 s | 4.37 s | 8.74 s |
00069 * | 8192 | 68 ms | 1.09 s | 4.37 s | 8.74 s | 17.5 s |
00070 * | 16384 | 137 ms | 2.18 s | 8.74 s | 17.5 s | 35 s* |
00071 *
00072 * *Note: 35s exceeds practical limits; use shorter timeouts for safety.
00073 *
00074 * @par Hardware Requirements
00075 * | Parameter | Value | Notes |
00076 * |-----|-----|-----|
00077 * | Clock Source | IWDTCCLK | 120 kHz dedicated oscillator |
00078 * | Base Address | 0x00088030 | 9-byte register block |
00079 * | Interrupt Vector | IWDTC_WUNI | IWDTC underflow (NMI mode) |
00080 * | Register Count | 5 | IWDTCRR, IWDTCR, IWDTCR, etc. |
00081 *
00082 * @par Production Safety Recommendation
00083 * The IWDTC is recommended for production systems because:
00084 * 1. Cannot be disabled by software (protects against runaway code)
00085 * 2. Survives main clock failure
00086 * 3. Configurable sleep mode behavior
00087 *
00088 * @par NASA Power of 10 Compliance
00089 * - Rule 1: [OK] No goto, setjmp, or recursion
00090 * - Rule 2: [OK] N/A (no loops in register definitions)
00091 * - Rule 3: [OK] No dynamic memory allocation
00092 * - Rule 4: [OK] All accessor functions are single-statement
00093 * - Rule 5: [OK] N/A (hardware layer)
00094 * - Rule 6: [OK] Minimal scope
00095 * - Rule 7: [OK] N/A (no return values to check)
00096 * - Rule 8: [OK] All constants use C23 typed enums
00097 * - Rule 9: [OK] No function pointers
00098 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00099 *
00100 * @par SOLID Principles
00101 * - **S**: Single responsibility - only IWDTC register definitions
00102 * - **O**: Open for extension via additional bit field enums
00103 * - **L**: N/A (no inheritance hierarchy)
00104 * - **I**: Minimal interface - only necessary accessor
00105 * - **D**: N/A (hardware register layer)
00106 *

```



```

00107 * @par Module Dependencies
00108 * - <stdint.h> - Fixed-width integer types
00109 *
00110 * @par Manual Reference
00111 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00112 * - Chapter 35: Independent Watchdog Timer (IWDT), pages 1559-1572
00113 *
00114 * @see rx72n_wdt_regs.h Software-controllable Watchdog Timer
00115 * @see rx_iwdt.h Higher-level IWDT API
00116 * @see rx72n_regs.h Main register include file
00117 *
00118 * @author Locked, Inc.
00119 * @date 2026-01-28
00120 * @copyright Copyright (c) 2026 Locked Inc.
00121 * SPDX-License-Identifier: MIT
00122 * @since Version 1.0.0
00123 */
00124
00125 #pragma once
00126
00127 #include <stddef.h>
00128 #include <stdint.h>
00129
00130 #ifdef __cplusplus
00131 extern "C" {
00132 #endif
00133
00134 /*
=====
00135 * Independent Watchdog Timer (IWDT)
00136 *
=====
00137 */
00138
00139 /**
00140 * @enum rx_iwdt_addresses_t
00141 * @brief IWDT base address constant
00142 *
00143 * @details
00144 * Base address for the Independent Watchdog Timer register block at 0x00088030.
00145 * The IWDT has a 9-byte register space with 5 registers and 2 reserved bytes.
00146 *
00147 * @par Memory Map
00148 * | Address | Register | Size | Description |
00149 * |-----|-----|-----|-----|
00150 * | 0x00088030 | IWDTRR | 1 | Refresh Register |
00151 * | 0x00088031 | Reserved | 1 | - |
00152 * | 0x00088032 | IWDTCR | 2 | Control Register |
00153 * | 0x00088034 | IWDTSR | 2 | Status Register |
00154 * | 0x00088036 | IWDTRCR | 1 | Reset Control Register |
00155 * | 0x00088037 | Reserved | 1 | - |
00156 * | 0x00088038 | IWDTCSTPR | 1 | Count Stop Control Register |
00157 *
00158 * @see iwdt() Accessor function
00159 * @since Version 1.0.0
00160 */
00161 typedef enum : uintptr_t {
00162 /**
00163 * @brief IWDT register base address (0x00088030)
00164 * @details Verified against RX72N Hardware Manual Ch35
00165 */
00166 k_iwdt_base_addr = 0x00088030,
00167 } rx_iwdt_addresses_t;
00168
00169 /**
00170 * @enum iwdt_reserved_sizes_t
00171 * @brief IWDT register reserved field sizes
00172 * @details Defines reserved byte counts for structure padding to match hardware layout.
00173 * @since Version 1.0.0
00174 */
00175 typedef enum : uint8_t {
00176 k_iwdt_reserved_after_iwdtrr_bytes = 1, /**< Reserved byte after IWDTRR @ 0x01 */
00177 k_iwdt_reserved_after_iwdtrcr_bytes = 1, /**< Reserved byte after IWDTRCR @ 0x07 */
00178 } iwdt_reserved_sizes_t;
00179
00180 /**
00181 * @struct rx_iwdt_regs_t
00182 * @brief IWDT Register Map Structure
00183 *
00184 * @details
00185 * Independent Watchdog Timer (IWDT) registers for safety-critical system
00186 * monitoring. Uses dedicated 120 kHz oscillator that operates independently
00187 * of the main system clock.
00188 *
00189 * @par Memory Layout Table
00190 * | Offset | Size | Field | Type | Description |
00191 * |-----|-----|-----|-----|-----|

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```

00192 * | 0x00 | 1 | iwdtrr | uint8_t | Refresh Register |
00193 * | 0x01 | 1 | reserved0 | - | Reserved |
00194 * | 0x02 | 2 | iwdtcr | uint16_t | Control Register |
00195 * | 0x04 | 2 | iwdtsr | uint16_t | Status Register |
00196 * | 0x06 | 1 | iwdtrcr | uint8_t | Reset Control Register |
00197 * | 0x07 | 1 | reserved1 | - | Reserved |
00198 * | 0x08 | 1 | iwdtcstpr | uint8_t | Count Stop Control Register |
00199 * | **Total** | **9** | | | (packed, no additional padding) |
00200 *
00201 * @par IWDT Characteristics
00202 * - **Cannot be stopped once started** - This is a safety feature
00203 * - **Clock-independent** - Continues operating if main clock fails
00204 * - **Sleep mode control** - Can stop counting during sleep (IWDTCSNTPR)
00205 *
00206 * @par Initialization Example
00207 * @code{.c}
00208 * // Configure IWDT for 1-second timeout (production configuration)
00209 * volatile rx_iwdt_regs_t* iwdt_regs = iwdt();
00210 *
00211 * // 1. Configure control register BEFORE first refresh
00212 * // TOPS=16384 cycles, CKS=div128, RPSS=100%, RPES=0% (no window)
00213 * iwdt_regs->iwdtcr = k_iwdt_tops_16384 | k_iwdt_cks_div_128 |
00214 * k_iwdt_rpss_100 | k_iwdt_rpes_0;
00215 *
00216 * // 2. Configure to generate reset on timeout (production)
00217 * iwdt_regs->iwdtrcr = k_iwdt_rstirqs_reset;
00218 *
00219 * // 3. Configure sleep mode behavior
00220 * iwdt_regs->iwdtcstpr = k_iwdt_slcstp_continue; // Keep counting in sleep
00221 *
00222 * // 4. Start IWDT by performing first refresh
00223 * // WARNING: IWDT cannot be stopped after this!
00224 * iwdt_regs->iwdtrr = k_iwdt_refresh_start; // Write 0x00
00225 * iwdt_regs->iwdtrr = k_iwdt_refresh_end; // Write 0xFF - IWDT NOW RUNNING
00226 * @endcode
00227 *
00228 * @par Refresh Example (in main loop)
00229 * @code{.c}
00230 * void app_main_loop(void) {
00231 * while (1) {
00232 * // Perform application tasks
00233 * process_commands();
00234 * update_motors();
00235 * check_safety();
00236 *
00237 * // Refresh IWDT (must complete both writes atomically)
00238 * iwdt()->iwdtrr = k_iwdt_refresh_start; // 0x00
00239 * iwdt()->iwdtrr = k_iwdt_refresh_end; // 0xFF
00240 *
00241 * // Wait for next cycle
00242 * tx_thread_sleep(100); // 100ms loop
00243 * }
00244 * }
00245 * @endcode
00246 *
00247 * @note Unlike WDT, IWDT has IWDTCSNTPR for sleep mode control
00248 *
00249 * @invariant sizeof(rx_iwdt_regs_t) == 9 bytes
00250 * @invariant All offsets match RX72N Hardware Manual Ch35
00251 *
00252 * @warning **IWDT cannot be stopped once started** - ensure configuration is correct
00253 * @warning Configuration must be complete before first refresh
00254 * @attention Recommended for production systems due to clock independence
00255 *
00256 * @see iwdt() Accessor function
00257 * @see iwdt_refresh_sequence_t Refresh values
00258 * @see iwdt_iwdtcr_bits_t Control register bits
00259 * @since Version 1.0.0
00260 */
00261 typedef struct __attribute__((packed)) {
00262 /**
00263 * @brief IWDT Refresh Register (IWDTRR) @ offset 0x00
00264 * @details Write 0x00 then 0xFF to refresh the watchdog counter.
00265 * First refresh also starts the IWDT (cannot be stopped after).
00266 * @see iwdt_refresh_sequence_t Refresh values
00267 * @warning First refresh starts IWDT permanently until reset
00268 */
00269 volatile uint8_t iwdtrr;
00270
00271 uint8_t reserved0[k_iwdt_reserved_after_iwdtrr_bytes]; /**< Reserved @ 0x01 */
00272 /**
00273 * @brief IWDT Control Register (IWDTCR) @ offset 0x02
00274 * @details Configures timeout period, clock division, and window parameters.
00275 * Must be configured before first refresh (before starting IWDT).
00276 * @see iwdt_iwdtcr_bits_t Bit definitions
00277 */
00278

```

```

00279 volatile uint16_t iwdtcr;
00280
00281 /**
00282 * @brief IWDT Status Register (IWDTSR) @ offset 0x04
00283 * @details Contains current counter value and error flags.
00284 * @see iwdt_iwdtsr_bits_t Bit definitions
00285 */
00286 volatile uint16_t iwdtsr;
00287
00288 /**
00289 * @brief IWDT Reset Control Register (IWDTRCR) @ offset 0x06
00290 * @details Selects action on timeout: reset or NMI.
00291 * @see iwdt_iwdtrcr_bits_t Bit definitions
00292 */
00293 volatile uint8_t iwdtrcr;
00294
00295 uint8_t reserved1[k_iwdt_reserved_after_iwdtrcr_bytes]; /**< Reserved @ 0x07 */
00296
00297 /**
00298 * @brief IWDT Count Stop Control Register (IWDTCSTPR) @ offset 0x08
00299 * @details Controls whether IWDT counter continues during sleep modes.
00300 * @par Bit 7 (SLCSTP):
00301 * - 0: Continue counting during sleep (recommended)
00302 * - 1: Stop counting during sleep
00303 * @see iwdt_iwdtcstpr_bits_t Bit definitions
00304 */
00305 volatile uint8_t iwdtcstpr;
00306 } rx_iwdt_regs_t;
00307
00308 /**
00309 * @brief Get pointer to IWDT registers
00310 *
00311 * @details
00312 * Returns a volatile pointer to the IWDT register structure at address
00313 * 0x00088030. The IWDT is recommended for production systems due to its
00314 * clock-independent operation.
00315 *
00316 * @return Volatile pointer to IWDT register structure
00317 * @retval Non-NULL Always returns valid pointer (hardware address)
00318 *
00319 * @pre None (hardware register always accessible)
00320 * @post Pointer is valid for the lifetime of program execution
00321 *
00322 * @note Thread Safety: Safe - returns constant hardware address
00323 * @note This function is always inlined for zero overhead
00324 *
00325 * @par Example:
00326 * @code{.c}
00327 * // Refresh IWDT
00328 * iwdt()->iwdtrr = k_iwdt_refresh_start;
00329 * iwdt()->iwdtrr = k_iwdt_refresh_end;
00330 *
00331 * // Check counter value
00332 * uint16_t counter = iwdt()->iwdtsr & k_iwdt_sr_cntval_mask;
00333 * @endcode
00334 *
00335 * @see k_iwdt_base_addr Base address constant
00336 * @see rx_iwdt_init() Higher-level initialization
00337 * @since Version 1.0.0
00338 */
00339 static inline volatile rx_iwdt_regs_t* iwdt(void)
00340 {
00341 return (volatile rx_iwdt_regs_t*)k_iwdt_base_addr;
00342 }
00343
00344 /**
00345 * @enum iwdt_refresh_sequence_t
00346 * @brief IWDT Refresh Register (IWDTRR) Write Sequence Values
00347 *
00348 * @details
00349 * The IWDT counter is refreshed by writing a specific two-byte sequence:
00350 * 1. Write 0x00 (k_iwdt_refresh_start)
00351 * 2. Write 0xFF (k_iwdt_refresh_end)
00352 *
00353 * The first refresh also **starts** the IWDT, which cannot be stopped until reset.
00354 *
00355 * @par Refresh Example
00356 * @code{.c}
00357 * // Refresh IWDT (atomic - disable interrupts if needed)
00358 * iwdt()->iwdtrr = k_iwdt_refresh_start; // Step 1: Write 0x00
00359 * iwdt()->iwdtrr = k_iwdt_refresh_end; // Step 2: Write 0xFF
00360 * @endcode
00361 *
00362 * @warning First refresh starts IWDT permanently
00363 * @warning Both writes must complete in sequence - partial refresh is invalid
00364 * @see rx_iwdt_regs_t::iwdtrr Refresh register
00365 * @since Version 1.0.0

```

```

00366 */
00367 typedef enum : uint8_t {
00368 /**
00369 * @brief First write value for refresh sequence (0x00)
00370 * @details Must be written first, followed immediately by k_iwdt_refresh_end.
00371 */
00372 k_iwdt_refresh_start = 0x00,
00373
00374 /**
00375 * @brief Second write value for refresh sequence (0xFF)
00376 * @details Must be written immediately after k_iwdt_refresh_start.
00377 * Counter reloads to initial value on completion.
00378 */
00379 k_iwdt_refresh_end = 0xFF,
00380 } iwdt_refresh_sequence_t;
00381
00382 /**
00383 * @enum iwdt_iwdtcr_shifts_t
00384 * @brief IWDT Control Register (IWDTCR) Bit Field Positions
00385 *
00386 * @details
00387 * Bit positions for IWDTCR register fields. Used to construct field values.
00388 *
00389 * @par Register Layout (16-bit)
00390 * | Bits | Field | Description |
00391 * |-----|-----|-----|
00392 * | 1:0 | TOPS | Timeout period selection |
00393 * | 3:2 | - | Reserved |
00394 * | 7:4 | CKS | Clock division ratio |
00395 * | 9:8 | RPES | Window end position |
00396 * | 11:10 | - | Reserved |
00397 * | 13:12 | RPSS | Window start position |
00398 * | 15:14 | - | Reserved |
00399 *
00400 * @since Version 1.0.0
00401 */
00402 typedef enum : uint8_t {
00403 k_iwdt_iwdtcr_cks_shift = 4, /**< CKS field position (bits 7:4) */
00404 k_iwdt_iwdtcr_rpes_shift = 8, /**< RPES field position (bits 9:8) */
00405 k_iwdt_iwdtcr_rpss_shift = 12, /**< RPSS field position (bits 13:12) */
00406 } iwdt_iwdtcr_shifts_t;
00407
00408 /**
00409 * @enum iwdt_iwdtcr_bits_t
00410 * @brief IWDT Control Register (IWDTCR) Bit Field Values
00411 *
00412 * @details
00413 * Configuration values for IWDT timeout period, clock division, and
00414 * window protection. These values can be OR'd together to configure IWDTCR.
00415 *
00416 * @par Timeout Calculation
00417 * @f[
00418 * T_{timeout} = \frac{TOPS \times CKS}{120,000}
00419 * @f]
00420 *
00421 * @par Timeout Examples (IWDTCCLK = 120 kHz)
00422 * | Configuration | Timeout |
00423 * |-----|-----|
00424 * | TOPS_1024 + CKS_DIV_1 | 8.5 ms |
00425 * | TOPS_16384 + CKS_DIV_128 | 17.5 s |
00426 * | TOPS_16384 + CKS_DIV_64 | 8.7 s |
00427 * | TOPS_4096 + CKS_DIV_32 | ~1.1 s |
00428 *
00429 * @par Production Configuration Example
00430 * @code{.c}
00431 * // 1-second timeout, no window protection
00432 * iwdt()->iwdtcr = k_iwdt_tops_16384 | k_iwdt_cks_div_128 |
00433 * k_iwdt_rpss_100 | k_iwdt_rpes_0;
00434 * @endcode
00435 *
00436 * @see rx_iwdt_regs_t::iwdtcr Control register
00437 * @since Version 1.0.0
00438 */
00439 typedef enum : uint16_t {
00440 /**
=====
00441 * Timeout Period Select (TOPS) - bits 1:0
00442 * Counter cycles before timeout (14-bit counter max values)
00443 *
=====
00444 */
00445 /**
00446 * @brief 1024 cycles timeout (~8.5ms at 120kHz, no division)
00447 * @details Counter loads 0x03FF (1023). Shortest timeout period.
00448 */
00449 k_iwdt_tops_1024 = 0x0000,

```

```

00450
00451 /**
00452 * @brief 4096 cycles timeout (~34ms at 120kHz, no division)
00453 * @details Counter loads 0x0FFF (4095).
00454 */
00455 k_iwtdt_tops_4096 = 0x0001,
00456
00457 /**
00458 * @brief 8192 cycles timeout (~68ms at 120kHz, no division)
00459 * @details Counter loads 0x1FFF (8191).
00460 */
00461 k_iwtdt_tops_8192 = 0x0002,
00462
00463 /**
00464 * @brief 16384 cycles timeout (~137ms at 120kHz, no division)
00465 * @details Counter loads 0x3FFF (16383). Longest timeout period.
00466 */
00467 k_iwtdt_tops_16384 = 0x0003,
00468
00469 /*
=====
00470 * Clock Division Ratio (CKS) - bits 7:4
00471 * IWDTCLK (120 kHz) divided by this value feeds the counter
00472 *
=====
*/
00473
00474 /**
00475 * @brief IWDTCLK divided by 1 (fastest, shortest timeout)
00476 * @details CKS[3:0] = 0000. 120 kHz counter clock.
00477 */
00478 k_iwtdt_cks_div_1 = (0x00 « k_iwtdt_iwdtcr_cks_shift),
00479
00480 /**
00481 * @brief IWDTCLK divided by 16
00482 * @details CKS[3:0] = 0010. 7.5 kHz counter clock.
00483 */
00484 k_iwtdt_cks_div_16 = (0x02 « k_iwtdt_iwdtcr_cks_shift),
00485
00486 /**
00487 * @brief IWDTCLK divided by 32
00488 * @details CKS[3:0] = 0011. 3.75 kHz counter clock.
00489 */
00490 k_iwtdt_cks_div_32 = (0x03 « k_iwtdt_iwdtcr_cks_shift),
00491
00492 /**
00493 * @brief IWDTCLK divided by 64
00494 * @details CKS[3:0] = 0100. 1.875 kHz counter clock.
00495 */
00496 k_iwtdt_cks_div_64 = (0x04 « k_iwtdt_iwdtcr_cks_shift),
00497
00498 /**
00499 * @brief IWDTCLK divided by 128 (recommended for ~1s timeout)
00500 * @details CKS[3:0] = 1111. 937.5 Hz counter clock.
00501 * With TOPS_16384: ~17.5 second timeout.
00502 */
00503 k_iwtdt_cks_div_128 = (0x0F « k_iwtdt_iwdtcr_cks_shift),
00504
00505 /**
00506 * @brief IWDTCLK divided by 256 (slowest, longest timeout)
00507 * @details CKS[3:0] = 0101. 468.75 Hz counter clock.
00508 */
00509 k_iwtdt_cks_div_256 = (0x05 « k_iwtdt_iwdtcr_cks_shift),
00510
00511 /*
=====
00512 * Window End Position (RPES) - bits 9:8
00513 * Specifies when refresh window CLOSES (refresh becomes prohibited)
00514 *
=====
*/
00515
00516 /**
00517 * @brief Window closes at 75% counter remaining
00518 * @details Refresh prohibited when counter < 75% of initial value.
00519 */
00520 k_iwtdt_rpes_75 = (0x00 « k_iwtdt_iwdtcr_rpes_shift),
00521
00522 /**
00523 * @brief Window closes at 50% counter remaining
00524 */
00525 k_iwtdt_rpes_50 = (0x01 « k_iwtdt_iwdtcr_rpes_shift),
00526
00527 /**
00528 * @brief Window closes at 25% counter remaining
00529 */
00530 k_iwtdt_rpes_25 = (0x02 « k_iwtdt_iwdtcr_rpes_shift),

```

```

00531
00532 /**
00533 * @brief No window end (refresh allowed until timeout)
00534 * @details Recommended for most applications.
00535 */
00536 k_iwdt_rpes_0 = (0x03 « k_iwdt_iwdtcr_rpes_shift),
00537
00538 /*
=====
00539 * Window Start Position (RPSS) - bits 13:12
00540 * Specifies when refresh window OPENS (refresh becomes allowed)
00541 *
=====
*/
00542
00543 /**
00544 * @brief Window opens at 25% counter remaining
00545 * @details Strictest early-refresh detection.
00546 */
00547 k_iwdt_rpss_25 = (0x00 « k_iwdt_iwdtcr_rpss_shift),
00548
00549 /**
00550 * @brief Window opens at 50% counter remaining
00551 */
00552 k_iwdt_rpss_50 = (0x01 « k_iwdt_iwdtcr_rpss_shift),
00553
00554 /**
00555 * @brief Window opens at 75% counter remaining
00556 */
00557 k_iwdt_rpss_75 = (0x02 « k_iwdt_iwdtcr_rpss_shift),
00558
00559 /**
00560 * @brief No window start (refresh allowed immediately)
00561 * @details Recommended for most applications (no early-refresh detection).
00562 */
00563 k_iwdt_rpss_100 = (0x03 « k_iwdt_iwdtcr_rpss_shift),
00564 } iwdt_iwdtcr_bits_t;
00565
00566 /**
00567 * @enum iwdt_iwdtsr_bits_t
00568 * @brief IWDTSR Status Register (IWDTSR) Bit Field Values
00569 *
00570 * @details
00571 * Contains the current down-counter value and error flags.
00572 *
00573 * @par Register Layout (16-bit)
00574 * | Bits | Field | R/W | Description |
00575 * |-----|-----|-----|-----|
00576 * | 13:0 | CNTVAL | R | Current counter value |
00577 * | 14 | UNDF | R/W | Underflow flag (write 0 to clear) |
00578 * | 15 | REFEF | R/W | Refresh error flag (write 0 to clear) |
00579 *
00580 * @par Error Checking Example
00581 * @code{.c}
00582 * uint16_t status = iwdt()->iwdtsr;
00583 * if (status & k_iwdt_sr_undf) {
00584 * rx_log_warn("IWDTSR", "Reset was caused by IWDTSR timeout");
00585 * }
00586 * @endcode
00587
00588 * @see rx_iwdt_regs_t::iwdtsr Status register
00589 * @since Version 1.0.0
00590 */
00591 typedef enum : uint16_t {
00592 /**
00593 * @brief Counter value mask (bits 13:0)
00594 * @details Extracts 14-bit down-counter value.
00595 */
00596 k_iwdt_sr_cntval_mask = 0x3FFF,
00597
00598 /**
00599 * @brief Underflow flag bit position
00600 */
00601 k_iwdt_sr_undf_shift = 14,
00602
00603 /**
00604 * @brief Refresh error flag bit position
00605 */
00606 k_iwdt_sr_refef_shift = 15,
00607
00608 /**
00609 * @brief Underflow flag - Bit 14
00610 * @details Set when counter reaches 0 (timeout). Write 0 to clear.
00611 */
00612 k_iwdt_sr_undf = (1U « 14),
00613
00614 /**

```

```

00615 * @brief Refresh error flag - Bit 15
00616 * @details Set when refresh occurs outside valid window. Write 0 to clear.
00617 */
00618 k_iwtdt_sr_ref = (1U « 15),
00619 } iwtdt_iwtdtsr_bits_t;
00620
00621 /**
00622 * @enum iwtdt_iwtdtrcr_bits_t
00623 * @brief IWDT Reset Control Register (IWDTRCR) Bit Field Values
00624 *
00625 * @details
00626 * Controls the action taken on IWDT timeout.
00627 *
00628 * @par Register Layout (8-bit)
00629 * | Bit | Field | R/W | Description |
00630 * |-----|-----|-----|-----|
00631 * | 6:0 | - | - | Reserved |
00632 * | 7 | RSTIRQS | R/W | 0=NMI, 1=Reset on timeout |
00633 *
00634 * @par Configuration Example
00635 * @code{.c}
00636 * // Production: Generate reset on timeout (recommended)
00637 * iwtdt()->iwtdtrcr = k_iwtdt_rstirqs_reset;
00638 * @endcode
00639 *
00640 * @see rx_iwtdt_regs_t::iwtdtrcr Reset control register
00641 * @since Version 1.0.0
00642 */
00643 typedef enum : uint8_t {
00644 /**
00645 * @brief RSTIRQS bit position
00646 */
00647 k_iwtdt_rstirqs_pos = 7,
00648
00649 /**
00650 * @brief RSTIRQS bit mask
00651 */
00652 k_iwtdt_rcr_rstirqs_mask = (1U « 7),
00653
00654 /**
00655 * @brief Generate internal reset on timeout (RSTIRQS=1)
00656 * @details Recommended for production systems.
00657 */
00658 k_iwtdt_rstirqs_reset = (1U « 7),
00659
00660 /**
00661 * @brief Generate NMI on timeout (RSTIRQS=0)
00662 * @details Useful for debugging or logging before reset.
00663 */
00664 k_iwtdt_rstirqs_nmi = 0x00,
00665 } iwtdt_iwtdtrcr_bits_t;
00666
00667 /**
00668 * @enum iwtdt_iwtdtcstpr_shifts_t
00669 * @brief IWDT CSTPR bit field positions
00670 * @since Version 1.0.0
00671 */
00672 typedef enum : uint8_t {
00673 k_iwtdt_cstpr_slcstp_pos = 7, /**< Sleep Mode Count Stop bit position */
00674 } iwtdt_iwtdtcstpr_shifts_t;
00675
00676 /**
00677 * @enum iwtdt_iwtdtcstpr_bits_t
00678 * @brief IWDT Count Stop Control Register (IWDTCSTPR) Bit Field Values
00679 *
00680 * @details
00681 * Controls whether the IWDT counter continues during sleep modes.
00682 * This register is unique to IWDT (WDT does not have this capability).
00683 *
00684 * @par Register Layout (8-bit)
00685 * | Bit | Field | R/W | Description |
00686 * |-----|-----|-----|-----|
00687 * | 6:0 | - | - | Reserved |
00688 * | 7 | SLCSTP | R/W | 0=Continue counting, 1=Stop in sleep |
00689 *
00690 * @par Sleep Mode Behavior
00691 * - **Continue (k_iwtdt_slcstp_continue)**: IWDT keeps running in sleep.
00692 * Ensures system wakes up if sleep takes too long. **Recommended.**
00693 * - **Stop (k_iwtdt_slcstp_stop)**: IWDT pauses during sleep.
00694 * Use only if sleep duration is unpredictable and legitimate.
00695 *
00696 * @par Configuration Example
00697 * @code{.c}
00698 * // Recommended: Keep counting during sleep
00699 * iwtdt()->iwtdtcstpr = k_iwtdt_slcstp_continue;
00700 *
00701 * // Alternative: Stop counting during sleep

```

```

00702 * iwdt()->iwdtcstpr = k_iwdt_slcstp_stop;
00703 * @endcode
00704 *
00705 * @see rx_iwdt_regs_t::iwdtcstpr Count stop control register
00706 * @since Version 1.0.0
00707 */
00708 typedef enum : uint8_t {
00709 /**
00710 * @brief SLCSTP bit mask
00711 */
00712 k_iwdt_cstpr_slcstp_mask = (1U « 7),
00713
00714 /**
00715 * @brief Stop counting during sleep (SLCSTP=1)
00716 * @details IWDT counter pauses when MCU enters sleep mode.
00717 * @warning Use with caution - may mask stuck-in-sleep conditions
00718 */
00719 k_iwdt_slcstp_stop = (1U « 7),
00720
00721 /**
00722 * @brief Continue counting during sleep (SLCSTP=0)
00723 * @details IWDT counter continues in sleep mode. **Recommended.**
00724 * Ensures system recovery if sleep is not properly exited.
00725 */
00726 k_iwdt_slcstp_continue = 0x00,
00727 } iwdt_iwdtcstpr_bits_t;
00728
00729 /*
=====
00730 * Static Assertions - Verify Register Layout at Compile Time
00731 *
=====
00732 */
00733
00734 /* The k_iwdt_base_addr enum declaration above is itself the contract for the
00735 * IWDT base address (per Hardware Manual). A static_assert that re-states the
00736 * same literal here would duplicate the magic number without adding a real
00737 * cross-check, so we rely on the enum declaration alone. */
00738
00739 /* Verify register structure layout */
00740 static_assert(sizeof(rx_iwdt_regs_t) == 9, "IWDT register structure size incorrect");
00741 static_assert(offsetof(rx_iwdt_regs_t, iwdtrr) == 0x00, "IWDTRR offset incorrect");
00742 static_assert(offsetof(rx_iwdt_regs_t, iwdtcr) == 0x02, "IWDTCR offset incorrect");
00743 static_assert(offsetof(rx_iwdt_regs_t, iwdtsr) == 0x04, "IWDTSR offset incorrect");
00744 static_assert(offsetof(rx_iwdt_regs_t, iwdtrcr) == 0x06, "IWDTRCR offset incorrect");
00745 static_assert(offsetof(rx_iwdt_regs_t, iwdtcstpr) == 0x08, "IWDTCSTPR offset incorrect");
00746
00747 #ifdef __cplusplus
00748 }
00749 #endif

```

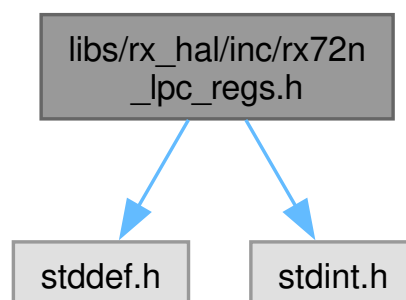
## 8.31 libs/rx\_hal/inc/rx72n\_lpc\_regs.h File Reference

RX72N Low Power Consumption Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n\_lpc\_regs.h:



This graph shows which files directly or indirectly include this file:





- struct rx\_dps\_regs\_t  
Deep Standby Register Block (non-contiguous).

- ```

enum rx_lpc_addresses_t : uintptr_t {
    k_opccr_addr = 0x000800A0U , k_rstckcr_addr = 0x000800A1U , k_dpsbycr_base_addr =
0x0008C280U , k_dpsbycr_addr = 0x0008C280U ,
    k_dpsier0_addr = 0x0008C282U , k_dpsier1_addr = 0x0008C283U , k_dpsier2_addr = 0x0008C284U , k_dpsier3_addr = 0x0008C285U ,
    k_dpsifr0_addr = 0x0008C286U , k_dpsifr1_addr = 0x0008C287U , k_dpsifr2_addr = 0x0008C288U , k_dpsifr3_addr = 0x0008C289U ,
    k_dpsiegr0_addr = 0x0008C28AU , k_dpsiegr1_addr = 0x0008C28BU , k_dpsiegr2_addr =
0x0008C28CU , k_dpsiegr3_addr = 0x0008C28DU ,
    k_dpsbkr_base_addr = 0x0008C2A0U }

    Low Power Consumption register addresses.
• enum rx_lpc_sizes_t : uint8_t { k_dpsbkr_size = 32U , k_dps_regs_count = 4U }

    Low Power Consumption register and buffer sizes.
• enum rx_opccr_bits_t : uint8_t {
    k_opccr_opcm_highspeed = 0x00U , k_opccr_opcm_lowspeed1 = 0x06U , k_opccr_opcm_lowspeed2
= 0x07U , k_opccr_opcm_mask = 0x07U ,
    k_opccr_opcmtsf = 0x10U }

    OPCCR bit definitions.
• enum rx_rstckcr_bits_t : uint8_t { k_rstckcr_rstcksel_hoco = 0x01U , k_rstckcr_rstcksel_main
= 0x02U , k_rstckcr_rstcksel_mask = 0x07U , k_rstckcr_rstcken = 0x80U }

    RSTCKCR bit definitions.
• enum rx_dpsbycr_bits_t : uint8_t {
    k_dpsbycr_deepcut_ram_usb_on = 0x00U , k_dpsbycr_deepcut_ram_usb_off = 0x01U ,
    k_dpsbycr_deepcut_lvd_off = 0x03U , k_dpsbycr_deepcut_mask = 0x03U ,
    k_dpsbycr_iokeep = 0x40U , k_dpsbycr_dpsby = 0x80U }

    DPSBYCR bit definitions.
• enum rx_dpsier0_bits_t : uint8_t {
    k_dpsier0_dirq0e = 0x01U , k_dpsier0_dirq1e = 0x02U , k_dpsier0_dirq2e = 0x04U ,
    k_dpsier0_dirq3e = 0x08U ,
    k_dpsier0_dirq4e = 0x10U , k_dpsier0_dirq5e = 0x20U , k_dpsier0_dirq6e = 0x40U ,
    k_dpsier0_dirq7e = 0x80U }

    DPSIER0 bit definitions (IRQ0-7 wake-up enable).
• enum rx_dpsier1_bits_t : uint8_t {
    k_dpsier1_dirq8e = 0x01U , k_dpsier1_dirq9e = 0x02U , k_dpsier1_dirq10e = 0x04U ,
    k_dpsier1_dirq11e = 0x08U ,
    k_dpsier1_dirq12e = 0x10U , k_dpsier1_dirq13e = 0x20U , k_dpsier1_dirq14e = 0x40U ,
    k_dpsier1_dirq15e = 0x80U }

    DPSIER1 bit definitions (IRQ8-15 wake-up enable).
• enum rx_dpsier2_bits_t : uint8_t {
    k_dpsier2_dlvd1ie = 0x01U , k_dpsier2_dlvd2ie = 0x02U , k_dpsier2_drtciie = 0x04U ,
    k_dpsier2_drtcaie = 0x08U ,
    k_dpsier2_dnmie = 0x10U , k_dpsier2_driicdie = 0x20U , k_dpsier2_driiccie = 0x40U ,
    k_dpsier2_dusbie = 0x80U }

```

- DPSIER2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up enable).
- enum `rx_dpsier3_bits_t` : `uint8_t` { `k_dpsier3_dcanie` = 0x01U }
- DPSIER3 bit definitions (CAN wake-up enable).
- enum `rx_dpsifr0_bits_t` : `uint8_t` {
`k_dpsifr0_dirq0f` = 0x01U , `k_dpsifr0_dirq1f` = 0x02U , `k_dpsifr0_dirq2f` = 0x04U ,
`k_dpsifr0_dirq3f` = 0x08U ,
`k_dpsifr0_dirq4f` = 0x10U , `k_dpsifr0_dirq5f` = 0x20U , `k_dpsifr0_dirq6f` = 0x40U ,
`k_dpsifr0_dirq7f` = 0x80U }
- DPSIFR0 bit definitions (IRQ0-7 wake-up flags).
- enum `rx_dpsifr1_bits_t` : `uint8_t` {
`k_dpsifr1_dirq8f` = 0x01U , `k_dpsifr1_dirq9f` = 0x02U , `k_dpsifr1_dirq10f` = 0x04U ,
`k_dpsifr1_dirq11f` = 0x08U ,
`k_dpsifr1_dirq12f` = 0x10U , `k_dpsifr1_dirq13f` = 0x20U , `k_dpsifr1_dirq14f` = 0x40U ,
`k_dpsifr1_dirq15f` = 0x80U }
- DPSIFR1 bit definitions (IRQ8-15 wake-up flags).
- enum `rx_dpsifr2_bits_t` : `uint8_t` {
`k_dpsifr2_dlvdlif` = 0x01U , `k_dpsifr2_dlvdl2if` = 0x02U , `k_dpsifr2_drtciif` = 0x04U ,
`k_dpsifr2_drtcaif` = 0x08U ,
`k_dpsifr2_dnmif` = 0x10U , `k_dpsifr2_driicdif` = 0x20U , `k_dpsifr2_driiccif` = 0x40U ,
`k_dpsifr2_dusbif` = 0x80U }
- DPSIFR2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up flags).
- enum `rx_dpsifr3_bits_t` : `uint8_t` { `k_dpsifr3_dcanif` = 0x01U }
- DPSIFR3 bit definitions (CAN wake-up flag).
- enum `rx_dpsiegr0_bits_t` : `uint8_t` {
`k_dpsiegr0_dirq0eg` = 0x01U , `k_dpsiegr0_dirq1eg` = 0x02U , `k_dpsiegr0_dirq2eg` = 0x04U ,
`k_dpsiegr0_dirq3eg` = 0x08U ,
`k_dpsiegr0_dirq4eg` = 0x10U , `k_dpsiegr0_dirq5eg` = 0x20U , `k_dpsiegr0_dirq6eg` = 0x40U ,
`k_dpsiegr0_dirq7eg` = 0x80U }
- DPSIEGR0 bit definitions (IRQ0-7 edge select).
- enum `rx_dpsiegr1_bits_t` : `uint8_t` {
`k_dpsiegr1_dirq8eg` = 0x01U , `k_dpsiegr1_dirq9eg` = 0x02U , `k_dpsiegr1_dirq10eg` = 0x04U ,
`k_dpsiegr1_dirq11eg` = 0x08U ,
`k_dpsiegr1_dirq12eg` = 0x10U , `k_dpsiegr1_dirq13eg` = 0x20U , `k_dpsiegr1_dirq14eg` = 0x40U ,
`k_dpsiegr1_dirq15eg` = 0x80U }
- DPSIEGR1 bit definitions (IRQ8-15 edge select).
- enum `rx_dpsiegr2_bits_t` : `uint8_t` {
`k_dpsiegr2_dlvdl1eg` = 0x01U , `k_dpsiegr2_dlvdl2eg` = 0x02U , `k_dpsiegr2_dnmieg` = 0x10U ,
`k_dpsiegr2_driicdeg` = 0x20U ,
`k_dpsiegr2_driicceg` = 0x40U }
- DPSIEGR2 bit definitions (LVD, NMI, I2C edge select).
- enum `rx_dpsiegr3_bits_t` : `uint8_t` { `k_dpsiegr3_dcanieg` = 0x01U }
- DPSIEGR3 bit definitions (CAN edge select).

Functions

- static volatile `uint8_t` * `opccr_reg` (void)
 Get pointer to OPCCR register.
- static volatile `uint8_t` * `rstckcr_reg` (void)
 Get pointer to RSTCKCR register.
- static volatile `rx_dps_regs_t` * `dps_regs` (void)
 Get pointer to Deep Standby registers.
- static volatile `uint8_t` * `dpsbkr` (void)
 Get pointer to Deep Standby Backup RAM.

8.31.1 Detailed Description

RX72N Low Power Consumption Register Definitions.

Low Power Consumption (LPC) register definitions for operating power control, deep software standby mode, and wake-up interrupt configuration on the RX72N microcontroller. This header complements [rx72n_system_regs.h](#) which contains the main standby (SBYCR) and module stop (MSTPCRA-D) registers.

Register Groups

Address Range	Description
0x000800A0 - 0x000800A1	Operating power control (OPCCR/RSTCKCR)
0x0008C280 - 0x0008C28D	Deep standby control & interrupt
0x0008C2A0 - 0x0008C2BF	Deep standby backup RAM (32 bytes)

Low Power Modes

The RX72N supports several low power modes:

- Sleep Mode: CPU stopped, peripherals run (lowest latency wakeup)
- All-Module Clock Stop Mode: CPU and all modules stopped
- Software Standby Mode: Most oscillators stopped, RAM retained
- Deep Software Standby Mode: Lowest power, backup RAM only

Operating Power Control Modes

Mode	ICLK Max	Description
High-speed (000b)	240 MHz	Full performance
Low-speed 1 (110b)	1 MHz	Low power, PLL off
Low-speed 2 (111b)	264 kHz	Lowest power, HOCO off

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 3: No dynamic memory allocation
- Rule 8: All constants use C23 typed enums
- Rule 10: Compiles with -Wall -Wextra -Werror

Manual References

- Chapter 11: Low Power Consumption
- Table 11.1: Low Power Consumption Functions
- Table 11.2: Mode Transitions

Verification Status

VERIFIED (2026-01-29) - All register addresses verified against RX72N Manual Rev.1.20, Chapter 11.

See also

[rx72n_system_regs.h](#) Main system registers (SBYCR, MSTPCRA-D)

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_lpc_regs.h](#).

8.31.2 Data Structure Documentation

8.31.2.1 struct rx_dps_regs_t

Deep Standby Register Block (non-contiguous).

The deep standby registers are NOT all contiguous in memory:

- 0x0008C280: DPSBYCR
- 0x0008C281: Reserved
- 0x0008C282-0x0008C28D: DPSIER, DPSIFR, DPSIEGR registers

This structure covers the contiguous portion from 0x0008C280 onwards.

Invariant

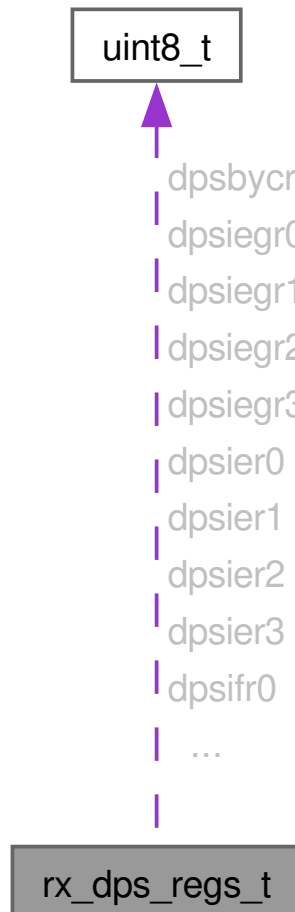
`sizeof(rx_dps_regs_t) == 14 bytes`

Since

Version 1.0.0

Definition at line 537 of file [rx72n_lpc_regs.h](#).

Collaboration diagram for rx_dps_regs_t:



Data Fields

volatile uint8_t	dpsbycr	Deep Standby Control @ 0x00 (0x0008C280)
volatile uint8_t	dpsiegr0	Deep Standby Int Edge 0 @ 0x0A (0x0008C28A)
volatile uint8_t	dpsiegr1	Deep Standby Int Edge 1 @ 0x0B (0x0008C28B)
volatile uint8_t	dpsiegr2	Deep Standby Int Edge 2 @ 0x0C (0x0008C28C)
volatile uint8_t	dpsiegr3	Deep Standby Int Edge 3 @ 0x0D (0x0008C28D)
volatile uint8_t	dpsier0	Deep Standby Int Enable 0 @ 0x02 (0x0008C282)
volatile uint8_t	dpsier1	Deep Standby Int Enable 1 @ 0x03 (0x0008C283)
volatile uint8_t	dpsier2	Deep Standby Int Enable 2 @ 0x04 (0x0008C284)
volatile uint8_t	dpsier3	Deep Standby Int Enable 3 @ 0x05 (0x0008C285)
volatile uint8_t	dpsifr0	Deep Standby Int Flag 0 @ 0x06 (0x0008C286)
volatile uint8_t	dpsifr1	Deep Standby Int Flag 1 @ 0x07 (0x0008C287)

volatile uint8_t	dpsifr2	Deep Standby Int Flag 2 @ 0x08 (0x0008C288)
volatile uint8_t	dpsifr3	Deep Standby Int Flag 3 @ 0x09 (0x0008C289)
uint8_t	reserved1	Reserved @ 0x01

8.31.3 Enumeration Type Documentation

8.31.3.1 rx_dpsbycr_bits_t

enum [rx_dpsbycr_bits_t](#) : uint8_t

DPSBYCR bit definitions.

Controls deep software standby mode entry and power supply configuration.

DEEPCUT[1:0]

- Power Supply Control]

Value	Standby RAM	USB Resume	LVD	Notes
00b	Powered	Enabled	Active	USB wake-up OK
01b	Not powered	Disabled	Active	RAM data lost
11b	Not powered	Disabled	Off	Lowest power

Note

10b is a prohibited setting for DEEPCUT[1:0].

Since

Version 1.0.0

Enumerator

k_dpsbycr_deepcut_ram_usb_on	Standby RAM & USB powered
k_dpsbycr_deepcut_ram_usb_off	Standby RAM & USB not powered
k_dpsbycr_deepcut_lvd_off	LVD stopped, lowest power
k_dpsbycr_deepcut_mask	DEEPCUT[1:0] bit mask
k_dpsbycr_iokeep	1 = Retain I/O state after deep standby exit
k_dpsbycr_dpsby	1 = Enter deep standby (with SBYCR.SSBY=1)

Definition at line 215 of file [rx72n_lpc_regs.h](#).

```

00215         : uint8_t {
00216     /* DEEPCUT[1:0] - Deep Cut Power Control (bits 1:0) */
00217     k_dpsbycr_deepcut_ram_usb_on = 0x00U, /**< Standby RAM & USB powered */
00218     k_dpsbycr_deepcut_ram_usb_off = 0x01U, /**< Standby RAM & USB not powered */
00219     k_dpsbycr_deepcut_lvd_off    = 0x03U, /**< LVD stopped, lowest power */
00220     k_dpsbycr_deepcut_mask      = 0x03U, /**< DEEPCUT[1:0] bit mask */
00221
00222     /* IOKEEP - I/O Port Retention (bit 6) */
00223     k_dpsbycr_iokeep = 0x40U, /**< 1 = Retain I/O state after deep standby exit */
00224
00225     /* DPSBY - Deep Software Standby (bit 7) */
00226     k_dpsbycr_dpsby = 0x80U, /**< 1 = Enter deep standby (with SBYCR.SSBY=1) */
00227 } rx_dpsbycr_bits_t;

```

8.31.3.2 rx`dpsiegr0`bits``t`

```
enum rx_dpsiegr0_bits_t : uint8_t
```

DPSIEGR0 bit definitions (IRQ0-7 edge select).

Selects edge polarity for IRQ0-DS to IRQ7-DS wake-up detection. 0 = Falling edge, 1 = Rising edge.

Since

Version 1.0.0

Enumerator

k_dpsiegr0_dirq0eg	IRQ0-DS edge select
k_dpsiegr0_dirq1eg	IRQ1-DS edge select
k_dpsiegr0_dirq2eg	IRQ2-DS edge select
k_dpsiegr0_dirq3eg	IRQ3-DS edge select
k_dpsiegr0_dirq4eg	IRQ4-DS edge select
k_dpsiegr0_dirq5eg	IRQ5-DS edge select
k_dpsiegr0_dirq6eg	IRQ6-DS edge select
k_dpsiegr0_dirq7eg	IRQ7-DS edge select

Definition at line 436 of file `rx72n_lpc_regs.h`.

```
00436         : uint8_t {
00437     k_dpsiegr0_dirq0eg = 0x01U, /**< IRQ0-DS edge select */
00438     k_dpsiegr0_dirq1eg = 0x02U, /**< IRQ1-DS edge select */
00439     k_dpsiegr0_dirq2eg = 0x04U, /**< IRQ2-DS edge select */
00440     k_dpsiegr0_dirq3eg = 0x08U, /**< IRQ3-DS edge select */
00441     k_dpsiegr0_dirq4eg = 0x10U, /**< IRQ4-DS edge select */
00442     k_dpsiegr0_dirq5eg = 0x20U, /**< IRQ5-DS edge select */
00443     k_dpsiegr0_dirq6eg = 0x40U, /**< IRQ6-DS edge select */
00444     k_dpsiegr0_dirq7eg = 0x80U, /**< IRQ7-DS edge select */
00445 } rx_dpsiegr0_bits_t;
```

8.31.3.3 rx`dpsiegr1`bits``t`

```
enum rx_dpsiegr1_bits_t : uint8_t
```

DPSIEGR1 bit definitions (IRQ8-15 edge select).

Selects edge polarity for IRQ8-DS to IRQ15-DS wake-up detection. 0 = Falling edge, 1 = Rising edge.

Since

Version 1.0.0

Enumerator

k_dpsiegr1_dirq8eg	IRQ8-DS edge select
k_dpsiegr1_dirq9eg	IRQ9-DS edge select

k_dpsiegr1_dirq10eg	IRQ10-DS edge select
k_dpsiegr1_dirq11eg	IRQ11-DS edge select
k_dpsiegr1_dirq12eg	IRQ12-DS edge select
k_dpsiegr1_dirq13eg	IRQ13-DS edge select
k_dpsiegr1_dirq14eg	IRQ14-DS edge select
k_dpsiegr1_dirq15eg	IRQ15-DS edge select

Definition at line 462 of file [rx72n_lpc_regs.h](#).

```

00462         : uint8_t {
00463     k_dpsiegr1_dirq8eg  = 0x01U, /**< IRQ8-DS edge select */
00464     k_dpsiegr1_dirq9eg  = 0x02U, /**< IRQ9-DS edge select */
00465     k_dpsiegr1_dirq10eg = 0x04U, /**< IRQ10-DS edge select */
00466     k_dpsiegr1_dirq11eg = 0x08U, /**< IRQ11-DS edge select */
00467     k_dpsiegr1_dirq12eg = 0x10U, /**< IRQ12-DS edge select */
00468     k_dpsiegr1_dirq13eg = 0x20U, /**< IRQ13-DS edge select */
00469     k_dpsiegr1_dirq14eg = 0x40U, /**< IRQ14-DS edge select */
00470     k_dpsiegr1_dirq15eg = 0x80U, /**< IRQ15-DS edge select */
00471 } rx_dpsiegr1_bits_t;

```

8.31.3.4 rx_dpsiegr2_bits_t

```
enum rx_dpsiegr2_bits_t : uint8_t
```

DPSIEGR2 bit definitions (LVD, NMI, I2C edge select).

Selects edge polarity for various peripheral wake-up detection. For LVD: 0 = VCC fall ($VCC < V_{det}$), 1 = VCC rise ($VCC \geq V_{det}$). For pins: 0 = Falling edge, 1 = Rising edge.

Since

Version 1.0.0

Enumerator

k_dpsiegr2_dlv1eg	LVD1 edge: 0=fall, 1=rise
k_dpsiegr2_dlv2eg	LVD2 edge: 0=fall, 1=rise
k_dpsiegr2_dnmieg	NMI edge select
k_dpsiegr2_driicdeg	SDA2-DS edge select
k_dpsiegr2_driicceg	SCL2-DS edge select

Definition at line 489 of file [rx72n_lpc_regs.h](#).

```

00489         : uint8_t {
00490     k_dpsiegr2_dlv1eg  = 0x01U, /**< LVD1 edge: 0=fall, 1=rise */
00491     k_dpsiegr2_dlv2eg  = 0x02U, /**< LVD2 edge: 0=fall, 1=rise */
00492     k_dpsiegr2_dnmieg  = 0x10U, /**< NMI edge select */
00493     k_dpsiegr2_driicdeg = 0x20U, /**< SDA2-DS edge select */
00494     k_dpsiegr2_driicceg = 0x40U, /**< SCL2-DS edge select */
00495 } rx_dpsiegr2_bits_t;

```


8.31.3.5 rx_dpsiegr3_bits_t

```
enum rx_dpsiegr3_bits_t : uint8_t
```

DPSIEGR3 bit definitions (CAN edge select).

Selects edge polarity for CAN receive wake-up detection. 0 = Falling edge, 1 = Rising edge.

Since

Version 1.0.0

Enumerator

k_dpsiegr3_dcanieg	CRX1-DS edge select
--------------------	---------------------

Definition at line 512 of file rx72n_lpc_regs.h.

```
00512         : uint8_t {
00513     k_dpsiegr3_dcanieg = 0x01U, /**< CRX1-DS edge select */
00514 } rx_dpsiegr3_bits_t;
```

8.31.3.6 rx_dpsier0_bits_t

```
enum rx_dpsier0_bits_t : uint8_t
```

DPSIER0 bit definitions (IRQ0-7 wake-up enable).

Enables external IRQ0-DS to IRQ7-DS pins as wake-up sources from deep software standby mode.

Since

Version 1.0.0

Enumerator

k_dpsier0_dirq0e	IRQ0-DS pin enable
k_dpsier0_dirq1e	IRQ1-DS pin enable
k_dpsier0_dirq2e	IRQ2-DS pin enable
k_dpsier0_dirq3e	IRQ3-DS pin enable
k_dpsier0_dirq4e	IRQ4-DS pin enable
k_dpsier0_dirq5e	IRQ5-DS pin enable
k_dpsier0_dirq6e	IRQ6-DS pin enable
k_dpsier0_dirq7e	IRQ7-DS pin enable

Definition at line 244 of file rx72n_lpc_regs.h.

```
00244         : uint8_t {
00245     k_dpsier0_dirq0e = 0x01U, /**< IRQ0-DS pin enable */
00246     k_dpsier0_dirq1e = 0x02U, /**< IRQ1-DS pin enable */
00247     k_dpsier0_dirq2e = 0x04U, /**< IRQ2-DS pin enable */
00248     k_dpsier0_dirq3e = 0x08U, /**< IRQ3-DS pin enable */
00249     k_dpsier0_dirq4e = 0x10U, /**< IRQ4-DS pin enable */
00250     k_dpsier0_dirq5e = 0x20U, /**< IRQ5-DS pin enable */
00251     k_dpsier0_dirq6e = 0x40U, /**< IRQ6-DS pin enable */
00252     k_dpsier0_dirq7e = 0x80U, /**< IRQ7-DS pin enable */
00253 } rx_dpsier0_bits_t;
```

8.31.3.7 rx_dpsier1_bits_t

enum [rx_dpsier1_bits_t](#) : uint8_t

DPSIER1 bit definitions (IRQ8-15 wake-up enable).

Enables external IRQ8-DS to IRQ15-DS pins as wake-up sources from deep software standby mode.

Since

Version 1.0.0

Enumerator

k_dpsier1_dirq8e	IRQ8-DS pin enable
k_dpsier1_dirq9e	IRQ9-DS pin enable
k_dpsier1_dirq10e	IRQ10-DS pin enable
k_dpsier1_dirq11e	IRQ11-DS pin enable
k_dpsier1_dirq12e	IRQ12-DS pin enable
k_dpsier1_dirq13e	IRQ13-DS pin enable
k_dpsier1_dirq14e	IRQ14-DS pin enable
k_dpsier1_dirq15e	IRQ15-DS pin enable

Definition at line 270 of file [rx72n_lpc_regs.h](#).

```
00270         : uint8_t {
00271     k_dpsier1_dirq8e = 0x01U, /**< IRQ8-DS pin enable */
00272     k_dpsier1_dirq9e = 0x02U, /**< IRQ9-DS pin enable */
00273     k_dpsier1_dirq10e = 0x04U, /**< IRQ10-DS pin enable */
00274     k_dpsier1_dirq11e = 0x08U, /**< IRQ11-DS pin enable */
00275     k_dpsier1_dirq12e = 0x10U, /**< IRQ12-DS pin enable */
00276     k_dpsier1_dirq13e = 0x20U, /**< IRQ13-DS pin enable */
00277     k_dpsier1_dirq14e = 0x40U, /**< IRQ14-DS pin enable */
00278     k_dpsier1_dirq15e = 0x80U, /**< IRQ15-DS pin enable */
00279 } rx_dpsier1_bits_t;
```

8.31.3.8 rx_dpsier2_bits_t

enum [rx_dpsier2_bits_t](#) : uint8_t

DPSIER2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up enable).

Enables various peripheral wake-up sources from deep software standby mode.

Since

Version 1.0.0

Enumerator

k_dpsier2_dlvd1ie	LVD1 (Voltage Monitor 1) enable
k_dpsier2_dlvd2ie	LVD2 (Voltage Monitor 2) enable

k_dpsier2_drtciie	RTC periodic interrupt enable
k_dpsier2_drtcaie	RTC alarm interrupt enable
k_dpsier2_dnmie	NMI pin enable (write-once)
k_dpsier2_driicdie	SDA2-DS (I2C data) enable
k_dpsier2_driiccie	SCL2-DS (I2C clock) enable
k_dpsier2_dusbie	USB suspend/resume enable

Definition at line 295 of file [rx72n_lpc_regs.h](#).

```

00295         : uint8_t {
00296     k_dpsier2_dlvdl1ie = 0x01U, /**< LVD1 (Voltage Monitor 1) enable */
00297     k_dpsier2_dlvdl2ie = 0x02U, /**< LVD2 (Voltage Monitor 2) enable */
00298     k_dpsier2_drtciie = 0x04U, /**< RTC periodic interrupt enable */
00299     k_dpsier2_drtcaie = 0x08U, /**< RTC alarm interrupt enable */
00300     k_dpsier2_dnmie   = 0x10U, /**< NMI pin enable (write-once) */
00301     k_dpsier2_driicdie = 0x20U, /**< SDA2-DS (I2C data) enable */
00302     k_dpsier2_driiccie = 0x40U, /**< SCL2-DS (I2C clock) enable */
00303     k_dpsier2_dusbie   = 0x80U, /**< USB suspend/resume enable */
00304 } rx_dpsier2_bits_t;

```

8.31.3.9 rx_dpsier3_bits_t

```
enum rx_dpsier3_bits_t : uint8_t
```

DPSIER3 bit definitions (CAN wake-up enable).

Enables CAN receive pin as wake-up source from deep software standby mode.

Since

Version 1.0.0

Enumerator

k_dpsier3_dcanie	CRX1-DS (CAN1 receive) enable
------------------	-------------------------------

Definition at line 320 of file [rx72n_lpc_regs.h](#).

```

00320         : uint8_t {
00321     k_dpsier3_dcanie = 0x01U, /**< CRX1-DS (CAN1 receive) enable */
00322 } rx_dpsier3_bits_t;

```

8.31.3.10 rx_dpsifr0_bits_t

```
enum rx_dpsifr0_bits_t : uint8_t
```

DPSIFR0 bit definitions (IRQ0-7 wake-up flags).

Indicates which IRQ0-DS to IRQ7-DS pin triggered wake-up from deep software standby mode. Write 0 after reading 1 to clear.

Since

Version 1.0.0

Enumerator

k_dpsifr0_dirq0f	IRQ0-DS wake-up flag
k_dpsifr0_dirq1f	IRQ1-DS wake-up flag
k_dpsifr0_dirq2f	IRQ2-DS wake-up flag
k_dpsifr0_dirq3f	IRQ3-DS wake-up flag
k_dpsifr0_dirq4f	IRQ4-DS wake-up flag
k_dpsifr0_dirq5f	IRQ5-DS wake-up flag
k_dpsifr0_dirq6f	IRQ6-DS wake-up flag
k_dpsifr0_dirq7f	IRQ7-DS wake-up flag

Definition at line 339 of file [rx72n_lpc_regs.h](#).

```

00339     : uint8_t {
00340     k_dpsifr0_dirq0f = 0x01U, /**< IRQ0-DS wake-up flag */
00341     k_dpsifr0_dirq1f = 0x02U, /**< IRQ1-DS wake-up flag */
00342     k_dpsifr0_dirq2f = 0x04U, /**< IRQ2-DS wake-up flag */
00343     k_dpsifr0_dirq3f = 0x08U, /**< IRQ3-DS wake-up flag */
00344     k_dpsifr0_dirq4f = 0x10U, /**< IRQ4-DS wake-up flag */
00345     k_dpsifr0_dirq5f = 0x20U, /**< IRQ5-DS wake-up flag */
00346     k_dpsifr0_dirq6f = 0x40U, /**< IRQ6-DS wake-up flag */
00347     k_dpsifr0_dirq7f = 0x80U, /**< IRQ7-DS wake-up flag */
00348 } rx_dpsifr0_bits_t;

```

8.31.3.11 rx`dpsifr1`bits` t

```
enum rx_dpsifr1_bits_t : uint8_t
```

DPSIFR1 bit definitions (IRQ8-15 wake-up flags).

Indicates which IRQ8-DS to IRQ15-DS pin triggered wake-up from deep software standby mode. Write 0 after reading 1 to clear.

Since

Version 1.0.0

Enumerator

k_dpsifr1_dirq8f	IRQ8-DS wake-up flag
k_dpsifr1_dirq9f	IRQ9-DS wake-up flag
k_dpsifr1_dirq10f	IRQ10-DS wake-up flag
k_dpsifr1_dirq11f	IRQ11-DS wake-up flag
k_dpsifr1_dirq12f	IRQ12-DS wake-up flag
k_dpsifr1_dirq13f	IRQ13-DS wake-up flag
k_dpsifr1_dirq14f	IRQ14-DS wake-up flag
k_dpsifr1_dirq15f	IRQ15-DS wake-up flag

Definition at line 365 of file [rx72n_lpc_regs.h](#).

```

00365         : uint8_t {
00366     k_dpsifr1_dirq8f = 0x01U, /**< IRQ8-DS wake-up flag */
00367     k_dpsifr1_dirq9f = 0x02U, /**< IRQ9-DS wake-up flag */
00368     k_dpsifr1_dirq10f = 0x04U, /**< IRQ10-DS wake-up flag */
00369     k_dpsifr1_dirq11f = 0x08U, /**< IRQ11-DS wake-up flag */
00370     k_dpsifr1_dirq12f = 0x10U, /**< IRQ12-DS wake-up flag */
00371     k_dpsifr1_dirq13f = 0x20U, /**< IRQ13-DS wake-up flag */
00372     k_dpsifr1_dirq14f = 0x40U, /**< IRQ14-DS wake-up flag */
00373     k_dpsifr1_dirq15f = 0x80U, /**< IRQ15-DS wake-up flag */
00374 } rx_dpsifr1_bits_t;

```

8.31.3.12 rx`dpsifr2`bits` t

```
enum rx_dpsifr2_bits_t : uint8_t
```

DPSIFR2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up flags).

Indicates which peripheral triggered wake-up from deep software standby mode. Write 0 after reading 1 to clear.

Since

Version 1.0.0

Enumerator

k_dpsifr2_dlvdlif	LVD1 wake-up flag
k_dpsifr2_dlvdl2if	LVD2 wake-up flag
k_dpsifr2_drtciif	RTC periodic interrupt flag
k_dpsifr2_drtcaif	RTC alarm interrupt flag
k_dpsifr2_dnmif	NMI wake-up flag
k_dpsifr2_driicdif	SDA2-DS wake-up flag
k_dpsifr2_driiccif	SCL2-DS wake-up flag
k_dpsifr2_dusbif	USB suspend/resume flag

Definition at line 391 of file [rx72n_lpc_regs.h](#).

```

00391         : uint8_t {
00392     k_dpsifr2_dlvdlif = 0x01U, /**< LVD1 wake-up flag */
00393     k_dpsifr2_dlvdl2if = 0x02U, /**< LVD2 wake-up flag */
00394     k_dpsifr2_drtciif = 0x04U, /**< RTC periodic interrupt flag */
00395     k_dpsifr2_drtcaif = 0x08U, /**< RTC alarm interrupt flag */
00396     k_dpsifr2_dnmif = 0x10U, /**< NMI wake-up flag */
00397     k_dpsifr2_driicdif = 0x20U, /**< SDA2-DS wake-up flag */
00398     k_dpsifr2_driiccif = 0x40U, /**< SCL2-DS wake-up flag */
00399     k_dpsifr2_dusbif = 0x80U, /**< USB suspend/resume flag */
00400 } rx_dpsifr2_bits_t;

```

8.31.3.13 rx`dpsifr3`bits` t

```
enum rx_dpsifr3_bits_t : uint8_t
```

DPSIFR3 bit definitions (CAN wake-up flag).

Indicates CAN receive triggered wake-up from deep software standby mode. Write 0 after reading 1 to clear.

Since

Version 1.0.0

Enumerator

k_dpsifr3_dcanif	CRX1-DS wake-up flag
------------------	----------------------

Definition at line 417 of file rx72n_lpc_regs.h.

```
00417         : uint8_t {
00418     k_dpsifr3_dcanif = 0x01U, /**< CRX1-DS wake-up flag */
00419 } rx_dpsifr3_bits_t;
```

8.31.3.14 rx_lpc_addresses_t

enum [rx_lpc_addresses_t](#) : uintptr_t

Low Power Consumption register addresses.

Addresses for operating power control and deep standby registers. These registers control low power mode behavior and wake-up sources.

Memory Map

Address	Register	Size	Description
0x000800A0	OPCCR	8-bit	Operating Power Control
0x000800A1	RSTCKCR	8-bit	Sleep Mode Return Clock Switch
0x0008C280	DPSBYCR	8-bit	Deep Standby Control
0x0008C282-5	DPSIER0-3	8-bit	Deep Standby Int Enable
0x0008C286-9	DPSIFR0-3	8-bit	Deep Standby Int Flags
0x0008C28A-D	DPSIEGR0-3	8-bit	Deep Standby Int Edge
0x0008C2A0-BF	DPSBKRY	8-bit	Deep Standby Backup RAM

Since

Version 1.0.0

Enumerator

k_opccr_addr	Operating Power Control Register
k_rstckcr_addr	Sleep Mode Return Clock Switch Register
k_dpsbycr_base_addr	Deep Standby Control base
k_dpsbycr_addr	Deep Standby Control Register
k_dpsier0_addr	Deep Standby Interrupt Enable 0
k_dpsier1_addr	Deep Standby Interrupt Enable 1
k_dpsier2_addr	Deep Standby Interrupt Enable 2

k_dpsier3_addr	Deep Standby Interrupt Enable 3
k_dpsifr0_addr	Deep Standby Interrupt Flag 0
k_dpsifr1_addr	Deep Standby Interrupt Flag 1
k_dpsifr2_addr	Deep Standby Interrupt Flag 2
k_dpsifr3_addr	Deep Standby Interrupt Flag 3
k_dpsiegr0_addr	Deep Standby Interrupt Edge 0
k_dpsiegr1_addr	Deep Standby Interrupt Edge 1
k_dpsiegr2_addr	Deep Standby Interrupt Edge 2
k_dpsiegr3_addr	Deep Standby Interrupt Edge 3
k_dpsbkr_base_addr	Deep Standby Backup RAM base

Definition at line 91 of file [rx72n_lpc_regs.h](#).

```

00091         : uintptr_t {
00092     /* Operating Power Control Registers @ 0x000800Ax */
00093     k_opccr_addr = 0x000800A0U, /**< Operating Power Control Register */
00094     k_rstckcr_addr = 0x000800A1U, /**< Sleep Mode Return Clock Switch Register */
00095
00096     /* Deep Standby Registers @ 0x0008C28x */
00097     k_dpsbycr_base_addr = 0x0008C280U, /**< Deep Standby Control base */
00098     k_dpsbycr_addr = 0x0008C280U, /**< Deep Standby Control Register */
00099     k_dpsier0_addr = 0x0008C282U, /**< Deep Standby Interrupt Enable 0 */
00100     k_dpsier1_addr = 0x0008C283U, /**< Deep Standby Interrupt Enable 1 */
00101     k_dpsier2_addr = 0x0008C284U, /**< Deep Standby Interrupt Enable 2 */
00102     k_dpsier3_addr = 0x0008C285U, /**< Deep Standby Interrupt Enable 3 */
00103     k_dpsifr0_addr = 0x0008C286U, /**< Deep Standby Interrupt Flag 0 */
00104     k_dpsifr1_addr = 0x0008C287U, /**< Deep Standby Interrupt Flag 1 */
00105     k_dpsifr2_addr = 0x0008C288U, /**< Deep Standby Interrupt Flag 2 */
00106     k_dpsifr3_addr = 0x0008C289U, /**< Deep Standby Interrupt Flag 3 */
00107     k_dpsiegr0_addr = 0x0008C28AU, /**< Deep Standby Interrupt Edge 0 */
00108     k_dpsiegr1_addr = 0x0008C28BU, /**< Deep Standby Interrupt Edge 1 */
00109     k_dpsiegr2_addr = 0x0008C28CU, /**< Deep Standby Interrupt Edge 2 */
00110     k_dpsiegr3_addr = 0x0008C28DU, /**< Deep Standby Interrupt Edge 3 */
00111
00112     /* Deep Standby Backup RAM @ 0x0008C2A0-0x0008C2BF (32 bytes) */
00113     k_dpsbkr_base_addr = 0x0008C2A0U, /**< Deep Standby Backup RAM base */
00114 } rx_lpc_addresses_t;

```

8.31.3.15 rx_lpc_sizes_t

```
enum rx_lpc_sizes_t : uint8_t
```

Low Power Consumption register and buffer sizes.

Since

Version 1.0.0

Enumerator

k_dpsbkr_size	Deep standby backup RAM size (bytes)
k_dps_regs_count	Number of DPSIERx/DPSIFRx/DPSIEGRx registers

Definition at line 122 of file [rx72n_lpc_regs.h](#).

```

00122     : uint8_t {
00123     k_dpsbkr_size = 32U, /**< Deep standby backup RAM size (bytes) */
00124     k_dps_regs_count = 4U, /**< Number of DPSIERx/DPSIFRx/DPSIEGRx registers */
00125 } rx_lpc_sizes_t;

```

8.31.3.16 rx_opccr_bits_t

```
enum rx_opccr_bits_t : uint8_t
```

OPCCR bit definitions.

Operating Power Control Register controls operating power mode for normal operation, sleep mode, and all-module clock stop mode.

Operating Power Control Modes (OPCM[2:0

)]

Value	Mode	ICLK Max	Notes
000b	High-speed	240 MHz	Full performance
110b	Low-speed 1	1 MHz	PLL must be stopped
111b	Low-speed 2	264 kHz	HOCO/PLL must be stopped

Since

Version 1.0.0

Enumerator

k_opccr_opcm_highspeed	High-speed operating mode (240 MHz max)
k_opccr_opcm_lowspeed1	Low-speed operating mode 1 (1 MHz max)
k_opccr_opcm_lowspeed2	Low-speed operating mode 2 (264 kHz max)
k_opccr_opcm_mask	OPCM[2:0] bit mask
k_opccr_opcmtsf	1 = Mode transition in progress

Definition at line 149 of file rx72n_lpc_regs.h.

```
00149         : uint8_t {
00150     /* OPCM[2:0] - Operating Power Control Mode Select (bits 2:0) */
00151     k_opccr_opcm_highspeed = 0x00U, /**< High-speed operating mode (240 MHz max) */
00152     k_opccr_opcm_lowspeed1 = 0x06U, /**< Low-speed operating mode 1 (1 MHz max) */
00153     k_opccr_opcm_lowspeed2 = 0x07U, /**< Low-speed operating mode 2 (264 kHz max) */
00154     k_opccr_opcm_mask      = 0x07U, /**< OPCM[2:0] bit mask */
00155
00156     /* OPCMTSF - Operating Power Control Mode Transition Status Flag (bit 4) */
00157     k_opccr_opcmtsf = 0x10U, /**< 1 = Mode transition in progress */
00158 } rx_opccr_bits_t;
```

8.31.3.17 rx_rstckcr_bits_t

```
enum rx_rstckcr_bits_t : uint8_t
```

RSTCKCR bit definitions.

Controls clock source switching at the time of release from sleep mode. When enabled, the selected clock source is automatically started upon wake-up from sleep mode.

RSTCKSEL[2:0

- Clock Source Select]

Value	Clock Source
001b	HOCO (High-speed on-chip)
010b	Main clock oscillator

Since

Version 1.0.0

Enumerator

k_rstckcr_rstcksel_hoco	HOCO selected for wake-up
k_rstckcr_rstcksel_main	Main oscillator selected for wake-up
k_rstckcr_rstcksel_mask	RSTCKSEL[2:0] bit mask
k_rstckcr_rstcken	1 = Enable clock source switching

Definition at line 182 of file rx72n_lpc_regs.h.

```

00182      : uint8_t {
00183      /* RSTCKSEL[2:0] - Sleep Mode Return Clock Source Select (bits 2:0) */
00184      k_rstckcr_rstcksel_hoco = 0x01U, /**< HOCO selected for wake-up */
00185      k_rstckcr_rstcksel_main = 0x02U, /**< Main oscillator selected for wake-up */
00186      k_rstckcr_rstcksel_mask = 0x07U, /**< RSTCKSEL[2:0] bit mask */
00187
00188      /* RSTCKEN - Sleep Mode Return Clock Source Switching Enable (bit 7) */
00189      k_rstckcr_rstcken = 0x80U, /**< 1 = Enable clock source switching */
00190 } rx_rstckcr_bits_t;

```

8.31.4 Function Documentation

8.31.4.1 dps_regs()

```
volatile rx_dps_regs_t * dps_regs (
    void ) [inline], [static]
```

Get pointer to Deep Standby registers.

Returns

Volatile pointer to Deep Standby register structure

Provides access to DPSBYCR, DPSIERx, DPSIFRx, and DPSIEGRx registers.

Example - Configure IRQ0 as wake-up source:

```

volatile rx_dps_regs_t* dps = dps_regs();

dps->dpsier0 = k_dpsier0_dirq0e;    // Enable IRQ0-DS wake-up
dps->dpsiegr0 = k_dpsiegr0_dirq0eg; // Rising edge trigger
dps->dpsifr0 = 0x00;                // Clear flags before entering

dps->dpsbycr = k_dpsbycr_dpsby;     // Enable deep standby

```

Since

Version 1.0.0

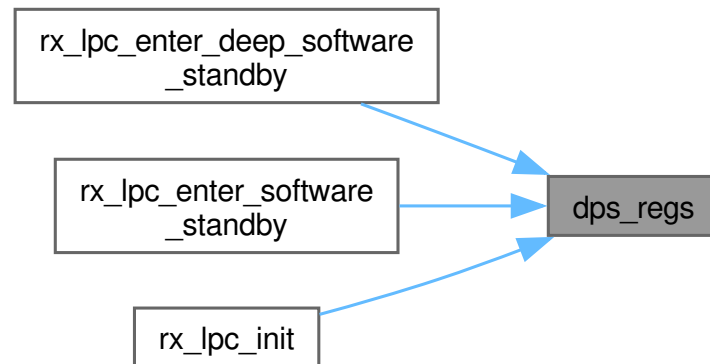
Definition at line 622 of file [rx72n_lpc_regs.h](#).

```
00623 {
00624     return (volatile rx_dps_regs_t*)k_dpsbycr_base_addr;
00625 }
```

References [k_dpsbycr_base_addr](#).

Referenced by [rx_lpc_enter_deep_software_standby\(\)](#), [rx_lpc_enter_software_standby\(\)](#), and [rx_lpc_init\(\)](#).

Here is the caller graph for this function:



8.31.4.2 `dpsbkr()`

```
volatile uint8_t * dpsbkr (
    void ) [inline], [static]
```

Get pointer to Deep Standby Backup RAM.

Returns

Volatile pointer to 32-byte backup RAM array

DPSBKRY (y=0-31) is 32 bytes of RAM retained during deep software standby mode. Use this to preserve critical state across deep sleep.

Warning

Backup RAM is undefined after power-on; initialize before use.

Example - Save/restore state across deep standby:

```
// Before entering deep standby
volatile uint8_t* bkram = dpsbkr();
bkram[0] = magic_byte_1;
bkram[1] = magic_byte_2;
bkram[2] = (uint8_t)(counter » 8);
bkram[3] = (uint8_t)(counter & 0xFF);

// After waking from deep standby
if (bkram[0] == magic_byte_1 && bkram[1] == magic_byte_2) {
    counter = ((uint16_t)bkram[2] « 8) | bkram[3];
}
```

Since

Version 1.0.0

Definition at line 655 of file [rx72n_lpc_regs.h](#).

```
00656 {  
00657     return (volatile uint8_t*)k_dpsbkr_base_addr;  
00658 }
```

References [k_dpsbkr_base_addr](#).

8.31.4.3 opccr_reg()

```
volatile uint8_t * opccr_reg (  
    void ) [inline], [static]
```

Get pointer to OPCCR register.

Returns

Volatile pointer to Operating Power Control Register

OPCCR controls operating power mode for power optimization.

Example - Enter Low-Speed Mode 1:

```
// Switch to LOCO, then enter low-speed mode 1  
*opccr_reg() = k_opccr_opcm_lowspeed1;  
while (*opccr_reg() & k_opccr_opcmsts) { } // Wait for transition
```

Since

Version 1.0.0

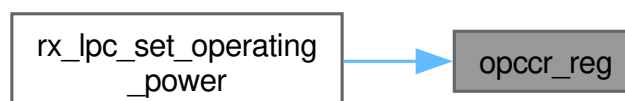
Definition at line 576 of file [rx72n_lpc_regs.h](#).

```
00577 {  
00578     return (volatile uint8_t*)k_opccr_addr;  
00579 }
```

References [k_opccr_addr](#).

Referenced by [rx_lpc_set_operating_power\(\)](#).

Here is the caller graph for this function:



8.31.4.4 rstckcr_reg()

```
volatile uint8_t * rstckcr_reg (
    void ) [inline], [static]
```

Get pointer to RSTCKCR register.

Returns

Volatile pointer to Sleep Mode Return Clock Source Register

RSTCKCR controls automatic clock source switching on sleep mode exit.

Example - Auto-switch to HOCO on wake-up:

```
*rstckcr_reg() = k_rstckcr_rstcken | k_rstckcr_rstcksel_hoco;
```

Since

Version 1.0.0

Definition at line 596 of file [rx72n_lpc_regs.h](#).

```
00597 {
00598     return (volatile uint8_t*)k_rstckcr_addr;
00599 }
```

References [k_rstckcr_addr](#).

8.32 rx72n_lpc_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_lpc_regs.h
00003  * @brief RX72N Low Power Consumption Register Definitions
00004  *
00005  * @details
00006  * Low Power Consumption (LPC) register definitions for operating power control,
00007  * deep software standby mode, and wake-up interrupt configuration on the RX72N
00008  * microcontroller. This header complements rx72n_system_regs.h which contains
00009  * the main standby (SBYCR) and module stop (MSTPCRA-D) registers.
00010  *
00011  * @par Register Groups
00012  * | Address Range | Description |
00013  * |-----|-----|
00014  * | 0x000800A0 - 0x000800A1 | Operating power control (OPCCR/RSTCKCR) |
00015  * | 0x0008C280 - 0x0008C28D | Deep standby control & interrupt |
00016  * | 0x0008C2A0 - 0x0008C2BF | Deep standby backup RAM (32 bytes) |
00017  *
00018  * @par Low Power Modes
00019  * The RX72N supports several low power modes:
00020  * - **Sleep Mode**: CPU stopped, peripherals run (lowest latency wakeup)
00021  * - **All-Module Clock Stop Mode**: CPU and all modules stopped
00022  * - **Software Standby Mode**: Most oscillators stopped, RAM retained
00023  * - **Deep Software Standby Mode**: Lowest power, backup RAM only
00024  *
00025  * @par Operating Power Control Modes
00026  * | Mode | ICLK Max | Description |
00027  * |-----|-----|-----|
00028  * | High-speed (000b) | 240 MHz | Full performance |
00029  * | Low-speed 1 (110b) | 1 MHz | Low power, PLL off |
00030  * | Low-speed 2 (111b) | 264 kHz | Lowest power, HOCO off |
00031  *
00032  * @par NASA Power of 10 Compliance
00033  * - Rule 1: No goto, setjmp, or recursion
```

```

00034 * - Rule 3: No dynamic memory allocation
00035 * - Rule 8: All constants use C23 typed enums
00036 * - Rule 10: Compiles with -Wall -Wextra -Werror
00037 *
00038 * @par Manual References
00039 * - Chapter 11: Low Power Consumption
00040 * - Table 11.1: Low Power Consumption Functions
00041 * - Table 11.2: Mode Transitions
00042 *
00043 * @par Verification Status
00044 * VERIFIED (2026-01-29) - All register addresses verified against
00045 * RX72N Manual Rev.1.20, Chapter 11.
00046 *
00047 * @see rx72n_system_regs.h Main system registers (SBYCR, MSTPCRA-D)
00048 *
00049 * @author Locked, Inc.
00050 * @date 2026-01-29
00051 * @copyright Copyright (c) 2026 Locked Inc.
00052 * SPDX-License-Identifier: MIT
00053 * @since Version 1.0.0
00054 */
00055
00056 #pragma once
00057
00058 #include <stddef.h>
00059 #include <stdint.h>
00060
00061 #ifdef __cplusplus
00062 extern "C" {
00063 #endif
00064
00065 /*
=====
00066 * Address Definitions
00067 *
=====
00068 */
00069
00070 /**
00071 * @enum rx_lpc_addresses_t
00072 * @brief Low Power Consumption register addresses
00073 *
00074 * @details
00075 * Addresses for operating power control and deep standby registers.
00076 * These registers control low power mode behavior and wake-up sources.
00077 *
00078 * @par Memory Map
00079 * | Address | Register | Size | Description |
00080 * |-----|-----|-----|-----|
00081 * | 0x000800A0 | OPCCR | 8-bit | Operating Power Control |
00082 * | 0x000800A1 | RSTCKCR | 8-bit | Sleep Mode Return Clock Switch |
00083 * | 0x0008C280 | DPSBYCR | 8-bit | Deep Standby Control |
00084 * | 0x0008C282-5 | DPSIER0-3 | 8-bit | Deep Standby Int Enable |
00085 * | 0x0008C286-9 | DPSIFR0-3 | 8-bit | Deep Standby Int Flags |
00086 * | 0x0008C28A-D | DPSIEGR0-3 | 8-bit | Deep Standby Int Edge |
00087 * | 0x0008C2A0-BF | DPSBKRy | 8-bit | Deep Standby Backup RAM |
00088 *
00089 * @since Version 1.0.0
00090 */
00091 typedef enum : uintptr_t {
00092     /* Operating Power Control Registers @ 0x000800Ax */
00093     k_opccr_addr = 0x000800A0U, /**< Operating Power Control Register */
00094     k_rstckcr_addr = 0x000800A1U, /**< Sleep Mode Return Clock Switch Register */
00095
00096     /* Deep Standby Registers @ 0x0008C28x */
00097     k_dpsbycr_base_addr = 0x0008C280U, /**< Deep Standby Control base */
00098     k_dpsbycr_addr = 0x0008C280U, /**< Deep Standby Control Register */
00099     k_dpsier0_addr = 0x0008C282U, /**< Deep Standby Interrupt Enable 0 */
00100     k_dpsier1_addr = 0x0008C283U, /**< Deep Standby Interrupt Enable 1 */
00101     k_dpsier2_addr = 0x0008C284U, /**< Deep Standby Interrupt Enable 2 */
00102     k_dpsier3_addr = 0x0008C285U, /**< Deep Standby Interrupt Enable 3 */
00103     k_dpsifr0_addr = 0x0008C286U, /**< Deep Standby Interrupt Flag 0 */
00104     k_dpsifr1_addr = 0x0008C287U, /**< Deep Standby Interrupt Flag 1 */
00105     k_dpsifr2_addr = 0x0008C288U, /**< Deep Standby Interrupt Flag 2 */
00106     k_dpsifr3_addr = 0x0008C289U, /**< Deep Standby Interrupt Flag 3 */
00107     k_dpsiegr0_addr = 0x0008C28AU, /**< Deep Standby Interrupt Edge 0 */
00108     k_dpsiegr1_addr = 0x0008C28BU, /**< Deep Standby Interrupt Edge 1 */
00109     k_dpsiegr2_addr = 0x0008C28CU, /**< Deep Standby Interrupt Edge 2 */
00110     k_dpsiegr3_addr = 0x0008C28DU, /**< Deep Standby Interrupt Edge 3 */
00111
00112     /* Deep Standby Backup RAM @ 0x0008C2A0-0x0008C2BF (32 bytes) */
00113     k_dpsbkr_base_addr = 0x0008C2A0U, /**< Deep Standby Backup RAM base */
00114 } rx_lpc_addresses_t;
00115
00116 /**
00117 * @enum rx_lpc_sizes_t
00118 * @brief Low Power Consumption register and buffer sizes

```

```

00119 *
00120 * @since Version 1.0.0
00121 */
00122 typedef enum : uint8_t {
00123     k_dpsbkr_size = 32U, /**< Deep standby backup RAM size (bytes) */
00124     k_dps_regs_count = 4U, /**< Number of DPSIERx/DPSIFRx/DPSIEGRx registers */
00125 } rx_lpc_sizes_t;
00126
00127 /*
=====
00128 * OPCCR - Operating Power Control Register (@ 0x000800A0)
00129 *
=====
00130 */
00131
00132 /**
00133  * @enum rx_opccr_bits_t
00134  * @brief OPCCR bit definitions
00135  *
00136  * @details
00137  * Operating Power Control Register controls operating power mode for
00138  * normal operation, sleep mode, and all-module clock stop mode.
00139  *
00140  * @par Operating Power Control Modes (OPCM[2:0])
00141  * | Value | Mode | ICLK Max | Notes |
00142  * |-----|-----|-----|-----|
00143  * | 000b | High-speed | 240 MHz | Full performance |
00144  * | 110b | Low-speed 1 | 1 MHz | PLL must be stopped |
00145  * | 111b | Low-speed 2 | 264 kHz | HOCO/PLL must be stopped |
00146  *
00147  * @since Version 1.0.0
00148  */
00149 typedef enum : uint8_t {
00150     /* OPCM[2:0] - Operating Power Control Mode Select (bits 2:0) */
00151     k_opccr_opcm_highspeed = 0x00U, /**< High-speed operating mode (240 MHz max) */
00152     k_opccr_opcm_lowspeed1 = 0x06U, /**< Low-speed operating mode 1 (1 MHz max) */
00153     k_opccr_opcm_lowspeed2 = 0x07U, /**< Low-speed operating mode 2 (264 kHz max) */
00154     k_opccr_opcm_mask = 0x07U, /**< OPCM[2:0] bit mask */
00155
00156     /* OPCMTSF - Operating Power Control Mode Transition Status Flag (bit 4) */
00157     k_opccr_opcmtsf = 0x10U, /**< 1 = Mode transition in progress */
00158 } rx_opccr_bits_t;
00159
00160 /*
=====
00161 * RSTCKCR - Sleep Mode Return Clock Source Switching Register (@ 0x000800A1)
00162 *
=====
00163 */
00164
00165 /**
00166  * @enum rx_rstckcr_bits_t
00167  * @brief RSTCKCR bit definitions
00168  *
00169  * @details
00170  * Controls clock source switching at the time of release from sleep mode.
00171  * When enabled, the selected clock source is automatically started
00172  * upon wake-up from sleep mode.
00173  *
00174  * @par RSTCKSEL[2:0] - Clock Source Select
00175  * | Value | Clock Source |
00176  * |-----|-----|
00177  * | 001b | HOCO (High-speed on-chip) |
00178  * | 010b | Main clock oscillator |
00179  *
00180  * @since Version 1.0.0
00181  */
00182 typedef enum : uint8_t {
00183     /* RSTCKSEL[2:0] - Sleep Mode Return Clock Source Select (bits 2:0) */
00184     k_rstckcr_rstcksel_hoco = 0x01U, /**< HOCO selected for wake-up */
00185     k_rstckcr_rstcksel_main = 0x02U, /**< Main oscillator selected for wake-up */
00186     k_rstckcr_rstcksel_mask = 0x07U, /**< RSTCKSEL[2:0] bit mask */
00187
00188     /* RSTCKEN - Sleep Mode Return Clock Source Switching Enable (bit 7) */
00189     k_rstckcr_rstcken = 0x80U, /**< 1 = Enable clock source switching */
00190 } rx_rstckcr_bits_t;
00191
00192 /*
=====
00193 * DPSBYCR - Deep Standby Control Register (@ 0x0008C280)
00194 *
=====
00195 */
00196
00197 /**
00198  * @enum rx_dpsbycr_bits_t
00199  * @brief DPSBYCR bit definitions

```

```

00200 *
00201 * @details
00202 * Controls deep software standby mode entry and power supply configuration.
00203 *
00204 * @par DEEPCUT[1:0] - Power Supply Control
00205 * | Value | Standby RAM | USB Resume | LVD | Notes |
00206 * |-----|-----|-----|-----|-----|
00207 * | 00b | Powered | Enabled | Active | USB wake-up OK |
00208 * | 01b | Not powered | Disabled | Active | RAM data lost |
00209 * | 11b | Not powered | Disabled | Off | Lowest power |
00210 *
00211 * @note 10b is a prohibited setting for DEEPCUT[1:0].
00212 *
00213 * @since Version 1.0.0
00214 */
00215 typedef enum : uint8_t {
00216     /* DEEPCUT[1:0] - Deep Cut Power Control (bits 1:0) */
00217     k_dpsbycr_deepcut_ram_usb_on = 0x00U, /**< Standby RAM & USB powered */
00218     k_dpsbycr_deepcut_ram_usb_off = 0x01U, /**< Standby RAM & USB not powered */
00219     k_dpsbycr_deepcut_lvd_off = 0x03U, /**< LVD stopped, lowest power */
00220     k_dpsbycr_deepcut_mask = 0x03U, /**< DEEPCUT[1:0] bit mask */
00221
00222     /* IOKEEP - I/O Port Retention (bit 6) */
00223     k_dpsbycr_iokeep = 0x40U, /**< 1 = Retain I/O state after deep standby exit */
00224
00225     /* DPSBY - Deep Software Standby (bit 7) */
00226     k_dpsbycr_dpsby = 0x80U, /**< 1 = Enter deep standby (with SBYCR.SSBY=1) */
00227 } rx_dpsbycr_bits_t;
00228
00229 /*
=====
00230 * DPSIER0 - Deep Standby Interrupt Enable Register 0 (@ 0x0008C282)
00231 *
=====
00232 */
00233
00234 /**
00235 * @enum rx_dpsier0_bits_t
00236 * @brief DPSIER0 bit definitions (IRQ0-7 wake-up enable)
00237 *
00238 * @details
00239 * Enables external IRQ0-DS to IRQ7-DS pins as wake-up sources from
00240 * deep software standby mode.
00241 *
00242 * @since Version 1.0.0
00243 */
00244 typedef enum : uint8_t {
00245     k_dpsier0_dirq0e = 0x01U, /**< IRQ0-DS pin enable */
00246     k_dpsier0_dirq1e = 0x02U, /**< IRQ1-DS pin enable */
00247     k_dpsier0_dirq2e = 0x04U, /**< IRQ2-DS pin enable */
00248     k_dpsier0_dirq3e = 0x08U, /**< IRQ3-DS pin enable */
00249     k_dpsier0_dirq4e = 0x10U, /**< IRQ4-DS pin enable */
00250     k_dpsier0_dirq5e = 0x20U, /**< IRQ5-DS pin enable */
00251     k_dpsier0_dirq6e = 0x40U, /**< IRQ6-DS pin enable */
00252     k_dpsier0_dirq7e = 0x80U, /**< IRQ7-DS pin enable */
00253 } rx_dpsier0_bits_t;
00254
00255 /*
=====
00256 * DPSIER1 - Deep Standby Interrupt Enable Register 1 (@ 0x0008C283)
00257 *
=====
00258 */
00259
00260 /**
00261 * @enum rx_dpsier1_bits_t
00262 * @brief DPSIER1 bit definitions (IRQ8-15 wake-up enable)
00263 *
00264 * @details
00265 * Enables external IRQ8-DS to IRQ15-DS pins as wake-up sources from
00266 * deep software standby mode.
00267 *
00268 * @since Version 1.0.0
00269 */
00270 typedef enum : uint8_t {
00271     k_dpsier1_dirq8e = 0x01U, /**< IRQ8-DS pin enable */
00272     k_dpsier1_dirq9e = 0x02U, /**< IRQ9-DS pin enable */
00273     k_dpsier1_dirq10e = 0x04U, /**< IRQ10-DS pin enable */
00274     k_dpsier1_dirq11e = 0x08U, /**< IRQ11-DS pin enable */
00275     k_dpsier1_dirq12e = 0x10U, /**< IRQ12-DS pin enable */
00276     k_dpsier1_dirq13e = 0x20U, /**< IRQ13-DS pin enable */
00277     k_dpsier1_dirq14e = 0x40U, /**< IRQ14-DS pin enable */
00278     k_dpsier1_dirq15e = 0x80U, /**< IRQ15-DS pin enable */
00279 } rx_dpsier1_bits_t;
00280
00281 /*
=====

```

```

00282 * DPSIER2 - Deep Standby Interrupt Enable Register 2 (@ 0x0008C284)
00283 *
=====
00284 */
00285
00286 /**
00287  * @enum rx_dpsier2_bits_t
00288  * @brief DPSIER2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up enable)
00289  *
00290  * @details
00291  * Enables various peripheral wake-up sources from deep software standby mode.
00292  *
00293  * @since Version 1.0.0
00294  */
00295 typedef enum : uint8_t {
00296     k_dpsier2_dlvdl1e = 0x01U, /**< LVD1 (Voltage Monitor 1) enable */
00297     k_dpsier2_dlvdl2e = 0x02U, /**< LVD2 (Voltage Monitor 2) enable */
00298     k_dpsier2_drtciie = 0x04U, /**< RTC periodic interrupt enable */
00299     k_dpsier2_drtcaie = 0x08U, /**< RTC alarm interrupt enable */
00300     k_dpsier2_dnmie   = 0x10U, /**< NMI pin enable (write-once) */
00301     k_dpsier2_driicdie = 0x20U, /**< SDA2-DS (I2C data) enable */
00302     k_dpsier2_driiccie = 0x40U, /**< SCL2-DS (I2C clock) enable */
00303     k_dpsier2_dusbie   = 0x80U, /**< USB suspend/resume enable */
00304 } rx_dpsier2_bits_t;
00305
00306 /*
=====
00307 * DPSIER3 - Deep Standby Interrupt Enable Register 3 (@ 0x0008C285)
00308 *
=====
00309 */
00310
00311 /**
00312  * @enum rx_dpsier3_bits_t
00313  * @brief DPSIER3 bit definitions (CAN wake-up enable)
00314  *
00315  * @details
00316  * Enables CAN receive pin as wake-up source from deep software standby mode.
00317  *
00318  * @since Version 1.0.0
00319  */
00320 typedef enum : uint8_t {
00321     k_dpsier3_dcanie = 0x01U, /**< CRX1-DS (CAN1 receive) enable */
00322 } rx_dpsier3_bits_t;
00323
00324 /*
=====
00325 * DPSIFR0 - Deep Standby Interrupt Flag Register 0 (@ 0x0008C286)
00326 *
=====
00327 */
00328
00329 /**
00330  * @enum rx_dpsifr0_bits_t
00331  * @brief DPSIFR0 bit definitions (IRQ0-7 wake-up flags)
00332  *
00333  * @details
00334  * Indicates which IRQ0-DS to IRQ7-DS pin triggered wake-up from
00335  * deep software standby mode. Write 0 after reading 1 to clear.
00336  *
00337  * @since Version 1.0.0
00338  */
00339 typedef enum : uint8_t {
00340     k_dpsifr0_dirq0f = 0x01U, /**< IRQ0-DS wake-up flag */
00341     k_dpsifr0_dirq1f = 0x02U, /**< IRQ1-DS wake-up flag */
00342     k_dpsifr0_dirq2f = 0x04U, /**< IRQ2-DS wake-up flag */
00343     k_dpsifr0_dirq3f = 0x08U, /**< IRQ3-DS wake-up flag */
00344     k_dpsifr0_dirq4f = 0x10U, /**< IRQ4-DS wake-up flag */
00345     k_dpsifr0_dirq5f = 0x20U, /**< IRQ5-DS wake-up flag */
00346     k_dpsifr0_dirq6f = 0x40U, /**< IRQ6-DS wake-up flag */
00347     k_dpsifr0_dirq7f = 0x80U, /**< IRQ7-DS wake-up flag */
00348 } rx_dpsifr0_bits_t;
00349
00350 /*
=====
00351 * DPSIFR1 - Deep Standby Interrupt Flag Register 1 (@ 0x0008C287)
00352 *
=====
00353 */
00354
00355 /**
00356  * @enum rx_dpsifr1_bits_t
00357  * @brief DPSIFR1 bit definitions (IRQ8-15 wake-up flags)
00358  *
00359  * @details
00360  * Indicates which IRQ8-DS to IRQ15-DS pin triggered wake-up from
00361  * deep software standby mode. Write 0 after reading 1 to clear.

```



```

00362 *
00363 * @since Version 1.0.0
00364 */
00365 typedef enum : uint8_t {
00366     k_dpsifr1_dirq8f = 0x01U, /**< IRQ8-DS wake-up flag */
00367     k_dpsifr1_dirq9f = 0x02U, /**< IRQ9-DS wake-up flag */
00368     k_dpsifr1_dirq10f = 0x04U, /**< IRQ10-DS wake-up flag */
00369     k_dpsifr1_dirq11f = 0x08U, /**< IRQ11-DS wake-up flag */
00370     k_dpsifr1_dirq12f = 0x10U, /**< IRQ12-DS wake-up flag */
00371     k_dpsifr1_dirq13f = 0x20U, /**< IRQ13-DS wake-up flag */
00372     k_dpsifr1_dirq14f = 0x40U, /**< IRQ14-DS wake-up flag */
00373     k_dpsifr1_dirq15f = 0x80U, /**< IRQ15-DS wake-up flag */
00374 } rx_dpsifr1_bits_t;
00375
00376 /*
=====
00377 * DPSIFR2 - Deep Standby Interrupt Flag Register 2 (@ 0x0008C288)
00378 *
=====
00379 */
00380
00381 /**
00382 * @enum rx_dpsifr2_bits_t
00383 * @brief DPSIFR2 bit definitions (LVD, RTC, NMI, I2C, USB wake-up flags)
00384 *
00385 * @details
00386 * Indicates which peripheral triggered wake-up from deep software
00387 * standby mode. Write 0 after reading 1 to clear.
00388 *
00389 * @since Version 1.0.0
00390 */
00391 typedef enum : uint8_t {
00392     k_dpsifr2_dlvdlif = 0x01U, /**< LVD1 wake-up flag */
00393     k_dpsifr2_dlvdlif = 0x02U, /**< LVD2 wake-up flag */
00394     k_dpsifr2_drtciif = 0x04U, /**< RTC periodic interrupt flag */
00395     k_dpsifr2_drtcaif = 0x08U, /**< RTC alarm interrupt flag */
00396     k_dpsifr2_dnmif = 0x10U, /**< NMI wake-up flag */
00397     k_dpsifr2_driicdif = 0x20U, /**< SDA2-DS wake-up flag */
00398     k_dpsifr2_driicdif = 0x40U, /**< SCL2-DS wake-up flag */
00399     k_dpsifr2_dusbif = 0x80U, /**< USB suspend/resume flag */
00400 } rx_dpsifr2_bits_t;
00401
00402 /*
=====
00403 * DPSIFR3 - Deep Standby Interrupt Flag Register 3 (@ 0x0008C289)
00404 *
=====
00405 */
00406
00407 /**
00408 * @enum rx_dpsifr3_bits_t
00409 * @brief DPSIFR3 bit definitions (CAN wake-up flag)
00410 *
00411 * @details
00412 * Indicates CAN receive triggered wake-up from deep software standby mode.
00413 * Write 0 after reading 1 to clear.
00414 *
00415 * @since Version 1.0.0
00416 */
00417 typedef enum : uint8_t {
00418     k_dpsifr3_dcanif = 0x01U, /**< CRX1-DS wake-up flag */
00419 } rx_dpsifr3_bits_t;
00420
00421 /*
=====
00422 * DPSIEGR0 - Deep Standby Interrupt Edge Register 0 (@ 0x0008C28A)
00423 *
=====
00424 */
00425
00426 /**
00427 * @enum rx_dpsiegr0_bits_t
00428 * @brief DPSIEGR0 bit definitions (IRQ0-7 edge select)
00429 *
00430 * @details
00431 * Selects edge polarity for IRQ0-DS to IRQ7-DS wake-up detection.
00432 * 0 = Falling edge, 1 = Rising edge.
00433 *
00434 * @since Version 1.0.0
00435 */
00436 typedef enum : uint8_t {
00437     k_dpsiegr0_dirq0eg = 0x01U, /**< IRQ0-DS edge select */
00438     k_dpsiegr0_dirq1eg = 0x02U, /**< IRQ1-DS edge select */
00439     k_dpsiegr0_dirq2eg = 0x04U, /**< IRQ2-DS edge select */
00440     k_dpsiegr0_dirq3eg = 0x08U, /**< IRQ3-DS edge select */
00441     k_dpsiegr0_dirq4eg = 0x10U, /**< IRQ4-DS edge select */
00442     k_dpsiegr0_dirq5eg = 0x20U, /**< IRQ5-DS edge select */

```

```

00443 k_dpsiegr0_dirq6eg = 0x40U, /**< IRQ6-DS edge select */
00444 k_dpsiegr0_dirq7eg = 0x80U, /**< IRQ7-DS edge select */
00445 } rx_dpsiegr0_bits_t;
00446
00447 /*
=====
00448 * DPSIEGR1 - Deep Standby Interrupt Edge Register 1 (@ 0x0008C28B)
00449 *
=====
00450 */
00451
00452 /**
00453 * @enum rx_dpsiegr1_bits_t
00454 * @brief DPSIEGR1 bit definitions (IRQ8-15 edge select)
00455 *
00456 * @details
00457 * Selects edge polarity for IRQ8-DS to IRQ15-DS wake-up detection.
00458 * 0 = Falling edge, 1 = Rising edge.
00459 *
00460 * @since Version 1.0.0
00461 */
00462 typedef enum : uint8_t {
00463 k_dpsiegr1_dirq8eg = 0x01U, /**< IRQ8-DS edge select */
00464 k_dpsiegr1_dirq9eg = 0x02U, /**< IRQ9-DS edge select */
00465 k_dpsiegr1_dirq10eg = 0x04U, /**< IRQ10-DS edge select */
00466 k_dpsiegr1_dirq11eg = 0x08U, /**< IRQ11-DS edge select */
00467 k_dpsiegr1_dirq12eg = 0x10U, /**< IRQ12-DS edge select */
00468 k_dpsiegr1_dirq13eg = 0x20U, /**< IRQ13-DS edge select */
00469 k_dpsiegr1_dirq14eg = 0x40U, /**< IRQ14-DS edge select */
00470 k_dpsiegr1_dirq15eg = 0x80U, /**< IRQ15-DS edge select */
00471 } rx_dpsiegr1_bits_t;
00472
00473 /*
=====
00474 * DPSIEGR2 - Deep Standby Interrupt Edge Register 2 (@ 0x0008C28C)
00475 *
=====
00476 */
00477
00478 /**
00479 * @enum rx_dpsiegr2_bits_t
00480 * @brief DPSIEGR2 bit definitions (LVD, NMI, I2C edge select)
00481 *
00482 * @details
00483 * Selects edge polarity for various peripheral wake-up detection.
00484 * For LVD: 0 = VCC fall (VCC < Vdet), 1 = VCC rise (VCC >= Vdet).
00485 * For pins: 0 = Falling edge, 1 = Rising edge.
00486 *
00487 * @since Version 1.0.0
00488 */
00489 typedef enum : uint8_t {
00490 k_dpsiegr2_dlv1eg = 0x01U, /**< LVD1 edge: 0=fall, 1=rise */
00491 k_dpsiegr2_dlv2eg = 0x02U, /**< LVD2 edge: 0=fall, 1=rise */
00492 k_dpsiegr2_dnmieg = 0x10U, /**< NMI edge select */
00493 k_dpsiegr2_driicdeg = 0x20U, /**< SDA2-DS edge select */
00494 k_dpsiegr2_driicceg = 0x40U, /**< SCL2-DS edge select */
00495 } rx_dpsiegr2_bits_t;
00496
00497 /*
=====
00498 * DPSIEGR3 - Deep Standby Interrupt Edge Register 3 (@ 0x0008C28D)
00499 *
=====
00500 */
00501
00502 /**
00503 * @enum rx_dpsiegr3_bits_t
00504 * @brief DPSIEGR3 bit definitions (CAN edge select)
00505 *
00506 * @details
00507 * Selects edge polarity for CAN receive wake-up detection.
00508 * 0 = Falling edge, 1 = Rising edge.
00509 *
00510 * @since Version 1.0.0
00511 */
00512 typedef enum : uint8_t {
00513 k_dpsiegr3_dcanieg = 0x01U, /**< CRX1-DS edge select */
00514 } rx_dpsiegr3_bits_t;
00515
00516 /*
=====
00517 * Deep Standby Register Structure (@ 0x0008C280-0x0008C28D)
00518 *
=====
00519 */
00520
00521 /**

```

```

00522 * @struct rx_dps_regs_t
00523 * @brief Deep Standby Register Block (non-contiguous)
00524 *
00525 * @details
00526 * The deep standby registers are NOT all contiguous in memory:
00527 * - 0x0008C280: DPSBYCR
00528 * - 0x0008C281: Reserved
00529 * - 0x0008C282-0x0008C28D: DPSIER, DPSIFR, DPSIEGR registers
00530 *
00531 * This structure covers the contiguous portion from 0x0008C280 onwards.
00532 *
00533 * @invariant sizeof(rx_dps_regs_t) == 14 bytes
00534 *
00535 * @since Version 1.0.0
00536 */
00537 typedef struct __attribute__((packed)) {
00538     volatile uint8_t dpsbycr; /**< Deep Standby Control @ 0x00 (0x0008C280) */
00539     uint8_t reserved1; /**< Reserved @ 0x01 */
00540     volatile uint8_t dpsier0; /**< Deep Standby Int Enable 0 @ 0x02 (0x0008C282) */
00541     volatile uint8_t dpsier1; /**< Deep Standby Int Enable 1 @ 0x03 (0x0008C283) */
00542     volatile uint8_t dpsier2; /**< Deep Standby Int Enable 2 @ 0x04 (0x0008C284) */
00543     volatile uint8_t dpsier3; /**< Deep Standby Int Enable 3 @ 0x05 (0x0008C285) */
00544     volatile uint8_t dpsifr0; /**< Deep Standby Int Flag 0 @ 0x06 (0x0008C286) */
00545     volatile uint8_t dpsifr1; /**< Deep Standby Int Flag 1 @ 0x07 (0x0008C287) */
00546     volatile uint8_t dpsifr2; /**< Deep Standby Int Flag 2 @ 0x08 (0x0008C288) */
00547     volatile uint8_t dpsifr3; /**< Deep Standby Int Flag 3 @ 0x09 (0x0008C289) */
00548     volatile uint8_t dpsiegr0; /**< Deep Standby Int Edge 0 @ 0x0A (0x0008C28A) */
00549     volatile uint8_t dpsiegr1; /**< Deep Standby Int Edge 1 @ 0x0B (0x0008C28B) */
00550     volatile uint8_t dpsiegr2; /**< Deep Standby Int Edge 2 @ 0x0C (0x0008C28C) */
00551     volatile uint8_t dpsiegr3; /**< Deep Standby Int Edge 3 @ 0x0D (0x0008C28D) */
00552 } rx_dps_regs_t;
00553
00554 /*
=====
00555 * Register Accessor Functions
00556 *
=====
00557 */
00558
00559 /**
00560 * @brief Get pointer to OPCCR register
00561 *
00562 * @return Volatile pointer to Operating Power Control Register
00563 *
00564 * @details
00565 * OPCCR controls operating power mode for power optimization.
00566 *
00567 * @par Example - Enter Low-Speed Mode 1:
00568 * @code
00569 * // Switch to LOCO, then enter low-speed mode 1
00570 * *opccr_reg() = k_opccr_opcm_lowspeed1;
00571 * while (*opccr_reg() & k_opccr_opcmstf) { } // Wait for transition
00572 * @endcode
00573 *
00574 * @since Version 1.0.0
00575 */
00576 static inline volatile uint8_t* opccr_reg(void)
00577 {
00578     return (volatile uint8_t*)k_opccr_addr;
00579 }
00580
00581 /**
00582 * @brief Get pointer to RSTCKCR register
00583 *
00584 * @return Volatile pointer to Sleep Mode Return Clock Source Register
00585 *
00586 * @details
00587 * RSTCKCR controls automatic clock source switching on sleep mode exit.
00588 *
00589 * @par Example - Auto-switch to HOCO on wake-up:
00590 * @code
00591 * *rstckcr_reg() = k_rstckcr_rstcken | k_rstckcr_rstcksel_hoco;
00592 * @endcode
00593 *
00594 * @since Version 1.0.0
00595 */
00596 static inline volatile uint8_t* rstckcr_reg(void)
00597 {
00598     return (volatile uint8_t*)k_rstckcr_addr;
00599 }
00600
00601 /**
00602 * @brief Get pointer to Deep Standby registers
00603 *
00604 * @return Volatile pointer to Deep Standby register structure
00605 *
00606 * @details

```

```

00607 * Provides access to DPSBYCR, DPSIERx, DPSIFRx, and DPSIEGRx registers.
00608 *
00609 * @par Example - Configure IRQ0 as wake-up source:
00610 * @code
00611 * volatile rx_dps_regs_t* dps = dps_regs();
00612 *
00613 * dps->dpsier0 = k_dpsier0_dirq0e;    // Enable IRQ0-DS wake-up
00614 * dps->dpsiegr0 = k_dpsiegr0_dirq0eg; // Rising edge trigger
00615 * dps->dpsifr0 = 0x00;                // Clear flags before entering
00616 *
00617 * dps->dpsbycr = k_dpsbycr_dpsby;     // Enable deep standby
00618 * @endcode
00619 *
00620 * @since Version 1.0.0
00621 */
00622 static inline volatile rx_dps_regs_t* dps_regs(void)
00623 {
00624     return (volatile rx_dps_regs_t*)k_dpsbycr_base_addr;
00625 }
00626
00627 /**
00628 * @brief Get pointer to Deep Standby Backup RAM
00629 *
00630 * @return Volatile pointer to 32-byte backup RAM array
00631 *
00632 * @details
00633 * DPSBKRY (y=0-31) is 32 bytes of RAM retained during deep software
00634 * standby mode. Use this to preserve critical state across deep sleep.
00635 *
00636 * @warning Backup RAM is undefined after power-on; initialize before use.
00637 *
00638 * @par Example - Save/restore state across deep standby:
00639 * @code
00640 * // Before entering deep standby
00641 * volatile uint8_t* bkram = dpsbkr();
00642 * bkram[0] = magic_byte_1;
00643 * bkram[1] = magic_byte_2;
00644 * bkram[2] = (uint8_t)(counter » 8);
00645 * bkram[3] = (uint8_t)(counter & 0xFF);
00646 *
00647 * // After waking from deep standby
00648 * if (bkram[0] == magic_byte_1 && bkram[1] == magic_byte_2) {
00649 *     counter = ((uint16_t)bkram[2] « 8) | bkram[3];
00650 * }
00651 * @endcode
00652 *
00653 * @since Version 1.0.0
00654 */
00655 static inline volatile uint8_t* dpsbkr(void)
00656 {
00657     return (volatile uint8_t*)k_dpsbkr_base_addr;
00658 }
00659
00660 /*
=====
00661 * Static Assertions - Verify Register Layout at Compile Time
00662 *
=====
00663 */
00664
00665 /* Verify OPCCR/RSTCKCR addresses (Ch11 Section 11.2.6/11.2.7) */
00666 static_assert(k_opccr_addr == 0x000800A0, "OPCCR address incorrect (expected 0x000800A0)");
00667 static_assert(k_rstckcr_addr == 0x000800A1, "RSTCKCR address incorrect (expected 0x000800A1)");
00668
00669 /* Verify Deep Standby Control Register address (Ch11 Section 11.2.8) */
00670 static_assert(k_dpsbycr_addr == 0x0008C280, "DPSBYCR address incorrect (expected 0x0008C280)");
00671
00672 /* Verify DPSIER addresses (Ch11 Sections 11.2.9-11.2.12) */
00673 static_assert(k_dpsier0_addr == 0x0008C282, "DPSIER0 address incorrect (expected 0x0008C282)");
00674 static_assert(k_dpsier1_addr == 0x0008C283, "DPSIER1 address incorrect (expected 0x0008C283)");
00675 static_assert(k_dpsier2_addr == 0x0008C284, "DPSIER2 address incorrect (expected 0x0008C284)");
00676 static_assert(k_dpsier3_addr == 0x0008C285, "DPSIER3 address incorrect (expected 0x0008C285)");
00677
00678 /* Verify DPSIFR addresses (Ch11 Sections 11.2.13-11.2.16) */
00679 static_assert(k_dpsifr0_addr == 0x0008C286, "DPSIFR0 address incorrect (expected 0x0008C286)");
00680 static_assert(k_dpsifr1_addr == 0x0008C287, "DPSIFR1 address incorrect (expected 0x0008C287)");
00681 static_assert(k_dpsifr2_addr == 0x0008C288, "DPSIFR2 address incorrect (expected 0x0008C288)");
00682 static_assert(k_dpsifr3_addr == 0x0008C289, "DPSIFR3 address incorrect (expected 0x0008C289)");
00683
00684 /* Verify DPSIEGR addresses (Ch11 Sections 11.2.17-11.2.20) */
00685 static_assert(k_dpsiegr0_addr == 0x0008C28A, "DPSIEGR0 address incorrect (expected 0x0008C28A)");
00686 static_assert(k_dpsiegr1_addr == 0x0008C28B, "DPSIEGR1 address incorrect (expected 0x0008C28B)");
00687 static_assert(k_dpsiegr2_addr == 0x0008C28C, "DPSIEGR2 address incorrect (expected 0x0008C28C)");
00688 static_assert(k_dpsiegr3_addr == 0x0008C28D, "DPSIEGR3 address incorrect (expected 0x0008C28D)");
00689
00690 /* Verify Deep Standby Backup RAM address (Ch11 Section 11.2.21) */
00691 static_assert(k_dpsbkr_base_addr == 0x0008C2A0,

```

```

00692     "DPSBKR base address incorrect (expected 0x0008C2A0)");
00693
00694 /* Verify structure size and offsets */
00695 static_assert(sizeof(rx_dps_regs_t) == 14, "rx_dps_regs_t size incorrect (expected 14 bytes)");
00696 static_assert(offsetof(rx_dps_regs_t, dpsbycr) == 0, "DPSBYCR offset incorrect (expected 0)");
00697 static_assert(offsetof(rx_dps_regs_t, dpsier0) == 2, "DPSIER0 offset incorrect (expected 2)");
00698 static_assert(offsetof(rx_dps_regs_t, dpsifr0) == 6, "DPSIFR0 offset incorrect (expected 6)");
00699 static_assert(offsetof(rx_dps_regs_t, dpsiegr0) == 10, "DPSIEGR0 offset incorrect (expected 10)");
00700 static_assert(offsetof(rx_dps_regs_t, dpsiegr3) == 13, "DPSIEGR3 offset incorrect (expected 13)");
00701
00702 /* Verify address relationships */
00703 static_assert((k_dpsier0_addr - k_dpsbycr_addr) == 2, "DPSIER0 must be at offset 2 from DPSBYCR");
00704 static_assert((k_dpsifr0_addr - k_dpsbycr_addr) == 6, "DPSIFR0 must be at offset 6 from DPSBYCR");
00705 static_assert((k_dpsiegr0_addr - k_dpsbycr_addr) == 10,
00706               "DPSIEGR0 must be at offset 10 from DPSBYCR");
00707 static_assert((k_dpsbkr_base_addr - k_dpsbycr_addr) == 0x20,
00708               "DPSBKR must be at offset 0x20 from DPSBYCR");
00709
00710 #ifdef __cplusplus
00711 }
00712 #endif

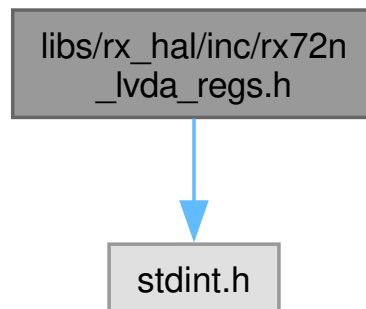
```

8.33 libs/rx_hal/inc/rx72n_lvda_regs.h File Reference

RX72N Voltage Detection Circuit (LVDA) Register Definitions.

#include <stdint.h>

Include dependency graph for rx72n_lvda_regs.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_lvda_addresses_t` : `uintptr_t` {
`k_lvd1cr1_addr` = 0x000800E0 , `k_lvd1sr_addr` = 0x000800E1 , `k_lvd2cr1_addr` = 0x000800E2
, `k_lvd2sr_addr` = 0x000800E3 ,
`k_lvcmpr_addr` = 0x0008C297 , `k_lvd1vr_addr` = 0x0008C298 , `k_lvd1cr0_addr` = 0x0008C29A
, `k_lvd2cr0_addr` = 0x0008C29B }
LVDA register addresses.

- enum `rx_lvd_cr1_bits_t` : `uint8_t` {
`k_lvd_idtsel_rise` = 0x00 , `k_lvd_idtsel_drop` = 0x01 , `k_lvd_idtsel_both` = 0x02 ,
`k_lvd_irqsel_nmi` = (0 << 2) ,
`k_lvd_irqsel_maskable` = (1 << 2) , `k_lvd_idtsel_mask` = 0x03 , `k_lvd_irqsel_mask` = 0x04 }
LVDA register bits.

LVD1CR1/LVD2CR1 register bit definitions.

- enum `rx_lvd_sr_bits_t` : `uint8_t` {
`k_lvd_det_clear` = 0x00 , `k_lvd_det_detected` = 0x01 , `k_lvd_mon_below` = 0x00 ,
`k_lvd_mon_above` = 0x02 ,
`k_lvd_det_mask` = 0x01 , `k_lvd_mon_mask` = 0x02 }

LVD1SR/LVD2SR register bit definitions.

- enum `rx_lvcmpr_bits_t` : `uint8_t` {
`k_lvd1e_enable` = (1 << 5) , `k_lvd1e_disable` = (0 << 5) , `k_lvd2e_enable` = (1 << 6) ,
`k_lvd2e_disable` = (0 << 6) ,
`k_lvd1e_mask` = 0x20 , `k_lvd2e_mask` = 0x40 }

LVCMPCR register bit definitions.

- enum `rx_lvdlvlr_bits_t` : `uint8_t` {
`k_lvd1lvl_2v99` = 0x09 , `k_lvd1lvl_2v92` = 0x0A , `k_lvd1lvl_2v85` = 0x0B , `k_lvd2lvl_2v99` = 0x90 ,
`k_lvd2lvl_2v92` = 0xA0 , `k_lvd2lvl_2v85` = 0xB0 , `k_lvd1lvl_mask` = 0x0F , `k_lvd2lvl_mask` = 0xF0 ,
`k_lvdlvlr_reset_value` = 0xBB }

LVDLVLR register bit definitions.

- enum `rx_lvd_cr0_bits_t` : `uint8_t` {
`k_lvd_rie_disable` = (0 << 0) , `k_lvd_rie_enable` = (1 << 0) , `k_lvd_dfdis_filter_on` = (0 << 1) ,
`k_lvd_dfdis_filter_off` = (1 << 1) ,
`k_lvd_cmpe_disable` = (0 << 2) , `k_lvd_cmpe_enable` = (1 << 2) , `k_lvd_fsamp_loco_div2` = (0 << 4) ,
`k_lvd_fsamp_loco_div4` = (1 << 4) ,
`k_lvd_fsamp_loco_div8` = (2 << 4) , `k_lvd_fsamp_loco_div16` = (3 << 4) , `k_lvd_ri_interrupt` = (0 << 6) ,
`k_lvd_ri_reset` = (1 << 6) ,
`k_lvd_rn_vcc_above` = (0 << 7) , `k_lvd_rn_after_assert` = (1 << 7) , `k_lvd_rie_mask` = 0x01 ,
`k_lvd_dfdis_mask` = 0x02 ,
`k_lvd_cmpe_mask` = 0x04 , `k_lvd_fsamp_mask` = 0x30 , `k_lvd_ri_mask` = 0x40 ,
`k_lvd_rn_mask` = 0x80 }

LVD1CR0/LVD2CR0 register bit definitions.

- enum `rx_lvd_interrupts_t` : `uint8_t` { `k_lvd1_irq` = 88 , `k_lvd2_irq` = 89 }

LVD interrupt vector numbers.

Functions

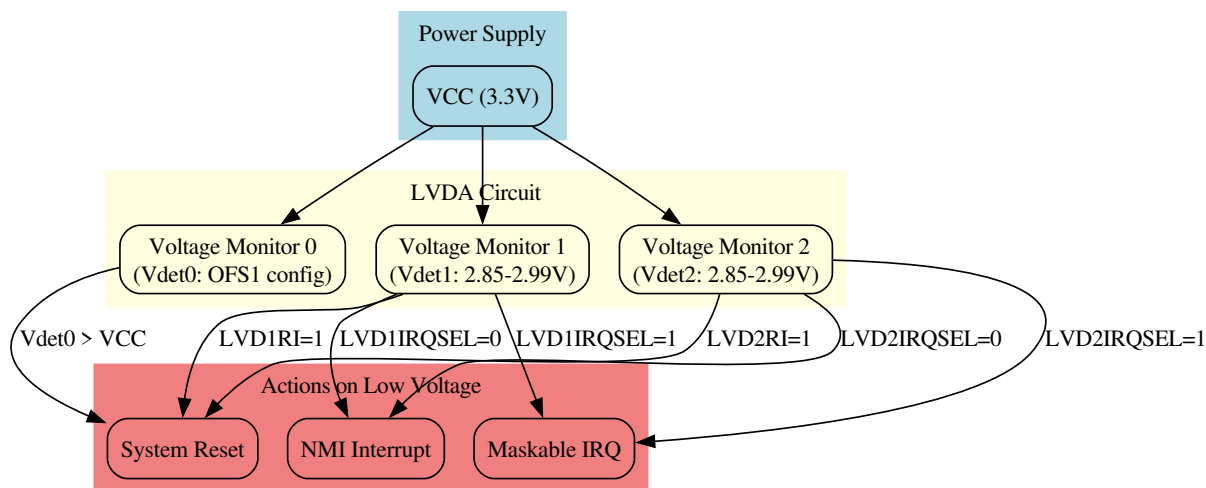
- static volatile `uint8_t` * `lvd1cr1_reg` (void)
Get pointer to LVD1CR1 register.
- static volatile `uint8_t` * `lvd1sr_reg` (void)
Get pointer to LVD1SR register.
- static volatile `uint8_t` * `lvd2cr1_reg` (void)
Get pointer to LVD2CR1 register.
- static volatile `uint8_t` * `lvd2sr_reg` (void)
Get pointer to LVD2SR register.
- static volatile `uint8_t` * `lvcmpr_reg` (void)
Get pointer to LVCMPCR register.
- static volatile `uint8_t` * `lvdlvlr_reg` (void)
Get pointer to LVDLVLR register.
- static volatile `uint8_t` * `lvd1cr0_reg` (void)
Get pointer to LVD1CR0 register.
- static volatile `uint8_t` * `lvd2cr0_reg` (void)
Get pointer to LVD2CR0 register.

8.33.1 Detailed Description

RX72N Voltage Detection Circuit (LVDA) Register Definitions.

Register definitions for the LVDA module which provides brownout protection and voltage monitoring for safety-critical motor control. The LVDA can trigger resets or interrupts when VCC drops below configurable thresholds.

Voltage Monitoring System



STAR Project LVDA Usage

Monitor	Threshold	Action	Use Case
Vdet0	OFS1 conf	Reset	Emergency brownout protection
Vdet1	2.92V	Interrupt	Low voltage warning
Vdet2	2.85V	Reset	Critical brownout shutdown

Voltage Detection Levels

Level	Voltage (typ)	Hysteresis	Notes
Vdet1↔_1	2.99V	~50mV	Early warning
Vdet1↔_2	2.92V	~50mV	Low voltage threshold
Vdet1↔_3	2.85V	~50mV	Critical threshold

Hardware Requirements

Parameter	Value	Notes
LVD1/2 Regs	0x000800E0	4-byte block
LVCMPCR etc	0x0008C297	5-byte block (non-contiguous)
Enable Time	td(E-A)	Wait after enabling LVDnE
Clock	LOCO (240 kHz)	For digital filter

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops in register definitions)
- Rule 3: No dynamic memory allocation
- Rule 4: All accessor functions are single-statement
- Rule 5: N/A (hardware layer)
- Rule 6: Minimal scope - inline accessors only
- Rule 7: N/A (no return values to check)
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

Module Dependencies

- `<stdint.h>` - Fixed-width integer types

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 8: Voltage Detection Circuit (LVDA), pages 316-334

See also

[rx72n_regs.h](#) Main register include file

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Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_lvda_regs.h](#).

8.33.2 Enumeration Type Documentation

8.33.2.1 rx`lvcmpr`bits``t`

enum [rx_lvcmpr_bits_t](#) : uint8_t

LVCMPCR register bit definitions.

Enable bits for voltage detection circuits 1 and 2.

Note

VCC must be at least 80mV above detection level when enabling

Since

Version 1.0.0

Enumerator

k_lvd1e_enable	Enable voltage detection 1 circuit
k_lvd1e_disable	Disable voltage detection 1 circuit
k_lvd2e_enable	Enable voltage detection 2 circuit
k_lvd2e_disable	Disable voltage detection 2 circuit
k_lvd1e_mask	LVD1E field mask (bit 5)
k_lvd2e_mask	LVD2E field mask (bit 6)

Definition at line 392 of file [rx72n_lvda_regs.h](#).

```

00392         : uint8_t {
00393     k_lvd1e_enable = (1 « 5), /**< Enable voltage detection 1 circuit */
00394     k_lvd1e_disable = (0 « 5), /**< Disable voltage detection 1 circuit */
00395     k_lvd2e_enable = (1 « 6), /**< Enable voltage detection 2 circuit */
00396     k_lvd2e_disable = (0 « 6), /**< Disable voltage detection 2 circuit */
00397
00398     /* Field masks */
00399     k_lvd1e_mask = 0x20, /**< LVD1E field mask (bit 5) */
00400     k_lvd2e_mask = 0x40, /**< LVD2E field mask (bit 6) */
00401 } rx_lvcmpr_bits_t;
```

8.33.2.2 rx`lvd`cr0`bits``t`

enum [rx_lvd_cr0_bits_t](#) : uint8_t

LVD1CR0/LVD2CR0 register bit definitions.

Control bits for interrupt/reset enable, digital filter, and mode selection.

Digital Filter Sampling Clock (FSAMP)

Value	Clock	Sample Time
00	LOCO/2 (120 kHz)	~17 us
01	LOCO/4 (60 kHz)	~33 us
10	LOCO/8 (30 kHz)	~67 us
11	LOCO/16 (15 kHz)	~133 us

Since

Version 1.0.0

Enumerator

k_lvd_rie_disable	Disable interrupt/reset
k_lvd_rie_enable	Enable interrupt/reset
k_lvd_dfdis_filter_on	Digital filter enabled
k_lvd_dfdis_filter_off	Digital filter disabled
k_lvd_cmpe_disable	Output disabled
k_lvd_cmpe_enable	Output enabled
k_lvd_fsamp_loco_div2	LOCO/2 (120 kHz)
k_lvd_fsamp_loco_div4	LOCO/4 (60 kHz)
k_lvd_fsamp_loco_div8	LOCO/8 (30 kHz)
k_lvd_fsamp_loco_div16	LOCO/16 (15 kHz)
k_lvd_ri_interrupt	Generate interrupt on Vdet passage
k_lvd_ri_reset	Generate reset when voltage falls below Vdet
k_lvd_rn_vcc_above	Negate after VCC > Vdet detected
k_lvd_rn_after_assert	Negate after stabilization time from assert
k_lvd_rie_mask	RIE field mask (bit 0)
k_lvd_dfdis_mask	DFDIS field mask (bit 1)
k_lvd_cmpe_mask	CMPE field mask (bit 2)
k_lvd_fsamp_mask	FSAMP field mask (bits 5:4)
k_lvd_ri_mask	RI field mask (bit 6)
k_lvd_rn_mask	RN field mask (bit 7)

Definition at line 458 of file [rx72n_lvda_regs.h](#).

```

00458         : uint8_t {
00459     /* Interrupt/Reset enable (RIE) */
00460     k_lvd_rie_disable = (0 « 0), /**< Disable interrupt/reset */
00461     k_lvd_rie_enable  = (1 « 0), /**< Enable interrupt/reset */
00462
00463     /* Digital filter disable (DFDIS) */
00464     k_lvd_dfdis_filter_on = (0 « 1), /**< Digital filter enabled */
00465     k_lvd_dfdis_filter_off = (1 « 1), /**< Digital filter disabled */
00466
00467     /* Comparison result output enable (CMPE) */
00468     k_lvd_cmpe_disable = (0 « 2), /**< Output disabled */
00469     k_lvd_cmpe_enable  = (1 « 2), /**< Output enabled */
00470
00471     /* Digital filter sampling clock select (FSAMP) */
00472     k_lvd_fsamp_loco_div2 = (0 « 4), /**< LOCO/2 (120 kHz) */
00473     k_lvd_fsamp_loco_div4 = (1 « 4), /**< LOCO/4 (60 kHz) */

```

```

00474 k_lvd_fsamp_loco_div8 = (2 « 4), /**< LOCO/8 (30 kHz) */
00475 k_lvd_fsamp_loco_div16 = (3 « 4), /**< LOCO/16 (15 kHz) */
00476
00477 /* Mode select (RI) */
00478 k_lvd_ri_interrupt = (0 « 6), /**< Generate interrupt on Vdet passage */
00479 k_lvd_ri_reset = (1 « 6), /**< Generate reset when voltage falls below Vdet */
00480
00481 /* Reset negate select (RN) */
00482 k_lvd_rn_vcc_above = (0 « 7), /**< Negate after VCC > Vdet detected */
00483 k_lvd_rn_after_assert = (1 « 7), /**< Negate after stabilization time from assert */
00484
00485 /* Field masks */
00486 k_lvd_rie_mask = 0x01, /**< RIE field mask (bit 0) */
00487 k_lvd_dfdis_mask = 0x02, /**< DFDIS field mask (bit 1) */
00488 k_lvd_cmpe_mask = 0x04, /**< CMPE field mask (bit 2) */
00489 k_lvd_fsamp_mask = 0x30, /**< FSAMP field mask (bits 5:4) */
00490 k_lvd_ri_mask = 0x40, /**< RI field mask (bit 6) */
00491 k_lvd_rn_mask = 0x80, /**< RN field mask (bit 7) */
00492 } rx_lvd_cr0_bits_t;

```

8.33.2.3 rx_lvd_cr1_bits_t

```
enum rx_lvd_cr1_bits_t : uint8_t
```

LVD1CR1/LVD2CR1 register bit definitions.

Control bits for interrupt generation condition and type selection.

Interrupt Condition Select (IDTSEL)

Value	Condition
00	VCC >= Vdet (rise) detected
01	VCC < Vdet (drop) detected
10	Both drop and rise detected
11	Prohibited

Since

Version 1.0.0

Enumerator

k_lvd_idtsel_rise	Interrupt on VCC >= Vdet (rise)
k_lvd_idtsel_drop	Interrupt on VCC < Vdet (drop)
k_lvd_idtsel_both	Interrupt on both rise and drop
k_lvd_irqsel_nmi	Non-maskable interrupt
k_lvd_irqsel_maskable	Maskable interrupt
k_lvd_idtsel_mask	IDTSEL field mask (bits 1:0)
k_lvd_irqsel_mask	IRQSEL field mask (bit 2)

Definition at line 346 of file rx72n_lvda_regs.h.

```

00346 : uint8_t {
00347 /* Interrupt generation condition select (IDTSEL) */
00348 k_lvd_idtsel_rise = 0x00, /**< Interrupt on VCC >= Vdet (rise) */
00349 k_lvd_idtsel_drop = 0x01, /**< Interrupt on VCC < Vdet (drop) */

```

```

00350 k_lvd_idtsel_both = 0x02, /**< Interrupt on both rise and drop */
00351
00352 /* Interrupt type select (IRQSEL) */
00353 k_lvd_irqsel_nmi = (0 « 2), /**< Non-maskable interrupt */
00354 k_lvd_irqsel_maskable = (1 « 2), /**< Maskable interrupt */
00355
00356 /* Field masks */
00357 k_lvd_idtsel_mask = 0x03, /**< IDTSEL field mask (bits 1:0) */
00358 k_lvd_irqsel_mask = 0x04, /**< IRQSEL field mask (bit 2) */
00359 } rx_lvd_cr1_bits_t;

```

8.33.2.4 rx_lvd_interrupts_t

```
enum rx_lvd_interrupts_t : uint8_t
```

LVD interrupt vector numbers.

ICU interrupt vector numbers for voltage monitoring. These can be configured as NMI or maskable interrupts.

Since

Version 1.0.0

Enumerator

k_lvd1_irq	LVD1 - Voltage monitor 1 interrupt (Table 15.5 p.512)
k_lvd2_irq	LVD2 - Voltage monitor 2 interrupt (Table 15.5 p.512)

Definition at line 504 of file rx72n_lvda_regs.h.

```

00504 : uint8_t {
00505 k_lvd1_irq = 88, /**< LVD1 - Voltage monitor 1 interrupt (Table 15.5 p.512) */
00506 k_lvd2_irq = 89, /**< LVD2 - Voltage monitor 2 interrupt (Table 15.5 p.512) */
00507 } rx_lvd_interrupts_t;

```

8.33.2.5 rx_lvd_sr_bits_t

```
enum rx_lvd_sr_bits_t : uint8_t
```

LVD1SR/LVD2SR register bit definitions.

Status bits for voltage detection and monitoring.

Since

Version 1.0.0

Enumerator

k_lvd_det_clear	Clear detection flag (write)
k_lvd_det_detected	Vdet passage detected (read)
k_lvd_mon_below	VCC < Vdet (read)
k_lvd_mon_above	VCC >= Vdet or monitor disabled (read)

k_lvd_det_mask	DET flag mask (bit 0)
k_lvd_mon_mask	MON flag mask (bit 1)

Definition at line 370 of file [rx72n_lvda_regs.h](#).

```

00370      : uint8_t {
00371  k_lvd_det_clear    = 0x00, /**< Clear detection flag (write) */
00372  k_lvd_det_detected = 0x01, /**< Vdet passage detected (read) */
00373  k_lvd_mon_below    = 0x00, /**< VCC < Vdet (read) */
00374  k_lvd_mon_above    = 0x02, /**< VCC >= Vdet or monitor disabled (read) */
00375
00376  /** Field masks */
00377  k_lvd_det_mask = 0x01, /**< DET flag mask (bit 0) */
00378  k_lvd_mon_mask = 0x02, /**< MON flag mask (bit 1) */
00379 } rx_lvd_sr_bits_t;

```

8.33.2.6 rx`lvda`addresses`t

```
enum rx_lvda_addresses_t : uintptr_t
```

LVDA register addresses.

The LVDA registers are located in two non-contiguous memory regions:

- Control/Status registers at 0x000800E0-0x000800E3
- Configuration registers at 0x0008C297-0x0008C29B

Memory Map

Address	Register	Size	Description
0x000800E0	LVD1CR1	1	Voltage Monitor 1 Control Reg 1
0x000800E1	LVD1SR	1	Voltage Monitor 1 Status
0x000800E2	LVD2CR1	1	Voltage Monitor 2 Control Reg 1
0x000800E3	LVD2SR	1	Voltage Monitor 2 Status
0x0008C297	LVCMPCR	1	Voltage Monitor Circuit Control
0x0008C298	LVDLVLR	1	Voltage Detection Level Select
0x0008C29A	LVD1CR0	1	Voltage Monitor 1 Control Reg 0
0x0008C29B	LVD2CR0	1	Voltage Monitor 2 Control Reg 0

Note

Manual: Ch8.2 - Register Descriptions

Since

Version 1.0.0

Enumerator

k_lvd1cr1_addr	Voltage Monitor 1 Control Register 1
k_lvd1sr_addr	Voltage Monitor 1 Status Register
k_lvd2cr1_addr	Voltage Monitor 2 Control Register 1
k_lvd2sr_addr	Voltage Monitor 2 Status Register
k_lvcmpcr_addr	Voltage Monitor Circuit Control Register
k_lvdlvlr_addr	Voltage Detection Level Select Register
k_lvd1cr0_addr	Voltage Monitor 1 Control Register 0
k_lvd2cr0_addr	Voltage Monitor 2 Control Register 0

Definition at line 144 of file [rx72n_lvda_regs.h](#).

```

00144      : uintptr_t {
00145      /* First register block (0x000800E0-0x000800E3) */
00146      k_lvd1cr1_addr = 0x000800E0, /**< Voltage Monitor 1 Control Register 1 */
00147      k_lvd1sr_addr = 0x000800E1, /**< Voltage Monitor 1 Status Register */
00148      k_lvd2cr1_addr = 0x000800E2, /**< Voltage Monitor 2 Control Register 1 */
00149      k_lvd2sr_addr = 0x000800E3, /**< Voltage Monitor 2 Status Register */
00150
00151      /* Second register block (0x0008C297-0x0008C29B) */
00152      k_lvcmpcr_addr = 0x0008C297, /**< Voltage Monitor Circuit Control Register */
00153      k_lvdlvlr_addr = 0x0008C298, /**< Voltage Detection Level Select Register */
00154      /* Note: 0x0008C299 is reserved */
00155      k_lvd1cr0_addr = 0x0008C29A, /**< Voltage Monitor 1 Control Register 0 */
00156      k_lvd2cr0_addr = 0x0008C29B, /**< Voltage Monitor 2 Control Register 0 */
00157 } rx_lvda_addresses_t;

```

8.33.2.7 rx_lvdlvlr_bits_t

```
enum rx_lvdlvlr_bits_t : uint8_t
```

LVDLVLR register bit definitions.

Voltage detection level selection for monitors 1 and 2.

Voltage Levels (typical values at Ta = 25degC)

Setting	LVD1LVL/LVD2LVL	Voltage
↔ —↔ 1	1001b (0x09)	2.99V
↔ —↔ 2	1010b (0x0A)	2.92V
↔ —↔ 3	1011b (0x0B)	2.85V

Warning

Other values are prohibited

Note

Can only change when both LVD1E and LVD2E are 0

Since

Version 1.0.0

Enumerator

k_lvd1lvl_2v99	Vdet1 = 2.99V (highest)
k_lvd1lvl_2v92	Vdet1 = 2.92V
k_lvd1lvl_2v85	Vdet1 = 2.85V (lowest)
k_lvd2lvl_2v99	Vdet2 = 2.99V (highest)
k_lvd2lvl_2v92	Vdet2 = 2.92V
k_lvd2lvl_2v85	Vdet2 = 2.85V (lowest)
k_lvd1lvl_mask	LVD1LVL field mask (bits 3:0)
k_lvd2lvl_mask	LVD2LVL field mask (bits 7:4)
k_lvdlvlr_reset_value	Both at 2.85V after reset

Definition at line 422 of file [rx72n_lvda_regs.h](#).

```

00422     : uint8_t {
00423     /* LVD1 voltage levels (bits 3:0) */
00424     k_lvd1lvl_2v99 = 0x09, /**< Vdet1 = 2.99V (highest) */
00425     k_lvd1lvl_2v92 = 0x0A, /**< Vdet1 = 2.92V */
00426     k_lvd1lvl_2v85 = 0x0B, /**< Vdet1 = 2.85V (lowest) */
00427
00428     /* LVD2 voltage levels (bits 7:4) */
00429     k_lvd2lvl_2v99 = 0x90, /**< Vdet2 = 2.99V (highest) */
00430     k_lvd2lvl_2v92 = 0xA0, /**< Vdet2 = 2.92V */
00431     k_lvd2lvl_2v85 = 0xB0, /**< Vdet2 = 2.85V (lowest) */
00432
00433     /* Field masks */
00434     k_lvd1lvl_mask = 0x0F, /**< LVD1LVL field mask (bits 3:0) */
00435     k_lvd2lvl_mask = 0xF0, /**< LVD2LVL field mask (bits 7:4) */
00436
00437     /* Default after reset */
00438     k_lvdlvlr_reset_value = 0xBB, /**< Both at 2.85V after reset */
00439 } rx_lvdlvlr_bits_t;

```

8.33.3 Function Documentation

8.33.3.1 lvcmpcr_reg()

```
volatile uint8_t * lvcmpcr_reg (
    void ) [inline], [static]
```

Get pointer to LVCMPCR register.

Voltage Monitoring Circuit Control Register. Enables/disables voltage detection circuits 1 and 2.

Returns

Volatile pointer to LVCMPCR register

Register Bits:

- b5 LVD1E: Voltage detection 1 circuit enable
- b6 LVD2E: Voltage detection 2 circuit enable

See also

[rx_lvcmpr_bits_t](#) Bit definitions

Since

Version 1.0.0

Definition at line 256 of file [rx72n_lvda_regs.h](#).

```
00257 {
00258     return (volatile uint8_t*)k_lvcmpr_addr;
00259 }
```

References [k_lvcmpr_addr](#).

8.33.3.2 lvd1cr0_reg()

```
volatile uint8_t * lvd1cr0_reg (
    void ) [inline], [static]
```

Get pointer to LVD1CR0 register.

Voltage Monitoring 1 Circuit Control Register 0. Controls interrupt/reset enable, digital filter, and comparison output enable.

Returns

Volatile pointer to LVD1CR0 register

Register Bits:

- b0 LVD1RIE: Interrupt/reset enable
- b1 LVD1DFDIS: Digital filter disable (0=enabled, 1=disabled)
- b2 LVD1CMPE: Comparison result output enable
- b5:4 LVD1FSAMP[1:0]: Digital filter sampling clock select
- b6 LVD1RI: Mode select (0=interrupt, 1=reset)
- b7 LVD1RN: Reset negate select

See also

[rx_lvd1cr0_bits_t](#) Bit definitions

Since

Version 1.0.0

Definition at line 302 of file [rx72n_lvda_regs.h](#).

```
00303 {
00304     return (volatile uint8_t*)k_lvd1cr0_addr;
00305 }
```

References [k_lvd1cr0_addr](#).

8.33.3.3 lvd1cr1_reg()

```
volatile uint8_t * lvd1cr1_reg (
    void )    [inline], [static]
```

Get pointer to LVD1CR1 register.

Voltage Monitoring 1 Circuit Control Register 1. Controls interrupt generation condition and interrupt type selection.

Returns

Volatile pointer to LVD1CR1 register

Register Bits:

- b1:0 LVD1IDTSEL[1:0]: Interrupt generation condition select
- b2 LVD1IRQSEL: Interrupt type (0=NMI, 1=Maskable)

See also

`rx_lvd1cr1_bits_t` Bit definitions

Since

Version 1.0.0

Definition at line 180 of file `rx72n_lvda_regs.h`.

```
00181 {
00182     return (volatile uint8_t*)k_lvd1cr1_addr;
00183 }
```

References [k_lvd1cr1_addr](#).

8.33.3.4 lvd1sr_reg()

```
volatile uint8_t * lvd1sr_reg (
    void )    [inline], [static]
```

Get pointer to LVD1SR register.

Voltage Monitoring 1 Status Register. Contains detection flag and voltage monitor flag.

Returns

Volatile pointer to LVD1SR register

Register Bits:

- b0 LVD1DET: Voltage change detection flag (write 0 to clear)
- b1 LVD1MON: Voltage monitor flag (0=VCC<Vdet1, 1=VCC>=Vdet1)

See also

`rx_lvd1sr_bits_t` Bit definitions

Since

Version 1.0.0

Definition at line 201 of file `rx72n_lvda_regs.h`.

```
00202 {
00203     return (volatile uint8_t*)k_lvd1sr_addr;
00204 }
```

References [k_lvd1sr_addr](#).

8.33.3.5 lvd2cr0`reg()

```
volatile uint8_t * lvd2cr0_reg (
    void ) [inline], [static]
```

Get pointer to LVD2CR0 register.

Voltage Monitoring 2 Circuit Control Register 0. Controls interrupt/reset enable, digital filter, and comparison output enable.

Returns

Volatile pointer to LVD2CR0 register

See also

`rx_lvd2cr0_bits_t` Bit definitions

Since

Version 1.0.0

Definition at line 319 of file [rx72n_lvda_regs.h](#).

```
00320 {
00321     return (volatile uint8_t*)k_lvd2cr0_addr;
00322 }
```

References [k_lvd2cr0_addr](#).

8.33.3.6 lvd2cr1`reg()

```
volatile uint8_t * lvd2cr1_reg (
    void ) [inline], [static]
```

Get pointer to LVD2CR1 register.

Voltage Monitoring 2 Circuit Control Register 1. Controls interrupt generation condition and interrupt type selection.

Returns

Volatile pointer to LVD2CR1 register

See also

`rx_lvd2cr1_bits_t` Bit definitions

Since

Version 1.0.0

Definition at line 218 of file [rx72n_lvda_regs.h](#).

```
00219 {
00220     return (volatile uint8_t*)k_lvd2cr1_addr;
00221 }
```

References [k_lvd2cr1_addr](#).

8.33.3.7 lvd2sr_reg()

```
volatile uint8_t * lvd2sr_reg (  
    void )    [inline], [static]
```

Get pointer to LVD2SR register.

Voltage Monitoring 2 Status Register. Contains detection flag and voltage monitor flag.

Returns

Volatile pointer to LVD2SR register

See also

[rx_lvd2sr_bits_t](#) Bit definitions

Since

Version 1.0.0

Definition at line 235 of file [rx72n_lvda_regs.h](#).

```
00236 {  
00237     return (volatile uint8_t*)k_lvd2sr_addr;  
00238 }
```

References [k_lvd2sr_addr](#).

8.33.3.8 lvd1vlr_reg()

```
volatile uint8_t * lvd1vlr_reg (  
    void )    [inline], [static]
```

Get pointer to LVD1VLR register.

Voltage Detection Level Select Register. Sets the detection voltage thresholds for both voltage monitors.

Returns

Volatile pointer to LVD1VLR register

Register Bits:

- b3:0 LVD1LVL[3:0]: Voltage detection 1 level (2.85V/2.92V/2.99V)
- b7:4 LVD2LVL[3:0]: Voltage detection 2 level (2.85V/2.92V/2.99V)

See also

[rx_lvd1vlr_bits_t](#) Bit definitions

Since

Version 1.0.0

Definition at line 277 of file [rx72n_lvda_regs.h](#).

```
00278 {  
00279     return (volatile uint8_t*)k_lvd1vlr_addr;  
00280 }
```

References [k_lvd1vlr_addr](#).

8.34 rx72n_lvda_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_lvda_regs.h
00003  * @brief RX72N Voltage Detection Circuit (LVDA) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the LVDA module which provides brownout protection
00007  * and voltage monitoring for safety-critical motor control. The LVDA can
00008  * trigger resets or interrupts when VCC drops below configurable thresholds.
00009  *
00010  * @par Voltage Monitoring System
00011  * @dot
00012  * digraph lvda_system {
00013  *   rankdir=TB;
00014  *   node [shape=box, style=rounded];
00015  *
00016  *   subgraph cluster_power {
00017  *     label="Power Supply";
00018  *     style=filled;
00019  *     color=lightblue;
00020  *
00021  *     vcc [label="VCC (3.3V)"];
00022  *   }
00023  *
00024  *   subgraph cluster_lvda {
00025  *     label="LVDA Circuit";
00026  *     style=filled;
00027  *     color=lightyellow;
00028  *
00029  *     mon0 [label="Voltage Monitor 0\n(Vdet0: OFS1 config)"];
00030  *     mon1 [label="Voltage Monitor 1\n(Vdet1: 2.85-2.99V)"];
00031  *     mon2 [label="Voltage Monitor 2\n(Vdet2: 2.85-2.99V)"];
00032  *   }
00033  *
00034  *   subgraph cluster_actions {
00035  *     label="Actions on Low Voltage";
00036  *     style=filled;
00037  *     color=lightcoral;
00038  *
00039  *     reset [label="System Reset"];
00040  *     nmi [label="NMI Interrupt"];
00041  *     irq [label="Maskable IRQ"];
00042  *   }
00043  *
00044  *   vcc -> mon0;
00045  *   vcc -> mon1;
00046  *   vcc -> mon2;
00047  *   mon0 -> reset [label="Vdet0 > VCC"];
00048  *   mon1 -> reset [label="LVD1RI=1"];
00049  *   mon1 -> nmi [label="LVD1IRQSEL=0"];
00050  *   mon1 -> irq [label="LVD1IRQSEL=1"];
00051  *   mon2 -> reset [label="LVD2RI=1"];
00052  *   mon2 -> nmi [label="LVD2IRQSEL=0"];
00053  *   mon2 -> irq [label="LVD2IRQSEL=1"];
00054  * }
00055  * @enddot
00056  *
00057  * @par STAR Project LVDA Usage
00058  * | Monitor | Threshold | Action | Use Case |
00059  * |-----|-----|-----|-----|
00060  * | Vdet0 | OFS1 conf | Reset | Emergency brownout protection |
00061  * | Vdet1 | 2.92V | Interrupt | Low voltage warning |
00062  * | Vdet2 | 2.85V | Reset | Critical brownout shutdown |
00063  *
00064  * @par Voltage Detection Levels
00065  * | Level | Voltage (typ) | Hysteresis | Notes |
00066  * |-----|-----|-----|-----|
00067  * | Vdet1_1 | 2.99V | ~50mV | Early warning |
00068  * | Vdet1_2 | 2.92V | ~50mV | Low voltage threshold |
00069  * | Vdet1_3 | 2.85V | ~50mV | Critical threshold |
00070  *
00071  * @par Hardware Requirements
00072  * | Parameter | Value | Notes |
00073  * |-----|-----|-----|
00074  * | LVD1/2 Regs | 0x000800E0 | 4-byte block |
00075  * | LVCMPCR etc | 0x0008C297 | 5-byte block (non-contiguous)|
00076  * | Enable Time | td(E-A) | Wait after enabling LVDnE |
00077  * | Clock | LOCO (240 kHz) | For digital filter |
00078  *
00079  * @par NASA Power of 10 Compliance
00080  * - Rule 1: No goto, setjmp, or recursion
00081  * - Rule 2: N/A (no loops in register definitions)
00082  * - Rule 3: No dynamic memory allocation

```

```

00083 * - Rule 4: All accessor functions are single-statement
00084 * - Rule 5: N/A (hardware layer)
00085 * - Rule 6: Minimal scope - inline accessors only
00086 * - Rule 7: N/A (no return values to check)
00087 * - Rule 8: All constants use C23 typed enums
00088 * - Rule 9: No function pointers
00089 * - Rule 10: Compiles with -Wall -Wextra -Werror
00090 *
00091 * @par Module Dependencies
00092 * - <stdint.h> - Fixed-width integer types
00093 *
00094 * @par Manual References
00095 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00096 * - Chapter 8: Voltage Detection Circuit (LVDA), pages 316-334
00097 *
00098 * @see rx72n_regs.h Main register include file
00099 *
00100 * @author Locked, Inc.
00101 * @date 2026-01-29
00102 * @copyright Copyright (c) 2026 Locked Inc.
00103 * SPDX-License-Identifier: MIT
00104 * @since Version 1.0.0
00105 */
00106
00107 #pragma once
00108
00109 #include <stdint.h>
00110
00111 #ifdef __cplusplus
00112 extern "C" {
00113 #endif
00114
00115 /*
=====
00116 * LVDA - Voltage Detection Circuit for Brownout Protection
00117 *
=====
00118 */
00119
00120 /**
00121 * @enum rx_lvda_addresses_t
00122 * @brief LVDA register addresses
00123 *
00124 * @details
00125 * The LVDA registers are located in two non-contiguous memory regions:
00126 * - Control/Status registers at 0x000800E0-0x000800E3
00127 * - Configuration registers at 0x0008C297-0x0008C29B
00128 *
00129 * @par Memory Map
00130 * | Address | Register | Size | Description |
00131 * |-----|-----|-----|-----|
00132 * | 0x000800E0 | LVD1CR1 | 1 | Voltage Monitor 1 Control Reg 1 |
00133 * | 0x000800E1 | LVD1SR | 1 | Voltage Monitor 1 Status |
00134 * | 0x000800E2 | LVD2CR1 | 1 | Voltage Monitor 2 Control Reg 1 |
00135 * | 0x000800E3 | LVD2SR | 1 | Voltage Monitor 2 Status |
00136 * | 0x0008C297 | LVCMPCR | 1 | Voltage Monitor Circuit Control |
00137 * | 0x0008C298 | LVDLVL | 1 | Voltage Detection Level Select |
00138 * | 0x0008C29A | LVD1CR0 | 1 | Voltage Monitor 1 Control Reg 0 |
00139 * | 0x0008C29B | LVD2CR0 | 1 | Voltage Monitor 2 Control Reg 0 |
00140 *
00141 * @note Manual: Ch8.2 - Register Descriptions
00142 * @since Version 1.0.0
00143 */
00144 typedef enum : uintptr_t {
00145     /* First register block (0x000800E0-0x000800E3) */
00146     k_lvdlcr1_addr = 0x000800E0, /**< Voltage Monitor 1 Control Register 1 */
00147     k_lvdl1sr_addr = 0x000800E1, /**< Voltage Monitor 1 Status Register */
00148     k_lvdlcr2_addr = 0x000800E2, /**< Voltage Monitor 2 Control Register 1 */
00149     k_lvdl2sr_addr = 0x000800E3, /**< Voltage Monitor 2 Status Register */
00150
00151     /* Second register block (0x0008C297-0x0008C29B) */
00152     k_lvcmpcr_addr = 0x0008C297, /**< Voltage Monitor Circuit Control Register */
00153     k_lvdlvlr_addr = 0x0008C298, /**< Voltage Detection Level Select Register */
00154     /* Note: 0x0008C299 is reserved */
00155     k_lvdlcr0_addr = 0x0008C29A, /**< Voltage Monitor 1 Control Register 0 */
00156     k_lvdlcr2r0_addr = 0x0008C29B, /**< Voltage Monitor 2 Control Register 0 */
00157 } rx_lvda_addresses_t;
00158
00159 /*
=====
00160 * Register Accessor Functions
00161 *
=====
00162 */
00163
00164 /**
00165 * @brief Get pointer to LVD1CR1 register

```

```

00166 *
00167 * @details
00168 * Voltage Monitoring 1 Circuit Control Register 1. Controls interrupt
00169 * generation condition and interrupt type selection.
00170 *
00171 * @return Volatile pointer to LVD1CR1 register
00172 *
00173 * @par Register Bits:
00174 * - b1:0 LVD1IDTSEL[1:0]: Interrupt generation condition select
00175 * - b2 LVD1IRQSEL: Interrupt type (0=NMI, 1=Maskable)
00176 *
00177 * @see rx_lvd1cr1_bits_t Bit definitions
00178 * @since Version 1.0.0
00179 */
00180 static inline volatile uint8_t* lvd1cr1_reg(void)
00181 {
00182     return (volatile uint8_t*)k_lvd1cr1_addr;
00183 }
00184
00185 /**
00186 * @brief Get pointer to LVD1SR register
00187 *
00188 * @details
00189 * Voltage Monitoring 1 Status Register. Contains detection flag and
00190 * voltage monitor flag.
00191 *
00192 * @return Volatile pointer to LVD1SR register
00193 *
00194 * @par Register Bits:
00195 * - b0 LVD1DET: Voltage change detection flag (write 0 to clear)
00196 * - b1 LVD1MON: Voltage monitor flag (0=VCC<Vdet1, 1=VCC>=Vdet1)
00197 *
00198 * @see rx_lvd1sr_bits_t Bit definitions
00199 * @since Version 1.0.0
00200 */
00201 static inline volatile uint8_t* lvd1sr_reg(void)
00202 {
00203     return (volatile uint8_t*)k_lvd1sr_addr;
00204 }
00205
00206 /**
00207 * @brief Get pointer to LVD2CR1 register
00208 *
00209 * @details
00210 * Voltage Monitoring 2 Circuit Control Register 1. Controls interrupt
00211 * generation condition and interrupt type selection.
00212 *
00213 * @return Volatile pointer to LVD2CR1 register
00214 *
00215 * @see rx_lvd2cr1_bits_t Bit definitions
00216 * @since Version 1.0.0
00217 */
00218 static inline volatile uint8_t* lvd2cr1_reg(void)
00219 {
00220     return (volatile uint8_t*)k_lvd2cr1_addr;
00221 }
00222
00223 /**
00224 * @brief Get pointer to LVD2SR register
00225 *
00226 * @details
00227 * Voltage Monitoring 2 Status Register. Contains detection flag and
00228 * voltage monitor flag.
00229 *
00230 * @return Volatile pointer to LVD2SR register
00231 *
00232 * @see rx_lvd2sr_bits_t Bit definitions
00233 * @since Version 1.0.0
00234 */
00235 static inline volatile uint8_t* lvd2sr_reg(void)
00236 {
00237     return (volatile uint8_t*)k_lvd2sr_addr;
00238 }
00239
00240 /**
00241 * @brief Get pointer to LVCMPCR register
00242 *
00243 * @details
00244 * Voltage Monitoring Circuit Control Register. Enables/disables voltage
00245 * detection circuits 1 and 2.
00246 *
00247 * @return Volatile pointer to LVCMPCR register
00248 *
00249 * @par Register Bits:
00250 * - b5 LVD1E: Voltage detection 1 circuit enable
00251 * - b6 LVD2E: Voltage detection 2 circuit enable
00252 *

```

```

00253 * @see rx_lvcmpcr_bits_t Bit definitions
00254 * @since Version 1.0.0
00255 */
00256 static inline volatile uint8_t* lvcmpcr_reg(void)
00257 {
00258     return (volatile uint8_t*)k_lvcmpcr_addr;
00259 }
00260
00261 /**
00262 * @brief Get pointer to LVDLVLR register
00263 *
00264 * @details
00265 * Voltage Detection Level Select Register. Sets the detection voltage
00266 * thresholds for both voltage monitors.
00267 *
00268 * @return Volatile pointer to LVDLVLR register
00269 *
00270 * @par Register Bits:
00271 * - b3:0 LVD1LVL[3:0]: Voltage detection 1 level (2.85V/2.92V/2.99V)
00272 * - b7:4 LVD2LVL[3:0]: Voltage detection 2 level (2.85V/2.92V/2.99V)
00273 *
00274 * @see rx_lvdvlr_bits_t Bit definitions
00275 * @since Version 1.0.0
00276 */
00277 static inline volatile uint8_t* lvdvlr_reg(void)
00278 {
00279     return (volatile uint8_t*)k_lvdvlr_addr;
00280 }
00281
00282 /**
00283 * @brief Get pointer to LVD1CR0 register
00284 *
00285 * @details
00286 * Voltage Monitoring 1 Circuit Control Register 0. Controls interrupt/reset
00287 * enable, digital filter, and comparison output enable.
00288 *
00289 * @return Volatile pointer to LVD1CR0 register
00290 *
00291 * @par Register Bits:
00292 * - b0 LVD1RIE: Interrupt/reset enable
00293 * - b1 LVD1DFDIS: Digital filter disable (0=enabled, 1=disabled)
00294 * - b2 LVD1CMPE: Comparison result output enable
00295 * - b5:4 LVD1FSAMP[1:0]: Digital filter sampling clock select
00296 * - b6 LVD1RI: Mode select (0=interrupt, 1=reset)
00297 * - b7 LVD1RN: Reset negate select
00298 *
00299 * @see rx_lvd1cr0_bits_t Bit definitions
00300 * @since Version 1.0.0
00301 */
00302 static inline volatile uint8_t* lvd1cr0_reg(void)
00303 {
00304     return (volatile uint8_t*)k_lvd1cr0_addr;
00305 }
00306
00307 /**
00308 * @brief Get pointer to LVD2CR0 register
00309 *
00310 * @details
00311 * Voltage Monitoring 2 Circuit Control Register 0. Controls interrupt/reset
00312 * enable, digital filter, and comparison output enable.
00313 *
00314 * @return Volatile pointer to LVD2CR0 register
00315 *
00316 * @see rx_lvd2cr0_bits_t Bit definitions
00317 * @since Version 1.0.0
00318 */
00319 static inline volatile uint8_t* lvd2cr0_reg(void)
00320 {
00321     return (volatile uint8_t*)k_lvd2cr0_addr;
00322 }
00323
00324 /*
=====
00325 * Register Bit Definitions
00326 *
=====
00327 */
00328
00329 /**
00330 * @enum rx_lvd_cr1_bits_t
00331 * @brief LVD1CR1/LVD2CR1 register bit definitions
00332 *
00333 * @details
00334 * Control bits for interrupt generation condition and type selection.
00335 *
00336 * @par Interrupt Condition Select (IDTSEL)
00337 * | Value | Condition |

```

```

00338 * |-----|-----|
00339 * | 00 | VCC >= Vdet (rise) detected |
00340 * | 01 | VCC < Vdet (drop) detected |
00341 * | 10 | Both drop and rise detected |
00342 * | 11 | Prohibited |
00343 *
00344 * @since Version 1.0.0
00345 */
00346 typedef enum : uint8_t {
00347     /* Interrupt generation condition select (IDTSEL) */
00348     k_lvd_idtsel_rise = 0x00, /**< Interrupt on VCC >= Vdet (rise) */
00349     k_lvd_idtsel_drop = 0x01, /**< Interrupt on VCC < Vdet (drop) */
00350     k_lvd_idtsel_both = 0x02, /**< Interrupt on both rise and drop */
00351
00352     /* Interrupt type select (IRQSEL) */
00353     k_lvd_irqsel_nmi = (0 << 2), /**< Non-maskable interrupt */
00354     k_lvd_irqsel_maskable = (1 << 2), /**< Maskable interrupt */
00355
00356     /* Field masks */
00357     k_lvd_idtsel_mask = 0x03, /**< IDTSEL field mask (bits 1:0) */
00358     k_lvd_irqsel_mask = 0x04, /**< IRQSEL field mask (bit 2) */
00359 } rx_lvd_cr1_bits_t;
00360
00361 /**
00362 * @enum rx_lvd_sr_bits_t
00363 * @brief LVD1SR/LVD2SR register bit definitions
00364 *
00365 * @details
00366 * Status bits for voltage detection and monitoring.
00367 *
00368 * @since Version 1.0.0
00369 */
00370 typedef enum : uint8_t {
00371     k_lvd_det_clear = 0x00, /**< Clear detection flag (write) */
00372     k_lvd_det_detected = 0x01, /**< Vdet passage detected (read) */
00373     k_lvd_mon_below = 0x00, /**< VCC < Vdet (read) */
00374     k_lvd_mon_above = 0x02, /**< VCC >= Vdet or monitor disabled (read) */
00375
00376     /* Field masks */
00377     k_lvd_det_mask = 0x01, /**< DET flag mask (bit 0) */
00378     k_lvd_mon_mask = 0x02, /**< MON flag mask (bit 1) */
00379 } rx_lvd_sr_bits_t;
00380
00381 /**
00382 * @enum rx_lvcmpcr_bits_t
00383 * @brief LVCMPCR register bit definitions
00384 *
00385 * @details
00386 * Enable bits for voltage detection circuits 1 and 2.
00387 *
00388 * @note VCC must be at least 80mV above detection level when enabling
00389 *
00390 * @since Version 1.0.0
00391 */
00392 typedef enum : uint8_t {
00393     k_lvd1e_enable = (1 << 5), /**< Enable voltage detection 1 circuit */
00394     k_lvd1e_disable = (0 << 5), /**< Disable voltage detection 1 circuit */
00395     k_lvd2e_enable = (1 << 6), /**< Enable voltage detection 2 circuit */
00396     k_lvd2e_disable = (0 << 6), /**< Disable voltage detection 2 circuit */
00397
00398     /* Field masks */
00399     k_lvd1e_mask = 0x20, /**< LVD1E field mask (bit 5) */
00400     k_lvd2e_mask = 0x40, /**< LVD2E field mask (bit 6) */
00401 } rx_lvcmpcr_bits_t;
00402
00403 /**
00404 * @enum rx_lvdlvlr_bits_t
00405 * @brief LVDLVLR register bit definitions
00406 *
00407 * @details
00408 * Voltage detection level selection for monitors 1 and 2.
00409 *
00410 * @par Voltage Levels (typical values at Ta = 25degC)
00411 * | Setting | LVD1LVL/LVD2LVL | Voltage |
00412 * |-----|-----|-----|
00413 * | _1 | 1001b (0x09) | 2.99V |
00414 * | _2 | 1010b (0x0A) | 2.92V |
00415 * | _3 | 1011b (0x0B) | 2.85V |
00416 *
00417 * @warning Other values are prohibited
00418 * @note Can only change when both LVD1E and LVD2E are 0
00419 *
00420 * @since Version 1.0.0
00421 */
00422 typedef enum : uint8_t {
00423     /* LVD1 voltage levels (bits 3:0) */
00424     k_lvd1lvl_2v99 = 0x09, /**< Vdet1 = 2.99V (highest) */

```



```

00425 k_lvd1lvl_2v92 = 0x0A, /**< Vdet1 = 2.92V */
00426 k_lvd1lvl_2v85 = 0x0B, /**< Vdet1 = 2.85V (lowest) */
00427
00428 /* LVD2 voltage levels (bits 7:4) */
00429 k_lvd2lvl_2v99 = 0x90, /**< Vdet2 = 2.99V (highest) */
00430 k_lvd2lvl_2v92 = 0xA0, /**< Vdet2 = 2.92V */
00431 k_lvd2lvl_2v85 = 0xB0, /**< Vdet2 = 2.85V (lowest) */
00432
00433 /* Field masks */
00434 k_lvd1lvl_mask = 0x0F, /**< LVD1LVL field mask (bits 3:0) */
00435 k_lvd2lvl_mask = 0xF0, /**< LVD2LVL field mask (bits 7:4) */
00436
00437 /* Default after reset */
00438 k_lvd1vlr_reset_value = 0xBB, /**< Both at 2.85V after reset */
00439 } rx_lvd1vlr_bits_t;
00440
00441 /**
00442  * @enum rx_lvd_cr0_bits_t
00443  * @brief LVD1CR0/LVD2CR0 register bit definitions
00444  *
00445  * @details
00446  * Control bits for interrupt/reset enable, digital filter, and mode selection.
00447  *
00448  * @par Digital Filter Sampling Clock (FSAMP)
00449  * | Value | Clock | Sample Time |
00450  * |-----|-----|-----|
00451  * | 00 | LOCO/2 (120 kHz) | ~17 us |
00452  * | 01 | LOCO/4 (60 kHz) | ~33 us |
00453  * | 10 | LOCO/8 (30 kHz) | ~67 us |
00454  * | 11 | LOCO/16 (15 kHz) | ~133 us |
00455  *
00456  * @since Version 1.0.0
00457  */
00458 typedef enum : uint8_t {
00459     /* Interrupt/Reset enable (RIE) */
00460     k_lvd_rie_disable = (0 « 0), /**< Disable interrupt/reset */
00461     k_lvd_rie_enable = (1 « 0), /**< Enable interrupt/reset */
00462
00463     /* Digital filter disable (DFDIS) */
00464     k_lvd_dfdis_filter_on = (0 « 1), /**< Digital filter enabled */
00465     k_lvd_dfdis_filter_off = (1 « 1), /**< Digital filter disabled */
00466
00467     /* Comparison result output enable (CMPE) */
00468     k_lvd_cmpe_disable = (0 « 2), /**< Output disabled */
00469     k_lvd_cmpe_enable = (1 « 2), /**< Output enabled */
00470
00471     /* Digital filter sampling clock select (FSAMP) */
00472     k_lvd_fsamp_loco_div2 = (0 « 4), /**< LOCO/2 (120 kHz) */
00473     k_lvd_fsamp_loco_div4 = (1 « 4), /**< LOCO/4 (60 kHz) */
00474     k_lvd_fsamp_loco_div8 = (2 « 4), /**< LOCO/8 (30 kHz) */
00475     k_lvd_fsamp_loco_div16 = (3 « 4), /**< LOCO/16 (15 kHz) */
00476
00477     /* Mode select (RI) */
00478     k_lvd_ri_interrupt = (0 « 6), /**< Generate interrupt on Vdet passage */
00479     k_lvd_ri_reset = (1 « 6), /**< Generate reset when voltage falls below Vdet */
00480
00481     /* Reset negate select (RN) */
00482     k_lvd_rn_vcc_above = (0 « 7), /**< Negate after VCC > Vdet detected */
00483     k_lvd_rn_after_assert = (1 « 7), /**< Negate after stabilization time from assert */
00484
00485     /* Field masks */
00486     k_lvd_rie_mask = 0x01, /**< RIE field mask (bit 0) */
00487     k_lvd_dfdis_mask = 0x02, /**< DFDIS field mask (bit 1) */
00488     k_lvd_cmpe_mask = 0x04, /**< CMPE field mask (bit 2) */
00489     k_lvd_fsamp_mask = 0x30, /**< FSAMP field mask (bits 5:4) */
00490     k_lvd_ri_mask = 0x40, /**< RI field mask (bit 6) */
00491     k_lvd_rn_mask = 0x80, /**< RN field mask (bit 7) */
00492 } rx_lvd_cr0_bits_t;
00493
00494 /**
00495  * @enum rx_lvd_interrupts_t
00496  * @brief LVD interrupt vector numbers
00497  *
00498  * @details
00499  * ICU interrupt vector numbers for voltage monitoring. These can be
00500  * configured as NMI or maskable interrupts.
00501  *
00502  * @since Version 1.0.0
00503  */
00504 typedef enum : uint8_t {
00505     k_lvd1_irq = 88, /**< LVD1 - Voltage monitor 1 interrupt (Table 15.5 p.512) */
00506     k_lvd2_irq = 89, /**< LVD2 - Voltage monitor 2 interrupt (Table 15.5 p.512) */
00507 } rx_lvd_interrupts_t;
00508
00509 /*
=====
00510  * Static Assertions - Verify Register Addresses at Compile Time

```


Enumerations

- enum `rx_mpc_addresses_t` : `uintptr_t` { `k_mpc_base_addr` = 0x0008C100 }
- MPC base address (verified against RX72N Hardware Manual Ch23).
- enum `mpc_register_offsets_t` : `uint8_t` {
`k_mpc_offset_pwpr` = 0x1F , `k_mpc_offset_pfs` = 0x40 , `k_mpc_offset_port5_pfs` = 0x28
`k_mpc_offset_porta_pfs` = 0x50 ,
`k_mpc_offset_porte_pfs` = 0x70 , `k_mpc_offset_portj_pfs` = 0x90 , `k_mpc_pfs_total_bytes` }
 Byte offsets of key MPC registers from the MPC base address (0x0008C100).
- enum `mpc_reserved_sizes_t` : `uint8_t` {
`k_mpc_reserved1_bytes` = 1 , `k_mpc_reserved2_bytes` = 4 , `k_mpc_reserved3_bytes` = 16 ,
`k_mpc_reserved4_bytes` = 8 ,
`k_mpc_dscr2_count` = 24 , `k_port3_reserved_bytes` = 3 , `k_port5_reserved_bytes` = 1 ,
`k_port6_reserved_bytes` = 1 ,
`k_port7_reserved_bytes` = 1 , `k_portf_reserved1_bytes` = 2 , `k_portf_reserved2_bytes` = 2 ,
`k_portj_reserved1_bytes` = 1 ,
`k_portj_reserved2_bytes` = 2 }
 MPC register reserved field sizes.
- enum `mpc_pwpr_bits_t` : `uint8_t` { `k_mpc_pwpr_pfswe` = (1 << 6) , `k_mpc_pwpr_b0wi` = (1 << 7) }
- MPC Write Protect Register (PWPR) bit definitions.
- enum `pfs_bits_t` : `uint8_t` { `k_pfs_psel_mask` = 0x1F , `k_pfs_isel` = (1 << 6) , `k_pfs_asel` = (1 << 7) }
- Pin Function Select (PFS) register bit definitions.

Functions

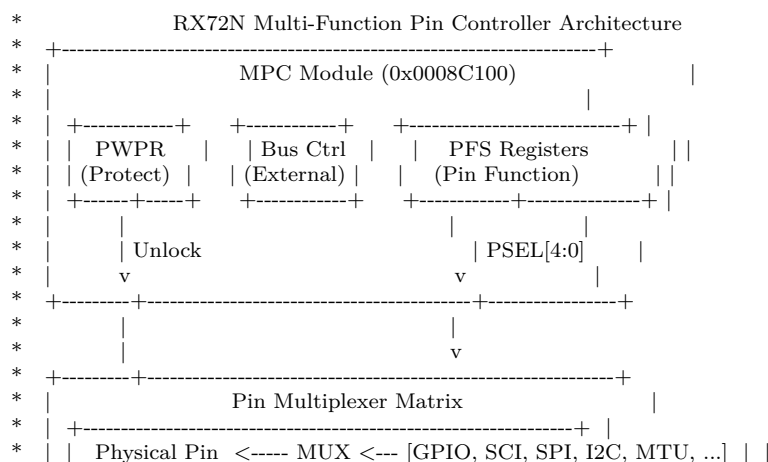
- static volatile `rx_mpc_regs_t * mpc` (void)
Get pointer to MPC registers.

8.35.1 Detailed Description

RX72N Multi-Function Pin Controller (MPC) Register Definitions.

Register definitions for the Multi-Function Pin Controller (MPC) peripheral on the RX72N microcontroller. MPC configures which peripheral function is connected to each GPIO pin (pin multiplexing).

MPC System Architecture



```

* | +-----+ |
* +-----+
*
* PFS Write Protection Sequence:
* +-----+ +-----+ +-----+ +-----+ +-----+
* | PWPR | -> | PWPR | -> | Write | -> | PWPR | -> | PWPR |
* | =0x00 | | =0x40 | | PnmPFS | | =0x00 | | =0x80 |
* | (B0WI=0) | | (PFSWE=1) | | | | (PFSWE=0) | | (B0WI=1) |
* +-----+ +-----+ +-----+ +-----+ +-----+
*
* Memory Map (Verified against RX72N Manual Ch23):
* +-----+
* | Address | Register |
* +-----+
* | 0x0008C100 | PFCSE (CS Output Enable) |
* | 0x0008C102 | PFCSS0/1 (CS Output Pin Select) |
* | 0x0008C104 | PFAOE0/1 (Address Output Enable) |
* | 0x0008C106 | PFBCR0-3 (External Bus Control) |
* | 0x0008C10E | PFENET (Ethernet Control) |
* | 0x0008C11F | PWPR (Write Protect Register) |
* | 0x0008C128 | DSCR2[24] (Drive Capacity Control) |
* | 0x0008C140 | P00PFS - PJ5PFS (Pin Function Select) |
* +-----+
*
* STAR Project Pin Assignments (144-pin LFQFP):
* +-----+
* | Pin | Port | PSEL | Function |
* +-----+
* | USB D+/- | (USB) | - | USB CDC Debug |
* | Pin 72 | PB7 | 0x0A | SCI9-TXD (Debug UART TX) |
* | Pin 74 | PB6 | 0x0A | SCI9-RXD (Debug UART RX) |
* | Pin 32 | PA0 | 0x0D | RSPI0-MOSIA (SPI COPI) |
* | Pin 33 | PA1 | 0x0D | RSPI0-MISOA (SPI CIPO) |
* | Pin 35 | PA3 | 0x0D | RSPI0-RSPCKA (SPI CLK) |
* | Pin 34 | PA4 | 0x0D | RSPI0-SSLA0-B (SPI CS) |
* | Pin 97 | PE5 | 0x1E | GTIOC0A (Motor 0 PWM+) |
* | Pin 96 | PE4 | 0x1E | GTIOC0B (Motor 0 PWM-) |
* | Pin 60 | P24 | 0x02 | MTCLKA (Encoder 0 Phase A) |
* | Pin 61 | P25 | 0x02 | MTCLKB (Encoder 0 Phase B) |
* +-----+
*

```

PFS Register Bit Layout

Each pin has a Pin Function Select (PFS) register:

```

* Bit 7 | Bit 6 | Bit 5 | Bits 4-0
* -----
* ASEL | ISEL | Reserved | PSEL[4:0]
* (Analog) | (IRQ) | | (Peripheral Select)
*

```

Common PSEL Values (varies by pin, check RX72N Manual Appendix)

PSEL	Function Examples
0x00	Hi-Z / General Purpose I/O
0x02	MTU3a clock inputs (MTCLKA-D)
0x1E	GPTW outputs (GTIOCnA/B)
0x0A	SCI (TXD, RXD, SCK, CTS)
0x0D	RSPI (COPI, CIPO, RSPCK, SSL)
0x0F	RIIC (SDA, SCL)

Example: Configure PB7 as SCI9 TXD

```
#include "rx72n_mpc_regs.h"

// Unlock PFS write protection
mpc()->pwpr = 0x00; // Clear B0WI bit
mpc()->pwpr = k_mpc_pwpr_pfswe; // Set PFSWE to enable PFS writes

// Configure PB7 for SCI9 TXD (PSEL = 0x0A)
mpc()->pb7pfs = 0x0A;

// Re-lock PFS write protection
mpc()->pwpr = 0x00; // Clear PFSWE
mpc()->pwpr = k_mpc_pwpr_b0wi; // Set B0WI to protect PFSWE

// Don't forget to also enable peripheral mode in PORT PMR register!
```

NASA Power of 10 Compliance

- Rule 3: No dynamic memory - all register access via static inline functions
- Rule 5: Compile-time assertions verify all register layout assumptions
- Rule 8: Uses typed enums (C23) for all constants, no magic numbers
- Rule 10: Clean build with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: MPC register definitions only, no driver logic
- Open/Closed: Accessors work with any pin configuration
- Interface Segregation: Minimal API - just register access functions

Manual Reference

RX72N Group User's Manual: Hardware, Chapter 23 (Multi-Function Pin Controller)

Verification Status

[PASS] VERIFIED (2026-01-28) - All register addresses and offsets verified against RX72N Manual Chapter 23 and Appendix pin function tables.

See also

[rx72n_port_regs.h](#) GPIO port direction and data registers

[rx_mpc.h](#) Higher-level MPC driver with PSEL constants

Date

2026-01-28

Copyright

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Definition in file [rx72n_mpc_regs.h](#).

8.36 rx72n_mpc_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_mpc_regs.h
00003  * @brief RX72N Multi-Function Pin Controller (MPC) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the Multi-Function Pin Controller (MPC) peripheral
00007  * on the RX72N microcontroller. MPC configures which peripheral function is
00008  * connected to each GPIO pin (pin multiplexing).
00009  *
00010  * @par MPC System Architecture
00011  * @verbatim
00012  *
00013  *      RX72N Multi-Function Pin Controller Architecture
00014  *
00015  *      +-----+
00016  *      | MPC Module (0x0008C100) |
00017  *      +-----+
00018  *      | PWSTR | Bus Ctrl | PFS Registers |
00019  *      | (Protect) | (External) | (Pin Function) |
00020  *      +-----+
00021  *      | Unlock | PSEL[4:0] |
00022  *      | v | v |
00023  *      +-----+
00024  *      | | v |
00025  *      +-----+
00026  *      | Pin Multiplexer Matrix |
00027  *      +-----+
00028  *      | Physical Pin <---- MUX <--- [GPIO, SCI, SPI, I2C, MTU, ...] | |
00029  *      +-----+
00030  *
00031  *
00032  * PFS Write Protection Sequence:
00033  *
00034  *      +-----+ +-----+ +-----+ +-----+
00035  *      | PWSTR | -> | PWSTR | -> | Write | -> | PWSTR | -> | PWSTR |
00036  *      | =0x00 | | =0x40 | | PnmPFS | | =0x00 | | =0x80 |
00037  *      | (B0WI=0) | | (PFSWE=1) | | | | (PFSWE=0) | | (B0WI=1) |
00038  *      +-----+ +-----+ +-----+ +-----+
00039  *
00040  * Memory Map (Verified against RX72N Manual Ch23):
00041  *
00042  *      +-----+
00043  *      | Address | Register |
00044  *      +-----+
00045  *      | 0x0008C100 | PFCSE (CS Output Enable) |
00046  *      | 0x0008C102 | PFCSS0/1 (CS Output Pin Select) |
00047  *      | 0x0008C104 | PFAOE0/1 (Address Output Enable) |
00048  *      | 0x0008C106 | PFBCR0-3 (External Bus Control) |
00049  *      | 0x0008C10E | PFENET (Ethernet Control) |
00050  *      | 0x0008C11F | PWSTR (Write Protect Register) |
00051  *      | 0x0008C128 | DSCR2[24] (Drive Capacity Control) |
00052  *      | 0x0008C140 | P00PFS - PJ5PFS (Pin Function Select) |
00053  *      +-----+
00054  *
00055  * STAR Project Pin Assignments (144-pin LQFP):
00056  *
00057  *      +-----+
00058  *      | Pin | Port | PSEL | Function |
00059  *      +-----+
00060  *      | USB D+/- (USB) | - | USB CDC Debug |
00061  *      | Pin 72 | PB7 | 0x0A | SCI9-TXD (Debug UART TX) |
00062  *      | Pin 74 | PB6 | 0x0A | SCI9-RXD (Debug UART RX) |
00063  *      | Pin 32 | PA0 | 0x0D | RSPi0-MOSIA (SPI COPI) |
00064  *      | Pin 33 | PA1 | 0x0D | RSPi0-MISOA (SPI CIPO) |
00065  *      | Pin 35 | PA3 | 0x0D | RSPi0-RSPCKA (SPI CLK) |
00066  *      | Pin 34 | PA4 | 0x0D | RSPi0-SSLA0-B (SPI CS) |
00067  *      | Pin 97 | PE5 | 0x1E | GTIOC0A (Motor 0 PWM+) |
00068  *      | Pin 96 | PE4 | 0x1E | GTIOC0B (Motor 0 PWM-) |
00069  *      | Pin 60 | P24 | 0x02 | MTCLKA (Encoder 0 Phase A) |
00070  *      | Pin 61 | P25 | 0x02 | MTCLKB (Encoder 0 Phase B) |
00071  *      +-----+
00072  * @endverbatim
00073  *
00074  * @par PFS Register Bit Layout
00075  * Each pin has a Pin Function Select (PFS) register:
00076  * @verbatim
00077  * Bit 7 | Bit 6 | Bit 5 | Bits 4-0
00078  * -----
00079  * ASEL | ISEL | Reserved | PSEL[4:0]
00080  * (Analog) (IRQ) | (Peripheral Select)
00081  * @endverbatim
00082  *
00083  * @par Common PSEL Values (varies by pin, check RX72N Manual Appendix)
00084  * | PSEL | Function Examples |

```

```

00083 * |-----|-----|
00084 * | 0x00 | Hi-Z / General Purpose I/O |
00085 * | 0x02 | MTU3a clock inputs (MTCLKA-D) |
00086 * | 0x1E | GPTW outputs (GTIOCnA/B) |
00087 * | 0x0A | SCI (TXD, RXD, SCK, CTS) |
00088 * | 0x0D | RSPI (COPI, CPO, RSPCK, SSL) |
00089 * | 0x0F | RIIC (SDA, SCL) |
00090 *
00091 * @par Example: Configure PB7 as SCI9 TXD
00092 * @code
00093 * #include "rx72n_mpc_regs.h"
00094 *
00095 * // Unlock PFS write protection
00096 * mpc()->pwpr = 0x00; // Clear B0WI bit
00097 * mpc()->pwpr = k_mpc_pwpr_pfswe; // Set PFSWE to enable PFS writes
00098 *
00099 * // Configure PB7 for SCI9 TXD (PSEL = 0x0A)
00100 * mpc()->pb7pfs = 0x0A;
00101 *
00102 * // Re-lock PFS write protection
00103 * mpc()->pwpr = 0x00; // Clear PFSWE
00104 * mpc()->pwpr = k_mpc_pwpr_b0wi; // Set B0WI to protect PFSWE
00105 *
00106 * // Don't forget to also enable peripheral mode in PORT PMR register!
00107 * @endcode
00108 *
00109 * @par NASA Power of 10 Compliance
00110 * - Rule 3: No dynamic memory - all register access via static inline functions
00111 * - Rule 5: Compile-time assertions verify all register layout assumptions
00112 * - Rule 8: Uses typed enums (C23) for all constants, no magic numbers
00113 * - Rule 10: Clean build with -Wall -Wextra -Werror
00114 *
00115 * @par SOLID Principles
00116 * - Single Responsibility: MPC register definitions only, no driver logic
00117 * - Open/Closed: Accessors work with any pin configuration
00118 * - Interface Segregation: Minimal API - just register access functions
00119 *
00120 * @par Manual Reference
00121 * RX72N Group User's Manual: Hardware, Chapter 23 (Multi-Function Pin Controller)
00122 *
00123 * @par Verification Status
00124 * [PASS] VERIFIED (2026-01-28) - All register addresses and offsets verified
00125 * against RX72N Manual Chapter 23 and Appendix pin function tables.
00126 *
00127 * @see rx72n_port_regs.h GPIO port direction and data registers
00128 * @see rx_mpc.h Higher-level MPC driver with PSEL constants
00129 *
00130 * @date 2026-01-28
00131 * @copyright Copyright (c) 2026 Locked Inc.
00132 * SPDX-License-Identifier: MIT
00133 */
00134
00135 #pragma once
00136
00137 #include <stddef.h>
00138 #include <stdint.h>
00139
00140 #ifdef __cplusplus
00141 extern "C" {
00142 #endif
00143
00144 /**
00145 * @defgroup mpc_regs MPC Register Definitions
00146 * @brief Multi-Function Pin Controller (MPC) hardware register interface
00147 * @{
00148 */
00149
00150 /**
=====
00151 * Multi-Function Pin Controller (MPC)
00152 *
=====
00153 */
00154
00155 /**
00156 * @enum rx_mpc_addresses_t
00157 * @brief MPC base address (verified against RX72N Hardware Manual Ch23)
00158 *
00159 * @details
00160 * The MPC module is located at a single base address. All registers are
00161 * accessed relative to this base.
00162 */
00163 typedef enum : uintptr_t {
00164     k_mpc_base_addr = 0x0008C100, /**< MPC register base address */
00165 } rx_mpc_addresses_t;
00166
00167 /**

```

```

00168 * @enum mpc_register_offsets_t
00169 * @brief Byte offsets of key MPC registers from the MPC base address (0x0008C100)
00170 *
00171 * @details
00172 * These constants are used in static_assert calls to verify the struct layout
00173 * matches the hardware register map. Each constant encodes the exact byte offset
00174 * from either the MPC base or the PFS block base (P00PFS), as noted.
00175 *
00176 * @note All values verified against RX72N Manual Chapter 23, Section 23.2
00177 * @since Version 1.0.0
00178 */
00179 typedef enum : uint8_t {
00180     k_mpc_offset_pwpr = 0x1F, /**< PWPR offset from MPC base (0x0008C11F) */
00181     k_mpc_offset_pfs = 0x40, /**< PFS block start offset from MPC base (P00PFS = 0x0008C140) */
00182     k_mpc_offset_port5_pfs = 0x28, /**< Port 5 PFS block offset from P00PFS */
00183     k_mpc_offset_porta_pfs = 0x50, /**< Port A PFS block offset from P00PFS */
00184     k_mpc_offset_porte_pfs = 0x70, /**< Port E PFS block offset from P00PFS */
00185     k_mpc_offset_portj_pfs = 0x90, /**< Port J PFS block offset from P00PFS */
00186     k_mpc_pfs_total_bytes =
00187         152, /**< Total size in bytes of the PFS register block (P00PFS through PJ5PFS with reserved padding) */
00188 } mpc_register_offsets_t;
00189
00190 /** @brief MPC register reserved field sizes */
00191 typedef enum : uint8_t {
00192     k_mpc_reserved1_bytes = 1, /**< Reserved after PFCSE (0x0008C101) */
00193     k_mpc_reserved2_bytes = 4, /**< Reserved after PFBCR3 (0x0008C10A-0x0008C10D) */
00194     k_mpc_reserved3_bytes = 16, /**< Reserved after PFENET (0x0008C10F-0x0008C11E) */
00195     k_mpc_reserved4_bytes = 8, /**< Reserved after PWPR (0x0008C120-0x0008C127) */
00196     k_mpc_dscr2_count = 24, /**< DSCR2 registers count (0x0008C128-0x0008C13F) */
00197
00198     /* Port reserved byte counts (for ports with <8 pins) */
00199     k_port3_reserved_bytes = 3, /**< Port 3: P35-P37 (P35 is NMI-only) */
00200     k_port5_reserved_bytes = 1, /**< Port 5: P53 (dedicated BCLK) */
00201     k_port6_reserved_bytes = 1, /**< Port 6: P65 (dedicated CKE) */
00202     k_port7_reserved_bytes = 1, /**< Port 7: P70 (does not exist) */
00203     k_portf_reserved1_bytes = 2, /**< Port F: PF3-PF4 */
00204     k_portf_reserved2_bytes = 2, /**< Port F: PF6-PF7 */
00205     k_portj_reserved1_bytes = 1, /**< Port J: PJ4 */
00206     k_portj_reserved2_bytes = 2, /**< Port J: PJ6-PJ7 */
00207 } mpc_reserved_sizes_t;
00208
00209 /**
00210  * @brief Pin Function Select Register (C bitfield)
00211  * @details
00212  * Each pin has a PFS register controlling peripheral function selection.
00213  *
00214  * This struct uses C bitfields (the `N` syntax) to pack multiple fields
00215  * into a single byte, matching the hardware register layout exactly:
00216  *
00217  * Bit layout in hardware register (1 byte):
00218  * ```
00219  * 7   6   5   4   3   2   1   0
00220  * +-----+
00221  * | ASEL | ISEL | Reserved | PSEL[4:0] |
00222  * +-----+
00223  * ```
00224  *
00225  * The syntax `volatile uint8_t name : bits` defines a bitfield where:
00226  * - `name` is the field name (or omitted for padding/reserved bits)
00227  * - `bits` is the number of bits the field occupies
00228  *
00229  * Example: `psel : 5` means psel occupies 5 bits (bits 0-4)
00230  *
00231  * This is standard C (not C++) and commonly used in embedded systems
00232  * for hardware register definitions where bits are packed together.
00233  */
00234 typedef struct {
00235     volatile uint8_t psel : 5; /**< Peripheral Select (bits 0-4) */
00236     uint8_t : 1; /**< Reserved bit (bit 5) - unnamed bitfield for padding */
00237     volatile uint8_t isel : 1; /**< Interrupt Input Select (bit 6) */
00238     volatile uint8_t asel : 1; /**< Analog Input Select (bit 7) */
00239 } rx_pfs_regs_t;
00240
00241 /**
00242  * @brief MPC Register Map
00243  * @details
00244  * Multi-Function Pin Controller (MPC) registers for pin function selection.
00245  * Controls which peripheral function each GPIO pin is assigned to.
00246  * Base address: 0x0008C100
00247  *
00248  * CRITICAL: This struct matches the exact hardware memory layout!
00249  * - 0x0008C100: Bus control registers (PFCSE, PFCSS, PFAOE, PFBCR, PFENET)
00250  * - 0x0008C11F: PWPR (Write Protect Register)
00251  * - 0x0008C140: PFS registers start (P00PFS)
00252  *
00253  * Package Support (144-pin LFQFP R5F572NNHxFB):
00254  * - Available ports: 0-9 (partial), A-E (full), F (partial), J (partial: PJ3, PJ5)

```



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00255 * - Not available on 144-pin: G, H
00256 * - Note: All ports are defined for compatibility with larger packages
00257 *
00258 * Complete Memory Layout:
00259 * - 0x0008C100: PFCSE (CS Output Enable Register)
00260 * - 0x0008C101: Reserved (1 byte)
00261 * - 0x0008C102-0x0008C103: PFCSS0, PFCSS1 (CS Output Pin Select)
00262 * - 0x0008C104-0x0008C105: PFAOE0, PFAOE1 (Address Output Enable)
00263 * - 0x0008C106-0x0008C109: PFBCR0-3 (External Bus Control)
00264 * - 0x0008C10A-0x0008C10D: Reserved (4 bytes)
00265 * - 0x0008C10E: PFENET (Ethernet Control)
00266 * - 0x0008C10F-0x0008C11E: Reserved (16 bytes)
00267 * - 0x0008C11F: PWPR (Write Protect Register)
00268 * - 0x0008C120-0x0008C127: Reserved (8 bytes)
00269 * - 0x0008C128-0x0008C13F: DSCR2 (Drive Capacity Control, 24 bytes)
00270 * - 0x0008C140+: PFS registers (P00PFS through PJPFS, 152 bytes)
00271 */
00272 typedef struct {
00273     /* Bus Control Registers (0x0008C100 - 0x0008C10E) */
00274     volatile uint8_t pfcae; /*< +0x00: CS Output Enable Register */
00275     uint8_t reserved1[k_mpc_reserved1_bytes]; /*< +0x01: Reserved */
00276     volatile uint8_t pfcss0; /*< +0x02: CS Output Pin Select 0 */
00277     volatile uint8_t pfcss1; /*< +0x03: CS Output Pin Select 1 */
00278     volatile uint8_t pfaoe0; /*< +0x04: Address Output Enable 0 */
00279     volatile uint8_t pfaoe1; /*< +0x05: Address Output Enable 1 */
00280     volatile uint8_t pfbcr0; /*< +0x06: External Bus Control 0 */
00281     volatile uint8_t pfbcr1; /*< +0x07: External Bus Control 1 */
00282     volatile uint8_t pfbcr2; /*< +0x08: External Bus Control 2 */
00283     volatile uint8_t pfbcr3; /*< +0x09: External Bus Control 3 */
00284     uint8_t reserved2[k_mpc_reserved2_bytes]; /*< +0x0A-0x0D: Reserved */
00285     volatile uint8_t pfenet; /*< +0x0E: Ethernet Control Register */
00286     uint8_t reserved3[k_mpc_reserved3_bytes]; /*< +0x0F-0x1E: Reserved */
00287
00288     /* Write Protection (0x0008C11F) */
00289     volatile uint8_t pwpr; /*< +0x1F: Write Protect Register (enable PFS writes) */
00290
00291     /* Reserved and Drive Capacity (0x0008C120 - 0x0008C13F) */
00292     uint8_t reserved4[k_mpc_reserved4_bytes]; /*< +0x20-0x27: Reserved */
00293     uint8_t dscr2[k_mpc_dscr2_count]; /*< +0x28-0x3F: Drive Capacity Control 2 */
00294
00295     /* Pin Function Select Registers start at 0x0008C140 (+0x40 from base) */
00296     volatile uint8_t p00pfs; /*< Port 0 Pin 0 Function Select */
00297     volatile uint8_t p01pfs; /*< Port 0 Pin 1 Function Select */
00298     volatile uint8_t p02pfs; /*< Port 0 Pin 2 Function Select */
00299     volatile uint8_t p03pfs; /*< Port 0 Pin 3 Function Select */
00300     volatile uint8_t p04pfs; /*< Port 0 Pin 4 Function Select */
00301     volatile uint8_t p05pfs; /*< Port 0 Pin 5 Function Select */
00302     volatile uint8_t p06pfs; /*< Port 0 Pin 6 Function Select */
00303     volatile uint8_t p07pfs; /*< Port 0 Pin 7 Function Select */
00304     volatile uint8_t p10pfs; /*< Port 1 Pin 0 Function Select */
00305     volatile uint8_t p11pfs; /*< Port 1 Pin 1 Function Select */
00306     volatile uint8_t p12pfs; /*< Port 1 Pin 2 Function Select */
00307     volatile uint8_t p13pfs; /*< Port 1 Pin 3 Function Select */
00308     volatile uint8_t p14pfs; /*< Port 1 Pin 4 Function Select */
00309     volatile uint8_t p15pfs; /*< Port 1 Pin 5 Function Select */
00310     volatile uint8_t p16pfs; /*< Port 1 Pin 6 Function Select */
00311     volatile uint8_t p17pfs; /*< Port 1 Pin 7 Function Select */
00312     volatile uint8_t p20pfs; /*< Port 2 Pin 0 Function Select */
00313     volatile uint8_t p21pfs; /*< Port 2 Pin 1 Function Select */
00314     volatile uint8_t p22pfs; /*< Port 2 Pin 2 Function Select */
00315     volatile uint8_t p23pfs; /*< Port 2 Pin 3 Function Select */
00316     volatile uint8_t p24pfs; /*< Port 2 Pin 4 Function Select */
00317     volatile uint8_t p25pfs; /*< Port 2 Pin 5 Function Select */
00318     volatile uint8_t p26pfs; /*< Port 2 Pin 6 Function Select */
00319     volatile uint8_t p27pfs; /*< Port 2 Pin 7 Function Select */
00320     volatile uint8_t p30pfs; /*< Port 3 Pin 0 Function Select */
00321     volatile uint8_t p31pfs; /*< Port 3 Pin 1 Function Select */
00322     volatile uint8_t p32pfs; /*< Port 3 Pin 2 Function Select */
00323     volatile uint8_t p33pfs; /*< Port 3 Pin 3 Function Select */
00324     volatile uint8_t p34pfs; /*< Port 3 Pin 4 Function Select */
00325     uint8_t port3_reserved[k_port3_reserved_bytes]; /*< Port 3 reserved (P35=NMI, P36-P37 N/A) */
00326     volatile uint8_t p40pfs; /*< Port 4 Pin 0 Function Select */
00327     volatile uint8_t p41pfs; /*< Port 4 Pin 1 Function Select */
00328     volatile uint8_t p42pfs; /*< Port 4 Pin 2 Function Select */
00329     volatile uint8_t p43pfs; /*< Port 4 Pin 3 Function Select */
00330     volatile uint8_t p44pfs; /*< Port 4 Pin 4 Function Select */
00331     volatile uint8_t p45pfs; /*< Port 4 Pin 5 Function Select */
00332     volatile uint8_t p46pfs; /*< Port 4 Pin 6 Function Select */
00333     volatile uint8_t p47pfs; /*< Port 4 Pin 7 Function Select */
00334     volatile uint8_t p50pfs; /*< Port 5 Pin 0 Function Select */
00335     volatile uint8_t p51pfs; /*< Port 5 Pin 1 Function Select */
00336     volatile uint8_t p52pfs; /*< Port 5 Pin 2 Function Select */
00337     uint8_t port5_reserved[k_port5_reserved_bytes]; /*< Port 5 reserved (P53=BCLK dedicated) */
00338     volatile uint8_t p54pfs; /*< Port 5 Pin 4 Function Select */
00339     volatile uint8_t p55pfs; /*< Port 5 Pin 5 Function Select */
00340     volatile uint8_t p56pfs; /*< Port 5 Pin 6 Function Select */
00341     volatile uint8_t p57pfs; /*< Port 5 Pin 7 Function Select */

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00429 volatile uint8_t ph1pfs;          /**< Port H Pin 1 Function Select */
00430 volatile uint8_t ph2pfs;          /**< Port H Pin 2 Function Select */
00431 volatile uint8_t ph3pfs;          /**< Port H Pin 3 Function Select */
00432 volatile uint8_t ph4pfs;          /**< Port H Pin 4 Function Select */
00433 volatile uint8_t ph5pfs;          /**< Port H Pin 5 Function Select */
00434 volatile uint8_t ph6pfs;          /**< Port H Pin 6 Function Select */
00435 volatile uint8_t ph7pfs;          /**< Port H Pin 7 Function Select */
00436 volatile uint8_t pj0pfs;          /**< Port J Pin 0 Function Select */
00437 volatile uint8_t pj1pfs;          /**< Port J Pin 1 Function Select */
00438 volatile uint8_t pj2pfs;          /**< Port J Pin 2 Function Select */
00439 volatile uint8_t pj3pfs;          /**< Port J Pin 3 Function Select */
00440 uint8_t portj_reserved1[k_portj_reserved1_bytes]; /**< Port J reserved (PJ4 N/A) */
00441 volatile uint8_t pj5pfs;          /**< Port J Pin 5 Function Select */
00442 uint8_t portj_reserved2[k_portj_reserved2_bytes]; /**< Port J reserved (PJ6-PJ7 N/A) */
00443 } rx_mpc_regs_t;
00444
00445 /**
00446  * @brief Get pointer to MPC registers
00447  *
00448  * @details
00449  * Returns a volatile pointer to the MPC register structure at address 0x0008C100.
00450  * Provides access to all MPC registers including PWPR and PFS registers.
00451  *
00452  * @return Volatile pointer to MPC register structure
00453  *
00454  * @par Example
00455  * @code
00456  * // Unlock, configure pin, re-lock
00457  * mpc()->pwpr = 0x00;          // Clear B0WI
00458  * mpc()->pwpr = k_mpc_pwpr_pfswe; // Enable PFS write
00459  * mpc()->pb7pfs = 0x0A;        // SCI9 TXD
00460  * mpc()->pwpr = 0x00;          // Clear PFSWE
00461  * mpc()->pwpr = k_mpc_pwpr_b0wi; // Protect PFSWE
00462  * @endcode
00463  */
00464 static inline volatile rx_mpc_regs_t* mpc(void)
00465 {
00466     return (volatile rx_mpc_regs_t*)k_mpc_base_addr;
00467 }
00468
00469 /**
00470  * @enum mpc_pwpr_bits_t
00471  * @brief MPC Write Protect Register (PWPR) bit definitions
00472  *
00473  * @details
00474  * The PWPR register protects PFS registers from accidental writes.
00475  * A specific unlock sequence is required to modify any PFS register.
00476  *
00477  * @par PWPR Register Layout
00478  * @verbatim
00479  * Bit | Name | Description
00480  * ----+-----+-----
00481  * 7 | B0WI | PFSWE Bit Write Disable (1=PFSWE protected)
00482  * 6 | PFSWE | PFS Write Enable (1=PFS writes allowed)
00483  * 5-0 | - | Reserved
00484  * @endverbatim
00485  *
00486  * @par Unlock Sequence
00487  * 1. Write 0x00 to PWPR (clear B0WI)
00488  * 2. Write 0x40 to PWPR (set PFSWE)
00489  * 3. Write to desired PFS register(s)
00490  * 4. Write 0x00 to PWPR (clear PFSWE)
00491  * 5. Write 0x80 to PWPR (set B0WI)
00492  *
00493  * @warning Failing to re-lock after configuration leaves PFS vulnerable!
00494  */
00495 typedef enum : uint8_t {
00496     k_mpc_pwpr_pfswe = (1 << 6), /**< Bit 6: PFS Write Enable (1=allow writes) */
00497     k_mpc_pwpr_b0wi = (1 << 7), /**< Bit 7: PFSWE Write Disable (1=protect PFSWE) */
00498 } mpc_pwpr_bits_t;
00499
00500 /**
00501  * @enum pfs_bits_t
00502  * @brief Pin Function Select (PFS) register bit definitions
00503  *
00504  * @details
00505  * Each GPIO pin has a PFS register controlling its function assignment.
00506  * The PSEL field selects which peripheral function the pin is connected to.
00507  *
00508  * @par PFS Register Layout
00509  * @verbatim
00510  * Bit | Name | Description
00511  * ----+-----+-----
00512  * 7 | ASEL | Analog Input Select (1=analog function)
00513  * 6 | ISEL | IRQ Input Select (1=IRQ input enabled)
00514  * 5 | Reserved | Always write 0
00515  * 4-0 | PSEL[4:0] | Peripheral function select (0x00-0x1F)

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```

00516 * @endverbatim
00517 *
00518 * @note PSEL values are pin-specific. Check RX72N Manual Appendix for
00519 * the function assignment table for each pin.
00520 */
00521 typedef enum : uint8_t {
00522     k_pfs_psel_mask = 0x1F, /**< Bits 0-4: Peripheral Select mask */
00523     k_pfs_isel      = (1 « 6), /**< Bit 6: Interrupt Input Select */
00524     k_pfs_asel      = (1 « 7), /**< Bit 7: Analog Input Select */
00525 } pfs_bits_t;
00526
00527 /*
=====
00528 * Compile-Time Register Layout Verification
00529 *
=====
00530 * These static assertions verify the struct layout matches hardware exactly.
00531 * If any assertion fails, the struct definition is WRONG and must be fixed!
00532 *
00533 * Reference: RX72N Group User's Manual: Hardware, Chapter 23, Section 23.2
00534 *
=====
00535 */
00536
00537 /**
00538 * @defgroup mpc_static_assert MPC Static Assertions
00539 * @brief Compile-time verification of MPC register layout
00540 * @details
00541 * These static assertions are verified at compile time to ensure the
00542 * C structure definitions match the actual hardware register layout.
00543 * A failed assertion indicates a bug in the register definitions that
00544 * must be fixed before the code can compile.
00545 * @{
00546 */
00547
00548 /* Verify critical register offsets from base (0x0008C100) */
00549 static_assert(offsetof(rx_mpc_regs_t, pfcse) == 0,
00550     "PFCSE must be at offset 0x00 (address 0x0008C100)");
00551 static_assert(offsetof(rx_mpc_regs_t, pwpr) == k_mpc_offset_pwpr,
00552     "PWPR must be at offset 0x1F (address 0x0008C11F)");
00553 static_assert(offsetof(rx_mpc_regs_t, p00pfs) == k_mpc_offset_pfs,
00554     "P00PFS must be at offset 0x40 (address 0x0008C140)");
00555
00556 /* Verify specific port offsets from P00PFS base */
00557 static_assert(offsetof(rx_mpc_regs_t, p50pfs) == k_mpc_offset_pfs + k_mpc_offset_port5_pfs,
00558     "P50PFS must be at offset 0x68 (Port 5 base)");
00559 static_assert(offsetof(rx_mpc_regs_t, pa0pfs) == k_mpc_offset_pfs + k_mpc_offset_porta_pfs,
00560     "PA0PFS must be at offset 0x90 (Port A base)");
00561 static_assert(offsetof(rx_mpc_regs_t, pe0pfs) == k_mpc_offset_pfs + k_mpc_offset_porte_pfs,
00562     "PE0PFS must be at offset 0xB0 (Port E base)");
00563 static_assert(offsetof(rx_mpc_regs_t, pj0pfs) == k_mpc_offset_pfs + k_mpc_offset_portj_pfs,
00564     "PJ0PFS must be at offset 0xD0 (Port J base)");
00565
00566 /* Verify struct total size (64 bytes control + 152 bytes PFS = 216 bytes) */
00567 static_assert(sizeof(rx_mpc_regs_t) == k_mpc_offset_pfs + k_mpc_pfs_total_bytes,
00568     "MPC struct size must be 64 + 152 = 216 bytes total");
00569
00570 /** @} */ /* End of mpc_static_assert group */
00571
00572 /** @} */ /* End of mpc_regs group */
00573
00574 /*
=====
00575 * Pin Mapping Examples
00576 *
=====
00577 * Example MPC configuration for common STAR peripherals.
00578 * See rx_mpc.h for PSEL constant definitions and helper functions.
00579
00580 * 1. Configure PB7 as SCI9 TXD (Debug UART):
00581 *     mpc()->pwpr = 0x00; // Clear B0WI
00582 *     mpc()->pwpr = k_mpc_pwpr_pfswe; // Enable PFS write
00583 *     mpc()->pb7pfs = 0x0A; // PSEL = 0x0A for SCI
00584 *     mpc()->pwpr = 0x00; // Disable PFS write
00585 *     mpc()->pwpr = k_mpc_pwpr_b0wi; // Protect PFSWE
00586
00587 * 2. Configure PE5 as GTIOC0A (Motor 0 PWM Phase):
00588 *     mpc()->pwpr = 0x00;
00589 *     mpc()->pwpr = k_mpc_pwpr_pfswe;
00590 *     mpc()->pe5pfs = 0x1E; // PSEL = 0x1E for GPTW (k_psel_gptw)
00591 *     mpc()->pwpr = 0x00;
00592 *     mpc()->pwpr = k_mpc_pwpr_b0wi;
00593
00594 * 3. Configure P24 as MTCLKA (Encoder 0 Phase A):
00595 *     mpc()->pwpr = 0x00;
00596 *     mpc()->pwpr = k_mpc_pwpr_pfswe;
00597 *     mpc()->p24pfs = 0x02; // PSEL = 0x02 for MTU CLK

```

```

00598 *      mpc()->pwpr = 0x00;
00599 *      mpc()->pwpr = k_mpc_pwpr_b0wi;
00600 *
00601 * 4. Configure PA4 as SSLA0-B (RPI5 SPI Chip Select):
00602 *      mpc()->pwpr = 0x00;
00603 *      mpc()->pwpr = k_mpc_pwpr_pfswe;
00604 *      mpc()->pa4pfs = 0x0D;           // PSEL = 0x0D for RSPi
00605 *      mpc()->pwpr = 0x00;
00606 *      mpc()->pwpr = k_mpc_pwpr_b0wi;
00607 *
00608 * * Note: After MPC configuration, also set PMR (Port Mode Register) bit to '1'
00609 * to enable peripheral mode. See rx_mpc.h for helper functions and named constants.
00610 */
00611
00612 #ifdef __cplusplus
00613 }
00614 #endif

```

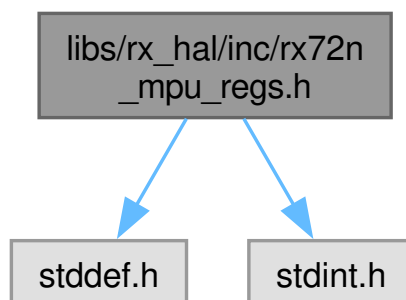
8.37 libs/rx72n/inc/rx72nmpuregs.h File Reference

RX72N Memory Protection Unit (MPU) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_mpu_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct `rx_mpu_region_regs_t`
MPU per-region register pair.

Enumerations

- enum `mpu_addresses_t` : `uintptr_t` {
`k_mpu_region_base_addr` = 0x00086400U , `k_mpu_control_base_addr` = 0x00086500U ,
`k_mpu_region_size` = 0x08U , `k_mpu_rspage0_addr` = 0x00086400U ,
`k_mpu_rspage1_addr` = 0x00086408U , `k_mpu_rspage2_addr` = 0x00086410U , `k_mpu_rspage3_addr`
= 0x00086418U , `k_mpu_rspage4_addr` = 0x00086420U ,
`k_mpu_rspage5_addr` = 0x00086428U , `k_mpu_rspage6_addr` = 0x00086430U , `k_mpu_rspage7_addr`
= 0x00086438U , `k_mpu_repage0_addr` = 0x00086404U ,
`k_mpu_repage1_addr` = 0x0008640CU , `k_mpu_repage2_addr` = 0x00086414U , `k_mpu_repage3_addr`
= 0x0008641CU , `k_mpu_repage4_addr` = 0x00086424U ,
`k_mpu_repage5_addr` = 0x0008642CU , `k_mpu_repage6_addr` = 0x00086434U , `k_mpu_repage7_addr`
= 0x0008643CU , `k_mpu_mpen_addr` = 0x00086500U ,
`k_mpu_mpbac_addr` = 0x00086504U , `k_mpu_mpeclr_addr` = 0x00086508U , `k_mpu_mpests_addr`
= 0x0008650CU , `k_mpu_mpdea_addr` = 0x00086514U ,
`k_mpu_mpsa_addr` = 0x00086520U , `k_mpu_mpop_s_addr` = 0x00086524U , `k_mpu_mpopi_addr`
= 0x00086526U , `k_mpu_mhiti_addr` = 0x00086528U ,
`k_mpu_mhitd_addr` = 0x0008652CU }

MPU module addresses.

- enum `mpu_region_t` : `uint8_t` {
`k_mpu_region_0` = 0U , `k_mpu_region_1` = 1U , `k_mpu_region_2` = 2U , `k_mpu_region_3`
= 3U ,
`k_mpu_region_4` = 4U , `k_mpu_region_5` = 5U , `k_mpu_region_6` = 6U , `k_mpu_region_7`
= 7U ,
`k_mpu_region_count` = 8U }

MPU region numbers.

- enum `mpu_page_t` : `uint32_t` { `k_mpu_page_size` = 16U , `k_mpu_page_shift` = 4U ,
`k_mpu_page_mask` = 0xFFFFFFFF0U , `k_mpu_page_max` = 0xFFFFFFFFU }

MPU page configuration constants.

- enum `rspage_bits_t` : `uint32_t` { `k_rspage_reserved_mask` = 0x0000000FU , `k_rspage_rspn_mask`
= 0xFFFFFFFF0U , `k_rspage_rspn_shift` = 4U }

Region Start Page Number Register (RSPAGEn) bit definitions.

- enum `repage_bits_t` : `uint32_t` {
`k_repage_v` = 0x00000001U , `k_repage_uac_r` = 0x00000002U , `k_repage_uac_w` =
0x00000004U , `k_repage_uac_x` = 0x00000008U ,
`k_repage_uac_mask` = 0x0000000EU , `k_repage_uac_shift` = 1U , `k_repage_repn_mask` =
0xFFFFFFFF0U , `k_repage_repn_shift` = 4U ,
`k_repage_uac_ro` = `k_repage_uac_r` , `k_repage_uac_rw` = `k_repage_uac_r` | `k_repage_uac_w` ,
`k_repage_uac_xo` = `k_repage_uac_x` , `k_repage_uac_rx` = `k_repage_uac_r` |
`k_repage_uac_x` ,
`k_repage_uac_rwx` = `k_repage_uac_r` | `k_repage_uac_w` | `k_repage_uac_x` }

Region End Page Number Register (REPAGEn) bit definitions.

- enum `mpen_bits_t` : `uint32_t` { `k_mpen_mpen` = 0x00000001U }

Memory Protection Enable Register (MPEN) bit definitions.

- enum `mpbac_bits_t` : `uint32_t` {
`k_mpbac_ubac_r` = 0x00000002U , `k_mpbac_ubac_w` = 0x00000004U , `k_mpbac_ubac_x` =
0x00000008U , `k_mpbac_ubac_mask` = 0x0000000EU ,
`k_mpbac_ubac_shift` = 1U }

Background Access Control Register (MPBAC) bit definitions.

- enum `mpeclr_bits_t` : `uint32_t` { `k_mpeclr_clr` = 0x00000001U }

Error Status Clearing Register (MPECLR) bit definitions.

- enum `mpests_bits_t` : `uint32_t` { `k_mpests_imper` = 0x00000001U , `k_mpests_dimper` =
0x00000002U , `k_mpests_drw` = 0x00000004U }

Memory Protection Error Status Register (MPESTS) bit definitions.

- enum `mpops_bits_t` : `uint16_t` { `k_mpop_s` = 0x0001U }

Region Search Operation Register (MPOPS) bit definitions.

- enum `mpopi_bits_t` : `uint16_t` { `k_mpopi_inv` = 0x0001U }

Region Invalidation Operation Register (MPOPI) bit definitions.

- enum `mhiti_bits_t` : `uint32_t` {
`k_mhiti_uhaci_r` = 0x00000002U , `k_mhiti_uhaci_w` = 0x00000004U , `k_mhiti_uhaci_x` = 0x00000008U , `k_mhiti_uhaci_mask` = 0x0000000EU ,
`k_mhiti_hiti0` = 0x00010000U , `k_mhiti_hiti1` = 0x00020000U , `k_mhiti_hiti2` = 0x00040000U ,
`k_mhiti_hiti3` = 0x00080000U ,
`k_mhiti_hiti4` = 0x00100000U , `k_mhiti_hiti5` = 0x00200000U , `k_mhiti_hiti6` = 0x00400000U ,
`k_mhiti_hiti7` = 0x00800000U ,
`k_mhiti_hiti_mask` = 0x00FF0000U , `k_mhiti_hiti_shift` = 16U }

Instruction Hit Region Register (MHITI) bit definitions.

- enum `mhitd_bits_t` : `uint32_t` {
`k_mhitd_uhacd_r` = 0x00000002U , `k_mhitd_uhacd_w` = 0x00000004U , `k_mhitd_uhacd_x` = 0x00000008U , `k_mhitd_uhacd_mask` = 0x0000000EU ,
`k_mhitd_hitd0` = 0x00010000U , `k_mhitd_hitd1` = 0x00020000U , `k_mhitd_hitd2` = 0x00040000U , `k_mhitd_hitd3` = 0x00080000U ,
`k_mhitd_hitd4` = 0x00100000U , `k_mhitd_hitd5` = 0x00200000U , `k_mhitd_hitd6` = 0x00400000U , `k_mhitd_hitd7` = 0x00800000U ,
`k_mhitd_hitd_mask` = 0x00FF0000U , `k_mhitd_hitd_shift` = 16U }

Data Hit Region Register (MHITD) bit definitions.

Functions

- static volatile `rx_mpu_region_regs_t` * `mpu_region` (`uint8_t` region)
 Get pointer to MPU region register pair.
- static volatile `uint32_t` * `mpu_rspage_reg` (`uint8_t` region)
 Get pointer to RSPAGE register for a region.
- static volatile `uint32_t` * `mpu_repage_reg` (`uint8_t` region)
 Get pointer to REPAGE register for a region.
- static volatile `uint32_t` * `mpu_mpen_reg` (`void`)
 Get pointer to MPEN (Memory Protection Enable) register.
- static volatile `uint32_t` * `mpu_mpbac_reg` (`void`)
 Get pointer to MPBAC (Background Access Control) register.
- static volatile `uint32_t` * `mpu_mpeclr_reg` (`void`)
 Get pointer to MPECLR (Error Status Clearing) register.
- static volatile `uint32_t` * `mpu_mpests_reg` (`void`)
 Get pointer to MPESTS (Error Status) register.
- static volatile `uint32_t` * `mpu_mpdea_reg` (`void`)
 Get pointer to MPDEA (Data Error Address) register.
- static volatile `uint32_t` * `mpu_mpsa_reg` (`void`)
 Get pointer to MPSA (Region Search Address) register.
- static volatile `uint16_t` * `mpu_mpops_reg` (`void`)
 Get pointer to MPOPS (Region Search Operation) register.
- static volatile `uint16_t` * `mpu_mpopi_reg` (`void`)
 Get pointer to MPOPI (Region Invalidation) register.
- static volatile `uint32_t` * `mpu_mhiti_reg` (`void`)
 Get pointer to MHITI (Instruction Hit Region) register.
- static volatile `uint32_t` * `mpu_mhitd_reg` (`void`)
 Get pointer to MHITD (Data Hit Region) register.
- static `uint32_t` `mpu_addr_to_page` (`uint32_t` addr)
 Convert address to page number.
- static `uint32_t` `mpu_page_to_reg` (`uint32_t` page)
 Convert page number to RSPAGE/REPAGE register value.

8.37.1 Detailed Description

RX72N Memory Protection Unit (MPU) Register Definitions.

Complete register definitions for the RX72N Memory Protection Unit. The MPU provides address-based access control for user-mode programs, protecting code and data regions from unauthorized access.

MPU Overview

- Covers entire 4 GB address space (0x00000000-0xFFFFFFFF)
- 8 independent configurable regions (Region 0-7)
- 16-byte page granularity (smallest unit of protection)
- 3 access types: Execute (X), Write (W), Read (R)
- Protection only active in user mode (supervisor mode bypasses)
- Background region provides default access for unprotected areas
- Overlapping regions use OR of permissions (permission wins)

Register Organization

Address Range	Description
0x00086400-0x043F	Region registers (RSPAGE0-7, REPAGE0-7)
0x00086500-0x052F	Control/status registers

Region Registers (8 regions x 8 bytes each = 64 bytes)

Offset	Register	Description
0x00	RSPAGEn	Region n Start Page Number (32-bit)
0x04	REPAGEn	Region n End Page Number (32-bit)

Control/Status Registers

Address	Size	Register	Description
0x00086500	32	MPEN	Memory Protection Enable
0x00086504	32	MPBAC	Background Access Control
0x00086508	32	MPECLR	Error Status Clearing
0x0008650C	32	MPESTS	Error Status
0x00086514	32	MPDEA	Data Error Address
0x00086520	32	MPSA	Region Search Address
0x00086524	16	MPOPS	Region Search Operation
0x00086526	16	MPOPI	Region Invalidation Operation

Address	Size	Register	Description
0x00086528	32	MHITI	Instruction Hit Region
0x0008652C	32	MHITD	Data Hit Region

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops)
- Rule 3: No dynamic memory allocation
- Rule 4: Short accessor functions
- Rule 5: Static assertions verify addresses
- Rule 6: Minimal scope
- Rule 7: N/A
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 17: Memory-Protection Unit (MPU), pages 657-676
- Section 17.2: Register Descriptions

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_mpu_regs.h](#).

8.37.2 Data Structure Documentation

8.37.2.1 struct rx_mpu_region_regs_t

MPU per-region register pair.

Memory-mapped register layout for one MPU region. Each region has 8 bytes (RSPAGE + REPAGE).

Note

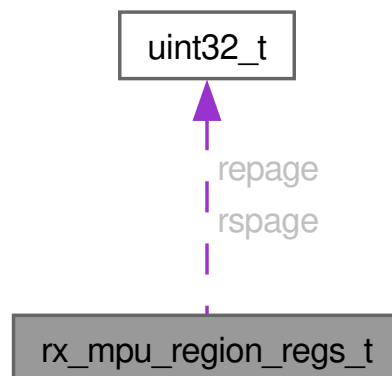
Structure size verified as 8 bytes.

Since

Version 1.0.0

Definition at line 623 of file [rx72n_mpu_regs.h](#).

Collaboration diagram for rx_mpu_region_regs_t:



Data Fields

volatile uint32_t	repag	Region End Page Number Register (REPAGEn). Bits [31:4] = End page number (inclusive) Bit [3] = Execute permission (UAC.X) Bit [2] = Write permission (UAC.W) Bit [1] = Read permission (UAC.R) Bit [0] = Valid bit (V)
volatile uint32_t	rspag	Region Start Page Number Register (RSPAGEn). Bits [31:4] = Start page number (address[31:4]) Bits [3:0] = Reserved (write 0)

8.37.3 Enumeration Type Documentation

8.37.3.1 mhitd`bits` t

```
enum mhitd_bits_t : uint32_t
```

Data Hit Region Register (MHITD) bit definitions.

After a data memory protection error or region search, this register indicates which region was hit and what permissions were set.

Register Layout (32-bit @ 0x0008652C)

Bits	Field	Description
0	-	Reserved (read as 0)
3:↔ 1	UHACD[2:↔ 0]	Hit region access control (RWX)
15:↔ 4	-	Reserved (read as 0)
23:16	HITD[7:↔ 0]	Region hit flags (1=hit in region)
31:24	-	Reserved (read as 0)

Note

HITD[7:0] = 0 means error in background region

Manual ref: Ch17 section 17.2.12

Since

Version 1.0.0

Enumerator

k_mhitd_uhacd_r	Read permission in hit region (bit 1).
k_mhitd_uhacd_w	Write permission in hit region (bit 2).
k_mhitd_uhacd_x	Execute permission in hit region (bit 3).
k_mhitd_uhacd_mask	Access control bits mask (bits 3:1).
k_mhitd_hitd0	Region 0 hit flag (bit 16).
k_mhitd_hitd1	Region 1 hit flag (bit 17).
k_mhitd_hitd2	Region 2 hit flag (bit 18).
k_mhitd_hitd3	Region 3 hit flag (bit 19).
k_mhitd_hitd4	Region 4 hit flag (bit 20).
k_mhitd_hitd5	Region 5 hit flag (bit 21).

k_mhited_hitd6	Region 6 hit flag (bit 22).
k_mhited_hitd7	Region 7 hit flag (bit 23).
k_mhited_hitd_mask	Hit region flags mask (bits 23:16).
k_mhited_hitd_shift	Hit region flags shift amount.

Definition at line 563 of file [rx72n_mpu_regs.h](#).

```

00563         : uint32_t {
00564     /** @brief Read permission in hit region (bit 1) */
00565     k_mhited_uhacd_r = 0x00000002U,
00566
00567     /** @brief Write permission in hit region (bit 2) */
00568     k_mhited_uhacd_w = 0x00000004U,
00569
00570     /** @brief Execute permission in hit region (bit 3) */
00571     k_mhited_uhacd_x = 0x00000008U,
00572
00573     /** @brief Access control bits mask (bits 3:1) */
00574     k_mhited_uhacd_mask = 0x0000000EU,
00575
00576     /** @brief Region 0 hit flag (bit 16) */
00577     k_mhited_hitd0 = 0x00010000U,
00578
00579     /** @brief Region 1 hit flag (bit 17) */
00580     k_mhited_hitd1 = 0x00020000U,
00581
00582     /** @brief Region 2 hit flag (bit 18) */
00583     k_mhited_hitd2 = 0x00040000U,
00584
00585     /** @brief Region 3 hit flag (bit 19) */
00586     k_mhited_hitd3 = 0x00080000U,
00587
00588     /** @brief Region 4 hit flag (bit 20) */
00589     k_mhited_hitd4 = 0x00100000U,
00590
00591     /** @brief Region 5 hit flag (bit 21) */
00592     k_mhited_hitd5 = 0x00200000U,
00593
00594     /** @brief Region 6 hit flag (bit 22) */
00595     k_mhited_hitd6 = 0x00400000U,
00596
00597     /** @brief Region 7 hit flag (bit 23) */
00598     k_mhited_hitd7 = 0x00800000U,
00599
00600     /** @brief Hit region flags mask (bits 23:16) */
00601     k_mhited_hitd_mask = 0x00FF0000U,
00602
00603     /** @brief Hit region flags shift amount */
00604     k_mhited_hitd_shift = 16U,
00605 } mhited_bits_t;

```

8.37.3.2 mhited_bits_t

```
enum mhited_bits_t : uint32_t
```

Instruction Hit Region Register (MHITI) bit definitions.

After an instruction memory protection error, this register indicates which region caused the error and what permissions were set.

Register Layout (32-bit @ 0x00086528)

Bits	Field	Description
0	-	Reserved (read as 0)

Bits	Field	Description
3:↵ 1	UHACI[2:↵ 0]	Hit region access control (RWX)
15:↵ 4	-	Reserved (read as 0)
23:16	HITI[7:↵ 0]	Region hit flags (1=hit in region)
31:24	-	Reserved (read as 0)

Note

HITI[7:0] = 0 means error in background region

Manual ref: Ch17 section 17.2.11

Since

Version 1.0.0

Enumerator

k_mhiti_uhaci_r	Read permission in hit region (bit 1).
k_mhiti_uhaci_w	Write permission in hit region (bit 2).
k_mhiti_uhaci_x	Execute permission in hit region (bit 3).
k_mhiti_uhaci_mask	Access control bits mask (bits 3:1).
k_mhiti_hiti0	Region 0 hit flag (bit 16).
k_mhiti_hiti1	Region 1 hit flag (bit 17).
k_mhiti_hiti2	Region 2 hit flag (bit 18).
k_mhiti_hiti3	Region 3 hit flag (bit 19).
k_mhiti_hiti4	Region 4 hit flag (bit 20).
k_mhiti_hiti5	Region 5 hit flag (bit 21).
k_mhiti_hiti6	Region 6 hit flag (bit 22).
k_mhiti_hiti7	Region 7 hit flag (bit 23).
k_mhiti_hiti_mask	Hit region flags mask (bits 23:16).
k_mhiti_hiti_shift	Hit region flags shift amount.

Definition at line 493 of file rx72n_mpu_regs.h.

```

00493         : uint32_t {
00494     /** @brief Read permission in hit region (bit 1) */
00495     k_mhiti_uhaci_r = 0x00000002U,
00496
00497     /** @brief Write permission in hit region (bit 2) */
00498     k_mhiti_uhaci_w = 0x00000004U,
00499
00500     /** @brief Execute permission in hit region (bit 3) */
00501     k_mhiti_uhaci_x = 0x00000008U,
00502
00503     /** @brief Access control bits mask (bits 3:1) */
00504     k_mhiti_uhaci_mask = 0x0000000EU,
00505
00506     /** @brief Region 0 hit flag (bit 16) */

```

```

00507 k_mhiti_hiti0 = 0x00010000U,
00508
00509 /** @brief Region 1 hit flag (bit 17) */
00510 k_mhiti_hiti1 = 0x00020000U,
00511
00512 /** @brief Region 2 hit flag (bit 18) */
00513 k_mhiti_hiti2 = 0x00040000U,
00514
00515 /** @brief Region 3 hit flag (bit 19) */
00516 k_mhiti_hiti3 = 0x00080000U,
00517
00518 /** @brief Region 4 hit flag (bit 20) */
00519 k_mhiti_hiti4 = 0x00100000U,
00520
00521 /** @brief Region 5 hit flag (bit 21) */
00522 k_mhiti_hiti5 = 0x00200000U,
00523
00524 /** @brief Region 6 hit flag (bit 22) */
00525 k_mhiti_hiti6 = 0x00400000U,
00526
00527 /** @brief Region 7 hit flag (bit 23) */
00528 k_mhiti_hiti7 = 0x00800000U,
00529
00530 /** @brief Hit region flags mask (bits 23:16) */
00531 k_mhiti_hiti_mask = 0x00FF0000U,
00532
00533 /** @brief Hit region flags shift amount */
00534 k_mhiti_hiti_shift = 16U,
00535 } mhiti_bits_t;

```

8.37.3.3 mpbac`bits`

```
enum mpbac_bits_t : uint32_t
```

Background Access Control Register (MPBAC) bit definitions.

Sets default access permissions for addresses not covered by any region. The background region covers the entire address space. Default is all access prohibited (0).

Register Layout (32-bit @ 0x00086504)

Bits	Field	Description
0	-	Reserved (read as 0, write 0)
1	UBAC.R	Read permission (0=no, 1=yes)
2	UBAC.W	Write permission (0=no, 1=yes)
3	UBAC.X	Execute permission (0=no, 1=yes)
31:↵ 4	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.4

Since

Version 1.0.0

Enumerator

k_mpbac_ubac_r	Read permission (bit 1) - Allow data read.
k_mpbac_ubac_w	Write permission (bit 2) - Allow data write.
k_mpbac_ubac_x	Execute permission (bit 3) - Allow instruction fetch.
k_mpbac_ubac_mask	Access control bits mask (UBAC[2:0] = bits 3:1).
k_mpbac_ubac_shift	Access control shift amount.

Definition at line 333 of file [rx72n_mpu_regs.h](#).

```

00333         : uint32_t {
00334     /** @brief Read permission (bit 1) - Allow data read */
00335     k_mpbac_ubac_r = 0x00000002U,
00336
00337     /** @brief Write permission (bit 2) - Allow data write */
00338     k_mpbac_ubac_w = 0x00000004U,
00339
00340     /** @brief Execute permission (bit 3) - Allow instruction fetch */
00341     k_mpbac_ubac_x = 0x00000008U,
00342
00343     /** @brief Access control bits mask (UBAC[2:0] = bits 3:1) */
00344     k_mpbac_ubac_mask = 0x0000000EU,
00345
00346     /** @brief Access control shift amount */
00347     k_mpbac_ubac_shift = 1U,
00348 } mpbac_bits_t;

```

8.37.3.4 mpeclr`bits``t`

```
enum mpeclr\_bits\_t : uint32_t
```

Error Status Clearing Register (MPECLR) bit definitions.

Writing 1 to CLR clears the IMPER, DMPER, and DRW bits in MPESTS. Always reads as 0.

Register Layout (32-bit @ 0x00086508)

Bits	Field	Description
0	CLR	Error Status Clear (write 1 to clr)
31:↵ 1	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.5

Since

Version 1.0.0

Enumerator

k_mpeclr_clr	Error Status Clear bit (bit 0) - Write 1 to clear MPESTS.
--------------	---

Definition at line 372 of file [rx72n_mpu_regs.h](#).

```
00372         : uint32_t {
00373     /** @brief Error Status Clear bit (bit 0) - Write 1 to clear MPESTS */
00374     k_mpeclr_clr = 0x00000001U,
00375 } mpeclr_bits_t;
```

8.37.3.5 mpen_bits_t

```
enum mpen_bits_t : uint32_t
```

Memory Protection Enable Register (MPEN) bit definitions.

Enables or disables memory protection globally. Protection takes effect when CPU transitions to user mode.

Register Layout (32-bit @ 0x00086500)

Bits	Field	Description
0	MPEN	Memory Protection Enable
31:↵ 1	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.3

Since

Version 1.0.0

Enumerator

k_mpen_mpen	Memory Protection Enable bit (bit 0).
-------------	---------------------------------------

Definition at line 302 of file [rx72n_mpu_regs.h](#).

```
00302         : uint32_t {
00303     /** @brief Memory Protection Enable bit (bit 0) */
00304     k_mpen_mpen = 0x00000001U,
00305 } mpen_bits_t;
```


8.37.3.6 mpests'bits't

```
enum mpests_bits_t : uint32_t
```

Memory Protection Error Status Register (MPESTS) bit definitions.

Indicates the type of memory protection error that occurred. Bits are cleared by writing 1 to MPECLR.↵
CLR.

Register Layout (32-bit @ 0x0008650C)

Bits	Field	Description
0	IMPER	Instruction MPE generated (1=error)
1	DMPER	Data MPE generated (1=error)
2	DRW	Data access type (0=read, 1=write)
31:↵ 3	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.6

Since

Version 1.0.0

Enumerator

k_mpests_imper	Instruction Memory Protection Error (bit 0).
k_mpests_dmper	Data Memory Protection Error (bit 1).
k_mpests_drw	Data Read/Write indicator (bit 2) - 0=read, 1=write.

Definition at line 401 of file rx72n_mpu_regs.h.

```
00401         : uint32_t {
00402     /** @brief Instruction Memory Protection Error (bit 0) */
00403     k_mpests_imper = 0x00000001U,
00404
00405     /** @brief Data Memory Protection Error (bit 1) */
00406     k_mpests_dmper = 0x00000002U,
00407
00408     /** @brief Data Read/Write indicator (bit 2) - 0=read, 1=write */
00409     k_mpests_drw = 0x00000004U,
00410 } mpests_bits_t;
```

8.37.3.7 mpopi'bits't

```
enum mpopi_bits_t : uint16_t
```

Region Invalidation Operation Register (MPOPI) bit definitions.

Writing 1 to INV clears the V (valid) bit in all REPAGEn registers, effectively disabling all region protections (only background remains). Always reads as 0.

Register Layout (16-bit @ 0x00086526)

Bits	Field	Description
0	INV	Region Invalidate Start (write 1)
15:↵ 1	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.10

Since

Version 1.0.0

Enumerator

k_mpopi_inv	Region Invalidate Start bit (bit 0).
-------------	--------------------------------------

Definition at line 462 of file rx72n_mpu_regs.h.

```
00462         : uint16_t {
00463     /** @brief Region Invalidate Start bit (bit 0) */
00464     k_mpopi_inv = 0x0001U,
00465 } mpopi_bits_t;
```

8.37.3.8 mpops`bits``t`

```
enum mpops_bits_t : uint16_t
```

Region Search Operation Register (MPOPS) bit definitions.

Writing 1 to S initiates a region search using the address in MPSA. Results are stored in MHITD. Always reads as 0.

Register Layout (16-bit @ 0x00086524)

Bits	Field	Description
0	S	Region Search Start (write 1)
15:↵ 1	-	Reserved (read as 0, write 0)

Note

Manual ref: Ch17 section 17.2.9

Since

Version 1.0.0

Enumerator

k_mpop_s	Region Search Start bit (bit 0).
----------	----------------------------------

Definition at line 434 of file rx72n_mpu_regs.h.

```
00434      : uint16_t {
00435      /** @brief Region Search Start bit (bit 0) */
00436      k_mpop_s = 0x0001U,
00437 } mpops_bits_t;
```

8.37.3.9 mpu_addresses_t

```
enum mpu_addresses_t : uintptr_t
```

MPU module addresses.

Base addresses for MPU registers. All addresses verified against RX72N Hardware Manual Ch17.

Note

Addresses verified 2026-01-29 against manual section 17.2

Since

Version 1.0.0

Enumerator

k_mpu_region_base_addr	MPU region registers base address.
k_mpu_control_base_addr	MPU control registers base address.
k_mpu_region_size	Region register block size (8 bytes per region: RSPAGE + REPAGE).
k_mpu_rspage0_addr	RSPAGE0 address
k_mpu_rspage1_addr	RSPAGE1 address
k_mpu_rspage2_addr	RSPAGE2 address
k_mpu_rspage3_addr	RSPAGE3 address
k_mpu_rspage4_addr	RSPAGE4 address
k_mpu_rspage5_addr	RSPAGE5 address
k_mpu_rspage6_addr	RSPAGE6 address
k_mpu_rspage7_addr	RSPAGE7 address
k_mpu_repage0_addr	REPAGE0 address
k_mpu_repage1_addr	REPAGE1 address
k_mpu_repage2_addr	REPAGE2 address
k_mpu_repage3_addr	REPAGE3 address

k_mpu_repage4_addr	REPAGE4 address
k_mpu_repage5_addr	REPAGE5 address
k_mpu_repage6_addr	REPAGE6 address
k_mpu_repage7_addr	REPAGE7 address
k_mpu_mpen_addr	Memory Protection Enable
k_mpu_mpbac_addr	Background Access Control
k_mpu_mpeclr_addr	Error Status Clearing
k_mpu_mpests_addr	Error Status
k_mpu_mpdea_addr	Data Error Address
k_mpu_mpsa_addr	Region Search Address
k_mpu_mpops_addr	Region Search Operation
k_mpu_mpopi_addr	Region Invalidation Operation
k_mpu_mhiti_addr	Instruction Hit Region
k_mpu_mhitd_addr	Data Hit Region

Definition at line 94 of file [rx72n_mpu_regs.h](#).

```

00094      : uintptr_t {
00095      /** @brief MPU region registers base address */
00096      k_mpu_region_base_addr = 0x00086400U,
00097
00098      /** @brief MPU control registers base address */
00099      k_mpu_control_base_addr = 0x00086500U,
00100
00101      /** @brief Region register block size (8 bytes per region: RSPAGE + REPAGE) */
00102      k_mpu_region_size = 0x08U,
00103
00104      /* Region Start Page Number Registers (RSPAGEn) */
00105      k_mpu_rspace0_addr = 0x00086400U, /**< RSPAGE0 address */
00106      k_mpu_rspace1_addr = 0x00086408U, /**< RSPAGE1 address */
00107      k_mpu_rspace2_addr = 0x00086410U, /**< RSPAGE2 address */
00108      k_mpu_rspace3_addr = 0x00086418U, /**< RSPAGE3 address */
00109      k_mpu_rspace4_addr = 0x00086420U, /**< RSPAGE4 address */
00110      k_mpu_rspace5_addr = 0x00086428U, /**< RSPAGE5 address */
00111      k_mpu_rspace6_addr = 0x00086430U, /**< RSPAGE6 address */
00112      k_mpu_rspace7_addr = 0x00086438U, /**< RSPAGE7 address */
00113
00114      /* Region End Page Number Registers (REPAGEn) */
00115      k_mpu_repage0_addr = 0x00086404U, /**< REPAGE0 address */
00116      k_mpu_repage1_addr = 0x0008640CU, /**< REPAGE1 address */
00117      k_mpu_repage2_addr = 0x00086414U, /**< REPAGE2 address */
00118      k_mpu_repage3_addr = 0x0008641CU, /**< REPAGE3 address */
00119      k_mpu_repage4_addr = 0x00086424U, /**< REPAGE4 address */
00120      k_mpu_repage5_addr = 0x0008642CU, /**< REPAGE5 address */
00121      k_mpu_repage6_addr = 0x00086434U, /**< REPAGE6 address */
00122      k_mpu_repage7_addr = 0x0008643CU, /**< REPAGE7 address */
00123
00124      /* Control/Status Registers */
00125      k_mpu_mpen_addr = 0x00086500U, /**< Memory Protection Enable */
00126      k_mpu_mpbac_addr = 0x00086504U, /**< Background Access Control */
00127      k_mpu_mpeclr_addr = 0x00086508U, /**< Error Status Clearing */
00128      k_mpu_mpests_addr = 0x0008650CU, /**< Error Status */
00129      k_mpu_mpdea_addr = 0x00086514U, /**< Data Error Address */
00130      k_mpu_mpsa_addr = 0x00086520U, /**< Region Search Address */
00131      k_mpu_mpops_addr = 0x00086524U, /**< Region Search Operation */
00132      k_mpu_mpopi_addr = 0x00086526U, /**< Region Invalidation Operation */
00133      k_mpu_mhiti_addr = 0x00086528U, /**< Instruction Hit Region */
00134      k_mpu_mhitd_addr = 0x0008652CU, /**< Data Hit Region */
00135 } mpu_addresses_t;

```

8.37.3.10 mpu_page_t

```
enum mpu_page_t : uint32_t
```

MPU page configuration constants.

The MPU divides the address space into 16-byte pages. Page number = address[31:4] (upper 28 bits).

Since

Version 1.0.0

Enumerator

k_mpu_page_size	Page size in bytes (16 bytes).
k_mpu_page_shift	Shift amount to convert address to page number.
k_mpu_page_mask	Mask for page number bits [31:4].
k_mpu_page_max	Maximum page number ($2^{28} - 1$).

Definition at line 166 of file rx72n_mpu_regs.h.

```

00166         : uint32_t {
00167     /** @brief Page size in bytes (16 bytes) */
00168     k_mpu_page_size = 16U,
00169
00170     /** @brief Shift amount to convert address to page number */
00171     k_mpu_page_shift = 4U,
00172
00173     /** @brief Mask for page number bits [31:4] */
00174     k_mpu_page_mask = 0xFFFFFFFFU,
00175
00176     /** @brief Maximum page number ( $2^{28} - 1$ ) */
00177     k_mpu_page_max = 0xFFFFFFFFU,
00178 } mpu_page_t;

```

8.37.3.11 mpu`region`_t

```
enum mpu_region_t : uint8_t
```

MPU region numbers.

Region indices for use with accessor functions.

Since

Version 1.0.0

Enumerator

k_mpu_region_0	MPU region 0
k_mpu_region_1	MPU region 1
k_mpu_region_2	MPU region 2
k_mpu_region_3	MPU region 3
k_mpu_region_4	MPU region 4
k_mpu_region_5	MPU region 5
k_mpu_region_6	MPU region 6

k_mpu_region_7	MPU region 7
k_mpu_region_count	Total number of MPU regions

Definition at line 144 of file rx72n_mpu_regs.h.

```

00144      : uint8_t {
00145 k_mpu_region_0  = 0U, /**< MPU region 0 */
00146 k_mpu_region_1  = 1U, /**< MPU region 1 */
00147 k_mpu_region_2  = 2U, /**< MPU region 2 */
00148 k_mpu_region_3  = 3U, /**< MPU region 3 */
00149 k_mpu_region_4  = 4U, /**< MPU region 4 */
00150 k_mpu_region_5  = 5U, /**< MPU region 5 */
00151 k_mpu_region_6  = 6U, /**< MPU region 6 */
00152 k_mpu_region_7  = 7U, /**< MPU region 7 */
00153 k_mpu_region_count = 8U, /**< Total number of MPU regions */
00154 } mpu_region_t;

```

8.37.3.12 repage_bits_t

```
enum repage_bits_t : uint32_t
```

Region End Page Number Register (REPAGEn) bit definitions.

Specifies the ending page number and access control for a protection region. The end page is INCLUDED in the protected region.

Register Layout (32-bit @ 0x00086404 + n*8)

Bits	Field	Description
0	V	Valid (0=invalid, 1=valid)
1	UAC.R	Read permission (0=no, 1=yes)
2	UAC.W	Write permission (0=no, 1=yes)
3	UAC.X	Execute permission (0=no, 1=yes)
31:↵ 4	REPN[27:↵ 0]	Region End Page Number

Note

Manual ref: Ch17 section 17.2.2

Since

Version 1.0.0

Enumerator

k_repage_v	Valid bit (bit 0) - Region enabled when set.
------------	--

k_repage_uac_r	Read permission (bit 1) - Allow data read.
k_repage_uac_w	Write permission (bit 2) - Allow data write.
k_repage_uac_x	Execute permission (bit 3) - Allow instruction fetch.
k_repage_uac_mask	Access control bits mask (UAC[2:0] = bits 3:1).
k_repage_uac_shift	Access control shift amount.
k_repage_repn_mask	End page number field mask (bits 31:4).
k_repage_repn_shift	End page number shift amount.
k_repage_uac_ro	Read-only region (R).
k_repage_uac_rw	Read-write region (RW).
k_repage_uac_xo	Execute-only region (X).
k_repage_uac_rx	Read-execute region (RX).
k_repage_uac_rwx	Full access region (RWX).

Definition at line 238 of file `rx72n_mpu_regs.h`.

```

00238         : uint32_t {
00239     /** @brief Valid bit (bit 0) - Region enabled when set */
00240     k_repage_v = 0x00000001U,
00241
00242     /** @brief Read permission (bit 1) - Allow data read */
00243     k_repage_uac_r = 0x00000002U,
00244
00245     /** @brief Write permission (bit 2) - Allow data write */
00246     k_repage_uac_w = 0x00000004U,
00247
00248     /** @brief Execute permission (bit 3) - Allow instruction fetch */
00249     k_repage_uac_x = 0x00000008U,
00250
00251     /** @brief Access control bits mask (UAC[2:0] = bits 3:1) */
00252     k_repage_uac_mask = 0x0000000EU,
00253
00254     /** @brief Access control shift amount */
00255     k_repage_uac_shift = 1U,
00256
00257     /** @brief End page number field mask (bits 31:4) */
00258     k_repage_repn_mask = 0xFFFFFFFF0U,
00259
00260     /** @brief End page number shift amount */
00261     k_repage_repn_shift = 4U,
00262
00263     /** Common permission combinations */
00264     /** @brief Read-only region (R) */
00265     k_repage_uac_ro = k_repage_uac_r,
00266
00267     /** @brief Read-write region (RW) */
00268     k_repage_uac_rw = k_repage_uac_r | k_repage_uac_w,
00269
00270     /** @brief Execute-only region (X) */
00271     k_repage_uac_xo = k_repage_uac_x,
00272
00273     /** @brief Read-execute region (RX) */
00274     k_repage_uac_rx = k_repage_uac_r | k_repage_uac_x,
00275
00276     /** @brief Full access region (RWX) */
00277     k_repage_uac_rwx = k_repage_uac_r | k_repage_uac_w | k_repage_uac_x,
00278 } repage_bits_t;

```

8.37.3.13 rspace`bits` t

```
enum rspace_bits_t : uint32_t
```

Region Start Page Number Register (RSPAGEn) bit definitions.

Specifies the starting page number for a protection region. The page number corresponds to address bits [31:4].

Register Layout (32-bit @ 0x00086400 + n*8)

Bits	Field	Description
3:↔ 0	-	Reserved (read as 0, write 0)
31:↔ 4	RSPN[27:↔ 0]	Region Start Page Number

Note

Manual ref: Ch17 section 17.2.1

Since

Version 1.0.0

Enumerator

k_rspage_reserved_mask	Reserved bits mask (bits 3:0).
k_rspage_rspn_mask	Start page number field mask (bits 31:4).
k_rspage_rspn_shift	Start page number shift amount.

Definition at line 202 of file [rx72n_mpu_regs.h](#).

```

00202     : uint32_t {
00203     /** @brief Reserved bits mask (bits 3:0) */
00204     k_rspage_reserved_mask = 0x0000000FU,
00205
00206     /** @brief Start page number field mask (bits 31:4) */
00207     k_rspage_rspn_mask = 0xFFFFFFFF0U,
00208
00209     /** @brief Start page number shift amount */
00210     k_rspage_rspn_shift = 4U,
00211 } rspage_bits_t;

```

8.37.4 Function Documentation

8.37.4.1 mpu_addr_to_page()

```
uint32_t mpu_addr_to_page (
    uint32_t addr) [inline], [static]
```

Convert address to page number.

Parameters

addr	Memory address
------	----------------

Returns

Page number (address[31:4])

Example:

```
uint32_t page = mpu_addr_to_page(0x00012340);
// page = 0x00001234
```

Since

Version 1.0.0

Definition at line 822 of file [rx72n_mpu_regs.h](#).

```
00823 {
00824     return (addr » k_mpu_page_shift) & k_mpu_page_max;
00825 }
```

References [k_mpu_page_max](#), and [k_mpu_page_shift](#).

8.37.4.2 mpu_mhيتدreg()

```
volatile uint32_t * mpu_mhيتدreg (
    void )    [inline], [static]
```

Get pointer to MHITD (Data Hit Region) register.

Returns

Volatile pointer to MHITD register (32-bit)

Since

Version 1.0.0

Definition at line 798 of file [rx72n_mpu_regs.h](#).

```
00799 {
00800     return (volatile uint32_t*)k_mpu_mhيتد_addr;
00801 }
```

References [k_mpu_mhيتد_addr](#).

8.37.4.3 mpu_mhيتreg()

```
volatile uint32_t * mpu_mhيتreg (
    void )    [inline], [static]
```

Get pointer to MHITI (Instruction Hit Region) register.

Returns

Volatile pointer to MHITI register (32-bit)

Since

Version 1.0.0

Definition at line 788 of file [rx72n_mpu_regs.h](#).

```
00789 {
00790     return (volatile uint32_t*)k_mpu_mhيت_addr;
00791 }
```

References [k_mpu_mhيت_addr](#).

8.37.4.4 mpu`mpbac`reg()

```
volatile uint32_t * mpu__mpbac_reg (  
    void )    [inline], [static]
```

Get pointer to MPBAC (Background Access Control) register.

Returns

Volatile pointer to MPBAC register (32-bit)

Since

Version 1.0.0

Definition at line 718 of file [rx72n_mpu_regs.h](#).

```
00719 {  
00720     return (volatile uint32_t*)k_mpu__mpbac_addr;  
00721 }
```

References [k_mpu__mpbac_addr](#).

8.37.4.5 mpu`mpdea`reg()

```
volatile uint32_t * mpu__mpdea_reg (  
    void )    [inline], [static]
```

Get pointer to MPDEA (Data Error Address) register.

Returns

Volatile pointer to MPDEA register (32-bit)

Since

Version 1.0.0

Definition at line 748 of file [rx72n_mpu_regs.h](#).

```
00749 {  
00750     return (volatile uint32_t*)k_mpu__mpdea_addr;  
00751 }
```

References [k_mpu__mpdea_addr](#).

8.37.4.6 mpu_mpeclr_reg()

```
volatile uint32_t * mpu_mpeclr_reg (  
    void )    [inline], [static]
```

Get pointer to MPECLR (Error Status Clearing) register.

Returns

Volatile pointer to MPECLR register (32-bit)

Since

Version 1.0.0

Definition at line 728 of file [rx72n_mpu_regs.h](#).

```
00729 {  
00730     return (volatile uint32_t*)k_mpu_mpeclr_addr;  
00731 }
```

References [k_mpu_mpeclr_addr](#).

8.37.4.7 mpu_mpen_reg()

```
volatile uint32_t * mpu_mpen_reg (  
    void )    [inline], [static]
```

Get pointer to MPEN (Memory Protection Enable) register.

Returns

Volatile pointer to MPEN register (32-bit)

Since

Version 1.0.0

Definition at line 708 of file [rx72n_mpu_regs.h](#).

```
00709 {  
00710     return (volatile uint32_t*)k_mpu_mpen_addr;  
00711 }
```

References [k_mpu_mpen_addr](#).

8.37.4.8 mpu`mpests`reg()

```
volatile uint32_t * mpu_mpests_reg (  
    void )    [inline], [static]
```

Get pointer to MPESTS (Error Status) register.

Returns

Volatile pointer to MPESTS register (32-bit)

Since

Version 1.0.0

Definition at line 738 of file [rx72n_mpu_regs.h](#).

```
00739 {  
00740     return (volatile uint32_t*)k_mpu_mpests_addr;  
00741 }
```

References [k_mpu_mpests_addr](#).

8.37.4.9 mpu`mpopi`reg()

```
volatile uint16_t * mpu_mpopi_reg (  
    void )    [inline], [static]
```

Get pointer to MPOPI (Region Invalidation) register.

Returns

Volatile pointer to MPOPI register (16-bit)

Since

Version 1.0.0

Definition at line 778 of file [rx72n_mpu_regs.h](#).

```
00779 {  
00780     return (volatile uint16_t*)k_mpu_mpopi_addr;  
00781 }
```

References [k_mpu_mpopi_addr](#).

8.37.4.10 mpu`mpops`reg()

```
volatile uint16_t * mpu__mpops_reg (  
    void )    [inline], [static]
```

Get pointer to MPOPS (Region Search Operation) register.

Returns

Volatile pointer to MPOPS register (16-bit)

Since

Version 1.0.0

Definition at line 768 of file [rx72n_mpu_regs.h](#).

```
00769 {  
00770     return (volatile uint16_t*)k_mpu__mpops_addr;  
00771 }
```

References [k_mpu__mpops_addr](#).

8.37.4.11 mpu`mpsa`reg()

```
volatile uint32_t * mpu__mpsa_reg (  
    void )    [inline], [static]
```

Get pointer to MPSA (Region Search Address) register.

Returns

Volatile pointer to MPSA register (32-bit)

Since

Version 1.0.0

Definition at line 758 of file [rx72n_mpu_regs.h](#).

```
00759 {  
00760     return (volatile uint32_t*)k_mpu__mpsa_addr;  
00761 }
```

References [k_mpu__mpsa_addr](#).

8.37.4.12 mpu_page_to_reg()

```
uint32_t mpu_page_to_reg (
    uint32_t page)    [inline], [static]
```

Convert page number to RSPAGE/REPAGE register value.

Parameters

page	Page number (28 bits)
------	-----------------------

Returns

Register value with page number in bits [31:4]

Note

For REPAGE, caller must OR in access control and valid bits.

Example:

```
uint32_t rspage_val = mpu_page_to_reg(0x00001234);
uint32_t repage_val = mpu_page_to_reg(0x00001234) | k_repage_uac_rw | k_repage_v;
```

Since

Version 1.0.0

Definition at line 843 of file [rx72n_mpu_regs.h](#).

```
00844 {
00845     return (page « k_mpu_page_shift) & k_rspage_rspn_mask;
00846 }
```

References [k_mpu_page_shift](#), and [k_rspage_rspn_mask](#).

8.37.4.13 mpu_region()

```
volatile rx\_mpu\_region\_regs\_t * mpu_region (
    uint8_t region)    [inline], [static]
```

Get pointer to MPU region register pair.

Parameters

region	Region number (0-7)
--------	---------------------

Returns

Volatile pointer to region register structure

Precondition

`region < k_mpu_region_count` (8)

Note

Thread Safety: Safe - returns constant hardware address

Example:

```
// Configure region 0 to protect RAM (0x00000000-0x0007FFFF) as read-write
volatile rx_mpu_region_regs_t* r0 = mpu_region(k_mpu_region_0);
r0->rspage = 0x00000000; // Start at page 0
r0->repage = (0x0007FFF0) | k_repage_uac_rw | k_repage_v; // End at page 0x7FFF, RW, valid
```

Since

Version 1.0.0

Definition at line 669 of file `rx72n_mpu_regs.h`.

```
00670 {
00671     return (volatile rx_mpu_region_regs_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size));
00672 }
```

References `k_mpu_region_base_addr`, and `k_mpu_region_size`.

8.37.4.14 mpu_repage_reg()

```
volatile uint32_t * mpu_repage_reg (
    uint8_t region) [inline], [static]
```

Get pointer to REPAGE register for a region.

Parameters

region	Region number (0-7)
--------	---------------------

Returns

Volatile pointer to REPAGE register (32-bit)

Since

Version 1.0.0

Definition at line 693 of file `rx72n_mpu_regs.h`.

```
00694 {
00695     return (volatile uint32_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size) + 4U);
00696 }
```

References `k_mpu_region_base_addr`, and `k_mpu_region_size`.

8.37.4.15 mpu_rspage_reg()

```
volatile uint32_t * mpu_rspage_reg (
    uint8_t region)    [inline], [static]
```

Get pointer to RSPAGE register for a region.

Parameters

region	Region number (0-7)
--------	---------------------

Returns

Volatile pointer to RSPAGE register (32-bit)

Since

Version 1.0.0

Definition at line 681 of file [rx72n_mpu_regs.h](#).

```
00682 {
00683     return (volatile uint32_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size));
00684 }
```

References [k_mpu_region_base_addr](#), and [k_mpu_region_size](#).

8.38 rx72n_mpu_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_mpu_regs.h
00003  * @brief RX72N Memory Protection Unit (MPU) Register Definitions
00004  *
00005  * @details
00006  * Complete register definitions for the RX72N Memory Protection Unit.
00007  * The MPU provides address-based access control for user-mode programs,
00008  * protecting code and data regions from unauthorized access.
00009  *
00010  * @par MPU Overview
00011  * - Covers entire 4 GB address space (0x00000000-0xFFFFFFFF)
00012  * - 8 independent configurable regions (Region 0-7)
00013  * - 16-byte page granularity (smallest unit of protection)
00014  * - 3 access types: Execute (X), Write (W), Read (R)
00015  * - Protection only active in user mode (supervisor mode bypasses)
00016  * - Background region provides default access for unprotected areas
00017  * - Overlapping regions use OR of permissions (permission wins)
00018  *
00019  * @par Register Organization
00020  * | Address Range | Description |
00021  * |-----|-----|
00022  * | 0x00086400-0x043F | Region registers (RSPAGE0-7, REPAGE0-7) |
00023  * | 0x00086500-0x052F | Control/status registers |
00024  *
00025  * @par Region Registers (8 regions x 8 bytes each = 64 bytes)
00026  * | Offset | Register | Description |
00027  * |-----|-----|-----|
00028  * | 0x00 | RSPAGEn | Region n Start Page Number (32-bit) |
00029  * | 0x04 | REPAGEn | Region n End Page Number (32-bit) |
00030  *
00031  * @par Control/Status Registers
00032  * | Address | Size | Register | Description |
00033  * |-----|-----|-----|-----|
00034  * | 0x00086500 | 32 | MPEN | Memory Protection Enable |
00035  * | 0x00086504 | 32 | MPBAC | Background Access Control |
00036  * | 0x00086508 | 32 | MPECLR | Error Status Clearing |
```



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00037 * | 0x0008650C | 32 | MPESTS | Error Status |
00038 * | 0x00086514 | 32 | MPDEA | Data Error Address |
00039 * | 0x00086520 | 32 | MPSA | Region Search Address |
00040 * | 0x00086524 | 16 | MPOPS | Region Search Operation |
00041 * | 0x00086526 | 16 | MPOPI | Region Invalidation Operation |
00042 * | 0x00086528 | 32 | MHITI | Instruction Hit Region |
00043 * | 0x0008652C | 32 | MHITD | Data Hit Region |
00044 *
00045 * @par NASA Power of 10 Compliance
00046 * - Rule 1: No goto, setjmp, or recursion
00047 * - Rule 2: N/A (no loops)
00048 * - Rule 3: No dynamic memory allocation
00049 * - Rule 4: Short accessor functions
00050 * - Rule 5: Static assertions verify addresses
00051 * - Rule 6: Minimal scope
00052 * - Rule 7: N/A
00053 * - Rule 8: All constants use C23 typed enums
00054 * - Rule 9: No function pointers
00055 * - Rule 10: Compiles with -Wall -Wextra -Werror
00056 *
00057 * @par Manual References
00058 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00059 * - Chapter 17: Memory-Protection Unit (MPU), pages 657-676
00060 * - Section 17.2: Register Descriptions
00061 *
00062 * @author Locked, Inc.
00063 * @date 2026-01-29
00064 * @copyright Copyright (c) 2026 Locked Inc.
00065 * SPDX-License-Identifier: MIT
00066 * @since Version 1.0.0
00067 */
00068
00069 #pragma once
00070
00071 #include <stddef.h>
00072 #include <stdint.h>
00073
00074 #ifdef __cplusplus
00075 extern "C" {
00076 #endif
00077
00078 /*
=====
00079 * MPU Base Addresses and Constants
00080 *
=====
00081 */
00082 /**
00083 * @enum mpu_addresses_t
00084 * @brief MPU module addresses
00085 *
00086 *
00087 * @details
00088 * Base addresses for MPU registers.
00089 * All addresses verified against RX72N Hardware Manual Ch17.
00090 *
00091 * @note Addresses verified 2026-01-29 against manual section 17.2
00092 * @since Version 1.0.0
00093 */
00094 typedef enum : uintptr_t {
00095     /** @brief MPU region registers base address */
00096     k_mpu_region_base_addr = 0x00086400U,
00097
00098     /** @brief MPU control registers base address */
00099     k_mpu_control_base_addr = 0x00086500U,
00100
00101     /** @brief Region register block size (8 bytes per region: RSPAGE + REPAGE) */
00102     k_mpu_region_size = 0x08U,
00103
00104     /** Region Start Page Number Registers (RSPAGEn) */
00105     k_mpu_rspage0_addr = 0x00086400U, /**< RSPAGE0 address */
00106     k_mpu_rspage1_addr = 0x00086408U, /**< RSPAGE1 address */
00107     k_mpu_rspage2_addr = 0x00086410U, /**< RSPAGE2 address */
00108     k_mpu_rspage3_addr = 0x00086418U, /**< RSPAGE3 address */
00109     k_mpu_rspage4_addr = 0x00086420U, /**< RSPAGE4 address */
00110     k_mpu_rspage5_addr = 0x00086428U, /**< RSPAGE5 address */
00111     k_mpu_rspage6_addr = 0x00086430U, /**< RSPAGE6 address */
00112     k_mpu_rspage7_addr = 0x00086438U, /**< RSPAGE7 address */
00113
00114     /** Region End Page Number Registers (REPAGEn) */
00115     k_mpu_repage0_addr = 0x00086404U, /**< REPAGE0 address */
00116     k_mpu_repage1_addr = 0x0008640CU, /**< REPAGE1 address */
00117     k_mpu_repage2_addr = 0x00086414U, /**< REPAGE2 address */
00118     k_mpu_repage3_addr = 0x0008641CU, /**< REPAGE3 address */
00119     k_mpu_repage4_addr = 0x00086424U, /**< REPAGE4 address */
00120     k_mpu_repage5_addr = 0x0008642CU, /**< REPAGE5 address */
00121     k_mpu_repage6_addr = 0x00086434U, /**< REPAGE6 address */

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00122 k_mpu_repage7_addr = 0x0008643CU, /**< REPAGE7 address */
00123
00124 /* Control/Status Registers */
00125 k_mpu_mpen_addr = 0x00086500U, /**< Memory Protection Enable */
00126 k_mpu_mpbac_addr = 0x00086504U, /**< Background Access Control */
00127 k_mpu_mpeclr_addr = 0x00086508U, /**< Error Status Clearing */
00128 k_mpu_mpests_addr = 0x0008650CU, /**< Error Status */
00129 k_mpu_mpdea_addr = 0x00086514U, /**< Data Error Address */
00130 k_mpu_mpsa_addr = 0x00086520U, /**< Region Search Address */
00131 k_mpu_mpops_addr = 0x00086524U, /**< Region Search Operation */
00132 k_mpu_mpopi_addr = 0x00086526U, /**< Region Invalidation Operation */
00133 k_mpu_mhiti_addr = 0x00086528U, /**< Instruction Hit Region */
00134 k_mpu_mhitd_addr = 0x0008652CU, /**< Data Hit Region */
00135 } mpu_addresses_t;
00136
00137 /**
00138  * @enum mpu_region_t
00139  * @brief MPU region numbers
00140  *
00141  * @details Region indices for use with accessor functions.
00142  * @since Version 1.0.0
00143  */
00144 typedef enum : uint8_t {
00145     k_mpu_region_0 = 0U, /**< MPU region 0 */
00146     k_mpu_region_1 = 1U, /**< MPU region 1 */
00147     k_mpu_region_2 = 2U, /**< MPU region 2 */
00148     k_mpu_region_3 = 3U, /**< MPU region 3 */
00149     k_mpu_region_4 = 4U, /**< MPU region 4 */
00150     k_mpu_region_5 = 5U, /**< MPU region 5 */
00151     k_mpu_region_6 = 6U, /**< MPU region 6 */
00152     k_mpu_region_7 = 7U, /**< MPU region 7 */
00153     k_mpu_region_count = 8U, /**< Total number of MPU regions */
00154 } mpu_region_t;
00155
00156 /**
00157  * @enum mpu_page_t
00158  * @brief MPU page configuration constants
00159  *
00160  * @details
00161  * The MPU divides the address space into 16-byte pages.
00162  * Page number = address[31:4] (upper 28 bits).
00163  *
00164  * @since Version 1.0.0
00165  */
00166 typedef enum : uint32_t {
00167     /** @brief Page size in bytes (16 bytes) */
00168     k_mpu_page_size = 16U,
00169
00170     /** @brief Shift amount to convert address to page number */
00171     k_mpu_page_shift = 4U,
00172
00173     /** @brief Mask for page number bits [31:4] */
00174     k_mpu_page_mask = 0xFFFFFFF0U,
00175
00176     /** @brief Maximum page number (228 - 1) */
00177     k_mpu_page_max = 0xFFFFFFFFU,
00178 } mpu_page_t;
00179
00180 /**
=====
00181  * RSPAGEn Register (Region Start Page Number)
00182  *
=====
00183  */
00184
00185 /**
00186  * @enum rspage_bits_t
00187  * @brief Region Start Page Number Register (RSPAGEn) bit definitions
00188  *
00189  * @details
00190  * Specifies the starting page number for a protection region.
00191  * The page number corresponds to address bits [31:4].
00192  *
00193  * @par Register Layout (32-bit @ 0x00086400 + n*8)
00194  * | Bits | Field | Description |
00195  * |-----|-----|-----|
00196  * | 3:0 | - | Reserved (read as 0, write 0) |
00197  * | 31:4 | RSPN[27:0] | Region Start Page Number |
00198  *
00199  * @note Manual ref: Ch17 section 17.2.1
00200  * @since Version 1.0.0
00201  */
00202 typedef enum : uint32_t {
00203     /** @brief Reserved bits mask (bits 3:0) */
00204     k_rspage_reserved_mask = 0x0000000FU,
00205
00206     /** @brief Start page number field mask (bits 31:4) */

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00207 k_rspage_rspn_mask = 0xFFFFFFFF0U,
00208
00209 /** @brief Start page number shift amount */
00210 k_rspage_rspn_shift = 4U,
00211 } rspage_bits_t;
00212
00213 /*
=====
00214 * REPAGEn Register (Region End Page Number)
00215 *
=====
00216 */
00217
00218 /**
00219 * @enum repage_bits_t
00220 * @brief Region End Page Number Register (REPAGEn) bit definitions
00221 *
00222 * @details
00223 * Specifies the ending page number and access control for a protection region.
00224 * The end page is INCLUDED in the protected region.
00225 *
00226 * @par Register Layout (32-bit @ 0x00086404 + n*8)
00227 * | Bits | Field | Description |
00228 * |-----|-----|-----|
00229 * | 0 | V | Valid (0=invalid, 1=valid) |
00230 * | 1 | UAC.R | Read permission (0=no, 1=yes) |
00231 * | 2 | UAC.W | Write permission (0=no, 1=yes) |
00232 * | 3 | UAC.X | Execute permission (0=no, 1=yes) |
00233 * | 31:4 | REPN[27:0] | Region End Page Number |
00234 *
00235 * @note Manual ref: Ch17 section 17.2.2
00236 * @since Version 1.0.0
00237 */
00238 typedef enum : uint32_t {
00239 /** @brief Valid bit (bit 0) - Region enabled when set */
00240 k_repage_v = 0x00000001U,
00241
00242 /** @brief Read permission (bit 1) - Allow data read */
00243 k_repage_uac_r = 0x00000002U,
00244
00245 /** @brief Write permission (bit 2) - Allow data write */
00246 k_repage_uac_w = 0x00000004U,
00247
00248 /** @brief Execute permission (bit 3) - Allow instruction fetch */
00249 k_repage_uac_x = 0x00000008U,
00250
00251 /** @brief Access control bits mask (UAC[2:0] = bits 3:1) */
00252 k_repage_uac_mask = 0x0000000EU,
00253
00254 /** @brief Access control shift amount */
00255 k_repage_uac_shift = 1U,
00256
00257 /** @brief End page number field mask (bits 31:4) */
00258 k_repage_repn_mask = 0xFFFFFFFF0U,
00259
00260 /** @brief End page number shift amount */
00261 k_repage_repn_shift = 4U,
00262
00263 /** Common permission combinations */
00264 /** @brief Read-only region (R) */
00265 k_repage_uac_ro = k_repage_uac_r,
00266
00267 /** @brief Read-write region (RW) */
00268 k_repage_uac_rw = k_repage_uac_r | k_repage_uac_w,
00269
00270 /** @brief Execute-only region (X) */
00271 k_repage_uac_xo = k_repage_uac_x,
00272
00273 /** @brief Read-execute region (RX) */
00274 k_repage_uac_rx = k_repage_uac_r | k_repage_uac_x,
00275
00276 /** @brief Full access region (RWX) */
00277 k_repage_uac_rwx = k_repage_uac_r | k_repage_uac_w | k_repage_uac_x,
00278 } repage_bits_t;
00279
00280 /*
=====
00281 * MPEN Register (Memory Protection Enable)
00282 *
=====
00283 */
00284
00285 /**
00286 * @enum mpen_bits_t
00287 * @brief Memory Protection Enable Register (MPEN) bit definitions
00288 *
00289 * @details

```

```

00290 * Enables or disables memory protection globally.
00291 * Protection takes effect when CPU transitions to user mode.
00292 *
00293 * @par Register Layout (32-bit @ 0x00086500)
00294 * | Bits | Field | Description |
00295 * |-----|-----|-----|
00296 * | 0 | MPEN | Memory Protection Enable |
00297 * | 31:1 | - | Reserved (read as 0, write 0) |
00298 *
00299 * @note Manual ref: Ch17 section 17.2.3
00300 * @since Version 1.0.0
00301 */
00302 typedef enum : uint32_t {
00303 /** @brief Memory Protection Enable bit (bit 0) */
00304 k_mpen_mpen = 0x00000001U,
00305 } mpen_bits_t;
00306
00307 /*
=====
00308 * MPBAC Register (Background Access Control)
00309 *
=====
00310 */
00311 /**
00312 * @enum mpbac_bits_t
00313 * @brief Background Access Control Register (MPBAC) bit definitions
00314 *
00315 * @details
00316 * Sets default access permissions for addresses not covered by any region.
00317 * The background region covers the entire address space.
00318 * Default is all access prohibited (0).
00319 *
00320 *
00321 * @par Register Layout (32-bit @ 0x00086504)
00322 * | Bits | Field | Description |
00323 * |-----|-----|-----|
00324 * | 0 | - | Reserved (read as 0, write 0) |
00325 * | 1 | UBAC.R | Read permission (0=no, 1=yes) |
00326 * | 2 | UBAC.W | Write permission (0=no, 1=yes) |
00327 * | 3 | UBAC.X | Execute permission (0=no, 1=yes) |
00328 * | 31:4 | - | Reserved (read as 0, write 0) |
00329 *
00330 * @note Manual ref: Ch17 section 17.2.4
00331 * @since Version 1.0.0
00332 */
00333 typedef enum : uint32_t {
00334 /** @brief Read permission (bit 1) - Allow data read */
00335 k_mpbac_ubac_r = 0x00000002U,
00336
00337 /** @brief Write permission (bit 2) - Allow data write */
00338 k_mpbac_ubac_w = 0x00000004U,
00339
00340 /** @brief Execute permission (bit 3) - Allow instruction fetch */
00341 k_mpbac_ubac_x = 0x00000008U,
00342
00343 /** @brief Access control bits mask (UBAC[2:0] = bits 3:1) */
00344 k_mpbac_ubac_mask = 0x0000000EU,
00345
00346 /** @brief Access control shift amount */
00347 k_mpbac_ubac_shift = 1U,
00348 } mpbac_bits_t;
00349
00350 /*
=====
00351 * MPECLR Register (Error Status Clearing)
00352 *
=====
00353 */
00354 /**
00355 * @enum mpeclr_bits_t
00356 * @brief Error Status Clearing Register (MPECLR) bit definitions
00357 *
00358 * @details
00359 * Writing 1 to CLR clears the IMPER, DMPER, and DRW bits in MPESTS.
00360 * Always reads as 0.
00361 *
00362 *
00363 * @par Register Layout (32-bit @ 0x00086508)
00364 * | Bits | Field | Description |
00365 * |-----|-----|-----|
00366 * | 0 | CLR | Error Status Clear (write 1 to clr) |
00367 * | 31:1 | - | Reserved (read as 0, write 0) |
00368 *
00369 * @note Manual ref: Ch17 section 17.2.5
00370 * @since Version 1.0.0
00371 */
00372 typedef enum : uint32_t {

```

```

00373 /** @brief Error Status Clear bit (bit 0) - Write 1 to clear MPESTS */
00374 k_mpeclr_clr = 0x00000001U,
00375 } mpeclr_bits_t;
00376
00377 /*
=====
00378 * MPESTS Register (Error Status)
00379 *
=====
00380 */
00381
00382 /**
00383 * @enum mpests_bits_t
00384 * @brief Memory Protection Error Status Register (MPESTS) bit definitions
00385 *
00386 * @details
00387 * Indicates the type of memory protection error that occurred.
00388 * Bits are cleared by writing 1 to MPECLR.CLR.
00389 *
00390 * @par Register Layout (32-bit @ 0x0008650C)
00391 * | Bits | Field | Description |
00392 * |-----|-----|-----|
00393 * | 0 | IMPER | Instruction MPE generated (1=error) |
00394 * | 1 | DMPER | Data MPE generated (1=error) |
00395 * | 2 | DRW | Data access type (0=read, 1=write) |
00396 * | 31:3 | - | Reserved (read as 0, write 0) |
00397 *
00398 * @note Manual ref: Ch17 section 17.2.6
00399 * @since Version 1.0.0
00400 */
00401 typedef enum : uint32_t {
00402 /** @brief Instruction Memory Protection Error (bit 0) */
00403 k_mpests_imper = 0x00000001U,
00404
00405 /** @brief Data Memory Protection Error (bit 1) */
00406 k_mpests_dmper = 0x00000002U,
00407
00408 /** @brief Data Read/Write indicator (bit 2) - 0=read, 1=write */
00409 k_mpests_drw = 0x00000004U,
00410 } mpests_bits_t;
00411
00412 /*
=====
00413 * MPOPS Register (Region Search Operation)
00414 *
=====
00415 */
00416
00417 /**
00418 * @enum mpops_bits_t
00419 * @brief Region Search Operation Register (MPOPS) bit definitions
00420 *
00421 * @details
00422 * Writing 1 to S initiates a region search using the address in MPSA.
00423 * Results are stored in MHITD. Always reads as 0.
00424 *
00425 * @par Register Layout (16-bit @ 0x00086524)
00426 * | Bits | Field | Description |
00427 * |-----|-----|-----|
00428 * | 0 | S | Region Search Start (write 1) |
00429 * | 15:1 | - | Reserved (read as 0, write 0) |
00430 *
00431 * @note Manual ref: Ch17 section 17.2.9
00432 * @since Version 1.0.0
00433 */
00434 typedef enum : uint16_t {
00435 /** @brief Region Search Start bit (bit 0) */
00436 k_mpops_s = 0x0001U,
00437 } mpops_bits_t;
00438
00439 /*
=====
00440 * MPOPI Register (Region Invalidation Operation)
00441 *
=====
00442 */
00443
00444 /**
00445 * @enum mpopi_bits_t
00446 * @brief Region Invalidation Operation Register (MPOPI) bit definitions
00447 *
00448 * @details
00449 * Writing 1 to INV clears the V (valid) bit in all REPAGEn registers,
00450 * effectively disabling all region protections (only background remains).
00451 * Always reads as 0.
00452 *
00453 * @par Register Layout (16-bit @ 0x00086526)

```

```

00454 * | Bits | Field | Description |
00455 * |-----|-----|-----|
00456 * | 0 | INV | Region Invalidate Start (write 1) |
00457 * | 15:1 | - | Reserved (read as 0, write 0) |
00458 *
00459 * @note Manual ref: Ch17 section 17.2.10
00460 * @since Version 1.0.0
00461 */
00462 typedef enum : uint16_t {
00463 /** @brief Region Invalidate Start bit (bit 0) */
00464 k_mpopi_inv = 0x0001U,
00465 } mpopi_bits_t;
00466
00467 /*
=====
00468 * MHITI Register (Instruction Hit Region)
00469 *
=====
00470 */
00471
00472 /**
00473 * @enum mhiti_bits_t
00474 * @brief Instruction Hit Region Register (MHITI) bit definitions
00475 *
00476 * @details
00477 * After an instruction memory protection error, this register indicates
00478 * which region caused the error and what permissions were set.
00479 *
00480 * @par Register Layout (32-bit @ 0x00086528)
00481 * | Bits | Field | Description |
00482 * |-----|-----|-----|
00483 * | 0 | - | Reserved (read as 0) |
00484 * | 3:1 | UHACI[2:0] | Hit region access control (RWX) |
00485 * | 15:4 | - | Reserved (read as 0) |
00486 * | 23:16 | HITI[7:0] | Region hit flags (1=hit in region) |
00487 * | 31:24 | - | Reserved (read as 0) |
00488 *
00489 * @note HITI[7:0] = 0 means error in background region
00490 * @note Manual ref: Ch17 section 17.2.11
00491 * @since Version 1.0.0
00492 */
00493 typedef enum : uint32_t {
00494 /** @brief Read permission in hit region (bit 1) */
00495 k_mhiti_uhaci_r = 0x00000002U,
00496
00497 /** @brief Write permission in hit region (bit 2) */
00498 k_mhiti_uhaci_w = 0x00000004U,
00499
00500 /** @brief Execute permission in hit region (bit 3) */
00501 k_mhiti_uhaci_x = 0x00000008U,
00502
00503 /** @brief Access control bits mask (bits 3:1) */
00504 k_mhiti_uhaci_mask = 0x0000000EU,
00505
00506 /** @brief Region 0 hit flag (bit 16) */
00507 k_mhiti_hiti0 = 0x00010000U,
00508
00509 /** @brief Region 1 hit flag (bit 17) */
00510 k_mhiti_hiti1 = 0x00020000U,
00511
00512 /** @brief Region 2 hit flag (bit 18) */
00513 k_mhiti_hiti2 = 0x00040000U,
00514
00515 /** @brief Region 3 hit flag (bit 19) */
00516 k_mhiti_hiti3 = 0x00080000U,
00517
00518 /** @brief Region 4 hit flag (bit 20) */
00519 k_mhiti_hiti4 = 0x00100000U,
00520
00521 /** @brief Region 5 hit flag (bit 21) */
00522 k_mhiti_hiti5 = 0x00200000U,
00523
00524 /** @brief Region 6 hit flag (bit 22) */
00525 k_mhiti_hiti6 = 0x00400000U,
00526
00527 /** @brief Region 7 hit flag (bit 23) */
00528 k_mhiti_hiti7 = 0x00800000U,
00529
00530 /** @brief Hit region flags mask (bits 23:16) */
00531 k_mhiti_hiti_mask = 0x00FF0000U,
00532
00533 /** @brief Hit region flags shift amount */
00534 k_mhiti_hiti_shift = 16U,
00535 } mhiti_bits_t;
00536
00537 /*
=====

```

```

00538 * MHITD Register (Data Hit Region)
00539 *
=====
00540 */
00541
00542 /**
00543  * @enum mhitd_bits_t
00544  * @brief Data Hit Region Register (MHITD) bit definitions
00545  *
00546  * @details
00547  * After a data memory protection error or region search, this register
00548  * indicates which region was hit and what permissions were set.
00549  *
00550  * @par Register Layout (32-bit @ 0x0008652C)
00551  * | Bits | Field | Description |
00552  * |-----|-----|-----|
00553  * | 0 | - | Reserved (read as 0) |
00554  * | 3:1 | UHACD[2:0] | Hit region access control (RWX) |
00555  * | 15:4 | - | Reserved (read as 0) |
00556  * | 23:16 | HITD[7:0] | Region hit flags (1=hit in region) |
00557  * | 31:24 | - | Reserved (read as 0) |
00558  *
00559  * @note HITD[7:0] = 0 means error in background region
00560  * @note Manual ref: Ch17 section 17.2.12
00561  * @since Version 1.0.0
00562  */
00563 typedef enum : uint32_t {
00564     /** @brief Read permission in hit region (bit 1) */
00565     k_mhitd_uhacd_r = 0x00000002U,
00566
00567     /** @brief Write permission in hit region (bit 2) */
00568     k_mhitd_uhacd_w = 0x00000004U,
00569
00570     /** @brief Execute permission in hit region (bit 3) */
00571     k_mhitd_uhacd_x = 0x00000008U,
00572
00573     /** @brief Access control bits mask (bits 3:1) */
00574     k_mhitd_uhacd_mask = 0x0000000EU,
00575
00576     /** @brief Region 0 hit flag (bit 16) */
00577     k_mhitd_hitd0 = 0x00010000U,
00578
00579     /** @brief Region 1 hit flag (bit 17) */
00580     k_mhitd_hitd1 = 0x00020000U,
00581
00582     /** @brief Region 2 hit flag (bit 18) */
00583     k_mhitd_hitd2 = 0x00040000U,
00584
00585     /** @brief Region 3 hit flag (bit 19) */
00586     k_mhitd_hitd3 = 0x00080000U,
00587
00588     /** @brief Region 4 hit flag (bit 20) */
00589     k_mhitd_hitd4 = 0x00100000U,
00590
00591     /** @brief Region 5 hit flag (bit 21) */
00592     k_mhitd_hitd5 = 0x00200000U,
00593
00594     /** @brief Region 6 hit flag (bit 22) */
00595     k_mhitd_hitd6 = 0x00400000U,
00596
00597     /** @brief Region 7 hit flag (bit 23) */
00598     k_mhitd_hitd7 = 0x00800000U,
00599
00600     /** @brief Hit region flags mask (bits 23:16) */
00601     k_mhitd_hitd_mask = 0x00FF0000U,
00602
00603     /** @brief Hit region flags shift amount */
00604     k_mhitd_hitd_shift = 16U,
00605 } mhitd_bits_t;
00606
00607 /*
=====
00608 * MPU Region Register Structure
00609 *
=====
00610 */
00611
00612 /**
00613  * @struct rx_mpu_region_regs_t
00614  * @brief MPU per-region register pair
00615  *
00616  * @details
00617  * Memory-mapped register layout for one MPU region.
00618  * Each region has 8 bytes (RSPAGE + REPAGE).
00619  *
00620  * @note Structure size verified as 8 bytes.
00621  * @since Version 1.0.0

```

```

00622 */
00623 typedef struct __attribute__((packed)) {
00624 /**
00625  * @brief Region Start Page Number Register (RSPAGEn)
00626  * @details
00627  * Bits [31:4] = Start page number (address[31:4])
00628  * Bits [3:0] = Reserved (write 0)
00629  */
00630 volatile uint32_t rspage;
00631
00632 /**
00633  * @brief Region End Page Number Register (REPAGEn)
00634  * @details
00635  * Bits [31:4] = End page number (inclusive)
00636  * Bit [3] = Execute permission (UAC.X)
00637  * Bit [2] = Write permission (UAC.W)
00638  * Bit [1] = Read permission (UAC.R)
00639  * Bit [0] = Valid bit (V)
00640  */
00641 volatile uint32_t repage;
00642 } rx_mpu_region_regs_t;
00643
00644 /*
=====
00645 * Register Accessor Functions - Region Registers
00646 *
=====
00647 */
00648
00649 /**
00650  * @brief Get pointer to MPU region register pair
00651  *
00652  * @param region Region number (0-7)
00653  * @return Volatile pointer to region register structure
00654  *
00655  * @pre region < k_mpu_region_count (8)
00656  *
00657  * @note Thread Safety: Safe - returns constant hardware address
00658  *
00659  * @par Example:
00660  * @code{.c}
00661  * // Configure region 0 to protect RAM (0x00000000-0x0007FFFF) as read-write
00662  * volatile rx_mpu_region_regs_t* r0 = mpu_region(k_mpu_region_0);
00663  * r0->rspage = 0x00000000; // Start at page 0
00664  * r0->repage = (0x0007FFF0) | k_repage_uac_rw | k_repage_v; // End at page 0x7FFF, RW, valid
00665  * @endcode
00666  *
00667  * @since Version 1.0.0
00668  */
00669 static inline volatile rx_mpu_region_regs_t* mpu_region(uint8_t region)
00670 {
00671     return (volatile rx_mpu_region_regs_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size));
00672 }
00673
00674 /**
00675  * @brief Get pointer to RSPAGE register for a region
00676  *
00677  * @param region Region number (0-7)
00678  * @return Volatile pointer to RSPAGE register (32-bit)
00679  * @since Version 1.0.0
00680  */
00681 static inline volatile uint32_t* mpu_rspage_reg(uint8_t region)
00682 {
00683     return (volatile uint32_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size));
00684 }
00685
00686 /**
00687  * @brief Get pointer to REPAGE register for a region
00688  *
00689  * @param region Region number (0-7)
00690  * @return Volatile pointer to REPAGE register (32-bit)
00691  * @since Version 1.0.0
00692  */
00693 static inline volatile uint32_t* mpu_repage_reg(uint8_t region)
00694 {
00695     return (volatile uint32_t*)(k_mpu_region_base_addr + (region * k_mpu_region_size) + 4U);
00696 }
00697
00698 /*
=====
00699 * Register Accessor Functions - Control Registers
00700 *
=====
00701 */
00702
00703 /**
00704  * @brief Get pointer to MPEN (Memory Protection Enable) register

```



```

00705 * @return Volatile pointer to MPEN register (32-bit)
00706 * @since Version 1.0.0
00707 */
00708 static inline volatile uint32_t* mpu_mpen_reg(void)
00709 {
00710     return (volatile uint32_t*)k_mpu_mpen_addr;
00711 }
00712
00713 /**
00714 * @brief Get pointer to MPBAC (Background Access Control) register
00715 * @return Volatile pointer to MPBAC register (32-bit)
00716 * @since Version 1.0.0
00717 */
00718 static inline volatile uint32_t* mpu_mpbac_reg(void)
00719 {
00720     return (volatile uint32_t*)k_mpu_mpbac_addr;
00721 }
00722
00723 /**
00724 * @brief Get pointer to MPECLR (Error Status Clearing) register
00725 * @return Volatile pointer to MPECLR register (32-bit)
00726 * @since Version 1.0.0
00727 */
00728 static inline volatile uint32_t* mpu_mpeclr_reg(void)
00729 {
00730     return (volatile uint32_t*)k_mpu_mpeclr_addr;
00731 }
00732
00733 /**
00734 * @brief Get pointer to MPESTS (Error Status) register
00735 * @return Volatile pointer to MPESTS register (32-bit)
00736 * @since Version 1.0.0
00737 */
00738 static inline volatile uint32_t* mpu_mpests_reg(void)
00739 {
00740     return (volatile uint32_t*)k_mpu_mpests_addr;
00741 }
00742
00743 /**
00744 * @brief Get pointer to MPDEA (Data Error Address) register
00745 * @return Volatile pointer to MPDEA register (32-bit)
00746 * @since Version 1.0.0
00747 */
00748 static inline volatile uint32_t* mpu_mpdea_reg(void)
00749 {
00750     return (volatile uint32_t*)k_mpu_mpdea_addr;
00751 }
00752
00753 /**
00754 * @brief Get pointer to MPSA (Region Search Address) register
00755 * @return Volatile pointer to MPSA register (32-bit)
00756 * @since Version 1.0.0
00757 */
00758 static inline volatile uint32_t* mpu_mpsa_reg(void)
00759 {
00760     return (volatile uint32_t*)k_mpu_mpsa_addr;
00761 }
00762
00763 /**
00764 * @brief Get pointer to MPOPS (Region Search Operation) register
00765 * @return Volatile pointer to MPOPS register (16-bit)
00766 * @since Version 1.0.0
00767 */
00768 static inline volatile uint16_t* mpu_mpops_reg(void)
00769 {
00770     return (volatile uint16_t*)k_mpu_mpops_addr;
00771 }
00772
00773 /**
00774 * @brief Get pointer to MPOPI (Region Invalidation) register
00775 * @return Volatile pointer to MPOPI register (16-bit)
00776 * @since Version 1.0.0
00777 */
00778 static inline volatile uint16_t* mpu_mpopi_reg(void)
00779 {
00780     return (volatile uint16_t*)k_mpu_mpopi_addr;
00781 }
00782
00783 /**
00784 * @brief Get pointer to MHITI (Instruction Hit Region) register
00785 * @return Volatile pointer to MHITI register (32-bit)
00786 * @since Version 1.0.0
00787 */
00788 static inline volatile uint32_t* mpu_mhiti_reg(void)
00789 {
00790     return (volatile uint32_t*)k_mpu_mhiti_addr;
00791 }

```

```

00792
00793 /**
00794  * @brief Get pointer to MHITD (Data Hit Region) register
00795  * @return Volatile pointer to MHITD register (32-bit)
00796  * @since Version 1.0.0
00797  */
00798 static inline volatile uint32_t* mpu_mhitd_reg(void)
00799 {
00800     return (volatile uint32_t*)k_mpu_mhitd_addr;
00801 }
00802
00803 /*
=====
00804  * Helper Functions - Page Number Conversion
00805  *
=====
00806  */
00807
00808 /**
00809  * @brief Convert address to page number
00810  *
00811  * @param addr Memory address
00812  * @return Page number (address[31:4])
00813  *
00814  * @par Example:
00815  * @code{.c}
00816  * uint32_t page = mpu_addr_to_page(0x00012340);
00817  * // page = 0x00001234
00818  * @endcode
00819  *
00820  * @since Version 1.0.0
00821  */
00822 static inline uint32_t mpu_addr_to_page(uint32_t addr)
00823 {
00824     return (addr » k_mpu_page_shift) & k_mpu_page_max;
00825 }
00826
00827 /**
00828  * @brief Convert page number to RSPAGE/REPAGE register value
00829  *
00830  * @param page Page number (28 bits)
00831  * @return Register value with page number in bits [31:4]
00832  *
00833  * @note For REPAGE, caller must OR in access control and valid bits.
00834  *
00835  * @par Example:
00836  * @code{.c}
00837  * uint32_t rspage_val = mpu_page_to_reg(0x00001234);
00838  * uint32_t repage_val = mpu_page_to_reg(0x00001234) | k_repage_uac_rw | k_repage_v;
00839  * @endcode
00840  *
00841  * @since Version 1.0.0
00842  */
00843 static inline uint32_t mpu_page_to_reg(uint32_t page)
00844 {
00845     return (page « k_mpu_page_shift) & k_rspage_rspn_mask;
00846 }
00847
00848 /*
=====
00849  * Static Assertions - Address Verification
00850  *
=====
00851  */
00852
00853 /**
00854  * @name Address Verification
00855  * @brief Compile-time verification of register addresses and offsets
00856  * @{
00857  */
00858
00859 /* Verify base addresses */
00860 static_assert(k_mpu_region_base_addr == 0x00086400U,
00861     "MPU region base address must be 0x00086400 per Ch17 manual");
00862
00863 static_assert(k_mpu_control_base_addr == 0x00086500U,
00864     "MPU control base address must be 0x00086500 per Ch17 manual");
00865
00866 /* Verify RSPAGE addresses */
00867 static_assert(k_mpu_rspage0_addr == 0x00086400U,
00868     "RSPAGE0 address must be 0x00086400 per Ch17 manual");
00869
00870 static_assert(k_mpu_rspage1_addr == 0x00086408U,
00871     "RSPAGE1 address must be 0x00086408 per Ch17 manual");
00872
00873 static_assert(k_mpu_rspage7_addr == 0x00086438U,
00874     "RSPAGE7 address must be 0x00086438 per Ch17 manual");

```

```

00875
00876 /* Verify REPAGE addresses */
00877 static_assert(k_mpu_repage0_addr == 0x00086404U,
00878     "REPAGE0 address must be 0x00086404 per Ch17 manual");
00879
00880 static_assert(k_mpu_repage1_addr == 0x0008640CU,
00881     "REPAGE1 address must be 0x0008640C per Ch17 manual");
00882
00883 static_assert(k_mpu_repage7_addr == 0x0008643CU,
00884     "REPAGE7 address must be 0x0008643C per Ch17 manual");
00885
00886 /* Verify region spacing */
00887 static_assert(k_mpu_region_size == 0x08U, "MPU region spacing must be 8 bytes");
00888
00889 static_assert(k_mpu_rspage1_addr - k_mpu_rspage0_addr == k_mpu_region_size,
00890     "RSPAGE0 to RSPAGE1 spacing must be 8 bytes");
00891
00892 static_assert(k_mpu_rspage7_addr - k_mpu_rspage0_addr == 7 * k_mpu_region_size,
00893     "RSPAGE0 to RSPAGE7 spacing must be 56 bytes");
00894
00895 /* Verify control register addresses */
00896 static_assert(k_mpu_mpen_addr == 0x00086500U, "MPEN address must be 0x00086500 per Ch17 manual");
00897
00898 static_assert(k_mpu_mpbac_addr == 0x00086504U, "MPBAC address must be 0x00086504 per Ch17 manual");
00899
00900 static_assert(k_mpu_mpeclr_addr == 0x00086508U,
00901     "MPECLR address must be 0x00086508 per Ch17 manual");
00902
00903 static_assert(k_mpu_mpests_addr == 0x0008650CU,
00904     "MPESTS address must be 0x0008650C per Ch17 manual");
00905
00906 static_assert(k_mpu_mpdea_addr == 0x00086514U, "MPDEA address must be 0x00086514 per Ch17 manual");
00907
00908 static_assert(k_mpu_mpsa_addr == 0x00086520U, "MPSA address must be 0x00086520 per Ch17 manual");
00909
00910 static_assert(k_mpu_mpops_addr == 0x00086524U, "MPOPS address must be 0x00086524 per Ch17 manual");
00911
00912 static_assert(k_mpu_mpopi_addr == 0x00086526U, "MPOPI address must be 0x00086526 per Ch17 manual");
00913
00914 static_assert(k_mpu_mhiti_addr == 0x00086528U, "MHITI address must be 0x00086528 per Ch17 manual");
00915
00916 static_assert(k_mpu_mhitd_addr == 0x0008652CU, "MHITD address must be 0x0008652C per Ch17 manual");
00917
00918 /* Verify structure layout */
00919 static_assert(sizeof(rx_mpu_region_regs_t) == k_mpu_region_size,
00920     "MPU region struct size must be 8 bytes");
00921
00922 static_assert(offsetof(rx_mpu_region_regs_t, rspage) == 0, "RSPAGE offset in struct must be 0");
00923
00924 static_assert(offsetof(rx_mpu_region_regs_t, repage) == 4, "REPAGE offset in struct must be 4");
00925
00926 /* The enum declarations above ARE the bit-position contract; assert the
00927  * RELATIONSHIPS between them rather than re-stating each literal value
00928  * (which would just duplicate the magic numbers and trip
00929  * readability-magic-numbers without adding a real check). */
00930 static_assert(k_repage_v == 0x01U, "REPAGE V bit must be bit 0");
00931 static_assert(k_repage_uac_r == (k_repage_v « 1U), "UAC.R must be REPAGE.V left-shift 1");
00932 static_assert(k_repage_uac_w == (k_repage_uac_r « 1U), "UAC.W must be UAC.R left-shift 1");
00933 static_assert(k_repage_uac_x == (k_repage_uac_w « 1U), "UAC.X must be UAC.W left-shift 1");
00934 static_assert(k_mpen_mpen == 0x01U, "MPEN.MPEN bit must be bit 0");
00935
00936 static_assert(k_mpests_imper == 0x01U, "MPESTS.IMPER bit must be bit 0");
00937
00938 static_assert(k_mpests_dmper == 0x02U, "MPESTS.DMPER bit must be bit 1");
00939
00940 static_assert(k_mpests_drw == 0x04U, "MPESTS.DRW bit must be bit 2");
00941
00942 /** @} */
00943
00944 #ifdef __cplusplus
00945 }
00946 #endif

```

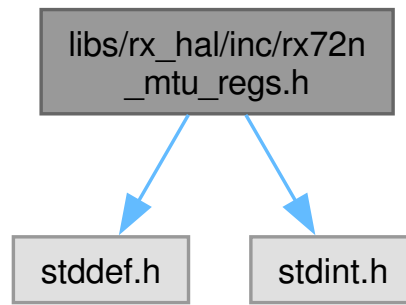
8.39 libs/rx`hal/inc/rx72n`mtu`regs.h File Reference

RX72N Multi-Function Timer Unit (MTU3a) register definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_mtu_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_mtu_channel_regs_t](#)
Standard MTU channel register map (MTU0-MTU2).
- struct [rx_mtu3_channel_regs_t](#)
MTU3 register map accessed via base address 0x000C1200.
- struct [rx_mtu12_phase_regs_t](#)
MTU1/MTU2 extended register map for 32-bit phase counting mode.
- struct [rx_mtu_tstr_regs_t](#)
MTU Start Register Map.

Enumerations

- enum [rx_mtu_addresses_t](#) : [uintptr_t](#) {
[k_mtu0_base_addr](#) = 0x000C1300 , [k_mtu1_base_addr](#) = 0x000C1380 , [k_mtu2_base_addr](#) = 0x000C1400 , [k_mtu3_base_addr](#) = 0x000C1200 ,
[k_mtu4_base_addr](#) = ([k_mtu3_base_addr](#) + 1U) , [k_mtu5u_base_addr](#) = 0x000C1C80 ,
[k_mtu5v_base_addr](#) = 0x000C1C90 , [k_mtu5w_base_addr](#) = 0x000C1CA0 ,
[k_mtu6_base_addr](#) = 0x000C1A00 , [k_mtu7_base_addr](#) = ([k_mtu6_base_addr](#) + 1U) ,
[k_mtu8_base_addr](#) = 0x000C1600 , [k_mtu_tstra_base_addr](#) = 0x000C1280 ,
[k_mtu_tstrb_base_addr](#) = 0x000C1A80 }
 MTU3a peripheral base addresses (RX72N memory map Chapter 4).
- enum [mtu_bit_values_t](#) : [uint8_t](#) { [k_mtu_bit_value_set](#) = 1 }
 Bit value constants for shift operations.
- enum [mtu3_padding_sizes_t](#) : [uint8_t](#) {
[k_mtu3_pad_after_tiorl](#) = 2U , [k_mtu3_pad_after_tier](#) = 7U , [k_mtu3_pad_after_tcnt](#) = 6U ,
[k_mtu3_pad_after_tgrb](#) = 8U ,
[k_mtu3_pad_after_tgrd](#) = 4U , [k_mtu3_pad_after_tsr](#) = 11U , [k_mtu3_pad_after_tbtm](#) = 19U , [k_mtu3_pad_after_tcr2](#) = 37U }
 Named padding array sizes for [rx_mtu3_channel_regs_t](#) reserved fields.
- enum [mtu12_phase_padding_sizes_t](#) : [uint8_t](#) { [k_mtu12p_pad_before_tcntlw](#) = 14U }
 Named padding array sizes for [rx_mtu12_phase_regs_t](#) reserved fields.

- enum `mtu_tcr_bits_t` : `uint8_t` {
`k_mtu_tcr_cclr_shift` = 5 , `k_mtu_tcr_tpssc_mask` = 0x07 , `k_mtu_tcr_ckege_mask` = 0x18 ,
`k_mtu_tcr_cclr_mask` = 0xE0 ,
`k_mtu_tcr_tpssc_1` = 0x00 , `k_mtu_tcr_tpssc_4` = 0x01 , `k_mtu_tcr_tpssc_16` = 0x02 ,
`k_mtu_tcr_tpssc_64` = 0x03 ,
`k_mtu_tcr_cclr_tgra` = (`k_mtu_bit_value_set` << `k_mtu_tcr_cclr_shift`) }

Timer Control Register (TCR) bit field definitions.

- enum `mtu_tmdr_bits_t` : `uint8_t` {
`k_mtu_tmdr_bfa_shift` = 4 , `k_mtu_tmdr_bfb_shift` = 5 , `k_mtu_tmdr_md_mask` = 0x0F ,
`k_mtu_tmdr_md_normal` = 0x00 ,
`k_mtu_tmdr_md_pwm1` = 0x02 , `k_mtu_tmdr_md_pwm2` = 0x03 , `k_mtu_tmdr_bfa` = (`k_mtu_bit_value_set` << `k_mtu_tmdr_bfa_shift`) , `k_mtu_tmdr_bfb` = (`k_mtu_bit_value_set` << `k_mtu_tmdr_bfb_shift`) }

Timer Mode Register (TMDR) bit field definitions.

- enum `mtu_tmdr3_bits_t` : `uint8_t` { `k_mtu_tmdr3_lwa_shift` = 0 , `k_mtu_tmdr3_phcksel_shift` = 1 , `k_mtu_tmdr3_lwa` , `k_mtu_tmdr3_phcksel` }

Timer Mode Register 3 (TMDR3) bits for MTU1/MTU2 32-bit cascaded mode.

- enum `mtu_tior_bits_t` : `uint8_t` {
`k_mtu_tior_ioa_mask` = 0x0F , `k_mtu_tior_iob_mask` = 0xF0 , `k_mtu_tior_init_low` = 0x02 ,
`k_mtu_tior_init_high` = 0x05 ,
`k_mtu_tior_toggle` = 0x03 }

Timer I/O Control Register (TIOR) bit field definitions.

- enum `mtu_tstr_bits_t` : `uint8_t` {
`k_mtu_tstr_cst0_shift` = 0 , `k_mtu_tstr_cst1_shift` = 1 , `k_mtu_tstr_cst2_shift` = 2 ,
`k_mtu_tstr_cst8_shift` = 3 ,
`k_mtu_tstr_cst3_shift` = 6 , `k_mtu_tstr_cst4_shift` = 7 , `k_mtu_tstr_cst0` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst0_shift`) , `k_mtu_tstr_cst1` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst1_shift`) ,
`k_mtu_tstr_cst2` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst2_shift`) , `k_mtu_tstr_cst3` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst3_shift`) , `k_mtu_tstr_cst4` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst4_shift`) , `k_mtu_tstr_cst8` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst8_shift`) }

Timer Start Register A (TSTRA) bit definitions - controls MTU0-4, MTU8.

- enum `mtu_tstrb_bits_t` : `uint8_t` { `k_mtu_tstr_cst6_shift` = 6 , `k_mtu_tstr_cst7_shift` = 7 ,
`k_mtu_tstr_cst6` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst6_shift`) , `k_mtu_tstr_cst7` = (`k_mtu_bit_value_set` << `k_mtu_tstr_cst7_shift`) }

Timer Start Register B (TSTRB) bit definitions - controls MTU6-7.

- enum `mtu_channel_reg_offsets_t` : `uint8_t` { `k_mtu_ch_off_tcr` = 0x00U , `k_mtu_ch_off_tcncnt` = 0x06U , `k_mtu_ch_off_tgra` = 0x08U , `k_mtu_ch_off_tgrb` = 0x0AU }

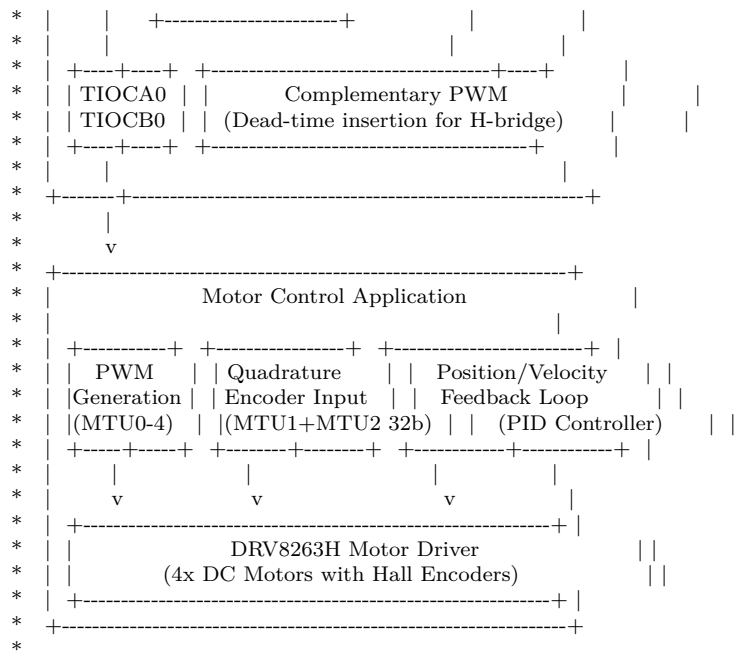
Byte offsets of registers within `rx_mtu_channel_regs_t`.

- enum `mtu3_reg_offsets_t` : `uint8_t` {
`k_mtu3_off_tcr` = 0x00U , `k_mtu3_off_tmdr1` = 0x02U , `k_mtu3_off_tiorh` = 0x04U ,
`k_mtu3_off_tiorl` = 0x05U ,
`k_mtu3_off_tier` = 0x08U , `k_mtu3_off_tcncnt` = 0x10U , `k_mtu3_off_tgra` = 0x18U ,
`k_mtu3_off_tgrb` = 0x1AU ,
`k_mtu3_off_tgrc` = 0x24U , `k_mtu3_off_tgrd` = 0x26U , `k_mtu3_off_tsr` = 0x2CU ,
`k_mtu3_off_tbtm` = 0x38U ,
`k_mtu3_off_tcr2` = 0x4CU , `k_mtu3_off_tgre` = 0x72U , `k_mtu3_struct_size` = 0x74U }

Byte offsets of MTU3 registers from base 0x000C1200.

- enum `mtu12_phase_reg_offsets_t` : `uint8_t` {
`k_mtu12p_off_tcr` = 0x00U , `k_mtu12p_off_tmdr` = 0x01U , `k_mtu12p_off_tcncnt` = 0x06U ,
`k_mtu12p_off_tgra` = 0x08U ,
`k_mtu12p_off_tgrb` = 0x0AU , `k_mtu12p_off_tmdr3` = 0x11U , `k_mtu12p_off_tcncntlw` = 0x20U ,
`k_mtu12p_off_tgralw` = 0x24U ,
`k_mtu12p_off_tgrblw` = 0x28U , `k_mtu12p_struct_size` = 0x2CU }

Byte offsets of registers within `rx_mtu12_phase_regs_t`.



MTU3a Channel Capabilities

Channel	Counter	PWM	Phase Count	Complementary	Buffer
MTU0	16-bit	Yes	No	No	Yes
MTU1	16/32	Yes	Yes	No	No
MTU2	16/32	Yes	Yes	No	No
MTU3	16-bit	Yes	No	Yes (w/MTU4)	Yes
MTU4	16-bit	Yes	No	Yes (w/MTU3)	Yes

| MTU5 | 16-bit | Waveform generator (special) | No | | MTU6 | 16-bit | Yes | No | Yes (w/MTU7)
 | Yes | | MTU7 | 16-bit | Yes | No | Yes (w/MTU6) | Yes | | MTU8 | 32-bit | Yes | No | No | Yes |

STAR Motor Encoder Configuration

The STAR robot uses 4x brushed DC gearmotors with Hall effect quadrature encoders (341 PPR per revolution). MTU1 and MTU2 are configured in 32-bit cascaded phase counting mode for high-resolution position tracking:

$$\text{Position (revolutions)} = \frac{\text{TCNTLW}}{341 \times 4}$$

With 32-bit counter at 341 PPR x 4 edges = 1364 counts/rev:

- Range: +/-3,146,879 revolutions before overflow
- At 210 RPM max: 10.4 days continuous operation without wraparound

Clock Configuration

MTU timers derive their clock from PCLKA (Peripheral Clock A):

$$f_{timer} = \frac{f_{PCLKA}}{prescaler} = \frac{120 \text{ MHz}}{prescaler}$$

Available prescaler values: 1, 4, 16, 64, 256, 1024

PWM Frequency Calculation

For PWM mode 1 with compare match on TGRA:

$$f_{PWM} = \frac{f_{PCLKA}}{prescaler \times (TGRA + 1)}$$

Example: 20 kHz PWM at 120 MHz PCLKA, prescaler=1:

$$TGRA = \frac{120 \text{ MHz}}{1 \times 20 \text{ kHz}} - 1 = 5999$$

Hardware Reference

- RX72N Group User's Manual: Hardware, Chapter 24 (MTU3a)
- Section 24.2: Register Descriptions
- Section 24.3: Operations (Timer modes, PWM, Phase counting)
- Section 24.4: Interrupts

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion in inline functions
- Rule 2: [OK] Fixed loop bounds (N/A - register definitions only)
- Rule 3: [OK] Static allocation only (all definitions are compile-time)
- Rule 5: [OK] Static assertions verify register layout at compile time
- Rule 8: [OK] C23 typed enums eliminate preprocessor for constants
- Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: MTU register definitions only, no driver logic
- Open/Closed: Extend via new enums; don't modify existing values
- Interface Segregation: Separate structs for different channel types
- Dependency Inversion: Higher-level drivers depend on these abstractions

Related Modules

- [rx_mtu.c](#): MTU HAL driver implementation
- [rx_mtu_encoder.h](#): Quadrature encoder abstraction
- [rx72n_gptw_regs.h](#): Alternative PWM timer (GPTW)
- [rx72n_port_regs.h](#): GPIO for timer I/O pins
- [rx72n_mpc.h](#): Pin function selection for MTU I/O

RX72N Hardware Manual Chapter 24 for complete MTU3a specification
DOXYGEN_ROADMAP.md for documentation standards

Locked, Inc. Contributors

2026-01-01

1.0.0

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8.40 rx72n.mtu.regs.h

```

00001 /**
00002  * @file rx72n_mtu_regs.h
00003  * @brief RX72N Multi-Function Timer Unit (MTU3a) register definitions
00004  *
00005  * @details
00006  * This file provides complete hardware register definitions for the RX72N's
00007  * Multi-Function Timer Unit version 3a (MTU3a). The MTU3a is a highly flexible
00008  * timer peripheral supporting PWM generation, pulse counting, phase counting
00009  * (quadrature decoding), and general-purpose timing functions.
00010  *
00011  * @par System Architecture
00012  * @verbatim
00013  * STAR Robot Motor Control System
00014  * =====
00015  *
00016  * +-----+
00017  * | MTU3a Timer Architecture |
00018  * |
00019  * | +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ |
00020  * | | MTU0 | | MTU1 | | MTU2 | | ... | MTU6 | | MTU7 | | |
00021  * | | 16-bit | | 16-bit | | 16-bit | | 16-bit | | 16-bit | | 16-bit | |
00022  * | | PWM | | Encoder | | Encoder | | Compl PWM | | Compl PWM | |
00023  * | +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ |
00024  * | | | | | | | | | | | | | | | | |
00025  * | | +-----+ +-----+ +-----+ +-----+ | |
00026  * | | 32-bit Cascaded Mode | | | | | | | | |
00027  * | | (MTU1+MTU2 for | | | | | | | |
00028  * | | Hall Encoders) | | | | | | | |
00029  * | | +-----+ | | | | | | | |
00030  * | | | | | | | | | | | | | |
00031  * | +-----+ +-----+ +-----+ +-----+ |
00032  * | | TIOCA0 | | Complementary PWM | | |
00033  * | | TIOCB0 | | (Dead-time insertion for H-bridge) | | |
00034  * | +-----+ +-----+ +-----+ +-----+ |
00035  * | | | | | | | | | | | | |
00036  * | +-----+ +-----+ +-----+ +-----+ |
00037  * | | | | | | | | | | | | |
00038  * | | v | | | | | | | | |
00039  * | +-----+ +-----+ +-----+ +-----+ |

```

```

00040 * | Motor Control Application |
00041 * | | | | |
00042 * | +-----+ +-----+ +-----+ |
00043 * | | PWM | | Quadrature | | Position/Velocity | |
00044 * | | Generation | | Encoder Input | | Feedback Loop | |
00045 * | | (MTU0-4) | | (MTU1+MTU2 32b) | | (PID Controller) | |
00046 * | +-----+ +-----+ +-----+ |
00047 * | | | | |
00048 * | | v | | v | | v | |
00049 * | +-----+ |
00050 * | | DRV8263H Motor Driver | |
00051 * | | (4x DC Motors with Hall Encoders) | |
00052 * | +-----+ |
00053 * | |
00054 * @endverbatim
00055 *
00056 * @par MTU3a Channel Capabilities
00057 * | Channel | Counter | PWM | Phase Count | Complementary | Buffer |
00058 * |-----|-----|-----|-----|-----|
00059 * | MTU0 | 16-bit | Yes | No | No | Yes |
00060 * | MTU1 | 16/32 | Yes | Yes | No | No |
00061 * | MTU2 | 16/32 | Yes | Yes | No | No |
00062 * | MTU3 | 16-bit | Yes | No | Yes (w/MTU4) | Yes |
00063 * | MTU4 | 16-bit | Yes | No | Yes (w/MTU3) | Yes |
00064 * | MTU5 | 16-bit | Waveform generator (special) | No |
00065 * | MTU6 | 16-bit | Yes | No | Yes (w/MTU7) | Yes |
00066 * | MTU7 | 16-bit | Yes | No | Yes (w/MTU6) | Yes |
00067 * | MTU8 | 32-bit | Yes | No | No | Yes |
00068 *
00069 * @par STAR Motor Encoder Configuration
00070 * The STAR robot uses 4x brushed DC gearmotors with Hall effect quadrature
00071 * encoders (341 PPR per revolution). MTU1 and MTU2 are configured in 32-bit
00072 * cascaded phase counting mode for high-resolution position tracking:
00073 *
00074 * @f[
00075 * \text{Position (revolutions)} = \frac{\text{TCNTLW}}{341 \times 4}
00076 * @f]
00077 *
00078 * With 32-bit counter at 341 PPR x 4 edges = 1364 counts/rev:
00079 * - Range: +/-3,146,879 revolutions before overflow
00080 * - At 210 RPM max: 10.4 days continuous operation without wraparound
00081 *
00082 * @par Clock Configuration
00083 * MTU timers derive their clock from PCLKA (Peripheral Clock A):
00084 * @f[
00085 * f_{timer} = \frac{f_{PCLKA}}{prescaler} = \frac{120 \text{ MHz}}{prescaler}
00086 * @f]
00087 *
00088 * Available prescaler values: 1, 4, 16, 64, 256, 1024
00089 *
00090 * @par PWM Frequency Calculation
00091 * For PWM mode 1 with compare match on TGRA:
00092 * @f[
00093 * f_{PWM} = \frac{f_{PCLKA}}{prescaler \times (TGRA + 1)}
00094 * @f]
00095 *
00096 * Example: 20 kHz PWM at 120 MHz PCLKA, prescaler=1:
00097 * @f[
00098 * TGRA = \frac{120 \text{ MHz}}{1 \times 20 \text{ kHz}} - 1 = 5999
00099 * @f]
00100 *
00101 * @par Hardware Reference
00102 * - RX72N Group User's Manual: Hardware, Chapter 24 (MTU3a)
00103 * - Section 24.2: Register Descriptions
00104 * - Section 24.3: Operations (Timer modes, PWM, Phase counting)
00105 * - Section 24.4: Interrupts
00106 *
00107 * @par NASA Power of 10 Compliance
00108 * - Rule 1: [OK] No goto, setjmp, or recursion in inline functions
00109 * - Rule 2: [OK] Fixed loop bounds (N/A - register definitions only)
00110 * - Rule 3: [OK] Static allocation only (all definitions are compile-time)
00111 * - Rule 5: [OK] Static assertions verify register layout at compile time
00112 * - Rule 8: [OK] C23 typed enums eliminate preprocessor for constants
00113 * - Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror
00114 *
00115 * @par SOLID Principles
00116 * - **Single Responsibility**: MTU register definitions only, no driver logic
00117 * - **Open/Closed**: Extend via new enums; don't modify existing values
00118 * - **Interface Segregation**: Separate structs for different channel types
00119 * - **Dependency Inversion**: Higher-level drivers depend on these abstractions
00120 *
00121 * @par Related Modules
00122 * - [rx_mtu.c](rx_mtu_8c.html): MTU HAL driver implementation
00123 * - [rx_mtu_encoder.h](rx_mtu_encoder_8h.html): Quadrature encoder abstraction
00124 * - [rx72n_gptw_regs.h](rx72n_gptw_regs_8h.html): Alternative PWM timer (GPTW)
00125 * - [rx72n_port_regs.h](rx72n_port_regs_8h.html): GPIO for timer I/O pins
00126 * - [rx72n_mpc.h](rx72n_mpc_8h.html): Pin function selection for MTU I/O

```

```

00127 *
00128 * @see RX72N Hardware Manual Chapter 24 for complete MTU3a specification
00129 * @see DOXYGEN_ROADMAP.md for documentation standards
00130 *
00131 * @author Locked, Inc. Contributors
00132 * @date 2026-01-01
00133 * @version 1.0.0
00134 * @copyright Copyright (c) 2026 Locked Inc.
00135 * SPDX-License-Identifier: MIT
00136 *
00137 * @defgroup mtu_regs MTU Register Definitions
00138 * @{
00139 */
00140
00141 #pragma once
00142
00143 #include <stddef.h>
00144 #include <stdint.h>
00145
00146 #ifdef __cplusplus
00147 extern "C" {
00148 #endif
00149
00150 /*
=====
00151 * Multi-Function Timer Unit (MTU3a) - Base Addresses
00152 *
=====
00153 */
00154 /**
00155 * @enum rx_mtu_addresses_t
00156 * @brief MTU3a peripheral base addresses (RX72N memory map Chapter 4)
00157 *
00158 * @details
00159 * Base addresses for all MTU3a timer channels. The MTU3a module occupies
00160 * memory region 0x000C1200-0x000C1CFF in the RX72N peripheral address space.
00161 *
00162 * @par Memory Map Overview
00163 * @verbatim
00164 *      Address Range      Channel    Notes
00165 *      -----
00166 *      0x000C1200-0x000C127F MTU3/4   Interleaved (complementary PWM pair)
00167 *      0x000C1280-0x000C12FF TSTRA    Timer Start Register A (MTU0-4, 8)
00168 *      0x000C1300-0x000C137F MTU0     General-purpose timer
00169 *      0x000C1380-0x000C13FF MTU1     Phase counting capable
00170 *      0x000C1400-0x000C147F MTU2     Phase counting capable
00171 *      0x000C1600-0x000C16FF MTU8     32-bit native counter
00172 *      0x000C1A00-0x000C1A7F MTU6/7   Interleaved (complementary PWM pair)
00173 *      0x000C1A80-0x000C1AFF TSTRB    Timer Start Register B (MTU6-7)
00174 *      0x000C1C80-0x000C1CFF MTU5     Waveform generator (U/V/W phases)
00175 * @endverbatim
00176 *
00177 * @par Hardware Design Notes
00178 * - MTU3/MTU4 share base address 0x000C1200 (interleaved registers)
00179 * - MTU6/MTU7 share base address 0x000C1A00 (interleaved registers)
00180 * - Interleaving enables synchronized complementary PWM generation
00181 * - MTU1/MTU2 can cascade to form 32-bit counter (TCNTLW)
00182 *
00183 * @par Code Example
00184 * @code
00185 * // Access MTU0 timer counter
00186 * volatile rx_mtu_channel_regs_t* mtu = mtu0();
00187 * uint16_t count = mtu->tcnt;
00188 *
00189 * // Configure MTU0 for 20 kHz PWM
00190 * mtu->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra; // PCLKA/1, clear on TGRA
00191 * mtu->tgra = 5999; // Period: 120 MHz / 6000 = 20 kHz
00192 * mtu->tgrb = 3000; // 50% duty cycle
00193 * @endcode
00194 *
00195 * @see RX72N Hardware Manual Chapter 4 (Address Map)
00196 * @see RX72N Hardware Manual Section 24.2 (MTU Register Descriptions)
00197 */
00198
00199 typedef enum : uintptr_t {
00200 /** @brief MTU0 base address (0x000C1300) - general purpose 16-bit timer */
00201 k_mtu0_base_addr = 0x000C1300,
00202
00203 /** @brief MTU1 base address (0x000C1380) - 16-bit with phase counting, cascadable to 32-bit */
00204 k_mtu1_base_addr = 0x000C1380,
00205
00206 /** @brief MTU2 base address (0x000C1400) - 16-bit with phase counting, cascadable to 32-bit */
00207 k_mtu2_base_addr = 0x000C1400,
00208
00209 /** @brief MTU3 base address (0x000C1200) - complementary PWM pair with MTU4 */
00210 k_mtu3_base_addr = 0x000C1200,
00211

```

```

00212 /**
00213  * @brief MTU4 base address (MTU3 + 1) - interleaved with MTU3
00214  * @details MTU4 registers are at odd offsets relative to MTU3 at even offsets.
00215  * This interleaving enables synchronized complementary PWM operation.
00216  */
00217 k_mtu4_base_addr = (k_mtu3_base_addr + 1U),
00218
00219 /** @brief MTU5U base address (0x000C1C80) - waveform generator U-phase */
00220 k_mtu5u_base_addr = 0x000C1C80,
00221
00222 /** @brief MTU5V base address (0x000C1C90) - waveform generator V-phase */
00223 k_mtu5v_base_addr = 0x000C1C90,
00224
00225 /** @brief MTU5W base address (0x000C1CA0) - waveform generator W-phase */
00226 k_mtu5w_base_addr = 0x000C1CA0,
00227
00228 /**
00229  * @brief MTU6 base address (0x000C1A00) - complementary PWM pair with MTU7
00230  * @details MTU6 and MTU7 share the same base address with interleaved registers.
00231  */
00232 k_mtu6_base_addr = 0x000C1A00,
00233
00234 /**
00235  * @brief MTU7 base address (MTU6 + 1) - interleaved with MTU6
00236  * @details MTU7 registers are at odd offsets relative to MTU6 at even offsets.
00237  * This interleaving enables synchronized complementary PWM operation.
00238  * (Manual: MTU7.TCR = 0x000C1A01 = MTU6 base + 1)
00239  */
00240 k_mtu7_base_addr = (k_mtu6_base_addr + 1U),
00241
00242 /** @brief MTU8 base address (0x000C1600) - native 32-bit counter */
00243 k_mtu8_base_addr = 0x000C1600,
00244
00245 /** @brief TSTRA base address (0x000C1280) - timer start control for MTU0-4, MTU8 */
00246 k_mtu_tstra_base_addr = 0x000C1280,
00247
00248 /** @brief TSTRB base address (0x000C1A80) - timer start control for MTU6-7 */
00249 k_mtu_tstrb_base_addr = 0x000C1A80,
00250 } rx_mtu_addresses_t;
00251
00252 /**
00253  * @brief Bit value constants for shift operations
00254  */
00255 * Named constants to eliminate magic numbers in bit shift expressions.
00256 */
00257 typedef enum : uint8_t {
00258     k_mtu_bit_value_set = 1, /**< Bit value to shift (replaces 1, 1U, 0x01 in shifts) */
00259 } mtu_bit_values_t;
00260
00261 /**
00262  * @struct rx_mtu_channel_regs_t
00263  * @brief Standard MTU channel register map (MTU0-MTU2)
00264  */
00265 * @details
00266 * Memory-mapped register structure for consecutive-layout 16-bit MTU channels.
00267 * This structure applies to MTU0, MTU1, and MTU2 only.
00268 * MTU6 and MTU7 have an interleaved register layout identical to MTU3/MTU4
00269 * and must be accessed via mtu6() which returns rx_mtu3_channel_regs_t*.
00270 */
00271 * @par Register Memory Layout (18 bytes total)
00272 * @verbatim
00273 *   Offset  Size  Register  Description
00274 *   -----
00275 *   0x00    1    TCR        Timer Control (prescaler, edge, clear source)
00276 *   0x01    1    TMDR       Timer Mode (normal, PWM1, PWM2, phase count)
00277 *   0x02    1    TIORH      Timer I/O Control H (TIOCA/TIOCB output mode)
00278 *   0x03    1    TIORL      Timer I/O Control L (TIOCC/TIOCD output mode)
00279 *   0x04    1    TIER       Timer Interrupt Enable (TGIA-D, overflow)
00280 *   0x05    1    TSR        Timer Status (interrupt flags, read to clear)
00281 *   0x06    2    TCNT       Timer Counter (16-bit, counts up/down)
00282 *   0x08    2    TGRA       Timer General Register A (compare/capture)
00283 *   0x0A    2    TGRB       Timer General Register B (compare/capture)
00284 *   0x0C    2    TGRC       Timer General Register C (compare/capture)
00285 *   0x0E    2    TGRD       Timer General Register D (compare/capture)
00286 * @endverbatim
00287 */
00288 * @par Timer Operating Modes
00289 * The TMDR register MD bits select the operating mode:
00290 * - Normal (0x00): Free-running counter with compare match
00291 * - PWM Mode 1 (0x02): TGRA=period, TGRB=duty cycle
00292 * - PWM Mode 2 (0x03): TGRC/TGRD as buffer registers
00293 * - Phase Counting (0x04-0x07): Quadrature encoder input
00294 */
00295 * @par PWM Generation Example
00296 * @code
00297 * // Configure MTU0 for 20 kHz PWM, 50% duty cycle
00298 * volatile rx_mtu_channel_regs_t* mtu = mtu0();

```

```

00299 *
00300 * // Stop timer during configuration
00301 * mtu_tstra()->tstr &= ~k_mtu_tstr_cst0;
00302 *
00303 * // Configure prescaler and clear source
00304 * mtu->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra;
00305 *
00306 * // Set PWM mode 1
00307 * mtu->tmdr = k_mtu_tmdr_md_pwm1;
00308 *
00309 * // Configure output: low initially, high on compare match
00310 * mtu->tiorh = k_mtu_tior_init_low;
00311 *
00312 * // Set period and duty cycle (120 MHz / 6000 = 20 kHz)
00313 * mtu->tgra = 5999; // Period
00314 * mtu->tgrb = 3000; // 50% duty (high for 3000 counts)
00315 *
00316 * // Clear counter and start timer
00317 * mtu->tcnt = 0;
00318 * mtu_tstra()->tstr |= k_mtu_tstr_cst0;
00319 * @endcode
00320 *
00321 * @invariant Structure size must be exactly 18 bytes (0x10 offset)
00322 * @invariant All registers are volatile for hardware access
00323 *
00324 * @see rx_mtu34_channel_regs_t for extended MTU3/MTU4 with TGR E/F
00325 * @see rx_mtu12_phase_regs_t for 32-bit phase counting mode
00326 * @see RX72N Hardware Manual Section 24.2 (Register Descriptions)
00327 */
00328 typedef struct {
00329 /**
00330  * @brief Timer Control Register (TCR) - prescaler, edge, clear source
00331  * @details Controls clock source selection and counter clear conditions.
00332  * @see mtu_tcr_bits_t for bit field definitions
00333  */
00334 volatile uint8_t tcr;
00335
00336 /**
00337  * @brief Timer Mode Register (TMDR) - operating mode selection
00338  * @details Selects normal, PWM, or phase counting mode.
00339  * @see mtu_tmdr_bits_t for mode values
00340  */
00341 volatile uint8_t tmdr;
00342
00343 /**
00344  * @brief Timer I/O Control Register H - TIOCA/TIOCB output control
00345  * @details Controls output pin behavior on compare match A/B.
00346  * @see mtu_tior_bits_t for output mode values
00347  */
00348 volatile uint8_t tiorh;
00349
00350 /**
00351  * @brief Timer I/O Control Register L - TIOCC/TIOCD output control
00352  * @details Controls output pin behavior on compare match C/D.
00353  * @see mtu_tior_bits_t for output mode values
00354  */
00355 volatile uint8_t tiorl;
00356
00357 /**
00358  * @brief Timer Interrupt Enable Register (TIER)
00359  * @details Enables interrupts for compare match and overflow events.
00360  */
00361 volatile uint8_t tier;
00362
00363 /**
00364  * @brief Timer Status Register (TSR) - interrupt flags
00365  * @details Contains interrupt flags; read to check, write 0 to clear.
00366  */
00367 * @warning Per RX72N HW manual R01UH0824EJ0111 Section 21.2 (MTU register
00368 * tables), MTU0 has NO TSR at this offset -- accessing this field
00369 * via the MTU0 base (0x000C1300) targets a reserved byte and the
00370 * result is undefined. This field is only valid on MTU channels
00371 * that define a TSR (e.g., MTU1, MTU2, MTU3/4 TSR copies); callers
00372 * must gate access by channel before reading/writing tsr.
00373 */
00374 volatile uint8_t tsr;
00375
00376 /**
00377  * @brief Timer Counter (TCNT) - 16-bit counter value
00378  * @details Current counter value. Cleared by compare match or overflow
00379  * depending on TCR.CCLR setting.
00380  */
00381 volatile uint16_t tcnt;
00382
00383 /**
00384  * @brief Timer General Register A (TGRA) - compare/capture
00385  * @details In PWM mode: defines period. In capture mode: stores captured value.

```

```

00386 */
00387 volatile uint16_t tgra;
00388
00389 /**
00390  * @brief Timer General Register B (TGRB) - compare/capture
00391  * @details In PWM mode: defines duty cycle (when to toggle output).
00392  */
00393 volatile uint16_t tgrb;
00394
00395 /**
00396  * @brief Timer General Register C (TGRC) - compare/capture
00397  * @details Additional compare register; can buffer TGRA in some modes.
00398  */
00399 volatile uint16_t tgrc;
00400
00401 /**
00402  * @brief Timer General Register D (TGRD) - compare/capture
00403  * @details Additional compare register; can buffer TGRB in some modes.
00404  */
00405 volatile uint16_t tgrd;
00406 } rx_mtu3_channel_regs_t;
00407
00408 /**
00409  * @enum mtu3_padding_sizes_t
00410  * @brief Named padding array sizes for rx_mtu3_channel_regs_t reserved fields
00411  *
00412  * @details
00413  * Each constant names the number of reserved bytes between two adjacent MTU3
00414  * registers in the hardware layout. The comments show which hardware registers
00415  * occupy those byte ranges (verified against RX72N Ch 5 I/O Register Address
00416  * Table).
00417  *
00418  * @since Version 1.0.0
00419  */
00420 typedef enum : uint8_t {
00421     /** @brief 2 reserved bytes at +0x06-0x07: MTU4.TIORH, MTU4.TIORL */
00422     k_mtu3_pad_after_tiorl = 2U,
00423
00424     /** @brief 7 reserved bytes at +0x09-0x0F: MTU4.TIER, MTU4.TOERA, reserved x2,
00425      *      MTU4.TGCRA, MTU4.TOCR1A, MTU4.TOCR2A */
00426     k_mtu3_pad_after_tier = 7U,
00427
00428     /** @brief 6 reserved bytes at +0x12-0x17: MTU4.TCNT(2), MTU4.TCDRA(2), MTU4.TDDRA(2) */
00429     k_mtu3_pad_after_tcnt = 6U,
00430
00431     /** @brief 8 reserved bytes at +0x1C-0x23: MTU4.TGRA(2), MTU4.TGRB(2),
00432      *      MTU4.TCNTSA(2), MTU4.TCBRA(2) */
00433     k_mtu3_pad_after_tgrb = 8U,
00434
00435     /** @brief 4 reserved bytes at +0x28-0x2B: MTU4.TGRC(2), MTU4.TGRD(2) */
00436     k_mtu3_pad_after_tgrd = 4U,
00437
00438     /** @brief 11 reserved bytes at +0x2D-0x37: MTU4.TSR, reserved x2, MTU4.TITCR1A,
00439      *      MTU4.TITCNT1A, MTU4.TBTERA, reserved, MTU4.TDERA, reserved, MTU4.TOLBRA,
00440      *      reserved */
00441     k_mtu3_pad_after_tsr = 11U,
00442
00443     /** @brief 19 reserved bytes at +0x39-0x4B: MTU4.TBTM, MTU4.TITMRA, MTU4.TITCR2A,
00444      *      MTU4.TITCNT2A, reserved, MTU4.TADCR(2), reserved(2), MTU4.TADCORA(2),
00445      *      MTU4.TADCORB(2), MTU4.TADCOBRA(2), MTU4.TADCOBRB(2) */
00446     k_mtu3_pad_after_tbtm = 19U,
00447
00448     /** @brief 37 reserved bytes at +0x4D-0x71: MTU4.TCR2, reserved x18, MTU4.TWCRA,
00449      *      reserved x15, MTU4.TMDR2A, reserved */
00450     k_mtu3_pad_after_tcr2 = 37U,
00451 } mtu3_padding_sizes_t;
00452
00453 /**
00454  * @struct rx_mtu3_channel_regs_t
00455  * @brief MTU3 register map accessed via base address 0x000C1200
00456  *
00457  * @details
00458  * Memory-mapped register structure for MTU3 channel, which supports
00459  * complementary PWM generation for H-bridge motor driver control along
00460  * with standard timer/counter and phase-counting encoder modes.
00461  *
00462  * The MTU3/MTU4 peripheral block places MTU3 and MTU4 registers at
00463  * INTERLEAVED addresses starting at 0x000C1200. MTU3 registers are NOT
00464  * at consecutive byte offsets -- there are large gaps between them where
00465  * MTU4-specific registers and shared MTU3/4 control registers reside.
00466  *
00467  * @warning This struct is ONLY valid for MTU3 (mtu3() base 0x000C1200).
00468  * MTU4 registers have DIFFERENT offsets relative to the MTU4 base
00469  * address (0x000C1201) and CANNOT use this struct. For MTU4,
00470  * compute each register address individually using the Ch 5 I/O
00471  * address table (e.g. MTU4.TIORH = 0x000C1206, not base+0x04).
00472  * See RX72N Hardware Manual Ch 5 I/O Register Address Table.

```

```

00473 *
00474 * @par Register Memory Layout (Ch 5 I/O Register Address Table, verified)
00475 * @verbatim
00476 *   Offset  Size  Register  Address      Description
00477 *   -----  ----  -
00478 *   0x00    1    tcr       0x000C1200    Timer Control (prescaler, clear source)
00479 *   0x01    -    [MTU4.TCR]              (reserved -- do not access via this struct)
00480 *   0x02    1    tmdr1     0x000C1202    Timer Mode Register 1 (normal/PWM/phase)
00481 *   0x03    -    [MTU4.TMDR1]             (reserved)
00482 *   0x04    1    tiorh     0x000C1204    Timer I/O Control H (TIOCA/TIOCB output)
00483 *   0x05    1    tiorl     0x000C1205    Timer I/O Control L (TIOCC/TIOCD output)
00484 *   0x06    -    [MTU4.TIORH]             (reserved)
00485 *   0x07    -    [MTU4.TIORL]             (reserved)
00486 *   0x08    1    tier       0x000C1208    Timer Interrupt Enable
00487 *   0x09-0x0F - [shared/MTU4]              (reserved -- TOERA, TGCRA, TOCR1A, TOCR2A)
00488 *   0x10    2    tcnt       0x000C1210    Timer Counter (16-bit)
00489 *   0x12-0x17 - [shared/MTU4]              (reserved -- MTU4.TCNT, TCDRA, TDDRA)
00490 *   0x18    2    tgra       0x000C1218    Timer General Register A (period/capture)
00491 *   0x1A    2    tgrb       0x000C121A    Timer General Register B (duty/capture)
00492 *   0x1C-0x23 - [shared/MTU4]              (reserved -- MTU4.TGRA/B, TCNTSA, TCBRA)
00493 *   0x24    2    tgrc       0x000C1224    Timer General Register C (buffer/capture)
00494 *   0x26    2    tgrd       0x000C1226    Timer General Register D (buffer/capture)
00495 *   0x28-0x2B - [MTU4.TGRC/D]             (reserved)
00496 *   0x2C    1    tsr        0x000C122C    Timer Status Register (interrupt flags)
00497 *   0x2D-0x37 - [shared/MTU4]              (reserved -- MTU4.TSR, TITCR1A, TBTERA, etc.)
00498 *   0x38    1    tbtm       0x000C1238    Timer Buffer Transfer Mode
00499 *   0x39-0x4B - [shared/MTU4]              (reserved -- MTU4.TBTM, TITMRA, MTU4.TADCR etc.)
00500 *   0x4C    1    tcr2       0x000C124C    Timer Control Register 2 (phase count edge)
00501 *   0x4D-0x71 - [shared/MTU4]              (reserved -- MTU4.TCR2, TWCRA, TMDR2A, etc.)
00502 *   0x72    2    tgref      0x000C1272    Timer General Register E (dead-time/extended)
00503 * @endverbatim
00504 *
00505 * @note MTU3 has NO TGRF register. Only MTU4 has TGRF (at 0x000C1276).
00506 *
00507 * @par Dead-Time Insertion via TGRE
00508 * TGRE provides dead-time values to prevent H-bridge shoot-through.
00509 * MTU4.TGRE and MTU4.TGRF must be set separately using direct address access.
00510 * @f[
00511 *  $t_{\text{dead}} = \frac{\text{text}\{\text{TGRE}\}}{\text{f}_{\text{PCLKA}}} = \frac{\text{text}\{\text{TGRE}\}}{120 \text{ MHz}}$ 
00512 * @f]
00513 *
00514 * Example: 500ns dead-time requires TGRE = 60 (at 120 MHz PCLKA)
00515 *
00516 * @par Complementary PWM Example
00517 * @code
00518 * // Configure MTU3/4 for complementary PWM with 500ns dead-time
00519 * volatile rx_mtu3_channel_regs_t* mtu = mtu3();
00520 *
00521 * // Stop both timers
00522 * mtu->tstra()->tstr &= ~(k_mtu_tstr_cst3 | k_mtu_tstr_cst4);
00523 *
00524 * // Configure for complementary PWM mode (via TMDR1)
00525 * mtu->tmdr1 = 0x0D; // Complementary PWM mode
00526 *
00527 * // Set period (TGRA) and duty cycle (TGRB)
00528 * mtu->tgra = 5999; // 20 kHz period at 120 MHz PCLKA
00529 * mtu->tgrb = 3000; // 50% duty cycle
00530 *
00531 * // Set dead-time for MTU3 (TGRE at 0x000C1272)
00532 * mtu->tgre = 60; // 500ns at 120 MHz
00533 * // MTU4.TGRF must be set via direct address (MTU4 uses different struct offsets):
00534 * *(volatile uint16_t*)0x000C1276U = 60; // MTU4.TGRF = 500ns dead-time
00535 *
00536 * // Enable buffer transfer on counter clear
00537 * mtu->tbtm = 0x03;
00538 *
00539 * // Start both timers synchronously
00540 * mtu_tstra()->tstr |= (k_mtu_tstr_cst3 | k_mtu_tstr_cst4);
00541 * @endcode
00542 *
00543 * @invariant Struct size is 0x74 bytes (116) covering MTU3 registers up to TGRE
00544 * @invariant Only valid when cast from mtu3() base address (0x000C1200)
00545 * @invariant MTU4 registers CANNOT use this struct (different relative offsets)
00546 *
00547 * @see rx_mtu_channel_regs_t for MTU0/1/2 standard (consecutive) channel struct
00548 * @see mtu6() for MTU6 access using this interleaved struct
00549 * @see RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a registers)
00550 * @see RX72N Hardware Manual Section 24.2 (MTU3/MTU4 Register Descriptions)
00551 */
00552 typedef struct {
00553 /** @brief TCR -- Timer Control Register (+0x00 = 0x000C1200) */
00554 volatile uint8_t tcr;
00555
00556 /** @brief Reserved -- MTU4.TCR at this address (+0x01 = 0x000C1201) */
00557 volatile uint8_t reserved_01;
00558
00559 /** @brief TMDR1 -- Timer Mode Register 1 (+0x02 = 0x000C1202) */

```



```

00560 volatile uint8_t tmdr1;
00561
00562 /** @brief Reserved -- MTU4.TMDR1 at this address (+0x03 = 0x000C1203) */
00563 volatile uint8_t reserved_03;
00564
00565 /** @brief TIORH -- Timer I/O Control H: TIOCA/TIOCB output modes (+0x04 = 0x000C1204) */
00566 volatile uint8_t tiorh;
00567
00568 /** @brief TIORL -- Timer I/O Control L: TIOCC/TIOCD output modes (+0x05 = 0x000C1205) */
00569 volatile uint8_t tiorl;
00570
00571 /** @brief Reserved -- MTU4.TIORH and MTU4.TIORL occupy +0x06 and +0x07 */
00572 volatile uint8_t reserved_06[k_mtu3_pad_after_tiorl];
00573
00574 /** @brief TIER -- Timer Interrupt Enable Register (+0x08 = 0x000C1208) */
00575 volatile uint8_t tier;
00576
00577 /**
00578  * @brief Reserved (+0x09-0x0F) -- MTU4.TIER, MTU.TOERA, reserved x2,
00579  *      MTU.TGCRA, MTU.TOCR1A, MTU.TOCR2A occupy these addresses
00580  */
00581 volatile uint8_t reserved_09[k_mtu3_pad_after_tier];
00582
00583 /** @brief TCNT -- Timer Counter 16-bit (+0x10 = 0x000C1210) */
00584 volatile uint16_t tcnt;
00585
00586 /**
00587  * @brief Reserved (+0x12-0x17) -- MTU4.TCNT(2), MTU.TCDRA(2),
00588  *      MTU.TDDRA(2) occupy these addresses
00589  */
00590 volatile uint8_t reserved_12[k_mtu3_pad_after_tcnt];
00591
00592 /** @brief TGRA -- Timer General Register A: period in PWM mode (+0x18 = 0x000C1218) */
00593 volatile uint16_t tgra;
00594
00595 /** @brief TGRB -- Timer General Register B: duty cycle in PWM mode (+0x1A = 0x000C121A) */
00596 volatile uint16_t tgrb;
00597
00598 /**
00599  * @brief Reserved (+0x1C-0x23) -- MTU4.TGRA(2), MTU4.TGRB(2),
00600  *      MTU.TCNTSA(2), MTU.TCBRA(2) occupy these addresses
00601  */
00602 volatile uint8_t reserved_1c[k_mtu3_pad_after_tgrb];
00603
00604 /** @brief TGRC -- Timer General Register C: buffer for TGRA (+0x24 = 0x000C1224) */
00605 volatile uint16_t tgrc;
00606
00607 /** @brief TGRD -- Timer General Register D: buffer for TGRB (+0x26 = 0x000C1226) */
00608 volatile uint16_t tgrd;
00609
00610 /** @brief Reserved (+0x28-0x2B) -- MTU4.TGRC(2), MTU4.TGRD(2) */
00611 volatile uint8_t reserved_28[k_mtu3_pad_after_tgrd];
00612
00613 /** @brief TSR -- Timer Status Register: interrupt flags, write 0 to clear (+0x2C = 0x000C122C) */
00614 volatile uint8_t tsr;
00615
00616 /**
00617  * @brief Reserved (+0x2D-0x37) -- MTU4.TSR, reserved x2, MTU.TITCR1A,
00618  *      MTU.TITCNT1A, MTU.TBTERA, reserved, MTU.TDERA, reserved,
00619  *      MTU.TOLBRA, reserved occupy these addresses
00620  */
00621 volatile uint8_t reserved_2d[k_mtu3_pad_after_tsr];
00622
00623 /** @brief TBTM -- Timer Buffer Transfer Mode Register (+0x38 = 0x000C1238) */
00624 volatile uint8_t tbtm;
00625
00626 /**
00627  * @brief Reserved (+0x39-0x4B) -- MTU4.TBTM, MTU.TITMRA, MTU.TITCR2A,
00628  *      MTU.TITCNT2A, reserved, MTU4.TADCR(2), reserved(2),
00629  *      MTU4.TADCORA(2), MTU4.TADCORB(2), MTU4.TADCOBRA(2),
00630  *      MTU4.TADCOBRB(2) occupy these addresses
00631  */
00632 volatile uint8_t reserved_39[k_mtu3_pad_after_tbtm];
00633
00634 /** @brief TCR2 -- Timer Control Register 2: phase count clock edge (+0x4C = 0x000C124C) */
00635 volatile uint8_t tcr2;
00636
00637 /**
00638  * @brief Reserved (+0x4D-0x71) -- MTU4.TCR2, reserved x18, MTU.TWCRA,
00639  *      reserved x15, MTU.TMDR2A, reserved occupy these addresses
00640  */
00641 volatile uint8_t reserved_4d[k_mtu3_pad_after_tcr2];
00642
00643 /**
00644  * @brief TGRE -- Timer General Register E: dead-time / extended TGR (+0x72 = 0x000C1272)
00645  * @details MTU3 has TGRE only, MTU4 has both TGRE (0x000C1274) and TGRF (0x000C1276).
00646  *      MTU3 does NOT have a TGRF register.

```



```

00647 */
00648 volatile uint16_t tgre;
00649 } rx_mtu3_channel_regs_t;
00650
00651 /**
00652  * @enum mtu12_phase_padding_sizes_t
00653  * @brief Named padding array sizes for rx_mtu12_phase_regs_t reserved fields
00654  *
00655  * @details
00656  * Each constant names the number of reserved bytes between two adjacent
00657  * registers in the 32-bit phase counting register layout.
00658  * Verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.
00659  *
00660  * @since Version 1.0.0
00661  */
00662 typedef enum : uint8_t {
00663     /** @brief 14 reserved bytes at offset 0x12-0x1F (gap before 32-bit long-word regs) */
00664     k_mtu12p_pad_before_tcntlw = 14U,
00665 } mtu12_phase_padding_sizes_t;
00666
00667 /**
00668  * @struct rx_mtu12_phase_regs_t
00669  * @brief MTU1/MTU2 extended register map for 32-bit phase counting mode
00670  *
00671  * @details
00672  * Extended register structure for MTU1 and MTU2 when operating in 32-bit
00673  * cascaded phase counting mode. This mode is ideal for quadrature encoder
00674  * applications requiring high-resolution position tracking over extended
00675  * ranges without software overflow handling.
00676  *
00677  * @par STAR Robot Encoder Configuration
00678  * The STAR robot uses 4x brushed DC gearmotors with Hall effect quadrature
00679  * encoders. MTU1 and MTU2 are cascaded to provide 32-bit position tracking:
00680  *
00681  * @verbatim
00682  * Encoder Signals          MTU1/MTU2 32-bit Cascaded Mode
00683  * -----
00684  *
00685  * Motor 1 (FL)          +-----+
00686  * +-----+          |          | 32-bit TCNTLW          |
00687  * | Hall A --->+> MTCLKA --->| [MTU2.TCNT | MTU1.TCNT]      |
00688  * | Hall B --->+> MTCLKB --->| Upper 16   Lower 16          |
00689  * +-----+          |          |
00690  * 341 PPR x 4 edges      | Position = TCNTLW / 1364        |
00691  * = 1364 counts/rev      | Range: +/-3.1M revolutions      |
00692  *          +-----+
00693  * @endverbatim
00694  *
00695  * @par 32-bit Cascaded Mode Benefits
00696  * - **Extended Range**: 232 counts vs 216 counts (65536x more range)
00697  * - **Atomic Access**: TCNTLW, TGRALW, TGRBLW are 32-bit atomic operations
00698  * - **No Race Conditions**: No need to disable interrupts during read
00699  * - **Hardware Implementation**: Zero software overhead for overflow tracking
00700  *
00701  * @par Position Calculation
00702  * For STAR robot Hall encoders (341 PPR, 4x quadrature = 1364 counts/rev):
00703  * @f[
00704  * \text{Position (revolutions)} = \frac{\text{TCNTLW}}{1364}
00705  * @f[
00706  * @f[
00707  * \text{Position (degrees)} = \frac{\text{TCNTLW}}{1364} \times 360\text{deg}
00708  * @f[
00709  *
00710  * @par Overflow Analysis
00711  * At maximum motor speed (210 RPM = 3.5 rev/sec):
00712  * @f[
00713  * t_{\text{overflow}} = \frac{2^{31}}{1364 \times 3.5} = 449,686 \text{ seconds} = 10.4 \text{ days}
00714  * @f[
00715  *
00716  * @par Register Memory Layout (44 bytes total)
00717  * @verbatim
00718  * Offset  Size  Register  Description
00719  * -----
00720  * 0x00    1    TCR       Timer Control (prescaler, edge)
00721  * 0x01    1    TMDR      Timer Mode (phase counting mode)
00722  * 0x02    1    TIORH     Timer I/O Control H
00723  * 0x03    1    TIORL     Timer I/O Control L
00724  * 0x04    1    TIER      Timer Interrupt Enable
00725  * 0x05    1    TSR       Timer Status (flags)
00726  * 0x06    2    TCNT      Timer Counter (16-bit, lower word)
00727  * 0x08    2    TGRA      Timer General Register A (16-bit)
00728  * 0x0A    2    TGRB      Timer General Register B (16-bit)
00729  * 0x0C    2    TGRC      Timer General Register C
00730  * 0x0E    2    TGRD      Timer General Register D
00731  * 0x10    1    Reserved  -
00732  * 0x11    1    TMDR3     32-bit mode control (LWA, PHCKSEL)
00733  * 0x12    14   Reserved  -

```

```

00734 * 0x20 4 TCNTLW 32-bit cascaded counter (atomic)
00735 * 0x24 4 TGRALW 32-bit general register A (atomic)
00736 * 0x28 4 TGRBLW 32-bit general register B (atomic)
00737 * @endverbatim
00738 *
00739 * @par 32-bit Mode Configuration Example
00740 * @code
00741 * // Configure MTU1+MTU2 for 32-bit quadrature encoder (Hall sensors)
00742 * volatile rx_mtu12_phase_regs_t* mtu = mtu1_phase();
00743 *
00744 * // Stop timers during configuration
00745 * mtu_tstra()->tstr &= ~(k_mtu_tstr_cst1 | k_mtu_tstr_cst2);
00746 *
00747 * // Configure for phase counting mode 4 (count on both edges)
00748 * mtu->tmdr = 0x04; // Phase counting mode 4
00749 *
00750 * // Enable 32-bit cascaded mode (LWA bit) with both-edge counting
00751 * mtu->tmdr3 = k_mtu_tmdr3_lwa | k_mtu_tmdr3_phcksel;
00752 *
00753 * // Clear 32-bit counter
00754 * mtu->tcntlw = 0;
00755 *
00756 * // Start both timers
00757 * mtu_tstra()->tstr |= (k_mtu_tstr_cst1 | k_mtu_tstr_cst2);
00758 *
00759 * // Read 32-bit position (atomic, no race condition)
00760 * int32_t position = (int32_t)mtu->tcntlw;
00761 * @endcode
00762 *
00763 * @invariant Structure size must be exactly 44 bytes (0x2C)
00764 * @invariant TMDR3 at offset 0x11 controls 32-bit cascaded mode
00765 * @invariant TCNTLW/TGRALW/TGRBLW are 32-bit aligned for atomic access
00766 *
00767 * @see mtu_tmdr3_bits_t for TMDR3 bit definitions
00768 * @see rx_mtu_channel_regs_t for standard 16-bit mode
00769 * @see RX72N Hardware Manual Section 24.2.25 (TMDR3 Register)
00770 * @see RX72N Hardware Manual Section 24.2.26 (32-bit Registers)
00771 * @see RX72N Hardware Manual Section 24.3.5 (Phase Counting Mode)
00772 */
00773 typedef struct {
00774 /* ----- Standard 16-bit registers (0x00-0x0F) ----- */
00775
00776 /**
00777 * @brief Timer Control Register (TCR) - prescaler and edge selection
00778 * @details For phase counting, TCR.CKEG selects which edge of MTCLKA/B to count.
00779 */
00780 volatile uint8_t tcr;
00781
00782 /**
00783 * @brief Timer Mode Register (TMDR) - phase counting mode select
00784 * @details Mode 4-7 select different phase counting configurations.
00785 * Mode 4: Count up/down on MTCLKA, direction from MTCLKB.
00786 */
00787 volatile uint8_t tmdr;
00788
00789 /** @brief Timer I/O Control Register H (typically unused in phase counting) */
00790 volatile uint8_t tiorh;
00791
00792 /** @brief Timer I/O Control Register L (typically unused in phase counting) */
00793 volatile uint8_t tiorl;
00794
00795 /**
00796 * @brief Timer Interrupt Enable Register (TIER)
00797 * @details Enable overflow/underflow interrupts for position limit detection.
00798 */
00799 volatile uint8_t tier;
00800
00801 /**
00802 * @brief Timer Status Register (TSR) - interrupt flags
00803 * @details Contains overflow/underflow flags for position monitoring.
00804 */
00805 volatile uint8_t tsr;
00806
00807 /**
00808 * @brief Timer Counter (TCNT) - lower 16 bits of cascaded counter
00809 * @details Use TCNTLW for atomic 32-bit access when LWA bit is set.
00810 */
00811 volatile uint16_t tcnt;
00812
00813 /** @brief Timer General Register A - 16-bit compare (use TGRALW for 32-bit) */
00814 volatile uint16_t tgra;
00815
00816 /** @brief Timer General Register B - 16-bit compare (use TGRBLW for 32-bit) */
00817 volatile uint16_t tgrb;
00818
00819 /** @brief Timer General Register C - 16-bit compare */
00820 volatile uint16_t tgrc;

```

```

00821
00822 /** @brief Timer General Register D - 16-bit compare */
00823 volatile uint16_t tgrd;
00824
00825 /* ----- Extended 32-bit mode registers (0x10-0x2B) ----- */
00826
00827 /** @brief Reserved byte at offset 0x10 */
00828 uint8_t reserved1;
00829
00830 /**
00831  * @brief Timer Mode Register 3 (TMDR3) - 32-bit mode control
00832  * @details
00833  * - Bit 0 (LWA): Long Word Access enable (32-bit cascaded mode)
00834  * - Bit 1 (PHCKSEL): Phase counting clock select (both edges)
00835  * @see mtu_tmdr3_bits_t for bit definitions
00836  */
00837 volatile uint8_t tmdr3;
00838
00839 /** @brief Reserved bytes at offset 0x12-0x1F (14 bytes, before 32-bit long-word regs) */
00840 uint8_t reserved2[k_mtu12p_pad_before_tcntlw];
00841
00842 /**
00843  * @brief 32-bit Cascaded Counter (TCNTLW) - atomic position value
00844  * @details
00845  * Combined MTU1.TCNT (lower 16 bits) and MTU2.TCNT (upper 16 bits).
00846  * Provides atomic 32-bit read/write without race conditions.
00847  * Sign-extended for bidirectional counting (use int32_t cast).
00848  *
00849  * @note Only valid when TMDR3.LWA = 1 (32-bit mode enabled)
00850  */
00851 volatile uint32_t tcntlw;
00852
00853 /**
00854  * @brief 32-bit General Register A (TGRALW) - atomic compare value
00855  * @details 32-bit compare register for position threshold interrupts.
00856  * @note Only valid when TMDR3.LWA = 1
00857  */
00858 volatile uint32_t tgralw;
00859
00860 /**
00861  * @brief 32-bit General Register B (TGRBLW) - atomic compare value
00862  * @details 32-bit compare register for position threshold interrupts.
00863  * @note Only valid when TMDR3.LWA = 1
00864  */
00865 volatile uint32_t tgrblw;
00866 } rx_mtu12_phase_regs_t;
00867
00868 /**
00869  * @brief MTU Start Register Map
00870  * @details
00871  * Timer start/stop control registers.
00872  * Base addresses:
00873  * - TSTRA: 0x000C1280 (MTU0-4, MTU8)
00874  * - TSTRB: 0x000C1A80 (MTU6-7)
00875  */
00876 typedef struct {
00877     volatile uint8_t tstr; /**< Timer Start Register (channel enable bits) */
00878 } rx_mtu_tstr_regs_t;
00879
00880 /*
=====
00881  * Inline Register Accessors
00882  *
=====
00883  */
00884
00885 /**
00886  * @brief Get pointer to MTU0 registers
00887  * @details
00888  * Returns a volatile pointer to MTU0's register structure at base address
00889  * 0x000C1300. MTU0 is a general-purpose 16-bit timer commonly used for
00890  * PWM generation or interval timing.
00891  *
00892  * @return Volatile pointer to MTU0 register structure
00893  *
00894  * @par Example
00895  * @code
00896  * // Read MTU0 counter value
00897  * uint16_t count = mtu0()->tcnt;
00898  *
00899  * // Configure MTU0 for 20 kHz PWM
00900  * mtu0()->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra;
00901  * mtu0()->tgra = 5999; // Period: 120 MHz / 6000 = 20 kHz
00902  * @endcode
00903  *
00904  * @note Thread-safe: returns constant hardware address
00905  */

```

```

00906 * @see rx_mtu_channel_regs_t for register layout
00907 */
00908 static inline volatile rx_mtu_channel_regs_t* mtu0(void)
00909 {
00910     return (volatile rx_mtu_channel_regs_t*)k_mtu0_base_addr;
00911 }
00912 /**
00913 * @brief Get pointer to MTU1 registers (16-bit mode)
00914 *
00915 * @details
00916 * Returns a volatile pointer to MTU1's standard 16-bit register structure.
00917 * For 32-bit cascaded phase counting mode, use mtu1_phase() instead.
00918 *
00919 * @return Volatile pointer to MTU1 register structure
00920 *
00921 * @see mtu1_phase() for 32-bit cascaded phase counting
00922 * @see rx_mtu_channel_regs_t for register layout
00923 */
00924 /* GCOVR_EXCL_START -- trivial hardware-address accessor exercised on
00925 * the real target only. Host-test builds use mock_rx_mtu_regs.h's
00926 * mtu1() override, so the real body never executes under coverage. */
00927 static inline volatile rx_mtu_channel_regs_t* mtu1(void)
00928 {
00929     return (volatile rx_mtu_channel_regs_t*)k_mtu1_base_addr;
00930 }
00931 /* GCOVR_EXCL_STOP */
00932 /**
00933 * @brief Get pointer to MTU2 registers (16-bit mode)
00934 *
00935 * @details
00936 * Returns a volatile pointer to MTU2's standard 16-bit register structure.
00937 * For 32-bit cascaded phase counting mode, use mtu2_phase() instead.
00938 *
00939 * @return Volatile pointer to MTU2 register structure
00940 *
00941 * @see mtu2_phase() for 32-bit cascaded phase counting
00942 * @see rx_mtu_channel_regs_t for register layout
00943 */
00944 /* GCOVR_EXCL_START -- trivial hardware-address accessor; host-test
00945 * builds use mock_rx_mtu_regs.h's override. */
00946 static inline volatile rx_mtu_channel_regs_t* mtu2(void)
00947 {
00948     return (volatile rx_mtu_channel_regs_t*)k_mtu2_base_addr;
00949 }
00950 /* GCOVR_EXCL_STOP */
00951 /**
00952 * @brief Get pointer to MTU1 extended registers for 32-bit phase counting mode
00953 *
00954 * @details
00955 * Provides access to MTU1's extended register set including TMDR3 and 32-bit
00956 * cascaded counter registers (TCNTLW, TGRALW, TGRBLW). Use this accessor when
00957 * configuring or reading 32-bit cascaded phase counting mode.
00958 *
00959 * For standard 16-bit phase counting, use mtu1() instead.
00960 *
00961 * @return Volatile pointer to MTU1 extended register structure
00962 *
00963 * @see rx_mtu12_phase_regs_t for complete register map
00964 * @see RX72N Hardware Manual Section 24.2.25 for TMDR3 register
00965 * @see RX72N Hardware Manual Section 24.2.26 for 32-bit registers
00966 */
00967 static inline volatile rx_mtu12_phase_regs_t* mtu1_phase(void)
00968 {
00969     return (volatile rx_mtu12_phase_regs_t*)k_mtu1_base_addr;
00970 }
00971 /**
00972 * @brief Get pointer to MTU2 extended registers for 32-bit phase counting mode
00973 *
00974 * @details
00975 * Provides access to MTU2's extended register set including TMDR3 and 32-bit
00976 * cascaded counter registers (TCNTLW, TGRALW, TGRBLW). Use this accessor when
00977 * configuring or reading 32-bit cascaded phase counting mode.
00978 *
00979 * For standard 16-bit phase counting, use mtu2() instead.
00980 *
00981 * @return Volatile pointer to MTU2 extended register structure
00982 *
00983 * @see rx_mtu12_phase_regs_t for complete register map
00984 * @see RX72N Hardware Manual Section 24.2.25 for TMDR3 register
00985 * @see RX72N Hardware Manual Section 24.2.26 for 32-bit registers
00986 */
00987 static inline volatile rx_mtu12_phase_regs_t* mtu2_phase(void)
00988 {
00989 }

```

```

00993     return (volatile rx_mtu12_phase_regs_t*)k_mtu2_base_addr;
00994 }
00995
00996 /**
00997  * @brief Get pointer to MTU3 registers via rx_mtu3_channel_regs_t struct
00998  *
00999  * @details
01000  * Returns a pointer to MTU3's register space cast to rx_mtu3_channel_regs_t.
01001  * The struct has been carefully padded to match the actual hardware layout
01002  * verified against the RX72N Ch 5 I/O Register Address Table.
01003  *
01004  * MTU3 and MTU4 registers are interleaved at 0x000C1200: MTU3 at
01005  * 0x000C1200, MTU4 at 0x000C1201, with shared MTU3/4 registers interspersed.
01006  * The struct fields (tcr, tmdr1, tiorh, tiorl, tier, tcnt, tgra, tgrb, tgrc,
01007  * tgrd, tsr, tbtm, tcr2, tgre) are placed at the correct byte offsets for
01008  * MTU3. The gap bytes in the struct are reserved/MTU4/shared-register space.
01009  *
01010  * @warning Do NOT use this pointer for MTU4 register access. MTU4 registers
01011  * are at different offsets relative to the MTU4 base (0x000C1201).
01012  * Access MTU4 registers using the individual address constants or
01013  * by computing addresses manually from k_mtu4_base_addr.
01014  *
01015  * @return Volatile pointer to MTU3 register structure (base 0x000C1200)
01016  *
01017  * @see rx_mtu3_channel_regs_t for complete register layout with addresses
01018  * @see RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)
01019  */
01020 static inline volatile rx_mtu3_channel_regs_t* mtu3(void)
01021 {
01022     return (volatile rx_mtu3_channel_regs_t*)k_mtu3_base_addr;
01023 }
01024
01025 /**
01026  * @brief Get raw base address pointer for MTU4 register access
01027  *
01028  * @details
01029  * Returns the MTU4 base address (0x000C1201) as a void pointer. MTU4 registers
01030  * are at DIFFERENT relative offsets than MTU3 registers, so the rx_mtu3_channel_regs_t
01031  * struct CANNOT be used for MTU4.
01032  *
01033  * MTU4 register addresses from the RX72N Ch 5 I/O Address Table:
01034  * - MTU4.TCR = 0x000C1201 (k_mtu4_base_addr + 0x00)
01035  * - MTU4.TMDR1 = 0x000C1203 (k_mtu4_base_addr + 0x02)
01036  * - MTU4.TIORH = 0x000C1206 (k_mtu4_base_addr + 0x05) -- NOTE: +0x05, not +0x04
01037  * - MTU4.TIORL = 0x000C1207 (k_mtu4_base_addr + 0x06)
01038  * - MTU4.TIER = 0x000C1209 (k_mtu4_base_addr + 0x08)
01039  * - MTU4.TCNT = 0x000C1212 (k_mtu4_base_addr + 0x11) -- NOTE: odd address
01040  * - MTU4.TGRA = 0x000C121C (k_mtu4_base_addr + 0x1B)
01041  * - MTU4.TGRB = 0x000C121E (k_mtu4_base_addr + 0x1D)
01042  * - MTU4.TGRC = 0x000C1228 (k_mtu4_base_addr + 0x27)
01043  * - MTU4.TGRD = 0x000C122A (k_mtu4_base_addr + 0x29)
01044  * - MTU4.TSR = 0x000C122D (k_mtu4_base_addr + 0x2C)
01045  * - MTU4.TBTM = 0x000C1239 (k_mtu4_base_addr + 0x38)
01046  * - MTU4.TCR2 = 0x000C124D (k_mtu4_base_addr + 0x4C)
01047  * - MTU4.TGRE = 0x000C1274 (k_mtu4_base_addr + 0x73)
01048  * - MTU4.TGRF = 0x000C1276 (k_mtu4_base_addr + 0x75) -- MTU3 has no TGRF
01049  *
01050  * Access MTU4 registers by casting the returned pointer to the appropriate
01051  * volatile integer pointer and applying the correct offset, e.g.:
01052  * *(volatile uint8_t*)((uintptr_t)mtu4() + 0x05U) = value; // MTU4.TIORH
01053  *
01054  * @return Volatile void pointer to MTU4 base address (0x000C1201)
01055  *
01056  * @see RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)
01057  */
01058 static inline volatile void* mtu4(void)
01059 {
01060     return (volatile void*)k_mtu4_base_addr;
01061 }
01062
01063 /**
01064  * @brief Get pointer to MTU6 registers (interleaved layout, identical to MTU3)
01065  *
01066  * @details
01067  * MTU6/MTU7 registers are interleaved at 0x000C1A00 in the same pattern as
01068  * MTU3/MTU4 at 0x000C1200. MTU6 registers are at even offsets; MTU7 registers
01069  * are at odd offsets. Use rx_mtu3_channel_regs_t to access MTU6 fields at
01070  * their correct byte offsets (TCR=+0x00, TMDR1=+0x02, TIORH=+0x04, etc.).
01071  *
01072  * @warning Do NOT cast this pointer to rx_mtu_channel_regs_t -- that struct
01073  * assumes consecutive layout (TMDR at +0x01) which is wrong for MTU6.
01074  *
01075  * @return Volatile pointer to MTU6 register structure (interleaved layout)
01076  *
01077  * @see rx_mtu3_channel_regs_t for register layout and field offsets
01078  * @see mtu7() for raw MTU7 base pointer (void*)
01079  */

```

```

01080 static inline volatile rx_mtu3_channel_regs_t* mtu6(void)
01081 {
01082     return (volatile rx_mtu3_channel_regs_t*)k_mtu6_base_addr;
01083 }
01084
01085 /**
01086  * @brief Get raw base address pointer for MTU7 register access
01087  *
01088  * @details
01089  * Returns the MTU7 base address (k_mtu6_base_addr + 1 = 0x000C1A01) as a
01090  * void pointer. MTU7 registers are at DIFFERENT relative offsets than MTU6
01091  * registers (same situation as MTU4 relative to MTU3), so no struct type
01092  * can be directly applied. Access MTU7 registers by computing addresses
01093  * manually from k_mtu7_base_addr using the Ch 5 I/O Address Table.
01094  *
01095  * @return Volatile void pointer to MTU7 base address (0x000C1A01)
01096  *
01097  * @see mtu6() for MTU6 access via rx_mtu3_channel_regs_t
01098  * @see RX72N Hardware Manual Ch 5 I/O Register Address Table (MTU3a)
01099  */
01100 static inline volatile void* mtu7(void)
01101 {
01102     return (volatile void*)k_mtu7_base_addr;
01103 }
01104
01105 /**
01106  * @brief Get pointer to MTU TSTRA registers (MTU0-4, MTU8)
01107  * @return Volatile pointer to MTU TSTRA register structure
01108  */
01109 /* GCOVR_EXCL_START -- trivial hardware-address accessor; host-test
01110  * builds use mock_rx_mtu_regs.h's override. */
01111 static inline volatile rx_mtu_tstr_regs_t* mtu_tstra(void)
01112 {
01113     return (volatile rx_mtu_tstr_regs_t*)k_mtu_tstra_base_addr;
01114 }
01115 /* GCOVR_EXCL_STOP */
01116
01117 /**
01118  * @brief Get pointer to MTU TSTRB registers (MTU6-7)
01119  * @return Volatile pointer to MTU TSTRB register structure
01120  */
01121 static inline volatile rx_mtu_tstr_regs_t* mtu_tstrb(void)
01122 {
01123     return (volatile rx_mtu_tstr_regs_t*)k_mtu_tstrb_base_addr;
01124 }
01125
01126 /**
01127  * @enum mtu_tcr_bits_t
01128  * @brief Timer Control Register (TCR) bit field definitions
01129  *
01130  * @details
01131  * The TCR register controls timer clock source selection, counting edge,
01132  * and counter clear source. Each MTU channel has its own TCR register.
01133  *
01134  * @par Register Bit Layout
01135  * @verbatim
01136  *   Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
01137  *   -----
01138  *   | CCLR[2:0] | CKEG[1:0] | TPSC[2:0] |
01139  *   | Counter Clear | Clock Edge | Timer Prescaler |
01140  *   -----
01141  * @endverbatim
01142  *
01143  * @par Prescaler Values (TPSC)
01144  * | Value | Prescaler | Frequency (120 MHz PCLKA) |
01145  * |-----|-----|-----|
01146  * | 0x00 | PCLKA/1 | 120 MHz |
01147  * | 0x01 | PCLKA/4 | 30 MHz |
01148  * | 0x02 | PCLKA/16 | 7.5 MHz |
01149  * | 0x03 | PCLKA/64 | 1.875 MHz |
01150  *
01151  * @par Counter Clear Source (CCLR)
01152  * | Value | Source | Description |
01153  * |-----|-----|-----|
01154  * | 0x01 | TGRA | Clear on TGRA compare match |
01155  * | 0x02 | TGRB | Clear on TGRB compare match |
01156  * | 0x03 | Sync | Synchronous clear (MTU3/4) |
01157  *
01158  * @see rx_mtu_channel_regs_t::tcr
01159  * @see RX72N Hardware Manual Section 24.2.1 (TCR Register)
01160  */
01161 typedef enum : uint8_t {
01162     /** @brief Counter Clear Source bit position (bits 5-7) */
01163     k_mtu_tcr_cclr_shift = 5,
01164
01165     /** @brief Timer Prescaler mask - bits 0-2 */
01166     k_mtu_tcr_tpsec_mask = 0x07,

```

```

01167
01168 /** @brief Clock Edge mask - bits 3-4 */
01169 k_mtu_tcr_ckege_mask = 0x18,
01170
01171 /** @brief Counter Clear Source mask - bits 5-7 */
01172 k_mtu_tcr_cclr_mask = 0xE0,
01173
01174 /** @brief Prescaler PCLKA/1 (120 MHz @ 120 MHz PCLKA) */
01175 k_mtu_tcr_tpsc_1 = 0x00,
01176
01177 /** @brief Prescaler PCLKA/4 (30 MHz @ 120 MHz PCLKA) */
01178 k_mtu_tcr_tpsc_4 = 0x01,
01179
01180 /** @brief Prescaler PCLKA/16 (7.5 MHz @ 120 MHz PCLKA) */
01181 k_mtu_tcr_tpsc_16 = 0x02,
01182
01183 /** @brief Prescaler PCLKA/64 (1.875 MHz @ 120 MHz PCLKA) */
01184 k_mtu_tcr_tpsc_64 = 0x03,
01185
01186 /** @brief Clear counter on TGRA compare match */
01187 k_mtu_tcr_cclr_tgra = (k_mtu_bit_value_set « k_mtu_tcr_cclr_shift),
01188 } mtu_tcr_bits_t;
01189
01190 /**
01191  * @enum mtu_tmdr_bits_t
01192  * @brief Timer Mode Register (TMDR) bit field definitions
01193  *
01194  * @details
01195  * The TMDR register selects the timer operating mode: normal operation,
01196  * PWM generation, phase counting, or complementary PWM.
01197  *
01198  * @par Register Bit Layout
01199  * @verbatim
01200  * Bit 7 Bit 6 Bit 5 Bit 4 Bit 3 Bit 2 Bit 1 Bit 0
01201  * -----
01202  * | - | - | BFB | BFA | MD[3:0] |
01203  * | Rsvrd | Rsvrd | BufferB | BufferA | Mode Select |
01204  * -----
01205  * @endverbatim
01206  *
01207  * @par Operating Modes (MD[3:0])
01208  * | Value | Mode | Description |
01209  * |-----|-----|-----|
01210  * | 0x00 | Normal | Free-running with compare match |
01211  * | 0x02 | PWM 1 | TGRA=period, TGRB=duty |
01212  * | 0x03 | PWM 2 | TGRC/D as buffer registers |
01213  * | 0x04 | Phase 1 | Up/down on MTCLKA, dir from MTCLKB |
01214  * | 0x05 | Phase 2 | Up on MTCLKA, down on MTCLKB |
01215  * | 0x06 | Phase 3 | Count on MTCLKA edges only |
01216  * | 0x07 | Phase 4 | Count on MTCLKA and MTCLKB edges |
01217  * | 0x0D | Compl PWM | Complementary PWM (MTU3/4, 6/7) |
01218  *
01219  * @par Buffer Transfer Mode
01220  * When BFA or BFB is set, buffer registers (TGRC->TGRA, TGRD->TGRB) are
01221  * automatically transferred on counter clear, enabling glitch-free
01222  * duty cycle updates during PWM operation.
01223  *
01224  * @see rx_mtu_channel_regs_t::tmdr
01225  * @see RX72N Hardware Manual Section 24.2.2 (TMDR Register)
01226  */
01227 typedef enum : uint8_t {
01228 /** @brief Buffer mode A bit position */
01229 k_mtu_tmdr_bfa_shift = 4,
01230
01231 /** @brief Buffer mode B bit position */
01232 k_mtu_tmdr_bfb_shift = 5,
01233
01234 /** @brief Mode select mask (bits 0-3) */
01235 k_mtu_tmdr_md_mask = 0x0F,
01236
01237 /** @brief Normal mode - free-running counter with compare match */
01238 k_mtu_tmdr_md_normal = 0x00,
01239
01240 /** @brief PWM mode 1 - TGRA defines period, TGRB defines duty cycle */
01241 k_mtu_tmdr_md_pwm1 = 0x02,
01242
01243 /** @brief PWM mode 2 - TGRC/TGRD as buffer registers for glitch-free updates */
01244 k_mtu_tmdr_md_pwm2 = 0x03,
01245
01246 /** @brief Enable TGRC->TGRA buffer transfer on counter clear */
01247 k_mtu_tmdr_bfa = (k_mtu_bit_value_set « k_mtu_tmdr_bfa_shift),
01248
01249 /** @brief Enable TGRD->TGRB buffer transfer on counter clear */
01250 k_mtu_tmdr_bfb = (k_mtu_bit_value_set « k_mtu_tmdr_bfb_shift),
01251 } mtu_tmdr_bits_t;
01252
01253 /**

```



```

01254 * @brief Timer Mode Register 3 (TMDR3) bits for MTU1/MTU2 32-bit cascaded mode
01255 * @details
01256 * TMDR3 register controls 32-bit cascaded counter operation for MTU1/MTU2 in
01257 * phase counting mode. Available at offset 0x11 in rx_mtu12_phase_regs_t.
01258 *
01259 * Key features:
01260 * - LWA: Enables 32-bit cascaded counter (TCNTLW, TGRALW, TGRBLW)
01261 * - PHCKSEL: Selects phase counting clock edge (both edges vs single edge)
01262 *
01263 * @see RX72N Hardware Manual Section 24.2.25 (TMDR3 Register)
01264 */
01265 typedef enum : uint8_t {
01266     k_mtu_tmdr3_lwa_shift = 0, /**< Long Word Access bit position */
01267     k_mtu_tmdr3_phcksel_shift = 1, /**< Phase Counting Clock Select bit position */
01268     k_mtu_tmdr3_lwa =
01269         (k_mtu_bit_value_set « k_mtu_tmdr3_lwa_shift), /**< Enable 32-bit cascaded counter (LWA) */
01270     k_mtu_tmdr3_phcksel =
01271         (k_mtu_bit_value_set
01272          « k_mtu_tmdr3_phcksel_shift), /**< Phase counting clock select (both edges when set) */
01273 } mtu_tmdr3_bits_t;
01274
01275 /**
01276 * @enum mtu_tior_bits_t
01277 * @brief Timer I/O Control Register (TIOR) bit field definitions
01278 *
01279 * @details
01280 * The TIOR register controls the behavior of timer output pins on compare
01281 * match events. TIORH controls TIOCA/TIOCB outputs, TIORL controls TIOCC/TIOCD.
01282 *
01283 * @par Register Bit Layout (TIORH)
01284 * @verbatim
01285 *   Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
01286 *   -----
01287 *   |   IOB[3:0]   |   IOA[3:0]   |
01288 *   | TIOCB Output Mode | TIOCA Output Mode |
01289 *   -----
01290 * @endverbatim
01291 *
01292 * @par Output Modes
01293 * | Value | Initial | On Match | PWM Application |
01294 * |-----|-----|-----|-----|
01295 * | 0x02 | Low    | High    | Active-high PWM |
01296 * | 0x05 | High   | Low     | Active-low PWM  |
01297 * | 0x03 | -      | Toggle  | 50% duty square |
01298 *
01299 * @par PWM Configuration Example
01300 * @code
01301 * // Configure TIOCA0 for active-high PWM output
01302 * mtu0()->tiorh = k_mtu_tior_init_low; // Low initially, high on match
01303 * @endcode
01304 *
01305 * @see rx_mtu_channel_regs_t::tiorh
01306 * @see rx_mtu_channel_regs_t::tiorl
01307 * @see RX72N Hardware Manual Section 24.2.4 (TIOR Register)
01308 */
01309 typedef enum : uint8_t {
01310     /** @brief I/O Control A mask - bits 0-3 (TIOCA output mode) */
01311     k_mtu_tior_ioa_mask = 0x0F,
01312
01313     /** @brief I/O Control B mask - bits 4-7 (TIOCB output mode) */
01314     k_mtu_tior_iob_mask = 0xF0,
01315
01316     /** @brief Initial output low, set high on compare match (active-high PWM) */
01317     k_mtu_tior_init_low = 0x02,
01318
01319     /** @brief Initial output high, set low on compare match (active-low PWM) */
01320     k_mtu_tior_init_high = 0x05,
01321
01322     /** @brief Toggle output on compare match (50% duty square wave) */
01323     k_mtu_tior_toggle = 0x03,
01324 } mtu_tior_bits_t;
01325
01326 /**
01327 * @enum mtu_tstr_bits_t
01328 * @brief Timer Start Register A (TSTRA) bit definitions - controls MTU0-4, MTU8
01329 *
01330 * @details
01331 * TSTRA (at 0x000C1280) controls the start/stop state of MTU channels 0-4 and 8.
01332 * Setting a CSTn bit starts the corresponding timer; clearing it stops the timer.
01333 *
01334 * @par Register Bit Layout (TSTRA)
01335 * @verbatim
01336 *   Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
01337 *   -----
01338 *   | CST4 | CST3 | -   | -   | CST8 | CST2 | CST1 | CST0 |
01339 *   | MTU4 | MTU3 | Rsvd | Rsvd | MTU8 | MTU2 | MTU1 | MTU0 |
01340 *   -----

```



```

01341 * @endverbatim
01342 *
01343 * @par Synchronized Start Example
01344 * @code
01345 * // Start MTU1 and MTU2 simultaneously for 32-bit cascaded mode
01346 * mtu_tstra()->tstr |= (k_mtu_tstr_cst1 | k_mtu_tstr_cst2);
01347 *
01348 * // Stop all timers for configuration
01349 * mtu_tstra()->tstr = 0;
01350 * @endcode
01351 *
01352 * @see rx_mtu_tstr_regs_t
01353 * @see mtu_tstra() for register accessor
01354 * @see RX72N Hardware Manual Section 24.2.15 (TSTRA Register)
01355 */
01356 typedef enum : uint8_t {
01357     /** @brief Counter Start 0 bit position (MTU0) */
01358     k_mtu_tstr_cst0_shift = 0,
01359
01360     /** @brief Counter Start 1 bit position (MTU1) */
01361     k_mtu_tstr_cst1_shift = 1,
01362
01363     /** @brief Counter Start 2 bit position (MTU2) */
01364     k_mtu_tstr_cst2_shift = 2,
01365
01366     /** @brief Counter Start 8 bit position (MTU8) */
01367     k_mtu_tstr_cst8_shift = 3,
01368
01369     /** @brief Counter Start 3 bit position (MTU3) */
01370     k_mtu_tstr_cst3_shift = 6,
01371
01372     /** @brief Counter Start 4 bit position (MTU4) */
01373     k_mtu_tstr_cst4_shift = 7,
01374
01375     /** @brief Start MTU0 counter (TSTRA bit 0) */
01376     k_mtu_bit_value_set « k_mtu_tstr_cst0_shift),
01377
01378     /** @brief Start MTU1 counter (TSTRA bit 1) */
01379     k_mtu_bit_value_set « k_mtu_tstr_cst1_shift),
01380
01381     /** @brief Start MTU2 counter (TSTRA bit 2) */
01382     k_mtu_bit_value_set « k_mtu_tstr_cst2_shift),
01383
01384     /** @brief Start MTU3 counter (TSTRA bit 6) */
01385     k_mtu_bit_value_set « k_mtu_tstr_cst3_shift),
01386
01387     /** @brief Start MTU4 counter (TSTRA bit 7) */
01388     k_mtu_bit_value_set « k_mtu_tstr_cst4_shift),
01389
01390     /** @brief Start MTU8 counter (TSTRA bit 3) */
01391     k_mtu_bit_value_set « k_mtu_tstr_cst8_shift),
01392 } mtu_tstr_bits_t;
01393
01394 /**
01395 * @enum mtu_tstrb_bits_t
01396 * @brief Timer Start Register B (TSTRB) bit definitions - controls MTU6-7
01397 *
01398 * @details
01399 * TSTRB (at 0x000C1A80) controls the start/stop state of MTU channels 6 and 7.
01400 * These channels are typically used together for complementary PWM generation.
01401 *
01402 * @par Register Bit Layout (TSTRB)
01403 * @verbatim
01404 *   Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
01405 *   -----
01406 *   | CST7 | CST6 |   -   |   -   |   -   |   -   |   -   |
01407 *   | MTU7 | MTU6 |           Reserved           |
01408 *   -----
01409 * @endverbatim
01410 *
01411 * @see rx_mtu_tstr_regs_t
01412 * @see mtu_tstrb() for register accessor
01413 * @see RX72N Hardware Manual Section 24.2.16 (TSTRB Register)
01414 */
01415 typedef enum : uint8_t {
01416     /** @brief Counter Start 6 bit position (MTU6) */
01417     k_mtu_tstr_cst6_shift = 6,
01418
01419     /** @brief Counter Start 7 bit position (MTU7) */
01420     k_mtu_tstr_cst7_shift = 7,
01421
01422     /** @brief Start MTU6 counter (TSTRB bit 6) */
01423     k_mtu_bit_value_set « k_mtu_tstr_cst6_shift),
01424
01425     /** @brief Start MTU7 counter (TSTRB bit 7) */
01426     k_mtu_bit_value_set « k_mtu_tstr_cst7_shift),
01427 } mtu_tstrb_bits_t;

```

```

01428
01429 /**
01430  * @enum mtu_channel_reg_offsets_t
01431  * @brief Byte offsets of registers within rx_mtu_channel_regs_t
01432  *
01433  * @details
01434  * Named constants used in compile-time layout verification.
01435  * Applies to standard MTU channels (MTU0, MTU1, MTU2, MTU6, MTU7).
01436  * Verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.
01437  *
01438  * @since Version 1.0.0
01439  */
01440 typedef enum : uint8_t {
01441     /** @brief TCR at offset 0x00 (first register in channel block) */
01442     k_mtu_ch_off_tcr = 0x00U,
01443     /** @brief TCNT at offset 0x06 */
01444     k_mtu_ch_off_tcncnt = 0x06U,
01445     /** @brief TGRA at offset 0x08 */
01446     k_mtu_ch_off_tgra = 0x08U,
01447     /** @brief TGRB at offset 0x0A */
01448     k_mtu_ch_off_tgrb = 0x0AU,
01449 } mtu_channel_reg_offsets_t;
01450
01451 /**
01452  * @enum mtu3_reg_offsets_t
01453  * @brief Byte offsets of MTU3 registers from base 0x000C1200
01454  *
01455  * @details
01456  * Named constants for compile-time layout verification of rx_mtu3_channel_regs_t.
01457  * The MTU3/MTU4 interleaved layout means offsets are sparse -- see the struct
01458  * definition for the full picture of which bytes belong to MTU4 vs. shared regs.
01459  * All values verified against RX72N Hardware Manual Ch 5 I/O Register Address Table.
01460  *
01461  * @since Version 1.0.0
01462  */
01463 typedef enum : uint8_t {
01464     /** @brief TCR at offset 0x00 = 0x000C1200 */
01465     k_mtu3_off_tcr = 0x00U,
01466     /** @brief TMDR1 at offset 0x02 = 0x000C1202 */
01467     k_mtu3_off_tmdr1 = 0x02U,
01468     /** @brief TIORH at offset 0x04 = 0x000C1204 */
01469     k_mtu3_off_tiorh = 0x04U,
01470     /** @brief TIORL at offset 0x05 = 0x000C1205 */
01471     k_mtu3_off_tiorl = 0x05U,
01472     /** @brief TIER at offset 0x08 = 0x000C1208 */
01473     k_mtu3_off_tier = 0x08U,
01474     /** @brief TCNT at offset 0x10 = 0x000C1210 */
01475     k_mtu3_off_tcncnt = 0x10U,
01476     /** @brief TGRA at offset 0x18 = 0x000C1218 */
01477     k_mtu3_off_tgra = 0x18U,
01478     /** @brief TGRB at offset 0x1A = 0x000C121A */
01479     k_mtu3_off_tgrb = 0x1AU,
01480     /** @brief TGRC at offset 0x24 = 0x000C1224 */
01481     k_mtu3_off_tgrc = 0x24U,
01482     /** @brief TGRD at offset 0x26 = 0x000C1226 */
01483     k_mtu3_off_tgrd = 0x26U,
01484     /** @brief TSR at offset 0x2C = 0x000C122C */
01485     k_mtu3_off_tsr = 0x2CU,
01486     /** @brief TBTM at offset 0x38 = 0x000C1238 */
01487     k_mtu3_off_tbtm = 0x38U,
01488     /** @brief TCR2 at offset 0x4C = 0x000C124C */
01489     k_mtu3_off_tcr2 = 0x4CU,
01490     /** @brief TGRE at offset 0x72 = 0x000C1272 */
01491     k_mtu3_off_tgre = 0x72U,
01492     /** @brief Total size of rx_mtu3_channel_regs_t = 0x74 bytes (116) */
01493     k_mtu3_struct_size = 0x74U,
01494 } mtu3_reg_offsets_t;
01495
01496 /**
01497  * @enum mtu12_phase_reg_offsets_t
01498  * @brief Byte offsets of registers within rx_mtu12_phase_regs_t
01499  *
01500  * @details
01501  * Named constants for compile-time layout verification of the 32-bit phase
01502  * counting register struct. Verified against RX72N Hardware Manual
01503  * Ch 5 I/O Register Address Table and Ch 24 Table 24.11.
01504  *
01505  * @since Version 1.0.0
01506  */
01507 typedef enum : uint8_t {
01508     /** @brief TCR at offset 0x00 */
01509     k_mtu12p_off_tcr = 0x00U,
01510     /** @brief TMDR at offset 0x01 */
01511     k_mtu12p_off_tmdr = 0x01U,
01512     /** @brief TCNT at offset 0x06 */
01513     k_mtu12p_off_tcncnt = 0x06U,
01514     /** @brief TGRA at offset 0x08 */

```

```

01515 k_mtu12p_off_tgra = 0x08U,
01516 /** @brief TGRB at offset 0x0A */
01517 k_mtu12p_off_tgrb = 0x0AU,
01518 /** @brief TMDR3 at offset 0x11 (32-bit mode enable register) */
01519 k_mtu12p_off_tmdr3 = 0x11U,
01520 /** @brief TCNTLW at offset 0x20 (32-bit counter in long-word mode) */
01521 k_mtu12p_off_tcntlw = 0x20U,
01522 /** @brief TGRALW at offset 0x24 (32-bit compare A in long-word mode) */
01523 k_mtu12p_off_tgralw = 0x24U,
01524 /** @brief TGRBLW at offset 0x28 (32-bit compare B in long-word mode) */
01525 k_mtu12p_off_tgrblw = 0x28U,
01526 /** @brief Total size of rx_mtu12_phase_regs_t = 0x2C bytes (44) */
01527 k_mtu12p_struct_size = 0x2CU,
01528 } mtu12_phase_reg_offsets_t;
01529
01530 /**
01531  * @enum mtu_periph_space_t
01532  * @brief Address space mask and expected base for MTU peripheral verification
01533  *
01534  * @details
01535  * Used in static_assert to verify MTU base addresses fall in the correct
01536  * peripheral address space (0x000Cxxxx), as required by the RX72N memory map.
01537  *
01538  * @since Version 1.0.0
01539  */
01540 typedef enum : uintptr_t {
01541     /** @brief Mask selecting the upper 16 bits of a 32-bit address */
01542     k_mtu_periph_space_mask = 0xFFFF0000U,
01543     /** @brief Expected upper 16 bits of all MTU base addresses */
01544     k_mtu_periph_space_base = 0x000C0000U,
01545 } mtu_periph_space_t;
01546
01547 /*
=====
01548 * Static Assertions - Compile-time Register Layout Verification
01549 *
=====
01550 *
01551 * These static assertions verify that the C struct layouts exactly match
01552 * the hardware register addresses defined in the RX72N User's Manual.
01553 * Any mismatch will cause a compile-time error, preventing runtime bugs.
01554 *
01555 * Reference: RX72N Group User's Manual: Hardware, Chapter 5 I/O Register
01556 * Address Table (MTU3a section), and Chapter 24 (MTU3a).
01557 *
01558 * All MTU3 offsets verified against Ch 5 I/O Register Address Table:
01559 * TCR=0x000C1200, TMDR1=0x000C1202, TIORH=0x000C1204, TIORL=0x000C1205,
01560 * TIER=0x000C1208, TCNT=0x000C1210, TGRA=0x000C1218, TGRB=0x000C121A,
01561 * TGRC=0x000C1224, TGRD=0x000C1226, TSR=0x000C122C, TBTM=0x000C1238,
01562 * TCR2=0x000C124C, TGRE=0x000C1272
01563 */
01564
01565 /** @name MTU Channel Register Offset Verification
01566  * @brief Verify rx_mtu_channel_regs_t matches hardware layout
01567  * @{
01568  */
01569 static_assert(offsetof(rx_mtu_channel_regs_t, tcr) == k_mtu_ch_off_tcr,
01570     "MTU TCR register offset incorrect");
01571 static_assert(offsetof(rx_mtu_channel_regs_t, tcnt) == k_mtu_ch_off_tcnt,
01572     "MTU TCNT register offset incorrect");
01573 static_assert(offsetof(rx_mtu_channel_regs_t, tgra) == k_mtu_ch_off_tgra,
01574     "MTU TGRA register offset incorrect");
01575 static_assert(offsetof(rx_mtu_channel_regs_t, tgrb) == k_mtu_ch_off_tgrb,
01576     "MTU TGRB register offset incorrect");
01577
01578 /* Verify MTU3 register offsets match Ch 5 I/O Register Address Table.
01579  * These offsets reflect the actual sparse/interleaved hardware layout, not
01580  * the previous wrong consecutive-byte layout.
01581  * (MTU3 base = 0x000C1200; each offset added to base gives the hardware addr) */
01582 static_assert(offsetof(rx_mtu3_channel_regs_t, tcr) == k_mtu3_off_tcr,
01583     "MTU3 TCR register offset incorrect -- expected 0x000C1200");
01584 static_assert(offsetof(rx_mtu3_channel_regs_t, tmdr1) == k_mtu3_off_tmdr1,
01585     "MTU3 TMDR1 register offset incorrect -- expected 0x000C1202");
01586 static_assert(offsetof(rx_mtu3_channel_regs_t, tiorh) == k_mtu3_off_tiorh,
01587     "MTU3 TIORH register offset incorrect -- expected 0x000C1204");
01588 static_assert(offsetof(rx_mtu3_channel_regs_t, tiorl) == k_mtu3_off_tiorl,
01589     "MTU3 TIORL register offset incorrect -- expected 0x000C1205");
01590 static_assert(offsetof(rx_mtu3_channel_regs_t, tier) == k_mtu3_off_tier,
01591     "MTU3 TIER register offset incorrect -- expected 0x000C1208");
01592 static_assert(offsetof(rx_mtu3_channel_regs_t, tcnt) == k_mtu3_off_tcnt,
01593     "MTU3 TCNT register offset incorrect -- expected 0x000C1210");
01594 static_assert(offsetof(rx_mtu3_channel_regs_t, tgra) == k_mtu3_off_tgra,
01595     "MTU3 TGRA register offset incorrect -- expected 0x000C1218");
01596 static_assert(offsetof(rx_mtu3_channel_regs_t, tgrb) == k_mtu3_off_tgrb,
01597     "MTU3 TGRB register offset incorrect -- expected 0x000C121A");
01598 static_assert(offsetof(rx_mtu3_channel_regs_t, tgrc) == k_mtu3_off_tgrc,
01599     "MTU3 TGRC register offset incorrect -- expected 0x000C1224");

```

```

01600 static_assert(offsetof(rx_mtu3_channel_regs_t, tgrd) == k_mtu3_off_tgrd,
01601                 "MTU3 TGRD register offset incorrect -- expected 0x000C1226");
01602 static_assert(offsetof(rx_mtu3_channel_regs_t, tsr) == k_mtu3_off_tsr,
01603                 "MTU3 TSR register offset incorrect -- expected 0x000C122C");
01604 static_assert(offsetof(rx_mtu3_channel_regs_t, tbtm) == k_mtu3_off_tbtm,
01605                 "MTU3 TBTM register offset incorrect -- expected 0x000C1238");
01606 static_assert(offsetof(rx_mtu3_channel_regs_t, tcr2) == k_mtu3_off_tcr2,
01607                 "MTU3 TCR2 register offset incorrect -- expected 0x000C124C");
01608 static_assert(offsetof(rx_mtu3_channel_regs_t, tgre) == k_mtu3_off_tgre,
01609                 "MTU3 TGRE register offset incorrect -- expected 0x000C1272");
01610 static_assert(sizeof(rx_mtu3_channel_regs_t) == k_mtu3_struct_size,
01611                 "MTU3 register structure size incorrect (expected 0x74 = 116 bytes)");
01612
01613 /* Verify MTU1/2 32-bit phase counting register offsets (Ch24 Table 24.11) */
01614 static_assert(offsetof(rx_mtu12_phase_regs_t, tcr) == k_mtu12p_off_tcr,
01615                 "MTU1/2 Phase TCR register offset incorrect");
01616 static_assert(offsetof(rx_mtu12_phase_regs_t, tmdr) == k_mtu12p_off_tmdr,
01617                 "MTU1/2 Phase TMDR register offset incorrect");
01618 static_assert(offsetof(rx_mtu12_phase_regs_t, tcnt) == k_mtu12p_off_tcnt,
01619                 "MTU1/2 Phase TCNT register offset incorrect");
01620 static_assert(offsetof(rx_mtu12_phase_regs_t, tgra) == k_mtu12p_off_tgra,
01621                 "MTU1/2 Phase TGRA register offset incorrect");
01622 static_assert(offsetof(rx_mtu12_phase_regs_t, tgrb) == k_mtu12p_off_tgrb,
01623                 "MTU1/2 Phase TGRB register offset incorrect");
01624 static_assert(offsetof(rx_mtu12_phase_regs_t, tmdr3) == k_mtu12p_off_tmdr3,
01625                 "MTU1/2 Phase TMDR3 register offset incorrect");
01626 static_assert(offsetof(rx_mtu12_phase_regs_t, tcntlw) == k_mtu12p_off_tcntlw,
01627                 "MTU1/2 Phase TCNTLW register offset incorrect");
01628 static_assert(offsetof(rx_mtu12_phase_regs_t, tgralw) == k_mtu12p_off_tgralw,
01629                 "MTU1/2 Phase TGRALW register offset incorrect");
01630 static_assert(offsetof(rx_mtu12_phase_regs_t, tgrblw) == k_mtu12p_off_tgrblw,
01631                 "MTU1/2 Phase TGRBLW register offset incorrect");
01632 static_assert(sizeof(rx_mtu12_phase_regs_t) == k_mtu12p_struct_size,
01633                 "MTU1/2 Phase register structure size incorrect (expected 44 bytes)");
01634
01635 /* Verify TSTR register structure */
01636 static_assert(sizeof(rx_mtu_tstr_regs_t) == 1, "MTU TSTR register structure size mismatch");
01637
01638 /* Verify base addresses are in correct peripheral space (0x000C1xxx) */
01639 static_assert((k_mtu0_base_addr & k_mtu_periph_space_mask) == k_mtu_periph_space_base,
01640                 "MTU0 base address not in MTU peripheral space");
01641 static_assert((k_mtu3_base_addr & k_mtu_periph_space_mask) == k_mtu_periph_space_base,
01642                 "MTU3 base address not in MTU peripheral space");
01643 static_assert((k_mtu6_base_addr & k_mtu_periph_space_mask) == k_mtu_periph_space_base,
01644                 "MTU6 base address not in MTU peripheral space");
01645
01646 /* Verify MTU3 and MTU4 are interleaved (MTU4 = MTU3 + 1 byte) */
01647 static_assert(k_mtu4_base_addr == (k_mtu3_base_addr + 1U),
01648                 "MTU3 and MTU4 must be interleaved (MTU4 = MTU3 + 1 byte)");
01649
01650 /* Verify MTU6 and MTU7 are interleaved (MTU7 = MTU6 + 1 byte) */
01651 static_assert(k_mtu7_base_addr == (k_mtu6_base_addr + 1U),
01652                 "MTU6 and MTU7 must be interleaved (MTU7 = MTU6 + 1 byte)");
01653 /** @} */ /* End of static assertion group */
01654
01655 /** @} */ /* End of mtu_regs defgroup */
01656
01657 #ifdef __cplusplus
01658 }
01659 #endif

```

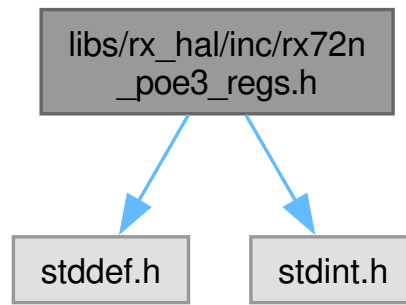
8.41 libs/rx/hal/inc/rx72n_poe3_regs.h File Reference

RX72N Port Output Enable 3 (POE3) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_poe3_regs.h:



Enumerations

- enum `poe3_addresses_t` : `uintptr_t` {
`k_poe3_base_addr` = 0x0008C4C0U , `k_poe3_icsr1_addr` = 0x0008C4C0U , `k_poe3_icsr2_addr` = 0x0008C4C4U , `k_poe3_icsr3_addr` = 0x0008C4C8U ,
`k_poe3_icsr4_addr` = 0x0008C4D6U , `k_poe3_icsr5_addr` = 0x0008C4D8U , `k_poe3_icsr6_addr` = 0x0008C4DCU , `k_poe3_ocr1_addr` = 0x0008C4C2U ,
`k_poe3_ocr2_addr` = 0x0008C4C6U , `k_poe3_spoer_addr` = 0x0008C4CAU , `k_poe3_poecr1_addr` = 0x0008C4CBU , `k_poe3_poecr2_addr` = 0x0008C4CCU ,
`k_poe3_poecr4_addr` = 0x0008C4D0U , `k_poe3_poecr5_addr` = 0x0008C4D2U , `k_poe3_alr1_addr` = 0x0008C4DAU , `k_poe3_m0selr1_addr` = 0x0008C4E4U ,
`k_poe3_m0selr2_addr` = 0x0008C4E5U , `k_poe3_m3selr_addr` = 0x0008C4E6U , `k_poe3_m4selr1_addr` = 0x0008C4E7U , `k_poe3_m4selr2_addr` = 0x0008C4E8U ,
`k_poe3_m6selr_addr` = 0x0008C4EAU }
 POE3 module addresses.

- enum `poe3_offsets_t` : `uint8_t` {
`k_poe3_offset_icsr1` = 0x00U , `k_poe3_offset_ocr1` = 0x02U , `k_poe3_offset_icsr2` = 0x04U ,
`k_poe3_offset_ocr2` = 0x06U ,
`k_poe3_offset_icsr3` = 0x08U , `k_poe3_offset_spoer` = 0x0AU , `k_poe3_offset_poecr1` = 0x0BU ,
`k_poe3_offset_poecr2` = 0x0CU ,
`k_poe3_offset_poecr4` = 0x10U , `k_poe3_offset_poecr5` = 0x12U , `k_poe3_offset_icsr4` = 0x16U ,
`k_poe3_offset_icsr5` = 0x18U ,
`k_poe3_offset_alr1` = 0x1AU , `k_poe3_offset_icsr6` = 0x1CU , `k_poe3_offset_m0selr1` = 0x24U ,
`k_poe3_offset_m0selr2` = 0x25U ,
`k_poe3_offset_m3selr` = 0x26U , `k_poe3_offset_m4selr1` = 0x27U , `k_poe3_offset_m4selr2` = 0x28U , `k_poe3_offset_m6selr` = 0x2AU }
 POE3 register offsets from base address.

- enum `icsr_poemode_t` : `uint16_t` {
`k_icsr_poemode_falling_edge` = 0x0000U , `k_icsr_poemode_lowlevel_pclk8` = 0x0001U ,
`k_icsr_poemode_lowlevel_pclk16` = 0x0002U , `k_icsr_poemode_lowlevel_pclk128` = 0x0003U ,
`k_icsr_poemode_mask` = 0x0003U }
 POE# Mode Select bits (POE0M/POE4M/POE8M/POE10M/POE11M).

- enum `icsr_bits_t` : `uint16_t` { `k_icsr_pie` = 0x0100U , `k_icsr_poef` = 0x1000U , `k_icsr3_poe8e` = 0x0200U }
 ICSR common bit definitions.

- enum `icsr6_bits_t` : `uint16_t` { `k_icsr6_ostste` = 0x0200U , `k_icsr6_oststf` = 0x1000U }
 ICSR6 bit definitions (Oscillation Stop Detection).

- enum `ocr_bits_t` : `uint16_t` { `k_ocr_oe` = 0x0100U , `k_ocr_oe` = 0x0200U , `k_ocr_osf` = 0x8000U }
 OCSR bit definitions (simultaneous conduction detection).

- enum `spoer_bits_t` : `uint8_t` { `k_spoer_mtuch34hiz` = 0x01U , `k_spoer_mtuch67hiz` = 0x02U ,
`k_spoer_mtuch0hiz` = 0x04U }
 SPOER bit definitions (software high-impedance control).

- enum `poe3r1_bits_t` : `uint8_t` { `k_poe3r1_mtu0aze` = 0x01U , `k_poe3r1_mtu0bze` = 0x02U , `k_poe3r1_mtu0cze` = 0x04U , `k_poe3r1_mtu0dze` = 0x08U }
POE3R1 bit definitions (MTU0 pin high-impedance enable).
- enum `poe3r2_bits_t` : `uint16_t` {
`k_poe3r2_mtu7bdze` = 0x0001U , `k_poe3r2_mtu7acze` = 0x0002U , `k_poe3r2_mtu6bdze` = 0x0004U , `k_poe3r2_mtu4bdze` = 0x0100U ,
`k_poe3r2_mtu4acze` = 0x0200U , `k_poe3r2_mtu3bdze` = 0x0400U }
POE3R2 bit definitions (MTU3/4 and MTU6/7 complementary PWM control).
- enum `poe3r4_bits_t` : `uint16_t` {
`k_poe3r4_ic2addmt34ze` = 0x0004U , `k_poe3r4_ic3addmt34ze` = 0x0008U , `k_poe3r4_ic4addmt34ze` = 0x0010U , `k_poe3r4_ic5addmt34ze` = 0x0020U ,
`k_poe3r4_ic1addmt67ze` = 0x0200U , `k_poe3r4_ic3addmt67ze` = 0x0800U , `k_poe3r4_ic4addmt67ze` = 0x1000U , `k_poe3r4_ic5addmt67ze` = 0x2000U }
POE3R4 bit definitions (add POE# flag conditions for MTU3/4 and MTU6/7).
- enum `poe3r5_bits_t` : `uint16_t` { `k_poe3r5_ic1addmt0ze` = 0x0002U , `k_poe3r5_ic2addmt0ze` = 0x0004U , `k_poe3r5_ic4addmt0ze` = 0x0010U , `k_poe3r5_ic5addmt0ze` = 0x0020U }
POE3R5 bit definitions (add POE# flag conditions for MTU0 pins).
- enum `alr1_bits_t` : `uint16_t` {
`k_alr1_olsg0a` = 0x0001U , `k_alr1_olsg0b` = 0x0002U , `k_alr1_olsg1a` = 0x0004U , `k_alr1_olsg1b` = 0x0008U ,
`k_alr1_olsg2a` = 0x0010U , `k_alr1_olsg2b` = 0x0020U , `k_alr1_olsen` = 0x0080U }
ALR1 bit definitions (active level for shoot-through detection).
- enum `poe3_mstpcr_t` : `uint32_t` { `k_poe3_mstpcra_bit` = (1U << 9U) }
POE3 module stop control bits.

Functions

- static volatile `uint16_t` * `poe3_icsr1_reg` (void)
Get pointer to ICSR1 register (POE0# control).
- static volatile `uint16_t` * `poe3_icsr2_reg` (void)
Get pointer to ICSR2 register (POE4# control).
- static volatile `uint16_t` * `poe3_icsr3_reg` (void)
Get pointer to ICSR3 register (POE8# control).
- static volatile `uint16_t` * `poe3_icsr4_reg` (void)
Get pointer to ICSR4 register (POE10# control).
- static volatile `uint16_t` * `poe3_icsr5_reg` (void)
Get pointer to ICSR5 register (POE11# control).
- static volatile `uint16_t` * `poe3_icsr6_reg` (void)
Get pointer to ICSR6 register (oscillation stop).
- static volatile `uint16_t` * `poe3_ocr1_reg` (void)
Get pointer to OCSR1 register (MTU3/4 shoot-through).
- static volatile `uint16_t` * `poe3_ocr2_reg` (void)
Get pointer to OCSR2 register (MTU6/7 shoot-through).
- static volatile `uint8_t` * `poe3_spoer_reg` (void)
Get pointer to SPOER register (software emergency stop).
- static volatile `uint8_t` * `poe3_poe3r1_reg` (void)
Get pointer to POE3R1 register (MTU0 pin control).
- static volatile `uint16_t` * `poe3_poe3r2_reg` (void)
Get pointer to POE3R2 register (MTU3/4 and MTU6/7 pin control).
- static volatile `uint16_t` * `poe3_poe3r4_reg` (void)
Get pointer to POE3R4 register (MTU3/4 and MTU6/7 additional POE# conditions).
- static volatile `uint16_t` * `poe3_poe3r5_reg` (void)

- Get pointer to POECR5 register (MTU0 additional POE# conditions).
- static volatile uint16_t * [poe3_alr1_reg](#) (void)
 - Get pointer to ALR1 register (active level).
- static volatile uint8_t * [poe3_m0selr1_reg](#) (void)
 - Get pointer to M0SELR1 register (MTU0 pin select 1).
- static volatile uint8_t * [poe3_m0selr2_reg](#) (void)
 - Get pointer to M0SELR2 register (MTU0 pin select 2).
- static volatile uint8_t * [poe3_m3selr_reg](#) (void)
 - Get pointer to M3SELR register (MTU3 pin select).
- static volatile uint8_t * [poe3_m4selr1_reg](#) (void)
 - Get pointer to M4SELR1 register (MTU4 pin select 1).
- static volatile uint8_t * [poe3_m4selr2_reg](#) (void)
 - Get pointer to M4SELR2 register (MTU4 pin select 2).
- static volatile uint8_t * [poe3_m6selr_reg](#) (void)
 - Get pointer to M6SELR register (MTU6 pin select).

8.41.1 Detailed Description

RX72N Port Output Enable 3 (POE3) Register Definitions.

Complete register definitions for the RX72N Port Output Enable 3 module. POE3 provides emergency output disable for MTU (Multi-Function Timer Pulse Unit) pins, enabling hardware-level motor protection.

POE3 Overview

- Emergency output disable for MTU0, MTU3/4, MTU6/7 pins
- Input signal detection (POE0#, POE4#, POE8#, POE10#, POE11#)
- Simultaneous conduction detection (shoot-through protection)
- Oscillation stop detection response
- Software-triggered high-impedance state

Target Pins for High-Impedance Control

Timer	Pins Controlled
MTU0	MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D
MTU3	MTIOC3B, MTIOC3D
MTU4	MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D
MTU6	MTIOC6B, MTIOC6D
MTU7	MTIOC7A, MTIOC7B, MTIOC7C, MTIOC7D

Register Organization (POE3 base: 0x0008C4C0)

Offset	Size	Register	Description
0x00	16	ICSR1	Input Level Control/Status 1 (POE0#)
0x02	16	OCSR1	Output Level Control/Status 1
0x04	16	ICSR2	Input Level Control/Status 2 (POE4#)
0x06	16	OCSR2	Output Level Control/Status 2
0x08	16	ICSR3	Input Level Control/Status 3 (POE8#)
0x0A	8	SPOER	Software Port Output Enable
0x0B	8	POECR1	Port Output Enable Control 1
0x0C	16	POECR2	Port Output Enable Control 2 (MTU3/4 b10:8, MTU6/7 b2:0)
0x10	16	POECR4	Port Output Enable Control 4 (ICxADD conditions MT34/MT67)
0x12	16	POECR5	Port Output Enable Control 5 (ICxADD conditions MT0)
0x16	16	ICSR4	Input Level Control/Status 4 (POE10#)
0x18	16	ICSR5	Input Level Control/Status 5 (POE11#)
0x1A	16	ALR1	Active Level Register 1
0x1C	16	ICSR6	Input Level Control/Status 6 (osc stop)
0x24	8	M0SELR1	MTU0 Pin Select 1
0x25	8	M0SELR2	MTU0 Pin Select 2
0x26	8	M3SELR	MTU3 Pin Select
0x27	8	M4SELR1	MTU4 Pin Select 1
0x28	8	M4SELR2	MTU4 Pin Select 2
0x2A	8	M6SELR	MTU6 Pin Select

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops)
- Rule 3: No dynamic memory allocation
- Rule 4: Short accessor functions
- Rule 5: Static assertions verify addresses
- Rule 6: Minimal scope
- Rule 7: N/A
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 25: Port Output Enable 3 (POE3a), pages 1199-1240
- Section 25.2: Register Descriptions

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_poe3_regs.h](#).

8.41.2 Enumeration Type Documentation

8.41.2.1 alr1`bits``t`enum [alr1_bits_t](#) : uint16_t

ALR1 bit definitions (active level for shoot-through detection).

Register Layout (16-bit @ 0x0008C4DA)

Bits	Field	Description
0	OLSG0A	Active level for MTIOC3B/4A (group 0, positive)
1	OLSG0B	Active level for MTIOC3D/4C (group 0, negative)
2	OLSG1A	Active level for MTIOC3B/4A (group 1, positive)
3	OLSG1B	Active level for MTIOC3D/4C (group 1, negative)
4	OLSG2A	Active level for MTIOC4B (group 2, positive)
5	OLSG2B	Active level for MTIOC4D (group 2, negative)
6	-	Reserved
7	OLSEN	Active level select enable
15:↵ 8	-	Reserved

When OLSEN=0, active level determined by MTU.TOCR registers. When OLSEN=1, active level determined by OLSGnA/B bits (0=low, 1=high).

Since

Version 1.0.0

Enumerator

k_alr1_olsg0a	Active level for group 0 positive
k_alr1_olsg0b	Active level for group 0 negative
k_alr1_olsg1a	Active level for group 1 positive
k_alr1_olsg1b	Active level for group 1 negative
k_alr1_olsg2a	Active level for group 2 positive
k_alr1_olsg2b	Active level for group 2 negative
k_alr1_olsen	Active level select enable

Definition at line 536 of file [rx72n_poe3_regs.h](#).

```

00536      : uint16_t {
00537  k_alr1_olsg0a = 0x0001U, /**< Active level for group 0 positive */
00538  k_alr1_olsg0b = 0x0002U, /**< Active level for group 0 negative */
00539  k_alr1_olsg1a = 0x0004U, /**< Active level for group 1 positive */
00540  k_alr1_olsg1b = 0x0008U, /**< Active level for group 1 negative */
00541  k_alr1_olsg2a = 0x0010U, /**< Active level for group 2 positive */
00542  k_alr1_olsg2b = 0x0020U, /**< Active level for group 2 negative */
00543  k_alr1_olsen  = 0x0080U, /**< Active level select enable */
00544 } alr1_bits_t;

```

8.41.2.2 icsr6_bits_t

enum [icsr6_bits_t](#) : uint16_t

ICSR6 bit definitions (Oscillation Stop Detection).

Register Layout (16-bit @ 0x0008C4DC)

Bits	Field	Description
8:↵ 0	-	Reserved
9	OSTSTE	Osc stop high-impedance enable
11:10	-	Reserved
12	OSTSTF	Osc stop flag
15:13	-	Reserved

Since

Version 1.0.0

Enumerator

k_icsr6_ostste	Oscillation Stop High-Impedance Enable (bit 9).
k_icsr6_oststf	Oscillation Stop Flag (bit 12).

Definition at line 278 of file [rx72n_poe3_regs.h](#).

```
00278         : uint16_t {
00279     /** @brief Oscillation Stop High-Impedance Enable (bit 9) */
00280     k_icsr6_ostste = 0x0200U,
00281
00282     /** @brief Oscillation Stop Flag (bit 12) */
00283     k_icsr6_oststf = 0x1000U,
00284 } icsr6_bits_t;
```

8.41.2.3 icsr_bits_t

enum [icsr_bits_t](#) : uint16_t

ICSR common bit definitions.

Register Layout (16-bit)

Bits	Field	Description
1:↔ 0	POEmM	POE# mode select
7:↔ 2	-	Reserved
8	PIEn	Port interrupt enable
9	POE8E	High-impedance enable (ICSR3 only)
11:10	-	Reserved
12	POEmF	POE# flag
15:13	-	Reserved

Since

Version 1.0.0

Enumerator

k_icsr_pie	Port Interrupt Enable (bit 8).
k_icsr_poef	POE# Flag (bit 12) - high-impedance request detected.
k_icsr3_poe8e	POE8 High-Impedance Enable (bit 9, ICSR3 only).

Definition at line 250 of file [rx72n_poe3_regs.h](#).

```
00250         : uint16_t {
00251     /** Common bits for ICSR1/2/3/4/5 */
00252     /** @brief Port Interrupt Enable (bit 8) */
00253     k_icsr_pie = 0x0100U,
00254
00255     /** @brief POE# Flag (bit 12) - high-impedance request detected */
00256     k_icsr_poef = 0x1000U,
00257
00258     /** ICSR3-specific bits */
00259     /** @brief POE8 High-Impedance Enable (bit 9, ICSR3 only) */
00260     k_icsr3_poe8e = 0x0200U,
00261 } icsr_bits_t;
```

8.41.2.4 icsr`poemode`

enum [icsr_poemode_t](#) : uint16_t

POE# Mode Select bits (POE0M/POE4M/POE8M/POE10M/POE11M).

Selects input detection mode for POE# pins. These bits can only be modified once after reset.

Note

Manual ref: Ch25 sections 25.2.1-25.2.5

Since

Version 1.0.0

Enumerator

k_icsr_poemode_falling_edge	Accept on falling edge of POE# input.
k_icsr_poemode_lowlevel_pclk8	Low-level sampling: 16 samples at PCLK/8.
k_icsr_poemode_lowlevel_pclk16	Low-level sampling: 16 samples at PCLK/16.
k_icsr_poemode_lowlevel_pclk128	Low-level sampling: 16 samples at PCLK/128.
k_icsr_poemode_mask	POEm mode select mask (bits 1:0).

Definition at line 216 of file [rx72n_poe3_regs.h](#).

```

00216         : uint16_t {
00217     /** @brief Accept on falling edge of POE# input */
00218     k_icsr_poemode_falling_edge = 0x0000U,
00219
00220     /** @brief Low-level sampling: 16 samples at PCLK/8 */
00221     k_icsr_poemode_lowlevel_pclk8 = 0x0001U,
00222
00223     /** @brief Low-level sampling: 16 samples at PCLK/16 */
00224     k_icsr_poemode_lowlevel_pclk16 = 0x0002U,
00225
00226     /** @brief Low-level sampling: 16 samples at PCLK/128 */
00227     k_icsr_poemode_lowlevel_pclk128 = 0x0003U,
00228
00229     /** @brief POEm mode select mask (bits 1:0) */
00230     k_icsr_poemode_mask = 0x0003U,
00231 } icsr_poemode_t;

```

8.41.2.5 ocsr`bits`

enum [ocsr_bits_t](#) : uint16_t

OCSR bit definitions (simultaneous conduction detection).

Register Layout (16-bit)

Bits	Field	Description
7:0	-	Reserved
8	OIE _n	Simultaneous conduction interrupt enable
9	OCE _n	Simultaneous conduction high-impedance enable
14:10	-	Reserved
15	OSF _n	Simultaneous conduction flag

Used to detect and respond to simultaneous active-level output on complementary PWM pin pairs (shoot-through detection).

Note

Manual ref: Ch25 sections 25.2.7, 25.2.8

Since

Version 1.0.0

Enumerator

k_ocrs_oie	Simultaneous Conduction Interrupt Enable (bit 8).
k_ocrs_oce	Simultaneous Conduction High-Impedance Enable (bit 9). Can only be modified once after reset.
k_ocrs_osf	Simultaneous Conduction Flag (bit 15).

Definition at line 311 of file [rx72n_poe3_regs.h](#).

```

00311     : uint16_t {
00312     /** @brief Simultaneous Conduction Interrupt Enable (bit 8) */
00313     k_ocrs_oie = 0x0100U,
00314
00315     /**
00316      * @brief Simultaneous Conduction High-Impedance Enable (bit 9)
00317      * @details Can only be modified once after reset.
00318      */
00319     k_ocrs_oce = 0x0200U,
00320
00321     /** @brief Simultaneous Conduction Flag (bit 15) */
00322     k_ocrs_osf = 0x8000U,
00323 } ocsr_bits_t;

```

8.41.2.6 poe3_addresses_t

```
enum poe3_addresses_t : uintptr_t
```

POE3 module addresses.

Base address and register addresses for POE3. All addresses verified against RX72N Hardware Manual Ch25.

Note

Addresses verified 2026-01-29 against manual section 25.2

Since

Version 1.0.0

Enumerator

k_poe3_base_addr	POE3 module base address.
k_poe3_icsr1_addr	ICSR1 address - Input Level Control/Status 1 (POE0#).
k_poe3_icsr2_addr	ICSR2 address - Input Level Control/Status 2 (POE4#).
k_poe3_icsr3_addr	ICSR3 address - Input Level Control/Status 3 (POE8#).
k_poe3_icsr4_addr	ICSR4 address - Input Level Control/Status 4 (POE10#).
k_poe3_icsr5_addr	ICSR5 address - Input Level Control/Status 5 (POE11#).
k_poe3_icsr6_addr	ICSR6 address - Input Level Control/Status 6 (osc stop).
k_poe3_ocr1_addr	OCSR1 address - Output Level Control/Status 1 (MTU3/4).
k_poe3_ocr2_addr	OCSR2 address - Output Level Control/Status 2 (MTU6/7).
k_poe3_spoer_addr	SPOER address - Software Port Output Enable Register.
k_poe3_poecr1_addr	POECR1 address - Port Output Enable Control 1 (MTU0).
k_poe3_poecr2_addr	POECR2 address - Port Output Enable Control 2 (MTU3/4).
k_poe3_poecr4_addr	POECR4 address - Port Output Enable Control 4 (MTU6/7).
k_poe3_poecr5_addr	POECR5 address - Port Output Enable Control 5 (additional).
k_poe3_alr1_addr	ALR1 address - Active Level Register 1.
k_poe3_m0selr1_addr	M0SELR1 address - MTU0 Pin Select 1.
k_poe3_m0selr2_addr	M0SELR2 address - MTU0 Pin Select 2.
k_poe3_m3selr_addr	M3SELR address - MTU3 Pin Select.
k_poe3_m4selr1_addr	M4SELR1 address - MTU4 Pin Select 1.
k_poe3_m4selr2_addr	M4SELR2 address - MTU4 Pin Select 2.
k_poe3_m6selr_addr	M6SELR address - MTU6 Pin Select.

Definition at line 99 of file [rx72n_poe3_regs.h](#).

```

00099      : uintptr_t {
00100      /** @brief POE3 module base address */
00101      k_poe3_base_addr = 0x0008C4C0U,
00102
00103      /* Input Level Control/Status Registers */
00104      /** @brief ICSR1 address - Input Level Control/Status 1 (POE0#) */
00105      k_poe3_icsr1_addr = 0x0008C4C0U,
00106
00107      /** @brief ICSR2 address - Input Level Control/Status 2 (POE4#) */
00108      k_poe3_icsr2_addr = 0x0008C4C4U,
00109
00110      /** @brief ICSR3 address - Input Level Control/Status 3 (POE8#) */
00111      k_poe3_icsr3_addr = 0x0008C4C8U,
00112
00113      /** @brief ICSR4 address - Input Level Control/Status 4 (POE10#) */
00114      k_poe3_icsr4_addr = 0x0008C4D6U,
00115
00116      /** @brief ICSR5 address - Input Level Control/Status 5 (POE11#) */
00117      k_poe3_icsr5_addr = 0x0008C4D8U,
00118
00119      /** @brief ICSR6 address - Input Level Control/Status 6 (osc stop) */
00120      k_poe3_icsr6_addr = 0x0008C4DCU,
00121
00122      /* Output Level Control/Status Registers */
00123      /** @brief OCSR1 address - Output Level Control/Status 1 (MTU3/4) */
00124      k_poe3_ocr1_addr = 0x0008C4C2U,
00125
00126      /** @brief OCSR2 address - Output Level Control/Status 2 (MTU6/7) */
00127      k_poe3_ocr2_addr = 0x0008C4C6U,
00128
00129      /* Control Registers */
00130      /** @brief SPOER address - Software Port Output Enable Register */

```

```

00131 k_poe3_spoer_addr = 0x0008C4CAU,
00132
00133 /** @brief POECR1 address - Port Output Enable Control 1 (MTU0) */
00134 k_poe3_poecr1_addr = 0x0008C4CBU,
00135
00136 /** @brief POECR2 address - Port Output Enable Control 2 (MTU3/4) */
00137 k_poe3_poecr2_addr = 0x0008C4CCU,
00138
00139 /** @brief POECR4 address - Port Output Enable Control 4 (MTU6/7) */
00140 k_poe3_poecr4_addr = 0x0008C4D0U,
00141
00142 /** @brief POECR5 address - Port Output Enable Control 5 (additional) */
00143 k_poe3_poecr5_addr = 0x0008C4D2U,
00144
00145 /** @brief ALR1 address - Active Level Register 1 */
00146 k_poe3_alr1_addr = 0x0008C4DAU,
00147
00148 /* Pin Select Registers */
00149 /** @brief M0SELR1 address - MTU0 Pin Select 1 */
00150 k_poe3_m0selr1_addr = 0x0008C4E4U,
00151
00152 /** @brief M0SELR2 address - MTU0 Pin Select 2 */
00153 k_poe3_m0selr2_addr = 0x0008C4E5U,
00154
00155 /** @brief M3SELR address - MTU3 Pin Select */
00156 k_poe3_m3selr_addr = 0x0008C4E6U,
00157
00158 /** @brief M4SELR1 address - MTU4 Pin Select 1 */
00159 k_poe3_m4selr1_addr = 0x0008C4E7U,
00160
00161 /** @brief M4SELR2 address - MTU4 Pin Select 2 */
00162 k_poe3_m4selr2_addr = 0x0008C4E8U,
00163
00164 /** @brief M6SELR address - MTU6 Pin Select */
00165 k_poe3_m6selr_addr = 0x0008C4EAU,
00166 } poe3_addresses_t;

```

8.41.2.7 poe3_mstpcr_t

enum [poe3_mstpcr_t](#) : uint32_t

POE3 module stop control bits.

POE3 is controlled by MSTPCRA.MSTPA9. Must be cleared (0) before accessing POE3 registers.

Note

Manual ref: Ch11 (Low Power Consumption) module stop tables

Since

Version 1.0.0

Enumerator

k_poe3_mstpcra_bit	MSTPCRA bit for POE3 (bit 9). 0: POE3 module is running 1: POE3 module is stopped (default)
--------------------	--

Definition at line 562 of file [rx72n_poe3_regs.h](#).

```

00562         : uint32_t {
00563     /**
00564      * @brief MSTPCRA bit for POE3 (bit 9)
00565      * @details
00566      * - 0: POE3 module is running
00567      * - 1: POE3 module is stopped (default)
00568      */
00569     k_poe3_mstpcra_bit = (1U « 9U),
00570 } poe3_mstpcr_t;

```

8.41.2.8 poe3'offsets't

```
enum poe3_offsets_t : uint8_t
```

POE3 register offsets from base address.

Offsets from POE3 base address (0x0008C4C0) for each register.

Since

Version 1.0.0

Enumerator

k_poe3_offset_icsr1	ICSR1 offset (16-bit)
k_poe3_offset_ocrs1	OCSR1 offset (16-bit)
k_poe3_offset_icsr2	ICSR2 offset (16-bit)
k_poe3_offset_ocrs2	OCSR2 offset (16-bit)
k_poe3_offset_icsr3	ICSR3 offset (16-bit)
k_poe3_offset_spoer	SPOER offset (8-bit)
k_poe3_offset_poecr1	POECR1 offset (8-bit)
k_poe3_offset_poecr2	POECR2 offset (8-bit)
k_poe3_offset_poecr4	POECR4 offset (16-bit)
k_poe3_offset_poecr5	POECR5 offset (16-bit)
k_poe3_offset_icsr4	ICSR4 offset (16-bit)
k_poe3_offset_icsr5	ICSR5 offset (16-bit)
k_poe3_offset_alr1	ALR1 offset (16-bit)
k_poe3_offset_icsr6	ICSR6 offset (16-bit)
k_poe3_offset_m0selr1	M0SELR1 offset (8-bit)
k_poe3_offset_m0selr2	M0SELR2 offset (8-bit)
k_poe3_offset_m3selr	M3SELR offset (8-bit)
k_poe3_offset_m4selr1	M4SELR1 offset (8-bit)
k_poe3_offset_m4selr2	M4SELR2 offset (8-bit)
k_poe3_offset_m6selr	M6SELR offset (8-bit)

Definition at line 177 of file [rx72n_poe3_regs.h](#).

```
00177      : uint8_t {
00178  k_poe3_offset_icsr1  = 0x00U, /**< ICSR1 offset (16-bit) */
00179  k_poe3_offset_ocrs1  = 0x02U, /**< OCSR1 offset (16-bit) */
00180  k_poe3_offset_icsr2  = 0x04U, /**< ICSR2 offset (16-bit) */
00181  k_poe3_offset_ocrs2  = 0x06U, /**< OCSR2 offset (16-bit) */
00182  k_poe3_offset_icsr3  = 0x08U, /**< ICSR3 offset (16-bit) */
00183  k_poe3_offset_spoer  = 0x0AU, /**< SPOER offset (8-bit) */
00184  k_poe3_offset_poecr1 = 0x0BU, /**< POECR1 offset (8-bit) */
00185  k_poe3_offset_poecr2 = 0x0CU, /**< POECR2 offset (8-bit) */
00186  k_poe3_offset_poecr4 = 0x10U, /**< POECR4 offset (16-bit) */
00187  k_poe3_offset_poecr5 = 0x12U, /**< POECR5 offset (16-bit) */
00188  k_poe3_offset_icsr4  = 0x16U, /**< ICSR4 offset (16-bit) */
00189  k_poe3_offset_icsr5  = 0x18U, /**< ICSR5 offset (16-bit) */
00190  k_poe3_offset_alr1   = 0x1AU, /**< ALR1 offset (16-bit) */
00191  k_poe3_offset_icsr6  = 0x1CU, /**< ICSR6 offset (16-bit) */
00192  k_poe3_offset_m0selr1 = 0x24U, /**< M0SELR1 offset (8-bit) */
00193  k_poe3_offset_m0selr2 = 0x25U, /**< M0SELR2 offset (8-bit) */
00194  k_poe3_offset_m3selr = 0x26U, /**< M3SELR offset (8-bit) */
00195  k_poe3_offset_m4selr1 = 0x27U, /**< M4SELR1 offset (8-bit) */
00196  k_poe3_offset_m4selr2 = 0x28U, /**< M4SELR2 offset (8-bit) */
00197  k_poe3_offset_m6selr = 0x2AU, /**< M6SELR offset (8-bit) */
00198 } poe3_offsets_t;
```


8.41.2.9 poecr1_bits_t

```
enum poecr1_bits_t : uint8_t
```

POECR1 bit definitions (MTU0 pin high-impedance enable).

Register Layout (8-bit @ 0x0008C4CB)

Bit	Field	Description
0	MTU0AZE	MTIOC0A high-impedance enable
1	MTU0BZE	MTIOC0B high-impedance enable
2	MTU0CZE	MTIOC0C high-impedance enable
3	MTU0DZE	MTIOC0D high-impedance enable
7:↔ 4	-	Reserved

Note

These bits can only be modified once after reset.

Since

Version 1.0.0

Enumerator

k_poecr1_mtu0aze	MTIOC0A high-impedance enable
k_poecr1_mtu0bze	MTIOC0B high-impedance enable
k_poecr1_mtu0cze	MTIOC0C high-impedance enable
k_poecr1_mtu0dze	MTIOC0D high-impedance enable

Definition at line 381 of file rx72n_poe3_regs.h.

```
00381      : uint8_t {
00382  k_poecr1_mtu0aze = 0x01U, /**< MTIOC0A high-impedance enable */
00383  k_poecr1_mtu0bze = 0x02U, /**< MTIOC0B high-impedance enable */
00384  k_poecr1_mtu0cze = 0x04U, /**< MTIOC0C high-impedance enable */
00385  k_poecr1_mtu0dze = 0x08U, /**< MTIOC0D high-impedance enable */
00386 } poecr1_bits_t;
```

8.41.2.10 poecr2_bits_t

```
enum poecr2_bits_t : uint16_t
```

POECR2 bit definitions (MTU3/4 and MTU6/7 complementary PWM control).

Register Layout (16-bit @ 0x0008C4CC)

Bits	Field	Description
1: 0	-	Reserved (MTU7BDZE, MTU7ACZE in b0,b1)
0	MTU7BDZE	MTIOC7B/7D high-impedance enable
1	MTU7ACZE	MTIOC7A/7C high-impedance enable
2	MTU6BDZE	MTIOC6B/6D high-impedance enable
7: 3	-	Reserved
8	MTU4BDZE	MTIOC4B/4D high-impedance enable
9	MTU4ACZE	MTIOC4A/4C high-impedance enable
10	MTU3BDZE	MTIOC3B/3D high-impedance enable
15:11	-	Reserved

Reset value: b10=1, b9=1, b8=1, b2=1, b1=1, b0=1 (all ZE bits set). Note 2 in the manual: Set MTU6/7 bits (b2:b0) to 0 when MTU6 or MTU7 is not used.

Note

These bits can only be modified once after reset.

Since

Version 1.0.0

Enumerator

k_poecr2_mtu7bdze	MTIOC7B/7D high-impedance enable (bit 0)
k_poecr2_mtu7acze	MTIOC7A/7C high-impedance enable (bit 1)
k_poecr2_mtu6bdze	MTIOC6B/6D high-impedance enable (bit 2)
k_poecr2_mtu4bdze	MTIOC4B/4D high-impedance enable (bit 8)
k_poecr2_mtu4acze	MTIOC4A/4C high-impedance enable (bit 9)
k_poecr2_mtu3bdze	MTIOC3B/3D high-impedance enable (bit 10)

Definition at line 418 of file rx72n_poe3_regs.h.

```

00418 : uint16_t {
00419 /* MTU6/7 hi-Z enable (lower byte, bits 2:0) */
00420 k_poecr2_mtu7bdze = 0x0001U, /**< MTIOC7B/7D high-impedance enable (bit 0) */
00421 k_poecr2_mtu7acze = 0x0002U, /**< MTIOC7A/7C high-impedance enable (bit 1) */
00422 k_poecr2_mtu6bdze = 0x0004U, /**< MTIOC6B/6D high-impedance enable (bit 2) */
00423 /* MTU3/4 hi-Z enable (upper byte, bits 10:8) */
00424 k_poecr2_mtu4bdze = 0x0100U, /**< MTIOC4B/4D high-impedance enable (bit 8) */
00425 k_poecr2_mtu4acze = 0x0200U, /**< MTIOC4A/4C high-impedance enable (bit 9) */
00426 k_poecr2_mtu3bdze = 0x0400U, /**< MTIOC3B/3D high-impedance enable (bit 10) */
00427 } poecr2_bits_t;

```

8.41.2.11 poecr4_bits_t

enum poecr4_bits_t : uint16_t

POECR4 bit definitions (add POE# flag conditions for MTU3/4 and MTU6/7).

Register Layout (16-bit @ 0x0008C4D0)

Bits	Field	Description
1: 0	-	Reserved (b0 reads 0, b1 reads 1, write as read)
2	IC2ADDMT34ZE	Add POE4F condition to MTU3/4 hi-Z
3	IC3ADDMT34ZE	Add POE8F condition to MTU3/4 hi-Z
4	IC4ADDMT34ZE	Add POE10F condition to MTU3/4 hi-Z
5	IC5ADDMT34ZE	Add POE11F condition to MTU3/4 hi-Z
8: 6	-	Reserved
9	IC1ADDMT67ZE	Add POE0F condition to MTU6/7 hi-Z
10	-	Reserved (reads 1, write 1)
11	IC3ADDMT67ZE	Add POE8F condition to MTU6/7 hi-Z
12	IC4ADDMT67ZE	Add POE10F condition to MTU6/7 hi-Z
13	IC5ADDMT67ZE	Add POE11F condition to MTU6/7 hi-Z
15:14	-	Reserved

POECR4 extends the high-impedance control conditions for MTU3/4 and MTU6/7 pins by adding optional POE# flag triggers.

Note

These bits can only be modified once after reset.

Since

Version 1.0.0

Enumerator

k_poecr4_ic2addmt34ze	Add POE4F condition to MTU3/4 hi-Z (bit 2)
k_poecr4_ic3addmt34ze	Add POE8F condition to MTU3/4 hi-Z (bit 3)
k_poecr4_ic4addmt34ze	Add POE10F condition to MTU3/4 hi-Z (bit 4)
k_poecr4_ic5addmt34ze	Add POE11F condition to MTU3/4 hi-Z (bit 5)
k_poecr4_ic1addmt67ze	Add POE0F condition to MTU6/7 hi-Z (bit 9)
k_poecr4_ic3addmt67ze	Add POE8F condition to MTU6/7 hi-Z (bit 11)
k_poecr4_ic4addmt67ze	Add POE10F condition to MTU6/7 hi-Z (bit 12)
k_poecr4_ic5addmt67ze	Add POE11F condition to MTU6/7 hi-Z (bit 13)

Definition at line 461 of file rx72n_poe3_regs.h.

```

00461     : uint16_t {
00462     /* MTU3/4 additional condition bits */
00463     k_poecr4_ic2addmt34ze = 0x0004U, /**< Add POE4F condition to MTU3/4 hi-Z (bit 2) */
00464     k_poecr4_ic3addmt34ze = 0x0008U, /**< Add POE8F condition to MTU3/4 hi-Z (bit 3) */
00465     k_poecr4_ic4addmt34ze = 0x0010U, /**< Add POE10F condition to MTU3/4 hi-Z (bit 4) */
00466     k_poecr4_ic5addmt34ze = 0x0020U, /**< Add POE11F condition to MTU3/4 hi-Z (bit 5) */
00467     /* MTU6/7 additional condition bits */
00468     k_poecr4_ic1addmt67ze = 0x0200U, /**< Add POE0F condition to MTU6/7 hi-Z (bit 9) */
00469     k_poecr4_ic3addmt67ze = 0x0800U, /**< Add POE8F condition to MTU6/7 hi-Z (bit 11) */
00470     k_poecr4_ic4addmt67ze = 0x1000U, /**< Add POE10F condition to MTU6/7 hi-Z (bit 12) */
00471     k_poecr4_ic5addmt67ze = 0x2000U, /**< Add POE11F condition to MTU6/7 hi-Z (bit 13) */
00472 } poecr4_bits_t;

```

8.41.2.12 poecr5_bits_t

```
enum poecr5_bits_t : uint16_t
```

POECR5 bit definitions (add POE# flag conditions for MTU0 pins).

Register Layout (16-bit @ 0x0008C4D2)

Bits	Field	Description
0	-	Reserved (reads 0, write 0)
1	IC1ADDMT0ZE	Add POE0F condition to MTU0 hi-Z
2	IC2ADDMT0ZE	Add POE4F condition to MTU0 hi-Z
3	-	Reserved (reads 1, write 1)
4	IC4ADDMT0ZE	Add POE10F condition to MTU0 hi-Z
5	IC5ADDMT0ZE	Add POE11F condition to MTU0 hi-Z
15:6	-	Reserved

POECR5 extends the high-impedance control conditions for MTU0 pins by adding optional POE# flag triggers beyond the default POE8F/SPOER conditions.

Note

These bits can only be modified once after reset.

Since

Version 1.0.0

Enumerator

k_poecr5_ic1addmt0ze	Add POE0F condition to MTU0 hi-Z (bit 1)
k_poecr5_ic2addmt0ze	Add POE4F condition to MTU0 hi-Z (bit 2)
k_poecr5_ic4addmt0ze	Add POE10F condition to MTU0 hi-Z (bit 4)
k_poecr5_ic5addmt0ze	Add POE11F condition to MTU0 hi-Z (bit 5)

Definition at line 501 of file rx72n_poe3_regs.h.

```
00501     : uint16_t {
00502   k_poecr5_ic1addmt0ze = 0x0002U, /**< Add POE0F condition to MTU0 hi-Z (bit 1) */
00503   k_poecr5_ic2addmt0ze = 0x0004U, /**< Add POE4F condition to MTU0 hi-Z (bit 2) */
00504   k_poecr5_ic4addmt0ze = 0x0010U, /**< Add POE10F condition to MTU0 hi-Z (bit 4) */
00505   k_poecr5_ic5addmt0ze = 0x0020U, /**< Add POE11F condition to MTU0 hi-Z (bit 5) */
00506 } poecr5_bits_t;
```

8.41.2.13 spoer_bits_t

enum [spoer_bits_t](#) : uint8_t

SPOER bit definitions (software high-impedance control).

Register Layout (8-bit @ 0x0008C4CA)

Bit	Field	Description
0	MTUCH34HIZ	MTU3/4 high-impedance enable
1	MTUCH67HIZ	MTU6/7 high-impedance enable
2	MTUCH0HIZ	MTU0 high-impedance enable
7:↔ 3	-	Reserved

Software-triggered emergency stop for MTU pins. Writing 1 puts outputs in high-impedance state immediately.

Note

Manual ref: Ch25 section 25.2.10

Since

Version 1.0.0

Enumerator

k_spoer_mtuch34hiz	MTU3/4 Pins High-Impedance Enable (bit 0).
k_spoer_mtuch67hiz	MTU6/7 Pins High-Impedance Enable (bit 1).
k_spoer_mtuch0hiz	MTU0 Pins High-Impedance Enable (bit 2).

Definition at line 349 of file [rx72n_poe3_regs.h](#).

```

00349         : uint8_t {
00350     /** @brief MTU3/4 Pins High-Impedance Enable (bit 0) */
00351     k_spoer_mtuch34hiz = 0x01U,
00352
00353     /** @brief MTU6/7 Pins High-Impedance Enable (bit 1) */
00354     k_spoer_mtuch67hiz = 0x02U,
00355
00356     /** @brief MTU0 Pins High-Impedance Enable (bit 2) */
00357     k_spoer_mtuch0hiz = 0x04U,
00358 } spoer_bits_t;

```

8.41.3 Function Documentation

8.41.3.1 poe3_alr1_reg()

```
volatile uint16_t * poe3_alr1_reg (
    void ) [inline], [static]
```

Get pointer to ALR1 register (active level).

Returns

Volatile pointer to ALR1 register (16-bit)

Since

Version 1.0.0

Definition at line 720 of file [rx72n_poe3_regs.h](#).

```
00721 {
00722     return (volatile uint16_t*)k_poe3_alr1_addr;
00723 }
```

References [k_poe3_alr1_addr](#).

8.41.3.2 poe3`icsr1`reg()

```
volatile uint16_t * poe3_icsr1_reg (
    void )    [inline], [static]
```

Get pointer to ICSR1 register (POE0# control).

Returns

Volatile pointer to ICSR1 register (16-bit)

Precondition

POE3 module stop bit (MSTPCRA.MSTPA9) must be cleared

Since

Version 1.0.0

Definition at line 583 of file [rx72n_poe3_regs.h](#).

```
00584 {
00585     return (volatile uint16_t*)k_poe3_icsr1_addr;
00586 }
```

References [k_poe3_icsr1_addr](#).

8.41.3.3 poe3`icsr2`reg()

```
volatile uint16_t * poe3_icsr2_reg (
    void )    [inline], [static]
```

Get pointer to ICSR2 register (POE4# control).

Returns

Volatile pointer to ICSR2 register (16-bit)

Since

Version 1.0.0

Definition at line 593 of file [rx72n_poe3_regs.h](#).

```
00594 {
00595     return (volatile uint16_t*)k_poe3_icsr2_addr;
00596 }
```

References [k_poe3_icsr2_addr](#).

8.41.3.4 poe3_icsr3_reg()

```
volatile uint16_t * poe3_icsr3_reg (  
    void )    [inline], [static]
```

Get pointer to ICSR3 register (POE8# control).

Returns

Volatile pointer to ICSR3 register (16-bit)

Since

Version 1.0.0

Definition at line 603 of file [rx72n_poe3_regs.h](#).

```
00604 {  
00605     return (volatile uint16_t*)k_poe3_icsr3_addr;  
00606 }
```

References [k_poe3_icsr3_addr](#).

8.41.3.5 poe3_icsr4_reg()

```
volatile uint16_t * poe3_icsr4_reg (  
    void )    [inline], [static]
```

Get pointer to ICSR4 register (POE10# control).

Returns

Volatile pointer to ICSR4 register (16-bit)

Since

Version 1.0.0

Definition at line 613 of file [rx72n_poe3_regs.h](#).

```
00614 {  
00615     return (volatile uint16_t*)k_poe3_icsr4_addr;  
00616 }
```

References [k_poe3_icsr4_addr](#).

8.41.3.6 `poe3_icsr5_reg()`

```
volatile uint16_t * poe3_icsr5_reg (  
    void )    [inline], [static]
```

Get pointer to ICSR5 register (POE11# control).

Returns

Volatile pointer to ICSR5 register (16-bit)

Since

Version 1.0.0

Definition at line 623 of file [rx72n_poe3_regs.h](#).

```
00624 {  
00625     return (volatile uint16_t*)k_poe3_icsr5_addr;  
00626 }
```

References [k_poe3_icsr5_addr](#).

8.41.3.7 `poe3_icsr6_reg()`

```
volatile uint16_t * poe3_icsr6_reg (  
    void )    [inline], [static]
```

Get pointer to ICSR6 register (oscillation stop).

Returns

Volatile pointer to ICSR6 register (16-bit)

Since

Version 1.0.0

Definition at line 633 of file [rx72n_poe3_regs.h](#).

```
00634 {  
00635     return (volatile uint16_t*)k_poe3_icsr6_addr;  
00636 }
```

References [k_poe3_icsr6_addr](#).

8.41.3.8 poe3_m0selr1_reg()

```
volatile uint8_t * poe3_m0selr1_reg (  
    void )    [inline], [static]
```

Get pointer to M0SELR1 register (MTU0 pin select 1).

Returns

Volatile pointer to M0SELR1 register (8-bit)

Since

Version 1.0.0

Definition at line 730 of file [rx72n_poe3_regs.h](#).

```
00731 {  
00732     return (volatile uint8_t*)k_poe3_m0selr1_addr;  
00733 }
```

References [k_poe3_m0selr1_addr](#).

8.41.3.9 poe3_m0selr2_reg()

```
volatile uint8_t * poe3_m0selr2_reg (  
    void )    [inline], [static]
```

Get pointer to M0SELR2 register (MTU0 pin select 2).

Returns

Volatile pointer to M0SELR2 register (8-bit)

Since

Version 1.0.0

Definition at line 740 of file [rx72n_poe3_regs.h](#).

```
00741 {  
00742     return (volatile uint8_t*)k_poe3_m0selr2_addr;  
00743 }
```

References [k_poe3_m0selr2_addr](#).

8.41.3.10 `poe3_m3selr_reg()`

```
volatile uint8_t * poe3_m3selr_reg (  
    void )    [inline], [static]
```

Get pointer to M3SELR register (MTU3 pin select).

Returns

Volatile pointer to M3SELR register (8-bit)

Since

Version 1.0.0

Definition at line 750 of file [rx72n_poe3_regs.h](#).

```
00751 {  
00752     return (volatile uint8_t*)k_poe3_m3selr_addr;  
00753 }
```

References [k_poe3_m3selr_addr](#).

8.41.3.11 `poe3_m4selr1_reg()`

```
volatile uint8_t * poe3_m4selr1_reg (  
    void )    [inline], [static]
```

Get pointer to M4SELR1 register (MTU4 pin select 1).

Returns

Volatile pointer to M4SELR1 register (8-bit)

Since

Version 1.0.0

Definition at line 760 of file [rx72n_poe3_regs.h](#).

```
00761 {  
00762     return (volatile uint8_t*)k_poe3_m4selr1_addr;  
00763 }
```

References [k_poe3_m4selr1_addr](#).

8.41.3.12 poe3_m4selr2_reg()

```
volatile uint8_t * poe3_m4selr2_reg (  
    void )    [inline], [static]
```

Get pointer to M4SELR2 register (MTU4 pin select 2).

Returns

Volatile pointer to M4SELR2 register (8-bit)

Since

Version 1.0.0

Definition at line 770 of file [rx72n_poe3_regs.h](#).

```
00771 {  
00772     return (volatile uint8_t*)k_poe3_m4selr2_addr;  
00773 }
```

References [k_poe3_m4selr2_addr](#).

8.41.3.13 poe3_m6selr_reg()

```
volatile uint8_t * poe3_m6selr_reg (  
    void )    [inline], [static]
```

Get pointer to M6SELR register (MTU6 pin select).

Returns

Volatile pointer to M6SELR register (8-bit)

Since

Version 1.0.0

Definition at line 780 of file [rx72n_poe3_regs.h](#).

```
00781 {  
00782     return (volatile uint8_t*)k_poe3_m6selr_addr;  
00783 }
```

References [k_poe3_m6selr_addr](#).

8.41.3.14 `poe3'ocsr1'reg()`

```
volatile uint16_t * poe3_ocsr1_reg (  
    void )    [inline], [static]
```

Get pointer to OCSR1 register (MTU3/4 shoot-through).

Returns

Volatile pointer to OCSR1 register (16-bit)

Since

Version 1.0.0

Definition at line 643 of file [rx72n_poe3_regs.h](#).

```
00644 {  
00645     return (volatile uint16_t*)k_poe3_ocsr1_addr;  
00646 }
```

References [k_poe3_ocsr1_addr](#).

8.41.3.15 `poe3'ocsr2'reg()`

```
volatile uint16_t * poe3_ocsr2_reg (  
    void )    [inline], [static]
```

Get pointer to OCSR2 register (MTU6/7 shoot-through).

Returns

Volatile pointer to OCSR2 register (16-bit)

Since

Version 1.0.0

Definition at line 653 of file [rx72n_poe3_regs.h](#).

```
00654 {  
00655     return (volatile uint16_t*)k_poe3_ocsr2_addr;  
00656 }
```

References [k_poe3_ocsr2_addr](#).

8.41.3.16 poe3_poecr1_reg()

```
volatile uint8_t * poe3_poecr1_reg (  
    void )    [inline], [static]
```

Get pointer to POECR1 register (MTU0 pin control).

Returns

Volatile pointer to POECR1 register (8-bit)

Since

Version 1.0.0

Definition at line 680 of file [rx72n_poe3_regs.h](#).

```
00681 {  
00682     return (volatile uint8_t*)k_poe3_poecr1_addr;  
00683 }
```

References [k_poe3_poecr1_addr](#).

8.41.3.17 poe3_poecr2_reg()

```
volatile uint16_t * poe3_poecr2_reg (  
    void )    [inline], [static]
```

Get pointer to POECR2 register (MTU3/4 and MTU6/7 pin control).

Returns

Volatile pointer to POECR2 register (16-bit)

Since

Version 1.0.0

Definition at line 690 of file [rx72n_poe3_regs.h](#).

```
00691 {  
00692     return (volatile uint16_t*)k_poe3_poecr2_addr;  
00693 }
```

References [k_poe3_poecr2_addr](#).

8.41.3.18 `poe3_poecr4_reg()`

```
volatile uint16_t * poe3_poecr4_reg (  
    void )    [inline], [static]
```

Get pointer to POECR4 register (MTU3/4 and MTU6/7 additional POE# conditions).

Returns

Volatile pointer to POECR4 register (16-bit)

Since

Version 1.0.0

Definition at line 700 of file [rx72n_poe3_regs.h](#).

```
00701 {  
00702     return (volatile uint16_t*)k_poe3_poecr4_addr;  
00703 }
```

References [k_poe3_poecr4_addr](#).

8.41.3.19 `poe3_poecr5_reg()`

```
volatile uint16_t * poe3_poecr5_reg (  
    void )    [inline], [static]
```

Get pointer to POECR5 register (MTU0 additional POE# conditions).

Returns

Volatile pointer to POECR5 register (16-bit)

Since

Version 1.0.0

Definition at line 710 of file [rx72n_poe3_regs.h](#).

```
00711 {  
00712     return (volatile uint16_t*)k_poe3_poecr5_addr;  
00713 }
```

References [k_poe3_poecr5_addr](#).

8.41.3.20 poe3`spoer`reg()

```
volatile uint8_t * poe3_spoer_reg (
    void ) [inline], [static]
```

Get pointer to SPOER register (software emergency stop).

Returns

Volatile pointer to SPOER register (8-bit)

Example - Emergency stop all MTU outputs:

```
// Immediately disable all MTU complementary PWM outputs
*poe3_spoer_reg() = k_spoer_mtuch0hiz | k_spoer_mtuch34hiz | k_spoer_mtuch67hiz;
```

Since

Version 1.0.0

Definition at line 670 of file rx72n_poe3_regs.h.

```
00671 {
00672     return (volatile uint8_t*)k_poe3_spoer_addr;
00673 }
```

References [k_poe3_spoer_addr](#).

8.42 rx72n`poe3`regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_poe3_regs.h
00003  * @brief RX72N Port Output Enable 3 (POE3) Register Definitions
00004  *
00005  * @details
00006  * Complete register definitions for the RX72N Port Output Enable 3 module.
00007  * POE3 provides emergency output disable for MTU (Multi-Function Timer Pulse Unit)
00008  * pins, enabling hardware-level motor protection.
00009  *
00010  * @par POE3 Overview
00011  * - Emergency output disable for MTU0, MTU3/4, MTU6/7 pins
00012  * - Input signal detection (POE0#, POE4#, POE8#, POE10#, POE11#)
00013  * - Simultaneous conduction detection (shoot-through protection)
00014  * - Oscillation stop detection response
00015  * - Software-triggered high-impedance state
00016  *
00017  * @par Target Pins for High-Impedance Control
00018  * | Timer | Pins Controlled |
00019  * |-----|-----|
00020  * | MTU0 | MTIOC0A, MTIOC0B, MTIOC0C, MTIOC0D |
00021  * | MTU3 | MTIOC3B, MTIOC3D |
00022  * | MTU4 | MTIOC4A, MTIOC4B, MTIOC4C, MTIOC4D |
00023  * | MTU6 | MTIOC6B, MTIOC6D |
00024  * | MTU7 | MTIOC7A, MTIOC7B, MTIOC7C, MTIOC7D |
00025  *
00026  * @par Register Organization (POE3 base: 0x0008C4C0)
00027  * | Offset | Size | Register | Description |
00028  * |-----|-----|-----|-----|
00029  * | 0x00 | 16 | ICSR1 | Input Level Control/Status 1 (POE0#) |
00030  * | 0x02 | 16 | OCSR1 | Output Level Control/Status 1 |
00031  * | 0x04 | 16 | ICSR2 | Input Level Control/Status 2 (POE4#) |
00032  * | 0x06 | 16 | OCSR2 | Output Level Control/Status 2 |
00033  * | 0x08 | 16 | ICSR3 | Input Level Control/Status 3 (POE8#) |
00034  * | 0x0A | 8 | SPOER | Software Port Output Enable |
00035  * | 0x0B | 8 | POECR1 | Port Output Enable Control 1 |
```

```

00036 * | 0x0C | 16 | POECR2 | Port Output Enable Control 2 (MTU3/4 b10:8, MTU6/7 b2:0) |
00037 * | 0x10 | 16 | POECR4 | Port Output Enable Control 4 (ICxADD conditions MT34/MT67) |
00038 * | 0x12 | 16 | POECR5 | Port Output Enable Control 5 (ICxADD conditions MT0) |
00039 * | 0x16 | 16 | ICSR4 | Input Level Control/Status 4 (POE10#) |
00040 * | 0x18 | 16 | ICSR5 | Input Level Control/Status 5 (POE11#) |
00041 * | 0x1A | 16 | ALR1 | Active Level Register 1 |
00042 * | 0x1C | 16 | ICSR6 | Input Level Control/Status 6 (osc stop) |
00043 * | 0x24 | 8 | M0SELR1 | MTU0 Pin Select 1 |
00044 * | 0x25 | 8 | M0SELR2 | MTU0 Pin Select 2 |
00045 * | 0x26 | 8 | M3SELR | MTU3 Pin Select |
00046 * | 0x27 | 8 | M4SELR1 | MTU4 Pin Select 1 |
00047 * | 0x28 | 8 | M4SELR2 | MTU4 Pin Select 2 |
00048 * | 0x2A | 8 | M6SELR | MTU6 Pin Select |
00049 *
00050 * @par NASA Power of 10 Compliance
00051 * - Rule 1: No goto, setjmp, or recursion
00052 * - Rule 2: N/A (no loops)
00053 * - Rule 3: No dynamic memory allocation
00054 * - Rule 4: Short accessor functions
00055 * - Rule 5: Static assertions verify addresses
00056 * - Rule 6: Minimal scope
00057 * - Rule 7: N/A
00058 * - Rule 8: All constants use C23 typed enums
00059 * - Rule 9: No function pointers
00060 * - Rule 10: Compiles with -Wall -Wextra -Werror
00061 *
00062 * @par Manual References
00063 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00064 * - Chapter 25: Port Output Enable 3 (POE3a), pages 1199-1240
00065 * - Section 25.2: Register Descriptions
00066 *
00067 * @author Locked, Inc.
00068 * @date 2026-01-29
00069 * @copyright Copyright (c) 2026 Locked Inc.
00070 * SPDX-License-Identifier: MIT
00071 * @since Version 1.0.0
00072 */
00073
00074 #pragma once
00075
00076 #include <stddef.h>
00077 #include <stdint.h>
00078
00079 #ifdef __cplusplus
00080 extern "C" {
00081 #endif
00082
00083 /*
=====
00084 * POE3 Base Address and Register Offsets
00085 *
=====
00086 */
00087 /**
00088 * @enum poe3_addresses_t
00089 * @brief POE3 module addresses
00090 *
00091 * @details
00092 * Base address and register addresses for POE3.
00093 * All addresses verified against RX72N Hardware Manual Ch25.
00094 *
00095 * @note Addresses verified 2026-01-29 against manual section 25.2
00096 * @since Version 1.0.0
00097 */
00098 typedef enum : uintptr_t {
00099     /** @brief POE3 module base address */
00100     k_poe3_base_addr = 0x0008C4C0U,
00101
00102     /** Input Level Control/Status Registers */
00103     /** @brief ICSR1 address - Input Level Control/Status 1 (POE0#) */
00104     k_poe3_icsr1_addr = 0x0008C4C0U,
00105
00106     /** @brief ICSR2 address - Input Level Control/Status 2 (POE4#) */
00107     k_poe3_icsr2_addr = 0x0008C4C4U,
00108
00109     /** @brief ICSR3 address - Input Level Control/Status 3 (POE8#) */
00110     k_poe3_icsr3_addr = 0x0008C4C8U,
00111
00112     /** @brief ICSR4 address - Input Level Control/Status 4 (POE10#) */
00113     k_poe3_icsr4_addr = 0x0008C4D6U,
00114
00115     /** @brief ICSR5 address - Input Level Control/Status 5 (POE11#) */
00116     k_poe3_icsr5_addr = 0x0008C4D8U,
00117
00118     /** @brief ICSR6 address - Input Level Control/Status 6 (osc stop) */
00119     k_poe3_icsr6_addr = 0x0008C4DCU,
00120

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00121
00122 /* Output Level Control/Status Registers */
00123 /** @brief OCSR1 address - Output Level Control/Status 1 (MTU3/4) */
00124 k_poe3_ocrs1_addr = 0x0008C4C2U,
00125
00126 /** @brief OCSR2 address - Output Level Control/Status 2 (MTU6/7) */
00127 k_poe3_ocrs2_addr = 0x0008C4C6U,
00128
00129 /* Control Registers */
00130 /** @brief SPOER address - Software Port Output Enable Register */
00131 k_poe3_spoer_addr = 0x0008C4CAU,
00132
00133 /** @brief POECR1 address - Port Output Enable Control 1 (MTU0) */
00134 k_poe3_poecr1_addr = 0x0008C4CBU,
00135
00136 /** @brief POECR2 address - Port Output Enable Control 2 (MTU3/4) */
00137 k_poe3_poecr2_addr = 0x0008C4CCU,
00138
00139 /** @brief POECR4 address - Port Output Enable Control 4 (MTU6/7) */
00140 k_poe3_poecr4_addr = 0x0008C4D0U,
00141
00142 /** @brief POECR5 address - Port Output Enable Control 5 (additional) */
00143 k_poe3_poecr5_addr = 0x0008C4D2U,
00144
00145 /** @brief ALR1 address - Active Level Register 1 */
00146 k_poe3_alr1_addr = 0x0008C4DAU,
00147
00148 /* Pin Select Registers */
00149 /** @brief M0SELR1 address - MTU0 Pin Select 1 */
00150 k_poe3_m0selr1_addr = 0x0008C4E4U,
00151
00152 /** @brief M0SELR2 address - MTU0 Pin Select 2 */
00153 k_poe3_m0selr2_addr = 0x0008C4E5U,
00154
00155 /** @brief M3SELR address - MTU3 Pin Select */
00156 k_poe3_m3selr_addr = 0x0008C4E6U,
00157
00158 /** @brief M4SELR1 address - MTU4 Pin Select 1 */
00159 k_poe3_m4selr1_addr = 0x0008C4E7U,
00160
00161 /** @brief M4SELR2 address - MTU4 Pin Select 2 */
00162 k_poe3_m4selr2_addr = 0x0008C4E8U,
00163
00164 /** @brief M6SELR address - MTU6 Pin Select */
00165 k_poe3_m6selr_addr = 0x0008C4EAU,
00166 } poe3_addresses_t;
00167
00168 /**
00169  * @enum poe3_offsets_t
00170  * @brief POE3 register offsets from base address
00171  *
00172  * @details
00173  * Offsets from POE3 base address (0x0008C4C0) for each register.
00174  *
00175  * @since Version 1.0.0
00176  */
00177 typedef enum : uint8_t {
00178     k_poe3_offset_icsr1 = 0x00U, /**< ICSR1 offset (16-bit) */
00179     k_poe3_offset_ocrs1 = 0x02U, /**< OCSR1 offset (16-bit) */
00180     k_poe3_offset_icsr2 = 0x04U, /**< ICSR2 offset (16-bit) */
00181     k_poe3_offset_ocrs2 = 0x06U, /**< OCSR2 offset (16-bit) */
00182     k_poe3_offset_icsr3 = 0x08U, /**< ICSR3 offset (16-bit) */
00183     k_poe3_offset_spoer = 0x0AU, /**< SPOER offset (8-bit) */
00184     k_poe3_offset_poecr1 = 0x0BU, /**< POECR1 offset (8-bit) */
00185     k_poe3_offset_poecr2 = 0x0CU, /**< POECR2 offset (8-bit) */
00186     k_poe3_offset_poecr4 = 0x10U, /**< POECR4 offset (16-bit) */
00187     k_poe3_offset_poecr5 = 0x12U, /**< POECR5 offset (16-bit) */
00188     k_poe3_offset_icsr4 = 0x16U, /**< ICSR4 offset (16-bit) */
00189     k_poe3_offset_icsr5 = 0x18U, /**< ICSR5 offset (16-bit) */
00190     k_poe3_offset_alr1 = 0x1AU, /**< ALR1 offset (16-bit) */
00191     k_poe3_offset_icsr6 = 0x1CU, /**< ICSR6 offset (16-bit) */
00192     k_poe3_offset_m0selr1 = 0x24U, /**< M0SELR1 offset (8-bit) */
00193     k_poe3_offset_m0selr2 = 0x25U, /**< M0SELR2 offset (8-bit) */
00194     k_poe3_offset_m3selr = 0x26U, /**< M3SELR offset (8-bit) */
00195     k_poe3_offset_m4selr1 = 0x27U, /**< M4SELR1 offset (8-bit) */
00196     k_poe3_offset_m4selr2 = 0x28U, /**< M4SELR2 offset (8-bit) */
00197     k_poe3_offset_m6selr = 0x2AU, /**< M6SELR offset (8-bit) */
00198 } poe3_offsets_t;
00199
00200 /*
=====
00201 * ICSR1/2/4/5 Register Bits (Input Level Control/Status)
00202 *
=====
00203 */
00204
00205 /**

```

```

00206 * @enum icshr_poemode_t
00207 * @brief POE# Mode Select bits (POE0M/POE4M/POE8M/POE10M/POE11M)
00208 *
00209 * @details
00210 * Selects input detection mode for POE# pins.
00211 * These bits can only be modified once after reset.
00212 *
00213 * @note Manual ref: Ch25 sections 25.2.1-25.2.5
00214 * @since Version 1.0.0
00215 */
00216 typedef enum : uint16_t {
00217     /** @brief Accept on falling edge of POE# input */
00218     k_icshr_poemode_falling_edge = 0x0000U,
00219
00220     /** @brief Low-level sampling: 16 samples at PCLK/8 */
00221     k_icshr_poemode_lowlevel_pclk8 = 0x0001U,
00222
00223     /** @brief Low-level sampling: 16 samples at PCLK/16 */
00224     k_icshr_poemode_lowlevel_pclk16 = 0x0002U,
00225
00226     /** @brief Low-level sampling: 16 samples at PCLK/128 */
00227     k_icshr_poemode_lowlevel_pclk128 = 0x0003U,
00228
00229     /** @brief POEm mode select mask (bits 1:0) */
00230     k_icshr_poemode_mask = 0x0003U,
00231 } icshr_poemode_t;
00232
00233 /**
00234 * @enum icshr_bits_t
00235 * @brief ICSR common bit definitions
00236 *
00237 * @par Register Layout (16-bit)
00238 * | Bits | Field | Description |
00239 * |-----|-----|-----|
00240 * | 1:0 | POEmM | POE# mode select |
00241 * | 7:2 | - | Reserved |
00242 * | 8 | PIEn | Port interrupt enable |
00243 * | 9 | POE8E | High-impedance enable (ICSR3 only) |
00244 * | 11:10 | - | Reserved |
00245 * | 12 | POEmF | POE# flag |
00246 * | 15:13 | - | Reserved |
00247 *
00248 * @since Version 1.0.0
00249 */
00250 typedef enum : uint16_t {
00251     /* Common bits for ICSR1/2/3/4/5 */
00252     /** @brief Port Interrupt Enable (bit 8) */
00253     k_icshr_pie = 0x0100U,
00254
00255     /** @brief POE# Flag (bit 12) - high-impedance request detected */
00256     k_icshr_poef = 0x1000U,
00257
00258     /* ICSR3-specific bits */
00259     /** @brief POE8 High-Impedance Enable (bit 9, ICSR3 only) */
00260     k_icshr3_poe8e = 0x0200U,
00261 } icshr_bits_t;
00262
00263 /**
00264 * @enum icshr6_bits_t
00265 * @brief ICSR6 bit definitions (Oscillation Stop Detection)
00266 *
00267 * @par Register Layout (16-bit @ 0x0008C4DC)
00268 * | Bits | Field | Description |
00269 * |-----|-----|-----|
00270 * | 8:0 | - | Reserved |
00271 * | 9 | OSTSTE | Osc stop high-impedance enable |
00272 * | 11:10 | - | Reserved |
00273 * | 12 | OSTSTF | Osc stop flag |
00274 * | 15:13 | - | Reserved |
00275 *
00276 * @since Version 1.0.0
00277 */
00278 typedef enum : uint16_t {
00279     /** @brief Oscillation Stop High-Impedance Enable (bit 9) */
00280     k_icshr6_ostste = 0x0200U,
00281
00282     /** @brief Oscillation Stop Flag (bit 12) */
00283     k_icshr6_oststf = 0x1000U,
00284 } icshr6_bits_t;
00285
00286 /*
=====
00287 * OCSR1/2 Register Bits (Output Level Control/Status)
00288 *
=====
00289 */
00290

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```

00291 /**
00292  * @enum ocsr_bits_t
00293  * @brief OCSR bit definitions (simultaneous conduction detection)
00294  *
00295  * @par Register Layout (16-bit)
00296  * | Bits | Field | Description |
00297  * |-----|-----|-----|
00298  * | 7:0 | - | Reserved |
00299  * | 8 | OIEn | Simultaneous conduction interrupt enable |
00300  * | 9 | OCEn | Simultaneous conduction high-impedance enable |
00301  * | 14:10 | - | Reserved |
00302  * | 15 | OSFn | Simultaneous conduction flag |
00303  *
00304  * @details
00305  * Used to detect and respond to simultaneous active-level output on
00306  * complementary PWM pin pairs (shoot-through detection).
00307  *
00308  * @note Manual ref: Ch25 sections 25.2.7, 25.2.8
00309  * @since Version 1.0.0
00310  */
00311 typedef enum : uint16_t {
00312     /** @brief Simultaneous Conduction Interrupt Enable (bit 8) */
00313     k_ocrs_oie = 0x0100U,
00314
00315     /**
00316      * @brief Simultaneous Conduction High-Impedance Enable (bit 9)
00317      * @details Can only be modified once after reset.
00318      */
00319     k_ocrs_oce = 0x0200U,
00320
00321     /** @brief Simultaneous Conduction Flag (bit 15) */
00322     k_ocrs_osf = 0x8000U,
00323 } ocsr_bits_t;
00324
00325 /*
=====
00326 * SPOER Register Bits (Software Port Output Enable)
00327 *
=====
00328 */
00329
00330 /**
00331  * @enum spoer_bits_t
00332  * @brief SPOER bit definitions (software high-impedance control)
00333  *
00334  * @par Register Layout (8-bit @ 0x0008C4CA)
00335  * | Bit | Field | Description |
00336  * |-----|-----|-----|
00337  * | 0 | MTUCH34HIZ | MTU3/4 high-impedance enable |
00338  * | 1 | MTUCH67HIZ | MTU6/7 high-impedance enable |
00339  * | 2 | MTUCH0HIZ | MTU0 high-impedance enable |
00340  * | 7:3 | - | Reserved |
00341  *
00342  * @details
00343  * Software-triggered emergency stop for MTU pins.
00344  * Writing 1 puts outputs in high-impedance state immediately.
00345  *
00346  * @note Manual ref: Ch25 section 25.2.10
00347  * @since Version 1.0.0
00348  */
00349 typedef enum : uint8_t {
00350     /** @brief MTU3/4 Pins High-Impedance Enable (bit 0) */
00351     k_spoer_mtuch34hiz = 0x01U,
00352
00353     /** @brief MTU6/7 Pins High-Impedance Enable (bit 1) */
00354     k_spoer_mtuch67hiz = 0x02U,
00355
00356     /** @brief MTU0 Pins High-Impedance Enable (bit 2) */
00357     k_spoer_mtuch0hiz = 0x04U,
00358 } spoer_bits_t;
00359
00360 /*
=====
00361 * POECR1 Register Bits (MTU0 Pin Control)
00362 *
=====
00363 */
00364
00365 /**
00366  * @enum poecr1_bits_t
00367  * @brief POECR1 bit definitions (MTU0 pin high-impedance enable)
00368  *
00369  * @par Register Layout (8-bit @ 0x0008C4CB)
00370  * | Bit | Field | Description |
00371  * |-----|-----|-----|
00372  * | 0 | MTU0AZE | MTIOC0A high-impedance enable |
00373  * | 1 | MTU0BZE | MTIOC0B high-impedance enable |

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00374 * | 2 | MTU0CZE | MTIOC0C high-impedance enable |
00375 * | 3 | MTU0DZE | MTIOC0D high-impedance enable |
00376 * | 7:4 | - | Reserved |
00377 *
00378 * @note These bits can only be modified once after reset.
00379 * @since Version 1.0.0
00380 */
00381 typedef enum : uint8_t {
00382     k_poecr1_mtu0aze = 0x01U, /**< MTIOC0A high-impedance enable */
00383     k_poecr1_mtu0bze = 0x02U, /**< MTIOC0B high-impedance enable */
00384     k_poecr1_mtu0cze = 0x04U, /**< MTIOC0C high-impedance enable */
00385     k_poecr1_mtu0dze = 0x08U, /**< MTIOC0D high-impedance enable */
00386 } poecr1_bits_t;
00387
00388 /*
=====
00389 * POECR2 Register Bits (MTU3/4 Pin Control)
00390 *
=====
00391 */
00392
00393 /**
00394 * @enum poecr2_bits_t
00395 * @brief POECR2 bit definitions (MTU3/4 and MTU6/7 complementary PWM control)
00396 *
00397 * @par Register Layout (16-bit @ 0x0008C4CC)
00398 * | Bits | Field | Description |
00399 * |-----|-----|-----|
00400 * | 1:0 | - | Reserved (MTU7BDZE, MTU7ACZE in b0,b1) |
00401 * | 0 | MTU7BDZE | MTIOC7B/7D high-impedance enable |
00402 * | 1 | MTU7ACZE | MTIOC7A/7C high-impedance enable |
00403 * | 2 | MTU6BDZE | MTIOC6B/6D high-impedance enable |
00404 * | 7:3 | - | Reserved |
00405 * | 8 | MTU4BDZE | MTIOC4B/4D high-impedance enable |
00406 * | 9 | MTU4ACZE | MTIOC4A/4C high-impedance enable |
00407 * | 10 | MTU3BDZE | MTIOC3B/3D high-impedance enable |
00408 * | 15:11 | - | Reserved |
00409 *
00410 * @details
00411 * Reset value: b10=1, b9=1, b8=1, b2=1, b1=1, b0=1 (all ZE bits set).
00412 * Note 2 in the manual: Set MTU6/7 bits (b2:b0) to 0 when MTU6 or MTU7
00413 * is not used.
00414 *
00415 * @note These bits can only be modified once after reset.
00416 * @since Version 1.0.0
00417 */
00418 typedef enum : uint16_t {
00419     /* MTU6/7 hi-Z enable (lower byte, bits 2:0) */
00420     k_poecr2_mtu7bdze = 0x0001U, /**< MTIOC7B/7D high-impedance enable (bit 0) */
00421     k_poecr2_mtu7acze = 0x0002U, /**< MTIOC7A/7C high-impedance enable (bit 1) */
00422     k_poecr2_mtu6bdze = 0x0004U, /**< MTIOC6B/6D high-impedance enable (bit 2) */
00423     /* MTU3/4 hi-Z enable (upper byte, bits 10:8) */
00424     k_poecr2_mtu4bdze = 0x0100U, /**< MTIOC4B/4D high-impedance enable (bit 8) */
00425     k_poecr2_mtu4acze = 0x0200U, /**< MTIOC4A/4C high-impedance enable (bit 9) */
00426     k_poecr2_mtu3bdze = 0x0400U, /**< MTIOC3B/3D high-impedance enable (bit 10) */
00427 } poecr2_bits_t;
00428
00429 /*
=====
00430 * POECR4 Register Bits (MTU6/7 Pin Control)
00431 *
=====
00432 */
00433
00434 /**
00435 * @enum poecr4_bits_t
00436 * @brief POECR4 bit definitions (add POE# flag conditions for MTU3/4 and MTU6/7)
00437 *
00438 * @par Register Layout (16-bit @ 0x0008C4D0)
00439 * | Bits | Field | Description |
00440 * |-----|-----|-----|
00441 * | 1:0 | - | Reserved (b0 reads 0, b1 reads 1, write as read) |
00442 * | 2 | IC2ADDMT34ZE | Add POE4F condition to MTU3/4 hi-Z |
00443 * | 3 | IC3ADDMT34ZE | Add POE8F condition to MTU3/4 hi-Z |
00444 * | 4 | IC4ADDMT34ZE | Add POE10F condition to MTU3/4 hi-Z |
00445 * | 5 | IC5ADDMT34ZE | Add POE11F condition to MTU3/4 hi-Z |
00446 * | 8:6 | - | Reserved |
00447 * | 9 | IC1ADDMT67ZE | Add POE0F condition to MTU6/7 hi-Z |
00448 * | 10 | - | Reserved (reads 1, write 1) |
00449 * | 11 | IC3ADDMT67ZE | Add POE8F condition to MTU6/7 hi-Z |
00450 * | 12 | IC4ADDMT67ZE | Add POE10F condition to MTU6/7 hi-Z |
00451 * | 13 | IC5ADDMT67ZE | Add POE11F condition to MTU6/7 hi-Z |
00452 * | 15:14 | - | Reserved |
00453 *
00454 * @details
00455 * POECR4 extends the high-impedance control conditions for MTU3/4 and MTU6/7
00456 * pins by adding optional POE# flag triggers.

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```

00457 *
00458 * @note These bits can only be modified once after reset.
00459 * @since Version 1.0.0
00460 */
00461 typedef enum : uint16_t {
00462     /* MTU3/4 additional condition bits */
00463     k_poecr4_ic2addmt34ze = 0x0004U, /**< Add POE4F condition to MTU3/4 hi-Z (bit 2) */
00464     k_poecr4_ic3addmt34ze = 0x0008U, /**< Add POE8F condition to MTU3/4 hi-Z (bit 3) */
00465     k_poecr4_ic4addmt34ze = 0x0010U, /**< Add POE10F condition to MTU3/4 hi-Z (bit 4) */
00466     k_poecr4_ic5addmt34ze = 0x0020U, /**< Add POE11F condition to MTU3/4 hi-Z (bit 5) */
00467     /* MTU6/7 additional condition bits */
00468     k_poecr4_ic1addmt67ze = 0x0200U, /**< Add POE0F condition to MTU6/7 hi-Z (bit 9) */
00469     k_poecr4_ic3addmt67ze = 0x0800U, /**< Add POE8F condition to MTU6/7 hi-Z (bit 11) */
00470     k_poecr4_ic4addmt67ze = 0x1000U, /**< Add POE10F condition to MTU6/7 hi-Z (bit 12) */
00471     k_poecr4_ic5addmt67ze = 0x2000U, /**< Add POE11F condition to MTU6/7 hi-Z (bit 13) */
00472 } poecr4_bits_t;
00473
00474 /*
=====
00475 * POECR5 Register Bits (Additional Pin Control)
00476 *
=====
00477 */
00478
00479 /**
00480 * @enum poecr5_bits_t
00481 * @brief POECR5 bit definitions (add POE# flag conditions for MTU0 pins)
00482 *
00483 * @par Register Layout (16-bit @ 0x0008C4D2)
00484 * | Bits | Field | Description |
00485 * |-----|-----|-----|
00486 * | 0 | - | Reserved (reads 0, write 0) |
00487 * | 1 | IC1ADDMT0ZE | Add POE0F condition to MTU0 hi-Z |
00488 * | 2 | IC2ADDMT0ZE | Add POE4F condition to MTU0 hi-Z |
00489 * | 3 | - | Reserved (reads 1, write 1) |
00490 * | 4 | IC4ADDMT0ZE | Add POE10F condition to MTU0 hi-Z |
00491 * | 5 | IC5ADDMT0ZE | Add POE11F condition to MTU0 hi-Z |
00492 * | 15:6 | - | Reserved |
00493 *
00494 * @details
00495 * POECR5 extends the high-impedance control conditions for MTU0 pins by
00496 * adding optional POE# flag triggers beyond the default POE8F/SPOER conditions.
00497 *
00498 * @note These bits can only be modified once after reset.
00499 * @since Version 1.0.0
00500 */
00501 typedef enum : uint16_t {
00502     k_poecr5_ic1addmt0ze = 0x0002U, /**< Add POE0F condition to MTU0 hi-Z (bit 1) */
00503     k_poecr5_ic2addmt0ze = 0x0004U, /**< Add POE4F condition to MTU0 hi-Z (bit 2) */
00504     k_poecr5_ic4addmt0ze = 0x0010U, /**< Add POE10F condition to MTU0 hi-Z (bit 4) */
00505     k_poecr5_ic5addmt0ze = 0x0020U, /**< Add POE11F condition to MTU0 hi-Z (bit 5) */
00506 } poecr5_bits_t;
00507
00508 /*
=====
00509 * ALR1 Register Bits (Active Level Setting)
00510 *
=====
00511 */
00512
00513 /**
00514 * @enum alr1_bits_t
00515 * @brief ALR1 bit definitions (active level for shoot-through detection)
00516 *
00517 * @par Register Layout (16-bit @ 0x0008C4DA)
00518 * | Bits | Field | Description |
00519 * |-----|-----|-----|
00520 * | 0 | OLSG0A | Active level for MTIOC3B/4A (group 0, positive) |
00521 * | 1 | OLSG0B | Active level for MTIOC3D/4C (group 0, negative) |
00522 * | 2 | OLSG1A | Active level for MTIOC3B/4A (group 1, positive) |
00523 * | 3 | OLSG1B | Active level for MTIOC3D/4C (group 1, negative) |
00524 * | 4 | OLSG2A | Active level for MTIOC4B (group 2, positive) |
00525 * | 5 | OLSG2B | Active level for MTIOC4D (group 2, negative) |
00526 * | 6 | - | Reserved |
00527 * | 7 | OLSEN | Active level select enable |
00528 * | 15:8 | - | Reserved |
00529 *
00530 * @details
00531 * When OLSEN=0, active level determined by MTU.TOCR registers.
00532 * When OLSEN=1, active level determined by OLSGnA/B bits (0=low, 1=high).
00533 *
00534 * @since Version 1.0.0
00535 */
00536 typedef enum : uint16_t {
00537     k_alr1_olsg0a = 0x0001U, /**< Active level for group 0 positive */
00538     k_alr1_olsg0b = 0x0002U, /**< Active level for group 0 negative */
00539     k_alr1_olsg1a = 0x0004U, /**< Active level for group 1 positive */

```

```

00540 k_alr1_olsg1b = 0x0008U, /**< Active level for group 1 negative */
00541 k_alr1_olsg2a = 0x0010U, /**< Active level for group 2 positive */
00542 k_alr1_olsg2b = 0x0020U, /**< Active level for group 2 negative */
00543 k_alr1_olsen = 0x0080U, /**< Active level select enable */
00544 } alr1_bits_t;
00545
00546 /*
=====
00547 * Module Stop Control
00548 *
=====
00549 */
00550
00551 /**
00552 * @enum poe3_mstpcr_t
00553 * @brief POE3 module stop control bits
00554 *
00555 * @details
00556 * POE3 is controlled by MSTPCRA.MSTPA9.
00557 * Must be cleared (0) before accessing POE3 registers.
00558 *
00559 * @note Manual ref: Ch11 (Low Power Consumption) module stop tables
00560 * @since Version 1.0.0
00561 */
00562 typedef enum : uint32_t {
00563 /**
00564 * @brief MSTPCRA bit for POE3 (bit 9)
00565 * @details
00566 * - 0: POE3 module is running
00567 * - 1: POE3 module is stopped (default)
00568 */
00569 k_poe3_mstpcra_bit = (1U « 9U),
00570 } poe3_mstpcr_t;
00571
00572 /*
=====
00573 * Register Accessor Functions
00574 *
=====
00575 */
00576
00577 /**
00578 * @brief Get pointer to ICSR1 register (POE0# control)
00579 * @return Volatile pointer to ICSR1 register (16-bit)
00580 * @pre POE3 module stop bit (MSTPCRA.MSTPA9) must be cleared
00581 * @since Version 1.0.0
00582 */
00583 static inline volatile uint16_t* poe3_icsr1_reg(void)
00584 {
00585     return (volatile uint16_t*)k_poe3_icsr1_addr;
00586 }
00587
00588 /**
00589 * @brief Get pointer to ICSR2 register (POE4# control)
00590 * @return Volatile pointer to ICSR2 register (16-bit)
00591 * @since Version 1.0.0
00592 */
00593 static inline volatile uint16_t* poe3_icsr2_reg(void)
00594 {
00595     return (volatile uint16_t*)k_poe3_icsr2_addr;
00596 }
00597
00598 /**
00599 * @brief Get pointer to ICSR3 register (POE8# control)
00600 * @return Volatile pointer to ICSR3 register (16-bit)
00601 * @since Version 1.0.0
00602 */
00603 static inline volatile uint16_t* poe3_icsr3_reg(void)
00604 {
00605     return (volatile uint16_t*)k_poe3_icsr3_addr;
00606 }
00607
00608 /**
00609 * @brief Get pointer to ICSR4 register (POE10# control)
00610 * @return Volatile pointer to ICSR4 register (16-bit)
00611 * @since Version 1.0.0
00612 */
00613 static inline volatile uint16_t* poe3_icsr4_reg(void)
00614 {
00615     return (volatile uint16_t*)k_poe3_icsr4_addr;
00616 }
00617
00618 /**
00619 * @brief Get pointer to ICSR5 register (POE11# control)
00620 * @return Volatile pointer to ICSR5 register (16-bit)
00621 * @since Version 1.0.0
00622 */

```

```

00623 static inline volatile uint16_t* poe3_icsr5_reg(void)
00624 {
00625     return (volatile uint16_t*)k_poe3_icsr5_addr;
00626 }
00627
00628 /**
00629  * @brief Get pointer to ICSR6 register (oscillation stop)
00630  * @return Volatile pointer to ICSR6 register (16-bit)
00631  * @since Version 1.0.0
00632  */
00633 static inline volatile uint16_t* poe3_icsr6_reg(void)
00634 {
00635     return (volatile uint16_t*)k_poe3_icsr6_addr;
00636 }
00637
00638 /**
00639  * @brief Get pointer to OCSR1 register (MTU3/4 shoot-through)
00640  * @return Volatile pointer to OCSR1 register (16-bit)
00641  * @since Version 1.0.0
00642  */
00643 static inline volatile uint16_t* poe3_ocrs1_reg(void)
00644 {
00645     return (volatile uint16_t*)k_poe3_ocrs1_addr;
00646 }
00647
00648 /**
00649  * @brief Get pointer to OCSR2 register (MTU6/7 shoot-through)
00650  * @return Volatile pointer to OCSR2 register (16-bit)
00651  * @since Version 1.0.0
00652  */
00653 static inline volatile uint16_t* poe3_ocrs2_reg(void)
00654 {
00655     return (volatile uint16_t*)k_poe3_ocrs2_addr;
00656 }
00657
00658 /**
00659  * @brief Get pointer to SPOER register (software emergency stop)
00660  * @return Volatile pointer to SPOER register (8-bit)
00661  *
00662  * @par Example - Emergency stop all MTU outputs:
00663  * @code{.c}
00664  * // Immediately disable all MTU complementary PWM outputs
00665  * *poe3_spoer_reg() = k_spoer_mtuch0hiz | k_spoer_mtuch34hiz | k_spoer_mtuch67hiz;
00666  * @endcode
00667  *
00668  * @since Version 1.0.0
00669  */
00670 static inline volatile uint8_t* poe3_spoer_reg(void)
00671 {
00672     return (volatile uint8_t*)k_poe3_spoer_addr;
00673 }
00674
00675 /**
00676  * @brief Get pointer to POECR1 register (MTU0 pin control)
00677  * @return Volatile pointer to POECR1 register (8-bit)
00678  * @since Version 1.0.0
00679  */
00680 static inline volatile uint8_t* poe3_poecr1_reg(void)
00681 {
00682     return (volatile uint8_t*)k_poe3_poecr1_addr;
00683 }
00684
00685 /**
00686  * @brief Get pointer to POECR2 register (MTU3/4 and MTU6/7 pin control)
00687  * @return Volatile pointer to POECR2 register (16-bit)
00688  * @since Version 1.0.0
00689  */
00690 static inline volatile uint16_t* poe3_poecr2_reg(void)
00691 {
00692     return (volatile uint16_t*)k_poe3_poecr2_addr;
00693 }
00694
00695 /**
00696  * @brief Get pointer to POECR4 register (MTU3/4 and MTU6/7 additional POE# conditions)
00697  * @return Volatile pointer to POECR4 register (16-bit)
00698  * @since Version 1.0.0
00699  */
00700 static inline volatile uint16_t* poe3_poecr4_reg(void)
00701 {
00702     return (volatile uint16_t*)k_poe3_poecr4_addr;
00703 }
00704
00705 /**
00706  * @brief Get pointer to POECR5 register (MTU0 additional POE# conditions)
00707  * @return Volatile pointer to POECR5 register (16-bit)
00708  * @since Version 1.0.0
00709  */

```



```

00710 static inline volatile uint16_t* poe3_poecr5_reg(void)
00711 {
00712     return (volatile uint16_t*)k_poe3_poecr5_addr;
00713 }
00714
00715 /**
00716  * @brief Get pointer to ALR1 register (active level)
00717  * @return Volatile pointer to ALR1 register (16-bit)
00718  * @since Version 1.0.0
00719  */
00720 static inline volatile uint16_t* poe3_alr1_reg(void)
00721 {
00722     return (volatile uint16_t*)k_poe3_alr1_addr;
00723 }
00724
00725 /**
00726  * @brief Get pointer to M0SELR1 register (MTU0 pin select 1)
00727  * @return Volatile pointer to M0SELR1 register (8-bit)
00728  * @since Version 1.0.0
00729  */
00730 static inline volatile uint8_t* poe3_m0selr1_reg(void)
00731 {
00732     return (volatile uint8_t*)k_poe3_m0selr1_addr;
00733 }
00734
00735 /**
00736  * @brief Get pointer to M0SELR2 register (MTU0 pin select 2)
00737  * @return Volatile pointer to M0SELR2 register (8-bit)
00738  * @since Version 1.0.0
00739  */
00740 static inline volatile uint8_t* poe3_m0selr2_reg(void)
00741 {
00742     return (volatile uint8_t*)k_poe3_m0selr2_addr;
00743 }
00744
00745 /**
00746  * @brief Get pointer to M3SELR register (MTU3 pin select)
00747  * @return Volatile pointer to M3SELR register (8-bit)
00748  * @since Version 1.0.0
00749  */
00750 static inline volatile uint8_t* poe3_m3selr_reg(void)
00751 {
00752     return (volatile uint8_t*)k_poe3_m3selr_addr;
00753 }
00754
00755 /**
00756  * @brief Get pointer to M4SELR1 register (MTU4 pin select 1)
00757  * @return Volatile pointer to M4SELR1 register (8-bit)
00758  * @since Version 1.0.0
00759  */
00760 static inline volatile uint8_t* poe3_m4selr1_reg(void)
00761 {
00762     return (volatile uint8_t*)k_poe3_m4selr1_addr;
00763 }
00764
00765 /**
00766  * @brief Get pointer to M4SELR2 register (MTU4 pin select 2)
00767  * @return Volatile pointer to M4SELR2 register (8-bit)
00768  * @since Version 1.0.0
00769  */
00770 static inline volatile uint8_t* poe3_m4selr2_reg(void)
00771 {
00772     return (volatile uint8_t*)k_poe3_m4selr2_addr;
00773 }
00774
00775 /**
00776  * @brief Get pointer to M6SELR register (MTU6 pin select)
00777  * @return Volatile pointer to M6SELR register (8-bit)
00778  * @since Version 1.0.0
00779  */
00780 static inline volatile uint8_t* poe3_m6selr_reg(void)
00781 {
00782     return (volatile uint8_t*)k_poe3_m6selr_addr;
00783 }
00784
00785 /*
=====
00786  * Static Assertions - Address Verification
00787  *
=====
00788  */
00789
00790 /**
00791  * @name Address Verification
00792  * @brief Compile-time verification of register addresses
00793  * @{
00794  */

```



```

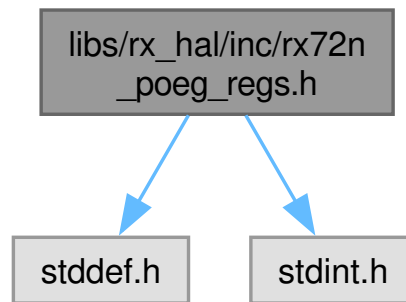
00795
00796 /* Verify base address */
00797 static_assert(k_poe3_base_addr == 0x0008C4C0U,
00798     "POE3 base address must be 0x0008C4C0 per Ch25 manual");
00799
00800 /* Verify ICSR register addresses */
00801 static_assert(k_poe3_icsr1_addr == 0x0008C4C0U, "ICSR1 address must be 0x0008C4C0 per Ch25 manual");
00802
00803 static_assert(k_poe3_icsr2_addr == 0x0008C4C4U, "ICSR2 address must be 0x0008C4C4 per Ch25 manual");
00804
00805 static_assert(k_poe3_icsr3_addr == 0x0008C4C8U, "ICSR3 address must be 0x0008C4C8 per Ch25 manual");
00806
00807 static_assert(k_poe3_icsr4_addr == 0x0008C4D6U, "ICSR4 address must be 0x0008C4D6 per Ch25 manual");
00808
00809 static_assert(k_poe3_icsr5_addr == 0x0008C4D8U, "ICSR5 address must be 0x0008C4D8 per Ch25 manual");
00810
00811 static_assert(k_poe3_icsr6_addr == 0x0008C4DCU, "ICSR6 address must be 0x0008C4DC per Ch25 manual");
00812
00813 /* Verify OCSR register addresses */
00814 static_assert(k_poe3_ocr1_addr == 0x0008C4C2U, "OCSR1 address must be 0x0008C4C2 per Ch25 manual");
00815
00816 static_assert(k_poe3_ocr2_addr == 0x0008C4C6U, "OCSR2 address must be 0x0008C4C6 per Ch25 manual");
00817
00818 /* Verify control register addresses */
00819 static_assert(k_poe3_spoer_addr == 0x0008C4CAU, "SPOER address must be 0x0008C4CA per Ch25 manual");
00820
00821 static_assert(k_poe3_poecr1_addr == 0x0008C4CBU,
00822     "POECR1 address must be 0x0008C4CB per Ch25 manual");
00823
00824 static_assert(k_poe3_poecr2_addr == 0x0008C4CCU,
00825     "POECR2 address must be 0x0008C4CC per Ch25 manual");
00826
00827 static_assert(k_poe3_poecr4_addr == 0x0008C4D0U,
00828     "POECR4 address must be 0x0008C4D0 per Ch25 manual");
00829
00830 static_assert(k_poe3_poecr5_addr == 0x0008C4D2U,
00831     "POECR5 address must be 0x0008C4D2 per Ch25 manual");
00832
00833 static_assert(k_poe3_alr1_addr == 0x0008C4DAU, "ALR1 address must be 0x0008C4DA per Ch25 manual");
00834
00835 /* Verify pin select register addresses */
00836 static_assert(k_poe3_m0selr1_addr == 0x0008C4E4U,
00837     "M0SELR1 address must be 0x0008C4E4 per Ch25 manual");
00838
00839 static_assert(k_poe3_m0selr2_addr == 0x0008C4E5U,
00840     "M0SELR2 address must be 0x0008C4E5 per Ch25 manual");
00841
00842 static_assert(k_poe3_m3selr_addr == 0x0008C4E6U,
00843     "M3SELR address must be 0x0008C4E6 per Ch25 manual");
00844
00845 static_assert(k_poe3_m4selr1_addr == 0x0008C4E7U,
00846     "M4SELR1 address must be 0x0008C4E7 per Ch25 manual");
00847
00848 static_assert(k_poe3_m4selr2_addr == 0x0008C4E8U,
00849     "M4SELR2 address must be 0x0008C4E8 per Ch25 manual");
00850
00851 static_assert(k_poe3_m6selr_addr == 0x0008C4EAU,
00852     "M6SELR address must be 0x0008C4EA per Ch25 manual");
00853
00854 /* Verify register offsets from base */
00855 static_assert(k_poe3_icsr1_addr - k_poe3_base_addr == k_poe3_offset_icsr1,
00856     "ICSR1 offset must match");
00857
00858 static_assert(k_poe3_ocr1_addr - k_poe3_base_addr == k_poe3_offset_ocr1,
00859     "OCSR1 offset must match");
00860
00861 static_assert(k_poe3_spoer_addr - k_poe3_base_addr == k_poe3_offset_spoer,
00862     "SPOER offset must match");
00863
00864 static_assert(k_poe3_alr1_addr - k_poe3_base_addr == k_poe3_offset_alr1, "ALR1 offset must match");
00865
00866 /** @} */
00867
00868 #ifdef __cplusplus
00869 }
00870 #endif

```

8.43 libs/rx_hal/inc/rx72n_poeg_regs.h File Reference

RX72N GPTW Port Output Enable (POEG) Register Definitions.

```
#include <stddef.h>
#include <stdint.h>
Include dependency graph for rx72n_poeg_regs.h:
```



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_poegg_regs_t](#)
POEG Group n Register Structure.

Enumerations

- enum [rx_poeg_addresses_t](#) : [uintptr_t](#) {
[k_poegga_base_addr](#) = 0x0009E000 , [k_poeggb_base_addr](#) = 0x0009E100 , [k_poeggc_base_addr](#) = 0x0009E200 , [k_poeggd_base_addr](#) = 0x0009E300 ,
[k_poeg_group_spacing](#) = 0x100 }
 POEG group base addresses.
- enum [rx_poegg_bit_shifts_t](#) : [uint8_t](#) {
[k_poeg_pidf_shift](#) = 0 , [k_poeg_iocf_shift](#) = 1 , [k_poeg_ostpf_shift](#) = 2 , [k_poeg_ssf_shift](#) = 3 ,
[k_poeg_pide_shift](#) = 4 , [k_poeg_ioce_shift](#) = 5 , [k_poeg_ostpe_shift](#) = 6 , [k_poeg_st_shift](#) = 16 ,
[k_poeg_inv_shift](#) = 28 , [k_poeg_nfen_shift](#) = 29 , [k_poeg_nfcs_shift](#) = 30 }
 POEGGn register bit field positions.
- enum [rx_poegg_bits_t](#) : [uint32_t](#) {
[k_poeg_pidf_detected](#) = (1U << [k_poeg_pidf_shift](#)) , [k_poeg_iocf_detected](#) = (1U << [k_poeg_iocf_shift](#)) , [k_poeg_ostpf_detected](#) = (1U << [k_poeg_ostpf_shift](#)) , [k_poeg_ssf_stop](#) = (1U << [k_poeg_ssf_shift](#)) ,
[k_poeg_pide_enable](#) = (1U << [k_poeg_pide_shift](#)) , [k_poeg_ioce_enable](#) = (1U << [k_poeg_ioce_shift](#)) , [k_poeg_ostpe_enable](#) = (1U << [k_poeg_ostpe_shift](#)) , [k_poeg_st_high](#) = (1U << [k_poeg_st_shift](#)) ,
[k_poeg_inv_invert](#) = (1U << [k_poeg_inv_shift](#)) , [k_poeg_nfen_enable](#) = (1U << [k_poeg_nfen_shift](#)) , [k_poeg_nfcs_pclkb_div1](#) = (0U << [k_poeg_nfcs_shift](#)) , [k_poeg_nfcs_pclkb_div8](#) = (1U << [k_poeg_nfcs_shift](#)) ,
[k_poeg_nfcs_pclkb_div32](#) = (2U << [k_poeg_nfcs_shift](#)) , [k_poeg_nfcs_pclkb_div128](#) = (3U << [k_poeg_nfcs_shift](#)) , [k_poeg_all_flags_mask](#) , [k_poeg_all_enable_mask](#) ,
[k_poeg_nfcs_mask](#) = (3U << [k_poeg_nfcs_shift](#)) }

POEGGn register bit definitions.

- enum `rx_poeg_interrupts_t` : `uint16_t` {
`k_poeg_irq_groupbl2_vector` = 107 , `k_poeg_grpbl2_bit_poeggai` = 7 , `k_poeg_grpbl2_bit_poeggbi` = 8 , `k_poeg_grpbl2_bit_poeggci` = 9 ,
`k_poeg_grpbl2_bit_poeggdi` = 10 }

POEG interrupt routing via GROUPBL2.

- enum `rx_poeg_group_regs_t` : `uintptr_t` { `k_poeg_grpbl2_addr` = 0x00087630 , `k_poeg_genbl2_addr` = 0x00087670 }

Memory-mapped addresses of the GROUPBL2 status/enable registers.

Functions

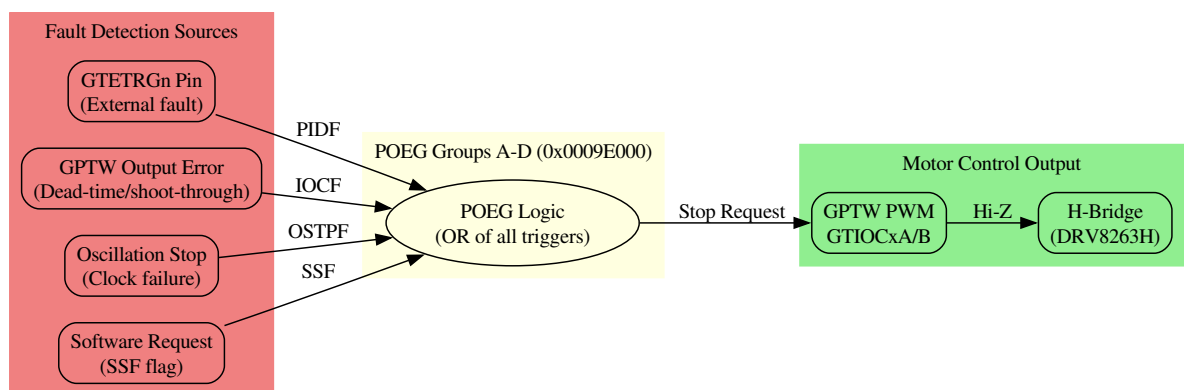
- static volatile `rx_poegg_regs_t` * `poegga` (void)
 Get pointer to POEG Group A registers.
- static volatile `rx_poegg_regs_t` * `poeggb` (void)
 Get pointer to POEG Group B registers.
- static volatile `rx_poegg_regs_t` * `poeggc` (void)
 Get pointer to POEG Group C registers.
- static volatile `rx_poegg_regs_t` * `poeggd` (void)
 Get pointer to POEG Group D registers.

8.43.1 Detailed Description

RX72N GPTW Port Output Enable (POEG) Register Definitions.

Register definitions for the POEG module which provides emergency PWM output disable functionality for the General PWM Timer (GPTW). The POEG is critical for motor control safety, allowing immediate PWM shutdown on fault detection.

Safety-Critical Emergency Stop System



STAR Project POEG Usage

Group	Assigned To	Trigger Source	Use Case
A	GPTW0 (Motor 0)	GTETRGA + Software	Front-left motor E-stop
B	GPTW1 (Motor 1)	GTETRGB + Software	Front-right motor E-stop
C	GPTW2 (Motor 2)	GTETRGC + Software	Back-right motor E-stop
D	GPTW3 (Motor 3)	GTETRGD + Software	Back-left motor E-stop

Key Features

- Emergency PWM output disable (immediate Hi-Z on fault)
- Multiple trigger sources: external pin, GPTW error, oscillation stop, software
- Digital noise filter on external trigger inputs
- Per-group independent control (4 groups for 4 motors)
- Interrupt generation on fault detection

Hardware Requirements

Parameter	Value	Notes
Base Address	0x0009E000	Group A base
Group Spacing	0x100 bytes	A=0x0000, B=0x0100, C=0x0200, D=0x0300
Module Stop	MSTPCRA.MSTPA7	Same as GPTW (shared)
Clock	PCLKB (60 MHz)	For noise filter sampling

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops in register definitions)
- Rule 3: No dynamic memory allocation
- Rule 4: All accessor functions are single-statement
- Rule 5: N/A (hardware layer)
- Rule 6: Minimal scope - inline accessors only
- Rule 7: N/A (no return values to check)
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Single responsibility - only POEG register definitions
- O: Open for extension via additional trigger configurations
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessor functions
- D: N/A (hardware register layer)

Module Dependencies

- `<stdint.h>` - Fixed-width integer types
- `<stddef.h>` - `offsetof` macro for static assertions

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 27: GPTW Port Output Enable (POEG), pages 1432-1441

See also

[rx72n_gptw_regs.h](#) General PWM Timer registers

[rx72n_regs.h](#) Main register include file

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Date

2026-01-29

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Since

Version 1.0.0

Definition in file [rx72n_poeg_regs.h](#).

8.43.2 Data Structure Documentation

8.43.2.1 struct rx72n_poeg_regs_t

POEG Group n Register Structure.

Each POEG group has a single 32-bit register (POEGGn) that controls emergency output disable for the associated GPTW channels. The register provides multiple trigger sources and status flags.

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	4	poeggn	uint32↵ _t	Group n Setting Register
Total	4			

Emergency Stop Trigger Sources

1. External Pin (PIDF): GTETRGN pin input detection
2. GPTW Output Error (IOCF): Dead-time error or shoot-through detection
3. Oscillation Stop (OSTPF): Main clock failure detection
4. Software (SSF): Direct software emergency stop command

Configuration Example

```
// Enable software emergency stop and external trigger for Group A
volatile rx_poegg_regs_t* poeg_a = poegga();

// Enable external pin detection (one-time write after reset)
poeg_a->poeggn = k_poeg_pide_enable;

// Later, trigger emergency stop via software
poeg_a->poeggn |= k_poeg_ssf_stop; // Immediate PWM disable

// Clear emergency stop (after fault resolved)
poeg_a->poeggn &= ~k_poeg_ssf_stop; // Resume PWM
```

Invariant

sizeof(rx_poegg_regs_t) == 4 bytes

Note

All triggers are OR'd together - any trigger stops output

Warning

Enable bits (PIDE, IOCE, OSTPE) can only be written once after reset

Attention

Module stop bit shared with GPTW (MSTPCRA.MSTPA7)

See also

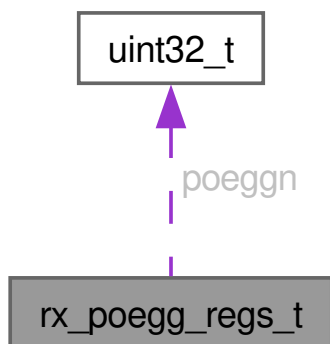
[poegga\(\)](#), [poeggb\(\)](#), [poeggc\(\)](#), [poeggd\(\)](#) Accessor functions
[rx_poegg_bits_t](#) Register bit definitions

Since

Version 1.0.0

Definition at line 226 of file [rx72n_poeg_regs.h](#).

Collaboration diagram for rx_poegg_regs_t:



Data Fields

volatile uint32_t	poeggn	POEG Group n Setting Register (POEGGn) @ offset 0x00. Controls emergency output disable triggers and status. <table><thead><tr><th>Bits</th><th>Field</th><th>R/W</th><th>Description</th></tr></thead><tbody><tr><td>0</td><td>PIDF</td><td>R/W*</td><td>Port Input Detection Flag</td></tr><tr><td>1</td><td>IOCF</td><td>R/W*</td><td>GPTW Output Stop Request Flag</td></tr><tr><td>2</td><td>OSTPF</td><td>R/W*</td><td>Oscillation Stop Detection Flag</td></tr><tr><td>3</td><td>SSF</td><td>R/W</td><td>Software Stop Flag</td></tr><tr><td>4</td><td>PIDE</td><td>R/W+</td><td>Port Input Detection Enable</td></tr><tr><td>5</td><td>IOCE</td><td>R/W+</td><td>GPTW Output Stop Request Enable</td></tr><tr><td>6</td><td>OSTPE</td><td>R/W+</td><td>Oscillation Stop Enable</td></tr><tr><td>15:7</td><td>-</td><td>R</td><td>Reserved (read as 0)</td></tr><tr><td>16</td><td>ST</td><td>R</td><td>GTETRGN Input Status</td></tr><tr><td>27:17</td><td>-</td><td>R</td><td>Reserved (read as 0)</td></tr><tr><td>28</td><td>INV</td><td>R/W</td><td>GTETRGN Input Inverting</td></tr><tr><td>29</td><td>NFEN</td><td>R/W</td><td>Noise Filter Enable</td></tr><tr><td>31:30</td><td>NFCS[1:0]</td><td>R/W</td><td>Noise Filter Clock Select</td></tr></tbody></table> Bit Layout (32 bits) Note * Only 0 can be written to clear flag + Can only be written once after reset See also rx_poegg_bits_t Bit definitions	Bits	Field	R/W	Description	0	PIDF	R/W*	Port Input Detection Flag	1	IOCF	R/W*	GPTW Output Stop Request Flag	2	OSTPF	R/W*	Oscillation Stop Detection Flag	3	SSF	R/W	Software Stop Flag	4	PIDE	R/W+	Port Input Detection Enable	5	IOCE	R/W+	GPTW Output Stop Request Enable	6	OSTPE	R/W+	Oscillation Stop Enable	15:7	-	R	Reserved (read as 0)	16	ST	R	GTETRGN Input Status	27:17	-	R	Reserved (read as 0)	28	INV	R/W	GTETRGN Input Inverting	29	NFEN	R/W	Noise Filter Enable	31:30	NFCS[1:0]	R/W	Noise Filter Clock Select
Bits	Field	R/W	Description																																																							
0	PIDF	R/W*	Port Input Detection Flag																																																							
1	IOCF	R/W*	GPTW Output Stop Request Flag																																																							
2	OSTPF	R/W*	Oscillation Stop Detection Flag																																																							
3	SSF	R/W	Software Stop Flag																																																							
4	PIDE	R/W+	Port Input Detection Enable																																																							
5	IOCE	R/W+	GPTW Output Stop Request Enable																																																							
6	OSTPE	R/W+	Oscillation Stop Enable																																																							
15:7	-	R	Reserved (read as 0)																																																							
16	ST	R	GTETRGN Input Status																																																							
27:17	-	R	Reserved (read as 0)																																																							
28	INV	R/W	GTETRGN Input Inverting																																																							
29	NFEN	R/W	Noise Filter Enable																																																							
31:30	NFCS[1:0]	R/W	Noise Filter Clock Select																																																							

8.43.3 Enumeration Type Documentation

8.43.3.1 rx_poeg_addresses_t

```
enum rx_poeg_addresses_t : uintptr_t
```

POEG group base addresses.

Base addresses for each POEG group (A-D). Each group controls emergency output disable for associated GPTW channels. Groups are spaced 0x100 bytes apart.

Memory Map

Address	Group	Controls	Description
0x0009E000	A	GPTW ch0-3	Emergency stop group A
0x0009E100	B	GPTW ch0-3	Emergency stop group B
0x0009E200	C	GPTW ch0-3	Emergency stop group C
0x0009E300	D	GPTW ch0-3	Emergency stop group D

Note

Manual: Ch27.2.1 - POEG Group n Setting Register

See also

[poegga\(\)](#) Accessor function for Group A

Since

Version 1.0.0

Enumerator

k_poegga_base_addr	POEG Group A base address (0x0009E000). Verified against RX72N Hardware Manual Ch27.2.1
k_poeggb_base_addr	POEG Group B base address (0x0009E100). Verified against RX72N Hardware Manual Ch27.2.1
k_poeggc_base_addr	POEG Group C base address (0x0009E200). Verified against RX72N Hardware Manual Ch27.2.1
k_poeggd_base_addr	POEG Group D base address (0x0009E300). Verified against RX72N Hardware Manual Ch27.2.1
k_poeg_group_spacing	Group spacing (0x100 bytes between groups). Each POEG group has a 256-byte register block

Definition at line 148 of file rx72n_poeg_regs.h.

```

00148         : uintptr_t {
00149     /**
00150      * @brief POEG Group A base address (0x0009E000)
00151      * @details Verified against RX72N Hardware Manual Ch27.2.1
00152      */
00153     k_poegga_base_addr = 0x0009E000,
00154
00155     /**
00156      * @brief POEG Group B base address (0x0009E100)
00157      * @details Verified against RX72N Hardware Manual Ch27.2.1
00158      */
00159     k_poeggb_base_addr = 0x0009E100,
00160
00161     /**
00162      * @brief POEG Group C base address (0x0009E200)
00163      * @details Verified against RX72N Hardware Manual Ch27.2.1
00164      */
00165     k_poeggc_base_addr = 0x0009E200,
00166
00167     /**
00168      * @brief POEG Group D base address (0x0009E300)
00169      * @details Verified against RX72N Hardware Manual Ch27.2.1
00170      */
00171     k_poeggd_base_addr = 0x0009E300,
00172
00173     /**
00174      * @brief Group spacing (0x100 bytes between groups)
00175      * @details Each POEG group has a 256-byte register block
00176      */
00177     k_poeg_group_spacing = 0x100,
00178 } rx_poeg_addresses_t;

```

8.43.3.2 rx`poeg`group`regs``t`

```
enum rx_poeg_group_regs_t : uintptr_t
```

Memory-mapped addresses of the GROUPBL2 status/enable registers.

Per RX72N HW manual R01UH0824EJ0111 Section 15.2 ("ICU Registers") the group interrupt registers are located in the ICU register block:

Register	Address	Purpose
GRPBL2	0x00087630	Group BL2 interrupt source status (R-only)
GENBL2	0x00087670	Group BL2 per-source enable (R/W)

Since

Version 1.0.0

Enumerator

k_poeg_grpbl2_addr	GRPBL2: sub-source status
k_poeg_genbl2_addr	GENBL2: sub-source enable

Definition at line 478 of file [rx72n_poeg_regs.h](#).

```
00478      : uintptr_t {
00479  k_poeg_grpbl2_addr = 0x00087630, /**< GRPBL2: sub-source status */
00480  k_poeg_genbl2_addr = 0x00087670, /**< GENBL2: sub-source enable */
00481 } rx_poeg_group_regs_t;
```

8.43.3.3 rx`poeg`interrupts``t`

```
enum rx_poeg_interrupts_t : uint16_t
```

POEG interrupt routing via GROUPBL2.

On RX72N, POEGGA/B/C/D interrupts are not dedicated vectors; they are sub-sources of the shared GROUPBL2 vector (IPR/IR/IER index 107). The individual source status and enables live in the GRPBL2 / GENBL2 registers at bits 7-10 respectively. Per RX72N HW manual R01UH0824EJ0111 Section 15.3 (Table 15.4) "Interrupt Vector Table".

Values previously encoded here (188-191) pointed at unassigned INTB slots and could never fire; corrected below.

Since

Version 1.0.0

Enumerator

k_poeg_irq_groupbl2_vector	GROUPBL2 shared ICU vector
----------------------------	----------------------------

k_poeg_grpbl2_bit_poeggai	GRPBL2/GENBL2 bit for POEGGAI
k_poeg_grpbl2_bit_poeggbi	GRPBL2/GENBL2 bit for POEGGBI
k_poeg_grpbl2_bit_poeggci	GRPBL2/GENBL2 bit for POEGGCI
k_poeg_grpbl2_bit_poeggdi	GRPBL2/GENBL2 bit for POEGGDI

Definition at line 455 of file [rx72n_poeg_regs.h](#).

```

00455      : uint16_t {
00456  k_poeg_irq_groupbl2_vector = 107, /**< GROUPBL2 shared ICU vector */
00457  k_poeg_grpbl2_bit_poeggai = 7,  /**< GRPBL2/GENBL2 bit for POEGGAI */
00458  k_poeg_grpbl2_bit_poeggbi = 8,  /**< GRPBL2/GENBL2 bit for POEGGBI */
00459  k_poeg_grpbl2_bit_poeggci = 9,  /**< GRPBL2/GENBL2 bit for POEGGCI */
00460  k_poeg_grpbl2_bit_poeggdi = 10, /**< GRPBL2/GENBL2 bit for POEGGDI */
00461 } rx_poeg_interrupts_t;

```

8.43.3.4 rx_poegg_bit_shifts_t

```
enum rx_poegg_bit_shifts_t : uint8_t
```

POEGGn register bit field positions.

Bit positions for POEGGn register fields. Used to construct field values.

Since

Version 1.0.0

Enumerator

k_poeg_pidf_shift	PIDF - Port Input Detection Flag (bit 0)
k_poeg_iocf_shift	IOCF - GPTW Output Stop Flag (bit 1)
k_poeg_ostpf_shift	OSTPF - Oscillation Stop Flag (bit 2)
k_poeg_ssf_shift	SSF - Software Stop Flag (bit 3)
k_poeg_pide_shift	PIDE - Port Input Detection Enable (bit 4)
k_poeg_ioce_shift	IOCE - GPTW Output Stop Enable (bit 5)
k_poeg_ostpe_shift	OSTPE - Oscillation Stop Enable (bit 6)
k_poeg_st_shift	ST - GTETRGN Input Status (bit 16)
k_poeg_inv_shift	INV - Input Inverting (bit 28)
k_poeg_nfen_shift	NFEN - Noise Filter Enable (bit 29)
k_poeg_nfcs_shift	NFCS[1:0] - Noise Filter Clock Select (bits 31:30)

Definition at line 359 of file [rx72n_poeg_regs.h](#).

```

00359      : uint8_t {
00360  k_poeg_pidf_shift = 0, /**< PIDF - Port Input Detection Flag (bit 0) */
00361  k_poeg_iocf_shift = 1, /**< IOCF - GPTW Output Stop Flag (bit 1) */
00362  k_poeg_ostpf_shift = 2, /**< OSTPF - Oscillation Stop Flag (bit 2) */
00363  k_poeg_ssf_shift = 3,  /**< SSF - Software Stop Flag (bit 3) */
00364  k_poeg_pide_shift = 4,  /**< PIDE - Port Input Detection Enable (bit 4) */
00365  k_poeg_ioce_shift = 5,  /**< IOCE - GPTW Output Stop Enable (bit 5) */
00366  k_poeg_ostpe_shift = 6, /**< OSTPE - Oscillation Stop Enable (bit 6) */
00367  k_poeg_st_shift = 16,  /**< ST - GTETRGN Input Status (bit 16) */
00368  k_poeg_inv_shift = 28, /**< INV - Input Inverting (bit 28) */
00369  k_poeg_nfen_shift = 29, /**< NFEN - Noise Filter Enable (bit 29) */
00370  k_poeg_nfcs_shift = 30, /**< NFCS[1:0] - Noise Filter Clock Select (bits 31:30) */
00371 } rx_poegg_bit_shifts_t;

```

8.43.3.5 rx`poegg`bits`r

enum [rx_poegg_bits_t](#) : uint32_t

POEGGn register bit definitions.

Configuration and status bit values for the POEG Group n Setting Register.

Flag Bits (Status/Control)

Bit	Name	Description
PIDF	b0	Port input detection flag (write 0 to clear)
IOCF	b1	GPTW output stop detected (write 0 to clear)
OSTPF	b2	Oscillation stop detected (write 0 to clear)
SSF	b3	Software stop flag (read/write)

Enable Bits (Write-Once After Reset)

Bit	Name	Description
PIDE	b4	Enable port input detection
IOCE	b5	Enable GPTW output stop detection
OSTPE	b6	Enable oscillation stop detection

Filter Configuration

NFCS	Clock	Sampling Delay
00	PCLKB/ $\frac{1}{1}$	~50 ns @ 60 MHz (3 samples)
01	PCLKB/ $\frac{1}{8}$	~400 ns @ 60 MHz (3 samples)
10	PCLKB/32	~1.6 μ s @ 60 MHz (3 samples)
11	PCLKB/128	~6.4 μ s @ 60 MHz (3 samples)

See also

[rx_poegg_regs_t::poeggn](#) Register definition

Since

Version 1.0.0

Enumerator

k_poeg_pidf_detected	Port input fault detected
k_poeg_iocf_detected	GPTW output error detected
k_poeg_ostpf_detected	Oscillation stop detected
k_poeg_ssf_stop	Software emergency stop active
k_poeg_pide_enable	Enable external pin detection
k_poeg_ioce_enable	Enable GPTW error detection
k_poeg_ostpe_enable	Enable oscillation stop detection
k_poeg_st_high	GTETRGN input is high
k_poeg_inv_invert	Invert external trigger input
k_poeg_nfen_enable	Enable digital noise filter
k_poeg_nfcs_pclkb_div1	Sample @ PCLKB/1 (~50ns)
k_poeg_nfcs_pclkb_div8	Sample @ PCLKB/8 (~400ns)
k_poeg_nfcs_pclkb_div32	Sample @ PCLKB/32 (~1.6us)
k_poeg_nfcs_pclkb_div128	Sample @ PCLKB/128 (~6.4us)
k_poeg_all_flags_mask	All flag bits
k_poeg_all_enable_mask	All enable bits
k_poeg_nfcs_mask	NFCS field mask

Definition at line 406 of file rx72n_poeg_regs.h.

```

00406      : uint32_t {
00407      /* Flag bits (lower byte) - write 0 to clear */
00408      k_poeg_pidf_detected = (1U « k_poeg_pidf_shift), /**< Port input fault detected */
00409      k_poeg_iocf_detected = (1U « k_poeg_iocf_shift), /**< GPTW output error detected */
00410      k_poeg_ostpf_detected = (1U « k_poeg_ostpf_shift), /**< Oscillation stop detected */
00411      k_poeg_ssf_stop      = (1U « k_poeg_ssf_shift), /**< Software emergency stop active */
00412
00413      /* Enable bits (write-once after reset) */
00414      k_poeg_pide_enable = (1U « k_poeg_pide_shift), /**< Enable external pin detection */
00415      k_poeg_ioce_enable = (1U « k_poeg_ioce_shift), /**< Enable GPTW error detection */
00416      k_poeg_ostpe_enable = (1U « k_poeg_ostpe_shift), /**< Enable oscillation stop detection */
00417
00418      /* Status bit (read-only) */
00419      k_poeg_st_high = (1U « k_poeg_st_shift), /**< GTETRGN input is high */
00420
00421      /* Input configuration */
00422      k_poeg_inv_invert = (1U « k_poeg_inv_shift), /**< Invert external trigger input */
00423      k_poeg_nfen_enable = (1U « k_poeg_nfen_shift), /**< Enable digital noise filter */
00424
00425      /* Noise filter clock select (NFCS[1:0]) */
00426      k_poeg_nfcs_pclkb_div1 = (0U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/1 (~50ns) */
00427      k_poeg_nfcs_pclkb_div8 = (1U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/8 (~400ns) */
00428      k_poeg_nfcs_pclkb_div32 = (2U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/32 (~1.6us) */
00429      k_poeg_nfcs_pclkb_div128 = (3U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/128 (~6.4us) */
00430
00431      /* Combined masks for convenience */
00432      k_poeg_all_flags_mask = (k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected |
00433          k_poeg_ssf_stop), /**< All flag bits */
00434      k_poeg_all_enable_mask =
00435          (k_poeg_pide_enable | k_poeg_ioce_enable | k_poeg_ostpe_enable), /**< All enable bits */
00436      k_poeg_nfcs_mask = (3U « k_poeg_nfcs_shift), /**< NFCS field mask */
00437 } rx_poegg_bits_t;

```

8.43.4 Function Documentation

8.43.4.1 poegga()

volatile rx_poegg_regs_t * poegga (

void) [inline], [static]

Get pointer to POEG Group A registers.

Returns a volatile pointer to POEG Group A register structure at address 0x0009E000. Use for emergency stop control of motor 0.

Returns

Volatile pointer to POEG Group A register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

Example:

```
// Emergency stop motor 0
poegga()->poeggn |= k_poeg_ssf_stop;
```

See also

[k_poegga_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 282 of file [rx72n_poeg_regs.h](#).

```
00283 {
00284     return (volatile rx_poegg_regs_t*)k_poegga_base_addr;
00285 }
```

References [k_poegga_base_addr](#).

8.43.4.2 poeggb()

```
volatile rx_poegg_regs_t * poeggb (  
    void )    [inline], [static]
```

Get pointer to POEG Group B registers.

Returns a volatile pointer to POEG Group B register structure at address 0x0009E100. Use for emergency stop control of motor 1.

Returns

Volatile pointer to POEG Group B register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)

Postcondition

Pointer valid for lifetime of program execution

See also

[k_poeggb_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 303 of file [rx72n_poeg_regs.h](#).

```
00304 {  
00305     return (volatile rx_poegg_regs_t*)k_poeggb_base_addr;  
00306 }
```

References [k_poeggb_base_addr](#).

8.43.4.3 poeggc()

```
volatile rx_poegg_regs_t * poeggc (  
    void )    [inline], [static]
```

Get pointer to POEG Group C registers.

Returns a volatile pointer to POEG Group C register structure at address 0x0009E200. Use for emergency stop control of motor 2.

Returns

Volatile pointer to POEG Group C register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)

Postcondition

Pointer valid for lifetime of program execution

See also

[k_poeggc_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 324 of file [rx72n_poeg_regs.h](#).

```
00325 {  
00326     return (volatile rx_poegg_regs_t*)k_poeggc_base_addr;  
00327 }
```

References [k_poeggc_base_addr](#).

8.43.4.4 poeggd()

```
volatile rx_poegg_regs_t * poeggd (  
    void )    [inline], [static]
```

Get pointer to POEG Group D registers.

Returns a volatile pointer to POEG Group D register structure at address 0x0009E300. Use for emergency stop control of motor 3.

Returns

Volatile pointer to POEG Group D register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)

Postcondition

Pointer valid for lifetime of program execution

See also

[k_poeggd_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 345 of file [rx72n_poeg_regs.h](#).

```
00346 {  
00347     return (volatile rx_poegg_regs_t*)k_poeggd_base_addr;  
00348 }
```

References [k_poeggd_base_addr](#).

8.44 rx72n_poeg_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_poeg_regs.h
00003  * @brief RX72N GPTW Port Output Enable (POEG) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the POEG module which provides emergency PWM output
00007  * disable functionality for the General PWM Timer (GPTW). The POEG is critical
00008  * for motor control safety, allowing immediate PWM shutdown on fault detection.
00009  *
00010  * @par Safety-Critical Emergency Stop System
00011  * @dot
00012  * digraph poeg_system {
00013  *   rankdir=LR;
00014  *   node [shape=box, style=rounded];
00015  *
00016  *   subgraph cluster_triggers {
00017  *     label="Fault Detection Sources";
00018  *     style=filled;
00019  *     color=lightcoral;
00020  *
00021  *     pin [label="GTETRGN Pin\n(External fault)"];
00022  *     gptw [label="GPTW Output Error\n(Dead-time/shoot-through)"];
00023  *     osc [label="Oscillation Stop\n(Clock failure)"];
00024  *     sw [label="Software Request\n(SSF flag)"];
00025  *   }
00026  *
00027  *   subgraph cluster_poeg {
00028  *     label="POEG Groups A-D (0x0009E000)";
00029  *     style=filled;
00030  *     color=lightyellow;
00031  *
00032  *     poeg [label="POEG Logic\n(OR of all triggers)" shape=ellipse];
00033  *   }
00034  *
00035  *   subgraph cluster_output {
00036  *     label="Motor Control Output";
00037  *     style=filled;
00038  *     color=lightgreen;
00039  *
00040  *     pwm [label="GPTW PWM\nGTIOCxA/B"];
00041  *     motor [label="H-Bridge\n(DRV8263H)"];
00042  *   }
00043  *
00044  *   pin -> poeg [label="PIDF"];
00045  *   gptw -> poeg [label="IOCF"];
00046  *   osc -> poeg [label="OSTPF"];
00047  *   sw -> poeg [label="SSF"];
00048  *   poeg -> pwm [label="Stop Request"];
00049  *   pwm -> motor [label="Hi-Z"];
00050  * }
00051  * @enddot
00052  *
00053  * @par STAR Project POEG Usage
00054  * | Group | Assigned To | Trigger Source | Use Case |
00055  * |-----|-----|-----|-----|
00056  * | A | GPTW0 (Motor 0) | GTETRGA + Software | Front-left motor E-stop |
00057  * | B | GPTW1 (Motor 1) | GTETRGB + Software | Front-right motor E-stop |
00058  * | C | GPTW2 (Motor 2) | GTETRG C + Software | Back-right motor E-stop |
00059  * | D | GPTW3 (Motor 3) | GTETRGD + Software | Back-left motor E-stop |
00060  *
00061  * @par Key Features
00062  * - Emergency PWM output disable (immediate Hi-Z on fault)
00063  * - Multiple trigger sources: external pin, GPTW error, oscillation stop, software
00064  * - Digital noise filter on external trigger inputs
00065  * - Per-group independent control (4 groups for 4 motors)
00066  * - Interrupt generation on fault detection
00067  *
00068  * @par Hardware Requirements
00069  * | Parameter | Value | Notes |
00070  * |-----|-----|-----|
00071  * | Base Address | 0x0009E000 | Group A base |
00072  * | Group Spacing | 0x100 bytes | A=0x0000, B=0x0100, C=0x0200, D=0x0300 |
00073  * | Module Stop | MSTPCRA.MSTPA7 | Same as GPTW (shared) |
00074  * | Clock | PCLKB (60 MHz) | For noise filter sampling |
00075  *
00076  * @par NASA Power of 10 Compliance
00077  * - Rule 1: No goto, setjmp, or recursion
00078  * - Rule 2: N/A (no loops in register definitions)
00079  * - Rule 3: No dynamic memory allocation
00080  * - Rule 4: All accessor functions are single-statement
00081  * - Rule 5: N/A (hardware layer)
00082  * - Rule 6: Minimal scope - inline accessors only

```

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00083 * - Rule 7: N/A (no return values to check)
00084 * - Rule 8: All constants use C23 typed enums
00085 * - Rule 9: No function pointers
00086 * - Rule 10: Compiles with -Wall -Wextra -Werror
00087 *
00088 * @par SOLID Principles
00089 * - **S**: Single responsibility - only POEG register definitions
00090 * - **O**: Open for extension via additional trigger configurations
00091 * - **I**: N/A (no inheritance hierarchy)
00092 * - **I**: Minimal interface - only necessary accessor functions
00093 * - **D**: N/A (hardware register layer)
00094 *
00095 * @par Module Dependencies
00096 * - <stdint.h> - Fixed-width integer types
00097 * - <stddef.h> - offsetof macro for static assertions
00098 *
00099 * @par Manual References
00100 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00101 * - Chapter 27: GPTW Port Output Enable (POEG), pages 1432-1441
00102 *
00103 * @see rx72n_gptw_regs.h General PWM Timer registers
00104 * @see rx72n_regs.h Main register include file
00105 *
00106 * @author Locked, Inc.
00107 * @date 2026-01-29
00108 * @copyright Copyright (c) 2026 Locked Inc.
00109 * SPDX-License-Identifier: MIT
00110 * @since Version 1.0.0
00111 */
00112
00113 #pragma once
00114
00115 #include <stddef.h>
00116 #include <stdint.h>
00117
00118 #ifdef __cplusplus
00119 extern "C" {
00120 #endif
00121
00122 /*
=====
00123 * POEG - GPTW Port Output Enable for Emergency Motor Stop
00124 *
=====
00125 */
00126
00127 /**
00128 * @enum rx_poeg_addresses_t
00129 * @brief POEG group base addresses
00130 *
00131 * @details
00132 * Base addresses for each POEG group (A-D). Each group controls emergency
00133 * output disable for associated GPTW channels. Groups are spaced 0x100 bytes
00134 * apart.
00135 *
00136 * @par Memory Map
00137 * | Address | Group | Controls | Description |
00138 * |-----|-----|-----|-----|
00139 * | 0x0009E000 | A | GPTW ch0-3 | Emergency stop group A |
00140 * | 0x0009E100 | B | GPTW ch0-3 | Emergency stop group B |
00141 * | 0x0009E200 | C | GPTW ch0-3 | Emergency stop group C |
00142 * | 0x0009E300 | D | GPTW ch0-3 | Emergency stop group D |
00143 *
00144 * @note Manual: Ch27.2.1 - POEG Group n Setting Register
00145 * @see poegga() Accessor function for Group A
00146 * @since Version 1.0.0
00147 */
00148 typedef enum : uintptr_t {
00149 /**
00150 * @brief POEG Group A base address (0x0009E000)
00151 * @details Verified against RX72N Hardware Manual Ch27.2.1
00152 */
00153 k_poegga_base_addr = 0x0009E000,
00154
00155 /**
00156 * @brief POEG Group B base address (0x0009E100)
00157 * @details Verified against RX72N Hardware Manual Ch27.2.1
00158 */
00159 k_poeggb_base_addr = 0x0009E100,
00160
00161 /**
00162 * @brief POEG Group C base address (0x0009E200)
00163 * @details Verified against RX72N Hardware Manual Ch27.2.1
00164 */
00165 k_poeggc_base_addr = 0x0009E200,
00166
00167 /**

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```

00168 * @brief POEG Group D base address (0x0009E300)
00169 * @details Verified against RX72N Hardware Manual Ch27.2.1
00170 */
00171 k_poeggd_base_addr = 0x0009E300,
00172
00173 /**
00174 * @brief Group spacing (0x100 bytes between groups)
00175 * @details Each POEG group has a 256-byte register block
00176 */
00177 k_poeg_group_spacing = 0x100,
00178 } rx_poeg_addresses_t;
00179
00180 /**
00181 * @struct rx_poegg_regs_t
00182 * @brief POEG Group n Register Structure
00183 *
00184 * @details
00185 * Each POEG group has a single 32-bit register (POEGGn) that controls
00186 * emergency output disable for the associated GPTW channels. The register
00187 * provides multiple trigger sources and status flags.
00188 *
00189 * @par Memory Layout Table
00190 * | Offset | Size | Field | Type | Description |
00191 * |-----|-----|-----|-----|-----|
00192 * | 0x00 | 4 | poeggn | uint32_t | Group n Setting Register |
00193 * | **Total** | **4** | | | |
00194 *
00195 * @par Emergency Stop Trigger Sources
00196 * 1. **External Pin (PIDF)**: GTETRGN pin input detection
00197 * 2. **GPTW Output Error (IOCF)**: Dead-time error or shoot-through detection
00198 * 3. **Oscillation Stop (OSTPF)**: Main clock failure detection
00199 * 4. **Software (SSF)**: Direct software emergency stop command
00200 *
00201 * @par Configuration Example
00202 * @code{.c}
00203 * // Enable software emergency stop and external trigger for Group A
00204 * volatile rx_poegg_regs_t* poeg_a = poegga();
00205 *
00206 * // Enable external pin detection (one-time write after reset)
00207 * poeg_a->poeggn = k_poeg_pide_enable;
00208 *
00209 * // Later, trigger emergency stop via software
00210 * poeg_a->poeggn |= k_poeg_ssf_stop; // Immediate PWM disable
00211 *
00212 * // Clear emergency stop (after fault resolved)
00213 * poeg_a->poeggn &= ~k_poeg_ssf_stop; // Resume PWM
00214 * @endcode
00215 *
00216 * @invariant sizeof(rx_poegg_regs_t) == 4 bytes
00217 *
00218 * @note All triggers are OR'd together - any trigger stops output
00219 * @warning Enable bits (PIDE, IOCE, OSTPE) can only be written once after reset
00220 * @attention Module stop bit shared with GPTW (MSTPCRA.MSTPA7)
00221 *
00222 * @see poegga(), poeggb(), poeggc(), poeggd() Accessor functions
00223 * @see rx_poegg_bits_t Register bit definitions
00224 * @since Version 1.0.0
00225 */
00226 typedef struct {
00227 /**
00228 * @brief POEG Group n Setting Register (POEGGn) @ offset 0x00
00229 *
00230 * @details
00231 * Controls emergency output disable triggers and status.
00232 *
00233 * @par Bit Layout (32-bit)
00234 * | Bits | Field | R/W | Description |
00235 * |-----|-----|-----|-----|
00236 * | 0 | PIDF | R/W* | Port Input Detection Flag |
00237 * | 1 | IOCF | R/W* | GPTW Output Stop Request Flag |
00238 * | 2 | OSTPF | R/W* | Oscillation Stop Detection Flag |
00239 * | 3 | SSF | R/W | Software Stop Flag |
00240 * | 4 | PIDE | R/W+ | Port Input Detection Enable |
00241 * | 5 | IOCE | R/W+ | GPTW Output Stop Request Enable |
00242 * | 6 | OSTPE | R/W+ | Oscillation Stop Enable |
00243 * | 15:7 | - | R | Reserved (read as 0) |
00244 * | 16 | ST | R | GTETRGN Input Status |
00245 * | 27:17 | - | R | Reserved (read as 0) |
00246 * | 28 | INV | R/W | GTETRGN Input Inverting |
00247 * | 29 | NFEN | R/W | Noise Filter Enable |
00248 * | 31:30 | NFCS[1:0] | R/W | Noise Filter Clock Select |
00249 *
00250 * @note * Only 0 can be written to clear flag
00251 * @note + Can only be written once after reset
00252 *
00253 * @see rx_poegg_bits_t Bit definitions
00254 */

```

```

00255  volatile uint32_t poeggn;
00256 } rx_poegg_regs_t;
00257
00258 /**
00259  * @brief Get pointer to POEG Group A registers
00260  *
00261  * @details
00262  * Returns a volatile pointer to POEG Group A register structure at
00263  * address 0x0009E000. Use for emergency stop control of motor 0.
00264  *
00265  * @return Volatile pointer to POEG Group A register structure
00266  * @retval Non-NULL Always returns valid pointer (hardware address)
00267  *
00268  * @pre GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)
00269  * @post Pointer valid for lifetime of program execution
00270  *
00271  * @note Thread Safety: Safe - returns constant hardware address
00272  *
00273  * @par Example:
00274  * @code{.c}
00275  * // Emergency stop motor 0
00276  * poegga()->poeggn |= k_poeg_ssf_stop;
00277  * @endcode
00278  *
00279  * @see k_poegga_base_addr Base address constant
00280  * @since Version 1.0.0
00281  */
00282 static inline volatile rx_poegg_regs_t* poegga(void)
00283 {
00284     return (volatile rx_poegg_regs_t*)k_poegga_base_addr;
00285 }
00286
00287 /**
00288  * @brief Get pointer to POEG Group B registers
00289  *
00290  * @details
00291  * Returns a volatile pointer to POEG Group B register structure at
00292  * address 0x0009E100. Use for emergency stop control of motor 1.
00293  *
00294  * @return Volatile pointer to POEG Group B register structure
00295  * @retval Non-NULL Always returns valid pointer (hardware address)
00296  *
00297  * @pre GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)
00298  * @post Pointer valid for lifetime of program execution
00299  *
00300  * @see k_poeggb_base_addr Base address constant
00301  * @since Version 1.0.0
00302  */
00303 static inline volatile rx_poegg_regs_t* poeggb(void)
00304 {
00305     return (volatile rx_poegg_regs_t*)k_poeggb_base_addr;
00306 }
00307
00308 /**
00309  * @brief Get pointer to POEG Group C registers
00310  *
00311  * @details
00312  * Returns a volatile pointer to POEG Group C register structure at
00313  * address 0x0009E200. Use for emergency stop control of motor 2.
00314  *
00315  * @return Volatile pointer to POEG Group C register structure
00316  * @retval Non-NULL Always returns valid pointer (hardware address)
00317  *
00318  * @pre GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)
00319  * @post Pointer valid for lifetime of program execution
00320  *
00321  * @see k_poeggc_base_addr Base address constant
00322  * @since Version 1.0.0
00323  */
00324 static inline volatile rx_poegg_regs_t* poeggc(void)
00325 {
00326     return (volatile rx_poegg_regs_t*)k_poeggc_base_addr;
00327 }
00328
00329 /**
00330  * @brief Get pointer to POEG Group D registers
00331  *
00332  * @details
00333  * Returns a volatile pointer to POEG Group D register structure at
00334  * address 0x0009E300. Use for emergency stop control of motor 3.
00335  *
00336  * @return Volatile pointer to POEG Group D register structure
00337  * @retval Non-NULL Always returns valid pointer (hardware address)
00338  *
00339  * @pre GPTW/POEG module enabled (MSTPCRA.MSTPA7 = 0)
00340  * @post Pointer valid for lifetime of program execution
00341  */

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```

00342 * @see k_poeggd_base_addr Base address constant
00343 * @since Version 1.0.0
00344 */
00345 static inline volatile rx_poeggs_regs_t* poeggs(void)
00346 {
00347     return (volatile rx_poeggs_regs_t*)k_poeggd_base_addr;
00348 }
00349
00350 /**
00351 * @enum rx_poeggs_bit_shifts_t
00352 * @brief POEGGS register bit field positions
00353 *
00354 * @details
00355 * Bit positions for POEGGS register fields. Used to construct field values.
00356 *
00357 * @since Version 1.0.0
00358 */
00359 typedef enum : uint8_t {
00360     k_poeg_pidf_shift = 0, /**< PIDF - Port Input Detection Flag (bit 0) */
00361     k_poeg_iocf_shift = 1, /**< IOCF - GPTW Output Stop Flag (bit 1) */
00362     k_poeg_ostpf_shift = 2, /**< OSTPF - Oscillation Stop Flag (bit 2) */
00363     k_poeg_ssf_shift = 3, /**< SSF - Software Stop Flag (bit 3) */
00364     k_poeg_pide_shift = 4, /**< PIDE - Port Input Detection Enable (bit 4) */
00365     k_poeg_ioce_shift = 5, /**< IOCE - GPTW Output Stop Enable (bit 5) */
00366     k_poeg_ostpe_shift = 6, /**< OSTPE - Oscillation Stop Enable (bit 6) */
00367     k_poeg_st_shift = 16, /**< ST - GTETRGS Input Status (bit 16) */
00368     k_poeg_inv_shift = 28, /**< INV - Input Inverting (bit 28) */
00369     k_poeg_nfcs_shift = 29, /**< NFEN - Noise Filter Enable (bit 29) */
00370     k_poeg_nfcs_shift = 30, /**< NFCS[1:0] - Noise Filter Clock Select (bits 31:30) */
00371 } rx_poeggs_bit_shifts_t;
00372
00373 /**
00374 * @enum rx_poeggs_bits_t
00375 * @brief POEGGS register bit definitions
00376 *
00377 * @details
00378 * Configuration and status bit values for the POEG Group n Setting Register.
00379 *
00380 * @par Flag Bits (Status/Control)
00381 * | Bit | Name | Description |
00382 * |-----|-----|-----|
00383 * | PIDF | b0 | Port input detection flag (write 0 to clear) |
00384 * | IOCF | b1 | GPTW output stop detected (write 0 to clear) |
00385 * | OSTPF | b2 | Oscillation stop detected (write 0 to clear) |
00386 * | SSF | b3 | Software stop flag (read/write) |
00387 *
00388 * @par Enable Bits (Write-Once After Reset)
00389 * | Bit | Name | Description |
00390 * |-----|-----|-----|
00391 * | PIDE | b4 | Enable port input detection |
00392 * | IOCE | b5 | Enable GPTW output stop detection |
00393 * | OSTPE | b6 | Enable oscillation stop detection |
00394 *
00395 * @par Filter Configuration
00396 * | NFCS | Clock | Sampling Delay |
00397 * |-----|-----|-----|
00398 * | 00 | PCLKB/1 | ~50 ns @ 60 MHz (3 samples) |
00399 * | 01 | PCLKB/8 | ~400 ns @ 60 MHz (3 samples) |
00400 * | 10 | PCLKB/32 | ~1.6 us @ 60 MHz (3 samples) |
00401 * | 11 | PCLKB/128 | ~6.4 us @ 60 MHz (3 samples) |
00402 *
00403 * @see rx_poeggs_regs_t::poeggs Register definition
00404 * @since Version 1.0.0
00405 */
00406 typedef enum : uint32_t {
00407     /* Flag bits (lower byte) - write 0 to clear */
00408     k_poeg_pidf_detected = (1U « k_poeg_pidf_shift), /**< Port input fault detected */
00409     k_poeg_iocf_detected = (1U « k_poeg_iocf_shift), /**< GPTW output error detected */
00410     k_poeg_ostpf_detected = (1U « k_poeg_ostpf_shift), /**< Oscillation stop detected */
00411     k_poeg_ssf_stop = (1U « k_poeg_ssf_shift), /**< Software emergency stop active */
00412
00413     /* Enable bits (write-once after reset) */
00414     k_poeg_pide_enable = (1U « k_poeg_pide_shift), /**< Enable external pin detection */
00415     k_poeg_ioce_enable = (1U « k_poeg_ioce_shift), /**< Enable GPTW error detection */
00416     k_poeg_ostpe_enable = (1U « k_poeg_ostpe_shift), /**< Enable oscillation stop detection */
00417
00418     /* Status bit (read-only) */
00419     k_poeg_st_high = (1U « k_poeg_st_shift), /**< GTETRGS input is high */
00420
00421     /* Input configuration */
00422     k_poeg_inv_invert = (1U « k_poeg_inv_shift), /**< Invert external trigger input */
00423     k_poeg_nfcs_enable = (1U « k_poeg_nfcs_shift), /**< Enable digital noise filter */
00424
00425     /* Noise filter clock select (NFCS[1:0]) */
00426     k_poeg_nfcs_pclkb_div1 = (0U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/1 (~50ns) */
00427     k_poeg_nfcs_pclkb_div8 = (1U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/8 (~400ns) */
00428     k_poeg_nfcs_pclkb_div32 = (2U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/32 (~1.6us) */

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00429 k_poeg_nfcs_pclkb_div128 = (3U « k_poeg_nfcs_shift), /**< Sample @ PCLKB/128 (~6.4us) */
00430
00431 /* Combined masks for convenience */
00432 k_poeg_all_flags_mask = (k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected |
00433 k_poeg_ssf_stop), /**< All flag bits */
00434 k_poeg_all_enable_mask =
00435 (k_poeg_pide_enable | k_poeg_ioce_enable | k_poeg_ostpe_enable), /**< All enable bits */
00436 k_poeg_nfcs_mask = (3U « k_poeg_nfcs_shift), /**< NFCS field mask */
00437 } rx_poegg_bits_t;
00438
00439 /**
00440 * @enum rx_poeg_interrupts_t
00441 * @brief POEG interrupt routing via GROUPBL2
00442 *
00443 * @details
00444 * On RX72N, POEGGA/B/C/D interrupts are not dedicated vectors; they are
00445 * sub-sources of the shared GROUPBL2 vector (IPR/IR/IER index 107). The
00446 * individual source status and enables live in the GRPBL2 / GENBL2
00447 * registers at bits 7-10 respectively. Per RX72N HW manual
00448 * R01UH0824EJ0111 Section 15.3 (Table 15.4) "Interrupt Vector Table".
00449 *
00450 * Values previously encoded here (188-191) pointed at unassigned INTB
00451 * slots and could never fire; corrected below.
00452 *
00453 * @since Version 1.0.0
00454 */
00455 typedef enum : uint16_t {
00456 k_poeg_irq_groupbl2_vector = 107, /**< GROUPBL2 shared ICU vector */
00457 k_poeg_grpbl2_bit_poeggai = 7, /**< GRPBL2/GENBL2 bit for POEGGAI */
00458 k_poeg_grpbl2_bit_poeggbi = 8, /**< GRPBL2/GENBL2 bit for POEGGBI */
00459 k_poeg_grpbl2_bit_poeggci = 9, /**< GRPBL2/GENBL2 bit for POEGGCI */
00460 k_poeg_grpbl2_bit_poeggdi = 10, /**< GRPBL2/GENBL2 bit for POEGGDI */
00461 } rx_poeg_interrupts_t;
00462
00463 /**
00464 * @enum rx_poeg_group_regs_t
00465 * @brief Memory-mapped addresses of the GROUPBL2 status/enable registers
00466 *
00467 * @details
00468 * Per RX72N HW manual R01UH0824EJ0111 Section 15.2 ("ICU Registers")
00469 * the group interrupt registers are located in the ICU register block:
00470 *
00471 * | Register | Address | Purpose |
00472 * |-----|-----|-----|
00473 * | GRPBL2 | 0x00087630 | Group BL2 interrupt source status (R-only) |
00474 * | GENBL2 | 0x00087670 | Group BL2 per-source enable (R/W) |
00475 *
00476 * @since Version 1.0.0
00477 */
00478 typedef enum : uintptr_t {
00479 k_poeg_grpbl2_addr = 0x00087630, /**< GRPBL2: sub-source status */
00480 k_poeg_genbl2_addr = 0x00087670, /**< GENBL2: sub-source enable */
00481 } rx_poeg_group_regs_t;
00482
00483 /*
=====
00484 * Static Assertions - Verify Register Layout at Compile Time
00485 *
=====
00486 */
00487
00488 /* Verify base addresses match Hardware Manual Ch27.2.1 */
00489 static_assert(k_poegga_base_addr == 0x0009E000, "POEGGA base address incorrect");
00490 static_assert(k_poeggb_base_addr == 0x0009E100, "POEGGB base address incorrect");
00491 static_assert(k_poeggc_base_addr == 0x0009E200, "POEGGC base address incorrect");
00492 static_assert(k_poeggd_base_addr == 0x0009E300, "POEGGD base address incorrect");
00493
00494 /* Verify group spacing */
00495 static_assert((k_poeggb_base_addr - k_poegga_base_addr) == k_poeg_group_spacing,
00496 "POEG group A-B spacing incorrect");
00497 static_assert((k_poeggc_base_addr - k_poeggb_base_addr) == k_poeg_group_spacing,
00498 "POEG group B-C spacing incorrect");
00499 static_assert((k_poeggd_base_addr - k_poeggc_base_addr) == k_poeg_group_spacing,
00500 "POEG group C-D spacing incorrect");
00501
00502 /* Verify register structure layout */
00503 static_assert(sizeof(rx_poegg_regs_t) == 4, "POEG register structure size incorrect");
00504 static_assert(offsetof(rx_poegg_regs_t, poeggn) == 0x00, "POEGGn offset incorrect");
00505
00506 /* Verify bit field positions */
00507 static_assert(k_poeg_pidf_detected == 0x00000001, "PIDF bit value incorrect");
00508 static_assert(k_poeg_iocf_detected == 0x00000002, "IOCF bit value incorrect");
00509 static_assert(k_poeg_ostpf_detected == 0x00000004, "OSTPF bit value incorrect");
00510 static_assert(k_poeg_ssf_stop == 0x00000008, "SSF bit value incorrect");
00511 static_assert(k_poeg_pide_enable == 0x00000010, "PIDE bit value incorrect");
00512 static_assert(k_poeg_ioce_enable == 0x00000020, "IOCE bit value incorrect");
00513 static_assert(k_poeg_ostpe_enable == 0x00000040, "OSTPE bit value incorrect");

```


Functions

- static volatile [rx_port_regs_t](#) * [port0](#) (void)
Get pointer to PORT0 registers.
- static volatile [rx_port_regs_t](#) * [port1](#) (void)
Get pointer to PORT1 registers.
- static volatile [rx_port_regs_t](#) * [port2](#) (void)
Get pointer to PORT2 registers.
- static volatile [rx_port_regs_t](#) * [port3](#) (void)
Get pointer to PORT3 registers.
- static volatile [rx_port_regs_t](#) * [port4](#) (void)
Get pointer to PORT4 registers.
- static volatile [rx_port_regs_t](#) * [port5](#) (void)
Get pointer to PORT5 registers.
- static volatile [rx_port_regs_t](#) * [port6](#) (void)
Get pointer to PORT6 registers.
- static volatile [rx_port_regs_t](#) * [port7](#) (void)
Get pointer to PORT7 registers.
- static volatile [rx_port_regs_t](#) * [port8](#) (void)
Get pointer to PORT8 registers.
- static volatile [rx_port_regs_t](#) * [port9](#) (void)
Get pointer to PORT9 registers.
- static volatile [rx_port_regs_t](#) * [porta](#) (void)
Get pointer to PORTA registers.
- static volatile [rx_port_regs_t](#) * [portb](#) (void)
Get pointer to PORTB registers.
- static volatile [rx_port_regs_t](#) * [portc](#) (void)
Get pointer to PORTC registers.
- static volatile [rx_port_regs_t](#) * [portd](#) (void)
Get pointer to PORTD registers.
- static volatile [rx_port_regs_t](#) * [porte](#) (void)
Get pointer to PORTE registers.
- static volatile [rx_port_regs_t](#) * [portf](#) (void)
Get pointer to PORTF registers.
- static volatile [rx_port_regs_t](#) * [portj](#) (void)
Get pointer to PORTJ registers.
- static volatile [uint16_t](#) * [port0_odr](#) (void)
Get pointer to PORT0 ODR register.
- static volatile [uint16_t](#) * [port1_odr](#) (void)
Get pointer to PORT1 ODR register.
- static volatile [uint16_t](#) * [port2_odr](#) (void)
Get pointer to PORT2 ODR register.
- static volatile [uint16_t](#) * [port3_odr](#) (void)
Get pointer to PORT3 ODR register.
- static volatile [uint16_t](#) * [port4_odr](#) (void)
Get pointer to PORT4 ODR register.
- static volatile [uint16_t](#) * [port5_odr](#) (void)
Get pointer to PORT5 ODR register.
- static volatile [uint16_t](#) * [port6_odr](#) (void)
Get pointer to PORT6 ODR register.
- static volatile [uint16_t](#) * [port7_odr](#) (void)

- Get pointer to PORT7 ODR register.
- static volatile uint16_t * [port8_odr](#) (void)
- Get pointer to PORT8 ODR register.
- static volatile uint16_t * [port9_odr](#) (void)
- Get pointer to PORT9 ODR register.
- static volatile uint16_t * [porta_odr](#) (void)
- Get pointer to PORTA ODR register.
- static volatile uint16_t * [portb_odr](#) (void)
- Get pointer to PORTB ODR register.
- static volatile uint16_t * [portc_odr](#) (void)
- Get pointer to PORTC ODR register.
- static volatile uint16_t * [portd_odr](#) (void)
- Get pointer to PORTD ODR register.
- static volatile uint16_t * [porte_odr](#) (void)
- Get pointer to PORTE ODR register.
- static volatile uint16_t * [portf_odr](#) (void)
- Get pointer to PORTF ODR register.
- static volatile uint16_t * [portj_odr](#) (void)
- Get pointer to PORTJ ODR register.

8.45.1 Detailed Description

RX72N PORT (GPIO) register definitions for digital pin control.

This file provides complete hardware register definitions for the RX72N's General Purpose I/O (GPIO) port peripheral. The PORT module controls digital pin direction, output state, input reading, and special modes like open-drain and pull-up resistors.

Critical Architecture Note

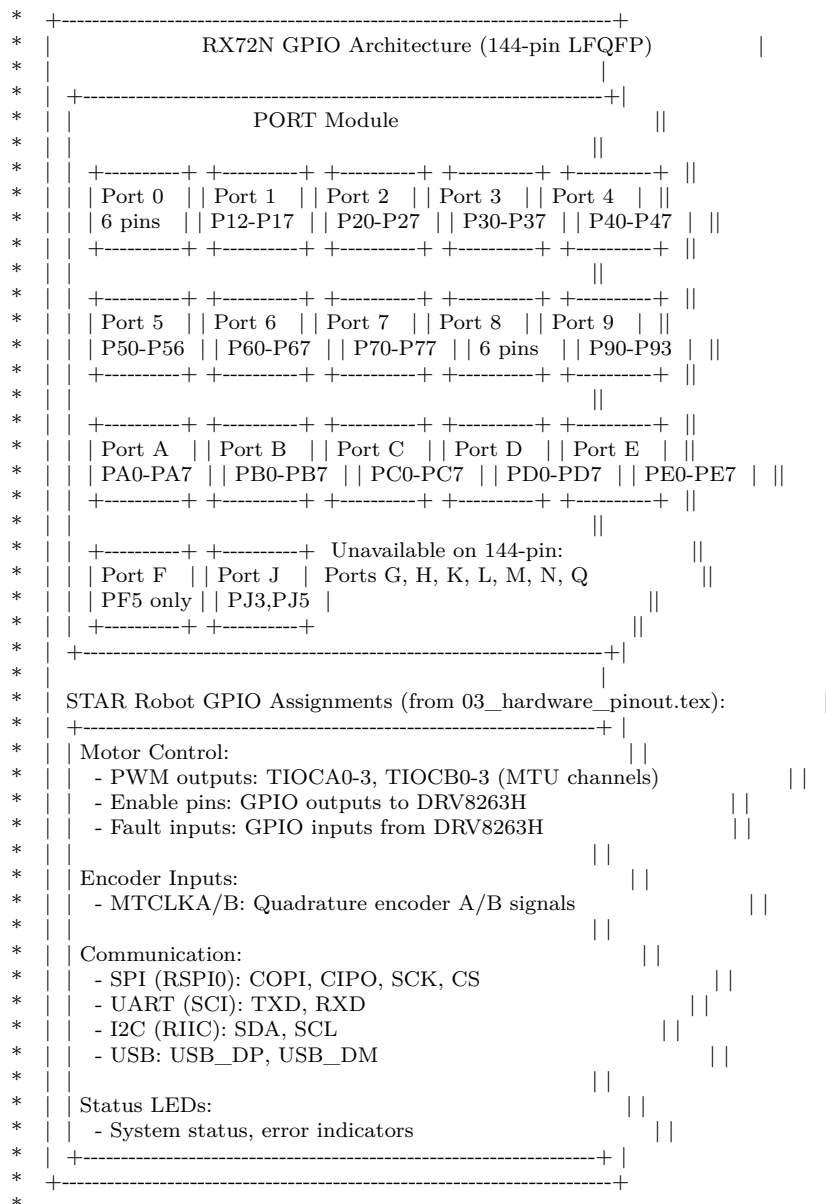
Warning

RX72N PORT registers are organized by REGISTER TYPE, not by PORT!

* Traditional MCU Layout	RX72N Layout (Type-Grouped)
* -----	* -----
* PORT0: PDR0, PODR0, ...	PDR: PDR0, PDR1, PDR2, ... (contiguous)
* PORT1: PDR1, PODR1, ...	PODR: PODR0, PODR1, PODR2, ... (contiguous)
* PORT2: PDR2, PODR2, ...	PIDR: PIDR0, PIDR1, PIDR2, ... (contiguous)
* ...	
* ...	

This struct provides a PORT-centric view with correct padding to handle the type-grouped hardware layout transparently.

System Architecture - STAR Robot GPIO Usage



Register Memory Map

Register	Base Address	Description
PDR	0x0008C000	Port Direction (0=input, 1=output)
PODR	0x0008C020	Port Output Data (output level)
PIDR	0x0008C040	Port Input Data (read pin state)
PMR	0x0008C060	Port Mode (0=GPIO, 1=peripheral)
ODR0/↔ 1	0x0008C080	Open Drain Control
PCR	0x0008C0C0	Pull-up Control
DSCR	0x0008C0E0	Drive Capacity Control

Register	Base Address	Description
DSCR2	0x0008C128	Drive Capacity Control 2

GPIO Configuration Example

```
// Configure PD0 as output (LED)
portd()->pmr &= ~(1 << 0); // GPIO mode (not peripheral)
portd()->pdr |= (1 << 0); // Output direction
portd()->podr |= (1 << 0); // Set high (LED on)

// Configure PA0 as input with pull-up (button)
porta()->pmr &= ~(1 << 0); // GPIO mode
porta()->pdr &= ~(1 << 0); // Input direction
porta()->pcr |= (1 << 0); // Enable pull-up

// Read button state
bool pressed = (porta()->pidr & (1 << 0)) == 0; // Active-low
```

Package Support

This file supports the 144-pin LQFP package (R5F572NNHxFB). Available ports and pins:

- Port 0: P00-P03, P05, P07 (6 pins)
- Port 1: P12-P17 (6 pins)
- Port 2: P20-P27 (full 8 pins)
- Port 3: P30-P37 (full 8 pins, P35 input-only)
- Port 4: P40-P47 (full 8 pins)
- Port 5: P50-P56 (7 pins)
- Port 6: P60-P67 (full 8 pins)
- Port 7: P70-P77 (full 8 pins)
- Port 8: P80-P83, P86-P87 (6 pins)
- Port 9: P90-P93 (4 pins)
- Port A: PA0-PA7 (full 8 pins)
- Port B: PB0-PB7 (full 8 pins)
- Port C: PC0-PC7 (full 8 pins)
- Port D: PD0-PD7 (full 8 pins)
- Port E: PE0-PE7 (full 8 pins)
- Port F: PF5 (1 pin)
- Port J: PJ3, PJ5 (2 pins)

Hardware Reference

- RX72N Group User's Manual: Hardware, Chapter 22 (I/O Ports)
- Section 22.3: Register Descriptions
- Section 22.4: Initialization

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 3: [OK] Static allocation only (all definitions compile-time)
- Rule 5: [OK] Static assertions verify register layout at compile time
- Rule 8: [OK] C23 typed enums eliminate preprocessor constants
- Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: GPIO register definitions only, no driver logic
- Open/Closed: Extend via new enums; don't modify existing values
- Interface Segregation: Single struct handles all port register types
- Dependency Inversion: Higher-level drivers depend on these abstractions

Related Modules

- [rx72n_mpc.h](#): Multi-function Pin Controller (pin function select)
- [rx72n_mtu_regs.h](#): Timer I/O pins
- [rx72n_sci_regs.h](#): UART pins
- [rx72n_rspi_regs.h](#): SPI pins

See also

RX72N Hardware Manual Chapter 22 for complete I/O Ports specification
DOXYGEN_ROADMAP.md for documentation standards

Author

Locked, Inc. Contributors

Date

2026-01-05

Version

1.0.0

Copyright

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Definition in file [rx72n_port_regs.h](#).

8.46 rx72n`port`regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_port_regs.h
00003  * @brief RX72N PORT (GPIO) register definitions for digital pin control
00004  *
00005  * @details
00006  * This file provides complete hardware register definitions for the RX72N's
00007  * General Purpose I/O (GPIO) port peripheral. The PORT module controls
00008  * digital pin direction, output state, input reading, and special modes
00009  * like open-drain and pull-up resistors.
00010  *
00011  * @par Critical Architecture Note
00012  * @warning RX72N PORT registers are organized by REGISTER TYPE, not by PORT!
00013  *
```

```

00014 * @verbatim
00015 *   Traditional MCU Layout          RX72N Layout (Type-Grouped)
00016 *   -----
00017 *   PORT0: PDR0, PODR0, ...        PDR: PDR0, PDR1, PDR2, ... (contiguous)
00018 *   PORT1: PDR1, PODR1, ...        PODR: PODR0, PODR1, PODR2, ... (contiguous)
00019 *   PORT2: PDR2, PODR2, ...        PIDR: PIDR0, PIDR1, PIDR2, ... (contiguous)
00020 *   ...
00021 * @endverbatim
00022 *
00023 * This struct provides a PORT-centric view with correct padding to handle
00024 * the type-grouped hardware layout transparently.
00025 *
00026 * @par System Architecture - STAR Robot GPIO Usage
00027 * @verbatim
00028 *   +-----+
00029 *   |               RX72N GPIO Architecture (144-pin LQFP)               |
00030 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00031 *   |               PORT Module               |
00032 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00033 *   |               |               |               |               |               |
00034 *   |   Port 0   |   Port 1   |   Port 2   |   Port 3   |   Port 4   |   |
00035 *   |   6 pins   |   P12-P17  |   P20-P27  |   P30-P37  |   P40-P47  |   |
00036 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00037 *   |               |               |               |               |               |
00038 *   |   Port 5   |   Port 6   |   Port 7   |   Port 8   |   Port 9   |   |
00039 *   |   P50-P56  |   P60-P67  |   P70-P77  |   6 pins  |   P90-P93  |   |
00040 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00041 *   |               |               |               |               |               |
00042 *   |   Port A   |   Port B   |   Port C   |   Port D   |   Port E   |   |
00043 *   |   PA0-PA7  |   PB0-PB7  |   PC0-PC7  |   PD0-PD7  |   PE0-PE7  |   |
00044 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00045 *   |               |               |               |               |               |
00046 *   |   Port F   |   Port J   |   Ports G, H, K, L, M, N, Q   |   |
00047 *   |   PF5 only |   PJ3,PJ5 |               |               |               |   |
00048 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00049 *   |               |               | Unavailable on 144-pin:   |   |
00050 *   |   Port F   |   Port J   |   Ports G, H, K, L, M, N, Q   |   |
00051 *   |   PF5 only |   PJ3,PJ5 |               |               |               |   |
00052 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00053 *   |               |               |               |               |               |
00054 *   | STAR Robot GPIO Assignments (from 03_hardware_pinout.tex): |
00055 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00056 *   | Motor Control: |               |               |               |               |
00057 *   | - PWM outputs: TIOCA0-3, TIOCB0-3 (MTU channels) |               |
00058 *   | - Enable pins: GPIO outputs to DRV8263H |               |
00059 *   | - Fault inputs: GPIO inputs from DRV8263H |               |
00060 *   | Encoder Inputs: |               |               |               |               |
00061 *   | - MTCLKA/B: Quadrature encoder A/B signals |               |
00062 *   | Communication: |               |               |               |               |
00063 *   | - SPI (RSPi0): COPI, CIPO, SCK, CS |               |
00064 *   | - UART (SCI): TXD, RXD |               |
00065 *   | - I2C (RIIC): SDA, SCL |               |
00066 *   | - USB: USB_DP, USB_DM |               |
00067 *   | Status LEDs: |               |               |               |               |
00068 *   | - System status, error indicators |               |
00069 *   |-----+-----+-----+-----+-----+-----+-----+-----+
00070 *   |               |               |               |               |               |
00071 *   |               |               |               |               |               |
00072 *   |               |               |               |               |               |
00073 *   |               |               |               |               |               |
00074 *   +-----+
00075 * @endverbatim
00076 *
00077 * @par Register Memory Map
00078 * | Register | Base Address | Description |
00079 * |-----|-----|-----|
00080 * | PDR      | 0x0008C000 | Port Direction (0=input, 1=output) |
00081 * | PODR     | 0x0008C020 | Port Output Data (output level) |
00082 * | PIDR     | 0x0008C040 | Port Input Data (read pin state) |
00083 * | PMR      | 0x0008C060 | Port Mode (0=GPIO, 1=peripheral) |
00084 * | ODR0/1   | 0x0008C080 | Open Drain Control |
00085 * | PCR      | 0x0008C0C0 | Pull-up Control |
00086 * | DSCR     | 0x0008C0E0 | Drive Capacity Control |
00087 * | DSCR2     | 0x0008C128 | Drive Capacity Control 2 |
00088 *
00089 * @par GPIO Configuration Example
00090 * @code
00091 * // Configure PD0 as output (LED)
00092 * portd()->pmr &= ~(1 << 0); // GPIO mode (not peripheral)
00093 * portd()->pdr |= (1 << 0); // Output direction
00094 * portd()->podr |= (1 << 0); // Set high (LED on)
00095 *
00096 * // Configure PA0 as input with pull-up (button)
00097 * porta()->pmr &= ~(1 << 0); // GPIO mode
00098 * porta()->pdr &= ~(1 << 0); // Input direction
00099 * porta()->pcr |= (1 << 0); // Enable pull-up
00100 *

```

```

00101 * // Read button state
00102 * bool pressed = (porta()->pidr & (1 « 0)) == 0; // Active-low
00103 * @endcode
00104 *
00105 * @par Package Support
00106 * This file supports the 144-pin LFQFP package (R5F572NNHxFB).
00107 * Available ports and pins:
00108 * - Port 0: P00-P03, P05, P07 (6 pins)
00109 * - Port 1: P12-P17 (6 pins)
00110 * - Port 2: P20-P27 (full 8 pins)
00111 * - Port 3: P30-P37 (full 8 pins, P35 input-only)
00112 * - Port 4: P40-P47 (full 8 pins)
00113 * - Port 5: P50-P56 (7 pins)
00114 * - Port 6: P60-P67 (full 8 pins)
00115 * - Port 7: P70-P77 (full 8 pins)
00116 * - Port 8: P80-P83, P86-P87 (6 pins)
00117 * - Port 9: P90-P93 (4 pins)
00118 * - Port A: PA0-PA7 (full 8 pins)
00119 * - Port B: PB0-PB7 (full 8 pins)
00120 * - Port C: PC0-PC7 (full 8 pins)
00121 * - Port D: PD0-PD7 (full 8 pins)
00122 * - Port E: PE0-PE7 (full 8 pins)
00123 * - Port F: PF5 (1 pin)
00124 * - Port J: PJ3, PJ5 (2 pins)
00125 *
00126 * @par Hardware Reference
00127 * - RX72N Group User's Manual: Hardware, Chapter 22 (I/O Ports)
00128 * - Section 22.3: Register Descriptions
00129 * - Section 22.4: Initialization
00130 *
00131 * @par NASA Power of 10 Compliance
00132 * - Rule 1: [OK] No goto, setjmp, or recursion
00133 * - Rule 3: [OK] Static allocation only (all definitions compile-time)
00134 * - Rule 5: [OK] Static assertions verify register layout at compile time
00135 * - Rule 8: [OK] C23 typed enums eliminate preprocessor constants
00136 * - Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror
00137 *
00138 * @par SOLID Principles
00139 * - **Single Responsibility**: GPIO register definitions only, no driver logic
00140 * - **Open/Closed**: Extend via new enums; don't modify existing values
00141 * - **Interface Segregation**: Single struct handles all port register types
00142 * - **Dependency Inversion**: Higher-level drivers depend on these abstractions
00143 *
00144 * @par Related Modules
00145 * - [rx72n_mpc.h](rx72n__mpc__8h.html): Multi-function Pin Controller (pin function select)
00146 * - [rx72n_mtu_regs.h](rx72n__mtu__regs__8h.html): Timer I/O pins
00147 * - [rx72n_sci_regs.h](rx72n__sci__regs__8h.html): UART pins
00148 * - [rx72n_rspi_regs.h](rx72n__rspi__regs__8h.html): SPI pins
00149 *
00150 * @see RX72N Hardware Manual Chapter 22 for complete I/O Ports specification
00151 * @see DOXYGEN_ROADMAP.md for documentation standards
00152 *
00153 * @author Locked, Inc. Contributors
00154 * @date 2026-01-05
00155 * @version 1.0.0
00156 * @copyright Copyright (c) 2026 Locked Inc.
00157 * SPDX-License-Identifier: MIT
00158 *
00159 * @defgroup port_regs PORT Register Definitions
00160 * @{
00161 */
00162
00163 #pragma once
00164
00165 #include <assert.h>
00166 #include <stddef.h>
00167 #include <stdint.h>
00168
00169 #ifdef __cplusplus
00170 extern "C" {
00171 #endif
00172
00173 /*
=====
00174 * Port Register Base Addresses
00175 *
=====
00176 */
00177 /**
00178 * @enum rx_port_reg_bases_t
00179 * @brief PORT register block base addresses (organized by register type)
00180 *
00181 * @details
00182 * The RX72N organizes port registers by type rather than by port number.
00183 * All PDR registers are contiguous, all PODR registers are contiguous, etc.
00184 * This enum provides base addresses for each register type block.
00185 */

```

```

00186 *
00187 * @par Memory Map Overview
00188 * @verbatim
00189 *   Address Range      Register Type   Description
00190 *   -----
00191 *   0x0008C000-0x0008C017 PDR[]       Port Direction Registers
00192 *   0x0008C020-0x0008C037 PODR[]      Port Output Data Registers
00193 *   0x0008C040-0x0008C057 PIDR[]      Port Input Data Registers
00194 *   0x0008C060-0x0008C077 PMR[]       Port Mode Registers
00195 *   0x0008C080-0x0008C0B7 ODR0/ODR1[] Open Drain Control (16-bit per port)
00196 *   0x0008C0C0-0x0008C0D7 PCR[]       Pull-up Control Registers
00197 *   0x0008C0E0-0x0008C0F7 DSCR[]      Drive Capacity Registers
00198 *   0x0008C128-0x0008C13F DSCR2[]     Drive Capacity 2 Registers
00199 * @endverbatim
00200 *
00201 * @par Register Access Pattern
00202 * To access a specific port's register, add the port offset to the base:
00203 * - PORT0.PDR = k_port_pdr_base + k_port_offset_0 = 0x0008C000
00204 * - PORTB.PDR = k_port_pdr_base + k_port_offset_b = 0x0008C00B
00205 * - PORTD.PODR = k_port_podr_base + k_port_offset_d = 0x0008C02D
00206 *
00207 * @see rx_port_offsets_t for port offset values
00208 * @see RX72N Hardware Manual Section 22.3 (Register Descriptions)
00209 */
00210 typedef enum : uintptr_t {
00211 /**
00212  * @brief PDR (Port Direction Register) base address
00213  * @details All PDRn registers at 0x0008C000 + port_offset.
00214  * PDR bit = 0: Input, PDR bit = 1: Output.
00215  */
00216 k_port_pdr_base = 0x0008C000,
00217 /**
00218  * @brief PODR (Port Output Data Register) base address
00219  * @details All PODRn registers at 0x0008C020 + port_offset.
00220  * PODR bit = 0: Output low, PODR bit = 1: Output high.
00221  */
00222 k_port_podr_base = 0x0008C020,
00223 /**
00224  * @brief PIDR (Port Input Data Register) base address
00225  * @details All PIDRn registers at 0x0008C040 + port_offset.
00226  * Read-only. Returns current pin logic level.
00227  */
00228 k_port_pidr_base = 0x0008C040,
00229 /**
00230  * @brief PMR (Port Mode Register) base address
00231  * @details All PMRn registers at 0x0008C060 + port_offset.
00232  * PMR bit = 0: GPIO mode, PMR bit = 1: Peripheral function.
00233  */
00234 k_port_pmr_base = 0x0008C060,
00235 /**
00236  * @brief ODR0 (Open Drain Control 0) base address
00237  * @details ODR0/ODR1 form a 16-bit register per port at 0x0008C080 + 2*port_offset.
00238  * ODR bit = 0: CMOS output, ODR bit = 1: Open-drain output.
00239  * @note ODR1 immediately follows ODR0; accessed as 16-bit odr field in struct.
00240  */
00241 k_port_odr0_base = 0x0008C080,
00242 /**
00243  * @brief PCR (Pull-up Control Register) base address
00244  * @details All PCRn registers at 0x0008C0C0 + port_offset.
00245  * PCR bit = 0: Pull-up disabled, PCR bit = 1: Pull-up enabled.
00246  */
00247 k_port_pcr_base = 0x0008C0C0,
00248 /**
00249  * @brief DSCR (Drive Capacity Control Register) base address
00250  * @details All DSCRn registers at 0x0008C0E0 + port_offset.
00251  * DSCR bit = 0: Normal drive, DSCR bit = 1: High drive.
00252  */
00253 k_port_dscr_base = 0x0008C0E0,
00254 /**
00255  * @brief DSCR2 (Drive Capacity Control Register 2) base address
00256  * @details All DSCR2n registers at 0x0008C128 + port_offset.
00257  * DSCR2 bit = 0: Normal drive (same as DSCR).
00258  * DSCR2 bit = 1: Enhanced high drive capacity.
00259  * @note Not available on PORT4.
00260  */
00261 k_port_dscr2_base = 0x0008C128,
00262 } rx_port_reg_bases_t;
00263
00264 /**
00265  * @enum rx_port_offsets_t

```



```

00273 * @brief Port number offsets for register access (144-pin LQFP)
00274 *
00275 * @details
00276 * Offset values to add to register base addresses for accessing specific ports.
00277 * The 144-pin LQFP package has broader port availability compared to the
00278 * 144-pin package. Ports G, H, K, L, M, N, Q are not bonded out.
00279 *
00280 * @par Available Pins by Port (144-pin LQFP)
00281 * | Port | Offset | Available Pins | Notes |
00282 * |-----|-----|-----|-----|
00283 * | 0 | 0x00 | P00-P03, P05, P07 | 6 pins (no P04, P06) |
00284 * | 1 | 0x01 | P12-P17 | 6 pins (no P10, P11) |
00285 * | 2 | 0x02 | P20-P27 | Full 8 pins |
00286 * | 3 | 0x03 | P30-P37 | Full 8 pins, P35 input-only |
00287 * | 4 | 0x04 | P40-P47 | Full 8 pins |
00288 * | 5 | 0x05 | P50-P56 | 7 pins (no P57) |
00289 * | 6 | 0x06 | P60-P67 | Full 8 pins |
00290 * | 7 | 0x07 | P70-P77 | Full 8 pins |
00291 * | 8 | 0x08 | P80-P83, P86-P87 | 6 pins (no P84, P85) |
00292 * | 9 | 0x09 | P90-P93 | 4 pins (no P94-P97) |
00293 * | A | 0x0A | PA0-PA7 | Full 8 pins |
00294 * | B | 0x0B | PB0-PB7 | Full 8 pins |
00295 * | C | 0x0C | PC0-PC7 | Full 8 pins |
00296 * | D | 0x0D | PD0-PD7 | Full 8 pins |
00297 * | E | 0x0E | PE0-PE7 | Full 8 pins |
00298 * | F | 0x0F | PF5 | 1 pin only |
00299 * | J | 0x12 | PJ3, PJ5 | 2 pins only |
00300 *
00301 * @par Usage Example
00302 * @code
00303 * // Calculate PORTB PDR address (uintptr_t preserves full pointer width on all hosts)
00304 * uintptr_t portb_pdr_addr = k_port_pdr_base + k_port_offset_b; // 0x0008C00B
00305 *
00306 * // Calculate PORTD PODR address
00307 * uintptr_t portd_podr_addr = k_port_podr_base + k_port_offset_d; // 0x0008C02D
00308 * @endcode
00309 *
00310 * @see rx_port_reg_bases_t for register base addresses
00311 * @see RX72N Hardware Manual Chapter 22.1.1 (Package Pin Availability)
00312 */
00313 typedef enum : uint8_t {
00314 /** @brief Port 0 offset - P00-P03, P05, P07 (6 pins) */
00315 k_port_offset_0 = 0x00,
00316
00317 /** @brief Port 1 offset - P12-P17 (6 pins) */
00318 k_port_offset_1 = 0x01,
00319
00320 /** @brief Port 2 offset - P20-P27 (8 pins) */
00321 k_port_offset_2 = 0x02,
00322
00323 /** @brief Port 3 offset - P30-P37 (8 pins, P35 input-only) */
00324 k_port_offset_3 = 0x03,
00325
00326 /** @brief Port 4 offset - P40-P47 (8 pins) */
00327 k_port_offset_4 = 0x04,
00328
00329 /** @brief Port 5 offset - P50-P56 (7 pins) */
00330 k_port_offset_5 = 0x05,
00331
00332 /** @brief Port 6 offset - P60-P67 (8 pins) */
00333 k_port_offset_6 = 0x06,
00334
00335 /** @brief Port 7 offset - P70-P77 (8 pins) */
00336 k_port_offset_7 = 0x07,
00337
00338 /** @brief Port 8 offset - P80-P83, P86-P87 (6 pins) */
00339 k_port_offset_8 = 0x08,
00340
00341 /** @brief Port 9 offset - P90-P93 (4 pins) */
00342 k_port_offset_9 = 0x09,
00343
00344 /** @brief Port A offset - PA0-PA7 (8 pins) */
00345 k_port_offset_a = 0x0A,
00346
00347 /** @brief Port B offset - Full: PB0-PB7 (8 pins) */
00348 k_port_offset_b = 0x0B,
00349
00350 /** @brief Port C offset - Full: PC0-PC7 (8 pins) */
00351 k_port_offset_c = 0x0C,
00352
00353 /** @brief Port D offset - Full: PD0-PD7 (8 pins) */
00354 k_port_offset_d = 0x0D,
00355
00356 /** @brief Port E offset - PE0-PE7 (8 pins) */
00357 k_port_offset_e = 0x0E,
00358
00359 /** @brief Port F offset - PF5 (1 pin) */

```

```

00360 k_port_offset_f = 0x0F,
00361
00362 /** @brief Port J offset - PJ3, PJ5 (2 pins) */
00363 k_port_offset_j = 0x12,
00364 } rx_port_offsets_t;
00365
00366 /*
=====
00367 * Port Register Structure
00368 *
=====
00369 */
00370
00371 /**
00372 * @struct rx_port_regs_t
00373 * @brief Port Register Map providing logical per-port view of RX72N GPIO
00374 *
00375 * @details
00376 * This structure provides a convenient PORT-centric view of the GPIO registers,
00377 * abstracting away the hardware's type-grouped layout. Padding bytes ensure
00378 * correct alignment between register types.
00379 *
00380 * @par Register Memory Layout (297 bytes total per port view)
00381 * @verbatim
00382 *   Offset  Size  Register  Description
00383 *   -----
00384 *   0x00    1    PDR       Port Direction (0=input, 1=output)
00385 *   0x01   31    [pad]     Reserved padding to PODR
00386 *   0x20    1    PODR      Port Output Data (0=low, 1=high)
00387 *   0x21   31    [pad]     Reserved padding to PIDR
00388 *   0x40    1    PIDR      Port Input Data (read-only, current pin level)
00389 *   0x41   31    [pad]     Reserved padding to PMR
00390 *   0x60    1    PMR       Port Mode (0=GPIO, 1=peripheral function)
00391 *   0x61   95    [pad]     Reserved (includes ODR space, see note below)
00392 *   0xC0    1    PCR       Pull-up Control (0=disabled, 1=enabled)
00393 *   0xC1   31    [pad]     Reserved padding to DSCR
00394 *   0xE0    1    DSCR      Drive Capacity (0=normal, 1=high drive)
00395 *   0xE1   71    [pad]     Reserved padding to DSCR2
00396 *   0x128   1    DSCR2     Drive Capacity 2 (enhanced high drive)
00397 * @endverbatim
00398 *
00399 * @note ODR (Open Drain Control) registers are NOT part of this structure.
00400 * They use word addressing and are accessed via separate port*_odr() functions
00401 * at hardware addresses: 0x0008C080 + (port_offset * 2)
00402 *
00403 * @par GPIO Configuration Sequence
00404 * Typical GPIO output configuration sequence:
00405 * 1. Set PMR = 0 (GPIO mode, not peripheral)
00406 * 2. Set PDR = 1 (output direction)
00407 * 3. Set PODR = desired output level
00408 * 4. Optionally set DSCR = 1 for high drive current
00409 *
00410 * Typical GPIO input configuration sequence:
00411 * 1. Set PMR = 0 (GPIO mode)
00412 * 2. Set PDR = 0 (input direction)
00413 * 3. Optionally set PCR = 1 for internal pull-up
00414 * 4. Read PIDR for current pin state
00415 *
00416 * @par LED Blink Example
00417 * @code
00418 * // Configure PD0 as output and blink LED
00419 * volatile rx_port_regs_t* pd = portd();
00420 *
00421 * pd->pmr &= ~(1 << 0); // GPIO mode
00422 * pd->pdr |= (1 << 0); // Output direction
00423 *
00424 * while (1) {
00425 *     pd->podr ^= (1 << 0); // Toggle LED
00426 *     delay_ms(500);
00427 * }
00428 * @endcode
00429 *
00430 * @par Button Read Example
00431 * @code
00432 * // Configure PA0 as input with pull-up and read button
00433 * volatile rx_port_regs_t* pa = porta();
00434 *
00435 * pa->pmr &= ~(1 << 0); // GPIO mode
00436 * pa->pdr &= ~(1 << 0); // Input direction
00437 * pa->pcr |= (1 << 0); // Enable pull-up (button connects to GND)
00438 *
00439 * bool pressed = ((pa->pidr & (1 << 0)) == 0); // Active-low
00440 * @endcode
00441 *
00442 * @invariant Structure size must be exactly 297 bytes (0x129)
00443 * @invariant All register offsets must match hardware layout exactly
00444 * @invariant Structure must be packed to ensure correct padding sizes

```

```

00445 *
00446 * @note Uses __attribute__((packed)) to prevent compiler padding optimizations
00447 *
00448 * @warning This structure assumes PORT0 padding layout (offset=0x00).
00449 * The type-grouped hardware layout places ODR at varying struct offsets
00450 * for different ports (0x80 + port_offset*2). ODR is accessed via
00451 * separate port*_odr() functions to ensure correct word addressing.
00452 *
00453 * @warning Do not use sizeof() for port address calculations; use offset enum
00454 *
00455 * @note PORT3 lacks DSCR (has only DSCR2).
00456 * @note PORT4 lacks both DSCR and DSCR2 (hardware limitation).
00457 * @note All other ports (0-2, 5, A-E, J) have both DSCR and DSCR2.
00458 *
00459 * @see rx_port_reg_bases_t for register base addresses
00460 * @see rx_port_offsets_t for port offset values
00461 * @see RX72N Hardware Manual Section 22.3 (Register Descriptions)
00462 */
00463 typedef struct __attribute__((packed)) {
00464 /**
00465  * @brief Port Direction Register (PDR) - +0x00
00466  * @details Configures each pin as input (0) or output (1).
00467  * @note Before writing PDR, ensure PMR = 0 for GPIO mode.
00468  */
00469 volatile uint8_t pdr;
00470
00471 /** @brief Reserved padding (0x01-0x1F) to align PODR at +0x20 */
00472 volatile uint8_t _pad1[0x1F];
00473
00474 /**
00475  * @brief Port Output Data Register (PODR) - +0x20
00476  * @details Sets output level: 0 = low, 1 = high.
00477  * @note Only effective when PDR bit = 1 (output direction).
00478  */
00479 volatile uint8_t podr;
00480
00481 /** @brief Reserved padding (0x21-0x3F) to align PIDR at +0x40 */
00482 volatile uint8_t _pad2[0x1F];
00483
00484 /**
00485  * @brief Port Input Data Register (PIDR) - +0x40
00486  * @details Read-only register returning current pin logic level.
00487  * @note Always readable regardless of PDR setting.
00488  */
00489 volatile uint8_t pidr;
00490
00491 /** @brief Reserved padding (0x41-0x5F) to align PMR at +0x60 */
00492 volatile uint8_t _pad3[0x1F];
00493
00494 /**
00495  * @brief Port Mode Register (PMR) - +0x60
00496  * @details Selects GPIO (0) or peripheral function (1) mode.
00497  * @note Set PMR = 0 before configuring GPIO direction/output.
00498  * @see MPC (Multi-function Pin Controller) for peripheral selection.
00499  */
00500 volatile uint8_t pmr;
00501
00502 /** @brief Reserved padding (0x61-0xBF) to align PCR at +0xC0 */
00503 volatile uint8_t _pad4[0x5F];
00504
00505 /**
00506  * @brief Pull-up Control Register (PCR) - +0xC0
00507  * @details Enables internal pull-up resistor (approx 50kOhm typical).
00508  * PCR bit = 0: Pull-up disabled.
00509  * PCR bit = 1: Pull-up enabled.
00510  * @note Only effective when PDR bit = 0 (input direction).
00511  */
00512 volatile uint8_t pcr;
00513
00514 /** @brief Reserved padding (0xC1-0xDF) to align DSCR at +0xE0 */
00515 volatile uint8_t _pad5[0x1F];
00516
00517 /**
00518  * @brief Drive Capacity Control Register (DSCR) - +0xE0
00519  * @details Configures output drive strength.
00520  * DSCR bit = 0: Normal drive capacity.
00521  * DSCR bit = 1: High drive capacity.
00522  * @note High drive useful for long traces or high capacitance loads.
00523  */
00524 volatile uint8_t dscr;
00525
00526 /** @brief Reserved padding (0xE1-0x127) to align DSCR2 at +0x128 */
00527 volatile uint8_t _pad6[0x47];
00528
00529 /**
00530  * @brief Drive Capacity Control Register 2 (DSCR2) - +0x128
00531  * @details Secondary drive capacity control for high-current applications.

```

```

00532 * DSCR2 bit = 0: Normal drive capacity (same as DSCR).
00533 * DSCR2 bit = 1: Enhanced high drive capacity.
00534 * @note Not available on PORT4 (hardware limitation).
00535 * @note PORT3 lacks DSCR but has DSCR2.
00536 */
00537 volatile uint8_t dscr2;
00538 } rx_port_regs_t;
00539
00540 /*
=====
00541 * Static Assertions - Compile-time Register Layout Verification
00542 *
=====
00543 *
00544 * These static assertions verify that the rx_port_regs_t struct layout
00545 * exactly matches the RX72N hardware register addresses. Any mismatch
00546 * will cause a compile-time error, preventing runtime GPIO bugs.
00547 *
00548 * Reference: RX72N Group User's Manual: Hardware, Chapter 22 (I/O Ports)
00549 * Section 22.3: Register Descriptions
00550 */
00551
00552 /** @name PORT Register Offset Verification
00553 * @brief Verify rx_port_regs_t matches hardware layout
00554 * @{
00555 */
00556 /* Note: struct size is 0x129 including DSCR2, ODR is accessed separately via word-addressed functions */
00557 static_assert(sizeof(rx_port_regs_t) == 0x129,
00558               "Port register struct size must be 0x129 (297 bytes)");
00559 static_assert(offsetof(rx_port_regs_t, pdr) == 0x00, "PDR must be at offset 0x00");
00560 static_assert(offsetof(rx_port_regs_t, podr) == 0x20, "PODR must be at offset 0x20");
00561 static_assert(offsetof(rx_port_regs_t, pidr) == 0x40, "PIDR must be at offset 0x40");
00562 static_assert(offsetof(rx_port_regs_t, pmr) == 0x60, "PMR must be at offset 0x60");
00563 static_assert(offsetof(rx_port_regs_t, pcr) == 0xC0, "PCR must be at offset 0xC0");
00564 static_assert(offsetof(rx_port_regs_t, dscr) == 0xE0, "DSCR must be at offset 0xE0");
00565 static_assert(offsetof(rx_port_regs_t, dscr2) == 0x128, "DSCR2 must be at offset 0x128");
00566 /** @} */ /* End of register offset verification group */
00567
00568 /** @name PORT Register Base Address Verification
00569 * @brief Verify register base addresses match hardware memory map
00570 * @{
00571 */
00572 static_assert((uint32_t)k_port_dscr2_base == 0x0008C128,
00573               "DSCR2 base address must be 0x0008C128 (DSCR + 0x48)");
00574 /** @} */ /* End of base address verification group */
00575
00576 /*
=====
00577 * Inline Register Accessors
00578 *
=====
00579 *
00580 * These inline functions provide type-safe access to each port's registers.
00581 * Using functions instead of macros provides:
00582 * - Type safety (returns proper volatile pointer)
00583 * - Debugger visibility (can set breakpoints)
00584 * - Namespace cleanliness (no macro pollution)
00585 */
00586
00587 /**
00588 * @brief Get pointer to PORT0 registers
00589 *
00590 * @details
00591 * Returns a volatile pointer to PORT0's register structure. PORT0 has
00592 * P00-P03, P05, P07 available on the 144-pin LQFP package (6 pins).
00593 *
00594 * @return Volatile pointer to PORT0 register structure
00595 *
00596 * @par Register Addresses
00597 * - PDR: 0x0008C000
00598 * - PODR: 0x0008C020
00599 * - PIDR: 0x0008C040
00600 * - PMR: 0x0008C060
00601 *
00602 * @par Available Pins (144-pin)
00603 * P00-P03, P05, P07 (6 pins)
00604 *
00605 * @note Thread-safe: returns constant hardware address
00606 * @see rx_port_regs_t for register layout
00607 */
00608 static inline volatile rx_port_regs_t* port0(void)
00609 {
00610     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_0);
00611 }
00612
00613 /**
00614 * @brief Get pointer to PORT1 registers

```

```

00615 * @return Volatile pointer to PORT1 register structure
00616 * @note PORT1 PDR = 0x0008C001, PODR = 0x0008C021, PIDR = 0x0008C041
00617 * @note 144-pin: Only P12-P17 available
00618 */
00619 static inline volatile rx_port_regs_t* port1(void)
00620 {
00621     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_1);
00622 }
00623
00624 /**
00625 * @brief Get pointer to PORT2 registers
00626 * @return Volatile pointer to PORT2 register structure
00627 * @note PORT2 PDR = 0x0008C002, PODR = 0x0008C022, PIDR = 0x0008C042
00628 * @note 144-pin: P20-P27 available
00629 */
00630 static inline volatile rx_port_regs_t* port2(void)
00631 {
00632     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_2);
00633 }
00634
00635 /**
00636 * @brief Get pointer to PORT3 registers
00637 * @return Volatile pointer to PORT3 register structure
00638 * @note PORT3 PDR = 0x0008C003, PODR = 0x0008C023, PIDR = 0x0008C043
00639 * @note 144-pin: P30-P37 available (P35 is input-only)
00640 */
00641 /* LCOV_EXCL_START */
00642 static inline volatile rx_port_regs_t* port3(void)
00643 {
00644     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_3);
00645 }
00646 /* LCOV_EXCL_STOP */
00647
00648 /**
00649 * @brief Get pointer to PORT4 registers
00650 * @return Volatile pointer to PORT4 register structure
00651 * @note PORT4 PDR = 0x0008C004, PODR = 0x0008C024, PIDR = 0x0008C044
00652 * @note 144-pin: P40-P47 available
00653 */
00654 /* LCOV_EXCL_START */
00655 static inline volatile rx_port_regs_t* port4(void)
00656 {
00657     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_4);
00658 }
00659 /* LCOV_EXCL_STOP */
00660
00661 /**
00662 * @brief Get pointer to PORT5 registers
00663 * @return Volatile pointer to PORT5 register structure
00664 * @note PORT5 PDR = 0x0008C005, PODR = 0x0008C025, PIDR = 0x0008C045
00665 * @note 144-pin: P50-P56 available (7 pins)
00666 */
00667 /* LCOV_EXCL_START */
00668 static inline volatile rx_port_regs_t* port5(void)
00669 {
00670     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_5);
00671 }
00672 /* LCOV_EXCL_STOP */
00673
00674 /**
00675 * @brief Get pointer to PORT6 registers
00676 * @return Volatile pointer to PORT6 register structure
00677 * @note PORT6 PDR = 0x0008C006, PODR = 0x0008C026, PIDR = 0x0008C046
00678 * @note 144-pin: P60-P67 available (8 pins)
00679 */
00680 /* LCOV_EXCL_START */
00681 static inline volatile rx_port_regs_t* port6(void)
00682 {
00683     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_6);
00684 }
00685 /* LCOV_EXCL_STOP */
00686
00687 /**
00688 * @brief Get pointer to PORT7 registers
00689 * @return Volatile pointer to PORT7 register structure
00690 * @note PORT7 PDR = 0x0008C007, PODR = 0x0008C027, PIDR = 0x0008C047
00691 * @note 144-pin: P70-P77 available (8 pins)
00692 */
00693 /* LCOV_EXCL_START */
00694 static inline volatile rx_port_regs_t* port7(void)
00695 {
00696     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_7);
00697 }
00698 /* LCOV_EXCL_STOP */
00699
00700 /**
00701 * @brief Get pointer to PORT8 registers

```

```

00702 * @return Volatile pointer to PORT8 register structure
00703 * @note PORT8 PDR = 0x0008C008, PODR = 0x0008C028, PIDR = 0x0008C048
00704 * @note 144-pin: P80-P83, P86-P87 available (6 pins)
00705 */
00706 /* LCOV_EXCL_START */
00707 static inline volatile rx_port_regs_t* port8(void)
00708 {
00709     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_8);
00710 }
00711 /* LCOV_EXCL_STOP */
00712 /**
00713 * @brief Get pointer to PORT9 registers
00714 * @return Volatile pointer to PORT9 register structure
00715 * @note PORT9 PDR = 0x0008C009, PODR = 0x0008C029, PIDR = 0x0008C049
00716 * @note 144-pin: P90-P93 available (4 pins)
00717 */
00718 /* LCOV_EXCL_START */
00719 static inline volatile rx_port_regs_t* port9(void)
00720 {
00721     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_9);
00722 }
00723 /* LCOV_EXCL_STOP */
00724 /**
00725 * @brief Get pointer to PORTA registers
00726 * @return Volatile pointer to PORTA register structure
00727 * @note PORTA PDR = 0x0008C00A, PODR = 0x0008C02A, PIDR = 0x0008C04A
00728 * @note 144-pin: PA0-PA7 available
00729 */
00730 static inline volatile rx_port_regs_t* porta(void)
00731 {
00732     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_a);
00733 }
00734 /**
00735 * @brief Get pointer to PORTB registers
00736 * @return Volatile pointer to PORTB register structure
00737 * @note PORTB PDR = 0x0008C00B, PODR = 0x0008C02B, PIDR = 0x0008C04B
00738 * @note 144-pin: PB0-PB7 available
00739 */
00740 static inline volatile rx_port_regs_t* portb(void)
00741 {
00742     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_b);
00743 }
00744 /**
00745 * @brief Get pointer to PORTC registers
00746 * @return Volatile pointer to PORTC register structure
00747 * @note PORTC PDR = 0x0008C00C, PODR = 0x0008C02C, PIDR = 0x0008C04C
00748 * @note 144-pin: PC0-PC7 available
00749 */
00750 static inline volatile rx_port_regs_t* portc(void)
00751 {
00752     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_c);
00753 }
00754 /**
00755 * @brief Get pointer to PORTD registers
00756 * @return Volatile pointer to PORTD register structure
00757 * @note PORTD PDR = 0x0008C00D, PODR = 0x0008C02D, PIDR = 0x0008C04D
00758 * @note 144-pin: PD0-PD7 available
00759 */
00760 static inline volatile rx_port_regs_t* portd(void)
00761 {
00762     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_d);
00763 }
00764 /**
00765 * @brief Get pointer to PORTE registers
00766 * @return Volatile pointer to PORTE register structure
00767 * @note PORTE PDR = 0x0008C00E, PODR = 0x0008C02E, PIDR = 0x0008C04E
00768 * @note 144-pin: PE0-PE7 available
00769 */
00770 static inline volatile rx_port_regs_t* porte(void)
00771 {
00772     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_e);
00773 }
00774 /**
00775 * @brief Get pointer to PORTF registers
00776 * @return Volatile pointer to PORTF register structure
00777 * @note PORTF PDR = 0x0008C00F, PODR = 0x0008C02F, PIDR = 0x0008C04F
00778 * @note 144-pin: PF5 available (1 pin)
00779 */
00780 /* LCOV_EXCL_START */
00781 static inline volatile rx_port_regs_t* portf(void)

```

```

00789 {
00790     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_f);
00791 }
00792 /* LCOV_EXCL_STOP */
00793
00794 /**
00795  * @brief Get pointer to PORTJ registers
00796  * @return Volatile pointer to PORTJ register structure
00797  * @note PORTJ PDR = 0x0008C012, PODR = 0x0008C032, PIDR = 0x0008C052
00798  * @note 144-pin: PJ3, PJ5 available (2 pins)
00799  */
00800 static inline volatile rx_port_regs_t* portj(void)
00801 {
00802     return (volatile rx_port_regs_t*)(k_port_pdr_base + k_port_offset_j);
00803 }
00804
00805 /*
=====
00806  * ODR (Open Drain) Register Accessors
00807  *
=====
00808  *
00809  * ODR registers use WORD (16-bit) addressing, not byte addressing like other
00810  * PORT registers. Each port's ODR is at: 0x0008C080 + (port_offset * 2)
00811  *
00812  * Hardware addresses (from RX72N manual Ch22):
00813  *   PORT0.ODR: 0x0008C080 + (0x00 * 2) = 0x0008C080
00814  *   PORT1.ODR: 0x0008C080 + (0x01 * 2) = 0x0008C082
00815  *   PORTA.ODR: 0x0008C080 + (0x0A * 2) = 0x0008C094
00816  *   PORTJ.ODR: 0x0008C080 + (0x12 * 2) = 0x0008C0A4
00817  *
00818  * These inline accessors ensure correct addressing for all ports.
00819  */
00820
00821 /**
00822  * @brief Get pointer to PORT0 ODR register
00823  * @return Volatile pointer to 16-bit ODR register (combines ODR0+ODR1)
00824  * @note PORT0.ODR = 0x0008C080
00825  */
00826 static inline volatile uint16_t* port0_odr(void)
00827 {
00828     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_0 * 2));
00829 }
00830
00831 /**
00832  * @brief Get pointer to PORT1 ODR register
00833  * @return Volatile pointer to 16-bit ODR register
00834  * @note PORT1.ODR = 0x0008C082 (word addressing: base + offset*2)
00835  */
00836 static inline volatile uint16_t* port1_odr(void)
00837 {
00838     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_1 * 2));
00839 }
00840
00841 /**
00842  * @brief Get pointer to PORT2 ODR register
00843  * @return Volatile pointer to 16-bit ODR register
00844  * @note PORT2.ODR = 0x0008C084
00845  */
00846 static inline volatile uint16_t* port2_odr(void)
00847 {
00848     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_2 * 2));
00849 }
00850
00851 /**
00852  * @brief Get pointer to PORT3 ODR register
00853  * @return Volatile pointer to 16-bit ODR register
00854  * @note PORT3.ODR = 0x0008C086
00855  */
00856 static inline volatile uint16_t* port3_odr(void)
00857 {
00858     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_3 * 2));
00859 }
00860
00861 /**
00862  * @brief Get pointer to PORT4 ODR register
00863  * @return Volatile pointer to 16-bit ODR register
00864  * @note PORT4.ODR = 0x0008C088
00865  */
00866 static inline volatile uint16_t* port4_odr(void)
00867 {
00868     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_4 * 2));
00869 }
00870
00871 /**
00872  * @brief Get pointer to PORT5 ODR register
00873  * @return Volatile pointer to 16-bit ODR register

```



```

00874 * @note PORT5.ODR = 0x0008C08A
00875 */
00876 static inline volatile uint16_t* port5_odr(void)
00877 {
00878     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_5 * 2));
00879 }
00880
00881 /**
00882 * @brief Get pointer to PORT6 ODR register
00883 * @return Volatile pointer to 16-bit ODR register
00884 * @note PORT6.ODR = 0x0008C08C (offset 0x06 * 2)
00885 */
00886 static inline volatile uint16_t* port6_odr(void)
00887 {
00888     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_6 * 2));
00889 }
00890
00891 /**
00892 * @brief Get pointer to PORT7 ODR register
00893 * @return Volatile pointer to 16-bit ODR register
00894 * @note PORT7.ODR = 0x0008C08E (offset 0x07 * 2)
00895 */
00896 static inline volatile uint16_t* port7_odr(void)
00897 {
00898     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_7 * 2));
00899 }
00900
00901 /**
00902 * @brief Get pointer to PORT8 ODR register
00903 * @return Volatile pointer to 16-bit ODR register
00904 * @note PORT8.ODR = 0x0008C090 (offset 0x08 * 2)
00905 */
00906 static inline volatile uint16_t* port8_odr(void)
00907 {
00908     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_8 * 2));
00909 }
00910
00911 /**
00912 * @brief Get pointer to PORT9 ODR register
00913 * @return Volatile pointer to 16-bit ODR register
00914 * @note PORT9.ODR = 0x0008C092 (offset 0x09 * 2)
00915 */
00916 static inline volatile uint16_t* port9_odr(void)
00917 {
00918     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_9 * 2));
00919 }
00920
00921 /**
00922 * @brief Get pointer to PORTA ODR register
00923 * @return Volatile pointer to 16-bit ODR register
00924 * @note PORTA.ODR = 0x0008C094 (offset 0x0A * 2)
00925 */
00926 static inline volatile uint16_t* porta_odr(void)
00927 {
00928     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_a * 2));
00929 }
00930
00931 /**
00932 * @brief Get pointer to PORTB ODR register
00933 * @return Volatile pointer to 16-bit ODR register
00934 * @note PORTB.ODR = 0x0008C096
00935 */
00936 static inline volatile uint16_t* portb_odr(void)
00937 {
00938     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_b * 2));
00939 }
00940
00941 /**
00942 * @brief Get pointer to PORTC ODR register
00943 * @return Volatile pointer to 16-bit ODR register
00944 * @note PORTC.ODR = 0x0008C098
00945 */
00946 static inline volatile uint16_t* portc_odr(void)
00947 {
00948     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_c * 2));
00949 }
00950
00951 /**
00952 * @brief Get pointer to PORTD ODR register
00953 * @return Volatile pointer to 16-bit ODR register
00954 * @note PORTD.ODR = 0x0008C09A
00955 */
00956 static inline volatile uint16_t* portd_odr(void)
00957 {
00958     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_d * 2));
00959 }
00960

```



```

00961 /**
00962  * @brief Get pointer to PORTE ODR register
00963  * @return Volatile pointer to 16-bit ODR register
00964  * @note PORTE.ODR = 0x0008C09C
00965  */
00966 static inline volatile uint16_t* porte_odr(void)
00967 {
00968     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_e * 2));
00969 }
00970
00971 /**
00972  * @brief Get pointer to PORTF ODR register
00973  * @return Volatile pointer to 16-bit ODR register
00974  * @note PORTF.ODR = 0x0008C09E (offset 0x0F * 2)
00975  */
00976 static inline volatile uint16_t* portf_odr(void)
00977 {
00978     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_f * 2));
00979 }
00980
00981 /**
00982  * @brief Get pointer to PORTJ ODR register
00983  * @return Volatile pointer to 16-bit ODR register
00984  * @note PORTJ.ODR = 0x0008C0A4 (offset 0x12 * 2)
00985  */
00986 static inline volatile uint16_t* portj_odr(void)
00987 {
00988     return (volatile uint16_t*)(k_port_odr0_base + (k_port_offset_j * 2));
00989 }
00990
00991 /** @} */ /* End of port_regs defgroup */
00992
00993 #ifdef __cplusplus
00994 }
00995 #endif

```

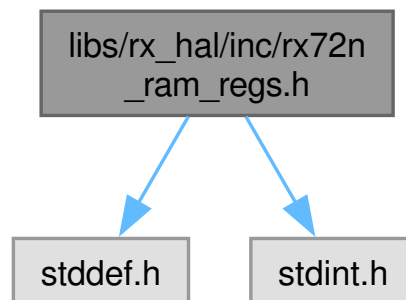
8.47 libs/rx_hal/inc/rx72n_ram_regs.h File Reference

RX72N RAM and Expansion RAM register definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_ram_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_ram_regs_t](#)
RAM control register structure.
- struct [rx_exram_regs_t](#)
Expansion RAM control register structure.

Enumerations

- enum `rx_ram_region_addr_t` : `uintptr_t` {
`k_rx_ram_base_addr` = 0x00000000 , `k_rx_ram_end_addr` = 0x0007FFFF , `k_rx_ram_size` = 0x00080000 , `k_rx_exram_base_addr` = 0x00800000 ,
`k_rx_exram_end_addr` = 0x0087FFFF , `k_rx_exram_size` = 0x00080000 }
RAM region base addresses and sizes.
- enum `rx_ram_reg_addr_t` : `uintptr_t` { `k_rx_ram_reg_base_addr` = 0x00081200 ,
`k_rx_exram_reg_base_addr` = 0x00081240 }
RAM control register base addresses.
- enum `rx_ram_reg_offset_t` : `uint8_t` { `k_rx_ram_offset_rammode` = 0x00 , `k_rx_ram_offset_ramsts` = 0x01 , `k_rx_ram_offset_ramprcr` = 0x04 , `k_rx_ram_offset_ramecad` = 0x08 }
RAM register offsets from base 0x00081200.
- enum `rx_exram_reg_offset_t` : `uint8_t` { `k_rx_exram_offset_exrammode` = 0x00 ,
`k_rx_exram_offset_exramsts` = 0x01 , `k_rx_exram_offset_exramprcr` = 0x04 , `k_rx_exram_offset_exramecad` = 0x08 }
Expansion RAM register offsets from base 0x00081240.
- enum `rx_rammode_bits_t` : `uint8_t` { `k_rx_rammode_parity_disabled` = 0x00 , `k_rx_rammode_parity_enabled` = 0x01 }
RAMMODE register bit definitions.
- enum `rx_ramsts_bits_t` : `uint8_t` { `k_rx_ramsts_ramerr_bit` = 0 , `k_rx_ramsts_ramerr_mask` = 0x01 }
RAMSTS register bit definitions.
- enum `rx_ramprcr_bits_t` : `uint8_t` {
`k_rx_ramprcr_ramprcr_bit` = 0 , `k_rx_ramprcr_ramprcr_mask` = 0x01 , `k_rx_ramprcr_key` = 0xF0 , `k_rx_ramprcr_unlock` = 0xF1 ,
`k_rx_ramprcr_lock` = 0xF0 }
RAMPRCR register bit definitions.
- enum `rx_exrammode_bits_t` : `uint8_t` { `k_rx_exrammode_parity_disabled` = 0x00 ,
`k_rx_exrammode_parity_enabled` = 0x01 }
EXRAMMODE register bit definitions.
- enum `rx_exramsts_bits_t` : `uint8_t` { `k_rx_exramsts_exramerr_bit` = 0 , `k_rx_exramsts_exramerr_mask` = 0x01 }
EXRAMSTS register bit definitions.
- enum `rx_exramprcr_bits_t` : `uint8_t` {
`k_rx_exramprcr_exramprcr_bit` = 0 , `k_rx_exramprcr_exramprcr_mask` = 0x01 , `k_rx_exramprcr_key` = 0xF0 , `k_rx_exramprcr_unlock` = 0xF1 ,
`k_rx_exramprcr_lock` = 0xF0 }
EXRAMPRCR register bit definitions.
- enum `rx_ram_mstpcrc_bits_t` : `uint32_t` { `k_rx_mstpcrc_ram_bit` = 0 , `k_rx_mstpcrc_exram_bit` = 2 }
MSTPCRC register bits for parity-RAM module stop control.

Functions

- static `rx_ram_regs_t * ram_regs` (void)
Get pointer to RAM control registers.
- static `rx_exram_regs_t * exram_regs` (void)
Get pointer to Expansion RAM control registers.

8.47.1 Detailed Description

RX72N RAM and Expansion RAM register definitions.

This header defines register structures and addresses for the two parity-checked static RAM regions on the RX72N MCU:

- RAM (512 KB at 0x00000000-0x0007FFFF): 1-bit parity checking, Memory Bus 1
- Expansion RAM (512 KB at 0x00800000-0x0087FFFF): 1-bit parity checking, Memory Bus 3

The ECC-protected 32 KB region (ECCRAM) has its own header, [rx72n_eccram_regs.h](#), since its control block and usage pattern differ substantially from the parity-only regions.

All register addresses are verified against the RX72N User's Manual Chapter 60.

Memory Map (from Manual Ch60, Table 60.1):

Type	Address Range	Size	Error Checking
RAM	0x00000000 - 0x0007FFFF	512 KB	1-bit parity detection
Expansion RAM	0x00800000 - 0x0087FFFF	512 KB	1-bit parity detection

Register Base Addresses:

- RAM registers: 0x00081200
- Expansion RAM registers: 0x00081240

Note

All RAM regions operate after reset without module stop.

Data is NOT retained during deep software standby mode.

When ICLK > 120 MHz, MEMWAIT must be set to 1 for Expansion RAM.

NASA Power of 10 Compliance:

- Rule 5: Static assertions verify all register addresses
- Rule 8: All constants use typed enums, no macros for values

See also

[rx72n_eccram_regs.h](#) ECC-protected RAM definitions

RX72N User's Manual Chapter 60 (RAM)

RX72N User's Manual Chapter 11 (Low Power Consumption) for MSTPCRC

RX72N User's Manual Section 9.2.2 (MEMWAIT register)

Since

Version 1.0.0

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Definition in file [rx72n_ram_regs.h](#).

8.47.2 Data Structure Documentation

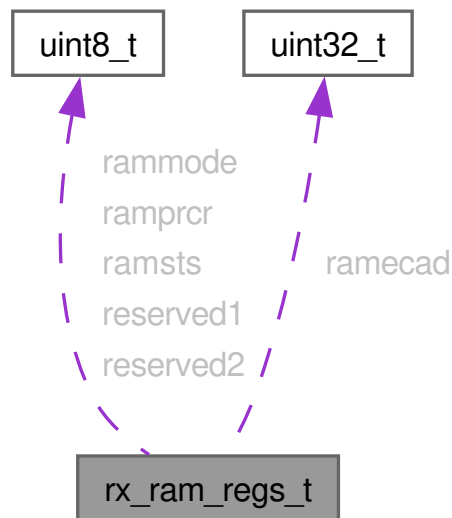
8.47.2.1 struct rx'ram'regs't

RAM control register structure.

Memory-mapped structure for RAM control registers at 0x00081200. Verified against Manual Ch60, sections 60.2.1-60.2.4.

Definition at line 272 of file [rx72n_ram_regs.h](#).

Collaboration diagram for rx_ram_regs_t:



Data Fields

volatile uint32_t	ramecad	0x08: RAM Error Address Capture Register
volatile uint8_t	rammode	0x00: RAM Operating Mode Control Register
volatile uint8_t	ramprcr	0x04: RAM Protection Register
volatile uint8_t	ramsts	0x01: RAM Error Status Register
volatile uint8_t	reserved1↔ [2]	0x02-0x03: Reserved
volatile uint8_t	reserved2↔ [3]	0x05-0x07: Reserved

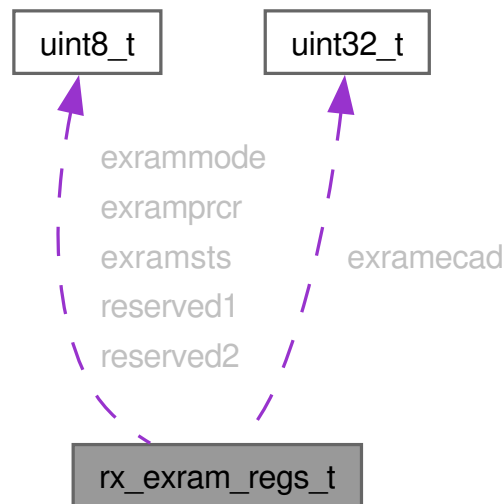
8.47.2.2 struct rx'exram'regs't

Expansion RAM control register structure.

Memory-mapped structure for Expansion RAM control registers at 0x00081240. Verified against Manual Ch60, sections 60.2.5-60.2.8.

Definition at line 289 of file [rx72n_ram_regs.h](#).

Collaboration diagram for rx_exram_regs_t:



Data Fields

volatile uint32_t	exramecad	0x08: Expansion RAM Error Address Capture Register
volatile uint8_t	exrammode	0x00: Expansion RAM Operating Mode Control Register
volatile uint8_t	exramprcr	0x04: Expansion RAM Protection Register
volatile uint8_t	exramsts	0x01: Expansion RAM Error Status Register
volatile uint8_t	reserved1↔ [2]	0x02-0x03: Reserved
volatile uint8_t	reserved2↔ [3]	0x05-0x07: Reserved

8.47.3 Enumeration Type Documentation

8.47.3.1 rx_exram_reg_offset_t

enum [rx_exram_reg_offset_t](#) : uint8_t

Expansion RAM register offsets from base 0x00081240.

Offsets verified against Manual Ch60, sections 60.2.5-60.2.8.

Enumerator

k_rx_exram_offset_exrammode	EXRAMMODE @ 0x00081240 - Expansion RAM Operating Mode Control Register.
k_rx_exram_offset_exramsts	EXRAMSTS @ 0x00081241 - Expansion RAM Error Status Register.
k_rx_exram_offset_exramprcr	EXRAMPRCR @ 0x00081244 - Expansion RAM Protection Register.
k_rx_exram_offset_exramecad	EXRAMECAD @ 0x00081248 - Expansion RAM Error Address Capture Register.

Definition at line 130 of file [rx72n_ram_regs.h](#).

```

00130         : uint8_t {
00131     /** @brief EXRAMMODE @ 0x00081240 - Expansion RAM Operating Mode Control Register */
00132     k_rx_exram_offset_exrammode = 0x00,
00133     /** @brief EXRAMSTS @ 0x00081241 - Expansion RAM Error Status Register */
00134     k_rx_exram_offset_exramsts = 0x01,
00135     /** @brief EXRAMPRCR @ 0x00081244 - Expansion RAM Protection Register */
00136     k_rx_exram_offset_exramprcr = 0x04,
00137     /** @brief EXRAMECAD @ 0x00081248 - Expansion RAM Error Address Capture Register */
00138     k_rx_exram_offset_exramecad = 0x08,
00139 } rx_exram_reg_offset_t;

```

8.47.3.2 rx_exrammode_bits_t

```
enum rx_exrammode_bits_t : uint8_t
```

EXRAMMODE register bit definitions.

Expansion RAM Operating Mode Select from Manual Ch60, section 60.2.5.

Enumerator

k_rx_exrammode_parity_disabled	Parity checking disabled (EXRAMMODE[1:0] = 00b).
k_rx_exrammode_parity_enabled	Parity checking enabled (EXRAMMODE[1:0] = 01b).

Definition at line 201 of file [rx72n_ram_regs.h](#).

```

00201         : uint8_t {
00202     /** @brief Parity checking disabled (EXRAMMODE[1:0] = 00b) */
00203     k_rx_exrammode_parity_disabled = 0x00,
00204     /** @brief Parity checking enabled (EXRAMMODE[1:0] = 01b) */
00205     k_rx_exrammode_parity_enabled = 0x01,
00206 } rx_exrammode_bits_t;

```

8.47.3.3 rx_exramprcr_bits_t

```
enum rx_exramprcr_bits_t : uint8_t
```

EXRAMPRCR register bit definitions.

Expansion RAM Protection Register from Manual Ch60, section 60.2.8. Write 0xF1 to enable writing to EXRAMMODE register.

Enumerator

k_rx_exramprcr_exramprcr_bit	EXRAMPRCR bit position.
k_rx_exramprcr_exramprcr_mask	EXRAMPRCR enable mask.
k_rx_exramprcr_key	Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register.
k_rx_exramprcr_unlock	Complete value to enable EXRAMMODE writes.
k_rx_exramprcr_lock	Complete value to disable EXRAMMODE writes.

Definition at line 230 of file [rx72n_ram_regs.h](#).

```

00230         : uint8_t {
00231     /** @brief EXRAMPRCR bit position */
00232     k_rx_exramprcr_exramprcr_bit = 0,
00233     /** @brief EXRAMPRCR enable mask */
00234     k_rx_exramprcr_exramprcr_mask = 0x01,
00235     /** @brief Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register */
00236     k_rx_exramprcr_key = 0xF0,
00237     /** @brief Complete value to enable EXRAMMODE writes */
00238     k_rx_exramprcr_unlock = 0xF1,
00239     /** @brief Complete value to disable EXRAMMODE writes */
00240     k_rx_exramprcr_lock = 0xF0,
00241 } rx_exramprcr_bits_t;

```

8.47.3.4 rx_exramsts_bits_t

```
enum rx_exramsts_bits_t : uint8_t
```

EXRAMSTS register bit definitions.

Expansion RAM Error Status Flag from Manual Ch60, section 60.2.6.

Enumerator

k_rx_exramsts_exramerr_bit	Expansion RAM error flag bit position.
k_rx_exramsts_exramerr_mask	Expansion RAM error flag mask.

Definition at line 215 of file [rx72n_ram_regs.h](#).

```
00215         : uint8_t {
00216     /** @brief Expansion RAM error flag bit position */
00217     k_rx_exramsts_exramerr_bit = 0,
00218     /** @brief Expansion RAM error flag mask */
00219     k_rx_exramsts_exramerr_mask = 0x01,
00220 } rx_exramsts_bits_t;
```

8.47.3.5 rx_ram_mstpcrc_bits_t

```
enum rx_ram_mstpcrc_bits_t : uint32_t
```

MSTPCRC register bits for parity-RAM module stop control.

Module stop control from Manual Ch60, section 60.4.1. Setting these bits stops clock supply to the respective RAM areas.

Note

The MSTPC6 bit for ECCRAM lives in [rx72n_eccram_regs.h](#).

Enumerator

k_rx_mstpcrc_ram_bit	MSTPC0 - RAM module stop bit (stops RAM clock).
k_rx_mstpcrc_exram_bit	MSTPC2 - Expansion RAM module stop bit.

Definition at line 253 of file [rx72n_ram_regs.h](#).

```
00253         : uint32_t {
00254     /** @brief MSTPC0 - RAM module stop bit (stops RAM clock) */
00255     k_rx_mstpcrc_ram_bit = 0,
00256     /** @brief MSTPC2 - Expansion RAM module stop bit */
00257     k_rx_mstpcrc_exram_bit = 2,
00258 } rx_ram_mstpcrc_bits_t;
```

8.47.3.6 rx_ram_reg_addr_t

```
enum rx_ram_reg_addr_t : uintptr_t
```

RAM control register base addresses.

Defines register base addresses for RAM control, verified against Ch60.

Enumerator

k_rx_ram_reg_base_addr	RAM control registers base address.
k_rx_exram_reg_base_addr	Expansion RAM control registers base address.

Definition at line 94 of file [rx72n_ram_regs.h](#).

```
00094 : uintptr_t {
00095 /** @brief RAM control registers base address */
00096 k_rx_ram_reg_base_addr = 0x00081200,
00097 /** @brief Expansion RAM control registers base address */
00098 k_rx_exram_reg_base_addr = 0x00081240,
00099 } rx_ram_reg_addr_t;
```

8.47.3.7 rx_ram_reg_offset_t

```
enum rx_ram_reg_offset_t : uint8_t
```

RAM register offsets from base 0x00081200.

Offsets verified against Manual Ch60, sections 60.2.1-60.2.4.

Enumerator

k_rx_ram_offset_rammode	RAMMODE @ 0x00081200 - RAM Operating Mode Control Register.
k_rx_ram_offset_ramsts	RAMSTS @ 0x00081201 - RAM Error Status Register.
k_rx_ram_offset_ramprcr	RAMPRCR @ 0x00081204 - RAM Protection Register.
k_rx_ram_offset_ramecad	RAMECAD @ 0x00081208 - RAM Error Address Capture Register.

Definition at line 112 of file [rx72n_ram_regs.h](#).

```
00112 : uint8_t {
00113 /** @brief RAMMODE @ 0x00081200 - RAM Operating Mode Control Register */
00114 k_rx_ram_offset_rammode = 0x00,
00115 /** @brief RAMSTS @ 0x00081201 - RAM Error Status Register */
00116 k_rx_ram_offset_ramsts = 0x01,
00117 /** @brief RAMPRCR @ 0x00081204 - RAM Protection Register */
00118 k_rx_ram_offset_ramprcr = 0x04,
00119 /** @brief RAMECAD @ 0x00081208 - RAM Error Address Capture Register */
00120 k_rx_ram_offset_ramecad = 0x08,
00121 } rx_ram_reg_offset_t;
```


8.47.3.8 rx_ram_region_addr_t

```
enum rx_ram_region_addr_t : uintptr_t
```

RAM region base addresses and sizes.

Defines the address ranges for all RAM regions on the RX72N. These are from Manual Ch60, Table 60.1.

Enumerator

k_rx_ram_base_addr	RAM start address (512 KB, Memory Bus 1).
k_rx_ram_end_addr	RAM end address (inclusive).
k_rx_ram_size	RAM size in bytes. 512 KB
k_rx_exram_base_addr	Expansion RAM start address (512 KB, Memory Bus 3).
k_rx_exram_end_addr	Expansion RAM end address (inclusive).
k_rx_exram_size	Expansion RAM size in bytes. 512 KB

Definition at line 67 of file rx72n_ram_regs.h.

```
00067     : uintptr_t {
00068     /** @brief RAM start address (512 KB, Memory Bus 1) */
00069     k_rx_ram_base_addr = 0x00000000,
00070     /** @brief RAM end address (inclusive) */
00071     k_rx_ram_end_addr = 0x0007FFFF,
00072     /** @brief RAM size in bytes */
00073     k_rx_ram_size = 0x00080000, /**< 512 KB */
00074
00075     /** @brief Expansion RAM start address (512 KB, Memory Bus 3) */
00076     k_rx_exram_base_addr = 0x00800000,
00077     /** @brief Expansion RAM end address (inclusive) */
00078     k_rx_exram_end_addr = 0x0087FFFF,
00079     /** @brief Expansion RAM size in bytes */
00080     k_rx_exram_size = 0x00080000, /**< 512 KB */
00081 } rx_ram_region_addr_t;
```

8.47.3.9 rx_rammode_bits_t

```
enum rx_rammode_bits_t : uint8_t
```

RAMMODE register bit definitions.

RAM Operating Mode Select bits from Manual Ch60, section 60.2.1.

Enumerator

k_rx_rammode_parity_disabled	Parity checking disabled (RAMMODE[1:0] = 00b).
k_rx_rammode_parity_enabled	Parity checking enabled (RAMMODE[1:0] = 01b).

Definition at line 152 of file rx72n_ram_regs.h.

```
00152     : uint8_t {
00153     /** @brief Parity checking disabled (RAMMODE[1:0] = 00b) */
00154     k_rx_rammode_parity_disabled = 0x00,
00155     /** @brief Parity checking enabled (RAMMODE[1:0] = 01b) */
00156     k_rx_rammode_parity_enabled = 0x01,
00157 } rx_rammode_bits_t;
```

8.47.3.10 rx`ramprcr`bits`t

```
enum rx_ramprcr_bits_t : uint8_t
```

RAMPRCR register bit definitions.

RAM Protection Register from Manual Ch60, section 60.2.4. Write 0xF1 to enable writing to RAMMODE register.

Enumerator

k_rx_ramprcr_ramprcr_bit	RAMPRCR bit position (enable RAMMODE writes).
k_rx_ramprcr_ramprcr_mask	RAMPRCR enable mask.
k_rx_ramprcr_key	Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register.
k_rx_ramprcr_unlock	Complete value to enable RAMMODE writes (key + RAMPRCR=1).
k_rx_ramprcr_lock	Complete value to disable RAMMODE writes (key + RAMPRCR=0).

Definition at line 181 of file [rx72n_ram_regs.h](#).

```
00181         : uint8_t {
00182     /** @brief RAMPRCR bit position (enable RAMMODE writes) */
00183     k_rx_ramprcr_ramprcr_bit = 0,
00184     /** @brief RAMPRCR enable mask */
00185     k_rx_ramprcr_ramprcr_mask = 0x01,
00186     /** @brief Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register */
00187     k_rx_ramprcr_key = 0xF0,
00188     /** @brief Complete value to enable RAMMODE writes (key + RAMPRCR=1) */
00189     k_rx_ramprcr_unlock = 0xF1,
00190     /** @brief Complete value to disable RAMMODE writes (key + RAMPRCR=0) */
00191     k_rx_ramprcr_lock = 0xF0,
00192 } rx_ramprcr_bits_t;
```

8.47.3.11 rx`ramsts`bits`t

```
enum rx_ramsts_bits_t : uint8_t
```

RAMSTS register bit definitions.

RAM Error Status Flag from Manual Ch60, section 60.2.2.

Enumerator

k_rx_ramsts_ramerr_bit	RAM error flag bit position.
k_rx_ramsts_ramerr_mask	RAM error flag mask.

Definition at line 166 of file [rx72n_ram_regs.h](#).

```
00166         : uint8_t {
00167     /** @brief RAM error flag bit position */
00168     k_rx_ramsts_ramerr_bit = 0,
00169     /** @brief RAM error flag mask */
00170     k_rx_ramsts_ramerr_mask = 0x01,
00171 } rx_ramsts_bits_t;
```

8.47.4 Function Documentation

8.47.4.1 exram_regs()

```
rx_exram_regs_t * exram_regs (  
    void )    [inline], [static]
```

Get pointer to Expansion RAM control registers.

Returns a pointer to the Expansion RAM control register structure at 0x00081240.

Returns

Pointer to [rx_exram_regs_t](#) structure

Example:

```
// Enable parity checking for Expansion RAM  
exram_regs()->exramprcr = k_rx_exramprcr_unlock; // Unlock  
exram_regs()->exrammode = k_rx_exrammode_parity_enabled;  
exram_regs()->exramprcr = k_rx_exramprcr_lock; // Lock
```

Definition at line 339 of file [rx72n_ram_regs.h](#).

```
00340 {  
00341     return (rx_exram_regs_t*)k_rx_exram_reg_base_addr;  
00342 }
```

References [k_rx_exram_reg_base_addr](#).

8.47.4.2 ram_regs()

```
rx_ram_regs_t * ram_regs (  
    void )    [inline], [static]
```

Get pointer to RAM control registers.

Returns a pointer to the RAM control register structure at 0x00081200.

Returns

Pointer to [rx_ram_regs_t](#) structure

Example:

```
// Enable parity checking for RAM  
ram_regs()->ramprcr = k_rx_ramprcr_unlock; // Unlock  
ram_regs()->rammode = k_rx_rammode_parity_enabled;  
ram_regs()->ramprcr = k_rx_ramprcr_lock; // Lock
```

Definition at line 318 of file [rx72n_ram_regs.h](#).

```
00319 {  
00320     return (rx_ram_regs_t*)k_rx_ram_reg_base_addr;  
00321 }
```

References [k_rx_ram_reg_base_addr](#).

8.48 rx72n_ram_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_ram_regs.h
00003  * @brief RX72N RAM and Expansion RAM register definitions
00004  *
00005  * @details
00006  * This header defines register structures and addresses for the two
00007  * parity-checked static RAM regions on the RX72N MCU:
00008  *
00009  * - **RAM** (512 KB at 0x00000000-0x0007FFFF): 1-bit parity checking, Memory Bus 1
00010  * - **Expansion RAM** (512 KB at 0x00800000-0x0087FFFF): 1-bit parity checking, Memory Bus 3
00011  *
00012  * The ECC-protected 32 KB region (ECCRAM) has its own header,
00013  * rx72n_eccram_regs.h, since its control block and usage pattern
00014  * differ substantially from the parity-only regions.
00015  *
00016  * All register addresses are verified against the RX72N User's Manual Chapter 60.
00017  *
00018  * @par Memory Map (from Manual Ch60, Table 60.1):
00019  * | Type | Address Range | Size | Error Checking |
00020  * |-----|-----|-----|-----|
00021  * | RAM | 0x00000000 - 0x0007FFFF | 512 KB | 1-bit parity detection |
00022  * | Expansion RAM | 0x00800000 - 0x0087FFFF | 512 KB | 1-bit parity detection |
00023  *
00024  * @par Register Base Addresses:
00025  * - RAM registers: 0x00081200
00026  * - Expansion RAM registers: 0x00081240
00027  *
00028  * @note All RAM regions operate after reset without module stop.
00029  * @note Data is NOT retained during deep software standby mode.
00030  * @note When ICLK > 120 MHz, MEMWAIT must be set to 1 for Expansion RAM.
00031  *
00032  * @par NASA Power of 10 Compliance:
00033  * - Rule 5: Static assertions verify all register addresses
00034  * - Rule 8: All constants use typed enums, no macros for values
00035  *
00036  * @see rx72n_eccram_regs.h ECC-protected RAM definitions
00037  * @see RX72N User's Manual Chapter 60 (RAM)
00038  * @see RX72N User's Manual Chapter 11 (Low Power Consumption) for MSTPCRC
00039  * @see RX72N User's Manual Section 9.2.2 (MEMWAIT register)
00040  *
00041  * @since Version 1.0.0
00042  * @copyright Copyright (c) 2026 Locked Inc.
00043  * SPDX-License-Identifier: MIT
00044  */
00045
00046 #pragma once
00047
00048 #ifdef __cplusplus
00049 extern "C" {
00050 #endif
00051
00052 #include <stdint.h>
00053 #include <stdint.h>
00054
00055 /**
=====
00056  * MEMORY REGION CONSTANTS
=====
00057  */
=====
00058
00059 /**
00060  * @enum rx_ram_region_addr_t
00061  * @brief RAM region base addresses and sizes
00062  *
00063  * @details
00064  * Defines the address ranges for all RAM regions on the RX72N.
00065  * These are from Manual Ch60, Table 60.1.
00066  */
00067 typedef enum : uintptr_t {
00068     /** @brief RAM start address (512 KB, Memory Bus 1) */
00069     k_rx_ram_base_addr = 0x00000000,
00070     /** @brief RAM end address (inclusive) */
00071     k_rx_ram_end_addr = 0x0007FFFF,
00072     /** @brief RAM size in bytes */
00073     k_rx_ram_size = 0x00080000, /** < 512 KB */
00074
00075     /** @brief Expansion RAM start address (512 KB, Memory Bus 3) */
00076     k_rx_exram_base_addr = 0x00800000,
00077     /** @brief Expansion RAM end address (inclusive) */
00078     k_rx_exram_end_addr = 0x0087FFFF,
00079     /** @brief Expansion RAM size in bytes */

```

```

00080 k_rx_exram_size = 0x00080000, /**< 512 KB */
00081 } rx_ram_region_addr_t;
00082
00083 /*
=====
00084 * REGISTER BASE ADDRESSES
00085 *
=====
*/
00086
00087 /**
00088 * @enum rx_ram_reg_addr_t
00089 * @brief RAM control register base addresses
00090 *
00091 * @details
00092 * Defines register base addresses for RAM control, verified against Ch60.
00093 */
00094 typedef enum : uintptr_t {
00095 /** @brief RAM control registers base address */
00096 k_rx_ram_reg_base_addr = 0x00081200,
00097 /** @brief Expansion RAM control registers base address */
00098 k_rx_exram_reg_base_addr = 0x00081240,
00099 } rx_ram_reg_addr_t;
00100
00101 /*
=====
00102 * RAM REGISTER OFFSETS
00103 *
=====
*/
00104
00105 /**
00106 * @enum rx_ram_reg_offset_t
00107 * @brief RAM register offsets from base 0x00081200
00108 *
00109 * @details
00110 * Offsets verified against Manual Ch60, sections 60.2.1-60.2.4.
00111 */
00112 typedef enum : uint8_t {
00113 /** @brief RAMMODE @ 0x00081200 - RAM Operating Mode Control Register */
00114 k_rx_ram_offset_rammode = 0x00,
00115 /** @brief RAMSTS @ 0x00081201 - RAM Error Status Register */
00116 k_rx_ram_offset_ramsts = 0x01,
00117 /** @brief RAMPRCR @ 0x00081204 - RAM Protection Register */
00118 k_rx_ram_offset_ramprcr = 0x04,
00119 /** @brief RAMECAD @ 0x00081208 - RAM Error Address Capture Register */
00120 k_rx_ram_offset_ramecad = 0x08,
00121 } rx_ram_reg_offset_t;
00122
00123 /**
00124 * @enum rx_exram_reg_offset_t
00125 * @brief Expansion RAM register offsets from base 0x00081240
00126 *
00127 * @details
00128 * Offsets verified against Manual Ch60, sections 60.2.5-60.2.8.
00129 */
00130 typedef enum : uint8_t {
00131 /** @brief EXRAMMODE @ 0x00081240 - Expansion RAM Operating Mode Control Register */
00132 k_rx_exram_offset_exrammode = 0x00,
00133 /** @brief EXRAMSTS @ 0x00081241 - Expansion RAM Error Status Register */
00134 k_rx_exram_offset_exramsts = 0x01,
00135 /** @brief EXRAMPRCR @ 0x00081244 - Expansion RAM Protection Register */
00136 k_rx_exram_offset_exramprcr = 0x04,
00137 /** @brief EXRAMECAD @ 0x00081248 - Expansion RAM Error Address Capture Register */
00138 k_rx_exram_offset_exramecad = 0x08,
00139 } rx_exram_reg_offset_t;
00140
00141 /*
=====
00142 * BIT FIELD DEFINITIONS
00143 *
=====
*/
00144
00145 /**
00146 * @enum rx_rammode_bits_t
00147 * @brief RAMMODE register bit definitions
00148 *
00149 * @details
00150 * RAM Operating Mode Select bits from Manual Ch60, section 60.2.1.
00151 */
00152 typedef enum : uint8_t {
00153 /** @brief Parity checking disabled (RAMMODE[1:0] = 00b) */
00154 k_rx_rammode_parity_disabled = 0x00,
00155 /** @brief Parity checking enabled (RAMMODE[1:0] = 01b) */
00156 k_rx_rammode_parity_enabled = 0x01,
00157 } rx_rammode_bits_t;

```

```

00158
00159 /**
00160  * @enum rx_ramsts_bits_t
00161  * @brief RAMSTS register bit definitions
00162  *
00163  * @details
00164  * RAM Error Status Flag from Manual Ch60, section 60.2.2.
00165  */
00166 typedef enum : uint8_t {
00167     /** @brief RAM error flag bit position */
00168     k_rx_ramsts_ramerr_bit = 0,
00169     /** @brief RAM error flag mask */
00170     k_rx_ramsts_ramerr_mask = 0x01,
00171 } rx_ramsts_bits_t;
00172
00173 /**
00174  * @enum rx_rampcr_bits_t
00175  * @brief RAMPRCR register bit definitions
00176  *
00177  * @details
00178  * RAM Protection Register from Manual Ch60, section 60.2.4.
00179  * Write 0xF1 to enable writing to RAMMODE register.
00180  */
00181 typedef enum : uint8_t {
00182     /** @brief RAMPRCR bit position (enable RAMMODE writes) */
00183     k_rx_rampcr_rampcr_bit = 0,
00184     /** @brief RAMPRCR enable mask */
00185     k_rx_rampcr_rampcr_mask = 0x01,
00186     /** @brief Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register */
00187     k_rx_rampcr_key = 0xF0,
00188     /** @brief Complete value to enable RAMMODE writes (key + RAMPRCR=1) */
00189     k_rx_rampcr_unlock = 0xF1,
00190     /** @brief Complete value to disable RAMMODE writes (key + RAMPRCR=0) */
00191     k_rx_rampcr_lock = 0xF0,
00192 } rx_rampcr_bits_t;
00193
00194 /**
00195  * @enum rx_exrammode_bits_t
00196  * @brief EXRAMMODE register bit definitions
00197  *
00198  * @details
00199  * Expansion RAM Operating Mode Select from Manual Ch60, section 60.2.5.
00200  */
00201 typedef enum : uint8_t {
00202     /** @brief Parity checking disabled (EXRAMMODE[1:0] = 00b) */
00203     k_rx_exrammode_parity_disabled = 0x00,
00204     /** @brief Parity checking enabled (EXRAMMODE[1:0] = 01b) */
00205     k_rx_exrammode_parity_enabled = 0x01,
00206 } rx_exrammode_bits_t;
00207
00208 /**
00209  * @enum rx_exramsts_bits_t
00210  * @brief EXRAMSTS register bit definitions
00211  *
00212  * @details
00213  * Expansion RAM Error Status Flag from Manual Ch60, section 60.2.6.
00214  */
00215 typedef enum : uint8_t {
00216     /** @brief Expansion RAM error flag bit position */
00217     k_rx_exramsts_exramerr_bit = 0,
00218     /** @brief Expansion RAM error flag mask */
00219     k_rx_exramsts_exramerr_mask = 0x01,
00220 } rx_exramsts_bits_t;
00221
00222 /**
00223  * @enum rx_exramprcr_bits_t
00224  * @brief EXRAMPRCR register bit definitions
00225  *
00226  * @details
00227  * Expansion RAM Protection Register from Manual Ch60, section 60.2.8.
00228  * Write 0xF1 to enable writing to EXRAMMODE register.
00229  */
00230 typedef enum : uint8_t {
00231     /** @brief EXRAMPRCR bit position */
00232     k_rx_exramprcr_exramprcr_bit = 0,
00233     /** @brief EXRAMPRCR enable mask */
00234     k_rx_exramprcr_exramprcr_mask = 0x01,
00235     /** @brief Key byte: KW[6:0] = 1111000b placed in bits[7:1] of register */
00236     k_rx_exramprcr_key = 0xF0,
00237     /** @brief Complete value to enable EXRAMMODE writes */
00238     k_rx_exramprcr_unlock = 0xF1,
00239     /** @brief Complete value to disable EXRAMMODE writes */
00240     k_rx_exramprcr_lock = 0xF0,
00241 } rx_exramprcr_bits_t;
00242
00243 /**
00244  * @enum rx_ram_mstpcr_bits_t

```

```

00245 * @brief MSTPCRC register bits for parity-RAM module stop control
00246 *
00247 * @details
00248 * Module stop control from Manual Ch60, section 60.4.1.
00249 * Setting these bits stops clock supply to the respective RAM areas.
00250 *
00251 * @note The MSTPC6 bit for ECCRAM lives in rx72n_eccram_regs.h.
00252 */
00253 typedef enum : uint32_t {
00254     /** @brief MSTPC0 - RAM module stop bit (stops RAM clock) */
00255     k_rx_mstpcrc_ram_bit = 0,
00256     /** @brief MSTPC2 - Expansion RAM module stop bit */
00257     k_rx_mstpcrc_exram_bit = 2,
00258 } rx_ram_mstpcrc_bits_t;
00259
00260 /*
=====
00261 * REGISTER STRUCTURES
00262 *
=====
*/
00263
00264 /**
00265  * @struct rx_ram_regs_t
00266  * @brief RAM control register structure
00267  *
00268  * @details
00269  * Memory-mapped structure for RAM control registers at 0x00081200.
00270  * Verified against Manual Ch60, sections 60.2.1-60.2.4.
00271  */
00272 typedef struct {
00273     volatile uint8_t rammode;    /**< 0x00: RAM Operating Mode Control Register */
00274     volatile uint8_t ramsts;     /**< 0x01: RAM Error Status Register */
00275     volatile uint8_t reserved1[2]; /**< 0x02-0x03: Reserved */
00276     volatile uint8_t ramprcr;    /**< 0x04: RAM Protection Register */
00277     volatile uint8_t reserved2[3]; /**< 0x05-0x07: Reserved */
00278     volatile uint32_t ramecad;   /**< 0x08: RAM Error Address Capture Register */
00279 } rx_ram_regs_t;
00280
00281 /**
00282  * @struct rx_exram_regs_t
00283  * @brief Expansion RAM control register structure
00284  *
00285  * @details
00286  * Memory-mapped structure for Expansion RAM control registers at 0x00081240.
00287  * Verified against Manual Ch60, sections 60.2.5-60.2.8.
00288  */
00289 typedef struct {
00290     volatile uint8_t exrammode;  /**< 0x00: Expansion RAM Operating Mode Control Register */
00291     volatile uint8_t exramsts;   /**< 0x01: Expansion RAM Error Status Register */
00292     volatile uint8_t reserved1[2]; /**< 0x02-0x03: Reserved */
00293     volatile uint8_t exramprcr;  /**< 0x04: Expansion RAM Protection Register */
00294     volatile uint8_t reserved2[3]; /**< 0x05-0x07: Reserved */
00295     volatile uint32_t exramecad; /**< 0x08: Expansion RAM Error Address Capture Register */
00296 } rx_exram_regs_t;
00297
00298 /*
=====
00299 * INLINE ACCESSOR FUNCTIONS
00300 *
=====
*/
00301
00302 /**
00303  * @brief Get pointer to RAM control registers
00304  *
00305  * @details
00306  * Returns a pointer to the RAM control register structure at 0x00081200.
00307  *
00308  * @return Pointer to rx_ram_regs_t structure
00309  *
00310  * @par Example:
00311  * @code
00312  * // Enable parity checking for RAM
00313  * ram_regs()->ramprcr = k_rx_ramprcr_unlock; // Unlock
00314  * ram_regs()->rammode = k_rx_rammode_parity_enabled;
00315  * ram_regs()->ramprcr = k_rx_ramprcr_lock; // Lock
00316  * @endcode
00317  */
00318 static inline rx_ram_regs_t* ram_regs(void)
00319 {
00320     return (rx_ram_regs_t*)k_rx_ram_reg_base_addr;
00321 }
00322
00323 /**
00324  * @brief Get pointer to Expansion RAM control registers
00325  */

```

```

00326 * @details
00327 * Returns a pointer to the Expansion RAM control register structure at 0x00081240.
00328 *
00329 * @return Pointer to rx_exram_regs_t structure
00330 *
00331 * @par Example:
00332 * @code
00333 * // Enable parity checking for Expansion RAM
00334 * exram_regs()->exramprcr = k_rx_exramprcr_unlock; // Unlock
00335 * exram_regs()->exrammode = k_rx_exrammode_parity_enabled;
00336 * exram_regs()->exramprcr = k_rx_exramprcr_lock; // Lock
00337 * @endcode
00338 */
00339 static inline rx_exram_regs_t* exram_regs(void)
00340 {
00341     return (rx_exram_regs_t*)k_rx_exram_reg_base_addr;
00342 }
00343
00344 /*
=====
00345 * STATIC ASSERTIONS - Address Verification
00346 *
=====
*/
00347
00348 /* RAM register structure size and offset verification */
00349 static_assert(sizeof(rx_ram_regs_t) == 12, "rx_ram_regs_t size must be 12 bytes");
00350 static_assert(offsetof(rx_ram_regs_t, rammode) == 0x00, "RAMMODE offset must be 0x00");
00351 static_assert(offsetof(rx_ram_regs_t, ramsts) == 0x01, "RAMSTS offset must be 0x01");
00352 static_assert(offsetof(rx_ram_regs_t, ramprcr) == 0x04, "RAMPRCR offset must be 0x04");
00353 static_assert(offsetof(rx_ram_regs_t, ramecad) == 0x08, "RAMECAD offset must be 0x08");
00354
00355 /* Expansion RAM register structure size and offset verification */
00356 static_assert(sizeof(rx_exram_regs_t) == 12, "rx_exram_regs_t size must be 12 bytes");
00357 static_assert(offsetof(rx_exram_regs_t, exrammode) == 0x00, "EXRAMMODE offset must be 0x00");
00358 static_assert(offsetof(rx_exram_regs_t, exramsts) == 0x01, "EXRAMSTS offset must be 0x01");
00359 static_assert(offsetof(rx_exram_regs_t, exramprcr) == 0x04, "EXRAMPRCR offset must be 0x04");
00360 static_assert(offsetof(rx_exram_regs_t, exramecad) == 0x08, "EXRAMECAD offset must be 0x08");
00361
00362 /* Base address verification */
00363 static_assert(k_rx_ram_reg_base_addr == 0x00081200, "RAM register base must be 0x00081200");
00364 static_assert(k_rx_exram_reg_base_addr == 0x00081240,
00365     "Expansion RAM register base must be 0x00081240");
00366
00367 /* Memory region verification */
00368 static_assert(k_rx_ram_base_addr == 0x00000000, "RAM base must be 0x00000000");
00369 static_assert(k_rx_ram_size == 0x00080000, "RAM size must be 512 KB (0x80000)");
00370 static_assert(k_rx_exram_base_addr == 0x00080000, "Expansion RAM base must be 0x00080000");
00371 static_assert(k_rx_exram_size == 0x00080000, "Expansion RAM size must be 512 KB (0x80000)");
00372
00373 /* Protection register key values */
00374 static_assert(k_rx_ramprcr_unlock == 0xF1, "RAMPRCR unlock value must be 0xF1");
00375 static_assert(k_rx_exramprcr_unlock == 0xF1, "EXRAMPRCR unlock value must be 0xF1");
00376
00377 #ifdef __cplusplus
00378 }
00379 #endif

```

8.49 libs/rx/hal/inc/rx72n_regs.h File Reference

RX72N Hardware Register Definitions - Main Aggregator Header.

```

#include <stdint.h>
#include "rx72n_system_regs.h"
#include "rx72n_port_regs.h"
#include "rx72n_adc_regs.h"
#include "rx72n_sci_regs.h"
#include "rx72n_riic_regs.h"
#include "rx72n_rspi_regs.h"
#include "rx72n_cmt_regs.h"
#include "rx72n_icu_regs.h"
#include "rx72n_iwdt_regs.h"
#include "rx72n_wdt_regs.h"
#include "rx72n_crc_regs.h"
#include "rx72n_cac_regs.h"

```

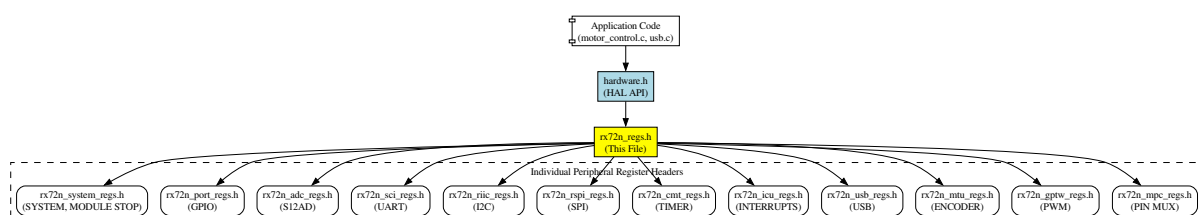

Include dependency graph for rx72n_regs.h:



RX72N Hardware Register Definitions - Main Aggregator Header.

Purpose: Simplifies HAL implementation by consolidating all register headers. Based On: Renesas RX72N Group User's Manual: Hardware (Rev.1.00, Feb 2019) Target MCU: R5F572NNHxFB (144-pin LFQFP, 4MB Flash, 512KB SRAM)

This file acts as a facade, grouping related peripheral register definitions:



8.49.1.3 Included Peripheral Register Files

Header File	Peripheral	Base Address	Purpose in STAR
rx72n_system_regs.h	System Control	0x00080000	Clock config, module stop control
rx72n_port_regs.h	GPIO Ports	0x0008C000	Digital I/O, motor control signals
rx72n_adc_regs.h	S12AD0/1	0x00089000	Current sensing, voltage measurement
rx72n_sci_regs.h	SCI0-12	0x0008A000	UART debug output (SCI9)
rx72n_riic_regs.h	RIIC0-2	0x00088300	I2C sensors (IMU)
rx72n_rspi_regs.h	RSPI0-2	0x000D0000	SPI: RPi5 comm (RSPI1 not used)
rx72n_cmt_regs.h	CMT0-3	0x00088000	ThreadX system tick (100 Hz)
rx72n_icu_regs.h	ICU	0x00087000	Interrupt routing and priorities
rx72n_iwdt_regs.h	IWDT	0x00088030	Independent watchdog (safety)
rx72n_wdt_regs.h	WDT	0x00088020	Watchdog timer
rx72n_crc_regs.h	CRC	0x00088280	Hardware CRC-32 (Protocol validation)
rx72n_usb_regs.h	USB0	0x000A0000	USB 2.0 full-speed device mode
rx72n_mtu_regs.h	MTU0-4	0x000C1200	Quadrature encoder counting
rx72n_gptw_regs.h	GPTW0-3	0x000C2000	High-resolution PWM (motors)
rx72n_mpc_regs.h	MPC	0x0008C100	Pin function multiplexing
rx72n_flash_regs.h	FLASH	0x007FE000	Flash programming, ROM cache
rx72n_ram_regs.h	RAM	0x00081200	RAM/EXRAM/ECCRAM control, parity, ECC
rx72n_clock.h	Clock Defs	-	Frequency constants (240 MHz)

8.49.1.4 RX72N Memory Map (Relevant Sections)

```

0x00000000 +-----+
| Internal ROM (4MB Flash) |
0x00400000 +-----+
| Reserved |
0x00800000 +-----+
| Peripheral Registers |
| CMT: 0x00088000 |
| ICU: 0x00087000 |
| ADC: 0x00089000 |
| SCI: 0x0008A000 |
| PORT: 0x0008C000 |
| RSPI: 0x000D0000 |
0x000D4000 +-----+
| Reserved |
0x00800000 +-----+
| Internal RAM (512KB SRAM) |
0x00880000 +-----+
| Reserved |
0xFFFFFFFF +-----+

```

8.49.1.5 Usage Guidelines

8.49.1.5.1 Recommended Practice

Application code should NOT include this file directly. Instead, use the HAL API provided by [hardware.h](#) which abstracts register access with type-safe functions.

```

// WRONG - Direct register access (bypasses HAL safety checks)
#include "rx72n_regs.h"
PORT3.PDR.BIT.B0 = 1; // No validation, no error checking

// CORRECT - Use HAL API (safe, validated, portable)
#include "hardware.h"
gpio_set_output(k_port_c_pin_0); // Type-safe, validated, error-checked

```

8.49.1.5.2 When Direct Register Access is Acceptable

Only use direct register access when:

1. Implementing new HAL drivers (inside lib/rx_hal/src/)
2. Low-level initialization code (clock setup, early boot)
3. Interrupt handlers requiring minimal latency
4. Performance-critical code after profiling shows HAL overhead is significant

8.49.1.5.3 Register Access Pattern (for HAL implementation)

```
#include "rx72n_regs.h"

// Example: Configure CMT0 for 10ms period @ 240 MHz
void timer_init(void) {
    // 1. Enable module clock
    SYSTEM.MSTPCRA.BIT.MSTPA15 = 0; // CMT0 module stop release

    // 2. Stop timer
    CMT0.CMSTR.BIT.STR = 0;

    // 3. Configure
    CMT0.CMCR.WORD = 0x0042; // PCLK/128, compare match interrupt
    CMT0.CMCOR = 18750; // 240MHz / 128 / 100Hz - 1

    // 4. Start timer
    CMT0.CMSTR.BIT.STR = 1;
}
```

8.49.1.6 Design Rationale

8.49.1.6.1 Why Separate Register Headers?

Approach	Pros	Cons
Single monolithic file	Easy to find everything	>10,000 lines, slow compilation, merge conflicts
Separate files (this approach)	Modular, fast compilation, maintainable	Requires aggregator (this file)

8.49.1.6.2 Benefits of This Architecture

1. Compile-time optimization: Only includes necessary peripheral headers
2. Maintainability: Each peripheral in separate file (~200-500 lines each)
3. Testability: Individual headers can be tested/validated independently
4. Team collaboration: Multiple developers can work on different peripherals without conflicts

8.49.1.7 Register Access Performance

Operation	Cycles @ 240 MHz	Time	Notes
Single register write	1-2	4-8 ns	Direct memory store
Single register read	1-2	4-8 ns	Direct memory load
Bit-field write	4-6	17-25 ns	Read-modify-write sequence
Protected register write	10-15	42-63 ns	Includes unlock/lock sequence

8.49.1.8 Safety Considerations

8.49.1.8.1 Write-Protected Registers

Some critical registers require unlock sequence before modification:

- System registers: Clock config, module stop control
- Port function selection: MPC registers
- Watchdog: WDT/IWDT configuration

Always use HAL functions for these registers to ensure correct unlock/lock sequence.

8.49.1.8.2 Reserved Bits

- Never write to reserved bits (may cause undefined behavior or peripheral malfunction)
- Read as 0, write as 0 for all reserved bits
- HAL functions handle this automatically

8.49.1.9 Dependencies

- Individual peripheral register headers (rx72n*_regs.h)
- <stdint.h> for fixed-width integer types (uint8_t, uint16_t, uint32_t)

NASA Power of 10 Compliance:

- Rule 1 [YES]: No goto, setjmp, recursion (pure header file with includes and definitions)
- Rule 2 [N/A]: No loops (header file only)
- Rule 3 [YES]: No dynamic allocation (register definitions are compile-time constants)
- Rule 4 [N/A]: No functions (header file only)
- Rule 5 [N/A]: No functions to validate (header file only)
- Rule 6 [YES]: All definitions at file scope (standard C practice)
- Rule 7 [N/A]: No function calls (header file only)
- Rule 8 [YES]: No macros except include guards
- Rule 9 [YES]: Register definitions use typed pointers (single level of dereferencing)
- Rule 10 [YES]: Compiles without warnings with -Wall -Wextra -Werror

SOLID Principles:

- S (Single Responsibility): Only aggregates register definitions (no logic, no functions)
- O (Open/Closed): New peripherals added by including new headers (no modification needed)
- L (Liskov Substitution): N/A (no polymorphism in C header files)
- I (Interface Segregation): Each peripheral in separate header (users include only what they need)
- D (Dependency Inversion): HAL ([hardware.h](#)) depends on this abstraction, not concrete implementations

Module Dependencies:

```
rx72n_regs.h (this file)
+--> rx72n_system_regs.h
+--> rx72n_port_regs.h
+--> rx72n_adc_regs.h
+--> rx72n_sci_regs.h
+--> rx72n_riic_regs.h
+--> rx72n_rspi_regs.h
+--> rx72n_cmt_regs.h
+--> rx72n_icu_regs.h
+--> rx72n_iwdt_regs.h
+--> rx72n_wdt_regs.h
+--> rx72n_crc_regs.h
+--> rx72n_usb_regs.h
+--> rx72n_mtu_regs.h
+--> rx72n_gptw_regs.h
+--> rx72n_mpc_regs.h
+--> rx72n_flash_regs.h
+--> rx72n_ram_regs.h
+--> rx72n_clock.h

Used by:
+--> hardware.h (HAL API implementation)
```

Author

Locked, Inc.

Date

2026-01-27

Copyright

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See also

[hardware.h](#) for type-safe HAL API (recommended for application code)
Renesas RX72N Group User's Manual: Hardware (Rev.1.00)

Definition in file [rx72n_regs.h](#).

8.50 rx72n_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_regs.h
00003  * @brief RX72N Hardware Register Definitions - Main Aggregator Header
00004  *
00005  * @details
00006  * ## Overview
00007  * Central aggregator header that includes all RX72N peripheral register definitions
00008  * required for STAR motor controller firmware. Provides a single include point for
00009  * accessing all hardware registers across the entire RX72N microcontroller.
00010  *
00011  * **Purpose**: Simplifies HAL implementation by consolidating all register headers.
00012  * **Based On**: Renesas RX72N Group User's Manual: Hardware (Rev.1.00, Feb 2019)
00013  * **Target MCU**: R5F572NNHxFB (144-pin LQFP, 4MB Flash, 512KB SRAM)
00014  *
00015  * ## Architecture
00016  *
00017  * This file acts as a facade, grouping related peripheral register definitions:
00018  *
00019  * @dot
00020  * digraph register_hierarchy {
00021  *   rankdir=TB;
00022  *   node [shape=box, style=rounded];
00023  *
00024  *   app [label="Application Code\n(motor_control.c, usb.c)", shape=component];
00025  *   hal [label="hardware.h\n(HAL API)", style="filled", fillcolor="lightblue"];
00026  *   regs [label="rx72n_regs.h\n(This File)", style="filled", fillcolor="yellow"];
00027  *
00028  *   subgraph cluster_peripherals {
00029  *     label="Individual Peripheral Register Headers";
00030  *     style=dashed;
00031  *
00032  *     sys [label="rx72n_system_regs.h\n(SYSTEM, MODULE STOP)"];
00033  *     port [label="rx72n_port_regs.h\n(GPIO)"];
00034  *     adc [label="rx72n_adc_regs.h\n(S12AD)"];
00035  *     sci [label="rx72n_sci_regs.h\n(UART)"];
00036  *     riic [label="rx72n_riic_regs.h\n(I2C)"];
00037  *     rspi [label="rx72n_rspi_regs.h\n(SPI)"];
00038  *     cmt [label="rx72n_cmt_regs.h\n(TIMER)"];
00039  *     icu [label="rx72n_icu_regs.h\n(INTERRUPTS)"];
00040  *     usb [label="rx72n_usb_regs.h\n(USB)"];
00041  *     mtu [label="rx72n_mtu_regs.h\n(ENCODER)"];
00042  *     gptw [label="rx72n_gptw_regs.h\n(PWM)"];
00043  *     mpc [label="rx72n_mpc_regs.h\n(PIN MUX)"];
00044  *   }
00045  *
00046  *   app -> hal;
00047  *   hal -> regs;
00048  *   regs -> sys;
00049  *   regs -> port;
00050  *   regs -> adc;
00051  *   regs -> sci;
00052  *   regs -> riic;
00053  *   regs -> rspi;
00054  *   regs -> cmt;
00055  *   regs -> icu;
00056  *   regs -> usb;
00057  *   regs -> mtu;
00058  *   regs -> gptw;
00059  *   regs -> mpc;
00060  * }
00061  * @enddot
00062  *
00063  * ## Included Peripheral Register Files
00064  *
00065  * | Header File | Peripheral | Base Address | Purpose in STAR |
00066  * |-----|-----|-----|-----|
00067  * | rx72n_system_regs.h | System Control | 0x00080000 | Clock config, module stop control |
00068  * | rx72n_port_regs.h | GPIO Ports | 0x0008C000 | Digital I/O, motor control signals |
00069  * | rx72n_adc_regs.h | S12AD0/1 | 0x00089000 | Current sensing, voltage measurement |
00070  * | rx72n_sci_regs.h | SCI0-12 | 0x0008A000 | UART debug output (SCI9) |
00071  * | rx72n_riic_regs.h | RIIC0-2 | 0x00088300 | I2C sensors (IMU) |
00072  * | rx72n_rspi_regs.h | RSPI0-2 | 0x000D0000 | SPI: RPi5 comm (RSPI1 not used) |
00073  * | rx72n_cmt_regs.h | CMT0-3 | 0x00088000 | ThreadX system tick (100 Hz) |
00074  * | rx72n_icu_regs.h | ICU | 0x00087000 | Interrupt routing and priorities |
00075  * | rx72n_iwdt_regs.h | IWDT | 0x00088030 | Independent watchdog (safety) |
00076  * | rx72n_wdt_regs.h | WDT | 0x00088020 | Watchdog timer |
00077  * | rx72n_crc_regs.h | CRC | 0x00088280 | Hardware CRC-32 (Protocol validation) |
00078  * | rx72n_usb_regs.h | USB0 | 0x000A0000 | USB 2.0 full-speed device mode |
00079  * | rx72n_mtu_regs.h | MTU0-4 | 0x000C1200 | Quadrature encoder counting |
00080  * | rx72n_gptw_regs.h | GPTW0-3 | 0x000C2000 | High-resolution PWM (motors) |
00081  * | rx72n_mpc_regs.h | MPC | 0x0008C100 | Pin function multiplexing |
00082  * | rx72n_flash_regs.h | FLASH | 0x007FE000 | Flash programming, ROM cache |

```

```

00083 * | rx72n_ram_regs.h | RAM | 0x00081200 | RAM/EXRAM/ECCRAM control, parity, ECC |
00084 * | rx72n_clock.h | Clock Defs | - | Frequency constants (240 MHz) |
00085 *
00086 * ## RX72N Memory Map (Relevant Sections)
00087 *
00088 * ```
00089 * 0x00000000 +-----+
00090 *          | Internal ROM (4MB Flash) |
00091 * 0x00400000 +-----+
00092 *          | Reserved |
00093 * 0x00800000 +-----+
00094 *          | Peripheral Registers |
00095 *          | CMT: 0x00088000 |
00096 *          | ICU: 0x00087000 |
00097 *          | ADC: 0x00089000 |
00098 *          | SCI: 0x0008A000 |
00099 *          | PORT: 0x0008C000 |
00100 *          | RSP: 0x000D0000 |
00101 * 0x000D4000 +-----+
00102 *          | Reserved |
00103 * 0x00800000 +-----+
00104 *          | Internal RAM (512KB SRAM) |
00105 * 0x00880000 +-----+
00106 *          | Reserved |
00107 * 0xFFFFFFFF +-----+
00108 * ```
00109 *
00110 * ## Usage Guidelines
00111 *
00112 * ### Recommended Practice
00113 * **Application code should NOT include this file directly.** Instead, use the HAL API
00114 * provided by `hardware.h` which abstracts register access with type-safe functions.
00115 *
00116 * @code{.c}
00117 * // WRONG - Direct register access (bypasses HAL safety checks)
00118 * #include "rx72n_regs.h"
00119 * PORT3.PDR.BIT.B0 = 1; // No validation, no error checking
00120 *
00121 * // CORRECT - Use HAL API (safe, validated, portable)
00122 * #include "hardware.h"
00123 * gpio_set_output(k_port_c_pin_0); // Type-safe, validated, error-checked
00124 * @endcode
00125 *
00126 * ### When Direct Register Access is Acceptable
00127 * Only use direct register access when:
00128 * 1. Implementing new HAL drivers (inside lib/rx_hal/src/)
00129 * 2. Low-level initialization code (clock setup, early boot)
00130 * 3. Interrupt handlers requiring minimal latency
00131 * 4. Performance-critical code after profiling shows HAL overhead is significant
00132 *
00133 * ### Register Access Pattern (for HAL implementation)
00134 * @code{.c}
00135 * #include "rx72n_regs.h"
00136 *
00137 * // Example: Configure CMT0 for 10ms period @ 240 MHz
00138 * void timer_init(void) {
00139 *     // 1. Enable module clock
00140 *     SYSTEM.MSTPCRA.BIT.MSTPA15 = 0; // CMT0 module stop release
00141 *
00142 *     // 2. Stop timer
00143 *     CMT0.CMSTR.BIT.STR = 0;
00144 *
00145 *     // 3. Configure
00146 *     CMT0.CMCR.WORD = 0x0042; // PCLK/128, compare match interrupt
00147 *     CMT0.CMCOR = 18750; // 240MHz / 128 / 100Hz - 1
00148 *
00149 *     // 4. Start timer
00150 *     CMT0.CMSTR.BIT.STR = 1;
00151 * }
00152 * @endcode
00153 *
00154 * ## Design Rationale
00155 *
00156 * ### Why Separate Register Headers?
00157 * | Approach | Pros | Cons |
00158 * |-----|-----|-----|
00159 * | **Single monolithic file** | Easy to find everything | >10,000 lines, slow compilation, merge conflicts |
00160 * | **Separate files (this approach)** | Modular, fast compilation, maintainable | Requires aggregator (this file) |
00161 *
00162 * ### Benefits of This Architecture
00163 * 1. **Compile-time optimization**: Only includes necessary peripheral headers
00164 * 2. **Maintainability**: Each peripheral in separate file (~200-500 lines each)
00165 * 3. **Testability**: Individual headers can be tested/validated independently
00166 * 4. **Team collaboration**: Multiple developers can work on different peripherals without conflicts
00167 *
00168 * ## Register Access Performance
00169 *

```

```

00170 * | Operation | Cycles @ 240 MHz | Time | Notes |
00171 * |-----|-----|-----|-----|
00172 * | Single register write | 1-2 | 4-8 ns | Direct memory store |
00173 * | Single register read | 1-2 | 4-8 ns | Direct memory load |
00174 * | Bit-field write | 4-6 | 17-25 ns | Read-modify-write sequence |
00175 * | Protected register write | 10-15 | 42-63 ns | Includes unlock/lock sequence |
00176 *
00177 * ## Safety Considerations
00178 *
00179 * ### Write-Protected Registers
00180 * Some critical registers require unlock sequence before modification:
00181 * - **System registers**: Clock config, module stop control
00182 * - **Port function selection**: MPC registers
00183 * - **Watchdog**: WDT/IWDT configuration
00184 *
00185 * **Always use HAL functions** for these registers to ensure correct unlock/lock sequence.
00186 *
00187 * ### Reserved Bits
00188 * - **Never write to reserved bits** (may cause undefined behavior or peripheral malfunction)
00189 * - **Read as 0, write as 0** for all reserved bits
00190 * - HAL functions handle this automatically
00191 *
00192 * ## Dependencies
00193 * - Individual peripheral register headers (rx72n_*_regs.h)
00194 * - `<stdint.h>` for fixed-width integer types (uint8_t, uint16_t, uint32_t)
00195 *
00196 * @par NASA Power of 10 Compliance:
00197 * - **Rule 1 [YES]**: No goto, setjmp, recursion (pure header file with includes and definitions)
00198 * - **Rule 2 [N/A]**: No loops (header file only)
00199 * - **Rule 3 [YES]**: No dynamic allocation (register definitions are compile-time constants)
00200 * - **Rule 4 [N/A]**: No functions (header file only)
00201 * - **Rule 5 [N/A]**: No functions to validate (header file only)
00202 * - **Rule 6 [YES]**: All definitions at file scope (standard C practice)
00203 * - **Rule 7 [N/A]**: No function calls (header file only)
00204 * - **Rule 8 [YES]**: No macros except include guards
00205 * - **Rule 9 [YES]**: Register definitions use typed pointers (single level of dereferencing)
00206 * - **Rule 10 [YES]**: Compiles without warnings with -Wall -Wextra -Werror
00207 *
00208 * @par SOLID Principles:
00209 * - **S (Single Responsibility)**: Only aggregates register definitions (no logic, no functions)
00210 * - **O (Open/Closed)**: New peripherals added by including new headers (no modification needed)
00211 * - **L (Liskov Substitution)**: N/A (no polymorphism in C header files)
00212 * - **I (Interface Segregation)**: Each peripheral in separate header (users include only what they need)
00213 * - **D (Dependency Inversion)**: HAL (hardware.h) depends on this abstraction, not concrete implementations
00214 *
00215 * @par Module Dependencies:
00216 * ```
00217 * rx72n_regs.h (this file)
00218 * +--> rx72n_system_regs.h
00219 * +--> rx72n_port_regs.h
00220 * +--> rx72n_adc_regs.h
00221 * +--> rx72n_sci_regs.h
00222 * +--> rx72n_riic_regs.h
00223 * +--> rx72n_rspi_regs.h
00224 * +--> rx72n_cmt_regs.h
00225 * +--> rx72n_icu_regs.h
00226 * +--> rx72n_iwdt_regs.h
00227 * +--> rx72n_wdt_regs.h
00228 * +--> rx72n_crc_regs.h
00229 * +--> rx72n_usb_regs.h
00230 * +--> rx72n_mtu_regs.h
00231 * +--> rx72n_gptw_regs.h
00232 * +--> rx72n_mpc_regs.h
00233 * +--> rx72n_flash_regs.h
00234 * +--> rx72n_ram_regs.h
00235 * +--> rx72n_clock.h
00236 *
00237 * Used by:
00238 * +--> hardware.h (HAL API implementation)
00239 * ```
00240 *
00241 * @author Locked, Inc.
00242 * @date 2026-01-27
00243 * @copyright Copyright (c) 2026 Locked Inc.
00244 * SPDX-License-Identifier: MIT
00245 *
00246 * @see hardware.h for type-safe HAL API (recommended for application code)
00247 * @see Renesas RX72N Group User's Manual: Hardware (Rev.1.00)
00248 */
00249
00250 #pragma once
00251
00252 #include <stdint.h>
00253
00254 #ifdef __cplusplus
00255 extern "C" {
00256 #endif

```



```
00257
00258 /*
=====
00259 * System Control Registers
00260 *
=====
00261 */
00262
00263 #include "rx72n_system_regs.h"
00264
00265 /*
=====
00266 * Port Registers (GPIO)
00267 *
=====
00268 */
00269
00270 #include "rx72n_port_regs.h"
00271
00272 /*
=====
00273 * 12-bit A/D Converter (S12ADFa) - For Current Sensing
00274 *
=====
00275 */
00276
00277 #include "rx72n_adc_regs.h"
00278
00279 /*
=====
00280 * Serial Communication Interface (SCI) - For UART/Debug
00281 *
=====
00282 */
00283
00284 #include "rx72n_sci_regs.h"
00285
00286 /*
=====
00287 * I2C Bus Interface (RIIC) - For I2C Communication
00288 *
=====
00289 */
00290
00291 #include "rx72n_riic_regs.h"
00292
00293 /*
=====
00294 * Renesas Serial Peripheral Interface (RSPI)
00295 * For SPI to RPi5 (RSPI1 not used in current design)
00296 *
=====
00297 */
00298
00299 #include "rx72n_rspi_regs.h"
00300
00301 /*
=====
00302 * Compare Match Timer (CMT) - For ThreadX System Tick
00303 *
=====
00304 */
00305
00306 #include "rx72n_cmt_regs.h"
00307
00308 /*
=====
00309 * Interrupt Controller (ICU)
00310 *
=====
00311 */
00312
00313 #include "rx72n_icu_regs.h"
00314
00315 /*
=====
00316 * Independent Watchdog Timer (IWDt)
00317 *
=====
00318 */
00319
00320 #include "rx72n_iwdt_regs.h"
00321
00322 /*
=====
00323 * Watchdog Timer (WDT)
00324 *
```

```

=====
00325 */
00326
00327 #include "rx72n_wdt_regs.h"
00328
00329 /*
=====
00330 * CRC Calculator - Hardware CRC-32 Acceleration
00331 *
=====
00332 */
00333
00334 #include "rx72n_crc_regs.h"
00335
00336 /*
=====
00337 * Clock Frequency Accuracy Measurement Circuit (CAC)
00338 *
=====
00339 */
00340
00341 #include "rx72n_cac_regs.h"
00342
00343 /*
=====
00344 * USB 2.0 Full-Speed Host/Function Module (USB0)
00345 *
=====
00346 * Restored on feat/usb0. rx72n_usb_regs.h carries the audited register map
00347 * for the 0x000A0000 USBb block (HUM Ch.40, manual p.1928).
00348 */
00349
00350 #include "rx72n_usb_regs.h"
00351
00352 /*
=====
00353 * Multi-Function Timer Unit (MTU) - Encoder Quadrature Counting
00354 *
=====
00355 */
00356
00357 #include "rx72n_mtu_regs.h"
00358
00359 /*
=====
00360 * General PWM Timer (GPTW) - Motor PWM Generation
00361 *
=====
00362 */
00363
00364 #include "rx72n_gptw_regs.h"
00365
00366 /*
=====
00367 * GPTW Port Output Enable (POEG) - Emergency Motor Stop
00368 *
=====
00369 */
00370
00371 #include "rx72n_poeg_regs.h"
00372
00373 /*
=====
00374 * 16-Bit Timer Pulse Unit (TPU) - Quadrature Encoder Phase Counting
00375 *
=====
00376 */
00377
00378 #include "rx72n_tpu_regs.h"
00379
00380 /*
=====
00381 * 8-Bit Timer (TMRb) - Four 8-bit Channels / Two 16-bit Cascaded Pairs
00382 *
=====
00383 */
00384
00385 #include "rx72n_tmr_regs.h"
00386
00387 /*
=====
00388 * Voltage Detection Circuit (LVDA) - Brownout Protection
00389 *
=====
00390 */
00391
00392 #include "rx72n_lvda_regs.h"

```

```

00393
00394 /*
=====
00395 * Multi-Function Pin Controller (MPC)
00396 *
=====
00397 */
00398
00399 #include "rx72n_mpc_regs.h"
00400
00401 /*
=====
00402 * Flash Memory (FLASH) - Programming/Erase and ROM Cache
00403 *
=====
00404 */
00405
00406 #include "rx72n_flash_regs.h"
00407
00408 /*
=====
00409 * RAM Control (RAM/EXRAM) - Parity-checked Memory Configuration
00410 *
=====
00411 */
00412
00413 #include "rx72n_ram_regs.h"
00414
00415 /*
=====
00416 * ECCRAM Control - ECC-protected RAM (SEC-DED)
00417 *
=====
00418 */
00419
00420 #include "rx72n_eccram_regs.h"
00421
00422 /*
=====
00423 * DMA Controller (DMAC) - High-throughput DMA Transfers
00424 *
=====
00425 */
00426
00427 #include "rx72n_dmac_regs.h"
00428
00429 /*
=====
00430 * Data Transfer Controller (DTC) - Lightweight Interrupt-driven Transfers
00431 *
=====
00432 */
00433
00434 #include "rx72n_dtc_regs.h"
00435
00436 /*
=====
00437 * Memory Protection Unit (MPU) - User Mode Memory Access Control
00438 *
=====
00439 */
00440
00441 #include "rx72n_mpu_regs.h"
00442
00443 /*
=====
00444 * Low Power Consumption (LPC) - Deep Standby and Operating Power Control
00445 *
=====
00446 */
00447
00448 #include "rx72n_lpc_regs.h"
00449
00450 /*
=====
00451 * Event Link Controller (ELC) - Hardware Event Routing
00452 *
=====
00453 */
00454
00455 #include "rx72n_elc_regs.h"
00456
00457 /*
=====
00458 * Clock Frequencies
00459 *
=====

```


ICCR2 (I2C Control Register 2) bit definitions.

- enum `riic_icsr1_bits_t` : `uint8_t` {
`k_riic_icsr1_hoa` = (1 << 7) , `k_riic_icsr1_did` = (1 << 5) , `k_riic_icsr1_gca` = (1 << 3) ,
`k_riic_icsr1_aas2` = (1 << 2) ,
`k_riic_icsr1_aas1` = (1 << 1) , `k_riic_icsr1_aas0` = (1 << 0) }

ICSR1 (I2C Status Register 1) bit definitions.

- enum `riic_icsr2_bits_t` : `uint8_t` {
`k_riic_icsr2_tdre` = (1 << 7) , `k_riic_icsr2_tend` = (1 << 6) , `k_riic_icsr2_rdrf` = (1 << 5) ,
`k_riic_icsr2_nackf` = (1 << 4) ,
`k_riic_icsr2_stop` = (1 << 3) , `k_riic_icsr2_start` = (1 << 2) , `k_riic_icsr2_al` = (1 << 1) ,
`k_riic_icsr2_tmof` = (1 << 0) }

ICSR2 (I2C Status Register 2) bit definitions.

Functions

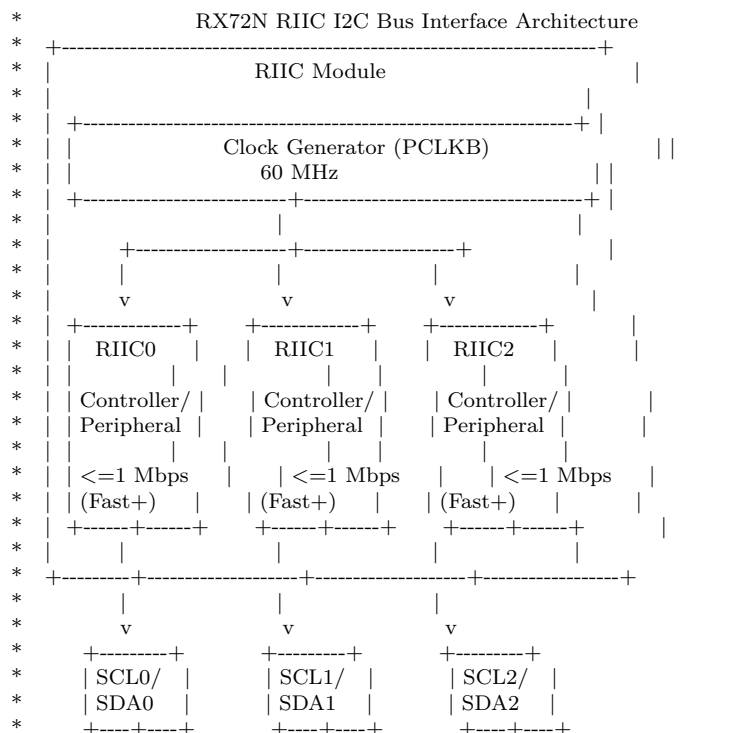
- static volatile `rx_riic_regs_t * riic0` (void)
Get pointer to RIIC0 registers.
- static volatile `rx_riic_regs_t * riic1` (void)
Get pointer to RIIC1 registers.
- static volatile `rx_riic_regs_t * riic2` (void)
Get pointer to RIIC2 registers.

8.51.1 Detailed Description

RX72N I2C Bus Interface (RIIC) Register Definitions.

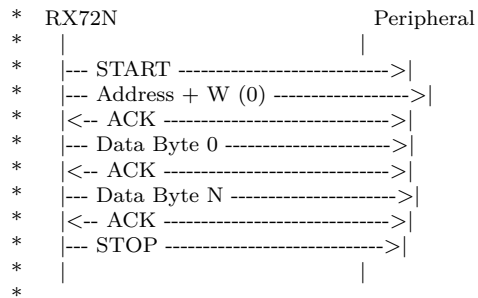
Register definitions for the I2C Bus Interface (RIIC) peripheral on the RX72N microcontroller. RIIC provides I2C communication with support for both controller and peripheral modes.

RIIC System Architecture



Mode	Speed	CKS	ICBRL	ICBRH	Actual Rate
Fast	400 kHz	1	19	16	400.0 kHz
Fast Plus	1 MHz	0	14	14	1.00 MHz

I2C Controller Write Transaction



Example: Initialize RIIC0 for 400 kHz Fast Mode

```

#include "rx72n_riic_regs.h"

// Reset RIIC0
riic0()->iccr1 = k_riic_iccr1_iicrst;

// Configure bit rate for 400 kHz
// PCLKB=60MHz, CKS=1 (/2), BRL=19, BRH=16
// f_SCL = 60MHz / ((19+16+2) * 2^1) = 810.8 kHz / 2 = 405.4 kHz
riic0()->icmr1 = 0x28; // CKS=1, BCWP=1
riic0()->icbrl = 0x13; // Low period = 19
riic0()->icbrh = 0x10; // High period = 16

// Enable RIIC
riic0()->iccr1 = k_riic_iccr1_ice;

// Start transaction: Set ST bit
riic0()->iccr2 = k_riic_iccr2_st;

// Wait for start condition
while (!(riic0()->icsr2 & k_riic_icsr2_start)) {}

// Send address + write bit
riic0()->icdr1 = (0x50 << 1) | 0; // EEPROM address 0x50, write

// Wait for transmit complete
while (!(riic0()->icsr2 & k_riic_icsr2_tdre)) {}

```

NASA Power of 10 Compliance

- Rule 3: No dynamic memory - all register access via static inline functions
- Rule 5: Compile-time assertions verify all register layout assumptions
- Rule 8: Uses typed enums (C23) for all constants, no magic numbers
- Rule 10: Clean build with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: RIIC register definitions only, no driver logic
- Open/Closed: Accessors work with any I2C configuration
- Interface Segregation: Minimal API - just register access functions

STAR Project I2C Usage

The STAR project does not currently use I2C, but RIIC is available for future sensor expansion (temperature sensors, IMU, etc.).

Manual Reference

RX72N Group User's Manual: Hardware, Chapter 42 (I2C-bus Interface)

Verification Status

[PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified against RX72N Manual Ch42 section 42.2

See also

[rx72n_mpc_regs.h](#) MPC pin configuration for I2C pins

Date

2026-01-28

Copyright

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Definition in file [rx72n_riic_regs.h](#).

8.52 rx72n_riic_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_riic_regs.h
00003  * @brief RX72N I2C Bus Interface (RIIC) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the I2C Bus Interface (RIIC) peripheral on the
00007  * RX72N microcontroller. RIIC provides I2C communication with
00008  * support for both controller and peripheral modes.
00009  *
00010  * @par RIIC System Architecture
00011  * @verbatim
00012  *
00013  *      RX72N RIIC I2C Bus Interface Architecture
00014  *      +-----+
00015  *      | RIIC Module |
00016  *      +-----+
00017  *      | Clock Generator (PCLKB) |
00018  *      | 60 MHz |
00019  *      +-----+
00020  *      |
00021  *      | +-----+ +-----+ +-----+
00022  *      | | v | | v | | v |
00023  *      | +-----+ +-----+ +-----+
00024  *      | RIIC0 | | RIIC1 | | RIIC2 |
00025  *      | | | | | | |
00026  *      | Controller/ | | Controller/ | | Controller/ |
00027  *      | Peripheral | | Peripheral | | Peripheral |
00028  *      | | | | | | |
00029  *      | <=1 Mbps | | <=1 Mbps | | <=1 Mbps |
00030  *      | (Fast+) | | (Fast+) | | (Fast+) |
00031  *

```



```

00032 * | +-----+ +-----+ +-----+ |
00033 * | | | | | | | |
00034 * | +-----+ +-----+ +-----+ |
00035 * | | | | | | | |
00036 * | v v v |
00037 * | +-----+ +-----+ +-----+ |
00038 * | | SCL0/ | | SCL1/ | | SCL2/ | |
00039 * | | SDA0 | | SDA1 | | SDA2 | |
00040 * | +-----+ +-----+ +-----+ |
00041 * | | | | | | | |
00042 * | v v v |
00043 * | +-----+ |
00044 * | | I2C Bus | |
00045 * | | +-----+ +-----+ +-----+ | |
00046 * | | |Sensor| |EEPROM| |RTC | |Other| | |
00047 * | | |0x48 | |0x50 | |0x68 | | ... | | |
00048 * | | +-----+ +-----+ +-----+ | |
00049 * | | +-----+ |
00050 *
00051 * Memory Map (Verified against RX72N Manual Ch42):
00052 * | +-----+ |
00053 * | | Address | Register | |
00054 * | | +-----+ |
00055 * | | 0x00088300 | RIIC0 base (ICCR1 at +0x00) | |
00056 * | | 0x00088320 | RIIC1 base (ICCR1 at +0x00) | |
00057 * | | 0x00088340 | RIIC2 base (ICCR1 at +0x00) | |
00058 * | | +-----+ |
00059 * | | Offset | Register (per channel) | |
00060 * | | 0x00 | ICCR1 (Control Register 1) | |
00061 * | | 0x01 | ICCR2 (Control Register 2) | |
00062 * | | 0x02 | ICMR1 (Mode Register 1) | |
00063 * | | 0x03 | ICMR2 (Mode Register 2) | |
00064 * | | 0x04 | ICMR3 (Mode Register 3) | |
00065 * | | 0x05 | ICFER (Function Enable) | |
00066 * | | 0x06 | ICSEER (Status Enable) | |
00067 * | | 0x07 | ICIER (Interrupt Enable) | |
00068 * | | 0x08 | ICSR1 (Status Register 1) | |
00069 * | | 0x09 | ICSR2 (Status Register 2) | |
00070 * | | 0x0A-0x0F | SARLn/SARUn (Peripheral Addresses) | |
00071 * | | 0x10 | ICBRL (Bit Rate Low) | |
00072 * | | 0x11 | ICBRH (Bit Rate High) | |
00073 * | | 0x12 | ICDRT (Transmit Data) | |
00074 * | | 0x13 | ICDRR (Receive Data) | |
00075 * | | +-----+ |
00076 * @endverbatim
00077 *
00078 * @par I2C Bit Rate Calculation
00079 * The bit rate is determined by ICBRL and ICBRH registers:
00080 *
00081 * @f
00082 * 
$$f_{\text{SCL}} = \frac{f_{\text{PCLKB}}}{(\text{ICBRL} + \text{ICBRH} + 2) \times 2^{\text{CKS}}}$$

00083 * @f
00084 *
00085 * Where:
00086 * - @f$ f_{\text{SCL}} @f$ = SCL clock frequency (target bit rate)
00087 * - @f$ f_{\text{PCLKB}} @f$ = Peripheral clock B (60 MHz)
00088 * - @f$ ICBRL @f$ = SCL low period setting (5 bits)
00089 * - @f$ ICBRH @f$ = SCL high period setting (5 bits)
00090 * - @f$ CKS @f$ = Internal clock divider (0-7)
00091 *
00092 * @par Common Speed Configurations (PCLKB = 60 MHz)
00093 * | Mode | Speed | CKS | ICBRL | ICBRH | Actual Rate |
00094 * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00095 * | Standard | 100 kHz | 3 | 18 | 16 | 100.0 kHz |
00096 * | Fast | 400 kHz | 1 | 19 | 16 | 400.0 kHz |
00097 * | Fast Plus | 1 MHz | 0 | 14 | 14 | 1.00 MHz |
00098 *
00099 * @par I2C Controller Write Transaction
00100 * @verbatim
00101 * RX72N Peripheral
00102 * |
00103 * |--- START ----->|
00104 * |--- Address + W (0) ----->|
00105 * |<-- ACK ----->|
00106 * |--- Data Byte 0 ----->|
00107 * |<-- ACK ----->|
00108 * |--- Data Byte N ----->|
00109 * |<-- ACK ----->|
00110 * |--- STOP ----->|
00111 * |
00112 * @endverbatim
00113 *
00114 * @par Example: Initialize RIIC0 for 400 kHz Fast Mode
00115 * @code
00116 * #include "rx72n_riic_regs.h"
00117 *
00118 * // Reset RIIC0

```

```

00119 * riic0()->iccr1 = k_riic_iccr1_iicrst;
00120 *
00121 * // Configure bit rate for 400 kHz
00122 * // PCLKB=60MHz, CKS=1 (/2), BRL=19, BRH=16
00123 * // f_SCL = 60MHz / ((19+16+2) * 2^1) = 810.8 kHz / 2 = 405.4 kHz
00124 * riic0()->icmr1 = 0x28; // CKS=1, BCWP=1
00125 * riic0()->icbrl = 0x13; // Low period = 19
00126 * riic0()->icbrh = 0x10; // High period = 16
00127 *
00128 * // Enable RIIC
00129 * riic0()->iccr1 = k_riic_iccr1_ice;
00130 *
00131 * // Start transaction: Set ST bit
00132 * riic0()->iccr2 = k_riic_iccr2_st;
00133 *
00134 * // Wait for start condition
00135 * while (!(riic0()->icsr2 & k_riic_icsr2_start)) {}
00136 *
00137 * // Send address + write bit
00138 * riic0()->icdrt = (0x50 « 1) | 0; // EEPROM address 0x50, write
00139 *
00140 * // Wait for transmit complete
00141 * while (!(riic0()->icsr2 & k_riic_icsr2_tdre)) {}
00142 * @endcode
00143 *
00144 * @par NASA Power of 10 Compliance
00145 * - Rule 3: No dynamic memory - all register access via static inline functions
00146 * - Rule 5: Compile-time assertions verify all register layout assumptions
00147 * - Rule 8: Uses typed enums (C23) for all constants, no magic numbers
00148 * - Rule 10: Clean build with -Wall -Wextra -Werror
00149 *
00150 * @par SOLID Principles
00151 * - Single Responsibility: RIIC register definitions only, no driver logic
00152 * - Open/Closed: Accessors work with any I2C configuration
00153 * - Interface Segregation: Minimal API - just register access functions
00154 *
00155 * @par STAR Project I2C Usage
00156 * The STAR project does not currently use I2C, but RIIC is available for
00157 * future sensor expansion (temperature sensors, IMU, etc.).
00158 *
00159 * @par Manual Reference
00160 * RX72N Group User's Manual: Hardware, Chapter 42 (I2C-bus Interface)
00161 *
00162 * @par Verification Status
00163 * [PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified
00164 * against RX72N Manual Ch42 section 42.2
00165 *
00166 * @see rx72n_mpc_regs.h MPC pin configuration for I2C pins
00167 *
00168 * @date 2026-01-28
00169 * @copyright Copyright (c) 2026 Locked Inc.
00170 * SPDX-License-Identifier: MIT
00171 */
00172
00173 #pragma once
00174
00175 #include <stddef.h>
00176 #include <stdint.h>
00177
00178 #ifdef __cplusplus
00179 extern "C" {
00180 #endif
00181
00182 /**
00183  * @defgroup riic_regs RIIC Register Definitions
00184  * @brief I2C Bus Interface (RIIC) hardware register interface
00185  * @{
00186  */
00187
00188 /*
=====
00189 * I2C Bus Interface (RIIC) - For I2C Communication
00190 *
=====
00191 */
00192
00193 /**
00194  * @enum rx_riic_addresses_t
00195  * @brief RIIC channel base addresses (verified against RX72N Hardware Manual Ch42)
00196  */
00197
00198 * @details
00199 * The RX72N has 3 independent RIIC channels. Each channel has a 32-byte
00200 * register block (though only 20 bytes are used).
00201
00202 * @par Memory Layout
00203 * @verbatim
00203 *   Address      | Channel | Pins

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```

00204 * -----+-----+-----
00205 * 0x00088300 | RIIC0 | P16/SCL0, P17/SDA0
00206 * 0x00088320 | RIIC1 | P21/SCL1, P20/SDA1
00207 * 0x00088340 | RIIC2 | P66/SCL2, P67/SDA2
00208 * @endverbatim
00209 *
00210 * @note Channel spacing is 0x20 (32 bytes) between adjacent channels.
00211 */
00212 typedef enum : uintptr_t {
00213     k_riic0_base_addr = 0x00088300, /**< RIIC0 register base address */
00214     k_riic1_base_addr = 0x00088320, /**< RIIC1 register base address */
00215     k_riic2_base_addr = 0x00088340, /**< RIIC2 register base address */
00216 } rx_riic_addresses_t;
00217
00218 /**
00219 * @struct rx_riic_regs_t
00220 * @brief RIIC Channel Register Map (20 bytes per channel)
00221 *
00222 * @details
00223 * I2C Bus Interface (RIIC) registers for I2C controller/peripheral communication.
00224 * All registers are 8-bit and byte-accessible.
00225 *
00226 * @par Register Layout (20 bytes)
00227 * @verbatim
00228 *   Offset | Size | Register | Description
00229 *   -----+-----+-----
00230 *   0x00 | 1 | ICCR1 | Control 1 (enable, reset)
00231 *   0x01 | 1 | ICCR2 | Control 2 (start, stop, restart)
00232 *   0x02 | 1 | ICMR1 | Mode 1 (clock divider)
00233 *   0x03 | 1 | ICMR2 | Mode 2 (timeout, SDA delay)
00234 *   0x04 | 1 | ICMR3 | Mode 3 (ACK/NACK control)
00235 *   0x05 | 1 | ICFER | Function Enable
00236 *   0x06 | 1 | ICSEI | Status Enable
00237 *   0x07 | 1 | ICIEI | Interrupt Enable
00238 *   0x08 | 1 | ICSR1 | Status 1 (ACK status)
00239 *   0x09 | 1 | ICSR2 | Status 2 (TX/RX flags)
00240 *   0x0A | 1 | SARL0 | Peripheral Address 0 Low
00241 *   0x0B | 1 | SARU0 | Peripheral Address 0 Upper
00242 *   0x0C | 1 | SARL1 | Peripheral Address 1 Low
00243 *   0x0D | 1 | SARU1 | Peripheral Address 1 Upper
00244 *   0x0E | 1 | SARL2 | Peripheral Address 2 Low
00245 *   0x0F | 1 | SARU2 | Peripheral Address 2 Upper
00246 *   0x10 | 1 | ICBRL | Bit Rate Low period
00247 *   0x11 | 1 | ICBRH | Bit Rate High period
00248 *   0x12 | 1 | ICDRT | Transmit Data
00249 *   0x13 | 1 | ICDRR | Receive Data
00250 * @endverbatim
00251 *
00252 * @par Initialization Sequence
00253 * 1. Set IICRST bit in ICCR1 (internal reset)
00254 * 2. Configure ICMR1/2/3 (clock, timeouts)
00255 * 3. Configure ICBRL/ICBRH (bit rate)
00256 * 4. Configure ICSEI (status detection)
00257 * 5. Configure ICIEI (interrupts)
00258 * 6. Clear IICRST and set ICE in ICCR1 (enable)
00259 *
00260 * @invariant sizeof(rx_riic_regs_t) == 20
00261 *
00262 * @see riic0(), riic1(), riic2() Accessor functions for each channel
00263 */
00264 typedef struct {
00265     /**
00266      * @brief I2C Bus Control Register 1
00267      * @details Enables RIIC module and controls internal reset and pin state.
00268      * | Bit | Name | Description |
00269      * |-----|-----|-----|
00270      * | 7 | ICE | RIIC Enable (1=enabled) |
00271      * | 6 | IICRST | Internal Reset (1=reset state) |
00272      * | 5 | CLO | Additional SCL Output (bus recovery clock) |
00273      * | 4 | SOWP | SCLO and SDA0 Write Protect |
00274      * | 3 | SCLO | SCL Output Control/Monitor |
00275      * | 2 | SDAO | SDA Output Control/Monitor |
00276      * | 1 | SCLI | SCL Line Input Monitor |
00277      * | 0 | SDAI | SDA Line Input Monitor |
00278      */
00279     volatile uint8_t iccr1;
00280
00281     /**
00282      * @brief I2C Bus Control Register 2
00283      * @details Controls start/stop/restart conditions and bus state.
00284      * | Bit | Name | Description |
00285      * |-----|-----|-----|
00286      * | 7 | BBSY | Bus Busy (read-only) |
00287      * | 6 | MST | Controller Mode (1=controller) |
00288      * | 5 | TRS | Transmit/Receive (1=transmit) |
00289      * | 3 | SP | Stop Condition Request |
00290      * | 2 | RS | Restart Condition Request |

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```

00291 * | 1 | ST | Start Condition Request |
00292 */
00293 volatile uint8_t iccr2;
00294
00295 /**
00296 * @brief I2C Bus Mode Register 1
00297 * @details Configures internal reference clock divider.
00298 * | Bits | Name | Description |
00299 * |-----|-----|
00300 * | 6-4 | CKS | Clock Select (0-7, divides PCLKB by 2^CKS) |
00301 * | 3 | BCWP | ICCR2 Write Protect (1=BC bits read-only) |
00302 * | 2-0 | BC | Bit Counter |
00303 */
00304 volatile uint8_t icmr1;
00305
00306 /**
00307 * @brief I2C Bus Mode Register 2
00308 * @details Configures timeout and SDA output delay.
00309 */
00310 volatile uint8_t icmr2;
00311
00312 /**
00313 * @brief I2C Bus Mode Register 3
00314 * @details Controls ACK/NACK transmission and wait state.
00315 * | Bit | Name | Description |
00316 * |-----|-----|
00317 * | 6 | WAIT | Wait insertion (1=insert wait) |
00318 * | 5 | RDRFS | RDRF Flag Set Timing |
00319 * | 4 | ACKWP | ACKBT Write Protect |
00320 * | 3 | ACKBT | ACK Bit to Transmit (0=ACK, 1=NACK) |
00321 */
00322 volatile uint8_t icmr3;
00323
00324 /**
00325 * @brief I2C Bus Function Enable Register
00326 * @details Enables various RIIC functions (timeout, noise filter, etc.).
00327 */
00328 volatile uint8_t icfer;
00329
00330 /**
00331 * @brief I2C Bus Status Enable Register
00332 * @details Selects which conditions set status flags.
00333 */
00334 volatile uint8_t icser;
00335
00336 /**
00337 * @brief I2C Bus Interrupt Enable Register
00338 * @details Enables interrupt generation for various conditions.
00339 */
00340 volatile uint8_t icier;
00341
00342 /**
00343 * @brief I2C Bus Status Register 1
00344 * @details Contains peripheral address detection and host/device-ID detection flags.
00345 * | Bit | Name | Description |
00346 * |-----|-----|
00347 * | 7 | HOA | Host Address Detection Flag |
00348 * | 5 | DID | Device-ID Address Detection Flag |
00349 * | 3 | GCA | General Call Address Detection Flag |
00350 * | 2 | AAS2 | Peripheral Address 2 Detection Flag |
00351 * | 1 | AAS1 | Peripheral Address 1 Detection Flag |
00352 * | 0 | AAS0 | Peripheral Address 0 Detection Flag |
00353 */
00354 volatile uint8_t icr1;
00355
00356 /**
00357 * @brief I2C Bus Status Register 2
00358 * @details Contains transmit/receive status flags and condition detection.
00359 * | Bit | Name | Description |
00360 * |-----|-----|
00361 * | 7 | TDRE | Transmit Data Empty |
00362 * | 6 | TEND | Transmit End |
00363 * | 5 | RDRF | Receive Data Full |
00364 * | 4 | NACKF | NACK Detection Flag |
00365 * | 3 | STOP | Stop Condition Detection |
00366 * | 2 | START | Start Condition Detection |
00367 * | 1 | AL | Arbitration-Lost Flag |
00368 * | 0 | TMOF | Timeout Detection Flag |
00369 */
00370 volatile uint8_t icr2;
00371
00372 volatile uint8_t sarl0; /**< Peripheral Address Register L0 (7-bit: bits 7-1) */
00373 volatile uint8_t saru0; /**< Peripheral Address Register U0 (10-bit: bits 9-8) */
00374 volatile uint8_t sarl1; /**< Peripheral Address Register L1 */
00375 volatile uint8_t saru1; /**< Peripheral Address Register U1 */
00376 volatile uint8_t sarl2; /**< Peripheral Address Register L2 */
00377 volatile uint8_t saru2; /**< Peripheral Address Register U2 */

```

```

00378
00379 /**
00380  * @brief I2C Bus Bit Rate Low-Level Width Setting Register
00381  * @details Sets the low period of SCL clock.
00382  * | Bits | Name | Description |
00383  * |-----|-----|-----|
00384  * | 4-0 | BRL | Low period count (affects SCL low time) |
00385  */
00386 volatile uint8_t icbrl;
00387
00388 /**
00389  * @brief I2C Bus Bit Rate High-Level Width Setting Register
00390  * @details Sets the high period of SCL clock.
00391  * | Bits | Name | Description |
00392  * |-----|-----|-----|
00393  * | 4-0 | BRH | High period count (affects SCL high time) |
00394  */
00395 volatile uint8_t icbrh;
00396
00397 /**
00398  * @brief I2C Bus Transmit Data Register
00399  * @details Write data here to transmit on I2C bus.
00400  * - First byte after START should be peripheral address + R/W bit
00401  * - Subsequent bytes are data
00402  */
00403 volatile uint8_t icdrt;
00404
00405 /**
00406  * @brief I2C Bus Receive Data Register
00407  * @details Read received data from here. Read clears RDRF flag.
00408  */
00409 volatile uint8_t icdrr;
00410 } rx_riic_regs_t;
00411
00412 /*
=====
00413 * Inline Register Accessor Functions
00414 *
=====
00415 * These functions provide type-safe access to RIIC hardware registers.
00416 * Using functions instead of macros enables:
00417 * - Type checking at compile time
00418 * - Debugger-friendly register access
00419 * - Consistent volatile semantics
00420 */
00421
00422 /**
00423  * @brief Get pointer to RIIC0 registers
00424  *
00425  * @details
00426  * Returns a volatile pointer to RIIC channel 0 registers at address 0x00088300.
00427  * RIIC0 pins are P16 (SCL0) and P17 (SDA0).
00428  *
00429  * @return Volatile pointer to RIIC0 register structure
00430  *
00431  * @par Example
00432  * @code
00433  * // Enable RIIC0
00434  * riic0()->iccr1 = k_riic_iccr1_ice;
00435  *
00436  * // Start condition
00437  * riic0()->iccr2 = k_riic_iccr2_st;
00438  * @endcode
00439  */
00440 static inline volatile rx_riic_regs_t* riic0(void)
00441 {
00442     return (volatile rx_riic_regs_t*)k_riic0_base_addr;
00443 }
00444
00445 /**
00446  * @brief Get pointer to RIIC1 registers
00447  *
00448  * @details
00449  * Returns a volatile pointer to RIIC channel 1 registers at address 0x00088320.
00450  * RIIC1 pins are P21 (SCL1) and P20 (SDA1).
00451  *
00452  * @return Volatile pointer to RIIC1 register structure
00453  */
00454 static inline volatile rx_riic_regs_t* riic1(void)
00455 {
00456     return (volatile rx_riic_regs_t*)k_riic1_base_addr;
00457 }
00458
00459 /**
00460  * @brief Get pointer to RIIC2 registers
00461  *
00462  * @details

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00463 * Returns a volatile pointer to RIIC channel 2 registers at address 0x00088340.
00464 * RIIC2 pins are P66 (SCL2) and P67 (SDA2).
00465 *
00466 * @return Volatile pointer to RIIC2 register structure
00467 */
00468 static inline volatile rx_riic_regs_t* riic2(void)
00469 {
00470     return (volatile rx_riic_regs_t*)k_riic2_base_addr;
00471 }
00472
00473 /**
00474 * @enum riic_iccr1_bits_t
00475 * @brief ICCR1 (I2C Control Register 1) bit definitions
00476 *
00477 * @details
00478 * Controls RIIC module enable/disable, internal reset, bus recovery clock,
00479 * and provides pin output control and input monitoring.
00480 *
00481 * @par Register Layout
00482 * @verbatim
00483 * Bit | Name | Description
00484 * ----+-----+-----
00485 * 7 | ICE | RIIC Enable (1=enabled, 0=disabled)
00486 * 6 | IICRST | Internal Reset (1=reset state)
00487 * 5 | CLO | Additional SCL Output (bus recovery clock)
00488 * 4 | SOWP | SCLO and SDAO Write Protect
00489 * 3 | SCLO | SCL Output Control/Monitor
00490 * 2 | SDAO | SDA Output Control/Monitor
00491 * 1 | SCLI | SCL Line Input Monitor (read-only)
00492 * 0 | SDAI | SDA Line Input Monitor (read-only)
00493 * @endverbatim
00494 */
00495 typedef enum : uint8_t {
00496     k_riic_iccr1_ice = (1 << 7), /**< Bit 7: I2C Bus Interface Enable */
00497     k_riic_iccr1_iicrst = (1 << 6), /**< Bit 6: Internal Reset (hold in reset) */
00498     k_riic_iccr1_clo = (1 << 5), /**< Bit 5: Additional SCL Output (bus recovery) */
00499     k_riic_iccr1_sowp = (1 << 4), /**< Bit 4: SCLO and SDAO Write Protect */
00500     k_riic_iccr1_sclo = (1 << 3), /**< Bit 3: SCL Output Control/Monitor */
00501     k_riic_iccr1_sdao = (1 << 2), /**< Bit 2: SDA Output Control/Monitor */
00502     k_riic_iccr1_scli = (1 << 1), /**< Bit 1: SCL Line Input Monitor (read-only) */
00503     k_riic_iccr1_sda_i = (1 << 0), /**< Bit 0: SDA Line Input Monitor (read-only) */
00504 } riic_iccr1_bits_t;
00505
00506 /**
00507 * @enum riic_iccr2_bits_t
00508 * @brief ICCR2 (I2C Control Register 2) bit definitions
00509 *
00510 * @details
00511 * Controls bus conditions (start, stop, restart) and provides bus status.
00512 *
00513 * @par Register Layout
00514 * @verbatim
00515 * Bit | Name | R/W | Description
00516 * ----+-----+-----+-----
00517 * 7 | BBSY | R | Bus Busy (1=bus busy, 0=bus free)
00518 * 6 | MST | R | Controller Mode (1=controller, 0=peripheral)
00519 * 5 | TRS | R | Transmit Mode (1=transmit, 0=receive)
00520 * 4 | - | - | Reserved
00521 * 3 | SP | W | Stop Condition Request (write 1 to issue)
00522 * 2 | RS | W | Restart Condition Request
00523 * 1 | ST | W | Start Condition Request
00524 * 0 | - | - | Reserved
00525 * @endverbatim
00526 *
00527 * @note ST, RS, SP bits are write-only and always read as 0.
00528 */
00529 typedef enum : uint8_t {
00530     k_riic_iccr2_bbsy = (1 << 7), /**< Bit 7: Bus Busy (read-only) */
00531     k_riic_iccr2_mst = (1 << 6), /**< Bit 6: Controller Mode (read-only) */
00532     k_riic_iccr2_trs = (1 << 5), /**< Bit 5: Transmit/Receive Select (1=transmit, read-only) */
00533     k_riic_iccr2_sp = (1 << 3), /**< Bit 3: Stop Condition Request */
00534     k_riic_iccr2_rs = (1 << 2), /**< Bit 2: Restart Condition Request */
00535     k_riic_iccr2_st = (1 << 1), /**< Bit 1: Start Condition Request */
00536 } riic_iccr2_bits_t;
00537
00538 /**
00539 * @enum riic_icsr1_bits_t
00540 * @brief ICSR1 (I2C Status Register 1) bit definitions
00541 *
00542 * @details
00543 * Contains peripheral address detection flags, general call detection,
00544 * device-ID detection, and host address detection (peripheral mode).
00545 *
00546 * @par Register Layout
00547 * @verbatim
00548 * Bit | Name | Description
00549 * ----+-----+-----

```

```

00550 * 7 | HOA | Host Address Detection Flag
00551 * 6 | - | Reserved
00552 * 5 | DID | Device-ID Address Detection Flag
00553 * 4 | - | Reserved
00554 * 3 | GCA | General Call Address Detection Flag
00555 * 2 | AAS2 | Peripheral Address 2 Detection Flag
00556 * 1 | AAS1 | Peripheral Address 1 Detection Flag
00557 * 0 | AAS0 | Peripheral Address 0 Detection Flag
00558 * @endverbatim
00559 *
00560 * @par Usage
00561 * @code
00562 * if (riic0()->icsr1 & k_riic_icsr1_aas0) {
00563 *     // Peripheral address 0 detected - we are being addressed
00564 * }
00565 * @endcode
00566 */
00567 typedef enum : uint8_t {
00568     k_riic_icsr1_hoa = (1 << 7), /**< Bit 7: Host Address Detection Flag */
00569     k_riic_icsr1_did = (1 << 5), /**< Bit 5: Device-ID Address Detection Flag */
00570     k_riic_icsr1_gca = (1 << 3), /**< Bit 3: General Call Address Detection Flag */
00571     k_riic_icsr1_aas2 = (1 << 2), /**< Bit 2: Peripheral Address 2 Detection Flag */
00572     k_riic_icsr1_aas1 = (1 << 1), /**< Bit 1: Peripheral Address 1 Detection Flag */
00573     k_riic_icsr1_aas0 = (1 << 0), /**< Bit 0: Peripheral Address 0 Detection Flag */
00574 } riic_icsr1_bits_t;
00575
00576 /**
00577 * @enum riic_icsr2_bits_t
00578 * @brief ICSR2 (I2C Status Register 2) bit definitions
00579 *
00580 * @details
00581 * Contains transmit/receive status flags and condition detection.
00582 *
00583 * @par Register Layout
00584 * @verbatim
00585 * Bit | Name | Description
00586 * ----+-----+-----
00587 * 7 | TDRE | Transmit Data Empty (1=can write ICDRT)
00588 * 6 | TEND | Transmit End (1=all data transmitted)
00589 * 5 | RDRF | Receive Data Full (1=data in ICDRR)
00590 * 4 | NACKF | NACK Detection Flag
00591 * 3 | STOP | Stop Condition Detected
00592 * 2 | START | Start Condition Detected
00593 * 1 | AL | Arbitration-Lost Flag
00594 * 0 | TMOF | Timeout Detection Flag
00595 * @endverbatim
00596 *
00597 * @par Typical Polling Sequence
00598 * @code
00599 * // Wait for transmit ready
00600 * while (!(riic0()->icsr2 & k_riic_icsr2_tdre)) {}
00601 *
00602 * // Wait for receive complete
00603 * while (!(riic0()->icsr2 & k_riic_icsr2_rdrf)) {}
00604 * @endcode
00605 */
00606 typedef enum : uint8_t {
00607     k_riic_icsr2_tdre = (1 << 7), /**< Bit 7: Transmit Data Empty */
00608     k_riic_icsr2_tend = (1 << 6), /**< Bit 6: Transmit End */
00609     k_riic_icsr2_rdrf = (1 << 5), /**< Bit 5: Receive Data Full */
00610     k_riic_icsr2_nackf = (1 << 4), /**< Bit 4: NACK Detection Flag */
00611     k_riic_icsr2_stop = (1 << 3), /**< Bit 3: Stop Condition Detection */
00612     k_riic_icsr2_start = (1 << 2), /**< Bit 2: Start Condition Detection */
00613     k_riic_icsr2_al = (1 << 1), /**< Bit 1: Arbitration-Lost Flag */
00614     k_riic_icsr2_tmofo = (1 << 0), /**< Bit 0: Timeout Detection Flag */
00615 } riic_icsr2_bits_t;
00616
00617 /*
=====
00618 * Static Assertions - Verify Register Layout at Compile Time
00619 *
=====
00620 * These assertions verify our register structures match the RX72N hardware
00621 * exactly. If any assertion fails, the struct definition must be corrected.
00622 *
00623 * Reference: RX72N Group User's Manual: Hardware, Chapter 42, Section 42.2
00624 *
=====
00625 */
00626
00627 /**
00628 * @defgroup riic_static_assert RIIC Static Assertions
00629 * @brief Compile-time verification of RIIC register layout
00630 * @details
00631 * These static assertions are verified at compile time to ensure the
00632 * C structure definitions match the actual hardware register layout.
00633 * A failed assertion indicates a bug in the register definitions that

```



```

00634 * must be fixed before the code can compile.
00635 * @{
00636 */
00637
00638 /* Verify base addresses match Hardware Manual Ch42 */
00639 static_assert(k_riic0_base_addr == 0x00088300, "RIIC0 base address incorrect");
00640 static_assert(k_riic1_base_addr == 0x00088320, "RIIC1 base address incorrect");
00641 static_assert(k_riic2_base_addr == 0x00088340, "RIIC2 base address incorrect");
00642
00643 /* Verify register structure size (20 bytes = 0x14) */
00644 static_assert(sizeof(rx_riic_regs_t) == 20,
00645               "RIIC register structure size incorrect (expected 0x14)");
00646
00647 /* Verify control register offsets */
00648 static_assert(offsetof(rx_riic_regs_t, iccr1) == 0x00, "ICCR1 offset incorrect");
00649 static_assert(offsetof(rx_riic_regs_t, iccr2) == 0x01, "ICCR2 offset incorrect");
00650
00651 /* Verify mode register offsets */
00652 static_assert(offsetof(rx_riic_regs_t, icmr1) == 0x02, "ICMR1 offset incorrect");
00653 static_assert(offsetof(rx_riic_regs_t, icmr2) == 0x03, "ICMR2 offset incorrect");
00654 static_assert(offsetof(rx_riic_regs_t, icmr3) == 0x04, "ICMR3 offset incorrect");
00655
00656 /* Verify function/status/interrupt register offsets */
00657 static_assert(offsetof(rx_riic_regs_t, icfer) == 0x05, "ICFER offset incorrect");
00658 static_assert(offsetof(rx_riic_regs_t, icser) == 0x06, "ICSER offset incorrect");
00659 static_assert(offsetof(rx_riic_regs_t, icier) == 0x07, "ICIER offset incorrect");
00660 static_assert(offsetof(rx_riic_regs_t, icser1) == 0x08, "ICSR1 offset incorrect");
00661 static_assert(offsetof(rx_riic_regs_t, icser2) == 0x09, "ICSR2 offset incorrect");
00662
00663 /* Verify peripheral address register offsets */
00664 static_assert(offsetof(rx_riic_regs_t, sarl0) == 0x0A, "SARL0 offset incorrect");
00665 static_assert(offsetof(rx_riic_regs_t, saru0) == 0x0B, "SARU0 offset incorrect");
00666 static_assert(offsetof(rx_riic_regs_t, sarl1) == 0x0C, "SARL1 offset incorrect");
00667 static_assert(offsetof(rx_riic_regs_t, saru1) == 0x0D, "SARU1 offset incorrect");
00668 static_assert(offsetof(rx_riic_regs_t, sarl2) == 0x0E, "SARL2 offset incorrect");
00669 static_assert(offsetof(rx_riic_regs_t, saru2) == 0x0F, "SARU2 offset incorrect");
00670
00671 /* Verify bit rate and data register offsets */
00672 static_assert(offsetof(rx_riic_regs_t, icbri) == 0x10, "ICBRI offset incorrect");
00673 static_assert(offsetof(rx_riic_regs_t, icbrih) == 0x11, "ICBRIH offset incorrect");
00674 static_assert(offsetof(rx_riic_regs_t, icdrt) == 0x12, "ICDRT offset incorrect");
00675 static_assert(offsetof(rx_riic_regs_t, icdrr) == 0x13, "ICDRT offset incorrect");
00676
00677 /* Verify channel spacing (0x20 = 32 bytes between channels) */
00678 static_assert((k_riic1_base_addr - k_riic0_base_addr) == 0x20, "RIIC0 to RIIC1 spacing incorrect");
00679 static_assert((k_riic2_base_addr - k_riic1_base_addr) == 0x20, "RIIC1 to RIIC2 spacing incorrect");
00680
00681 /** @} */ /* End of riic_static_assert group */
00682
00683 /** @} */ /* End of riic_regs group */
00684
00685 #ifdef __cplusplus
00686 }
00687 #endif

```

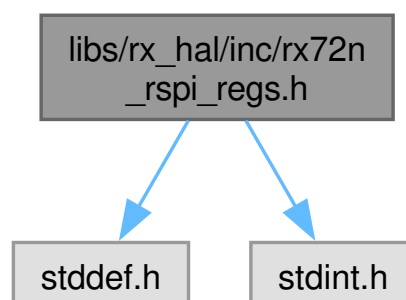
8.53 libs/rx_hal/inc/rx72n_rspi_regs.h File Reference

RX72N RSPI Serial Peripheral Interface Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_rspi_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_rspi_regs_t](#)
RSPI Register Map Structure (Complete - All 22 Registers).

Enumerations

- enum [rx_rspi_addresses_t](#) : [uintptr_t](#) { [k_rspi0_base_addr](#) = 0x000D0100 , [k_rspi1_base_addr](#) = 0x000D0140 , [k_rspi2_base_addr](#) = 0x000D0300 }
RSPI channel base addresses.
- enum [rspi_spcr_bits_t](#) : [uint8_t](#) {
[k_rspi_spcr_sprie](#) = (1 << 7) , [k_rspi_spcr_spe](#) = (1 << 6) , [k_rspi_spcr_sptie](#) = (1 << 5) ,
[k_rspi_spcr_speie](#) = (1 << 4) ,
[k_rspi_spcr_mstr](#) = (1 << 3) , [k_rspi_spcr_modfe](#) = (1 << 2) , [k_rspi_spcr_txmd](#) = (1 << 1) , [k_rspi_spcr_spms](#) = (1 << 0) }
RSPI Control Register (SPCR) Bit Definitions.
- enum [rspi_spsr_bits_t](#) : [uint8_t](#) {
[k_rspi_spsr_sprf](#) = (1 << 7) , [k_rspi_spsr_sptef](#) = (1 << 5) , [k_rspi_spsr_udrf](#) = (1 << 4) ,
[k_rspi_spsr_perf](#) = (1 << 3) ,
[k_rspi_spsr_modf](#) = (1 << 2) , [k_rspi_spsr_idlnf](#) = (1 << 1) , [k_rspi_spsr_ovrf](#) = (1 << 0) }
RSPI Status Register (SPSR) Bit Definitions.
- enum [rspi_sppcr_bits_t](#) : [uint8_t](#) { [k_rspi_sppcr_moife](#) = (1 << 5) , [k_rspi_sppcr_moifv](#) = (1 << 4) , [k_rspi_sppcr_splp2](#) = (1 << 1) , [k_rspi_sppcr_splp](#) = (1 << 0) }
RSPI Pin Control Register (SPPCR) Bit Definitions.
- enum [rspi_spdcr_bits_t](#) : [uint8_t](#) { [k_rspi_spdcr_sprtd](#) = (1 << 4) , [k_rspi_spdcr_splw](#) = (1 << 5) }
RSPI Data Control Register (SPDCR) Bit Definitions.

Functions

- static volatile [rx_rspi_regs_t](#) * [rspi0](#) (void)
Get pointer to RSPI0 registers (Raspberry Pi 5 communication).
- static volatile [rx_rspi_regs_t](#) * [rspi1](#) (void)
Get pointer to RSPI1 registers (not used in current design).
- static volatile [rx_rspi_regs_t](#) * [rspi2](#) (void)
Get pointer to RSPI2 registers (not used in current design).

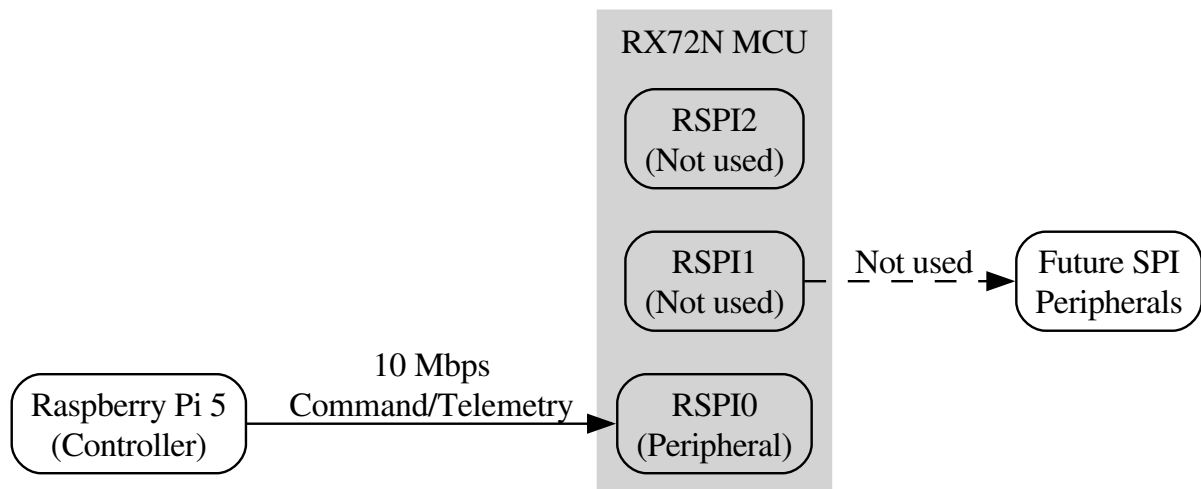
8.53.1 Detailed Description

RX72N RSPI Serial Peripheral Interface Register Definitions.

This file defines all 22 RSPI registers per the RX72N Hardware Manual (R01UH0824EJ0120 Rev.1.20, Ch44). All register addresses, offsets, and bit definitions have been verified against the manual.

System Architecture

The RSPI module provides high-speed serial communication between the RX72N and external devices. In the STAR project, RSPI is used for:



RSPI Channel Assignments

Channel	Base Address	Mode	Target	Speed	Purpose
RSPI0	0x000D0100	Peripheral	Raspberry Pi 5	10 Mbps	Command/telemetry
RSPI1	0x000D0140	Not used	-	-	Not used in current design
RSPI2	0x000D0300	Not used	-	-	Not used in current design

Key Features

- Full-duplex or simplex (transmit-only) synchronous communications
- Controller/Peripheral mode operation (using inclusive terminology)
- 8 to 32-bit configurable data frame length
- MSB-first or LSB-first transmission order
- Up to 8 command registers for sequential transfer mode
- Configurable clock polarity (CPOL) and phase (CPHA)
- DMA/DTC transfer support for high-throughput applications
- Mode fault detection for multi-controller configurations

Hardware Requirements

Parameter	RSPI0 (RPi5)	RSPI1 (Not used)
PCLKB	60 MHz	-
Max Bit Rate	30 Mbps	-
Configured	10 Mbps	-
Data Width	8 bits	-
CPOL/CPHA	0/↔ 0	-

Memory Map (per channel)

Offset	Size	Register	Description
0x00	1	SPCR	Control Register
0x01	1	SSLP	Chip Select Polarity Register
0x02	1	SPPCR	Pin Control Register
0x03	1	SPSR	Status Register
0x04	4	SPDR	Data Register (32-bit)
0x08	1	SPSCR	Sequence Control Register
0x09	1	SPSSR	Sequence Status Register
0x0A	1	SPBR	Bit Rate Register
0x0B	1	SPDCR	Data Control Register
0x0C	1	SPCKD	Clock Delay Register
0x0D	1	SSLND	SSL Negation Delay Register
0x0E	1	SPND	Next-Access Delay Register
0x0F	1	SPCR2	Control Register 2
0x10	2	SPCMD0	Command Register 0
0x12	2	SPCMD1	Command Register 1
0x14	2	SPCMD2	Command Register 2
0x16	2	SPCMD3	Command Register 3
0x18	2	SPCMD4	Command Register 4
0x1A	2	SPCMD5	Command Register 5
0x1C	2	SPCMD6	Command Register 6
0x1E	2	SPCMD7	Command Register 7
0x20	1	SPDCR2	Data Control Register 2

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
- Rule 2: [OK] N/A (no loops in register definitions)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessor functions are single-statement inline
- Rule 5: [OK] N/A (accessor functions have no assertions needed)

- Rule 6: [OK] Minimal scope - inline accessors only
- Rule 7: [OK] N/A (no return values to check in definitions)
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers in register definitions
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Single responsibility - only register definitions
- O: Open for extension via additional bit field enums
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessors
- D: N/A (hardware register layer, no dependencies to invert)

Module Dependencies

- `<stdint.h>` - Fixed-width integer types
- No other dependencies (hardware register layer)

Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 44: Serial Peripheral Interface (RSPI), pages 2349-2373
- Section 44.2: Register Descriptions, pages 2353-2373

See also

`rx_spi_comm.h` Higher-level SPI communication API

[rx72n_regs.h](#) Main register include file

`docs/sections/03_hardware_pinout.tex` STAR pin assignments

Author

Locked, Inc.

Date

2026-01-28

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_rspi_regs.h](#).

8.53.2 Data Structure Documentation

8.53.2.1 struct rx`rspi`regs`t

RSPI Register Map Structure (Complete - All 22 Registers).

Renesas Serial Peripheral Interface (RSPI) registers for SPI communication. Supports controller and peripheral modes with configurable clock and data format. Addresses verified against RX72N Hardware Manual (R01UH0824EJ0120 Rev.1.20).

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	1	spcr	uint8↔ __t	Control Register (enable, mode)
0x01	1	sslp	uint8↔ __t	Chip Select Polarity Register
0x02	1	sppcr	uint8↔ __t	Pin Control Register (loopback)
0x03	1	spsr	uint8↔ __t	Status Register (flags)
0x04	4	spdr	uint32↔ __t	Data Register (TX/RX)
0x08	1	spscr	uint8↔ __t	Sequence Control Register
0x09	1	spssr	uint8↔ __t	Sequence Status Register
0x0A	1	spbr	uint8↔ __t	Bit Rate Register
0x0B	1	spdcr	uint8↔ __t	Data Control Register
0x0C	1	spckd	uint8↔ __t	Clock Delay Register
0x0D	1	sslnd	uint8↔ __t	SSL Negation Delay Register
0x0E	1	spnd	uint8↔ __t	Next-Access Delay Register
0x0F	1	spcr2	uint8↔ __t	Control Register 2
0x10	2	spcmd0	uint16↔ __t	Command Register 0
0x12	2	spcmd1	uint16↔ __t	Command Register 1
0x14	2	spcmd2	uint16↔ __t	Command Register 2
0x16	2	spcmd3	uint16↔ __t	Command Register 3

Offset	Size	Field	Type	Description
0x18	2	spcmd4	uint16↔ __t	Command Register 4
0x1A	2	spcmd5	uint16↔ __t	Command Register 5
0x1C	2	spcmd6	uint16↔ __t	Command Register 6
0x1E	2	spcmd7	uint16↔ __t	Command Register 7
0x20	1	spdcr2	uint8↔ __t	Data Control Register 2
Total	33			(no padding due to packed attribute)

Sequential Transfer Mode

Up to 8 commands (SPCMD0-SPCMD7) can be executed sequentially in looped execution. Each command configures:

- SSL signal selection and assertion
- Bit rate division
- Clock polarity (CPOL) and phase (CPHA)
- Transfer data length (8-32 bits)
- MSB/LSB first transmission order
- Inter-transfer delay settings

Initialization Example (Peripheral Mode - RPi5 Communication)

```
// Configure RSPI0 for 10 Mbps peripheral mode (RPi5 is controller)
volatile rx_rspi_regs_t* spi = rspi0();

// 1. Disable RSPI before configuration
spi->spr = 0x00;

// 2. Configure pin polarity (SSL active low)
spi->sslp = 0x00;

// 3. Configure command register 0 for 8-bit transfers
// SPB[3:0]=0111 (8 bits), CPOL=0, CPHA=0, LSBF=0 (MSB first)
spi->spcmd0 = 0x0700;

// 4. Enable RSPI in peripheral mode with interrupts
spi->spr = k_rspi_spr_spe | k_rspi_spr_sprie | k_rspi_spr_sptie;
```

Initialization Example (Controller Mode - Not Used in Current Design)

```
// Configure RSPI1 for 5 Mbps controller mode (not used in current design)
volatile rx_rspi_regs_t* spi = rspi1();

// 1. Disable RSPI before configuration
spi->spr = 0x00;

// 2. Configure bit rate for 5 Mbps (PCLKB=60MHz, n=5: 60/(2*(5+1))=5MHz)
spi->spbr = 5;

// 3. Configure command register 0 for 16-bit transfers
// SPB[3:0]=1111 (16 bits), CPOL=0, CPHA=1
spi->spcmd0 = 0x0F02;

// 4. Enable RSPI in controller mode
spi->spr = k_rspi_spr_spe | k_rspi_spr_mstr | k_rspi_spr_sptie;
```

Data Transfer Example

```
// Full-duplex transfer (controller mode)
static rx_err_t rspi_transfer(volatile rx_rspi_regs_t* spi,
                             uint8_t tx_data, uint8_t* rx_data) {
    // Wait for TX buffer empty
    while (!(spi->spsr & k_rspi_spsr_sptef)) { }

    // Write data (starts transfer)
    spi->spdr = tx_data;

    // Wait for RX buffer full
    while (!(spi->spsr & k_rspi_spsr_sprf)) { }

    // Read received data
    *rx_data = (uint8_t)spi->spdr;

    return k_rx_ok;
}
```

Invariant

sizeof(rx_rspi_regs_t) == 33 bytes (verified by static assertion)

Structure is packed with no padding

All offsets match RX72N Hardware Manual Ch44

Note

Structure uses `__attribute__((packed))` to ensure no compiler padding

All fields are volatile for hardware register access

Warning

Do not modify reserved bits in control registers

Attention

RSPI must be disabled (SPE=0) before changing most configuration

See also

[rspi0\(\)](#), [rspi1\(\)](#), [rspi2\(\)](#) Accessor functions for each channel

[rspi_spcr_bits_t](#) Control register bit definitions

[rspi_spsr_bits_t](#) Status register bit definitions

Since

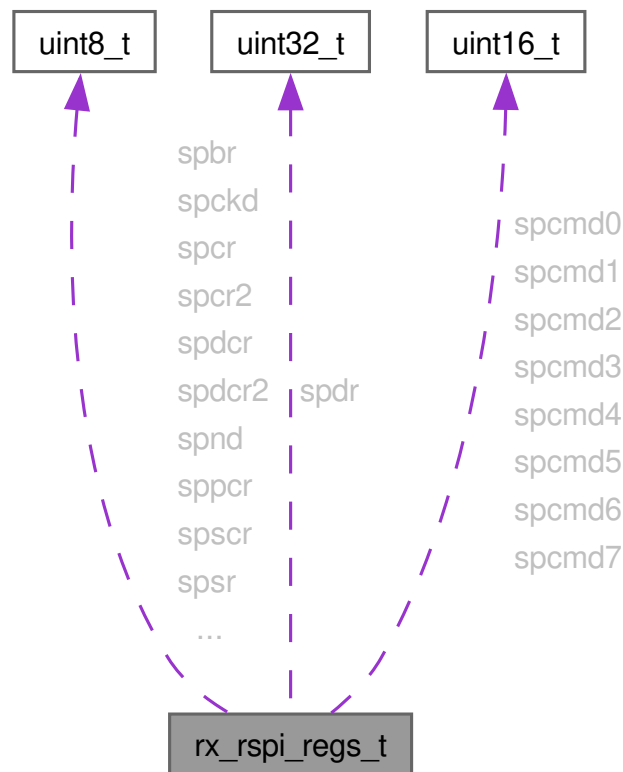
Version 1.0.0

Version

1.0.0

Definition at line 330 of file [rx72n_rspi_regs.h](#).

Collaboration diagram for rx_rspi_regs_t:



Data Fields

volatile uint8_t	spbr	SPI Bit Rate Register (SPBR) @ offset 0x0A. Sets SPI clock frequency: $f_{SPCK} = PCLKB / (2 \times (SPBR + 1))$ Calculation Examples (PCLKB = 60 MHz): • SPBR=0: 30 MHz (maximum) • SPBR=2: 10 MHz (RPi5 communication) • SPBR=5: 5 MHz (not used in current design) • SPBR=29: 1 MHz Note Only used in controller mode; ignored in peripheral mode
------------------	------	--

volatile uint8_t	spckd	SPI Clock Delay Register (SPCKD) @ offset 0x0C. Sets delay from SSL assertion to first SPCK edge. Delay: 1 to 8 SPCK cycles (value + 1)
volatile uint16_t	spcmd0	SPI Command Register 0 (SPCMD0) @ offset 0x10. Primary command configuration for transfer parameters. Bit Fields: • Bit 15 (SCKDEN): RSPCK delay enable • Bit 14 (SLNDEN): SSL negation delay enable • Bit 13 (SPNDEN): Next-access delay enable • Bit 12 (LSBF): LSB-first transfer (0=MSB first) • Bits 11:8 (SPB[3:0]): Data length (0111=8bit, 1111=16bit, etc.) • Bit 7 (SSLKP): SSL signal level keep (burst transfer) • Bits 6:4 (SSLA[2:0]): SSL signal assertion select (0=SSL0) • Bits 3:2 (BRDV[1:0]): Bit rate division (0=base, 1=/2, 2=/4, 3=/8) • Bit 1 (CPOL): Clock polarity (0=low when idle) • Bit 0 (CPHA): Clock phase (0=sample odd edge)
volatile uint16_t	spcmd1	SPI Command Register 1 @ offset 0x12
volatile uint16_t	spcmd2	SPI Command Register 2 @ offset 0x14
volatile uint16_t	spcmd3	SPI Command Register 3 @ offset 0x16
volatile uint16_t	spcmd4	SPI Command Register 4 @ offset 0x18
volatile uint16_t	spcmd5	SPI Command Register 5 @ offset 0x1A
volatile uint16_t	spcmd6	SPI Command Register 6 @ offset 0x1C
volatile uint16_t	spcmd7	SPI Command Register 7 @ offset 0x1E
volatile uint8_t	spr	SPI Control Register (SPCR) @ offset 0x00. Controls RSPi operation including enable, mode, and interrupts. Bit Fields: • Bit 7 (SPRIE): Receive interrupt enable • Bit 6 (SPE): SPI function enable • Bit 5 (SPTIE): Transmit interrupt enable • Bit 4 (SPEIE): Error interrupt enable • Bit 3 (MSTR): Controller/Peripheral mode (1=Controller) • Bit 2 (MODFEN): Mode fault detection enable • Bit 1 (TXMD): Transmit-only mode • Bit 0 (SPMS): SPI mode select See also rspi_spr_bits_t Bit definitions

volatile uint8_t	sPCR2	SPI Control Register 2 (SPCR2) @ offset 0x0F. Additional control including parity and idle interrupt.
volatile uint8_t	spDCR	SPI Data Control Register (SPDCR) @ offset 0x0B. Controls data access mode and receive data ready detection. See also rspi_spdcr_bits_t Bit definitions
volatile uint8_t	spDCR2	SPI Data Control Register 2 (SPDCR2) @ offset 0x20. Controls byte swap for 16/32-bit transfers.
volatile uint32_t	spDR	SPI Data Register (SPDR) @ offset 0x04. 32-bit TX/RX data register. Access width depends on SPDCR.SPLW. Access Modes: • Word access (SPLW=1): 32-bit read/write • Byte access (SPLW=0): 8-bit read/write (lower byte only) Note For 8-bit transfers, only lower byte is significant For 16-bit transfers, use appropriate data alignment
volatile uint8_t	spND	SPI Next-Access Delay Register (SPND) @ offset 0x0E. Sets delay from SSL negation to next SSL assertion. Delay: 1 to 8 SPCK cycles (value + 1)
volatile uint8_t	sPPCR	SPI Pin Control Register (SPPCR) @ offset 0x02. Controls loopback modes and COPI idle value. Bit Fields: • Bit 5 (MOIFE): Enable COPI idle value fixing • Bit 4 (MOIFV): COPI idle value (0=low, 1=high) • Bit 1 (SPLP2): Loopback mode 2 (no inversion) • Bit 0 (SPLP): Loopback mode (with inversion) See also rspi_sppcr_bits_t Bit definitions
volatile uint8_t	spSCR	SPI Sequence Control Register (SPSCR) @ offset 0x08. Controls command sequence length (0-7 = 1-8 commands). Value: 0 to 7 (number of commands - 1)

volatile uint8_t	spsr	SPI Status Register (SPSR) @ offset 0x03. Indicates RSPI status including buffer flags and error flags. Bit Fields: • Bit 7 (SPRF): Receive buffer full • Bit 5 (SPTEF): Transmit buffer empty • Bit 4 (UDRF): Underrun error • Bit 3 (PERF): Parity error • Bit 2 (MODF): Mode fault error • Bit 1 (IDLNF): Idle flag (0=idle) • Bit 0 (OVRF): Overrun error See also rspi_spsr_bits_t Bit definitions
volatile uint8_t	spssr	SPI Sequence Status Register (SPSSR) @ offset 0x09. Indicates current command sequence position.
volatile uint8_t	sslnd	SPI Chip Select Negation Delay Register (SSLND) @ offset 0x0D. Sets delay from last SPCK edge to SSL negation. Delay: 1 to 8 SPCK cycles (value + 1)
volatile uint8_t	sslp	SPI Chip Select Polarity Register (SSLP) @ offset 0x01. Sets polarity of SSL0-SSL3 signals (0=active low, 1=active high). Note Most SPI peripherals expect active-low chip select (default 0x00)

8.53.3 Enumeration Type Documentation

8.53.3.1 rspi`spcr`bits`t

```
enum rspi\_spcr\_bits\_t : uint8_t
```

RSPI Control Register (SPCR) Bit Definitions.

Controls RSPI operation including mode selection, interrupt enables, and function enable. The SPCR register must be configured before enabling RSPI with the SPE bit.

Register Layout (8-bit @ offset 0x00)

Bit	Name	R/W	Description
7	SPRIE	R/W	Receive interrupt enable

Bit	Name	R/W	Description
6	SPE	R/W	SPI function enable (1=enabled)
5	SPTIE	R/W	Transmit interrupt enable
4	SPEIE	R/W	Error interrupt enable
3	MSTR	R/W	Controller/Peripheral mode (1=Controller)
2	MODFEN	R/W	Mode fault detection enable
1	TXMD	R/W	Transmit-only mode (simplex)
0	SPMS	R/W	SPI mode select

Configuration Examples

```
// Peripheral mode with receive/transmit interrupts (RPi5 communication)
rspi0()->spcr = k_rspi_spcr_spe | k_rspi_spcr_sprie | k_rspi_spcr_sptie;

// Controller mode with transmit interrupt (not used in current design)
rspi1()->spcr = k_rspi_spcr_spe | k_rspi_spcr_mstr | k_rspi_spcr_sptie;

// Disable RSPI before reconfiguration
rspi0()->spcr = 0x00;
```

Warning

SPE must be cleared (0) before changing other bits

See also

[rx_rspi_regs_t::spcr](#) Register field in structure

Since

Version 1.0.0

Enumerator

k_rspi_spcr_sprie	Receive Interrupt Enable (SPRIE) - Bit 7. Enables interrupt when receive buffer is full (SPRF=1). Use for DMA trigger or ISR-driven receive processing.
k_rspi_spcr_spe	SPI Function Enable (SPE) - Bit 6. Enables RSPI operation. Must be set last after configuration. Warning Clear this bit before modifying other SPCR bits.
k_rspi_spcr_sptie	Transmit Interrupt Enable (SPTIE) - Bit 5. Enables interrupt when transmit buffer is empty (SPTEF=1). Use for DMA trigger or ISR-driven transmit processing.
k_rspi_spcr_speie	Error Interrupt Enable (SPEIE) - Bit 4. Enables interrupt on error conditions (overrun, underrun, mode fault). Recommended for robust error handling in production systems.

k_rspi_spcr_mstr	Controller/Peripheral Mode Select (MSTR) - Bit 3. Selects RSPI operating mode: 0: Peripheral mode (clock input from external controller) 1: Controller mode (clock output to external peripherals) Note RSPI0 uses peripheral mode (RPI5 is controller) RSPI1 not used in the current STAR design (controller mode)
k_rspi_spcr_modfe	Mode Fault Error Detection Enable (MODFEN) - Bit 2. Enables mode fault detection in multi-controller configurations. Not typically needed for single-controller systems.
k_rspi_spcr_txmd	Transmit-Only Mode (TXMD) - Bit 1. Enables simplex transmit-only operation (no receive). Useful for write-only peripherals like displays.
k_rspi_spcr_spms	SPI Mode Select (SPMS) - Bit 0. Selects 3-wire or 4-wire SPI mode.

Definition at line 612 of file [rx72n_rspi_regs.h](#).

```

00612         : uint8_t {
00613     /**
00614     * @brief Receive Interrupt Enable (SPRIE) - Bit 7
00615     * @details Enables interrupt when receive buffer is full (SPRF=1).
00616     * Use for DMA trigger or ISR-driven receive processing.
00617     */
00618     k_rspi_spcr_sprrie = (1 « 7),
00619
00620     /**
00621     * @brief SPI Function Enable (SPE) - Bit 6
00622     * @details Enables RSPI operation. Must be set last after configuration.
00623     * @warning Clear this bit before modifying other SPCR bits.
00624     */
00625     k_rspi_spcr_spe = (1 « 6),
00626
00627     /**
00628     * @brief Transmit Interrupt Enable (SPTIE) - Bit 5
00629     * @details Enables interrupt when transmit buffer is empty (SPTEF=1).
00630     * Use for DMA trigger or ISR-driven transmit processing.
00631     */
00632     k_rspi_spcr_sptie = (1 « 5),
00633
00634     /**
00635     * @brief Error Interrupt Enable (SPEIE) - Bit 4
00636     * @details Enables interrupt on error conditions (overrun, underrun, mode fault).
00637     * Recommended for robust error handling in production systems.
00638     */
00639     k_rspi_spcr_speie = (1 « 4),
00640
00641     /**
00642     * @brief Controller/Peripheral Mode Select (MSTR) - Bit 3
00643     * @details Selects RSPI operating mode:
00644     * - 0: Peripheral mode (clock input from external controller)
00645     * - 1: Controller mode (clock output to external peripherals)
00646     * @note RSPI0 uses peripheral mode (RPI5 is controller)
00647     * @note RSPI1 not used in the current STAR design (controller mode)
00648     */
00649     k_rspi_spcr_mstr = (1 « 3),
00650
00651     /**
00652     * @brief Mode Fault Error Detection Enable (MODFEN) - Bit 2
00653     * @details Enables mode fault detection in multi-controller configurations.
00654     * Not typically needed for single-controller systems.
00655     */
00656     k_rspi_spcr_modfe = (1 « 2),
00657
00658     /**
00659     * @brief Transmit-Only Mode (TXMD) - Bit 1
00660     * @details Enables simplex transmit-only operation (no receive).
00661     * Useful for write-only peripherals like displays.
00662     */
00663     k_rspi_spcr_txmd = (1 « 1),
00664
00665     /**
00666     * @brief SPI Mode Select (SPMS) - Bit 0

```

```

00667  * @details Selects 3-wire or 4-wire SPI mode.
00668  */
00669  k_rspi_spcr_spm = (1 « 0),
00670 } rspi_spcr_bits_t;

```

8.53.3.2 rspi_spcr_bits_t

```
enum rspi_spcr_bits_t : uint8_t
```

RSPI Data Control Register (SPDCR) Bit Definitions.

Controls data access modes including word/byte access to SPDR register and receive data ready detection timing.

Register Layout (8-bit @ offset 0x0B)

Bit	Name	R/W	Description
7	-	-	Reserved
6	SPBYT	R/W	Byte access (1=8-bit, 0=word/longword per SPLW)
5	SPLW	R/W	Longword access (0=word/16-bit, 1=longword/32-bit)
4	SPRDTD	R/W	Receive data ready timing (0=RX buf, 1=TX buf)
3:↵ 2	-	-	Reserved
1:↵ 0	SPFC	R/W	Number of frames (0=1, 1=2, 2=3, 3=4)

Access Mode Selection

The SPLW bit determines how the SPDR register is accessed (when SPBYT=0):

- Word access (SPLW=0): 16-bit read/write (for 8-16 bit SPI frames)
- Longword access (SPLW=1): 32-bit read/write (for 20/24/32 bit SPI frames)

Example

```

// Configure for 8-bit byte access (typical for RPi5 communication)
rspi0()->spdcr = 0x00; // Byte access

// Configure for 32-bit word access (higher throughput)
rspi1()->spdcr = k_rspi_spcr_splw; // Word access

```

See also

[rx_rspi_regs_t::spdcr](#) Register field in structure

Since

Version 1.0.0

Enumerator

k_rspi_spdcr_sprtdt	Receive Data Ready Detection (SPRDTD) - Bit 4. Selects whether SPDR reads return receive or transmit buffer: • 0: SPDR reads from receive buffer (normal operation) • 1: SPDR reads from transmit buffer
k_rspi_spdcr_splw	Longword Access Mode (SPLW) - Bit 5. SPDR access width selection (when SPBYT=0): • 0: Word access (16-bit) – use for 8-16 bit SPI frames • 1: Longword access (32-bit) – use for 20/24/32 bit SPI frames Note For 8-bit SPI (RPi5 communication), leave at 0 (default)

Definition at line 876 of file [rx72n_rspi_regs.h](#).

```

00876         : uint8_t {
00877     /**
00878     * @brief Receive Data Ready Detection (SPRDTD) - Bit 4
00879     * @details Selects whether SPDR reads return receive or transmit buffer:
00880     * - 0: SPDR reads from receive buffer (normal operation)
00881     * - 1: SPDR reads from transmit buffer
00882     */
00883     k_rspi_spdcr_sprtdt = (1 << 4),
00884
00885     /**
00886     * @brief Longword Access Mode (SPLW) - Bit 5
00887     * @details SPDR access width selection (when SPBYT=0):
00888     * - 0: Word access (16-bit) -- use for 8-16 bit SPI frames
00889     * - 1: Longword access (32-bit) -- use for 20/24/32 bit SPI frames
00890     * @note For 8-bit SPI (RPi5 communication), leave at 0 (default)
00891     */
00892     k_rspi_spdcr_splw = (1 << 5),
00893 } rspi_spdcr_bits_t;

```

8.53.3.3 rspi`spcr`bits`t

```
enum rspi\_sppcr\_bits\_t : uint8_t
```

RSPI Pin Control Register (SPPCR) Bit Definitions.

Controls RSPI pin behavior including loopback modes for testing and COPI (Controller Out Peripheral In) idle value control.

Register Layout (8-bit @ offset 0x02)

Bit	Name	R/W	Description
7	-	-	Reserved
6	-	-	Reserved

Bit	Name	R/W	Description
5	MOIFE	R/W	COPI idle fixed value enable
4	MOIFV	R/W	COPI idle fixed value (0=low, 1=high)
3↔ 2	-	-	Reserved
1	SPLP2	R/W	Loopback mode 2 (no inversion)
0	SPLP	R/W	Loopback mode (with inversion)

Loopback Modes (for Testing)

- SPLP: Internal loopback with data inversion (for self-test)
- SPLP2: Internal loopback without inversion (for data verification)

COPI Idle Control

When MOIFE is set, the COPI pin is driven to the value specified by MOIFV during idle periods. Useful for peripherals that sample the line between transfers.

Example

```
// Enable loopback mode for self-test
rspi0()->sppcr = k_rspi_sppcr_splp2; // Loopback without inversion

// Normal operation with COPI idle high
rspi1()->sppcr = k_rspi_sppcr_moife | k_rspi_sppcr_moifv;
```

See also

[rx_rspi_regs_t::sppcr](#) Register field in structure

Since

Version 1.0.0

Enumerator

k_rspi_sppcr_moife	COPI Idle Fixed Value Enable (MOIFE) - Bit 5. When set, COPI pin is driven to MOIFV value during idle.
k_rspi_sppcr_moifv	COPI Idle Fixed Value (MOIFV) - Bit 4. COPI idle level when MOIFE is enabled: • 0: Drive COPI low during idle • 1: Drive COPI high during idle
k_rspi_sppcr_splp2	Loopback Mode 2 (SPLP2) - Bit 1. Internal loopback without data inversion. CPIO receives exactly what COPI transmits. Useful for data path verification.
k_rspi_sppcr_splp	Loopback Mode (SPLP) - Bit 0. Internal loopback with data inversion. Useful for hardware self-test where inverted echo is expected.

Definition at line 811 of file [rx72n_rspi_regs.h](#).

```

00811         : uint8_t {
00812     /**
00813      * @brief COPI Idle Fixed Value Enable (MOIFE) - Bit 5
00814      * @details When set, COPI pin is driven to MOIFV value during idle.
00815      */
00816     k_rspi_sppcr_moife = (1 « 5),
00817
00818     /**
00819      * @brief COPI Idle Fixed Value (MOIFV) - Bit 4
00820      * @details COPI idle level when MOIFE is enabled:
00821      * - 0: Drive COPI low during idle
00822      * - 1: Drive COPI high during idle
00823      */
00824     k_rspi_sppcr_moifv = (1 « 4),
00825
00826     /**
00827      * @brief Loopback Mode 2 (SPLP2) - Bit 1
00828      * @details Internal loopback without data inversion. CIPO receives
00829      * exactly what COPI transmits. Useful for data path verification.
00830      */
00831     k_rspi_sppcr_splp2 = (1 « 1),
00832
00833     /**
00834      * @brief Loopback Mode (SPLP) - Bit 0
00835      * @details Internal loopback with data inversion. Useful for hardware
00836      * self-test where inverted echo is expected.
00837      */
00838     k_rspi_sppcr_splp = (1 « 0),
00839 } rspi_sppcr_bits_t;

```

8.53.3.4 rspi_spsr_bits_t

```
enum rspi\_spsr\_bits\_t : uint8_t
```

RSPI Status Register (SPSR) Bit Definitions.

Indicates RSPI status including buffer full/empty flags and error flags. Status flags are used to determine when data can be read/written and to detect error conditions.

Register Layout (8-bit @ offset 0x03)

Bit	Name	R/W	Description
7	SPRF	R	Receive buffer full (data ready to read)
6	-	-	Reserved
5	SPTEF	R	Transmit buffer empty (ready for new data)
4	UDRF	R/W	Underrun error (write 0 to clear)
3	PERF	R/W	Parity error (write 0 to clear)
2	MODF	R/W	Mode fault error (write 0 to clear)
1	IDLNF	R	Idle flag (0=idle, 1=transfer in progress)
0	OVRF	R/W	Overrun error (write 0 to clear)

Error Detection

- OVRF: Overrun error - receive buffer not read before next byte arrived
- MODF: Mode fault error - SSL signal error in multi-controller config
- PERF: Parity error - parity check failed during transmission
- UDRF: Underrun error - transmit buffer not written before needed

Polling Example

```

// Wait for transmit buffer empty before writing
while (!(rspl()->spsr & k_rspi_spsr_sptef)) {
    // Optionally yield to RTOS
}
rspl()->spdr = tx_data; // Write data

// Wait for receive buffer full before reading
while (!(rspl()->spsr & k_rspi_spsr_sprf)) { }
uint8_t rx_data = (uint8_t)rspl()->spdr; // Read data

// Check for errors
if (rspl()->spsr & (k_rspi_spsr_ovrf | k_rspi_spsr_udrf)) {
    // Handle error - clear by writing 0 to error bits
    rspl()->spsr = ~(k_rspi_spsr_ovrf | k_rspi_spsr_udrf);
}

```

Warning

Error flags must be cleared by writing 0 to the specific bit

See also

[rx_rspi_regs_t::spsr](#) Register field in structure

Since

Version 1.0.0

Enumerator

k_rspi_spsr_sprf	Receive Buffer Full Flag (SPRF) - Bit 7. Set when receive data is available in SPDR. Read SPDR to clear.
k_rspi_spsr_sptef	Transmit Buffer Empty Flag (SPTEF) - Bit 5. Set when transmit buffer can accept new data. Write SPDR to clear.
k_rspi_spsr_udrf	Underrun Error Flag (UDRF) - Bit 4. Set when transmit buffer was empty during transfer (controller mode). Write 0 to this bit to clear. Indicates software isn't feeding data fast enough.
k_rspi_spsr_perf	Parity Error Flag (PERF) - Bit 3. Set when parity check fails (if parity enabled). Write 0 to clear.
k_rspi_spsr_modf	Mode Fault Error Flag (MODF) - Bit 2. Set on SSL signal error in multi-controller configuration. Write 0 to clear. Typically not used in single-controller systems.
k_rspi_spsr_idlnf	Idle Flag (IDLNF) - Bit 1. Transfer status indicator: 0: RSPI is idle (no transfer in progress) 1: Transfer is in progress Note Read-only flag, useful for determining safe reconfiguration time
k_rspi_spsr_ovrf	Overrun Error Flag (OVRF) - Bit 0. Set when receive buffer overflows (new data arrived before old read). Write 0 to this bit to clear. Indicates software isn't reading data fast enough.

Definition at line 722 of file [rx72n_rspi_regs.h](#).

```

00722         : uint8_t {
00723     /**
00724      * @brief Receive Buffer Full Flag (SPRF) - Bit 7
00725      * @details Set when receive data is available in SPDR. Read SPDR to clear.
00726      */
00727     k_rspi_spsr_sprf = (1 « 7),
00728
00729     /**
00730      * @brief Transmit Buffer Empty Flag (SPTEF) - Bit 5
00731      * @details Set when transmit buffer can accept new data. Write SPDR to clear.
00732      */
00733     k_rspi_spsr_sptef = (1 « 5),
00734
00735     /**
00736      * @brief Underrun Error Flag (UDRF) - Bit 4
00737      * @details Set when transmit buffer was empty during transfer (controller mode).
00738      * Write 0 to this bit to clear. Indicates software isn't feeding data fast enough.
00739      */
00740     k_rspi_spsr_udrf = (1 « 4),
00741
00742     /**
00743      * @brief Parity Error Flag (PERF) - Bit 3
00744      * @details Set when parity check fails (if parity enabled). Write 0 to clear.
00745      */
00746     k_rspi_spsr_perf = (1 « 3),
00747
00748     /**
00749      * @brief Mode Fault Error Flag (MODF) - Bit 2
00750      * @details Set on SSL signal error in multi-controller configuration.
00751      * Write 0 to clear. Typically not used in single-controller systems.
00752      */
00753     k_rspi_spsr_modf = (1 « 2),
00754
00755     /**
00756      * @brief Idle Flag (IDLNF) - Bit 1
00757      * @details Transfer status indicator:
00758      * - 0: RSPI is idle (no transfer in progress)
00759      * - 1: Transfer is in progress
00760      * @note Read-only flag, useful for determining safe reconfiguration time
00761      */
00762     k_rspi_spsr_idlnf = (1 « 1),
00763
00764     /**
00765      * @brief Overrun Error Flag (OVRF) - Bit 0
00766      * @details Set when receive buffer overflows (new data arrived before old read).
00767      * Write 0 to this bit to clear. Indicates software isn't reading data fast enough.
00768      */
00769     k_rspi_spsr_ovrf = (1 « 0),
00770 } rspi_spsr_bits_t;

```

8.53.3.5 rx`rspi`addresses`t

```
enum rx_rspi_addresses_t : uintptr_t
```

RSPI channel base addresses.

Base addresses for the three RSPI channels on the RX72N. Each channel has a 64-byte register space (0x00-0x3F), though only 33 bytes are used. Addresses verified against RX72N Hardware Manual Ch44, Table 44.1.

Pin Naming Convention

Hardware signal names (MOSIA/MISOA/MOSIB/MISOB/RSPCKA/RSPCKB) are from the RX72N datasheet. This project uses COPI/CIPO terminology per OSHWA inclusive naming standards. Pin descriptions below show: Hardware Name (Project Name).

Channel Spacing

- RSPI0 -> RSPI1: 0x40 bytes (contiguous)
- RSPI1 -> RSPI2: 0x1C0 bytes (gap for other peripherals)

Usage Example

```
// Access RSPi0 registers directly
volatile rx_rspi_regs_t* spi = (volatile rx_rspi_regs_t*)k_rspi0_base_addr;
spi->spr = k_rspi_spr_spe; // Enable SPI

// Preferred: Use accessor function
rspi0()->spr = k_rspi_spr_spe;
```

Invariant

`k_rspi0_base_addr < k_rspi1_base_addr < k_rspi2_base_addr`

See also

[rspi0\(\)](#), [rspi1\(\)](#), [rspi2\(\)](#) Preferred accessor functions

Since

Version 1.0.0

Enumerator

<code>k_rspi0_base_addr</code>	RSPi0 register base address (0x000D0100). Used for RPi5 communication in peripheral mode at 10 Mbps. Pin Assignments: • RSPCKA: P27/RSPCKA (clock input from RPi5) • MOSIA: P26/MOSIA (RX72N hardware name) -> COPI (project name) - data from RPi5 • MISOA: P30/MISOA (RX72N hardware name) -> CIPO (project name) - data to RPi5 • SSLA0: P54/SSLA0 (chip select from RPi5)
<code>k_rspi1_base_addr</code>	RSPi1 register base address (0x000D0140). Not used in the current STAR design. Available for future SPI peripherals. Directly follows RSPi0 in memory (0x40 byte offset). Pin Assignments: • RSPCKB: PE5/RSPCKB (clock output to peripherals) • PE6/MOSIB (RX72N hardware pin name, project terminology:↔ COPI) - data out to peripheral • PE7/MISOB (RX72N hardware pin name, project terminology:↔ CIPO) - data in from peripheral • SSLB0-3: Individual chip selects (available for future peripherals)
<code>k_rspi2_base_addr</code>	RSPi2 register base address (0x000D0300). Not used in the current STAR design. Available for future expansion. Located at 0x1C0 bytes after RSPi1 (gap for other peripherals).

Definition at line 174 of file [rx72n_rspi_regs.h](#).

```
00174 : uintptr_t {
00175 /**
```

```

00176  * @brief RSPi0 register base address (0x000D0100)
00177  * @details
00178  * Used for RPi5 communication in peripheral mode at 10 Mbps.
00179  * @par Pin Assignments:
00180  * - RSPCKA: P27/RSPCKA (clock input from RPi5)
00181  * - MOSIA: P26/MOSIA (RX72N hardware name) -> COPI (project name) - data from RPi5
00182  * - MISOA: P30/MISOA (RX72N hardware name) -> CIPO (project name) - data to RPi5
00183  * - SSLA0: P54/SSLA0 (chip select from RPi5)
00184  */
00185 k_rspi0_base_addr = 0x000D0100,
00186
00187 /**
00188  * @brief RSPi1 register base address (0x000D0140)
00189  * @details
00190  * Not used in the current STAR design. Available for future SPI peripherals.
00191  * Directly follows RSPi0 in memory (0x40 byte offset).
00192  * @par Pin Assignments:
00193  * - RSPCKB: PE5/RSPCKB (clock output to peripherals)
00194  * - PE6/MOSIB (RX72N hardware pin name, project terminology: COPI) - data out to peripheral
00195  * - PE7/MISOB (RX72N hardware pin name, project terminology: CIPO) - data in from peripheral
00196  * - SSLB0-3: Individual chip selects (available for future peripherals)
00197  */
00198 k_rspi1_base_addr = 0x000D0140,
00199
00200 /**
00201  * @brief RSPi2 register base address (0x000D0300)
00202  * @details
00203  * Not used in the current STAR design. Available for future expansion.
00204  * Located at 0x1C0 bytes after RSPi1 (gap for other peripherals).
00205  */
00206 k_rspi2_base_addr = 0x000D0300,
00207 } rx_rspi_addresses_t;

```

8.53.4 Function Documentation

8.53.4.1 rspi0()

```
volatile rx_rspi_regs_t * rspi0 (
    void ) [inline], [static]
```

Get pointer to RSPi0 registers (Raspberry Pi 5 communication).

Returns a volatile pointer to the RSPi0 register structure at address 0x000D0100. RSPi0 is configured as a peripheral for communication with the Raspberry Pi 5 controller at 10 Mbps.

Returns

Volatile pointer to RSPi0 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

None (hardware register always accessible)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example:

```
volatile rx_rspi_regs_t* spi = rspi0();
spi->spr = k_rspi_spr_spe; // Enable RSPi0
```

See also

[k_rspi0_base_addr](#) Base address constant

[rx_spi_comm_init\(\)](#) Higher-level initialization

Since

Version 1.0.0

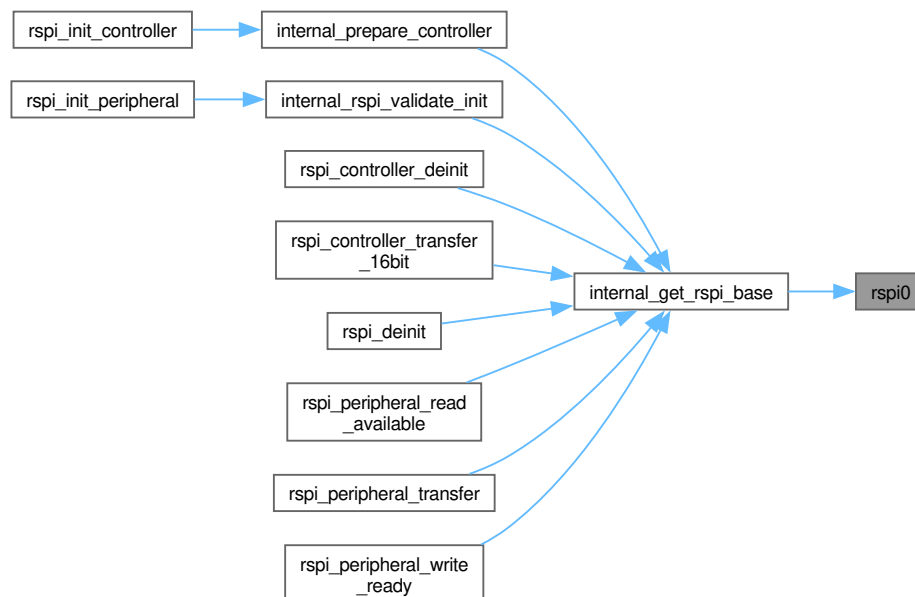
Definition at line 510 of file [rx72n_rspi_regs.h](#).

```
00511 {
00512     return (volatile rx_rspi_regs_t*)k_rspi0_base_addr;
00513 }
```

References [k_rspi0_base_addr](#).

Referenced by [internal_get_rspi_base\(\)](#).

Here is the caller graph for this function:



8.53.4.2 rspi1()

```
volatile rx_rspi_regs_t * rspi1 (  
    void )    [inline], [static]
```

Get pointer to RSPI1 registers (not used in current design).

Returns a volatile pointer to the RSPI1 register structure at address 0x000D0140. RSPI1 is not used in the current STAR design.

Returns

Volatile pointer to RSPI1 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

None (hardware register always accessible)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example:

```
volatile rx_rspi_regs_t* spi = rspi1();  
spi->spcr = k_rspi_spcr_spe | k_rspi_spcr_mstr; // Enable as controller
```

See also

[k_rspi_base_addr](#) Base address constant

[rspi0\(\)](#) Primary SPI channel (RPi5 communication)

Since

Version 1.0.0

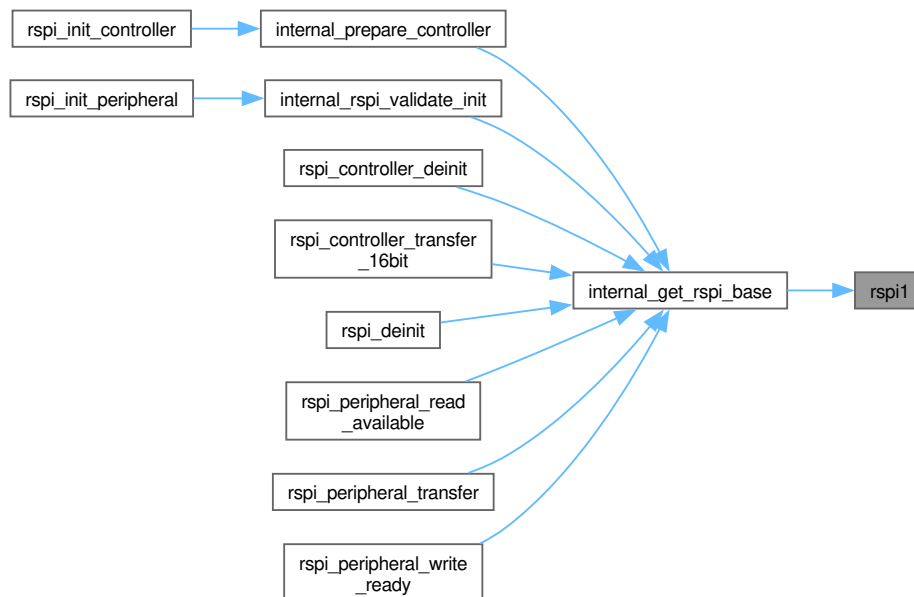
Definition at line 541 of file [rx72n_rspi_regs.h](#).

```
00542 {
00543     return (volatile rx_rspi_regs_t*)k_rspi1_base_addr;
00544 }
```

References [k_rspi1_base_addr](#).

Referenced by [internal_get_rspi_base\(\)](#).

Here is the caller graph for this function:



8.53.4.3 `rspi2()`

```
volatile rx_rspi_regs_t * rspi2 (
    void ) [inline], [static]
```

Get pointer to RSPI2 registers (not used in current design).

Returns a volatile pointer to the RSPI2 register structure at address 0x000D0300. RSPI2 is not used in the current STAR design.

Returns

Volatile pointer to RSPI2 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

None (hardware register always accessible)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

Currently unused in STAR project

See also

[k_rspi2_base_addr](#) Base address constant

Since

Version 1.0.0

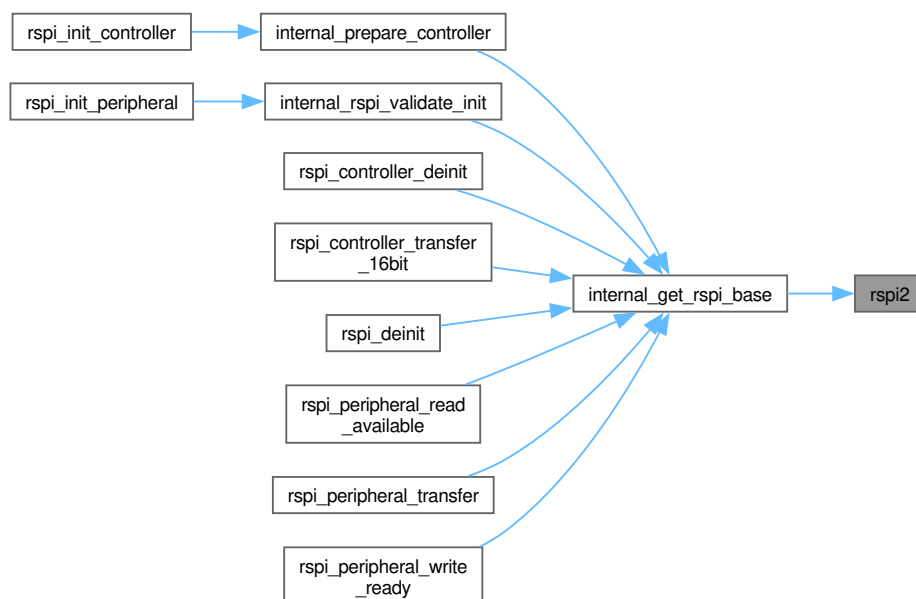
Definition at line 565 of file [rx72n_rspi_regs.h](#).

```
00566 {
00567     return (volatile rx_rspi_regs_t*)k_rspi2_base_addr;
00568 }
```

References [k_rspi2_base_addr](#).

Referenced by [internal_get_rspi_base\(\)](#).

Here is the caller graph for this function:



8.54 rx72n`rspi`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_rspi_regs.h
00003  * @brief RX72N RSPI Serial Peripheral Interface Register Definitions
00004  *
00005  * @details
00006  * This file defines all 22 RSPI registers per the RX72N Hardware Manual
00007  * (R01UH0824EJ0120 Rev.1.20, Ch44). All register addresses, offsets, and
00008  * bit definitions have been verified against the manual.
00009  *
00010  * @par System Architecture
00011  * The RSPI module provides high-speed serial communication between the RX72N
00012  * and external devices. In the STAR project, RSPI is used for:
00013  *
00014  * @dot
00015  * digraph rspi_architecture {
00016  *   rankdir=LR;
00017  *   node [shape=box, style=rounded];
00018  *
00019  *   subgraph cluster_rx72n {
00020  *     label="RX72N MCU";
00021  *     style=filled;
00022  *     color=lightgrey;
00023  *     rspi0 [label="RSPI0\n(Peripheral)"];
00024  *     rspi1 [label="RSPI1\n(Not used)"];
00025  *     rspi2 [label="RSPI2\n(Not used)"];
00026  *   }
00027  *
00028  *   rpi5 [label="Raspberry Pi 5\n(Controller)"];
00029  *   future [label="Future SPI\n(Peripherals)"];
00030  *
00031  *   rspi0 -.-> rpi5 [label="10 Mbps\n(Command/Telemetry)"];
00032  *   rspi1 -.-> future [label="Not used", style=dashed];
00033  * }
00034  * @enddot
00035  *
00036  * @par RSPI Channel Assignments
00037  * | Channel | Base Address | Mode | Target | Speed | Purpose |
00038  * |-----|-----|-----|-----|-----|-----|
00039  * | RSPI0 | 0x00D0100 | Peripheral | Raspberry Pi 5 | 10 Mbps | Command/telemetry |
00040  * | RSPI1 | 0x00D0140 | Not used | - | - | Not used in current design |
00041  * | RSPI2 | 0x00D0300 | Not used | - | - | Not used in current design |
00042  *
00043  * @par Key Features
00044  * - Full-duplex or simplex (transmit-only) synchronous communications
00045  * - Controller/Peripheral mode operation (using inclusive terminology)
00046  * - 8 to 32-bit configurable data frame length
00047  * - MSB-first or LSB-first transmission order
00048  * - Up to 8 command registers for sequential transfer mode
00049  * - Configurable clock polarity (CPOL) and phase (CPHA)
00050  * - DMA/DTC transfer support for high-throughput applications
00051  * - Mode fault detection for multi-controller configurations
00052  *
00053  * @par Hardware Requirements
00054  * | Parameter | RSPI0 (RPI5) | RSPI1 (Not used) |
00055  * |-----|-----|-----|
00056  * | PCLKB | 60 MHz | - |
00057  * | Max Bit Rate | 30 Mbps | - |
00058  * | Configured | 10 Mbps | - |
00059  * | Data Width | 8 bits | - |
00060  * | CPOL/CPHA | 0/0 | - |
00061  *
00062  * @par Memory Map (per channel)
00063  * | Offset | Size | Register | Description |
00064  * |-----|-----|-----|-----|
00065  * | 0x00 | 1 | SPCR | Control Register |
00066  * | 0x01 | 1 | SSLP | Chip Select Polarity Register |
00067  * | 0x02 | 1 | SPPCR | Pin Control Register |
00068  * | 0x03 | 1 | SPSR | Status Register |
00069  * | 0x04 | 4 | SPDR | Data Register (32-bit) |
00070  * | 0x08 | 1 | SPSCR | Sequence Control Register |
00071  * | 0x09 | 1 | SPSSR | Sequence Status Register |
00072  * | 0x0A | 1 | SPBR | Bit Rate Register |
00073  * | 0x0B | 1 | SPDCR | Data Control Register |
00074  * | 0x0C | 1 | SPCKD | Clock Delay Register |
00075  * | 0x0D | 1 | SSLND | SSL Negation Delay Register |
00076  * | 0x0E | 1 | SPND | Next-Access Delay Register |
00077  * | 0x0F | 1 | SPCR2 | Control Register 2 |
00078  * | 0x10 | 2 | SPCMD0 | Command Register 0 |
00079  * | 0x12 | 2 | SPCMD1 | Command Register 1 |
00080  * | 0x14 | 2 | SPCMD2 | Command Register 2 |
00081  * | 0x16 | 2 | SPCMD3 | Command Register 3 |
00082  * | 0x18 | 2 | SPCMD4 | Command Register 4 |

```

```

00083 * | 0x1A | 2 | SPCMD5 | Command Register 5 |
00084 * | 0x1C | 2 | SPCMD6 | Command Register 6 |
00085 * | 0x1E | 2 | SPCMD7 | Command Register 7 |
00086 * | 0x20 | 1 | SPDCR2 | Data Control Register 2 |
00087 *
00088 * @par NASA Power of 10 Compliance
00089 * - Rule 1: [OK] No goto, setjmp, or recursion in accessor functions
00090 * - Rule 2: [OK] N/A (no loops in register definitions)
00091 * - Rule 3: [OK] No dynamic memory allocation
00092 * - Rule 4: [OK] All accessor functions are single-statement inline
00093 * - Rule 5: [OK] N/A (accessor functions have no assertions needed)
00094 * - Rule 6: [OK] Minimal scope - inline accessors only
00095 * - Rule 7: [OK] N/A (no return values to check in definitions)
00096 * - Rule 8: [OK] All constants use C23 typed enums
00097 * - Rule 9: [OK] No function pointers in register definitions
00098 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00099 *
00100 * @par SOLID Principles
00101 * - **S**: Single responsibility - only register definitions
00102 * - **O**: Open for extension via additional bit field enums
00103 * - **I**: N/A (no inheritance hierarchy)
00104 * - **I**: Minimal interface - only necessary accessors
00105 * - **D**: N/A (hardware register layer, no dependencies to invert)
00106 *
00107 * @par Module Dependencies
00108 * - `<stdint.h>` - Fixed-width integer types
00109 * - No other dependencies (hardware register layer)
00110 *
00111 * @par Manual Reference
00112 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00113 * - Chapter 44: Serial Peripheral Interface (RSPi), pages 2349-2373
00114 * - Section 44.2: Register Descriptions, pages 2353-2373
00115 *
00116 * @see rx_spi_comm.h Higher-level SPI communication API
00117 * @see rx72n_regs.h Main register include file
00118 * @see docs/sections/03_hardware_pinout.tex STAR pin assignments
00119 *
00120 * @author Locked, Inc.
00121 * @date 2026-01-28
00122 * @copyright Copyright (c) 2026 Locked Inc.
00123 * SPDX-License-Identifier: MIT
00124 * @since Version 1.0.0
00125 */
00126
00127 #pragma once
00128
00129 #include <stddef.h>
00130 #include <stdint.h>
00131
00132 #ifdef __cplusplus
00133 extern "C" {
00134 #endif
00135
00136 /*
=====
00137 * Renesas Serial Peripheral Interface (RSPi) - For SPI Communication
00138 *
=====
00139 */
00140
00141 /**
00142 * @enum rx_rspi_addresses_t
00143 * @brief RSPi channel base addresses
00144 *
00145 * @details
00146 * Base addresses for the three RSPi channels on the RX72N. Each channel
00147 * has a 64-byte register space (0x00-0x3F), though only 33 bytes are used.
00148 * Addresses verified against RX72N Hardware Manual Ch44, Table 44.1.
00149 *
00150 * @par Pin Naming Convention
00151 * **Hardware signal names** (MOSIA/MISOA/MOSIB/MISOB/RSPCKA/RSPCKB) are from the
00152 * RX72N datasheet. This project uses **COPI/CIPO terminology** per OSHWA inclusive
00153 * naming standards. Pin descriptions below show: Hardware Name (Project Name).
00154 *
00155 * @par Channel Spacing
00156 * - RSPi0 -> RSPi1: 0x40 bytes (contiguous)
00157 * - RSPi1 -> RSPi2: 0x1C0 bytes (gap for other peripherals)
00158 *
00159 * @par Usage Example
00160 * @code
00161 * // Access RSPi0 registers directly
00162 * volatile rx_rspi_regs_t* spi = (volatile rx_rspi_regs_t*)k_rspi0_base_addr;
00163 * spi->spcr = k_rspi_spcr_spe; // Enable SPI
00164 *
00165 * // Preferred: Use accessor function
00166 * rspi0()->spcr = k_rspi_spcr_spe;
00167 * @endcode

```

```

00168 *
00169 * @invariant k_rspi0_base_addr < k_rspi1_base_addr < k_rspi2_base_addr
00170 *
00171 * @see rspi0(), rspi1(), rspi2() Preferred accessor functions
00172 * @since Version 1.0.0
00173 */
00174 typedef enum : uintptr_t {
00175 /**
00176  * @brief RSPi0 register base address (0x000D0100)
00177  * @details
00178  * Used for RPi5 communication in peripheral mode at 10 Mbps.
00179  * @par Pin Assignments:
00180  * - RSPCKA: P27/RSPCKA (clock input from RPi5)
00181  * - MOSIA: P26/MOSIA (RX72N hardware name) -> COPI (project name) - data from RPi5
00182  * - MISOA: P30/MISOA (RX72N hardware name) -> CIPO (project name) - data to RPi5
00183  * - SSLA0: P54/SSLA0 (chip select from RPi5)
00184  */
00185 k_rspi0_base_addr = 0x000D0100,
00186
00187 /**
00188  * @brief RSPi1 register base address (0x000D0140)
00189  * @details
00190  * Not used in the current STAR design. Available for future SPI peripherals.
00191  * Directly follows RSPi0 in memory (0x40 byte offset).
00192  * @par Pin Assignments:
00193  * - RSPCKB: PE5/RSPCKB (clock output to peripherals)
00194  * - PE6/MOSIB (RX72N hardware pin name, project terminology: COPI) - data out to peripheral
00195  * - PE7/MISOB (RX72N hardware pin name, project terminology: CIPO) - data in from peripheral
00196  * - SSLB0-3: Individual chip selects (available for future peripherals)
00197  */
00198 k_rspi1_base_addr = 0x000D0140,
00199
00200 /**
00201  * @brief RSPi2 register base address (0x000D0300)
00202  * @details
00203  * Not used in the current STAR design. Available for future expansion.
00204  * Located at 0x1C0 bytes after RSPi1 (gap for other peripherals).
00205  */
00206 k_rspi2_base_addr = 0x000D0300,
00207 } rx_rspi_addresses_t;
00208
00209 /**
00210  * @struct rx_rspi_regs_t
00211  * @brief RSPi Register Map Structure (Complete - All 22 Registers)
00212  *
00213  * @details
00214  * Renesas Serial Peripheral Interface (RSPi) registers for SPI communication.
00215  * Supports controller and peripheral modes with configurable clock and data format.
00216  * Addresses verified against RX72N Hardware Manual (R01UH0824EJ0120 Rev.1.20).
00217  *
00218  * @par Memory Layout Table
00219  * | Offset | Size | Field | Type | Description |
00220  * |-----|-----|-----|-----|-----|
00221  * | 0x00 | 1 | spcr | uint8_t | Control Register (enable, mode) |
00222  * | 0x01 | 1 | sslp | uint8_t | Chip Select Polarity Register |
00223  * | 0x02 | 1 | sppcr | uint8_t | Pin Control Register (loopback) |
00224  * | 0x03 | 1 | spsr | uint8_t | Status Register (flags) |
00225  * | 0x04 | 4 | spdr | uint32_t | Data Register (TX/RX) |
00226  * | 0x08 | 1 | spscr | uint8_t | Sequence Control Register |
00227  * | 0x09 | 1 | spssr | uint8_t | Sequence Status Register |
00228  * | 0x0A | 1 | spbr | uint8_t | Bit Rate Register |
00229  * | 0x0B | 1 | spdcr | uint8_t | Data Control Register |
00230  * | 0x0C | 1 | spckd | uint8_t | Clock Delay Register |
00231  * | 0x0D | 1 | sslnd | uint8_t | SSL Negation Delay Register |
00232  * | 0x0E | 1 | spnd | uint8_t | Next-Access Delay Register |
00233  * | 0x0F | 1 | spcr2 | uint8_t | Control Register 2 |
00234  * | 0x10 | 2 | spcmd0 | uint16_t | Command Register 0 |
00235  * | 0x12 | 2 | spcmd1 | uint16_t | Command Register 1 |
00236  * | 0x14 | 2 | spcmd2 | uint16_t | Command Register 2 |
00237  * | 0x16 | 2 | spcmd3 | uint16_t | Command Register 3 |
00238  * | 0x18 | 2 | spcmd4 | uint16_t | Command Register 4 |
00239  * | 0x1A | 2 | spcmd5 | uint16_t | Command Register 5 |
00240  * | 0x1C | 2 | spcmd6 | uint16_t | Command Register 6 |
00241  * | 0x1E | 2 | spcmd7 | uint16_t | Command Register 7 |
00242  * | 0x20 | 1 | spdcr2 | uint8_t | Data Control Register 2 |
00243  * | **Total** | **33** | | | (no padding due to packed attribute) |
00244  *
00245  * @par Sequential Transfer Mode
00246  * Up to 8 commands (SPCMD0-SPCMD7) can be executed sequentially in looped
00247  * execution. Each command configures:
00248  * - SSL signal selection and assertion
00249  * - Bit rate division
00250  * - Clock polarity (CPOL) and phase (CPHA)
00251  * - Transfer data length (8-32 bits)
00252  * - MSB/LSB first transmission order
00253  * - Inter-transfer delay settings
00254  *

```

```

00255 * @par Initialization Example (Peripheral Mode - RPi5 Communication)
00256 * @code{.c}
00257 * // Configure RSPi0 for 10 Mbps peripheral mode (RPi5 is controller)
00258 * volatile rx_rspi_regs_t* spi = rspi0();
00259 *
00260 * // 1. Disable RSPi before configuration
00261 * spi->spr = 0x00;
00262 *
00263 * // 2. Configure pin polarity (SSL active low)
00264 * spi->sslp = 0x00;
00265 *
00266 * // 3. Configure command register 0 for 8-bit transfers
00267 * //   SPB[3:0]=0111 (8 bits), CPOL=0, CPHA=0, LSBF=0 (MSB first)
00268 * spi->spcmd0 = 0x0700;
00269 *
00270 * // 4. Enable RSPi in peripheral mode with interrupts
00271 * spi->spr = k_rspi_spr_spe | k_rspi_spr_sptie | k_rspi_spr_sptie;
00272 * @endcode
00273 *
00274 * @par Initialization Example (Controller Mode - Not Used in Current Design)
00275 * @code{.c}
00276 * // Configure RSPi1 for 5 Mbps controller mode (not used in current design)
00277 * volatile rx_rspi_regs_t* spi = rspi1();
00278 *
00279 * // 1. Disable RSPi before configuration
00280 * spi->spr = 0x00;
00281 *
00282 * // 2. Configure bit rate for 5 Mbps (PCLKB=60MHz, n=5: 60/(2*(5+1))=5MHz)
00283 * spi->spbr = 5;
00284 *
00285 * // 3. Configure command register 0 for 16-bit transfers
00286 * //   SPB[3:0]=1111 (16 bits), CPOL=0, CPHA=1
00287 * spi->spcmd0 = 0x0F02;
00288 *
00289 * // 4. Enable RSPi in controller mode
00290 * spi->spr = k_rspi_spr_spe | k_rspi_spr_mstr | k_rspi_spr_sptie;
00291 * @endcode
00292 *
00293 * @par Data Transfer Example
00294 * @code{.c}
00295 * // Full-duplex transfer (controller mode)
00296 * static rx_err_t rspi_transfer(volatile rx_rspi_regs_t* spi,
00297 *                               uint8_t tx_data, uint8_t* rx_data) {
00298 *     // Wait for TX buffer empty
00299 *     while (!(spi->spsr & k_rspi_spsr_sptef)) { }
00300 *
00301 *     // Write data (starts transfer)
00302 *     spi->spdr = tx_data;
00303 *
00304 *     // Wait for RX buffer full
00305 *     while (!(spi->spsr & k_rspi_spsr_sprf)) { }
00306 *
00307 *     // Read received data
00308 *     *rx_data = (uint8_t)spi->spdr;
00309 *
00310 *     return k_rx_ok;
00311 * }
00312 * @endcode
00313 *
00314 * @invariant sizeof(rx_rspi_regs_t) == 33 bytes (verified by static assertion)
00315 * @invariant Structure is packed with no padding
00316 * @invariant All offsets match RX72N Hardware Manual Ch44
00317 *
00318 * @note Structure uses `__attribute__((packed))` to ensure no compiler padding
00319 * @note All fields are volatile for hardware register access
00320 * @warning Do not modify reserved bits in control registers
00321 * @attention RSPi must be disabled (SPE=0) before changing most configuration
00322 *
00323 * @see rspi0(), rspi1(), rspi2() Accessor functions for each channel
00324 * @see rspi_spr_bits_t Control register bit definitions
00325 * @see rspi_spsr_bits_t Status register bit definitions
00326 *
00327 * @since Version 1.0.0
00328 * @version 1.0.0
00329 */
00330 typedef struct __attribute__((packed)) {
00331     /**
00332      * @brief SPI Control Register (SPCR) @ offset 0x00
00333      * @details Controls RSPi operation including enable, mode, and interrupts.
00334      * @par Bit Fields:
00335      * - Bit 7 (SPRIE): Receive interrupt enable
00336      * - Bit 6 (SPE): SPI function enable
00337      * - Bit 5 (SPTIE): Transmit interrupt enable
00338      * - Bit 4 (SPEIE): Error interrupt enable
00339      * - Bit 3 (MSTR): Controller/Peripheral mode (1=Controller)
00340      * - Bit 2 (MODFEN): Mode fault detection enable
00341      * - Bit 1 (TXMD): Transmit-only mode

```

```

00342 * - Bit 0 (SPMS): SPI mode select
00343 * @see rspi_spcr_bits_t Bit definitions
00344 */
00345 volatile uint8_t spcr;
00346
00347 /**
00348 * @brief SPI Chip Select Polarity Register (SSLP) @ offset 0x01
00349 * @details Sets polarity of SSL0-SSL3 signals (0=active low, 1=active high).
00350 * @note Most SPI peripherals expect active-low chip select (default 0x00)
00351 */
00352 volatile uint8_t sslp;
00353
00354 /**
00355 * @brief SPI Pin Control Register (SPPCR) @ offset 0x02
00356 * @details Controls loopback modes and COPI idle value.
00357 * @par Bit Fields:
00358 * - Bit 5 (MOIFE): Enable COPI idle value fixing
00359 * - Bit 4 (MOIFV): COPI idle value (0=low, 1=high)
00360 * - Bit 1 (SPLP2): Loopback mode 2 (no inversion)
00361 * - Bit 0 (SPLP): Loopback mode (with inversion)
00362 * @see rspi_sppcr_bits_t Bit definitions
00363 */
00364 volatile uint8_t sppcr;
00365
00366 /**
00367 * @brief SPI Status Register (SPSR) @ offset 0x03
00368 * @details Indicates RSPI status including buffer flags and error flags.
00369 * @par Bit Fields:
00370 * - Bit 7 (SPRF): Receive buffer full
00371 * - Bit 5 (SPTEF): Transmit buffer empty
00372 * - Bit 4 (UDRF): Underrun error
00373 * - Bit 3 (PERF): Parity error
00374 * - Bit 2 (MODF): Mode fault error
00375 * - Bit 1 (IDLNF): Idle flag (0=idle)
00376 * - Bit 0 (OVRF): Overrun error
00377 * @see rspi_spsr_bits_t Bit definitions
00378 */
00379 volatile uint8_t spsr;
00380
00381 /**
00382 * @brief SPI Data Register (SPDR) @ offset 0x04
00383 * @details 32-bit TX/RX data register. Access width depends on SPDCR.SPLW.
00384 * @par Access Modes:
00385 * - Word access (SPLW=1): 32-bit read/write
00386 * - Byte access (SPLW=0): 8-bit read/write (lower byte only)
00387 * @note For 8-bit transfers, only lower byte is significant
00388 * @note For 16-bit transfers, use appropriate data alignment
00389 */
00390 volatile uint32_t spdr;
00391
00392 /**
00393 * @brief SPI Sequence Control Register (SPSCR) @ offset 0x08
00394 * @details Controls command sequence length (0-7 = 1-8 commands).
00395 * @par Value: 0 to 7 (number of commands - 1)
00396 */
00397 volatile uint8_t spscr;
00398
00399 /**
00400 * @brief SPI Sequence Status Register (SPSSR) @ offset 0x09
00401 * @details Indicates current command sequence position.
00402 */
00403 volatile uint8_t spssr;
00404
00405 /**
00406 * @brief SPI Bit Rate Register (SPBR) @ offset 0x0A
00407 * @details Sets SPI clock frequency:  $f_{SPCK} = PCLKB / (2 \times (SPBR + 1))$ 
00408 * @par Calculation Examples ( $PCLKB = 60 \text{ MHz}$ ):
00409 * - SPBR=0: 30 MHz (maximum)
00410 * - SPBR=2: 10 MHz (RPi5 communication)
00411 * - SPBR=5: 5 MHz (not used in current design)
00412 * - SPBR=29: 1 MHz
00413 * @note Only used in controller mode; ignored in peripheral mode
00414 */
00415 volatile uint8_t spbr;
00416
00417 /**
00418 * @brief SPI Data Control Register (SPDCR) @ offset 0x0B
00419 * @details Controls data access mode and receive data ready detection.
00420 * @see rspi_spdcr_bits_t Bit definitions
00421 */
00422 volatile uint8_t spdcr;
00423
00424 /**
00425 * @brief SPI Clock Delay Register (SPCKD) @ offset 0x0C
00426 * @details Sets delay from SSL assertion to first SPCK edge.
00427 * @par Delay: 1 to 8 SPCK cycles (value + 1)
00428 */

```

```

00429 volatile uint8_t spckd;
00430
00431 /**
00432  * @brief SPI Chip Select Negation Delay Register (SSLND) @ offset 0x0D
00433  * @details Sets delay from last SPCK edge to SSL negation.
00434  * @par Delay: 1 to 8 SPCK cycles (value + 1)
00435  */
00436 volatile uint8_t sslnd;
00437
00438 /**
00439  * @brief SPI Next-Access Delay Register (SPND) @ offset 0x0E
00440  * @details Sets delay from SSL negation to next SSL assertion.
00441  * @par Delay: 1 to 8 SPCK cycles (value + 1)
00442  */
00443 volatile uint8_t spnd;
00444
00445 /**
00446  * @brief SPI Control Register 2 (SPCR2) @ offset 0x0F
00447  * @details Additional control including parity and idle interrupt.
00448  */
00449 volatile uint8_t spcr2;
00450
00451 /**
00452  * @brief SPI Command Register 0 (SPCMD0) @ offset 0x10
00453  * @details Primary command configuration for transfer parameters.
00454  * @par Bit Fields:
00455  * - Bit 15 (SCKDEN): RSPCK delay enable
00456  * - Bit 14 (SLNDEN): SSL negation delay enable
00457  * - Bit 13 (SPNDEN): Next-access delay enable
00458  * - Bit 12 (LSBF): LSB-first transfer (0=MSB first)
00459  * - Bits 11:8 (SPB[3:0]): Data length (0111=8bit, 1111=16bit, etc.)
00460  * - Bit 7 (SSLKP): SSL signal level keep (burst transfer)
00461  * - Bits 6:4 (SSLA[2:0]): SSL signal assertion select (0=SSL0)
00462  * - Bits 3:2 (BRDV[1:0]): Bit rate division (0=base, 1=/2, 2=/4, 3=/8)
00463  * - Bit 1 (CPOL): Clock polarity (0=low when idle)
00464  * - Bit 0 (CPHA): Clock phase (0=sample odd edge)
00465  */
00466 volatile uint16_t spcmd0;
00467
00468 volatile uint16_t spcmd1; /**< SPI Command Register 1 @ offset 0x12 */
00469 volatile uint16_t spcmd2; /**< SPI Command Register 2 @ offset 0x14 */
00470 volatile uint16_t spcmd3; /**< SPI Command Register 3 @ offset 0x16 */
00471 volatile uint16_t spcmd4; /**< SPI Command Register 4 @ offset 0x18 */
00472 volatile uint16_t spcmd5; /**< SPI Command Register 5 @ offset 0x1A */
00473 volatile uint16_t spcmd6; /**< SPI Command Register 6 @ offset 0x1C */
00474 volatile uint16_t spcmd7; /**< SPI Command Register 7 @ offset 0x1E */
00475
00476 /**
00477  * @brief SPI Data Control Register 2 (SPDCR2) @ offset 0x20
00478  * @details Controls byte swap for 16/32-bit transfers.
00479  */
00480 volatile uint8_t spdcr2;
00481 } rx_rspi_regs_t;
00482
00483 /**
00484  * @brief Get pointer to RSPi0 registers (Raspberry Pi 5 communication)
00485  * @details
00486  * Returns a volatile pointer to the RSPi0 register structure at address
00487  * 0x000D0100. RSPi0 is configured as a peripheral for communication
00488  * with the Raspberry Pi 5 controller at 10 Mbps.
00489  *
00490  * @return Volatile pointer to RSPi0 register structure
00491  * @retval Non-NULL Always returns valid pointer (hardware address)
00492  *
00493  * @pre None (hardware register always accessible)
00494  * @post Pointer is valid for the lifetime of program execution
00495  *
00496  * @note Thread Safety: Safe - returns constant hardware address
00497  * @note This function is always inlined for zero overhead
00498  *
00499  * @par Example:
00500  * @code{.c}
00501  * volatile rx_rspi_regs_t* spi = rspi0();
00502  * spi->spcr = k_rspi_spcr_spe; // Enable RSPi0
00503  * @endcode
00504  *
00505  * @see k_rspi0_base_addr Base address constant
00506  * @see rx_spi_comm_init() Higher-level initialization
00507  * @since Version 1.0.0
00508  */
00509
00510 static inline volatile rx_rspi_regs_t* rspi0(void)
00511 {
00512     return (volatile rx_rspi_regs_t*)k_rspi0_base_addr;
00513 }
00514
00515 /**

```



```

00516 * @brief Get pointer to RSPi1 registers (not used in current design)
00517 *
00518 * @details
00519 * Returns a volatile pointer to the RSPi1 register structure at address
00520 * 0x000D0140. RSPi1 is not used in the current STAR design.
00521 *
00522 * @return Volatile pointer to RSPi1 register structure
00523 * @retval Non-NULL Always returns valid pointer (hardware address)
00524 *
00525 * @pre None (hardware register always accessible)
00526 * @post Pointer is valid for the lifetime of program execution
00527 *
00528 * @note Thread Safety: Safe - returns constant hardware address
00529 * @note This function is always inlined for zero overhead
00530 *
00531 * @par Example:
00532 * @code{.c}
00533 * volatile rx_rspi_regs_t* spi = rspi1();
00534 * spi->spr = k_rspi_spr_spe | k_rspi_spr_mstr; // Enable as controller
00535 * @endcode
00536 *
00537 * @see k_rspi1_base_addr Base address constant
00538 * @see rspi0() Primary SPI channel (RPi5 communication)
00539 * @since Version 1.0.0
00540 */
00541 static inline volatile rx_rspi_regs_t* rspi1(void)
00542 {
00543     return (volatile rx_rspi_regs_t*)k_rspi1_base_addr;
00544 }
00545
00546 /**
00547 * @brief Get pointer to RSPi2 registers (not used in current design)
00548 *
00549 * @details
00550 * Returns a volatile pointer to the RSPi2 register structure at address
00551 * 0x000D0300. RSPi2 is not used in the current STAR design.
00552 *
00553 * @return Volatile pointer to RSPi2 register structure
00554 * @retval Non-NULL Always returns valid pointer (hardware address)
00555 *
00556 * @pre None (hardware register always accessible)
00557 * @post Pointer is valid for the lifetime of program execution
00558 *
00559 * @note Thread Safety: Safe - returns constant hardware address
00560 * @note Currently unused in STAR project
00561 *
00562 * @see k_rspi2_base_addr Base address constant
00563 * @since Version 1.0.0
00564 */
00565 static inline volatile rx_rspi_regs_t* rspi2(void)
00566 {
00567     return (volatile rx_rspi_regs_t*)k_rspi2_base_addr;
00568 }
00569
00570 /*
=====
00571 * RSPi Bit Field Definitions (All Complete)
00572 *
=====
00573 */
00574 /**
00575 * @enum rspi_spr_bits_t
00576 * @brief RSPi Control Register (SPCR) Bit Definitions
00577 *
00578 * @details
00579 * Controls RSPi operation including mode selection, interrupt enables,
00580 * and function enable. The SPCR register must be configured before enabling
00581 * RSPi with the SPE bit.
00582 *
00583 * @par Register Layout (8-bit @ offset 0x00)
00584 * | Bit | Name | R/W | Description |
00585 * |-----|-----|-----|-----|
00586 * | 7 | SPRIE | R/W | Receive interrupt enable |
00587 * | 6 | SPE | R/W | SPI function enable (1=enabled) |
00588 * | 5 | SPTIE | R/W | Transmit interrupt enable |
00589 * | 4 | SPEIE | R/W | Error interrupt enable |
00590 * | 3 | MSTR | R/W | Controller/Peripheral mode (1=Controller) |
00591 * | 2 | MODFEN | R/W | Mode fault detection enable |
00592 * | 1 | TXMD | R/W | Transmit-only mode (simplex) |
00593 * | 0 | SPMS | R/W | SPI mode select |
00594 *
00595 * @par Configuration Examples
00596 * @code{.c}
00597 * // Peripheral mode with receive/transmit interrupts (RPi5 communication)
00598 * rspi0()->spr = k_rspi_spr_spe | k_rspi_spr_sprie | k_rspi_spr_sptie;
00599 *
00600 *

```



```

00601 * // Controller mode with transmit interrupt (not used in current design)
00602 * rspi0()->spr = k_rspi_spcr_spe | k_rspi_spcr_mstr | k_rspi_spcr_sptie;
00603 *
00604 * // Disable RSPi before reconfiguration
00605 * rspi0()->spr = 0x00;
00606 * @endcode
00607 *
00608 * @warning SPE must be cleared (0) before changing other bits
00609 * @see rx_rspi_regs_t::spcr Register field in structure
00610 * @since Version 1.0.0
00611 */
00612 typedef enum : uint8_t {
00613 /**
00614  * @brief Receive Interrupt Enable (SPRIE) - Bit 7
00615  * @details Enables interrupt when receive buffer is full (SPRF=1).
00616  * Use for DMA trigger or ISR-driven receive processing.
00617  */
00618 k_rspi_spcr_sprrie = (1 « 7),
00619
00620 /**
00621  * @brief SPI Function Enable (SPE) - Bit 6
00622  * @details Enables RSPi operation. Must be set last after configuration.
00623  * @warning Clear this bit before modifying other SPCR bits.
00624  */
00625 k_rspi_spcr_spe = (1 « 6),
00626
00627 /**
00628  * @brief Transmit Interrupt Enable (SPTIE) - Bit 5
00629  * @details Enables interrupt when transmit buffer is empty (SPTEF=1).
00630  * Use for DMA trigger or ISR-driven transmit processing.
00631  */
00632 k_rspi_spcr_sptie = (1 « 5),
00633
00634 /**
00635  * @brief Error Interrupt Enable (SPEIE) - Bit 4
00636  * @details Enables interrupts on error conditions (overrun, underrun, mode fault).
00637  * Recommended for robust error handling in production systems.
00638  */
00639 k_rspi_spcr_speie = (1 « 4),
00640
00641 /**
00642  * @brief Controller/Peripheral Mode Select (MSTR) - Bit 3
00643  * @details Selects RSPi operating mode:
00644  * - 0: Peripheral mode (clock input from external controller)
00645  * - 1: Controller mode (clock output to external peripherals)
00646  * @note RSPi0 uses peripheral mode (RPi5 is controller)
00647  * @note RSPi1 not used in the current STAR design (controller mode)
00648  */
00649 k_rspi_spcr_mstr = (1 « 3),
00650
00651 /**
00652  * @brief Mode Fault Error Detection Enable (MODFEN) - Bit 2
00653  * @details Enables mode fault detection in multi-controller configurations.
00654  * Not typically needed for single-controller systems.
00655  */
00656 k_rspi_spcr_modfe = (1 « 2),
00657
00658 /**
00659  * @brief Transmit-Only Mode (TXMD) - Bit 1
00660  * @details Enables simplex transmit-only operation (no receive).
00661  * Useful for write-only peripherals like displays.
00662  */
00663 k_rspi_spcr_txmd = (1 « 1),
00664
00665 /**
00666  * @brief SPI Mode Select (SPMS) - Bit 0
00667  * @details Selects 3-wire or 4-wire SPI mode.
00668  */
00669 k_rspi_spcr_spms = (1 « 0),
00670 } rspi_spcr_bits_t;
00671
00672 /**
00673  * @enum rspi_spsr_bits_t
00674  * @brief RSPi Status Register (SPSR) Bit Definitions
00675  *
00676  * @details
00677  * Indicates RSPi status including buffer full/empty flags and error flags.
00678  * Status flags are used to determine when data can be read/written and
00679  * to detect error conditions.
00680  *
00681  * @par Register Layout (8-bit @ offset 0x03)
00682  * | Bit | Name | R/W | Description |
00683  * |-----|-----|-----|-----|
00684  * | 7 | SPRF | R | Receive buffer full (data ready to read) |
00685  * | 6 | - | - | Reserved |
00686  * | 5 | SPTEF | R | Transmit buffer empty (ready for new data) |
00687  * | 4 | UDRF | R/W | Underrun error (write 0 to clear) |

```

```

00688 * | 3 | PERF | R/W | Parity error (write 0 to clear) |
00689 * | 2 | MODF | R/W | Mode fault error (write 0 to clear) |
00690 * | 1 | IDLNF | R | Idle flag (0=idle, 1=transfer in progress) |
00691 * | 0 | OVRF | R/W | Overrun error (write 0 to clear) |
00692 *
00693 * @par Error Detection
00694 * - **OVRF**: Overrun error - receive buffer not read before next byte arrived
00695 * - **MODF**: Mode fault error - SSL signal error in multi-controller config
00696 * - **PERF**: Parity error - parity check failed during transmission
00697 * - **UDRF**: Underrun error - transmit buffer not written before needed
00698 *
00699 * @par Polling Example
00700 * @code{.c}
00701 * // Wait for transmit buffer empty before writing
00702 * while (!(rspl()->spsr & k_rspi_spsr_sptef)) {
00703 *     // Optionally yield to RTOS
00704 * }
00705 * rspl()->spdr = tx_data; // Write data
00706 *
00707 * // Wait for receive buffer full before reading
00708 * while (!(rspl()->spsr & k_rspi_spsr_sprf)) { }
00709 * uint8_t rx_data = (uint8_t)rspl()->spdr; // Read data
00710 *
00711 * // Check for errors
00712 * if (rspl()->spsr & (k_rspi_spsr_ovrf | k_rspi_spsr_udrf)) {
00713 *     // Handle error - clear by writing 0 to error bits
00714 *     rspl()->spsr = ~(k_rspi_spsr_ovrf | k_rspi_spsr_udrf);
00715 * }
00716 * @endcode
00717 *
00718 * @warning Error flags must be cleared by writing 0 to the specific bit
00719 * @see rx_rspi_regs_t::spsr Register field in structure
00720 * @since Version 1.0.0
00721 */
00722 typedef enum : uint8_t {
00723     /**
00724      * @brief Receive Buffer Full Flag (SPRF) - Bit 7
00725      * @details Set when receive data is available in SPDR. Read SPDR to clear.
00726      */
00727     k_rspi_spsr_sprf = (1 « 7),
00728     /**
00729      * @brief Transmit Buffer Empty Flag (SPTEF) - Bit 5
00730      * @details Set when transmit buffer can accept new data. Write SPDR to clear.
00731      */
00732     k_rspi_spsr_sptef = (1 « 5),
00733     /**
00734      * @brief Underrun Error Flag (UDRF) - Bit 4
00735      * @details Set when transmit buffer was empty during transfer (controller mode).
00736      * Write 0 to this bit to clear. Indicates software isn't feeding data fast enough.
00737      */
00738     k_rspi_spsr_udrf = (1 « 4),
00739     /**
00740      * @brief Parity Error Flag (PERF) - Bit 3
00741      * @details Set when parity check fails (if parity enabled). Write 0 to clear.
00742      */
00743     k_rspi_spsr_perf = (1 « 3),
00744     /**
00745      * @brief Mode Fault Error Flag (MODF) - Bit 2
00746      * @details Set on SSL signal error in multi-controller configuration.
00747      * Write 0 to clear. Typically not used in single-controller systems.
00748      */
00749     k_rspi_spsr_modf = (1 « 2),
00750     /**
00751      * @brief Idle Flag (IDLNF) - Bit 1
00752      * @details Transfer status indicator:
00753      * - 0: RSPi is idle (no transfer in progress)
00754      * - 1: Transfer is in progress
00755      * @note Read-only flag, useful for determining safe reconfiguration time
00756      */
00757     k_rspi_spsr_idlnf = (1 « 1),
00758     /**
00759      * @brief Overrun Error Flag (OVRF) - Bit 0
00760      * @details Set when receive buffer overflows (new data arrived before old read).
00761      * Write 0 to this bit to clear. Indicates software isn't reading data fast enough.
00762      */
00763     k_rspi_spsr_ovrf = (1 « 0),
00764 } rspi_spsr_bits_t;
00765
00766 /**
00767 * @enum rspi_sppcr_bits_t
00768 * @brief RSPi Pin Control Register (SPPCR) Bit Definitions

```

```

00775 *
00776 * @details
00777 * Controls RSPI pin behavior including loopback modes for testing and
00778 * COPI (Controller Out Peripheral In) idle value control.
00779 *
00780 * @par Register Layout (8-bit @ offset 0x02)
00781 * | Bit | Name | R/W | Description |
00782 * |-----|-----|-----|-----|
00783 * | 7 | - | - | Reserved |
00784 * | 6 | - | - | Reserved |
00785 * | 5 | MOIFE | R/W | COPI idle fixed value enable |
00786 * | 4 | MOIFV | R/W | COPI idle fixed value (0=low, 1=high) |
00787 * | 3:2 | - | - | Reserved |
00788 * | 1 | SPLP2 | R/W | Loopback mode 2 (no inversion) |
00789 * | 0 | SPLP | R/W | Loopback mode (with inversion) |
00790 *
00791 * @par Loopback Modes (for Testing)
00792 * - **SPLP**: Internal loopback with data inversion (for self-test)
00793 * - **SPLP2**: Internal loopback without inversion (for data verification)
00794 *
00795 * @par COPI Idle Control
00796 * When MOIFE is set, the COPI pin is driven to the value specified by MOIFV
00797 * during idle periods. Useful for peripherals that sample the line between transfers.
00798 *
00799 * @par Example
00800 * @code{.c}
00801 * // Enable loopback mode for self-test
00802 * rspi0()->sppcr = k_rspi_sppcr_splp2; // Loopback without inversion
00803 *
00804 * // Normal operation with COPI idle high
00805 * rspi1()->sppcr = k_rspi_sppcr_moife | k_rspi_sppcr_moifv;
00806 * @endcode
00807 *
00808 * @see rx_rspi_regs_t::sppcr Register field in structure
00809 * @since Version 1.0.0
00810 */
00811 typedef enum : uint8_t {
00812 /**
00813 * @brief COPI Idle Fixed Value Enable (MOIFE) - Bit 5
00814 * @details When set, COPI pin is driven to MOIFV value during idle.
00815 */
00816 k_rspi_sppcr_moife = (1 << 5),
00817 /**
00818 * @brief COPI Idle Fixed Value (MOIFV) - Bit 4
00819 * @details COPI idle level when MOIFE is enabled:
00820 * - 0: Drive COPI low during idle
00821 * - 1: Drive COPI high during idle
00822 */
00823 k_rspi_sppcr_moifv = (1 << 4),
00824 /**
00825 * @brief Loopback Mode 2 (SPLP2) - Bit 1
00826 * @details Internal loopback without data inversion. CIPO receives
00827 * exactly what COPI transmits. Useful for data path verification.
00828 */
00829 k_rspi_sppcr_splp2 = (1 << 1),
00830 /**
00831 * @brief Loopback Mode (SPLP) - Bit 0
00832 * @details Internal loopback with data inversion. Useful for hardware
00833 * self-test where inverted echo is expected.
00834 */
00835 k_rspi_sppcr_splp = (1 << 0),
00836 } rspi_sppcr_bits_t;
00837
00838 k_rspi_sppcr_splp = (1 << 0),
00839 } rspi_sppcr_bits_t;
00840
00841 /**
00842 * @enum rspi_spdcr_bits_t
00843 * @brief RSPI Data Control Register (SPDCR) Bit Definitions
00844 *
00845 * @details
00846 * Controls data access modes including word/byte access to SPDR register
00847 * and receive data ready detection timing.
00848 *
00849 * @par Register Layout (8-bit @ offset 0x0B)
00850 * | Bit | Name | R/W | Description |
00851 * |-----|-----|-----|-----|
00852 * | 7 | - | - | Reserved |
00853 * | 6 | SPBYT | R/W | Byte access (1=8-bit, 0=word/longword per SPLW) |
00854 * | 5 | SPLW | R/W | Longword access (0=word/16-bit, 1=longword/32-bit) |
00855 * | 4 | SPRDTD | R/W | Receive data ready timing (0=RX buf, 1=TX buf) |
00856 * | 3:2 | - | - | Reserved |
00857 * | 1:0 | SPFC | R/W | Number of frames (0=1, 1=2, 2=3, 3=4) |
00858 *
00859 * @par Access Mode Selection
00860 * The SPLW bit determines how the SPDR register is accessed (when SPBYT=0):
00861 * - **Word access (SPLW=0)**: 16-bit read/write (for 8-16 bit SPI frames)

```

```

00862 * - **Longword access (SPLW=1)**: 32-bit read/write (for 20/24/32 bit SPI frames)
00863 *
00864 * @par Example
00865 * @code{.c}
00866 * // Configure for 8-bit byte access (typical for RPi5 communication)
00867 * rspi0()->spdcr = 0x00; // Byte access
00868 *
00869 * // Configure for 32-bit word access (higher throughput)
00870 * rspl1()->spdcr = k_rspl_splw; // Word access
00871 * @endcode
00872 *
00873 * @see rx_rspl_regs_t::spdcr Register field in structure
00874 * @since Version 1.0.0
00875 */
00876 typedef enum : uint8_t {
00877 /**
00878 * @brief Receive Data Ready Detection (SPRDTD) - Bit 4
00879 * @details Selects whether SPDR reads return receive or transmit buffer:
00880 * - 0: SPDR reads from receive buffer (normal operation)
00881 * - 1: SPDR reads from transmit buffer
00882 */
00883 k_rspl_spdcr_sprtdt = (1 « 4),
00884
00885 /**
00886 * @brief Longword Access Mode (SPLW) - Bit 5
00887 * @details SPDR access width selection (when SPBYT=0):
00888 * - 0: Word access (16-bit) -- use for 8-16 bit SPI frames
00889 * - 1: Longword access (32-bit) -- use for 20/24/32 bit SPI frames
00890 * @note For 8-bit SPI (RPi5 communication), leave at 0 (default)
00891 */
00892 k_rspl_spdcr_splw = (1 « 5),
00893 } rspl_spdcr_bits_t;
00894
00895 /*
=====
00896 * Static Assertions - Verify Register Layout at Compile Time
00897 *
=====
00898 */
00899
00900 /* Verify base addresses match Hardware Manual Ch44 */
00901 static_assert(k_rspl0_base_addr == 0x000D0100, "RSPI0 base address incorrect");
00902 static_assert(k_rspl1_base_addr == 0x000D0140, "RSPI1 base address incorrect");
00903 static_assert(k_rspl2_base_addr == 0x000D0300, "RSPI2 base address incorrect");
00904
00905 /* Verify register structure layout and offsets */
00906 static_assert(sizeof(rx_rspl_regs_t) == 33, "RSPI register structure size incorrect");
00907 static_assert(offsetof(rx_rspl_regs_t, spcr) == 0x00, "SPCR offset incorrect");
00908 static_assert(offsetof(rx_rspl_regs_t, sslp) == 0x01, "SSLP offset incorrect");
00909 static_assert(offsetof(rx_rspl_regs_t, sppcr) == 0x02, "SPPCR offset incorrect");
00910 static_assert(offsetof(rx_rspl_regs_t, spsr) == 0x03, "SPSR offset incorrect");
00911 static_assert(offsetof(rx_rspl_regs_t, spdr) == 0x04, "SPDR offset incorrect");
00912 static_assert(offsetof(rx_rspl_regs_t, spscr) == 0x08, "SPSCR offset incorrect");
00913 static_assert(offsetof(rx_rspl_regs_t, spssr) == 0x09, "SPSSR offset incorrect");
00914 static_assert(offsetof(rx_rspl_regs_t, spbr) == 0x0A, "SPBR offset incorrect");
00915 static_assert(offsetof(rx_rspl_regs_t, spdcr) == 0x0B, "SPDCR offset incorrect");
00916 static_assert(offsetof(rx_rspl_regs_t, spckd) == 0x0C, "SPCKD offset incorrect");
00917 static_assert(offsetof(rx_rspl_regs_t, sslnd) == 0x0D, "SSLND offset incorrect");
00918 static_assert(offsetof(rx_rspl_regs_t, spnd) == 0x0E, "SPND offset incorrect");
00919 static_assert(offsetof(rx_rspl_regs_t, spcr2) == 0x0F, "SPCR2 offset incorrect");
00920 static_assert(offsetof(rx_rspl_regs_t, spcmd0) == 0x10, "SPCMD0 offset incorrect");
00921 static_assert(offsetof(rx_rspl_regs_t, spcmd1) == 0x12, "SPCMD1 offset incorrect");
00922 static_assert(offsetof(rx_rspl_regs_t, spcmd2) == 0x14, "SPCMD2 offset incorrect");
00923 static_assert(offsetof(rx_rspl_regs_t, spcmd3) == 0x16, "SPCMD3 offset incorrect");
00924 static_assert(offsetof(rx_rspl_regs_t, spcmd4) == 0x18, "SPCMD4 offset incorrect");
00925 static_assert(offsetof(rx_rspl_regs_t, spcmd5) == 0x1A, "SPCMD5 offset incorrect");
00926 static_assert(offsetof(rx_rspl_regs_t, spcmd6) == 0x1C, "SPCMD6 offset incorrect");
00927 static_assert(offsetof(rx_rspl_regs_t, spcmd7) == 0x1E, "SPCMD7 offset incorrect");
00928 static_assert(offsetof(rx_rspl_regs_t, spdcr2) == 0x20, "SPDCR2 offset incorrect");
00929
00930 /* Verify channel spacing (RSPI0 to RSPI1 = 0x40 bytes, RSPI1 to RSPI2 = 0x1C0 bytes) */
00931 static_assert((k_rspl1_base_addr - k_rspl0_base_addr) == 0x40, "RSPI0 to RSPI1 spacing incorrect");
00932 static_assert((k_rspl2_base_addr - k_rspl1_base_addr) == 0x1C0, "RSPI1 to RSPI2 spacing incorrect");
00933
00934 #ifdef __cplusplus
00935 }
00936 #endif

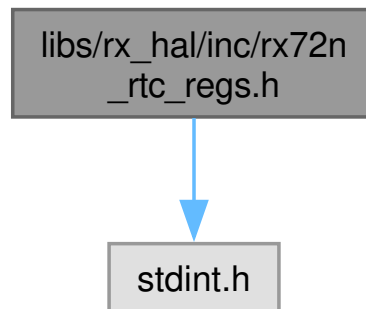
```

8.55 libs/rx/hal/inc/rx72n_rtc_regs.h File Reference

RX72N Real-Time Clock (RTC) Register Definitions.

```
#include <stdint.h>
```

Include dependency graph for rx72n_rtc_regs.h:



Enumerations

- `enum rtc_addresses_t : uintptr_t {`
`k_rtc_base_addr = 0x0008C400U , k_rtc_rcr3_addr = 0x0008C426U , k_rtc_rcr1_addr =`
`0x0008C422U , k_rtc_rcr2_addr = 0x0008C424U ,`
`k_rtc_rcr4_addr = 0x0008C428U }`
 RTC module register addresses.
- `enum rtc_offsets_t : uint8_t {`
`k_rtc_offset_r64cnt = 0x00U , k_rtc_offset_rseccnt = 0x02U , k_rtc_offset_rmincnt = 0x04U`
`, k_rtc_offset_rhrcnt = 0x06U ,`
`k_rtc_offset_rwkcnt = 0x08U , k_rtc_offset_rdaycnt = 0x0AU , k_rtc_offset_rmoncnt = 0x0CU ,`
`k_rtc_offset_ryrcnt = 0x0EU ,`
`k_rtc_offset_rsecar = 0x10U , k_rtc_offset_rminar = 0x12U , k_rtc_offset_rhrar = 0x14U ,`
`k_rtc_offset_rwkar = 0x16U ,`
`k_rtc_offset_rdayar = 0x18U , k_rtc_offset_rmonar = 0x1AU , k_rtc_offset_ryrar = 0x1CU ,`
`k_rtc_offset_ryraren = 0x1EU ,`
`k_rtc_offset_rcr1 = 0x22U , k_rtc_offset_rcr2 = 0x24U , k_rtc_offset_rcr3 = 0x26U ,`
`k_rtc_offset_rcr4 = 0x28U ,`
`k_rtc_offset_rfrh = 0x2AU , k_rtc_offset_rfri = 0x2CU , k_rtc_offset_radj = 0x2EU }`
 RTC register offsets from base address.
- `enum rcr3_bits_t : uint8_t { k_rcr3_rtcen_disable = 0x00U , k_rcr3_rtcen_enable = 0x01U ,`
`k_rcr3_rtcen_bit = 0x01U }`
 RTC Control Register 3 (RCR3) Bit Definitions.

Functions

- `static volatile uint8_t * rtc_rcr3_reg (void)`
 Get pointer to RTC Control Register 3 (RCR3).
- `static volatile uint8_t * rtc_rcr1_reg (void)`
 Get pointer to RTC Control Register 1 (RCR1).
- `static volatile uint8_t * rtc_rcr2_reg (void)`
 Get pointer to RTC Control Register 2 (RCR2).
- `static volatile uint8_t * rtc_rcr4_reg (void)`
 Get pointer to RTC Control Register 4 (RCR4).

8.55.1 Detailed Description

RX72N Real-Time Clock (RTC) Register Definitions.

Minimal RTC register definitions for sub-clock oscillator disable functionality. The RTC module is not used in the STAR project. These definitions exist solely to properly disable the sub-clock oscillator during system initialization.

Why This File Exists (Design Rationale)

The RX72N sub-clock oscillator (32.768 kHz) consumes power even when not actively used. To fully disable it, the RTC module's RCR3.RTCEN bit must be cleared to 0. This file provides the minimum register definitions needed for this operation without including the full RTC implementation.

CRITICAL ADDRESS VERIFICATION (2026-01-29)

All register addresses verified against RX72N Hardware Manual Ch33. Previous implementation had CRITICAL BUG - wrong struct offsets:

- Old code: RCR3 at offset 0x1A (0x0008C41A = RMONAR register!)
- Correct: RCR3 at offset 0x26 (0x0008C426 per manual)

Key Register Addresses (from Ch33)

Register	Address	Offset	Description
R64CNT	0x0008C400	0x00	64-Hz Counter
RSECCNT	0x0008C402	0x02	Second Counter (BCD)
RMINCNT	0x0008C404	0x04	Minute Counter (BCD)
RHRCNT	0x0008C406	0x06	Hour Counter (BCD)
RWKCNT	0x0008C408	0x08	Day-of-Week Counter
RDAYCNT	0x0008C40A	0x0A	Day Counter (BCD)
RMONCNT	0x0008C40C	0x0C	Month Counter (BCD)
RYRCNT	0x0008C40E	0x0E	Year Counter (BCD, 16-bit)
RSECAR	0x0008C410	0x10	Second Alarm
...	...	...	(alarm registers)
RYRAREN	0x0008C41E	0x1E	Year Alarm Enable
(gap)	0x0008C420	0x20	Reserved (2 bytes)
RCR1	0x0008C422	0x22	RTC Control Register 1
RCR2	0x0008C424	0x24	RTC Control Register 2
RCR3	0x0008C426	0x26	RTC Control Register 3
RCR4	0x0008C428	0x28	RTC Control Register 4

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] Single-statement accessor
- Rule 5: [OK] Static assertions verify addresses
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 33: Realtime Clock (RTCd), pages 1616-1670
- Section 33.2.11: RTC Control Register 3 (RCR3), page 1637

Note

This is an intentionally minimal implementation

Warning

Do not expand this file unless RTC functionality is actually needed

See also

[rx72n_system_regs.h](#) SOSCCR register for sub-clock oscillator control
[rx_clock_power_init.c](#) Clock initialization implementation

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_rtc_regs.h](#).

8.55.2 Enumeration Type Documentation

8.55.2.1 rcr3_bits_t

enum [rcr3_bits_t](#) : uint8_t

RTC Control Register 3 (RCR3) Bit Definitions.

Controls sub-clock oscillator input to the RTC module. This is the only RTC register actively used in the STAR project.

Register Layout (8-bit @ 0x0008C426)

Bit	Field	R/W	Description
0	RTCEN	R/W	Sub-Clock Oscillator Control for RTC
7:↔ 1	-	R	Reserved (read as 0)

RTCEN Bit Behavior

- 0 (k_rcr3_rtcen_disable): Sub-clock oscillator stopped for RTC.
- 1 (k_rcr3_rtcen_enable): Sub-clock oscillator running for RTC.

Note

Manual ref: Ch33 section 33.2.11 (page 1637)

Since

Version 1.0.0

Enumerator

k_rcr3_rtcen_disable	Disable sub-clock input to RTC (RTCEN=0). Use this value to disable RTC and save power. This is the STAR project default configuration.
k_rcr3_rtcen_enable	Enable sub-clock input to RTC (RTCEN=1). Use this value to enable RTC timekeeping. Not used in STAR project.
k_rcr3_rtcen_bit	RTCEN bit position (bit 0).

Definition at line 194 of file [rx72n_rtc_regs.h](#).

```

00194         : uint8_t {
00195     /**
00196      * @brief Disable sub-clock input to RTC (RTCEN=0)
00197      * @details Use this value to disable RTC and save power.
00198      * This is the STAR project default configuration.
00199      */
00200     k_rcr3_rtcen_disable = 0x00U,
00201
00202     /**
00203      * @brief Enable sub-clock input to RTC (RTCEN=1)
00204      * @details Use this value to enable RTC timekeeping.
00205      * Not used in STAR project.
00206      */
00207     k_rcr3_rtcen_enable = 0x01U,
00208
00209     /**
00210      * @brief RTCEN bit position (bit 0)
00211      */
00212     k_rcr3_rtcen_bit = 0x01U,
00213 } rcr3_bits_t;
```


8.55.2.2 rtc_addresses_t

enum `rtc_addresses_t` : `uintptr_t`

RTC module register addresses.

Addresses for RTC registers verified against RX72N Hardware Manual Ch33. Only addresses needed for sub-clock disable are defined.

Note

Addresses verified 2026-01-29 against manual section 33.2

Since

Version 1.0.0

Enumerator

<code>k_rtc_base_addr</code>	RTC register base address (0x0008C400). Manual ref: Ch33 section 33.2.1 (R64CNT at 0x0008C400)
<code>k_rtc_rcr3_addr</code>	RTC Control Register 3 (RCR3) address. Direct address for RCR3 at 0x0008C426 (offset 0x26 from base). Manual ref: Ch33 section 33.2.11 Note This is the ONLY RTC register actively used in STAR project
<code>k_rtc_rcr1_addr</code>	RTC Control Register 1 (RCR1) address. Manual ref: Ch33 section 33.2.9, address 0x0008C422
<code>k_rtc_rcr2_addr</code>	RTC Control Register 2 (RCR2) address. Manual ref: Ch33 section 33.2.10, address 0x0008C424
<code>k_rtc_rcr4_addr</code>	RTC Control Register 4 (RCR4) address. Manual ref: Ch33 section 33.2.12, address 0x0008C428

Definition at line 96 of file `rx72n_rtc_regs.h`.

```

00096         : uintptr_t {
00097     /**
00098      * @brief RTC register base address (0x0008C400)
00099      * @details Manual ref: Ch33 section 33.2.1 (R64CNT at 0x0008C400)
00100      */
00101     k_rtc_base_addr = 0x0008C400U,
00102
00103     /**
00104      * @brief RTC Control Register 3 (RCR3) address
00105      * @details
00106      * Direct address for RCR3 at 0x0008C426 (offset 0x26 from base).
00107      * Manual ref: Ch33 section 33.2.11
00108      * @note This is the ONLY RTC register actively used in STAR project
00109      */
00110     k_rtc_rcr3_addr = 0x0008C426U,
00111
00112     /**
00113      * @brief RTC Control Register 1 (RCR1) address
00114      * @details Manual ref: Ch33 section 33.2.9, address 0x0008C422
00115      */
00116     k_rtc_rcr1_addr = 0x0008C422U,
00117
00118     /**
00119      * @brief RTC Control Register 2 (RCR2) address
00120      * @details Manual ref: Ch33 section 33.2.10, address 0x0008C424
00121      */
00122     k_rtc_rcr2_addr = 0x0008C424U,
00123
00124     /**
00125      * @brief RTC Control Register 4 (RCR4) address
00126      * @details Manual ref: Ch33 section 33.2.12, address 0x0008C428
00127      */
00128     k_rtc_rcr4_addr = 0x0008C428U,
00129 } rtc_addresses_t;

```

8.55.2.3 rtc_offsets_t

```
enum rtc_offsets_t : uint8_t
```

RTC register offsets from base address.

Offsets verified against RX72N Hardware Manual Ch33 register table. Used for static assertions to verify struct layout.

Note

Offsets verified 2026-01-29 against manual

Since

Version 1.0.0

Enumerator

k_rtc_offset_r64cnt	64-Hz Counter offset
k_rtc_offset_rseccnt	Second Counter offset (note: 0x02, not 0x01)
k_rtc_offset_rmincnt	Minute Counter offset
k_rtc_offset_rhrcnt	Hour Counter offset
k_rtc_offset_rwkcnt	Day-of-Week Counter offset
k_rtc_offset_rdaycnt	Day Counter offset
k_rtc_offset_rmoncnt	Month Counter offset
k_rtc_offset_ryrcnt	Year Counter offset (16-bit)
k_rtc_offset_rsecar	Second Alarm offset
k_rtc_offset_rminar	Minute Alarm offset
k_rtc_offset_rhrar	Hour Alarm offset
k_rtc_offset_rwkar	Day-of-Week Alarm offset
k_rtc_offset_rdayar	Day Alarm offset
k_rtc_offset_rmonar	Month Alarm offset
k_rtc_offset_ryrar	Year Alarm offset (16-bit)
k_rtc_offset_ryraren	Year Alarm Enable offset
k_rtc_offset_rcr1	RTC Control Register 1 offset
k_rtc_offset_rcr2	RTC Control Register 2 offset
k_rtc_offset_rcr3	RTC Control Register 3 offset KEY
k_rtc_offset_rcr4	RTC Control Register 4 offset
k_rtc_offset_rfrh	Frequency Register H offset
k_rtc_offset_rfrl	Frequency Register L offset
k_rtc_offset_radj	Time Error Adjustment offset

Definition at line 142 of file rx72n_rtc_regs.h.

```
00142         : uint8_t {
00143   k_rtc_offset_r64cnt = 0x00U, /**< 64-Hz Counter offset */
00144   k_rtc_offset_rseccnt = 0x02U, /**< Second Counter offset (note: 0x02, not 0x01) */
00145   k_rtc_offset_rmincnt = 0x04U, /**< Minute Counter offset */
```

```

00146 k_rtc_offset_rhrcnt = 0x06U, /**< Hour Counter offset */
00147 k_rtc_offset_rwkcnt = 0x08U, /**< Day-of-Week Counter offset */
00148 k_rtc_offset_rdaycnt = 0x0AU, /**< Day Counter offset */
00149 k_rtc_offset_rmoncnt = 0x0CU, /**< Month Counter offset */
00150 k_rtc_offset_ryrcnt = 0x0EU, /**< Year Counter offset (16-bit) */
00151 k_rtc_offset_rsecar = 0x10U, /**< Second Alarm offset */
00152 k_rtc_offset_rminar = 0x12U, /**< Minute Alarm offset */
00153 k_rtc_offset_rhrar = 0x14U, /**< Hour Alarm offset */
00154 k_rtc_offset_rwkar = 0x16U, /**< Day-of-Week Alarm offset */
00155 k_rtc_offset_rdayar = 0x18U, /**< Day Alarm offset */
00156 k_rtc_offset_rmonar = 0x1AU, /**< Month Alarm offset */
00157 k_rtc_offset_ryrar = 0x1CU, /**< Year Alarm offset (16-bit) */
00158 k_rtc_offset_ryraren = 0x1EU, /**< Year Alarm Enable offset */
00159 k_rtc_offset_rcr1 = 0x22U, /**< RTC Control Register 1 offset */
00160 k_rtc_offset_rcr2 = 0x24U, /**< RTC Control Register 2 offset */
00161 k_rtc_offset_rcr3 = 0x26U, /**< RTC Control Register 3 offset **KEY** */
00162 k_rtc_offset_rcr4 = 0x28U, /**< RTC Control Register 4 offset */
00163 k_rtc_offset_rfrh = 0x2AU, /**< Frequency Register H offset */
00164 k_rtc_offset_rfrl = 0x2CU, /**< Frequency Register L offset */
00165 k_rtc_offset_radj = 0x2EU, /**< Time Error Adjustment offset */
00166 } rtc_offsets_t;

```

8.55.3 Function Documentation

8.55.3.1 rtc_rcr1_reg()

```

volatile uint8_t * rtc_rcr1_reg (
    void ) [inline], [static]

```

Get pointer to RTC Control Register 1 (RCR1).

Returns a volatile pointer to RCR1 at address 0x0008C422. Not used in STAR project but provided for completeness.

Returns

Volatile pointer to RCR1 register (8-bit)

Since

Version 1.0.0

Definition at line 266 of file [rx72n_rtc_regs.h](#).

```

00267 {
00268     return (volatile uint8_t*)k_rtc_rcr1_addr;
00269 }

```

References [k_rtc_rcr1_addr](#).

8.55.3.2 rtc_rcr2_reg()

```

volatile uint8_t * rtc_rcr2_reg (
    void ) [inline], [static]

```

Get pointer to RTC Control Register 2 (RCR2).

Returns a volatile pointer to RCR2 at address 0x0008C424. Not used in STAR project but provided for completeness.

Returns

Volatile pointer to RCR2 register (8-bit)

Since

Version 1.0.0

Definition at line 281 of file [rx72n_rtc_regs.h](#).

```
00282 {
00283     return (volatile uint8_t*)k_rtc_rcr2_addr;
00284 }
```

References [k_rtc_rcr2_addr](#).

8.55.3.3 rtc_rcr3_reg()

```
volatile uint8_t * rtc_rcr3_reg (
    void ) [inline], [static]
```

Get pointer to RTC Control Register 3 (RCR3).

Returns a volatile pointer to RCR3 at address 0x0008C426. This is the only RTC register used in STAR project (for sub-clock disable).

Address Verification

- Manual: Ch33 section 33.2.11 specifies RCR3 at 0x0008C426
- This accessor uses the direct address, NOT struct offset
- Static assertion verifies address at compile time

Returns

Volatile pointer to RCR3 register (8-bit)

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

RTC module stop bit (MSTPCRC.MSTPC27) should be cleared for access

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example (Sub-Clock Disable):

```
// During system initialization
*rtc_rcr3_reg() = k_rcr3_rtcen_disable; // Disable sub-clock to RTC
```

See also

[k_rtc_rcr3_addr](#) Address constant

[rcr3_bits_t](#) RCR3 control bits

Since

Version 1.0.0

Definition at line 251 of file [rx72n_rtc_regs.h](#).

```
00252 {
00253     return (volatile uint8_t*)k_rtc_rcr3_addr;
00254 }
```

References [k_rtc_rcr3_addr](#).

8.55.3.4 rtc_rcr4_reg()

```
volatile uint8_t * rtc_rcr4_reg (
    void ) [inline], [static]
```

Get pointer to RTC Control Register 4 (RCR4).

Returns a volatile pointer to RCR4 at address 0x0008C428. Not used in STAR project but provided for completeness.

Returns

Volatile pointer to RCR4 register (8-bit)

Since

Version 1.0.0

Definition at line 296 of file [rx72n_rtc_regs.h](#).

```
00297 {
00298     return (volatile uint8_t*)k_rtc_rcr4_addr;
00299 }
```

References [k_rtc_rcr4_addr](#).

8.56 rx72n_rtc_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_rtc_regs.h
00003  * @brief RX72N Real-Time Clock (RTC) Register Definitions
00004  *
00005  * @details
00006  * Minimal RTC register definitions for sub-clock oscillator disable functionality.
00007  * The RTC module is not used in the STAR project. These definitions exist
00008  * solely to properly disable the sub-clock oscillator during system initialization.
00009  *
00010  * @par Why This File Exists (Design Rationale)
00011  * The RX72N sub-clock oscillator (32.768 kHz) consumes power even when not
00012  * actively used. To fully disable it, the RTC module's RCR3.RTCEN bit must be
00013  * cleared to 0. This file provides the minimum register definitions needed for
00014  * this operation without including the full RTC implementation.
00015  *
00016  * @par CRITICAL ADDRESS VERIFICATION (2026-01-29)
00017  * All register addresses verified against RX72N Hardware Manual Ch33.
00018  * Previous implementation had CRITICAL BUG - wrong struct offsets:
00019  * - Old code: RCR3 at offset 0x1A (0x0008C41A = RMONAR register!)
00020  * - Correct: RCR3 at offset 0x26 (0x0008C426 per manual)
00021  *
00022  * @par Key Register Addresses (from Ch33)
00023  * | Register | Address | Offset | Description |
00024  * |-----|-----|-----|-----|
00025  * | R64CNT | 0x0008C400 | 0x00 | 64-Hz Counter |
00026  * | RSECCNT | 0x0008C402 | 0x02 | Second Counter (BCD) |
00027  * | RMINCNT | 0x0008C404 | 0x04 | Minute Counter (BCD) |
00028  * | RHRCNT | 0x0008C406 | 0x06 | Hour Counter (BCD) |
00029  * | RWKCNT | 0x0008C408 | 0x08 | Day-of-Week Counter |
00030  * | RDAYCNT | 0x0008C40A | 0x0A | Day Counter (BCD) |
00031  * | RMONCNT | 0x0008C40C | 0x0C | Month Counter (BCD) |
00032  * | RYRCNT | 0x0008C40E | 0x0E | Year Counter (BCD, 16-bit) |
00033  * | RSECAR | 0x0008C410 | 0x10 | Second Alarm |
00034  * | ... | ... | ... | (alarm registers) |
00035  * | RYRAREN | 0x0008C41E | 0x1E | Year Alarm Enable |
00036  * | (gap) | 0x0008C420 | 0x20 | Reserved (2 bytes) |
00037  * | RCR1 | 0x0008C422 | 0x22 | RTC Control Register 1 |
00038  * | RCR2 | 0x0008C424 | 0x24 | RTC Control Register 2 |
00039  * | RCR3 | 0x0008C426 | 0x26 | RTC Control Register 3 |
00040  * | RCR4 | 0x0008C428 | 0x28 | RTC Control Register 4 |
00041  *
00042  * @par NASA Power of 10 Compliance
00043  * - Rule 1: [OK] No goto, setjmp, or recursion
00044  * - Rule 2: [OK] N/A (no loops)
00045  * - Rule 3: [OK] No dynamic memory allocation
00046  * - Rule 4: [OK] Single-statement accessor
00047  * - Rule 5: [OK] Static assertions verify addresses
00048  * - Rule 6: [OK] Minimal scope
00049  * - Rule 7: [OK] N/A
00050  * - Rule 8: [OK] All constants use C23 typed enums
00051  * - Rule 9: [OK] No function pointers
00052  * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00053  *
00054  * @par Manual References
00055  * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00056  * - Chapter 33: Realtime Clock (RTCd), pages 1616-1670
00057  * - Section 33.2.11: RTC Control Register 3 (RCR3), page 1637
00058  *
00059  * @note This is an intentionally minimal implementation
00060  * @warning Do not expand this file unless RTC functionality is actually needed
00061  *
00062  * @see rx72n_system_regs.h SOSCCR register for sub-clock oscillator control
00063  * @see rx_clock_power_init.c Clock initialization implementation
00064  *
00065  * @author Locked, Inc.
00066  * @date 2026-01-29
00067  * @copyright Copyright (c) 2026 Locked Inc.
00068  * SPDX-License-Identifier: MIT
00069  * @since Version 1.0.0
00070  */
00071
00072 #pragma once
00073
00074 #include <stdint.h>
00075
00076 #ifdef __cplusplus
00077 extern "C" {
00078 #endif
00079
00080 /*
=====
00081  * RTC Register Addresses

```

```

00082 *
=====
00083 */
00084
00085 /**
00086  * @enum rtc_addresses_t
00087  * @brief RTC module register addresses
00088  *
00089  * @details
00090  * Addresses for RTC registers verified against RX72N Hardware Manual Ch33.
00091  * Only addresses needed for sub-clock disable are defined.
00092  *
00093  * @note Addresses verified 2026-01-29 against manual section 33.2
00094  * @since Version 1.0.0
00095  */
00096 typedef enum : uintptr_t {
00097     /**
00098      * @brief RTC register base address (0x0008C400)
00099      * @details Manual ref: Ch33 section 33.2.1 (R64CNT at 0x0008C400)
00100      */
00101     k_rtc_base_addr = 0x0008C400U,
00102
00103     /**
00104      * @brief RTC Control Register 3 (RCR3) address
00105      * @details
00106      * Direct address for RCR3 at 0x0008C426 (offset 0x26 from base).
00107      * Manual ref: Ch33 section 33.2.11
00108      * @note This is the ONLY RTC register actively used in STAR project
00109      */
00110     k_rtc_rcr3_addr = 0x0008C426U,
00111
00112     /**
00113      * @brief RTC Control Register 1 (RCR1) address
00114      * @details Manual ref: Ch33 section 33.2.9, address 0x0008C422
00115      */
00116     k_rtc_rcr1_addr = 0x0008C422U,
00117
00118     /**
00119      * @brief RTC Control Register 2 (RCR2) address
00120      * @details Manual ref: Ch33 section 33.2.10, address 0x0008C424
00121      */
00122     k_rtc_rcr2_addr = 0x0008C424U,
00123
00124     /**
00125      * @brief RTC Control Register 4 (RCR4) address
00126      * @details Manual ref: Ch33 section 33.2.12, address 0x0008C428
00127      */
00128     k_rtc_rcr4_addr = 0x0008C428U,
00129 } rtc_addresses_t;
00130
00131 /**
00132  * @enum rtc_offsets_t
00133  * @brief RTC register offsets from base address
00134  *
00135  * @details
00136  * Offsets verified against RX72N Hardware Manual Ch33 register table.
00137  * Used for static assertions to verify struct layout.
00138  *
00139  * @note Offsets verified 2026-01-29 against manual
00140  * @since Version 1.0.0
00141  */
00142 typedef enum : uint8_t {
00143     k_rtc_offset_r64cnt = 0x00U, /**< 64-Hz Counter offset */
00144     k_rtc_offset_rsecnt = 0x02U, /**< Second Counter offset (note: 0x02, not 0x01) */
00145     k_rtc_offset_rmincnt = 0x04U, /**< Minute Counter offset */
00146     k_rtc_offset_rhrcnt = 0x06U, /**< Hour Counter offset */
00147     k_rtc_offset_rwkcnt = 0x08U, /**< Day-of-Week Counter offset */
00148     k_rtc_offset_rdaycnt = 0x0AU, /**< Day Counter offset */
00149     k_rtc_offset_rmoncnt = 0x0CU, /**< Month Counter offset */
00150     k_rtc_offset_ryrcnt = 0x0EU, /**< Year Counter offset (16-bit) */
00151     k_rtc_offset_rsecar = 0x10U, /**< Second Alarm offset */
00152     k_rtc_offset_rminar = 0x12U, /**< Minute Alarm offset */
00153     k_rtc_offset_rhrar = 0x14U, /**< Hour Alarm offset */
00154     k_rtc_offset_rwkar = 0x16U, /**< Day-of-Week Alarm offset */
00155     k_rtc_offset_rdayar = 0x18U, /**< Day Alarm offset */
00156     k_rtc_offset_rmonar = 0x1AU, /**< Month Alarm offset */
00157     k_rtc_offset_ryrar = 0x1CU, /**< Year Alarm offset (16-bit) */
00158     k_rtc_offset_ryraren = 0x1EU, /**< Year Alarm Enable offset */
00159     k_rtc_offset_rcr1 = 0x22U, /**< RTC Control Register 1 offset */
00160     k_rtc_offset_rcr2 = 0x24U, /**< RTC Control Register 2 offset */
00161     k_rtc_offset_rcr3 = 0x26U, /**< RTC Control Register 3 offset **KEY** */
00162     k_rtc_offset_rcr4 = 0x28U, /**< RTC Control Register 4 offset */
00163     k_rtc_offset_rfh = 0x2AU, /**< Frequency Register H offset */
00164     k_rtc_offset_rfl = 0x2CU, /**< Frequency Register L offset */
00165     k_rtc_offset_radj = 0x2EU, /**< Time Error Adjustment offset */
00166 } rtc_offsets_t;
00167

```

```

00168 /*
=====
00169 * RCR3 Register Bits
00170 *
=====
00171 */
00172
00173 /**
00174 * @enum rcr3_bits_t
00175 * @brief RTC Control Register 3 (RCR3) Bit Definitions
00176 *
00177 * @details
00178 * Controls sub-clock oscillator input to the RTC module. This is the
00179 * only RTC register actively used in the STAR project.
00180 *
00181 * @par Register Layout (8-bit @ 0x0008C426)
00182 * | Bit | Field | R/W | Description |
00183 * |-----|-----|-----|-----|
00184 * | 0 | RTCEN | R/W | Sub-Clock Oscillator Control for RTC |
00185 * | 7:1 | - | R | Reserved (read as 0) |
00186 *
00187 * @par RTCEN Bit Behavior
00188 * - **0 (k_rcr3_rtcent_disable)**: Sub-clock oscillator stopped for RTC.
00189 * - **1 (k_rcr3_rtcent_enable)**: Sub-clock oscillator running for RTC.
00190 *
00191 * @note Manual ref: Ch33 section 33.2.11 (page 1637)
00192 * @since Version 1.0.0
00193 */
00194 typedef enum : uint8_t {
00195 /**
00196 * @brief Disable sub-clock input to RTC (RTCEN=0)
00197 * @details Use this value to disable RTC and save power.
00198 * This is the STAR project default configuration.
00199 */
00200 k_rcr3_rtcent_disable = 0x00U,
00201
00202 /**
00203 * @brief Enable sub-clock input to RTC (RTCEN=1)
00204 * @details Use this value to enable RTC timekeeping.
00205 * Not used in STAR project.
00206 */
00207 k_rcr3_rtcent_enable = 0x01U,
00208
00209 /**
00210 * @brief RTCEN bit position (bit 0)
00211 */
00212 k_rcr3_rtcent_bit = 0x01U,
00213 } rcr3_bits_t;
00214
00215 /*
=====
00216 * Register Accessor Functions
00217 *
=====
00218 */
00219
00220 /**
00221 * @brief Get pointer to RTC Control Register 3 (RCR3)
00222 *
00223 * @details
00224 * Returns a volatile pointer to RCR3 at address 0x0008C426.
00225 * This is the only RTC register used in STAR project (for sub-clock disable).
00226 *
00227 * @par Address Verification
00228 * - Manual: Ch33 section 33.2.11 specifies RCR3 at 0x0008C426
00229 * - This accessor uses the direct address, NOT struct offset
00230 * - Static assertion verifies address at compile time
00231 *
00232 * @return Volatile pointer to RCR3 register (8-bit)
00233 * @retval Non-NULL Always returns valid pointer (hardware address)
00234 *
00235 * @pre RTC module stop bit (MSTPCRC.MSTPC27) should be cleared for access
00236 * @post Pointer is valid for the lifetime of program execution
00237 *
00238 * @note Thread Safety: Safe - returns constant hardware address
00239 * @note This function is always inlined for zero overhead
00240 *
00241 * @par Example (Sub-Clock Disable):
00242 * @code{.c}
00243 * // During system initialization
00244 * *rtc_rcr3_reg() = k_rcr3_rtcent_disable; // Disable sub-clock to RTC
00245 * @endcode
00246 *
00247 * @see k_rtc_rcr3_addr Address constant
00248 * @see rcr3_bits_t RCR3 control bits
00249 * @since Version 1.0.0
00250 */

```



```

00251 static inline volatile uint8_t* rtc_rcr3_reg(void)
00252 {
00253     return (volatile uint8_t*)k_rtc_rcr3_addr;
00254 }
00255
00256 /**
00257  * @brief Get pointer to RTC Control Register 1 (RCR1)
00258  *
00259  * @details
00260  * Returns a volatile pointer to RCR1 at address 0x0008C422.
00261  * Not used in STAR project but provided for completeness.
00262  *
00263  * @return Volatile pointer to RCR1 register (8-bit)
00264  * @since Version 1.0.0
00265  */
00266 static inline volatile uint8_t* rtc_rcr1_reg(void)
00267 {
00268     return (volatile uint8_t*)k_rtc_rcr1_addr;
00269 }
00270
00271 /**
00272  * @brief Get pointer to RTC Control Register 2 (RCR2)
00273  *
00274  * @details
00275  * Returns a volatile pointer to RCR2 at address 0x0008C424.
00276  * Not used in STAR project but provided for completeness.
00277  *
00278  * @return Volatile pointer to RCR2 register (8-bit)
00279  * @since Version 1.0.0
00280  */
00281 static inline volatile uint8_t* rtc_rcr2_reg(void)
00282 {
00283     return (volatile uint8_t*)k_rtc_rcr2_addr;
00284 }
00285
00286 /**
00287  * @brief Get pointer to RTC Control Register 4 (RCR4)
00288  *
00289  * @details
00290  * Returns a volatile pointer to RCR4 at address 0x0008C428.
00291  * Not used in STAR project but provided for completeness.
00292  *
00293  * @return Volatile pointer to RCR4 register (8-bit)
00294  * @since Version 1.0.0
00295  */
00296 static inline volatile uint8_t* rtc_rcr4_reg(void)
00297 {
00298     return (volatile uint8_t*)k_rtc_rcr4_addr;
00299 }
00300
00301 /*
00302  * Static Assertions - Address Verification
00303  */
00304 /*
00305  *
00306  */
00307 /**
00308  * @name Address Verification
00309  * @brief Compile-time verification of register addresses
00310  * @{
00311  */
00312 /* Verify RCR3 address is correct (critical for sub-clock disable) */
00313 static_assert(k_rtc_rcr3_addr == 0x0008C426U, "RCR3 address must be 0x0008C426 per Ch33 manual");
00314
00315 /* Verify base address */
00316 static_assert(k_rtc_base_addr == 0x0008C400U,
00317     "RTC base address must be 0x0008C400 per Ch33 manual");
00318
00319 /* Verify RCR3 offset from base */
00320 static_assert(k_rtc_rcr3_addr - k_rtc_base_addr == k_rtc_offset_rcr3,
00321     "RCR3 offset must be 0x26 from base");
00322
00323 /* Verify other control register addresses */
00324 static_assert(k_rtc_rcr1_addr == 0x0008C422U, "RCR1 address must be 0x0008C422 per Ch33 manual");
00325
00326 static_assert(k_rtc_rcr2_addr == 0x0008C424U, "RCR2 address must be 0x0008C424 per Ch33 manual");
00327
00328 static_assert(k_rtc_rcr4_addr == 0x0008C428U, "RCR4 address must be 0x0008C428 per Ch33 manual");
00329
00330 /* Verify control register spacing (each 2 bytes apart) */
00331 static_assert(k_rtc_rcr2_addr - k_rtc_rcr1_addr == 2U, "RCR1 to RCR2 spacing must be 2 bytes");
00332
00333 static_assert(k_rtc_rcr3_addr - k_rtc_rcr2_addr == 2U, "RCR2 to RCR3 spacing must be 2 bytes");
00334
00335 static_assert(k_rtc_rcr4_addr - k_rtc_rcr3_addr == 2U, "RCR3 to RCR4 spacing must be 2 bytes");

```

```

00336
00337 /** @} */
00338
00339 #ifdef __cplusplus
00340 }
00341 #endif

```

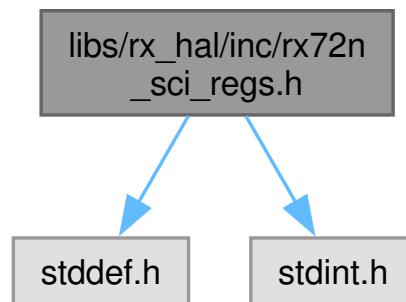
8.57 libs/rx_hal/inc/rx72n_sci_regs.h File Reference

RX72N Serial Communication Interface (SCI) register definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_sci_regs.h:



This graph shows which files directly or indirectly include this file:



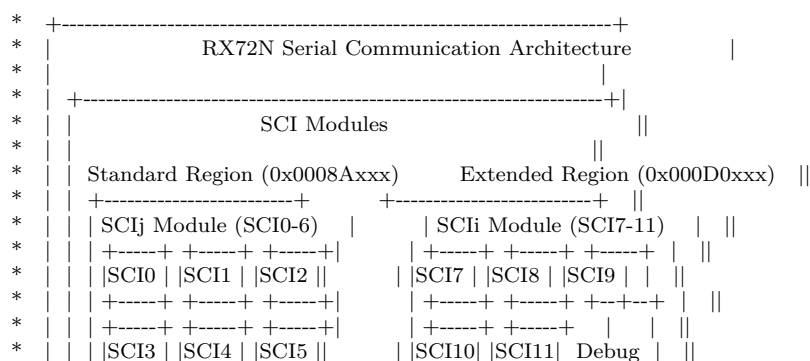
Data Structures

- struct [rx_sci_regs_t](#)
SCI register map covering UART and SPI mode registers (14 bytes).

Enumerations

- enum [rx_sci_addresses_t](#) : [uintptr_t](#) {
[k_sci0_base_addr](#) = 0x0008A000 , [k_sci1_base_addr](#) = 0x0008A020 , [k_sci2_base_addr](#) = 0x0008A040 , [k_sci3_base_addr](#) = 0x0008A060 ,
[k_sci4_base_addr](#) = 0x0008A080 , [k_sci5_base_addr](#) = 0x0008A0A0 , [k_sci6_base_addr](#) = 0x0008A0C0 , [k_sci7_base_addr](#) = 0x000D00E0 ,
[k_sci8_base_addr](#) = 0x000D0000 , [k_sci9_base_addr](#) = 0x000D0020 , [k_sci10_base_addr](#) = 0x000D0040 , [k_sci11_base_addr](#) = 0x000D0060 ,
[k_sci12_base_addr](#) = 0x0008B300 }
 SCI channel base addresses (verified against RX72N Hardware Manual).
- enum [rx_sci_ssr_flags_t](#) : [uint8_t](#) {
[k_sci_ssr_tdre_flag](#) = 0x80U , [k_sci_ssr_rdrf_flag](#) = 0x40U , [k_sci_ssr_orer_flag](#) = 0x20U ,
[k_sci_ssr_fer_flag](#) = 0x10U ,
[k_sci_ssr_per_flag](#) = 0x08U , [k_sci_ssr_tend_flag](#) = 0x04U , [k_sci_ssr_error_mask](#) = 0x38U
 }
 SCI Serial Status Register (SSR) flag masks.
- enum [sci_channel_limits_t](#) : [uint8_t](#) { [k_sci_channel_max](#) = 13 }
 SCI channel count.

- static volatile `rx_sci_regs_t * sci0` (void)
Get pointer to SCI0 registers.
- static volatile `rx_sci_regs_t * sci1` (void)
Get pointer to SCI1 registers.
- static volatile `rx_sci_regs_t * sci2` (void)
Get pointer to SCI2 registers.
- static volatile `rx_sci_regs_t * sci3` (void)
Get pointer to SCI3 registers.
- static volatile `rx_sci_regs_t * sci4` (void)
Get pointer to SCI4 registers.
- static volatile `rx_sci_regs_t * sci5` (void)
Get pointer to SCI5 registers.
- static volatile `rx_sci_regs_t * sci6` (void)
Get pointer to SCI6 registers.
- static volatile `rx_sci_regs_t * sci7` (void)
Get pointer to SCI7 registers.
- static volatile `rx_sci_regs_t * sci8` (void)
Get pointer to SCI8 registers.
- static volatile `rx_sci_regs_t * sci9` (void)
Get pointer to SCI9 registers (Debug UART).
- static volatile `rx_sci_regs_t * sci10` (void)
Get pointer to SCI10 registers.
- static volatile `rx_sci_regs_t * sci11` (void)
Get pointer to SCI11 registers.
- static volatile `rx_sci_regs_t * sci12` (void)
Get pointer to SCI12 registers.
- static volatile `rx_sci_regs_t * sci_get_channel` (uint8_t channel)
Get SCI register base for a channel.



```

* | | +---+ +---+ +---+ | | +---+ +---+ UART | |
* | | +---+ | | +-----+ | |
* | | |SCI6 | | | | | |
* | | +---+ | | +-----+ | |
* | | +-----+ | | |SCIh Module (SCI12) | |
* | | | @ 0x0008B300 | | |
* | | | +---+ | | |
* | | | |SCI12 | | |
* | | | +---+ | | |
* | | +-----+ | |
* | +-----+ |
* | |
* | STAR Robot SCI Usage: |
* | +-----+ |
* | | Debug UART (SCI9): | |
* | | +---+ +---+ +---+ +---+ | |
* | | |RX72N |--->| CY7C65213 |--->| USB |--->| Host | | |
* | | |SCI9 | | USB-UART | | Cable | | PC | | |
* | | |TXD9/RXD9|<---| Bridge |<---| |<---| Console | | |
* | | +---+ +---+ +---+ +---+ | |
* | | | | | | | |
* | | Settings: 115200 baud, 8N1, no flow control | |
* | | Purpose: Logging, debugging, firmware updates | |
* | +-----+ |
* |

```

SCI Channel Summary

Channel	Base Address	Region	STAR Usage
SCI0	0x0008A000	Standard	Available
SCI1	0x0008A020	Standard	Available
SCI2	0x0008A040	Standard	Available
SCI3	0x0008A060	Standard	Available
SCI4	0x0008A080	Standard	Available
SCI5	0x0008A0A0	Standard	Available
SCI6	0x0008A0C0	Standard	Available
SCI7	0x000D00E0	Extended	Available
SCI8	0x000D0000	Extended	Available
SCI9	0x000D0020	Extended	Debug UART
SCI10	0x000D0040	Extended	Available
SCI11	0x000D0060	Extended	Available
SCI12	0x0008B300	Standard	Available

Baud Rate Calculation

For asynchronous mode with n=0 (CKS=00, PCLK/1):

$$\text{BRR} = \frac{f_{\text{PCLK}}}{32 \times B} - 1$$

Where f_PCLK depends on the channel module (manual section 5):

- SCIj (SCI0-SCI6) and SCIh (SCI12): f_PCLK = f_PCLKB

- SCi (SCI7-SCI11, extended region): $f_PCLK = f_PCLKA$

Example: 115200 baud with $f_PCLKB = 60\text{ MHz}$, $n=0$:

$$BRR = \frac{60\text{ MHz}}{32 \times 115200} - 1 = 16.28 - 1 = 15.28 \approx 15$$

UART Configuration Example

```
// Initialize SCI9 for 115200 baud, 8N1
volatile rx_sci_regs_t* uart = sci9();

// Disable TX/RX during configuration
uart->scr = 0x00;

// Set mode: async, 8-bit, no parity, 1 stop bit, PCLKA/1
uart->smr = 0x00;

// Enable noise filter and bit rate modulation
uart->semr = 0x04; // NFEN = 1

// Set baud rate: 115200 at 120 MHz PCLKA
uart->brr = 32;

// Clear errors and enable TX/RX
uart->ssr = 0x84; // Clear error flags
uart->scr = 0x30; // TE=1, RE=1 (enable TX and RX)
```

Hardware Reference

- RX72N Group User's Manual: Hardware, Chapter 41 (SCI)
- Section 41.2: Register Descriptions
- Section 41.3: UART Mode Operation
- Section 41.6: Baud Rate Generator

Verification Status

[PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified against RX72N Manual Ch41 Section 41.2

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 3: [OK] Static allocation only (all definitions compile-time)
- Rule 5: [OK] Static assertions verify register layout at compile time
- Rule 8: [OK] C23 typed enums eliminate preprocessor constants
- Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: SCI register definitions only, no driver logic
- Open/Closed: Extend via new enums; don't modify existing values
- Interface Segregation: Basic UART registers; advanced features separate
- Dependency Inversion: Higher-level UART drivers depend on these abstractions

Related Modules

- [uart.c](#): UART HAL driver implementation
- [rx72n_port_regs.h](#): GPIO for TXD/RXD pins
- [rx72n_mpc.h](#): Pin function selection for SCI pins
- [rx72n_icu_regs.h](#): SCI interrupt vectors

See also

RX72N Hardware Manual Chapter 41 for complete SCI specification
DOXYGEN_ROADMAP.md for documentation standards

Author

Locked, Inc. Contributors

Date

2026-01-28

Version

1.0.0

Copyright

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Definition in file [rx72n_sci_regs.h](#).

8.58 rx72n_sci_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_sci_regs.h
00003  * @brief RX72N Serial Communication Interface (SCI) register definitions
00004  *
00005  * @details
00006  * This file provides complete hardware register definitions for the RX72N's
00007  * Serial Communication Interface (SCI) peripheral. The SCI module supports
00008  * multiple serial communication protocols including UART (asynchronous),
00009  * clock-synchronous serial, smart card interface, and I2C modes.
00010  *
00011  * @par System Architecture - STAR Robot Serial Communication
00012  * @verbatim
00013  * +-----+
00014  * |               RX72N Serial Communication Architecture               |
00015  * |               |
00016  * | +-----+ |
00017  * | |               SCI Modules               | |
00018  * | |               | |
00019  * | | Standard Region (0x0008Axxx)   Extended Region (0x000D0xxx)   | |
00020  * | | +-----+ +-----+ | |
00021  * | | |SCIj Module (SCI0-6) | |SCIi Module (SCI7-11) | | |
00022  * | | +-----+ +-----+ | | +-----+ +-----+ | |
00023  * | | |SCI0 |SCI1 |SCI2 || |SCI7 |SCI8 |SCI9 | | |
00024  * | | +-----+ +-----+ | | +-----+ +-----+ | |
00025  * | | +-----+ +-----+ | | +-----+ +-----+ | |
00026  * | | |SCI3 |SCI4 |SCI5 || |SCI10|SCI11| Debug | |
```

```

00027 * | | | +---+ +---+ +---+ | | +---+ +---+ UART | ||
00028 * | | | +---+ | | +-----+ | ||
00029 * | | | |SCI6 | | | | | ||
00030 * | | | +---+ | | +-----+ | ||
00031 * | | | +-----+ | | | SCIh Module (SCI12) | ||
00032 * | | | | @ 0x0008B300 | | | ||
00033 * | | | | +---+ | | | ||
00034 * | | | | |SCI12 | | | ||
00035 * | | | | +---+ | | | ||
00036 * | | | +-----+ | | | ||
00037 * | | | +-----+ | | | ||
00038 * | | | | | ||
00039 * | | | STAR Robot SCI Usage: | | |
00040 * | | | +-----+ | | |
00041 * | | | | Debug UART (SCI9): | | |
00042 * | | | | +---+ +---+ +---+ +---+ | | |
00043 * | | | | |RX72N |--->| CY7C65213 |--->| USB |--->| Host | | |
00044 * | | | | |SCI9 | |USB-UART | |Cable | |PC | | |
00045 * | | | | |TXD9/RXD9|<---| Bridge |<---| Console | | |
00046 * | | | | +---+ +---+ +---+ +---+ | | |
00047 * | | | | | | | | | | | | | |
00048 * | | | | Settings: 115200 baud, 8N1, no flow control | | |
00049 * | | | | Purpose: Logging, debugging, firmware updates | | |
00050 * | | | +-----+ | | |
00051 * | | | +-----+ | | |
00052 * | | | @endverbatim
00053 *
00054 * @par SCI Channel Summary
00055 * | Channel | Base Address | Region | STAR Usage |
00056 * |-----|-----|-----|-----|
00057 * | SCI0 | 0x0008A000 | Standard | Available |
00058 * | SCI1 | 0x0008A020 | Standard | Available |
00059 * | SCI2 | 0x0008A040 | Standard | Available |
00060 * | SCI3 | 0x0008A060 | Standard | Available |
00061 * | SCI4 | 0x0008A080 | Standard | Available |
00062 * | SCI5 | 0x0008A0A0 | Standard | Available |
00063 * | SCI6 | 0x0008A0C0 | Standard | Available |
00064 * | SCI7 | 0x000D00E0 | Extended | Available |
00065 * | SCI8 | 0x000D0000 | Extended | Available |
00066 * | SCI9 | 0x000D0020 | Extended | **Debug UART** |
00067 * | SCI10 | 0x000D0040 | Extended | Available |
00068 * | SCI11 | 0x000D0060 | Extended | Available |
00069 * | SCI12 | 0x0008B300 | Standard | Available |
00070 *
00071 * @par Baud Rate Calculation
00072 * For asynchronous mode with n=0 (CKS=00, PCLK/1):
00073 * @f
00074 * \text{BRR} = \frac{f_{PCLK}}{32 \times B} - 1
00075 * @f
00076 *
00077 * Where f_PCLK depends on the channel module (manual section 5):
00078 * - SCIj (SCI0-SCI6) and SCIh (SCI12): f_PCLK = f_PCLKB
00079 * - SCIi (SCI7-SCI11, extended region): f_PCLK = f_PCLKA
00080 *
00081 * Example: 115200 baud with f_PCLKB = 60 MHz, n=0:
00082 * @f
00083 * \text{BRR} = \frac{60 \text{ MHz}}{32 \times 115200} - 1 = 16.28 - 1 = 15.28 \approx 15
00084 * @f
00085 *
00086 * @par UART Configuration Example
00087 * @code
00088 * // Initialize SCI9 for 115200 baud, 8N1
00089 * volatile rx_sci_regs_t* uart = sci9();
00090 *
00091 * // Disable TX/RX during configuration
00092 * uart->scr = 0x00;
00093 *
00094 * // Set mode: async, 8-bit, no parity, 1 stop bit, PCLKA/1
00095 * uart->smr = 0x00;
00096 *
00097 * // Enable noise filter and bit rate modulation
00098 * uart->semr = 0x04; // NFEN = 1
00099 *
00100 * // Set baud rate: 115200 at 120 MHz PCLKA
00101 * uart->brr = 32;
00102 *
00103 * // Clear errors and enable TX/RX
00104 * uart->ssr = 0x84; // Clear error flags
00105 * uart->scr = 0x30; // TE=1, RE=1 (enable TX and RX)
00106 * @endcode
00107 *
00108 * @par Hardware Reference
00109 * - RX72N Group User's Manual: Hardware, Chapter 41 (SCI)
00110 * - Section 41.2: Register Descriptions
00111 * - Section 41.3: UART Mode Operation
00112 * - Section 41.6: Baud Rate Generator
00113 *

```

```

00114 * @par Verification Status
00115 * [PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified
00116 * against RX72N Manual Ch41 Section 41.2
00117 *
00118 * @par NASA Power of 10 Compliance
00119 * - Rule 1: [OK] No goto, setjmp, or recursion
00120 * - Rule 3: [OK] Static allocation only (all definitions compile-time)
00121 * - Rule 5: [OK] Static assertions verify register layout at compile time
00122 * - Rule 8: [OK] C23 typed enums eliminate preprocessor constants
00123 * - Rule 10: [OK] Header compiles cleanly with -Wall -Wextra -Werror
00124 *
00125 * @par SOLID Principles
00126 * - Single Responsibility: SCI register definitions only, no driver logic
00127 * - Open/Closed: Extend via new enums; don't modify existing values
00128 * - Interface Segregation: Basic UART registers; advanced features separate
00129 * - Dependency Inversion: Higher-level UART drivers depend on these abstractions
00130 *
00131 * @par Related Modules
00132 * - [uart.c](uart_8c.html): UART HAL driver implementation
00133 * - [rx72n_port_regs.h](rx72n__port__regs_8h.html): GPIO for TXD/RXD pins
00134 * - [rx72n_mpc.h](rx72n__mpc_8h.html): Pin function selection for SCI pins
00135 * - [rx72n_icu_regs.h](rx72n__icu__regs_8h.html): SCI interrupt vectors
00136 *
00137 * @see RX72N Hardware Manual Chapter 41 for complete SCI specification
00138 * @see DOXYGEN_ROADMAP.md for documentation standards
00139 *
00140 * @author Locked, Inc. Contributors
00141 * @date 2026-01-28
00142 * @version 1.0.0
00143 * @copyright Copyright (c) 2026 Locked Inc.
00144 * SPDX-License-Identifier: MIT
00145 *
00146 * @defgroup sci_regs SCI Register Definitions
00147 * @{
00148 */
00149
00150 #pragma once
00151
00152 #include <stddef.h>
00153 #include <stdint.h>
00154
00155 #ifdef __cplusplus
00156 extern "C" {
00157 #endif
00158
00159 /*
=====
00160 * Serial Communication Interface (SCI) - Base Addresses
00161 *
=====
00162 */
00163
00164 /**
00165 * @enum rx_sci_addresses_t
00166 * @brief SCI channel base addresses (verified against RX72N Hardware Manual)
00167 *
00168 * @details
00169 * Base addresses for all 13 SCI channels. The RX72N organizes SCI channels
00170 * into three modules in different memory regions:
00171 *
00172 * @par Memory Map Overview
00173 * @verbatim
00174 *   Address Range      Module  Channels  Notes
00175 *   -----
00176 *   0x0008A000-0x0008A0DF SCIj  SCI0-6   Standard region, 0x20 spacing
00177 *   0x000D0000-0x000D00FF SCIl  SCI7-11  Extended region, varying spacing
00178 *   0x0008B300-0x0008B31F SCIlh SCI12    Standard region, separate
00179 * @endverbatim
00180 *
00181 * @par Channel Spacing
00182 * - SCIj (SCI0-6): 0x20 bytes between channels
00183 * - SCIl (SCI7-11): Non-uniform spacing (see individual addresses)
00184 * - SCIlh (SCI12): Standalone at 0x0008B300
00185 *
00186 * @par STAR Robot Usage
00187 * SCI9 is designated as the debug UART, connected to the CY7C65213 USB-UART
00188 * bridge IC for host PC communication at 115200 baud.
00189 *
00190 * @see RX72N Hardware Manual Section 41.2 (Register Addresses)
00191 */
00192 typedef enum : uintptr_t {
00193     /** @brief SCI0 base address - standard region (SCIj module) */
00194     k_sci0_base_addr = 0x0008A000,
00195
00196     /** @brief SCI1 base address - standard region (SCIj module) */
00197     k_sci1_base_addr = 0x0008A020,
00198

```



```

00199 /** @brief SCI2 base address - standard region (SCIj module) */
00200 k_sci2_base_addr = 0x0008A040,
00201
00202 /** @brief SCI3 base address - standard region (SCIj module) */
00203 k_sci3_base_addr = 0x0008A060,
00204
00205 /** @brief SCI4 base address - standard region (SCIj module) */
00206 k_sci4_base_addr = 0x0008A080,
00207
00208 /** @brief SCI5 base address - standard region (SCIj module) */
00209 k_sci5_base_addr = 0x0008A0A0,
00210
00211 /** @brief SCI6 base address - standard region (SCIj module) */
00212 k_sci6_base_addr = 0x0008A0C0,
00213
00214 /** @brief SCI7 base address - extended region (SCIi module) */
00215 k_sci7_base_addr = 0x000D00E0,
00216
00217 /** @brief SCI8 base address - extended region (SCIi module) */
00218 k_sci8_base_addr = 0x000D0000,
00219
00220 /**
00221  * @brief SCI9 base address - extended region (SCIi module)
00222  * @note **STAR Debug UART** - Connected to CY7C65213 USB-UART bridge
00223  */
00224 k_sci9_base_addr = 0x000D0020,
00225
00226 /** @brief SCI10 base address - extended region (SCIi module) */
00227 k_sci10_base_addr = 0x000D0040,
00228
00229 /** @brief SCI11 base address - extended region (SCIi module) */
00230 k_sci11_base_addr = 0x000D0060,
00231
00232 /** @brief SCI12 base address - standard region (SCIh module) */
00233 k_sci12_base_addr = 0x0008B300,
00234 } rx_sci_addresses_t;
00235
00236 /**
00237  * @struct rx_sci_regs_t
00238  * @brief SCI register map covering UART and SPI mode registers (14 bytes)
00239  *
00240  * @details
00241  * Memory-mapped register structure for the RX72N SCI peripheral. Covers
00242  * registers from SMR (0x00) through SPMR (0x0D) to support both asynchronous
00243  * UART and clock-synchronous SPI modes.
00244  *
00245  * @par Register Memory Layout (14 bytes total)
00246  * @verbatim
00247  *   Offset  Size  Register  Description
00248  *   -----
00249  *   0x00    1    SMR       Serial Mode (data format, clock source)
00250  *   0x01    1    BRR       Bit Rate (baud rate divisor)
00251  *   0x02    1    SCR       Serial Control (TX/RX enable, interrupts)
00252  *   0x03    1    TDR       Transmit Data (write-only data buffer)
00253  *   0x04    1    SSR       Serial Status (flags, error conditions)
00254  *   0x05    1    RDR       Receive Data (read-only data buffer)
00255  *   0x06    1    SCMR      Smart Card Mode (SDIR for bit order)
00256  *   0x07    1    SEMR      Serial Extended Mode (noise filter, modulation)
00257  *   0x08    1    SNFR      Noise Filter Setting
00258  *   0x09    1    SIMR1     I2C Mode Register 1
00259  *   0x0A    1    SIMR2     I2C Mode Register 2
00260  *   0x0B    1    SIMR3     I2C Mode Register 3
00261  *   0x0C    1    SISR      I2C Status Register
00262  *   0x0D    1    SPMR      SPI Mode Register (CKPH, CKPOL)
00263  * @endverbatim
00264  *
00265  * @par UART Transmission Sequence
00266  * @verbatim
00267  *   1. Wait for SSR.TDRE = 1 (transmit data register empty)
00268  *   2. Write data byte to TDR
00269  *   3. SSR.TDRE automatically clears to 0
00270  *   4. Data shifts out serially via TXD pin
00271  *   5. SSR.TEND = 1 when transmission complete
00272  * @endverbatim
00273  *
00274  * @par UART Reception Sequence
00275  * @verbatim
00276  *   1. Data arrives via RXD pin
00277  *   2. SSR.RDRF = 1 when byte received
00278  *   3. Read data byte from RDR
00279  *   4. SSR.RDRF automatically clears to 0
00280  *   5. Check SSR.ORER/FER/PER for errors
00281  * @endverbatim
00282  *
00283  * @par Transmit Character Example
00284  * @code
00285  * void uart_putc(volatile rx_sci_regs_t* uart, char c)

```

```

00286 * {
00287 * // Wait for transmit buffer empty
00288 * while ((uart->ssr & 0x80) == 0) {
00289 * // SSR.TDRE bit 7
00290 * }
00291 * // Write character to transmit
00292 * uart->tdr = (uint8_t)c;
00293 * }
00294 * @endcode
00295 *
00296 * @par Receive Character Example
00297 * @code
00298 * char uart_getc(volatile rx_sci_regs_t* uart)
00299 * {
00300 * // Wait for receive data ready
00301 * while ((uart->ssr & 0x40) == 0) {
00302 * // SSR.RDRF bit 6
00303 * }
00304 * // Read received character
00305 * return (char)uart->rdr;
00306 * }
00307 * @endcode
00308 *
00309 * @invariant Structure size must be exactly 14 bytes
00310 * @invariant All registers are 8-bit (byte-accessible)
00311 *
00312 * @note Offsets 0x00-0x07 are unchanged from the original 8-byte structure.
00313 * Existing UART code works unmodified. Offsets 0x08-0x0D add I2C mode
00314 * and SPI mode register access needed for clock-synchronous SPI.
00315 *
00316 * @see rx_sci_addresses_t for channel base addresses
00317 * @see RX72N Hardware Manual Section 41.2 (Register Descriptions)
00318 */
00319 typedef struct {
00320 /**
00321 * @brief Serial Mode Register (SMR) - data format configuration
00322 * @details
00323 * - Bits 7: CM - Clock mode (0=async, 1=sync)
00324 * - Bits 6: CHR - Character length (0=8-bit, 1=7-bit)
00325 * - Bits 5: PE - Parity enable (0=disable, 1=enable)
00326 * - Bits 4: PM - Parity mode (0=even, 1=odd)
00327 * - Bits 3: STOP - Stop bits (0=1 bit, 1=2 bits)
00328 * - Bits 2: MP - Multi-processor mode
00329 * - Bits 1-0: CKS - Clock select (00=PCLK/1, 01=PCLK/4, 10=PCLK/16, 11=PCLK/64)
00330 */
00331 volatile uint8_t smr;
00332
00333 /**
00334 * @brief Bit Rate Register (BRR) - baud rate divisor
00335 * @details
00336 * Async mode: Baud rate = PCLK / (64 * 2^(2n-1) * (BRR + 1))
00337 * Sync mode: Bit rate = PCLK / (4 * (BRR + 1)) for CKS=0
00338 * PCLK = PCLKB (60 MHz) for SCI0-6/12; PCLKA (120 MHz) for SCI7-11.
00339 *
00340 * Common async values at 60 MHz PCLKB (SCI0-6/12), CKS=0:
00341 * - 9600 baud: BRR = 194
00342 * - 38400 baud: BRR = 47
00343 * - 115200 baud: BRR = 15
00344 *
00345 * Common sync values at 60 MHz PCLKB (SCI0-6/12), CKS=0:
00346 * - BRR=1: 7.5 MHz SPI clock
00347 * - BRR=3: 3.75 MHz SPI clock
00348 */
00349 volatile uint8_t brr;
00350
00351 /**
00352 * @brief Serial Control Register (SCR) - TX/RX enable and interrupts
00353 * @details
00354 * - Bit 7: TIE - Transmit interrupt enable
00355 * - Bit 6: RIE - Receive interrupt enable
00356 * - Bit 5: TE - Transmit enable
00357 * - Bit 4: RE - Receive enable
00358 * - Bit 3: MPIE - Multi-processor interrupt enable
00359 * - Bit 2: TEIE - Transmit end interrupt enable
00360 * - Bits 1-0: CKE - Clock enable (external clock source)
00361 */
00362 volatile uint8_t scr;
00363
00364 /**
00365 * @brief Transmit Data Register (TDR) - write-only data buffer
00366 * @details Write data here to transmit. Wait for SSR.TDRE = 1 first.
00367 */
00368 volatile uint8_t tdr;
00369
00370 /**
00371 * @brief Serial Status Register (SSR) - status flags and errors
00372 * @details

```

```

00373 * - Bit 7: TDRE - Transmit data register empty (1=ready to write)
00374 * - Bit 6: RDRF - Receive data register full (1=data available)
00375 * - Bit 5: ORER - Overrun error (1=error occurred)
00376 * - Bit 4: FER - Framing error (1=error occurred)
00377 * - Bit 3: PER - Parity error (1=error occurred)
00378 * - Bit 2: TEND - Transmit end (1=transmission complete)
00379 * - Bit 1: MPB - Multi-processor bit
00380 * - Bit 0: MPBT - Multi-processor bit transfer
00381 *
00382 * @note Write 0 to error bits to clear them (do not write 1).
00383 */
00384 volatile uint8_t ssr;
00385
00386 /**
00387 * @brief Receive Data Register (RDR) - read-only data buffer
00388 * @details Read received data here. Wait for SSR.RDRF = 1 first.
00389 */
00390 volatile uint8_t rdr;
00391
00392 /**
00393 * @brief Smart Card Mode Register (SCMR) - smart card and bit order settings
00394 * @details
00395 * - Bit 7: BCP2 - Base clock pulse 2 (smart card clock cycles per ETU)
00396 * - Bits 6-5: Reserved (write 1)
00397 * - Bit 4: CHR1 - Character length 1 (0=9/8 bit, 1=8/7 bit; use with SMR.CHR)
00398 * - Bit 3: SDIR - Data transfer direction (0=LSB first, 1=MSB first)
00399 * - Bit 2: SINV - Smart card data inversion (0=no inversion, 1=inverted)
00400 * - Bit 1: Reserved (write 1)
00401 * - Bit 0: SMIF - Smart card interface mode (0=disabled, 1=enabled)
00402 *
00403 * @note For SPI mode, set SDIR=1 for MSB-first transfers.
00404 */
00405 volatile uint8_t scmr;
00406
00407 /**
00408 * @brief Serial Extended Mode Register (SEMR) - advanced settings
00409 * @details
00410 * - Bit 7: RXDESEL - RXD extended input control
00411 * - Bit 6: BGDM - Baud rate generator double-speed mode
00412 * - Bit 5: NFEN - Digital noise filter enable
00413 * - Bit 4: ABCS - Asynchronous mode base clock select
00414 * - Bit 3: ABCSE - Asynchronous mode extended base clock select
00415 * - Bit 2: BRME - Bit rate modulation enable
00416 * - Bit 1: Reserved
00417 * - Bit 0: ACS0 - Asynchronous mode clock source select 0
00418 */
00419 volatile uint8_t semr;
00420
00421 /**
00422 * @brief Noise Filter Setting Register (SNFR)
00423 * @details
00424 * - Bits 2-0: NFCS - Noise filter clock select
00425 */
00426 volatile uint8_t snfr;
00427
00428 /**
00429 * @brief I2C Mode Register 1 (SIMR1)
00430 * @details Used in I2C mode. Not used for UART or SPI.
00431 */
00432 volatile uint8_t simr1;
00433
00434 /**
00435 * @brief I2C Mode Register 2 (SIMR2)
00436 * @details Used in I2C mode. Not used for UART or SPI.
00437 */
00438 volatile uint8_t simr2;
00439
00440 /**
00441 * @brief I2C Mode Register 3 (SIMR3)
00442 * @details Used in I2C mode. Not used for UART or SPI.
00443 */
00444 volatile uint8_t simr3;
00445
00446 /**
00447 * @brief I2C Status Register (SISR)
00448 * @details Used in I2C mode. Not used for UART or SPI.
00449 */
00450 volatile uint8_t sISR;
00451
00452 /**
00453 * @brief SPI Mode Register (SPMR) - SPI clock polarity and phase
00454 * @details
00455 * - Bit 7: CKPH - Clock phase (inverted from standard CPHA in some Renesas docs)
00456 * - Bit 6: CKPOL - Clock polarity (0=idle LOW, 1=idle HIGH)
00457 * - Bit 4: MFF - Mode fault flag
00458 * - Bit 3: Reserved
00459 * - Bit 2: MSS - Controller/peripheral select (0=controller, 1=peripheral)

```

```

00460  * - Bit 1: CTSE - CTS enable (0=disabled, 1=CTS output enabled)
00461  * - Bit 0: SSE - SS pin enable
00462  */
00463  volatile uint8_t spmr;
00464 } rx_sci_regs_t;
00465
00466 /**
00467  * @enum rx_sci_ssr_flags_t
00468  * @brief SCI Serial Status Register (SSR) flag masks
00469  *
00470  * @details
00471  * Bit masks for the SSR register per RX72N HUM 34.2.7. Error flags
00472  * (ORER, FER, PER) are sticky and must be cleared by read-modify-write
00473  * with a 0 written to the bit position (writing 1 has no effect).
00474  *
00475  * @note Bit positions match both the SCIA/b/c/d/e/f (standard) and
00476  * SCIG/h/i (extended) SSR layouts -- the same mask works for all SCI
00477  * channels on the RX72N.
00478  */
00479 typedef enum : uint8_t {
00480     k_sci_ssr_tdre_flag = 0x80U, /**< Transmit data register empty (HUM 34.2.7 bit 7) */
00481     k_sci_ssr_rdrf_flag = 0x40U, /**< Receive data register full (bit 6) */
00482     k_sci_ssr_orer_flag = 0x20U, /**< Overrun error (bit 5) */
00483     k_sci_ssr_fer_flag = 0x10U, /**< Framing error (bit 4) */
00484     k_sci_ssr_per_flag = 0x08U, /**< Parity error (bit 3) */
00485     k_sci_ssr_tend_flag = 0x04U, /**< Transmit end (bit 2) */
00486     k_sci_ssr_error_mask = 0x38U, /**< ORER|FER|PER -- receive error flags */
00487 } rx_sci_ssr_flags_t;
00488
00489 /**
00490  * @brief Get pointer to SCI0 registers
00491  * @return Volatile pointer to SCI0 register structure
00492  */
00493 static inline volatile rx_sci_regs_t* sci0(void)
00494 {
00495     return (volatile rx_sci_regs_t*)k_sci0_base_addr;
00496 }
00497
00498 /**
00499  * @brief Get pointer to SCI1 registers
00500  * @return Volatile pointer to SCI1 register structure
00501  */
00502 static inline volatile rx_sci_regs_t* sci1(void)
00503 {
00504     return (volatile rx_sci_regs_t*)k_sci1_base_addr;
00505 }
00506
00507 /**
00508  * @brief Get pointer to SCI2 registers
00509  * @return Volatile pointer to SCI2 register structure
00510  */
00511 static inline volatile rx_sci_regs_t* sci2(void)
00512 {
00513     return (volatile rx_sci_regs_t*)k_sci2_base_addr;
00514 }
00515
00516 /**
00517  * @brief Get pointer to SCI3 registers
00518  * @return Volatile pointer to SCI3 register structure
00519  */
00520 static inline volatile rx_sci_regs_t* sci3(void)
00521 {
00522     return (volatile rx_sci_regs_t*)k_sci3_base_addr;
00523 }
00524
00525 /**
00526  * @brief Get pointer to SCI4 registers
00527  * @return Volatile pointer to SCI4 register structure
00528  */
00529 static inline volatile rx_sci_regs_t* sci4(void)
00530 {
00531     return (volatile rx_sci_regs_t*)k_sci4_base_addr;
00532 }
00533
00534 /**
00535  * @brief Get pointer to SCI5 registers
00536  * @return Volatile pointer to SCI5 register structure
00537  */
00538 static inline volatile rx_sci_regs_t* sci5(void)
00539 {
00540     return (volatile rx_sci_regs_t*)k_sci5_base_addr;
00541 }
00542
00543 /**
00544  * @brief Get pointer to SCI6 registers
00545  * @return Volatile pointer to SCI6 register structure
00546  */

```

```

00547 static inline volatile rx_sci_regs_t* sci6(void)
00548 {
00549     return (volatile rx_sci_regs_t*)k_sci6_base_addr;
00550 }
00551
00552 /**
00553  * @brief Get pointer to SCI7 registers
00554  * @return Volatile pointer to SCI7 register structure
00555  */
00556 static inline volatile rx_sci_regs_t* sci7(void)
00557 {
00558     return (volatile rx_sci_regs_t*)k_sci7_base_addr;
00559 }
00560
00561 /**
00562  * @brief Get pointer to SCI8 registers
00563  * @return Volatile pointer to SCI8 register structure
00564  */
00565 static inline volatile rx_sci_regs_t* sci8(void)
00566 {
00567     return (volatile rx_sci_regs_t*)k_sci8_base_addr;
00568 }
00569
00570 /**
00571  * @brief Get pointer to SCI9 registers (Debug UART)
00572  * @return Volatile pointer to SCI9 register structure
00573  */
00574 static inline volatile rx_sci_regs_t* sci9(void)
00575 {
00576     return (volatile rx_sci_regs_t*)k_sci9_base_addr;
00577 }
00578
00579 /**
00580  * @brief Get pointer to SCI10 registers
00581  * @return Volatile pointer to SCI10 register structure
00582  */
00583 static inline volatile rx_sci_regs_t* sci10(void)
00584 {
00585     return (volatile rx_sci_regs_t*)k_sci10_base_addr;
00586 }
00587
00588 /**
00589  * @brief Get pointer to SCI11 registers
00590  * @return Volatile pointer to SCI11 register structure
00591  */
00592 static inline volatile rx_sci_regs_t* sci11(void)
00593 {
00594     return (volatile rx_sci_regs_t*)k_sci11_base_addr;
00595 }
00596
00597 /**
00598  * @brief Get pointer to SCI12 registers
00599  * @return Volatile pointer to SCI12 register structure
00600  */
00601 static inline volatile rx_sci_regs_t* sci12(void)
00602 {
00603     return (volatile rx_sci_regs_t*)k_sci12_base_addr;
00604 }
00605
00606 /*
00607  * Multi-Channel Support
00608  */
00609 /*
00610  */
00611 /**
00612  * @brief SCI channel count
00613  */
00614 typedef enum : uint8_t {
00615     k_sci_channel_max = 13, /**< Total SCI channels (0-12) */
00616 } sci_channel_limits_t;
00617
00618 #if defined(USE MOCK SCI REGS)
00619 volatile rx_sci_regs_t* sci_get_channel(uint8_t channel);
00620 #else
00621 /**
00622  * @brief Get SCI register base for a channel
00623  * @param[in] channel SCI channel number (0-12)
00624  * @return Pointer to SCI registers, or nullptr if invalid channel
00625  */
00626 static inline volatile rx_sci_regs_t* sci_get_channel(uint8_t channel)
00627 {
00628     switch (channel) {
00629         case 0:
00630             return sci0();
00631         case 1:

```

```

00632     return sci1();
00633 case 2:
00634     return sci2();
00635 case 3:
00636     return sci3();
00637 case 4:
00638     return sci4();
00639 case 5:
00640     return sci5();
00641 case 6:
00642     return sci6();
00643 case 7:
00644     return sci7();
00645 case 8:
00646     return sci8();
00647 case 9:
00648     return sci9();
00649 case 10:
00650     return sci10();
00651 case 11:
00652     return sci11();
00653 case 12:
00654     return sci12();
00655 default:
00656     return nullptr; /* Invalid channel */
00657 }
00658 }
00659 #endif
00660
00661 /*
=====
00662 * Static Assertions - Compile-time Register Layout Verification
00663 *
=====
00664 *
00665 * These static assertions verify that the rx_sci_regs_t struct layout and
00666 * channel base addresses exactly match the RX72N hardware. Any mismatch
00667 * will cause a compile-time error, preventing runtime UART bugs.
00668 *
00669 * Reference: RX72N Group User's Manual: Hardware, Chapter 41 (SCI)
00670 * Section 41.2: Register Descriptions
00671 */
00672
00673 /** @name SCI Register Offset Verification
00674 * @brief Verify rx_sci_regs_t matches hardware layout
00675 * @{
00676 */
00677 static_assert(sizeof(rx_sci_regs_t) == 14, "SCI register structure must be 14 bytes");
00678 static_assert(offsetof(rx_sci_regs_t, smr) == 0x00, "SCI SMR register offset incorrect");
00679 static_assert(offsetof(rx_sci_regs_t, brr) == 0x01, "SCI BRR register offset incorrect");
00680 static_assert(offsetof(rx_sci_regs_t, scr) == 0x02, "SCI SCR register offset incorrect");
00681 static_assert(offsetof(rx_sci_regs_t, tdr) == 0x03, "SCI TDR register offset incorrect");
00682 static_assert(offsetof(rx_sci_regs_t, ssr) == 0x04, "SCI SSR register offset incorrect");
00683 static_assert(offsetof(rx_sci_regs_t, rdr) == 0x05, "SCI RDR register offset incorrect");
00684 static_assert(offsetof(rx_sci_regs_t, scmr) == 0x06, "SCI SCMR register offset incorrect");
00685 static_assert(offsetof(rx_sci_regs_t, semr) == 0x07, "SCI SEMR register offset incorrect");
00686 static_assert(offsetof(rx_sci_regs_t, snfr) == 0x08, "SCI SNFR register offset incorrect");
00687 static_assert(offsetof(rx_sci_regs_t, simr1) == 0x09, "SCI SIMR1 register offset incorrect");
00688 static_assert(offsetof(rx_sci_regs_t, simr2) == 0x0A, "SCI SIMR2 register offset incorrect");
00689 static_assert(offsetof(rx_sci_regs_t, simr3) == 0x0B, "SCI SIMR3 register offset incorrect");
00690 static_assert(offsetof(rx_sci_regs_t, sistr) == 0x0C, "SCI SISR register offset incorrect");
00691 static_assert(offsetof(rx_sci_regs_t, spmr) == 0x0D, "SCI SPMR register offset incorrect");
00692 /** @} */
00693
00694 /** @name SCI Base Address Verification
00695 * @brief Verify channel base addresses match RX72N Manual Ch41
00696 * @{
00697 */
00698 static_assert(k_sci0_base_addr == 0x0008A000, "SCI0 base address incorrect");
00699 static_assert(k_sci1_base_addr == 0x0008A020, "SCI1 base address incorrect");
00700 static_assert(k_sci2_base_addr == 0x0008A040, "SCI2 base address incorrect");
00701 static_assert(k_sci3_base_addr == 0x0008A060, "SCI3 base address incorrect");
00702 static_assert(k_sci4_base_addr == 0x0008A080, "SCI4 base address incorrect");
00703 static_assert(k_sci5_base_addr == 0x0008A0A0, "SCI5 base address incorrect");
00704 static_assert(k_sci6_base_addr == 0x0008A0C0, "SCI6 base address incorrect");
00705 static_assert(k_sci7_base_addr == 0x000D00E0, "SCI7 base address incorrect");
00706 static_assert(k_sci8_base_addr == 0x000D0000, "SCI8 base address incorrect");
00707 static_assert(k_sci9_base_addr == 0x000D0020, "SCI9 base address incorrect");
00708 static_assert(k_sci10_base_addr == 0x000D0040, "SCI10 base address incorrect");
00709 static_assert(k_sci11_base_addr == 0x000D0060, "SCI11 base address incorrect");
00710 static_assert(k_sci12_base_addr == 0x0008B300, "SCI12 base address incorrect");
00711 /** @} */
00712
00713 /** @name SCI Memory Region Verification
00714 * @brief Verify base addresses are in correct peripheral regions
00715 * @{
00716 */

```

```

00717 static_assert((k_sci0_base_addr & 0xFFFF0000) == 0x00080000,
00718                "SCI0 base address not in SCI standard peripheral space");
00719 static_assert((k_sci8_base_addr & 0xFFFF0000) == 0x000D0000,
00720                "SCI8 base address not in SCI extended peripheral space");
00721 static_assert((k_sci12_base_addr & 0xFFFF0000) == 0x00080000,
00722                "SCI12 base address not in SCI standard peripheral space");
00723 static_assert((k_sci1_base_addr - k_sci0_base_addr) == 0x20,
00724                "SCI1 channel spacing incorrect (expected 0x20 bytes)");
00725 /** @} */
00726
00727 /** @} */ /* End of sci_regs defgroup */
00728
00729 #ifdef __cplusplus
00730 }
00731 #endif

```

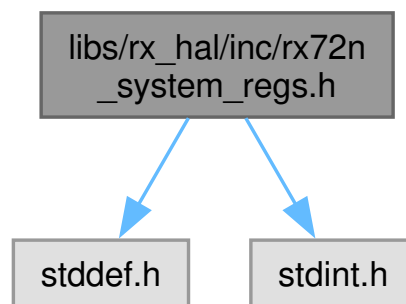
8.59 libs/rx_hal/inc/rx72n_system_regs.h File Reference

RX72N System Control Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_system_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_system_regs_t](#)
System Control Register Map (0x00080000 - 0x00080041).
- struct [rx_rstsr01_regs_t](#)
Reset Status Registers 0 and 1 (contiguous).

Enumerations

- enum `rx_system_addresses_t` : `uintptr_t` { `k_system_base_addr` = 0x00080000 , `k_prccr_addr` = 0x000803FE }

System register base addresses.

- enum `system_reserved_sizes_t` : `uint8_t` {
`k_system_reserved_02_05` = 4 , `k_system_reserved_0a_0b` = 2 , `k_system_reserved_0e_0f` = 2 , `k_system_reserved_2b_2f` = 5 ,
`k_system_reserved_31` = 1 , `k_system_reserved_38_3b` = 4 , `k_system_reserved_3d` = 1 }

System register reserved field sizes.

- enum `mdmonr_bits_t` : `uint16_t` { `k_mdmonr_md` = 0x0001U , `k_mdmonr_md_mask` = 0x0001U }

Mode Monitor Register (MDMONR) bit definitions.

- enum `syscr0_bits_t` : `uint16_t` {
`k_syscr0_rome` = 0x0001U , `k_syscr0_exbe` = 0x0002U , `k_syscr0_key` = 0x5A00U ,
`k_syscr0_key_mask` = 0xFF00U ,
`k_syscr0_rom_enabled` = (0x5A00U | 0x0001U) , `k_syscr0_rom_disabled` = 0x5A00U ,
`k_syscr0_extbus_enable` = (0x5A00U | 0x0003U) }

System Control Register 0 (SYSCR0) bit definitions.

- enum `syscr1_bits_t` : `uint16_t` {
`k_syscr1_rame` = 0x0001U , `k_syscr1_reserved_mask` = 0x003EU , `k_syscr1_eccrame` = 0x0040U ,
`k_syscr1_sbyrame` = 0x0080U ,
`k_syscr1_all_ram_enabled` = 0x00FFU , `k_syscr1_disable_eccram` = 0x00BFU , `k_syscr1_disable_sbyram` = 0x007FU }

System Control Register 1 (SYSCR1) bit definitions.

- enum `ckocr_bits_t` : `uint16_t` {
`k_ckocr_ckosel_loco` = (0U << 8) , `k_ckocr_ckosel_hoco` = (1U << 8) , `k_ckocr_ckosel_main` = (2U << 8) , `k_ckocr_ckosel_subclock` = (3U << 8) ,
`k_ckocr_ckosel_pll` = (4U << 8) , `k_ckocr_ckosel_ppll` = (6U << 8) , `k_ckocr_ckosel_mask` = (7U << 8) , `k_ckocr_ckodiv_1` = (0U << 12) ,
`k_ckocr_ckodiv_2` = (1U << 12) , `k_ckocr_ckodiv_4` = (2U << 12) , `k_ckocr_ckodiv_8` = (3U << 12) , `k_ckocr_ckodiv_16` = (4U << 12) ,
`k_ckocr_ckodiv_mask` = (7U << 12) , `k_ckocr_ckostp` = (1U << 15) , `k_ckocr_output_disabled` = `k_ckocr_ckostp` , `k_ckocr_pll_div4` }

Clock Output Control Register (CKOCR) bit definitions.

- enum `sckcr_divider_t` : `uint8_t` {
`k_clock_div_1` = 0 , `k_clock_div_2` = 1 , `k_clock_div_4` = 2 , `k_clock_div_8` = 3 ,
`k_clock_div_16` = 4 , `k_clock_div_32` = 5 , `k_clock_div_64` = 6 , `k_bck_div_3` = 9 }

Clock divider values for SCKCR fields.

- enum `sckcr_bits_t` : `uint32_t` {
`k_sckcr_pckd_shift` = 0 , `k_sckcr_pckc_shift` = 4 , `k_sckcr_pckb_shift` = 8 , `k_sckcr_pcka_shift` = 12 ,
`k_sckcr_bck_shift` = 16 , `k_sckcr_ick_shift` = 24 , `k_sckcr_fck_shift` = 28 , `k_sckcr_pckd_mask` = 0xFUL << `k_sckcr_pckd_shift` ,
`k_sckcr_pckc_mask` = 0xFUL << `k_sckcr_pckc_shift` , `k_sckcr_pckb_mask` = 0xFUL << `k_sckcr_pckb_shift` , `k_sckcr_pcka_mask` = 0xFUL << `k_sckcr_pcka_shift` , `k_sckcr_bck_mask` = 0xFUL << `k_sckcr_bck_shift` ,
`k_sckcr_ick_mask` = 0xFUL << `k_sckcr_ick_shift` , `k_sckcr_fck_mask` = 0xFUL << `k_sckcr_fck_shift` , `k_sckcr_pstop0` = (1UL << 22) , `k_sckcr_pstop1` = (1UL << 23) ,
`k_sckcr_star_240mhz` }

System Clock Control Register (SCKCR) bit definitions.

- enum `sckcr3_cksel_t` : `uint16_t` {
`k_sckcr3_cksel_loco` = 0x0000 , `k_sckcr3_cksel_hoco` = 0x0100 , `k_sckcr3_cksel_main` = 0x0200 ,
`k_sckcr3_cksel_subclock` = 0x0300 ,
`k_sckcr3_cksel_pll` = 0x0400 , `k_sckcr3_cksel_mask` = 0x0700 }

System clock source selection (SCKCR3 CKSEL field).

- enum `pllcr_bits_t` : `uint16_t` {
`k_pllcr_plidiv_1` = 0x0000 , `k_pllcr_plidiv_2` = 0x0001 , `k_pllcr_plidiv_3` = 0x0002 ,
`k_pllcr_plidiv_mask` = 0x0003 ,
`k_pllcr_src_main` = 0x0000 , `k_pllcr_src_hoco` = 0x0010 , `k_pllcr_src_mask` = 0x0010 ,
`k_pllcr_stc_shift` = 8 ,
`k_pllcr_stc_mask` = 0x3F00 , `k_pllcr_star_24mhz_to_240mhz` , `k_pllcr_12mhz_to_240mhz` }
PLL Control Register (PLLCR) bit definitions.
- enum `pllcr2_bits_t` : `uint8_t` { `k_pllcr2_pll_running` = 0x00 , `k_pllcr2_pll_stopped` = 0x01 }
PLL Control Register 2 (PLLCR2) bit definitions.
- enum `rx_module_stop_bits_a_t` : `uint8_t` { `k_mstpa_cmt23` = 14 }
Module stop bits for MSTPCRA register.
- enum `rx_module_stop_bits_b_t` : `uint8_t` { `k_mstpb_usb0` = 19 , `k_mstpb_crc` = 23 }
Module stop bits for MSTPCRB register.
- enum `rx_rstsr_addresses_t` : `uintptr_t` { `k_rstsr0_addr` = 0x0008C290 , `k_rstsr1_addr` = 0x0008C291 , `k_rstsr2_addr` = 0x000800C0 }
Reset status register addresses.
- enum `rstsr0_bits_t` : `uint8_t` {
`k_rstsr0_porf` = 1U , `k_rstsr0_lvd0rf` = 2U , `k_rstsr0_lvd1rf` = 4U , `k_rstsr0_lvd2rf` = 8U ,
`k_rstsr0_dpsrstf` = 128U }
- enum `rstsr1_bits_t` : `uint8_t` { `k_rstsr1_cwsf` = 1U }
- enum `rstsr2_bits_t` : `uint8_t` { `k_rstsr2_iwdtrf` = 1U , `k_rstsr2_wdtrf` = 2U , `k_rstsr2_swrf` = 4U }
- enum `rx_swrr_addresses_t` : `uintptr_t` { `k_swrr_addr` = 0x000800C2 }
Software Reset Register (SWRR) address.
- enum `rx_swrr_values_t` : `uint16_t` { `k_swrr_reset_key` = 0xA501 }
Software Reset Register value.
- enum `rx_memwait_addresses_t` : `uintptr_t` { `k_memwait_addr` = 0x0008101C }
MEMWAIT register address.
- enum `rx_memwait_bits_t` : `uint8_t` { `k_memwait_no_wait` = 0x00 , `k_memwait_one_wait` = 0x01 }
MEMWAIT register bit values.
- enum `rx_ppll_addresses_t` : `uintptr_t` { `k_ppllcr_addr` = 0x00080048 , `k_ppllcr2_addr` = 0x0008004A , `k_ppllcr3_addr` = 0x0008004B }
PPLL register addresses.
- enum `rx_ppll_config_t` : `uint16_t` { `k_ppll_config_48mhz` = 0x1300 }
PPLL configuration values.
- enum `rx_ppllcr3_config_t` : `uint8_t` { `k_ppllcr3_div5` = 0x04 }
PPLLCR3 frequency divider values.
- enum `rx_ppll_control_t` : `uint8_t` { `k_ppll_enabled` = 0x00 , `k_ppll_disabled` = 0x01 }
PPLL control values.
- enum `rx_ppll_flags_t` : `uint8_t` { `k_ppll_stable_flag` = 0x20 }
PPLL stabilization flag in OSCOVFSR.
- enum `rx_system_addr_refs_t` : `uintptr_t` {
`k_ref_system_base` = 0x00080000 , `k_ref_prcr` = 0x000803FE , `k_ref_rstsr0` = 0x0008C290 ,
`k_ref_rstsr1` = 0x0008C291 ,
`k_ref_rstsr2` = 0x000800C0 , `k_ref_swrr` = 0x000800C2 , `k_ref_memwait` = 0x0008101C ,
`k_ref_ppllcr` = 0x00080048 ,
`k_ref_ppllcr2` = 0x0008004A , `k_ref_ppllcr3` = 0x0008004B }
Expected hardware addresses per RX72N Hardware Manual.
- enum `rx_system_struct_layout_t` : `uint8_t` {
`k_sys_regs_size` = 0x42 , `k_sys_reg_mdmonr_off` = 0x00 , `k_sys_reg_syscr0_off` = 0x06 ,
`k_sys_reg_syscr1_off` = 0x08 ,
`k_sys_reg_sbyscr_off` = 0x0C , `k_sys_reg_mstpcra_off` = 0x10 , `k_sys_reg_mstpcrb_off` = 0x14

```
, k_sys_reg_mstpcrc_off = 0x18 ,
k_sys_reg_mstpcrd_off = 0x1C , k_sys_reg_sckcr_off = 0x20 , k_sys_reg_sckcr2_off = 0x24 ,
k_sys_reg_sckcr3_off = 0x26 ,
k_sys_reg_pllcr_off = 0x28 , k_sys_reg_pllcr2_off = 0x2A , k_sys_reg_bckcr_off = 0x30 ,
k_sys_reg_mosccr_off = 0x32 ,
k_sys_reg_sosccr_off = 0x33 , k_sys_reg_lococr_off = 0x34 , k_sys_reg_ilocr_off = 0x35 ,
k_sys_reg_hococr_off = 0x36 ,
k_sys_reg_hococr2_off = 0x37 , k_sys_reg_oscovfsr_off = 0x3C , k_sys_reg_ckocr_off = 0x3E
, k_sys_reg_ostdcr_off = 0x40 ,
k_sys_reg_ostdsr_off = 0x41 }
```

Expected size and member offsets for `rx_system_regs_t`.

- enum `rx_rstsr01_layout_t` : `uint8_t` { `k_rstsr01_struct_size` = 2 , `k_rstsr0_member_off` = 0 , `k_rstsr1_member_off` = 1 }

Expected size and member offsets for `rx_rstsr01_regs_t`.

- enum `rx_prcr_offset_t` : `uint16_t` { `k_prcr_from_sys_base` = 0x03FE }

PRCR register offset from system register base address.

- enum `rx_clock_verify_t` : `uint8_t` { `k_nibble_all_ones` = 0x0F , `k_pllcr_stc_for_x10` }

Reference values for clock field-width and PLL multiplier assertions.

Functions

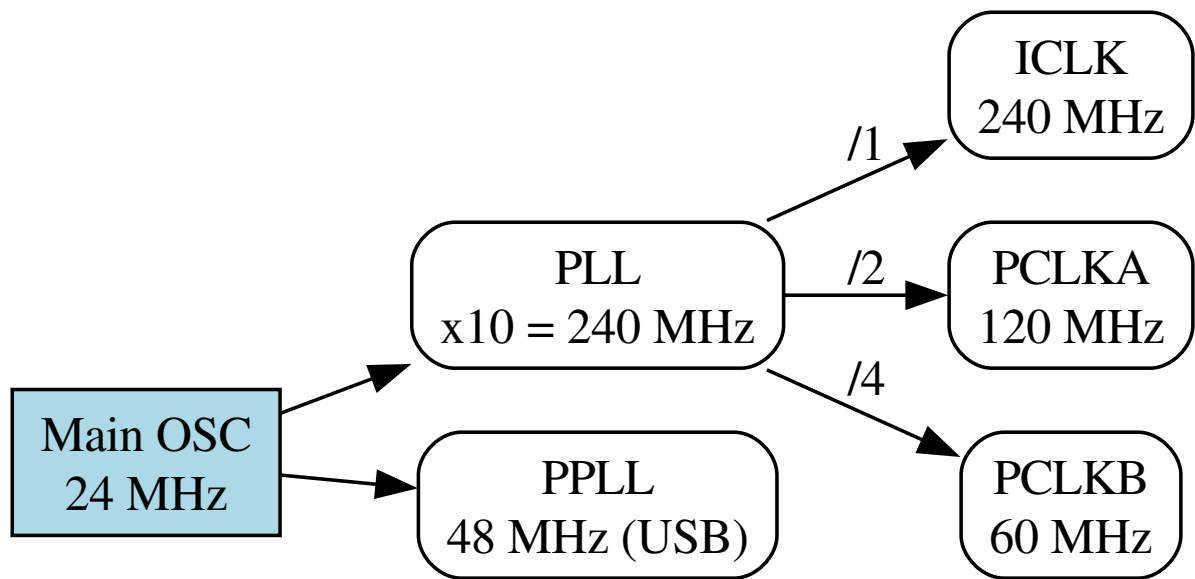
- static volatile `rx_system_regs_t` * `system_regs` (void)
Get pointer to system registers.
- static volatile `uint16_t` * `prcr_reg` (void)
Get pointer to PRCR (Protection Register).
- static volatile `rx_rstsr01_regs_t` * `rstsr01` (void)
Get pointer to RSTSR0/RSTSR1 registers.
- static volatile `uint8_t` * `rstsr2` (void)
Get pointer to RSTSR2 register.
- static volatile `uint16_t` * `swrr_reg` (void)
Get pointer to SWRR (Software Reset Register).
- static volatile `uint8_t` * `memwait_reg` (void)
Get pointer to MEMWAIT register.
- static volatile `uint16_t` * `ppllcr_reg` (void)
Get pointer to PPLLCR register.
- static volatile `uint8_t` * `ppllcr2_reg` (void)
Get pointer to PPLLCR2 register.
- static volatile `uint8_t` * `ppllcr3_reg` (void)
Get pointer to PPLLCR3 register.

8.59.1 Detailed Description

RX72N System Control Register Definitions.

System control registers for clock generation, power management, oscillator control, and module stop control on the RX72N microcontroller. This is the foundation for system initialization and power optimization.

System Architecture



STAR Project Clock Configuration

Clock	Frequency	Divider	Purpose
ICLK	240 MHz	/↵ 1	CPU instruction clock
PCLKA	120 MHz	/↵ 2	High-speed peripherals
PCLKB	60 MHz	/↵ 4	Peripherals (CMT, WDT)
PCLKC	60 MHz	/↵ 4	Peripherals
PCLKD	60 MHz	/↵ 4	ADC
FCLK	60 MHz	/↵ 4	Flash operations
BCLK	120 MHz	/↵ 2	External bus
UCLK	48 MHz	PPLL	USB

Register Map Overview

IMPORTANT: System registers are NOT in a single contiguous block!

Address Range	Description
0x00080000 - 0x00080041	Main system registers (this struct)
0x00080048 - 0x0008004A	PPLL registers

Address Range	Description
0x000803FE	PRCR (Protection Register)
0x000800C0	RSTSR2 (Reset Status 2)
0x0008101C	MEMWAIT
0x0008C290 - 0x0008C291	RSTSR0/1 (Reset Status 0/1)

Module Stop Control (MSTPCR)

Peripherals can be stopped to save power. Bit=1 stops the module:

- MSTPCRA: CMT, DTC, DMAC
- MSTPCRB: USB, CRC, SCI, RSPI
- MSTPCRC: RAM, Flash
- MSTPCRD: ADC, DAC, MTU, GPTW

Register Protection (PRCR)

Critical registers require PRCR unlock before modification:

- PRC0: Clock generation (SCKCR, PLLCR, oscillator control)
- PRC1: Module stop control (MSTPCRA-D)
- PRC3: Low voltage detection

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessors are single-statement
- Rule 5: [OK] N/A (hardware layer)
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Single responsibility - only system register definitions
- O: Open for extension via additional register accessors
- L: N/A
- I: Separate accessors for non-contiguous registers
- D: N/A (hardware register layer)

Module Dependencies

- `<stddef.h>` - offsetof macro for static assertions
- `<stdint.h>` - Fixed-width integer types

Manual References

- Chapter 5: I/O Registers (address map)
- Chapter 9: Clock Generation Circuit
- Chapter 13: Register Write Protection
- Chapter 11: Low Power Consumption

Verification Status

[PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified against RX72N Manual Ch05. Critical bug fixed: PRCR was incorrectly embedded in the contiguous struct at offset 0x0A, but it's actually at 0x000803FE.

See also

[rx72n_clock.h](#) Clock-specific definitions

[rx_register_protection.h](#) PRCR unlock/lock API

[rx_clock_power_init.c](#) Clock initialization implementation

Author

Locked, Inc.

Date

2026-01-28

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_system_regs.h](#).

8.59.2 Data Structure Documentation

8.59.2.1 struct rx_system_regs_t

System Control Register Map (0x00080000 - 0x00080041).

System control registers for clock configuration, power management, and module stop control. This structure covers the main contiguous block of system registers.

Memory Layout Table

Offset	Size	Field	Description
0x00	2	mdmonr	Mode Monitor Register
0x02	4	reserved	-
0x06	2	syscr0	System Control Register 0
0x08	2	syscr1	System Control Register 1
0x0A	2	reserved	-
0x0C	2	sbycr	Standby Control Register
0x0E	2	reserved	-
0x10	4	mstpcra	Module Stop Control Register A
0x14	4	mstpcrb	Module Stop Control Register B
0x18	4	mstpcrc	Module Stop Control Register C
0x1C	4	mstpcrd	Module Stop Control Register D
0x20	4	sckcr	System Clock Control Register
0x24	2	sckcr2	System Clock Control Register 2
0x26	2	sckcr3	System Clock Control Register 3
0x28	2	pllcr	PLL Control Register
0x2A	1	pllcr2	PLL Control Register 2
0x2B	5	reserved	-
0x30	1	bckcr	External Bus Clock Control
0x31	1	reserved	-
0x32	1	mosccr	Main Oscillator Control
0x33	1	sosccr	Sub-Clock Oscillator Control
0x34	1	lococr	LOCO Control
0x35	1	ilococr	IWDT Oscillator Control
0x36	1	hococr	HOCO Control
0x37	1	hococr2	HOCO Control 2
0x38	4	reserved	-
0x3C	1	oscovfsr	Oscillation Stabilization Flag
0x3D	1	reserved	-
0x3E	2	ckocr	Clock Output Control Register
0x40	1	ostdcr	Oscillation Stop Detection Control
0x41	1	ostdsr	Oscillation Stop Detection Status

Offset	Size	Field	Description
Total	66		

Clock Initialization Example

```
// Initialize system to 240 MHz from 24 MHz main oscillator
volatile rx_system_regs_t* sys = system_regs();

// 1. Unlock register protection
*prcr_reg() = k_rx_prcr_unlock_prc0_prc1;

// 2. Enable main oscillator
sys->mosccr = 0x00; // Start main oscillator

// 3. Wait for oscillator stabilization
while (!(sys->oscovfsr & 0x01)) { }

// 4. Configure PLL (24 MHz x 10 = 240 MHz)
sys->pllcr = 0x0913; // STC=9 (x10), PLIDIV=1 (/1)
sys->pllcr2 = 0x00; // Enable PLL

// 5. Wait for PLL stabilization
while (!(sys->oscovfsr & 0x04)) { }

// 6. Set MEMWAIT for >120 MHz operation
*memwait_reg() = k_memwait_one_wait;

// 7. Switch to PLL
sys->sckcr3 = 0x0400; // CKSEL = PLL

// 8. Re-lock protection
*prcr_reg() = k_rx_prcr_lock;
```

Module Stop Example

```
// Enable CMT0 module (clear bit to enable)
*prcr_reg() = k_rx_prcr_unlock_prc1; // Unlock MSTPCR
system_regs()->mstpcra &= ~(1UL « 15); // Enable CMT0
*prcr_reg() = k_rx_prcr_lock;

// Enable USB0 module
*prcr_reg() = k_rx_prcr_unlock_prc1;
system_regs()->mstpcrb &= ~(1UL « k_mstpb_usb0);
*prcr_reg() = k_rx_prcr_lock;
```

Note

PRCR (Protection Register) is NOT in this struct - it's at 0x000803FE. Use [prcr_reg\(\)](#) accessor instead.

Invariant

sizeof(rx_system_regs_t) == 66 bytes (0x42)
All register offsets match RX72N Manual Ch05

Warning

Most registers require PRCR unlock before modification

Attention

MEMWAIT must be set before ICLK > 120 MHz

See also

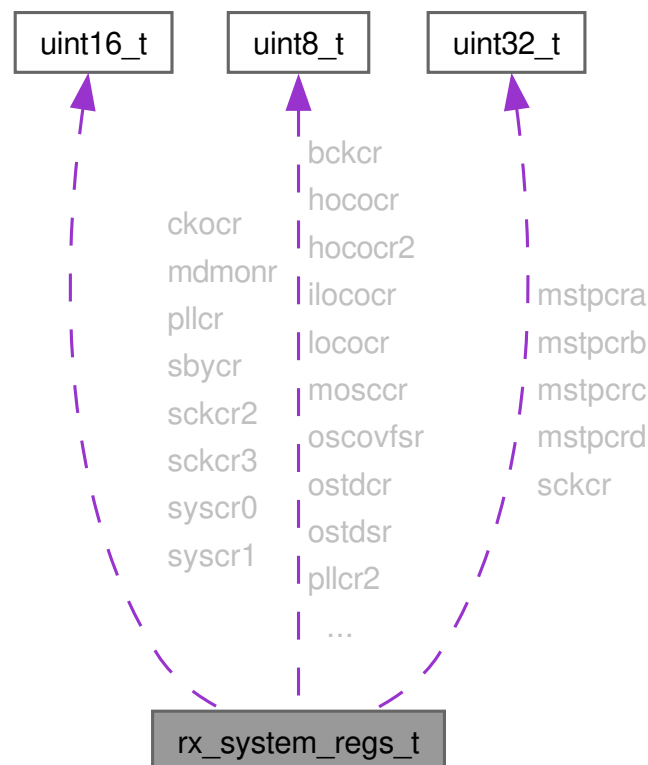
[system_regs\(\)](#) Accessor function
[prcr_reg\(\)](#) Protection register accessor

Since

Version 1.0.0

Definition at line 730 of file [rx72n_system_regs.h](#).

Collaboration diagram for rx_system_regs_t:



Data Fields

volatile uint8_t	bckcr	External Bus Clock Control Register (BCKCR) @ 0x30. Controls external bus clock output
volatile uint16_t	ckocr	Clock Output Control Register (CKOCR) @ 0x3E. Controls CLKOUT pin output source and divider See also ckocr_bits_t for bit field definitions

volatile uint8_t	hococr	High-Speed On-Chip Oscillator Control (HOCOOCR) @ 0x36. Controls HOCO
volatile uint8_t	hococr2	High-Speed On-Chip Oscillator Control 2 (HOCOOCR2) @ 0x37. HOCO frequency selection
volatile uint8_t	ilococr	IWDT-Dedicated Oscillator Control (ILOCOOCR) @ 0x35. Controls IWDT oscillator (120 kHz)
volatile uint8_t	lococr	Low-Speed On-Chip Oscillator Control (LOCOOCR) @ 0x34. Controls LOCO (240 kHz)
volatile uint16_t	mdmonr	Mode Monitor Register (MDMONR) @ 0x00. Read-only - indicates current operating mode
volatile uint8_t	mosccr	Main Clock Oscillator Control Register (MOSCCR) @ 0x32. Controls main oscillator (24 MHz crystal)
volatile uint32_t	mstpcra	Module Stop Control Register A (MST-PCRA) @ 0x10. Controls CMT, DTC, DMAC module power. Bit=1 stops module.
volatile uint32_t	mstpcrb	Module Stop Control Register B (MST-PCRB) @ 0x14. Controls USB, CRC, SCI, RSPI module power. Bit=1 stops module.
volatile uint32_t	mstpcrc	Module Stop Control Register C (MST-PCRC) @ 0x18. Controls RAM, Flash module power
volatile uint32_t	mstpcrd	Module Stop Control Register D (MST-PCRD) @ 0x1C. Controls ADC, DAC, MTU, GPTW module power. Bit=1 stops module.

volatile uint8_t	oscovfsr	Oscillation Stabilization Flag Register (OSCOVFSR) @ 0x3C. Indicates when oscillators are stable (1=stable, 0=not stable): • Bit 0 (MOOVF): Main clock oscillation stable • Bit 1 (SOOVF): Sub-clock oscillation stable • Bit 2 (PLOVF): PLL clock oscillation stable • Bit 3 (HCOVF): HOCO clock oscillation stable • Bit 4 (ILCOVF): IWDG-dedicated clock oscillation stable • Bit 5 (PPLOVF): PPLL clock oscillation stable • Bits 7:6: Reserved
volatile uint8_t	ostdcr	Oscillation Stop Detection Control (OSTDCR) @ 0x40. Enables oscillation stop detection
volatile uint8_t	ostdsr	Oscillation Stop Detection Status (OSTDSR) @ 0x41. Indicates oscillation stop detection status
volatile uint16_t	pllcr	PLL Control Register (PLLCR) @ 0x28. Sets PLL multiplication factor and input divider
volatile uint8_t	pllcr2	PLL Control Register 2 (PLLCR2) @ 0x2A. PLL enable (0=enabled, 1=disabled)
uint8_t	reserved0↵ [k_system_reserved_02_05]	Reserved @ 0x02-0x05
uint8_t	reserved1↵ [k_system_reserved_0a_0b]	Reserved @ 0x0A-0x0B
uint8_t	reserved2↵ [k_system_reserved_0e_0f]	Reserved @ 0x0E-0x0F
uint8_t	reserved3↵ [k_system_reserved_2b_2f]	Reserved @ 0x2B-0x2F
uint8_t	reserved4 [k_system_reserved_31]	Reserved @ 0x31
uint8_t	reserved5↵ [k_system_reserved_38_3b]	Reserved @ 0x38-0x3B
uint8_t	reserved6 [k_system_reserved_3d]	Reserved @ 0x3D
volatile uint16_t	sbycr	Standby Control Register (SBYCR) @ 0x0C. Controls standby mode (SSBY bit)

volatile uint32_t	sckcr	System Clock Control Register (SCKCR) @ 0x20. Sets division ratios for ICLK, PCLKA, PCLKB, PCLKC, PCLKD, FCLK, BCLK
volatile uint16_t	sckcr2	System Clock Control Register 2 (SCKCR2) @ 0x24. Sets UCK (USB clock) division ratio
volatile uint16_t	sckcr3	System Clock Control Register 3 (SCKCR3) @ 0x26. Selects system clock source (CKSEL bits)
volatile uint8_t	sosccr	Sub-Clock Oscillator Control Register (SOSCCR) @ 0x33. Controls sub-clock oscillator (32.768 kHz)
volatile uint16_t	syscr0	System Control Register 0 (SYSCR0) @ 0x06. Controls ROM/RAM enable and external bus settings
volatile uint16_t	syscr1	System Control Register 1 (SYSCR1) @ 0x08. Controls RAM wait states

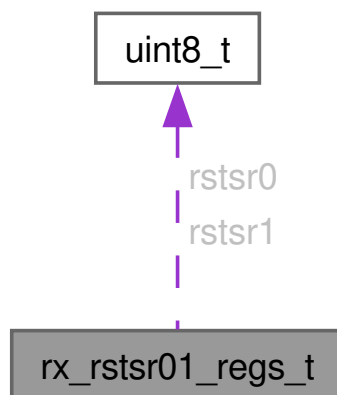
8.59.2.2 struct rx_rstsr01_regs_t

Reset Status Registers 0 and 1 (contiguous).

RSTSR0 and RSTSR1 are adjacent in memory at 0x0008C290-0x0008C291.

Definition at line 973 of file [rx72n_system_regs.h](#).

Collaboration diagram for rx_rstsr01_regs_t:



Data Fields

volatile uint8_t	rstsr0	Reset Status Register 0 (voltage, power-on)
volatile uint8_t	rstsr1	Reset Status Register 1 (cold/warm start)

8.59.3 Enumeration Type Documentation

8.59.3.1 ckocr_bits_t

```
enum ckocr_bits_t : uint16_t
```

Clock Output Control Register (CKOCR) bit definitions.

Controls the clock output on CLKOUT pin. Can output various system clocks divided by configurable ratio.

Register Layout (16-bit @ offset 0x3E)

Bit	Field	Description
7:↔ 0	-	Reserved
10:↔ 8	CKOSEL	Clock output source select
11	-	Reserved
14:12	CKODIV	Clock output divider
15	CKOSTP	Clock output stop (0=run, 1=stop)

Clock Output Sources (CKOSEL)

Value	Source	Description
0	LOCO	Low-speed on-chip oscillator
1	HOCO	High-speed on-chip oscillator
2	Main clock	Main oscillator
3	Sub-clock	Sub-clock oscillator
4	PLL	PLL output

Clock Output Dividers (CKODIV[2:0
at bits 14:12])

Value	Divider	Output Frequency
000	$\frac{f}{1}$	f (source frequency)
001	$\frac{f}{2}$	$\frac{f}{2}$
010	$\frac{f}{4}$	$\frac{f}{4}$
011	$\frac{f}{8}$	$\frac{f}{8}$
100	/16	f/16
other	–	PROHIBITED

Clock Output Sources (CKOSEL[2:0
at bits 10:8])

Value	Source
000	LOCO
001	HOCO
010	Main clock
011	Sub-clock
100	PLL circuit
110	PPLL circuit
other	PROHIBITED

Warning

Clock output disabled by default (CKOSTP = 1)
CKODIV values 5, 6, 7 are PROHIBITED (max divisor is /16)

Note

Requires PRCR unlock (PRC0) before modification

See also

Manual Ch09 section 9.2.21 CKOCR

Since

Version 1.0.0

Enumerator

k_ckocr_ksel_loco	LOCO (low-speed on-chip osc)
k_ckocr_ksel_hoco	HOCO (high-speed on-chip osc)
k_ckocr_ksel_main	Main oscillator
k_ckocr_ksel_subclock	Sub-clock oscillator
k_ckocr_ksel_pll	PLL output
k_ckocr_ksel_ppll	PPLL output (110b, not 101b)
k_ckocr_ksel_mask	CKOSEL field mask
k_ckocr_ckodiv_1	Divide by 1
k_ckocr_ckodiv_2	Divide by 2
k_ckocr_ckodiv_4	Divide by 4
k_ckocr_ckodiv_8	Divide by 8
k_ckocr_ckodiv_16	Divide by 16 (maximum valid divisor)
k_ckocr_ckodiv_mask	CKODIV field mask
k_ckocr_ckostp	Clock output stop (1=stopped/low, 0=running)
k_ckocr_output_disabled	Default: output disabled
k_ckocr_pll_div4	PLL/4 output (60 MHz @ 240 MHz PLL)

Definition at line 356 of file [rx72n_system_regs.h](#).

```

00356         : uint16_t {
00357     /* Clock output source select (CKOSEL, bits 10:8) */
00358     k_ckocr_ksel_loco    = (0U « 8), /**< LOCO (low-speed on-chip osc) */
00359     k_ckocr_ksel_hoco    = (1U « 8), /**< HOCO (high-speed on-chip osc) */
00360     k_ckocr_ksel_main    = (2U « 8), /**< Main oscillator */
00361     k_ckocr_ksel_subclock = (3U « 8), /**< Sub-clock oscillator */
00362     k_ckocr_ksel_pll     = (4U « 8), /**< PLL output */
00363     k_ckocr_ksel_ppll    = (6U « 8), /**< PPLL output (110b, not 101b) */
00364     k_ckocr_ksel_mask    = (7U « 8), /**< CKOSEL field mask */
00365
00366     /* Clock output divider (CKODIV, bits 14:12) -- max is /16, values 5-7 PROHIBITED */
00367     k_ckocr_ckodiv_1     = (0U « 12), /**< Divide by 1 */
00368     k_ckocr_ckodiv_2     = (1U « 12), /**< Divide by 2 */
00369     k_ckocr_ckodiv_4     = (2U « 12), /**< Divide by 4 */
00370     k_ckocr_ckodiv_8     = (3U « 12), /**< Divide by 8 */
00371     k_ckocr_ckodiv_16    = (4U « 12), /**< Divide by 16 (maximum valid divisor) */
00372     k_ckocr_ckodiv_mask  = (7U « 12), /**< CKODIV field mask */
00373
00374     /* Clock output stop (CKOSTP, bit 15) */
00375     k_ckocr_ckostp       = (1U « 15), /**< Clock output stop (1=stopped/low, 0=running) */
00376
00377     /* Convenience values */
00378     k_ckocr_output_disabled = k_ckocr_ckostp, /**< Default: output disabled */
00379     k_ckocr_pll_div4      =
00380         (k_ckocr_ksel_pll | k_ckocr_ckodiv_4), /**< PLL/4 output (60 MHz @ 240 MHz PLL) */
00381 } ckocr_bits_t;

```

8.59.3.2 mdmonr`bits`

```
enum mdmonr_bits_t : uint16_t
```

Mode Monitor Register (MDMONR) bit definitions.

Read-only register indicating MCU mode pin status at reset. Located at address 0x00080000.

Register Layout (16-bit @ offset 0x00)

Bit	Field	Description
0	MD	MD Pin Status Flag (read-only)
15↔ 1	-	Reserved (read as 0)

MD Pin Modes (at reset)

- MD = 0: Boot mode (low level on MD pin)
- MD = 1: Single-chip mode (high level on MD pin)

Note

Manual ref: Ch03 section 3.2.1

Since

Version 1.0.0

Enumerator

k_mdmonr_md	MD pin status (0=low/boot, 1=high/single-chip)
k_mdmonr_md_mask	MD bit mask

Definition at line 210 of file [rx72n_system_regs.h](#).

```
00210      : uint16_t {
00211  k_mdmonr_md      = 0x0001U, /**< MD pin status (0=low/boot, 1=high/single-chip) */
00212  k_mdmonr_md_mask = 0x0001U, /**< MD bit mask */
00213 } mdmonr_bits_t;
```

8.59.3.3 pllcr2_bits_t

```
enum pllcr2_bits_t : uint8_t
```

PLL Control Register 2 (PLLCR2) bit definitions.

Controls PLL enable/disable. Note inverted logic: 0 = enabled, 1 = stopped.

Register Layout (8-bit @ offset 0x2A)

Bit	Field	Description
0	PLEN	PLL enable (0=run, 1=stopped)
7↔ 1	-	Reserved

Warning

Inverted logic! 0 = PLL running, 1 = PLL stopped

Note

Must wait for OSCOVFSR.PLOVF before switching clock to PLL

Requires PRCR unlock (PRC0) before modification

See also

Manual Ch09 section 9.2.4

Since

Version 1.0.0

Enumerator

k_pllcr2_pll_running	PLL enabled (running)
k_pllcr2_pll_stopped	PLL disabled (stopped)

Definition at line 623 of file [rx72n_system_regs.h](#).

```
00623      : uint8_t {
00624  k_pllcr2_pll_running = 0x00, /**< PLL enabled (running) */
00625  k_pllcr2_pll_stopped = 0x01, /**< PLL disabled (stopped) */
00626 } pllcr2_bits_t;
```

8.59.3.4 pllcr_bits_t

```
enum pllcr\_bits\_t : uint16_t
```

PLL Control Register (PLLCR) bit definitions.

Controls PLL input divider, output multiplier, and source selection. PLL output frequency = (Input / PLIDIV) x (STC + 1)

Register Layout (16-bit @ offset 0x28)

Bits	Field	Description
1:↔ 0	PLIDIV	PLL input divider (00=/1, 01=/2, 10=/3)
3:↔ 2	-	Reserved
4	PLLSRCSEL	PLL source (0=main osc, 1=HOCO)
7:↔ 5	-	Reserved

Bits	Field	Description
13:8	STC	PLL multiplier (value N -> x(N+1))
15:14	-	Reserved

PLIDIV Encoding (bits 1:0)

Value	Divider	Note
00	1	No division
01	2	
10	3	
11	–	PROHIBITED

STAR Project PLL Configuration

Input: 24 MHz main oscillator Target: 240 MHz PLL output Calculation: $24 \text{ MHz} / 1 \times 10 = 240 \text{ MHz}$

- PLIDIV = 00 (divide by 1) -> 24 MHz
- STC = 0x09 (multiply by 10) -> 240 MHz
- PLLSRCSEL = 0 (main oscillator source)
- PLLCR = 0x0900

Warning

PLL must be stopped (PLLCR2.PLEN = 1) before changing PLLCR

Note

Requires PRCR unlock (PRC0) before modification

See also

Manual Ch09 section 9.2.5 PLLCR

Since

Version 1.0.0

Enumerator

k_pllcr_plidiv_1	Divide PLL input by 1 (binary 00)
k_pllcr_plidiv_2	Divide PLL input by 2 (binary 01)
k_pllcr_plidiv_3	Divide PLL input by 3 (binary 10); value 11 is PROHIBITED
k_pllcr_plidiv_mask	PLIDIV field mask
k_pllcr_src_main	Main oscillator as PLL source (STAR config)
k_pllcr_src_hoco	HOCO as PLL source
k_pllcr_src_mask	PLLSRCSEL field mask
k_pllcr_stc_shift	
k_pllcr_stc_mask	
k_pllcr_star_24mhz_to_240mhz	STAR: 24 MHz x 10 = 240 MHz
k_pllcr_12mhz_to_240mhz	Alternative: 12 MHz x 10 = 120 MHz (NOT 240 MHz!)

Definition at line 574 of file rx72n_system_regs.h.

```

00574         : uint16_t {
00575     /* PLL input divider (PLIDIV, bits 1:0) */
00576     k_pllcr_plidiv_1 = 0x0000, /**< Divide PLL input by 1 (binary 00) */
00577     k_pllcr_plidiv_2 = 0x0001, /**< Divide PLL input by 2 (binary 01) */
00578     k_pllcr_plidiv_3 = 0x0002, /**< Divide PLL input by 3 (binary 10); value 11 is PROHIBITED */
00579     k_pllcr_plidiv_mask = 0x0003, /**< PLIDIV field mask */
00580
00581     /* PLL source select (PLLSRCSEL, bit 4) */
00582     k_pllcr_src_main = 0x0000, /**< Main oscillator as PLL source (STAR config) */
00583     k_pllcr_src_hoco = 0x0010, /**< HOCO as PLL source */
00584     k_pllcr_src_mask = 0x0010, /**< PLLSRCSEL field mask */
00585
00586     /* PLL multiplier field (STC, bits 13:8) - value N means multiply by (N+1) */
00587     k_pllcr_stc_shift = 8,
00588     k_pllcr_stc_mask = 0x3F00,
00589
00590     /* Convenience values for common configurations */
00591     k_pllcr_star_24mhz_to_240mhz =
00592         (k_pllcr_plidiv_1 | /* 24 MHz input (no division) */
00593          k_pllcr_src_main | /* Main oscillator source */
00594          (19U « k_pllcr_stc_shift) /* STC=19 -> x10.0 (manual p345: 0b010011) */
00595         ), /**< STAR: 24 MHz x 10 = 240 MHz */
00596
00597     k_pllcr_12mhz_to_240mhz =
00598         (k_pllcr_plidiv_1 | /* 12 MHz input (no division) */
00599          k_pllcr_src_main | /* Main oscillator source */
00600          (19U « k_pllcr_stc_shift) /* STC=19 -> x10.0 (manual p345); = 0x1300 */
00601         ), /**< Alternative: 12 MHz x 10 = 120 MHz (NOT 240 MHz!) */
00602 } pllcr_bits_t;

```

8.59.3.5 rstsr0`bits``t`

```
enum rstsr0_bits_t : uint8_t
```

Enumerator

k_rstsr0_porf	Power-On Reset Detect Flag
k_rstsr0_lvd0rf	Voltage-Monitoring 0 Reset Detect Flag

k_rstsr0_lvd1rf	Voltage-Monitoring 1 Reset Detect Flag
k_rstsr0_lvd2rf	Voltage-Monitoring 2 Reset Detect Flag
k_rstsr0_dpsrstf	Deep Software Standby Reset Flag

Definition at line 998 of file [rx72n_system_regs.h](#).

```
00998      : uint8_t {
00999  k_rstsr0_porf = 1U, /**< Power-On Reset Detect Flag */
01000  k_rstsr0_lvd0rf = 2U, /**< Voltage-Monitoring 0 Reset Detect Flag */
01001  k_rstsr0_lvd1rf = 4U, /**< Voltage-Monitoring 1 Reset Detect Flag */
01002  k_rstsr0_lvd2rf = 8U, /**< Voltage-Monitoring 2 Reset Detect Flag */
01003  k_rstsr0_dpsrstf = 128U, /**< Deep Software Standby Reset Flag */
01004 } rstsr0_bits_t;
```

8.59.3.6 rstsr1_bits_t

```
enum rstsr1\_bits\_t : uint8_t
```

Enumerator

k_rstsr1_cwsf	Cold/Warm Start Determination Flag
---------------	------------------------------------

Definition at line 1007 of file [rx72n_system_regs.h](#).

```
01007      : uint8_t {
01008  k_rstsr1_cwsf = 1U, /**< Cold/Warm Start Determination Flag */
01009 } rstsr1_bits_t;
```

8.59.3.7 rstsr2_bits_t

```
enum rstsr2\_bits\_t : uint8_t
```

Enumerator

k_rstsr2_iwdtrf	Independent Watchdog Timer Reset Detect Flag
k_rstsr2_wdtrf	Watchdog Timer Reset Detect Flag
k_rstsr2_swrf	Software Reset Detect Flag

Definition at line 1012 of file [rx72n_system_regs.h](#).

```
01012      : uint8_t {
01013  k_rstsr2_iwdtrf = 1U, /**< Independent Watchdog Timer Reset Detect Flag */
01014  k_rstsr2_wdtrf = 2U, /**< Watchdog Timer Reset Detect Flag */
01015  k_rstsr2_swrf = 4U, /**< Software Reset Detect Flag */
01016 } rstsr2_bits_t;
```

8.59.3.8 rx_clock_verify_t

```
enum rx\_clock\_verify\_t : uint8_t
```

Reference values for clock field-width and PLL multiplier assertions.

Since

Version 1.0.0

Enumerator

k_nibble_all_ones	All bits set in 4-bit field (verifies 4-bit mask width)
k_pllcr_stc_for_x10	PLLCR STC field value for x10.0 (manual p345: 0b010011=19; formula (STC+1)/2=mult)

Definition at line 1296 of file [rx72n_system_regs.h](#).

```
01296         : uint8_t {
01297   k_nibble_all_ones = 0x0F, /**< All bits set in 4-bit field (verifies 4-bit mask width) */
01298   k_pllcr_stc_for_x10 =
01299     19, /**< PLLCR STC field value for x10.0 (manual p345: 0b010011=19; formula (STC+1)/2=mult) */
01300 } rx_clock_verify_t;
```

8.59.3.9 rx`memwait`addresses`t

```
enum rx_memwait_addresses_t : uintptr_t
```

MEMWAIT register address.

Memory Wait Cycle Setting Register controls wait cycles for flash and ECCRAM. MANDATORY: Must be set to 1 when ICLK > 120 MHz. Reference: RX72N manual Ch09 section 9.2.2, page 341, line 962.

Enumerator

k_memwait_addr	Memory Wait Cycle Setting Register
----------------	------------------------------------

Definition at line 1080 of file [rx72n_system_regs.h](#).

```
01080         : uintptr_t {
01081   k_memwait_addr = 0x0008101C, /**< Memory Wait Cycle Setting Register */
01082 } rx_memwait_addresses_t;
```

8.59.3.10 rx`memwait`bits`t

```
enum rx_memwait_bits_t : uint8_t
```

MEMWAIT register bit values.

- 0: No wait cycle (ICLK <= 120 MHz)
- 1: One wait cycle (ICLK > 120 MHz) - REQUIRED for 240 MHz operation

Enumerator

k_memwait_no_wait	No wait cycle (ICLK <= 120 MHz)
k_memwait_one_wait	One wait cycle (ICLK > 120 MHz) - MANDATORY

Definition at line 1090 of file [rx72n_system_regs.h](#).

```
01090         : uint8_t {
01091   k_memwait_no_wait = 0x00, /**< No wait cycle (ICLK <= 120 MHz) */
01092   k_memwait_one_wait = 0x01, /**< One wait cycle (ICLK > 120 MHz) - MANDATORY */
01093 } rx_memwait_bits_t;
```

8.59.3.11 rx`module`stop`bits`a`b`it

```
enum rx_module_stop_bits_a_t : uint8_t
```

Module stop bits for MSTPCRA register.

Enumerator

k_mstpa_cmt23	CMT2/CMT3 module stop bit in MSTPCRA
---------------	--------------------------------------

Definition at line 939 of file [rx72n_system_regs.h](#).

```
00939         : uint8_t {
00940   k_mstpa_cmt23 = 14, /**< CMT2/CMT3 module stop bit in MSTPCRA */
00941 } rx_module_stop_bits_a_t;
```

8.59.3.12 rx`module`stop`bits`b`b`it

```
enum rx_module_stop_bits_b_t : uint8_t
```

Module stop bits for MSTPCRB register.

Enumerator

k_mstpb_usb0	USB0 module stop bit in MSTPCRB
k_mstpb_crc	CRC module stop bit in MSTPCRB

Definition at line 944 of file [rx72n_system_regs.h](#).

```
00944         : uint8_t {
00945   k_mstpb_usb0 = 19, /**< USB0 module stop bit in MSTPCRB */
00946   k_mstpb_crc = 23, /**< CRC module stop bit in MSTPCRB */
00947 } rx_module_stop_bits_b_t;
```

8.59.3.13 rx`ppll`addresses`b`b`it

```
enum rx_ppll_addresses_t : uintptr_t
```

PPLL register addresses.

PPLL (PLL for specific purposes) generates the 48 MHz USB clock (UCLK). Must be configured for USB functionality. Reference: RX72N manual Ch09, PPLL frequency synthesizer section.

Enumerator

k_pllcr_addr	PPLL Control Register @ 0x00080048
k_pllcr2_addr	PPLL Control Register 2 @ 0x0008004A
k_pllcr3_addr	PPLL Control Register 3 @ 0x0008004B (PPLCK divider)

Definition at line 1117 of file [rx72n_system_regs.h](#).

```
01117         : uintptr_t {
01118   k_pllcr_addr = 0x00080048, /**< PPLL Control Register @ 0x00080048 */
01119   k_pllcr2_addr = 0x0008004A, /**< PPLL Control Register 2 @ 0x0008004A */
01120   k_pllcr3_addr = 0x0008004B, /**< PPLL Control Register 3 @ 0x0008004B (PPLCK divider) */
01121 } rx_ppll_addresses_t;
```

8.59.3.14 rx_ppll_config_t

```
enum rx_ppll_config_t : uint16_t
```

PPLL configuration values.

PPLL generates 240 MHz internal clock from 24 MHz main oscillator. PPLLCR3 then divides by 5 to produce 48 MHz USB clock (UCLK). Calculation: 24 MHz * 10 = 240 MHz (PPLL output); 240 / 5 = 48 MHz (UCLK)

- PPLIDIV[1:0] = 00 (divide input by 1, binary 00)
- PPLSTC[5:0] = 13h (multiply by x10.0, STC=19=0b010011 at bits 13:8; formula (STC+1)/2=mult)
- PPLSRCSEL = 0 (main clock source, shared with PLL via PLLCR.PLLSRCSEL)
- PPLLCR = 0x1300

Per manual Ch09 section 9.2.23 (p366): PPLSTC minimum valid value is 0b010011 (x10.0). Settings below x10.0 are prohibited – PPLL output must be 120-240 MHz.

Enumerator

k_ppll_config_48mhz	PPLLCR: PPLSTC=19 (x10.0), PPLIDIV=00 (/1): 24MHz*10=240MHz
---------------------	---

Definition at line 1137 of file rx72n_system_regs.h.

```
01137      : uint16_t {
01138  k_ppll_config_48mhz = 0x1300, /**< PPLLCR: PPLSTC=19 (x10.0), PPLIDIV=00 (/1): 24MHz*10=240MHz */
01139 } rx_ppll_config_t;
```

8.59.3.15 rx_ppll_control_t

```
enum rx_ppll_control_t : uint8_t
```

PPLL control values.

Enumerator

k_ppll_enabled	PPLLCR2: Enable PPLL
k_ppll_disabled	PPLLCR2: Disable PPLL

Definition at line 1157 of file rx72n_system_regs.h.

```
01157      : uint8_t {
01158  k_ppll_enabled = 0x00, /**< PPLLCR2: Enable PPLL */
01159  k_ppll_disabled = 0x01, /**< PPLLCR2: Disable PPLL */
01160 } rx_ppll_control_t;
```

8.59.3.16 rx_ppll_flags_t

```
enum rx_ppll_flags_t : uint8_t
```

PPLL stabilization flag in OSCOVFSR.

OSCOVFSR bit layout (manual Ch09 section 9.2.14):

- b0 MOOVF: Main clock oscillation stabilization flag
- b1 SOOVF: Sub-clock oscillation stabilization flag
- b2 PLOVF: PLL clock oscillation stabilization flag
- b3 HCOVF: HOCO clock oscillation stabilization flag
- b4 ILCOVF: IWDG-dedicated clock oscillation stabilization flag
- b5 PPLOVF: PPLL clock oscillation stabilization flag

Enumerator

k_ppll_stable_flag	OSCOVFSR: PPLOVF at bit 5 (0x20); NOT bit 3
--------------------	---

Definition at line 1173 of file rx72n_system_regs.h.

```
01173 : uint8_t {
01174 k_ppll_stable_flag = 0x20, /**< OSCOVFSR: PPLOVF at bit 5 (0x20); NOT bit 3 */
01175 } rx_ppll_flags_t;
```

8.59.3.17 rx_pllcr3_config_t

```
enum rx_pllcr3_config_t : uint8_t
```

PPLLCR3 frequency divider values.

PPLLCR3 PPLCK[3:0] selects the division ratio applied to the PPLL output before it reaches the USB clock (UCLK) pin via PACKCR.UPLLSEL=1. Per manual Ch09 section 9.2.25 (p368): 0001=/2, 0010=/3, 0011=/4, 0100=/5 (settings other than these are prohibited) Set AFTER PPLL is operating and stable (PPLOVF=1), while USB is stopped.

Enumerator

k_pllcr3_div5	PPLLCR3: PPLCK=0100=/5, 240MHz/5=48MHz USB (manual p368)
---------------	--

Definition at line 1150 of file rx72n_system_regs.h.

```
01150 : uint8_t {
01151 k_pllcr3_div5 = 0x04, /**< PPLLCR3: PPLCK=0100=/5, 240MHz/5=48MHz USB (manual p368) */
01152 } rx_pllcr3_config_t;
```

8.59.3.18 rx`prcr`offset``t`

```
enum rx_prcr_offset_t : uint16_t
```

PRCR register offset from system register base address.

PRCR is not part of `rx_system_regs_t`; it sits at base+0x3FE.

Since

Version 1.0.0

Enumerator

k_prcr_from_sys_base	PRCR offset from system base (NOT in system struct)
----------------------	---

Definition at line 1287 of file `rx72n_system_regs.h`.

```
01287      : uint16_t {
01288  k_prcr_from_sys_base = 0x03FE, /**< PRCR offset from system base (NOT in system struct) */
01289 } rx_prcr_offset_t;
```

8.59.3.19 rx`rstsr01`layout``t`

```
enum rx_rstsr01_layout_t : uint8_t
```

Expected size and member offsets for `rx_rstsr01_regs_t`.

Since

Version 1.0.0

Enumerator

k_rstsr01_struct_size	Total size of <code>rx_rstsr01_regs_t</code> (2 bytes)
k_rstsr0_member_off	Byte offset of rstsr0 member
k_rstsr1_member_off	Byte offset of rstsr1 member

Definition at line 1275 of file `rx72n_system_regs.h`.

```
01275      : uint8_t {
01276  k_rstsr01_struct_size = 2, /**< Total size of rx_rstsr01_regs_t (2 bytes) */
01277  k_rstsr0_member_off  = 0, /**< Byte offset of rstsr0 member */
01278  k_rstsr1_member_off  = 1, /**< Byte offset of rstsr1 member */
01279 } rx_rstsr01_layout_t;
```


8.59.3.20 rx_rstsr_addresses_t

```
enum rx_rstsr_addresses_t : uintptr_t
```

Reset status register addresses.

IMPORTANT: RSTSR registers are NOT contiguous in memory!

- RSTSR0/1 are adjacent at 0x0008C290-0x0008C291
- RSTSR2 is at a separate address 0x000800C0 Verified against RX72N Group User's Manual Hardware (R01UH0824EJ0120 Rev.1.20)

Enumerator

k_rstsr0_addr	Reset Status Register 0 address (page 286)
k_rstsr1_addr	Reset Status Register 1 address (page 288)
k_rstsr2_addr	Reset Status Register 2 address (page 289)

Definition at line 962 of file [rx72n_system_regs.h](#).

```
00962     : uintptr_t {
00963   k_rstsr0_addr = 0x0008C290, /**< Reset Status Register 0 address (page 286) */
00964   k_rstsr1_addr = 0x0008C291, /**< Reset Status Register 1 address (page 288) */
00965   k_rstsr2_addr = 0x000800C0, /**< Reset Status Register 2 address (page 289) */
00966 } rx_rstsr_addresses_t;
```

8.59.3.21 rx_swrr_addresses_t

```
enum rx_swrr_addresses_t : uintptr_t
```

Software Reset Register (SWRR) address.

Writing 0xA501 to this register triggers an immediate software reset. Reference: RX72N manual Ch06 section 6.2.4, page 290.

Enumerator

k_swrr_addr	Software Reset Register address (16-bit)
-------------	--

Definition at line 1029 of file [rx72n_system_regs.h](#).

```
01029     : uintptr_t {
01030   k_swrr_addr = 0x000800C2, /**< Software Reset Register address (16-bit) */
01031 } rx_swrr_addresses_t;
```

8.59.3.22 rx_swrr_values_t

```
enum rx_swrr_values_t : uint16_t
```

Software Reset Register value.

Write this value to SWRR to trigger a software reset. The MCU will reset after the internal reset time (tRESW2) elapses.

Warning

This immediately resets the MCU - ensure all critical state is saved.

Enumerator

k_swrr_reset_key	Write this value to trigger software reset
------------------	--

Definition at line 1040 of file [rx72n_system_regs.h](#).

```
01040      : uint16_t {
01041    k_swrr_reset_key = 0xA501, /**< Write this value to trigger software reset */
01042  } rx_swrr_values_t;
```

8.59.3.23 rx_system_addr_refs_t

```
enum rx_system_addr_refs_t : uintptr_t
```

Expected hardware addresses per RX72N Hardware Manual.

Each member documents the address mandated by the hardware manual chapter cited in the comment. Used in static_assert to cross-check address enum definitions against the manual reference values.

Since

Version 1.0.0

Enumerator

k_ref_system_base	System register base address (Ch05, Sec 5.2.2)
k_ref_prcr	PRCR address – outside system struct (Ch05)
k_ref_rstsr0	RSTSR0 address (Ch05)
k_ref_rstsr1	RSTSR1 address (Ch05)
k_ref_rstsr2	RSTSR2 address (Ch05)
k_ref_swrr	SWRR address (Ch06)
k_ref_memwait	MEMWAIT address (Ch09)
k_ref_pllcr	PPLLCR address (Ch09)
k_ref_pllcr2	PPLLCR2 address (Ch09)
k_ref_pllcr3	PPLLCR3 address (Ch09 section 9.2.25, p368)

Definition at line 1221 of file [rx72n_system_regs.h](#).

```
01221      : uintptr_t {
01222    k_ref_system_base = 0x00080000, /**< System register base address (Ch05, Sec 5.2.2) */
01223    k_ref_prcr        = 0x000803FE, /**< PRCR address -- outside system struct (Ch05) */
01224    k_ref_rstsr0      = 0x0008C290, /**< RSTSR0 address (Ch05) */
01225    k_ref_rstsr1      = 0x0008C291, /**< RSTSR1 address (Ch05) */
01226    k_ref_rstsr2      = 0x000800C0, /**< RSTSR2 address (Ch05) */
01227    k_ref_swrr        = 0x000800C2, /**< SWRR address (Ch06) */
01228    k_ref_memwait     = 0x0008101C, /**< MEMWAIT address (Ch09) */
01229    k_ref_pllcr       = 0x00080048, /**< PPLLCR address (Ch09) */
01230    k_ref_pllcr2      = 0x0008004A, /**< PPLLCR2 address (Ch09) */
01231    k_ref_pllcr3      = 0x0008004B, /**< PPLLCR3 address (Ch09 section 9.2.25, p368) */
01232  } rx_system_addr_refs_t;
```

8.59.3.24 rx_system_addresses_t

```
enum rx_system_addresses_t : uintptr_t
```

System register base addresses.

Base addresses for system control registers. Note that system registers are spread across multiple non-contiguous memory regions.

Address Map

Address	Register/Block	Description
0x00080000	System Base	Main system register block
0x000803FE	PRCR	Register protection control

Warning

PRCR is NOT in the main system block - use [prcr_reg\(\)](#) accessor

See also

[system_regs\(\)](#) Main system register accessor

[prcr_reg\(\)](#) PRCR register accessor

Since

Version 1.0.0

Enumerator

k_system_base_addr	System register base address (0x00080000). Start of main 66-byte system register block (0x00-0x41)
k_pcr_addr	PRCR (Protection Register) address (0x000803FE). Located separately from main system block. Used to protect clock, module stop, and voltage detection registers from accidental writes. Warning Not part of rx_system_regs_t structure

Definition at line 148 of file [rx72n_system_regs.h](#).

```
00148     : uintptr_t {
00149     /**
00150      * @brief System register base address (0x00080000)
00151      * @details Start of main 66-byte system register block (0x00-0x41)
00152      */
00153     k_system_base_addr = 0x00080000,
00154
00155     /**
00156      * @brief PRCR (Protection Register) address (0x000803FE)
00157      * @details Located separately from main system block. Used to protect
00158      * clock, module stop, and voltage detection registers from accidental writes.
00159      * @warning Not part of rx_system_regs_t structure
00160      */
00161     k_pcr_addr = 0x000803FE,
00162 } rx_system_addresses_t;
```

8.59.3.25 rx_system_struct_layout_t

```
enum rx_system_struct_layout_t : uint8_t
```

Expected size and member offsets for [rx_system_regs_t](#).

Authoritative per RX72N Hardware Manual Ch05, Section 5.2.2 address map. Used in static_assert to verify the compiler produces the correct layout.

Since

Version 1.0.0

Enumerator

k_sys_regs_size	Total size of rx_system_regs_t (66 bytes, 0x00-0x41)
k_sys_reg_mdmmonr_off	Offset of MDMONR (mode monitor)
k_sys_reg_syscr0_off	Offset of SYSCR0
k_sys_reg_syscr1_off	Offset of SYSCR1
k_sys_reg_sbycr_off	Offset of SBYCR (standby control)
k_sys_reg_mstpcra_off	Offset of MSTPCRA
k_sys_reg_mstpcrb_off	Offset of MSTPCRB
k_sys_reg_mstpcrc_off	Offset of MSTPCRC
k_sys_reg_mstpcrd_off	Offset of MSTPCRD
k_sys_reg_sckcr_off	Offset of SCKCR (system clock control)
k_sys_reg_sckcr2_off	Offset of SCKCR2
k_sys_reg_sckcr3_off	Offset of SCKCR3 (clock source select)
k_sys_reg_pllcr_off	Offset of PLLCR
k_sys_reg_pllcr2_off	Offset of PLLCR2 (PLL enable)
k_sys_reg_bckcr_off	Offset of BCKCR (external bus clock)
k_sys_reg_mosccr_off	Offset of MOSCCR (main oscillator control)
k_sys_reg_sosccr_off	Offset of SOSCCR (sub-clock control)
k_sys_reg_lococr_off	Offset of LOCOCR (LOCO control)
k_sys_reg_ilococr_off	Offset of ILOCOCR (IWDT oscillator control)
k_sys_reg_hococr_off	Offset of HOCOCR (HOCO control)
k_sys_reg_hococr2_off	Offset of HOCOCR2 (HOCO frequency)
k_sys_reg_oscovfsr_off	Offset of OSCOVFSR (oscillation stable flags)
k_sys_reg_ckocr_off	Offset of CKOCR (clock output control)
k_sys_reg_ostdcr_off	Offset of OSTDCR (oscillation stop detect control)
k_sys_reg_ostdsr_off	Offset of OSTDSR (oscillation stop detect status)

Definition at line 1242 of file [rx72n_system_regs.h](#).

```
01242         : uint8_t {
01243     k_sys_regs_size      = 0x42, /**< Total size of rx_system_regs_t (66 bytes, 0x00-0x41) */
01244     k_sys_reg_mdmmonr_off = 0x00, /**< Offset of MDMONR (mode monitor) */
01245     k_sys_reg_syscr0_off  = 0x06, /**< Offset of SYSCR0 */
01246     k_sys_reg_syscr1_off  = 0x08, /**< Offset of SYSCR1 */
01247     k_sys_reg_sbycr_off   = 0x0C, /**< Offset of SBYCR (standby control) */
01248     k_sys_reg_mstpcra_off = 0x10, /**< Offset of MSTPCRA */
```

```

01249 k_sys_reg_mstpcrb_off = 0x14, /**< Offset of MSTPCRB */
01250 k_sys_reg_mstpcrc_off = 0x18, /**< Offset of MSTPCRC */
01251 k_sys_reg_mstpcrd_off = 0x1C, /**< Offset of MSTPCRD */
01252 k_sys_reg_sckcr_off = 0x20, /**< Offset of SCKCR (system clock control) */
01253 k_sys_reg_sckcr2_off = 0x24, /**< Offset of SCKCR2 */
01254 k_sys_reg_sckcr3_off = 0x26, /**< Offset of SCKCR3 (clock source select) */
01255 k_sys_reg_pllcr_off = 0x28, /**< Offset of PLLCR */
01256 k_sys_reg_pllcr2_off = 0x2A, /**< Offset of PLLCR2 (PLL enable) */
01257 k_sys_reg_bckcr_off = 0x30, /**< Offset of BCKCR (external bus clock) */
01258 k_sys_reg_mosccr_off = 0x32, /**< Offset of MOSCCR (main oscillator control) */
01259 k_sys_reg_sosccr_off = 0x33, /**< Offset of SOSCCR (sub-clock control) */
01260 k_sys_reg_lococr_off = 0x34, /**< Offset of LOCOCR (LOCO control) */
01261 k_sys_reg_ilococr_off = 0x35, /**< Offset of ILOCOCR (IWDT oscillator control) */
01262 k_sys_reg_hococr_off = 0x36, /**< Offset of HOCOcr (HOCO control) */
01263 k_sys_reg_hococr2_off = 0x37, /**< Offset of HOCOcr2 (HOCO frequency) */
01264 k_sys_reg_oscovfsr_off = 0x3C, /**< Offset of OSCOVFSR (oscillation stable flags) */
01265 k_sys_reg_ckocr_off = 0x3E, /**< Offset of CKOCR (clock output control) */
01266 k_sys_reg_ostdcr_off = 0x40, /**< Offset of OSTDCR (oscillation stop detect control) */
01267 k_sys_reg_ostdsr_off = 0x41, /**< Offset of OSTDSR (oscillation stop detect status) */
01268 } rx_system_struct_layout_t;

```

8.59.3.26 sckcr3`cksel` t

```
enum sckcr3_cksel_t : uint16_t
```

System clock source selection (SCKCR3 CKSEL field).

Selects which oscillator/PLL output is used as the system clock source. This is the clock that feeds into SCKCR dividers.

Clock Sources (CKSEL[2:0
at bits 10:8 of SCKCR3])

Value	Source	Typical Frequency	Use Case
0	LOCO	240 kHz (typ)	Low power mode
1	HOCO	16-20 MHz (config)	Boot, low power
2	Main clock	24 MHz (STAR)	External crystal
3	Sub-clock	32.768 kHz	RTC, ultra low power
4	PLL	240 MHz (STAR)	Normal operation
other	—	PROHIBITED	—

Note

PPLL is NOT a valid SCKCR3 CKSEL option. PPLL is used only as the USB clock source, selected via PACKCR.UPLLSEL.

STAR Project Settings

- Boot: LOCO (default after reset)
- Normal operation: PLL @ 240 MHz

Warning

Switching clock sources requires waiting for oscillator stabilization

Note

Requires PRCR unlock (PRC0) before modification

See also

Manual Ch09 section 9.2.4 SCKCR3

Since

Version 1.0.0

Enumerator

k_sckcr3_cksel_loco	LOCO (Low-speed On-Chip Oscillator, 240 kHz)
k_sckcr3_cksel_hoco	HOCO (High-speed On-Chip Oscillator, 16-20 MHz)
k_sckcr3_cksel_main	Main clock oscillator (STAR: 24 MHz crystal)
k_sckcr3_cksel_subclock	Sub-clock oscillator (32.768 kHz)
k_sckcr3_cksel_pll	PLL circuit (STAR: 240 MHz). Values 5-7 are prohibited.
k_sckcr3_cksel_mask	CKSEL field mask (bits 10:8)

Definition at line 525 of file [rx72n_system_regs.h](#).

```

00525     : uint16_t {
00526 k_sckcr3_cksel_loco    = 0x0000, /**< LOCO (Low-speed On-Chip Oscillator, 240 kHz) */
00527 k_sckcr3_cksel_hoco    = 0x0100, /**< HOCO (High-speed On-Chip Oscillator, 16-20 MHz) */
00528 k_sckcr3_cksel_main    = 0x0200, /**< Main clock oscillator (STAR: 24 MHz crystal) */
00529 k_sckcr3_cksel_subclock = 0x0300, /**< Sub-clock oscillator (32.768 kHz) */
00530 k_sckcr3_cksel_pll     = 0x0400, /**< PLL circuit (STAR: 240 MHz). Values 5-7 are prohibited. */
00531 k_sckcr3_cksel_mask    = 0x0700, /**< CKSEL field mask (bits 10:8) */
00532 } sckcr3_cksel_t;

```

8.59.3.27 sckcr`bits``t`

```
enum sckcr\_bits\_t : uint32_t
```

System Clock Control Register (SCKCR) bit definitions.

32-bit register controlling all main system clock dividers. Each field is 4 bits wide and uses the encoding from [sckcr_divider_t](#).

Register Layout (32-bit @ offset 0x20)

Bits	Field	Clock	STAR Project Setting
3:↔ 0	PCKD	PCLKD	/4 (60 MHz from 240)
7:↔ 4	PCKC	PCLKC	/4 (60 MHz from 240)
11:↔ 8	PCKB	PCLKB	/4 (60 MHz from 240)

Bits	Field	Clock	STAR Project Setting
15:12	PCKA	PCLKA	/2 (120 MHz from 240)
19:16	BCK	BCLK	/2 (120 MHz from 240)
21:20	-	Reserved	-
22	PSTOP0	PCLKA/B	0 (running)
23	PSTOP1	USB	0 (running)
27:24	ICK	ICLK	/1 (240 MHz)
31:28	FCK	FCLK	/4 (60 MHz from 240)

STAR Project Configuration

With 24 MHz main oscillator x PLL(x10) = 240 MHz:

- ICLK = 240 MHz (CPU instruction clock)
- PCLKA = 120 MHz (high-speed peripherals)
- PCLKB = 60 MHz (CMT, WDT, etc.)
- PCLKC = 60 MHz (general peripherals)
- PCLKD = 60 MHz (ADC)
- FCLK = 60 MHz (Flash memory)
- BCLK = 120 MHz (External bus)

Warning

ICLK > 120 MHz requires MEMWAIT = 1 before changing SCKCR

Note

Requires PRCR unlock (PRC0) before modification

See also

Manual Ch09 section 9.2.2

Since

Version 1.0.0

Enumerator

k_sckcr_pckd_shift	
k_sckcr_pckc_shift	
k_sckcr_pckb_shift	
k_sckcr_pcka_shift	
k_sckcr_bck_shift	
k_sckcr_ick_shift	

k_sckcr_fck_shift	
k_sckcr_pckd_mask	
k_sckcr_pckc_mask	
k_sckcr_pckb_mask	
k_sckcr_pcka_mask	
k_sckcr_bck_mask	
k_sckcr_ick_mask	
k_sckcr_fck_mask	
k_sckcr_pstop0	SDCLK pin output control (0=enabled, 1=disabled/fixed-high)
k_sckcr_pstop1	BCLK pin output control (0=enabled, 1=disabled/fixed-high)
k_sckcr_star_240mhz	Complete SCKCR value for STAR 240 MHz configuration

Definition at line 459 of file `rx72n_system_regs.h`.

```

00459         : uint32_t {
00460     /* Field shifts (for manual bit manipulation if needed) */
00461     k_sckcr_pckd_shift = 0,
00462     k_sckcr_pckc_shift = 4,
00463     k_sckcr_pckb_shift = 8,
00464     k_sckcr_pcka_shift = 12,
00465     k_sckcr_bck_shift  = 16,
00466     k_sckcr_ick_shift  = 24,
00467     k_sckcr_fck_shift  = 28,
00468
00469     /* Field masks */
00470     k_sckcr_pckd_mask = 0xFUL « k_sckcr_pckd_shift,
00471     k_sckcr_pckc_mask = 0xFUL « k_sckcr_pckc_shift,
00472     k_sckcr_pckb_mask = 0xFUL « k_sckcr_pckb_shift,
00473     k_sckcr_pcka_mask = 0xFUL « k_sckcr_pcka_shift,
00474     k_sckcr_bck_mask  = 0xFUL « k_sckcr_bck_shift,
00475     k_sckcr_ick_mask  = 0xFUL « k_sckcr_ick_shift,
00476     k_sckcr_fck_mask  = 0xFUL « k_sckcr_fck_shift,
00477
00478     /* PSTOP bits (pin output control) */
00479     k_sckcr_pstop0 = (1UL « 22), /**< SDCLK pin output control (0=enabled, 1=disabled/fixed-high) */
00480     k_sckcr_pstop1 = (1UL « 23), /**< BCLK pin output control (0=enabled, 1=disabled/fixed-high) */
00481
00482     /* STAR Project configuration value (24 MHz x 10 = 240 MHz) */
00483     k_sckcr_star_240mhz =
00484     (((uint32_t)k_clock_div_4 « k_sckcr_pckd_shift) | /* PCLKD = 240/4 = 60 MHz */
00485     ((uint32_t)k_clock_div_4 « k_sckcr_pckc_shift) | /* PCLKC = 240/4 = 60 MHz */
00486     ((uint32_t)k_clock_div_4 « k_sckcr_pckb_shift) | /* PCLKB = 240/4 = 60 MHz */
00487     ((uint32_t)k_clock_div_2 « k_sckcr_pcka_shift) | /* PCLKA = 240/2 = 120 MHz */
00488     ((uint32_t)k_clock_div_2 « k_sckcr_bck_shift) | /* BCLK = 240/2 = 120 MHz */
00489     ((uint32_t)k_clock_div_1 « k_sckcr_ick_shift) | /* ICLK = 240/1 = 240 MHz */
00490     ((uint32_t)k_clock_div_4 « k_sckcr_fck_shift) | /* FCLK = 240/4 = 60 MHz */
00491     0UL, /* PSTOP0/1 = 0 (clocks running) */
00492     ), /**< Complete SCKCR value for STAR 240 MHz configuration */
00493 } sckcr_bits_t;

```

8.59.3.28 sckcr_divider_t

```
enum sckcr_divider_t : uint8_t
```

Clock divider values for SCKCR fields.

All SCKCR clock divider fields use the same encoding. Value N means divide by 2^N . For example, 0b0010 (2) means divide by 4.

Divider Encoding (for PCKD, PCKC, PCKB, PCKA, ICK, FCK fields)

Bits	Divider	Example (240 MHz input)
0000	$\swarrow$ 1	240 MHz
0001	$\swarrow$ 2	120 MHz
0010	$\swarrow$ 4	60 MHz
0011	$\swarrow$ 8	30 MHz
0100	/16	15 MHz
0101	/32	7.5 MHz
0110	/64	3.75 MHz
other	–	PROHIBITED

Note

The BCK field also accepts 1001 (x1/3) as an additional valid setting.

Warning

Values 0111 and 1000 are PROHIBITED for all SCKCR fields.

See also

Manual Ch09 section 9.2.1 Table 9.1, section 9.2.2

Since

Version 1.0.0

Enumerator

k_clock_div_1	Divide by 1 (no division)
k_clock_div_2	Divide by 2
k_clock_div_4	Divide by 4
k_clock_div_8	Divide by 8
k_clock_div_16	Divide by 16
k_clock_div_32	Divide by 32
k_clock_div_64	Divide by 64
k_bck_div_3	Divide by 3 (ONLY valid for BCK[3:0] field, binary 1001)

Definition at line 408 of file [rx72n_system_regs.h](#).

```

00408      : uint8_t {
00409  k_clock_div_1 = 0, /**< Divide by 1 (no division) */
00410  k_clock_div_2 = 1, /**< Divide by 2 */
00411  k_clock_div_4 = 2, /**< Divide by 4 */
00412  k_clock_div_8 = 3, /**< Divide by 8 */
00413  k_clock_div_16 = 4, /**< Divide by 16 */
00414  k_clock_div_32 = 5, /**< Divide by 32 */
00415  k_clock_div_64 = 6, /**< Divide by 64 */
00416  /* NOTE: /128 does NOT exist in SCKCR. Maximum divisor is /64 (0110 = 6). */
00417  /* Values 7 and 8 are PROHIBITED per RX72N Manual Ch09 section 9.2.2. */
00418  /* BCK field additionally accepts k_bck_div_3 = 9 (binary 1001 = x1/3). */
00419  k_bck_div_3 = 9, /**< Divide by 3 (ONLY valid for BCK[3:0] field, binary 1001) */
00420 } sckcr_divider_t;

```

8.59.3.29 syscr0`bits`

enum [syscr0_bits_t](#) : uint16_t

System Control Register 0 (SYSCR0) bit definitions.

Controls on-chip ROM enable and external bus enable. Requires key code 0x5A in upper byte for writes. Located at address 0x00080006.

Register Layout (16-bit @ offset 0x06)

Bit	Field	Description
0	ROME	On-Chip ROM Enable
1	EXBE	External Bus Enable
7↔ 2	-	Reserved (write 0)
15↔ 8	KEY[7↔ 0]	Key code (write 0x5A to modify)

Operating Modes

ROME	EXBE	Mode
1	0	Single-chip mode
1	1	On-chip ROM enabled extended mode
0	1	On-chip ROM disabled extended mode
0	0	Single-chip mode (ROM disabled)

Warning

Once ROME is set to 0, it cannot be reverted to 1
Do not modify EXBE while accessing external bus

Note

Manual ref: Ch03 section 3.2.2

Since

Version 1.0.0

Enumerator

k_syscr0_rome	On-Chip ROM Enable (0=disabled, 1=enabled)
k_syscr0_exbe	External Bus Enable (0=disabled, 1=enabled)
k_syscr0_key	Key code for SYSCR0 write (must be in upper byte)
k_syscr0_key_mask	Key code mask
k_syscr0_rom_enabled	Enable ROM (write value)
k_syscr0_rom_disabled	Disable ROM (write value) - IRREVERSIBLE
k_syscr0_extbus_enable	Enable ROM + ExtBus (write value)

Definition at line 246 of file rx72n_system_regs.h.

```

00246      : uint16_t {
00247  k_syscr0_rome    = 0x0001U, /**< On-Chip ROM Enable (0=disabled, 1=enabled) */
00248  k_syscr0_exbe    = 0x0002U, /**< External Bus Enable (0=disabled, 1=enabled) */
00249  k_syscr0_key     = 0x5A00U, /**< Key code for SYSCR0 write (must be in upper byte) */
00250  k_syscr0_key_mask = 0xFF00U, /**< Key code mask */
00251
00252  /** Convenience values for common write operations */
00253  k_syscr0_rom_enabled = (0x5A00U | 0x0001U), /**< Enable ROM (write value) */
00254  k_syscr0_rom_disabled = 0x5A00U, /**< Disable ROM (write value) - IRREVERSIBLE */
00255  k_syscr0_extbus_enable = (0x5A00U | 0x0003U), /**< Enable ROM + ExtBus (write value) */
00256 } syscr0_bits_t;

```

8.59.3.30 syscr1`bits`

enum syscr1_bits_t : uint16_t

System Control Register 1 (SYSCR1) bit definitions.

Controls RAM, ECCRAM, and Standby RAM enable. Located at address 0x00080008.

Register Layout (16-bit @ offset 0x08)

Bit	Field	Description
0	RAME	RAM Enable
5:↵ 1	-	Reserved (read/write as 1)
6	ECCRAME	ECCRAM Enable
7	SBYRAME	Standby RAM Enable
15:↵ 8	-	Reserved (write 0)

Default Value: 0x00FF (all RAM enabled)

- RAME = 1 (RAM enabled)
- Reserved bits 5:1 = 1
- ECCRAME = 1 (ECCRAM enabled)
- SBYRAME = 1 (Standby RAM enabled)

Warning

Do not write 0 to RAME/ECCRAM/SBYRAME during access to that memory

Note

Manual ref: Ch03 section 3.2.3

Since

Version 1.0.0

Enumerator

k_syscr1_rame	RAM Enable (0=disabled, 1=enabled)
k_syscr1_reserved_mask	Reserved bits 5:1 (read/write as 1)
k_syscr1_eccrame	ECCRAM Enable (0=disabled, 1=enabled)
k_syscr1_sbyrame	Standby RAM Enable (0=disabled, 1=enabled)
k_syscr1_all_ram_enabled	Default: all RAM types enabled
k_syscr1_disable_eccram	Disable ECCRAM (RAME+SBYRAME enabled)
k_syscr1_disable_sbyram	Disable StandbyRAM (RAME+ECCRAM enabled)

Definition at line 286 of file [rx72n_system_regs.h](#).

```

00286         : uint16_t {
00287     k_syscr1_rame      = 0x0001U, /**< RAM Enable (0=disabled, 1=enabled) */
00288     k_syscr1_reserved_mask = 0x003EU, /**< Reserved bits 5:1 (read/write as 1) */
00289     k_syscr1_eccrame    = 0x0040U, /**< ECCRAM Enable (0=disabled, 1=enabled) */
00290     k_syscr1_sbyrame    = 0x0080U, /**< Standby RAM Enable (0=disabled, 1=enabled) */
00291
00292     /* Convenience values */
00293     k_syscr1_all_ram_enabled = 0x00FFU, /**< Default: all RAM types enabled */
00294     k_syscr1_disable_eccram = 0x00BFU, /**< Disable ECCRAM (RAME+SBYRAME enabled) */
00295     k_syscr1_disable_sbyram = 0x007FU, /**< Disable StandbyRAM (RAME+ECCRAM enabled) */
00296 } syscr1_bits_t;

```

8.59.3.31 system`reserved`sizes`t

```
enum system\_reserved\_sizes\_t : uint8_t
```

System register reserved field sizes.

Defines reserved byte counts for gaps in the system register map. These ensure correct structure alignment with hardware.

Since

Version 1.0.0

Enumerator

k_system_reserved_02_05	Reserved @ 0x02-0x05 (between MDMONR and SYSCR0)
k_system_reserved_0a_0b	Reserved @ 0x0A-0x0B (between SYSCR1 and SBYCR)

<code>k_system_reserved_0e_0f</code>	Reserved @ 0x0E-0x0F (between SBYCR and MSTPCRA)
<code>k_system_reserved_2b_2f</code>	Reserved @ 0x2B-0x2F (between PLLCR2 and BCKCR)
<code>k_system_reserved_31</code>	Reserved @ 0x31 (between BCKCR and MOSCCR)
<code>k_system_reserved_38_3b</code>	Reserved @ 0x38-0x3B (between HOCOCR2 and OSCOVFSR)
<code>k_system_reserved_3d</code>	Reserved @ 0x3D (between OSCOVFSR and CKOCR)

Definition at line 174 of file [rx72n_system_regs.h](#).

```

00174     : uint8_t {
00175     k_system_reserved_02_05 = 4, /**< Reserved @ 0x02-0x05 (between MDMONR and SYSCR0) */
00176     k_system_reserved_0a_0b = 2, /**< Reserved @ 0x0A-0x0B (between SYSCR1 and SBYCR) */
00177     k_system_reserved_0e_0f = 2, /**< Reserved @ 0x0E-0x0F (between SBYCR and MSTPCRA) */
00178     k_system_reserved_2b_2f = 5, /**< Reserved @ 0x2B-0x2F (between PLLCR2 and BCKCR) */
00179     k_system_reserved_31    = 1, /**< Reserved @ 0x31 (between BCKCR and MOSCCR) */
00180     k_system_reserved_38_3b = 4, /**< Reserved @ 0x38-0x3B (between HOCOCR2 and OSCOVFSR) */
00181     k_system_reserved_3d    = 1, /**< Reserved @ 0x3D (between OSCOVFSR and CKOCR) */
00182 } system_reserved_sizes_t;

```

8.59.4 Function Documentation

8.59.4.1 `memwait_reg()`

```
volatile uint8_t * memwait_reg (
    void ) [inline], [static]
```

Get pointer to MEMWAIT register.

Returns

Volatile pointer to MEMWAIT register

Note

CRITICAL: Must set to 1 before increasing ICLK above 120 MHz

Definition at line 1100 of file [rx72n_system_regs.h](#).

```

01101 {
01102     return (volatile uint8_t*)k_memwait_addr;
01103 }

```

References [k_memwait_addr](#).

8.59.4.2 `ppllcr2_reg()`

```
volatile uint8_t * ppllcr2_reg (
    void ) [inline], [static]
```

Get pointer to PPLLCR2 register.

Returns

Volatile pointer to PPLLCR2 register

Definition at line 1190 of file [rx72n_system_regs.h](#).

```

01191 {
01192     return (volatile uint8_t*)k_ppllcr2_addr;
01193 }

```

References [k_ppllcr2_addr](#).

8.59.4.3 ppllcr3_reg()

```
volatile uint8_t * ppllcr3_reg (  
    void )    [inline], [static]
```

Get pointer to PPLLCR3 register.

Returns

Volatile pointer to PPLLCR3 register

Note

Set PPLCK[3:0] while PLL is running and USB module is stopped.

Definition at line 1200 of file [rx72n_system_regs.h](#).

```
01201 {  
01202     return (volatile uint8_t*)k_ppllcr3_addr;  
01203 }
```

References [k_ppllcr3_addr](#).

8.59.4.4 ppllcr_reg()

```
volatile uint16_t * ppllcr_reg (  
    void )    [inline], [static]
```

Get pointer to PPLLCR register.

Returns

Volatile pointer to PPLLCR register

Definition at line 1181 of file [rx72n_system_regs.h](#).

```
01182 {  
01183     return (volatile uint16_t*)k_ppllcr_addr;  
01184 }
```

References [k_ppllcr_addr](#).

8.59.4.5 prcr_reg()

```
volatile uint16_t * prcr_reg (  
    void )    [inline], [static]
```

Get pointer to PRCR (Protection Register).

Returns

Volatile pointer to 16-bit PRCR register

PRCR is at address 0x000803FE, which is NOT contiguous with the main system register block (0x00080000-0x00080041). This accessor provides the correct address for register protection control.

Manual Reference:

RX72N Group User's Manual: Hardware, Chapter 13 (Register Write Protection)

- Address: 0x000803FE (verified against manual)
- Size: 16-bit
- Key: 0xA5 in upper byte required for any write

Note

Use `k_rx_prcr_unlock_*` and `k_rx_prcr_lock` values from `rx_register_protection.h` for unlock/↔ lock operations.

Example:

```
#include "rx_register_protection.h"
*prcr_reg() = k_rx_prcr_unlock_prc1; // Unlock MSTPCR group
system_regs()->mstpcra &= ~(1UL « 15); // Enable CMT0
*prcr_reg() = k_rx_prcr_lock; // Re-lock protection
```

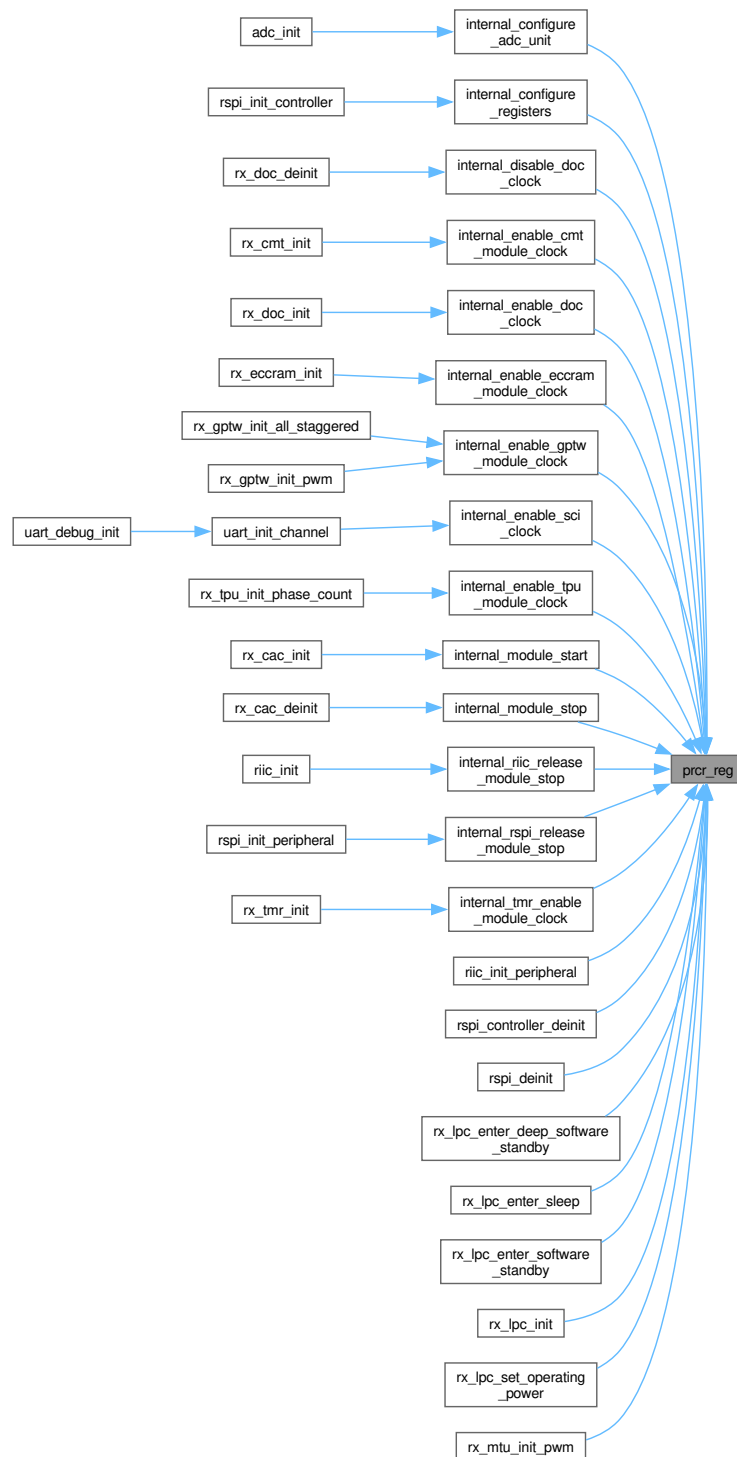
Definition at line 933 of file `rx72n_system_regs.h`.

```
00934 {
00935     return (volatile uint16_t*)k_prcr_addr;
00936 }
```

References `k_prcr_addr`.

Referenced by `internal_configure_adc_unit()`, `internal_configure_registers()`, `internal_disable_doc_clock()`, `internal_enable_cmt_module_clock()`, `internal_enable_doc_clock()`, `internal_enable_eccram_module_clock()`, `internal_enable_gptw_module_clock()`, `internal_enable_sci_clock()`, `internal_enable_tpu_module_clock()`, `internal_module_start()`, `internal_module_stop()`, `internal_riic_release_module_stop()`, `internal_rspi_release_module_stop()`, `internal_tmr_enable_module_clock()`, `riic_init_peripheral()`, `rspi_controller_deinit()`, `rspi_deinit()`, `rx_lpc_enter_deep_software_standby()`, `rx_lpc_enter_sleep()`, `rx_lpc_enter_software_standby()`, `rx_lpc_init()`, `rx_lpc_set_operating_power()`, and `rx_mtu_init_pwm()`.

Here is the caller graph for this function:



8.59.4.6 rstsr01()

```
volatile rx\_rstsr01\_regs\_t *rstsr01 (
    void ) [inline], [static]
```

Get pointer to RSTSR0/RSTSR1 registers.

Returns

Volatile pointer to RSTSR0/1 register structure

Definition at line 982 of file [rx72n_system_regs.h](#).

```
00983 {
00984     return (volatile rx_rstsr01_regs_t*)k_rstsr0_addr;
00985 }
```

References [k_rstsr0_addr](#).

8.59.4.7 rstsr2()

```
volatile uint8_t * rstsr2 (
    void )    [inline], [static]
```

Get pointer to RSTSR2 register.

Returns

Volatile pointer to RSTSR2 register

Note

RSTSR2 is at a separate address from RSTSR0/1

Definition at line 992 of file [rx72n_system_regs.h](#).

```
00993 {
00994     return (volatile uint8_t*)k_rstsr2_addr;
00995 }
```

References [k_rstsr2_addr](#).

8.59.4.8 swrr_reg()

```
volatile uint16_t * swrr_reg (
    void )    [inline], [static]
```

Get pointer to SWRR (Software Reset Register).

Returns

Volatile pointer to 16-bit SWRR register

SWRR is at address 0x000800C2. Writing 0xA501 triggers a software reset.

Warning

Writing to this register causes immediate MCU reset!

Manual Reference:

RX72N Group User's Manual: Hardware, Chapter 6 (Resets), page 290.

Example:

```
// Trigger a software reset (MCU will restart)
*swrr_reg() = k_swrr_reset_key;
// Code never reaches here - MCU resets immediately
```

Definition at line 1063 of file [rx72n_system_regs.h](#).

```
01064 {
01065     return (volatile uint16_t*)k_swrr_addr;
01066 }
```

References [k_swrr_addr](#).

8.59.4.9 system_regs()

```
volatile rx_system_regs_t * system_regs (  
    void )    [inline], [static]
```

Get pointer to system registers.

Returns

Volatile pointer to system register structure

Note

Named `system_regs()` instead of `system()` to avoid naming conflicts

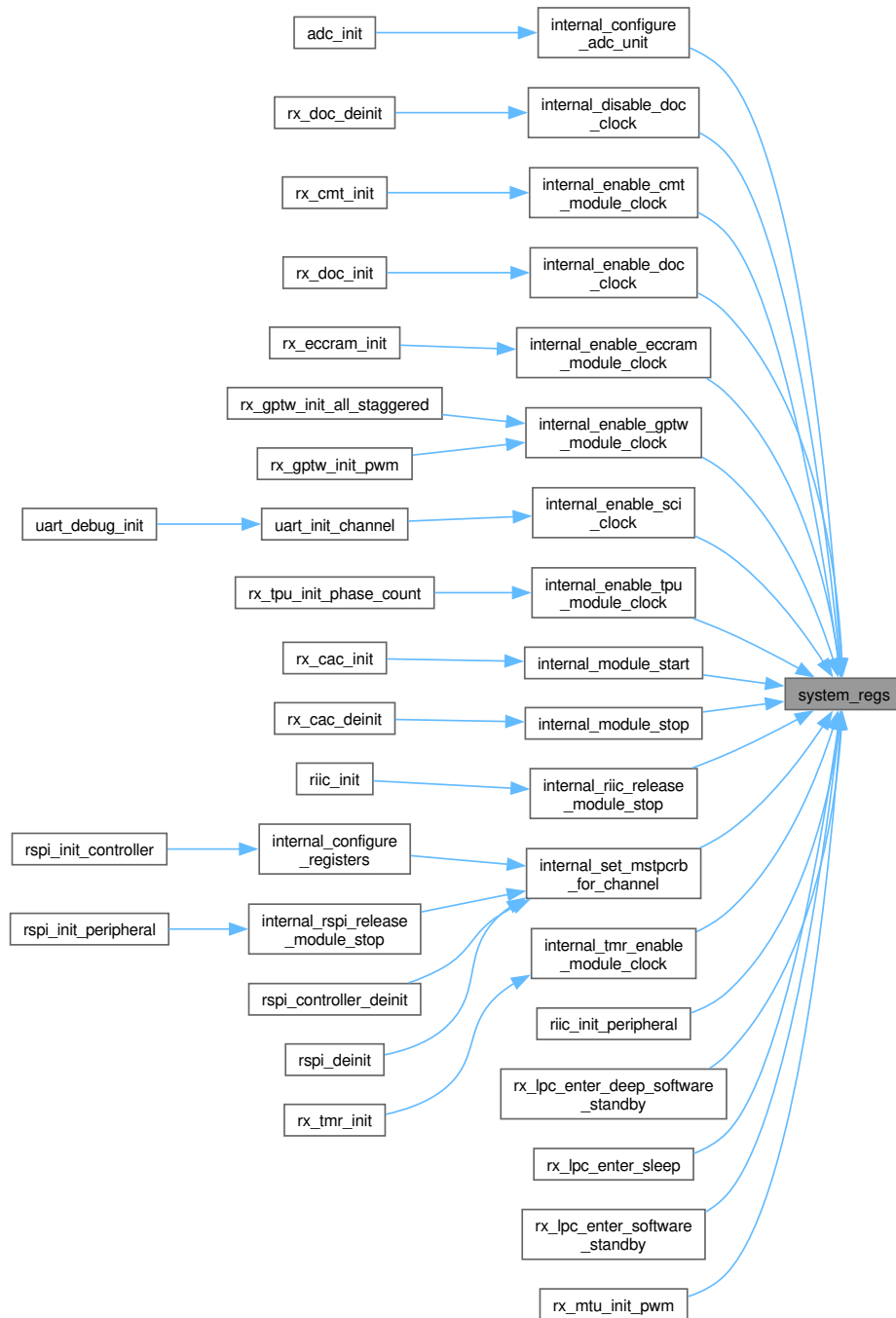
Definition at line 896 of file `rx72n_system_regs.h`.

```
00897 {  
00898     return (volatile rx_system_regs_t*)k_system_base_addr;  
00899 }
```

References `k_system_base_addr`.

Referenced by `internal_configure_adc_unit()`, `internal_disable_doc_clock()`, `internal_enable_cmt_module_clock()`, `internal_enable_doc_clock()`, `internal_enable_eccram_module_clock()`, `internal_enable_gptw_module_clock()`, `internal_enable_sci_clock()`, `internal_enable_tpu_module_clock()`, `internal_module_start()`, `internal_module_stop()`, `internal_riic_release_module_stop()`, `internal_set_mstpcrb_for_channel()`, `internal_tmr_enable_module_clock()`, `riic_init_peripheral()`, `rx_lpc_enter_deep_software_standby()`, `rx_lpc_enter_sleep()`, `rx_lpc_enter_software_standby()` and `rx_mtu_init_pwm()`.

Here is the caller graph for this function:



8.60 rx72n\system\regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_system_regs.h
00003  * @brief RX72N System Control Register Definitions
00004  *
00005  * @details
00006  * System control registers for clock generation, power management, oscillator
00007  * control, and module stop control on the RX72N microcontroller. This is the
00008  * foundation for system initialization and power optimization.
00009  *
00010  * @par System Architecture
00011  * @dot
00012  * digraph system_clocks {
00013  *   rankdir=LR;
  
```

```

00014 * node [shape=box, style=rounded];
00015 *
00016 * mosc [label="Main OSC\n24 MHz" style="filled" fillcolor="lightblue"];
00017 * pll [label="PLL\nx10 = 240 MHz"];
00018 * ppll [label="PPLL\n48 MHz (USB)"];
00019 *
00020 * iclk [label="ICLK\n240 MHz"];
00021 * pclka [label="PCLKA\n120 MHz"];
00022 * pclkb [label="PCLKB\n60 MHz"];
00023 *
00024 * mosc -> pll;
00025 * mosc -> ppll;
00026 * pll -> iclk [label="/1"];
00027 * pll -> pclka [label="/2"];
00028 * pll -> pclkb [label="/4"];
00029 * }
00030 * @enddot
00031 *
00032 * @par STAR Project Clock Configuration
00033 * | Clock | Frequency | Divider | Purpose |
00034 * |-----|-----|-----|-----|
00035 * | ICLK | 240 MHz | /1 | CPU instruction clock |
00036 * | PCLKA | 120 MHz | /2 | High-speed peripherals |
00037 * | PCLKB | 60 MHz | /4 | Peripherals (CMT, WDT) |
00038 * | PCLKC | 60 MHz | /4 | Peripherals |
00039 * | PCLKD | 60 MHz | /4 | ADC |
00040 * | FCLK | 60 MHz | /4 | Flash operations |
00041 * | BCLK | 120 MHz | /2 | External bus |
00042 * | UCLK | 48 MHz | PPLL | USB |
00043 *
00044 * @par Register Map Overview
00045 * **IMPORTANT**: System registers are NOT in a single contiguous block!
00046 *
00047 * | Address Range | Description |
00048 * |-----|-----|
00049 * | 0x00080000 - 0x00080041 | Main system registers (this struct) |
00050 * | 0x00080048 - 0x000800A | PPLL registers |
00051 * | 0x000803FE | PRCR (Protection Register) |
00052 * | 0x000800C0 | RSTSR2 (Reset Status 2) |
00053 * | 0x0008101C | MEMWAIT |
00054 * | 0x0008C290 - 0x0008C291 | RSTSR0/1 (Reset Status 0/1) |
00055 *
00056 * @par Module Stop Control (MSTPCR)
00057 * Peripherals can be stopped to save power. Bit=1 stops the module:
00058 * - **MSTPCRA**: CMT, DTC, DMAC
00059 * - **MSTPCRB**: USB, CRC, SCI, RSPI
00060 * - **MSTPCRC**: RAM, Flash
00061 * - **MSTPCRD**: ADC, DAC, MTU, GPTW
00062 *
00063 * @par Register Protection (PRCR)
00064 * Critical registers require PRCR unlock before modification:
00065 * - **PRC0**: Clock generation (SCKCR, PLLCR, oscillator control)
00066 * - **PRC1**: Module stop control (MSTPCRA-D)
00067 * - **PRC3**: Low voltage detection
00068 *
00069 * @par NASA Power of 10 Compliance
00070 * - Rule 1: [OK] No goto, setjmp, or recursion
00071 * - Rule 2: [OK] N/A (no loops)
00072 * - Rule 3: [OK] No dynamic memory allocation
00073 * - Rule 4: [OK] All accessors are single-statement
00074 * - Rule 5: [OK] N/A (hardware layer)
00075 * - Rule 6: [OK] Minimal scope
00076 * - Rule 7: [OK] N/A
00077 * - Rule 8: [OK] All constants use C23 typed enums
00078 * - Rule 9: [OK] No function pointers
00079 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00080 *
00081 * @par SOLID Principles
00082 * - **S**: Single responsibility - only system register definitions
00083 * - **O**: Open for extension via additional register accessors
00084 * - **L**: N/A
00085 * - **I**: Separate accessors for non-contiguous registers
00086 * - **D**: N/A (hardware register layer)
00087 *
00088 * @par Module Dependencies
00089 * - `<stddef.h>` - offsetof macro for static assertions
00090 * - `<stdint.h>` - Fixed-width integer types
00091 *
00092 * @par Manual References
00093 * - Chapter 5: I/O Registers (address map)
00094 * - Chapter 9: Clock Generation Circuit
00095 * - Chapter 13: Register Write Protection
00096 * - Chapter 11: Low Power Consumption
00097 *
00098 * @par Verification Status
00099 * [PASS] VERIFIED (2026-01-28) - All base addresses and register offsets verified
00100 * against RX72N Manual Ch05. Critical bug fixed: PRCR was incorrectly embedded

```

```

00101 * in the contiguous struct at offset 0x0A, but it's actually at 0x000803FE.
00102 *
00103 * @see rx72n_clock.h Clock-specific definitions
00104 * @see rx_register_protection.h PRCR unlock/lock API
00105 * @see rx_clock_power_init.c Clock initialization implementation
00106 *
00107 * @author Locked, Inc.
00108 * @date 2026-01-28
00109 * @copyright Copyright (c) 2026 Locked Inc.
00110 * SPDX-License-Identifier: MIT
00111 * @since Version 1.0.0
00112 */
00113
00114 #pragma once
00115
00116 #include <stddef.h>
00117 #include <stdint.h>
00118
00119 #ifdef __cplusplus
00120 extern "C" {
00121 #endif
00122
00123 /*
=====
00124 * System Control Registers (Contiguous Block @ 0x00080000-0x00080041)
00125 *
=====
00126 */
00127
00128 /**
00129 * @enum rx_system_addresses_t
00130 * @brief System register base addresses
00131 *
00132 * @details
00133 * Base addresses for system control registers. Note that system registers
00134 * are spread across multiple non-contiguous memory regions.
00135 *
00136 * @par Address Map
00137 * | Address | Register/Block | Description |
00138 * |-----|-----|-----|
00139 * | 0x00080000 | System Base | Main system register block |
00140 * | 0x000803FE | PRCR | Register protection control |
00141 *
00142 * @warning PRCR is NOT in the main system block - use prcr_reg() accessor
00143 *
00144 * @see system_regs() Main system register accessor
00145 * @see prcr_reg() PRCR register accessor
00146 * @since Version 1.0.0
00147 */
00148 typedef enum : uintptr_t {
00149 /**
00150 * @brief System register base address (0x00080000)
00151 * @details Start of main 66-byte system register block (0x00-0x41)
00152 */
00153 k_system_base_addr = 0x00080000,
00154
00155 /**
00156 * @brief PRCR (Protection Register) address (0x000803FE)
00157 * @details Located separately from main system block. Used to protect
00158 * clock, module stop, and voltage detection registers from accidental writes.
00159 * @warning Not part of rx_system_regs_t structure
00160 */
00161 k_prcr_addr = 0x000803FE,
00162 } rx_system_addresses_t;
00163
00164 /**
00165 * @enum system_reserved_sizes_t
00166 * @brief System register reserved field sizes
00167 *
00168 * @details
00169 * Defines reserved byte counts for gaps in the system register map.
00170 * These ensure correct structure alignment with hardware.
00171 *
00172 * @since Version 1.0.0
00173 */
00174 typedef enum : uint8_t {
00175 k_system_reserved_02_05 = 4, /**< Reserved @ 0x02-0x05 (between MDMONR and SYSCR0) */
00176 k_system_reserved_0a_0b = 2, /**< Reserved @ 0x0A-0x0B (between SYSCR1 and SBYCR) */
00177 k_system_reserved_0e_0f = 2, /**< Reserved @ 0x0E-0x0F (between SBYCR and MSTPCRA) */
00178 k_system_reserved_2b_2f = 5, /**< Reserved @ 0x2B-0x2F (between PLLCR2 and BCKCR) */
00179 k_system_reserved_31 = 1, /**< Reserved @ 0x31 (between BCKCR and MOSCCR) */
00180 k_system_reserved_38_3b = 4, /**< Reserved @ 0x38-0x3B (between HOCOCR2 and OSCOVFSR) */
00181 k_system_reserved_3d = 1, /**< Reserved @ 0x3D (between OSCOVFSR and CKOCR) */
00182 } system_reserved_sizes_t;
00183
00184 /*
=====

```

```

00185 * Ch03 Operating Modes - MDMONR, SYSCR0, SYSCR1 Bit Definitions
00186 *
=====
00187 */
00188
00189 /**
00190  * @enum mdmonr_bits_t
00191  * @brief Mode Monitor Register (MDMONR) bit definitions
00192  *
00193  * @details
00194  * Read-only register indicating MCU mode pin status at reset.
00195  * Located at address 0x00080000.
00196  *
00197  * @par Register Layout (16-bit @ offset 0x00)
00198  * | Bit | Field | Description |
00199  * |-----|-----|-----|
00200  * | 0 | MD | MD Pin Status Flag (read-only) |
00201  * | 15:1 | - | Reserved (read as 0) |
00202  *
00203  * @par MD Pin Modes (at reset)
00204  * - MD = 0: Boot mode (low level on MD pin)
00205  * - MD = 1: Single-chip mode (high level on MD pin)
00206  *
00207  * @note Manual ref: Ch03 section 3.2.1
00208  * @since Version 1.0.0
00209  */
00210 typedef enum : uint16_t {
00211     k_mdmonr_md = 0x0001U, /**< MD pin status (0=low/boot, 1=high/single-chip) */
00212     k_mdmonr_md_mask = 0x0001U, /**< MD bit mask */
00213 } mdmonr_bits_t;
00214
00215 /**
00216  * @enum syscr0_bits_t
00217  * @brief System Control Register 0 (SYSCR0) bit definitions
00218  *
00219  * @details
00220  * Controls on-chip ROM enable and external bus enable.
00221  * Requires key code 0x5A in upper byte for writes.
00222  * Located at address 0x00080006.
00223  *
00224  * @par Register Layout (16-bit @ offset 0x06)
00225  * | Bit | Field | Description |
00226  * |-----|-----|-----|
00227  * | 0 | ROME | On-Chip ROM Enable |
00228  * | 1 | EXBE | External Bus Enable |
00229  * | 7:2 | - | Reserved (write 0) |
00230  * | 15:8 | KEY[7:0] | Key code (write 0x5A to modify) |
00231  *
00232  * @par Operating Modes
00233  * | ROME | EXBE | Mode |
00234  * |-----|-----|-----|
00235  * | 1 | 0 | Single-chip mode |
00236  * | 1 | 1 | On-chip ROM enabled extended mode |
00237  * | 0 | 1 | On-chip ROM disabled extended mode |
00238  * | 0 | 0 | Single-chip mode (ROM disabled) |
00239  *
00240  * @warning Once ROME is set to 0, it cannot be reverted to 1
00241  * @warning Do not modify EXBE while accessing external bus
00242  *
00243  * @note Manual ref: Ch03 section 3.2.2
00244  * @since Version 1.0.0
00245  */
00246 typedef enum : uint16_t {
00247     k_syscr0_rome = 0x0001U, /**< On-Chip ROM Enable (0=disabled, 1=enabled) */
00248     k_syscr0_exbe = 0x0002U, /**< External Bus Enable (0=disabled, 1=enabled) */
00249     k_syscr0_key = 0x5A00U, /**< Key code for SYSCR0 write (must be in upper byte) */
00250     k_syscr0_key_mask = 0xFF00U, /**< Key code mask */
00251
00252     /* Convenience values for common write operations */
00253     k_syscr0_rom_enabled = (0x5A00U | 0x0001U), /**< Enable ROM (write value) */
00254     k_syscr0_rom_disabled = 0x5A00U, /**< Disable ROM (write value) - IRREVERSIBLE */
00255     k_syscr0_extbus_enabled = (0x5A00U | 0x0003U), /**< Enable ROM + ExtBus (write value) */
00256 } syscr0_bits_t;
00257
00258 /**
00259  * @enum syscr1_bits_t
00260  * @brief System Control Register 1 (SYSCR1) bit definitions
00261  *
00262  * @details
00263  * Controls RAM, ECCRAM, and Standby RAM enable.
00264  * Located at address 0x00080008.
00265  *
00266  * @par Register Layout (16-bit @ offset 0x08)
00267  * | Bit | Field | Description |
00268  * |-----|-----|-----|
00269  * | 0 | RAME | RAM Enable |
00270  * | 5:1 | - | Reserved (read/write as 1) |

```

```

00271 * | 6 | ECCRAME | ECCRAM Enable |
00272 * | 7 | SBYRAME | Standby RAM Enable |
00273 * | 15:8 | - | Reserved (write 0) |
00274 *
00275 * @par Default Value: 0x00FF (all RAM enabled)
00276 * - RAME = 1 (RAM enabled)
00277 * - Reserved bits 5:1 = 1
00278 * - ECCRAME = 1 (ECCRAM enabled)
00279 * - SBYRAME = 1 (Standby RAM enabled)
00280 *
00281 * @warning Do not write 0 to RAME/ECCRAME/SBYRAME during access to that memory
00282 *
00283 * @note Manual ref: Ch03 section 3.2.3
00284 * @since Version 1.0.0
00285 */
00286 typedef enum : uint16_t {
00287     k_syscr1_rame = 0x0001U, /**< RAM Enable (0=disabled, 1=enabled) */
00288     k_syscr1_reserved_mask = 0x003EU, /**< Reserved bits 5:1 (read/write as 1) */
00289     k_syscr1_eccrame = 0x0040U, /**< ECCRAM Enable (0=disabled, 1=enabled) */
00290     k_syscr1_sbyrame = 0x0080U, /**< Standby RAM Enable (0=disabled, 1=enabled) */
00291
00292     /* Convenience values */
00293     k_syscr1_all_ram_enabled = 0x00FFU, /**< Default: all RAM types enabled */
00294     k_syscr1_disable_eccram = 0x00BFU, /**< Disable ECCRAM (RAME+SBYRAME enabled) */
00295     k_syscr1_disable_sbyram = 0x007FU, /**< Disable StandbyRAM (RAME+ECCRAM enabled) */
00296 } syscr1_bits_t;
00297
00298 /*
=====
00299 * Ch09 Clock Generation - CKOCR, SCKCR, SCKCR3, PLLCR Bit Definitions
00300 *
=====
00301 */
00302
00303 /**
00304 * @enum ckocr_bits_t
00305 * @brief Clock Output Control Register (CKOCR) bit definitions
00306 *
00307 * @details
00308 * Controls the clock output on CLKOUT pin. Can output various system clocks
00309 * divided by configurable ratio.
00310 *
00311 * @par Register Layout (16-bit @ offset 0x3E)
00312 * | Bit | Field | Description |
00313 * |-----|-----|-----|
00314 * | 7:0 | - | Reserved |
00315 * | 10:8 | CKOSEL | Clock output source select |
00316 * | 11 | - | Reserved |
00317 * | 14:12 | CKODIV | Clock output divider |
00318 * | 15 | CKOSTP | Clock output stop (0=run, 1=stop) |
00319 *
00320 * @par Clock Output Sources (CKOSEL)
00321 * | Value | Source | Description |
00322 * |-----|-----|-----|
00323 * | 0 | LOCO | Low-speed on-chip oscillator |
00324 * | 1 | HOCO | High-speed on-chip oscillator |
00325 * | 2 | Main clock | Main oscillator |
00326 * | 3 | Sub-clock | Sub-clock oscillator |
00327 * | 4 | PLL | PLL output |
00328 *
00329 * @par Clock Output Dividers (CKODIV[2:0] at bits 14:12)
00330 * | Value | Divider | Output Frequency |
00331 * |-----|-----|-----|
00332 * | 000 | /1 | f (source frequency) |
00333 * | 001 | /2 | f/2 |
00334 * | 010 | /4 | f/4 |
00335 * | 011 | /8 | f/8 |
00336 * | 100 | /16 | f/16 |
00337 * | other | -- | PROHIBITED |
00338 *
00339 * @par Clock Output Sources (CKOSEL[2:0] at bits 10:8)
00340 * | Value | Source |
00341 * |-----|-----|
00342 * | 000 | LOCO |
00343 * | 001 | HOCO |
00344 * | 010 | Main clock |
00345 * | 011 | Sub-clock |
00346 * | 100 | PLL circuit |
00347 * | 110 | PLL circuit |
00348 * | other | PROHIBITED |
00349 *
00350 * @warning Clock output disabled by default (CKOSTP = 1)
00351 * @warning CKODIV values 5, 6, 7 are PROHIBITED (max divisor is /16)
00352 * @note Requires PRCR unlock (PRC0) before modification
00353 * @see Manual Ch09 section 9.2.21 CKOCR
00354 * @since Version 1.0.0
00355 */

```

```

00356 typedef enum : uint16_t {
00357     /* Clock output source select (CKOSEL, bits 10:8) */
00358     k_ckocr_ckosel_loco = (0U « 8), /**< LOCO (low-speed on-chip osc) */
00359     k_ckocr_ckosel_hoco = (1U « 8), /**< HOCO (high-speed on-chip osc) */
00360     k_ckocr_ckosel_main = (2U « 8), /**< Main oscillator */
00361     k_ckocr_ckosel_subclock = (3U « 8), /**< Sub-clock oscillator */
00362     k_ckocr_ckosel_pll = (4U « 8), /**< PLL output */
00363     k_ckocr_ckosel_ppll = (6U « 8), /**< PPLL output (110b, not 101b) */
00364     k_ckocr_ckosel_mask = (7U « 8), /**< CKOSEL field mask */
00365
00366     /* Clock output divider (CKODIV, bits 14:12) -- max is /16, values 5-7 PROHIBITED */
00367     k_ckocr_ckodiv_1 = (0U « 12), /**< Divide by 1 */
00368     k_ckocr_ckodiv_2 = (1U « 12), /**< Divide by 2 */
00369     k_ckocr_ckodiv_4 = (2U « 12), /**< Divide by 4 */
00370     k_ckocr_ckodiv_8 = (3U « 12), /**< Divide by 8 */
00371     k_ckocr_ckodiv_16 = (4U « 12), /**< Divide by 16 (maximum valid divisor) */
00372     k_ckocr_ckodiv_mask = (7U « 12), /**< CKODIV field mask */
00373
00374     /* Clock output stop (CKOSTP, bit 15) */
00375     k_ckocr_ckostp = (1U « 15), /**< Clock output stop (1=stopped/low, 0=running) */
00376
00377     /* Convenience values */
00378     k_ckocr_output_disabled = k_ckocr_ckostp, /**< Default: output disabled */
00379     k_ckocr_pll_div4 =
00380         (k_ckocr_ckosel_pll | k_ckocr_ckodiv_4), /**< PLL/4 output (60 MHz @ 240 MHz PLL) */
00381 } ckocr_bits_t;
00382
00383 /**
00384  * @enum skcr_divider_t
00385  * @brief Clock divider values for SCKCR fields
00386  *
00387  * @details
00388  * All SCKCR clock divider fields use the same encoding. Value N means divide
00389  * by 2N. For example, 0b0010 (2) means divide by 4.
00390  *
00391  * @par Divider Encoding (for PCKD, PCKC, PCKB, PCKA, ICK, FCK fields)
00392  * | Bits | Divider | Example (240 MHz input) |
00393  * |-----|-----|-----|
00394  * | 0000 | /1      | 240 MHz                  |
00395  * | 0001 | /2      | 120 MHz                  |
00396  * | 0010 | /4      | 60 MHz                   |
00397  * | 0011 | /8      | 30 MHz                   |
00398  * | 0100 | /16     | 15 MHz                   |
00399  * | 0101 | /32     | 7.5 MHz                  |
00400  * | 0110 | /64     | 3.75 MHz                 |
00401  * | other | --      | PROHIBITED              |
00402  *
00403  * @note The BCK field also accepts 1001 (x1/3) as an additional valid setting.
00404  * @warning Values 0111 and 1000 are PROHIBITED for all SCKCR fields.
00405  * @see Manual Ch09 section 9.2.1 Table 9.1, section 9.2.2
00406  * @since Version 1.0.0
00407  */
00408 typedef enum : uint8_t {
00409     k_clock_div_1 = 0, /**< Divide by 1 (no division) */
00410     k_clock_div_2 = 1, /**< Divide by 2 */
00411     k_clock_div_4 = 2, /**< Divide by 4 */
00412     k_clock_div_8 = 3, /**< Divide by 8 */
00413     k_clock_div_16 = 4, /**< Divide by 16 */
00414     k_clock_div_32 = 5, /**< Divide by 32 */
00415     k_clock_div_64 = 6, /**< Divide by 64 */
00416     /* NOTE: /128 does NOT exist in SCKCR. Maximum divisor is /64 (0110 = 6). */
00417     /* Values 7 and 8 are PROHIBITED per RX72N Manual Ch09 section 9.2.2. */
00418     /* BCK field additionally accepts k_bck_div_3 = 9 (binary 1001 = x1/3). */
00419     k_bck_div_3 = 9, /**< Divide by 3 (ONLY valid for BCK[3:0] field, binary 1001) */
00420 } skcr_divider_t;
00421
00422 /**
00423  * @enum skcr_bits_t
00424  * @brief System Clock Control Register (SCKCR) bit definitions
00425  *
00426  * @details
00427  * 32-bit register controlling all main system clock dividers. Each field is
00428  * 4 bits wide and uses the encoding from skcr_divider_t.
00429  *
00430  * @par Register Layout (32-bit @ offset 0x20)
00431  * | Bits | Field | Clock | STAR Project Setting |
00432  * |-----|-----|-----|-----|
00433  * | 3:0  | PCKD  | PCLKD | /4 (60 MHz from 240) |
00434  * | 7:4  | PCKC  | PCLKC | /4 (60 MHz from 240) |
00435  * | 11:8  | PCKB  | PCLKB | /4 (60 MHz from 240) |
00436  * | 15:12 | PCKA  | PCLKA | /2 (120 MHz from 240) |
00437  * | 19:16 | BCK   | BCLK  | /2 (120 MHz from 240) |
00438  * | 21:20 | -     | Reserved | - |
00439  * | 22    | PSTOP0 | PCLKA/B | 0 (running) |
00440  * | 23    | PSTOP1 | USB     | 0 (running) |
00441  * | 27:24 | ICK   | ICLK   | /1 (240 MHz) |
00442  * | 31:28 | FCK   | FCLK   | /4 (60 MHz from 240) |

```



```

00443 *
00444 * @par STAR Project Configuration
00445 * With 24 MHz main oscillator x PLL(x10) = 240 MHz:
00446 * - ICLK = 240 MHz (CPU instruction clock)
00447 * - PCLKA = 120 MHz (high-speed peripherals)
00448 * - PCLKB = 60 MHz (CMT, WDT, etc.)
00449 * - PCLKC = 60 MHz (general peripherals)
00450 * - PCLKD = 60 MHz (ADC)
00451 * - FCLK = 60 MHz (Flash memory)
00452 * - BCLK = 120 MHz (External bus)
00453 *
00454 * @warning ICLK > 120 MHz requires MEMWAIT = 1 before changing SCKCR
00455 * @note Requires PRCR unlock (PRC0) before modification
00456 * @see Manual Ch09 section 9.2.2
00457 * @since Version 1.0.0
00458 */
00459 typedef enum : uint32_t {
00460     /* Field shifts (for manual bit manipulation if needed) */
00461     k_sckcr_pckd_shift = 0,
00462     k_sckcr_pckc_shift = 4,
00463     k_sckcr_pckb_shift = 8,
00464     k_sckcr_pcka_shift = 12,
00465     k_sckcr_bck_shift = 16,
00466     k_sckcr_ick_shift = 24,
00467     k_sckcr_fck_shift = 28,
00468
00469     /* Field masks */
00470     k_sckcr_pckd_mask = 0xFUL « k_sckcr_pckd_shift,
00471     k_sckcr_pckc_mask = 0xFUL « k_sckcr_pckc_shift,
00472     k_sckcr_pckb_mask = 0xFUL « k_sckcr_pckb_shift,
00473     k_sckcr_pcka_mask = 0xFUL « k_sckcr_pcka_shift,
00474     k_sckcr_bck_mask = 0xFUL « k_sckcr_bck_shift,
00475     k_sckcr_ick_mask = 0xFUL « k_sckcr_ick_shift,
00476     k_sckcr_fck_mask = 0xFUL « k_sckcr_fck_shift,
00477
00478     /* PSTOP bits (pin output control) */
00479     k_sckcr_pstop0 = (1UL « 22), /*< SDCLK pin output control (0=enabled, 1=disabled/fixed-high) */
00480     k_sckcr_pstop1 = (1UL « 23), /*< BCLK pin output control (0=enabled, 1=disabled/fixed-high) */
00481
00482     /* STAR Project configuration value (24 MHz x 10 = 240 MHz) */
00483     k_sckcr_star_240mhz =
00484         (((uint32_t)k_clock_div_4 « k_sckcr_pckd_shift) | /* PCLKD = 240/4 = 60 MHz */
00485          ((uint32_t)k_clock_div_4 « k_sckcr_pckc_shift) | /* PCLKC = 240/4 = 60 MHz */
00486          ((uint32_t)k_clock_div_4 « k_sckcr_pckb_shift) | /* PCLKB = 240/4 = 60 MHz */
00487          ((uint32_t)k_clock_div_2 « k_sckcr_pcka_shift) | /* PCLKA = 240/2 = 120 MHz */
00488          ((uint32_t)k_clock_div_2 « k_sckcr_bck_shift) | /* BCLK = 240/2 = 120 MHz */
00489          ((uint32_t)k_clock_div_1 « k_sckcr_ick_shift) | /* ICLK = 240/1 = 240 MHz */
00490          ((uint32_t)k_clock_div_4 « k_sckcr_fck_shift) | /* FCLK = 240/4 = 60 MHz */
00491          0UL /* PSTOP0/1 = 0 (clocks running) */
00492     ), /*< Complete SCKCR value for STAR 240 MHz configuration */
00493 } sckcr_bits_t;
00494
00495 /**
00496  * @enum sckcr3_ksel_t
00497  * @brief System clock source selection (SCKCR3 CKSEL field)
00498  *
00499  * @details
00500  * Selects which oscillator/PLL output is used as the system clock source.
00501  * This is the clock that feeds into SCKCR dividers.
00502  *
00503  * @par Clock Sources (CKSEL[2:0] at bits 10:8 of SCKCR3)
00504  * | Value | Source | Typical Frequency | Use Case |
00505  * |-----|-----|-----|-----|
00506  * | 0 | LOCO | 240 kHz (typ) | Low power mode |
00507  * | 1 | HOCO | 16-20 MHz (config) | Boot, low power |
00508  * | 2 | Main clock | 24 MHz (STAR) | External crystal |
00509  * | 3 | Sub-clock | 32.768 kHz | RTC, ultra low power |
00510  * | 4 | PLL | 240 MHz (STAR) | Normal operation |
00511  * | other | -- | PROHIBITED | -- |
00512  *
00513  * @note PPLL is NOT a valid SCKCR3 CKSEL option. PPLL is used only as
00514  * the USB clock source, selected via PACKCR.UPLLSEL.
00515  *
00516  * @par STAR Project Settings
00517  * - Boot: LOCO (default after reset)
00518  * - Normal operation: PLL @ 240 MHz
00519  *
00520  * @warning Switching clock sources requires waiting for oscillator stabilization
00521  * @note Requires PRCR unlock (PRC0) before modification
00522  * @see Manual Ch09 section 9.2.4 SCKCR3
00523  * @since Version 1.0.0
00524  */
00525 typedef enum : uint16_t {
00526     k_sckcr3_ksel_loco = 0x0000, /*< LOCO (Low-speed On-Chip Oscillator, 240 kHz) */
00527     k_sckcr3_ksel_hoco = 0x0100, /*< HOCO (High-speed On-Chip Oscillator, 16-20 MHz) */
00528     k_sckcr3_ksel_main = 0x0200, /*< Main clock oscillator (STAR: 24 MHz crystal) */
00529     k_sckcr3_ksel_subclock = 0x0300, /*< Sub-clock oscillator (32.768 kHz) */

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```

00530 k_sckcr3_cksel_pll = 0x0400, /**< PLL circuit (STAR: 240 MHz). Values 5-7 are prohibited. */
00531 k_sckcr3_cksel_mask = 0x0700, /**< CKSEL field mask (bits 10:8) */
00532 } sckcr3_cksel_t;
00533
00534 /**
00535  * @enum pllcr_bits_t
00536  * @brief PLL Control Register (PLLCR) bit definitions
00537  *
00538  * @details
00539  * Controls PLL input divider, output multiplier, and source selection.
00540  * PLL output frequency = (Input / PLIDIV) x (STC + 1)
00541  *
00542  * @par Register Layout (16-bit @ offset 0x28)
00543  * | Bits | Field | Description |
00544  * |-----|-----|-----|
00545  * | 1:0 | PLIDIV | PLL input divider (00=/1, 01=/2, 10=/3) |
00546  * | 3:2 | - | Reserved |
00547  * | 4 | PLLSRCSEL | PLL source (0=main osc, 1=HOCO) |
00548  * | 7:5 | - | Reserved |
00549  * | 13:8 | STC | PLL multiplier (value N -> x(N+1)) |
00550  * | 15:14 | - | Reserved |
00551  *
00552  * @par PLIDIV Encoding (bits 1:0)
00553  * | Value | Divider | Note |
00554  * |-----|-----|-----|
00555  * | 00 | /1 | No division |
00556  * | 01 | /2 | |
00557  * | 10 | /3 | |
00558  * | 11 | -- | PROHIBITED |
00559  *
00560  * @par STAR Project PLL Configuration
00561  * Input: 24 MHz main oscillator
00562  * Target: 240 MHz PLL output
00563  * Calculation: 24 MHz / 1 x 10 = 240 MHz
00564  * - PLIDIV = 00 (divide by 1) -> 24 MHz
00565  * - STC = 0x09 (multiply by 10) -> 240 MHz
00566  * - PLLSRCSEL = 0 (main oscillator source)
00567  * - PLLCR = 0x0900
00568  *
00569  * @warning PLL must be stopped (PLLCR2.PLEN = 1) before changing PLLCR
00570  * @note Requires PRCR unlock (PRC0) before modification
00571  * @see Manual Ch09 section 9.2.5 PLLCR
00572  * @since Version 1.0.0
00573  */
00574 typedef enum : uint16_t {
00575     /** PLL input divider (PLIDIV, bits 1:0) */
00576     k_pllcr_plidiv_1 = 0x0000, /**< Divide PLL input by 1 (binary 00) */
00577     k_pllcr_plidiv_2 = 0x0001, /**< Divide PLL input by 2 (binary 01) */
00578     k_pllcr_plidiv_3 = 0x0002, /**< Divide PLL input by 3 (binary 10); value 11 is PROHIBITED */
00579     k_pllcr_plidiv_mask = 0x0003, /**< PLIDIV field mask */
00580
00581     /** PLL source select (PLLSRCSEL, bit 4) */
00582     k_pllcr_src_main = 0x0000, /**< Main oscillator as PLL source (STAR config) */
00583     k_pllcr_src_hoco = 0x0010, /**< HOCO as PLL source */
00584     k_pllcr_src_mask = 0x0010, /**< PLLSRCSEL field mask */
00585
00586     /** PLL multiplier field (STC, bits 13:8) - value N means multiply by (N+1) */
00587     k_pllcr_stc_shift = 8,
00588     k_pllcr_stc_mask = 0x3F00,
00589
00590     /** Convenience values for common configurations */
00591     k_pllcr_star_24mhz_to_240mhz =
00592         (k_pllcr_plidiv_1 |
00593          k_pllcr_src_main |
00594          (19U « k_pllcr_stc_shift) /** STC=19 -> x10.0 (manual p345: 0b010011) */
00595         ),
00596     k_pllcr_12mhz_to_240mhz =
00597         (k_pllcr_plidiv_1 |
00598          k_pllcr_src_main |
00599          (19U « k_pllcr_stc_shift) /** STC=19 -> x10.0 (manual p345); = 0x1300 */
00600         ),
00601     /** Alternative: 12 MHz x 10 = 120 MHz (NOT 240 MHz!) */
00602 } pllcr_bits_t;
00603
00604 /**
00605  * @enum pllcr2_bits_t
00606  * @brief PLL Control Register 2 (PLLCR2) bit definitions
00607  *
00608  * @details
00609  * Controls PLL enable/disable. Note inverted logic: 0 = enabled, 1 = stopped.
00610  *
00611  * @par Register Layout (8-bit @ offset 0x2A)
00612  * | Bit | Field | Description |
00613  * |-----|-----|-----|
00614  * | 0 | PLEN | PLL enable (0=run, 1=stopped) |
00615  * | 7:1 | - | Reserved |
00616  */

```

```

00617 * @warning Inverted logic! 0 = PLL running, 1 = PLL stopped
00618 * @note Must wait for OSCOVFSR.PLOVF before switching clock to PLL
00619 * @note Requires PRCR unlock (PRC0) before modification
00620 * @see Manual Ch09 section 9.2.4
00621 * @since Version 1.0.0
00622 */
00623 typedef enum : uint8_t {
00624     k_pllcr2_pll_running = 0x00, /**< PLL enabled (running) */
00625     k_pllcr2_pll_stopped = 0x01, /**< PLL disabled (stopped) */
00626 } pllcr2_bits_t;
00627
00628 /**
00629  * @struct rx_system_regs_t
00630  * @brief System Control Register Map (0x00080000 - 0x00080041)
00631  *
00632  * @details
00633  * System control registers for clock configuration, power management,
00634  * and module stop control. This structure covers the main contiguous
00635  * block of system registers.
00636  *
00637  * @par Memory Layout Table
00638  * | Offset | Size | Field | Description |
00639  * |-----|-----|-----|-----|
00640  * | 0x00 | 2 | mdmonr | Mode Monitor Register |
00641  * | 0x02 | 4 | reserved | - |
00642  * | 0x06 | 2 | syscr0 | System Control Register 0 |
00643  * | 0x08 | 2 | syscr1 | System Control Register 1 |
00644  * | 0x0A | 2 | reserved | - |
00645  * | 0x0C | 2 | sbycr | Standby Control Register |
00646  * | 0x0E | 2 | reserved | - |
00647  * | 0x10 | 4 | mstpcra | Module Stop Control Register A |
00648  * | 0x14 | 4 | mstpcrb | Module Stop Control Register B |
00649  * | 0x18 | 4 | mstpcrc | Module Stop Control Register C |
00650  * | 0x1C | 4 | mstpcrd | Module Stop Control Register D |
00651  * | 0x20 | 4 | sckcr | System Clock Control Register |
00652  * | 0x24 | 2 | sckcr2 | System Clock Control Register 2 |
00653  * | 0x26 | 2 | sckcr3 | System Clock Control Register 3 |
00654  * | 0x28 | 2 | pllcr | PLL Control Register |
00655  * | 0x2A | 1 | pllcr2 | PLL Control Register 2 |
00656  * | 0x2B | 5 | reserved | - |
00657  * | 0x30 | 1 | bckcr | External Bus Clock Control |
00658  * | 0x31 | 1 | reserved | - |
00659  * | 0x32 | 1 | mosccr | Main Oscillator Control |
00660  * | 0x33 | 1 | sosccr | Sub-Clock Oscillator Control |
00661  * | 0x34 | 1 | lococr | LOCO Control |
00662  * | 0x35 | 1 | ilococr | IWDTC Oscillator Control |
00663  * | 0x36 | 1 | hococr | HOCO Control |
00664  * | 0x37 | 1 | hococr2 | HOCO Control 2 |
00665  * | 0x38 | 4 | reserved | - |
00666  * | 0x3C | 1 | oscovfsr | Oscillation Stabilization Flag |
00667  * | 0x3D | 1 | reserved | - |
00668  * | 0x3E | 2 | ckocr | Clock Output Control Register |
00669  * | 0x40 | 1 | ostcdr | Oscillation Stop Detection Control |
00670  * | 0x41 | 1 | ostdsr | Oscillation Stop Detection Status |
00671  * | **Total** | **66** | |
00672  *
00673  * @par Clock Initialization Example
00674  * @code{.c}
00675  * // Initialize system to 240 MHz from 24 MHz main oscillator
00676  * volatile rx_system_regs_t* sys = system_regs();
00677  *
00678  * // 1. Unlock register protection
00679  * *prcr_reg() = k_rx_prcr_unlock_prc0_prc1;
00680  *
00681  * // 2. Enable main oscillator
00682  * sys->mosccr = 0x00; // Start main oscillator
00683  *
00684  * // 3. Wait for oscillator stabilization
00685  * while (!(sys->oscovfsr & 0x01)) { }
00686  *
00687  * // 4. Configure PLL (24 MHz x 10 = 240 MHz)
00688  * sys->pllcr = 0x0913; // STC=9 (x10), PLIDIV=1 (/1)
00689  * sys->pllcr2 = 0x00; // Enable PLL
00690  *
00691  * // 5. Wait for PLL stabilization
00692  * while (!(sys->oscovfsr & 0x04)) { }
00693  *
00694  * // 6. Set MEMWAIT for >120 MHz operation
00695  * *memwait_reg() = k_memwait_one_wait;
00696  *
00697  * // 7. Switch to PLL
00698  * sys->sckcr3 = 0x0400; // CKSEL = PLL
00699  *
00700  * // 8. Re-lock protection
00701  * *prcr_reg() = k_rx_prcr_lock;
00702  * @endcode
00703  *

```

```

00704 * @par Module Stop Example
00705 * @code{.c}
00706 * // Enable CMT0 module (clear bit to enable)
00707 * *prcr_reg() = k_rx_prcr_unlock_prc1; // Unlock MSTPCR
00708 * system_regs()->mstpcra &= ~(1UL « 15); // Enable CMT0
00709 * *prcr_reg() = k_rx_prcr_lock;
00710 *
00711 * // Enable USB0 module
00712 * *prcr_reg() = k_rx_prcr_unlock_prc1;
00713 * system_regs()->mstpcrb &= ~(1UL « k_mstpb_usb0);
00714 * *prcr_reg() = k_rx_prcr_lock;
00715 * @endcode
00716 *
00717 * @note PRCR (Protection Register) is NOT in this struct - it's at 0x000803FE.
00718 *     Use prcr_reg() accessor instead.
00719 *
00720 * @invariant sizeof(rx_system_regs_t) == 66 bytes (0x42)
00721 * @invariant All register offsets match RX72N Manual Ch05
00722 *
00723 * @warning Most registers require PRCR unlock before modification
00724 * @attention MEMWAIT must be set before ICLK > 120 MHz
00725 *
00726 * @see system_regs() Accessor function
00727 * @see prcr_reg() Protection register accessor
00728 * @since Version 1.0.0
00729 */
00730 typedef struct __attribute__((packed)) {
00731 /**
00732  * @brief Mode Monitor Register (MDMONR) @ 0x00
00733  * @details Read-only - indicates current operating mode
00734  */
00735 volatile uint16_t mdmonr;
00736 uint8_t reserved0[k_system_reserved_02_05]; /**< Reserved @ 0x02-0x05 */
00737
00738 /**
00739  * @brief System Control Register 0 (SYSCR0) @ 0x06
00740  * @details Controls ROM/RAM enable and external bus settings
00741  */
00742 volatile uint16_t syscr0;
00743
00744 /**
00745  * @brief System Control Register 1 (SYSCR1) @ 0x08
00746  * @details Controls RAM wait states
00747  */
00748 volatile uint16_t syscr1;
00749 uint8_t reserved1[k_system_reserved_0a_0b]; /**< Reserved @ 0x0A-0x0B */
00750
00751 /**
00752  * @brief Standby Control Register (SBYCR) @ 0x0C
00753  * @details Controls standby mode (SSBY bit)
00754  */
00755 volatile uint16_t sbycr;
00756 uint8_t reserved2[k_system_reserved_0e_0f]; /**< Reserved @ 0x0E-0x0F */
00757
00758 /**
00759  * @brief Module Stop Control Register A (MSTPCRA) @ 0x10
00760  * @details Controls CMT, DTC, DMAC module power. Bit=1 stops module.
00761  */
00762 volatile uint32_t mstpcra;
00763
00764 /**
00765  * @brief Module Stop Control Register B (MSTPCRB) @ 0x14
00766  * @details Controls USB, CRC, SCI, RSPI module power. Bit=1 stops module.
00767  */
00768 volatile uint32_t mstpcrb;
00769
00770 /**
00771  * @brief Module Stop Control Register C (MSTPCRC) @ 0x18
00772  * @details Controls RAM, Flash module power
00773  */
00774 volatile uint32_t mstpcrc;
00775
00776 /**
00777  * @brief Module Stop Control Register D (MSTPCRD) @ 0x1C
00778  * @details Controls ADC, DAC, MTU, GPTW module power. Bit=1 stops module.
00779  */
00780 volatile uint32_t mstpcrd;
00781
00782 /**
00783  * @brief System Clock Control Register (SCKCR) @ 0x20
00784  * @details Sets division ratios for ICLK, PCLKA, PCLKB, PCLKC, PCLKD, FCLK, BCLK
00785  */
00786 volatile uint32_t sckcr;
00787
00788 /**
00789  * @brief System Clock Control Register 2 (SCKCR2) @ 0x24
00790  * @details Sets UCK (USB clock) division ratio

```

```

00791  */
00792  volatile uint16_t sckcr2;
00793
00794  /**
00795   * @brief System Clock Control Register 3 (SCKCR3) @ 0x26
00796   * @details Selects system clock source (CKSEL bits)
00797   */
00798  volatile uint16_t sckcr3;
00799
00800  /**
00801   * @brief PLL Control Register (PLLCR) @ 0x28
00802   * @details Sets PLL multiplication factor and input divider
00803   */
00804  volatile uint16_t pllcr;
00805
00806  /**
00807   * @brief PLL Control Register 2 (PLLCR2) @ 0x2A
00808   * @details PLL enable (0=enabled, 1=disabled)
00809   */
00810  volatile uint8_t pllcr2;
00811  uint8_t          reserved3[k_system_reserved_2b_2f]; /**< Reserved @ 0x2B-0x2F */
00812
00813  /**
00814   * @brief External Bus Clock Control Register (BCKCR) @ 0x30
00815   * @details Controls external bus clock output
00816   */
00817  volatile uint8_t bckcr;
00818  uint8_t          reserved4[k_system_reserved_31]; /**< Reserved @ 0x31 */
00819
00820  /**
00821   * @brief Main Clock Oscillator Control Register (MOSCCR) @ 0x32
00822   * @details Controls main oscillator (24 MHz crystal)
00823   */
00824  volatile uint8_t mosccr;
00825
00826  /**
00827   * @brief Sub-Clock Oscillator Control Register (SOSCCR) @ 0x33
00828   * @details Controls sub-clock oscillator (32.768 kHz)
00829   */
00830  volatile uint8_t sosccr;
00831
00832  /**
00833   * @brief Low-Speed On-Chip Oscillator Control (LOCOCR) @ 0x34
00834   * @details Controls LOCO (240 kHz)
00835   */
00836  volatile uint8_t lococr;
00837
00838  /**
00839   * @brief IWDt-Dedicated Oscillator Control (ILOCOCR) @ 0x35
00840   * @details Controls IWDt oscillator (120 kHz)
00841   */
00842  volatile uint8_t ilococr;
00843
00844  /**
00845   * @brief High-Speed On-Chip Oscillator Control (HOCOCCR) @ 0x36
00846   * @details Controls HOCO
00847   */
00848  volatile uint8_t hococr;
00849
00850  /**
00851   * @brief High-Speed On-Chip Oscillator Control 2 (HOCOCCR2) @ 0x37
00852   * @details HOCO frequency selection
00853   */
00854  volatile uint8_t hococr2;
00855  uint8_t          reserved5[k_system_reserved_38_3b]; /**< Reserved @ 0x38-0x3B */
00856
00857  /**
00858   * @brief Oscillation Stabilization Flag Register (OSCOVFSR) @ 0x3C
00859   * @details Indicates when oscillators are stable (1=stable, 0=not stable):
00860   * - Bit 0 (MOOVF): Main clock oscillation stable
00861   * - Bit 1 (SOOVF): Sub-clock oscillation stable
00862   * - Bit 2 (PLOVF): PLL clock oscillation stable
00863   * - Bit 3 (HCOVF): HOCO clock oscillation stable
00864   * - Bit 4 (ILCOVF): IWDt-dedicated clock oscillation stable
00865   * - Bit 5 (PPLOVF): PPLL clock oscillation stable
00866   * - Bits 7:6: Reserved
00867   */
00868  volatile uint8_t oscovfsr;
00869  uint8_t          reserved6[k_system_reserved_3d]; /**< Reserved @ 0x3D */
00870
00871  /**
00872   * @brief Clock Output Control Register (CKOCR) @ 0x3E
00873   * @details Controls CLKOUT pin output source and divider
00874   * @see ckocr_bits_t for bit field definitions
00875   */
00876  volatile uint16_t ckocr;
00877

```

```

00878 /**
00879  * @brief Oscillation Stop Detection Control (OSTDCR) @ 0x40
00880  * @details Enables oscillation stop detection
00881  */
00882 volatile uint8_t ostdcr;
00883
00884 /**
00885  * @brief Oscillation Stop Detection Status (OSTDSR) @ 0x41
00886  * @details Indicates oscillation stop detection status
00887  */
00888 volatile uint8_t ostdsr;
00889 } rx_system_regs_t;
00890
00891 /**
00892  * @brief Get pointer to system registers
00893  * @return Volatile pointer to system register structure
00894  * @note Named system_regs() instead of system() to avoid naming conflicts
00895  */
00896 static inline volatile rx_system_regs_t* system_regs(void)
00897 {
00898     return (volatile rx_system_regs_t*)k_system_base_addr;
00899 }
00900
00901 /*
=====
00902  * Protection Register (PRCR) - At separate address 0x000803FE
00903  *
=====
00904  */
00905
00906 /**
00907  * @brief Get pointer to PRCR (Protection Register)
00908  *
00909  * @return Volatile pointer to 16-bit PRCR register
00910  *
00911  * @details
00912  * PRCR is at address 0x000803FE, which is NOT contiguous with the main
00913  * system register block (0x00080000-0x00080041). This accessor provides
00914  * the correct address for register protection control.
00915  *
00916  * @par Manual Reference:
00917  * RX72N Group User's Manual: Hardware, Chapter 13 (Register Write Protection)
00918  * - Address: 0x000803FE (verified against manual)
00919  * - Size: 16-bit
00920  * - Key: 0xA5 in upper byte required for any write
00921  *
00922  * @note Use k_rx_prcr_unlock_* and k_rx_prcr_lock values from
00923  *       rx_register_protection.h for unlock/lock operations.
00924  *
00925  * @par Example:
00926  * @code
00927  * #include "rx_register_protection.h"
00928  * *prcr_reg() = k_rx_prcr_unlock_prc1; // Unlock MSTPCR group
00929  * system_regs()->mstpcra &= ~(1UL « 15); // Enable CMT0
00930  * *prcr_reg() = k_rx_prcr_lock; // Re-lock protection
00931  * @endcode
00932  */
00933 static inline volatile uint16_t* prcr_reg(void)
00934 {
00935     return (volatile uint16_t*)k_prcr_addr;
00936 }
00937
00938 /** @brief Module stop bits for MSTPCRA register */
00939 typedef enum : uint8_t {
00940     k_mstpa_cmt23 = 14, /**< CMT2/CMT3 module stop bit in MSTPCRA */
00941 } rx_module_stop_bits_a_t;
00942
00943 /** @brief Module stop bits for MSTPCRB register */
00944 typedef enum : uint8_t {
00945     k_mstpb_usb0 = 19, /**< USB0 module stop bit in MSTPCRB */
00946     k_mstpb_crc = 23, /**< CRC module stop bit in MSTPCRB */
00947 } rx_module_stop_bits_b_t;
00948
00949 /*
=====
00950  * Reset Status Registers (RSTSR)
00951  *
=====
00952  */
00953
00954 /**
00955  * @brief Reset status register addresses
00956  * @details
00957  * IMPORTANT: RSTSR registers are NOT contiguous in memory!
00958  * - RSTSR0/1 are adjacent at 0x0008C290-0x0008C291
00959  * - RSTSR2 is at a separate address 0x000800C0
00960  * Verified against RX72N Group User's Manual Hardware (R01UH0824EJ0120 Rev.1.20)

```

```

00961 */
00962 typedef enum : uintptr_t {
00963     k_rstsr0_addr = 0x0008C290, /**< Reset Status Register 0 address (page 286) */
00964     k_rstsr1_addr = 0x0008C291, /**< Reset Status Register 1 address (page 288) */
00965     k_rstsr2_addr = 0x000800C0, /**< Reset Status Register 2 address (page 289) */
00966 } rx_rstsr_addresses_t;
00967
00968 /**
00969  * @brief Reset Status Registers 0 and 1 (contiguous)
00970  * @details
00971  * RSTSR0 and RSTSR1 are adjacent in memory at 0x0008C290-0x0008C291.
00972  */
00973 typedef struct {
00974     volatile uint8_t rstsr0; /**< Reset Status Register 0 (voltage, power-on) */
00975     volatile uint8_t rstsr1; /**< Reset Status Register 1 (cold/warm start) */
00976 } rx_rstsr01_regs_t;
00977
00978 /**
00979  * @brief Get pointer to RSTSR0/RSTSR1 registers
00980  * @return Volatile pointer to RSTSR0/1 register structure
00981  */
00982 static inline volatile rx_rstsr01_regs_t* rstsr01(void)
00983 {
00984     return (volatile rx_rstsr01_regs_t*)k_rstsr0_addr;
00985 }
00986
00987 /**
00988  * @brief Get pointer to RSTSR2 register
00989  * @return Volatile pointer to RSTSR2 register
00990  * @note RSTSR2 is at a separate address from RSTSR0/1
00991  */
00992 static inline volatile uint8_t* rstsr2(void)
00993 {
00994     return (volatile uint8_t*)k_rstsr2_addr;
00995 }
00996
00997 /** RSTSR0 bit definitions (page 286) */
00998 typedef enum : uint8_t {
00999     k_rstsr0_porf = 1U, /**< Power-On Reset Detect Flag */
01000     k_rstsr0_lvd0rf = 2U, /**< Voltage-Monitoring 0 Reset Detect Flag */
01001     k_rstsr0_lvd1rf = 4U, /**< Voltage-Monitoring 1 Reset Detect Flag */
01002     k_rstsr0_lvd2rf = 8U, /**< Voltage-Monitoring 2 Reset Detect Flag */
01003     k_rstsr0_dpsrstf = 128U, /**< Deep Software Standby Reset Flag */
01004 } rstsr0_bits_t;
01005
01006 /** RSTSR1 bit definitions (page 288) */
01007 typedef enum : uint8_t {
01008     k_rstsr1_cwsf = 1U, /**< Cold/Warm Start Determination Flag */
01009 } rstsr1_bits_t;
01010
01011 /** RSTSR2 bit definitions (page 289) */
01012 typedef enum : uint8_t {
01013     k_rstsr2_iwdtrf = 1U, /**< Independent Watchdog Timer Reset Detect Flag */
01014     k_rstsr2_wdtrf = 2U, /**< Watchdog Timer Reset Detect Flag */
01015     k_rstsr2_swrf = 4U, /**< Software Reset Detect Flag */
01016 } rstsr2_bits_t;
01017
01018 /*
=====
01019  * Software Reset Register (SWRR) - Ch06 page 290
01020  *
=====
01021  */
01022
01023 /**
01024  * @brief Software Reset Register (SWRR) address
01025  * @details
01026  * Writing 0xA501 to this register triggers an immediate software reset.
01027  * Reference: RX72N manual Ch06 section 6.2.4, page 290.
01028  */
01029 typedef enum : uintptr_t {
01030     k_swrr_addr = 0x000800C2, /**< Software Reset Register address (16-bit) */
01031 } rx_swrr_addresses_t;
01032
01033 /**
01034  * @brief Software Reset Register value
01035  * @details
01036  * Write this value to SWRR to trigger a software reset.
01037  * The MCU will reset after the internal reset time (tRESW2) elapses.
01038  * @warning This immediately resets the MCU - ensure all critical state is saved.
01039  */
01040 typedef enum : uint16_t {
01041     k_swrr_reset_key = 0xA501, /**< Write this value to trigger software reset */
01042 } rx_swrr_values_t;
01043
01044 /**
01045  * @brief Get pointer to SWRR (Software Reset Register)

```



```

01046 * @return Volatile pointer to 16-bit SWRR register
01047 *
01048 * @details
01049 * SWRR is at address 0x000800C2. Writing 0xA501 triggers a software reset.
01050 *
01051 * @warning Writing to this register causes immediate MCU reset!
01052 *
01053 * @par Manual Reference:
01054 * RX72N Group User's Manual: Hardware, Chapter 6 (Resets), page 290.
01055 *
01056 * @par Example:
01057 * @code
01058 * // Trigger a software reset (MCU will restart)
01059 * *swrr_reg() = k_swrr_reset_key;
01060 * // Code never reaches here - MCU resets immediately
01061 * @endcode
01062 */
01063 static inline volatile uint16_t* swrr_reg(void)
01064 {
01065     return (volatile uint16_t*)k_swrr_addr;
01066 }
01067
01068 /*
=====
01069 * Memory Wait Cycle Setting Register (MEMWAIT)
01070 *
=====
01071 */
01072
01073 /**
01074 * @brief MEMWAIT register address
01075 * @details
01076 * Memory Wait Cycle Setting Register controls wait cycles for flash and ECCRAM.
01077 * MANDATORY: Must be set to 1 when ICLK > 120 MHz.
01078 * Reference: RX72N manual Ch09 section 9.2.2, page 341, line 962.
01079 */
01080 typedef enum : uintptr_t {
01081     k_memwait_addr = 0x0008101C, /**< Memory Wait Cycle Setting Register */
01082 } rx_memwait_addresses_t;
01083
01084 /**
01085 * @brief MEMWAIT register bit values
01086 * @details
01087 * - 0: No wait cycle (ICLK <= 120 MHz)
01088 * - 1: One wait cycle (ICLK > 120 MHz) - REQUIRED for 240 MHz operation
01089 */
01090 typedef enum : uint8_t {
01091     k_memwait_no_wait = 0x00, /**< No wait cycle (ICLK <= 120 MHz) */
01092     k_memwait_one_wait = 0x01, /**< One wait cycle (ICLK > 120 MHz) - MANDATORY */
01093 } rx_memwait_bits_t;
01094
01095 /**
01096 * @brief Get pointer to MEMWAIT register
01097 * @return Volatile pointer to MEMWAIT register
01098 * @note CRITICAL: Must set to 1 before increasing ICLK above 120 MHz
01099 */
01100 static inline volatile uint8_t* memwait_reg(void)
01101 {
01102     return (volatile uint8_t*)k_memwait_addr;
01103 }
01104
01105 /*
=====
01106 * PPLL Control Registers (for USB clock)
01107 *
=====
01108 */
01109
01110 /**
01111 * @brief PPLL register addresses
01112 * @details
01113 * PPLL (PLL for specific purposes) generates the 48 MHz USB clock (UCLK).
01114 * Must be configured for USB functionality.
01115 * Reference: RX72N manual Ch09, PPLL frequency synthesizer section.
01116 */
01117 typedef enum : uintptr_t {
01118     k_pllcr_addr = 0x00080048, /**< PPLL Control Register @ 0x00080048 */
01119     k_pllcr2_addr = 0x0008004A, /**< PPLL Control Register 2 @ 0x0008004A */
01120     k_pllcr3_addr = 0x0008004B, /**< PPLL Control Register 3 @ 0x0008004B (PPLCK divider) */
01121 } rx_pll_addresses_t;
01122
01123 /**
01124 * @brief PPLL configuration values
01125 * @details
01126 * PPLL generates 240 MHz internal clock from 24 MHz main oscillator.
01127 * PPLLCR3 then divides by 5 to produce 48 MHz USB clock (UCLK).
01128 * Calculation: 24 MHz * 10 = 240 MHz (PPLL output); 240 / 5 = 48 MHz (UCLK)

```



```

01129 * - PPLIDIV[1:0] = 00 (divide input by 1, binary 00)
01130 * - PPLSTC[5:0] = 13h (multiply by x10.0, STC=19=0b010011 at bits 13:8; formula (STC+1)/2=mult)
01131 * - PPLSRCSEL = 0 (main clock source, shared with PLL via PLLCR.PLLSRCSEL)
01132 * - PLLCR = 0x1300
01133 *
01134 * Per manual Ch09 section 9.2.23 (p366): PPLSTC minimum valid value is 0b010011 (x10.0).
01135 * Settings below x10.0 are prohibited -- PLL output must be 120-240 MHz.
01136 */
01137 typedef enum : uint16_t {
01138     k_pll_config_48mhz = 0x1300, /**< PLLCR: PPLSTC=19 (x10.0), PPLIDIV=00 (/1): 24MHz*10=240MHz */
01139 } rx_pll_config_t;
01140
01141 /**
01142  * @brief PLLCR3 frequency divider values
01143  * @details
01144  * PLLCR3 PPLCK[3:0] selects the division ratio applied to the PLL output
01145  * before it reaches the USB clock (UCLK) pin via PACKCR.UPLSEL=1.
01146  * Per manual Ch09 section 9.2.25 (p368):
01147  * 0001=/2, 0010=/3, 0011=/4, 0100=/5 (settings other than these are prohibited)
01148  * Set AFTER PLL is operating and stable (PPLOVF=1), while USB is stopped.
01149 */
01150 typedef enum : uint8_t {
01151     k_pllcr3_div5 = 0x04, /**< PLLCR3: PPLCK=0100=/5, 240MHz/5=48MHz USB (manual p368) */
01152 } rx_pllcr3_config_t;
01153
01154 /**
01155  * @brief PLL control values
01156 */
01157 typedef enum : uint8_t {
01158     k_pll_enabled = 0x00, /**< PLLCR2: Enable PLL */
01159     k_pll_disabled = 0x01, /**< PLLCR2: Disable PLL */
01160 } rx_pll_control_t;
01161
01162 /**
01163  * @brief PLL stabilization flag in OSCOVFSR
01164  * @details
01165  * OSCOVFSR bit layout (manual Ch09 section 9.2.14):
01166  * - b0 MOOVF: Main clock oscillation stabilization flag
01167  * - b1 SOOVF: Sub-clock oscillation stabilization flag
01168  * - b2 PLOVF: PLL clock oscillation stabilization flag
01169  * - b3 HCOVF: HOCO clock oscillation stabilization flag
01170  * - b4 ILCOVF: IWDI-dedicated clock oscillation stabilization flag
01171  * - b5 PPLOVF: PLL clock oscillation stabilization flag
01172 */
01173 typedef enum : uint8_t {
01174     k_pll_stable_flag = 0x20, /**< OSCOVFSR: PPLOVF at bit 5 (0x20); NOT bit 3 */
01175 } rx_pll_flags_t;
01176
01177 /**
01178  * @brief Get pointer to PLLCR register
01179  * @return Volatile pointer to PLLCR register
01180 */
01181 static inline volatile uint16_t* pllcr_reg(void)
01182 {
01183     return (volatile uint16_t*)k_pllcr_addr;
01184 }
01185
01186 /**
01187  * @brief Get pointer to PLLCR2 register
01188  * @return Volatile pointer to PLLCR2 register
01189 */
01190 static inline volatile uint8_t* pllcr2_reg(void)
01191 {
01192     return (volatile uint8_t*)k_pllcr2_addr;
01193 }
01194
01195 /**
01196  * @brief Get pointer to PLLCR3 register
01197  * @return Volatile pointer to PLLCR3 register
01198  * @note Set PPLCK[3:0] while PLL is running and USB module is stopped.
01199 */
01200 static inline volatile uint8_t* pllcr3_reg(void)
01201 {
01202     return (volatile uint8_t*)k_pllcr3_addr;
01203 }
01204
01205 /*
=====
01206 * Reference Constants for Static Assertions
01207 * Named constants eliminate magic-number literals in the static_assert checks
01208 * below. Values are authoritative per RX72N Hardware Manual.
01209 *
=====
01210 */
01211
01212 /**
01213  * @enum rx_system_addr_refs_t

```

```

01214 * @brief Expected hardware addresses per RX72N Hardware Manual
01215 * @details
01216 * Each member documents the address mandated by the hardware manual chapter
01217 * cited in the comment. Used in static_assert to cross-check address enum
01218 * definitions against the manual reference values.
01219 * @since Version 1.0.0
01220 */
01221 typedef enum : uintptr_t {
01222     k_ref_system_base = 0x00080000, /**< System register base address (Ch05, Sec 5.2.2) */
01223     k_ref_prcr        = 0x000803FE, /**< PRCR address -- outside system struct (Ch05) */
01224     k_ref_rstsr0       = 0x0008C290, /**< RSTSR0 address (Ch05) */
01225     k_ref_rstsr1       = 0x0008C291, /**< RSTSR1 address (Ch05) */
01226     k_ref_rstsr2       = 0x000800C0, /**< RSTSR2 address (Ch05) */
01227     k_ref_swrr         = 0x000800C2, /**< SWRR address (Ch06) */
01228     k_ref_memwait      = 0x0008101C, /**< MEMWAIT address (Ch09) */
01229     k_ref_pllcr        = 0x00080048, /**< PLLCR address (Ch09) */
01230     k_ref_pllcr2       = 0x0008004A, /**< PLLCR2 address (Ch09) */
01231     k_ref_pllcr3       = 0x0008004B, /**< PLLCR3 address (Ch09 section 9.2.25, p368) */
01232 } rx_system_addr_refs_t;
01233
01234 /**
01235 * @enum rx_system_struct_layout_t
01236 * @brief Expected size and member offsets for rx_system_regs_t
01237 * @details
01238 * Authoritative per RX72N Hardware Manual Ch05, Section 5.2.2 address map.
01239 * Used in static_assert to verify the compiler produces the correct layout.
01240 * @since Version 1.0.0
01241 */
01242 typedef enum : uint8_t {
01243     k_sys_regs_size      = 0x42, /**< Total size of rx_system_regs_t (66 bytes, 0x00-0x41) */
01244     k_sys_reg_mdmonr_off = 0x00, /**< Offset of MDMONR (mode monitor) */
01245     k_sys_reg_syscr0_off = 0x06, /**< Offset of SYSCR0 */
01246     k_sys_reg_syscr1_off = 0x08, /**< Offset of SYSCR1 */
01247     k_sys_reg_sbycr_off  = 0x0C, /**< Offset of SBYCR (standby control) */
01248     k_sys_reg_mstpcra_off = 0x10, /**< Offset of MSTPCRA */
01249     k_sys_reg_mstpcrb_off = 0x14, /**< Offset of MSTPCRB */
01250     k_sys_reg_mstpcrc_off = 0x18, /**< Offset of MSTPCRC */
01251     k_sys_reg_mstpcrd_off = 0x1C, /**< Offset of MSTPCRD */
01252     k_sys_reg_sckcr_off  = 0x20, /**< Offset of SCKCR (system clock control) */
01253     k_sys_reg_sckcr2_off = 0x24, /**< Offset of SCKCR2 */
01254     k_sys_reg_sckcr3_off = 0x26, /**< Offset of SCKCR3 (clock source select) */
01255     k_sys_reg_pllcr_off  = 0x28, /**< Offset of PLLCR */
01256     k_sys_reg_pllcr2_off = 0x2A, /**< Offset of PLLCR2 (PLL enable) */
01257     k_sys_reg_bckcr_off  = 0x30, /**< Offset of BCKCR (external bus clock) */
01258     k_sys_reg_mosccr_off = 0x32, /**< Offset of MOSCCR (main oscillator control) */
01259     k_sys_reg_sosccr_off = 0x33, /**< Offset of SOSCCR (sub-clock control) */
01260     k_sys_reg_lococr_off = 0x34, /**< Offset of LOCOCR (LOCO control) */
01261     k_sys_reg_ilococr_off = 0x35, /**< Offset of ILOCOCR (IWDG oscillator control) */
01262     k_sys_reg_hococr_off = 0x36, /**< Offset of HOCOCR (HOCO control) */
01263     k_sys_reg_hococr2_off = 0x37, /**< Offset of HOCOCR2 (HOCO frequency) */
01264     k_sys_reg_oscovfsr_off = 0x3C, /**< Offset of OSCOVFSR (oscillation stable flags) */
01265     k_sys_reg_ckocr_off  = 0x3E, /**< Offset of CKOCR (clock output control) */
01266     k_sys_reg_ostdcr_off = 0x40, /**< Offset of OSTDCR (oscillation stop detect control) */
01267     k_sys_reg_ostdsr_off = 0x41, /**< Offset of OSTDSR (oscillation stop detect status) */
01268 } rx_system_struct_layout_t;
01269
01270 /**
01271 * @enum rx_rstsr01_layout_t
01272 * @brief Expected size and member offsets for rx_rstsr01_regs_t
01273 * @since Version 1.0.0
01274 */
01275 typedef enum : uint8_t {
01276     k_rstsr01_struct_size = 2, /**< Total size of rx_rstsr01_regs_t (2 bytes) */
01277     k_rstsr0_member_off  = 0, /**< Byte offset of rstsr0 member */
01278     k_rstsr1_member_off  = 1, /**< Byte offset of rstsr1 member */
01279 } rx_rstsr01_layout_t;
01280
01281 /**
01282 * @enum rx_prcr_offset_t
01283 * @brief PRCR register offset from system register base address
01284 * @details PRCR is not part of rx_system_regs_t; it sits at base+0x3FE.
01285 * @since Version 1.0.0
01286 */
01287 typedef enum : uint16_t {
01288     k_prcr_from_sys_base = 0x03FE, /**< PRCR offset from system base (NOT in system struct) */
01289 } rx_prcr_offset_t;
01290
01291 /**
01292 * @enum rx_clock_verify_t
01293 * @brief Reference values for clock field-width and PLL multiplier assertions
01294 * @since Version 1.0.0
01295 */
01296 typedef enum : uint8_t {
01297     k_nibble_all_ones = 0x0F, /**< All bits set in 4-bit field (verifies 4-bit mask width) */
01298     k_pllcr_stc_for_x10 =
01299         19, /**< PLLCR STC field value for x10.0 (manual p345: 0b010011=19; formula (STC+1)/2=mult) */
01300 } rx_clock_verify_t;

```

```

01301
01302 /*
=====
01303 * Static Assertions - Verify Register Layout at Compile Time
01304 * Verified against RX72N Manual Ch05, Section 5.2.2 Address Map
01305 *
=====
01306 */
01307
01308 /* Verify base addresses match Hardware Manual Ch05 */
01309 static_assert((uintptr_t)k_system_base_addr == (uintptr_t)k_ref_system_base,
01310               "System register base address incorrect");
01311 static_assert((uintptr_t)k_prcr_addr == (uintptr_t)k_ref_prcr,
01312               "PRCR address incorrect (must be 0x000803FE, NOT in system struct)");
01313
01314 /* Verify system register structure layout - offsets per RX72N Manual Ch05 */
01315 static_assert(sizeof(rx_system_regs_t) == k_sys_regs_size,
01316               "System register structure size incorrect (expected 0x42)");
01317 static_assert(offsetof(rx_system_regs_t, mdmonr) == k_sys_reg_mdmonr_off,
01318               "MDMONR offset incorrect (expected 0x00)");
01319 static_assert(offsetof(rx_system_regs_t, syscr0) == k_sys_reg_syscr0_off,
01320               "SYSCR0 offset incorrect (expected 0x06)");
01321 static_assert(offsetof(rx_system_regs_t, syscr1) == k_sys_reg_syscr1_off,
01322               "SYSCR1 offset incorrect (expected 0x08)");
01323 static_assert(offsetof(rx_system_regs_t, sbycr) == k_sys_reg_sbycr_off,
01324               "SBYCR offset incorrect (expected 0x0C)");
01325 static_assert(offsetof(rx_system_regs_t, mstpcra) == k_sys_reg_mstpcra_off,
01326               "MSTPCRA offset incorrect (expected 0x10)");
01327 static_assert(offsetof(rx_system_regs_t, mstpcrb) == k_sys_reg_mstpcrb_off,
01328               "MSTPCRB offset incorrect (expected 0x14)");
01329 static_assert(offsetof(rx_system_regs_t, mstpcrc) == k_sys_reg_mstpcrc_off,
01330               "MSTPCRC offset incorrect (expected 0x18)");
01331 static_assert(offsetof(rx_system_regs_t, mstpcrd) == k_sys_reg_mstpcrd_off,
01332               "MSTPCRD offset incorrect (expected 0x1C)");
01333 static_assert(offsetof(rx_system_regs_t, sckcr) == k_sys_reg_sckcr_off,
01334               "SCKCR offset incorrect (expected 0x20)");
01335 static_assert(offsetof(rx_system_regs_t, sckcr2) == k_sys_reg_sckcr2_off,
01336               "SCKCR2 offset incorrect (expected 0x24)");
01337 static_assert(offsetof(rx_system_regs_t, sckcr3) == k_sys_reg_sckcr3_off,
01338               "SCKCR3 offset incorrect (expected 0x26)");
01339 static_assert(offsetof(rx_system_regs_t, pllcr) == k_sys_reg_pllcr_off,
01340               "PLLCR offset incorrect (expected 0x28)");
01341 static_assert(offsetof(rx_system_regs_t, pllcr2) == k_sys_reg_pllcr2_off,
01342               "PLLCR2 offset incorrect (expected 0x2A)");
01343 static_assert(offsetof(rx_system_regs_t, bckcr) == k_sys_reg_bckcr_off,
01344               "BCKCR offset incorrect (expected 0x30)");
01345 static_assert(offsetof(rx_system_regs_t, mosccr) == k_sys_reg_mosccr_off,
01346               "MOSCCR offset incorrect (expected 0x32)");
01347 static_assert(offsetof(rx_system_regs_t, sosccr) == k_sys_reg_sosccr_off,
01348               "SOSCCR offset incorrect (expected 0x33)");
01349 static_assert(offsetof(rx_system_regs_t, lococr) == k_sys_reg_lococr_off,
01350               "LOCOCR offset incorrect (expected 0x34)");
01351 static_assert(offsetof(rx_system_regs_t, ilococr) == k_sys_reg_ilococr_off,
01352               "ILOCOCR offset incorrect (expected 0x35)");
01353 static_assert(offsetof(rx_system_regs_t, hococr) == k_sys_reg_hococr_off,
01354               "HOCOCC offset incorrect (expected 0x36)");
01355 static_assert(offsetof(rx_system_regs_t, hococr2) == k_sys_reg_hococr2_off,
01356               "HOCOCC2 offset incorrect (expected 0x37)");
01357 static_assert(offsetof(rx_system_regs_t, oscovfsr) == k_sys_reg_oscovfsr_off,
01358               "OSCOVFSR offset incorrect (expected 0x3C)");
01359 static_assert(offsetof(rx_system_regs_t, ckocr) == k_sys_reg_ckocr_off,
01360               "CKOCR offset incorrect (expected 0x3E)");
01361 static_assert(offsetof(rx_system_regs_t, ostdcr) == k_sys_reg_ostdcr_off,
01362               "OSTDCR offset incorrect (expected 0x40)");
01363 static_assert(offsetof(rx_system_regs_t, ostdsr) == k_sys_reg_ostdsr_off,
01364               "OSTDSR offset incorrect (expected 0x41)");
01365
01366 /* Verify RSTSR addresses match Hardware Manual Ch05 */
01367 static_assert((uintptr_t)k_rstsr0_addr == (uintptr_t)k_ref_rstsr0, "RSTSR0 address incorrect");
01368 static_assert((uintptr_t)k_rstsr1_addr == (uintptr_t)k_ref_rstsr1, "RSTSR1 address incorrect");
01369 static_assert((uintptr_t)k_rstsr2_addr == (uintptr_t)k_ref_rstsr2, "RSTSR2 address incorrect");
01370 static_assert(sizeof(rx_rstsr01_regs_t) == k_rstsr01_struct_size,
01371               "RSTSR01 structure size incorrect");
01372 static_assert(offsetof(rx_rstsr01_regs_t, rstsr0) == k_rstsr0_member_off,
01373               "RSTSR0 offset incorrect");
01374 static_assert(offsetof(rx_rstsr01_regs_t, rstsr1) == k_rstsr1_member_off,
01375               "RSTSR1 offset incorrect");
01376
01377 /* Verify SWRR address matches Hardware Manual Ch06 */
01378 static_assert((uintptr_t)k_swrr_addr == (uintptr_t)k_ref_swrr, "SWRR address incorrect");
01379
01380 /* Verify MEMWAIT address match Hardware Manual Ch09 */
01381 static_assert((uintptr_t)k_memwait_addr == (uintptr_t)k_ref_memwait, "MEMWAIT address incorrect");
01382
01383 /* Verify PPLL addresses match Hardware Manual Ch09 */
01384 static_assert((uintptr_t)k_ppllcr_addr == (uintptr_t)k_ref_ppllcr, "PPLLCR address incorrect");
01385 static_assert((uintptr_t)k_ppllcr2_addr == (uintptr_t)k_ref_ppllcr2, "PPLLCR2 address incorrect");

```

```

01386 static_assert((uintptr_t)k_pllcr3_addr == (uintptr_t)k_ref_pllcr3, "PLLCR3 address incorrect");
01387
01388 /* Verify PRCR is NOT embedded in system struct (critical check) */
01389 static_assert((uintptr_t)(k_prcr_addr - k_system_base_addr) == (uintptr_t)k_prcr_from_sys_base,
01390               "PRCR must be at offset 0x3FE from system base, not embedded in struct");
01391
01392 /* Verify SCKCR clock divider field positions and masks */
01393 static_assert((k_sckcr_pckd_mask » k_sckcr_pckd_shift) == (uint32_t)k_nibble_all_ones,
01394               "SCKCR PCKD field mask incorrect");
01395 static_assert((k_sckcr_ick_mask » k_sckcr_ick_shift) == (uint32_t)k_nibble_all_ones,
01396               "SCKCR ICK field mask incorrect");
01397 static_assert((k_sckcr_fck_mask » k_sckcr_fck_shift) == (uint32_t)k_nibble_all_ones,
01398               "SCKCR FCK field mask incorrect");
01399
01400 /* Verify STAR project SCKCR configuration value */
01401 static_assert((k_sckcr_star_240mhz & k_sckcr_ick_mask) ==
01402               ((uint32_t)k_clock_div_1 « k_sckcr_ick_shift),
01403               "STAR SCKCR: ICLK should be /1 (240 MHz)");
01404 static_assert((k_sckcr_star_240mhz & k_sckcr_pcka_mask) ==
01405               ((uint32_t)k_clock_div_2 « k_sckcr_pcka_shift),
01406               "STAR SCKCR: PCLKA should be /2 (120 MHz)");
01407 static_assert((k_sckcr_star_240mhz & k_sckcr_pckb_mask) ==
01408               ((uint32_t)k_clock_div_4 « k_sckcr_pckb_shift),
01409               "STAR SCKCR: PCLKB should be /4 (60 MHz)");
01410 static_assert((k_sckcr_star_240mhz & k_sckcr_fck_mask) ==
01411               ((uint32_t)k_clock_div_4 « k_sckcr_fck_shift),
01412               "STAR SCKCR: FCLK should be /4 (60 MHz)");
01413
01414 /* Verify PLLCR multiplier encoding */
01415 static_assert((k_pllcr_star_24mhz_to_240mhz & k_pllcr_stc_mask) ==
01416               ((uint32_t)k_pllcr_stc_for_x10 « k_pllcr_stc_shift),
01417               "STAR PLLCR: STC should be 9 (multiply by 10)");
01418 static_assert((k_pllcr_star_24mhz_to_240mhz & k_pllcr_pldiv_mask) == k_pllcr_pldiv_1,
01419               "STAR PLLCR: PLIDIV should be /1");
01420 static_assert((k_pllcr_star_24mhz_to_240mhz & k_pllcr_src_mask) == k_pllcr_src_main,
01421               "STAR PLLCR: Source should be main oscillator");
01422
01423 /* Verify clock source select values */
01424 static_assert((k_sckcr3_cksel_pll & k_sckcr3_cksel_mask) == k_sckcr3_cksel_pll,
01425               "SCKCR3 PLL select value incorrect");
01426 static_assert((k_sckcr3_cksel_main & k_sckcr3_cksel_mask) == k_sckcr3_cksel_main,
01427               "SCKCR3 main clock select value incorrect");
01428
01429 /* Verify CKOCR bit positions */
01430 static_assert((k_ckocr_ckosel_pll & k_ckocr_ckosel_mask) == k_ckocr_ckosel_pll,
01431               "CKOCR PLL select value incorrect");
01432 static_assert((k_ckocr_ckodiv_4 & k_ckocr_ckodiv_mask) == k_ckocr_ckodiv_4,
01433               "CKOCR divide-by-4 value incorrect");
01434
01435 #ifdef __cplusplus
01436 }
01437 #endif

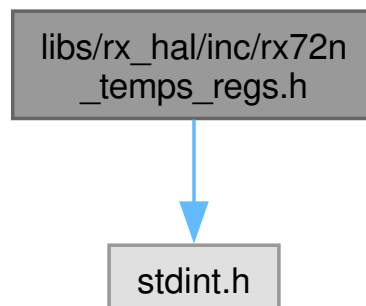
```

8.61 libs/rx_hal/inc/rx72n_temps_regs.h File Reference

RX72N Temperature Sensor (TEMPS) Register Definitions.

#include <stdint.h>

Include dependency graph for rx72n_temps_regs.h:



Enumerations

- enum `temps_addresses_t` : `uintptr_t` { `k_temps_tscr_addr` = 0x0008C500U , `k_temps_tscdr_addr` = 0xFE7F7D7CU }
- Temperature Sensor register addresses.
- enum `tscr_bits_t` : `uint8_t` {
`k_tscr_tsen_bit` = 0x80U , `k_tscr_tsoe_bit` = 0x10U , `k_tscr_disabled` = 0x00U ,
`k_tscr_sensor_only` = 0x80U ,
`k_tscr_sensor_with_output` = 0x90U , `k_tscr_reserved_mask` = 0x6FU }
- Temperature Sensor Control Register (TSCR) bit definitions.
- enum `temps_timing_t` : `uint16_t` { `k_temps_tstbl_us` = 30U , `k_temps_tostbl_us` = 0U }
- Temperature sensor timing constants.
- enum `temps_calibration_t` : `uint16_t` { `k_temps_cal_temp_celsius` = 125U , `k_temps_adc_full_scale` = 4096U , `k_temps_avcc_mv` = 3300U , `k_temps_slope_uv_per_c` = 4100U }
- Temperature sensor calibration constants.
- enum `temps_mstpcr_t` : `uint32_t` { `k_temps_mstpcrb_bit` = (1U << 8U) }
- Temperature sensor module stop control bits.

Functions

- static volatile `uint8_t` * `temps_tscr_reg` (void)
Get pointer to Temperature Sensor Control Register (TSCR).
- static volatile const `uint32_t` * `temps_tscdr_reg` (void)
Get pointer to Temperature Sensor Calibration Data Register (TSCDR).

8.61.1 Detailed Description

RX72N Temperature Sensor (TEMPS) Register Definitions.

Complete register definitions for the RX72N internal temperature sensor. The temperature sensor outputs a voltage proportional to die temperature, which can be converted to temperature via the 12-bit ADC unit 1.

Temperature Measurement Overview

The RX72N has a built-in temperature sensor that:

1. Outputs voltage proportional to temperature
2. Connects to 12-bit A/D converter unit 1
3. Has factory-programmed calibration data (CAL125 @ 125degC)

Temperature Calculation Formula

$T = (V_s - V_1) / \text{Slope} + 125$ [degC] Where:

- V_s : Measured voltage (from ADC)
- V_1 : Calibration voltage at 125degC = $3.3 \times \text{TSCDR} / 4096$ [V]
- Slope: Temperature slope from Table 63.57 (typ. -4.1 mV/degC)

Register Summary

Register	Address	Size	Description
TSCR	0x0008C500	8	Temperature Sensor Control Register
TSCDR	0xFE7F7D7C	32	Calibration Data (read-only, ROM)

Usage Sequence (from Ch58 manual)

1. Clear module stop bit (MSTPCRB.MSTPC8 = 0)
2. Configure ADC unit 1 for temperature sensor input
3. Set TSCR.TSEN = 1 (start temperature sensor)
4. Wait 30 us for reference voltage stabilization (tTSTBL)
5. Set TSCR.TSOE = 1 (enable output to ADC)
6. Wait 0 us for output stabilization (tOSTBL)
7. Start ADC conversion
8. Read ADC result from ADTSDR register
9. Calculate temperature using formula above

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] Single-statement accessors
- Rule 5: [OK] Static assertions verify addresses
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 58: Temperature Sensor (TEMPS), pages 2963-2969
- Section 58.2.1: TSCR register @ 0x0008C500
- Section 58.2.2: TSCDR register @ 0xFE7F7D7C
- Section 63 Table 63.57: Temperature slope characteristics

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_temps_regs.h](#).

8.61.2 Enumeration Type Documentation

8.61.2.1 temps_addresses_t

enum [temps_addresses_t](#) : uintptr_t

Temperature Sensor register addresses.

Addresses verified against RX72N Hardware Manual Ch58.

Note

Addresses verified 2026-01-29 against manual section 58.2

Since

Version 1.0.0

Enumerator

k_temps_tscr_addr	Temperature Sensor Control Register (TSCR) address. 8-bit register at 0x0008C500 controlling sensor enable and output. Manual ref: Ch58 section 58.2.1
k_temps_tsedr_addr	Temperature Sensor Calibration Data Register (TSEDR) address. 32-bit read-only register at 0xFE7F7D7C containing factory calibration. Located in on-chip ROM area (TEMPSCONST). Manual ref: Ch58 section 58.2.2 Note Only readable when SYSCR0.ROME = 1 (on-chip ROM enabled)

Definition at line 89 of file [rx72n_temps_regs.h](#).

```

00089         : uintptr_t {
00090     /**
00091      * @brief Temperature Sensor Control Register (TSCR) address
00092      * @details
00093      * 8-bit register at 0x0008C500 controlling sensor enable and output.
00094      * Manual ref: Ch58 section 58.2.1
00095      */
00096     k_temps_tscr_addr = 0x0008C500U,
00097
00098     /**
00099      * @brief Temperature Sensor Calibration Data Register (TSEDR) address
00100      * @details
00101      * 32-bit read-only register at 0xFE7F7D7C containing factory calibration.
00102      * Located in on-chip ROM area (TEMPSCONST).
00103      * Manual ref: Ch58 section 58.2.2
00104      * @note Only readable when SYSCR0.ROME = 1 (on-chip ROM enabled)
00105      */
00106     k_temps_tsedr_addr = 0xFE7F7D7CU,
00107 } temps_addresses_t;

```

8.61.2.2 temps_calibration_t

enum [temps_calibration_t](#) : uint16_t

Temperature sensor calibration constants.

Constants for temperature calculation using factory calibration data.

Temperature Calculation

The calibration data (TSCDR) was measured at:

- $T_j = 125\text{degC}$ (junction temperature)
- $AVCC1 = 3.3\text{V}$ (ADC reference voltage)

Note

Manual ref: Ch58 section 58.3.1

Since

Version 1.0.0

Enumerator

k_temps_cal_temp_celsius	Calibration temperature (125degC). Factory calibration performed at this temperature
k_temps_adc_full_scale	ADC full scale value (12-bit = 4096).
k_temps_avcc_mv	ADC reference voltage in millivolts (3300 mV = 3.3V).
k_temps_slope_uv_per_c	Typical temperature slope in uV/degC (absolute value). From Table 63.57: typ. -4.1 mV/degC = -4100 uV/degC Note Actual value varies by chip; two-point calibration recommended

Definition at line 221 of file [rx72n_temps_regs.h](#).

```

00221     : uint16_t {
00222     /**
00223      * @brief Calibration temperature (125degC)
00224      * @details Factory calibration performed at this temperature
00225      */
00226     k_temps_cal_temp_celsius = 125U,
00227     /**
00228      * @brief ADC full scale value (12-bit = 4096)
00229      */
00230     k_temps_adc_full_scale = 4096U,
00231     /**
00232      * @brief ADC reference voltage in millivolts (3300 mV = 3.3V)
00233      */
00234     k_temps_avcc_mv = 3300U,
00235     /**
00236      * @brief Typical temperature slope in uV/degC (absolute value)
00237      * @details From Table 63.57: typ. -4.1 mV/degC = -4100 uV/degC
00238      * @note Actual value varies by chip; two-point calibration recommended
00239      */
00240     k_temps_slope_uv_per_c = 4100U,
00241     } temps_calibration_t;

```

8.61.2.3 temps_mstpcr_t

```
enum temps\_mstpcr\_t : uint32_t
```

Temperature sensor module stop control bits.

The temperature sensor is controlled by MSTPCRB.MSTPC8. Must be cleared (0) before accessing TEMPS registers.

Note

Manual ref: Ch11 (Low Power Consumption) module stop tables

Since

Version 1.0.0

Enumerator

k_temps_mstpcrb_bit	MSTPCRB bit for temperature sensor (bit 8). 0: Temperature sensor module is running 1: Temperature sensor module is stopped (default) Manual ref: Ch11 section 11.2.3 MSTPCRB, bit b8 = MSTPB8 (Temperature Sensor)
---------------------	--

Definition at line 262 of file [rx72n_temps_regs.h](#).

```

00262         : uint32_t {
00263     /**
00264     * @brief MSTPCRB bit for temperature sensor (bit 8)
00265     * @details
00266     * - 0: Temperature sensor module is running
00267     * - 1: Temperature sensor module is stopped (default)
00268     * Manual ref: Ch11 section 11.2.3 MSTPCRB, bit b8 = MSTPB8 (Temperature Sensor)
00269     */
00270     k_temps_mstpcrb_bit = (1U « 8U),
00271 } temps_mstpcr_t;

```

8.61.2.4 temps_timing_t

enum [temps_timing_t](#) : uint16_t

Temperature sensor timing constants.

Timing requirements from Ch58 Table 58.2 for proper sensor operation.

Note

Manual ref: Ch58 section 58.3.4, Table 58.2

Since

Version 1.0.0

Enumerator

k_temps_tstbl_us	Reference voltage stabilization wait time (tTSTBL). Minimum 30 us wait after setting TSEN = 1
k_temps_tostbl_us	Output stabilization wait time (tOSTBL). Minimum 0 us wait after setting TSOE = 1

Definition at line 187 of file [rx72n_temps_regs.h](#).

```

00187         : uint16_t {
00188     /**
00189     * @brief Reference voltage stabilization wait time (tTSTBL)
00190     * @details Minimum 30 us wait after setting TSEN = 1
00191     */
00192     k_temps_tstbl_us = 30U,
00193
00194     /**
00195     * @brief Output stabilization wait time (tOSTBL)
00196     * @details Minimum 0 us wait after setting TSOE = 1
00197     */
00198     k_temps_tostbl_us = 0U,
00199 } temps_timing_t;

```

8.61.2.5 tscr_bits_t

```
enum tscr_bits_t : uint8_t
```

Temperature Sensor Control Register (TSCR) bit definitions.

Controls the temperature sensor enable state and output to ADC.

Register Layout (8-bit @ 0x0008C500)

Bit	Field	R/W	Reset	Description
7	TSEN	R/W	0	Temperature Sensor Enable
6↔ 5	-	R/W	0	Reserved (write 0)
4	TSOE	R/W	0	Temperature Sensor Output Enable
3↔ 0	-	R/W	0	Reserved (write 0)

Note

Manual ref: Ch58 section 58.2.1 (page 2964)

Since

Version 1.0.0

Enumerator

k_tscr_tsen_bit	TSEN bit position (bit 7). Temperature Sensor Enable • 0: Stops the temperature sensor • 1: Starts the temperature sensor
k_tscr_tsoe_bit	TSOE bit position (bit 4). Temperature Sensor Output Enable • 0: Disables output to 12-bit ADC unit 1 • 1: Enables output to 12-bit ADC unit 1
k_tscr_disabled	Temperature sensor disabled, output disabled (default).
k_tscr_sensor_only	Temperature sensor enabled, output disabled. Note Use this during tTSTBL wait (30 us stabilization)

k_tscr_sensor_with_output	Temperature sensor enabled with output to ADC. Note Set this after tTSTBL wait before starting ADC conversion
k_tscr_reserved_mask	Reserved bits mask (bits 6:5, 3:0 must be 0).

Definition at line 132 of file rx72n_temps_regs.h.

```

00132         : uint8_t {
00133     /**
00134      * @brief TSEN bit position (bit 7)
00135      * @details Temperature Sensor Enable
00136      * - 0: Stops the temperature sensor
00137      * - 1: Starts the temperature sensor
00138      */
00139     k_tscr_tsen_bit = 0x80U,
00140
00141     /**
00142      * @brief TSOE bit position (bit 4)
00143      * @details Temperature Sensor Output Enable
00144      * - 0: Disables output to 12-bit ADC unit 1
00145      * - 1: Enables output to 12-bit ADC unit 1
00146      */
00147     k_tscr_tsoe_bit = 0x10U,
00148
00149     /**
00150      * @brief Temperature sensor disabled, output disabled (default)
00151      */
00152     k_tscr_disabled = 0x00U,
00153
00154     /**
00155      * @brief Temperature sensor enabled, output disabled
00156      * @note Use this during tTSTBL wait (30 us stabilization)
00157      */
00158     k_tscr_sensor_only = 0x80U,
00159
00160     /**
00161      * @brief Temperature sensor enabled with output to ADC
00162      * @note Set this after tTSTBL wait before starting ADC conversion
00163      */
00164     k_tscr_sensor_with_output = 0x90U,
00165
00166     /**
00167      * @brief Reserved bits mask (bits 6:5, 3:0 must be 0)
00168      */
00169     k_tscr_reserved_mask = 0x6FU,
00170 } tscr_bits_t;

```

8.61.3 Function Documentation

8.61.3.1 temps_tscdr_reg()

```
volatile const uint32_t * temps_tscdr_reg (
    void ) [inline], [static]
```

Get pointer to Temperature Sensor Calibration Data Register (TSCDR).

Returns a volatile pointer to TSCDR at address 0xFE7F7D7C. This is a 32-bit read-only register containing factory calibration data.

Calibration Data Format

The lower 12 bits contain the ADC value measured at 125degC and 3.3V:

- Bits [11:0]: ADC value (0-4095)
- Bits [31:12]: Reserved (read as 0)

Converting to Voltage

$$V1 = 3.3 \times (\text{TSCDR} \& 0\text{xFFF}) / 4096 \text{ [V]}$$

Returns

Volatile pointer to TSCDR register (32-bit, read-only)

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

On-chip ROM must be enabled (SYSCR0.ROME = 1)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Warning

This register is read-only; writes have no effect

Example (Read Calibration):

```
// Read calibration data
uint32_t cal125 = *temps_tscdr_reg() & 0xFFFF;

// Calculate reference voltage at 125degC
// V1 = 3.3 x cal125 / 4096 (in fixed-point or float)
```

See also

[k_temps_tscdr_addr](#) Address constant

Since

Version 1.0.0

Definition at line 353 of file [rx72n_temps_regs.h](#).

```
00354 {
00355     return (volatile const uint32_t*)k_temps_tscdr_addr;
00356 }
```

References [k_temps_tscdr_addr](#).

8.61.3.2 temps_tscr_reg()

```
volatile uint8_t * temps_tscr_reg (
    void ) [inline], [static]
```

Get pointer to Temperature Sensor Control Register (TSCR).

Returns a volatile pointer to TSCR at address 0x0008C500.

Returns

Volatile pointer to TSCR register (8-bit)

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

Temperature sensor module stop (MSTPCRB.MSTPC8) must be cleared

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example (Enable Temperature Sensor):

```
// 1. Start temperature sensor
*temps_tscr_reg() = k_tscr_sensor_only;

// 2. Wait for stabilization (30 us)
delay_us(k_temps_tstbl_us);

// 3. Enable output to ADC
*temps_tscr_reg() = k_tscr_sensor_with_output;

// 4. Start ADC conversion (see S12AD1 registers)
```

See also

[k_temps_tscr_addr](#) Address constant

[tscr_bits_t](#) Control bits

Since

Version 1.0.0

Definition at line 311 of file [rx72n_temps_regs.h](#).

```
00312 {
00313     return (volatile uint8_t*)k_temps_tscr_addr;
00314 }
```

References [k_temps_tscr_addr](#).

8.62 rx72n_temps_regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_temps_regs.h
00003  * @brief RX72N Temperature Sensor (TEMPS) Register Definitions
00004  *
00005  * @details
00006  * Complete register definitions for the RX72N internal temperature sensor.
00007  * The temperature sensor outputs a voltage proportional to die temperature,
00008  * which can be converted to temperature via the 12-bit ADC unit 1.
00009  *
00010  * @par Temperature Measurement Overview
00011  * The RX72N has a built-in temperature sensor that:
00012  * 1. Outputs voltage proportional to temperature
00013  * 2. Connects to 12-bit A/D converter unit 1
00014  * 3. Has factory-programmed calibration data (CAL125 @ 125degC)
00015  *
00016  * @par Temperature Calculation Formula
00017  *  $T = (V_s - V_1) / \text{Slope} + 125 \text{ [degC]}$ 
00018  * Where:
00019  * -  $V_s$ : Measured voltage (from ADC)
00020  * -  $V_1$ : Calibration voltage at 125degC =  $3.3 \times \text{TSCDR} / 4096 \text{ [V]}$ 
00021  * - Slope: Temperature slope from Table 63.57 (typ. -4.1 mV/degC)
00022  *
00023  * @par Register Summary
00024  * | Register | Address | Size | Description |
00025  * |-----|-----|-----|-----|
00026  * | TSCR | 0x0008C500 | 8 | Temperature Sensor Control Register |
00027  * | TSCDR | 0xFE7F7D7C | 32 | Calibration Data (read-only, ROM) |
00028  *
00029  * @par Usage Sequence (from Ch58 manual)
00030  * 1. Clear module stop bit (MSTPCRB.MSTPC8 = 0)
00031  * 2. Configure ADC unit 1 for temperature sensor input
00032  * 3. Set TSCR.TSEN = 1 (start temperature sensor)
00033  * 4. Wait 30 us for reference voltage stabilization (tTSTBL)
00034  * 5. Set TSCR.TSOE = 1 (enable output to ADC)
00035  * 6. Wait 0 us for output stabilization (tOSTBL)
00036  * 7. Start ADC conversion
00037  * 8. Read ADC result from ADTSDR register
00038  * 9. Calculate temperature using formula above
00039  *
00040  * @par NASA Power of 10 Compliance
00041  * - Rule 1: [OK] No goto, setjmp, or recursion
00042  * - Rule 2: [OK] N/A (no loops)
00043  * - Rule 3: [OK] No dynamic memory allocation
00044  * - Rule 4: [OK] Single-statement accessors
00045  * - Rule 5: [OK] Static assertions verify addresses
00046  * - Rule 6: [OK] Minimal scope
00047  * - Rule 7: [OK] N/A
00048  * - Rule 8: [OK] All constants use C23 typed enums
00049  * - Rule 9: [OK] No function pointers
00050  * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00051  *
00052  * @par Manual References
00053  * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00054  * - Chapter 58: Temperature Sensor (TEMPS), pages 2963-2969
00055  * - Section 58.2.1: TSCR register @ 0x0008C500
00056  * - Section 58.2.2: TSCDR register @ 0xFE7F7D7C
00057  * - Section 63 Table 63.57: Temperature slope characteristics
00058  *
00059  * @author Locked, Inc.
00060  * @date 2026-01-29
00061  * @copyright Copyright (c) 2026 Locked Inc.
00062  * SPDX-License-Identifier: MIT
00063  * @since Version 1.0.0
00064  */
00065
00066 #pragma once
00067
00068 #include <stdint.h>
00069
00070 #ifdef __cplusplus
00071 extern "C" {
00072 #endif
00073
00074 /**
=====
00075  * Temperature Sensor Register Addresses
00076  *
=====
00077  */
00078
00079 /**
00080  * @enum temps_addresses_t

```

```

00081 * @brief Temperature Sensor register addresses
00082 *
00083 * @details
00084 * Addresses verified against RX72N Hardware Manual Ch58.
00085 *
00086 * @note Addresses verified 2026-01-29 against manual section 58.2
00087 * @since Version 1.0.0
00088 */
00089 typedef enum : uintptr_t {
00090 /**
00091  * @brief Temperature Sensor Control Register (TSCR) address
00092  * @details
00093  * 8-bit register at 0x0008C500 controlling sensor enable and output.
00094  * Manual ref: Ch58 section 58.2.1
00095  */
00096 k_temps_tscr_addr = 0x0008C500U,
00097
00098 /**
00099  * @brief Temperature Sensor Calibration Data Register (TSCDR) address
00100  * @details
00101  * 32-bit read-only register at 0xFE7F7D7C containing factory calibration.
00102  * Located in on-chip ROM area (TEMPSCONST).
00103  * Manual ref: Ch58 section 58.2.2
00104  * @note Only readable when SYSCR0.ROME = 1 (on-chip ROM enabled)
00105  */
00106 k_temps_tscdr_addr = 0xFE7F7D7CU,
00107 } temps_addresses_t;
00108
00109 /*
=====
00110 * TSCR Register Bits
00111 *
=====
00112 */
00113
00114 /**
00115 * @enum tscr_bits_t
00116 * @brief Temperature Sensor Control Register (TSCR) bit definitions
00117 *
00118 * @details
00119 * Controls the temperature sensor enable state and output to ADC.
00120 *
00121 * @par Register Layout (8-bit @ 0x0008C500)
00122 * | Bit | Field | R/W | Reset | Description |
00123 * |-----|-----|-----|-----|-----|
00124 * | 7 | TSEN | R/W | 0 | Temperature Sensor Enable |
00125 * | 6:5 | - | R/W | 0 | Reserved (write 0) |
00126 * | 4 | TSOE | R/W | 0 | Temperature Sensor Output Enable |
00127 * | 3:0 | - | R/W | 0 | Reserved (write 0) |
00128 *
00129 * @note Manual ref: Ch58 section 58.2.1 (page 2964)
00130 * @since Version 1.0.0
00131 */
00132 typedef enum : uint8_t {
00133 /**
00134  * @brief TSEN bit position (bit 7)
00135  * @details Temperature Sensor Enable
00136  * - 0: Stops the temperature sensor
00137  * - 1: Starts the temperature sensor
00138  */
00139 k_tscr_tsen_bit = 0x80U,
00140
00141 /**
00142  * @brief TSOE bit position (bit 4)
00143  * @details Temperature Sensor Output Enable
00144  * - 0: Disables output to 12-bit ADC unit 1
00145  * - 1: Enables output to 12-bit ADC unit 1
00146  */
00147 k_tscr_tsoe_bit = 0x10U,
00148
00149 /**
00150  * @brief Temperature sensor disabled, output disabled (default)
00151  */
00152 k_tscr_disabled = 0x00U,
00153
00154 /**
00155  * @brief Temperature sensor enabled, output disabled
00156  * @note Use this during tTSTBL wait (30 us stabilization)
00157  */
00158 k_tscr_sensor_only = 0x80U,
00159
00160 /**
00161  * @brief Temperature sensor enabled with output to ADC
00162  * @note Set this after tTSTBL wait before starting ADC conversion
00163  */
00164 k_tscr_sensor_with_output = 0x90U,
00165

```

```

00166 /**
00167  * @brief Reserved bits mask (bits 6:5, 3:0 must be 0)
00168  */
00169 k_tscr_reserved_mask = 0x6FU,
00170 } tscr_bits_t;
00171
00172 /*
=====
00173 * Temperature Sensor Timing Constants
00174 *
=====
00175 */
00176
00177 /**
00178  * @enum temps_timing_t
00179  * @brief Temperature sensor timing constants
00180  *
00181  * @details
00182  * Timing requirements from Ch58 Table 58.2 for proper sensor operation.
00183  *
00184  * @note Manual ref: Ch58 section 58.3.4, Table 58.2
00185  * @since Version 1.0.0
00186  */
00187 typedef enum : uint16_t {
00188 /**
00189  * @brief Reference voltage stabilization wait time (tTSTBL)
00190  * @details Minimum 30 us wait after setting TSEN = 1
00191  */
00192 k_temps_tstbl_us = 30U,
00193
00194 /**
00195  * @brief Output stabilization wait time (tOSTBL)
00196  * @details Minimum 0 us wait after setting TSOE = 1
00197  */
00198 k_temps_tostbl_us = 0U,
00199 } temps_timing_t;
00200
00201 /*
=====
00202 * Temperature Sensor Calibration Constants
00203 *
=====
00204 */
00205
00206 /**
00207  * @enum temps_calibration_t
00208  * @brief Temperature sensor calibration constants
00209  *
00210  * @details
00211  * Constants for temperature calculation using factory calibration data.
00212  *
00213  * @par Temperature Calculation
00214  * The calibration data (TSCDR) was measured at:
00215  * - Tj = 125degC (junction temperature)
00216  * - AVCC1 = 3.3V (ADC reference voltage)
00217  *
00218  * @note Manual ref: Ch58 section 58.3.1
00219  * @since Version 1.0.0
00220  */
00221 typedef enum : uint16_t {
00222 /**
00223  * @brief Calibration temperature (125degC)
00224  * @details Factory calibration performed at this temperature
00225  */
00226 k_temps_cal_temp_celsius = 125U,
00227
00228 /**
00229  * @brief ADC full scale value (12-bit = 4096)
00230  */
00231 k_temps_adc_full_scale = 4096U,
00232
00233 /**
00234  * @brief ADC reference voltage in millivolts (3300 mV = 3.3V)
00235  */
00236 k_temps_avcc_mv = 3300U,
00237
00238 /**
00239  * @brief Typical temperature slope in uV/degC (absolute value)
00240  * @details From Table 63.57: typ. -4.1 mV/degC = -4100 uV/degC
00241  * @note Actual value varies by chip; two-point calibration recommended
00242  */
00243 k_temps_slope_uv_per_c = 4100U,
00244 } temps_calibration_t;
00245
00246 /*
=====
00247 * Module Stop Control

```



```

00248 *
=====
00249 */
00250
00251 /**
00252  * @enum temps_mstpcr_t
00253  * @brief Temperature sensor module stop control bits
00254  *
00255  * @details
00256  * The temperature sensor is controlled by MSTPCRB.MSTPC8.
00257  * Must be cleared (0) before accessing TEMPS registers.
00258  *
00259  * @note Manual ref: Ch11 (Low Power Consumption) module stop tables
00260  * @since Version 1.0.0
00261  */
00262 typedef enum : uint32_t {
00263     /**
00264     * @brief MSTPCRB bit for temperature sensor (bit 8)
00265     * @details
00266     * - 0: Temperature sensor module is running
00267     * - 1: Temperature sensor module is stopped (default)
00268     * Manual ref: Ch11 section 11.2.3 MSTPCRB, bit b8 = MSTPB8 (Temperature Sensor)
00269     */
00270     k_temps_mstpcrb_bit = (1U « 8U),
00271 } temps_mstpcr_t;
00272
00273 /*
=====
00274 * Register Accessor Functions
00275 *
=====
00276 */
00277
00278 /**
00279  * @brief Get pointer to Temperature Sensor Control Register (TSCR)
00280  *
00281  * @details
00282  * Returns a volatile pointer to TSCR at address 0x0008C500.
00283  *
00284  * @return Volatile pointer to TSCR register (8-bit)
00285  * @retval Non-NULL Always returns valid pointer (hardware address)
00286  *
00287  * @pre Temperature sensor module stop (MSTPCRB.MSTPC8) must be cleared
00288  * @post Pointer is valid for the lifetime of program execution
00289  *
00290  * @note Thread Safety: Safe - returns constant hardware address
00291  * @note This function is always inlined for zero overhead
00292  *
00293  * @par Example (Enable Temperature Sensor):
00294  * @code{.c}
00295  * // 1. Start temperature sensor
00296  * *temps_tscr_reg() = k_tscr_sensor_only;
00297  *
00298  * // 2. Wait for stabilization (30 us)
00299  * delay_us(k_temps_tstbl_us);
00300  *
00301  * // 3. Enable output to ADC
00302  * *temps_tscr_reg() = k_tscr_sensor_with_output;
00303  *
00304  * // 4. Start ADC conversion (see S12AD1 registers)
00305  * @endcode
00306  *
00307  * @see k_temps_tscr_addr Address constant
00308  * @see tscr_bits_t Control bits
00309  * @since Version 1.0.0
00310  */
00311 static inline volatile uint8_t* temps_tscr_reg(void)
00312 {
00313     return (volatile uint8_t*)k_temps_tscr_addr;
00314 }
00315
00316 /**
00317  * @brief Get pointer to Temperature Sensor Calibration Data Register (TSCDR)
00318  *
00319  * @details
00320  * Returns a volatile pointer to TSCDR at address 0xFE7F7D7C.
00321  * This is a 32-bit read-only register containing factory calibration data.
00322  *
00323  * @par Calibration Data Format
00324  * The lower 12 bits contain the ADC value measured at 125degC and 3.3V:
00325  * - Bits [11:0]: ADC value (0-4095)
00326  * - Bits [31:12]: Reserved (read as 0)
00327  *
00328  * @par Converting to Voltage
00329  *  $V1 = 3.3 \times (TSCDR \& 0xFFF) / 4096$  [V]
00330  *
00331  * @return Volatile pointer to TSCDR register (32-bit, read-only)

```

```

00332 * @retval Non-NULL Always returns valid pointer (hardware address)
00333 *
00334 * @pre On-chip ROM must be enabled (SYSCR0.ROME = 1)
00335 * @post Pointer is valid for the lifetime of program execution
00336 *
00337 * @note Thread Safety: Safe - returns constant hardware address
00338 * @note This function is always inlined for zero overhead
00339 * @warning This register is read-only; writes have no effect
00340 *
00341 * @par Example (Read Calibration):
00342 * @code{.c}
00343 * // Read calibration data
00344 * uint32_t cal125 = *temps_tscdr_reg() & 0xFFFF;
00345 *
00346 * // Calculate reference voltage at 125degC
00347 * // V1 = 3.3 x cal125 / 4096 (in fixed-point or float)
00348 * @endcode
00349 *
00350 * @see k_temps_tscdr_addr Address constant
00351 * @since Version 1.0.0
00352 */
00353 static inline volatile const uint32_t* temps_tscdr_reg(void)
00354 {
00355     return (volatile const uint32_t*)k_temps_tscdr_addr;
00356 }
00357
00358 /*
=====
00359 * Static Assertions - Address Verification
00360 *
=====
00361 */
00362 /**
00363  * @name Address Verification
00364  * @brief Compile-time verification of register addresses
00365  * @{
00366  */
00367
00368 /* The enum declarations above ARE the contract per Ch58 manual.
00369 * Compile-time relationships between them (rather than against re-stated
00370 * literals) catch reordering / accidental-edit bugs without re-asserting
00371 * the magic numbers a second time. */
00372
00373 /* TSEN and TSOE must be distinct single-bit masks within byte */
00374 static_assert(k_tscr_tsen_bit != k_tscr_tsoe_bit, "TSEN and TSOE must differ");
00375 static_assert((k_tscr_tsen_bit & k_tscr_tsoe_bit) == 0U,
00376     "TSEN and TSOE must be non-overlapping bits");
00377
00378 /* Verify combined sensor+output value */
00379 static_assert(k_tscr_sensor_with_output == (k_tscr_tsen_bit | k_tscr_tsoe_bit),
00380     "Sensor with output must be TSEN|TSOE");
00381
00382 /** @} */
00383
00384 #ifdef __cplusplus
00385 }
00386 #endif
00387

```

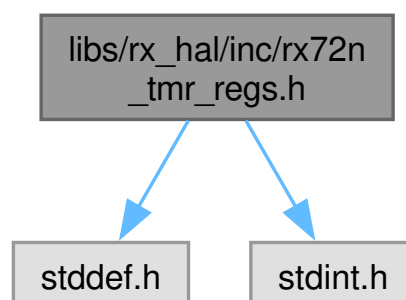
8.63 libs/rx_hal/inc/rx72n_tmr_regs.h File Reference

RX72N 8-Bit Timer (TMRb) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_tmr_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_tmr_channel_regs_t](#)
TMR single-channel register block.
- struct [rx_tmr_cascade_regs_t](#)
TMR cascaded 16-bit register view (TMR01 or TMR23).

Enumerations

- enum [rx_tmr_addresses_t](#) : `uintptr_t` {
`k_tmr0_base_addr = 0x00088200` , `k_tmr1_base_addr = 0x00088201` , `k_tmr2_base_addr = 0x00088210` , `k_tmr3_base_addr = 0x00088211` ,
`k_tmr01_cascade_base_addr = 0x00088200` , `k_tmr23_cascade_base_addr = 0x00088210` ,
`k_tmr_unit_spacing = 0x10` }
TMR channel base addresses.
- enum [rx_tmr_counts_t](#) : `uint8_t` { `k_tmr_channel_count = 4` }
TMR channel count constant.
- enum [rx_tmr_tcr_bits_t](#) : `uint8_t` { `k_tmr_tcr_cmieb = (1U << 7)` , `k_tmr_tcr_cmiea = (1U << 6)` , `k_tmr_tcr_ovie = (1U << 5)` }
TCR bit positions and single-bit masks.
- enum [rx_tmr_tcr_cclr_t](#) : `uint8_t` { `k_tmr_tcr_cclr_disabled = (0U << 3)` , `k_tmr_tcr_cclr_cmp_match_a = (1U << 3)` , `k_tmr_tcr_cclr_cmp_match_b = (2U << 3)` , `k_tmr_tcr_cclr_external_sig = (3U << 3)` }
TCR CCLR[1:0] counter clear source.
- enum [rx_tmr_tcr_masks_t](#) : `uint8_t` { `k_tmr_tcr_cclr_mask = 0x18` }
TCR register field masks.
- enum [rx_tmr_tcsr_bits_t](#) : `uint8_t` { `k_tmr_tcsr_adte = (1U << 4)` }
TCSR bit positions (even channels TMR0/TMR2).
- enum [rx_tmr_tcsr_output_t](#) : `uint8_t` { `k_tmr_tcsr_output_no_change = 0x0` , `k_tmr_tcsr_output_low = 0x1` , `k_tmr_tcsr_output_high = 0x2` , `k_tmr_tcsr_output_toggle = 0x3` }
TCSR OSA/OSB output select values (identical encoding).
- enum [rx_tmr_tcsr_shift_t](#) : `uint8_t` { `k_tmr_tcsr_osa_shift = 0` , `k_tmr_tcsr_osb_shift = 2` }
TCSR output-select field shifts.
- enum [rx_tmr_tcsr_masks_t](#) : `uint8_t` { `k_tmr_tcsr_osa_mask = 0x03` , `k_tmr_tcsr_osb_mask = 0x0C` , `k_tmr_tcsr_odd_reserved = 0x10` }
TCSR field masks.
- enum [rx_tmr_tccr_bits_t](#) : `uint8_t` { `k_tmr_tccr_tmris = (1U << 7)` }
TCCR bit positions.
- enum [rx_tmr_tccr_shift_t](#) : `uint8_t` { `k_tmr_tccr_cks_shift = 0` , `k_tmr_tccr_css_shift = 3` }
TCCR field shifts.
- enum [rx_tmr_tccr_masks_t](#) : `uint8_t` { `k_tmr_tccr_cks_mask = 0x07` , `k_tmr_tccr_css_mask = 0x18` }
TCCR field masks.

- enum `rx_tmr_tccr_css_t`: `uint8_t` { `k_tmr_tccr_css_external` = (0U << 3), `k_tmr_tccr_css_internal` = (1U << 3), `k_tmr_tccr_css_cascade` = (3U << 3) }
TCCR CSS[1:0] clock source select (per Table 30.5).
- enum `rx_tmr_tccr_cks_internal_t`: `uint8_t` {
`k_tmr_tccr_cks_prohibited` = 0x0, `k_tmr_tccr_cks_pclk_div1` = 0x1, `k_tmr_tccr_cks_pclk_div2` = 0x2, `k_tmr_tccr_cks_pclk_div8` = 0x3,
`k_tmr_tccr_cks_pclk_div32` = 0x4, `k_tmr_tccr_cks_pclk_div64` = 0x5, `k_tmr_tccr_cks_pclk_div1024` = 0x6, `k_tmr_tccr_cks_pclk_div8192` = 0x7 }
TCCR CKS[2:0] divider values when CSS = internal clock.
- enum `rx_tmr_tccr_cks_external_t`: `uint8_t` { `k_tmr_tccr_cks_ext_rising` = 0x1, `k_tmr_tccr_cks_ext_falling` = 0x2, `k_tmr_tccr_cks_ext_both` = 0x3 }
TCCR CKS[2:0] edge selection when CSS = external clock.
- enum `rx_tmr_tcstr_bits_t`: `uint8_t` { `k_tmr_tcstr_tcs` = (1U << 0) }
TCSTR bit positions.
- enum `rx_tmr_mstpcra_t`: `uint32_t` { `k_tmr_mstpcra_mstpa4` = (1U << 4), `k_tmr_mstpcra_mstpa5` = (1U << 5) }
MSTPCRA bits that control the TMR module stop state.
- enum `rx_tmr_intb_source_t`: `uint8_t` {
`k_tmr_intb_cmia0` = 3, `k_tmr_intb_cmib0` = 4, `k_tmr_intb_ovi0` = 5, `k_tmr_intb_cmia1` = 6,
`k_tmr_intb_cmib1` = 7, `k_tmr_intb_ovi1` = 8, `k_tmr_intb_cmia2` = 9, `k_tmr_intb_cmib2` = 10,
`k_tmr_intb_ovi2` = 11, `k_tmr_intb_cmia3` = 12, `k_tmr_intb_cmib3` = 13, `k_tmr_intb_ovi3` = 14 }
TMR Software Configurable Interrupt B source numbers.

Functions

- static volatile `rx_tmr_channel_regs_t * tmr0` (void)
Get pointer to TMR0 channel register block (unit 0 even channel).
- static volatile `rx_tmr_channel_regs_t * tmr1` (void)
Get pointer to TMR1 channel register block (unit 0 odd channel).
- static volatile `rx_tmr_channel_regs_t * tmr2` (void)
Get pointer to TMR2 channel register block (unit 1 even channel).
- static volatile `rx_tmr_channel_regs_t * tmr3` (void)
Get pointer to TMR3 channel register block (unit 1 odd channel).
- static volatile `rx_tmr_cascade_regs_t * tmr01_cascade` (void)
Get pointer to TMR01 cascaded 16-bit register view.
- static volatile `rx_tmr_cascade_regs_t * tmr23_cascade` (void)
Get pointer to TMR23 cascaded 16-bit register view.

8.63.1 Detailed Description

RX72N 8-Bit Timer (TMRb) Register Definitions.

Register definitions for the TMR peripheral which provides 4 channels of 8-bit timers (TMR0..TMR3) organized into two units. Each unit contains two channels that may operate independently (two 8-bit counters) or cascaded (one 16-bit counter, TMR01 or TMR23).

Channel and Unit Layout

Unit	Channels	Module Stop Bit	Cascade Name
0	TMR0, TMR1	MSTPCRA.MSTPA5	TMR01
1	TMR2, TMR3	MSTPCRA.MSTPA4	TMR23

Key Features

- 8-bit up-counter with two compare-match registers (TCORA, TCORB)
- Optional 16-bit cascaded mode (even + odd channel concatenated)
- Counter clear on compare match A, compare match B, or external reset
- 8 clock sources per channel (external edge/level, PCLK dividers, cascade)
- Compare match A/B and overflow interrupts via SELECTB
- Dedicated TMO output pin per channel for hardware pulse generation

Register Map Overview (Unit 0 / Unit 1)

Even Ch Addr	Odd Ch Addr	Register	Size	Description
0x00088200	0x00088201	TCR	8	Timer Control
0x00088202	0x00088203	TCSR	8	Timer Control/Status
0x00088204	0x00088205	TCORA	8	Time Constant A
0x00088206	0x00088207	TCORB	8	Time Constant B
0x00088208	0x00088209	TCNT	8	Timer Counter
0x0008820A	0x0008820B	TCCR	8	Timer Counter Control
0x0008820C	0x0008820D	TCSTR	8	Timer Counter Start (ELC)

Unit 1 uses the same layout offset +0x10 (TMR2 base 0x00088210, TMR3 0x00088211).

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops in register definitions)
- Rule 3: No dynamic memory allocation
- Rule 4: All accessor functions are single-statement
- Rule 5: N/A (hardware layer)
- Rule 6: Minimal scope - inline accessors only
- Rule 7: N/A (no return values to check)
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0111 Rev.1.11)

- Chapter 30: 8-Bit Timer (TMRb), pages 1553-1580
- Chapter 15 Table 15.3: Software Configurable Interrupt B sources

See also

[rx_tmr.h](#) TMR HAL driver API

[rx72n_regs.h](#) Main register include file

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx72n_tmr_regs.h](#).

8.63.2 Data Structure Documentation

8.63.2.1 struct rx_tmr_channel_regs_t

TMR single-channel register block.

Register block for one TMR channel accessed via byte strides of 2. Because even and odd channels share the same 16-byte block at different parity offsets, accessing TMRn directly requires striding by 2 bytes per field; this structure therefore represents ONE channel with the interleaved odd byte reserved.

The layout follows the byte addressing in Table 30.1:

- TMR0.TCR @ 0x88200, TMR1.TCR @ 0x88201
- TMR0.TCSR @ 0x88202, TMR1.TCSR @ 0x88203
- TMR0.TCORA @ 0x88204, TMR1.TCORA @ 0x88205
- TMR0.TCORB @ 0x88206, TMR1.TCORB @ 0x88207
- TMR0.TCNT @ 0x88208, TMR1.TCNT @ 0x88209
- TMR0.TCCR @ 0x8820A, TMR1.TCCR @ 0x8820B
- TMR0.TCSTR @ 0x8820C, TMR1.TCSTR @ 0x8820D

Pointer returned by [tmr0\(\)](#)/tmr2() points to the even byte; pointer returned by [tmr1\(\)](#)/tmr3() points to the odd byte. Each struct field therefore accesses the channel's own byte.

Invariant

`sizeof(rx_tmr_channel_regs_t) == 14` (spans byte 0 through 12)

See also

[tmr0\(\)](#) [tmr1\(\)](#) [tmr2\(\)](#) [tmr3\(\)](#) Accessor functions

[rx_tmr_cascade_regs_t](#) 16-bit cascade view

Since

Version 1.0.0

Definition at line 153 of file [rx72n_tmr_regs.h](#).

Collaboration diagram for `rx_tmr_channel_regs_t`:



Data Fields

volatile <code>uint8_t</code>	<code>tccr</code>	Timer Counter Control (TCCR) @ +0x0A
volatile <code>uint8_t</code>	<code>tccr_pair</code>	Paired channel TCCR @ +0x0B
volatile <code>uint8_t</code>	<code>tcnt</code>	Timer Counter (TCNT) @ +0x08
volatile <code>uint8_t</code>	<code>tcnt_pair</code>	Paired channel TCNT @ +0x09
volatile <code>uint8_t</code>	<code>tcora</code>	Time Constant A (TCORA) @ +0x04

volatile uint8_t	tcora_pair	Paired channel TCORA @ +0x05
volatile uint8_t	tcorb	Time Constant B (TCORB) @ +0x06
volatile uint8_t	tcorb_pair	Paired channel TCORB @ +0x07
volatile uint8_t	tcr	Timer Control Register (TCR) @ +0x00
volatile uint8_t	tcr_pair	Paired channel TCR @ +0x01
volatile uint8_t	tcsr	Timer Control/Status (TCSR) @ +0x02
volatile uint8_t	tcsr_pair	Paired channel TCSR @ +0x03
volatile uint8_t	tcstr	Timer Counter Start (TCSTR) @ +0x0C
volatile uint8_t	tcstr_pair	Paired channel TCSTR @ +0x0D

8.63.2.2 struct rx_tmr_cascade_regs_t

TMR cascaded 16-bit register view (TMR01 or TMR23).

When two TMR channels in the same unit are cascaded, TCORA, TCORB, TCNT, and TCCR may be accessed as 16-bit values with the even channel forming the upper byte and the odd channel forming the lower byte.

Only TCORA/TCORB/TCNT/TCCR have defined 16-bit semantics (manual Table 30.4); the other bytes are kept per-channel via the per-channel register block accessed through [tmr0\(\)](#)/[tmr1\(\)](#)/[tmr2\(\)](#)/[tmr3\(\)](#).

Invariant

```
sizeof(rx_tmr_cascade_regs_t) == 12
```

See also

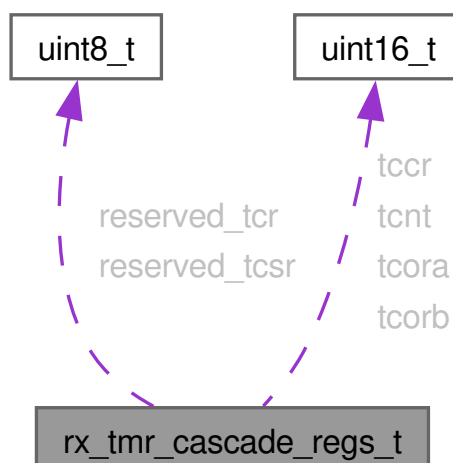
[tmr01_cascade\(\)](#) [tmr23_cascade\(\)](#) Accessor functions

Since

Version 1.0.0

Definition at line 188 of file [rx72n_tmr_regs.h](#).

Collaboration diagram for rx_tmr_cascade_regs_t:



Data Fields

volatile uint8_t	reserved_tcr↔ [2]	TCR bytes (per-channel only) @ +0x00..0x01
volatile uint8_t	reserved_tcsr↔ [2]	TCSR bytes (per-channel only) @ +0x02..0x03
volatile uint16_t	tccr	16-bit cascaded TCCR @ +0x0A
volatile uint16_t	tcnt	16-bit cascaded TCNT @ +0x08
volatile uint16_t	tcora	16-bit cascaded TCORA @ +0x04
volatile uint16_t	tcorb	16-bit cascaded TCORB @ +0x06

8.63.3 Enumeration Type Documentation

8.63.3.1 rx_tmr_addresses_t

```
enum rx_tmr_addresses_t : uintptr_t
```

TMR channel base addresses.

Base addresses for each TMR channel (even channels at even byte offsets, odd channels at the following odd byte). The cascade (16-bit) address is the even-channel base; 16-bit bus accesses at those addresses transfer the even channel as the upper 8 bits and the odd channel as the lower 8.

Note

Manual: Ch30.1.1 Table 30.1 - TMR Base Address Map

See also

[tmr0\(\)](#) [tmr1\(\)](#) [tmr2\(\)](#) [tmr3\(\)](#) Accessor functions

Since

Version 1.0.0

Enumerator

k_tmr0_base_addr	TMR0 even channel (unit 0)
k_tmr1_base_addr	TMR1 odd channel (unit 0)
k_tmr2_base_addr	TMR2 even channel (unit 1)
k_tmr3_base_addr	TMR3 odd channel (unit 1)
k_tmr01_cascade_base_addr	Unit 0 cascade (TMR0 upper, TMR1 lower) - 16-bit access base.
k_tmr23_cascade_base_addr	Unit 1 cascade (TMR2 upper, TMR3 lower) - 16-bit access base.
k_tmr_unit_spacing	Byte spacing between units (TMR2 - TMR0).

Definition at line 93 of file [rx72n_tmr_regs.h](#).

```
00093 : uintptr_t {
00094 k_tmr0_base_addr = 0x00088200, /**< TMR0 even channel (unit 0) */
```

```

00095 k_tmr1_base_addr = 0x00088201, /**< TMR1 odd channel (unit 0) */
00096 k_tmr2_base_addr = 0x00088210, /**< TMR2 even channel (unit 1) */
00097 k_tmr3_base_addr = 0x00088211, /**< TMR3 odd channel (unit 1) */
00098
00099 /** @brief Unit 0 cascade (TMR0 upper, TMR1 lower) - 16-bit access base */
00100 k_tmr01_cascade_base_addr = 0x00088200,
00101
00102 /** @brief Unit 1 cascade (TMR2 upper, TMR3 lower) - 16-bit access base */
00103 k_tmr23_cascade_base_addr = 0x00088210,
00104
00105 /** @brief Byte spacing between units (TMR2 - TMR0) */
00106 k_tmr_unit_spacing = 0x10,
00107 } rx_tmr_addresses_t;

```

8.63.3.2 rx_tmr_counts_t

```
enum rx_tmr_counts_t : uint8_t
```

TMR channel count constant.

Since

Version 1.0.0

Enumerator

k_tmr_channel_count	Total TMR channels (TMR0..TMR3)
---------------------	---------------------------------

Definition at line 114 of file [rx72n_tmr_regs.h](#).

```

00114 : uint8_t {
00115 k_tmr_channel_count = 4, /**< Total TMR channels (TMR0..TMR3) */
00116 } rx_tmr_counts_t;

```

8.63.3.3 rx_tmr_intb_source_t

```
enum rx_tmr_intb_source_t : uint8_t
```

TMR Software Configurable Interrupt B source numbers.

TMR compare-match and overflow interrupts are routed through ICU SELECTB. These source numbers are written into an SLIBRn register (n in 128..207) to map a TMR event onto the corresponding interrupt vector.

Note

Manual: Ch15 Table 15.3 - Software Configurable Interrupt B sources

Since

Version 1.0.0

Enumerator

k_tmr_intb_cmia0	TMR0 TCORA compare match
------------------	--------------------------

k_tmr_intb_cmib0	TMR0 TCORB compare match
k_tmr_intb_ovi0	TMR0 TCNT overflow
k_tmr_intb_cmia1	TMR1 TCORA compare match
k_tmr_intb_cmib1	TMR1 TCORB compare match
k_tmr_intb_ovi1	TMR1 TCNT overflow
k_tmr_intb_cmia2	TMR2 TCORA compare match
k_tmr_intb_cmib2	TMR2 TCORB compare match
k_tmr_intb_ovi2	TMR2 TCNT overflow
k_tmr_intb_cmia3	TMR3 TCORA compare match
k_tmr_intb_cmib3	TMR3 TCORB compare match
k_tmr_intb_ovi3	TMR3 TCNT overflow

Definition at line 576 of file [rx72n_tmr_regs.h](#).

```

00576      : uint8_t {
00577  k_tmr_intb_cmia0 = 3, /**< TMR0 TCORA compare match */
00578  k_tmr_intb_cmib0 = 4, /**< TMR0 TCORB compare match */
00579  k_tmr_intb_ovi0 = 5, /**< TMR0 TCNT overflow */
00580  k_tmr_intb_cmia1 = 6, /**< TMR1 TCORA compare match */
00581  k_tmr_intb_cmib1 = 7, /**< TMR1 TCORB compare match */
00582  k_tmr_intb_ovi1 = 8, /**< TMR1 TCNT overflow */
00583  k_tmr_intb_cmia2 = 9, /**< TMR2 TCORA compare match */
00584  k_tmr_intb_cmib2 = 10, /**< TMR2 TCORB compare match */
00585  k_tmr_intb_ovi2 = 11, /**< TMR2 TCNT overflow */
00586  k_tmr_intb_cmia3 = 12, /**< TMR3 TCORA compare match */
00587  k_tmr_intb_cmib3 = 13, /**< TMR3 TCORB compare match */
00588  k_tmr_intb_ovi3 = 14, /**< TMR3 TCNT overflow */
00589 } rx_tmr_intb_source_t;

```

8.63.3.4 rx_tmr_mstpcra_t

```
enum rx_tmr_mstpcra_t : uint32_t
```

MSTPCRA bits that control the TMR module stop state.

Unit 0 (TMR0/TMR1) is gated by MSTPA5. Unit 1 (TMR2/TMR3) is gated by MSTPA4.

Note

Manual: Ch11 Module Stop Control Register A (MSTPCRA)

Since

Version 1.0.0

Enumerator

k_tmr_mstpcra_mstpa4	Unit 1 (TMR2/TMR3) stop
k_tmr_mstpcra_mstpa5	Unit 0 (TMR0/TMR1) stop

Definition at line 554 of file [rx72n_tmr_regs.h](#).

```

00554      : uint32_t {
00555  k_tmr_mstpcra_mstpa4 = (1U « 4), /**< Unit 1 (TMR2/TMR3) stop */
00556  k_tmr_mstpcra_mstpa5 = (1U « 5), /**< Unit 0 (TMR0/TMR1) stop */
00557 } rx_tmr_mstpcra_t;

```

8.63.3.5 rx_tmr_tccr_bits_t

```
enum rx_tmr_tccr_bits_t : uint8_t
```

TCCR bit positions.

Bit Layout (8-bit)

Bits	Field	R/W	Description
7	TMRIS	R/W	External reset detection (0=edge, 1=level)
6↔ 5	-	R/W	Reserved (write 0)
4↔ 3	CSS[1:↔ 0]	R/W	Clock source select
2↔ 0	CKS[2:↔ 0]	R/W	Clock divider select

Note

Manual: Ch30.2.5 - Timer Counter Control Register (TCCR)

Since

Version 1.0.0

Enumerator

k_tmr_tccr_tmris	Timer reset detection condition
------------------	---------------------------------

Definition at line 418 of file rx72n_tmr_regs.h.

```
00418     : uint8_t {
00419   k_tmr_tccr_tmris = (1U « 7), /**< Timer reset detection condition */
00420 } rx_tmr_tccr_bits_t;
```

8.63.3.6 rx_tmr_tccr_cks_external_t

```
enum rx_tmr_tccr_cks_external_t : uint8_t
```

TCCR CKS[2:0] edge selection when CSS = external clock.

External Edge Selection (Table 30.5)

CKS	Edge
000b	Clock prohibited

CKS	Edge
001b	Count at rising edge
010b	Count at falling edge
011b	Count at both rising and falling

Since

Version 1.0.0

Enumerator

k_tmr_tccr_cks_ext_rising	001b: rising-edge count
k_tmr_tccr_cks_ext_falling	010b: falling-edge count
k_tmr_tccr_cks_ext_both	011b: both-edge count

Definition at line 508 of file [rx72n_tmr_regs.h](#).

```

00508      : uint8_t {
00509  k_tmr_tccr_cks_ext_rising = 0x1, /**< 001b: rising-edge count      */
00510  k_tmr_tccr_cks_ext_falling = 0x2, /**< 010b: falling-edge count   */
00511  k_tmr_tccr_cks_ext_both   = 0x3, /**< 011b: both-edge count     */
00512 } rx_tmr_tccr_cks_external_t;
```

8.63.3.7 rx_tmr_tccr_cks_internal_t

enum [rx_tmr_tccr_cks_internal_t](#) : uint8_t

TCCR CKS[2:0] divider values when CSS = internal clock.

Internal Clock Dividers (Table 30.5)

CKS	Source
000b	Clock prohibited
001b	PCLK
010b	PCLK/ $\leftarrow$ 2
011b	PCLK/ $\leftarrow$ 8
100b	PCLK/32
101b	PCLK/64
110b	PCLK/1024
111b	PCLK/8192

Note

Manual table order differs from register encoding. Values below match the exact bit pattern written to CKS[2:0].

Since

Version 1.0.0

Enumerator

k_tmr_tccr_cks_prohibited	000b: clock input prohibited
k_tmr_tccr_cks_pclk_div1	001b: PCLK / 1
k_tmr_tccr_cks_pclk_div2	010b: PCLK / 2
k_tmr_tccr_cks_pclk_div8	011b: PCLK / 8
k_tmr_tccr_cks_pclk_div32	100b: PCLK / 32
k_tmr_tccr_cks_pclk_div64	101b: PCLK / 64
k_tmr_tccr_cks_pclk_div1024	110b: PCLK / 1024
k_tmr_tccr_cks_pclk_div8192	111b: PCLK / 8192

Definition at line 483 of file [rx72n_tmr_regs.h](#).

```

00483      : uint8_t {
00484  k_tmr_tccr_cks_prohibited  = 0x0, /**< 000b: clock input prohibited */
00485  k_tmr_tccr_cks_pclk_div1   = 0x1, /**< 001b: PCLK / 1          */
00486  k_tmr_tccr_cks_pclk_div2   = 0x2, /**< 010b: PCLK / 2          */
00487  k_tmr_tccr_cks_pclk_div8   = 0x3, /**< 011b: PCLK / 8          */
00488  k_tmr_tccr_cks_pclk_div32  = 0x4, /**< 100b: PCLK / 32         */
00489  k_tmr_tccr_cks_pclk_div64  = 0x5, /**< 101b: PCLK / 64         */
00490  k_tmr_tccr_cks_pclk_div1024 = 0x6, /**< 110b: PCLK / 1024        */
00491  k_tmr_tccr_cks_pclk_div8192 = 0x7, /**< 111b: PCLK / 8192        */
00492 } rx_tmr_tccr_cks_internal_t;

```

8.63.3.8 rx_tmr_tccr_css_t

```
enum rx\_tmr\_tccr\_css\_t : uint8_t
```

TCCR CSS[1:0] clock source select (per Table 30.5).

Source Meaning

Value	Channel	Source
00b	any	External count clock on TMCI pin (edge selected by CKS)
01b	any	Internal clock (PCLK divider selected by CKS)
10b	any	Setting prohibited
11b	TMR0/ 2	Count on odd-channel TCNT overflow (cascade lower->up)
11b	TMR1/ 3	Count on even-channel TCORA compare match (cascade)

Since

Version 1.0.0

Enumerator

k_tmr_tccr_css_external	External count clock on TMCI pin
k_tmr_tccr_css_internal	Internal PCLK source (use CKS)
k_tmr_tccr_css_cascade	Cascade from paired channel

Definition at line 457 of file [rx72n_tmr_regs.h](#).

```
00457      : uint8_t {
00458  k_tmr_tccr_css_external = (0U « 3), /**< External count clock on TMCI pin */
00459  k_tmr_tccr_css_internal = (1U « 3), /**< Internal PCLK source (use CKS) */
00460  k_tmr_tccr_css_cascade  = (3U « 3), /**< Cascade from paired channel */
00461 } rx_tmr_tccr_css_t;
```

8.63.3.9 rx_tmr_tccr_masks_t

```
enum rx_tmr_tccr_masks_t : uint8_t
```

TCCR field masks.

Since

Version 1.0.0

Enumerator

k_tmr_tccr_cks_mask	CKS[2:0] field mask (bits 2:0)
k_tmr_tccr_css_mask	CSS[1:0] field mask (bits 4:3)

Definition at line 437 of file [rx72n_tmr_regs.h](#).

```
00437      : uint8_t {
00438  k_tmr_tccr_cks_mask = 0x07, /**< CKS[2:0] field mask (bits 2:0) */
00439  k_tmr_tccr_css_mask = 0x18, /**< CSS[1:0] field mask (bits 4:3) */
00440 } rx_tmr_tccr_masks_t;
```

8.63.3.10 rx_tmr_tccr_shift_t

```
enum rx_tmr_tccr_shift_t : uint8_t
```

TCCR field shifts.

Since

Version 1.0.0

Enumerator

k_tmr_tccr_cks_shift	CKS[2:0] starts at bit 0
----------------------	--------------------------

k_tmr_tccr_css_shift	CSS[1:0] starts at bit 3
----------------------	--------------------------

Definition at line 427 of file rx72n_tmr_regs.h.

```
00427      : uint8_t {
00428  k_tmr_tccr_cks_shift = 0, /**< CKS[2:0] starts at bit 0 */
00429  k_tmr_tccr_css_shift = 3, /**< CSS[1:0] starts at bit 3 */
00430 } rx_tmr_tccr_shift_t;
```

8.63.3.11 rx_tmr_tcr_bits_t

enum rx_tmr_tcr_bits_t : uint8_t

TCR bit positions and single-bit masks.

Bit Layout (8-bit)

Bits	Field	R/W	Description
7	CMIEB	R/W	TCORB compare match interrupt enable
6	CMIEA	R/W	TCORA compare match interrupt enable
5	OVIE	R/W	TCNT overflow interrupt enable
4:↔ 3	CCLR[1:↔ 0]	R/W	Counter clear source select
2:↔ 0	-	R/W	Reserved (must be written 0)

Note

Manual: Ch30.2.4 - Timer Control Register (TCR)

Since

Version 1.0.0

Enumerator

k_tmr_tcr_cmieb	TCORB compare match interrupt enable
k_tmr_tcr_cmiea	TCORA compare match interrupt enable
k_tmr_tcr_ovie	Overflow interrupt enable

Definition at line 306 of file rx72n_tmr_regs.h.

```
00306      : uint8_t {
00307  k_tmr_tcr_cmieb = (1U « 7), /**< TCORB compare match interrupt enable */
00308  k_tmr_tcr_cmiea = (1U « 6), /**< TCORA compare match interrupt enable */
00309  k_tmr_tcr_ovie  = (1U « 5), /**< Overflow interrupt enable */
00310 } rx_tmr_tcr_bits_t;
```


8.63.3.12 rx_tmr_tcr_cclr_t

```
enum rx_tmr_tcr_cclr_t : uint8_t
```

TCR CCLR[1:0] counter clear source.

Note

Manual: Ch30.2.4 Table - CCLR field

Since

Version 1.0.0

Enumerator

k_tmr_tcr_cclr_disabled	00b: Counter clear disabled
k_tmr_tcr_cclr_cmp_match_a	01b: Clear on TCORA match
k_tmr_tcr_cclr_cmp_match_b	10b: Clear on TCORB match
k_tmr_tcr_cclr_external_sig	11b: Clear on external reset

Definition at line 319 of file [rx72n_tmr_regs.h](#).

```
00319 : uint8_t {
00320 k_tmr_tcr_cclr_disabled = (0U « 3), /**< 00b: Counter clear disabled */
00321 k_tmr_tcr_cclr_cmp_match_a = (1U « 3), /**< 01b: Clear on TCORA match */
00322 k_tmr_tcr_cclr_cmp_match_b = (2U « 3), /**< 10b: Clear on TCORB match */
00323 k_tmr_tcr_cclr_external_sig = (3U « 3), /**< 11b: Clear on external reset */
00324 } rx_tmr_tcr_cclr_t;
```

8.63.3.13 rx_tmr_tcr_masks_t

```
enum rx_tmr_tcr_masks_t : uint8_t
```

TCR register field masks.

Since

Version 1.0.0

Enumerator

k_tmr_tcr_cclr_mask	CCLR[1:0] field mask (bits 4:3)
---------------------	---------------------------------

Definition at line 331 of file [rx72n_tmr_regs.h](#).

```
00331 : uint8_t {
00332 k_tmr_tcr_cclr_mask = 0x18, /**< CCLR[1:0] field mask (bits 4:3) */
00333 } rx_tmr_tcr_masks_t;
```

8.63.3.14 rx_tmr_tcsr_bits_t

```
enum rx_tmr_tcsr_bits_t : uint8_t
```

TCSR bit positions (even channels TMR0/TMR2).

Bit Layout (8-bit, even channels)

Bits	Field	R/W	Description
7↔ 5	-	R	Undefined at reset
4	ADTE	R/W	A/D trigger enable (TMR0/TMR2 only)
3↔ 2	OSB[1↔ 0]	R/W	TMO output select on TCORB compare match
1↔ 0	OSA[1↔ 0]	R/W	TMO output select on TCORA compare match

Note

For odd channels (TMR1/TMR3), bit 4 is reserved and reads as 1.

Manual: Ch30.2.6 - Timer Control/Status Register (TCSR)

Since

Version 1.0.0

Enumerator

k_tmr_tcsr_adte	A/D trigger enable (even channels only)
-----------------	---

Definition at line 356 of file rx72n_tmr_regs.h.

```
00356         : uint8_t {
00357   k_tmr_tcsr_adte = (1U « 4), /**< A/D trigger enable (even channels only) */
00358 } rx_tmr_tcsr_bits_t;
```

8.63.3.15 rx_tmr_tcsr_masks_t

```
enum rx_tmr_tcsr_masks_t : uint8_t
```

TCSR field masks.

Since

Version 1.0.0

Enumerator

k_tmr_tcsr_osa_mask	OSA[1:0] field mask (bits 1:0)
k_tmr_tcsr_osb_mask	OSB[1:0] field mask (bits 3:2)
k_tmr_tcsr_odd_reserved	Odd-channel bit 4 reserved, reads as 1

Definition at line 392 of file [rx72n_tmr_regs.h](#).

```
00392      : uint8_t {
00393  k_tmr_tcsr_osa_mask    = 0x03, /**< OSA[1:0] field mask (bits 1:0) */
00394  k_tmr_tcsr_osb_mask    = 0x0C, /**< OSB[1:0] field mask (bits 3:2) */
00395  k_tmr_tcsr_odd_reserved = 0x10, /**< Odd-channel bit 4 reserved, reads as 1 */
00396 } rx_tmr_tcsr_masks_t;
```

8.63.3.16 rx_tmr_tcsr_output_t

```
enum rx_tmr_tcsr_output_t : uint8_t
```

TCSR OSA/OSB output select values (identical encoding).

Controls the TMO pin behavior on compare match. Both OSA and OSB use this 2-bit encoding; shift the value left by 0 for OSA and 2 for OSB.

Since

Version 1.0.0

Enumerator

k_tmr_tcsr_output_no_change	00b: No change
k_tmr_tcsr_output_low	01b: Drive low on compare match
k_tmr_tcsr_output_high	10b: Drive high on compare match
k_tmr_tcsr_output_toggle	11b: Toggle on compare match

Definition at line 370 of file [rx72n_tmr_regs.h](#).

```
00370      : uint8_t {
00371  k_tmr_tcsr_output_no_change = 0x0, /**< 00b: No change */
00372  k_tmr_tcsr_output_low      = 0x1, /**< 01b: Drive low on compare match */
00373  k_tmr_tcsr_output_high     = 0x2, /**< 10b: Drive high on compare match */
00374  k_tmr_tcsr_output_toggle   = 0x3, /**< 11b: Toggle on compare match */
00375 } rx_tmr_tcsr_output_t;
```

8.63.3.17 rx_tmr_tcsr_shift_t

```
enum rx_tmr_tcsr_shift_t : uint8_t
```

TCSR output-select field shifts.

Since

Version 1.0.0

Enumerator

k_tmr_tcsr_osa_shift	OSA[1:0] starts at bit 0
k_tmr_tcsr_osb_shift	OSB[1:0] starts at bit 2

Definition at line 382 of file [rx72n_tmr_regs.h](#).

```
00382      : uint8_t {
00383  k_tmr_tcsr_osa_shift = 0, /**< OSA[1:0] starts at bit 0 */
00384  k_tmr_tcsr_osb_shift = 2, /**< OSB[1:0] starts at bit 2 */
00385 } rx_tmr_tcsr_shift_t;
```

8.63.3.18 rx_tmr_tcsr_bits_t

```
enum rx_tmr_tcsr_bits_t : uint8_t
```

TCSTR bit positions.

Bit Layout (8-bit)

Bits	Field	R/W	Description
7:1	-	R/W	Reserved
0	TCS	R/W	Count state from ELC (0=stopped, 1=running)

Note

Manual: Ch30.2.7 - Timer Counter Start Register (TCSTR)

TCS is only valid when ELC event controller drives the channel; software should write 0 to stop and leave 1 for ELC to set.

Since

Version 1.0.0

Enumerator

k_tmr_tcsr_tcs	Timer counter status (ELC)
----------------	----------------------------

Definition at line 534 of file [rx72n_tmr_regs.h](#).

```
00534      : uint8_t {
00535  k_tmr_tcsr_tcs = (1U << 0), /**< Timer counter status (ELC) */
00536 } rx_tmr_tcsr_bits_t;
```

8.63.4 Function Documentation

8.63.4.1 tmr0()

```
volatile rx_tmr_channel_regs_t * tmr0 (
    void ) [inline], [static]
```

Get pointer to TMR0 channel register block (unit 0 even channel).

Returns

Volatile pointer to TMR0 register block at 0x00088200

Precondition

TMR unit 0 module clock enabled (MSTPCRA.MSTPA5 = 0)

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

See also

[k_tmr0_base_addr](#)

Since

Version 1.0.0

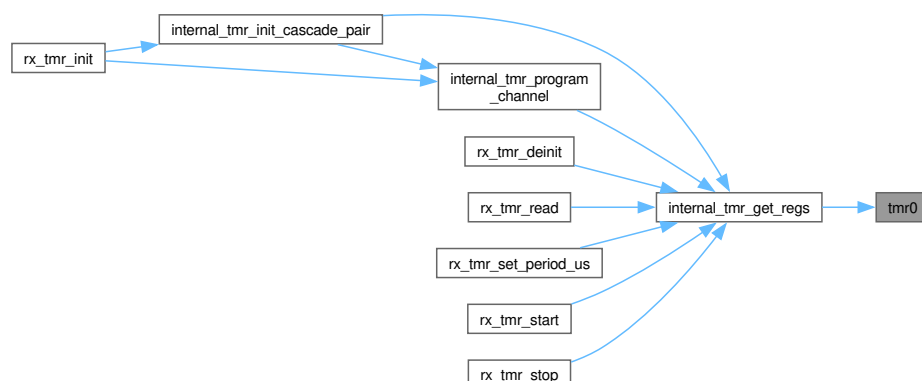
Definition at line 211 of file [rx72n_tmr_regs.h](#).

```
00212 {
00213     return (volatile rx_tmr_channel_regs_t*)k_tmr0_base_addr;
00214 }
```

References [k_tmr0_base_addr](#).

Referenced by [internal_tmr_get_regs\(\)](#).

Here is the caller graph for this function:



8.63.4.2 tmr01_cascade()

```
volatile rx\_tmr\_cascade\_regs\_t * tmr01_cascade (  
    void )    [inline], [static]
```

Get pointer to TMR01 cascaded 16-bit register view.

Returns

Volatile pointer to TMR01 cascade block at 0x00088200

Precondition

TMR unit 0 module clock enabled

Cascade mode configured via TMR1.TCCR (CSS/CKS = overflow of TMR0)

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

Since

Version 1.0.0

Definition at line 267 of file [rx72n_tmr_regs.h](#).

```
00268 {  
00269     return (volatile rx\_tmr\_cascade\_regs\_t*)k_tmr01_cascade_base_addr;  
00270 }
```

References [k_tmr01_cascade_base_addr](#).

8.63.4.3 tmr1()

```
volatile rx\_tmr\_channel\_regs\_t * tmr1 (  
    void )    [inline], [static]
```

Get pointer to TMR1 channel register block (unit 0 odd channel).

Returns

Volatile pointer to TMR1 register block at 0x00088201

Precondition

TMR unit 0 module clock enabled (MSTPCRA.MSTPA5 = 0)

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

See also

[k_tmr1_base_addr](#)

Since

Version 1.0.0

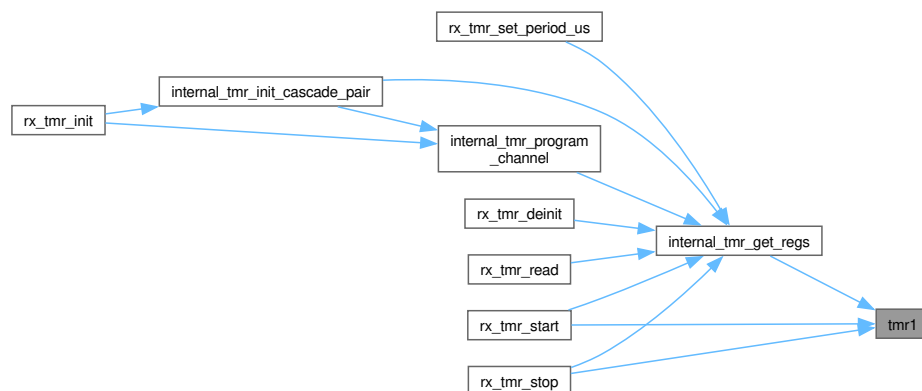
Definition at line 225 of file [rx72n_tmr_regs.h](#).

```
00226 {
00227     return (volatile rx_tmr_channel_regs_t*)k_tmr1_base_addr;
00228 }
```

References [k_tmr1_base_addr](#).

Referenced by [internal_tmr_get_regs\(\)](#), [rx_tmr_start\(\)](#), and [rx_tmr_stop\(\)](#).

Here is the caller graph for this function:

8.63.4.4 `tmr2()`

```
volatile rx_tmr_channel_regs_t * tmr2 (
    void ) [inline], [static]
```

Get pointer to TMR2 channel register block (unit 1 even channel).

Returns

Volatile pointer to TMR2 register block at 0x00088210

Precondition

TMR unit 1 module clock enabled (MSTPCRA.MSTPA4 = 0)

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

See also

[k_tmr2_base_addr](#)

Since

Version 1.0.0

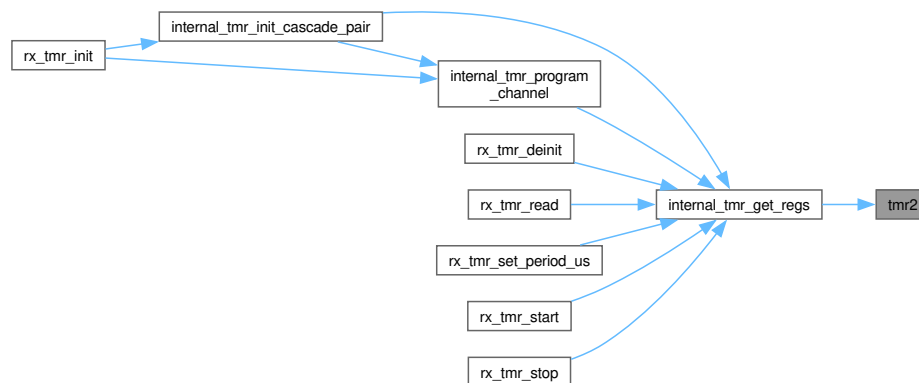
Definition at line 239 of file [rx72n_tmr_regs.h](#).

```
00240 {
00241     return (volatile rx_tmr_channel_regs_t*)k_tmr2_base_addr;
00242 }
```

References [k_tmr2_base_addr](#).

Referenced by [internal_tmr_get_regs\(\)](#).

Here is the caller graph for this function:



8.63.4.5 tmr23_cascade()

```
volatile rx_tmr_cascade_regs_t * tmr23_cascade (  
    void )    [inline], [static]
```

Get pointer to TMR23 cascaded 16-bit register view.

Returns

Volatile pointer to TMR23 cascade block at 0x00088210

Precondition

TMR unit 1 module clock enabled

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

Since

Version 1.0.0

Definition at line 280 of file [rx72n_tmr_regs.h](#).

```
00281 {  
00282     return (volatile rx_tmr_cascade_regs_t*)k_tmr23_cascade_base_addr;  
00283 }
```

References [k_tmr23_cascade_base_addr](#).

8.63.4.6 tmr3()

```
volatile rx_tmr_channel_regs_t * tmr3 (  
    void )    [inline], [static]
```

Get pointer to TMR3 channel register block (unit 1 odd channel).

Returns

Volatile pointer to TMR3 register block at 0x00088211

Precondition

TMR unit 1 module clock enabled (MSTPCRA.MSTPA4 = 0)

Postcondition

Returned pointer is valid for the lifetime of the program

Note

Thread Safety: Safe - returns a constant hardware address

See also

[k_tmr3_base_addr](#)

Since

Version 1.0.0

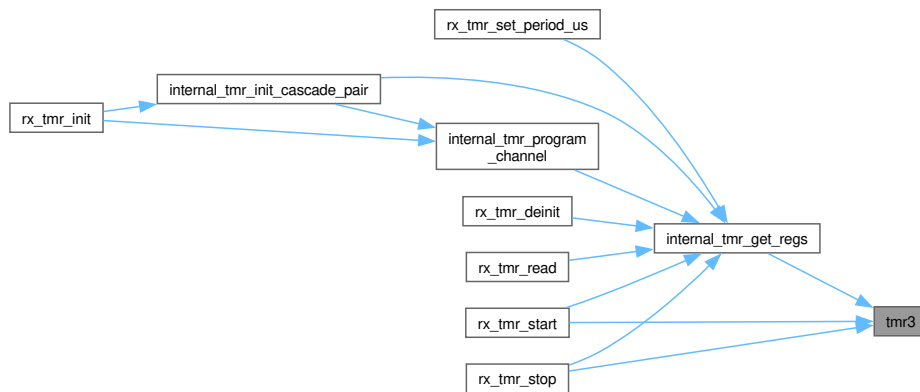
Definition at line 253 of file [rx72n_tmr_regs.h](#).

```
00254 {
00255     return (volatile rx_tmr_channel_regs_t*)k_tmr3_base_addr;
00256 }
```

References [k_tmr3_base_addr](#).

Referenced by [internal_tmr_get_regs\(\)](#), [rx_tmr_start\(\)](#), and [rx_tmr_stop\(\)](#).

Here is the caller graph for this function:



8.64 rx72n_tmr_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_tmr_regs.h
00003  * @brief RX72N 8-Bit Timer (TMRb) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the TMR peripheral which provides 4 channels of
00007  * 8-bit timers (TMR0..TMR3) organized into two units. Each unit contains two
00008  * channels that may operate independently (two 8-bit counters) or cascaded
00009  * (one 16-bit counter, TMR01 or TMR23).
00010  *
00011  * @par Channel and Unit Layout
00012  * | Unit | Channels | Module Stop Bit | Cascade Name |
00013  * |-----|-----|-----|-----|
00014  * | 0 | TMR0, TMR1 | MSTPCRA.MSTPA5 | TMR01 |
```

```

00015 * | 1 | TMR2, TMR3 | MSTPCRA.MSTPA4 | TMR23 |
00016 *
00017 * @par Key Features
00018 * - 8-bit up-counter with two compare-match registers (TCORA, TCORB)
00019 * - Optional 16-bit cascaded mode (even + odd channel concatenated)
00020 * - Counter clear on compare match A, compare match B, or external reset
00021 * - 8 clock sources per channel (external edge/level, PCLK dividers, cascade)
00022 * - Compare match A/B and overflow interrupts via SELECTB
00023 * - Dedicated TMO output pin per channel for hardware pulse generation
00024 *
00025 * @par Register Map Overview (Unit 0 / Unit 1)
00026 * | Even Ch Addr | Odd Ch Addr | Register | Size | Description |
00027 * |-----|-----|-----|-----|-----|
00028 * | 0x00088200 | 0x00088201 | TCR | 8 | Timer Control |
00029 * | 0x00088202 | 0x00088203 | TCSR | 8 | Timer Control/Status |
00030 * | 0x00088204 | 0x00088205 | TCORA | 8 | Time Constant A |
00031 * | 0x00088206 | 0x00088207 | TCORB | 8 | Time Constant B |
00032 * | 0x00088208 | 0x00088209 | TCNT | 8 | Timer Counter |
00033 * | 0x0008820A | 0x0008820B | TCCR | 8 | Timer Counter Control |
00034 * | 0x0008820C | 0x0008820D | TCSTR | 8 | Timer Counter Start (ELC) |
00035 *
00036 * Unit 1 uses the same layout offset +0x10 (TMR2 base 0x00088210, TMR3 0x00088211).
00037 *
00038 * @par NASA Power of 10 Compliance
00039 * - Rule 1: No goto, setjmp, or recursion
00040 * - Rule 2: N/A (no loops in register definitions)
00041 * - Rule 3: No dynamic memory allocation
00042 * - Rule 4: All accessor functions are single-statement
00043 * - Rule 5: N/A (hardware layer)
00044 * - Rule 6: Minimal scope - inline accessors only
00045 * - Rule 7: N/A (no return values to check)
00046 * - Rule 8: All constants use C23 typed enums
00047 * - Rule 9: No function pointers
00048 * - Rule 10: Compiles with -Wall -Wextra -Werror
00049 *
00050 * @par Manual References
00051 * RX72N Group User's Manual: Hardware (R01UH0824EJ0111 Rev.1.11)
00052 * - Chapter 30: 8-Bit Timer (TMRb), pages 1553-1580
00053 * - Chapter 15 Table 15.3: Software Configurable Interrupt B sources
00054 *
00055 * @see rx_tmr.h TMR HAL driver API
00056 * @see rx72n_regs.h Main register include file
00057 *
00058 * @author Locked, Inc.
00059 * @date 2026-04-21
00060 * @copyright Copyright (c) 2026 Locked Inc.
00061 * SPDX-License-Identifier: MIT
00062 * @since Version 1.0.0
00063 */
00064
00065 #pragma once
00066
00067 #include <stddef.h>
00068 #include <stdint.h>
00069
00070 #ifdef __cplusplus
00071 extern "C" {
00072 #endif
00073
00074 /*
=====
00075 * TMR Base Addresses
00076 *
=====
00077 */
00078
00079 /**
00080 * @enum rx_tmr_addresses_t
00081 * @brief TMR channel base addresses
00082 *
00083 * @details
00084 * Base addresses for each TMR channel (even channels at even byte offsets,
00085 * odd channels at the following odd byte). The cascade (16-bit) address is
00086 * the even-channel base; 16-bit bus accesses at those addresses transfer
00087 * the even channel as the upper 8 bits and the odd channel as the lower 8.
00088 *
00089 * @note Manual: Ch30.1.1 Table 30.1 - TMR Base Address Map
00090 * @see tmr0() tmr1() tmr2() tmr3() Accessor functions
00091 * @since Version 1.0.0
00092 */
00093 typedef enum : uintptr_t {
00094     k_tmr0_base_addr = 0x00088200, /**< TMR0 even channel (unit 0) */
00095     k_tmr1_base_addr = 0x00088201, /**< TMR1 odd channel (unit 0) */
00096     k_tmr2_base_addr = 0x00088210, /**< TMR2 even channel (unit 1) */
00097     k_tmr3_base_addr = 0x00088211, /**< TMR3 odd channel (unit 1) */
00098
00099     /** @brief Unit 0 cascade (TMR0 upper, TMR1 lower) - 16-bit access base */

```

```

00100 k_tmr01_cascade_base_addr = 0x00088200,
00101
00102 /** @brief Unit 1 cascade (TMR2 upper, TMR3 lower) - 16-bit access base */
00103 k_tmr23_cascade_base_addr = 0x00088210,
00104
00105 /** @brief Byte spacing between units (TMR2 - TMR0) */
00106 k_tmr_unit_spacing = 0x10,
00107 } rx_tmr_addresses_t;
00108
00109 /**
00110  * @enum rx_tmr_counts_t
00111  * @brief TMR channel count constant
00112  * @since Version 1.0.0
00113  */
00114 typedef enum : uint8_t {
00115     k_tmr_channel_count = 4, /**< Total TMR channels (TMR0..TMR3) */
00116 } rx_tmr_counts_t;
00117
00118 /*
00119  * TMR Channel Register Structure
00120  */
00121
00122 /**
00123  * @struct rx_tmr_channel_regs_t
00124  * @brief TMR single-channel register block
00125  *
00126  * @details
00127  * Register block for one TMR channel accessed via byte strides of 2. Because
00128  * even and odd channels share the same 16-byte block at different parity
00129  * offsets, accessing TMRn directly requires striding by 2 bytes per field;
00130  * this structure therefore represents ONE channel with the interleaved odd
00131  * byte reserved.
00132  *
00133  * The layout follows the byte addressing in Table 30.1:
00134  * - TMR0.TCR @ 0x88200, TMR1.TCR @ 0x88201
00135  * - TMR0.TCSR @ 0x88202, TMR1.TCSR @ 0x88203
00136  * - TMR0.TCOR @ 0x88204, TMR1.TCOR @ 0x88205
00137  * - TMR0.TCORB @ 0x88206, TMR1.TCORB @ 0x88207
00138  * - TMR0.TCNT @ 0x88208, TMR1.TCNT @ 0x88209
00139  * - TMR0.TCCR @ 0x8820A, TMR1.TCCR @ 0x8820B
00140  * - TMR0.TCSTR @ 0x8820C, TMR1.TCSTR @ 0x8820D
00141  *
00142  * Pointer returned by tmr0()/tmr2() points to the even byte; pointer returned
00143  * by tmr1()/tmr3() points to the odd byte. Each struct field therefore
00144  * accesses the channel's own byte.
00145  *
00146  * @invariant sizeof(rx_tmr_channel_regs_t) == 14 (spans byte 0 through 12)
00147  *
00148  * @see tmr0() tmr1() tmr2() tmr3() Accessor functions
00149  * @see rx_tmr_cascade_regs_t 16-bit cascade view
00150  * @since Version 1.0.0
00151  */
00152
00153 typedef struct {
00154     volatile uint8_t tcr; /**< Timer Control Register (TCR) @ +0x00 */
00155     volatile uint8_t tcr_pair; /**< Paired channel TCR @ +0x01 */
00156     volatile uint8_t tcsr; /**< Timer Control/Status (TCSR) @ +0x02 */
00157     volatile uint8_t tcsr_pair; /**< Paired channel TCSR @ +0x03 */
00158     volatile uint8_t tcora; /**< Time Constant A (TCORA) @ +0x04 */
00159     volatile uint8_t tcora_pair; /**< Paired channel TCORA @ +0x05 */
00160     volatile uint8_t tcorb; /**< Time Constant B (TCORB) @ +0x06 */
00161     volatile uint8_t tcorb_pair; /**< Paired channel TCORB @ +0x07 */
00162     volatile uint8_t tcnt; /**< Timer Counter (TCNT) @ +0x08 */
00163     volatile uint8_t tcnt_pair; /**< Paired channel TCNT @ +0x09 */
00164     volatile uint8_t tccr; /**< Timer Counter Control (TCCR) @ +0x0A */
00165     volatile uint8_t tccr_pair; /**< Paired channel TCCR @ +0x0B */
00166     volatile uint8_t tcstr; /**< Timer Counter Start (TCSTR) @ +0x0C */
00167     volatile uint8_t tcstr_pair; /**< Paired channel TCSTR @ +0x0D */
00168 } rx_tmr_channel_regs_t;
00169
00170 /**
00171  * @struct rx_tmr_cascade_regs_t
00172  * @brief TMR cascaded 16-bit register view (TMR01 or TMR23)
00173  *
00174  * @details
00175  * When two TMR channels in the same unit are cascaded, TCORA, TCORB, TCNT,
00176  * and TCCR may be accessed as 16-bit values with the even channel forming
00177  * the upper byte and the odd channel forming the lower byte.
00178  *
00179  * Only TCORA/TCORB/TCNT/TCCR have defined 16-bit semantics (manual Table
00180  * 30.4); the other bytes are kept per-channel via the per-channel register
00181  * block accessed through tmr0()/tmr1()/tmr2()/tmr3().
00182  *
00183  * @invariant sizeof(rx_tmr_cascade_regs_t) == 12
00184  */

```

```

00185 * @see tmr01_cascade() tmr23_cascade() Accessor functions
00186 * @since Version 1.0.0
00187 */
00188 typedef struct {
00189     volatile uint8_t reserved_tcr[2]; /**< TCR bytes (per-channel only) @ +0x00..0x01 */
00190     volatile uint8_t reserved_tcsr[2]; /**< TCSR bytes (per-channel only) @ +0x02..0x03 */
00191     volatile uint16_t tcora; /**< 16-bit cascaded TCORA @ +0x04 */
00192     volatile uint16_t tcorb; /**< 16-bit cascaded TCORB @ +0x06 */
00193     volatile uint16_t tcnt; /**< 16-bit cascaded TCNT @ +0x08 */
00194     volatile uint16_t tccr; /**< 16-bit cascaded TCCR @ +0x0A */
00195 } rx_tmr_cascade_regs_t;
00196
00197 /*
=====
00198 * Inline Accessor Functions
00199 *
=====
00200 */
00201
00202 /**
00203 * @brief Get pointer to TMR0 channel register block (unit 0 even channel)
00204 * @return Volatile pointer to TMR0 register block at 0x00088200
00205 * @pre TMR unit 0 module clock enabled (MSTPCRA.MSTPA5 = 0)
00206 * @post Returned pointer is valid for the lifetime of the program
00207 * @note Thread Safety: Safe - returns a constant hardware address
00208 * @see k_tmr0_base_addr
00209 * @since Version 1.0.0
00210 */
00211 static inline volatile rx_tmr_channel_regs_t* tmr0(void)
00212 {
00213     return (volatile rx_tmr_channel_regs_t*)k_tmr0_base_addr;
00214 }
00215
00216 /**
00217 * @brief Get pointer to TMR1 channel register block (unit 0 odd channel)
00218 * @return Volatile pointer to TMR1 register block at 0x00088201
00219 * @pre TMR unit 0 module clock enabled (MSTPCRA.MSTPA5 = 0)
00220 * @post Returned pointer is valid for the lifetime of the program
00221 * @note Thread Safety: Safe - returns a constant hardware address
00222 * @see k_tmr1_base_addr
00223 * @since Version 1.0.0
00224 */
00225 static inline volatile rx_tmr_channel_regs_t* tmr1(void)
00226 {
00227     return (volatile rx_tmr_channel_regs_t*)k_tmr1_base_addr;
00228 }
00229
00230 /**
00231 * @brief Get pointer to TMR2 channel register block (unit 1 even channel)
00232 * @return Volatile pointer to TMR2 register block at 0x00088210
00233 * @pre TMR unit 1 module clock enabled (MSTPCRA.MSTPA4 = 0)
00234 * @post Returned pointer is valid for the lifetime of the program
00235 * @note Thread Safety: Safe - returns a constant hardware address
00236 * @see k_tmr2_base_addr
00237 * @since Version 1.0.0
00238 */
00239 static inline volatile rx_tmr_channel_regs_t* tmr2(void)
00240 {
00241     return (volatile rx_tmr_channel_regs_t*)k_tmr2_base_addr;
00242 }
00243
00244 /**
00245 * @brief Get pointer to TMR3 channel register block (unit 1 odd channel)
00246 * @return Volatile pointer to TMR3 register block at 0x00088211
00247 * @pre TMR unit 1 module clock enabled (MSTPCRA.MSTPA4 = 0)
00248 * @post Returned pointer is valid for the lifetime of the program
00249 * @note Thread Safety: Safe - returns a constant hardware address
00250 * @see k_tmr3_base_addr
00251 * @since Version 1.0.0
00252 */
00253 static inline volatile rx_tmr_channel_regs_t* tmr3(void)
00254 {
00255     return (volatile rx_tmr_channel_regs_t*)k_tmr3_base_addr;
00256 }
00257
00258 /**
00259 * @brief Get pointer to TMR01 cascaded 16-bit register view
00260 * @return Volatile pointer to TMR01 cascade block at 0x00088200
00261 * @pre TMR unit 0 module clock enabled
00262 * @pre Cascade mode configured via TMR1.TCCR (CSS/CKS = overflow of TMR0)
00263 * @post Returned pointer is valid for the lifetime of the program
00264 * @note Thread Safety: Safe - returns a constant hardware address
00265 * @since Version 1.0.0
00266 */
00267 static inline volatile rx_tmr_cascade_regs_t* tmr01_cascade(void)
00268 {
00269     return (volatile rx_tmr_cascade_regs_t*)k_tmr01_cascade_base_addr;

```

```

00270 }
00271
00272 /**
00273  * @brief Get pointer to TMR23 cascaded 16-bit register view
00274  * @return Volatile pointer to TMR23 cascade block at 0x00088210
00275  * @pre TMR unit 1 module clock enabled
00276  * @post Returned pointer is valid for the lifetime of the program
00277  * @note Thread Safety: Safe - returns a constant hardware address
00278  * @since Version 1.0.0
00279  */
00280 static inline volatile rx_tmr_cascade_regs_t* tmr23_cascade(void)
00281 {
00282     return (volatile rx_tmr_cascade_regs_t*)k_tmr23_cascade_base_addr;
00283 }
00284
00285 /*
=====
00286  * TCR - Timer Control Register Bit Definitions
00287  *
=====
00288  */
00289
00290 /**
00291  * @enum rx_tmr_tcr_bits_t
00292  * @brief TCR bit positions and single-bit masks
00293  *
00294  * @par Bit Layout (8-bit)
00295  * | Bits | Field | R/W | Description |
00296  * |-----|-----|-----|-----|
00297  * | 7 | CMIEB | R/W | TCORB compare match interrupt enable |
00298  * | 6 | CMIEA | R/W | TCORA compare match interrupt enable |
00299  * | 5 | OVIE | R/W | TCNT overflow interrupt enable |
00300  * | 4:3 | CCLR[1:0] | R/W | Counter clear source select |
00301  * | 2:0 | - | R/W | Reserved (must be written 0) |
00302  *
00303  * @note Manual: Ch30.2.4 - Timer Control Register (TCR)
00304  * @since Version 1.0.0
00305  */
00306 typedef enum : uint8_t {
00307     k_tmr_tcr_cmieb = (1U « 7), /**< TCORB compare match interrupt enable */
00308     k_tmr_tcr_cmiea = (1U « 6), /**< TCORA compare match interrupt enable */
00309     k_tmr_tcr_ovie = (1U « 5), /**< Overflow interrupt enable */
00310 } rx_tmr_tcr_bits_t;
00311
00312 /**
00313  * @enum rx_tmr_tcr_cclr_t
00314  * @brief TCR CCLR[1:0] counter clear source
00315  *
00316  * @note Manual: Ch30.2.4 Table - CCLR field
00317  * @since Version 1.0.0
00318  */
00319 typedef enum : uint8_t {
00320     k_tmr_tcr_cclr_disabled = (0U « 3), /**< 00b: Counter clear disabled */
00321     k_tmr_tcr_cclr_cmp_match_a = (1U « 3), /**< 01b: Clear on TCORA match */
00322     k_tmr_tcr_cclr_cmp_match_b = (2U « 3), /**< 10b: Clear on TCORB match */
00323     k_tmr_tcr_cclr_external_sig = (3U « 3), /**< 11b: Clear on external reset */
00324 } rx_tmr_tcr_cclr_t;
00325
00326 /**
00327  * @enum rx_tmr_tcr_masks_t
00328  * @brief TCR register field masks
00329  * @since Version 1.0.0
00330  */
00331 typedef enum : uint8_t {
00332     k_tmr_tcr_cclr_mask = 0x18, /**< CCLR[1:0] field mask (bits 4:3) */
00333 } rx_tmr_tcr_masks_t;
00334
00335 /*
=====
00336  * TCSR - Timer Control/Status Register Bit Definitions
00337  *
=====
00338  */
00339
00340 /**
00341  * @enum rx_tmr_tcsr_bits_t
00342  * @brief TCSR bit positions (even channels TMR0/TMR2)
00343  *
00344  * @par Bit Layout (8-bit, even channels)
00345  * | Bits | Field | R/W | Description |
00346  * |-----|-----|-----|-----|
00347  * | 7:5 | - | R | Undefined at reset |
00348  * | 4 | ADTE | R/W | A/D trigger enable (TMR0/TMR2 only) |
00349  * | 3:2 | OSB[1:0] | R/W | TMO output select on TCORB compare match |
00350  * | 1:0 | OSA[1:0] | R/W | TMO output select on TCORA compare match |
00351  *
00352  * @note For odd channels (TMR1/TMR3), bit 4 is reserved and reads as 1.

```

```

00353 * @note Manual: Ch30.2.6 - Timer Control/Status Register (TCSR)
00354 * @since Version 1.0.0
00355 */
00356 typedef enum : uint8_t {
00357     k_tmr_tcsr_adte = (1U « 4), /**< A/D trigger enable (even channels only) */
00358 } rx_tmr_tcsr_bits_t;
00359
00360 /**
00361 * @enum rx_tmr_tcsr_output_t
00362 * @brief TCSR OSA/OSB output select values (identical encoding)
00363 *
00364 * @details
00365 * Controls the TMO pin behavior on compare match. Both OSA and OSB use this
00366 * 2-bit encoding; shift the value left by 0 for OSA and 2 for OSB.
00367 *
00368 * @since Version 1.0.0
00369 */
00370 typedef enum : uint8_t {
00371     k_tmr_tcsr_output_no_change = 0x0, /**< 00b: No change */
00372     k_tmr_tcsr_output_low       = 0x1, /**< 01b: Drive low on compare match */
00373     k_tmr_tcsr_output_high      = 0x2, /**< 10b: Drive high on compare match */
00374     k_tmr_tcsr_output_toggle    = 0x3, /**< 11b: Toggle on compare match */
00375 } rx_tmr_tcsr_output_t;
00376
00377 /**
00378 * @enum rx_tmr_tcsr_shift_t
00379 * @brief TCSR output-select field shifts
00380 * @since Version 1.0.0
00381 */
00382 typedef enum : uint8_t {
00383     k_tmr_tcsr_osa_shift = 0, /**< OSA[1:0] starts at bit 0 */
00384     k_tmr_tcsr_osb_shift = 2, /**< OSB[1:0] starts at bit 2 */
00385 } rx_tmr_tcsr_shift_t;
00386
00387 /**
00388 * @enum rx_tmr_tcsr_masks_t
00389 * @brief TCSR field masks
00390 * @since Version 1.0.0
00391 */
00392 typedef enum : uint8_t {
00393     k_tmr_tcsr_osa_mask = 0x03, /**< OSA[1:0] field mask (bits 1:0) */
00394     k_tmr_tcsr_osb_mask = 0x0C, /**< OSB[1:0] field mask (bits 3:2) */
00395     k_tmr_tcsr_odd_reserved = 0x10, /**< Odd-channel bit 4 reserved, reads as 1 */
00396 } rx_tmr_tcsr_masks_t;
00397
00398 /*
=====
00399 * TCCR - Timer Counter Control Register Bit Definitions
00400 *
=====
00401 */
00402
00403 /**
00404 * @enum rx_tmr_tccr_bits_t
00405 * @brief TCCR bit positions
00406 *
00407 * @par Bit Layout (8-bit)
00408 * | Bits | Field | R/W | Description |
00409 * |-----|-----|-----|-----|
00410 * | 7 | TMRIS | R/W | External reset detection (0=edge, 1=level) |
00411 * | 6:5 | - | R/W | Reserved (write 0) |
00412 * | 4:3 | CSS[1:0] | R/W | Clock source select |
00413 * | 2:0 | CKS[2:0] | R/W | Clock divider select |
00414 *
00415 * @note Manual: Ch30.2.5 - Timer Counter Control Register (TCCR)
00416 * @since Version 1.0.0
00417 */
00418 typedef enum : uint8_t {
00419     k_tmr_tccr_tmrisk = (1U « 7), /**< Timer reset detection condition */
00420 } rx_tmr_tccr_bits_t;
00421
00422 /**
00423 * @enum rx_tmr_tccr_shift_t
00424 * @brief TCCR field shifts
00425 * @since Version 1.0.0
00426 */
00427 typedef enum : uint8_t {
00428     k_tmr_tccr_cks_shift = 0, /**< CKS[2:0] starts at bit 0 */
00429     k_tmr_tccr_css_shift = 3, /**< CSS[1:0] starts at bit 3 */
00430 } rx_tmr_tccr_shift_t;
00431
00432 /**
00433 * @enum rx_tmr_tccr_masks_t
00434 * @brief TCCR field masks
00435 * @since Version 1.0.0
00436 */
00437 typedef enum : uint8_t {

```

```

00438 k_tmr_tccr_cks_mask = 0x07, /**< CKS[2:0] field mask (bits 2:0) */
00439 k_tmr_tccr_css_mask = 0x18, /**< CSS[1:0] field mask (bits 4:3) */
00440 } rx_tmr_tccr_masks_t;
00441
00442 /**
00443  * @enum rx_tmr_tccr_css_t
00444  * @brief TCCR CSS[1:0] clock source select (per Table 30.5)
00445
00446  * @par Source Meaning
00447  * | Value | Channel | Source |
00448  * |-----|-----|-----|
00449  * | 00b | any | External count clock on TMCI pin (edge selected by CKS) |
00450  * | 01b | any | Internal clock (PCLK divider selected by CKS) |
00451  * | 10b | any | Setting prohibited |
00452  * | 11b | TMR0/2 | Count on odd-channel TCNT overflow (cascade lower->up) |
00453  * | 11b | TMR1/3 | Count on even-channel TCORA compare match (cascade) |
00454
00455  * @since Version 1.0.0
00456  */
00457 typedef enum : uint8_t {
00458     k_tmr_tccr_css_external = (0U « 3), /**< External count clock on TMCI pin */
00459     k_tmr_tccr_css_internal = (1U « 3), /**< Internal PCLK source (use CKS) */
00460     k_tmr_tccr_css_cascade = (3U « 3), /**< Cascade from paired channel */
00461 } rx_tmr_tccr_css_t;
00462
00463 /**
00464  * @enum rx_tmr_tccr_cks_internal_t
00465  * @brief TCCR CKS[2:0] divider values when CSS = internal clock
00466
00467  * @par Internal Clock Dividers (Table 30.5)
00468  * | CKS | Source |
00469  * |-----|-----|
00470  * | 000b | Clock prohibited |
00471  * | 001b | PCLK |
00472  * | 010b | PCLK/2 |
00473  * | 011b | PCLK/8 |
00474  * | 100b | PCLK/32 |
00475  * | 101b | PCLK/64 |
00476  * | 110b | PCLK/1024 |
00477  * | 111b | PCLK/8192 |
00478
00479  * @note Manual table order differs from register encoding. Values below
00480  * match the exact bit pattern written to CKS[2:0].
00481  * @since Version 1.0.0
00482  */
00483 typedef enum : uint8_t {
00484     k_tmr_tccr_cks_prohibited = 0x0, /**< 000b: clock input prohibited */
00485     k_tmr_tccr_cks_pclk_div1 = 0x1, /**< 001b: PCLK / 1 */
00486     k_tmr_tccr_cks_pclk_div2 = 0x2, /**< 010b: PCLK / 2 */
00487     k_tmr_tccr_cks_pclk_div8 = 0x3, /**< 011b: PCLK / 8 */
00488     k_tmr_tccr_cks_pclk_div32 = 0x4, /**< 100b: PCLK / 32 */
00489     k_tmr_tccr_cks_pclk_div64 = 0x5, /**< 101b: PCLK / 64 */
00490     k_tmr_tccr_cks_pclk_div1024 = 0x6, /**< 110b: PCLK / 1024 */
00491     k_tmr_tccr_cks_pclk_div8192 = 0x7, /**< 111b: PCLK / 8192 */
00492 } rx_tmr_tccr_cks_internal_t;
00493
00494 /**
00495  * @enum rx_tmr_tccr_cks_external_t
00496  * @brief TCCR CKS[2:0] edge selection when CSS = external clock
00497
00498  * @par External Edge Selection (Table 30.5)
00499  * | CKS | Edge |
00500  * |-----|-----|
00501  * | 000b | Clock prohibited |
00502  * | 001b | Count at rising edge |
00503  * | 010b | Count at falling edge |
00504  * | 011b | Count at both rising and falling |
00505
00506  * @since Version 1.0.0
00507  */
00508 typedef enum : uint8_t {
00509     k_tmr_tccr_cks_ext_rising = 0x1, /**< 001b: rising-edge count */
00510     k_tmr_tccr_cks_ext_falling = 0x2, /**< 010b: falling-edge count */
00511     k_tmr_tccr_cks_ext_both = 0x3, /**< 011b: both-edge count */
00512 } rx_tmr_tccr_cks_external_t;
00513
00514 /**
=====
00515  * TCSTR - Timer Counter Start Register Bit Definitions
00516  *
=====
00517  */
00518
00519 /**
00520  * @enum rx_tmr_tcstr_bits_t
00521  * @brief TCSTR bit positions
00522  */

```



```

00523 * @par Bit Layout (8-bit)
00524 * | Bits | Field | R/W | Description |
00525 * |-----|-----|-----|-----|
00526 * | 7:1 | - | R/W | Reserved |
00527 * | 0 | TCS | R/W | Count state from ELC (0=stopped, 1=running)|
00528 *
00529 * @note Manual: Ch30.2.7 - Timer Counter Start Register (TCSTR)
00530 * @note TCS is only valid when ELC event controller drives the channel;
00531 * software should write 0 to stop and leave 1 for ELC to set.
00532 * @since Version 1.0.0
00533 */
00534 typedef enum : uint8_t {
00535     k_tmr_tcstr_tcs = (1U « 0), /**< Timer counter status (ELC) */
00536 } rx_tmr_tcstr_bits_t;
00537
00538 /*
=====
00539 * Module Stop Control Bits
00540 *
=====
00541 */
00542
00543 /**
00544 * @enum rx_tmr_mstpcra_t
00545 * @brief MSTPCRA bits that control the TMR module stop state
00546 *
00547 * @details
00548 * Unit 0 (TMR0/TMR1) is gated by MSTPA5.
00549 * Unit 1 (TMR2/TMR3) is gated by MSTPA4.
00550 *
00551 * @note Manual: Ch11 Module Stop Control Register A (MSTPCRA)
00552 * @since Version 1.0.0
00553 */
00554 typedef enum : uint32_t {
00555     k_tmr_mstpcra_mstpa4 = (1U « 4), /**< Unit 1 (TMR2/TMR3) stop */
00556     k_tmr_mstpcra_mstpa5 = (1U « 5), /**< Unit 0 (TMR0/TMR1) stop */
00557 } rx_tmr_mstpcra_t;
00558
00559 /*
=====
00560 * SELECTB Interrupt Source Numbers
00561 *
=====
00562 */
00563
00564 /**
00565 * @enum rx_tmr_intb_source_t
00566 * @brief TMR Software Configurable Interrupt B source numbers
00567 *
00568 * @details
00569 * TMR compare-match and overflow interrupts are routed through ICU SELECTB.
00570 * These source numbers are written into an SLIBRn register (n in 128..207)
00571 * to map a TMR event onto the corresponding interrupt vector.
00572 *
00573 * @note Manual: Ch15 Table 15.3 - Software Configurable Interrupt B sources
00574 * @since Version 1.0.0
00575 */
00576 typedef enum : uint8_t {
00577     k_tmr_intb_cmia0 = 3, /**< TMR0 TCORA compare match */
00578     k_tmr_intb_cmib0 = 4, /**< TMR0 TCORB compare match */
00579     k_tmr_intb_ovi0 = 5, /**< TMR0 TCNT overflow */
00580     k_tmr_intb_cmia1 = 6, /**< TMR1 TCORA compare match */
00581     k_tmr_intb_cmib1 = 7, /**< TMR1 TCORB compare match */
00582     k_tmr_intb_ovi1 = 8, /**< TMR1 TCNT overflow */
00583     k_tmr_intb_cmia2 = 9, /**< TMR2 TCORA compare match */
00584     k_tmr_intb_cmib2 = 10, /**< TMR2 TCORB compare match */
00585     k_tmr_intb_ovi2 = 11, /**< TMR2 TCNT overflow */
00586     k_tmr_intb_cmia3 = 12, /**< TMR3 TCORA compare match */
00587     k_tmr_intb_cmib3 = 13, /**< TMR3 TCORB compare match */
00588     k_tmr_intb_ovi3 = 14, /**< TMR3 TCNT overflow */
00589 } rx_tmr_intb_source_t;
00590
00591 /*
=====
00592 * Static Assertions - Verify Register Layout at Compile Time
00593 *
=====
00594 */
00595
00596 static_assert(k_tmr0_base_addr == 0x00088200, "TMR0 base address incorrect");
00597 static_assert(k_tmr1_base_addr == 0x00088201, "TMR1 base address incorrect");
00598 static_assert(k_tmr2_base_addr == 0x00088210, "TMR2 base address incorrect");
00599 static_assert(k_tmr3_base_addr == 0x00088211, "TMR3 base address incorrect");
00600 static_assert((k_tmr2_base_addr - k_tmr0_base_addr) == k_tmr_unit_spacing,
00601              "TMR unit spacing incorrect");
00602
00603 static_assert(sizeof(rx_tmr_channel_regs_t) == 14, "TMR channel register block size incorrect");

```

```

00604 static_assert(offsetof(rx_tmr_channel_regs_t, tcr) == 0x00, "TCR offset incorrect");
00605 static_assert(offsetof(rx_tmr_channel_regs_t, tcsr) == 0x02, "TCSR offset incorrect");
00606 static_assert(offsetof(rx_tmr_channel_regs_t, tcora) == 0x04, "TCORA offset incorrect");
00607 static_assert(offsetof(rx_tmr_channel_regs_t, tcorb) == 0x06, "TCORB offset incorrect");
00608 static_assert(offsetof(rx_tmr_channel_regs_t, tcnt) == 0x08, "TCNT offset incorrect");
00609 static_assert(offsetof(rx_tmr_channel_regs_t, tccr) == 0x0A, "TCCR offset incorrect");
00610 static_assert(offsetof(rx_tmr_channel_regs_t, tcstr) == 0x0C, "TCSTR offset incorrect");
00611
00612 static_assert(sizeof(rx_tmr_cascade_regs_t) == 12, "TMR cascade register block size incorrect");
00613 static_assert(offsetof(rx_tmr_cascade_regs_t, tcora) == 0x04, "Cascade TCORA offset incorrect");
00614 static_assert(offsetof(rx_tmr_cascade_regs_t, tcorb) == 0x06, "Cascade TCORB offset incorrect");
00615 static_assert(offsetof(rx_tmr_cascade_regs_t, tcnt) == 0x08, "Cascade TCNT offset incorrect");
00616 static_assert(offsetof(rx_tmr_cascade_regs_t, tccr) == 0x0A, "Cascade TCCR offset incorrect");
00617
00618 static_assert(k_tmr_tcr_cmieb == 0x80, "TCR CMIEB bit value incorrect");
00619 static_assert(k_tmr_tcr_cmiea == 0x40, "TCR CMIEA bit value incorrect");
00620 static_assert(k_tmr_tcr_ovie == 0x20, "TCR OVIE bit value incorrect");
00621 static_assert(k_tmr_tcr_cclr_cmp_match_a == 0x08, "TCR CCLR match A incorrect");
00622 static_assert(k_tmr_tcr_cclr_cmp_match_b == 0x10, "TCR CCLR match B incorrect");
00623
00624 static_assert(k_tmr_tccr_tmrisc == 0x80, "TCCR TMRIS bit value incorrect");
00625 static_assert(k_tmr_tccr_css_internal == 0x08, "TCCR CSS internal value incorrect");
00626 static_assert(k_tmr_tccr_css_cascade == 0x18, "TCCR CSS cascade value incorrect");
00627
00628 static_assert(k_tmr_tcstr_tcs == 0x01, "TCSTR TCS bit value incorrect");
00629 static_assert(k_tmr_mstpcra_mstpa5 == 0x20, "MSTPCRA.MSTPA5 bit value incorrect");
00630 static_assert(k_tmr_mstpcra_mstpa4 == 0x10, "MSTPCRA.MSTPA4 bit value incorrect");
00631
00632 #ifdef __cplusplus
00633 }
00634 #endif

```

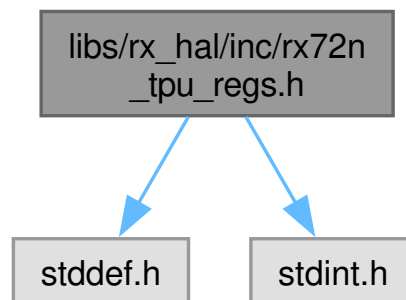
8.65 libs/rx_hal/inc/rx72n_tpu_regs.h File Reference

RX72N 16-Bit Timer Pulse Unit (TPU) Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_tpu_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_tpu_control_regs_t](#)
TPU Control Block Register Structure.
- struct [rx_tpu_ext_regs_t](#)
TPU Extended Channel Register Structure (TPU0 and TPU3).
- struct [rx_tpu_regs_t](#)
TPU Basic Channel Register Structure (TPU1, TPU2, TPU4, TPU5).

Enumerations

- enum `rx_tpu_addresses_t` : `uintptr_t` {
`k_tpu_control_base_addr` = 0x00088100 , `k_tpu0_base_addr` = 0x00088110 , `k_tpu1_base_addr` = 0x00088120 , `k_tpu2_base_addr` = 0x00088130 ,
`k_tpu3_base_addr` = 0x00088140 , `k_tpu4_base_addr` = 0x00088150 , `k_tpu5_base_addr` = 0x00088160 , `k_tpu_channel_spacing` = 0x10 }
TPU base addresses.
- enum `rx_tpu_channel_count_t` : `uint8_t` { `k_tpu_channel_count` = 6 }
TPU channel count constant.
- enum `rx_tpu_tstr_bits_t` : `uint8_t` {
`k_tpu_tstr_cst0` = (1U << 0) , `k_tpu_tstr_cst1` = (1U << 1) , `k_tpu_tstr_cst2` = (1U << 2) ,
`k_tpu_tstr_cst3` = (1U << 3) ,
`k_tpu_tstr_cst4` = (1U << 4) , `k_tpu_tstr_cst5` = (1U << 5) }
TSTR (Timer Start Register) bit definitions.
- enum `rx_tpu_tsyrr_bits_t` : `uint8_t` {
`k_tpu_tsyrr_sync0` = (1U << 0) , `k_tpu_tsyrr_sync1` = (1U << 1) , `k_tpu_tsyrr_sync2` = (1U << 2) ,
`k_tpu_tsyrr_sync3` = (1U << 3) ,
`k_tpu_tsyrr_sync4` = (1U << 4) , `k_tpu_tsyrr_sync5` = (1U << 5) }
TSYR (Timer Synchronous Register) bit definitions.
- enum `rx_tpu_tcr_shift_t` : `uint8_t` { `k_tpu_tcr_tpssc_shift` = 0 , `k_tpu_tcr_ckege_shift` = 3 ,
`k_tpu_tcr_cclr_shift` = 5 }
TCR (Timer Control Register) bit field positions.
- enum `rx_tpu_tcr_cclr_t` : `uint8_t` { `k_tpu_tcr_cclr_disabled` = (0U << 5) , `k_tpu_tcr_cclr_tgra_match` = (1U << 5) ,
`k_tpu_tcr_cclr_tgrb_match` = (2U << 5) , `k_tpu_tcr_cclr_sync_clear` = (3U << 5) }
TCR CCLR[2:0] counter clear source values (for basic channels).
- enum `rx_tpu_tcr_masks_t` : `uint8_t` { `k_tpu_tcr_tpssc_mask` = 0x07 , `k_tpu_tcr_ckege_mask` = 0x18 ,
`k_tpu_tcr_cclr_mask` = 0xE0 }
TCR register field masks.
- enum `rx_tpu_tmdr_md_t` : `uint8_t` {
`k_tpu_tmdr_md_normal` = 0x00 , `k_tpu_tmdr_md_pwm1` = 0x02 , `k_tpu_tmdr_md_pwm2` = 0x03 ,
`k_tpu_tmdr_md_phase_count_1` = 0x04 ,
`k_tpu_tmdr_md_phase_count_2` = 0x05 , `k_tpu_tmdr_md_phase_count_3` = 0x06 ,
`k_tpu_tmdr_md_phase_count_4` = 0x07 }
TMDR MD[3:0] mode select values.
- enum `rx_tpu_tmdr_shift_t` : `uint8_t` {
`k_tpu_tmdr_md_shift` = 0 , `k_tpu_tmdr_bfa_shift` = 4 , `k_tpu_tmdr_bfb_shift` = 5 ,
`k_tpu_tmdr_icselb_shift` = 6 ,
`k_tpu_tmdr_icseld_shift` = 7 }
TMDR bit field positions.
- enum `rx_tpu_tmdr_masks_t` : `uint8_t` { `k_tpu_tmdr_md_mask` = 0x0F }
TMDR register field masks.
- enum `rx_tpu_tier_bits_t` : `uint8_t` {
`k_tpu_tier_tgiea` = (1U << 0) , `k_tpu_tier_tgieb` = (1U << 1) , `k_tpu_tier_tgiec` = (1U << 2) ,
`k_tpu_tier_tgied` = (1U << 3) ,
`k_tpu_tier_tciev` = (1U << 4) , `k_tpu_tier_tcieu` = (1U << 5) , `k_tpu_tier_ttge` = (1U << 7) }
TIER (Timer Interrupt Enable Register) bit definitions.
- enum `rx_tpu_tsr_bits_t` : `uint8_t` {
`k_tpu_tsr_tgfa` = (1U << 0) , `k_tpu_tsr_tgfb` = (1U << 1) , `k_tpu_tsr_tgfc` = (1U << 2) ,
`k_tpu_tsr_tgfd` = (1U << 3) ,
`k_tpu_tsr_tcfv` = (1U << 4) , `k_tpu_tsr_tcfu` = (1U << 5) , `k_tpu_tsr_tcfd` = (1U << 7) }
TSR (Timer Status Register) bit definitions.
- enum `rx_tpu_tsr_reserved_t` : `uint8_t` { `k_tpu_tsr_reserved_bit6` = (1U << 6) }

TSR reserved bit value (bit 6 must be written as 1).

- enum `rx_tpu_nfr_bits_t` : `uint8_t` { `k_tpu_nfr_nfaen` = (1U << 0) , `k_tpu_nfr_nfben` = (1U << 1) , `k_tpu_nfr_nfcen` = (1U << 2) , `k_tpu_nfr_nfdn` = (1U << 3) }

NFCR (Noise Filter Control Register) bit definitions.

- enum `rx_tpu_nfr_clock_t` : `uint8_t` { `k_tpu_nfr_nfcs_pclk_div1` = (0U << 4) , `k_tpu_nfr_nfcs_pclk_div8` = (1U << 4) , `k_tpu_nfr_nfcs_pclk_div32` = (2U << 4) , `k_tpu_nfr_nfcs_count_clock` = (3U << 4) }

NFCR NFCS[1:0] noise filter clock select values.

- enum `rx_tpu_nfr_masks_t` : `uint8_t` { `k_tpu_nfr_nfcs_mask` = 0x30 , `k_tpu_nfr_enable_mask` = 0x0F }

NFCR register field masks.

- enum `rx_tpu_mstpcra_t` : `uint32_t` { `k_tpu_mstpcra_mstpa13` = (1U << 13) }

MSTPCRA bit for TPU module stop control.

- enum `rx_tpu_intb_source_t` : `uint8_t` {
`k_tpu_intb_tgi0a` = 15 , `k_tpu_intb_tgi0b` = 16 , `k_tpu_intb_tgi0c` = 17 , `k_tpu_intb_tgi0d` = 18 ,
`k_tpu_intb_tci0v` = 19 , `k_tpu_intb_tgi1a` = 20 , `k_tpu_intb_tgi1b` = 21 , `k_tpu_intb_tci1v` = 22 ,
`k_tpu_intb_tci1u` = 23 , `k_tpu_intb_tgi2a` = 24 , `k_tpu_intb_tgi2b` = 25 , `k_tpu_intb_tci2v` = 26 ,
`k_tpu_intb_tci2u` = 27 , `k_tpu_intb_tgi3a` = 28 , `k_tpu_intb_tgi3b` = 29 , `k_tpu_intb_tgi3c` = 30 ,
`k_tpu_intb_tgi3d` = 31 , `k_tpu_intb_tci3v` = 32 , `k_tpu_intb_tgi4a` = 33 , `k_tpu_intb_tgi4b` = 34 ,
`k_tpu_intb_tci4v` = 35 , `k_tpu_intb_tci4u` = 36 , `k_tpu_intb_tgi5a` = 37 , `k_tpu_intb_tgi5b` = 38 ,
`k_tpu_intb_tci5v` = 39 , `k_tpu_intb_tci5u` = 40 }

TPU Software Configurable Interrupt B source numbers.

Functions

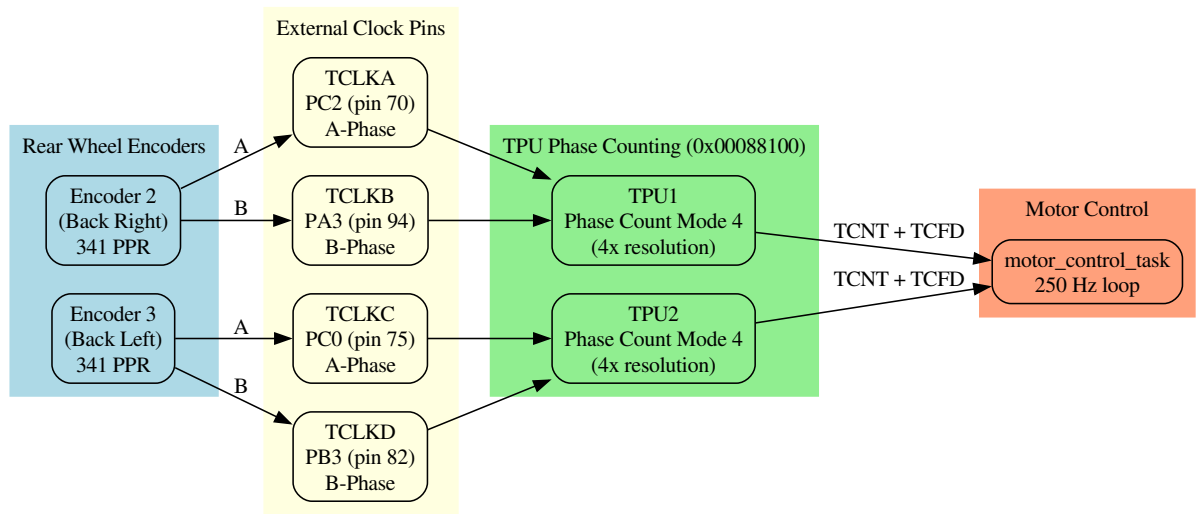
- static volatile `rx_tpu_control_regs_t` * `tpu_control` (void)
 Get pointer to TPU control block registers.
- static volatile `rx_tpu_ext_regs_t` * `tpu0` (void)
 Get pointer to TPU0 channel registers (extended).
- static volatile `rx_tpu_regs_t` * `tpu1` (void)
 Get pointer to TPU1 channel registers (basic, phase counting).
- static volatile `rx_tpu_regs_t` * `tpu2` (void)
 Get pointer to TPU2 channel registers (basic, phase counting).
- static volatile `rx_tpu_ext_regs_t` * `tpu3` (void)
 Get pointer to TPU3 channel registers (extended).
- static volatile `rx_tpu_regs_t` * `tpu4` (void)
 Get pointer to TPU4 channel registers (basic, phase counting).
- static volatile `rx_tpu_regs_t` * `tpu5` (void)
 Get pointer to TPU5 channel registers (basic, phase counting).

8.65.1 Detailed Description

RX72N 16-Bit Timer Pulse Unit (TPU) Register Definitions.

Register definitions for the TPU module which provides 6 channels of 16-bit timers with quadrature encoder phase counting capability. In STAR, the TPU is used for rear wheel encoder decoding via hardware phase counting mode.

STAR Project Quadrature Encoder System



TPU Channel Assignments in STAR

Channel	Type	Use in STAR	Phase Count	Clock Pins
TPU0	Extended	Not used	No	-
TPU1	Basic	Back-right encoder	Yes (mode 4)	TCLKA + TCLKB
TPU2	Basic	Back-left encoder	Yes (mode 4)	TCLKC + TCLKD
TPU3	Extended	Not used	No	-
TPU4	Basic	Not used (paired 2)	Yes	TCLKC + TCLKD
TPU5	Basic	Not used (paired 1)	Yes	TCLKA + TCLKB

Phase Counting Mode 4 (4x Resolution)

Counts on all edges of both A-phase and B-phase inputs, providing 4x the base encoder resolution.
 For 341 PPR encoders: $341 * 4 = 1364$ counts/rev.

A-Phase Edge	B-Phase Level	Direction
Rising	Low	Up
Rising	High	Down
Falling	Low	Down
Falling	High	Up

| (same for B-phase edges with A-phase level swapped) |

Key Features

- 6 channels of 16-bit timers (TPU0-TPU5)
- Hardware quadrature phase counting on TPU1/2/4/5
- 4 phase counting modes (1x, 2x, 3x, 4x resolution)
- TSR.TCFD direction flag for up/down detection
- Overflow/underflow interrupt support
- Noise filter on I/O pins
- Synchronous operation between channels

Hardware Requirements

Parameter	Value	Notes
Control Base	0x00088100	TSTR, TSYR, NFCR registers
Channel Spacing	0x10 bytes	TPU0=0x110, TPU1=0x120, etc.
Module Stop	MSTPCRA.MSTPA13	0=run, 1=stop
Clock	PCLKB (60 MHz)	For noise filter and prescaler

Register Map Overview

Address	Register	Size	Description
0x00088100	TSTR	8	Timer Start Register
0x00088101	TSYR	8	Timer Synchronous Register
0x00088108-0D	NFCRn	8x6	Noise Filter Control (0-5)
0x00088110-1F	TPU0 registers	16	Extended channel (4 TGR)
0x00088120-2F	TPU1 registers	16	Basic channel (2 TGR)
0x00088130-3F	TPU2 registers	16	Basic channel (2 TGR)
0x00088140-4F	TPU3 registers	16	Extended channel (4 TGR)
0x00088150-5F	TPU4 registers	16	Basic channel (2 TGR)
0x00088160-6F	TPU5 registers	16	Basic channel (2 TGR)

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: N/A (no loops in register definitions)
- Rule 3: No dynamic memory allocation
- Rule 4: All accessor functions are single-statement
- Rule 5: N/A (hardware layer)
- Rule 6: Minimal scope - inline accessors only
- Rule 7: N/A (no return values to check)
- Rule 8: All constants use C23 typed enums
- Rule 9: No function pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Single responsibility - only TPU register definitions
- O: Open for extension via additional channel configurations
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessor functions
- D: N/A (hardware register layer)

Module Dependencies

- `<stdint.h>` - Fixed-width integer types
- `<stddef.h>` - `offsetof` macro for static assertions

Manual References

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 28: 16-Bit Timer Pulse Unit (TPUa), pages 1442-1508

See also

[rx72n_mtu_regs.h](#) MTU registers (front wheel encoders)

[rx72n_regs.h](#) Main register include file

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Date

2026-02-10

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Since

Version 1.0.0

Definition in file [rx72n_tpu_regs.h](#).

8.65.2 Data Structure Documentation

8.65.2.1 struct rx_tpu_control_regs_t

TPU Control Block Register Structure.

Contains the shared control registers for all TPU channels: the Timer Start Register (TSTR), Timer Synchronous Register (TSYR), and per-channel Noise Filter Control Registers (NFCR).

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	1	tstr	uint8↔ _t	Timer Start Register
0x01	1	tsyr	uint8↔ _t	Timer Synchronous Register
0x02	6	reserved	uint8↔ t[]	Reserved
0x08	6	nfcrl [0..5]	uint8↔ t[]	Noise Filter Control (per ch)
Total	14			

Invariant

```
sizeof(rx_tpu_control_regs_t) == 14 bytes
```

See also

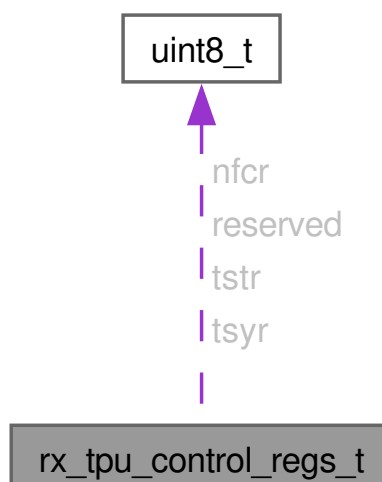
[tpu_control\(\)](#) Accessor function
[rx_tpu_tstr_bits_t](#) TSTR bit definitions
[rx_tpu_tsyr_bits_t](#) TSYR bit definitions

Since

Version 1.0.0

Definition at line 283 of file [rx72n_tpu_regs.h](#).

Collaboration diagram for rx_tpu_control_regs_t:



Data Fields

volatile uint8_t	nfc ^r [6]	Noise Filter Control Registers (NFCR) @ offset 0x08-0x0D. Per-channel noise filter configuration. Index 0 = TPU0.NFCR through index 5 = TPU5.NFCR. Set only while TCNT is stopped. Bit Layout (8-bit, per channel) <table><tr><th>Bits</th><th>Field</th><th>R/W</th><th>Description</th></tr><tr><td>0</td><td>NFAEN</td><td>R/W</td><td>Noise filter enable TIOCA pin</td></tr><tr><td>1</td><td>NFBEN</td><td>R/W</td><td>Noise filter enable TIOCB pin</td></tr><tr><td>2</td><td>NFCEN</td><td>R/W</td><td>Noise filter enable TIOCC (0,3)</td></tr><tr><td>3</td><td>NFDEN</td><td>R/W</td><td>Noise filter enable TIOCD (0,3)</td></tr><tr><td>5:↔ 4</td><td>NFCS[1:↔ 0]</td><td>R/W</td><td>Noise filter clock select</td></tr><tr><td>7:↔ 6</td><td>-</td><td>R</td><td>Reserved (read as 0)</td></tr></table> Warning Set NFCR only while TCNT is stopped (CSTn = 0) See also rx_tpu_nfc_bits_t Bit definitions	Bits	Field	R/W	Description	0	NFAEN	R/W	Noise filter enable TIOCA pin	1	NFBEN	R/W	Noise filter enable TIOCB pin	2	NFCEN	R/W	Noise filter enable TIOCC (0,3)	3	NFDEN	R/W	Noise filter enable TIOCD (0,3)	5:↔ 4	NFCS[1:↔ 0]	R/W	Noise filter clock select	7:↔ 6	-	R	Reserved (read as 0)				
Bits	Field	R/W	Description																															
0	NFAEN	R/W	Noise filter enable TIOCA pin																															
1	NFBEN	R/W	Noise filter enable TIOCB pin																															
2	NFCEN	R/W	Noise filter enable TIOCC (0,3)																															
3	NFDEN	R/W	Noise filter enable TIOCD (0,3)																															
5:↔ 4	NFCS[1:↔ 0]	R/W	Noise filter clock select																															
7:↔ 6	-	R	Reserved (read as 0)																															
volatile uint8_t	reserved↔ [6]	Reserved bytes @ offset 0x02-0x07.																																
volatile uint8_t	tstr	Timer Start Register (TSTR) @ offset 0x00. Starts or stops TCNT operation for TPU0 through TPU5. Each bit independently controls one channel's counter. Bit Layout (8-bit) <table><tr><th>Bits</th><th>Field</th><th>R/W</th><th>Description</th></tr><tr><td>0</td><td>CST0</td><td>R/W</td><td>TPU0 counter start (1=run)</td></tr><tr><td>1</td><td>CST1</td><td>R/W</td><td>TPU1 counter start (1=run)</td></tr><tr><td>2</td><td>CST2</td><td>R/W</td><td>TPU2 counter start (1=run)</td></tr><tr><td>3</td><td>CST3</td><td>R/W</td><td>TPU3 counter start (1=run)</td></tr><tr><td>4</td><td>CST4</td><td>R/W</td><td>TPU4 counter start (1=run)</td></tr><tr><td>5</td><td>CST5</td><td>R/W</td><td>TPU5 counter start (1=run)</td></tr><tr><td>7:↔ 6</td><td>-</td><td>R/W</td><td>Reserved (write 0)</td></tr></table> See also rx_tpu_tstr_bits_t Bit definitions	Bits	Field	R/W	Description	0	CST0	R/W	TPU0 counter start (1=run)	1	CST1	R/W	TPU1 counter start (1=run)	2	CST2	R/W	TPU2 counter start (1=run)	3	CST3	R/W	TPU3 counter start (1=run)	4	CST4	R/W	TPU4 counter start (1=run)	5	CST5	R/W	TPU5 counter start (1=run)	7:↔ 6	-	R/W	Reserved (write 0)
Bits	Field	R/W	Description																															
0	CST0	R/W	TPU0 counter start (1=run)																															
1	CST1	R/W	TPU1 counter start (1=run)																															
2	CST2	R/W	TPU2 counter start (1=run)																															
3	CST3	R/W	TPU3 counter start (1=run)																															
4	CST4	R/W	TPU4 counter start (1=run)																															
5	CST5	R/W	TPU5 counter start (1=run)																															
7:↔ 6	-	R/W	Reserved (write 0)																															

volatile uint8_t	tsyr	Timer Synchronous Register (TSYR) @ offset 0x01. Selects independent or synchronous operation for TCNT of TPU0 through TPU5. When synchronous, channels share counter writes. Bit Layout (8-bit)<table><thead><tr><th>Bits</th><th>Field</th><th>R/W</th><th>Description</th></tr></thead><tbody><tr><td>0</td><td>SYNC0</td><td>R/W</td><td>TPU0 synchronous (1=enabled)</td></tr><tr><td>1</td><td>SYNC1</td><td>R/W</td><td>TPU1 synchronous (1=enabled)</td></tr><tr><td>2</td><td>SYNC2</td><td>R/W</td><td>TPU2 synchronous (1=enabled)</td></tr><tr><td>3</td><td>SYNC3</td><td>R/W</td><td>TPU3 synchronous (1=enabled)</td></tr><tr><td>4</td><td>SYNC4</td><td>R/W</td><td>TPU4 synchronous (1=enabled)</td></tr><tr><td>5</td><td>SYNC5</td><td>R/W</td><td>TPU5 synchronous (1=enabled)</td></tr><tr><td>7↔6</td><td>-</td><td>R/W</td><td>Reserved (write 0)</td></tr></tbody></table> See also rx_tpu_tsy_bits_t Bit definitions	Bits	Field	R/W	Description	0	SYNC0	R/W	TPU0 synchronous (1=enabled)	1	SYNC1	R/W	TPU1 synchronous (1=enabled)	2	SYNC2	R/W	TPU2 synchronous (1=enabled)	3	SYNC3	R/W	TPU3 synchronous (1=enabled)	4	SYNC4	R/W	TPU4 synchronous (1=enabled)	5	SYNC5	R/W	TPU5 synchronous (1=enabled)	7↔6	-	R/W	Reserved (write 0)
Bits	Field	R/W	Description																															
0	SYNC0	R/W	TPU0 synchronous (1=enabled)																															
1	SYNC1	R/W	TPU1 synchronous (1=enabled)																															
2	SYNC2	R/W	TPU2 synchronous (1=enabled)																															
3	SYNC3	R/W	TPU3 synchronous (1=enabled)																															
4	SYNC4	R/W	TPU4 synchronous (1=enabled)																															
5	SYNC5	R/W	TPU5 synchronous (1=enabled)																															
7↔6	-	R/W	Reserved (write 0)																															

8.65.2.2 struct rx_tpu_ext_regs_t

TPU Extended Channel Register Structure (TPU0 and TPU3).

Register layout for TPU0 and TPU3 which have two I/O control registers (TIORH + TIORL) and four timer general registers (TGRA through TGRD). These channels do NOT support phase counting mode.

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	1	tcr	uint8↔ _t	Timer Control Register
0x01	1	tmdr	uint8↔ _t	Timer Mode Register
0x02	1	tiorh	uint8↔ _t	Timer I/O Control Register H
0x03	1	tiorl	uint8↔ _t	Timer I/O Control Register L
0x04	1	tier	uint8↔ _t	Timer Interrupt Enable Register
0x05	1	tsr	uint8↔ _t	Timer Status Register
0x06	2	tcnt	uint16↔ _t	Timer Counter
0x08	2	tgra	uint16↔ _t	Timer General Register A

Offset	Size	Field	Type	Description
0x0A	2	tgrb	uint16↔ _t	Timer General Register B
0x0C	2	tgrc	uint16↔ _t	Timer General Register C
0x0E	2	tgrd	uint16↔ _t	Timer General Register D
Total	16			

Invariant

```
sizeof(rx_tpu_ext_regs_t) == 16 bytes
```

Note

TPU0 and TPU3 do NOT support phase counting mode

Buffer operation available: TGRA+TGRC and TGRB+TGRD pairs

See also

[tpu0\(\)](#) Accessor for TPU0

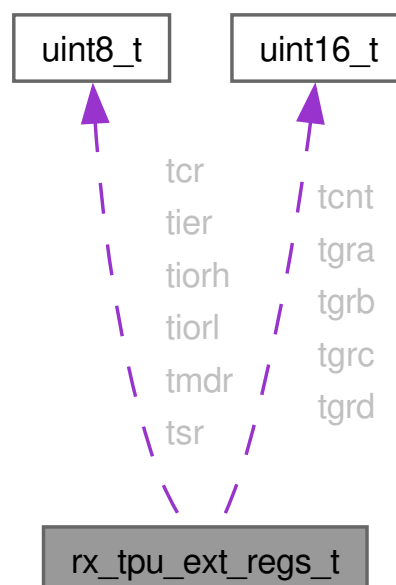
[tpu3\(\)](#) Accessor for TPU3

Since

Version 1.0.0

Definition at line 395 of file [rx72n_tpu_regs.h](#).

Collaboration diagram for rx_tpu_ext_regs_t:



Data Fields

volatile uint16_t	tcnt	Timer Counter (TCNT) @ +0x06, 16-bit
volatile uint8_t	tcr	Timer Control Register (TCR) @ +0x00
volatile uint16_t	tgra	Timer General Register A (TGRA) @ +0x08
volatile uint16_t	tgrb	Timer General Register B (TGRB) @ +0x0A
volatile uint16_t	tgrc	Timer General Register C (TGRC) @ +0x0C
volatile uint16_t	tgrd	Timer General Register D (TGRD) @ +0x0E
volatile uint8_t	tier	Timer Interrupt Enable Register (TIER) @ +0x04
volatile uint8_t	tiorh	Timer I/O Control Register H (TIORH) @ +0x02
volatile uint8_t	tiorl	Timer I/O Control Register L (TIORL) @ +0x03
volatile uint8_t	tmdr	Timer Mode Register (TMDR) @ +0x01
volatile uint8_t	tsr	Timer Status Register (TSR) @ +0x05

8.65.2.3 struct rx'tpu'regs't

TPU Basic Channel Register Structure (TPU1, TPU2, TPU4, TPU5).

Register layout for TPU1/2/4/5 which have a single I/O control register (TIO) and two timer general registers (TGRA and TGRB). These channels support phase counting mode for quadrature encoder decoding.

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	1	tcr	uint8_t	Timer Control Register
0x01	1	tmdr	uint8_t	Timer Mode Register
0x02	1	tior	uint8_t	Timer I/O Control Register
0x03	1	reserved	uint8_t	Reserved
0x04	1	tier	uint8_t	Timer Interrupt Enable Register
0x05	1	tsr	uint8_t	Timer Status Register
0x06	2	tcnt	uint16_t	Timer Counter (up/down in phase)
0x08	2	tgra	uint16_t	Timer General Register A
0x0A	2	tgrb	uint16_t	Timer General Register B
0x0C	4	reserved2	uint8_t[]	Reserved

Offset	Size	Field	Type	Description
Total	16			

Invariant

`sizeof(rx_tpu_regs_t) == 16 bytes`

Phase Counting Mode Usage

```
// Configure TPU1 for phase counting mode 4 (4x resolution)
volatile rx_tpu_regs_t* ch = tpu1();

// 1. Stop counter first
tpu_control()->tstr &= (uint8_t)~k_tpu_tstr_cst1;

// 2. Set phase counting mode 4 in TMDR
ch->tmdr = k_tpu_tmdr_md_phase_count_4;

// 3. Start counter
tpu_control()->tstr |= k_tpu_tstr_cst1;

// 4. Read counter and direction
uint16_t count = ch->tcnt;
bool counting_up = (ch->tsr & k_tpu_tsr_tcfid) != 0;
```

See also

[tpu1\(\)](#) Accessor for back-right encoder

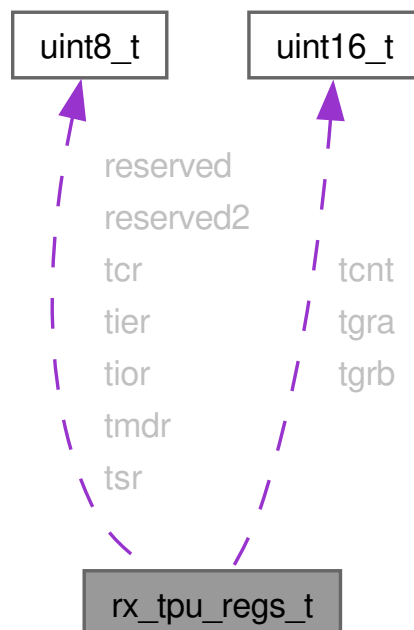
[tpu2\(\)](#) Accessor for back-left encoder

Since

Version 1.0.0

Definition at line 458 of file [rx72n_tpu_regs.h](#).

Collaboration diagram for `rx_tpu_regs_t`:



Data Fields

volatile uint8_t	reserved	Reserved @ +0x03
volatile uint8_t	reserved2↵ [4]	Reserved @ +0x0C-0x0F
volatile uint16_t	tcnt	Timer Counter (TCNT) @ +0x06, up/down in phase mode
volatile uint8_t	tcr	Timer Control Register (TCR) @ +0x00
volatile uint16_t	tgra	Timer General Register A (TGRA) @ +0x08
volatile uint16_t	tgrb	Timer General Register B (TGRB) @ +0x0A
volatile uint8_t	tier	Timer Interrupt Enable Register (TIER) @ +0x04
volatile uint8_t	tior	Timer I/O Control Register (TIOR) @ +0x02
volatile uint8_t	tmdr	Timer Mode Register (TMDR) @ +0x01
volatile uint8_t	tsr	Timer Status Register (TSR) @ +0x05

8.65.3 Enumeration Type Documentation

8.65.3.1 rx'tpu'addresses't

```
enum rx_tpu_addresses_t : uintptr_t
```

TPU base addresses.

Base addresses for the TPU control block and individual channel register blocks. The control block contains TSTR, TSYR, and NFCR registers shared across all channels. Each channel has a 16-byte register block.

Memory Map

Address	Block	Description
0x00088100	Control	TSTR, TSYR, NFCR (shared across channels)
0x00088110	TPU0	Extended channel (TIORH/L, TGRA-TGRD)
0x00088120	TPU1	Basic channel (TIOR, TGRA-TGRB)
0x00088130	TPU2	Basic channel (TIOR, TGRA-TGRB)
0x00088140	TPU3	Extended channel (TIORH/L, TGRA-TGRD)
0x00088150	TPU4	Basic channel (TIOR, TGRA-TGRB)
0x00088160	TPU5	Basic channel (TIOR, TGRA-TGRB)

Note

Manual: Ch28.2 - Register Descriptions

See also

[tpu_control\(\)](#) Accessor for control block

[tpu1\(\)](#) Accessor for TPU1 channel (back-right encoder)

Since

Version 1.0.0

Enumerator

k_tpu_control_base_addr	TPU control block base address (0x00088100). Contains TSTR, TSYR at +0x00/+0x01 and NFCR[0-5] at +0x08
k_tpu0_base_addr	TPU0 channel base address (0x00088110) - Extended channel. Extended channel with TIORH/TIOL and TGRA through TGRD
k_tpu1_base_addr	TPU1 channel base address (0x00088120) - Basic channel. Back-right encoder, phase counting capable (TCLKA/TCLKB)
k_tpu2_base_addr	TPU2 channel base address (0x00088130) - Basic channel. Back-left encoder, phase counting capable (TCLKC/TCLKD)
k_tpu3_base_addr	TPU3 channel base address (0x00088140) - Extended channel. Extended channel with TIORH/TIOL and TGRA through TGRD
k_tpu4_base_addr	TPU4 channel base address (0x00088150) - Basic channel. Phase counting capable, paired with TPU2 (TCLKC/TCLKD)
k_tpu5_base_addr	TPU5 channel base address (0x00088160) - Basic channel. Phase counting capable, paired with TPU1 (TCLKA/TCLKB)
k_tpu_channel_spacing	Channel register block spacing (0x10 bytes between channels). Each TPU channel occupies a 16-byte register block

Definition at line 194 of file [rx72n_tpu_regs.h](#).

```

00194         : uintptr_t {
00195     /**
00196      * @brief TPU control block base address (0x00088100)
00197      * @details Contains TSTR, TSYR at +0x00/+0x01 and NFCR[0-5] at +0x08
00198      */
00199     k_tpu_control_base_addr = 0x00088100,
00200
00201     /**
00202      * @brief TPU0 channel base address (0x00088110) - Extended channel
00203      * @details Extended channel with TIORH/TIOL and TGRA through TGRD
00204      */
00205     k_tpu0_base_addr = 0x00088110,
00206
00207     /**
00208      * @brief TPU1 channel base address (0x00088120) - Basic channel
00209      * @details Back-right encoder, phase counting capable (TCLKA/TCLKB)
00210      */
00211     k_tpu1_base_addr = 0x00088120,
00212
00213     /**
00214      * @brief TPU2 channel base address (0x00088130) - Basic channel
00215      * @details Back-left encoder, phase counting capable (TCLKC/TCLKD)
00216      */
00217     k_tpu2_base_addr = 0x00088130,
00218
00219     /**
00220      * @brief TPU3 channel base address (0x00088140) - Extended channel
00221      * @details Extended channel with TIORH/TIOL and TGRA through TGRD
00222      */
00223     k_tpu3_base_addr = 0x00088140,
00224
00225     /**
00226      * @brief TPU4 channel base address (0x00088150) - Basic channel
00227      * @details Phase counting capable, paired with TPU2 (TCLKC/TCLKD)
00228      */
00229     k_tpu4_base_addr = 0x00088150,
00230
00231     /**
00232      * @brief TPU5 channel base address (0x00088160) - Basic channel
00233      * @details Phase counting capable, paired with TPU1 (TCLKA/TCLKB)
00234      */
00235     k_tpu5_base_addr = 0x00088160,
00236
00237     /**
00238      * @brief Channel register block spacing (0x10 bytes between channels)
00239      * @details Each TPU channel occupies a 16-byte register block
00240      */

```

```
00241 k_tpu_channel_spacing = 0x10,
00242 } rx_tpu_addresses_t;
```

8.65.3.2 rx'tpu'channel'count't

```
enum rx_tpu_channel_count_t : uint8_t
```

TPU channel count constant.

Since

Version 1.0.0

Enumerator

k_tpu_channel_count	Total TPU channels (TPU0-TPU5)
---------------------	--------------------------------

Definition at line 254 of file [rx72n_tpu_regs.h](#).

```
00254      : uint8_t {
00255 k_tpu_channel_count = 6, /**< Total TPU channels (TPU0-TPU5) */
00256 } rx_tpu_channel_count_t;
```

8.65.3.3 rx'tpu'intb'source't

```
enum rx_tpu_intb_source_t : uint8_t
```

TPU Software Configurable Interrupt B source numbers.

TPU interrupts are routed through the ICU Software Configurable Interrupt B mechanism. These source numbers are written to SLIBRn registers (n=128-207) to map them to interrupt vector numbers 128-207.

For phase counting mode, the most relevant sources are:

- TCIXV (overflow): TCNT wrapped 0xFFFF -> 0x0000 while counting up
- TCIXU (underflow): TCNT wrapped 0x0000 -> 0xFFFF while counting down

Configuration Example

```
// Map TPU1 overflow to vector 128 (INTB128)
ICU.SLIBR128.BYTE = k_tpu_intb_tci1v;
// Map TPU1 underflow to vector 129 (INTB129)
ICU.SLIBR129.BYTE = k_tpu_intb_tci1u;
```

Note

Manual: Ch15 Table 15.3 - Interrupt Sources for Software Configurable Interrupt B, and Ch28.4 - Interrupt Sources

Warning

These are NOT vector numbers - they are source numbers for SLIBR

Since

Version 1.0.0

Enumerator

k_tpu_intb_tgi0a	TPU0 TGRA compare match/input capture
k_tpu_intb_tgi0b	TPU0 TGRB compare match/input capture
k_tpu_intb_tgi0c	TPU0 TGRC compare match/input capture
k_tpu_intb_tgi0d	TPU0 TGRD compare match/input capture
k_tpu_intb_tci0v	TPU0 overflow
k_tpu_intb_tgi1a	TPU1 TGRA compare match/input capture
k_tpu_intb_tgi1b	TPU1 TGRB compare match/input capture
k_tpu_intb_tci1v	TPU1 overflow (phase counting up wrap)
k_tpu_intb_tci1u	TPU1 underflow (phase counting down wrap)
k_tpu_intb_tgi2a	TPU2 TGRA compare match/input capture
k_tpu_intb_tgi2b	TPU2 TGRB compare match/input capture
k_tpu_intb_tci2v	TPU2 overflow (phase counting up wrap)
k_tpu_intb_tci2u	TPU2 underflow (phase counting down wrap)
k_tpu_intb_tgi3a	TPU3 TGRA compare match/input capture
k_tpu_intb_tgi3b	TPU3 TGRB compare match/input capture
k_tpu_intb_tgi3c	TPU3 TGRC compare match/input capture
k_tpu_intb_tgi3d	TPU3 TGRD compare match/input capture
k_tpu_intb_tci3v	TPU3 overflow
k_tpu_intb_tgi4a	TPU4 TGRA compare match/input capture
k_tpu_intb_tgi4b	TPU4 TGRB compare match/input capture
k_tpu_intb_tci4v	TPU4 overflow
k_tpu_intb_tci4u	TPU4 underflow (phase counting)
k_tpu_intb_tgi5a	TPU5 TGRA compare match/input capture
k_tpu_intb_tgi5b	TPU5 TGRB compare match/input capture
k_tpu_intb_tci5v	TPU5 overflow
k_tpu_intb_tci5u	TPU5 underflow (phase counting)

Definition at line 1030 of file [rx72n_tpu_regs.h](#).

```

01030         : uint8_t {
01031     /* TPU0 interrupt sources */
01032     k_tpu_intb_tgi0a = 15, /**< TPU0 TGRA compare match/input capture */
01033     k_tpu_intb_tgi0b = 16, /**< TPU0 TGRB compare match/input capture */
01034     k_tpu_intb_tgi0c = 17, /**< TPU0 TGRC compare match/input capture */
01035     k_tpu_intb_tgi0d = 18, /**< TPU0 TGRD compare match/input capture */
01036     k_tpu_intb_tci0v = 19, /**< TPU0 overflow */
01037
01038     /* TPU1 interrupt sources (back-right encoder) */
01039     k_tpu_intb_tgi1a = 20, /**< TPU1 TGRA compare match/input capture */
01040     k_tpu_intb_tgi1b = 21, /**< TPU1 TGRB compare match/input capture */
01041     k_tpu_intb_tci1v = 22, /**< TPU1 overflow (phase counting up wrap) */
01042     k_tpu_intb_tci1u = 23, /**< TPU1 underflow (phase counting down wrap) */
01043 
```

```

01044 /* TPU2 interrupt sources (back-left encoder) */
01045 k_tpu_intb_tgi2a = 24, /**< TPU2 TGRA compare match/input capture */
01046 k_tpu_intb_tgi2b = 25, /**< TPU2 TGRB compare match/input capture */
01047 k_tpu_intb_tci2v = 26, /**< TPU2 overflow (phase counting up wrap) */
01048 k_tpu_intb_tci2u = 27, /**< TPU2 underflow (phase counting down wrap) */
01049
01050 /* TPU3 interrupt sources */
01051 k_tpu_intb_tgi3a = 28, /**< TPU3 TGRA compare match/input capture */
01052 k_tpu_intb_tgi3b = 29, /**< TPU3 TGRB compare match/input capture */
01053 k_tpu_intb_tgi3c = 30, /**< TPU3 TGRC compare match/input capture */
01054 k_tpu_intb_tgi3d = 31, /**< TPU3 TGRD compare match/input capture */
01055 k_tpu_intb_tci3v = 32, /**< TPU3 overflow */
01056
01057 /* TPU4 interrupt sources */
01058 k_tpu_intb_tgi4a = 33, /**< TPU4 TGRA compare match/input capture */
01059 k_tpu_intb_tgi4b = 34, /**< TPU4 TGRB compare match/input capture */
01060 k_tpu_intb_tci4v = 35, /**< TPU4 overflow */
01061 k_tpu_intb_tci4u = 36, /**< TPU4 underflow (phase counting) */
01062
01063 /* TPU5 interrupt sources */
01064 k_tpu_intb_tgi5a = 37, /**< TPU5 TGRA compare match/input capture */
01065 k_tpu_intb_tgi5b = 38, /**< TPU5 TGRB compare match/input capture */
01066 k_tpu_intb_tci5v = 39, /**< TPU5 overflow */
01067 k_tpu_intb_tci5u = 40, /**< TPU5 underflow (phase counting) */
01068 } rx_tpu_intb_source_t;

```

8.65.3.4 rx'tpu'mstpcra't

```
enum rx_tpu_mstpcra_t : uint32_t
```

MSTPCRA bit for TPU module stop control.

The TPU module is controlled by MSTPA13 in the Module Stop Control Register A (MSTPCRA). Write 0 to enable, 1 to stop.

Since

Version 1.0.0

Enumerator

k_tpu_mstpcra_mstpa13	MSTPA13: TPU module stop (1=stop)
-----------------------	-----------------------------------

Definition at line 995 of file [rx72n_tpu_regs.h](#).

```

00995 : uint32_t {
00996 k_tpu_mstpcra_mstpa13 = (1U « 13), /**< MSTPA13: TPU module stop (1=stop) */
00997 } rx_tpu_mstpcra_t;

```

8.65.3.5 rx'tpu'nfc'r'bits't

```
enum rx_tpu_nfc_r_bits_t : uint8_t
```

NFCR (Noise Filter Control Register) bit definitions.

Controls digital noise filtering on TIOC I/O pins. Each channel has independent filter enable bits for each pin and a shared clock select.

Note

NFCEN/NFDEN are reserved on TPU1/2/4/5 (only TPU0/3 have C/D pins)

Set NFCR only while TCNT is stopped (CSTn = 0)

Manual: Ch28.2.10 - Noise Filter Control Register (NFCR)

Since

Version 1.0.0

Enumerator

k_tpu_nfcr_nfaen	Noise filter enable for TIOCA pin
k_tpu_nfcr_nfben	Noise filter enable for TIOCB pin
k_tpu_nfcr_nfcen	Noise filter enable for TIOCC (TPU0/3)
k_tpu_nfcr_nfden	Noise filter enable for TIOCD (TPU0/3)

Definition at line 938 of file [rx72n_tpu_regs.h](#).

```
00938      : uint8_t {
00939  k_tpu_nfcr_nfaen = (1U « 0), /**< Noise filter enable for TIOCA pin */
00940  k_tpu_nfcr_nfben = (1U « 1), /**< Noise filter enable for TIOCB pin */
00941  k_tpu_nfcr_nfcen = (1U « 2), /**< Noise filter enable for TIOCC (TPU0/3) */
00942  k_tpu_nfcr_nfden = (1U « 3), /**< Noise filter enable for TIOCD (TPU0/3) */
00943 } rx_tpu_nfcr_bits_t;
```

8.65.3.6 rx`tpu`nfcr`clock`

enum [rx_tpu_nfcr_clock_t](#) : uint8_t

NFCR NFCS[1:0] noise filter clock select values.

Selects the sampling clock for the digital noise filter. The filter requires 3 consecutive samples at the selected level to pass.

Filter Delay at 60 MHz PCLKB

NFCS	Clock	3-Sample Delay
00	PCLKB/ $\leftarrow$ 1	~50 ns
01	PCLKB/ $\leftarrow$ 8	~400 ns
10	PCLKB/32	~1.6 us
11	Count clk	Depends on prescaler setting

Since

Version 1.0.0

Enumerator

k_tpu_nfcr_nfcs_pclk_div1	Sample @ PCLKB/1 (~50ns)
k_tpu_nfcr_nfcs_pclk_div8	Sample @ PCLKB/8 (~400ns)
k_tpu_nfcr_nfcs_pclk_div32	Sample @ PCLKB/32 (~1.6us)
k_tpu_nfcr_nfcs_count_clock	Sample @ count clock source

Definition at line 963 of file [rx72n_tpu_regs.h](#).

```
00963      : uint8_t {
00964  k_tpu_nfcr_nfcs_pclk_div1 = (0U « 4), /**< Sample @ PCLKB/1 (~50ns) */
00965  k_tpu_nfcr_nfcs_pclk_div8 = (1U « 4), /**< Sample @ PCLKB/8 (~400ns) */
00966  k_tpu_nfcr_nfcs_pclk_div32 = (2U « 4), /**< Sample @ PCLKB/32 (~1.6us) */
00967  k_tpu_nfcr_nfcs_count_clock = (3U « 4), /**< Sample @ count clock source */
00968 } rx_tpu_nfcr_clock_t;
```

8.65.3.7 rx'tpu'nfcr'masks't

```
enum rx_tpu_nfcr_masks_t : uint8_t
```

NFCR register field masks.

Since

Version 1.0.0

Enumerator

k_tpu_nfcr_nfcs_mask	NFCS[1:0] field mask (bits 5:4)
k_tpu_nfcr_enable_mask	All noise filter enable bits (bits 3:0)

Definition at line 975 of file [rx72n_tpu_regs.h](#).

```
00975      : uint8_t {
00976  k_tpu_nfcr_nfcs_mask = 0x30, /**< NFCS[1:0] field mask (bits 5:4) */
00977  k_tpu_nfcr_enable_mask = 0x0F, /**< All noise filter enable bits (bits 3:0) */
00978 } rx_tpu_nfcr_masks_t;
```

8.65.3.8 rx'tpu'tcr'cclr't

```
enum rx_tpu_tcr_cclr_t : uint8_t
```

TCR CCLR[2:0] counter clear source values (for basic channels).

Selects the event that clears TCNT. In phase counting mode, only TGRA and TGRB compare match clearing is available.

Note

Bit 7 is reserved for TPU1/2/4/5 (basic channels)

See also

rx_tpu_tcr_shift_t::k_tpu_tcr_cclr_shift Field position

Since

Version 1.0.0

Enumerator

k_tpu_tcr_cclr_disabled	TCNT free-running (no clear)
k_tpu_tcr_cclr_tgra_match	Clear on TGRA compare match
k_tpu_tcr_cclr_tgrb_match	Clear on TGRB compare match
k_tpu_tcr_cclr_sync_clear	Synchronous clear

Definition at line 750 of file [rx72n_tpu_regs.h](#).

```
00750      : uint8_t {
00751  k_tpu_tcr_cclr_disabled = (0U « 5), /**< TCNT free-running (no clear) */
00752  k_tpu_tcr_cclr_tgra_match = (1U « 5), /**< Clear on TGRA compare match */
00753  k_tpu_tcr_cclr_tgrb_match = (2U « 5), /**< Clear on TGRB compare match */
00754  k_tpu_tcr_cclr_sync_clear = (3U « 5), /**< Synchronous clear */
00755 } rx_tpu_tcr_cclr_t;
```

8.65.3.9 rx`tpu`tcr`masks`t

```
enum rx_tpu_tcr_masks_t : uint8_t
```

TCR register field masks.

Since

Version 1.0.0

Enumerator

k_tpu_tcr_tpsc_mask	TPSC[2:0] field mask (bits 2:0)
k_tpu_tcr_ckege_mask	CKEG[1:0] field mask (bits 4:3)
k_tpu_tcr_cclr_mask	CCLR[2:0] field mask (bits 7:5)

Definition at line 762 of file [rx72n_tpu_regs.h](#).

```
00762     : uint8_t {
00763   k_tpu_tcr_tpsc_mask = 0x07, /**< TPSC[2:0] field mask (bits 2:0) */
00764   k_tpu_tcr_ckege_mask = 0x18, /**< CKEG[1:0] field mask (bits 4:3) */
00765   k_tpu_tcr_cclr_mask = 0xE0, /**< CCLR[2:0] field mask (bits 7:5) */
00766 } rx_tpu_tcr_masks_t;
```

8.65.3.10 rx`tpu`tcr`shift`t

```
enum rx_tpu_tcr_shift_t : uint8_t
```

TCR (Timer Control Register) bit field positions.

Bit positions for the Timer Control Register fields: prescaler select, clock edge select, and counter clear source select.

Note

In phase counting mode, TPSC and CKEG settings are ignored

See also

[rx_tpu_tcr_cclr_t](#) Counter clear source values

Since

Version 1.0.0

Enumerator

k_tpu_tcr_tpsc_shift	TPSC[2:0] - Timer Prescaler Select (bits 2:0)
k_tpu_tcr_ckege_shift	CKEG[1:0] - Input Clock Edge Select (bits 4:3)
k_tpu_tcr_cclr_shift	CCLR[2:0] - Counter Clear Source (bits 7:5)

Definition at line 732 of file [rx72n_tpu_regs.h](#).

```
00732     : uint8_t {
00733   k_tpu_tcr_tpsc_shift = 0, /**< TPSC[2:0] - Timer Prescaler Select (bits 2:0) */
00734   k_tpu_tcr_ckege_shift = 3, /**< CKEG[1:0] - Input Clock Edge Select (bits 4:3) */
00735   k_tpu_tcr_cclr_shift = 5, /**< CCLR[2:0] - Counter Clear Source (bits 7:5) */
00736 } rx_tpu_tcr_shift_t;
```

8.65.3.11 rx_tpu_tier_bits_t

```
enum rx_tpu_tier_bits_t : uint8_t
```

TIER (Timer Interrupt Enable Register) bit definitions.

Controls interrupt generation for timer compare match, overflow, and underflow events. In phase counting mode, TCIEV (overflow) and TCIEU (underflow) are the most relevant interrupts.

Bit Layout (8-bit)

Bit	Field	Description
0	TGIEA	TGRA compare match/input capture IRQ
1	TGIEB	TGRB compare match/input capture IRQ
2	TGIEC	TGRC compare match/input capture (0,3)
3	TGIED	TGRD compare match/input capture (0,3)
4	TCIEV	Overflow interrupt enable
5	TCIEU	Underflow interrupt enable (1,2,4,5)
6	-	Reserved (read as 1, write 1)
7	TTGE	A/D conversion start request enable

Note

TGIEC/TGIED reserved on TPU1/2/4/5 (read as 0, write 0)

TCIEU reserved on TPU0/TPU3 (read as 0, write 0)

TTGE reserved on TPU5 (read as 0, write 0)

Manual: Ch28.2.4 - Timer Interrupt Enable Register (TIER)

Since

Version 1.0.0

Enumerator

k_tpu_tier_tgiea	TGRA interrupt enable
k_tpu_tier_tgieb	TGRB interrupt enable
k_tpu_tier_tgiec	TGRC interrupt enable (TPU0/3 only)
k_tpu_tier_tgied	TGRD interrupt enable (TPU0/3 only)
k_tpu_tier_tciev	Overflow interrupt enable (all channels)
k_tpu_tier_tcieu	Underflow interrupt enable (TPU1/2/4/5)
k_tpu_tier_ttge	A/D conversion start request enable

Definition at line 857 of file [rx72n_tpu_regs.h](#).

```
00857     : uint8_t {
00858   k_tpu_tier_tgiea = (1U « 0), /**< TGRA interrupt enable */
00859   k_tpu_tier_tgieb = (1U « 1), /**< TGRB interrupt enable */
00860   k_tpu_tier_tgiec = (1U « 2), /**< TGRC interrupt enable (TPU0/3 only) */
00861   k_tpu_tier_tgied = (1U « 3), /**< TGRD interrupt enable (TPU0/3 only) */
00862   k_tpu_tier_tciev = (1U « 4), /**< Overflow interrupt enable (all channels) */
00863   k_tpu_tier_tcieu = (1U « 5), /**< Underflow interrupt enable (TPU1/2/4/5) */
00864   k_tpu_tier_ttge  = (1U « 7), /**< A/D conversion start request enable */
00865 } rx_tpu_tier_bits_t;
```

8.65.3.12 rx`tpu`tmdr`masks`h

```
enum rx_tpu_tmdr_masks_t : uint8_t
```

TMDR register field masks.

Since

Version 1.0.0

Enumerator

k_tpu_tmdr_md_mask	MD[3:0] field mask (bits 3:0)
--------------------	-------------------------------

Definition at line 821 of file rx72n_tpu_regs.h.

```
00821      : uint8_t {
00822  k_tpu_tmdr_md_mask = 0x0F, /**< MD[3:0] field mask (bits 3:0) */
00823 } rx_tpu_tmdr_masks_t;
```

8.65.3.13 rx`tpu`tmdr`md`h

```
enum rx_tpu_tmdr_md_t : uint8_t
```

TMDR MD[3:0] mode select values.

Selects the operating mode for the TPU channel. Phase counting modes (1-4) are only available on TPU1, TPU2, TPU4, and TPU5.

Phase Counting Mode Comparison

Mode	Resolution	Count Edges
1	1x	A-phase rising only
2	1x	B-phase rising only
3	2x	A-phase rising + falling
4	4x	All edges of both A and B phases

Note

Phase counting mode cannot be set for TPU0 and TPU3

Manual: Ch28.2.2 - Timer Mode Register (TMDR)

Since

Version 1.0.0

Enumerator

k_tpu_tmdr_md_normal	Normal operation
k_tpu_tmdr_md_pwm1	PWM mode 1
k_tpu_tmdr_md_pwm2	PWM mode 2
k_tpu_tmdr_md_phase_count_1	Phase counting mode 1 (1x)
k_tpu_tmdr_md_phase_count_2	Phase counting mode 2 (1x)
k_tpu_tmdr_md_phase_count_3	Phase counting mode 3 (2x)
k_tpu_tmdr_md_phase_count_4	Phase counting mode 4 (4x) - STAR default

Definition at line 793 of file [rx72n_tpu_regs.h](#).

```

00793         : uint8_t {
00794     k_tpu_tmdr_md_normal      = 0x00, /**< Normal operation */
00795     k_tpu_tmdr_md_pwm1       = 0x02, /**< PWM mode 1 */
00796     k_tpu_tmdr_md_pwm2       = 0x03, /**< PWM mode 2 */
00797     k_tpu_tmdr_md_phase_count_1 = 0x04, /**< Phase counting mode 1 (1x) */
00798     k_tpu_tmdr_md_phase_count_2 = 0x05, /**< Phase counting mode 2 (1x) */
00799     k_tpu_tmdr_md_phase_count_3 = 0x06, /**< Phase counting mode 3 (2x) */
00800     k_tpu_tmdr_md_phase_count_4 = 0x07, /**< Phase counting mode 4 (4x) - STAR default */
00801 } rx_tpu_tmdr_md_t;

```

8.65.3.14 rx_tpu_tmdr_shift_t

```
enum rx_tpu_tmdr_shift_t : uint8_t
```

TMDR bit field positions.

Since

Version 1.0.0

Enumerator

k_tpu_tmdr_md_shift	MD[3:0] - Mode Select (bits 3:0)
k_tpu_tmdr_bfa_shift	BFA - Buffer Operation A (bit 4, TPU0/3 only)
k_tpu_tmdr_bfb_shift	BFB - Buffer Operation B (bit 5, TPU0/3 only)
k_tpu_tmdr_icselb_shift	ICSELB - TGRB input capture select (bit 6)
k_tpu_tmdr_icseld_shift	ICSELD - TGRD input capture select (bit 7, TPU0/3)

Definition at line 808 of file [rx72n_tpu_regs.h](#).

```

00808         : uint8_t {
00809     k_tpu_tmdr_md_shift      = 0, /**< MD[3:0] - Mode Select (bits 3:0) */
00810     k_tpu_tmdr_bfa_shift     = 4, /**< BFA - Buffer Operation A (bit 4, TPU0/3 only) */
00811     k_tpu_tmdr_bfb_shift     = 5, /**< BFB - Buffer Operation B (bit 5, TPU0/3 only) */
00812     k_tpu_tmdr_icselb_shift  = 6, /**< ICSELB - TGRB input capture select (bit 6) */
00813     k_tpu_tmdr_icseld_shift  = 7, /**< ICSELD - TGRD input capture select (bit 7, TPU0/3) */
00814 } rx_tpu_tmdr_shift_t;

```


8.65.3.15 rx`tpu`tsr`bits``t`

```
enum rx_tpu_tsr_bits_t : uint8_t
```

TSR (Timer Status Register) bit definitions.

Status flags for timer events and counting direction. The TCFD flag is critical for phase counting mode - it indicates whether TCNT is currently counting up (forward) or down (reverse).

Bit Layout (8-bit)

Bit	Field	Description
0	TGFA	TGRA compare match/input capture flag
1	TGFB	TGRB compare match/input capture flag
2	TGFC	TGRC compare match/input capture (0,3)
3	TGFD	TGRD compare match/input capture (0,3)
4	TCFV	Overflow flag (all channels)
5	TCFU	Underflow flag (TPU1/2/4/5 only)
6	-	Reserved (read as 1, write 1)
7	TCFD	Counting Direction Flag (TPU1/2/4/5)

Note

Flags (bits 0-5) are cleared by writing 0 to them

TCFD is read-only: 1=counting up, 0=counting down

Bit 6 reset value is 1 (must write 1 to preserve)

TGFC/TGFD reserved on TPU1/2/4/5 (read as 0, write 0)

TCFU/TCFD reserved on TPU0/TPU3 (read as 0/1 respectively)

Manual: Ch28.2.5 - Timer Status Register (TSR)

Since

Version 1.0.0

Enumerator

k_tpu_tsr_tgfa	TGRA compare match/input capture occurred
k_tpu_tsr_tgfb	TGRB compare match/input capture occurred
k_tpu_tsr_tgfc	TGRC compare match/input capture (TPU0/3)
k_tpu_tsr_tgfd	TGRD compare match/input capture (TPU0/3)
k_tpu_tsr_tcfv	Overflow flag (TCNT wrapped 0xFFFF->0x0000)
k_tpu_tsr_tcfu	Underflow flag (TCNT wrapped 0x0000->0xFFFF)
k_tpu_tsr_tcfv	Counting direction: 1=up, 0=down (read-only)

Definition at line 901 of file [rx72n_tpu_regs.h](#).

```
00901     : uint8_t {
00902   k_tpu_tsr_tgfa = (1U « 0), /**< TGRA compare match/input capture occurred */
00903   k_tpu_tsr_tgfb = (1U « 1), /**< TGRB compare match/input capture occurred */
00904   k_tpu_tsr_tgfc = (1U « 2), /**< TGRC compare match/input capture (TPU0/3) */
00905   k_tpu_tsr_tgfd = (1U « 3), /**< TGRD compare match/input capture (TPU0/3) */
00906   k_tpu_tsr_tcfv = (1U « 4), /**< Overflow flag (TCNT wrapped 0xFFFF->0x0000) */
00907   k_tpu_tsr_tcfu = (1U « 5), /**< Underflow flag (TCNT wrapped 0x0000->0xFFFF) */
00908   k_tpu_tsr_tcfv = (1U « 7), /**< Counting direction: 1=up, 0=down (read-only) */
00909 } rx_tpu_tsr_bits_t;
```

8.65.3.16 rx_tpu_tsr_reserved_t

```
enum rx_tpu_tsr_reserved_t : uint8_t
```

TSR reserved bit value (bit 6 must be written as 1).

Since

Version 1.0.0

Enumerator

k_tpu_tsr_reserved_bit6	Bit 6: read as 1, write 1
-------------------------	---------------------------

Definition at line 916 of file [rx72n_tpu_regs.h](#).

```
00916         : uint8_t {
00917   k_tpu_tsr_reserved_bit6 = (1U « 6), /**< Bit 6: read as 1, write 1 */
00918 } rx_tpu_tsr_reserved_t;
```

8.65.3.17 rx_tpu_tstr_bits_t

```
enum rx_tpu_tstr_bits_t : uint8_t
```

TSTR (Timer Start Register) bit definitions.

Each CSTn bit independently starts (1) or stops (0) the corresponding TPU_n channel counter. Multiple channels can be started simultaneously by OR-ing the appropriate bits.

Note

Manual: Ch28.2.8 - Timer Start Register (TSTR)

See also

[rx_tpu_control_regs_t::tstr](#) Register field

Since

Version 1.0.0

Enumerator

k_tpu_tstr_cst0	TPU0 counter start (1=run, 0=stop)
k_tpu_tstr_cst1	TPU1 counter start (back-right encoder)
k_tpu_tstr_cst2	TPU2 counter start (back-left encoder)
k_tpu_tstr_cst3	TPU3 counter start
k_tpu_tstr_cst4	TPU4 counter start
k_tpu_tstr_cst5	TPU5 counter start

Definition at line 684 of file [rx72n_tpu_regs.h](#).

```
00684         : uint8_t {
00685   k_tpu_tstr_cst0 = (1U « 0), /**< TPU0 counter start (1=run, 0=stop) */
00686   k_tpu_tstr_cst1 = (1U « 1), /**< TPU1 counter start (back-right encoder) */
00687   k_tpu_tstr_cst2 = (1U « 2), /**< TPU2 counter start (back-left encoder) */
00688   k_tpu_tstr_cst3 = (1U « 3), /**< TPU3 counter start */
00689   k_tpu_tstr_cst4 = (1U « 4), /**< TPU4 counter start */
00690   k_tpu_tstr_cst5 = (1U « 5), /**< TPU5 counter start */
00691 } rx_tpu_tstr_bits_t;
```

8.65.3.18 rx`tpu`tsyr`bits``t`

```
enum rx_tpu_tsyr_bits_t : uint8_t
```

TSYR (Timer Synchronous Register) bit definitions.

Each SYNCn bit enables (1) or disables (0) synchronous operation for the corresponding TPU_n channel. When multiple channels are set to synchronous, their TCNTs can be written simultaneously.

Note

Manual: Ch28.2.9 - Timer Synchronous Register (TSYR)

See also

[rx_tpu_control_regs_t::tsyr](#) Register field

Since

Version 1.0.0

Enumerator

k_tpu_tsyr_sync0	TPU0 synchronous operation enable
k_tpu_tsyr_sync1	TPU1 synchronous operation enable
k_tpu_tsyr_sync2	TPU2 synchronous operation enable
k_tpu_tsyr_sync3	TPU3 synchronous operation enable
k_tpu_tsyr_sync4	TPU4 synchronous operation enable
k_tpu_tsyr_sync5	TPU5 synchronous operation enable

Definition at line 706 of file [rx72n_tpu_regs.h](#).

```
00706      : uint8_t {
00707  k_tpu_tsyr_sync0 = (1U << 0), /**< TPU0 synchronous operation enable */
00708  k_tpu_tsyr_sync1 = (1U << 1), /**< TPU1 synchronous operation enable */
00709  k_tpu_tsyr_sync2 = (1U << 2), /**< TPU2 synchronous operation enable */
00710  k_tpu_tsyr_sync3 = (1U << 3), /**< TPU3 synchronous operation enable */
00711  k_tpu_tsyr_sync4 = (1U << 4), /**< TPU4 synchronous operation enable */
00712  k_tpu_tsyr_sync5 = (1U << 5), /**< TPU5 synchronous operation enable */
00713 } rx_tpu_tsyr_bits_t;
```

8.65.4 Function Documentation

8.65.4.1 tpu0()

```
volatile rx_tpu_ext_regs_t * tpu0 (
    void ) [inline], [static]
```

Get pointer to TPU0 channel registers (extended).

Returns a volatile pointer to TPU0 extended channel registers at address 0x00088110. TPU0 has TIORH/TIORL and TGRA through TGRD.

Returns

Volatile pointer to TPU0 extended register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

TPU0 does NOT support phase counting mode

See also

[k_tpu0_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 528 of file [rx72n_tpu_regs.h](#).

```
00529 {
00530     return (volatile rx_tpu_ext_regs_t*)k_tpu0_base_addr;
00531 }
```

References [k_tpu0_base_addr](#).

8.65.4.2 tpu1()

```
volatile rx_tpu_regs_t * tpu1 (
    void )    [inline], [static]
```

Get pointer to TPU1 channel registers (basic, phase counting).

Returns a volatile pointer to TPU1 basic channel registers at address 0x00088120. TPU1 supports phase counting mode using TCLKA (A-phase) and TCLKB (B-phase) external clock pins. Used for back-right encoder.

Returns

Volatile pointer to TPU1 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

Example:

```
// Read back-right encoder count and direction
uint16_t count = tpu1()->tcnt;
bool forward = (tpu1()->tsr & k_tpu_tsr_tcfld) != 0;
```

See also

[k_tpu1_base_addr](#) Base address constant

Since

Version 1.0.0

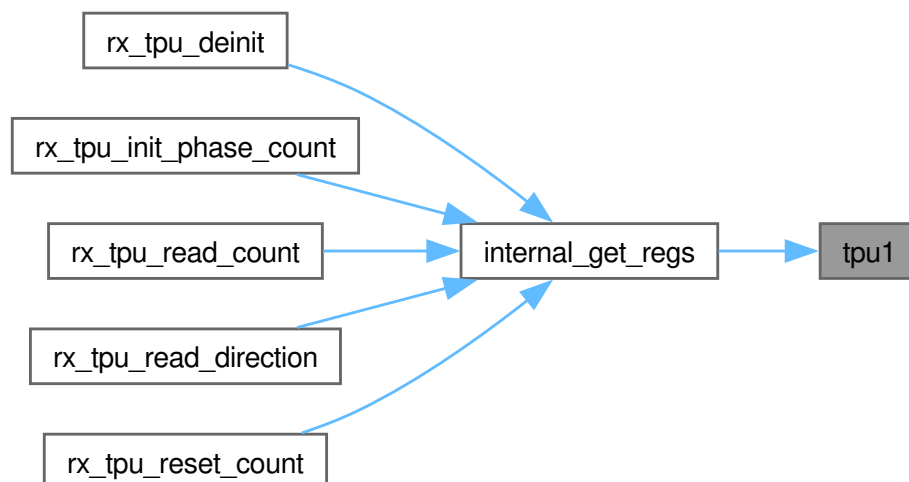
Definition at line 561 of file [rx72n_tpu_regs.h](#).

```
00562 {
00563     return (volatile rx_tpu_regs_t*)k_tpu1_base_addr;
00564 }
```

References [k_tpu1_base_addr](#).

Referenced by [internal_get_regs\(\)](#).

Here is the caller graph for this function:



8.65.4.3 tpu2()

```
volatile rx_tpu_regs_t * tpu2 (
    void )    [inline], [static]
```

Get pointer to TPU2 channel registers (basic, phase counting).

Returns a volatile pointer to TPU2 basic channel registers at address 0x00088130. TPU2 supports phase counting mode using TCLKC (A-phase) and TCLKD (B-phase) external clock pins. Used for back-left encoder.

Returns

Volatile pointer to TPU2 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

See also

[k_tpu2_base_addr](#) Base address constant

Since

Version 1.0.0

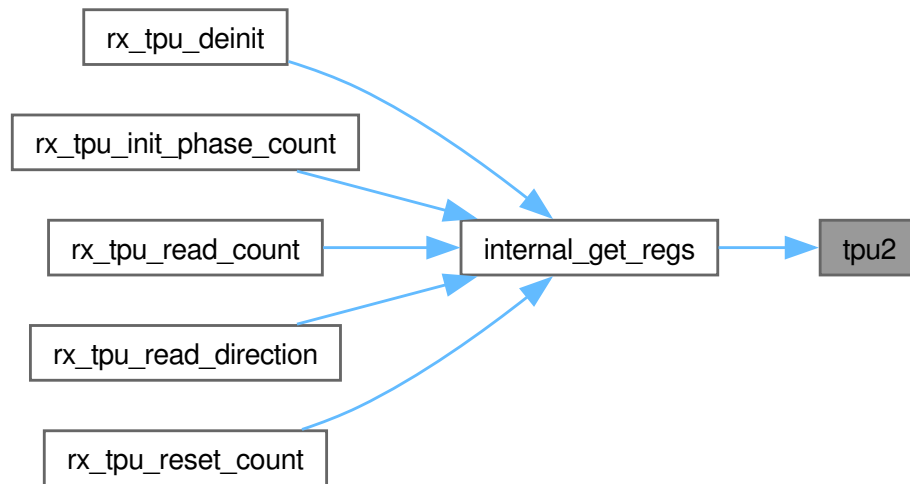
Definition at line 588 of file [rx72n_tpu_regs.h](#).

```
00589 {
00590     return (volatile rx_tpu_regs_t*)k_tpu2_base_addr;
00591 }
```

References [k_tpu2_base_addr](#).

Referenced by [internal_get_regs\(\)](#).

Here is the caller graph for this function:



8.65.4.4 tpu3()

```
volatile rx_tpu_ext_regs_t * tpu3 (
    void )    [inline], [static]
```

Get pointer to TPU3 channel registers (extended).

Returns a volatile pointer to TPU3 extended channel registers at address 0x00088140. TPU3 has TIORH/TIORL and TGRA through TGRD.

Returns

Volatile pointer to TPU3 extended register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address
TPU3 does NOT support phase counting mode

See also

[k_tpu3_base_addr](#) Base address constant

Since

Version 1.0.0

Definition at line 613 of file [rx72n_tpu_regs.h](#).

```
00614 {
00615     return (volatile rx_tpu_ext_regs_t*)k_tpu3_base_addr;
00616 }
```

References [k_tpu3_base_addr](#).

8.65.4.5 tpu4()

```
volatile rx_tpu_regs_t * tpu4 (
    void )    [inline], [static]
```

Get pointer to TPU4 channel registers (basic, phase counting).

Returns a volatile pointer to TPU4 basic channel registers at address 0x00088150. TPU4 supports phase counting mode using TCLKC/TCLKD pins (paired with TPU2).

Returns

Volatile pointer to TPU4 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

See also

[k_tpu4_base_addr](#) Base address constant

Since

Version 1.0.0

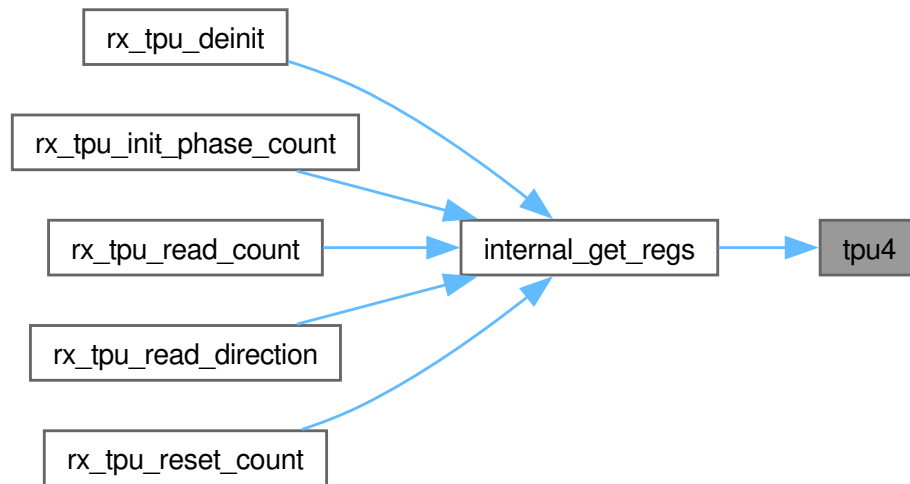
Definition at line 637 of file [rx72n_tpu_regs.h](#).

```
00638 {
00639     return (volatile rx_tpu_regs_t*)k_tpu4_base_addr;
00640 }
```

References [k_tpu4_base_addr](#).

Referenced by [internal_get_regs\(\)](#).

Here is the caller graph for this function:



8.65.4.6 tpu5()

```
volatile rx_tpu_regs_t * tpu5 (
    void ) [inline], [static]
```

Get pointer to TPU5 channel registers (basic, phase counting).

Returns a volatile pointer to TPU5 basic channel registers at address 0x00088160. TPU5 supports phase counting mode using TCLKA/TCLKB pins (paired with TPU1).

Returns

Volatile pointer to TPU5 register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

See also

[k_tpu5_base_addr](#) Base address constant

Since

Version 1.0.0

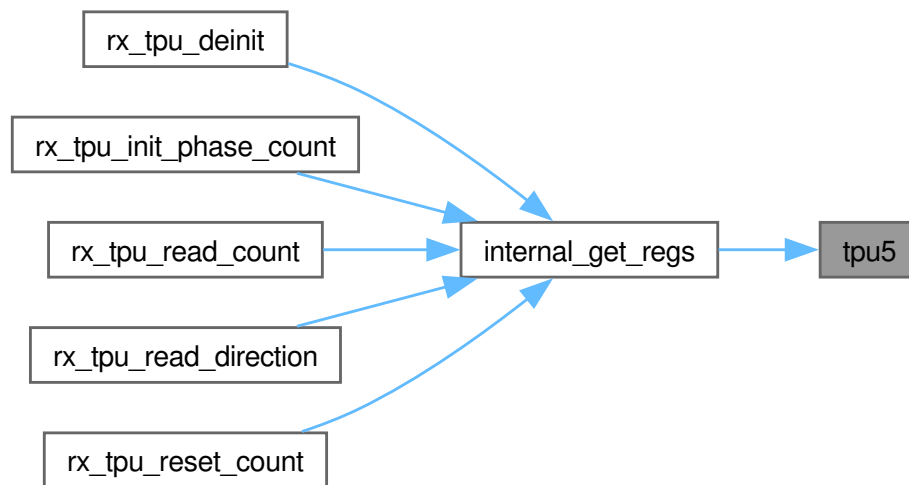
Definition at line 661 of file [rx72n_tpu_regs.h](#).

```
00662 {
00663     return (volatile rx_tpu_regs_t*)k_tpu5_base_addr;
00664 }
```

References [k_tpu5_base_addr](#).

Referenced by [internal_get_regs\(\)](#).

Here is the caller graph for this function:



8.65.4.7 tpu_control()

```
volatile rx_tpu_control_regs_t * tpu_control (
    void )    [inline], [static]
```

Get pointer to TPU control block registers.

Returns a volatile pointer to the TPU control register block at address 0x00088100. Contains TSTR (timer start), TSYR (synchronous), and NFCR (noise filter) registers shared across all channels.

Returns

Volatile pointer to TPU control register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)

Postcondition

Pointer valid for lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

Example:

```
// Start TPU1 and TPU2 counters simultaneously  
tpu_control()->tstr |= (k_tpu_tstr_cst1 | k_tpu_tstr_cst2);
```

See also

[k_tpu_control_base_addr](#) Base address constant

Since

Version 1.0.0

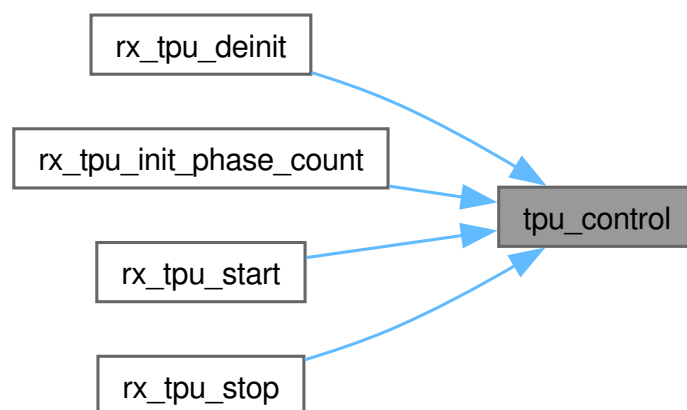
Definition at line 503 of file [rx72n_tpu_regs.h](#).

```
00504 {  
00505     return (volatile rx_tpu_control_regs_t*)k_tpu_control_base_addr;  
00506 }
```

References [k_tpu_control_base_addr](#).

Referenced by [rx_tpu_deinit\(\)](#), [rx_tpu_init_phase_count\(\)](#), [rx_tpu_start\(\)](#), and [rx_tpu_stop\(\)](#).

Here is the caller graph for this function:



8.66 rx72n`tpu`regs.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx72n_tpu_regs.h
00003  * @brief RX72N 16-Bit Timer Pulse Unit (TPU) Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the TPU module which provides 6 channels of 16-bit
00007  * timers with quadrature encoder phase counting capability. In STAR, the TPU
00008  * is used for rear wheel encoder decoding via hardware phase counting mode.
00009  *
00010  * @par STAR Project Quadrature Encoder System
00011  * @dot
00012  * digraph tpu_system {
00013  *   rankdir=LR;
00014  *   node [shape=box, style=rounded];
00015  *
00016  *   subgraph cluster_encoders {
00017  *     label="Rear Wheel Encoders";
00018  *     style=filled;
00019  *     color=lightblue;
00020  *
00021  *     enc2 [label="Encoder 2\n(Back Right)\n341 PPR"];
00022  *     enc3 [label="Encoder 3\n(Back Left)\n341 PPR"];
00023  *   }
00024  *
00025  *   subgraph cluster_pins {
00026  *     label="External Clock Pins";
00027  *     style=filled;
00028  *     color=lightyellow;
00029  *
00030  *     tclka [label="TCLKA\nPC2 (pin 70)\nA-Phase"];
00031  *     tclkb [label="TCLKB\nPA3 (pin 94)\nB-Phase"];
00032  *     tclkc [label="TCLKC\nPC0 (pin 75)\nA-Phase"];
00033  *     tclkd [label="TCLKD\nPB3 (pin 82)\nB-Phase"];
00034  *   }
00035  *
00036  *   subgraph cluster_tpu {
00037  *     label="TPU Phase Counting (0x00088100)";
00038  *     style=filled;
00039  *     color=lightgreen;
00040  *
00041  *     tpu1 [label="TPU1\nPhase Count Mode 4\n(4x resolution)"];
00042  *     tpu2 [label="TPU2\nPhase Count Mode 4\n(4x resolution)"];
00043  *   }
00044  *
00045  *   subgraph cluster_output {
00046  *     label="Motor Control";
00047  *     style=filled;
00048  *     color=lightsalmon;
00049  *
00050  *     ctrl [label="motor_control_task\n250 Hz loop"];
00051  *   }
00052  *
00053  *   enc2 -> tclka [label="A"];
00054  *   enc2 -> tclkb [label="B"];
00055  *   enc3 -> tclkc [label="A"];
00056  *   enc3 -> tclkd [label="B"];
00057  *   tclka -> tpu1;
00058  *   tclkb -> tpu1;
00059  *   tclkc -> tpu2;
00060  *   tclkd -> tpu2;
00061  *   tpu1 -> ctrl [label="TCNT + TCFD"];
00062  *   tpu2 -> ctrl [label="TCNT + TCFD"];
00063  * }
00064  * @enddot
00065  *
00066  * @par TPU Channel Assignments in STAR
00067  * | Channel | Type   | Use in STAR | Phase Count | Clock Pins |
00068  * |-----|-----|-----|-----|-----|
00069  * | TPU0    | Extended | Not used    | No          | -          |
00070  * | TPU1    | Basic   | Back-right encoder | Yes (mode 4) | TCLKA + TCLKB |
00071  * | TPU2    | Basic   | Back-left encoder  | Yes (mode 4) | TCLKC + TCLKD |
00072  * | TPU3    | Extended | Not used    | No          | -          |
00073  * | TPU4    | Basic   | Not used (paired 2) | Yes          | TCLKC + TCLKD |
00074  * | TPU5    | Basic   | Not used (paired 1) | Yes          | TCLKA + TCLKB |
00075  *
00076  * @par Phase Counting Mode 4 (4x Resolution)
00077  * Counts on all edges of both A-phase and B-phase inputs, providing 4x the
00078  * base encoder resolution. For 341 PPR encoders: 341 * 4 = 1364 counts/rev.
00079  *
00080  * | A-Phase Edge | B-Phase Level | Direction |
00081  * |-----|-----|-----|
00082  * | Rising      | Low          | Up        |

```

```

00083 * | Rising      | High      | Down      |
00084 * | Falling    | Low       | Down      |
00085 * | Falling    | High      | Up         |
00086 * | (same for B-phase edges with A-phase level swapped) |
00087 *
00088 * @par Key Features
00089 * - 6 channels of 16-bit timers (TPU0-TPU5)
00090 * - Hardware quadrature phase counting on TPU1/2/4/5
00091 * - 4 phase counting modes (1x, 2x, 3x, 4x resolution)
00092 * - TSR.TCFD direction flag for up/down detection
00093 * - Overflow/underflow interrupt support
00094 * - Noise filter on I/O pins
00095 * - Synchronous operation between channels
00096 *
00097 * @par Hardware Requirements
00098 * | Parameter      | Value          | Notes          |
00099 * |-----|-----|-----|
00100 * | Control Base   | 0x00088100     | TSTR, TSYR, NFCR registers |
00101 * | Channel Spacing| 0x10 bytes     | TPU0=0x110, TPU1=0x120, etc. |
00102 * | Module Stop    | MSTPCRA.MSTPA13 | 0=run, 1=stop |
00103 * | Clock          | PCLKB (60 MHz) | For noise filter and prescaler |
00104 *
00105 * @par Register Map Overview
00106 * | Address      | Register      | Size | Description          |
00107 * |-----|-----|-----|-----|
00108 * | 0x00088100   | TSTR          | 8    | Timer Start Register |
00109 * | 0x00088101   | TSYR          | 8    | Timer Synchronous Register |
00110 * | 0x00088108-0D | NFCRn        | 8x6  | Noise Filter Control (0-5) |
00111 * | 0x00088110-1F | TPU0 registers | 16   | Extended channel (4 TGR) |
00112 * | 0x00088120-2F | TPU1 registers | 16   | Basic channel (2 TGR) |
00113 * | 0x00088130-3F | TPU2 registers | 16   | Basic channel (2 TGR) |
00114 * | 0x00088140-4F | TPU3 registers | 16   | Extended channel (4 TGR) |
00115 * | 0x00088150-5F | TPU4 registers | 16   | Basic channel (2 TGR) |
00116 * | 0x00088160-6F | TPU5 registers | 16   | Basic channel (2 TGR) |
00117 *
00118 * @par NASA Power of 10 Compliance
00119 * - Rule 1: No goto, setjmp, or recursion
00120 * - Rule 2: N/A (no loops in register definitions)
00121 * - Rule 3: No dynamic memory allocation
00122 * - Rule 4: All accessor functions are single-statement
00123 * - Rule 5: N/A (hardware layer)
00124 * - Rule 6: Minimal scope - inline accessors only
00125 * - Rule 7: N/A (no return values to check)
00126 * - Rule 8: All constants use C23 typed enums
00127 * - Rule 9: No function pointers
00128 * - Rule 10: Compiles with -Wall -Wextra -Werror
00129 *
00130 * @par SOLID Principles
00131 * - **S**: Single responsibility - only TPU register definitions
00132 * - **O**: Open for extension via additional channel configurations
00133 * - **I**: N/A (no inheritance hierarchy)
00134 * - **I**: Minimal interface - only necessary accessor functions
00135 * - **D**: N/A (hardware register layer)
00136 *
00137 * @par Module Dependencies
00138 * - <stdint.h> - Fixed-width integer types
00139 * - <stddef.h> - offsetof macro for static assertions
00140 *
00141 * @par Manual References
00142 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00143 * - Chapter 28: 16-Bit Timer Pulse Unit (TPUa), pages 1442-1508
00144 *
00145 * @see rx72n_mtu_regs.h MTU registers (front wheel encoders)
00146 * @see rx72n_regs.h Main register include file
00147 *
00148 * @author Locked, Inc.
00149 * @date 2026-02-10
00150 * @copyright Copyright (c) 2026 Locked Inc.
00151 * SPDX-License-Identifier: MIT
00152 * @since Version 1.0.0
00153 */
00154
00155 #pragma once
00156
00157 #include <stddef.h>
00158 #include <stdint.h>
00159
00160 #ifdef __cplusplus
00161 extern "C" {
00162 #endif
00163
00164 /*
=====
00165 * TPU - 16-Bit Timer Pulse Unit for Quadrature Encoder Phase Counting
00166 *
=====
00167 */

```

```

00168
00169 /**
00170  * @enum rx_tpu_addresses_t
00171  * @brief TPU base addresses
00172  *
00173  * @details
00174  * Base addresses for the TPU control block and individual channel register
00175  * blocks. The control block contains TSTR, TSyr, and NfCR registers shared
00176  * across all channels. Each channel has a 16-byte register block.
00177  *
00178  * @par Memory Map
00179  * | Address | Block | Description |
00180  * |-----|-----|-----|
00181  * | 0x00088100 | Control | TSTR, TSyr, NfCR (shared across channels) |
00182  * | 0x00088110 | TPU0 | Extended channel (TIORH/L, TGRA-TGRD) |
00183  * | 0x00088120 | TPU1 | Basic channel (TIOR, TGRA-TGRB) |
00184  * | 0x00088130 | TPU2 | Basic channel (TIOR, TGRA-TGRB) |
00185  * | 0x00088140 | TPU3 | Extended channel (TIORH/L, TGRA-TGRD) |
00186  * | 0x00088150 | TPU4 | Basic channel (TIOR, TGRA-TGRB) |
00187  * | 0x00088160 | TPU5 | Basic channel (TIOR, TGRA-TGRB) |
00188  *
00189  * @note Manual: Ch28.2 - Register Descriptions
00190  * @see tpu_control() Accessor for control block
00191  * @see tpu1() Accessor for TPU1 channel (back-right encoder)
00192  * @since Version 1.0.0
00193  */
00194 typedef enum : uintptr_t {
00195 /**
00196  * @brief TPU control block base address (0x00088100)
00197  * @details Contains TSTR, TSyr at +0x00/+0x01 and NfCR[0-5] at +0x08
00198  */
00199 k_tpu_control_base_addr = 0x00088100,
00200
00201 /**
00202  * @brief TPU0 channel base address (0x00088110) - Extended channel
00203  * @details Extended channel with TIORH/TIOrL and TGRA through TGRD
00204  */
00205 k_tpu0_base_addr = 0x00088110,
00206
00207 /**
00208  * @brief TPU1 channel base address (0x00088120) - Basic channel
00209  * @details Back-right encoder, phase counting capable (TCLKA/TCLKB)
00210  */
00211 k_tpu1_base_addr = 0x00088120,
00212
00213 /**
00214  * @brief TPU2 channel base address (0x00088130) - Basic channel
00215  * @details Back-left encoder, phase counting capable (TCLKC/TCLKD)
00216  */
00217 k_tpu2_base_addr = 0x00088130,
00218
00219 /**
00220  * @brief TPU3 channel base address (0x00088140) - Extended channel
00221  * @details Extended channel with TIORH/TIOrL and TGRA through TGRD
00222  */
00223 k_tpu3_base_addr = 0x00088140,
00224
00225 /**
00226  * @brief TPU4 channel base address (0x00088150) - Basic channel
00227  * @details Phase counting capable, paired with TPU2 (TCLKC/TCLKD)
00228  */
00229 k_tpu4_base_addr = 0x00088150,
00230
00231 /**
00232  * @brief TPU5 channel base address (0x00088160) - Basic channel
00233  * @details Phase counting capable, paired with TPU1 (TCLKA/TCLKB)
00234  */
00235 k_tpu5_base_addr = 0x00088160,
00236
00237 /**
00238  * @brief Channel register block spacing (0x10 bytes between channels)
00239  * @details Each TPU channel occupies a 16-byte register block
00240  */
00241 k_tpu_channel_spacing = 0x10,
00242 } rx_tpu_addresses_t;
00243
00244 /**
=====
00245  * Control Block Registers (TSTR, TSyr, NfCR)
00246  *
=====
00247  */
00248
00249 /**
00250  * @enum rx_tpu_channel_count_t
00251  * @brief TPU channel count constant
00252  * @since Version 1.0.0

```

```

00253 */
00254 typedef enum : uint8_t {
00255     k_tpu_channel_count = 6, /**< Total TPU channels (TPU0-TPU5) */
00256 } rx_tpu_channel_count_t;
00257
00258 /**
00259  * @struct rx_tpu_control_regs_t
00260  * @brief TPU Control Block Register Structure
00261  *
00262  * @details
00263  * Contains the shared control registers for all TPU channels: the Timer
00264  * Start Register (TSTR), Timer Synchronous Register (TSYR), and per-channel
00265  * Noise Filter Control Registers (NFCR).
00266  *
00267  * @par Memory Layout Table
00268  * | Offset | Size | Field | Type | Description |
00269  * |-----|-----|-----|-----|-----|
00270  * | 0x00 | 1 | tstr | uint8_t | Timer Start Register |
00271  * | 0x01 | 1 | tsyr | uint8_t | Timer Synchronous Register |
00272  * | 0x02 | 6 | reserved | uint8_t[] | Reserved |
00273  * | 0x08 | 6 | nfc[0..5] | uint8_t[] | Noise Filter Control (per ch) |
00274  * | **Total** | **14** | | | |
00275  *
00276  * @invariant sizeof(rx_tpu_control_regs_t) == 14 bytes
00277  *
00278  * @see tpu_control() Accessor function
00279  * @see rx_tpu_tstr_bits_t TSTR bit definitions
00280  * @see rx_tpu_tsyr_bits_t TSYR bit definitions
00281  * @since Version 1.0.0
00282  */
00283 typedef struct {
00284     /**
00285      * @brief Timer Start Register (TSTR) @ offset 0x00
00286      *
00287      * @details
00288      * Starts or stops TCNT operation for TPU0 through TPU5. Each bit
00289      * independently controls one channel's counter.
00290      *
00291      * @par Bit Layout (8-bit)
00292      * | Bits | Field | R/W | Description |
00293      * |-----|-----|-----|-----|
00294      * | 0 | CST0 | R/W | TPU0 counter start (1=run) |
00295      * | 1 | CST1 | R/W | TPU1 counter start (1=run) |
00296      * | 2 | CST2 | R/W | TPU2 counter start (1=run) |
00297      * | 3 | CST3 | R/W | TPU3 counter start (1=run) |
00298      * | 4 | CST4 | R/W | TPU4 counter start (1=run) |
00299      * | 5 | CST5 | R/W | TPU5 counter start (1=run) |
00300      * | 7:6 | - | R/W | Reserved (write 0) |
00301      *
00302      * @see rx_tpu_tstr_bits_t Bit definitions
00303      */
00304     volatile uint8_t tstr;
00305
00306     /**
00307      * @brief Timer Synchronous Register (TSYR) @ offset 0x01
00308      *
00309      * @details
00310      * Selects independent or synchronous operation for TCNT of TPU0
00311      * through TPU5. When synchronous, channels share counter writes.
00312      *
00313      * @par Bit Layout (8-bit)
00314      * | Bits | Field | R/W | Description |
00315      * |-----|-----|-----|-----|
00316      * | 0 | SYNC0 | R/W | TPU0 synchronous (1=enabled) |
00317      * | 1 | SYNC1 | R/W | TPU1 synchronous (1=enabled) |
00318      * | 2 | SYNC2 | R/W | TPU2 synchronous (1=enabled) |
00319      * | 3 | SYNC3 | R/W | TPU3 synchronous (1=enabled) |
00320      * | 4 | SYNC4 | R/W | TPU4 synchronous (1=enabled) |
00321      * | 5 | SYNC5 | R/W | TPU5 synchronous (1=enabled) |
00322      * | 7:6 | - | R/W | Reserved (write 0) |
00323      *
00324      * @see rx_tpu_tsyr_bits_t Bit definitions
00325      */
00326     volatile uint8_t tsyr;
00327
00328     /**
00329      * @brief Reserved bytes @ offset 0x02-0x07
00330      */
00331     volatile uint8_t reserved[6];
00332
00333     /**
00334      * @brief Noise Filter Control Registers (NFCR) @ offset 0x08-0x0D
00335      *
00336      * @details
00337      * Per-channel noise filter configuration. Index 0 = TPU0.NFCR through
00338      * index 5 = TPU5.NFCR. Set only while TCNT is stopped.
00339      */

```

```

00340 * @par Bit Layout (8-bit, per channel)
00341 * | Bits | Field | R/W | Description |
00342 * |-----|-----|-----|-----|
00343 * | 0 | NFAEN | R/W | Noise filter enable TIOCA pin |
00344 * | 1 | NFBEN | R/W | Noise filter enable TIOCB pin |
00345 * | 2 | NFCEN | R/W | Noise filter enable TIOCC (0,3) |
00346 * | 3 | NFDEN | R/W | Noise filter enable TIOCD (0,3) |
00347 * | 5:4 | NFCS[1:0] | R/W | Noise filter clock select |
00348 * | 7:6 | - | R | Reserved (read as 0) |
00349 *
00350 * @warning Set NFCR only while TCNT is stopped (CSTn = 0)
00351 * @see rx_tpu_nfc_bits_t Bit definitions
00352 */
00353 volatile uint8_t nfc[6];
00354 } rx_tpu_control_regs_t;
00355
00356 /*
=====
00357 * Per-Channel Register Structures
00358 *
=====
00359 */
00360
00361 /**
00362 * @struct rx_tpu_ext_regs_t
00363 * @brief TPU Extended Channel Register Structure (TPU0 and TPU3)
00364 *
00365 * @details
00366 * Register layout for TPU0 and TPU3 which have two I/O control registers
00367 * (TIORH + TIORL) and four timer general registers (TGRA through TGRD).
00368 * These channels do NOT support phase counting mode.
00369 *
00370 * @par Memory Layout Table
00371 * | Offset | Size | Field | Type | Description |
00372 * |-----|-----|-----|-----|-----|
00373 * | 0x00 | 1 | tcr | uint8_t | Timer Control Register |
00374 * | 0x01 | 1 | tmdr | uint8_t | Timer Mode Register |
00375 * | 0x02 | 1 | tiorh | uint8_t | Timer I/O Control Register H |
00376 * | 0x03 | 1 | tiorl | uint8_t | Timer I/O Control Register L |
00377 * | 0x04 | 1 | tier | uint8_t | Timer Interrupt Enable Register |
00378 * | 0x05 | 1 | tsr | uint8_t | Timer Status Register |
00379 * | 0x06 | 2 | tcnt | uint16_t | Timer Counter |
00380 * | 0x08 | 2 | tgra | uint16_t | Timer General Register A |
00381 * | 0x0A | 2 | tgrb | uint16_t | Timer General Register B |
00382 * | 0x0C | 2 | tgrc | uint16_t | Timer General Register C |
00383 * | 0x0E | 2 | tgrd | uint16_t | Timer General Register D |
00384 * | **Total** | **16** | | |
00385 *
00386 * @invariant sizeof(rx_tpu_ext_regs_t) == 16 bytes
00387 *
00388 * @note TPU0 and TPU3 do NOT support phase counting mode
00389 * @note Buffer operation available: TGRA+TGRC and TGRB+TGRD pairs
00390 *
00391 * @see tpu0() Accessor for TPU0
00392 * @see tpu3() Accessor for TPU3
00393 * @since Version 1.0.0
00394 */
00395 typedef struct {
00396 volatile uint8_t tcr; /**< Timer Control Register (TCR) @ +0x00 */
00397 volatile uint8_t tmdr; /**< Timer Mode Register (TMDR) @ +0x01 */
00398 volatile uint8_t tiorh; /**< Timer I/O Control Register H (TIORH) @ +0x02 */
00399 volatile uint8_t tiorl; /**< Timer I/O Control Register L (TIORL) @ +0x03 */
00400 volatile uint8_t tier; /**< Timer Interrupt Enable Register (TIER) @ +0x04 */
00401 volatile uint8_t tsr; /**< Timer Status Register (TSR) @ +0x05 */
00402 volatile uint16_t tcnt; /**< Timer Counter (TCNT) @ +0x06, 16-bit */
00403 volatile uint16_t tgra; /**< Timer General Register A (TGRA) @ +0x08 */
00404 volatile uint16_t tgrb; /**< Timer General Register B (TGRB) @ +0x0A */
00405 volatile uint16_t tgrc; /**< Timer General Register C (TGRC) @ +0x0C */
00406 volatile uint16_t tgrd; /**< Timer General Register D (TGRD) @ +0x0E */
00407 } rx_tpu_ext_regs_t;
00408
00409 /**
00410 * @struct rx_tpu_regs_t
00411 * @brief TPU Basic Channel Register Structure (TPU1, TPU2, TPU4, TPU5)
00412 *
00413 * @details
00414 * Register layout for TPU1/2/4/5 which have a single I/O control register
00415 * (TIOR) and two timer general registers (TGRA and TGRB). These channels
00416 * support phase counting mode for quadrature encoder decoding.
00417 *
00418 * @par Memory Layout Table
00419 * | Offset | Size | Field | Type | Description |
00420 * |-----|-----|-----|-----|-----|
00421 * | 0x00 | 1 | tcr | uint8_t | Timer Control Register |
00422 * | 0x01 | 1 | tmdr | uint8_t | Timer Mode Register |
00423 * | 0x02 | 1 | tior | uint8_t | Timer I/O Control Register |
00424 * | 0x03 | 1 | reserved | uint8_t | Reserved |

```



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00425 * | 0x04 | 1 | tier | uint8_t | Timer Interrupt Enable Register |
00426 * | 0x05 | 1 | tsr | uint8_t | Timer Status Register |
00427 * | 0x06 | 2 | tcnt | uint16_t | Timer Counter (up/down in phase) |
00428 * | 0x08 | 2 | tgra | uint16_t | Timer General Register A |
00429 * | 0x0A | 2 | tgrb | uint16_t | Timer General Register B |
00430 * | 0x0C | 4 | reserved2 | uint8_t[] | Reserved |
00431 * | **Total** | **16** | | | |
00432 *
00433 * @invariant sizeof(rx_tpu_regs_t) == 16 bytes
00434 *
00435 * @par Phase Counting Mode Usage
00436 * @code{.c}
00437 * // Configure TPU1 for phase counting mode 4 (4x resolution)
00438 * volatile rx_tpu_regs_t* ch = tpu1();
00439 *
00440 * // 1. Stop counter first
00441 * tpu_control()->tstr &= (uint8_t)~k_tpu_tstr_cst1;
00442 *
00443 * // 2. Set phase counting mode 4 in TMDR
00444 * ch->tmdr = k_tpu_tmdr_md_phase_count_4;
00445 *
00446 * // 3. Start counter
00447 * tpu_control()->tstr |= k_tpu_tstr_cst1;
00448 *
00449 * // 4. Read counter and direction
00450 * uint16_t count = ch->tcnt;
00451 * bool counting_up = (ch->tsr & k_tpu_tsr_tcf) != 0;
00452 * @endcode
00453 *
00454 * @see tpu1() Accessor for back-right encoder
00455 * @see tpu2() Accessor for back-left encoder
00456 * @since Version 1.0.0
00457 */
00458 typedef struct {
00459     volatile uint8_t tcr; /**< Timer Control Register (TCR) @ +0x00 */
00460     volatile uint8_t tmdr; /**< Timer Mode Register (TMDR) @ +0x01 */
00461     volatile uint8_t tior; /**< Timer I/O Control Register (TIOR) @ +0x02 */
00462     volatile uint8_t reserved; /**< Reserved @ +0x03 */
00463     volatile uint8_t tier; /**< Timer Interrupt Enable Register (TIER) @ +0x04 */
00464     volatile uint8_t tsr; /**< Timer Status Register (TSR) @ +0x05 */
00465     volatile uint16_t tcnt; /**< Timer Counter (TCNT) @ +0x06, up/down in phase mode */
00466     volatile uint16_t tgra; /**< Timer General Register A (TGRA) @ +0x08 */
00467     volatile uint16_t tgrb; /**< Timer General Register B (TGRB) @ +0x0A */
00468     volatile uint8_t reserved2[4]; /**< Reserved @ +0x0C-0x0F */
00469 } rx_tpu_regs_t;
00470
00471 /*
=====
00472 * Inline Accessor Functions
00473 *
=====
00474 */
00475
00476 /**
00477 * @brief Get pointer to TPU control block registers
00478 *
00479 * @details
00480 * Returns a volatile pointer to the TPU control register block at address
00481 * 0x00088100. Contains TSTR (timer start), TSYSR (synchronous), and
00482 * NFCR (noise filter) registers shared across all channels.
00483 *
00484 * @return Volatile pointer to TPU control register structure
00485 * @retval Non-NULL Always returns valid pointer (hardware address)
00486 *
00487 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00488 * @post Pointer valid for lifetime of program execution
00489 *
00490 * @note Thread Safety: Safe - returns constant hardware address
00491 *
00492 * @par Example:
00493 * @code{.c}
00494 * // Start TPU1 and TPU2 counters simultaneously
00495 * tpu_control()->tstr |= (k_tpu_tstr_cst1 | k_tpu_tstr_cst2);
00496 * @endcode
00497 *
00498 * @see k_tpu_control_base_addr Base address constant
00499 * @since Version 1.0.0
00500 */
00501 /* GCOVR_EXCL_START -- trivial hardware-address accessor; host-test
00502 * builds use mock_rx72n_tpu_regs.h's override. */
00503 static inline volatile rx_tpu_control_regs_t* tpu_control(void)
00504 {
00505     return (volatile rx_tpu_control_regs_t*)k_tpu_control_base_addr;
00506 }
00507 /* GCOVR_EXCL_STOP */
00508
00509 /**

```

```

00510 * @brief Get pointer to TPU0 channel registers (extended)
00511 *
00512 * @details
00513 * Returns a volatile pointer to TPU0 extended channel registers at address
00514 * 0x00088110. TPU0 has TIORH/TIOL and TGRA through TGRD.
00515 *
00516 * @return Volatile pointer to TPU0 extended register structure
00517 * @retval Non-NULL Always returns valid pointer (hardware address)
00518 *
00519 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00520 * @post Pointer valid for lifetime of program execution
00521 *
00522 * @note Thread Safety: Safe - returns constant hardware address
00523 * @note TPU0 does NOT support phase counting mode
00524 *
00525 * @see k_tpu0_base_addr Base address constant
00526 * @since Version 1.0.0
00527 */
00528 static inline volatile rx_tpu_ext_regs_t* tpu0(void)
00529 {
00530     return (volatile rx_tpu_ext_regs_t*)k_tpu0_base_addr;
00531 }
00532
00533 /**
00534 * @brief Get pointer to TPU1 channel registers (basic, phase counting)
00535 *
00536 * @details
00537 * Returns a volatile pointer to TPU1 basic channel registers at address
00538 * 0x00088120. TPU1 supports phase counting mode using TCLKA (A-phase)
00539 * and TCLKB (B-phase) external clock pins. Used for back-right encoder.
00540 *
00541 * @return Volatile pointer to TPU1 register structure
00542 * @retval Non-NULL Always returns valid pointer (hardware address)
00543 *
00544 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00545 * @post Pointer valid for lifetime of program execution
00546 *
00547 * @note Thread Safety: Safe - returns constant hardware address
00548 *
00549 * @par Example:
00550 * @code{.c}
00551 * // Read back-right encoder count and direction
00552 * uint16_t count = tpu1()->tcnt;
00553 * bool forward = (tpu1()->tsr & k_tpu_tsr_tcfld) != 0;
00554 * @endcode
00555 *
00556 * @see k_tpu1_base_addr Base address constant
00557 * @since Version 1.0.0
00558 */
00559 /* GCOVR_EXCL_START -- trivial hardware-address accessor; host-test
00560 * builds use mock_rx72n_tpu_regs.h's override. */
00561 static inline volatile rx_tpu_regs_t* tpu1(void)
00562 {
00563     return (volatile rx_tpu_regs_t*)k_tpu1_base_addr;
00564 }
00565 /* GCOVR_EXCL_STOP */
00566
00567 /**
00568 * @brief Get pointer to TPU2 channel registers (basic, phase counting)
00569 *
00570 * @details
00571 * Returns a volatile pointer to TPU2 basic channel registers at address
00572 * 0x00088130. TPU2 supports phase counting mode using TCLKC (A-phase)
00573 * and TCLKD (B-phase) external clock pins. Used for back-left encoder.
00574 *
00575 * @return Volatile pointer to TPU2 register structure
00576 * @retval Non-NULL Always returns valid pointer (hardware address)
00577 *
00578 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00579 * @post Pointer valid for lifetime of program execution
00580 *
00581 * @note Thread Safety: Safe - returns constant hardware address
00582 *
00583 * @see k_tpu2_base_addr Base address constant
00584 * @since Version 1.0.0
00585 */
00586 /* GCOVR_EXCL_START -- trivial hardware-address accessor; host-test
00587 * builds use mock_rx72n_tpu_regs.h's override. */
00588 static inline volatile rx_tpu_regs_t* tpu2(void)
00589 {
00590     return (volatile rx_tpu_regs_t*)k_tpu2_base_addr;
00591 }
00592 /* GCOVR_EXCL_STOP */
00593
00594 /**
00595 * @brief Get pointer to TPU3 channel registers (extended)
00596 *

```

```

00597 * @details
00598 * Returns a volatile pointer to TPU3 extended channel registers at address
00599 * 0x00088140. TPU3 has TIORH/TIORL and TGRA through TGRD.
00600 *
00601 * @return Volatile pointer to TPU3 extended register structure
00602 * @retval Non-NULL Always returns valid pointer (hardware address)
00603 *
00604 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00605 * @post Pointer valid for lifetime of program execution
00606 *
00607 * @note Thread Safety: Safe - returns constant hardware address
00608 * @note TPU3 does NOT support phase counting mode
00609 *
00610 * @see k_tpu3_base_addr Base address constant
00611 * @since Version 1.0.0
00612 */
00613 static inline volatile rx_tpu_ext_regs_t* tpu3(void)
00614 {
00615     return (volatile rx_tpu_ext_regs_t*)k_tpu3_base_addr;
00616 }
00617
00618 /**
00619 * @brief Get pointer to TPU4 channel registers (basic, phase counting)
00620 *
00621 * @details
00622 * Returns a volatile pointer to TPU4 basic channel registers at address
00623 * 0x00088150. TPU4 supports phase counting mode using TCLKC/TCLKD pins
00624 * (paired with TPU2).
00625 *
00626 * @return Volatile pointer to TPU4 register structure
00627 * @retval Non-NULL Always returns valid pointer (hardware address)
00628 *
00629 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00630 * @post Pointer valid for lifetime of program execution
00631 *
00632 * @note Thread Safety: Safe - returns constant hardware address
00633 *
00634 * @see k_tpu4_base_addr Base address constant
00635 * @since Version 1.0.0
00636 */
00637 static inline volatile rx_tpu_regs_t* tpu4(void)
00638 {
00639     return (volatile rx_tpu_regs_t*)k_tpu4_base_addr;
00640 }
00641
00642 /**
00643 * @brief Get pointer to TPU5 channel registers (basic, phase counting)
00644 *
00645 * @details
00646 * Returns a volatile pointer to TPU5 basic channel registers at address
00647 * 0x00088160. TPU5 supports phase counting mode using TCLKA/TCLKB pins
00648 * (paired with TPU1).
00649 *
00650 * @return Volatile pointer to TPU5 register structure
00651 * @retval Non-NULL Always returns valid pointer (hardware address)
00652 *
00653 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00654 * @post Pointer valid for lifetime of program execution
00655 *
00656 * @note Thread Safety: Safe - returns constant hardware address
00657 *
00658 * @see k_tpu5_base_addr Base address constant
00659 * @since Version 1.0.0
00660 */
00661 static inline volatile rx_tpu_regs_t* tpu5(void)
00662 {
00663     return (volatile rx_tpu_regs_t*)k_tpu5_base_addr;
00664 }
00665
00666 /*
=====
00667 * TSTR - Timer Start Register Bit Definitions
00668 *
=====
00669 */
00670
00671 /**
00672 * @enum rx_tpu_tstr_bits_t
00673 * @brief TSTR (Timer Start Register) bit definitions
00674 *
00675 * @details
00676 * Each CSTn bit independently starts (1) or stops (0) the corresponding
00677 * TPU channel counter. Multiple channels can be started simultaneously
00678 * by OR-ing the appropriate bits.
00679 *
00680 * @note Manual: Ch28.2.8 - Timer Start Register (TSTR)
00681 * @see rx_tpu_control_regs_t::tstr Register field

```

```

00682 * @since Version 1.0.0
00683 */
00684 typedef enum : uint8_t {
00685     k_tpu_tstr_cst0 = (1U « 0), /**< TPU0 counter start (1=run, 0=stop) */
00686     k_tpu_tstr_cst1 = (1U « 1), /**< TPU1 counter start (back-right encoder) */
00687     k_tpu_tstr_cst2 = (1U « 2), /**< TPU2 counter start (back-left encoder) */
00688     k_tpu_tstr_cst3 = (1U « 3), /**< TPU3 counter start */
00689     k_tpu_tstr_cst4 = (1U « 4), /**< TPU4 counter start */
00690     k_tpu_tstr_cst5 = (1U « 5), /**< TPU5 counter start */
00691 } rx_tpu_tstr_bits_t;
00692
00693 /**
00694 * @enum rx_tpu_tsyrr_bits_t
00695 * @brief TSRR (Timer Synchronous Register) bit definitions
00696 *
00697 * @details
00698 * Each SYNCn bit enables (1) or disables (0) synchronous operation for the
00699 * corresponding TPU channel. When multiple channels are set to synchronous,
00700 * their TCNTs can be written simultaneously.
00701 *
00702 * @note Manual: Ch28.2.9 - Timer Synchronous Register (TSRR)
00703 * @see rx_tpu_control_regs_t::tsyr Register field
00704 * @since Version 1.0.0
00705 */
00706 typedef enum : uint8_t {
00707     k_tpu_tsyrr_sync0 = (1U « 0), /**< TPU0 synchronous operation enable */
00708     k_tpu_tsyrr_sync1 = (1U « 1), /**< TPU1 synchronous operation enable */
00709     k_tpu_tsyrr_sync2 = (1U « 2), /**< TPU2 synchronous operation enable */
00710     k_tpu_tsyrr_sync3 = (1U « 3), /**< TPU3 synchronous operation enable */
00711     k_tpu_tsyrr_sync4 = (1U « 4), /**< TPU4 synchronous operation enable */
00712     k_tpu_tsyrr_sync5 = (1U « 5), /**< TPU5 synchronous operation enable */
00713 } rx_tpu_tsyrr_bits_t;
00714
00715 /*
=====
00716 * TCR - Timer Control Register Bit Definitions
00717 *
=====
00718 */
00719 /**
00720 * @enum rx_tpu_tcr_shift_t
00721 * @brief TCR (Timer Control Register) bit field positions
00722 *
00723 * @details
00724 * Bit positions for the Timer Control Register fields: prescaler select,
00725 * clock edge select, and counter clear source select.
00726 *
00727 * @note In phase counting mode, TPSC and CKEG settings are ignored
00728 * @see rx_tpu_tcr_cclr_t Counter clear source values
00729 * @since Version 1.0.0
00730 */
00731 /*
00732 typedef enum : uint8_t {
00733     k_tpu_tcr_tpssc_shift = 0, /**< TPSC[2:0] - Timer Prescaler Select (bits 2:0) */
00734     k_tpu_tcr_ckege_shift = 3, /**< CKEG[1:0] - Input Clock Edge Select (bits 4:3) */
00735     k_tpu_tcr_cclr_shift = 5, /**< CCLR[2:0] - Counter Clear Source (bits 7:5) */
00736 } rx_tpu_tcr_shift_t;
00737
00738 /**
00739 * @enum rx_tpu_tcr_cclr_t
00740 * @brief TCR CCLR[2:0] counter clear source values (for basic channels)
00741 *
00742 * @details
00743 * Selects the event that clears TCNT. In phase counting mode, only TGRA
00744 * and TGRB compare match clearing is available.
00745 *
00746 * @note Bit 7 is reserved for TPU1/2/4/5 (basic channels)
00747 * @see rx_tpu_tcr_shift_t::k_tpu_tcr_cclr_shift Field position
00748 * @since Version 1.0.0
00749 */
00750 typedef enum : uint8_t {
00751     k_tpu_tcr_cclr_disabled = (0U « 5), /**< TCNT free-running (no clear) */
00752     k_tpu_tcr_cclr_tgra_match = (1U « 5), /**< Clear on TGRA compare match */
00753     k_tpu_tcr_cclr_tgrb_match = (2U « 5), /**< Clear on TGRB compare match */
00754     k_tpu_tcr_cclr_sync_clear = (3U « 5), /**< Synchronous clear */
00755 } rx_tpu_tcr_cclr_t;
00756
00757 /**
00758 * @enum rx_tpu_tcr_masks_t
00759 * @brief TCR register field masks
00760 * @since Version 1.0.0
00761 */
00762 typedef enum : uint8_t {
00763     k_tpu_tcr_tpssc_mask = 0x07, /**< TPSC[2:0] field mask (bits 2:0) */
00764     k_tpu_tcr_ckege_mask = 0x18, /**< CKEG[1:0] field mask (bits 4:3) */
00765     k_tpu_tcr_cclr_mask = 0xE0, /**< CCLR[2:0] field mask (bits 7:5) */
00766 } rx_tpu_tcr_masks_t;

```

```

00767
00768 /*
=====
00769 * TMDR - Timer Mode Register Bit Definitions
00770 *
=====
00771 */
00772
00773 /**
00774 * @enum rx_tpu_tmdr_md_t
00775 * @brief TMDR MD[3:0] mode select values
00776 *
00777 * @details
00778 * Selects the operating mode for the TPU channel. Phase counting modes
00779 * (1-4) are only available on TPU1, TPU2, TPU4, and TPU5.
00780 *
00781 * @par Phase Counting Mode Comparison
00782 * | Mode | Resolution | Count Edges |
00783 * |-----|-----|-----|
00784 * | 1 | 1x | A-phase rising only |
00785 * | 2 | 1x | B-phase rising only |
00786 * | 3 | 2x | A-phase rising + falling |
00787 * | 4 | 4x | All edges of both A and B phases |
00788 *
00789 * @note Phase counting mode cannot be set for TPU0 and TPU3
00790 * @note Manual: Ch28.2.2 - Timer Mode Register (TMDR)
00791 * @since Version 1.0.0
00792 */
00793 typedef enum : uint8_t {
00794     k_tpu_tmdr_md_normal = 0x00, /**< Normal operation */
00795     k_tpu_tmdr_md_pwm1 = 0x02, /**< PWM mode 1 */
00796     k_tpu_tmdr_md_pwm2 = 0x03, /**< PWM mode 2 */
00797     k_tpu_tmdr_md_phase_count_1 = 0x04, /**< Phase counting mode 1 (1x) */
00798     k_tpu_tmdr_md_phase_count_2 = 0x05, /**< Phase counting mode 2 (1x) */
00799     k_tpu_tmdr_md_phase_count_3 = 0x06, /**< Phase counting mode 3 (2x) */
00800     k_tpu_tmdr_md_phase_count_4 = 0x07, /**< Phase counting mode 4 (4x) - STAR default */
00801 } rx_tpu_tmdr_md_t;
00802
00803 /**
00804 * @enum rx_tpu_tmdr_shift_t
00805 * @brief TMDR bit field positions
00806 * @since Version 1.0.0
00807 */
00808 typedef enum : uint8_t {
00809     k_tpu_tmdr_md_shift = 0, /**< MD[3:0] - Mode Select (bits 3:0) */
00810     k_tpu_tmdr_bfa_shift = 4, /**< BFA - Buffer Operation A (bit 4, TPU0/3 only) */
00811     k_tpu_tmdr_bfb_shift = 5, /**< BFB - Buffer Operation B (bit 5, TPU0/3 only) */
00812     k_tpu_tmdr_icselb_shift = 6, /**< ICSELB - TGRB input capture select (bit 6) */
00813     k_tpu_tmdr_icseld_shift = 7, /**< ICSELD - TGRD input capture select (bit 7, TPU0/3) */
00814 } rx_tpu_tmdr_shift_t;
00815
00816 /**
00817 * @enum rx_tpu_tmdr_masks_t
00818 * @brief TMDR register field masks
00819 * @since Version 1.0.0
00820 */
00821 typedef enum : uint8_t {
00822     k_tpu_tmdr_md_mask = 0x0F, /**< MD[3:0] field mask (bits 3:0) */
00823 } rx_tpu_tmdr_masks_t;
00824
00825 /*
=====
00826 * TIER - Timer Interrupt Enable Register Bit Definitions
00827 *
=====
00828 */
00829
00830 /**
00831 * @enum rx_tpu_tier_bits_t
00832 * @brief TIER (Timer Interrupt Enable Register) bit definitions
00833 *
00834 * @details
00835 * Controls interrupt generation for timer compare match, overflow, and
00836 * underflow events. In phase counting mode, TCIEV (overflow) and TCIEU
00837 * (underflow) are the most relevant interrupts.
00838 *
00839 * @par Bit Layout (8-bit)
00840 * | Bit | Field | Description |
00841 * |-----|-----|-----|
00842 * | 0 | TGIEA | TGRA compare match/input capture IRQ |
00843 * | 1 | TGIEB | TGRB compare match/input capture IRQ |
00844 * | 2 | TGIEC | TGRD compare match/input capture (0,3) |
00845 * | 3 | TGIED | TGRD compare match/input capture (0,3) |
00846 * | 4 | TCIEV | Overflow interrupt enable |
00847 * | 5 | TCIEU | Underflow interrupt enable (1,2,4,5) |
00848 * | 6 | - | Reserved (read as 1, write 1) |
00849 * | 7 | TTGE | A/D conversion start request enable |

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```

00850 *
00851 * @note TGIEC/TGIED reserved on TPU1/2/4/5 (read as 0, write 0)
00852 * @note TCIEU reserved on TPU0/TPU3 (read as 0, write 0)
00853 * @note TTGE reserved on TPU5 (read as 0, write 0)
00854 * @note Manual: Ch28.2.4 - Timer Interrupt Enable Register (TIER)
00855 * @since Version 1.0.0
00856 */
00857 typedef enum : uint8_t {
00858     k_tpu_tier_tgiea = (1U « 0), /**< TGRA interrupt enable */
00859     k_tpu_tier_tgieb = (1U « 1), /**< TGRB interrupt enable */
00860     k_tpu_tier_tgiec = (1U « 2), /**< TGRC interrupt enable (TPU0/3 only) */
00861     k_tpu_tier_tgied = (1U « 3), /**< TGRD interrupt enable (TPU0/3 only) */
00862     k_tpu_tier_tciev = (1U « 4), /**< Overflow interrupt enable (all channels) */
00863     k_tpu_tier_tcieu = (1U « 5), /**< Underflow interrupt enable (TPU1/2/4/5) */
00864     k_tpu_tier_ttge = (1U « 7), /**< A/D conversion start request enable */
00865 } rx_tpu_tier_bits_t;
00866
00867 /*
=====
00868 * TSR - Timer Status Register Bit Definitions
00869 *
=====
00870 */
00871
00872 /**
00873 * @enum rx_tpu_tsr_bits_t
00874 * @brief TSR (Timer Status Register) bit definitions
00875 *
00876 * @details
00877 * Status flags for timer events and counting direction. The TCFD flag
00878 * is critical for phase counting mode - it indicates whether TCNT is
00879 * currently counting up (forward) or down (reverse).
00880 *
00881 * @par Bit Layout (8-bit)
00882 * | Bit | Field | Description |
00883 * |-----|-----|-----|
00884 * | 0 | TGFA | TGRA compare match/input capture flag |
00885 * | 1 | TGFB | TGRB compare match/input capture flag |
00886 * | 2 | TGFC | TGRC compare match/input capture (0,3) |
00887 * | 3 | TGFD | TGRD compare match/input capture (0,3) |
00888 * | 4 | TCFV | Overflow flag (all channels) |
00889 * | 5 | TCFU | Underflow flag (TPU1/2/4/5 only) |
00890 * | 6 | - | Reserved (read as 1, write 1) |
00891 * | 7 | TCFD | Counting Direction Flag (TPU1/2/4/5) |
00892 *
00893 * @note Flags (bits 0-5) are cleared by writing 0 to them
00894 * @note TCFD is read-only: 1=counting up, 0=counting down
00895 * @note Bit 6 reset value is 1 (must write 1 to preserve)
00896 * @note TGFC/TGFD reserved on TPU1/2/4/5 (read as 0, write 0)
00897 * @note TCFU/TCFD reserved on TPU0/TPU3 (read as 0/1 respectively)
00898 * @note Manual: Ch28.2.5 - Timer Status Register (TSR)
00899 * @since Version 1.0.0
00900 */
00901 typedef enum : uint8_t {
00902     k_tpu_tsr_tgfa = (1U « 0), /**< TGRA compare match/input capture occurred */
00903     k_tpu_tsr_tgfb = (1U « 1), /**< TGRB compare match/input capture occurred */
00904     k_tpu_tsr_tgfc = (1U « 2), /**< TGRC compare match/input capture (TPU0/3) */
00905     k_tpu_tsr_tgfd = (1U « 3), /**< TGRD compare match/input capture (TPU0/3) */
00906     k_tpu_tsr_tcfv = (1U « 4), /**< Overflow flag (TCNT wrapped 0xFFFF->0x0000) */
00907     k_tpu_tsr_tcfu = (1U « 5), /**< Underflow flag (TCNT wrapped 0x0000->0xFFFF) */
00908     k_tpu_tsr_tcfu = (1U « 7), /**< Counting direction: 1=up, 0=down (read-only) */
00909 } rx_tpu_tsr_bits_t;
00910
00911 /**
00912 * @enum rx_tpu_tsr_reserved_t
00913 * @brief TSR reserved bit value (bit 6 must be written as 1)
00914 * @since Version 1.0.0
00915 */
00916 typedef enum : uint8_t {
00917     k_tpu_tsr_reserved_bit6 = (1U « 6), /**< Bit 6: read as 1, write 1 */
00918 } rx_tpu_tsr_reserved_t;
00919
00920 /*
=====
00921 * NFCR - Noise Filter Control Register Bit Definitions
00922 *
=====
00923 */
00924
00925 /**
00926 * @enum rx_tpu_nfc_bits_t
00927 * @brief NFCR (Noise Filter Control Register) bit definitions
00928 *
00929 * @details
00930 * Controls digital noise filtering on TIOC I/O pins. Each channel has
00931 * independent filter enable bits for each pin and a shared clock select.
00932 *

```

```

00933 * @note NFCEN/NFDEN are reserved on TPU1/2/4/5 (only TPU0/3 have C/D pins)
00934 * @note Set NFCR only while TCNT is stopped (CSTn = 0)
00935 * @note Manual: Ch28.2.10 - Noise Filter Control Register (NFCR)
00936 * @since Version 1.0.0
00937 */
00938 typedef enum : uint8_t {
00939     k_tpu_nfr_nfaen = (1U « 0), /**< Noise filter enable for TIOCA pin */
00940     k_tpu_nfr_nfben = (1U « 1), /**< Noise filter enable for TIOCB pin */
00941     k_tpu_nfr_nfcen = (1U « 2), /**< Noise filter enable for TIOCC (TPU0/3) */
00942     k_tpu_nfr_nfden = (1U « 3), /**< Noise filter enable for TIOCD (TPU0/3) */
00943 } rx_tpu_nfr_bits_t;
00944
00945 /**
00946 * @enum rx_tpu_nfr_clock_t
00947 * @brief NFCR NFCS[1:0] noise filter clock select values
00948 *
00949 * @details
00950 * Selects the sampling clock for the digital noise filter. The filter
00951 * requires 3 consecutive samples at the selected level to pass.
00952 *
00953 * @par Filter Delay at 60 MHz PCLKB
00954 * | NFCS | Clock      | 3-Sample Delay          |
00955 * |-----|-----|-----|
00956 * | 00   | PCLKB/1   | ~50 ns                  |
00957 * | 01   | PCLKB/8   | ~400 ns                 |
00958 * | 10   | PCLKB/32  | ~1.6 us                 |
00959 * | 11   | Count clk | Depends on prescaler setting |
00960 *
00961 * @since Version 1.0.0
00962 */
00963 typedef enum : uint8_t {
00964     k_tpu_nfr_nfcs_pclk_div1  = (0U « 4), /**< Sample @ PCLKB/1 (~50ns) */
00965     k_tpu_nfr_nfcs_pclk_div8  = (1U « 4), /**< Sample @ PCLKB/8 (~400ns) */
00966     k_tpu_nfr_nfcs_pclk_div32 = (2U « 4), /**< Sample @ PCLKB/32 (~1.6us) */
00967     k_tpu_nfr_nfcs_count_clock = (3U « 4), /**< Sample @ count clock source */
00968 } rx_tpu_nfr_clock_t;
00969
00970 /**
00971 * @enum rx_tpu_nfr_masks_t
00972 * @brief NFCR register field masks
00973 * @since Version 1.0.0
00974 */
00975 typedef enum : uint8_t {
00976     k_tpu_nfr_nfcs_mask  = 0x30, /**< NFCS[1:0] field mask (bits 5:4) */
00977     k_tpu_nfr_enable_mask = 0x0F, /**< All noise filter enable bits (bits 3:0) */
00978 } rx_tpu_nfr_masks_t;
00979
00980 /*
=====
00981 * Module Stop Bit Definition
00982 *
=====
00983 */
00984
00985 /**
00986 * @enum rx_tpu_mstpcra_t
00987 * @brief MSTPCRA bit for TPU module stop control
00988 *
00989 * @details
00990 * The TPU module is controlled by MSTPA13 in the Module Stop Control
00991 * Register A (MSTPCRA). Write 0 to enable, 1 to stop.
00992 *
00993 * @since Version 1.0.0
00994 */
00995 typedef enum : uint32_t {
00996     k_tpu_mstpcra_mstpa13 = (1U « 13), /**< MSTPA13: TPU module stop (1=stop) */
00997 } rx_tpu_mstpcra_t;
00998
00999 /*
=====
01000 * TPU Interrupt Source Numbers (Software Configurable Interrupt B)
01001 *
=====
01002 */
01003
01004 /**
01005 * @enum rx_tpu_intb_source_t
01006 * @brief TPU Software Configurable Interrupt B source numbers
01007 *
01008 * @details
01009 * TPU interrupts are routed through the ICU Software Configurable Interrupt B
01010 * mechanism. These source numbers are written to SLIBRn registers (n=128-207)
01011 * to map them to interrupt vector numbers 128-207.
01012 *
01013 * For phase counting mode, the most relevant sources are:
01014 * - TCIXV (overflow): TCNT wrapped 0xFFFF -> 0x0000 while counting up
01015 * - TCIXU (underflow): TCNT wrapped 0x0000 -> 0xFFFF while counting down

```



```

01016 *
01017 * @par Configuration Example
01018 * @code{.c}
01019 * // Map TPU1 overflow to vector 128 (INTB128)
01020 * ICU.SLIBR128.BYTE = k_tpu_intb_tci1v;
01021 * // Map TPU1 underflow to vector 129 (INTB129)
01022 * ICU.SLIBR129.BYTE = k_tpu_intb_tci1u;
01023 * @endcode
01024 *
01025 * @note Manual: Ch15 Table 15.3 - Interrupt Sources for Software Configurable
01026 *       Interrupt B, and Ch28.4 - Interrupt Sources
01027 * @warning These are NOT vector numbers - they are source numbers for SLIBR
01028 * @since Version 1.0.0
01029 */
01030 typedef enum : uint8_t {
01031     /* TPU0 interrupt sources */
01032     k_tpu_intb_tgi0a = 15, /**< TPU0 TGRA compare match/input capture */
01033     k_tpu_intb_tgi0b = 16, /**< TPU0 TGRB compare match/input capture */
01034     k_tpu_intb_tgi0c = 17, /**< TPU0 TGRC compare match/input capture */
01035     k_tpu_intb_tgi0d = 18, /**< TPU0 TGRD compare match/input capture */
01036     k_tpu_intb_tci0v = 19, /**< TPU0 overflow */
01037
01038     /* TPU1 interrupt sources (back-right encoder) */
01039     k_tpu_intb_tgi1a = 20, /**< TPU1 TGRA compare match/input capture */
01040     k_tpu_intb_tgi1b = 21, /**< TPU1 TGRB compare match/input capture */
01041     k_tpu_intb_tci1v = 22, /**< TPU1 overflow (phase counting up wrap) */
01042     k_tpu_intb_tci1u = 23, /**< TPU1 underflow (phase counting down wrap) */
01043
01044     /* TPU2 interrupt sources (back-left encoder) */
01045     k_tpu_intb_tgi2a = 24, /**< TPU2 TGRA compare match/input capture */
01046     k_tpu_intb_tgi2b = 25, /**< TPU2 TGRB compare match/input capture */
01047     k_tpu_intb_tci2v = 26, /**< TPU2 overflow (phase counting up wrap) */
01048     k_tpu_intb_tci2u = 27, /**< TPU2 underflow (phase counting down wrap) */
01049
01050     /* TPU3 interrupt sources */
01051     k_tpu_intb_tgi3a = 28, /**< TPU3 TGRA compare match/input capture */
01052     k_tpu_intb_tgi3b = 29, /**< TPU3 TGRB compare match/input capture */
01053     k_tpu_intb_tgi3c = 30, /**< TPU3 TGRC compare match/input capture */
01054     k_tpu_intb_tgi3d = 31, /**< TPU3 TGRD compare match/input capture */
01055     k_tpu_intb_tci3v = 32, /**< TPU3 overflow */
01056
01057     /* TPU4 interrupt sources */
01058     k_tpu_intb_tgi4a = 33, /**< TPU4 TGRA compare match/input capture */
01059     k_tpu_intb_tgi4b = 34, /**< TPU4 TGRB compare match/input capture */
01060     k_tpu_intb_tci4v = 35, /**< TPU4 overflow */
01061     k_tpu_intb_tci4u = 36, /**< TPU4 underflow (phase counting) */
01062
01063     /* TPU5 interrupt sources */
01064     k_tpu_intb_tgi5a = 37, /**< TPU5 TGRA compare match/input capture */
01065     k_tpu_intb_tgi5b = 38, /**< TPU5 TGRB compare match/input capture */
01066     k_tpu_intb_tci5v = 39, /**< TPU5 overflow */
01067     k_tpu_intb_tci5u = 40, /**< TPU5 underflow (phase counting) */
01068 } rx_tpu_intb_source_t;
01069
01070 /*
=====
01071 * Static Assertions - Verify Register Layout at Compile Time
01072 *
=====
01073 */
01074
01075 /* --- Base Address Relationship Verification (Manual Ch28.2) ---
01076 * The k_tpu*_base_addr enum declarations above are themselves the contract for
01077 * each channel's absolute address. We assert the RELATIONSHIPS between them
01078 * (uniform channel spacing, control block adjacency) rather than re-stating
01079 * each literal -- which would just duplicate the magic numbers. */
01080 static_assert(k_tpu0_base_addr - k_tpu_control_base_addr == k_tpu_channel_spacing,
01081     "TPU0 must follow control block by one channel spacing");
01082 static_assert((k_tpu1_base_addr - k_tpu0_base_addr) == k_tpu_channel_spacing,
01083     "TPU0-TPU1 spacing incorrect");
01084 static_assert((k_tpu2_base_addr - k_tpu1_base_addr) == k_tpu_channel_spacing,
01085     "TPU1-TPU2 spacing incorrect");
01086 static_assert((k_tpu3_base_addr - k_tpu2_base_addr) == k_tpu_channel_spacing,
01087     "TPU2-TPU3 spacing incorrect");
01088 static_assert((k_tpu4_base_addr - k_tpu3_base_addr) == k_tpu_channel_spacing,
01089     "TPU3-TPU4 spacing incorrect");
01090 static_assert((k_tpu5_base_addr - k_tpu4_base_addr) == k_tpu_channel_spacing,
01091     "TPU4-TPU5 spacing incorrect");
01092
01093 /* --- Control Block Structure Verification --- */
01094 static_assert(sizeof(rx_tpu_control_regs_t) == 14, "TPU control register structure size incorrect");
01095 static_assert(offsetof(rx_tpu_control_regs_t, tstr) == 0x00, "TSTR offset incorrect");
01096 static_assert(offsetof(rx_tpu_control_regs_t, tsyr) == 0x01, "TSYR offset incorrect");
01097 static_assert(offsetof(rx_tpu_control_regs_t, nfcrr) == 0x08, "NFCR array offset incorrect");
01098
01099 /* --- Extended Channel Structure (TPU0/TPU3) Verification --- */
01100 static_assert(sizeof(rx_tpu_ext_regs_t) == 16,

```



```

01101         "TPU extended channel register structure size incorrect");
01102 static_assert(offsetof(rx_tpu_ext_regs_t, tcr) == 0x00, "TCR offset incorrect (extended)");
01103 static_assert(offsetof(rx_tpu_ext_regs_t, tmdr) == 0x01, "TMDR offset incorrect (extended)");
01104 static_assert(offsetof(rx_tpu_ext_regs_t, tiorh) == 0x02, "TIO RH offset incorrect");
01105 static_assert(offsetof(rx_tpu_ext_regs_t, tiorl) == 0x03, "TIO RL offset incorrect");
01106 static_assert(offsetof(rx_tpu_ext_regs_t, tier) == 0x04, "TIER offset incorrect (extended)");
01107 static_assert(offsetof(rx_tpu_ext_regs_t, tsr) == 0x05, "TSR offset incorrect (extended)");
01108 static_assert(offsetof(rx_tpu_ext_regs_t, tcnt) == 0x06, "TCNT offset incorrect (extended)");
01109 static_assert(offsetof(rx_tpu_ext_regs_t, tgra) == 0x08, "TGRA offset incorrect (extended)");
01110 static_assert(offsetof(rx_tpu_ext_regs_t, tgrb) == 0x0A, "TGRB offset incorrect (extended)");
01111 static_assert(offsetof(rx_tpu_ext_regs_t, tgrc) == 0x0C, "TGRC offset incorrect");
01112 static_assert(offsetof(rx_tpu_ext_regs_t, tgrd) == 0x0E, "TGRD offset incorrect");
01113
01114 /* --- Basic Channel Structure (TPU1/2/4/5) Verification --- */
01115 static_assert(sizeof(rx_tpu_regs_t) == 16, "TPU basic channel register structure size incorrect");
01116 static_assert(offsetof(rx_tpu_regs_t, tcr) == 0x00, "TCR offset incorrect (basic)");
01117 static_assert(offsetof(rx_tpu_regs_t, tmdr) == 0x01, "TMDR offset incorrect (basic)");
01118 static_assert(offsetof(rx_tpu_regs_t, tior) == 0x02, "TIO R offset incorrect");
01119 static_assert(offsetof(rx_tpu_regs_t, tier) == 0x04, "TIER offset incorrect (basic)");
01120 static_assert(offsetof(rx_tpu_regs_t, tsr) == 0x05, "TSR offset incorrect (basic)");
01121 static_assert(offsetof(rx_tpu_regs_t, tcnt) == 0x06, "TCNT offset incorrect (basic)");
01122 static_assert(offsetof(rx_tpu_regs_t, tgra) == 0x08, "TGRA offset incorrect (basic)");
01123 static_assert(offsetof(rx_tpu_regs_t, tgrb) == 0x0A, "TGRB offset incorrect (basic)");
01124
01125 /* --- Bit Field Relationship Verification ---
01126 * The k_tpu_*_bits enum declarations above use (1U « N) form, which is itself
01127 * the bit-position contract per HW Manual Ch28. Re-stating each literal value
01128 * (== 0x01, == 0x80, etc.) would duplicate the magic number; assert the
01129 * RELATIONSHIPS between sibling bits and field values instead. */
01130 static_assert(k_tpu_tstr_cst5 == (k_tpu_tstr_cst0 « 5U),
01131     "TSTR CST5 must be CST0 left-shift 5 (six channel-start bits 0..5)");
01132 static_assert(k_tpu_tsr_tcfu == (k_tpu_tsr_tcfv « 1U),
01133     "TSR TCFU must follow TCFV (underflow follows overflow)");
01134 static_assert(k_tpu_tsr_tcfv == (k_tpu_tsr_tcfv « 3U),
01135     "TSR TCFD direction flag is three bits above TCFV (bit 7 vs bit 4)");
01136 static_assert(k_tpu_tier_tcieu == (k_tpu_tier_tciev « 1U),
01137     "TIER TCIEU must follow TCIEV (underflow IE follows overflow IE)");
01138 static_assert(k_tpu_tmdr_md_phase_count_4 ==
01139     (k_tpu_tmdr_md_phase_count_1 + (k_tpu_tmdr_md_pwm2 - k_tpu_tmdr_md_pwm1) * 3U),
01140     "Phase counting mode 4 sits 3 codes above mode 1");
01141
01142 /* --- NFCR Relationship Verification --- */
01143 static_assert(k_tpu_nfr_nfaen == (k_tpu_nfr_nfben « 1U),
01144     "NFCR NFAEN must precede NFBEN by 1 bit");
01145 static_assert(k_tpu_nfr_nfcs_pclk_div32 == k_tpu_nfr_nfcs_mask - k_tpu_nfr_nfcs_pclk_div8,
01146     "NFCS field encodes div32 = mask - div8 (NFCS[1:0] = 2)");
01147
01148 #ifdef __cplusplus
01149 }
01150 #endif

```

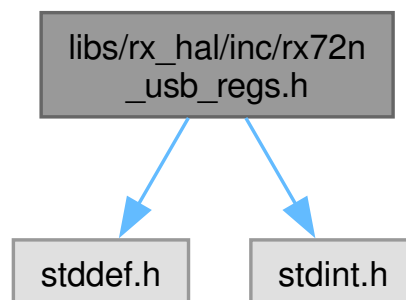
8.67 libs/rx_hal/inc/rx72n_usb_regs.h File Reference

RX72N USB 2.0 Full-Speed Host/Function Module Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_usb_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct `rx_usb_regs_t`
USB0 Register Map (Base: 0x000A0000).

Enumerations

- enum `usb_register_offsets_t` : `uint16_t` {
`k_usb_offset_syscfg` = 0x00 , `k_usb_offset_syssts0` = 0x04 , `k_usb_offset_dvstctr0` = 0x08 ,
`k_usb_offset_cfifo` = 0x14 ,
`k_usb_offset_d0fifo` = 0x18 , `k_usb_offset_d1fifo` = 0x1C , `k_usb_offset_cffosel` = 0x20 ,
`k_usb_offset_cffoctr` = 0x22 ,
`k_usb_offset_d0ffosel` = 0x28 , `k_usb_offset_d0ffoctr` = 0x2A , `k_usb_offset_d1ffosel` = 0x2C ,
`k_usb_offset_d1ffoctr` = 0x2E ,
`k_usb_offset_intenb0` = 0x30 , `k_usb_offset_intenb1` = 0x32 , `k_usb_offset_brdyenb` = 0x36 ,
`k_usb_offset_nrdyenb` = 0x38 ,
`k_usb_offset_bempenb` = 0x3A , `k_usb_offset_softcfg` = 0x3C , `k_usb_offset_intsts0` = 0x40 ,
`k_usb_offset_intsts1` = 0x42 ,
`k_usb_offset_brdysts` = 0x46 , `k_usb_offset_nrdysts` = 0x48 , `k_usb_offset_bempsts` = 0x4A ,
`k_usb_offset_frmnum` = 0x4C ,
`k_usb_offset_dvchgr` = 0x4E , `k_usb_offset_usbaddr` = 0x50 , `k_usb_offset_usbreq` = 0x54 ,
`k_usb_offset_usbval` = 0x56 ,
`k_usb_offset_usbindx` = 0x58 , `k_usb_offset_usbleng` = 0x5A , `k_usb_offset_dcpcfg` = 0x5C ,
`k_usb_offset_dcpmaxp` = 0x5E ,
`k_usb_offset_dcpctr` = 0x60 , `k_usb_offset_pipesel` = 0x64 , `k_usb_offset_pipecfg` = 0x68 ,
`k_usb_offset_pipemaxp` = 0x6C ,
`k_usb_offset_pipeperi` = 0x6E , `k_usb_offset_pipe1ctr` = 0x70 , `k_usb_offset_pipe9ctr` = 0x80 ,
`k_usb_offset_pipe1tre` = 0x90 ,
`k_usb_offset_pipe5trn` = 0xA2 , `k_usb_offset_devadd0` = 0xD0 , `k_usb_offset_devadd5` = 0xDA ,
`k_usb_offset_physlew` = 0xF0 }

USB register offsets (verified against manual Chapter 40).

- enum `usb_reserved_sizes_t` : `uint8_t` {
`k_usb_reserved_02_04_bytes` = 2 , `k_usb_reserved_06_08_bytes` = 2 , `k_usb_reserved_0a_14_bytes` = 10 ,
`k_usb_reserved_24_28_bytes` = 4 ,
`k_usb_reserved_34_36_bytes` = 2 , `k_usb_reserved_3e_40_bytes` = 2 , `k_usb_reserved_44_46_bytes` = 2 ,
`k_usb_reserved_52_54_bytes` = 2 ,
`k_usb_reserved_62_64_bytes` = 2 , `k_usb_reserved_66_68_bytes` = 2 , `k_usb_reserved_6a_6c_bytes` = 2 ,
`k_usb_reserved_82_90_bytes` = 14 ,
`k_usb_reserved_a4_d0_bytes` = 44 , `k_usb_reserved_dc_f0_bytes` = 20 }

Reserved field sizes for USB register structure.

- enum `rx_usb_addresses_t` : `uintptr_t` { `k_usb0_base_addr` = 0x000A0000 , `k_usb_dpusr0r_addr` = 0x000A0400 }

USB0 peripheral base address (verified against RX72N Hardware Manual Ch40).

- enum `usb_syscfg_bits_t` : `uint16_t` {
`k_usb_syscfg_usbe` = (1U << 0) , `k_usb_syscfg_dprpu` = (1U << 4) , `k_usb_syscfg_drpd` = (1U << 5) ,
`k_usb_syscfg_dcfm` = (1U << 6) ,
`k_usb_syscfg_scke` = (1U << 10) }
- enum `usb_syssts0_bits_t` : `uint16_t` {
`k_usb_syssts0_inst_mask` = 0x0003 , `k_usb_syssts0_inst_se0` = 0x0000 , `k_usb_syssts0_inst_fs_j` = 0x0001 ,
`k_usb_syssts0_inst_fs_k` = 0x0002 ,
`k_usb_syssts0_inst_sel` = 0x0003 , `k_usb_syssts0_idmon` = (1U << 2) , `k_usb_syssts0_softa` = (1U << 5) ,
`k_usb_syssts0_hact` = (1U << 6) ,
`k_usb_syssts0_ovcmon_mask` = (3U << 14) }

- enum `usb_dvstctr0_bits_t` : `uint16_t` {
`k_usb_dvstctr0_rhst_mask` = 0x0007 , `k_usb_dvstctr0_rhst_undecided` = 0x0000 ,
`k_usb_dvstctr0_rhst_ls` = 0x0001 , `k_usb_dvstctr0_rhst_fs` = 0x0002 ,
`k_usb_dvstctr0_rhst_reset` = 0x0004 , `k_usb_dvstctr0_uact` = (1U << 4) , `k_usb_dvstctr0_resume`
= (1U << 5) , `k_usb_dvstctr0_usbrst` = (1U << 6) ,
`k_usb_dvstctr0_rwupe` = (1U << 7) , `k_usb_dvstctr0_wkup` = (1U << 8) , `k_usb_dvstctr0_vbusen`
= (1U << 9) , `k_usb_dvstctr0_exicen` = (1U << 10) ,
`k_usb_dvstctr0_hnpbtoa` = (1U << 11) }
- enum `usb_intenb0_bits_t` : `uint16_t` {
`k_usb_intenb0_brbye` = (1U << 8) , `k_usb_intenb0_nrdye` = (1U << 9) , `k_usb_intenb0_bempe`
= (1U << 10) , `k_usb_intenb0_ctre` = (1U << 11) ,
`k_usb_intenb0_dvse` = (1U << 12) , `k_usb_intenb0_sofe` = (1U << 13) , `k_usb_intenb0_rsme`
= (1U << 14) , `k_usb_intenb0_vbse` = (1U << 15) }
- enum `usb_intsts0_bits_t` : `uint16_t` {
`k_usb_intsts0_ctsq_mask` = 0x0007 , `k_usb_intsts0_ctsq_idle` = 0x0000 , `k_usb_intsts0_ctsq_rd_data`
= 0x0001 , `k_usb_intsts0_ctsq_rd_status` = 0x0002 ,
`k_usb_intsts0_ctsq_wr_data` = 0x0003 , `k_usb_intsts0_ctsq_wr_status` = 0x0004 ,
`k_usb_intsts0_ctsq_wr_nd` = 0x0005 , `k_usb_intsts0_ctsq_seq_err` = 0x0006 ,
`k_usb_intsts0_valid` = (1U << 3) , `k_usb_intsts0_dvsq_mask` = (7U << 4) , `k_usb_intsts0_dvsq_powered`
= (0U << 4) , `k_usb_intsts0_dvsq_default` = (1U << 4) ,
`k_usb_intsts0_dvsq_address` = (2U << 4) , `k_usb_intsts0_dvsq_configured` = (3U << 4) ,
`k_usb_intsts0_dvsq_suspend` = (4U << 4) , `k_usb_intsts0_vbsts` = (1U << 7) ,
`k_usb_intsts0_brby` = (1U << 8) , `k_usb_intsts0_nrdy` = (1U << 9) , `k_usb_intsts0_bemp` =
(1U << 10) , `k_usb_intsts0_crtt` = (1U << 11) ,
`k_usb_intsts0_dvst` = (1U << 12) , `k_usb_intsts0_sofr` = (1U << 13) , `k_usb_intsts0_resm`
= (1U << 14) , `k_usb_intsts0_vbint` = (1U << 15) }
- enum `usb_dcpcfg_bits_t` : `uint16_t` { `k_usb_dcpcfg_dir` = (1U << 4) , `k_usb_dcpcfg_shtnak`
= (1U << 7) }
- enum `usb_dcpcetr_bits_t` : `uint16_t` {
`k_usb_dcpcetr_pid_mask` = 0x0003 , `k_usb_dcpcetr_pid_nak` = 0x0000 , `k_usb_dcpcetr_pid_buf`
= 0x0001 , `k_usb_dcpcetr_pid_stall` = 0x0002 ,
`k_usb_dcpcetr_ccpl` = (1U << 2) , `k_usb_dcpcetr_pbusy` = (1U << 5) , `k_usb_dcpcetr_sqmon`
= (1U << 6) , `k_usb_dcpcetr_sqset` = (1U << 7) ,
`k_usb_dcpcetr_sqclr` = (1U << 8) , `k_usb_dcpcetr_sureqclr` = (1U << 11) , `k_usb_dcpcetr_sureq`
= (1U << 14) , `k_usb_dcpcetr_bsts` = (1U << 15) }
- enum `usb_pipe_bits_t` : `uint16_t` {
`k_usb_pipe_bit_0` = (1U << 0) , `k_usb_pipe_bit_1` = (1U << 1) , `k_usb_pipe_bit_2` = (1U
<< 2) , `k_usb_pipe_bit_3` = (1U << 3) ,
`k_usb_pipe_bit_4` = (1U << 4) , `k_usb_pipe_bit_5` = (1U << 5) , `k_usb_pipe_bit_6` = (1U
<< 6) , `k_usb_pipe_bit_7` = (1U << 7) ,
`k_usb_pipe_bit_8` = (1U << 8) , `k_usb_pipe_bit_9` = (1U << 9) , `k_usb_pipe_all` = 0x03FF
}

USB pipe interrupt control bits.

- enum `usb_pipecfg_bits_t` : `uint16_t` {
`k_usb_pipecfg_epnum_mask` = 0x000F , `k_usb_pipecfg_dir` = (1U << 4) , `k_usb_pipecfg_shtnak`
= (1U << 7) , `k_usb_pipecfg_dblb` = (1U << 9) ,
`k_usb_pipecfg_bfre` = (1U << 10) , `k_usb_pipecfg_type_mask` = (3U << 14) , `k_usb_pipecfg_type_bulk`
= (1U << 14) , `k_usb_pipecfg_type_int` = (2U << 14) ,
`k_usb_pipecfg_type_iso` = (3U << 14) }
- enum `usb_pipectr_bits_t` : `uint16_t` {
`k_usb_pipectr_pid_mask` = 0x0003 , `k_usb_pipectr_pid_nak` = 0x0000 , `k_usb_pipectr_pid_buf`
= 0x0001 , `k_usb_pipectr_pid_stall` = 0x0002 ,
`k_usb_pipectr_pbusy` = (1U << 5) , `k_usb_pipectr_sqmon` = (1U << 6) , `k_usb_pipectr_sqset`
= (1U << 7) , `k_usb_pipectr_sqclr` = (1U << 8) ,
`k_usb_pipectr_aclrm` = (1U << 9) , `k_usb_pipectr_atrepm` = (1U << 10) , `k_usb_pipectr_inbufm`
= (1U << 14) , `k_usb_pipectr_bsts` = (1U << 15) }

PIPEnCTR bits for pipes 1-5 (manual p.1976).

- enum `usb_pipe69ctr_bits_t` : `uint16_t` {
`k_usb_pipe69ctr_pid_mask` = 0x0003 , `k_usb_pipe69ctr_pid_nak` = 0x0000 , `k_usb_pipe69ctr_pid_buf` = 0x0001 , `k_usb_pipe69ctr_pid_stall` = 0x0002 ,
`k_usb_pipe69ctr_pbusy` = (1U << 5) , `k_usb_pipe69ctr_sqmon` = (1U << 6) , `k_usb_pipe69ctr_sqset` = (1U << 7) , `k_usb_pipe69ctr_sqclr` = (1U << 8) ,
`k_usb_pipe69ctr_aclrm` = (1U << 9) , `k_usb_pipe69ctr_bsts` = (1U << 15) }
PIPEnCTR bits for pipes 6-9 (manual p.1981).
- enum `usb_pipetre_bits_t` : `uint16_t` { `k_usb_pipetre_trclr` = (1U << 8) , `k_usb_pipetre_trenb` = (1U << 9) }
- enum `usb_cfifosel_bits_t` : `uint16_t` {
`k_usb_cfifosel_curpipe_mask` = 0x000F , `k_usb_cfifosel_curpipe_dcp` = 0x0000 , `k_usb_cfifosel_isel` = (1U << 5) , `k_usb_cfifosel_bigend` = (1U << 8) ,
`k_usb_cfifosel_mbw_mask` = (1U << 10) , `k_usb_cfifosel_mbw_8` = (0U << 10) ,
`k_usb_cfifosel_mbw_16` = (1U << 10) , `k_usb_cfifosel_rew` = (1U << 14) ,
`k_usb_cfifosel_rcnt` = (1U << 15) }
CFIFOSEL bits (manual p.1942).
- enum `usb_dnfifosel_bits_t` : `uint16_t` {
`k_usb_dnfifosel_curpipe_mask` = 0x000F , `k_usb_dnfifosel_curpipe_dcp` = 0x0000 ,
`k_usb_dnfifosel_bigend` = (1U << 8) , `k_usb_dnfifosel_mbw_mask` = (1U << 10) ,
`k_usb_dnfifosel_mbw_8` = (0U << 10) , `k_usb_dnfifosel_mbw_16` = (1U << 10) ,
`k_usb_dnfifosel_drege` = (1U << 12) , `k_usb_dnfifosel_dclrm` = (1U << 13) ,
`k_usb_dnfifosel_rew` = (1U << 14) , `k_usb_dnfifosel_rcnt` = (1U << 15) }
DOFIFOSEL / D1FIFOSEL bits (manual p.1943-1944).
- enum `usb_fifoctr_bits_t` : `uint16_t` { `k_usb_fifoctr_dtlm_mask` = 0x01FF , `k_usb_fifoctr_frdy` = (1U << 13) , `k_usb_fifoctr_bclr` = (1U << 14) , `k_usb_fifoctr_bval` = (1U << 15) }
- enum `usb_dpusr0r_bits_t` : `uint32_t` {
`k_usb_dpusr0r_srpc0` = (1UL << 0) , `k_usb_dpusr0r_rpue0` = (1UL << 1) , `k_usb_dpusr0r_fixphy0` = (1UL << 4) , `k_usb_dpusr0r_dp0` = (1UL << 16) ,
`k_usb_dpusr0r_dm0` = (1UL << 17) , `k_usb_dpusr0r_dovca0` = (1UL << 20) , `k_usb_dpusr0r_dovcb0` = (1UL << 21) , `k_usb_dpusr0r_dvbsts0` = (1UL << 23) }
- enum `usb_physlew_bits_t` : `uint32_t` {
`k_usb_physlew_slewr00` = (1UL << 0) , `k_usb_physlew_slewr01` = (1UL << 1) ,
`k_usb_physlew_slewf00` = (1UL << 2) , `k_usb_physlew_slewf01` = (1UL << 3) ,
`k_usb_physlew_rx72n` = 0x00000005 }
- enum `rx_usb_interrupt_vector_t` : `uint8_t` { `k_vect_usb0_d0fifo` = 34 , `k_vect_usb0_d1fifo` = 35 , `k_vect_usb0_usbi` = 144 , `k_vect_usb0_usbr` = 90 }
- enum `rx_usb_sli_source_t` : `uint8_t` { `k_usb0_usbi_sli_src` = 62 }
- enum `usb_cdc_endpoints_t` : `uint8_t` {
`k_usb_cdc_ep_ctrl` = 0 , `k_usb_cdc_ep_bulk_in` = 1 , `k_usb_cdc_ep_bulk_out` = 2 ,
`k_usb_cdc_ep_interrupt_in` = 3 ,
`k_usb_cdc_max_packet_fs` = 64 }

Functions

- static volatile `rx_usb_regs_t` * `usb0` (void)
- static volatile `uint32_t` * `usb_dpusr0r` (void)

Get pointer to USB deep-standby transceiver control register (DPUSR0R), 32-bit, at absolute 0x000A0400.

8.67.1 Detailed Description

RX72N USB 2.0 Full-Speed Host/Function Module Register Definitions.

Register definitions for the USB 2.0 Full-Speed Host/Function Module (USB0) verified against RX72N Group User's Manual: Hardware (R01UH0824EJ0111), Chapter 40 "USB 2.0 FS Host/Function Module (USBb)".

Verification Status (2026-04-20):

- All register addresses verified against manual Chapter 40
- 45 registers defined with correct offsets
- 55+ static assertions verify all offsets at compile time
- Removed non-existent registers (BUSWAIT, PLLSTA, TESTMODE, D0FBCFG, D1FBCFG, PHYSET, PIPEBUF, LPCTRL, LPSTS - these do NOT exist on RX72N)
- Renamed UFRMNUM to DVCHGR (correct register name per manual)
- CRITICAL FIX (2026-02-03): Changed FIFO registers from uint32_t to uint16_t per manual Ch40 p.1941. CFIFO/D0FIFO/D1FIFO are 16-bit WORD registers, NOT 32-bit. Added padding to maintain correct struct offsets.
- FIX (2026-04-20): Added missing DVSTCTR0 bits VBUSEN(b9), EXICEN(b10), HNPBTOA(b11) per manual p.1940.
- FIX (2026-04-20): Split usb_fifo_sel_bits_t into [usb_cfifosel_bits_t](#) and [usb_dnfifosel_bits_t](#). CFIFOSEL has ISEL(b5) but NOT DREQE/DCLRM. D0/D1FIFOSEL have DREQE(b12)/↔ DCLRM(b13) but NOT ISEL. Per manual p.1942-1944.

Date

2026-04-20

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Definition in file [rx72n_usb_regs.h](#).

8.67.2 Data Structure Documentation

8.67.2.1 struct rx_usb_regs_t

USB0 Register Map (Base: 0x000A0000).

Note

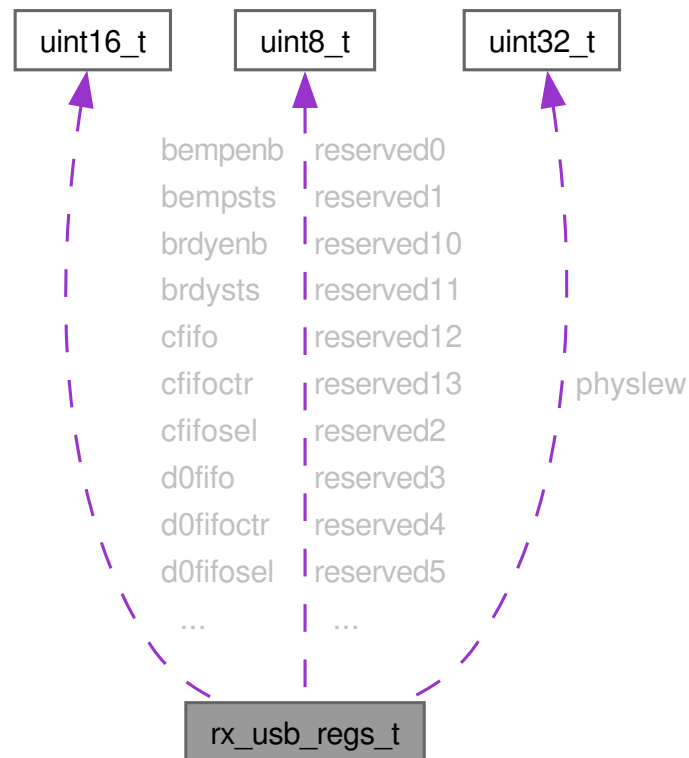
The following registers do NOT exist on RX72N (removed from struct): BUSWAIT, PLLSTA, TESTMODE, D0FBCFG, D1FBCFG, PHYSET, PIPEBUF, LPCTRL, LPSTS

FIFO registers (CFIFO, D0FIFO, D1FIFO) are 16-bit WORD registers per manual Ch40 p.1941. Access width controlled by CFIFOSEL.MBW/D0FIFOSEL.MBW:

- MBW=1 ([k_usb_cfifosel_mbw_16](#)): 16-bit word access (b15-b0)
- MBW=0 ([k_usb_cfifosel_mbw_8](#)): 8-bit byte access (b7-b0)

Definition at line [122](#) of file [rx72n_usb_regs.h](#).

Collaboration diagram for rx_usb_regs_t:



Data Fields

volatile uint16_t	bempenb	
volatile uint16_t	bempsts	
volatile uint16_t	brdyenb	
volatile uint16_t	brdysts	
volatile uint16_t	cfifo	Common FIFO port @ 0x14 (16-bit WORD, manual p.1941)
volatile uint16_t	cfifoctr	
volatile uint16_t	cfifosel	
volatile uint16_t	d0fifo	D0 FIFO port @ 0x18 (16-bit WORD, manual p.1941)
volatile uint16_t	d0fifotr	
volatile uint16_t	d0fifosel	
volatile uint16_t	d1fifo	D1 FIFO port @ 0x1C (16-bit WORD, manual p.1941)
volatile uint16_t	d1fifotr	
volatile uint16_t	d1fifosel	
volatile uint16_t	dcpcfg	
volatile uint16_t	dcpctr	
volatile uint16_t	dcpmaxp	
volatile uint16_t	devadd0	
volatile uint16_t	devadd1	
volatile uint16_t	devadd2	
volatile uint16_t	devadd3	

volatile uint16_t	devadd4	
volatile uint16_t	devadd5	
volatile uint16_t	dvchgr	Device State Change (NOT ufrm-num!)
volatile uint16_t	dvstctr0	
volatile uint16_t	frmnum	
volatile uint16_t	intenb0	
volatile uint16_t	intenb1	
volatile uint16_t	intsts0	
volatile uint16_t	intsts1	
volatile uint16_t	nrdyenb	
volatile uint16_t	nrdsts	
volatile uint32_t	physlew	PHY Cross-Point Adjustment @0xF0 (32-bit R/W per hirakuni45 rw32_t)
volatile uint16_t	pipe1ctr	
volatile uint16_t	pipe1tre	
volatile uint16_t	pipe1trn	
volatile uint16_t	pipe2ctr	
volatile uint16_t	pipe2tre	
volatile uint16_t	pipe2trn	
volatile uint16_t	pipe3ctr	
volatile uint16_t	pipe3tre	
volatile uint16_t	pipe3trn	
volatile uint16_t	pipe4ctr	
volatile uint16_t	pipe4tre	
volatile uint16_t	pipe4trn	
volatile uint16_t	pipe5ctr	
volatile uint16_t	pipe5tre	
volatile uint16_t	pipe5trn	
volatile uint16_t	pipe6ctr	
volatile uint16_t	pipe7ctr	
volatile uint16_t	pipe8ctr	
volatile uint16_t	pipe9ctr	
volatile uint16_t	pipecfg	
volatile uint16_t	pipemaxp	
volatile uint16_t	pipeperi	
volatile uint16_t	pipesel	
uint8_t	reserved0↵ [k_usb_reserved_02_04_bytes]	
uint8_t	reserved1↵ [k_usb_reserved_06_08_bytes]	
uint8_t	reserved10↵ [k_usb_reserved_6a_6c_bytes]	

uint8_t	reserved11↵ [k_usb_reserved_82_90_bytes]	
uint8_t	reserved12↵ [k_usb_reserved_a4_d0_bytes]	
uint8_t	reserved13↵ [k_usb_reserved_dc_f0_bytes]	
uint8_t	reserved2↵ [k_usb_reserved_0a_14_bytes]	
uint8_t	reserved3↵ [k_usb_reserved_24_28_bytes]	
uint8_t	reserved4↵ [k_usb_reserved_34_36_bytes]	
uint8_t	reserved5↵ [k_usb_reserved_3e_40_bytes]	
uint8_t	reserved6↵ [k_usb_reserved_44_46_bytes]	
uint8_t	reserved7↵ [k_usb_reserved_52_54_bytes]	
uint8_t	reserved8↵ [k_usb_reserved_62_64_bytes]	
uint8_t	reserved9↵ [k_usb_reserved_66_68_bytes]	
uint16_t	reserved_cfifo	Padding @ 0x16 to maintain D0↵ FIFO offset
uint16_t	reserved_d0fifo	Padding @ 0x1A
uint16_t	reserved_d1fifo	Padding @ 0x1E
volatile uint16_t	sofcfg	
volatile uint16_t	syscfg	
volatile uint16_t	syssts0	
volatile uint16_t	usbaddr	
volatile uint16_t	usbidx	
volatile uint16_t	usbleng	
volatile uint16_t	usbreq	
volatile uint16_t	usbval	

8.67.3 Enumeration Type Documentation

8.67.3.1 rx'usb'addresses't

```
enum rx_usb_addresses_t : uintptr_t
```

USB0 peripheral base address (verified against RX72N Hardware Manual Ch40).

Memory-mapped base address for the USB 2.0 Full-Speed Host/Function Module (USB0). Cast to a volatile pointer via `usb0()` to access the register map.

The underlying type is `uintptr_t` to correctly express that this value is a memory address, not a plain integer. On RX72N (32-bit) `uintptr_t == uint32_t`, so there is no runtime difference.

Memory Layout

```

*   Address      | Peripheral
*   -----+-----
*   0x000A0000   | USB0 register block (see rx_usb_regs_t)
*
```

See also

[rx_usb_regs_t](#) Register map structure accessed via this address

[usb0\(\)](#) Inline accessor that casts this address to a pointer

Since

Version 1.0.0

Enumerator

k_usb0_base_addr	USB0 register block base address (manual p.1928, Ch40)
k_usb_dpusr0r_addr	

Definition at line 230 of file [rx72n_usb_regs.h](#).

```

00230         : uintptr_t {
00231     k_usb0_base_addr = 0x000A0000, /**< USB0 register block base address (manual p.1928, Ch40) */
00232     /* Deep-standby USB transceiver control / pin monitor register. Shared
00233      * with USB-common (outside the USB0 rx_usb_regs_t block). 32-bit R/W
00234      * at absolute 0x000A0400 per hirakuni45 RX600/usb.hpp and Renesas FSP
00235      * iodef.h. Used to release the PHY from "output fixed" state after
00236      * deep-standby wakeup or first boot. */
00237     k_usb_dpusr0r_addr = 0x000A0400,
00238 } rx_usb_addresses_t;
```

8.67.3.2 rx`usb`interrupt`vector`t

```
enum rx_usb_interrupt_vector_t : uint8_t
```

Enumerator

k_vect_usb0_d0fifo	
k_vect_usb0_d1fifo	
k_vect_usb0_usbi	
k_vect_usb0_usbr	

Definition at line 525 of file [rx72n_usb_regs.h](#).

```

00525         : uint8_t {
00526     k_vect_usb0_d0fifo = 34,
00527     k_vect_usb0_d1fifo = 35,
00528     k_vect_usb0_usbi   = 144,
00529     k_vect_usb0_usbr   = 90,
00530 } rx_usb_interrupt_vector_t;
```

8.67.3.3 rx'usb'sli'source't

```
enum rx_usb_sli_source_t : uint8_t
```

Enumerator

k_usb0_usbi_sli_src	USBIO source code for ICU.SLIBR[144..207]
---------------------	---

Definition at line 534 of file [rx72n_usb_regs.h](#).

```
00534      : uint8_t {
00535  k_usb0_usbi_sli_src = 62, /**< USBIO source code for ICU.SLIBR[144..207] */
00536 } rx_usb_sli_source_t;
```

8.67.3.4 usb'cdc'endpoints't

```
enum usb_cdc_endpoints_t : uint8_t
```

Enumerator

k_usb_cdc_ep_ctrl	
k_usb_cdc_ep_bulk_in	
k_usb_cdc_ep_bulk_out	
k_usb_cdc_ep_interrupt_in	
k_usb_cdc_max_packet_fs	

Definition at line 539 of file [rx72n_usb_regs.h](#).

```
00539      : uint8_t {
00540  k_usb_cdc_ep_ctrl      = 0,
00541  k_usb_cdc_ep_bulk_in   = 1,
00542  k_usb_cdc_ep_bulk_out  = 2,
00543  k_usb_cdc_ep_interrupt_in = 3,
00544  k_usb_cdc_max_packet_fs = 64,
00545 } usb_cdc_endpoints_t;
```

8.67.3.5 usb'cffi'osel'bits't

```
enum usb_cffi_osel_bits_t : uint16_t
```

CFIFOSEL bits (manual p.1942).

Note

CFIFOSEL has ISEL(b5) for direction select.

CFIFOSEL does NOT have DREQE or DCLRM (those are D0/D1 only).

Enumerator

k_usb_cffi_osel_curpipe_mask	b3:b0: Port access pipe
k_usb_cffi_osel_curpipe_dcp	

k_usb_cfifosel_isel	b5: Access direction (0=read,1=write)
k_usb_cfifosel_bigend	b8: Endian select
k_usb_cfifosel_mbw_mask	b10: MBW bit
k_usb_cfifosel_mbw_8	8-bit access
k_usb_cfifosel_mbw_16	16-bit access
k_usb_cfifosel_rew	b14: Read pointer rewind
k_usb_cfifosel_rcnt	b15: Read count mode

Definition at line 444 of file [rx72n_usb_regs.h](#).

```

00444      : uint16_t {
00445  k_usb_cfifosel_curpipe_mask = 0x000F, /**< b3:b0: Port access pipe */
00446  k_usb_cfifosel_curpipe_dcp = 0x0000,
00447  k_usb_cfifosel_isel      = (1U « 5), /**< b5: Access direction (0=read,1=write) */
00448  k_usb_cfifosel_bigend    = (1U « 8), /**< b8: Endian select */
00449  k_usb_cfifosel_mbw_mask  = (1U « 10), /**< b10: MBW bit */
00450  k_usb_cfifosel_mbw_8     = (0U « 10), /**< 8-bit access */
00451  k_usb_cfifosel_mbw_16    = (1U « 10), /**< 16-bit access */
00452  k_usb_cfifosel_rew       = (1U « 14), /**< b14: Read pointer rewind */
00453  k_usb_cfifosel_rcnt      = (1U « 15), /**< b15: Read count mode */
00454 } usb_cfifosel_bits_t;

```

8.67.3.6 usb`dcpcfg`bits`*t*

enum [usb_dcpcfg_bits_t](#) : uint16_t

Enumerator

k_usb_dcpcfg_dir	b4: Transfer direction (0=receive,1=transmit)
k_usb_dcpcfg_shtnak	b7: Pipe disabled at end of transfer

Definition at line 334 of file [rx72n_usb_regs.h](#).

```

00334      : uint16_t {
00335  k_usb_dcpcfg_dir = (1U « 4), /**< b4: Transfer direction (0=receive,1=transmit) */
00336  k_usb_dcpcfg_shtnak = (1U « 7), /**< b7: Pipe disabled at end of transfer */
00337 } usb_dcpcfg_bits_t;

```

8.67.3.7 usb`dcpctr`bits`*t*

enum [usb_dcpctr_bits_t](#) : uint16_t

Enumerator

k_usb_dcpctr_pid_mask	b1:b0: Response PID
k_usb_dcpctr_pid_nak	
k_usb_dcpctr_pid_buf	
k_usb_dcpctr_pid_stall	
k_usb_dcpctr_ccpl	b2: Control transfer end enable
k_usb_dcpctr_pbusy	b5: Pipe busy flag
k_usb_dcpctr_sqmon	b6: Sequence toggle bit monitor

k_usb_dcpctr_sqset	b7: Sequence toggle bit set
k_usb_dcpctr_sqclr	b8: Sequence toggle bit clear
k_usb_dcpctr_sureqclr	b11: SUREQ bit clear
k_usb_dcpctr_sureq	b14: Setup token transmission
k_usb_dcpctr_bsts	b15: Buffer status flag

Definition at line 340 of file [rx72n_usb_regs.h](#).

```

00340         : uint16_t {
00341     k_usb_dcpctr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00342     k_usb_dcpctr_pid_nak = 0x0000,
00343     k_usb_dcpctr_pid_buf = 0x0001,
00344     k_usb_dcpctr_pid_stall = 0x0002,
00345     k_usb_dcpctr_ccpl = (1U « 2), /**< b2: Control transfer end enable */
00346     k_usb_dcpctr_pbushy = (1U « 5), /**< b5: Pipe busy flag */
00347     k_usb_dcpctr_sqmon = (1U « 6), /**< b6: Sequence toggle bit monitor */
00348     k_usb_dcpctr_sqset = (1U « 7), /**< b7: Sequence toggle bit set */
00349     k_usb_dcpctr_sqclr = (1U « 8), /**< b8: Sequence toggle bit clear */
00350     k_usb_dcpctr_sureqclr = (1U « 11), /**< b11: SUREQ bit clear */
00351     k_usb_dcpctr_sureq = (1U « 14), /**< b14: Setup token transmission */
00352     k_usb_dcpctr_bsts = (1U « 15), /**< b15: Buffer status flag */
00353 } usb_dcpctr_bits_t;

```

8.67.3.8 usb_dnfifosel_bits_t

enum [usb_dnfifosel_bits_t](#) : uint16_t

D0FIFOSEL / D1FIFOSEL bits (manual p.1943-1944).

Note

D0/D1FIFOSEL have DREQE(b12) and DCLRM(b13) for DMA/DTC.

D0/D1FIFOSEL do NOT have ISEL (direction select is always host RX).

Enumerator

k_usb_dnfifosel_curpipe_mask	b3:b0: Port access pipe
k_usb_dnfifosel_curpipe_dcp	
k_usb_dnfifosel_bigend	b8: Endian select
k_usb_dnfifosel_mbw_mask	b10: MBW bit
k_usb_dnfifosel_mbw_8	8-bit access
k_usb_dnfifosel_mbw_16	16-bit access
k_usb_dnfifosel_dreqe	b12: DMA/DTC transfer request enable
k_usb_dnfifosel_dclrm	b13: Auto buffer clear mode
k_usb_dnfifosel_rew	b14: Read pointer rewind
k_usb_dnfifosel_rcnt	b15: Read count mode

Definition at line 462 of file [rx72n_usb_regs.h](#).

```

00462         : uint16_t {
00463     k_usb_dnfifosel_curpipe_mask = 0x000F, /**< b3:b0: Port access pipe */
00464     k_usb_dnfifosel_curpipe_dcp = 0x0000,
00465     k_usb_dnfifosel_bigend = (1U « 8), /**< b8: Endian select */
00466     k_usb_dnfifosel_mbw_mask = (1U « 10), /**< b10: MBW bit */
00467     k_usb_dnfifosel_mbw_8 = (0U « 10), /**< 8-bit access */
00468     k_usb_dnfifosel_mbw_16 = (1U « 10), /**< 16-bit access */
00469     k_usb_dnfifosel_dreqe = (1U « 12), /**< b12: DMA/DTC transfer request enable */
00470     k_usb_dnfifosel_dclrm = (1U « 13), /**< b13: Auto buffer clear mode */
00471     k_usb_dnfifosel_rew = (1U « 14), /**< b14: Read pointer rewind */
00472     k_usb_dnfifosel_rcnt = (1U « 15), /**< b15: Read count mode */
00473 } usb_dnfifosel_bits_t;

```

8.67.3.9 usb_dpusr0r_bits_t

```
enum usb_dpusr0r_bits_t : uint32_t
```

Enumerator

k_usb_dpusr0r_srpc0	USB0 single-ended receiver power control
k_usb_dpusr0r_rpue0	USB0 DP pull-up resistor enable (deep-standby)
k_usb_dpusr0r_fixphy0	USB0 PHY output fixed (1=fixed, 0=normal)
k_usb_dpusr0r_dp0	USB0 D+ line monitor
k_usb_dpusr0r_dm0	USB0 D- line monitor
k_usb_dpusr0r_dovca0	USB0 overcurrent flag A
k_usb_dpusr0r_dovcb0	USB0 overcurrent flag B
k_usb_dpusr0r_dvbsts0	USB0 VBUS status

Definition at line 486 of file [rx72n_usb_regs.h](#).

```
00486         : uint32_t {
00487   k_usb_dpusr0r_srpc0  = (1UL << 0), /**< USB0 single-ended receiver power control */
00488   k_usb_dpusr0r_rpue0  = (1UL << 1), /**< USB0 DP pull-up resistor enable (deep-standby) */
00489   k_usb_dpusr0r_fixphy0 = (1UL << 4), /**< USB0 PHY output fixed (1=fixed, 0=normal) */
00490   k_usb_dpusr0r_dp0    = (1UL << 16), /**< USB0 D+ line monitor */
00491   k_usb_dpusr0r_dm0    = (1UL << 17), /**< USB0 D- line monitor */
00492   k_usb_dpusr0r_dovca0 = (1UL << 20), /**< USB0 overcurrent flag A */
00493   k_usb_dpusr0r_dovcb0 = (1UL << 21), /**< USB0 overcurrent flag B */
00494   k_usb_dpusr0r_dvbsts0 = (1UL << 23), /**< USB0 VBUS status */
00495 } usb_dpusr0r_bits_t;
```

8.67.3.10 usb_dvstctr0_bits_t

```
enum usb_dvstctr0_bits_t : uint16_t
```

Enumerator

k_usb_dvstctr0_rhst_mask	b2:b0: Reset handshake status
k_usb_dvstctr0_rhst_undecided	
k_usb_dvstctr0_rhst_ls	
k_usb_dvstctr0_rhst_fs	
k_usb_dvstctr0_rhst_reset	
k_usb_dvstctr0_uact	b4: USB bus enable
k_usb_dvstctr0_resume	b5: Resume output
k_usb_dvstctr0_usbrst	b6: USB bus reset output
k_usb_dvstctr0_rwupe	b7: Wakeup detection enable
k_usb_dvstctr0_wkup	b8: Wakeup output
k_usb_dvstctr0_vbusen	b9: VBUS output enable
k_usb_dvstctr0_exicen	b10: External IC output enable
k_usb_dvstctr0_hnpbtoa	b11: HNP control (B to A transition)

Definition at line 277 of file [rx72n_usb_regs.h](#).

```
00277         : uint16_t {
00278   k_usb_dvstctr0_rhst_mask  = 0x0007, /**< b2:b0: Reset handshake status */
```

```

00279 k_usb_dvstctr0_rhst_undecided = 0x0000,
00280 k_usb_dvstctr0_rhst_ls       = 0x0001,
00281 k_usb_dvstctr0_rhst_fs      = 0x0002,
00282 k_usb_dvstctr0_rhst_reset    = 0x0004,
00283 k_usb_dvstctr0_uact          = (1U « 4), /**< b4: USB bus enable */
00284 k_usb_dvstctr0_resume        = (1U « 5), /**< b5: Resume output */
00285 k_usb_dvstctr0_usbrst        = (1U « 6), /**< b6: USB bus reset output */
00286 k_usb_dvstctr0_rwupe         = (1U « 7), /**< b7: Wakeup detection enable */
00287 k_usb_dvstctr0_wkup          = (1U « 8), /**< b8: Wakeup output */
00288 k_usb_dvstctr0_vbusen        = (1U « 9), /**< b9: VBUS output enable */
00289 k_usb_dvstctr0_exicen        = (1U « 10), /**< b10: External IC output enable */
00290 k_usb_dvstctr0_hnpbtoa       = (1U « 11), /**< b11: HNP control (B to A transition) */
00291 } usb_dvstctr0_bits_t;

```

8.67.3.11 usb`fifotr`bits`

```
enum usb_fifoctr_bits_t : uint16_t
```

Enumerator

k_usb_fifoctr_dtlh_mask	b8:b0: Data length
k_usb_fifoctr_frdy	b13: FIFO ready
k_usb_fifoctr_bclr	b14: CPU buffer clear
k_usb_fifoctr_bval	b15: Buffer valid flag

Definition at line 476 of file [rx72n_usb_regs.h](#).

```

00476 : uint16_t {
00477 k_usb_fifoctr_dtlh_mask = 0x01FF, /**< b8:b0: Data length */
00478 k_usb_fifoctr_frdy      = (1U « 13), /**< b13: FIFO ready */
00479 k_usb_fifoctr_bclr      = (1U « 14), /**< b14: CPU buffer clear */
00480 k_usb_fifoctr_bval      = (1U « 15), /**< b15: Buffer valid flag */
00481 } usb_fifoctr_bits_t;

```

8.67.3.12 usb`intenb0`bits`

```
enum usb_intenb0_bits_t : uint16_t
```

Enumerator

k_usb_intenb0_brbye	b8: Buffer ready interrupt enable
k_usb_intenb0_nrdye	b9: Buffer not ready interrupt enable
k_usb_intenb0_bempe	b10: Buffer empty interrupt enable
k_usb_intenb0_ctre	b11: Control transfer stage transition interrupt enable
k_usb_intenb0_dvse	b12: Device state transition interrupt enable
k_usb_intenb0_sofe	b13: Frame number refresh interrupt enable
k_usb_intenb0_rsme	b14: Resume interrupt enable
k_usb_intenb0_vbse	b15: VBUS interrupt enable

Definition at line 294 of file [rx72n_usb_regs.h](#).

```

00294 : uint16_t {
00295 k_usb_intenb0_brbye = (1U « 8), /**< b8: Buffer ready interrupt enable */
00296 k_usb_intenb0_nrdye = (1U « 9), /**< b9: Buffer not ready interrupt enable */
00297 k_usb_intenb0_bempe = (1U « 10), /**< b10: Buffer empty interrupt enable */
00298 k_usb_intenb0_ctre  = (1U « 11), /**< b11: Control transfer stage transition interrupt enable */
00299 k_usb_intenb0_dvse  = (1U « 12), /**< b12: Device state transition interrupt enable */
00300 k_usb_intenb0_sofe  = (1U « 13), /**< b13: Frame number refresh interrupt enable */
00301 k_usb_intenb0_rsme  = (1U « 14), /**< b14: Resume interrupt enable */
00302 k_usb_intenb0_vbse  = (1U « 15), /**< b15: VBUS interrupt enable */
00303 } usb_intenb0_bits_t;

```

8.67.3.13 usb_intsts0_bits_t

```
enum usb\_intsts0\_bits\_t : uint16_t
```

Enumerator

k_usb_intsts0_ctsq_mask	b2:b0: Control transfer stage
k_usb_intsts0_ctsq_idle	
k_usb_intsts0_ctsq_rd_data	
k_usb_intsts0_ctsq_rd_status	
k_usb_intsts0_ctsq_wr_data	
k_usb_intsts0_ctsq_wr_status	
k_usb_intsts0_ctsq_wr_nd	
k_usb_intsts0_ctsq_seq_err	
k_usb_intsts0_valid	b3: Setup packet received
k_usb_intsts0_dvsq_mask	b6:b4: Device state
k_usb_intsts0_dvsq_powered	
k_usb_intsts0_dvsq_default	
k_usb_intsts0_dvsq_address	
k_usb_intsts0_dvsq_configured	
k_usb_intsts0_dvsq_suspend	
k_usb_intsts0_vbsts	b7: VBUS input status
k_usb_intsts0_brdy	b8: Buffer ready
k_usb_intsts0_nrdy	b9: Buffer not ready
k_usb_intsts0_bemp	b10: Buffer empty
k_usb_intsts0_ctrtr	b11: Control transfer stage transition
k_usb_intsts0_dvst	b12: Device state transition
k_usb_intsts0_sofr	b13: Frame number refresh
k_usb_intsts0_resm	b14: Resume
k_usb_intsts0_vbint	b15: VBUS

Definition at line 306 of file [rx72n_usb_regs.h](#).

```
00306         : uint16_t {
00307     k_usb_intsts0_ctsq_mask    = 0x0007, /**< b2:b0: Control transfer stage */
00308     k_usb_intsts0_ctsq_idle    = 0x0000,
00309     k_usb_intsts0_ctsq_rd_data = 0x0001,
00310     k_usb_intsts0_ctsq_rd_status = 0x0002,
00311     k_usb_intsts0_ctsq_wr_data = 0x0003,
00312     k_usb_intsts0_ctsq_wr_status = 0x0004,
00313     k_usb_intsts0_ctsq_wr_nd  = 0x0005,
00314     k_usb_intsts0_ctsq_seq_err = 0x0006,
00315     k_usb_intsts0_valid       = (1U « 3), /**< b3: Setup packet received */
00316     k_usb_intsts0_dvsq_mask    = (7U « 4), /**< b6:b4: Device state */
00317     k_usb_intsts0_dvsq_powered = (0U « 4),
00318     k_usb_intsts0_dvsq_default = (1U « 4),
00319     k_usb_intsts0_dvsq_address = (2U « 4),
00320     k_usb_intsts0_dvsq_configured = (3U « 4),
00321     k_usb_intsts0_dvsq_suspend = (4U « 4),
00322     k_usb_intsts0_vbsts        = (1U « 7), /**< b7: VBUS input status */
00323     k_usb_intsts0_brdy         = (1U « 8), /**< b8: Buffer ready */
00324     k_usb_intsts0_nrdy         = (1U « 9), /**< b9: Buffer not ready */
00325     k_usb_intsts0_bemp         = (1U « 10), /**< b10: Buffer empty */
00326     k_usb_intsts0_ctrtr        = (1U « 11), /**< b11: Control transfer stage transition */
00327     k_usb_intsts0_dvst         = (1U « 12), /**< b12: Device state transition */
00328     k_usb_intsts0_sofr         = (1U « 13), /**< b13: Frame number refresh */
00329     k_usb_intsts0_resm         = (1U « 14), /**< b14: Resume */
```

```
00330 k_usb_intsts0_vbint      = (1U « 15), /**< b15: VBUS */
00331 } usb_intsts0_bits_t;
```

8.67.3.14 usb_physlew_bits_t

```
enum usb_physlew_bits_t : uint32_t
```

Enumerator

k_usb_physlew_slewr00	Rising-edge slew rate, lane 0
k_usb_physlew_slewr01	Rising-edge slew rate, lane 1
k_usb_physlew_slewf00	Falling-edge slew rate, lane 0
k_usb_physlew_slewf01	Falling-edge slew rate, lane 1
k_usb_physlew_rx72n	Recommended RX72N USB0 value (tinyusb + FSP)

Definition at line 501 of file [rx72n_usb_regs.h](#).

```
00501      : uint32_t {
00502 k_usb_physlew_slewr00 = (1UL « 0), /**< Rising-edge slew rate, lane 0 */
00503 k_usb_physlew_slewr01 = (1UL « 1), /**< Rising-edge slew rate, lane 1 */
00504 k_usb_physlew_slewf00 = (1UL « 2), /**< Falling-edge slew rate, lane 0 */
00505 k_usb_physlew_slewf01 = (1UL « 3), /**< Falling-edge slew rate, lane 1 */
00506 k_usb_physlew_rx72n   = 0x00000005, /**< Recommended RX72N USB0 value (tinyusb + FSP) */
00507 } usb_physlew_bits_t;
```

8.67.3.15 usb_pipe69ctr_bits_t

```
enum usb_pipe69ctr_bits_t : uint16_t
```

PIPEnCTR bits for pipes 6-9 (manual p.1981).

Pipes 6-9 are interrupt-only pipes with a reduced control register layout. Bits ATREPM(b10) and INBUFM(b14) are reserved (not present) compared to the pipes 1-5 layout in [usb_pipectr_bits_t](#).

Enumerator

k_usb_pipe69ctr_pid_mask	b1:b0: Response PID
k_usb_pipe69ctr_pid_nak	
k_usb_pipe69ctr_pid_buf	
k_usb_pipe69ctr_pid_stall	
k_usb_pipe69ctr_pbusy	b5: Pipe busy flag
k_usb_pipe69ctr_sqmon	b6: Sequence toggle bit monitor
k_usb_pipe69ctr_sqset	b7: Sequence toggle bit set
k_usb_pipe69ctr_sqclr	b8: Sequence toggle bit clear
k_usb_pipe69ctr_aclrm	b9: Auto buffer clear mode
k_usb_pipe69ctr_bsts	b15: Buffer status flag

Definition at line 419 of file [rx72n_usb_regs.h](#).

```
00419      : uint16_t {
00420 k_usb_pipe69ctr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00421 k_usb_pipe69ctr_pid_nak = 0x0000,
00422 k_usb_pipe69ctr_pid_buf = 0x0001,
```



```

00423 k_usb_pipe69ctr_pid_stall = 0x0002,
00424 k_usb_pipe69ctr_pbusy    = (1U « 5), /**< b5: Pipe busy flag */
00425 k_usb_pipe69ctr_sqmon    = (1U « 6), /**< b6: Sequence toggle bit monitor */
00426 k_usb_pipe69ctr_sqset    = (1U « 7), /**< b7: Sequence toggle bit set */
00427 k_usb_pipe69ctr_sqclr    = (1U « 8), /**< b8: Sequence toggle bit clear */
00428 k_usb_pipe69ctr_aclrm    = (1U « 9), /**< b9: Auto buffer clear mode */
00429 k_usb_pipe69ctr_bsts     = (1U « 15), /**< b15: Buffer status flag */
00430 } usb_pipe69ctr_bits_t;

```

8.67.3.16 usb_pipe_bits_t

```
enum usb_pipe_bits_t : uint16_t
```

USB pipe interrupt control bits.

Bit definitions for BRDYENB/BRDYSTS (buffer ready), BEMPENB/BEMPSTS (buffer empty), and NRDYENB/NRDYSTS (not ready) registers. Each bit corresponds to one pipe (bit 0 = DCP, bits 1-9 = pipes 1-9).

Enumerator

k_usb_pipe_bit_0	DCP (Default Control Pipe)
k_usb_pipe_bit_1	Data pipe 1
k_usb_pipe_bit_2	Data pipe 2
k_usb_pipe_bit_3	Data pipe 3
k_usb_pipe_bit_4	Data pipe 4
k_usb_pipe_bit_5	Data pipe 5
k_usb_pipe_bit_6	Data pipe 6
k_usb_pipe_bit_7	Data pipe 7
k_usb_pipe_bit_8	Data pipe 8
k_usb_pipe_bit_9	Data pipe 9
k_usb_pipe_all	All 10 pipes (bits 9:0)

Definition at line 363 of file rx72n_usb_regs.h.

```

00363 : uint16_t {
00364 k_usb_pipe_bit_0 = (1U « 0), /**< DCP (Default Control Pipe) */
00365 k_usb_pipe_bit_1 = (1U « 1), /**< Data pipe 1 */
00366 k_usb_pipe_bit_2 = (1U « 2), /**< Data pipe 2 */
00367 k_usb_pipe_bit_3 = (1U « 3), /**< Data pipe 3 */
00368 k_usb_pipe_bit_4 = (1U « 4), /**< Data pipe 4 */
00369 k_usb_pipe_bit_5 = (1U « 5), /**< Data pipe 5 */
00370 k_usb_pipe_bit_6 = (1U « 6), /**< Data pipe 6 */
00371 k_usb_pipe_bit_7 = (1U « 7), /**< Data pipe 7 */
00372 k_usb_pipe_bit_8 = (1U « 8), /**< Data pipe 8 */
00373 k_usb_pipe_bit_9 = (1U « 9), /**< Data pipe 9 */
00374 k_usb_pipe_all  = 0x03FF, /**< All 10 pipes (bits 9:0) */
00375 } usb_pipe_bits_t;

```

8.67.3.17 `usb_pipecfg_bits_t`

```
enum usb\_pipecfg\_bits\_t : uint16_t
```

Enumerator

<code>k_usb_pipecfg_epnum_mask</code>	b3:b0: Endpoint number
<code>k_usb_pipecfg_dir</code>	b4: Transfer direction (0=receive,1=transmit)
<code>k_usb_pipecfg_shtnak</code>	b7: Pipe disabled at end of transfer
<code>k_usb_pipecfg_dblb</code>	b9: Double buffer mode (pipes 1-5 only)
<code>k_usb_pipecfg_bfre</code>	b10: BRDY interrupt operation specification
<code>k_usb_pipecfg_type_mask</code>	b15:b14: Transfer type
<code>k_usb_pipecfg_type_bulk</code>	
<code>k_usb_pipecfg_type_int</code>	
<code>k_usb_pipecfg_type_iso</code>	

Definition at line 378 of file [rx72n_usb_regs.h](#).

```
00378      : uint16_t {
00379  k_usb_pipecfg_epnum_mask = 0x000F,    /**< b3:b0: Endpoint number */
00380  k_usb_pipecfg_dir      = (1U « 4),    /**< b4: Transfer direction (0=receive,1=transmit) */
00381  k_usb_pipecfg_shtnak   = (1U « 7),    /**< b7: Pipe disabled at end of transfer */
00382  k_usb_pipecfg_dblb     = (1U « 9),    /**< b9: Double buffer mode (pipes 1-5 only) */
00383  k_usb_pipecfg_bfre     = (1U « 10),   /**< b10: BRDY interrupt operation specification */
00384  k_usb_pipecfg_type_mask = (3U « 14),   /**< b15:b14: Transfer type */
00385  k_usb_pipecfg_type_bulk = (1U « 14),
00386  k_usb_pipecfg_type_int  = (2U « 14),
00387  k_usb_pipecfg_type_iso  = (3U « 14),
00388 } usb_pipecfg_bits_t;
```

8.67.3.18 `usb_pipectr_bits_t`

```
enum usb\_pipectr\_bits\_t : uint16_t
```

PIPEnCTR bits for pipes 1-5 (manual p.1976).

Note

Pipes 1-5 have all bits including ATREPM(b10) and INBUFM(b14).

Pipes 6-9 have a reduced set – see [usb_pipe69ctr_bits_t](#).

Enumerator

<code>k_usb_pipectr_pid_mask</code>	b1:b0: Response PID
<code>k_usb_pipectr_pid_nak</code>	
<code>k_usb_pipectr_pid_buf</code>	
<code>k_usb_pipectr_pid_stall</code>	
<code>k_usb_pipectr_pbusy</code>	b5: Pipe busy flag
<code>k_usb_pipectr_sqmon</code>	b6: Sequence toggle bit monitor
<code>k_usb_pipectr_sqset</code>	b7: Sequence toggle bit set
<code>k_usb_pipectr_sqclr</code>	b8: Sequence toggle bit clear

k_usb_pipectr_aclrm	b9: Auto buffer clear mode
k_usb_pipectr_atrepm	b10: Auto response mode (pipes 1-5 only)
k_usb_pipectr_inbufm	b14: Transmit buffer monitor (pipes 1-5 only)
k_usb_pipectr_bsts	b15: Buffer status flag

Definition at line 396 of file [rx72n_usb_regs.h](#).

```

00396      : uint16_t {
00397  k_usb_pipectr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00398  k_usb_pipectr_pid_nak  = 0x0000,
00399  k_usb_pipectr_pid_buf  = 0x0001,
00400  k_usb_pipectr_pid_stall = 0x0002,
00401  k_usb_pipectr_pbusy    = (1U « 5), /**< b5: Pipe busy flag */
00402  k_usb_pipectr_sqmon    = (1U « 6), /**< b6: Sequence toggle bit monitor */
00403  k_usb_pipectr_sqset    = (1U « 7), /**< b7: Sequence toggle bit set */
00404  k_usb_pipectr_sqclr    = (1U « 8), /**< b8: Sequence toggle bit clear */
00405  k_usb_pipectr_aclrm    = (1U « 9), /**< b9: Auto buffer clear mode */
00406  k_usb_pipectr_atrepm   = (1U « 10), /**< b10: Auto response mode (pipes 1-5 only) */
00407  k_usb_pipectr_inbufm   = (1U « 14), /**< b14: Transmit buffer monitor (pipes 1-5 only) */
00408  k_usb_pipectr_bsts     = (1U « 15), /**< b15: Buffer status flag */
00409 } usb_pipectr_bits_t;

```

8.67.3.19 usb_pipetre_bits_t

```
enum usb_pipetre_bits_t : uint16_t
```

Enumerator

k_usb_pipetre_trclr	b8: Transaction counter clear
k_usb_pipetre_trenb	b9: Transaction counter enable

Definition at line 433 of file [rx72n_usb_regs.h](#).

```

00433      : uint16_t {
00434  k_usb_pipetre_trclr = (1U « 8), /**< b8: Transaction counter clear */
00435  k_usb_pipetre_trenb = (1U « 9), /**< b9: Transaction counter enable */
00436 } usb_pipetre_bits_t;

```

8.67.3.20 usb_register_offsets_t

```
enum usb_register_offsets_t : uint16_t
```

USB register offsets (verified against manual Chapter 40).

Enumerator

k_usb_offset_syscfg	
k_usb_offset_syssts0	
k_usb_offset_dvstctr0	
k_usb_offset_cfifo	
k_usb_offset_d0fifo	
k_usb_offset_d1fifo	
k_usb_offset_cfifosel	
k_usb_offset_cfioctr	

k_usb_offset_d0fifosel	
k_usb_offset_d0fifotr	
k_usb_offset_d1fifosel	
k_usb_offset_d1fifotr	
k_usb_offset_intenb0	
k_usb_offset_intenb1	
k_usb_offset_brdyenb	
k_usb_offset_nrdyenb	
k_usb_offset_bempenb	
k_usb_offset_sofcfg	
k_usb_offset_intsts0	
k_usb_offset_intsts1	
k_usb_offset_brdysts	
k_usb_offset_nrdysts	
k_usb_offset_bempsts	
k_usb_offset_frmnum	
k_usb_offset_dvchgr	
k_usb_offset_usbaddr	
k_usb_offset_usbreq	
k_usb_offset_usbval	
k_usb_offset_usbindx	
k_usb_offset_usbleng	
k_usb_offset_dcpcfg	
k_usb_offset_dcpmaxp	
k_usb_offset_dcpctr	
k_usb_offset_pipesel	
k_usb_offset_pipecfg	
k_usb_offset_pipemaxp	
k_usb_offset_pipeperi	
k_usb_offset_pipe1ctr	
k_usb_offset_pipe9ctr	
k_usb_offset_pipe1tre	
k_usb_offset_pipe5trn	
k_usb_offset_devadd0	
k_usb_offset_devadd5	
k_usb_offset_physlew	

Definition at line 43 of file [rx72n_usb_regs.h](#).

```

00043     : uint16_t {
00044 k_usb_offset_syscfg    = 0x00,
00045 k_usb_offset_syssts0   = 0x04,
00046 k_usb_offset_dvstctr0  = 0x08,
00047 k_usb_offset_cfifo     = 0x14,
00048 k_usb_offset_d0fifo    = 0x18,
00049 k_usb_offset_d1fifo    = 0x1C,
00050 k_usb_offset_cfifosel  = 0x20,
00051 k_usb_offset_cfifotr   = 0x22,

```

```

00052 k_usb_offset_d0fifosel = 0x28,
00053 k_usb_offset_d0fifoctr = 0x2A,
00054 k_usb_offset_d1fifosel = 0x2C,
00055 k_usb_offset_d1fifoctr = 0x2E,
00056 k_usb_offset_intenb0  = 0x30,
00057 k_usb_offset_intenb1  = 0x32,
00058 k_usb_offset_brdyenb  = 0x36,
00059 k_usb_offset_nrdyenb   = 0x38,
00060 k_usb_offset_bempenb   = 0x3A,
00061 k_usb_offset_sofcfg    = 0x3C,
00062 k_usb_offset_intsts0    = 0x40,
00063 k_usb_offset_intsts1   = 0x42,
00064 k_usb_offset_brdysts   = 0x46,
00065 k_usb_offset_nrdysts   = 0x48,
00066 k_usb_offset_bempsts   = 0x4A,
00067 k_usb_offset_frmnum    = 0x4C,
00068 k_usb_offset_dvchgr     = 0x4E,
00069 k_usb_offset_usbaddr   = 0x50,
00070 k_usb_offset_usbreql    = 0x54,
00071 k_usb_offset_usbval     = 0x56,
00072 k_usb_offset_usbindx   = 0x58,
00073 k_usb_offset_usbleng    = 0x5A,
00074 k_usb_offset_dcpcfg     = 0x5C,
00075 k_usb_offset_dcpmaxp    = 0x5E,
00076 k_usb_offset_dcpctr     = 0x60,
00077 k_usb_offset_pipesel    = 0x64,
00078 k_usb_offset_pipecfg    = 0x68,
00079 k_usb_offset_pipemaxp   = 0x6C,
00080 k_usb_offset_pipeperi    = 0x6E,
00081 k_usb_offset_pipe1ctr   = 0x70,
00082 k_usb_offset_pipe9ctr   = 0x80,
00083 k_usb_offset_pipe1tre    = 0x90,
00084 k_usb_offset_pipe5trn   = 0xA2,
00085 k_usb_offset_devadd0    = 0xD0,
00086 k_usb_offset_devadd5    = 0xDA,
00087 k_usb_offset_physlew    = 0xF0,
00088 } usb_register_offsets_t;

```

8.67.3.21 usb_reserved_sizes_t

```
enum usb_reserved_sizes_t : uint8_t
```

Reserved field sizes for USB register structure.

Enumerator

k_usb_reserved_02_04_bytes	
k_usb_reserved_06_08_bytes	
k_usb_reserved_0a_14_bytes	
k_usb_reserved_24_28_bytes	
k_usb_reserved_34_36_bytes	
k_usb_reserved_3e_40_bytes	
k_usb_reserved_44_46_bytes	
k_usb_reserved_52_54_bytes	
k_usb_reserved_62_64_bytes	
k_usb_reserved_66_68_bytes	
k_usb_reserved_6a_6c_bytes	
k_usb_reserved_82_90_bytes	
k_usb_reserved_a4_d0_bytes	
k_usb_reserved_dc_f0_bytes	

Definition at line 93 of file [rx72n_usb_regs.h](#).

```

00093         : uint8_t {
00094     k_usb_reserved_02_04_bytes = 2,
00095     k_usb_reserved_06_08_bytes = 2,
00096     k_usb_reserved_0a_14_bytes = 10,
00097     k_usb_reserved_24_28_bytes = 4,
00098     k_usb_reserved_34_36_bytes = 2,
00099     k_usb_reserved_3e_40_bytes = 2,
00100     k_usb_reserved_44_46_bytes = 2,
00101     k_usb_reserved_52_54_bytes = 2,
00102     k_usb_reserved_62_64_bytes = 2,
00103     k_usb_reserved_66_68_bytes = 2,
00104     k_usb_reserved_6a_6c_bytes = 2,
00105     k_usb_reserved_82_90_bytes = 14,
00106     k_usb_reserved_a4_d0_bytes = 44,
00107     k_usb_reserved_dc_f0_bytes = 20,
00108 } usb_reserved_sizes_t;

```

8.67.3.22 usb_syscfg_bits_t

```
enum usb_syscfg_bits_t : uint16_t
```

Enumerator

k_usb_syscfg_usbe	b0: USB operation enable
k_usb_syscfg_dprpu	b4: D+ pull-up resistor (function mode)
k_usb_syscfg_drpd	b5: D+/D- pull-down resistors (host mode)
k_usb_syscfg_dcfm	b6: Controller function select (0=function,1=host)
k_usb_syscfg_scke	b10: USB clock enable

Definition at line 255 of file `rx72n_usb_regs.h`.

```

00255         : uint16_t {
00256     k_usb_syscfg_usbe = (1U « 0), /**< b0: USB operation enable */
00257     k_usb_syscfg_dprpu = (1U « 4), /**< b4: D+ pull-up resistor (function mode) */
00258     k_usb_syscfg_drpd = (1U « 5), /**< b5: D+/D- pull-down resistors (host mode) */
00259     k_usb_syscfg_dcfm = (1U « 6), /**< b6: Controller function select (0=function,1=host) */
00260     k_usb_syscfg_scke = (1U « 10), /**< b10: USB clock enable */
00261 } usb_syscfg_bits_t;

```

8.67.3.23 usb_syssts0_bits_t

```
enum usb_syssts0_bits_t : uint16_t
```

Enumerator

k_usb_syssts0_lnst_mask	b1:b0: USB data line status
k_usb_syssts0_lnst_se0	
k_usb_syssts0_lnst_fs_j	
k_usb_syssts0_lnst_fs_k	
k_usb_syssts0_lnst_se1	
k_usb_syssts0_idmon	b2: ID pin monitor
k_usb_syssts0_sofea	b5: SOF active
k_usb_syssts0_htact	b6: Host sequencer active
k_usb_syssts0_ovcmon_mask	b15:b14: Overcurrent monitor

Definition at line 264 of file [rx72n_usb_regs.h](#).

```
00264     : uint16_t {
00265     k_usb_syssts0_linst_mask = 0x0003, /**< b1:b0: USB data line status */
00266     k_usb_syssts0_linst_se0 = 0x0000,
00267     k_usb_syssts0_linst_fs_j = 0x0001,
00268     k_usb_syssts0_linst_fs_k = 0x0002,
00269     k_usb_syssts0_linst_sel = 0x0003,
00270     k_usb_syssts0_idmon    = (1U « 2), /**< b2: ID pin monitor */
00271     k_usb_syssts0_sofea    = (1U « 5), /**< b5: SOF active */
00272     k_usb_syssts0_htact    = (1U « 6), /**< b6: Host sequencer active */
00273     k_usb_syssts0_ovcmon_mask = (3U « 14), /**< b15:b14: Overcurrent monitor */
00274 } usb_syssts0_bits_t;
```

8.67.4 Function Documentation

8.67.4.1 usb0()

```
volatile rx_usb_regs_t * usb0 (
    void ) [inline], [static]
```

Definition at line 240 of file [rx72n_usb_regs.h](#).

```
00241 {
00242     return (volatile rx_usb_regs_t*)k_usb0_base_addr;
00243 }
```

References [k_usb0_base_addr](#).

8.67.4.2 usb`dpusr0r`()

```
volatile uint32_t * usb_dpusr0r (
    void ) [inline], [static]
```

Get pointer to USB deep-standby transceiver control register (DPUSR0R), 32-bit, at absolute 0x000↵A0400.

Definition at line 249 of file [rx72n_usb_regs.h](#).

```
00250 {
00251     return (volatile uint32_t*)k_usb_dpusr0r_addr;
00252 }
```

References [k_usb_dpusr0r_addr](#).

8.68 rx72n`usb`regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_usb_regs.h
00003  * @brief RX72N USB 2.0 Full-Speed Host/Function Module Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the USB 2.0 Full-Speed Host/Function Module (USB0)
00007  * verified against RX72N Group User's Manual: Hardware (R01UH0824EJ0111),
00008  * Chapter 40 "USB 2.0 FS Host/Function Module (USBb)".
00009  *
00010  * @par Verification Status (2026-04-20):
00011  * - All register addresses verified against manual Chapter 40
00012  * - 45 registers defined with correct offsets
00013  * - 55+ static assertions verify all offsets at compile time
00014  * - Removed non-existent registers (BUSWAIT, PLLSTA, TESTMODE, D0FBCFG,
00015  *   D1FBCFG, PHYSET, PIPEBUF, LPCTRL, LPSTS - these do NOT exist on RX72N)
00016  * - Renamed UFRMNUM to DVCHGR (correct register name per manual)
00017  * - CRITICAL FIX (2026-02-03): Changed FIFO registers from uint32_t to uint16_t
```

```

00018 * per manual Ch40 p.1941. CFIFO/D0FIFO/D1FIFO are 16-bit WORD registers,
00019 * NOT 32-bit. Added padding to maintain correct struct offsets.
00020 * - FIX (2026-04-20): Added missing DVSTCTR0 bits VBUSEN(b9), EXICEN(b10),
00021 * HNPBTOA(b11) per manual p.1940.
00022 * - FIX (2026-04-20): Split usb_fifoel_bits_t into usb_cffosel_bits_t and
00023 * usb_dnfifosel_bits_t. CFIFOSEL has ISEL(b5) but NOT DREQE/DCLRM.
00024 * D0/D1FIFOSEL have DREQE(b12)/DCLRM(b13) but NOT ISEL. Per manual p.1942-1944.
00025 *
00026 * @date 2026-04-20
00027 * @copyright Copyright (c) 2026 Locked Inc.
00028 * SPDX-License-Identifier: MIT
00029 */
00030
00031 #pragma once
00032
00033 #include <stdint.h>
00034 #include <stdint.h>
00035
00036 #ifdef __cplusplus
00037 extern "C" {
00038 #endif
00039
00040 /**
00041  * @brief USB register offsets (verified against manual Chapter 40)
00042  */
00043 typedef enum : uint16_t {
00044     k_usb_offset_syscfg = 0x00,
00045     k_usb_offset_syssts0 = 0x04,
00046     k_usb_offset_dvstctr0 = 0x08,
00047     k_usb_offset_cfifo = 0x14,
00048     k_usb_offset_d0fifo = 0x18,
00049     k_usb_offset_d1fifo = 0x1C,
00050     k_usb_offset_cffosel = 0x20,
00051     k_usb_offset_cffoctr = 0x22,
00052     k_usb_offset_d0fifosel = 0x28,
00053     k_usb_offset_d0fifoctr = 0x2A,
00054     k_usb_offset_d1fifosel = 0x2C,
00055     k_usb_offset_d1fifoctr = 0x2E,
00056     k_usb_offset_intenb0 = 0x30,
00057     k_usb_offset_intenb1 = 0x32,
00058     k_usb_offset_brdyenb = 0x36,
00059     k_usb_offset_nrdyenb = 0x38,
00060     k_usb_offset_bempenb = 0x3A,
00061     k_usb_offset_sofcfg = 0x3C,
00062     k_usb_offset_intsts0 = 0x40,
00063     k_usb_offset_intsts1 = 0x42,
00064     k_usb_offset_brdsts = 0x46,
00065     k_usb_offset_nrdsts = 0x48,
00066     k_usb_offset_bempsts = 0x4A,
00067     k_usb_offset_frmnum = 0x4C,
00068     k_usb_offset_dvchgr = 0x4E,
00069     k_usb_offset_usbaddr = 0x50,
00070     k_usb_offset_usbreq = 0x54,
00071     k_usb_offset_usbval = 0x56,
00072     k_usb_offset_usbindx = 0x58,
00073     k_usb_offset_usbleng = 0x5A,
00074     k_usb_offset_dcpcfg = 0x5C,
00075     k_usb_offset_dcpmaxp = 0x5E,
00076     k_usb_offset_dcpctr = 0x60,
00077     k_usb_offset_pipesel = 0x64,
00078     k_usb_offset_pipecfg = 0x68,
00079     k_usb_offset_pipemaxp = 0x6C,
00080     k_usb_offset_pipeperi = 0x6E,
00081     k_usb_offset_pipe1ctr = 0x70,
00082     k_usb_offset_pipe9ctr = 0x80,
00083     k_usb_offset_pipe1tre = 0x90,
00084     k_usb_offset_pipe5trn = 0xA2,
00085     k_usb_offset_devadd0 = 0xD0,
00086     k_usb_offset_devadd5 = 0xDA,
00087     k_usb_offset_physlew = 0xF0,
00088 } usb_register_offsets_t;
00089
00090 /**
00091  * @brief Reserved field sizes for USB register structure
00092  */
00093 typedef enum : uint8_t {
00094     k_usb_reserved_02_04_bytes = 2,
00095     k_usb_reserved_06_08_bytes = 2,
00096     k_usb_reserved_0a_14_bytes = 10,
00097     k_usb_reserved_24_28_bytes = 4,
00098     k_usb_reserved_34_36_bytes = 2,
00099     k_usb_reserved_3e_40_bytes = 2,
00100     k_usb_reserved_44_46_bytes = 2,
00101     k_usb_reserved_52_54_bytes = 2,
00102     k_usb_reserved_62_64_bytes = 2,
00103     k_usb_reserved_66_68_bytes = 2,
00104     k_usb_reserved_6a_6c_bytes = 2,

```



```

00105 k_usb_reserved_82_90_bytes = 14,
00106 k_usb_reserved_a4_d0_bytes = 44,
00107 k_usb_reserved_dc_f0_bytes = 20,
00108 } usb_reserved_sizes_t;
00109
00110 /**
00111 * @struct rx_usb_regs_t
00112 * @brief USB0 Register Map (Base: 0x000A0000)
00113 *
00114 * @note The following registers do NOT exist on RX72N (removed from struct):
00115 * @BUSWAIT, PLLSTA, TESTMODE, D0FBCFG, D1FBCFG, PHYSET, PIPEBUF, LPCTRL, LPSTS
00116 *
00117 * @note FIFO registers (CFIFO, D0FIFO, D1FIFO) are 16-bit WORD registers per
00118 * manual Ch40 p.1941. Access width controlled by CFIFOSEL.MBW/D0FIFOSEL.MBW:
00119 * - MBW=1 (k_usb_cfifosel_mbw_16): 16-bit word access (b15-b0)
00120 * - MBW=0 (k_usb_cfifosel_mbw_8): 8-bit byte access (b7-b0)
00121 */
00122 typedef struct {
00123     volatile uint16_t syscfg;
00124     uint8_t reserved0[k_usb_reserved_02_04_bytes];
00125     volatile uint16_t syssts0;
00126     uint8_t reserved1[k_usb_reserved_06_08_bytes];
00127     volatile uint16_t dvstctr0;
00128     uint8_t reserved2[k_usb_reserved_0a_14_bytes];
00129     volatile uint16_t cfifo; /**< Common FIFO port @ 0x14 (16-bit WORD, manual p.1941) */
00130     uint16_t reserved_cfifo; /**< Padding @ 0x16 to maintain D0FIFO offset */
00131     volatile uint16_t d0ffifo; /**< D0 FIFO port @ 0x18 (16-bit WORD, manual p.1941) */
00132     uint16_t reserved_d0ffifo; /**< Padding @ 0x1A */
00133     volatile uint16_t d1ffifo; /**< D1 FIFO port @ 0x1C (16-bit WORD, manual p.1941) */
00134     uint16_t reserved_d1ffifo; /**< Padding @ 0x1E */
00135     volatile uint16_t cfifosel;
00136     volatile uint16_t cfifoctr;
00137     uint8_t reserved3[k_usb_reserved_24_28_bytes];
00138     volatile uint16_t d0ffifosel;
00139     volatile uint16_t d0ffifoctr;
00140     volatile uint16_t d1ffifosel;
00141     volatile uint16_t d1ffifoctr;
00142     volatile uint16_t intenb0;
00143     volatile uint16_t intenb1;
00144     uint8_t reserved4[k_usb_reserved_34_36_bytes];
00145     volatile uint16_t brdyenb;
00146     volatile uint16_t nrdyenb;
00147     volatile uint16_t bempenb;
00148     volatile uint16_t sofcfg;
00149     uint8_t reserved5[k_usb_reserved_3e_40_bytes];
00150     volatile uint16_t intsts0;
00151     volatile uint16_t intsts1;
00152     uint8_t reserved6[k_usb_reserved_44_46_bytes];
00153     volatile uint16_t brdysts;
00154     volatile uint16_t nrdysts;
00155     volatile uint16_t bempsts;
00156     volatile uint16_t frmmnum;
00157     volatile uint16_t dvchgr; /**< Device State Change (NOT ufrmmnum!) */
00158     volatile uint16_t usbaddr;
00159     uint8_t reserved7[k_usb_reserved_52_54_bytes];
00160     volatile uint16_t usbreq;
00161     volatile uint16_t usbval;
00162     volatile uint16_t usbidx;
00163     volatile uint16_t usbleng;
00164     volatile uint16_t dcpcfg;
00165     volatile uint16_t dcpmaxp;
00166     volatile uint16_t dcpctr;
00167     uint8_t reserved8[k_usb_reserved_62_64_bytes];
00168     volatile uint16_t pipesel;
00169     uint8_t reserved9[k_usb_reserved_66_68_bytes];
00170     volatile uint16_t pipecfg;
00171     uint8_t reserved10[k_usb_reserved_6a_6c_bytes];
00172     volatile uint16_t pipemaxp;
00173     volatile uint16_t pipeperi;
00174     volatile uint16_t pipe1ctr;
00175     volatile uint16_t pipe2ctr;
00176     volatile uint16_t pipe3ctr;
00177     volatile uint16_t pipe4ctr;
00178     volatile uint16_t pipe5ctr;
00179     volatile uint16_t pipe6ctr;
00180     volatile uint16_t pipe7ctr;
00181     volatile uint16_t pipe8ctr;
00182     volatile uint16_t pipe9ctr;
00183     uint8_t reserved11[k_usb_reserved_82_90_bytes];
00184     volatile uint16_t pipe1tre;
00185     volatile uint16_t pipe1trn;
00186     volatile uint16_t pipe2tre;
00187     volatile uint16_t pipe2trn;
00188     volatile uint16_t pipe3tre;
00189     volatile uint16_t pipe3trn;
00190     volatile uint16_t pipe4tre;
00191     volatile uint16_t pipe4trn;

```

```

00192 volatile uint16_t pipe5tre;
00193 volatile uint16_t pipe5trn;
00194 uint8_t reserved12[k_usb_reserved_a4_d0_bytes];
00195 volatile uint16_t devadd0;
00196 volatile uint16_t devadd1;
00197 volatile uint16_t devadd2;
00198 volatile uint16_t devadd3;
00199 volatile uint16_t devadd4;
00200 volatile uint16_t devadd5;
00201 uint8_t reserved13[k_usb_reserved_dc_f0_bytes];
00202 volatile uint32_t
00203 physlew; /**< PHY Cross-Point Adjustment @0xF0 (32-bit R/W per hirakuni45 rw32_t) */
00204 } rx_usb_regs_t;
00205
00206 /**
00207  * @enum rx_usb_addresses_t
00208  * @brief USB0 peripheral base address (verified against RX72N Hardware Manual Ch40)
00209  *
00210  * @details
00211  * Memory-mapped base address for the USB 2.0 Full-Speed Host/Function Module
00212  * (USB0). Cast to a volatile pointer via usb0() to access the register map.
00213  *
00214  * The underlying type is uintptr_t to correctly express that this value is a
00215  * memory address, not a plain integer. On RX72N (32-bit) uintptr_t == uint32_t,
00216  * so there is no runtime difference.
00217  *
00218  * @par Memory Layout
00219  * @verbatim
00220  *   Address      | Peripheral
00221  *   -----+-----
00222  *   0x000A0000    | USB0 register block (see rx_usb_regs_t)
00223  * @endverbatim
00224  *
00225  * @see rx_usb_regs_t Register map structure accessed via this address
00226  * @see usb0() Inline accessor that casts this address to a pointer
00227  *
00228  * @since Version 1.0.0
00229  */
00230 typedef enum : uintptr_t {
00231     k_usb0_base_addr = 0x000A0000, /**< USB0 register block base address (manual p.1928, Ch40) */
00232     /** Deep-standby USB transceiver control / pin monitor register. Shared
00233      * with USB-common (outside the USB0 rx_usb_regs_t block). 32-bit R/W
00234      * at absolute 0x000A0400 per hirakuni45 RX600/usb.hpp and Renesas FSP
00235      * iodef.h. Used to release the PHY from "output fixed" state after
00236      * deep-standby wakeup or first boot. */
00237     k_usb_dpusr0r_addr = 0x000A0400,
00238 } rx_usb_addresses_t;
00239
00240 static inline volatile rx_usb_regs_t* usb0(void)
00241 {
00242     return (volatile rx_usb_regs_t*)k_usb0_base_addr;
00243 }
00244
00245 /**
00246  * @brief Get pointer to USB deep-standby transceiver control register
00247  * (DPUSR0R), 32-bit, at absolute 0x000A0400.
00248  */
00249 static inline volatile uint32_t* usb_dpusr0r(void)
00250 {
00251     return (volatile uint32_t*)k_usb_dpusr0r_addr;
00252 }
00253
00254 /** SYSCFG bits (manual p.1931) */
00255 typedef enum : uint16_t {
00256     k_usb_syscfg_usbe = (1U << 0), /**< b0: USB operation enable */
00257     k_usb_syscfg_dprpu = (1U << 4), /**< b4: D+ pull-up resistor (function mode) */
00258     k_usb_syscfg_drpd = (1U << 5), /**< b5: D+/D- pull-down resistors (host mode) */
00259     k_usb_syscfg_dcfm = (1U << 6), /**< b6: Controller function select (0=function,1=host) */
00260     k_usb_syscfg_scke = (1U << 10), /**< b10: USB clock enable */
00261 } usb_syscfg_bits_t;
00262
00263 /** SYSSTS0 bits (manual p.1934) */
00264 typedef enum : uint16_t {
00265     k_usb_syssts0_lnst_mask = 0x0003, /**< b1:b0: USB data line status */
00266     k_usb_syssts0_lnst_se0 = 0x0000,
00267     k_usb_syssts0_lnst_fs_j = 0x0001,
00268     k_usb_syssts0_lnst_fs_k = 0x0002,
00269     k_usb_syssts0_lnst_sel = 0x0003,
00270     k_usb_syssts0_idmon = (1U << 2), /**< b2: ID pin monitor */
00271     k_usb_syssts0_sofea = (1U << 5), /**< b5: SOF active */
00272     k_usb_syssts0_htact = (1U << 6), /**< b6: Host sequencer active */
00273     k_usb_syssts0_ovcmon_mask = (3U << 14), /**< b15:b14: Overcurrent monitor */
00274 } usb_syssts0_bits_t;
00275
00276 /** DVSTCTR0 bits (manual p.1937) */
00277 typedef enum : uint16_t {
00278     k_usb_dvstctr0_rhst_mask = 0x0007, /**< b2:b0: Reset handshake status */

```

```

00279 k_usb_dvstctr0_rhst_undecided = 0x0000,
00280 k_usb_dvstctr0_rhst_ls      = 0x0001,
00281 k_usb_dvstctr0_rhst_fs     = 0x0002,
00282 k_usb_dvstctr0_rhst_reset  = 0x0004,
00283 k_usb_dvstctr0_uact        = (1U « 4), /**< b4: USB bus enable */
00284 k_usb_dvstctr0_resume      = (1U « 5), /**< b5: Resume output */
00285 k_usb_dvstctr0_usbrst      = (1U « 6), /**< b6: USB bus reset output */
00286 k_usb_dvstctr0_rwupe       = (1U « 7), /**< b7: Wakeup detection enable */
00287 k_usb_dvstctr0_wkup        = (1U « 8), /**< b8: Wakeup output */
00288 k_usb_dvstctr0_vbusen      = (1U « 9), /**< b9: VBUS output enable */
00289 k_usb_dvstctr0_exicen      = (1U « 10), /**< b10: External IC output enable */
00290 k_usb_dvstctr0_hnpbtoa     = (1U « 11), /**< b11: HNP control (B to A transition) */
00291 } usb_dvstctr0_bits_t;
00292
00293 /* INTENB0 bits (manual p.1944) */
00294 typedef enum : uint16_t {
00295     k_usb_intenb0_brbye = (1U « 8), /**< b8: Buffer ready interrupt enable */
00296     k_usb_intenb0_nrdye = (1U « 9), /**< b9: Buffer not ready interrupt enable */
00297     k_usb_intenb0_bempe = (1U « 10), /**< b10: Buffer empty interrupt enable */
00298     k_usb_intenb0_ctre  = (1U « 11), /**< b11: Control transfer stage transition interrupt enable */
00299     k_usb_intenb0_dvse  = (1U « 12), /**< b12: Device state transition interrupt enable */
00300     k_usb_intenb0_sofe  = (1U « 13), /**< b13: Frame number refresh interrupt enable */
00301     k_usb_intenb0_rsme  = (1U « 14), /**< b14: Resume interrupt enable */
00302     k_usb_intenb0_vbse  = (1U « 15), /**< b15: VBUS interrupt enable */
00303 } usb_intenb0_bits_t;
00304
00305 /* INTSTS0 bits (manual p.1951) */
00306 typedef enum : uint16_t {
00307     k_usb_intsts0_ctsq_mask = 0x0007, /**< b2:b0: Control transfer stage */
00308     k_usb_intsts0_ctsq_idle = 0x0000,
00309     k_usb_intsts0_ctsq_rd_data = 0x0001,
00310     k_usb_intsts0_ctsq_rd_status = 0x0002,
00311     k_usb_intsts0_ctsq_wr_data = 0x0003,
00312     k_usb_intsts0_ctsq_wr_status = 0x0004,
00313     k_usb_intsts0_ctsq_wr_nd = 0x0005,
00314     k_usb_intsts0_ctsq_seq_err = 0x0006,
00315     k_usb_intsts0_valid = (1U « 3), /**< b3: Setup packet received */
00316     k_usb_intsts0_dvsq_mask = (7U « 4), /**< b6:b4: Device state */
00317     k_usb_intsts0_dvsq_powered = (0U « 4),
00318     k_usb_intsts0_dvsq_default = (1U « 4),
00319     k_usb_intsts0_dvsq_address = (2U « 4),
00320     k_usb_intsts0_dvsq_configured = (3U « 4),
00321     k_usb_intsts0_dvsq_suspend = (4U « 4),
00322     k_usb_intsts0_vbsts = (1U « 7), /**< b7: VBUS input status */
00323     k_usb_intsts0_brdy = (1U « 8), /**< b8: Buffer ready */
00324     k_usb_intsts0_nrdy = (1U « 9), /**< b9: Buffer not ready */
00325     k_usb_intsts0_bempe = (1U « 10), /**< b10: Buffer empty */
00326     k_usb_intsts0_ctrtr = (1U « 11), /**< b11: Control transfer stage transition */
00327     k_usb_intsts0_dvst = (1U « 12), /**< b12: Device state transition */
00328     k_usb_intsts0_sofr = (1U « 13), /**< b13: Frame number refresh */
00329     k_usb_intsts0_resm = (1U « 14), /**< b14: Resume */
00330     k_usb_intsts0_vbint = (1U « 15), /**< b15: VBUS */
00331 } usb_intsts0_bits_t;
00332
00333 /* DCPCFG bits (manual p.1966) */
00334 typedef enum : uint16_t {
00335     k_usb_dcpcfg_dir = (1U « 4), /**< b4: Transfer direction (0=receive,1=transmit) */
00336     k_usb_dcpcfg_shtnak = (1U « 7), /**< b7: Pipe disabled at end of transfer */
00337 } usb_dcpcfg_bits_t;
00338
00339 /* DCPCTR bits (manual p.1968) */
00340 typedef enum : uint16_t {
00341     k_usb_dcpctr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00342     k_usb_dcpctr_pid_nak = 0x0000,
00343     k_usb_dcpctr_pid_buf = 0x0001,
00344     k_usb_dcpctr_pid_stall = 0x0002,
00345     k_usb_dcpctr_ccpl = (1U « 2), /**< b2: Control transfer end enable */
00346     k_usb_dcpctr_pbuse = (1U « 5), /**< b5: Pipe busy flag */
00347     k_usb_dcpctr_sqmon = (1U « 6), /**< b6: Sequence toggle bit monitor */
00348     k_usb_dcpctr_sqset = (1U « 7), /**< b7: Sequence toggle bit set */
00349     k_usb_dcpctr_sqclr = (1U « 8), /**< b8: Sequence toggle bit clear */
00350     k_usb_dcpctr_sureqclr = (1U « 11), /**< b11: SUREQ bit clear */
00351     k_usb_dcpctr_sureq = (1U « 14), /**< b14: Setup token transmission */
00352     k_usb_dcpctr_bsts = (1U « 15), /**< b15: Buffer status flag */
00353 } usb_dcpctr_bits_t;
00354
00355 /**
00356  * @brief USB pipe interrupt control bits
00357  *
00358  * @details
00359  * Bit definitions for BRDYENB/BRDYSTS (buffer ready), BEMPENB/BEMPSTS
00360  * (buffer empty), and NRDYENB/NRDYSTS (not ready) registers.
00361  * Each bit corresponds to one pipe (bit 0 = DCP, bits 1-9 = pipes 1-9).
00362  */
00363 typedef enum : uint16_t {
00364     k_usb_pipe_bit_0 = (1U « 0), /**< DCP (Default Control Pipe) */
00365     k_usb_pipe_bit_1 = (1U « 1), /**< Data pipe 1 */

```

```

00366 k_usb_pipe_bit_2 = (1U « 2), /**< Data pipe 2 */
00367 k_usb_pipe_bit_3 = (1U « 3), /**< Data pipe 3 */
00368 k_usb_pipe_bit_4 = (1U « 4), /**< Data pipe 4 */
00369 k_usb_pipe_bit_5 = (1U « 5), /**< Data pipe 5 */
00370 k_usb_pipe_bit_6 = (1U « 6), /**< Data pipe 6 */
00371 k_usb_pipe_bit_7 = (1U « 7), /**< Data pipe 7 */
00372 k_usb_pipe_bit_8 = (1U « 8), /**< Data pipe 8 */
00373 k_usb_pipe_bit_9 = (1U « 9), /**< Data pipe 9 */
00374 k_usb_pipe_all = 0x03FF, /**< All 10 pipes (bits 9:0) */
00375 } usb_pipe_bits_t;
00376
00377 /* PIPECFG bits (manual p.1972) */
00378 typedef enum : uint16_t {
00379     k_usb_pipecfg_epnum_mask = 0x000F, /**< b3:b0: Endpoint number */
00380     k_usb_pipecfg_dir = (1U « 4), /**< b4: Transfer direction (0=receive,1=transmit) */
00381     k_usb_pipecfg_shtnak = (1U « 7), /**< b7: Pipe disabled at end of transfer */
00382     k_usb_pipecfg_dblb = (1U « 9), /**< b9: Double buffer mode (pipes 1-5 only) */
00383     k_usb_pipecfg_bfre = (1U « 10), /**< b10: BRDY interrupt operation specification */
00384     k_usb_pipecfg_type_mask = (3U « 14), /**< b15:b14: Transfer type */
00385     k_usb_pipecfg_type_bulk = (1U « 14),
00386     k_usb_pipecfg_type_int = (2U « 14),
00387     k_usb_pipecfg_type_iso = (3U « 14),
00388 } usb_pipecfg_bits_t;
00389
00390 /**
00391  * @brief PIPEnCTR bits for pipes 1-5 (manual p.1976)
00392  *
00393  * @note Pipes 1-5 have all bits including ATREPM(b10) and INBUFM(b14).
00394  * @note Pipes 6-9 have a reduced set -- see usb_pipe69ctr_bits_t.
00395  */
00396 typedef enum : uint16_t {
00397     k_usb_pipectr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00398     k_usb_pipectr_pid_nak = 0x0000,
00399     k_usb_pipectr_pid_buf = 0x0001,
00400     k_usb_pipectr_pid_stall = 0x0002,
00401     k_usb_pipectr_pbusy = (1U « 5), /**< b5: Pipe busy flag */
00402     k_usb_pipectr_sqmon = (1U « 6), /**< b6: Sequence toggle bit monitor */
00403     k_usb_pipectr_sqset = (1U « 7), /**< b7: Sequence toggle bit set */
00404     k_usb_pipectr_sqclr = (1U « 8), /**< b8: Sequence toggle bit clear */
00405     k_usb_pipectr_aclrm = (1U « 9), /**< b9: Auto buffer clear mode */
00406     k_usb_pipectr_atrepm = (1U « 10), /**< b10: Auto response mode (pipes 1-5 only) */
00407     k_usb_pipectr_inbufm = (1U « 14), /**< b14: Transmit buffer monitor (pipes 1-5 only) */
00408     k_usb_pipectr_bsts = (1U « 15), /**< b15: Buffer status flag */
00409 } usb_pipectr_bits_t;
00410
00411 /**
00412  * @brief PIPEnCTR bits for pipes 6-9 (manual p.1981)
00413  *
00414  * @details
00415  * Pipes 6-9 are interrupt-only pipes with a reduced control register layout.
00416  * Bits ATREPM(b10) and INBUFM(b14) are reserved (not present) compared to
00417  * the pipes 1-5 layout in usb_pipectr_bits_t.
00418  */
00419 typedef enum : uint16_t {
00420     k_usb_pipe69ctr_pid_mask = 0x0003, /**< b1:b0: Response PID */
00421     k_usb_pipe69ctr_pid_nak = 0x0000,
00422     k_usb_pipe69ctr_pid_buf = 0x0001,
00423     k_usb_pipe69ctr_pid_stall = 0x0002,
00424     k_usb_pipe69ctr_pbusy = (1U « 5), /**< b5: Pipe busy flag */
00425     k_usb_pipe69ctr_sqmon = (1U « 6), /**< b6: Sequence toggle bit monitor */
00426     k_usb_pipe69ctr_sqset = (1U « 7), /**< b7: Sequence toggle bit set */
00427     k_usb_pipe69ctr_sqclr = (1U « 8), /**< b8: Sequence toggle bit clear */
00428     k_usb_pipe69ctr_aclrm = (1U « 9), /**< b9: Auto buffer clear mode */
00429     k_usb_pipe69ctr_bsts = (1U « 15), /**< b15: Buffer status flag */
00430 } usb_pipe69ctr_bits_t;
00431
00432 /* PIPEnTRE bits for pipes 1-5 (manual p.1984) */
00433 typedef enum : uint16_t {
00434     k_usb_pipetre_trclr = (1U « 8), /**< b8: Transaction counter clear */
00435     k_usb_pipetre_trenb = (1U « 9), /**< b9: Transaction counter enable */
00436 } usb_pipetre_bits_t;
00437
00438 /**
00439  * @brief CFIFOSEL bits (manual p.1942)
00440  *
00441  * @note CFIFOSEL has ISEL(b5) for direction select.
00442  * @note CFIFOSEL does NOT have DREQE or DCLRM (those are D0/D1 only).
00443  */
00444 typedef enum : uint16_t {
00445     k_usb_cfifosel_curpipe_mask = 0x000F, /**< b3:b0: Port access pipe */
00446     k_usb_cfifosel_curpipe_dcp = 0x0000,
00447     k_usb_cfifosel_isel = (1U « 5), /**< b5: Access direction (0=read,1=write) */
00448     k_usb_cfifosel_bigend = (1U « 8), /**< b8: Endian select */
00449     k_usb_cfifosel_mbw_mask = (1U « 10), /**< b10: MBW bit */
00450     k_usb_cfifosel_mbw_8 = (0U « 10), /**< 8-bit access */
00451     k_usb_cfifosel_mbw_16 = (1U « 10), /**< 16-bit access */
00452     k_usb_cfifosel_rew = (1U « 14), /**< b14: Read pointer rewind */

```

```

00453 k_usb_cfifosel_rcnt      = (1U « 15), /**< b15: Read count mode */
00454 } usb_cfifosel_bits_t;
00455
00456 /**
00457  * @brief D0/FIFOSEL / D1/FIFOSEL bits (manual p.1943-1944)
00458  *
00459  * @note D0/D1/FIFOSEL have DREQE(b12) and DCLRM(b13) for DMA/DTC.
00460  * @note D0/D1/FIFOSEL do NOT have ISEL (direction select is always host RX).
00461  */
00462 typedef enum : uint16_t {
00463     k_usb_dnifosel_curpipe_mask = 0x000F, /**< b3:b0: Port access pipe */
00464     k_usb_dnifosel_curpipe_dcp = 0x0000,
00465     k_usb_dnifosel_bigend      = (1U « 8), /**< b8: Endian select */
00466     k_usb_dnifosel_mbw_mask   = (1U « 10), /**< b10: MBW bit */
00467     k_usb_dnifosel_mbw_8      = (0U « 10), /**< 8-bit access */
00468     k_usb_dnifosel_mbw_16     = (1U « 10), /**< 16-bit access */
00469     k_usb_dnifosel_dreque     = (1U « 12), /**< b12: DMA/DTC transfer request enable */
00470     k_usb_dnifosel_dclrm      = (1U « 13), /**< b13: Auto buffer clear mode */
00471     k_usb_dnifosel_rew        = (1U « 14), /**< b14: Read pointer rewind */
00472     k_usb_dnifosel_rcnt      = (1U « 15), /**< b15: Read count mode */
00473 } usb_dnifosel_bits_t;
00474
00475 /* FIFOCTR bits (manual p.1945 CFIFOCTR, p.1946 D0/D1/FIFOCTR) */
00476 typedef enum : uint16_t {
00477     k_usb_fifoctr_dtl_n_mask = 0x01FF, /**< b8:b0: Data length */
00478     k_usb_fifoctr_frdy       = (1U « 13), /**< b13: FIFO ready */
00479     k_usb_fifoctr_bclr       = (1U « 14), /**< b14: CPU buffer clear */
00480     k_usb_fifoctr_bval       = (1U « 15), /**< b15: Buffer valid flag */
00481 } usb_fifoctr_bits_t;
00482
00483 /* DPUSR0R bits (Deep-Standby USB Transceiver Control / Pin Monitor,
00484  * 32-bit, absolute 0x000A0400; fields from hirakuni45 RX600/usb.hpp
00485  * and RX72N HW manual Ch40). */
00486 typedef enum : uint32_t {
00487     k_usb_dpusr0r_srpc0      = (1UL « 0), /**< USB0 single-ended receiver power control */
00488     k_usb_dpusr0r_rpu0       = (1UL « 1), /**< USB0 DP pull-up resistor enable (deep-standby) */
00489     k_usb_dpusr0r_fixphy0    = (1UL « 4), /**< USB0 PHY output fixed (1=fixed, 0=normal) */
00490     k_usb_dpusr0r_dp0        = (1UL « 16), /**< USB0 D+ line monitor */
00491     k_usb_dpusr0r_dm0        = (1UL « 17), /**< USB0 D- line monitor */
00492     k_usb_dpusr0r_dovca0     = (1UL « 20), /**< USB0 overcurrent flag A */
00493     k_usb_dpusr0r_dovcb0     = (1UL « 21), /**< USB0 overcurrent flag B */
00494     k_usb_dpusr0r_dvbsts0    = (1UL « 23), /**< USB0 VBUS status */
00495 } usb_dpusr0r_bits_t;
00496
00497 /* PHYSLEW bits (PHY Cross-Point Adjustment, 32-bit, USB0 offset 0xF0).
00498  * RX72N manual Ch40 + tinyusb renesas/usb_a DCD: for RX72N USB0 the
00499  * recommended slew-rate setting is SLEWR00=1, SLEWF00=1 => value 0x5.
00500  * Other RX parts leave this register at its reset default. */
00501 typedef enum : uint32_t {
00502     k_usb_physlew_slewr00    = (1UL « 0), /**< Rising-edge slew rate, lane 0 */
00503     k_usb_physlew_slewr01    = (1UL « 1), /**< Rising-edge slew rate, lane 1 */
00504     k_usb_physlew_slewf00    = (1UL « 2), /**< Falling-edge slew rate, lane 0 */
00505     k_usb_physlew_slewf01    = (1UL « 3), /**< Falling-edge slew rate, lane 1 */
00506     k_usb_physlew_rx72n      = 0x00000005, /**< Recommended RX72N USB0 value (tinyusb + FSP) */
00507 } usb_physlew_bits_t;
00508
00509 /* Interrupt vectors.
00510  *
00511  * D0/FIFO0 / D1/FIFO0 / USBR0 are FIXED ICU vectors on RX72N (Renesas FSP
00512  * iodef.h: VECT_USB0_D0/FIFO0=34, D1/FIFO0=35, USBR0=90).
00513  *
00514  * USBI0 is NOT a fixed vector on RX72N. It is a Group-B software
00515  * configurable interrupt (SELECTB) with source code 62 (RX72N HW manual
00516  * Ch15 Table 15.3, cross-checked against hirakuni45/RX RX72N/icu.hpp
00517  * SELECTB::USBI0 = 62). To use it, firmware must pick any vector slot
00518  * in the SELECTB range (144..207, per SLIBRn register range in the HW
00519  * manual) and program ICU.SLIBRn = k_usb0_usbi_sli_src. We pick 144 as
00520  * the first free SELECTB slot.
00521  *
00522  * Earlier revisions of this enum hard-coded k_vect_usb0_usbi = 36, which
00523  * is a reserved gap in the RX72N fixed vector table -- IER/IPR/IR writes
00524  * landed on an inert slot and the USBI ISR never fired. */
00525 typedef enum : uint8_t {
00526     k_vect_usb0_d0fifo = 34,
00527     k_vect_usb0_d1fifo = 35,
00528     k_vect_usb0_usbi   = 144,
00529     k_vect_usb0_usbr   = 90,
00530 } rx_usb_interrupt_vector_t;
00531
00532 /* SELECTB source codes (written to ICU.SLIBRn to map a peripheral source
00533  * onto a SELECTB vector slot). Manual Ch15 Table 15.3. */
00534 typedef enum : uint8_t {
00535     k_usb0_usbi_sli_src = 62, /**< USBI0 source code for ICU.SLIBR[144..207] */
00536 } rx_usb_sli_source_t;
00537
00538 /* CDC endpoints */
00539 typedef enum : uint8_t {

```



```

00540 k_usb_cdc_ep_ctrl      = 0,
00541 k_usb_cdc_ep_bulk_in   = 1,
00542 k_usb_cdc_ep_bulk_out  = 2,
00543 k_usb_cdc_ep_interrupt_in = 3,
00544 k_usb_cdc_max_packet_fs = 64,
00545 } usb_cdc_endpoints_t;
00546
00547 /* Static assertions -- every register offset verified against manual Ch40 */
00548 static_assert(k_usb0_base_addr == 0x000A0000, "USB0 base");
00549 static_assert(offsetof(rx_usb_regs_t, syscfg) == 0x00, "SYSCFG");
00550 static_assert(offsetof(rx_usb_regs_t, syssts0) == 0x04, "SYSSTS0");
00551 static_assert(offsetof(rx_usb_regs_t, dvstctr0) == 0x08, "DVSTCTR0");
00552 static_assert(offsetof(rx_usb_regs_t, cfifo) == 0x14, "CFIFO");
00553 static_assert(offsetof(rx_usb_regs_t, d0fifo) == 0x18, "D0FIFO");
00554 static_assert(offsetof(rx_usb_regs_t, d1fifo) == 0x1C, "D1FIFO");
00555 static_assert(offsetof(rx_usb_regs_t, cfifosel) == 0x20, "CFIFOSEL");
00556 static_assert(offsetof(rx_usb_regs_t, cfifoctr) == 0x22, "CFIFOCTR");
00557 static_assert(offsetof(rx_usb_regs_t, d0fifosel) == 0x28, "D0FIFOSEL");
00558 static_assert(offsetof(rx_usb_regs_t, d0fifoctr) == 0x2A, "D0FIFOCTR");
00559 static_assert(offsetof(rx_usb_regs_t, d1fifosel) == 0x2C, "D1FIFOSEL");
00560 static_assert(offsetof(rx_usb_regs_t, d1fifoctr) == 0x2E, "D1FIFOCTR");
00561 static_assert(offsetof(rx_usb_regs_t, intenb0) == 0x30, "INTENB0");
00562 static_assert(offsetof(rx_usb_regs_t, intenb1) == 0x32, "INTENB1");
00563 static_assert(offsetof(rx_usb_regs_t, brdyenb) == 0x36, "BRDYENB");
00564 static_assert(offsetof(rx_usb_regs_t, nrdyenb) == 0x38, "NRDYENB");
00565 static_assert(offsetof(rx_usb_regs_t, bempenb) == 0x3A, "BEMPENB");
00566 static_assert(offsetof(rx_usb_regs_t, sofcfg) == 0x3C, "SOFCFG");
00567 static_assert(offsetof(rx_usb_regs_t, intsts0) == 0x40, "INTSTS0");
00568 static_assert(offsetof(rx_usb_regs_t, intsts1) == 0x42, "INTSTS1");
00569 static_assert(offsetof(rx_usb_regs_t, brdysts) == 0x46, "BRDYSTS");
00570 static_assert(offsetof(rx_usb_regs_t, nrdysts) == 0x48, "NRDYSTS");
00571 static_assert(offsetof(rx_usb_regs_t, bempsts) == 0x4A, "BEMPSTS");
00572 static_assert(offsetof(rx_usb_regs_t, frmnum) == 0x4C, "FRMNUM");
00573 static_assert(offsetof(rx_usb_regs_t, dvchgr) == 0x4E, "DVCHGR");
00574 static_assert(offsetof(rx_usb_regs_t, usbaddr) == 0x50, "USBADDR");
00575 static_assert(offsetof(rx_usb_regs_t, usbreq) == 0x54, "USBREQ");
00576 static_assert(offsetof(rx_usb_regs_t, usbval) == 0x56, "USBVAL");
00577 static_assert(offsetof(rx_usb_regs_t, usbindx) == 0x58, "USBINDX");
00578 static_assert(offsetof(rx_usb_regs_t, usbleng) == 0x5A, "USBLENG");
00579 static_assert(offsetof(rx_usb_regs_t, dcpcfg) == 0x5C, "DCPCFG");
00580 static_assert(offsetof(rx_usb_regs_t, dcpmaxp) == 0x5E, "DCPMAXP");
00581 static_assert(offsetof(rx_usb_regs_t, dcpctr) == 0x60, "DCPCTR");
00582 static_assert(offsetof(rx_usb_regs_t, pipesel) == 0x64, "PIPESEL");
00583 static_assert(offsetof(rx_usb_regs_t, pipecfg) == 0x68, "PIPECFG");
00584 static_assert(offsetof(rx_usb_regs_t, pipemaxp) == 0x6C, "PIPEMAXP");
00585 static_assert(offsetof(rx_usb_regs_t, pipeperi) == 0x6E, "PIPEPERI");
00586 static_assert(offsetof(rx_usb_regs_t, pipe1ctr) == 0x70, "PIPE1CTR");
00587 static_assert(offsetof(rx_usb_regs_t, pipe9ctr) == 0x80, "PIPE9CTR");
00588 static_assert(offsetof(rx_usb_regs_t, pipe1tre) == 0x90, "PIPE1TRE");
00589 static_assert(offsetof(rx_usb_regs_t, pipe5trn) == 0xA2, "PIPE5TRN");
00590 static_assert(offsetof(rx_usb_regs_t, devadd0) == 0xD0, "DEVADD0");
00591 static_assert(offsetof(rx_usb_regs_t, devadd5) == 0xDA, "DEVADD5");
00592 static_assert(offsetof(rx_usb_regs_t, physlew) == 0xF0, "PHYSLEW");
00593
00594 #ifdef __cplusplus
00595 }
00596 #endif

```

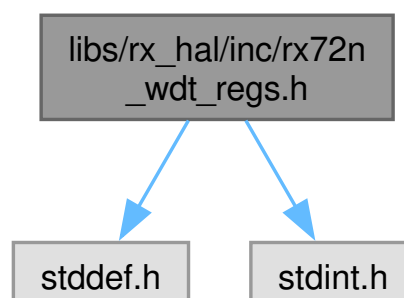
8.69 libs/rx_hal/inc/rx72n_wdt_regs.h File Reference

RX72N WDT Watchdog Timer Register Definitions.

```
#include <stddef.h>
```

```
#include <stdint.h>
```

Include dependency graph for rx72n_wdt_regs.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_wdt_regs_t](#)
WDT Register Map Structure.

Enumerations

- enum [rx_wdt_addresses_t](#) : [uintptr_t](#) { [k_wdt_base_addr](#) = 0x00088020 }
WDT base address constant.
- enum [wdt_reserved_sizes_t](#) : [uint8_t](#) { [k_wdt_reserved_after_wdtrr_bytes](#) = 1 , [k_wdt_reserved_after_wdtrcr_bytes](#) = 1 }
WDT register reserved field sizes.
- enum [wdt_refresh_sequence_t](#) : [uint8_t](#) { [k_wdt_refresh_start](#) = 0x00 , [k_wdt_refresh_end](#) = 0xFF }
WDT Refresh Register (WDTRR) Write Sequence Values.
- enum [wdt_wdtrcr_pos_t](#) : [uint8_t](#) { [k_wdt_cr_cks_pos](#) = 4 , [k_wdt_cr_rpes_pos](#) = 8 , [k_wdt_cr_rpss_pos](#) = 12 }
WDT Control Register (WDTCR) Bit Field Positions.
- enum [wdt_wdtrcr_bits_t](#) : [uint16_t](#) {
[k_wdt_tops_1024](#) = 0x0000 , [k_wdt_tops_4096](#) = 0x0001 , [k_wdt_tops_8192](#) = 0x0002 ,
[k_wdt_tops_16384](#) = 0x0003 ,
[k_wdt_cks_div_4](#) = (0x01 << [k_wdt_cr_cks_pos](#)) , [k_wdt_cks_div_64](#) = (0x04 << [k_wdt_cr_cks_pos](#)) , [k_wdt_cks_div_128](#) = (0x0F << [k_wdt_cr_cks_pos](#)) , [k_wdt_cks_div_512](#) = (0x06 << [k_wdt_cr_cks_pos](#)) ,
[k_wdt_cks_div_2048](#) = (0x07 << [k_wdt_cr_cks_pos](#)) , [k_wdt_cks_div_8192](#) = (0x08 << [k_wdt_cr_cks_pos](#)) , [k_wdt_rpes_75](#) = (0x00 << [k_wdt_cr_rpes_pos](#)) , [k_wdt_rpes_50](#) = (0x01 << [k_wdt_cr_rpes_pos](#)) ,
[k_wdt_rpes_25](#) = (0x02 << [k_wdt_cr_rpes_pos](#)) , [k_wdt_rpes_0](#) = (0x03 << [k_wdt_cr_rpes_pos](#)) , [k_wdt_rpss_25](#) = (0x00 << [k_wdt_cr_rpss_pos](#)) , [k_wdt_rpss_50](#) = (0x01 << [k_wdt_cr_rpss_pos](#)) ,
[k_wdt_rpss_75](#) = (0x02 << [k_wdt_cr_rpss_pos](#)) , [k_wdt_rpss_100](#) = (0x03 << [k_wdt_cr_rpss_pos](#)) }
WDT Control Register (WDTCR) Bit Field Values.
- enum [wdt_wdtsr_pos_t](#) : [uint8_t](#) { [k_wdt_sr_undff_pos](#) = 14 , [k_wdt_sr_refef_pos](#) = 15 }
WDT Status Register (WDTSR) Bit Field Positions.
- enum [wdt_wdtsr_bits_t](#) : [uint16_t](#) { [k_wdt_sr_cntval_mask](#) = 0x3FFF , [k_wdt_sr_undff](#) = (1U << [k_wdt_sr_undff_pos](#)) , [k_wdt_sr_refef](#) = (1U << [k_wdt_sr_refef_pos](#)) }
WDT Status Register (WDTSR) Bit Field Values.
- enum [wdt_wdtrcr_bits_t](#) : [uint8_t](#) { [k_wdt_rcr_rstirqs_pos](#) = 7 , [k_wdt_rcr_rstirqs_mask](#) = (1U << [k_wdt_rcr_rstirqs_pos](#)) , [k_wdt_rstirqs_reset](#) = (1U << [k_wdt_rcr_rstirqs_pos](#)) , [k_wdt_rstirqs_nmi](#) = 0x00 }
WDT Reset Control Register (WDTRCR) Bit Field Values.

Functions

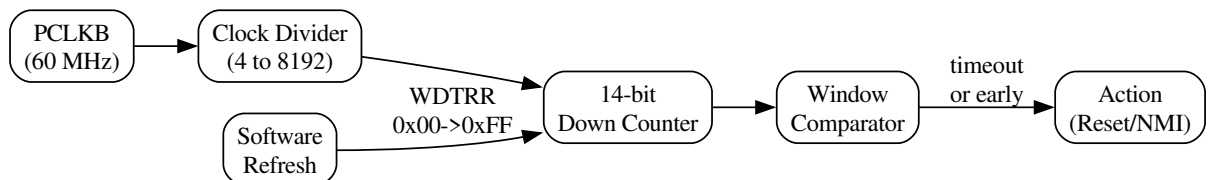
- static volatile [rx_wdt_regs_t](#) * [wdt](#) (void)
Get pointer to WDT registers.

8.69.1 Detailed Description

RX72N WDT Watchdog Timer Register Definitions.

Register definitions for the Watchdog Timer (WDT) module on the RX72N microcontroller. The WDT provides system monitoring to detect software runaway conditions and reset the system if necessary.

System Architecture



WDT vs IWDTC Comparison

Feature	WDT (This Module)	IWDTC
Clock Source	PCLKB (60 MHz)	Dedicated 120 kHz
Stop in Software	Yes	No (once started)
Debug/Sleep Ctrl	No	Yes (IWDTCSTPR)
Best For	Development, debugging	Production safety
Timeout Range	4 us to 2.2 s	8.5 ms to 17.5 s

Key Features

- 14-bit down counter using PCLKB peripheral clock
- Configurable timeout period: 1024 to 16384 cycles
- Configurable clock division: 4 to 8192
- Optional window protection to detect too-early refresh
- Can be stopped in software (unlike IWDTC)
- Generates reset or NMI on timeout (configurable)

Timeout Calculation

$$T_{timeout} = \frac{TOPS \times CKS}{f_{PCLKB}}$$

Example with PCLKB = 60 MHz, TOPS = 16384, CKS = 8192:

$$T_{timeout} = \frac{16384 \times 8192}{60,000,000} = 2.24 \text{ seconds}$$

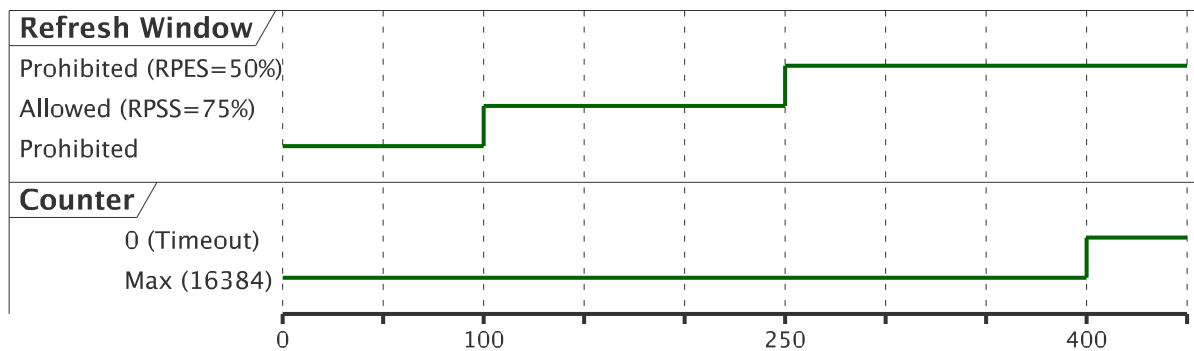
Timeout Period Table (PCLKB = 60 MHz)

TOPS	CKS/ $\frac{PCLKB}{4}$	CKS/64	CKS/128	CKS/8192
1024	17 μ s	1.1 ms	2.2 ms	140 ms
4096	68 μ s	4.4 ms	8.7 ms	559 ms
8192	137 μ s	8.7 ms	17.5 ms	1.12 s
16384	273 μ s	17.5 ms	34.9 ms	2.24 s

Window Protection

The WDT supports window protection to detect both late refresh (timeout) and early refresh (software error). The refresh window is defined by:

- RPSS: Window start position (when refresh becomes allowed)
- RPES: Window end position (when refresh becomes prohibited)



Hardware Requirements

Parameter	Value	Notes
Clock Source	PCLKB	60 MHz in STAR configuration
Base Address	0x00088020	8-byte register block
Interrupt Vector	WDT_WOVI	Watchdog overflow (NMI mode)
Register Count	4	WDTRR, WDTCR, WDTSR, WDTRCR

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] N/A (no loops in register definitions)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All accessor functions are single-statement
- Rule 5: [OK] N/A (hardware layer)
- Rule 6: [OK] Minimal scope
- Rule 7: [OK] N/A (no return values to check)
- Rule 8: [OK] All constants use C23 typed enums
- Rule 9: [OK] No function pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Single responsibility - only WDT register definitions
- O: Open for extension via additional bit field enums
- L: N/A (no inheritance hierarchy)
- I: Minimal interface - only necessary accessor
- D: N/A (hardware register layer)

Module Dependencies

- `<stddef.h>` - `offsetof` macro for static assertions
- `<stdint.h>` - Fixed-width integer types

Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)

- Chapter 34: Watchdog Timer (WDT), pages 1547-1558

See also

[rx72n_iwdt_regs.h](#) Independent Watchdog Timer (production use)

[rx_wdt.h](#) Higher-level WDT API

[rx72n_regs.h](#) Main register include file

Author

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Date

2026-01-28

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Since

Version 1.0.0

Definition in file [rx72n_wdt_regs.h](#).

8.69.2 Data Structure Documentation

8.69.2.1 struct rx_wdt_regs_t

WDT Register Map Structure.

Watchdog Timer (WDT) registers for software-controllable watchdog functionality. Uses PCLKB clock and can be stopped in software (unlike IWDT).

Memory Layout Table

Offset	Size	Field	Type	Description
0x00	1	wdtrr	uint8↔ _t	Refresh Register
0x01	1	reserved0	-	Reserved
0x02	2	wdtcr	uint16↔ _t	Control Register
0x04	2	wdtsr	uint16↔ _t	Status Register
0x06	1	wdtrcr	uint8↔ _t	Reset Control Register
0x07	1	reserved1	-	Reserved
Total	8			(packed, no additional padding)

WDT Refresh Sequence

The watchdog is refreshed by writing a specific sequence to WDTRR:

1. Write 0x00 to WDTRR
2. Write 0xFF to WDTRR

This two-step sequence prevents accidental refresh from single-byte errors.

Initialization Example

```
// Configure WDT for 2-second timeout with window protection
volatile rx_wdt_regs_t* wdt_regs = wdt();

// 1. Configure control register (before starting)
//   TOPS=16384 cycles, CKS=div8192, RPSS=100%, RPES=0%
wdt_regs->wdtcr = k_wdt_tops_16384 | k_wdt_cks_div_8192 |
    k_wdt_rpss_100 | k_wdt_rpes_0;

// 2. Configure to generate reset on timeout
wdt_regs->wdtrcr = k_wdt_rstirqs_reset;

// 3. Start WDT by performing first refresh
wdt_regs->wdtrr = k_wdt_refresh_start; // Write 0x00
wdt_regs->wdtrr = k_wdt_refresh_end;   // Write 0xFF
```

Refresh Example (in main loop)

```

void app_main_loop(void) {
    while (1) {
        // Perform application tasks
        process_commands();
        update_motors();

        // Refresh watchdog (must complete both writes atomically)
        wdt()->wdtrr = k_wdt_refresh_start; // 0x00
        wdt()->wdtrr = k_wdt_refresh_end;   // 0xFF

        // Wait for next cycle
        tx_thread_sleep(10);
    }
}

```

Note

WDT has 4 registers only. WDTCSTPR (count stop) exists for IWDT but NOT for WDT.

Invariant

sizeof(rx_wdt_regs_t) == 8 bytes
 All offsets match RX72N Hardware Manual Ch34

Warning

Refresh must complete both writes (0x00 then 0xFF) atomically
 Do not refresh outside the valid window if window protection is enabled

See also

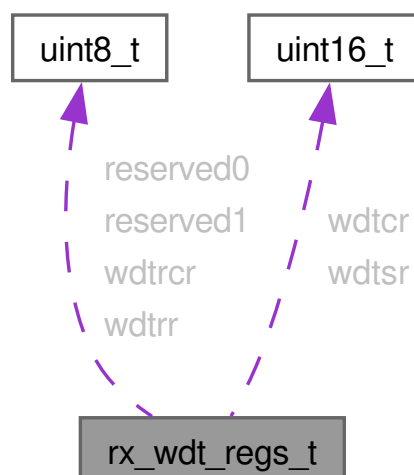
[wdt\(\)](#) Accessor function
[wdt_refresh_sequence_t](#) Refresh values
[wdt_wdtrcr_bits_t](#) Control register bits

Since

Version 1.0.0

Definition at line 266 of file [rx72n_wdt_regs.h](#).

Collaboration diagram for rx_wdt_regs_t:



Data Fields

uint8_t	reserved0↵ [k_wdt_reserved_after_wdtrr_bytes]	Reserved @ 0x01
uint8_t	reserved1↵ [k_wdt_reserved_after_wdtrcr_bytes]	Reserved @ 0x07
volatile uint16_t	wdtcr	WDT Control Register (WDTCR) @ offset 0x02. Configures timeout period, clock division, and window parameters. Must be configured before starting the WDT (before first refresh). Bit Fields: • Bits 1:0 (TOPS): Timeout period selection • Bits 7:4 (CKS):↵ Clock division ratio • Bits 9:8 (RPES):↵ Window end position • Bits 13:12 (RPSS): Window start position See also wdt_wdtrcr_bits_t Bit definitions
volatile uint8_t	wdtrcr	WDT Reset Control Register (WDTRCR) @ offset 0x06. Selects action on timeout: reset or NMI. Bit 7 (RSTIRQS): • 0: Generate NMI on timeout • 1: Generate internal reset on timeout See also wdt_wdtrcr_bits_t Bit definitions

volatile uint8_t	wdtrr	WDT Refresh Register (WDTRR) @ offset 0x00. Write 0x00 then 0xFF to refresh the watchdog counter. Both writes must complete within the valid window if window protection is enabled. See also wdt_refresh_sequence_t Refresh values Warning Single byte writes are invalid - must write both values in sequence
volatile uint16_t	wdtsr	WDT Status Register (WDTSR) @ offset 0x04. Contains current counter value and error flags. Bit Fields: • Bits 13:0 (CNTVAL): Current down-counter value (read-only) • Bit 14 (UNDFE): Under-flow flag (timeout occurred) • Bit 15 (REFEF): Refresh error flag (window violation) See also wdt_wdtsr_bits_t Bit definitions

8.69.3 Enumeration Type Documentation

8.69.3.1 rx'wdt'addresses't

enum [rx_wdt_addresses_t](#) : uintptr_t

WDT base address constant.

Base address for the Watchdog Timer register block at 0x00088020. The WDT has an 8-byte register space with 4 registers and 2 reserved bytes.

Memory Map

Address	Register	Size	Description
0x00088020	WDTRR	1	Refresh Register
0x00088021	Reserved	1	-
0x00088022	WDTCR	2	Control Register
0x00088024	WDTSR	2	Status Register
0x00088026	WDTRCR	1	Reset Control Register
0x00088027	Reserved	1	-

See also

[wdt\(\)](#) Accessor function

Since

Version 1.0.0

Enumerator

<code>k_wdt_base_addr</code>	WDT register base address (0x00088020). Verified against RX72N Hardware Manual Ch34
------------------------------	---

Definition at line 172 of file [rx72n_wdt_regs.h](#).

```

00172      : uintptr_t {
00173      /**
00174       * @brief WDT register base address (0x00088020)
00175       * @details Verified against RX72N Hardware Manual Ch34
00176       */
00177      k_wdt_base_addr = 0x00088020,
00178 } rx_wdt_addresses_t;
```

8.69.3.2 wdt`refresh`sequence`t

```
enum wdt\_refresh\_sequence\_t : uint8_t
```

WDT Refresh Register (WDTRR) Write Sequence Values.

The WDT counter is refreshed by writing a specific two-byte sequence:

1. Write 0x00 (`k_wdt_refresh_start`)
2. Write 0xFF (`k_wdt_refresh_end`)

This sequence must complete within the valid refresh window if window protection is enabled, otherwise a refresh error (REFEF) will occur.

Refresh Example

```

// Refresh watchdog (atomic - disable interrupts if needed)
wdt()->wdtrr = k_wdt_refresh_start; // Step 1: Write 0x00
wdt()->wdtrr = k_wdt_refresh_end;   // Step 2: Write 0xFF
```

Warning

Both writes must complete in sequence - partial refresh is invalid
Do not insert other operations between the two writes

See also

[rx_wdt_regs_t::wdtrr](#) Refresh register

Since

Version 1.0.0

Enumerator

<code>k_wdt_refresh_start</code>	First write value for refresh sequence (0x00). Must be written first, followed immediately by <code>k_wdt_refresh_end</code> .
<code>k_wdt_refresh_end</code>	Second write value for refresh sequence (0xFF). Must be written immediately after <code>k_wdt_refresh_start</code> . Counter reloads to initial value on completion.

Definition at line 375 of file [rx72n_wdt_regs.h](#).

```

00375      : uint8_t {
00376  /**
00377   * @brief First write value for refresh sequence (0x00)
00378   * @details Must be written first, followed immediately by k_wdt_refresh_end.
00379   */
00380   k_wdt_refresh_start = 0x00,
00381
00382  /**
00383   * @brief Second write value for refresh sequence (0xFF)
00384   * @details Must be written immediately after k_wdt_refresh_start.
00385   * Counter reloads to initial value on completion.
00386   */
00387   k_wdt_refresh_end = 0xFF,
00388 } wdt_refresh_sequence_t;
```

8.69.3.3 `wdt_reserved_sizes_t`

```
enum wdt\_reserved\_sizes\_t : uint8_t
```

WDT register reserved field sizes.

Defines reserved byte counts for structure padding to match hardware layout.

Since

Version 1.0.0

Enumerator

<code>k_wdt_reserved_after_wdtrr_bytes</code>	Reserved byte after WDTRR @ 0x01
<code>k_wdt_reserved_after_wdtrcr_bytes</code>	Reserved byte after WDTRCR @ 0x07

Definition at line 186 of file [rx72n_wdt_regs.h](#).

```

00186      : uint8_t {
00187   k_wdt_reserved_after_wdtrr_bytes = 1, /**< Reserved byte after WDTRR @ 0x01 */
00188   k_wdt_reserved_after_wdtrcr_bytes = 1, /**< Reserved byte after WDTRCR @ 0x07 */
00189 } wdt_reserved_sizes_t;
```


8.69.3.4 wdt_wdter_bits_t

enum [wdt_wdter_bits_t](#) : uint16_t

WDT Control Register (WDTCR) Bit Field Values.

Configuration values for WDT timeout period, clock division, and window protection. These values can be OR'd together to configure WDTCR.

Timeout Period Calculation

$$T_{timeout} = \frac{TOPS \times CKS}{f_{PCLKB}}$$

Timeout Examples (PCLKB = 60 MHz)

Configuration	Timeout
TOPS_1024 + CKS_DIV_4	17 us
TOPS_16384 + CKS_DIV_8192	2.24 s
TOPS_4096 + CKS_DIV_2048	140 ms

Window Protection

Window protection prevents both late refresh (timeout) and early refresh (software error). Configure RPSS for start and RPES for end of window.

Configuration Example

```
// 2-second timeout, no window protection
wdt()->wdtcr = k_wdt_tops_16384 | k_wdt_cks_div_8192 |
    k_wdt_rpss_100 | k_wdt_rpes_0;

// 140ms timeout with 25%-75% refresh window
wdt()->wdtcr = k_wdt_tops_4096 | k_wdt_cks_div_2048 |
    k_wdt_rpss_75 | k_wdt_rpes_25;
```

Warning

Only 6 CKS settings are valid - others are prohibited

See also

[rx_wdt_regs_t::wdtcr](#) Control register

Since

Version 1.0.0

Enumerator

k_wdt_tops_1024	1024 cycles timeout (counter loads 0x03FF) Shortest timeout period. Use with larger CKS for practical timeouts.
k_wdt_tops_4096	4096 cycles timeout (counter loads 0x0FFF)
k_wdt_tops_8192	8192 cycles timeout (counter loads 0x1FFF)
k_wdt_tops_16384	16384 cycles timeout (counter loads 0x3FFF) Longest timeout period. Combined with CKS_DIV_8192 gives ~2.24s at 60MHz.
k_wdt_cks_div_4	PCLKB divided by 4 (fastest, shortest timeout). CKS[3:0] = 0001. At 60 MHz: 15 MHz counter clock.
k_wdt_cks_div_64	PCLKB divided by 64. CKS[3:0] = 0100. At 60 MHz: 937.5 kHz counter clock.
k_wdt_cks_div_128	PCLKB divided by 128. CKS[3:0] = 1111. At 60 MHz: 468.75 kHz counter clock.
k_wdt_cks_div_512	PCLKB divided by 512. CKS[3:0] = 0110. At 60 MHz: 117.19 kHz counter clock.
k_wdt_cks_div_2048	PCLKB divided by 2048. CKS[3:0] = 0111. At 60 MHz: 29.3 kHz counter clock.
k_wdt_cks_div_8192	PCLKB divided by 8192 (slowest, longest timeout). CKS[3:0] = 1000. At 60 MHz: 7.32 kHz counter clock.
k_wdt_rpes_75	Window closes at 75% counter remaining. Refresh prohibited when counter < 75% of initial value. Provides early detection of runaway code.
k_wdt_rpes_50	Window closes at 50% counter remaining.
k_wdt_rpes_25	Window closes at 25% counter remaining.
k_wdt_rpes_0	No window end (refresh always allowed until timeout). Use with RPSS_100 to disable window protection entirely.
k_wdt_rpss_25	Window opens at 25% counter remaining. Refresh allowed only when counter < 25% of initial value. Strictest early-refresh detection.
k_wdt_rpss_50	Window opens at 50% counter remaining.
k_wdt_rpss_75	Window opens at 75% counter remaining.
k_wdt_rpss_100	No window start (refresh allowed immediately). Use with RPES_0 to disable window protection entirely.

Definition at line 455 of file [rx72n_wdt_regs.h](#).

```

00455         : uint16_t {
00456     /*
=====
00457     * Timeout Period Select (TOPS) - bits 1:0
00458     * Counter cycles before timeout (14-bit counter max values)
00459     *
=====
00460     */
00461     /**
00462     * @brief 1024 cycles timeout (counter loads 0x03FF)
00463     * @details Shortest timeout period. Use with larger CKS for practical timeouts.
00464     */
00465     k_wdt_tops_1024 = 0x0000,
00466     /**
00467     * @brief 4096 cycles timeout (counter loads 0x0FFF)
00468     */
00469     k_wdt_tops_4096 = 0x0001,
00470
00471 
```

```

00472 /**
00473  * @brief 8192 cycles timeout (counter loads 0x1FFF)
00474  */
00475 k_wdt_tops_8192 = 0x0002,
00476
00477 /**
00478  * @brief 16384 cycles timeout (counter loads 0x3FFF)
00479  * @details Longest timeout period. Combined with CKS_DIV_8192 gives ~2.24s at 60MHz.
00480  */
00481 k_wdt_tops_16384 = 0x0003,
00482
00483 /*
=====
00484  * Clock Division Ratio (CKS) - bits 7:4
00485  * PCLKB divided by this value feeds the counter
00486  * NOTE: Only 6 valid settings per Manual Table 34.2
00487  *
=====
00488  */
00489 /**
00490  * @brief PCLKB divided by 4 (fastest, shortest timeout)
00491  * @details CKS[3:0] = 0001. At 60 MHz: 15 MHz counter clock.
00492  */
00493 k_wdt_cks_div_4 = (0x01 « k_wdt_cr_cks_pos),
00494
00495 /**
00496  * @brief PCLKB divided by 64
00497  * @details CKS[3:0] = 0100. At 60 MHz: 937.5 kHz counter clock.
00498  */
00499 k_wdt_cks_div_64 = (0x04 « k_wdt_cr_cks_pos),
00500
00501 /**
00502  * @brief PCLKB divided by 128
00503  * @details CKS[3:0] = 1111. At 60 MHz: 468.75 kHz counter clock.
00504  */
00505 k_wdt_cks_div_128 = (0x0F « k_wdt_cr_cks_pos),
00506
00507 /**
00508  * @brief PCLKB divided by 512
00509  * @details CKS[3:0] = 0110. At 60 MHz: 117.19 kHz counter clock.
00510  */
00511 k_wdt_cks_div_512 = (0x06 « k_wdt_cr_cks_pos),
00512
00513 /**
00514  * @brief PCLKB divided by 2048
00515  * @details CKS[3:0] = 0111. At 60 MHz: 29.3 kHz counter clock.
00516  */
00517 k_wdt_cks_div_2048 = (0x07 « k_wdt_cr_cks_pos),
00518
00519 /**
00520  * @brief PCLKB divided by 8192 (slowest, longest timeout)
00521  * @details CKS[3:0] = 1000. At 60 MHz: 7.32 kHz counter clock.
00522  */
00523 k_wdt_cks_div_8192 = (0x08 « k_wdt_cr_cks_pos),
00524
00525 /*
=====
00526  * Window End Position (RPES) - bits 9:8
00527  * Specifies when refresh window CLOSES (refresh becomes prohibited)
00528  * Value = counter percentage below which refresh is not allowed
00529  *
=====
00530  */
00531 /**
00532  * @brief Window closes at 75% counter remaining
00533  * @details Refresh prohibited when counter < 75% of initial value.
00534  * Provides early detection of runaway code.
00535  */
00536 k_wdt_rpes_75 = (0x00 « k_wdt_cr_rpes_pos),
00537
00538 /**
00539  * @brief Window closes at 50% counter remaining
00540  */
00541 k_wdt_rpes_50 = (0x01 « k_wdt_cr_rpes_pos),
00542
00543 /**
00544  * @brief Window closes at 25% counter remaining
00545  */
00546 k_wdt_rpes_25 = (0x02 « k_wdt_cr_rpes_pos),
00547
00548 /**
00549  * @brief No window end (refresh always allowed until timeout)
00550  * @details Use with RPSS_100 to disable window protection entirely.
00551  */
00552 k_wdt_rpes_0 = (0x03 « k_wdt_cr_rpes_pos),

```

```

00553
00554  /*
=====
00555  * Window Start Position (RPSS) - bits 13:12
00556  * Specifies when refresh window OPENS (refresh becomes allowed)
00557  * Value = counter percentage below which refresh is allowed
00558  *
=====
00559  */
00560  /**
00561  * @brief Window opens at 25% counter remaining
00562  * @details Refresh allowed only when counter < 25% of initial value.
00563  * Strictest early-refresh detection.
00564  */
00565  k_wdt_rpss_25 = (0x00 « k_wdt_cr_rpss_pos),
00566
00567  /**
00568  * @brief Window opens at 50% counter remaining
00569  */
00570  k_wdt_rpss_50 = (0x01 « k_wdt_cr_rpss_pos),
00571
00572  /**
00573  * @brief Window opens at 75% counter remaining
00574  */
00575  k_wdt_rpss_75 = (0x02 « k_wdt_cr_rpss_pos),
00576
00577  /**
00578  * @brief No window start (refresh allowed immediately)
00579  * @details Use with RPES_0 to disable window protection entirely.
00580  */
00581  k_wdt_rpss_100 = (0x03 « k_wdt_cr_rpss_pos),
00582  } wdt_wdter_bits_t;

```

8.69.3.5 wdt_wdter_pos_t

```
enum wdt_wdter_pos_t : uint8_t
```

WDT Control Register (WDTCR) Bit Field Positions.

Bit positions for WDTCR register fields. Used to construct field values.

Register Layout (16-bit)

Bits	Field	Description
1:↔ 0	TOPS	Timeout period selection
3:↔ 2	-	Reserved
7:↔ 4	CKS	Clock division ratio
9:↔ 8	RPES	Window end position
11:10	-	Reserved
13:12	RPSS	Window start position
15:14	-	Reserved

Since

Version 1.0.0

Enumerator

k_wdt_cr_cks_pos	CKS field position (bits 7:4)
k_wdt_cr_rpes_pos	RPES field position (bits 9:8)
k_wdt_cr_rpss_pos	RPSS field position (bits 13:12)

Definition at line 410 of file [rx72n_wdt_regs.h](#).

```
00410      : uint8_t {
00411  k_wdt_cr_cks_pos = 4, /**< CKS field position (bits 7:4) */
00412  k_wdt_cr_rpes_pos = 8, /**< RPES field position (bits 9:8) */
00413  k_wdt_cr_rpss_pos = 12, /**< RPSS field position (bits 13:12) */
00414 } wdt_wdtrcr_pos_t;
```

8.69.3.6 wdt_wdtrcr_bits_t

```
enum wdt\_wdtrcr\_bits\_t : uint8_t
```

WDT Reset Control Register (WDTRCR) Bit Field Values.

Controls the action taken on WDT timeout or window violation. Choice between generating an internal reset or Non-Maskable Interrupt (NMI).

Register Layout (8-bit)

Bit	Field	R/W	Description
6:↔ 0	-	-	Reserved
7	RSTIRQS	R/W	0=NMI, 1=Reset on timeout

Mode Selection

- Reset Mode (k_wdt_rstirqs_reset): Recommended for production. Ensures system recovers even if NMI handler is corrupted.
- NMI Mode (k_wdt_rstirqs_nmi): Useful for debugging. Allows saving diagnostic info before reset.

Configuration Example

```
// Production: Generate reset on timeout
wdt()->wdtrcr = k_wdt_rstirqs_reset;

// Debug: Generate NMI to capture state before reset
wdt()->wdtrcr = k_wdt_rstirqs_nmi;
```

See also

[rx_wdt_regs_t::wdtrcr](#) Reset control register

Since

Version 1.0.0

Enumerator

k_wdt_rcr_rstirqs_pos	RSTIRQS bit position.
k_wdt_rcr_rstirqs_mask	RSTIRQS bit mask.
k_wdt_rstirqs_reset	Generate internal reset on timeout (RSTIRQS=1). Recommended for production. System resets automatically when watchdog times out, ensuring recovery.
k_wdt_rstirqs_nmi	Generate NMI on timeout (RSTIRQS=0). Useful for debugging. NMI handler can log diagnostics before manually resetting. Requires NMI handler implementation.

Definition at line 695 of file [rx72n_wdt_regs.h](#).

```

00695         : uint8_t {
00696     /**
00697      * @brief RSTIRQS bit position
00698      */
00699     k_wdt_rcr_rstirqs_pos = 7,
00700
00701     /**
00702      * @brief RSTIRQS bit mask
00703      */
00704     k_wdt_rcr_rstirqs_mask = (1U « k_wdt_rcr_rstirqs_pos),
00705
00706     /**
00707      * @brief Generate internal reset on timeout (RSTIRQS=1)
00708      * @details Recommended for production. System resets automatically
00709      * when watchdog times out, ensuring recovery.
00710      */
00711     k_wdt_rstirqs_reset = (1U « k_wdt_rcr_rstirqs_pos),
00712
00713     /**
00714      * @brief Generate NMI on timeout (RSTIRQS=0)
00715      * @details Useful for debugging. NMI handler can log diagnostics
00716      * before manually resetting. Requires NMI handler implementation.
00717      */
00718     k_wdt_rstirqs_nmi = 0x00,
00719 } wdt_wdtrcr_bits_t;

```

8.69.3.7 wdt_wdtsr_bits_t

```
enum wdt\_wdtsr\_bits\_t : uint16_t
```

WDT Status Register (WDTSR) Bit Field Values.

Contains the current down-counter value and error flags. The counter value can be read at any time; flags indicate timeout or window violation.

Register Layout (16-bit)

Bits	Field	R/W	Description
13:↔ 0	CNTVAL	R	Current counter value
14	UNDFE	R/W	Underflow flag (write 0 to clear)
15	REFEF	R/W	Refresh error flag (write 0 to clear)

Reading Counter Example

```
// Get current counter value
uint16_t counter = wdt()->wdtsr & k_wdt_sr_cntval_mask;

// Calculate remaining time (example with CKS_DIV_8192, PCLKB=60MHz)
// Time per count = 8192 / 60MHz = 136.5 us
float remaining_ms = counter * 0.1365f;
```

Error Checking Example

```
// Check for errors after reset
uint16_t status = wdt()->wdtsr;
if (status & k_wdt_sr_undff) {
    // Watchdog timeout occurred (system was reset)
    rx_log_warn("WDT", "Watchdog reset detected");
    wdt()->wdtsr = ~k_wdt_sr_undff; // Clear flag
}
if (status & k_wdt_sr_refef) {
    // Window violation occurred
    rx_log_error("WDT", "Watchdog window violation");
    wdt()->wdtsr = ~k_wdt_sr_refef; // Clear flag
}
```

See also

[rx_wdt_regs_t::wdtsr](#) Status register

Since

Version 1.0.0

Enumerator

k_wdt_sr_cntval_mask	Counter value mask (bits 13:0). Extracts 14-bit down-counter value. Value decrements from initial (TOPS value - 1) toward 0. Refresh reloads to initial value.
k_wdt_sr_undff	Underflow flag - Bit 14. Set when counter reaches 0 (timeout). In reset mode, this flag indicates the reset was caused by WDT. Write 0 to clear.
k_wdt_sr_refef	Refresh error flag - Bit 15. Set when refresh occurs outside the valid window (too early). Indicates software error - code refreshed before window opened. Write 0 to clear.

Definition at line 639 of file [rx72n_wdt_regs.h](#).

```
00639         : uint16_t {
00640     /**
00641      * @brief Counter value mask (bits 13:0)
00642      * @details Extracts 14-bit down-counter value. Value decrements from
00643      * initial (TOPS value - 1) toward 0. Refresh reloads to initial value.
00644      */
00645     k_wdt_sr_cntval_mask = 0x3FFF,
00646
00647     /**
00648      * @brief Underflow flag - Bit 14
00649      * @details Set when counter reaches 0 (timeout). In reset mode, this flag
00650      * indicates the reset was caused by WDT. Write 0 to clear.
00651      */
00652     k_wdt_sr_undff = (1U « k_wdt_sr_undff_pos),
00653
00654     /**
00655      * @brief Refresh error flag - Bit 15
00656      * @details Set when refresh occurs outside the valid window (too early).
00657      * Indicates software error - code refreshed before window opened.
00658      * Write 0 to clear.
00659      */
00660     k_wdt_sr_refef = (1U « k_wdt_sr_refef_pos),
00661 } wdt_wdtsr_bits_t;
```

8.69.3.8 wdt_wdtsr_pos_t

```
enum wdt_wdtsr_pos_t : uint8_t
```

WDT Status Register (WDTSR) Bit Field Positions.

Bit positions for status flags in the WDTSR register.

Since

Version 1.0.0

Enumerator

k_wdt_sr_undff_pos	Underflow flag bit position
k_wdt_sr_refef_pos	Refresh error flag bit position

Definition at line 590 of file [rx72n_wdt_regs.h](#).

```
00590      : uint8_t {
00591  k_wdt_sr_undff_pos = 14, /**< Underflow flag bit position */
00592  k_wdt_sr_refef_pos = 15, /**< Refresh error flag bit position */
00593 } wdt_wdtsr_pos_t;
```

8.69.4 Function Documentation

8.69.4.1 wdt()

```
volatile rx_wdt_regs_t * wdt (
    void ) [inline], [static]
```

Get pointer to WDT registers.

Returns a volatile pointer to the WDT register structure at address 0x00088020. The WDT is a software-controllable watchdog timer that uses the PCLKB peripheral clock.

Returns

Volatile pointer to WDT register structure

Return values

Non-NULL	Always returns valid pointer (hardware address)
----------	---

Precondition

None (hardware register always accessible)

Postcondition

Pointer is valid for the lifetime of program execution

Note

Thread Safety: Safe - returns constant hardware address

This function is always inlined for zero overhead

Example:

```
// Refresh watchdog
wdt()->wdtrr = k_wdt_refresh_start;
wdt()->wdtrr = k_wdt_refresh_end;

// Check counter value
uint16_t counter = wdt()->wdtsr & k_wdt_sr_cntval_mask;
```

See also

[k_wdt_base_addr](#) Base address constant

[rx_wdt_init\(\)](#) Higher-level initialization

Since

Version 1.0.0

Definition at line 346 of file [rx72n_wdt_regs.h](#).

```
00347 {
00348     return (volatile rx_wdt_regs_t*)k_wdt_base_addr;
00349 }
```

References [k_wdt_base_addr](#).

8.70 rx72n_wdt_regs.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx72n_wdt_regs.h
00003  * @brief RX72N WDT Watchdog Timer Register Definitions
00004  *
00005  * @details
00006  * Register definitions for the Watchdog Timer (WDT) module on the RX72N
00007  * microcontroller. The WDT provides system monitoring to detect software
00008  * runaway conditions and reset the system if necessary.
00009  *
00010  * @par System Architecture
00011  * @dot
00012  * digraph wdt_architecture {
00013  *     rankdir=LR;
00014  *     node [shape=box, style=rounded];
00015  *
00016  *     pclk [label="PCLKB\n(60 MHz)"];
00017  *     divider [label="Clock Divider\n(4 to 8192)"];
00018  *     counter [label="14-bit\nDown Counter"];
00019  *     window [label="Window\nComparator"];
00020  *     action [label="Action\n(Reset/NMI)"];
00021  *
00022  *     sw [label="Software\nRefresh"];
00023  *
00024  *     pclk --> divider;
00025  *     divider --> counter;
00026  *     counter --> window;
00027  *     window --> action [label="timeout\nor early"];
00028  *     sw --> counter [label="WDTRR\n0x00->0xFF"];
00029  * }
00030  * @enddot
00031  *
00032  * @par WDT vs IWDT Comparison
00033  * | Feature          | WDT (This Module) | IWDT
```

```

00034 * |-----|-----|-----|
00035 * | Clock Source | PCLKB (60 MHz) | Dedicated 120 kHz |
00036 * | Stop in Software | Yes | No (once started) |
00037 * | Debug/Sleep Ctrl | No | Yes (IWDT CSTPR) |
00038 * | Best For | Development, debugging | Production safety |
00039 * | Timeout Range | 4 us to 2.2 s | 8.5 ms to 17.5 s |
00040 *
00041 * @par Key Features
00042 * - 14-bit down counter using PCLKB peripheral clock
00043 * - Configurable timeout period: 1024 to 16384 cycles
00044 * - Configurable clock division: 4 to 8192
00045 * - Optional window protection to detect too-early refresh
00046 * - Can be stopped in software (unlike IWDT)
00047 * - Generates reset or NMI on timeout (configurable)
00048 *
00049 * @par Timeout Calculation
00050 * @f[
00051 *  $T_{\text{timeout}} = \frac{\text{TOPS} \times \text{CKS}}{f_{\text{PCLKB}}}$ 
00052 * @f]
00053 *
00054 * Example with PCLKB = 60 MHz, TOPS = 16384, CKS = 8192:
00055 * @f[
00056 *  $T_{\text{timeout}} = \frac{16384 \times 8192}{60,000,000} = 2.24 \text{ seconds}$ 
00057 * @f]
00058 *
00059 * @par Timeout Period Table (PCLKB = 60 MHz)
00060 * | TOPS | CKS/4 | CKS/64 | CKS/128 | CKS/8192 |
00061 * |-----|-----|-----|-----|
00062 * | 1024 | 17 us | 1.1 ms | 2.2 ms | 140 ms |
00063 * | 4096 | 68 us | 4.4 ms | 8.7 ms | 559 ms |
00064 * | 8192 | 137 us | 8.7 ms | 17.5 ms | 1.12 s |
00065 * | 16384 | 273 us | 17.5 ms | 34.9 ms | 2.24 s |
00066 *
00067 * @par Window Protection
00068 * The WDT supports window protection to detect both late refresh (timeout)
00069 * and early refresh (software error). The refresh window is defined by:
00070 * - **RPSS**: Window start position (when refresh becomes allowed)
00071 * - **RPES**: Window end position (when refresh becomes prohibited)
00072 *
00073 * @startuml
00074 * robust "Refresh Window" as RW
00075 * robust "Counter" as CNT
00076 *
00077 * @0
00078 * CNT is "Max (16384)"
00079 * RW is "Prohibited"
00080 *
00081 * @100
00082 * RW is "Allowed (RPSS=75%)"
00083 *
00084 * @250
00085 * RW is "Prohibited (RPES=50%)"
00086 *
00087 * @400
00088 * CNT is "0 (Timeout)"
00089 * @enduml
00090 *
00091 * @par Hardware Requirements
00092 * | Parameter | Value | Notes |
00093 * |-----|-----|-----|
00094 * | Clock Source | PCLKB | 60 MHz in STAR configuration |
00095 * | Base Address | 0x00088020 | 8-byte register block |
00096 * | Interrupt Vector | WDT_WOVI | Watchdog overflow (NMI mode) |
00097 * | Register Count | 4 | WDTRR, WDTCR, WDTSR, WDTRCR |
00098 *
00099 * @par NASA Power of 10 Compliance
00100 * - Rule 1: [OK] No goto, setjmp, or recursion
00101 * - Rule 2: [OK] N/A (no loops in register definitions)
00102 * - Rule 3: [OK] No dynamic memory allocation
00103 * - Rule 4: [OK] All accessor functions are single-statement
00104 * - Rule 5: [OK] N/A (hardware layer)
00105 * - Rule 6: [OK] Minimal scope
00106 * - Rule 7: [OK] N/A (no return values to check)
00107 * - Rule 8: [OK] All constants use C23 typed enums
00108 * - Rule 9: [OK] No function pointers
00109 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00110 *
00111 * @par SOLID Principles
00112 * - **S**: Single responsibility - only WDT register definitions
00113 * - **O**: Open for extension via additional bit field enums
00114 * - **L**: N/A (no inheritance hierarchy)
00115 * - **I**: Minimal interface - only necessary accessor
00116 * - **D**: N/A (hardware register layer)
00117 *
00118 * @par Module Dependencies
00119 * - `<stddef.h>` - offsetof macro for static assertions
00120 * - `<stdint.h>` - Fixed-width integer types

```

```

00121 *
00122 * @par Manual Reference
00123 * RX72N Group User's Manual: Hardware (R01UH0824EJ0120 Rev.1.20)
00124 * - Chapter 34: Watchdog Timer (WDT), pages 1547-1558
00125 *
00126 * @see rx72n_iwdt_regs.h Independent Watchdog Timer (production use)
00127 * @see rx_wdt.h Higher-level WDT API
00128 * @see rx72n_regs.h Main register include file
00129 *
00130 * @author Locked, Inc.
00131 * @date 2026-01-28
00132 * @copyright Copyright (c) 2026 Locked Inc.
00133 * SPDX-License-Identifier: MIT
00134 * @since Version 1.0.0
00135 */
00136
00137 #pragma once
00138
00139 #include <stddef.h>
00140 #include <stdint.h>
00141
00142 #ifdef __cplusplus
00143 extern "C" {
00144 #endif
00145
00146 /*
=====
00147 * Watchdog Timer (WDT)
=====
00148 */
00149
00150 /**
00151  * @enum rx_wdt_addresses_t
00152  * @brief WDT base address constant
00153  *
00154  * @details
00155  * Base address for the Watchdog Timer register block at 0x00088020.
00156  * The WDT has an 8-byte register space with 4 registers and 2 reserved bytes.
00157  *
00158  * @par Memory Map
00159  * | Address | Register | Size | Description |
00160  * |-----|-----|-----|-----|
00161  * | 0x00088020 | WDTRR | 1 | Refresh Register |
00162  * | 0x00088021 | Reserved | 1 | - |
00163  * | 0x00088022 | WDTCSR | 2 | Control Register |
00164  * | 0x00088024 | WDTSR | 2 | Status Register |
00165  * | 0x00088026 | WDTRCR | 1 | Reset Control Register |
00166  * | 0x00088027 | Reserved | 1 | - |
00167  *
00168  * @see wdt() Accessor function
00169  * @since Version 1.0.0
00170  */
00171
00172 typedef enum : uintptr_t {
00173     /**
00174      * @brief WDT register base address (0x00088020)
00175      * @details Verified against RX72N Hardware Manual Ch34
00176      */
00177     k_wdt_base_addr = 0x00088020,
00178 } rx_wdt_addresses_t;
00179
00180 /**
00181  * @enum wdt_reserved_sizes_t
00182  * @brief WDT register reserved field sizes
00183  * @details Defines reserved byte counts for structure padding to match hardware layout.
00184  * @since Version 1.0.0
00185  */
00186 typedef enum : uint8_t {
00187     k_wdt_reserved_after_wdtrr_bytes = 1, /**< Reserved byte after WDTRR @ 0x01 */
00188     k_wdt_reserved_after_wdtrcr_bytes = 1, /**< Reserved byte after WDTRCR @ 0x07 */
00189 } wdt_reserved_sizes_t;
00190
00191 /**
00192  * @struct rx_wdt_regs_t
00193  * @brief WDT Register Map Structure
00194  *
00195  * @details
00196  * Watchdog Timer (WDT) registers for software-controllable watchdog functionality.
00197  * Uses PCLKB clock and can be stopped in software (unlike IWDT).
00198  *
00199  * @par Memory Layout Table
00200  * | Offset | Size | Field | Type | Description |
00201  * |-----|-----|-----|-----|-----|
00202  * | 0x00 | 1 | wdtrr | uint8_t | Refresh Register |
00203  * | 0x01 | 1 | reserved0 | - | Reserved |
00204  * | 0x02 | 2 | wdtrcr | uint16_t | Control Register |
00205  * | 0x04 | 2 | wdtsr | uint16_t | Status Register |

```

```

00206 * | 0x06 | 1 | wdtrcr | uint8_t | Reset Control Register |
00207 * | 0x07 | 1 | reserved1 | - | Reserved |
00208 * | **Total** | **8** | | | (packed, no additional padding) |
00209 *
00210 * @par WDT Refresh Sequence
00211 * The watchdog is refreshed by writing a specific sequence to WDTRR:
00212 * 1. Write 0x00 to WDTRR
00213 * 2. Write 0xFF to WDTRR
00214 *
00215 * This two-step sequence prevents accidental refresh from single-byte errors.
00216 *
00217 * @par Initialization Example
00218 * @code{.c}
00219 * // Configure WDT for 2-second timeout with window protection
00220 * volatile rx_wdt_regs_t* wdt_regs = wdt();
00221 *
00222 * // 1. Configure control register (before starting)
00223 * // TOPS=16384 cycles, CKS=div8192, RPSS=100%, RPES=0%
00224 * wdt_regs->wdtcr = k_wdt_tops_16384 | k_wdt_cks_div_8192 |
00225 * k_wdt_rpss_100 | k_wdt_rpes_0;
00226 *
00227 * // 2. Configure to generate reset on timeout
00228 * wdt_regs->wdtrcr = k_wdt_rstirqs_reset;
00229 *
00230 * // 3. Start WDT by performing first refresh
00231 * wdt_regs->wdtrr = k_wdt_refresh_start; // Write 0x00
00232 * wdt_regs->wdtrr = k_wdt_refresh_end; // Write 0xFF
00233 * @endcode
00234 *
00235 * @par Refresh Example (in main loop)
00236 * @code{.c}
00237 * void app_main_loop(void) {
00238 *     while (1) {
00239 *         // Perform application tasks
00240 *         process_commands();
00241 *         update_motors();
00242 *
00243 *         // Refresh watchdog (must complete both writes atomically)
00244 *         wdt()->wdtrr = k_wdt_refresh_start; // 0x00
00245 *         wdt()->wdtrr = k_wdt_refresh_end; // 0xFF
00246 *
00247 *         // Wait for next cycle
00248 *         tx_thread_sleep(10);
00249 *     }
00250 * }
00251 * @endcode
00252 *
00253 * @note WDT has 4 registers only. WDTCSTPR (count stop) exists for IWDT but NOT for WDT.
00254 *
00255 * @invariant sizeof(rx_wdt_regs_t) == 8 bytes
00256 * @invariant All offsets match RX72N Hardware Manual Ch34
00257 *
00258 * @warning Refresh must complete both writes (0x00 then 0xFF) atomically
00259 * @warning Do not refresh outside the valid window if window protection is enabled
00260 *
00261 * @see wdt() Accessor function
00262 * @see wdt_refresh_sequence_t Refresh values
00263 * @see wdt_wdtrcr_bits_t Control register bits
00264 * @since Version 1.0.0
00265 */
00266 typedef struct __attribute__((packed)) {
00267 /**
00268 * @brief WDT Refresh Register (WDTRR) @ offset 0x00
00269 * @details Write 0x00 then 0xFF to refresh the watchdog counter.
00270 * Both writes must complete within the valid window if window protection is enabled.
00271 * @see wdt_refresh_sequence_t Refresh values
00272 * @warning Single byte writes are invalid - must write both values in sequence
00273 */
00274 volatile uint8_t wdtrr;
00275
00276 uint8_t reserved0[k_wdt_reserved_after_wdtrr_bytes]; /**< Reserved @ 0x01 */
00277
00278 /**
00279 * @brief WDT Control Register (WDTCR) @ offset 0x02
00280 * @details Configures timeout period, clock division, and window parameters.
00281 * Must be configured before starting the WDT (before first refresh).
00282 * @par Bit Fields:
00283 * - Bits 1:0 (TOPS): Timeout period selection
00284 * - Bits 7:4 (CKS): Clock division ratio
00285 * - Bits 9:8 (RPES): Window end position
00286 * - Bits 13:12 (RPSS): Window start position
00287 * @see wdt_wdtrcr_bits_t Bit definitions
00288 */
00289 volatile uint16_t wdtcr;
00290
00291 /**
00292 * @brief WDT Status Register (WDTSR) @ offset 0x04

```

```

00293 * @details Contains current counter value and error flags.
00294 * @par Bit Fields:
00295 * - Bits 13:0 (CNTVAL): Current down-counter value (read-only)
00296 * - Bit 14 (UNDFE): Underflow flag (timeout occurred)
00297 * - Bit 15 (REFEF): Refresh error flag (window violation)
00298 * @see wdt_wdtsr_bits_t Bit definitions
00299 */
00300 volatile uint16_t wdtsr;
00301
00302 /**
00303 * @brief WDT Reset Control Register (WDTRCR) @ offset 0x06
00304 * @details Selects action on timeout: reset or NMI.
00305 * @par Bit 7 (RSTIRQS):
00306 * - 0: Generate NMI on timeout
00307 * - 1: Generate internal reset on timeout
00308 * @see wdt_wdtrcr_bits_t Bit definitions
00309 */
00310 volatile uint8_t wdtrcr;
00311
00312 uint8_t reserved1[k_wdt_reserved_after_wdtrcr_bytes]; /**< Reserved @ 0x07 */
00313 } rx_wdt_regs_t;
00314
00315 /**
00316 * @brief Get pointer to WDT registers
00317 *
00318 * @details
00319 * Returns a volatile pointer to the WDT register structure at address
00320 * 0x00088020. The WDT is a software-controllable watchdog timer that
00321 * uses the PCLKB peripheral clock.
00322 *
00323 * @return Volatile pointer to WDT register structure
00324 * @retval Non-NULL Always returns valid pointer (hardware address)
00325 *
00326 * @pre None (hardware register always accessible)
00327 * @post Pointer is valid for the lifetime of program execution
00328 *
00329 * @note Thread Safety: Safe - returns constant hardware address
00330 * @note This function is always inlined for zero overhead
00331 *
00332 * @par Example:
00333 * @code{.c}
00334 * // Refresh watchdog
00335 * wdt()->wdtrr = k_wdt_refresh_start;
00336 * wdt()->wdtrr = k_wdt_refresh_end;
00337 *
00338 * // Check counter value
00339 * uint16_t counter = wdt()->wdtsr & k_wdt_sr_cntval_mask;
00340 * @endcode
00341 *
00342 * @see k_wdt_base_addr Base address constant
00343 * @see rx_wdt_init() Higher-level initialization
00344 * @since Version 1.0.0
00345 */
00346 static inline volatile rx_wdt_regs_t* wdt(void)
00347 {
00348     return (volatile rx_wdt_regs_t*)k_wdt_base_addr;
00349 }
00350
00351 /**
00352 * @enum wdt_refresh_sequence_t
00353 * @brief WDT Refresh Register (WDTRR) Write Sequence Values
00354 *
00355 * @details
00356 * The WDT counter is refreshed by writing a specific two-byte sequence:
00357 * 1. Write 0x00 (k_wdt_refresh_start)
00358 * 2. Write 0xFF (k_wdt_refresh_end)
00359 *
00360 * This sequence must complete within the valid refresh window if window
00361 * protection is enabled, otherwise a refresh error (REFEF) will occur.
00362 *
00363 * @par Refresh Example
00364 * @code{.c}
00365 * // Refresh watchdog (atomic - disable interrupts if needed)
00366 * wdt()->wdtrr = k_wdt_refresh_start; // Step 1: Write 0x00
00367 * wdt()->wdtrr = k_wdt_refresh_end;   // Step 2: Write 0xFF
00368 * @endcode
00369 *
00370 * @warning Both writes must complete in sequence - partial refresh is invalid
00371 * @warning Do not insert other operations between the two writes
00372 * @see rx_wdt_regs_t::wdtrr Refresh register
00373 * @since Version 1.0.0
00374 */
00375 typedef enum : uint8_t {
00376     /**
00377      * @brief First write value for refresh sequence (0x00)
00378      * @details Must be written first, followed immediately by k_wdt_refresh_end.
00379      */

```

```

00380 k_wdt_refresh_start = 0x00,
00381
00382 /**
00383  * @brief Second write value for refresh sequence (0xFF)
00384  * @details Must be written immediately after k_wdt_refresh_start.
00385  * Counter reloads to initial value on completion.
00386  */
00387 k_wdt_refresh_end = 0xFF,
00388 } wdt_refresh_sequence_t;
00389
00390 /**
00391  * @enum wdt_wdtr_pos_t
00392  * @brief WDT Control Register (WDTCR) Bit Field Positions
00393  *
00394  * @details
00395  * Bit positions for WDTCR register fields. Used to construct field values.
00396  *
00397  * @par Register Layout (16-bit)
00398  * | Bits | Field | Description |
00399  * |-----|-----|-----|
00400  * | 1:0 | TOPS | Timeout period selection |
00401  * | 3:2 | - | Reserved |
00402  * | 7:4 | CKS | Clock division ratio |
00403  * | 9:8 | RPES | Window end position |
00404  * | 11:10 | - | Reserved |
00405  * | 13:12 | RPSS | Window start position |
00406  * | 15:14 | - | Reserved |
00407  *
00408  * @since Version 1.0.0
00409  */
00410 typedef enum : uint8_t {
00411     k_wdt_cr_cks_pos = 4, /**< CKS field position (bits 7:4) */
00412     k_wdt_cr_rpes_pos = 8, /**< RPES field position (bits 9:8) */
00413     k_wdt_cr_rpss_pos = 12, /**< RPSS field position (bits 13:12) */
00414 } wdt_wdtr_pos_t;
00415
00416 /**
00417  * @enum wdt_wdtr_bits_t
00418  * @brief WDT Control Register (WDTCR) Bit Field Values
00419  *
00420  * @details
00421  * Configuration values for WDT timeout period, clock division, and
00422  * window protection. These values can be OR'd together to configure WDTCR.
00423  *
00424  * @par Timeout Period Calculation
00425  * @f[
00426     T_{timeout} = \frac{TOPS \times CKS}{f_{PCLKB}}
00427     @f]
00428  *
00429  * @par Timeout Examples (PCLKB = 60 MHz)
00430  * | Configuration | Timeout |
00431  * |-----|-----|
00432  * | TOPS_1024 + CKS_DIV_4 | 17 us |
00433  * | TOPS_16384 + CKS_DIV_8192 | 2.24 s |
00434  * | TOPS_4096 + CKS_DIV_2048 | 140 ms |
00435  *
00436  * @par Window Protection
00437  * Window protection prevents both late refresh (timeout) and early refresh
00438  * (software error). Configure RPSS for start and RPES for end of window.
00439  *
00440  * @par Configuration Example
00441  * @code{.c}
00442  * // 2-second timeout, no window protection
00443  * wdt()->wdtr = k_wdt_tops_16384 | k_wdt_cks_div_8192 |
00444  *               k_wdt_rpss_100 | k_wdt_rpes_0;
00445  *
00446  * // 140ms timeout with 25%-75% refresh window
00447  * wdt()->wdtr = k_wdt_tops_4096 | k_wdt_cks_div_2048 |
00448  *               k_wdt_rpss_75 | k_wdt_rpes_25;
00449  * @endcode
00450  *
00451  * @warning Only 6 CKS settings are valid - others are prohibited
00452  * @see rx_wdt_regs_t::wdtr Control register
00453  * @since Version 1.0.0
00454  */
00455 typedef enum : uint16_t {
00456     /**
00457      * Timeout Period Select (TOPS) - bits 1:0
00458      * Counter cycles before timeout (14-bit counter max values)
00459      */
00460
00461     /**
00462      * @brief 1024 cycles timeout (counter loads 0x03FF)
00463      * @details Shortest timeout period. Use with larger CKS for practical timeouts.

```

```

00464  */
00465  k_wdt_tops_1024 = 0x0000,
00466
00467  /**
00468   * @brief 4096 cycles timeout (counter loads 0x0FFF)
00469   */
00470  k_wdt_tops_4096 = 0x0001,
00471
00472  /**
00473   * @brief 8192 cycles timeout (counter loads 0x1FFF)
00474   */
00475  k_wdt_tops_8192 = 0x0002,
00476
00477  /**
00478   * @brief 16384 cycles timeout (counter loads 0x3FFF)
00479   * @details Longest timeout period. Combined with CKS_DIV_8192 gives ~2.24s at 60MHz.
00480   */
00481  k_wdt_tops_16384 = 0x0003,
00482
00483  /*
=====
00484  * Clock Division Ratio (CKS) - bits 7:4
00485  * PCLKB divided by this value feeds the counter
00486  * NOTE: Only 6 valid settings per Manual Table 34.2
00487  *
=====
00488  */
00489  /**
00490   * @brief PCLKB divided by 4 (fastest, shortest timeout)
00491   * @details CKS[3:0] = 0001. At 60 MHz: 15 MHz counter clock.
00492   */
00493  k_wdt_cks_div_4 = (0x01 « k_wdt_cr_cks_pos),
00494
00495  /**
00496   * @brief PCLKB divided by 64
00497   * @details CKS[3:0] = 0100. At 60 MHz: 937.5 kHz counter clock.
00498   */
00499  k_wdt_cks_div_64 = (0x04 « k_wdt_cr_cks_pos),
00500
00501  /**
00502   * @brief PCLKB divided by 128
00503   * @details CKS[3:0] = 1111. At 60 MHz: 468.75 kHz counter clock.
00504   */
00505  k_wdt_cks_div_128 = (0x0F « k_wdt_cr_cks_pos),
00506
00507  /**
00508   * @brief PCLKB divided by 512
00509   * @details CKS[3:0] = 0110. At 60 MHz: 117.19 kHz counter clock.
00510   */
00511  k_wdt_cks_div_512 = (0x06 « k_wdt_cr_cks_pos),
00512
00513  /**
00514   * @brief PCLKB divided by 2048
00515   * @details CKS[3:0] = 0111. At 60 MHz: 29.3 kHz counter clock.
00516   */
00517  k_wdt_cks_div_2048 = (0x07 « k_wdt_cr_cks_pos),
00518
00519  /**
00520   * @brief PCLKB divided by 8192 (slowest, longest timeout)
00521   * @details CKS[3:0] = 1000. At 60 MHz: 7.32 kHz counter clock.
00522   */
00523  k_wdt_cks_div_8192 = (0x08 « k_wdt_cr_cks_pos),
00524
00525  /*
=====
00526  * Window End Position (RPES) - bits 9:8
00527  * Specifies when refresh window CLOSES (refresh becomes prohibited)
00528  * Value = counter percentage below which refresh is not allowed
00529  *
=====
00530  */
00531  /**
00532   * @brief Window closes at 75% counter remaining
00533   * @details Refresh prohibited when counter < 75% of initial value.
00534   * Provides early detection of runaway code.
00535   */
00536  k_wdt_rpes_75 = (0x00 « k_wdt_cr_rpes_pos),
00537
00538  /**
00539   * @brief Window closes at 50% counter remaining
00540   */
00541  k_wdt_rpes_50 = (0x01 « k_wdt_cr_rpes_pos),
00542
00543  /**
00544   * @brief Window closes at 25% counter remaining

```

```

00545 */
00546 k_wdt_rpes_25 = (0x02 « k_wdt_cr_rpes_pos),
00547
00548 /**
00549  * @brief No window end (refresh always allowed until timeout)
00550  * @details Use with RPSS_100 to disable window protection entirely.
00551  */
00552 k_wdt_rpes_0 = (0x03 « k_wdt_cr_rpes_pos),
00553
00554 */
=====
00555 * Window Start Position (RPSS) - bits 13:12
00556 * Specifies when refresh window OPENS (refresh becomes allowed)
00557 * Value = counter percentage below which refresh is allowed
00558
=====
*/
00559
00560 /**
00561  * @brief Window opens at 25% counter remaining
00562  * @details Refresh allowed only when counter < 25% of initial value.
00563  * Strictest early-refresh detection.
00564  */
00565 k_wdt_rpss_25 = (0x00 « k_wdt_cr_rpss_pos),
00566
00567 /**
00568  * @brief Window opens at 50% counter remaining
00569  */
00570 k_wdt_rpss_50 = (0x01 « k_wdt_cr_rpss_pos),
00571
00572 /**
00573  * @brief Window opens at 75% counter remaining
00574  */
00575 k_wdt_rpss_75 = (0x02 « k_wdt_cr_rpss_pos),
00576
00577 /**
00578  * @brief No window start (refresh allowed immediately)
00579  * @details Use with RPES_0 to disable window protection entirely.
00580  */
00581 k_wdt_rpss_100 = (0x03 « k_wdt_cr_rpss_pos),
00582 } wdt_wdtsr_bits_t;
00583
00584 /**
00585  * @enum wdt_wdtsr_pos_t
00586  * @brief WDT Status Register (WDTSR) Bit Field Positions
00587  * @details Bit positions for status flags in the WDTSR register.
00588  * @since Version 1.0.0
00589  */
00590 typedef enum : uint8_t {
00591     k_wdt_sr_undff_pos = 14, /**< Underflow flag bit position */
00592     k_wdt_sr_refef_pos = 15, /**< Refresh error flag bit position */
00593 } wdt_wdtsr_pos_t;
00594
00595 /**
00596  * @enum wdt_wdtsr_bits_t
00597  * @brief WDT Status Register (WDTSR) Bit Field Values
00598  *
00599  * @details
00600  * Contains the current down-counter value and error flags. The counter value
00601  * can be read at any time; flags indicate timeout or window violation.
00602  *
00603  * @par Register Layout (16-bit)
00604  * | Bits | Field | R/W | Description |
00605  * |-----|-----|-----|-----|
00606  * | 13:0 | CNTVAL | R | Current counter value |
00607  * | 14 | UNDF | R/W | Underflow flag (write 0 to clear) |
00608  * | 15 | REFEF | R/W | Refresh error flag (write 0 to clear) |
00609  *
00610  * @par Reading Counter Example
00611  * @code{.c}
00612  * // Get current counter value
00613  * uint16_t counter = wdt()->wdtsr & k_wdt_sr_cntval_mask;
00614  *
00615  * // Calculate remaining time (example with CKS_DIV_8192, PCLKB=60MHz)
00616  * // Time per count = 8192 / 60MHz = 136.5 us
00617  * float remaining_ms = counter * 0.1365f;
00618  * @endcode
00619  *
00620  * @par Error Checking Example
00621  * @code{.c}
00622  * // Check for errors after reset
00623  * uint16_t status = wdt()->wdtsr;
00624  * if (status & k_wdt_sr_undff) {
00625  *     // Watchdog timeout occurred (system was reset)
00626  *     rx_log_warn("WDT", "Watchdog reset detected");
00627  *     wdt()->wdtsr = ~k_wdt_sr_undff; // Clear flag
00628  * }

```



```

00629 * if (status & k_wdt_sr_refef) {
00630 *     // Window violation occurred
00631 *     rx_log_error("WDT", "Watchdog window violation");
00632 *     wdt()->wdtsr = ~k_wdt_sr_refef; // Clear flag
00633 * }
00634 * @endcode
00635 *
00636 * @see rx_wdt_regs_t::wdtsr Status register
00637 * @since Version 1.0.0
00638 */
00639 typedef enum : uint16_t {
00640 /**
00641 * @brief Counter value mask (bits 13:0)
00642 * @details Extracts 14-bit down-counter value. Value decrements from
00643 * initial (TOPS value - 1) toward 0. Refresh reloads to initial value.
00644 */
00645 k_wdt_sr_cntval_mask = 0x3FFF,
00646
00647 /**
00648 * @brief Underflow flag - Bit 14
00649 * @details Set when counter reaches 0 (timeout). In reset mode, this flag
00650 * indicates the reset was caused by WDT. Write 0 to clear.
00651 */
00652 k_wdt_sr_undff = (1U « k_wdt_sr_undff_pos),
00653
00654 /**
00655 * @brief Refresh error flag - Bit 15
00656 * @details Set when refresh occurs outside the valid window (too early).
00657 * Indicates software error - code refreshed before window opened.
00658 * Write 0 to clear.
00659 */
00660 k_wdt_sr_refef = (1U « k_wdt_sr_refef_pos),
00661 } wdt_wdtsr_bits_t;
00662
00663 /**
00664 * @enum wdt_wdtrcr_bits_t
00665 * @brief WDT Reset Control Register (WDTRCR) Bit Field Values
00666 *
00667 * @details
00668 * Controls the action taken on WDT timeout or window violation.
00669 * Choice between generating an internal reset or Non-Maskable Interrupt (NMI).
00670 *
00671 * @par Register Layout (8-bit)
00672 * | Bit | Field | R/W | Description |
00673 * |-----|-----|-----|-----|
00674 * | 6:0 | - | - | Reserved |
00675 * | 7 | RSTIRQS | R/W | 0=NMI, 1=Reset on timeout |
00676 *
00677 * @par Mode Selection
00678 * - **Reset Mode (k_wdt_rstirqs_reset)**: Recommended for production.
00679 * Ensures system recovers even if NMI handler is corrupted.
00680 * - **NMI Mode (k_wdt_rstirqs_nmi)**: Useful for debugging.
00681 * Allows saving diagnostic info before reset.
00682 *
00683 * @par Configuration Example
00684 * @code{.c}
00685 * // Production: Generate reset on timeout
00686 * wdt()->wdtrcr = k_wdt_rstirqs_reset;
00687 *
00688 * // Debug: Generate NMI to capture state before reset
00689 * wdt()->wdtrcr = k_wdt_rstirqs_nmi;
00690 * @endcode
00691 *
00692 * @see rx_wdt_regs_t::wdtrcr Reset control register
00693 * @since Version 1.0.0
00694 */
00695 typedef enum : uint8_t {
00696 /**
00697 * @brief RSTIRQS bit position
00698 */
00699 k_wdt_rcr_rstirqs_pos = 7,
00700
00701 /**
00702 * @brief RSTIRQS bit mask
00703 */
00704 k_wdt_rcr_rstirqs_mask = (1U « k_wdt_rcr_rstirqs_pos),
00705
00706 /**
00707 * @brief Generate internal reset on timeout (RSTIRQS=1)
00708 * @details Recommended for production. System resets automatically
00709 * when watchdog times out, ensuring recovery.
00710 */
00711 k_wdt_rstirqs_reset = (1U « k_wdt_rcr_rstirqs_pos),
00712
00713 /**
00714 * @brief Generate NMI on timeout (RSTIRQS=0)
00715 * @details Useful for debugging. NMI handler can log diagnostics

```

```

00716  * before manually resetting. Requires NMI handler implementation.
00717  */
00718  k_wdt_rstirqs_nmi = 0x00,
00719 } wdt_wdtrcr_bits_t;
00720
00721 /*
=====
00722 * Static Assertions - Verify Register Layout at Compile Time
00723 *
=====
00724 */
00725
00726 /* The k_wdt_base_addr enum declaration above is itself the contract for the
00727 * WDT base address (per Hardware Manual). A static_assert that re-states the
00728 * same literal here would just duplicate the magic number without adding a
00729 * real cross-check, so we rely on the enum declaration alone. */
00730
00731 /* Verify register structure layout (WDT has 4 registers, 8 bytes total) */
00732 static_assert(sizeof(rx_wdt_regs_t) == 8, "WDT register structure size incorrect");
00733 static_assert(offsetof(rx_wdt_regs_t, wdtrr) == 0x00, "WDTRR offset incorrect");
00734 static_assert(offsetof(rx_wdt_regs_t, wdtrcr) == 0x02, "WDTRCR offset incorrect");
00735 static_assert(offsetof(rx_wdt_regs_t, wdtsr) == 0x04, "WDTSR offset incorrect");
00736 static_assert(offsetof(rx_wdt_regs_t, wdtrcr) == 0x06, "WDTRCR offset incorrect");
00737
00738 #ifdef __cplusplus
00739 }
00740 #endif

```

8.71 libs/rx_hal/inc/rx_cac.h File Reference

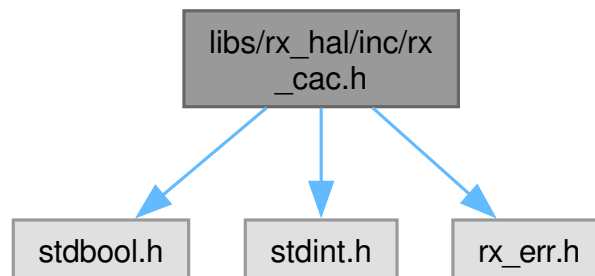
Clock Frequency Accuracy Measurement Circuit (CAC) Driver API for RX72N.

```

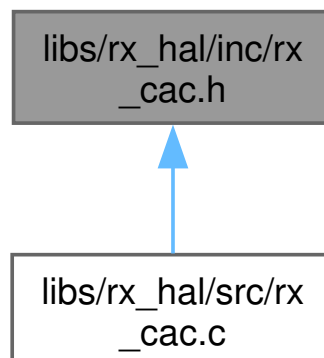
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_cac.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_cac_config_t](#)
Configuration parameters for [rx_cac_init\(\)](#).

Enumerations

- enum [rx_cac_clock_t](#) : `uint8_t` {
 [k_cac_clock_main](#) = 0 , [k_cac_clock_sub](#) = 1 , [k_cac_clock_hoco](#) = 2 , [k_cac_clock_loco](#) = 3 ,
 [k_cac_clock_pclkb](#) = 4 , [k_cac_clock_iwdtclk](#) = 5 , [k_cac_clock_uclk](#) = 6 , [k_cac_clock_clkout25](#)
 = 7 }
Clock source selector for CAC measured/reference inputs.
- enum [rx_cac_ref_div_t](#) : `uint8_t` { [k_cac_ref_div_32](#) = 0 , [k_cac_ref_div_128](#) = 1 ,
 [k_cac_ref_div_1024](#) = 2 , [k_cac_ref_div_8192](#) = 3 }
Reference-clock divider (CACR2.RCDS[1:0]) per RX72N HW Manual p.394.
- enum [rx_cac_target_div_t](#) : `uint8_t` { [k_cac_target_div_1](#) = 0 , [k_cac_target_div_4](#) = 1 ,
 [k_cac_target_div_8](#) = 2 , [k_cac_target_div_32](#) = 3 }
Measured-clock divider (CACR1.TCSS[1:0]) per RX72N HW Manual p.393.

Functions

- `rx_err_t rx_cac_init (const rx\_cac\_config\_t *config)`
Initialize the CAC module with the supplied configuration.
- `rx_err_t rx_cac_start (void)`
Begin CAC measurement (CACR0.CFME = 1).
- `rx_err_t rx_cac_stop (void)`
Stop CAC measurement (CACR0.CFME = 0).
- `bool rx_cac_check (uint32_t *out_count)`
Sample the CAC status and counter buffer, clearing acknowledged flags.
- `rx_err_t rx_cac_deinit (void)`
Tear down the CAC module and put it back in module-stop.

8.71.1 Detailed Description

Clock Frequency Accuracy Measurement Circuit (CAC) Driver API for RX72N.

Public driver API for the RX72N Clock Frequency Accuracy Measurement Circuit (CAC). The CAC compares a measurement-target clock against a reference clock by counting target-clock edges between reference-clock edges, then comparing the count against upper/lower limits set in CAULVR/CALLVR. If the count is outside the window, the FERRF flag is set and (when enabled) a group-interrupt is raised via GROUPBL0.

STAR Usage

STAR uses CAC to watchdog the main-oscillator (MOSC, 24 MHz crystal) against the internal HOCO (16 MHz). A drifted crystal - from aging, mechanical damage, or PCB leakage - would silently corrupt motor-control timing; the FERRF ISR instead triggers motor e-stop.

Typical Configuration

```
rx_cac_config_t cfg = {
    .measured_clock = k_cac_clock_main,    // 24 MHz crystal (MOSC)
    .reference_clock = k_cac_clock_hoco,   // 16 MHz HOCO
    .reference_div   = k_cac_ref_div_1024, // 16 MHz / 1024 = 15.625 kHz
    .caulvr         = 1560,                // Upper count bound
    .callvr         = 1500,                // Lower count bound
    .enable_ferrie  = true,                // Raise IRQ on drift
};
(void)rx_cac_init(&cfg);
(void)rx_cac_start();
```

NASA Power of 10 Compliance

- Rule 1: [OK] No goto / setjmp / recursion
- Rule 2: [OK] All loops bounded
- Rule 3: [OK] No dynamic allocation (static state only)
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] ≥ 2 checks per function (

Precondition

/

Postcondition

)

- Rule 6: [OK] File-scope statics, minimal scope
- Rule 7: [OK] rx_err_t returns, checked via RX_RETURN_ON_ERROR
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [OK] Single-level pointers
- Rule 10: [OK] -Wall -Wextra -Werror clean

See also

[rx72n_cac_regs.h](#) CAC register definitions
RX72N HW Manual R01UH0824EJ0111 Chapter 10

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_cac.h](#).

8.71.2 Data Structure Documentation

8.71.2.1 struct rx'cac'config't

Configuration parameters for [rx_cac_init\(\)](#).

Fully describes one CAC measurement campaign. Callers fill in every field before invoking [rx_cac_init\(\)](#); the driver does not look at partial or uninitialised members.

Example: MOSC watchdog against HOCO

```
rx_cac_config_t cfg = {
    .measured_clock = k_cac_clock_main,
    .reference_clock = k_cac_clock_hoco,
    .target_div     = k_cac_target_div_1,
    .reference_div   = k_cac_ref_div_1024,
    .caulvr         = 1560,
    .callvr         = 1500,
    .enable_ferrie  = true,
    .enable_mendie  = false,
    .enable_ovfie   = true,
};
```

Invariant

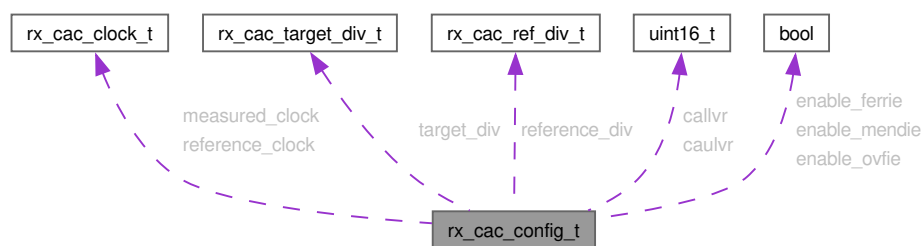
callvr < caulvr (driver rejects otherwise)

Since

Version 1.0.0

Definition at line 157 of file [rx_cac.h](#).

Collaboration diagram for rx_cac_config_t:



Data Fields

uint16_t	callvr	Lower-limit counter value - CALLVR. FERRF asserts when (counter < callvr) at sampling time.
uint16_t	caulvr	Upper-limit counter value - CAULVR. FERRF asserts when (counter > caulvr) at sampling time.

bool	enable_ferrie	Enable frequency-error interrupt (CAICR.FERRIE). When true, the CAC raises an IRQ on every transition into the "counter outside [callvr, caulvr]" state. Must be used together with a GROUPBL0 IRQ handler wired to bit 26 (FERRF).
bool	enable_mendie	Enable measurement-end interrupt (CAICR.MENDIE).
bool	enable_ovfie	Enable counter-overflow interrupt (CAICR.OVFIE).
rx_cac_clock_t	measured_clock	Measured (target) clock source - CACR1.FMCS[2:0]. The clock whose frequency is being evaluated. All eight sources defined by rx_cac_clock_t are legal here.
rx_cac_clock_t	reference_clock	Reference clock source - CACR2.RSCS[2:0]. The clock the measurement is compared against. Only the first six entries of rx_cac_clock_t are legal (Main..IWDTCCLK); passing k_cac_clock_uclk or k_cac_clock_clkout25 causes rx_cac_init() to return k_err_invalid_arg .
rx_cac_ref_div_t	reference_div	Divider applied to the reference clock - CACR2.RCDS[1:0]. Selects how many reference-clock edges accumulate into a single measurement window (32 / 128 / 1024 / 8192).
rx_cac_target_div_t	target_div	Divider applied to the measured clock - CACR1.TCSS[1:0]. Defaults to k_cac_target_div_1 for highest-resolution counts.

8.71.3 Enumeration Type Documentation

8.71.3.1 `rx_cac_clock_t`

```
enum rx\_cac\_clock\_t : uint8_t
```

Clock source selector for CAC measured/reference inputs.

Maps directly onto the FMCS[2:0] / RSCS[2:0] field values defined in CACR1 / CACR2 (RX72N HW Manual p.393 / p.394). Not every source is legal as a reference - RSCS supports values 000-101 only (Main, Sub, HOCO, LOCO, PCLKB, IWDTCCLK); the extra values (UCLK, CLKOUT25M) are measured-only.

Since

Version 1.0.0

Enumerator

k_cac_clock_main	Main-clock oscillator (MOSC, external crystal)
k_cac_clock_sub	Sub-clock oscillator (32.768 kHz)
k_cac_clock_hoco	High-speed on-chip oscillator
k_cac_clock_loco	Low-speed on-chip oscillator
k_cac_clock_pclkb	Peripheral clock B

k_cac_clock_iwdtclk	IWDT-dedicated oscillator
k_cac_clock_uclk	USB clock (measured-only, not a valid RSCS)
k_cac_clock_clkout25	25 MHz CLKOUT (measured-only, not a valid RSCS)

Definition at line 83 of file [rx_cac.h](#).

```

00083      : uint8_t {
00084  k_cac_clock_main    = 0, /**< Main-clock oscillator (MOSC, external crystal) */
00085  k_cac_clock_sub     = 1, /**< Sub-clock oscillator (32.768 kHz) */
00086  k_cac_clock_hoco    = 2, /**< High-speed on-chip oscillator */
00087  k_cac_clock_loco    = 3, /**< Low-speed on-chip oscillator */
00088  k_cac_clock_pclkb   = 4, /**< Peripheral clock B */
00089  k_cac_clock_iwdtclk = 5, /**< IWDT-dedicated oscillator */
00090  k_cac_clock_uclk    = 6, /**< USB clock (measured-only, not a valid RSCS) */
00091  k_cac_clock_clkout25 = 7, /**< 25 MHz CLKOUT (measured-only, not a valid RSCS) */
00092 } rx_cac_clock_t;

```

8.71.3.2 rx_cac_ref_div_t

```
enum rx\_cac\_ref\_div\_t : uint8_t
```

Reference-clock divider (CACR2.RCDS[1:0]) per RX72N HW Manual p.394.

Since

Version 1.0.0

Enumerator

k_cac_ref_div_32	Reference clock / 32
k_cac_ref_div_128	Reference clock / 128
k_cac_ref_div_1024	Reference clock / 1024
k_cac_ref_div_8192	Reference clock / 8192

Definition at line 99 of file [rx_cac.h](#).

```

00099      : uint8_t {
00100  k_cac_ref_div_32    = 0, /**< Reference clock / 32 */
00101  k_cac_ref_div_128   = 1, /**< Reference clock / 128 */
00102  k_cac_ref_div_1024  = 2, /**< Reference clock / 1024 */
00103  k_cac_ref_div_8192  = 3, /**< Reference clock / 8192 */
00104 } rx_cac_ref_div_t;

```

8.71.3.3 rx_cac_target_div_t

```
enum rx\_cac\_target\_div\_t : uint8_t
```

Measured-clock divider (CACR1.TCSS[1:0]) per RX72N HW Manual p.393.

TCSS is a target-clock divider (not a digital filter). Use /1 for the fastest allowed count rate and larger dividers only when the measurement target clock exceeds the counter bandwidth.

Since

Version 1.0.0

Enumerator

k_cac_target_div_1	Target clock / 1
k_cac_target_div_4	Target clock / 4
k_cac_target_div_8	Target clock / 8
k_cac_target_div_32	Target clock / 32

Definition at line 117 of file [rx_cac.h](#).

```

00117      : uint8_t {
00118  k_cac_target_div_1 = 0, /**< Target clock / 1 */
00119  k_cac_target_div_4 = 1, /**< Target clock / 4 */
00120  k_cac_target_div_8 = 2, /**< Target clock / 8 */
00121  k_cac_target_div_32 = 3, /**< Target clock / 32 */
00122 } rx_cac_target_div_t;

```

8.71.4 Function Documentation

8.71.4.1 rx_cac_check()

```

bool rx_cac_check (
    uint32_t * out_count)

```

Sample the CAC status and counter buffer, clearing acknowledged flags.

Reads CASTR, optionally writes back FERRE / MENDE / OVFF acknowledgement bits via CAICR, and returns the last-captured counter buffer via out_count. If out_count is nullptr the counter is not returned; the function still reads/acks CASTR.

Parameters

out	out_count	Optional pointer that receives CACNTBR on success.
-----	-----------	--

Returns

bool

Return values

true	FERRE was set at read time (frequency outside limits)
false	FERRE was not set

Precondition

[rx_cac_init\(\)](#) returned k_rx_ok (driver returns false otherwise)

Postcondition

CASTR.FERRF / MENDF / OVFF are cleared (write-1-clear via CAICR)

Note

Safe to call from ISR context provided out_count is ISR-local.

Since

Version 1.0.0

Sample the CAC status and counter buffer, clearing acknowledged flags.

Reads CASTR (status), latches the current CACNTBR (counter buffer) value if out_count is non-NULL, then issues a write-1-clear to the IRQ status flags in CAICR (preserving the IRQ enables). Returns whether the FERRF (frequency-error) flag was set since the last check.

Note

Safe from any context including ISR (single register read + write, no blocking).

Definition at line 314 of file [rx_cac.c](#).

```
00315 {
00316     if (!s_initialized) {
00317         return false;
00318     }
00319     volatile rx_cac_regs_t* regs = cac();
00320     uint8_t status = regs->castr;
00321     if (out_count != nullptr) {
00322         *out_count = (uint32_t)regs->cacntbr;
00323     }
00324     /* Preserve enabled IRQ settings while issuing write-1-clear acknowledgements. */
00325     uint8_t ack =
00326         (uint8_t)(regs->caicr & (uint8_t)(k_cac_caicr_ferrie_mask | k_cac_caicr_mendie_mask |
00327                                         k_cac_caicr_ovfie_mask));
00328     ack |= k_cac_caicr_clear_all_flags;
00329     regs->caicr = ack;
00330     return (bool)((status & k_cac_castr_ferrf_mask) != 0U);
00331 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacntbr](#), [rx_cac_regs_t::caicr](#), [rx_cac_regs_t::castr](#), [k_cac_caicr_clear_all_flags](#), [k_cac_caicr_ferrie_mask](#), [k_cac_caicr_mendie_mask](#), [k_cac_caicr_ovfie_mask](#), [k_cac_castr_ferrf_mask](#), and [s_initialized](#).

Here is the call graph for this function:



8.71.4.2 rx_cac_deinit()

```
rx_err_t rx_cac_deinit (
    void ) [nodiscard]
```

Tear down the CAC module and put it back in module-stop.

Clears CACR0.CFME, disables all interrupt enables, clears status flags, then asserts MSTPCRC.↔ MSTPC19 to power the module down. Init state is reset so a subsequent [rx_cac_init\(\)](#) call is permitted.

Returns

rx_err_t Error code

Return values

k_rx_ok	Module stopped and powered down
k_rx_err_not_initialized	rx_cac_init() not called

Precondition

[rx_cac_init\(\)](#) returned k_rx_ok

Postcondition

CACR0.CFME = 0 and MSTPCRC.MSTPC19 = 1

Since

Version 1.0.0

Tear down the CAC module and put it back in module-stop.

Stops the measurement (CFME=0), clears CACR1 / CACR2 (resetting the reference clock and tolerance), clears all IRQ flags via CAICR, then gates the CAC module clock through [internal_module_stop\(\)](#). Marks the driver as uninitialized.

Note

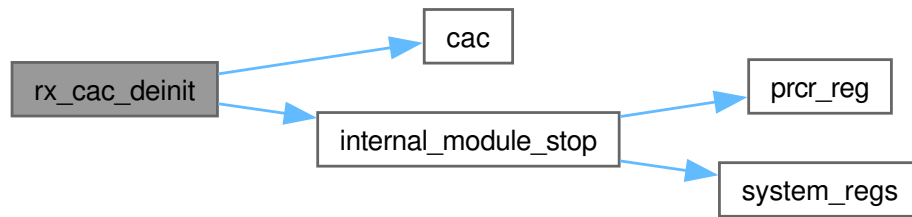
Not thread-safe. Serialize all rx_cac_* operations.

Definition at line 344 of file [rx_cac.c](#).

```
00345 {
00346     if (!s_initialized) {
00347         rx_log_error(s_tag, "rx_cac_deinit: not initialized");
00348         return k_rx_err_not_initialized;
00349     }
00350     volatile rx_cac_regs_t* regs = cac();
00351     regs->cacr0 = k_cac_cacr0_cfme_stop;
00352     regs->cacr1 = 0U;
00353     regs->cacr2 = 0U;
00354     regs->caicr = k_cac_caicr_clear_all_flags;
00355     internal_module_stop();
00356     s_initialized = false;
00357     return k_rx_ok;
00358 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [rx_cac_regs_t::cacr1](#), [rx_cac_regs_t::cacr2](#), [rx_cac_regs_t::caicr](#), [internal_module_stop\(\)](#), [k_cac_cacr0_cfme_stop](#), [k_cac_caicr_clear_all_flags](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.71.4.3 rx'cac'init()

```
rx_err_t rx_cac_init (
    const rx_cac_config_t * config)  [nodiscard]
```

Initialize the CAC module with the supplied configuration.

Performs the full CAC bring-up sequence mandated by the RX72N HW Manual Ch10.3 "Operation":

1. Unlock PRCR.PRC1 and clear MSTPCRC.MSTPC19, then re-lock PRCR.
2. With CACR0.CFME = 0, program CACR1 (FMCS, TCSS, EDGES, CAIE), CACR2 (RSCS, RCDS, RPS, DFS), CAULVR, CALLVR, and CAICR enables.
3. Clear any stale CASTR flags via CAICR write-1-clear.
4. Leave CFME = 0 (caller must invoke [rx_cac_start\(\)](#) to begin measuring).

Parameters

in	config	Pointer to configuration struct (must not be nullptr)
----	--------	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Initialization succeeded, CAC ready
k_rx_err_null_ptr	config is nullptr
k_rx_err_invalid_arg	Invalid clock/divider, or callvr >= caulvr
k_rx_err_invalid_state	CAC already initialized (call deinit first)

Precondition

config != nullptr

config->callvr < config->caulvr

config->reference_clock is in [k_cac_clock_main .. k_cac_clock_iwdtlk]

config->measured_clock is in [k_cac_clock_main .. k_cac_clock_clkout25]

config->target_div is in [k_cac_target_div_1 .. k_cac_target_div_32]

config->reference_div is in [k_cac_ref_div_32 .. k_cac_ref_div_8192]

Postcondition

CAC module is powered on (MSTPCRC.MSTPC19 = 0)
CACR0.CFME = 0 (measurement not yet running)
All configured limits and interrupt enables are loaded

Note

Not thread-safe. Call only during single-threaded initialization.

Since

Version 1.0.0

Definition at line 225 of file [rx_cac.c](#).

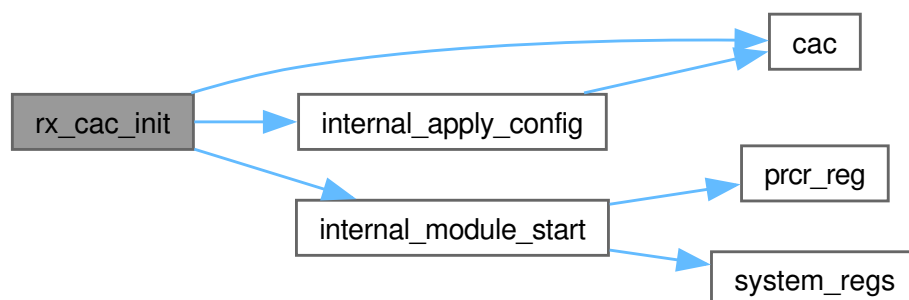
```

00226 {
00227     RX_CHECK_NULL_PTR(config, s_tag, "rx_cac_init: config is nullptr");
00228     if (s_initialized) {
00229         rx_log_error(s_tag, "rx_cac_init: already initialized");
00230         return k_rx_err_invalid_state;
00231     }
00232     if (config->callvr >= config->caulvr) {
00233         rx_log_error(s_tag, "rx_cac_init: callvr must be < caulvr");
00234         return k_rx_err_invalid_arg;
00235     }
00236     if ((uint8_t)config->reference_clock > k_cac_clock_iwdtclk) {
00237         rx_log_error(s_tag, "rx_cac_init: reference_clock out of range");
00238         return k_rx_err_invalid_arg;
00239     }
00240     if ((uint8_t)config->measured_clock > k_cac_clock_clkout25) {
00241         rx_log_error(s_tag, "rx_cac_init: measured_clock out of range");
00242         return k_rx_err_invalid_arg;
00243     }
00244     if ((uint8_t)config->target_div > k_cac_target_div_32) {
00245         rx_log_error(s_tag, "rx_cac_init: target_div out of range");
00246         return k_rx_err_invalid_arg;
00247     }
00248     if ((uint8_t)config->reference_div > k_cac_ref_div_8192) {
00249         rx_log_error(s_tag, "rx_cac_init: reference_div out of range");
00250         return k_rx_err_invalid_arg;
00251     }
00252     internal_module_start();
00253     volatile rx_cac_regs_t* regs = cac();
00254     regs->cacr0 = k_cac_cacr0_cfme_stop;
00255     internal_apply_config(config);
00256     s_initialized = true;
00257     return k_rx_ok;
00258 }

```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [rx_cac_config_t::callvr](#), [rx_cac_config_t::caulvr](#), [internal_apply_config\(\)](#), [internal_module_start\(\)](#), [k_cac_cacr0_cfme_stop](#), [k_cac_clock_clkout25](#), [k_cac_clock_iwdtclk](#), [k_cac_ref_div_8192](#), [k_cac_target_div_32](#), [rx_cac_config_t::measured_clock](#), [rx_cac_config_t::reference_clock](#), [rx_cac_config_t::reference_div](#), [s_initialized](#), [s_tag](#), and [rx_cac_config_t::target_div](#).

Here is the call graph for this function:



8.71.4.4 rx'cac'start()

```
rx_err_t rx_cac_start (
    void ) [nodiscard]
```

Begin CAC measurement (CACR0.CFME = 1).

Returns

rx_err_t Error code

Return values

k_rx_ok	Measurement started
k_rx_err_not_initialized	rx_cac_init() not called

Precondition

[rx_cac_init\(\)](#) returned k_rx_ok

Postcondition

CACR0.CFME = 1

Since

Version 1.0.0

Begin CAC measurement (CACR0.CFME = 1).

Sets CACR0.CFME=1 to enable comparison of the measurement clock against the reference clock configured by [rx_cac_init\(\)](#). Subsequent calls to [rx_cac_check\(\)](#) will report whether the measurement is within the configured tolerance window and return the latched count.

Note

Not thread-safe. Serialize all rx_cac_* operations.

Definition at line 271 of file [rx_cac.c](#).

```
00272 {
00273     if (!s_initialized) {
00274         rx_log_error(s_tag, "rx_cac_start: not initialized");
00275         return k_rx_err_not_initialized;
00276     }
00277     cac()->cacr0 = k_cac_cacr0_cfme_start;
00278     return k_rx_ok;
00279 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [k_cac_cacr0_cfme_start](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.71.4.5 rx_cac_stop()

```
rx_err_t rx_cac_stop (
    void ) [nodiscard]
```

Stop CAC measurement (CACR0.CFME = 0).

Returns

rx_err_t Error code

Return values

k_rx_ok	Measurement stopped
k_rx_err_not_initialized	rx_cac_init() not called

Precondition

[rx_cac_init\(\)](#) returned k_rx_ok

Postcondition

CACR0.CFME = 0

Since

Version 1.0.0

Stop CAC measurement (CACR0.CFME = 0).

Clears CACR0.CFME, halting the measurement counter. The measurement configuration (reference clock, tolerance, IRQ enables) is preserved so a subsequent [rx_cac_start\(\)](#) resumes from a clean state. Useful when entering low-power modes or when switching reference clocks.

Note

Not thread-safe. Serialize all rx_cac_* operations.

Definition at line 292 of file [rx_cac.c](#).

```
00293 {
00294     if (!s_initialized) {
00295         rx_log_error(s_tag, "rx_cac_stop: not initialized");
00296         return k_rx_err_not_initialized;
00297     }
00298     cac()->cacr0 = k_cac_cacr0_cfme_stop;
00299     return k_rx_ok;
00300 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [k_cac_cacr0_cfme_stop](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.72 rx_cac.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_cac.h
00003  * @brief Clock Frequency Accuracy Measurement Circuit (CAC) Driver API for RX72N
00004  *
00005  * @details
00006  * Public driver API for the RX72N Clock Frequency Accuracy Measurement
00007  * Circuit (CAC). The CAC compares a measurement-target clock against a
00008  * reference clock by counting target-clock edges between reference-clock
00009  * edges, then comparing the count against upper/lower limits set in
00010  * CAULVR/CALLVR. If the count is outside the window, the FERRF flag is set
00011  * and (when enabled) a group-interrupt is raised via GROUPBL0.
00012  *
00013  * @par STAR Usage
00014  * STAR uses CAC to watchdog the main-oscillator (MOSC, 24 MHz crystal)
00015  * against the internal HOCO (16 MHz). A drifted crystal - from aging,
00016  * mechanical damage, or PCB leakage - would silently corrupt motor-control
00017  * timing; the FERRF ISR instead triggers motor e-stop.
00018  *
00019  * @par Typical Configuration
00020  * @code{.c}
00021  * rx_cac_config_t cfg = {
00022  *     .measured_clock = k_cac_clock_main,    // 24 MHz crystal (MOSC)
00023  *     .reference_clock = k_cac_clock_hoco,    // 16 MHz HOCO
00024  *     .reference_div   = k_cac_ref_div_1024,  // 16 MHz / 1024 = 15.625 kHz
00025  *     .caulvr          = 1560,                // Upper count bound
00026  *     .callvr          = 1500,                // Lower count bound
00027  *     .enable_ferrie   = true,                // Raise IRQ on drift
00028  * };
00029  * (void)rx_cac_init(&cfg);
00030  * (void)rx_cac_start();
00031  * @endcode
00032  *
00033  * @par NASA Power of 10 Compliance
00034  * - Rule 1: [OK] No goto / setjmp / recursion
00035  * - Rule 2: [OK] All loops bounded
00036  * - Rule 3: [OK] No dynamic allocation (static state only)
00037  * - Rule 4: [OK] All functions < 60 lines
00038  * - Rule 5: [OK] >= 2 checks per function (@pre/@post)
00039  * - Rule 6: [OK] File-scope statics, minimal scope
00040  * - Rule 7: [OK] rx_err_t returns, checked via RX_RETURN_ON_ERROR
00041  * - Rule 8: [OK] C23 typed enums for all constants
00042  * - Rule 9: [OK] Single-level pointers
00043  * - Rule 10: [OK] -Wall -Wextra -Werror clean
00044  *
00045  * @see rx72n_cac_regs.h CAC register definitions
00046  * @see RX72N HW Manual R01UH0824EJ0111 Chapter 10
00047  *
00048  * @author Locked, Inc.
00049  * @date 2026-04-21
00050  * @copyright Copyright (c) 2026 Locked Inc.
00051  * SPDX-License-Identifier: MIT
00052  * @since Version 1.0.0
00053  */
00054
00055 #pragma once
00056
00057 #include <stdbool.h>
00058 #include <stdint.h>
00059
00060 #include "rx_err.h"
00061
00062 #ifdef __cplusplus
00063 extern "C" {
00064 #endif
00065
00066 /**
=====
00067  * Configuration Enums
00068  *
=====
00069  */
00070
00071 /**
00072  * @enum rx_cac_clock_t
00073  * @brief Clock source selector for CAC measured/reference inputs
00074  *
00075  * @details
00076  * Maps directly onto the FMCS[2:0] / RSCS[2:0] field values defined in
00077  * CACR1 / CACR2 (RX72N HW Manual p.393 / p.394). Not every source is legal
00078  * as a reference - RSCS supports values 000-101 only (Main, Sub, HOCO, LOCO,
00079  * PCLKB, IWDTCCLK); the extra values (UCLK, CLKOUT25M) are measured-only.
00080  */

```

```

00081 * @since Version 1.0.0
00082 */
00083 typedef enum : uint8_t {
00084     k_cac_clock_main    = 0, /**< Main-clock oscillator (MOSC, external crystal) */
00085     k_cac_clock_sub     = 1, /**< Sub-clock oscillator (32.768 kHz) */
00086     k_cac_clock_hoco    = 2, /**< High-speed on-chip oscillator */
00087     k_cac_clock_loco    = 3, /**< Low-speed on-chip oscillator */
00088     k_cac_clock_pclkb   = 4, /**< Peripheral clock B */
00089     k_cac_clock_iwdtclk = 5, /**< IWDI-dedicated oscillator */
00090     k_cac_clock_uclk    = 6, /**< USB clock (measured-only, not a valid RSCS) */
00091     k_cac_clock_clkout25 = 7, /**< 25 MHz CLKOUT (measured-only, not a valid RSCS) */
00092 } rx_cac_clock_t;
00093
00094 /**
00095  * @enum rx_cac_ref_div_t
00096  * @brief Reference-clock divider (CACR2.RCDS[1:0]) per RX72N HW Manual p.394
00097  * @since Version 1.0.0
00098  */
00099 typedef enum : uint8_t {
00100     k_cac_ref_div_32    = 0, /**< Reference clock / 32 */
00101     k_cac_ref_div_128   = 1, /**< Reference clock / 128 */
00102     k_cac_ref_div_1024  = 2, /**< Reference clock / 1024 */
00103     k_cac_ref_div_8192  = 3, /**< Reference clock / 8192 */
00104 } rx_cac_ref_div_t;
00105
00106 /**
00107  * @enum rx_cac_target_div_t
00108  * @brief Measured-clock divider (CACR1.TCSS[1:0]) per RX72N HW Manual p.393
00109  *
00110  * @details
00111  * TCSS is a target-clock divider (not a digital filter). Use /1 for the
00112  * fastest allowed count rate and larger dividers only when the measurement
00113  * target clock exceeds the counter bandwidth.
00114  */
00115 * @since Version 1.0.0
00116 */
00117 typedef enum : uint8_t {
00118     k_cac_target_div_1  = 0, /**< Target clock / 1 */
00119     k_cac_target_div_4  = 1, /**< Target clock / 4 */
00120     k_cac_target_div_8  = 2, /**< Target clock / 8 */
00121     k_cac_target_div_32 = 3, /**< Target clock / 32 */
00122 } rx_cac_target_div_t;
00123
00124 /*
00125  * Configuration Struct
00126  */
00127 =====
00128
00129 /**
00130  * @struct rx_cac_config_t
00131  * @brief Configuration parameters for rx_cac_init()
00132  *
00133  * @details
00134  * Fully describes one CAC measurement campaign. Callers fill in every field
00135  * before invoking rx_cac_init(); the driver does not look at partial or
00136  * uninitialised members.
00137  *
00138  * @par Example: MOSC watchdog against HOCO
00139  * @code{.c}
00140  * rx_cac_config_t cfg = {
00141  *     .measured_clock = k_cac_clock_main,
00142  *     .reference_clock = k_cac_clock_hoco,
00143  *     .target_div     = k_cac_target_div_1,
00144  *     .reference_div   = k_cac_ref_div_1024,
00145  *     .caulvr         = 1560,
00146  *     .callvr         = 1500,
00147  *     .enable_ferrie  = true,
00148  *     .enable_mendie  = false,
00149  *     .enable_ovfie   = true,
00150  * };
00151  * @endcode
00152  *
00153  * @invariant caulvr < caulvr (driver rejects otherwise)
00154  */
00155 * @since Version 1.0.0
00156 */
00157 typedef struct {
00158     /**
00159      * @brief Measured (target) clock source - CACR1.FMCS[2:0]
00160      * @details The clock whose frequency is being evaluated. All eight
00161      * sources defined by rx_cac_clock_t are legal here.
00162      */
00163     rx_cac_clock_t measured_clock;
00164 }
00165 /**

```



```

00166 * @brief Reference clock source - CACR2.RSCS[2:0]
00167 * @details The clock the measurement is compared against. Only the
00168 * first six entries of rx_cac_clock_t are legal (Main..IWDTCCLK); passing
00169 * k_cac_clock_uclk or k_cac_clock_clkout25 causes rx_cac_init() to
00170 * return k_rx_err_invalid_arg.
00171 */
00172 rx_cac_clock_t reference_clock;
00173
00174 /**
00175 * @brief Divider applied to the measured clock - CACR1.TCSS[1:0]
00176 * @details Defaults to k_cac_target_div_1 for highest-resolution counts.
00177 */
00178 rx_cac_target_div_t target_div;
00179
00180 /**
00181 * @brief Divider applied to the reference clock - CACR2.RCDS[1:0]
00182 * @details Selects how many reference-clock edges accumulate into a
00183 * single measurement window (32 / 128 / 1024 / 8192).
00184 */
00185 rx_cac_ref_div_t reference_div;
00186
00187 /**
00188 * @brief Upper-limit counter value - CAULVR
00189 * @details FERRF asserts when (counter > caulvr) at sampling time.
00190 */
00191 uint16_t caulvr;
00192
00193 /**
00194 * @brief Lower-limit counter value - CALLVR
00195 * @details FERRF asserts when (counter < callvr) at sampling time.
00196 */
00197 uint16_t callvr;
00198
00199 /**
00200 * @brief Enable frequency-error interrupt (CAICR.FERRIE)
00201 * @details When true, the CAC raises an IRQ on every transition into the
00202 * "counter outside [callvr, caulvr]" state. Must be used together with
00203 * a GROUPBL0 IRQ handler wired to bit 26 (FERRF).
00204 */
00205 bool enable_ferrie;
00206
00207 /**
00208 * @brief Enable measurement-end interrupt (CAICR.MENDIE)
00209 */
00210 bool enable_mendie;
00211
00212 /**
00213 * @brief Enable counter-overflow interrupt (CAICR.OVFIE)
00214 */
00215 bool enable_ovfie;
00216 } rx_cac_config_t;
00217
00218 /*
=====
00219 * Public API
00220 *
=====
00221 */
00222
00223 /**
00224 * @brief Initialize the CAC module with the supplied configuration
00225 *
00226 * @details
00227 * Performs the full CAC bring-up sequence mandated by the RX72N HW Manual
00228 * Ch10.3 "Operation":
00229 * 1. Unlock PRCR.PRC1 and clear MSTPCRC.MSTPC19, then re-lock PRCR.
00230 * 2. With CACR0.CFME = 0, program CACR1 (FMCS, TCSS, EDGES, CAIE),
00231 * CACR2 (RSCS, RCDS, RPS, DFS), CAULVR, CALLVR, and CAICR enables.
00232 * 3. Clear any stale CASTR flags via CAICR write-1-clear.
00233 * 4. Leave CFME = 0 (caller must invoke rx_cac_start() to begin measuring).
00234 */
00235 * @param[in] config Pointer to configuration struct (must not be nullptr)
00236 *
00237 * @return rx_err_t Error code
00238 * @retval k_rx_ok Initialization succeeded, CAC ready
00239 * @retval k_rx_err_null_ptr config is nullptr
00240 * @retval k_rx_err_invalid_arg Invalid clock/divider, or callvr >= caulvr
00241 * @retval k_rx_err_invalid_state CAC already initialized (call deinit first)
00242 *
00243 * @pre config != nullptr
00244 * @pre config->callvr < config->caulvr
00245 * @pre config->reference_clock is in [k_cac_clock_main .. k_cac_clock_iwdtclk]
00246 * @pre config->measured_clock is in [k_cac_clock_main .. k_cac_clock_clkout25]
00247 * @pre config->target_div is in [k_cac_target_div_1 .. k_cac_target_div_32]
00248 * @pre config->reference_div is in [k_cac_ref_div_32 .. k_cac_ref_div_8192]
00249 *
00250 * @post CAC module is powered on (MSTPCRC.MSTPC19 = 0)

```

```

00251 * @post CACR0.CFME = 0 (measurement not yet running)
00252 * @post All configured limits and interrupt enables are loaded
00253 *
00254 * @note Not thread-safe. Call only during single-threaded initialization.
00255 * @since Version 1.0.0
00256 */
00257 [[nodiscard]] rx_err_t rx_cac_init(const rx_cac_config_t* config);
00258
00259 /**
00260 * @brief Begin CAC measurement (CACR0.CFME = 1)
00261 *
00262 * @return rx_err_t Error code
00263 * @retval k_rx_ok Measurement started
00264 * @retval k_rx_err_not_initialized rx_cac_init() not called
00265 *
00266 * @pre rx_cac_init() returned k_rx_ok
00267 * @post CACR0.CFME = 1
00268 *
00269 * @since Version 1.0.0
00270 */
00271 [[nodiscard]] rx_err_t rx_cac_start(void);
00272
00273 /**
00274 * @brief Stop CAC measurement (CACR0.CFME = 0)
00275 *
00276 * @return rx_err_t Error code
00277 * @retval k_rx_ok Measurement stopped
00278 * @retval k_rx_err_not_initialized rx_cac_init() not called
00279 *
00280 * @pre rx_cac_init() returned k_rx_ok
00281 * @post CACR0.CFME = 0
00282 *
00283 * @since Version 1.0.0
00284 */
00285 [[nodiscard]] rx_err_t rx_cac_stop(void);
00286
00287 /**
00288 * @brief Sample the CAC status and counter buffer, clearing acknowledged flags
00289 *
00290 * @details
00291 * Reads CASTR, optionally writes back FERRF / MENDF / OVFF acknowledgement
00292 * bits via CAICR, and returns the last-captured counter buffer via
00293 * @p out_count. If @p out_count is nullptr the counter is not returned; the
00294 * function still reads/acks CASTR.
00295 *
00296 * @param[out] out_count Optional pointer that receives CACNTBR on success.
00297 *
00298 * @return bool
00299 * @retval true FERRF was set at read time (frequency outside limits)
00300 * @retval false FERRF was not set
00301 *
00302 * @pre rx_cac_init() returned k_rx_ok (driver returns false otherwise)
00303 * @post CASTR.FERRF / MENDF / OVFF are cleared (write-1-clear via CAICR)
00304 *
00305 * @note Safe to call from ISR context provided @p out_count is ISR-local.
00306 * @since Version 1.0.0
00307 */
00308 bool rx_cac_check(uint32_t* out_count);
00309
00310 /**
00311 * @brief Tear down the CAC module and put it back in module-stop
00312 *
00313 * @details
00314 * Clears CACR0.CFME, disables all interrupt enables, clears status flags,
00315 * then asserts MSTPCRC.MSTPC19 to power the module down. Init state is
00316 * reset so a subsequent rx_cac_init() call is permitted.
00317 *
00318 * @return rx_err_t Error code
00319 * @retval k_rx_ok Module stopped and powered down
00320 * @retval k_rx_err_not_initialized rx_cac_init() not called
00321 *
00322 * @pre rx_cac_init() returned k_rx_ok
00323 * @post CACR0.CFME = 0 and MSTPCRC.MSTPC19 = 1
00324 *
00325 * @since Version 1.0.0
00326 */
00327 [[nodiscard]] rx_err_t rx_cac_deinit(void);
00328
00329 #ifdef __cplusplus
00330 }
00331 #endif

```

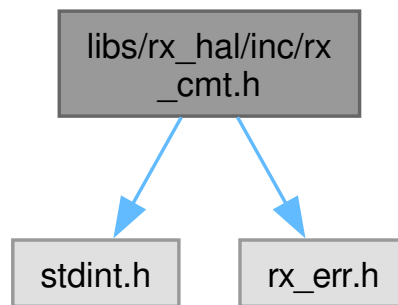
8.73 libs/rx_hal/inc/rx_cmt.h File Reference

CMT (Compare Match Timer) Driver API for RX72N - Periodic Interrupt Generation.

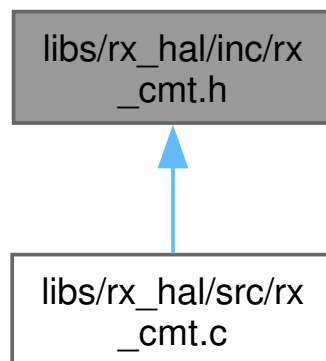
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_cmt.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_cmt_config_t](#)
CMT channel configuration structure.

Typedefs

- typedef void(* [rx_cmt_callback_t](#)) (void *user_data)
CMT interrupt callback function type.

Enumerations

- enum [rx_cmt_channel_t](#) : uint8_t { [k_cmt_channel_0](#) = 0 , [k_cmt_channel_1](#) = 1 , [k_cmt_channel_2](#) = 2 , [k_cmt_channel_3](#) = 3 }
- CMT channel identifiers.

Functions

- `rx_err_t rx_cmt_init (rx_cmt_channel_t channel, const rx_cmt_config_t *config)`
Initialize CMT channel for periodic interrupts.
- `rx_err_t rx_cmt_start (rx_cmt_channel_t channel)`
Start CMT timer.
- `rx_err_t rx_cmt_stop (rx_cmt_channel_t channel)`
Stop CMT timer.
- `rx_err_t rx_cmt_get_count (rx_cmt_channel_t channel, uint16_t *count)`
Get CMT counter value.
- `rx_err_t rx_cmt_deinit (rx_cmt_channel_t channel)`
Deinitialize CMT channel.

8.73.1 Detailed Description

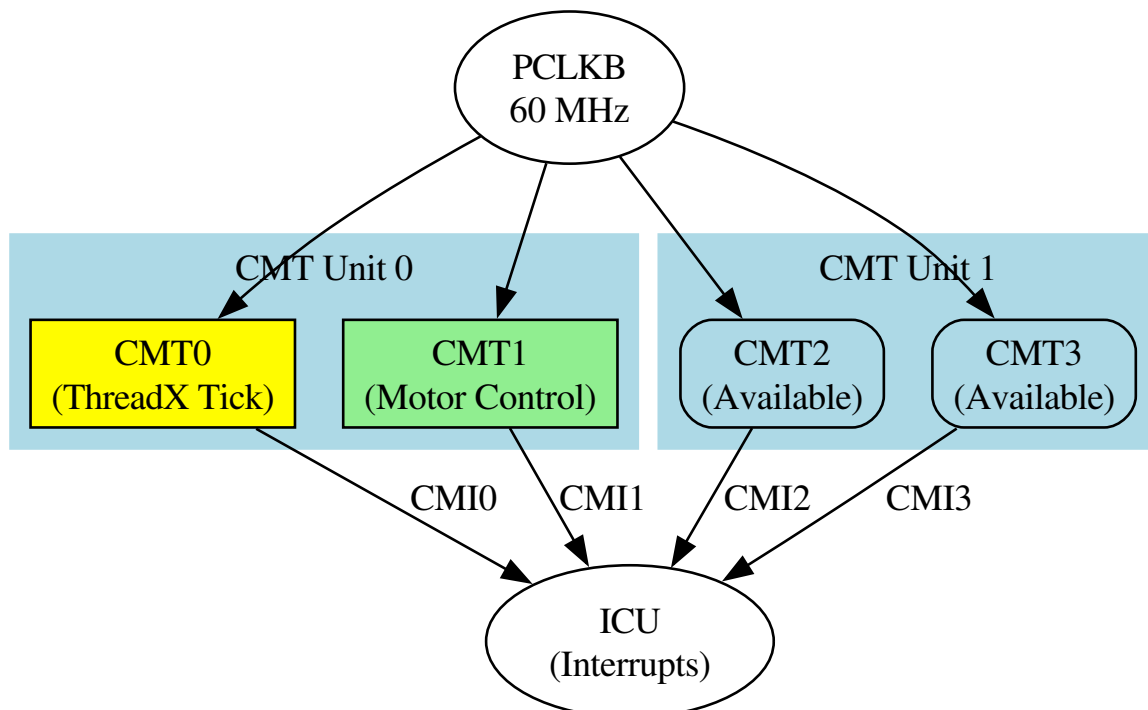
CMT (Compare Match Timer) Driver API for RX72N - Periodic Interrupt Generation.

8.73.2 Overview

The Compare Match Timer (CMT) driver provides periodic interrupt generation for the RX72N microcontroller. CMT is used for system timing, motor control loops, and other periodic tasks requiring precise timing.

8.73.2.1 CMT Hardware Overview

The RX72N has 4 CMT channels organized in 2 units:



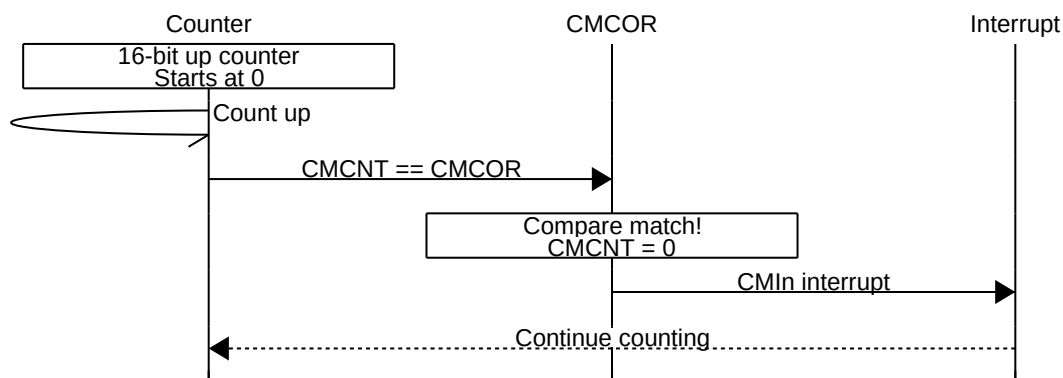
8.73.2.2 Channel Allocation

Channel	Usage	Frequency	Description
CMT0	ThreadX	100 Hz	System tick (RESERVED)
CMT1	Motor	100-1000 Hz	Motor control loop
CMT2	Available	User-defined	General purpose
CMT3	Available	User-defined	General purpose

Warning

CMT0 is reserved for ThreadX system tick. Do not use CMT0 with this API.

8.73.2.3 CMT Operation Principle



8.73.2.4 Clock Divider Selection

Divider	CKS Value	Clock Rate	Min Freq	Max Freq	Notes
PCLKB/ $\frac{60}{8}$	0b00	7.5 MHz	115 Hz	7.5 MHz	High resolution
PCLKB/32	0b01	1.875 MHz	29 Hz	1.875 MHz	Medium resolution
PCLKB/128	0b10	468.75 kHz	7.2 Hz	468 kHz	Recommended
PCLKB/512	0b11	117.2 kHz	1.8 Hz	117 kHz	Low frequency

Calculation example for 250 Hz motor control loop:

PCLKB = 60 MHz

Divider = 128 -> Counter clock = 468.75 kHz

Period = $468750 / 250 = 1875$ counts

8.73.2.5 Memory-Mapped Register Layout

Register	Offset	Description
CMSTR0	0x0000	Start register for CMT0/CMT1
CMSTR1	0x0010	Start register for CMT2/CMT3
CMCRn	+0x00	Control register (clock select, interrupt enable)
CMCNTn	+0x02	16-bit counter register
CMCORn	+0x04	16-bit compare match register

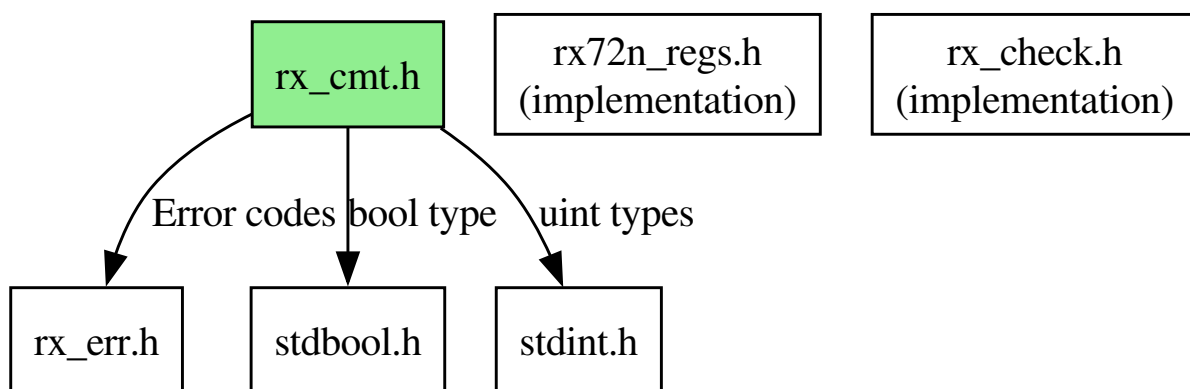
8.73.2.6 NASA Power of 10 Compliance

Rule	Status	Implementation
Rule 1: Control flow	↔ [PASS]↔	No goto, setjmp, or recursion
Rule 2: Loop bounds	↔ [PASS]↔	Bounded divider search (4 iterations max)
Rule 3: No heap	↔ [PASS]↔	Static callback arrays only
Rule 4: Function length	↔ [PASS]↔	All functions under 60 lines
Rule 5: Assertions	↔ [PASS]↔	Parameter validation in all functions
Rule 6: Data scope	↔ [PASS]↔	File-scope static with s_ prefix
Rule 7: Return checks	↔ [PASS]↔	All internal calls checked
Rule 8: Preprocessor	↔ [PASS]↔	C23 typed enums for all constants
Rule 9: Pointers	↔ [PASS]↔	Single-level only (callback + user_data)
Rule 10: Warnings	↔ [PASS]↔	-Wall -Wextra -Werror clean

8.73.2.7 SOLID Principles

Principle	Application
S - Single Responsibility	CMT driver handles only compare match timers
O - Open/Closed	Extended via callback registration
L - Liskov Substitution	All channels use identical API
I - Interface Segregation	Simple 5-function API
D - Dependency Inversion	Callback injection for user code

8.73.2.8 Module Dependencies



See also

[rx72n_cmt_regs.h](#) CMT register definitions
[timer.h](#) Timer abstraction (uses CMT internally)
[hardware_init.h](#) Boot sequence context

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_cmt.h](#).

8.73.3 Data Structure Documentation

8.73.3.1 struct rx_cmt_config_t

CMT channel configuration structure.

Configuration parameters for initializing a CMT channel. All fields must be set before calling [rx_cmt_init\(\)](#).

8.73.3.2 Memory Layout

Offset	Field	Size	Description
0x00	frequency_hz	4	Target interrupt frequency
0x04	callback	4	Interrupt callback pointer
0x08	user_data	4	User data for callback
0x0C	priority	1	Interrupt priority level

Total size: 13 bytes (16 with padding)

8.73.3.3 Frequency Calculation

The driver automatically selects the best clock divider for the requested frequency. Valid range depends on PCLKB (60 MHz):

- Minimum: ~1.8 Hz (with /512 divider)
- Maximum: ~7.5 MHz (with /8 divider)

8.73.3.4 Example Configuration

```
rx_cmt_config_t config = {
    .frequency_hz = 250,           // 250 Hz motor control
    .callback = motor_tick_handler,
    .user_data = &motor_context,
    .priority = 8                 // Medium priority
};
```

See also

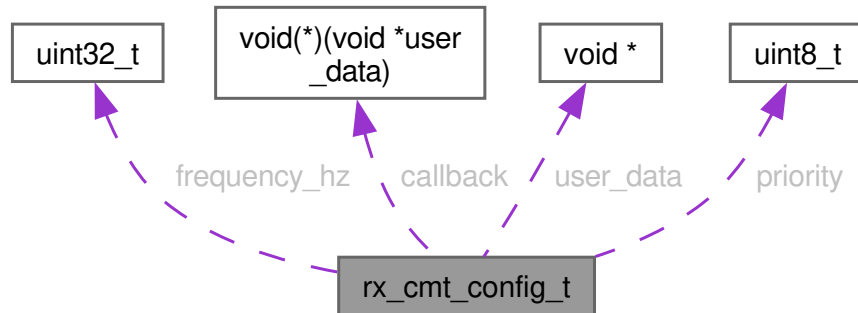
[rx_cmt_init\(\)](#) Uses this configuration

Since

Version 1.0.0

Definition at line 292 of file [rx_cmt.h](#).

Collaboration diagram for rx_cmt_config_t:



Data Fields

rx_cmt_callback_t	callback	Interrupt callback function. Function called on each compare match interrupt. May be NULL if no callback needed (timer runs but no notification). See also rx_cmt_callback_t Callback function signature
uint32_t	frequency_hz	Desired interrupt frequency in Hz. Target frequency for periodic interrupts. Driver selects optimal divider. Valid range: 1-7500000 Hz (depends on PCLKB). Note Actual frequency may differ slightly due to integer division.
uint8_t	priority	Interrupt priority level (0-15). ICU interrupt priority. Higher values = higher priority. • 0: Lowest priority (may be masked) • 15: Highest priority Note Motor control typically uses priority 8-12. Warning Priority 0 disables the interrupt.

<code>void *</code>	<code>user_data</code>	User data passed to callback. Opaque pointer passed to callback on each invocation. Typically used to pass context or state to the callback handler. Note May be NULL if callback doesn't need user data.
---------------------	------------------------	--

8.73.4 Typedef Documentation

8.73.4.1 `rx_cmt_callback_t`

```
typedef void(* rx_cmt_callback_t) (void *user_data)
```

CMT interrupt callback function type.

User-provided function called from CMT interrupt context on each compare match. Keep callbacks short and non-blocking to avoid interrupt latency issues.

8.73.4.2 Callback Execution Context

- Interrupt level: IPL equal to configured priority
- Stack: Interrupt stack (not task stack)
- Timing: Called every compare match period
- Duration: Keep under 10 us for real-time performance

8.73.4.3 Allowed Operations

Operation	Allowed	Notes
Set flag	↔ [PASS]↔	Signal task to process
Write GPIO	↔ [PASS]↔	Toggle debug pin
Queue message	↔ [PASS]↔	Use ThreadX queue
Blocking call	↔ [FAIL]↔	Causes interrupt overrun
Memory allocation	↔ [FAIL]↔	NASA Rule 3 violation

Operation	Allowed	Notes
Long computation	↔ [FAIL]↔	Delays other interrupts

Parameters

in	user_data	User data pointer provided during rx_cmt_init() May be NULL if no user data needed.
----	-----------	---

Note

Callback executes in interrupt context. Keep it fast!

Warning

Do not call blocking functions from callback.

Do not disable interrupts for extended periods.

Example:

```
static volatile bool s_motor_tick = false;

void motor_timer_callback(void* user_data) {
    (void)user_data;
    s_motor_tick = true; // Signal motor task
}
```

See also

[rx_cmt_config_t::callback](#) Callback registration

Since

Version 1.0.0

Definition at line 249 of file [rx_cmt.h](#).

8.73.5 Enumeration Type Documentation

8.73.5.1 rx`cmt`channel`_t

```
enum rx\_cmt\_channel\_t : uint8_t
```

CMT channel identifiers.

Identifies the 4 available CMT channels. Channels are grouped into units:

- Unit 0: CMT0, CMT1 (shared CMSTR0)
- Unit 1: CMT2, CMT3 (shared CMSTR1)

8.73.5.2 Channel Assignment

Channel	Purpose	Notes
k_cmt_channel_0	ThreadX tick	RESERVED - Do not use
k_cmt_channel_1	Motor control	100-1000 Hz typical
k_cmt_channel_2	User defined	Available
k_cmt_channel_3	User defined	Available

See also

[rx_cmt_init\(\)](#) Channel selection for initialization

Since

Version 1.0.0

Enumerator

k_cmt_channel_0	CMT0 - Reserved for ThreadX system tick (100 Hz)
k_cmt_channel_1	CMT1 - Motor control loop (user configurable)
k_cmt_channel_2	CMT2 - Available for general use
k_cmt_channel_3	CMT3 - Available for general use

Definition at line 196 of file [rx_cmt.h](#).

```

00196     : uint8_t {
00197   k_cmt_channel_0 = 0, /**< CMT0 - Reserved for ThreadX system tick (100 Hz) */
00198   k_cmt_channel_1 = 1, /**< CMT1 - Motor control loop (user configurable) */
00199   k_cmt_channel_2 = 2, /**< CMT2 - Available for general use */
00200   k_cmt_channel_3 = 3, /**< CMT3 - Available for general use */
00201 } rx_cmt_channel_t;

```

8.73.6 Function Documentation

8.73.6.1 rx_cmt_deinit()

```

rx_err_t rx_cmt_deinit (
    const rx\_cmt\_channel\_t channel) [nodiscard]

```

Deinitialize CMT channel.

Stops timer, disables interrupts, clears callback, and marks channel as uninitialized. Channel can be reinitialized with [rx_cmt_init\(\)](#) after this.

Parameters

in	channel	CMT channel to deinitialize Valid range: k_cmt_channel_0 to k_cmt_channel_3
----	---------	---

Returns

rx_err_t Deinitialization status

Return values

k_rx_ok	Channel deinitialized successfully
k_rx_err_invalid_arg	Invalid channel number
k_rx_err_hw_error	Hardware failed to stop

Postcondition

Timer stopped
Interrupt disabled
Callback cleared
Channel marked as uninitialized

Note

Does not re-enable module stop bit (CMT module remains powered).

Thread Safety:

Not thread-safe. Ensure no callbacks executing during deinit.

See also

[rx_cmt_init\(\)](#) Reinitialize channel

Since

Version 1.0.0

Deinitialize CMT channel.

Stops the CMT counter, disables the ICU interrupt for the channel, clears the stored callback and user data, and marks the channel as uninitialized. After this call the channel may be re-initialized with [rx_cmt_init\(\)](#).

Algorithm steps:

1. Validate channel range
2. Stop timer via [rx_cmt_stop\(\)](#) and propagate any error
3. Compute ICU vector, IER index, and IER bit for the channel
4. Clear the IER bit to disable the interrupt
5. Clear callback, user_data, and initialized flag

Precondition

channel must be in range [0, 3]

Caller should ensure no other task is using the channel callback

Postcondition

Counter halted (CMSTR bit cleared)

ICU interrupt disabled for channel

s_cmt_initialized[channel] set to false

s_cmt_callback[channel] and s_cmt_user_data[channel] set to nullptr

Note

Safe to call on an uninitialized channel (rx_cmt_stop tolerates it)

Module clock is NOT re-asserted - hardware remains clocked for re-use

Thread Safety:

Not thread-safe. Ensure no ISR or other thread is using the channel.

Example:

```
rx_err_t err = rx_cmt_deinit(k_cmt_channel_1);
if (err != k_rx_ok) {
    rx_log_error("APP", "CMT deinit failed");
}
```

See also

[rx_cmt_init\(\)](#) Re-initialize after deinit

[rx_cmt_stop\(\)](#) Stop without full deinit

Since

Version 1.0.0

Definition at line 1390 of file [rx_cmt.c](#).

```
01391 {
01392     if ((uint8_t)channel >= k_cmt_max_channels) {
01393         return k_rx_err_invalid_arg;
01394     }
01395
01396     /* Stop timer and propagate any errors */
01397     rx_err_t err = rx_cmt_stop(channel);
01398     if (err != k_rx_ok) {
01399         return err;
01400     }
01401
01402     /* Disable interrupt only for channels with fixed ICU vectors (CMI0/CMI1).
01403      * CMI2/CMI3 are SELECTB sources; interrupt wiring for those is not
01404      * implemented (see internal_configure_cmt_interrupt() for details). */
01405     if (channel == k_cmt_channel_0 || channel == k_cmt_channel_1) {
01406         const uint8_t vector = k_vect_cmt0_cmi0 + channel;
01407         const uint8_t ier_index = vector / k_icu_ier_bits_per_reg;
01408         const uint8_t ier_bit = vector % k_icu_ier_bits_per_reg;
01409         const uint8_t ier_mask = (uint8_t)(k_cmt_bit_mask_lsb « ier_bit);
01410         icu()->ier[ier_index] &= (uint8_t)~ier_mask;
01411     }
```

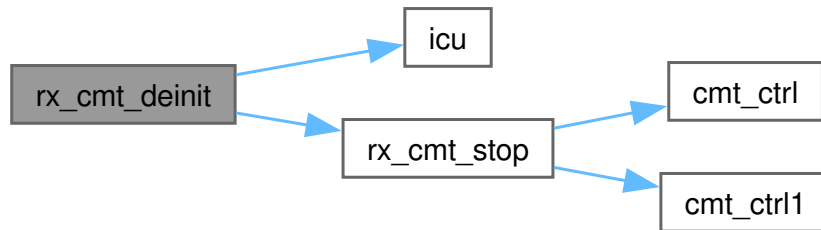
```

01412
01413  /* Clear callback */
01414  s_cmt_callback[channel] = nullptr;
01415  s_cmt_user_data[channel] = nullptr;
01416  s_cmt_initialized[channel] = false;
01417
01418  rx_log_info(s_tag, "CMT deinitialized");
01419
01420  return k_rx_ok;
01421 }

```

References [icu\(\)](#), [rx_icu_regs_t::ier](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_max_channels](#), [k_icu_ier_bits_per_reg](#), [k_vect_cmt0_cmi0](#), [rx_cmt_stop\(\)](#), [s_cmt_callback](#), [s_cmt_initialized](#), [s_cmt_user_data](#), and [s_tag](#).

Here is the call graph for this function:



8.73.6.2 rx'cmt'get'count()

```

rx_err_t rx_cmt_get_count (
    const rx_cmt_channel_t channel,
    uint16_t * count) [nodiscard]

```

Get CMT counter value.

Reads the current 16-bit counter value. Useful for timing measurements or debugging. Counter resets to 0 on each compare match.

Parameters

in	channel	CMT channel to read Valid range: k_cmt_channel_0 to k_cmt_channel_3
out	count	Pointer to store counter value (0-65535)

Returns

rx_err_t Read status

Return values

k_rx_ok	Counter value read successfully
k_rx_err_null_ptr	count pointer is nullptr
k_rx_err_invalid_arg	Invalid channel number
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized via [rx_cmt_init\(\)](#)
count != nullptr

Postcondition

*count contains current counter value

Note

Counter may change immediately after reading if timer running.

Thread Safety:

Safe to call from multiple threads (atomic 16-bit read).

See also

[rx_cmt_stop\(\)](#) Stop timer to freeze counter

Since

Version 1.0.0

Get CMT counter value.

Returns the instantaneous value of the 16-bit CMCNT register for the specified channel. The counter increments at PCLKB divided by the configured divider (/8, /32, /128, or /512) and resets to 0 on compare match. Useful for measuring elapsed time within a period.

Precondition

Channel must be initialized via [rx_cmt_init\(\)](#)
count must point to valid uint16_t storage

Postcondition

*count set to CMCNT register value at time of read
Counter continues running (read is non-destructive)

Note

Value may change between consecutive reads if counter is running
For precise time measurements disable interrupts around consecutive reads

Thread Safety:

Read-only; safe to call from multiple contexts, but value may be stale.

Example:

```
uint16_t ticks = 0;
rx_err_t err = rx_cmt_get_count(k_cmt_channel_1, &ticks);
if (err == k_rx_ok) {
    // ticks = current counter position within the period
}
```

See also

[rx_cmt_init\(\)](#) Initialize channel and set frequency
[rx_cmt_start\(\)](#) Start the counter

Since

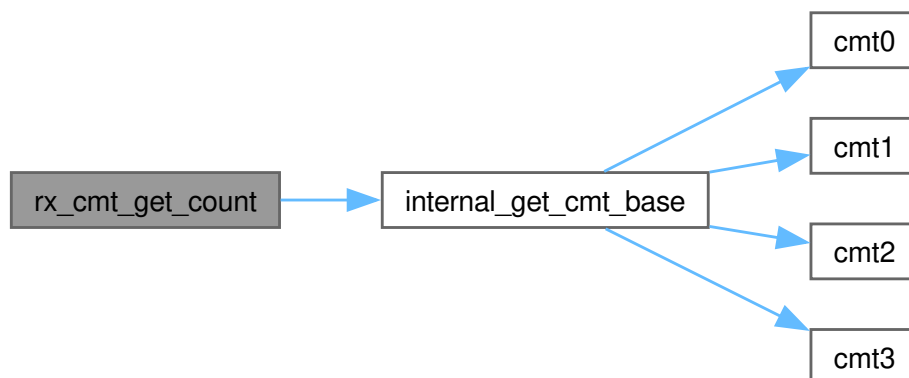
Version 1.0.0

Definition at line 1326 of file [rx_cmt.c](#).

```
01327 {
01328     const volatile rx_cmt_channel_regs_t* cmt = nullptr;
01329
01330     RX_CHECK_NULL_PTR(count, s_tag, "count pointer is nullptr");
01331
01332     if ((uint8_t)channel >= k_cmt_max_channels || !s_cmt_initialized[channel]) {
01333         return k_rx_err_invalid_state;
01334     }
01335
01336     cmt = internal_get_cmt_base(channel);
01337     if (cmt == nullptr) {
01338         return k_rx_err_invalid_arg;
01339     }
01340
01341     *count = cmt->cmcnt;
01342     return k_rx_ok;
01343 }
```

References [rx_cmt_channel_regs_t::cmcnt](#), [internal_get_cmt_base\(\)](#), [k_cmt_max_channels](#), [s_cmt_initialized](#), and [s_tag](#).

Here is the call graph for this function:



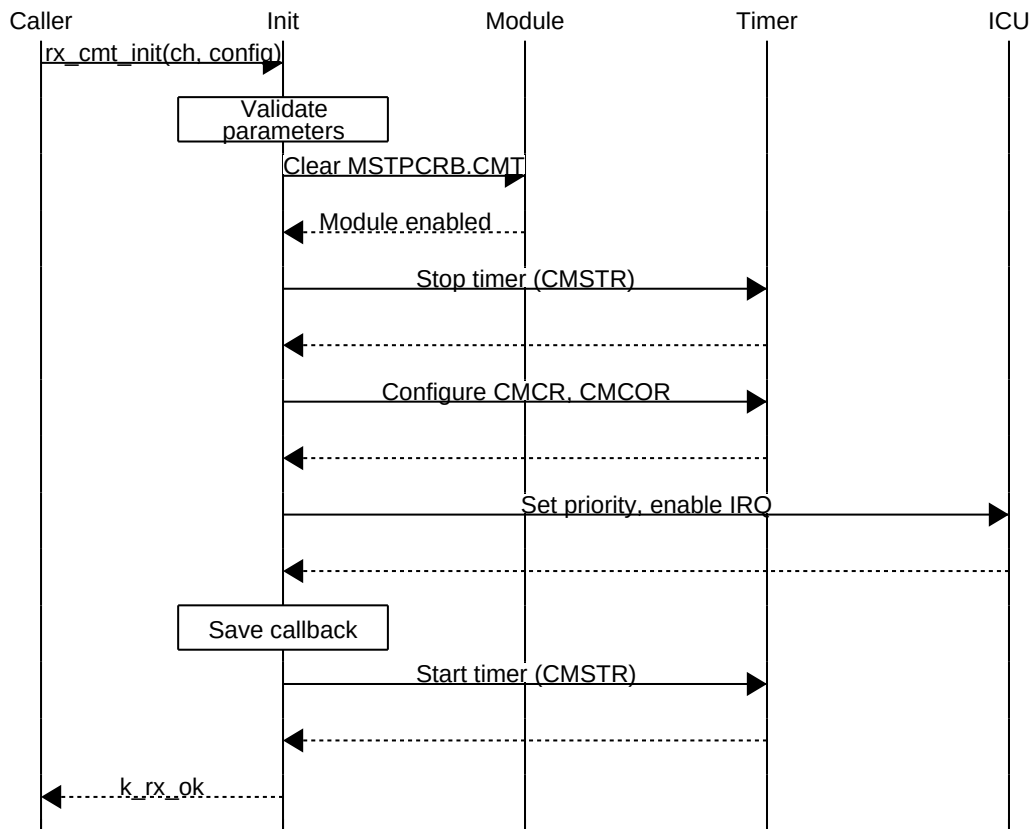
8.73.6.3 rx'cmt'init()

```
rx_err_t rx_cmt_init (
    const rx_cmt_channel_t channel,
    const rx_cmt_config_t * config) [nodiscard]
```

Initialize CMT channel for periodic interrupts.

Configures a CMT channel for periodic interrupt generation at the specified frequency. The timer starts immediately after initialization.

8.73.6.4 Initialization Sequence



8.73.6.5 Divider Auto-Selection

The driver automatically selects the smallest divider that produces a period fitting in the 16-bit compare register:

Frequency	Divider	Period	Actual Freq
1000 Hz	/32	1875	1000.00 Hz
250 Hz	/128	1875	250.00 Hz
100 Hz	/128	4687	100.02 Hz
10 Hz	/512	11719	10.00 Hz

Parameters

in	channel	CMT channel to initialize (1-3, CMT0 reserved) Valid range: k_cmt_channel_1 to k_cmt_channel_3
in	config	Pointer to configuration structure (must not be NULL)

Returns

`rx_err_t` Initialization status

Return values

<code>k_rx_ok</code>	Channel initialized and started successfully
<code>k_rx_err_null_ptr</code>	config is nullptr
<code>k_rx_err_invalid_arg</code>	Invalid channel, frequency out of range, or invalid priority
<code>k_rx_err_conflict</code>	Channel is CMT0 (reserved for ThreadX)

Precondition

Module stop cleared for CMT (or will be cleared by this function)

Channel not currently in use (call `rx_cmt_deinit()` first if reinitializing)

config->frequency_hz > 0 and within achievable range

config->priority in range [1, 15] (0 disables interrupt)

Postcondition

CMT channel running at specified frequency

Callback registered and will be called on each compare match

Interrupt enabled at specified priority

Note

Timer starts automatically. Use `rx_cmt_stop()` to pause.

CMT0 is reserved for ThreadX - returns `k_rx_err_conflict`.

Warning

Do not initialize CMT0 - it's managed by ThreadX.

High frequencies (>100 kHz) may cause excessive interrupt load.

Thread Safety:

Not thread-safe. Call from single thread only, typically during init.

Performance:

- Execution time: ~50 us @ 240 MHz
- Stack usage: ~64 bytes

Example - Motor Control Timer:

```
static void motor_tick(void* user_data) {
    motor_context_t* ctx = (motor_context_t*)user_data;
    ctx->tick_count++;
}

void motor_timer_init(void) {
    static motor_context_t s_motor_ctx;

    rx_cmt_config_t config = {
        .frequency_hz = 250,           // 250 Hz control loop
        .callback = motor_tick,
        .user_data = &s_motor_ctx,
        .priority = 10                 // High priority
    };

    rx_err_t err = rx_cmt_init(k_cmt_channel_1, &config);
    if (err != k_rx_ok) {
        rx_log_error("MOTOR", "Failed to init CMT");
    }
}
```

See also

[rx_cmt_stop\(\)](#) Pause timer
[rx_cmt_start\(\)](#) Resume timer
[rx_cmt_deinit\(\)](#) Release channel

Since

Version 1.0.0

Initialize CMT channel for periodic interrupts.

Configures a CMT (Compare Match Timer) channel to fire a user callback at the requested frequency. Validates parameters, selects the optimal clock divider, enables the module clock, stops the timer for safe configuration, programs timer registers, configures the ICU interrupt routing, saves the callback, and starts the timer.

Algorithm steps:

1. Validate channel and config parameters via [internal_validate_cmt_init_params\(\)](#)
2. Obtain CMT channel register base via [internal_get_cmt_base\(\)](#)
3. Calculate optimal divider and CMCOR value via [internal_calculate_cmt_params\(\)](#)
4. Enable CMT module clock (clear MSTPCRB bit) via [internal_enable_cmt_module_clock\(\)](#)
5. Stop the timer to allow safe register writes via [rx_cmt_stop\(\)](#)
6. Program CMCR (divider + CMIE) and CMCOR via [internal_configure_cmt_timer_registers\(\)](#)
7. Configure ICU interrupt vector priority and enable via [internal_configure_cmt_interrupt\(\)](#)
8. Save callback pointer and mark channel initialized via [internal_save_cmt_callback\(\)](#)
9. Start the timer via [rx_cmt_start\(\)](#)

Clock divider selection: The driver iterates dividers /8, /32, /128, /512 and picks the first that produces a period fitting in 16 bits:

```
period = (PCLKB / divider) / frequency_hz
if (0 < period <= 65535): use this divider
```

Precondition

config must point to a valid `rx_cmt_config_t` structure
config->frequency_hz must be > 0 and $\leq \text{PCLKB} / 8$ (7.5 MHz at 60 MHz PCLKB)
channel must not be `k_cmt_channel_0`
System clocks must be initialized ($\text{PCLKB} = 60 \text{ MHz}$)

Postcondition

Timer running at requested frequency
config->callback registered and called from ISR at each compare match
`s_cmt_initialized[channel]` set to true
ICU interrupt for the channel enabled at specified priority

Note

Not thread-safe - call once per channel during system initialization
Callback executes in interrupt context; keep it short and non-blocking

Warning

CMT0 is reserved for ThreadX; do not use `k_cmt_channel_0`

Thread Safety:

Not thread-safe. Call once per channel during initialization before starting the RTOS scheduler or with interrupts disabled.

Performance:

- Execution time: $\sim 5 \text{ us}$ (mostly register writes)
- Stack usage: 48 bytes

Example:

```
static void my_timer_cb(void* user_data)
{
    // Called at 1 kHz from interrupt context
    (void)user_data;
    toggle_led();
}

const rx_cmt_config_t cfg = {
    .frequency_hz = 1000, // 1 kHz
    .priority     = 5,
    .callback     = my_timer_cb,
    .user_data    = nullptr,
};

rx_err_t err = rx_cmt_init(k_cmt_channel_1, &cfg);
if (err != k_rx_ok) {
    rx_log_error("APP", "CMT init failed");
}
```

See also

[rx_cmt_deinit\(\)](#) Stop and release channel
[rx_cmt_start\(\)](#) Start a previously stopped channel
[rx_cmt_stop\(\)](#) Stop the timer without deinitializing
[rx_cmt_config_t](#) Configuration structure definition

Since

Version 1.0.0

Definition at line 1051 of file [rx_cmt.c](#).

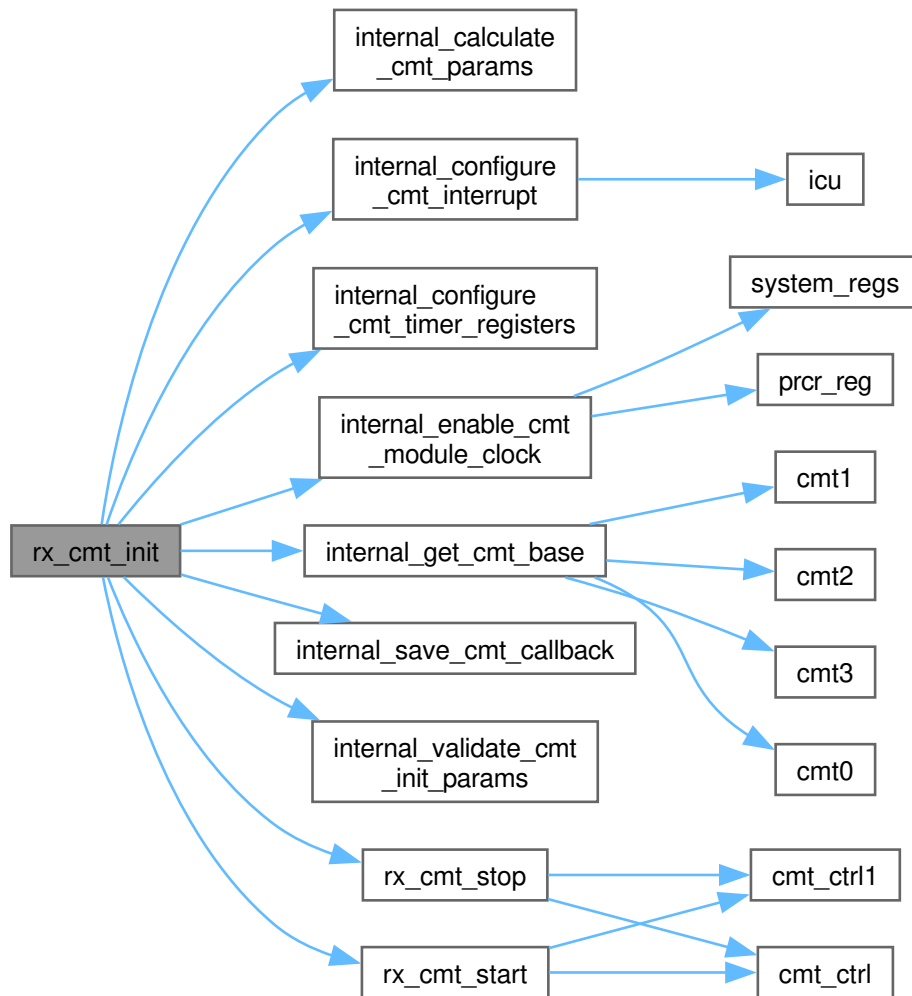
```

01052 {
01053     /* Validate parameters */
01054     rx_err_t err = internal_validate_cmt_init_params(channel, config);
01055     if (err != k_rx_ok) {
01056         return err;
01057     }
01058
01059     volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(channel);
01060     if (cmt == nullptr) {
01061         return k_rx_err_invalid_arg;
01062     }
01063
01064     /* Calculate divider and compare value */
01065     uint8_t divider;
01066     uint16_t cmcor;
01067     err = internal_calculate_cmt_params(config->frequency_hz, &divider, &cmcor);
01068     if (err != k_rx_ok) {
01069         return err;
01070     }
01071
01072     /* Enable CMT module (clear module stop bit) */
01073     internal_enable_cmt_module_clock();
01074
01075     /* Stop timer before configuration */
01076     err = rx_cmt_stop(channel);
01077     if (err != k_rx_ok) {
01078         return err;
01079     }
01080
01081     /* Configure timer registers */
01082     internal_configure_cmt_timer_registers(cmt, divider, cmcor);
01083
01084     /* Configure interrupt */
01085     err = internal_configure_cmt_interrupt(
01086         (rx_cmt_interrupt_config_t){.channel = channel, .priority = config->priority});
01087     if (err != k_rx_ok) {
01088         return err;
01089     }
01090
01091     /* Save callback and mark initialized */
01092     internal_save_cmt_callback(channel, config);
01093
01094     /* Start timer */
01095     err = rx_cmt_start(channel);
01096     if (err != k_rx_ok) {
01097         return err;
01098     }
01099
01100     rx_log_info(s_tag, "CMT initialized");
01101
01102     return k_rx_ok;
01103 }

```

References [rx_cmt_config_t::frequency_hz](#), [internal_calculate_cmt_params\(\)](#), [internal_configure_cmt_interrupt\(\)](#), [internal_configure_cmt_timer_registers\(\)](#), [internal_enable_cmt_module_clock\(\)](#), [internal_get_cmt_base\(\)](#), [internal_save_cmt_callback\(\)](#), [internal_validate_cmt_init_params\(\)](#), [rx_cmt_config_t::priority](#), [rx_cmt_start\(\)](#), [rx_cmt_stop\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.73.6.6 rx'cmt'start()

```
rx_err_t rx_cmt_start (
    const rx_cmt_channel_t channel) [nodiscard]
```

Start CMT timer.

Starts the timer counter and interrupt generation. The timer is automatically started during [rx_cmt_init\(\)](#), so this function is only needed after calling [rx_cmt_stop\(\)](#) to resume operation.

Parameters

in	channel	CMT channel to start Valid range: k_cmt_channel_0 to k_cmt_channel_3
----	---------	--

Returns

`rx_err_t` Start status

Return values

<code>k_rx_ok</code>	Timer started successfully
<code>k_rx_err_invalid_arg</code>	Invalid channel number
<code>k_rx_err_invalid_state</code>	Channel not initialized
<code>k_rx_err_hw_error</code>	Hardware failed to start (CMSTR readback failed)

Precondition

Channel initialized via `rx_cmt_init()`

Postcondition

Timer counter incrementing

Interrupts generating at configured frequency

Note

Counter resumes from current value (not reset to 0).

Thread Safety:

Not thread-safe. Call from single thread or protect with mutex.

See also

`rx_cmt_stop()` Stop timer

`rx_cmt_init()` Initialize channel

Since

Version 1.0.0

Start CMT timer.

Sets the appropriate bit in the CMSTR0 or CMSTR1 register to start the CMT counter running. After setting the bit the register is read back to verify the hardware accepted the write. Channels 0/1 share CMSTR0; channels 2/3 share CMSTR1. Channels 0 and 1 use STR0/STR1 bits respectively, as do channels 2 and 3.

Precondition

channel must be initialized via `rx_cmt_init()`

channel must be in range [0, 3]

Postcondition

CMSTR bit for channel is set
Counter begins incrementing from its current value

Note

Usually called automatically by `rx_cmt_init()`; call explicitly only after `rx_cmt_stop()`
Performs a readback check to verify the hardware write succeeded

Thread Safety:

Not thread-safe; do not call concurrently with stop/deinit on the same channel.

Example:

```
// Pause then resume timer
rx_cmt_stop(k_cmt_channel_1);
// ... do critical work ...
rx_cmt_start(k_cmt_channel_1);
```

See also

`rx_cmt_stop()` Stop the counter
`rx_cmt_init()` Initialize channel (calls start internally)

Since

Version 1.0.0

Definition at line 1142 of file `rx_cmt.c`.

```
01143 {
01144     if ((uint8_t)channel >= k_cmt_max_channels) {
01145         return k_rx_err_invalid_arg;
01146     }
01147
01148     if (!s_cmt_initialized[channel]) {
01149         return k_rx_err_invalid_state;
01150     }
01151
01152     /* Set corresponding bit in CMSTR register */
01153     switch (channel) {
01154         case k_cmt_channel_0: {
01155             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01156             cmt_ctrl()->cmstr0 |= bit_mask;
01157             const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01158             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01159                 return k_rx_err_hw_error;
01160             }
01161             break;
01162         }
01163         case k_cmt_channel_1: {
01164             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01165             cmt_ctrl1()->cmstr0 |= bit_mask;
01166             const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01167             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01168                 return k_rx_err_hw_error;
01169             }
01170             break;
01171         }
01172         case k_cmt_channel_2: {
01173             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01174             cmt_ctrl1()->cmstr0 |= bit_mask;
01175             const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01176             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
```

```

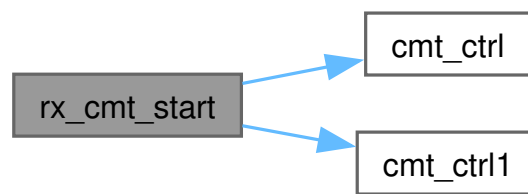
01177     return k_rx_err_hw_error;
01178 }
01179 break;
01180 }
01181 case k_cmt_channel_3: {
01182     const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01183     cmt_ctrl1()->cmstr0 |= bit_mask;
01184     const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01185     if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01186         return k_rx_err_hw_error;
01187     }
01188     break;
01189 }
01190 default:
01191     return k_rx_err_invalid_arg;
01192 }
01193 }
01194 return k_rx_ok;
01195 }

```

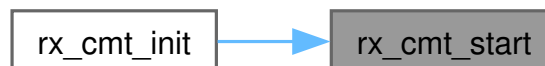
References [rx_cmt_control_regs_t::cmstr0](#), [cmt_ctrl\(\)](#), [cmt_ctrl1\(\)](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_channel_2](#), [k_cmt_channel_3](#), [k_cmt_cmstr_str0](#), [k_cmt_cmstr_str1](#), [k_cmt_max_channels](#), [k_cmt_value_zero](#), and [s_cmt_initialized](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.73.6.7 rx_cmt_stop()

```

rx_err_t rx_cmt_stop (
    const rx_cmt_channel_t channel) [nodiscard]

```

Stop CMT timer.

Stops the timer counter and interrupt generation. Counter value is preserved and can be read with [rx_cmt_get_count\(\)](#). Use [rx_cmt_start\(\)](#) to resume.

Parameters

in	channel	CMT channel to stop Valid range: <code>k_cmt_channel_0</code> to <code>k_cmt_channel_3</code>
----	---------	---

Returns

`rx_err_t` Stop status

Return values

<code>k_rx_ok</code>	Timer stopped successfully
<code>k_rx_err_invalid_arg</code>	Invalid channel number
<code>k_rx_err_hw_error</code>	Hardware failed to stop (CMSTR readback failed)

Postcondition

Timer counter frozen at current value

No further interrupts until [rx_cmt_start\(\)](#) called

Note

Counter value preserved. Call [rx_cmt_start\(\)](#) to resume counting.

Thread Safety:

Not thread-safe. Call from single thread or protect with mutex.

See also

[rx_cmt_start\(\)](#) Resume timer

[rx_cmt_get_count\(\)](#) Read frozen counter value

Since

Version 1.0.0

Stop CMT timer.

Clears the appropriate bit in CMSTR0 or CMSTR1 to halt the CMT counter. After clearing the bit the register is read back to verify the hardware accepted the write. The channel remains initialized; call [rx_cmt_start\(\)](#) to resume counting.

Note that [rx_cmt_stop\(\)](#) does NOT require the channel to be initialized (`s_cmt_initialized` check is omitted intentionally so that [rx_cmt_init\(\)](#) can call it safely before marking the channel as initialized).

Precondition

channel must be in range [0, 3]

Interrupts may still fire if one is already pending when stop is called

Postcondition

CMSTR bit for channel is cleared

Counter stops incrementing (current CMCNT value preserved)

Note

Does not clear the initialized flag - channel can be restarted

Called by `rx_cmt_init()` before programming registers (safe on uninitialized channel)

Thread Safety:

Not thread-safe; do not call concurrently with start/init/deinit on the same channel.

Example:

```
// Temporarily suspend timer
rx_cmt_stop(k_cmt_channel_2);
reconfigure_something();
rx_cmt_start(k_cmt_channel_2);
```

See also

`rx_cmt_start()` Resume counting

`rx_cmt_deinit()` Stop and release channel resources

Since

Version 1.0.0

Definition at line 1237 of file `rx_cmt.c`.

```
01238 {
01239     if ((uint8_t)channel >= k_cmt_max_channels) {
01240         return k_rx_err_invalid_arg;
01241     }
01242
01243     /* Clear corresponding bit in CMSTR register */
01244     switch (channel) {
01245     case k_cmt_channel_0: {
01246         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01247         cmt_ctrl()->cmstr0 &= (uint16_t)~bit_mask;
01248         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01249         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01250             return k_rx_err_hw_error;
01251         }
01252         break;
01253     }
01254     case k_cmt_channel_1: {
01255         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01256         cmt_ctrl()->cmstr0 &= (uint16_t)~bit_mask;
01257         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01258         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01259             return k_rx_err_hw_error;
01260         }
01261         break;
01262     }
01263     case k_cmt_channel_2: {
01264         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01265         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01266         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01267         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01268             return k_rx_err_hw_error;
01269         }
01270         break;
01271     }
01272     case k_cmt_channel_3: {
01273         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01274         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01275         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01276         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01277             return k_rx_err_hw_error;
01278         }
01279         break;
01280     }
```

```

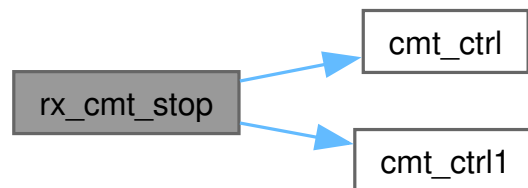
01281     default:
01282     return k_rx_err_invalid_arg;
01283 }
01284
01285 return k_rx_ok;
01286 }

```

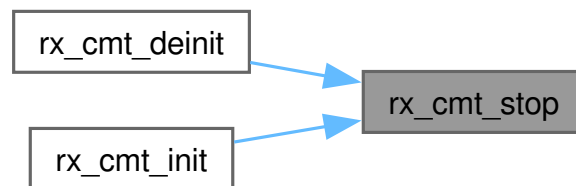
References [rx_cmt_control_regs_t::cmstr0](#), [cmt_ctrl\(\)](#), [cmt_ctrl1\(\)](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_channel_2](#), [k_cmt_channel_3](#), [k_cmt_cmstr_str0](#), [k_cmt_cmstr_str1](#), [k_cmt_max_channels](#), and [k_cmt_value_zero](#).

Referenced by [rx_cmt_deinit\(\)](#), and [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.74 rx_cmt.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_cmt.h
00003  * @brief CMT (Compare Match Timer) Driver API for RX72N - Periodic Interrupt Generation
00004  *
00005  * @details
00006  * # Overview
00007  *
00008  * The Compare Match Timer (CMT) driver provides periodic interrupt generation for the
00009  * RX72N microcontroller. CMT is used for system timing, motor control loops, and
00010  * other periodic tasks requiring precise timing.
00011  *
00012  * ## CMT Hardware Overview
00013  *
00014  * The RX72N has **4 CMT channels** organized in 2 units:
00015  *
00016  * @dot
00017  * digraph cmt_architecture {
00018  *     rankdir=TB;
00019  *     node [shape=box, style=rounded];
00020  *
00021  *     subgraph cluster_unit0 {
00022  *         label="CMT Unit 0";
00023  *         style=filled;
00024  *         color=lightblue;
00025  *         cmt0 [label="CMT0\n(ThreadX Tick)", fillcolor=yellow, style=filled];
00026  *         cmt1 [label="CMT1\n(Motor Control)", fillcolor=lightgreen, style=filled];
00027  *     }
00028  *
00029  *     subgraph cluster_unit1 {
00030  *         label="CMT Unit 1";

```

```

00031 * style=filled;
00032 * color=lightblue;
00033 * cmt2 [label="CMT2\n(Available)"];
00034 * cmt3 [label="CMT3\n(Available)"];
00035 * }
00036 *
00037 * pclk [label="PCLK\n60 MHz", shape=ellipse];
00038 * icu [label="ICU\n(Interrupts)", shape=ellipse];
00039 *
00040 * pclk -> cmt0;
00041 * pclk -> cmt1;
00042 * pclk -> cmt2;
00043 * pclk -> cmt3;
00044 *
00045 * cmt0 -> icu [label="CMT0"];
00046 * cmt1 -> icu [label="CMT1"];
00047 * cmt2 -> icu [label="CMT2"];
00048 * cmt3 -> icu [label="CMT3"];
00049 * }
00050 * @enddot
00051 *
00052 * ## Channel Allocation
00053 *
00054 * | Channel | Usage | Frequency | Description |
00055 * |-----|-----|-----|-----|
00056 * | **CMT0** | ThreadX | 100 Hz | System tick (RESERVED) |
00057 * | **CMT1** | Motor | 100-1000 Hz | Motor control loop |
00058 * | **CMT2** | Available | User-defined | General purpose |
00059 * | **CMT3** | Available | User-defined | General purpose |
00060 *
00061 * @warning **CMT0 is reserved** for ThreadX system tick. Do not use CMT0 with this API.
00062 *
00063 * ## CMT Operation Principle
00064 *
00065 * @msc
00066 * width=600;
00067 * Counter, CMCOR, Interrupt;
00068 *
00069 * Counter box Counter [label="16-bit up counter\nStarts at 0"];
00070 * Counter -> Counter [label="Count up"];
00071 * Counter => CMCOR [label="CMCNT == CMCOR"];
00072 * CMCOR box CMCOR [label="Compare match!\nCMCNT = 0"];
00073 * CMCOR => Interrupt [label="CMIn interrupt"];
00074 * Interrupt » Counter [label="Continue counting"];
00075 * @endmsc
00076 *
00077 * ## Clock Divider Selection
00078 *
00079 * | Divider | CKS Value | Clock Rate | Min Freq | Max Freq | Notes |
00080 * |-----|-----|-----|-----|-----|-----|
00081 * | PCLKB/8 | 0b00 | 7.5 MHz | 115 Hz | 7.5 MHz | High resolution |
00082 * | PCLKB/32 | 0b01 | 1.875 MHz | 29 Hz | 1.875 MHz | Medium resolution |
00083 * | PCLKB/128 | 0b10 | 468.75 kHz | 7.2 Hz | 468 kHz | **Recommended** |
00084 * | PCLKB/512 | 0b11 | 117.2 kHz | 1.8 Hz | 117 kHz | Low frequency |
00085 *
00086 * **Calculation example for 250 Hz motor control loop:**
00087 * ```
00088 * PCLKB = 60 MHz
00089 * Divider = 128 -> Counter clock = 468.75 kHz
00090 * Period = 468750 / 250 = 1875 counts
00091 * ```
00092 *
00093 * ## Memory-Mapped Register Layout
00094 *
00095 * | Register | Offset | Description |
00096 * |-----|-----|-----|
00097 * | CMSTR0 | 0x0000 | Start register for CMT0/CMT1 |
00098 * | CMSTR1 | 0x0010 | Start register for CMT2/CMT3 |
00099 * | CMCRn | +0x00 | Control register (clock select, interrupt enable) |
00100 * | CMCNtn | +0x02 | 16-bit counter register |
00101 * | CMCORn | +0x04 | 16-bit compare match register |
00102 *
00103 * ## NASA Power of 10 Compliance
00104 *
00105 * | Rule | Status | Implementation |
00106 * |-----|-----|-----|
00107 * | **Rule 1: Control flow** | [PASS] | No goto, setjmp, or recursion |
00108 * | **Rule 2: Loop bounds** | [PASS] | Bounded divider search (4 iterations max) |
00109 * | **Rule 3: No heap** | [PASS] | Static callback arrays only |
00110 * | **Rule 4: Function length** | [PASS] | All functions under 60 lines |
00111 * | **Rule 5: Assertions** | [PASS] | Parameter validation in all functions |
00112 * | **Rule 6: Data scope** | [PASS] | File-scope static with s_ prefix |
00113 * | **Rule 7: Return checks** | [PASS] | All internal calls checked |
00114 * | **Rule 8: Preprocessor** | [PASS] | C23 typed enums for all constants |
00115 * | **Rule 9: Pointers** | [PASS] | Single-level only (callback + user_data) |
00116 * | **Rule 10: Warnings** | [PASS] | -Wall -Wextra -Werror clean |
00117 *

```

```

00118 * ## SOLID Principles
00119 *
00120 * | Principle | Application |
00121 * |-----|-----|
00122 * **S - Single Responsibility** | CMT driver handles only compare match timers |
00123 * **O - Open/Closed** | Extended via callback registration |
00124 * **L - Liskov Substitution** | All channels use identical API |
00125 * **I - Interface Segregation** | Simple 5-function API |
00126 * **D - Dependency Inversion** | Callback injection for user code |
00127 *
00128 * ## Module Dependencies
00129 *
00130 * @dot
00131 * digraph dependencies {
00132 *     rankdir=TB;
00133 *     node [shape=box];
00134 *
00135 *     rx_cmt [label="rx_cmt.h", fillcolor=lightgreen, style=filled];
00136 *     rx_err [label="rx_err.h"];
00137 *     stdbool [label="stdbool.h"];
00138 *     stdint [label="stdint.h"];
00139 *     rx72n_regs [label="rx72n_regs.h\n(implementation)"];
00140 *     rx_check [label="rx_check.h\n(implementation)"];
00141 *
00142 *     rx_cmt -> rx_err [label="Error codes"];
00143 *     rx_cmt -> stdbool [label="bool type"];
00144 *     rx_cmt -> stdint [label="uint types"];
00145 * }
00146 * @enddot
00147 *
00148 * @see rx72n_cmt_regs.h CMT register definitions
00149 * @see timer.h Timer abstraction (uses CMT internally)
00150 * @see hardware_init.h Boot sequence context
00151 *
00152 * @author Locked, Inc.
00153 * @date 2026-01-27
00154 * @copyright Copyright (c) 2026 Locked Inc.
00155 * SPDX-License-Identifier: MIT
00156 * @since Version 1.0.0
00157 */
00158
00159 #pragma once
00160
00161 #include <stdint.h>
00162
00163 #include "rx_err.h"
00164
00165 #ifdef __cplusplus
00166 extern "C" {
00167 #endif
00168
00169 /*
=====
00170 * Type Definitions
00171 *
=====
00172 */
00173 /**
00174 * @enum rx_cmt_channel_t
00175 * @brief CMT channel identifiers
00176 *
00177 * @details
00178 * Identifies the 4 available CMT channels. Channels are grouped into units:
00179 * - Unit 0: CMT0, CMT1 (shared CMSTR0)
00180 * - Unit 1: CMT2, CMT3 (shared CMSTR1)
00181 *
00182 * ## Channel Assignment
00183 *
00184 * | Channel | Purpose | Notes |
00185 * |-----|-----|-----|
00186 * | k_cmt_channel_0 | ThreadX tick | **RESERVED - Do not use** |
00187 * | k_cmt_channel_1 | Motor control | 100-1000 Hz typical |
00188 * | k_cmt_channel_2 | User defined | Available |
00189 * | k_cmt_channel_3 | User defined | Available |
00190 *
00191 * @see rx_cmt_init() Channel selection for initialization
00192 *
00193 * @since Version 1.0.0
00194 */
00195
00196 typedef enum : uint8_t {
00197     k_cmt_channel_0 = 0, /**< CMT0 - Reserved for ThreadX system tick (100 Hz) */
00198     k_cmt_channel_1 = 1, /**< CMT1 - Motor control loop (user configurable) */
00199     k_cmt_channel_2 = 2, /**< CMT2 - Available for general use */
00200     k_cmt_channel_3 = 3, /**< CMT3 - Available for general use */
00201 } rx_cmt_channel_t;
00202

```

```

00203 /**
00204  * @brief CMT interrupt callback function type
00205  *
00206  * @details
00207  * User-provided function called from CMT interrupt context on each compare match.
00208  * Keep callbacks short and non-blocking to avoid interrupt latency issues.
00209  *
00210  * ## Callback Execution Context
00211  *
00212  * - **Interrupt level:** IPL equal to configured priority
00213  * - **Stack:** Interrupt stack (not task stack)
00214  * - **Timing:** Called every compare match period
00215  * - **Duration:** Keep under 10 us for real-time performance
00216  *
00217  * ## Allowed Operations
00218  *
00219  * | Operation | Allowed | Notes |
00220  * |-----|-----|-----|
00221  * | Set flag | [PASS] | Signal task to process |
00222  * | Write GPIO | [PASS] | Toggle debug pin |
00223  * | Queue message | [PASS] | Use ThreadX queue |
00224  * | Blocking call | [FAIL] | Causes interrupt overrun |
00225  * | Memory allocation | [FAIL] | NASA Rule 3 violation |
00226  * | Long computation | [FAIL] | Delays other interrupts |
00227  *
00228  * @param[in] user_data User data pointer provided during rx_cmt_init()
00229  * May be NULL if no user data needed.
00230  *
00231  * @note Callback executes in interrupt context. Keep it fast!
00232  * @warning Do not call blocking functions from callback.
00233  * @warning Do not disable interrupts for extended periods.
00234  *
00235  * @par Example:
00236  * @code{.c}
00237  * static volatile bool s_motor_tick = false;
00238  *
00239  * void motor_timer_callback(void* user_data) {
00240  *     (void)user_data;
00241  *     s_motor_tick = true; // Signal motor task
00242  * }
00243  * @endcode
00244  *
00245  * @see rx_cmt_config_t::callback Callback registration
00246  *
00247  * @since Version 1.0.0
00248  */
00249 typedef void (*rx_cmt_callback_t)(void* user_data);
00250
00251 /**
00252  * @struct rx_cmt_config_t
00253  * @brief CMT channel configuration structure
00254  *
00255  * @details
00256  * Configuration parameters for initializing a CMT channel. All fields must be
00257  * set before calling rx_cmt_init().
00258  *
00259  * ## Memory Layout
00260  *
00261  * | Offset | Field | Size | Description |
00262  * |-----|-----|-----|-----|
00263  * | 0x00 | frequency_hz | 4 | Target interrupt frequency |
00264  * | 0x04 | callback | 4 | Interrupt callback pointer |
00265  * | 0x08 | user_data | 4 | User data for callback |
00266  * | 0x0C | priority | 1 | Interrupt priority level |
00267  *
00268  * **Total size:** 13 bytes (16 with padding)
00269  *
00270  * ## Frequency Calculation
00271  *
00272  * The driver automatically selects the best clock divider for the requested
00273  * frequency. Valid range depends on PCLKB (60 MHz):
00274  * - Minimum: ~1.8 Hz (with /512 divider)
00275  * - Maximum: ~7.5 MHz (with /8 divider)
00276  *
00277  * ## Example Configuration
00278  *
00279  * @code{.c}
00280  * rx_cmt_config_t config = {
00281  *     .frequency_hz = 250, // 250 Hz motor control
00282  *     .callback = motor_tick_handler,
00283  *     .user_data = &motor_context,
00284  *     .priority = 8 // Medium priority
00285  * };
00286  * @endcode
00287  *
00288  * @see rx_cmt_init() Uses this configuration
00289  */

```



```

00290 * @since Version 1.0.0
00291 */
00292 typedef struct {
00293 /**
00294  * @brief Desired interrupt frequency in Hz
00295  * @details
00296  * Target frequency for periodic interrupts. Driver selects optimal divider.
00297  * Valid range: 1-7500000 Hz (depends on PCLKB).
00298  * @note Actual frequency may differ slightly due to integer division.
00299  */
00300 uint32_t frequency_hz;
00301
00302 /**
00303  * @brief Interrupt callback function
00304  * @details
00305  * Function called on each compare match interrupt. May be NULL if no
00306  * callback needed (timer runs but no notification).
00307  * @see rx_cmt_callback_t Callback function signature
00308  */
00309 rx_cmt_callback_t callback;
00310
00311 /**
00312  * @brief User data passed to callback
00313  * @details
00314  * Opaque pointer passed to callback on each invocation. Typically used
00315  * to pass context or state to the callback handler.
00316  * @note May be NULL if callback doesn't need user data.
00317  */
00318 void* user_data;
00319
00320 /**
00321  * @brief Interrupt priority level (0-15)
00322  * @details
00323  * ICU interrupt priority. Higher values = higher priority.
00324  * - 0: Lowest priority (may be masked)
00325  * - 15: Highest priority
00326  * @note Motor control typically uses priority 8-12.
00327  * @warning Priority 0 disables the interrupt.
00328  */
00329 uint8_t priority;
00330 } rx_cmt_config_t;
00331
00332 /*
=====
00333 * Public API
00334 *
=====
00335 */
00336
00337 /**
00338  * @brief Initialize CMT channel for periodic interrupts
00339  *
00340  * @details
00341  * Configures a CMT channel for periodic interrupt generation at the specified
00342  * frequency. The timer starts immediately after initialization.
00343  *
00344  * ## Initialization Sequence
00345  *
00346  * @msc
00347  * width=600;
00348  * Caller, Init, Module, Timer, ICU;
00349  *
00350  * Caller => Init [label="rx_cmt_init(ch, config)"];
00351  * Init box Init [label="Validate parameters"];
00352  * Init => Module [label="Clear MSTPCRB.CMT"];
00353  * Module » Init [label="Module enabled"];
00354  * Init => Timer [label="Stop timer (CMSTR)"];
00355  * Timer » Init;
00356  * Init => Timer [label="Configure CMCR, CMCOR"];
00357  * Timer » Init;
00358  * Init => ICU [label="Set priority, enable IRQ"];
00359  * ICU » Init;
00360  * Init box Init [label="Save callback"];
00361  * Init => Timer [label="Start timer (CMSTR)"];
00362  * Timer » Init;
00363  * Init » Caller [label="k_rx_ok"];
00364  * @endmsc
00365  *
00366  * ## Divider Auto-Selection
00367  *
00368  * The driver automatically selects the smallest divider that produces a
00369  * period fitting in the 16-bit compare register:
00370  *
00371  * | Frequency | Divider | Period | Actual Freq |
00372  * |-----|-----|-----|-----|
00373  * | 1000 Hz | /32 | 1875 | 1000.00 Hz |
00374  * | 250 Hz | /128 | 1875 | 250.00 Hz |

```

```

00375 * | 100 Hz | /128 | 4687 | 100.02 Hz |
00376 * | 10 Hz | /512 | 11719 | 10.00 Hz |
00377 *
00378 * @param[in] channel CMT channel to initialize (1-3, CMT0 reserved)
00379 *         Valid range: k_cmt_channel_1 to k_cmt_channel_3
00380 * @param[in] config Pointer to configuration structure (must not be NULL)
00381 *
00382 * @return rx_err_t Initialization status
00383 *
00384 * @retval k_rx_ok Channel initialized and started successfully
00385 * @retval k_rx_err_null_ptr config is nullptr
00386 * @retval k_rx_err_invalid_arg Invalid channel, frequency out of range, or invalid priority
00387 * @retval k_rx_err_conflict Channel is CMT0 (reserved for ThreadX)
00388 *
00389 * @pre Module stop cleared for CMT (or will be cleared by this function)
00390 * @pre Channel not currently in use (call rx_cmt_deinit() first if reinitializing)
00391 * @pre config->frequency_hz > 0 and within achievable range
00392 * @pre config->priority in range [1, 15] (0 disables interrupt)
00393 *
00394 * @post CMT channel running at specified frequency
00395 * @post Callback registered and will be called on each compare match
00396 * @post Interrupt enabled at specified priority
00397 *
00398 * @note Timer starts automatically. Use rx_cmt_stop() to pause.
00399 * @note CMT0 is reserved for ThreadX - returns k_rx_err_conflict.
00400 *
00401 * @warning Do not initialize CMT0 - it's managed by ThreadX.
00402 * @warning High frequencies (>100 kHz) may cause excessive interrupt load.
00403 *
00404 * @par Thread Safety:
00405 * Not thread-safe. Call from single thread only, typically during init.
00406 *
00407 * @par Performance:
00408 * - Execution time: ~50 us @ 240 MHz
00409 * - Stack usage: ~64 bytes
00410 *
00411 * @par Example - Motor Control Timer:
00412 * @code{.c}
00413 * static void motor_tick(void* user_data) {
00414 *     motor_context_t* ctx = (motor_context_t*)user_data;
00415 *     ctx->tick_count++;
00416 * }
00417 *
00418 * void motor_timer_init(void) {
00419 *     static motor_context_t s_motor_ctx;
00420 *
00421 *     rx_cmt_config_t config = {
00422 *         .frequency_hz = 250,        // 250 Hz control loop
00423 *         .callback = motor_tick,
00424 *         .user_data = &s_motor_ctx,
00425 *         .priority = 10              // High priority
00426 *     };
00427 *
00428 *     rx_err_t err = rx_cmt_init(k_cmt_channel_1, &config);
00429 *     if (err != k_rx_ok) {
00430 *         rx_log_error("MOTOR", "Failed to init CMT");
00431 *     }
00432 * }
00433 * @endcode
00434 *
00435 * @see rx_cmt_stop() Pause timer
00436 * @see rx_cmt_start() Resume timer
00437 * @see rx_cmt_deinit() Release channel
00438 *
00439 * @since Version 1.0.0
00440 *
00441 * @callgraph
00442 * @callergraph
00443 */
00444 [[nodiscard]] rx_err_t rx_cmt_init(rx_cmt_channel_t channel, const rx_cmt_config_t* config);
00445
00446 /**
00447 * @brief Start CMT timer
00448 *
00449 * @details
00450 * Starts the timer counter and interrupt generation. The timer is automatically
00451 * started during rx_cmt_init(), so this function is only needed after calling
00452 * rx_cmt_stop() to resume operation.
00453 *
00454 * @param[in] channel CMT channel to start
00455 *         Valid range: k_cmt_channel_0 to k_cmt_channel_3
00456 *
00457 * @return rx_err_t Start status
00458 *
00459 * @retval k_rx_ok Timer started successfully
00460 * @retval k_rx_err_invalid_arg Invalid channel number
00461 * @retval k_rx_err_invalid_state Channel not initialized

```

```

00462 * @retval k_rx_err_hw_error Hardware failed to start (CMSTR readback failed)
00463 *
00464 * @pre Channel initialized via rx_cmt_init()
00465 *
00466 * @post Timer counter incrementing
00467 * @post Interrupts generating at configured frequency
00468 *
00469 * @note Counter resumes from current value (not reset to 0).
00470 *
00471 * @par Thread Safety:
00472 * Not thread-safe. Call from single thread or protect with mutex.
00473 *
00474 * @see rx_cmt_stop() Stop timer
00475 * @see rx_cmt_init() Initialize channel
00476 *
00477 * @since Version 1.0.0
00478 */
00479 [[nodiscard]] rx_err_t rx_cmt_start(rx_cmt_channel_t channel);
00480
00481 /**
00482 * @brief Stop CMT timer
00483 *
00484 * @details
00485 * Stops the timer counter and interrupt generation. Counter value is preserved
00486 * and can be read with rx_cmt_get_count(). Use rx_cmt_start() to resume.
00487 *
00488 * @param[in] channel CMT channel to stop
00489 *             Valid range: k_cmt_channel_0 to k_cmt_channel_3
00490 *
00491 * @return rx_err_t Stop status
00492 *
00493 * @retval k_rx_ok Timer stopped successfully
00494 * @retval k_rx_err_invalid_arg Invalid channel number
00495 * @retval k_rx_err_hw_error Hardware failed to stop (CMSTR readback failed)
00496 *
00497 * @post Timer counter frozen at current value
00498 * @post No further interrupts until rx_cmt_start() called
00499 *
00500 * @note Counter value preserved. Call rx_cmt_start() to resume counting.
00501 *
00502 * @par Thread Safety:
00503 * Not thread-safe. Call from single thread or protect with mutex.
00504 *
00505 * @see rx_cmt_start() Resume timer
00506 * @see rx_cmt_get_count() Read frozen counter value
00507 *
00508 * @since Version 1.0.0
00509 */
00510 [[nodiscard]] rx_err_t rx_cmt_stop(rx_cmt_channel_t channel);
00511
00512 /**
00513 * @brief Get CMT counter value
00514 *
00515 * @details
00516 * Reads the current 16-bit counter value. Useful for timing measurements
00517 * or debugging. Counter resets to 0 on each compare match.
00518 *
00519 * @param[in] channel CMT channel to read
00520 *             Valid range: k_cmt_channel_0 to k_cmt_channel_3
00521 * @param[out] count Pointer to store counter value (0-65535)
00522 *
00523 * @return rx_err_t Read status
00524 *
00525 * @retval k_rx_ok Counter value read successfully
00526 * @retval k_rx_err_null_ptr count pointer is nullptr
00527 * @retval k_rx_err_invalid_arg Invalid channel number
00528 * @retval k_rx_err_invalid_state Channel not initialized
00529 *
00530 * @pre Channel initialized via rx_cmt_init()
00531 * @pre count != nullptr
00532 *
00533 * @post *count contains current counter value
00534 *
00535 * @note Counter may change immediately after reading if timer running.
00536 *
00537 * @par Thread Safety:
00538 * Safe to call from multiple threads (atomic 16-bit read).
00539 *
00540 * @see rx_cmt_stop() Stop timer to freeze counter
00541 *
00542 * @since Version 1.0.0
00543 */
00544 [[nodiscard]] rx_err_t rx_cmt_get_count(rx_cmt_channel_t channel, uint16_t* count);
00545
00546 /**
00547 * @brief Deinitialize CMT channel
00548 *

```

```

00549 * @details
00550 * Stops timer, disables interrupts, clears callback, and marks channel as
00551 * uninitialized. Channel can be reinitialized with rx_cmt_init() after this.
00552 *
00553 * @param[in] channel CMT channel to deinitialize
00554 *             Valid range: k_cmt_channel_0 to k_cmt_channel_3
00555 *
00556 * @return rx_err_t Deinitialization status
00557 *
00558 * @retval k_rx_ok Channel deinitialized successfully
00559 * @retval k_rx_err_invalid_arg Invalid channel number
00560 * @retval k_rx_err_hw_error Hardware failed to stop
00561 *
00562 * @post Timer stopped
00563 * @post Interrupt disabled
00564 * @post Callback cleared
00565 * @post Channel marked as uninitialized
00566 *
00567 * @note Does not re-enable module stop bit (CMT module remains powered).
00568 *
00569 * @par Thread Safety:
00570 * Not thread-safe. Ensure no callbacks executing during deinit.
00571 *
00572 * @see rx_cmt_init() Reinitialize channel
00573 *
00574 * @since Version 1.0.0
00575 */
00576 [[nodiscard]] rx_err_t rx_cmt_deinit(rx_cmt_channel_t channel);
00577
00578 #ifdef __cplusplus
00579 }
00580 #endif

```

8.75 libs/rx_hal/inc/rx_doc.h File Reference

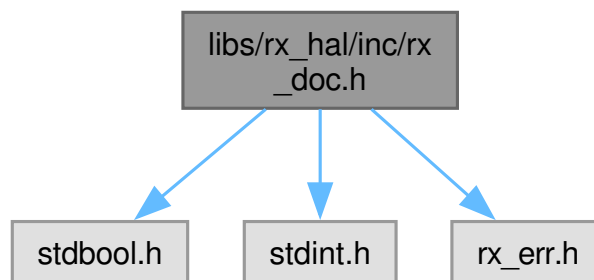
RX72N DOC (Data Operation Circuit) HAL Public API.

```

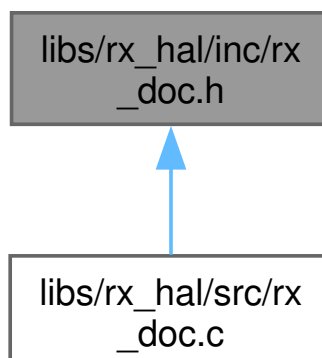
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"

```

Include dependency graph for rx_doc.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_doc_mode_t`: `uint8_t` { `k_rx_doc_mode_compare` = 0, `k_rx_doc_mode_compare_neq` = 1, `k_rx_doc_mode_add` = 2, `k_rx_doc_mode_subtract` = 3 }
Selects the DOC operating mode.

Functions

- `rx_err_t rx_doc_init(rx_doc_mode_t mode)`
Initialize DOC in the selected operating mode.
- `rx_err_t rx_doc_set_reference(uint16_t reference)`
Load the DOC reference value (DODIR) used for the next operation.
- `rx_err_t rx_doc_compare(uint16_t value, bool *matched)`
Execute a compare operation against the loaded reference.
- `rx_err_t rx_doc_add(uint16_t addend, uint16_t *sum, bool *overflow)`
Execute a 16-bit add using the DOC.
- `rx_err_t rx_doc_subtract(uint16_t subtrahend, uint16_t *difference, bool *borrow)`
Execute a 16-bit subtract using the DOC.
- `rx_err_t rx_doc_deinit(void)`
Deinitialize DOC and gate its clock.

8.75.1 Detailed Description

RX72N DOC (Data Operation Circuit) HAL Public API.

Higher-level driver API for the RX72N Data Operation Circuit. The DOC is a fixed-function 16-bit compute accelerator supporting four operations:

- Compare equal (`k_rx_doc_mode_compare`): flag set when `DODIR == DODSR`
- Compare not-equal (`k_rx_doc_mode_compare_neq`): flag set when `DODIR != DODSR`
- Add (`k_rx_doc_mode_add`): `DODSR += DODIR`, flag on carry past `0xFFFF`
- Subtract (`k_rx_doc_mode_subtract`): `DODSR -= DODIR`, flag on borrow below `0x0000`

8.75.1.1 STAR Use Case

Low-cost integrity check on small safety-critical values (PID gains, shared telemetry words, firmware identifiers). Significantly cheaper than a CRC when the payload is only 16 bits.

8.75.1.2 Usage Pattern

```
// 1. Initialize in compare-equal mode
rx_err_t err = rx_doc_init(k_rx_doc_mode_compare);

// 2. Load the expected reference value
err = rx_doc_set_reference(0xCAFE);

// 3. Run the operation
bool matched = false;
err = rx_doc_compare(0xCAFE, &matched); // matched = true

// 4. Later, switch modes for a running-sum check
rx_doc_init(k_rx_doc_mode_add);
rx_doc_set_reference(0x0000);
uint16_t sum = 0;
bool overflow = false;
rx_doc_add(0x1234, &sum, &overflow); // sum = 0x1234, overflow = false

// 5. Power down
rx_doc_deinit();
```

8.75.1.3 Thread Safety

The DOC is a single-instance peripheral. All functions in this API are not thread-safe; callers must serialize access via an RTOS mutex or by confining DOC use to a single task.

NASA Power of 10 Compliance

- Rule 1: No goto / setjmp / recursion
- Rule 2: No unbounded loops
- Rule 3: No dynamic allocation
- Rule 4: All functions ≤ 60 lines
- Rule 5: Minimum 2 validation checks per function
- Rule 6: File-scope state is static, narrowest possible
- Rule 7: All return values checked or cast to void
- Rule 8: C23 typed enums for all constants
- Rule 9: Single-level pointers
- Rule 10: Compiles with -Wall -Wextra -Werror

See also

[rx72n_doc_regs.h](#) Register-layer definitions

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_doc.h](#).

8.75.2 Enumeration Type Documentation

8.75.2.1 rx'doc'mode't

enum [rx_doc_mode_t](#) : uint8_t

Selects the DOC operating mode.

Determines which DOC hardware mode (OMS + DCSEL) is configured by [rx_doc_init\(\)](#).

Mode value	OMS	DCSEL	Hardware behavior
k_rx_doc_mode_compare	00	1	DOPCF set when DODIR == DODSR
k_rx_doc_mode_compare_neq	00	0	DOPCF set when DODIR != DODSR
k_rx_doc_mode_add	01	x	DODSR += DODIR; DOPCF on carry
k_rx_doc_mode_subtract	10	x	DODSR -= DODIR; DOPCF on borrow

Since

Version 1.0.0

Enumerator

k_rx_doc_mode_compare	Compare equal (flag on match)
k_rx_doc_mode_compare_neq	Compare not-equal (flag on mismatch)
k_rx_doc_mode_add	16-bit addition
k_rx_doc_mode_subtract	16-bit subtraction

Definition at line 104 of file [rx_doc.h](#).

```

00104     : uint8_t {
00105   k_rx_doc_mode_compare    = 0, /**< Compare equal (flag on match)    */
00106   k_rx_doc_mode_compare_neq = 1, /**< Compare not-equal (flag on mismatch) */
00107   k_rx_doc_mode_add        = 2, /**< 16-bit addition                */
00108   k_rx_doc_mode_subtract   = 3, /**< 16-bit subtraction            */
00109 } rx_doc_mode_t;
```

8.75.3 Function Documentation

8.75.3.1 rx'doc'add()

```

rx_err_t rx_doc_add (
    uint16_t addend,
    uint16_t * sum,
    bool * overflow)
```

Execute a 16-bit add using the DOC.

Writes addend to DODSR, which triggers DODSR = DODSR_before + DODIR. Reads back DODSR for the sum, reads DOPCF to detect carry (sum overflowed past 0xFFFF), then clears DOPCF.

To chain adds (running sum), call [rx_doc_set_reference\(\)](#) once with the initial accumulator and then call [rx_doc_add\(\)](#) repeatedly; each call uses DODIR as the first operand and addend as the second.

Parameters

in	addend	16-bit value to add to the reference
out	sum	16-bit result (DODSR after the operation)
out	overflow	true if the add carried past 0xFFFF, false otherwise

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success
<code>k_rx_err_null_ptr</code>	sum or overflow is nullptr
<code>k_rx_err_not_initialized</code>	rx_doc_init() has not been called
<code>k_rx_err_invalid_state</code>	Driver is initialized but not in add mode

Precondition

Driver initialized in `k_rx_doc_mode_add`

Postcondition

`*sum == (DODIR + addend) mod 65536`
`*overflow` reflects DOPCF after the operation
DOPCF is cleared

Since

Version 1.0.0

Execute a 16-bit add using the DOC.

Writes the addend into DODSR; the DOC then computes `sum = DODIR + addend` with overflow flagged in DOPCF. Reads the result back through DODSR (post-operation register) and reports the overflow flag, then clears DOPCF before return.

Note

Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 333 of file [rx_doc.c](#).

```

00334 {
00335     RX_CHECK_NULL_PTR(sum, s_tag, "rx_doc_add: sum is NULL");
00336     RX_CHECK_NULL_PTR(overflow, s_tag, "rx_doc_add: overflow is NULL");
00337
00338     if (s_doc_init_state != k_doc_initialized) {
00339         rx_log_error(s_tag, "rx_doc_add: driver not initialized");
00340         return k_rx_err_not_initialized;
00341     }
00342     if (s_doc_mode != k_rx_doc_mode_add) {
00343         rx_log_error(s_tag, "rx_doc_add: driver not in add mode");
00344         return k_rx_err_invalid_state;

```



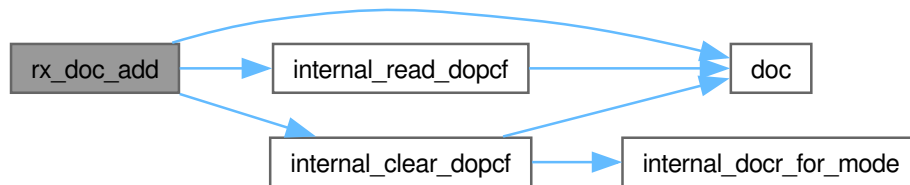
```

00345 }
00346
00347 doc()->dodsr = addend;
00348 *sum        = doc()->dodsr;
00349 *overflow    = internal_read_dopcf();
00350 internal_clear_dopcf();
00351 return k_rx_ok;
00352 }

```

References [doc\(\)](#), [rx_doc_regs_t::dodsr](#), [internal_clear_dopcf\(\)](#), [internal_read_dopcf\(\)](#), [k_doc_initialized](#), [k_rx_doc_mode_add](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.75.3.2 rx`doc`compare()

```

rx_err_t rx_doc_compare (
    uint16_t value,
    bool * matched)

```

Execute a compare operation against the loaded reference.

Writes value to DODSR, which triggers the DOC to evaluate the configured compare condition. Reads DOPCF to report whether the condition fired, then clears DOPCF.

Interpretation depends on the mode selected at init time:

- `k_rx_doc_mode_compare`: `*matched = true` when `DODIR == value`
- `k_rx_doc_mode_compare_neq`: `*matched = true` when `DODIR != value`

Parameters

in	value	16-bit value to compare against the DODIR reference
out	matched	Set to true if the compare condition fired, false otherwise

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success
<code>k_rx_err_null_ptr</code>	matched is nullptr

k_rx_err_not_initialized	rx_doc_init() has not been called
k_rx_err_invalid_state	Driver is initialized but not in a compare mode

Precondition

Driver initialized in a compare mode (compare or compare_neq)
DODIR loaded via [rx_doc_set_reference\(\)](#)

Postcondition

*matched reflects the configured condition
DOPCF is cleared

Since

Version 1.0.0

Execute a compare operation against the loaded reference.

Writes value into DODSR, latches the comparator flag (DOPCF), and reports whether the configured compare condition is satisfied. In k_rx_doc_mode_compare the flag is set on equality; in k_rx_doc_mode_compare_neq the flag is set on inequality. The comparator flag is cleared internally before return so successive compares are independent.

Note

Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 303 of file [rx_doc.c](#).

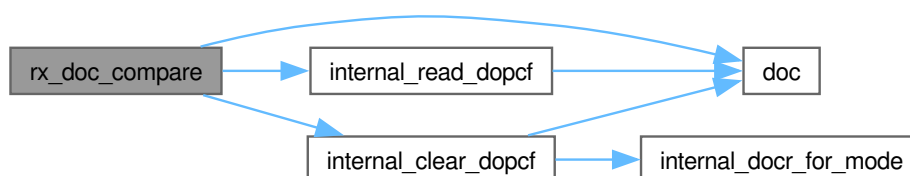
```

00304 {
00305     RX_CHECK_NULL_PTR(matched, s_tag, "rx_doc_compare: matched is NULL");
00306
00307     if (s_doc_init_state != k_doc_initialized) {
00308         rx_log_error(s_tag, "rx_doc_compare: driver not initialized");
00309         return k_rx_err_not_initialized;
00310     }
00311     if ((s_doc_mode != k_rx_doc_mode_compare) && (s_doc_mode != k_rx_doc_mode_compare_neq)) {
00312         rx_log_error(s_tag, "rx_doc_compare: driver not in a compare mode");
00313         return k_rx_err_invalid_state;
00314     }
00315
00316     doc()->dodsr = value;
00317     *matched = internal_read_dopcf();
00318     internal_clear_dopcf();
00319     return k_rx_ok;
00320 }

```

References [doc\(\)](#), [rx_doc_regs_t::dodsr](#), [internal_clear_dopcf\(\)](#), [internal_read_dopcf\(\)](#), [k_doc_initialized](#), [k_rx_doc_mode_compare](#), [k_rx_doc_mode_compare_neq](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.75.3.3 rx'doc'deinit()

```
rx_err_t rx_doc_deinit (
    void )
```

Deinitialize DOC and gate its clock.

Clears DOCR (disables DOPCIE, deselects the mode), sets MSTPCRB.MSTPB6 under PRCR.PRC1 unlock to stop the DOC module, and marks the driver uninitialized.

Safe to call without a prior [rx_doc_init\(\)](#) (returns k_rx_ok and does nothing).

Returns

rx_err_t Error code

Return values

k_rx_ok	Always
---------	--------

Postcondition

DOCR == 0
MSTPCRB.MSTPB6 == 1 (DOC module stopped)
Driver is marked uninitialized

Since

Version 1.0.0

Deinitialize DOC and gate its clock.

Idempotent: returns k_rx_ok immediately if the driver was not initialized. Otherwise clears DOC->DOCR (mode register), disables the DOC module clock through [internal_disable_doc_clock\(\)](#), and marks the driver as uninitialized.

Note

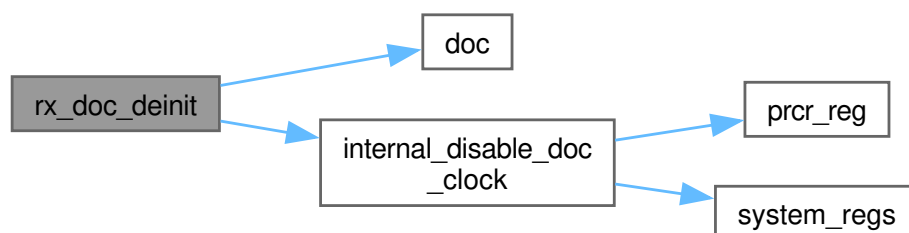
Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 397 of file [rx_doc.c](#).

```
00398 {
00399     if (s_doc_init_state != k_doc_initialized) {
00400         return k_rx_ok;
00401     }
00402
00403     doc()->docr = 0U;
00404     internal_disable_doc_clock();
00405
00406     s_doc_init_state = k_doc_not_initialized;
00407     return k_rx_ok;
00408 }
```

References [doc\(\)](#), [rx_doc_regs_t::docr](#), [internal_disable_doc_clock\(\)](#), [k_doc_initialized](#), [k_doc_not_initialized](#), and [s_doc_init_state](#).

Here is the call graph for this function:



8.75.3.4 rx_doc_init()

```
rx_err_t rx_doc_init (
    rx_doc_mode_t mode)
```

Initialize DOC in the selected operating mode.

Powers on the DOC (clears MSTPCRB.MSTPB6 under PRCR.PRC1 unlock), then configures DOCR with the requested operation mode, clears any stale DOPCF, and marks the driver initialized.

Subsequent operations expect the driver to remain in this mode until [rx_doc_init\(\)](#) is called again with a different mode (re-init is allowed and is the supported way to switch modes).

Parameters

in	mode	Operating mode to configure
----	------	-----------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_invalid_arg	mode is not a valid rx_doc_mode_t value

Precondition

PRCR accessor ([prcr_reg\(\)](#)) is addressable
MSTPCRB is writable once PRCR.PRC1 is unlocked

Postcondition

DOC module stop bit cleared (MSTPCRB.MSTPB6 = 0)
DOCR configured for the requested mode, DOPCIE = 0, DOPCF cleared
Driver is marked initialized

Note

Not thread-safe. Caller must serialize DOC access.

See also

[rx_doc_deinit\(\)](#)

Since

Version 1.0.0

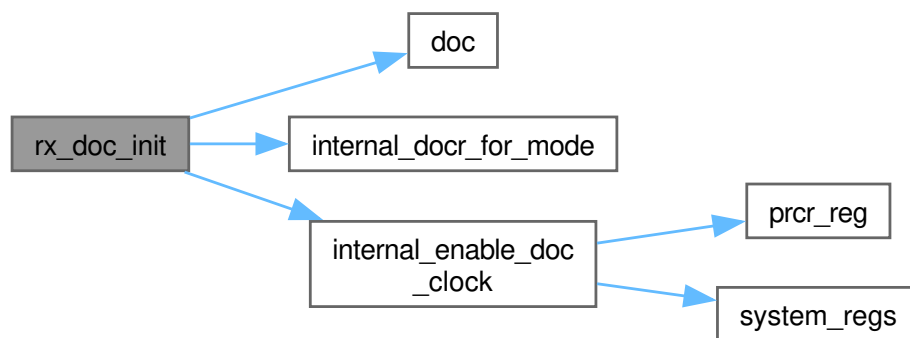
Definition at line 250 of file [rx_doc.c](#).

```

00251 {
00252     uint8_t      mode_bits = 0U;
00253     const rx_err_t err      = internal_docr_for_mode(mode, &mode_bits);
00254     if (err != k_rx_ok) {
00255         rx_log_error(s_tag, "rx_doc_init: invalid mode");
00256         return err;
00257     }
00258
00259     internal_enable_doc_clock();
00260     doc()->docr = (uint8_t)(mode_bits | k_doc_dopcfcl_clear);
00261
00262     s_doc_mode      = mode;
00263     s_doc_init_state = k_doc_initialized;
00264     return k_rx_ok;
00265 }
```

References [doc\(\)](#), [rx_doc_regs_t::docr](#), [internal_docr_for_mode\(\)](#), [internal_enable_doc_clock\(\)](#), [k_doc_dopcfcl_clear](#), [k_doc_initialized](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.75.3.5 rx_doc_set_reference()

```

rx_err_t rx_doc_set_reference (
    uint16_t reference)
```

Load the DOC reference value (DODIR) used for the next operation.

Parameters

in	reference	16-bit reference / operand value
----	-----------	----------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
---------	---------

k_rx_err_not_initialized	rx_doc_init() has not been called
--------------------------	---

Precondition

Driver must be initialized via [rx_doc_init\(\)](#)

Postcondition

DODIR == reference

Note

Writing DODIR does not trigger the operation – only writing DODSR does.

Since

Version 1.0.0

Load the DOC reference value (DODIR) used for the next operation.

Writes the supplied 16-bit reference value into the Data Operation Circuit Input register (DODIR) that subsequent [rx_doc_compare\(\)](#) / [rx_doc_add\(\)](#) / [rx_doc_subtract\(\)](#) calls operate against. Validates driver init state but does not validate or constrain the value range (any 16-bit value is meaningful in compare/add/subtract modes).

Note

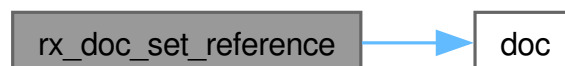
Not thread-safe. The DOC peripheral is global; serialize access at the application layer.

Definition at line 280 of file [rx_doc.c](#).

```
00281 {
00282   RX_VALIDATE_INIT(s_doc_init_state == k_doc_initialized,
00283                   s_tag,
00284                   "rx_doc_set_reference: driver not initialized");
00285
00286   doc()->dodir = reference;
00287   return k_rx_ok;
00288 }
```

References [doc\(\)](#), [rx_doc_regs_t::dodir](#), [k_doc_initialized](#), [s_doc_init_state](#), and [s_tag](#).

Here is the call graph for this function:



8.75.3.6 rx'doc'subtract()

```
rx_err_t rx_doc_subtract (
    uint16_t subtrahend,
    uint16_t * difference,
    bool * borrow)
```

Execute a 16-bit subtract using the DOC.

Writes subtrahend to DODSR, which triggers $DODSR = DODSR_before - DODIR$. Reads back DODSR for the difference, reads DOPCF to detect borrow (difference would have gone below 0x0000), then clears DOPCF.

Parameters

in	subtrahend	16-bit value to subtract from the reference
out	difference	16-bit result (DODSR after the operation)
out	borrow	true if the subtract borrowed below 0x0000, false otherwise

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	difference or borrow is nullptr
k_rx_err_not_initialized	rx_doc_init() has not been called
k_rx_err_invalid_state	Driver is initialized but not in subtract mode

Precondition

Driver initialized in k_rx_doc_mode_subtract

Postcondition

*difference == $(DODIR - subtrahend) \bmod 65536$

*borrow reflects DOPCF after the operation

DOPCF is cleared

Since

Version 1.0.0

Execute a 16-bit subtract using the DOC.

Writes subtrahend into DODSR; the DOC computes $difference = DODIR - subtrahend$ with borrow flagged in DOPCF. Reads the result back through DODSR and reports the borrow flag, then clears DOPCF before return.

Note

Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 365 of file [rx_doc.c](#).

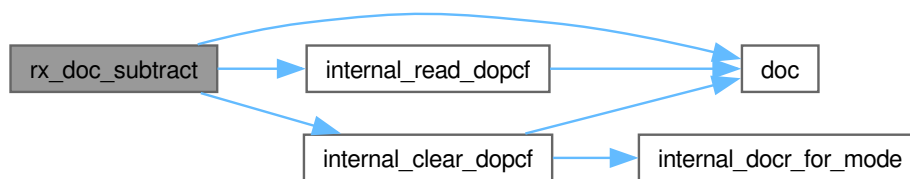
```

00366 {
00367     RX_CHECK_NULL_PTR(difference, s_tag, "rx_doc_subtract: difference is NULL");
00368     RX_CHECK_NULL_PTR(borrow, s_tag, "rx_doc_subtract: borrow is NULL");
00369
00370     if (s_doc_init_state != k_doc_initialized) {
00371         rx_log_error(s_tag, "rx_doc_subtract: driver not initialized");
00372         return k_rx_err_not_initialized;
00373     }
00374     if (s_doc_mode != k_rx_doc_mode_subtract) {
00375         rx_log_error(s_tag, "rx_doc_subtract: driver not in subtract mode");
00376         return k_rx_err_invalid_state;
00377     }
00378
00379     doc()->dodsr = subtrahend;
00380     *difference = doc()->dodsr;
00381     *borrow     = internal_read_dopcf();
00382     internal_clear_dopcf();
00383     return k_rx_ok;
00384 }

```

References [doc\(\)](#), [rx_doc_regs_t::dodsr](#), [internal_clear_dopcf\(\)](#), [internal_read_dopcf\(\)](#), [k_doc_initialized](#), [k_rx_doc_mode_subtract](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.76 rx_doc.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_doc.h
00003  * @brief RX72N DOC (Data Operation Circuit) HAL Public API
00004  *
00005  * @details
00006  * Higher-level driver API for the RX72N Data Operation Circuit. The DOC is a
00007  * fixed-function 16-bit compute accelerator supporting four operations:
00008  *
00009  * - **Compare equal** (k_rx_doc_mode_compare): flag set when DODIR == DODSR
00010  * - **Compare not-equal** (k_rx_doc_mode_compare_neq): flag set when DODIR != DODSR
00011  * - **Add** (k_rx_doc_mode_add): DODSR += DODIR, flag on carry past 0xFFFF
00012  * - **Subtract** (k_rx_doc_mode_subtract): DODSR -= DODIR, flag on borrow below 0x0000
00013  *
00014  * ## STAR Use Case
00015  *
00016  * Low-cost integrity check on small safety-critical values (PID gains, shared
00017  * telemetry words, firmware identifiers). Significantly cheaper than a CRC
00018  * when the payload is only 16 bits.
00019  *
00020  * ## Usage Pattern
00021  *
00022  * @code{.c}
00023  * // 1. Initialize in compare-equal mode
00024  * rx_err_t err = rx_doc_init(k_rx_doc_mode_compare);
00025  *
00026  * // 2. Load the expected reference value
00027  * err = rx_doc_set_reference(0xCAFE);
00028  *
00029  * // 3. Run the operation
00030  * bool matched = false;
00031  * err = rx_doc_compare(0xCAFE, &matched); // matched = true

```



```

00032 *
00033 * // 4. Later, switch modes for a running-sum check
00034 * rx_doc_init(k_rx_doc_mode_add);
00035 * rx_doc_set_reference(0x0000);
00036 * uint16_t sum = 0;
00037 * bool overflow = false;
00038 * rx_doc_add(0x1234, &sum, &overflow);    // sum = 0x1234, overflow = false
00039 *
00040 * // 5. Power down
00041 * rx_doc_deinit();
00042 * @endcode
00043 *
00044 * ## Thread Safety
00045 *
00046 * The DOC is a single-instance peripheral. All functions in this API are
00047 * **not** thread-safe; callers must serialize access via an RTOS mutex or by
00048 * confining DOC use to a single task.
00049 *
00050 * @par NASA Power of 10 Compliance
00051 * - Rule 1: No goto / setjmp / recursion
00052 * - Rule 2: No unbounded loops
00053 * - Rule 3: No dynamic allocation
00054 * - Rule 4: All functions <= 60 lines
00055 * - Rule 5: Minimum 2 validation checks per function
00056 * - Rule 6: File-scope state is static, narrowest possible
00057 * - Rule 7: All return values checked or cast to void
00058 * - Rule 8: C23 typed enums for all constants
00059 * - Rule 9: Single-level pointers
00060 * - Rule 10: Compiles with -Wall -Wextra -Werror
00061 *
00062 * @see rx72n_doc_regs.h Register-layer definitions
00063 *
00064 * @author Locked, Inc.
00065 * @date 2026-04-21
00066 * @copyright Copyright (c) 2026 Locked Inc.
00067 * SPDX-License-Identifier: MIT
00068 * @since Version 1.0.0
00069 */
00070
00071 #pragma once
00072
00073 #include <stdbool.h>
00074 #include <stdint.h>
00075
00076 #include "rx_err.h"
00077
00078 #ifdef __cplusplus
00079 extern "C" {
00080 #endif
00081
00082 /*
=====
00083 * DOC Operating Mode
00084 *
=====
00085 */
00086
00087 /**
00088 * @enum rx_doc_mode_t
00089 * @brief Selects the DOC operating mode
00090 *
00091 * @details
00092 * Determines which DOC hardware mode (OMS + DCSEL) is configured by
00093 * rx_doc_init().
00094 *
00095 * | Mode value | OMS | DCSEL | Hardware behavior |
00096 * |-----|-----|-----|-----|
00097 * | k_rx_doc_mode_compare | 00 | 1 | DOPCF set when DODIR == DODSR |
00098 * | k_rx_doc_mode_compare_neq | 00 | 0 | DOPCF set when DODIR != DODSR |
00099 * | k_rx_doc_mode_add | 01 | x | DODSR += DODIR; DOPCF on carry |
00100 * | k_rx_doc_mode_subtract | 10 | x | DODSR -= DODIR; DOPCF on borrow |
00101 *
00102 * @since Version 1.0.0
00103 */
00104 typedef enum : uint8_t {
00105     k_rx_doc_mode_compare = 0, /**< Compare equal (flag on match) */
00106     k_rx_doc_mode_compare_neq = 1, /**< Compare not-equal (flag on mismatch) */
00107     k_rx_doc_mode_add = 2, /**< 16-bit addition */
00108     k_rx_doc_mode_subtract = 3, /**< 16-bit subtraction */
00109 } rx_doc_mode_t;
00110
00111 /*
=====
00112 * Public API
00113 *
=====
00114 */

```

```

00115
00116 /**
00117  * @brief Initialize DOC in the selected operating mode
00118  *
00119  * @details
00120  * Powers on the DOC (clears MSTPCRB.MSTPB6 under PRCR.PRC1 unlock), then
00121  * configures DOCR with the requested operation mode, clears any stale DOPCF,
00122  * and marks the driver initialized.
00123  *
00124  * Subsequent operations expect the driver to remain in this mode until
00125  * rx_doc_init() is called again with a different mode (re-init is allowed
00126  * and is the supported way to switch modes).
00127  *
00128  * @param[in] mode Operating mode to configure
00129  *
00130  * @return rx_err_t Error code
00131  * @retval k_rx_ok Success
00132  * @retval k_rx_err_invalid_arg mode is not a valid rx_doc_mode_t value
00133  *
00134  * @pre PRCR accessor (prcr_reg()) is addressable
00135  * @pre MSTPCRB is writable once PRCR.PRC1 is unlocked
00136  * @post DOC module stop bit cleared (MSTPCRB.MSTPB6 = 0)
00137  * @post DOCR configured for the requested mode, DOPCF = 0, DOPCF cleared
00138  * @post Driver is marked initialized
00139  *
00140  * @note Not thread-safe. Caller must serialize DOC access.
00141  *
00142  * @see rx_doc_deinit()
00143  * @since Version 1.0.0
00144  */
00145 rx_err_t rx_doc_init(rx_doc_mode_t mode);
00146
00147 /**
00148  * @brief Load the DOC reference value (DODIR) used for the next operation
00149  *
00150  * @param[in] reference 16-bit reference / operand value
00151  *
00152  * @return rx_err_t Error code
00153  * @retval k_rx_ok Success
00154  * @retval k_rx_err_not_initialized rx_doc_init() has not been called
00155  *
00156  * @pre Driver must be initialized via rx_doc_init()
00157  * @post DODIR == reference
00158  *
00159  * @note Writing DODIR does not trigger the operation -- only writing DODSR does.
00160  * @since Version 1.0.0
00161  */
00162 rx_err_t rx_doc_set_reference(uint16_t reference);
00163
00164 /**
00165  * @brief Execute a compare operation against the loaded reference
00166  *
00167  * @details
00168  * Writes @p value to DODSR, which triggers the DOC to evaluate the configured
00169  * compare condition. Reads DOPCF to report whether the condition fired, then
00170  * clears DOPCF.
00171  *
00172  * Interpretation depends on the mode selected at init time:
00173  * - k_rx_doc_mode_compare: *matched = true when DODIR == value
00174  * - k_rx_doc_mode_compare_neq: *matched = true when DODIR != value
00175  *
00176  * @param[in] value 16-bit value to compare against the DODIR reference
00177  * @param[out] matched Set to true if the compare condition fired, false otherwise
00178  *
00179  * @return rx_err_t Error code
00180  * @retval k_rx_ok Success
00181  * @retval k_rx_err_null_ptr matched is nullptr
00182  * @retval k_rx_err_not_initialized rx_doc_init() has not been called
00183  * @retval k_rx_err_invalid_state Driver is initialized but not in a compare mode
00184  *
00185  * @pre Driver initialized in a compare mode (compare or compare_neq)
00186  * @pre DODIR loaded via rx_doc_set_reference()
00187  * @post *matched reflects the configured condition
00188  * @post DOPCF is cleared
00189  *
00190  * @since Version 1.0.0
00191  */
00192 rx_err_t rx_doc_compare(uint16_t value, bool* matched);
00193
00194 /**
00195  * @brief Execute a 16-bit add using the DOC
00196  *
00197  * @details
00198  * Writes @p addend to DODSR, which triggers DODSR = DODSR_before + DODIR.
00199  * Reads back DODSR for the sum, reads DOPCF to detect carry (sum overflowed
00200  * past 0xFFFF), then clears DOPCF.
00201  *

```

```

00202 * To chain adds (running sum), call rx_doc_set_reference() once with the
00203 * initial accumulator and then call rx_doc_add() repeatedly; each call uses
00204 * DODIR as the first operand and @p addend as the second.
00205 *
00206 * @param[in] addend 16-bit value to add to the reference
00207 * @param[out] sum 16-bit result (DODSR after the operation)
00208 * @param[out] overflow true if the add carried past 0xFFFF, false otherwise
00209 *
00210 * @return rx_err_t Error code
00211 * @retval k_rx_ok Success
00212 * @retval k_rx_err_null_ptr sum or overflow is nullptr
00213 * @retval k_rx_err_not_initialized rx_doc_init() has not been called
00214 * @retval k_rx_err_invalid_state Driver is initialized but not in add mode
00215 *
00216 * @pre Driver initialized in k_rx_doc_mode_add
00217 * @post *sum == (DODIR + addend) mod 65536
00218 * @post *overflow reflects DOPCF after the operation
00219 * @post DOPCF is cleared
00220 *
00221 * @since Version 1.0.0
00222 */
00223 rx_err_t rx_doc_add(uint16_t addend, uint16_t* sum, bool* overflow);
00224
00225 /**
00226 * @brief Execute a 16-bit subtract using the DOC
00227 *
00228 * @details
00229 * Writes @p subtrahend to DODSR, which triggers DODSR = DODSR_before - DODIR.
00230 * Reads back DODSR for the difference, reads DOPCF to detect borrow (difference
00231 * would have gone below 0x0000), then clears DOPCF.
00232 *
00233 * @param[in] subtrahend 16-bit value to subtract from the reference
00234 * @param[out] difference 16-bit result (DODSR after the operation)
00235 * @param[out] borrow true if the subtract borrowed below 0x0000, false otherwise
00236 *
00237 * @return rx_err_t Error code
00238 * @retval k_rx_ok Success
00239 * @retval k_rx_err_null_ptr difference or borrow is nullptr
00240 * @retval k_rx_err_not_initialized rx_doc_init() has not been called
00241 * @retval k_rx_err_invalid_state Driver is initialized but not in subtract mode
00242 *
00243 * @pre Driver initialized in k_rx_doc_mode_subtract
00244 * @post *difference == (DODIR - subtrahend) mod 65536
00245 * @post *borrow reflects DOPCF after the operation
00246 * @post DOPCF is cleared
00247 *
00248 * @since Version 1.0.0
00249 */
00250 rx_err_t rx_doc_subtract(uint16_t subtrahend, uint16_t* difference, bool* borrow);
00251
00252 /**
00253 * @brief Deinitialize DOC and gate its clock
00254 *
00255 * @details
00256 * Clears DOCR (disables DOPCIE, deselects the mode), sets MSTPCRB.MSTPB6 under
00257 * PRCR.PRC1 unlock to stop the DOC module, and marks the driver uninitialized.
00258 *
00259 * Safe to call without a prior rx_doc_init() (returns k_rx_ok and does nothing).
00260 *
00261 * @return rx_err_t Error code
00262 * @retval k_rx_ok Always
00263 *
00264 * @post DOCR == 0
00265 * @post MSTPCRB.MSTPB6 == 1 (DOC module stopped)
00266 * @post Driver is marked uninitialized
00267 *
00268 * @since Version 1.0.0
00269 */
00270 rx_err_t rx_doc_deinit(void);
00271
00272 #ifdef __cplusplus
00273 }
00274 #endif

```

8.77 libs/rx`hal/inc/rx`eccram.h File Reference

ECC-Protected RAM (ECCRAM) Driver API.

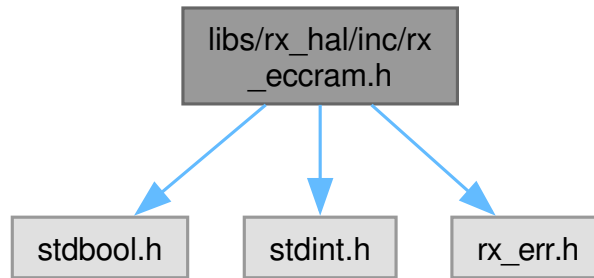
```

#include <stdbool.h>
#include <stdint.h>

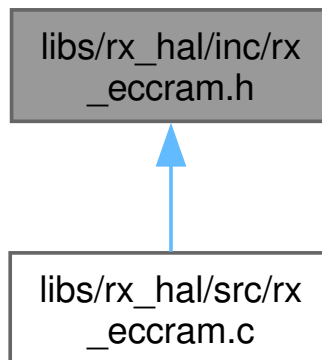
```

```
#include "rx_err.h"
```

Include dependency graph for rx_eccram.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_eccram_status_t](#)
Snapshot of ECCRAM error flags and captured addresses.

Typedefs

- typedef void(* [rx_eccram_on_1bit_fn_t](#)) (uintptr_t addr, void *ctx)
1-bit error ISR callback signature
- typedef void(* [rx_eccram_on_2bit_fn_t](#)) (uintptr_t addr, void *ctx)
2-bit error ISR callback signature

Enumerations

- enum [rx_eccram_mode_t](#) : uint8_t { [k_eccram_mode_disabled](#) = 0 , [k_eccram_mode_correct_only](#) = 1 , [k_eccram_mode_correct_and_detect](#) = 2 , [k_eccram_mode_detect_only](#) = 3 }
ECCRAM operating modes exposed to application code.

Functions

- `rx_err_t rx_eccram_init (rx_eccram_mode_t mode)`
Initialise ECCRAM in the requested operating mode.
- `rx_err_t rx_eccram_get_error_status (rx_eccram_status_t *out)`
Read a snapshot of ECCRAM error status and captured addresses.
- `rx_err_t rx_eccram_clear_errors (void)`
Clear 1-bit and 2-bit error status flags.
- `rx_err_t rx_eccram_register_error_isr (rx_eccram_on_1bit_fn_t on_1bit, rx_eccram_on_2bit_fn_t on_2bit, void *ctx)`
Register user callbacks for ECC error ISRs.
- `uintptr_t rx_eccram_region_start (void)`
Return the inclusive start address of the ECCRAM data region.
- `uintptr_t rx_eccram_region_end (void)`
Return the inclusive end address of the ECCRAM data region.

8.77.1 Detailed Description

ECC-Protected RAM (ECCRAM) Driver API.

Public driver interface for the 32 KB ECC-protected RAM region on the RX72N. Provides mode configuration, error-status readout, error clearing, ISR registration for 1-bit/2-bit errors, and region introspection (start/end addresses for use by linker scripts).

STAR Use Case

Bit-flip-resistant storage for safety-critical mutable state:

- Motor PID integrator accumulators
- Last-good encoder counts
- nanopb frame sequence numbers
- IWDT kick counter
- `shared_data_t` handoff between `motor_control_task` and `comm_task`

Single-bit errors are silently corrected by hardware; double-bit errors trigger the registered `on_2bit` handler, which captures the failing address and typically e-stops the system instead of acting on garbage.

Initialization Order

1. Call `rx_eccram_init(k_eccram_mode_correct_and_detect)` during `hardware_init` before any code places data in ECCRAM.
2. The driver clears `MSTPCRC.MSTPC6`, enables ECC generation without error checking, zeroes the entire 32 KB region with 32-bit stores so every ECC syndrome matches its data, then switches to the requested check mode.
3. Call `rx_eccram_register_error_isr()` to install application handlers (optional – the driver installs safe defaults).

Linker Integration

The driver provides `rx_eccram_region_start()` and `rx_eccram_region_end()` so linker scripts can place a dedicated `.eccram` output section between these addresses. See `star-rx72n-firmware/linker.lds`.

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_eccram.h](#).

8.77.2 Data Structure Documentation

8.77.2.1 struct rx_eccram_status_t

Snapshot of ECCRAM error flags and captured addresses.

Returned by [rx_eccram_get_error_status\(\)](#) so callers can introspect ECC health without touching registers directly.

Invariant

one_bit_addr is meaningful only when one_bit_error is true

two_bit_addr is meaningful only when two_bit_error is true

See also

[rx_eccram_get_error_status\(\)](#)

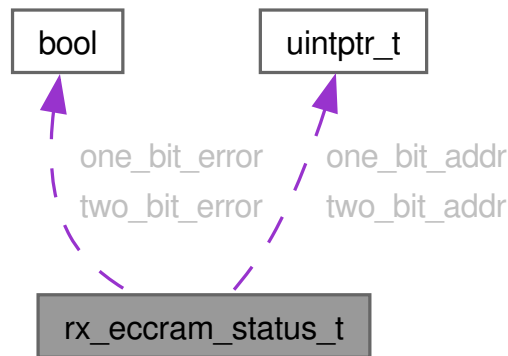
[rx_eccram_clear_errors\(\)](#)

Since

Version 1.0.0

Definition at line 103 of file [rx_eccram.h](#).

Collaboration diagram for rx_eccram_status_t:



Data Fields

uintptr_t	one_bit_addr	Captured address of last 1-bit error (from ECCRAM1ECAD).
bool	one_bit_error	true if ECCRAM1STS.ECC1ERR is set
uintptr_t	two_bit_addr	Captured address of last 2-bit error (from ECCRAM2ECAD).
bool	two_bit_error	true if ECCRAM2STS.ECC2ERR is set

8.77.3 Typedef Documentation

8.77.3.1 rx_eccram_on_1bit_fn_t

```
typedef void(* rx_eccram_on_1bit_fn_t) (uintptr_t addr, void *ctx)
```

1-bit error ISR callback signature

Parameters

in	addr	Captured failing address (from ECCRAM1ECAD)
in	ctx	Opaque context pointer supplied at registration

Note

Invoked from ISR context. Keep it short; log and return.

Since

Version 1.0.0

Definition at line 123 of file [rx_eccram.h](#).

8.77.3.2 rx_eccram_on_2bit_fn_t

```
typedef void(* rx_eccram_on_2bit_fn_t) (uintptr_t addr, void *ctx)
```

2-bit error ISR callback signature

Parameters

in	addr	Captured failing address (from ECCRAM2ECAD)
in	ctx	Opaque context pointer supplied at registration

Note

Invoked from ISR context. Typical implementation triggers e-stop via `shared_data` rather than continuing.

Since

Version 1.0.0

Definition at line 135 of file [rx_eccram.h](#).

8.77.4 Enumeration Type Documentation

8.77.4.1 rx_eccram_mode_t

```
enum rx_eccram_mode_t : uint8_t
```

ECCRAM operating modes exposed to application code.

Maps the raw hardware ECCRAMMODE field (RX72N HW Manual Ch60, section 60.2.9, page 2981) into the four behaviours the application actually cares about. The "detect only" configuration is emulated by installing a 1-bit handler that treats any 1-bit event as a fault rather than a silent correction – the RX72N hardware always corrects 1-bit errors on the data delivered to the CPU when ECC is enabled.

See also

RX72N HW Manual Chapter 60, section 60.2.9, page 2981

Since

Version 1.0.0

Enumerator

k_eccram_mode_disabled	ECC disabled – contents are unchecked (debug/test only).
k_eccram_mode_correct_only	1-bit correction only – 2-bit errors go undetected
k_eccram_mode_correct_and_detect	1-bit correction + 2-bit detection (recommended default)

k_eccram_mode_detect_only	Detect-only: surface both 1-bit and 2-bit errors via ISRs.
---------------------------	--

Definition at line 77 of file [rx_eccram.h](#).

```

00077      : uint8_t {
00078      /** @brief ECC disabled -- contents are unchecked (debug/test only) */
00079      k_eccram_mode_disabled = 0,
00080      /** @brief 1-bit correction only -- 2-bit errors go undetected */
00081      k_eccram_mode_correct_only = 1,
00082      /** @brief 1-bit correction + 2-bit detection (recommended default) */
00083      k_eccram_mode_correct_and_detect = 2,
00084      /** @brief Detect-only: surface both 1-bit and 2-bit errors via ISRs */
00085      k_eccram_mode_detect_only = 3,
00086  } rx_eccram_mode_t;

```

8.77.5 Function Documentation

8.77.5.1 rx_eccram_clear_errors()

```

rx_err_t rx_eccram_clear_errors (
    void ) [nodiscard]

```

Clear 1-bit and 2-bit error status flags.

Writes k_rx_eccram1sts_clear and k_rx_eccram2sts_clear to acknowledge any outstanding error flags. Does not touch the captured addresses: those remain latched until the next error.

Returns

rx_err_t Error code

Return values

k_rx_ok	Flags cleared.
---------	----------------

Precondition

[rx_eccram_init\(\)](#) has been called.

ECCRAMPRCR protection does not guard ECCRAM1STS/ECCRAM2STS (verified in Manual section 60.2.13 – both registers are directly writable without PRCR unlock).

Postcondition

ECCRAM1STS.ECC1ERR == 0 and ECCRAM2STS.ECC2ERR == 0.

Note

Thread Safety: Call from ISR or task context; writes are atomic.

Since

Version 1.0.0

Definition at line 433 of file [rx_eccram.c](#).

```
00434 {
00435     volatile rx_eccram_regs_t* regs = eccram_regs();
00436     regs->eccram1sts = k_rx_eccram1sts_clear;
00437     regs->eccram2sts = k_rx_eccram2sts_clear;
00438     return k_rx_ok;
00439 }
```

References [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [k_rx_eccram1sts_clear](#), and [k_rx_eccram2sts_clear](#).

Here is the call graph for this function:



8.77.5.2 rx'eccram'get'error'status()

```
rx_err_t rx_eccram_get_error_status (
    rx_eccram_status_t * out) [nodiscard]
```

Read a snapshot of ECCRAM error status and captured addresses.

Reads ECCRAM1STS, ECCRAM2STS, ECCRAM1ECAD, and ECCRAM2ECAD into a plain struct. No registers are modified – call [rx_eccram_clear_errors\(\)](#) to acknowledge faults.

Parameters

out	out	Pointer to status struct populated on success.
-----	-----	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Status read successfully.
<code>k_rx_err_null_ptr</code>	out is nullptr.

Precondition

[rx_eccram_init\(\)](#) has been called.

out is a valid writable pointer.

Postcondition

out fields reflect the register state at call time.
No hardware state is modified by this call.

Note

Thread Safety: Safe for concurrent readers (pure register reads).

Since

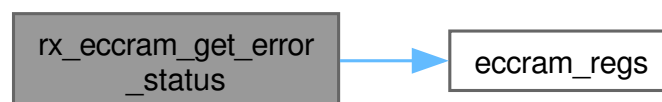
Version 1.0.0

Definition at line 416 of file [rx_eccram.c](#).

```
00417 {
00418     RX_CHECK_NULL_PTR(out, s_tag, "out is nullptr");
00419
00420     volatile rx_eccram_regs_t* regs = eccram_regs();
00421
00422     const uint8_t s1 = regs->eccram1sts;
00423     const uint8_t s2 = regs->eccram2sts;
00424
00425     out->one_bit_error = (bool)((s1 & k_rx_eccram1sts_ecc1err_mask) != 0U);
00426     out->two_bit_error = (bool)((s2 & k_rx_eccram2sts_ecc2err_mask) != 0U);
00427     out->one_bit_addr = (uintptr_t)regs->eccram1ecad;
00428     out->two_bit_addr = (uintptr_t)regs->eccram2ecad;
00429
00430     return k_rx_ok;
00431 }
```

References [rx_eccram_regs_t::eccram1ecad](#), [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2ecad](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [k_rx_eccram1sts_ecc1err_mask](#), [k_rx_eccram2sts_ecc2err_mask](#), [rx_eccram_status_t::one_bit_addr](#), [rx_eccram_status_t::one_bit_error](#), [s_tag](#), [rx_eccram_status_t::two_bit_addr](#), and [rx_eccram_status_t::two_bit_error](#).

Here is the call graph for this function:



8.77.5.3 rx_eccram_init()

```
rx_err_t rx_eccram_init (
    rx_eccram_mode_t mode) [nodiscard]
```

Initialise ECCRAM in the requested operating mode.

Performs the full initialisation sequence required by the RX72N HW manual:

1. Clear MSTPCRC.MSTPC6 under system PRCR unlock so that the ECCRAM control block is clocked (Manual section 60.4.1, page 2994).
2. Unlock ECCRAMPRCR and set ECCRAMMODE = k_rx_eccrammode_ecc_no_check so ECC bits are generated but nothing is checked.

3. Zero all 32 KB of ECCRAM using aligned 32-bit stores (Manual section 60.3.3: uninitialised ECC syndromes would falsely flag).
4. Switch ECCRAMMODE to the mode that backs the requested mode argument and re-lock ECCRAMPRCR.
5. For any mode other than `k_eccram_mode_disabled`, enable the 1-bit status-update latch (`ECCRAM1STSEN`).

Parameters

in	mode	Desired operating mode (see rx_eccram_mode_t)
----	------	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Initialisation successful.
<code>k_rx_err_invalid_arg</code>	mode is outside rx_eccram_mode_t range.
<code>k_rx_err_hw_init_failed</code>	Hardware mode read-back did not match.

Precondition

System clock is configured before this call.

Interrupts are globally disabled OR this function is called before any task can generate ECCRAM traffic.

Postcondition

`MSTPCRC.MSTPC6 == 0` (ECCRAM control block clocked).

All 32 KB of ECCRAM are zeroed with matching ECC syndromes.

`ECCRAMMODE` reflects mode.

Note

Thread Safety: Call once from `hardware_init()`. Not reentrant.

Warning

This function must run before any data is placed in the `.eccram` linker section – it wipes the region.

NASA Power of 10 Compliance:

- Rule 2: Zero-fill loop bounded by `k_rx_eccram_region_size_bytes / 4`.
- Rule 5: 2 preconditions, 3 postconditions.
- Rule 7: Every internal helper's return value is checked.

See also

[rx_eccram_register_error_isr\(\)](#)
[rx_eccram_get_error_status\(\)](#)

Since

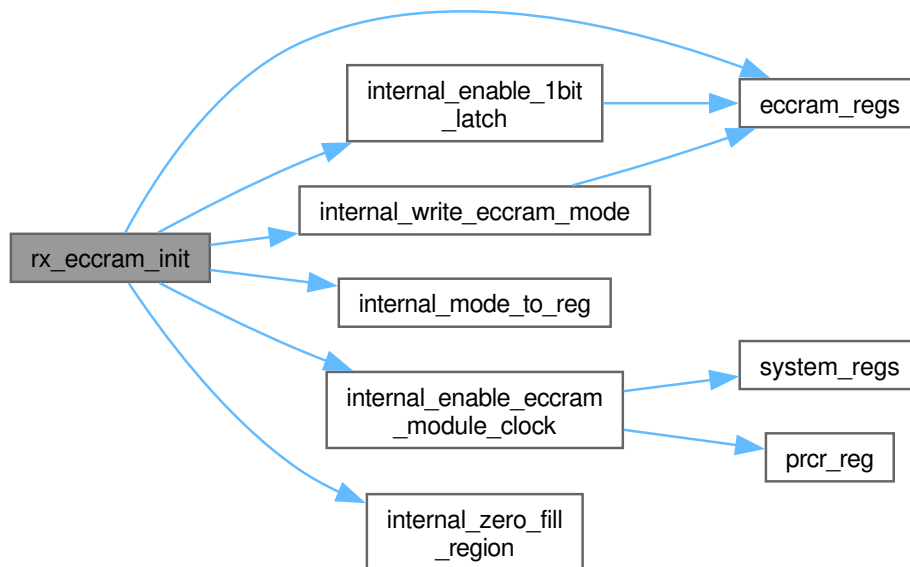
Version 1.0.0

Definition at line 368 of file [rx_eccram.c](#).

```
00369 {
00370     if (mode > k_eccram_mode_detect_only) {
00371         rx_log_error(s_tag, "invalid mode");
00372         return k_rx_err_invalid_arg;
00373     }
00374
00375     s_initialized = false;
00376     s_on_1bit     = NULL;
00377     s_on_2bit     = NULL;
00378     s_isr_ctx     = NULL;
00379     s_last_1bit_addr = 0;
00380     s_last_2bit_addr = 0;
00381
00382     internal_enable_eccram_module_clock();
00383
00384     internal_write_eccram_mode(k_rx_eccrammode_ecc_no_check);
00385
00386     internal_zero_fill_region();
00387
00388     const uint8_t final_mode_reg = internal_mode_to_reg(mode);
00389     internal_write_eccram_mode(final_mode_reg);
00390
00391     volatile rx_eccram_regs_t* regs = eccram_regs();
00392     const uint8_t actual = regs->eccrammode & k_rx_eccrammode_rammod_mask;
00393     /* HW-only failure: a stuck ECCRAMMODE bit can only happen on real silicon
00394      * (e.g. ECCRAMPRCR did not unlock, or the control block lost clock). The
00395      * host-side mock memory always reflects writes verbatim, so the true branch
00396      * is unreachable in unit tests. The behavior is still documented in the
00397      * header (@retval k_rx_err_hw_init_failed) and validated on hardware via
00398      * post-bringup smoke tests. */
00399     if (actual != (final_mode_reg & k_rx_eccrammode_rammod_mask)) { /* GCOVR_EXCL_BR_LINE */
00400         rx_log_error(s_tag, "ECCRAMMODE read-back mismatch"); /* GCOVR_EXCL_LINE */
00401         return k_rx_err_hw_init_failed; /* GCOVR_EXCL_LINE */
00402     }
00403
00404     regs->eccram1sts = k_rx_eccram1sts_clear;
00405     regs->eccram2sts = k_rx_eccram2sts_clear;
00406
00407     if (mode != k_eccram_mode_disabled) {
00408         internal_enable_1bit_latch();
00409     }
00410
00411     s_initialized = true;
00412     rx_log_info(s_tag, "ECCRAM initialised");
00413     return k_rx_ok;
00414 }
```

References [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [rx_eccram_regs_t::eccrammode](#), [internal_enable_1bit_latch\(\)](#), [internal_enable_eccram_module_clock\(\)](#), [internal_mode_to_reg\(\)](#), [internal_write_eccram_mode\(\)](#), [internal_zero_fill_region\(\)](#), [k_eccram_mode_detect_only](#), [k_eccram_mode_disabled](#), [k_rx_eccram1sts_clear](#), [k_rx_eccram2sts_clear](#), [k_rx_eccrammode_ecc_no_check](#), [k_rx_eccrammode_rammod_mask](#), [s_initialized](#), [s_isr_ctx](#), [s_last_1bit_addr](#), [s_last_2bit_addr](#), [s_on_1bit](#), [s_on_2bit](#), and [s_tag](#).

Here is the call graph for this function:



8.77.5.4 rx_eccram_region_end()

```
uintptr_t rx_eccram_region_end (
    void )
```

Return the inclusive end address of the ECCRAM data region.

Returns

Physical address 0x00FFFFFF as a uintptr_t.

Precondition

None.

Postcondition

Return value equals k_rx_eccram_region_end_addr.

Return value - rx_eccram_region_start() + 1 equals 32 KB.

Note

Thread Safety: Returns a compile-time constant; always safe.

See also

[rx_eccram_region_start\(\)](#)

RX72N HW Manual Chapter 60, Table 60.1, page 2977

Since

Version 1.0.0

Definition at line 465 of file [rx_eccram.c](#).

```
00466 {
00467     return k_rx_eccram_region_end_addr;
00468 }
```

References [k_rx_eccram_region_end_addr](#).

8.77.5.5 rx'eccram'region'start()

```
uintptr_t rx_eccram_region_start (  
    void )
```

Return the inclusive start address of the ECCRAM data region.

Exposes the hardware-defined base of the ECC-protected region so that linker scripts and runtime allocators can place objects inside it without hardcoding the literal address.

Returns

Physical address 0x00FF8000 as a `uintptr_t`.

Precondition

None.

Postcondition

Return value equals `k_rx_eccram_region_base_addr`.

Return value is word-aligned.

Note

Thread Safety: Returns a compile-time constant; always safe.

See also

[rx_eccram_region_end\(\)](#)

RX72N HW Manual Chapter 60, Table 60.1, page 2977

Since

Version 1.0.0

Definition at line 460 of file [rx_eccram.c](#).

```
00461 {  
00462     return k_rx_eccram_region_base_addr;  
00463 }
```

References [k_rx_eccram_region_base_addr](#).

8.77.5.6 rx_eccram_register_error_isr()

```
rx_err_t rx_eccram_register_error_isr (
    rx_eccram_on_1bit_fn_t on_1bit,
    rx_eccram_on_2bit_fn_t on_2bit,
    void * ctx) [nodiscard]
```

Register user callbacks for ECC error ISRs.

Installs 1-bit and/or 2-bit error callbacks that the driver's ISR trampoline will invoke on the respective RAM-error vectors. Passing nullptr for either callback leaves the previously installed handler in place, allowing callers to register them independently.

Dependency-Injection Rationale (NASA Rule 9 deviation)

Function pointers are permitted here because they are the exact pattern STAR uses to allow test mocks and application-level policy to substitute in without modifying the driver. See CLAUDE.md "SOLID Principles for C / Dependency Inversion".

Parameters

in	on_1bit	Handler invoked from ISR on 1-bit error (may be nullptr).
in	on_2bit	Handler invoked from ISR on 2-bit error (may be nullptr).
in	ctx	Opaque pointer passed through to the handlers.

Returns

rx_err_t Error code

Return values

k_rx_ok	Handlers installed.
k_rx_err_invalid_state	rx_eccram_init() has not yet been called.

Precondition

[rx_eccram_init\(\)](#) has been called.

Handlers, if non-null, are safe to call from interrupt context.

Postcondition

The driver's ISR trampolines dispatch to the supplied handlers.

ctx is retained until this function is called again.

Note

Thread Safety: Install once during init before interrupts are enabled, or guard calls at the task level.

Since

Version 1.0.0

Definition at line 441 of file rx_eccram.c.

```
00444 {
00445     if (!s_initialized) {
00446         rx_log_error(s_tag, "not initialised");
00447         return k_rx_err_invalid_state;
00448     }
00449
00450     if (on_1bit != NULL) {
00451         s_on_1bit = on_1bit;
00452     }
00453     if (on_2bit != NULL) {
00454         s_on_2bit = on_2bit;
00455     }
00456     s_isr_ctx = ctx;
00457     return k_rx_ok;
00458 }
```

References [s_initialized](#), [s_isr_ctx](#), [s_on_1bit](#), [s_on_2bit](#), and [s_tag](#).

8.78 rx_eccram.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_eccram.h
00003  * @brief ECC-Protected RAM (ECCRAM) Driver API
00004  *
00005  * @details
00006  * Public driver interface for the 32 KB ECC-protected RAM region on the
00007  * RX72N. Provides mode configuration, error-status readout, error
00008  * clearing, ISR registration for 1-bit/2-bit errors, and region
00009  * introspection (start/end addresses for use by linker scripts).
00010  *
00011  * @par STAR Use Case
00012  * Bit-flip-resistant storage for safety-critical mutable state:
00013  * - Motor PID integrator accumulators
00014  * - Last-good encoder counts
00015  * - nanopb frame sequence numbers
00016  * - IWDt kick counter
00017  * - shared_data_t handoff between motor_control_task and comm_task
00018  *
00019  * Single-bit errors are silently corrected by hardware; double-bit errors
00020  * trigger the registered on_2bit handler, which captures the failing
00021  * address and typically e-stops the system instead of acting on garbage.
00022  *
00023  * @par Initialization Order
00024  * 1. Call rx_eccram_init(k_eccram_mode_correct_and_detect) during
00025  *    hardware_init before any code places data in ECCRAM.
00026  * 2. The driver clears MSTPCRC.MSTPC6, enables ECC generation without
00027  *    error checking, zeroes the entire 32 KB region with 32-bit stores
00028  *    so every ECC syndrome matches its data, then switches to the
00029  *    requested check mode.
00030  * 3. Call rx_eccram_register_error_isr() to install application
00031  *    handlers (optional -- the driver installs safe defaults).
00032  *
00033  * @par Linker Integration
00034  * The driver provides `rx_eccram_region_start()` and
00035  * `rx_eccram_region_end()` so linker scripts can place a dedicated
00036  * `.eccram` output section between these addresses. See
00037  * `star-rx72n-firmware/linker.ld`.
00038  *
00039  * @author Locked, Inc.
00040  * @date 2026-04-21
00041  * @copyright Copyright (c) 2026 Locked Inc.
00042  * SPDX-License-Identifier: MIT
00043  * @since Version 1.0.0
00044  */
00045
00046 #pragma once
00047
00048 #include <stdbool.h>
00049 #include <stdint.h>
00050
00051 #include "rx_err.h"
```

```

00052
00053 #ifdef __cplusplus
00054 extern "C" {
00055 #endif
00056
00057 /*
=====
00058  * Types
00059  *
=====
00060  */
00061
00062 /**
00063  * @enum rx_eccram_mode_t
00064  * @brief ECCRAM operating modes exposed to application code
00065  *
00066  * @details
00067  * Maps the raw hardware ECCRAMMODE field (RX72N HW Manual Ch60,
00068  * section 60.2.9, page 2981) into the four behaviours the application
00069  * actually cares about. The "detect only" configuration is emulated by
00070  * installing a 1-bit handler that treats any 1-bit event as a fault
00071  * rather than a silent correction -- the RX72N hardware always corrects
00072  * 1-bit errors on the data delivered to the CPU when ECC is enabled.
00073  *
00074  * @see RX72N HW Manual Chapter 60, section 60.2.9, page 2981
00075  * @since Version 1.0.0
00076  */
00077 typedef enum : uint8_t {
00078     /** @brief ECC disabled -- contents are unchecked (debug/test only) */
00079     k_eccram_mode_disabled = 0,
00080     /** @brief 1-bit correction only -- 2-bit errors go undetected */
00081     k_eccram_mode_correct_only = 1,
00082     /** @brief 1-bit correction + 2-bit detection (recommended default) */
00083     k_eccram_mode_correct_and_detect = 2,
00084     /** @brief Detect-only: surface both 1-bit and 2-bit errors via ISRs */
00085     k_eccram_mode_detect_only = 3,
00086 } rx_eccram_mode_t;
00087
00088 /**
00089  * @struct rx_eccram_status_t
00090  * @brief Snapshot of ECCRAM error flags and captured addresses
00091  *
00092  * @details
00093  * Returned by rx_eccram_get_error_status() so callers can introspect
00094  * ECC health without touching registers directly.
00095  *
00096  * @invariant one_bit_addr is meaningful only when one_bit_error is true
00097  * @invariant two_bit_addr is meaningful only when two_bit_error is true
00098  *
00099  * @see rx_eccram_get_error_status()
00100  * @see rx_eccram_clear_errors()
00101  * @since Version 1.0.0
00102  */
00103 typedef struct {
00104     /** @brief true if ECCRAM1STS.ECC1ERR is set */
00105     bool one_bit_error;
00106     /** @brief true if ECCRAM2STS.ECC2ERR is set */
00107     bool two_bit_error;
00108     /** @brief Captured address of last 1-bit error (from ECCRAM1ECAD) */
00109     uintptr_t one_bit_addr;
00110     /** @brief Captured address of last 2-bit error (from ECCRAM2ECAD) */
00111     uintptr_t two_bit_addr;
00112 } rx_eccram_status_t;
00113
00114 /**
00115  * @brief 1-bit error ISR callback signature
00116  *
00117  * @param[in] addr Captured failing address (from ECCRAM1ECAD)
00118  * @param[in] ctx Opaque context pointer supplied at registration
00119  *
00120  * @note Invoked from ISR context. Keep it short; log and return.
00121  * @since Version 1.0.0
00122  */
00123 typedef void (*rx_eccram_on_1bit_fn_t)(uintptr_t addr, void* ctx);
00124
00125 /**
00126  * @brief 2-bit error ISR callback signature
00127  *
00128  * @param[in] addr Captured failing address (from ECCRAM2ECAD)
00129  * @param[in] ctx Opaque context pointer supplied at registration
00130  *
00131  * @note Invoked from ISR context. Typical implementation triggers
00132  *       e-stop via shared_data rather than continuing.
00133  * @since Version 1.0.0
00134  */
00135 typedef void (*rx_eccram_on_2bit_fn_t)(uintptr_t addr, void* ctx);
00136

```

```

00137 /*
=====
00138 * Public API
00139 *
=====
00140 */
00141
00142 /**
00143  * @brief Initialise ECCRAM in the requested operating mode
00144  *
00145  * @details
00146  * Performs the full initialisation sequence required by the RX72N HW
00147  * manual:
00148  * 1. Clear MSTPCRC.MSTPC6 under system PRCR unlock so that the ECCRAM
00149  *    control block is clocked (Manual section 60.4.1, page 2994).
00150  * 2. Unlock ECCRAMPRCR and set ECCRAMMODE = k_rx_eccrammode_ecc_no_check
00151  *    so ECC bits are generated but nothing is checked.
00152  * 3. Zero all 32 KB of ECCRAM using aligned 32-bit stores (Manual
00153  *    section 60.3.3: uninitialised ECC syndromes would falsely flag).
00154  * 4. Switch ECCRAMMODE to the mode that backs the requested
00155  *    @p mode argument and re-lock ECCRAMPRCR.
00156  * 5. For any mode other than k_eccram_mode_disabled, enable the 1-bit
00157  *    status-update latch (ECCRAM1STSEN).
00158  *
00159  * @param[in] mode Desired operating mode (see rx_eccram_mode_t)
00160  *
00161  * @return rx_err_t Error code
00162  * @retval k_rx_ok Initialisation successful.
00163  * @retval k_rx_err_invalid_arg @p mode is outside rx_eccram_mode_t range.
00164  * @retval k_rx_err_hw_init_failed Hardware mode read-back did not match.
00165  *
00166  * @pre System clock is configured before this call.
00167  * @pre Interrupts are globally disabled OR this function is called
00168  *    before any task can generate ECCRAM traffic.
00169  * @post MSTPCRC.MSTPC6 == 0 (ECCRAM control block clocked).
00170  * @post All 32 KB of ECCRAM are zeroed with matching ECC syndromes.
00171  * @post ECCRAMMODE reflects @p mode.
00172  *
00173  * @note Thread Safety: Call once from hardware_init(). Not reentrant.
00174  * @warning This function must run before any data is placed in the
00175  *    `eccram' linker section -- it wipes the region.
00176  *
00177  * @par NASA Power of 10 Compliance:
00178  * - Rule 2: Zero-fill loop bounded by k_rx_eccram_region_size_bytes / 4.
00179  * - Rule 5: 2 preconditions, 3 postconditions.
00180  * - Rule 7: Every internal helper's return value is checked.
00181  *
00182  * @see rx_eccram_register_error_isr()
00183  * @see rx_eccram_get_error_status()
00184  * @since Version 1.0.0
00185  */
00186 [[nodiscard]] rx_err_t rx_eccram_init(rx_eccram_mode_t mode);
00187
00188 /**
00189  * @brief Read a snapshot of ECCRAM error status and captured addresses
00190  *
00191  * @details
00192  * Reads ECCRAM1STS, ECCRAM2STS, ECCRAM1ECAD, and ECCRAM2ECAD into a
00193  * plain struct. No registers are modified -- call rx_eccram_clear_errors()
00194  * to acknowledge faults.
00195  *
00196  * @param[out] out Pointer to status struct populated on success.
00197  *
00198  * @return rx_err_t Error code
00199  * @retval k_rx_ok Status read successfully.
00200  * @retval k_rx_err_null_ptr @p out is nullptr.
00201  *
00202  * @pre rx_eccram_init() has been called.
00203  * @pre @p out is a valid writable pointer.
00204  * @post @p out fields reflect the register state at call time.
00205  * @post No hardware state is modified by this call.
00206  *
00207  * @note Thread Safety: Safe for concurrent readers (pure register reads).
00208  *
00209  * @since Version 1.0.0
00210  */
00211 [[nodiscard]] rx_err_t rx_eccram_get_error_status(rx_eccram_status_t* out);
00212
00213 /**
00214  * @brief Clear 1-bit and 2-bit error status flags
00215  *
00216  * @details
00217  * Writes k_rx_eccram1sts_clear and k_rx_eccram2sts_clear to acknowledge
00218  * any outstanding error flags. Does not touch the captured addresses:
00219  * those remain latched until the next error.
00220  *
00221  * @return rx_err_t Error code

```

```

00222 * @retval k_rx_ok Flags cleared.
00223 *
00224 * @pre rx_eccram_init() has been called.
00225 * @pre ECCRAMPRCR protection does not guard ECCRAM1STS/ECCRAM2STS
00226 * (verified in Manual section 60.2.13 -- both registers are
00227 * directly writable without PRCR unlock).
00228 * @post ECCRAM1STS.ECC1ERR == 0 and ECCRAM2STS.ECC2ERR == 0.
00229 *
00230 * @note Thread Safety: Call from ISR or task context; writes are atomic.
00231 *
00232 * @since Version 1.0.0
00233 */
00234 [[nodiscard]] rx_err_t rx_eccram_clear_errors(void);
00235
00236 /**
00237 * @brief Register user callbacks for ECC error ISRs
00238 *
00239 * @details
00240 * Installs 1-bit and/or 2-bit error callbacks that the driver's ISR
00241 * trampoline will invoke on the respective RAM-error vectors. Passing
00242 * nullptr for either callback leaves the previously installed handler
00243 * in place, allowing callers to register them independently.
00244 *
00245 * @par Dependency-Injection Rationale (NASA Rule 9 deviation)
00246 * Function pointers are permitted here because they are the exact
00247 * pattern STAR uses to allow test mocks and application-level policy
00248 * to substitute in without modifying the driver. See CLAUDE.md "SOLID
00249 * Principles for C / Dependency Inversion".
00250 *
00251 * @param[in] on_1bit Handler invoked from ISR on 1-bit error (may be nullptr).
00252 * @param[in] on_2bit Handler invoked from ISR on 2-bit error (may be nullptr).
00253 * @param[in] ctx Opaque pointer passed through to the handlers.
00254 *
00255 * @return rx_err_t Error code
00256 * @retval k_rx_ok Handlers installed.
00257 * @retval k_rx_err_invalid_state rx_eccram_init() has not yet been called.
00258 *
00259 * @pre rx_eccram_init() has been called.
00260 * @pre Handlers, if non-null, are safe to call from interrupt context.
00261 * @post The driver's ISR trampolines dispatch to the supplied handlers.
00262 * @post @p ctx is retained until this function is called again.
00263 *
00264 * @note Thread Safety: Install once during init before interrupts are
00265 * enabled, or guard calls at the task level.
00266 *
00267 * @since Version 1.0.0
00268 */
00269 [[nodiscard]] rx_err_t rx_eccram_register_error_isr(rx_eccram_on_1bit_fn_t on_1bit,
00270 rx_eccram_on_2bit_fn_t on_2bit,
00271 void* ctx);
00272
00273 /**
00274 * @brief Return the inclusive start address of the ECCRAM data region
00275 *
00276 * @details
00277 * Exposes the hardware-defined base of the ECC-protected region so that
00278 * linker scripts and runtime allocators can place objects inside it
00279 * without hardcoding the literal address.
00280 *
00281 * @return Physical address 0x00FF8000 as a uintptr_t.
00282 *
00283 * @pre None.
00284 * @post Return value equals k_rx_eccram_region_base_addr.
00285 * @post Return value is word-aligned.
00286 *
00287 * @note Thread Safety: Returns a compile-time constant; always safe.
00288 *
00289 * @see rx_eccram_region_end()
00290 * @see RX72N HW Manual Chapter 60, Table 60.1, page 2977
00291 * @since Version 1.0.0
00292 */
00293 uintptr_t rx_eccram_region_start(void);
00294
00295 /**
00296 * @brief Return the inclusive end address of the ECCRAM data region
00297 *
00298 * @return Physical address 0x00FFFFFF as a uintptr_t.
00299 *
00300 * @pre None.
00301 * @post Return value equals k_rx_eccram_region_end_addr.
00302 * @post Return value - rx_eccram_region_start() + 1 equals 32 KB.
00303 *
00304 * @note Thread Safety: Returns a compile-time constant; always safe.
00305 *
00306 * @see rx_eccram_region_start()
00307 * @see RX72N HW Manual Chapter 60, Table 60.1, page 2977
00308 * @since Version 1.0.0

```

```

00309 */
00310 uintptr_t rx_eccram_region_end(void);
00311
00312 #ifdef __cplusplus
00313 }
00314 #endif

```

8.79 libs/rx_hal/inc/rx_gptw.h File Reference

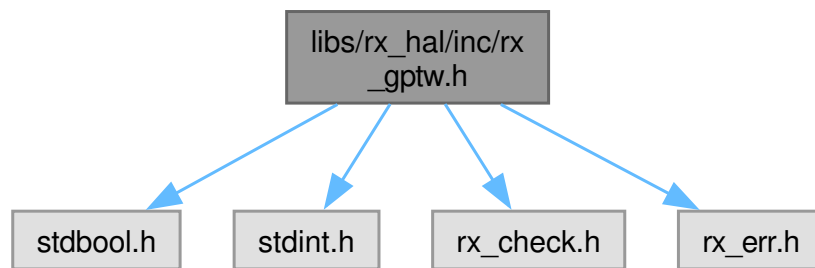
GPTW PWM Driver for RX72N Motor Control.

```

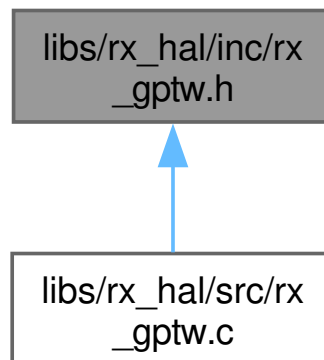
#include <stdbool.h>
#include <stdint.h>
#include "rx_check.h"
#include "rx_err.h"

```

Include dependency graph for rx_gptw.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_gptw_channel_id_t](#)
Type-safe GPTW channel wrapper (prevents accidental argument swaps).
- struct [rx_gptw_output_id_t](#)
Type-safe GPTW output wrapper (prevents accidental argument swaps).
- struct [rx_gptw_config_t](#)
GPTW PWM Configuration Structure.

Enumerations

- enum `rx_gptw_channel_t` : `uint8_t` { `k_gptw_channel_0` , `k_gptw_channel_1` , `k_gptw_channel_2` , `k_gptw_channel_3` }
GPTW Channel Identifiers for 4-Motor Control System.
- enum `rx_gptw_output_t` : `uint8_t` { `k_gptw_output_a` , `k_gptw_output_b` }
GPTW Output Channel Selection for Complementary PWM.
- enum `rx_gptw_wave_mode_t` : `uint8_t` { `k_gptw_wave_saw_pwm` , `k_gptw_wave_tri_pwm1` , `k_gptw_wave_tri_pwm2` , `k_gptw_wave_tri_pwm3` }
GPTW PWM Waveform Mode Selection.
- enum `rx_gptw_duty_calc_constants_t` : `uint8_t` { `k_gptw_duty_50_pct_num` = 50 , `k_gptw_duty_75_pct_num` = 75 , `k_gptw_duty_pct_denom` = 100 }
Named constants for duty cycle percentage calculations.
- enum `rx_gptw_count_t` : `uint8_t` { `k_rx_gptw_channel_count` = 4 }
Initialize all 4 GPTW channels with 90 degree phase staggering.

Functions

- static `rx_gptw_channel_id_t rx_gptw_channel_id` (`rx_gptw_channel_t` value)
Helper to construct a GPTW channel wrapper.
- static `rx_gptw_output_id_t rx_gptw_output_id` (`rx_gptw_output_t` value)
Helper to construct a GPTW output wrapper.
- `rx_err_t rx_gptw_init_all_staggered` (`const rx_gptw_config_t *configs`[`k_rx_gptw_channel_count`]
)
Initialize all 4 GPTW channels with 90-degree phase staggering.
- `rx_err_t rx_gptw_init_pwm` (`rx_gptw_channel_t` channel, `const rx_gptw_config_t *config`)
Initialize a single GPTW channel for PWM output.
- `rx_err_t rx_gptw_set_duty` (`rx_gptw_channel_id_t` channel, `rx_gptw_output_id_t` output, `float` duty_percent)
Set PWM duty cycle (floating-point percentage).
- `rx_err_t rx_gptw_set_duty_raw` (`rx_gptw_channel_t` channel, `rx_gptw_output_t` output, `uint32_t` duty_count)
Set PWM duty cycle using raw counter value.
- `rx_err_t rx_gptw_get_duty` (`rx_gptw_channel_t` channel, `rx_gptw_output_t` output, `float *duty_percent`)
Get current duty cycle as a floating-point percentage.
- `rx_err_t rx_gptw_get_period` (`rx_gptw_channel_t` channel, `uint32_t *period_count`)
Get the 32-bit PWM period count for a GPTW channel.
- `rx_err_t rx_gptw_enable_output` (`rx_gptw_channel_t` channel, `rx_gptw_output_t` output, `bool` enable)
Enable or disable a PWM output pin.
- `rx_err_t rx_gptw_start` (`rx_gptw_channel_t` channel)
Start the GPTW timer counter for PWM generation.
- `rx_err_t rx_gptw_stop` (`rx_gptw_channel_t` channel)
Stop the GPTW timer counter.
- `rx_err_t rx_gptw_deinit` (`rx_gptw_channel_t` channel)
Deinitialize a GPTW channel and release all resources.

8.79.1 Detailed Description

GPTW PWM Driver for RX72N Motor Control.

Production-grade PWM driver for the RX72N General PWM Timer (GPTW) peripheral, optimized for brushed DC motor control with H-bridge drivers. This module provides high-resolution 32-bit PWM generation with complementary outputs, deadtime insertion, and phase-staggered operation for 4-motor systems.

8.79.1.1 Purpose and Design Rationale

The GPTW peripheral is chosen over MTU for motor control because:

1. 32-bit resolution - Higher precision than MTU's 16-bit (0.000015% vs 0.0015%)
2. Dedicated motor pins - Port E pins optimized for PWM output
3. Hardware deadtime - Prevents H-bridge shoot-through without CPU overhead
4. Phase staggering - Reduces peak current draw and EMI

8.79.1.2 Key Features

PWM Generation:

- 4 independent channels (GPTW0-GPTW3) for 4 motors
- 32-bit counter resolution for precise duty cycle control
- Sawtooth (edge-aligned) or triangle (center-aligned) waveforms
- Complementary A/B outputs per channel for H-bridge control
- Configurable deadtime insertion (0-65535 ns)

Phase Staggering:

- 90-degree phase offset between channels
- Channel 0: 0deg, Channel 1: 90deg, Channel 2: 180deg, Channel 3: 270deg
- Spreads current draw across PWM period
- Reduces peak power supply current by ~75%
- Reduces conducted EMI by preventing simultaneous switching

Motor Control Integration:

- Direct interface with DRV8263H H-bridge drivers
- IN2/IN1 control mode support (sign-magnitude)
- Glitch-free duty cycle updates on period boundary
- Fast raw duty cycle API for tight control loops

8.79.1.3 Hardware Architecture

Pin Assignments (from hardware_pinout.h):

Motor	Channel	Output A	Output B	Function
Motor 0	GPTW0	PE5/GTIOC0A	PE2/GTIOC0B	Front-Left
Motor 1	GPTW1	PE4/GTIOC1A	PE1/GTIOC1B	Front-Right
Motor 2	GPTW2	PE3/GTIOC2A	PE0/GTIOC2B	Rear-Left
Motor 3	GPTW3	PE7/GTIOC3A	PE6/GTIOC3B	Rear-Right

Register Base Addresses:

Register Set	Base Address	Description
GPTW Common	0x000C2000	Shared control registers
GPTW0	0x000C2100	Channel 0 registers
GPTW1	0x000C2180	Channel 1 registers
GPTW2	0x000C2200	Channel 2 registers
GPTW3	0x000C2280	Channel 3 registers

Clock Configuration:

Parameter	Value	Notes
Clock Source	PCLKA	Peripheral Clock A
Clock Frequency	120 MHz	After system clock configuration
Counter Resolution	8.33 ns	1/120MHz
PWM Frequency	20 kHz	Typical for motor control
Period Counts	6000	120MHz / 20kHz
Duty Resolution	0.0167%	1/6000

8.79.1.4 PWM Waveform Modes

Triangle (Center-Aligned)

Counter: 0->Period->0->Period
 Update: At crest/trough
 Best for: Motor control

Sawtooth (Edge-Aligned)

Counter: 0->Period->0
 Update: At trough
 Best for: Simple PWM

8.79.1.5 Performance Characteristics

Operation	Execution Time	Memory Usage	Notes
rx_gptw_init_all_staggered()	~100 us	0 heap	One-time setup
rx_gptw_set_duty()	~5 us	0	Float calculation
rx_gptw_set_duty_raw()	~1 us	0	Direct register write
rx_gptw_get_duty()	~3 us	0	Register read + calc
rx_gptw_enable_output()	~1 us	0	Single register write

Memory Footprint:

- Static variables: ~64 bytes (initialization flags, period cache)
- Code size: ~2 KB (all functions)
- Stack usage: < 64 bytes (deepest call)
- Heap usage: 0 bytes (zero dynamic allocation)

8.79.1.6 Thread Safety and RTOS Integration

Thread-Safe Operations:

- [rx_gptw_set_duty\(\)](#) and [rx_gptw_set_duty_raw\(\)](#) are atomic register writes
- Safe to call from motor control ISR or ThreadX tasks
- No mutex required for duty cycle updates

Non-Thread-Safe Operations:

- [rx_gptw_init_all_staggered\(\)](#) - Call once during system initialization
- [rx_gptw_deinit\(\)](#) - Call only when no motor control active

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	GPTW peripheral required
Clock	PCLKA @ 120 MHz	System clock configuration
Pins	Port E (PE0-PE7)	Configured via MPC
Power	3.3V I/O	LVC MOS compatible

Module Dependencies:

This module depends on:

- [rx_err.h](#) - Error code definitions
- [rx_check.h](#) - Parameter validation macros
- [rx72n_gptw_regs.h](#) - GPTW register definitions
- [rx72n_mpc_regs.h](#) - Pin mux configuration

This module is used by:

- [rx_motor.c](#) - Motor control driver
- [rx_drv8263.c](#) - H-bridge driver interface
- [motor_control_task.c](#) - Motor control task

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	↔ [OK]↔	No goto/setjmp/recursion
2. Fixed loop bounds	↔ [OK]↔	Loops bounded by <code>k_gptw_channel_3</code> (4 iterations max)
3. No dynamic allocation	↔ [OK]↔	Zero malloc/free, all static
4. Small functions	↔ [OK]↔	All functions < 60 lines
5. Assertions (≥ 2 per function)	↔ [OK]↔	Minimum 2 validation checks per function
6. Narrow scope	↔ [OK]↔	File-scope statics, minimal globals
7. Check return values	↔ [OK]↔	All functions return <code>rx_err_t</code>
8. Limited preprocessor	↔ [OK]↔	C23 typed enums, minimal macros
9. Pointer restrictions	↔ [OK]↔	Single-level pointers only
10. Compiler warnings	↔ [OK]↔	Compiles with <code>-Wall -Wextra -Werror</code>

SOLID Principles:

Single Responsibility (S):

- This module handles ONLY GPTW PWM generation
- Motor control logic is in `rx_motor.c`
- H-bridge configuration is in `rx_drv8263.c`

Open/Closed (O):

- Extensible via `rx_gptw_config_t` structure

- New waveform modes can be added without modifying existing code
- Frequency and deadtime configurable at runtime

Liskov Substitution (L):

- All functions return `rx_err_t` with consistent semantics
- Channel and output parameters are validated uniformly

Interface Segregation (I):

- Separate functions for float vs raw duty cycle
- Init, set, get, enable are independent operations
- Wrapper types prevent parameter swapping

Dependency Inversion (D):

- Depends on abstract error codes, not specific implementations
- Motor control depends on this interface, not hardware registers directly

See also

`rx_motor.h` Motor control driver using GPTW

`rx_drv8263.h` H-bridge driver interface

[rx72n_gptw_regs.h](#) Register definitions

RX72N Hardware Manual Section 26 - General PWM Timer

`docs/sections/03_hardware_pinout.tex` Pin assignments

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_gptw.h](#).

8.79.2 Data Structure Documentation

8.79.2.1 struct rx'gptw'channel'id't

Type-safe GPTW channel wrapper (prevents accidental argument swaps).

Wraps [rx_gptw_channel_t](#) in a struct to leverage the C type system for catching argument transposition errors at compile time. Without this wrapper, swapping channel and output arguments (both `uint8_t`) would compile silently. Construct via [rx_gptw_channel_id\(\)](#) helper function.

Invariant

value must be a valid [rx_gptw_channel_t](#) value (`k_gptw_channel_0` through `k_gptw_channel_3`)

```
rx_gptw_channel_id_t ch = rx_gptw_channel_id(k_gptw_channel_0);
```

See also

[rx_gptw_channel_id\(\)](#) Constructor function

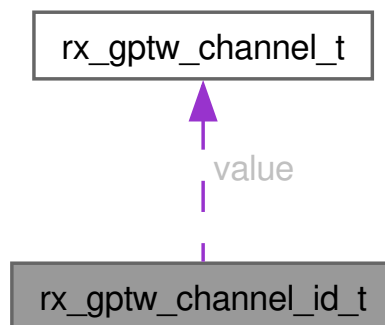
[rx_gptw_output_id_t](#) Companion wrapper for output parameter

Since

Version 1.0.0

Definition at line 512 of file [rx_gptw.h](#).

Collaboration diagram for `rx_gptw_channel_id_t`:



Data Fields

rx_gptw_channel_t	value	GPTW channel number (0-3)
-----------------------------------	-------	---------------------------

8.79.2.2 struct rx_gptw_output_id_t

Type-safe GPTW output wrapper (prevents accidental argument swaps).

Wraps [rx_gptw_output_t](#) in a struct to leverage the C type system for catching argument transposition errors at compile time. Without this wrapper, swapping channel and output arguments (both `uint8_t`) would compile silently. Construct via [rx_gptw_output_id\(\)](#) helper function.

Invariant

value must be `k_gptw_output_a` or `k_gptw_output_b`

```
rx_gptw_output_id_t out = rx_gptw_output_id(k_gptw_output_a);
```

See also

[rx_gptw_output_id\(\)](#) Constructor function

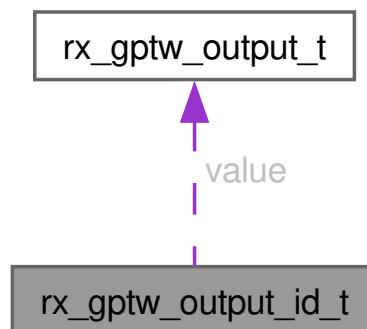
[rx_gptw_channel_id_t](#) Companion wrapper for channel parameter

Since

Version 1.0.0

Definition at line 536 of file [rx_gptw.h](#).

Collaboration diagram for `rx_gptw_output_id_t`:



Data Fields

rx_gptw_output_t	value	GPTW output identifier (A or B)
----------------------------------	-------	---------------------------------

8.79.2.3 struct rx_gptw_config_t

GPTW PWM Configuration Structure.

Contains all configuration parameters for initializing a GPTW channel for PWM output. This structure is passed to [rx_gptw_init_pwm\(\)](#) or [rx_gptw_init_all_staggered\(\)](#) to configure the timer.

8.79.2.4 Configuration Guidelines

Frequency Selection:

- Motor control: 15-25 kHz (above audible range, efficient switching)
- LED dimming: 100-1000 Hz (no flicker, simple filtering)
- Servo control: 50-400 Hz (standard RC servo range)

Deadtime Selection:

- Minimum: 100 ns (propagation delay margin)
- Typical: 500-1000 ns (good balance)
- Maximum: 5000 ns (very conservative, reduces efficiency)

Configuration Parameter Table:

Parameter	Min	Max	Default	Unit	Notes
frequency_hz	100	1000000	20000	Hz	Limited by PCLKA/2
deadtime_ns	0	65535	1000	ns	0 disables deadtime
wave_mode	0	3	0	enum	See rx_gptw_wave_mode_t
enable_complementary	false	true	true	bool	For H-bridge
invert_polarity	false	true	false	bool	Active-low output

Memory Layout:

Offset	Size	Field	Type	Alignment
0	4	frequency_hz	uint32_t	4
4	2	deadtime_ns	uint16_t	2
6	1	wave_mode	uint8_t	1
7	1	enable_complementary	bool	1
8	1	invert_polarity	bool	1
9-11	3	(padding)	-	-
Total	12	-	-	4

Usage Example (Motor Control):

```
// Standard motor control configuration
rx_gptw_config_t motor_config = {
    .frequency_hz = 20000,      // 20 kHz (inaudible)
    .deadtime_ns = 1000,       // 1 us deadtime
    .wave_mode = k_gptw_wave_saw_pwm, // Edge-aligned
    .enable_complementary = true, // H-bridge requires this
    .invert_polarity = false    // Active-high outputs
};

rx_err_t err = rx_gptw_init_all_staggered(&motor_config);
if (err != k_rx_ok) {
    rx_log_error("GPTW", "Failed to initialize: %d", err);
}
```

Usage Example (LED Dimming):

```
// LED PWM configuration (no complementary needed)
rx_gptw_config_t led_config = {
    .frequency_hz = 1000,      // 1 kHz (no flicker)
    .deadtime_ns = 0,         // No deadtime for LEDs
    .wave_mode = k_gptw_wave_saw_pwm,
    .enable_complementary = false, // Single output only
    .invert_polarity = false
};

rx_err_t err = rx_gptw_init_pwm(k_gptw_channel_0, &led_config);
```

Invariant

frequency_hz > 0 (enforced by rx_gptw_init_*)
 frequency_hz <= PCLKA / 2 (Nyquist limit)
 wave_mode in [0, 3] (valid enum range)

Note

Structure is copied during initialization; no need to keep alive

Warning

Invalid configurations rejected with k_rx_err_invalid_arg

See also

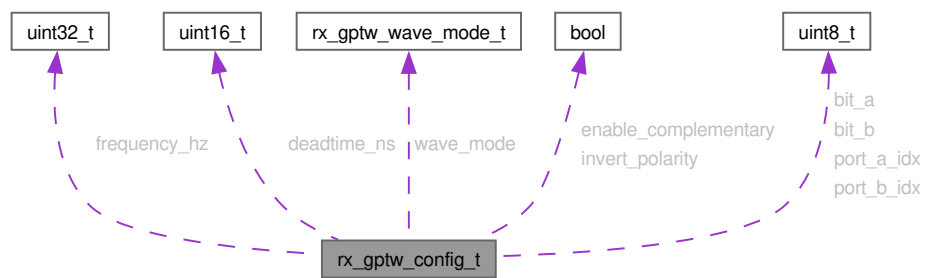
[rx_gptw_init_pwm\(\)](#) Single channel initialization
[rx_gptw_init_all_staggered\(\)](#) 4-channel staggered initialization
[rx_gptw_wave_mode_t](#) Waveform mode options

Since

Version 1.0.0

Definition at line 697 of file [rx_gptw.h](#).

Collaboration diagram for rx_gptw_config_t:



Data Fields

uint8_t	bit_a	
uint8_t	bit_b	
uint16_t	deadtime_ns	Deadtime in nanoseconds. Deadtime inserted between complementary output transitions to prevent shoot-through (simultaneous conduction of high-side and low-side switches). Set to 0 to disable deadtime insertion. Deadtime Calculation: $\text{Deadtime Counts} = \frac{\text{deadtime_ns} \times f_{PCLKA}}{10^9}$ At 120 MHz: 1000 ns = 120 counts Valid Range: [0, 65535 ns] Recommended: 1000 ns (1 us) for DRV8263H Warning Too short may cause shoot-through; too long reduces efficiency

bool	enable_complementary	Enable complementary outputs. When true, Output B is the inverted version of Output A with deadtime inserted. Required for H-bridge motor control. • true: Both A and B outputs active (complementary) • false: Only Output A active, Output B held low Warning When complementary mode is enabled, GTCCRB is tied to GTCCRA by the hardware. Writing GTCCRB directly has no effect; the output B duty is always the complement of output A. For independent A/B control (e.g., DRV8263H IN2/IN1 mode), set this to false.
uint32_t	frequency_hz	PWM frequency in Hz. Sets the PWM switching frequency. For motor control, 15-25 kHz is typical (above audible range). Higher frequencies reduce current ripple but increase switching losses. Frequency Calculation: $\text{Period Count} = \frac{f_{PCLKA}}{f_{PWM}} - 1$ Where $f_{PCLKA} = 120 \text{ MHz}$ Valid Range: [100, 1000000 Hz] Recommended: 20000 Hz for motor control Note Actual frequency may differ slightly due to integer division

bool	invert_polarity	Invert PWM polarity. When true, inverts the PWM output polarity. • false: HIGH during duty cycle, LOW otherwise (active-high) • true: LOW during duty cycle, HIGH otherwise (active-low) Note Use for active-low driver inputs or inverted LED connections
uint8_t	port_a_idx	Pin coordinates for GTIOC*A and GTIOC*B outputs. The lib does not know the board's pin map. Caller MUST set these to the actual pads where GTIOC_A and GTIOC_B should appear. Source the values from a board-level pin map header (e.g. src/inc/hardware_config.h). port*_idx is the standard RX72N port index: 1, 2, 8, etc., with PORTA=10, PORTB=11, PORTC=12, PORTD=13, PORTE=14, PORTF=15. bit_* is the bit position within that port (0..7).
uint8_t	port_b_idx	
rx_gptw_wave_mode_t	wave_mode	Waveform mode (sawtooth or triangle). Selects counter behavior and PWM alignment. • Sawtooth: Edge-aligned, simpler, single update point • Triangle: Center-aligned, symmetric ripple, better for motors See also rx_gptw_wave_mode_t for detailed mode descriptions

8.79.3 Enumeration Type Documentation

8.79.3.1 rx`gptw`channel` t

enum [rx_gptw_channel_t](#) : uint8_t

GPTW Channel Identifiers for 4-Motor Control System.

Identifies the four GPTW timer channels used for motor PWM generation. Each channel controls one motor through complementary A/B outputs connected to a DRV8263H H-bridge driver.

8.79.3.2 Channel Architecture

Each GPTW channel provides:

- 32-bit up/down counter
- Two compare/capture registers (GTCCRA, GTCCRB)
- Two output pins (GTIOCnA, GTIOCnB)
- Independent deadtime insertion
- Synchronized start/stop capability

Channel-to-Motor Mapping:

Channel	Motor	Position	Register Base	Phase Offset
GPTW0	Motor 0	Front-Left	0x000C2100	0deg
GPTW1	Motor 1	Front-Right	0x000C2180	90deg
GPTW2	Motor 2	Rear-Left	0x000C2200	180deg
GPTW3	Motor 3	Rear-Right	0x000C2280	270deg

Pin Assignments:

Channel	Output A	Output B	MPC Setting
k_gptw_channel_0	PE5/GTIOC0A	PE2/GTIOC0B	PSEL = 0x1E
k_gptw_channel_1	PE4/GTIOC1A	PE1/GTIOC1B	PSEL = 0x1E
k_gptw_channel_2	PE3/GTIOC2A	PE0/GTIOC2B	PSEL = 0x1E
k_gptw_channel_3	PE7/GTIOC3A	PE6/GTIOC3B	PSEL = 0x1E

Usage Example:

```
// Initialize all channels with staggered phases
rx_gptw_config_t config = {
    .frequency_hz = 20000,
    .deadtime_ns = 1000,
    .wave_mode = k_gptw_wave_saw_pwm,
    .enable_complementary = true,
    .invert_polarity = false
};
rx_gptw_init_all_staggered(&config);

// Set duty cycle for motor 0 (front-left)
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
    rx_gptw_output_id(k_gptw_output_a),
    50.0F); // 50% duty cycle
```

Invariant

Channel values are contiguous: 0, 1, 2, 3

All channels use same register structure (0x80 byte spacing)

Channel count equals motor count (4)

Note

C23 typed enum with uint8_t underlying type (1-byte values)

Channel 4 and 5 exist in hardware but are not used in STAR

See also

[rx_gptw_channel_id\(\)](#) Type-safe wrapper constructor

[rx_gptw_output_t](#) Output selection within channel

[rx72n_gptw_regs.h](#) Register structure definitions

Since

Version 1.0.0

Enumerator

<code>k_gptw_channel_0</code>	GPTW0 channel for Motor 0 (Front-Left). Register base: 0x000C2100. Outputs: PE5/GTIOC0A, PE2/GTIOC0B. Phase offset: 0deg in staggered mode
<code>k_gptw_channel_1</code>	GPTW1 channel for Motor 1 (Front-Right). Register base: 0x000C2180. Outputs: PE4/GTIOC1A, PE1/GTIOC1B. Phase offset: 90deg in staggered mode
<code>k_gptw_channel_2</code>	GPTW2 channel for Motor 2 (Rear-Left). Register base: 0x000C2200. Outputs: PE3/GTIOC2A, PE0/GTIOC2B. Phase offset: 180deg in staggered mode
<code>k_gptw_channel_3</code>	GPTW3 channel for Motor 3 (Rear-Right). Register base: 0x000C2280. Outputs: PE7/GTIOC3A, PE6/GTIOC3B. Phase offset: 270deg in staggered mode

Definition at line 278 of file [rx_gptw.h](#).

```

00278     : uint8_t {
00279     k_gptw_channel_0 =
00280     0, /**< GPTW0 channel for Motor 0 (Front-Left). Register base: 0x000C2100. Outputs: PE5/GTIOC0A,
00281     PE2/GTIOC0B. Phase offset: 0deg in staggered mode */
00281     k_gptw_channel_1 =
00282     1, /**< GPTW1 channel for Motor 1 (Front-Right). Register base: 0x000C2180. Outputs: PE4/GTIOC1A,
00283     PE1/GTIOC1B. Phase offset: 90deg in staggered mode */
00283     k_gptw_channel_2 =
00284     2, /**< GPTW2 channel for Motor 2 (Rear-Left). Register base: 0x000C2200. Outputs: PE3/GTIOC2A,
00285     PE0/GTIOC2B. Phase offset: 180deg in staggered mode */
00285     k_gptw_channel_3 =
00286     3, /**< GPTW3 channel for Motor 3 (Rear-Right). Register base: 0x000C2280. Outputs: PE7/GTIOC3A,
00287     PE6/GTIOC3B. Phase offset: 270deg in staggered mode */
00287 } rx_gptw_channel_t;
```

8.79.3.3 rx_gptw_count_t

```
enum rx_gptw_count_t : uint8_t
```

Initialize all 4 GPTW channels with 90 degree phase staggering.

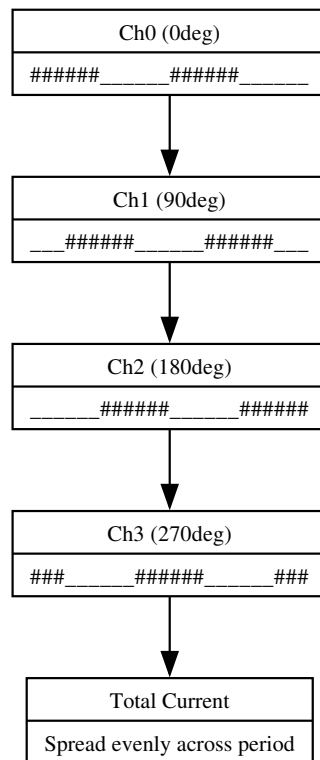
Configures GPTW0-3 for synchronized PWM operation with phase offsets that spread current draw across the PWM period. This is the recommended initialization function for 4-motor systems as it reduces peak current and EMI compared to synchronized PWM.

8.79.3.4 Phase Staggering Architecture

Phase staggering offsets the start of each channel's PWM period:

- Channel 0: 0deg (reference phase)
- Channel 1: 90deg (period / 4 offset)
- Channel 2: 180deg (period / 2 offset)
- Channel 3: 270deg (3 x period / 4 offset)

Phase Staggered PWM (4 channels at 50% duty)



8.79.3.5 Benefits of Phase Staggering

1. **Reduced Peak Current:** Instead of all motors switching simultaneously, current draw is spread across the period, reducing peak by up to 75%
2. **Lower EMI:** Staggered edges reduce conducted emissions
3. **Smaller Capacitors:** Lower peak current allows smaller power supply capacitors
4. **Better Power Quality:** More constant current draw improves voltage regulation

8.79.3.6 Algorithm Steps

1. Validate configuration parameters (NULL check, frequency range, wave mode)
2. Calculate period count from frequency and PCLKA
3. Enable GPTW module clock (MSTPCRA.MSTPA7 = 0)
4. Configure MPC for Port E pins (GTIOC function select)
5. For each channel (0-3):
 - a. Configure timer mode register (GTCR)
 - b. Set period register (GTPR)
 - c. Calculate and set phase offset (GTCNT initial value)
 - d. Configure output mode and polarity (GTIOR)
 - e. Set deadtime registers (GTDVU, GTDVD)
 - f. Set initial duty cycle to 0% (GTCCRA, GTCCRB)
6. Start all channels simultaneously (GTSTR)

Parameters

in	config	Common configuration for all channels
		• config->frequency_hz: PWM frequency [100, 1000000] Hz • config->deadtime_ns: Deadtime [0, 65535] ns • config->wave_mode: Waveform mode (sawtooth or triangle) • config->enable_complementary: Enable A/B complementary outputs • config->invert_polarity: Invert output polarity

Returns

rx_err_t Error code indicating success or failure reason

Return values

k_rx_ok	All 4 channels initialized successfully
k_rx_err_null_ptr	config pointer is nullptr
k_rx_err_invalid_arg	frequency_hz is 0 or wave_mode out of range
k_rx_err_hw_error	GPTW hardware access failed (NULL register pointer)

Precondition

System clock configured (PCLKA = 120 MHz)
 Port E pins not used by other peripherals

Postcondition

All 4 GPTW channels running with 90deg phase offsets
 All duty cycles initialized to 0% (motors stopped)
 MPC configured for GTIOC function on Port E pins

Invariant

Phase relationship maintained: $\text{Ch1} = \text{Ch0} + 90\text{deg}$, $\text{Ch2} = \text{Ch0} + 180\text{deg}$, $\text{Ch3} = \text{Ch0} + 270\text{deg}$

Note

Thread-safe: Call only once during system initialization

Propagates errors from [internal_calculate_period\(\)](#), [internal_configure_gptw_hardware\(\)](#)

Warning

Calling this function while motors are running will cause glitches

Attention

This function should be used INSTEAD of [rx_gptw_init_pwm\(\)](#) for motor control

Performance:

- Execution time: ~ 100 us at 240 MHz
- Memory usage: 0 heap, ~ 64 bytes stack

Example (Complete Motor System Initialization):

```
#include "rx_gptw.h"
#include "rx_drv8263.h"
#include "rx_motor.h"

rx_err_t motor_system_init(void)
{
    // Configure GPTW for 20 kHz PWM with 1 us deadtime
    rx_gptw_config_t pwm_config = {
        .frequency_hz = 20000,
        .deadtime_ns = 1000,
        .wave_mode = k_gptw_wave_saw_pwm,
        .enable_complementary = true,
        .invert_polarity = false
    };

    rx_err_t err = rx_gptw_init_all_staggered(&pwm_config);
    if (err != k_rx_ok) {
        rx_log_error("MOTOR", "GPTW init failed: %d", err);
        return err;
    }

    // Initialize H-bridge drivers
    err = rx_drv8263_init_all();
    if (err != k_rx_ok) {
        rx_log_error("MOTOR", "DRV8263H init failed: %d", err);
        return err;
    }

    rx_log_info("MOTOR", "Motor system initialized (4 channels, 20kHz, staggered)");
    return k_rx_ok;
}
```


Example (Verify Phase Offsets):

```
// Diagnostic: read counter values to verify phase relationship
void verify_phase_stagger(void)
{
    uint32_t cnt0, cnt1, cnt2, cnt3;
    uint32_t period;

    rx_gptw_get_period(k_gptw_channel_0, &period);

    // Read all counters (will have some jitter due to sequential reads)
    cnt0 = GPTW0->GTCNT;
    cnt1 = GPTW1->GTCNT;
    cnt2 = GPTW2->GTCNT;
    cnt3 = GPTW3->GTCNT;

    // Expected: cnt1 ~ cnt0 + period/4, cnt2 ~ cnt0 + period/2, etc.
    rx_log_debug("GPTW", "Period=%lu, Offsets: %lu, %lu, %lu, %lu",
        period, cnt0, cnt1 - cnt0, cnt2 - cnt0, cnt3 - cnt0);
}
```

See also

[rx_gptw_init_pwm\(\)](#) Single channel initialization (not recommended for motors)

[rx_gptw_set_duty\(\)](#) Set duty cycle after initialization

[rx_gptw_deinit\(\)](#) Cleanup function

[rx_drv8263_init_all\(\)](#) H-bridge driver initialization

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 1: [OK] No goto/setjmp/recursion
- Rule 2: [OK] Loop bounded by k_gptw_channel_3 (4 iterations)
- Rule 5: [OK] 2 preconditions, 3 postconditions
- Rule 7: [OK] All internal function returns checked

Enumerator

k_rx_gptw_channel_count	
-------------------------	--

Definition at line 964 of file [rx_gptw.h](#).

```
00964         : uint8_t {
00965     k_rx_gptw_channel_count = 4,
00966 } rx_gptw_count_t;
```

8.79.3.7 rx_gptw_duty_calc_constants_t

```
enum rx_gptw_duty_calc_constants_t : uint8_t
```

Named constants for duty cycle percentage calculations.

Provides named numerators and a common denominator for computing raw duty counts as integer fractions of the period. Using named constants instead of magic literals satisfies NASA Power of 10 Rule 8 (no magic numbers).

Example:

```
// 50% duty: period * 50 / 100
uint32_t half = (period * k_gptw_duty_50_pct_num) / k_gptw_duty_pct_denom;
```

Invariant

All numerator constants must be $\leq$ k_gptw_duty_pct_denom (non-negative)

k_gptw_duty_pct_denom must be non-zero (denominator for percentage calculation)

See also

[rx_gptw_set_duty_raw\(\)](#) Uses these constants in documentation examples

[rx_gptw_get_period\(\)](#) Provides the period value for the calculation

Since

Version 1.0.0

Enumerator

k_gptw_duty_50_pct_num	Numerator for 50% duty cycle calculation
k_gptw_duty_75_pct_num	Numerator for 75% duty cycle calculation
k_gptw_duty_pct_denom	Denominator for percentage duty calculations

Definition at line 486 of file [rx_gptw.h](#).

```
00486      : uint8_t {
00487   k_gptw_duty_50_pct_num = 50, /**< Numerator for 50% duty cycle calculation */
00488   k_gptw_duty_75_pct_num = 75, /**< Numerator for 75% duty cycle calculation */
00489   k_gptw_duty_pct_denom = 100, /**< Denominator for percentage duty calculations */
00490 } rx_gptw_duty_calc_constants_t;
```

8.79.3.8 rx_gptw_output_t

```
enum rx\_gptw\_output\_t : uint8_t
```

GPTW Output Channel Selection for Complementary PWM.

Selects between the two outputs (A and B) available on each GPTW channel. Each output has an independent duty cycle and is controlled separately.

8.79.3.9 Independent Output Mode (STAR Configuration)

For the DRV8263H motor driver in IN2/IN1 mode, the GPTW outputs operate independently (not in complementary mode). Each output has its own compare register and duty cycle:

- Output A: Static direction signal (0% or 100% duty)
- Output B: Variable-duty PWM for speed control

The outputs are NOT inverted versions of each other and no hardware deadtime insertion is used. This is safe because the DRV8263H has integrated shoot-through protection on its internal H-bridge FETs.

8.79.3.10 IN2/IN1 Mode (Sign-Magnitude)

For DRV8263H H-bridge control in IN2/IN1 mode:

- Output A: Connected to IN2 (direction) input - controls direction
- Output B: Connected to IN1 (PWM) input - controls PWM duty

H-Bridge Connection:

Output	DRV8263H Pin	Function	Signal Type
Output A (GTIOCnA)	IN2	Direction control	Static HIGH/LOW
Output B (GTIOCnB)	IN1	PWM speed control	PWM waveform

Usage Example:

```
// Set forward direction at 75% duty for motor 0
// Output A = HIGH (forward direction)
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
    rx_gptw_output_id(k_gptw_output_a),
    100.0F); // IN2 = HIGH (direction)

// Output B = PWM at 75% (motor speed)
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
    rx_gptw_output_id(k_gptw_output_b),
    75.0F); // IN1 = 75% PWM

// Reverse direction: set Output A LOW
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
    rx_gptw_output_id(k_gptw_output_a),
    0.0F); // IN2 = LOW (reverse)
```

Invariant

Output values are 0 (A) or 1 (B) only
Both outputs available on all channels

Note

C23 typed enum with uint8_t underlying type

Warning

In independent mode, each output is controlled separately; the DRV8263H handles shoot-through protection internally

See also

[rx_gptw_output_id\(\)](#) Type-safe wrapper constructor
[rx_gptw_channel_t](#) Channel selection
[rx_drv8263.h](#) H-bridge driver interface

Since

Version 1.0.0

Enumerator

k_gptw_output_a	GTIOCA output pin. Connected to DRV8263H IN2 (direction) in IN2/IN1 mode. Controls motor direction: HIGH=forward, LOW=reverse
k_gptw_output_b	GTIOCB output pin. Connected to DRV8263H IN1 (PWM) in IN2/IN1 mode. PWM duty cycle controls motor speed (0-100%)

Definition at line 354 of file [rx_gptw.h](#).

```

00354     : uint8_t {
00355     k_gptw_output_a =
00356     0, /**< GTIOCA output pin. Connected to DRV8263H IN2 (direction) in IN2/IN1 mode. Controls motor direction:
        HIGH=forward, LOW=reverse */
00357     k_gptw_output_b =
00358     1, /**< GTIOCB output pin. Connected to DRV8263H IN1 (PWM) in IN2/IN1 mode. PWM duty cycle controls motor
        speed (0-100%) */
00359 } rx_gptw_output_t;

```

8.79.3.11 rx_gptw_wave_mode_t

```
enum rx\_gptw\_wave\_mode\_t : uint8_t
```

GPTW PWM Waveform Mode Selection.

Selects the PWM waveform generation mode. The mode determines counter behavior, output alignment, and duty cycle update timing. Motor control typically uses sawtooth (edge-aligned) or triangle (center-aligned) modes.

8.79.3.12 Waveform Comparison

Sawtooth (Edge-Aligned):

- Counter counts up from 0 to period, then resets to 0
- PWM edges aligned to period start
- Simpler timing, lower switching losses
- Asymmetric current ripple

Triangle (Center-Aligned):

- Counter counts up to period, then down to 0
- PWM pulses centered in period
- Symmetric current ripple (better for motors)
- Reduced EMI due to staggered edges

Sawtooth Mode
Counter: 0..P repeats
Update: trough only
PWM edges: aligned to start
Best for: LED dimming

Triangle Mode
Counter: 0..P..0 repeats
Update: crest and/or trough
PWM edges: centered
Best for: motor control

Mode Selection Guide:

Mode	Counter Pattern	Update Timing	Use Case
k_gptw_wave_saw_pwm	0->P->0 (reset)	Trough	Simple PWM, LEDs
k_gptw_wave_tri_pwm1	0->P->0 (fold)	Trough	Motor control (standard)
k_gptw_wave_tri_pwm2	0->P->0 (fold)	Crest+↔ Trough	Motor control (fast update)
k_gptw_wave_tri_pwm3	0->P->0 (fold)	64-bit at trough	High-precision motor

Mathematical Model:

Sawtooth PWM Frequency:

$$f_{PWM} = \frac{f_{PCLKA}}{(GTPR + 1)}$$

Triangle PWM Frequency (half the sawtooth frequency for same period):

$$f_{PWM} = \frac{f_{PCLKA}}{2 \times (GTPR + 1)}$$

Where:

- $f_{PCLKA} = 120$ MHz (peripheral clock)
- $GTPR$ = Period register value

Usage Example:

```
// Configure for motor control with triangle wave (center-aligned)
rx_gptw_config_t config = {
    .frequency_hz = 20000,    // 20 kHz PWM
    .deadtime_ns = 1000,     // 1 us deadtime
    .wave_mode = k_gptw_wave_tri_pwm1, // Triangle, trough update
    .enable_complementary = true,
    .invert_polarity = false
};

// For simple LED dimming, use sawtooth
rx_gptw_config_t led_config = {
    .frequency_hz = 1000,    // 1 kHz (no audible noise)
    .wave_mode = k_gptw_wave_saw_pwm, // Sawtooth (simpler)
    .enable_complementary = false,
    .invert_polarity = false
};
```

Invariant

Mode values map directly to GTCR.MD bit field (0-3)
 All modes support complementary output with deadtime

Note

C23 typed enum with uint8_t underlying type
 Triangle modes have half the effective PWM frequency for same period

See also

RX72N Hardware Manual Section 26.2.12 (GTCR.MD bits)
[rx_gptw_config_t](#) Configuration structure using this enum

Since

Version 1.0.0

Enumerator

k_gptw_wave_saw_pwm	Sawtooth-wave PWM (edge-aligned). Counter: 0->Period->reset. Single update point at trough. Best for simple PWM applications. GTCR.MD = 0b000
k_gptw_wave_tri_pwm1	Triangle-wave PWM mode 1 (center-aligned). Counter: 0->Period->0. Update at trough only. Best for motor control with standard update rate. GTCR.MD = 0b100
k_gptw_wave_tri_pwm2	Triangle-wave PWM mode 2 (center-aligned). Counter: 0->Period->0. Update at both crest and trough. Best for motor control requiring fast duty updates. GTCR.MD = 0b101

<code>k_gptw_wave_tri_pwm3</code>	Triangle-wave PWM mode 3 (center-aligned). Counter: 0->Period->0. 64-bit buffer transfer at trough. Best for high-precision motor control. GTCR.MD = 0b110
-----------------------------------	--

Definition at line 452 of file [rx_gptw.h](#).

```

00452         : uint8_t {
00453     k_gptw_wave_saw_pwm =
00454     0, /**< Sawtooth-wave PWM (edge-aligned). Counter: 0->Period->reset. Single update point at trough. Best for simple
        PWM applications. GTCR.MD = 0b000 */
00455     k_gptw_wave_tri_pwm1 =
00456     1, /**< Triangle-wave PWM mode 1 (center-aligned). Counter: 0->Period->0. Update at trough only. Best for motor
        control with standard update rate. GTCR.MD = 0b100 */
00457     k_gptw_wave_tri_pwm2 =
00458     2, /**< Triangle-wave PWM mode 2 (center-aligned). Counter: 0->Period->0. Update at both crest and trough. Best
        for motor control requiring fast duty updates. GTCR.MD = 0b101 */
00459     k_gptw_wave_tri_pwm3 =
00460     3, /**< Triangle-wave PWM mode 3 (center-aligned). Counter: 0->Period->0. 64-bit buffer transfer at trough. Best for
        high-precision motor control. GTCR.MD = 0b110 */
00461 } rx_gptw_wave_mode_t;

```

8.79.4 Function Documentation

8.79.4.1 rx_gptw_channel_id()

```

rx\_gptw\_channel\_id\_t rx_gptw_channel_id (
    rx\_gptw\_channel\_t value)    [inline], [static]

```

Helper to construct a GPTW channel wrapper.

Provides type-safe wrapper construction with NASA Power of 10 Rule 5 validation (minimum 2 checks per function):

- Pre-condition: Input value is within valid channel range [0-3]
- Post-condition: Wrapper correctly stores the input value

Parameters

in	value	GPTW channel enum value (<code>k_gptw_channel_0</code> to <code>k_gptw_channel_3</code>)
----	-------	--

Returns

Wrapped channel ID for type-safe function calls

Precondition

value must be a valid [rx_gptw_channel_t](#) in range [`k_gptw_channel_0`, `k_gptw_channel_3`]
 Caller must pass a compile-time known or previously validated channel value

Postcondition

Returned wrapper .value field equals the input value
 Wrapper is immediately usable in any API accepting [rx_gptw_channel_id_t](#)

Note

Reentrant and lock-free; contains no shared mutable state

See also

[rx_gptw_output_id\(\)](#) Companion wrapper constructor for output parameter

[rx_gptw_channel_id_t](#) Wrapper type constructed by this function

Since

Version 1.0.0

Definition at line 564 of file [rx_gptw.h](#).

```
00565 {
00566     /* Pre-condition: channel must be in valid range */
00567     /* Note: value >= k_gptw_channel_0 is always true for uint8_t since k_gptw_channel_0 == 0 */
00568     RX_ASSERT(value <= k_gptw_channel_3, "GPTW channel out of range");
00569
00570     const rx_gptw_channel_id_t id = {.value = value};
00571
00572     return id;
00573 }
```

References [k_gptw_channel_3](#).

8.79.4.2 rx_gptw_deinit()

```
rx_err_t rx_gptw_deinit (
    const rx_gptw_channel_t channel)    [nodiscard]
```

Deinitialize a GPTW channel and release all resources.

Performs a full shutdown sequence: stops the timer counter, disables both PWM outputs, resets the compare and period registers, and marks the channel as uninitialized. After deinit, the channel must be re-initialized via [rx_gptw_init_pwm\(\)](#) before use.

Parameters

in	channel	GPTW channel number (valid: 0-3)
----	---------	----------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, channel fully deinitialized
k_rx_err_invalid_arg	Channel out of range

Precondition

Channel should be initialized (safe to call if already deinitialized)
Motor outputs should be in a safe state before deinit

Postcondition

Timer stopped, outputs disabled, channel marked uninitialized
All channel registers reset to default values

Note

Not thread-safe. Caller must provide synchronization.

Example:

```
// Full shutdown of GPTW channel 0
rx_err_t err = rx_gptw_deinit(k_gptw_channel_0);
if (err == k_rx_ok) {
    // Channel released, must call rx_gptw_init_pwm() before reuse
}
```

See also

[rx_gptw_init_pwm\(\)](#) Re-initialize after deinit
[rx_gptw_stop\(\)](#) Stop timer without full deinit

Since

Version 1.0.0

Deinitialize a GPTW channel and release all resources.

Stops the timer, disables both PWM outputs (A and B), and resets the counter to zero. Call only when motor control is no longer active.

Precondition

Motor using this channel must be stopped before calling
channel must be in range [k_gptw_channel_0, k_gptw_channel_3]

Postcondition

Timer counter stopped (GTCR.CST = 0)
Both outputs disabled (GTIOR.OAE = 0, GTIOR.OBE = 0)
Counter register zeroed (GTCNT = k_gptw_period_zero)

Invariant

Write protection re-locked (GTWP) on all exit paths

Note

Not thread-safe: modifies hardware registers directly

Does not clear `s_gptw_initialized[]` or `s_gptw_period[]`

Warning

Ensure motor is at rest before calling to prevent abrupt stop

```
rx_err_t err = rx_gptw_deinit(k_gptw_channel_0);
```

See also

[rx_gptw_init_all_staggered\(\)](#) Re-initialize after deinit

[rx_gptw_stop\(\)](#) Stop without full teardown

Since

Version 1.0.0

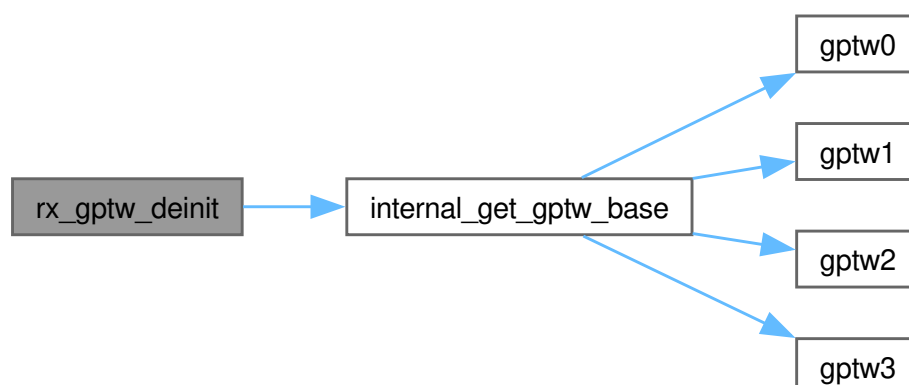
Definition at line 1846 of file `rx_gptw.c`.

```

01847 {
01848     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01849         return k_rx_err_invalid_arg;
01850     }
01851
01852     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01853     if (gptw == nullptr) {
01854         return k_rx_err_invalid_arg;
01855     }
01856
01857     /* Unlock write protection */
01858     gptw->gtwp = k_gptw_gtwp_unlock;
01859
01860     /* Stop counter */
01861     gptw->gtcr &= ~k_gptw_gtcr_cst;
01862
01863     /* Disable both outputs (OAE and OBE) */
01864     gptw->gtior &= ~(k_gptw_gtior_oae | k_gptw_gtior_obo);
01865
01866     /* Reset counter to zero */
01867     gptw->gtcnt = k_gptw_period_zero;
01868
01869     /* Re-lock write protection */
01870     gptw->gtwp = k_gptw_gtwp_lock;
01871
01872     return k_rx_ok;
01873 }
```

References `rx_gptw_channel_regs_t::gtcnt`, `rx_gptw_channel_regs_t::gtcr`, `rx_gptw_channel_regs_t::gtior`, `rx_gptw_channel_regs_t::gtwp`, `internal_get_gptw_base()`, `k_gptw_gtcr_cst`, `k_gptw_gtior_oae`, `k_gptw_gtior_obo`, `k_gptw_gtwp_lock`, `k_gptw_gtwp_unlock`, `k_gptw_max_channels`, and `k_gptw_period_zero`.

Here is the call graph for this function:



8.79.4.3 rx`gptw`enable`output()

```
rx_err_t rx_gptw_enable_output (
    const rx_gptw_channel_t channel,
    const rx_gptw_output_t output,
    bool enable) [nodiscard]
```

Enable or disable a PWM output pin.

Controls the GTONCR (General Timer Output Negation Control Register) bits that gate the PWM signal to the physical pin. When disabled, the pin is driven to its inactive level (determined by polarity configuration).

Parameters

in	channel	GPTW channel number (valid: 0-3)
in	output	Output channel (k_gptw_output_a or k_gptw_output_b)
in	enable	true to drive PWM on pin, false to hold pin inactive

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, output enable state updated
k_rx_err_invalid_arg	Channel or output out of range
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel must be initialized via rx_gptw_init_pwm()

Output must be k_gptw_output_a or k_gptw_output_b

Postcondition

GTONCR bit updated for the specified output

Pin either drives PWM signal or holds inactive level

Note

Not thread-safe. Caller must provide synchronization.

Example:

```
// Enable PWM output on channel 0, output A
rx_err_t err = rx_gptw_enable_output(k_gptw_channel_0, k_gptw_output_a, true);
if (err != k_rx_ok) {
    // Handle output enable failure
}
```

See also

[rx_gptw_start\(\)](#) Start timer (separate from output enable)

[rx_gptw_stop\(\)](#) Stop timer (separate from output enable)

Since

Version 1.0.0

Enable or disable a PWM output pin.

Implementation of [rx_gptw_enable_output\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Controls the Output Enable bits (OAE/OBE) in the GTIOR register. When disabled, the corresponding output pin is held in its initial state (typically LOW).

8.79.4.4 Register Modification

Output	Bit Modified	Effect
Output A	GTIOR.OAE	Enable/disable GTIOCnA pin
Output B	GTIOR.OBE	Enable/disable GTIOCnB pin

Precondition

Channel initialized via [rx_gptw_init_*](#)()

Postcondition

GTIOR.OAE or GTIOR.OBE modified

Write protection re-enabled

Note

Not thread-safe: Modifies hardware registers

Timer continues running even when output disabled

Warning

Disable outputs before motor emergency stop to ensure safe state

See also

[rx_gptw.h](#) Complete API documentation

Definition at line 1286 of file [rx_gptw.c](#).

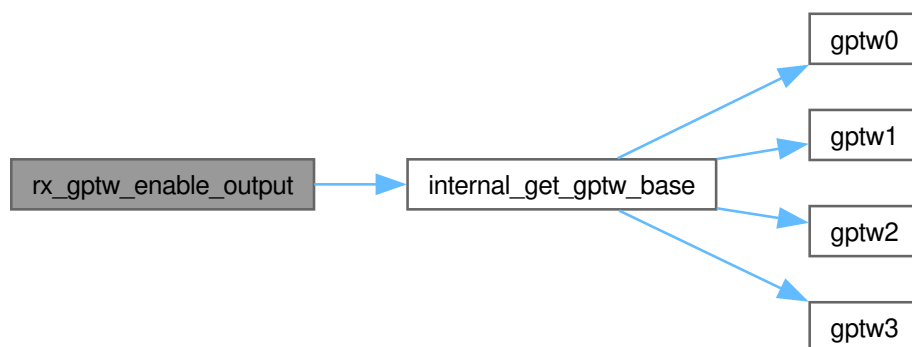
```

01287 {
01288     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01289         return k_rx_err_invalid_state;
01290     }
01291
01292     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01293     if (gptw == nullptr) {
01294         return k_rx_err_invalid_arg;
01295     }
01296
01297     /* Unlock for modification */
01298     gptw->gtwp = k_gptw_gtwp_unlock;
01299
01300     /* Modify GTIOR to enable/disable output */
01301     switch (output) {
01302     case k_gptw_output_a:
01303         if (enable) {
01304             gptw->gtior |= k_gptw_gtior_oae;
01305         } else {
01306             gptw->gtior &= ~k_gptw_gtior_oae;
01307         }
01308         break;
01309     case k_gptw_output_b:
01310         if (enable) {
01311             gptw->gtior |= k_gptw_gtior_obe;
01312         } else {
01313             gptw->gtior &= ~k_gptw_gtior_obe;
01314         }
01315         break;
01316     default:
01317         gptw->gtwp = k_gptw_gtwp_lock;
01318         return k_rx_err_invalid_arg;
01319     }
01320
01321     gptw->gtwp = k_gptw_gtwp_lock;
01322
01323     return k_rx_ok;
01324 }

```

References [rx_gptw_channel_regs_t::gtior](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtior_oae](#), [k_gptw_gtior_obe](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), [k_gptw_max_channels](#), [k_gptw_output_a](#), [k_gptw_output_b](#), and [s_gptw_initialized](#).

Here is the call graph for this function:



8.79.4.5 rx_gptw_get_duty()

```

rx_err_t rx_gptw_get_duty (
    const rx_gptw_channel_t channel,

```

```
const rx_gptw_output_t output,
float * duty_percent) [nodiscard]
```

Get current duty cycle as a floating-point percentage.

Reads the current compare register value for the specified output and converts it to a percentage of the period count. The value reflects the most recently committed duty cycle (may differ from the buffered value pending a period boundary update).

Parameters

in	channel	GPTW channel number (valid: 0-3)
in	output	Output channel (k_gptw_output_a or k_gptw_output_b)
out	duty_percent	Pointer to store duty cycle in percent [0.0, 100.0]

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, duty_percent written
k_rx_err_null_ptr	duty_percent is nullptr
k_rx_err_invalid_arg	Channel or output out of range
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel must be initialized via `rx_gptw_init_pwm()`
duty_percent must be a valid non-null pointer

Postcondition

duty_percent contains current duty cycle percentage
No hardware state modified (read-only operation)

Note

Not thread-safe. Caller must provide synchronization.

Example:

```
float duty = 0.0F;
rx_err_t err = rx_gptw_get_duty(k_gptw_channel_0, k_gptw_output_a, &duty);
if (err == k_rx_ok) {
    // duty now contains current duty cycle percentage [0.0, 100.0]
}
```

See also

[rx_gptw_set_duty\(\)](#) Set duty cycle by percentage
[rx_gptw_get_period\(\)](#) Get period count for manual calculation

Since

Version 1.0.0

Get current duty cycle as a floating-point percentage.

Reads the current compare register value (GTCCRA or GTCCRB) from hardware and converts it to a percentage using the cached period from `s_gptw_period[]`.

Algorithm:

$$\text{duty_percent} = \frac{\text{duty_count} \times 100}{\text{period}}$$

Precondition

Channel must be initialized via `rx_gptw_init_*`()
`duty_percent` must point to valid float storage

Postcondition

`*duty_percent` contains duty cycle in `[0.0, 100.0]` on success
Hardware registers unchanged

Note

Returns actual register value; may differ from the last set value if buffer transfer has not yet occurred at period boundary

Not thread-safe: concurrent `rx_gptw_set_duty` calls may cause torn reads

```
float duty;  
rx_err_t err = rx_gptw_get_duty(k_gptw_channel_0, k_gptw_output_a, &duty);
```

See also

[rx_gptw_set_duty\(\)](#) Set duty using floating-point percentage
[rx_gptw_get_period\(\)](#) Retrieve period count

Since

Version 1.0.0

Definition at line 1188 of file [rx_gptw.c](#).

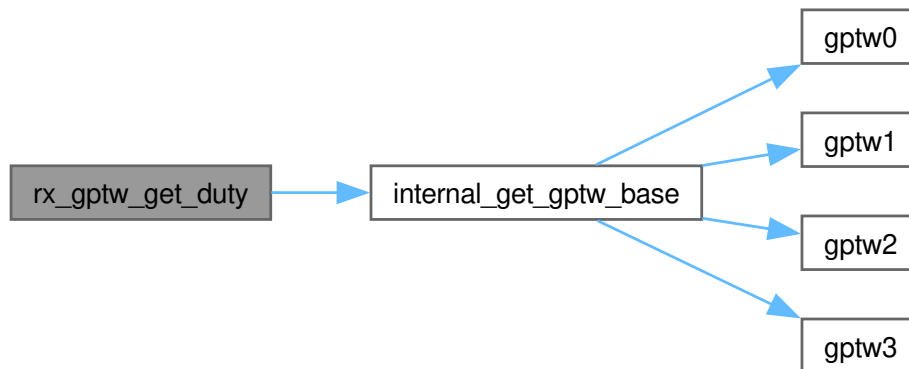
```

01191 {
01192     RX_CHECK_NULL_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
01193
01194     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01195         return k_rx_err_invalid_state;
01196     }
01197
01198     const volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01199     if (gptw == nullptr) {
01200         return k_rx_err_invalid_arg;
01201     }
01202
01203     const uint32_t period = s_gptw_period[channel];
01204     uint32_t duty_count = k_gptw_period_zero;
01205
01206     switch (output) {
01207         case k_gptw_output_a:
01208             duty_count = gptw->gtccra;
01209             break;
01210         case k_gptw_output_b:
01211             duty_count = gptw->gtccrb;
01212             break;
01213         default:
01214             return k_rx_err_invalid_arg;
01215     }
01216
01217     *duty_percent = ((float)duty_count * (float)k_gptw_duty_max) / (float)period;
01218
01219     return k_rx_ok;
01220 }

```

References [rx_gptw_channel_regs_t::gtccra](#), [rx_gptw_channel_regs_t::gtccrb](#), [internal_get_gptw_base\(\)](#), [k_gptw_duty_max](#), [k_gptw_max_channels](#), [k_gptw_output_a](#), [k_gptw_output_b](#), [k_gptw_period_zero](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

Here is the call graph for this function:



8.79.4.6 rx_gptw_get_period()

```

rx_err_t rx_gptw_get_period (
    const rx_gptw_channel_t channel,
    uint32_t * period_count) [nodiscard]

```

Get the 32-bit PWM period count for a GPTW channel.

Reads the GTPR (General Timer Period Register) value for the specified channel. This value determines the PWM frequency and is needed to calculate raw duty counts: $\text{duty_count} = (\text{duty_percent} / 100) * \text{period_count}$.

Parameters

in	channel	GPTW channel number (valid: 0-3)
out	period_count	Pointer to store 32-bit period count

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, period_count written
k_rx_err_null_ptr	period_count is nullptr
k_rx_err_invalid_arg	Channel out of range
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel must be initialized via [rx_gptw_init_pwm\(\)](#)
period_count must be a valid non-null pointer

Postcondition

period_count contains the GTPR register value
No hardware state modified (read-only operation)

Note

Not thread-safe. Caller must provide synchronization.

Example:

```
uint32_t period = 0;
rx_err_t err = rx_gptw_get_period(k_gptw_channel_0, &period);
if (err == k_rx_ok) {
    // Calculate raw duty count for 50% duty cycle
    uint32_t duty_count = (period * k_gptw_duty_50_pct_num) / k_gptw_duty_pct_denom;
}
```

See also

[rx_gptw_set_duty_raw\(\)](#) Uses period_count for raw duty calculation

Since

Version 1.0.0

Get the 32-bit PWM period count for a GPTW channel.

Implementation of [rx_gptw_get_period\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Returns the cached period value from s_gptw_period[], which matches the GTPR register value set during initialization.

Precondition

Channel initialized via `rx_gptw_init_*`()
`period_count` points to valid `uint32_t` storage

Postcondition

`*period_count` contains period value on success

Note

Thread-safe: Reads from cached value, not hardware
 Useful for calculating raw duty counts from percentages

See also

[rx_gptw.h](#) Complete API documentation

Definition at line 1243 of file `rx_gptw.c`.

```
01244 {
01245   RX_CHECK_NULL_PTR(period_count, s_tag, "period_count pointer is nullptr");
01246
01247   if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01248       return k_rx_err_invalid_state;
01249   }
01250
01251   *period_count = s_gptw_period[channel];
01252   return k_rx_ok;
01253 }
```

References [k_gptw_max_channels](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

8.79.4.7 `rx_gptw_init_all_staggered()`

```
rx_err_t rx_gptw_init_all_staggered (
    const rx\_gptw\_config\_t * configs[k_rx_gptw_channel_count]) [nodiscard]
```

Initialize all 4 GPTW channels with 90-degree phase staggering.

Implementation of `rx_gptw_init_all_staggered()` - see [rx_gptw.h](#) for complete documentation with diagrams, examples, and NASA compliance notes.

8.79.4.8 Implementation Overview

1. Validate inputs: NULL check, frequency > 0, valid `wave_mode`
2. Enable module clock: Clear `MSTPCRA.MSTPA7`
3. Stop all channels: Atomic stop via `GTSTPA` register
4. Calculate period: Common period for all channels
5. Configure each channel: Loop calls [internal_configure_channel_staggered\(\)](#)
6. Start all channels: Atomic start via `GTSTRA` register

8.79.4.9 Phase Staggering Benefits

- Reduces peak current by spreading PWM edges across period
- Lowers EMI from simultaneous switching transients
- Allows smaller power supply capacitors

See also

[rx_gptw.h](#) Complete API documentation with diagrams and examples

Definition at line 1747 of file [rx_gptw.c](#).

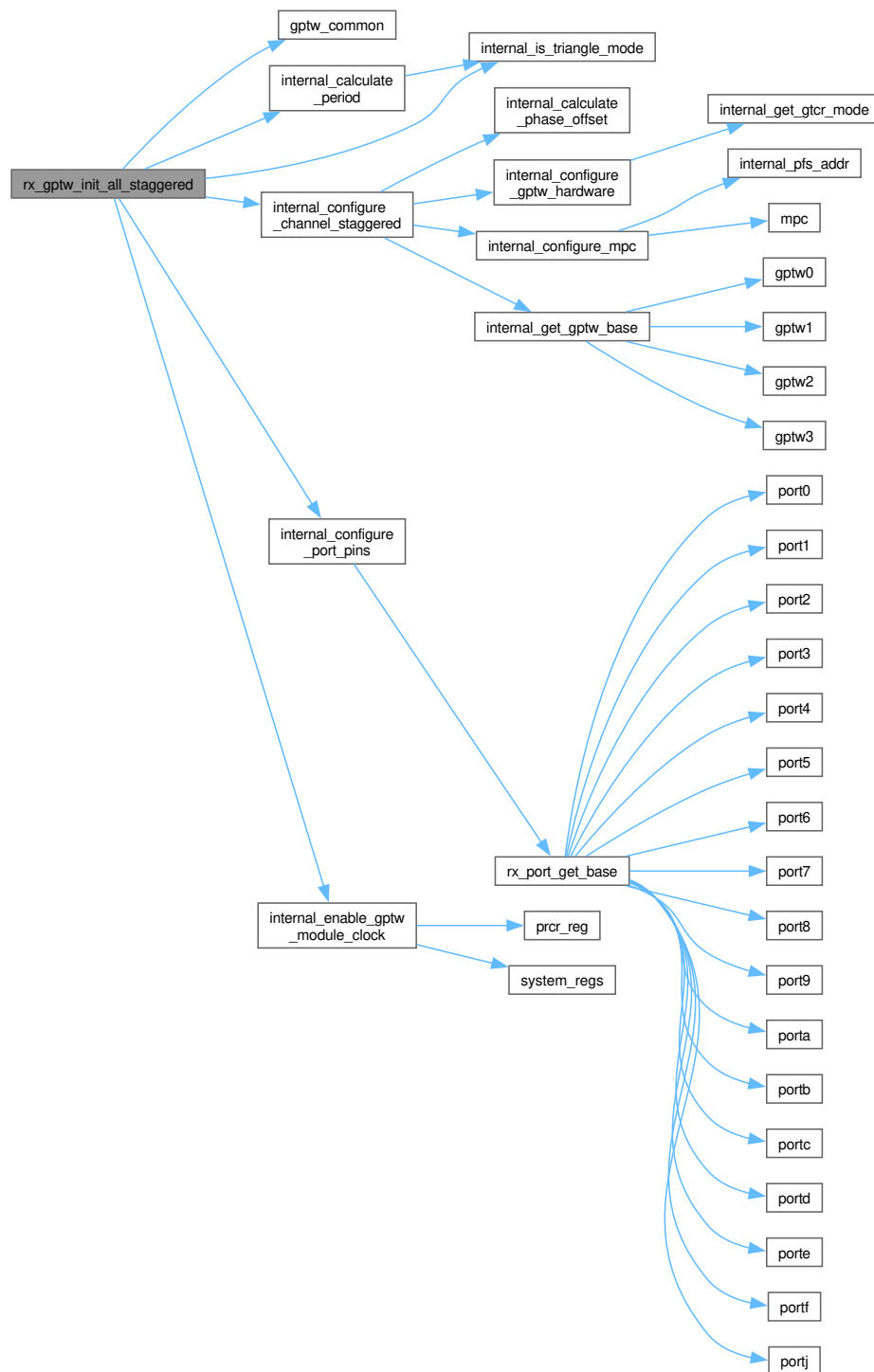
```

01748 {
01749     RX_CHECK_NULL_PTR(configs, s_tag, "configs array pointer is nullptr");
01750     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01751         RX_CHECK_NULL_PTR(configs[i], s_tag, "configs[i] pointer is nullptr");
01752     }
01753
01754     /* All channels must share frequency / wave_mode (otherwise the staggered
01755      * phasing math doesn't work). Validate once on configs[0]. */
01756     if (configs[0]->frequency_hz == 0) {
01757         rx_log_error(s_tag, "frequency_hz is zero");
01758         return k_rx_err_invalid_arg;
01759     }
01760     switch (configs[0]->wave_mode) {
01761         case k_gptw_wave_saw_pwm:
01762         case k_gptw_wave_tri_pwm1:
01763         case k_gptw_wave_tri_pwm2:
01764         case k_gptw_wave_tri_pwm3:
01765             break;
01766         default:
01767             rx_log_error(s_tag, "Invalid wave_mode");
01768             return k_rx_err_invalid_arg;
01769     }
01770
01771     rx_log_info(s_tag, "Initializing all GPTW channels (staggered)");
01772
01773     internal_enable_gptw_module_clock();
01774
01775     if (gptw_common() != nullptr) {
01776         gptw_common()->gtstpa = k_gptw_all_channels_mask; /* Stop Ch0-3 */
01777     }
01778
01779     uint32_t period;
01780     rx_err_t err =
01781         internal_calculate_period(configs[0]->frequency_hz, configs[0]->wave_mode, &period);
01782     if (err != k_rx_ok) {
01783         return err;
01784     }
01785
01786     const bool is_triangle = internal_is_triangle_mode(configs[0]->wave_mode);
01787
01788     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01789         err =
01790             internal_configure_channel_staggered((rx_gptw_channel_t)i, configs[i], period, is_triangle);
01791         if (err != k_rx_ok) {
01792             return err;
01793         }
01794     }
01795
01796     /* Atomic start of all 4 channels. */
01797     if (gptw_common() == nullptr) {
01798         return k_rx_err_hw_error;
01799     }
01800     rx_log_info(s_tag, "Global start of GPTW channels");
01801     gptw_common()->gtstra = k_gptw_all_channels_mask;
01802
01803     /* Now that valid PWM is running on every channel, route the outputs
01804      * to their pads (PMR=1). Doing this after start avoids exposing the
01805      * downstream H-bridges to the GTIOR reset/initial state during the
01806      * brief window between mux and counter start. */
01807     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01808         internal_configure_port_pins(configs[i]);
01809     }
01810
01811     return k_rx_ok;
01812 }

```

References `gptw_common()`, `rx_gptw_common_regs_t::gtstpa`, `rx_gptw_common_regs_t::gtstra`, `internal_calculate_period()`, `internal_configure_channel_staggered()`, `internal_configure_port_pins()`, `internal_enable_gptw_module_clock()`, `internal_is_triangle_mode()`, `k_gptw_all_channels_mask`, `k_gptw_max_channels`, `k_gptw_wave_saw_pwm`, `k_gptw_wave_tri_pwm1`, `k_gptw_wave_tri_pwm2`, `k_gptw_wave_tri_pwm3`, `k_rx_gptw_channel_count`, and `s_tag`.

Here is the call graph for this function:



8.79.4.10 rx_gptw_init_pwm()

```
rx_err_t rx_gptw_init_pwm (
    const rx_gptw_channel_t channel,
    const rx_gptw_config_t * config)  [nodiscard]
```

Initialize a single GPTW channel for PWM output.

Configures the specified GPTW channel for PWM mode (sawtooth or triangle wave). Automatically configures MPC for the pin alternate functions, sets up dead-time insertion, and starts PWM generation at 0% duty cycle. The GTCNT counter is reset to zero during initialization.

Parameters

in	channel	GPTW channel number (valid: 0-3)
in	config	Pointer to PWM configuration struct (must be non-null)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, channel initialized and timer running
k_rx_err_invalid_arg	Channel out of range or config is null
k_rx_err_invalid_state	GPTW module not enabled in MSTPCR

Precondition

GPTW module clock must be enabled

config must point to a valid [rx_gptw_config_t](#)

Postcondition

Channel timer running at configured frequency with 0% duty

GTCNT reset to zero (affects phase staggering if called per-channel)

Note

Not thread-safe. Call from single-threaded init context.

Warning

Resets GTCNT to 0. If phase staggering is required, use [rx_gptw_init_all_staggered\(\)](#) instead, or configure phase offsets after all channels are initialized.

See also

[rx_gptw_init_all_staggered\(\)](#) Phase-staggered multi-channel init
[rx_gptw_deinit\(\)](#) Release channel resources

Since

Version 1.0.0

Initialize a single GPTW channel for PWM output.

Configures one GPTW channel for PWM generation. This function performs complete hardware setup including module clock enable, MPC configuration, port pin setup, timer configuration, and automatic timer start. For 4-motor systems, prefer [rx_gptw_init_all_staggered\(\)](#) instead.

8.79.4.11 Initialization Sequence

1. Validate config pointer ([RX_CHECK_NULL_PTR](#))
2. Validate channel range [0-3]
3. Get channel register base address
4. Calculate period from frequency and wave mode
5. Enable GPTW module clock ([MSTPCRA.MSTPA7](#))
6. Stop timer if already running
7. Configure GPTW hardware registers ([GTCR](#), [GTIOR](#), [GTPR](#), etc.)
8. Configure MPC for Port E alternate function
9. Configure port pins ([PMR](#), [PDR](#))
10. Store period value and set initialized flag
11. Start timer

Precondition

System clock configured ([PCLKA](#) = 120 MHz)
Port E pins not used by other peripherals

Postcondition

GPTW channel configured and running
Duty cycle initialized to 0%
MPC and port pins configured

Invariant

[s_gptw_initialized\[channel\]](#) == true on success
[s_gptw_period\[channel\]](#) contains valid period on success

Note

Not thread-safe: Call once during system initialization

Timer starts automatically - use [rx_gptw_stop\(\)](#) to halt if needed

Warning

Calling while channel is running will cause glitches

Attention

For 4-motor systems, use [rx_gptw_init_all_staggered\(\)](#) instead

Performance:

- Execution time: ~50 us at 240 MHz
- Memory: 0 bytes heap, ~32 bytes stack

Example (Single Channel for LED PWM):

```
rx_gptw_config_t led_config = {
    .frequency_hz = 1000,
    .deadtime_ns = 0,
    .wave_mode = k_gptw_wave_saw_pwm,
    .enable_complementary = false,
    .invert_polarity = false
};

rx_err_t err = rx_gptw_init_pwm(k_gptw_channel_0, &led_config);
if (err == k_rx_ok) {
    rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
        rx_gptw_output_id(k_gptw_output_a), 50.0f);
}
```

See also

[rx_gptw_init_all_staggered\(\)](#) Preferred for 4-motor systems

[rx_gptw_deinit\(\)](#) Cleanup function

[rx_gptw_config_t](#) Configuration structure

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] nullptr check, channel range check
- Rule 7: [OK] All internal function returns checked

Definition at line 958 of file [rx_gptw.c](#).

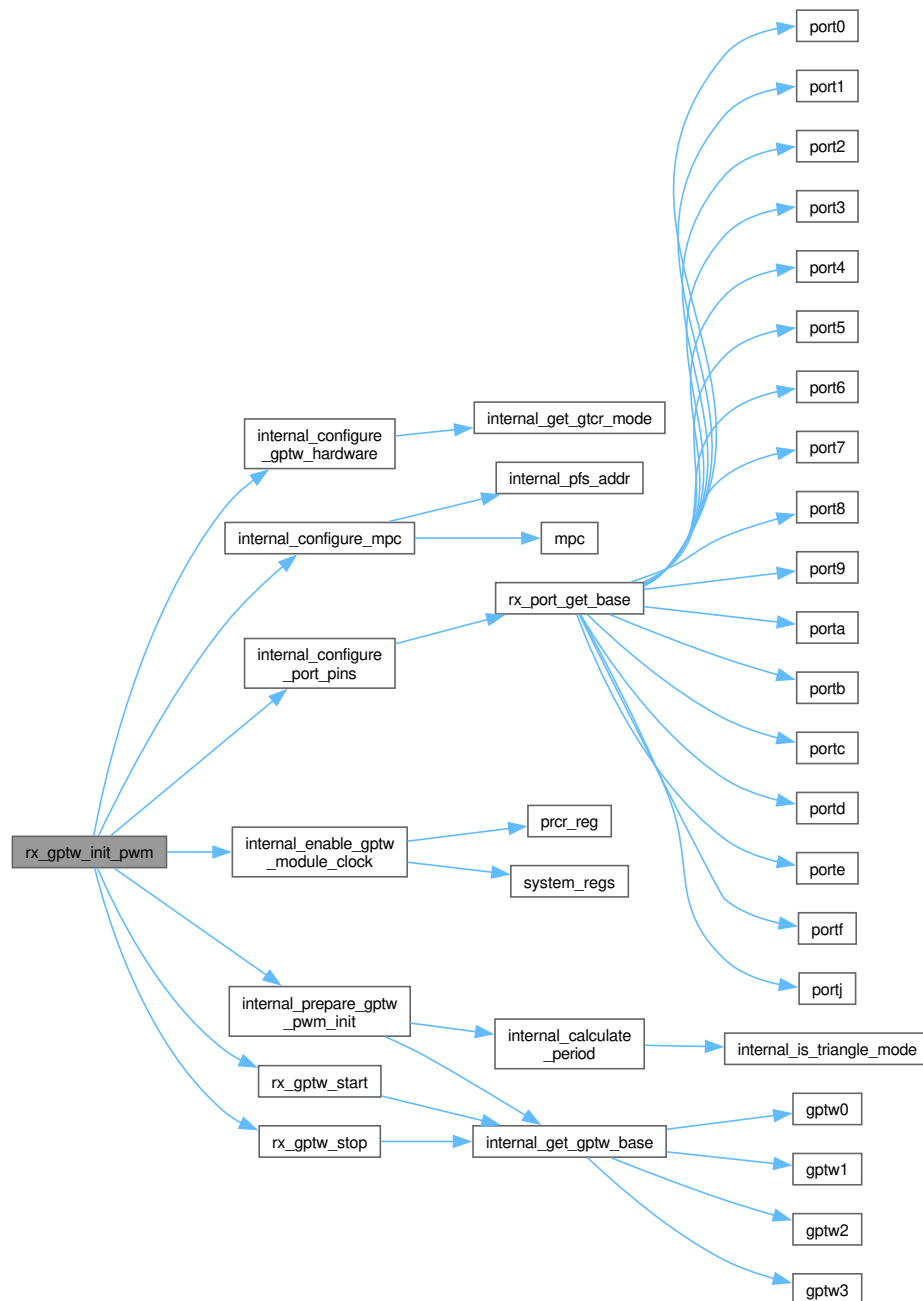
```

00959 {
00960     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00961
00962     volatile rx_gptw_channel_regs_t* gptw = nullptr;
00963     uint32_t period = 0U;
00964     rx_err_t err = internal_prepare_gptw_pwm_init(channel, config, &gptw, &period);
00965     if (err != k_rx_ok) {
00966         return err;
00967     }
00968
00969     rx_log_info(s_tag, "Initializing GPTW");
00970
00971     /* Enable GPTW module clock */
00972     internal_enable_gptw_module_clock();
00973
00974     /* Stop timer before configuration */
00975     err = rx_gptw_stop(channel);
00976     if (err != k_rx_ok) {
00977         rx_log_error(s_tag, "Failed to stop GPTW");
00978         return err;
00979     }
00980
00981     /* Configure GPTW hardware registers */
00982     err = internal_configure_gptw_hardware(gptw, config, period);
00983     if (err != k_rx_ok) {
00984         rx_log_error(s_tag, "Failed to configure GPTW hardware");
00985         return err;
00986     }
00987
00988     /* Write the PFS (peripheral function select) for the GTIOC pads. Pin
00989      * is still in GPIO mode at this point -- PMR is set later. */
00990     err = internal_configure_mpc(config);
00991     if (err != k_rx_ok) {
00992         rx_log_error(s_tag, "Failed to configure MPC");
00993         return err;
00994     }
00995
00996     /* Save period for duty cycle calculations */
00997     s_gptw_period[channel] = period;
00998     s_gptw_initialized[channel] = true;
00999
01000     /* Start timer FIRST -- the GTIOC output must be producing valid PWM
01001      * before we route it to the pad. Otherwise the brief window between
01002      * PMR=1 and counter start exposes downstream hardware (e.g. DRV8263H
01003      * H-bridge inputs) to whatever GTIOR's reset/initial state happens to
01004      * be, which has been observed to latch driver faults. */
01005     err = rx_gptw_start(channel);
01006     if (err != k_rx_ok) {
01007         rx_log_error(s_tag, "Failed to start GPTW");
01008         return err;
01009     }
01010
01011     /* Now route the running PWM to the pad: PDR=1 (output) and PMR=1
01012      * (peripheral mode). The pin transitions from GPIO straight to a
01013      * fully-running, valid PWM signal -- no glitch window. */
01014     internal_configure_port_pins(config);
01015
01016     rx_log_info(s_tag, "GPTW initialized successfully");
01017
01018     return k_rx_ok;
01019 }

```

References [internal_configure_gptw_hardware\(\)](#), [internal_configure_mpc\(\)](#), [internal_configure_port_pins\(\)](#), [internal_enable_gptw_module_clock\(\)](#), [internal_prepare_gptw_pwm_init\(\)](#), [rx_gptw_start\(\)](#), [rx_gptw_stop\(\)](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

Here is the call graph for this function:



8.79.4.12 rx`gptw`output`id()

```

rx_gptw_output_id_t rx_gptw_output_id (
    rx_gptw_output_t value)  [inline], [static]
  
```

Helper to construct a GPTW output wrapper.

Provides type-safe wrapper construction with NASA Power of 10 Rule 5 validation (minimum 2 checks per function):

- Pre-condition: Input value is within valid output range [A-B]
- Post-condition: Wrapper correctly stores the input value

Parameters

in	value	GPTW output enum value (k_gptw_output_a or k_gptw_output_b)
----	-------	---

Returns

Wrapped output ID for type-safe function calls

Precondition

value must be a valid [rx_gptw_output_t](#) (k_gptw_output_a or k_gptw_output_b)

Caller must pass a compile-time known or previously validated output value

Postcondition

Returned wrapper .value field equals the input value

Wrapper is immediately usable in any API accepting [rx_gptw_output_id_t](#)

Note

Reentrant and lock-free; contains no shared mutable state

See also

[rx_gptw_channel_id\(\)](#) Companion wrapper constructor for channel parameter

[rx_gptw_output_id_t](#) Wrapper type constructed by this function

Since

Version 1.0.0

Definition at line 599 of file [rx_gptw.h](#).

```
00600 {
00601     /* Pre-condition: output must be in valid range */
00602     /* Note: value >= k_gptw_output_a is always true for uint8_t since k_gptw_output_a == 0 */
00603     RX_ASSERT(value <= k_gptw_output_b, "GPTW output out of range");
00604
00605     const rx_gptw_output_id_t id = {.value = value};
00606
00607     return id;
00608 }
```

References [k_gptw_output_b](#).

8.79.4.13 rx_gptw_set_duty()

```
rx_err_t rx_gptw_set_duty (
    const rx_gptw_channel_id_t channel,
    const rx_gptw_output_id_t output,
    const float duty_percent) [nodiscard]
```

Set PWM duty cycle (floating-point percentage).

Updates the duty cycle for the specified GPTW channel and output. The new duty cycle is written to the compare register buffer and takes effect on the next PWM period boundary, ensuring glitch-free operation.

8.79.4.14 Algorithm Steps

1. Validate channel and output wrapper values
2. Check channel is initialized
3. Clamp duty_percent to [0.0, 100.0] range
4. Calculate raw count: $\text{duty_count} = (\text{duty_percent} / 100.0) \times \text{period_count}$
5. Write to compare register (GTCCRA or GTCCRB)
6. Register buffer transfers to active on next period boundary

Duty Cycle Calculation:

$$\text{duty_count} = \left\lfloor \frac{\text{duty_percent}}{100} \times \text{period_count} \right\rfloor$$

For 20 kHz PWM with 120 MHz PCLKA:

- Period count = 6000
- 50% duty = 3000 counts
- Resolution = 0.0167% per count

Parameters

in	channel	GPTW channel wrapped in rx_gptw_channel_id_t for type safety • Use <code>rx_gptw_channel_id(k_gptw_channel_0)</code> through <code>rx_gptw_channel_id(k_gptw_channel_3)</code>
in	output	Output selection wrapped in rx_gptw_output_id_t for type safety • Use <code>rx_gptw_output_id(k_gptw_output_a)</code> or <code>rx_gptw_output_id(k_gptw_output_b)</code>
in	duty_percent	Duty cycle in percent [0.0, 100.0] • Values < 0.0 are clamped to 0.0 • Values > 100.0 are clamped to 100.0 • NaN and infinity are treated as invalid

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Duty cycle updated successfully
<code>k_rx_err_invalid_arg</code>	Channel or output out of range, or duty is NaN/infinity
<code>k_rx_err_not_initialized</code>	Channel not initialized via <code>rx_gptw_init_*</code> ()

Precondition

Channel initialized via `rx_gptw_init_all_staggered()` or `rx_gptw_init_pwm()`
 duty_percent is a valid float (not NaN or infinity)

Postcondition

Compare register updated with new duty value
 Duty takes effect on next period boundary (no glitch)

Invariant

Actual duty always in [0%, 100%] regardless of input

Note

Thread-safe: Atomic register write, safe from ISR or multiple threads
 Use `rx_gptw_set_duty_raw()` in tight control loops to avoid float overhead

Warning

Floating-point calculation adds ~4 us overhead vs raw version

Performance:

- Execution time: ~5 us at 240 MHz (includes float multiply)
- Memory usage: 0 bytes heap, ~16 bytes stack

Example (Set Motor Speed):

```
// Set motor 0 forward at 75% speed
// Output A = direction (100% = forward)
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
  rx_gptw_output_id(k_gptw_output_a),
  100.0F);

// Output B = speed (75% duty)
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
  rx_gptw_output_id(k_gptw_output_b),
  75.0F);
```

Example (PID Control Loop):

```
void motor_control_task(void)
{
    while (true) {
        float setpoint = get_velocity_setpoint();
        float measured = read_encoder_velocity();
        float output;

        rx_pid_compute(&pid, setpoint, measured, 0.01F, &output);

        // Convert PID output (-100 to +100) to duty cycle
        float duty = fabsf(output);
        bool forward = (output >= 0.0F);

        // Set direction
        rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                        rx_gptw_output_id(k_gptw_output_a),
                        forward ? 100.0F : 0.0F);

        // Set speed
        rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                        rx_gptw_output_id(k_gptw_output_b),
                        duty);

        tx_thread_sleep(1); // 100 Hz control loop
    }
}
```

See also

[rx_gptw_set_duty_raw\(\)](#) Raw count version for tight loops
[rx_gptw_get_duty\(\)](#) Read current duty cycle
[rx_gptw_channel_id\(\)](#) Wrapper constructor for channel
[rx_gptw_output_id\(\)](#) Wrapper constructor for output

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 2 postconditions
- Rule 7: [OK] Returns error code, caller must check

Set PWM duty cycle (floating-point percentage).

Converts floating-point duty percentage [0.0, 100.0] to raw count value and delegates to [rx_gptw_set_duty_raw\(\)](#) for hardware register update. The compare register (GTCCRA or GTCCRB) is written and the change takes effect on the next PWM period boundary (glitch-free via buffer mode).

Algorithm:

$$\text{duty_count} = \left\lfloor \frac{\text{duty_percent} \times \text{period}}{100} \right\rfloor$$

Precondition

Channel must be initialized via [rx_gptw_init_*](#)()
duty_percent must be in [0.0, 100.0]

Postcondition

Compare register written; change takes effect next period boundary
 Motor output drive updated on next PWM cycle

Note

Not thread-safe: concurrent writes to same output cause undefined behavior

```
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                rx_gptw_output_id(k_gptw_output_a), 75.0f);
```

See also

[rx_gptw_set_duty_raw\(\)](#) Set duty using raw count value
[rx_gptw_get_duty\(\)](#) Read current duty cycle

Since

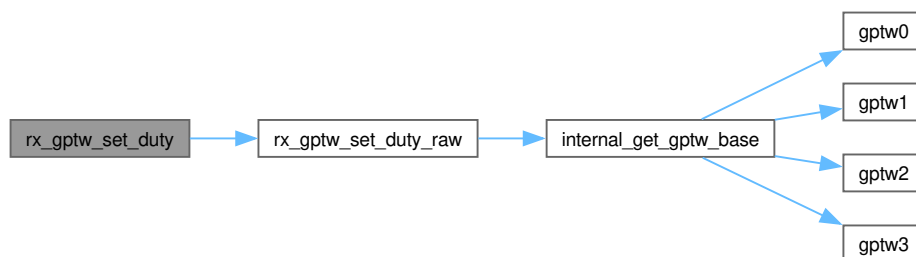
Version 1.0.0

Definition at line 1055 of file [rx_gptw.c](#).

```
01058 {
01059     const rx_gptw_channel_t channel_value = channel.value;
01060     const rx_gptw_output_t output_value = output.value;
01061
01062     if (((int32_t)channel_value >= (int32_t)k_gptw_max_channels) ||
01063         !s_gptw_initialized[channel_value]) {
01064         return k_rx_err_invalid_state;
01065     }
01066
01067     if (duty_percent < (float)k_gptw_duty_min || duty_percent > (float)k_gptw_duty_max) {
01068         rx_log_error(s_tag, "Invalid duty cycle");
01069         return k_rx_err_invalid_arg;
01070     }
01071
01072     /* Convert percentage to count value */
01073     const uint32_t period = s_gptw_period[channel_value];
01074     const float period_f = (float)period;
01075     const float duty_count_f = (duty_percent * period_f) / (float)k_gptw_duty_divisor;
01076     const uint32_t duty_count = (uint32_t)duty_count_f;
01077
01078     return rx_gptw_set_duty_raw(channel_value, output_value, duty_count);
01079 }
```

References [k_gptw_duty_divisor](#), [k_gptw_duty_max](#), [k_gptw_duty_min](#), [k_gptw_max_channels](#), [rx_gptw_set_duty_raw\(\)](#), [s_gptw_initialized](#), [s_gptw_period](#), [s_tag](#), [rx_gptw_channel_id_t::value](#), and [rx_gptw_output_id_t::value](#).

Here is the call graph for this function:



8.79.4.15 rx_gptw_set_duty_raw()

```
rx_err_t rx_gptw_set_duty_raw (
    const rx_gptw_channel_t channel,
    rx_gptw_output_t output,
    uint32_t duty_count) [nodiscard]
```

Set PWM duty cycle using raw counter value.

Updates duty cycle using a raw counter value for efficiency in tight control loops where floating-point overhead should be avoided. The count is written to the compare register buffer and takes effect on the next PWM period boundary, ensuring glitch-free operation.

Parameters

in	channel	GPTW channel (valid: 0-3)
in	output	Output channel (k_gptw_output_a or k_gptw_output_b)
in	duty_count	Duty cycle count (valid range: 0 to period_count)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, duty cycle updated
k_rx_err_invalid_arg	Channel, output, or count out of range
k_rx_err_invalid_state	Channel not initialized via rx_gptw_init_pwm()

Precondition

Channel must be initialized via rx_gptw_init_pwm()

duty_count must not exceed the period count for the channel

Postcondition

Compare register buffer updated with new duty count

Active register updated on next period boundary

Note

Thread-safe: single 32-bit register write is inherently atomic on RX72N

Performance:

Constant time: single 32-bit register write after validation.

Example:

```
// Read current period, compute 75% duty, then set raw count
uint32_t period = 0;
rx_err_t err = rx_gptw_get_period(k_gptw_channel_0, &period);
if (err == k_rx_ok) {
    const uint32_t duty_count = (period * k_gptw_duty_75_pct_num) / k_gptw_duty_pct_denom;
    err = rx_gptw_set_duty_raw(k_gptw_channel_0, k_gptw_output_b, duty_count);
}
```

See also

[rx_gptw_set_duty\(\)](#) Percentage-based version with float conversion

[rx_gptw_get_period\(\)](#) Query period count for duty calculation

Since

Version 1.0.0

Set PWM duty cycle using raw counter value.

Directly writes duty_count to the compare register (GTCCRA or GTCCRB). Buffer mode (enabled during init via GTBER) ensures the update takes effect on the next PWM period boundary without glitches. Values exceeding the period are clamped to period (100% duty).

Register Access:

- Output A: Writes to GTCCRA
- Output B: Writes to GTCCRB

Precondition

Channel must be initialized via rx_gptw_init_*

duty_count should be in [0, period] for meaningful duty; higher values clamped

Postcondition

GTCCRA or GTCCRB written with clamped duty_count

Change takes effect at next PWM period boundary

Note

Not thread-safe: concurrent writes to the same register cause undefined behavior

Values exceeding period are silently clamped to period

```
uint32_t period;
rx_gptw_get_period(k_gptw_channel_0, &period);
rx_gptw_set_duty_raw(k_gptw_channel_0, k_gptw_output_a, period / 2U); // 50%
```

See also

[rx_gptw_set_duty\(\)](#) Set duty using floating-point percentage

[rx_gptw_get_period\(\)](#) Retrieve period count for duty calculation

Since

Version 1.0.0

Definition at line 1117 of file [rx_gptw.c](#).

```

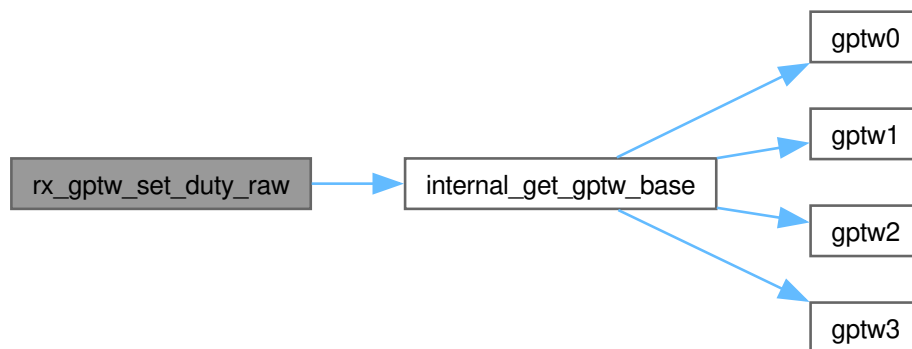
01118 {
01119     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01120         return k_rx_err_invalid_state;
01121     }
01122
01123     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01124     if (gptw == nullptr) {
01125         return k_rx_err_invalid_arg;
01126     }
01127
01128     /* Clamp to period */
01129     const uint32_t period = s_gptw_period[channel];
01130     if (duty_count > period) {
01131         duty_count = period;
01132     }
01133
01134     /* GTCCRA/GTCCRB are protected by GTWP.WP on RX72N. Unlock, write, re-lock.
01135      * With GTBER=0 (default after our fixed init), the written value takes
01136      * effect immediately rather than being double-buffered to the next period. */
01137     gptw->gtwp = k_gptw_gtwp_unlock;
01138     switch (output) {
01139     case k_gptw_output_a:
01140         gptw->gtccra = duty_count;
01141         break;
01142     case k_gptw_output_b:
01143         gptw->gtccrb = duty_count;
01144         break;
01145     default:
01146         gptw->gtwp = k_gptw_gtwp_lock;
01147         return k_rx_err_invalid_arg;
01148     }
01149     gptw->gtwp = k_gptw_gtwp_lock;
01150
01151     return k_rx_ok;
01152 }

```

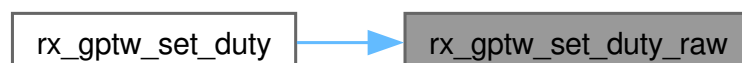
References [rx_gptw_channel_regs_t::gtccra](#), [rx_gptw_channel_regs_t::gtccrb](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), [k_gptw_max_channels](#), [k_gptw_output_a](#), [k_gptw_output_b](#), [s_gptw_initialized](#), and [s_gptw_period](#).

Referenced by [rx_gptw_set_duty\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.79.4.16 rx_gptw_start()

```
rx_err_t rx_gptw_start (
    const rx_gptw_channel_t channel) [nodiscard]
```

Start the GPTW timer counter for PWM generation.

Sets the GTCR.CST bit to begin counting. The timer is automatically started during [rx_gptw_init_pwm\(\)](#), but can be stopped via [rx_gptw_stop\(\)](#) and restarted with this function. Counting resumes from the current GTCNT value (not reset to zero).

Parameters

in	channel	GPTW channel number (valid: 0-3)
----	---------	----------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, timer started
k_rx_err_invalid_arg	Channel out of range

Precondition

Channel must be initialized via [rx_gptw_init_pwm\(\)](#)

Channel must currently be stopped (safe to call if already running)

Postcondition

Timer counter incrementing at peripheral clock rate

PWM outputs active (if enabled via [rx_gptw_enable_output\(\)](#))

Note

Not thread-safe. Caller must provide synchronization.

Example:

```
// Restart timer after a stop
rx_err_t err = rx_gptw_start(k_gptw_channel_0);
if (err != k_rx_ok) {
    // Handle start failure
}
```

See also

[rx_gptw_stop\(\)](#) Stop the timer counter

[rx_gptw_init_pwm\(\)](#) Initializes and starts timer automatically

Since

Version 1.0.0

Start the GPTW timer counter for PWM generation.

Implementation of [rx_gptw_start\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Sets the Count Start (CST) bit in GTCR to begin counter operation. The timer will count according to the configured waveform mode (sawtooth or triangle) and generate PWM output on the enabled pins.

8.79.4.17 Register Access

1. Unlock write protection (GTWP)
2. Set GTCR.CST bit
3. Lock write protection (GTWP)

Precondition

Channel configured via `rx_gptw_init_*`() (recommended)

Postcondition

Timer counting actively

PWM output generated on enabled pins

Note

Does NOT require channel to be "initialized" (no flag check)

Timer automatically started during `rx_gptw_init_*`()

Idempotent: Safe to call when already running

See also

[rx_gptw_stop\(\)](#) Stop the timer

[rx_gptw.h](#) Complete API documentation

Definition at line 1356 of file `rx_gptw.c`.

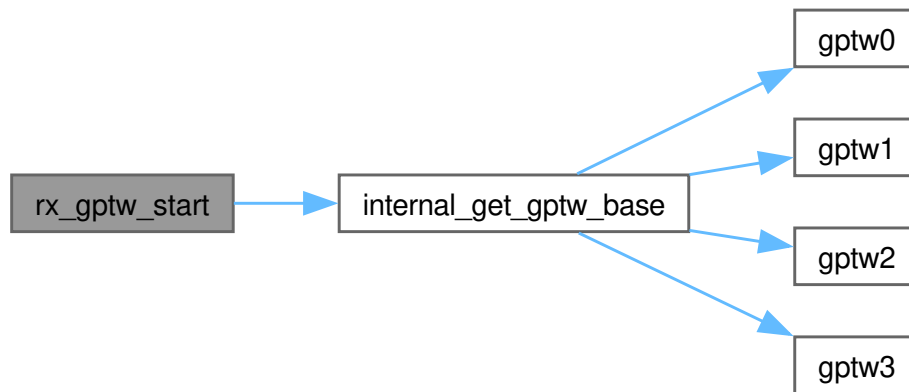
```

01357 {
01358     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01359         return k_rx_err_invalid_arg;
01360     }
01361
01362     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01363     if (gptw == nullptr) {
01364         return k_rx_err_invalid_arg;
01365     }
01366
01367     /* Unlock and set CST bit in GTCR */
01368     gptw->gtwp = k_gptw_gtwp_unlock;
01369     gptw->gtcr |= k_gptw_gtcr_cst;
01370     gptw->gtwp = k_gptw_gtwp_lock;
01371
01372     return k_rx_ok;
01373 }
```

References `rx_gptw_channel_regs_t::gtcr`, `rx_gptw_channel_regs_t::gtwp`, `internal_get_gptw_base()`, `k_gptw_gtcr_cst`, `k_gptw_gtwp_lock`, `k_gptw_gtwp_unlock`, and `k_gptw_max_channels`.

Referenced by `rx_gptw_init_pwm()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.79.4.18 rx_gptw_stop()

```
rx_err_t rx_gptw_stop (
    const rx_gptw_channel_t channel) [nodiscard]
```

Stop the GPTW timer counter.

Clears the GTCR.CST bit to halt counting. PWM outputs hold their last state when the timer stops. The GTCNT value is preserved and counting resumes from that value when `rx_gptw_start()` is called.

Parameters

in	channel	GPTW channel number (valid: 0-3)
----	---------	----------------------------------

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, timer stopped
<code>k_rx_err_invalid_arg</code>	Channel out of range

Precondition

Channel must be initialized via `rx_gptw_init_pwm()`

Channel should be running (safe to call if already stopped)

Postcondition

Timer counter halted at current GTCNT value
PWM outputs frozen at last driven state

Note

Not thread-safe. Caller must provide synchronization.

Warning

Outputs hold last state; explicitly set duty to 0 before stopping if low output is required.

Example:

```
// Stop timer and ensure outputs are low
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                rx_gptw_output_id(k_gptw_output_b), 0.0F);
rx_err_t err = rx_gptw_stop(k_gptw_channel_0);
```

See also

[rx_gptw_start\(\)](#) Resume timer counter
[rx_gptw_deinit\(\)](#) Full shutdown including output disable

Since

Version 1.0.0

Stop the GPTW timer counter.

Implementation of [rx_gptw_stop\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Clears the Count Start (CST) bit in GTCR to halt counter operation. The timer counter value (GTCNT) is preserved and PWM outputs hold their current state.

8.79.4.19 Register Access

1. Unlock write protection (GTWP)
2. Clear GTCR.CST bit
3. Lock write protection (GTWP)

Precondition

None (can be called on uninitialized channel)

Postcondition

Timer stopped (GTCR.CST = 0)
 Counter value preserved (GTCNT unchanged)
 Output pins hold last state

Note

Does NOT require channel to be "initialized" (no flag check)
 Idempotent: Safe to call when already stopped
 For safety shutdown, also disable outputs via [rx_gptw_enable_output\(\)](#)

Warning

Outputs may remain HIGH when stopped - disable outputs for safe state

See also

[rx_gptw_start\(\)](#) Resume the timer
[rx_gptw_enable_output\(\)](#) Disable outputs for safe state
[rx_gptw.h](#) Complete API documentation

Definition at line 1408 of file [rx_gptw.c](#).

```

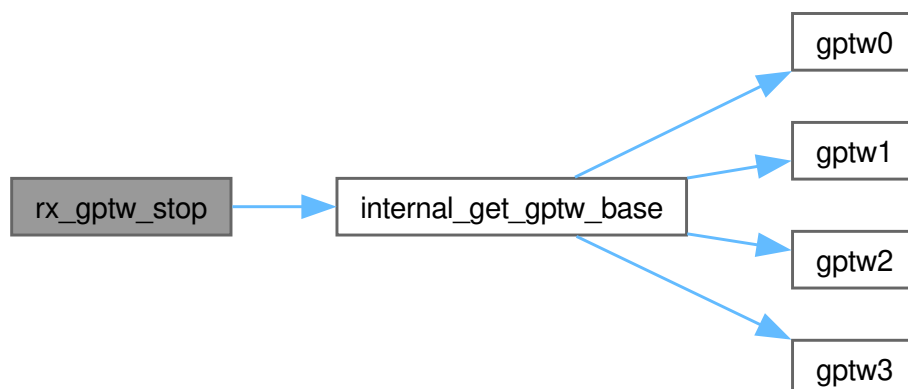
01409 {
01410     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01411         return k_rx_err_invalid_arg;
01412     }
01413
01414     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01415     if (gptw == nullptr) {
01416         return k_rx_err_invalid_arg;
01417     }
01418
01419     /* Unlock and clear CST bit in GTCR */
01420     gptw->gtwp = k_gptw_gtwp_unlock;
01421     gptw->gtcr &= ~k_gptw_gtcr_cst;
01422     gptw->gtwp = k_gptw_gtwp_lock;
01423
01424     return k_rx_ok;
01425 }

```

References [rx_gptw_channel_regs_t::gtcr](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtcr_cst](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), and [k_gptw_max_channels](#).

Referenced by [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.80 rx_gptw.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_gptw.h
00003  * @brief GPTW PWM Driver for RX72N Motor Control
00004  *
00005  * @details
00006  * **Production-grade PWM driver** for the RX72N General PWM Timer (GPTW) peripheral,
00007  * optimized for brushed DC motor control with H-bridge drivers. This module provides
00008  * high-resolution 32-bit PWM generation with complementary outputs, deadtime insertion,
00009  * and phase-staggered operation for 4-motor systems.
00010  *
00011  * ## Purpose and Design Rationale
00012  *
00013  * The GPTW peripheral is chosen over MTU for motor control because:
00014  * 1. **32-bit resolution** - Higher precision than MTU's 16-bit (0.000015% vs 0.0015%)
00015  * 2. **Dedicated motor pins** - Port E pins optimized for PWM output
00016  * 3. **Hardware deadtime** - Prevents H-bridge shoot-through without CPU overhead
00017  * 4. **Phase staggering** - Reduces peak current draw and EMI
00018  *
00019  * ## Key Features
00020  *
00021  * **PWM Generation**:
00022  * - 4 independent channels (GPTW0-GPTW3) for 4 motors
00023  * - 32-bit counter resolution for precise duty cycle control
00024  * - Sawtooth (edge-aligned) or triangle (center-aligned) waveforms
00025  * - Complementary A/B outputs per channel for H-bridge control
00026  * - Configurable deadtime insertion (0-65535 ns)
00027  *
00028  * **Phase Staggering**:
00029  * - 90-degree phase offset between channels
00030  * - Channel 0: 0deg, Channel 1: 90deg, Channel 2: 180deg, Channel 3: 270deg
00031  * - Spreads current draw across PWM period
00032  * - Reduces peak power supply current by ~75%
00033  * - Reduces conducted EMI by preventing simultaneous switching
00034  *
00035  * **Motor Control Integration**:
00036  * - Direct interface with DRV8263H H-bridge drivers
00037  * - IN2/IN1 control mode support (sign-magnitude)
00038  * - Glitch-free duty cycle updates on period boundary
00039  * - Fast raw duty cycle API for tight control loops
00040  *
00041  * ## Hardware Architecture
00042  *
00043  * @par Pin Assignments (from hardware_pinout.h):
00044  *
00045  * | Motor | Channel | Output A | Output B | Function |
00046  * |-----|-----|-----|-----|-----|
00047  * | Motor 0 | GPTW0 | PE5/GTIOC0A | PE2/GTIOC0B | Front-Left |
00048  * | Motor 1 | GPTW1 | PE4/GTIOC1A | PE1/GTIOC1B | Front-Right |
00049  * | Motor 2 | GPTW2 | PE3/GTIOC2A | PE0/GTIOC2B | Rear-Left |
00050  * | Motor 3 | GPTW3 | PE7/GTIOC3A | PE6/GTIOC3B | Rear-Right |
00051  *
00052  * @par Register Base Addresses:
00053  *
00054  * | Register Set | Base Address | Description |
00055  * |-----|-----|-----|
00056  * | GPTW Common | 0x000C2000 | Shared control registers |
00057  * | GPTW0 | 0x000C2100 | Channel 0 registers |
00058  * | GPTW1 | 0x000C2180 | Channel 1 registers |
00059  * | GPTW2 | 0x000C2200 | Channel 2 registers |
00060  * | GPTW3 | 0x000C2280 | Channel 3 registers |
00061  *
00062  * @par Clock Configuration:
00063  *
00064  * | Parameter | Value | Notes |
00065  * |-----|-----|-----|
00066  * | Clock Source | PCLKA | Peripheral Clock A |
00067  * | Clock Frequency | 120 MHz | After system clock configuration |
00068  * | Counter Resolution | 8.33 ns | 1/120MHz |
00069  * | PWM Frequency | 20 kHz | Typical for motor control |
00070  * | Period Counts | 6000 | 120MHz / 20kHz |
00071  * | Duty Resolution | 0.0167% | 1/6000 |
00072  *
00073  * ## PWM Waveform Modes
00074  *
00075  * @dot
00076  * digraph waveform_modes {
00077  *   rankdir=LR;
00078  *   node [shape=box, style=rounded];
00079  *
00080  *   subgraph cluster_saw {
00081  *     label="Sawtooth (Edge-Aligned)";
00082  *     saw_desc [label="Counter: 0->Period->0\nUpdate: At trough\nBest for: Simple PWM"];

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00083 * }
00084 *
00085 * subgraph cluster_tri {
00086 *     label="Triangle (Center-Aligned)";
00087 *     tri_desc [label="Counter: 0->Period->0->Period\nUpdate: At crest/trough\nBest for: Motor control"];
00088 * }
00089 * }
00090 * @enddot
00091 *
00092 * ## Performance Characteristics
00093 *
00094 * | Operation | Execution Time | Memory Usage | Notes |
00095 * |-----|-----|-----|-----|
00096 * | rx_gptw_init_all_staggered() | ~100 us | 0 heap | One-time setup |
00097 * | rx_gptw_set_duty() | ~5 us | 0 | Float calculation |
00098 * | rx_gptw_set_duty_raw() | ~1 us | 0 | Direct register write |
00099 * | rx_gptw_get_duty() | ~3 us | 0 | Register read + calc |
00100 * | rx_gptw_enable_output() | ~1 us | 0 | Single register write |
00101 *
00102 * **Memory Footprint**:
00103 * - Static variables: ~64 bytes (initialization flags, period cache)
00104 * - Code size: ~2 KB (all functions)
00105 * - Stack usage: < 64 bytes (deepest call)
00106 * - Heap usage: 0 bytes (zero dynamic allocation)
00107 *
00108 * ## Thread Safety and RTOS Integration
00109 *
00110 * **Thread-Safe Operations**:
00111 * - rx_gptw_set_duty() and rx_gptw_set_duty_raw() are atomic register writes
00112 * - Safe to call from motor control ISR or ThreadX tasks
00113 * - No mutex required for duty cycle updates
00114 *
00115 * **Non-Thread-Safe Operations**:
00116 * - rx_gptw_init_all_staggered() - Call once during system initialization
00117 * - rx_gptw_deinit() - Call only when no motor control active
00118 *
00119 * @par Hardware Requirements:
00120 *
00121 * | Component | Requirement | Notes |
00122 * |-----|-----|-----|
00123 * | MCU | Renesas RX72N | GPTW peripheral required |
00124 * | Clock | PCLKA @ 120 MHz | System clock configuration |
00125 * | Pins | Port E (PE0-PE7) | Configured via MPC |
00126 * | Power | 3.3V I/O | LVCMOS compatible |
00127 *
00128 * @par Module Dependencies:
00129 *
00130 * **This module depends on**:
00131 * - `rx_err.h` - Error code definitions
00132 * - `rx_check.h` - Parameter validation macros
00133 * - `rx72n_gptw_regs.h` - GPTW register definitions
00134 * - `rx72n_mpc_regs.h` - Pin mux configuration
00135 *
00136 * **This module is used by**:
00137 * - `rx_motor.c` - Motor control driver
00138 * - `rx_drv8263.c` - H-bridge driver interface
00139 * - `motor_control_task.c` - Motor control task
00140 *
00141 * @par NASA Power of 10 Compliance:
00142 *
00143 * | Rule | Status | Implementation |
00144 * |-----|-----|-----|
00145 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion |
00146 * | 2. Fixed loop bounds | [OK] | Loops bounded by k_gptw_channel_3 (4 iterations max) |
00147 * | 3. No dynamic allocation | [OK] | Zero malloc/free, all static |
00148 * | 4. Small functions | [OK] | All functions < 60 lines |
00149 * | 5. Assertions (>=2 per function) | [OK] | Minimum 2 validation checks per function |
00150 * | 6. Narrow scope | [OK] | File-scope statics, minimal globals |
00151 * | 7. Check return values | [OK] | All functions return rx_err_t |
00152 * | 8. Limited preprocessor | [OK] | C23 typed enums, minimal macros |
00153 * | 9. Pointer restrictions | [OK] | Single-level pointers only |
00154 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00155 *
00156 * @par SOLID Principles:
00157 *
00158 * **Single Responsibility (S)**:
00159 * - This module handles ONLY GPTW PWM generation
00160 * - Motor control logic is in rx_motor.c
00161 * - H-bridge configuration is in rx_drv8263.c
00162 *
00163 * **Open/Closed (O)**:
00164 * - Extensible via rx_gptw_config_t structure
00165 * - New waveform modes can be added without modifying existing code
00166 * - Frequency and deadtime configurable at runtime
00167 *
00168 * **Liskov Substitution (L)**:
00169 * - All functions return rx_err_t with consistent semantics

```



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00170 * - Channel and output parameters are validated uniformly
00171 *
00172 * **Interface Segregation (I)**:
00173 * - Separate functions for float vs raw duty cycle
00174 * - Init, set, get, enable are independent operations
00175 * - Wrapper types prevent parameter swapping
00176 *
00177 * **Dependency Inversion (D)**:
00178 * - Depends on abstract error codes, not specific implementations
00179 * - Motor control depends on this interface, not hardware registers directly
00180 *
00181 * @see rx_motor.h Motor control driver using GPTW
00182 * @see rx_drv8263.h H-bridge driver interface
00183 * @see rx72n_gptw_regs.h Register definitions
00184 * @see RX72N Hardware Manual Section 26 - General PWM Timer
00185 * @see docs/sections/03_hardware_pinout.tex Pin assignments
00186 *
00187 * @author Locked, Inc.
00188 * @date 2026-01-27
00189 * @copyright Copyright (c) 2026 Locked Inc.
00190 * SPDX-License-Identifier: MIT
00191 * @since Version 1.0.0
00192 */
00193
00194 #pragma once
00195
00196 #include <stdbool.h>
00197 #include <stdint.h>
00198
00199 #include "rx_check.h"
00200 #include "rx_err.h"
00201
00202 #ifdef __cplusplus
00203 extern "C" {
00204 #endif
00205
00206 /*
=====
00207 * Type Definitions
00208 *
=====
00209 */
00210
00211 /**
00212 * @enum rx_gptw_channel_t
00213 * @brief GPTW Channel Identifiers for 4-Motor Control System
00214 *
00215 * @details
00216 * Identifies the four GPTW timer channels used for motor PWM generation.
00217 * Each channel controls one motor through complementary A/B outputs connected
00218 * to a DRV8263H H-bridge driver.
00219 *
00220 * ## Channel Architecture
00221 *
00222 * Each GPTW channel provides:
00223 * - 32-bit up/down counter
00224 * - Two compare/capture registers (GTCCRA, GTCCRB)
00225 * - Two output pins (GTIOCnA, GTIOCnB)
00226 * - Independent deadtime insertion
00227 * - Synchronized start/stop capability
00228 *
00229 * @par Channel-to-Motor Mapping:
00230 *
00231 * | Channel | Motor | Position | Register Base | Phase Offset |
00232 * |-----|-----|-----|-----|-----|
00233 * | GPTW0 | Motor 0 | Front-Left | 0x000C2100 | 0deg |
00234 * | GPTW1 | Motor 1 | Front-Right | 0x000C2180 | 90deg |
00235 * | GPTW2 | Motor 2 | Rear-Left | 0x000C2200 | 180deg |
00236 * | GPTW3 | Motor 3 | Rear-Right | 0x000C2280 | 270deg |
00237 *
00238 * @par Pin Assignments:
00239 *
00240 * | Channel | Output A | Output B | MPC Setting |
00241 * |-----|-----|-----|-----|
00242 * | k_gptw_channel_0 | PE5/GTIOC0A | PE2/GTIOC0B | PSEL = 0x1E |
00243 * | k_gptw_channel_1 | PE4/GTIOC1A | PE1/GTIOC1B | PSEL = 0x1E |
00244 * | k_gptw_channel_2 | PE3/GTIOC2A | PE0/GTIOC2B | PSEL = 0x1E |
00245 * | k_gptw_channel_3 | PE7/GTIOC3A | PE6/GTIOC3B | PSEL = 0x1E |
00246 *
00247 * @par Usage Example:
00248 * @code{.c}
00249 * // Initialize all channels with staggered phases
00250 * rx_gptw_config_t config = {
00251 *     .frequency_hz = 20000,
00252 *     .deadtime_ns = 1000,
00253 *     .wave_mode = k_gptw_wave_saw_pwm,
00254 *     .enable_complementary = true,

```

```

00255 *     .invert_polarity = false
00256 * };
00257 * rx_gptw_init_all_staggered(&config);
00258 *
00259 * // Set duty cycle for motor 0 (front-left)
00260 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
00261 *                 rx_gptw_output_id(k_gptw_output_a),
00262 *                 50.0F); // 50% duty cycle
00263 * @endcode
00264 *
00265 * @invariant Channel values are contiguous: 0, 1, 2, 3
00266 * @invariant All channels use same register structure (0x80 byte spacing)
00267 * @invariant Channel count equals motor count (4)
00268 *
00269 * @note C23 typed enum with uint8_t underlying type (1-byte values)
00270 * @note Channel 4 and 5 exist in hardware but are not used in STAR
00271 *
00272 * @see rx_gptw_channel_id() Type-safe wrapper constructor
00273 * @see rx_gptw_output_t Output selection within channel
00274 * @see rx72n_gptw_regs.h Register structure definitions
00275 *
00276 * @since Version 1.0.0
00277 */
00278 typedef enum : uint8_t {
00279     k_gptw_channel_0 =
00280         0, /**< GPTW0 channel for Motor 0 (Front-Left). Register base: 0x000C2100. Outputs: PE5/GTIOC0A,
00281            PE2/GTIOC0B. Phase offset: 0deg in staggered mode */
00282     k_gptw_channel_1 =
00283         1, /**< GPTW1 channel for Motor 1 (Front-Right). Register base: 0x000C2180. Outputs: PE4/GTIOC1A,
00284            PE1/GTIOC1B. Phase offset: 90deg in staggered mode */
00285     k_gptw_channel_2 =
00286         2, /**< GPTW2 channel for Motor 2 (Rear-Left). Register base: 0x000C2200. Outputs: PE3/GTIOC2A,
00287            PE0/GTIOC2B. Phase offset: 180deg in staggered mode */
00288     k_gptw_channel_3 =
00289         3, /**< GPTW3 channel for Motor 3 (Rear-Right). Register base: 0x000C2280. Outputs: PE7/GTIOC3A,
00290            PE6/GTIOC3B. Phase offset: 270deg in staggered mode */
00291 } rx_gptw_channel_t;
00292
00293 /**
00294 * @enum rx_gptw_output_t
00295 * @brief GPTW Output Channel Selection for Complementary PWM
00296 *
00297 * @details
00298 * Selects between the two outputs (A and B) available on each GPTW channel.
00299 * Each output has an independent duty cycle and is controlled separately.
00300 *
00301 * ## Independent Output Mode (STAR Configuration)
00302 *
00303 * For the DRV8263H motor driver in IN2/IN1 mode, the GPTW outputs operate
00304 * independently** (not in complementary mode). Each output has its own
00305 * compare register and duty cycle:
00306 * - Output A: Static direction signal (0% or 100% duty)
00307 * - Output B: Variable-duty PWM for speed control
00308 *
00309 * The outputs are NOT inverted versions of each other and no hardware
00310 * deadtime insertion is used. This is safe because the DRV8263H has
00311 * integrated shoot-through protection on its internal H-bridge FETs.
00312 *
00313 * ## IN2/IN1 Mode (Sign-Magnitude)
00314 *
00315 * For DRV8263H H-bridge control in IN2/IN1 mode:
00316 * - Output A: Connected to IN2 (direction) input - controls direction
00317 * - Output B: Connected to IN1 (PWM) input - controls PWM duty
00318 *
00319 * @par H-Bridge Connection:
00320 *
00321 * | Output | DRV8263H Pin | Function | Signal Type |
00322 * |-----|-----|-----|-----|
00323 * | Output A (GTIOCnA) | IN2 | Direction control | Static HIGH/LOW |
00324 * | Output B (GTIOCnB) | IN1 | PWM speed control | PWM waveform |
00325 *
00326 * @par Usage Example:
00327 * @code{.c}
00328 * // Set forward direction at 75% duty for motor 0
00329 * // Output A = HIGH (forward direction)
00330 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
00331 *                 rx_gptw_output_id(k_gptw_output_a),
00332 *                 100.0F); // IN2 = HIGH (direction)
00333 *
00334 * // Output B = PWM at 75% (motor speed)
00335 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
00336 *                 rx_gptw_output_id(k_gptw_output_b),
00337 *                 75.0F); // IN1 = 75% PWM
00338 *
00339 * // Reverse direction: set Output A LOW
00340 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
00341 *                 rx_gptw_output_id(k_gptw_output_a),

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00338 *          0.0F); // IN2 = LOW (reverse)
00339 * @endcode
00340 *
00341 * @invariant Output values are 0 (A) or 1 (B) only
00342 * @invariant Both outputs available on all channels
00343 *
00344 * @note C23 typed enum with uint8_t underlying type
00345 * @warning In independent mode, each output is controlled separately; the
00346 *          DRV8263H handles shoot-through protection internally
00347 *
00348 * @see rx_gptw_output_id() Type-safe wrapper constructor
00349 * @see rx_gptw_channel_t Channel selection
00350 * @see rx_drv8263.h H-bridge driver interface
00351 *
00352 * @since Version 1.0.0
00353 */
00354 typedef enum : uint8_t {
00355     k_gptw_output_a =
00356         0, /**< GTIOCA output pin. Connected to DRV8263H IN2 (direction) in IN2/IN1 mode. Controls motor direction:
            HIGH=forward, LOW=reverse */
00357     k_gptw_output_b =
00358         1, /**< GTIOCB output pin. Connected to DRV8263H IN1 (PWM) in IN2/IN1 mode. PWM duty cycle controls motor
            speed (0-100%) */
00359 } rx_gptw_output_t;
00360
00361 /**
00362 * @enum rx_gptw_wave_mode_t
00363 * @brief GPTW PWM Waveform Mode Selection
00364 *
00365 * @details
00366 * Selects the PWM waveform generation mode. The mode determines counter behavior,
00367 * output alignment, and duty cycle update timing. Motor control typically uses
00368 * sawtooth (edge-aligned) or triangle (center-aligned) modes.
00369 *
00370 * ## Waveform Comparison
00371 *
00372 * **Sawtooth (Edge-Aligned)**:
00373 * - Counter counts up from 0 to period, then resets to 0
00374 * - PWM edges aligned to period start
00375 * - Simpler timing, lower switching losses
00376 * - Asymmetric current ripple
00377 *
00378 * **Triangle (Center-Aligned)**:
00379 * - Counter counts up to period, then down to 0
00380 * - PWM pulses centered in period
00381 * - Symmetric current ripple (better for motors)
00382 * - Reduced EMI due to staggered edges
00383 *
00384 * @dot
00385 * digraph waveform_comparison {
00386 *     rankdir=TB;
00387 *     node [shape=record, fontsize=10];
00388 *
00389 *     sawtooth [label="{Sawtooth Mode|Counter: 0..P repeats|Update: trough only|PWM edges: aligned to start|Best for:
            LED dimming}"];
00390 *     triangle [label="{Triangle Mode|Counter: 0..P.0 repeats|Update: crest and/or trough|PWM edges: centered|Best for:
            motor control}"];
00391 *
00392 *     sawtooth -> triangle [style=invis];
00393 * }
00394 * @enddot
00395 *
00396 * @par Mode Selection Guide:
00397 *
00398 * | Mode | Counter Pattern | Update Timing | Use Case |
00399 * |-----|-----|-----|-----|
00400 * | k_gptw_wave_saw_pwm | 0->P->0 (reset) | Trough | Simple PWM, LEDs |
00401 * | k_gptw_wave_tri_pwm1 | 0->P->0 (fold) | Trough | Motor control (standard) |
00402 * | k_gptw_wave_tri_pwm2 | 0->P->0 (fold) | Crest+Trough | Motor control (fast update) |
00403 * | k_gptw_wave_tri_pwm3 | 0->P->0 (fold) | 64-bit at trough | High-precision motor |
00404 *
00405 * @par Mathematical Model:
00406 *
00407 * **Sawtooth PWM Frequency**:
00408 * @f[
00409 *     f_{PWM} = \frac{f_{PCLKA}}{(GTPR + 1)}
00410 * @f]
00411 *
00412 * **Triangle PWM Frequency** (half the sawtooth frequency for same period):
00413 * @f[
00414 *     f_{PWM} = \frac{f_{PCLKA}}{2 \times (GTPR + 1)}
00415 * @f]
00416 *
00417 * Where:
00418 * - @f f_{PCLKA} @f = 120 MHz (peripheral clock)
00419 * - @f GTPR @f = Period register value
00420 *

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```

00421 * @par Usage Example:
00422 * @code{.c}
00423 * // Configure for motor control with triangle wave (center-aligned)
00424 * rx_gptw_config_t config = {
00425 *     .frequency_hz = 20000,    // 20 kHz PWM
00426 *     .deadtime_ns = 1000,     // 1 us deadtime
00427 *     .wave_mode = k_gptw_wave_tri_pwm1, // Triangle, trough update
00428 *     .enable_complementary = true,
00429 *     .invert_polarity = false
00430 * };
00431 *
00432 * // For simple LED dimming, use sawtooth
00433 * rx_gptw_config_t led_config = {
00434 *     .frequency_hz = 1000,     // 1 kHz (no audible noise)
00435 *     .wave_mode = k_gptw_wave_saw_pwm, // Sawtooth (simpler)
00436 *     .enable_complementary = false,
00437 *     .invert_polarity = false
00438 * };
00439 * @endcode
00440 *
00441 * @invariant Mode values map directly to GTCR.MD bit field (0-3)
00442 * @invariant All modes support complementary output with deadtime
00443 *
00444 * @note C23 typed enum with uint8_t underlying type
00445 * @note Triangle modes have half the effective PWM frequency for same period
00446 *
00447 * @see RX72N Hardware Manual Section 26.2.12 (GTCR.MD bits)
00448 * @see rx_gptw_config_t Configuration structure using this enum
00449 *
00450 * @since Version 1.0.0
00451 */
00452 typedef enum : uint8_t {
00453     k_gptw_wave_saw_pwm =
00454         0, /**< Sawtooth-wave PWM (edge-aligned). Counter: 0->Period->reset. Single update point at trough. Best for simple
PWM applications. GTCR.MD = 0b000 */
00455     k_gptw_wave_tri_pwm1 =
00456         1, /**< Triangle-wave PWM mode 1 (center-aligned). Counter: 0->Period->0. Update at trough only. Best for motor
control with standard update rate. GTCR.MD = 0b100 */
00457     k_gptw_wave_tri_pwm2 =
00458         2, /**< Triangle-wave PWM mode 2 (center-aligned). Counter: 0->Period->0. Update at both crest and trough. Best
for motor control requiring fast duty updates. GTCR.MD = 0b101 */
00459     k_gptw_wave_tri_pwm3 =
00460         3, /**< Triangle-wave PWM mode 3 (center-aligned). Counter: 0->Period->0. 64-bit buffer transfer at trough. Best for
high-precision motor control. GTCR.MD = 0b110 */
00461 } rx_gptw_wave_mode_t;
00462
00463 /**
00464 * @enum rx_gptw_duty_calc_constants_t
00465 * @brief Named constants for duty cycle percentage calculations
00466 *
00467 * @details
00468 * Provides named numerators and a common denominator for computing raw duty
00469 * counts as integer fractions of the period. Using named constants instead of
00470 * magic literals satisfies NASA Power of 10 Rule 8 (no magic numbers).
00471 *
00472 * @par Example:
00473 * @code{.c}
00474 * // 50% duty: period * 50 / 100
00475 * uint32_t half = (period * k_gptw_duty_50_pct_num) / k_gptw_duty_pct_denom;
00476 * @endcode
00477 *
00478 * @invariant All numerator constants must be <= k_gptw_duty_pct_denom (non-negative)
00479 * @invariant k_gptw_duty_pct_denom must be non-zero (denominator for percentage calculation)
00480 *
00481 * @see rx_gptw_set_duty_raw() Uses these constants in documentation examples
00482 * @see rx_gptw_get_period() Provides the period value for the calculation
00483 *
00484 * @since Version 1.0.0
00485 */
00486 typedef enum : uint8_t {
00487     k_gptw_duty_50_pct_num = 50, /**< Numerator for 50% duty cycle calculation */
00488     k_gptw_duty_75_pct_num = 75, /**< Numerator for 75% duty cycle calculation */
00489     k_gptw_duty_pct_denom = 100, /**< Denominator for percentage duty calculations */
00490 } rx_gptw_duty_calc_constants_t;
00491
00492 /**
00493 * @struct rx_gptw_channel_id_t
00494 * @brief Type-safe GPTW channel wrapper (prevents accidental argument swaps)
00495 *
00496 * @details
00497 * Wraps rx_gptw_channel_t in a struct to leverage the C type system for
00498 * catching argument transposition errors at compile time. Without this wrapper,
00499 * swapping channel and output arguments (both uint8_t) would compile silently.
00500 * Construct via rx_gptw_channel_id() helper function.
00501 *
00502 * @invariant value must be a valid rx_gptw_channel_t value (k_gptw_channel_0 through k_gptw_channel_3)
00503 */

```

```

00504 * @code
00505 * rx_gptw_channel_id_t ch = rx_gptw_channel_id(k_gptw_channel_0);
00506 * @endcode
00507 *
00508 * @see rx_gptw_channel_id() Constructor function
00509 * @see rx_gptw_output_id_t Companion wrapper for output parameter
00510 * @since Version 1.0.0
00511 */
00512 typedef struct {
00513     rx_gptw_channel_t value; /**< GPTW channel number (0-3) */
00514 } rx_gptw_channel_id_t;
00515
00516 /**
00517 * @struct rx_gptw_output_id_t
00518 * @brief Type-safe GPTW output wrapper (prevents accidental argument swaps)
00519 *
00520 * @details
00521 * Wraps rx_gptw_output_t in a struct to leverage the C type system for
00522 * catching argument transposition errors at compile time. Without this wrapper,
00523 * swapping channel and output arguments (both uint8_t) would compile silently.
00524 * Construct via rx_gptw_output_id() helper function.
00525 *
00526 * @invariant value must be k_gptw_output_a or k_gptw_output_b
00527 *
00528 * @code
00529 * rx_gptw_output_id_t out = rx_gptw_output_id(k_gptw_output_a);
00530 * @endcode
00531 *
00532 * @see rx_gptw_output_id() Constructor function
00533 * @see rx_gptw_channel_id_t Companion wrapper for channel parameter
00534 * @since Version 1.0.0
00535 */
00536 typedef struct {
00537     rx_gptw_output_t value; /**< GPTW output identifier (A or B) */
00538 } rx_gptw_output_id_t;
00539
00540 /**
00541 * @brief Helper to construct a GPTW channel wrapper
00542 *
00543 * @details Provides type-safe wrapper construction with NASA Power of 10 Rule 5
00544 * validation (minimum 2 checks per function):
00545 * - Pre-condition: Input value is within valid channel range [0-3]
00546 * - Post-condition: Wrapper correctly stores the input value
00547 *
00548 * @param[in] value GPTW channel enum value (k_gptw_channel_0 to k_gptw_channel_3)
00549 *
00550 * @return Wrapped channel ID for type-safe function calls
00551 *
00552 * @pre value must be a valid rx_gptw_channel_t in range [k_gptw_channel_0, k_gptw_channel_3]
00553 * @pre Caller must pass a compile-time known or previously validated channel value
00554 * @post Returned wrapper .value field equals the input value
00555 * @post Wrapper is immediately usable in any API accepting rx_gptw_channel_id_t
00556 *
00557 * @note Reentrant and lock-free; contains no shared mutable state
00558 *
00559 * @see rx_gptw_output_id() Companion wrapper constructor for output parameter
00560 * @see rx_gptw_channel_id_t Wrapper type constructed by this function
00561 *
00562 * @since Version 1.0.0
00563 */
00564 static inline rx_gptw_channel_id_t rx_gptw_channel_id(rx_gptw_channel_t value)
00565 {
00566     /* Pre-condition: channel must be in valid range */
00567     /* Note: value >= k_gptw_channel_0 is always true for uint8_t since k_gptw_channel_0 == 0 */
00568     RX_ASSERT(value <= k_gptw_channel_3, "GPTW channel out of range");
00569
00570     const rx_gptw_channel_id_t id = {.value = value};
00571
00572     return id;
00573 }
00574
00575 /**
00576 * @brief Helper to construct a GPTW output wrapper
00577 *
00578 * @details Provides type-safe wrapper construction with NASA Power of 10 Rule 5
00579 * validation (minimum 2 checks per function):
00580 * - Pre-condition: Input value is within valid output range [A-B]
00581 * - Post-condition: Wrapper correctly stores the input value
00582 *
00583 * @param[in] value GPTW output enum value (k_gptw_output_a or k_gptw_output_b)
00584 *
00585 * @return Wrapped output ID for type-safe function calls
00586 *
00587 * @pre value must be a valid rx_gptw_output_t (k_gptw_output_a or k_gptw_output_b)
00588 * @pre Caller must pass a compile-time known or previously validated output value
00589 * @post Returned wrapper .value field equals the input value
00590 * @post Wrapper is immediately usable in any API accepting rx_gptw_output_id_t

```

```

00591 *
00592 * @note Reentrant and lock-free; contains no shared mutable state
00593 *
00594 * @see rx_gptw_channel_id() Companion wrapper constructor for channel parameter
00595 * @see rx_gptw_output_id_t Wrapper type constructed by this function
00596 *
00597 * @since Version 1.0.0
00598 */
00599 static inline rx_gptw_output_id_t rx_gptw_output_id(rx_gptw_output_t value)
00600 {
00601     /* Pre-condition: output must be in valid range */
00602     /* Note: value >= k_gptw_output_a is always true for uint8_t since k_gptw_output_a == 0 */
00603     RX_ASSERT(value <= k_gptw_output_b, "GPTW output out of range");
00604
00605     const rx_gptw_output_id_t id = {.value = value};
00606
00607     return id;
00608 }
00609
00610 /**
00611 * @struct rx_gptw_config_t
00612 * @brief GPTW PWM Configuration Structure
00613 *
00614 * @details
00615 * Contains all configuration parameters for initializing a GPTW channel
00616 * for PWM output. This structure is passed to rx_gptw_init_pwm() or
00617 * rx_gptw_init_all_staggered() to configure the timer.
00618 *
00619 * ## Configuration Guidelines
00620 *
00621 * **Frequency Selection**:
00622 * - Motor control: 15-25 kHz (above audible range, efficient switching)
00623 * - LED dimming: 100-1000 Hz (no flicker, simple filtering)
00624 * - Servo control: 50-400 Hz (standard RC servo range)
00625 *
00626 * **Deadtime Selection**:
00627 * - Minimum: 100 ns (propagation delay margin)
00628 * - Typical: 500-1000 ns (good balance)
00629 * - Maximum: 5000 ns (very conservative, reduces efficiency)
00630 *
00631 * @par Configuration Parameter Table:
00632 *
00633 * | Parameter | Min | Max | Default | Unit | Notes |
00634 * |-----|-----|-----|-----|-----|-----|
00635 * | frequency_hz | 100 | 1000000 | 20000 | Hz | Limited by PCLKA/2 |
00636 * | deadtime_ns | 0 | 65535 | 1000 | ns | 0 disables deadtime |
00637 * | wave_mode | 0 | 3 | 0 | enum | See rx_gptw_wave_mode_t |
00638 * | enable_complementary | false | true | true | bool | For H-bridge |
00639 * | invert_polarity | false | true | false | bool | Active-low output |
00640 *
00641 * @par Memory Layout:
00642 *
00643 * | Offset | Size | Field | Type | Alignment |
00644 * |-----|-----|-----|-----|-----|
00645 * | 0 | 4 | frequency_hz | uint32_t | 4 |
00646 * | 4 | 2 | deadtime_ns | uint16_t | 2 |
00647 * | 6 | 1 | wave_mode | uint8_t | 1 |
00648 * | 7 | 1 | enable_complementary | bool | 1 |
00649 * | 8 | 1 | invert_polarity | bool | 1 |
00650 * | 9-11 | 3 | (padding) | - | - |
00651 * | **Total** | **12** | - | - | **4** |
00652 *
00653 * @par Usage Example (Motor Control):
00654 * @code{.c}
00655 * // Standard motor control configuration
00656 * rx_gptw_config_t motor_config = {
00657 *     .frequency_hz = 20000, // 20 kHz (inaudible)
00658 *     .deadtime_ns = 1000, // 1 us deadtime
00659 *     .wave_mode = k_gptw_wave_saw_pwm, // Edge-aligned
00660 *     .enable_complementary = true, // H-bridge requires this
00661 *     .invert_polarity = false // Active-high outputs
00662 * };
00663 *
00664 * rx_err_t err = rx_gptw_init_all_staggered(&motor_config);
00665 * if (err != k_rx_ok) {
00666 *     rx_log_error("GPTW", "Failed to initialize: %d", err);
00667 * }
00668 * @endcode
00669 *
00670 * @par Usage Example (LED Dimming):
00671 * @code{.c}
00672 * // LED PWM configuration (no complementary needed)
00673 * rx_gptw_config_t led_config = {
00674 *     .frequency_hz = 1000, // 1 kHz (no flicker)
00675 *     .deadtime_ns = 0, // No deadtime for LEDs
00676 *     .wave_mode = k_gptw_wave_saw_pwm,
00677 *     .enable_complementary = false, // Single output only

```

```

00678 *   .invert_polarity = false
00679 * };
00680 *
00681 * rx_err_t err = rx_gptw_init_pwm(k_gptw_channel_0, &led_config);
00682 * @endcode
00683 *
00684 * @invariant frequency_hz > 0 (enforced by rx_gptw_init_*)
00685 * @invariant frequency_hz <= PCLKA / 2 (Nyquist limit)
00686 * @invariant wave_mode in [0, 3] (valid enum range)
00687 *
00688 * @note Structure is copied during initialization; no need to keep alive
00689 * @warning Invalid configurations rejected with k_rx_err_invalid_arg
00690 *
00691 * @see rx_gptw_init_pwm() Single channel initialization
00692 * @see rx_gptw_init_all_staggered() 4-channel staggered initialization
00693 * @see rx_gptw_wave_mode_t Waveform mode options
00694 *
00695 * @since Version 1.0.0
00696 */
00697 typedef struct {
00698 /**
00699  * @brief PWM frequency in Hz
00700  * @details
00701  * Sets the PWM switching frequency. For motor control, 15-25 kHz is typical
00702  * (above audible range). Higher frequencies reduce current ripple but increase
00703  * switching losses.
00704  *
00705  * @par Frequency Calculation:
00706  * @f[
00707  * \text{Period Count} = \frac{f_{\text{PCLKA}}}{f_{\text{PWM}}} - 1
00708  * @f]
00709  *
00710  * Where @f$ f_{\text{PCLKA}} @f$ = 120 MHz
00711  *
00712  * @par Valid Range: [100, 1000000] Hz
00713  * @par Recommended: 20000 Hz for motor control
00714  * @note Actual frequency may differ slightly due to integer division
00715  */
00716 uint32_t frequency_hz;
00717
00718 /**
00719  * @brief Deadtime in nanoseconds
00720  * @details
00721  * Deadtime inserted between complementary output transitions to prevent
00722  * shoot-through (simultaneous conduction of high-side and low-side switches).
00723  * Set to 0 to disable deadtime insertion.
00724  *
00725  * @par Deadtime Calculation:
00726  * @f[
00727  * \text{Deadtime Counts} = \frac{\text{deadtime\_ns}}{10^9} \times f_{\text{PCLKA}}
00728  * @f]
00729  *
00730  * At 120 MHz: 1000 ns = 120 counts
00731  *
00732  * @par Valid Range: [0, 65535] ns
00733  * @par Recommended: 1000 ns (1 us) for DRV8263H
00734  * @warning Too short may cause shoot-through; too long reduces efficiency
00735  */
00736 uint16_t deadtime_ns;
00737
00738 /**
00739  * @brief Waveform mode (sawtooth or triangle)
00740  * @details
00741  * Selects counter behavior and PWM alignment.
00742  * - Sawtooth: Edge-aligned, simpler, single update point
00743  * - Triangle: Center-aligned, symmetric ripple, better for motors
00744  *
00745  * @see rx_gptw_wave_mode_t for detailed mode descriptions
00746  */
00747 rx_gptw_wave_mode_t wave_mode;
00748
00749 /**
00750  * @brief Enable complementary outputs
00751  * @details
00752  * When true, Output B is the inverted version of Output A with deadtime
00753  * inserted. Required for H-bridge motor control.
00754  *
00755  * - true: Both A and B outputs active (complementary)
00756  * - false: Only Output A active, Output B held low
00757  *
00758  * @warning When complementary mode is enabled, GTCCRB is tied to GTCCRA
00759  * by the hardware. Writing GTCCRB directly has no effect; the
00760  * output B duty is always the complement of output A. For
00761  * independent A/B control (e.g., DRV8263H IN2/IN1 mode), set
00762  * this to false.
00763  */
00764 bool enable_complementary;

```



```

00765
00766 /**
00767  * @brief Invert PWM polarity
00768  * @details
00769  * When true, inverts the PWM output polarity.
00770  * - false: HIGH during duty cycle, LOW otherwise (active-high)
00771  * - true: LOW during duty cycle, HIGH otherwise (active-low)
00772  *
00773  * @note Use for active-low driver inputs or inverted LED connections
00774  */
00775 bool invert_polarity;
00776
00777 /**
00778  * @brief Pin coordinates for GTIOC*A and GTIOC*B outputs.
00779  *
00780  * @details
00781  * The lib does not know the board's pin map. Caller MUST set these to
00782  * the actual pads where GTIOC_A and GTIOC_B should appear. Source the
00783  * values from a board-level pin map header (e.g. src/inc/hardware_config.h).
00784  *
00785  * `port_*_idx` is the standard RX72N port index: 1, 2, 8, etc., with
00786  * PORTA=10, PORTB=11, PORTC=12, PORTD=13, PORTE=14, PORTF=15.
00787  * `bit_*` is the bit position within that port (0..7).
00788  */
00789 uint8_t port_a_idx;
00790 uint8_t bit_a;
00791 uint8_t port_b_idx;
00792 uint8_t bit_b;
00793 } rx_gptw_config_t;
00794
00795 /**
=====
00796  * Public API
00797  *
=====
00798  */
00799
00800 /**
00801  * @brief Initialize all 4 GPTW channels with 90 degree phase staggering
00802  *
00803  * @details
00804  * Configures GPTW0-3 for synchronized PWM operation with phase offsets that
00805  * spread current draw across the PWM period. This is the **recommended
00806  * initialization function** for 4-motor systems as it reduces peak current
00807  * and EMI compared to synchronized PWM.
00808  *
00809  * ## Phase Staggering Architecture
00810  *
00811  * Phase staggering offsets the start of each channel's PWM period:
00812  * - **Channel 0**: 0deg (reference phase)
00813  * - **Channel 1**: 90deg (period / 4 offset)
00814  * - **Channel 2**: 180deg (period / 2 offset)
00815  * - **Channel 3**: 270deg (3 x period / 4 offset)
00816  *
00817  * @dot
00818  * digraph phase_stagger {
00819  *   rankdir=TB;
00820  *   node [shape=record, fontsize=10];
00821  *
00822  *   title [label="Phase Staggered PWM (4 channels at 50% duty)", shape=none];
00823  *
00824  *   ch0 [label="{Ch0 (0deg)|#####}"];
00825  *   ch1 [label="{Ch1 (90deg)|#####}"];
00826  *   ch2 [label="{Ch2 (180deg)|#####}"];
00827  *   ch3 [label="{Ch3 (270deg)|#####}"];
00828  *   current [label="{Total Current|Spread evenly across period}"];
00829  *
00830  *   title -> ch0 [style=invis];
00831  *   ch0 -> ch1 -> ch2 -> ch3 -> current;
00832  * }
00833  * @enddot
00834  *
00835  * ## Benefits of Phase Staggering
00836  *
00837  * 1. **Reduced Peak Current**: Instead of all motors switching simultaneously,
00838  *   current draw is spread across the period, reducing peak by up to 75%
00839  * 2. **Lower EMI**: Staggered edges reduce conducted emissions
00840  * 3. **Smaller Capacitors**: Lower peak current allows smaller power supply capacitors
00841  * 4. **Better Power Quality**: More constant current draw improves voltage regulation
00842  *
00843  * ## Algorithm Steps
00844  *
00845  * 1. Validate configuration parameters (NULL check, frequency range, wave mode)
00846  * 2. Calculate period count from frequency and PCLKA
00847  * 3. Enable GPTW module clock (MSTPCRA.MSTPA7 = 0)
00848  * 4. Configure MPC for Port E pins (GTIOC function select)
00849  * 5. For each channel (0-3):

```



```

00850 *   a. Configure timer mode register (GTCR)
00851 *   b. Set period register (GTPR)
00852 *   c. Calculate and set phase offset (GTCNT initial value)
00853 *   d. Configure output mode and polarity (GTIOR)
00854 *   e. Set deadtime registers (GTDVU, GTDVD)
00855 *   f. Set initial duty cycle to 0% (GTCCRA, GTCCRB)
00856 * 6. Start all channels simultaneously (GTSTR)
00857
00858 * @param[in] config Common configuration for all channels
00859 * - config->frequency_hz: PWM frequency [100, 1000000] Hz
00860 * - config->deadtime_ns: Deadtime [0, 65535] ns
00861 * - config->wave_mode: Waveform mode (sawtooth or triangle)
00862 * - config->enable_complementary: Enable A/B complementary outputs
00863 * - config->invert_polarity: Invert output polarity
00864
00865 * @return rx_err_t Error code indicating success or failure reason
00866 * @retval k_rx_ok All 4 channels initialized successfully
00867 * @retval k_rx_err_null_ptr config pointer is nullptr
00868 * @retval k_rx_err_invalid_arg frequency_hz is 0 or wave_mode out of range
00869 * @retval k_rx_err_hw_error GPTW hardware access failed (NULL register pointer)
00870
00871 * @pre System clock configured (PCLKA = 120 MHz)
00872 * @pre Port E pins not used by other peripherals
00873
00874 * @post All 4 GPTW channels running with 90deg phase offsets
00875 * @post All duty cycles initialized to 0% (motors stopped)
00876 * @post MPC configured for GTIOC function on Port E pins
00877
00878 * @invariant Phase relationship maintained: Ch1 = Ch0 + 90deg, Ch2 = Ch0 + 180deg, Ch3 = Ch0 + 270deg
00879
00880 * @note Thread-safe: Call only once during system initialization
00881 * @note Propagates errors from internal_calculate_period(), internal_configure_gptw_hardware()
00882
00883 * @warning Calling this function while motors are running will cause glitches
00884 * @attention This function should be used INSTEAD of rx_gptw_init_pwm() for motor control
00885
00886 * @par Performance:
00887 * - Execution time: ~100 us at 240 MHz
00888 * - Memory usage: 0 heap, ~64 bytes stack
00889
00890 * @par Example (Complete Motor System Initialization):
00891 * @code{.c}
00892 * #include "rx_gptw.h"
00893 * #include "rx_drv8263.h"
00894 * #include "rx_motor.h"
00895
00896 * rx_err_t motor_system_init(void)
00897 * {
00898 *     // Configure GPTW for 20 kHz PWM with 1 us deadtime
00899 *     rx_gptw_config_t pwm_config = {
00900 *         .frequency_hz = 20000,
00901 *         .deadtime_ns = 1000,
00902 *         .wave_mode = k_gptw_wave_saw_pwm,
00903 *         .enable_complementary = true,
00904 *         .invert_polarity = false
00905 *     };
00906
00907 *     rx_err_t err = rx_gptw_init_all_staggered(&pwm_config);
00908 *     if (err != k_rx_ok) {
00909 *         rx_log_error("MOTOR", "GPTW init failed: %d", err);
00910 *         return err;
00911 *     }
00912
00913 *     // Initialize H-bridge drivers
00914 *     err = rx_drv8263_init_all();
00915 *     if (err != k_rx_ok) {
00916 *         rx_log_error("MOTOR", "DRV8263H init failed: %d", err);
00917 *         return err;
00918 *     }
00919
00920 *     rx_log_info("MOTOR", "Motor system initialized (4 channels, 20kHz, staggered)");
00921 *     return k_rx_ok;
00922 * }
00923 * @endcode
00924
00925 * @par Example (Verify Phase Offsets):
00926 * @code{.c}
00927 * // Diagnostic: read counter values to verify phase relationship
00928 * void verify_phase_stagger(void)
00929 * {
00930 *     uint32_t cnt0, cnt1, cnt2, cnt3;
00931 *     uint32_t period;
00932
00933 *     rx_gptw_get_period(k_gptw_channel_0, &period);
00934
00935 *     // Read all counters (will have some jitter due to sequential reads)
00936 *     cnt0 = GPTW0->GTCNT;

```

```

00937 *   cnt1 = GPTW1->GTCNT;
00938 *   cnt2 = GPTW2->GTCNT;
00939 *   cnt3 = GPTW3->GTCNT;
00940 *
00941 *   // Expected: cnt1 ~ cnt0 + period/4, cnt2 ~ cnt0 + period/2, etc.
00942 *   rx_log_debug("GPTW", "Period=%lu, Offsets: %lu, %lu, %lu",
00943 *               period, cnt0, cnt1 - cnt0, cnt2 - cnt0, cnt3 - cnt0);
00944 * }
00945 * @endcode
00946 *
00947 * @see rx_gptw_init_pwm() Single channel initialization (not recommended for motors)
00948 * @see rx_gptw_set_duty() Set duty cycle after initialization
00949 * @see rx_gptw_deinit() Cleanup function
00950 * @see rx_drv8263_init_all() H-bridge driver initialization
00951 *
00952 * @since Version 1.0.0
00953 *
00954 * @par NASA Power of 10 Compliance:
00955 * - Rule 1: [OK] No goto/setjmp/recursion
00956 * - Rule 2: [OK] Loop bounded by k_gptw_channel_3 (4 iterations)
00957 * - Rule 5: [OK] 2 preconditions, 3 postconditions
00958 * - Rule 7: [OK] All internal function returns checked
00959 *
00960 * @callgraph
00961 * @callergraph
00962 */
00963 /* Public count constant matching the private k_gptw_max_channels in rx_gptw.c. */
00964 typedef enum : uint8_t {
00965     k_rx_gptw_channel_count = 4,
00966 } rx_gptw_count_t;
00967
00968 [[nodiscard]] rx_err_t
00969 rx_gptw_init_all_staggered(const rx_gptw_config_t* configs[k_rx_gptw_channel_count]);
00970
00971 /**
00972 * @brief Initialize a single GPTW channel for PWM output
00973 *
00974 * @details
00975 * Configures the specified GPTW channel for PWM mode (sawtooth or triangle wave).
00976 * Automatically configures MPC for the pin alternate functions, sets up dead-time
00977 * insertion, and starts PWM generation at 0% duty cycle. The GTCNT counter is
00978 * reset to zero during initialization.
00979 *
00980 * @param[in] channel GPTW channel number (valid: 0-3)
00981 * @param[in] config Pointer to PWM configuration struct (must be non-null)
00982 *
00983 * @return rx_err_t Error code
00984 * @retval k_rx_ok Success, channel initialized and timer running
00985 * @retval k_rx_err_invalid_arg Channel out of range or config is null
00986 * @retval k_rx_err_invalid_state GPTW module not enabled in MSTPCR
00987 *
00988 * @pre GPTW module clock must be enabled
00989 * @pre config must point to a valid rx_gptw_config_t
00990 * @post Channel timer running at configured frequency with 0% duty
00991 * @post GTCNT reset to zero (affects phase staggering if called per-channel)
00992 *
00993 * @note Not thread-safe. Call from single-threaded init context.
00994 * @warning Resets GTCNT to 0. If phase staggering is required, use
00995 *         rx_gptw_init_all_staggered() instead, or configure phase offsets
00996 *         after all channels are initialized.
00997 *
00998 * @see rx_gptw_init_all_staggered() Phase-staggered multi-channel init
00999 * @see rx_gptw_deinit() Release channel resources
01000 *
01001 * @since Version 1.0.0
01002 */
01003 [[nodiscard]] rx_err_t rx_gptw_init_pwm(rx_gptw_channel_t channel, const rx_gptw_config_t* config);
01004
01005 /**
01006 * @brief Set PWM duty cycle (floating-point percentage)
01007 *
01008 * @details
01009 * Updates the duty cycle for the specified GPTW channel and output. The new duty
01010 * cycle is written to the compare register buffer and takes effect on the next
01011 * PWM period boundary, ensuring glitch-free operation.
01012 *
01013 * ## Algorithm Steps
01014 *
01015 * 1. Validate channel and output wrapper values
01016 * 2. Check channel is initialized
01017 * 3. Clamp duty_percent to [0.0, 100.0] range
01018 * 4. Calculate raw count: duty_count = (duty_percent / 100.0) x period_count
01019 * 5. Write to compare register (GTCCRA or GTCCRB)
01020 * 6. Register buffer transfers to active on next period boundary
01021 *
01022 * @par Duty Cycle Calculation:
01023 * @f[

```

```

01024 * \text{duty\_count} = \left\lfloor \frac{\text{duty\_percent}}{100} \right\rfloor \times \text{period\_count} \right\rfloor
01025 * @f]
01026 *
01027 * For 20 kHz PWM with 120 MHz PCLKA:
01028 * - Period count = 6000
01029 * - 50% duty = 3000 counts
01030 * - Resolution = 0.0167% per count
01031 *
01032 * @param[in] channel GPTW channel wrapped in rx_gptw_channel_id_t for type safety
01033 * - Use rx_gptw_channel_id(k_gptw_channel_0) through rx_gptw_channel_id(k_gptw_channel_3)
01034 * @param[in] output Output selection wrapped in rx_gptw_output_id_t for type safety
01035 * - Use rx_gptw_output_id(k_gptw_output_a) or rx_gptw_output_id(k_gptw_output_b)
01036 * @param[in] duty_percent Duty cycle in percent [0.0, 100.0]
01037 * - Values < 0.0 are clamped to 0.0
01038 * - Values > 100.0 are clamped to 100.0
01039 * - NaN and infinity are treated as invalid
01040 *
01041 * @return rx_err_t Error code
01042 * @retval k_rx_ok Duty cycle updated successfully
01043 * @retval k_rx_err_invalid_arg Channel or output out of range, or duty is NaN/infinity
01044 * @retval k_rx_err_not_initialized Channel not initialized via rx_gptw_init_*( )
01045 *
01046 * @pre Channel initialized via rx_gptw_init_all_staggered() or rx_gptw_init_pwm()
01047 * @pre duty_percent is a valid float (not NaN or infinity)
01048 *
01049 * @post Compare register updated with new duty value
01050 * @post Duty takes effect on next period boundary (no glitch)
01051 *
01052 * @invariant Actual duty always in [0%, 100%] regardless of input
01053 *
01054 * @note Thread-safe: Atomic register write, safe from ISR or multiple threads
01055 * @note Use rx_gptw_set_duty_raw() in tight control loops to avoid float overhead
01056 *
01057 * @warning Floating-point calculation adds ~4 us overhead vs raw version
01058 *
01059 * @par Performance:
01060 * - Execution time: ~5 us at 240 MHz (includes float multiply)
01061 * - Memory usage: 0 bytes heap, ~16 bytes stack
01062 *
01063 * @par Example (Set Motor Speed):
01064 * @code{.c}
01065 * // Set motor 0 forward at 75% speed
01066 * // Output A = direction (100% = forward)
01067 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01068 *                 rx_gptw_output_id(k_gptw_output_a),
01069 *                 100.0F);
01070 *
01071 * // Output B = speed (75% duty)
01072 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01073 *                 rx_gptw_output_id(k_gptw_output_b),
01074 *                 75.0F);
01075 * @endcode
01076 *
01077 * @par Example (PID Control Loop):
01078 * @code{.c}
01079 * void motor_control_task(void)
01080 * {
01081 *     while (true) {
01082 *         float setpoint = get_velocity_setpoint();
01083 *         float measured = read_encoder_velocity();
01084 *         float output;
01085 *
01086 *         rx_pid_compute(&pid, setpoint, measured, 0.01F, &output);
01087 *
01088 *         // Convert PID output (-100 to +100) to duty cycle
01089 *         float duty = fabsf(output);
01090 *         bool forward = (output >= 0.0F);
01091 *
01092 *         // Set direction
01093 *         rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01094 *                         rx_gptw_output_id(k_gptw_output_a),
01095 *                         forward ? 100.0F : 0.0F);
01096 *
01097 *         // Set speed
01098 *         rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01099 *                         rx_gptw_output_id(k_gptw_output_b),
01100 *                         duty);
01101 *
01102 *         tx_thread_sleep(1); // 100 Hz control loop
01103 *     }
01104 * }
01105 * @endcode
01106 *
01107 * @see rx_gptw_set_duty_raw() Raw count version for tight loops
01108 * @see rx_gptw_get_duty() Read current duty cycle
01109 * @see rx_gptw_channel_id() Wrapper constructor for channel
01110 * @see rx_gptw_output_id() Wrapper constructor for output

```

```

01111 *
01112 * @since Version 1.0.0
01113 *
01114 * @par NASA Power of 10 Compliance:
01115 * - Rule 5: [OK] 2 preconditions, 2 postconditions
01116 * - Rule 7: [OK] Returns error code, caller must check
01117 */
01118 [[nodiscard]] rx_err_t
01119 rx_gptw_set_duty(rx_gptw_channel_id_t channel, rx_gptw_output_id_t output, float duty_percent);
01120
01121 /**
01122 * @brief Set PWM duty cycle using raw counter value
01123 *
01124 * @details
01125 * Updates duty cycle using a raw counter value for efficiency in tight control
01126 * loops where floating-point overhead should be avoided. The count is written
01127 * to the compare register buffer and takes effect on the next PWM period
01128 * boundary, ensuring glitch-free operation.
01129 *
01130 * @param[in] channel GPTW channel (valid: 0-3)
01131 * @param[in] output Output channel (k_gptw_output_a or k_gptw_output_b)
01132 * @param[in] duty_count Duty cycle count (valid range: 0 to period_count)
01133 *
01134 * @return rx_err_t Error code
01135 * @retval k_rx_ok Success, duty cycle updated
01136 * @retval k_rx_err_invalid_arg Channel, output, or count out of range
01137 * @retval k_rx_err_invalid_state Channel not initialized via rx_gptw_init_pwm()
01138 *
01139 * @pre Channel must be initialized via rx_gptw_init_pwm()
01140 * @pre duty_count must not exceed the period count for the channel
01141 * @post Compare register buffer updated with new duty count
01142 * @post Active register updated on next period boundary
01143 *
01144 * @note Thread-safe: single 32-bit register write is inherently atomic on RX72N
01145 *
01146 * @par Performance:
01147 * Constant time: single 32-bit register write after validation.
01148 *
01149 * @par Example:
01150 * @code{.c}
01151 * // Read current period, compute 75% duty, then set raw count
01152 * uint32_t period = 0;
01153 * rx_err_t err = rx_gptw_get_period(k_gptw_channel_0, &period);
01154 * if (err == k_rx_ok) {
01155 *     const uint32_t duty_count = (period * k_gptw_duty_75_pct_num) / k_gptw_duty_pct_denom;
01156 *     err = rx_gptw_set_duty_raw(k_gptw_channel_0, k_gptw_output_b, duty_count);
01157 * }
01158 * @endcode
01159 *
01160 * @see rx_gptw_set_duty() Percentage-based version with float conversion
01161 * @see rx_gptw_get_period() Query period count for duty calculation
01162 *
01163 * @since Version 1.0.0
01164 */
01165 [[nodiscard]] rx_err_t
01166 rx_gptw_set_duty_raw(rx_gptw_channel_t channel, rx_gptw_output_t output, uint32_t duty_count);
01167
01168 /**
01169 * @brief Get current duty cycle as a floating-point percentage
01170 *
01171 * @details
01172 * Reads the current compare register value for the specified output and
01173 * converts it to a percentage of the period count. The value reflects the
01174 * most recently committed duty cycle (may differ from the buffered value
01175 * pending a period boundary update).
01176 *
01177 * @param[in] channel GPTW channel number (valid: 0-3)
01178 * @param[in] output Output channel (k_gptw_output_a or k_gptw_output_b)
01179 * @param[out] duty_percent Pointer to store duty cycle in percent [0.0, 100.0]
01180 *
01181 * @return rx_err_t Error code
01182 * @retval k_rx_ok Success, duty_percent written
01183 * @retval k_rx_err_null_ptr duty_percent is nullptr
01184 * @retval k_rx_err_invalid_arg Channel or output out of range
01185 * @retval k_rx_err_invalid_state Channel not initialized
01186 *
01187 * @pre Channel must be initialized via rx_gptw_init_pwm()
01188 * @pre duty_percent must be a valid non-null pointer
01189 * @post duty_percent contains current duty cycle percentage
01190 * @post No hardware state modified (read-only operation)
01191 *
01192 * @note Not thread-safe. Caller must provide synchronization.
01193 *
01194 * @par Example:
01195 * @code{.c}
01196 * float duty = 0.0F;
01197 * rx_err_t err = rx_gptw_get_duty(k_gptw_channel_0, k_gptw_output_a, &duty);

```

```

01198 * if (err == k_rx_ok) {
01199 *     // duty now contains current duty cycle percentage [0.0, 100.0]
01200 * }
01201 * @endcode
01202 *
01203 * @see rx_gptw_set_duty() Set duty cycle by percentage
01204 * @see rx_gptw_get_period() Get period count for manual calculation
01205 *
01206 * @since Version 1.0.0
01207 */
01208 [[nodiscard]] rx_err_t
01209 rx_gptw_get_duty(rx_gptw_channel_t channel, rx_gptw_output_t output, float* duty_percent);
01210 /**
01211 * @brief Get the 32-bit PWM period count for a GPTW channel
01212 *
01213 * @details
01214 * Reads the GTPR (General Timer Period Register) value for the specified
01215 * channel. This value determines the PWM frequency and is needed to
01216 * calculate raw duty counts: duty_count = (duty_percent / 100) * period_count.
01217 *
01218 * @param[in] channel GPTW channel number (valid: 0-3)
01219 * @param[out] period_count Pointer to store 32-bit period count
01220 *
01221 * @return rx_err_t Error code
01222 * @retval k_rx_ok Success, period_count written
01223 * @retval k_rx_err_null_ptr period_count is nullptr
01224 * @retval k_rx_err_invalid_arg Channel out of range
01225 * @retval k_rx_err_invalid_state Channel not initialized
01226 *
01227 * @pre Channel must be initialized via rx_gptw_init_pwm()
01228 * @pre period_count must be a valid non-null pointer
01229 * @post period_count contains the GTPR register value
01230 * @post No hardware state modified (read-only operation)
01231 *
01232 * @note Not thread-safe. Caller must provide synchronization.
01233 *
01234 * @par Example:
01235 * @code{.c}
01236 * uint32_t period = 0;
01237 * rx_err_t err = rx_gptw_get_period(k_gptw_channel_0, &period);
01238 * if (err == k_rx_ok) {
01239 *     // Calculate raw duty count for 50% duty cycle
01240 *     uint32_t duty_count = (period * k_gptw_duty_50_pct_num) / k_gptw_duty_pct_denom;
01241 * }
01242 * @endcode
01243 *
01244 * @see rx_gptw_set_duty_raw() Uses period_count for raw duty calculation
01245 *
01246 * @since Version 1.0.0
01247 */
01248 [[nodiscard]] rx_err_t rx_gptw_get_period(rx_gptw_channel_t channel, uint32_t* period_count);
01250 /**
01251 * @brief Enable or disable a PWM output pin
01252 *
01253 * @details
01254 * Controls the GTONCR (General Timer Output Negation Control Register) bits
01255 * that gate the PWM signal to the physical pin. When disabled, the pin is
01256 * driven to its inactive level (determined by polarity configuration).
01257 *
01258 * @param[in] channel GPTW channel number (valid: 0-3)
01259 * @param[in] output Output channel (k_gptw_output_a or k_gptw_output_b)
01260 * @param[in] enable true to drive PWM on pin, false to hold pin inactive
01261 *
01262 * @return rx_err_t Error code
01263 * @retval k_rx_ok Success, output enable state updated
01264 * @retval k_rx_err_invalid_arg Channel or output out of range
01265 * @retval k_rx_err_invalid_state Channel not initialized
01266 *
01267 * @pre Channel must be initialized via rx_gptw_init_pwm()
01268 * @pre Output must be k_gptw_output_a or k_gptw_output_b
01269 * @post GTONCR bit updated for the specified output
01270 * @post Pin either drives PWM signal or holds inactive level
01271 *
01272 * @note Not thread-safe. Caller must provide synchronization.
01273 *
01274 * @par Example:
01275 * @code{.c}
01276 * // Enable PWM output on channel 0, output A
01277 * rx_err_t err = rx_gptw_enable_output(k_gptw_channel_0, k_gptw_output_a, true);
01278 * if (err != k_rx_ok) {
01279 *     // Handle output enable failure
01280 * }
01281 * @endcode
01282 *
01283 * @see rx_gptw_start() Start timer (separate from output enable)

```

```

01285 * @see rx_gptw_stop() Stop timer (separate from output enable)
01286 *
01287 * @since Version 1.0.0
01288 */
01289 [[nodiscard]] rx_err_t
01290 rx_gptw_enable_output(rx_gptw_channel_t channel, rx_gptw_output_t output, bool enable);
01291
01292 /**
01293 * @brief Start the GPTW timer counter for PWM generation
01294 *
01295 * @details
01296 * Sets the GTCR.CST bit to begin counting. The timer is automatically
01297 * started during rx_gptw_init_pwm(), but can be stopped via rx_gptw_stop()
01298 * and restarted with this function. Counting resumes from the current
01299 * GTCNT value (not reset to zero).
01300 *
01301 * @param[in] channel GPTW channel number (valid: 0-3)
01302 *
01303 * @return rx_err_t Error code
01304 * @retval k_rx_ok Success, timer started
01305 * @retval k_rx_err_invalid_arg Channel out of range
01306 *
01307 * @pre Channel must be initialized via rx_gptw_init_pwm()
01308 * @pre Channel must currently be stopped (safe to call if already running)
01309 * @post Timer counter incrementing at peripheral clock rate
01310 * @post PWM outputs active (if enabled via rx_gptw_enable_output())
01311 *
01312 * @note Not thread-safe. Caller must provide synchronization.
01313 *
01314 * @par Example:
01315 * @code{.c}
01316 * // Restart timer after a stop
01317 * rx_err_t err = rx_gptw_start(k_gptw_channel_0);
01318 * if (err != k_rx_ok) {
01319 *     // Handle start failure
01320 * }
01321 * @endcode
01322 *
01323 * @see rx_gptw_stop() Stop the timer counter
01324 * @see rx_gptw_init_pwm() Initializes and starts timer automatically
01325 *
01326 * @since Version 1.0.0
01327 */
01328 [[nodiscard]] rx_err_t rx_gptw_start(rx_gptw_channel_t channel);
01329
01330 /**
01331 * @brief Stop the GPTW timer counter
01332 *
01333 * @details
01334 * Clears the GTCR.CST bit to halt counting. PWM outputs hold their last
01335 * state when the timer stops. The GTCNT value is preserved and counting
01336 * resumes from that value when rx_gptw_start() is called.
01337 *
01338 * @param[in] channel GPTW channel number (valid: 0-3)
01339 *
01340 * @return rx_err_t Error code
01341 * @retval k_rx_ok Success, timer stopped
01342 * @retval k_rx_err_invalid_arg Channel out of range
01343 *
01344 * @pre Channel must be initialized via rx_gptw_init_pwm()
01345 * @pre Channel should be running (safe to call if already stopped)
01346 * @post Timer counter halted at current GTCNT value
01347 * @post PWM outputs frozen at last driven state
01348 *
01349 * @note Not thread-safe. Caller must provide synchronization.
01350 * @warning Outputs hold last state; explicitly set duty to 0 before
01351 *         stopping if low output is required.
01352 *
01353 * @par Example:
01354 * @code{.c}
01355 * // Stop timer and ensure outputs are low
01356 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01357 *                  rx_gptw_output_id(k_gptw_output_b), 0.0F);
01358 * rx_err_t err = rx_gptw_stop(k_gptw_channel_0);
01359 * @endcode
01360 *
01361 * @see rx_gptw_start() Resume timer counter
01362 * @see rx_gptw_deinit() Full shutdown including output disable
01363 *
01364 * @since Version 1.0.0
01365 */
01366 [[nodiscard]] rx_err_t rx_gptw_stop(rx_gptw_channel_t channel);
01367
01368 /**
01369 * @brief Deinitialize a GPTW channel and release all resources
01370 *
01371 * @details

```

```

01372 * Performs a full shutdown sequence: stops the timer counter, disables both
01373 * PWM outputs, resets the compare and period registers, and marks the channel
01374 * as uninitialized. After deinit, the channel must be re-initialized via
01375 * rx_gptw_init_pwm() before use.
01376 *
01377 * @param[in] channel GPTW channel number (valid: 0-3)
01378 *
01379 * @return rx_err_t Error code
01380 * @retval k_rx_ok Success, channel fully deinitialized
01381 * @retval k_rx_err_invalid_arg Channel out of range
01382 *
01383 * @pre Channel should be initialized (safe to call if already deinitialized)
01384 * @pre Motor outputs should be in a safe state before deinit
01385 * @post Timer stopped, outputs disabled, channel marked uninitialized
01386 * @post All channel registers reset to default values
01387 *
01388 * @note Not thread-safe. Caller must provide synchronization.
01389 *
01390 * @par Example:
01391 * @code{.c}
01392 * // Full shutdown of GPTW channel 0
01393 * rx_err_t err = rx_gptw_deinit(k_gptw_channel_0);
01394 * if (err == k_rx_ok) {
01395 *     // Channel released, must call rx_gptw_init_pwm() before reuse
01396 * }
01397 * @endcode
01398 *
01399 * @see rx_gptw_init_pwm() Re-initialize after deinit
01400 * @see rx_gptw_stop() Stop timer without full deinit
01401 *
01402 * @since Version 1.0.0
01403 */
01404 [[nodiscard]] rx_err_t rx_gptw_deinit(rx_gptw_channel_t channel);
01405
01406 #ifdef __cplusplus
01407 }
01408 #endif

```

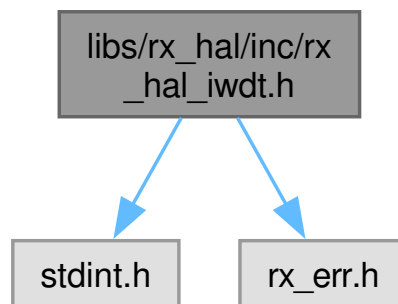
8.81 libs/rx_hal/inc/rx_hal_iwdt.h File Reference

Independent Watchdog Timer (IWDT) HAL Driver for RX72N.

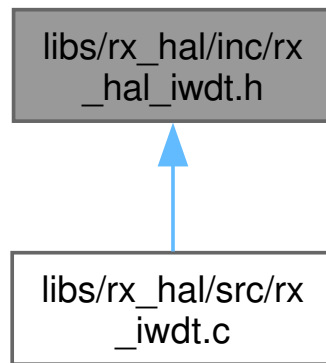
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_hal_iwdt.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_iwdt_timeout_t` : `uint16_t` {
`k_iwdt_timeout_128ms`, `k_iwdt_timeout_512ms`, `k_iwdt_timeout_1000ms`, `k_iwdt_timeout_2048ms`,
`k_iwdt_timeout_8192ms`, `k_iwdt_timeout_16384ms` }
 IWDT Timeout Period Options.
- enum `rx_iwdt_reset_cause_t` : `uint8_t` { `k_iwdt_reset_none` , `k_iwdt_reset_underflow` ,
`k_iwdt_reset_refresh_error` }
 IWDT Reset Cause Information.
- enum `rx_iwdt_limits_t` : `uint8_t` { `k_iwdt_max_tasks` = 8 }
 Maximum number of tasks that can be monitored.

Functions

- `rx_err_t rx_hal_iwdt_init` (`uint32_t timeout_ms`)
 Initialize the Independent Watchdog Timer.
- `void rx_hal_iwdt_feed` (`void`)
 Feed (refresh) the watchdog timer.
- `bool rx_hal_iwdt_was_reset` (`void`)
 Check if last reset was caused by watchdog timeout.
- `rx_iwdt_reset_cause_t rx_hal_iwdt_get_reset_cause` (`void`)
 Get detailed reset cause from IWDT.
- `void rx_hal_iwdt_clear_status` (`void`)
 Clear watchdog status flags.
- `rx_err_t rx_hal_iwdt_register_task` (`const char *task_name`, `uint32_t timeout_ms`)
 Register a task for heartbeat monitoring.
- `void rx_hal_iwdt_task_heartbeat` (`const char *task_name`)
 Record a task heartbeat.
- `rx_err_t rx_hal_iwdt_check_tasks` (`void`)
 Check for task deadlocks.
- `const char * rx_hal_iwdt_get_failed_task` (`void`)
 Get the name of the last failed task (if any).

8.81.1 Detailed Description

Independent Watchdog Timer (IWDT) HAL Driver for RX72N.

Hardware Abstraction Layer for the RX72N Independent Watchdog Timer (IWDT) peripheral. This driver provides low-level register access and task-level monitoring for safety-critical system recovery.

8.81.1.1 Purpose and Design Rationale

The IWDT provides true clock-independent watchdog functionality:

- Dedicated 120 kHz on-chip oscillator (IWDTCCLK)
- Operates even if main system clock fails
- Cannot be disabled once started (hardware safety feature)
- Triggers system reset on timeout

This module is the HAL layer providing direct hardware access. For higher-level watchdog features (state-dependent timeouts, comprehensive task monitoring), see rx_core/inc/rx_iwdt.h.

8.81.1.2 Key Features

Hardware Watchdog:

- Internal 120 kHz clock source (IWDTCCLK)
- Configurable timeout: 128ms to 16,384ms
- Window mode (optional, disabled by default)
- Reset or NMI on timeout
- Refresh error detection

Task Monitoring:

- Register up to 8 tasks for heartbeat monitoring
- Per-task timeout configuration
- Failed task identification for diagnostics

8.81.1.3 Hardware Architecture

Clock and Timing:

Parameter	Value	Notes
Clock Source	IWDTCCLK	Internal RC oscillator
Clock Frequency	120 kHz	+/-1% accuracy
Clock Period	8.33 us	1/120kHz
Counter Width	14-bit	Down-counter
Min Timeout	128 ms	$2^{14} / (120k / 1)$
Max Timeout	16,384 ms	$2^{14} / (120k / 128)$

Register Map:

Register	Address	Description
IWDTRR	0x00088030	Refresh register
IWDTR\{r}	0x00088032	Control register
IWDTSR	0x00088034	Status register
IWDTR\{r}	0x00088036	Reset control register
IWDTCSTPR	0x00088038	Count stop register

Timeout Calculation:

$$T_{timeout} = \frac{2^{14} \times \text{TOPS}}{f_{IWDTCCLK}}$$

Where TOPS is the timeout period setting (1, 4, 16, 32, 64, or 128).

8.81.1.4 Sub-clock Pin Handling

The IWDTC uses the internal IWDTCCLK, not the sub-clock oscillator. When sub-clock is unused:

- XCIN: Connect to VSS (ground) via pull-down resistor
- XCOUT: Leave open (unconnected)

System initialization disables sub-clock via SOSC\{r}.SOSTP = 1.

8.81.1.5 Performance Characteristics

Operation	Execution Time	Notes
rx_hal_iwdt_init()	~20 us	One-time configuration
rx_hal_iwdt_feed()	~1 us	Two register writes
rx_hal_iwdt_task_heartbeat()	~3 us	Timestamp update
rx_hal_iwdt_check_tasks()	~5 us x N tasks	Linear scan

Memory Footprint:

- Static variables: ~256 bytes (task tracking)
- Code size: ~1 KB
- Stack usage: < 32 bytes
- Heap usage: 0 bytes

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	IWDT peripheral
Clock	IWDTCLK	120 kHz internal
XCIN Pin	VSS via resistor	When sub-clock unused
XCOU Pin	Open	When sub-clock unused

Module Dependencies:

This module depends on:

- `rx_err.h` - Error code definitions
- [rx72n_iwdt_regs.h](#) - IWDT register definitions

This module is used by:

- `main.c` - System initialization
- All RTOS tasks - Heartbeat reporting
- `rx_core/inc/rx_iwdt.h` - Higher-level watchdog features

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	↔ [OK]↔	No goto/setjmp/recursion
2. Fixed loop bounds	↔ [OK]↔	Loops bounded by <code>k_iwdt_max_tasks</code> (8)
3. No dynamic allocation	↔ [OK]↔	Static task array only
4. Small functions	↔ [OK]↔	All < 60 lines
5. Assertions	↔ [OK]↔	Minimum 2 checks per function

Rule	Status	Implementation
6. Narrow scope	↔ [OK]↔	File-scope statics
7. Check return values	↔ [OK]↔	All return rx_err_t
8. Limited preprocessor	↔ [OK]↔	C23 typed enums only
9. Pointer restrictions	↔ [OK]↔	Single-level pointers
10. Compiler warnings	↔ [OK]↔	-Wall -Wextra -Werror

SOLID Principles:

Single Responsibility (S):

- HAL layer: Hardware register access only
- Task monitoring separate from hardware watchdog

Open/Closed (O):

- Timeout values configurable
- Task monitoring extensible

Dependency Inversion (D):

- Higher-level rx_core/inc/rx_iwdt.h depends on this HAL interface

See also

rx_core/inc/rx_iwdt.h Higher-level watchdog with state management

[rx72n_iwdt_regs.h](#) Register definitions

RX72N Hardware Manual Chapter 35 - Independent Watchdog Timer

Author

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Date

2026-01-27

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Since

Version 1.0.0

Definition in file [rx_hal_iwdt.h](#).

8.81.2 Enumeration Type Documentation

8.81.2.1 rx_iwdt_limits_t

```
enum rx_iwdt_limits_t : uint8_t
```

Maximum number of tasks that can be monitored.

Enumerator

k_iwdt_max_tasks	Support up to 8 monitored tasks
------------------	---------------------------------

Definition at line 403 of file [rx_hal_iwdt.h](#).

```
00403         : uint8_t {
00404     k_iwdt_max_tasks = 8, /**< Support up to 8 monitored tasks */
00405 } rx_iwdt_limits_t;
```

8.81.2.2 rx_iwdt_reset_cause_t

```
enum rx_iwdt_reset_cause_t : uint8_t
```

IWDT Reset Cause Information.

Identifies the specific cause of an IWDT-triggered reset. This information is read from the IWDTSR status register and can be used for post-mortem diagnostics and logging.

8.81.2.3 Reset Cause Detection

At startup, read the reset cause before calling [rx_hal_iwdt_init\(\)](#):

```
void startup_diagnostics(void)
{
    rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();

    switch (cause) {
    case k_iwdt_reset_none:
        uart_debug_puts("Normal startup\\r\\n");
        break;
    case k_iwdt_reset_underflow:
        uart_debug_puts("[WARN] Watchdog timeout - task deadlock?\\r\\n");
        log_failed_task(rx_hal_iwdt_get_failed_task());
        break;
    case k_iwdt_reset_refresh_error:
        uart_debug_puts("[ERROR] Watchdog refresh error - code bug\\r\\n");
        break;
    }

    rx_hal_iwdt_clear_status();
}
```

Reset Cause Table:

Cause	IWDTSR Bits	Meaning	Typical Issue
None	0x0000	Not a watchdog reset	Power-on, external reset
Underflow	UNDF=1	Counter reached zero	Task deadlock, infinite loop

Cause	IWDTSR Bits	Meaning	Typical Issue
Refresh Error	REFEF=1	Invalid refresh	Window violation, wrong sequence

Invariant

Values map to IWDTSR bit patterns

`k_iwdt_reset_none` (0) is the default for non-watchdog resets

Note

C23 typed enum with `uint8_t` underlying type

See also

[`rx\_hal\_iwdt\_get\_reset\_cause\(\)`](#) Read the reset cause

[`rx\_hal\_iwdt\_clear\_status\(\)`](#) Clear status flags after reading

Since

Version 1.0.0

Enumerator

<code>k_iwdt_reset_none</code>	No watchdog reset occurred. System reset was from power-on, external reset pin, or other source. <code>IWDTSR.UNDFF=0</code> , <code>REFEF=0</code>
<code>k_iwdt_reset_underflow</code>	Reset due to counter underflow. Counter reached zero before <code>rx_hal_iwdt_feed()</code> was called. Indicates task deadlock or infinite loop. <code>IWDTSR.UNDFF=1</code>
<code>k_iwdt_reset_refresh_error</code>	Reset due to invalid refresh. Either wrong refresh sequence (not 0x00, 0xFF) or refresh during window-prohibited period. <code>IWDTSR.REFEF=1</code>

Definition at line 276 of file [rx_hal_iwdt.h](#).

```

00276         : uint8_t {
00277     k_iwdt_reset_none =
00278         0, /**< No watchdog reset occurred. System reset was from power-on, external reset pin, or other source.
IWDTSR.UNDFF=0, REFEF=0 */
00279     k_iwdt_reset_underflow =
00280         1, /**< Reset due to counter underflow. Counter reached zero before rx_hal_iwdt_feed() was called. Indicates task
deadlock or infinite loop. IWDTSR.UNDFF=1 */
00281     k_iwdt_reset_refresh_error =
00282         2, /**< Reset due to invalid refresh. Either wrong refresh sequence (not 0x00, 0xFF) or refresh during
window-prohibited period. IWDTSR.REFEF=1 */
00283 } rx_iwdt_reset_cause_t;
```

8.81.2.4 rx_iwdt_timeout_t

```
enum rx_iwdt_timeout_t : uint16_t
```

IWDT Timeout Period Options.

Pre-defined timeout periods for the IWDT. These values correspond to hardware divider settings that produce specific timeout durations from the 120 kHz `IWDTCLK`.

8.81.2.5 Timeout Selection Guide

Timeout	Use Case	Notes
128ms	High-frequency control loops (> 50 Hz)	Very responsive
512ms	Normal control loops (10-50 Hz)	Fast detection
1000ms	Default - typical applications	Good balance
2048ms	Low-frequency tasks (1-10 Hz)	Tolerant
8192ms	Slow background operations	Long timeout
16384ms	Initialization/maintenance	Maximum

Timeout Calculation:

Timeout periods are derived from:

$$T = \frac{\text{Cycle Count}}{f_{IWDTCLK}} = \frac{\text{Cycle Count}}{120 \text{ kHz}}$$

Usage Example:

```
// Use predefined timeout for 1 second
rx_hal_iwdt_init(k_iwdt_timeout_1000ms);

// Or use custom value (rounded up to nearest supported)
rx_hal_iwdt_init(500); // Rounds to 512ms
```

Invariant

All values are in milliseconds

Values correspond to hardware divider settings

Note

C23 typed enum with uint16_t underlying type

Actual timeout may vary +/-1% due to oscillator tolerance

Since

Version 1.0.0

Enumerator

k_iwdt_timeout_128ms	Minimum timeout (128ms). Use for high-frequency control loops requiring fast failure detection. TOPS=1 divider
k_iwdt_timeout_512ms	Short timeout (512ms). Use for normal control loops (10-50 Hz). Provides good responsiveness. TOPS=4 divider

k_iwdt_timeout_1000ms	Default timeout (1 second). Recommended for typical applications. Good balance between detection speed and tolerance. TOPS~8
k_iwdt_timeout_2048ms	Medium timeout (2 seconds). Use for low-frequency tasks that may have variable execution time. TOPS=16 divider
k_iwdt_timeout_8192ms	Long timeout (8 seconds). Use for slow background operations or maintenance tasks. TOPS=64 divider
k_iwdt_timeout_16384ms	Maximum timeout (~16 seconds). Use only for initialization phases or very slow operations. TOPS=128 divider

Definition at line 209 of file [rx_hal_iwdt.h](#).

```

00209     : uint16_t {
00210     k_iwdt_timeout_128ms =
00211         128, /**< Minimum timeout (128ms). Use for high-frequency control loops requiring fast failure detection. TOPS=1
            divider */
00212     k_iwdt_timeout_512ms =
00213         512, /**< Short timeout (512ms). Use for normal control loops (10-50 Hz). Provides good responsiveness. TOPS=4
            divider */
00214     k_iwdt_timeout_1000ms =
00215         1000, /**< Default timeout (1 second). **Recommended** for typical applications. Good balance between detection
            speed and tolerance. TOPS=8 */
00216     k_iwdt_timeout_2048ms =
00217         2048, /**< Medium timeout (2 seconds). Use for low-frequency tasks that may have variable execution time. TOPS=16
            divider */
00218     k_iwdt_timeout_8192ms =
00219         8192, /**< Long timeout (8 seconds). Use for slow background operations or maintenance tasks. TOPS=64 divider */
00220     k_iwdt_timeout_16384ms =
00221         16384, /**< Maximum timeout (~16 seconds). Use only for initialization phases or very slow operations. TOPS=128
            divider */
00222 } rx_iwdt_timeout_t;

```

8.81.3 Function Documentation

8.81.3.1 rx_hal_iwdt_clear_status()

```
void rx_hal_iwdt_clear_status (
    void )
```

Clear watchdog status flags.

Clears the underflow and refresh error flags in IWDTSR. Call after reading the reset cause to prepare for next reset detection.

Note

Flags are typically cleared once at startup after logging the cause.

Clears the UNDF (underflow) and REFE (refresh error) flags in the IWDT Status Register. These flags are "write-0-to-clear" type - writing 0 to a flag clears it, writing 1 has no effect.

Clear Sequence

- * 1. Read IWDTSR (get current status)
- * 2. AND with ~(UNDF | REFE) to clear both flags
- * 3. Write back to IWDTSR
- *
- * Note: Read-modify-write is safe because flags are write-0-to-clear
- *

Precondition

Status has been read via [rx_hal_iwdt_was_reset\(\)](#) or [rx_hal_iwdt_get_reset_cause\(\)](#)

Postcondition

UNDFE flag cleared

REFE flag cleared

Note

On host builds, function is a no-op

Call after reading status at startup to prepare for next reset

Example:

```
// Startup diagnostic sequence
if (rx_hal_iwdt_was_reset()) {
    rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
    log_watchdog_reset(cause);
}
// Clear flags so next reset can be detected
rx_hal_iwdt_clear_status();
```

See also

[rx_hal_iwdt_was_reset\(\)](#) Check reset status

[rx_hal_iwdt_get_reset_cause\(\)](#) Get specific cause

Since

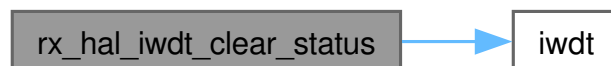
Version 1.0.0

Definition at line 1037 of file [rx_iwdt.c](#).

```
01038 {
01039 #ifdef __RX__
01040 /*
01041  * Clear status flags by writing 0 to the flag bits
01042  *
01043  * Bits 14-15 are write-0-to-clear
01044  */
01045 iwdt()->iwdtsr &= (uint16_t) ~(uint16_t)(k_iwdt_sr_undff | k_iwdt_sr_refef);
01046
01047 #else
01048 /* Host-side stub - no operation */
01049 #endif
01050 }
```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undff](#).

Here is the call graph for this function:



8.81.3.2 rx_hal_iwdt_feed()

```
void rx_hal_iwdt_feed (
    void )
```

Feed (refresh) the watchdog timer.

Resets the watchdog counter to prevent timeout. This function must be called periodically, within the configured timeout period.

The refresh sequence writes 0x00 followed by 0xFF to the IWDTRR register. This sequence must complete atomically without interruption.

Note

Call this function from your main control loop.

Calling more frequently than needed has no negative effect.

Do NOT call from interrupt handlers unless you understand the implications for your system's safety.

Example placement:

```
// Motor control task at 100 Hz (10ms period)
// With 1000ms timeout, we have 100 chances to feed
void motor_control_task_entry(ULONG input) {
    while (1) {
        read_sensors();
        update_pid();
        set_motor_output();
        rx_hal_iwdt_feed(); // Feed at end of each iteration
        tx_thread_sleep(1); // 10ms at 100Hz tick
    }
}
```

Feed (refresh) the watchdog timer.

Performs the IWDT refresh sequence by writing the magic values 0x00 then 0xFF to the IWDTRR register. This resets the down-counter to its initial value, preventing timeout and system reset.

Refresh Sequence (Atomic Operation)

```
* +-----+
* | 1. Disable interrupts (save PSW) |
* | 2. Write 0x00 to IWDTRR         |
* | 3. Write 0xFF to IWDTRR         |
* | 4. Restore interrupt state      |
* +-----+
* \\CRITICAL: Steps 2 and 3 must not be interrupted!
*
```

Timing Requirements

- * - Execution time: ~2 us including interrupt disable/enable
- * - Must be called faster than configured timeout period
- * - Typical call rate: Every 100-500 ms depending on timeout
- *

Precondition

IWDTRR must be initialized via [rx_hal_iwdt_init\(\)](#)

Called at rate faster than configured timeout

Postcondition

IWDT counter reset to initial value
 Interrupt state restored to previous value

Note

Safe to call from ISRs (uses atomic disable/restore)
 On host builds, function is a no-op

Warning

Incomplete refresh sequence causes refresh error -> system reset
 Missing calls to this function cause timeout -> system reset

Example:

```
// Main supervision loop
while (1) {
    // Check all tasks for deadlock
    rx_err_t err = rx_hal_iwdt_check_tasks();
    if (err == k_rx_ok) {
        // All tasks healthy, feed hardware watchdog
        rx_hal_iwdt_feed();
    }
    // If task timeout detected, DON'T feed - let watchdog reset
    tx_thread_sleep(100); // 100 ms check interval
}
```

See also

[rx_hal_iwdt_init\(\)](#) Start the watchdog
[rx_hal_iwdt_check_tasks\(\)](#) Verify task health before feeding

Since

Version 1.0.0

Definition at line 821 of file [rx_iwdt.c](#).

```
00822 {
00823 #ifdef __RX__
00824 /*
00825  * IWDT Refresh Sequence (Section 35.3.3)
00826  *
00827  * Write 0x00 followed by 0xFF to IWDTRR.
00828  * This sequence resets the down-counter to its initial value.
00829  *
00830  * CRITICAL: This sequence must not be interrupted.
00831  * An incomplete sequence will trigger a refresh error.
00832  *
00833  * Uses inline assembly to save/restore PSW (Program Status Word) because
00834  * there is no C library function to manipulate interrupt state on RX architecture.
00835  */
00836
00837 /* Disable interrupts during refresh for atomicity */
00838 uint32_t psw;
00839
00840 __asm__ volatile("mvfc psw, %0" : "=r"(psw));
00841 __asm__ volatile("clrpsw i");
00842
00843 /* Perform refresh sequence */
00844 iwdt()->iwdtrr = k_iwdt_refresh_start; /* Write 0x00 */
00845 iwdt()->iwdtrr = k_iwdt_refresh_end; /* Write 0xFF */
00846
```

```

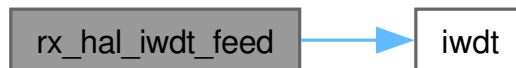
00847  /* Restore interrupt state */
00848  __asm__ volatile("mvtc %0, psw" :: "r"(psw));
00849
00850  #else
00851  /* Host-side stub - no operation */
00852  #endif
00853  }

```

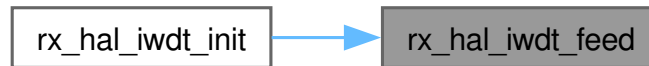
References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtrr](#), [k_iwdt_refresh_end](#), and [k_iwdt_refresh_start](#).

Referenced by [rx_hal_iwdt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.81.3.3 rx_hal_iwdt_get_reset_cause()

```

rx_iwdt_reset_cause_t rx_hal_iwdt_get_reset_cause (
    void )

```

Get detailed reset cause from IWDT.

Provides more detailed information about what caused a watchdog reset:

- Underflow: Counter reached zero (normal timeout)
- Refresh error: Invalid refresh sequence or window violation

Returns

Reset cause enumeration

Note

This provides more detail than [rx_hal_iwdt_was_reset\(\)](#) for diagnostics.

Get detailed reset cause from IWDT.

Reads the IWDT Status Register to determine the specific cause of the last watchdog reset. Refresh errors take priority over underflow since they indicate a software bug in the refresh sequence.

Reset Cause Priority

- * 1. Refresh Error (REFEF=1): Invalid refresh sequence
- * - Incomplete 0x00/0xFF write
- * - Write occurred outside window (if window mode enabled)
- * 2. Underflow (UNDF=1): Counter reached zero
- * - rx_hal_iwdt_feed() not called within timeout period
- * - Software hang, deadlock, or infinite loop
- * 3. None: No watchdog reset occurred
- * - Power-on reset, external reset, or other cause
- *

Precondition

Can be called before [rx_hal_iwdt_init\(\)](#)

Status flags preserved across reset

Postcondition

Status flags NOT cleared (call [rx_hal_iwdt_clear_status\(\)](#) to clear)

Note

On host builds, always returns `k_iwdt_reset_none`

Refresh error has priority over underflow in return value

Example:

```
rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
switch (cause) {
    case k_iwdt_reset_refresh_error:
        uart_debug_puts("ERROR: Watchdog refresh sequence error!\r\n");
        break;
    case k_iwdt_reset_underflow:
        uart_debug_puts("WARN: Watchdog timeout - software hang detected\r\n");
        break;
    case k_iwdt_reset_none:
    default:
        uart_debug_puts("Normal startup\r\n");
        break;
}
rx_hal_iwdt_clear_status(); // Clear flags after reading
```

See also

[rx_hal_iwdt_was_reset\(\)](#) Quick check for any watchdog reset

[rx_hal_iwdt_clear_status\(\)](#) Clear status flags

Since

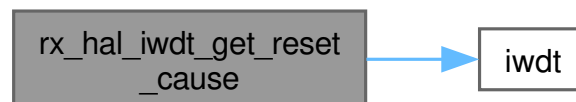
Version 1.0.0

Definition at line 976 of file [rx_iwdt.c](#).

```
00977 {
00978 #ifdef __RX__
00979     const uint16_t status = iwdt()->iwdtsr;
00980
00981     if (status & k_iwdt_sr_refef) {
00982         return k_iwdt_reset_refresh_error;
00983     }
00984
00985     if (status & k_iwdt_sr_undff) {
00986         return k_iwdt_reset_underflow;
00987     }
00988
00989     return k_iwdt_reset_none;
00990
00991 #else
00992     return k_iwdt_reset_none;
00993 #endif
00994 }
```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_reset_none](#), [k_iwdt_reset_refresh_error](#), [k_iwdt_reset_underflow](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undff](#).

Here is the call graph for this function:



8.81.3.4 rx_hal_iwdt_init()

```
rx_err_t rx_hal_iwdt_init (
    const uint32_t timeout_ms) [nodiscard]
```

Initialize the Independent Watchdog Timer.

Configures IWDT with the specified timeout period. Once started, the watchdog cannot be stopped and must be fed periodically to prevent reset.

Configuration:

- Clock: 120 kHz internal IWDT-dedicated oscillator (IWDTCLK)
- Window mode: Disabled (refresh allowed anytime)
- Action: System reset on timeout
- Count stop in sleep: Disabled (continues counting)

Parameters

in	timeout_ms	Desired timeout in milliseconds. Actual timeout will be rounded up to nearest supported value (128, 512, 1024, 2048, 4096, 8192, or 16384 ms).
----	------------	--

Returns

`k_rx_ok` on success
`k_rx_err_invalid_arg` if timeout is out of supported range
`k_rx_err_invalid_state` if IWDT is already running

Note

This function starts the watchdog - `rx_hal_iwdt_feed()` must be called within the timeout period to prevent reset.

On RX72N, IWDT starts counting after the first refresh operation.

Configures and starts the hardware IWDT with the specified timeout period. Once started, the watchdog cannot be stopped - periodic calls to `rx_hal_iwdt_feed()` are required to prevent system reset.

Initialization Sequence

- * 1. Validate not already initialized
- * 2. Validate timeout_ms within range
- * 3. Look up timeout configuration in table
- * 4. Configure IWDT\|r (timeout period, clock divisor, window)
- * 5. Configure IWDT\|r (reset action - not NMI)
- * 6. Configure IWDT\|r (continue counting in sleep)
- * 7. Perform first refresh (starts the watchdog)
- * 8. Set initialized flag
- *

Point of No Return

- * After step 7 (first refresh), the IWDT is running and CANNOT be stopped.
- * Any failure after this point leaves system in watchdog-protected state.
- *

Precondition

IWDT must not be already initialized
 timeout_ms in range [`k_iwdt_timeout_min_ms`, `k_iwdt_timeout_max_ms`]
 System startup, single-threaded context

Postcondition

IWDT running with selected timeout period
`s_iwdt_initialized` set to `k_iwdt_initialized`
 Periodic `rx_hal_iwdt_feed()` calls now required

Note

On host builds (non-RX), skips hardware access but sets initialized flag
 Actual timeout $\geq$ requested (selected from lookup table)

Warning

Once initialized, IWDT cannot be disabled (requires hard reset)

Must call `rx_hal_iwdt_feed()` faster than timeout period

Example:

```
// Initialize with 2 second timeout
rx_err_t err = rx_hal_iwdt_init(2000);
if (err != k_rx_ok) {
    // Handle error - system cannot start safely
    while (1) { // halt }
}

// Now must feed periodically
while (1) {
    // ... main loop ...
    rx_hal_iwdt_feed(); // Must occur every < 2 seconds
    tx_thread_sleep(100);
}
```

See also

[rx_hal_iwdt_feed\(\)](#) Reset watchdog counter

[rx_hal_iwdt.h](#) Public API and configuration details

Since

Version 1.0.0

Definition at line 686 of file `rx_iwdt.c`.

```
00687 {
00688 #ifdef __RX__
00689 if (s_iwdt_initialized == k_iwdt_initialized) {
00690     return k_rx_err_invalid_state;
00691 }
00692
00693 if (timeout_ms < k_iwdt_timeout_min_ms) {
00694     return k_rx_err_invalid_arg;
00695 }
00696
00697 if (timeout_ms > k_iwdt_timeout_max_ms) {
00698     return k_rx_err_invalid_arg;
00699 }
00700
00701 /* Find best timeout configuration */
00702 const iwdt_timeout_entry_t* config = nullptr;
00703 rx_err_t err = internal_find_timeout_config(timeout_ms, &config);
00704 if (err != k_rx_ok) {
00705     return err;
00706 }
00707
00708 /*
00709  * Configure IWDTCR Control Register (IWDTCR)
00710  *
00711  * Bits [1:0] TOPS - Timeout Period Select
00712  * Bits [7:4] CKS - Clock Division Ratio Select
00713  * Bits [9:8] RPES - Window End Position (0x03 = 0%, window disabled)
00714  * Bits [13:12] RPSS - Window Start Position (0x00 = 100%, full window)
00715  */
00716 err = internal_configure_iwdt_control_register(config);
00717 if (err != k_rx_ok) {
00718     return err;
00719 }
00720
00721 /*
00722  * Configure Reset Control Register (IWDTRCR)
00723  *
00724  * Bit 7 RSTIRQS - Reset Interrupt Request Select
00725  * 0 = Non-maskable interrupt request (NMI)
00726  * 1 = Reset
00727  */
```



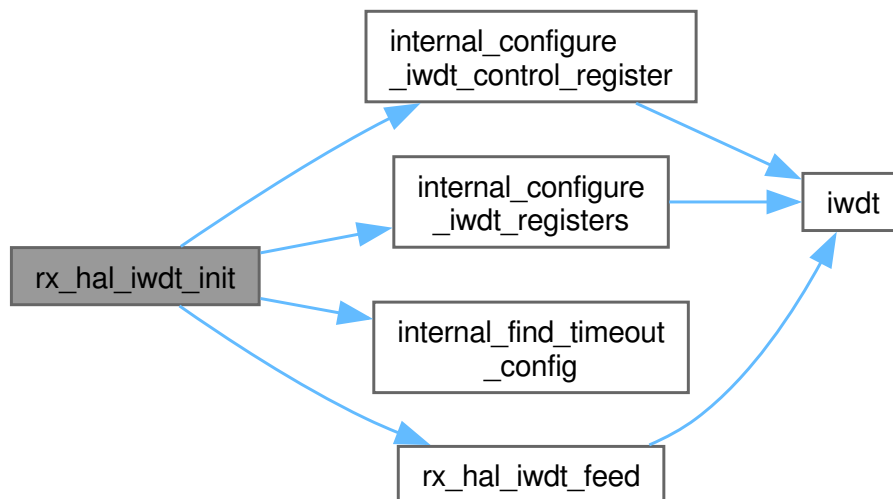
```

00728  * We use reset (1) for safety - NMI could be masked or ignored.
00729  */
00730  err = internal_configure_iwdt_registers();
00731  if (err != k_rx_ok) {
00732      return err;
00733  }
00734
00735  /*
00736  * Configure Count Stop Control Register (IWDTCSCTPR)
00737  *
00738  * Bit 7 SLCSTP - Sleep Mode Count Stop Control
00739  * 0 = Count continues during sleep
00740  * 1 = Count stops during sleep
00741  *
00742  * We continue counting during sleep for safety.
00743  */
00744  /*
00745  * Start the IWDT by performing first refresh
00746  *
00747  * The IWDT starts counting after the first refresh sequence.
00748  * After this, the watchdog CANNOT be stopped.
00749  */
00750  rx_hal_iwdt_feed();
00751
00752  s_iwdt_initialized = k_iwdt_initialized;
00753  return k_rx_ok;
00754
00755 #else
00756  /* Host-side stub for unit testing */
00757  (void)timeout_ms;
00758  s_iwdt_initialized = k_iwdt_initialized;
00759  return k_rx_ok;
00760 #endif
00761 }

```

References [internal_configure_iwdt_control_register\(\)](#), [internal_configure_iwdt_registers\(\)](#), [internal_find_timeout_config\(\)](#), [k_iwdt_initialized](#), [k_iwdt_timeout_max_ms](#), [k_iwdt_timeout_min_ms](#), [rx_hal_iwdt_feed\(\)](#), and [s_iwdt_initialized](#).

Here is the call graph for this function:



8.81.3.5 rx_hal_iwdt_was_reset()

```

bool rx_hal_iwdt_was_reset (
    void ) [nodiscard]

```

Check if last reset was caused by watchdog timeout.

Reads the IWDT status register to determine if the previous system reset was triggered by the watchdog timer.

Returns

true if watchdog caused the last reset
false if reset was from another source (power-on, external, etc.)

Note

Call this early in startup (before clearing status flags) to log or handle watchdog-induced resets appropriately.

Check if last reset was caused by watchdog timeout.

Reads the IWDTSR Status Register (IWDTSR) to determine if the previous system reset was caused by watchdog timeout (underflow) or refresh error. This function should be called early in startup for diagnostics.

Status Register Flags

- * IWDTSR Bit 14 (UNDF) - Underflow Flag:
 - * - 1: Counter reached 0 (timeout occurred)
 - * - 0: No underflow
- * IWDTSR Bit 15 (REFEF) - Refresh Error Flag:
 - * - 1: Invalid refresh sequence detected
 - * - 0: No refresh error

Precondition

Can be called before `rx_hal_iwdt_init()`
Status flags preserved across reset (hardware feature)

Postcondition

Status flags NOT cleared (call `rx_hal_iwdt_clear_status()` to clear)

Note

On host builds, always returns false
Call `rx_hal_iwdt_get_reset_cause()` for specific cause

Example:

```
// Check at startup
if (rx_hal_iwdt_was_reset()) {
    uart_debug_puts("[WARN] Recovered from watchdog reset!\r\n");
    const char* failed = rx_hal_iwdt_get_failed_task();
    if (failed != nullptr) {
        uart_debug_printf("Failed task: %s\r\n", failed);
    }
}
```

See also

[rx_hal_iwdt_get_reset_cause\(\)](#) Get specific reset cause
[rx_hal_iwdt_clear_status\(\)](#) Clear status flags after reading
[rx_hal_iwdt_get_failed_task\(\)](#) Get name of task that caused timeout

Since

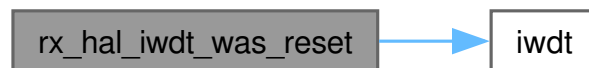
Version 1.0.0

Definition at line 901 of file [rx_iwdt.c](#).

```
00902 {
00903 #ifdef __RX__
00904 /*
00905  * Check IWDT Status Register for underflow or refresh error
00906  *
00907  * Bit 14 UNDF - Underflow Flag
00908  * 1 = Counter underflow occurred
00909  *
00910  * Bit 15 REFEF - Refresh Error Flag
00911  * 1 = Refresh error occurred
00912  */
00913 const uint16_t status = iwdt()->iwdtsr;
00914
00915 return ((status & k_iwdt_sr_undf) != 0) || ((status & k_iwdt_sr_refef) != 0);
00916
00917 #else
00918 return false;
00919 #endif
00920 }
```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undf](#).

Here is the call graph for this function:



8.82 rx_hal_iwdt.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_hal_iwdt.h
00003  * @brief Independent Watchdog Timer (IWDT) HAL Driver for RX72N
00004  *
00005  * @details
00006  * **Hardware Abstraction Layer** for the RX72N Independent Watchdog Timer (IWDT)
00007  * peripheral. This driver provides low-level register access and task-level
00008  * monitoring for safety-critical system recovery.
00009  *
00010  * ## Purpose and Design Rationale
00011  *
00012  * The IWDT provides **true clock-independent** watchdog functionality:
00013  * - Dedicated 120 kHz on-chip oscillator (IWDTCLK)
00014  * - Operates even if main system clock fails
00015  * - Cannot be disabled once started (hardware safety feature)
00016  * - Triggers system reset on timeout
00017  *
00018  * This module is the **HAL layer** providing direct hardware access. For
00019  * higher-level watchdog features (state-dependent timeouts, comprehensive
00020  * task monitoring), see `rx_core/inc/rx_iwdt.h`.
00021  *
00022  * ## Key Features
00023  *
00024  * **Hardware Watchdog**:
00025  * - Internal 120 kHz clock source (IWDTCLK)
00026  * - Configurable timeout: 128ms to 16,384ms
```

```

00027 * - Window mode (optional, disabled by default)
00028 * - Reset or NMI on timeout
00029 * - Refresh error detection
00030 *
00031 * Task Monitoring:
00032 * - Register up to 8 tasks for heartbeat monitoring
00033 * - Per-task timeout configuration
00034 * - Failed task identification for diagnostics
00035 *
00036 * Hardware Architecture
00037 *
00038 * @par Clock and Timing:
00039 *
00040 * | Parameter | Value | Notes |
00041 * |-----|-----|-----|
00042 * | Clock Source | IWDTCCLK | Internal RC oscillator |
00043 * | Clock Frequency | 120 kHz | +/-1% accuracy |
00044 * | Clock Period | 8.33 us | 1/120kHz |
00045 * | Counter Width | 14-bit | Down-counter |
00046 * | Min Timeout | 128 ms | 214 / (120k / 1) |
00047 * | Max Timeout | 16,384 ms | 214 / (120k / 128) |
00048 *
00049 * @par Register Map:
00050 *
00051 * | Register | Address | Description |
00052 * |-----|-----|-----|
00053 * | IWDTRR | 0x00088030 | Refresh register |
00054 * | IWDTR\|r | 0x00088032 | Control register |
00055 * | IWDTSR | 0x00088034 | Status register |
00056 * | IWDTR\|r | 0x00088036 | Reset control register |
00057 * | IWDTCSTPR | 0x00088038 | Count stop register |
00058 *
00059 * @par Timeout Calculation:
00060 *
00061 * @f
00062 *  $T_{\text{timeout}} = \frac{2^{14}}{\text{TOPS}} \times f_{\text{IWDTCCLK}}$ 
00063 * @f
00064 *
00065 * Where TOPS is the timeout period setting (1, 4, 16, 32, 64, or 128).
00066 *
00067 * Sub-clock Pin Handling
00068 *
00069 * The IWDTC uses the internal IWDTCCLK, not the sub-clock oscillator.
00070 * When sub-clock is unused:
00071 * - XCIN: Connect to VSS (ground) via pull-down resistor
00072 * - XCOUT: Leave open (unconnected)
00073 *
00074 * System initialization disables sub-clock via SOSC\|r.SOSTP = 1.
00075 *
00076 * Performance Characteristics
00077 *
00078 * | Operation | Execution Time | Notes |
00079 * |-----|-----|-----|
00080 * | rx_hal_iwdt_init() | ~20 us | One-time configuration |
00081 * | rx_hal_iwdt_feed() | ~1 us | Two register writes |
00082 * | rx_hal_iwdt_task_heartbeat() | ~3 us | Timestamp update |
00083 * | rx_hal_iwdt_check_tasks() | ~5 us x N tasks | Linear scan |
00084 *
00085 * Memory Footprint:
00086 * - Static variables: ~256 bytes (task tracking)
00087 * - Code size: ~1 KB
00088 * - Stack usage: < 32 bytes
00089 * - Heap usage: 0 bytes
00090 *
00091 * @par Hardware Requirements:
00092 *
00093 * | Component | Requirement | Notes |
00094 * |-----|-----|-----|
00095 * | MCU | Renesas RX72N | IWDTC peripheral |
00096 * | Clock | IWDTCCLK | 120 kHz internal |
00097 * | XCIN Pin | VSS via resistor | When sub-clock unused |
00098 * | XCOUT Pin | Open | When sub-clock unused |
00099 *
00100 * @par Module Dependencies:
00101 *
00102 * This module depends on:
00103 * - `rx_err.h` - Error code definitions
00104 * - `rx72n_iwdt_regs.h` - IWDTC register definitions
00105 *
00106 * This module is used by:
00107 * - `main.c` - System initialization
00108 * - All RTOS tasks - Heartbeat reporting
00109 * - `rx_core/inc/rx_iwdt.h` - Higher-level watchdog features
00110 *
00111 * @par NASA Power of 10 Compliance:
00112 *
00113 * | Rule | Status | Implementation |

```

```

00114 * |-----|-----|-----|
00115 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion |
00116 * | 2. Fixed loop bounds | [OK] | Loops bounded by k_iwtdt_max_tasks (8) |
00117 * | 3. No dynamic allocation | [OK] | Static task array only |
00118 * | 4. Small functions | [OK] | All < 60 lines |
00119 * | 5. Assertions | [OK] | Minimum 2 checks per function |
00120 * | 6. Narrow scope | [OK] | File-scope statics |
00121 * | 7. Check return values | [OK] | All return rx_err_t |
00122 * | 8. Limited preprocessor | [OK] | C23 typed enums only |
00123 * | 9. Pointer restrictions | [OK] | Single-level pointers |
00124 * | 10. Compiler warnings | [OK] | -Wall -Wextra -Werror |
00125 *
00126 * @par SOLID Principles:
00127 *
00128 * **Single Responsibility (S)**:
00129 * - HAL layer: Hardware register access only
00130 * - Task monitoring separate from hardware watchdog
00131 *
00132 * **Open/Closed (O)**:
00133 * - Timeout values configurable
00134 * - Task monitoring extensible
00135 *
00136 * **Dependency Inversion (D)**:
00137 * - Higher-level `rx_core/inc/rx_iwtdt.h` depends on this HAL interface
00138 *
00139 * @see rx_core/inc/rx_iwtdt.h Higher-level watchdog with state management
00140 * @see rx72n_iwtdt_regs.h Register definitions
00141 * @see RX72N Hardware Manual Chapter 35 - Independent Watchdog Timer
00142 *
00143 * @author Locked, Inc.
00144 * @date 2026-01-27
00145 * @copyright Copyright (c) 2026 Locked Inc.
00146 * SPDX-License-Identifier: MIT
00147 * @since Version 1.0.0
00148 */
00149
00150 #pragma once
00151
00152 #include <stdint.h>
00153
00154 #include "rx_err.h"
00155
00156 #ifdef __cplusplus
00157 extern "C" {
00158 #endif
00159
00160 /*
=====
00161 * Configuration Constants
00162 *
=====
00163 */
00164
00165 /**
00166 * @enum rx_iwtdt_timeout_t
00167 * @brief IWDT Timeout Period Options
00168 *
00169 * @details
00170 * Pre-defined timeout periods for the IWDT. These values correspond to
00171 * hardware divider settings that produce specific timeout durations from
00172 * the 120 kHz IWDTCLK.
00173 *
00174 * ## Timeout Selection Guide
00175 *
00176 * | Timeout | Use Case | Notes |
00177 * |-----|-----|-----|
00178 * | 128ms | High-frequency control loops (> 50 Hz) | Very responsive |
00179 * | 512ms | Normal control loops (10-50 Hz) | Fast detection |
00180 * | 1000ms | **Default** - typical applications | Good balance |
00181 * | 2048ms | Low-frequency tasks (1-10 Hz) | Tolerant |
00182 * | 8192ms | Slow background operations | Long timeout |
00183 * | 16384ms | Initialization/maintenance | Maximum |
00184 *
00185 * @par Timeout Calculation:
00186 *
00187 * Timeout periods are derived from:
00188 * @f[
00189 * T = \frac{\text{Cycle Count}}{f_{\text{IWDTCLK}}} = \frac{\text{Cycle Count}}{120\,\text{kHz}}
00190 * @f]
00191 *
00192 * @par Usage Example:
00193 * @code{.c}
00194 * // Use predefined timeout for 1 second
00195 * rx_hal_iwtdt_init(k_iwtdt_timeout_1000ms);
00196 *
00197 * // Or use custom value (rounded up to nearest supported)
00198 * rx_hal_iwtdt_init(500); // Rounds to 512ms

```

```

00199 * @endcode
00200 *
00201 * @invariant All values are in milliseconds
00202 * @invariant Values correspond to hardware divider settings
00203 *
00204 * @note C23 typed enum with uint16_t underlying type
00205 * @note Actual timeout may vary +/-1% due to oscillator tolerance
00206 *
00207 * @since Version 1.0.0
00208 */
00209 typedef enum : uint16_t {
00210     k_iwdt_timeout_128ms =
00211         128, /**< Minimum timeout (128ms). Use for high-frequency control loops requiring fast failure detection. TOPS=1
00212         divider */
00213         k_iwdt_timeout_512ms =
00214         512, /**< Short timeout (512ms). Use for normal control loops (10-50 Hz). Provides good responsiveness. TOPS=4
00215         divider */
00216         k_iwdt_timeout_1000ms =
00217         1000, /**< Default timeout (1 second). **Recommended** for typical applications. Good balance between detection
00218         speed and tolerance. TOPS=8 */
00219         k_iwdt_timeout_2048ms =
00220         2048, /**< Medium timeout (2 seconds). Use for low-frequency tasks that may have variable execution time. TOPS=16
00221         divider */
00222         k_iwdt_timeout_8192ms =
00223         8192, /**< Long timeout (8 seconds). Use for slow background operations or maintenance tasks. TOPS=64 divider */
00224         k_iwdt_timeout_16384ms =
00225         16384, /**< Maximum timeout (~16 seconds). Use only for initialization phases or very slow operations. TOPS=128
00226         divider */
00227 } rx_iwdt_timeout_t;
00228
00229 /**
00230 * @enum rx_iwdt_reset_cause_t
00231 * @brief IWDT Reset Cause Information
00232 *
00233 * @details
00234 * Identifies the specific cause of an IWDT-triggered reset. This information
00235 * is read from the IWDTSR status register and can be used for post-mortem
00236 * diagnostics and logging.
00237 *
00238 * ## Reset Cause Detection
00239 *
00240 * At startup, read the reset cause before calling rx_hal_iwdt_init():
00241 *
00242 * @code{.c}
00243 * void startup_diagnostics(void)
00244 * {
00245 *     rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
00246 *
00247 *     switch (cause) {
00248 *     case k_iwdt_reset_none:
00249 *         uart_debug_puts("Normal startup\\r\\n");
00250 *         break;
00251 *     case k_iwdt_reset_underflow:
00252 *         uart_debug_puts("[WARN] Watchdog timeout - task deadlock?\\r\\n");
00253 *         log_failed_task(rx_hal_iwdt_get_failed_task());
00254 *         break;
00255 *     case k_iwdt_reset_refresh_error:
00256 *         uart_debug_puts("[ERROR] Watchdog refresh error - code bug\\r\\n");
00257 *         break;
00258 *     }
00259 *
00260 *     rx_hal_iwdt_clear_status();
00261 * }
00262 * @endcode
00263 *
00264 * @par Reset Cause Table:
00265 *
00266 * | Cause | IWDTSR Bits | Meaning | Typical Issue |
00267 * |-----|-----|-----|-----|
00268 * | None | 0x0000 | Not a watchdog reset | Power-on, external reset |
00269 * | Underflow | UNDF=1 | Counter reached zero | Task deadlock, infinite loop |
00270 * | Refresh Error | REFEF=1 | Invalid refresh | Window violation, wrong sequence |
00271 *
00272 * @invariant Values map to IWDTSR bit patterns
00273 * @invariant k_iwdt_reset_none (0) is the default for non-watchdog resets
00274 *
00275 * @note C23 typed enum with uint8_t underlying type
00276 * @see rx_hal_iwdt_get_reset_cause() Read the reset cause
00277 * @see rx_hal_iwdt_clear_status() Clear status flags after reading
00278 *
00279 * @since Version 1.0.0
00280 */
00281 typedef enum : uint8_t {
00282     k_iwdt_reset_none =
00283         0, /**< No watchdog reset occurred. System reset was from power-on, external reset pin, or other source.
00284         IWDTSR.UNDF=0, REFEF=0 */
00285     k_iwdt_reset_underflow =

```

```

00280 1, /**< Reset due to counter underflow. Counter reached zero before rx_hal_iwdt_feed() was called. Indicates task
00281      k_iwdt_reset_refresh_error =
00282 2, /**< Reset due to invalid refresh. Either wrong refresh sequence (not 0x00, 0xFF) or refresh during
00283      window-prohibited period. IWDTSR.REFEF=1 */
00283 } rx_iwdt_reset_cause_t;
00284
00285 /*
=====
00286 * Initialization
00287 *
=====
00288 */
00289
00290 /**
00291  * @brief Initialize the Independent Watchdog Timer
00292  *
00293  * Configures IWDTSR with the specified timeout period. Once started, the
00294  * watchdog cannot be stopped and must be fed periodically to prevent reset.
00295  *
00296  * Configuration:
00297  * - Clock: 120 kHz internal IWDTSR-dedicated oscillator (IWDTSRCLK)
00298  * - Window mode: Disabled (refresh allowed anytime)
00299  * - Action: System reset on timeout
00300  * - Count stop in sleep: Disabled (continues counting)
00301  *
00302  * @param[in] timeout_ms Desired timeout in milliseconds.
00303  *               Actual timeout will be rounded up to nearest
00304  *               supported value (128, 512, 1024, 2048, 4096,
00305  *               8192, or 16384 ms).
00306  *
00307  * @return k_rx_ok on success
00308  * @return k_rx_err_invalid_arg if timeout is out of supported range
00309  * @return k_rx_err_invalid_state if IWDTSR is already running
00310  *
00311  * @note This function starts the watchdog - rx_hal_iwdt_feed() must be called
00312  *        within the timeout period to prevent reset.
00313  * @note On RX72N, IWDTSR starts counting after the first refresh operation.
00314  */
00315 [[nodiscard]] rx_err_t rx_hal_iwdt_init(uint32_t timeout_ms);
00316
00317 /*
=====
00318 * Runtime Operations
00319 *
=====
00320 */
00321
00322 /**
00323  * @brief Feed (refresh) the watchdog timer
00324  *
00325  * Resets the watchdog counter to prevent timeout. This function must be
00326  * called periodically, within the configured timeout period.
00327  *
00328  * The refresh sequence writes 0x00 followed by 0xFF to the IWDTSR register.
00329  * This sequence must complete atomically without interruption.
00330  *
00331  * @note Call this function from your main control loop.
00332  * @note Calling more frequently than needed has no negative effect.
00333  * @note Do NOT call from interrupt handlers unless you understand the
00334  *        implications for your system's safety.
00335  *
00336  * Example placement:
00337  * @code
00338  * // Motor control task at 100 Hz (10ms period)
00339  * // With 1000ms timeout, we have 100 chances to feed
00340  * void motor_control_task_entry(ULONG input) {
00341  *     while (1) {
00342  *         read_sensors();
00343  *         update_pid();
00344  *         set_motor_output();
00345  *         rx_hal_iwdt_feed(); // Feed at end of each iteration
00346  *         tx_thread_sleep(1); // 10ms at 100Hz tick
00347  *     }
00348  * }
00349  * @endcode
00350  */
00351 void rx_hal_iwdt_feed(void);
00352
00353 /*
=====
00354 * Diagnostics
00355 *
=====
00356 */
00357
00358 /**

```

```

00359 * @brief Check if last reset was caused by watchdog timeout
00360 *
00361 * Reads the IWDTCR status register to determine if the previous system
00362 * reset was triggered by the watchdog timer.
00363 *
00364 * @return true if watchdog caused the last reset
00365 * @return false if reset was from another source (power-on, external, etc.)
00366 *
00367 * @note Call this early in startup (before clearing status flags) to
00368 *       log or handle watchdog-induced resets appropriately.
00369 */
00370 [[nodiscard]] bool rx_hal_iwdt_was_reset(void);
00371
00372 /**
00373 * @brief Get detailed reset cause from IWDTCR
00374 *
00375 * Provides more detailed information about what caused a watchdog reset:
00376 * - Underflow: Counter reached zero (normal timeout)
00377 * - Refresh error: Invalid refresh sequence or window violation
00378 *
00379 * @return Reset cause enumeration
00380 *
00381 * @note This provides more detail than rx_hal_iwdt_was_reset() for diagnostics.
00382 */
00383 rx_iwdt_reset_cause_t rx_hal_iwdt_get_reset_cause(void);
00384
00385 /**
00386 * @brief Clear watchdog status flags
00387 *
00388 * Clears the underflow and refresh error flags in IWDTCR.
00389 * Call after reading the reset cause to prepare for next reset detection.
00390 *
00391 * @note Flags are typically cleared once at startup after logging the cause.
00392 */
00393 void rx_hal_iwdt_clear_status(void);
00394
00395 /*
=====
00396 * Task-Level Monitoring (Issue 20: Enhanced Watchdog)
00397 *
=====
00398 */
00399
00400 /**
00401 * @brief Maximum number of tasks that can be monitored
00402 */
00403 typedef enum : uint8_t {
00404     k_iwdt_max_tasks = 8, /**< Support up to 8 monitored tasks */
00405 } rx_iwdt_limits_t;
00406
00407 /**
00408 * @brief Register a task for heartbeat monitoring
00409 *
00410 * Registers a task with the watchdog system to enable deadlock detection.
00411 * Each task must call rx_hal_iwdt_task_heartbeat() periodically to indicate
00412 * it is still running.
00413 *
00414 * @param[in] task_name Name of the task (for logging)
00415 * @param[in] timeout_ms Maximum time between heartbeats before considering
00416 *                       the task dead (typically 3x the task period)
00417 *
00418 * @return k_rx_ok on success
00419 * @return k_rx_err_invalid_arg if task_name is nullptr
00420 * @return k_rx_err_no_mem if maximum tasks already registered
00421 *
00422 * @note Call during task initialization before entering main loop
00423 *
00424 * Example:
00425 * @code
00426 * // Motor_Controller runs at 250 Hz (4ms period)
00427 * // Set timeout to 3x period = 12ms
00428 * rx_hal_iwdt_register_task("Motor_Controller", 12);
00429 * @endcode
00430 */
00431 [[nodiscard]] rx_err_t rx_hal_iwdt_register_task(const char* task_name, uint32_t timeout_ms);
00432
00433 /**
00434 * @brief Record a task heartbeat
00435 *
00436 * Updates the last heartbeat timestamp for the calling task. Tasks must
00437 * call this function periodically (within their registered timeout) to
00438 * prove they are still alive.
00439 *
00440 * @param[in] task_name Name of the task (must match registration)
00441 *
00442 * @note Call this at the end of each task iteration
00443 * @note If any task misses 3+ heartbeats, watchdog will eventually trigger

```



```

00444 *
00445 * Example:
00446 * @code
00447 * void comm_manager_entry(ULONG input) {
00448 *     rx_hal_iwdt_register_task("Comm_Manager", 30);
00449 *     while (1) {
00450 *         process_ingress();
00451 *         process_egress();
00452 *         rx_hal_iwdt_task_heartbeat("Comm_Manager");
00453 *         tx_thread_sleep(1);
00454 *     }
00455 * }
00456 * @endcode
00457 */
00458 void rx_hal_iwdt_task_heartbeat(const char* task_name);
00459 /**
00460 * @brief Check for task deadlocks
00461 *
00462 * Examines all registered tasks to see if any have exceeded their heartbeat
00463 * timeout. Should be called periodically (e.g., from lowest priority task).
00464 *
00465 * @return k_rx_ok if all tasks healthy
00466 * @return k_rx_err_timeout if one or more tasks have deadlocked
00467 *
00468 * @note This function logs which task(s) failed before returning
00469 * @note The hardware IWDT will eventually reset the system if deadlock persists
00470 */
00471 [[nodiscard]] rx_err_t rx_hal_iwdt_check_tasks(void);
00472 /**
00473 * @brief Get the name of the last failed task (if any)
00474 *
00475 * Returns the name of the task that failed to heartbeat before the last
00476 * watchdog reset. Useful for post-mortem diagnostics.
00477 *
00478 * @return Task name if reset was due to task deadlock, NULL otherwise
00479 *
00480 * @note This information persists across reset in a special RAM region
00481 * @note Call early in startup to retrieve failure information
00482 */
00483 const char* rx_hal_iwdt_get_failed_task(void);
00484 #ifdef __cplusplus
00485 }
00486 #endif /* STAR_RX_IWDT_H */

```

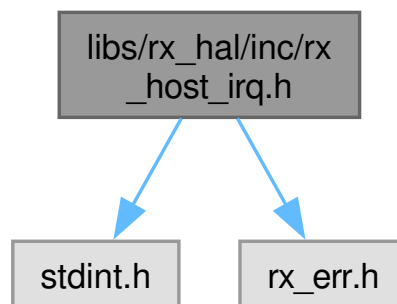
8.83 libs/rx_hal/inc/rx_host_irq.h File Reference

HOST_IRQ GPIO Output Driver for RPi5 Data-Ready Signaling.

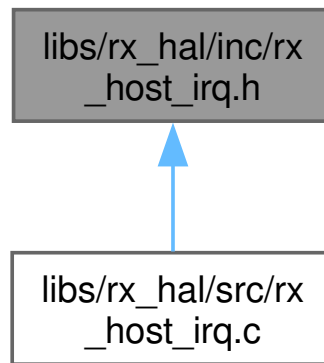
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_host_irq.h:



This graph shows which files directly or indirectly include this file:



Functions

- `rx_err_t rx_host_irq_init (void)`
Initialize HOST_IRQ pin as GPIO output (deasserted / HIGH).
- `rx_err_t rx_host_irq_deinit (void)`
Deinitialize HOST_IRQ pin (revert to input).
- `rx_err_t rx_host_irq_assert (void)`
Assert HOST_IRQ (drive LOW to signal data ready).
- `rx_err_t rx_host_irq_deassert (void)`
Deassert HOST_IRQ (drive HIGH, idle state).
- `rx_err_t rx_host_irq_is_asserted (bool *asserted)`
Check if HOST_IRQ is currently asserted.

8.83.1 Detailed Description

HOST_IRQ GPIO Output Driver for RPi5 Data-Ready Signaling.

Manages the HOST_IRQ output pin (P67, pin 98) used to signal the Raspberry Pi 5 host controller that the RX72N has data ready for SPI transfer. The signal is active-low (assert LOW to request attention, deassert HIGH when idle).

8.83.1.1 Hardware Configuration

Parameter	Value
Pin	P67 (PORT6, pin 7, package pin 98)
Direction	Output (RX72N -> RPi5)
Active Level	Active-low (asserted = LOW)
Default State	HIGH (deasserted / idle)
Drive	CMOS push-pull

8.83.1.2 Signaling Protocol

```
RX72N has telemetry data ready
|
rx_host_irq_assert()  -> P67 driven LOW
|
RPI5 detects falling edge on its GPIO
|
RPI5 initiates SPI read
|
rx_host_irq_deassert()  -> P67 driven HIGH
```

NASA Power of 10 Compliance:

- Rule 1: No goto, setjmp, recursion
- Rule 3: No dynamic memory (static variables only)
- Rule 4: All functions < 60 lines
- Rule 5: 2+ validation checks per function
- Rule 7: All return values checked
- Rule 8: C23 typed enums, no magic numbers
- Rule 10: -Wall -Wextra -Werror clean

See also

`rx_spi_comm.h` SPI communication driver
`hardware_config.h` Pin assignments (`host_irq_pins` group)

Author

Locked, Inc.

Date

2026-02-10

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Since

Version 1.0.0

Definition in file [rx_host_irq.h](#).

8.83.2 Function Documentation

8.83.2.1 `rx_host_irq_assert()`

```
rx_err_t rx_host_irq_assert (
    void ) [nodiscard]
```

Assert HOST_IRQ (drive LOW to signal data ready).

Drives P67 LOW to signal the RPi5 that the RX72N has data ready for SPI transfer. The RPi5 should respond by initiating an SPI read.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, pin driven LOW
<code>k_rx_err_invalid_state</code>	Not initialized

Precondition

[rx_host_irq_init\(\)](#) called successfully

Postcondition

P67 driven LOW (active)

Note

Thread-safe: single register write is atomic on RX72N

See also

[rx_host_irq_deassert\(\)](#) Release the signal

Since

Version 1.0.0

Assert HOST_IRQ (drive LOW to signal data ready).

Drives the P67 output LOW to signal the host SoC that the RX72N has data ready or wants attention. HOST_IRQ is an active-low, level-triggered line; the host will continue to see "asserted" until [rx_host_irq_deassert\(\)](#) drives it back HIGH. The function is a single register read-modify-write on PORT6.PODR; no interrupts are touched.

Note

Safe to call from task context. Not protected against concurrent calls; serialize at the application layer.

Definition at line 148 of file [rx_host_irq.c](#).

```
00149 {
00150     if (!s_initialized) {
00151         return k_rx_err_invalid_state;
00152     }
00153
00154     volatile rx_port_regs_t* p6 = port6();
00155     p6->podr &= (uint8_t)~k_host_irq_pin_mask;
00156
00157     return k_rx_ok;
00158 }
```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::podr](#), [port6\(\)](#), and [s_initialized](#).

Here is the call graph for this function:



8.83.2.2 rx_host_irq_deassert()

```
rx_err_t rx_host_irq_deassert (
    void ) [nodiscard]
```

Deassert HOST_IRQ (drive HIGH, idle state).

Drives P67 HIGH (idle). Call after the RPi5 has completed its SPI read.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, pin driven HIGH
<code>k_rx_err_invalid_state</code>	Not initialized

Precondition

[rx_host_irq_init\(\)](#) called successfully

Postcondition

P67 driven HIGH (idle)

Note

Thread-safe: single register write is atomic on RX72N

See also

[rx_host_irq_assert\(\)](#) Assert the signal

Since

Version 1.0.0

Deassert HOST_IRQ (drive HIGH, idle state).

Drives the P67 output HIGH to release the active-low HOST_IRQ line back to its idle state. Single register read-modify-write on PORT6.PODR; no interrupts touched.

Note

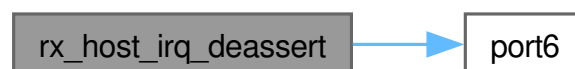
Safe to call from task context. Not protected against concurrent calls; serialize at the application layer.

Definition at line 171 of file [rx_host_irq.c](#).

```
00172 {
00173     if (!s_initialized) {
00174         return k_rx_err_invalid_state;
00175     }
00176
00177     volatile rx_port_regs_t* p6 = port6();
00178     p6->podr |= k_host_irq_pin_mask;
00179
00180     return k_rx_ok;
00181 }
```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::podr](#), [port6\(\)](#), and [s_initialized](#).

Here is the call graph for this function:



8.83.2.3 rx_host_irq_deinit()

```
rx_err_t rx_host_irq_deinit (
    void ) [nodiscard]
```

Deinitialize HOST_IRQ pin (revert to input).

Reverts P67 to input mode (safe default) and deasserts the output.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success
<code>k_rx_err_invalid_state</code>	Not initialized

Precondition

[rx_host_irq_init\(\)](#) called successfully

Postcondition

P67 reverted to input, no longer driving

Since

Version 1.0.0

Deinitialize HOST_IRQ pin (revert to input).

Symmetric counterpart to [rx_host_irq_init\(\)](#). Drives P67 high one last time so the line is left in its idle (de-asserted) state, then changes the pin direction back to input, gating off any spurious downstream effects. Clears the `s_initialized` flag so subsequent assert/deassert calls return `k_rx_err_invalid_state`.

Note

Not thread-safe. Serialize all `rx_host_irq_*` operations at the application layer.

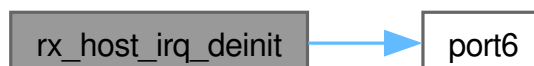
Definition at line 114 of file [rx_host_irq.c](#).

```

00115 {
00116     if (!s_initialized) {
00117         rx_log_error(s_tag, "Not initialized");
00118         return k_rx_err_invalid_state;
00119     }
00120
00121     volatile rx_port_regs_t* p6 = port6();
00122
00123     /* Deassert before reverting to input */
00124     p6->podr |= k_host_irq_pin_mask;
00125
00126     /* Revert to input (safe default) */
00127     p6->pdr &= (uint8_t)~k_host_irq_pin_mask;
00128
00129     s_initialized = false;
00130
00131     rx_log_info(s_tag, "HOST_IRQ P67 deinitialized (input)");
00132     return k_rx_ok;
00133 }
```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::podr](#), [port6\(\)](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.83.2.4 rx_host_irq_init()

```
rx_err_t rx_host_irq_init (
    void ) [nodiscard]
```

Initialize HOST_IRQ pin as GPIO output (deasserted / HIGH).

Configures P67 as a GPIO output in CMOS push-pull mode. Sets the initial output level to HIGH (deasserted / idle). The MPC function select is set to GPIO (not IRQ input).

Initialization steps:

1. Clear PMR bit 7 (GPIO mode, not peripheral)
2. Set PODR bit 7 HIGH (deasserted state)
3. Set PDR bit 7 (output direction)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, P67 configured as output
k_rx_err_invalid_state	Already initialized

Precondition

P67 not in use by another peripheral (RIIC2 etc.)

Postcondition

P67 configured as GPIO output, driven HIGH

Note

Not thread-safe. Call during system init only.

See also

[rx_host_irq_deinit\(\)](#) Disable the output

Since

Version 1.0.0

Definition at line 77 of file [rx_host_irq.c](#).

```

00078 {
00079     if (s_initialized) {
00080         rx_log_error(s_tag, "Already initialized");
00081         return k_rx_err_invalid_state;
00082     }
00083
00084     volatile rx_port_regs_t* p6 = port6();
00085
00086     /* Step 1: GPIO mode (clear PMR bit 7) */
00087     p6->pmr &= (uint8_t)~k_host_irq_pin_mask;
00088
00089     /* Step 2: Set output HIGH (deasserted / idle) before enabling output */
00090     p6->podr |= k_host_irq_pin_mask;
00091
00092     /* Step 3: Set direction to output */
00093     p6->pdr |= k_host_irq_pin_mask;
00094
00095     s_initialized = true;
00096
00097     rx_log_info(s_tag, "HOST_IRQ P67 initialized (output, idle HIGH)");
00098     return k_rx_ok;
00099 }

```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_port_regs_t::podr](#), [port6\(\)](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.83.2.5 rx_host_irq_is_asserted()

```

rx_err_t rx_host_irq_is_asserted (
    bool * asserted) [nodiscard]

```

Check if HOST_IRQ is currently asserted.

Returns the current state of the HOST_IRQ output.

Parameters

out	asserted	Pointer to store state (true = asserted / LOW)
-----	----------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	asserted is nullptr

Precondition

`rx_host_irq_init()` called

Postcondition

asserted set to current output state

Note

Thread-safe (reads volatile register)

Since

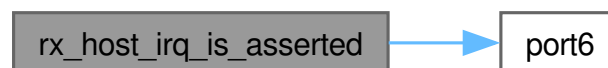
Version 1.0.0

Definition at line 183 of file `rx_host_irq.c`.

```
00184 {
00185     if (asserted == nullptr) {
00186         return k_rx_err_null_ptr;
00187     }
00188
00189     volatile rx_port_regs_t* p6 = port6();
00190
00191     /* Active-low: pin LOW means asserted */
00192     *asserted = ((p6->podr & k_host_irq_pin_mask) == k_host_irq_active);
00193
00194     return k_rx_ok;
00195 }
```

References `k_host_irq_active`, `k_host_irq_pin_mask`, `rx_port_regs_t::podr`, and `port6()`.

Here is the call graph for this function:



8.84 rx_host_irq.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_host_irq.h
00003  * @brief HOST_IRQ GPIO Output Driver for RPi5 Data-Ready Signaling
00004  *
00005  * @details
00006  * Manages the HOST_IRQ output pin (P67, pin 98) used to signal the Raspberry
00007  * Pi 5 host controller that the RX72N has data ready for SPI transfer. The
00008  * signal is active-low (assert LOW to request attention, deassert HIGH when
00009  * idle).
00010  *
00011  * ## Hardware Configuration
00012  *
00013  * | Parameter | Value |
00014  * |-----|-----|
00015  * | Pin | P67 (PORT6, pin 7, package pin 98) |
00016  * | Direction | Output (RX72N -> RPi5) |
00017  * | Active Level | Active-low (asserted = LOW) |
00018  * | Default State | HIGH (deasserted / idle) |
00019  * | Drive | CMOS push-pull |
00020  *
00021  * ## Signaling Protocol
00022  *
00023  * ```
```

```

00024 * RX72N has telemetry data ready
00025 * |
00026 * rx_host_irq_assert() -> P67 driven LOW
00027 * |
00028 * RPi5 detects falling edge on its GPIO
00029 * |
00030 * RPi5 initiates SPI read
00031 * |
00032 * rx_host_irq_deassert() -> P67 driven HIGH
00033 * ```
00034 *
00035 * @par NASA Power of 10 Compliance:
00036 * - Rule 1: No goto, setjmp, recursion
00037 * - Rule 3: No dynamic memory (static variables only)
00038 * - Rule 4: All functions < 60 lines
00039 * - Rule 5: 2+ validation checks per function
00040 * - Rule 7: All return values checked
00041 * - Rule 8: C23 typed enums, no magic numbers
00042 * - Rule 10: -Wall -Wextra -Werror clean
00043 *
00044 * @see rx_spi_comm.h SPI communication driver
00045 * @see hardware_config.h Pin assignments (host_irq_pins group)
00046 *
00047 * @author Locked, Inc.
00048 * @date 2026-02-10
00049 * @copyright Copyright (c) 2026 Locked Inc.
00050 * SPDX-License-Identifier: MIT
00051 * @since Version 1.0.0
00052 */
00053
00054 #pragma once
00055
00056 #include <stdint.h>
00057
00058 #include "rx_err.h"
00059
00060 #ifdef __cplusplus
00061 extern "C" {
00062 #endif
00063
00064 /*
=====
00065 * Public API
00066 *
=====
00067 */
00068
00069 /**
00070 * @brief Initialize HOST_IRQ pin as GPIO output (deasserted / HIGH)
00071 *
00072 * @details
00073 * Configures P67 as a GPIO output in CMOS push-pull mode. Sets the initial
00074 * output level to HIGH (deasserted / idle). The MPC function select is set
00075 * to GPIO (not IRQ input).
00076 *
00077 * Initialization steps:
00078 * 1. Clear PMR bit 7 (GPIO mode, not peripheral)
00079 * 2. Set PODR bit 7 HIGH (deasserted state)
00080 * 3. Set PDR bit 7 (output direction)
00081 *
00082 * @return rx_err_t Error code
00083 * @retval k_rx_ok Success, P67 configured as output
00084 * @retval k_rx_err_invalid_state Already initialized
00085 *
00086 * @pre P67 not in use by another peripheral (RIIC2 etc.)
00087 * @post P67 configured as GPIO output, driven HIGH
00088 *
00089 * @note Not thread-safe. Call during system init only.
00090 *
00091 * @see rx_host_irq_deinit() Disable the output
00092 * @since Version 1.0.0
00093 */
00094 [[nodiscard]] rx_err_t rx_host_irq_init(void);
00095
00096 /**
00097 * @brief Deinitialize HOST_IRQ pin (revert to input)
00098 *
00099 * @details
00100 * Reverts P67 to input mode (safe default) and deasserts the output.
00101 *
00102 * @return rx_err_t Error code
00103 * @retval k_rx_ok Success
00104 * @retval k_rx_err_invalid_state Not initialized
00105 *
00106 * @pre rx_host_irq_init() called successfully
00107 * @post P67 reverted to input, no longer driving
00108 */

```

```

00109 * @since Version 1.0.0
00110 */
00111 [[nodiscard]] rx_err_t rx_host_irq_deinit(void);
00112
00113 /**
00114 * @brief Assert HOST_IRQ (drive LOW to signal data ready)
00115 *
00116 * @details
00117 * Drives P67 LOW to signal the RPi5 that the RX72N has data ready for
00118 * SPI transfer. The RPi5 should respond by initiating an SPI read.
00119 *
00120 * @return rx_err_t Error code
00121 * @retval k_rx_ok Success, pin driven LOW
00122 * @retval k_rx_err_invalid_state Not initialized
00123 *
00124 * @pre rx_host_irq_init() called successfully
00125 * @post P67 driven LOW (active)
00126 *
00127 * @note Thread-safe: single register write is atomic on RX72N
00128 *
00129 * @see rx_host_irq_deassert() Release the signal
00130 * @since Version 1.0.0
00131 */
00132 [[nodiscard]] rx_err_t rx_host_irq_assert(void);
00133
00134 /**
00135 * @brief Deassert HOST_IRQ (drive HIGH, idle state)
00136 *
00137 * @details
00138 * Drives P67 HIGH (idle). Call after the RPi5 has completed its SPI read.
00139 *
00140 * @return rx_err_t Error code
00141 * @retval k_rx_ok Success, pin driven HIGH
00142 * @retval k_rx_err_invalid_state Not initialized
00143 *
00144 * @pre rx_host_irq_init() called successfully
00145 * @post P67 driven HIGH (idle)
00146 *
00147 * @note Thread-safe: single register write is atomic on RX72N
00148 *
00149 * @see rx_host_irq_assert() Assert the signal
00150 * @since Version 1.0.0
00151 */
00152 [[nodiscard]] rx_err_t rx_host_irq_deassert(void);
00153
00154 /**
00155 * @brief Check if HOST_IRQ is currently asserted
00156 *
00157 * @details
00158 * Returns the current state of the HOST_IRQ output.
00159 *
00160 * @param[out] asserted Pointer to store state (true = asserted / LOW)
00161 *
00162 * @return rx_err_t Error code
00163 * @retval k_rx_ok Success
00164 * @retval k_rx_err_null_ptr asserted is nullptr
00165 *
00166 * @pre rx_host_irq_init() called
00167 * @post asserted set to current output state
00168 *
00169 * @note Thread-safe (reads volatile register)
00170 *
00171 * @since Version 1.0.0
00172 */
00173 [[nodiscard]] rx_err_t rx_host_irq_is_asserted(bool* asserted);
00174
00175 #ifdef __cplusplus
00176 }
00177 #endif

```

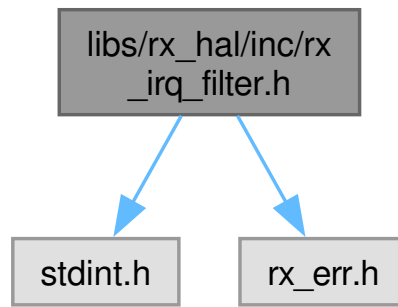
8.85 libs/rx/hal/inc/rx_irq_filter.h File Reference

IRQ Digital Filter Driver API for RX72N.

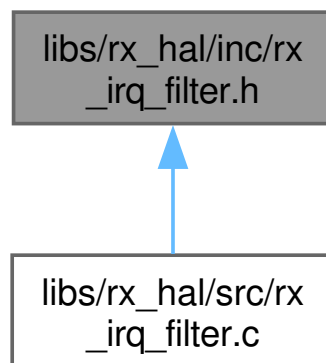
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_irq_filter.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_irq_filter_limits_t` : `uint8_t` { `k_rx_irq_max` = 15 }
IRQ filter limits for RX72N.
- enum `rx_irq_filter_clk_t` : `uint8_t` { `k_irq_filter_pclk_1` = 0 , `k_irq_filter_pclk_8` = 1 , `k_irq_filter_pclk_32` = 2 , `k_irq_filter_pclk_64` = 3 }
IRQ filter clock divisor options for sampling rate control.

Functions

- `rx_err_t rx_irq_filter_enable` (`uint8_t` irq_num, `rx_irq_filter_clk_t` filter_clk)
Enable hardware digital filter on an IRQ pin.
- `rx_err_t rx_irq_filter_disable` (`uint8_t` irq_num)
Disable hardware digital filter on an IRQ pin.
- `rx_err_t rx_irq_filter_is_enabled` (`uint8_t` irq_num, `bool` *enabled)
Check if digital filter is enabled on an IRQ pin.
- `rx_err_t rx_irq_filter_get_clock` (`uint8_t` irq_num, `rx_irq_filter_clk_t` *filter_clk)
Get current filter clock divisor setting for an IRQ.

Variables

- static const `uint8_t` `s_irq_max` = 15U
Maximum valid IRQ number (0-15).

Filter Response Times (PCLKB = 60 MHz)

Divisor	Sample Period	Response Time	Noise Rejection
PCLK/ $\leftarrow$ 1	16.7 ns	50 ns	Minimal
PCLK/ $\leftarrow$ 8	133 ns	400 ns	Fast signals
PCLK/32	533 ns	1.6 μ s	Moderate
PCLK/64	1.07 μ s	3.2 μ s	Buttons/switches

STAR Project Usage

IRQ Pin	Function	Recommended Filter	Notes
IRQ0	E-STOP Button	PCLK/64	Max filtering for safety
IRQ5	USB VBUS	PCLK/32	Moderate noise rejection
IRQ8	Expansion	PCLK/32	General-purpose interrupt

Hardware Requirements

Resource	Requirement	Notes
PCLKB	Active	Filter clock source
ICU	Enabled	Not in module stop

Memory Usage

Category	Size	Notes
Code	~200 bytes	All functions
Static data	0 bytes	No static variables
Stack	8 bytes	Per function call

Thread Safety

Not thread-safe. ICU register access is not atomic. Use mutex protection if filter configuration changes at runtime. Recommended: configure filters once during initialization before starting RTOS.

Usage Example - Basic

```
#include "rx_irq_filter.h"

void configure_interrupts(void)
{
    // Enable maximum filtering on E-STOP button (safety-critical)
    rx_irq_filter_enable(0, k_irq_filter_pclk_64);

    // Enable moderate filtering on VBUS detection
    rx_irq_filter_enable(5, k_irq_filter_pclk_32);

    // Configure ICU for edge detection (separate from filtering)
    // ...
}
```

Usage Example - Dynamic Configuration

```
void adjust_filter_for_mode(bool high_noise_environment)
{
    if (high_noise_environment) {
        // Increase filtering in noisy environments
        rx_irq_filter_enable(8, k_irq_filter_pclk_64);
    } else {
        // Reduce filtering for faster response
        rx_irq_filter_enable(8, k_irq_filter_pclk_8);
    }
}

bool check_filter_config(uint8_t irq_num)
{
    bool enabled;
    rx_irq_filter_clk_t clock;

    rx_irq_filter_is_enabled(irq_num, &enabled);
    rx_irq_filter_get_clock(irq_num, &clock);

    return enabled && (clock == k_irq_filter_pclk_64);
}
```

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, longjmp, recursion
- Rule 2: [OK] No loops (simple register operations)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 30 lines
- Rule 5: [OK] Input validation on all functions
- Rule 7: [OK] All return values are error codes
- Rule 8: [OK] C23 typed enums for filter clock options
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: Only IRQ filter configuration
- Interface Segregation: Minimal 4-function API

See also

[rx72n_icu_regs.h](#) ICU register definitions (IRQFLTE, IRQFLTC)
 RX72N Hardware Manual, Chapter 13 - Interrupt Controller Unit

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_irq_filter.h](#).

8.85.2 Enumeration Type Documentation

8.85.2.1 rx`irq`filter`clk`t

enum [rx_irq_filter_clk_t](#) : uint8_t

IRQ filter clock divisor options for sampling rate control.

Selects the clock divisor for IRQ digital filter sampling. Higher divisors result in slower sampling, providing more noise rejection but also longer response times to valid transitions.

The digital filter requires 3 consecutive samples at the same level to recognize a valid transition. Total response time = 3 x sample period.

Response Time Calculation

$$T_{response} = 3 \times \frac{1}{f_{PCLKB}/divisor} = \frac{3 \times divisor}{f_{PCLKB}}$$

Filter Characteristics (PCLKB = 60 MHz)

Divisor	Sample Period	Response Time	Best For
/↵ 1	16.7 ns	50 ns	High-speed signals
/↵ 8	133 ns	400 ns	Encoder pulses
/32	533 ns	1.6 us	Sensor interrupts

Divisor	Sample Period	Response Time	Best For
/64	1.07 us	3.2 us	Buttons, switches

Noise Rejection Capability

Noise pulses shorter than 2 x sample period are rejected. For PCLK/64 with PCLKB=60MHz, noise pulses < 2.1us are filtered out.

Note

Register encoding: 2 bits per IRQ in IRQFLTC register

Warning

Lower divisors provide faster response but less noise immunity

See also

[rx_irq_filter_enable\(\)](#) Set filter clock divisor

[rx_irq_filter_get_clock\(\)](#) Read current divisor setting

Since

Version 1.0.0

Enumerator

k_irq_filter_pclk_1	PCLKB/1 - Fastest sampling, minimal filtering. Sample period: 16.7 ns @ 60 MHz PCLKB Response time: 50 ns (3 samples) Noise rejection: < 33 ns pulses filtered Use Case: High-frequency signals requiring minimal delay Warning Provides almost no noise filtering
k_irq_filter_pclk_8	PCLKB/8 - Fast sampling, light filtering. Sample period: 133 ns @ 60 MHz PCLKB Response time: 400 ns (3 samples) Noise rejection: < 267 ns pulses filtered Use Case: Encoder signals, fast sensors

k_irq_filter_pclk_32	PCLKB/32 - Moderate sampling, good filtering. Sample period: 533 ns @ 60 MHz PCLKB Response time: 1.6 us (3 samples) Noise rejection: < 1.07 us pulses filtered Use Case: External interrupts, VBUS detection, expansion inputs
k_irq_filter_pclk_64	PCLKB/64 - Slowest sampling, maximum filtering. Sample period: 1.07 us @ 60 MHz PCLKB Response time: 3.2 us (3 samples) Noise rejection: < 2.1 us pulses filtered Use Case: Mechanical buttons, E-STOP, user input switches Note Recommended for all human-activated inputs

Definition at line 224 of file [rx_irq_filter.h](#).

```

00224         : uint8_t {
00225     /**
00226      * @brief PCLKB/1 - Fastest sampling, minimal filtering
00227      * @details
00228      * Sample period: 16.7 ns @ 60 MHz PCLKB
00229      * Response time: 50 ns (3 samples)
00230      * Noise rejection: < 33 ns pulses filtered
00231      * @par Use Case: High-frequency signals requiring minimal delay
00232      * @warning Provides almost no noise filtering
00233      */
00234     k_irq_filter_pclk_1 = 0,
00235
00236     /**
00237      * @brief PCLKB/8 - Fast sampling, light filtering
00238      * @details
00239      * Sample period: 133 ns @ 60 MHz PCLKB
00240      * Response time: 400 ns (3 samples)
00241      * Noise rejection: < 267 ns pulses filtered
00242      * @par Use Case: Encoder signals, fast sensors
00243      */
00244     k_irq_filter_pclk_8 = 1,
00245
00246     /**
00247      * @brief PCLKB/32 - Moderate sampling, good filtering
00248      * @details
00249      * Sample period: 533 ns @ 60 MHz PCLKB
00250      * Response time: 1.6 us (3 samples)
00251      * Noise rejection: < 1.07 us pulses filtered
00252      * @par Use Case: External interrupts, VBUS detection, expansion inputs
00253      */
00254     k_irq_filter_pclk_32 = 2,
00255
00256     /**
00257      * @brief PCLKB/64 - Slowest sampling, maximum filtering
00258      * @details
00259      * Sample period: 1.07 us @ 60 MHz PCLKB
00260      * Response time: 3.2 us (3 samples)
00261      * Noise rejection: < 2.1 us pulses filtered
00262      * @par Use Case: Mechanical buttons, E-STOP, user input switches
00263      * @note Recommended for all human-activated inputs
00264      */
00265     k_irq_filter_pclk_64 = 3,
00266 } rx_irq_filter_clk_t;

```

8.85.2.2 rx`irq`filter`limits`

```
enum rx_irq_filter_limits_t : uint8_t
```

IRQ filter limits for RX72N.

The RX72N ICU supports external IRQ pins IRQ0-IRQ15 (16 total).

Enumerator

k_rx_irq_max	
--------------	--

Definition at line 183 of file [rx_irq_filter.h](#).

```
00183     : uint8_t {
00184     k_rx_irq_max = 15,
00185 } rx_irq_filter_limits_t;
```

8.85.3 Function Documentation

8.85.3.1 rx_irq_filter_disable()

```
rx_err_t rx_irq_filter_disable (
    uint8_t irq_num) [nodiscard]
```

Disable hardware digital filter on an IRQ pin.

Disables the ICU digital filter for the specified IRQ pin. After disabling, the raw unfiltered input signal is passed through to the interrupt detector.

Algorithm Steps

1. Validate irq_num is in range [0, 15]
2. Calculate register index
3. Clear enable bit in IRQFLTE register

Parameters

in	irq_num	IRQ number to configure
		Valid range: [0, 15]

Returns

Error code indicating success or failure

Return values

k_rx_ok	Filter successfully disabled
k_rx_err_invalid_arg	irq_num > 15

Precondition

None (can be called regardless of current filter state)

Postcondition

IRQFLTE bit cleared for this IRQ
Raw input passed to interrupt detector

Note

Clock divisor setting is preserved (unchanged)
Safe to call even if filter was already disabled

Thread Safety

Not thread-safe. ICU register modification is not atomic.

Example

```
// Disable filter for faster response
rx_irq_filter_disable(5);
```

See also

[rx_irq_filter_enable\(\)](#) Enable filter

Since

Version 1.0.0

Clears the filter enable bit in IRQFLTE for the specified IRQ. After disabling, raw unfiltered input passes to interrupt detector.

Note

Clock divisor setting preserved for potential re-enable

See also

[rx_irq_filter.h](#) Full API documentation

Since

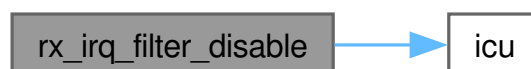
Version 1.0.0

Definition at line 236 of file [rx_irq_filter.c](#).

```
00237 {
00238     if (irq_num > k_rx_irq_max) {
00239         return k_rx_err_invalid_arg;
00240     }
00241
00242     #ifdef __RX__
00243     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00244     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00245
00246     /* Clear filter enable bit */
00247     icu()->irqfite[reg_idx] &= (uint8_t) ~(1 « bit_pos);
00248
00249     #endif
00250
00251     return k_rx_ok;
00252 }
```

References [icu\(\)](#), [rx_icu_regs_t::irqfite](#), [k_irq_filter_irqs_per_reg](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.85.3.2 rx_irq_filter_enable()

```
rx_err_t rx_irq_filter_enable (
    uint8_t irq_num,
    rx_irq_filter_clk_t filter_clk)  [nodiscard]
```

Enable hardware digital filter on an IRQ pin.

Enables the ICU digital filter for the specified IRQ pin with the given clock divisor. The filter samples the input and requires 3 consecutive samples at the same level to recognize a valid edge transition.

Algorithm Steps

1. Validate irq_num is in range [0, 15]
2. Validate filter_clk is valid divisor
3. Calculate register index (0 for IRQ0-7, 1 for IRQ8-15)
4. Set clock divisor in IRQFLTC register
5. Set enable bit in IRQFLTE register

Parameters

in	irq_num	IRQ number to configure - Valid range: [0, 15] - IRQ0-7 use IRQFLTE[0]/IRQFLTC[0] - IRQ8-15 use IRQFLTE[1]/IRQFLTC[1]
in	filter_clk	Clock divisor for filter sampling - k_irq_filter_pclk_1: Minimal filtering - k_irq_filter_pclk_64: Maximum filtering

Returns

Error code indicating success or failure

Return values

k_rx_ok	Filter successfully enabled
k_rx_err_invalid_arg	irq_num > 15 or filter_clk > k_irq_filter_pclk_64

Precondition

PCLKB clock must be running (filter clock source)
ICU module not in module stop

Postcondition

IRQFLTE bit set for this IRQ
IRQFLTC bits set to specified divisor
Filter immediately active

Note

Filter applies to all edge types (rising, falling, both)
Filter continues operating during software standby
Clock divisor is set before enable to avoid glitches

Warning

Enabling filter adds latency to interrupt response

Thread Safety

Not thread-safe. ICU register modification is not atomic.

Performance

- Execution time: ~200 ns @ 240 MHz
- No blocking operations

Example - E-STOP Button

```
// Maximum filtering for mechanical button debounce
rx_err_t err = rx_irq_filter_enable(0, k_irq_filter_pclk_64);
if (err != k_rx_ok) {
    // Handle error
}
```

See also

[rx_irq_filter_disable\(\)](#) Disable filter
[rx_irq_filter_clk_t](#) Clock divisor options

Since

Version 1.0.0

Configures the ICU filter enable (IRQFLTE) and filter clock (IRQFLTC) registers for the specified IRQ. Sets clock divisor first, then enables the filter to avoid sampling at incorrect rate during transition.

Register Modification Sequence

1. IRQFLTC: Clear and set 2-bit clock field for this IRQ
2. IRQFLTE: Set enable bit for this IRQ

Note

On host (non-RX) builds, validates parameters but skips register access

See also

[rx_irq_filter.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 170 of file [rx_irq_filter.c](#).

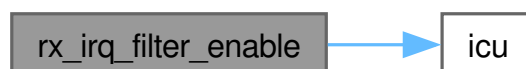
```

00171 {
00172     if (irq_num > k_rx_irq_max) {
00173         return k_rx_err_invalid_arg;
00174     }
00175
00176     if (filter_clk > k_irq_filter_pclk_64) {
00177         return k_rx_err_invalid_arg;
00178     }
00179
00180     #ifdef __RX__
00181     /*
00182      * Determine which register to use:
00183      * - IRQ0-7:  IRQFLTE[0], IRQFLTC[0]
00184      * - IRQ8-15: IRQFLTE[1], IRQFLTC[1]
00185      */
00186     uint8_t reg_idx  = irq_num / k_irq_filter_irqs_per_reg;
00187     uint8_t bit_pos  = irq_num % k_irq_filter_irqs_per_reg;
00188     uint8_t clock_pos = bit_pos * k_irq_filter_clock_bits;
00189
00190     /*
00191      * Set filter clock divisor first (before enabling)
00192      *
00193      * IRQFLTC[n] has 2 bits per IRQ:
00194      * - Bits [1:0]:  IRQ0/8 clock select
00195      * - Bits [3:2]:  IRQ1/9 clock select
00196      * - Bits [5:4]:  IRQ2/10 clock select
00197      * - etc.
00198      */
00199     uint16_t clock_mask = (uint16_t)(k_irq_filter_clock_mask « clock_pos);
00200     uint16_t clock_value = (uint16_t)(filter_clk « clock_pos);
00201
00202     icu()->irqftc[reg_idx] = (icu()->irqftc[reg_idx] & ~clock_mask) | clock_value;
00203
00204     /*
00205      * Enable filter for this IRQ
00206      *
00207      * IRQFLTE[n] has 1 bit per IRQ:
00208      * - Bit 0: IRQ0/8 filter enable
00209      * - Bit 1: IRQ1/9 filter enable
00210      * - etc.
00211      */
00212     icu()->irqfte[reg_idx] |= (uint8_t)(1 « bit_pos);
00213
00214     #else
00215     /* Host-side stub for unit testing */
00216     (void)filter_clk;
00217     #endif
00218
00219     return k_rx_ok;
00220 }

```

References [icu\(\)](#), [rx_icu_regs_t::irqftc](#), [rx_icu_regs_t::irqfte](#), [k_irq_filter_clock_bits](#), [k_irq_filter_clock_mask](#), [k_irq_filter_irqs_per_reg](#), [k_irq_filter_pclk_64](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.85.3.3 rx'irq'filter'get'clock()

```
rx_err_t rx_irq_filter_get_clock (
    uint8_t irq_num,
    rx_irq_filter_clk_t * filter_clk) [nodiscard]
```

Get current filter clock divisor setting for an IRQ.

Reads the IRQFLTC register to retrieve the current clock divisor setting for the specified IRQ's digital filter.

Parameters

in	irq_num	IRQ number to query Valid range: [0, 15]
out	filter_clk	Pointer to receive clock divisor setting - Returns rx_irq_filter_clk_t value - Must not be nullptr

Returns

Error code indicating success or failure

Return values

k_rx_ok	Clock setting successfully retrieved
k_rx_err_invalid_arg	irq_num > 15 or filter_clk pointer is nullptr

Precondition

filter_clk pointer must be non-NULL

Postcondition

*filter_clk contains current divisor setting
No registers modified (read-only operation)

Note

Returns current setting even if filter is disabled

Thread Safety

Thread-safe for reading. Register read is atomic.

Example

```

rx_irq_filter_clk_t current_clock;
rx_err_t err = rx_irq_filter_get_clock(0, &current_clock);
if (err == k_rx_ok) {
    if (current_clock == k_irq_filter_pclk_64) {
        // Maximum filtering configured
    }
}

```

See also

[rx_irq_filter_enable\(\)](#) Set clock divisor
[rx_irq_filter_clk_t](#) Clock divisor values

Since

Version 1.0.0

Reads the 2-bit clock divisor field from IRQFLTC register for the specified IRQ. Returns the divisor setting even if filter is disabled.

Note

On host builds, always returns k_irq_filter_pclk_1

See also

[rx_irq_filter.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 305 of file [rx_irq_filter.c](#).

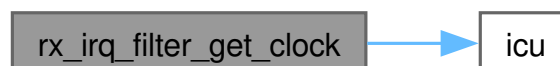
```

00306 {
00307     if (irq_num > k_rx_irq_max) {
00308         return k_rx_err_invalid_arg;
00309     }
00310
00311     if (filter_clk == nullptr) {
00312         return k_rx_err_invalid_arg;
00313     }
00314
00315     #ifdef __RX__
00316     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00317     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00318     uint8_t clock_pos = bit_pos * k_irq_filter_clock_bits;
00319
00320     uint16_t clock_value = (icu()->irqfltc[reg_idx] » clock_pos) & k_irq_filter_clock_mask;
00321     *filter_clk = (rx_irq_filter_clk_t)clock_value;
00322
00323     #else
00324     *filter_clk = k_irq_filter_pclk_1;
00325     #endif
00326
00327     return k_rx_ok;
00328 }

```

References [icu\(\)](#), [rx_icu_regs_t::irqfltc](#), [k_irq_filter_clock_bits](#), [k_irq_filter_clock_mask](#), [k_irq_filter_irqs_per_reg](#), [k_irq_filter_pclk_1](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.85.3.4 rx'irq'filter'is'enabled()

```
rx_err_t rx_irq_filter_is_enabled (
    uint8_t irq_num,
    bool * enabled) [nodiscard]
```

Check if digital filter is enabled on an IRQ pin.

Reads the IRQFLTE register to determine if the digital filter is currently enabled for the specified IRQ.

Parameters

in	irq_num	IRQ number to query Valid range: [0, 15]
out	enabled	Pointer to receive enable status - true: Filter is enabled - false: Filter is disabled - Must not be nullptr

Returns

Error code indicating success or failure

Return values

k_rx_ok	Status successfully retrieved
k_rx_err_invalid_arg	irq_num > 15 or enabled pointer is nullptr

Precondition

enabled pointer must be non-NULL

Postcondition

*enabled contains current filter state

No registers modified (read-only operation)

Thread Safety

Thread-safe for reading. Register read is atomic.

Example

```
bool filter_active;
rx_err_t err = rx_irq_filter_is_enabled(0, &filter_active);
if (err == k_rx_ok && filter_active) {
    // Filter is active on IRQ0
}
```

See also

[rx_irq_filter_enable\(\)](#) Enable filter
[rx_irq_filter_get_clock\(\)](#) Get clock setting

Since

Version 1.0.0

Reads the IRQFLTE register bit for the specified IRQ to determine current filter enable state.

Note

On host builds, always returns false (filter disabled)

See also

[rx_irq_filter.h](#) Full API documentation

Since

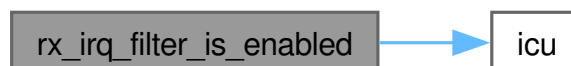
Version 1.0.0

Definition at line 268 of file [rx_irq_filter.c](#).

```
00269 {
00270     if (irq_num > k_rx_irq_max) {
00271         return k_rx_err_invalid_arg;
00272     }
00273
00274     if (enabled == nullptr) {
00275         return k_rx_err_invalid_arg;
00276     }
00277
00278     #ifdef __RX__
00279     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00280     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00281
00282     *enabled = (icu()->irqfte[reg_idx] & (1 « bit_pos)) != 0;
00283
00284     #else
00285     *enabled = false;
00286     #endif
00287
00288     return k_rx_ok;
00289 }
```

References [icu\(\)](#), [rx_icu_regs_t::irqfte](#), [k_irq_filter_irqs_per_reg](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.85.4 Variable Documentation

8.85.4.1 s_irq_max

```
const uint8_t s_irq_max = 15U [static]
```

Maximum valid IRQ number (0-15).

The RX72N ICU supports 16 external interrupt pins, numbered IRQ0 through IRQ15. This constant defines the maximum valid IRQ number for validation.

Value: 15

Valid IRQ Range: [0, 15

]

Note

Static const rather than enum due to use in comparison operations

Since

Version 1.0.0

Definition at line 282 of file [rx_irq_filter.h](#).

8.86 rx_irq_filter.h

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_irq_filter.h
00003  * @brief IRQ Digital Filter Driver API for RX72N
00004  *
00005  * @details
00006  * Provides hardware digital noise filtering for external interrupt (IRQ) pins
00007  * on the RX72N microcontroller. The filter helps eliminate electrical noise,
00008  * switch bounce, and transient glitches that could cause spurious interrupts.
00009  *
00010  * The RX72N ICU provides hardware digital filters on all 16 IRQ pins (IRQ0-15).
00011  * Each filter samples the input at a configurable rate and requires a stable
00012  * level for 3 consecutive samples to register as a valid transition.
00013  *
00014  * @par System Architecture
00015  * @verbatim
00016  *
00017  *      +-----+
00018  *      | External Inputs |
00019  *      | +-----+ +-----+ +-----+ +-----+ |
00020  *      | | E-STOP | | VBUS | | Expansion | | Hall | ... (16) | |
00021  *      | | Button | | Detect | | Input | | Encoder | |
00022  *      | +-----+ +-----+ +-----+ +-----+ |
00023  *      +-----+
00024  *      | | | | |
00025  *      +-----+
00026  *      | ICU Digital Filter Block |
00027  *      | +-----+ |
00028  *      | | IRQ0 --> [3-Sample Filter] --> IRQn Interrupt | |
00029  *      | | IRQ1 --> [3-Sample Filter] --> IRQn Interrupt | |
```

```

00030 * | | ... | |
00031 * | | IRQ15 -> [3-Sample Filter] --> IRQn Interrupt | |
00032 * | +-----+ | |
00033 * | | | |
00034 * | Control Registers: | |
00035 * | +-----+ +-----+ | |
00036 * | | IRQFLTE[2] | | IRQFLTC[2] | |
00037 * | | (Enable) | | (Clock) | |
00038 * | +-----+ +-----+ | |
00039 * | +-----+ |
00040 * @endverbatim
00041 *
00042 * @par Digital Filter Operation
00043 * @verbatim
00044 * Input Signal with Noise:
00045 * _____
00046 * | | | | | | | |
00047 * | | | | | | | |
00048 * | | | | | | | |
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00114 * | | | | | | | |
00115 * | | | | | | | |
00116 * | | | | | | | |

```

```

00117 *      // Increase filtering in noisy environments
00118 *      rx_irq_filter_enable(8, k_irq_filter_pclk_64);
00119 *      } else {
00120 *      // Reduce filtering for faster response
00121 *      rx_irq_filter_enable(8, k_irq_filter_pclk_8);
00122 *      }
00123 *  }
00124 *
00125 * bool check_filter_config(uint8_t irq_num)
00126 * {
00127 *     bool enabled;
00128 *     rx_irq_filter_clk_t clock;
00129 *
00130 *     rx_irq_filter_is_enabled(irq_num, &enabled);
00131 *     rx_irq_filter_get_clock(irq_num, &clock);
00132 *
00133 *     return enabled && (clock == k_irq_filter_pclk_64);
00134 * }
00135 * @endcode
00136 *
00137 * @par NASA Power of 10 Compliance
00138 * - **Rule 1**: [OK] No goto, setjmp, longjmp, recursion
00139 * - **Rule 2**: [OK] No loops (simple register operations)
00140 * - **Rule 3**: [OK] No dynamic memory allocation
00141 * - **Rule 4**: [OK] All functions < 30 lines
00142 * - **Rule 5**: [OK] Input validation on all functions
00143 * - **Rule 7**: [OK] All return values are error codes
00144 * - **Rule 8**: [OK] C23 typed enums for filter clock options
00145 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00146 *
00147 * @par SOLID Principles
00148 * - **Single Responsibility**: Only IRQ filter configuration
00149 * - **Interface Segregation**: Minimal 4-function API
00150 *
00151 * @see rx72n_icu_regs.h ICU register definitions (IRQFLTE, IRQFLTC)
00152 * @see RX72N Hardware Manual, Chapter 13 - Interrupt Controller Unit
00153 *
00154 * @author Locked, Inc.
00155 * @date 2026-01-29
00156 * @copyright Copyright (c) 2026 Locked Inc.
00157 * SPDX-License-Identifier: MIT
00158 * @since Version 1.0.0
00159 */
00160
00161 #pragma once
00162
00163 #include <stdint.h>
00164
00165 #include "rx_err.h"
00166
00167 #ifdef __cplusplus
00168 extern "C" {
00169 #endif
00170
00171 /*
=====
00172 * Configuration Constants
00173 *
=====
00174 */
00175 /**
00176 * @enum rx_irq_filter_limits_t
00177 * @brief IRQ filter limits for RX72N
00178 *
00179 * @details
00180 * The RX72N ICU supports external IRQ pins IRQ0-IRQ15 (16 total).
00181 */
00182 typedef enum : uint8_t {
00183     k_rx_irq_max = 15,
00184 } rx_irq_filter_limits_t;
00185
00186 /**
00187 * @enum rx_irq_filter_clk_t
00188 * @brief IRQ filter clock divisor options for sampling rate control
00189 *
00190 * @details
00191 * Selects the clock divisor for IRQ digital filter sampling. Higher divisors
00192 * result in slower sampling, providing more noise rejection but also longer
00193 * response times to valid transitions.
00194 *
00195 * The digital filter requires 3 consecutive samples at the same level to
00196 * recognize a valid transition. Total response time = 3 x sample period.
00197 *
00198 * @par Response Time Calculation
00199 * @f[
00200 * T_{response} = 3 \times \frac{1}{f_{PCLKB} / divisor} = \frac{3 \times divisor}{f_{PCLKB}}
00201 */

```

```

00202 * @f]
00203 *
00204 * @par Filter Characteristics (PCLKB = 60 MHz)
00205 * | Divisor | Sample Period | Response Time | Best For |
00206 * |-----|-----|-----|-----|
00207 * | /1 | 16.7 ns | 50 ns | High-speed signals |
00208 * | /8 | 133 ns | 400 ns | Encoder pulses |
00209 * | /32 | 533 ns | 1.6 us | Sensor interrupts |
00210 * | /64 | 1.07 us | 3.2 us | Buttons, switches |
00211 *
00212 * @par Noise Rejection Capability
00213 * Noise pulses shorter than 2 x sample period are rejected. For PCLK/64 with
00214 * PCLKB=60MHz, noise pulses < 2.1us are filtered out.
00215 *
00216 * @note Register encoding: 2 bits per IRQ in IRQFLTC register
00217 * @warning Lower divisors provide faster response but less noise immunity
00218 *
00219 * @see rx_irq_filter_enable() Set filter clock divisor
00220 * @see rx_irq_filter_get_clock() Read current divisor setting
00221 *
00222 * @since Version 1.0.0
00223 */
00224 typedef enum : uint8_t {
00225 /**
00226  * @brief PCLKB/1 - Fastest sampling, minimal filtering
00227  * @details
00228  * Sample period: 16.7 ns @ 60 MHz PCLKB
00229  * Response time: 50 ns (3 samples)
00230  * Noise rejection: < 33 ns pulses filtered
00231  * @par Use Case: High-frequency signals requiring minimal delay
00232  * @warning Provides almost no noise filtering
00233  */
00234 k_irq_filter_pclk_1 = 0,
00235
00236 /**
00237  * @brief PCLKB/8 - Fast sampling, light filtering
00238  * @details
00239  * Sample period: 133 ns @ 60 MHz PCLKB
00240  * Response time: 400 ns (3 samples)
00241  * Noise rejection: < 267 ns pulses filtered
00242  * @par Use Case: Encoder signals, fast sensors
00243  */
00244 k_irq_filter_pclk_8 = 1,
00245
00246 /**
00247  * @brief PCLKB/32 - Moderate sampling, good filtering
00248  * @details
00249  * Sample period: 533 ns @ 60 MHz PCLKB
00250  * Response time: 1.6 us (3 samples)
00251  * Noise rejection: < 1.07 us pulses filtered
00252  * @par Use Case: External interrupts, VBUS detection, expansion inputs
00253  */
00254 k_irq_filter_pclk_32 = 2,
00255
00256 /**
00257  * @brief PCLKB/64 - Slowest sampling, maximum filtering
00258  * @details
00259  * Sample period: 1.07 us @ 60 MHz PCLKB
00260  * Response time: 3.2 us (3 samples)
00261  * Noise rejection: < 2.1 us pulses filtered
00262  * @par Use Case: Mechanical buttons, E-STOP, user input switches
00263  * @note Recommended for all human-activated inputs
00264  */
00265 k_irq_filter_pclk_64 = 3,
00266 } rx_irq_filter_clk_t;
00267
00268 /**
00269  * @var s_irq_max
00270  * @brief Maximum valid IRQ number (0-15)
00271  *
00272  * @details
00273  * The RX72N ICU supports 16 external interrupt pins, numbered IRQ0 through
00274  * IRQ15. This constant defines the maximum valid IRQ number for validation.
00275  *
00276  * @par Value: 15
00277  * @par Valid IRQ Range: [0, 15]
00278  *
00279  * @note Static const rather than enum due to use in comparison operations
00280  * @since Version 1.0.0
00281  */
00282 static const uint8_t s_irq_max = 15U;
00283
00284 /*
=====
00285 * Public API
00286 *
=====

```



```

00287 */
00288
00289 /**
00290  * @brief Enable hardware digital filter on an IRQ pin
00291  *
00292  * @details
00293  * Enables the ICU digital filter for the specified IRQ pin with the given
00294  * clock divisor. The filter samples the input and requires 3 consecutive
00295  * samples at the same level to recognize a valid edge transition.
00296  *
00297  * @par Algorithm Steps
00298  * 1. Validate irq_num is in range [0, 15]
00299  * 2. Validate filter_clk is valid divisor
00300  * 3. Calculate register index (0 for IRQ0-7, 1 for IRQ8-15)
00301  * 4. Set clock divisor in IRQFLTC register
00302  * 5. Set enable bit in IRQFLTE register
00303  *
00304  * @param[in] irq_num IRQ number to configure
00305  *             - Valid range: [0, 15]
00306  *             - IRQ0-7 use IRQFLTE[0]/IRQFLTC[0]
00307  *             - IRQ8-15 use IRQFLTE[1]/IRQFLTC[1]
00308  * @param[in] filter_clk Clock divisor for filter sampling
00309  *             - k_irq_filter_pclk_1: Minimal filtering
00310  *             - k_irq_filter_pclk_64: Maximum filtering
00311  *
00312  * @return Error code indicating success or failure
00313  * @retval k_rx_ok Filter successfully enabled
00314  * @retval k_rx_err_invalid_arg irq_num > 15 or filter_clk > k_irq_filter_pclk_64
00315  *
00316  * @pre PCLKB clock must be running (filter clock source)
00317  * @pre ICU module not in module stop
00318  *
00319  * @post IRQFLTE bit set for this IRQ
00320  * @post IRQFLTC bits set to specified divisor
00321  * @post Filter immediately active
00322  *
00323  * @note Filter applies to all edge types (rising, falling, both)
00324  * @note Filter continues operating during software standby
00325  * @note Clock divisor is set before enable to avoid glitches
00326  *
00327  * @warning Enabling filter adds latency to interrupt response
00328  *
00329  * @par Thread Safety
00330  * Not thread-safe. ICU register modification is not atomic.
00331  *
00332  * @par Performance
00333  * - Execution time: ~200 ns @ 240 MHz
00334  * - No blocking operations
00335  *
00336  * @par Example - E-STOP Button
00337  * @code{.c}
00338  * // Maximum filtering for mechanical button debounce
00339  * rx_err_t err = rx_irq_filter_enable(0, k_irq_filter_pclk_64);
00340  * if (err != k_rx_ok) {
00341  *     // Handle error
00342  * }
00343  * @endcode
00344  *
00345  * @see rx_irq_filter_disable() Disable filter
00346  * @see rx_irq_filter_clk_t Clock divisor options
00347  *
00348  * @since Version 1.0.0
00349  * @callgraph
00350  */
00351 [[nodiscard]] rx_err_t rx_irq_filter_enable(uint8_t irq_num, rx_irq_filter_clk_t filter_clk);
00352
00353 /**
00354  * @brief Disable hardware digital filter on an IRQ pin
00355  *
00356  * @details
00357  * Disables the ICU digital filter for the specified IRQ pin. After disabling,
00358  * the raw unfiltered input signal is passed through to the interrupt detector.
00359  *
00360  * @par Algorithm Steps
00361  * 1. Validate irq_num is in range [0, 15]
00362  * 2. Calculate register index
00363  * 3. Clear enable bit in IRQFLTE register
00364  *
00365  * @param[in] irq_num IRQ number to configure
00366  *             - Valid range: [0, 15]
00367  *
00368  * @return Error code indicating success or failure
00369  * @retval k_rx_ok Filter successfully disabled
00370  * @retval k_rx_err_invalid_arg irq_num > 15
00371  *
00372  * @pre None (can be called regardless of current filter state)
00373  *

```

```

00374 * @post IRQFLTE bit cleared for this IRQ
00375 * @post Raw input passed to interrupt detector
00376 *
00377 * @note Clock divisor setting is preserved (unchanged)
00378 * @note Safe to call even if filter was already disabled
00379 *
00380 * @par Thread Safety
00381 * Not thread-safe. ICU register modification is not atomic.
00382 *
00383 * @par Example
00384 * @code{.c}
00385 * // Disable filter for faster response
00386 * rx_irq_filter_disable(5);
00387 * @endcode
00388 *
00389 * @see rx_irq_filter_enable() Enable filter
00390 *
00391 * @since Version 1.0.0
00392 * @callgraph
00393 */
00394 [[nodiscard]] rx_err_t rx_irq_filter_disable(uint8_t irq_num);
00395
00396 /**
00397 * @brief Check if digital filter is enabled on an IRQ pin
00398 *
00399 * @details
00400 * Reads the IRQFLTE register to determine if the digital filter is
00401 * currently enabled for the specified IRQ.
00402 *
00403 * @param[in] irq_num IRQ number to query
00404 *             - Valid range: [0, 15]
00405 * @param[out] enabled Pointer to receive enable status
00406 *                    - true: Filter is enabled
00407 *                    - false: Filter is disabled
00408 *                    - Must not be nullptr
00409 *
00410 * @return Error code indicating success or failure
00411 * @retval k_rx_ok Status successfully retrieved
00412 * @retval k_rx_err_invalid_arg irq_num > 15 or enabled pointer is nullptr
00413 *
00414 * @pre enabled pointer must be non-NULL
00415 *
00416 * @post *enabled contains current filter state
00417 * @post No registers modified (read-only operation)
00418 *
00419 * @par Thread Safety
00420 * Thread-safe for reading. Register read is atomic.
00421 *
00422 * @par Example
00423 * @code{.c}
00424 * bool filter_active;
00425 * rx_err_t err = rx_irq_filter_is_enabled(0, &filter_active);
00426 * if (err == k_rx_ok && filter_active) {
00427 *     // Filter is active on IRQ0
00428 * }
00429 * @endcode
00430 *
00431 * @see rx_irq_filter_enable() Enable filter
00432 * @see rx_irq_filter_get_clock() Get clock setting
00433 *
00434 * @since Version 1.0.0
00435 * @callgraph
00436 */
00437 [[nodiscard]] rx_err_t rx_irq_filter_is_enabled(uint8_t irq_num, bool* enabled);
00438
00439 /**
00440 * @brief Get current filter clock divisor setting for an IRQ
00441 *
00442 * @details
00443 * Reads the IRQFLTC register to retrieve the current clock divisor
00444 * setting for the specified IRQ's digital filter.
00445 *
00446 * @param[in] irq_num IRQ number to query
00447 *             - Valid range: [0, 15]
00448 * @param[out] filter_clk Pointer to receive clock divisor setting
00449 *                    - Returns rx_irq_filter_clk_t value
00450 *                    - Must not be nullptr
00451 *
00452 * @return Error code indicating success or failure
00453 * @retval k_rx_ok Clock setting successfully retrieved
00454 * @retval k_rx_err_invalid_arg irq_num > 15 or filter_clk pointer is nullptr
00455 *
00456 * @pre filter_clk pointer must be non-NULL
00457 *
00458 * @post *filter_clk contains current divisor setting
00459 * @post No registers modified (read-only operation)
00460 *

```

```

00461 * @note Returns current setting even if filter is disabled
00462 *
00463 * @par Thread Safety
00464 * Thread-safe for reading. Register read is atomic.
00465 *
00466 * @par Example
00467 * @code{.c}
00468 * rx_irq_filter_clk_t current_clock;
00469 * rx_err_t err = rx_irq_filter_get_clock(0, &current_clock);
00470 * if (err == k_rx_ok) {
00471 *     if (current_clock == k_irq_filter_pclk_64) {
00472 *         // Maximum filtering configured
00473 *     }
00474 * }
00475 * @endcode
00476 *
00477 * @see rx_irq_filter_enable() Set clock divisor
00478 * @see rx_irq_filter_clk_t Clock divisor values
00479 *
00480 * @since Version 1.0.0
00481 * @callgraph
00482 */
00483 [[nodiscard]] rx_err_t rx_irq_filter_get_clock(uint8_t irq_num, rx_irq_filter_clk_t* filter_clk);
00484
00485 #ifdef __cplusplus
00486 }
00487 #endif

```

8.87 libs/rx_hal/inc/rx_lpc.h File Reference

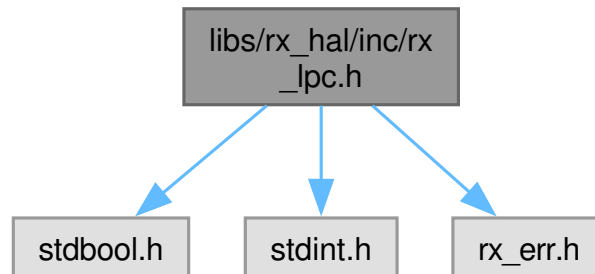
Low Power Consumption (LPC) HAL Driver for RX72N.

```
#include <stdbool.h>
```

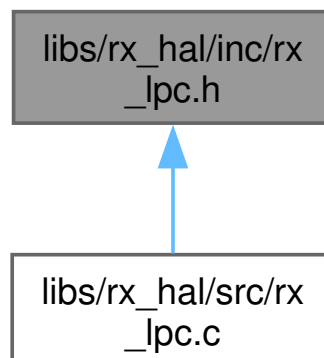
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_lpc.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_lpc_opcc_mode_t` : `uint8_t` { `k_lpc_opcc_high_speed` = 0 , `k_lpc_opcc_low_speed_1` = 1 , `k_lpc_opcc_low_speed_2` = 2 }
Operating Power Control (OPCCR) mode selector.
- enum `rx_lpc_deep_power_t` : `uint8_t` { `k_lpc_deep_ram_usb_on` = 0 , `k_lpc_deep_ram_usb_off` = 1 , `k_lpc_deep_lvd_off` = 2 }
Deep-standby power supply configuration (DPSBYCR.DEEP_CUT).
- enum `rx_lpc_last_mode_t` : `uint8_t` { `k_lpc_mode_none` = 0 , `k_lpc_mode_sleep` = 1 , `k_lpc_mode_software_standby` = 2 , `k_lpc_mode_deep_software_standby` = 3 }
Last low-power mode entered (test / diagnostic hook).
- enum `rx_lpc_wake_flags_t` : `uint32_t` {
`k_lpc_wake_irq0` = 0x00000001U , `k_lpc_wake_irq1` = 0x00000002U , `k_lpc_wake_irq2` = 0x00000004U , `k_lpc_wake_irq3` = 0x00000008U ,
`k_lpc_wake_irq4` = 0x00000010U , `k_lpc_wake_irq5` = 0x00000020U , `k_lpc_wake_irq6` = 0x00000040U , `k_lpc_wake_irq7` = 0x00000080U ,
`k_lpc_wake_irq8` = 0x00000100U , `k_lpc_wake_irq9` = 0x00000200U , `k_lpc_wake_irq10` = 0x00000400U , `k_lpc_wake_irq11` = 0x00000800U ,
`k_lpc_wake_irq12` = 0x00001000U , `k_lpc_wake_irq13` = 0x00002000U , `k_lpc_wake_irq14` = 0x00004000U , `k_lpc_wake_irq15` = 0x00008000U ,
`k_lpc_wake_lvd1` = 0x00010000U , `k_lpc_wake_lvd2` = 0x00020000U , `k_lpc_wake_rtc_periodic` = 0x00040000U , `k_lpc_wake_rtc_alarm` = 0x00080000U ,
`k_lpc_wake_nmi` = 0x00100000U , `k_lpc_wake_iic_sda` = 0x00200000U , `k_lpc_wake_iic_scl` = 0x00400000U , `k_lpc_wake_usb` = 0x00800000U ,
`k_lpc_wake_can1_rx` = 0x01000000U , `k_lpc_wake_all_mask` = 0x01FFFFFFU }
Wake-source bits for deep software standby.

Functions

- `rx_err_t rx_lpc_init` (void)
Initialize the Low Power Consumption driver.
- `rx_err_t rx_lpc_set_operating_power` (`rx_lpc_opcc_mode_t` mode)
Select the operating power control (OPCCR) mode.
- `rx_err_t rx_lpc_enter_sleep` (void)
Enter Sleep mode (CPU stop, peripherals run).
- `rx_err_t rx_lpc_enter_software_standby` (void)
Enter Software Standby mode.
- `rx_err_t rx_lpc_enter_deep_software_standby` (`uint32_t` wake_mask, `rx_lpc_deep_power_t` deep_power, bool keep_io)
Enter Deep Software Standby mode.
- bool `rx_lpc_was_deep_standby_wake` (void)
Query whether the last reset was caused by a deep-standby wake.
- `rx_err_t rx_lpc_get_wake_flags` (`uint32_t` *flags)
Read the deep-standby wake flags (DPSIFR0..DPSIFR3).

8.87.1 Detailed Description

Low Power Consumption (LPC) HAL Driver for RX72N.

Hardware Abstraction Layer for the RX72N Low Power Consumption subsystem. Provides a small, auditable API for entering the three RX72N low-power modes and for selecting the operating power control mode that governs active-run current.

8.87.1.1 Supported Modes (RX72N Manual Chapter 11)

Mode	CPU	Peripherals	RAM	Wake Latency	Exit Vector
Sleep	Stop	Run	Retained	ns	Continue next PC
Software Standby	Stop	Stop	Retained	us	Continue next PC
Deep Software Standby	Stop	Stop	Lost*	ms	Reset vector

* * Deep Software Standby: On-chip RAM contents are lost, but the 32-byte
 * Deep Standby Backup Register (DPSBKR) is retained. See rx72n_lpc_regs.h
 * for the DPSBKR accessor.
 *

8.87.1.2 Operating Power Control (OPCCR) Modes

Mode	ICLK max	Notes
High-speed (default)	240 MHz	Full performance
Low-speed 1	1 MHz	PLL/HOCO may be stopped
Low-speed 2	264 kHz	HOCO must be stopped (LOCO/Main only)

Note

RX72N only implements three OPCCR modes (high-speed, low-speed 1, low-speed 2). The "middle-speed" and "sub-oscillator-speed" modes from the RX100/RX200 family (SOPCCR) are not available on RX72N and therefore have no corresponding API enumerators in this driver.

8.87.1.3 Wake Source Selection (Deep Software Standby)

Wake sources for deep software standby are specified as a 32-bit bitmask passed to [rx_lpc_enter_deep_software_standby\(\)](#). The bitmask layout mirrors DPSIER0..DPSIER3 as a single little-endian word:

* bits [7: 0] - IRQ0-DS .. IRQ7-DS (DPSIER0)
 * bits [15: 8] - IRQ8-DS .. IRQ15-DS (DPSIER1)
 * bits [23:16] - LVD1/LVD2/RTC-P/RTC-A/NMI/IIC-SDA/IIC-SCL/USB (DPSIER2)
 * bits [31:24] - CAN1-RX (bit 24 only) (DPSIER3)
 *

The same encoding is returned by [rx_lpc_get_wake_flags\(\)](#) when reading the DPSIFRx flag registers after a deep-standby wake.

8.87.1.4 Hardware Register References

Register	Address	Manual Section	Purpose
SBYCR	0x0008000C	11.2.1 (p.405)	Standby control (SSBY, OPE)
OPCCR	0x000800A0	11.2.6 (p.414)	Operating power control

Register	Address	Manual Section	Purpose
RSTCKCR	0x000800A1	11.2.7 (p.417)	Sleep-mode return clock
DPSBYCR	0x0008C280	11.2.8 (p.418)	Deep standby control
DPSIER0-3	0x0008C282	11.2.9-12	Deep standby IRQ enable
DPSIFR0-3	0x0008C286	11.2.13-16	Deep standby wake flags
DPSIEGR0-3	0x0008C28A	11.2.17-20	Deep standby edge select
DPSBKR	0x0008C2A0	11.2.21 (p.433)	Deep standby backup RAM (32B)

8.87.1.5 Power Protection (PRCR)

All writes to SBYCR, OPCCR, RSTCKCR, and DPSBYCR require PRC1 to be unlocked via PRCR. This driver handles PRCR lock/unlock internally.

8.87.1.6 NASA Power of 10 Compliance

Rule	Status	Implementation
1	↔ [OK]↔	No goto/setjmp/recursion
2	↔ [OK]↔	All loops bounded (OPCMTSF poll has bounded retry)
3	↔ [OK]↔	Zero dynamic allocation
4	↔ [OK]↔	All functions < 60 lines
5	↔ [OK]↔	>= 2 pre/post conditions per function
6	↔ [OK]↔	Narrow scope, file-scope statics (s_*)
7	↔ [OK]↔	All return values checked
8	↔ [OK]↔	C23 typed enums, minimal preprocessor
9	↔ [OK]↔	Single-level pointers, no function pointers
10	↔ [OK]↔	Compiles with -Wall -Wextra -Werror

See also

[rx72n_lpc_regs.h](#) Register layout and bit definitions
[rx72n_system_regs.h](#) SBYCR / PRCR / MSTPCR_x accessors
RX72N Hardware Manual Chapter 11 - Low Power Consumption

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_lpc.h](#).

8.87.2 Enumeration Type Documentation

8.87.2.1 rx`lpc`deep`power`t

```
enum rx\_lpc\_deep\_power\_t : uint8_t
```

Deep-standby power supply configuration (DPSBYCR.DEEPCUT).

Selects how aggressively the deep-standby mode cuts power. Mirrors DPSBYCR.DEEPCUT[1:0] (RX72N Manual Section 11.2.8, page 418).

Note

The 10b DEEPCUT combination is reserved and is not exposed here.

Since

Version 1.0.0

Enumerator

<code>k_lpc_deep_ram_usb_on</code>	DEEPCUT=00b: standby RAM + USB powered
<code>k_lpc_deep_ram_usb_off</code>	DEEPCUT=01b: standby RAM + USB powered off

k_lpc_deep_lvd_off	DEEPCUT=11b: LVD also off (lowest power)
--------------------	--

Definition at line 152 of file [rx_lpc.h](#).

```
00152         : uint8_t {
00153     k_lpc_deep_ram_usb_on  = 0, /**< DEEPCUT=00b: standby RAM + USB powered */
00154     k_lpc_deep_ram_usb_off = 1, /**< DEEPCUT=01b: standby RAM + USB powered off */
00155     k_lpc_deep_lvd_off    = 2, /**< DEEPCUT=11b: LVD also off (lowest power) */
00156 } rx_lpc_deep_power_t;
```

8.87.2.2 rx_lpc_last_mode_t

```
enum rx_lpc_last_mode_t : uint8_t
```

Last low-power mode entered (test / diagnostic hook).

Recorded by rx_lpc_enter_*() prior to executing the WAIT instruction so that host-side unit tests can verify the driver reached the correct code path without executing a real WAIT.

Since

Version 1.0.0

Enumerator

k_lpc_mode_none	No low-power entry attempted yet
k_lpc_mode_sleep	Sleep mode (WAIT, SBYCR.SSBY=0)
k_lpc_mode_software_standby	Software standby (SBYCR.SSBY=1, DPSBY=0)
k_lpc_mode_deep_software_standby	Deep software standby (SSBY=1, DPSBY=1)

Definition at line 169 of file [rx_lpc.h](#).

```
00169         : uint8_t {
00170     k_lpc_mode_none      = 0, /**< No low-power entry attempted yet */
00171     k_lpc_mode_sleep     = 1, /**< Sleep mode (WAIT, SBYCR.SSBY=0) */
00172     k_lpc_mode_software_standby = 2, /**< Software standby (SBYCR.SSBY=1, DPSBY=0) */
00173     k_lpc_mode_deep_software_standby = 3, /**< Deep software standby (SSBY=1, DPSBY=1) */
00174 } rx_lpc_last_mode_t;
```

8.87.2.3 rx_lpc_opcc_mode_t

```
enum rx_lpc_opcc_mode_t : uint8_t
```

Operating Power Control (OPCCR) mode selector.

Selects the active-run power mode used while the CPU is running. The RX72N supports exactly three OPCCR modes (RX72N Manual Section 11.2.6, page 414). The underlying OPCM[2:0] field values are encoded directly.

Mode Transitions

Writing to OPCCR triggers an internal transition; the hardware sets OPCCR.OPCMTSF=1 until the transition completes. This driver polls that flag with a bounded retry count.

Warning

Low-speed mode 2 requires HOCO to be stopped. The caller is responsible for configuring the clock tree before invoking [rx_lpc_set_operating_power\(\)](#).

Since

Version 1.0.0

Enumerator

k_lpc_opcc_high_speed	High-speed mode (OPCM=000b, ICLK <= 240 MHz)
k_lpc_opcc_low_speed_1	Low-speed mode 1 (OPCM=110b, ICLK <= 1 MHz)
k_lpc_opcc_low_speed_2	Low-speed mode 2 (OPCM=111b, ICLK <= 264 kHz)

Definition at line 134 of file [rx_lpc.h](#).

```
00134         : uint8_t {
00135     k_lpc_opcc_high_speed = 0, /**< High-speed mode (OPCM=000b, ICLK <= 240 MHz) */
00136     k_lpc_opcc_low_speed_1 = 1, /**< Low-speed mode 1 (OPCM=110b, ICLK <= 1 MHz) */
00137     k_lpc_opcc_low_speed_2 = 2, /**< Low-speed mode 2 (OPCM=111b, ICLK <= 264 kHz) */
00138 } rx_lpc_opcc_mode_t;
```

8.87.2.4 rx`lpc`wake`flags``t`

```
enum rx\_lpc\_wake\_flags\_t : uint32_t
```

Wake-source bits for deep software standby.

32-bit bitmask layout used by [rx_lpc_enter_deep_software_standby\(\)](#) and returned by [rx_lpc_get_wake_flags\(\)](#). The layout mirrors DPSIER0..DPSIER3 (and the corresponding DPSIFR0..DPSIFR3 flag registers) as a single little-endian word.

Since

Version 1.0.0

Enumerator

k_lpc_wake_irq0	IRQ0-DS pin
k_lpc_wake_irq1	IRQ1-DS pin
k_lpc_wake_irq2	IRQ2-DS pin
k_lpc_wake_irq3	IRQ3-DS pin
k_lpc_wake_irq4	IRQ4-DS pin
k_lpc_wake_irq5	IRQ5-DS pin
k_lpc_wake_irq6	IRQ6-DS pin
k_lpc_wake_irq7	IRQ7-DS pin
k_lpc_wake_irq8	IRQ8-DS pin
k_lpc_wake_irq9	IRQ9-DS pin

k_lpc_wake_irq10	IRQ10-DS pin
k_lpc_wake_irq11	IRQ11-DS pin
k_lpc_wake_irq12	IRQ12-DS pin
k_lpc_wake_irq13	IRQ13-DS pin
k_lpc_wake_irq14	IRQ14-DS pin
k_lpc_wake_irq15	IRQ15-DS pin
k_lpc_wake_lvd1	LVD1 (Voltage Monitor 1)
k_lpc_wake_lvd2	LVD2 (Voltage Monitor 2)
k_lpc_wake_rtc_periodic	RTC periodic interrupt
k_lpc_wake_rtc_alarm	RTC alarm interrupt
k_lpc_wake_nmi	NMI pin (write-once bit)
k_lpc_wake_iic_sda	SDA2-DS pin
k_lpc_wake_iic_scl	SCL2-DS pin
k_lpc_wake_usb	USB suspend/resume
k_lpc_wake_can1_rx	CRX1-DS pin (CAN1 RX)
k_lpc_wake_all_mask	Mask of all defined wake sources

Definition at line 193 of file `rx_lpc.h`.

```

00193         : uint32_t {
00194     /* DPSIER0 - IRQ0-DS .. IRQ7-DS */
00195     k_lpc_wake_irq0 = 0x00000001U, /**< IRQ0-DS pin */
00196     k_lpc_wake_irq1 = 0x00000002U, /**< IRQ1-DS pin */
00197     k_lpc_wake_irq2 = 0x00000004U, /**< IRQ2-DS pin */
00198     k_lpc_wake_irq3 = 0x00000008U, /**< IRQ3-DS pin */
00199     k_lpc_wake_irq4 = 0x00000010U, /**< IRQ4-DS pin */
00200     k_lpc_wake_irq5 = 0x00000020U, /**< IRQ5-DS pin */
00201     k_lpc_wake_irq6 = 0x00000040U, /**< IRQ6-DS pin */
00202     k_lpc_wake_irq7 = 0x00000080U, /**< IRQ7-DS pin */
00203
00204     /* DPSIER1 - IRQ8-DS .. IRQ15-DS */
00205     k_lpc_wake_irq8 = 0x00000100U, /**< IRQ8-DS pin */
00206     k_lpc_wake_irq9 = 0x00000200U, /**< IRQ9-DS pin */
00207     k_lpc_wake_irq10 = 0x00000400U, /**< IRQ10-DS pin */
00208     k_lpc_wake_irq11 = 0x00000800U, /**< IRQ11-DS pin */
00209     k_lpc_wake_irq12 = 0x00001000U, /**< IRQ12-DS pin */
00210     k_lpc_wake_irq13 = 0x00002000U, /**< IRQ13-DS pin */
00211     k_lpc_wake_irq14 = 0x00004000U, /**< IRQ14-DS pin */
00212     k_lpc_wake_irq15 = 0x00008000U, /**< IRQ15-DS pin */
00213
00214     /* DPSIER2 - LVD / RTC / NMI / IIC / USB */
00215     k_lpc_wake_lvd1 = 0x00010000U, /**< LVD1 (Voltage Monitor 1) */
00216     k_lpc_wake_lvd2 = 0x00020000U, /**< LVD2 (Voltage Monitor 2) */
00217     k_lpc_wake_rtc_periodic = 0x00040000U, /**< RTC periodic interrupt */
00218     k_lpc_wake_rtc_alarm = 0x00080000U, /**< RTC alarm interrupt */
00219     k_lpc_wake_nmi = 0x00100000U, /**< NMI pin (write-once bit) */
00220     k_lpc_wake_iic_sda = 0x00200000U, /**< SDA2-DS pin */
00221     k_lpc_wake_iic_scl = 0x00400000U, /**< SCL2-DS pin */
00222     k_lpc_wake_usb = 0x00800000U, /**< USB suspend/resume */
00223
00224     /* DPSIER3 - CAN1 RX */
00225     k_lpc_wake_can1_rx = 0x01000000U, /**< CRX1-DS pin (CAN1 RX) */
00226
00227     /** Mask of all defined wake sources */
00228     k_lpc_wake_all_mask = 0x01FFFFFFU,
00229 } rx_lpc_wake_flags_t;

```

8.87.3 Function Documentation

8.87.3.1 rx_lpc_enter_deep_software_standby()

```

rx_err_t rx_lpc_enter_deep_software_standby (
    uint32_t wake_mask,

```

```

    rx_lpc_deep_power_t deep_power,
    bool keep_io) [nodiscard]

```

Enter Deep Software Standby mode.

Configures DPSIER0..DPSIER3 from wake_mask, sets DPSBYCR.DEEPCUT, optionally sets DPSBYCR.IOKEEP, sets SBYCR.SSBY=1 and DPSBYCR.DPSBY=1 under PRC1 protection, then executes WAIT.

On deep-standby wake the MCU restarts from the reset vector. This function therefore does not return in that case on real hardware; the return path only exists for host-side unit tests. After reset, rx_lpc_was_deep_standby_wake() returns true and rx_lpc_get_wake_flags() reports which source caused the wake.

Parameters

in	wake_mask	Bitwise OR of rx_lpc_wake_flags_t values; must not be 0
in	deep_power	Deep-cut power configuration (DEEPCUT[1:0])
in	keep_io	If true, set DPSBYCR.IOKEEP=1 so I/O state is retained across the wake-up until the application clears it

Returns

rx_err_t Error code (returned only on host builds / on error)

Return values

k_rx_ok	Configuration accepted (host build)
k_rx_err_invalid_arg	wake_mask == 0 or contains undefined bits; deep_power not one of rx_lpc_deep_power_t
k_rx_err_not_initialized	rx_lpc_init() has not been called

Precondition

rx_lpc_init() has been called

wake_mask != 0 and (wake_mask & ~k_lpc_wake_all_mask) == 0

Corresponding pins / peripherals configured by the caller

Postcondition

DPSIER0..DPSIER3 reflect the enabled wake sources

DPSBYCR.DEEPCUT/IOKEEP/DPSBY configured per arguments

s_last_mode == k_lpc_mode_deep_software_standby (test-visible)

Warning

This function does not return on real hardware. Any cleanup required after deep standby must be performed at the reset vector / application startup path.

Since

Version 1.0.0

Definition at line 403 of file [rx_lpc.c](#).

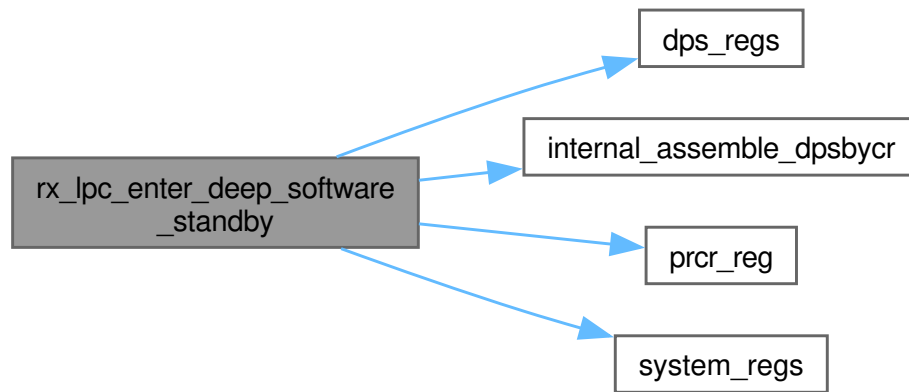
```

00404 {
00405     if (s_initialized != k_lpc_initialized) {
00406         return k_rx_err_not_initialized;
00407     }
00408     if (wake_mask == 0U) {
00409         return k_rx_err_invalid_arg;
00410     }
00411     if ((wake_mask & ~k_lpc_wake_all_mask) != 0U) {
00412         return k_rx_err_invalid_arg;
00413     }
00414     if ((deep_power != k_lpc_deep_ram_usb_on) && (deep_power != k_lpc_deep_ram_usb_off) &&
00415         (deep_power != k_lpc_deep_lvd_off)) {
00416         return k_rx_err_invalid_arg;
00417     }
00418
00419     const uint8_t dpsbycr_value = internal_assemble_dpsbycr(deep_power, keep_io);
00420
00421     #ifdef __RX__
00422         volatile uint16_t* prcr = prcr_reg();
00423         volatile rx_system_regs_t* sys = system_regs();
00424         volatile rx_dps_regs_t* dps = dps_regs();
00425
00426         *prcr = k_rx_prcr_unlock_prc1;
00427
00428         /* 1. Enable wake sources */
00429         dps->dpsier0 = (uint8_t)((wake_mask » 0U) & 0xFFU);
00430         dps->dpsier1 = (uint8_t)((wake_mask » 8U) & 0xFFU);
00431         dps->dpsier2 = (uint8_t)((wake_mask » 16U) & 0xFFU);
00432         dps->dpsier3 = (uint8_t)((wake_mask » 24U) & 0xFFU);
00433
00434         /* 2. Clear any stale wake flags */
00435         dps->dpsifr0 = 0U;
00436         dps->dpsifr1 = 0U;
00437         dps->dpsifr2 = 0U;
00438         dps->dpsifr3 = 0U;
00439
00440         /* 3. Program DPSBYCR (DEEPCUT, IOKEEP, DPSBY=1) */
00441         dps->dpsbycr = dpsbycr_value;
00442
00443         /* 4. Set SBYCR.SSBY so WAIT enters standby-family */
00444         sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssbby);
00445
00446         *prcr = k_rx_prcr_lock;
00447     #else
00448         (void)dpsbycr_value;
00449         internal_program_wake_enables(wake_mask);
00450     #endif
00451
00452     s_last_mode = k_lpc_mode_deep_software_standby;
00453
00454     #ifdef __RX__
00455         __asm__ volatile("wait");
00456         /* On real hardware, execution never reaches here - wake restarts at reset */
00457     #endif
00458
00459     return k_rx_ok;
00460 }

```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsbycr](#), [rx_dps_regs_t::dpsier0](#), [rx_dps_regs_t::dpsier1](#), [rx_dps_regs_t::dpsier2](#), [rx_dps_regs_t::dpsier3](#), [rx_dps_regs_t::dpsifr0](#), [rx_dps_regs_t::dpsifr1](#), [rx_dps_regs_t::dpsifr2](#), [rx_dps_regs_t::dpsifr3](#), [internal_assemble_dpsbycr\(\)](#), [k_lpc_deep_lvd_off](#), [k_lpc_deep_ram_usb_off](#), [k_lpc_deep_ram_usb_on](#), [k_lpc_initialized](#), [k_lpc_mode_deep_software_standby](#), [k_lpc_wake_all_mask](#), [k_sbycr_ssbby](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.87.3.2 rx_lpc_enter_sleep()

```
rx_err_t rx_lpc_enter_sleep (
    void ) [nodiscard]
```

Enter Sleep mode (CPU stop, peripherals run).

Clears SBYCR.SSBY under PRC1 protection, then executes the WAIT instruction. On RX72N, WAIT with SSBY=0 enters sleep mode. Execution resumes from the instruction after WAIT when any unmasked interrupt fires.

Returns

rx_err_t Error code

Return values

k_rx_ok	Returned from sleep by an interrupt
k_rx_err_not_initialized	rx_lpc_init() has not been called

Precondition

[rx_lpc_init\(\)](#) has been called

At least one interrupt source is armed and enabled in ICU

Postcondition

SBYCR.SSBY == 0 on exit

s_last_mode == k_lpc_mode_sleep (test-visible)

Note

On host builds the WAIT instruction is replaced by a no-op.

Since

Version 1.0.0

Enter Sleep mode (CPU stop, peripherals run).

Configures SBYCR for Sleep mode (SSBY=0) under PRCR.PRC1 protection, then issues a WAIT instruction to halt the CPU. Peripheral clocks continue running, so any enabled interrupt resumes execution at the instruction following WAIT. Records the last entered mode for post-mortem diagnostics. On the unit-test target the WAIT is skipped.

Note

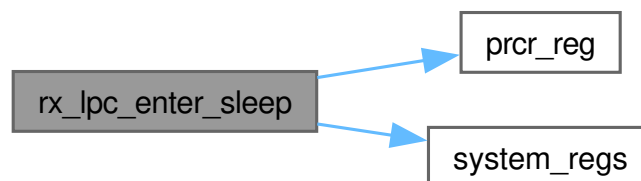
Not thread-safe. Serialize all rx_lpc_* operations.

Definition at line 338 of file [rx_lpc.c](#).

```
00339 {
00340     if (s_initialized != k_lpc_initialized) {
00341         return k_rx_err_not_initialized;
00342     }
00343
00344     #ifdef __RX__
00345     volatile uint16_t* prcr = prcr_reg();
00346     volatile rx_system_regs_t* sys = system_regs();
00347
00348     *prcr = k_rx_prcr_unlock_prc1;
00349     sys->sbycr = (uint16_t)(sys->sbycr & (uint16_t)~k_sbycr_ssbby);
00350     *prcr = k_rx_prcr_lock;
00351     #endif
00352
00353     s_last_mode = k_lpc_mode_sleep;
00354
00355     #ifdef __RX__
00356     __asm__ volatile("wait");
00357     #endif
00358
00359     return k_rx_ok;
00360 }
```

References [k_lpc_initialized](#), [k_lpc_mode_sleep](#), [k_sbycr_ssbby](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.87.3.3 rx_lpc_enter_software_standby()

```
rx_err_t rx_lpc_enter_software_standby (
    void ) [nodiscard]
```

Enter Software Standby mode.

Sets SBYCR.SSBY=1 and clears DPSBYCR.DPSBY=0 under PRC1 protection, then executes WAIT. On exit the MCU resumes at the instruction after WAIT (not the reset vector). RAM contents are retained.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Returned from software standby
<code>k_rx_err_not_initialized</code>	rx_lpc_init() has not been called

Precondition

[rx_lpc_init\(\)](#) has been called

A wake source (NMI, IRQn, RTC, IWDG, LVD) is enabled in the ICU

Postcondition

SBYCR.SSBY == 1 on entry, hardware auto-clears on wake

`s_last_mode` == `k_lpc_mode_software_standby` (test-visible)

Note

Typical current draw ~3 uA at 3.3 V (RX72N datasheet Table 58.xx).

Since

Version 1.0.0

Enter Software Standby mode.

Configures SBYCR.SSBY=1 and DPSBYCR.DPSBY=0 under PRCR.PRC1 so that the upcoming WAIT instruction enters Software Standby (not Sleep, not Deep Standby). Most peripheral clocks are gated off; only configured wake sources (level-sensitive IRQ pins, RTC, watchdog) can resume execution. On the unit-test target the WAIT is skipped.

Note

Not thread-safe. Serialize all `rx_lpc_*` operations.

Definition at line 374 of file [rx_lpc.c](#).

```

00375 {
00376     if (s_initialized != k_lpc_initialized) {
00377         return k_rx_err_not_initialized;
00378     }
00379 }
00380 #ifdef __RX__
00381 volatile uint16_t* prcr = prcr_reg();
00382 volatile rx_system_regs_t* sys = system_regs();
00383 volatile rx_dps_regs_t* dps = dps_regs();
00384
00385 *prcr = k_rx_prcr_unlock_prc1;
00386 /* Clear DPSBY so WAIT enters software standby, not deep standby */
00387 dps->dpsbycr = (uint8_t)(dps->dpsbycr & (uint8_t)~k_dpsbycr_dpsby);
00388 /* Set SSBY so WAIT enters standby, not sleep */
00389 sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssb);
00390 *prcr = k_rx_prcr_lock;
00391 #endif
00392
00393 s_last_mode = k_lpc_mode_software_standby;
00394
```

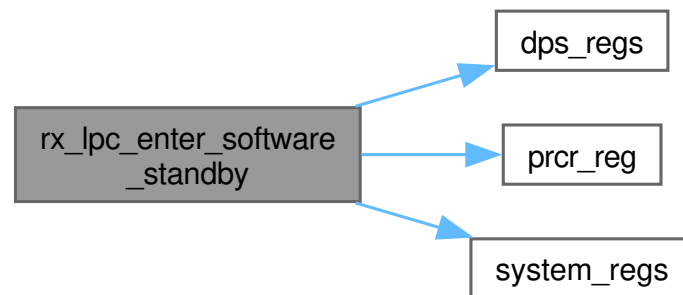
```

00395 #ifdef __RX__
00396     __asm__ volatile("wait");
00397 #endif
00398
00399     return k_rx_ok;
00400 }

```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsbycr](#), [k_dpsbycr_dpsby](#), [k_lpc_initialized](#), [k_lpc_mode_software_standby](#), [k_sbycr_ssby](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.87.3.4 rx_lpc_get_wake_flags()

```

rx_err_t rx_lpc_get_wake_flags (
    uint32_t * flags) [nodiscard]

```

Read the deep-standby wake flags (DPSIFR0..DPSIFR3).

Returns a 32-bit bitmask describing which wake sources triggered the most recent deep-standby exit. The layout matches [rx_lpc_wake_flags_t](#). Reading does not clear the flags; the driver clears them in [rx_lpc_init\(\)](#).

Parameters

out	flags	Pointer to receive the wake-flag bitmask
-----	-------	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Flags read successfully
<code>k_rx_err_invalid_arg</code>	flags is NULL
<code>k_rx_err_not_initialized</code>	rx_lpc_init() has not been called

Precondition

[rx_lpc_init\(\)](#) has been called
 flags != NULL

Postcondition

*flags contains a bitwise OR of [rx_lpc_wake_flags_t](#) values

Since

Version 1.0.0

Read the deep-standby wake flags (DPSIFR0..DPSIFR3).

Copies the cached `s_latched_wake_flags` into the caller-supplied destination. The bitfield has one bit per wake source; the layout is defined by `rx_lpc_wake_t`. Useful after wake to determine which event resumed execution.

Note

Not safe from ISR (acquires no lock but reads driver state that could be torn during init).

Definition at line 503 of file [rx_lpc.c](#).

```
00504 {
00505     if (flags == NULL) {
00506         return k_rx_err_invalid_arg;
00507     }
00508     if (s_initialized != k_lpc_initialized) {
00509         return k_rx_err_not_initialized;
00510     }
00511
00512     *flags = s_latched_wake_flags;
00513     return k_rx_ok;
00514 }
```

References [k_lpc_initialized](#), [s_initialized](#), and [s_latched_wake_flags](#).

8.87.3.5 rx`lpc`init()

```
rx_err_t rx_lpc_init (
    void ) [nodiscard]
```

Initialize the Low Power Consumption driver.

Clears the driver's internal state, latches whether the most recent reset was a deep-standby wake-up (so application code can read it back via [rx_lpc_was_deep_standby_wake\(\)](#)), and pre-clears any pending deep-standby wake flags (DPSIFR0..DPSIFR3) that may have been set by the preceding deep-standby exit.

This function does not change the operating power mode; applications should call [rx_lpc_set_operating_power\(\)](#) separately if they want to switch away from the default high-speed mode.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Driver initialized successfully
----------------------	---------------------------------

Precondition

May be called exactly once per reset
System clocks must already be configured

Postcondition

Driver state tracking is reset
Deep-standby wake flags are cleared

Note

Safe to call from single-threaded startup only.

Since

Version 1.0.0

Definition at line 242 of file [rx_lpc.c](#).

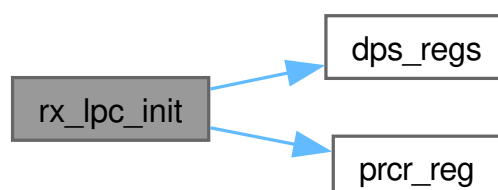
```

00243 {
00244     /* Latch whatever the boot path told us about the previous reset */
00245     #ifdef UNIT_TEST
00246         s_was_deep_standby_wake = s_inject_was_wake;
00247         s_latched_wake_flags    = s_inject_pending_flags;
00248         s_inject_was_wake       = false;
00249         s_inject_pending_flags  = 0U;
00250     #else
00251         s_was_deep_standby_wake = false;
00252         s_latched_wake_flags    = 0U;
00253     #endif
00254
00255     #ifdef __RX__
00256         /* Read DPSIFR0..DPSIFR3, OR them into the latched flags, then clear them */
00257         volatile rx_dps_regs_t* dps = dps_regs();
00258         uint32_t flags = 0U;
00259         flags |= ((uint32_t)dps->dpsifr0) « 0U;
00260         flags |= ((uint32_t)dps->dpsifr1) « 8U;
00261         flags |= ((uint32_t)dps->dpsifr2) « 16U;
00262         flags |= ((uint32_t)dps->dpsifr3) « 24U;
00263         s_latched_wake_flags |= flags;
00264
00265         /* Write 0 to each flag register to clear any pending wake bits */
00266         volatile uint16_t* prcr = prcr_reg();
00267         *prcr                    = k_rx_prcr_unlock_prc1;
00268         dps->dpsifr0              = 0U;
00269         dps->dpsifr1              = 0U;
00270         dps->dpsifr2              = 0U;
00271         dps->dpsifr3              = 0U;
00272         *prcr                    = k_rx_prcr_lock;
00273     #endif
00274
00275     s_last_mode = k_lpc_mode_none;
00276     s_initialized = k_lpc_initialized;
00277     return k_rx_ok;
00278 }

```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsifr0](#), [rx_dps_regs_t::dpsifr1](#), [rx_dps_regs_t::dpsifr2](#), [rx_dps_regs_t::dpsifr3](#), [k_lpc_initialized](#), [k_lpc_mode_none](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [s_latched_wake_flags](#), and [s_was_deep_standby_wake](#).

Here is the call graph for this function:



8.87.3.6 rx`lpc`set`operating`power()

```
rx_err_t rx_lpc_set_operating_power (
    rx_lpc_opcc_mode_t mode) [nodiscard]
```

Select the operating power control (OPCCR) mode.

Writes OPCCR.OPCM[2:0] under PRC1 protection and polls OPCCR.OPCMTSF=0 for transition completion (bounded retry).

Parameters

in	mode	Target operating power mode
----	------	-----------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Mode transition completed
k_rx_err_invalid_arg	mode not one of rx_lpc_opcc_mode_t
k_rx_err_not_initialized	rx_lpc_init() has not been called
k_rx_err_hw_timeout	OPCMTSF did not clear within bounded retries

Precondition

[rx_lpc_init\(\)](#) has been called

Clock tree pre-conditions for the target mode are satisfied by the caller

Postcondition

OPCCR.OPCM[2:0] reflects the requested mode

OPCCR.OPCMTSF == 0

Warning

Low-speed mode 2 requires the caller to stop HOCO first (RX72N Manual Section 11.4.4). Low-speed mode 1 requires PLL to be stopped.

Since

Version 1.0.0

Select the operating power control (OPCCR) mode.

Programs the OPCCR register with the OPCM bits derived from the requested mode (high-speed / middle-speed / low-power, etc.). Performs the standard PRCR.PRC1 unlock-write-lock dance, then bounded-polls the OPCMTSF transition-status flag for completion (max k_opcmtsfs_poll_max iterations). On the unit-test target the register write is skipped.

Note

Not thread-safe. Serialize all `rx_lpc_*` operations.

Definition at line 293 of file `rx_lpc.c`.

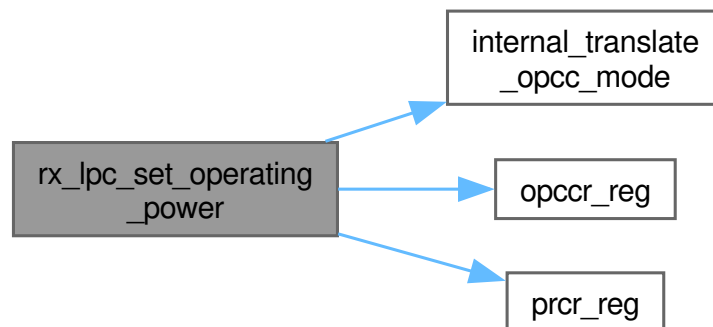
```

00294 {
00295     if (s_initialized != k_lpc_initialized) {
00296         return k_rx_err_not_initialized;
00297     }
00298
00299     uint8_t opcm_bits = 0U;
00300     rx_err_t err = internal_translate_opcc_mode(mode, &opcm_bits);
00301     if (err != k_rx_ok) {
00302         return err;
00303     }
00304
00305     #ifdef __RX__
00306     volatile uint16_t* prcr = prcr_reg();
00307     volatile uint8_t* ocr = opccr_reg();
00308
00309     *prcr = k_rx_prcr_unlock_prc1;
00310     *ocr = opcm_bits;
00311     *prcr = k_rx_prcr_lock;
00312
00313     /* Bounded poll for OPCMTSF=0 (Manual 11.2.6, page 414) */
00314     for (uint16_t i = 0U; i < k_opcmtsf_poll_max; i++) {
00315         if ((*ocr & k_opccr_opcmtsf) == 0U) {
00316             return k_rx_ok;
00317         }
00318     }
00319     return k_rx_err_hw_timeout;
00320 #else
00321     (void)opcm_bits;
00322     return k_rx_ok;
00323 #endif
00324 }

```

References `internal_translate_opcc_mode()`, `k_lpc_initialized`, `k_opccr_opcmtsf`, `k_opcmtsf_poll_max`, `opccr_reg()`, `prcr_reg()`, and `s_initialized`.

Here is the call graph for this function:



8.87.3.7 rx_lpc_was_deep_standby_wake()

```

bool rx_lpc_was_deep_standby_wake (
    void ) [nodiscard]

```

Query whether the last reset was caused by a deep-standby wake.

The information is latched by `rx_lpc_init()` from the internal state it inherits from the reset path. On real hardware the caller is expected to have inspected `RSTSR0/1/2` (see `rx_rstsr` accessors) and, if appropriate, to have called the test-only setter `rx_lpc_test_set_deep_standby_wake()` or equivalent boot-path bridge. The default for a cold power-on is false.

Returns

true if the most recent reset was a deep-standby wake-up
false otherwise

Precondition

[rx_lpc_init\(\)](#) has been called

Postcondition

Return value is stable until the next reset

Since

Version 1.0.0

Query whether the last reset was caused by a deep-standby wake.

Returns the cached `s_was_deep_standby_wake` flag latched at boot time. Allows application code to distinguish between a power-on reset and a resume from Deep Software Standby and take different paths (e.g., skip full re-initialization when waking).

Returns

true if the most recent reset was a Deep Software Standby wake; false otherwise (or before [rx_lpc_init\(\)](#) ran).

Precondition

None (function is safe to call at any time after init or before).

Postcondition

No state mutated.

Return value is stable across multiple calls in the same boot session.

Note

Safe from any context including ISR; pure read of static storage.

See also

[rx_lpc_get_wake_flags\(\)](#) Detailed wake-source bitfield.

Since

Version 1.0.0

Definition at line 486 of file [rx_lpc.c](#).

```
00487 {  
00488     return s_was_deep_standby_wake;  
00489 }
```

References [s_was_deep_standby_wake](#).

8.88 rx_lpc.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_lpc.h
00003  * @brief Low Power Consumption (LPC) HAL Driver for RX72N
00004  *
00005  * @details
00006  * **Hardware Abstraction Layer** for the RX72N Low Power Consumption subsystem.
00007  * Provides a small, auditable API for entering the three RX72N low-power modes
00008  * and for selecting the operating power control mode that governs active-run
00009  * current.
00010  *
00011  * ## Supported Modes (RX72N Manual Chapter 11)
00012  *
00013  * | Mode | CPU | Peripherals | RAM | Wake Latency | Exit Vector |
00014  * |-----|-----|-----|-----|-----|-----|
00015  * | Sleep | Stop | Run | Retained | ns | Continue next PC |
00016  * | Software Standby | Stop | Stop | Retained | us | Continue next PC |
00017  * | Deep Software Standby | Stop | Stop | Lost* | ms | Reset vector |
00018  *
00019  * @verbatim
00020  * * Deep Software Standby: On-chip RAM contents are lost, but the 32-byte
00021  * * Deep Standby Backup Register (DPSBKR) is retained. See rx72n_lpc_regs.h
00022  * * for the DPSBKR accessor.
00023  * @endverbatim
00024  *
00025  * ## Operating Power Control (OPCCR) Modes
00026  *
00027  * | Mode | ICLK max | Notes |
00028  * |-----|-----|-----|
00029  * | High-speed (default) | 240 MHz | Full performance |
00030  * | Low-speed 1 | 1 MHz | PLL/HOCO may be stopped |
00031  * | Low-speed 2 | 264 kHz | HOCO must be stopped (LOCO/Main only) |
00032  *
00033  * @note RX72N only implements three OPCCR modes (high-speed, low-speed 1,
00034  * low-speed 2). The "middle-speed" and "sub-oscillator-speed" modes
00035  * from the RX100/RX200 family (SOPCCR) are **not available** on RX72N
00036  * and therefore have no corresponding API enumerators in this driver.
00037  *
00038  * ## Wake Source Selection (Deep Software Standby)
00039  *
00040  * * Wake sources for deep software standby are specified as a 32-bit bitmask
00041  * * passed to rx_lpc_enter_deep_software_standby(). The bitmask layout mirrors
00042  * * DPSIER0..DPSIER3 as a single little-endian word:
00043  *
00044  * @verbatim
00045  * * bits [ 7: 0] - IRQ0-DS .. IRQ7-DS (DPSIER0)
00046  * * bits [15: 8] - IRQ8-DS .. IRQ15-DS (DPSIER1)
00047  * * bits [23:16] - LVD1/LVD2/RTC-P/RTC-A/NMI/IIC-SDA/IIC-SCL/USB (DPSIER2)
00048  * * bits [31:24] - CAN1-RX (bit 24 only) (DPSIER3)
00049  * @endverbatim
00050  *
00051  * * The same encoding is returned by rx_lpc_get_wake_flags() when reading the
00052  * * DPSIFRx flag registers after a deep-standby wake.
00053  *
00054  * ## Hardware Register References
00055  *
00056  * | Register | Address | Manual Section | Purpose |
00057  * |-----|-----|-----|-----|
00058  * | SBYCR | 0x0008000C | 11.2.1 (p.405) | Standby control (SSBY, OPE) |
00059  * | OPCCR | 0x000800A0 | 11.2.6 (p.414) | Operating power control |
00060  * | RSTCKCR | 0x000800A1 | 11.2.7 (p.417) | Sleep-mode return clock |
00061  * | DPSBYCR | 0x0008C280 | 11.2.8 (p.418) | Deep standby control |
00062  * | DPSIER0-3 | 0x0008C282 | 11.2.9-12 | Deep standby IRQ enable |
00063  * | DPSIFR0-3 | 0x0008C286 | 11.2.13-16 | Deep standby wake flags |
00064  * | DPSIEGR0-3 | 0x0008C28A | 11.2.17-20 | Deep standby edge select |
00065  * | DPSBKR | 0x0008C2A0 | 11.2.21 (p.433) | Deep standby backup RAM (32B) |
00066  *
00067  * ## Power Protection (PRCR)
00068  *
00069  * * All writes to SBYCR, OPCCR, RSTCKCR, and DPSBYCR require PRC1 to be
00070  * * unlocked via PRCR. This driver handles PRCR lock/unlock internally.
00071  *
00072  * ## NASA Power of 10 Compliance
00073  *
00074  * | Rule | Status | Implementation |
00075  * |-----|-----|-----|
00076  * | 1 | [OK] | No goto/setjmp/recursion |
00077  * | 2 | [OK] | All loops bounded (OPCMTSF poll has bounded retry) |
00078  * | 3 | [OK] | Zero dynamic allocation |
00079  * | 4 | [OK] | All functions < 60 lines |
00080  * | 5 | [OK] | >= 2 pre/post conditions per function |
00081  * | 6 | [OK] | Narrow scope, file-scope statics (s_*) |
00082  * | 7 | [OK] | All return values checked |

```

```

00083 * | 8 | [OK] | C23 typed enums, minimal preprocessor |
00084 * | 9 | [OK] | Single-level pointers, no function pointers |
00085 * | 10 | [OK] | Compiles with -Wall -Wextra -Werror |
00086 *
00087 * @see rx72n_lpc_regs.h Register layout and bit definitions
00088 * @see rx72n_system_regs.h SBYCR / PRCR / MSTPCRx accessors
00089 * @see RX72N Hardware Manual Chapter 11 - Low Power Consumption
00090 *
00091 * @author Locked, Inc.
00092 * @date 2026-04-21
00093 * @copyright Copyright (c) 2026 Locked Inc.
00094 * SPDX-License-Identifier: MIT
00095 * @since Version 1.0.0
00096 */
00097
00098 #pragma once
00099
00100 #include <stdbool.h>
00101 #include <stdint.h>
00102
00103 #include "rx_err.h"
00104
00105 #ifdef __cplusplus
00106 extern "C" {
00107 #endif
00108
00109 /*
=====
00110 * Configuration Types
00111 *
=====
00112 */
00113
00114 /**
00115 * @enum rx_lpc_opcc_mode_t
00116 * @brief Operating Power Control (OPCCR) mode selector
00117 *
00118 * @details
00119 * Selects the active-run power mode used while the CPU is running. The
00120 * RX72N supports exactly three OPCCR modes (RX72N Manual Section 11.2.6,
00121 * page 414). The underlying OPCM[2:0] field values are encoded directly.
00122 *
00123 * @par Mode Transitions
00124 * Writing to OPCCR triggers an internal transition; the hardware sets
00125 * OPCCR.OPCMTSF=1 until the transition completes. This driver polls that
00126 * flag with a bounded retry count.
00127 *
00128 * @warning Low-speed mode 2 requires HOCO to be stopped. The caller is
00129 * responsible for configuring the clock tree before invoking
00130 * rx_lpc_set_operating_power().
00131 *
00132 * @since Version 1.0.0
00133 */
00134 typedef enum : uint8_t {
00135     k_lpc_opcc_high_speed = 0, /**< High-speed mode (OPCM=000b, ICLK <= 240 MHz) */
00136     k_lpc_opcc_low_speed_1 = 1, /**< Low-speed mode 1 (OPCM=110b, ICLK <= 1 MHz) */
00137     k_lpc_opcc_low_speed_2 = 2, /**< Low-speed mode 2 (OPCM=111b, ICLK <= 264 kHz) */
00138 } rx_lpc_opcc_mode_t;
00139
00140 /**
00141 * @enum rx_lpc_deep_power_t
00142 * @brief Deep-standby power supply configuration (DPSBYCR.DEEPCUT)
00143 *
00144 * @details
00145 * Selects how aggressively the deep-standby mode cuts power. Mirrors
00146 * DPSBYCR.DEEPCUT[1:0] (RX72N Manual Section 11.2.8, page 418).
00147 *
00148 * @note The 10b DEEPCUT combination is reserved and is not exposed here.
00149 *
00150 * @since Version 1.0.0
00151 */
00152 typedef enum : uint8_t {
00153     k_lpc_deep_ram_usb_on = 0, /**< DEEPCUT=00b: standby RAM + USB powered */
00154     k_lpc_deep_ram_usb_off = 1, /**< DEEPCUT=01b: standby RAM + USB powered off */
00155     k_lpc_deep_lvd_off = 2, /**< DEEPCUT=11b: LVD also off (lowest power) */
00156 } rx_lpc_deep_power_t;
00157
00158 /**
00159 * @enum rx_lpc_last_mode_t
00160 * @brief Last low-power mode entered (test / diagnostic hook)
00161 *
00162 * @details
00163 * Recorded by rx_lpc_enter_*( ) prior to executing the WAIT instruction so
00164 * that host-side unit tests can verify the driver reached the correct code
00165 * path without executing a real WAIT.
00166 *
00167 * @since Version 1.0.0

```

```

00168 */
00169 typedef enum : uint8_t {
00170     k_lpc_mode_none           = 0, /**< No low-power entry attempted yet */
00171     k_lpc_mode_sleep         = 1, /**< Sleep mode (WAIT, SBYCR.SSBY=0) */
00172     k_lpc_mode_software_standby = 2, /**< Software standby (SBYCR.SSBY=1, DPSBY=0) */
00173     k_lpc_mode_deep_software_standby = 3, /**< Deep software standby (SSBY=1, DPSBY=1) */
00174 } rx_lpc_last_mode_t;
00175
00176 /*
=====
00177  * Wake-source Bitmask (Deep Software Standby)
00178  *
=====
00179  */
00180 /**
00181  * @enum rx_lpc_wake_flags_t
00182  * @brief Wake-source bits for deep software standby
00183  *
00184  * @details
00185  * 32-bit bitmask layout used by rx_lpc_enter_deep_software_standby() and
00186  * returned by rx_lpc_get_wake_flags(). The layout mirrors DPSIER0..DPSIER3
00187  * (and the corresponding DPSIFR0..DPSIFR3 flag registers) as a single
00188  * little-endian word.
00189  *
00190  * @since Version 1.0.0
00191  */
00192 /*
00193 typedef enum : uint32_t {
00194     /* DPSIER0 - IRQ0-DS .. IRQ7-DS */
00195     k_lpc_wake_irq0 = 0x00000001U, /**< IRQ0-DS pin */
00196     k_lpc_wake_irq1 = 0x00000002U, /**< IRQ1-DS pin */
00197     k_lpc_wake_irq2 = 0x00000004U, /**< IRQ2-DS pin */
00198     k_lpc_wake_irq3 = 0x00000008U, /**< IRQ3-DS pin */
00199     k_lpc_wake_irq4 = 0x00000010U, /**< IRQ4-DS pin */
00200     k_lpc_wake_irq5 = 0x00000020U, /**< IRQ5-DS pin */
00201     k_lpc_wake_irq6 = 0x00000040U, /**< IRQ6-DS pin */
00202     k_lpc_wake_irq7 = 0x00000080U, /**< IRQ7-DS pin */
00203
00204     /* DPSIER1 - IRQ8-DS .. IRQ15-DS */
00205     k_lpc_wake_irq8 = 0x00000100U, /**< IRQ8-DS pin */
00206     k_lpc_wake_irq9 = 0x00000200U, /**< IRQ9-DS pin */
00207     k_lpc_wake_irq10 = 0x00000400U, /**< IRQ10-DS pin */
00208     k_lpc_wake_irq11 = 0x00000800U, /**< IRQ11-DS pin */
00209     k_lpc_wake_irq12 = 0x00001000U, /**< IRQ12-DS pin */
00210     k_lpc_wake_irq13 = 0x00002000U, /**< IRQ13-DS pin */
00211     k_lpc_wake_irq14 = 0x00004000U, /**< IRQ14-DS pin */
00212     k_lpc_wake_irq15 = 0x00008000U, /**< IRQ15-DS pin */
00213
00214     /* DPSIER2 - LVD / RTC / NMI / IIC / USB */
00215     k_lpc_wake_lvd1 = 0x00010000U, /**< LVD1 (Voltage Monitor 1) */
00216     k_lpc_wake_lvd2 = 0x00020000U, /**< LVD2 (Voltage Monitor 2) */
00217     k_lpc_wake_rtc_periodic = 0x00040000U, /**< RTC periodic interrupt */
00218     k_lpc_wake_rtc_alarm = 0x00080000U, /**< RTC alarm interrupt */
00219     k_lpc_wake_nmi = 0x00100000U, /**< NMI pin (write-once bit) */
00220     k_lpc_wake_iic_sda = 0x00200000U, /**< SDA2-DS pin */
00221     k_lpc_wake_iic_scl = 0x00400000U, /**< SCL2-DS pin */
00222     k_lpc_wake_usb = 0x00800000U, /**< USB suspend/resume */
00223
00224     /* DPSIER3 - CAN1 RX */
00225     k_lpc_wake_can1_rx = 0x01000000U, /**< CRX1-DS pin (CAN1 RX) */
00226
00227     /** Mask of all defined wake sources */
00228     k_lpc_wake_all_mask = 0x01FFFFFFU,
00229 } rx_lpc_wake_flags_t;
00230
00231 /*
=====
00232  * Initialization
00233  *
=====
00234  */
00235 /**
00236  * @brief Initialize the Low Power Consumption driver
00237  *
00238  * @details
00239  * Clears the driver's internal state, latches whether the most recent reset
00240  * was a deep-standby wake-up (so application code can read it back via
00241  * rx_lpc_was_deep_standby_wake()), and pre-clears any pending deep-standby
00242  * wake flags (DPSIFR0..DPSIFR3) that may have been set by the preceding
00243  * deep-standby exit.
00244  *
00245  * This function does not change the operating power mode; applications
00246  * should call rx_lpc_set_operating_power() separately if they want to switch
00247  * away from the default high-speed mode.
00248  *
00249  * @return rx_err_t Error code
00250  */

```



```

00251 * @retval k_rx_ok Driver initialized successfully
00252 *
00253 * @pre May be called exactly once per reset
00254 * @pre System clocks must already be configured
00255 *
00256 * @post Driver state tracking is reset
00257 * @post Deep-standby wake flags are cleared
00258 *
00259 * @note Safe to call from single-threaded startup only.
00260 *
00261 * @since Version 1.0.0
00262 */
00263 [[nodiscard]] rx_err_t rx_lpc_init(void);
00264
00265 /*
=====
00266 * Operating Power Control (Active Run)
00267 *
=====
00268 */
00269
00270 /**
00271 * @brief Select the operating power control (OPCCR) mode
00272 *
00273 * @details
00274 * Writes OPCCR.OPCM[2:0] under PRC1 protection and polls OPCCR.OPCMTSF=0
00275 * for transition completion (bounded retry).
00276 *
00277 * @param[in] mode Target operating power mode
00278 *
00279 * @return rx_err_t Error code
00280 * @retval k_rx_ok Mode transition completed
00281 * @retval k_rx_err_invalid_arg mode not one of rx_lpc_opcc_mode_t
00282 * @retval k_rx_err_not_initialized rx_lpc_init() has not been called
00283 * @retval k_rx_err_hw_timeout OPCMTSF did not clear within bounded retries
00284 *
00285 * @pre rx_lpc_init() has been called
00286 * @pre Clock tree pre-conditions for the target mode are satisfied by the caller
00287 *
00288 * @post OPCCR.OPCM[2:0] reflects the requested mode
00289 * @post OPCCR.OPCMTSF == 0
00290 *
00291 * @warning Low-speed mode 2 requires the caller to stop HOCO first (RX72N
00292 * Manual Section 11.4.4). Low-speed mode 1 requires PLL to be
00293 * stopped.
00294 *
00295 * @since Version 1.0.0
00296 */
00297 [[nodiscard]] rx_err_t rx_lpc_set_operating_power(rx_lpc_opcc_mode_t mode);
00298
00299 /*
=====
00300 * Mode Entry
00301 *
=====
00302 */
00303
00304 /**
00305 * @brief Enter Sleep mode (CPU stop, peripherals run)
00306 *
00307 * @details
00308 * Clears SBYCR.SSBY under PRC1 protection, then executes the WAIT
00309 * instruction. On RX72N, WAIT with SSBY=0 enters sleep mode. Execution
00310 * resumes from the instruction after WAIT when any unmasked interrupt
00311 * fires.
00312 *
00313 * @return rx_err_t Error code
00314 * @retval k_rx_ok Returned from sleep by an interrupt
00315 * @retval k_rx_err_not_initialized rx_lpc_init() has not been called
00316 *
00317 * @pre rx_lpc_init() has been called
00318 * @pre At least one interrupt source is armed and enabled in ICU
00319 *
00320 * @post SBYCR.SSBY == 0 on exit
00321 * @post s_last_mode == k_lpc_mode_sleep (test-visible)
00322 *
00323 * @note On host builds the WAIT instruction is replaced by a no-op.
00324 *
00325 * @since Version 1.0.0
00326 */
00327 [[nodiscard]] rx_err_t rx_lpc_enter_sleep(void);
00328
00329 /**
00330 * @brief Enter Software Standby mode
00331 *
00332 * @details
00333 * Sets SBYCR.SSBY=1 and clears DPSBYCR.DPSBY=0 under PRC1 protection,

```

```

00334 * then executes WAIT. On exit the MCU resumes at the instruction after
00335 * WAIT (not the reset vector). RAM contents are retained.
00336 *
00337 * @return rx_err_t Error code
00338 * @retval k_rx_ok Returned from software standby
00339 * @retval k_rx_err_not_initialized rx_lpc_init() has not been called
00340 *
00341 * @pre rx_lpc_init() has been called
00342 * @pre A wake source (NMI, IRQn, RTC, IWDG, LVD) is enabled in the ICU
00343 *
00344 * @post SBYCR.SSBY == 1 on entry, hardware auto-clears on wake
00345 * @post s_last_mode == k_lpc_mode_software_standby (test-visible)
00346 *
00347 * @note Typical current draw ~3 uA at 3.3 V (RX72N datasheet Table 58.xx).
00348 *
00349 * @since Version 1.0.0
00350 */
00351 [[nodiscard]] rx_err_t rx_lpc_enter_software_standby(void);
00352
00353 /**
00354 * @brief Enter Deep Software Standby mode
00355 *
00356 * @details
00357 * Configures DPSIER0..DPSIER3 from @p wake_mask, sets DPSBYCR.DEEPCUT,
00358 * optionally sets DPSBYCR.IOKEEP, sets SBYCR.SSBY=1 and DPSBYCR.DPSBY=1
00359 * under PRC1 protection, then executes WAIT.
00360 *
00361 * On deep-standby wake the MCU restarts from the reset vector. This
00362 * function therefore does not return in that case on real hardware;
00363 * the return path only exists for host-side unit tests. After reset,
00364 * rx_lpc_was_deep_standby_wake() returns true and rx_lpc_get_wake_flags()
00365 * reports which source caused the wake.
00366 *
00367 * @param[in] wake_mask Bitwise OR of rx_lpc_wake_flags_t values; must not be 0
00368 * @param[in] deep_power Deep-cut power configuration (DEEPCUT[1:0])
00369 * @param[in] keep_io If true, set DPSBYCR.IOKEEP=1 so I/O state is retained
00370 * across the wake-up until the application clears it
00371 *
00372 * @return rx_err_t Error code (returned only on host builds / on error)
00373 * @retval k_rx_ok Configuration accepted (host build)
00374 * @retval k_rx_err_invalid_arg wake_mask == 0 or contains undefined bits; deep_power not one of
rx_lpc_deep_power_t
00375 * @retval k_rx_err_not_initialized rx_lpc_init() has not been called
00376 *
00377 * @pre rx_lpc_init() has been called
00378 * @pre wake_mask != 0 and (wake_mask & ~k_lpc_wake_all_mask) == 0
00379 * @pre Corresponding pins / peripherals configured by the caller
00380 *
00381 * @post DPSIER0..DPSIER3 reflect the enabled wake sources
00382 * @post DPSBYCR.DEEPCUT/IOKEEP/DPSBY configured per arguments
00383 * @post s_last_mode == k_lpc_mode_deep_software_standby (test-visible)
00384 *
00385 * @warning This function does not return on real hardware. Any cleanup
00386 * required after deep standby must be performed at the reset
00387 * vector / application startup path.
00388 *
00389 * @since Version 1.0.0
00390 */
00391 [[nodiscard]] rx_err_t rx_lpc_enter_deep_software_standby(uint32_t wake_mask,
00392 rx_lpc_deep_power_t deep_power,
00393 bool keep_io);
00394
00395 /
00396 =====
00397 * Post-Wake Diagnostics
00398 =====
00399 */
00400 /**
00401 * @brief Query whether the last reset was caused by a deep-standby wake
00402 *
00403 * @details
00404 * The information is latched by rx_lpc_init() from the internal state it
00405 * inherits from the reset path. On real hardware the caller is expected to
00406 * have inspected RSTSR0/1/2 (see rx_rstsr accessors) and, if appropriate,
00407 * to have called the test-only setter rx_lpc_test_set_deep_standby_wake()
00408 * or equivalent boot-path bridge. The default for a cold power-on is false.
00409 *
00410 * @return true if the most recent reset was a deep-standby wake-up
00411 * @return false otherwise
00412 *
00413 * @pre rx_lpc_init() has been called
00414 * @post Return value is stable until the next reset
00415 *
00416 * @since Version 1.0.0
00417 */

```

```

00418 [[nodiscard]] bool rx_lpc_was_deep_standby_wake(void);
00419
00420 /**
00421  * @brief Read the deep-standby wake flags (DPSIFR0..DPSIFR3)
00422  *
00423  * @details
00424  * Returns a 32-bit bitmask describing which wake sources triggered the most
00425  * recent deep-standby exit. The layout matches rx_lpc_wake_flags_t. Reading
00426  * does not clear the flags; the driver clears them in rx_lpc_init().
00427  *
00428  * @param[out] flags Pointer to receive the wake-flag bitmask
00429  *
00430  * @return rx_err_t Error code
00431  * @retval k_rx_ok Flags read successfully
00432  * @retval k_rx_err_invalid_arg flags is NULL
00433  * @retval k_rx_err_not_initialized rx_lpc_init() has not been called
00434  *
00435  * @pre rx_lpc_init() has been called
00436  * @pre flags != NULL
00437  *
00438  * @post *flags contains a bitwise OR of rx_lpc_wake_flags_t values
00439  *
00440  * @since Version 1.0.0
00441  */
00442 [[nodiscard]] rx_err_t rx_lpc_get_wake_flags(uint32_t* flags);
00443
00444 /*
=====
00445 * Test-Only Helpers (UNIT_TEST builds)
00446 *
=====
00447 */
00448
00449 #ifdef UNIT_TEST
00450 /**
00451  * @brief Reset all driver state (unit-test only)
00452  *
00453  * @details
00454  * Clears initialization flag, last-mode tracker, simulated DPSBYCR / SBYCR
00455  * shadow registers (in host builds), and simulated DPSIFRx flags. Intended
00456  * to be called from setUp() of every Unity test.
00457  *
00458  * @post Driver is returned to the exact state it has immediately after reset
00459  * @post rx_lpc_init() must be called again before any other API call
00460  *
00461  * @since Version 1.0.0
00462  */
00463 void rx_lpc_test_reset(void);
00464
00465 /**
00466  * @brief Retrieve the last low-power mode the driver was asked to enter
00467  *
00468  * @details
00469  * Host-side tests use this to verify the driver reached the intended mode
00470  * entry path without executing a real WAIT instruction.
00471  *
00472  * @return rx_lpc_last_mode_t Last mode entered (k_lpc_mode_none if never)
00473  *
00474  * @since Version 1.0.0
00475  */
00476 rx_lpc_last_mode_t rx_lpc_test_get_last_mode(void);
00477
00478 /**
00479  * @brief Force the deep-standby-wake flag (unit-test only)
00480  *
00481  * @details
00482  * Emulates the boot-path bridge from RSTSR* to rx_lpc_was_deep_standby_wake().
00483  * Must be called before rx_lpc_init() to be latched on the next init.
00484  *
00485  * @param[in] was_wake true to simulate a deep-standby-origin reset
00486  *
00487  * @since Version 1.0.0
00488  */
00489 void rx_lpc_test_set_deep_standby_wake(bool was_wake);
00490
00491 /**
00492  * @brief Inject simulated wake flags (unit-test only)
00493  *
00494  * @details
00495  * Emulates DPSIFR0..DPSIFR3 contents as observed at reset. The value is
00496  * latched on the next rx_lpc_init() and cleared afterwards, mirroring the
00497  * real hardware (which requires the caller to clear DPSIFR* on restart).
00498  *
00499  * @param[in] flags 32-bit wake-flag bitmask to inject
00500  *
00501  * @since Version 1.0.0
00502  */

```

```

00503 void rx_lpc_test_set_pending_wake_flags(uint32_t flags);
00504 #endif /* UNIT_TEST */
00505
00506 #ifdef __cplusplus
00507 }
00508 #endif

```

8.89 libs/rx_hal/inc/rx_mpc.h File Reference

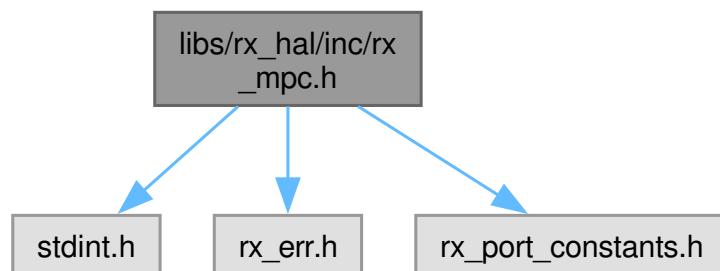
Multi-Function Pin Controller (MPC) Driver API for RX72N.

```
#include <stdint.h>
```

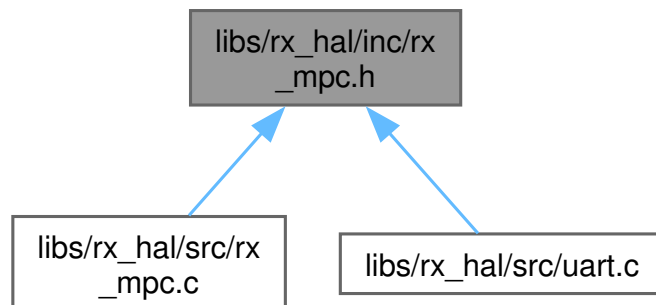
```
#include "rx_err.h"
```

```
#include "rx_port_constants.h"
```

Include dependency graph for rx_mpc.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_mpc_peripheral_config_t](#)
MPC peripheral configuration structure for pin function assignment.

Enumerations

- enum [rx_pin_function_t](#) : uint8_t { [k_pin_function_gpio](#) = 0x00 , [k_pin_function_periph](#) = 0x01 }
- enum [rx_pin_psel_t](#) : uint8_t {
[k_psel_mtu_ioc](#) = 0x01 , [k_psel_mtu_clk](#) = 0x02 , [k_psel_mtu_phase](#) = 0x02 ,
[k_psel_tpu_clk_portc](#) = 0x03 ,
[k_psel_tpu_clk_alt](#) = 0x04 , [k_psel_sci_tx](#) = 0x0A , [k_psel_sci_rx](#) = 0x0A , [k_psel_riic_scl](#) = 0x0F ,
[k_psel_riic_sda](#) = 0x0F , [k_psel_rspi_clk](#) = 0x0D , [k_psel_rspi_copi](#) = 0x0D , [k_psel_rspi_cipo](#) = 0x0D ,
[k_psel_adc](#) = 0x00 , [k_psel_usb_vbus](#) = 0x11 , [k_psel_gptw](#) = 0x1E }
Peripheral Select (PSEL) codes for common STAR project functions.

Functions

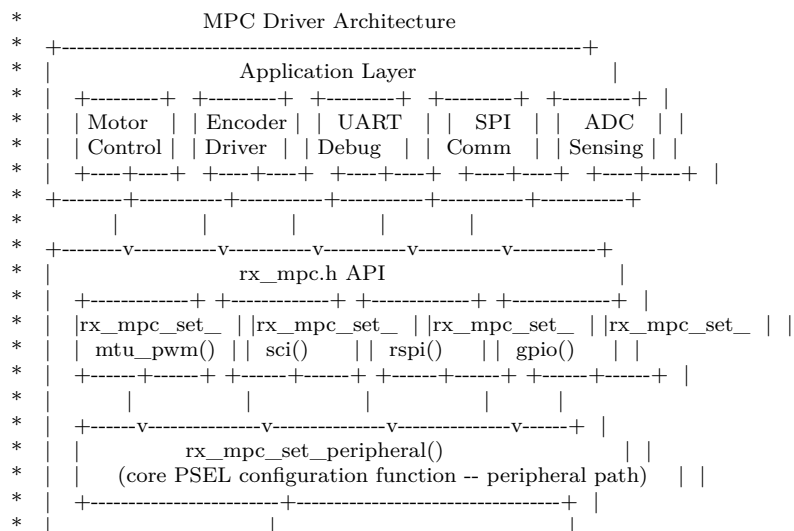
- `rx_err_t rx_mpc_set_gpio` (`rx_port_pin_t pin`)
Configure pin for GPIO (General Purpose I/O) function.
- `rx_err_t rx_mpc_set_peripheral` (`const rx_mpc_peripheral_config_t *config`)
Configure pin for peripheral function with specified PSEL code.
- `rx_err_t rx_mpc_set_mtu_pwm` (`rx_port_pin_t pin`)
Configure pin for MTU3a timer PWM output (MTIOC).
- `rx_err_t rx_mpc_set_mtu_encoder` (`rx_port_pin_t pin`)
Configure pin for MTU3a encoder/phase counter input (MTCLK).
- `rx_err_t rx_mpc_set_tpu_encoder` (`rx_port_pin_t pin`)
Configure pin for TPU encoder/phase counter input (TCLK).
- `rx_err_t rx_mpc_set_adc` (`rx_port_pin_t pin`)
Configure pin for ADC analog input.
- `rx_err_t rx_mpc_set_sci` (`rx_port_pin_t pin`)
Configure pin for SCI (Serial Communication Interface) UART function.
- `rx_err_t rx_mpc_set_riic` (`rx_port_pin_t pin`)
Configure pin for RIIC (I2C) bus function.
- `rx_err_t rx_mpc_set_rspi` (`rx_port_pin_t pin`)
Configure pin for RSPI (SPI) bus function.
- `rx_err_t rx_mpc_set_gptw` (`rx_port_pin_t pin`)
Configure pin for GPTW (General PWM Timer) complementary output.
- `rx_err_t rx_mpc_set_usb_vbus` (`rx_port_pin_t pin`)
Configure pin for USB VBUS detection function.
- `rx_err_t rx_mpc_set_irq` (`rx_port_pin_t pin`)
Configure pin for IRQ (external interrupt) function.

8.89.1 Detailed Description

Multi-Function Pin Controller (MPC) Driver API for RX72N.

High-level driver for configuring RX72N GPIO pin multiplexing. The MPC module controls which peripheral function is connected to each physical pin through Pin Function Select (PFS) registers.

System Architecture

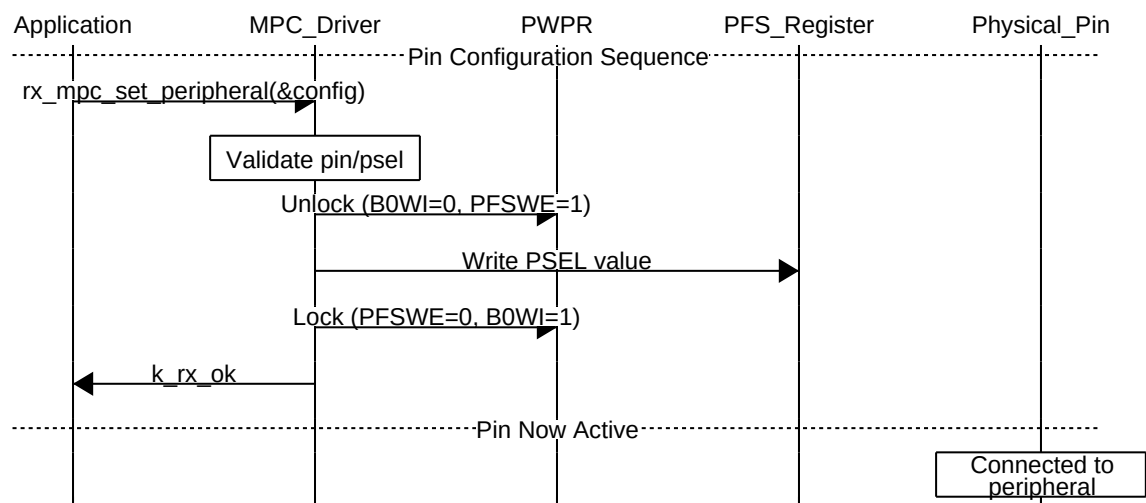


```

* | Note: rx_mpc_set_irq() and rx_mpc_set_adc() bypass          |
* | rx_mpc_set_peripheral() and write ISEL/ASEL bits directly  |
* | via internal_write_pfs() (ISEL=0x40, ASEL=0x80 exceed PSEL  |
* | 5-bit field; they are separate bits in the PFS register).  |
* +-----+-----+
* |                                     |
* +-----+-----+
* | rx72n_mpc_regs.h                  |
* | +-----+ +-----+ +-----+ |
* | | PWPR | | PFS Regs | | mpc() accessor | |
* | | (Protect) |---->| P00PFS- |<----| (0x0008C100) | |
* | | | | | PJ5PFS | | | | | |
* | +-----+ +-----+ +-----+ |
* +-----+-----+
* |                                     |
* +-----+-----+
* | Physical Pins                      |
* | Pin (function) --> Peripheral    |
* | (Configured via PFS register PSEL field) |
* | @see hardware_config.h for application-level pin assignments |
* +-----+-----+

```

Pin Configuration Flow



Pin Assignments

Application-level pin assignments are defined in `hardware_config.h`. This driver provides the MPC register interface for any valid pin/PSEL combination. See the RX72N Hardware Manual pin function tables for valid (pin, PSEL) pairs.

See also

`hardware_config.h` Application-level pin assignments

Hardware Requirements

Resource	Requirement	Notes
MPC Base	0x0008C100	Fixed hardware address
PCLKB	Active	Required for register access
MSTPCRA	Bit 9 = 0	MPC module stop cleared

Performance Characteristics

Operation	Typical Time	Notes
Pin config	~1 us	PWPR unlock/lock overhead
Bulk config (10 pins)	~10 us	Sequential calls

Memory Usage

Category	Size	Notes
Code	~800 bytes	All functions
Static data	4 bytes	Tag string only
Stack (per call)	16 bytes	Local variables

Thread Safety

Not thread-safe. The MPC PWPR unlock/lock sequence is non-atomic. If multiple threads call MPC functions concurrently, race conditions can occur where one thread's unlock is clobbered by another's lock. Use external mutex protection or configure all pins during single-threaded initialization before starting RTOS.

Initialization Order

MPC configuration should occur early in system initialization:

1. Clock configuration (PCLKB must be running)
2. Module stop clear (if MPC in module stop)
3. MPC pin configuration < This module
4. GPIO direction configuration (PMR, PDR registers)
5. Peripheral driver initialization

Usage Example

```

#include "rx_mpc.h"
#include "rx_port_constants.h"

void hardware_init_pins(void)
{
    rx_err_t err;

    // Configure motor PWM pins (GPTW channel 0)
    const rx_mpc_peripheral_config_t pwm_config = {
        .pin = k_port_e_pin_5, // PE5 = GPTW0A
        .psel = k_psel_gptw // GPTW function (0x1E)
    };
    err = rx_mpc_set_peripheral(&pwm_config);
    if (err != k_rx_ok) {
        // Handle error
    }

    // Configure encoder pins (MTU phase counting)
    err = rx_mpc_set_mtu_encoder(k_port_2_pin_4); // P24 = MTCLKA
    err = rx_mpc_set_mtu_encoder(k_port_2_pin_5); // P25 = MTCLKB

    // Configure debug UART (SCI9)
    err = rx_mpc_set_sci(k_port_b_pin_7); // PB7 = TXD9
    err = rx_mpc_set_sci(k_port_b_pin_6); // PB6 = RXD9

    // Configure SPI communication (RSPI0)
    err = rx_mpc_set_rspi(k_port_a_pin_0); // PA0 = COPI
    err = rx_mpc_set_rspi(k_port_a_pin_1); // PA1 = CIPO
    err = rx_mpc_set_rspi(k_port_a_pin_3); // PA3 = CLK
    err = rx_mpc_set_rspi(k_port_a_pin_4); // PA4 = SSL0
}

```

Error Handling Example

```

rx_err_t configure_pin_safe(rx_port_pin_t pin, uint8_t psel)
{
    // Validate PSEL range before calling
    if (psel > 0x1F) {
        rx_log_error("MPC", "PSEL exceeds 5-bit maximum");
        return k_rx_err_invalid_arg;
    }

    rx_mpc_peripheral_config_t config = {
        .pin = pin,
        .psel = psel
    };

    rx_err_t err = rx_mpc_set_peripheral(&config);
    if (err != k_rx_ok) {
        rx_log_error("MPC", "Pin configuration failed");
        // Configuration failed - pin remains in previous state
        return err;
    }

    // Remember: Also need to set PMR bit to enable peripheral mode!
    return k_rx_ok;
}

```

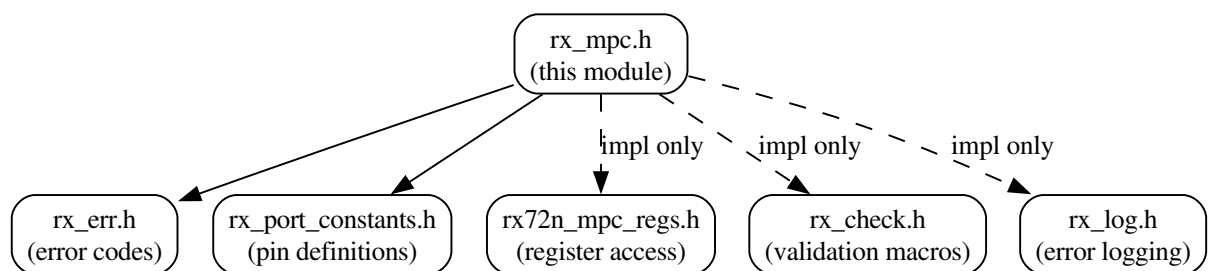
NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, longjmp, or recursion
- Rule 2: [OK] All loops have fixed upper bounds (switch/case only)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] Minimum 2 assertions per function (parameter validation)
- Rule 6: [OK] Variables declared at smallest scope
- Rule 7: [OK] All return values checked
- Rule 8: [OK] C23 typed enums for all constants, no magic numbers
- Rule 9: [WARN] Function pointers not used (simple direct calls)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- Single Responsibility: MPC driver handles only pin mux configuration
- Open/Closed: New peripherals added via [rx_mpc_set_peripheral\(\)](#) without modifying existing code
- Interface Segregation: Convenience functions (`set_sci`, `set_rspi`) separate from low-level `set_peripheral()`
- Dependency Inversion: Higher-level drivers depend on this abstraction, not on raw register writes

Module Dependencies



See also

[rx72n_mpc_regs.h](#) Low-level MPC register definitions[rx_port_constants.h](#) Pin enumeration definitions[docs/sections/03_hardware_pinout.tex](#) Complete STAR pinout documentation

Author

Locked, Inc.

Date

2026-01-29

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Since

Version 1.0.0

Definition in file [rx_mpc.h](#).

8.89.2 Data Structure Documentation

8.89.2.1 struct rx_mpc_peripheral_config_t

MPC peripheral configuration structure for pin function assignment.

This structure groups related parameters to prevent accidental parameter swapping when calling `rx_mpc_set_peripheral()`. Using a struct provides:

- Type safety (compiler catches wrong parameter types)
- Named parameters (self-documenting code)
- Extensibility (can add fields without changing function signature)

Memory Layout

Offset	Size	Field	Type	Alignment
0	2	pin	rx_port_pin_t (uint16_t)	2
2	1	psel	uint8_t	1
3	1	(padding)	-	-
Total	4 bytes			

Field Relationships

- `pin` determines which PFS register to modify
- `psel` determines what value to write to that register
- The (pin, psel) pair must be valid per RX72N hardware manual appendix

Usage Example - Basic Configuration

```
// Configure PE5 for GPTW0A (motor PWM output)
rx_mpc_peripheral_config_t pwm_config = {
    .pin = k_port_e_pin_5, // PE5
    .psel = k_psel_gptw    // GPTW function select (0x1E)
};
rx_err_t err = rx_mpc_set_peripheral(&pwm_config);
```

Usage Example - Multiple Pins

```
// Configure all SPI pins using array
const rx_mpc_peripheral_config_t spi_pins[] = {
    { .pin = k_port_a_pin_0, .psel = 0x0D }, // COPI
    { .pin = k_port_a_pin_1, .psel = 0x0D }, // CIP0
    { .pin = k_port_a_pin_3, .psel = 0x0D }, // CLK
    { .pin = k_port_a_pin_4, .psel = 0x0D }, // SSL0
};

for (uint8_t i = 0; i < 4; i++) {
    rx_err_t err = rx_mpc_set_peripheral(&spi_pins[i]);
    if (err != k_rx_ok) {
        // Handle error
    }
}
```

Invariant

pin must be a valid rx_port_pin_t value from rx_port_constants.h

psel must be in range [0, 31] (5-bit field)

(pin, psel) combination must be valid per hardware manual

See also

[rx_mpc_set_peripheral\(\)](#) Function that uses this structure

[rx_port_constants.h](#) Pin enumeration definitions

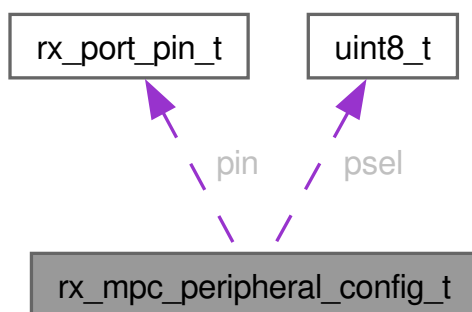
[rx_pin_psel_t](#) Common PSEL values for STAR project

Since

Version 1.0.0

Definition at line 308 of file [rx_mpc.h](#).

Collaboration diagram for rx_mpc_peripheral_config_t:



Data Fields

rx_port_pin_t	pin	GPIO pin to configure for peripheral function. Encoded pin identifier from rx_port_constants.h combining port and pin number. Use the k_port_X_pin_Y constants for type safety. Valid Values Any valid rx_port_pin_t constant, e.g.: • k_port_a_pin_0 through k_port_a_pin_7 • k_port_b_pin_0 through k_port_b_pin_7 • k_port_e_pin_0 through k_port_e_pin_7 • etc. Note On 144-pin package, ports G, H are not available Warning Using invalid pin values returns k_rx_err_invalid_arg																											
uint8_t	psel	Peripheral function select code (PSEL field value). Value written to the PSEL[4:0] field of the PFS register. This determines which peripheral function the pin is connected to. Valid Range: [0, 31 (0x00 - 0x1F)] <table border="1"> <thead> <tr> <th>PSEL</th><th>Function</th><th>Typical Pins</th></tr> </thead> <tbody> <tr> <td>0x00</td><td>Hi-Z/GPIO</td><td>All</td></tr> <tr> <td>0x01</td><td>MTU I/O compare (MTIOC)</td><td>P14-P17, P24-P27</td></tr> <tr> <td>0x02</td><td>MTU clock input (MTCLK)</td><td>P24, P25, PC0, PC1</td></tr> <tr> <td>0x03</td><td>MTU phase counter</td><td>P24, P25</td></tr> <tr> <td>0x1E</td><td>GPTW output (GTIOC)</td><td>PE3, PE7, P23, P22, etc.</td></tr> <tr> <td>0x0A</td><td>SCI (TXD/RXD)</td><td>PB6, PB7</td></tr> <tr> <td>0x0D</td><td>RSPI (CLK/COPI/CIPO/↔SSL)</td><td>PA0-PA4</td></tr> <tr> <td>0x0F</td><td>RIIC (SCL/SDA)</td><td>P12, P13</td></tr> </tbody> </table> Common Values (varies by pin, check hardware manual): Warning Wrong PSEL value for a pin may cause hardware malfunction See also rx_pin_psel_t Predefined PSEL constants	PSEL	Function	Typical Pins	0x00	Hi-Z/GPIO	All	0x01	MTU I/O compare (MTIOC)	P14-P17, P24-P27	0x02	MTU clock input (MTCLK)	P24, P25, PC0, PC1	0x03	MTU phase counter	P24, P25	0x1E	GPTW output (GTIOC)	PE3, PE7, P23, P22, etc.	0x0A	SCI (TXD/RXD)	PB6, PB7	0x0D	RSPI (CLK/COPI/CIPO/↔SSL)	PA0-PA4	0x0F	RIIC (SCL/SDA)	P12, P13
PSEL	Function	Typical Pins																											
0x00	Hi-Z/GPIO	All																											
0x01	MTU I/O compare (MTIOC)	P14-P17, P24-P27																											
0x02	MTU clock input (MTCLK)	P24, P25, PC0, PC1																											
0x03	MTU phase counter	P24, P25																											
0x1E	GPTW output (GTIOC)	PE3, PE7, P23, P22, etc.																											
0x0A	SCI (TXD/RXD)	PB6, PB7																											
0x0D	RSPI (CLK/COPI/CIPO/↔SSL)	PA0-PA4																											
0x0F	RIIC (SCL/SDA)	P12, P13																											

8.89.3 Enumeration Type Documentation

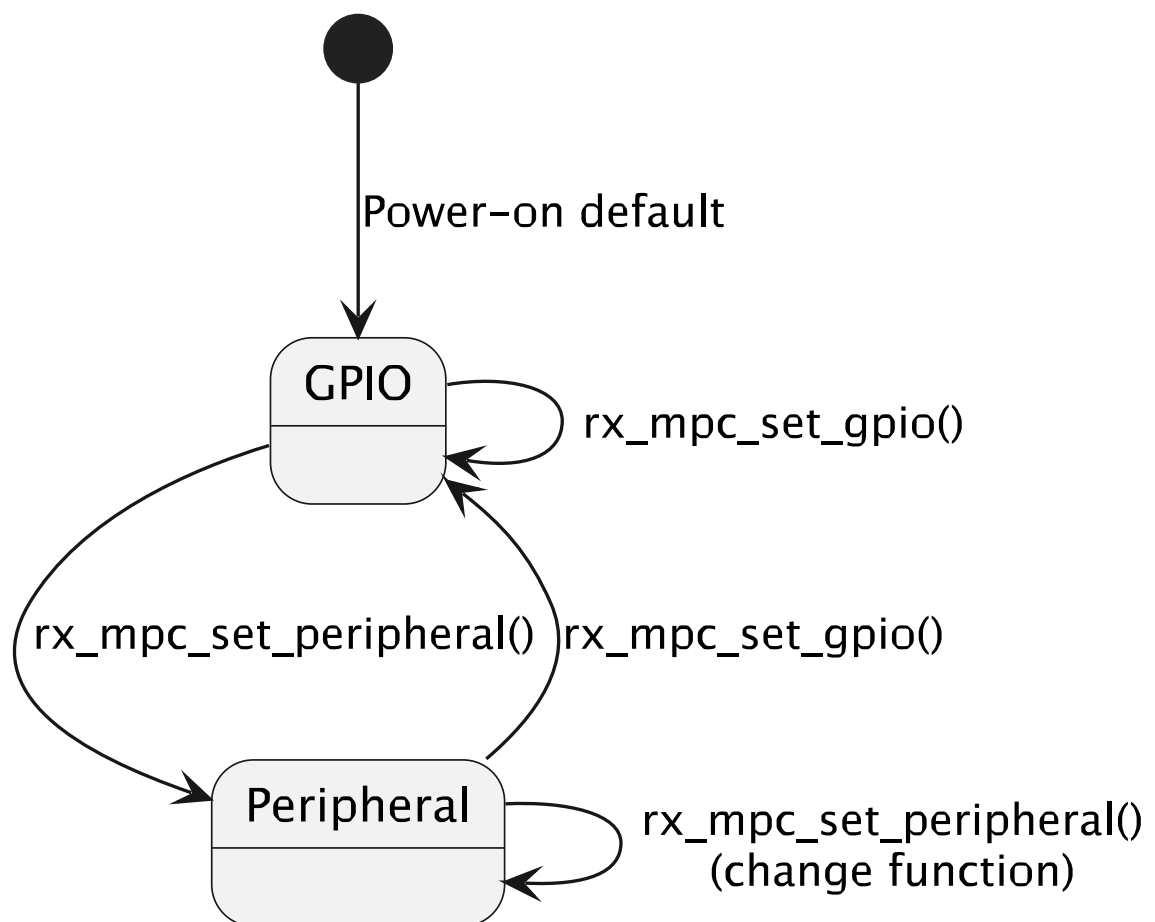
8.89.3.1 rx_pin_function_t

enum [rx_pin_function_t](#) : uint8_t

High-level pin function mode selection.

Specifies whether a pin is configured for GPIO operation or peripheral function mode. This is a conceptual abstraction - the actual hardware configuration is done through PFS register PSEL field and PORT PMR bit.

Pin Configuration State Machine



Note

This enum is for documentation/API clarity. Actual hardware uses PSEL field values directly.

See also

[rx_mpc_set_gpio\(\)](#) Configure pin for GPIO mode

[rx_mpc_set_peripheral\(\)](#) Configure pin for peripheral mode

Since

Version 1.0.0

Enumerator

k_pin_function_gpio	GPIO mode - pin controlled by PORT registers. Pin is disconnected from all peripheral functions. Direction and data controlled by PORT PDR and PODR registers. This is the power-on default for all pins. PFS Register Value: 0x00 (PSEL=0, ISEL=0, ASEL=0)
k_pin_function_periph	Peripheral function mode - pin connected to peripheral. Pin is connected to a peripheral function as determined by the PSEL field. The PORT PMR bit must also be set to 1 to actually enable peripheral mode. PFS Register Value: PSEL field set to peripheral code Note This is a placeholder value; actual PSEL varies by function

Definition at line 382 of file [rx_mpc.h](#).

```

00382         : uint8_t {
00383     /**
00384     * @brief GPIO mode - pin controlled by PORT registers
00385     * @details
00386     * Pin is disconnected from all peripheral functions. Direction and
00387     * data controlled by PORT PDR and PODR registers. This is the
00388     * power-on default for all pins.
00389     * @par PFS Register Value: 0x00 (PSEL=0, ISEL=0, ASEL=0)
00390     */
00391     k_pin_function_gpio = 0x00,
00392
00393     /**
00394     * @brief Peripheral function mode - pin connected to peripheral
00395     * @details
00396     * Pin is connected to a peripheral function as determined by the
00397     * PSEL field. The PORT PMR bit must also be set to 1 to actually
00398     * enable peripheral mode.
00399     * @par PFS Register Value: PSEL field set to peripheral code
00400     * @note This is a placeholder value; actual PSEL varies by function
00401     */
00402     k_pin_function_periph = 0x01,
00403 } rx_pin_function_t;

```

8.89.3.2 rx_pin_psel_t

```
enum rx_pin_psel_t : uint8_t
```

Peripheral Select (PSEL) codes for common STAR project functions.

These values are written to the PSEL[4:0] field of PFS registers to select which peripheral function a pin is connected to. Values are pin-specific - these constants represent the most common values used in the STAR project.

PSEL Field in PFS Register

```

* Bit:  7    6    5    4 3 2 1 0
*      +-----+-----+-----+-----+
*      | ASEL | ISEL | Rsvd |  PSEL[4:0]  |
*      +-----+-----+-----+-----+
*                               <-- These values -->
*

```

PSEL Values by Peripheral (STAR Project)

Peripheral	PSEL	Pins Used	Function
MTU3a MTIOC	0x01	P14-P17, P24-P27	PWM output
MTU3a MTCLK	0x02	P24, P25, PC0, PC1	Clock/encoder input
MTU3a Phase	0x03	P24, P25	Phase counting mode
GPTW GTIOC	0x1E	PE3, PE7, P23, P22, etc.	PWM output
SCI TXD/RXD	0x0A	PB6, PB7	UART data
RSPI signals	0x0D	PA0-PA4	SPI bus
RIIC SDA/SCL	0x0F	P12, P13	I2C bus

Warning

PSEL values are pin-specific! A value that works for one pin may select a completely different function on another pin. Always verify against the RX72N Hardware Manual pin function tables.

Usage Example

```

// Configure PB7 for SCI9 TXD
rx_mpc_peripheral_config_t config = {
    .pin = k_port_b_pin_7,
    .psel = k_psel_sci_tx // 0x0A
};
rx_mpc_set_peripheral(&config);

```

See also

[rx_mpc_set_peripheral\(\)](#) Use these values with this function

RX72N Hardware Manual, Appendix for complete pin function tables

Since

Version 1.0.0

Enumerator

k_psel_mtu_ioc	MTU I/O Compare/PWM output (MTIOC). Connects pin to MTU3a compare match output for PWM generation. Typical Pins: P14-P17 (MTU3), P24-P27 (MTU4) Application: Motor PWM outputs when using MTU3a timer
k_psel_mtu_clk	MTU clock input (MTCLKA/B/C/D). Connects pin to MTU3a external clock input. On RX72N, PSEL = 0x02 (0b000010) is the generic "MTCLK" function and is the same value whether the MTU is configured for normal external-clock counting or for phase counting (quadrature encoder) mode. The counting mode is selected by MTU.TMDR, not by PSEL – callers must not pick a different PSEL just because they want phase counting. Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package: • Table 23.5 (P1x pins): row PSEL = 000010b -> MTCLKA on P14, MTCLKB on P15 • Table 23.6 (P2x pins): row PSEL = 000010b -> MTCLKC on P22, MTCLKD on P23, MTCLKA on P24, MTCLKB on P25 • Table 23.17 (PAx pins): row PSEL = 000010b -> MTCLKC on PA1, MTCLKD on PA3, MTCLKA on PA4, MTCLKB on PA6 • Table 23.19 (PCx pins): row PSEL = 000010b -> MTCLKC on PC4, MTCLKD on PC5, MTCLKA on PC6, MTCLKB on PC7 Every MTCLK pin on this package resolves to PSEL = 0b00'0010 = 0x02. Typical Pins: P24/P25 (MTCLKA/B), PA1/PC5 (MTCLKC/D) Application: Encoder pulse counting, quadrature phase counting

k_psel_mtu_phase	MTU encoder phase counting input (alias for k_psel_mtu_clk). Same PSEL value on RX72N as k_psel_mtu_clk – both MTU external clock counting and MTU phase counting route through PSEL=0x02, and the distinction is selected later via MTU.TMDR (phase counting mode bits). The split constants exist only so calling code reads clearly; do not assume they are different values. Typical Pins: P24/P25 pair, PA1/PC5 pair (same as k_psel_mtu_clk) Application: Quadrature encoder interface with direction detection Note Previously 0x03 – that was wrong and caused MTU encoders to stay at TCNT=0x0000 because PSEL=0x03 on MTU-candidate pins selects an unrelated function (TPU on Port C pins, or nothing elsewhere). See chapter 23 of the RX72N Group Hardware Manual (R01UH0824EJ0111), Tables 23.5, 23.6, 23.17, 23.19.
k_psel_tpu_clk_portc	TPU external clock input on Port C pins (TCLKA/B/C/D on PC0-PC3). RX72N assigns TPU external clock to alternate function PSEL=0x03 on Port C pins only. Non-Port-C candidates for the same TCLK signal use a different PSEL (see k_psel_tpu_clk_alt). Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)", Table 23.19 (PCx pins). For the 144-pin LFQFP package the row PSEL = 000011b maps to: PC0 -> TCLKC, PC1 -> TCLKD, PC2 -> TCLKA, PC3 -> TCLKB All four TPU TCLK inputs on Port C resolve to PSEL = 0b00'0011 = 0x03. Typical Pins: PC2 (TCLKA), PC3 (TCLKB), PC0 (TCLKC), PC1 (TCLKD) Application: Rear-wheel quadrature encoder Phase A inputs on Port C Note rx_mpc_set_tpu_encoder() picks between this and k_psel_tpu_clk_alt automatically based on the port number of the pin argument.

k_psel_tpu_clk_alt	TPU external clock input on non-Port-C pins (TCLKA/B/C/D). RX72N assigns TPU external clock to alternate function PSEL=0x04 on Port 1, Port A, and Port B candidates for the TCLK signals. This differs from the Port C candidates (k_psel_tpu_clk_portc = 0x03) and the difference is silicon-level – the same TCLK signal is routed through different PFS alternate-function slots on different ports. Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package the row PSEL = 000100b maps to: • Table 23.5 (P1x pins): P14 -> TCLKA, P15 -> TCLKB, P16 -> TCLKC, P17 -> TCLKD • Table 23.17 (PAx pins): PA3 -> TCLKB • Table 23.18 (PBx pins): PB2 -> TCLKC, PB3 -> TCLKD All non-Port-C TPU TCLK inputs resolve to PSEL = 0b00'0100 = 0x04. Typical Pins: PA3 (TCLKB), PB3 (TCLKD) – STAR rear-wheel Phase B Application: Rear-wheel quadrature encoder Phase B inputs Note Before this split, rx_mpc_set_tpu_encoder() used PSEL=0x03 for every TPU pin, which silently misconfigured PA3/PB3 (the two non-Port-C pins on rear wheels) so one half of each rear-wheel encoder never incremented.
k_psel_sci_tx	SCI transmit data output (TXD). Connects pin to SCI channel transmit data output. Typical Pins: PB7 (SCI9-TXD for debug UART) Application: UART transmit, debug console output Note Same PSEL value used for both TX and RX on SCI pins

k_psel_sci_rx	SCI receive data input (RXD). Connects pin to SCI channel receive data input. Typical Pins: PB6 (SCI9-RXD for debug UART) Application: UART receive, debug console input Note Same PSEL value as TX (pin determines actual function)
k_psel_riic_scl	RIIC clock line (SCL). Connects pin to RIIC (I2C) clock line with open-drain output. Typical Pins: P12 (RIIC0-SCL) Application: I2C bus clock for sensors/peripherals Note RIIC automatically configures open-drain when enabled
k_psel_riic_sda	RIIC data line (SDA). Connects pin to RIIC (I2C) data line with open-drain output. Typical Pins: P13 (RIIC0-SDA) Application: I2C bus data for sensors/peripherals Note Same PSEL value as SCL (pin determines actual function)
k_psel_rspi_clk	RSPI clock (RSPCK). Connects pin to RSPI serial clock output (controller mode) or input (peripheral mode). Typical Pins: PA5 (RSPI0-RSPCKA); see RX72N HW manual R01UH0824EJ0111 Ch 23 Application: SPI bus clock for RPi5 communication

k_psel_rspi_copi	RSPI Controller Out Peripheral In (COPI, formerly MOSI). Connects pin to RSPI data output line in controller mode. Typical Pins: PA6 (RSPI0-MOSIA / COPI); see RX72N HW manual R01UH0824EJ0111 0 Application: SPI transmit data to peripherals Note Uses OSHWA-approved inclusive terminology
k_psel_rspi_cipo	RSPI Controller In Peripheral Out (CIPO, formerly MISO). Connects pin to RSPI data input line in controller mode. Typical Pins: PA7 (RSPI0-MISOA / CIPO); see RX72N HW manual R01UH0824EJ0111 0 Application: SPI receive data from peripherals Note Uses OSHWA-approved inclusive terminology
k_psel_adc	ADC analog input mode. Disables digital input buffer for analog signal measurement. Actually sets ASEL bit in PFS, not PSEL field. Typical Pins: P40-P47 (AN000-AN007) Application: Motor current sensing, analog input measurement Note Use rx_mpc_set_adc() for proper configuration (sets ASEL bit)

k_psel_usb_vbus	USB VBUS detection input. Connects pin to USB VBUS detection for 5V presence sensing. Required for USB enumeration and power monitoring. Typical Pins: P1.6 (USB0_VBUS on RX72N) Application: USB CDC debug interface, RPi5 communication Note Active-high: pin reads 1 when 5V present on USB bus Requires additional USB PHY and controller configuration Since Version 1.0.0
k_psel_gptw	GPTW complementary PWM output (GTIOC). Connects pin to GPTW (General PWM Timer) complementary PWM output with dead-time insertion for motor control. GPTW provides 4 channels (GPTW0-GPTW3) with phase-staggered PWM outputs. Typical Pins: • P2.3, P1.7 (GPTW0 - Motor 0 IN2/IN1) • P2.2, PC.3 (GPTW1 - Motor 1 IN2/IN1) • PE.3, P8.6 (GPTW2 - Motor 2 IN2/IN1) • PE.7, PC.6 (GPTW3 - Motor 3 IN2/IN1) Application: Motor control with direction/PWM signals for DRV8263H Feature: 90-degree phase staggering reduces peak current draw Since Version 1.0.0

Definition at line 453 of file [rx_mpc.h](#).

```

00453         : uint8_t {
00454     /**
00455     * @brief MTU I/O Compare/PWM output (MTIOC)
00456     * @details
00457     * Connects pin to MTU3a compare match output for PWM generation.
00458     * @par Typical Pins: P14-P17 (MTU3), P24-P27 (MTU4)
00459     * @par Application: Motor PWM outputs when using MTU3a timer
00460     */
00461     k_psel_mtu_ioc = 0x01,
00462
00463     /**
00464     * @brief MTU clock input (MTCLKA/B/C/D)
00465     * @details
00466     * Connects pin to MTU3a external clock input. On RX72N, PSEL = 0x02
00467     * (0b000010) is the generic "MTCLK" function and is the same value whether
00468     * the MTU is configured for normal external-clock counting or for phase

```

```

00469 * counting (quadrature encoder) mode. The counting mode is selected by
00470 * MTU.TMDR, not by PSEL -- callers must not pick a different PSEL just
00471 * because they want phase counting.
00472 *
00473 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00474 * "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package:
00475 * - Table 23.5 (P1x pins): row PSEL = 000010b -> MTCLKA on P14,
00476 *   MTCLKB on P15
00477 * - Table 23.6 (P2x pins): row PSEL = 000010b -> MTCLKC on P22,
00478 *   MTCLKD on P23, MTCLKA on P24, MTCLKB on P25
00479 * - Table 23.17 (PAx pins): row PSEL = 000010b -> MTCLKC on PA1,
00480 *   MTCLKD on PA3, MTCLKA on PA4, MTCLKB on PA6
00481 * - Table 23.19 (PCx pins): row PSEL = 000010b -> MTCLKC on PC4,
00482 *   MTCLKD on PC5, MTCLKA on PC6, MTCLKB on PC7
00483 * Every MTCLK pin on this package resolves to PSEL = 0b00'0010 = 0x02.
00484 *
00485 * @par Typical Pins: P24/P25 (MTCLKA/B), PA1/PC5 (MTCLKC/D)
00486 * @par Application: Encoder pulse counting, quadrature phase counting
00487 */
00488 k_psel_mtu_clk = 0x02,
00489
00490 /**
00491 * @brief MTU encoder phase counting input (alias for k_psel_mtu_clk)
00492 * @details
00493 * Same PSEL value on RX72N as k_psel_mtu_clk -- both MTU external clock
00494 * counting and MTU phase counting route through PSEL=0x02, and the
00495 * distinction is selected later via MTU.TMDR (phase counting mode bits).
00496 * The split constants exist only so calling code reads clearly; do not
00497 * assume they are different values.
00498 *
00499 * @par Typical Pins: P24/P25 pair, PA1/PC5 pair (same as k_psel_mtu_clk)
00500 * @par Application: Quadrature encoder interface with direction detection
00501 *
00502 * @note Previously 0x03 -- that was wrong and caused MTU encoders to stay
00503 *       at TCNT=0x0000 because PSEL=0x03 on MTU-candidate pins selects an
00504 *       unrelated function (TPU on Port C pins, or nothing elsewhere).
00505 *       See chapter 23 of the RX72N Group Hardware Manual
00506 *       (R01UH0824EJ0111), Tables 23.5, 23.6, 23.17, 23.19.
00507 */
00508 k_psel_mtu_phase = 0x02,
00509
00510 /**
00511 * @brief TPU external clock input on Port C pins (TCLKA/B/C/D on PC0-PC3)
00512 * @details
00513 * RX72N assigns TPU external clock to alternate function PSEL=0x03 on
00514 * Port C pins only. Non-Port-C candidates for the same TCLK signal use a
00515 * different PSEL (see k_psel_tpu_clk_alt).
00516 *
00517 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00518 * "Multi-Function Pin Controller (MPC)", Table 23.19 (PCx pins). For the
00519 * 144-pin LFQFP package the row PSEL = 000011b maps to:
00520 * PC0 -> TCLKC, PC1 -> TCLKD, PC2 -> TCLKA, PC3 -> TCLKB
00521 * All four TPU TCLK inputs on Port C resolve to PSEL = 0b00'0011 = 0x03.
00522 *
00523 * @par Typical Pins: PC2 (TCLKA), PC3 (TCLKB), PC0 (TCLKC), PC1 (TCLKD)
00524 * @par Application: Rear-wheel quadrature encoder Phase A inputs on Port C
00525 *
00526 * @note rx_mpc_set_tpu_encoder() picks between this and k_psel_tpu_clk_alt
00527 *       automatically based on the port number of the pin argument.
00528 */
00529 k_psel_tpu_clk_portc = 0x03,
00530
00531 /**
00532 * @brief TPU external clock input on non-Port-C pins (TCLKA/B/C/D)
00533 * @details
00534 * RX72N assigns TPU external clock to alternate function PSEL=0x04 on
00535 * Port 1, Port A, and Port B candidates for the TCLK signals. This differs
00536 * from the Port C candidates (k_psel_tpu_clk_portc = 0x03) and the
00537 * difference is silicon-level -- the same TCLK signal is routed through
00538 * different PFS alternate-function slots on different ports.
00539 *
00540 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00541 * "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package
00542 * the row PSEL = 000100b maps to:
00543 * - Table 23.5 (P1x pins): P14 -> TCLKA, P15 -> TCLKB,
00544 *   P16 -> TCLKC, P17 -> TCLKD
00545 * - Table 23.17 (PAx pins): PA3 -> TCLKB
00546 * - Table 23.18 (PBx pins): PB2 -> TCLKC, PB3 -> TCLKD
00547 * All non-Port-C TPU TCLK inputs resolve to PSEL = 0b00'0100 = 0x04.
00548 *
00549 * @par Typical Pins: PA3 (TCLKB), PB3 (TCLKD) -- STAR rear-wheel Phase B
00550 * @par Application: Rear-wheel quadrature encoder Phase B inputs
00551 *
00552 * @note Before this split, rx_mpc_set_tpu_encoder() used PSEL=0x03 for
00553 *       every TPU pin, which silently misconfigured PA3/PB3 (the two
00554 *       non-Port-C pins on rear wheels) so one half of each rear-wheel
00555 *       encoder never incremented.

```

```

00556 */
00557 k_psel_tpu_clk_alt = 0x04,
00558
00559 /**
00560  * @brief SCI transmit data output (TXD)
00561  * @details
00562  * Connects pin to SCI channel transmit data output.
00563  * @par Typical Pins: PB7 (SCI9-TXD for debug UART)
00564  * @par Application: UART transmit, debug console output
00565  * @note Same PSEL value used for both TX and RX on SCI pins
00566  */
00567 k_psel_sci_tx = 0x0A,
00568
00569 /**
00570  * @brief SCI receive data input (RXD)
00571  * @details
00572  * Connects pin to SCI channel receive data input.
00573  * @par Typical Pins: PB6 (SCI9-RXD for debug UART)
00574  * @par Application: UART receive, debug console input
00575  * @note Same PSEL value as TX (pin determines actual function)
00576  */
00577 k_psel_sci_rx = 0x0A,
00578
00579 /**
00580  * @brief RIIC clock line (SCL)
00581  * @details
00582  * Connects pin to RIIC (I2C) clock line with open-drain output.
00583  * @par Typical Pins: P12 (RIIC0-SCL)
00584  * @par Application: I2C bus clock for sensors/peripherals
00585  * @note RIIC automatically configures open-drain when enabled
00586  */
00587 k_psel_riic_scl = 0x0F,
00588
00589 /**
00590  * @brief RIIC data line (SDA)
00591  * @details
00592  * Connects pin to RIIC (I2C) data line with open-drain output.
00593  * @par Typical Pins: P13 (RIIC0-SDA)
00594  * @par Application: I2C bus data for sensors/peripherals
00595  * @note Same PSEL value as SCL (pin determines actual function)
00596  */
00597 k_psel_riic_sda = 0x0F,
00598
00599 /**
00600  * @brief RSPI clock (RSPCK)
00601  * @details
00602  * Connects pin to RSPI serial clock output (controller mode) or
00603  * input (peripheral mode).
00604  * @par Typical Pins: PA5 (RSPI0-RSPCKA); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17 (MPC PA
pin function select)
00605  * @par Application: SPI bus clock for RPi5 communication
00606  */
00607 k_psel_rspi_clk = 0x0D,
00608
00609 /**
00610  * @brief RSPI Controller Out Peripheral In (COPI, formerly MOSI)
00611  * @details
00612  * Connects pin to RSPI data output line in controller mode.
00613  * @par Typical Pins: PA6 (RSPI0-MOSIA / COPI); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17
(MPC PAn pin function select)
00614  * @par Application: SPI transmit data to peripherals
00615  * @note Uses OSHWA-approved inclusive terminology
00616  */
00617 k_psel_rspi_copi = 0x0D,
00618
00619 /**
00620  * @brief RSPI Controller In Peripheral Out (CIPO, formerly MISO)
00621  * @details
00622  * Connects pin to RSPI data input line in controller mode.
00623  * @par Typical Pins: PA7 (RSPI0-MISOA / CIPO); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17
(MPC PAn pin function select)
00624  * @par Application: SPI receive data from peripherals
00625  * @note Uses OSHWA-approved inclusive terminology
00626  */
00627 k_psel_rspi_cipo = 0x0D,
00628
00629 /**
00630  * @brief ADC analog input mode
00631  * @details
00632  * Disables digital input buffer for analog signal measurement.
00633  * Actually sets ASEL bit in PFS, not PSEL field.
00634  * @par Typical Pins: P40-P47 (AN000-AN007)
00635  * @par Application: Motor current sensing, analog input measurement
00636  * @note Use rx_mpc_set_adc() for proper configuration (sets ASEL bit)
00637  */
00638 k_psel_adc = 0x00,
00639

```

```

00640 /**
00641  * @brief USB VBUS detection input
00642  *
00643  * @details
00644  * Connects pin to USB VBUS detection for 5V presence sensing.
00645  * Required for USB enumeration and power monitoring.
00646  *
00647  * @par Typical Pins: P1.6 (USB0_VBUS on RX72N)
00648  * @par Application: USB CDC debug interface, RPi5 communication
00649  *
00650  * @note Active-high: pin reads 1 when 5V present on USB bus
00651  * @note Requires additional USB PHY and controller configuration
00652  *
00653  * @since Version 1.0.0
00654  */
00655 k_psel_usb_vbus = 0x11,
00656
00657 /**
00658  * @brief GPTW complementary PWM output (GTIOC)
00659  *
00660  * @details
00661  * Connects pin to GPTW (General PWM Timer) complementary PWM output
00662  * with dead-time insertion for motor control. GPTW provides 4 channels
00663  * (GPTW0-GPTW3) with phase-staggered PWM outputs.
00664  *
00665  * @par Typical Pins:
00666  * - P2.3, P1.7 (GPTW0 - Motor 0 IN2/IN1)
00667  * - P2.2, PC.3 (GPTW1 - Motor 1 IN2/IN1)
00668  * - PE.3, P8.6 (GPTW2 - Motor 2 IN2/IN1)
00669  * - PE.7, PC.6 (GPTW3 - Motor 3 IN2/IN1)
00670  *
00671  * @par Application: Motor control with direction/PWM signals for DRV8263H
00672  * @par Feature: 90-degree phase staggering reduces peak current draw
00673  *
00674  * @since Version 1.0.0
00675  */
00676 k_psel_gptw = 0x1E, /* 0b011110: per RX72N HW manual Tables 23.4, 23.6,
00677  * 23.10, 23.14, 23.16 -- the GTIOC* function on every
00678  * port (1, 2, 8, A, C, D, E) uses PSEL=0x1E. The old
00679  * value 0x14 selected an unrelated function. */
00680
00681 } rx_pin_psel_t;

```

8.89.4 Function Documentation

8.89.4.1 rx_mpc_set_adc()

```

rx_err_t rx_mpc_set_adc (
    const rx_port_pin_t pin) [nodiscard]

```

Configure pin for ADC analog input.

Configures a pin for analog-to-digital converter input by setting the ASEL bit in the PFS register. This disables the digital input buffer to prevent noise and current leakage during analog measurement.

ADC Pin Configuration

```

*   PFS Register for ADC:
*   +-----+-----+-----+-----+
*   | ASEL | ISEL | Rsvd | PSEL[4:0] |
*   |  1  |  0  |  0  |   0x00   |
*   +-----+-----+-----+-----+
*   ^
*   +-- Set to 1 for analog input
*

```


Common ADC Pins (STAR Project)

Pin	ADC Channel	Typical Use
P40	AN000	Motor 0 current sense
P41	AN001	Motor 1 current sense
P42	AN002	Motor 2 current sense
P43	AN003	Motor 3 current sense
P44	AN004	Analog input (unused)
P45	AN005	Temperature sensor

Parameters

in	pin	GPIO pin identifier for ADC input
		• Must be a pin connected to ADC channel • Typically P40-P47 for AN000-AN007

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for ADC input
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must be connected to ADC channel (not all pins support ADC)

PCLKB clock must be running

Postcondition

PFS register has ASEL = 1 (digital buffer disabled)

Pin ready for analog signal input

Note

Digital input buffer disabled - do not use as GPIO input

Also requires S12ADC configuration for actual conversion

Warning

Using a non-ADC pin with this function sets ASEL but has no effect on analog capability (pin hardware determines ADC availability)

Example - Current Sensing Setup

```
// Configure ADC pins for motor current sensing
rx_mpc_set_adc(k_port_4_pin_0); // AN000 = Motor 0 current
rx_mpc_set_adc(k_port_4_pin_1); // AN001 = Motor 1 current
rx_mpc_set_adc(k_port_4_pin_2); // AN002 = Motor 2 current
rx_mpc_set_adc(k_port_4_pin_3); // AN003 = Motor 3 current

// Now configure S12ADC for conversion
// (see adc.h)
```

See also

[rx72n_adc_regs.h](#) ADC register definitions

Since

Version 1.0.0

Configures a pin for analog-to-digital conversion by setting the ASEL bit in the PFS register. This disables the digital input buffer to prevent noise during analog measurement.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with analog capability

Postcondition

PFS register contains k_pfs_asel (0x80) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe
Sets ASEL bit (0x80) rather than PSEL field
Also configure S12ADC for actual conversion

```
// Configure P40 (AN000) for ADC analog input
rx_err_t err = rx_mpc_set_adc(RX_PORT_PIN(4, 0));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 1009 of file rx_mpc.c.

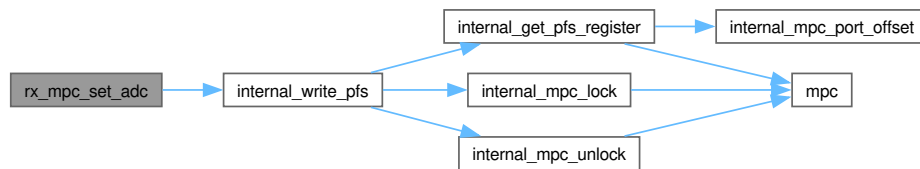
```

01010 {
01011  /* Extract port and pin number from type-safe enum */
01012  const uint8_t port  = rx_port_from_pin(pin);
01013  const uint8_t pin_num = rx_pin_from_pin(pin);
01014
01015  /* Validate the decoded port and pin are in valid range */
01016  if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
01017    rx_log_error(s_tag, "Invalid port or pin number");
01018    return k_rx_err_invalid_arg;
01019  }
01020
01021  /* ADC mode: Set ASEL = 1 to disable digital I/O
01022   * Common ADC pins: P40-P47 (AN000-AN007)
01023   */
01024  return internal_write_pfs(port, pin_num, k_pfs_asel);
01025 }

```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_pfs_asel](#), and [s_tag](#).

Here is the call graph for this function:



8.89.4.2 rx_mpc_set_gpio()

```

rx_err_t rx_mpc_set_gpio (
    const rx_port_pin_t pin) [nodiscard]

```

Configure pin for GPIO (General Purpose I/O) function.

Sets the pin to GPIO mode by writing 0x00 to its PFS register (PSEL=0, ISEL=0, ASEL=0). This disconnects the pin from all peripheral functions and allows it to be controlled by PORT PDR/PODR registers.

Algorithm Steps

1. Extract port number and pin number from encoded pin parameter
2. Validate port is in valid range [0, J]
3. Validate pin number is in valid range [0, 7]
4. Calculate PFS register address for this pin
5. Unlock PWPR write protection (B0WI=0, PFSWE=1)
6. Write 0x00 to PFS register (GPIO mode)
7. Lock PWPR write protection (PFSWE=0, B0WI=1)
8. Return success

Parameters

in	pin	GPIO pin identifier (<code>rx_port_pin_t</code> from <code>rx_port_constants.h</code>) - Encodes both port (A-J) and pin number (0-7) - Use <code>k_port_X_pin_Y</code> constants for type safety - Example: <code>k_port_e_pin_5</code> for PE5
----	-----	--

Returns

Error code indicating success or failure

Return values

<code>k_rx_ok</code>	Pin successfully configured for GPIO mode
<code>k_rx_err_invalid_arg</code>	Port number exceeds valid range (> Port J); Pin number exceeds 7; Port not available on package (G/H on 144-pin)

Precondition

PCLKB clock must be running (MPC register access requires it)

MPC module stop bit must be cleared (`MSTPCRA.MSTPA9 = 0`)

Postcondition

PFS register for specified pin contains 0x00

Pin is disconnected from all peripheral functions

PWPR is locked (`B0WI=1`) after operation

Invariant

MPC base address 0x0008C100 remains constant

Other pins' PFS registers are not modified

Note

Not thread-safe. Use mutex protection if calling from multiple threads.

This only sets PFS register. To complete GPIO setup, also configure PORT PMR (peripheral mode = 0) and PDR (direction) registers.

Warning

Calling on a pin that doesn't exist on the package returns error but does not cause hardware fault.

Performance

- Execution time: ~1 us @ 240 MHz (PWPR unlock/lock overhead)
- No loops or blocking operations

Thread Safety

Not thread-safe. The PWPR unlock/lock sequence is non-atomic.

Example - Single Pin

```
#include "rx_mpc.h"
#include "rx_port_constants.h"

// Configure PE5 for GPIO (was previously PWM output)
rx_err_t err = rx_mpc_set_gpio(k_port_e_pin_5);
if (err != k_rx_ok) {
    rx_log_error("INIT", "Failed to set PE5 to GPIO mode");
    return err;
}

// Now configure PORT registers for GPIO operation
// (set PMR=0, configure PDR for direction, etc.)
```

Example - Error Handling

```
// Attempt to configure invalid pin
rx_err_t err = rx_mpc_set_gpio(k_port_g_pin_0); // Port G not on 144-pin
if (err == k_rx_err_invalid_arg) {
    rx_log_warn("MPC", "Pin not available on this package");
}
```

See also

[rx_mpc_set_peripheral\(\)](#) Configure pin for peripheral function
[rx_port_constants.h](#) Pin enumeration definitions

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 5: [OK] 2 preconditions (clock, module stop), 3 postconditions
- Rule 7: [OK] All internal function returns checked

Sets the specified pin to GPIO mode by writing 0x00 to its PFS register. This disconnects the pin from all peripheral functions and allows it to be controlled by the PORT registers (PDR, PODR, PIDR).

Implementation Details

1. Extract port and pin number from encoded rx_port_pin_t
2. Validate port is in range [0, J]
3. Validate pin is in range [0, 7]
4. Write 0x00 to PFS register (via internal_write_pfs)

Precondition

PCLKB clock running, MPC not in module stop
pin must be a valid rx_port_pin_t constant

Postcondition

PFS register = 0x00 (GPIO mode)
PWPR locked after operation

Note

Thread safety: Not thread-safe
Also set PORT PMR bit to 0 for complete GPIO configuration

```
// Configure P30 (port 3, pin 0) for GPIO use
rx_err_t err = rx_mpc_set_gpio(RX_PORT_PIN(3, 0));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

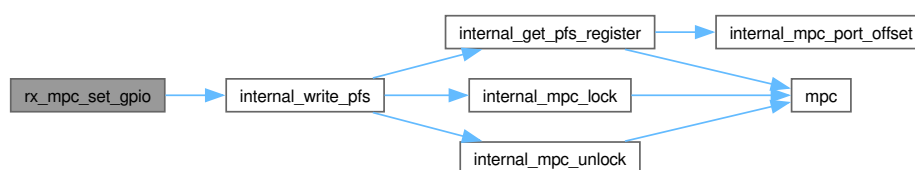
Version 1.0.0

Definition at line 744 of file [rx_mpc.c](#).

```
00745 {
00746     /* Extract port and pin number from rx_port_pin_t */
00747     /* Pre-condition: port and pin must be in valid range */
00748     const uint8_t port = rx_port_from_pin(pin);
00749     const uint8_t pin_num = rx_pin_from_pin(pin);
00750
00751     RX_CHECK_RANGE_TAG(port, k_mpc_port_start, k_mpc_port_j, k_rx_err_invalid_arg, s_tag);
00752     RX_CHECK_RANGE_TAG(pin_num, k_mpc_min_pin, k_mpc_max_pin, k_rx_err_invalid_arg, s_tag);
00753
00754     /* GPIO mode: PSEL = 0, ISEL = 0, ASEL = 0 */
00755     return internal_write_pfs(port, pin_num, k_mpc_pfs_gpio);
00756 }
```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_min_pin](#), [k_mpc_pfs_gpio](#), [k_mpc_port_j](#), [k_mpc_port_start](#), and [s_tag](#).

Here is the call graph for this function:



8.89.4.3 rx`mpc`set`gptw()

```
rx_err_t rx_mpc_set_gptw (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for GPTW (General PWM Timer) complementary output.

Configures a pin for GPTW operation by setting PSEL = 0x1E. GPTW provides complementary PWM output with dead-time insertion for motor control applications.

GPTW channels support:

- 20 kHz PWM frequency (ultrasonic, inaudible)
- 1 us dead-time (prevents DRV8263H shoot-through)
- 90-degree phase staggering (reduces peak current)
- IN2/IN1 mode (direction and PWM signals per DRV8263H)

STAR Project GPTW Pin Allocation

Pin	GPTW Ch	Signal	Motor	Description
P2.3	GPTW0	GTIOC0A	0	Motor 0 Direction (IN2)
P1.7	GPTW0	GTIOC0B	0	Motor 0 PWM (IN1)
P2.2	GPTW1	GTIOC1A	1	Motor 1 Direction (IN2)
PC. ↔ 3	GPTW1	GTETRGC	1	Motor 1 PWM (IN1) - GTETRGC at 0x1E on PC3 per Table 23.19
PE. ↔ 3	GPTW2	GTIOC2A	2	Motor 2 Direction (IN2)
P8.6	GPTW2	GTIOC2B	2	Motor 2 PWM (IN1)
PE. ↔ 7	GPTW3	GTIOC3A	3	Motor 3 Direction (IN2)
PC. ↔ 6	GPTW3	GTIOC3A	3	Motor 3 PWM (IN1) - GTIOC3A at 0x1E on PC6 per Table 23.19

Parameters

in	pin	GPIO pin identifier for GPTW function Must be a pin that supports GPTW output (see manual)
----	-----	--

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for GPTW
k_rx_err_null_ptr	Config pointer is NULL (should never happen)
k_rx_err_invalid_arg	Invalid port or pin number (≥ 16)
k_rx_err_hw_error	PWPR write-protect unlock failed

Precondition

Pin must support GPTW function (check RX72N hardware manual)

PCLKB clock must be running (MPC registers require clock)

Must be called during single-threaded initialization

Postcondition

PFS register configured with PSEL = 0x1E

PWPR register locked (write protection re-enabled)

Pin configured for GPTW PSEL; caller MUST subsequently set the corresponding PORTn.PMR bit (this function does NOT touch PMR) before GPTW can drive the pin.

Note

Thread Safety: Not thread-safe. Call during initialization only.

Configure both direction (A/IN2) and PWM (B/IN1) pins for each motor

GPTW channel configuration required for actual PWM operation

Warning

Motor safety: Configure pins before enabling GPTW channels

Example - Configure All Motor PWM Pins

```
// Configure 8 GPTW pins (4 motors x 2 pins = IN2 + IN1)
const rx_port_pin_t gptw_pins[] = {
    k_rx_p2_3, k_rx_p1_7, // Motor 0 IN2/IN1
    k_rx_p2_2, k_rx_pc_3, // Motor 1 IN2/IN1
    k_rx_pe_3, k_rx_p8_6, // Motor 2 IN2/IN1
    k_rx_pe_7, k_rx_pc_6  // Motor 3 IN2/IN1
};

for (uint8_t i = 0; i < 8; i++) {
    rx_err_t err = rx_mpc_set_gptw(gptw_pins[i]);
    if (err != k_rx_ok) {
        rx_log_error("GPIO", "GPTW pin config failed");
        return err;
    }
}

// Now safe to initialize GPTW channels
rx_gptw_init_all_staggered(&config);
```


See also

[rx_mpc_set_peripheral\(\)](#) Core pin configuration function
[rx_gptw_init_all_staggered\(\)](#) GPTW timer initialization
 RX72N Manual Chapter 23 - Multi-Function Pin Controller
 RX72N Manual Chapter 29 - General PWM Timer

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, recursion (delegates to rx_mpc_set_peripheral)
- Rule 3: [OK] No dynamic allocation (stack-only config struct)
- Rule 4: [OK] Function is 6 lines (well under 60 line limit)
- Rule 7: [OK] Return value from rx_mpc_set_peripheral checked by caller

Configure pin for GPTW (General PWM Timer) complementary output.

Convenience wrapper that configures a pin for GPTW operation using PSEL = k_psel_gptw (0x1E). GPTW provides complementary PWM output with dead-time insertion for motor control.

Precondition

PCLKB clock must be running, MPC not in module stop
 pin must encode a valid port/pin combination with GPTW multiplexing

Postcondition

PFS register contains k_psel_gptw (0x1E) for the specified pin
 PWPR locked after operation

Note

Thread safety: Not thread-safe
 Used for 4 GPTW channels (0-3) with 8 total pins (IN2 + IN1 per motor)
 Phase staggering configured separately via rx_gptw driver

```
// Configure motor 2 GPTW IN2 and IN1 pins (Port E)
rx_err_t err = rx_mpc_set_gptw(RX_PORT_PIN(0xE, 3)); // PE3 GTIOC2A (IN2)
if (err == k_rx_ok) {
    err = rx_mpc_set_gptw(RX_PORT_PIN(8, 6)); // P8.6 GTIOC2B (IN1)
}
```

See also

[rx_mpc.h](#) Full API documentation with usage examples
[rx_gptw_init_all_staggered\(\)](#) GPTW channel configuration

Since

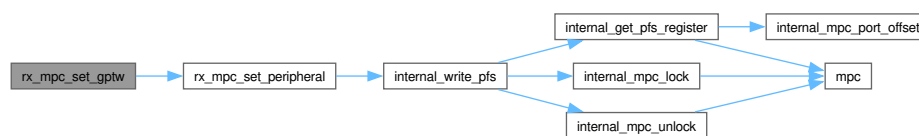
Version 1.0.0

Definition at line 1234 of file rx_mpc.c.

```
01235 {
01236     /* GPTW pins use PSEL = 0x1E
01237      * Supports 4 GPTW channels (0-3) for motor IN2/IN1 control */
01238     const rx_mpc_peripheral_config_t config = {
01239         .pin = pin,
01240         .psel = k_psel_gptw /* 0x1E - defined in rx_pin_psel_t enum */
01241     };
01242
01243     return rx_mpc_set_peripheral(&config);
01244 }
```

References [k_psel_gptw](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.89.4.4 rx_mpc_set_irq()

```
rx_err_t rx_mpc_set_irq (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for IRQ (external interrupt) function.

Sets pin function to IRQ mode via MPC (Multi-Function Pin Controller). This enables hardware edge detection on IRQ0-IRQ15 pins for use with the Interrupt Controller Unit (ICU).

PFS Register for IRQ Function:

- IRQ pins: ISEL bit (bit 6) = 0x40 written directly to PFS register
- This is NOT a PSEL value; [rx_mpc_set_peripheral\(\)](#) is NOT used
- Verify in RX72N Manual Section 20.3 (MPC), PFS register layout

Supported Pins:

- P00-P07: IRQ8-IRQ15
- Other IRQ pins per RX72N Manual Table 20.7

STAR Project IRQ Usage (HC-SR04 Ultrasonic Sensors)

Pin	IRQ Number	Sensor	Location
P03	IRQ11	Sonar 0	Front-Left
P02	IRQ10	Sonar 1	Front-Right
P01	IRQ9	Sonar 2	Back-Left
P00	IRQ8	Sonar 3	Back-Right

Parameters

in	pin	GPIO pin identifier (must support IRQ function) Use k_rx_p0_X constants for IRQ8-15
----	-----	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Pin configured for IRQ function
k_rx_err_invalid_arg	Invalid port or pin number (propagated from internal_write_pfs())

Note

[internal_mpc_unlock\(\)](#) and [internal_mpc_lock\(\)](#) do not propagate errors; only [internal_write_pfs\(\)](#) can return k_rx_err_invalid_arg on invalid port/pin values.

Precondition

Pin must have IRQ multiplexing capability (P00-P07 for IRQ8-15)
PCLKB running (MPC requires clock)
Must be called during single-threaded initialization

Postcondition

Pin ISEL bit set (0x40) in PFS register (IRQ input mode)
PSEL field remains 0 (no conflicting peripheral function)
PWPR register locked
Pin ready for ICU edge detection configuration

Note

Thread Safety: Not thread-safe. Call during initialization only.
Call this BEFORE enabling ICU interrupt

Warning

Does NOT configure ICU registers (edge detect, priority, etc.)
 Configure ICU separately for edge detection and priority as required

Example - IRQ Pin Setup

```
// Configure P03 for IRQ11 (external interrupt input)
rx_err_t err = rx_mpc_set_irq(k_rx_p0_3);
if (err != k_rx_ok) {
    rx_log_error("IRQ", "IRQ pin configuration failed");
    return err;
}

// Now configure ICU for edge detection (caller-specific step)
```

Invariant

MPC base address remains constant throughout operation
 Only target pin's PFS register is modified (other pins' PFS unchanged)

See also

[rx_mpc_set_gpio\(\)](#) Set pin back to GPIO mode
 RX72N Manual Section 20.3 (MPC), PFS register bit definitions
 RX72N Manual Chapter 15 (ICU - Interrupt Controller Unit)

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 5: [OK] 3 preconditions, 4 postconditions
- Rule 7: [OK] Caller checks return value ([[nodiscard]])
- Rule 8: [OK] k_pfs_isel from [pfs_bits_t](#) enum (no magic numbers)

Sets pin function to IRQ mode by writing the ISEL bit (bit 6, value 0x40) directly to the PFS register via [internal_write_pfs\(\)](#). This bypasses [rx_mpc_set_peripheral\(\)](#) because ISEL=0x40 exceeds the 5-bit PSEL field maximum (k_mpc_psel_max = 31).

Mirrors the approach used by [rx_mpc_set_adc\(\)](#) for ASEL (bit 7): both functions write their respective bit directly rather than going through the PSEL validation path.

Algorithm Steps:

1. Extract port number and pin number from encoded pin parameter
2. Validate port in range [0, J]
3. Validate pin in range [0, 7]
4. Write ISEL bit (0x40) to PFS register via [internal_write_pfs\(\)](#)

Precondition

Pin must have IRQ multiplexing capability (P00-P07 for IRQ8-15)
PCLKB clock must be running

Postcondition

Pin ISEL bit (0x40) set in PFS register
PSEL field remains 0 (no peripheral function conflict)

Note

Thread safety: Not thread-safe
Typical pins: P00-P07 for IRQ8-IRQ15 on RX72N
Used by HC-SR04 ultrasonic sensors for echo pulse measurement

```
// Configure P00 for IRQ8 (HC-SR04 echo input)
rx_err_t err = rx_mpc_set_irq(RX_PORT_PIN(0, 0));
if (err != k_rx_ok) {
    // handle error
}
// Then configure ICU for rising/falling edge detection
```

See also

[rx_mpc_set_adc\(\)](#) Same pattern: writes ASEL bit (0x80) directly
[rx_mpc.h](#) Full API documentation with usage examples
[rx_hcsr04_icu_configure\(\)](#) Configure ICU after calling this function

Since

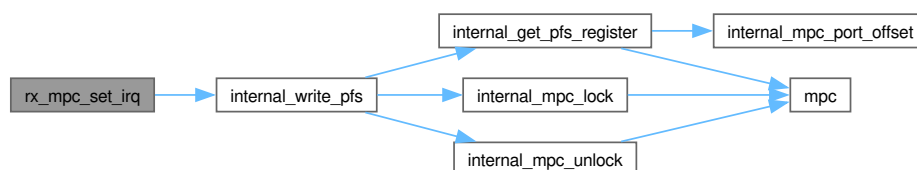
Version 1.0.0

Definition at line 1336 of file [rx_mpc.c](#).

```
01337 {
01338     /* Extract port and pin number from type-safe enum */
01339     const uint8_t port = rx_port_from_pin(pin);
01340     const uint8_t pin_num = rx_pin_from_pin(pin);
01341
01342     /* Validate the decoded port and pin are in valid range */
01343     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
01344         rx_log_error(s_tag, "Invalid port or pin number for IRQ function");
01345         return k_rx_err_invalid_arg;
01346     }
01347
01348     /* IRQ mode: Set ISEL = 1 (bit 6) to enable IRQ input function.
01349     * This is NOT a PSEL value; k_pfs_isel is 0x40 which exceeds the 5-bit
01350     * PSEL field. Write directly like rx_mpc_set_adc() does for ASEL.
01351     * Typical IRQ pins: P00-P07 (IRQ8-IRQ15) for HC-SR04 echo measurement. */
01352     return internal_write_pfs(port, pin_num, k_pfs_isel);
01353 }
```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_pfs_isel](#), and [s_tag](#).

Here is the call graph for this function:



8.89.4.5 rx_mpc_set_mtu_encoder()

```
rx_err_t rx_mpc_set_mtu_encoder (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for MTU3a encoder/phase counter input (MTCLK).

Configures a pin for MTU3a phase counting mode input, used with quadrature encoders. Sets PSEL = k_psel_mtu_phase (0x03) for phase counting with direction detection.

Quadrature Encoder Interface

```
* Encoder Output      MTU3a Phase Counting
* +-----+          +-----+
* | Phase A |----->| MTCLKA (Count Up) |
* | Phase B |----->| MTCLKB (Count Down) |
* | (Index Z) |----->| (Optional reset) |
* +-----+          +-----+
*                               Counter increments/decrements
*                               based on phase relationship
*
```

Common Encoder Pins (STAR Project)

Pin Pair	MTU Input	Encoder
P24/P25	MTCLKA/MTCLKB	Motor 0
PC0/PC1	MTCLKC/MTCLKD	Motor 1

Parameters

in	pin	GPIO pin identifier for encoder input
		- Must be a pin that supports MTCLK function - Configure in pairs (A+B) for quadrature

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for encoder input
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support MTU clock/phase function

PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x03 (MTU phase)

Pin ready for quadrature encoder input

Note

Configure both Phase A and Phase B pins for proper operation

Also requires MTU3a timer configuration in phase counting mode

Example - Quadrature Encoder Setup

```
// Configure encoder pins for Motor 0
rx_err_t err;

err = rx_mpc_set_mtu_encoder(k_port_2_pin_4); // P24 = Phase A
if (err != k_rx_ok) return err;

err = rx_mpc_set_mtu_encoder(k_port_2_pin_5); // P25 = Phase B
if (err != k_rx_ok) return err;

// Now configure MTU1 for phase counting mode
// (see rx_mtu_encoder.h)
```

See also

[rx_mpc_set_peripheral\(\)](#) Underlying implementation

[rx_mtu_encoder.h](#) Encoder driver using MTU phase counting

Since

Version 1.0.0

Convenience wrapper that configures a pin for MTU external clock / phase counting mode using PSEL = k_psel_mtu_phase (0x02). Used for quadrature encoders on pins that expose MTCLKA/B/C/D.

Precondition

PCLKB clock must be running, MPC not in module stop

pin must encode a valid port/pin combination supported by MTCLK input

Postcondition

PFS register contains k_psel_mtu_phase (0x02) for the specified pin

WPWR locked after operation

Note

Thread safety: Not thread-safe

Configure both Phase A and Phase B pins for proper operation

```
// Configure P24 and P25 for MTU phase counting (MTCLKA/B)
rx_err_t err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 4));
if (err == k_rx_ok) {
    err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 5));
}
```

See also

[rx_mpc.h](#) Full API documentation

[rx_mtu_encoder.h](#) Encoder driver using MTU phase counting

Since

Version 1.0.0

Definition at line 906 of file [rx_mpc.c](#).

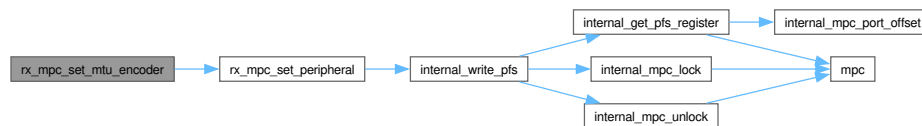
```

00907 {
00908     /* MTU external clock / phase counting pins use PSEL = 0x02 on every
00909     * RX72N MTCLK candidate pin. See RX72N Group Hardware Manual
00910     * (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)",
00911     * Tables 23.5 / 23.6 / 23.17 / 23.19 -- every row that selects MTCLKA,
00912     * MTCLKB, MTCLKC, or MTCLKD on a 144-pin LFQFP pin is PSEL = 000010b.
00913     * The old value 0x03 was wrong -- it caused MTU TCNT to stay at 0x0000
00914     * because no MTCLK source was actually connected through the PFS mux.
00915     *
00916     * Common STAR pins:
00917     * P24/P25 (Motor 0 front-right MTCLKA/B)
00918     * PA1/PC5 (Motor 1 front-left MTCLKC/D)
00919     */
00920     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_phase};
00921
00922     return rx_mpc_set_peripheral(&config);
00923 }

```

References [k_psel_mtu_phase](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.89.4.6 rx_mpc_set_mtu_pwm()

```

rx_err_t rx_mpc_set_mtu_pwm (
    const rx_port_pin_t pin)    [nodiscard]

```

Configure pin for MTU3a timer PWM output (MTIOC).

Convenience function that configures a pin for MTU3a I/O compare output, used for PWM generation. Internally calls [rx_mpc_set_peripheral\(\)](#) with PSEL = k_psel_mtu_ioc (0x01).

Common MTU PWM Pins

Pin	MTU Channel	Output	Typical Use
P14	MTU3	MTIOC3A	Motor 0 Phase A
P15	MTU3	MTIOC3B	Motor 0 Phase B
P16	MTU3	MTIOC3C	Motor 0 Phase C

Pin	MTU Channel	Output	Typical Use
P17	MTU3	MTIOC3D	Motor 0 Phase D
P24	MTU4	MTIOC4A	Motor 1 Phase A
P25	MTU4	MTIOC4B	Motor 1 Phase B
P26	MTU4	MTIOC4C	Motor 1 Phase C
P27	MTU4	MTIOC4D	Motor 1 Phase D

Parameters

in	pin	GPIO pin identifier for MTU PWM output - Must be a pin that supports MTIOC function - Use k_port_X_pin_Y constants
----	-----	---

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for MTU PWM
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support MTU I/O compare function (check hardware manual)

PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x01 (MTU IOC)

Pin ready for MTU3a compare match output

Note

Also requires MTU3a timer configuration and PORT PMR=1

STAR project uses GPTW for motor PWM; see [rx_mpc_set_gptw\(\)](#) which uses PSEL = k_psel_gptw (0x1E)

Example

```
// Configure P14-P17 for MTU3 PWM outputs
rx_mpc_set_mtu_pwm(k_port_1_pin_4); // MTIOC3A
rx_mpc_set_mtu_pwm(k_port_1_pin_5); // MTIOC3B
rx_mpc_set_mtu_pwm(k_port_1_pin_6); // MTIOC3C
rx_mpc_set_mtu_pwm(k_port_1_pin_7); // MTIOC3D
```

See also

[rx_mpc_set_peripheral\(\)](#) Underlying implementation
[rx_mtu.h](#) MTU timer driver

Since

Version 1.0.0

Convenience wrapper that configures a pin for MTU I/O compare output using PSEL = k_psel_mtu_ioc (0x01). Commonly used for motor PWM.

Precondition

PCLKB clock must be running, MPC not in module stop
 pin must encode a valid port/pin combination supported by MTU MTIOC

Postcondition

PFS register contains k_psel_mtu_ioc (0x01) for the specified pin
 PWPR locked after operation

Note

Thread safety: Not thread-safe

STAR project uses GPTW for motor PWM (see [rx_mpc_set_gptw\(\)](#), PSEL = k_psel_gptw (0x1E)), not MTU

```
// Configure P14 for MTU3 PWM output (MTIOCA)
rx_err_t err = rx_mpc_set_mtu_pwm(RX_PORT_PIN(1, 4));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 863 of file [rx_mpc.c](#).

```
00864 {
00865     /* MTU PWM (MTIOC) pins typically use PSEL = 0x01
00866     * Common pins:
00867     * - P14-P17: MTU3 MTIOCA/B/C/D
00868     * - P24-P27: MTU4 MTIOCA/B/C/D
00869     */
00870     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_ioc};
00871
00872     return rx_mpc_set_peripheral(&config);
00873 }
```

References [k_psel_mtu_ioc](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.89.4.7 rx_mpc_set_peripheral()

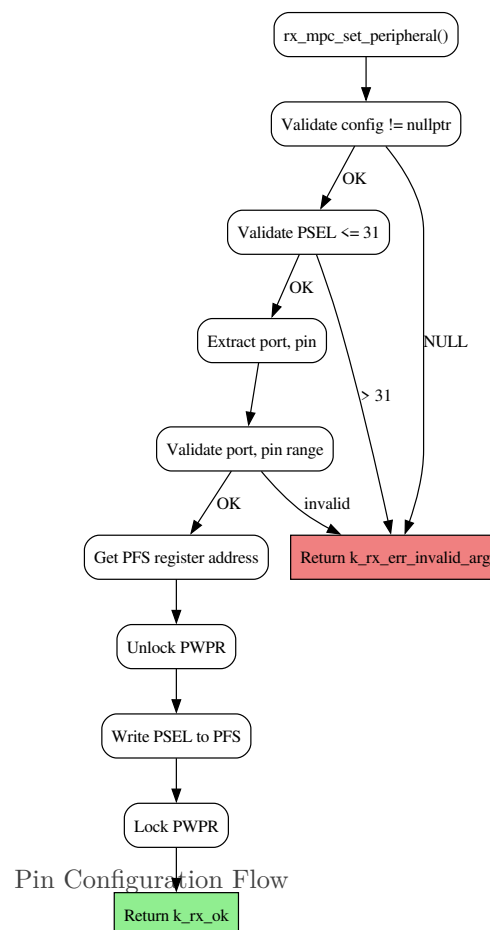
```
rx_err_t rx_mpc_set_peripheral (
    const rx_mpc_peripheral_config_t * config) [nodiscard]
```

Configure pin for peripheral function with specified PSEL code.

Core function for pin multiplexer configuration. Writes the specified PSEL value to the pin's PFS register, connecting the pin to a peripheral function. Uses a configuration structure to prevent parameter swapping.

Algorithm Steps

1. Validate config pointer is not nullptr
2. Validate PSEL value is in range [0, 31]
3. Extract port number and pin number from config->pin
4. Validate extracted port and pin are in valid ranges
5. Calculate PFS register address for this pin
6. Unlock PWPR write protection (B0WI=0, PFSWE=1)
7. Write config->psel to PFS register
8. Lock PWPR write protection (PFSWE=0, B0WI=1)
9. Return success



Parameters

in	config	Pointer to MPC peripheral configuration structure • config->pin: Target pin (rx_port_pin_t constant) • config->psel: PSEL value to write (0-31) • Must not be nullptr • Must remain valid only during function call
----	--------	--

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for peripheral function
k_rx_err_invalid_arg	config pointer is nullptr; config->psel exceeds 31 (5-bit field max); Port number exceeds valid range; Pin number exceeds 7; Port not available on package

Precondition

config pointer must be non-NULL
 config->psel must be in range [0, 31]
 config->pin must be a valid rx_port_pin_t value
 PCLKB clock must be running

Postcondition

PFS register for specified pin contains config->psel value
 Pin is connected to peripheral function specified by PSEL
 PWPR is locked (B0WI=1) after operation

Invariant

config structure is not modified (const pointer)
 Other pins' PFS registers are not modified

Note

Not thread-safe. Use mutex protection if calling from multiple threads.
 This only sets PFS register. For peripheral operation, also set PORT PMR bit to 1 to enable peripheral mode.
 PSEL values are pin-specific. Verify correct value in hardware manual.

Warning

Wrong PSEL value may connect pin to unexpected peripheral!
 Not all PSEL values are valid for all pins.

Performance

- Execution time: ~1 us @ 240 MHz
- No loops or blocking operations

Thread Safety

Not thread-safe. The PWPR unlock/lock sequence is non-atomic.

Example - Motor PWM Configuration

```
#include "rx_mpc.h"
#include "rx_port_constants.h"

// Configure PE5 for GPTW0A (Motor 0 PWM Phase+)
rx_mpc_peripheral_config_t pwm_config = {
    .pin = k_port_e_pin_5, // PE5
    .psel = k_psel_gptw // GPTW function (0x1E)
};

rx_err_t err = rx_mpc_set_peripheral(&pwm_config);
if (err != k_rx_ok) {
    rx_log_error("MOTOR", "PWM pin configuration failed");
    return err;
}

// Don't forget: Set PORT PMR bit to enable peripheral mode!
```

Example - Batch Configuration

```
// Configure multiple pins in a loop
const rx_mpc_peripheral_config_t pins[] = {
    { k_port_e_pin_5, k_psel_gptw }, // GPTW0A (0x1E)
    { k_port_e_pin_4, k_psel_gptw }, // GPTW0B (0x1E)
    { k_port_2_pin_4, 0x02 }, // MTCLKA
    { k_port_2_pin_5, 0x02 }, // MTCLKB
};

for (size_t i = 0; i < sizeof(pins)/sizeof(pins[0]); i++) {
    rx_err_t err = rx_mpc_set_peripheral(&pins[i]);
    if (err != k_rx_ok) {
        rx_log_error("MPC", "Pin config failed");
        return err;
    }
}
```

See also

[rx_mpc_set_gpio\(\)](#) Configure pin for GPIO mode
[rx_mpc_peripheral_config_t](#) Configuration structure definition
[rx_pin_psel_t](#) Common PSEL value constants

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 5: [OK] 4 preconditions, 3 postconditions
- Rule 7: [OK] All return values checked

Core function for peripheral pin configuration. Writes the specified PSEL value to the pin's PFS register, connecting it to a peripheral.

Implementation Details

1. Validate config pointer is not nullptr
2. Validate PSEL value is in range [0, 31]
3. Extract port and pin from config->pin
4. Validate extracted values
5. Write PSEL to PFS register (via internal_write_pfs)

Precondition

config must be non-NULL with valid fields
PCLKB clock running

Postcondition

PFS register contains config->psel
PWPR locked after operation

Note

Thread safety: Not thread-safe
Also set PORT PMR bit to 1 for peripheral mode

```
// Configure PA0 for RSPI clock (PSEL = 0x0D)
const rx_mpc_peripheral_config_t cfg = {
    .pin = RX_PORT_PIN(0xA, 0),
    .psel = 0x0D,
};
rx_err_t err = rx_mpc_set_peripheral(&cfg);
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 798 of file [rx_mpc.c](#).

```

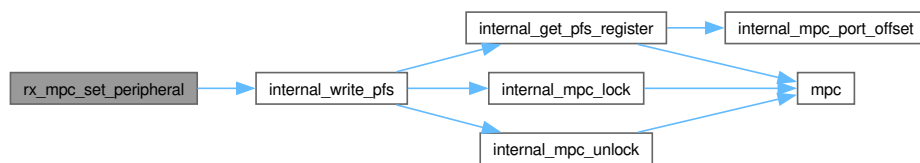
00799 {
00800     if (config == nullptr) {
00801         rx_log_error(s_tag, "Invalid configuration pointer");
00802         return k_rx_err_invalid_arg;
00803     }
00804
00805     if (config->psel > k_mpc_psel_max) {
00806         rx_log_error(s_tag, "Invalid PSEL value exceeds maximum");
00807         rx_log_error_val(s_tag, " psel=", config->psel);
00808         rx_log_error_val(s_tag, " k_mpc_psel_max=", (uint8_t)k_mpc_psel_max);
00809         return k_rx_err_invalid_arg;
00810     }
00811
00812     /* Extract port and pin number from rx_port_pin_t */
00813     const uint8_t port = rx_port_from_pin(config->pin);
00814     const uint8_t pin_num = rx_pin_from_pin(config->pin);
00815
00816     /* Validate the decoded port and pin are in valid range */
00817     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
00818         rx_log_error(s_tag, "Invalid port or pin number");
00819         rx_log_error_val(s_tag, " port=", port);
00820         rx_log_error_val(s_tag, " pin=", pin_num);
00821         return k_rx_err_invalid_arg;
00822     }
00823
00824     /* Peripheral mode: PSEL = specified, ISEL = 0, ASEL = 0.
00825     *
00826     * NOTE: PORT.PMR bit is intentionally NOT set here. Setting PMR=1 before
00827     * the peripheral itself (e.g. GPTW timer) is fully configured exposes
00828     * the pin to undefined/glitchy output that can latch a fault in
00829     * downstream silicon (e.g. DRV8263H motor driver). Caller is responsible
00830     * for setting PMR=1 AFTER the peripheral is fully initialized. */
00831     return internal_write_pfs(port, pin_num, config->psel);
00832 }

```

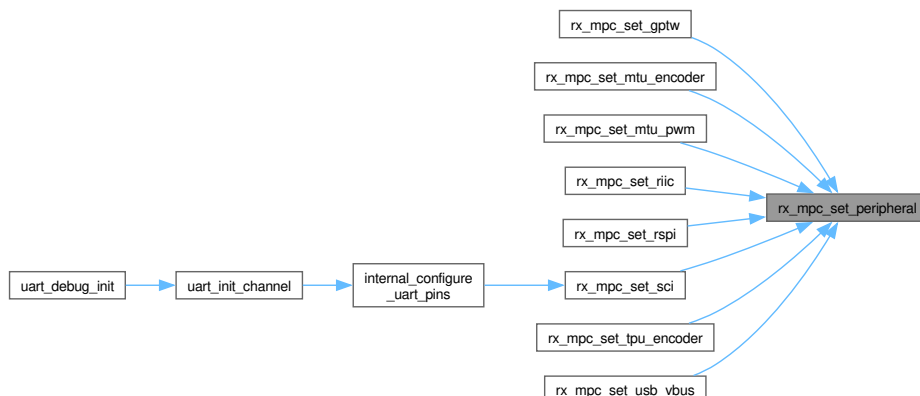
References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_mpc_psel_max](#), [rx_mpc_peripheral_config_t::rx_mpc_peripheral_config_t::psel](#), and [s_tag](#).

Referenced by [rx_mpc_set_gptw\(\)](#), [rx_mpc_set_mtu_encoder\(\)](#), [rx_mpc_set_mtu_pwm\(\)](#), [rx_mpc_set_riic\(\)](#), [rx_mpc_set_rspi\(\)](#), [rx_mpc_set_sci\(\)](#), [rx_mpc_set_tpu_encoder\(\)](#), and [rx_mpc_set_usb_vbus\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.89.4.8 rx_mpc_set_riic()

```
rx_err_t rx_mpc_set_riic (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for RIIC (I2C) bus function.

Configures a pin for RIIC (Renesas I2C) operation by setting PSEL = 0x0F. Works for both SCL (clock) and SDA (data) pins. RIIC hardware automatically configures the open-drain output when the channel is enabled.

I2C Bus Pins

Pin	RIIC Channel	Function	Notes
P12	RIIC0	SCL0	Clock line
P13	RIIC0	SDA0	Data line

Parameters

in	pin	GPIO pin identifier for RIIC function
		Must be a pin that supports RIIC SCL/SDA

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for RIIC I2C
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support RIIC function
PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x0F
Pin ready for I2C operation

Note

Configure both SCL and SDA pins
External pull-up resistors required (typically 4.7kOhm)
Also requires RIIC channel configuration

Example - I2C Bus Setup

```
// Configure I2C pins (RIIC0)
rx_mpc_set_riic(k_port_1_pin_2); // P12 = SCL0
rx_mpc_set_riic(k_port_1_pin_3); // P13 = SDA0

// Now configure RIIC0
// (see riic.h)
```

See also

[rx72n_riic_regs.h](#) RIIC register definitions

Since

Version 1.0.0

Convenience wrapper that configures a pin for RIIC operation using PSEL = k_psel_riic_scl (0x0F). Works for both SCL and SDA pins.

Unlike [rx_mpc_set_gptw\(\)](#), this function also sets PORT.PMR=1 for the pin before returning – RIIC cannot drive SDA/SCL while PMR=0 and the immediate-vs-deferred PMR trade-off differs from the motor-PWM case:

Peripheral	When to raise PMR	Reason
GPTW/PWM	AFTER timer is running	Push-pull drive: an un-
	(rx_gptw_init_pwm handles	initialized output could
	this after rx_gptw_start)	momentarily glitch
		DRV8263H H-bridge inputs
		to Vm/2 during the
		GTIOR reset state window.
RIIC/I2C	IMMEDIATELY after PFS	Open-drain + external
	(here, inside this wrapper)	pull-ups: idle bus is
		both lines high (3.3V);
		routing an un-initialized
		RIIC just leaves the
		pull-ups floating the
		lines high. No hazard.

Without PMR=1 on the RIIC pins, rx72n peripheral bits still toggle internally but the pad stays disconnected from RIIC; every Start condition times out with k_rx_err_timeout (0x108) at the first [riic_write\(\)](#)/[riic_read\(\)](#) call. Discovered 2026-04-22: production imu_task saw tight "[ERROR] [RIIC] Start condition timeout" loops while imu_test/main.c (which sets PMR=1 at line 227 before calling its own [riic_init\(\)](#)) talked to BNO055 + BMP280 cleanly on the same board.

Precondition

PCLKB clock must be running, MPC not in module stop

pin must encode a valid port/pin combination with RIIC multiplexing

Postcondition

PFS register contains `k_psel_riic_scl` (0x0F) for the specified pin
 PORT.PMR bit for the pin is set (pin routed to RIIC peripheral)
 PWPR locked after operation

Note

Thread safety: Not thread-safe
 Configure both SCL and SDA pins; external pull-ups required

```
// Configure P12 (SCL) and P13 (SDA) for RIIC0
rx_err_t err = rx_mpc_set_riic(RX_PORT_PIN(1, 2));
if (err == k_rx_ok) {
    err = rx_mpc_set_riic(RX_PORT_PIN(1, 3));
}
```

See also

[rx_mpc.h](#) Full API documentation
[rx_mpc_set_gptw](#) Contrast: deferred-PMR case
[star-rx72n-firmware/imu_test/main.c::riic1_init](#) Bench reference

Since

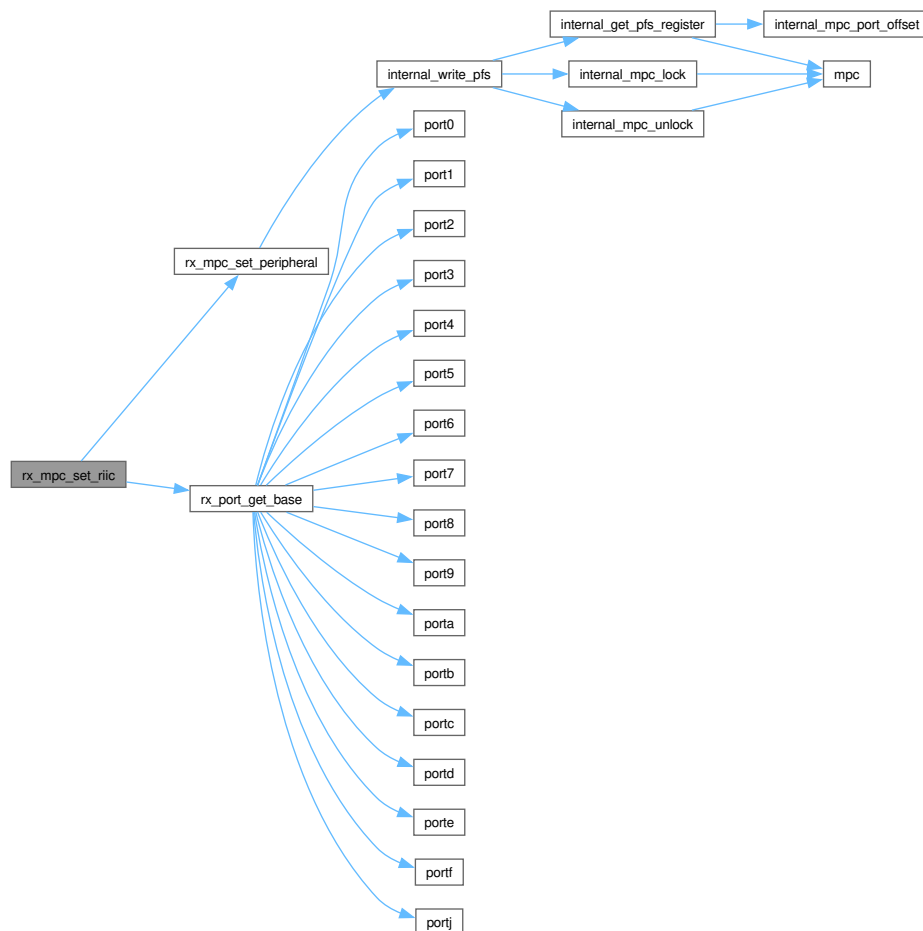
Version 1.0.0

Definition at line 1126 of file [rx_mpc.c](#).

```
01127 {
01128     /* RIIC pins use PSEL = 0x0F
01129     * SCL and SDA use the same PSEL code
01130     */
01131     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_riic_scl};
01132
01133     const rx_err_t err = rx_mpc_set_peripheral(&config);
01134     if (err != k_rx_ok) {
01135         return err;
01136     }
01137
01138     /* Set PORT.PMR=1 so SCL/SDA actually route to the RIIC peripheral.
01139     *
01140     * Unlike GPTW (where PMR must be deferred until after the timer is
01141     * running to avoid exposing DRV8263H inputs to GTIOR reset state),
01142     * I2C pins are open-drain with external pull-ups: driving them high
01143     * through the peripheral mux is the idle-bus state and perfectly safe.
01144     *
01145     * Without this bit set, P20/P21 remain GPIO (PMR=0) and RIIC1 can
01146     * never toggle the lines -- every Start condition times out with
01147     * k_rx_err_timeout (0x108). The bench-proven imu_test/main.c sets
01148     * PMR identically at rx72n-firmware/imu_test/main.c:227. */
01149     const uint8_t port_idx = rx_port_from_pin(pin);
01150     const uint8_t bit = rx_pin_from_pin(pin);
01151     volatile rx_port_regs_t* const port = rx_port_get_base(port_idx);
01152     if (port == nullptr) {
01153         return k_rx_err_invalid_arg;
01154     }
01155     port->pmr |= (uint8_t)(1U << bit);
01156     return k_rx_ok;
01157 }
```

References [k_psel_riic_scl](#), [rx_port_regs_t::pmr](#), [rx_mpc_set_peripheral\(\)](#), and [rx_port_get_base\(\)](#).

Here is the call graph for this function:



8.89.4.9 rx'mpc'set'rspi()

```
rx_err_t rx_mpc_set_rspi (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for RSPI (SPI) bus function.

Configures a pin for RSPI (Renesas SPI) operation by setting PSEL = 0x0D. Works for all SPI signals: RSPCK (clock), COPI (data out), CIPO (data in), and SSLn (chip select).

STAR Project SPI Configuration (RPi5 Communication)

Note: RSPI0 Signal names (MOSIA/MISOA) are RX72N datasheet hardware names. Function names (COPI/CIPO) are project terminology per OSHWA standards.

Pin	RSPI0 Signal	Function	Description
PA0	MOSIA	COPI	Controller Out, Peripheral In
PA1	MISOA	CIPO	Controller In, Peripheral Out
PA3	RSPCKA	CLK	SPI clock
PA4	SSLA0	CS	Chip select (active low)

SPI Bus Topology

```

*   RX72N (Controller)           RPi5 (Controller)
*   +-----+                   +-----+
*   | RSPi0      |             | SPI      |
*   | PA0 (COPI)  <-----+-----> CIPO    |
*   | PA1 (CIPO)  <-----+-----> COPI    |
*   | PA3 (CLK)   <-----+-----> CLK     |
*   | PA4 (CS)    <-----+-----> CS      |
*   +-----+                   +-----+
*   Note: RX72N is SPI peripheral, RPi5 is controller
*

```

Parameters

in	pin	GPIO pin identifier for RSPi function
		Must be a pin that supports RSPi signals

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for RSPi SPI
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support RSPi function

PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x0D

Pin ready for SPI operation

Note

Configure all 4 SPI pins (CLK, COPI, CIPO, CS)

Also requires RSPi channel configuration

Uses OSHWA-approved inclusive terminology (COPI/CIPO)

Example - SPI Bus Setup

```

// Configure SPI pins for RPi5 communication (RSPi0)
rx_mpc_set_rspi(k_port_a_pin_0); // PA0 = COPI
rx_mpc_set_rspi(k_port_a_pin_1); // PA1 = CIPO
rx_mpc_set_rspi(k_port_a_pin_3); // PA3 = CLK
rx_mpc_set_rspi(k_port_a_pin_4); // PA4 = CS (CS)

// Now configure RSPi0
// (see rspi.h)

```

See also

[rx72n_rspi_regs.h](#) RSPI register definitions
[rx_spi_comm.h](#) SPI communication driver

Since

Version 1.0.0

Convenience wrapper that configures a pin for RSPI operation using `PSEL = k_psel_rspi_clk (0x0D)`. Works for all SPI signals (CLK, COPI, CIPO, SSL).

Precondition

PCLKB clock must be running, MPC not in module stop
 pin must encode a valid port/pin combination with RSPI multiplexing

Postcondition

PFS register contains `k_psel_rspi_clk (0x0D)` for the specified pin
 PWPR locked after operation

Note

Thread safety: Not thread-safe
 Configure all 4 SPI pins (CLK, COPI, CIPO, CS)
 Uses OSHWA-approved inclusive terminology

```
// Configure PA0-PA4 for RSPI0 (CLK, COPI, CIPO, SSL0, SSL1)
rx_err_t err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 0)); // CLK
if (err == k_rx_ok) {
    err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 1)); // COPI
}
```

See also

[rx_mpc.h](#) Full API documentation
[rx_spi_comm.h](#) SPI communication driver

Since

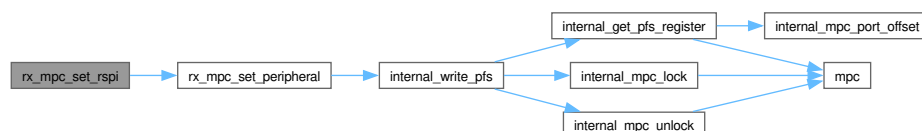
Version 1.0.0

Definition at line 1191 of file [rx_mpc.c](#).

```
01192 {
01193     /* RSPI pins use PSEL = 0x0D
01194      * All RSPI signals (CLK, COPI, CIPO, SSL) use same PSEL
01195      */
01196     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_rspi_clk};
01197
01198     return rx_mpc_set_peripheral(&config);
01199 }
```

References [k_psel_rspi_clk](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.89.4.10 rx_mpc_set_sci()

```
rx_err_t rx_mpc_set_sci (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for SCI (Serial Communication Interface) UART function.

Configures a pin for SCI UART operation by setting PSEL = 0x0A. Works for both TXD (transmit) and RXD (receive) pins - the same PSEL value is used, with the specific function determined by the pin's hardware connection.

STAR Project UART Configuration (Debug Console)

Pin	SCI Channel	Function	Notes
PB7	SCI9	TXD9	Debug output
PB6	SCI9	RXD9	Debug input

Parameters

in	pin	GPIO pin identifier for SCI function
		Must be a pin that supports SCI TXD/RXD

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for SCI UART
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support SCI function (check hardware manual)

PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x0A

Pin ready for UART operation

Note

Configure both TXD and RXD pins for bidirectional UART

Also requires SCI channel configuration and PORT PMR=1

Example - Debug UART Setup

```
// Configure debug UART pins (SCI9)
rx_mpc_set_sci(k_port_b_pin_7); // PB7 = TXD9
rx_mpc_set_sci(k_port_b_pin_6); // PB6 = RXD9

// Now configure SCI9 for UART mode
// (see uart.h)
```

See also

[rx72n_sci_regs.h](#) SCI register definitions

Since

Version 1.0.0

Convenience wrapper that configures a pin for SCI UART operation using PSEL = k_psel_sci_tx (0x0A). Works for both TXD and RXD pins.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with SCI multiplexing

Postcondition

PFS register contains k_psel_sci_tx (0x0A) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe
Configure both TXD and RXD pins for bidirectional UART

```
// Configure PB7 for SCI9 TXD
rx_err_t err = rx_mpc_set_sci(RX_PORT_PIN(0xB, 7));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

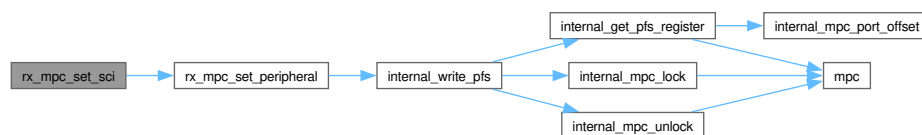
Definition at line 1056 of file `rx_mpc.c`.

```
01057 {
01058     /* SCI pins use PSEL = 0x0A
01059      * TX and RX use the same PSEL code
01060      */
01061     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_sci_tx};
01062
01063     return rx_mpc_set_peripheral(&config);
01064 }
```

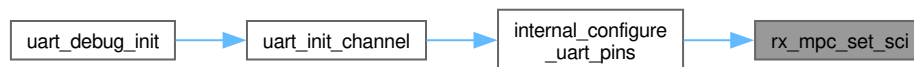
References `k_psel_sci_tx`, and `rx_mpc_set_peripheral()`.

Referenced by `internal_configure_uart_pins()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.89.4.11 `rx_mpc_set_tpu_encoder()`

```
rx_err_t rx_mpc_set_tpu_encoder (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for TPU encoder/phase counter input (TCLK).

Configures a pin for TPU phase counting mode input, used with quadrature encoders on the rear wheels. Sets PSEL = `k_psel_mtu_phase` (0x03), which is the generic "timer phase counting input" function on the RX72N. The routing to TPU (vs MTU) depends on which timer module has phase counting mode enabled.

This function exists as a separate API from `rx_mpc_set_mtu_encoder()` for documentation clarity, even though both write the same PSEL value.

TPU Encoder Pins (STAR Project - Rear Wheels)

Pin Pair	TPU Input	Encoder
PC2/PA3	TCLKA/TCLKB	Motor 2 (Rear Left)
PC0/PB3	TCLKC/TCLKD	Motor 3 (Rear Right)

Parameters

in	pin	GPIO pin identifier for TPU encoder input
		- Must be a pin that supports TCLK function - Configure in pairs (A+B) for quadrature

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for TPU encoder input
k_rx_err_invalid_arg	Invalid port or pin number

Precondition

Pin must support TPU clock/phase function
PCLKB clock must be running

Postcondition

PFS register contains PSEL = 0x03 (timer phase counting)
Pin ready for quadrature encoder input via TPU

Note

Configure both Phase A and Phase B pins for proper operation
Also requires TPU timer configuration in phase counting mode

Example - Rear Wheel Encoder Setup

```
rx_err_t err;

// Motor 2 (Rear Left) - TPU1
err = rx_mpc_set_tpu_encoder(k_rx_pc_2); // PC2 = TCLKA (Phase A)
if (err != k_rx_ok) return err;
err = rx_mpc_set_tpu_encoder(k_rx_pa_3); // PA3 = TCLKB (Phase B)
if (err != k_rx_ok) return err;

// Motor 3 (Rear Right) - TPU2
err = rx_mpc_set_tpu_encoder(k_rx_pc_0); // PC0 = TCLKC (Phase A)
if (err != k_rx_ok) return err;
err = rx_mpc_set_tpu_encoder(k_rx_pb_3); // PB3 = TCLKD (Phase B)
if (err != k_rx_ok) return err;
```

See also

[rx_mpc_set_mtu_encoder\(\)](#) Front wheel encoder pin configuration
[rx_mpc_set_peripheral\(\)](#) Underlying implementation
[rx_encoder_tpu.h](#) TPU encoder driver using phase counting

Since

Version 1.0.0

Convenience wrapper that configures a pin for TPU external clock input (TCLKA/B/C/D). Picks the correct PSEL value based on which port the pin is on, because RX72N uses two different alternate-function slots for the SAME TCLK signal depending on candidate:

- Port C pins (PC0/PC1/PC2/PC3) use PSEL = 0x03 (`k_psel_tpu_clk_portc`)
- Port 1 / Port A / Port B pins use PSEL = 0x04 (`k_psel_tpu_clk_alt`)

Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)". TCLK rows in Table 23.19 (Port C) use PSEL = 000011b, while TCLK rows in Tables 23.5 (P1x), 23.17 (PAx), and 23.18 (PBx) use PSEL = 000100b. Before this split, every TPU pin was configured with PSEL=0x03, which silently misconfigured the non-Port-C halves of STAR's rear-wheel encoders (PA3 for Motor 2 Phase B, PB3 for Motor 3 Phase B) so those edges were never counted.

Precondition

PCLKB clock must be running, MPC not in module stop
 pin must encode a valid port/pin combination supported by TCLK input

Postcondition

PFS register contains the correct TPU-TCLK PSEL for that pin's port
 PWPR locked after operation

Note

Thread safety: Not thread-safe
 Configure both Phase A and Phase B pins for proper operation

```
// Configure PC2 (TCLKA) and PA3 (TCLKB) for Motor 2 rear-left encoder
rx_err_t err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xC, 2)); // -> 0x03
if (err == k_rx_ok) {
    err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xA, 3));      // -> 0x04
}
```

See also

[rx_mpc.h](#) Full API documentation
[rx_encoder_tpu.h](#) TPU encoder driver using phase counting

Since

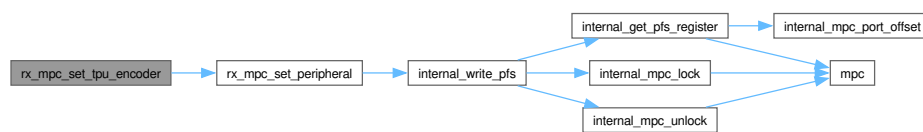
Version 1.0.0

Definition at line 967 of file [rx_mpc.c](#).

```
00968 {
00969     /* Pick the right PSEL for the TCLK function on this pin's port. */
00970     const uint8_t port = rx_port_from_pin(pin);
00971     const uint8_t psel =
00972         (port == k_mpc_port_c) ? (uint8_t)k_psel_tpu_clk_portc : (uint8_t)k_psel_tpu_clk_alt;
00973
00974     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = psel};
00975     return rx_mpc_set_peripheral(&config);
00976 }
```

References [k_mpc_port_c](#), [k_psel_tpu_clk_alt](#), [k_psel_tpu_clk_portc](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.89.4.12 rx_mpc_set_usb_vbus()

```
rx_err_t rx_mpc_set_usb_vbus (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for USB VBUS detection function.

Configures a pin for USB VBUS detection by setting PSEL = 0x11. VBUS detection is used for USB enumeration and power presence sensing.

The VBUS detect pin monitors the 5V power rail on the USB bus:

- Pin reads HIGH (1) when USB cable connected and 5V present
- Pin reads LOW (0) when USB cable disconnected or unpowered

Required for USB CDC communication with RPi5 host.

STAR Project USB Configuration

Pin	Function	Description
P1.↔ 6	USB0_VBUS	VBUS detect (5V presence)

Parameters

in	pin	GPIO pin identifier for USB VBUS detect Typically P1.6 for USB0_VBUS on RX72N
----	-----	---

Returns

Error code indicating success or failure

Return values

k_rx_ok	Pin successfully configured for USB VBUS
k_rx_err_null_ptr	Config pointer is NULL (should never happen)
k_rx_err_invalid_arg	Invalid port or pin number
k_rx_err_hw_error	PWPR write-protect unlock failed

Precondition

Pin must support USB VBUS function (P1.6 on RX72N)

PCLKB clock must be running

Must be called during single-threaded initialization

Postcondition

PFS register configured with PSEL = 0x11

PWPR register locked

Pin configured for USB VBUS PSEL; caller MUST subsequently set the corresponding PORTn.
PMR bit (this function does NOT touch PMR) before the USB peripheral sees the VBUS signal.

Note

Thread Safety: Not thread-safe. Call during initialization only.

Active-high detection (pin = 1 when 5V present)

Also requires USB PHY and USB0 controller configuration

Example - USB VBUS Setup

```
// Configure USB VBUS detect pin (P1.6)
rx_err_t err = rx_mpc_set_usb_vbus(k_rx_p1_6);
if (err != k_rx_ok) {
    rx_log_error("USB", "VBUS pin configuration failed");
    return err;
}

// Now configure USB0 controller
err = usb_init(USB_MODE_CDC);
```

See also

[rx_mpc_set_peripheral\(\)](#) Core pin configuration function

[rx_usb.h](#) USB controller driver

RX72N Manual Chapter 23 - Multi-Function Pin Controller

RX72N Manual Chapter 31 - USB Controller

Since

Version 1.0.0

Configure pin for USB VBUS detection function.

Convenience wrapper that configures a pin for USB VBUS detection using PSEL = 0x11. Required for USB CDC enumeration and power monitoring.

Precondition

PCLKB clock must be running, MPC not in module stop

pin must encode a valid port/pin combination with USB VBUS multiplexing

Postcondition

PFS register contains k_psel_usb_vbus (0x11) for the specified pin

PWPR locked after operation

Note

Thread safety: Not thread-safe

Active-high VBUS detect (pin = 1 when 5V present)

Also requires USB PHY and controller configuration

```
// Configure P16 for USB0_VBUS detection
rx_err_t err = rx_mpc_set_usb_vbus(RX_PORT_PIN(1, 6));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation with usage examples

[rx_usb.h](#) USB controller driver

Since

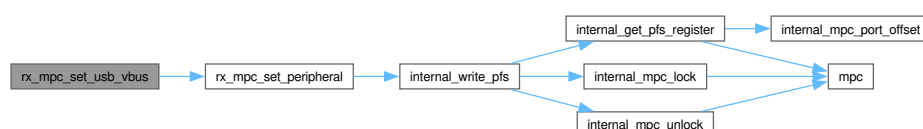
Version 1.0.0

Definition at line 1278 of file [rx_mpc.c](#).

```
01279 {
01280     /* USB VBUS detect uses PSEL = 0x11
01281      * Typical pin: P1.6 for USB0_VBUS on RX72N */
01282     const rx_mpc_peripheral_config_t config = {
01283         .pin = pin,
01284         .psel = k_psel_usb_vbus /* 0x11 - defined in rx_pin_psel_t enum */
01285     };
01286
01287     return rx_mpc_set_peripheral(&config);
01288 }
```

References [k_psel_usb_vbus](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:




```

00083 * | MPC Base | 0x0008C100 | Fixed hardware address |
00084 * | PCLKB | Active | Required for register access |
00085 * | MSTPCRA | Bit 9 = 0 | MPC module stop cleared |
00086 *
00087 * @par Performance Characteristics
00088 * | Operation | Typical Time | Notes |
00089 * |-----|-----|-----|
00090 * | Pin config | ~1 us | PWPR unlock/lock overhead |
00091 * | Bulk config (10 pins) | ~10 us | Sequential calls |
00092 *
00093 * @par Memory Usage
00094 * | Category | Size | Notes |
00095 * |-----|-----|-----|
00096 * | Code | ~800 bytes | All functions |
00097 * | Static data | 4 bytes | Tag string only |
00098 * | Stack (per call) | 16 bytes | Local variables |
00099 *
00100 * @par Thread Safety
00101 * **Not thread-safe.** The MPC PWPR unlock/lock sequence is non-atomic.
00102 * If multiple threads call MPC functions concurrently, race conditions can
00103 * occur where one thread's unlock is clobbered by another's lock. Use
00104 * external mutex protection or configure all pins during single-threaded
00105 * initialization before starting RTOS.
00106 *
00107 * @par Initialization Order
00108 * MPC configuration should occur early in system initialization:
00109 * 1. Clock configuration (PCLKB must be running)
00110 * 2. Module stop clear (if MPC in module stop)
00111 * 3. **MPC pin configuration** < This module
00112 * 4. GPIO direction configuration (PMR, PDR registers)
00113 * 5. Peripheral driver initialization
00114 *
00115 * @par Usage Example
00116 * @code{.c}
00117 * #include "rx_mpc.h"
00118 * #include "rx_port_constants.h"
00119 *
00120 * void hardware_init_pins(void)
00121 * {
00122 *     rx_err_t err;
00123 *
00124 *     // Configure motor PWM pins (GPTW channel 0)
00125 *     const rx_mpc_peripheral_config_t pwm_config = {
00126 *         .pin = k_port_e_pin_5, // PE5 = GPTW0A
00127 *         .psel = k_psel_gptw // GPTW function (0x1E)
00128 *     };
00129 *     err = rx_mpc_set_peripheral(&pwm_config);
00130 *     if (err != k_rx_ok) {
00131 *         // Handle error
00132 *     }
00133 *
00134 *     // Configure encoder pins (MTU phase counting)
00135 *     err = rx_mpc_set_mtu_encoder(k_port_2_pin_4); // P24 = MTCLKA
00136 *     err = rx_mpc_set_mtu_encoder(k_port_2_pin_5); // P25 = MTCLKB
00137 *
00138 *     // Configure debug UART (SCI9)
00139 *     err = rx_mpc_set_sci(k_port_b_pin_7); // PB7 = TXD9
00140 *     err = rx_mpc_set_sci(k_port_b_pin_6); // PB6 = RXD9
00141 *
00142 *     // Configure SPI communication (RSPI0)
00143 *     err = rx_mpc_set_rspi(k_port_a_pin_0); // PA0 = COPI
00144 *     err = rx_mpc_set_rspi(k_port_a_pin_1); // PA1 = CIPO
00145 *     err = rx_mpc_set_rspi(k_port_a_pin_3); // PA3 = CLK
00146 *     err = rx_mpc_set_rspi(k_port_a_pin_4); // PA4 = SSL0
00147 * }
00148 * @endcode
00149 *
00150 * @par Error Handling Example
00151 * @code{.c}
00152 * rx_err_t configure_pin_safe(rx_port_pin_t pin, uint8_t psel)
00153 * {
00154 *     // Validate PSEL range before calling
00155 *     if (psel > 0x1F) {
00156 *         rx_log_error("MPC", "PSEL exceeds 5-bit maximum");
00157 *         return k_rx_err_invalid_arg;
00158 *     }
00159 *
00160 *     rx_mpc_peripheral_config_t config = {
00161 *         .pin = pin,
00162 *         .psel = psel
00163 *     };
00164 *
00165 *     rx_err_t err = rx_mpc_set_peripheral(&config);
00166 *     if (err != k_rx_ok) {
00167 *         rx_log_error("MPC", "Pin configuration failed");
00168 *         // Configuration failed - pin remains in previous state
00169 *         return err;

```

```

00170 *   }
00171 *
00172 *   // Remember: Also need to set PMR bit to enable peripheral mode!
00173 *   return k_rx_ok;
00174 * }
00175 * @endcode
00176 *
00177 * @par NASA Power of 10 Compliance
00178 * - **Rule 1**: [OK] No goto, setjmp, longjmp, or recursion
00179 * - **Rule 2**: [OK] All loops have fixed upper bounds (switch/case only)
00180 * - **Rule 3**: [OK] No dynamic memory allocation
00181 * - **Rule 4**: [OK] All functions < 60 lines
00182 * - **Rule 5**: [OK] Minimum 2 assertions per function (parameter validation)
00183 * - **Rule 6**: [OK] Variables declared at smallest scope
00184 * - **Rule 7**: [OK] All return values checked
00185 * - **Rule 8**: [OK] C23 typed enums for all constants, no magic numbers
00186 * - **Rule 9**: [WARN] Function pointers not used (simple direct calls)
00187 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00188 *
00189 * @par SOLID Principles
00190 * - **Single Responsibility**: MPC driver handles only pin mux configuration
00191 * - **Open/Closed**: New peripherals added via rx_mpc_set_peripheral() without
00192 *   modifying existing code
00193 * - **Interface Segregation**: Convenience functions (set_sci, set_rspi) separate
00194 *   from low-level set_peripheral()
00195 * - **Dependency Inversion**: Higher-level drivers depend on this abstraction,
00196 *   not on raw register writes
00197 *
00198 * @par Module Dependencies
00199 * @dot
00200 * digraph mpc_deps {
00201 *   rankdir=TB;
00202 *   node [shape=box, style=rounded];
00203 *
00204 *   rx_mpc [label="rx_mpc.h\n(this module)"];
00205 *   rx_err [label="rx_err.h\n(error codes)"];
00206 *   rx_port [label="rx_port_constants.h\n(pin definitions)"];
00207 *   rx_mpc_regs [label="rx72n_mpc_regs.h\n(register access)"];
00208 *   rx_check [label="rx_check.h\n(validation macros)"];
00209 *   rx_log [label="rx_log.h\n(error logging)"];
00210 *
00211 *   rx_mpc -> rx_err;
00212 *   rx_mpc -> rx_port;
00213 *   rx_mpc -> rx_mpc_regs [style=dashed, label="impl only"];
00214 *   rx_mpc -> rx_check [style=dashed, label="impl only"];
00215 *   rx_mpc -> rx_log [style=dashed, label="impl only"];
00216 * }
00217 * @enddot
00218 *
00219 * @see rx72n_mpc_regs.h Low-level MPC register definitions
00220 * @see rx_port_constants.h Pin enumeration definitions
00221 * @see docs/sections/03_hardware_pinout.tex Complete STAR pinout documentation
00222 *
00223 * @author Locked, Inc.
00224 * @date 2026-01-29
00225 * @copyright Copyright (c) 2026 Locked Inc.
00226 * SPDX-License-Identifier: MIT
00227 * @since Version 1.0.0
00228 */
00229
00230 #pragma once
00231
00232 #include <stdint.h>
00233
00234 #include "rx_err.h"
00235 #include "rx_port_constants.h"
00236
00237 #ifdef __cplusplus
00238 extern "C" {
00239 #endif
00240
00241 /*
=====
00242 * Configuration Structures
00243 *
=====
*/
00244
00245 /**
00246 * @struct rx_mpc_peripheral_config_t
00247 * @brief MPC peripheral configuration structure for pin function assignment
00248 *
00249 * @details
00250 * This structure groups related parameters to prevent accidental parameter
00251 * swapping when calling rx_mpc_set_peripheral(). Using a struct provides:
00252 * - Type safety (compiler catches wrong parameter types)
00253 * - Named parameters (self-documenting code)

```



```

00255 * - Extensibility (can add fields without changing function signature)
00256 *
00257 * @par Memory Layout
00258 * | Offset | Size | Field | Type | Alignment |
00259 * |-----|-----|-----|-----|-----|
00260 * | 0 | 2 | pin | rx_port_pin_t (uint16_t) | 2 |
00261 * | 2 | 1 | psel | uint8_t | 1 |
00262 * | 3 | 1 | (padding) | - | - |
00263 * | **Total** | **4 bytes** | | | |
00264 *
00265 * @par Field Relationships
00266 * - `pin` determines which PFS register to modify
00267 * - `psel` determines what value to write to that register
00268 * - The (pin, psel) pair must be valid per RX72N hardware manual appendix
00269 *
00270 * @par Usage Example - Basic Configuration
00271 * @code{.c}
00272 * // Configure PE5 for GPTW0A (motor PWM output)
00273 * rx_mpc_peripheral_config_t pwm_config = {
00274 *     .pin = k_port_e_pin_5, // PE5
00275 *     .psel = k_psel_gptw // GPTW function select (0x1E)
00276 * };
00277 * rx_err_t err = rx_mpc_set_peripheral(&pwm_config);
00278 * @endcode
00279 *
00280 * @par Usage Example - Multiple Pins
00281 * @code{.c}
00282 * // Configure all SPI pins using array
00283 * const rx_mpc_peripheral_config_t spi_pins[] = {
00284 *     { .pin = k_port_a_pin_0, .psel = 0x0D }, // COPI
00285 *     { .pin = k_port_a_pin_1, .psel = 0x0D }, // CIPO
00286 *     { .pin = k_port_a_pin_3, .psel = 0x0D }, // CLK
00287 *     { .pin = k_port_a_pin_4, .psel = 0x0D }, // SSL0
00288 * };
00289 *
00290 * for (uint8_t i = 0; i < 4; i++) {
00291 *     rx_err_t err = rx_mpc_set_peripheral(&spi_pins[i]);
00292 *     if (err != k_rx_ok) {
00293 *         // Handle error
00294 *     }
00295 * }
00296 * @endcode
00297 *
00298 * @invariant pin must be a valid rx_port_pin_t value from rx_port_constants.h
00299 * @invariant psel must be in range [0, 31] (5-bit field)
00300 * @invariant (pin, psel) combination must be valid per hardware manual
00301 *
00302 * @see rx_mpc_set_peripheral() Function that uses this structure
00303 * @see rx_port_constants.h Pin enumeration definitions
00304 * @see rx_pin_psel_t Common PSEL values for STAR project
00305 *
00306 * @since Version 1.0.0
00307 */
00308 typedef struct {
00309 /**
00310 * @brief GPIO pin to configure for peripheral function
00311 * @details
00312 * Encoded pin identifier from rx_port_constants.h combining port and
00313 * pin number. Use the k_port_X_pin_Y constants for type safety.
00314 * @par Valid Values
00315 * Any valid rx_port_pin_t constant, e.g.:
00316 * - k_port_a_pin_0 through k_port_a_pin_7
00317 * - k_port_b_pin_0 through k_port_b_pin_7
00318 * - k_port_e_pin_0 through k_port_e_pin_7
00319 * - etc.
00320 * @note On 144-pin package, ports G, H are not available
00321 * @warning Using invalid pin values returns k_rx_err_invalid_arg
00322 */
00323 rx_port_pin_t pin;
00324 /**
00325 * @brief Peripheral function select code (PSEL field value)
00326 * @details
00327 * Value written to the PSEL[4:0] field of the PFS register. This
00328 * determines which peripheral function the pin is connected to.
00329 *
00330 * @par Valid Range: [0, 31] (0x00 - 0x1F)
00331 *
00332 * @par Common Values (varies by pin, check hardware manual):
00333 * | PSEL | Function | Typical Pins |
00334 * |-----|-----|-----|
00335 * | 0x00 | Hi-Z/GPIO | All |
00336 * | 0x01 | MTU I/O compare (MTIOC) | P14-P17, P24-P27 |
00337 * | 0x02 | MTU clock input (MTCLK) | P24, P25, PC0, PC1 |
00338 * | 0x03 | MTU phase counter | P24, P25 |
00339 * | 0x1E | GPTW output (GTIOC) | PE3, PE7, P23, P22, etc. |
00340 * | 0x0A | SCI (TXD/RXD) | PB6, PB7 |

```

```

00342 * | 0x0D | RSPI (CLK/COPI/CIPO/SSL) | PA0-PA4 |
00343 * | 0x0F | RIIC (SCL/SDA) | P12, P13 |
00344 *
00345 * @warning Wrong PSEL value for a pin may cause hardware malfunction
00346 * @see rx_pin_psel_t Predefined PSEL constants
00347 */
00348 uint8_t psel;
00349 } rx_mpc_peripheral_config_t;
00350
00351 /*
=====
00352 * Pin Function Codes
00353 *
=====
00354 */
00355 /**
00356  * @enum rx_pin_function_t
00357  * @brief High-level pin function mode selection
00358  *
00359  * @details
00360  * Specifies whether a pin is configured for GPIO operation or peripheral
00361  * function mode. This is a conceptual abstraction - the actual hardware
00362  * configuration is done through PFS register PSEL field and PORT PMR bit.
00363  *
00364  * @par Pin Configuration State Machine
00365  * @startuml
00366  * [*] --> GPIO : Power-on default
00367  * GPIO --> Peripheral : rx_mpc_set_peripheral()
00368  * Peripheral --> GPIO : rx_mpc_set_gpio()
00369  * GPIO --> GPIO : rx_mpc_set_gpio()
00370  * Peripheral --> Peripheral : rx_mpc_set_peripheral()\n(change function)
00371  * @enduml
00372  *
00373  * @note This enum is for documentation/API clarity. Actual hardware uses
00374  *       PSEL field values directly.
00375  *
00376  * @see rx_mpc_set_gpio() Configure pin for GPIO mode
00377  * @see rx_mpc_set_peripheral() Configure pin for peripheral mode
00378  *
00379  * @since Version 1.0.0
00380  */
00381 typedef enum : uint8_t {
00382 /**
00383  * @brief GPIO mode - pin controlled by PORT registers
00384  * @details
00385  * Pin is disconnected from all peripheral functions. Direction and
00386  * data controlled by PORT PDR and PODR registers. This is the
00387  * power-on default for all pins.
00388  * @par PFS Register Value: 0x00 (PSEL=0, ISEL=0, ASEL=0)
00389  */
00390 k_pin_function_gpio = 0x00,
00391
00392 /**
00393  * @brief Peripheral function mode - pin connected to peripheral
00394  * @details
00395  * Pin is connected to a peripheral function as determined by the
00396  * PSEL field. The PORT PMR bit must also be set to 1 to actually
00397  * enable peripheral mode.
00398  * @par PFS Register Value: PSEL field set to peripheral code
00399  * @note This is a placeholder value; actual PSEL varies by function
00400  */
00401 k_pin_function_periph = 0x01,
00402 } rx_pin_function_t;
00403
00404 /**
00405  * @enum rx_pin_psel_t
00406  * @brief Peripheral Select (PSEL) codes for common STAR project functions
00407  *
00408  * @details
00409  * These values are written to the PSEL[4:0] field of PFS registers to select
00410  * which peripheral function a pin is connected to. Values are pin-specific -
00411  * these constants represent the most common values used in the STAR project.
00412  *
00413  * @par PSEL Field in PFS Register
00414  * @verbatim
00415  * Bit:  7      6      5      4  3  2  1  0
00416  * +-----+-----+-----+-----+
00417  * | ASEL | ISEL | Rsvd | PSEL[4:0] |
00418  * +-----+-----+-----+-----+
00419  *                                     <-- These values -->
00420  * @endverbatim
00421  *
00422  * @par PSEL Values by Peripheral (STAR Project)
00423  * | Peripheral | PSEL | Pins Used | Function |
00424  * |-----|-----|-----|-----|
00425  * | MTU3a MTIOC | 0x01 | P14-P17, P24-P27 | PWM output |

```

```

00427 * | MTU3a MTCLK | 0x02 | P24, P25, PC0, PC1 | Clock/encoder input |
00428 * | MTU3a Phase | 0x03 | P24, P25 | Phase counting mode |
00429 * | GPTW GTIOC | 0x1E | PE3, PE7, P23, P22, etc. | PWM output |
00430 * | SCI TXD/RXD | 0x0A | PB6, PB7 | UART data |
00431 * | RSPI signals | 0x0D | PA0-PA4 | SPI bus |
00432 * | RIIC SDA/SCL | 0x0F | P12, P13 | I2C bus |
00433 *
00434 * @warning PSEL values are pin-specific! A value that works for one pin
00435 *         may select a completely different function on another pin.
00436 *         Always verify against the RX72N Hardware Manual pin function tables.
00437 *
00438 * @par Usage Example
00439 * @code{.c}
00440 * // Configure PB7 for SCI9 TXD
00441 * rx_mpc_peripheral_config_t config = {
00442 *     .pin = k_port_b_pin_7,
00443 *     .psel = k_psel_sci_tx // 0x0A
00444 * };
00445 * rx_mpc_set_peripheral(&config);
00446 * @endcode
00447 *
00448 * @see rx_mpc_set_peripheral() Use these values with this function
00449 * @see RX72N Hardware Manual, Appendix for complete pin function tables
00450 *
00451 * @since Version 1.0.0
00452 */
00453 typedef enum : uint8_t {
00454 /**
00455  * @brief MTU I/O Compare/PWM output (MTIOC)
00456  * @details
00457  * Connects pin to MTU3a compare match output for PWM generation.
00458  * @par Typical Pins: P14-P17 (MTU3), P24-P27 (MTU4)
00459  * @par Application: Motor PWM outputs when using MTU3a timer
00460  */
00461 k_psel_mtu_ioc = 0x01,
00462
00463 /**
00464  * @brief MTU clock input (MTCLKA/B/C/D)
00465  * @details
00466  * Connects pin to MTU3a external clock input. On RX72N, PSEL = 0x02
00467  * (0b000010) is the generic "MTCLK" function and is the same value whether
00468  * the MTU is configured for normal external-clock counting or for phase
00469  * counting (quadrature encoder) mode. The counting mode is selected by
00470  * MTU.TMDR, not by PSEL -- callers must not pick a different PSEL just
00471  * because they want phase counting.
00472  *
00473  * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00474  * "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package:
00475  * - Table 23.5 (P1x pins): row PSEL = 000010b -> MTCLKA on P14,
00476  *   MTCLKB on P15
00477  * - Table 23.6 (P2x pins): row PSEL = 000010b -> MTCLKC on P22,
00478  *   MTCLKD on P23, MTCLKA on P24, MTCLKB on P25
00479  * - Table 23.17 (PAx pins): row PSEL = 000010b -> MTCLKC on PA1,
00480  *   MTCLKD on PA3, MTCLKA on PA4, MTCLKB on PA6
00481  * - Table 23.19 (PCx pins): row PSEL = 000010b -> MTCLKC on PC4,
00482  *   MTCLKD on PC5, MTCLKA on PC6, MTCLKB on PC7
00483  * Every MTCLK pin on this package resolves to PSEL = 0b00'0010 = 0x02.
00484  *
00485  * @par Typical Pins: P24/P25 (MTCLKA/B), PA1/PC5 (MTCLKC/D)
00486  * @par Application: Encoder pulse counting, quadrature phase counting
00487  */
00488 k_psel_mtu_clk = 0x02,
00489
00490 /**
00491  * @brief MTU encoder phase counting input (alias for k_psel_mtu_clk)
00492  * @details
00493  * Same PSEL value on RX72N as k_psel_mtu_clk -- both MTU external clock
00494  * counting and MTU phase counting route through PSEL=0x02, and the
00495  * distinction is selected later via MTU.TMDR (phase counting mode bits).
00496  * The split constants exist only so calling code reads clearly; do not
00497  * assume they are different values.
00498  *
00499  * @par Typical Pins: P24/P25 pair, PA1/PC5 pair (same as k_psel_mtu_clk)
00500  * @par Application: Quadrature encoder interface with direction detection
00501  *
00502  * @note Previously 0x03 -- that was wrong and caused MTU encoders to stay
00503  *       at TCNT=0x0000 because PSEL=0x03 on MTU-candidate pins selects an
00504  *       unrelated function (TPU on Port C pins, or nothing elsewhere).
00505  *       See chapter 23 of the RX72N Group Hardware Manual
00506  *       (R01UH0824EJ0111), Tables 23.5, 23.6, 23.17, 23.19.
00507  */
00508 k_psel_mtu_phase = 0x02,
00509
00510 /**
00511  * @brief TPU external clock input on Port C pins (TCLKA/B/C/D on PC0-PC3)
00512  * @details
00513  * RX72N assigns TPU external clock to alternate function PSEL=0x03 on

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```

00514 * Port C pins only. Non-Port-C candidates for the same TCLK signal use a
00515 * different PSEL (see k_psel_tpu_clk_alt).
00516 *
00517 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00518 * "Multi-Function Pin Controller (MPC)", Table 23.19 (PCx pins). For the
00519 * 144-pin LFQFP package the row PSEL = 000011b maps to:
00520 * PC0 -> TCLKC, PC1 -> TCLKD, PC2 -> TCLKA, PC3 -> TCLKB
00521 * All four TPU TCLK inputs on Port C resolve to PSEL = 0b00'0011 = 0x03.
00522 *
00523 * @par Typical Pins: PC2 (TCLKA), PC3 (TCLKB), PC0 (TCLKC), PC1 (TCLKD)
00524 * @par Application: Rear-wheel quadrature encoder Phase A inputs on Port C
00525 *
00526 * @note rx_mpc_set_tpu_encoder() picks between this and k_psel_tpu_clk_alt
00527 * automatically based on the port number of the pin argument.
00528 */
00529 k_psel_tpu_clk_portc = 0x03,
00530
00531 /**
00532 * @brief TPU external clock input on non-Port-C pins (TCLKA/B/C/D)
00533 * @details
00534 * RX72N assigns TPU external clock to alternate function PSEL=0x04 on
00535 * Port 1, Port A, and Port B candidates for the TCLK signals. This differs
00536 * from the Port C candidates (k_psel_tpu_clk_portc = 0x03) and the
00537 * difference is silicon-level -- the same TCLK signal is routed through
00538 * different PFS alternate-function slots on different ports.
00539 *
00540 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00541 * "Multi-Function Pin Controller (MPC)". For the 144-pin LFQFP package
00542 * the row PSEL = 000100b maps to:
00543 * - Table 23.5 (P1x pins): P14 -> TCLKA, P15 -> TCLKB,
00544 * P16 -> TCLKC, P17 -> TCLKD
00545 * - Table 23.17 (PAx pins): PA3 -> TCLKB
00546 * - Table 23.18 (PBx pins): PB2 -> TCLKC, PB3 -> TCLKD
00547 * All non-Port-C TPU TCLK inputs resolve to PSEL = 0b00'0100 = 0x04.
00548 *
00549 * @par Typical Pins: PA3 (TCLKB), PB3 (TCLKD) -- STAR rear-wheel Phase B
00550 * @par Application: Rear-wheel quadrature encoder Phase B inputs
00551 *
00552 * @note Before this split, rx_mpc_set_tpu_encoder() used PSEL=0x03 for
00553 * every TPU pin, which silently misconfigured PA3/PB3 (the two
00554 * non-Port-C pins on rear wheels) so one half of each rear-wheel
00555 * encoder never incremented.
00556 */
00557 k_psel_tpu_clk_alt = 0x04,
00558
00559 /**
00560 * @brief SCI transmit data output (TXD)
00561 * @details
00562 * Connects pin to SCI channel transmit data output.
00563 * @par Typical Pins: PB7 (SCI9-TXD for debug UART)
00564 * @par Application: UART transmit, debug console output
00565 * @note Same PSEL value used for both TX and RX on SCI pins
00566 */
00567 k_psel_sci_tx = 0x0A,
00568
00569 /**
00570 * @brief SCI receive data input (RXD)
00571 * @details
00572 * Connects pin to SCI channel receive data input.
00573 * @par Typical Pins: PB6 (SCI9-RXD for debug UART)
00574 * @par Application: UART receive, debug console input
00575 * @note Same PSEL value as TX (pin determines actual function)
00576 */
00577 k_psel_sci_rx = 0x0A,
00578
00579 /**
00580 * @brief RIIC clock line (SCL)
00581 * @details
00582 * Connects pin to RIIC (I2C) clock line with open-drain output.
00583 * @par Typical Pins: P12 (RIIC0-SCL)
00584 * @par Application: I2C bus clock for sensors/peripherals
00585 * @note RIIC automatically configures open-drain when enabled
00586 */
00587 k_psel_riic_scl = 0x0F,
00588
00589 /**
00590 * @brief RIIC data line (SDA)
00591 * @details
00592 * Connects pin to RIIC (I2C) data line with open-drain output.
00593 * @par Typical Pins: P13 (RIIC0-SDA)
00594 * @par Application: I2C bus data for sensors/peripherals
00595 * @note Same PSEL value as SCL (pin determines actual function)
00596 */
00597 k_psel_riic_sda = 0x0F,
00598
00599 /**
00600 * @brief RSPi clock (RSPCK)

```

```

00601  * @details
00602  * Connects pin to RSPI serial clock output (controller mode) or
00603  * input (peripheral mode).
00604  * @par Typical Pins: PA5 (RSPI0-RSPCKA); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17 (MPC PAn
pin function select)
00605  * @par Application: SPI bus clock for RPi5 communication
00606  */
00607  k_psel_rspi_clk = 0x0D,
00608
00609  /**
00610  * @brief RSPI Controller Out Peripheral In (COPI, formerly MOSI)
00611  * @details
00612  * Connects pin to RSPI data output line in controller mode.
00613  * @par Typical Pins: PA6 (RSPI0-MOSIA / COPI); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17
(MPC PAn pin function select)
00614  * @par Application: SPI transmit data to peripherals
00615  * @note Uses OSHWA-approved inclusive terminology
00616  */
00617  k_psel_rspi_copi = 0x0D,
00618
00619  /**
00620  * @brief RSPI Controller In Peripheral Out (CIPO, formerly MISO)
00621  * @details
00622  * Connects pin to RSPI data input line in controller mode.
00623  * @par Typical Pins: PA7 (RSPI0-MISOA / CIPO); see RX72N HW manual R01UH0824EJ0111 Ch 23 Table 23.17
(MPC PAn pin function select)
00624  * @par Application: SPI receive data from peripherals
00625  * @note Uses OSHWA-approved inclusive terminology
00626  */
00627  k_psel_rspi_cipo = 0x0D,
00628
00629  /**
00630  * @brief ADC analog input mode
00631  * @details
00632  * Disables digital input buffer for analog signal measurement.
00633  * Actually sets ASEL bit in PFS, not PSEL field.
00634  * @par Typical Pins: P40-P47 (AN000-AN007)
00635  * @par Application: Motor current sensing, analog input measurement
00636  * @note Use rx_mpc_set_adc() for proper configuration (sets ASEL bit)
00637  */
00638  k_psel_adc = 0x00,
00639
00640  /**
00641  * @brief USB VBUS detection input
00642  *
00643  * @details
00644  * Connects pin to USB VBUS detection for 5V presence sensing.
00645  * Required for USB enumeration and power monitoring.
00646  *
00647  * @par Typical Pins: P1.6 (USB0_VBUS on RX72N)
00648  * @par Application: USB CDC debug interface, RPi5 communication
00649  *
00650  * @note Active-high: pin reads 1 when 5V present on USB bus
00651  * @note Requires additional USB PHY and controller configuration
00652  *
00653  * @since Version 1.0.0
00654  */
00655  k_psel_usb_vbus = 0x11,
00656
00657  /**
00658  * @brief GPTW complementary PWM output (GTIOC)
00659  *
00660  * @details
00661  * Connects pin to GPTW (General PWM Timer) complementary PWM output
00662  * with dead-time insertion for motor control. GPTW provides 4 channels
00663  * (GPTW0-GPTW3) with phase-staggered PWM outputs.
00664  *
00665  * @par Typical Pins:
00666  * - P2.3, P1.7 (GPTW0 - Motor 0 IN2/IN1)
00667  * - P2.2, PC.3 (GPTW1 - Motor 1 IN2/IN1)
00668  * - PE.3, P8.6 (GPTW2 - Motor 2 IN2/IN1)
00669  * - PE.7, PC.6 (GPTW3 - Motor 3 IN2/IN1)
00670  *
00671  * @par Application: Motor control with direction/PWM signals for DRV8263H
00672  * @par Feature: 90-degree phase staggering reduces peak current draw
00673  *
00674  * @since Version 1.0.0
00675  */
00676  k_psel_gptw = 0x1E, /* 0b0111110: per RX72N HW manual Tables 23.4, 23.6,
00677  * 23.10, 23.14, 23.16 -- the GTIOC* function on every
00678  * port (1, 2, 8, A, C, D, E) uses PSEL=0x1E. The old
00679  * value 0x14 selected an unrelated function. */
00680
00681 } rx_pin_psel_t;
00682
00683 /*
=====

```

```

00684 * Public API
00685 *
=====
00686 */
00687 /**
00688  * @brief Configure pin for GPIO (General Purpose I/O) function
00689  *
00690  * @details
00691  * Sets the pin to GPIO mode by writing 0x00 to its PFS register (PSEL=0,
00692  * ISEL=0, ASEL=0). This disconnects the pin from all peripheral functions
00693  * and allows it to be controlled by PORT PDR/PODR registers.
00694  *
00695  * @par Algorithm Steps
00696  * 1. Extract port number and pin number from encoded pin parameter
00697  * 2. Validate port is in valid range [0, J]
00698  * 3. Validate pin number is in valid range [0, 7]
00699  * 4. Calculate PFS register address for this pin
00700  * 5. Unlock PWPR write protection (B0WI=0, PFSWE=1)
00701  * 6. Write 0x00 to PFS register (GPIO mode)
00702  * 7. Lock PWPR write protection (PFSWE=0, B0WI=1)
00703  * 8. Return success
00704  *
00705  * @param[in] pin GPIO pin identifier (rx_port_pin_t from rx_port_constants.h)
00706  * - Encodes both port (A-J) and pin number (0-7)
00707  * - Use k_port_X_pin_Y constants for type safety
00708  * - Example: k_port_e_pin_5 for PE5
00709  *
00710  * @return Error code indicating success or failure
00711  * @retval k_rx_ok Pin successfully configured for GPIO mode
00712  * @retval k_rx_err_invalid_arg Port number exceeds valid range (> Port J); Pin number exceeds 7; Port not available on
00713  * package (G/H on 144-pin)
00714  *
00715  * @pre PCLKB clock must be running (MPC register access requires it)
00716  * @pre MPC module stop bit must be cleared (MSTPCRA.MSTPA9 = 0)
00717  *
00718  * @post PFS register for specified pin contains 0x00
00719  * @post Pin is disconnected from all peripheral functions
00720  * @post PWPR is locked (B0WI=1) after operation
00721  *
00722  * @invariant MPC base address 0x0008C100 remains constant
00723  * @invariant Other pins' PFS registers are not modified
00724  *
00725  * @note Not thread-safe. Use mutex protection if calling from multiple threads.
00726  * @note This only sets PFS register. To complete GPIO setup, also configure
00727  * PORT PMR (peripheral mode = 0) and PDR (direction) registers.
00728  *
00729  * @warning Calling on a pin that doesn't exist on the package returns error
00730  * but does not cause hardware fault.
00731  *
00732  * @par Performance
00733  * - Execution time: ~1 us @ 240 MHz (PWPR unlock/lock overhead)
00734  * - No loops or blocking operations
00735  *
00736  * @par Thread Safety
00737  * Not thread-safe. The PWPR unlock/lock sequence is non-atomic.
00738  *
00739  * @par Example - Single Pin
00740  * @code{.c}
00741  * #include "rx_mpc.h"
00742  * #include "rx_port_constants.h"
00743  *
00744  * // Configure PE5 for GPIO (was previously PWM output)
00745  * rx_err_t err = rx_mpc_set_gpio(k_port_e_pin_5);
00746  * if (err != k_rx_ok) {
00747  *     rx_log_error("INIT", "Failed to set PE5 to GPIO mode");
00748  *     return err;
00749  * }
00750  *
00751  * // Now configure PORT registers for GPIO operation
00752  * // (set PMR=0, configure PDR for direction, etc.)
00753  * @endcode
00754  *
00755  * @par Example - Error Handling
00756  * @code{.c}
00757  * // Attempt to configure invalid pin
00758  * rx_err_t err = rx_mpc_set_gpio(k_port_g_pin_0); // Port G not on 144-pin
00759  * if (err == k_rx_err_invalid_arg) {
00760  *     rx_log_warn("MPC", "Pin not available on this package");
00761  * }
00762  * @endcode
00763  *
00764  * @see rx_mpc_set_peripheral() Configure pin for peripheral function
00765  * @see rx_port_constants.h Pin enumeration definitions
00766  *
00767  * @since Version 1.0.0
00768  *

```



```

00769 * @par NASA Power of 10 Compliance
00770 * - Rule 5: [OK] 2 preconditions (clock, module stop), 3 postconditions
00771 * - Rule 7: [OK] All internal function returns checked
00772 *
00773 * @callgraph
00774 * @callergraph
00775 */
00776 [[nodiscard]] rx_err_t rx_mpc_set_gpio(rx_port_pin_t pin);
00777
00778 /**
00779 * @brief Configure pin for peripheral function with specified PSEL code
00780 *
00781 * @details
00782 * Core function for pin multiplexer configuration. Writes the specified
00783 * PSEL value to the pin's PFS register, connecting the pin to a peripheral
00784 * function. Uses a configuration structure to prevent parameter swapping.
00785 *
00786 * @par Algorithm Steps
00787 * 1. Validate config pointer is not nullptr
00788 * 2. Validate PSEL value is in range [0, 31]
00789 * 3. Extract port number and pin number from config->pin
00790 * 4. Validate extracted port and pin are in valid ranges
00791 * 5. Calculate PFS register address for this pin
00792 * 6. Unlock PWPR write protection (B0WI=0, PFSWE=1)
00793 * 7. Write config->psel to PFS register
00794 * 8. Lock PWPR write protection (PFSWE=0, B0WI=1)
00795 * 9. Return success
00796 *
00797 * @par Pin Configuration Flow
00798 * @dot
00799 * digraph config_flow {
00800 *   rankdir=TB;
00801 *   node [shape=box, style=rounded];
00802 *
00803 *   start [label="rx_mpc_set_peripheral()"];
00804 *   validate_ptr [label="Validate config != nullptr"];
00805 *   validate_psel [label="Validate PSEL <= 31"];
00806 *   extract [label="Extract port, pin"];
00807 *   validate_pin [label="Validate port, pin range"];
00808 *   get_pfs [label="Get PFS register address"];
00809 *   unlock [label="Unlock PWPR"];
00810 *   write [label="Write PSEL to PFS"];
00811 *   lock [label="Lock PWPR"];
00812 *   success [label="Return k_rx_ok", fillcolor=lightgreen, style=filled];
00813 *   error [label="Return k_rx_err_invalid_arg", fillcolor=lightcoral, style=filled];
00814 *
00815 *   start -> validate_ptr;
00816 *   validate_ptr -> validate_psel [label="OK"];
00817 *   validate_ptr -> error [label="NULL"];
00818 *   validate_psel -> extract [label="OK"];
00819 *   validate_psel -> error [label="> 31"];
00820 *   extract -> validate_pin;
00821 *   validate_pin -> get_pfs [label="OK"];
00822 *   validate_pin -> error [label="invalid"];
00823 *   get_pfs -> unlock;
00824 *   unlock -> write;
00825 *   write -> lock;
00826 *   lock -> success;
00827 * }
00828 * @enddot
00829 *
00830 * @param[in] config Pointer to MPC peripheral configuration structure
00831 * - config->pin: Target pin (rx_port_pin_t constant)
00832 * - config->psel: PSEL value to write (0-31)
00833 * - Must not be nullptr
00834 * - Must remain valid only during function call
00835 *
00836 * @return Error code indicating success or failure
00837 * @retval k_rx_ok Pin successfully configured for peripheral function
00838 * @retval k_rx_err_invalid_arg config pointer is nullptr; config->psel exceeds 31 (5-bit field max); Port number exceeds
00839 * valid range; Pin number exceeds 7; Port not available on package
00840 *
00841 * @pre config pointer must be non-NULL
00842 * @pre config->psel must be in range [0, 31]
00843 * @pre config->pin must be a valid rx_port_pin_t value
00844 * @pre PCLKB clock must be running
00845 *
00846 * @post PFS register for specified pin contains config->psel value
00847 * @post Pin is connected to peripheral function specified by PSEL
00848 * @post PWPR is locked (B0WI=1) after operation
00849 *
00850 * @invariant config structure is not modified (const pointer)
00851 * @invariant Other pins' PFS registers are not modified
00852 *
00853 * @note Not thread-safe. Use mutex protection if calling from multiple threads.
00854 * @note This only sets PFS register. For peripheral operation, also set
00855 * PORT PMR bit to 1 to enable peripheral mode.

```

```

00855 * @note PSEL values are pin-specific. Verify correct value in hardware manual.
00856 *
00857 * @warning Wrong PSEL value may connect pin to unexpected peripheral!
00858 * @warning Not all PSEL values are valid for all pins.
00859 *
00860 * @par Performance
00861 * - Execution time: ~1 us @ 240 MHz
00862 * - No loops or blocking operations
00863 *
00864 * @par Thread Safety
00865 * Not thread-safe. The PWPR unlock/lock sequence is non-atomic.
00866 *
00867 * @par Example - Motor PWM Configuration
00868 * @code{.c}
00869 * #include "rx_mpc.h"
00870 * #include "rx_port_constants.h"
00871 *
00872 * // Configure PE5 for GPTW0A (Motor 0 PWM Phase+)
00873 * rx_mpc_peripheral_config_t pwm_config = {
00874 *     .pin = k_port_e_pin_5, // PE5
00875 *     .psel = k_psel_gptw // GPTW function (0x1E)
00876 * };
00877 *
00878 * rx_err_t err = rx_mpc_set_peripheral(&pwm_config);
00879 * if (err != k_rx_ok) {
00880 *     rx_log_error("MOTOR", "PWM pin configuration failed");
00881 *     return err;
00882 * }
00883 *
00884 * // Don't forget: Set PORT PMR bit to enable peripheral mode!
00885 * @endcode
00886 *
00887 * @par Example - Batch Configuration
00888 * @code{.c}
00889 * // Configure multiple pins in a loop
00890 * const rx_mpc_peripheral_config_t pins[] = {
00891 *     { k_port_e_pin_5, k_psel_gptw }, // GPTW0A (0x1E)
00892 *     { k_port_e_pin_4, k_psel_gptw }, // GPTW0B (0x1E)
00893 *     { k_port_2_pin_4, 0x02 }, // MTCLKA
00894 *     { k_port_2_pin_5, 0x02 }, // MTCLKB
00895 * };
00896 *
00897 * for (size_t i = 0; i < sizeof(pins)/sizeof(pins[0]); i++) {
00898 *     rx_err_t err = rx_mpc_set_peripheral(&pins[i]);
00899 *     if (err != k_rx_ok) {
00900 *         rx_log_error("MPC", "Pin config failed");
00901 *         return err;
00902 *     }
00903 * }
00904 * @endcode
00905 *
00906 * @see rx_mpc_set_gpio() Configure pin for GPIO mode
00907 * @see rx_mpc_peripheral_config_t Configuration structure definition
00908 * @see rx_pin_psel_t Common PSEL value constants
00909 *
00910 * @since Version 1.0.0
00911 *
00912 * @par NASA Power of 10 Compliance
00913 * - Rule 5: [OK] 4 preconditions, 3 postconditions
00914 * - Rule 7: [OK] All return values checked
00915 *
00916 * @callgraph
00917 * @callergraph
00918 */
00919 [[nodiscard]] rx_err_t rx_mpc_set_peripheral(const rx_mpc_peripheral_config_t* config);
00920
00921 /**
00922 * @brief Configure pin for MTU3a timer PWM output (MTIOC)
00923 *
00924 * @details
00925 * Convenience function that configures a pin for MTU3a I/O compare output,
00926 * used for PWM generation. Internally calls rx_mpc_set_peripheral() with
00927 * PSEL = k_psel_mtu_ioc (0x01).
00928 *
00929 * @par Common MTU PWM Pins
00930 * | Pin | MTU Channel | Output | Typical Use |
00931 * |-----|-----|-----|-----|
00932 * | P14 | MTU3 | MTIOC3A | Motor 0 Phase A |
00933 * | P15 | MTU3 | MTIOC3B | Motor 0 Phase B |
00934 * | P16 | MTU3 | MTIOC3C | Motor 0 Phase C |
00935 * | P17 | MTU3 | MTIOC3D | Motor 0 Phase D |
00936 * | P24 | MTU4 | MTIOC4A | Motor 1 Phase A |
00937 * | P25 | MTU4 | MTIOC4B | Motor 1 Phase B |
00938 * | P26 | MTU4 | MTIOC4C | Motor 1 Phase C |
00939 * | P27 | MTU4 | MTIOC4D | Motor 1 Phase D |
00940 *
00941 * @param[in] pin GPIO pin identifier for MTU PWM output

```



```

00942 *          - Must be a pin that supports MTIOC function
00943 *          - Use k_port_X_pin_Y constants
00944 *
00945 * @return Error code indicating success or failure
00946 * @retval k_rx_ok Pin successfully configured for MTU PWM
00947 * @retval k_rx_err_invalid_arg Invalid port or pin number
00948 *
00949 * @pre Pin must support MTU I/O compare function (check hardware manual)
00950 * @pre PCLKB clock must be running
00951 *
00952 * @post PFS register contains PSEL = 0x01 (MTU IOC)
00953 * @post Pin ready for MTU3a compare match output
00954 *
00955 * @note Also requires MTU3a timer configuration and PORT PMR=1
00956 * @note STAR project uses GPTW for motor PWM; see rx_mpc_set_gptw() which uses PSEL = k_psel_gptw (0x1E)
00957 *
00958 * @par Example
00959 * @code{.c}
00960 * // Configure P14-P17 for MTU3 PWM outputs
00961 * rx_mpc_set_mtu_pwm(k_port_1_pin_4); // MTIOC3A
00962 * rx_mpc_set_mtu_pwm(k_port_1_pin_5); // MTIOC3B
00963 * rx_mpc_set_mtu_pwm(k_port_1_pin_6); // MTIOC3C
00964 * rx_mpc_set_mtu_pwm(k_port_1_pin_7); // MTIOC3D
00965 * @endcode
00966 *
00967 * @see rx_mpc_set_peripheral() Underlying implementation
00968 * @see rx_mtu.h MTU timer driver
00969 *
00970 * @since Version 1.0.0
00971 * @callgraph
00972 */
00973 [[nodiscard]] rx_err_t rx_mpc_set_mtu_pwm(rx_port_pin_t pin);
00974
00975 /**
00976 * @brief Configure pin for MTU3a encoder/phase counter input (MTCLK)
00977 *
00978 * @details
00979 * Configures a pin for MTU3a phase counting mode input, used with
00980 * quadrature encoders. Sets PSEL = k_psel_mtu_phase (0x03) for
00981 * phase counting with direction detection.
00982 *
00983 * @par Quadrature Encoder Interface
00984 * @verbatim
00985 *      Encoder Output      MTU3a Phase Counting
00986 *      +-----+          +-----+
00987 *      | Phase A |----->| MTCLKA (Count Up) |
00988 *      | Phase B |----->| MTCLKB (Count Down) |
00989 *      | (Index Z) |----->| (Optional reset) |
00990 *      +-----+          +-----+
00991 *                               Counter increments/decrements
00992 *                               based on phase relationship
00993 * @endverbatim
00994 *
00995 * @par Common Encoder Pins (STAR Project)
00996 * | Pin Pair | MTU Input | Encoder |
00997 * |-----|-----|
00998 * | P24/P25 | MTCLKA/MTCLKB | Motor 0 |
00999 * | PC0/PC1 | MTCLKC/MTCLKD | Motor 1 |
01000 *
01001 * @param[in] pin GPIO pin identifier for encoder input
01002 *          - Must be a pin that supports MTCLK function
01003 *          - Configure in pairs (A+B) for quadrature
01004 *
01005 * @return Error code indicating success or failure
01006 * @retval k_rx_ok Pin successfully configured for encoder input
01007 * @retval k_rx_err_invalid_arg Invalid port or pin number
01008 *
01009 * @pre Pin must support MTU clock/phase function
01010 * @pre PCLKB clock must be running
01011 *
01012 * @post PFS register contains PSEL = 0x03 (MTU phase)
01013 * @post Pin ready for quadrature encoder input
01014 *
01015 * @note Configure both Phase A and Phase B pins for proper operation
01016 * @note Also requires MTU3a timer configuration in phase counting mode
01017 *
01018 * @par Example - Quadrature Encoder Setup
01019 * @code{.c}
01020 * // Configure encoder pins for Motor 0
01021 * rx_err_t err;
01022 *
01023 * err = rx_mpc_set_mtu_encoder(k_port_2_pin_4); // P24 = Phase A
01024 * if (err != k_rx_ok) return err;
01025 *
01026 * err = rx_mpc_set_mtu_encoder(k_port_2_pin_5); // P25 = Phase B
01027 * if (err != k_rx_ok) return err;
01028 *

```

```

01029 * // Now configure MTU1 for phase counting mode
01030 * // (see rx_mtu_encoder.h)
01031 * @endcode
01032 *
01033 * @see rx_mpc_set_peripheral() Underlying implementation
01034 * @see rx_mtu_encoder.h Encoder driver using MTU phase counting
01035 *
01036 * @since Version 1.0.0
01037 * @callgraph
01038 */
01039 [[nodiscard]] rx_err_t rx_mpc_set_mtu_encoder(rx_port_pin_t pin);
01040
01041 /**
01042 * @brief Configure pin for TPU encoder/phase counter input (TCLK)
01043 *
01044 * @details
01045 * Configures a pin for TPU phase counting mode input, used with
01046 * quadrature encoders on the rear wheels. Sets PSEL = k_psel_mtu_phase
01047 * (0x03), which is the generic "timer phase counting input" function
01048 * on the RX72N. The routing to TPU (vs MTU) depends on which timer
01049 * module has phase counting mode enabled.
01050 *
01051 * This function exists as a separate API from rx_mpc_set_mtu_encoder()
01052 * for documentation clarity, even though both write the same PSEL value.
01053 *
01054 * @par TPU Encoder Pins (STAR Project - Rear Wheels)
01055 * | Pin Pair | TPU Input | Encoder |
01056 * |-----|-----|-----|
01057 * | PC2/PA3 | TCLKA/TCLKB | Motor 2 (Rear Left) |
01058 * | PC0/PB3 | TCLKC/TCLKD | Motor 3 (Rear Right) |
01059 *
01060 * @param[in] pin GPIO pin identifier for TPU encoder input
01061 * - Must be a pin that supports TCLK function
01062 * - Configure in pairs (A+B) for quadrature
01063 *
01064 * @return Error code indicating success or failure
01065 * @retval k_rx_ok Pin successfully configured for TPU encoder input
01066 * @retval k_rx_err_invalid_arg Invalid port or pin number
01067 *
01068 * @pre Pin must support TPU clock/phase function
01069 * @pre PCLKB clock must be running
01070 *
01071 * @post PFS register contains PSEL = 0x03 (timer phase counting)
01072 * @post Pin ready for quadrature encoder input via TPU
01073 *
01074 * @note Configure both Phase A and Phase B pins for proper operation
01075 * @note Also requires TPU timer configuration in phase counting mode
01076 *
01077 * @par Example - Rear Wheel Encoder Setup
01078 * @code{.c}
01079 * rx_err_t err;
01080 *
01081 * // Motor 2 (Rear Left) - TPU1
01082 * err = rx_mpc_set_tpu_encoder(k_rx_pc_2); // PC2 = TCLKA (Phase A)
01083 * if (err != k_rx_ok) return err;
01084 * err = rx_mpc_set_tpu_encoder(k_rx_pa_3); // PA3 = TCLKB (Phase B)
01085 * if (err != k_rx_ok) return err;
01086 *
01087 * // Motor 3 (Rear Right) - TPU2
01088 * err = rx_mpc_set_tpu_encoder(k_rx_pc_0); // PC0 = TCLKC (Phase A)
01089 * if (err != k_rx_ok) return err;
01090 * err = rx_mpc_set_tpu_encoder(k_rx_pb_3); // PB3 = TCLKD (Phase B)
01091 * if (err != k_rx_ok) return err;
01092 * @endcode
01093 *
01094 * @see rx_mpc_set_mtu_encoder() Front wheel encoder pin configuration
01095 * @see rx_mpc_set_peripheral() Underlying implementation
01096 * @see rx_encoder_tpu.h TPU encoder driver using phase counting
01097 *
01098 * @since Version 1.0.0
01099 * @callgraph
01100 */
01101 [[nodiscard]] rx_err_t rx_mpc_set_tpu_encoder(rx_port_pin_t pin);
01102
01103 /**
01104 * @brief Configure pin for ADC analog input
01105 *
01106 * @details
01107 * Configures a pin for analog-to-digital converter input by setting
01108 * the ASEL bit in the PFS register. This disables the digital input
01109 * buffer to prevent noise and current leakage during analog measurement.
01110 *
01111 * @par ADC Pin Configuration
01112 * @verbatim
01113 * PFS Register for ADC:
01114 * +-----+-----+-----+-----+
01115 * | ASEL | ISEL | Rsvd | PSEL[4:0] |

```

```

01116 * | 1 | 0 | 0 | 0x00 |
01117 * +-----+-----+-----+
01118 * ^
01119 * +-- Set to 1 for analog input
01120 * @endverbatim
01121 *
01122 * @par Common ADC Pins (STAR Project)
01123 * | Pin | ADC Channel | Typical Use |
01124 * |-----|-----|
01125 * | P40 | AN000 | Motor 0 current sense |
01126 * | P41 | AN001 | Motor 1 current sense |
01127 * | P42 | AN002 | Motor 2 current sense |
01128 * | P43 | AN003 | Motor 3 current sense |
01129 * | P44 | AN004 | Analog input (unused) |
01130 * | P45 | AN005 | Temperature sensor |
01131 *
01132 * @param[in] pin GPIO pin identifier for ADC input
01133 * - Must be a pin connected to ADC channel
01134 * - Typically P40-P47 for AN000-AN007
01135 *
01136 * @return Error code indicating success or failure
01137 * @retval k_rx_ok Pin successfully configured for ADC input
01138 * @retval k_rx_err_invalid_arg Invalid port or pin number
01139 *
01140 * @pre Pin must be connected to ADC channel (not all pins support ADC)
01141 * @pre PCLKB clock must be running
01142 *
01143 * @post PFS register has ASEL = 1 (digital buffer disabled)
01144 * @post Pin ready for analog signal input
01145 *
01146 * @note Digital input buffer disabled - do not use as GPIO input
01147 * @note Also requires S12ADC configuration for actual conversion
01148 *
01149 * @warning Using a non-ADC pin with this function sets ASEL but has no effect
01150 * on analog capability (pin hardware determines ADC availability)
01151 *
01152 * @par Example - Current Sensing Setup
01153 * @code{.c}
01154 * // Configure ADC pins for motor current sensing
01155 * rx_mpc_set_adc(k_port_4_pin_0); // AN000 = Motor 0 current
01156 * rx_mpc_set_adc(k_port_4_pin_1); // AN001 = Motor 1 current
01157 * rx_mpc_set_adc(k_port_4_pin_2); // AN002 = Motor 2 current
01158 * rx_mpc_set_adc(k_port_4_pin_3); // AN003 = Motor 3 current
01159 *
01160 * // Now configure S12ADC for conversion
01161 * // (see adc.h)
01162 * @endcode
01163 *
01164 * @see rx72n_adc_regs.h ADC register definitions
01165 *
01166 * @since Version 1.0.0
01167 * @callgraph
01168 */
01169 [[nodiscard]] rx_err_t rx_mpc_set_adc(rx_port_pin_t pin);
01170
01171 /**
01172 * @brief Configure pin for SCI (Serial Communication Interface) UART function
01173 *
01174 * @details
01175 * Configures a pin for SCI UART operation by setting PSEL = 0x0A.
01176 * Works for both TXD (transmit) and RXD (receive) pins - the same
01177 * PSEL value is used, with the specific function determined by the
01178 * pin's hardware connection.
01179 *
01180 * @par STAR Project UART Configuration (Debug Console)
01181 * | Pin | SCI Channel | Function | Notes |
01182 * |-----|-----|-----|
01183 * | PB7 | SCI9 | TXD9 | Debug output |
01184 * | PB6 | SCI9 | RXD9 | Debug input |
01185 *
01186 * @param[in] pin GPIO pin identifier for SCI function
01187 * - Must be a pin that supports SCI TXD/RXD
01188 *
01189 * @return Error code indicating success or failure
01190 * @retval k_rx_ok Pin successfully configured for SCI UART
01191 * @retval k_rx_err_invalid_arg Invalid port or pin number
01192 *
01193 * @pre Pin must support SCI function (check hardware manual)
01194 * @pre PCLKB clock must be running
01195 *
01196 * @post PFS register contains PSEL = 0x0A
01197 * @post Pin ready for UART operation
01198 *
01199 * @note Configure both TXD and RXD pins for bidirectional UART
01200 * @note Also requires SCI channel configuration and PORT PMR=1
01201 *
01202 * @par Example - Debug UART Setup

```

```

01203 * @code{.c}
01204 * // Configure debug UART pins (SCI9)
01205 * rx_mpc_set_sci(k_port_b_pin_7); // PB7 = TXD9
01206 * rx_mpc_set_sci(k_port_b_pin_6); // PB6 = RXD9
01207 *
01208 * // Now configure SCI9 for UART mode
01209 * // (see uart.h)
01210 * @endcode
01211 *
01212 * @see rx72n_sci_regs.h SCI register definitions
01213 *
01214 * @since Version 1.0.0
01215 * @callgraph
01216 */
01217 [[nodiscard]] rx_err_t rx_mpc_set_sci(rx_port_pin_t pin);
01218
01219 /**
01220 * @brief Configure pin for RIIC (I2C) bus function
01221 *
01222 * @details
01223 * Configures a pin for RIIC (Renesas I2C) operation by setting PSEL = 0x0F.
01224 * Works for both SCL (clock) and SDA (data) pins. RIIC hardware automatically
01225 * configures the open-drain output when the channel is enabled.
01226 *
01227 * @par I2C Bus Pins
01228 * | Pin | RIIC Channel | Function | Notes |
01229 * |-----|-----|-----|-----|
01230 * | P12 | RIIC0 | SCL0 | Clock line |
01231 * | P13 | RIIC0 | SDA0 | Data line |
01232 *
01233 * @param[in] pin GPIO pin identifier for RIIC function
01234 *           - Must be a pin that supports RIIC SCL/SDA
01235 *
01236 * @return Error code indicating success or failure
01237 * @retval k_rx_ok Pin successfully configured for RIIC I2C
01238 * @retval k_rx_err_invalid_arg Invalid port or pin number
01239 *
01240 * @pre Pin must support RIIC function
01241 * @pre PCLKB clock must be running
01242 *
01243 * @post PFS register contains PSEL = 0x0F
01244 * @post Pin ready for I2C operation
01245 *
01246 * @note Configure both SCL and SDA pins
01247 * @note External pull-up resistors required (typically 4.7kOhm)
01248 * @note Also requires RIIC channel configuration
01249 *
01250 * @par Example - I2C Bus Setup
01251 * @code{.c}
01252 * // Configure I2C pins (RIIC0)
01253 * rx_mpc_set_riic(k_port_1_pin_2); // P12 = SCL0
01254 * rx_mpc_set_riic(k_port_1_pin_3); // P13 = SDA0
01255 *
01256 * // Now configure RIIC0
01257 * // (see riic.h)
01258 * @endcode
01259 *
01260 * @see rx72n_riic_regs.h RIIC register definitions
01261 *
01262 * @since Version 1.0.0
01263 * @callgraph
01264 */
01265 [[nodiscard]] rx_err_t rx_mpc_set_riic(rx_port_pin_t pin);
01266
01267 /**
01268 * @brief Configure pin for RSPI (SPI) bus function
01269 *
01270 * @details
01271 * Configures a pin for RSPI (Renesas SPI) operation by setting PSEL = 0x0D.
01272 * Works for all SPI signals: RSPCK (clock), COPI (data out),
01273 * CIPO (data in), and SSLn (chip select).
01274 *
01275 * @par STAR Project SPI Configuration (RPi5 Communication)
01276 * **Note:** RSPI0 Signal names (MOSIA/MISOA) are RX72N datasheet hardware names.
01277 * Function names (COPI/CIPO) are project terminology per OSHWA standards.
01278 *
01279 * | Pin | RSPI0 Signal | Function | Description |
01280 * |-----|-----|-----|-----|
01281 * | PA0 | MOSIA | COPI | Controller Out, Peripheral In |
01282 * | PA1 | MISOA | CIPO | Controller In, Peripheral Out |
01283 * | PA3 | RSPCKA | CLK | SPI clock |
01284 * | PA4 | SSLA0 | CS | Chip select (active low) |
01285 *
01286 * @par SPI Bus Topology
01287 * @verbatim
01288 *      RX72N (Controller)          RPi5 (Controller)
01289 *      +-----+                  +-----+

```

```

01290 * | RSPi0 | | SPI |
01291 * | PA0 (COPI) <---+-----+---> CIPO |
01292 * | PA1 (CIPO) <---+-----+--- COPI |
01293 * | PA3 (CLK) <---+-----+--- CLK |
01294 * | PA4 (CS) <---+-----+--- CS |
01295 * +-----+ +-----+
01296 * Note: RX72N is SPI peripheral, RPi5 is controller
01297 * @endverbatim
01298 *
01299 * @param[in] pin GPIO pin identifier for RSPi function
01300 * - Must be a pin that supports RSPi signals
01301 *
01302 * @return Error code indicating success or failure
01303 * @retval k_rx_ok Pin successfully configured for RSPi SPI
01304 * @retval k_rx_err_invalid_arg Invalid port or pin number
01305 *
01306 * @pre Pin must support RSPi function
01307 * @pre PCLKB clock must be running
01308 *
01309 * @post PFS register contains PSEL = 0x0D
01310 * @post Pin ready for SPI operation
01311 *
01312 * @note Configure all 4 SPI pins (CLK, COPI, CIPO, CS)
01313 * @note Also requires RSPi channel configuration
01314 * @note Uses OSHWA-approved inclusive terminology (COPI/CIPO)
01315 *
01316 * @par Example - SPI Bus Setup
01317 * @code{.c}
01318 * // Configure SPI pins for RPi5 communication (RSPi0)
01319 * rx_mpc_set_rsipi(k_port_a_pin_0); // PA0 = COPI
01320 * rx_mpc_set_rsipi(k_port_a_pin_1); // PA1 = CIPO
01321 * rx_mpc_set_rsipi(k_port_a_pin_3); // PA3 = CLK
01322 * rx_mpc_set_rsipi(k_port_a_pin_4); // PA4 = CS (CS)
01323 *
01324 * // Now configure RSPi0
01325 * // (see rsipi.h)
01326 * @endcode
01327 *
01328 * @see rx72n_rsipi_regs.h RSPi register definitions
01329 * @see rx_spi_comm.h SPI communication driver
01330 *
01331 * @since Version 1.0.0
01332 * @callgraph
01333 */
01334 [[nodiscard]] rx_err_t rx_mpc_set_rsipi(rx_port_pin_t pin);
01335
01336 /**
01337 * @brief Configure pin for GPTW (General PWM Timer) complementary output
01338 *
01339 * @details
01340 * Configures a pin for GPTW operation by setting PSEL = 0x1E.
01341 * GPTW provides complementary PWM output with dead-time insertion
01342 * for motor control applications.
01343 *
01344 * GPTW channels support:
01345 * - 20 kHz PWM frequency (ultrasonic, inaudible)
01346 * - 1 us dead-time (prevents DRV8263H shoot-through)
01347 * - 90-degree phase staggering (reduces peak current)
01348 * - IN2/IN1 mode (direction and PWM signals per DRV8263H)
01349 *
01350 * @par STAR Project GPTW Pin Allocation
01351 * | Pin | GPTW Ch | Signal | Motor | Description |
01352 * |-----|-----|-----|-----|-----|
01353 * | P2.3 | GPTW0 | GTIOC0A | 0 | Motor 0 Direction (IN2) |
01354 * | P1.7 | GPTW0 | GTIOC0B | 0 | Motor 0 PWM (IN1) |
01355 * | P2.2 | GPTW1 | GTIOC1A | 1 | Motor 1 Direction (IN2) |
01356 * | PC.3 | GPTW1 | GTETRGC | 1 | Motor 1 PWM (IN1) - GTETRGC at 0x1E on PC3 per Table 23.19 |
01357 * | PE.3 | GPTW2 | GTIOC2A | 2 | Motor 2 Direction (IN2) |
01358 * | P8.6 | GPTW2 | GTIOC2B | 2 | Motor 2 PWM (IN1) |
01359 * | PE.7 | GPTW3 | GTIOC3A | 3 | Motor 3 Direction (IN2) |
01360 * | PC.6 | GPTW3 | GTIOC3A | 3 | Motor 3 PWM (IN1) - GTIOC3A at 0x1E on PC6 per Table 23.19 |
01361 *
01362 * @param[in] pin GPIO pin identifier for GPTW function
01363 * - Must be a pin that supports GPTW output (see manual)
01364 *
01365 * @return Error code indicating success or failure
01366 * @retval k_rx_ok Pin successfully configured for GPTW
01367 * @retval k_rx_err_null_ptr Config pointer is NULL (should never happen)
01368 * @retval k_rx_err_invalid_arg Invalid port or pin number (>= 16)
01369 * @retval k_rx_err_hw_error PWPR write-protect unlock failed
01370 *
01371 * @pre Pin must support GPTW function (check RX72N hardware manual)
01372 * @pre PCLKB clock must be running (MPC registers require clock)
01373 * @pre Must be called during single-threaded initialization
01374 *
01375 * @post PFS register configured with PSEL = 0x1E
01376 * @post PWPR register locked (write protection re-enabled)

```

```

01377 * @post Pin configured for GPTW PSEL; caller MUST subsequently set the
01378 *     corresponding PORTn.PMR bit (this function does NOT touch PMR)
01379 *     before GPTW can drive the pin.
01380 *
01381 * @note Thread Safety: Not thread-safe. Call during initialization only.
01382 * @note Configure both direction (A/IN2) and PWM (B/IN1) pins for each motor
01383 * @note GPTW channel configuration required for actual PWM operation
01384 *
01385 * @warning Motor safety: Configure pins before enabling GPTW channels
01386 *
01387 * @par Example - Configure All Motor PWM Pins
01388 * @code{.c}
01389 * // Configure 8 GPTW pins (4 motors x 2 pins = IN2 + IN1)
01390 * const rx_port_pin_t gptw_pins[] = {
01391 *     k_rx_p2_3, k_rx_p1_7, // Motor 0 IN2/IN1
01392 *     k_rx_p2_2, k_rx_pc_3, // Motor 1 IN2/IN1
01393 *     k_rx_pe_3, k_rx_p8_6, // Motor 2 IN2/IN1
01394 *     k_rx_pe_7, k_rx_pc_6  // Motor 3 IN2/IN1
01395 * };
01396 *
01397 * for (uint8_t i = 0; i < 8; i++) {
01398 *     rx_err_t err = rx_mpc_set_gptw(gptw_pins[i]);
01399 *     if (err != k_rx_ok) {
01400 *         rx_log_error("GPIO", "GPTW pin config failed");
01401 *         return err;
01402 *     }
01403 * }
01404 *
01405 * // Now safe to initialize GPTW channels
01406 * rx_gptw_init_all_staggered(&config);
01407 * @endcode
01408 *
01409 * @see rx_mpc_set_peripheral() Core pin configuration function
01410 * @see rx_gptw_init_all_staggered() GPTW timer initialization
01411 * @see RX72N Manual Chapter 23 - Multi-Function Pin Controller
01412 * @see RX72N Manual Chapter 29 - General PWM Timer
01413 *
01414 * @since Version 1.0.0
01415 *
01416 * @par NASA Power of 10 Compliance
01417 * - Rule 1: [OK] No goto, setjmp, recursion (delegates to rx_mpc_set_peripheral)
01418 * - Rule 3: [OK] No dynamic allocation (stack-only config struct)
01419 * - Rule 4: [OK] Function is 6 lines (well under 60 line limit)
01420 * - Rule 7: [OK] Return value from rx_mpc_set_peripheral checked by caller
01421 */
01422 [[nodiscard]] rx_err_t rx_mpc_set_gptw(rx_port_pin_t pin);
01423
01424 /**
01425 * @brief Configure pin for USB VBUS detection function
01426 *
01427 * @details
01428 * Configures a pin for USB VBUS detection by setting PSEL = 0x11.
01429 * VBUS detection is used for USB enumeration and power presence sensing.
01430 *
01431 * The VBUS detect pin monitors the 5V power rail on the USB bus:
01432 * - Pin reads HIGH (1) when USB cable connected and 5V present
01433 * - Pin reads LOW (0) when USB cable disconnected or unpowered
01434 *
01435 * Required for USB CDC communication with RPi5 host.
01436 *
01437 * @par STAR Project USB Configuration
01438 * | Pin | Function | Description |
01439 * |-----|-----|-----|
01440 * | P1.6 | USB0_VBUS | VBUS detect (5V presence) |
01441 *
01442 * @param[in] pin GPIO pin identifier for USB VBUS detect
01443 *     Typically P1.6 for USB0_VBUS on RX72N
01444 *
01445 * @return Error code indicating success or failure
01446 * @retval k_rx_ok Pin successfully configured for USB VBUS
01447 * @retval k_rx_err_null_ptr Config pointer is NULL (should never happen)
01448 * @retval k_rx_err_invalid_arg Invalid port or pin number
01449 * @retval k_rx_err_hw_error PWPR write-protect unlock failed
01450 *
01451 * @pre Pin must support USB VBUS function (P1.6 on RX72N)
01452 * @pre PCLKB clock must be running
01453 * @pre Must be called during single-threaded initialization
01454 *
01455 * @post PFS register configured with PSEL = 0x11
01456 * @post PWPR register locked
01457 * @post Pin configured for USB VBUS PSEL; caller MUST subsequently set
01458 *     the corresponding PORTn.PMR bit (this function does NOT touch
01459 *     PMR) before the USB peripheral sees the VBUS signal.
01460 *
01461 * @note Thread Safety: Not thread-safe. Call during initialization only.
01462 * @note Active-high detection (pin = 1 when 5V present)
01463 * @note Also requires USB PHY and USB0 controller configuration

```

```

01464 *
01465 * @par Example - USB VBUS Setup
01466 * @code{.c}
01467 * // Configure USB VBUS detect pin (P1.6)
01468 * rx_err_t err = rx_mpc_set_usb_vbus(k_rx_p1_6);
01469 * if (err != k_rx_ok) {
01470 *     rx_log_error("USB", "VBUS pin configuration failed");
01471 *     return err;
01472 * }
01473 *
01474 * // Now configure USB0 controller
01475 * err = usb_init(USB_MODE_CDC);
01476 * @endcode
01477 *
01478 * @see rx_mpc_set_peripheral() Core pin configuration function
01479 * @see rx_usb.h USB controller driver
01480 * @see RX72N Manual Chapter 23 - Multi-Function Pin Controller
01481 * @see RX72N Manual Chapter 31 - USB Controller
01482 *
01483 * @since Version 1.0.0
01484 */
01485 [[nodiscard]] rx_err_t rx_mpc_set_usb_vbus(rx_port_pin_t pin);
01486
01487 /**
01488 * @brief Configure pin for IRQ (external interrupt) function
01489 *
01490 * @details
01491 * Sets pin function to IRQ mode via MPC (Multi-Function Pin Controller).
01492 * This enables hardware edge detection on IRQ0-IRQ15 pins for use with
01493 * the Interrupt Controller Unit (ICU).
01494 *
01495 * **PFS Register for IRQ Function:**
01496 * - IRQ pins: ISEL bit (bit 6) = 0x40 written directly to PFS register
01497 * - This is NOT a PSEL value; rx_mpc_set_peripheral() is NOT used
01498 * - Verify in RX72N Manual Section 20.3 (MPC), PFS register layout
01499 *
01500 * **Supported Pins:**
01501 * - P00-P07: IRQ8-IRQ15
01502 * - Other IRQ pins per RX72N Manual Table 20.7
01503 *
01504 * @par STAR Project IRQ Usage (HC-SR04 Ultrasonic Sensors)
01505 * | Pin | IRQ Number | Sensor | Location |
01506 * |-----|-----|-----|-----|
01507 * | P03 | IRQ11 | Sonar 0 | Front-Left |
01508 * | P02 | IRQ10 | Sonar 1 | Front-Right |
01509 * | P01 | IRQ9 | Sonar 2 | Back-Left |
01510 * | P00 | IRQ8 | Sonar 3 | Back-Right |
01511 *
01512 * @param[in] pin GPIO pin identifier (must support IRQ function)
01513 *           Use k_rx_p0_X constants for IRQ8-15
01514 *
01515 * @return rx_err_t Error code
01516 * @retval k_rx_ok Pin configured for IRQ function
01517 * @retval k_rx_err_invalid_arg Invalid port or pin number (propagated from internal_write_pfs())
01518 *
01519 * @note internal_mpc_unlock() and internal_mpc_lock() do not propagate errors; only
01520 *       internal_write_pfs() can return k_rx_err_invalid_arg on invalid port/pin values.
01521 *
01522 * @pre Pin must have IRQ multiplexing capability (P00-P07 for IRQ8-15)
01523 * @pre PCLKB running (MPC requires clock)
01524 * @pre Must be called during single-threaded initialization
01525 *
01526 * @post Pin ISEL bit set (0x40) in PFS register (IRQ input mode)
01527 * @post PSEL field remains 0 (no conflicting peripheral function)
01528 * @post PWPR register locked
01529 * @post Pin ready for ICU edge detection configuration
01530 *
01531 * @note Thread Safety: Not thread-safe. Call during initialization only.
01532 * @note Call this BEFORE enabling ICU interrupt
01533 *
01534 * @warning Does NOT configure ICU registers (edge detect, priority, etc.)
01535 * @warning Configure ICU separately for edge detection and priority as required
01536 *
01537 * @par Example - IRQ Pin Setup
01538 * @code{.c}
01539 * // Configure P03 for IRQ11 (external interrupt input)
01540 * rx_err_t err = rx_mpc_set_irq(k_rx_p0_3);
01541 * if (err != k_rx_ok) {
01542 *     rx_log_error("IRQ", "IRQ pin configuration failed");
01543 *     return err;
01544 * }
01545 *
01546 * // Now configure ICU for edge detection (caller-specific step)
01547 * @endcode
01548 *
01549 * @invariant MPC base address remains constant throughout operation
01550 * @invariant Only target pin's PFS register is modified (other pins' PFS unchanged)

```



```

01551 *
01552 * @see rx_mpc_set_gpio() Set pin back to GPIO mode
01553 * @see RX72N Manual Section 20.3 (MPC), PFS register bit definitions
01554 * @see RX72N Manual Chapter 15 (ICU - Interrupt Controller Unit)
01555 *
01556 * @since Version 1.0.0
01557 *
01558 * @par NASA Power of 10 Compliance
01559 * - Rule 5: [OK] 3 preconditions, 4 postconditions
01560 * - Rule 7: [OK] Caller checks return value ([[nodiscard]])
01561 * - Rule 8: [OK] k_pfs_isel from pfs_bits_t enum (no magic numbers)
01562 *
01563 * @callgraph
01564 * @callergraph
01565 */
01566 [[nodiscard]] rx_err_t rx_mpc_set_irq(rx_port_pin_t pin);
01567
01568 #ifdef __cplusplus
01569 }
01570 #endif

```

8.91 libs/rx_hal/inc/rx_mtu.h File Reference

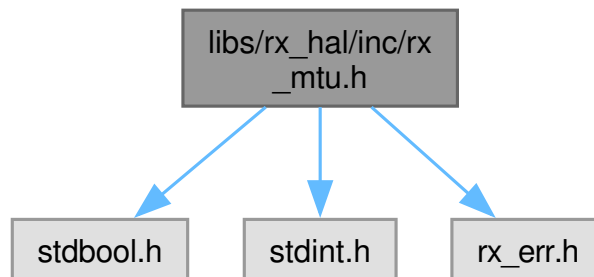
MTU PWM Driver for RX72N Multi-Function Timer Unit.

```
#include <stdbool.h>
```

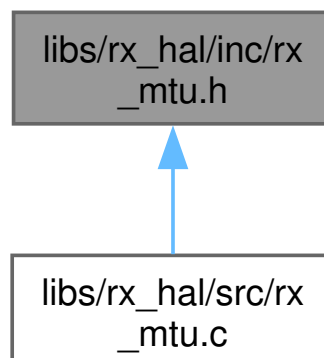
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_mtu.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_mtu_config_t](#)
MTU PWM Configuration Structure.

Enumerations

- enum `rx_mtu_channel_t` : `uint8_t` {
`k_mtu_channel_0` , `k_mtu_channel_1` , `k_mtu_channel_2` , `k_mtu_channel_3` ,
`k_mtu_channel_4` , `k_mtu_channel_6` , `k_mtu_channel_7` }
MTU Channel Identifiers.
- enum `rx_mtu_output_t` : `uint8_t` { `k_mtu_output_a` , `k_mtu_output_b` , `k_mtu_output_c` ,
`k_mtu_output_d` }
MTU Output Channel Selection.

Functions

- `rx_err_t rx_mtu_init_pwm` (`rx_mtu_channel_t` channel, `const rx_mtu_config_t *config`)
Initialize MTU channel for PWM output.
- `rx_err_t rx_mtu_set_duty` (`rx_mtu_channel_t` channel, `rx_mtu_output_t` output, `float duty_percent`)
Set PWM duty cycle.
- `rx_err_t rx_mtu_set_duty_raw` (`rx_mtu_channel_t` channel, `rx_mtu_output_t` output, `uint16_t duty_count`)
Set PWM duty cycle (raw count value).
- `rx_err_t rx_mtu_get_duty` (`rx_mtu_channel_t` channel, `rx_mtu_output_t` output, `float *duty_percent`)
Get current duty cycle.
- `rx_err_t rx_mtu_get_period` (`rx_mtu_channel_t` channel, `uint16_t *period_count`)
Get PWM period count.
- `rx_err_t rx_mtu_enable_output` (`rx_mtu_channel_t` channel, `rx_mtu_output_t` output, `bool enable`)
Enable or disable PWM output.
- `rx_err_t rx_mtu_start` (`rx_mtu_channel_t` channel)
Start MTU timer.
- `rx_err_t rx_mtu_stop` (`rx_mtu_channel_t` channel)
Stop MTU timer.
- `rx_err_t rx_mtu_deinit` (`rx_mtu_channel_t` channel)
Deinitialize MTU channel.

8.91.1 Detailed Description

MTU PWM Driver for RX72N Multi-Function Timer Unit.

Production-grade PWM driver for the RX72N Multi-Function Timer Unit (MTU3) peripheral. This module provides 16-bit PWM generation for motor control, quadrature encoder input, and general-purpose timing applications.

8.91.1.1 Purpose and Design Rationale

The MTU (Multi-Function Timer Unit) is a versatile timer peripheral that complements the GPTW module:

Feature	MTU3	GPTW
Counter Width	16-bit	32-bit
Channels	8 (MTU0-4, MTU6-7)	4 (GPTW0-3)
Outputs per Channel	Up to 4 (A/B/C/D)	2 (A/B)
Phase Counting	Yes (encoder input)	Limited
Complementary PWM	Yes (with deadtime)	Yes
Primary Use	Encoders, PWM	Motor PWM

STAR Project Usage:

- MTU1/MTU2: Quadrature encoder input (see `rx_mtu_encoder.h`)
- GPTW0-3: Motor PWM output (see [rx_gptw.h](#)) - recommended for motors

8.91.1.2 Key Features

PWM Generation:

- 8 timer channels with independent configuration
- PWM mode 1: Center-aligned (triangle wave) for symmetric current ripple
- PWM mode 2: Edge-aligned (sawtooth wave) for simple applications
- Complementary outputs with configurable deadtime
- Up to 4 outputs per channel (A/B/C/D) on some channels

Encoder Interface (MTU1/MTU2):

- Phase counting mode for quadrature encoders
- 16-bit pulse count with overflow detection
- See `rx_mtu_encoder.h` for encoder-specific functions

Timing Features:

- Input capture for pulse width measurement
- Output compare for waveform generation
- Cascade operation for extended range

8.91.1.3 Hardware Architecture

Register Base Addresses:

Channel	Base Address	Outputs	Notes
MTU0	0x000C1300	A,B,C,D	4 outputs
MTU1	0x000C1380	A,B	Encoder input
MTU2	0x000C1400	A,B	Encoder input
MTU3	0x000C1200	A,B,C,D	4 outputs, pairs with MTU4
MTU4	0x000C1201	A,B,C,D	4 outputs, pairs with MTU3
MTU6	0x000C1A00	A,B,C,D	4 outputs, pairs with MTU7
MTU7	0x000C1A01	A,B,C,D	4 outputs, pairs with MTU6

Note: MTU5 (U/V/W sub-channels at 0x000C1C80/90/A0) is not supported by this HAL.

Clock Configuration:

Parameter	Value	Notes
Clock Source	PCLKA	Peripheral Clock A
Clock Frequency	120 MHz	After system clock configuration
Counter Resolution	8.33 ns	1/120MHz
Max Period (16-bit)	546 us	65535 counts at 120 MHz
Min PWM Frequency	1831 Hz	1 / 546 us

8.91.1.4 PWM Modes

PWM Mode 1 (Center-Aligned):

- Counter counts up to period, then down to 0
- Symmetric current ripple (better for motors)
- Effective PWM frequency = $PCLKA / (2 \times \text{period})$

PWM Mode 2 (Edge-Aligned):

- Counter counts up to period, then resets
- Asymmetric ripple
- Effective PWM frequency = $PCLKA / \text{period}$

PWM Frequency Calculation:

For center-aligned (PWM mode 1):

$$f_{PWM} = \frac{f_{PCLKA}}{2 \times (TGRA + 1)}$$

For edge-aligned (PWM mode 2):

$$f_{PWM} = \frac{f_{PCLKA}}{TGRA + 1}$$

8.91.1.5 Performance Characteristics

Operation	Execution Time	Memory Usage	Notes
rx_mtu_init_pwm()	~50 us	0 heap	One-time setup
rx_mtu_set_duty()	~5 us	0	Float calculation
rx_mtu_set_duty_raw()	~1 us	0	Direct register write
rx_mtu_enable_output()	~1 us	0	Single register write

Memory Footprint:

- Static variables: ~32 bytes (initialization flags)
- Code size: ~1.5 KB (all functions)
- Stack usage: < 48 bytes (deepest call)
- Heap usage: 0 bytes (zero dynamic allocation)

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	MTU3 peripheral required
Clock	PCLKA @ 120 MHz	System clock configuration
Power	3.3V I/O	LVC MOS compatible

Module Dependencies:

This module depends on:

- [rx_err.h](#) - Error code definitions
- [rx72n_mtu_regs.h](#) - MTU register definitions

This module is used by:

- [rx_mtu_encoder.c](#) - Encoder interface (MTU1/MTU2)
- Application code for auxiliary PWM outputs

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	↔ [OK]↔	No goto/setjmp/recursion
2. Fixed loop bounds	↔ [OK]↔	Loops bounded by channel count (8 max)
3. No dynamic allocation	↔ [OK]↔	Zero malloc/free, all static
4. Small functions	↔ [OK]↔	All functions < 60 lines
5. Assertions (>=2 per function)	↔ [OK]↔	Minimum 2 validation checks per function
6. Narrow scope	↔ [OK]↔	File-scope statics, minimal globals
7. Check return values	↔ [OK]↔	All functions return rx_err_t
8. Limited preprocessor	↔ [OK]↔	C23 typed enums, minimal macros
9. Pointer restrictions	↔ [OK]↔	Single-level pointers only

Rule	Status	Implementation
10. Compiler warnings	↔ [OK]↔	Compiles with -Wall -Wextra -Werror

SOLID Principles:

Single Responsibility (S):

- This module handles ONLY MTU PWM configuration
- Encoder functions are in rx_mtu_encoder.h

Open/Closed (O):

- Extensible via [rx_mtu_config_t](#) structure
- New modes can be added without modifying existing code

Liskov Substitution (L):

- All functions return rx_err_t with consistent semantics
- Channel and output parameters are validated uniformly

Interface Segregation (I):

- PWM functions separate from encoder functions
- Float and raw duty cycle APIs separate

Dependency Inversion (D):

- Depends on abstract error codes
- Can be mocked for unit testing

See also

[rx_gptw.h](#) GPTW driver (preferred for motor PWM)
[rx_mtu_encoder.h](#) Encoder interface using MTU phase counting
[rx72n_mtu_regs.h](#) Register definitions
 RX72N Hardware Manual Section 24 - Multi-Function Timer Unit 3

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Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_mtu.h](#).

8.91.2 Data Structure Documentation

8.91.2.1 struct rx`mtu`config`t

MTU PWM Configuration Structure.

Contains all configuration parameters for initializing an MTU channel for PWM output. This structure is passed to `rx_mtu_init_pwm()` to configure the timer.

8.91.2.2 Configuration Guidelines

Frequency Limits (16-bit counter at 120 MHz):

- Maximum frequency: ~ 1.83 MHz (period = 65 counts)
- Minimum frequency: ~ 1831 Hz (period = 65535 counts)
- Recommended for motors: 15-25 kHz

Resolution Trade-off: Higher frequency $\rightarrow$ lower resolution (fewer counts per period)

- 20 kHz: 6000 counts $\rightarrow$ 0.017% resolution
- 100 kHz: 1200 counts $\rightarrow$ 0.083% resolution

Configuration Parameter Table:

Parameter	Min	Max	Default	Unit	Notes
frequency_hz	1831	1830000	20000	Hz	16-bit limit
deadtime_ns	0	65535	1000	ns	0 disables
enable_complementary	false	true	true	bool	A/B pairs
invert_polarity	false	true	false	bool	Active-low

Memory Layout:

Offset	Size	Field	Type	Alignment
0	4	frequency_hz	uint32_t	4
4	2	deadtime_ns	uint16_t	2
6	1	enable_complementary	bool	1

Offset	Size	Field	Type	Alignment
7	1	invert_polarity	bool	1
Total	8	-	-	4

Usage Example:

```
// Standard PWM configuration for auxiliary output
rx_mtu_config_t config = {
    .frequency_hz = 10000,           // 10 kHz
    .deadtime_ns = 500,             // 500 ns deadtime
    .enable_complementary = true,    // A/B complementary
    .invert_polarity = false        // Active-high
};

rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
if (err != k_rx_ok) {
    rx_log_error("MTU", "Init failed: %d", err);
}
```

Invariant

frequency_hz must produce period ≤ 65535 counts
 Structure is copied during init (no need to keep alive)

Note

For motor control, prefer [rx_gptw.h](#) with 32-bit resolution

Warning

Invalid configurations rejected with k_rx_err_invalid_arg

See also

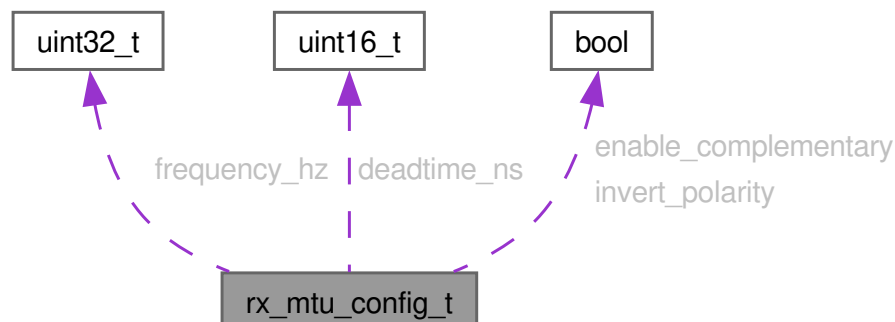
[rx_mtu_init_pwm\(\)](#) Channel initialization function
[rx_gptw_config_t](#) GPTW config (32-bit, recommended for motors)

Since

Version 1.0.0

Definition at line 404 of file [rx_mtu.h](#).

Collaboration diagram for rx_mtu_config_t:



Data Fields

uint16_t	deadtime_ns	Deadtime in nanoseconds. Deadtime inserted between complementary output transitions. Prevents shoot-through in H-bridge configurations. Set to 0 to disable deadtime insertion. Valid Range: [0, 65535 ns] Recommended: 500-1000 ns
bool	enable_complementary	Enable complementary outputs. When true, output pairs (A/B and C/D) are complementary: • Output B = inverted Output A (with deadtime) • Output D = inverted Output C (with deadtime) Required for H-bridge motor control applications.
uint32_t	frequency_hz	PWM frequency in Hz. Sets the PWM switching frequency. Limited by 16-bit counter: • Min: 1831 Hz (period = 65535 at 120 MHz) • Max: ~1.83 MHz (period = 65 at 120 MHz) Valid Range: [1831, 1830000 Hz] Recommended: 10000-20000 Hz for PWM applications
bool	invert_polarity	Invert PWM polarity. When true, inverts the PWM output polarity: • false: HIGH during duty cycle (active-high) • true: LOW during duty cycle (active-low)

8.91.3 Enumeration Type Documentation

8.91.3.1 rx_mtu_channel_t

```
enum rx_mtu_channel_t : uint8_t
```

MTU Channel Identifiers.

Identifies the MTU timer channels available on the RX72N. Note that MTU5 does not exist in this MCU family. Channels are grouped for specific functions:

8.91.3.2 Channel Groups and Functions

General Purpose PWM (MTU0, MTU3, MTU4, MTU6, MTU7):

- Up to 4 outputs per channel (A, B, C, D)
- Complementary PWM with deadtime
- MTU3+MTU4 and MTU6+MTU7 can be paired for synchronized operation

Encoder Input (MTU1, MTU2):

- Phase counting mode for quadrature encoders
- 16-bit pulse count with direction detection
- Used by rx_mtu_encoder.h for motor position feedback

Channel Specifications:

Channel	Base Address	Outputs	Phase Count	Paired With
MTU0	0x000C1300	A,B,C,D	No	-
MTU1	0x000C1380	A,B	Yes	MTU2
MTU2	0x000C1400	A,B	Yes	MTU1
MTU3	0x000C1200	A,B,C,D	No	MTU4
MTU4	0x000C1201	A,B,C,D	No	MTU3
MTU6	0x000C1A00	A,B,C,D	No	MTU7
MTU7	0x000C1A01	A,B,C,D	No	MTU6

STAR Project Usage:

Channel	Application	Configuration
MTU1	Encoder 0-1 (Motor 0, Motor 1)	Phase counting mode
MTU2	Encoder 2-3 (Motor 2, Motor 3)	Phase counting mode
MTU0,3,4,6,7	Available	General purpose PWM

Usage Example:

```
// Initialize MTU0 for PWM output
rx_mtu_config_t config = {
    .frequency_hz = 10000,
    .deadtime_ns = 500,
    .enable_complementary = true,
    .invert_polarity = false
};

rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
if (err != k_rx_ok) {
    rx_log_error("MTU", "Init failed: %d", err);
}
```

Invariant

Channel 5 is not defined (MTU5 hardware exists but is not supported by this HAL)
 Channel values match hardware register offsets

Note

C23 typed enum with uint8_t underlying type
 Value 5 is intentionally skipped (MTU5 U/V/W sub-channels not supported by this HAL)

See also

rx_mtu_encoder.h Encoder functions using MTU1/MTU2
 RX72N Hardware Manual Section 24 - MTU3 channels

Since

Version 1.0.0

Enumerator

k_mtu_channel_0	MTU0: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1300. Standalone operation
k_mtu_channel_1	MTU1: Encoder input channel. 2 outputs (A,B). Base: 0x000C1380. Phase counting with MTU2
k_mtu_channel_2	MTU2: Encoder input channel. 2 outputs (A,B). Base: 0x000C1400. Phase counting with MTU1
k_mtu_channel_3	MTU3: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1200. Pairs with MTU4
k_mtu_channel_4	MTU4: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1201. Pairs with MTU3
k_mtu_channel_6	MTU6: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A00. Pairs with MTU7
k_mtu_channel_7	MTU7: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A01. Pairs with MTU6

Definition at line 262 of file [rx_mtu.h](#).

```

00262         : uint8_t {
00263     k_mtu_channel_0 =
00264         0, /**< MTU0: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1300. Standalone operation */
00265     k_mtu_channel_1 =
00266         1, /**< MTU1: Encoder input channel. 2 outputs (A,B). Base: 0x000C1380. Phase counting with MTU2 */
00267     k_mtu_channel_2 =
00268         2, /**< MTU2: Encoder input channel. 2 outputs (A,B). Base: 0x000C1400. Phase counting with MTU1 */
00269     k_mtu_channel_3 =
00270         3, /**< MTU3: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1200. Pairs with MTU4 */
00271     k_mtu_channel_4 =
00272         4, /**< MTU4: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1201. Pairs with MTU3 */
00273     k_mtu_channel_6 =
00274         6, /**< MTU6: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A00. Pairs with MTU7 */
00275     k_mtu_channel_7 =
00276         7, /**< MTU7: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A01. Pairs with MTU6 */
00277 } rx_mtu_channel_t;
```

8.91.3.3 rx_mtu_output_t

enum `rx_mtu_output_t` : `uint8_t`

MTU Output Channel Selection.

Selects which output pin to configure for PWM or compare match output. MTU channels can have 2 or 4 outputs depending on the channel.

8.91.3.4 Output Availability by Channel

Channel	Output A	Output B	Output C	Output D
MTU0	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔
MTU1	↔ [OK]↔	↔ [OK]↔	-	-
MTU2	↔ [OK]↔	↔ [OK]↔	-	-
MTU3	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔
MTU4	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔
MTU6	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔
MTU7	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔	↔ [OK]↔

8.91.3.5 Complementary Pairs

When complementary output is enabled:

- A and B form a complementary pair
- C and D form a complementary pair (on channels with 4 outputs)
- Deadtime is inserted between transitions

Usage Example:

```
// Set 50% duty on output A of MTU0
rx_mtu_set_duty(k_mtu_channel_0, k_mtu_output_a, 50.0f);

// In complementary mode, output B is automatically inverted
// with deadtime inserted between transitions
```

Invariant

Output C and D only valid for channels with 4 outputs
 Output values are 0-3 (4 possible outputs)

Note

C23 typed enum with `uint8_t` underlying type

Warning

Using output C/D on MTU1/MTU2 will return `k_rx_err_invalid_arg`

See also

[rx_mtu_channel_t](#) Channel selection
[rx_mtu_set_duty\(\)](#) Set duty cycle on specific output

Since

Version 1.0.0

Enumerator

<code>k_mtu_output_a</code>	MTIOCA output pin. Available on all channels. Compare register: TGRA. Primary PWM output
<code>k_mtu_output_b</code>	MTIOCB output pin. Available on all channels. Compare register: TGRB. Complementary to A when enabled
<code>k_mtu_output_c</code>	MTIOCC output pin. MTU0,3,4,6,7 only. Compare register: TGRC. Secondary PWM output
<code>k_mtu_output_d</code>	MTIODC output pin. MTU0,3,4,6,7 only. Compare register: TGRD. Complementary to C when enabled

Definition at line 326 of file [rx_mtu.h](#).

```

00326         : uint8_t {
00327     k_mtu_output_a =
00328         0, /**< MTIOCA output pin. Available on all channels. Compare register: TGRA. Primary PWM output */
00329     k_mtu_output_b =
00330         1, /**< MTIOCB output pin. Available on all channels. Compare register: TGRB. Complementary to A when enabled */
00331     k_mtu_output_c =
00332         2, /**< MTIOCC output pin. MTU0,3,4,6,7 only. Compare register: TGRC. Secondary PWM output */
00333     k_mtu_output_d =
00334         3, /**< MTIODC output pin. MTU0,3,4,6,7 only. Compare register: TGRD. Complementary to C when enabled */
00335 } rx_mtu_output_t;
```

8.91.4 Function Documentation

8.91.4.1 rx`mtu`deinit()

```

rx_err_t rx_mtu_deinit (
    const rx\_mtu\_channel\_t channel) [nodiscard]
```

Deinitialize MTU channel.

Stops timer, disables outputs, and releases resources.

Parameters

in	channel	MTU channel (0-4, 6-7)
----	---------	------------------------

Returns

`k_rx_ok` on success

`k_rx_err_invalid_arg` if channel is invalid

Deinitialize MTU channel.

Performs an orderly shutdown of the specified MTU channel:

1. Stop the timer counter ([rx_mtu_stop\(\)](#))
2. Disable all four outputs (A, B, C, D) via [rx_mtu_enable_output\(\)](#)
3. Clear the cached period to 0
4. Clear the initialized flag

After this call the channel may be safely re-initialized with different parameters via [rx_mtu_init_pwm\(\)](#).

Precondition

Motor or load driven by this channel must be stopped externally before call
channel must be a valid MTU channel (0-4, 6-7)

Postcondition

Timer stopped and outputs disabled
`s_mtu_initialized[channel] == false`
`s_mtu_period[channel] == k_mtu_period_zero`

Note

Not thread-safe: do not call while another task is accessing this channel
Idempotent with respect to re-initialization (can call init again after)

Warning

Ensure motor is at rest before calling to avoid abrupt stop

```
rx_err_t err = rx_mtu_deinit(k_mtu_channel_0);
```

See also

[rx_mtu_init_pwm\(\)](#) Re-initialize after deinit
[rx_mtu_stop\(\)](#) Stop timer without full teardown

Since

Version 1.0.0

Definition at line 1087 of file [rx_mtu.c](#).

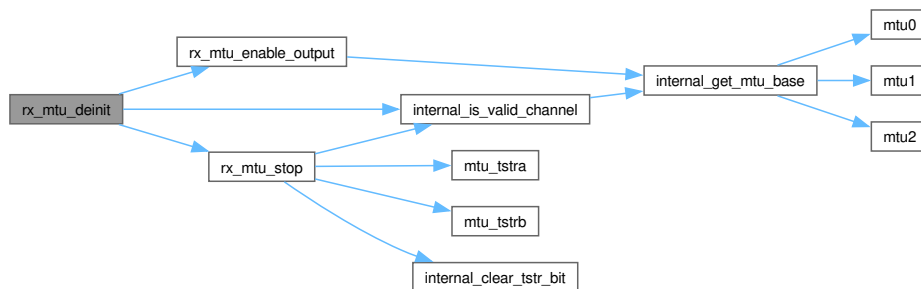
```

01088 {
01089     if (!internal_is_valid_channel(channel)) {
01090         return k_rx_err_invalid_arg;
01091     }
01092
01093     /* Stop timer */
01094     rx_err_t err = rx_mtu_stop(channel);
01095     if (err != k_rx_ok) {
01096         return err;
01097     }
01098
01099     /* Disable all outputs */
01100     err = rx_mtu_enable_output(channel, k_mtu_output_a, false);
01101     if (err != k_rx_ok) {
01102         return err;
01103     }
01104     err = rx_mtu_enable_output(channel, k_mtu_output_b, false);
01105     if (err != k_rx_ok) {
01106         return err;
01107     }
01108     err = rx_mtu_enable_output(channel, k_mtu_output_c, false);
01109     if (err != k_rx_ok) {
01110         return err;
01111     }
01112     err = rx_mtu_enable_output(channel, k_mtu_output_d, false);
01113     if (err != k_rx_ok) {
01114         return err;
01115     }
01116
01117     /* Mark as uninitialized */
01118     s_mtu_initialized[channel] = false;
01119     s_mtu_period[channel] = k_mtu_period_zero;
01120
01121     rx_log_info(s_tag, "MTU channel deinitialized");
01122
01123     return k_rx_ok;
01124 }

```

References [internal_is_valid_channel\(\)](#), [k_mtu_output_a](#), [k_mtu_output_b](#), [k_mtu_output_c](#), [k_mtu_output_d](#), [k_mtu_period_zero](#), [rx_mtu_enable_output\(\)](#), [rx_mtu_stop\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.91.4.2 rx_mtu_enable_output()

```
rx_err_t rx_mtu_enable_output (
    const rx_mtu_channel_t channel,
    rx_mtu_output_t output,
    const bool enable)  [nodiscard]
```

Enable or disable PWM output.

Parameters

in	channel	MTU channel (0-4, 6-7)
in	output	Output channel (A/B/C/D)
in	enable	True to enable output, false to disable

Returns

k_rx_ok on success
k_rx_err_invalid_arg if channel or output is invalid
k_rx_err_invalid_state if channel not initialized

Enable or disable PWM output.

Controls whether the MTIOCA/B/C/D pin actively drives the PWM waveform or is held in its initial state (off). Modifies the TIORH or TIORL register nibble corresponding to the requested output without disturbing other outputs.

Output	Register	Nibble
A	TIORH	Low (bits 3:0)
B	TIORH	High (bits 7:4)
C	TIORL	Low (bits 3:0)
D	TIORL	High (bits 7:4)

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
output must be a valid [rx_mtu_output_t](#) enum value

Postcondition

TIORH or TIORL nibble updated; timer continues running
Other outputs on the same channel are unchanged

Note

Not thread-safe: caller must prevent concurrent register access
Timer keeps running; only output pin drive changes

Warning

Outputs may remain HIGH when disabled, depending on last compare state

```
rx_mtu_enable_output(k_mtu_channel_0, k_mtu_output_a, false); // disable
```

See also

[rx_mtu_init_pwm\(\)](#) Initialize channel before calling
[rx_mtu_stop\(\)](#) Halt the entire timer

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Version 1.0.0

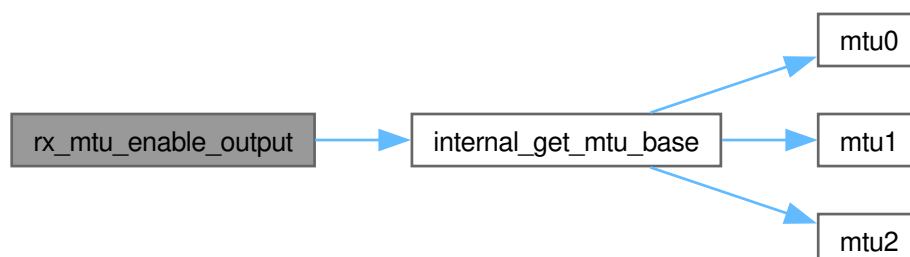
Definition at line 883 of file [rx_mtu.c](#).

```
00884 {
00885     volatile rx_mtu_channel_regs_t* const mtu =
00886         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00887     if (mtu == nullptr) {
00888         return k_rx_err_invalid_arg;
00889     }
00890
00891     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00892
00893     /* Enable/disable output by modifying TIOR registers */
00894     const uint8_t tior_value = enable ? k_mtu_tior_init_low : k_mtu_tior_disabled;
00895
00896     switch (output) {
00897     case k_mtu_output_a:
00898         mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_high_clear_low) |
00899                               (tior_value « k_mtu_tior_low_shift));
00900         break;
00901     case k_mtu_output_b:
00902         mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_low_clear_high) |
00903                               (tior_value « k_mtu_tior_high_shift));
00904         break;
00905     case k_mtu_output_c:
00906         mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_high_clear_low) |
00907                               (tior_value « k_mtu_tior_low_shift));
00908         break;
00909     case k_mtu_output_d:
00910         mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_low_clear_high) |
00911                               (tior_value « k_mtu_tior_high_shift));
00912         break;
00913     default:
00914         return k_rx_err_invalid_arg;
00915     }
00916
00917     return k_rx_ok;
00918 }
```

References [internal_get_mtu_base\(\)](#), [k_mtu_output_a](#), [k_mtu_output_b](#), [k_mtu_output_c](#), [k_mtu_output_d](#), [k_mtu_tior_disabled](#), [k_mtu_tior_high_shift](#), [k_mtu_tior_init_low](#), [k_mtu_tior_keep_high_clear_low](#), [k_mtu_tior_keep_low_clear_high](#), [k_mtu_tior_low_shift](#), [s_mtu_initialized](#), [s_tag](#), [rx_mtu_channel_regs_t::tiorh](#), and [rx_mtu_channel_regs_t::tiorl](#).

Referenced by [rx_mtu_deinit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.91.4.3 rx_mtu_get_duty()

```

rx_err_t rx_mtu_get_duty (
    const rx_mtu_channel_t channel,
    const rx_mtu_output_t output,
    float * duty_percent) [nodiscard]
  
```

Get current duty cycle.

Parameters

in	channel	MTU channel (0-4, 6-7)
in	output	Output channel (A/B/C/D)
out	duty_percent	Pointer to store duty cycle in percent

Returns

k_rx_ok on success
 k_rx_err_null_ptr if duty_percent is nullptr
 k_rx_err_invalid_arg if channel or output is invalid
 k_rx_err_invalid_state if channel not initialized

Get current duty cycle.

Reads the TGR register for the specified output and converts the raw count to a percentage using the cached period value.

$$\text{duty_percent} = \frac{\text{duty_count} \times 100}{\text{period}}$$

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
 duty_percent must point to valid float storage

Postcondition

*duty_percent contains duty cycle in [0.0, 100.0] on success
 Hardware TGR register unchanged

Note

Not thread-safe: concurrent `set_duty` calls may cause torn reads

```
float duty;
rx_err_t err = rx_mtu_get_duty(k_mtu_channel_0, k_mtu_output_b, &duty);
```

See also

[rx_mtu_set_duty\(\)](#) Set duty using percentage

[rx_mtu_get_period\(\)](#) Retrieve period count

Since

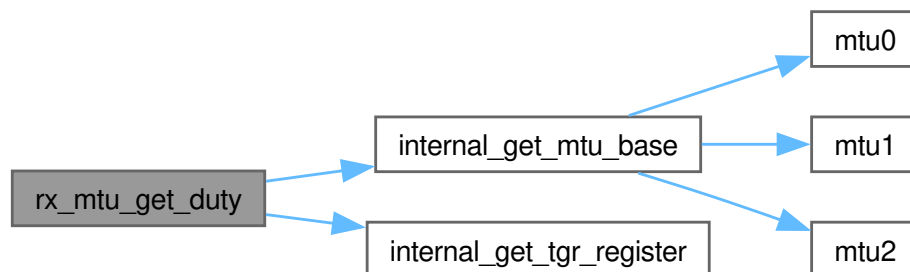
Version 1.0.0

Definition at line 775 of file [rx_mtu.c](#).

```
00776 {
00777     RX_VALIDATE_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
00778
00779     volatile rx_mtu_channel_regs_t* const mtu =
00780         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00781     if (mtu == nullptr) {
00782         return k_rx_err_invalid_arg;
00783     }
00784
00785     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00786
00787     const volatile uint16_t* const tgr = internal_get_tgr_register(mtu, output);
00788     if (tgr == nullptr) {
00789         return k_rx_err_invalid_arg;
00790     }
00791
00792     const uint16_t period = s_mtu_period[channel];
00793     const uint16_t duty_count = *tgr;
00794
00795     *duty_percent = (float)(duty_count * (float)k_mtu_duty_max) / period;
00796
00797     return k_rx_ok;
00798 }
```

References [internal_get_mtu_base\(\)](#), [internal_get_tgr_register\(\)](#), [k_mtu_duty_max](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.91.4.4 rx_mtu_get_period()

```
rx_err_t rx_mtu_get_period (
    const rx_mtu_channel_t channel,
    uint16_t * period_count)  [nodiscard]
```

Get PWM period count.

Returns the period register value (useful for raw duty cycle calculations).

Parameters

in	channel	MTU channel (0-4, 6-7)
out	period_count	Pointer to store period count

Returns

- k_rx_ok on success
- k_rx_err_null_ptr if period_count is nullptr
- k_rx_err_invalid_arg if channel is invalid
- k_rx_err_invalid_state if channel not initialized

Get PWM period count.

Returns the cached period value (TGRA) from initialization. Useful for calculating raw duty counts from percentages without floating-point.

Precondition

- Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
- period_count must point to valid uint16_t storage

Postcondition

- *period_count contains the TGRA register value on success
- Hardware registers unchanged

Note

- Thread-safe: reads from static cache, no hardware access

```
uint16_t period;
rx_err_t err = rx_mtu_get_period(k_mtu_channel_0, &period);
if (err == k_rx_ok) {
    uint16_t half_duty = period / 2U; // 50%
    rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, half_duty);
}
```

See also

- [rx_mtu.h](#) Complete API documentation

- [rx_mtu_set_duty_raw\(\)](#) Use period count to set duty

Since

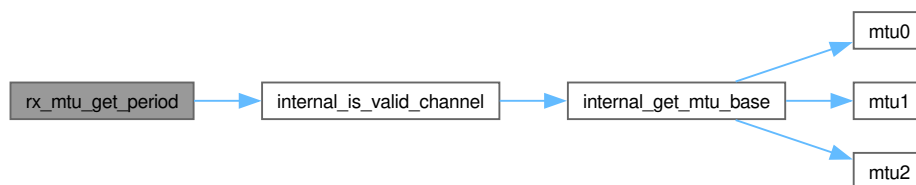
Version 1.0.0

Definition at line 831 of file [rx_mtu.c](#).

```
00832 {
00833   RX_VALIDATE_PTR(period_count, s_tag, "period_count pointer is nullptr");
00834
00835   if (!internal_is_valid_channel(channel)) {
00836     return k_rx_err_invalid_arg;
00837   }
00838
00839   RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00840
00841   *period_count = s_mtu_period[channel];
00842   return k_rx_ok;
00843 }
```

References [internal_is_valid_channel\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.91.4.5 rx`mtu`init`pwm()

```
rx_err_t rx_mtu_init_pwm (
    const rx_mtu_channel_t channel,
    const rx_mtu_config_t * config) [nodiscard]
```

Initialize MTU channel for PWM output.

Configures MTU channel for PWM mode 1 (center-aligned, triangle wave). Enables timer and starts PWM generation at 0% duty cycle.

Parameters

in	channel	MTU channel (0-4, 6-7)
in	config	PWM configuration

Returns

k_rx_ok on success
 k_rx_err_invalid_arg if channel or config is invalid
 k_rx_err_invalid_state if MTU module not enabled

Initialize MTU channel for PWM output.

Configures an MTU channel for PWM Mode 1 (triangle wave, center-aligned) operation. Performs complete hardware setup including module clock enable, timer configuration, and automatic start.

8.91.4.6 Initialization Sequence

1. Validate config pointer
2. Get channel register base address
3. Calculate period from frequency
4. Enable MTU module clock (clear MSTPA bit)
5. Stop timer if running
6. Configure TCR (clock source, clear condition)
7. Configure TMDR (PWM Mode 1)
8. Configure TIORH/TIOLR (output behavior)
9. Set TGRA (period), clear TGRB/TGRC/TGRD (duty = 0)
10. Clear TCNT counter
11. Store period and set initialized flag
12. Start timer

8.91.4.7 PWM Mode 1 Configuration

Register	Value	Effect
TCR	TPSC=1, CCLR=TGRA	PCLKA/1, clear on TGRA match
TMDR	MD=PWM1	Triangle wave mode
TIORH/ L	Init low	Outputs toggle on compare match

Precondition

System clock configured (PCLKA = 120 MHz)
config pointer valid with valid frequency

Postcondition

MTU channel configured and timer running
All duty cycles initialized to 0%
s_mtu_initialized[channel] = true

Note

Timer starts automatically
Call once during system initialization

Warning

Not thread-safe: Configure before starting motor tasks

Example:

```
rx_mtu_config_t config = {
    .frequency_hz = 20000 // 20 kHz
};
rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
```

See also

[rx_mtu.h](#) Complete API documentation

[rx_mtu_deinit\(\)](#) Cleanup function

Since

Version 1.0.0

Definition at line 554 of file [rx_mtu.c](#).

```
00555 {
00556     RX_VALIDATE_PTR(config, s_tag, "config pointer is nullptr");
00557
00558     volatile rx_mtu_channel_regs_t* const mtu =
00559         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00560     if (mtu == nullptr) {
00561         return k_rx_err_invalid_arg;
00562     }
00563
00564     /* Calculate period from frequency */
00565     uint16_t period;
00566     rx_err_t err = internal_calculate_period(config->frequency_hz, &period);
00567     if (err != k_rx_ok) {
00568         return err;
00569     }
00570
00571     rx_log_info(s_tag, "Initializing MTU");
00572
00573     /* Enable MTU module (clear module stop bit) */
00574     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00575
00576     system_regs()->mstpcra &= ~(k_mtu_bit_one « k_mtu_mstp_a_mtu0_4);
00577
00578     *prcr_reg() = k_rx_prcr_lock;
00579
00580     /* Stop timer before configuration */
00581     err = rx_mtu_stop(channel);
00582     if (err != k_rx_ok) {
00583         return err;
00584     }
00585
00586     /* Configure timer control register
00587      * - PCLKA/1 (120 MHz)
00588      * - Clear counter on TGRA compare match
00589      */
00590     mtu->tcr = k_mtu_tcr_tpsc_1 | k_mtu_tcr_cclr_tgra;
00591
00592     /* Configure PWM mode 1 (triangle wave, center-aligned) */
00593     mtu->tmdr = k_mtu_tmdr_md_pwm1;
00594
00595     /* Configure I/O control registers for PWM outputs
00596      * TIORH: Controls MTIOCA (output A) and MTIOCB (output B)
00597      * TIORL: Controls MTIOCC (output C) and MTIOCD (output D)
00598      *
00599      * For PWM mode 1:
00600      * - Initial low, high on up-count compare, low on down-count compare
00601      */
00602     mtu->tiorh = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCA */
00603                 (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCB */
00604     mtu->tiorl = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCC */
00605                 (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCD */
00606
00607     /* Set period (TGRA = top of triangle wave) */
00608     mtu->tgra = period;
00609
00610     /* Set initial duty cycle to 0% for all outputs */
00611     mtu->tgrb = k_mtu_period_zero; /* MTIOCB duty */
00612     mtu->tgrc = k_mtu_period_zero; /* MTIOCC duty */
00613     mtu->tgrd = k_mtu_period_zero; /* MTIOCD duty */
00614 }
```

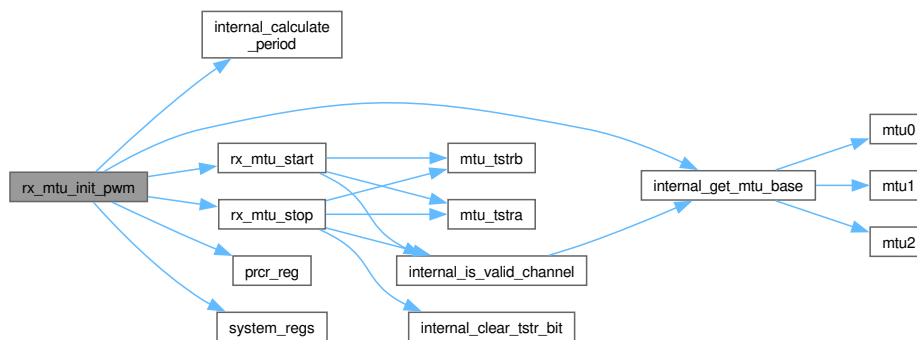
```

00615  /* Clear counter */
00616  mtu->tcnt = k_mtu_period_zero;
00617
00618  /* Save period for duty cycle calculations */
00619  s_mtu_period[channel] = period;
00620  s_mtu_initialized[channel] = true;
00621
00622  /* Start timer */
00623  err = rx_mtu_start(channel);
00624  if (err != k_rx_ok) {
00625      return err;
00626  }
00627
00628  rx_log_info(s_tag, "MTU initialized successfully");
00629
00630  return k_rx_ok;
00631 }

```

References [rx_mtu_config_t::frequency_hz](#), [internal_calculate_period\(\)](#), [internal_get_mtu_base\(\)](#), [k_mtu_bit_one](#), [k_mtu_mstp_a_mtu0_4](#), [k_mtu_period_zero](#), [k_mtu_tcr_cclr_tgra](#), [k_mtu_tcr_tpsc_1](#), [k_mtu_tior_high_shift](#), [k_mtu_tior_init_low](#), [k_mtu_tior_low_shift](#), [k_mtu_tmdr_md_pwm1](#), [rx_system_regs_t::mstp_cra](#), [prcr_reg\(\)](#), [rx_mtu_start\(\)](#), [rx_mtu_stop\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), [s_tag](#), [system_regs\(\)](#), [rx_mtu_channel_regs_t::tcnt](#), [rx_mtu_channel_regs_t::tcr](#), [rx_mtu_channel_regs_t::tgra](#), [rx_mtu_channel_regs_t::tgrb](#), [rx_mtu_channel_regs_t::tgrc](#), [rx_mtu_channel_regs_t::tgrd](#), [rx_mtu_channel_regs_t::tiorh](#), [rx_mtu_channel_regs_t::tiorl](#), and [rx_mtu_channel_regs_t::tmdr](#).

Here is the call graph for this function:



8.91.4.8 rx_mtu_set_duty()

```

rx_err_t rx_mtu_set_duty (
    const rx_mtu_channel_t channel,
    rx_mtu_output_t output,
    const float duty_percent) [nodiscard]

```

Set PWM duty cycle.

Updates duty cycle for specified output channel. Change takes effect on next PWM period for glitch-free operation.

Parameters

in	channel	MTU channel (0-4, 6-7)
in	output	Output channel (A/B/C/D)
in	duty_percent	Duty cycle in percent (0.0 - 100.0)

Returns

`k_rx_ok` on success
`k_rx_err_invalid_arg` if channel, output, or duty is invalid
`k_rx_err_invalid_state` if channel not initialized

Set PWM duty cycle.

Converts a percentage value to raw timer counts using the cached period and writes it to the appropriate TGR register. The update is buffered and takes effect at the next PWM period boundary (glitch-free).

Algorithm:

1. Validate channel and initialization state
2. Validate `duty_percent` in range [0.0, 100.0]
3. Retrieve cached period from `s_mtu_period[channel]`
4. Compute `duty_count = (duty_percent * period) / 100`
5. Write `duty_count` to TGR register via [internal_set_duty_raw_mtu\(\)](#)

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
`duty_percent` must be in [0.0, 100.0]

Postcondition

TGR register updated with new duty count
Change takes effect at next PWM period boundary

Note

Not thread-safe: caller must prevent concurrent access

```
rx_err_t err = rx_mtu_set_duty(k_mtu_channel_0, k_mtu_output_b, 50.0f);
```

See also

[rx_mtu_set_duty_raw\(\)](#) Set duty using raw count value
[rx_mtu_get_duty\(\)](#) Read current duty cycle

Since

Version 1.0.0

Definition at line 668 of file [rx_mtu.c](#).

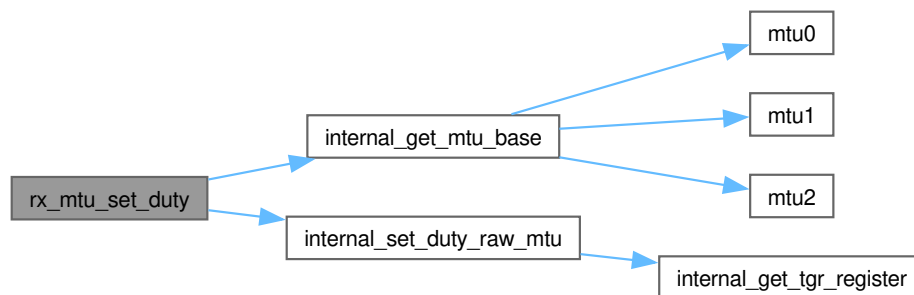
```

00669 {
00670     volatile rx_mtu_channel_regs_t* const mtu =
00671         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00672     if (mtu == nullptr) {
00673         return k_rx_err_invalid_arg;
00674     }
00675     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00676     if (duty_percent < (float)k_mtu_duty_min || duty_percent > (float)k_mtu_duty_max) {
00677         rx_log_error(s_tag, "Invalid duty cycle");
00678         return k_rx_err_invalid_arg;
00679     }
00680     /* Convert percentage to count value */
00681     const uint16_t period = s_mtu_period[channel];
00682     const uint16_t duty_count = (uint16_t)((duty_percent * period) / (float)k_mtu_duty_divisor);
00683     return internal_set_duty_raw_mtu(mtu, output, duty_count);
00684 }

```

References [internal_get_mtu_base\(\)](#), [internal_set_duty_raw_mtu\(\)](#), [k_mtu_duty_divisor](#), [k_mtu_duty_max](#), [k_mtu_duty_min](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.91.4.9 rx_mtu_set_duty_raw()

```

rx_err_t rx_mtu_set_duty_raw (
    const rx_mtu_channel_t channel,
    rx_mtu_output_t output,
    uint16_t duty_count) [nodiscard]

```

Set PWM duty cycle (raw count value).

Updates duty cycle using raw counter value for efficiency. Useful in tight control loops where floating-point calculation overhead should be avoided.

Parameters

in	channel	MTU channel (0-4, 6-7)
in	output	Output channel (A/B/C/D)
in	duty_count	Duty cycle count (0 - period_count)

Returns

[k_rx_ok](#) on success
[k_rx_err_invalid_arg](#) if channel, output, or count is invalid
[k_rx_err_invalid_state](#) if channel not initialized

Set PWM duty cycle (raw count value).

Directly sets the compare register (TGR) for the specified output to `duty_count` timer ticks. Values larger than the channel period are clamped to the period value (100% duty). The update is buffered.

Use this function when the caller has already converted a duty percentage to counts (e.g. via [rx_mtu_get_period\(\)](#)) to avoid redundant floating-point multiplication.

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
`duty_count` should be in `[0, period]`; values above period are clamped

Postcondition

TGR register written with (possibly clamped) `duty_count`
 Change takes effect at next PWM period boundary

Note

Not thread-safe: caller must prevent concurrent access

```
uint16_t period;
rx_mtu_get_period(k_mtu_channel_0, &period);
rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, period / 2U); // 50%
```

See also

[rx_mtu_set_duty\(\)](#) Set duty using floating-point percentage
[rx_mtu_get_period\(\)](#) Retrieve period for count calculation

Since

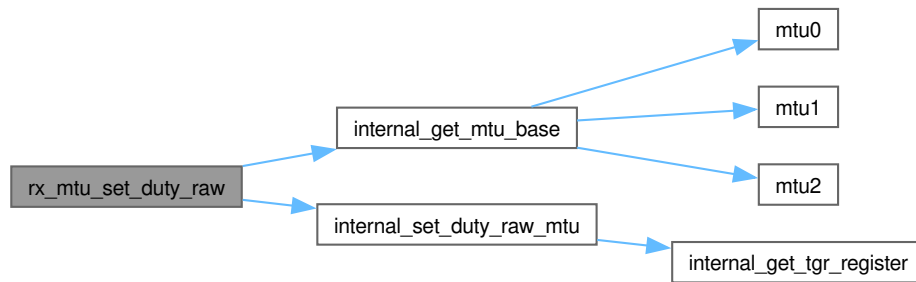
Version 1.0.0

Definition at line 724 of file [rx_mtu.c](#).

```
00725 {
00726     volatile rx_mtu_channel_regs_t* const mtu =
00727         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00728     if (mtu == nullptr) {
00729         return k_rx_err_invalid_arg;
00730     }
00731
00732     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00733
00734     /* Clamp to period */
00735     const uint16_t period = s_mtu_period[channel];
00736     if (duty_count > period) {
00737         duty_count = period;
00738     }
00739
00740     return internal_set_duty_raw_mtu(mtu, output, duty_count);
00741 }
```

References [internal_get_mtu_base\(\)](#), [internal_set_duty_raw_mtu\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.91.4.10 rx'mtu'start()

```
rx_err_t rx_mtu_start (
    const rx_mtu_channel_t channel) [nodiscard]
```

Start MTU timer.

Starts the timer counter for PWM generation. Timer is automatically started during init, but can be stopped/restarted.

Parameters

in	channel	MTU channel (0-4, 6-7)
----	---------	------------------------

Returns

k_rx_ok on success
k_rx_err_invalid_arg if channel is invalid

Start MTU timer.

Sets the Count Start (CSTn) bit in the appropriate TSTR register to begin counter operation. Channels 0-4 use TSTRA, channels 6-7 use TSTRB.

8.91.4.11 TSTR Bit Mapping

Channel	Register	Bit
MTU0	TSTRA	CST0
MTU1	TSTRA	CST1
MTU2	TSTRA	CST2
MTU3	TSTRA	CST3
MTU4	TSTRA	CST4
MTU6	TSTRB	CST6
MTU7	TSTRB	CST7

Precondition

Channel initialized via [rx_mtu_init_pwm\(\)](#)

Postcondition

Timer counting, PWM output active

Note

Timer automatically started during init

Idempotent: Safe to call when already running

See also

[rx_mtu_stop\(\)](#) Stop the timer

Since

Version 1.0.0

Definition at line 952 of file [rx_mtu.c](#).

```

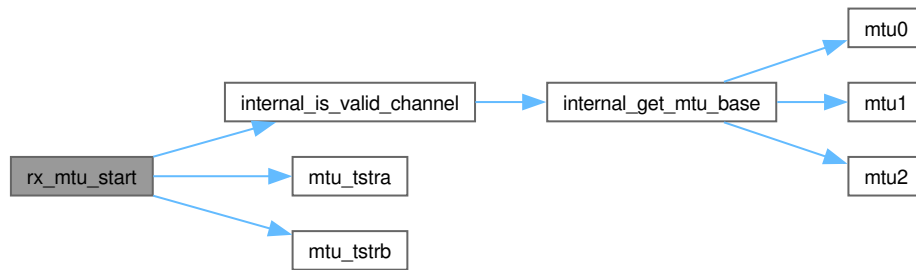
00953 {
00954     if (!internal_is_valid_channel(channel)) {
00955         return k_rx_err_invalid_arg;
00956     }
00957
00958     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00959
00960     /* Set corresponding bit in TSTR register */
00961     switch (channel) {
00962         case k_mtu_channel_0:
00963             mtu_tstra()->tstr |= k_mtu_tstr_cst0;
00964             break;
00965         case k_mtu_channel_1:
00966             mtu_tstra()->tstr |= k_mtu_tstr_cst1;
00967             break;
00968         case k_mtu_channel_2:
00969             mtu_tstra()->tstr |= k_mtu_tstr_cst2;
00970             break;
00971         case k_mtu_channel_3:
00972             mtu_tstra()->tstr |= k_mtu_tstr_cst3;
00973             break;
00974         case k_mtu_channel_4:
00975             mtu_tstra()->tstr |= k_mtu_tstr_cst4;
00976             break;
00977         case k_mtu_channel_6:
00978             mtu_tstrb()->tstr |= k_mtu_tstr_cst6;
00979             break;
00980         case k_mtu_channel_7:
00981             mtu_tstrb()->tstr |= k_mtu_tstr_cst7;
00982             break;
00983         default:
00984             return k_rx_err_invalid_arg;
00985     }
00986
00987     return k_rx_ok;
00988 }

```

References [internal_is_valid_channel\(\)](#), [k_mtu_channel_0](#), [k_mtu_channel_1](#), [k_mtu_channel_2](#), [k_mtu_channel_3](#), [k_mtu_channel_4](#), [k_mtu_channel_6](#), [k_mtu_channel_7](#), [k_mtu_tstr_cst0](#), [k_mtu_tstr_cst1](#), [k_mtu_tstr_cst2](#), [k_mtu_tstr_cst3](#), [k_mtu_tstr_cst4](#), [k_mtu_tstr_cst6](#), [k_mtu_tstr_cst7](#), [mtu_tstra\(\)](#), [mtu_tstrb\(\)](#), [s_mtu_initialized](#), [s_tag](#), and [rx_mtu_tstr_regs_t::tstr](#).

Referenced by [rx_mtu_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.91.4.12 rx_mtu_stop()

```
rx_err_t rx_mtu_stop (
    const rx_mtu_channel_t channel) [nodiscard]
```

Stop MTU timer.

Stops the timer counter. PWM outputs will hold their last state.

Parameters

in	channel	MTU channel (0-4, 6-7)
----	---------	------------------------

Returns

k_rx_ok on success

k_rx_err_invalid_arg if channel is invalid

Stop MTU timer.

Clears the Count Start (CSTn) bit in the appropriate TSTR register to halt counter operation. Counter value and output state are preserved.

Precondition

None (can be called on uninitialized channel)

Postcondition

Timer stopped, counter preserved

Output pins hold current state

Note

Does NOT check initialization flag (safe for use during init)

Idempotent: Safe to call when already stopped

Warning

Outputs may remain HIGH - disable outputs for safe state

See also

[rx_mtu_start\(\)](#) Resume the timer

[rx_mtu_enable_output\(\)](#) Disable outputs

Since

Version 1.0.0

Definition at line 1014 of file [rx_mtu.c](#).

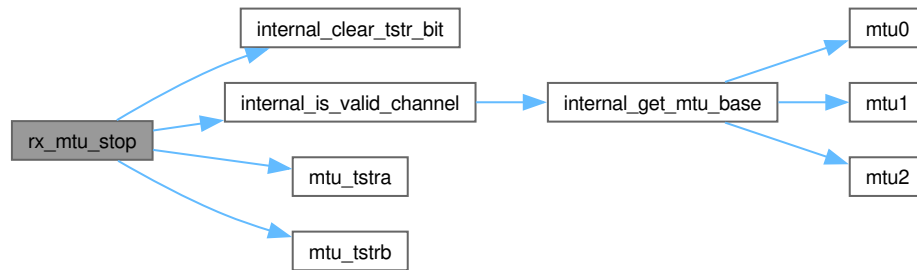
```

01015 {
01016     if (internal_is_valid_channel(channel)) {
01017         return k_rx_err_invalid_arg;
01018     }
01019
01020     /* Clear corresponding bit in TSTR register */
01021     rx_err_t err;
01022     switch (channel) {
01023         case k_mtu_channel_0:
01024             err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst0);
01025             break;
01026         case k_mtu_channel_1:
01027             err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst1);
01028             break;
01029         case k_mtu_channel_2:
01030             err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst2);
01031             break;
01032         case k_mtu_channel_3:
01033             err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst3);
01034             break;
01035         case k_mtu_channel_4:
01036             err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst4);
01037             break;
01038         case k_mtu_channel_6:
01039             err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst6);
01040             break;
01041         case k_mtu_channel_7:
01042             err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst7);
01043             break;
01044         default:
01045             return k_rx_err_invalid_arg;
01046     }
01047
01048     return err;
01049 }
```

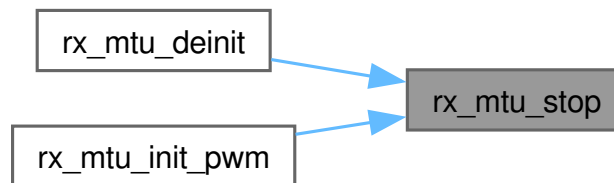
References [internal_clear_tstr_bit\(\)](#), [internal_is_valid_channel\(\)](#), [k_mtu_channel_0](#), [k_mtu_channel_1](#), [k_mtu_channel_2](#), [k_mtu_channel_3](#), [k_mtu_channel_4](#), [k_mtu_channel_6](#), [k_mtu_channel_7](#), [k_mtu_tstr_cst0](#), [k_mtu_tstr_cst1](#), [k_mtu_tstr_cst2](#), [k_mtu_tstr_cst3](#), [k_mtu_tstr_cst4](#), [k_mtu_tstr_cst6](#), [k_mtu_tstr_cst7](#), [mtu_tstra\(\)](#), and [mtu_tstrb\(\)](#).

Referenced by [rx_mtu_deinit\(\)](#), and [rx_mtu_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.92 rx_mtu.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_mtu.h
00003  * @brief MTU PWM Driver for RX72N Multi-Function Timer Unit
00004  *
00005  * @details
00006  * **Production-grade PWM driver** for the RX72N Multi-Function Timer Unit (MTU3)
00007  * peripheral. This module provides 16-bit PWM generation for motor control,
00008  * quadrature encoder input, and general-purpose timing applications.
00009  *
00010  * ## Purpose and Design Rationale
00011  *
00012  * The MTU (Multi-Function Timer Unit) is a versatile timer peripheral that
00013  * complements the GPTW module:
00014  *
00015  * | Feature | MTU3 | GPTW |
00016  * |-----|-----|-----|
00017  * | Counter Width | 16-bit | 32-bit |
00018  * | Channels | 8 (MTU0-4, MTU6-7) | 4 (GPTW0-3) |
00019  * | Outputs per Channel | Up to 4 (A/B/C/D) | 2 (A/B) |
00020  * | Phase Counting | Yes (encoder input) | Limited |
00021  * | Complementary PWM | Yes (with deadtime) | Yes |
00022  * | Primary Use | Encoders, PWM | Motor PWM |
00023  *
00024  * **STAR Project Usage**:
00025  * - MTU1/MTU2: Quadrature encoder input (see rx_mtu_encoder.h)
00026  * - GPTW0-3: Motor PWM output (see rx_gptw.h) - **recommended for motors**
00027  *
00028  * ## Key Features
00029  *
00030  * **PWM Generation**:
00031  * - 8 timer channels with independent configuration
00032  * - PWM mode 1: Center-aligned (triangle wave) for symmetric current ripple
00033  * - PWM mode 2: Edge-aligned (sawtooth wave) for simple applications
00034  * - Complementary outputs with configurable deadtime
00035  * - Up to 4 outputs per channel (A/B/C/D) on some channels
00036  *
00037  * **Encoder Interface** (MTU1/MTU2):
00038  * - Phase counting mode for quadrature encoders
00039  * - 16-bit pulse count with overflow detection
00040  * - See rx_mtu_encoder.h for encoder-specific functions
00041  *
00042  * **Timing Features**:
00043  * - Input capture for pulse width measurement
00044  * - Output compare for waveform generation
00045  * - Cascade operation for extended range
00046  *
00047  * ## Hardware Architecture
00048  *

```



```

00049 * @par Register Base Addresses:
00050 *
00051 * | Channel | Base Address | Outputs | Notes |
00052 * |-----|-----|-----|-----|
00053 * | MTU0 | 0x000C1300 | A,B,C,D | 4 outputs |
00054 * | MTU1 | 0x000C1380 | A,B | Encoder input |
00055 * | MTU2 | 0x000C1400 | A,B | Encoder input |
00056 * | MTU3 | 0x000C1200 | A,B,C,D | 4 outputs, pairs with MTU4 |
00057 * | MTU4 | 0x000C1201 | A,B,C,D | 4 outputs, pairs with MTU3 |
00058 * | MTU6 | 0x000C1A00 | A,B,C,D | 4 outputs, pairs with MTU7 |
00059 * | MTU7 | 0x000C1A01 | A,B,C,D | 4 outputs, pairs with MTU6 |
00060 *
00061 * Note: MTU5 (U/V/W sub-channels at 0x000C1C80/90/A0) is not supported by this HAL.
00062 *
00063 * @par Clock Configuration:
00064 *
00065 * | Parameter | Value | Notes |
00066 * |-----|-----|-----|
00067 * | Clock Source | PCLKA | Peripheral Clock A |
00068 * | Clock Frequency | 120 MHz | After system clock configuration |
00069 * | Counter Resolution | 8.33 ns | 1/120MHz |
00070 * | Max Period (16-bit) | 546 us | 65535 counts at 120 MHz |
00071 * | Min PWM Frequency | 1831 Hz | 1 / 546 us |
00072 *
00073 * ## PWM Modes
00074 *
00075 * **PWM Mode 1 (Center-Aligned)**:
00076 * - Counter counts up to period, then down to 0
00077 * - Symmetric current ripple (better for motors)
00078 * - Effective PWM frequency = PCLKA / (2 x period)
00079 *
00080 * **PWM Mode 2 (Edge-Aligned)**:
00081 * - Counter counts up to period, then resets
00082 * - Asymmetric ripple
00083 * - Effective PWM frequency = PCLKA / period
00084 *
00085 * @par PWM Frequency Calculation:
00086 *
00087 * For center-aligned (PWM mode 1):
00088 * @f[
00089 *  $f_{\text{PWM}} = \frac{f_{\text{PCLKA}}}{2 \times (\text{TGRA} + 1)}$ 
00090 * @f]
00091 *
00092 * For edge-aligned (PWM mode 2):
00093 * @f[
00094 *  $f_{\text{PWM}} = \frac{f_{\text{PCLKA}}}{\text{TGRA} + 1}$ 
00095 * @f]
00096 *
00097 * ## Performance Characteristics
00098 *
00099 * | Operation | Execution Time | Memory Usage | Notes |
00100 * |-----|-----|-----|-----|
00101 * | rx_mtu_init_pwm() | ~50 us | 0 heap | One-time setup |
00102 * | rx_mtu_set_duty() | ~5 us | 0 | Float calculation |
00103 * | rx_mtu_set_duty_raw() | ~1 us | 0 | Direct register write |
00104 * | rx_mtu_enable_output() | ~1 us | 0 | Single register write |
00105 *
00106 * **Memory Footprint**:
00107 * - Static variables: ~32 bytes (initialization flags)
00108 * - Code size: ~1.5 KB (all functions)
00109 * - Stack usage: < 48 bytes (deepest call)
00110 * - Heap usage: 0 bytes (zero dynamic allocation)
00111 *
00112 * @par Hardware Requirements:
00113 *
00114 * | Component | Requirement | Notes |
00115 * |-----|-----|-----|
00116 * | MCU | Renesas RX72N | MTU3 peripheral required |
00117 * | Clock | PCLKA @ 120 MHz | System clock configuration |
00118 * | Power | 3.3V I/O | LVCMOS compatible |
00119 *
00120 * @par Module Dependencies:
00121 *
00122 * **This module depends on**:
00123 * - `rx_err.h` - Error code definitions
00124 * - `rx72n_mtu_regs.h` - MTU register definitions
00125 *
00126 * **This module is used by**:
00127 * - `rx_mtu_encoder.c` - Encoder interface (MTU1/MTU2)
00128 * - Application code for auxiliary PWM outputs
00129 *
00130 * @par NASA Power of 10 Compliance:
00131 *
00132 * | Rule | Status | Implementation |
00133 * |-----|-----|-----|
00134 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion |
00135 * | 2. Fixed loop bounds | [OK] | Loops bounded by channel count (8 max) |

```

```

00136 * | 3. No dynamic allocation | [OK] | Zero malloc/free, all static |
00137 * | 4. Small functions | [OK] | All functions < 60 lines |
00138 * | 5. Assertions (>=2 per function) | [OK] | Minimum 2 validation checks per function |
00139 * | 6. Narrow scope | [OK] | File-scope statics, minimal globals |
00140 * | 7. Check return values | [OK] | All functions return rx_err_t |
00141 * | 8. Limited preprocessor | [OK] | C23 typed enums, minimal macros |
00142 * | 9. Pointer restrictions | [OK] | Single-level pointers only |
00143 * | 10. Compiler warnings | [OK] | Compiles with -Wall -Wextra -Werror |
00144 *
00145 * @par SOLID Principles:
00146 *
00147 * **Single Responsibility (S)**:
00148 * - This module handles ONLY MTU PWM configuration
00149 * - Encoder functions are in rx_mtu_encoder.h
00150 *
00151 * **Open/Closed (O)**:
00152 * - Extensible via rx_mtu_config_t structure
00153 * - New modes can be added without modifying existing code
00154 *
00155 * **Liskov Substitution (L)**:
00156 * - All functions return rx_err_t with consistent semantics
00157 * - Channel and output parameters are validated uniformly
00158 *
00159 * **Interface Segregation (I)**:
00160 * - PWM functions separate from encoder functions
00161 * - Float and raw duty cycle APIs separate
00162 *
00163 * **Dependency Inversion (D)**:
00164 * - Depends on abstract error codes
00165 * - Can be mocked for unit testing
00166 *
00167 * @see rx_gptw.h GPTW driver (preferred for motor PWM)
00168 * @see rx_mtu_encoder.h Encoder interface using MTU phase counting
00169 * @see rx72n_mtu_regs.h Register definitions
00170 * @see RX72N Hardware Manual Section 24 - Multi-Function Timer Unit 3
00171 *
00172 * @author Locked, Inc.
00173 * @date 2026-01-27
00174 * @copyright Copyright (c) 2026 Locked Inc.
00175 * SPDX-License-Identifier: MIT
00176 * @since Version 1.0.0
00177 */
00178
00179 #pragma once
00180
00181 #include <stdbool.h>
00182 #include <stdint.h>
00183
00184 #include "rx_err.h"
00185
00186 #ifdef __cplusplus
00187 extern "C" {
00188 #endif
00189
00190 /*
=====
00191 * Type Definitions
00192 *
=====
00193 */
00194
00195 /**
00196 * @enum rx_mtu_channel_t
00197 * @brief MTU Channel Identifiers
00198 *
00199 * @details
00200 * Identifies the MTU timer channels available on the RX72N. Note that MTU5
00201 * does not exist in this MCU family. Channels are grouped for specific functions:
00202 *
00203 * ## Channel Groups and Functions
00204 *
00205 * **General Purpose PWM** (MTU0, MTU3, MTU4, MTU6, MTU7):
00206 * - Up to 4 outputs per channel (A, B, C, D)
00207 * - Complementary PWM with deadline
00208 * - MTU3+MTU4 and MTU6+MTU7 can be paired for synchronized operation
00209 *
00210 * **Encoder Input** (MTU1, MTU2):
00211 * - Phase counting mode for quadrature encoders
00212 * - 16-bit pulse count with direction detection
00213 * - Used by rx_mtu_encoder.h for motor position feedback
00214 *
00215 * @par Channel Specifications:
00216 *
00217 * | Channel | Base Address | Outputs | Phase Count | Paired With |
00218 * |-----|-----|-----|-----|-----|
00219 * | MTU0 | 0x000C1300 | A,B,C,D | No | - |
00220 * | MTU1 | 0x000C1380 | A,B | Yes | MTU2 |

```

```

00221 * | MTU2 | 0x000C1400 | A,B | Yes | MTU1 |
00222 * | MTU3 | 0x000C1200 | A,B,C,D | No | MTU4 |
00223 * | MTU4 | 0x000C1201 | A,B,C,D | No | MTU3 |
00224 * | MTU6 | 0x000C1A00 | A,B,C,D | No | MTU7 |
00225 * | MTU7 | 0x000C1A01 | A,B,C,D | No | MTU6 |
00226 *
00227 * @par STAR Project Usage:
00228 *
00229 * | Channel | Application | Configuration |
00230 * |-----|-----|-----|
00231 * | MTU1 | Encoder 0-1 (Motor 0, Motor 1) | Phase counting mode |
00232 * | MTU2 | Encoder 2-3 (Motor 2, Motor 3) | Phase counting mode |
00233 * | MTU0,3,4,6,7 | Available | General purpose PWM |
00234 *
00235 * @par Usage Example:
00236 * @code{.c}
00237 * // Initialize MTU0 for PWM output
00238 * rx_mtu_config_t config = {
00239 *     .frequency_hz = 10000,
00240 *     .deadtime_ns = 500,
00241 *     .enable_complementary = true,
00242 *     .invert_polarity = false
00243 * };
00244 *
00245 * rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
00246 * if (err != k_rx_ok) {
00247 *     rx_log_error("MTU", "Init failed: %d", err);
00248 * }
00249 * @endcode
00250 *
00251 * @invariant Channel 5 is not defined (MTU5 hardware exists but is not supported by this HAL)
00252 * @invariant Channel values match hardware register offsets
00253 *
00254 * @note C23 typed enum with uint8_t underlying type
00255 * @note Value 5 is intentionally skipped (MTU5 U/V/W sub-channels not supported by this HAL)
00256 *
00257 * @see rx_mtu_encoder.h Encoder functions using MTU1/MTU2
00258 * @see RX72N Hardware Manual Section 24 - MTU3 channels
00259 *
00260 * @since Version 1.0.0
00261 */
00262 typedef enum : uint8_t {
00263     k_mtu_channel_0 =
00264         0, /**< MTU0: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1300. Standalone operation */
00265     k_mtu_channel_1 =
00266         1, /**< MTU1: Encoder input channel. 2 outputs (A,B). Base: 0x000C1380. Phase counting with MTU2 */
00267     k_mtu_channel_2 =
00268         2, /**< MTU2: Encoder input channel. 2 outputs (A,B). Base: 0x000C1400. Phase counting with MTU1 */
00269     k_mtu_channel_3 =
00270         3, /**< MTU3: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1200. Pairs with MTU4 */
00271     k_mtu_channel_4 =
00272         4, /**< MTU4: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1201. Pairs with MTU3 */
00273     k_mtu_channel_6 =
00274         6, /**< MTU6: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A00. Pairs with MTU7 */
00275     k_mtu_channel_7 =
00276         7, /**< MTU7: General purpose timer. 4 outputs (A,B,C,D). Base: 0x000C1A01. Pairs with MTU6 */
00277 } rx_mtu_channel_t;
00278
00279 /**
00280 * @enum rx_mtu_output_t
00281 * @brief MTU Output Channel Selection
00282 *
00283 * @details
00284 * Selects which output pin to configure for PWM or compare match output.
00285 * MTU channels can have 2 or 4 outputs depending on the channel.
00286 *
00287 * ## Output Availability by Channel
00288 *
00289 * | Channel | Output A | Output B | Output C | Output D |
00290 * |-----|-----|-----|-----|-----|
00291 * | MTU0 | [OK] | [OK] | [OK] | [OK] |
00292 * | MTU1 | [OK] | [OK] | - | - |
00293 * | MTU2 | [OK] | [OK] | - | - |
00294 * | MTU3 | [OK] | [OK] | [OK] | [OK] |
00295 * | MTU4 | [OK] | [OK] | [OK] | [OK] |
00296 * | MTU6 | [OK] | [OK] | [OK] | [OK] |
00297 * | MTU7 | [OK] | [OK] | [OK] | [OK] |
00298 *
00299 * ## Complementary Pairs
00300 *
00301 * When complementary output is enabled:
00302 * - A and B form a complementary pair
00303 * - C and D form a complementary pair (on channels with 4 outputs)
00304 * - Deadtime is inserted between transitions
00305 *
00306 * @par Usage Example:
00307 * @code{.c}

```

```

00308 * // Set 50% duty on output A of MTU0
00309 * rx_mtu_set_duty(k_mtu_channel_0, k_mtu_output_a, 50.0f);
00310 *
00311 * // In complementary mode, output B is automatically inverted
00312 * // with deadtime inserted between transitions
00313 * @endcode
00314 *
00315 * @invariant Output C and D only valid for channels with 4 outputs
00316 * @invariant Output values are 0-3 (4 possible outputs)
00317 *
00318 * @note C23 typed enum with uint8_t underlying type
00319 * @warning Using output C/D on MTU1/MTU2 will return k_rx_err_invalid_arg
00320 *
00321 * @see rx_mtu_channel_t Channel selection
00322 * @see rx_mtu_set_duty() Set duty cycle on specific output
00323 *
00324 * @since Version 1.0.0
00325 */
00326 typedef enum : uint8_t {
00327     k_mtu_output_a =
00328         0, /**< MTIOCA output pin. Available on all channels. Compare register: TGRA. Primary PWM output */
00329     k_mtu_output_b =
00330         1, /**< MTIOCB output pin. Available on all channels. Compare register: TGRB. Complementary to A when enabled */
00331     k_mtu_output_c =
00332         2, /**< MTIOCC output pin. MTU0,3,4,6,7 only. Compare register: TGRC. Secondary PWM output */
00333     k_mtu_output_d =
00334         3, /**< MTIOCD output pin. MTU0,3,4,6,7 only. Compare register: TGRD. Complementary to C when enabled */
00335 } rx_mtu_output_t;
00336
00337 /**
00338 * @struct rx_mtu_config_t
00339 * @brief MTU PWM Configuration Structure
00340 *
00341 * @details
00342 * Contains all configuration parameters for initializing an MTU channel
00343 * for PWM output. This structure is passed to rx_mtu_init_pwm() to
00344 * configure the timer.
00345 *
00346 * ## Configuration Guidelines
00347 *
00348 * **Frequency Limits** (16-bit counter at 120 MHz):
00349 * - Maximum frequency: ~1.83 MHz (period = 65 counts)
00350 * - Minimum frequency: ~1831 Hz (period = 65535 counts)
00351 * - Recommended for motors: 15-25 kHz
00352 *
00353 * **Resolution Trade-off**:
00354 * Higher frequency -> lower resolution (fewer counts per period)
00355 * - 20 kHz: 6000 counts -> 0.017% resolution
00356 * - 100 kHz: 1200 counts -> 0.083% resolution
00357 *
00358 * @par Configuration Parameter Table:
00359 *
00360 * | Parameter | Min | Max | Default | Unit | Notes |
00361 * |-----|-----|-----|-----|-----|-----|
00362 * | frequency_hz | 1831 | 1830000 | 20000 | Hz | 16-bit limit |
00363 * | deadtime_ns | 0 | 65535 | 1000 | ns | 0 disables |
00364 * | enable_complementary | false | true | true | bool | A/B pairs |
00365 * | invert_polarity | false | true | false | bool | Active-low |
00366 *
00367 * @par Memory Layout:
00368 *
00369 * | Offset | Size | Field | Type | Alignment |
00370 * |-----|-----|-----|-----|-----|
00371 * | 0 | 4 | frequency_hz | uint32_t | 4 |
00372 * | 4 | 2 | deadtime_ns | uint16_t | 2 |
00373 * | 6 | 1 | enable_complementary | bool | 1 |
00374 * | 7 | 1 | invert_polarity | bool | 1 |
00375 * | **Total** | **8** | - | - | **4** |
00376 *
00377 * @par Usage Example:
00378 * @code{.c}
00379 * // Standard PWM configuration for auxiliary output
00380 * rx_mtu_config_t config = {
00381 *     .frequency_hz = 10000, // 10 kHz
00382 *     .deadtime_ns = 500, // 500 ns deadtime
00383 *     .enable_complementary = true, // A/B complementary
00384 *     .invert_polarity = false // Active-high
00385 * };
00386 *
00387 * rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
00388 * if (err != k_rx_ok) {
00389 *     rx_log_error("MTU", "Init failed: %d", err);
00390 * }
00391 * @endcode
00392 *
00393 * @invariant frequency_hz must produce period <= 65535 counts
00394 * @invariant Structure is copied during init (no need to keep alive)

```

```

00395 *
00396 * @note For motor control, prefer rx_gptw.h with 32-bit resolution
00397 * @warning Invalid configurations rejected with k_rx_err_invalid_arg
00398 *
00399 * @see rx_mtu_init_pwm() Channel initialization function
00400 * @see rx_gptw_config_t GPTW config (32-bit, recommended for motors)
00401 *
00402 * @since Version 1.0.0
00403 */
00404 typedef struct {
00405 /**
00406  * @brief PWM frequency in Hz
00407  * @details
00408  * Sets the PWM switching frequency. Limited by 16-bit counter:
00409  * - Min: 1831 Hz (period = 65535 at 120 MHz)
00410  * - Max: ~1.83 MHz (period = 65 at 120 MHz)
00411  *
00412  * @par Valid Range: [1831, 1830000] Hz
00413  * @par Recommended: 10000-20000 Hz for PWM applications
00414  */
00415 uint32_t frequency_hz;
00416
00417 /**
00418  * @brief Deadtime in nanoseconds
00419  * @details
00420  * Deadtime inserted between complementary output transitions.
00421  * Prevents shoot-through in H-bridge configurations.
00422  * Set to 0 to disable deadtime insertion.
00423  *
00424  * @par Valid Range: [0, 65535] ns
00425  * @par Recommended: 500-1000 ns
00426  */
00427 uint16_t deadtime_ns;
00428
00429 /**
00430  * @brief Enable complementary outputs
00431  * @details
00432  * When true, output pairs (A/B and C/D) are complementary:
00433  * - Output B = inverted Output A (with deadtime)
00434  * - Output D = inverted Output C (with deadtime)
00435  *
00436  * Required for H-bridge motor control applications.
00437  */
00438 bool enable_complementary;
00439
00440 /**
00441  * @brief Invert PWM polarity
00442  * @details
00443  * When true, inverts the PWM output polarity:
00444  * - false: HIGH during duty cycle (active-high)
00445  * - true: LOW during duty cycle (active-low)
00446  */
00447 bool invert_polarity;
00448 } rx_mtu_config_t;
00449
00450 /*
=====
00451 * Public API
00452 *
=====
00453 */
00454
00455 /**
00456  * @brief Initialize MTU channel for PWM output
00457  *
00458  * Configures MTU channel for PWM mode 1 (center-aligned, triangle wave).
00459  * Enables timer and starts PWM generation at 0% duty cycle.
00460  *
00461  * @param[in] channel MTU channel (0-4, 6-7)
00462  * @param[in] config PWM configuration
00463  *
00464  * @return k_rx_ok on success
00465  * @return k_rx_err_invalid_arg if channel or config is invalid
00466  * @return k_rx_err_invalid_state if MTU module not enabled
00467  */
00468 [[nodiscard]] rx_err_t rx_mtu_init_pwm(rx_mtu_channel_t channel, const rx_mtu_config_t* config);
00469
00470 /**
00471  * @brief Set PWM duty cycle
00472  *
00473  * Updates duty cycle for specified output channel.
00474  * Change takes effect on next PWM period for glitch-free operation.
00475  *
00476  * @param[in] channel MTU channel (0-4, 6-7)
00477  * @param[in] output Output channel (A/B/C/D)
00478  * @param[in] duty_percent Duty cycle in percent (0.0 - 100.0)
00479  */

```

```

00480 * @return k_rx_ok on success
00481 * @return k_rx_err_invalid_arg if channel, output, or duty is invalid
00482 * @return k_rx_err_invalid_state if channel not initialized
00483 */
00484 [[nodiscard]] rx_err_t
00485 rx_mtu_set_duty(rx_mtu_channel_t channel, rx_mtu_output_t output, float duty_percent);
00486
00487 /**
00488 * @brief Set PWM duty cycle (raw count value)
00489 *
00490 * Updates duty cycle using raw counter value for efficiency.
00491 * Useful in tight control loops where floating-point calculation overhead
00492 * should be avoided.
00493 *
00494 * @param[in] channel MTU channel (0-4, 6-7)
00495 * @param[in] output Output channel (A/B/C/D)
00496 * @param[in] duty_count Duty cycle count (0 - period_count)
00497 *
00498 * @return k_rx_ok on success
00499 * @return k_rx_err_invalid_arg if channel, output, or count is invalid
00500 * @return k_rx_err_invalid_state if channel not initialized
00501 */
00502 [[nodiscard]] rx_err_t
00503 rx_mtu_set_duty_raw(rx_mtu_channel_t channel, rx_mtu_output_t output, uint16_t duty_count);
00504
00505 /**
00506 * @brief Get current duty cycle
00507 *
00508 * @param[in] channel MTU channel (0-4, 6-7)
00509 * @param[in] output Output channel (A/B/C/D)
00510 * @param[out] duty_percent Pointer to store duty cycle in percent
00511 *
00512 * @return k_rx_ok on success
00513 * @return k_rx_err_null_ptr if duty_percent is nullptr
00514 * @return k_rx_err_invalid_arg if channel or output is invalid
00515 * @return k_rx_err_invalid_state if channel not initialized
00516 */
00517 [[nodiscard]] rx_err_t
00518 rx_mtu_get_duty(rx_mtu_channel_t channel, rx_mtu_output_t output, float* duty_percent);
00519
00520 /**
00521 * @brief Get PWM period count
00522 *
00523 * Returns the period register value (useful for raw duty cycle calculations).
00524 *
00525 * @param[in] channel MTU channel (0-4, 6-7)
00526 * @param[out] period_count Pointer to store period count
00527 *
00528 * @return k_rx_ok on success
00529 * @return k_rx_err_null_ptr if period_count is nullptr
00530 * @return k_rx_err_invalid_arg if channel is invalid
00531 * @return k_rx_err_invalid_state if channel not initialized
00532 */
00533 [[nodiscard]] rx_err_t rx_mtu_get_period(rx_mtu_channel_t channel, uint16_t* period_count);
00534
00535 /**
00536 * @brief Enable or disable PWM output
00537 *
00538 * @param[in] channel MTU channel (0-4, 6-7)
00539 * @param[in] output Output channel (A/B/C/D)
00540 * @param[in] enable True to enable output, false to disable
00541 *
00542 * @return k_rx_ok on success
00543 * @return k_rx_err_invalid_arg if channel or output is invalid
00544 * @return k_rx_err_invalid_state if channel not initialized
00545 */
00546 [[nodiscard]] rx_err_t
00547 rx_mtu_enable_output(rx_mtu_channel_t channel, rx_mtu_output_t output, bool enable);
00548
00549 /**
00550 * @brief Start MTU timer
00551 *
00552 * Starts the timer counter for PWM generation.
00553 * Timer is automatically started during init, but can be stopped/restarted.
00554 *
00555 * @param[in] channel MTU channel (0-4, 6-7)
00556 *
00557 * @return k_rx_ok on success
00558 * @return k_rx_err_invalid_arg if channel is invalid
00559 */
00560 [[nodiscard]] rx_err_t rx_mtu_start(rx_mtu_channel_t channel);
00561
00562 /**
00563 * @brief Stop MTU timer
00564 *
00565 * Stops the timer counter. PWM outputs will hold their last state.
00566 */

```

```

00567 * @param[in] channel MTU channel (0-4, 6-7)
00568 *
00569 * @return k_rx_ok on success
00570 * @return k_rx_err_invalid_arg if channel is invalid
00571 */
00572 [[nodiscard]] rx_err_t rx_mtu_stop(rx_mtu_channel_t channel);
00573 /**
00574 * @brief Deinitialize MTU channel
00575 *
00576 * Stops timer, disables outputs, and releases resources.
00577 *
00578 * @param[in] channel MTU channel (0-4, 6-7)
00579 *
00580 * @return k_rx_ok on success
00581 * @return k_rx_err_invalid_arg if channel is invalid
00582 */
00583 [[nodiscard]] rx_err_t rx_mtu_deinit(rx_mtu_channel_t channel);
00584
00585 #ifdef __cplusplus
00586 }
00587 #endif

```

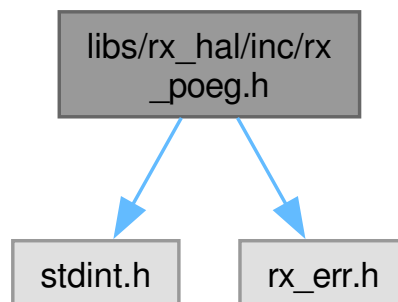
8.93 libs/rx_hal/inc/rx_poeg.h File Reference

POEG Motor Fault Protection Driver API.

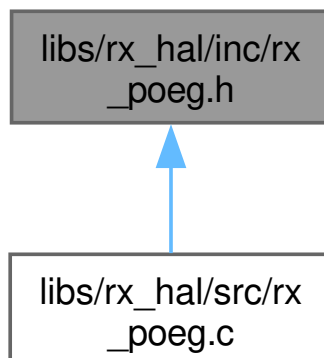
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_poeg.h:



This graph shows which files directly or indirectly include this file:



Enumerations

- enum `rx_poeg_motor_count_t` : `uint8_t` { `k_poeg_motor_count` = 4 }
Number of motors with POEG fault protection.
- enum `rx_poeg_isr_priority_t` : `uint8_t` { `k_poeg_isr_priority` = 14 }
ICU priority for POEG fault interrupts.

Functions

- `rx_err_t rx_poeg_init` (void)
Initialize POEG fault protection for all 4 motor channels.
- `rx_err_t rx_poeg_get_fault_status` (`uint8_t` motor_index, `bool *fault_active`)
Check if a motor has an active POEG fault.
- `rx_err_t rx_poeg_clear_fault` (`uint8_t` motor_index)
Clear POEG fault and re-enable motor output.
- `rx_err_t rx_poeg_software_stop` (`uint8_t` motor_index)
Trigger software emergency stop on a specific motor.
- `rx_err_t rx_poeg_clear_software_stop` (`uint8_t` motor_index)
Release software emergency stop on a specific motor.

8.93.1 Detailed Description

POEG Motor Fault Protection Driver API.

Provides hardware-level emergency PWM output disable via the POEG module. When a DRV8263H motor driver asserts its nFAULT output (active low), the corresponding GTETRGR pin triggers POEG to immediately disable GPTW PWM outputs, placing the H-bridge in a safe Hi-Z state.

System Architecture

```

* DRV8263H nFAULT (active low)
*   |
*   v
* GTETRGR pin --> POEG (PIDF flag) --> GPTW output disable (Hi-Z)
*   |                               |
*   v                               v
* ICU interrupt --> ISR --> shared_data fault flags
*

```

Motor-to-POEG Group Mapping

Motor	nFAULT Pin	GTETRGR	POEG Group	Vector
0	P15	A	A	188
1	PA6	B	B	189
2	PC4	C	C	190
3	P14	D	D	191

Two-Layer Protection

1. Hardware (automatic): POEG disables GPTW outputs within nanoseconds of nFAULT assertion. No firmware intervention required for the stop.
2. Firmware (ISR): POEG interrupt logs fault, sets `shared_data` flags, and prevents motor re-enable until fault is resolved.

See also

[rx72n_poeg_regs.h](#) POEG register definitions
[rx72n_gptw_regs.h](#) GPTW register definitions (GTINTAD)
[shared_data.h](#) Fault flag storage

Author

Locked, Inc.

Date

2026-02-10

Copyright

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Since

Version 1.0.0

Definition in file [rx_poeg.h](#).

8.93.2 Enumeration Type Documentation

8.93.2.1 `rx_poeg_isr_priority_t`

enum [rx_poeg_isr_priority_t](#) : `uint8_t`

ICU priority for POEG fault interrupts.

Priority 14 (near-maximum) ensures motor faults are handled immediately. Only NMI (level 15) has higher priority.

Since

Version 1.0.0

Enumerator

k_poeg_isr_priority	High priority for safety-critical fault ISR
---------------------	---

Definition at line 81 of file [rx_poeg.h](#).

```
00081      : uint8_t {
00082  k_poeg_isr_priority = 14, /**< High priority for safety-critical fault ISR */
00083 } rx_poeg_isr_priority_t;
```

8.93.2.2 rx_poeg_motor_count_t

```
enum rx_poeg_motor_count_t : uint8_t
```

Number of motors with POEG fault protection.

Since

Version 1.0.0

Enumerator

k_poeg_motor_count	4 motors, one POEG group each (A-D)
--------------------	-------------------------------------

Definition at line 67 of file [rx_poeg.h](#).

```
00067      : uint8_t {
00068  k_poeg_motor_count = 4, /**< 4 motors, one POEG group each (A-D) */
00069 } rx_poeg_motor_count_t;
```

8.93.3 Function Documentation

8.93.3.1 rx_poeg_clear_fault()

```
rx_err_t rx_poeg_clear_fault (
    uint8_t motor_index)
```

Clear POEG fault and re-enable motor output.

Attempts to clear the PIDF flag for the specified motor's POEG group. The flag can only be cleared when the underlying fault condition has been resolved (i.e., DRV8263H nFAULT deasserted / GTETRQ pin high).

After clearing, GPTW output re-enables at the end of the next GPTW counting cycle (delay of up to 1 PWM period, typically 50us at 20kHz).

Parameters

in	motor_index	Motor index (0-3)
----	-------------	-------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Fault cleared successfully
k_rx_err_invalid_arg	motor_index >= 4
k_rx_err_busy	Fault condition still active (nFAULT still asserted)

Precondition

Underlying fault resolved (DRV8263H nFAULT deasserted)

Postcondition

If successful, PIDF flag cleared and GPTW output will re-enable
shared_data fault_flags[motor_index] cleared

Warning

Returns k_rx_err_busy if fault source is still active

Note

Motor control task should verify estop is cleared before resuming PWM

Since

Version 1.0.0

Clear POEG fault and re-enable motor output.

Reads POEGGn, masks off the PIDF / IOCF / OSTPF latched-fault bits, and writes the result back. Refuses to clear (returns k_rx_err_busy) while the underlying nFAULT pin is still active (POEGGn.ST==0 indicates nFAULT still asserted under INV=1). After a successful clear, also clears any driver-fault E-Stop in shared_data when applicable, so the system can resume motor control if the underlying physical fault has been cleared.

Note

Not thread-safe with concurrent access to shared_data E-Stop fields. Typically called from the safety supervisor task.

Definition at line 327 of file rx_poeg.c.

```
00328 {
00329     if (motor_index >= k_poeg_motor_count) {
00330         return k_rx_err_invalid_arg;
00331     }
00332
00333     volatile rx_poegg_regs_t* poeg = s_poeg_groups[motor_index];
00334
00335     /* Check if fault source is still active (ST bit shows pin state) */
00336     /* With INV=1, ST=0 means nFAULT is still asserted (fault active) */
```

```

00337 if ((poeg->poeggn & k_poeg_st_high) == 0) {
00338     rx_log_warn_val(s_tag, "Cannot clear: nFAULT still active motor", (uint32_t)motor_index);
00339     return k_rx_err_busy;
00340 }
00341
00342 /* Clear PIDF and IOCF flags by writing 0 to them */
00343 uint32_t reg_val = poeg->poeggn;
00344 reg_val &= ~(k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected);
00345 poeg->poeggn = reg_val;
00346
00347 /* Clear e-stop if this was the cause */
00348 if (shared_data_get_estop_reason() == k_estop_reason_driver_fault) {
00349     (void)shared_data_clear_estop();
00350 }
00351
00352 rx_log_info_val(s_tag, "Fault cleared motor", (uint32_t)motor_index);
00353 return k_rx_ok;
00354 }

```

References [k_poeg_iocf_detected](#), [k_poeg_motor_count](#), [k_poeg_ostpf_detected](#), [k_poeg_pidf_detected](#), [k_poeg_st_high](#), [rx_poegg_regs_t::poeggn](#), [s_poeg_groups](#), and [s_tag](#).

8.93.3.2 rx_poeg_clear_software_stop()

```

rx_err_t rx_poeg_clear_software_stop (
    uint8_t motor_index)

```

Release software emergency stop on a specific motor.

Clears the SSF (Software Stop Flag) in the POEGGn register. GPTW output re-enables at end of next counting cycle if no other fault flags (PIDF, IOCF, OSTPF) are active.

Parameters

in	motor_index	Motor index (0-3)
----	-------------	-------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Software stop released
k_rx_err_invalid_arg	motor_index >= 4

Postcondition

SSF flag cleared in POEGGn register

See also

[rx_poeg_software_stop\(\)](#) Activate software stop

Since

Version 1.0.0

Release software emergency stop on a specific motor.

Reads POEGGn, clears the SSF bit, and writes the result back. This re-enables the GPTW outputs of the targeted motor (subject to any hardware-latched fault flags still being cleared first).

Note

Safe from any context including ISR; single read-modify-write.

Definition at line 388 of file [rx_poeg.c](#).

```
00389 {
00390     if (motor_index >= k_poeg_motor_count) {
00391         return k_rx_err_invalid_arg;
00392     }
00393
00394     uint32_t reg_val = s_poeg_groups[motor_index]->poeggn;
00395     reg_val &= ~k_poeg_ssf_stop;
00396     s_poeg_groups[motor_index]->poeggn = reg_val;
00397     return k_rx_ok;
00398 }
```

References [k_poeg_motor_count](#), [k_poeg_ssf_stop](#), and [s_poeg_groups](#).

8.93.3.3 rx_poeg_get_fault_status()

```
rx_err_t rx_poeg_get_fault_status (
    uint8_t motor_index,
    bool * fault_active)
```

Check if a motor has an active POEG fault.

Reads the POEGGn register for the specified motor's POEG group and returns whether the PIDF (port input detection flag) is set.

Parameters

in	motor_index	Motor index (0-3)
out	fault_active	true if POEG fault is active

Returns

rx_err_t Error code

Return values

k_rx_ok	Status read successfully
k_rx_err_null_ptr	fault_active is nullptr
k_rx_err_invalid_arg	motor_index >= 4

Precondition

POEG initialized via [rx_poeg_init\(\)](#)

Postcondition

No side effects on hardware state

Note

Thread Safety: Safe to call from any context (read-only register access)

Since

Version 1.0.0

Check if a motor has an active POEG fault.

Reads POEGn for the specified motor's POEG group and checks the OR of the three fault-source bits: PIDF (port input fault), IOCF (output comparator fault), and OSTPF (over-stop signal fault). Reports whether any of them are currently latched.

Note

Safe from any context including ISR; single register read.

Definition at line 297 of file [rx_poeg.c](#).

```
00298 {
00299   RX_CHECK_NULL_PTR(fault_active, s_tag, "fault_active is nullptr");
00300
00301   if (motor_index >= k_poeg_motor_count) {
00302       return k_rx_err_invalid_arg;
00303   }
00304
00305   const uint32_t status = s_poeg_groups[motor_index]->poeggn;
00306   *fault_active =
00307       ((status & (k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected)) != 0);
00308
00309   return k_rx_ok;
00310 }
```

References [k_poeg_iocf_detected](#), [k_poeg_motor_count](#), [k_poeg_ostpf_detected](#), [k_poeg_pidf_detected](#), [s_poeg_groups](#), and [s_tag](#).

8.93.3.4 rx_poeg_init()

```
rx_err_t rx_poeg_init (
    void )
```

Initialize POEG fault protection for all 4 motor channels.

Configures POEG Groups A-D for hardware motor fault protection:

1. Enables external pin detection (PIDE) on each group
2. Enables noise filter (NFEN) with PCLKB/8 sampling (~400ns)
3. Inverts input (INV) since DRV8263H nFAULT is active-low

4. Links each GPTW channel to its POEG group via GTINTAD
5. Configures ICU interrupts for vectors 188-191 at priority 14

After initialization, a DRV8263H fault will:

- Immediately disable GPTW PWM output (hardware, ~nanoseconds)
- Fire POEG ISR which sets fault flags in `shared_data`

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	All 4 POEG groups initialized successfully
<code>k_rx_err_hw_init_failed</code>	POEG register write verification failed

Precondition

GPTW module enabled (`MSTPCRA.MSTPA7 = 0`)
GTETRGR pins configured via MPC: `PDR=0` (input), `PSEL=0x1E` (GTETRGR function select)
`shared_data` initialized

Postcondition

POEG Groups A-D enabled with pin detection + noise filter
GPTW0-3 GTINTAD linked to POEG Groups A-D respectively
ICU vectors 188-191 enabled at priority 14
Any existing `nFAULT` assertion will trigger immediately

Note

Thread Safety: Call once during `hardware_init`, before motor tasks start

Warning

PIDE enable bit is write-once after reset – cannot be disabled

NASA Power of 10 Compliance:

- Rule 1: No goto, setjmp, recursion
- Rule 2: Loop bounded by `k_poeg_motor_count` (4)
- Rule 3: No dynamic allocation
- Rule 5: 3 preconditions, 4 postconditions
- Rule 7: Return value checked
- Rule 8: All constants as C23 typed enums

See also

[rx_poeg_clear_fault\(\)](#) Clear fault after resolution
[rx_poeg_get_fault_status\(\)](#) Query fault state

Since

Version 1.0.0

Definition at line 260 of file [rx_poeg.c](#).

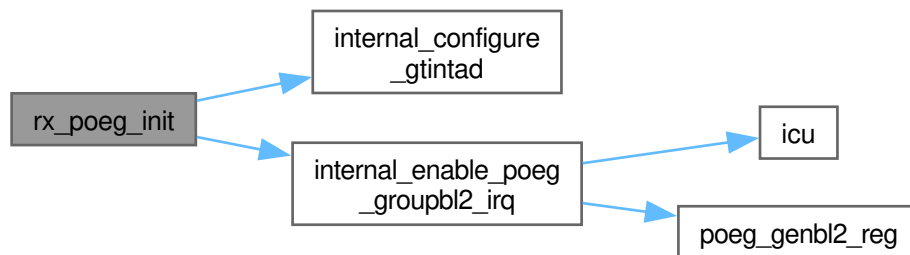
```

00261 {
00262     rx_log_info(s_tag, "Initializing POEG fault protection");
00263
00264     for (uint8_t i = 0; i < k_poeg_motor_count; i++) {
00265         /* Configure POEG group: enable pin detection, invert, noise filter */
00266         volatile rx_poegg_regs_t* poeg = s_poeg_groups[i];
00267         poeg->poeggn = k_poeg_init_value;
00268
00269         /* Verify write-once enable bits were accepted */
00270         if ((poeg->poeggn & k_poeg_pide_enable) == 0) {
00271             rx_log_error_val(s_tag, "POEG PIDE write failed motor", (uint32_t)i);
00272             return k_rx_err_hw_init_failed;
00273         }
00274
00275         /* Link GPTW channel to POEG group via GTINTAD */
00276         internal_configure_gtintad(i);
00277     }
00278
00279     /* Enable shared GROUPBL2 ICU vector (one dispatch for all POEG groups) */
00280     internal_enable_poeg_groupbl2_irq(k_poeg_isr_priority);
00281
00282     rx_log_info(s_tag, "POEG fault protection active (4 groups on GROUPBL2, priority 14)");
00283     return k_rx_ok;
00284 }

```

References [internal_configure_gtintad\(\)](#), [internal_enable_poeg_groupbl2_irq\(\)](#), [k_poeg_init_value](#), [k_poeg_isr_priority](#), [k_poeg_motor_count](#), [k_poeg_pide_enable](#), [rx_poegg_regs_t::poeggn](#), [s_poeg_groups](#), and [s_tag](#).

Here is the call graph for this function:



8.93.3.5 rx_poeg_software_stop()

```

rx_err_t rx_poeg_software_stop (
    uint8_t motor_index)

```

Trigger software emergency stop on a specific motor.

Sets the SSF (Software Stop Flag) in the POEGGn register, immediately disabling GPTW PWM output for the specified motor. This provides a firmware-controlled emergency stop independent of hardware faults.

Parameters

in	motor_index	Motor index (0-3)
----	-------------	-------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Software stop activated
k_rx_err_invalid_arg	motor_index >= 4

Postcondition

GPTW output immediately disabled for specified motor
SSF flag set in POEGGn register

See also

[rx_poeg_clear_software_stop\(\)](#) Release software stop

Since

Version 1.0.0

Trigger software emergency stop on a specific motor.

Sets POEGGn.SSF=1, force-disabling the GPTW outputs of the targeted motor as if a hardware fault had occurred. Does not touch the latched fault flags; once a Software Stop is set, the user must call [rx_poeg_clear_software_stop\(\)](#) to release.

Note

Safe from any context including ISR; single register write. Used by the safety supervisor for software-initiated E-Stop.

Definition at line 368 of file [rx_poeg.c](#).

```
00369 {  
00370     if (motor_index >= k_poeg_motor_count) {  
00371         return k_rx_err_invalid_arg;  
00372     }  
00373  
00374     s_poeg_groups[motor_index]->poeggn |= k_poeg_ssf_stop;  
00375     return k_rx_ok;  
00376 }
```

References [k_poeg_motor_count](#), [k_poeg_ssf_stop](#), and [s_poeg_groups](#).

8.94 rx_poeg.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_poeg.h
00003  * @brief POEG Motor Fault Protection Driver API
00004  *
00005  * @details
00006  * Provides hardware-level emergency PWM output disable via the POEG module.
00007  * When a DRV8263H motor driver asserts its nFAULT output (active low), the
00008  * corresponding GTETRGR pin triggers POEG to immediately disable GPTW PWM
00009  * outputs, placing the H-bridge in a safe Hi-Z state.
00010  *
00011  * @par System Architecture
00012  * @verbatim
00013  *   DRV8263H nFAULT (active low)
00014  *   |
00015  *   v
00016  *   GTETRGR pin --> POEG (PIDF flag) --> GPTW output disable (Hi-Z)
00017  *   |                                     |
00018  *   v                                     v
00019  *   ICU interrupt --> ISR --> shared_data fault flags
00020  * @endverbatim
00021  *
00022  * @par Motor-to-POEG Group Mapping
00023  * | Motor | nFAULT Pin | GTETRGR | POEG Group | Vector |
00024  * |-----|-----|-----|-----|-----|
00025  * | 0     | P15       | A       | A          | 188    |
00026  * | 1     | PA6       | B       | B          | 189    |
00027  * | 2     | PC4       | C       | C          | 190    |
00028  * | 3     | P14       | D       | D          | 191    |
00029  *
00030  * @par Two-Layer Protection
00031  * 1. **Hardware (automatic)**: POEG disables GPTW outputs within nanoseconds
00032  *   of nFAULT assertion. No firmware intervention required for the stop.
00033  * 2. **Firmware (ISR)**: POEG interrupt logs fault, sets shared_data flags,
00034  *   and prevents motor re-enable until fault is resolved.
00035  *
00036  * @see rx72n_poeg_regs.h POEG register definitions
00037  * @see rx72n_gptw_regs.h GPTW register definitions (GTINTAD)
00038  * @see shared_data.h Fault flag storage
00039  *
00040  * @author Locked, Inc.
00041  * @date 2026-02-10
00042  * @copyright Copyright (c) 2026 Locked Inc.
00043  * SPDX-License-Identifier: MIT
00044  * @since Version 1.0.0
00045  */
00046
00047 #pragma once
00048
00049 #include <stdint.h>
00050
00051 #include "rx_err.h"
00052
00053 #ifdef __cplusplus
00054 extern "C" {
00055 #endif
00056
00057 /**
00058  * Constants
00059  */
00060 */
00061
00062 /**
00063  * @enum rx_poeg_motor_count_t
00064  * @brief Number of motors with POEG fault protection
00065  * @since Version 1.0.0
00066  */
00067 typedef enum : uint8_t {
00068     k_poeg_motor_count = 4, /**< 4 motors, one POEG group each (A-D) */
00069 } rx_poeg_motor_count_t;
00070
00071 /**
00072  * @enum rx_poeg_isr_priority_t
00073  * @brief ICU priority for POEG fault interrupts
00074  *
00075  * @details
00076  * Priority 14 (near-maximum) ensures motor faults are handled immediately.
00077  * Only NMI (level 15) has higher priority.
00078  *
00079  * @since Version 1.0.0
00080  */

```

```

00081 typedef enum : uint8_t {
00082     k_poeg_isr_priority = 14, /**< High priority for safety-critical fault ISR */
00083 } rx_poeg_isr_priority_t;
00084
00085 /*
=====
00086 * Public API
00087 *
=====
00088 */
00089
00090 /**
00091 * @brief Initialize POEG fault protection for all 4 motor channels
00092 *
00093 * @details
00094 * Configures POEG Groups A-D for hardware motor fault protection:
00095 * 1. Enables external pin detection (PIDE) on each group
00096 * 2. Enables noise filter (NFEN) with PCLKB/8 sampling (~400ns)
00097 * 3. Inverts input (INV) since DRV8263H nFAULT is active-low
00098 * 4. Links each GPTW channel to its POEG group via GTINTAD
00099 * 5. Configures ICU interrupts for vectors 188-191 at priority 14
00100 *
00101 * After initialization, a DRV8263H fault will:
00102 * - Immediately disable GPTW PWM output (hardware, ~nanoseconds)
00103 * - Fire POEG ISR which sets fault flags in shared_data
00104 *
00105 * @return rx_err_t Error code
00106 * @retval k_rx_ok All 4 POEG groups initialized successfully
00107 * @retval k_rx_err_hw_init_failed POEG register write verification failed
00108 *
00109 * @pre GPTW module enabled (MSTPCRA.MSTPA7 = 0)
00110 * @pre GTETRGR pins configured via MPC: PDR=0 (input), PSEL=0x1E (GTETRGR function select)
00111 * @pre shared_data initialized
00112 *
00113 * @post POEG Groups A-D enabled with pin detection + noise filter
00114 * @post GPTW0-3 GTINTAD linked to POEG Groups A-D respectively
00115 * @post ICU vectors 188-191 enabled at priority 14
00116 * @post Any existing nFAULT assertion will trigger immediately
00117 *
00118 * @note Thread Safety: Call once during hardware_init, before motor tasks start
00119 * @warning PIDE enable bit is write-once after reset -- cannot be disabled
00120 *
00121 * @par NASA Power of 10 Compliance:
00122 * - Rule 1: No goto, setjmp, recursion
00123 * - Rule 2: Loop bounded by k_poeg_motor_count (4)
00124 * - Rule 3: No dynamic allocation
00125 * - Rule 5: 3 preconditions, 4 postconditions
00126 * - Rule 7: Return value checked
00127 * - Rule 8: All constants as C23 typed enums
00128 *
00129 * @see rx_poeg_clear_fault() Clear fault after resolution
00130 * @see rx_poeg_get_fault_status() Query fault state
00131 * @since Version 1.0.0
00132 */
00133 rx_err_t rx_poeg_init(void);
00134
00135 /**
00136 * @brief Check if a motor has an active POEG fault
00137 *
00138 * @details
00139 * Reads the POEGGn register for the specified motor's POEG group and
00140 * returns whether the PIDF (port input detection flag) is set.
00141 *
00142 * @param[in] motor_index Motor index (0-3)
00143 * @param[out] fault_active true if POEG fault is active
00144 *
00145 * @return rx_err_t Error code
00146 * @retval k_rx_ok Status read successfully
00147 * @retval k_rx_err_null_ptr fault_active is nullptr
00148 * @retval k_rx_err_invalid_arg motor_index >= 4
00149 *
00150 * @pre POEG initialized via rx_poeg_init()
00151 * @post No side effects on hardware state
00152 *
00153 * @note Thread Safety: Safe to call from any context (read-only register access)
00154 *
00155 * @since Version 1.0.0
00156 */
00157 rx_err_t rx_poeg_get_fault_status(uint8_t motor_index, bool* fault_active);
00158
00159 /**
00160 * @brief Clear POEG fault and re-enable motor output
00161 *
00162 * @details
00163 * Attempts to clear the PIDF flag for the specified motor's POEG group.
00164 * The flag can only be cleared when the underlying fault condition has
00165 * been resolved (i.e., DRV8263H nFAULT deasserted / GTETRGR pin high).

```

```

00166 *
00167 * After clearing, GPTW output re-enables at the end of the next GPTW
00168 * counting cycle (delay of up to 1 PWM period, typically 50us at 20kHz).
00169 *
00170 * @param[in] motor_index Motor index (0-3)
00171 *
00172 * @return rx_err_t Error code
00173 * @retval k_rx_ok Fault cleared successfully
00174 * @retval k_rx_err_invalid_arg motor_index >= 4
00175 * @retval k_rx_err_busy Fault condition still active (nFAULT still asserted)
00176 *
00177 * @pre Underlying fault resolved (DRV8263H nFAULT deasserted)
00178 * @post If successful, PIDF flag cleared and GPTW output will re-enable
00179 * @post shared_data fault_flags[motor_index] cleared
00180 *
00181 * @warning Returns k_rx_err_busy if fault source is still active
00182 * @note Motor control task should verify estop is cleared before resuming PWM
00183 *
00184 * @since Version 1.0.0
00185 */
00186 rx_err_t rx_poeg_clear_fault(uint8_t motor_index);
00187
00188 /**
00189 * @brief Trigger software emergency stop on a specific motor
00190 *
00191 * @details
00192 * Sets the SSF (Software Stop Flag) in the POEGGn register, immediately
00193 * disabling GPTW PWM output for the specified motor. This provides a
00194 * firmware-controlled emergency stop independent of hardware faults.
00195 *
00196 * @param[in] motor_index Motor index (0-3)
00197 *
00198 * @return rx_err_t Error code
00199 * @retval k_rx_ok Software stop activated
00200 * @retval k_rx_err_invalid_arg motor_index >= 4
00201 *
00202 * @post GPTW output immediately disabled for specified motor
00203 * @post SSF flag set in POEGGn register
00204 *
00205 * @see rx_poeg_clear_software_stop() Release software stop
00206 * @since Version 1.0.0
00207 */
00208 rx_err_t rx_poeg_software_stop(uint8_t motor_index);
00209
00210 /**
00211 * @brief Release software emergency stop on a specific motor
00212 *
00213 * @details
00214 * Clears the SSF (Software Stop Flag) in the POEGGn register.
00215 * GPTW output re-enables at end of next counting cycle if no other
00216 * fault flags (PIDF, IOCF, OSTPF) are active.
00217 *
00218 * @param[in] motor_index Motor index (0-3)
00219 *
00220 * @return rx_err_t Error code
00221 * @retval k_rx_ok Software stop released
00222 * @retval k_rx_err_invalid_arg motor_index >= 4
00223 *
00224 * @post SSF flag cleared in POEGGn register
00225 *
00226 * @see rx_poeg_software_stop() Activate software stop
00227 * @since Version 1.0.0
00228 */
00229 rx_err_t rx_poeg_clear_software_stop(uint8_t motor_index);
00230
00231 #ifdef __cplusplus
00232 }
00233 #endif

```

8.95 libs/rx_hal/inc/rx_port_utils.h File Reference

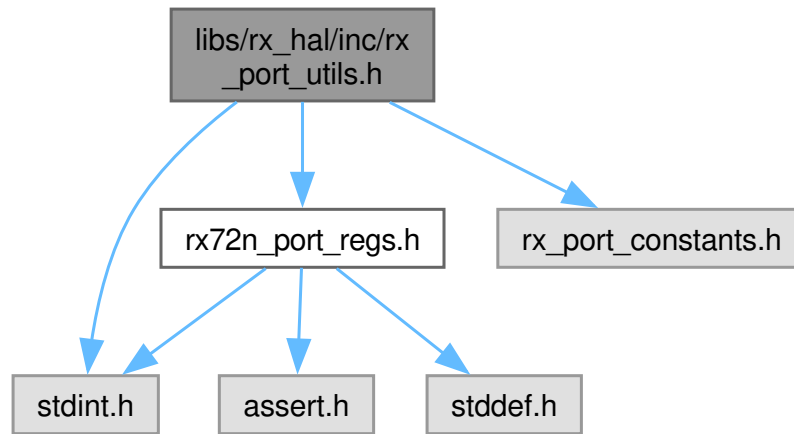
PORT utility functions for RX72N GPIO register access.

```

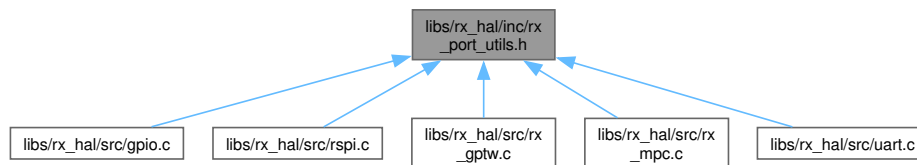
#include <stdint.h>
#include "rx72n_port_regs.h"
#include "rx_port_constants.h"

```

Include dependency graph for rx_port_utils.h:



This graph shows which files directly or indirectly include this file:



Functions

- static volatile `rx_port_regs_t * rx_port_get_base (uint8_t port)`
Map RX72N port number to PORT register base address.

8.95.1 Detailed Description

PORT utility functions for RX72N GPIO register access.

This file provides centralized utility functions for working with RX72N PORT (GPIO) registers. The primary function is mapping port numbers to register base addresses, which is critical for all GPIO operations in the STAR system.

8.95.1.1 Purpose and Design Rationale

The RX72N PORT peripheral has a unique type-grouped register layout where all PDR (direction) registers are contiguous, all PODR (output) registers are contiguous, etc. This differs from traditional MCUs where each port's registers are grouped together. These utilities abstract this complexity.

Key design decisions:

- Single source of truth: Port-to-address mapping centralized here
- Inline functions: Zero runtime overhead, optimized to direct register access
- Type safety: Uses C23 typed enums from `rx_port_constants.h`
- Package-specific: Only supports ports available on 144-pin LFQFP package

8.95.1.2 System Architecture Context

PORT utilities are foundational infrastructure used throughout the system:

```
Application Layer (hardware_pinout.h)
    v uses port constants
HAL Layer (rx_gpio.c, rx_mtu_encoder.c)
    v uses rx_port_get_base()
PORT Utils (THIS FILE - rx_port_utils.h)
    v maps to
Hardware Registers (rx72n_port_regs.h)
```

All GPIO configuration and motor control initialization depends on this mapping to access the correct hardware registers.

8.95.1.3 Hardware Requirements

Requirement	Specification
MCU	Renesas RX72N
Package	144-pin LFQFP
Available Ports	0-9, A-F, J (see rx_port_constants.h for per-port pin counts)
CPU Frequency	240 MHz
Memory Access	Direct register I/O

8.95.1.4 Performance Characteristics

- Execution time: 1-2 CPU cycles (inlined switch statement)
- Memory usage: Zero runtime overhead (compile-time constant folding)
- Code size: Minimal (optimized to direct address loading)
- Stack usage: Zero (inline function)

8.95.1.5 Usage Example

```
#include "rx_port_utils.h"
#include "rx_port_constants.h"

// Configure PORT D pin 0 as output (LED)
volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
if (port_d != nullptr) {
    port_d->pmr &= ~(1 << 0); // GPIO mode
    port_d->pdr |= (1 << 0); // Output direction
    port_d->podr |= (1 << 0); // Set high
}

// Read PORT A pin 0 state (button input)
volatile rx_port_regs_t* port_a = rx_port_get_base(k_rx_port_a);
if (port_a != nullptr) {
    uint8_t button_state = (port_a->pdr >> 0) & 0x01;
}
```

Module Dependencies

- [rx72n_port_regs.h](#): PORT register structure definitions
- [rx_port_constants.h](#): Port number constants (k_rx_port_*)

NASA Power of 10 Compliance

Rule	Compliance	Details
1	↔ [OK]↔	No goto, setjmp, recursion - simple switch statement
2	↔ [OK]↔	No loops (switch is O(1) jump table)
3	↔ [OK]↔	No dynamic allocation (inline function)
4	↔ [OK]↔	Function is 43 lines (under 60 line limit)
5	↔ [OK]↔	Implicit validation: returns NULL for invalid port
6	↔ [OK]↔	Minimal scope (single parameter, local to function)
7	↔ [OK]↔	Return value must be checked (NULL validation required)
8	↔ [OK]↔	Uses C23 typed enums (k_rx_port_*), no magic numbers
9	↔ [OK]↔	Single level of dereferencing for register access
10	↔ [OK]↔	Compiles with -Wall -Wextra -Werror

SOLID Principles

Single Responsibility: This utility has one job - map port numbers to register base addresses. It does not configure pins, read/write values, or manage pin modes. Those responsibilities belong to higher-level modules.

Open/Closed: Extensible for new package sizes (144-pin, 176-pin) by adding new case statements without modifying existing mappings. Closed to modification of existing port addresses (hardware-defined).

Liskov Substitution: All port register pointers are identical types (rx_port_regs_t*), ensuring consistent register access patterns across all ports.

Interface Segregation: Provides minimal interface - single function for single purpose. Callers only depend on what they need (port-to-address).

Dependency Inversion: High-level GPIO drivers depend on this abstraction, not on hardcoded addresses. Easy to mock for unit testing.

See also

[docs/sections/03_hardware_pinout.tex](#) Complete GPIO pin assignments
[rx72n_port_regs.h](#) PORT register structure and memory map
[rx_port_constants.h](#) Port number constant definitions

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Definition in file [rx_port_utils.h](#).

8.95.2 Function Documentation

8.95.2.1 `rx_port_get_base()`

```
volatile rx\_port\_regs\_t * rx_port_get_base (  
    uint8_t port) [inline], [static]
```

Map RX72N port number to PORT register base address.

Converts a port number constant (`k_rx_port_*`) to the corresponding volatile PORT register structure pointer for hardware access. This is the single source of truth for port-to-address mapping in the entire STAR firmware.

Algorithm steps:

1. Receive port number as input parameter
2. Switch on port number (compiler optimizes to jump table)
3. For valid port: call corresponding `portX()` inline accessor
4. For invalid port: return NULL to signal error
5. Caller must validate non-NULL before dereferencing

Implementation details:

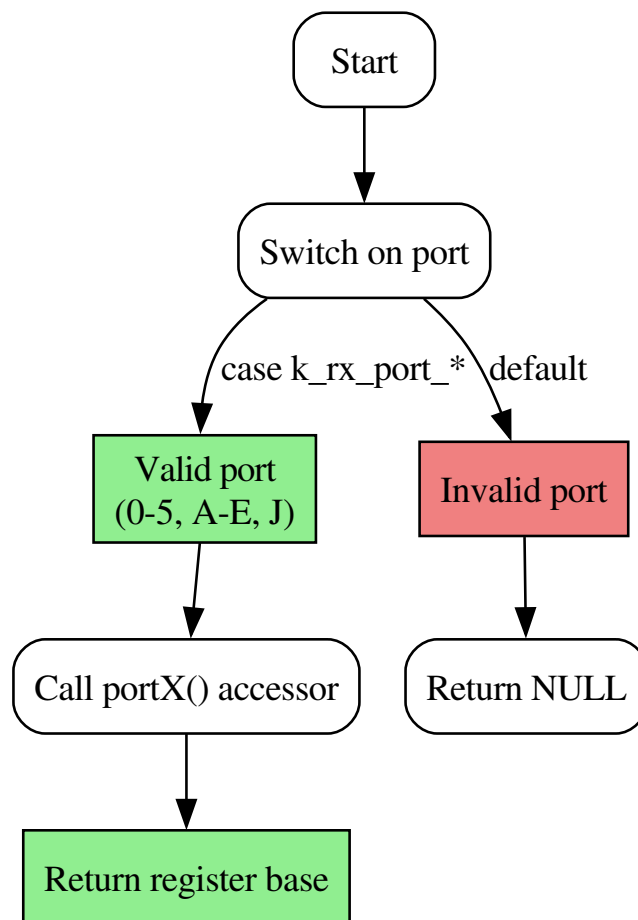
- Each case calls an inline accessor (`port0()`, `porta()`, etc.) defined in [rx72n_port_regs.h](#) which returns the typed register base address
- Compiler optimizes switch to direct jump table (O(1) lookup)
- Function is inline, so entire call optimizes to direct address load

- Invalid port numbers safely return NULL instead of accessing garbage memory

Package-specific behavior: Supports all ports physically available on 144-pin LQFP package:

- PORT0-9: Available (see rx_port_constants.h for per-port pin counts)
- PORTA-F: Available (PORTF has PF5 only)
- PORTJ: Available (PJ3, PJ5)
- Ports G-I, K-Q: Not available on 144-pin, will return nullptr

Control flow:



Parameters

in	port	Port number constant from <code>rx_port_constants.h</code> • Type: <code>uint8_t</code> (C23 typed enum <code>rx_port_num_t</code>) • Valid range: <code>k_rx_port_0</code> through <code>k_rx_port_j</code> • Units: N/A (enumerated constant) • Constraints: Must use <code>k_rx_port_*</code> constants, not raw integers • Invalid values: Any value not in <code>[0-5, 0x0A-0x0E, 0x13]</code> returns <code>nullptr</code>
----	------	--

Returns

Pointer to PORT register base structure, or `nullptr` if port invalid

Return values

Non-NULL	Valid port - pointer to volatile <code>rx_port_regs_t</code> structure with PDR, PODR, PIDR, PMR, etc. registers
NULL	Invalid port number - port not available on 144-pin LFQFP package, or invalid parameter value

Return Value Details:

Port Input	Return Value	Condition
<code>k_rx_port_0</code>	<code>&PORT0</code> registers	Port 0 (P00-P03, P05, P07)
<code>k_rx_port_1</code>	<code>&PORT1</code> registers	Port 1 (P12-P17)
<code>k_rx_port_2</code>	<code>&PORT2</code> registers	Port 2 (P20-P27)
<code>k_rx_port_3</code>	<code>&PORT3</code> registers	Port 3 (P30-P37)
<code>k_rx_port_4</code>	<code>&PORT4</code> registers	Port 4 (P40-P47)
<code>k_rx_port_5</code>	<code>&PORT5</code> registers	Port 5 (P50-P56)
<code>k_rx_port_6</code>	<code>&PORT6</code> registers	Port 6 (P60-P67)
<code>k_rx_port_7</code>	<code>&PORT7</code> registers	Port 7 (P70-P77)
<code>k_rx_port_8</code>	<code>&PORT8</code> registers	Port 8 (P80-P83, P86-P87)

Port Input	Return Value	Condition
<code>k_rx_port_9</code>	<code>&PORT9</code> registers	Port 9 (P90-P93)
<code>k_rx_port_a</code>	<code>&PORTA</code> registers	Port A (PA0-PA7)
<code>k_rx_port_b</code>	<code>&PORTB</code> registers	Port B (PB0-PB7)
<code>k_rx_port_c</code>	<code>&PORTC</code> registers	Port C (PC0-PC7)
<code>k_rx_port_d</code>	<code>&PORTD</code> registers	Port D (PD0-PD7)
<code>k_rx_port_e</code>	<code>&PORTE</code> registers	Port E (PE0-PE7)
<code>k_rx_port_f</code>	<code>&PORTF</code> registers	Port F (PF5 only)
<code>k_rx_port_j</code>	<code>&PORTJ</code> registers	Port J (PJ3, PJ5)
Any other value	NULL	Port not on 144-pin package

Precondition

None - function is always safe to call

Parameter should use `k_rx_port_*` constants (not validated at runtime)

Postcondition

If return value is non-NULL, pointer is valid volatile register address

If return value is nullptr, caller must not dereference (validation required)

No hardware state modified (read-only address calculation)

Invariant

Return value is either NULL or valid PORT register base address

Function has no side effects (pure address mapping)

Note

Thread Safety: Fully thread-safe and reentrant. No shared state, no hardware access, pure computation. Multiple threads can call simultaneously without synchronization.

Re-entrancy: Fully reentrant. No static variables, no globals. Safe for use in interrupt handlers and multi-threaded contexts.

Performance: Optimizes to 1-2 CPU cycles when inlined. Switch compiles to jump table (O(1) lookup). At 240 MHz: ~4-8 nanoseconds.

Memory: Zero stack usage (inlined). Zero heap usage. Zero static storage. Entire function optimizes to immediate address load.

Warning

NULL validation required: Caller MUST check return value before dereferencing. Failing to validate will cause nullptr fault if invalid port number is passed.

Attention

Package-specific: Only supports 144-pin LFQFP. Porting to 176-pin packages requires adding additional case statements for ports G-I, K-Q.

Example - Basic Usage:

```
#include "rx_port_utils.h"
#include "rx_port_constants.h"

// Configure PORT D as output
volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
if (port_d != nullptr) {
    port_d->pdr = 0xFF; // All pins output
    port_d->podr = 0x00; // All pins low
}
```

Example - Error Handling:

```
// Attempt to access unavailable port (returns NULL)
volatile rx_port_regs_t* port_g = rx_port_get_base(0x10); // Port G
if (port_g == nullptr) {
    // Handle error - port not available on 144-pin package
    rx_log_error("GPIO", "Port G not available on this package");
    return k_rx_err_invalid_arg;
}
```

Example - Motor Control Initialization:

```
// Initialize motor driver enable pins (PORT E)
volatile rx_port_regs_t* port_e = rx_port_get_base(k_rx_port_e);
if (port_e != nullptr) {
    port_e->pmr &= ~0x0F; // Pins 0-3 GPIO mode
    port_e->pdr |= 0x0F; // Pins 0-3 output
    port_e->podr &= ~0x0F; // Motors disabled (active high)
}
```

Example - LED Status Indicator:

```
// Toggle status LED on PORT D pin 0
volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
if (port_d != nullptr) {
    port_d->podr ^= (1 << 0); // Toggle PD0
}
```

See also

[port0\(\)](#) through [portj\(\)](#) Inline register accessors in [rx72n_port_regs.h](#)
[rx_port_constants.h](#) Port number constant definitions
[docs/sections/03_hardware_pinout.tex](#) Complete pin assignment reference

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 1 [OK] Simple control flow (switch only, no goto/recursion)
- Rule 2 [OK] No loops (O(1) switch statement)
- Rule 3 [OK] No dynamic allocation
- Rule 4 [OK] Function is 43 lines (under 60 line limit)
- Rule 5 [OK] Implicit validation: NULL return signals invalid input
- Rule 6 [OK] Minimal scope (single parameter, local to function)
- Rule 7 [OK] Return value MUST be validated by caller
- Rule 8 [OK] Uses C23 typed enums (k_rx_port_*), zero magic numbers
- Rule 9 [OK] Single level of pointer dereferencing for register access
- Rule 10 [OK] Compiles clean with -Wall -Wextra -Werror

Definition at line 322 of file [rx_port_utils.h](#).

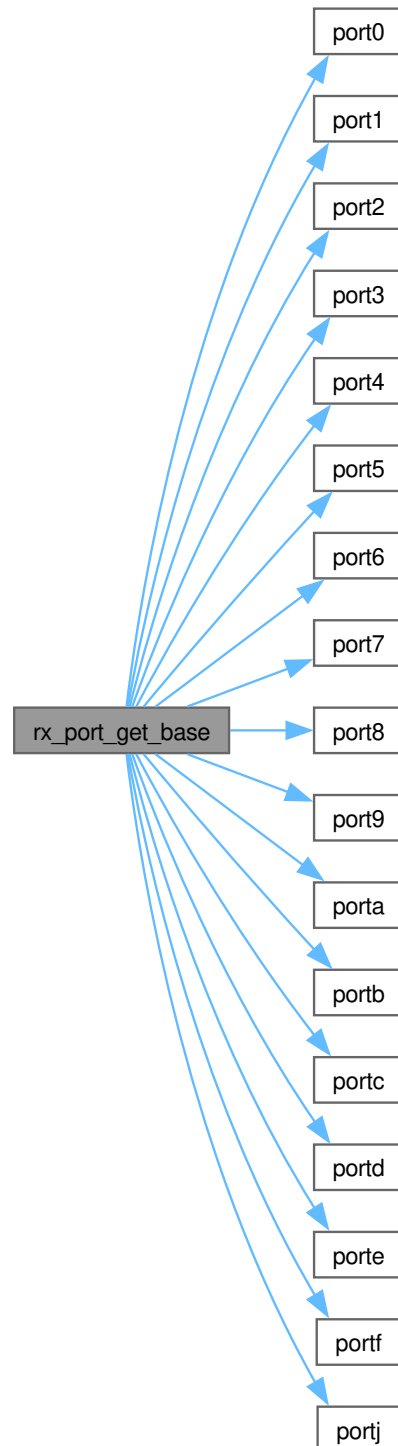
```

00323 {
00324     switch (port) {
00325     case k_rx_port_0: {
00326         return port0();
00327     }
00328     case k_rx_port_1: {
00329         return port1();
00330     }
00331     case k_rx_port_2: {
00332         return port2();
00333     }
00334     case k_rx_port_3: {
00335         return port3();
00336     }
00337     case k_rx_port_4: {
00338         return port4();
00339     }
00340     case k_rx_port_5: {
00341         return port5();
00342     }
00343     case k_rx_port_6: {
00344         return port6();
00345     }
00346     case k_rx_port_7: {
00347         return port7();
00348     }
00349     case k_rx_port_8: {
00350         return port8();
00351     }
00352     case k_rx_port_9: {
00353         return port9();
00354     }
00355     case k_rx_port_a: {
00356         return porta();
00357     }
00358     case k_rx_port_b: {
00359         return portb();
00360     }
00361     case k_rx_port_c: {
00362         return portc();
00363     }
00364     case k_rx_port_d: {
00365         return portd();
00366     }
00367     case k_rx_port_e: {
00368         return porte();
00369     }
00370     case k_rx_port_f: {
00371         return portf();
00372     }
00373     case k_rx_port_j: {
00374         return portj();
00375     }
00376     default: {
00377         return nullptr; /* Invalid port */
00378     }
00379 }
00380 }
```

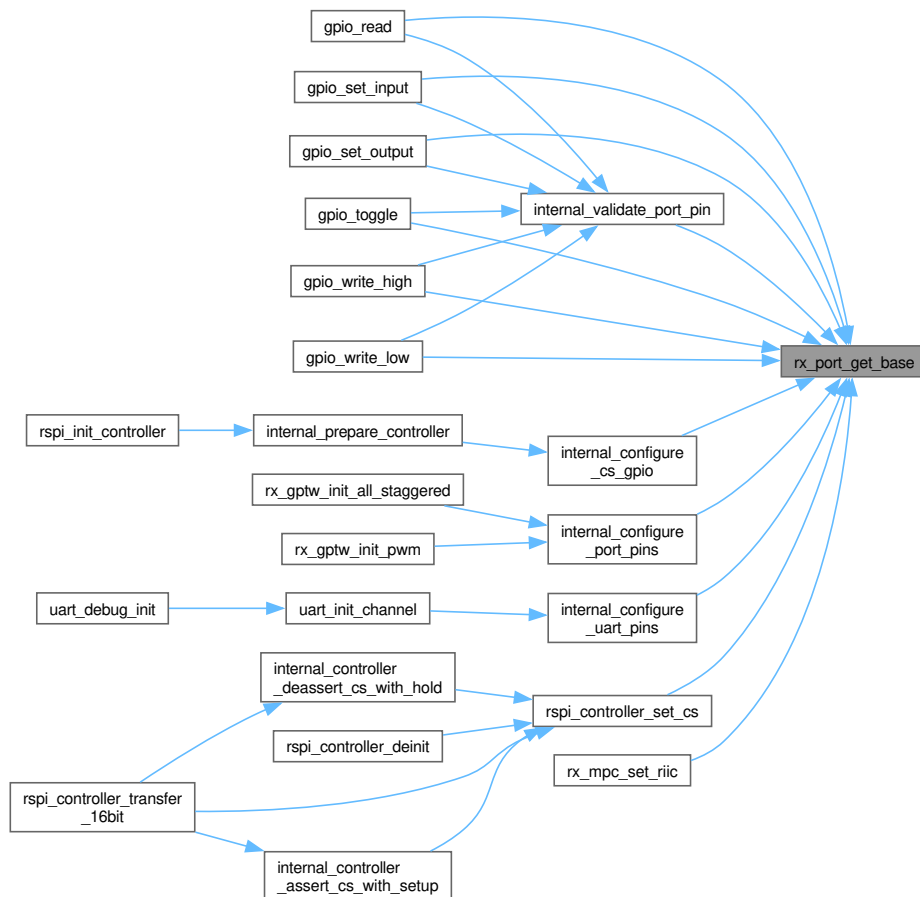
References [port0\(\)](#), [port1\(\)](#), [port2\(\)](#), [port3\(\)](#), [port4\(\)](#), [port5\(\)](#), [port6\(\)](#), [port7\(\)](#), [port8\(\)](#), [port9\(\)](#), [porta\(\)](#), [portb\(\)](#), [portc\(\)](#), [portd\(\)](#), [porte\(\)](#), [portf\(\)](#), and [portj\(\)](#).

Referenced by [gpio_read\(\)](#), [gpio_set_input\(\)](#), [gpio_set_output\(\)](#), [gpio_toggle\(\)](#), [gpio_write_high\(\)](#), [gpio_write_low\(\)](#), [internal_configure_cs_gpio\(\)](#), [internal_configure_port_pins\(\)](#), [internal_configure_uart_pins\(\)](#), [internal_validate_port_pin\(\)](#), [rspi_controller_set_cs\(\)](#), and [rx_mpc_set_riic\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.96 rx_port_utils.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_port_utils.h
00003  * @brief PORT utility functions for RX72N GPIO register access
00004  *
00005  * @details
00006  * This file provides centralized utility functions for working with RX72N PORT
00007  * (GPIO) registers. The primary function is mapping port numbers to register
00008  * base addresses, which is critical for all GPIO operations in the STAR system.
00009  *
00010  * ## Purpose and Design Rationale
00011  *
00012  * The RX72N PORT peripheral has a unique type-grouped register layout where
00013  * all PDR (direction) registers are contiguous, all PODR (output) registers
00014  * are contiguous, etc. This differs from traditional MCUs where each port's
00015  * registers are grouped together. These utilities abstract this complexity.
00016  *
00017  * Key design decisions:
00018  * - **Single source of truth**: Port-to-address mapping centralized here
00019  * - **Inline functions**: Zero runtime overhead, optimized to direct register access
00020  * - **Type safety**: Uses C23 typed enums from rx_port_constants.h
00021  * - **Package-specific**: Only supports ports available on 144-pin LFQFP package
00022  *
00023  * ## System Architecture Context
00024  *
00025  * PORT utilities are foundational infrastructure used throughout the system:
00026  * ```
00027  * Application Layer (hardware_pinout.h)
00028  *   v uses port constants
00029  * HAL Layer (rx_gpio.c, rx_mtu_encoder.c)
00030  *   v uses rx_port_get_base()
00031  * PORT Utils (THIS FILE - rx_port_utils.h)
00032  *   v maps to
00033  * Hardware Registers (rx72n_port_regs.h)
00034  * ```
00035  */

```

```

00036 * All GPIO configuration and motor control initialization depends on this
00037 * mapping to access the correct hardware registers.
00038 *
00039 * ## Hardware Requirements
00040 *
00041 * | Requirement          | Specification          |
00042 * |-----|-----|
00043 * | **MCU**              | Renesas RX72N          |
00044 * | **Package**          | 144-pin LFQFP          |
00045 * | **Available Ports**  | 0-9, A-F, J (see rx_port_constants.h for per-port pin counts) |
00046 * | **CPU Frequency**    | 240 MHz                |
00047 * | **Memory Access**    | Direct register I/O    |
00048 *
00049 * ## Performance Characteristics
00050 *
00051 * - **Execution time**: 1-2 CPU cycles (inlined switch statement)
00052 * - **Memory usage**: Zero runtime overhead (compile-time constant folding)
00053 * - **Code size**: Minimal (optimized to direct address loading)
00054 * - **Stack usage**: Zero (inline function)
00055 *
00056 * ## Usage Example
00057 *
00058 * @code
00059 * #include "rx_port_utils.h"
00060 * #include "rx_port_constants.h"
00061 *
00062 * // Configure PORT D pin 0 as output (LED)
00063 * volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
00064 * if (port_d != nullptr) {
00065 *     port_d->pmr &= ~(1 << 0); // GPIO mode
00066 *     port_d->pdr |= (1 << 0); // Output direction
00067 *     port_d->podr |= (1 << 0); // Set high
00068 * }
00069 *
00070 * // Read PORT A pin 0 state (button input)
00071 * volatile rx_port_regs_t* port_a = rx_port_get_base(k_rx_port_a);
00072 * if (port_a != nullptr) {
00073 *     uint8_t button_state = (port_a->pdr >> 0) & 0x01;
00074 * }
00075 * @endcode
00076 *
00077 * @par Module Dependencies
00078 * - rx72n_port_regs.h: PORT register structure definitions
00079 * - rx_port_constants.h: Port number constants (k_rx_port_*)
00080 *
00081 * @par NASA Power of 10 Compliance
00082 *
00083 * | Rule | Compliance | Details |
00084 * |-----|-----|
00085 * | 1 | [OK] | No goto, setjmp, recursion - simple switch statement |
00086 * | 2 | [OK] | No loops (switch is O(1) jump table) |
00087 * | 3 | [OK] | No dynamic allocation (inline function) |
00088 * | 4 | [OK] | Function is 43 lines (under 60 line limit) |
00089 * | 5 | [OK] | Implicit validation: returns NULL for invalid port |
00090 * | 6 | [OK] | Minimal scope (single parameter, local to function) |
00091 * | 7 | [OK] | Return value must be checked (NULL validation required) |
00092 * | 8 | [OK] | Uses C23 typed enums (k_rx_port_*), no magic numbers |
00093 * | 9 | [OK] | Single level of dereferencing for register access |
00094 * | 10 | [OK] | Compiles with -Wall -Wextra -Werror |
00095 *
00096 * @par SOLID Principles
00097 *
00098 * **Single Responsibility**: This utility has one job - map port numbers to
00099 * register base addresses. It does not configure pins, read/write values, or
00100 * manage pin modes. Those responsibilities belong to higher-level modules.
00101 *
00102 * **Open/Closed**: Extensible for new package sizes (144-pin, 176-pin) by
00103 * adding new case statements without modifying existing mappings. Closed to
00104 * modification of existing port addresses (hardware-defined).
00105 *
00106 * **Liskov Substitution**: All port register pointers are identical types
00107 * (rx_port_regs_t*), ensuring consistent register access patterns across
00108 * all ports.
00109 *
00110 * **Interface Segregation**: Provides minimal interface - single function for
00111 * single purpose. Callers only depend on what they need (port-to-address).
00112 *
00113 * **Dependency Inversion**: High-level GPIO drivers depend on this abstraction,
00114 * not on hardcoded addresses. Easy to mock for unit testing.
00115 *
00116 * @see docs/sections/03_hardware_pinout.tex Complete GPIO pin assignments
00117 * @see rx72n_port_regs.h PORT register structure and memory map
00118 * @see rx_port_constants.h Port number constant definitions
00119 *
00120 * @author Locked, Inc.
00121 * @date 2026-01-27
00122 * @copyright Copyright (c) 2026 Locked Inc.

```



```

00123 * SPDX-License-Identifier: MIT
00124 */
00125
00126 #pragma once
00127
00128 #include <stdint.h>
00129
00130 #include "rx72n_port_regs.h"
00131 #include "rx_port_constants.h"
00132
00133 #ifdef __cplusplus
00134 extern "C" {
00135 #endif
00136
00137 /**
00138  * @brief Map RX72N port number to PORT register base address
00139  *
00140  * @details
00141  * Converts a port number constant (k_rx_port_*) to the corresponding volatile
00142  * PORT register structure pointer for hardware access. This is the **single
00143  * source of truth** for port-to-address mapping in the entire STAR firmware.
00144  *
00145  * **Algorithm steps**:
00146  * 1. Receive port number as input parameter
00147  * 2. Switch on port number (compiler optimizes to jump table)
00148  * 3. For valid port: call corresponding portX() inline accessor
00149  * 4. For invalid port: return NULL to signal error
00150  * 5. Caller must validate non-NULL before dereferencing
00151  *
00152  * **Implementation details**:
00153  * - Each case calls an inline accessor (port0(), porta(), etc.) defined in
00154  *   rx72n_port_regs.h which returns the typed register base address
00155  * - Compiler optimizes switch to direct jump table (O(1) lookup)
00156  * - Function is inline, so entire call optimizes to direct address load
00157  * - Invalid port numbers safely return NULL instead of accessing garbage memory
00158  *
00159  * **Package-specific behavior**:
00160  * Supports all ports physically available on 144-pin LFQFP package:
00161  * - PORT0-9: Available (see rx_port_constants.h for per-port pin counts)
00162  * - PORTA-F: Available (PORTF has PF5 only)
00163  * - PORTJ: Available (PJ3, PJ5)
00164  * - Ports G-I, K-Q: Not available on 144-pin, will return nullptr
00165  *
00166  * **Control flow**:
00167  * @dot
00168  * digraph port_mapping {
00169  *   rankdir=TB;
00170  *   node [shape=box, style=rounded];
00171  *
00172  *   start [label="Start"];
00173  *   switch_port [label="Switch on port"];
00174  *   valid [label="Valid port\n(0-5, A-E, J)", fillcolor=lightgreen, style=filled];
00175  *   invalid [label="Invalid port", fillcolor=lightcoral, style=filled];
00176  *   call_accessor [label="Call portX() accessor"];
00177  *   return_null [label="Return NULL"];
00178  *   return_ptr [label="Return register base", fillcolor=lightgreen, style=filled];
00179  *
00180  *   start -> switch_port;
00181  *   switch_port -> valid [label="case k_rx_port_*"];
00182  *   switch_port -> invalid [label="default"];
00183  *   valid -> call_accessor;
00184  *   call_accessor -> return_ptr;
00185  *   invalid -> return_null;
00186  * }
00187  * @enddot
00188
00189 * @param[in] port Port number constant from rx_port_constants.h
00190 * - **Type**: uint8_t (C23 typed enum rx_port_num_t)
00191 * - **Valid range**: k_rx_port_0 through k_rx_port_j
00192 * - **Units**: N/A (enumerated constant)
00193 * - **Constraints**: Must use k_rx_port_* constants, not raw integers
00194 * - **Invalid values**: Any value not in [0-5, 0x0A-0x0E, 0x13] returns nullptr
00195 *
00196 * @return Pointer to PORT register base structure, or nullptr if port invalid
00197 * @retval Non-NULL Valid port - pointer to volatile rx_port_regs_t structure
00198 *         with PDR, PODR, PIDR, PMR, etc. registers
00199 * @retval NULL Invalid port number - port not available on 144-pin LFQFP
00200 *         package, or invalid parameter value
00201 *
00202 * @par Return Value Details:
00203 *
00204 * | Port Input | Return Value | Condition |
00205 * |-----|-----|-----|
00206 * | k_rx_port_0 | &PORT0 registers | Port 0 (P00-P03, P05, P07) |
00207 * | k_rx_port_1 | &PORT1 registers | Port 1 (P12-P17) |
00208 * | k_rx_port_2 | &PORT2 registers | Port 2 (P20-P27) |
00209 * | k_rx_port_3 | &PORT3 registers | Port 3 (P30-P37) |

```

```

00210 * | k_rx_port_4 | &PORT4 registers | Port 4 (P40-P47) |
00211 * | k_rx_port_5 | &PORT5 registers | Port 5 (P50-P56) |
00212 * | k_rx_port_6 | &PORT6 registers | Port 6 (P60-P67) |
00213 * | k_rx_port_7 | &PORT7 registers | Port 7 (P70-P77) |
00214 * | k_rx_port_8 | &PORT8 registers | Port 8 (P80-P83, P86-P87) |
00215 * | k_rx_port_9 | &PORT9 registers | Port 9 (P90-P93) |
00216 * | k_rx_port_a | &PORTA registers | Port A (PA0-PA7) |
00217 * | k_rx_port_b | &PORTB registers | Port B (PB0-PB7) |
00218 * | k_rx_port_c | &PORTC registers | Port C (PC0-PC7) |
00219 * | k_rx_port_d | &PORTD registers | Port D (PD0-PD7) |
00220 * | k_rx_port_e | &PORTE registers | Port E (PE0-PE7) |
00221 * | k_rx_port_f | &PORTF registers | Port F (PF5 only) |
00222 * | k_rx_port_j | &PORTJ registers | Port J (PJ3, PJ5) |
00223 * | Any other value | NULL | Port not on 144-pin package |
00224 *
00225 * @pre None - function is always safe to call
00226 * @pre Parameter should use k_rx_port_* constants (not validated at runtime)
00227 *
00228 * @post If return value is non-NULL, pointer is valid volatile register address
00229 * @post If return value is nullptr, caller must not dereference (validation required)
00230 * @post No hardware state modified (read-only address calculation)
00231 *
00232 * @invariant Return value is either NULL or valid PORT register base address
00233 * @invariant Function has no side effects (pure address mapping)
00234 *
00235 * @note **Thread Safety**: Fully thread-safe and reentrant. No shared state,
00236 * no hardware access, pure computation. Multiple threads can call
00237 * simultaneously without synchronization.
00238 *
00239 * @note **Re-entrancy**: Fully reentrant. No static variables, no globals.
00240 * Safe for use in interrupt handlers and multi-threaded contexts.
00241 *
00242 * @note **Performance**: Optimizes to 1-2 CPU cycles when inlined. Switch
00243 * compiles to jump table (O(1) lookup). At 240 MHz: ~4-8 nanoseconds.
00244 *
00245 * @note **Memory**: Zero stack usage (inlined). Zero heap usage. Zero static
00246 * storage. Entire function optimizes to immediate address load.
00247 *
00248 * @warning **NULL validation required**: Caller MUST check return value before
00249 * dereferencing. Failing to validate will cause nullptr fault if
00250 * invalid port number is passed.
00251 *
00252 * @attention **Package-specific**: Only supports 144-pin LQFP. Porting to
00253 * 176-pin packages requires adding additional case statements
00254 * for ports G-I, K-Q.
00255 *
00256 * @par Example - Basic Usage:
00257 * @code
00258 * #include "rx_port_utils.h"
00259 * #include "rx_port_constants.h"
00260 *
00261 * // Configure PORT D as output
00262 * volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
00263 * if (port_d != nullptr) {
00264 *     port_d->pdr = 0xFF; // All pins output
00265 *     port_d->podr = 0x00; // All pins low
00266 * }
00267 * @endcode
00268 *
00269 * @par Example - Error Handling:
00270 * @code
00271 * // Attempt to access unavailable port (returns NULL)
00272 * volatile rx_port_regs_t* port_g = rx_port_get_base(0x10); // Port G
00273 * if (port_g == nullptr) {
00274 *     // Handle error - port not available on 144-pin package
00275 *     rx_log_error("GPIO", "Port G not available on this package");
00276 *     return k_rx_err_invalid_arg;
00277 * }
00278 * @endcode
00279 *
00280 * @par Example - Motor Control Initialization:
00281 * @code
00282 * // Initialize motor driver enable pins (PORT E)
00283 * volatile rx_port_regs_t* port_e = rx_port_get_base(k_rx_port_e);
00284 * if (port_e != nullptr) {
00285 *     port_e->pmr &= ~0x0F; // Pins 0-3 GPIO mode
00286 *     port_e->pdr |= 0x0F; // Pins 0-3 output
00287 *     port_e->podr &= ~0x0F; // Motors disabled (active high)
00288 * }
00289 * @endcode
00290 *
00291 * @par Example - LED Status Indicator:
00292 * @code
00293 * // Toggle status LED on PORT D pin 0
00294 * volatile rx_port_regs_t* port_d = rx_port_get_base(k_rx_port_d);
00295 * if (port_d != nullptr) {
00296 *     port_d->podr ^= (1 << 0); // Toggle PD0

```

```

00297 * }
00298 * @endcode
00299 *
00300 * @see port0() through portj() Inline register accessors in rx72n_port_regs.h
00301 * @see rx_port_constants.h Port number constant definitions
00302 * @see docs/sections/03_hardware_pinout.tex Complete pin assignment reference
00303 *
00304 * @since Version 1.0.0
00305 *
00306 * @par NASA Power of 10 Compliance
00307 * - **Rule 1** [OK] Simple control flow (switch only, no goto/recursion)
00308 * - **Rule 2** [OK] No loops (O(1) switch statement)
00309 * - **Rule 3** [OK] No dynamic allocation
00310 * - **Rule 4** [OK] Function is 43 lines (under 60 line limit)
00311 * - **Rule 5** [OK] Implicit validation: NULL return signals invalid input
00312 * - **Rule 6** [OK] Minimal scope (single parameter, local to function)
00313 * - **Rule 7** [OK] Return value MUST be validated by caller
00314 * - **Rule 8** [OK] Uses C23 typed enums (k_rx_port_*), zero magic numbers
00315 * - **Rule 9** [OK] Single level of pointer dereferencing for register access
00316 * - **Rule 10** [OK] Compiles clean with -Wall -Wextra -Werror
00317 *
00318 * @callgraph
00319 * @callergraph
00320 */
00321 /* LCOV_EXCL_START */
00322 static inline volatile rx_port_regs_t* rx_port_get_base(uint8_t port)
00323 {
00324     switch (port) {
00325         case k_rx_port_0: {
00326             return port0();
00327         }
00328         case k_rx_port_1: {
00329             return port1();
00330         }
00331         case k_rx_port_2: {
00332             return port2();
00333         }
00334         case k_rx_port_3: {
00335             return port3();
00336         }
00337         case k_rx_port_4: {
00338             return port4();
00339         }
00340         case k_rx_port_5: {
00341             return port5();
00342         }
00343         case k_rx_port_6: {
00344             return port6();
00345         }
00346         case k_rx_port_7: {
00347             return port7();
00348         }
00349         case k_rx_port_8: {
00350             return port8();
00351         }
00352         case k_rx_port_9: {
00353             return port9();
00354         }
00355         case k_rx_port_a: {
00356             return porta();
00357         }
00358         case k_rx_port_b: {
00359             return portb();
00360         }
00361         case k_rx_port_c: {
00362             return portc();
00363         }
00364         case k_rx_port_d: {
00365             return portd();
00366         }
00367         case k_rx_port_e: {
00368             return porte();
00369         }
00370         case k_rx_port_f: {
00371             return portf();
00372         }
00373         case k_rx_port_j: {
00374             return portj();
00375         }
00376         default: {
00377             return nullptr; /* Invalid port */
00378         }
00379     }
00380 }
00381 /* LCOV_EXCL_STOP */
00382
00383 #ifdef __cplusplus

```

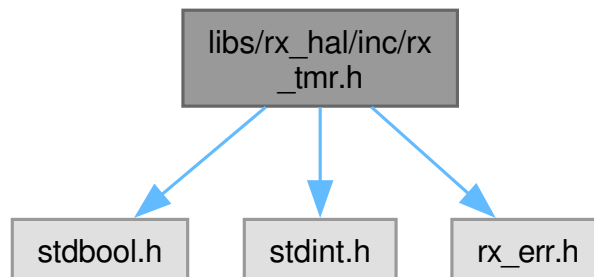
```
00384 }
00385 #endif
```

8.97 libs/rx_hal/inc/rx_tmr.h File Reference

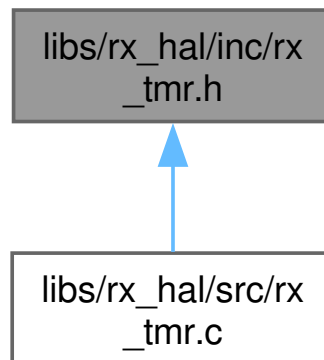
TMR HAL Driver for RX72N 8-Bit Timer (TMRb).

```
#include <stdbool.h>
#include <stdint.h>
#include "rx_err.h"
```

Include dependency graph for rx_tmr.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_tmr_config_t](#)
TMR channel configuration.

Typedefs

- typedef void(* [rx_tmr_isr_callback_t](#)) ([rx_tmr_channel_t](#) channel)
User callback for compare-match / overflow events.

Enumerations

- enum `rx_tmr_channel_t` : `uint8_t` { `k_tmr_channel_0` = 0 , `k_tmr_channel_1` = 1 , `k_tmr_channel_2` = 2 , `k_tmr_channel_3` = 3 }
TMR channel identifiers.
- enum `rx_tmr_mode_t` : `uint8_t` { `k_tmr_mode_8bit_independent` = 0 , `k_tmr_mode_16bit_cascade` = 1 }
TMR operating mode (independent 8-bit vs cascaded 16-bit).
- enum `rx_tmr_clock_source_t` : `uint8_t` { `k_tmr_clock_pclk_div_1` = 0 , `k_tmr_clock_pclk_div_2` = 1 , `k_tmr_clock_pclk_div_8` = 2 , `k_tmr_clock_pclk_div_32` = 3 , `k_tmr_clock_pclk_div_64` = 4 , `k_tmr_clock_pclk_div_1024` = 5 , `k_tmr_clock_pclk_div_8192` = 6 , `k_tmr_clock_external_rising` = 7 , `k_tmr_clock_external_falling` = 8 , `k_tmr_clock_external_both` = 9 }
TMR clock source selection.
- enum `rx_tmr_counter_clear_t` : `uint8_t` { `k_tmr_clear_disabled` = 0 , `k_tmr_clear_cmp_match_a` = 1 , `k_tmr_clear_cmp_match_b` = 2 , `k_tmr_clear_external_reset` = 3 }
Counter clear source (CCLR field of TCR).
- enum `rx_tmr_output_t` : `uint8_t` { `k_tmr_output_no_change` = 0 , `k_tmr_output_low` = 1 , `k_tmr_output_high` = 2 , `k_tmr_output_toggle` = 3 }
TMO pin behaviour on compare-match (OSA/OSB encoding).
- enum `rx_tmr_irq_source_t` : `uint8_t` { `k_tmr_irq_none` = 0x00 , `k_tmr_irq_cmp_match_a` = 0x01 , `k_tmr_irq_cmp_match_b` = 0x02 , `k_tmr_irq_overflow` = 0x04 }
TMR interrupt source bitmask (may be OR-combined).

Functions

- `rx_err_t rx_tmr_init` (`const rx_tmr_config_t *config`)
Initialise a TMR channel (or cascaded pair).
- `rx_err_t rx_tmr_start` (`rx_tmr_channel_t channel`)
Start counting on a TMR channel.
- `rx_err_t rx_tmr_stop` (`rx_tmr_channel_t channel`)
Stop a TMR channel counter.
- `rx_err_t rx_tmr_read` (`rx_tmr_channel_t channel`, `uint16_t *value`)
Read the current TMR counter value.
- `rx_err_t rx_tmr_set_period_us` (`rx_tmr_channel_t channel`, `uint32_t period_us`)
Set the compare-match period in microseconds.
- `rx_err_t rx_tmr_register_compare_match_isr` (`rx_tmr_channel_t channel`, `rx_tmr_isr_callback_t callback`)
Register an interrupt callback for compare-match / overflow events.
- `rx_err_t rx_tmr_deinit` (`rx_tmr_channel_t channel`)
Deinitialise a TMR channel.

8.97.1 Detailed Description

TMR HAL Driver for RX72N 8-Bit Timer (TMRb).

Hardware abstraction layer for the RX72N TMR peripheral, providing four 8-bit timers (TMR0..TMR3) organized as two units of two channels. Each unit can be used either as two independent 8-bit timers or as a single 16-bit cascaded timer (TMR01 or TMR23).

8.97.1.1 Architecture

Application Layer: periodic tasks, soft PWM, low-speed triggers
 HAL Layer: rx_tmr.h/c (THIS MODULE - **register** access)
 Register Layer: rx72n_tmr_regs.h (memory-mapped I/O)

8.97.1.2 Unit and Cascade Layout

Unit	Channels	Module Stop	Cascade Name
0	TMR0, TMR1	MSTPCRA.MSTPA5	TMR01
1	TMR2, TMR3	MSTPCRA.MSTPA4	TMR23

In a cascaded pair, the even channel (TMR0/TMR2) holds the upper 8 bits and the odd channel (TMR1/TMR3) holds the lower 8 bits.

8.97.1.3 Clock Sources

The TMR counts on either an external signal at the TMCI pin or an internal PCLK-divided signal:

Source	Divider	Tick @ PCLKB=60 MHz	8-bit max period
Internal	PCLK/1	16.67 ns	4.27 us
Internal	PCLK/2	33.33 ns	8.53 us
Internal	PCLK/8	133.3 ns	34.13 us
Internal	PCLK/32	533.3 ns	136.5 us
Internal	PCLK/64	1.067 us	273.1 us
Internal	PCLK/1024	17.07 us	4.369 ms
Internal	PCLK/8192	136.5 us	34.95 ms
External	rising	external	external
External	falling	external	external
External	both	external	external

16-bit cascaded mode extends the max period by 256x.

8.97.1.4 Interrupts

TMR compare-match A/B and overflow interrupts are routed via the ICU SELECTB (Software Configurable Interrupt B) mechanism. The driver provides a callback registration hook so higher layers can install handlers without touching the interrupt controller directly.

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: All loops bounded
- Rule 3: No dynamic memory allocation
- Rule 4: All functions < 60 lines
- Rule 5: Minimum 2 validation checks per function
- Rule 6: Variables at smallest scope
- Rule 7: All return values checked
- Rule 8: C23 typed enums, minimal macros
- Rule 9: Function pointers allowed (DIP) for ISR registration
- Rule 10: Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Only TMR hardware access (no scheduling, no signal logic)
- O: Extensible via [rx_tmr_config_t](#)
- L: Consistent rx_err_t return semantics
- I: Focused API: init, start/stop, read, set period, register ISR, deinit
- D: Depends on register abstractions, not raw addresses

See also

[rx72n_tmr_regs.h](#) TMR register definitions

[rx_tpu.h](#) TPU HAL (16-bit pulse unit)

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_tmr.h](#).

8.97.2 Data Structure Documentation

8.97.2.1 struct rx_tmr_config_t

TMR channel configuration.

Passed to `rx_tmr_init()` to configure one channel (independent mode) or one cascaded pair (16-bit mode). In cascade mode, `channel` must be an even channel (`k_tmr_channel_0` or `k_tmr_channel_2`) and the paired odd channel is configured automatically.

Example - 1 ms periodic interrupt on TMR0

```
rx_tmr_config_t cfg = {
    .channel      = k_tmr_channel_0,
    .mode         = k_tmr_mode_8bit_independent,
    .clock_source = k_tmr_clock_pclk_div_1024,
    .counter_clear = k_tmr_clear_cmp_match_a,
    .output_a     = k_tmr_output_no_change,
    .output_b     = k_tmr_output_no_change,
    .irq_mask     = k_tmr_irq_cmp_match_a,
};
rx_err_t err = rx_tmr_init(&cfg);
```

Invariant

`channel` is one of `k_tmr_channel_0..3`

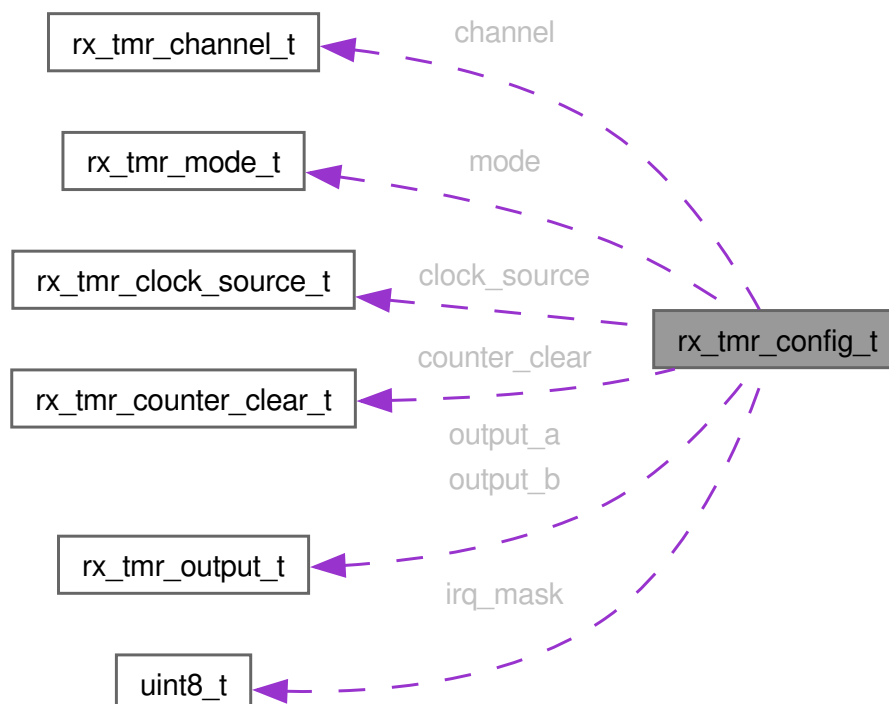
`clock_source` is a valid `rx_tmr_clock_source_t` value

Since

Version 1.0.0

Definition at line 235 of file `rx_tmr.h`.

Collaboration diagram for `rx_tmr_config_t`:



Data Fields

rx_tmr_channel_t	channel	TMR channel (cascade mode requires an even channel).
rx_tmr_clock_source_t	clock_source	Clock source (internal divider or external edge).
rx_tmr_counter_clear_t	counter_clear	Counter clear trigger.
uint8_t	irq_mask	Interrupt enable bitmask (OR of rx_tmr_irq_source_t values). Set to k_tmr_irq_none to leave all TMR interrupts disabled.
rx_tmr_mode_t	mode	Operating mode (8-bit independent or 16-bit cascade).
rx_tmr_output_t	output_a	TMO pin behaviour on TCORA compare match.
rx_tmr_output_t	output_b	TMO pin behaviour on TCORB compare match.

8.97.3 Typedef Documentation

8.97.3.1 rx`tmr`isr`callback`[t](#)

```
typedef void(* rx_tmr_isr_callback_t) (rx\_tmr\_channel\_t channel)
```

User callback for compare-match / overflow events.

Parameters

in	channel	TMR channel that raised the event
----	---------	-----------------------------------

Note

Called from interrupt context; keep implementations short.

Since

Version 1.0.0

Definition at line 270 of file [rx_tmr.h](#).

8.97.4 Enumeration Type Documentation

8.97.4.1 rx`tmr`channel`[t](#)

```
enum rx\_tmr\_channel\_t : uint8_t
```

TMR channel identifiers.

Four 8-bit TMR channels are available. Channels within the same unit (TMR0+TMR1 in unit 0, TMR2+TMR3 in unit 1) may be cascaded into a 16-bit counter; see [rx_tmr_mode_t](#) and [rx_tmr_init\(\)](#).

Since

Version 1.0.0

Enumerator

k_tmr_channel_0	TMR0 - unit 0 even channel
k_tmr_channel_1	TMR1 - unit 0 odd channel
k_tmr_channel_2	TMR2 - unit 1 even channel
k_tmr_channel_3	TMR3 - unit 1 odd channel

Definition at line 114 of file [rx_tmr.h](#).

```
00114      : uint8_t {
00115  k_tmr_channel_0 = 0, /**< TMR0 - unit 0 even channel */
00116  k_tmr_channel_1 = 1, /**< TMR1 - unit 0 odd channel */
00117  k_tmr_channel_2 = 2, /**< TMR2 - unit 1 even channel */
00118  k_tmr_channel_3 = 3, /**< TMR3 - unit 1 odd channel */
00119 } rx_tmr_channel_t;
```

8.97.4.2 rx_tmr_clock_source_t

```
enum rx\_tmr\_clock\_source\_t : uint8_t
```

TMR clock source selection.

Combines the CSS (clock source) and CKS (clock select) fields of TCCR into a single configuration value. Internal sources use PCLK dividers; external sources use the TMCI input pin with selectable edge detection.

Since

Version 1.0.0

Enumerator

k_tmr_clock_pclk_div_1	Internal PCLK / 1
k_tmr_clock_pclk_div_2	Internal PCLK / 2
k_tmr_clock_pclk_div_8	Internal PCLK / 8
k_tmr_clock_pclk_div_32	Internal PCLK / 32
k_tmr_clock_pclk_div_64	Internal PCLK / 64
k_tmr_clock_pclk_div_1024	Internal PCLK / 1024
k_tmr_clock_pclk_div_8192	Internal PCLK / 8192
k_tmr_clock_external_rising	External TMCI rising-edge count
k_tmr_clock_external_falling	External TMCI falling-edge count

<code>k_tmr_clock_external_both</code>	External TMCI both-edge count
--	-------------------------------

Definition at line 149 of file [rx_tmr.h](#).

```
00149      : uint8_t {
00150  k_tmr_clock_pclk_div_1    = 0, /**< Internal PCLK / 1 */
00151  k_tmr_clock_pclk_div_2    = 1, /**< Internal PCLK / 2 */
00152  k_tmr_clock_pclk_div_8    = 2, /**< Internal PCLK / 8 */
00153  k_tmr_clock_pclk_div_32   = 3, /**< Internal PCLK / 32 */
00154  k_tmr_clock_pclk_div_64   = 4, /**< Internal PCLK / 64 */
00155  k_tmr_clock_pclk_div_1024 = 5, /**< Internal PCLK / 1024 */
00156  k_tmr_clock_pclk_div_8192 = 6, /**< Internal PCLK / 8192 */
00157  k_tmr_clock_external_rising = 7, /**< External TMCI rising-edge count */
00158  k_tmr_clock_external_falling = 8, /**< External TMCI falling-edge count */
00159  k_tmr_clock_external_both  = 9, /**< External TMCI both-edge count */
00160 } rx_tmr_clock_source_t;
```

8.97.4.3 rx`tmr`counter`clear`

```
enum rx_tmr_counter_clear_t : uint8_t
```

Counter clear source (CCLR field of TCR).

Since

Version 1.0.0

Enumerator

<code>k_tmr_clear_disabled</code>	Counter never cleared by TMR itself
<code>k_tmr_clear_cmp_match_a</code>	Clear on TCORA compare match
<code>k_tmr_clear_cmp_match_b</code>	Clear on TCORB compare match
<code>k_tmr_clear_external_reset</code>	Clear on external reset input

Definition at line 168 of file [rx_tmr.h](#).

```
00168      : uint8_t {
00169  k_tmr_clear_disabled    = 0, /**< Counter never cleared by TMR itself */
00170  k_tmr_clear_cmp_match_a = 1, /**< Clear on TCORA compare match */
00171  k_tmr_clear_cmp_match_b = 2, /**< Clear on TCORB compare match */
00172  k_tmr_clear_external_reset = 3, /**< Clear on external reset input */
00173 } rx_tmr_counter_clear_t;
```

8.97.4.4 rx`tmr`irq`source`

```
enum rx_tmr_irq_source_t : uint8_t
```

TMR interrupt source bitmask (may be OR-combined).

Since

Version 1.0.0

Enumerator

k_tmr_irq_none	No interrupts enabled
k_tmr_irq_cmp_match_a	Enable CMIA (TCORA compare match)
k_tmr_irq_cmp_match_b	Enable CMIB (TCORB compare match)
k_tmr_irq_overflow	Enable OVI (TCNT overflow)

Definition at line 194 of file [rx_tmr.h](#).

```
00194      : uint8_t {
00195      k_tmr_irq_none      = 0x00, /**< No interrupts enabled          */
00196      k_tmr_irq_cmp_match_a = 0x01, /**< Enable CMIA (TCORA compare match) */
00197      k_tmr_irq_cmp_match_b = 0x02, /**< Enable CMIB (TCORB compare match) */
00198      k_tmr_irq_overflow    = 0x04, /**< Enable OVI (TCNT overflow)      */
00199 } rx_tmr_irq_source_t;
```

8.97.4.5 rx_tmr_mode_t

```
enum rx_tmr_mode_t : uint8_t
```

TMR operating mode (independent 8-bit vs cascaded 16-bit).

In 16-bit cascade mode, the even channel (TMR0/TMR2) provides the upper byte and the odd channel (TMR1/TMR3) provides the lower byte. When initialising cascade mode, pass the even channel number; the driver configures both channels accordingly.

Since

Version 1.0.0

Enumerator

k_tmr_mode_8bit_independent	Two independent 8-bit counters per unit
k_tmr_mode_16bit_cascade	Cascaded 16-bit counter (TMR01 or TMR23)

Definition at line 133 of file [rx_tmr.h](#).

```
00133      : uint8_t {
00134      k_tmr_mode_8bit_independent = 0, /**< Two independent 8-bit counters per unit */
00135      k_tmr_mode_16bit_cascade    = 1, /**< Cascaded 16-bit counter (TMR01 or TMR23) */
00136 } rx_tmr_mode_t;
```

8.97.4.6 rx_tmr_output_t

```
enum rx_tmr_output_t : uint8_t
```

TMO pin behaviour on compare-match (OSA/OSB encoding).

Since

Version 1.0.0

Enumerator

k_tmr_output_no_change	Pin unchanged on match
k_tmr_output_low	Drive pin low on match
k_tmr_output_high	Drive pin high on match
k_tmr_output_toggle	Toggle pin on match

Definition at line 181 of file [rx_tmr.h](#).

```
00181      : uint8_t {
00182  k_tmr_output_no_change = 0, /**< Pin unchanged on match */
00183  k_tmr_output_low      = 1, /**< Drive pin low on match */
00184  k_tmr_output_high     = 2, /**< Drive pin high on match */
00185  k_tmr_output_toggle   = 3, /**< Toggle pin on match */
00186 } rx_tmr_output_t;
```

8.97.5 Function Documentation

8.97.5.1 rx'tmr'deinit()

```
rx_err_t rx_tmr_deinit (
    rx_tmr_channel_t channel) [nodiscard]
```

Deinitialise a TMR channel.

Stops the counter, clears TCR/TCSR/TCCR, zeroes TCNT, removes any installed ISR callback, and marks the channel as uninitialised. The module stop bit is NOT asserted because the paired channel in the same unit may still be active.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Channel released
k_rx_err_invalid_arg	Channel out of range

Precondition

(none - safe to call on uninitialised channel)

Postcondition

Channel registers cleared
Internal init flag cleared

Note

Thread Safety: Not thread-safe.

Since

Version 1.0.0

Definition at line 625 of file [rx_tmr.c](#).

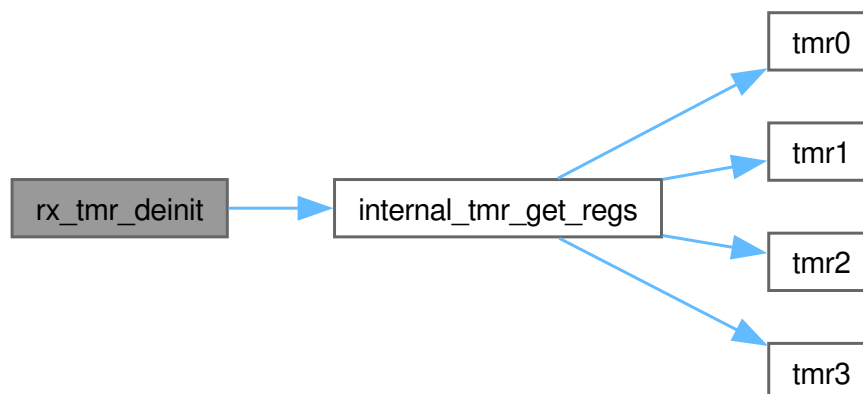
```

00626 {
00627     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00628         return k_rx_err_invalid_arg;
00629     }
00630
00631     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00632     regs->tccr = 0U;
00633     regs->tcr = 0U;
00634     regs->tcsr = 0U;
00635     regs->tcora = 0U;
00636     regs->tcorb = 0U;
00637     regs->tcnt = 0U;
00638
00639     const uint8_t idx = (uint8_t)channel;
00640     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00641         const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00642         volatile rx_tmr_channel_regs_t* odd_regs = internal_tmr_get_regs(odd);
00643         odd_regs->tccr = 0U;
00644         odd_regs->tcr = 0U;
00645         odd_regs->tcsr = 0U;
00646         odd_regs->tcora = 0U;
00647         odd_regs->tcorb = 0U;
00648         odd_regs->tcnt = 0U;
00649         s_tmr_initialized[(uint8_t)odd] = false;
00650         s_tmr_mode[(uint8_t)odd] = k_tmr_mode_8bit_independent;
00651         s_tmr_clock_bits[(uint8_t)odd] = 0U;
00652         s_tmr_callback[(uint8_t)odd] = NULL;
00653     }
00654
00655     s_tmr_initialized[idx] = false;
00656     s_tmr_mode[idx] = k_tmr_mode_8bit_independent;
00657     s_tmr_clock_bits[idx] = 0U;
00658     s_tmr_callback[idx] = NULL;
00659     return k_rx_ok;
00660 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_mode_8bit_independent](#), [s_tmr_callback](#), [s_tmr_clock_bits](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [rx_tmr_channel_regs_t::tcnt](#), [rx_tmr_channel_regs_t::tcora](#), [rx_tmr_channel_regs_t::tcorb](#), [rx_tmr_channel_regs_t::tcr](#), and [rx_tmr_channel_regs_t::tcsr](#).

Here is the call graph for this function:



8.97.5.2 rx'tmr'init()

```
rx_err_t rx_tmr_init (  
    const rx_tmr_config_t * config) [nodiscard]
```

Initialise a TMR channel (or cascaded pair).

Enables the module stop clock for the channel's unit (MSTPA5 for unit 0, MSTPA4 for unit 1), stops the counter, writes TCR/TCSR/TCCR/TCORA/ TCORB according to the supplied configuration, and clears TCNT to zero. In cascade mode, both channels of the unit are configured in one call.

Parameters

in	config	Pointer to configuration (non-NULL)
----	--------	-------------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	config is NULL
k_rx_err_invalid_arg	Invalid channel, clock, clear, output, or a cascade request on an odd channel
k_rx_err_invalid_state	Channel already initialised

Precondition

System clocks configured (PCLKB running)

In cascade mode, config->channel is k_tmr_channel_0 or k_tmr_channel_2 (even channel)

Postcondition

Channel counter stopped at 0

Module clock enabled (MSTPCRA bit cleared)

Internal init flag set

Note

Not thread-safe; call from main-task context at startup.

See also

[rx_tmr_start\(\)](#) Begin counting

[rx_tmr_deinit\(\)](#) Release the channel

Since

Version 1.0.0

Definition at line 464 of file [rx_tmr.c](#).

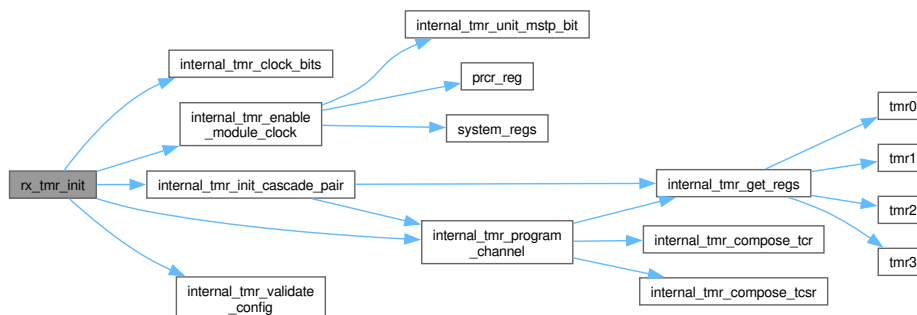
```

00465 {
00466     /* Pre-condition 1: non-null config */
00467     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is NULL");
00468
00469     /* Pre-condition 2: enum ranges */
00470     rx_err_t err = internal_tmr_validate_config(config);
00471     if (err != k_rx_ok) {
00472         rx_log_error(s_tag, "Invalid TMR config");
00473         return err;
00474     }
00475
00476     const uint8_t idx = (uint8_t)config->channel;
00477
00478     /* Pre-condition 3: not already initialised */
00479     if (s_tmr_initialized[idx]) {
00480         rx_log_error(s_tag, "Channel already initialised");
00481         return k_rx_err_invalid_state;
00482     }
00483
00484     /* Compose clock bits (validated via enum range above) */
00485     uint8_t clock_bits = 0U;
00486     err = internal_tmr_clock_bits(config->clock_source, &clock_bits);
00487     if (err != k_rx_ok) { /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range clock_source */
00488         return err; /* GCOVR_EXCL_LINE */
00489     } /* GCOVR_EXCL_BR_LINE */
00490
00491     /* 1. Enable unit module clock */
00492     internal_tmr_enable_module_clock(config->channel);
00493
00494     /* 2. Program the primary channel */
00495     internal_tmr_program_channel(config->channel, config);
00496
00497     /* 3. For cascade mode, program the paired odd channel */
00498     if (config->mode == k_tmr_mode_16bit_cascade) {
00499         internal_tmr_init_cascade_pair(config);
00500     }
00501
00502     /* 4. Cache state for the primary channel */
00503     s_tmr_mode[idx] = config->mode;
00504     s_tmr_clock_bits[idx] = clock_bits;
00505     s_tmr_clock_source[idx] = config->clock_source;
00506     s_tmr_initialized[idx] = true;
00507
00508     rx_log_info_val(s_tag, "TMR channel initialised", (uint32_t)config->channel);
00509     return k_rx_ok;
00510 }

```

References [rx_tmr_config_t::channel](#), [rx_tmr_config_t::clock_source](#), [internal_tmr_clock_bits\(\)](#), [internal_tmr_enable_module_clock\(\)](#), [internal_tmr_init_cascade_pair\(\)](#), [internal_tmr_program_channel\(\)](#), [internal_tmr_validate_config\(\)](#), [k_tmr_mode_16bit_cascade](#), [rx_tmr_config_t::mode](#), [s_tag](#), [s_tmr_clock_bits](#), [s_tmr_clock_source](#), [s_tmr_initialized](#), and [s_tmr_mode](#).

Here is the call graph for this function:



8.97.5.3 rx'tmr'read()

```
rx_err_t rx_tmr_read (  
    rx_tmr_channel_t channel,  
    uint16_t * value) [nodiscard]
```

Read the current TMR counter value.

For 8-bit mode, *value holds the 8-bit TCNT zero-extended to uint16_t. For 16-bit cascade mode, *value holds the full 16-bit value (even channel = upper byte, odd channel = lower byte).

Parameters

in	channel	TMR channel (cascade: pass the even channel)
out	value	Pointer receiving counter value (non-NULL)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	value is NULL
k_rx_err_invalid_arg	Channel out of range
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised

Postcondition

*value contains counter at time of read

Note

Thread Safety: Safe - single volatile read.

Since

Version 1.0.0

Definition at line 551 of file [rx_tmr.c](#).

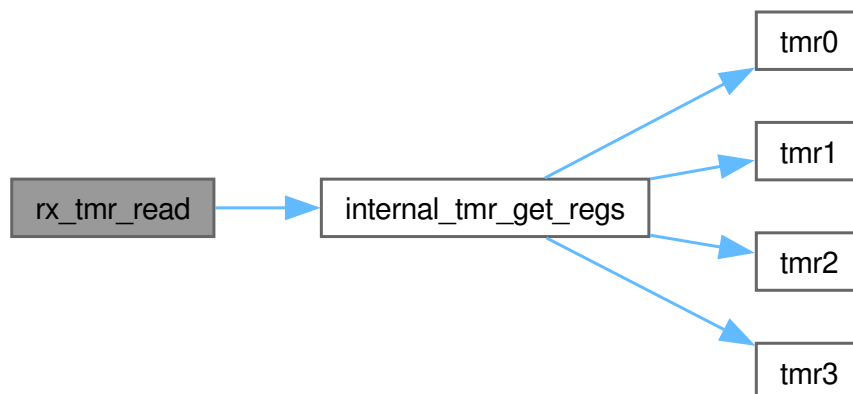
```

00552 {
00553     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is NULL");
00554     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00555         return k_rx_err_invalid_arg;
00556     }
00557     const uint8_t idx = (uint8_t)channel;
00558     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00559
00560     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00561         const uint8_t upper = internal_tmr_get_regs(channel)->tcnt;
00562         const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00563         const uint8_t lower = internal_tmr_get_regs(odd)->tcnt;
00564         *value = (uint16_t)(((uint16_t)upper << (uint16_t)k_tmr_byte_shift) | (uint16_t)lower);
00565     } else {
00566         *value = (uint16_t)internal_tmr_get_regs(channel)->tcnt;
00567     }
00568     return k_rx_ok;
00569 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_byte_shift](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [s_tag](#), [s_tmr_initialized](#), [s_tmr_mode](#), and [rx_tmr_channel_regs_t::tcnt](#).

Here is the call graph for this function:



8.97.5.4 rx_tmr_register_compare_match_isr()

```

rx_err_t rx_tmr_register_compare_match_isr (
    rx_tmr_channel_t channel,
    rx_tmr_isr_callback_t callback) [nodiscard]

```

Register an interrupt callback for compare-match / overflow events.

Installs a user callback invoked from the ICU SELECTB dispatcher when any of the channel's enabled TMR interrupts (CMIA, CMIB, or OVI) fire. The driver's ISR shim clears the source flag before calling back. Pass NULL to deregister.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

in	callback	User callback, or NULL to deregister
----	----------	--------------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_invalid_arg	Channel out of range

Precondition

Channel initialised with desired irq_mask

Postcondition

callback is installed (or cleared) atomically

Note

Thread Safety: Safe - function-pointer write is atomic on RX.

Since

Version 1.0.0

Definition at line 616 of file [rx_tmr.c](#).

```

00617 {
00618     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00619         return k_rx_err_invalid_arg;
00620     }
00621     s_tmr_callback[(uint8_t)channel] = callback;
00622     return k_rx_ok;
00623 }
```

References [k_tmr_channel_count_internal](#), and [s_tmr_callback](#).

8.97.5.5 rx`tmr`set`period`us()

```

rx_err_t rx_tmr_set_period_us (
    rx_tmr_channel_t channel,
    uint32_t period_us) [nodiscard]
```

Set the compare-match period in microseconds.

Recomputes and writes TCORA so that a TCORA compare match fires every period_us microseconds using the channel's configured clock source. TCORB is untouched. The caller must ensure the period fits into the channel counter width (8-bit single channel max = ~35 ms at PCLK/8192; 16-bit cascade max ~ = 8.9 s at PCLK/8192).

The underlying formula is $tcora = \text{round}(\text{period_us} * \text{PCLK_Hz} / (\text{divider} * 1e6)) - 1$, where divider is 1/2/8/32/64/1024/8192 according to the configured clock_source. External clock sources are not supported.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
in	period_us	Desired period in microseconds (1..max for mode)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, TCORA updated
k_rx_err_invalid_arg	Channel out of range, period=0, period too large for the selected divider, or external clock source
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised with an internal-clock clock_source

Postcondition

TCORA holds the rounded tick count - 1

Note

Thread Safety: Safe - single register write.

Since

Version 1.0.0

Definition at line 571 of file [rx_tmr.c](#).

```

00572 {
00573     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00574         return k_rx_err_invalid_arg;
00575     }
00576     const uint8_t idx = (uint8_t)channel;
00577     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00578
00579     if (period_us == 0U) {
00580         return k_rx_err_invalid_arg;
00581     }
00582
00583     const uint32_t divider = internal_tmr_clock_divider(s_tmr_clock_source[idx]);
00584     if (divider == 0U) {
00585         /* External clock selected - cannot compute period */
00586         return k_rx_err_invalid_arg;
00587     }
00588
00589     /* tick_hz = PCLKB / divider; ticks = period_us * tick_hz / 1e6
00590      * compare = ticks - 1 (TCORA fires when TCNT reaches compare value) */
00591     const uint64_t ticks =

```

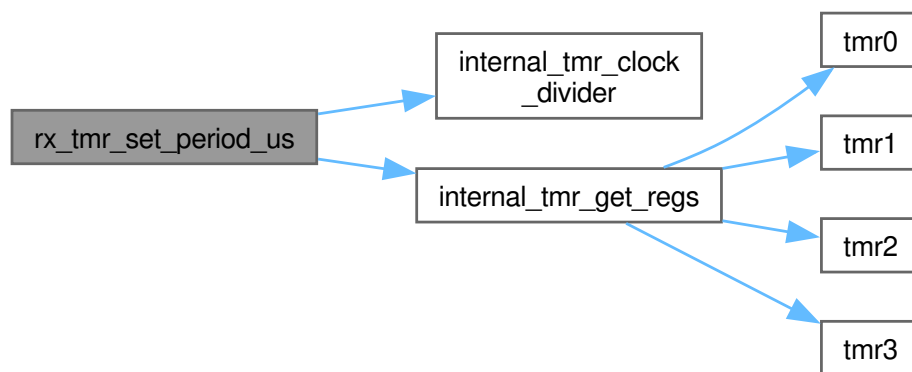
```

00592 ((uint64_t)period_us * (uint64_t)k_pclkb_hz) / ((uint64_t)divider * (uint64_t)k_tmr_us_per_sec);
00593 if (ticks == 0U) {
00594     return k_rx_err_invalid_arg;
00595 }
00596 const uint64_t compare = ticks - 1U;
00597
00598 const uint32_t max_compare =
00599     (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) ? k_tmr_max_tcora_16bit : k_tmr_max_tcora;
00600 if (compare > (uint64_t)max_compare) {
00601     rx_log_error_val(s_tag, "Period too large for divider", period_us);
00602     return k_rx_err_invalid_arg;
00603 }
00604
00605 if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00606     const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00607     internal_tmr_get_regs(channel)->tcora =
00608         (uint8_t)((compare » (uint64_t)k_tmr_byte_shift) & (uint64_t)k_tmr_byte_mask);
00609     internal_tmr_get_regs(odd)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00610 } else {
00611     internal_tmr_get_regs(channel)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00612 }
00613 return k_rx_ok;
00614 }

```

References [internal_tmr_clock_divider\(\)](#), [internal_tmr_get_regs\(\)](#), [k_pclkb_hz](#), [k_tmr_byte_mask](#), [k_tmr_byte_shift](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_max_tcora](#), [k_tmr_max_tcora_16bit](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_us_per_sec](#), [s_tag](#), [s_tmr_clock_source](#), [s_tmr_initialized](#), [s_tmr_mode](#), and [rx_tmr_channel_regs_t::tcora](#).

Here is the call graph for this function:



8.97.5.6 rx_tmr_start()

```

rx_err_t rx_tmr_start (
    rx_tmr_channel_t channel) [nodiscard]

```

Start counting on a TMR channel.

Writes the configured clock-source bits into TCCR so the counter begins advancing on each selected clock edge. Channel must have been initialised via [rx_tmr_init\(\)](#).

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Counter started
<code>k_rx_err_invalid_arg</code>	Channel out of range
<code>k_rx_err_invalid_state</code>	Channel not initialised

Precondition

Channel initialised via [rx_tmr_init\(\)](#)

Postcondition

TCCR programmed with clock source (counter running)

Note

Thread Safety: Safe - single register write.

Since

Version 1.0.0

Definition at line 512 of file [rx_tmr.c](#).

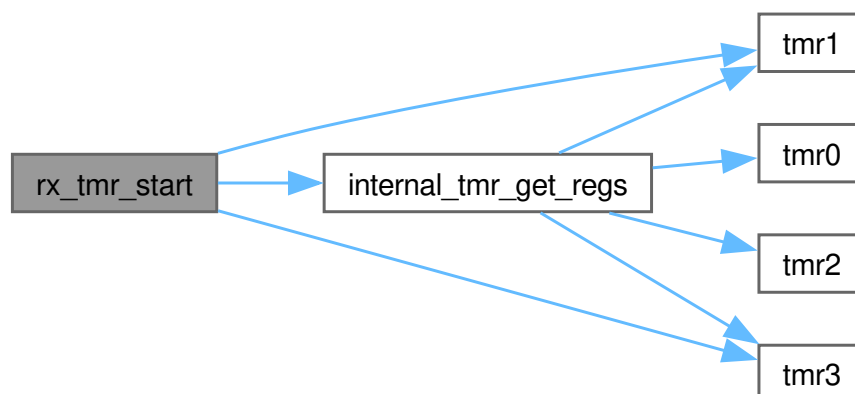
```

00513 {
00514     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00515         return k_rx_err_invalid_arg;
00516     }
00517     const uint8_t idx = (uint8_t)channel;
00518     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00519
00520     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00521     regs->tccr = s_tmr_clock_bits[idx];
00522
00523     /* In cascade mode the paired odd channel also needs its cascade bits
00524      * kept after a stop/start cycle. Cascade is only valid on even channels
00525      * (enforced by validate_config), so the ternary below is exhaustive. */
00526     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00527         volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00528         odd_regs->tccr = k_tmr_tccr_css_cascade;
00529     }
00530     return k_rx_ok;
00531 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_tccr_css_cascade](#), [s_tag](#), [s_tmr_clock_bits](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [tmr1\(\)](#), and [tmr3\(\)](#).

Here is the call graph for this function:



8.97.5.7 rx_tmr_stop()

```
rx_err_t rx_tmr_stop (
    rx_tmr_channel_t channel) [nodiscard]
```

Stop a TMR channel counter.

Writes 0 into TCCR CKS/CSS to disable the clock input. The TCNT value is preserved.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Counter stopped
k_rx_err_invalid_arg	Channel out of range
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised

Postcondition

Counter no longer advances

TCNT preserved

Note

Thread Safety: Safe - single register write.

Since

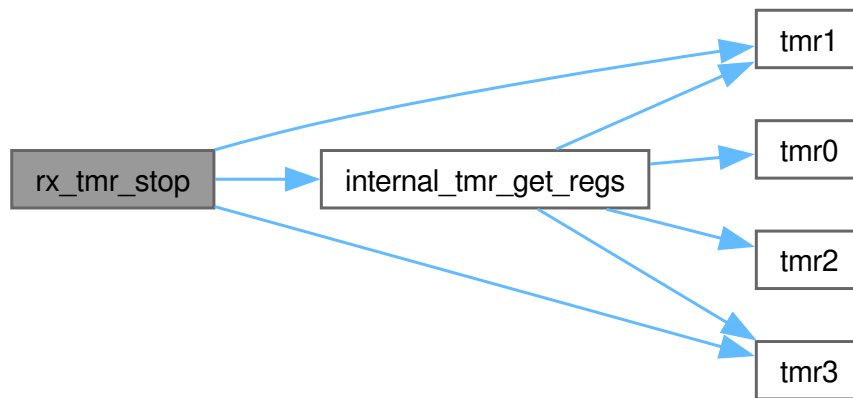
Version 1.0.0

Definition at line 533 of file [rx_tmr.c](#).

```
00534 {
00535     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00536         return k_rx_err_invalid_arg;
00537     }
00538     const uint8_t idx = (uint8_t)channel;
00539     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00540
00541     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00542     regs->tccr = 0U;
00543
00544     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00545         volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00546         odd_regs->tccr = 0U;
00547     }
00548     return k_rx_ok;
00549 }
```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [s_tag](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [tmr1\(\)](#), and [tmr3\(\)](#).

Here is the call graph for this function:



8.98 rx_tmr.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_tmr.h
00003  * @brief TMR HAL Driver for RX72N 8-Bit Timer (TMRb)
00004  *
00005  * @details
00006  * Hardware abstraction layer for the RX72N TMR peripheral, providing
00007  * four 8-bit timers (TMR0..TMR3) organized as two units of two channels.
00008  * Each unit can be used either as two independent 8-bit timers or as a
00009  * single 16-bit cascaded timer (TMR01 or TMR23).
00010  *
00011  * ## Architecture
00012  *
00013  * ```
00014  * Application Layer:  periodic tasks, soft PWM, low-speed triggers
00015  *                      |
00016  * HAL Layer:          rx_tmr.h/c (THIS MODULE - register access)
00017  *                      |
00018  * Register Layer:     rx72n_tmr_regs.h (memory-mapped I/O)
00019  * ```
00020  *
00021  * ## Unit and Cascade Layout
00022  *
00023  * | Unit | Channels | Module Stop | Cascade Name |
00024  * |-----|-----|-----|-----|
00025  * | 0 | TMR0, TMR1 | MSTPCRA.MSTPA5 | TMR01 |
00026  * | 1 | TMR2, TMR3 | MSTPCRA.MSTPA4 | TMR23 |
00027  *
00028  * In a cascaded pair, the even channel (TMR0/TMR2) holds the upper 8 bits
00029  * and the odd channel (TMR1/TMR3) holds the lower 8 bits.
00030  *
00031  * ## Clock Sources
00032  *
00033  * The TMR counts on either an external signal at the TMCI pin or an
00034  * internal PCLK-divided signal:
00035  *
00036  * | Source | Divider | Tick @ PCLKB=60 MHz | 8-bit max period |
00037  * |-----|-----|-----|-----|
00038  * | Internal | PCLK/1 | 16.67 ns | 4.27 us |
00039  * | Internal | PCLK/2 | 33.33 ns | 8.53 us |
00040  * | Internal | PCLK/8 | 133.3 ns | 34.13 us |
00041  * | Internal | PCLK/32 | 533.3 ns | 136.5 us |
00042  * | Internal | PCLK/64 | 1.067 us | 273.1 us |
00043  * | Internal | PCLK/1024 | 17.07 us | 4.369 ms |
00044  * | Internal | PCLK/8192 | 136.5 us | 34.95 ms |
00045  * | External | rising | external | external |
00046  * | External | falling | external | external |
00047  * | External | both | external | external |
00048  *
00049  * 16-bit cascaded mode extends the max period by 256x.
00050  *
00051  * ## Interrupts
00052  *
00053  * TMR compare-match A/B and overflow interrupts are routed via the ICU
00054  * SELECTB (Software Configurable Interrupt B) mechanism. The driver
00055  * provides a callback registration hook so higher layers can install
00056  * handlers without touching the interrupt controller directly.
00057  *
00058  * @par NASA Power of 10 Compliance

```



```

00059 * - Rule 1: No goto, setjmp, or recursion
00060 * - Rule 2: All loops bounded
00061 * - Rule 3: No dynamic memory allocation
00062 * - Rule 4: All functions < 60 lines
00063 * - Rule 5: Minimum 2 validation checks per function
00064 * - Rule 6: Variables at smallest scope
00065 * - Rule 7: All return values checked
00066 * - Rule 8: C23 typed enums, minimal macros
00067 * - Rule 9: Function pointers allowed (DIP) for ISR registration
00068 * - Rule 10: Compiles with -Wall -Wextra -Werror
00069 *
00070 * @par SOLID Principles
00071 * - **S**: Only TMR hardware access (no scheduling, no signal logic)
00072 * - **O**: Extensible via rx_tmr_config_t
00073 * - **I**: Consistent rx_err_t return semantics
00074 * - **I**: Focused API: init, start/stop, read, set period, register ISR, deinit
00075 * - **D**: Depends on register abstractions, not raw addresses
00076 *
00077 * @see rx72n_tmr_regs.h TMR register definitions
00078 * @see rx_tpu.h TPU HAL (16-bit pulse unit)
00079 *
00080 * @author Locked, Inc.
00081 * @date 2026-04-21
00082 * @copyright Copyright (c) 2026 Locked Inc.
00083 * SPDX-License-Identifier: MIT
00084 * @since Version 1.0.0
00085 */
00086
00087 #pragma once
00088
00089 #include <stdbool.h>
00090 #include <stdint.h>
00091
00092 #include "rx_err.h"
00093
00094 #ifdef __cplusplus
00095 extern "C" {
00096 #endif
00097
00098 /*
=====
00099  * Enumerations
00100  *
=====
00101  */
00102
00103 /**
00104  * @enum rx_tmr_channel_t
00105  * @brief TMR channel identifiers
00106  *
00107  * @details
00108  * Four 8-bit TMR channels are available. Channels within the same unit
00109  * (TMR0+TMR1 in unit 0, TMR2+TMR3 in unit 1) may be cascaded into a
00110  * 16-bit counter; see #rx_tmr_mode_t and rx_tmr_init().
00111  *
00112  * @since Version 1.0.0
00113  */
00114 typedef enum : uint8_t {
00115     k_tmr_channel_0 = 0, /**< TMR0 - unit 0 even channel */
00116     k_tmr_channel_1 = 1, /**< TMR1 - unit 0 odd channel */
00117     k_tmr_channel_2 = 2, /**< TMR2 - unit 1 even channel */
00118     k_tmr_channel_3 = 3, /**< TMR3 - unit 1 odd channel */
00119 } rx_tmr_channel_t;
00120
00121 /**
00122  * @enum rx_tmr_mode_t
00123  * @brief TMR operating mode (independent 8-bit vs cascaded 16-bit)
00124  *
00125  * @details
00126  * In 16-bit cascade mode, the even channel (TMR0/TMR2) provides the upper
00127  * byte and the odd channel (TMR1/TMR3) provides the lower byte. When
00128  * initialising cascade mode, pass the even channel number; the driver
00129  * configures both channels accordingly.
00130  *
00131  * @since Version 1.0.0
00132  */
00133 typedef enum : uint8_t {
00134     k_tmr_mode_8bit_independent = 0, /**< Two independent 8-bit counters per unit */
00135     k_tmr_mode_16bit_cascade = 1, /**< Cascaded 16-bit counter (TMR01 or TMR23) */
00136 } rx_tmr_mode_t;
00137
00138 /**
00139  * @enum rx_tmr_clock_source_t
00140  * @brief TMR clock source selection
00141  *
00142  * @details
00143  * Combines the CSS (clock source) and CKS (clock select) fields of TCCR

```

```

00144 * into a single configuration value. Internal sources use PCLK dividers;
00145 * external sources use the TMCI input pin with selectable edge detection.
00146 *
00147 * @since Version 1.0.0
00148 */
00149 typedef enum : uint8_t {
00150     k_tmr_clock_pclk_div_1      = 0, /**< Internal PCLK / 1 */
00151     k_tmr_clock_pclk_div_2      = 1, /**< Internal PCLK / 2 */
00152     k_tmr_clock_pclk_div_8      = 2, /**< Internal PCLK / 8 */
00153     k_tmr_clock_pclk_div_32     = 3, /**< Internal PCLK / 32 */
00154     k_tmr_clock_pclk_div_64     = 4, /**< Internal PCLK / 64 */
00155     k_tmr_clock_pclk_div_1024   = 5, /**< Internal PCLK / 1024 */
00156     k_tmr_clock_pclk_div_8192   = 6, /**< Internal PCLK / 8192 */
00157     k_tmr_clock_external_rising = 7, /**< External TMCI rising-edge count */
00158     k_tmr_clock_external_falling = 8, /**< External TMCI falling-edge count */
00159     k_tmr_clock_external_both   = 9, /**< External TMCI both-edge count */
00160 } rx_tmr_clock_source_t;
00161
00162 /**
00163 * @enum rx_tmr_counter_clear_t
00164 * @brief Counter clear source (CCLR field of TCR)
00165 *
00166 * @since Version 1.0.0
00167 */
00168 typedef enum : uint8_t {
00169     k_tmr_clear_disabled        = 0, /**< Counter never cleared by TMR itself */
00170     k_tmr_clear_cmp_match_a     = 1, /**< Clear on TCORA compare match */
00171     k_tmr_clear_cmp_match_b     = 2, /**< Clear on TCORB compare match */
00172     k_tmr_clear_external_reset  = 3, /**< Clear on external reset input */
00173 } rx_tmr_counter_clear_t;
00174
00175 /**
00176 * @enum rx_tmr_output_t
00177 * @brief TMO pin behaviour on compare-match (OSA/OSB encoding)
00178 *
00179 * @since Version 1.0.0
00180 */
00181 typedef enum : uint8_t {
00182     k_tmr_output_no_change      = 0, /**< Pin unchanged on match */
00183     k_tmr_output_low           = 1, /**< Drive pin low on match */
00184     k_tmr_output_high          = 2, /**< Drive pin high on match */
00185     k_tmr_output_toggle        = 3, /**< Toggle pin on match */
00186 } rx_tmr_output_t;
00187
00188 /**
00189 * @enum rx_tmr_irq_source_t
00190 * @brief TMR interrupt source bitmask (may be OR-combined)
00191 *
00192 * @since Version 1.0.0
00193 */
00194 typedef enum : uint8_t {
00195     k_tmr_irq_none              = 0x00, /**< No interrupts enabled */
00196     k_tmr_irq_cmp_match_a       = 0x01, /**< Enable CMIA (TCORA compare match) */
00197     k_tmr_irq_cmp_match_b       = 0x02, /**< Enable CMIB (TCORB compare match) */
00198     k_tmr_irq_overflow          = 0x04, /**< Enable OVI (TCNT overflow) */
00199 } rx_tmr_irq_source_t;
00200
00201 /*
=====
00202 * Configuration Structure
00203 *
=====
00204 */
00205
00206 /**
00207 * @struct rx_tmr_config_t
00208 * @brief TMR channel configuration
00209 *
00210 * @details
00211 * Passed to rx_tmr_init() to configure one channel (independent mode) or
00212 * one cascaded pair (16-bit mode). In cascade mode, #channel must be an
00213 * even channel (k_tmr_channel_0 or k_tmr_channel_2) and the paired odd
00214 * channel is configured automatically.
00215 *
00216 * @par Example - 1 ms periodic interrupt on TMR0
00217 * @code{.c}
00218 * rx_tmr_config_t cfg = {
00219 *     .channel      = k_tmr_channel_0,
00220 *     .mode         = k_tmr_mode_8bit_independent,
00221 *     .clock_source = k_tmr_clock_pclk_div_1024,
00222 *     .counter_clear = k_tmr_clear_cmp_match_a,
00223 *     .output_a     = k_tmr_output_no_change,
00224 *     .output_b     = k_tmr_output_no_change,
00225 *     .irq_mask     = k_tmr_irq_cmp_match_a,
00226 * };
00227 * rx_err_t err = rx_tmr_init(&cfg);
00228 * @endcode

```

```

00229 *
00230 * @invariant channel is one of k_tmr_channel_0..3
00231 * @invariant clock_source is a valid rx_tmr_clock_source_t value
00232 *
00233 * @since Version 1.0.0
00234 */
00235 typedef struct {
00236     /** @brief TMR channel (cascade mode requires an even channel) */
00237     rx_tmr_channel_t channel;
00238
00239     /** @brief Operating mode (8-bit independent or 16-bit cascade) */
00240     rx_tmr_mode_t mode;
00241
00242     /** @brief Clock source (internal divider or external edge) */
00243     rx_tmr_clock_source_t clock_source;
00244
00245     /** @brief Counter clear trigger */
00246     rx_tmr_counter_clear_t counter_clear;
00247
00248     /** @brief TMO pin behaviour on TCORA compare match */
00249     rx_tmr_output_t output_a;
00250
00251     /** @brief TMO pin behaviour on TCORB compare match */
00252     rx_tmr_output_t output_b;
00253
00254     /**
00255      * @brief Interrupt enable bitmask (OR of rx_tmr_irq_source_t values)
00256      * @details Set to k_tmr_irq_none to leave all TMR interrupts disabled.
00257      */
00258     uint8_t irq_mask;
00259 } rx_tmr_config_t;
00260
00261 /**
00262  * @typedef rx_tmr_isr_callback_t
00263  * @brief User callback for compare-match / overflow events
00264  *
00265  * @param[in] channel TMR channel that raised the event
00266  *
00267  * @note Called from interrupt context; keep implementations short.
00268  * @since Version 1.0.0
00269  */
00270 typedef void (*rx_tmr_isr_callback_t)(rx_tmr_channel_t channel);
00271
00272 /*
=====
00273 * Public API
00274 *
=====
00275 */
00276
00277 /**
00278  * @brief Initialise a TMR channel (or cascaded pair)
00279  *
00280  * @details
00281  * Enables the module stop clock for the channel's unit (MSTPA5 for unit
00282  * 0, MSTPA4 for unit 1), stops the counter, writes TCR/TCSR/TCCR/TCORA/
00283  * TCORB according to the supplied configuration, and clears TCNT to zero.
00284  * In cascade mode, both channels of the unit are configured in one call.
00285  *
00286  * @param[in] config Pointer to configuration (non-NULL)
00287  *
00288  * @return rx_err_t Error code
00289  * @retval k_rx_ok Success
00290  * @retval k_rx_err_null_ptr config is NULL
00291  * @retval k_rx_err_invalid_arg Invalid channel, clock, clear, output, or
00292  * a cascade request on an odd channel
00293  * @retval k_rx_err_invalid_state Channel already initialised
00294  *
00295  * @pre System clocks configured (PCLKB running)
00296  * @pre In cascade mode, config->channel is k_tmr_channel_0 or
00297  * k_tmr_channel_2 (even channel)
00298  *
00299  * @post Channel counter stopped at 0
00300  * @post Module clock enabled (MSTPCRA bit cleared)
00301  * @post Internal init flag set
00302  *
00303  * @note Not thread-safe; call from main-task context at startup.
00304  *
00305  * @see rx_tmr_start() Begin counting
00306  * @see rx_tmr_deinit() Release the channel
00307  * @since Version 1.0.0
00308  */
00309 [[nodiscard]] rx_err_t rx_tmr_init(const rx_tmr_config_t* config);
00310
00311 /**
00312  * @brief Start counting on a TMR channel
00313  *

```

```

00314 * @details
00315 * Writes the configured clock-source bits into TCCR so the counter begins
00316 * advancing on each selected clock edge. Channel must have been initialised
00317 * via rx_tmr_init().
00318 *
00319 * @param[in] channel TMR channel (cascade: pass the even channel)
00320 *
00321 * @return rx_err_t Error code
00322 * @retval k_rx_ok Counter started
00323 * @retval k_rx_err_invalid_arg Channel out of range
00324 * @retval k_rx_err_invalid_state Channel not initialised
00325 *
00326 * @pre Channel initialised via rx_tmr_init()
00327 * @post TCCR programmed with clock source (counter running)
00328 *
00329 * @note Thread Safety: Safe - single register write.
00330 * @since Version 1.0.0
00331 */
00332 [[nodiscard]] rx_err_t rx_tmr_start(rx_tmr_channel_t channel);
00333
00334 /**
00335 * @brief Stop a TMR channel counter
00336 *
00337 * @details
00338 * Writes 0 into TCCR CKS/CSS to disable the clock input. The TCNT value
00339 * is preserved.
00340 *
00341 * @param[in] channel TMR channel (cascade: pass the even channel)
00342 *
00343 * @return rx_err_t Error code
00344 * @retval k_rx_ok Counter stopped
00345 * @retval k_rx_err_invalid_arg Channel out of range
00346 * @retval k_rx_err_invalid_state Channel not initialised
00347 *
00348 * @pre Channel initialised
00349 * @post Counter no longer advances
00350 * @post TCNT preserved
00351 *
00352 * @note Thread Safety: Safe - single register write.
00353 * @since Version 1.0.0
00354 */
00355 [[nodiscard]] rx_err_t rx_tmr_stop(rx_tmr_channel_t channel);
00356
00357 /**
00358 * @brief Read the current TMR counter value
00359 *
00360 * @details
00361 * For 8-bit mode, *value holds the 8-bit TCNT zero-extended to uint16_t.
00362 * For 16-bit cascade mode, *value holds the full 16-bit value
00363 * (even channel = upper byte, odd channel = lower byte).
00364 *
00365 * @param[in] channel TMR channel (cascade: pass the even channel)
00366 * @param[out] value Pointer receiving counter value (non-NULL)
00367 *
00368 * @return rx_err_t Error code
00369 * @retval k_rx_ok Success
00370 * @retval k_rx_err_null_ptr value is NULL
00371 * @retval k_rx_err_invalid_arg Channel out of range
00372 * @retval k_rx_err_invalid_state Channel not initialised
00373 *
00374 * @pre Channel initialised
00375 * @post *value contains counter at time of read
00376 *
00377 * @note Thread Safety: Safe - single volatile read.
00378 * @since Version 1.0.0
00379 */
00380 [[nodiscard]] rx_err_t rx_tmr_read(rx_tmr_channel_t channel, uint16_t* value);
00381
00382 /**
00383 * @brief Set the compare-match period in microseconds
00384 *
00385 * @details
00386 * Recomputes and writes TCORA so that a TCORA compare match fires every
00387 * @p period_us microseconds using the channel's configured clock source.
00388 * TCORB is untouched. The caller must ensure the period fits into the
00389 * channel counter width (8-bit single channel max = ~35 ms at PCLK/8192;
00390 * 16-bit cascade max ~= 8.9 s at PCLK/8192).
00391 *
00392 * The underlying formula is
00393 * <tt>tcora = round(period_us * PCLK_Hz / (divider * 1e6)) - 1</tt>,
00394 * where divider is 1/2/8/32/64/1024/8192 according to the configured
00395 * clock_source. External clock sources are not supported.
00396 *
00397 * @param[in] channel TMR channel (cascade: pass the even channel)
00398 * @param[in] period_us Desired period in microseconds (1..max for mode)
00399 *
00400 * @return rx_err_t Error code

```

```

00401 * @retval k_rx_ok Success, TCORA updated
00402 * @retval k_rx_err_invalid_arg Channel out of range, period=0, period too
00403 *      large for the selected divider, or external clock source
00404 * @retval k_rx_err_invalid_state Channel not initialised
00405 *
00406 * @pre Channel initialised with an internal-clock clock_source
00407 * @post TCORA holds the rounded tick count - 1
00408 *
00409 * @note Thread Safety: Safe - single register write.
00410 * @since Version 1.0.0
00411 */
00412 [[nodiscard]] rx_err_t rx_tmr_set_period_us(rx_tmr_channel_t channel, uint32_t period_us);
00413
00414 /**
00415 * @brief Register an interrupt callback for compare-match / overflow events
00416 *
00417 * @details
00418 * Installs a user callback invoked from the ICU SELECTB dispatcher when
00419 * any of the channel's enabled TMR interrupts (CMIA, CMIB, or OVI) fire.
00420 * The driver's ISR shim clears the source flag before calling back.
00421 * Pass NULL to deregister.
00422 *
00423 * @param[in] channel TMR channel (cascade: pass the even channel)
00424 * @param[in] callback User callback, or NULL to deregister
00425 *
00426 * @return rx_err_t Error code
00427 * @retval k_rx_ok Success
00428 * @retval k_rx_err_invalid_arg Channel out of range
00429 *
00430 * @pre Channel initialised with desired irq_mask
00431 * @post callback is installed (or cleared) atomically
00432 *
00433 * @note Thread Safety: Safe - function-pointer write is atomic on RX.
00434 * @since Version 1.0.0
00435 */
00436 [[nodiscard]] rx_err_t rx_tmr_register_compare_match_isr(rx_tmr_channel_t channel,
00437                                                         rx_tmr_isr_callback_t callback);
00438
00439 /**
00440 * @brief Deinitialise a TMR channel
00441 *
00442 * @details
00443 * Stops the counter, clears TCR/TCSR/TCCR, zeroes TCNT, removes any
00444 * installed ISR callback, and marks the channel as uninitialised. The
00445 * module stop bit is NOT asserted because the paired channel in the same
00446 * unit may still be active.
00447 *
00448 * @param[in] channel TMR channel (cascade: pass the even channel)
00449 *
00450 * @return rx_err_t Error code
00451 * @retval k_rx_ok Channel released
00452 * @retval k_rx_err_invalid_arg Channel out of range
00453 *
00454 * @pre (none - safe to call on uninitialised channel)
00455 * @post Channel registers cleared
00456 * @post Internal init flag cleared
00457 *
00458 * @note Thread Safety: Not thread-safe.
00459 * @since Version 1.0.0
00460 */
00461 [[nodiscard]] rx_err_t rx_tmr_deinit(rx_tmr_channel_t channel);
00462
00463 #ifdef __cplusplus
00464 }
00465 #endif

```

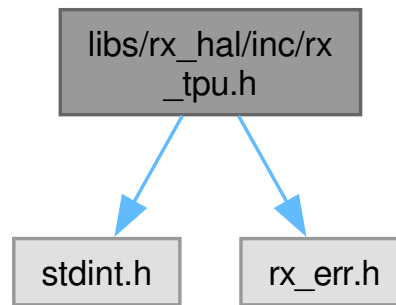
8.99 libs/rx`hal/inc/rx`tpu.h File Reference

TPU HAL Driver for RX72N 16-Bit Timer Pulse Unit (Phase Counting).

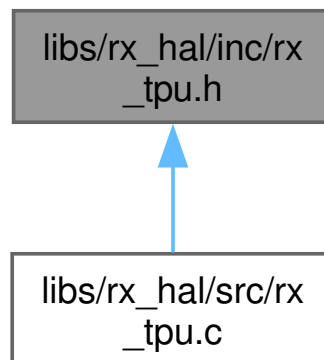
```
#include <stdint.h>
```

```
#include "rx_err.h"
```

Include dependency graph for rx_tpu.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_tpu_config_t](#)
TPU phase counting configuration.

Enumerations

- enum [rx_tpu_channel_t](#) : `uint8_t { k\_tpu\_channel\_1 = 1 , k\_tpu\_channel\_2 = 2 , k\_tpu\_channel\_4 = 4 , k\_tpu\_channel\_5 = 5 }`
TPU channel identifiers for phase counting.
- enum [rx_tpu_phase_mode_t](#) : `uint8_t { k\_tpu\_phase\_mode\_1 = 1 , k\_tpu\_phase\_mode\_2 = 2 , k\_tpu\_phase\_mode\_3 = 3 , k\_tpu\_phase\_mode\_4 = 4 }`
TPU phase counting mode selection.

Functions

- `rx_err_t rx\_tpu\_init\_phase\_count (const rx\_tpu\_config\_t *config)`
Initialize TPU channel for phase counting mode.
- `rx_err_t rx\_tpu\_start (rx\_tpu\_channel\_t channel)`
Start TPU channel counter.
- `rx_err_t rx\_tpu\_stop (rx\_tpu\_channel\_t channel)`
Stop TPU channel counter.
- `rx_err_t rx\_tpu\_read\_count (rx\_tpu\_channel\_t channel, uint16_t *count)`
Read TPU channel 16-bit counter value.
- `rx_err_t rx\_tpu\_read\_direction (rx\_tpu\_channel\_t channel, bool *counting_up)`
Read TPU channel counting direction.
- `rx_err_t rx\_tpu\_reset\_count (rx\_tpu\_channel\_t channel)`
Reset TPU channel counter to zero.
- `rx_err_t rx\_tpu\_deinit (rx\_tpu\_channel\_t channel)`
Deinitialize TPU channel.

8.99.1 Detailed Description

TPU HAL Driver for RX72N 16-Bit Timer Pulse Unit (Phase Counting).

Hardware abstraction layer for the RX72N TPU peripheral, focused on phase counting mode for quadrature encoder input. This driver provides low-level channel configuration, counter access, and direction reading.

8.99.1.1 STAR Project Usage

The TPU is used for rear wheel encoder decoding. Front wheels use the MTU peripheral (see [rx_mtu.h](#) / [rx_mtu_encoder.h](#)).

Encoder	Peripheral	Channel	Phase A Pin	Phase B Pin
0 (FL)	MTU1	MTU1	P24/MTCLKA (pin 33)	P25/MTCLKB (pin 32)
1 (FR)	MTU2	MTU2	PA1/MTCLKC (pin 96)	PC5/MTCLKD (pin 62)
2 (BR)	TPU1	TPU1	PC2/TCLKA (pin 70)	PA3/TCLKB (pin 94)
3 (BL)	TPU2	TPU2	PC0/TCLKC (pin 75)	PB3/TCLKD (pin 82)

8.99.1.2 Architecture

```

Application Layer: motor_control_task.c (250 Hz PID loop)
                  |
Encoder Layer:    rx_encoder_tpu.c (overflow handling, velocity)
                  |
HAL Layer:        rx_tpu.h/c (THIS MODULE - register access)
                  |
Register Layer:   rx72n_tpu_regs.h (memory-mapped I/O)

```

8.99.1.3 Key Features

- Phase counting mode 4 (4x resolution, all edges)
- 16-bit up/down counter with direction flag
- Hardware noise filter on input pins
- Module stop control for power management

Performance

Operation	Time	Notes
rx_tpu_init_phase()	~20 us	One-time setup
rx_tpu_read_count()	~0.5 us	Single register read
rx_tpu_read_direction()	~0.5 us	Single register read
rx_tpu_start()	~0.5 us	Single register write

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: All loops bounded
- Rule 3: No dynamic memory allocation
- Rule 4: All functions < 60 lines
- Rule 5: Minimum 2 validation checks per function
- Rule 6: Variables at smallest scope
- Rule 7: All return values checked
- Rule 8: C23 typed enums, minimal macros
- Rule 9: Single-level pointers only
- Rule 10: Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Only TPU hardware access (no encoder logic)
- O: Extensible via [rx_tpu_config_t](#)
- L: Consistent rx_err_t return semantics
- I: Focused API: init, start, stop, read, reset, deinit
- D: Depends on register abstractions, not raw addresses

See also

[rx72n_tpu_regs.h](#) TPU register definitions
[rx_mtu.h](#) MTU HAL (front wheel encoders)
[rx_mtu_encoder.h](#) MTU encoder interface

Author

Locked, Inc.

Date

2026-02-10

Copyright

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Since

Version 1.0.0

Definition in file [rx_tpu.h](#).

8.99.2 Data Structure Documentation

8.99.2.1 struct rx'tpu'config't

TPU phase counting configuration.

Configuration parameters for initializing a TPU channel in phase counting mode. Passed to [rx_tpu_init_phase_count\(\)](#).

Example

```
rx_tpu_config_t config = {
    .channel    = k_tpu_channel_1,
    .phase_mode = k_tpu_phase_mode_4,
};
rx_err_t err = rx_tpu_init_phase_count(&config);
```

Invariant

channel must be 1, 2, 4, or 5 (phase counting capable)
phase_mode must be 1-4

See also

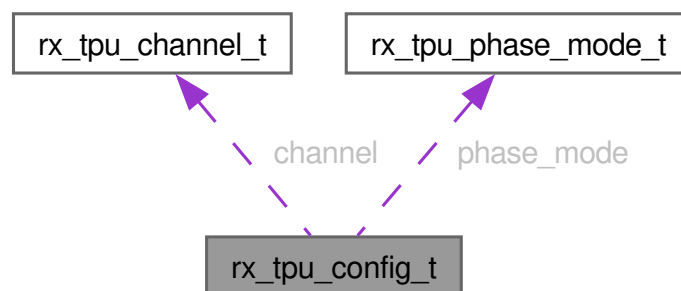
[rx_tpu_init_phase_count\(\)](#) Initialization function

Since

Version 1.0.0

Definition at line 167 of file [rx_tpu.h](#).

Collaboration diagram for rx_tpu_config_t:



Data Fields

rx_tpu_channel_t	channel	TPU channel to configure. Must be a phase-counting-capable channel (1, 2, 4, or 5).
----------------------------------	---------	--

<code>rx_tpu_phase_mode_t</code>	<code>phase_mode</code>	Phase counting mode (1-4). Selects the counting resolution. Mode 4 recommended for maximum resolution (4 counts per encoder pulse).
----------------------------------	-------------------------	--

8.99.3 Enumeration Type Documentation

8.99.3.1 `rx_tpu_channel_t`

enum `rx_tpu_channel_t` : `uint8_t`

TPU channel identifiers for phase counting.

Identifies TPU channels that support phase counting mode. Only TPU1/2/4/5 can do phase counting; TPU0 and TPU3 cannot.

STAR Channel Assignments

Value	Channel	Use	Clock Pins
1	TPU1	Back-right encoder	TCLKA + TCLKB
2	TPU2	Back-left encoder	TCLKC + TCLKD
4	TPU4	Available	TCLKC + TCLKD
5	TPU5	Available	TCLKA + TCLKB

Since

Version 1.0.0

Enumerator

<code>k_tpu_channel_1</code>	TPU1: back-right encoder (TCLKA/TCLKB)
<code>k_tpu_channel_2</code>	TPU2: back-left encoder (TCLKC/TCLKD)
<code>k_tpu_channel_4</code>	TPU4: available (TCLKC/TCLKD, paired with TPU2)
<code>k_tpu_channel_5</code>	TPU5: available (TCLKA/TCLKB, paired with TPU1)

Definition at line 112 of file `rx_tpu.h`.

```

00112         : uint8_t {
00113   k_tpu_channel_1 = 1, /**< TPU1: back-right encoder (TCLKA/TCLKB) */
00114   k_tpu_channel_2 = 2, /**< TPU2: back-left encoder (TCLKC/TCLKD) */
00115   k_tpu_channel_4 = 4, /**< TPU4: available (TCLKC/TCLKD, paired with TPU2) */
00116   k_tpu_channel_5 = 5, /**< TPU5: available (TCLKA/TCLKB, paired with TPU1) */
00117 } rx_tpu_channel_t;

```

8.99.3.2 rx'tpu'phase'mode't

enum [rx_tpu_phase_mode_t](#) : uint8_t

TPU phase counting mode selection.

Selects the phase counting resolution. Mode 4 (4x) provides the highest resolution by counting all edges of both A and B phases.

Mode Comparison

Mode	Resolution	Edges Counted	Counts/PPR
1	1x	A-phase rising only	1
2	1x	B-phase rising only	1
3	2x	A-phase rising and falling	2
4	4x	All edges of A and B phases	4

Since

Version 1.0.0

Enumerator

k_tpu_phase_mode_1	1x resolution (A rising only)
k_tpu_phase_mode_2	1x resolution (B rising only)
k_tpu_phase_mode_3	2x resolution (A both edges)
k_tpu_phase_mode_4	4x resolution (all edges) - STAR default

Definition at line 137 of file [rx_tpu.h](#).

```

00137     : uint8_t {
00138   k_tpu_phase_mode_1 = 1, /**< 1x resolution (A rising only) */
00139   k_tpu_phase_mode_2 = 2, /**< 1x resolution (B rising only) */
00140   k_tpu_phase_mode_3 = 3, /**< 2x resolution (A both edges) */
00141   k_tpu_phase_mode_4 = 4, /**< 4x resolution (all edges) - STAR default */
00142 } rx_tpu_phase_mode_t;
```

8.99.4 Function Documentation

8.99.4.1 rx'tpu'deinit()

rx_err_t rx_tpu_deinit (
 const [rx_tpu_channel_t](#) channel) [nodiscard]

Deinitialize TPU channel.

Stops the channel counter, resets mode to normal, and marks the channel as uninitialized. Does NOT disable the TPU module clock (other channels may still be active).

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
----	---------	-----------------------------

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, channel released
<code>k_rx_err_invalid_arg</code>	Invalid channel

Precondition

Channel was previously initialized

Postcondition

Counter stopped and zeroed
Mode register reset to normal operation
Channel marked as uninitialized

Note

Thread Safety: Not thread-safe, call during shutdown only

See also

[rx_tpu_init_phase_count\(\)](#) Re-initialize after deinit

Since

Version 1.0.0

Deinitialize TPU channel.

Performs an orderly shutdown of the specified TPU channel:

1. Validate channel and get array index
2. Stop counter (clear CSTn in TSTR)
3. Reset TMDR to normal operation mode (0x00)
4. Disable all interrupts (TIER = 0x40; bit 6 reserved/must-write-1)
5. Clear counter (TCNT = 0)
6. Mark channel as uninitialized in `s_tpu_initialized[]`

This function does NOT disable the TPU module clock (MSTPCRA.MSTPA13), as other channels may still be in use.

Precondition

channel must be a valid phase-counting TPU channel (1, 2, 4, or 5)
 Motor driven by this encoder must be stopped before calling

Postcondition

Counter stopped and zeroed
 TMDR reset to normal mode
 s_tpu_initialized[idx] == false

Note

Not thread-safe: do not access channel concurrently during deinit
 Channel may be partially initialized; deinit succeeds regardless

```
rx_err_t err = rx_tpu_deinit(k_tpu_channel_1);
```

See also

[rx_tpu_init_phase_count\(\)](#) Re-initialize after deinit
[rx_tpu_stop\(\)](#) Stop counter without full teardown

Since

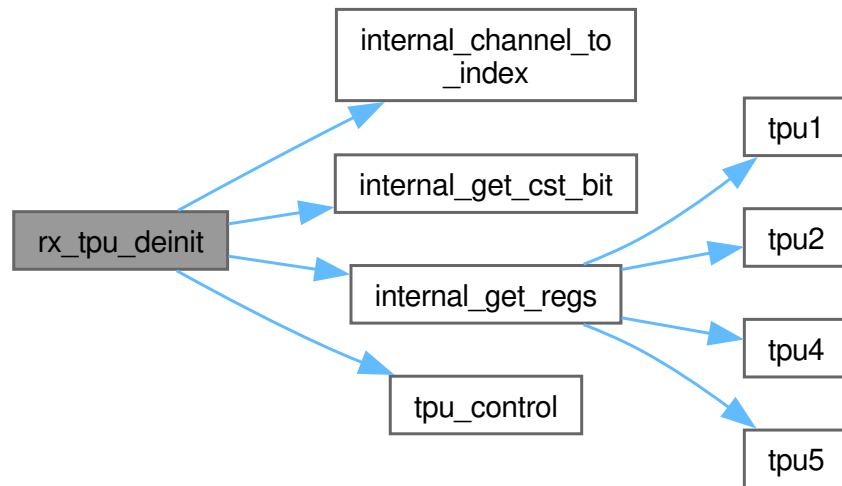
Version 1.0.0

Definition at line 777 of file [rx_tpu.c](#).

```
00778 {
00779     /* Pre-condition 1: validate channel */
00780     uint8_t idx = 0U;
00781     const rx_err_t err = internal_channel_to_index(channel, &idx);
00782     if (err != k_rx_ok) {
00783         return err;
00784     }
00785
00786     /* Get register pointer (no init check - allow deinit of partially init'd channel) */
00787     volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00788
00789     /* 1. Stop counter */
00790     const uint8_t cst_bit = internal_get_cst_bit(channel);
00791     tpu_control()->tstr &= (uint8_t)~cst_bit;
00792
00793     /* 2. Reset mode to normal operation */
00794     regs->tmdr = k_tpu_tmdr_reset;
00795
00796     /* 3. Disable all interrupts */
00797     regs->tier = k_tpu_tier_all_disabled;
00798
00799     /* 4. Clear counter */
00800     regs->tcnt = k_tpu_tcnt_zero;
00801
00802     /* 5. Mark channel as uninitialized */
00803     s_tpu_initialized[idx] = false;
00804
00805     rx_log_info_val(s_tag, "TPU channel deinitialized", (uint32_t)channel);
00806
00807     return k_rx_ok;
00808 }
```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tcnt_zero](#), [k_tpu_tier_all_disabled](#), [k_tpu_tmdr_reset](#), [s_tag](#), [s_tpu_initialized](#), [rx_tpu_regs_t::tcnt](#), [rx_tpu_regs_t::tier](#), [rx_tpu_regs_t::tmdr](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.99.4.2 rx_tpu_init_phase_count()

```
rx_err_t rx_tpu_init_phase_count (
    const rx_tpu_config_t * config) [nodiscard]
```

Initialize TPU channel for phase counting mode.

Configures a TPU channel for hardware quadrature encoder phase counting. This function:

1. Enables the TPU module (clears MSTPA13)
2. Stops the channel counter
3. Sets phase counting mode in TMDR
4. Clears the counter to zero
5. Enables overflow/underflow interrupts (if needed later)

After initialization, call [rx_tpu_start\(\)](#) to begin counting.

Parameters

in	config	Phase counting configuration
----	--------	------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, channel configured
k_rx_err_null_ptr	config is nullptr
k_rx_err_invalid_arg	Invalid channel (not 1/2/4/5); Invalid phase mode (not 1-4)

<code>k_rx_err_invalid_state</code>	Channel already initialized
-------------------------------------	-----------------------------

Precondition

TPU module clock enabled (or function enables it)
External clock pins (TCLKA-D) configured as peripheral I/O

Postcondition

Channel configured in phase counting mode
Counter cleared to zero
Counter NOT started (call [rx_tpu_start\(\)](#))

Note

Thread Safety: Not thread-safe, call during init only

See also

[rx_tpu_start\(\)](#) Start counting after init
[rx_tpu_deinit\(\)](#) Release channel resources

Since

Version 1.0.0

Implementation of [rx_tpu_init_phase_count\(\)](#) - see [rx_tpu.h](#) for complete API documentation.

Register Configuration Sequence

1. Unlock PRCR, clear MSTPCRA.MSTPA13, lock PRCR
2. Clear CSTn in TSTR (stop counter)
3. Write phase counting mode to TMDR.MD
4. Write 0x00 to TCR (free-running, external clock)
5. Write 0x00 to TIER (no interrupts)
6. Write 0x0000 to TCNT (clear counter)

See also

[rx_tpu.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 433 of file `rx_tpu.c`.

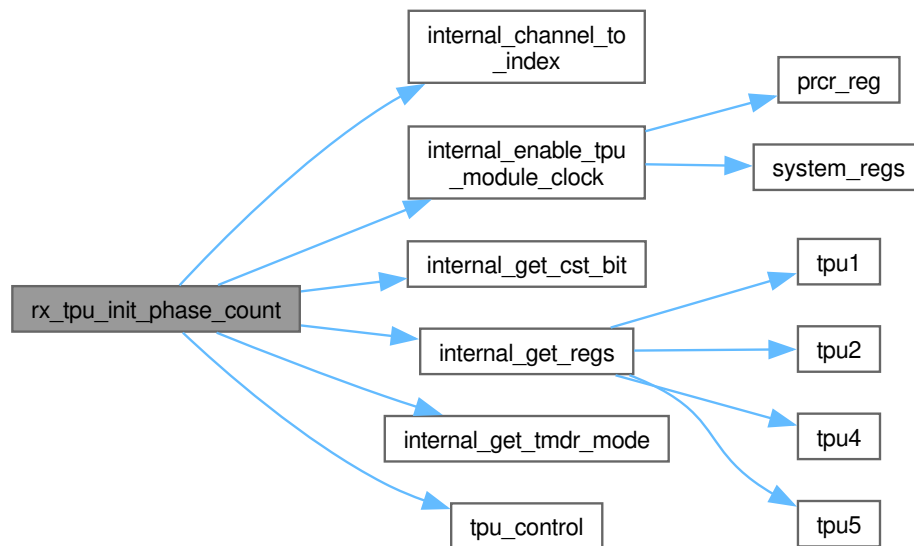
```

00434 {
00435     /* Pre-condition 1: validate config pointer (NASA Rule 5) */
00436     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
00437
00438     /* Pre-condition 2: validate channel is phase-counting capable */
00439     uint8_t idx = 0U;
00440     const rx_err_t err = internal_channel_to_index(config->channel, &idx);
00441     if (err != k_rx_ok) {
00442         return err;
00443     }
00444
00445     /* Pre-condition 3: validate phase mode range */
00446     const uint8_t tmdr_mode = internal_get_tmdr_mode(config->phase_mode);
00447     if (tmdr_mode == k_tpu_tmdr_md_normal) {
00448         rx_log_error_val(s_tag, "Invalid phase mode", (uint32_t)config->phase_mode);
00449         return k_rx_err_invalid_arg;
00450     }
00451
00452     /* Pre-condition 4: check not already initialized */
00453     if (s_tpu_initialized[idx]) {
00454         rx_log_error(s_tag, "Channel already initialized");
00455         return k_rx_err_invalid_state;
00456     }
00457
00458     /* 1. Enable TPU module clock */
00459     internal_enable_tpu_module_clock();
00460
00461     /* 2. Get register pointer */
00462     volatile rx_tpu_regs_t* regs = internal_get_regs(config->channel);
00463
00464     /* 3. Stop counter */
00465     const uint8_t cst_bit = internal_get_cst_bit(config->channel);
00466     tpu_control()->tstr &= (uint8_t)~cst_bit;
00467
00468     /* 4. Configure TCR - free-running (TPSC/CKEG ignored in phase count mode) */
00469     regs->tcr = k_tpu_tcr_phase_count;
00470
00471     /* 5. Set phase counting mode in TMDR */
00472     regs->tmdr = tmdr_mode;
00473
00474     /* 6. Disable all interrupts */
00475     regs->tier = k_tpu_tier_all_disabled;
00476
00477     /* 7. Clear counter */
00478     regs->tcnt = k_tpu_tcnt_zero;
00479
00480     /* 8. Mark channel as initialized */
00481     s_tpu_initialized[idx] = true;
00482
00483     rx_log_info(s_tag, "TPU phase count initialized");
00484     rx_log_info_val(s_tag, " channel", (uint32_t)config->channel);
00485     rx_log_info_val(s_tag, " mode", (uint32_t)config->phase_mode);
00486
00487     return k_rx_ok;
00488 }

```

References `rx_tpu_config_t::channel`, `internal_channel_to_index()`, `internal_enable_tpu_module_clock()`, `internal_get_cst_bit()`, `internal_get_regs()`, `internal_get_tmdr_mode()`, `k_tpu_tcnt_zero`, `k_tpu_tcr_phase_count`, `k_tpu_tier_all_disabled`, `k_tpu_tmdr_md_normal`, `rx_tpu_config_t::phase_mode`, `s_tag`, `s_tpu_initialized`, `rx_tpu_regs_t::tcnt`, `rx_tpu_regs_t::tcr`, `rx_tpu_regs_t::tier`, `rx_tpu_regs_t::tmdr`, `tpu_control()`, and `rx_tpu_control_regs_t::tstr`.

Here is the call graph for this function:



8.99.4.3 rx`tpu`read`count()

```

rx_err_t rx_tpu_read_count (
    const rx_tpu_channel_t channel,
    uint16_t * count) [nodiscard]

```

Read TPU channel 16-bit counter value.

Reads the current TCNT register value. In phase counting mode this is an up/down counter that tracks encoder position modulo 65536.

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
out	count	Pointer to store 16-bit counter value (0-65535)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, count written
k_rx_err_null_ptr	count is nullptr
k_rx_err_invalid_arg	Invalid channel
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized and running

Postcondition

count contains current TCNT value

Note

Thread Safety: Safe - single volatile register read

See also

[rx_tpu_read_direction\(\)](#) Read counting direction

Since

Version 1.0.0

Read TPU channel 16-bit counter value.

Performs a single volatile read of the TCNT register for the specified channel. In 4x phase counting mode, the counter increments or decrements on every edge of both A and B encoder channels.

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)
count must point to valid uint16_t storage

Postcondition

*count contains the TCNT register value at the time of the read
Hardware registers unchanged

Note

Not thread-safe: counter may wrap between multiple reads
Wraps naturally at 0 and 65535 (16-bit overflow)

```
uint16_t count;  
rx_err_t err = rx_tpu_read_count(k_tpu_channel_1, &count);
```

See also

[rx_tpu_reset_count\(\)](#) Clear counter to zero
[rx_tpu_read_direction\(\)](#) Read counting direction

Since

Version 1.0.0

Definition at line 613 of file rx_tpu.c.

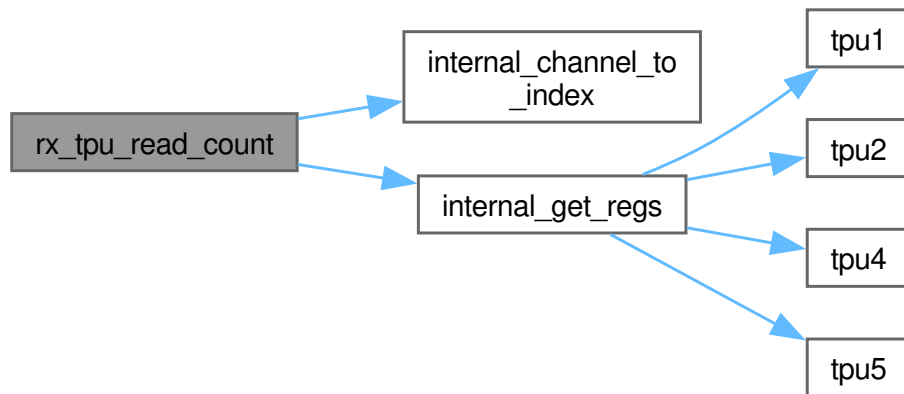
```

00614 {
00615     /* Pre-condition 1: validate output pointer */
00616     RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
00617
00618     /* Pre-condition 2: validate channel */
00619     uint8_t idx = 0U;
00620     const rx_err_t err = internal_channel_to_index(channel, &idx);
00621     if (err != k_rx_ok) {
00622         return err;
00623     }
00624
00625     /* Pre-condition 3: check initialized */
00626     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00627
00628     /* Read TCNT register */
00629     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00630
00631     *count = regs->tcnt;
00632
00633     return k_rx_ok;
00634 }

```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tcnt](#).

Here is the call graph for this function:



8.99.4.4 rx_tpu_read_direction()

```

rx_err_t rx_tpu_read_direction (
    const rx_tpu_channel_t channel,
    bool * counting_up) [nodiscard]

```

Read TPU channel counting direction.

Reads the TCFD bit in the TSR register to determine whether the counter is currently counting up (forward) or down (reverse).

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
----	---------	-----------------------------

out	counting_up	Pointer to store direction flag (true = counting up/forward, false = counting down/reverse)
-----	-------------	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, direction written
k_rx_err_null_ptr	counting_up is nullptr
k_rx_err_invalid_arg	Invalid channel
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized and running in phase counting mode

Postcondition

counting_up reflects TSR.TCFD bit state

Note

Thread Safety: Safe - single volatile register read

See also

[rx_tpu_read_count\(\)](#) Read counter value

Since

Version 1.0.0

Read TPU channel counting direction.

Reads the Timer Counter Flag Direction (TCFD) bit in the TSR register. TCFD reflects the last count event direction:

- TCFD = 1: Counter incremented on the last edge (counting up / forward)
- TCFD = 0: Counter decremented on the last edge (counting down / reverse)

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)

counting_up must point to valid bool storage

Postcondition

*counting_up reflects TSR.TCFD at the time of the read
Hardware registers unchanged

Note

Not thread-safe: value may change between reads if encoder is moving
Direction is unreliable if counter is stationary (no recent edge)

```
bool forward;
rx_err_t err = rx_tpu_read_direction(k_tpu_channel_1, &forward);
```

See also

[rx_tpu_read_count\(\)](#) Read counter position
[k_tpu_tsr_tcf](#) TSR direction bit mask

Since

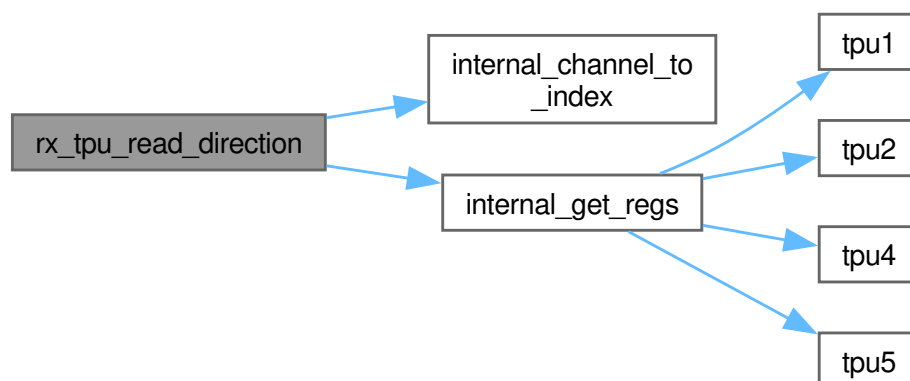
Version 1.0.0

Definition at line 666 of file [rx_tpu.c](#).

```
00667 {
00668     /* Pre-condition 1: validate output pointer */
00669     RX_CHECK_NULL_PTR(counting_up, s_tag, "Direction pointer is nullptr");
00670
00671     /* Pre-condition 2: validate channel */
00672     uint8_t idx = 0U;
00673     const rx_err_t err = internal_channel_to_index(channel, &idx);
00674     if (err != k_rx_ok) {
00675         return err;
00676     }
00677
00678     /* Pre-condition 3: check initialized */
00679     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00680
00681     /* Read TSR.TCFD bit */
00682     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00683
00684     *counting_up = (bool)((regs->tsr & k_tpu_tsr_tcf) != 0U);
00685
00686     return k_rx_ok;
00687 }
```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tsr_tcf](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tsr](#).

Here is the call graph for this function:



8.99.4.5 rx_tpu_reset_count()

```
rx_err_t rx_tpu_reset_count (  
    const rx_tpu_channel_t channel)  [nodiscard]
```

Reset TPU channel counter to zero.

Writes zero to the TCNT register. Should be called while the counter is stopped for reliable operation, though the hardware permits writing TCNT while running.

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
----	---------	-----------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, counter zeroed
k_rx_err_invalid_arg	Invalid channel
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized

Postcondition

TCNT = 0

Note

Thread Safety: Not thread-safe (read-modify-write possible issues)

Since

Version 1.0.0

Reset TPU channel counter to zero.

Writes k_tpu_tcnt_zero (0x0000) to the TCNT register, resetting the encoder position counter. The counter continues running from the new zero baseline.

Typical use: zero encoder position at a known reference point (home).

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)

Channel should be stopped ([rx_tpu_stop\(\)](#)) before resetting for atomic 0

Postcondition

TCNT register written to 0x0000

Counter resumes from zero if running

Note

Not thread-safe: write races with live counter updates from encoder

May cause a single spurious velocity estimate if called while running

```
rx_tpu_stop(k_tpu_channel_1);
rx_tpu_reset_count(k_tpu_channel_1);
rx_tpu_start(k_tpu_channel_1);
```

See also

[rx_tpu_stop\(\)](#) Stop counter before resetting for deterministic result

[rx_tpu_read_count\(\)](#) Verify counter after reset

Since

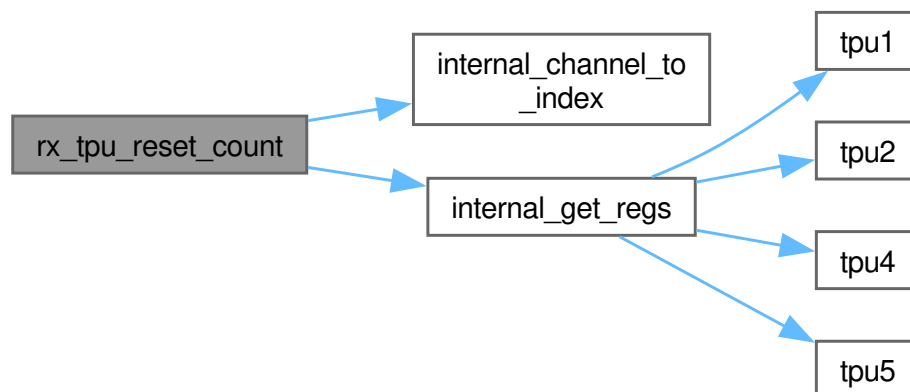
Version 1.0.0

Definition at line 721 of file [rx_tpu.c](#).

```
00722 {
00723     /* Pre-condition 1: validate channel */
00724     uint8_t idx = 0U;
00725     const rx_err_t err = internal_channel_to_index(channel, &idx);
00726     if (err != k_rx_ok) {
00727         return err;
00728     }
00729
00730     /* Pre-condition 2: check initialized */
00731     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00732
00733     /* Write 0 to TCNT */
00734     volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00735
00736     regs->tcnt = k_tpu_tcnt_zero;
00737
00738     return k_rx_ok;
00739 }
```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tcnt_zero](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tcnt](#).

Here is the call graph for this function:



8.99.4.6 rx_tpu_start()

```
rx_err_t rx_tpu_start (  
    const rx_tpu_channel_t channel)  [nodiscard]
```

Start TPU channel counter.

Sets the CSTn bit in TSTR to begin counter operation. In phase counting mode, the counter increments/decrements based on the external clock phase relationship.

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
----	---------	-----------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, counter started
k_rx_err_invalid_arg	Invalid channel
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized via [rx_tpu_init_phase_count\(\)](#)

Postcondition

CSTn bit set, counter running

Note

Thread Safety: Safe from any context (atomic register write)

See also

[rx_tpu_stop\(\)](#) Stop counter

Since

Version 1.0.0

Start TPU channel counter.

Sets the Count Start (CSTn) bit in the TSTR register to begin counter operation. Once started, the TCNT register increments or decrements on each external encoder clock edge according to the configured phase mode.

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)
 External encoder clock pins must be connected and valid

Postcondition

TSTR.CSTn bit set, counter begins counting encoder pulses
 TCNT updates in real time from encoder edges

Note

Not thread-safe: modifies shared TSTR register
 Idempotent: safe to call when already running

```
rx_err_t err = rx_tpu_start(k_tpu_channel_1);
```

See also

[rx_tpu_stop\(\)](#) Stop the counter
[rx_tpu_read_count\(\)](#) Read current counter value

Since

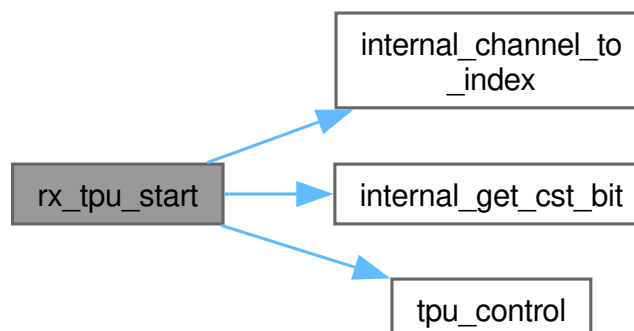
Version 1.0.0

Definition at line 518 of file [rx_tpu.c](#).

```
00519 {
00520     /* Pre-condition 1: validate channel */
00521     uint8_t idx = 0U;
00522     rx_err_t const err = internal_channel_to_index(channel, &idx);
00523     if (err != k_rx_ok) {
00524         return err;
00525     }
00526     /* Pre-condition 2: check initialized */
00527     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00528     /* Set CSTn bit in TSTR to start counter */
00529     const uint8_t cst_bit = internal_get_cst_bit(channel);
00530     tpu_control()->tstr |= cst_bit;
00531     return k_rx_ok;
00532 }
00533
00534
00535 }
```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [s_tag](#), [s_tpu_initialized](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.99.4.7 rx_tpu_stop()

```
rx_err_t rx_tpu_stop (  
    const rx_tpu_channel_t channel)  [nodiscard]
```

Stop TPU channel counter.

Clears the CSTn bit in TSTR to stop counter operation. The counter value is preserved.

Parameters

in	channel	TPU channel (1, 2, 4, or 5)
----	---------	-----------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, counter stopped
k_rx_err_invalid_arg	Invalid channel
k_rx_err_invalid_state	Channel not initialized

Precondition

Channel initialized

Postcondition

CSTn bit cleared, counter stopped

Counter value preserved

Note

Thread Safety: Safe from any context (atomic register write)

See also

[rx_tpu_start\(\)](#) Restart counter

Since

Version 1.0.0

Stop TPU channel counter.

Clears the Count Start (CSTn) bit in the TSTR register to halt counter operation. The TCNT value is preserved and encoder edges no longer update it. The channel remains initialized and can be restarted with [rx_tpu_start\(\)](#).

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)
 TSTR register must be accessible (module clock enabled)

Postcondition

TSTR.CSTn bit cleared, counter halted
 TCNT value preserved at last count

Note

Not thread-safe: modifies shared TSTR register
 Idempotent: safe to call when already stopped

```
rx_err_t err = rx_tpu_stop(k_tpu_channel_1);
```

See also

[rx_tpu_start\(\)](#) Resume counter operation
[rx_tpu_reset_count\(\)](#) Clear counter to zero

Since

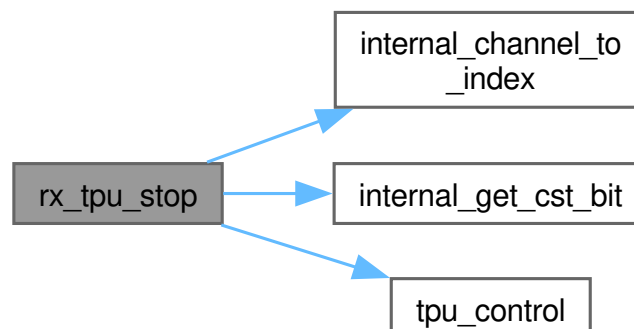
Version 1.0.0

Definition at line 565 of file [rx_tpu.c](#).

```
00566 {
00567  /* Pre-condition 1: validate channel */
00568  uint8_t idx = 0U;
00569  rx_err_t const err = internal_channel_to_index(channel, &idx);
00570  if (err != k_rx_ok) {
00571      return err;
00572  }
00573
00574  /* Pre-condition 2: check initialized */
00575  RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00576
00577  /* Clear CSTn bit in TSTR to stop counter */
00578  const uint8_t cst_bit = internal_get_cst_bit(channel);
00579  tpu_control()->tstr &= (uint8_t)~cst_bit;
00580
00581  return k_rx_ok;
00582 }
```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [s_tag](#), [s_tpu_initialized](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.100 rx_tpu.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_tpu.h
00003  * @brief TPU HAL Driver for RX72N 16-Bit Timer Pulse Unit (Phase Counting)
00004  *
00005  * @details
00006  * Hardware abstraction layer for the RX72N TPU peripheral, focused on
00007  * phase counting mode for quadrature encoder input. This driver provides
00008  * low-level channel configuration, counter access, and direction reading.
00009  *
00010  * ## STAR Project Usage
00011  *
00012  * The TPU is used for rear wheel encoder decoding. Front wheels use the MTU
00013  * peripheral (see rx_mtu.h / rx_mtu_encoder.h).
00014  *
00015  * | Encoder | Peripheral | Channel | Phase A Pin | Phase B Pin |
00016  * |-----|-----|-----|-----|-----|
00017  * | 0 (FL) | MTU1 | MTU1 | P24/MTCLKA (pin 33) | P25/MTCLKB (pin 32) |
00018  * | 1 (FR) | MTU2 | MTU2 | PA1/MTCLKC (pin 96) | PC5/MTCLKD (pin 62) |
00019  * | 2 (BR) | TPU1 | TPU1 | PC2/TCLKA (pin 70) | PA3/TCLKB (pin 94) |
00020  * | 3 (BL) | TPU2 | TPU2 | PC0/TCLKC (pin 75) | PB3/TCLKD (pin 82) |
00021  *
00022  * ## Architecture
00023  *
00024  * ```
00025  * Application Layer: motor_control_task.c (250 Hz PID loop)
00026  * |
00027  * Encoder Layer: rx_encoder_tpu.c (overflow handling, velocity)
00028  * |
00029  * HAL Layer: rx_tpu.h/c (THIS MODULE - register access)
00030  * |
00031  * Register Layer: rx72n_tpu_regs.h (memory-mapped I/O)
00032  * ```
00033  *
00034  * ## Key Features
00035  *
00036  * - Phase counting mode 4 (4x resolution, all edges)
00037  * - 16-bit up/down counter with direction flag
00038  * - Hardware noise filter on input pins
00039  * - Module stop control for power management
00040  *
00041  * @par Performance
00042  * | Operation | Time | Notes |
00043  * |-----|-----|-----|
00044  * | rx_tpu_init_phase() | ~20 us | One-time setup |
00045  * | rx_tpu_read_count() | ~0.5 us | Single register read |
00046  * | rx_tpu_read_direction() | ~0.5 us | Single register read |
00047  * | rx_tpu_start() | ~0.5 us | Single register write |
00048  *
00049  * @par NASA Power of 10 Compliance
00050  * - Rule 1: No goto, setjmp, or recursion
00051  * - Rule 2: All loops bounded
00052  * - Rule 3: No dynamic memory allocation
00053  * - Rule 4: All functions < 60 lines
00054  * - Rule 5: Minimum 2 validation checks per function
00055  * - Rule 6: Variables at smallest scope
00056  * - Rule 7: All return values checked
00057  * - Rule 8: C23 typed enums, minimal macros
00058  * - Rule 9: Single-level pointers only
00059  * - Rule 10: Compiles with -Wall -Wextra -Werror
00060  *
00061  * @par SOLID Principles
00062  * - **S**: Only TPU hardware access (no encoder logic)
00063  * - **O**: Extensible via rx_tpu_config_t
00064  * - **L**: Consistent rx_err_t return semantics
00065  * - **I**: Focused API: init, start, stop, read, reset, deinit
00066  * - **D**: Depends on register abstractions, not raw addresses
00067  *
00068  * @see rx72n_tpu_regs.h TPU register definitions
00069  * @see rx_mtu.h MTU HAL (front wheel encoders)
00070  * @see rx_mtu_encoder.h MTU encoder interface
00071  *
00072  * @author Locked, Inc.
00073  * @date 2026-02-10
00074  * @copyright Copyright (c) 2026 Locked Inc.
00075  * SPDX-License-Identifier: MIT
00076  * @since Version 1.0.0
00077  */
00078
00079 #pragma once
00080
00081 #include <stdint.h>
00082

```

```

00083 #include "rx_err.h"
00084
00085 #ifdef __cplusplus
00086 extern "C" {
00087 #endif
00088
00089 /*
=====
00090 * Type Definitions
00091 *
=====
00092 */
00093
00094 /**
00095  * @enum rx_tpu_channel_t
00096  * @brief TPU channel identifiers for phase counting
00097  *
00098  * @details
00099  * Identifies TPU channels that support phase counting mode. Only TPU1/2/4/5
00100  * can do phase counting; TPU0 and TPU3 cannot.
00101  *
00102  * @par STAR Channel Assignments
00103  * | Value | Channel | Use | Clock Pins |
00104  * |-----|-----|-----|-----|
00105  * | 1 | TPU1 | Back-right encoder | TCLKA + TCLKB |
00106  * | 2 | TPU2 | Back-left encoder | TCLKC + TCLKD |
00107  * | 4 | TPU4 | Available | TCLKC + TCLKD |
00108  * | 5 | TPU5 | Available | TCLKA + TCLKB |
00109  *
00110  * @since Version 1.0.0
00111  */
00112 typedef enum : uint8_t {
00113     k_tpu_channel_1 = 1, /**< TPU1: back-right encoder (TCLKA/TCLKB) */
00114     k_tpu_channel_2 = 2, /**< TPU2: back-left encoder (TCLKC/TCLKD) */
00115     k_tpu_channel_4 = 4, /**< TPU4: available (TCLKC/TCLKD, paired with TPU2) */
00116     k_tpu_channel_5 = 5, /**< TPU5: available (TCLKA/TCLKB, paired with TPU1) */
00117 } rx_tpu_channel_t;
00118
00119 /**
00120  * @enum rx_tpu_phase_mode_t
00121  * @brief TPU phase counting mode selection
00122  *
00123  * @details
00124  * Selects the phase counting resolution. Mode 4 (4x) provides the highest
00125  * resolution by counting all edges of both A and B phases.
00126  *
00127  * @par Mode Comparison
00128  * | Mode | Resolution | Edges Counted | Counts/PPR |
00129  * |-----|-----|-----|-----|
00130  * | 1 | 1x | A-phase rising only | 1 |
00131  * | 2 | 1x | B-phase rising only | 1 |
00132  * | 3 | 2x | A-phase rising and falling | 2 |
00133  * | 4 | 4x | All edges of A and B phases | 4 |
00134  *
00135  * @since Version 1.0.0
00136  */
00137 typedef enum : uint8_t {
00138     k_tpu_phase_mode_1 = 1, /**< 1x resolution (A rising only) */
00139     k_tpu_phase_mode_2 = 2, /**< 1x resolution (B rising only) */
00140     k_tpu_phase_mode_3 = 3, /**< 2x resolution (A both edges) */
00141     k_tpu_phase_mode_4 = 4, /**< 4x resolution (all edges) - STAR default */
00142 } rx_tpu_phase_mode_t;
00143
00144 /**
00145  * @struct rx_tpu_config_t
00146  * @brief TPU phase counting configuration
00147  *
00148  * @details
00149  * Configuration parameters for initializing a TPU channel in phase
00150  * counting mode. Passed to rx_tpu_init_phase_count().
00151  *
00152  * @par Example
00153  * @code{.c}
00154  * rx_tpu_config_t config = {
00155  *     .channel = k_tpu_channel_1,
00156  *     .phase_mode = k_tpu_phase_mode_4,
00157  * };
00158  * rx_err_t err = rx_tpu_init_phase_count(&config);
00159  * @endcode
00160  *
00161  * @invariant channel must be 1, 2, 4, or 5 (phase counting capable)
00162  * @invariant phase_mode must be 1-4
00163  *
00164  * @see rx_tpu_init_phase_count() Initialization function
00165  * @since Version 1.0.0
00166  */
00167 typedef struct {

```

```

00168 /**
00169  * @brief TPU channel to configure
00170  * @details Must be a phase-counting-capable channel (1, 2, 4, or 5).
00171  */
00172 rx_tpu_channel_t channel;
00173
00174 /**
00175  * @brief Phase counting mode (1-4)
00176  * @details Selects the counting resolution. Mode 4 recommended for
00177  * maximum resolution (4 counts per encoder pulse).
00178  */
00179 rx_tpu_phase_mode_t phase_mode;
00180 } rx_tpu_config_t;
00181
00182 /*
=====
00183 * Public API
00184 *
=====
00185 */
00186
00187 /**
00188  * @brief Initialize TPU channel for phase counting mode
00189  *
00190  * @details
00191  * Configures a TPU channel for hardware quadrature encoder phase counting.
00192  * This function:
00193  * 1. Enables the TPU module (clears MSTPA13)
00194  * 2. Stops the channel counter
00195  * 3. Sets phase counting mode in TMDR
00196  * 4. Clears the counter to zero
00197  * 5. Enables overflow/underflow interrupts (if needed later)
00198  *
00199  * After initialization, call rx_tpu_start() to begin counting.
00200  *
00201  * @param[in] config Phase counting configuration
00202  *
00203  * @return rx_err_t Error code
00204  * @retval k_rx_ok Success, channel configured
00205  * @retval k_rx_err_null_ptr config is nullptr
00206  * @retval k_rx_err_invalid_arg Invalid channel (not 1/2/4/5); Invalid phase mode (not 1-4)
00207  * @retval k_rx_err_invalid_state Channel already initialized
00208  *
00209  * @pre TPU module clock enabled (or function enables it)
00210  * @pre External clock pins (TCLKA-D) configured as peripheral I/O
00211  *
00212  * @post Channel configured in phase counting mode
00213  * @post Counter cleared to zero
00214  * @post Counter NOT started (call rx_tpu_start())
00215  *
00216  * @note Thread Safety: Not thread-safe, call during init only
00217  *
00218  * @see rx_tpu_start() Start counting after init
00219  * @see rx_tpu_deinit() Release channel resources
00220  * @since Version 1.0.0
00221  */
00222 [[nodiscard]] rx_err_t rx_tpu_init_phase_count(const rx_tpu_config_t* config);
00223
00224 /**
00225  * @brief Start TPU channel counter
00226  *
00227  * @details
00228  * Sets the CSTn bit in TSTR to begin counter operation.
00229  * In phase counting mode, the counter increments/decrements based
00230  * on the external clock phase relationship.
00231  *
00232  * @param[in] channel TPU channel (1, 2, 4, or 5)
00233  *
00234  * @return rx_err_t Error code
00235  * @retval k_rx_ok Success, counter started
00236  * @retval k_rx_err_invalid_arg Invalid channel
00237  * @retval k_rx_err_invalid_state Channel not initialized
00238  *
00239  * @pre Channel initialized via rx_tpu_init_phase_count()
00240  * @post CSTn bit set, counter running
00241  *
00242  * @note Thread Safety: Safe from any context (atomic register write)
00243  *
00244  * @see rx_tpu_stop() Stop counter
00245  * @since Version 1.0.0
00246  */
00247 [[nodiscard]] rx_err_t rx_tpu_start(rx_tpu_channel_t channel);
00248
00249 /**
00250  * @brief Stop TPU channel counter
00251  *
00252  * @details

```

```

00253 * Clears the CSTn bit in TSTR to stop counter operation.
00254 * The counter value is preserved.
00255 *
00256 * @param[in] channel TPU channel (1, 2, 4, or 5)
00257 *
00258 * @return rx_err_t Error code
00259 * @retval k_rx_ok Success, counter stopped
00260 * @retval k_rx_err_invalid_arg Invalid channel
00261 * @retval k_rx_err_invalid_state Channel not initialized
00262 *
00263 * @pre Channel initialized
00264 * @post CSTn bit cleared, counter stopped
00265 * @post Counter value preserved
00266 *
00267 * @note Thread Safety: Safe from any context (atomic register write)
00268 *
00269 * @see rx_tpu_start() Restart counter
00270 * @since Version 1.0.0
00271 */
00272 [[nodiscard]] rx_err_t rx_tpu_stop(rx_tpu_channel_t channel);
00273
00274 /**
00275 * @brief Read TPU channel 16-bit counter value
00276 *
00277 * @details
00278 * Reads the current TCNT register value. In phase counting mode this
00279 * is an up/down counter that tracks encoder position modulo 65536.
00280 *
00281 * @param[in] channel TPU channel (1, 2, 4, or 5)
00282 * @param[out] count Pointer to store 16-bit counter value (0-65535)
00283 *
00284 * @return rx_err_t Error code
00285 * @retval k_rx_ok Success, count written
00286 * @retval k_rx_err_null_ptr count is nullptr
00287 * @retval k_rx_err_invalid_arg Invalid channel
00288 * @retval k_rx_err_invalid_state Channel not initialized
00289 *
00290 * @pre Channel initialized and running
00291 * @post count contains current TCNT value
00292 *
00293 * @note Thread Safety: Safe - single volatile register read
00294 *
00295 * @see rx_tpu_read_direction() Read counting direction
00296 * @since Version 1.0.0
00297 */
00298 [[nodiscard]] rx_err_t rx_tpu_read_count(rx_tpu_channel_t channel, uint16_t* count);
00299
00300 /**
00301 * @brief Read TPU channel counting direction
00302 *
00303 * @details
00304 * Reads the TCFD bit in the TSR register to determine whether the
00305 * counter is currently counting up (forward) or down (reverse).
00306 *
00307 * @param[in] channel TPU channel (1, 2, 4, or 5)
00308 * @param[out] counting_up Pointer to store direction flag
00309 * (true = counting up/forward, false = counting down/reverse)
00310 *
00311 * @return rx_err_t Error code
00312 * @retval k_rx_ok Success, direction written
00313 * @retval k_rx_err_null_ptr counting_up is nullptr
00314 * @retval k_rx_err_invalid_arg Invalid channel
00315 * @retval k_rx_err_invalid_state Channel not initialized
00316 *
00317 * @pre Channel initialized and running in phase counting mode
00318 * @post counting_up reflects TSR.TCFD bit state
00319 *
00320 * @note Thread Safety: Safe - single volatile register read
00321 *
00322 * @see rx_tpu_read_count() Read counter value
00323 * @since Version 1.0.0
00324 */
00325 [[nodiscard]] rx_err_t rx_tpu_read_direction(rx_tpu_channel_t channel, bool* counting_up);
00326
00327 /**
00328 * @brief Reset TPU channel counter to zero
00329 *
00330 * @details
00331 * Writes zero to the TCNT register. Should be called while the counter
00332 * is stopped for reliable operation, though the hardware permits writing
00333 * TCNT while running.
00334 *
00335 * @param[in] channel TPU channel (1, 2, 4, or 5)
00336 *
00337 * @return rx_err_t Error code
00338 * @retval k_rx_ok Success, counter zeroed
00339 * @retval k_rx_err_invalid_arg Invalid channel

```

```

00340 * @retval k_rx_err_invalid_state Channel not initialized
00341 *
00342 * @pre Channel initialized
00343 * @post TCNT = 0
00344 *
00345 * @note Thread Safety: Not thread-safe (read-modify-write possible issues)
00346 *
00347 * @since Version 1.0.0
00348 */
00349 [[nodiscard]] rx_err_t rx_tpu_reset_count(rx_tpu_channel_t channel);
00350
00351 /**
00352 * @brief Deinitialize TPU channel
00353 *
00354 * @details
00355 * Stops the channel counter, resets mode to normal, and marks the
00356 * channel as uninitialized. Does NOT disable the TPU module clock
00357 * (other channels may still be active).
00358 *
00359 * @param[in] channel TPU channel (1, 2, 4, or 5)
00360 *
00361 * @return rx_err_t Error code
00362 * @retval k_rx_ok Success, channel released
00363 * @retval k_rx_err_invalid_arg Invalid channel
00364 *
00365 * @pre Channel was previously initialized
00366 * @post Counter stopped and zeroed
00367 * @post Mode register reset to normal operation
00368 * @post Channel marked as uninitialized
00369 *
00370 * @note Thread Safety: Not thread-safe, call during shutdown only
00371 *
00372 * @see rx_tpu_init_phase_count() Re-initialize after deinit
00373 * @since Version 1.0.0
00374 */
00375 [[nodiscard]] rx_err_t rx_tpu_deinit(rx_tpu_channel_t channel);
00376
00377 #ifdef __cplusplus
00378 }
00379 #endif

```

8.101 libs/rx/hal/src/adc.c File Reference

ADC Driver for RX72N S12ADFa - 12-bit Analog-to-Digital Conversion.

```

#include <string.h>
#include "hardware.h"
#include "rx72n_regs.h"
#include "rx_register_protection.h"
Include dependency graph for adc.c:

```



Enumerations

- enum `adc_constants_t` : `uint8_t` { `k_adc_max_units` = 2 , `k_adc_max_channels` = 8 , `k_adc_timeout_ms` = 100 }
ADC hardware configuration limits.
- enum `adc_bit_constants_t` : `uint8_t` { `k_adc_bit_one` = 1U }
Bit manipulation constants for ADC register operations.
- enum `adc_module_stop_bits_t` : `uint8_t` { `k_adc_mstpra_s12ad1` = 16 , `k_adc_mstpra_s12ad0` = 17 }
ADC module stop bit positions in MSTPCRA register.

- enum `adc_adcer_bits_t` : `uint16_t` {
`k_adc_adcer_adprc_mask` , `k_adc_adcer_adprc_shift` = 1U , `k_adc_adcer_adprc_12bit` = 0U
, `k_adc_adcer_adprc_10bit` = 1U ,
`k_adc_adcer_adprc_8bit` = 2U }
- ADC Control Extended Register (ADCER) bit definitions.
- enum `adc_adcsr_bits_t` : `uint16_t` { `k_adc_adcsr_idle` = 0U , `k_adc_adcsr_adst` = 32768U }
- ADC Control/Status Register (ADCSR) bit definitions.
- enum `adc_channel_boundaries_t` : `uint8_t` { `k_adc_channel_adansa0_max` = 15 ,
`k_adc_channel_adansa1_base` = 16 }
- ADC channel selection register boundaries.
- enum `adc_misc_constants_t` : `uint16_t` { `k_adc_timeout_multiplier` = 1000 , `k_adc_timeout_expired`
= 0 , `k_adc_reference_voltage_mv` = 3300 }
- ADC timeout and voltage reference constants.
- enum `adc_raw_value_t` : `uint16_t` { `k_adc_raw_value_reset` = 0 }
- Raw ADC value reset/default constants.

Functions

- static volatile `rx_s12ad_regs_t` * `internal_get_adc_base` (const `uint8_t` unit)
- Get ADC register base address from unit number.
- static `rx_err_t` `internal_configure_adc_unit` (const `uint8_t` unit, volatile `rx_s12ad_regs_t` *adc,
const `uint8_t` bits)
- Configure ADC unit control registers and enable module clock.
- static `rx_err_t` `internal_validate_unit_channel` (const `uint8_t` unit, `uint8_t` channel)
- Validate ADC unit and channel numbers.
- static `rx_err_t` `internal_read_channel_value` (volatile `rx_s12ad_regs_t` *adc, const `adc_channel_t`
channel, `uint16_t` *value)
- Read raw ADC value from a specific channel data register.
- `rx_err_t` `adc_init` (const `adc_unit_t` unit, const `adc_channel_t` channel, const `adc_resolution_t`
bits)
- Initialize ADC unit and enable a specific channel.
- `rx_err_t` `adc_read` (const `adc_unit_t` unit, const `adc_channel_t` channel, `uint16_t` *value)
- Read raw ADC conversion value from specified channel.
- `rx_err_t` `adc_read_voltage_mv` (const `adc_unit_t` unit, `adc_channel_t` channel, `adc_resolution_t`
bits, `uint32_t` *voltage_mv)
- Read ADC value and convert to millivolts.

Variables

- static const char *const `s_tag` = "ADC"
- Log tag for ADC module messages.
- static bool `s_adc_unit_initialized` [`k_adc_max_units`] = {false, false}
- Tracks initialization state for each ADC unit.
- static `adc_resolution_t` `s_adc_unit_resolution` [`k_adc_max_units`]
- Stores configured resolution for each ADC unit.

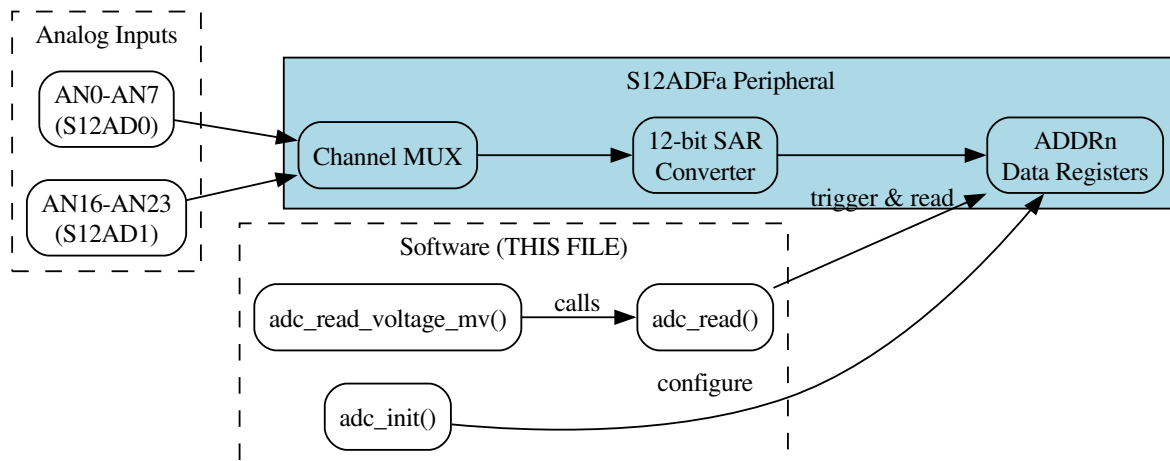
8.101.1 Detailed Description

ADC Driver for RX72N S12ADFa - 12-bit Analog-to-Digital Conversion.

8.101.1.1 Overview

This file provides the hardware abstraction layer (HAL) for the S12ADFa (12-bit Analog-to-Digital Converter) peripheral on the Renesas RX72N microcontroller. The S12ADFa supports dual ADC units with 8 channels each, configurable resolution (8/10/12-bit), and single-scan conversion mode.

8.101.1.2 Hardware Architecture

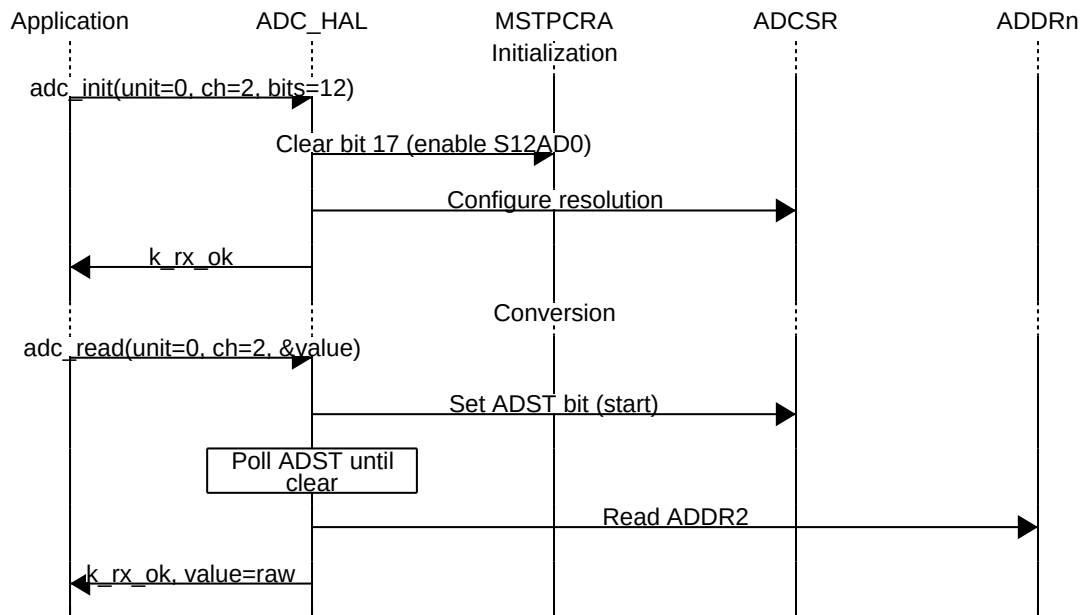


8.101.1.3 S12ADFa Features Used

Feature	Status	Notes
Dual ADC units (S12AD0, S12AD1)	↔ [OK]↔	Both supported
12/10/8-bit resolution	↔ [OK]↔	Configurable per unit
Single-scan mode	↔ [OK]↔	Software-triggered
Continuous scan mode	↔ [X]↔	Not implemented
Group scan mode	↔ [X]↔	Not implemented
A/D interrupts	↔ [X]↔	Polling only
Self-diagnosis	↔ [X]↔	Not implemented

Feature	Status	Notes
Double trigger	↔ [X]↔	Not implemented

8.101.1.4 Register Access Pattern



8.101.1.5 Conversion Timing

Resolution	Conversion Time	Sample Rate (max)
12-bit	~1.0 us	~1 MHz
10-bit	~0.9 us	~1.1 MHz
8-bit	~0.8 us	~1.25 MHz

Note: Times are approximate at PCLKB = 60 MHz.

8.101.1.6 Voltage Calculation

The `adc_read_voltage_mv()` function converts raw ADC values to millivolts:

$$V_{mV} = \frac{V_{raw} \times V_{ref}}{2^{bits} - 1}$$

Where:

- V_{raw} = Raw ADC value (0 to $2^{bits}-1$)
- V_{ref} = Reference voltage (3300 mV)
- $bits$ = Resolution (8, 10, or 12)

8.101.1.7 Thread Safety

Function	Thread-Safe	Notes
adc_init()	No	Call during initialization only
adc_read()	No	Caller must synchronize
adc_read_voltage_mv()	No	Caller must synchronize

Each ADC unit can be accessed independently, but concurrent access to the same unit requires external synchronization.

8.101.1.8 Memory Usage

Section	Size	Contents
.text	~800 bytes	Function code
.rodata	~80 bytes	Log strings, tag
.bss	6 bytes	s_adc_unit_initialized[2], s_adc_unit_resolution[2]

8.101.1.9 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	↔ [OK]↔	No goto, setjmp, recursion
2. Fixed loop bounds	↔ [OK]↔	Timeout loop has fixed maximum
3. No dynamic memory	↔ [OK]↔	Static arrays only
4. Functions < 60 lines	↔ [OK]↔	Helper functions for decomposition
5. Use assertions	↔ [OK]↔	RX_CHECK_NULL_PTR, RX_CHECK_RANGE_TAG
6. Data at smallest scope	↔ [OK]↔	Local variables, static helpers
7. Check return values	↔ [OK]↔	RX_RETURN_ON_ERROR macro
8. Limit preprocessor	↔ [OK]↔	Only includes

Rule	Status	Implementation
9. Restrict pointers	↔ [OK]↔	Hardware register pointers only
10. Compile warnings	↔ [OK]↔	-Wall -Wextra -Werror clean

See also

[rx72n_adc_regs.h](#) ADC register definitions
[hardware.h](#) ADC unit and channel types
[rx_register_protection.h](#) Module stop register access

Author

Locked, Inc.

Date

2026-01-01

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Since

Version 1.0.0

Definition in file [adc.c](#).

8.101.2 Enumeration Type Documentation

8.101.2.1 adc`adcer`bits`t

```
enum adc\_adcer\_bits\_t : uint16_t
```

ADC Control Extended Register (ADCER) bit definitions.

Defines the bit fields for the ADCER (A/D Control Extended Register) which controls ADC resolution and data format. The ADPRC field (bits 1:0) sets the A/D conversion resolution.

8.101.2.2 ADCER Register Layout (bits 1:0 - ADPRC)

ADPRC[2:↵ 1]	Resolution	Max Value	Conversion Time
00	12-bit	4095	~1.0 us
01	10-bit	1023	~0.9 us
10	8-bit	255	~0.8 us
11	Reserved	-	-

See also

[internal_configure_adc_unit\(\)](#) Uses these to set resolution

Since

Version 1.0.0

Enumerator

k_adc_adcer_adprc_mask	ADPRC field mask (2-bit, unshifted) per RX72N HW manual R01↵UH0824EJ0111 Section 56.2.13
k_adc_adcer_adprc_shift	ADPRC field bit position: bits 2:1 of ADCER
k_adc_adcer_adprc_12bit	ADPRC = 00: 12-bit resolution (default)
k_adc_adcer_adprc_10bit	ADPRC = 01: 10-bit resolution
k_adc_adcer_adprc_8bit	ADPRC = 10: 8-bit resolution

Definition at line 252 of file [adc.c](#).

```

00252     : uint16_t {
00253     k_adc_adcer_adprc_mask =
00254     3U, /**< ADPRC field mask (2-bit, unshifted) per RX72N HW manual R01UH0824EJ0111 Section 56.2.13 */
00255     k_adc_adcer_adprc_shift = 1U, /**< ADPRC field bit position: bits 2:1 of ADCER */
00256     k_adc_adcer_adprc_12bit = 0U, /**< ADPRC = 00: 12-bit resolution (default) */
00257     k_adc_adcer_adprc_10bit = 1U, /**< ADPRC = 01: 10-bit resolution */
00258     k_adc_adcer_adprc_8bit  = 2U, /**< ADPRC = 10: 8-bit resolution */
00259 } adc_adcer_bits_t;

```

8.101.2.3 adc`adcsr`bits`t

```
enum adc\_adcsr\_bits\_t : uint16_t
```

ADC Control/Status Register (ADCSR) bit definitions.

Defines bit values for the ADCSR (A/D Control Status Register) which controls conversion start and indicates conversion status.

8.101.2.4 ADCSR Bit 15 (ADST) - A/D Conversion Start

Value	Read Meaning	Write Meaning
0	Conversion stopped/complete	(No effect)
1	Conversion in progress	Start conversion

See also

[adc_read\(\)](#) Uses ADST to trigger and poll conversion

Since

Version 1.0.0

Enumerator

k_adc_adcsr_idle	Idle state: ADST=0 (conversion complete)
k_adc_adcsr_adst	Start bit: ADST=1 (bit 15 = 0x8000 = 32768)

Definition at line 280 of file [adc.c](#).

```
00280      : uint16_t {
00281  k_adc_adcsr_idle = 0U,    /**< Idle state: ADST=0 (conversion complete) */
00282  k_adc_adcsr_adst = 32768U, /**< Start bit: ADST=1 (bit 15 = 0x8000 = 32768) */
00283 } adc_adcsr_bits_t;
```

8.101.2.5 adc`bit`constants`

```
enum adc\_bit\_constants\_t : uint8_t
```

Bit manipulation constants for ADC register operations.

Single-bit value used for shift operations when setting/clearing individual bits in ADC control and channel selection registers.

Since

Version 1.0.0

Enumerator

k_adc_bit_one	Single-bit value for shift operations (1 << n)
---------------	--

Definition at line 197 of file [adc.c](#).

```
00197      : uint8_t {
00198  k_adc_bit_one = 1U, /**< Single-bit value for shift operations (1 « n) */
00199 } adc_bit_constants_t;
```

8.101.2.6 adc`channel`boundaries`t

enum [adc_channel_boundaries_t](#) : uint8_t

ADC channel selection register boundaries.

Defines the channel boundaries for ADANSA0 and ADANSA1 registers. Channels 0-15 are selected via ADANSA0, channels 16+ via ADANSA1.

For S12AD0: Physical channels AN0-AN7 (channels 0-7) For S12AD1: Physical channels AN16-AN23 (channels 0-7 mapped)

See also

[adc_init\(\)](#) Uses these for channel enable logic

Since

Version 1.0.0

Enumerator

k_adc_channel_adansa0_max	Maximum channel for ADANSA0 register
k_adc_channel_adansa1_base	Base channel for ADANSA1 register

Definition at line 300 of file [adc.c](#).

```
00300      : uint8_t {
00301  k_adc_channel_adansa0_max = 15, /**< Maximum channel for ADANSA0 register */
00302  k_adc_channel_adansa1_base = 16, /**< Base channel for ADANSA1 register */
00303 } adc_channel_boundaries_t;
```

8.101.2.7 adc`constants`t

enum [adc_constants_t](#) : uint8_t

ADC hardware configuration limits.

Defines the hardware limits and configuration constants for the S12ADFa peripheral on the RX72N microcontroller. These values are derived from the RX72N hardware manual Chapter 41 (S12ADFa).

See also

[internal_get_adc_base\(\)](#) Uses [k_adc_max_units](#) for validation

[s_adc_unit_initialized](#) Array sized by [k_adc_max_units](#)

Since

Version 1.0.0

Enumerator

k_adc_max_units	Number of ADC units (S12AD0, S12AD1)
k_adc_max_channels	Channels per unit (0-7 for S12AD0, 16-23 for S12AD1)
k_adc_timeout_ms	Maximum wait time for conversion completion (ms)

Definition at line 181 of file [adc.c](#).

```
00181      : uint8_t {
00182  k_adc_max_units    = 2,  /**< Number of ADC units (S12AD0, S12AD1) */
00183  k_adc_max_channels = 8,  /**< Channels per unit (0-7 for S12AD0, 16-23 for S12AD1) */
00184  k_adc_timeout_ms   = 100, /**< Maximum wait time for conversion completion (ms) */
00185 } adc_constants_t;
```

8.101.2.8 adc`misc`constants`t

enum [adc_misc_constants_t](#) : uint16_t

ADC timeout and voltage reference constants.

Miscellaneous constants for timeout handling and voltage calculations.

8.101.2.9 Timeout Calculation

Total timeout iterations = k_adc_timeout_ms * k_adc_timeout_multiplier = 100 * 1000 = 100,000 iterations

At ~10 cycles per iteration, this provides ~4ms timeout at 240 MHz.

8.101.2.10 Voltage Reference

The RX72N STAR board uses 3.3V as the ADC reference voltage (AVCC). The k_adc_reference_voltage_mv constant (3300 mV) is used in the voltage calculation formula.

See also

[adc_read\(\)](#) Uses timeout constants

[adc_read_voltage_mv\(\)](#) Uses reference voltage

Since

Version 1.0.0

Enumerator

k_adc_timeout_multiplier	Loop iterations per ms of timeout
--------------------------	-----------------------------------

k_adc_timeout_expired	Sentinel value: timeout counter reached zero
k_adc_reference_voltage_mv	ADC reference voltage: 3.3V = 3300 mV

Definition at line 332 of file [adc.c](#).

```
00332      : uint16_t {
00333   k_adc_timeout_multiplier = 1000, /**< Loop iterations per ms of timeout */
00334   k_adc_timeout_expired    = 0,    /**< Sentinel value: timeout counter reached zero */
00335   k_adc_reference_voltage_mv = 3300, /**< ADC reference voltage: 3.3V = 3300 mV */
00336 } adc_misc_constants_t;
```

8.101.2.11 adc`module`stop`bits`

```
enum adc_module_stop_bits_t : uint8_t
```

ADC module stop bit positions in MSTPCRA register.

Defines the bit positions in the MSTPCRA (Module Stop Control Register A) that control the clock supply to the S12AD0 and S12AD1 modules. These bits must be cleared (clock enabled) before accessing ADC registers.

8.101.2.12 Module Stop Control

Bit	Module	0 = Clock Enabled	1 = Clock Stopped
17	S12AD0	ADC unit 0 active	ADC unit 0 stopped
16	S12AD1	ADC unit 1 active	ADC unit 1 stopped

Note

Accessing ADC registers while module clock is stopped causes bus error
PRCR register must be unlocked before modifying MSTPCRA

See also

[internal_configure_adc_unit\(\)](#) Uses these to enable module clock
[rx_register_protection.h](#) PRCR unlock/lock functions

Since

Version 1.0.0

Enumerator

k_adc_mstpra_s12ad1	MSTPCRA bit 16: S12AD1 module stop control
k_adc_mstpra_s12ad0	MSTPCRA bit 17: S12AD0 module stop control

Definition at line 225 of file [adc.c](#).

```
00225      : uint8_t {
00226   k_adc_mstpra_s12ad1 = 16, /**< MSTPCRA bit 16: S12AD1 module stop control */
00227   k_adc_mstpra_s12ad0 = 17, /**< MSTPCRA bit 17: S12AD0 module stop control */
00228 } adc_module_stop_bits_t;
```

8.101.2.13 adc`raw`value`t

```
enum adc\_raw\_value\_t : uint16_t
```

Raw ADC value reset/default constants.

Default raw value used to initialize the raw_value variable before a conversion result is written. Using a named constant makes the intent explicit and avoids magic numeric literals.

See also

[adc_read_voltage_mv\(\)](#) Initializes raw_value with [k_adc_raw_value_reset](#)

Since

Version 1.0.0

Enumerator

k_adc_raw_value_reset	Reset/default raw ADC value before conversion
---------------------------------------	---

Definition at line [351](#) of file [adc.c](#).

```
00351      : uint16_t {
00352   k_adc_raw_value_reset = 0, /**< Reset/default raw ADC value before conversion */
00353 } adc_raw_value_t;
```

8.101.3 Function Documentation

8.101.3.1 adc`init`()

```
rx_err_t adc_init (
    const adc\_unit\_t unit,
    const adc\_channel\_t channel,
    const adc\_resolution\_t bits) [nodiscard]
```

Initialize ADC unit and enable a specific channel.

Initialize ADC unit and channel.

Configures the S12ADFa peripheral for A/D conversion. On first call for each unit, enables the module clock, configures resolution, and initializes the hardware. Subsequent calls to the same unit enable additional channels.

8.101.3.2 Initialization Behavior

Scenario	Action	Resolution	Result
First init for unit	Full init	Set to 'bits'	k_rx_ok
Subsequent init, same bits	Enable channel only	Unchanged	k_rx_ok

Scenario	Action	Resolution	Result
Subsequent init, different bits	Error	Unchanged	k_rx_err_invalid_arg

Precondition

- None (can be first call for unit)
- Resolution must match if unit already initialized

Postcondition

- Module clock enabled (if first init)
- Channel enabled in ADANSA register
- Unit marked as initialized

Note

- NOT thread-safe: Call from initialization code only
- Performance: $\sim 1\text{-}2$ us for first init, ~ 0.5 us for subsequent

Example:

```
// Initialize ADC unit 0, channel 2 with 12-bit resolution
rx_err_t err = adc_init(k_adc_unit_0, k_adc_channel_2, k_adc_resolution_12bit);
if (err != k_rx_ok) {
    rx_log_error("SENSOR", "ADC init failed: %d", err);
    return err;
}
```

See also

- [adc_read\(\)](#) Read after initialization
- [adc_read_voltage_mv\(\)](#) Read as voltage

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 3 postconditions

Definition at line 708 of file [adc.c](#).

```
00709 {
00710     /* Validate parameters */
00711     rx_err_t err = internal_validate_unit_channel(unit, channel);
00712     RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00713
00714     /* Validate resolution */
00715     if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00716         bits != k_adc_resolution_12bit) {
00717         rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00718         return k_rx_err_invalid_arg;
00719     }
00720 }
```

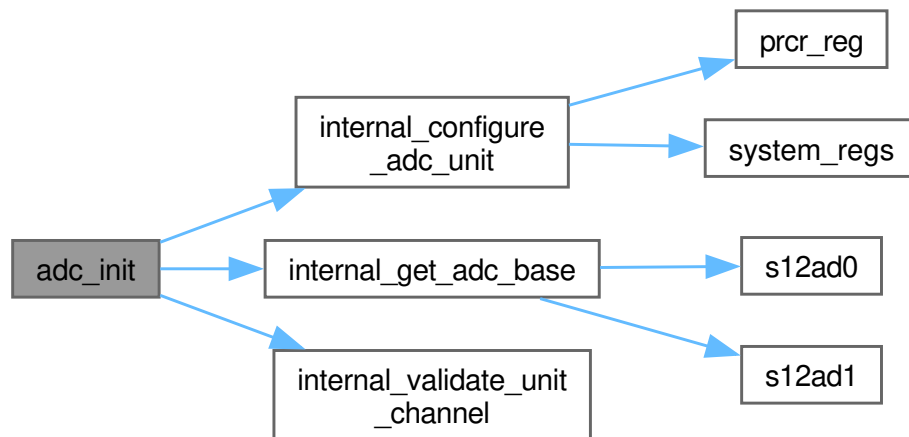
```

00721  /* Get ADC base */
00722  volatile rx_s12ad_regs_t* const adc = internal_get_adc_base(unit);
00723  if (adc == nullptr) {
00724      return k_rx_err_invalid_arg;
00725  }
00726
00727  /* Initialize ADC unit if not already initialized */
00728  if (!s_adc_unit_initialized[unit]) {
00729      /* Store resolution for validation on subsequent calls */
00730      s_adc_unit_resolution[unit] = bits;
00731
00732      err = internal_configure_adc_unit(unit, adc, bits);
00733      if (err != k_rx_ok) {
00734          return err;
00735      }
00736  } else {
00737      /* Unit already initialized - verify resolution matches */
00738      if (bits != s_adc_unit_resolution[unit]) {
00739          rx_log_error(s_tag, "ADC resolution mismatch on subsequent init");
00740          return k_rx_err_invalid_arg;
00741      }
00742  }
00743
00744  /* Enable the specified channel */
00745  if ((uint8_t)channel <= k_adc_channel_adansa0_max) {
00746      adc->adansa0 |= ((uint16_t)k_adc_bit_one << channel);
00747  } else {
00748      adc->adansa1 |= ((uint16_t)k_adc_bit_one << ((uint8_t)channel - k_adc_channel_adansa1_base));
00749  }
00750
00751  rx_log_debug(s_tag, "ADC channel enabled");
00752
00753  return k_rx_ok;
00754 }

```

References [rx_s12ad_regs_t::adansa0](#), [rx_s12ad_regs_t::adansa1](#), [internal_configure_adc_unit\(\)](#), [internal_get_adc_base\(\)](#), [internal_validate_unit_channel\(\)](#), [k_adc_bit_one](#), [k_adc_channel_adansa0_max](#), [k_adc_channel_adansa1_base](#), [s_adc_unit_initialized](#), [s_adc_unit_resolution](#), and [s_tag](#).

Here is the call graph for this function:



8.101.3.3 adc`read()

```

rx_err_t adc_read (
    const adc_unit_t unit,
    const adc_channel_t channel,
    uint16_t * value) [nodiscard]

```

Read raw ADC conversion value from specified channel.

Read ADC value.

Performs a single-scan A/D conversion and returns the raw digital value. This is a blocking operation that polls the ADST bit until conversion completes or timeout occurs.

8.101.3.4 Conversion Sequence

1. Validate parameters and initialization state
2. Start conversion by setting ADCSR.ADST bit
3. Poll ADCSR.ADST until cleared (conversion complete)
4. Read result from appropriate ADDRn register

8.101.3.5 Output Value Ranges

Resolution	Range	Full-Scale
12-bit	0-4095	3.3V
10-bit	0-1023	3.3V
8-bit	0-255	3.3V

Precondition

ADC unit must be initialized via `adc_init()`
value must point to valid `uint16_t` storage

Postcondition

*value contains raw ADC reading (if `k_rx_ok`)
ADC ready for next conversion

Note

NOT thread-safe: Caller must synchronize if shared
Blocking: Waits up to ~100ms for conversion
Performance: ~1-10 us typical (depends on conversion time)

Warning

Do not call from ISR (blocking wait)

Example:

```
uint16_t raw_value;  
rx_err_t err = adc_read(k_adc_unit_0, k_adc_channel_2, &raw_value);  
if (err == k_rx_ok) {  
    // Example: 12-bit resolution (0-4095 raw -> 0-3.3 V)  
    static const float s_vref_volts = 3.3F;  
    static const float s_adc_full_scale_12bit = 4095.0F;  
    float voltage_v = (raw_value * s_vref_volts) / s_adc_full_scale_12bit;  
}
```

See also

[adc_init\(\)](#) Must be called first
[adc_read_voltage_mv\(\)](#) Convenience function for voltage

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 2: [OK] Timeout loop has fixed upper bound
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 814 of file [adc.c](#).

```

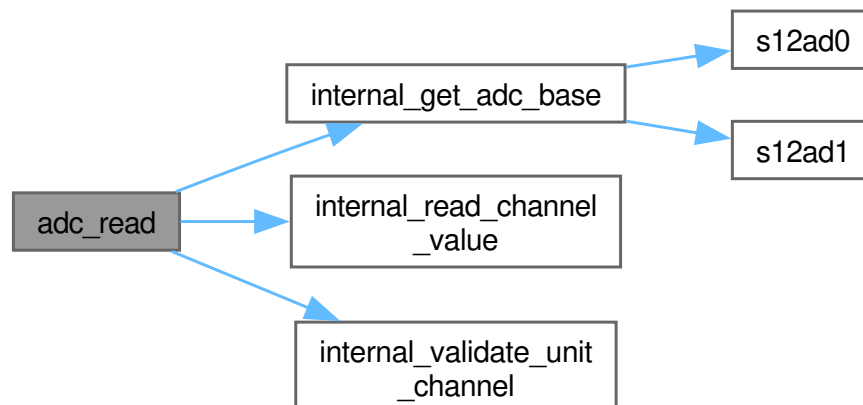
00815 {
00816     /* Validate parameters */
00817     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00818
00819     const rx_err_t err = internal_validate_unit_channel(unit, channel);
00820     RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00821
00822     /* Check if unit is initialized */
00823     if (!s_adc_unit_initialized[unit]) {
00824         rx_log_error(s_tag, "ADC unit not initialized");
00825         return k_rx_err_invalid_state;
00826     }
00827
00828     /* Get ADC base */
00829     volatile rx_s12ad_regs_t* const adc = internal_get_adc_base(unit);
00830     if (adc == nullptr) {
00831         return k_rx_err_invalid_arg;
00832     }
00833
00834     /* Start single-scan conversion */
00835     adc->adcsr = k_adc_adcsr_adst;
00836
00837     /* Wait for conversion to complete (poll ADCSR.ADST bit) */
00838     uint32_t timeout = k_adc_timeout_ms * k_adc_timeout_multiplier;
00839     while ((adc->adcsr & k_adc_adcsr_adst) != k_adc_adcsr_idle && timeout > k_adc_timeout_expired) {
00840         timeout--;
00841     }
00842
00843     if (timeout == k_adc_timeout_expired) {
00844         rx_log_error(s_tag, "ADC conversion timeout");
00845         return k_rx_err_timeout;
00846     }
00847
00848     /* Read conversion result from appropriate data register */
00849     return internal_read_channel_value(adc, channel, value);
00850 }

```

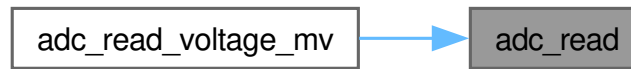
References [rx_s12ad_regs_t::adcsr](#), [internal_get_adc_base\(\)](#), [internal_read_channel_value\(\)](#), [internal_validate_unit_channel\(\)](#), [k_adc_adcsr_adst](#), [k_adc_adcsr_idle](#), [k_adc_timeout_expired](#), [k_adc_timeout_ms](#), [k_adc_timeout_multiplier](#), [s_adc_unit_initialized](#), and [s_tag](#).

Referenced by [adc_read_voltage_mv\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.101.3.6 adc_read_voltage_mv()

```

rx_err_t adc_read_voltage_mv (
    const adc_unit_t unit,
    adc_channel_t channel,
    adc_resolution_t bits,
    uint32_t * voltage_mv) [nodiscard]
  
```

Read ADC value and convert to millivolts.

Performs an A/D conversion and scales the result to millivolts. This is a convenience function that calls [adc_read\(\)](#) and applies the voltage calculation formula.

8.101.3.7 Voltage Calculation Formula

$$V_{mV} = \frac{V_{raw} \times 3300}{2^{bits} - 1}$$

8.101.3.8 Example Calculations

Resolution	Raw Value	Voltage (mV)	Calculation
12-bit	2048	1650	2048 x 3300 / 4095
10-bit	512	1650	512 x 3300 / 1023
8-bit	128	1655	128 x 3300 / 255

Precondition

ADC unit must be initialized via [adc_init\(\)](#)
 bits must match the resolution used in [adc_init\(\)](#)

Postcondition

*voltage_mv contains voltage in millivolts (0-3300)
 ADC ready for next conversion

Note

NOT thread-safe: Caller must synchronize if shared

Blocking: Waits for conversion (see `adc_read`)

Reference: Uses 3.3V reference (`k_adc_reference_voltage_mv = 3300`)

Example:

```
// Read motor current sense (DRV8263H IPROPI output)
uint32_t voltage_mv;
rx_err_t err = adc_read_voltage_mv(k_adc_unit_0, k_adc_channel_0,
                                   k_adc_resolution_12bit, &voltage_mv);
if (err == k_rx_ok) {
    // voltage_mv: 0-3300 mV
    // For DRV8263H-Q1 IPROPI (202 uA/A mirror, 5100 Ohm sense):
    // I_motor = V_IPROPI / k_ipropi_divisor
    // (where k_ipropi_divisor = 1.0302 = 202e-6 * 5100)
    // Fixed-point: scale = 10000, divisor = 10302 (1.0302 * 10000)
    enum : uint32_t { k_scale = 10000, k_divisor = 10302 };
    uint32_t current_ma = (voltage_mv * k_scale) / k_divisor;
    rx_log_info("ADC", "Current: %lu mA", current_ma);
}
```

See also

[adc_read\(\)](#) Lower-level raw value read

[adc_init\(\)](#) Initialize before reading

Since

Version 1.0.0

NASA Power of 10 Compliance:

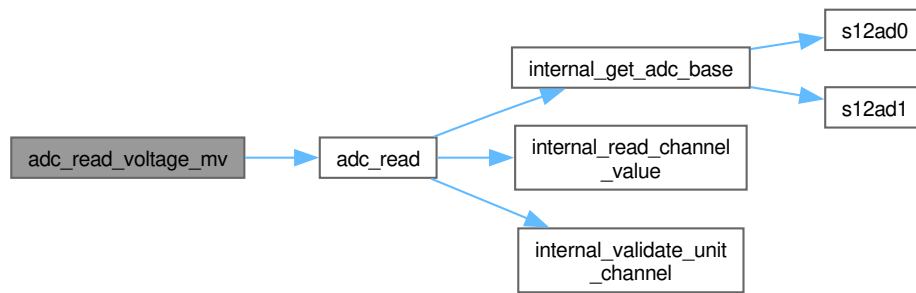
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 912 of file `adc.c`.

```
00916 {
00917     /* Parameter order: unit, channel, resolution bits. */
00918     RX_CHECK_NULL_PTR(voltage_mv, s_tag, "Voltage pointer is nullptr");
00919
00920     if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00921         bits != k_adc_resolution_12bit) {
00922         rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00923         return k_rx_err_invalid_arg;
00924     }
00925
00926     /* Validate that passed resolution matches unit's configured resolution */
00927     if (unit > k_adc_unit_1) {
00928         rx_log_error(s_tag, "Invalid ADC unit");
00929         return k_rx_err_invalid_arg;
00930     }
00931
00932     if (bits != s_adc_unit_resolution[unit]) {
00933         rx_log_error(s_tag, "ADC resolution mismatch in voltage read");
00934         return k_rx_err_invalid_arg;
00935     }
00936
00937     /* Read raw ADC value */
00938     uint16_t raw_value = k_adc_raw_value_reset;
00939     const rx_err_t err = adc_read(unit, channel, &raw_value);
00940     RX_RETURN_ON_ERROR(err, s_tag, "ADC read failed");
00941
00942     /* Calculate voltage (using ADC reference voltage) */
00943     const uint32_t max_value = ((uint32_t)k_adc_bit_one << bits) - k_adc_bit_one;
00944     *voltage_mv = ((uint32_t)raw_value * k_adc_reference_voltage_mv) / max_value;
00945
00946     return k_rx_ok;
00947 }
```

References [adc_read\(\)](#), [k_adc_bit_one](#), [k_adc_raw_value_reset](#), [k_adc_reference_voltage_mv](#), [k_adc_unit_1](#), [s_adc_unit_resolution](#), and [s_tag](#).

Here is the call graph for this function:



8.101.3.9 internal_configure_adc_unit()

```

rx_err_t internal_configure_adc_unit (
    const uint8_t unit,
    volatile rx_s12ad_regs_t * adc,
    const uint8_t bits) [static]
  
```

Configure ADC unit control registers and enable module clock.

Performs one-time initialization of an ADC unit:

1. Unlocks PRCR to allow MSTPCRA modification
2. Clears module stop bit in MSTPCRA (enables clock)
3. Locks PRCR
4. Configures ADCER register for desired resolution
5. Marks unit as initialized

8.101.3.10 Register Modification Sequence

```

PRCR = k_rx_prcr_unlock_prc1_prc3 // Enable MSTPCRA write
MSTPCRA &= ~(1 « module_stop_bit) // Clear module stop bit
PRCR = k_rx_prcr_lock // Disable protected writes
ADCER = (ADCER & ~mask) | resolution_bits // Set resolution
  
```

Precondition

adc must be valid pointer from [internal_get_adc_base\(\)](#)

Unit must not already be initialized

Postcondition

Module clock enabled (MSTPCRA bit cleared)

ADCER configured with requested resolution

s_adc_unit_initialized[unit] = true

Note

NOT thread-safe: Protected register access

Performance: ~1 us @ 240 MHz

PRCR unlock window is minimized

Warning

Must not be called while ADC is converting
 PRCR access requires proper unlock sequence

See also

[adc_init\(\)](#) Public API that calls this function
[rx_register_protection.h](#) PRCR constants

Since

Version 1.0.0

Definition at line 497 of file [adc.c](#).

```

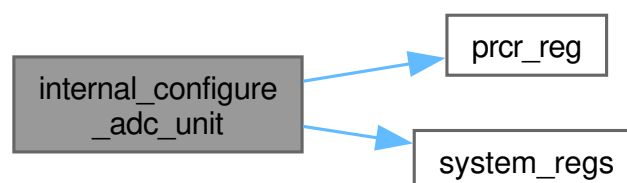
00498 {
00499     RX_CHECK_NULL_PTR(adc, s_tag, "ADC register pointer is nullptr");
00500     RX_CHECK_RANGE_TAG(unit, k_adc_unit_0, k_adc_unit_1, k_rx_err_invalid_arg, s_tag);
00501
00502     if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00503         bits != k_adc_resolution_12bit) {
00504         rx_log_error(s_tag, "Invalid ADC resolution");
00505         return k_rx_err_invalid_arg;
00506     }
00507
00508     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00509
00510     if (unit == k_adc_unit_0) {
00511         system_regs()->mstpcra &= ~(uint32_t)k_adc_bit_one « k_adc_mstpra_s12ad0;
00512     } else {
00513         system_regs()->mstpcra &= ~(uint32_t)k_adc_bit_one « k_adc_mstpra_s12ad1;
00514     }
00515
00516     *prcr_reg() = k_rx_prcr_lock;
00517
00518     uint16_t adcer = adc->adcer;
00519     adcer &= (uint16_t) ~(uint16_t)(k_adc_adcer_adprc_mask « k_adc_adcer_adprc_shift);
00520     if (bits == k_adc_resolution_8bit) {
00521         adcer |= (uint16_t)(k_adc_adcer_adprc_8bit « k_adc_adcer_adprc_shift);
00522     } else if (bits == k_adc_resolution_10bit) {
00523         adcer |= (uint16_t)(k_adc_adcer_adprc_10bit « k_adc_adcer_adprc_shift);
00524     } else {
00525         adcer |= (uint16_t)(k_adc_adcer_adprc_12bit « k_adc_adcer_adprc_shift);
00526     }
00527     adc->adcer = adcer;
00528
00529     s_adc_unit_initialized[unit] = true;
00530     rx_log_debug(s_tag, "ADC unit initialized");
00531
00532     return k_rx_ok;
00533 }

```

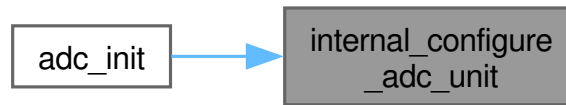
References [rx_s12ad_regs_t::adcer](#), [k_adc_adcer_adprc_10bit](#), [k_adc_adcer_adprc_12bit](#), [k_adc_adcer_adprc_8bit](#), [k_adc_adcer_adprc_mask](#), [k_adc_adcer_adprc_shift](#), [k_adc_bit_one](#), [k_adc_mstpra_s12ad0](#), [k_adc_mstpra_s12ad1](#), [k_adc_unit_0](#), [k_adc_unit_1](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), [s_adc_unit_initialized](#), [s_tag](#), and [system_regs\(\)](#).

Referenced by [adc_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.101.3.11 internal_get_adc_base()

```
volatile rx\_s12ad\_regs\_t * internal_get_adc_base (
    const uint8_t unit)  [static]
```

Get ADC register base address from unit number.

Returns a pointer to the register structure for the specified ADC unit. The RX72N has two ADC units with registers at different base addresses:

Unit	Peripheral	Base Address
0	S12AD0	0x00089000
1	S12AD1	0x00089100

Precondition

None (pure lookup function)

Postcondition

No state changes

Note

Thread-safe: Pure function with no shared state

Performance: O(1), ~10 cycles

See also

[s12ad0\(\)](#) Accessor for S12AD0 registers

[s12ad1\(\)](#) Accessor for S12AD1 registers

Since

Version 1.0.0

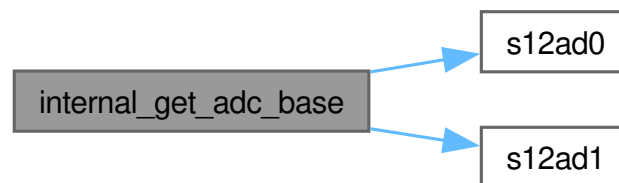
Definition at line 440 of file [adc.c](#).

```
00441 {
00442     switch (unit) {
00443         case k_adc_unit_0: {
00444             return s12ad0();
00445         }
00446         case k_adc_unit_1: {
00447             return s12ad1();
00448         }
00449         default: {
00450             return nullptr;
00451         }
00452     }
00453 }
```

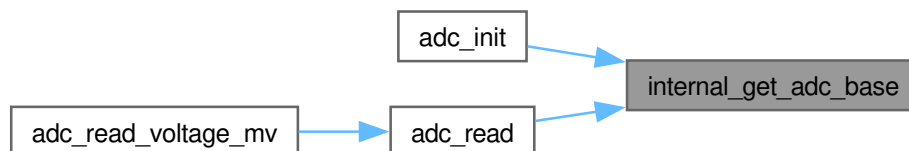
References [k_adc_unit_0](#), [k_adc_unit_1](#), [s12ad0\(\)](#), and [s12ad1\(\)](#).

Referenced by [adc_init\(\)](#), and [adc_read\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.101.3.12 internal`read`channel`value()

```
rx_err_t internal_read_channel_value (
    volatile rx_s12ad_regs_t * adc,
    const adc_channel_t channel,
    uint16_t * value) [static]
```

Read raw ADC value from a specific channel data register.

Helper function that reads the conversion result from the appropriate ADDRn register based on channel number. Extracted from [adc_read\(\)](#) to support NASA Power of 10 Rule 4 (functions under 60 lines).

8.101.3.13 Channel to Register Mapping

Channel	Register	Offset
0	ADDR0	0x020
1	ADDR1	0x022
2	ADDR2	0x024
3	ADDR3	0x026
4	ADDR4	0x028
5	ADDR5	0x02A
6	ADDR6	0x02C
7	ADDR7	0x02E

Precondition

adc must be valid pointer from [internal_get_adc_base\(\)](#)

ADC conversion must be complete (ADST bit cleared)

Postcondition

*value contains raw ADC reading (12/10/8-bit depending on config)

No hardware state changes

Note

Thread-safe: Read-only register access

Performance: O(1), ~30 cycles

See also

[adc_read\(\)](#) Calls this after conversion complete

Since

Version 1.0.0

Definition at line 617 of file [adc.c](#).

```

00620 {
00621     RX_CHECK_NULL_PTR(adc, s_tag, "ADC register pointer is nullptr");
00622     RX_CHECK_NULL_PTR(value, s_tag, "value pointer is nullptr");
00623
00624     switch (channel) {
00625         case k_adc_channel_0:
00626             *value = adc->addr0;
00627             break;
00628         case k_adc_channel_1:
00629             *value = adc->addr1;
00630             break;
00631         case k_adc_channel_2:
00632             *value = adc->addr2;
00633             break;
00634         case k_adc_channel_3:
00635             *value = adc->addr3;
00636             break;
00637         case k_adc_channel_4:

```

```

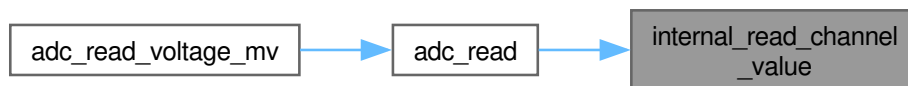
00638     *value = adc->addr4;
00639     break;
00640     case k_adc_channel_5:
00641     *value = adc->addr5;
00642     break;
00643     case k_adc_channel_6:
00644     *value = adc->addr6;
00645     break;
00646     case k_adc_channel_7:
00647     *value = adc->addr7;
00648     break;
00649     default:
00650     rx_log_error(s_tag, "Unsupported channel");
00651     return k_rx_err_invalid_arg;
00652 }
00653
00654 return k_rx_ok;
00655 }

```

References [rx_sl2ad_regs_t::addr0](#), [rx_sl2ad_regs_t::addr1](#), [rx_sl2ad_regs_t::addr2](#), [rx_sl2ad_regs_t::addr3](#), [rx_sl2ad_regs_t::addr4](#), [rx_sl2ad_regs_t::addr5](#), [rx_sl2ad_regs_t::addr6](#), [rx_sl2ad_regs_t::addr7](#), [k_adc_channel_0](#), [k_adc_channel_1](#), [k_adc_channel_2](#), [k_adc_channel_3](#), [k_adc_channel_4](#), [k_adc_channel_5](#), [k_adc_channel_6](#), [k_adc_channel_7](#), and [s_tag](#).

Referenced by [adc_read\(\)](#).

Here is the caller graph for this function:



8.101.3.14 internal_validate_unit_channel()

```

rx_err_t internal_validate_unit_channel (
    const uint8_t unit,
    uint8_t channel) [static]

```

Validate ADC unit and channel numbers.

Performs range validation for ADC unit and channel parameters before any hardware access. This prevents undefined behavior from invalid register access.

8.101.3.15 Validation Rules

Parameter	Valid Range	Error Code
unit	0-1 (k_adc_unit_0 to k_adc_unit_1)	k_rx_err_invalid_arg
channel	0-7 (k_adc_channel_0 to k_adc_channel_7)	k_rx_err_invalid_arg

Precondition

None (pure validation function)

Postcondition

No state changes

Note

Thread-safe: Pure function with no shared state

Performance: $O(1)$, ~ 20 cycles

Logs error message on validation failure

See also

[adc_init\(\)](#) Uses for parameter validation

[adc_read\(\)](#) Uses for parameter validation

Since

Version 1.0.0

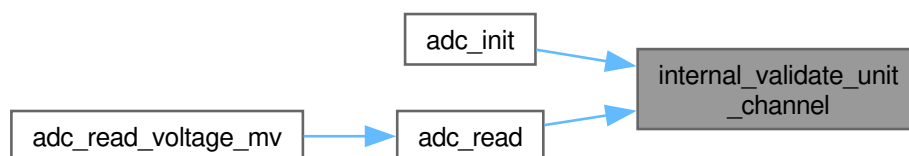
Definition at line 564 of file [adc.c](#).

```
00565 {  
00566     /* Validate unit */  
00567     if (unit > k_adc_unit_1) {  
00568         rx_log_error(s_tag, "Invalid ADC unit");  
00569         return k_rx_err_invalid_arg;  
00570     }  
00571  
00572     /* Validate channel */  
00573     if (channel > k_adc_channel_7) {  
00574         rx_log_error(s_tag, "Invalid ADC channel");  
00575         return k_rx_err_invalid_arg;  
00576     }  
00577  
00578     return k_rx_ok;  
00579 }
```

References [k_adc_channel_7](#), [k_adc_unit_1](#), and [s_tag](#).

Referenced by [adc_init\(\)](#), and [adc_read\(\)](#).

Here is the caller graph for this function:



8.101.4 Variable Documentation

8.101.4.1 s`adc`unit`initialized

`bool s_adc_unit_initialized[k_adc_max_units] = {false, false}` [static]

Tracks initialization state for each ADC unit.

Array indexed by ADC unit number (0 or 1). Set to true when `adc_init()` successfully initializes the unit. Used to prevent double-initialization and to validate that a unit is ready before reading.

Index	Unit	Default
0	S12AD0	false
1	S12AD1	false

Warning

NOT thread-safe - initialization should be done from single thread

See also

`adc_init()` Sets to true on successful initialization

`adc_read()` Checks before allowing read operation

Definition at line 386 of file `adc.c`.

00386 {false, false};

Referenced by `adc_init()`, `adc_read()`, and `internal_configure_adc_unit()`.

8.101.4.2 s`adc`unit`resolution

`adc_resolution_t s_adc_unit_resolution[k_adc_max_units]` [static]

Initial value:

= {k_adc_resolution_12bit,
k_adc_resolution_12bit}

Stores configured resolution for each ADC unit.

Array indexed by ADC unit number (0 or 1). Records the resolution configured during first `adc_init()` call for each unit. Subsequent init calls must use the same resolution (enforced at runtime).

Index	Unit	Default	Valid Values
0	S12AD0	12-bit	8, 10, 12
1	S12AD1	12-bit	8, 10, 12

Note

Resolution is set only on first init and cannot be changed

See also

`adc_init()` Sets on first initialization, validates on subsequent

`adc_read_voltage_mv()` Uses for voltage calculation

Definition at line 407 of file `adc.c`.

00407 {k_adc_resolution_12bit,
00408 k_adc_resolution_12bit};

Referenced by `adc_init()`, and `adc_read_voltage_mv()`.

8.101.4.3 s_tag

```
const char* const s_tag = "ADC" [static]
```

Log tag for ADC module messages.

Used with rx_log_* functions for consistent message tagging.

Definition at line 365 of file [adc.c](#).

Referenced by [adc_init\(\)](#), [adc_read\(\)](#), [adc_read_voltage_mv\(\)](#), [gpio_read\(\)](#), [gpio_set_input\(\)](#), [gpio_set_output\(\)](#), [gpio_toggle\(\)](#), [gpio_write_high\(\)](#), [gpio_write_low\(\)](#), [internal_calculate_bit_rate\(\)](#), [internal_calculate_cmt_params\(\)](#), [internal_calculate_period\(\)](#), [internal_calculate_period\(\)](#), [internal_calculate_spbr\(\)](#), [internal_channel_to_index\(\)](#), [internal_configure_adc_unit\(\)](#), [internal_configure_cs_gpio\(\)](#), [internal_configure_gptw_registers\(\)](#), [internal_controller_do_16bit_transfer\(\)](#), [internal_get_pfs_register\(\)](#), [internal_prepare_gptw_pwm_init\(\)](#), [internal_read_byte\(\)](#), [internal_read_channel_value\(\)](#), [internal_riic_bit_bang_read\(\)](#), [internal_riic_log_pre_recovery\(\)](#), [internal_riic_program_bit_rate\(\)](#), [internal_riic_read_phase\(\)](#), [internal_riic_validate_init\(\)](#), [internal_riic_write_phase\(\)](#), [internal_rspi_validate_init\(\)](#), [internal_send_start\(\)](#), [internal_send_stop\(\)](#), [internal_validate_cmt_init_params\(\)](#), [internal_validate_controller_args\(\)](#), [internal_validate_port_pin\(\)](#), [internal_validate_unit_channel\(\)](#), [internal_wait_bus_ready\(\)](#), [internal_write_address\(\)](#), [internal_write_byte\(\)](#), [poeg_groupbl2_isr\(\)](#), [riic_deinit_peripheral\(\)](#), [riic_init\(\)](#), [riic_init_peripheral\(\)](#), [riic_peripheral_read\(\)](#), [riic_peripheral_write\(\)](#), [riic_read\(\)](#), [riic_write\(\)](#), [riic_write_read\(\)](#), [rspi_controller_deinit\(\)](#), [rspi_controller_set_cs\(\)](#), [rspi_controller_transfer_16bit\(\)](#), [rspi_deinit\(\)](#), [rspi_init_peripheral\(\)](#), [rspi_peripheral_read_available\(\)](#), [rspi_peripheral_transfer\(\)](#), [rspi_peripheral_write_ready\(\)](#), [rx_cac_deinit\(\)](#), [rx_cac_init\(\)](#), [rx_cac_start\(\)](#), [rx_cac_stop\(\)](#), [rx_cmt_deinit\(\)](#), [rx_cmt_get_count\(\)](#), [rx_cmt_init\(\)](#), [rx_doc_add\(\)](#), [rx_doc_compare\(\)](#), [rx_doc_init\(\)](#), [rx_doc_set_reference\(\)](#), [rx_doc_subtract\(\)](#), [rx_eccram_get_error_status\(\)](#), [rx_eccram_init\(\)](#), [rx_eccram_register_error_isr\(\)](#), [rx_gptw_get_duty\(\)](#), [rx_gptw_get_period\(\)](#), [rx_gptw_init_all_staggered\(\)](#), [rx_gptw_init_pwm\(\)](#), [rx_gptw_set_duty\(\)](#), [rx_hal_iwdt_check_tasks\(\)](#), [rx_hal_iwdt_register_task\(\)](#), [rx_hal_iwdt_task_heartbeat\(\)](#), [rx_host_irq_deinit\(\)](#), [rx_host_irq_init\(\)](#), [rx_mpc_set_adc\(\)](#), [rx_mpc_set_gpio\(\)](#), [rx_mpc_set_irq\(\)](#), [rx_mpc_set_peripheral\(\)](#), [rx_mtu_deinit\(\)](#), [rx_mtu_enable_output\(\)](#), [rx_mtu_get_duty\(\)](#), [rx_mtu_get_period\(\)](#), [rx_mtu_init_pwm\(\)](#), [rx_mtu_set_duty\(\)](#), [rx_mtu_set_duty_raw\(\)](#), [rx_mtu_start\(\)](#), [rx_poeg_clear_fault\(\)](#), [rx_poeg_get_fault_status\(\)](#), [rx_poeg_init\(\)](#), [rx_tmr_init\(\)](#), [rx_tmr_read\(\)](#), [rx_tmr_set_period_us\(\)](#), [rx_tmr_start\(\)](#), [rx_tmr_stop\(\)](#), [rx_tpu_deinit\(\)](#), [rx_tpu_init_phase_count\(\)](#), [rx_tpu_read_count\(\)](#), [rx_tpu_read_direction\(\)](#), [rx_tpu_reset_count\(\)](#), [rx_tpu_start\(\)](#), [rx_tpu_stop\(\)](#), [timer_get_count\(\)](#), [timer_init\(\)](#), and [timer_stop\(\)](#).

8.102 adc.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file adc.c
00003  * @brief ADC Driver for RX72N S12ADFa - 12-bit Analog-to-Digital Conversion
00004  *
00005  * @details
00006  * ## Overview
00007  *
00008  * This file provides the hardware abstraction layer (HAL) for the S12ADFa
00009  * (12-bit Analog-to-Digital Converter) peripheral on the Renesas RX72N
00010  * microcontroller. The S12ADFa supports dual ADC units with 8 channels each,
00011  * configurable resolution (8/10/12-bit), and single-scan conversion mode.
00012  *
00013  * ## Hardware Architecture
00014  *
00015  * @dot
00016  * digraph adc_architecture {
00017  *   rankdir=LR;
00018  *   node [shape=box, style=rounded];
00019  *
00020  *   subgraph cluster_analog {
00021  *     label="Analog Inputs";
00022  *     style=dashed;
00023  *     an0 [label="AN0-AN7\n(S12AD0)"];
```

```

00024 *   an1 [label="AN16-AN23\n(S12AD1)"];
00025 * }
00026 *
00027 * subgraph cluster_adc {
00028 *   label="S12ADFa Peripheral";
00029 *   style=filled;
00030 *   fillcolor=lightblue;
00031 *   mux [label="Channel MUX"];
00032 *   sar [label="12-bit SAR\nConverter"];
00033 *   regs [label="ADDRn\nData Registers"];
00034 * }
00035 *
00036 * subgraph cluster_sw {
00037 *   label="Software (THIS FILE)";
00038 *   style=dashed;
00039 *   init [label="adc_init()"];
00040 *   read [label="adc_read()"];
00041 *   voltage [label="adc_read_voltage_mv()"];
00042 * }
00043 *
00044 * an0 -> mux;
00045 * an1 -> mux;
00046 * mux -> sar;
00047 * sar -> regs;
00048 * init -> regs [label="configure"];
00049 * read -> regs [label="trigger & read"];
00050 * voltage -> read [label="calls"];
00051 * }
00052 * @enddot
00053 *
00054 * ## S12ADFa Features Used
00055 *
00056 * | Feature | Status | Notes |
00057 * |-----|-----|-----|
00058 * | Dual ADC units (S12AD0, S12AD1) | [OK] | Both supported |
00059 * | 12/10/8-bit resolution | [OK] | Configurable per unit |
00060 * | Single-scan mode | [OK] | Software-triggered |
00061 * | Continuous scan mode | [X] | Not implemented |
00062 * | Group scan mode | [X] | Not implemented |
00063 * | A/D interrupts | [X] | Polling only |
00064 * | Self-diagnosis | [X] | Not implemented |
00065 * | Double trigger | [X] | Not implemented |
00066 *
00067 * ## Register Access Pattern
00068 *
00069 * @msc
00070 * Application, ADC_HAL, MSTPCRA, ADCSR, ADDRn;
00071 *
00072 * ... [label="Initialization"];
00073 * Application => ADC_HAL [label="adc_init(unit=0, ch=2, bits=12)"];
00074 * ADC_HAL => MSTPCRA [label="Clear bit 17 (enable S12AD0)"];
00075 * ADC_HAL => ADCSR [label="Configure resolution"];
00076 * ADC_HAL => Application [label="k_rx_ok"];
00077 *
00078 * ... [label="Conversion"];
00079 * Application => ADC_HAL [label="adc_read(unit=0, ch=2, &value)"];
00080 * ADC_HAL => ADCSR [label="Set ADST bit (start)"];
00081 * ADC_HAL box ADC_HAL [label="Poll ADST until clear"];
00082 * ADC_HAL => ADDRn [label="Read ADDR2"];
00083 * ADC_HAL => Application [label="k_rx_ok, value=raw"];
00084 * @endmsc
00085 *
00086 * ## Conversion Timing
00087 *
00088 * | Resolution | Conversion Time | Sample Rate (max) |
00089 * |-----|-----|-----|
00090 * | 12-bit | ~1.0 us | ~1 MHz |
00091 * | 10-bit | ~0.9 us | ~1.1 MHz |
00092 * | 8-bit | ~0.8 us | ~1.25 MHz |
00093 *
00094 * Note: Times are approximate at PCLKB = 60 MHz.
00095 *
00096 * ## Voltage Calculation
00097 *
00098 * The adc_read_voltage_mv() function converts raw ADC values to millivolts:
00099 *
00100 * @f[
00101 * 
$$V_{\text{mV}} = \frac{V_{\text{raw}}}{2^{\text{bits}} - 1} \times V_{\text{ref}}$$

00102 * @f]
00103 *
00104 * Where:
00105 * - @f$ V_{\text{raw}} @f$ = Raw ADC value (0 to 2bits-1)
00106 * - @f$ V_{\text{ref}} @f$ = Reference voltage (3300 mV)
00107 * - @f$ bits @f$ = Resolution (8, 10, or 12)
00108 *
00109 * ## Thread Safety
00110 *

```

```

00111 * | Function | Thread-Safe | Notes |
00112 * |-----|-----|-----|
00113 * | adc_init() | No | Call during initialization only |
00114 * | adc_read() | No | Caller must synchronize |
00115 * | adc_read_voltage_mv() | No | Caller must synchronize |
00116 *
00117 * Each ADC unit can be accessed independently, but concurrent access to the
00118 * same unit requires external synchronization.
00119 *
00120 * ## Memory Usage
00121 *
00122 * | Section | Size | Contents |
00123 * |-----|-----|-----|
00124 * | .text | ~800 bytes | Function code |
00125 * | .rodata | ~80 bytes | Log strings, tag |
00126 * | .bss | 6 bytes | s_adc_unit_initialized[2], s_adc_unit_resolution[2] |
00127 *
00128 * ## NASA Power of 10 Compliance
00129 *
00130 * | Rule | Status | Implementation |
00131 * |-----|-----|-----|
00132 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00133 * | 2. Fixed loop bounds | [OK] | Timeout loop has fixed maximum |
00134 * | 3. No dynamic memory | [OK] | Static arrays only |
00135 * | 4. Functions < 60 lines | [OK] | Helper functions for decomposition |
00136 * | 5. Use assertions | [OK] | RX_CHECK_NULL_PTR, RX_CHECK_RANGE_TAG |
00137 * | 6. Data at smallest scope | [OK] | Local variables, static helpers |
00138 * | 7. Check return values | [OK] | RX_RETURN_ON_ERROR macro |
00139 * | 8. Limit preprocessor | [OK] | Only includes |
00140 * | 9. Restrict pointers | [OK] | Hardware register pointers only |
00141 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror clean |
00142 *
00143 * @see rx72n_adc_regs.h ADC register definitions
00144 * @see hardware.h ADC unit and channel types
00145 * @see rx_register_protection.h Module stop register access
00146 *
00147 * @author Locked, Inc.
00148 * @date 2026-01-01
00149 * @copyright Copyright (c) 2026 Locked Inc.
00150 * SPDX-License-Identifier: MIT
00151 * @since Version 1.0.0
00152 */
00153
00154 #ifdef __RX__
00155
00156 #include <string.h>
00157
00158 #include "hardware.h"
00159 #include "rx72n_regs.h"
00160 #include "rx_register_protection.h"
00161
00162 /*
=====
00163 * Constants
00164 *
=====
00165 */
00166 /**
00167  * @enum adc_constants_t
00168  * @brief ADC hardware configuration limits
00169  *
00170  * @details
00171  * Defines the hardware limits and configuration constants for the S12ADFa
00172  * peripheral on the RX72N microcontroller. These values are derived from
00173  * the RX72N hardware manual Chapter 41 (S12ADFa).
00174  *
00175  * @see internal_get_adc_base() Uses k_adc_max_units for validation
00176  * @see s_adc_unit_initialized Array sized by k_adc_max_units
00177  *
00178  * @since Version 1.0.0
00179  */
00180
00181 typedef enum : uint8_t {
00182     k_adc_max_units = 2, /**< Number of ADC units (S12AD0, S12AD1) */
00183     k_adc_max_channels = 8, /**< Channels per unit (0-7 for S12AD0, 16-23 for S12AD1) */
00184     k_adc_timeout_ms = 100, /**< Maximum wait time for conversion completion (ms) */
00185 } adc_constants_t;
00186
00187 /**
00188  * @enum adc_bit_constants_t
00189  * @brief Bit manipulation constants for ADC register operations
00190  *
00191  * @details
00192  * Single-bit value used for shift operations when setting/clearing
00193  * individual bits in ADC control and channel selection registers.
00194  *
00195  * @since Version 1.0.0

```

```

00196 */
00197 typedef enum : uint8_t {
00198     k_adc_bit_one = 1U, /**< Single-bit value for shift operations (1 « n) */
00199 } adc_bit_constants_t;
00200
00201 /**
00202  * @enum adc_module_stop_bits_t
00203  * @brief ADC module stop bit positions in MSTPCRA register
00204  *
00205  * @details
00206  * Defines the bit positions in the MSTPCRA (Module Stop Control Register A)
00207  * that control the clock supply to the S12AD0 and S12AD1 modules. These bits
00208  * must be cleared (clock enabled) before accessing ADC registers.
00209  *
00210  * ## Module Stop Control
00211  *
00212  * | Bit | Module | 0 = Clock Enabled | 1 = Clock Stopped |
00213  * |-----|-----|-----|-----|
00214  * | 17 | S12AD0 | ADC unit 0 active | ADC unit 0 stopped |
00215  * | 16 | S12AD1 | ADC unit 1 active | ADC unit 1 stopped |
00216  *
00217  * @note Accessing ADC registers while module clock is stopped causes bus error
00218  * @note PRCR register must be unlocked before modifying MSTPCRA
00219  *
00220  * @see internal_configure_adc_unit() Uses these to enable module clock
00221  * @see rx_register_protection.h PRCR unlock/lock functions
00222  *
00223  * @since Version 1.0.0
00224  */
00225 typedef enum : uint8_t {
00226     k_adc_mstpra_s12ad1 = 16, /**< MSTPCRA bit 16: S12AD1 module stop control */
00227     k_adc_mstpra_s12ad0 = 17, /**< MSTPCRA bit 17: S12AD0 module stop control */
00228 } adc_module_stop_bits_t;
00229
00230 /**
00231  * @enum adc_adcer_bits_t
00232  * @brief ADC Control Extended Register (ADCER) bit definitions
00233  *
00234  * @details
00235  * Defines the bit fields for the ADCER (A/D Control Extended Register) which
00236  * controls ADC resolution and data format. The ADPRC field (bits 1:0) sets
00237  * the A/D conversion resolution.
00238  *
00239  * ## ADCER Register Layout (bits 1:0 - ADPRC)
00240  *
00241  * | ADPRC[2:1] | Resolution | Max Value | Conversion Time |
00242  * |-----|-----|-----|-----|
00243  * | 00 | 12-bit | 4095 | ~1.0 us |
00244  * | 01 | 10-bit | 1023 | ~0.9 us |
00245  * | 10 | 8-bit | 255 | ~0.8 us |
00246  * | 11 | Reserved | - | - |
00247  *
00248  * @see internal_configure_adc_unit() Uses these to set resolution
00249  *
00250  * @since Version 1.0.0
00251  */
00252 typedef enum : uint16_t {
00253     k_adc_adcer_adprc_mask =
00254         3U, /**< ADPRC field mask (2-bit, unshifted) per RX72N HW manual R01UH0824EJ0111 Section 56.2.13 */
00255     k_adc_adcer_adprc_shift = 1U, /**< ADPRC field bit position: bits 2:1 of ADCER */
00256     k_adc_adcer_adprc_12bit = 0U, /**< ADPRC = 00: 12-bit resolution (default) */
00257     k_adc_adcer_adprc_10bit = 1U, /**< ADPRC = 01: 10-bit resolution */
00258     k_adc_adcer_adprc_8bit = 2U, /**< ADPRC = 10: 8-bit resolution */
00259 } adc_adcer_bits_t;
00260
00261 /**
00262  * @enum adc_adcsr_bits_t
00263  * @brief ADC Control/Status Register (ADCSR) bit definitions
00264  *
00265  * @details
00266  * Defines bit values for the ADCSR (A/D Control Status Register) which
00267  * controls conversion start and indicates conversion status.
00268  *
00269  * ## ADCSR Bit 15 (ADST) - A/D Conversion Start
00270  *
00271  * | Value | Read Meaning | Write Meaning |
00272  * |-----|-----|-----|
00273  * | 0 | Conversion stopped/complete | (No effect) |
00274  * | 1 | Conversion in progress | Start conversion |
00275  *
00276  * @see adc_read() Uses ADST to trigger and poll conversion
00277  *
00278  * @since Version 1.0.0
00279  */
00280 typedef enum : uint16_t {
00281     k_adc_adcsr_idle = 0U, /**< Idle state: ADST=0 (conversion complete) */
00282     k_adc_adcsr_adst = 32768U, /**< Start bit: ADST=1 (bit 15 = 0x8000 = 32768) */

```

```

00283 } adc_adcsr_bits_t;
00284
00285 /**
00286  * @enum adc_channel_boundaries_t
00287  * @brief ADC channel selection register boundaries
00288  *
00289  * @details
00290  * Defines the channel boundaries for ADANSA0 and ADANSA1 registers.
00291  * Channels 0-15 are selected via ADANSA0, channels 16+ via ADANSA1.
00292  *
00293  * For S12AD0: Physical channels AN0-AN7 (channels 0-7)
00294  * For S12AD1: Physical channels AN16-AN23 (channels 0-7 mapped)
00295  *
00296  * @see adc_init() Uses these for channel enable logic
00297  *
00298  * @since Version 1.0.0
00299  */
00300 typedef enum : uint8_t {
00301     k_adc_channel_adansa0_max = 15, /**< Maximum channel for ADANSA0 register */
00302     k_adc_channel_adansa1_base = 16, /**< Base channel for ADANSA1 register */
00303 } adc_channel_boundaries_t;
00304
00305 /* ADC channel indices (k_adc_channel_0 through k_adc_channel_7) defined in hardware.h */
00306
00307 /**
00308  * @enum adc_misc_constants_t
00309  * @brief ADC timeout and voltage reference constants
00310  *
00311  * @details
00312  * Miscellaneous constants for timeout handling and voltage calculations.
00313  *
00314  * ## Timeout Calculation
00315  *
00316  * Total timeout iterations = k_adc_timeout_ms * k_adc_timeout_multiplier
00317  *                           = 100 * 1000 = 100,000 iterations
00318  *
00319  * At ~10 cycles per iteration, this provides ~4ms timeout at 240 MHz.
00320  *
00321  * ## Voltage Reference
00322  *
00323  * The RX72N STAR board uses 3.3V as the ADC reference voltage (AVCC).
00324  * The k_adc_reference_voltage_mv constant (3300 mV) is used in the
00325  * voltage calculation formula.
00326  *
00327  * @see adc_read() Uses timeout constants
00328  * @see adc_read_voltage_mv() Uses reference voltage
00329  *
00330  * @since Version 1.0.0
00331  */
00332 typedef enum : uint16_t {
00333     k_adc_timeout_multiplier = 1000, /**< Loop iterations per ms of timeout */
00334     k_adc_timeout_expired = 0, /**< Sentinel value: timeout counter reached zero */
00335     k_adc_reference_voltage_mv = 3300, /**< ADC reference voltage: 3.3V = 3300 mV */
00336 } adc_misc_constants_t;
00337
00338 /**
00339  * @enum adc_raw_value_t
00340  * @brief Raw ADC value reset/default constants
00341  *
00342  * @details
00343  * Default raw value used to initialize the raw_value variable before a
00344  * conversion result is written. Using a named constant makes the intent
00345  * explicit and avoids magic numeric literals.
00346  *
00347  * @see adc_read_voltage_mv() Initializes raw_value with k_adc_raw_value_reset
00348  *
00349  * @since Version 1.0.0
00350  */
00351 typedef enum : uint16_t {
00352     k_adc_raw_value_reset = 0, /**< Reset/default raw ADC value before conversion */
00353 } adc_raw_value_t;
00354
00355 /*
=====
00356  * Static Variables
00357  *
=====
00358  */
00359
00360 /**
00361  * @var s_tag
00362  * @brief Log tag for ADC module messages
00363  * @details Used with rx_log_* functions for consistent message tagging.
00364  */
00365 static const char* const s_tag = "ADC";
00366
00367 /**

```

```

00368 * @var s_adc_unit_initialized
00369 * @brief Tracks initialization state for each ADC unit
00370 *
00371 * @details
00372 * Array indexed by ADC unit number (0 or 1). Set to true when adc_init()
00373 * successfully initializes the unit. Used to prevent double-initialization
00374 * and to validate that a unit is ready before reading.
00375 *
00376 * | Index | Unit | Default |
00377 * |-----|-----|-----|
00378 * | 0 | S12AD0 | false |
00379 * | 1 | S12AD1 | false |
00380 *
00381 * @warning NOT thread-safe - initialization should be done from single thread
00382 *
00383 * @see adc_init() Sets to true on successful initialization
00384 * @see adc_read() Checks before allowing read operation
00385 */
00386 static bool s_adc_unit_initialized[k_adc_max_units] = {false, false};
00387
00388 /**
00389 * @var s_adc_unit_resolution
00390 * @brief Stores configured resolution for each ADC unit
00391 *
00392 * @details
00393 * Array indexed by ADC unit number (0 or 1). Records the resolution
00394 * configured during first adc_init() call for each unit. Subsequent
00395 * init calls must use the same resolution (enforced at runtime).
00396 *
00397 * | Index | Unit | Default | Valid Values |
00398 * |-----|-----|-----|-----|
00399 * | 0 | S12AD0 | 12-bit | 8, 10, 12 |
00400 * | 1 | S12AD1 | 12-bit | 8, 10, 12 |
00401 *
00402 * @note Resolution is set only on first init and cannot be changed
00403 *
00404 * @see adc_init() Sets on first initialization, validates on subsequent
00405 * @see adc_read_voltage_mv() Uses for voltage calculation
00406 */
00407 static adc_resolution_t s_adc_unit_resolution[k_adc_max_units] = {k_adc_resolution_12bit,
00408                                                                    k_adc_resolution_12bit};
00409
00410 /*
00411 * Internal Helper Functions
00412 *
00413 */
00414
00415 /**
00416 * @brief Get ADC register base address from unit number
00417 *
00418 * @details
00419 * Returns a pointer to the register structure for the specified ADC unit.
00420 * The RX72N has two ADC units with registers at different base addresses:
00421 *
00422 * | Unit | Peripheral | Base Address |
00423 * |-----|-----|-----|
00424 * | 0 | S12AD0 | 0x00089000 |
00425 * | 1 | S12AD1 | 0x00089100 |
00426 *
00427 *
00428 *
00429 * @pre None (pure lookup function)
00430 * @post No state changes
00431 *
00432 * @note Thread-safe: Pure function with no shared state
00433 * @note Performance: O(1), ~10 cycles
00434 *
00435 * @see s12ad0() Accessor for S12AD0 registers
00436 * @see s12ad1() Accessor for S12AD1 registers
00437 *
00438 * @since Version 1.0.0
00439 */
00440 static volatile rx_s12ad_regs_t* internal_get_adc_base(const uint8_t unit)
00441 {
00442     switch (unit) {
00443         case k_adc_unit_0: {
00444             return s12ad0();
00445         }
00446         case k_adc_unit_1: {
00447             return s12ad1();
00448         }
00449         default: {
00450             return nullptr;
00451         }
00452     }

```

```

00453 }
00454
00455 /**
00456  * @brief Configure ADC unit control registers and enable module clock
00457  *
00458  * @details
00459  * Performs one-time initialization of an ADC unit:
00460  * 1. Unlocks PRCR to allow MSTPCRA modification
00461  * 2. Clears module stop bit in MSTPCRA (enables clock)
00462  * 3. Locks PRCR
00463  * 4. Configures ADCER register for desired resolution
00464  * 5. Marks unit as initialized
00465  *
00466  * ## Register Modification Sequence
00467  *
00468  * @code
00469  * PRCR = k_rx_prcr_unlock_prc1_prc3 // Enable MSTPCRA write
00470  * MSTPCRA &= ~(1 « module_stop_bit) // Clear module stop bit
00471  * PRCR = k_rx_prcr_lock // Disable protected writes
00472  * ADCER = (ADCER & ~mask) | resolution_bits // Set resolution
00473  * @endcode
00474  *
00475  *
00476  *
00477  * @pre adc must be valid pointer from internal_get_adc_base()
00478  * @pre Unit must not already be initialized
00479  *
00480  * @post Module clock enabled (MSTPCRA bit cleared)
00481  * @post ADCER configured with requested resolution
00482  * @post s_adc_unit_initialized[unit] = true
00483  *
00484  * @note NOT thread-safe: Protected register access
00485  * @note Performance: ~1 us @ 240 MHz
00486  * @note PRCR unlock window is minimized
00487  *
00488  * @warning Must not be called while ADC is converting
00489  * @warning PRCR access requires proper unlock sequence
00490  *
00491  * @see adc_init() Public API that calls this function
00492  * @see rx_register_protection.h PRCR constants
00493  *
00494  * @since Version 1.0.0
00495  */
00496 static rx_err_t
00497 internal_configure_adc_unit(const uint8_t unit, volatile rx_s12ad_regs_t* adc, const uint8_t bits)
00498 {
00499     RX_CHECK_NULL_PTR(adc, s_tag, "ADC register pointer is nullptr");
00500     RX_CHECK_RANGE_TAG(unit, k_adc_unit_0, k_adc_unit_1, k_rx_err_invalid_arg, s_tag);
00501
00502     if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00503         bits != k_adc_resolution_12bit) {
00504         rx_log_error(s_tag, "Invalid ADC resolution");
00505         return k_rx_err_invalid_arg;
00506     }
00507
00508     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00509
00510     if (unit == k_adc_unit_0) {
00511         system_regs()->mstpcra &= ~((uint32_t)k_adc_bit_one « k_adc_mstpra_s12ad0);
00512     } else {
00513         system_regs()->mstpcra &= ~((uint32_t)k_adc_bit_one « k_adc_mstpra_s12ad1);
00514     }
00515
00516     *prcr_reg() = k_rx_prcr_lock;
00517
00518     uint16_t adcer = adc->adcer;
00519     adcer &= (uint16_t) ~(uint16_t)(k_adc_adcer_adprc_mask « k_adc_adcer_adprc_shift);
00520     if (bits == k_adc_resolution_8bit) {
00521         adcer |= (uint16_t)(k_adc_adcer_adprc_8bit « k_adc_adcer_adprc_shift);
00522     } else if (bits == k_adc_resolution_10bit) {
00523         adcer |= (uint16_t)(k_adc_adcer_adprc_10bit « k_adc_adcer_adprc_shift);
00524     } else {
00525         adcer |= (uint16_t)(k_adc_adcer_adprc_12bit « k_adc_adcer_adprc_shift);
00526     }
00527     adc->adcer = adcer;
00528
00529     s_adc_unit_initialized[unit] = true;
00530     rx_log_debug(s_tag, "ADC unit initialized");
00531
00532     return k_rx_ok;
00533 }
00534
00535 /**
00536  * @brief Validate ADC unit and channel numbers
00537  *
00538  * @details
00539  * Performs range validation for ADC unit and channel parameters before

```



```

00540 * any hardware access. This prevents undefined behavior from invalid
00541 * register access.
00542 *
00543 * ## Validation Rules
00544 *
00545 * | Parameter | Valid Range | Error Code |
00546 * |-----|-----|-----|
00547 * | unit | 0-1 (k_adc_unit_0 to k_adc_unit_1) | k_rx_err_invalid_arg |
00548 * | channel | 0-7 (k_adc_channel_0 to k_adc_channel_7) | k_rx_err_invalid_arg |
00549 *
00550 *
00551 *
00552 * @pre None (pure validation function)
00553 * @post No state changes
00554 *
00555 * @note Thread-safe: Pure function with no shared state
00556 * @note Performance: O(1), ~20 cycles
00557 * @note Logs error message on validation failure
00558 *
00559 * @see adc_init() Uses for parameter validation
00560 * @see adc_read() Uses for parameter validation
00561 *
00562 * @since Version 1.0.0
00563 */
00564 static rx_err_t internal_validate_unit_channel(const uint8_t unit, uint8_t channel)
00565 {
00566     /* Validate unit */
00567     if (unit > k_adc_unit_1) {
00568         rx_log_error(s_tag, "Invalid ADC unit");
00569         return k_rx_err_invalid_arg;
00570     }
00571
00572     /* Validate channel */
00573     if (channel > k_adc_channel_7) {
00574         rx_log_error(s_tag, "Invalid ADC channel");
00575         return k_rx_err_invalid_arg;
00576     }
00577
00578     return k_rx_ok;
00579 }
00580
00581 /**
00582 * @brief Read raw ADC value from a specific channel data register
00583 *
00584 * @details
00585 * Helper function that reads the conversion result from the appropriate ADDRn
00586 * register based on channel number. Extracted from adc_read() to support
00587 * NASA Power of 10 Rule 4 (functions under 60 lines).
00588 *
00589 * ## Channel to Register Mapping
00590 *
00591 * | Channel | Register | Offset |
00592 * |-----|-----|-----|
00593 * | 0 | ADDR0 | 0x020 |
00594 * | 1 | ADDR1 | 0x022 |
00595 * | 2 | ADDR2 | 0x024 |
00596 * | 3 | ADDR3 | 0x026 |
00597 * | 4 | ADDR4 | 0x028 |
00598 * | 5 | ADDR5 | 0x02A |
00599 * | 6 | ADDR6 | 0x02C |
00600 * | 7 | ADDR7 | 0x02E |
00601 *
00602 *
00603 *
00604 * @pre adc must be valid pointer from internal_get_adc_base()
00605 * @pre ADC conversion must be complete (ADST bit cleared)
00606 *
00607 * @post *value contains raw ADC reading (12/10/8-bit depending on config)
00608 * @post No hardware state changes
00609 *
00610 * @note Thread-safe: Read-only register access
00611 * @note Performance: O(1), ~30 cycles
00612 *
00613 * @see adc_read() Calls this after conversion complete
00614 *
00615 * @since Version 1.0.0
00616 */
00617 static rx_err_t internal_read_channel_value(volatile rx_s12ad_regs_t* adc,
00618                                             const adc_channel_t channel,
00619                                             uint16_t* value)
00620 {
00621     RX_CHECK_NULL_PTR(adc, s_tag, "ADC register pointer is nullptr");
00622     RX_CHECK_NULL_PTR(value, s_tag, "value pointer is nullptr");
00623
00624     switch (channel) {
00625     case k_adc_channel_0:
00626         *value = adc->addr0;

```

```

00627     break;
00628     case k_adc_channel_1:
00629         *value = adc->addr1;
00630         break;
00631     case k_adc_channel_2:
00632         *value = adc->addr2;
00633         break;
00634     case k_adc_channel_3:
00635         *value = adc->addr3;
00636         break;
00637     case k_adc_channel_4:
00638         *value = adc->addr4;
00639         break;
00640     case k_adc_channel_5:
00641         *value = adc->addr5;
00642         break;
00643     case k_adc_channel_6:
00644         *value = adc->addr6;
00645         break;
00646     case k_adc_channel_7:
00647         *value = adc->addr7;
00648         break;
00649     default:
00650         rx_log_error(s_tag, "Unsupported channel");
00651         return k_rx_err_invalid_arg;
00652 }
00653
00654 return k_rx_ok;
00655 }
00656
00657 /*
=====
00658 * Public API Implementation
00659 *
=====
00660 */
00661
00662 /**
00663  * @brief Initialize ADC unit and enable a specific channel
00664  *
00665  * @details
00666  *   Configures the S12ADFa peripheral for A/D conversion. On first call for
00667  *   each unit, enables the module clock, configures resolution, and initializes
00668  *   the hardware. Subsequent calls to the same unit enable additional channels.
00669  *
00670  * ## Initialization Behavior
00671  *
00672  * | Scenario | Action | Resolution | Result |
00673  * |-----|-----|-----|-----|
00674  * | First init for unit | Full init | Set to 'bits' | k_rx_ok |
00675  * | Subsequent init, same bits | Enable channel only | Unchanged | k_rx_ok |
00676  * | Subsequent init, different bits | Error | Unchanged | k_rx_err_invalid_arg |
00677  *
00678  *
00679  *
00680  * @pre None (can be first call for unit)
00681  * @pre Resolution must match if unit already initialized
00682  *
00683  * @post Module clock enabled (if first init)
00684  * @post Channel enabled in ADANSA register
00685  * @post Unit marked as initialized
00686  *
00687  * @note NOT thread-safe: Call from initialization code only
00688  * @note Performance: ~1-2 us for first init, ~0.5 us for subsequent
00689  *
00690  * @par Example:
00691  * @code
00692  * // Initialize ADC unit 0, channel 2 with 12-bit resolution
00693  * rx_err_t err = adc_init(k_adc_unit_0, k_adc_channel_2, k_adc_resolution_12bit);
00694  * if (err != k_rx_ok) {
00695  *     rx_log_error("SENSOR", "ADC init failed: %d", err);
00696  *     return err;
00697  * }
00698  * @endcode
00699  *
00700  * @see adc_read() Read after initialization
00701  * @see adc_read_voltage_mv() Read as voltage
00702  *
00703  * @since Version 1.0.0
00704  *
00705  * @par NASA Power of 10 Compliance:
00706  * - Rule 5: [OK] 2 preconditions, 3 postconditions
00707  */
00708 rx_err_t adc_init(const adc_unit_t unit, const adc_channel_t channel, const adc_resolution_t bits)
00709 {
00710     /* Validate parameters */
00711     rx_err_t err = internal_validate_unit_channel(unit, channel);

```

```

00712 RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00713
00714 /* Validate resolution */
00715 if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00716     bits != k_adc_resolution_12bit) {
00717     rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00718     return k_rx_err_invalid_arg;
00719 }
00720
00721 /* Get ADC base */
00722 volatile rx_s12ad_regs_t* const adc = internal_get_adc_base(unit);
00723 if (adc == nullptr) {
00724     return k_rx_err_invalid_arg;
00725 }
00726
00727 /* Initialize ADC unit if not already initialized */
00728 if (!s_adc_unit_initialized[unit]) {
00729     /* Store resolution for validation on subsequent calls */
00730     s_adc_unit_resolution[unit] = bits;
00731
00732     err = internal_configure_adc_unit(unit, adc, bits);
00733     if (err != k_rx_ok) {
00734         return err;
00735     }
00736 } else {
00737     /* Unit already initialized - verify resolution matches */
00738     if (bits != s_adc_unit_resolution[unit]) {
00739         rx_log_error(s_tag, "ADC resolution mismatch on subsequent init");
00740         return k_rx_err_invalid_arg;
00741     }
00742 }
00743
00744 /* Enable the specified channel */
00745 if ((uint8_t)channel <= k_adc_channel_adansa0_max) {
00746     adc->adansa0 |= ((uint16_t)k_adc_bit_one << channel);
00747 } else {
00748     adc->adansa1 |= ((uint16_t)k_adc_bit_one << ((uint8_t)channel - k_adc_channel_adansa1_base));
00749 }
00750
00751 rx_log_debug(s_tag, "ADC channel enabled");
00752
00753 return k_rx_ok;
00754 }
00755
00756 /**
00757  * @brief Read raw ADC conversion value from specified channel
00758  *
00759  * @details
00760  * Performs a single-scan A/D conversion and returns the raw digital value.
00761  * This is a blocking operation that polls the ADST bit until conversion
00762  * completes or timeout occurs.
00763  *
00764  * ## Conversion Sequence
00765  *
00766  * 1. Validate parameters and initialization state
00767  * 2. Start conversion by setting ADCSR.ADST bit
00768  * 3. Poll ADCSR.ADST until cleared (conversion complete)
00769  * 4. Read result from appropriate ADDRn register
00770  *
00771  * ## Output Value Ranges
00772  *
00773  * | Resolution | Range | Full-Scale |
00774  * |-----|-----|-----|
00775  * | 12-bit | 0-4095 | 3.3V |
00776  * | 10-bit | 0-1023 | 3.3V |
00777  * | 8-bit | 0-255 | 3.3V |
00778  *
00779  *
00780  *
00781  * @pre ADC unit must be initialized via adc_init()
00782  * @pre value must point to valid uint16_t storage
00783  *
00784  * @post *value contains raw ADC reading (if k_rx_ok)
00785  * @post ADC ready for next conversion
00786  *
00787  * @note NOT thread-safe: Caller must synchronize if shared
00788  * @note Blocking: Waits up to ~100ms for conversion
00789  * @note Performance: ~1-10 us typical (depends on conversion time)
00790  *
00791  * @warning Do not call from ISR (blocking wait)
00792  *
00793  * @par Example:
00794  * @code
00795  * uint16_t raw_value;
00796  * rx_err_t err = adc_read(k_adc_unit_0, k_adc_channel_2, &raw_value);
00797  * if (err == k_rx_ok) {
00798  *     // Example: 12-bit resolution (0-4095 raw -> 0-3.3 V)

```

```

00799 * static const float s_vref_volts = 3.3F;
00800 * static const float s_adc_full_scale_12bit = 4095.0F;
00801 * float voltage_mv = (raw_value * s_vref_volts) / s_adc_full_scale_12bit;
00802 * }
00803 * @endcode
00804 *
00805 * @see adc_init() Must be called first
00806 * @see adc_read_voltage_mv() Convenience function for voltage
00807 *
00808 * @since Version 1.0.0
00809 *
00810 * @par NASA Power of 10 Compliance:
00811 * - Rule 2: [OK] Timeout loop has fixed upper bound
00812 * - Rule 5: [OK] 2 preconditions, 2 postconditions
00813 */
00814 rx_err_t adc_read(const adc_unit_t unit, const adc_channel_t channel, uint16_t* value)
00815 {
00816     /* Validate parameters */
00817     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00818
00819     const rx_err_t err = internal_validate_unit_channel(unit, channel);
00820     RX_RETURN_ON_ERROR(err, s_tag, "Unit/channel validation failed");
00821
00822     /* Check if unit is initialized */
00823     if (!s_adc_unit_initialized[unit]) {
00824         rx_log_error(s_tag, "ADC unit not initialized");
00825         return k_rx_err_invalid_state;
00826     }
00827
00828     /* Get ADC base */
00829     volatile rx_sl2ad_regs_t* const adc = internal_get_adc_base(unit);
00830     if (adc == nullptr) {
00831         return k_rx_err_invalid_arg;
00832     }
00833
00834     /* Start single-scan conversion */
00835     adc->adcsr = k_adc_adcsr_adst;
00836
00837     /* Wait for conversion to complete (poll ADCSR.ADST bit) */
00838     uint32_t timeout = k_adc_timeout_ms * k_adc_timeout_multiplier;
00839     while ((adc->adcsr & k_adc_adcsr_adst) != k_adc_adcsr_idle && timeout > k_adc_timeout_expired) {
00840         timeout--;
00841     }
00842
00843     if (timeout == k_adc_timeout_expired) {
00844         rx_log_error(s_tag, "ADC conversion timeout");
00845         return k_rx_err_timeout;
00846     }
00847
00848     /* Read conversion result from appropriate data register */
00849     return internal_read_channel_value(adc, channel, value);
00850 }
00851 /**
00852 * @brief Read ADC value and convert to millivolts
00853 *
00854 * @details
00855 * Performs an A/D conversion and scales the result to millivolts. This is
00856 * a convenience function that calls adc_read() and applies the voltage
00857 * calculation formula.
00858 *
00859 * ## Voltage Calculation Formula
00860 *
00861 * @f[
00862 * V_{mV} = \frac{V_{raw}}{2^{\{bits\}} - 1} \times 3300
00863 * @f]
00864 *
00865 * ## Example Calculations
00866 *
00867 * | Resolution | Raw Value | Voltage (mV) | Calculation |
00868 * |-----|-----|-----|-----|
00869 * | 12-bit | 2048 | 1650 | 2048 x 3300 / 4095 |
00870 * | 10-bit | 512 | 1650 | 512 x 3300 / 1023 |
00871 * | 8-bit | 128 | 1655 | 128 x 3300 / 255 |
00872 *
00873 *
00874 *
00875 *
00876 * @pre ADC unit must be initialized via adc_init()
00877 * @pre bits must match the resolution used in adc_init()
00878 *
00879 * @post *voltage_mv contains voltage in millivolts (0-3300)
00880 * @post ADC ready for next conversion
00881 *
00882 * @note NOT thread-safe: Caller must synchronize if shared
00883 * @note Blocking: Waits for conversion (see adc_read)
00884 * @note Reference: Uses 3.3V reference (k_adc_reference_voltage_mv = 3300)
00885 *

```

```

00886 * @par Example:
00887 * @code
00888 * // Read motor current sense (DRV8263H IPROPI output)
00889 * uint32_t voltage_mv;
00890 * rx_err_t err = adc_read_voltage_mv(k_adc_unit_0, k_adc_channel_0,
00891 *                                   k_adc_resolution_12bit, &voltage_mv);
00892 * if (err == k_rx_ok) {
00893 *     // voltage_mv: 0-3300 mV
00894 *     // For DRV8263H-Q1 IPROPI (202 uA/A mirror, 5100 Ohm sense):
00895 *     // I_motor = V_IPROPI / k_ipropi_divisor
00896 *     // (where k_ipropi_divisor = 1.0302 = 202e-6 * 5100)
00897 *     // Fixed-point: scale = 10000, divisor = 10302 (1.0302 * 10000)
00898 *     enum : uint32_t { k_scale = 10000, k_divisor = 10302 };
00899 *     uint32_t current_ma = (voltage_mv * k_scale) / k_divisor;
00900 *     rx_log_info("ADC", "Current: %lu mA", current_ma);
00901 * }
00902 * @endcode
00903 *
00904 * @see adc_read() Lower-level raw value read
00905 * @see adc_init() Initialize before reading
00906 *
00907 * @since Version 1.0.0
00908 *
00909 * @par NASA Power of 10 Compliance:
00910 * - Rule 5: [OK] 2 preconditions, 2 postconditions
00911 */
00912 rx_err_t adc_read_voltage_mv(const adc_unit_t unit,
00913                             adc_channel_t channel,
00914                             adc_resolution_t bits,
00915                             uint32_t* voltage_mv)
00916 {
00917     /* Parameter order: unit, channel, resolution bits. */
00918     RX_CHECK_NULL_PTR(voltage_mv, s_tag, "Voltage pointer is nullptr");
00919
00920     if (bits != k_adc_resolution_8bit && bits != k_adc_resolution_10bit &&
00921         bits != k_adc_resolution_12bit) {
00922         rx_log_error(s_tag, "Invalid resolution (must be 8, 10, or 12 bits)");
00923         return k_rx_err_invalid_arg;
00924     }
00925
00926     /* Validate that passed resolution matches unit's configured resolution */
00927     if (unit > k_adc_unit_1) {
00928         rx_log_error(s_tag, "Invalid ADC unit");
00929         return k_rx_err_invalid_arg;
00930     }
00931
00932     if (bits != s_adc_unit_resolution[unit]) {
00933         rx_log_error(s_tag, "ADC resolution mismatch in voltage read");
00934         return k_rx_err_invalid_arg;
00935     }
00936
00937     /* Read raw ADC value */
00938     uint16_t raw_value = k_adc_raw_value_reset;
00939     const rx_err_t err = adc_read(unit, channel, &raw_value);
00940     RX_RETURN_ON_ERROR(err, s_tag, "ADC read failed");
00941
00942     /* Calculate voltage (using ADC reference voltage) */
00943     const uint32_t max_value = ((uint32_t)k_adc_bit_one << bits) - k_adc_bit_one;
00944     *voltage_mv = ((uint32_t)raw_value * k_adc_reference_voltage_mv) / max_value;
00945
00946     return k_rx_ok;
00947 }
00948
00949 #endif /* __RX__ */

```

8.103 libs/rx`hal/src/gpio.c File Reference

GPIO Driver for RX72N - General Purpose Input/Output Control.

```

#include <stddef.h>
#include "hardware.h"
#include "rx72n_regs.h"
#include "rx_port_constants.h"
#include "rx_port_utils.h"

```

Include dependency graph for gpio.c:



Enumerations

- enum [gpio_bit_constants_t](#) : uint8_t { [k_gpio_bit_set](#) = 1 , [k_gpio_bit_clear](#) = 0 }
GPIO register bit manipulation constants.

Functions

- static rx_err_t [internal_validate_port_pin](#) (const uint8_t port, const uint8_t pin)
Validate port and pin numbers against RX72N hardware limits.
- rx_err_t [gpio_set_output](#) (const rx_port_pin_t pin)
Configure GPIO pin as digital output.
- rx_err_t [gpio_set_input](#) (const rx_port_pin_t pin)
Configure GPIO pin as digital input.
- rx_err_t [gpio_write_high](#) (const rx_port_pin_t pin)
Set GPIO output pin to logic high (3.3V).
- rx_err_t [gpio_write_low](#) (const rx_port_pin_t pin)
Set GPIO output pin to logic low (0V).
- rx_err_t [gpio_toggle](#) (const rx_port_pin_t pin)
Toggle GPIO output pin state (high<->low).
- rx_err_t [gpio_read](#) (const rx_port_pin_t pin, bool *value)
Read GPIO input pin state.

Variables

- static const char *const [s_tag](#) = "GPIO"
Log tag for this module.

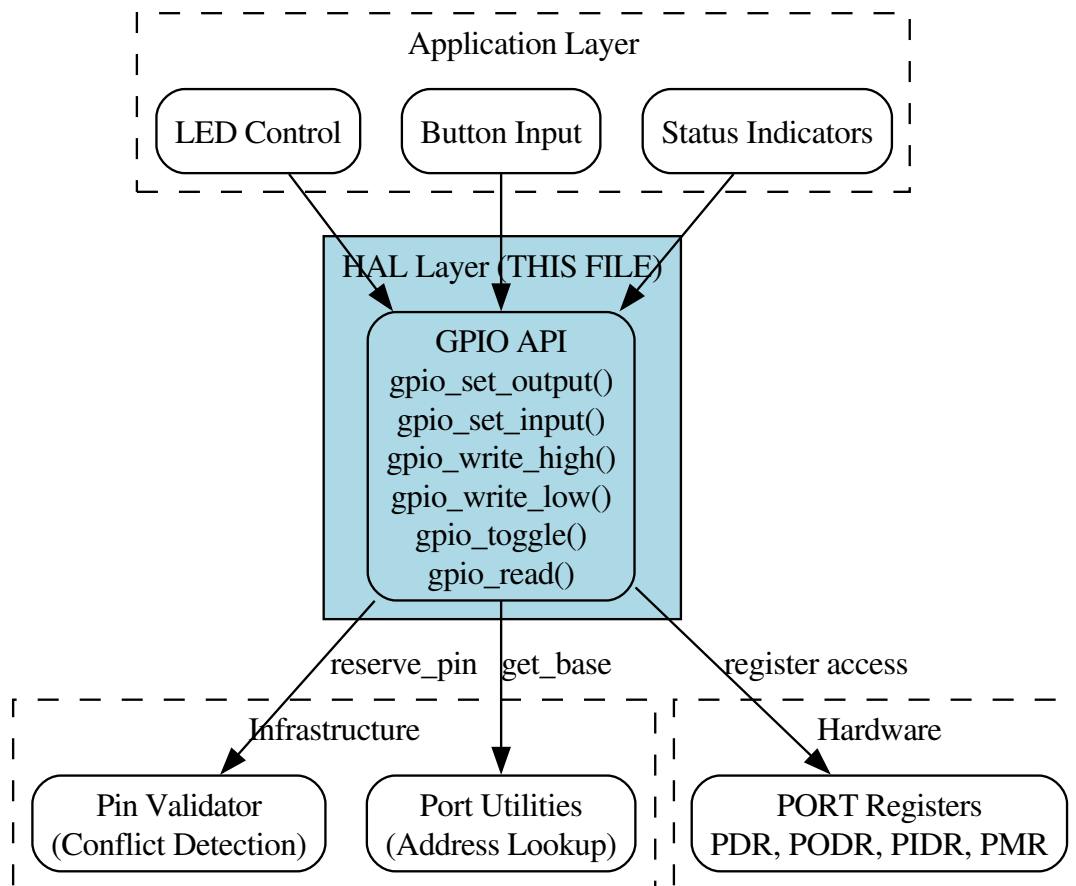
8.103.1 Detailed Description

GPIO Driver for RX72N - General Purpose Input/Output Control.

8.103.1.1 Overview

This file provides the hardware abstraction layer (HAL) for GPIO operations on the Renesas RX72N microcontroller. It handles pin direction configuration, digital output control, and digital input reading with integrated error handling, logging, and pin validation through the pin validator interface.

8.103.1.2 System Architecture



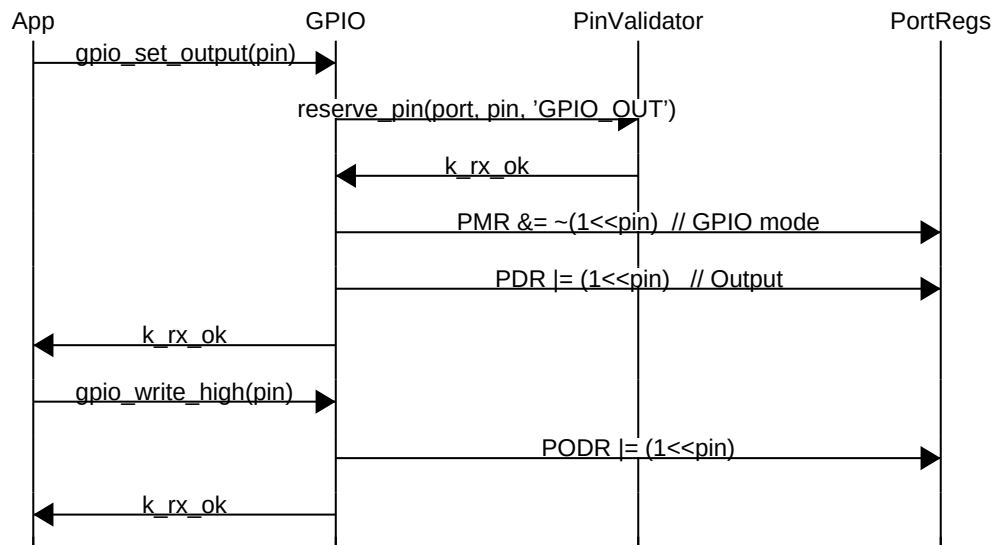
8.103.1.3 RX72N GPIO Register Architecture

Each GPIO port on the RX72N has multiple control registers:

Register	Name	Purpose
PDR	Port Direction Register	0=input, 1=output
PODR	Port Output Data Register	Output level (0=low, 1=high)
PIDR	Port Input Data Register	Read current pin level
PMR	Port Mode Register	0=GPIO, 1=peripheral function
PCR	Pull-up Control Register	Enable internal pull-up
ODR	Open-Drain Control Register	Enable open-drain mode

This driver primarily uses PDR, PODR, PIDR, and PMR.

8.103.1.4 Pin Configuration Sequence



8.103.1.5 Thread Safety

Function	Thread-Safe	Notes
gpio_set_output()	Partial	Pin reservation is mutex-protected
gpio_set_input()	Partial	Pin reservation is mutex-protected
gpio_write_high()	No	Direct register write
gpio_write_low()	No	Direct register write
gpio_toggle()	No	Read-modify-write, not atomic
gpio_read()	Yes	Read-only operation

For thread-safe output operations, caller must provide synchronization.

8.103.1.6 Performance Characteristics

Operation	Cycles	Time @ 240 MHz	Notes
<code>gpio_set_output</code>	~200-500	~1-2 us	Includes pin reservation
<code>gpio_set_input</code>	~200-500	~1-2 us	Includes pin reservation
<code>gpio_write_high</code>	~20-30	~100 ns	Direct register
<code>gpio_write_low</code>	~20-30	~100 ns	Direct register
<code>gpio_toggle</code>	~30-40	~150 ns	Read-modify-write
<code>gpio_read</code>	~20-30	~100 ns	Direct register read

8.103.1.7 Memory Usage

Section	Size	Contents
.text	~600 bytes	Function code
.rodata	~100 bytes	Log strings
.bss	0 bytes	No module-level state

8.103.1.8 NASA Power of 10 Compliance

Rule	Status	Implementation
1. Simplify control flow	↔ [OK]↔	No goto, setjmp, recursion
2. Fixed loop bounds	↔ [OK]↔	No loops in this module
3. No dynamic memory	↔ [OK]↔	All static, no malloc
4. Functions < 60 lines	↔ [OK]↔	All functions under 30 lines
5. Use assertions	↔ [OK]↔	RX_CHECK_NULL_PTR, validation
6. Data at smallest scope	↔ [OK]↔	Local variables only
7. Check return values	↔ [OK]↔	RX_RETURN_ON_ERROR macro
8. Limit preprocessor	↔ [OK]↔	Only includes
9. Restrict pointers	↔ [OK]↔	Only hardware register pointers
10. Compile warnings	↔ [OK]↔	-Wall -Wextra -Werror clean

8.103.1.9 SOLID Principles

Principle	Implementation
Single Responsibility	GPIO operations only (no PWM, no ADC)
Open/Closed	Extended via rx_port_pin_t encoding

Principle	Implementation
Liskov Substitution	N/A (not interface-based)
Interface Segregation	Separate functions for each operation
Dependency Inversion	Uses <code>pin_interface_t</code> for validation

See also

[rx_port_constants.h](#) Port and pin definitions

[rx_port_utils.h](#) Port address lookup utilities

[rx72n_regs.h](#) Register definitions

[rx_pin_validator.h](#) Pin conflict detection

Author

Locked, Inc.

Date

2026-01-01

Copyright

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Since

Version 1.0.0

Definition in file [gpio.c](#).

8.103.2 Enumeration Type Documentation

8.103.2.1 `gpio_bit_constants_t`

enum [gpio_bit_constants_t](#) : `uint8_t`

GPIO register bit manipulation constants.

Defines constants used for bit manipulation in GPIO register operations. Using named constants instead of magic numbers improves code readability and makes the intent of bit operations clear.

8.103.2.2 Usage Examples

```
// Set bit 5 in PDR register (configure as output)
port->pdr |= (k_gpio_bit_set « 5);

// Clear bit 3 in PMR register (configure as GPIO, not peripheral)
port->pmr &= ~(k_gpio_bit_set « 3);

// Check if bit 2 is set in PIDR register
if ((port->pidr & (k_gpio_bit_set « 2)) != k_gpio_bit_clear) {
    // Pin is high
}
```

See also

[gpio_set_output\(\)](#) Uses these for PDR/PMR manipulation

[gpio_read\(\)](#) Uses these for PIDR comparison

Since

Version 1.0.0

Enumerator

k_gpio_bit_set	Value for setting a single bit (1 << n pattern)
k_gpio_bit_clear	Value for comparison when checking if bit is cleared

Definition at line 201 of file [gpio.c](#).

```
00201      : uint8_t {
00202  k_gpio_bit_set    = 1, /**< Value for setting a single bit (1 « n pattern) */
00203  k_gpio_bit_clear  = 0, /**< Value for comparison when checking if bit is cleared */
00204 } gpio_bit_constants_t;
```

8.103.3 Function Documentation

8.103.3.1 gpio`read()

```
rx_err_t gpio_read (
    const rx_port_pin_t pin,
    bool * value)    [nodiscard]
```

Read GPIO input pin state.

Read current logic level of GPIO pin.

Reads the current logic level of a GPIO pin. Uses the PIDR (Port Input Data Register) which reflects the actual voltage level on the pin, regardless of whether the pin is configured as input or output.

8.103.3.2 Register Operation

```
value = (PIDR & (1 « pin)) != 0
```

8.103.3.3 Input vs Output Reading

- Input pin: PIDR reflects external voltage on pin
- Output pin: PIDR reflects actual pin state (may differ from PODR if pin is shorted or overdriven)

8.103.3.4 Voltage Thresholds (RX72N)

Level	Voltage Range	Returned Value
Low	0V to ~0.8V	false
High	~2.0V to VCC	true
Undefined	0.8V to 2.0V	Unpredictable

Precondition

value must be non-nullptr

Pin should be configured as input for external signals

Postcondition

*value contains current pin level (true=high, false=low)

No hardware state changes

Note

Thread-safe: Read-only operation

Performance: ~100 ns @ 240 MHz

ISR-safe: Can be called from interrupt handlers

Warning

Floating inputs may return unpredictable values

Use pull-up/pull-down for stable readings on open pins

Example:

```
// Read button state (active low with pull-up)
bool button_pressed;
rx_err_t err = gpio_read(RX_PORT_PIN(2, 0), &button_pressed);

if (err == k_rx_ok) {
    // Invert because button is active low
    if (!button_pressed) {
        rx_log_info("BTN", "Button pressed");
        handle_button_press();
    }
}
```

Example: Debounced button read

```
bool debounced_read(rx_port_pin_t pin) {
    bool first, second;
    gpio_read(pin, &first);
    tx_thread_sleep(1); // 1 tick delay
    gpio_read(pin, &second);
    return (first == second) ? first : false; // Only return if stable
}
```

See also

[gpio_set_input\(\)](#) Configure pin as input first
[gpio_write_high\(\)](#) Set output high
[gpio_write_low\(\)](#) Set output low

Since

Version 1.0.0

NASA Power of 10 Compliance:

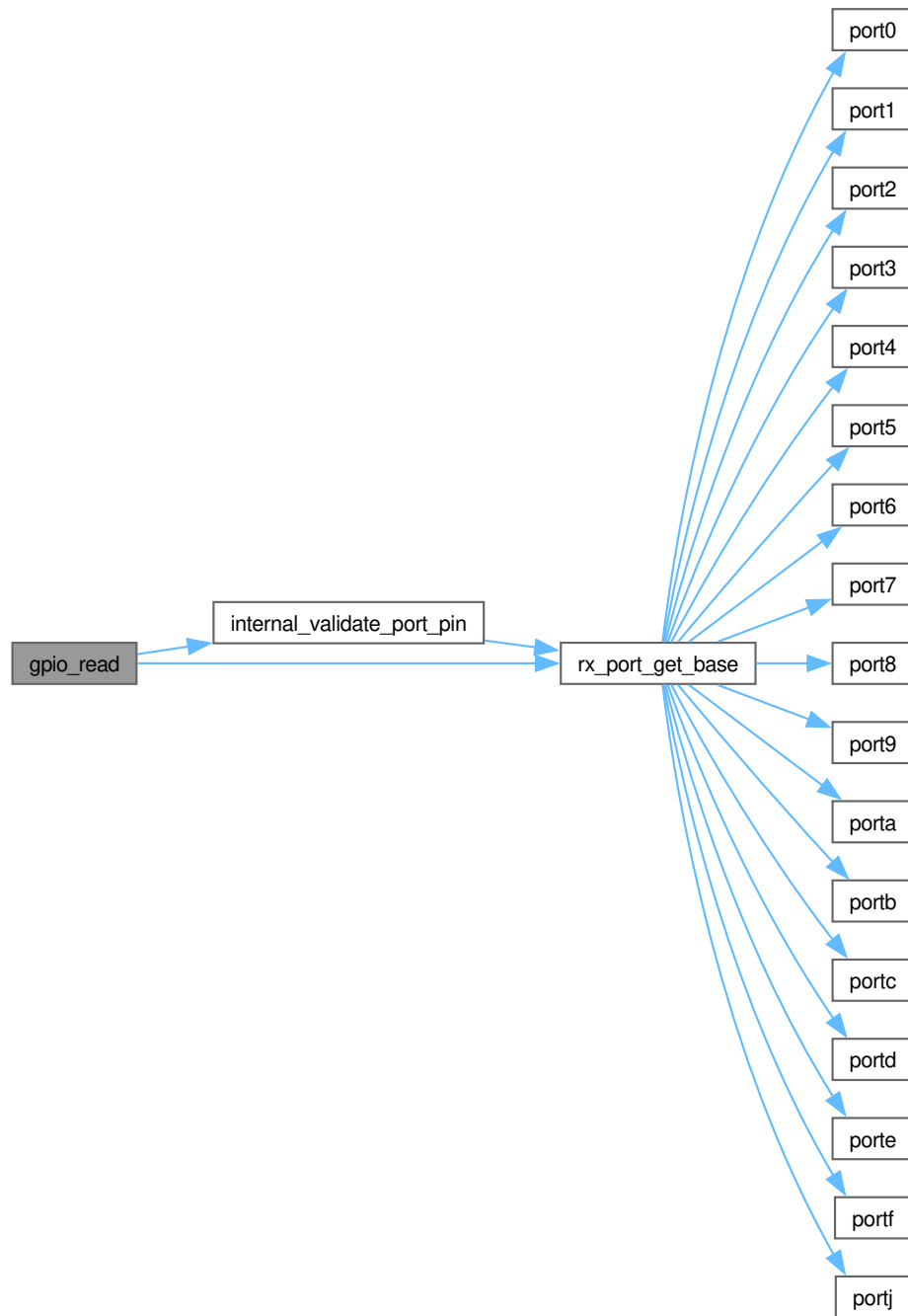
- Rule 5: [OK] 2 preconditions, 2 postconditions

Definition at line 803 of file [gpio.c](#).

```
00804 {
00805     /* Check null pointer */
00806     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00807
00808     /* Extract port and pin number from rx_port_pin_t */
00809     const uint8_t port = rx_port_from_pin(pin);
00810     const uint8_t pin_num = rx_pin_from_pin(pin);
00811
00812     /* Validate parameters */
00813     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00814     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00815
00816     /* Get port base address */
00817     const volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00818
00819     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00820     *value = (port_base->pidr & pin_mask) != k_gpio_bit_clear;
00821
00822     return k_rx_ok;
00823 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_clear](#), [k_gpio_bit_set](#), [rx_port_regs_t::pidr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.5 gpio_set_input()

```
rx_err_t gpio_set_input (
    const rx_port_pin_t pin) [nodiscard]
```

Configure GPIO pin as digital input.

Configures a GPIO pin for digital input mode. This function performs:

1. Parameter validation (port and pin number)
2. Pin reservation via the global pin validator (conflict detection)
3. PMR register clear (select GPIO mode instead of peripheral function)
4. PDR register clear (configure as input direction)

8.103.3.6 Register Operations

```
PMR &= ~(1 « pin) // Clear bit: Select GPIO mode (not peripheral)
PDR &= ~(1 « pin) // Clear bit: Configure as input
```

8.103.3.7 Pull-up Configuration

This function does NOT configure internal pull-up resistors. To enable pull-ups, additional configuration of the PCR register is required.

8.103.3.8 Reading Input State

After configuration, use [gpio_read\(\)](#) to read the current pin state. The PIDR register is read for input state, not PODR.

Precondition

- Pin must be a valid GPIO-capable pin on RX72N
- System infrastructure should be initialized (for pin validation)

Postcondition

- Pin is configured as GPIO input
- PMR bit for pin is cleared (GPIO mode)
- PDR bit for pin is cleared (input direction)
- Pin is reserved in validator (if available)

Note

- Thread-safe: Pin reservation uses mutex
- Performance: ~1-2 us @ 240 MHz (includes reservation)
- Pull-ups: Not enabled by this function (use PCR register)

Warning

- Floating inputs may read unpredictable values - use pull-up/pull-down
- Do not call from ISR (pin reservation may block)

Example:

```
// Configure button pin as input
rx_err_t err = gpio_set_input(RX_PORT_PIN(2, 0)); // Port 2, Pin 0
if (err != k_rx_ok) {
    rx_log_error("BTN", "Failed to configure button pin");
    return err;
}

// Read button state
bool pressed;
err = gpio_read(RX_PORT_PIN(2, 0), &pressed);
if (err == k_rx_ok && pressed) {
    rx_log_info("BTN", "Button pressed");
}
```

See also

[gpio_set_output\(\)](#) Configure pin as output
[gpio_read\(\)](#) Read input pin state
[RX_PORT_PIN\(\)](#) Macro to create rx_port_pin_t

Since

Version 1.0.0

NASA Power of 10 Compliance:

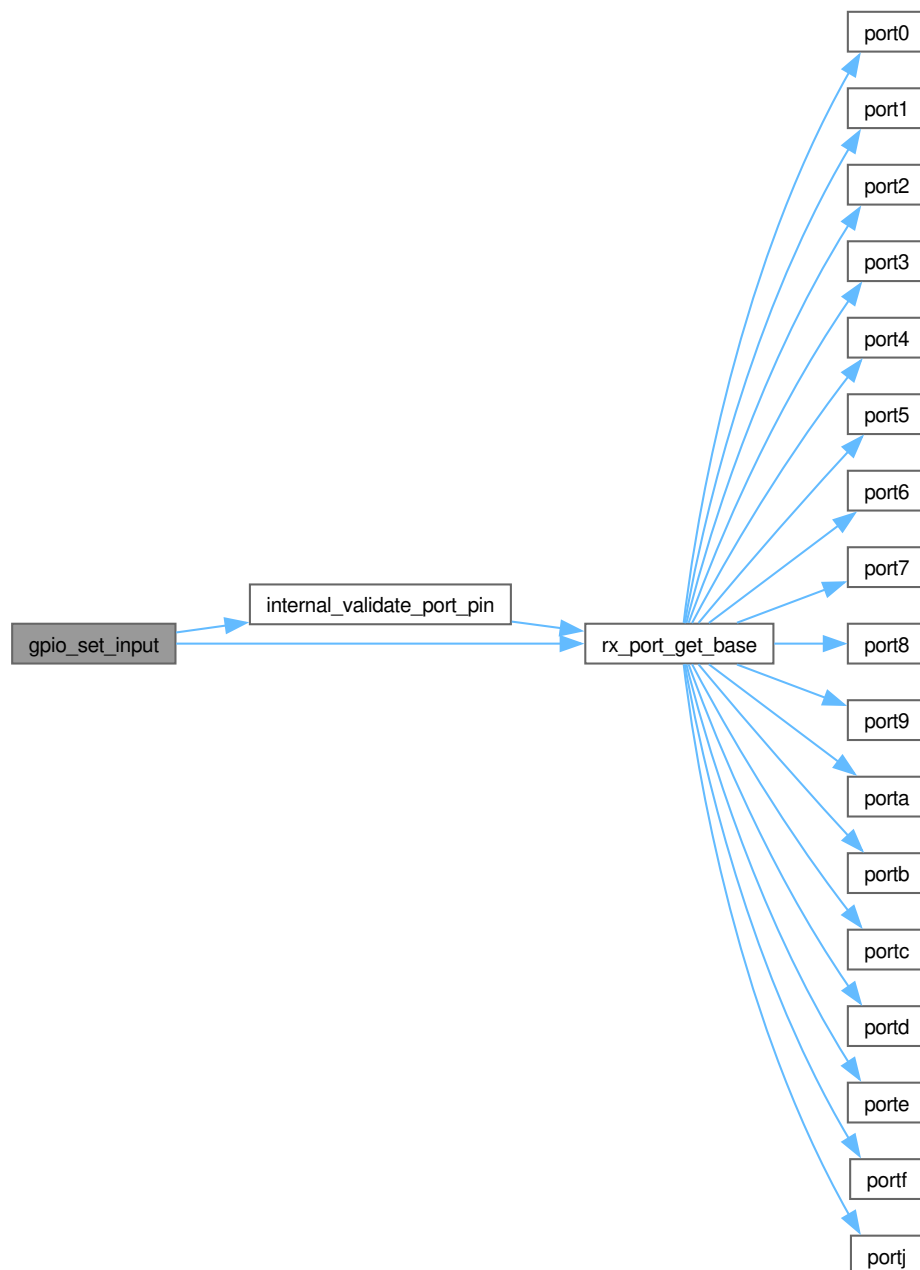
- Rule 5: [OK] 2 preconditions, 4 postconditions

Definition at line 481 of file [gpio.c](#).

```
00482 {
00483     /* Extract port and pin number from rx_port_pin_t */
00484     const uint8_t port = rx_port_from_pin(pin);
00485     const uint8_t pin_num = rx_pin_from_pin(pin);
00486
00487     /* Validate parameters */
00488     rx_err_t err = internal_validate_port_pin(port, pin_num);
00489     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00490
00491     /* Reserve pin through global pin validator */
00492     const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00493     if (pin_iface != nullptr) {
00494         err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_IN");
00495         if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00496             /* Allow conflict (pin already reserved), but fail on other errors */
00497             RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00498         }
00499     }
00500
00501     /* Get port base address */
00502     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00503
00504     /* Set as GPIO mode */
00505     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00506     port_base->pmr &= (uint8_t)~pin_mask;
00507
00508     /* Set as input */
00509     port_base->pdr &= (uint8_t)~pin_mask;
00510
00511     rx_log_debug(s_tag, "Pin configured as input");
00512
00513     return k_rx_ok;
00514 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_set](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.9 gpio`set`output()

```
rx_err_t gpio_set_output (
    const rx_port_pin_t pin) [nodiscard]
```

Configure GPIO pin as digital output.

Configures a GPIO pin for digital output mode. This function performs:

1. Parameter validation (port and pin number)
2. Pin reservation via the global pin validator (conflict detection)
3. PMR register clear (select GPIO mode instead of peripheral function)
4. PDR register set (configure as output direction)

8.103.3.10 Register Operations

```
PMR &= ~(1 « pin) // Clear bit: Select GPIO mode (not peripheral)
PDR |= (1 « pin) // Set bit: Configure as output
```

8.103.3.11 Pin Reservation

The function attempts to reserve the pin through `rx_infrastructure_get_pin_interface()`. If the pin is already reserved (`k_rx_err_gpio_conflict`), the function continues without error - this allows reconfiguration of already-owned pins.

8.103.3.12 Initial Output State

This function does NOT set the initial output level. After calling `gpio_set_output()`, call `gpio_write_high()` or `gpio_write_low()` to set the desired level.

Precondition

- Pin must be a valid GPIO-capable pin on RX72N
- System infrastructure should be initialized (for pin validation)

Postcondition

- Pin is configured as GPIO output
- PMR bit for pin is cleared (GPIO mode)
- PDR bit for pin is set (output direction)
- Pin is reserved in validator (if available)

Note

- Thread-safe: Pin reservation uses mutex, register writes are not atomic
- Performance: ~1-2 us @ 240 MHz (includes reservation)
- Idempotent: Safe to call multiple times on same pin

Warning

- Output level is undefined after this call - set with `gpio_write_*`
- Do not call from ISR (pin reservation may block)

Example:

```
// Configure LED pin as output
rx_err_t err = gpio_set_output(RX_PORT_PIN(0, 3)); // Port 0, Pin 3
if (err != k_rx_ok) {
    rx_log_error("LED", "Failed to configure LED pin");
    return err;
}

// Set initial state to low (LED off)
gpio_write_low(RX_PORT_PIN(0, 3));

// Toggle LED
gpio_toggle(RX_PORT_PIN(0, 3));
```

See also

[gpio_set_input\(\)](#) Configure pin as input
[gpio_write_high\(\)](#) Set output high after configuration
[gpio_write_low\(\)](#) Set output low after configuration
[RX_PORT_PIN\(\)](#) Macro to create rx_port_pin_t

Since

Version 1.0.0

NASA Power of 10 Compliance:

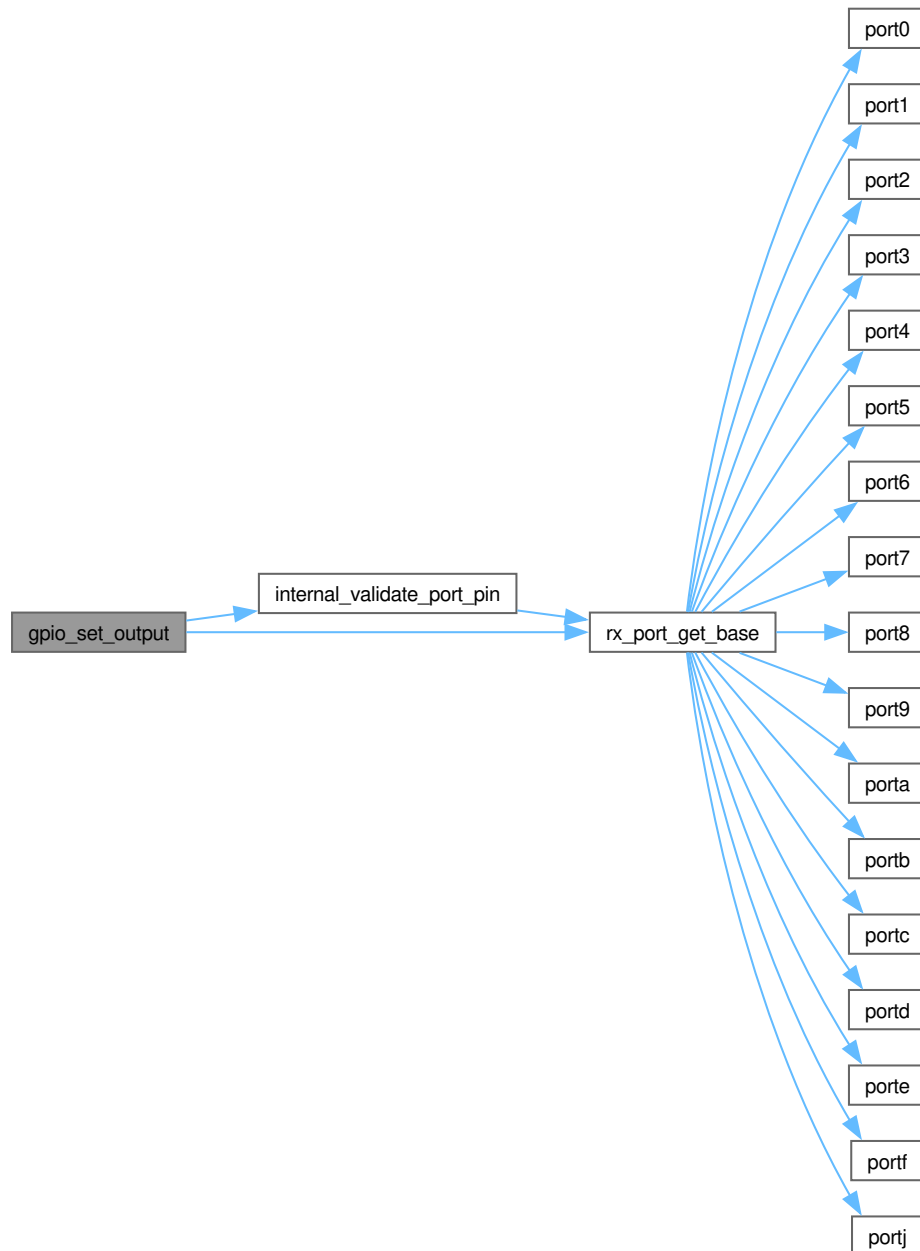
- Rule 5: [OK] 2 preconditions, 4 postconditions

Definition at line 375 of file [gpio.c](#).

```
00376 {
00377  /* Extract port and pin number from rx_port_pin_t */
00378  const uint8_t port = rx_port_from_pin(pin);
00379  const uint8_t pin_num = rx_pin_from_pin(pin);
00380
00381  /* Validate parameters */
00382  rx_err_t err = internal_validate_port_pin(port, pin_num);
00383  RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00384
00385  /* Reserve pin through global pin validator */
00386  const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00387  if (pin_iface != nullptr) {
00388      err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_OUT");
00389      if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00390          /* Allow conflict (pin already reserved), but fail on other errors */
00391          RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00392      }
00393  }
00394
00395  /* Get port base address */
00396  volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00397
00398  const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00399
00400  /* Set as GPIO mode (not peripheral) */
00401  port_base->pmr &= (uint8_t)~pin_mask;
00402
00403  /* Set as output */
00404  port_base->pdr |= pin_mask;
00405
00406  rx_log_debug(s_tag, "Pin configured as output");
00407
00408  return k_rx_ok;
00409 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_set](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.13 gpio_toggle()

```
rx_err_t gpio_toggle (
    const rx_port_pin_t pin) [nodiscard]
```

Toggle GPIO output pin state (high<->low).

Toggle GPIO output pin state (high to low, or low to high).

Inverts the current output state of a GPIO pin. If the pin is currently high, it becomes low. If currently low, it becomes high. Uses XOR operation for efficient toggling.

8.103.3.14 Register Operation

```
PODR ^= (1 « pin) // XOR bit: Toggle output
```

8.103.3.15 Atomicity Warning

This operation is a read-modify-write sequence and is NOT atomic. If an interrupt modifies the same port register between the read and write, the interrupt's change may be lost.

For thread-safe toggling, use explicit `write_high/write_low` with external synchronization, or disable interrupts around the toggle.

Precondition

Pin must be configured as output via [gpio_set_output\(\)](#)

Port and pin numbers must be valid

Postcondition

PODR bit for pin is inverted

Pin output level is opposite of previous state

Note

NOT thread-safe: Read-modify-write operation

NOT atomic: Can race with ISRs on same port

Performance: ~150 ns @ 240 MHz

Warning

Race condition possible with ISRs modifying same port

Use `gpio_write_high/low` with sync for critical applications

Example:

```
// Blink LED in loop
while (running) {
    gpio_toggle(RX_PORT_PIN(0, 3)); // Toggle LED
    tx_thread_sleep(50);             // 50ms delay (20 Hz blink)
}
```

See also

[gpio_write_high\(\)](#) Set output high explicitly

[gpio_write_low\(\)](#) Set output low explicitly

[gpio_read\(\)](#) Read current output state

Since

Version 1.0.0

Definition at line 706 of file [gpio.c](#).

```
00707 {
00708     /* Extract port and pin number from rx_port_pin_t */
00709     const uint8_t port = rx_port_from_pin(pin);
00710     const uint8_t pin_num = rx_pin_from_pin(pin);
00711
00712     /* Validate parameters */
00713     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00714     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00715
00716     /* Get port base address */
00717     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00718
00719     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00720     port_base->podr ^= pin_mask;
00721
00722     return k_rx_ok;
00723 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_set](#), [rx_port_regs_t::podr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.16 gpio`write`high()

```
rx_err_t gpio_write_high (
    const rx_port_pin_t pin) [nodiscard]
```

Set GPIO output pin to logic high (3.3V).

Set GPIO output pin to logic high (VDD).

Sets a GPIO output pin to logic high level. On the RX72N, this means the pin outputs VCC (typically 3.3V). The pin must be configured as an output before calling this function.

8.103.3.17 Register Operation

```
PODR |= (1 « pin) // Set bit: Output high
```

8.103.3.18 Timing

The output level changes on the next peripheral clock cycle after the register write completes. GPIO toggle rate is limited by bus speed.

Precondition

- Pin must be configured as output via [gpio_set_output\(\)](#)
- Port and pin numbers must be valid

Postcondition

- PODR bit for pin is set (output high)
- Pin outputs VCC level (typically 3.3V)

Note

- NOT thread-safe: Direct register write without synchronization
- Performance: ~100 ns @ 240 MHz
- ISR-safe: Can be called from interrupt handlers

Warning

- No check that pin is configured as output
- Calling on input pin has no visible effect

Example:

```
// Turn on LED (active high)
rx_err_t err = gpio_write_high(RX_PORT_PIN(0, 3));
if (err != k_rx_ok) {
    // Handle error
}
```

See also

- [gpio_write_low\(\)](#) Set output low
- [gpio_toggle\(\)](#) Toggle output state
- [gpio_set_output\(\)](#) Configure pin as output first

Since

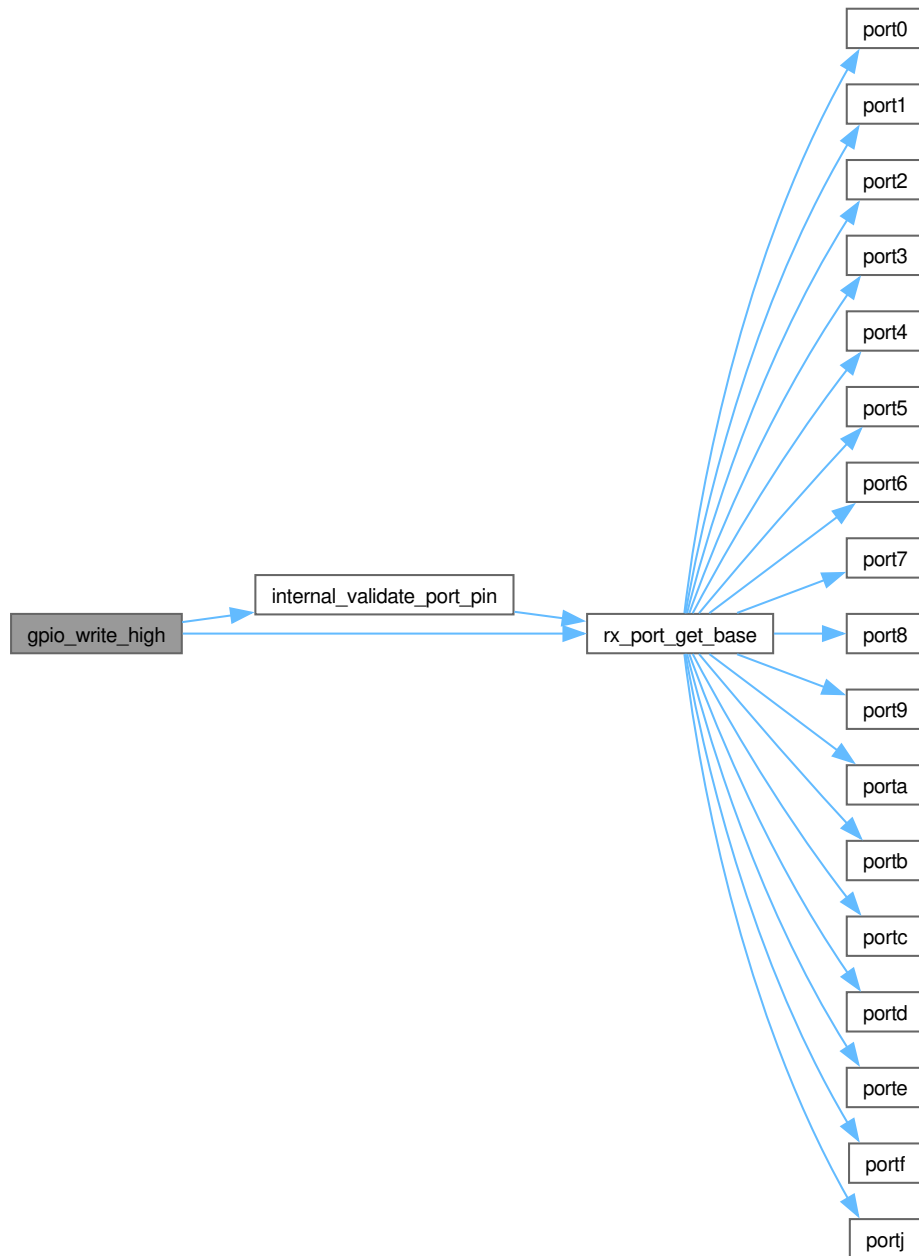
Version 1.0.0

Definition at line 565 of file [gpio.c](#).

```
00566 {
00567     /* Extract port and pin number from rx_port_pin_t */
00568     const uint8_t port    = rx_port_from_pin(pin);
00569     const uint8_t pin_num = rx_pin_from_pin(pin);
00570
00571     /* Validate parameters */
00572     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00573     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00574
00575     /* Get port base address */
00576     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00577
00578     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00579     port_base->podr |= pin_mask;
00580
00581     return k_rx_ok;
00582 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_set](#), [rx_port_regs_t::podr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.19 gpio_write_low()

```
rx_err_t gpio_write_low (
    const rx_port_pin_t pin) [nodiscard]
```

Set GPIO output pin to logic low (0V).

Set GPIO output pin to logic low (GND).

Sets a GPIO output pin to logic low level. On the RX72N, this means the pin outputs GND (0V). The pin must be configured as an output before calling this function.

8.103.3.20 Register Operation

```
PODR &= ~(1 « pin) // Clear bit: Output low
```

8.103.3.21 Open-Drain Mode

If the pin is configured for open-drain mode (ODR register), setting low actively pulls the pin to GND. Setting high makes the pin float (high-impedance), requiring an external pull-up.

Precondition

- Pin must be configured as output via [gpio_set_output\(\)](#)
- Port and pin numbers must be valid

Postcondition

- PODR bit for pin is cleared (output low)
- Pin outputs GND level (0V)

Note

- NOT thread-safe: Direct register write without synchronization
- Performance: ~100 ns @ 240 MHz
- ISR-safe: Can be called from interrupt handlers

Warning

- No check that pin is configured as output
- Calling on input pin has no visible effect

Example:

```
// Turn off LED (active high)
rx_err_t err = gpio_write_low(RX_PORT_PIN(0, 3));
if (err != k_rx_ok) {
    // Handle error
}
```

See also

- [gpio_write_high\(\)](#) Set output high
- [gpio_toggle\(\)](#) Toggle output state
- [gpio_set_output\(\)](#) Configure pin as output first

Since

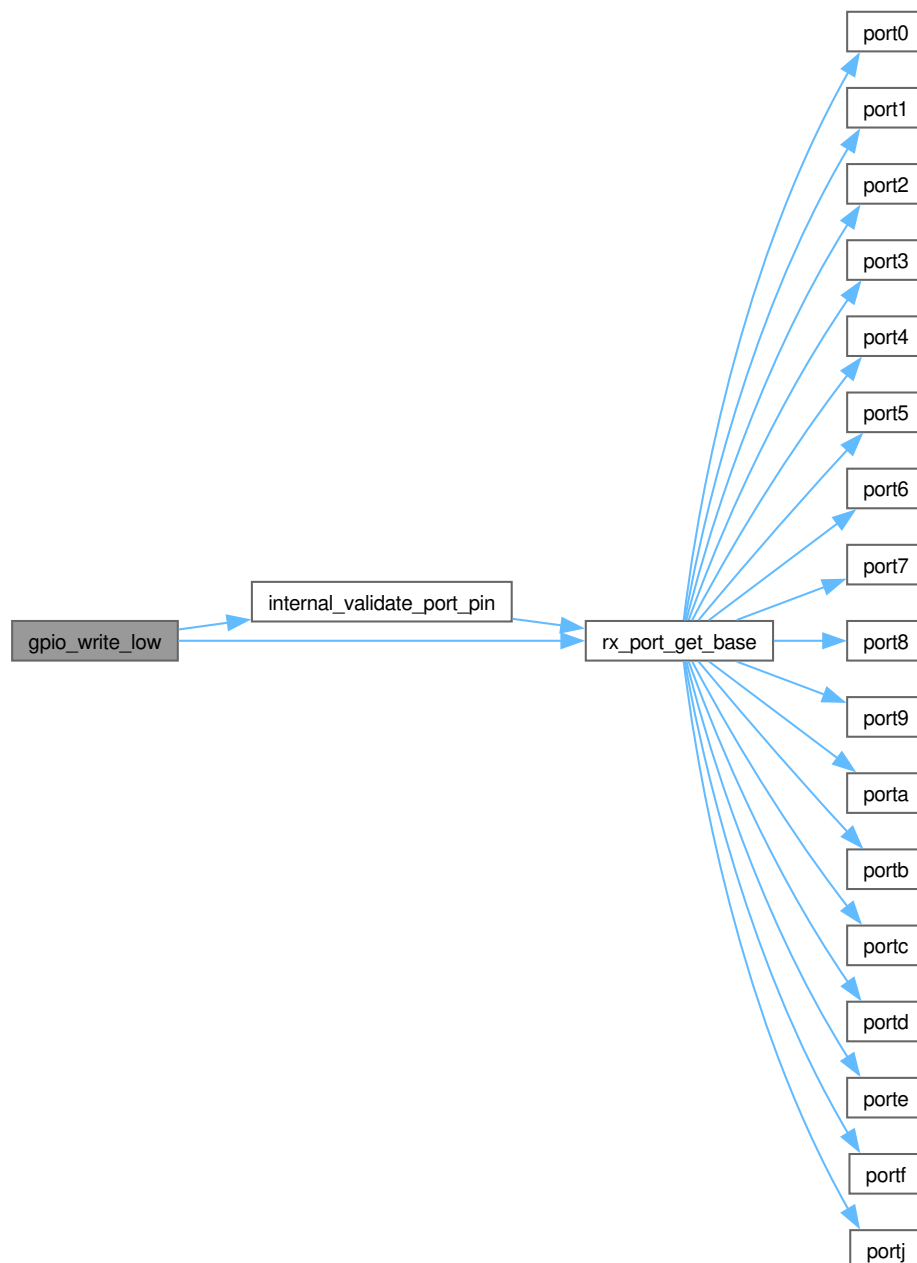
Version 1.0.0

Definition at line 634 of file [gpio.c](#).

```
00635 {
00636     /* Extract port and pin number from rx_port_pin_t */
00637     const uint8_t port = rx_port_from_pin(pin);
00638     const uint8_t pin_num = rx_pin_from_pin(pin);
00639
00640     /* Validate parameters */
00641     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00642     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00643
00644     /* Get port base address */
00645     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00646
00647     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00648     port_base->podr &= (uint8_t)~pin_mask;
00649
00650     return k_rx_ok;
00651 }
```

References [internal_validate_port_pin\(\)](#), [k_gpio_bit_set](#), [rx_port_regs_t::podr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.103.3.22 internal_validate_port_pin()

```

rx_err_t internal_validate_port_pin (
    const uint8_t port,
    const uint8_t pin) [static]
  
```

Validate port and pin numbers against RX72N hardware limits.

Performs validation of GPIO port and pin numbers before any register access. This function is called by all public GPIO functions to ensure safe operation and prevent out-of-bounds hardware register access.

8.103.3.23 Validation Algorithm

1. Check port validity via [rx_port_get_base\(\)](#) (returns NULL for invalid ports)
2. Check pin number against k_rx_pin_max (maximum valid pin number per port)

8.103.3.24 Port Validation

Uses `rx_port_get_base()` which returns NULL for any port number that does not correspond to a physical port on the RX72N. Valid port numbers are 0-9 (decimal) and 0xA-0x10 (hexadecimal, corresponding to ports A through G). Passing an out-of-range port is safely caught here before any register dereference occurs.

8.103.3.25 Pin Validation

Each RX72N GPIO port exposes up to 8 pins numbered 0 through 7. The `k_rx_pin_max` constant defines the inclusive upper bound (7). A lower-bound check against zero is omitted because `pin` is a `uint8_t`, making `pin < 0` structurally impossible; omitting it avoids a -Wtype-limits compiler warning.

8.103.3.26 Why Both Checks Are Required

An invalid port with a valid pin, or a valid port with an invalid pin, must both be rejected. Either condition alone would result in incorrect hardware register access, undefined behavior, or corrupted peripheral state.

Precondition

`port` must be within the RX72N valid port range (0-9, 0xA-0x10); values outside this range are rejected before any register access

`pin` must be within the valid pin range for the specified port (0 to `k_rx_pin_max`); values above `k_rx_pin_max` are rejected with an error

Postcondition

Returns `k_rx_ok` only when both port and pin are simultaneously valid; no partial-success return is possible

No hardware state is modified; this function is purely read-only with respect to peripheral registers and module-level variables

Note

Thread-safe: No shared mutable state accessed; safe to call concurrently

Performance: O(1), approximately 50 cycles at 240 MHz

An error log message is emitted for each failed validation to aid debugging

Example:

```
// Typical usage inside a GPIO public function
const uint8_t port = rx_port_from_pin(pin);
const uint8_t pin_num = rx_pin_from_pin(pin);
rx_err_t err = internal_validate_port_pin(port, pin_num);
RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
```

See also

`rx_port_get_base()` Port lookup function used internally for port validation

`k_rx_pin_max` Maximum valid pin index (inclusive upper bound)

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions, 2 postconditions

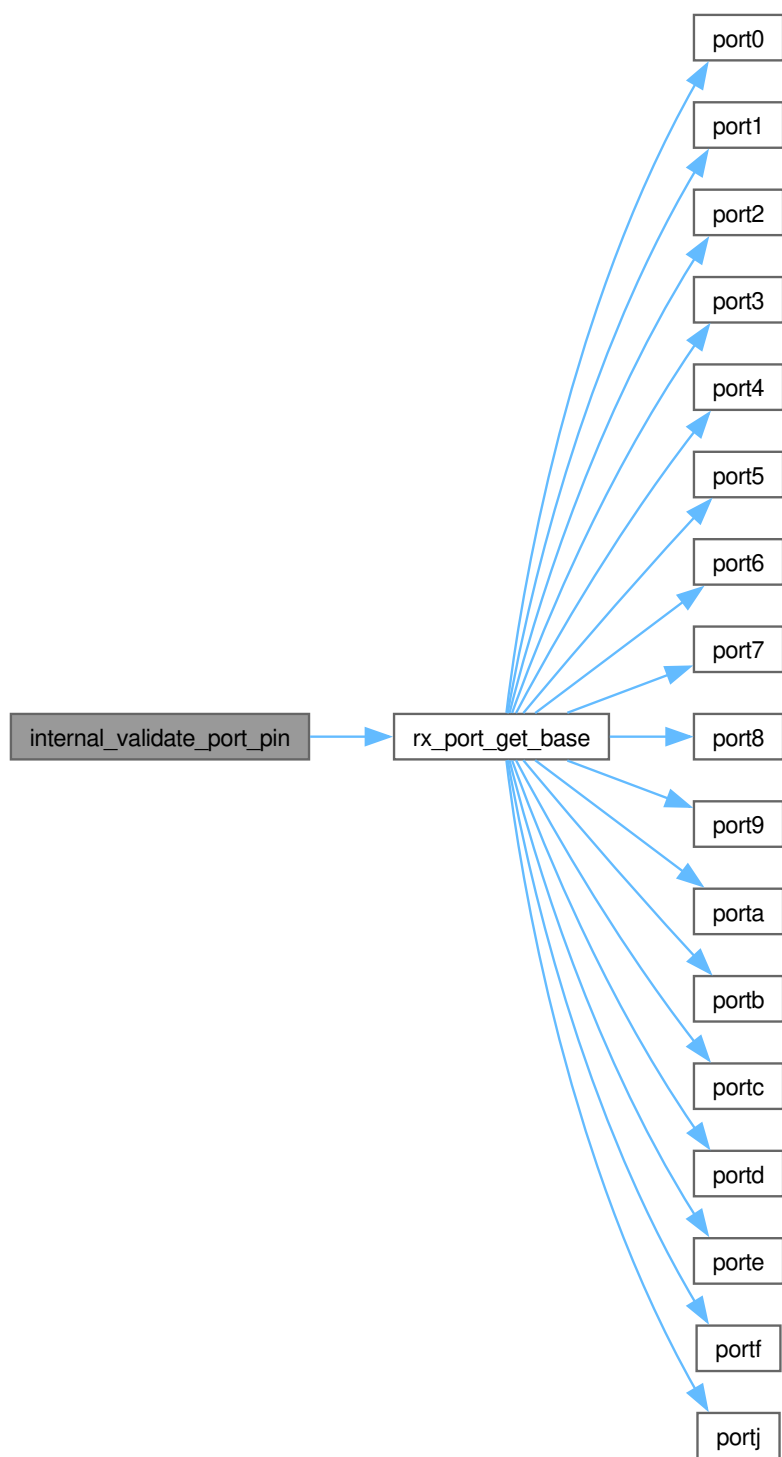
Definition at line 280 of file [gpio.c](#).

```
00281 {
00282     /* Validate port */
00283     const volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00284     if (port_base == nullptr) {
00285         rx_log_error(s_tag, "Invalid port number");
00286         return k_rx_err_gpio_invalid_port;
00287     }
00288
00289     /* Validate pin (lower bound check omitted - pin is uint8_t, k_rx_pin_min == 0,
00290      * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
00291     if (pin > k_rx_pin_max) {
00292         rx_log_error(s_tag, "Invalid pin number");
00293         return k_rx_err_gpio_invalid_pin;
00294     }
00295
00296     return k_rx_ok;
00297 }
```

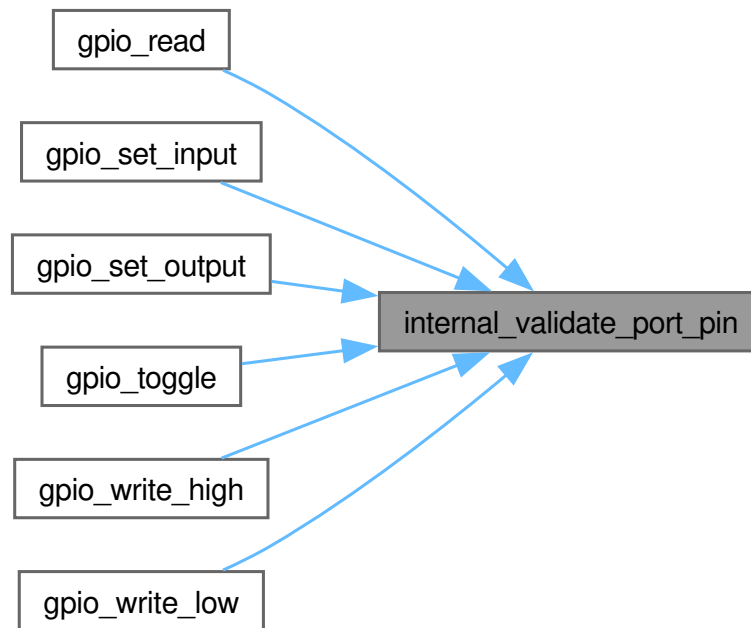
References [rx_port_get_base\(\)](#), and [s_tag](#).

Referenced by [gpio_read\(\)](#), [gpio_set_input\(\)](#), [gpio_set_output\(\)](#), [gpio_toggle\(\)](#), [gpio_write_high\(\)](#), and [gpio_write_low\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.103.4 Variable Documentation

8.103.4.1 s_tag

```
const char* const s_tag = "GPIO" [static]
```

Log tag for this module.

Definition at line 207 of file [gpio.c](#).

8.104 gpio.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file gpio.c
00003  * @brief GPIO Driver for RX72N - General Purpose Input/Output Control
00004  *
00005  * @details
00006  * ## Overview
00007  *
00008  * This file provides the hardware abstraction layer (HAL) for GPIO operations on
00009  * the Renesas RX72N microcontroller. It handles pin direction configuration,
00010  * digital output control, and digital input reading with integrated error handling,
00011  * logging, and pin validation through the pin validator interface.
00012  *
00013  * ## System Architecture
00014  *
00015  * @dot
00016  * digraph gpio_architecture {
00017  *   rankdir=TB;
00018  *   node [shape=box, style=rounded];
00019  *
00020  *   subgraph cluster_app {
00021  *     label="Application Layer";
00022  *     style=dashed;
00023  *     led [label="LED Control"];
00024  *     button [label="Button Input"];
00025  *     status [label="Status Indicators"];
00026  *   }
00027  *

```

```

00028 * subgraph cluster_hal {
00029 *   label="HAL Layer (THIS FILE)";
00030 *   style=filled;
00031 *   fillcolor=lightblue;
00032 *   gpio_api [label="GPIO
API\ngpio_set_output()\ngpio_set_input()\ngpio_write_high()\ngpio_write_low()\ngpio_toggle()\ngpio_read()"];
00033 * }
00034 *
00035 * subgraph cluster_infra {
00036 *   label="Infrastructure";
00037 *   style=dashed;
00038 *   pin_validator [label="Pin Validator\n(Conflict Detection)"];
00039 *   port_utils [label="Port Utilities\n(Address Lookup)"];
00040 * }
00041 *
00042 * subgraph cluster_hw {
00043 *   label="Hardware";
00044 *   style=dashed;
00045 *   port_regs [label="PORT Registers\nPDR, PODR, PIDR, PMR"];
00046 * }
00047 *
00048 * led -> gpio_api;
00049 * button -> gpio_api;
00050 * status -> gpio_api;
00051 * gpio_api -> pin_validator [label="reserve_pin"];
00052 * gpio_api -> port_utils [label="get_base"];
00053 * gpio_api -> port_regs [label="register access"];
00054 * }
00055 * @enddot
00056 *
00057 * ## RX72N GPIO Register Architecture
00058 *
00059 * Each GPIO port on the RX72N has multiple control registers:
00060 *
00061 * | Register | Name | Purpose |
00062 * |-----|-----|-----|
00063 * | PDR | Port Direction Register | 0=input, 1=output |
00064 * | PODR | Port Output Data Register | Output level (0=low, 1=high) |
00065 * | PIDR | Port Input Data Register | Read current pin level |
00066 * | PMR | Port Mode Register | 0=GPIO, 1=peripheral function |
00067 * | PCR | Pull-up Control Register | Enable internal pull-up |
00068 * | ODR | Open-Drain Control Register | Enable open-drain mode |
00069 *
00070 * This driver primarily uses PDR, PODR, PIDR, and PMR.
00071 *
00072 * ## Pin Configuration Sequence
00073 *
00074 * @msc
00075 * App, GPIO, PinValidator, PortRegs;
00076 *
00077 * App => GPIO [label="gpio_set_output(pin)"];
00078 * GPIO => PinValidator [label="reserve_pin(port, pin, 'GPIO_OUT')"];
00079 * PinValidator => GPIO [label="k_rx_ok"];
00080 * GPIO => PortRegs [label="PMR &= ~(1«pin) // GPIO mode"];
00081 * GPIO => PortRegs [label="PDR |= (1«pin) // Output"];
00082 * GPIO => App [label="k_rx_ok"];
00083 *
00084 * App => GPIO [label="gpio_write_high(pin)"];
00085 * GPIO => PortRegs [label="PODR |= (1«pin)"];
00086 * GPIO => App [label="k_rx_ok"];
00087 * @endmsc
00088 *
00089 * ## Thread Safety
00090 *
00091 * | Function | Thread-Safe | Notes |
00092 * |-----|-----|-----|
00093 * | gpio_set_output() | Partial | Pin reservation is mutex-protected |
00094 * | gpio_set_input() | Partial | Pin reservation is mutex-protected |
00095 * | gpio_write_high() | No | Direct register write |
00096 * | gpio_write_low() | No | Direct register write |
00097 * | gpio_toggle() | No | Read-modify-write, not atomic |
00098 * | gpio_read() | Yes | Read-only operation |
00099 *
00100 * For thread-safe output operations, caller must provide synchronization.
00101 *
00102 * ## Performance Characteristics
00103 *
00104 * | Operation | Cycles | Time @ 240 MHz | Notes |
00105 * |-----|-----|-----|-----|
00106 * | gpio_set_output | ~200-500 | ~1-2 us | Includes pin reservation |
00107 * | gpio_set_input | ~200-500 | ~1-2 us | Includes pin reservation |
00108 * | gpio_write_high | ~20-30 | ~100 ns | Direct register |
00109 * | gpio_write_low | ~20-30 | ~100 ns | Direct register |
00110 * | gpio_toggle | ~30-40 | ~150 ns | Read-modify-write |
00111 * | gpio_read | ~20-30 | ~100 ns | Direct register read |
00112 *
00113 * ## Memory Usage

```

```

00114 *
00115 * | Section | Size | Contents |
00116 * |-----|-----|-----|
00117 * | .text | ~600 bytes | Function code |
00118 * | .rodata | ~100 bytes | Log strings |
00119 * | .bss | 0 bytes | No module-level state |
00120 *
00121 * ## NASA Power of 10 Compliance
00122 *
00123 * | Rule | Status | Implementation |
00124 * |-----|-----|-----|
00125 * | 1. Simplify control flow | [OK] | No goto, setjmp, recursion |
00126 * | 2. Fixed loop bounds | [OK] | No loops in this module |
00127 * | 3. No dynamic memory | [OK] | All static, no malloc |
00128 * | 4. Functions < 60 lines | [OK] | All functions under 30 lines |
00129 * | 5. Use assertions | [OK] | RX_CHECK_NULL_PTR, validation |
00130 * | 6. Data at smallest scope | [OK] | Local variables only |
00131 * | 7. Check return values | [OK] | RX_RETURN_ON_ERROR macro |
00132 * | 8. Limit preprocessor | [OK] | Only includes |
00133 * | 9. Restrict pointers | [OK] | Only hardware register pointers |
00134 * | 10. Compile warnings | [OK] | -Wall -Wextra -Werror clean |
00135 *
00136 * ## SOLID Principles
00137 *
00138 * | Principle | Implementation |
00139 * |-----|-----|
00140 * | **Single Responsibility** | GPIO operations only (no PWM, no ADC) |
00141 * | **Open/Closed** | Extended via rx_port_pin_t encoding |
00142 * | **Liskov Substitution** | N/A (not interface-based) |
00143 * | **Interface Segregation** | Separate functions for each operation |
00144 * | **Dependency Inversion** | Uses pin_interface_t for validation |
00145 *
00146 * @see rx_port_constants.h Port and pin definitions
00147 * @see rx_port_utils.h Port address lookup utilities
00148 * @see rx72n_regs.h Register definitions
00149 * @see rx_pin_validator.h Pin conflict detection
00150 *
00151 * @author Locked, Inc.
00152 * @date 2026-01-01
00153 * @copyright Copyright (c) 2026 Locked Inc.
00154 * SPDX-License-Identifier: MIT
00155 * @since Version 1.0.0
00156 */
00157
00158 #ifdef __RX__
00159
00160 #include <stddef.h>
00161
00162 #include "hardware.h"
00163 #include "rx72n_regs.h"
00164 #include "rx_port_constants.h"
00165 #include "rx_port_utils.h"
00166
00167 /*
=====
00168 * Constants
00169 *
=====
00170 */
00171
00172 /**
00173 * @enum gpio_bit_constants_t
00174 * @brief GPIO register bit manipulation constants
00175 *
00176 * @details
00177 * Defines constants used for bit manipulation in GPIO register operations.
00178 * Using named constants instead of magic numbers improves code readability
00179 * and makes the intent of bit operations clear.
00180 *
00181 * ## Usage Examples
00182 *
00183 * @code
00184 * // Set bit 5 in PDR register (configure as output)
00185 * port->pdr |= (k_gpio_bit_set « 5);
00186 *
00187 * // Clear bit 3 in PMR register (configure as GPIO, not peripheral)
00188 * port->pmr &= ~(k_gpio_bit_set « 3);
00189 *
00190 * // Check if bit 2 is set in PIDR register
00191 * if ((port->pidr & (k_gpio_bit_set « 2)) != k_gpio_bit_clear) {
00192 * // Pin is high
00193 * }
00194 * @endcode
00195 *
00196 * @see gpio_set_output() Uses these for PDR/PMR manipulation
00197 * @see gpio_read() Uses these for PIDR comparison
00198 *

```

```

00199 * @since Version 1.0.0
00200 */
00201 typedef enum : uint8_t {
00202     k_gpio_bit_set    = 1, /**< Value for setting a single bit (1 « n pattern) */
00203     k_gpio_bit_clear = 0, /**< Value for comparison when checking if bit is cleared */
00204 } gpio_bit_constants_t;
00205
00206 /** @brief Log tag for this module */
00207 static const char* const s_tag = "GPIO";
00208
00209 /*
=====
00210 * Internal Helper Functions
00211 *
=====
00212 */
00213
00214 /**
00215  * @brief Validate port and pin numbers against RX72N hardware limits
00216  *
00217  * @details
00218  * Performs validation of GPIO port and pin numbers before any register access.
00219  * This function is called by all public GPIO functions to ensure safe operation
00220  * and prevent out-of-bounds hardware register access.
00221  *
00222  * ## Validation Algorithm
00223  *
00224  * 1. Check port validity via rx_port_get_base() (returns NULL for invalid ports)
00225  * 2. Check pin number against k_rx_pin_max (maximum valid pin number per port)
00226  *
00227  * ## Port Validation
00228  *
00229  * Uses rx_port_get_base() which returns NULL for any port number that does not
00230  * correspond to a physical port on the RX72N. Valid port numbers are 0-9
00231  * (decimal) and 0xA-0x10 (hexadecimal, corresponding to ports A through G).
00232  * Passing an out-of-range port is safely caught here before any register
00233  * dereference occurs.
00234  *
00235  * ## Pin Validation
00236  *
00237  * Each RX72N GPIO port exposes up to 8 pins numbered 0 through 7. The
00238  * k_rx_pin_max constant defines the inclusive upper bound (7). A lower-bound
00239  * check against zero is omitted because pin is a uint8_t, making pin < 0
00240  * structurally impossible; omitting it avoids a -Wtype-limits compiler warning.
00241  *
00242  * ## Why Both Checks Are Required
00243  *
00244  * An invalid port with a valid pin, or a valid port with an invalid pin, must
00245  * both be rejected. Either condition alone would result in incorrect hardware
00246  * register access, undefined behavior, or corrupted peripheral state.
00247  *
00248  *
00249  *
00250  * @pre port must be within the RX72N valid port range (0-9, 0xA-0x10);
00251  *       values outside this range are rejected before any register access
00252  * @pre pin must be within the valid pin range for the specified port (0 to
00253  *       k_rx_pin_max); values above k_rx_pin_max are rejected with an error
00254  * @post Returns k_rx_ok only when both port and pin are simultaneously valid;
00255  *       no partial-success return is possible
00256  * @post No hardware state is modified; this function is purely read-only with
00257  *       respect to peripheral registers and module-level variables
00258  *
00259  * @note Thread-safe: No shared mutable state accessed; safe to call concurrently
00260  * @note Performance: O(1), approximately 50 cycles at 240 MHz
00261  * @note An error log message is emitted for each failed validation to aid debugging
00262  *
00263  * @par Example:
00264  * @code
00265  * // Typical usage inside a GPIO public function
00266  * const uint8_t port  = rx_port_from_pin(pin);
00267  * const uint8_t pin_num = rx_pin_from_pin(pin);
00268  * rx_err_t err = internal_validate_port_pin(port, pin_num);
00269  * RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00270  * @endcode
00271  *
00272  * @see rx_port_get_base() Port lookup function used internally for port validation
00273  * @see k_rx_pin_max Maximum valid pin index (inclusive upper bound)
00274  *
00275  * @since Version 1.0.0
00276  *
00277  * @par NASA Power of 10 Compliance:
00278  * - Rule 5: [OK] 2 preconditions, 2 postconditions
00279  */
00280 static rx_err_t internal_validate_port_pin(const uint8_t port, const uint8_t pin)
00281 {
00282     /* Validate port */
00283     const volatile rx_port_regs_t* port_base = rx_port_get_base(port);

```

```

00284 if (port_base == nullptr) {
00285     rx_log_error(s_tag, "Invalid port number");
00286     return k_rx_err_gpio_invalid_port;
00287 }
00288
00289 /* Validate pin (lower bound check omitted - pin is uint8_t, k_rx_pin_min == 0,
00290 * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
00291 if (pin > k_rx_pin_max) {
00292     rx_log_error(s_tag, "Invalid pin number");
00293     return k_rx_err_gpio_invalid_pin;
00294 }
00295
00296 return k_rx_ok;
00297 }
00298
00299 /*
=====
00300 * Public API Implementation
00301 *
=====
00302 */
00303
00304 /**
00305 * @brief Configure GPIO pin as digital output
00306 *
00307 * @details
00308 * Configures a GPIO pin for digital output mode. This function performs:
00309 * 1. Parameter validation (port and pin number)
00310 * 2. Pin reservation via the global pin validator (conflict detection)
00311 * 3. PMR register clear (select GPIO mode instead of peripheral function)
00312 * 4. PDR register set (configure as output direction)
00313 *
00314 * ## Register Operations
00315 *
00316 * ```
00317 * PMR &= ~(1 « pin) // Clear bit: Select GPIO mode (not peripheral)
00318 * PDR |= (1 « pin) // Set bit: Configure as output
00319 * ```
00320 *
00321 * ## Pin Reservation
00322 *
00323 * The function attempts to reserve the pin through rx_infrastructure_get_pin_interface().
00324 * If the pin is already reserved (k_rx_err_gpio_conflict), the function continues
00325 * without error - this allows reconfiguration of already-owned pins.
00326 *
00327 * ## Initial Output State
00328 *
00329 * This function does NOT set the initial output level. After calling gpio_set_output(),
00330 * call gpio_write_high() or gpio_write_low() to set the desired level.
00331 *
00332 *
00333 *
00334 * @pre Pin must be a valid GPIO-capable pin on RX72N
00335 * @pre System infrastructure should be initialized (for pin validation)
00336 *
00337 * @post Pin is configured as GPIO output
00338 * @post PMR bit for pin is cleared (GPIO mode)
00339 * @post PDR bit for pin is set (output direction)
00340 * @post Pin is reserved in validator (if available)
00341 *
00342 * @note Thread-safe: Pin reservation uses mutex, register writes are not atomic
00343 * @note Performance: ~1-2 us @ 240 MHz (includes reservation)
00344 * @note Idempotent: Safe to call multiple times on same pin
00345 *
00346 * @warning Output level is undefined after this call - set with gpio_write_*
00347 * @warning Do not call from ISR (pin reservation may block)
00348 *
00349 * @par Example:
00350 * @code
00351 * // Configure LED pin as output
00352 * rx_err_t err = gpio_set_output(RX_PORT_PIN(0, 3)); // Port 0, Pin 3
00353 * if (err != k_rx_ok) {
00354 *     rx_log_error("LED", "Failed to configure LED pin");
00355 *     return err;
00356 * }
00357 *
00358 * // Set initial state to low (LED off)
00359 * gpio_write_low(RX_PORT_PIN(0, 3));
00360 *
00361 * // Toggle LED
00362 * gpio_toggle(RX_PORT_PIN(0, 3));
00363 * @endcode
00364 *
00365 * @see gpio_set_input() Configure pin as input
00366 * @see gpio_write_high() Set output high after configuration
00367 * @see gpio_write_low() Set output low after configuration
00368 * @see RX_PORT_PIN() Macro to create rx_port_pin_t

```

```

00369 *
00370 * @since Version 1.0.0
00371 *
00372 * @par NASA Power of 10 Compliance:
00373 * - Rule 5: [OK] 2 preconditions, 4 postconditions
00374 */
00375 rx_err_t gpio_set_output(const rx_port_pin_t pin)
00376 {
00377     /* Extract port and pin number from rx_port_pin_t */
00378     const uint8_t port = rx_port_from_pin(pin);
00379     const uint8_t pin_num = rx_pin_from_pin(pin);
00380
00381     /* Validate parameters */
00382     rx_err_t err = internal_validate_port_pin(port, pin_num);
00383     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00384
00385     /* Reserve pin through global pin validator */
00386     const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00387     if (pin_iface != nullptr) {
00388         err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_OUT");
00389         if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00390             /* Allow conflict (pin already reserved), but fail on other errors */
00391             RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00392         }
00393     }
00394
00395     /* Get port base address */
00396     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00397
00398     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set << pin_num);
00399
00400     /* Set as GPIO mode (not peripheral) */
00401     port_base->pmr &= (uint8_t)~pin_mask;
00402
00403     /* Set as output */
00404     port_base->pdr |= pin_mask;
00405
00406     rx_log_debug(s_tag, "Pin configured as output");
00407
00408     return k_rx_ok;
00409 }
00410
00411 /**
00412 * @brief Configure GPIO pin as digital input
00413 *
00414 * @details
00415 * Configures a GPIO pin for digital input mode. This function performs:
00416 * 1. Parameter validation (port and pin number)
00417 * 2. Pin reservation via the global pin validator (conflict detection)
00418 * 3. PMR register clear (select GPIO mode instead of peripheral function)
00419 * 4. PDR register clear (configure as input direction)
00420 *
00421 * ## Register Operations
00422 *
00423 * ```
00424 * PMR &= ~(1 << pin) // Clear bit: Select GPIO mode (not peripheral)
00425 * PDR &= ~(1 << pin) // Clear bit: Configure as input
00426 * ```
00427 *
00428 * ## Pull-up Configuration
00429 *
00430 * This function does NOT configure internal pull-up resistors. To enable
00431 * pull-ups, additional configuration of the PCR register is required.
00432 *
00433 * ## Reading Input State
00434 *
00435 * After configuration, use gpio_read() to read the current pin state.
00436 * The PIDR register is read for input state, not PODR.
00437 *
00438 *
00439 *
00440 * @pre Pin must be a valid GPIO-capable pin on RX72N
00441 * @pre System infrastructure should be initialized (for pin validation)
00442 *
00443 * @post Pin is configured as GPIO input
00444 * @post PMR bit for pin is cleared (GPIO mode)
00445 * @post PDR bit for pin is cleared (input direction)
00446 * @post Pin is reserved in validator (if available)
00447 *
00448 * @note Thread-safe: Pin reservation uses mutex
00449 * @note Performance: ~1-2 us @ 240 MHz (includes reservation)
00450 * @note Pull-ups: Not enabled by this function (use PCR register)
00451 *
00452 * @warning Floating inputs may read unpredictable values - use pull-up/pull-down
00453 * @warning Do not call from ISR (pin reservation may block)
00454 *
00455 * @par Example:

```

```

00456 * @code
00457 * // Configure button pin as input
00458 * rx_err_t err = gpio_set_input(RX_PORT_PIN(2, 0)); // Port 2, Pin 0
00459 * if (err != k_rx_ok) {
00460 *     rx_log_error("BTN", "Failed to configure button pin");
00461 *     return err;
00462 * }
00463 *
00464 * // Read button state
00465 * bool pressed;
00466 * err = gpio_read(RX_PORT_PIN(2, 0), &pressed);
00467 * if (err == k_rx_ok && pressed) {
00468 *     rx_log_info("BTN", "Button pressed");
00469 * }
00470 * @endcode
00471 *
00472 * @see gpio_set_output() Configure pin as output
00473 * @see gpio_read() Read input pin state
00474 * @see RX_PORT_PIN() Macro to create rx_port_pin_t
00475 *
00476 * @since Version 1.0.0
00477 *
00478 * @par NASA Power of 10 Compliance:
00479 * - Rule 5: [OK] 2 preconditions, 4 postconditions
00480 */
00481 rx_err_t gpio_set_input(const rx_port_pin_t pin)
00482 {
00483     /* Extract port and pin number from rx_port_pin_t */
00484     const uint8_t port = rx_port_from_pin(pin);
00485     const uint8_t pin_num = rx_pin_from_pin(pin);
00486
00487     /* Validate parameters */
00488     rx_err_t err = internal_validate_port_pin(port, pin_num);
00489     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00490
00491     /* Reserve pin through global pin validator */
00492     const rx_pin_interface_t* const pin_iface = rx_infrastructure_get_pin_interface();
00493     if (pin_iface != nullptr) {
00494         err = pin_iface->reserve_pin(pin_iface->ctx, port, pin_num, "GPIO_IN");
00495         if (err != k_rx_ok && err != k_rx_err_gpio_conflict) {
00496             /* Allow conflict (pin already reserved), but fail on other errors */
00497             RX_RETURN_ON_ERROR(err, s_tag, "Pin reservation failed");
00498         }
00499     }
00500
00501     /* Get port base address */
00502     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00503
00504     /* Set as GPIO mode */
00505     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00506     port_base->pmr &= (uint8_t)~pin_mask;
00507
00508     /* Set as input */
00509     port_base->pdr &= (uint8_t)~pin_mask;
00510
00511     rx_log_debug(s_tag, "Pin configured as input");
00512
00513     return k_rx_ok;
00514 }
00515
00516 /**
00517 * @brief Set GPIO output pin to logic high (3.3V)
00518 *
00519 * @details
00520 * Sets a GPIO output pin to logic high level. On the RX72N, this means
00521 * the pin outputs VCC (typically 3.3V). The pin must be configured as
00522 * an output before calling this function.
00523 *
00524 * ## Register Operation
00525 *
00526 * ```
00527 * PODR |= (1 « pin) // Set bit: Output high
00528 * ```
00529 *
00530 * ## Timing
00531 *
00532 * The output level changes on the next peripheral clock cycle after the
00533 * register write completes. GPIO toggle rate is limited by bus speed.
00534 *
00535 *
00536 *
00537 * @pre Pin must be configured as output via gpio_set_output()
00538 * @pre Port and pin numbers must be valid
00539 *
00540 * @post PODR bit for pin is set (output high)
00541 * @post Pin outputs VCC level (typically 3.3V)
00542 *

```

```

00543 * @note NOT thread-safe: Direct register write without synchronization
00544 * @note Performance: ~100 ns @ 240 MHz
00545 * @note ISR-safe: Can be called from interrupt handlers
00546 *
00547 * @warning No check that pin is configured as output
00548 * @warning Calling on input pin has no visible effect
00549 *
00550 * @par Example:
00551 * @code
00552 * // Turn on LED (active high)
00553 * rx_err_t err = gpio_write_high(RX_PORT_PIN(0, 3));
00554 * if (err != k_rx_ok) {
00555 * // Handle error
00556 * }
00557 * @endcode
00558 *
00559 * @see gpio_write_low() Set output low
00560 * @see gpio_toggle() Toggle output state
00561 * @see gpio_set_output() Configure pin as output first
00562 *
00563 * @since Version 1.0.0
00564 */
00565 rx_err_t gpio_write_high(const rx_port_pin_t pin)
00566 {
00567 /* Extract port and pin number from rx_port_pin_t */
00568 const uint8_t port = rx_port_from_pin(pin);
00569 const uint8_t pin_num = rx_pin_from_pin(pin);
00570
00571 /* Validate parameters */
00572 const rx_err_t err = internal_validate_port_pin(port, pin_num);
00573 RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00574
00575 /* Get port base address */
00576 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00577
00578 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00579 port_base->podr |= pin_mask;
00580
00581 return k_rx_ok;
00582 }
00583
00584 /**
00585 * @brief Set GPIO output pin to logic low (0V)
00586 *
00587 * @details
00588 * Sets a GPIO output pin to logic low level. On the RX72N, this means
00589 * the pin outputs GND (0V). The pin must be configured as an output
00590 * before calling this function.
00591 *
00592 * ## Register Operation
00593 *
00594 * ```
00595 * PODR &= ~(1 « pin) // Clear bit: Output low
00596 * ```
00597 *
00598 * ## Open-Drain Mode
00599 *
00600 * If the pin is configured for open-drain mode (ODR register), setting
00601 * low actively pulls the pin to GND. Setting high makes the pin float
00602 * (high-impedance), requiring an external pull-up.
00603 *
00604 *
00605 *
00606 * @pre Pin must be configured as output via gpio_set_output()
00607 * @pre Port and pin numbers must be valid
00608 *
00609 * @post PODR bit for pin is cleared (output low)
00610 * @post Pin outputs GND level (0V)
00611 *
00612 * @note NOT thread-safe: Direct register write without synchronization
00613 * @note Performance: ~100 ns @ 240 MHz
00614 * @note ISR-safe: Can be called from interrupt handlers
00615 *
00616 * @warning No check that pin is configured as output
00617 * @warning Calling on input pin has no visible effect
00618 *
00619 * @par Example:
00620 * @code
00621 * // Turn off LED (active high)
00622 * rx_err_t err = gpio_write_low(RX_PORT_PIN(0, 3));
00623 * if (err != k_rx_ok) {
00624 * // Handle error
00625 * }
00626 * @endcode
00627 *
00628 * @see gpio_write_high() Set output high
00629 * @see gpio_toggle() Toggle output state

```



```

00630 * @see gpio_set_output() Configure pin as output first
00631 *
00632 * @since Version 1.0.0
00633 */
00634 rx_err_t gpio_write_low(const rx_port_pin_t pin)
00635 {
00636     /* Extract port and pin number from rx_port_pin_t */
00637     const uint8_t port = rx_port_from_pin(pin);
00638     const uint8_t pin_num = rx_pin_from_pin(pin);
00639
00640     /* Validate parameters */
00641     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00642     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00643
00644     /* Get port base address */
00645     volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00646
00647     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00648     port_base->podr &= (uint8_t)~pin_mask;
00649
00650     return k_rx_ok;
00651 }
00652
00653 /**
00654 * @brief Toggle GPIO output pin state (high<->low)
00655 *
00656 * @details
00657 * Inverts the current output state of a GPIO pin. If the pin is currently
00658 * high, it becomes low. If currently low, it becomes high. Uses XOR
00659 * operation for efficient toggling.
00660 *
00661 * ## Register Operation
00662 *
00663 * ```
00664 * PODR ^= (1 « pin) // XOR bit: Toggle output
00665 * ```
00666 *
00667 * ## Atomicity Warning
00668 *
00669 * This operation is a read-modify-write sequence and is NOT atomic.
00670 * If an interrupt modifies the same port register between the read
00671 * and write, the interrupt's change may be lost.
00672 *
00673 * For thread-safe toggling, use explicit write_high/write_low with
00674 * external synchronization, or disable interrupts around the toggle.
00675 *
00676 *
00677 *
00678 * @pre Pin must be configured as output via gpio_set_output()
00679 * @pre Port and pin numbers must be valid
00680 *
00681 * @post PODR bit for pin is inverted
00682 * @post Pin output level is opposite of previous state
00683 *
00684 * @note NOT thread-safe: Read-modify-write operation
00685 * @note NOT atomic: Can race with ISRs on same port
00686 * @note Performance: ~150 ns @ 240 MHz
00687 *
00688 * @warning Race condition possible with ISRs modifying same port
00689 * @warning Use gpio_write_high/low with sync for critical applications
00690 *
00691 * @par Example:
00692 * @code
00693 * // Blink LED in loop
00694 * while (running) {
00695 *     gpio_toggle(RX_PORT_PIN(0, 3)); // Toggle LED
00696 *     tx_thread_sleep(50);           // 50ms delay (20 Hz blink)
00697 * }
00698 * @endcode
00699 *
00700 * @see gpio_write_high() Set output high explicitly
00701 * @see gpio_write_low() Set output low explicitly
00702 * @see gpio_read() Read current output state
00703 *
00704 * @since Version 1.0.0
00705 */
00706 rx_err_t gpio_toggle(const rx_port_pin_t pin)
00707 {
00708     /* Extract port and pin number from rx_port_pin_t */
00709     const uint8_t port = rx_port_from_pin(pin);
00710     const uint8_t pin_num = rx_pin_from_pin(pin);
00711
00712     /* Validate parameters */
00713     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00714     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00715
00716     /* Get port base address */

```

```

00717 volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00718
00719 const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00720 port_base->podr ^= pin_mask;
00721
00722 return k_rx_ok;
00723 }
00724
00725 /**
00726  * @brief Read GPIO input pin state
00727  *
00728  * @details
00729  * Reads the current logic level of a GPIO pin. Uses the PIDR (Port Input Data
00730  * Register) which reflects the actual voltage level on the pin, regardless
00731  * of whether the pin is configured as input or output.
00732  *
00733  * ## Register Operation
00734  *
00735  * ```
00736  * value = (PIDR & (1 « pin)) != 0
00737  * ```
00738  *
00739  * ## Input vs Output Reading
00740  *
00741  * - **Input pin**: PIDR reflects external voltage on pin
00742  * - **Output pin**: PIDR reflects actual pin state (may differ from PODR
00743  *   if pin is shorted or overdriven)
00744  *
00745  * ## Voltage Thresholds (RX72N)
00746  *
00747  * | Level | Voltage Range | Returned Value |
00748  * |-----|-----|-----|
00749  * | Low | 0V to ~0.8V | false |
00750  * | High | ~2.0V to VCC | true |
00751  * | Undefined | 0.8V to 2.0V | Unpredictable |
00752  *
00753  *
00754  *
00755  * @pre value must be non-nullptr
00756  * @pre Pin should be configured as input for external signals
00757  *
00758  * @post *value contains current pin level (true=high, false=low)
00759  * @post No hardware state changes
00760  *
00761  * @note Thread-safe: Read-only operation
00762  * @note Performance: ~100 ns @ 240 MHz
00763  * @note ISR-safe: Can be called from interrupt handlers
00764  *
00765  * @warning Floating inputs may return unpredictable values
00766  * @warning Use pull-up/pull-down for stable readings on open pins
00767  *
00768  * @par Example:
00769  * @code
00770  * // Read button state (active low with pull-up)
00771  * bool button_pressed;
00772  * rx_err_t err = gpio_read(RX_PORT_PIN(2, 0), &button_pressed);
00773  *
00774  * if (err == k_rx_ok) {
00775  *     // Invert because button is active low
00776  *     if (!button_pressed) {
00777  *         rx_log_info("BTN", "Button pressed");
00778  *         handle_button_press();
00779  *     }
00780  * }
00781  * @endcode
00782  *
00783  * @par Example: Debounced button read
00784  * @code
00785  * bool debounced_read(rx_port_pin_t pin) {
00786  *     bool first, second;
00787  *     gpio_read(pin, &first);
00788  *     tx_thread_sleep(1); // 1 tick delay
00789  *     gpio_read(pin, &second);
00790  *     return (first == second) ? first : false; // Only return if stable
00791  * }
00792  * @endcode
00793  *
00794  * @see gpio_set_input() Configure pin as input first
00795  * @see gpio_write_high() Set output high
00796  * @see gpio_write_low() Set output low
00797  *
00798  * @since Version 1.0.0
00799  *
00800  * @par NASA Power of 10 Compliance:
00801  * - Rule 5: [OK] 2 preconditions, 2 postconditions
00802  */
00803 rx_err_t gpio_read(const rx_port_pin_t pin, bool* value)

```

```

00804 {
00805     /* Check null pointer */
00806     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is nullptr");
00807
00808     /* Extract port and pin number from rx_port_pin_t */
00809     const uint8_t port = rx_port_from_pin(pin);
00810     const uint8_t pin_num = rx_pin_from_pin(pin);
00811
00812     /* Validate parameters */
00813     const rx_err_t err = internal_validate_port_pin(port, pin_num);
00814     RX_RETURN_ON_ERROR(err, s_tag, "Port/pin validation failed");
00815
00816     /* Get port base address */
00817     const volatile rx_port_regs_t* port_base = rx_port_get_base(port);
00818
00819     const uint8_t pin_mask = (uint8_t)(k_gpio_bit_set « pin_num);
00820     *value = (port_base->pidr & pin_mask) != k_gpio_bit_clear;
00821
00822     return k_rx_ok;
00823 }
00824
00825 #endif /* __RX__ */

```

8.105 libs/rx`hal/src/riic.c File Reference

RIIC (I2C) Driver Implementation for RX72N.

```

#include <string.h>
#include "hardware.h"
#include "rx72n_regs.h"
#include "rx_register_protection.h"

```

Include dependency graph for riic.c:



Data Structures

- struct [riic_recovery_pins_t](#)
Channel-specific port + pin info for bit-bang bus recovery.
- struct [riic_saved_cfg_t](#)
Wait for I2C bus to become idle (not busy).

Enumerations

- enum [riic_constants_t](#) : uint32_t {
[k_riic_max_channels](#) = 3 , [k_riic_timeout_us](#) = 25000 , [k_riic_timeout_zero](#) = 0 ,
[k_riic_length_zero](#) = 0 ,
[k_riic_register_clear](#) = 0 , [k_riic_last_index_offset](#) = 1 , [k_riic_max_transfer_length](#) = 256 }
RIIC channel count, timeout, and transfer limit constants.
- enum [riic_channel_num_t](#) : uint8_t { [k_riic_channel_0](#) = 0 , [k_riic_channel_1](#) = 1 ,
[k_riic_channel_2](#) = 2 }
RIIC channel identifiers for switch statement dispatch.
- enum [riic_module_stop_bits_t](#) : uint8_t { [k_riic_mstpb_riic0](#) = 21 , [k_riic_mstpb_riic1](#) = 20 ,
[k_riic_mstpc_riic2](#) = 17 , [k_riic_mstpb_bit_value](#) = 1 }
RIIC module stop bit positions in MSTPCRB (RIIC0/1) and MSTPCRC (RIIC2).
- enum [riic_frequency_t](#) : uint32_t { [k_riic_freq_100khz](#) = 100000 , [k_riic_freq_400khz](#) = 400000 ,
[k_riic_freq_1mhz](#) = 1000000 }
Supported I2C bus frequencies (Hz).

- enum `riic_bit_rate_t` : `uint8_t` {
`k_riic_cks_100khz` = 4U , `k_riic_icbrh_100k` = 14U , `k_riic_icbtl_100k` = 17U ,
`k_riic_cks_400khz` = 2U ,
`k_riic_icbrh_400k` = 8U , `k_riic_icbtl_400k` = 19U , `k_riic_cks_1mhz` = 0U , `k_riic_icbrh_1m`
= 15U ,
`k_riic_icbtl_1m` = 29U , `k_riic_cks_shift` = 4U }
Bit rate calculation constants for ICBRL/ICBRH registers.
- enum `riic_icmr1_values_t` : `uint8_t` { `k_riic_icmr1_controller_7bit` }
ICMR1 (Mode Register 1) configuration values.
- enum `riic_icmr2_values_t` : `uint8_t` { `k_riic_icmr2_default` = 0 }
ICMR2 (Mode Register 2) configuration values.
- enum `riic_icmr3_bits_t` : `uint8_t` {
`k_riic_icmr3_ackbt_pos` = 3 , `k_riic_icmr3_ackbt_mask` = (1U << `k_riic_icmr3_ackbt_pos`)
, `k_riic_icmr3_ackwp_pos` = 4 , `k_riic_icmr3_ackwp_mask` = (1U << `k_riic_icmr3_ackwp_pos`)
, `k_riic_icmr3_wait_pos` = 6 , `k_riic_icmr3_wait_mask` = (1U << `k_riic_icmr3_wait_pos`) ,
`k_riic_icmr3_init` = `k_riic_icmr3_ackwp_mask` }
ICMR3 (Mode Register 3) ACK/NACK control bit definitions.
- enum `riic_address_bits_t` : `uint8_t` { `k_riic_addr_shift` = 1 , `k_riic_addr_write_bit` = 0 ,
`k_riic_addr_read_bit` = 1 , `k_riic_addr_max_7bit` = 127 }
I2C 7-bit address formatting constants.
- enum `riic_recovery_constants_t` : `uint32_t` {
`k_riic_recov_cycles` = 9U , `k_riic_recov_half_us` = 5U , `k_riic_recov_cpu_mhz` = 240U ,
`k_riic_recov_stop_us` = 5U ,
`k_riic_port2_odr0_addr` = 0x0008C084U }
Bit-bang bus-recovery timing and ODR register addresses.
- enum `riic_channel_mode_t` : `uint8_t` { `k_riic_mode_uninitialized` = 0x00u , `k_riic_mode_controller`
= 0x01u , `k_riic_mode_peripheral` = 0x02u }
Per-channel RIIC initialization mode.
- enum `riic_peripheral_reg_constants_t` : `uint8_t` { `k_riic_icmr1_peripheral_7bit` = 0x00 ,
`k_riic_icser_sar0e` = 0x01 , `k_riic_sarl_addr_shift` = 1 , `k_riic_saru0_7bit` = 0x00 }
Register bit-field constants for peripheral mode configuration.
- enum `riic_peripheral_size_constants_t` : `uint16_t` { `k_riic_periph_timeout_us` = 1000 }
Size and timeout constants for peripheral mode operations.

Functions

- static `uint8_t` `internal_riic_channel_from_base` (const volatile `rx_riic_regs_t` *riic)
Resolve a RIIC register pointer to its channel number.
- static void `internal_riic_busy_wait_us` (uint16_t us, uint16_t cpu_mhz)
Cycle-counted busy-wait for bit-bang timing.
- static void `internal_riic_bit_bang_recover` (uint8_t channel)
Bit-bang 9 SCL edges + manual STOP to recover an externally-stuck bus.
- static void `internal_riic_force_fsm_reset` (volatile `rx_riic_regs_t` *riic)
Raise PMR so the RIIC peripheral takes the pads back.
- static void `internal_riic_bit_bang_handback` (uint8_t channel)
- static volatile `rx_riic_regs_t` * `internal_get_riic_base` (const uint8_t channel)
Get RIIC channel register base address from channel number.
- static `rx_err_t` `internal_calculate_bit_rate` (const uint32_t frequency_hz, uint8_t *icbtl, uint8_t *icbrh, uint8_t *cks_field)
Calculate RIIC bit rate register values for target frequency.
- static bool `internal_riic_poll_bbsy_clear` (volatile `rx_riic_regs_t` *const riic)

- Poll BBSY for up to `k_riic_timeout_us` iterations.

 - static void `internal_riic_save_config` (const volatile `rx_riic_regs_t` *const riic, `riic_saved_cfg_t` *const saved)

Save config registers wiped by an IICRST pulse.

 - static void `internal_riic_iicrst_with_restore` (volatile `rx_riic_regs_t` *const riic, const `riic_saved_cfg_t` *const saved)

Pulse IICRST and restore config from a saved snapshot.

 - static void `internal_riic_release_lines` (volatile `rx_riic_regs_t` *const riic)

Drive the SOWP/SCLO/SDAO release dance after an IICRST pulse.

 - static void `internal_riic_log_pre_recovery` (const volatile `rx_riic_regs_t` *const riic)

Log debug-level pre-recovery bus state.

 - static void `internal_riic_bit_bang_recovery_sequence` (volatile `rx_riic_regs_t` *const riic, const `riic_saved_cfg_t` *const saved)

Bit-bang last-resort bus recovery + peripheral reset + handback.

 - static `rx_err_t` `internal_wait_bus_ready` (volatile `rx_riic_regs_t` *riic)
 - static `rx_err_t` `internal_send_start` (volatile `rx_riic_regs_t` *riic)

Generate I2C START condition on the bus.

 - static `rx_err_t` `internal_send_stop` (volatile `rx_riic_regs_t` *riic)

Generate I2C STOP condition on the bus.

 - static `rx_err_t` `internal_write_byte` (volatile `rx_riic_regs_t` *riic, const `uint8_t` data)

Transmit single byte over I2C bus.

 - static `rx_err_t` `internal_write_address_for_read` (volatile `rx_riic_regs_t` *riic, const `uint8_t` addr, const `uint8_t` addr_byte)

Send the address|R byte that opens a controller-receive transaction.

 - static `rx_err_t` `internal_read_byte` (volatile `rx_riic_regs_t` *riic, `uint8_t` *data, const bool send_ack, const bool last_byte)

Receive single byte from I2C bus.

 - static `rx_err_t` `internal_riic_write_phase` (volatile `rx_riic_regs_t` *riic, const `i2c_device_addr_t` device_addr, const `uint8_t` *write_data, const `uint16_t` write_length)

Execute I2C write phase for combined write-read transfer.

 - static `rx_err_t` `internal_riic_read_phase` (volatile `rx_riic_regs_t` *riic, const `i2c_device_addr_t` device_addr, `uint8_t` *read_data, const `uint16_t` read_length)

Execute I2C read phase for combined write-read transfer.

 - static `rx_err_t` `internal_riic_validate_init` (const `riic_channel_t` channel, const `uint32_t` frequency_hz, volatile `rx_riic_regs_t` **riic_out)

Initialize RIIC (I2C) channel for controller mode operation.

 - static void `internal_riic_release_module_stop` (const `uint8_t` channel_value)

Release the module-stop bit for the requested RIIC channel.

 - static `rx_err_t` `internal_riic_program_bit_rate` (volatile `rx_riic_regs_t` *const riic, const `uint32_t` frequency_hz)

Program ICBRL/ICBRH/ICMR1..3 for the requested SCL frequency.

 - `rx_err_t` `riic_init` (const `riic_channel_t` channel, const `uint32_t` frequency_hz)

Initialize RIIC channel for I2C controller communication.

 - `rx_err_t` `riic_write` (const `riic_channel_t` channel, const `i2c_device_addr_t` device_addr, const `uint8_t` *data, const `uint16_t` length)

Write data to an I2C peripheral device.

 - `rx_err_t` `riic_read` (const `riic_channel_t` channel, const `i2c_device_addr_t` device_addr, `uint8_t` *data, const `uint16_t` length)

Read data from an I2C peripheral device.

 - `rx_err_t` `riic_write_read` (const `riic_channel_t` channel, const `i2c_device_addr_t` device_addr, const `uint8_t` *write_data, const `uint16_t` write_length, `uint8_t` *read_data, const `uint16_t` read_length)

- Combined write-then-read I2C transaction (register read).
- `rx_err_t riic_init_peripheral` (const `riic_channel_t` channel, const `i2c_device_addr_t` device_addr)
 - Initialize RIIC channel in peripheral (device) mode.
- `rx_err_t riic_peripheral_read` (const `riic_channel_t` channel, `uint8_t` *data, const `uint16_t` max_length, `uint16_t` *bytes_read)
 - Read bytes written by the I2C controller to this peripheral.
- `rx_err_t riic_peripheral_write` (const `riic_channel_t` channel, const `uint8_t` *data, const `uint16_t` length)
 - Write bytes for the I2C controller to read from this peripheral.
- `rx_err_t riic_deinit_peripheral` (const `riic_channel_t` channel)
 - Deinitialize an RIIC channel that was initialized in peripheral mode.

Variables

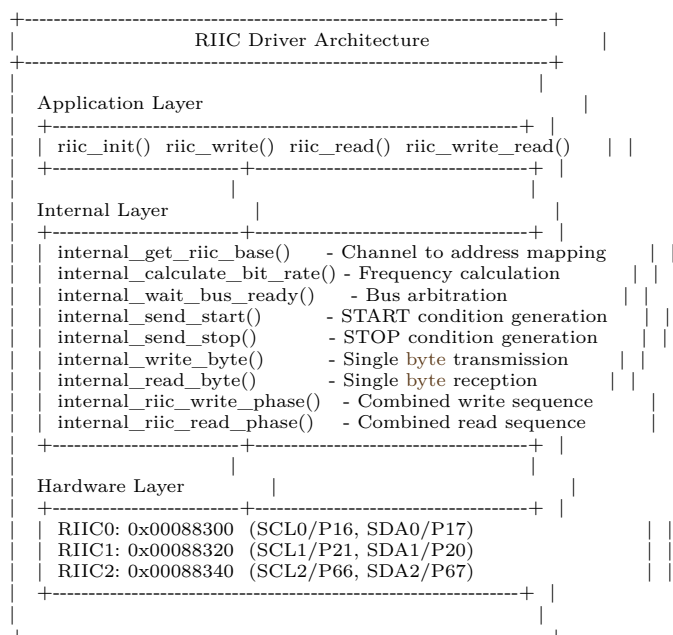
- static const char *const `s_tag` = "RIIC"
 - Logging tag for RIIC driver messages.
- static const `riic_recovery_pins_t` `k_riic_recovery_pins` [`k_riic_max_channels`]
 - Per-channel recovery pin lookup.
- static `riic_channel_mode_t` `s_riic_channel_mode` [`k_riic_max_channels`]
 - Per-channel RIIC mode tracking array.

8.105.1 Detailed Description

RIIC (I2C) Driver Implementation for RX72N.

Production-grade implementation of the I2C Bus Interface (RIIC) peripheral driver for the RX72N microcontroller. This driver provides controller-mode I2C communication with support for standard (100 kHz), fast (400 kHz), and fast-plus (1 MHz) modes.

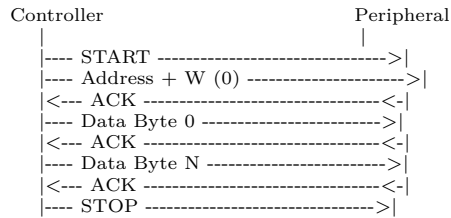
8.105.1.1 Module Architecture



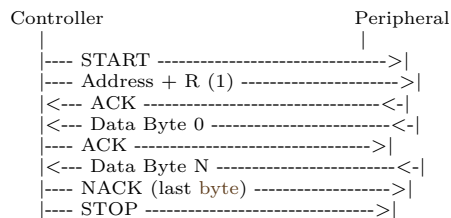
8.105.1.2 I2C Protocol Implementation

This driver implements the I2C protocol in controller mode:

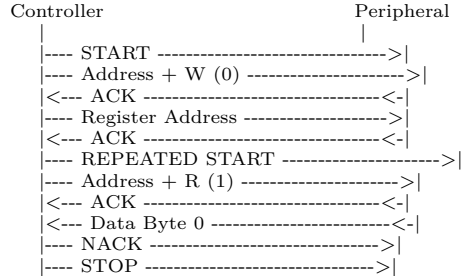
8.105.1.2.1 Write Transaction



8.105.1.2.2 Read Transaction



8.105.1.2.3 Write-Read Transaction (Register Read)



8.105.1.3 Bit Rate Calculation

The I2C clock frequency is calculated from PCLKB (60 MHz) as:

$$f_{SCL} = \frac{f_{PCLKB}}{(ICBRL + ICBRH + 2) \times 2^{CKS}}$$

For simplified calculation with 50% duty cycle (ICBRL = ICBRH):

$$\text{divisor} = \frac{f_{PCLKB}}{f_{SCL} \times 3}$$

Mode	Speed	Divisor	Actual Rate
Standard	100 kHz	200	100.0 kHz
Fast	400 kHz	50	400.0 kHz
Fast Plus	1 MHz	20	1.00 MHz

8.105.1.4 Memory Map (Chapter 42 - RIIC)

Channel	Base Address	Pin SCL	Pin SDA	Module Stop Bit
RIIC0	0x00088300	P16	P17	MSTPB21 (MSTPCRB b21)
RIIC1	0x00088320	P21	P20	MSTPB20 (MSTPCRB b20)
RIIC2	0x00088340	P66	P67	MSTPC17 (MSTPCRC b17)

8.105.1.5 Register Offsets (per channel)

Offset	Register	Description
0x00	ICCR1	Control Register 1
0x01	ICCR2	Control Register 2
0x02	ICMR1	Mode Register 1
0x03	ICMR2	Mode Register 2
0x04	ICMR3	Mode Register 3
0x10	ICBRL	Bit Rate Low period
0x11	ICBRH	Bit Rate High period
0x12	ICDRT	Transmit Data Register
0x13	ICDRR	Receive Data Register

8.105.1.6 Error Handling

The driver provides comprehensive error detection:

- `k'rx'err'null'ptr`: nullptr passed to function
- `k'rx'err'invalid'arg`: Invalid channel, address, or frequency
- `k'rx'err'invalid'state`: Channel not initialized
- `k'rx'err'timeout`: Bus busy or operation timeout
- `k'rx'err'nack`: Peripheral did not acknowledge

8.105.1.7 NASA Power of 10 Compliance

Rule	Implementation
Rule 1	No goto, setjmp, or recursion - sequential error handling
Rule 2	Bounded loops - all loops use k_riic_timeout_us timeout
Rule 3	No dynamic memory - static channel state array
Rule 4	Short functions - each < 60 lines
Rule 5	Validation - RX_CHECK_NULL_PTR, RX_CHECK_RANGE_TAG
Rule 6	Smallest scope - local variables declared at point of use
Rule 7	All return values checked - RX_RETURN_ON_ERROR
Rule 8	Limited preprocessor - only RX_CHECK macros
Rule 9	Function pointers via interface pattern (not used here)
Rule 10	Compiles with -Wall -Wextra -Werror

8.105.1.8 SOLID Principles Compliance

Principle	Implementation
S Single Responsibility	Driver handles only RIIC I2C operations
O Open/Closed	Supports multiple frequencies via parameter
L Liskov Substitution	Consistent error codes across all functions
I Interface Segregation	Separate read/write/write-read APIs
D Dependency Inversion	Uses hardware.h abstraction for registers

Thread Safety

This module is NOT thread-safe. Each RIIC channel should be accessed by only one thread at a time. If multiple threads need I2C access, provide external synchronization (mutex/semaphore) at the application level.

Example: Read Temperature Sensor

```
// Initialize RIIC0 at 400 kHz (Fast mode)
riic_channel_t channel = {.value = 0};
rx_err_t err = riic_init(channel, 400000);
if (err != k_rx_ok) {
    handle_error(err);
    return;
}

// Read 2 bytes from temperature sensor at address 0x48
i2c_device_addr_t sensor_addr = {.value = 0x48};
uint8_t temp_reg = 0x00; // Temperature register
uint8_t temp_data[2];

err = riic_write_read(channel, sensor_addr, &temp_reg, 1, temp_data, 2);
if (err == k_rx_ok) {
    int16_t raw_temp = (temp_data[0] << 8) | temp_data[1];
    float celsius = raw_temp * 0.0625f; // LM75 resolution
}
```

See also

[rx72n_riic_regs.h](#) RIIC register definitions

[hardware.h](#) Hardware abstraction layer

Manual Reference

RX72N Group User's Manual: Hardware, Chapter 42 (I2C-bus Interface)

Date

2026-01-01

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Definition in file [riic.c](#).

8.105.2 Data Structure Documentation

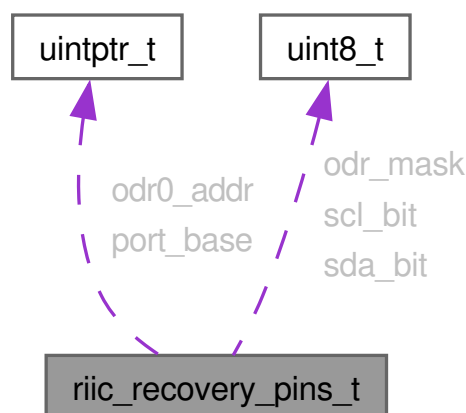
8.105.2.1 struct riic_recovery_pins_t

Channel-specific port + pin info for bit-bang bus recovery.

Indexed by riic channel number. A zero port_base entry means no recovery table entry has been provided for that channel (skipped at runtime). Only RIIC1 is currently populated because only the IMU uses RIIC; add RIIC0/RIIC2 entries when those channels are activated.

Definition at line [604](#) of file [riic.c](#).

Collaboration diagram for riic_recovery_pins_t:



Data Fields

uintptr_t	odr0_addr	ODR register address for N-ch open-drain selection
uint8_t	odr_mask	ODR bit mask for N-ch open-drain on SCL + SDA pin pair
uintptr_t	port_base	rx_port_regs_t* base address for the port holding SCL+SDA
uint8_t	scl_bit	SCL pin bit position within the port (0-7)
uint8_t	sda_bit	SDA pin bit position within the port (0-7)

8.105.2.2 struct riic'saved'cfg't

Wait for I2C bus to become idle (not busy).

Polls the BBSY (Bus Busy) bit in ICCR2 until the bus is free or timeout occurs. This function must be called before starting a new I2C transaction to ensure no other controller is using the bus (multi-controller arbitration).

8.105.2.3 Bus Busy Detection

The BBSY bit is set when:

- A START condition is detected on the bus
- The controller has issued a START condition

The BBSY bit is cleared when:

- A STOP condition is detected on the bus
- Bus arbitration is lost

8.105.2.4 Timeout Calculation

At 100 kHz (slowest mode), maximum byte transfer time:

- 9 bits per byte (8 data + 1 ACK) x 10 us = 90 us
- For 256 bytes (max transfer): 256 x 90 us = 23 ms

The 10 ms timeout (k_riic_timeout_us) provides margin for typical operations while catching true bus-stuck conditions.

Precondition

riic != nullptr

RIIC channel must be initialized and enabled

Postcondition

Bus is idle if `k_rx_ok` returned

Note

This is a blocking function - polls in tight loop.

Timeout is ~10ms at `k_riic_timeout_us` iterations.

Warning

In multi-controller systems, bus may become busy again immediately after this function returns - caller should proceed quickly.

NASA Power of 10 Compliance

- Rule 2: [OK] Bounded loop with `k_riic_timeout_us` maximum iterations
- Rule 5: [OK] nullptr check via `RX_CHECK_NULL_PTR`

Snapshot of RIIC config registers preserved across IICRST pulses

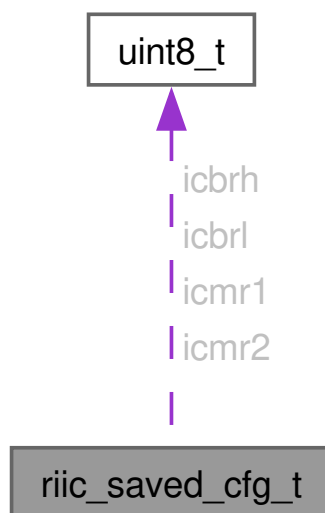
IICRST resets ICBRH/ICBRL/ICMR1/ICMR2/ICMR3 to power-on defaults. The recovery sequence saves these before pulsing IICRST and restores them after, so the post-recovery RIIC keeps the configured baud rate and mode bits.

Since

Version 1.0.0

Definition at line [1133](#) of file [riic.c](#).

Collaboration diagram for `riic_saved_cfg_t`:



Data Fields

uint8_t	icbrh	ICBRH (high-period bit-rate divisor).
uint8_t	icbrl	ICBRL (low-period bit-rate divisor).
uint8_t	icmr1	ICMR1 (mode register 1).
uint8_t	icmr2	ICMR2 (mode register 2).

8.105.3 Enumeration Type Documentation

8.105.3.1 riic`address`bits` t

enum [riic_address_bits_t](#) : uint8_t

I2C 7-bit address formatting constants.

I2C uses a 7-bit address plus a read/write direction bit in the first byte after START. The address is left-shifted by 1 and OR'd with the direction bit.

Address Byte Format

```

Bit:  7  6  5  4  3  2  1  0
      +-----+ +- R/W (0=Write, 1=Read)
      +- 7-bit Address

```

Address Calculation

- Write to 0x50: (0x50 << 1) | 0 = 0xA0
- Read from 0x50: (0x50 << 1) | 1 = 0xA1

Invariant

`k_riic_addr_max_7bit == 127` (I2C spec: addresses 0x00-0x7F)

```

// Format a write address byte for device at 0x50
riic_device_addr_t dev = {.value = 0x50};
uint8_t addr_byte = (uint8_t)((dev.value « k_riic_addr_shift) | k_riic_addr_write_bit);

```

See also

[internal_write_byte\(\)](#) Uses these constants to format address bytes

[riic_write\(\)](#) Passes formatted address bytes during write transactions

Enumerator

<code>k_riic_addr_shift</code>	Left shift amount for 7-bit address
<code>k_riic_addr_write_bit</code>	R/W bit value for write operation
<code>k_riic_addr_read_bit</code>	R/W bit value for read operation
<code>k_riic_addr_max_7bit</code>	Maximum valid 7-bit address (0x7F)

Definition at line 545 of file [riic.c](#).

```

00545     : uint8_t {
00546     k_riic_addr_shift    = 1, /**< Left shift amount for 7-bit address */
00547     k_riic_addr_write_bit = 0, /**< R/W bit value for write operation */
00548     k_riic_addr_read_bit  = 1, /**< R/W bit value for read operation */
00549     k_riic_addr_max_7bit  = 127, /**< Maximum valid 7-bit address (0x7F) */
00550 } riic_address_bits_t;

```

8.105.3.2 riic`bit`rate`t

```
enum riic_bit_rate_t : uint8_t
```

Bit rate calculation constants for ICBRL/ICBRH registers.

These constants are used in `internal_calculate_bit_rate()` to compute the correct ICBRL and ICBRH register values for a target frequency.

Calculation Formula

For 50% duty cycle (ICBRL = ICBRH = divisor):

$$\text{divisor} = \frac{f_{PCLKB}}{f_{target} \times 3}$$

Where the factor of 3 accounts for:

- SCL high period (ICBRH + 1) cycles
- SCL low period (ICBRL + 1) cycles
- Internal synchronization overhead

Enumerator

k_riic_cks_100khz	ICMR1.CKS[2:0] = 100b for 100 kbps @ 60 MHz PCLK
k_riic_icbrh_100k	ICBRH count, HUM Table 42.5
k_riic_icbtl_100k	ICBRL count, HUM Table 42.5
k_riic_cks_400khz	ICMR1.CKS[2:0] = 010b for 400 kbps @ 60 MHz PCLK
k_riic_icbrh_400k	ICBRH count, HUM Table 42.5
k_riic_icbtl_400k	ICBRL count, HUM Table 42.5
k_riic_cks_1mhz	ICMR1.CKS[2:0] = 000b for 1 Mbps @ 60 MHz PCLK
k_riic_icbrh_1m	ICBRH count, HUM Table 42.5
k_riic_icbtl_1m	ICBRL count, HUM Table 42.5
k_riic_cks_shift	ICMR1 bit position of CKS[0] (CKS is bits [6:4])

Definition at line 379 of file `riic.c`.

```
00379     : uint8_t {
00380     k_riic_cks_100khz = 4U, /**< ICMR1.CKS[2:0] = 100b for 100 kbps @ 60 MHz PCLK */
00381     k_riic_icbrh_100k = 14U, /**< ICBRH count, HUM Table 42.5 */
00382     k_riic_icbtl_100k = 17U, /**< ICBRL count, HUM Table 42.5 */
00383     k_riic_cks_400khz = 2U, /**< ICMR1.CKS[2:0] = 010b for 400 kbps @ 60 MHz PCLK */
00384     k_riic_icbrh_400k = 8U, /**< ICBRH count, HUM Table 42.5 */
00385     k_riic_icbtl_400k = 19U, /**< ICBRL count, HUM Table 42.5 */
00386     k_riic_cks_1mhz = 0U, /**< ICMR1.CKS[2:0] = 000b for 1 Mbps @ 60 MHz PCLK */
00387     k_riic_icbrh_1m = 15U, /**< ICBRH count, HUM Table 42.5 */
00388     k_riic_icbtl_1m = 29U, /**< ICBRL count, HUM Table 42.5 */
00389     k_riic_cks_shift = 4U, /**< ICMR1 bit position of CKS[0] (CKS is bits [6:4]) */
00390 } riic_bit_rate_t;
```

8.105.3.3 riic`channel`mode`t

```
enum riic_channel_mode_t : uint8_t
```

Per-channel RIIC initialization mode.

Tracks whether each RIIC channel has been initialized, and if so, whether it was initialized for controller or peripheral operation. This prevents peripheral-mode APIs from running on a controller-initialized channel and vice-versa, satisfying the Liskov Substitution Principle for RIIC API implementations.

Invariant

Set only by [riic_init\(\)](#) (controller) and [riic_init_peripheral\(\)](#) (peripheral)

Reset to k_riic_mode_uninitialized by [riic_deinit_peripheral\(\)](#) on success

Since

Version 1.0.0

Enumerator

k_riic_mode_uninitialized	Channel not yet initialized
k_riic_mode_controller	Initialized by riic_init() – controller mode
k_riic_mode_peripheral	Initialized by riic_init_peripheral() – peripheral mode

Definition at line 872 of file [riic.c](#).

```
00872      : uint8_t {
00873  k_riic_mode_uninitialized = 0x00u, /**< Channel not yet initialized */
00874  k_riic_mode_controller   = 0x01u, /**< Initialized by riic_init() -- controller mode */
00875  k_riic_mode_peripheral   = 0x02u, /**< Initialized by riic_init_peripheral() -- peripheral mode */
00876 } riic_channel_mode_t;
```

8.105.3.4 riic`channel`num`t

```
enum riic_channel_num_t : uint8_t
```

RIIC channel identifiers for switch statement dispatch.

Provides named constants for channel selection in [internal_get_riic_base\(\)](#). Using enums instead of raw integers ensures type safety and improves code readability in switch statements.

Channel to Pin Mapping

Channel	SCL Pin	SDA Pin	Base Address
RIIC0	P16	P17	0x00088300
RIIC1	P21	P20	0x00088320
RIIC2	P66	P67	0x00088340

Invariant

Values are contiguous starting from 0 and match hardware channel indices

```
// Select channel 0 base address
volatile rx_riic_regs_t* regs = internal_get_riic_base(k_riic_channel_0);
```

See also

[internal_get_riic_base\(\)](#) Maps channel number to hardware register pointer
[riic_constants_t](#) For [k_riic_max_channels](#) bound used with these values

Enumerator

k_riic_channel_0	RIIC channel 0 (P16/SCL0, P17/SDA0)
k_riic_channel_1	RIIC channel 1 (P21/SCL1, P20/SDA1)
k_riic_channel_2	RIIC channel 2 (P66/SCL2, P67/SDA2)

Definition at line 285 of file [riic.c](#).

```
00285         : uint8_t {
00286   k_riic_channel_0 = 0, /**< RIIC channel 0 (P16/SCL0, P17/SDA0) */
00287   k_riic_channel_1 = 1, /**< RIIC channel 1 (P21/SCL1, P20/SDA1) */
00288   k_riic_channel_2 = 2, /**< RIIC channel 2 (P66/SCL2, P67/SDA2) */
00289 } riic_channel_num_t;
```

8.105.3.5 `riic_constants_t`

```
enum riic\_constants\_t : uint32_t
```

RIIC channel count, timeout, and transfer limit constants.

General-purpose constants for RIIC driver operation including channel limits, timeout values, and transfer size constraints.

Timeout Selection

The 10ms timeout provides sufficient margin for:

- Standard mode: 100 kHz = 10us/bit, 90us/byte -> 10ms = ~111 bytes
- Fast mode: 400 kHz = 2.5us/bit, 22.5us/byte -> 10ms = ~444 bytes
- Fast Plus: 1 MHz = 1us/bit, 9us/byte -> 10ms = ~1111 bytes

Invariant

`k_riic_max_channels == 3` (hardware limitation)
`k_riic_max_transfer_length == 256` (single-byte length limit)

Enumerator

<code>k_riic_max_channels</code>	Total RIIC channels: RIIC0, RIIC1, RIIC2
----------------------------------	--

k_riic_timeout_us	Busy-loop iteration budget for I2C flag waits. Each iteration reads ICSR2/ICCR2 + decrements a volatile counter (~10 cycles @ 240 MHz ~ = 40 ns), so 25k iters ~ = 1 ms wall time – enough to cover a byte at 100 kHz (90 us) with 10x slack for clock stretching. Kept below the 900 ms IWDT window even if a 256-byte transfer stacks up 256 per-byte timeouts back-to-back.
k_riic_timeout_zero	Timeout counter expired sentinel value
k_riic_length_zero	Zero-length transfer (invalid) sentinel
k_riic_register_clear	Zero value used to clear hardware registers
k_riic_last_index_offset	Offset to calculate last byte index from length
k_riic_max_transfer_length	Maximum bytes per transfer operation

Definition at line 242 of file [riic.c](#).

```

00242         : uint32_t {
00243     k_riic_max_channels      = 3,    /**< Total RIIC channels: RIIC0, RIIC1, RIIC2 */
00244     k_riic_timeout_us       = 25000, /**< Busy-loop iteration budget for I2C flag waits.
00245                                     * Each iteration reads ICSR2/ICCR2 + decrements a
00246                                     * volatile counter (~10 cycles @ 240 MHz ~ = 40 ns),
00247                                     * so 25k iters ~ = 1 ms wall time -- enough to
00248                                     * cover a byte at 100 kHz (90 us) with 10x slack
00249                                     * for clock stretching. Kept below the 900 ms IWDT
00250                                     * window even if a 256-byte transfer stacks up
00251                                     * 256 per-byte timeouts back-to-back. */
00252     k_riic_timeout_zero     = 0,    /**< Timeout counter expired sentinel value */
00253     k_riic_length_zero      = 0,    /**< Zero-length transfer (invalid) sentinel */
00254     k_riic_register_clear   = 0,    /**< Zero value used to clear hardware registers */
00255     k_riic_last_index_offset = 1,    /**< Offset to calculate last byte index from length */
00256     k_riic_max_transfer_length = 256, /**< Maximum bytes per transfer operation */
00257 } riic_constants_t;

```

8.105.3.6 riic_frequency_t

```
enum riic\_frequency\_t : uint32_t
```

Supported I2C bus frequencies (Hz).

I2C standard defines three speed grades. This driver supports all three modes with appropriate bit rate register configuration.

Timing Characteristics (at PCLKB = 60 MHz)

Mode	Frequency	Bit Time	Byte Time	Max Cable
Standard	100 kHz	10 us	90 us	~1 meter
Fast	400 kHz	2.5 us	22.5 us	~0.5 m
Fast Plus	1 MHz	1 us	9 us	~0.2 m

Invariant

Only these three discrete values are accepted by [riic_init\(\)](#)

```

// Initialize channel 0 at standard (100 kHz) mode
riic_channel_t ch = {.value = 0};
rx_err_t err = riic_init(ch, k_riic_freq_100khz);

```

See also

[riic_init\(\)](#) Validates frequency_hz against these enum values

[internal_calculate_bit_rate\(\)](#) Uses frequency to compute ICBRL/ICBRH

Note

Fast Plus mode requires pull-up resistors < 1kOhm for proper rise time.

Enumerator

k_riic_freq_100khz	Standard mode: 100 kHz (Sm)
k_riic_freq_400khz	Fast mode: 400 kHz (Fm)
k_riic_freq_1mhz	Fast mode plus: 1 MHz (Fm+)

Definition at line 345 of file [riic.c](#).

```
00345      : uint32_t {
00346  k_riic_freq_100khz = 100000, /**< Standard mode: 100 kHz (Sm) */
00347  k_riic_freq_400khz = 400000, /**< Fast mode: 400 kHz (Fm) */
00348  k_riic_freq_1mhz   = 1000000, /**< Fast mode plus: 1 MHz (Fm+) */
00349 } riic_frequency_t;
```

8.105.3.7 riic_icmr1_values_t

```
enum riic_icmr1_values_t : uint8_t
```

ICMR1 (Mode Register 1) configuration values.

ICMR1 configures the controller/peripheral mode and addressing type. For STAR project, we use controller mode with 7-bit addressing.

ICMR1 Register Layout (Bits 7-4)

Bit	Name	Description
6-4	CKS	Clock divider (0-7)
3	BCWP	BC write protect
2-0	BC	Bit counter

Enumerator

k_riic_icmr1_controller_7bit	CKS=0, BCWP=1, 7-bit addressing; MTWP=0 keeps MST/↵ TRS read-only so the hardware START/STOP FSM drives them, per HUM section 42.2.3. MTWP=1 was tried and caused "Start condition failed" on every transaction because iccr2=MST TRS pre-writes the mode bits before START fires and the FSM gets desynced.
------------------------------	--

Definition at line 407 of file [riic.c](#).

```
00407      : uint8_t {
00408  k_riic_icmr1_controller_7bit =
00409      0x08, /**< CKS=0, BCWP=1, 7-bit addressing; MTWP=0 keeps MST/TRS read-only so the hardware START/STOP
               FSM drives them, per HUM section 42.2.3. MTWP=1 was tried and caused "Start condition failed" on every transaction
               because iccr2=MST|TRS pre-writes the mode bits before START fires and the FSM gets desynced. */
00410 } riic_icmr1_values_t;
```

8.105.3.8 riic`icmr2`values`t

```
enum riic_icmr2_values_t : uint8_t
```

ICMR2 (Mode Register 2) configuration values.

ICMR2 configures hardware timeout detection and SDA output hold-time delay. The STAR project disables hardware timeout (TMOE = 0) because bus-busy detection is handled in software by [internal_wait_bus_ready\(\)](#) using the `k_riic_timeout_us` countdown. The SDA output delay is also left at zero (no additional hold time) since the pull-up resistors on the target PCB provide adequate signal integrity at all supported speeds.

8.105.3.9 ICMR2 Register Bit Layout

Bit	Name	Function	Value Used
7	TMOH	Timeout detection (high period)	0 (off)
6	TMOL	Timeout detection (low period)	0 (off)
5	TMOE	Timeout enable	0 (off)
4:↔ 3	SDDL	SDA output hold-time delay	0 (none)
2	DLCS	SDA delay clock source	0
1:↔ 0	-	Reserved	0

Invariant

`k_riic_icmr2_default == 0`; writing 0x00 leaves all ICMR2 control bits in their hardware-reset state (timeouts off, no SDA delay)

```
// Apply default ICMR2 configuration during RIIC channel initialization
volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_0);
riic->icmr2 = k_riic_icmr2_default;
```

See also

[riic_init\(\)](#) Applies this value during channel initialization

[k_riic_timeout_us](#) Software timeout used instead of hardware TMOE

Enumerator

<code>k_riic_icmr2_default</code>	No hardware timeout, no SDA delay
-----------------------------------	-----------------------------------

Definition at line 447 of file [riic.c](#).

```
00447 : uint8_t {
00448 k_riic_icmr2_default = 0, /**< No hardware timeout, no SDA delay */
00449 } riic_icmr2_values_t;
```

8.105.3.10 riic_icmr3_bits_t

```
enum riic_icmr3_bits_t : uint8_t
```

ICMR3 (Mode Register 3) ACK/NACK control bit definitions.

ICMR3 controls ACK/NACK transmission during receive operations. The key field is ACKBT (Acknowledgement Bit), which determines whether the controller sends an ACK or NACK after each received byte.

8.105.3.11 ACK/NACK Signalling Rules (I2C Specification)

- Set ACKBT=0 (clear bit) to send ACK: instructs the peripheral to continue transmitting; used for all bytes except the last in a read transfer.
- Set ACKBT=1 (set bit) to send NACK: instructs the peripheral to stop transmitting; must be applied before clocking in the final byte so that the NACK is sent automatically at the correct clock edge.

8.105.3.12 ICMR3 Register Bit Layout (relevant bits)

Bit	Name	Function
4	ACKWP	ACKBT write protect (1=writable)
3	ACKBT	ACK/NACK bit to transmit

Invariant

k_riic_icmr3_ackbt_pos == 3 matches the RX72N hardware bit position for ACKBT in ICMR3; changing this breaks ACK/NACK control

k_riic_icmr3_ackbt_mask == (1U << 3) == 0x08; derived from pos

k_riic_icmr3_ackwp_pos == 4 matches the RX72N hardware bit position for ACKWP; changing this silently breaks ACKBT writes

k_riic_icmr3_ackwp_mask == (1U << 4) == 0x10; derived from pos

Note

NACK must be sent before reading the last byte to signal to the peripheral that the transfer is complete per I2C specification.

ACKWP=1 is REQUIRED before any ACKBT write. Without it, writes to ACKBT are silently ignored – the controller keeps ACKing every byte, the peripheral never sees NACK, and STOP times out with the bus locked. ACKWP=1 is set in [riic_init\(\)](#) as part of k_riic_icmr3_init.

```
// Prepare to NACK the last byte in a read transfer (ACKWP must already be 1)
volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_0);
riic->icmr3 |= k_riic_icmr3_ackbt_mask; // Set ACKBT=1 -> send NACK
```

```
// Re-enable ACK for subsequent transfers
riic->icmr3 &= (uint8_t)~k_riic_icmr3_ackbt_mask; // Set ACKBT=0 -> send ACK
```

See also

- [internal_read_byte\(\)](#) Uses these constants to control ACK/NACK output
- [riic_read\(\)](#) Orchestrates NACK signalling for multi-byte reads

Since

Version 1.0.0

Enumerator

<code>k_riic_icmr3_ackbt_pos</code>	ACKBT bit position (bit 3)
<code>k_riic_icmr3_ackbt_mask</code>	ACKBT bit mask (0x08)
<code>k_riic_icmr3_ackwp_pos</code>	ACKWP bit position (bit 4)
<code>k_riic_icmr3_ackwp_mask</code>	ACKWP bit mask (0x10)
<code>k_riic_icmr3_wait_pos</code>	WAIT bit position (bit 6)
<code>k_riic_icmr3_wait_mask</code>	WAIT bit mask (0x40)
<code>k_riic_icmr3_init</code>	ICMR3 init: ACKWP=1, ACKBT=0

Definition at line 504 of file [riic.c](#).

```

00504      : uint8_t {
00505  k_riic_icmr3_ackbt_pos = 3,          /**< ACKBT bit position (bit 3) */
00506  k_riic_icmr3_ackbt_mask = (1U « k_riic_icmr3_ackbt_pos), /**< ACKBT bit mask (0x08) */
00507  k_riic_icmr3_ackwp_pos = 4,          /**< ACKWP bit position (bit 4) */
00508  k_riic_icmr3_ackwp_mask = (1U « k_riic_icmr3_ackwp_pos), /**< ACKWP bit mask (0x10) */
00509  k_riic_icmr3_wait_pos = 6,           /**< WAIT bit position (bit 6) */
00510  k_riic_icmr3_wait_mask = (1U « k_riic_icmr3_wait_pos), /**< WAIT bit mask (0x40) */
00511  k_riic_icmr3_init      = k_riic_icmr3_ackwp_mask,      /**< ICMR3 init: ACKWP=1, ACKBT=0 */
00512 } riic_icmr3_bits_t;

```

8.105.3.13 riic`module`stop`bits`

enum [riic_module_stop_bits_t](#) : uint8_t

RIIC module stop bit positions in MSTPCRB (RIIC0/1) and MSTPCRC (RIIC2).

The RX72N uses a module stop mechanism to reduce power consumption. RIIC0 and RIIC1 module stop bits reside in MSTPCRB; RIIC2 resides in MSTPCRC (manual Ch11, p408-410). RIIC channels must be enabled (bit cleared) before use.

Module Enable Procedure

1. Unlock register protection: `PRCR = k_rx_prcr_unlock_prc1_prc3`
2. RIIC0: `MSTPCRB &= ~(1 << k_riic_mstpb_riic0)` RIIC1: `MSTPCRB &= ~(1 << k_riic_mstpb_riic1)` RIIC2: `MSTPCRC &= ~(1 << k_riic_mstpc_riic2)`
3. Lock register protection: `PRCR = k_rx_prcr_lock`

See also

`rx_register_protection.h` for PRCR register access

Enumerator

<code>k_riic_mstpb_riic0</code>	MSTPCRB bit 21: RIIC0 module stop control (manual p408)
<code>k_riic_mstpb_riic1</code>	MSTPCRB bit 20: RIIC1 module stop control (manual p408)
<code>k_riic_mstpc_riic2</code>	MSTPCRC bit 17: RIIC2 module stop control (manual p410)
<code>k_riic_mstpb_bit_value</code>	Single bit value for bit manipulation

Definition at line 310 of file `riic.c`.

```
00310      : uint8_t {
00311  k_riic_mstpb_riic0    = 21, /**< MSTPCRB bit 21: RIIC0 module stop control (manual p408) */
00312  k_riic_mstpb_riic1    = 20, /**< MSTPCRB bit 20: RIIC1 module stop control (manual p408) */
00313  k_riic_mstpc_riic2    = 17, /**< MSTPCRC bit 17: RIIC2 module stop control (manual p410) */
00314  k_riic_mstpb_bit_value = 1,  /**< Single bit value for bit manipulation */
00315 } riic_module_stop_bits_t;
```

8.105.3.14 riic peripheral reg constants

enum `riic_peripheral_reg_constants_t` : `uint8_t`

Register bit-field constants for peripheral mode configuration.

Bit field values used when configuring RIIC for peripheral (device) mode. In peripheral mode the RX72N responds to an external I2C controller (RPi5). All values fit in `uint8_t` since they are 8-bit register constants.

See also

`riic_init_peripheral()`

Since

Version 1.0.0

Enumerator

<code>k_riic_icmr1_peripheral_7bit</code>	CKS=0, MS=0 (peripheral mode), 7-bit
<code>k_riic_icser_sar0e</code>	ICSER bit 0: enable SAR0 address matching
<code>k_riic_sarl_addr_shift</code>	7-bit addr stored in SARL[7:1]
<code>k_riic_saru0_7bit</code>	SARU0 = 0 for 7-bit addressing

Definition at line 2801 of file `riic.c`.

```
02801      : uint8_t {
02802  k_riic_icmr1_peripheral_7bit = 0x00, /**< CKS=0, MS=0 (peripheral mode), 7-bit */
02803  k_riic_icser_sar0e          = 0x01, /**< ICSEr bit 0: enable SAR0 address matching */
02804  k_riic_sarl_addr_shift      = 1,    /**< 7-bit addr stored in SARL[7:1] */
02805  k_riic_saru0_7bit           = 0x00, /**< SARU0 = 0 for 7-bit addressing */
02806 } riic_peripheral_reg_constants_t;
```

8.105.3.15 riic`peripheral`size`constants`

```
enum riic_peripheral_size_constants_t : uint16_t
```

Size and timeout constants for peripheral mode operations.

Timeout values for peripheral mode read/write. `k_riic_peripheral_transfer_limit` is declared in [hardware.h](#) (public API) so callers can statically size their buffers; only the timeout lives here.

See also

```
riic_peripheral_read()
riic_peripheral_write()
```

Since

Version 1.0.0

Enumerator

<code>k_riic_periph_timeout_us</code>	1ms timeout for TDRE poll in write
---------------------------------------	------------------------------------

Definition at line 2821 of file [riic.c](#).

```
02821      : uint16_t {
02822 k_riic_periph_timeout_us = 1000, /**< 1ms timeout for TDRE poll in write */
02823 } riic_peripheral_size_constants_t;
```

8.105.3.16 riic`recovery`constants`

```
enum riic_recovery_constants_t : uint32_t
```

Bit-bang bus-recovery timing and ODR register addresses.

When a peripheral on the bus holds SDA/SCL low, IICRST alone cannot clear a stuck BBSY because the peripheral reset does not toggle external SCL edges. [internal_riic_bit_bang_recover\(\)](#) temporarily drops PMR, drives SCL low-high 9 times to walk any stuck peripheral through a byte + ACK, generates a manual STOP, then hands the pads back to the RIIC peripheral.

ODR0 register addresses per port (RX72N HW manual Table 22.2): Port 2 ODR0 @ 0x0008C084 – pin pair bits (P20 -> bit1, P21 -> bit3)

Enumerator

<code>k_riic_recov_cycles</code>	9 SCL edges: flush one byte + ACK
<code>k_riic_recov_half_us</code>	5 us half-period -> ~100 kHz SCL
<code>k_riic_recov_cpu_mhz</code>	ICLK frequency for busy-wait math
<code>k_riic_recov_stop_us</code>	STOP condition SDA-low-then-high delay
<code>k_riic_port2_odr0_addr</code>	Port 2 ODR0 (P20/P21 N-ch open-drain)

Definition at line 586 of file [riic.c](#).

```
00586      : uint32_t {
00587 k_riic_recov_cycles    = 9U,      /**< 9 SCL edges: flush one byte + ACK */
00588 k_riic_recov_half_us  = 5U,      /**< 5 us half-period -> ~100 kHz SCL */
00589 k_riic_recov_cpu_mhz  = 240U,    /**< ICLK frequency for busy-wait math */
00590 k_riic_recov_stop_us  = 5U,      /**< STOP condition SDA-low-then-high delay */
00591 k_riic_port2_odr0_addr = 0x0008C084U, /**< Port 2 ODR0 (P20/P21 N-ch open-drain) */
00592 } riic_recovery_constants_t;
```

8.105.4 Function Documentation

8.105.4.1 internal_calculate_bit_rate()

```
rx_err_t internal_calculate_bit_rate (
    const uint32_t frequency_hz,
    uint8_t * icbtl,
    uint8_t * icbrh,
    uint8_t * cks_field) [static]
```

Calculate RIIC bit rate register values for target frequency.

Computes the ICBRL and ICBRH register values to achieve the desired I2C bus frequency. Uses a simplified calculation assuming 50% duty cycle (equal high and low periods).

8.105.4.2 Calculation Algorithm

The RX72N RIIC bit rate formula (from Chapter 42):

$$f_{SCL} = \frac{f_{PCLKB}}{(ICBRL + 1) + (ICBRH + 1) + \text{sync}}$$

Simplified for 50% duty cycle (ICBRL = ICBRH):

$$\text{divisor} = \frac{f_{PCLKB}}{f_{target} \times 3}$$

8.105.4.3 Example Calculations (PCLKB = 60 MHz)

Target	Calculation	Divisor	Actual Rate
100 kHz	60M / (100k x 3)	200	100.0 kHz
400 kHz	60M / (400k x 3)	50	400.0 kHz
1 MHz	60M / (1M x 3)	20	1.00 MHz

Precondition

```
icbtl != nullptr
icbrh != nullptr
frequency_hz is a supported I2C frequency
```

Postcondition

```
*icbtl contains the low-period divisor (1-255)
*icbrh contains the high-period divisor (1-255)
*icbtl == *icbrh (50% duty cycle)
```


Note

Uses PCLKB (60 MHz) as the source clock per RX72N specification.
For asymmetric duty cycle, manually set ICBRL != ICBRH after init.

Warning

Frequencies outside 100 kHz - 1 MHz may produce out-of-range divisors.

See also

[riic_init\(\)](#) Uses this function during channel initialization
[k_pclkb_hz](#) PCLKB frequency constant from [hardware.h](#)

Definition at line 1042 of file [riic.c](#).

```

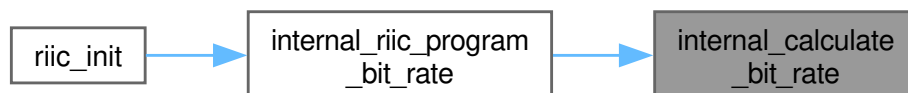
01046 {
01047     if (icbrl == nullptr || icbrh == nullptr || cks_field == nullptr) {
01048         return k_rx_err_null_ptr;
01049     }
01050
01051     /* Look up the HUM Table 42.5 row for the requested rate. Only the three
01052      * discrete rates the riic_channel_freq_t enum accepts are supported; we
01053      * do not compute CKS/ICBRH/ICBRL from a formula because the formula in
01054      * the HUM depends on SCL rise/fall times and the table values already
01055      * bake in the standard-mode / fast-mode / fast-mode-plus tr/tf budgets. */
01056     if (frequency_hz == (uint32_t)k_riic_freq_100khz) {
01057         *cks_field = (uint8_t)k_riic_cks_100khz;
01058         *icbrh     = (uint8_t)k_riic_icbrh_100k;
01059         *icbrl     = (uint8_t)k_riic_icbrl_100k;
01060         return k_rx_ok;
01061     }
01062     if (frequency_hz == (uint32_t)k_riic_freq_400khz) {
01063         *cks_field = (uint8_t)k_riic_cks_400khz;
01064         *icbrh     = (uint8_t)k_riic_icbrh_400k;
01065         *icbrl     = (uint8_t)k_riic_icbrl_400k;
01066         return k_rx_ok;
01067     }
01068     if (frequency_hz == (uint32_t)k_riic_freq_1mhz) {
01069         *cks_field = (uint8_t)k_riic_cks_1mhz;
01070         *icbrh     = (uint8_t)k_riic_icbrh_1m;
01071         *icbrl     = (uint8_t)k_riic_icbrl_1m;
01072         return k_rx_ok;
01073     }
01074
01075     rx_log_error(s_tag, "Invalid frequency (expected 100 kHz, 400 kHz, or 1 MHz)");
01076     return k_rx_err_invalid_arg;
01077 }

```

References [k_riic_cks_100khz](#), [k_riic_cks_1mhz](#), [k_riic_cks_400khz](#), [k_riic_freq_100khz](#), [k_riic_freq_1mhz](#), [k_riic_freq_400khz](#), [k_riic_icbrh_100k](#), [k_riic_icbrh_1m](#), [k_riic_icbrh_400k](#), [k_riic_icbrl_100k](#), [k_riic_icbrl_1m](#), [k_riic_icbrl_400k](#), and [s_tag](#).

Referenced by [internal_riic_program_bit_rate\(\)](#).

Here is the caller graph for this function:



8.105.4.4 `internal_get_riic_base()`

```
volatile rx_riic_regs_t * internal_get_riic_base (
    const uint8_t channel) [static]
```

Get RIIC channel register base address from channel number.

Maps an RIIC channel number (0-2) to its corresponding hardware register base address using a compile-time-dispatched switch statement. This function is the single point of truth for the channel-to-address mapping and provides a clean abstraction between the logical channel index used throughout the driver and the physical memory-mapped peripheral addresses defined by the RX72N hardware.

8.105.4.5 Channel-to-Address Mapping

The three RIIC channels are located at fixed, evenly spaced addresses in the RX72N peripheral address space (0x00088300, 0x00088320, 0x00088340). Each block is 32 bytes in size and contains all control, status, bit-rate, and data registers for one channel.

8.105.4.6 Memory Map

Channel	Accessor	Base Address	Register Block Size
0	riic0()	0x00088300	32 bytes
1	riic1()	0x00088320	32 bytes
2	riic2()	0x00088340	32 bytes

8.105.4.7 Why Return NULL for Invalid Channel

Returning nullptr for an out-of-range channel allows every caller to perform a single null-pointer check with `RX_CHECK_NULL_PTR` rather than duplicating range validation logic. This pattern is consistent with [rx_port_get_base\(\)](#) in the GPIO driver.

Precondition

- channel must be less than `k_riic_max_channels` (3); values ≥ 3 yield nullptr
- caller must have verified that the corresponding RIIC module stop bit in `MSTPCRB` has been cleared (module enabled) before using the returned pointer

Postcondition

- returned pointer is non-NULL if and only if channel is 0, 1, or 2
- returned pointer, when non-NULL, is aligned to the RIIC peripheral register block boundary as required by the RX72N hardware specification

Note

- This function does NOT validate the channel - the caller is responsible for ensuring the value is in range before calling
- The return type is volatile to prevent the compiler from optimizing away hardware register reads or writes through the returned pointer

Performance:

O(1) - simple switch statement with a direct return per case

Example:

```
// Look up the register base for RIIC channel 1 and send a START condition
volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_1);
if (riic == nullptr) {
    return k_rx_err_invalid_arg;
}
riic->iccr2 |= k_riic_iccr2_st; // Issue START condition
```

See also

[riic0\(\)](#) Inline accessor for RIIC channel 0 base address

[riic1\(\)](#) Inline accessor for RIIC channel 1 base address

[riic2\(\)](#) Inline accessor for RIIC channel 2 base address

[k_riic_max_channels](#) Upper bound on valid channel numbers

Since

Version 1.0.0

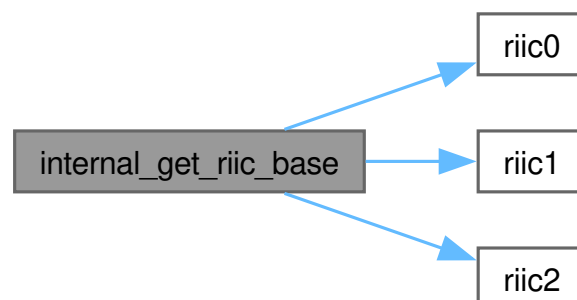
Definition at line 979 of file [riic.c](#).

```
00980 {
00981     switch (channel) {
00982     case k_riic_channel_0: {
00983         return riic0();
00984     }
00985     case k_riic_channel_1: {
00986         return riic1();
00987     }
00988     case k_riic_channel_2: {
00989         return riic2();
00990     }
00991     default: {
00992         return nullptr;
00993     }
00994 }
00995 }
```

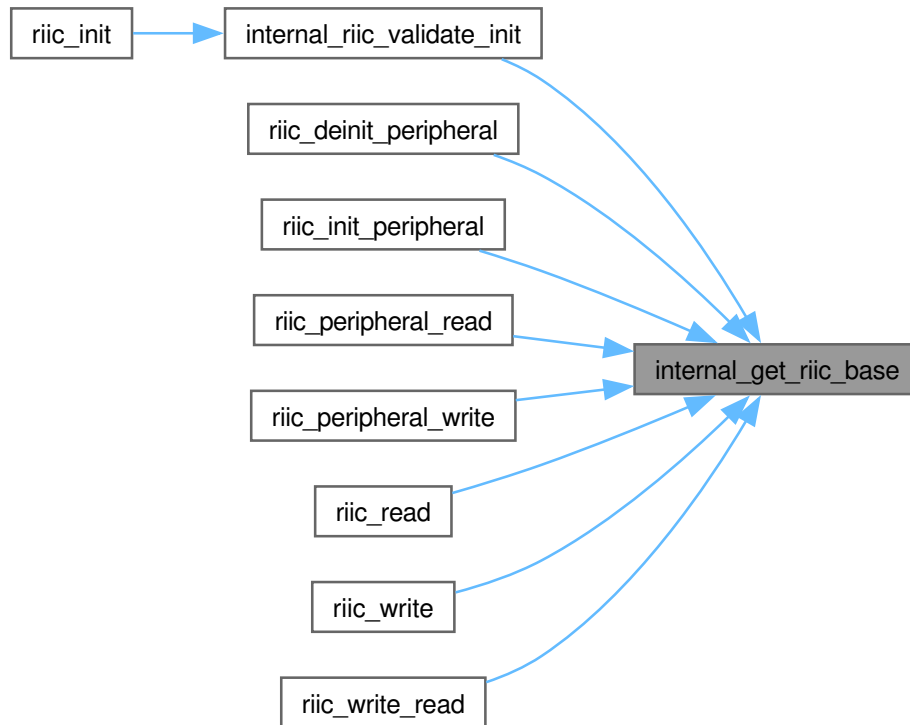
References [k_riic_channel_0](#), [k_riic_channel_1](#), [k_riic_channel_2](#), [riic0\(\)](#), [riic1\(\)](#), and [riic2\(\)](#).

Referenced by [internal_riic_validate_init\(\)](#), [riic_deinit_peripheral\(\)](#), [riic_init_peripheral\(\)](#), [riic_peripheral_read\(\)](#), [riic_peripheral_write\(\)](#), [riic_read\(\)](#), [riic_write\(\)](#), and [riic_write_read\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.8 internal_read_byte()

```

rx_err_t internal_read_byte (
    volatile rx_riic_regs_t * riic,
    uint8_t * data,
    const bool send_ack,
    const bool last_byte) [static]

```

Receive single byte from I2C bus.

Reads one byte from the I2C bus and sends ACK or NACK as specified. The ACK/NACK response is configured before reading the byte from the receive register.

8.105.4.9 Reception Sequence

1. Wait for RDRF (Receive Data Register Full) flag
2. Configure ACK/NACK response in ICMR3.ACKBT
3. Read byte from ICDRR (Receive Data Register)
4. Hardware generates ACK/NACK on 9th clock
5. RDRF flag is cleared by read

8.105.4.10 ACK vs NACK Decision

Situation	Response	ACKBT	Purpose
More bytes to read	ACK	0	Continue transfer
Last byte	NACK	1	Signal end of read

8.105.4.11 I2C Read Byte Timing

```

SCL ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
    || || || || || || || ||
    ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
    D7 D6 D5 D4 D3 D2 D1 D0 ACK/NACK
SDA -----+
    Peripheral drives    +- Controller drives ACK/NACK

```

Precondition

```

riic != nullptr
data != nullptr
Controller is in receive mode (address + read bit sent)

```

Postcondition

```

*data contains the received byte
ACK or NACK has been sent to peripheral
RDRF flag is cleared

```

Note

ACK/NACK is configured BEFORE reading ICDRR.
 NACK on last byte is REQUIRED per I2C specification.

Warning

Failure to send NACK on last byte may cause peripheral to continue driving data, corrupting next transaction.

Example: Reading Multiple Bytes

```

for (uint16_t i = 0; i < length; i++) {
    bool send_ack = (i < length - 1); // ACK all except last
    err = internal_read_byte(riic, &buffer[i], send_ack);
    if (err != k_rx_ok) break;
}

```

See also

[internal_write_byte\(\)](#) Companion function for transmitting bytes

Definition at line 1713 of file [riic.c](#).

```

01717 {
01718     /* Wait for receive data full */
01719     uint32_t timeout = k_riic_timeout_us;
01720
01721     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01722     RX_CHECK_NULL_PTR(data, s_tag, "data pointer is nullptr");
01723
01724     while (!(riic->icsr2 & k_riic_icsr2_rdrf) && timeout > k_riic_timeout_zero) {
01725         timeout--;
01726     }
01727
01728     if (timeout == k_riic_timeout_zero) {
01729         rx_log_error(s_tag, "Read timeout");
01730         return k_rx_err_timeout;
01731     }
01732
01733     /* Configure ACK/NACK for the byte we are about to release from ICDRR.
01734      * The peripheral clock-stretches SCL between the 8th clock and the ACK
01735      * clock while RDRF is set, so setting ACKBT now takes effect on the
01736      * upcoming ACK/NACK bit. */
01737     if (!send_ack) {
01738         riic->icmr3 |= k_riic_icmr3_ackbt_mask; /* ACKBT = 1 (NACK) */
01739     } else {
01740         riic->icmr3 &= ~(uint8_t)k_riic_icmr3_ackbt_mask; /* ACKBT = 0 (ACK) */
01741     }
01742
01743     /* For the final byte of a receive, queue SP before reading ICDRR so the
01744      * RIIC generates NACK + STOP atomically after the 9th clock. Per RX72N
01745      * HW manual Figure 42.10 the SP request must be set between RDRF and
01746      * the final ICDRR read so the STOP is chained to the 9th clock's NACK.
01747      *
01748      * Do NOT queue SP after the final ICDRR read: the 9th clock will have
01749      * already fired, SCL is released, and a fresh SP produces no STOP edge.
01750      * That was the bug behind "Stop condition timeout" on 1-byte reads. */
01751     if (last_byte) {
01752         riic->icsr2 &= (uint8_t)~(uint8_t)k_riic_icsr2_stop;
01753         riic->iccr2 |= k_riic_iccr2_sp;
01754     }
01755
01756     /* Read data -- releases SCL for the ACK/NACK clock. For the last byte,
01757      * SP was queued above so hardware generates STOP immediately after the
01758      * NACK on the 9th clock.
01759      *
01760      * Do NOT instead issue SP AFTER this ICDRR read. Per RX72N HW manual
01761      * Figure 42.10, once the 9th clock has fired the peripheral drops the
01762      * auto clock-stretch and releases SCL; the STOP condition must be
01763      * chained into that release, not requested after the bus has already
01764      * returned to idle. Calling internal_send_stop() after the read will
01765      * time out waiting for ICSR2.STOP to assert because no STOP edge is
01766      * ever produced. */
01767     *data = riic->icdrr;
01768
01769     if (last_byte) {
01770         timeout = k_riic_timeout_us;
01771         while (!(riic->icsr2 & k_riic_icsr2_stop) && timeout > k_riic_timeout_zero) {
01772             timeout--;
01773         }
01774         if (timeout == k_riic_timeout_zero) {
01775             rx_log_error(s_tag, "STOP after final ICDRR timeout");
01776             return k_rx_err_timeout;
01777         }
01778         /* Clear STOP + NACKF so the next transaction starts from a clean
01779          * status register. ICMR3.ACKBT left at 1 is harmless -- it gets
01780          * re-written by the next internal_read_byte. */
01781         riic->icsr2 &= (uint8_t)~(uint8_t)(k_riic_icsr2_stop | k_riic_icsr2_nackf);
01782     }
01783
01784     return k_rx_ok;
01785 }

```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icdrr](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icsr2](#), [k_riic_iccr2_sp](#), [k_riic_icmr3_ackbt_mask](#), [k_riic_icsr2_nackf](#), [k_riic_icsr2_rdrf](#), [k_riic_icsr2_stop](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), and [s_tag](#).

8.105.4.12 internal`riic`bit`bang`handback()

```
void internal_riic_bit_bang_handback (
    uint8_t channel) [static]
```

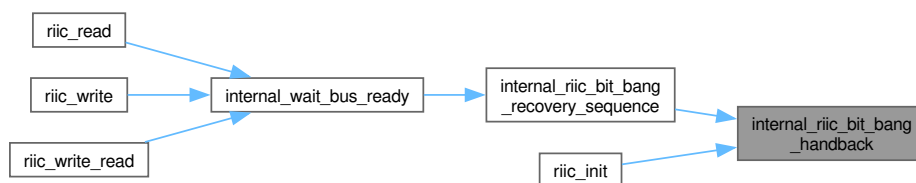
Definition at line 840 of file [riic.c](#).

```
00841 {
00842     if (channel >= (uint8_t)k_riic_max_channels) {
00843         return;
00844     }
00845     const riic_recovery_pins_t* const pins = &k_riic_recovery_pins[channel];
00846     if (pins->port_base == 0U) {
00847         return;
00848     }
00849     volatile rx_port_regs_t* const port = (volatile rx_port_regs_t*)pins->port_base;
00850     const uint8_t scl_mask = (uint8_t)(1U « pins->scl_bit);
00851     const uint8_t sda_mask = (uint8_t)(1U « pins->sda_bit);
00852     port->pmr |= (uint8_t)(scl_mask | sda_mask);
00853 }
00854 }
```

References [k_riic_max_channels](#), [k_riic_recovery_pins](#), [rx_port_regs_t::pmr](#), [riic_recovery_pins_t::port_base](#), [riic_recovery_pins_t::scl_bit](#), and [riic_recovery_pins_t::sda_bit](#).

Referenced by [internal_riic_bit_bang_recovery_sequence\(\)](#), and [riic_init\(\)](#).

Here is the caller graph for this function:



8.105.4.13 internal`riic`bit`bang`recover()

```
void internal_riic_bit_bang_recover (
    uint8_t channel) [static]
```

Bit-bang 9 SCL edges + manual STOP to recover an externally-stuck bus.

Called from `internal_wait_bus_ready` when BBSY stays latched even after an IICRST pulse. That combination means an I2C peripheral on the bus (BNO055 or BMP280 in the STAR system) got interrupted mid-byte – e.g. the RX72N reset during a previous transaction – and is still holding SDA low waiting to finish sending the remaining bits of its byte + ACK.

IICRST only resets the RIIC controller side; it does not toggle SCL on the wire, so it cannot unstick an external peripheral. The only software path that clears this is to physically wiggle SCL from the MCU pad – which requires handing the pad from RIIC back to GPIO long enough to clock out a full byte + ACK + STOP.

Sequence:

1. Drop PMR -> pads are GPIO again, RIIC loses them.
2. Enable N-ch open-drain so we cannot drive-fight a peripheral still holding the line low.
3. Drive both lines high, set as output. External pull-ups dominate.

4. 9x { SCL low; wait; SCL high; wait } – one byte + ACK worth of clocks, enough to clock out any peripheral stuck mid-byte.
5. Manual STOP: SDA low -> high with SCL high – tells any peripheral still listening ”transaction over, idle the bus”.
6. Tristate pads, clear open-drain. Leave PMR=0. The caller raises PMR back to 1 AFTER it has reset the RIIC peripheral (IICRST + SOWP dance), so the peripheral hands the pads back already in a ”driving high” state. Raising PMR while the peripheral still has leftover SCLO=SDAO=0 from the previous transaction would glitch the bus low again immediately.

Only RIIC channels with a populated entry in `k_riic_recovery_pins[]` are handled; other channels return silently.

Precondition

MPC PFS values for SCL/SDA were set during channel init and have not been disturbed (we only toggle PMR, not PFS).

Postcondition

PMR for SCL/SDA is CLEARED (pads stay in GPIO mode, tristated). Caller must reset the RIIC peripheral and then raise PMR.

PDR bits for SCL/SDA are cleared (tristated).

NASA Power of 10 Compliance

- Rule 2: bounded loop (`k_riic_recov_cycles = 9`)
- Rule 5: channel bounds check + `port_base != 0` validation

See also

[internal_riic_bit_bang_handback\(\)](#) Companion that raises PMR once the peripheral has been reset to a clean SCLO=SDAO=1 state.

Definition at line 717 of file `riic.c`.

```

00718 {
00719     if (channel >= (uint8_t)k_riic_max_channels) {
00720         return;
00721     }
00722     const riic_recovery_pins_t* const pins = &k_riic_recovery_pins[channel];
00723     if (pins->port_base == 0U) {
00724         return; /* No recovery table entry for this channel. */
00725     }
00726     volatile rx_port_regs_t* const port = (volatile rx_port_regs_t*)pins->port_base;
00727     volatile uint8_t* const odr0 = (volatile uint8_t*)pins->odr0_addr;
00728     const uint8_t scl_mask = (uint8_t)(1U << pins->scl_bit);
00729     const uint8_t sda_mask = (uint8_t)(1U << pins->sda_bit);
00730     const uint8_t both = (uint8_t)(scl_mask | sda_mask);
00731     /* Hand pads back to GPIO so we can bit-bang. PFS values are preserved
00732      * (we only toggle PMR, not PFS/PWPR). */
00733     port->pmr &= (uint8_t)~both;
00734     /* N-ch open-drain: avoid push-pull fight with a peripheral still holding low. */
00735     *odr0 |= pins->odr_mask;
00736     /* Drive both high, then set as output. Pull-ups bring the bus up. */
00737     port->podr |= both;
00738     port->pdr |= both;
00739 }

```



```

00745 for (uint16_t i = 0; i < (uint16_t)k_riic_recov_cycles; i++) {
00746     port->podr &= (uint8_t)~scl_mask;
00747     internal_riic_busy_wait_us((uint16_t)k_riic_recov_half_us, (uint16_t)k_riic_recov_cpu_mhz);
00748     port->podr |= scl_mask;
00749     internal_riic_busy_wait_us((uint16_t)k_riic_recov_half_us, (uint16_t)k_riic_recov_cpu_mhz);
00750 }
00751
00752 /* Manual STOP: SDA low (SCL already high), then SDA high. */
00753 port->podr &= (uint8_t)~sda_mask;
00754 internal_riic_busy_wait_us((uint16_t)k_riic_recov_stop_us, (uint16_t)k_riic_recov_cpu_mhz);
00755 port->podr |= sda_mask;
00756 internal_riic_busy_wait_us((uint16_t)k_riic_recov_stop_us, (uint16_t)k_riic_recov_cpu_mhz);
00757
00758 /* Tristate pads + clear open-drain. DO NOT raise PMR here -- see
00759  * internal_riic_bit_bang_handback() for the reason. */
00760 port->pdr &= (uint8_t)~both;
00761 *odr0 &= (uint8_t)~pins->odr_mask;
00762 }

```

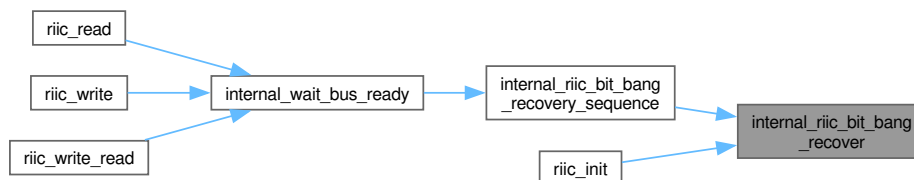
References [internal_riic_busy_wait_us\(\)](#), [k_riic_max_channels](#), [k_riic_recov_cpu_mhz](#), [k_riic_recov_cycles](#), [k_riic_recov_half_us](#), [k_riic_recov_stop_us](#), [k_riic_recovery_pins](#), [riic_recovery_pins_t::odr0_addr](#), [riic_recovery_pins_t::odr_mask](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_port_regs_t::podr](#), [riic_recovery_pins_t::port_base](#), [riic_recovery_pins_t::scl_bit](#), and [riic_recovery_pins_t::sda_bit](#).

Referenced by [internal_riic_bit_bang_recovery_sequence\(\)](#), and [riic_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.14 internal_riic`bit`bang`recovery`sequence()

```

void internal_riic_bit_bang_recovery_sequence (
    volatile rx_riic_regs_t *const riic,
    const riic_saved_cfg_t *const saved) [static]

```

Bit-bang last-resort bus recovery + peripheral reset + handback.

Walks the bus through 9 SCL edges + manual STOP via GPIO bit-bang to release a peripheral holding SDA low from an aborted transaction across an MCU reboot (verified on BNO055 against the production STAR PCB). Critical ordering preserved: bit-bang leaves PMR=0 (pads still GPIO), peripheral reset runs while pads are GPIO, then handback raises PMR=1 with the SOWP dance already done so the lines stay high-Z.

Parameters

in	riic	RIIC peripheral base (must be non-null)
in	saved	Pre-recovery config snapshot (used for restore)

Precondition

riic and saved non-null; bus has refused to clear via IICRST alone

Postcondition

Pads handed back to the peripheral with config restored

Since

Version 1.0.0

Definition at line 1289 of file [riic.c](#).

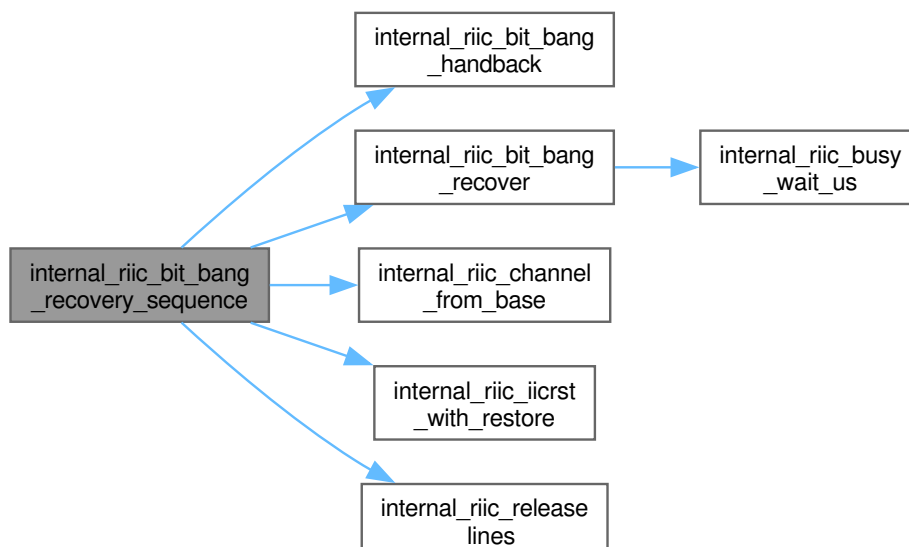
```

01291 {
01292     const uint8_t channel = internal_riic_channel_from_base(riic);
01293     rx_log_warn(s_tag, "IICRST insufficient - bit-bang bus recovery");
01294     internal_riic_bit_bang_recover(channel); /* leaves PMR=0 */
01295
01296     internal_riic_iicrst_with_restore(riic, saved);
01297     internal_riic_release_lines(riic);
01298
01299     internal_riic_bit_bang_handback(channel);
01300
01301     if (channel == (uint8_t)k_riic_channel_1) {
01302         rx_log_debug_val(s_tag,
01303             " post-recover PIDR=0x",
01304             ((volatile const rx_port_regs_t*)0x0008C002U)->pidr);
01305     }
01306 }
```

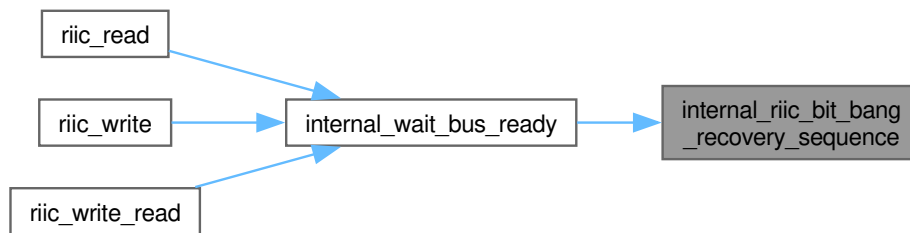
References [internal_riic_bit_bang_handback\(\)](#), [internal_riic_bit_bang_recover\(\)](#), [internal_riic_channel_from_base\(\)](#), [internal_riic_iicrst_with_restore\(\)](#), [internal_riic_release_lines\(\)](#), [k_riic_channel_1](#), and [s_tag](#).

Referenced by [internal_wait_bus_ready\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.15 internal`riic`busy`wait`us()

```

void internal_riic_busy_wait_us (
    uint16_t us,
    uint16_t cpu_mhz)  [inline], [static]
  
```

Cycle-counted busy-wait for bit-bang timing.

Volatile counter loop calibrated for the current ICLK frequency. Mirrors hardware_init.c::internal_↔ busy_wait_us – duplicated here so [riic.c](#) has no cross-module dependency on src/ helpers.

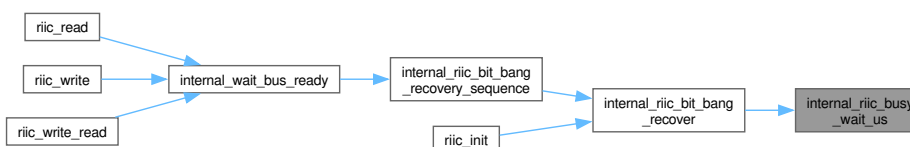
Definition at line 660 of file [riic.c](#).

```

00661 {
00662     volatile uint32_t cycles = (uint32_t)us * (uint32_t)cpu_mhz;
00663     while (cycles > 0U) {
00664         cycles--;
00665     }
00666 }
  
```

Referenced by [internal_riic_bit_bang_recover\(\)](#).

Here is the caller graph for this function:



8.105.4.16 internal`riic`channel`from`base()

```

uint8_t internal_riic_channel_from_base (
    const volatile rx_riic_regs_t * riic)  [static]
  
```

Resolve a RIIC register pointer to its channel number.

Compares the pointer against the three RIIC base addresses. Returns k_riic_max_channels (sentinel) if the pointer does not match any channel, which callers treat as "skip bit-bang recovery".

Definition at line 636 of file [riic.c](#).

```

00637 {
00638     const uintptr_t base = (uintptr_t)riic;
00639     if (base == (uintptr_t)k_riic0_base_addr) {
00640         return (uint8_t)k_riic_channel_0;
00641     }
  
```

```

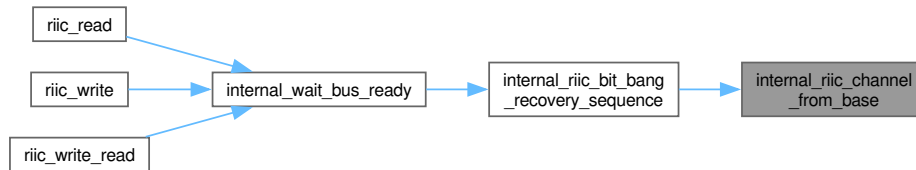
00642 if (base == (uintptr_t)k_riic1_base_addr) {
00643     return (uint8_t)k_riic_channel_1;
00644 }
00645 if (base == (uintptr_t)k_riic2_base_addr) {
00646     return (uint8_t)k_riic_channel_2;
00647 }
00648 return (uint8_t)k_riic_max_channels;
00649 }

```

References [k_riic0_base_addr](#), [k_riic1_base_addr](#), [k_riic2_base_addr](#), [k_riic_channel_0](#), [k_riic_channel_1](#), [k_riic_channel_2](#), and [k_riic_max_channels](#).

Referenced by [internal_riic_bit_bang_recovery_sequence\(\)](#).

Here is the caller graph for this function:



8.105.4.17 internal_riic_force_fsm_reset()

```

void internal_riic_force_fsm_reset (
    volatile rx_riic_regs_t * riic) [static]

```

Raise PMR so the RIIC peripheral takes the pads back.

Companion to [internal_riic_bit_bang_recover\(\)](#). Call this AFTER the RIIC peripheral has been IICRST-pulsed and put through the SOWP / SCLO / SDAO release dance so its internal output latches are high. Raising PMR while those latches are still 0 would glitch the bus low again immediately and could re-confuse the peripheral we just unstuck.

Precondition

[internal_riic_bit_bang_recover\(\)](#) was just called on the same channel.

The RIIC peripheral has been reset (IICRST cycle + SOWP release).

Postcondition

PMR bits for SCL/SDA are set, pads routed to RIIC peripheral.

Pulse IICRST on a controller-initialized RIIC channel and restore config.

Per RX72N HW manual section 42.2.1, writing 1 to ICCR1.IICRST while ICCR1.ICE=0 performs an internal reset of the RIIC peripheral's FSM without touching the pin mux. That drops any half-completed transaction state – BBSY, MST, TRS, pending SP/RS/ST request bits, NACKF, latched AL, the "dummy read kicked reception" flag – back to idle, which is what we need before a new transaction when an earlier transaction on the same peripheral may have left the FSM partially advanced. The config registers (ICBRH/L, ICMR1..3) are wiped by IICRST, so we restore them from the live values (which were set in [riic_init\(\)](#) and must already be valid since ICE was on before the pulse).

Do NOT substitute this for [riic_init\(\)](#): this helper assumes MSTPCR and MPC/PFS are already set up and that ICE=1 was previously toggled to latch the peripheral out of module-stop. It is a fast pre-transaction FSM scrub, not a cold boot.

Currently unused in the main transaction path – kept as a `[[maybe_unused]]` helper because it is the right escape hatch when the FSM has been observed to get stuck mid-transaction and the bit-bang recovery in [internal_wait_bus_ready\(\)](#) is too heavyweight (it tears down and rebuilds pad direction via PMR, which adds ~100 us versus ~2 us here).

Precondition

ICE was set to 1 earlier (channel previously passed `riic_init`).

Postcondition

ICCR1.ICE = 1, ICCR1.IICRST = 0, ICMR1..3 / ICBRH/L restored, ICSR1 / ICSR2 cleared.

Definition at line 811 of file [riic.c](#).

```

00812 {
00813     const uint8_t saved_icbrh = riic->icbrh;
00814     const uint8_t saved_icbhl = riic->icbhl;
00815     const uint8_t saved_icmr1 = riic->icmr1;
00816     const uint8_t saved_icmr2 = riic->icmr2;
00817
00818     riic->iccr1 = k_riic_iccr1_iicrst;
00819     riic->iccr1 = k_riic_register_clear;
00820
00821     riic->icbrh = saved_icbrh;
00822     riic->icbhl = saved_icbhl;
00823     riic->icmr1 = saved_icmr1;
00824     riic->icmr2 = saved_icmr2;
00825     riic->icmr3 = k_riic_icmr3_init;
00826
00827     riic->icsr1 = k_riic_register_clear;
00828     riic->icsr2 = k_riic_register_clear;
00829
00830     riic->iccr1 = k_riic_iccr1_ice;
00831
00832     uint8_t iccr1 = riic->iccr1;
00833     iccr1 &= (uint8_t) ~(uint8_t) k_riic_iccr1_sowp;
00834     iccr1 |= (uint8_t) (k_riic_iccr1_sclo | k_riic_iccr1_sdao);
00835     riic->iccr1 = iccr1;
00836     iccr1 |= (uint8_t) k_riic_iccr1_sowp;
00837     riic->iccr1 = iccr1;
00838 }
```

References [rx_riic_regs_t::icbrh](#), [rx_riic_regs_t::icbhl](#), [rx_riic_regs_t::iccr1](#), [rx_riic_regs_t::icmr1](#), [rx_riic_regs_t::icmr2](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icsr1](#), [rx_riic_regs_t::icsr2](#), [k_riic_iccr1_ice](#), [k_riic_iccr1_iicrst](#), [k_riic_iccr1_sclo](#), [k_riic_iccr1_sdao](#), [k_riic_iccr1_sowp](#), [k_riic_icmr3_init](#), and [k_riic_register_clear](#).

8.105.4.18 `internal_riic_iicrst_with_restore()`

```

void internal_riic_iicrst_with_restore (
    volatile rx\_riic\_regs\_t *const riic,
    const riic\_saved\_cfg\_t *const saved) [static]
```

Pulse IICRST and restore config from a saved snapshot.

Asserts IICRST=1 then deasserts back to configuration state, restores ICBRH/ICBRL/ICMR1/ICMR2 plus ICMR3=`k_riic_icmr3_init` (IICRST also wipes ICMR3, and ACKWP=1 must be restored so reads can NACK the final byte). Clears stale ICSR1/ICSR2 flags before re-enabling ICE=1.

Parameters

in	riic	RIIC peripheral base (must be non-null)
in	saved	Pre-recovery config snapshot

Precondition

[riic](#) and [saved](#) non-null

Postcondition

ICE=1 with saved config and clean status registers

Since

Version 1.0.0

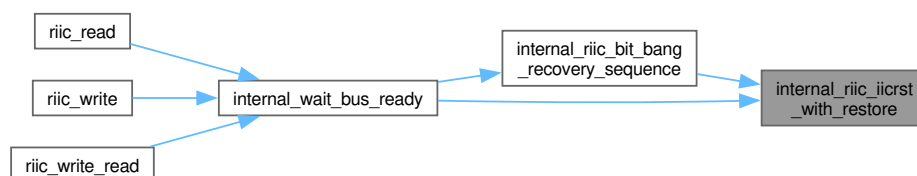
Definition at line 1206 of file [riic.c](#).

```
01208 {
01209     riic->iccr1 = k_riic_iccr1_iicrst;
01210     riic->iccr1 = k_riic_register_clear;
01211
01212     riic->icbrh = saved->icbrh;
01213     riic->icbrl = saved->icbrl;
01214     riic->icmr1 = saved->icmr1;
01215     riic->icmr2 = saved->icmr2;
01216     riic->icmr3 = k_riic_icmr3_init;
01217
01218     riic->icsr1 = k_riic_register_clear;
01219     riic->icsr2 = k_riic_register_clear;
01220
01221     riic->iccr1 = k_riic_iccr1_ice;
01222 }
```

References [riic_saved_cfg_t::icbrh](#), [rx_riic_regs_t::icbrh](#), [riic_saved_cfg_t::icbrl](#), [rx_riic_regs_t::icbrl](#), [rx_riic_regs_t::iccr1](#), [riic_saved_cfg_t::icmr1](#), [rx_riic_regs_t::icmr1](#), [riic_saved_cfg_t::icmr2](#), [rx_riic_regs_t::icmr2](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icsr1](#), [rx_riic_regs_t::icsr2](#), [k_riic_iccr1_ice](#), [k_riic_iccr1_iicrst](#), [k_riic_icmr3_init](#), and [k_riic_register_clear](#).

Referenced by [internal_riic_bit_bang_recovery_sequence\(\)](#), and [internal_wait_bus_ready\(\)](#).

Here is the caller graph for this function:



8.105.4.19 internal_riic_log_pre_recovery()

```
void internal_riic_log_pre_recovery (
    const volatile rx\_riic\_regs\_t *const riic) [static]
```

Log debug-level pre-recovery bus state.

Parameters

in	riic	RIIC peripheral base (must be non-null)
----	------	---

Note

PIDR struct access lets [rx72n_port_regs.h](#) static_asserts catch layout drift at compile time. Compiled out when LOG_LEVEL < k_log_debug.

Since

Version 1.0.0

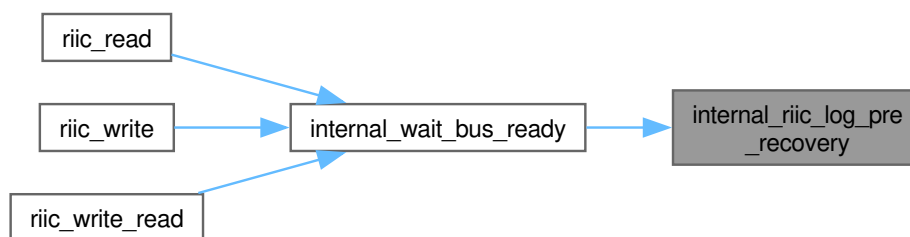
Definition at line 1261 of file [riic.c](#).

```
01262 {
01263   rx_log_debug_val(s_tag, " PIDR =0x", ((volatile const rx_port_regs_t*)0x0008C002U)->pidr);
01264   rx_log_debug_val(s_tag, " ICCR2=0x", riic->iccr2);
01265   rx_log_debug_val(s_tag, " ICSR2=0x", riic->icsr2);
01266 }
```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icsr2](#), and [s_tag](#).

Referenced by [internal_wait_bus_ready\(\)](#).

Here is the caller graph for this function:



8.105.4.20 `internal`riic`poll`bbsy`clear()`

```
bool internal_riic_poll_bbsy_clear (
    volatile rx\_riic\_regs\_t *const riic)  [static]
```

Poll BBSY for up to `k_riic_timeout_us` iterations.

Tight-loop watcher for the Bus Busy bit in ICCR2. Returns true when BBSY clears within the timeout, false otherwise.

Parameters

in	riic	RIIC peripheral base (must be non-null)
----	------	---

Returns

bool true if BBSY observed clear, false on timeout

Precondition

riic non-null

Postcondition

Returns false on timeout

Note

Caller decides recovery action (warn + IICRST or bit-bang).

Since

Version 1.0.0

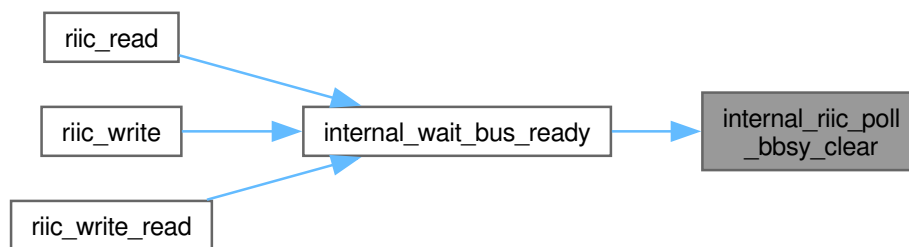
Definition at line 1158 of file [riic.c](#).

```
01159 {
01160     uint32_t timeout = k_riic_timeout_us;
01161     while ((riic->iccr2 & k_riic_iccr2_bbsy) && timeout > k_riic_timeout_zero) {
01162         timeout--;
01163     }
01164     return timeout > k_riic_timeout_zero;
01165 }
```

References [rx_riic_regs_t::iccr2](#), [k_riic_iccr2_bbsy](#), [k_riic_timeout_us](#), and [k_riic_timeout_zero](#).

Referenced by [internal_wait_bus_ready\(\)](#).

Here is the caller graph for this function:



8.105.4.21 internal_riic_program_bit_rate()

```
rx_err_t internal_riic_program_bit_rate (
    volatile rx\_riic\_regs\_t *const riic,
    const uint32_t frequency_hz) [static]
```

Program ICBRL/ICBRH/ICMR1..3 for the requested SCL frequency.

Looks up CKS / ICBRH / ICBRL via [internal_calculate_bit_rate\(\)](#), then writes the bit-rate registers and registers. CKS is in ICMR1[6:4] and must be written alongside MTWP/BCWP/BC in a single ICMR1 write. ICMR3 = ACKWP | 0 – ACKWP=1 is mandatory so [internal_read_byte\(\)](#) can NACK the final received byte (otherwise STOP after every read times out).

Parameters

in	riic	RIIC peripheral base (must be non-null)
in	frequency_hz	Validated SCL frequency

Returns

`rx_err_t` Error code from [internal_calculate_bit_rate\(\)](#)

Precondition

RIIC peripheral is in IICRST or post-IICRST configuration state

Postcondition

ICBRL/ICBRH/ICMR1/ICMR2/ICMR3 programmed

Since

Version 1.0.0

Definition at line 2213 of file [riic.c](#).

```

02215 {
02216     uint8_t      icbrl    = 0;
02217     uint8_t      icbrh    = 0;
02218     uint8_t      cks_field = 0;
02219     const rx_err_t err     = internal_calculate_bit_rate(frequency_hz, &icbrl, &icbrh, &cks_field);
02220     RX_RETURN_ON_ERROR(err, s_tag, "Bit rate calculation failed");
02221
02222     riic->icbrl = icbrl;
02223     riic->icbrh = icbrh;
02224     riic->icmr1 = (uint8_t)(k_riic_icmr1_controller_7bit | (uint8_t)(cks_field « k_riic_cks_shift));
02225     riic->icmr2 = k_riic_icmr2_default;
02226     riic->icmr3 = k_riic_icmr3_init;
02227
02228     return k_rx_ok;
02229 }
```

References [rx_riic_regs_t::icbrh](#), [rx_riic_regs_t::icbrl](#), [rx_riic_regs_t::icmr1](#), [rx_riic_regs_t::icmr2](#), [rx_riic_regs_t::icmr3](#), [internal_calculate_bit_rate\(\)](#), [k_riic_cks_shift](#), [k_riic_icmr1_controller_7bit](#), [k_riic_icmr2_default](#), [k_riic_icmr3_init](#), and [s_tag](#).

Referenced by [riic_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.22 internal_riic_read_phase()

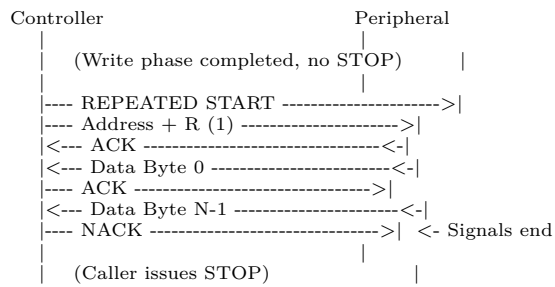
```

rx_err_t internal_riic_read_phase (
    volatile rx_riic_regs_t * riic,
    const i2c_device_addr_t device_addr,
    uint8_t * read_data,
    const uint16_t read_length) [static]
```

Execute I2C read phase for combined write-read transfer.

Performs the read portion of a write-read (register read) transaction. Issues a repeated START condition (without releasing the bus), then reads data bytes from the peripheral.

8.105.4.23 Transaction Sequence



8.105.4.24 Repeated START vs STOP+START

The repeated START (Sr) is critical for atomic register reads:

- Prevents other controllers from arbitrating between address write and data read
- Ensures peripheral's internal address pointer is still valid
- Required by many I2C peripherals (EEPROMs, sensors, etc.)

8.105.4.25 ACK/NACK Protocol

Per I2C specification, the controller must NACK the last byte to signal end of read. This allows the peripheral to release SDA for the STOP condition.

Precondition

```

riic points to valid, initialized RIIC channel
Write phase has been completed (bus is active)
device_addr.value <= 127
read_data != nullptr
read_length > 0
  
```

Postcondition

```

On success: read_data contains received bytes
On failure: STOP issued, bus released
  
```

Note

```

Does NOT issue STOP on success - caller must issue STOP.
On error, STOP is always issued for bus cleanup.
NACK is automatically sent on last byte.
  
```

See also

[internal_riic_write_phase\(\)](#) Precedes this function in write-read
[riic_write_read\(\)](#) Public API that combines both phases

Definition at line 1921 of file [riic.c](#).

```

01925 {
01926     /* Send repeated start condition */
01927     riic->iccr2 |= k_riic_iccr2_rs;
01928
01929     uint32_t timeout = k_riic_timeout_us;
01930     while (!((riic->icsr2 & k_riic_icsr2_start) && timeout > k_riic_timeout_zero) {
01931         timeout--;
01932     }
01933
01934     if (timeout == k_riic_timeout_zero) {
01935         rx_err_t stop_err = internal_send_stop(riic);
01936         (void)stop_err; /* Best-effort cleanup on timeout */
01937         rx_log_error(s_tag, "Repeated start timeout");
01938         return k_rx_err_timeout;
01939     }
01940
01941     riic->icsr2 &= (uint8_t) ~(uint8_t)k_riic_icsr2_start;
01942
01943     /* Explicit ICCR2 write to clear any pending SP/RS/ST request bits ahead of
01944      * the addr|R write. MST and TRS are read-only when ICMR1.MTWP=0, so this
01945      * is a no-op for them; TRS will auto-clear to 0 on the 9th clock of the
01946      * addr|R byte per HUM 42.2.1. */
01947     riic->iccr2 = k_riic_iccr2_mst;
01948
01949     /* Send device address (read) -- use the receive-mode helper, NOT
01950      * internal_write_byte: TEND/TDRE-reassertion don't fire after addr|R in
01951      * controller-receive mode; RDRF does. This leaves the address-ACK RDRF
01952      * asserted but does NOT do the dummy read (that responsibility is the
01953      * caller's so it can pre-set ACKBT for 1-byte transfers). */
01954     rx_err_t err = internal_write_address_for_read(
01955         riic,
01956         (uint8_t)((device_addr.value « k_riic_addr_shift) | k_riic_addr_read_bit));
01957     if (err != k_rx_ok) {
01958         rx_err_t stop_err = internal_send_stop(riic);
01959         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01960         return err;
01961     }
01962
01963     /* For a 1-byte read, set ACKBT=1 BEFORE the dummy read so the sole data
01964      * byte's 9th-clock ACK slot carries NACK (HUM Fig 42.10 "1- or 2-byte
01965      * receive" path). The dummy read below releases SCL and starts clocking
01966      * byte 0; the 9th clock of byte 0 samples ACKBT at that moment. For
01967      * multi-byte reads we leave ACKBT=0 here and flip it to 1 inside the
01968      * loop on iteration N-2 (HUM Fig 42.11 "3 bytes or more" path) -- the
01969      * read of byte N-2 is what releases SCL for byte N-1 to clock in, and
01970      * byte N-1's 9th clock then picks up ACKBT=1. Setting ACKBT=1 on
01971      * iteration N-1 is ALWAYS TOO LATE: by the time we enter that iteration
01972      * RDRF has already asserted, meaning byte N-1's 9th clock has already
01973      * fired and its ACK slot has already been transmitted with whatever
01974      * ACKBT was in the prior iteration (typically 0 = ACK). That leaves
01975      * the peripheral thinking "keep streaming" and holding SDA for another
01976      * byte, which is why the subsequent SP never chains a STOP edge and we
01977      * see "Stop condition timeout" at the end of every multi-byte read. */
01978     if (read_length == 1U) {
01979         riic->icmr3 |= k_riic_icmr3_ackbt_mask;
01980     }
01981
01982     (void)riic->icdrr; /* dummy read kicks reception */
01983
01984     /* Receive loop. For each iteration:
01985      * - Wait for RDRF (byte i now in ICDRR, byte i's 9th clock fired).
01986      * - If we are on the penultimate byte (i == N-2), flip ACKBT=1 so
01987      *   byte N-1's 9th clock carries NACK. Nothing to do on iteration
01988      *   N-1 except clear STOP / queue SP ahead of the final ICDRR read.
01989      * - Issue the final STOP request between reading byte N-2 and the
01990      *   final ICDRR read.
01991      * - Read ICDRR: copies byte i data and (on non-last iterations)
01992      *   releases SCL for byte i+1 to clock in. On the last iteration
01993      *   there is no byte i+1; with SP queued the 9th-clock NACK is
01994      *   followed by a STOP edge.
01995      *
01996      * Do NOT call internal_send_stop() after this loop on success: SP was
01997      * already queued in the final iteration and the hardware generates the
01998      * STOP edge chained to byte N-1's 9th clock. A second SP on an idle
01999      * bus either times out or spuriously re-raises BBSY. */
02000     for (uint16_t i = 0; i < read_length; i++) {
02001         uint32_t timeout = k_riic_timeout_us;
02002         while (!((riic->icsr2 & k_riic_icsr2_rdrf) && timeout > k_riic_timeout_zero) {

```

```

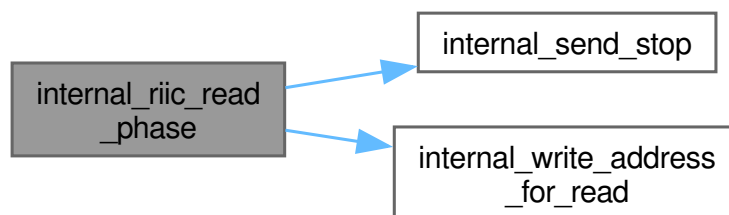
02003     timeout--;
02004 }
02005 if (timeout == k_riic_timeout_zero) {
02006     rx_log_error_val(s_tag, "Read byte RDRF timeout i=", (uint8_t)i);
02007     rx_err_t stop_err = internal_send_stop(riic);
02008     (void)stop_err;
02009     return k_rx_err_timeout;
02010 }
02011
02012 if (read_length >= 2U && i == (uint16_t)(read_length - 2U)) {
02013     /* Penultimate byte: NACK will fall on byte N-1's 9th clock. */
02014     riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02015 }
02016
02017 if (i == (uint16_t)(read_length - k_riic_last_index_offset)) {
02018     /* Last iteration: queue STOP before the final read so the NACK on
02019      * byte N-1's 9th clock is chained to a STOP edge by hardware. */
02020     riic->icsr2 &= (uint8_t) ~(uint8_t) k_riic_icsr2_stop;
02021     riic->iccr2 |= k_riic_iccr2_sp;
02022 }
02023
02024 read_data[i] = riic->icdrr;
02025 }
02026
02027 /* Wait for the hardware-generated STOP to complete. */
02028 uint32_t stop_timeout = k_riic_timeout_us;
02029 while (!(riic->icsr2 & k_riic_icsr2_stop) && stop_timeout > k_riic_timeout_zero) {
02030     stop_timeout--;
02031 }
02032 if (stop_timeout == k_riic_timeout_zero) {
02033     rx_log_error(s_tag, "STOP after final ICDRR timeout");
02034     return k_rx_err_timeout;
02035 }
02036
02037 /* Clear STOP / NACKF for the next transaction. ACKBT stays latched at 1
02038  * from the penultimate-iteration write; riic_init / recovery / the
02039  * pre-transaction ICSR2 clear all restore ICMR3 as needed. */
02040 riic->icsr2 &= (uint8_t) ~(uint8_t) (k_riic_icsr2_stop | k_riic_icsr2_nackf);
02041
02042 return k_rx_ok;
02043 }

```

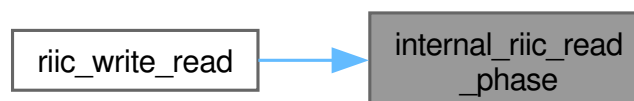
References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icdrr](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icsr2](#), [internal_send_stop\(\)](#), [internal_write_address_for_read\(\)](#), [k_riic_addr_read_bit](#), [k_riic_addr_shift](#), [k_riic_iccr2_mst](#), [k_riic_iccr2_rs](#), [k_riic_iccr2_sp](#), [k_riic_icmr3_ackbt_mask](#), [k_riic_icsr2_nackf](#), [k_riic_icsr2_rdrf](#), [k_riic_icsr2_start](#), [k_riic_icsr2_stop](#), [k_riic_last_index_offset](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), [s_tag](#), and [i2c_device_addr_t::value](#).

Referenced by [riic_write_read\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.26 internal`riic`release`lines()

```
void internal_riic_release_lines (
    volatile rx_riic_regs_t *const riic) [static]
```

Drive the SOWP/SCLO/SDAO release dance after an IICRST pulse.

Forces SCLO=SDAO=1 via SOWP write-protect-clear so the open-drain drivers release the bus (high-Z) on handback to the peripheral.

Parameters

in	riic	RIIC peripheral base (must be non-null)
----	------	---

Precondition

riic non-null and ICE=1

Postcondition

SOWP restored to write-protected

Since

Version 1.0.0

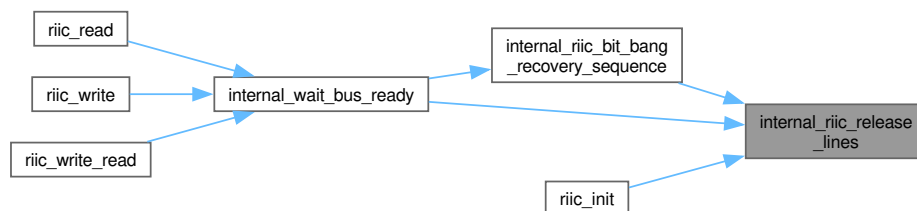
Definition at line 1239 of file [riic.c](#).

```
01240 {
01241     uint8_t iccr1 = riic->iccr1;
01242     iccr1 &= (uint8_t) ~(uint8_t) k_riic_iccr1_sowp;
01243     iccr1 |= (uint8_t)(k_riic_iccr1_sclo | k_riic_iccr1_sdao);
01244     riic->iccr1 = iccr1;
01245     iccr1 |= (uint8_t)k_riic_iccr1_sowp;
01246     riic->iccr1 = iccr1;
01247 }
```

References [rx_riic_regs_t::iccr1](#), [k_riic_iccr1_sclo](#), [k_riic_iccr1_sdao](#), and [k_riic_iccr1_sowp](#).

Referenced by [internal_riic_bit_bang_recovery_sequence\(\)](#), [internal_wait_bus_ready\(\)](#), and [riic_init\(\)](#).

Here is the caller graph for this function:



8.105.4.27 `internal_riic_release_module_stop()`

```
void internal_riic_release_module_stop (
    const uint8_t channel_value) [static]
```

Release the module-stop bit for the requested RIIC channel.

Wraps the protected-register PRCR unlock/lock around the MSTPCRB/CRC write that releases the channel's clock gate. Channel-to-MSTP-bit mapping mirrors the RX72N HW manual MSTP table.

Parameters

in	channel_value	Validated channel index
----	---------------	-------------------------

Precondition

`channel_value < k_riic_max_channels`

Postcondition

Channel module clock is running

Since

Version 1.0.0

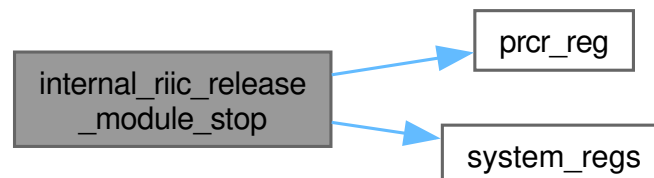
Definition at line 2178 of file [riic.c](#).

```
02179 {
02180     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
02181
02182     if (channel_value == k_riic_channel_0) {
02183         system_regs()->mstpcrb &= ~((uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic0);
02184     } else if (channel_value == k_riic_channel_1) {
02185         system_regs()->mstpcrb &= ~((uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic1);
02186     } else {
02187         system_regs()->mstpcrc &= ~((uint32_t)k_riic_mstpb_bit_value « k_riic_mstpc_riic2);
02188     }
02189
02190     *prcr_reg() = k_rx_prcr_lock;
02191 }
```

References [k_riic_channel_0](#), [k_riic_channel_1](#), [k_riic_mstpb_bit_value](#), [k_riic_mstpb_riic0](#), [k_riic_mstpb_riic1](#), [k_riic_mstpc_riic2](#), [rx_system_regs_t::mstpcrb](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [riic_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.28 internal`riic`save`config()

```
void internal_riic_save_config (
    const volatile rx_riic_regs_t *const riic,
    riic_saved_cfg_t *const saved) [static]
```

Save config registers wiped by an IICRST pulse.

Parameters

in	riic	RIIC peripheral base (must be non-null)
out	saved	Snapshot of ICBRH/ICBRL/ICMR1/ICMR2

Precondition

riic and saved non-null

Postcondition

saved holds the pre-reset register values

Since

Version 1.0.0

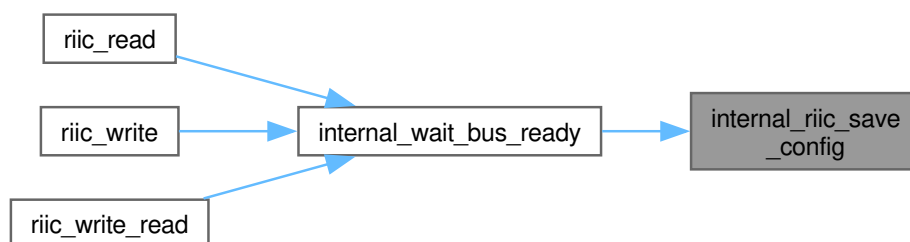
Definition at line 1179 of file [riic.c](#).

```
01181 {
01182     saved->icbrh = riic->icbrh;
01183     saved->icbtl = riic->icbtl;
01184     saved->icmr1 = riic->icmr1;
01185     saved->icmr2 = riic->icmr2;
01186 }
```

References [riic_saved_cfg_t::icbrh](#), [rx_riic_regs_t::icbrh](#), [riic_saved_cfg_t::icbtl](#), [rx_riic_regs_t::icbtl](#), [riic_saved_cfg_t::icmr1](#), [rx_riic_regs_t::icmr1](#), [riic_saved_cfg_t::icmr2](#), and [rx_riic_regs_t::icmr2](#).

Referenced by [internal_wait_bus_ready\(\)](#).

Here is the caller graph for this function:



8.105.4.29 `internal_riic_validate_init()`

```

rx_err_t internal_riic_validate_init (
    const riic_channel_t channel,
    const uint32_t frequency_hz,
    volatile rx_riic_regs_t ** riic_out) [static]

```

Initialize RIIC (I2C) channel for controller mode operation.

Configures an RIIC channel for I2C controller mode communication at the specified bus frequency. This function must be called before any I2C transfers on the channel.

8.105.4.30 Initialization Sequence

1. Validate channel number and frequency
2. Enable RIIC module clock (clear MSTPCRB bit)
3. Reset RIIC internal state (IICRST)
4. Calculate and configure bit rate registers
5. Configure controller mode, 7-bit addressing
6. Enable I2C bus interface (ICE)
7. Mark channel as initialized

8.105.4.31 Module Stop Control

The RIIC peripheral is disabled by default to save power. This function enables the module clock by clearing the appropriate MSTPCRB bit:

Channel	MSTPCRB Bit	Default State
RIIC0	Bit 21	Stopped
RIIC1	Bit 20	Stopped
RIIC2	Bit 19	Stopped

8.105.4.32 Supported Frequencies

Frequency	Mode	Use Case
100 kHz	Standard	Legacy devices
400 kHz	Fast	Most I2C peripherals
1 MHz	Fast Plus	High-speed sensors

Precondition

`channel.value < 3`

`frequency_hz` is one of: 100000, 400000, 1000000

Postcondition

Channel is enabled and ready for I2C transfers

```
s_riic_channel_mode[channel.value] == k_riic_mode_controller
```

Note

This function modifies protected registers (MSTPCRB) using PRCR unlock.

Re-initializing an already initialized channel is allowed.

Pin configuration (MPC) must be done separately if needed.

Warning

Using unsupported frequencies may cause communication failures.

Example

```
// Initialize RIIC0 for Fast mode (400 kHz)
riic_channel_t channel = {.value = 0};
rx_err_t err = riic_init(channel, 400000);
if (err != k_rx_ok) {
    rx_log_error("I2C", "Failed to initialize RIIC0");
}
```

See also

[riic_write\(\)](#) Write data after initialization

[riic_read\(\)](#) Read data after initialization

[riic_write_read\(\)](#) Combined write-read after initialization

Since

Version 1.0.0

Validate `riic_init` arguments and resolve peripheral base

Checks that channel is in range, frequency is one of the three supported buckets (100k/400k/1M), and resolves the channel to its peripheral base address.

Parameters

in	channel	Channel handle
in	frequency_hz	Requested SCL frequency
out	riic_out	On success, set to the peripheral base

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Inputs valid, <code>*riic_out</code> non-null
<code>k_rx_err_invalid_arg</code>	Bounds violation or unknown channel

Precondition

`riic_out` non-null

Since

Version 1.0.0

Definition at line 2142 of file [riic.c](#).

```

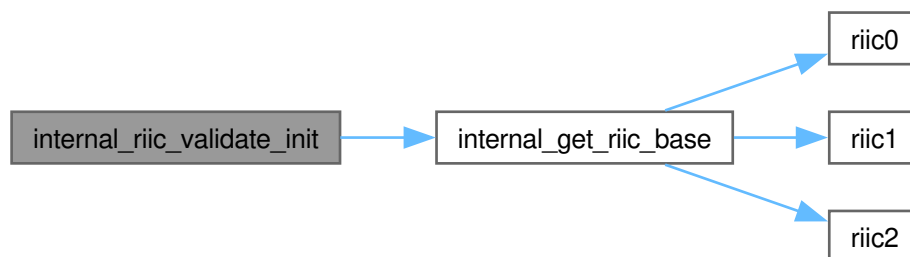
02145 {
02146     if (channel.value >= k_riic_max_channels) {
02147         rx_log_error(s_tag, "Invalid RIIC channel");
02148         return k_rx_err_invalid_arg;
02149     }
02150     if (frequency_hz != k_riic_freq_100khz && frequency_hz != k_riic_freq_400khz &&
02151         frequency_hz != k_riic_freq_1mhz) {
02152         rx_log_error(s_tag, "Invalid I2C frequency (use 100000, 400000, or 1000000)");
02153         return k_rx_err_invalid_arg;
02154     }
02155     *riic_out = internal_get_riic_base(channel.value);
02156     if (*riic_out == nullptr) {
02157         return k_rx_err_invalid_arg;
02158     }
02159     return k_rx_ok;
02160 }

```

References [internal_get_riic_base\(\)](#), [k_riic_freq_100khz](#), [k_riic_freq_1mhz](#), [k_riic_freq_400khz](#), [k_riic_max_channels](#), [s_tag](#), and [riic_channel_t::value](#).

Referenced by [riic_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.33 internal`riic`write`phase()

```

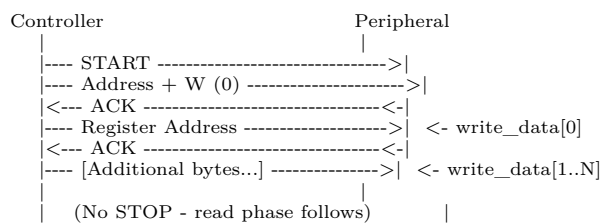
rx_err_t internal_riic_write_phase (
    volatile rx_riic_regs_t * riic,
    const i2c_device_addr_t device_addr,
    const uint8_t * write_data,
    const uint16_t write_length) [static]

```

Execute I2C write phase for combined write-read transfer.

Performs the write portion of a write-read (register read) transaction. This typically writes a register address to the peripheral, which sets an internal pointer for the subsequent read phase.

8.105.4.34 Transaction Sequence



8.105.4.35 Error Handling

On any error (timeout or NACK), this function issues a STOP condition to release the bus before returning the error code.

Precondition

- riic points to valid, initialized RIIC channel
- device_addr.value <= 127
- write_data != nullptr
- write_length > 0
- Bus is idle (not busy)

Postcondition

- On success: Bus is active, internal address pointer set
- On failure: STOP issued, bus released

Note

- Does NOT issue STOP on success - caller must continue with read phase or issue STOP.
- On error, STOP is always issued for bus cleanup.

See also

[internal_riic_read_phase\(\)](#) Continues the transaction with read

[riic_write_read\(\)](#) Public API that combines both phases

Definition at line 1832 of file [riic.c](#).

```

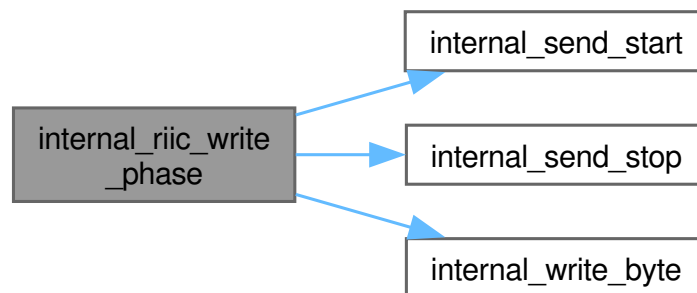
01836 {
01837     /* Set controller transmit mode */
01838     riic->iccr2 = k_riic_iccr2_mst | k_riic_iccr2_trs;
01839
01840     /* Send start condition */
01841     rx_err_t err = internal_send_start(riic);
01842     RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
01843
01844     /* Send device address (write) */
01845     err = internal_write_byte(riic, (device_addr.value « k_riic_addr_shift) | k_riic_addr_write_bit);
01846     if (err != k_rx_ok) {
01847         rx_err_t stop_err = internal_send_stop(riic);
01848         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01849         return err;
01850     }
01851
01852     /* Send write data */
01853     for (uint16_t i = 0; i < write_length; i++) {
01854         err = internal_write_byte(riic, write_data[i]);
01855         if (err != k_rx_ok) {
01856             rx_err_t stop_err = internal_send_stop(riic);
01857             (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01858             return err;
01859         }
01860     }
01861
01862     return k_rx_ok;
01863 }

```

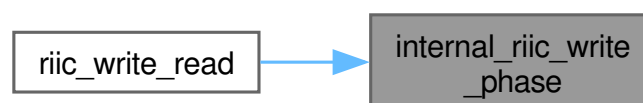
References [rx_riic_regs_t::iccr2](#), [internal_send_start\(\)](#), [internal_send_stop\(\)](#), [internal_write_byte\(\)](#), [k_riic_addr_shift](#), [k_riic_addr_write_bit](#), [k_riic_iccr2_mst](#), [k_riic_iccr2_trs](#), [s_tag](#), and [i2c_device_addr_t::value](#).

Referenced by [riic_write_read\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.36 internal_send_start()

```
rx_err_t internal_send_start (
    volatile rx_riic_regs_t * riic) [static]
```

Generate I2C START condition on the bus.

Issues a START condition by setting the ST bit in ICCR2, then waits for hardware confirmation via the START flag in ICSR2. The START condition signals all peripherals that a transaction is beginning.

8.105.4.37 START Condition Timing

SDA ----+
 +-----
 SCL -----+
 +-----
 | |
 | |
 +- SCL goes low (start of first bit)
 +- SDA goes low while SCL high (START)

8.105.4.38 Hardware Sequence

1. Set ST bit in ICCR2 (request START)
2. Hardware pulls SDA low while SCL is high
3. Hardware sets START flag in ICSR2
4. Software clears START flag to acknowledge

Precondition

riic != nullptr
 Bus must be idle (BBSY = 0) - use [internal_wait_bus_ready\(\)](#) first
 RIIC channel must be enabled (ICE = 1)

Postcondition

START flag is cleared in ICSR2
 Bus is now busy (BBSY = 1)

Note

Clears the START detection flag after successful START.
 After START, caller must send address byte within timeout.

Warning

If START fails, caller must issue STOP to release bus.

See also

[internal_send_stop\(\)](#) Companion function for ending transactions

[internal_wait_bus_ready\(\)](#) Should be called before START

Definition at line 1389 of file [riic.c](#).

```

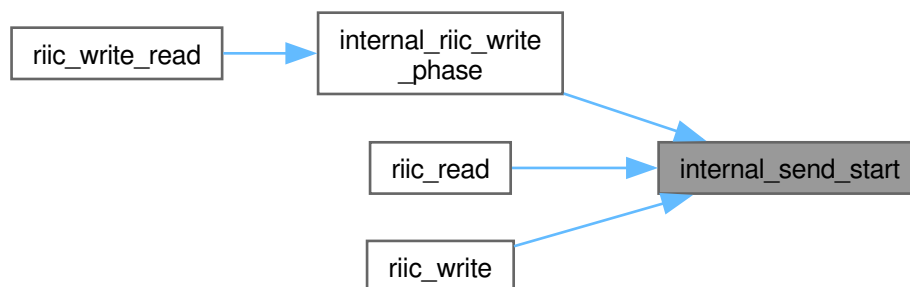
01390 {
01391     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01392
01393     /* Issue start condition */
01394     riic->iccr2 |= k_riic_iccr2_st;
01395
01396     /* Wait for start condition to be issued */
01397     uint32_t timeout = k_riic_timeout_us;
01398     while (!(riic->icsr2 & k_riic_icsr2_start) && timeout > k_riic_timeout_zero) {
01399         timeout--;
01400     }
01401
01402     if (timeout == k_riic_timeout_zero) {
01403         rx_log_error(s_tag, "Start condition timeout");
01404         return k_rx_err_timeout;
01405     }
01406
01407     /* Clear start flag */
01408     riic->icsr2 &= (uint8_t) ~(uint8_t) k_riic_icsr2_start;
01409
01410     return k_rx_ok;
01411 }

```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icsr2](#), [k_riic_iccr2_st](#), [k_riic_icsr2_start](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), and [s_tag](#).

Referenced by [internal_riic_write_phase\(\)](#), [riic_read\(\)](#), and [riic_write\(\)](#).

Here is the caller graph for this function:



8.105.4.39 `internal_send_stop()`

```

rx_err_t internal_send_stop (
    volatile rx\_riic\_regs\_t * riic) [static]

```

Generate I2C STOP condition on the bus.

Issues a STOP condition by setting the SP bit in ICCR2, then waits for hardware confirmation via the STOP flag in ICSR2. The STOP condition releases the bus for other controllers.

8.105.4.40 STOP Condition Timing

```

SDA -----+-----
           |
SCL ----+---+-----
       |   |
       |   +- SDA goes high while SCL high (STOP)
       +- SCL goes high (end of last bit/ACK)

```

8.105.4.41 Hardware Sequence

1. Set SP bit in ICCR2 (request STOP)
2. Hardware drives SCL high
3. Hardware drives SDA high while SCL is high
4. Hardware sets STOP flag in ICSR2
5. Software clears STOP flag to acknowledge

8.105.4.42 Error Recovery

STOP is also used for error recovery - if a transaction fails at any point, issuing STOP releases the bus and resets the peripheral state machine.

Precondition

riic != nullptr
A transaction is in progress (bus is busy)

Postcondition

STOP flag is cleared in ICSR2
Bus is idle (BBSY = 0)
Peripheral state machine returns to idle

Note

Always call STOP at the end of a transaction or to recover from errors.
STOP after NACK in read transactions signals end of transfer to peripheral.

Best-Effort Cleanup

In error paths, STOP is called to release the bus even if the original error should be preserved. The calling function should:

```
if (err != k_rx_ok) {  
    rx_err_t stop_err = internal_send_stop(riic);  
    (void)stop_err; // Preserve original error  
    return err;  
}
```

See also

[internal_send_start\(\)](#) Companion function for starting transactions

Definition at line 1470 of file [riic.c](#).

```

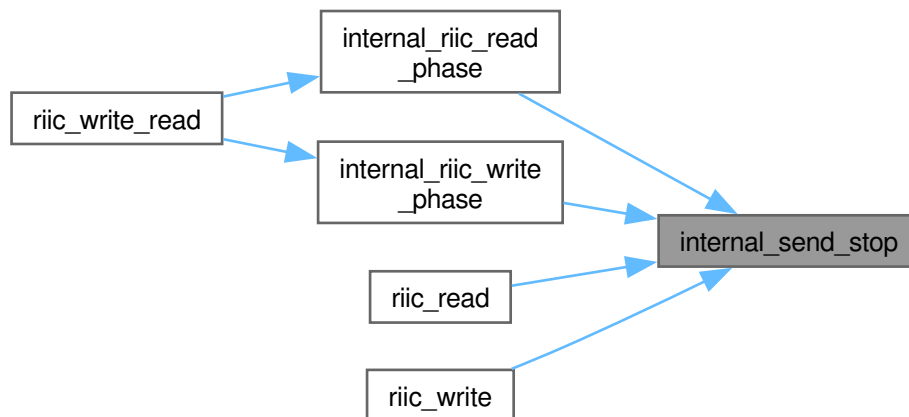
01471 {
01472   RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01473
01474   /* Issue stop condition */
01475   riic->iccr2 |= k_riic_iccr2_sp;
01476
01477   /* Wait for stop condition to be issued */
01478   uint32_t timeout = k_riic_timeout_us;
01479   while (!(riic->icsr2 & k_riic_icsr2_stop) && timeout > k_riic_timeout_zero) {
01480     timeout--;
01481   }
01482
01483   if (timeout == k_riic_timeout_zero) {
01484     rx_log_error(s_tag, "Stop condition timeout");
01485     return k_rx_err_timeout;
01486   }
01487
01488   /* Clear stop flag */
01489   riic->icsr2 &= (uint8_t) ~(uint8_t)k_riic_icsr2_stop;
01490
01491   return k_rx_ok;
01492 }

```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icsr2](#), [k_riic_iccr2_sp](#), [k_riic_icsr2_stop](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), and [s_tag](#).

Referenced by [internal_riic_read_phase\(\)](#), [internal_riic_write_phase\(\)](#), [riic_read\(\)](#), and [riic_write\(\)](#).

Here is the caller graph for this function:



8.105.4.43 internal_wait_bus_ready()

```

rx_err_t internal_wait_bus_ready (
    volatile rx\_riic\_regs\_t * riic) [static]

```

Definition at line 1308 of file [riic.c](#).

```

01309 {
01310   RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01311
01312   if (internal_riic_poll_bbsy_clear(riic)) {
01313     return k_rx_ok;
01314   }
01315
01316   /* BBSY can latch stuck when the peripheral comes up observing lines that
01317    * are mid-transition. Per RX72N HW manual section 42.2.1 the only
01318    * software-side clear is an IICRST pulse -- which wipes the config

```



```

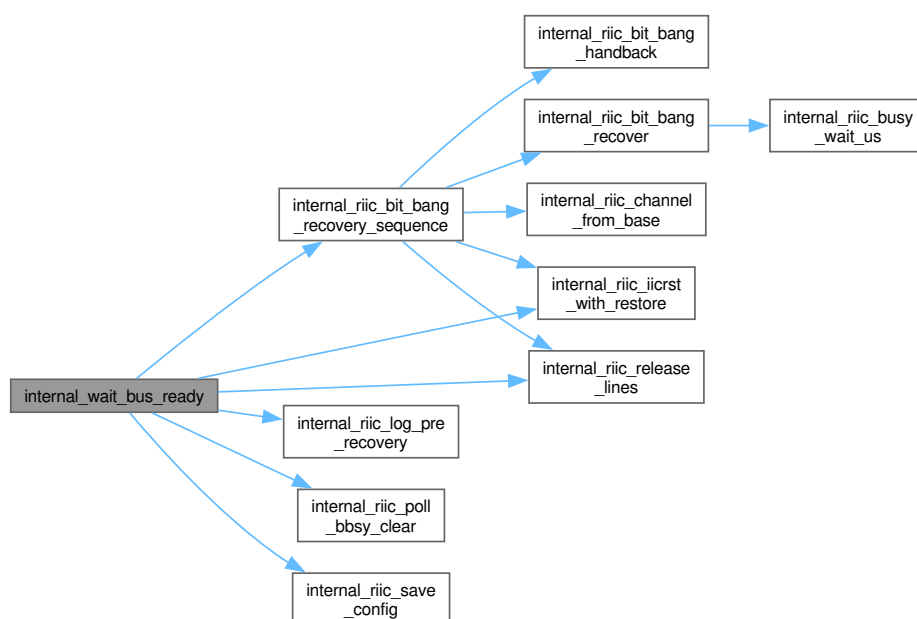
01319  /* registers back to reset, so we must restore them after. */
01320  rx_log_warn(s_tag, "BBSY stuck - pulsing IICRST to recover");
01321  internal_riic_log_pre_recovery(riic);
01322
01323  riic_saved_cfg_t saved;
01324  internal_riic_save_config(riic, &saved);
01325
01326  internal_riic_iicrst_with_restore(riic, &saved);
01327  internal_riic_release_lines(riic);
01328
01329  /* Second wait. If BBSY still set after a full IICRST cycle, the bus is
01330   * genuinely stuck and IICRST alone cannot clear it because the
01331   * peripheral reset does not toggle external SCL edges. */
01332  if (internal_riic_poll_bbsy_clear(riic)) {
01333      return k_rx_ok;
01334  }
01335
01336  internal_riic_bit_bang_recovery_sequence(riic, &saved);
01337
01338  if (!internal_riic_poll_bbsy_clear(riic)) {
01339      rx_log_error(s_tag, "I2C bus busy timeout (after bit-bang recovery)");
01340      return k_rx_err_timeout;
01341  }
01342
01343  return k_rx_ok;
01344 }

```

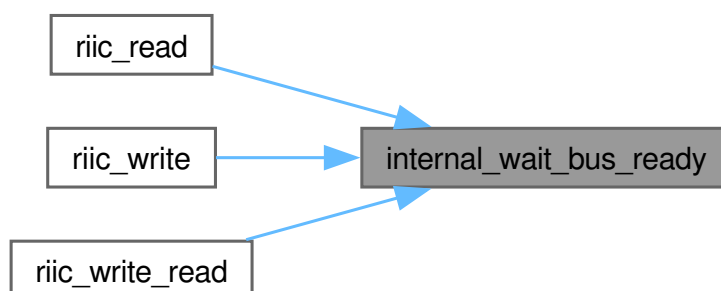
References [internal_riic_bit_bang_recovery_sequence\(\)](#), [internal_riic_iicrst_with_restore\(\)](#), [internal_riic_log_pre_recovery\(\)](#), [internal_riic_poll_bbsy_clear\(\)](#), [internal_riic_release_lines\(\)](#), [internal_riic_save_config\(\)](#), and [s_tag](#).

Referenced by [riic_read\(\)](#), [riic_write\(\)](#), and [riic_write_read\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.105.4.44 `internal_write_address_for_read()`

```
rx_err_t internal_write_address_for_read (
    volatile rx_riic_regs_t * riic,
    const uint8_t addr_byte) [static]
```

Send the address|R byte that opens a controller-receive transaction.

After the controller switches to receive mode (ICCR2 = MST, TRS=0) and has issued START or repeated-START, the address byte with R bit must be sent. `internal_write_byte()` cannot be used here: per RX72N HW manual section 42.2.1 the TRS bit auto-clears to 0 on the rising edge of the 9th SCL of an addr|R byte, which moves the peripheral into receive mode and sets RDRF automatically – TEND and TDRE-reassertion no longer fire. Waiting for transmit-side flags in that state hangs forever; this helper waits for RDRF | NACKF instead.

Precondition

Bus is active; ICCR2 = MST (TRS=0); start condition already issued

Postcondition

On success: RDRF=1 in ICSR2, ICDRR holds the first data byte

See also

[internal_write_byte\(\)](#) Standard transmit-mode byte writer

[internal_read_byte\(\)](#) Reads byte after RDRF asserts

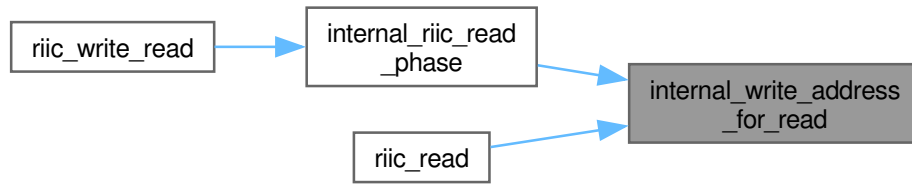
Definition at line 1611 of file [riic.c](#).

```
01613 {
01614     uint32_t timeout = k_riic_timeout_us;
01615     while (!((riic->icr2 & k_riic_icr2_tdre) && timeout > k_riic_timeout_zero) {
01616         timeout--;
01617     }
01618     if (timeout == k_riic_timeout_zero) {
01619         rx_log_error(s_tag, "TDRE timeout before addr|R");
01620         return k_rx_err_timeout;
01621     }
01622
01623     riic->icdrt = addr_byte;
01624
01625     /* In controller-receive mode, RDRF fires once the first data byte has been
01626      * clocked in after the address ACK. NACKF fires if the peripheral didn't
01627      * ACK the address. Wait for either. */
01628     const uint8_t wait_mask = (uint8_t)(k_riic_icr2_rdrf | k_riic_icr2_nackf);
01629     timeout = k_riic_timeout_us;
01630     while (!((riic->icr2 & wait_mask) && timeout > k_riic_timeout_zero) {
01631         timeout--;
01632     }
01633     if (timeout == k_riic_timeout_zero) {
01634         rx_log_error(s_tag, "addr|R: neither RDRF nor NACKF asserted");
01635         return k_rx_err_timeout;
01636     }
01637
01638     if (riic->icr2 & k_riic_icr2_nackf) {
01639         riic->icr2 &= (uint8_t)~(uint8_t)k_riic_icr2_nackf;
01640         rx_log_error(s_tag, "Peripheral NACKed address|R");
01641         return k_rx_err_nack;
01642     }
01643
01644     /* Caller is responsible for the dummy ICDRR read that kicks off actual
01645      * data reception. Doing it here unconditionally would be wrong for the
01646      * single-byte read case where ACKBT=1 must be set BEFORE the dummy read
01647      * so byte 0 is NACKed (RX72N HW manual Figure 42.10, 1-byte path). The
01648      * caller sets ACKBT per the transfer length, then does the dummy read. */
01649     return k_rx_ok;
01650 }
```

References [rx_riic_regs_t::icdrt](#), [rx_riic_regs_t::icr2](#), [k_riic_icr2_nackf](#), [k_riic_icr2_rdrf](#), [k_riic_icr2_tdre](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), and [s_tag](#).

Referenced by [internal_riic_read_phase\(\)](#), and [riic_read\(\)](#).

Here is the caller graph for this function:



8.105.4.45 internal`write`byte()

```

rx_err_t internal_write_byte (
    volatile rx_riic_regs_t * riic,
    const uint8_t data) [static]
  
```

Transmit single byte over I2C bus.

Writes one byte to the I2C bus and waits for acknowledgment from the peripheral. This function handles both address bytes and data bytes.

8.105.4.46 Transmission Sequence

1. Wait for TDRE (Transmit Data Register Empty) flag
2. Write byte to ICDRT (Transmit Data Register)
3. Hardware shifts out 8 bits MSB-first
4. Hardware generates 9th clock for ACK
5. Peripheral drives SDA low (ACK) or leaves high (NACK)
6. Check NACKF flag to determine response

8.105.4.47 I2C Byte Timing

```

SCL  ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
      || || || || || || || ||
      ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
      D7 D6 D5 D4 D3 D2 D1 D0 ACK
SDA  -----+
      Data bits (8)      +- ACK (low)
  
```

8.105.4.48 NACK Handling

NACK is received when:

- Address byte: No peripheral at that address
- Data byte: Peripheral buffer full or error condition
- Last byte: Normal end-of-read signaling (expected)

Precondition

riic != nullptr (not checked - caller must ensure)
 START condition has been issued
 Previous byte (if any) has been acknowledged

Postcondition

On success, byte has been transmitted and ACK received
 On NACK, NACKF flag is cleared for next operation

Note

Does NOT send START or STOP - caller manages transaction framing.
 First byte after START should be address with R/W bit.

Warning

On NACK, caller should issue STOP to release bus.

Address Byte Format

For address bytes, caller must format as: $(\text{addr} \ll 1) \mid \text{direction}$

- Write: $(0x50 \ll 1) \mid 0 = 0xA0$
- Read: $(0x50 \ll 1) \mid 1 = 0xA1$

See also

[internal_read_byte\(\)](#) Companion function for receiving bytes

Definition at line 1548 of file [riic.c](#).

```

01549 {
01550     /* Wait for transmit data empty */
01551     uint32_t timeout = k_riic_timeout_us;
01552     while (!(riic->icsr2 & k_riic_icsr2_tdre) && timeout > k_riic_timeout_zero) {
01553         timeout--;
01554     }
01555
01556     if (timeout == k_riic_timeout_zero) {
01557         rx_log_error(s_tag, "Write timeout");
01558         return k_rx_err_timeout;
01559     }
01560
01561     /* Write data */
01562     riic->icdrt = data;
01563
01564     /* Wait for transmit END (byte + ACK clock complete). Per RX72N HW manual
01565     * section 42.2.7, TEND asserts on the rising edge of the 9th clock after
01566     * the ACK/NACK response has been sampled, so NACKF is valid only once
01567     * TEND=1. Waiting for TDRE to clear is wrong: TDRE clears the instant
01568     * ICDRT is written, so the wait would exit in zero iterations and NACKF
01569     * would be checked before the byte has actually been transmitted. */
01570     timeout = k_riic_timeout_us;
01571     while (!(riic->icsr2 & k_riic_icsr2_tend) && timeout > k_riic_timeout_zero) {
01572         timeout--;
01573     }
01574
01575     if (timeout == k_riic_timeout_zero) {
01576         rx_log_error(s_tag, "TEND timeout (byte never finished transmitting)");
01577         return k_rx_err_timeout;
01578     }
01579

```

```

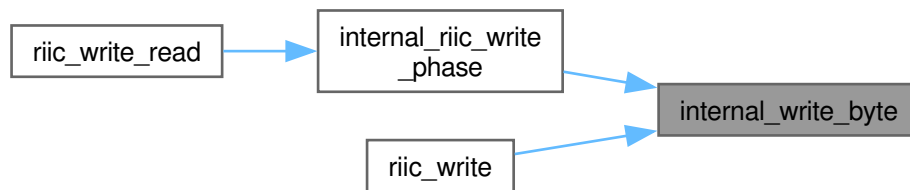
01580  /* Check for NACK -- valid only now that TEND=1. */
01581  if (riic->icsr2 & k_riic_icsr2_nackf) {
01582      riic->icsr2 &= (uint8_t) ~(uint8_t)k_riic_icsr2_nackf;
01583      rx_log_error(s_tag, "NACK received");
01584      return k_rx_err_nack;
01585  }
01586
01587  return k_rx_ok;
01588 }

```

References [rx_riic_regs_t::icdrt](#), [rx_riic_regs_t::icsr2](#), [k_riic_icsr2_nackf](#), [k_riic_icsr2_tdre](#), [k_riic_icsr2_tend](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), and [s_tag](#).

Referenced by [internal_riic_write_phase\(\)](#), and [riic_write\(\)](#).

Here is the caller graph for this function:



8.105.4.49 riic`deinit`peripheral()

```

rx_err_t riic_deinit_peripheral (
    const riic\_channel\_t channel) [nodiscard]

```

Deinitialize an RIIC channel that was initialized in peripheral mode.

Deinitialize an RIIC channel previously initialized in peripheral mode.

Disables the peripheral address match (ICSER.SAR0E), clears the SARL0/SARU0 address registers, resets the RIIC module, and marks the channel as uninitialized. After this call, [riic_init\(\)](#) or [riic_init_peripheral\(\)](#) may be called again on the same channel.

Precondition

channel.value must be in range [0, k_riic_max_channels - 1]

Channel must have been initialized via [riic_init_peripheral\(\)](#)

Postcondition

s_riic_channel_mode[channel.value] == k_riic_mode_uninitialized on success

RIIC peripheral address match disabled; hardware in reset state

Note

Only available on RX72N target (guarded by RX preprocessor)

Not thread-safe; provide external synchronization if needed

See also

[riic_init_peripheral\(\)](#) Complementary initialization function

Since

Version 1.0.0

Definition at line 3100 of file [riic.c](#).

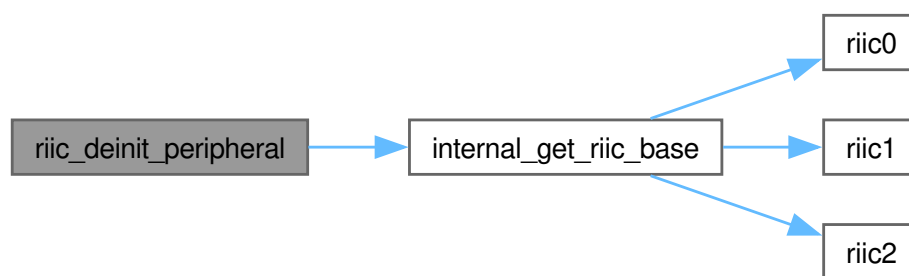
```

03101 {
03102     if (channel.value >= k_riic_max_channels) {
03103         return k_rx_err_invalid_arg;
03104     }
03105
03106     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03107         return k_rx_err_invalid_state;
03108     }
03109
03110     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03111     if (riic == nullptr) {
03112         return k_rx_err_invalid_arg;
03113     }
03114
03115     /* Disable peripheral address match (clear SAR0E in IC SER) */
03116     riic->icser = (uint8_t)(riic->icser & ~(uint8_t)k_riic_icser_sar0e);
03117
03118     /* Clear peripheral address registers */
03119     riic->sar0 = k_riic_register_clear;
03120     riic->sar0 = k_riic_register_clear;
03121
03122     /* Reset the RIIC module */
03123     riic->iccr1 = k_riic_iccr1_iicrst;
03124     riic->iccr1 = k_riic_register_clear;
03125
03126     /* Mark channel as uninitialized */
03127     s_riic_channel_mode[channel.value] = k_riic_mode_uninitialized;
03128
03129     rx_log_debug(s_tag, "RIIC peripheral mode deinitialized");
03130
03131     return k_rx_ok;
03132 }

```

References [rx_riic_regs_t::iccr1](#), [rx_riic_regs_t::icser](#), [internal_get_riic_base\(\)](#), [k_riic_iccr1_iicrst](#), [k_riic_icser_sar0e](#), [k_riic_max_channels](#), [k_riic_mode_peripheral](#), [k_riic_mode_uninitialized](#), [k_riic_register_clear](#), [s_riic_channel_mode](#), [s_tag](#), [rx_riic_regs_t::sar0](#), [rx_riic_regs_t::sar0](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.50 riic_init()

```

rx_err_t riic_init (
    riic_channel_t channel,
    uint32_t frequency_hz) [nodiscard]

```

Initialize RIIC channel for I2C controller communication.

Configures the specified RIIC peripheral channel for I2C controller mode at the requested clock frequency. Performs full hardware initialization including module stop release, pin function selection (SCL/SDA via MPC), internal reset sequence, bit rate register (BRH/BRL) calculation from PCLKB, and interrupt configuration.

8.105.4.51 Algorithm

1. Validate channel number (0-2) and frequency
2. Release RIIC module from module stop (MSTPCRB)
3. Assert internal reset (ICCR1.IICRST = 1)
4. Configure SCL/SDA pins via MPC register
5. Set bit rate registers (BRH/BRL) from PCLKB and frequency_hz
6. Configure noise filter, timeout detection
7. Release internal reset (ICCR1.IICRST = 0)
8. Enable I2C bus (ICCR1.ICE = 1)

8.105.4.52 Supported Frequencies

Frequency	Mode	BRL/BRH	Max Bus Length
100000 Hz	Standard	Auto-calculated	1 m
400000 Hz	Fast	Auto-calculated	0.3 m
1000000 Hz	Fast-plus	Auto-calculated	0.1 m

Parameters

in	channel	RIIC channel wrapper • channel.value range: 0-2 • Use {.value = N} designated initializer
in	frequency_hz	I2C clock frequency in Hz • Valid values: 100000, 400000, 1000000 • Must match connected device capabilities

Returns

rx_err_t Error code indicating success or failure

Return values

k_rx_ok	Channel successfully initialized
k_rx_err_invalid_arg	Channel > 2 or unsupported frequency
k_rx_err_invalid_state	Channel already initialized

Precondition

Channel must not be already initialized (call `riic_deinit` first to reinitialize)
 System clocks must be configured (PCLKB running at expected frequency)

Postcondition

RIIC channel configured and ready for I2C transactions
 SCL/SDA pins configured with open-drain output and internal pull-ups

Note

Thread Safety: NOT thread-safe. Initialize each channel from a single thread during system startup.

Example: Initialize IMU I2C Bus

```
riic_channel_t ch = {.value = 0};
rx_err_t err = riic_init(ch, 400000); // 400 kHz fast mode
if (err != k_rx_ok) {
    rx_log_error_val("I2C", "RIIC0 init failed", err);
    return;
}
rx_log_info("I2C", "RIIC0 initialized at 400 kHz");
```

See also

[riic_write\(\)](#) Write data to I2C device
[riic_read\(\)](#) Read data from I2C device
[riic_write_read\(\)](#) Combined write-then-read transaction

Since

Version 1.0.0

Definition at line 2231 of file [riic.c](#).

```
02232 {
02233     volatile rx_riic_regs_t* riic = nullptr;
02234     const rx_err_t check = internal_riic_validate_init(channel, frequency_hz, &riic);
02235     if (check != k_rx_ok) {
02236         return check;
02237     }
02238
02239     internal_riic_release_module_stop(channel.value);
02240
02241     /* Unconditional bit-bang bus recovery BEFORE the peripheral observes the
02242      * bus. Per RX72N HW manual section 42.2.1, the RIIC comes out of
02243      * module-stop + IICRST with SCLO=SDAO=0 internally -- it drives the bus
02244      * low until the SOWP dance below writes SCLO=SDAO=1. hardware_init has
02245      * already raised PMR for SCL/SDA by the time riic_init runs, so without
02246      * this step the low-drive glitches the external bus and can latch
02247      * BBSY=1 as soon as ICE=1. Channels without an entry in
02248      * k_riic_recovery_pins[] (RIIC0, RIIC2) are no-ops. */
02249     internal_riic_bit_bang_recover(channel.value); /* leaves PMR=0 */
02250
02251     /* Reset RIIC */
02252     riic->iccr1 = k_riic_iccr1_iicrst;
02253     riic->iccr1 = k_riic_register_clear;
02254
02255     RX_RETURN_ON_ERROR(internal_riic_program_bit_rate(riic, frequency_hz),
02256                        s_tag,
02257                        "RIIC bit-rate program failed");
02258
02259     /* Enable I2C bus interface */
02260     riic->iccr1 = k_riic_iccr1_ice;
```



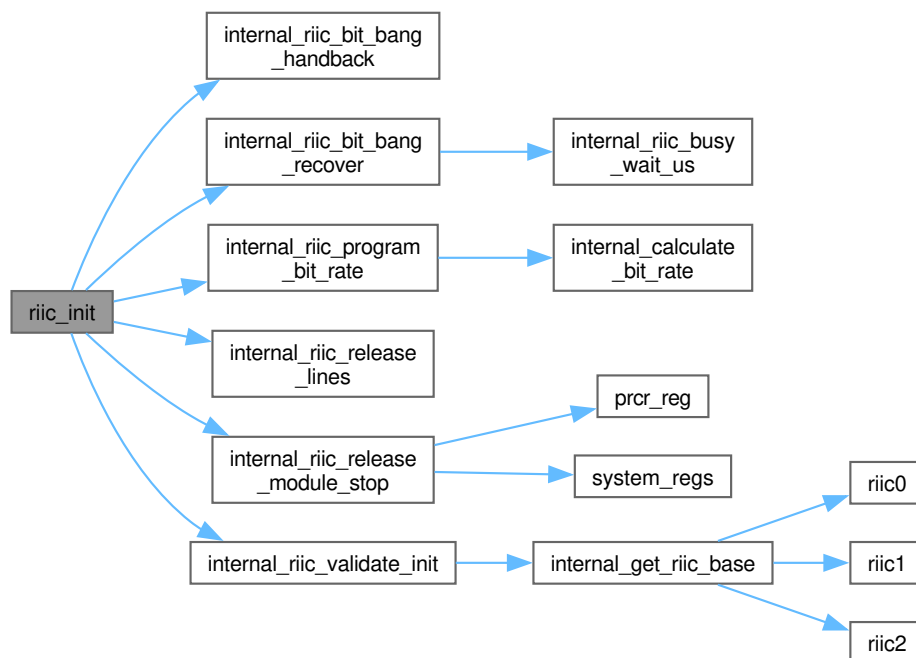
```

02261
02262 /* Post-enable bus release: per RX72N HW manual section 42.2.1 the RIIC
02263 * leaves SCL0/SDAO=0 after ICE=1 -- which drives both lines LOW -- until
02264 * it detects its first STOP on the wire. */
02265 internal_riic_release_lines(riic);
02266
02267 /* Pair to the unconditional bit-bang above: now that SCL0=SDAO=1 in the
02268 * peripheral (set by the SOWP dance), it is safe to route the pads back
02269 * to RIIC. */
02270 internal_riic_bit_bang_handback(channel.value);
02271
02272 s_riic_channel_mode[channel.value] = k_riic_mode_controller;
02273 rx_log_debug(s_tag, "RIIC channel initialized");
02274
02275 return k_rx_ok;
02276 }

```

References [rx_riic_regs_t::iccr1](#), [internal_riic_bit_bang_handback\(\)](#), [internal_riic_bit_bang_recover\(\)](#), [internal_riic_program_bit_rate\(\)](#), [internal_riic_release_lines\(\)](#), [internal_riic_release_module_stop\(\)](#), [internal_riic_validate_init\(\)](#), [k_riic_iccr1_ice](#), [k_riic_iccr1_iicrst](#), [k_riic_mode_controller](#), [k_riic_register_clear](#), [s_riic_channel_mode](#), [s_tag](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.53 riic`init`peripheral()

```

rx_err_t riic_init_peripheral (
    const riic_channel_t channel,
    const i2c_device_addr_t device_addr) [nodiscard]

```

Initialize RIIC channel in peripheral (device) mode.

Configures the RIIC peripheral to act as an I2C peripheral device, responding to a fixed 7-bit I2C address issued by a controller (RPI5). Sets up the peripheral address register (SARL0/SARU0), enables address match detection (ICSER.SAR0E), and leaves the module in peripheral mode (MS=0 in ICMR1).

8.105.4.54 Initialization Sequence

1. Validate channel number and device address
2. Enable RIIC module clock (clear MSTPCRB bit)
3. Reset RIIC internal state (IICRST)
4. Configure peripheral address in SARL0/SARU0
5. Enable SAR0 address match detection (ICSER.SAR0E)
6. Configure peripheral mode (MS=0), 7-bit addressing
7. Enable I2C bus interface (ICE)
8. Mark channel as initialized

Precondition

channel.value must be in range [0, k_riic_max_channels - 1]
 device_addr.value must be <= k_riic_addr_max_7bit (0x7F)

Postcondition

RIIC peripheral responds to device_addr on the I2C bus
 s_riic_channel_mode[channel.value] == k_riic_mode_peripheral

Note

Only available on RX72N target (guarded by RX preprocessor)
 Not thread-safe during initialization

See also

[riic_peripheral_read\(\)](#) Read data written by I2C controller
[riic_peripheral_write\(\)](#) Provide data to I2C controller read

Since

Version 1.0.0

Definition at line 2860 of file [riic.c](#).

```

02861 {
02862     if (channel.value >= k_riic_max_channels) {
02863         return k_rx_err_invalid_arg;
02864     }
02865     if (device_addr.value > k_riic_addr_max_7bit) {
02866         rx_log_error(s_tag, "Invalid peripheral device address");
02867         return k_rx_err_invalid_arg;
02868     }
02869     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02870     if (riic == nullptr) {
02871         return k_rx_err_invalid_arg;
02872     }
02873     /* Enable RIIC module (clear module stop bit) */
02874     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;

```

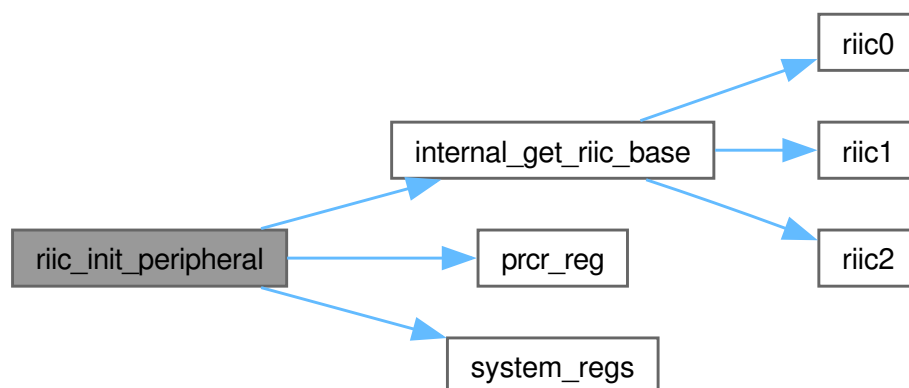
```

02878
02879 if (channel.value == k_riic_channel_0) {
02880     system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic0);
02881 } else if (channel.value == k_riic_channel_1) {
02882     system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic1);
02883 } else {
02884     system_regs()->mstpcrc &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpc_riic2);
02885 }
02886
02887 *prcr_reg() = k_rx_prcr_lock;
02888
02889 /* Reset RIIC */
02890 riic->iccr1 = k_riic_iccr1_iicrst;
02891 riic->iccr1 = k_riic_register_clear;
02892
02893 /* Set peripheral (device) address in SARL0/SARU0
02894  * SARL0[7:1] = device_addr.value (7-bit address)
02895  * SARU0 = 0 for 7-bit mode */
02896 riic->sarl0 = (uint8_t)(device_addr.value « k_riic_sarl_addr_shift);
02897 riic->saru0 = k_riic_saru0_7bit;
02898
02899 /* Enable SAR0 address match detection */
02900 riic->icser = k_riic_icser_sar0e;
02901
02902 /* Configure for peripheral mode (MS=0), 7-bit addressing, CKS=0 */
02903 riic->icmr1 = k_riic_icmr1_peripheral_7bit;
02904 riic->icmr2 = k_riic_icmr2_default;
02905 riic->icmr3 = k_riic_icmr2_default;
02906
02907 /* Enable I2C bus interface */
02908 riic->iccr1 = k_riic_iccr1_ice;
02909
02910 /* Mark channel as initialized in peripheral mode */
02911 s_riic_channel_mode[channel.value] = k_riic_mode_peripheral;
02912
02913 rx_log_debug(s_tag, "RIIC peripheral mode initialized");
02914
02915 return k_rx_ok;
02916 }

```

References [rx_riic_regs_t::iccr1](#), [rx_riic_regs_t::icmr1](#), [rx_riic_regs_t::icmr2](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icser](#), [internal_get_riic_base\(\)](#), [k_riic_addr_max_7bit](#), [k_riic_channel_0](#), [k_riic_channel_1](#), [k_riic_iccr1_ice](#), [k_riic_iccr1_iicrst](#), [k_riic_icmr1_peripheral_7bit](#), [k_riic_icmr2_default](#), [k_riic_icser_sar0e](#), [k_riic_max_channels](#), [k_riic_mode_peripheral](#), [k_riic_mstpb_bit_value](#), [k_riic_mstpb_riic0](#), [k_riic_mstpb_riic1](#), [k_riic_mstpc_riic2](#), [k_riic_register_clear](#), [k_riic_sarl_addr_shift](#), [k_riic_saru0_7bit](#), [rx_system_regs_t::mstpcrb](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), [s_riic_channel_mode](#), [s_tag](#), [rx_riic_regs_t::sarl0](#), [rx_riic_regs_t::saru0](#), [system_regs\(\)](#), [i2c_device_addr_t::value](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.55 riic`peripheral`read()

```

rx_err_t riic_peripheral_read (
    const riic\_channel\_t channel,

```

```
uint8_t * data,  
const uint16_t max_length,  
uint16_t * bytes_read) [nodiscard]
```

Read bytes written by the I2C controller to this peripheral.

Polls ICSR2.RDRF (Receive Data Register Full) to detect incoming bytes from the I2C controller (RPi5). Reads up to max_length bytes into the caller-supplied buffer. Returns immediately with bytes_read == 0 if no data is currently available (non-blocking). Each byte read clears the RDRF flag automatically on the RX72N.

8.105.4.56 Read Algorithm

1. Validate pointers, channel, and max_length
2. Clear *bytes_read to 0
3. Poll ICSR2.RDRF; if set, read ICDRR into data[*bytes_read], increment count
4. Repeat until RDRF clear or max_length reached
5. Return k_rx_ok with actual count in *bytes_read

Precondition

channel must be initialized via [riic_init_peripheral\(\)](#)
data != nullptr with at least max_length bytes of capacity
max_length >= 1 && max_length <= k_riic_peripheral_transfer_limit

Postcondition

*bytes_read contains the number of bytes placed in data (0 to max_length)
data[0..(*bytes_read)-1] contain valid received bytes on k_rx_ok

Note

Non-blocking: returns immediately with available data (bytes_read may be 0)
Only available on RX72N target (guarded by RX preprocessor)
Not thread-safe, caller must provide external synchronization

See also

[riic_init_peripheral\(\)](#) Initialize channel first
[riic_peripheral_write\(\)](#) Provide read data to controller

Since

Version 1.0.0

Definition at line 2952 of file [riic.c](#).

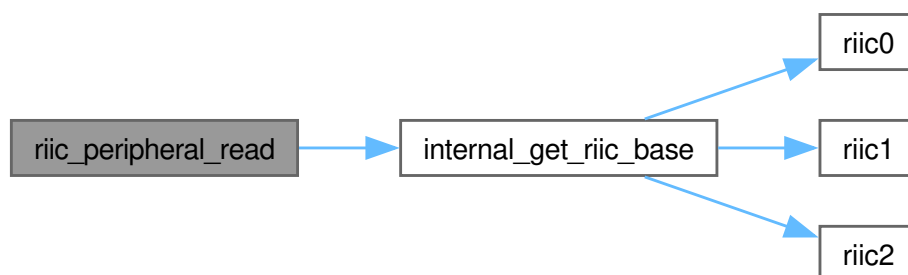
```

02956 {
02957     if (data == nullptr || bytes_read == nullptr) {
02958         return k_rx_err_null_ptr;
02959     }
02960
02961     if (channel.value >= k_riic_max_channels) {
02962         return k_rx_err_invalid_arg;
02963     }
02964
02965     if (max_length == 0) {
02966         return k_rx_err_invalid_arg;
02967     }
02968
02969     if (max_length > k_riic_peripheral_transfer_limit) {
02970         rx_log_error(s_tag, "Peripheral read max_length exceeds limit");
02971         return k_rx_err_invalid_arg;
02972     }
02973
02974     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
02975         return k_rx_err_invalid_state;
02976     }
02977
02978     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02979     if (riic == nullptr) {
02980         return k_rx_err_invalid_arg;
02981     }
02982
02983     *bytes_read = 0;
02984
02985     /* Poll ICSR2.RDRF (bit 1) for each available byte - non-blocking */
02986     while (*bytes_read < max_length) {
02987         if (!(riic->icsr2 & k_riic_icsr2_rdrf)) {
02988             break; /* No more data available right now */
02989         }
02990         data[*bytes_read] = riic->icdrr;
02991         (*bytes_read)++;
02992     }
02993
02994     return k_rx_ok;
02995 }

```

References [rx_riic_regs_t::icdrr](#), [rx_riic_regs_t::icsr2](#), [internal_get_riic_base\(\)](#), [k_riic_icsr2_rdrf](#), [k_riic_max_channels](#), [k_riic_mode_peripheral](#), [k_riic_peripheral_transfer_limit](#), [s_riic_channel_mode](#), [s_tag](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.57 riic`peripheral`write()

```

rx_err_t riic_peripheral_write (
    const riic_channel_t channel,
    const uint8_t * data,
    const uint16_t length) [nodiscard]

```

Write bytes for the I2C controller to read from this peripheral.

Loads data bytes into the RIIC transmit data register (ICDRT) one at a time. For each byte, polls ICSR2.TDRE (Transmit Data Register Empty) until the register is available or a 1 ms timeout expires, then writes the byte. The controller clock-shifts bytes out automatically after each write.

8.105.4.58 Write Algorithm

1. Validate pointer, channel, and length
2. For each byte in data[0..length-1]: a. Poll ICSR2.TDRE with k_riic_periph_timeout_us count-down b. If timeout, return k_rx_err_timeout c. Write data[i] to ICDRT
3. Return k_rx_ok

Precondition

channel must be initialized via [riic_init_peripheral\(\)](#)
 data != nullptr with at least length valid bytes
 length >= 1

Postcondition

On k_rx_ok: all length bytes have been written to ICDRT
 On k_rx_err_timeout: partial write may have occurred (i bytes written)

Note

Only available on RX72N target (guarded by RX preprocessor)
 Not thread-safe, caller must provide external synchronization

See also

[riic_init_peripheral\(\)](#) Initialize channel first
[riic_peripheral_read\(\)](#) Receive data from controller

Since

Version 1.0.0

Definition at line 3032 of file [riic.c](#).

```

03033 {
03034     if (data == nullptr) {
03035         return k_rx_err_null_ptr;
03036     }
03037
03038     if (channel.value >= k_riic_max_channels) {
03039         return k_rx_err_invalid_arg;
03040     }
03041
03042     if (length == 0 || length > k_riic_peripheral_transfer_limit) {
03043         rx_log_error(
03044             s_tag,
03045             "riic_peripheral_write: length out of range (1..k_riic_peripheral_transfer_limit)");
03046         return k_rx_err_invalid_arg;
03047     }
03048
03049     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03050         return k_rx_err_invalid_state;
03051     }
03052
03053     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03054     if (riic == nullptr) {
03055         return k_rx_err_invalid_arg;
03056     }
03057

```

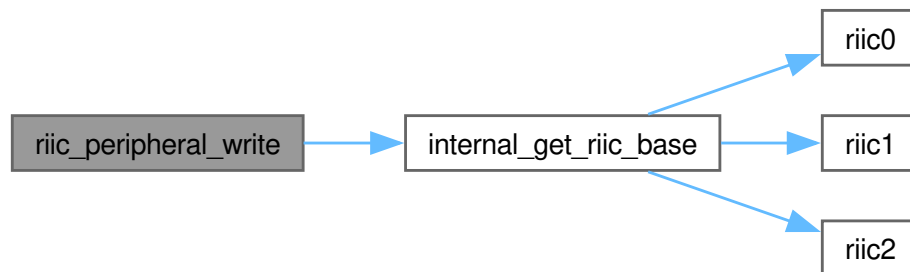
```

03058 for (uint16_t i = 0; i < length; i++) {
03059     /* Wait for ICSR2.TDRE (transmit data register empty) */
03060     uint32_t timeout = k_riic_periph_timeout_us;
03061     while (!(riic->icsr2 & k_riic_icsr2_tdre) && timeout > k_riic_timeout_zero) {
03062         timeout--;
03063     }
03064     if (timeout == k_riic_timeout_zero) {
03065         return k_rx_err_timeout;
03066     }
03067     riic->icdrt = data[i];
03070 }
03071 rx_log_debug(s_tag, "RIIC peripheral write complete");
03073 return k_rx_ok;
03075 }

```

References [rx_riic_regs_t::icdrt](#), [rx_riic_regs_t::icsr2](#), [internal_get_riic_base\(\)](#), [k_riic_icsr2_tdre](#), [k_riic_max_channels](#), [k_riic_mode_peripheral](#), [k_riic_periph_timeout_us](#), [k_riic_peripheral_transfer_limit](#), [k_riic_timeout_zero](#), [s_riic_channel_mode](#), [s_tag](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.59 riic_read()

```

rx_err_t riic_read (
    const riic_channel_t channel,
    const i2c_device_addr_t device_addr,
    uint8_t * data,
    const uint16_t length) [nodiscard]

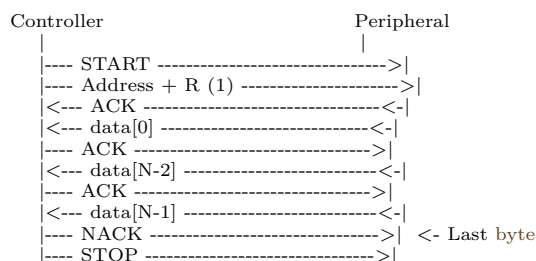
```

Read data from an I2C peripheral device.

Read data from I2C device on specified RIIC channel.

Performs a complete I2C read transaction: START, address+R, receive data bytes (ACK all except last), NACK last byte, STOP. This function is suitable for reading device status, sensor data, or sequential memory reads where the device auto-increments its internal pointer.

8.105.4.60 Transaction Sequence



8.105.4.61 When to Use `riic_read()` vs `riic_write_read()`

Function	Use When
<code>riic_read()</code>	Device auto-increments address, or reading status
<code>riic_write_read()</code>	Need to specify register address first

8.105.4.62 Common Use Cases

- Read device status register (if device supports direct read)
- Continue reading after previous write set address pointer
- Read from devices with auto-increment address mode

Precondition

`riic_init()` called for this channel
`data != nullptr`
 Buffer size $\geq$ length
 length ≥ 1 && length ≤ 256
 device_addr.value ≤ 127

Postcondition

On success: data contains received bytes, bus released
 On failure: Bus released via STOP

Note

This is a blocking function - returns only after transfer complete.
 NACK is automatically sent before reading last byte.

Warning

Most peripherals require `riic_write_read()` to specify which register to read from. Use this function only when the device supports direct reads or address auto-increment.

Example: Read Temperature Sensor (Direct Read)

```

// Read 2 bytes from temperature sensor at address 0x48
// (sensor auto-reports temperature on read)
uint8_t temp_data[2];
riic_channel_t channel = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};

rx_err_t err = riic_read(channel, addr, temp_data, 2);
if (err == k_rx_ok) {
    int16_t raw = (temp_data[0] « 8) | temp_data[1];
    float celsius = raw * 0.0625f;
}

```


See also

- [riic_init\(\)](#) Initialize channel before use
- [riic_write\(\)](#) Write-only transaction
- [riic_write_read\(\)](#) Combined write-then-read (register read)

Definition at line 2507 of file [riic.c](#).

```

02511 {
02512     RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02513     RX_CHECK_RANGE_TAG(channel.value,
02514                         k_riic_channel_0,
02515                         (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02516                         k_rx_err_invalid_arg,
02517                         s_tag);
02518
02519     if (device_addr.value > k_riic_addr_max_7bit) {
02520         rx_log_error(s_tag, "Invalid device address");
02521         return k_rx_err_invalid_arg;
02522     }
02523
02524     if (length == k_riic_length_zero) {
02525         rx_log_error(s_tag, "Read length cannot be zero");
02526         return k_rx_err_invalid_arg;
02527     }
02528
02529     if (length > k_riic_max_transfer_length) {
02530         rx_log_error(s_tag, "Read length exceeds maximum");
02531         return k_rx_err_invalid_arg;
02532     }
02533
02534     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02535     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02536         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02537         return k_rx_err_invalid_state;
02538     }
02539
02540     /* Get RIIC base */
02541     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02542
02543     /* Wait for bus ready */
02544     rx_err_t err = internal_wait_bus_ready(riic);
02545     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02546
02547     /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02548     riic->icsr2 = k_riic_register_clear;
02549
02550     /* Set controller receive mode */
02551     riic->iccr2 = k_riic_iccr2_mst;
02552
02553     /* Send start condition */
02554     err = internal_send_start(riic);
02555     RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02556
02557     /* Send device address (read) -- use the receive-mode helper, NOT
02558     * internal_write_byte: TEND/TDRE-reassertion don't fire after addr|R in
02559     * controller-receive mode; RDRF does. */
02560     err = internal_write_address_for_read(
02561         riic,
02562         (uint8_t)((device_addr.value « k_riic_addr_shift) | k_riic_addr_read_bit));
02563     if (err != k_rx_ok) {
02564         rx_err_t stop_err = internal_send_stop(riic);
02565         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02566         return err;
02567     }
02568
02569     /* 1-byte read: pre-set ACKBT=1 before the dummy read so the sole byte's
02570     * 9th-clock ACK slot carries NACK (HUM Fig 42.10). */
02571     if (length == 1U) {
02572         riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02573     }
02574
02575     (void)riic->icdrr; /* dummy read kicks reception */
02576
02577     /* Mirror of the read loop in internal_riic_read_phase -- see that
02578     * function for the why-ACKBT-on-N-2 rationale. */
02579     for (uint16_t i = 0; i < length; i++) {
02580         uint32_t timeout = k_riic_timeout_us;
02581         while (!(riic->icsr2 & k_riic_icsr2_rdrf) && timeout > k_riic_timeout_zero) {
02582             timeout--;
02583         }
02584         if (timeout == k_riic_timeout_zero) {
02585             rx_log_error_val(s_tag, "Read byte RDRF timeout i=", (uint8_t)i);
02586             rx_err_t stop_err = internal_send_stop(riic);

```

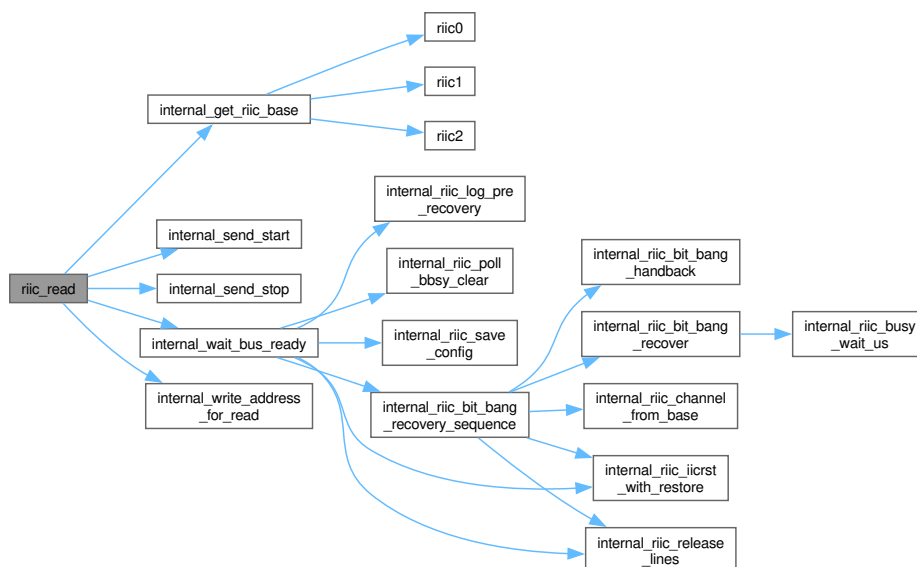
```

02587     (void)stop_err;
02588     return k_rx_err_timeout;
02589 }
02590
02591 if (length >= 2U && i == (uint16_t)(length - 2U)) {
02592     riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02593 }
02594
02595 if (i == (uint16_t)(length - k_riic_last_index_offset)) {
02596     riic->icsr2 &= (uint8_t) ~(uint8_t)k_riic_icsr2_stop;
02597     riic->iccr2 |= k_riic_iccr2_sp;
02598 }
02599
02600 data[i] = riic->icdrr;
02601 }
02602
02603 uint32_t stop_timeout = k_riic_timeout_us;
02604 while (!(riic->icsr2 & k_riic_icsr2_stop) && stop_timeout > k_riic_timeout_zero) {
02605     stop_timeout--;
02606 }
02607 if (stop_timeout == k_riic_timeout_zero) {
02608     rx_log_error(s_tag, "STOP after final ICDRR timeout");
02609     return k_rx_err_timeout;
02610 }
02611 riic->icsr2 &= (uint8_t) ~(uint8_t)(k_riic_icsr2_stop | k_riic_icsr2_nackf);
02612
02613 return k_rx_ok;
02614 }

```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icdrr](#), [rx_riic_regs_t::icmr3](#), [rx_riic_regs_t::icsr2](#), [internal_get_riic_base\(\)](#), [internal_send_start\(\)](#), [internal_send_stop\(\)](#), [internal_wait_bus_ready\(\)](#), [internal_write_address_for_read\(\)](#), [k_riic_addr_max_7bit](#), [k_riic_addr_read_bit](#), [k_riic_addr_shift](#), [k_riic_channel_0](#), [k_riic_iccr2_mst](#), [k_riic_iccr2_sp](#), [k_riic_icmr3_ackbt_mask](#), [k_riic_icsr2_nackf](#), [k_riic_icsr2_rdrf](#), [k_riic_icsr2_stop](#), [k_riic_last_index_offset](#), [k_riic_length_zero](#), [k_riic_max_channels](#), [k_riic_max_transfer_length](#), [k_riic_mode_controller](#), [k_riic_register_clear](#), [k_riic_timeout_us](#), [k_riic_timeout_zero](#), [s_riic_channel_mode](#), [s_tag](#), [i2c_device_addr_t::value](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.63 riic_write()

```

rx_err_t riic_write (
    const riic_channel_t channel,
    const i2c_device_addr_t device_addr,

```

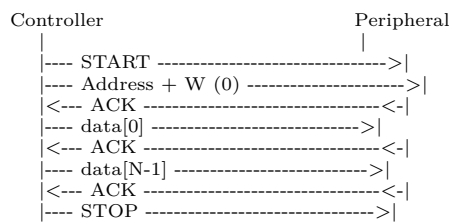
```
const uint8_t * data,
const uint16_t length) [nodiscard]
```

Write data to an I2C peripheral device.

Write data to I2C device on specified RIIC channel.

Performs a complete I2C write transaction: START, address+W, data bytes, STOP. This function is suitable for writing configuration registers, sending commands, or storing data in EEPROM.

8.105.4.64 Transaction Sequence



8.105.4.65 Common Use Cases

Use Case	Data Format
Config register write	[reg_addr, value]
Multi-byte register	[reg_addr, val0, val1, ...]
EEPROM page write	[addr_hi, addr_lo, data...]
Sensor command	[command_↔ byte]

8.105.4.66 Error Handling

On any error (timeout or NACK), the function issues a STOP condition to release the bus before returning. The original error code is preserved.

Precondition

```
riic_init() called for this channel
data != nullptr
length >= 1 && length <= 256
device_addr.value <= 127
```

Postcondition

```
On success: All bytes transferred, bus released
On failure: Bus released via STOP
```

Note

This is a blocking function - returns only after transfer complete.
 STOP is always issued (success or failure) to release the bus.

Warning

NACK on address usually means device not present or wrong address.
 NACK on data may indicate device buffer full or write-protected.

Example: Write Configuration Register

```
// Write value 0x42 to register 0x05 of device at address 0x48
uint8_t config_data[2] = {0x05, 0x42}; // [reg_addr, value]
riic_channel_t channel = {.value = 0};
i2c_device_addr_t addr = {.value = 0x48};

rx_err_t err = riic_write(channel, addr, config_data, 2);
if (err != k_rx_ok) {
    handle_i2c_error(err);
}
```

See also

[riic_init\(\)](#) Initialize channel before use
[riic_read\(\)](#) Read-only transaction
[riic_write_read\(\)](#) Combined write-then-read transaction

Definition at line 2349 of file [riic.c](#).

```
02353 {
02354     RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02355     RX_CHECK_RANGE_TAG(channel.value,
02356                         k_riic_channel_0,
02357                         (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02358                         k_rx_err_invalid_arg,
02359                         s_tag);
02360
02361     if (device_addr.value > k_riic_addr_max_7bit) {
02362         rx_log_error(s_tag, "Invalid device address");
02363         return k_rx_err_invalid_arg;
02364     }
02365
02366     if (length == k_riic_length_zero) {
02367         rx_log_error(s_tag, "Write length cannot be zero");
02368         return k_rx_err_invalid_arg;
02369     }
02370
02371     if (length > k_riic_max_transfer_length) {
02372         rx_log_error(s_tag, "Write length exceeds maximum");
02373         return k_rx_err_invalid_arg;
02374     }
02375
02376     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02377     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02378         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02379         return k_rx_err_invalid_state;
02380     }
02381
02382     /* Get RIIC base */
02383     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02384
02385     /* Wait for bus ready */
02386     rx_err_t err = internal_wait_bus_ready(riic);
02387     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02388
02389     /* Pre-clear any status flags left over from a previous transaction (e.g.
02390     * NACKF, STOP, AL) that normal bus activity does not clear. Per RX72N
02391     * HW manual section 42.2.7, these are sticky software-clear flags that
02392     * IICRST wipes but normal transactions do not. Without this pre-clear a
02393     * stale NACKF survives into the next write and internal_write_byte
```

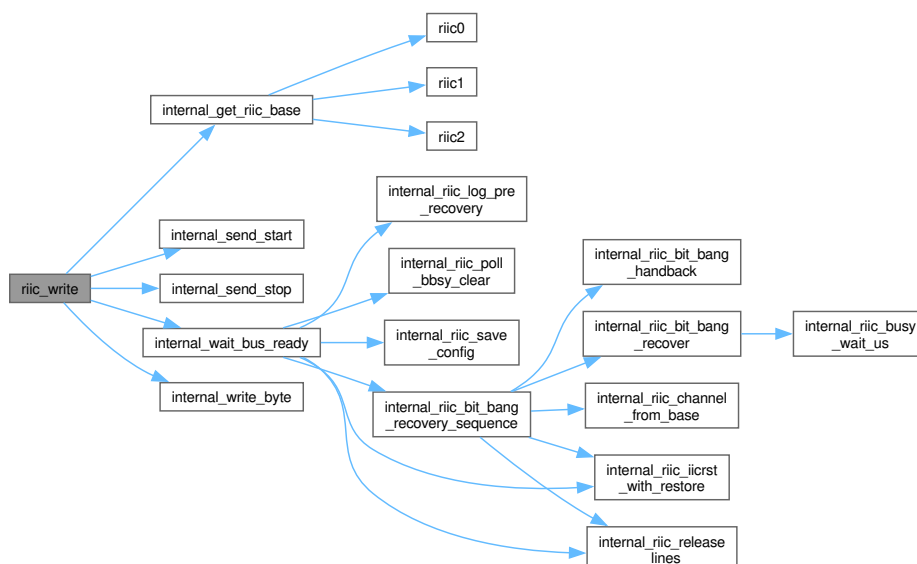
```

02394  * returns k_rx_err_nack on a byte that was actually ACKed; a stale AL
02395  * silently clears MST on the next START. */
02396  riic->icr2 = k_riic_register_clear;
02397
02398  /* Set controller transmit mode */
02399  riic->iccr2 = k_riic_iccr2_mst | k_riic_iccr2_trs;
02400
02401  /* Send start condition */
02402  err = internal_send_start(riic);
02403  RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02404
02405  /* Send device address (write) */
02406  err = internal_write_byte(riic, (device_addr.value « k_riic_addr_shift) | k_riic_addr_write_bit);
02407  if (err != k_rx_ok) {
02408      rx_err_t stop_err = internal_send_stop(riic);
02409      (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02410      return err;
02411  }
02412
02413  /* Send data bytes */
02414  for (uint16_t i = 0; i < length; i++) {
02415      err = internal_write_byte(riic, data[i]);
02416      if (err != k_rx_ok) {
02417          rx_err_t stop_err = internal_send_stop(riic);
02418          (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02419          return err;
02420      }
02421  }
02422
02423  /* Send stop condition */
02424  err = internal_send_stop(riic);
02425  RX_RETURN_ON_ERROR(err, s_tag, "Stop condition failed");
02426
02427  return k_rx_ok;
02428 }

```

References [rx_riic_regs_t::iccr2](#), [rx_riic_regs_t::icr2](#), [internal_get_riic_base\(\)](#), [internal_send_start\(\)](#), [internal_send_stop\(\)](#), [internal_wait_bus_ready\(\)](#), [internal_write_byte\(\)](#), [k_riic_addr_max_7bit](#), [k_riic_addr_shift](#), [k_riic_addr_write_bit](#), [k_riic_channel_0](#), [k_riic_iccr2_mst](#), [k_riic_iccr2_trs](#), [k_riic_last_index_offset](#), [k_riic_length_zero](#), [k_riic_max_channels](#), [k_riic_max_transfer_length](#), [k_riic_mode_controller](#), [k_riic_register_clear](#), [s_riic_channel_mode](#), [s_tag](#), [i2c_device_addr_t::value](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.4.67 riic`write`read()

```

rx_err_t riic_write_read (
    const riic\_channel\_t channel,

```

```

const i2c_device_addr_t device_addr,
const uint8_t * write_data,
const uint16_t write_length,
uint8_t * read_data,
const uint16_t read_length) [nodiscard]

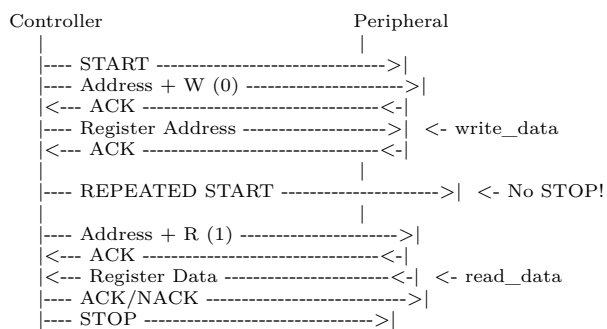
```

Combined write-then-read I2C transaction (register read).

Write then read from I2C device using repeated START (combined transaction).

Performs a complete I2C write-read transaction using a repeated START condition. This is the most common pattern for reading registers from I2C peripherals: write the register address, then read the data.

8.105.4.68 Transaction Sequence (Register Read)



8.105.4.69 Why Repeated START?

The repeated START (Sr) is critical for atomic register access:

Approach	Bus Release	Risk
STOP + START	Yes	Another controller could interrupt
Repeated START	No	Transaction is atomic

Many I2C peripherals require repeated START and may not work correctly with separate write-then-read transactions.

8.105.4.70 Common Use Cases

Peripheral Type	write`data	read`length	Purpose
Temperature	[reg_↔ addr]	2	Read temp register
EEPROM	[addr_hi, addr_lo]	N	Read N bytes
Accelerometer	[reg_↔ addr]	6	Read XYZ axes

Peripheral Type	write`data	read`length	Purpose
RTC	[seconds_↔ reg]	7	Read time

Precondition

```

riic_init() called for this channel
write_data != nullptr
read_data != nullptr
write_length >= 1 && write_length <= 256
read_length >= 1 && read_length <= 256
device_addr.value <= 127

```

Postcondition

On success: read_data contains register contents, bus released
On failure: Bus released via STOP

Note

This is a blocking function - returns only after transfer complete.
Uses repeated START (no bus release between write and read phases).
NACK is automatically sent before reading last byte.

Example: Read WHO_AM_I Register from IMU

```

// Read WHO_AM_I register (0x0F) from MPU6050 at address 0x68
riic_channel_t channel = {.value = 0};
i2c_device_addr_t imu_addr = {.value = 0x68};

uint8_t reg_addr = 0x0F; // WHO_AM_I register
uint8_t who_am_i;

rx_err_t err = riic_write_read(channel, imu_addr,
                               &reg_addr, 1, // Write: register address
                               &who_am_i, 1); // Read: 1 byte

if (err == k_rx_ok && who_am_i == 0x68) {
    // IMU detected and responding correctly
}

```

Example: Read 6 Bytes of Accelerometer Data

```

// Read ACCEL_XOUT_H through ACCEL_ZOUT_L (6 bytes starting at 0x3B)
uint8_t reg_addr = 0x3B;
uint8_t accel_data[6];

rx_err_t err = riic_write_read(channel, imu_addr,
                               &reg_addr, 1,
                               accel_data, 6);

if (err == k_rx_ok) {
    int16_t accel_x = (accel_data[0] « 8) | accel_data[1];
    int16_t accel_y = (accel_data[2] « 8) | accel_data[3];
    int16_t accel_z = (accel_data[4] « 8) | accel_data[5];
}

```

See also

[riic_init\(\)](#) Initialize channel before use
[riic_write\(\)](#) Write-only transaction
[riic_read\(\)](#) Read-only transaction (no address write)

Definition at line 2721 of file [riic.c](#).

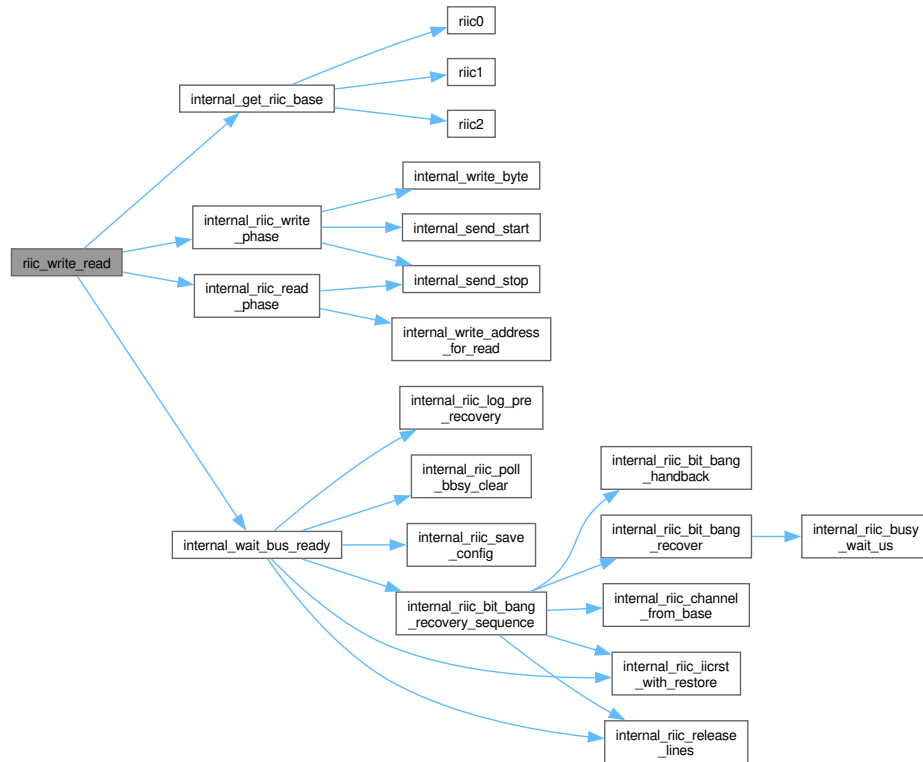
```

02727 {
02728     RX_CHECK_NULL_PTR(write_data, s_tag, "Write data pointer is nullptr");
02729     RX_CHECK_NULL_PTR(read_data, s_tag, "Read data pointer is nullptr");
02730     RX_CHECK_RANGE_TAG(channel.value,
02731         k_riic_channel_0,
02732         (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02733         k_rx_err_invalid_arg,
02734         s_tag);
02735
02736     if (device_addr.value > k_riic_addr_max_7bit) {
02737         rx_log_error(s_tag, "Invalid device address");
02738         return k_rx_err_invalid_arg;
02739     }
02740
02741     if (write_length == k_riic_length_zero || read_length == k_riic_length_zero) {
02742         rx_log_error(s_tag, "Read/write length cannot be zero");
02743         return k_rx_err_invalid_arg;
02744     }
02745
02746     if (write_length > k_riic_max_transfer_length || read_length > k_riic_max_transfer_length) {
02747         rx_log_error(s_tag, "Read/write length exceeds maximum");
02748         return k_rx_err_invalid_arg;
02749     }
02750
02751     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02752     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02753         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02754         return k_rx_err_invalid_state;
02755     }
02756
02757     /* Get RIIC base */
02758     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02759
02760     /* Wait for bus ready */
02761     rx_err_t err = internal_wait_bus_ready(riic);
02762     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02763
02764     /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02765     riic->icsr2 = k_riic_register_clear;
02766
02767     /* Perform write phase */
02768     err = internal_riic_write_phase(riic, device_addr, write_data, write_length);
02769     if (err != k_rx_ok) {
02770         return err;
02771     }
02772
02773     /* Perform read phase. On success the final byte's NACK + STOP is
02774      * issued atomically inside internal_read_byte(), so no separate
02775      * internal_send_stop() call is needed here. */
02776     err = internal_riic_read_phase(riic, device_addr, read_data, read_length);
02777     if (err != k_rx_ok) {
02778         return err;
02779     }
02780
02781     return k_rx_ok;
02782 }

```

References [rx_riic_regs_t::icsr2](#), [internal_get_riic_base\(\)](#), [internal_riic_read_phase\(\)](#), [internal_riic_write_phase\(\)](#), [internal_wait_bus_ready\(\)](#), [k_riic_addr_max_7bit](#), [k_riic_channel_0](#), [k_riic_last_index_offset](#), [k_riic_length_zero](#), [k_riic_max_channels](#), [k_riic_max_transfer_length](#), [k_riic_mode_controller](#), [k_riic_register_clear](#), [s_riic_channel_mode](#), [s_tag](#), [i2c_device_addr_t::value](#), and [riic_channel_t::value](#).

Here is the call graph for this function:



8.105.5 Variable Documentation

8.105.5.1 k'riic'recovery'pins

const [riic_recovery_pins_t](#) k_riic_recovery_pins[k_riic_max_channels] [static]

Initial value:

```

= {
[k_riic_channel_0] = {0U, 0U, 0U, 0U, 0U},
[k_riic_channel_1] =
{0x0008C002U, k_riic_port2_odr0_addr, 1U, 0U, (uint8_t)((1U « 1) | (1U « 3))},
[k_riic_channel_2] = {0U, 0U, 0U, 0U, 0U},
}

```

Per-channel recovery pin lookup.

Mirrors hardware_init.c::internal_riic1_bus_recover for the RIIC1 channel; zero-filled entries are skipped. See STAR PCB schematic + docs/sections/03_hardware_pinout.tex for the canonical pin assignment.

Definition at line 620 of file [riic.c](#).

```

00620                                     {
00621 [k_riic_channel_0] = {0U, 0U, 0U, 0U, 0U},
00622 [k_riic_channel_1] =
00623 {0x0008C002U, k_riic_port2_odr0_addr, 1U, 0U, (uint8_t)((1U « 1) | (1U « 3))},
00624 [k_riic_channel_2] = {0U, 0U, 0U, 0U, 0U},
00625 };

```

Referenced by [internal_riic_bit_bang_handback\(\)](#), and [internal_riic_bit_bang_recover\(\)](#).

8.105.5.2 s'riic'channel'mode

`riic_channel_mode_t s_riic_channel_mode[k_riic_max_channels]` [static]

Initial value:

```
= {k_riic_mode_uninitialized,
   k_riic_mode_uninitialized,
   k_riic_mode_uninitialized}
```

Per-channel RIIC mode tracking array.

Tracks the operational mode of each RIIC channel: uninitialized, controller, or peripheral. Used to enforce that controller APIs are only called on controller-initialized channels and vice-versa.

Index	Channel	Initial State
0	RIIC0	k_riic_mode_uninitialized
1	RIIC1	k_riic_mode_uninitialized
2	RIIC2	k_riic_mode_uninitialized

Invariant

Array size == k_riic_max_channels (3)

Warning

Do not modify directly – only `riic_init()`, `riic_init_peripheral()`, and `riic_deinit_peripheral()` may write to this array.

Note

Not thread-safe; provide external synchronization if needed.

Since

Version 1.0.0

Definition at line 900 of file `riic.c`.

```
00900                                     {k_riic_mode_uninitialized,
00901                                     k_riic_mode_uninitialized,
00902                                     k_riic_mode_uninitialized};
```

Referenced by `riic_deinit_peripheral()`, `riic_init()`, `riic_init_peripheral()`, `riic_peripheral_read()`, `riic_peripheral_write()`, `riic_read()`, `riic_write()`, and `riic_write_read()`.

8.105.5.3 s'tag

`const char* const s_tag = "RIIC"` [static]

Logging tag for RIIC driver messages.

Used with `rx_log_error()` and `rx_log_debug()` for consistent log output formatting. All RIIC driver log messages are prefixed with "RIIC".

Note

Not thread-safe, but read-only after initialization.

Definition at line 570 of file `riic.c`.

8.106 riic.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file riic.c
00003  * @brief RIIC (I2C) Driver Implementation for RX72N
00004  *
00005  * @details
00006  * **Production-grade implementation** of the I2C Bus Interface (RIIC) peripheral
00007  * driver for the RX72N microcontroller. This driver provides controller-mode
00008  * I2C communication with support for standard (100 kHz), fast (400 kHz), and
00009  * fast-plus (1 MHz) modes.
00010  *
00011  * ## Module Architecture
00012  *
00013  * ```
00014  * +-----+
00015  * |                RIIC Driver Architecture                |
00016  * +-----+
00017  * |
00018  * | Application Layer                                     |
00019  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00020  * | | riic_init() riic_write() riic_read() riic_write_read() | | |
00021  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00022  * | |
00023  * | Internal Layer                                       |
00024  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00025  * | | internal_get_riic_base() - Channel to address mapping | | |
00026  * | | internal_calculate_bit_rate() - Frequency calculation | | |
00027  * | | internal_wait_bus_ready() - Bus arbitration          | | |
00028  * | | internal_send_start() - START condition generation  | | |
00029  * | | internal_send_stop() - STOP condition generation     | | |
00030  * | | internal_write_byte() - Single byte transmission     | | |
00031  * | | internal_read_byte() - Single byte reception         | | |
00032  * | | internal_riic_write_phase() - Combined write sequence | | |
00033  * | | internal_riic_read_phase() - Combined read sequence  | | |
00034  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00035  * | |
00036  * | Hardware Layer                                       |
00037  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00038  * | | RIIC0: 0x00088300 (SCL0/P16, SDA0/P17)                | | |
00039  * | | RIIC1: 0x00088320 (SCL1/P21, SDA1/P20)                | | |
00040  * | | RIIC2: 0x00088340 (SCL2/P66, SDA2/P67)                | | |
00041  * | +-----+ +-----+ +-----+ +-----+ +-----+ |
00042  * | |
00043  * +-----+
00044  * ```
00045  *
00046  * ## I2C Protocol Implementation
00047  *
00048  * This driver implements the I2C protocol in controller mode:
00049  *
00050  * ### Write Transaction
00051  *
00052  * Controller                                     Peripheral
00053  * |----- START ----->|
00054  * |----- Address + W (0) ----->|
00055  * |<--- ACK -----<|
00056  * |----- Data Byte 0 ----->|
00057  * |<--- ACK -----<|
00058  * |----- Data Byte N ----->|
00059  * |<--- ACK -----<|
00060  * |----- STOP ----->|
00061  *
00062  *
00063  *
00064  * ### Read Transaction
00065  *
00066  * Controller                                     Peripheral
00067  * |----- START ----->|
00068  * |----- Address + R (1) ----->|
00069  * |<--- ACK -----<|
00070  * |<--- Data Byte 0 -----<|
00071  * |----- ACK ----->|
00072  * |<--- Data Byte N -----<|
00073  * |----- NACK (last byte) ----->|
00074  * |----- STOP ----->|
00075  *
00076  *
00077  *
00078  * ### Write-Read Transaction (Register Read)
00079  *
00080  * Controller                                     Peripheral
00081  * |----- START ----->|
00082  *

```

```

00083 * |---- Address + W (0) ----->|
00084 * |<--- ACK -----<-|
00085 * |---- Register Address ----->|
00086 * |<--- ACK -----<-|
00087 * |---- REPEATED START ----->|
00088 * |---- Address + R (1) ----->|
00089 * |<--- ACK -----<-|
00090 * |<--- Data Byte 0 -----<-|
00091 * |---- NACK ----->|
00092 * |---- STOP ----->|
00093 * ```
00094 *
00095 * ## Bit Rate Calculation
00096 *
00097 * The I2C clock frequency is calculated from PCLKB (60 MHz) as:
00098 *
00099 * @f[
00100 * f_{SCL} = \frac{f_{PCLKB}}{(ICBRL + ICBRH + 2) \times 2^{CKS}}
00101 * @f]
00102 *
00103 * For simplified calculation with 50% duty cycle (ICBRL = ICBRH):
00104 *
00105 * @f[
00106 * \text{divisor} = \frac{f_{PCLKB}}{f_{SCL} \times 3}
00107 * @f]
00108 *
00109 * | Mode | Speed | Divisor | Actual Rate |
00110 * |-----|-----|-----|-----|
00111 * | Standard | 100 kHz | 200 | 100.0 kHz |
00112 * | Fast | 400 kHz | 50 | 400.0 kHz |
00113 * | Fast Plus | 1 MHz | 20 | 1.00 MHz |
00114 *
00115 * ## Memory Map (Chapter 42 - RIIC)
00116 *
00117 * | Channel | Base Address | Pin SCL | Pin SDA | Module Stop Bit |
00118 * |-----|-----|-----|-----|-----|
00119 * | RIIC0 | 0x00088300 | P16 | P17 | MSTPB21 (MSTPCRB b21) |
00120 * | RIIC1 | 0x00088320 | P21 | P20 | MSTPB20 (MSTPCRB b20) |
00121 * | RIIC2 | 0x00088340 | P66 | P67 | MSTPC17 (MSTPCRC b17) |
00122 *
00123 * ## Register Offsets (per channel)
00124 *
00125 * | Offset | Register | Description |
00126 * |-----|-----|-----|
00127 * | 0x00 | ICCR1 | Control Register 1 |
00128 * | 0x01 | ICCR2 | Control Register 2 |
00129 * | 0x02 | ICMR1 | Mode Register 1 |
00130 * | 0x03 | ICMR2 | Mode Register 2 |
00131 * | 0x04 | ICMR3 | Mode Register 3 |
00132 * | 0x10 | ICBRL | Bit Rate Low period |
00133 * | 0x11 | ICBRH | Bit Rate High period |
00134 * | 0x12 | ICDRT | Transmit Data Register |
00135 * | 0x13 | ICDRR | Receive Data Register |
00136 *
00137 * ## Error Handling
00138 *
00139 * The driver provides comprehensive error detection:
00140 * - **k_rx_err_null_ptr**: nullptr passed to function
00141 * - **k_rx_err_invalid_arg**: Invalid channel, address, or frequency
00142 * - **k_rx_err_invalid_state**: Channel not initialized
00143 * - **k_rx_err_timeout**: Bus busy or operation timeout
00144 * - **k_rx_err_nack**: Peripheral did not acknowledge
00145 *
00146 * ## NASA Power of 10 Compliance
00147 *
00148 * | Rule | Implementation |
00149 * |-----|-----|
00150 * | Rule 1 | No goto, setjmp, or recursion - sequential error handling |
00151 * | Rule 2 | Bounded loops - all loops use k_riic_timeout_us timeout |
00152 * | Rule 3 | No dynamic memory - static channel state array |
00153 * | Rule 4 | Short functions - each < 60 lines |
00154 * | Rule 5 | Validation - RX_CHECK_NULL_PTR, RX_CHECK_RANGE_TAG |
00155 * | Rule 6 | Smallest scope - local variables declared at point of use |
00156 * | Rule 7 | All return values checked - RX_RETURN_ON_ERROR |
00157 * | Rule 8 | Limited preprocessor - only RX_CHECK macros |
00158 * | Rule 9 | Function pointers via interface pattern (not used here) |
00159 * | Rule 10 | Compiles with -Wall -Wextra -Werror |
00160 *
00161 * ## SOLID Principles Compliance
00162 *
00163 * | Principle | Implementation |
00164 * |-----|-----|
00165 * | **S** Single Responsibility | Driver handles only RIIC I2C operations |
00166 * | **O** Open/Closed | Supports multiple frequencies via parameter |
00167 * | **L** Liskov Substitution | Consistent error codes across all functions |
00168 * | **I** Interface Segregation | Separate read/write/write-read APIs |
00169 * | **D** Dependency Inversion | Uses hardware.h abstraction for registers |

```

```

00170 *
00171 * @par Thread Safety
00172 * This module is NOT thread-safe. Each RIIC channel should be accessed
00173 * by only one thread at a time. If multiple threads need I2C access, provide
00174 * external synchronization (mutex/semaphore) at the application level.
00175 *
00176 * @par Example: Read Temperature Sensor
00177 * @code
00178 * // Initialize RIIC0 at 400 kHz (Fast mode)
00179 * riic_channel_t channel = {.value = 0};
00180 * rx_err_t err = riic_init(channel, 400000);
00181 * if (err != k_rx_ok) {
00182 *     handle_error(err);
00183 *     return;
00184 * }
00185 *
00186 * // Read 2 bytes from temperature sensor at address 0x48
00187 * i2c_device_addr_t sensor_addr = {.value = 0x48};
00188 * uint8_t temp_reg = 0x00; // Temperature register
00189 * uint8_t temp_data[2];
00190 *
00191 * err = riic_write_read(channel, sensor_addr, &temp_reg, 1, temp_data, 2);
00192 * if (err == k_rx_ok) {
00193 *     int16_t raw_temp = (temp_data[0] << 8) | temp_data[1];
00194 *     float celsius = raw_temp * 0.0625f; // LM75 resolution
00195 * }
00196 * @endcode
00197 *
00198 * @see rx72n_riic_regs.h RIIC register definitions
00199 * @see hardware.h Hardware abstraction layer
00200 *
00201 * @par Manual Reference
00202 * RX72N Group User's Manual: Hardware, Chapter 42 (I2C-bus Interface)
00203 *
00204 * @date 2026-01-01
00205 * @copyright Copyright (c) 2026 Locked Inc.
00206 * SPDX-License-Identifier: MIT
00207 */
00208
00209 #ifdef __RX__
00210
00211 #include <string.h>
00212
00213 #include "hardware.h"
00214 #include "rx72n_regs.h"
00215 #include "rx_register_protection.h"
00216
00217 /*
=====
00218 * Constants
00219 *
=====
00220 * All constants use C23 typed enums per STAR coding standards.
00221 * No magic numbers - every value has a named constant with documentation.
00222 *
=====
00223 */
00224 /**
00225 * @enum riic_constants_t
00226 * @brief RIIC channel count, timeout, and transfer limit constants
00227 *
00228 * @details
00229 * General-purpose constants for RIIC driver operation including channel
00230 * limits, timeout values, and transfer size constraints.
00231 *
00232 * @par Timeout Selection
00233 * The 10ms timeout provides sufficient margin for:
00234 * - Standard mode: 100 kHz = 10us/bit, 90us/byte -> 10ms = ~111 bytes
00235 * - Fast mode: 400 kHz = 2.5us/bit, 22.5us/byte -> 10ms = ~444 bytes
00236 * - Fast Plus: 1 MHz = 1us/bit, 9us/byte -> 10ms = ~1111 bytes
00237 *
00238 * @invariant k_riic_max_channels == 3 (hardware limitation)
00239 * @invariant k_riic_max_transfer_length == 256 (single-byte length limit)
00240 */
00241
00242 typedef enum : uint32_t {
00243     k_riic_max_channels = 3, /**< Total RIIC channels: RIIC0, RIIC1, RIIC2 */
00244     k_riic_timeout_us = 25000, /**< Busy-loop iteration budget for I2C flag waits.
00245     * Each iteration reads ICSR2/ICCR2 + decrements a
00246     * volatile counter (~10 cycles @ 240 MHz ~ 40 ns),
00247     * so 25k iters ~ 1 ms wall time -- enough to
00248     * cover a byte at 100 kHz (90 us) with 10x slack
00249     * for clock stretching. Kept below the 900 ms IWDWT
00250     * window even if a 256-byte transfer stacks up
00251     * 256 per-byte timeouts back-to-back. */
00252     k_riic_timeout_zero = 0, /**< Timeout counter expired sentinel value */
00253     k_riic_length_zero = 0, /**< Zero-length transfer (invalid) sentinel */

```

```

00254 k_riic_register_clear    = 0,    /**< Zero value used to clear hardware registers */
00255 k_riic_last_index_offset = 1,    /**< Offset to calculate last byte index from length */
00256 k_riic_max_transfer_length = 256, /**< Maximum bytes per transfer operation */
00257 } riic_constants_t;
00258
00259 /**
00260 * @enum riic_channel_num_t
00261 * @brief RIIC channel identifiers for switch statement dispatch
00262 *
00263 * @details
00264 * Provides named constants for channel selection in internal_get_riic_base().
00265 * Using enums instead of raw integers ensures type safety and improves
00266 * code readability in switch statements.
00267
00268 * @par Channel to Pin Mapping
00269 * | Channel | SCL Pin | SDA Pin | Base Address |
00270 * |-----|-----|-----|-----|
00271 * | RIIC0   | P16     | P17     | 0x00088300   |
00272 * | RIIC1   | P21     | P20     | 0x00088320   |
00273 * | RIIC2   | P66     | P67     | 0x00088340   |
00274 *
00275 * @invariant Values are contiguous starting from 0 and match hardware channel indices
00276
00277 * @code
00278 * // Select channel 0 base address
00279 * volatile rx_riic_regs_t* regs = internal_get_riic_base(k_riic_channel_0);
00280 * @endcode
00281 *
00282 * @see internal_get_riic_base() Maps channel number to hardware register pointer
00283 * @see riic_constants_t For k_riic_max_channels bound used with these values
00284 */
00285 typedef enum : uint8_t {
00286     k_riic_channel_0 = 0, /**< RIIC channel 0 (P16/SCL0, P17/SDA0) */
00287     k_riic_channel_1 = 1, /**< RIIC channel 1 (P21/SCL1, P20/SDA1) */
00288     k_riic_channel_2 = 2, /**< RIIC channel 2 (P66/SCL2, P67/SDA2) */
00289 } riic_channel_num_t;
00290
00291 /**
00292 * @enum riic_module_stop_bits_t
00293 * @brief RIIC module stop bit positions in MSTPCRB (RIIC0/1) and MSTPCRC (RIIC2)
00294 *
00295 * @details
00296 * The RX72N uses a module stop mechanism to reduce power consumption.
00297 * RIIC0 and RIIC1 module stop bits reside in MSTPCRB; RIIC2 resides in
00298 * MSTPCRC (manual Ch11, p408-410). RIIC channels must be enabled (bit
00299 * cleared) before use.
00300 *
00301 * @par Module Enable Procedure
00302 * 1. Unlock register protection: PRCR = k_rx_prcr_unlock_prc1_prc3
00303 * 2. RIIC0: MSTPCRB &= ~(1 « k_riic_mstpb_riic0)
00304 *    RIIC1: MSTPCRB &= ~(1 « k_riic_mstpb_riic1)
00305 *    RIIC2: MSTPCRC &= ~(1 « k_riic_mstpc_riic2)
00306 * 3. Lock register protection: PRCR = k_rx_prcr_lock
00307 *
00308 * @see rx_register_protection.h for PRCR register access
00309 */
00310 typedef enum : uint8_t {
00311     k_riic_mstpb_riic0 = 21, /**< MSTPCRB bit 21: RIIC0 module stop control (manual p408) */
00312     k_riic_mstpb_riic1 = 20, /**< MSTPCRB bit 20: RIIC1 module stop control (manual p408) */
00313     k_riic_mstpc_riic2 = 17, /**< MSTPCRC bit 17: RIIC2 module stop control (manual p410) */
00314     k_riic_mstpb_bit_value = 1, /**< Single bit value for bit manipulation */
00315 } riic_module_stop_bits_t;
00316
00317 /**
00318 * @enum riic_frequency_t
00319 * @brief Supported I2C bus frequencies (Hz)
00320 *
00321 * @details
00322 * I2C standard defines three speed grades. This driver supports all three
00323 * modes with appropriate bit rate register configuration.
00324 *
00325 * @par Timing Characteristics (at PCLKB = 60 MHz)
00326 * | Mode      | Frequency | Bit Time | Byte Time | Max Cable |
00327 * |-----|-----|-----|-----|-----|
00328 * | Standard  | 100 kHz   | 10 us    | 90 us     | ~1 meter  |
00329 * | Fast      | 400 kHz   | 2.5 us   | 22.5 us   | ~0.5 m    |
00330 * | Fast Plus | 1 MHz     | 1 us     | 9 us      | ~0.2 m    |
00331 *
00332 * @invariant Only these three discrete values are accepted by riic_init()
00333 *
00334 * @code
00335 * // Initialize channel 0 at standard (100 kHz) mode
00336 * riic_channel_t ch = {.value = 0};
00337 * rx_err_t err = riic_init(ch, k_riic_freq_100khz);
00338 * @endcode
00339 *
00340 * @see riic_init() Validates frequency_hz against these enum values

```

```

00341 * @see internal_calculate_bit_rate() Uses frequency to compute ICBRL/ICBRH
00342 *
00343 * @note Fast Plus mode requires pull-up resistors < 1kOhm for proper rise time.
00344 */
00345 typedef enum : uint32_t {
00346     k_riic_freq_100khz = 100000, /**< Standard mode: 100 kHz (Sm) */
00347     k_riic_freq_400khz = 400000, /**< Fast mode: 400 kHz (Fm) */
00348     k_riic_freq_1mhz = 1000000, /**< Fast mode plus: 1 MHz (Fm+) */
00349 } riic_frequency_t;
00350
00351 /**
00352 * @enum riic_bit_rate_t
00353 * @brief Bit rate calculation constants for ICBRL/ICBRH registers
00354 *
00355 * @details
00356 * These constants are used in internal_calculate_bit_rate() to compute
00357 * the correct ICBRL and ICBRH register values for a target frequency.
00358 *
00359 * @par Calculation Formula
00360 * For 50% duty cycle (ICBRL = ICBRH = divisor):
00361 * @f[
00362 * \text{divisor} = \frac{f_{\text{PCLKB}}}{f_{\text{target}}} \times 3
00363 * @f]
00364 *
00365 * Where the factor of 3 accounts for:
00366 * - SCL high period (ICBRH + 1) cycles
00367 * - SCL low period (ICBRL + 1) cycles
00368 * - Internal synchronization overhead
00369 */
00370 /* ICMR1 CKS field and ICBRH/ICBRL values for the three supported transfer
00371 * rates at PCLKB = 60 MHz. Values straight out of RX72N HW manual Table
00372 * 42.5 "Examples of ICBRH/ICBRL Settings for Transfer Rate", PCLK=60 MHz
00373 * column. Low 5 bits of ICBRH/ICBRL are the actual counts; high 3 bits are
00374 * reserved (read-as-1, write-ignored). Using CKS=0 with a computed divisor
00375 * silently truncated >31 divisors to 5 bits and produced MHz-scale SCL on
00376 * a 100 kHz bus, which most peripherals mostly tolerate but which BNO055
00377 * reads do not (the addr|R ACK succeeds but the first SCL of data
00378 * reception glitches the RIIC FSM and MST drops mid-read). */
00379 typedef enum : uint8_t {
00380     k_riic_cks_100khz = 4U, /**< ICMR1.CKS[2:0] = 100b for 100 kbps @ 60 MHz PCLK */
00381     k_riic_icbrh_100k = 14U, /**< ICBRH count, HUM Table 42.5 */
00382     k_riic_icbrl_100k = 17U, /**< ICBRL count, HUM Table 42.5 */
00383     k_riic_cks_400khz = 2U, /**< ICMR1.CKS[2:0] = 010b for 400 kbps @ 60 MHz PCLK */
00384     k_riic_icbrh_400k = 8U, /**< ICBRH count, HUM Table 42.5 */
00385     k_riic_icbrl_400k = 19U, /**< ICBRL count, HUM Table 42.5 */
00386     k_riic_cks_1mhz = 0U, /**< ICMR1.CKS[2:0] = 000b for 1 Mbps @ 60 MHz PCLK */
00387     k_riic_icbrh_1m = 15U, /**< ICBRH count, HUM Table 42.5 */
00388     k_riic_icbrl_1m = 29U, /**< ICBRL count, HUM Table 42.5 */
00389     k_riic_cks_shift = 4U, /**< ICMR1 bit position of CKS[0] (CKS is bits [6:4]) */
00390 } riic_bit_rate_t;
00391
00392 /**
00393 * @enum riic_icmr1_values_t
00394 * @brief ICMR1 (Mode Register 1) configuration values
00395 *
00396 * @details
00397 * ICMR1 configures the controller/peripheral mode and addressing type.
00398 * For STAR project, we use controller mode with 7-bit addressing.
00399 *
00400 * @par ICMR1 Register Layout (Bits 7-4)
00401 * | Bit | Name | Description |
00402 * |-----|-----|
00403 * | 6-4 | CKS | Clock divider (0-7) |
00404 * | 3 | BCWP | BC write protect |
00405 * | 2-0 | BC | Bit counter |
00406 */
00407 typedef enum : uint8_t {
00408     k_riic_icmr1_controller_7bit =
00409         0x08, /**< CKS=0, BCWP=1, 7-bit addressing; MTWP=0 keeps MST/TRS read-only so the hardware START/STOP
00410             FSM drives them, per HUM section 42.2.3. MTWP=1 was tried and caused "Start condition failed" on every transaction
00411             because iccr2=MST|TRS pre-writes the mode bits before START fires and the FSM gets desynced. */
00412 } riic_icmr1_values_t;
00413
00414 /**
00415 * @enum riic_icmr2_values_t
00416 * @brief ICMR2 (Mode Register 2) configuration values
00417 *
00418 * @details
00419 * ICMR2 configures hardware timeout detection and SDA output hold-time delay.
00420 * The STAR project disables hardware timeout (TMOE = 0) because bus-busy
00421 * detection is handled in software by internal_wait_bus_ready() using the
00422 * k_riic_timeout_us countdown. The SDA output delay is also left at zero
00423 * (no additional hold time) since the pull-up resistors on the target PCB
00424 * provide adequate signal integrity at all supported speeds.
00425 *
00426 * ## ICMR2 Register Bit Layout

```

```

00426 * | Bit | Name | Function | Value Used |
00427 * |-----|-----|-----|-----|
00428 * | 7 | TMOH | Timeout detection (high period) | 0 (off) |
00429 * | 6 | TMOL | Timeout detection (low period) | 0 (off) |
00430 * | 5 | TMOE | Timeout enable | 0 (off) |
00431 * | 4:3 | SDDL | SDA output hold-time delay | 0 (none) |
00432 * | 2 | DLCS | SDA delay clock source | 0 |
00433 * | 1:0 | - | Reserved | 0 |
00434 *
00435 * @invariant k_riic_icmr2_default == 0; writing 0x00 leaves all ICMR2 control
00436 * bits in their hardware-reset state (timeouts off, no SDA delay)
00437 *
00438 * @code
00439 * // Apply default ICMR2 configuration during RIIC channel initialization
00440 * volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_0);
00441 * riic->icmr2 = k_riic_icmr2_default;
00442 * @endcode
00443 *
00444 * @see riic_init() Applies this value during channel initialization
00445 * @see k_riic_timeout_us Software timeout used instead of hardware TMOE
00446 */
00447 typedef enum : uint8_t {
00448     k_riic_icmr2_default = 0, /**< No hardware timeout, no SDA delay */
00449 } riic_icmr2_values_t;
00450
00451 /**
00452 * @enum riic_icmr3_bits_t
00453 * @brief ICMR3 (Mode Register 3) ACK/NACK control bit definitions
00454 *
00455 * @details
00456 * ICMR3 controls ACK/NACK transmission during receive operations. The key
00457 * field is ACKBT (Acknowledge Bit), which determines whether the controller
00458 * sends an ACK or NACK after each received byte.
00459 *
00460 * ## ACK/NACK Signalling Rules (I2C Specification)
00461 *
00462 * - Set ACKBT=0 (clear bit) to send ACK: instructs the peripheral to continue
00463 * transmitting; used for all bytes except the last in a read transfer.
00464 * - Set ACKBT=1 (set bit) to send NACK: instructs the peripheral to stop
00465 * transmitting; must be applied before clocking in the final byte so that
00466 * the NACK is sent automatically at the correct clock edge.
00467 *
00468 * ## ICMR3 Register Bit Layout (relevant bits)
00469 *
00470 * | Bit | Name | Function |
00471 * |-----|-----|-----|
00472 * | 4 | ACKWP | ACKBT write protect (1=writeable) |
00473 * | 3 | ACKBT | ACK/NACK bit to transmit |
00474 *
00475 * @invariant k_riic_icmr3_ackbt_pos == 3 matches the RX72N hardware bit
00476 * position for ACKBT in ICMR3; changing this breaks ACK/NACK control
00477 * @invariant k_riic_icmr3_ackbt_mask == (1U < 3) == 0x08; derived from pos
00478 * @invariant k_riic_icmr3_ackwp_pos == 4 matches the RX72N hardware bit
00479 * position for ACKWP; changing this silently breaks ACKBT writes
00480 * @invariant k_riic_icmr3_ackwp_mask == (1U < 4) == 0x10; derived from pos
00481 *
00482 * @note NACK must be sent before reading the last byte to signal to the
00483 * peripheral that the transfer is complete per I2C specification.
00484 *
00485 * @note ACKWP=1 is REQUIRED before any ACKBT write. Without it, writes to
00486 * ACKBT are silently ignored -- the controller keeps ACKing every byte,
00487 * the peripheral never sees NACK, and STOP times out with the bus
00488 * locked. ACKWP=1 is set in riic_init() as part of k_riic_icmr3_init.
00489 *
00490 * @code
00491 * // Prepare to NACK the last byte in a read transfer (ACKWP must already be 1)
00492 * volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_0);
00493 * riic->icmr3 |= k_riic_icmr3_ackbt_mask; // Set ACKBT=1 -> send NACK
00494 *
00495 * // Re-enable ACK for subsequent transfers
00496 * riic->icmr3 &= (uint8_t)~k_riic_icmr3_ackbt_mask; // Set ACKBT=0 -> send ACK
00497 * @endcode
00498 *
00499 * @see internal_read_byte() Uses these constants to control ACK/NACK output
00500 * @see riic_read() Orchestrates NACK signalling for multi-byte reads
00501 *
00502 * @since Version 1.0.0
00503 */
00504 typedef enum : uint8_t {
00505     k_riic_icmr3_ackbt_pos = 3, /**< ACKBT bit position (bit 3) */
00506     k_riic_icmr3_ackbt_mask = (1U < k_riic_icmr3_ackbt_pos), /**< ACKBT bit mask (0x08) */
00507     k_riic_icmr3_ackwp_pos = 4, /**< ACKWP bit position (bit 4) */
00508     k_riic_icmr3_ackwp_mask = (1U < k_riic_icmr3_ackwp_pos), /**< ACKWP bit mask (0x10) */
00509     k_riic_icmr3_wait_pos = 6, /**< WAIT bit position (bit 6) */
00510     k_riic_icmr3_wait_mask = (1U < k_riic_icmr3_wait_pos), /**< WAIT bit mask (0x40) */
00511     k_riic_icmr3_init = k_riic_icmr3_ackwp_mask, /**< ICMR3 init: ACKWP=1, ACKBT=0 */
00512 } riic_icmr3_bits_t;

```



```

00513
00514 /**
00515  * @enum riic_address_bits_t
00516  * @brief I2C 7-bit address formatting constants
00517  *
00518  * @details
00519  * I2C uses a 7-bit address plus a read/write direction bit in the first
00520  * byte after START. The address is left-shifted by 1 and OR'd with the
00521  * direction bit.
00522  *
00523  * @par Address Byte Format
00524  * ```
00525  * Bit: 7 6 5 4 3 2 1 0
00526  *      +-----+ +- R/W (0=Write, 1=Read)
00527  *      +- 7-bit Address
00528  * ```
00529  *
00530  * @par Address Calculation
00531  * - Write to 0x50: (0x50 « 1) | 0 = 0xA0
00532  * - Read from 0x50: (0x50 « 1) | 1 = 0xA1
00533  *
00534  * @invariant k_riic_addr_max_7bit == 127 (I2C spec: addresses 0x00-0x7F)
00535  *
00536  * @code
00537  * // Format a write address byte for device at 0x50
00538  * riic_device_addr_t dev = {.value = 0x50};
00539  * uint8_t addr_byte = (uint8_t)((dev.value « k_riic_addr_shift) | k_riic_addr_write_bit);
00540  * @endcode
00541  *
00542  * @see internal_write_byte() Uses these constants to format address bytes
00543  * @see riic_write() Passes formatted address bytes during write transactions
00544  */
00545 typedef enum : uint8_t {
00546     k_riic_addr_shift = 1, /**< Left shift amount for 7-bit address */
00547     k_riic_addr_write_bit = 0, /**< R/W bit value for write operation */
00548     k_riic_addr_read_bit = 1, /**< R/W bit value for read operation */
00549     k_riic_addr_max_7bit = 127, /**< Maximum valid 7-bit address (0x7F) */
00550 } riic_address_bits_t;
00551
00552 /*
=====
00553  * Static Variables
00554  *
=====
00555  * Module-level state variables. These track initialization status and
00556  * provide logging context. Protected by the s_ prefix per STAR coding standards.
00557  *
=====
00558  */
00559
00560 /**
00561  * @var s_tag
00562  * @brief Logging tag for RIIC driver messages
00563  *
00564  * @details
00565  * Used with rx_log_error() and rx_log_debug() for consistent log output
00566  * formatting. All RIIC driver log messages are prefixed with "RIIC".
00567  *
00568  * @note Not thread-safe, but read-only after initialization.
00569  */
00570 static const char* const s_tag = "RIIC";
00571
00572 /**
00573  * @enum riic_recovery_constants_t
00574  * @brief Bit-bang bus-recovery timing and ODR register addresses
00575  *
00576  * @details
00577  * When a peripheral on the bus holds SDA/SCL low, IICRST alone cannot clear a
00578  * stuck BBSY because the peripheral reset does not toggle external SCL
00579  * edges. internal_riic_bit_bang_recover() temporarily drops PMR, drives
00580  * SCL low-high 9 times to walk any stuck peripheral through a byte + ACK,
00581  * generates a manual STOP, then hands the pads back to the RIIC peripheral.
00582  *
00583  * ODR0 register addresses per port (RX72N HW manual Table 22.2):
00584  * Port 2 ODR0 @ 0x0008C084 -- pin pair bits (P20 -> bit1, P21 -> bit3)
00585  */
00586 typedef enum : uint32_t {
00587     k_riic_recov_cycles = 9U, /**< 9 SCL edges: flush one byte + ACK */
00588     k_riic_recov_half_us = 5U, /**< 5 us half-period -> ~100 kHz SCL */
00589     k_riic_recov_cpu_mhz = 240U, /**< ICLK frequency for busy-wait math */
00590     k_riic_recov_stop_us = 5U, /**< STOP condition SDA-low-then-high delay */
00591     k_riic_port2_odr0_addr = 0x0008C084U, /**< Port 2 ODR0 (P20/P21 N-ch open-drain) */
00592 } riic_recovery_constants_t;
00593
00594 /**
00595  * @struct riic_recovery_pins_t
00596  * @brief Channel-specific port + pin info for bit-bang bus recovery

```

```

00597 *
00598 * @details
00599 * Indexed by riic channel number. A zero port_base entry means no recovery
00600 * table entry has been provided for that channel (skipped at runtime).
00601 * Only RIIC1 is currently populated because only the IMU uses RIIC; add
00602 * RIIC0/RIIC2 entries when those channels are activated.
00603 */
00604 typedef struct {
00605     uintptr_t port_base; /**< rx_port_regs_t* base address for the port holding SCL+SDA */
00606     uintptr_t odr0_addr; /**< ODR register address for N-ch open-drain selection */
00607     uint8_t scl_bit; /**< SCL pin bit position within the port (0-7) */
00608     uint8_t sda_bit; /**< SDA pin bit position within the port (0-7) */
00609     uint8_t odr_mask; /**< ODR bit mask for N-ch open-drain on SCL + SDA pin pair */
00610 } riic_recovery_pins_t;
00611
00612 /**
00613 * @brief Per-channel recovery pin lookup
00614 *
00615 * @details
00616 * Mirrors hardware_init.c::internal_riic1_bus_recover for the RIIC1
00617 * channel; zero-filled entries are skipped. See STAR PCB schematic +
00618 * docs/sections/03_hardware_pinout.tex for the canonical pin assignment.
00619 */
00620 static const riic_recovery_pins_t k_riic_recovery_pins[k_riic_max_channels] = {
00621     [k_riic_channel_0] = {0U, 0U, 0U, 0U, 0U},
00622     [k_riic_channel_1] =
00623         {0x0008C002U, k_riic_port2_odr0_addr, 1U, 0U, (uint8_t)((1U << 1) | (1U << 3))},
00624     [k_riic_channel_2] = {0U, 0U, 0U, 0U, 0U},
00625 };
00626
00627 /**
00628 * @brief Resolve a RIIC register pointer to its channel number
00629 *
00630 * @details
00631 * Compares the pointer against the three RIIC base addresses. Returns
00632 * k_riic_max_channels (sentinel) if the pointer does not match any channel,
00633 * which callers treat as "skip bit-bang recovery".
00634 */
00635 static uint8_t internal_riic_channel_from_base(const volatile rx_riic_regs_t* riic)
00636 {
00637     const uintptr_t base = (uintptr_t)riic;
00638     if (base == (uintptr_t)k_riic0_base_addr) {
00639         return (uint8_t)k_riic_channel_0;
00640     }
00641     if (base == (uintptr_t)k_riic1_base_addr) {
00642         return (uint8_t)k_riic_channel_1;
00643     }
00644     if (base == (uintptr_t)k_riic2_base_addr) {
00645         return (uint8_t)k_riic_channel_2;
00646     }
00647     return (uint8_t)k_riic_max_channels;
00648 }
00649
00650 /**
00651 * @brief Cycle-counted busy-wait for bit-bang timing
00652 *
00653 * @details
00654 * Volatile counter loop calibrated for the current ICLK frequency. Mirrors
00655 * hardware_init.c::internal_busy_wait_us -- duplicated here so riic.c has
00656 * no cross-module dependency on src/ helpers.
00657 */
00658 static inline void internal_riic_busy_wait_us(uint16_t us, uint16_t cpu_mhz)
00659 {
00660     volatile uint32_t cycles = (uint32_t)us * (uint32_t)cpu_mhz;
00661     while (cycles > 0U) {
00662         cycles--;
00663     }
00664 }
00665
00666 /**
00667 * @brief Bit-bang 9 SCL edges + manual STOP to recover an externally-stuck bus
00668 *
00669 * @details
00670 * Called from internal_wait_bus_ready when BBSY stays latched even after
00671 * an IICRST pulse. That combination means an I2C peripheral on the bus
00672 * (BNO055 or BMP280 in the STAR system) got interrupted mid-byte -- e.g.
00673 * the RX72N reset during a previous transaction -- and is still holding
00674 * SDA low waiting to finish sending the remaining bits of its byte + ACK.
00675 * IICRST only resets the RIIC controller side; it does not toggle SCL on
00676 * the wire, so it cannot unstick an external peripheral. The only
00677 * software path that clears this is to physically wiggle SCL from the
00678 * MCU pad -- which requires handing the pad from RIIC back to GPIO long
00679 * enough to clock out a full byte + ACK + STOP.
00680 */

```

```

00684 * Sequence:
00685 * 1. Drop PMR -> pads are GPIO again, RIIC loses them.
00686 * 2. Enable N-ch open-drain so we cannot drive-fight a peripheral still
00687 * holding the line low.
00688 * 3. Drive both lines high, set as output. External pull-ups dominate.
00689 * 4. 9x { SCL low; wait; SCL high; wait } -- one byte + ACK worth of
00690 * clocks, enough to clock out any peripheral stuck mid-byte.
00691 * 5. Manual STOP: SDA low -> high with SCL high -- tells any peripheral
00692 * still listening "transaction over, idle the bus".
00693 * 6. Tristate pads, clear open-drain. Leave PMR=0. The caller raises
00694 * PMR back to 1 AFTER it has reset the RIIC peripheral (IICRST +
00695 * SOWP dance), so the peripheral hands the pads back already in a
00696 * "driving high" state. Raising PMR while the peripheral still has
00697 * leftover SCLO=SDAO=0 from the previous transaction would glitch
00698 * the bus low again immediately.
00699 *
00700 * Only RIIC channels with a populated entry in k_riic_recovery_pins[]
00701 * are handled; other channels return silently.
00702 *
00703 *
00704 * @pre MPC PFS values for SCL/SDA were set during channel init and have
00705 * not been disturbed (we only toggle PMR, not PFS).
00706 * @post PMR for SCL/SDA is CLEARED (pads stay in GPIO mode, tristated).
00707 * Caller must reset the RIIC peripheral and then raise PMR.
00708 * @post PDR bits for SCL/SDA are cleared (tristated).
00709 *
00710 * @par NASA Power of 10 Compliance
00711 * - Rule 2: bounded loop (k_riic_recov_cycles = 9)
00712 * - Rule 5: channel bounds check + port_base != 0 validation
00713 *
00714 * @see internal_riic_bit_bang_handback() Companion that raises PMR once
00715 * the peripheral has been reset to a clean SCLO=SDAO=1 state.
00716 */
00717 static void internal_riic_bit_bang_recover(uint8_t channel)
00718 {
00719     if (channel >= (uint8_t)k_riic_max_channels) {
00720         return;
00721     }
00722     const riic_recovery_pins_t* const pins = &k_riic_recovery_pins[channel];
00723     if (pins->port_base == 0U) {
00724         return; /* No recovery table entry for this channel. */
00725     }
00726
00727     volatile rx_port_regs_t* const port = (volatile rx_port_regs_t*)pins->port_base;
00728     volatile uint8_t* const odr0 = (volatile uint8_t*)pins->odr0_addr;
00729
00730     const uint8_t scl_mask = (uint8_t)(1U « pins->scl_bit);
00731     const uint8_t sda_mask = (uint8_t)(1U « pins->sda_bit);
00732     const uint8_t both = (uint8_t)(scl_mask | sda_mask);
00733
00734     /* Hand pads back to GPIO so we can bit-bang. PFS values are preserved
00735      * (we only toggle PMR, not PFS/PWPR). */
00736     port->pmr &= (uint8_t)~both;
00737
00738     /* N-ch open-drain: avoid push-pull fight with a peripheral still holding low. */
00739     *odr0 |= pins->odr_mask;
00740
00741     /* Drive both high, then set as output. Pull-ups bring the bus up. */
00742     port->podr |= both;
00743     port->pdr |= both;
00744
00745     for (uint16_t i = 0; i < (uint16_t)k_riic_recov_cycles; i++) {
00746         port->podr &= (uint8_t)~scl_mask;
00747         internal_riic_busy_wait_us((uint16_t)k_riic_recov_half_us, (uint16_t)k_riic_recov_cpu_mhz);
00748         port->podr |= scl_mask;
00749         internal_riic_busy_wait_us((uint16_t)k_riic_recov_half_us, (uint16_t)k_riic_recov_cpu_mhz);
00750     }
00751
00752     /* Manual STOP: SDA low (SCL already high), then SDA high. */
00753     port->podr &= (uint8_t)~sda_mask;
00754     internal_riic_busy_wait_us((uint16_t)k_riic_recov_stop_us, (uint16_t)k_riic_recov_cpu_mhz);
00755     port->podr |= sda_mask;
00756     internal_riic_busy_wait_us((uint16_t)k_riic_recov_stop_us, (uint16_t)k_riic_recov_cpu_mhz);
00757
00758     /* Tristate pads + clear open-drain. DO NOT raise PMR here -- see
00759      * internal_riic_bit_bang_handback() for the reason. */
00760     port->pdr &= (uint8_t)~both;
00761     *odr0 &= (uint8_t)~pins->odr_mask;
00762 }
00763
00764 /**
00765 * @brief Raise PMR so the RIIC peripheral takes the pads back
00766 *
00767 * @details
00768 * Companion to internal_riic_bit_bang_recover(). Call this AFTER the
00769 * RIIC peripheral has been IICRST-pulsed and put through the SOWP / SCLO
00770 * / SDAO release dance so its internal output latches are high. Raising

```

```

00771 * PMR while those latches are still 0 would glitch the bus low again
00772 * immediately and could re-confuse the peripheral we just unstuck.
00773 *
00774 *
00775 * @pre internal_riic_bit_bang_recover() was just called on the same channel.
00776 * @pre The RIIC peripheral has been reset (IICRST cycle + SOWP release).
00777 * @post PMR bits for SCL/SDA are set, pads routed to RIIC peripheral.
00778 */
00779 /**
00780 * @brief Pulse IICRST on a controller-initialized RIIC channel and restore config.
00781 *
00782 * @details
00783 * Per RX72N HW manual section 42.2.1, writing 1 to ICCR1.IICRST while
00784 * ICCR1.ICE=0 performs an internal reset of the RIIC peripheral's FSM
00785 * without touching the pin mux. That drops any half-completed transaction
00786 * state -- BBSY, MST, TRS, pending SP/RS/ST request bits, NACKF, latched
00787 * AL, the "dummy read kicked reception" flag -- back to idle, which is
00788 * what we need before a new transaction when an earlier transaction on
00789 * the same peripheral may have left the FSM partially advanced. The
00790 * config registers (ICBRH/L, ICMR1..3) are wiped by IICRST, so we
00791 * restore them from the live values (which were set in riic_init() and
00792 * must already be valid since ICE was on before the pulse).
00793 *
00794 * Do NOT substitute this for riic_init(): this helper assumes MSTPCR and
00795 * MPC/PFS are already set up and that ICE=1 was previously toggled to
00796 * latch the peripheral out of module-stop. It is a fast pre-transaction
00797 * FSM scrub, not a cold boot.
00798 *
00799 * Currently unused in the main transaction path -- kept as a [[maybe_unused]]
00800 * helper because it is the right escape hatch when the FSM has been
00801 * observed to get stuck mid-transaction and the bit-bang recovery in
00802 * internal_wait_bus_ready() is too heavyweight (it tears down and rebuilds
00803 * pad direction via PMR, which adds ~100 us versus ~2 us here).
00804 *
00805 *
00806 * @pre ICE was set to 1 earlier (channel previously passed riic_init).
00807 * @post ICCR1.ICE = 1, ICCR1.IICRST = 0, ICMR1..3 / ICBRH/L restored,
00808 *       ICSR1 / ICSR2 cleared.
00809 */
00810 [[maybe_unused]]
00811 static void internal_riic_force_fsm_reset(volatile rx_riic_regs_t* riic)
00812 {
00813     const uint8_t saved_icbrh = riic->icbrh;
00814     const uint8_t saved_icbrl = riic->icbrl;
00815     const uint8_t saved_icmr1 = riic->icmr1;
00816     const uint8_t saved_icmr2 = riic->icmr2;
00817
00818     riic->iccr1 = k_riic_iccr1_iicrst;
00819     riic->iccr1 = k_riic_register_clear;
00820
00821     riic->icbrh = saved_icbrh;
00822     riic->icbrl = saved_icbrl;
00823     riic->icmr1 = saved_icmr1;
00824     riic->icmr2 = saved_icmr2;
00825     riic->icmr3 = k_riic_icmr3_init;
00826
00827     riic->icsr1 = k_riic_register_clear;
00828     riic->icsr2 = k_riic_register_clear;
00829
00830     riic->iccr1 = k_riic_iccr1_ice;
00831
00832     uint8_t iccr1 = riic->iccr1;
00833     iccr1 &= (uint8_t) ~(uint8_t) k_riic_iccr1_sowp;
00834     iccr1 |= (uint8_t) (k_riic_iccr1_sclo | k_riic_iccr1_sdao);
00835     riic->iccr1 = iccr1;
00836     iccr1 |= (uint8_t) k_riic_iccr1_sowp;
00837     riic->iccr1 = iccr1;
00838 }
00839
00840 static void internal_riic_bit_bang_handback(uint8_t channel)
00841 {
00842     if (channel >= (uint8_t) k_riic_max_channels) {
00843         return;
00844     }
00845     const riic_recovery_pins_t* const pins = &k_riic_recovery_pins[channel];
00846     if (pins->port_base == 0U) {
00847         return;
00848     }
00849
00850     volatile rx_port_regs_t* const port = (volatile rx_port_regs_t*) pins->port_base;
00851     const uint8_t scl_mask = (uint8_t) (1U << pins->scl_bit);
00852     const uint8_t sda_mask = (uint8_t) (1U << pins->sda_bit);
00853     port->pmr |= (uint8_t) (scl_mask | sda_mask);
00854 }
00855
00856 /**
00857 * @enum riic_channel_mode_t

```

```

00858 * @brief Per-channel RIIC initialization mode
00859 *
00860 * @details
00861 * Tracks whether each RIIC channel has been initialized, and if so, whether
00862 * it was initialized for controller or peripheral operation. This prevents
00863 * peripheral-mode APIs from running on a controller-initialized channel and
00864 * vice-versa, satisfying the Liskov Substitution Principle for RIIC API
00865 * implementations.
00866 *
00867 * @invariant Set only by riic_init() (controller) and riic_init_peripheral() (peripheral)
00868 * @invariant Reset to k_riic_mode_uninitialized by riic_deinit_peripheral() on success
00869 *
00870 * @since Version 1.0.0
00871 */
00872 typedef enum : uint8_t {
00873     k_riic_mode_uninitialized = 0x00u, /**< Channel not yet initialized */
00874     k_riic_mode_controller    = 0x01u, /**< Initialized by riic_init() -- controller mode */
00875     k_riic_mode_peripheral    = 0x02u, /**< Initialized by riic_init_peripheral() -- peripheral mode */
00876 } riic_channel_mode_t;
00877
00878 /**
00879 * @var s_riic_channel_mode
00880 * @brief Per-channel RIIC mode tracking array
00881 *
00882 * @details
00883 * Tracks the operational mode of each RIIC channel: uninitialized, controller,
00884 * or peripheral. Used to enforce that controller APIs are only called on
00885 * controller-initialized channels and vice-versa.
00886 *
00887 * | Index | Channel | Initial State |
00888 * |-----|-----|-----|
00889 * | 0     | RIIC0   | k_riic_mode_uninitialized |
00890 * | 1     | RIIC1   | k_riic_mode_uninitialized |
00891 * | 2     | RIIC2   | k_riic_mode_uninitialized |
00892 *
00893 * @invariant Array size == k_riic_max_channels (3)
00894 * @warning Do not modify directly -- only riic_init(), riic_init_peripheral(),
00895 *         and riic_deinit_peripheral() may write to this array.
00896 * @note Not thread-safe; provide external synchronization if needed.
00897 *
00898 * @since Version 1.0.0
00899 */
00900 static riic_channel_mode_t s_riic_channel_mode[k_riic_max_channels] = {k_riic_mode_uninitialized,
00901                                                                    k_riic_mode_uninitialized,
00902                                                                    k_riic_mode_uninitialized};
00903
00904 /*
=====
00905 * Internal Helper Functions
00906 *
=====
00907 * These functions provide low-level I2C operations used by the public API.
00908 * All internal functions follow NASA Power of 10 rules for safety-critical code.
00909 *
=====
00910 */
00911
00912 /**
00913 * @brief Get RIIC channel register base address from channel number
00914 *
00915 * @details
00916 * Maps an RIIC channel number (0-2) to its corresponding hardware register
00917 * base address using a compile-time-dispatched switch statement. This function
00918 * is the single point of truth for the channel-to-address mapping and provides
00919 * a clean abstraction between the logical channel index used throughout the
00920 * driver and the physical memory-mapped peripheral addresses defined by the
00921 * RX72N hardware.
00922 *
00923 * ## Channel-to-Address Mapping
00924 *
00925 * The three RIIC channels are located at fixed, evenly spaced addresses in the
00926 * RX72N peripheral address space (0x00088300, 0x00088320, 0x00088340). Each
00927 * block is 32 bytes in size and contains all control, status, bit-rate, and
00928 * data registers for one channel.
00929 *
00930 * ## Memory Map
00931 *
00932 * | Channel | Accessor | Base Address | Register Block Size |
00933 * |-----|-----|-----|-----|
00934 * | 0       | riic0()  | 0x00088300   | 32 bytes |
00935 * | 1       | riic1()  | 0x00088320   | 32 bytes |
00936 * | 2       | riic2()  | 0x00088340   | 32 bytes |
00937 *
00938 * ## Why Return NULL for Invalid Channel
00939 *
00940 * Returning nullptr for an out-of-range channel allows every caller to perform
00941 * a single null-pointer check with RX_CHECK_NULL_PTR rather than duplicating

```

```

00942 * range validation logic. This pattern is consistent with rx_port_get_base()
00943 * in the GPIO driver.
00944 *
00945 *
00946 *
00947 * @pre channel must be less than k_riic_max_channels (3); values >= 3 yield nullptr
00948 * @pre caller must have verified that the corresponding RIIC module stop bit in
00949 * MSTPCR_B has been cleared (module enabled) before using the returned pointer
00950 * @post returned pointer is non-NULL if and only if channel is 0, 1, or 2
00951 * @post returned pointer, when non-NULL, is aligned to the RIIC peripheral register
00952 * block boundary as required by the RX72N hardware specification
00953 *
00954 * @note This function does NOT validate the channel - the caller is responsible
00955 * for ensuring the value is in range before calling
00956 * @note The return type is volatile to prevent the compiler from optimizing away
00957 * hardware register reads or writes through the returned pointer
00958 *
00959 * @par Performance:
00960 * O(1) - simple switch statement with a direct return per case
00961 *
00962 * @par Example:
00963 * @code
00964 * // Look up the register base for RIIC channel 1 and send a START condition
00965 * volatile rx_riic_regs_t* riic = internal_get_riic_base(k_riic_channel_1);
00966 * if (riic == nullptr) {
00967 *     return k_rx_err_invalid_arg;
00968 * }
00969 * riic->iccr2 |= k_riic_iccr2_st; // Issue START condition
00970 * @endcode
00971 *
00972 * @see riic0() Inline accessor for RIIC channel 0 base address
00973 * @see riic1() Inline accessor for RIIC channel 1 base address
00974 * @see riic2() Inline accessor for RIIC channel 2 base address
00975 * @see k_riic_max_channels Upper bound on valid channel numbers
00976 *
00977 * @since Version 1.0.0
00978 */
00979 static volatile rx_riic_regs_t* internal_get_riic_base(const uint8_t channel)
00980 {
00981     switch (channel) {
00982         case k_riic_channel_0: {
00983             return riic0();
00984         }
00985         case k_riic_channel_1: {
00986             return riic1();
00987         }
00988         case k_riic_channel_2: {
00989             return riic2();
00990         }
00991         default: {
00992             return nullptr;
00993         }
00994     }
00995 }
00996
00997 /**
00998 * @brief Calculate RIIC bit rate register values for target frequency
00999 *
01000 * @details
01001 * Computes the ICBRL and ICBRH register values to achieve the desired I2C
01002 * bus frequency. Uses a simplified calculation assuming 50% duty cycle
01003 * (equal high and low periods).
01004 *
01005 * ### Calculation Algorithm
01006 *
01007 * The RX72N RIIC bit rate formula (from Chapter 42):
01008 * @f[
01009 * f_{SCL} = \frac{f_{PCLKB}}{(ICBRL + 1) + (ICBRH + 1) + \text{sync}}
01010 * @f]
01011 *
01012 * Simplified for 50% duty cycle (ICBRL = ICBRH):
01013 * @f[
01014 * \text{divisor} = \frac{f_{PCLKB}}{f_{target}} \times 3
01015 * @f]
01016 *
01017 * ### Example Calculations (PCLKB = 60 MHz)
01018 *
01019 * | Target | Calculation | Divisor | Actual Rate |
01020 * |-----|-----|-----|-----|
01021 * | 100 kHz | 60M / (100k x 3) | 200 | 100.0 kHz |
01022 * | 400 kHz | 60M / (400k x 3) | 50 | 400.0 kHz |
01023 * | 1 MHz | 60M / (1M x 3) | 20 | 1.00 MHz |
01024 *
01025 *
01026 *
01027 * @pre icbri != nullptr
01028 * @pre icbrh != nullptr

```

```

01029 * @pre frequency_hz is a supported I2C frequency
01030 * @post *icbri contains the low-period divisor (1-255)
01031 * @post *icbrh contains the high-period divisor (1-255)
01032 * @post *icbri == *icbrh (50% duty cycle)
01033 *
01034 * @note Uses PCLKB (60 MHz) as the source clock per RX72N specification.
01035 * @note For asymmetric duty cycle, manually set ICBRL != ICBRH after init.
01036 *
01037 * @warning Frequencies outside 100 kHz - 1 MHz may produce out-of-range divisors.
01038 *
01039 * @see riic_init() Uses this function during channel initialization
01040 * @see k_pclkb_hz PCLKB frequency constant from hardware.h
01041 */
01042 static rx_err_t internal_calculate_bit_rate(const uint32_t frequency_hz,
01043      uint8_t* icbri,
01044      uint8_t* icbrh,
01045      uint8_t* cks_field)
01046 {
01047     if (icbri == nullptr || icbrh == nullptr || cks_field == nullptr) {
01048         return k_rx_err_null_ptr;
01049     }
01050
01051     /* Look up the HUM Table 42.5 row for the requested rate. Only the three
01052      * discrete rates the riic_channel_freq_t enum accepts are supported; we
01053      * do not compute CKS/ICBRH/ICBRL from a formula because the formula in
01054      * the HUM depends on SCL rise/fall times and the table values already
01055      * bake in the standard-mode / fast-mode / fast-mode-plus tr/tf budgets. */
01056     if (frequency_hz == (uint32_t)k_riic_freq_100khz) {
01057         *cks_field = (uint8_t)k_riic_cks_100khz;
01058         *icbrh = (uint8_t)k_riic_icbrh_100k;
01059         *icbri = (uint8_t)k_riic_icbri_100k;
01060         return k_rx_ok;
01061     }
01062     if (frequency_hz == (uint32_t)k_riic_freq_400khz) {
01063         *cks_field = (uint8_t)k_riic_cks_400khz;
01064         *icbrh = (uint8_t)k_riic_icbrh_400k;
01065         *icbri = (uint8_t)k_riic_icbri_400k;
01066         return k_rx_ok;
01067     }
01068     if (frequency_hz == (uint32_t)k_riic_freq_1mhz) {
01069         *cks_field = (uint8_t)k_riic_cks_1mhz;
01070         *icbrh = (uint8_t)k_riic_icbrh_1m;
01071         *icbri = (uint8_t)k_riic_icbri_1m;
01072         return k_rx_ok;
01073     }
01074
01075     rx_log_error(s_tag, "Invalid frequency (expected 100 kHz, 400 kHz, or 1 MHz)");
01076     return k_rx_err_invalid_arg;
01077 }
01078
01079 /**
01080 * @brief Wait for I2C bus to become idle (not busy)
01081 *
01082 * @details
01083 * Polls the BBSY (Bus Busy) bit in ICCR2 until the bus is free or timeout
01084 * occurs. This function must be called before starting a new I2C transaction
01085 * to ensure no other controller is using the bus (multi-controller arbitration).
01086 *
01087 * ## Bus Busy Detection
01088 *
01089 * The BBSY bit is set when:
01090 * - A START condition is detected on the bus
01091 * - The controller has issued a START condition
01092 *
01093 * The BBSY bit is cleared when:
01094 * - A STOP condition is detected on the bus
01095 * - Bus arbitration is lost
01096 *
01097 * ## Timeout Calculation
01098 *
01099 * At 100 kHz (slowest mode), maximum byte transfer time:
01100 * - 9 bits per byte (8 data + 1 ACK) x 10 us = 90 us
01101 * - For 256 bytes (max transfer): 256 x 90 us = 23 ms
01102 *
01103 * The 10 ms timeout (k_riic_timeout_us) provides margin for typical
01104 * operations while catching true bus-stuck conditions.
01105 *
01106 *
01107 *
01108 * @pre riic != nullptr
01109 * @pre RIIC channel must be initialized and enabled
01110 * @post Bus is idle if k_rx_ok returned
01111 *
01112 * @note This is a blocking function - polls in tight loop.
01113 * @note Timeout is ~10ms at k_riic_timeout_us iterations.
01114 *
01115 * @warning In multi-controller systems, bus may become busy again immediately

```



```

01116 *      after this function returns - caller should proceed quickly.
01117 *
01118 * @par NASA Power of 10 Compliance
01119 * - Rule 2: [OK] Bounded loop with k_riic_timeout_us maximum iterations
01120 * - Rule 5: [OK] nullptr check via RX_CHECK_NULL_PTR
01121 */
01122 /**
01123 * @brief Snapshot of RIIC config registers preserved across IICRST pulses
01124 *
01125 * @details
01126 * IICRST resets ICBRH/ICBRL/ICMR1/ICMR2/ICMR3 to power-on defaults. The
01127 * recovery sequence saves these before pulsing IICRST and restores them
01128 * after, so the post-recovery RIIC keeps the configured baud rate and
01129 * mode bits.
01130 *
01131 * @since Version 1.0.0
01132 */
01133 typedef struct {
01134     uint8_t icbrh; /**< @brief ICBRH (high-period bit-rate divisor) */
01135     uint8_t icbrl; /**< @brief ICBRL (low-period bit-rate divisor) */
01136     uint8_t icmr1; /**< @brief ICMR1 (mode register 1) */
01137     uint8_t icmr2; /**< @brief ICMR2 (mode register 2) */
01138 } riic_saved_cfg_t;
01139
01140 /**
01141 * @brief Poll BBSY for up to k_riic_timeout_us iterations
01142 *
01143 * @details
01144 * Tight-loop watcher for the Bus Busy bit in ICCR2. Returns true when
01145 * BBSY clears within the timeout, false otherwise.
01146 *
01147 * @param[in] riic RIIC peripheral base (must be non-null)
01148 *
01149 * @return bool true if BBSY observed clear, false on timeout
01150 *
01151 * @pre riic non-null
01152 * @post Returns false on timeout
01153 *
01154 * @note Caller decides recovery action (warn + IICRST or bit-bang).
01155 *
01156 * @since Version 1.0.0
01157 */
01158 static bool internal_riic_poll_bbsy_clear(volatile rx_riic_regs_t* const riic)
01159 {
01160     uint32_t timeout = k_riic_timeout_us;
01161     while ((riic->iccr2 & k_riic_iccr2_bbsy) && timeout > k_riic_timeout_zero) {
01162         timeout--;
01163     }
01164     return timeout > k_riic_timeout_zero;
01165 }
01166
01167 /**
01168 * @brief Save config registers wiped by an IICRST pulse
01169 *
01170 * @param[in] riic RIIC peripheral base (must be non-null)
01171 * @param[out] saved Snapshot of ICBRH/ICBRL/ICMR1/ICMR2
01172 *
01173 * @pre riic and saved non-null
01174 * @post saved holds the pre-reset register values
01175 *
01176 * @since Version 1.0.0
01177 */
01178 static void internal_riic_save_config(const volatile rx_riic_regs_t* const riic,
01179                                     riic_saved_cfg_t* const saved)
01180 {
01181     saved->icbrh = riic->icbrh;
01182     saved->icbrl = riic->icbrl;
01183     saved->icmr1 = riic->icmr1;
01184     saved->icmr2 = riic->icmr2;
01185 }
01186
01187 /**
01188 * @brief Pulse IICRST and restore config from a saved snapshot
01189 *
01190 * @details
01191 * Asserts IICRST=1 then deasserts back to configuration state, restores
01192 * ICBRH/ICBRL/ICMR1/ICMR2 plus ICMR3=k_riic_icmr3_init (IICRST also wipes
01193 * ICMR3, and ACKWP=1 must be restored so reads can NACK the final byte).
01194 * Clears stale ICSR1/ICSR2 flags before re-enabling ICE=1.
01195 *
01196 * @param[in] riic RIIC peripheral base (must be non-null)
01197 * @param[in] saved Pre-recovery config snapshot
01198 *
01199 * @pre riic and saved non-null
01200 * @post ICE=1 with saved config and clean status registers

```



```

01203 *
01204 * @since Version 1.0.0
01205 */
01206 static void internal_riic_iicrst_with_restore(volatile rx_riic_regs_t* const riic,
01207                                             const riic_saved_cfg_t* const saved)
01208 {
01209     riic->iccr1 = k_riic_iccr1_iicrst;
01210     riic->iccr1 = k_riic_register_clear;
01211
01212     riic->icbrh = saved->icbrh;
01213     riic->icbrl = saved->icbrl;
01214     riic->icmr1 = saved->icmr1;
01215     riic->icmr2 = saved->icmr2;
01216     riic->icmr3 = k_riic_icmr3_init;
01217
01218     riic->icsr1 = k_riic_register_clear;
01219     riic->icsr2 = k_riic_register_clear;
01220
01221     riic->iccr1 = k_riic_iccr1_ice;
01222 }
01223
01224 /**
01225  * @brief Drive the SOWP/SCLO/SDAO release dance after an IICRST pulse
01226  *
01227  * @details
01228  * Forces SCLO=SDAO=1 via SOWP write-protect-clear so the open-drain
01229  * drivers release the bus (high-Z) on handback to the peripheral.
01230  *
01231  * @param[in] riic RIIC peripheral base (must be non-null)
01232  *
01233  *
01234  * @pre riic non-null and ICE=1
01235  * @post SOWP restored to write-protected
01236  *
01237  * @since Version 1.0.0
01238  */
01239 static void internal_riic_release_lines(volatile rx_riic_regs_t* const riic)
01240 {
01241     uint8_t iccr1 = riic->iccr1;
01242     iccr1 &= (uint8_t) ~(uint8_t) k_riic_iccr1_sowp;
01243     iccr1 |= (uint8_t) (k_riic_iccr1_sclo | k_riic_iccr1_sdao);
01244     riic->iccr1 = iccr1;
01245     iccr1 |= (uint8_t) k_riic_iccr1_sowp;
01246     riic->iccr1 = iccr1;
01247 }
01248
01249 /**
01250  * @brief Log debug-level pre-recovery bus state
01251  *
01252  * @param[in] riic RIIC peripheral base (must be non-null)
01253  *
01254  *
01255  * @note PIDR struct access lets rx72n_port_regs.h static_asserts catch
01256  *       layout drift at compile time. Compiled out when
01257  *       LOG_LEVEL < k_log_debug.
01258  *
01259  * @since Version 1.0.0
01260  */
01261 static void internal_riic_log_pre_recovery(const volatile rx_riic_regs_t* const riic)
01262 {
01263     rx_log_debug_val(s_tag, " PIDR =0x", ((volatile const rx_port_regs_t*)0x0008C002U)->pidr);
01264     rx_log_debug_val(s_tag, " ICCR2=0x", riic->iccr2);
01265     rx_log_debug_val(s_tag, " ICSR2=0x", riic->icsr2);
01266 }
01267
01268 /**
01269  * @brief Bit-bang last-resort bus recovery + peripheral reset + handback
01270  *
01271  * @details
01272  * Walks the bus through 9 SCL edges + manual STOP via GPIO bit-bang to
01273  * release a peripheral holding SDA low from an aborted transaction
01274  * across an MCU reboot (verified on BNO055 against the production STAR
01275  * PCB). Critical ordering preserved: bit-bang leaves PMR=0 (pads still
01276  * GPIO), peripheral reset runs while pads are GPIO, then handback
01277  * raises PMR=1 with the SOWP dance already done so the lines stay
01278  * high-Z.
01279  *
01280  * @param[in] riic RIIC peripheral base (must be non-null)
01281  * @param[in] saved Pre-recovery config snapshot (used for restore)
01282  *
01283  *
01284  * @pre riic and saved non-null; bus has refused to clear via IICRST alone
01285  * @post Pads handed back to the peripheral with config restored
01286  *
01287  * @since Version 1.0.0
01288  */
01289 static void internal_riic_bit_bang_recovery_sequence(volatile rx_riic_regs_t* const riic,

```

```

01290                                     const riic_saved_cfg_t* const saved)
01291 {
01292     const uint8_t channel = internal_riic_channel_from_base(riic);
01293     rx_log_warn(s_tag, "IICRST insufficient - bit-bang bus recovery");
01294     internal_riic_bit_bang_recover(channel); /* leaves PMR=0 */
01295
01296     internal_riic_iicrst_with_restore(riic, saved);
01297     internal_riic_release_lines(riic);
01298
01299     internal_riic_bit_bang_handback(channel);
01300
01301     if (channel == (uint8_t)k_riic_channel_1) {
01302         rx_log_debug_val(s_tag,
01303             " post-recover PIDR=0x",
01304             ((volatile const rx_port_regs_t*)0x0008C002U)->pidr);
01305     }
01306 }
01307
01308 static rx_err_t internal_wait_bus_ready(volatile rx_riic_regs_t* riic)
01309 {
01310     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01311
01312     if (internal_riic_poll_bbsy_clear(riic)) {
01313         return k_rx_ok;
01314     }
01315
01316     /* BBSY can latch stuck when the peripheral comes up observing lines that
01317      * are mid-transition. Per RX72N HW manual section 42.2.1 the only
01318      * software-side clear is an IICRST pulse -- which wipes the config
01319      * registers back to reset, so we must restore them after. */
01320     rx_log_warn(s_tag, "BBSY stuck - pulsing IICRST to recover");
01321     internal_riic_log_pre_recovery(riic);
01322
01323     riic_saved_cfg_t saved;
01324     internal_riic_save_config(riic, &saved);
01325
01326     internal_riic_iicrst_with_restore(riic, &saved);
01327     internal_riic_release_lines(riic);
01328
01329     /* Second wait. If BBSY still set after a full IICRST cycle, the bus is
01330      * genuinely stuck and IICRST alone cannot clear it because the
01331      * peripheral reset does not toggle external SCL edges. */
01332     if (internal_riic_poll_bbsy_clear(riic)) {
01333         return k_rx_ok;
01334     }
01335
01336     internal_riic_bit_bang_recovery_sequence(riic, &saved);
01337
01338     if (internal_riic_poll_bbsy_clear(riic)) {
01339         rx_log_error(s_tag, "I2C bus busy timeout (after bit-bang recovery)");
01340         return k_rx_err_timeout;
01341     }
01342
01343     return k_rx_ok;
01344 }
01345
01346 /**
01347  * @brief Generate I2C START condition on the bus
01348  *
01349  * @details
01350  * Issues a START condition by setting the ST bit in ICCR2, then waits for
01351  * hardware confirmation via the START flag in ICSR2. The START condition
01352  * signals all peripherals that a transaction is beginning.
01353  *
01354  * ### START Condition Timing
01355  *
01356  * ```
01357  * SDA  -----+
01358  *          +-----+
01359  * SCL  -----+
01360  *          +-----+
01361  *          |       |
01362  *          |       +-- SCL goes low (start of first bit)
01363  *          +-- SDA goes low while SCL high (START)
01364  * ```
01365  *
01366  * ### Hardware Sequence
01367  *
01368  * 1. Set ST bit in ICCR2 (request START)
01369  * 2. Hardware pulls SDA low while SCL is high
01370  * 3. Hardware sets START flag in ICSR2
01371  * 4. Software clears START flag to acknowledge
01372  *
01373  *
01374  *
01375  * @pre riic != nullptr
01376  * @pre Bus must be idle (BBSY = 0) - use internal_wait_bus_ready() first

```

```

01377 * @pre RIIC channel must be enabled (ICE = 1)
01378 * @post START flag is cleared in ICSR2
01379 * @post Bus is now busy (BBSY = 1)
01380 *
01381 * @note Clears the START detection flag after successful START.
01382 * @note After START, caller must send address byte within timeout.
01383 *
01384 * @warning If START fails, caller must issue STOP to release bus.
01385 *
01386 * @see internal_send_stop() Companion function for ending transactions
01387 * @see internal_wait_bus_ready() Should be called before START
01388 */
01389 static rx_err_t internal_send_start(volatile rx_riic_regs_t* riic)
01390 {
01391     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01392
01393     /* Issue start condition */
01394     riic->iccr2 |= k_riic_iccr2_st;
01395
01396     /* Wait for start condition to be issued */
01397     uint32_t timeout = k_riic_timeout_us;
01398     while (!(riic->iccr2 & k_riic_iccr2_start) && timeout > k_riic_timeout_zero) {
01399         timeout--;
01400     }
01401
01402     if (timeout == k_riic_timeout_zero) {
01403         rx_log_error(s_tag, "Start condition timeout");
01404         return k_rx_err_timeout;
01405     }
01406
01407     /* Clear start flag */
01408     riic->iccr2 &= ~(uint8_t)k_riic_iccr2_start;
01409
01410     return k_rx_ok;
01411 }
01412 /**
01413  * @brief Generate I2C STOP condition on the bus
01414  *
01415  * @details
01416  * Issues a STOP condition by setting the SP bit in ICCR2, then waits for
01417  * hardware confirmation via the STOP flag in ICSR2. The STOP condition
01418  * releases the bus for other controllers.
01419  *
01420  * ## STOP Condition Timing
01421  *
01422  * ```
01423  * SDA -----+-----
01424  *             |
01425  * SCL ---+-----+-----
01426  *       |   |
01427  *       |   +- SDA goes high while SCL high (STOP)
01428  *       +- SCL goes high (end of last bit/ACK)
01429  * ```
01430  *
01431  * ## Hardware Sequence
01432  *
01433  * 1. Set SP bit in ICCR2 (request STOP)
01434  * 2. Hardware drives SCL high
01435  * 3. Hardware drives SDA high while SCL is high
01436  * 4. Hardware sets STOP flag in ICSR2
01437  * 5. Software clears STOP flag to acknowledge
01438  *
01439  * ## Error Recovery
01440  *
01441  * STOP is also used for error recovery - if a transaction fails at any
01442  * point, issuing STOP releases the bus and resets the peripheral state
01443  * machine.
01444  *
01445  *
01446  *
01447  *
01448  * @pre riic != nullptr
01449  * @pre A transaction is in progress (bus is busy)
01450  * @post STOP flag is cleared in ICSR2
01451  * @post Bus is idle (BBSY = 0)
01452  * @post Peripheral state machine returns to idle
01453  *
01454  * @note Always call STOP at the end of a transaction or to recover from errors.
01455  * @note STOP after NACK in read transactions signals end of transfer to peripheral.
01456  *
01457  * @par Best-Effort Cleanup
01458  * In error paths, STOP is called to release the bus even if the original
01459  * error should be preserved. The calling function should:
01460  *
01461  * @code
01461  * if (err != k_rx_ok) {
01462  *     rx_err_t stop_err = internal_send_stop(riic);
01463  *     (void)stop_err; // Preserve original error

```

```

01464 *   return err;
01465 * }
01466 * @endcode
01467 *
01468 * @see internal_send_start() Companion function for starting transactions
01469 */
01470 static rx_err_t internal_send_stop(volatile rx_riic_regs_t* riic)
01471 {
01472     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01473     /* Issue stop condition */
01474     riic->iccr2 |= k_riic_iccr2_sp;
01475     /* Wait for stop condition to be issued */
01476     uint32_t timeout = k_riic_timeout_us;
01477     while (!(riic->iccr2 & k_riic_iccr2_stop) && timeout > k_riic_timeout_zero) {
01478         timeout--;
01479     }
01480     if (timeout == k_riic_timeout_zero) {
01481         rx_log_error(s_tag, "Stop condition timeout");
01482         return k_rx_err_timeout;
01483     }
01484     /* Clear stop flag */
01485     riic->iccr2 &= ~(uint8_t)k_riic_iccr2_stop;
01486     return k_rx_ok;
01487 }
01488 /**
01489 * @brief Transmit single byte over I2C bus
01490 *
01491 * @details
01492 * Writes one byte to the I2C bus and waits for acknowledgment from the
01493 * peripheral. This function handles both address bytes and data bytes.
01494 *
01495 * ## Transmission Sequence
01496 *
01497 * 1. Wait for TDRE (Transmit Data Register Empty) flag
01498 * 2. Write byte to ICDRT (Transmit Data Register)
01499 * 3. Hardware shifts out 8 bits MSB-first
01500 * 4. Hardware generates 9th clock for ACK
01501 * 5. Peripheral drives SDA low (ACK) or leaves high (NACK)
01502 * 6. Check NACKF flag to determine response
01503 *
01504 * ## I2C Byte Timing
01505 *
01506 * ```
01507 * SCL  ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
01508 *      || || || || || || || ||
01509 *      ++ ++ ++ ++ ++ ++ ++ ++ ++
01510 *      D7 D6 D5 D4 D3 D2 D1 D0 ACK
01511 * SDA  -----+
01512 *      Data bits (8)      +- ACK (low)
01513 * ```
01514 *
01515 * ## NACK Handling
01516 *
01517 * NACK is received when:
01518 * - Address byte: No peripheral at that address
01519 * - Data byte: Peripheral buffer full or error condition
01520 * - Last byte: Normal end-of-read signaling (expected)
01521 *
01522 * @pre riic != nullptr (not checked - caller must ensure)
01523 * @pre START condition has been issued
01524 * @pre Previous byte (if any) has been acknowledged
01525 * @post On success, byte has been transmitted and ACK received
01526 * @post On NACK, NACKF flag is cleared for next operation
01527 *
01528 * @note Does NOT send START or STOP - caller manages transaction framing.
01529 * @note First byte after START should be address with R/W bit.
01530 *
01531 * @warning On NACK, caller should issue STOP to release bus.
01532 *
01533 * @par Address Byte Format
01534 * For address bytes, caller must format as: (addr « 1) | direction
01535 * - Write: (0x50 « 1) | 0 = 0xA0
01536 * - Read: (0x50 « 1) | 1 = 0xA1
01537 *
01538 * @see internal_read_byte() Companion function for receiving bytes
01539 */
01540 static rx_err_t internal_write_byte(volatile rx_riic_regs_t* riic, const uint8_t data)
01541 {
01542     /* Wait for transmit data empty */

```

```

01551 uint32_t timeout = k_riic_timeout_us;
01552 while (!(riic->icr2 & k_riic_icr2_tdre) && timeout > k_riic_timeout_zero) {
01553     timeout--;
01554 }
01555
01556 if (timeout == k_riic_timeout_zero) {
01557     rx_log_error(s_tag, "Write timeout");
01558     return k_rx_err_timeout;
01559 }
01560
01561 /* Write data */
01562 riic->icdrt = data;
01563
01564 /* Wait for transmit END (byte + ACK clock complete). Per RX72N HW manual
01565  * section 42.2.7, TEND asserts on the rising edge of the 9th clock after
01566  * the ACK/NACK response has been sampled, so NACKF is valid only once
01567  * TEND=1. Waiting for TDRE to clear is wrong: TDRE clears the instant
01568  * ICDRT is written, so the wait would exit in zero iterations and NACKF
01569  * would be checked before the byte has actually been transmitted. */
01570 timeout = k_riic_timeout_us;
01571 while (!(riic->icr2 & k_riic_icr2_tend) && timeout > k_riic_timeout_zero) {
01572     timeout--;
01573 }
01574
01575 if (timeout == k_riic_timeout_zero) {
01576     rx_log_error(s_tag, "TEND timeout (byte never finished transmitting)");
01577     return k_rx_err_timeout;
01578 }
01579
01580 /* Check for NACK -- valid only now that TEND=1. */
01581 if (riic->icr2 & k_riic_icr2_nackf) {
01582     riic->icr2 &= (uint8_t) ~(uint8_t) k_riic_icr2_nackf;
01583     rx_log_error(s_tag, "NACK received");
01584     return k_rx_err_nack;
01585 }
01586
01587 return k_rx_ok;
01588 }
01589
01590 /**
01591  * @brief Send the address|R byte that opens a controller-receive transaction
01592  *
01593  * @details
01594  * After the controller switches to receive mode (ICR2 = MST, TRS=0) and
01595  * has issued START or repeated-START, the address byte with R bit must be
01596  * sent. internal_write_byte() cannot be used here: per RX72N HW manual
01597  * section 42.2.1 the TRS bit auto-clears to 0 on the rising edge of the
01598  * 9th SCL of an addr|R byte, which moves the peripheral into receive mode
01599  * and sets RDRF automatically -- TEND and TDRE-reassertion no longer fire.
01600  * Waiting for transmit-side flags in that state hangs forever; this helper
01601  * waits for RDRF | NACKF instead.
01602  *
01603  *
01604  *
01605  * @pre Bus is active; ICCR2 = MST (TRS=0); start condition already issued
01606  * @post On success: RDRF=1 in ICSR2, ICDRR holds the first data byte
01607  *
01608  * @see internal_write_byte() Standard transmit-mode byte writer
01609  * @see internal_read_byte() Reads byte after RDRF asserts
01610  */
01611 static rx_err_t internal_write_address_for_read(volatile rx_riic_regs_t* riic,
01612                                                 const uint8_t addr_byte)
01613 {
01614     uint32_t timeout = k_riic_timeout_us;
01615     while (!(riic->icr2 & k_riic_icr2_tdre) && timeout > k_riic_timeout_zero) {
01616         timeout--;
01617     }
01618     if (timeout == k_riic_timeout_zero) {
01619         rx_log_error(s_tag, "TDRE timeout before addr|R");
01620         return k_rx_err_timeout;
01621     }
01622
01623     riic->icdrt = addr_byte;
01624
01625     /* In controller-receive mode, RDRF fires once the first data byte has been
01626      * clocked in after the address ACK. NACKF fires if the peripheral didn't
01627      * ACK the address. Wait for either. */
01628     const uint8_t wait_mask = (uint8_t)(k_riic_icr2_rdrf | k_riic_icr2_nackf);
01629     timeout = k_riic_timeout_us;
01630     while (!(riic->icr2 & wait_mask) && timeout > k_riic_timeout_zero) {
01631         timeout--;
01632     }
01633     if (timeout == k_riic_timeout_zero) {
01634         rx_log_error(s_tag, "addr|R: neither RDRF nor NACKF asserted");
01635         return k_rx_err_timeout;
01636     }
01637 }

```

```

01638 if (riic->icsr2 & k_riic_icsr2_nackf) {
01639     riic->icsr2 &= (uint8_t) ~(uint8_t) k_riic_icsr2_nackf;
01640     rx_log_error(s_tag, "Peripheral NACKed address|R");
01641     return k_rx_err_nack;
01642 }
01643
01644 /* Caller is responsible for the dummy ICDRR read that kicks off actual
01645 * data reception. Doing it here unconditionally would be wrong for the
01646 * single-byte read case where ACKBT=1 must be set BEFORE the dummy read
01647 * so byte 0 is NACKed (RX72N HW manual Figure 42.10, 1-byte path). The
01648 * caller sets ACKBT per the transfer length, then does the dummy read. */
01649 return k_rx_ok;
01650 }
01651
01652 /**
01653 * @brief Receive single byte from I2C bus
01654 *
01655 * @details
01656 * Reads one byte from the I2C bus and sends ACK or NACK as specified.
01657 * The ACK/NACK response is configured before reading the byte from the
01658 * receive register.
01659 *
01660 * ## Reception Sequence
01661 *
01662 * 1. Wait for RDRF (Receive Data Register Full) flag
01663 * 2. Configure ACK/NACK response in ICMR3.ACKBT
01664 * 3. Read byte from ICDRR (Receive Data Register)
01665 * 4. Hardware generates ACK/NACK on 9th clock
01666 * 5. RDRF flag is cleared by read
01667 *
01668 * ## ACK vs NACK Decision
01669 *
01670 * | Situation          | Response | ACKBT | Purpose |
01671 * |-----|-----|-----|-----|
01672 * | More bytes to read | ACK      | 0     | Continue transfer |
01673 * | Last byte          | NACK     | 1     | Signal end of read |
01674 *
01675 * ## I2C Read Byte Timing
01676 *
01677 * ```
01678 *   SCL  ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
01679 *         || || || || || || || ||
01680 *         ++ ++ ++ ++ ++ ++ ++ ++ ++ ++
01681 *         D7 D6 D5 D4 D3 D2 D1 D0 ACK/NACK
01682 *   SDA  -----+
01683 *         Peripheral drives  +- Controller drives ACK/NACK
01684 *   ```
01685 *
01686 *
01687 *
01688 * @pre riic != nullptr
01689 * @pre data != nullptr
01690 * @pre Controller is in receive mode (address + read bit sent)
01691 * @post *data contains the received byte
01692 * @post ACK or NACK has been sent to peripheral
01693 * @post RDRF flag is cleared
01694 *
01695 * @note ACK/NACK is configured BEFORE reading ICDRR.
01696 * @note NACK on last byte is REQUIRED per I2C specification.
01697 *
01698 * @warning Failure to send NACK on last byte may cause peripheral
01699 *          to continue driving data, corrupting next transaction.
01700 *
01701 * @par Example: Reading Multiple Bytes
01702 * @code
01703 * for (uint16_t i = 0; i < length; i++) {
01704 *     bool send_ack = (i < length - 1); // ACK all except last
01705 *     err = internal_read_byte(riic, &buffer[i], send_ack);
01706 *     if (err != k_rx_ok) break;
01707 * }
01708 * @endcode
01709 *
01710 * @see internal_write_byte() Companion function for transmitting bytes
01711 */
01712 [[maybe_unused]]
01713 static rx_err_t internal_read_byte(volatile rx_riic_regs_t* riic,
01714                                   uint8_t* data,
01715                                   const bool send_ack,
01716                                   const bool last_byte)
01717 {
01718     /* Wait for receive data full */
01719     uint32_t timeout = k_riic_timeout_us;
01720
01721     RX_CHECK_NULL_PTR(riic, s_tag, "RIIC peripheral base is nullptr");
01722     RX_CHECK_NULL_PTR(data, s_tag, "data pointer is nullptr");
01723
01724     while (!(riic->icsr2 & k_riic_icsr2_rdrf) && timeout > k_riic_timeout_zero) {

```

```

01725     timeout--;
01726 }
01727
01728 if (timeout == k_riic_timeout_zero) {
01729     rx_log_error(s_tag, "Read timeout");
01730     return k_rx_err_timeout;
01731 }
01732
01733 /* Configure ACK/NACK for the byte we are about to release from ICDRR.
01734 * The peripheral clock-stretches SCL between the 8th clock and the ACK
01735 * clock while RDRF is set, so setting ACKBT now takes effect on the
01736 * upcoming ACK/NACK bit. */
01737 if (!send_ack) {
01738     riic->icmr3 |= k_riic_icmr3_ackbt_mask; /* ACKBT = 1 (NACK) */
01739 } else {
01740     riic->icmr3 &= (uint8_t) ~(uint8_t) k_riic_icmr3_ackbt_mask; /* ACKBT = 0 (ACK) */
01741 }
01742
01743 /* For the final byte of a receive, queue SP before reading ICDRR so the
01744 * RIIC generates NACK + STOP atomically after the 9th clock. Per RX72N
01745 * HW manual Figure 42.10 the SP request must be set between RDRF and
01746 * the final ICDRR read so the STOP is chained to the 9th clock's NACK.
01747 *
01748 * Do NOT queue SP after the final ICDRR read: the 9th clock will have
01749 * already fired, SCL is released, and a fresh SP produces no STOP edge.
01750 * That was the bug behind "Stop condition timeout" on 1-byte reads. */
01751 if (last_byte) {
01752     riic->icsr2 &= (uint8_t) ~(uint8_t) k_riic_icsr2_stop;
01753     riic->iccr2 |= k_riic_iccr2_sp;
01754 }
01755
01756 /* Read data -- releases SCL for the ACK/NACK clock. For the last byte,
01757 * SP was queued above so hardware generates STOP immediately after the
01758 * NACK on the 9th clock.
01759 *
01760 * Do NOT instead issue SP AFTER this ICDRR read. Per RX72N HW manual
01761 * Figure 42.10, once the 9th clock has fired the peripheral drops the
01762 * auto clock-stretch and releases SCL; the STOP condition must be
01763 * chained into that release, not requested after the bus has already
01764 * returned to idle. Calling internal_send_stop() after the read will
01765 * time out waiting for ICSR2.STOP to assert because no STOP edge is
01766 * ever produced. */
01767 *data = riic->icdrr;
01768
01769 if (last_byte) {
01770     timeout = k_riic_timeout_us;
01771     while (!(riic->icsr2 & k_riic_icsr2_stop) && timeout > k_riic_timeout_zero) {
01772         timeout--;
01773     }
01774     if (timeout == k_riic_timeout_zero) {
01775         rx_log_error(s_tag, "STOP after final ICDRR timeout");
01776         return k_rx_err_timeout;
01777     }
01778     /* Clear STOP + NACKF so the next transaction starts from a clean
01779     * status register. ICMR3.ACKBT left at 1 is harmless -- it gets
01780     * re-written by the next internal_read_byte. */
01781     riic->icsr2 &= (uint8_t) ~(uint8_t) (k_riic_icsr2_stop | k_riic_icsr2_nackf);
01782 }
01783
01784 return k_rx_ok;
01785 }
01786
01787 /**
01788 * @brief Execute I2C write phase for combined write-read transfer
01789 *
01790 * @details
01791 * Performs the write portion of a write-read (register read) transaction.
01792 * This typically writes a register address to the peripheral, which sets
01793 * an internal pointer for the subsequent read phase.
01794 *
01795 * ### Transaction Sequence
01796 *
01797 * ```
01798 * Controller                                Peripheral
01799 * |----- START ----->|
01800 * |----- Address + W (0) ----->|
01801 * |<--- ACK -----<|
01802 * |----- Register Address ----->| <- write_data[0]
01803 * |<--- ACK -----<|
01804 * |----- [Additional bytes...] ----->| <- write_data[1..N]
01805 * |-----|
01806 * |----- (No STOP - read phase follows) -----|
01807 * ```
01808 *
01809 * ### Error Handling
01810 *
01811 */

```

```

01812 * On any error (timeout or NACK), this function issues a STOP condition
01813 * to release the bus before returning the error code.
01814 *
01815 *
01816 *
01817 * @pre riic points to valid, initialized RIIC channel
01818 * @pre device_addr.value <= 127
01819 * @pre write_data != nullptr
01820 * @pre write_length > 0
01821 * @pre Bus is idle (not busy)
01822 * @post On success: Bus is active, internal address pointer set
01823 * @post On failure: STOP issued, bus released
01824 *
01825 * @note Does NOT issue STOP on success - caller must continue with read
01826 *       phase or issue STOP.
01827 * @note On error, STOP is always issued for bus cleanup.
01828 *
01829 * @see internal_riic_read_phase() Continues the transaction with read
01830 * @see riic_write_read() Public API that combines both phases
01831 */
01832 static rx_err_t internal_riic_write_phase(volatile rx_riic_regs_t* riic,
01833                                           const i2c_device_addr_t device_addr,
01834                                           const uint8_t* write_data,
01835                                           const uint16_t write_length)
01836 {
01837     /* Set controller transmit mode */
01838     riic->iccr2 = k_riic_iccr2_mst | k_riic_iccr2_trs;
01839
01840     /* Send start condition */
01841     rx_err_t err = internal_send_start(riic);
01842     RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
01843
01844     /* Send device address (write) */
01845     err = internal_write_byte(riic, (device_addr.value << k_riic_addr_shift) | k_riic_addr_write_bit);
01846     if (err != k_rx_ok) {
01847         rx_err_t stop_err = internal_send_stop(riic);
01848         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01849         return err;
01850     }
01851
01852     /* Send write data */
01853     for (uint16_t i = 0; i < write_length; i++) {
01854         err = internal_write_byte(riic, write_data[i]);
01855         if (err != k_rx_ok) {
01856             rx_err_t stop_err = internal_send_stop(riic);
01857             (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01858             return err;
01859         }
01860     }
01861
01862     return k_rx_ok;
01863 }
01864
01865 /**
01866  * @brief Execute I2C read phase for combined write-read transfer
01867  *
01868  * @details
01869  * Performs the read portion of a write-read (register read) transaction.
01870  * Issues a repeated START condition (without releasing the bus), then
01871  * reads data bytes from the peripheral.
01872  *
01873  * ## Transaction Sequence
01874  *
01875  * ```
01876  * Controller                                Peripheral
01877  * |                                           |
01878  * | (Write phase completed, no STOP)         |
01879  * |                                           |
01880  * |---- REPEATED START ----->|
01881  * |---- Address + R (1) ----->|
01882  * |<--- ACK -----<|
01883  * |<--- Data Byte 0 -----<|
01884  * |---- ACK ----->|
01885  * |<--- Data Byte N-1 -----<|
01886  * |---- NACK ----->| <- Signals end
01887  * |                                           |
01888  * | (Caller issues STOP)                   |
01889  * ```
01890  *
01891  * ## Repeated START vs STOP+START
01892  *
01893  * The repeated START (Sr) is critical for atomic register reads:
01894  * - Prevents other controllers from arbitrating between write and data read
01895  * - Ensures peripheral's internal address pointer is still valid
01896  * - Required by many I2C peripherals (EEPROMs, sensors, etc.)
01897  *
01898  * ## ACK/NACK Protocol

```



```

01899 *
01900 * Per I2C specification, the controller must NACK the last byte to signal
01901 * end of read. This allows the peripheral to release SDA for the STOP
01902 * condition.
01903 *
01904 *
01905 *
01906 * @pre riic points to valid, initialized RIIC channel
01907 * @pre Write phase has been completed (bus is active)
01908 * @pre device_addr.value <= 127
01909 * @pre read_data != nullptr
01910 * @pre read_length > 0
01911 * @post On success: read_data contains received bytes
01912 * @post On failure: STOP issued, bus released
01913 *
01914 * @note Does NOT issue STOP on success - caller must issue STOP.
01915 * @note On error, STOP is always issued for bus cleanup.
01916 * @note NACK is automatically sent on last byte.
01917 *
01918 * @see internal_riic_write_phase() Precedes this function in write-read
01919 * @see riic_write_read() Public API that combines both phases
01920 */
01921 static rx_err_t internal_riic_read_phase(volatile rx_riic_regs_t* riic,
01922                                         const i2c_device_addr_t device_addr,
01923                                         uint8_t* read_data,
01924                                         const uint16_t read_length)
01925 {
01926     /* Send repeated start condition */
01927     riic->iccr2 |= k_riic_iccr2_rs;
01928
01929     uint32_t timeout = k_riic_timeout_us;
01930     while (!(riic->icrsr2 & k_riic_icrsr2_start) && timeout > k_riic_timeout_zero) {
01931         timeout--;
01932     }
01933
01934     if (timeout == k_riic_timeout_zero) {
01935         rx_err_t stop_err = internal_send_stop(riic);
01936         (void)stop_err; /* Best-effort cleanup on timeout */
01937         rx_log_error(s_tag, "Repeated start timeout");
01938         return k_rx_err_timeout;
01939     }
01940
01941     riic->icrsr2 &= (uint8_t) ~(uint8_t) k_riic_icrsr2_start;
01942
01943     /* Explicit ICCR2 write to clear any pending SP/RS/ST request bits ahead of
01944      * the addr|R write. MST and TRS are read-only when ICMR1.MTWP=0, so this
01945      * is a no-op for them; TRS will auto-clear to 0 on the 9th clock of the
01946      * addr|R byte per HUM 42.2.1. */
01947     riic->iccr2 = k_riic_iccr2_mst;
01948
01949     /* Send device address (read) -- use the receive-mode helper, NOT
01950      * internal_write_byte: TEND/TDRE-reassertion don't fire after addr|R in
01951      * controller-receive mode; RDRF does. This leaves the address-ACK RDRF
01952      * asserted but does NOT do the dummy read (that responsibility is the
01953      * caller's so it can pre-set ACKBT for 1-byte transfers). */
01954     rx_err_t err = internal_write_address_for_read(
01955         riic,
01956         (uint8_t)((device_addr.value « k_riic_addr_shift) | k_riic_addr_read_bit));
01957     if (err != k_rx_ok) {
01958         rx_err_t stop_err = internal_send_stop(riic);
01959         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
01960         return err;
01961     }
01962
01963     /* For a 1-byte read, set ACKBT=1 BEFORE the dummy read so the sole data
01964      * byte's 9th-clock ACK slot carries NACK (HUM Fig 42.10 "1- or 2-byte
01965      * receive" path). The dummy read below releases SCL and starts clocking
01966      * byte 0; the 9th clock of byte 0 samples ACKBT at that moment. For
01967      * multi-byte reads we leave ACKBT=0 here and flip it to 1 inside the
01968      * loop on iteration N-2 (HUM Fig 42.11 "3 bytes or more" path) -- the
01969      * read of byte N-2 is what releases SCL for byte N-1 to clock in, and
01970      * byte N-1's 9th clock then picks up ACKBT=1. Setting ACKBT=1 on
01971      * iteration N-1 is ALWAYS TOO LATE: by the time we enter that iteration
01972      * RDRF has already asserted, meaning byte N-1's 9th clock has already
01973      * fired and its ACK slot has already been transmitted with whatever
01974      * ACKBT was in the prior iteration (typically 0 = ACK). That leaves
01975      * the peripheral thinking "keep streaming" and holding SDA for another
01976      * byte, which is why the subsequent SP never chains a STOP edge and we
01977      * see "Stop condition timeout" at the end of every multi-byte read. */
01978     if (read_length == 1U) {
01979         riic->icmr3 |= k_riic_icmr3_ackbt_mask;
01980     }
01981
01982     (void)riic->icdrr; /* dummy read kicks reception */
01983
01984     /* Receive loop. For each iteration:
01985      * - Wait for RDRF (byte i now in ICDRR, byte i's 9th clock fired).

```

```

01986 * - If we are on the penultimate byte (i == N-2), flip ACKBT=1 so
01987 * byte N-1's 9th clock carries NACK. Nothing to do on iteration
01988 * N-1 except clear STOP / queue SP ahead of the final ICDRR read.
01989 * - Issue the final STOP request between reading byte N-2 and the
01990 * final ICDRR read.
01991 * - Read ICDRR: copies byte i data and (on non-last iterations)
01992 * releases SCL for byte i+1 to clock in. On the last iteration
01993 * there is no byte i+1; with SP queued the 9th-clock NACK is
01994 * followed by a STOP edge.
01995 *
01996 * Do NOT call internal_send_stop() after this loop on success: SP was
01997 * already queued in the final iteration and the hardware generates the
01998 * STOP edge chained to byte N-1's 9th clock. A second SP on an idle
01999 * bus either times out or spuriously re-raises BBSY. */
02000 for (uint16_t i = 0; i < read_length; i++) {
02001     uint32_t timeout = k_riic_timeout_us;
02002     while (!(riic->icmr2 & k_riic_icmr2_rdrf) && timeout > k_riic_timeout_zero) {
02003         timeout--;
02004     }
02005     if (timeout == k_riic_timeout_zero) {
02006         rx_log_error_val(s_tag, "Read byte RDRF timeout i=", (uint8_t)i);
02007         rx_err_t stop_err = internal_send_stop(riic);
02008         (void)stop_err;
02009         return k_rx_err_timeout;
02010     }
02011     if (read_length >= 2U && i == (uint16_t)(read_length - 2U)) {
02012         /* Penultimate byte: NACK will fall on byte N-1's 9th clock. */
02013         riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02014     }
02015     if (i == (uint16_t)(read_length - k_riic_last_index_offset)) {
02016         /* Last iteration: queue STOP before the final read so the NACK on
02017          * byte N-1's 9th clock is chained to a STOP edge by hardware. */
02018         riic->icmr2 &= (uint8_t) ~(uint8_t)k_riic_icmr2_stop;
02019         riic->iccr2 |= k_riic_iccr2_sp;
02020     }
02021     read_data[i] = riic->icdrr;
02022 }
02023
02024 /* Wait for the hardware-generated STOP to complete. */
02025 uint32_t stop_timeout = k_riic_timeout_us;
02026 while (!(riic->icmr2 & k_riic_icmr2_stop) && stop_timeout > k_riic_timeout_zero) {
02027     stop_timeout--;
02028 }
02029 if (stop_timeout == k_riic_timeout_zero) {
02030     rx_log_error(s_tag, "STOP after final ICDRR timeout");
02031     return k_rx_err_timeout;
02032 }
02033
02034 /* Clear STOP / NACKF for the next transaction. ACKBT stays latched at 1
02035 * from the penultimate-iteration write; riic_init / recovery / the
02036 * pre-transaction ICSR2 clear all restore ICMR3 as needed. */
02037 riic->icmr2 &= (uint8_t) ~(uint8_t)(k_riic_icmr2_stop | k_riic_icmr2_nackf);
02038
02039 return k_rx_ok;
02040 }
02041
02042 /*
02043 =====
02044 * Public API Implementation
02045 =====
02046
02047 * These functions provide the user-facing I2C interface. All functions
02048 * validate inputs, check initialization state, and provide comprehensive
02049 * error reporting.
02050 *
02051 =====
02052 */
02053
02054 /**
02055 * @brief Initialize RIIC (I2C) channel for controller mode operation
02056 *
02057 * @details
02058 * Configures an RIIC channel for I2C controller mode communication at the
02059 * specified bus frequency. This function must be called before any I2C
02060 * transfers on the channel.
02061 *
02062 * ## Initialization Sequence
02063 *
02064 * 1. Validate channel number and frequency
02065 * 2. Enable RIIC module clock (clear MSTPCRB bit)
02066 * 3. Reset RIIC internal state (ICRST)
02067 * 4. Calculate and configure bit rate registers
02068 * 5. Configure controller mode, 7-bit addressing
02069 * 6. Enable I2C bus interface (ICE)

```

```

02070 * 7. Mark channel as initialized
02071 *
02072 * ## Module Stop Control
02073 *
02074 * The RIIC peripheral is disabled by default to save power. This function
02075 * enables the module clock by clearing the appropriate MSTPCRB bit:
02076 *
02077 * | Channel | MSTPCRB Bit | Default State |
02078 * |-----|-----|-----|
02079 * | RIIC0 | Bit 21 | Stopped |
02080 * | RIIC1 | Bit 20 | Stopped |
02081 * | RIIC2 | Bit 19 | Stopped |
02082 *
02083 * ## Supported Frequencies
02084 *
02085 * | Frequency | Mode | Use Case |
02086 * |-----|-----|-----|
02087 * | 100 kHz | Standard | Legacy devices |
02088 * | 400 kHz | Fast | Most I2C peripherals |
02089 * | 1 MHz | Fast Plus | High-speed sensors |
02090 *
02091 *
02092 *
02093 * @pre channel.value < 3
02094 * @pre frequency_hz is one of: 100000, 400000, 1000000
02095 * @post Channel is enabled and ready for I2C transfers
02096 * @post s_riic_channel_mode[channel.value] == k_riic_mode_controller
02097 *
02098 * @note This function modifies protected registers (MSTPCRB) using PRCR unlock.
02099 * @note Re-initializing an already initialized channel is allowed.
02100 * @note Pin configuration (MPC) must be done separately if needed.
02101 *
02102 * @warning Using unsupported frequencies may cause communication failures.
02103 *
02104 * @par Example
02105 * @code
02106 * // Initialize RIIC0 for Fast mode (400 kHz)
02107 * riic_channel_t channel = {.value = 0};
02108 * rx_err_t err = riic_init(channel, 400000);
02109 * if (err != k_rx_ok) {
02110 *     rx_log_error("I2C", "Failed to initialize RIIC0");
02111 * }
02112 * @endcode
02113 *
02114 * @see riic_write() Write data after initialization
02115 * @see riic_read() Read data after initialization
02116 * @see riic_write_read() Combined write-read after initialization
02117 *
02118 * @since Version 1.0.0
02119 *
02120 * @callgraph
02121 */
02122 /**
02123 * @brief Validate riic_init arguments and resolve peripheral base
02124 *
02125 * @details
02126 * Checks that channel is in range, frequency is one of the three
02127 * supported buckets (100k/400k/1M), and resolves the channel to its
02128 * peripheral base address.
02129 *
02130 * @param[in] channel Channel handle
02131 * @param[in] frequency_hz Requested SCL frequency
02132 * @param[out] riic_out On success, set to the peripheral base
02133 *
02134 * @return rx_err_t Error code
02135 * @retval k_rx_ok Inputs valid, *riic_out non-null
02136 * @retval k_rx_err_invalid_arg Bounds violation or unknown channel
02137 *
02138 * @pre riic_out non-null
02139 *
02140 * @since Version 1.0.0
02141 */
02142 static rx_err_t internal_riic_validate_init(const riic_channel_t channel,
02143                                             const uint32_t frequency_hz,
02144                                             volatile rx_riic_regs_t** riic_out)
02145 {
02146     if (channel.value >= k_riic_max_channels) {
02147         rx_log_error(s_tag, "Invalid RIIC channel");
02148         return k_rx_err_invalid_arg;
02149     }
02150     if (frequency_hz != k_riic_freq_100khz && frequency_hz != k_riic_freq_400khz &&
02151         frequency_hz != k_riic_freq_1mhz) {
02152         rx_log_error(s_tag, "Invalid I2C frequency (use 100000, 400000, or 1000000)");
02153         return k_rx_err_invalid_arg;
02154     }
02155     *riic_out = internal_get_riic_base(channel.value);
02156     if (*riic_out == nullptr) {

```

```

02157     return k_rx_err_invalid_arg;
02158 }
02159 return k_rx_ok;
02160 }
02161
02162 /**
02163  * @brief Release the module-stop bit for the requested RIIC channel
02164  *
02165  * @details
02166  * Wraps the protected-register PRCR unlock/lock around the MSTPCRB/CRC
02167  * write that releases the channel's clock gate. Channel-to-MSTP-bit
02168  * mapping mirrors the RX72N HW manual MSTP table.
02169  *
02170  * @param[in] channel_value Validated channel index
02171  *
02172  *
02173  * @pre channel_value < k_riic_max_channels
02174  * @post Channel module clock is running
02175  *
02176  * @since Version 1.0.0
02177  */
02178 static void internal_riic_release_module_stop(const uint8_t channel_value)
02179 {
02180     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
02181
02182     if (channel_value == k_riic_channel_0) {
02183         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic0;
02184     } else if (channel_value == k_riic_channel_1) {
02185         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic1;
02186     } else {
02187         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic2;
02188     }
02189
02190     *prcr_reg() = k_rx_prcr_lock;
02191 }
02192
02193 /**
02194  * @brief Program ICBRL/ICBRH/ICMR1..3 for the requested SCL frequency
02195  *
02196  * @details
02197  * Looks up CKS / ICBRL / ICBRH via internal_calculate_bit_rate(), then
02198  * writes the bit-rate registers and mode registers. CKS is in ICMR1[6:4]
02199  * and must be written alongside MTWP/BCWP/BC in a single ICMR1 write.
02200  * ICMR3 = ACKWP | 0 -- ACKWP=1 is mandatory so internal_read_byte() can
02201  * NACK the final received byte (otherwise STOP after every read times out).
02202  *
02203  * @param[in] riic RIIC peripheral base (must be non-null)
02204  * @param[in] frequency_hz Validated SCL frequency
02205  *
02206  * @return rx_err_t Error code from internal_calculate_bit_rate()
02207  *
02208  * @pre RIIC peripheral is in IICRST or post-IICRST configuration state
02209  * @post ICBRL/ICBRH/ICMR1/ICMR2/ICMR3 programmed
02210  *
02211  * @since Version 1.0.0
02212  */
02213 static rx_err_t internal_riic_program_bit_rate(volatile rx_riic_regs_t* const riic,
02214                                               const uint32_t frequency_hz)
02215 {
02216     uint8_t icbrl = 0;
02217     uint8_t icbrh = 0;
02218     uint8_t cks_field = 0;
02219     const rx_err_t err = internal_calculate_bit_rate(frequency_hz, &icbrl, &icbrh, &cks_field);
02220     RX_RETURN_ON_ERROR(err, s_tag, "Bit rate calculation failed");
02221
02222     riic->icbrl = icbrl;
02223     riic->icbrh = icbrh;
02224     riic->icmr1 = (uint8_t)(k_riic_icmr1_controller_7bit | (uint8_t)(cks_field « k_riic_cks_shift));
02225     riic->icmr2 = k_riic_icmr2_default;
02226     riic->icmr3 = k_riic_icmr3_init;
02227
02228     return k_rx_ok;
02229 }
02230
02231 rx_err_t riic_init(const riic_channel_t channel, const uint32_t frequency_hz)
02232 {
02233     volatile rx_riic_regs_t* riic = nullptr;
02234     const rx_err_t check = internal_riic_validate_init(channel, frequency_hz, &riic);
02235     if (check != k_rx_ok) {
02236         return check;
02237     }
02238
02239     internal_riic_release_module_stop(channel.value);
02240
02241     /* Unconditional bit-bang bus recovery BEFORE the peripheral observes the
02242      * bus. Per RX72N HW manual section 42.2.1, the RIIC comes out of
02243      * module-stop + IICRST with SCLO=SDAO=0 internally -- it drives the bus

```

```

02244 * low until the SOWP dance below writes SCLO=SDAO=1. hardware_init has
02245 * already raised PMR for SCL/SDA by the time riic_init runs, so without
02246 * this step the low-drive glitches the external bus and can latch
02247 * BBSY=1 as soon as ICE=1. Channels without an entry in
02248 * k_riic_recovery_pins[] (RIIC0, RIIC2) are no-ops. */
02249 internal_riic_bit_bang_recover(channel.value); /* leaves PMR=0 */
02250
02251 /* Reset RIIC */
02252 riic->iccr1 = k_riic_iccr1_icrst;
02253 riic->iccr1 = k_riic_register_clear;
02254
02255 RX_RETURN_ON_ERROR(internal_riic_program_bit_rate(riic, frequency_hz),
02256                     s_tag,
02257                     "RIIC bit-rate program failed");
02258
02259 /* Enable I2C bus interface */
02260 riic->iccr1 = k_riic_iccr1_ice;
02261
02262 /* Post-enable bus release: per RX72N HW manual section 42.2.1 the RIIC
02263 * leaves SCLO/SDAO=0 after ICE=1 -- which drives both lines LOW -- until
02264 * it detects its first STOP on the wire. */
02265 internal_riic_release_lines(riic);
02266
02267 /* Pair to the unconditional bit-bang above: now that SCLO=SDAO=1 in the
02268 * peripheral (set by the SOWP dance), it is safe to route the pads back
02269 * to RIIC. */
02270 internal_riic_bit_bang_handback(channel.value);
02271
02272 s_riic_channel_mode[channel.value] = k_riic_mode_controller;
02273 rx_log_debug(s_tag, "RIIC channel initialized");
02274
02275 return k_rx_ok;
02276 }
02277
02278 /**
02279 * @brief Write data to an I2C peripheral device
02280 *
02281 * @details
02282 * Performs a complete I2C write transaction: START, address+W, data bytes,
02283 * STOP. This function is suitable for writing configuration registers,
02284 * sending commands, or storing data in EEPROM.
02285 *
02286 * ## Transaction Sequence
02287 *
02288 * ```
02289 * Controller                                Peripheral
02290 * |----- START ----->|
02291 * |----- Address + W (0) ----->|
02292 * |<--- ACK -----<|
02293 * |----- data[0] ----->|
02294 * |<--- ACK -----<|
02295 * |----- data[N-1] ----->|
02296 * |<--- ACK -----<|
02297 * |----- STOP ----->|
02298 * ```
02299 *
02300 * ## Common Use Cases
02301 *
02302 * | Use Case | Data Format |
02303 * |-----|-----|
02304 * | Config register write | [reg_addr, value] |
02305 * | Multi-byte register | [reg_addr, val0, val1, ...] |
02306 * | EEPROM page write | [addr_hi, addr_lo, data...] |
02307 * | Sensor command | [command_byte] |
02308 *
02309 * ## Error Handling
02310 *
02311 * On any error (timeout or NACK), the function issues a STOP condition
02312 * to release the bus before returning. The original error code is preserved.
02313 *
02314 *
02315 *
02316 *
02317 * @pre riic_init() called for this channel
02318 * @pre data != nullptr
02319 * @pre length >= 1 && length <= 256
02320 * @pre device_addr.value <= 127
02321 * @post On success: All bytes transferred, bus released
02322 * @post On failure: Bus released via STOP
02323 *
02324 * @note This is a blocking function - returns only after transfer complete.
02325 * @note STOP is always issued (success or failure) to release the bus.
02326 *
02327 * @warning NACK on address usually means device not present or wrong address.
02328 * @warning NACK on data may indicate device buffer full or write-protected.
02329 *
02330 * @par Example: Write Configuration Register

```

```

02331 * @code
02332 * // Write value 0x42 to register 0x05 of device at address 0x48
02333 * uint8_t config_data[2] = {0x05, 0x42}; // [reg_addr, value]
02334 * riic_channel_t channel = {.value = 0};
02335 * i2c_device_addr_t addr = {.value = 0x48};
02336 *
02337 * rx_err_t err = riic_write(channel, addr, config_data, 2);
02338 * if (err != k_rx_ok) {
02339 *     handle_i2c_error(err);
02340 * }
02341 * @endcode
02342 *
02343 * @see riic_init() Initialize channel before use
02344 * @see riic_read() Read-only transaction
02345 * @see riic_write_read() Combined write-then-read transaction
02346 *
02347 * @callgraph
02348 */
02349 rx_err_t riic_write(const riic_channel_t channel,
02350                    const i2c_device_addr_t device_addr,
02351                    const uint8_t* data,
02352                    const uint16_t length)
02353 {
02354     RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02355     RX_CHECK_RANGE_TAG(channel.value,
02356                        k_riic_channel_0,
02357                        (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02358                        k_rx_err_invalid_arg,
02359                        s_tag);
02360
02361     if (device_addr.value > k_riic_addr_max_7bit) {
02362         rx_log_error(s_tag, "Invalid device address");
02363         return k_rx_err_invalid_arg;
02364     }
02365
02366     if (length == k_riic_length_zero) {
02367         rx_log_error(s_tag, "Write length cannot be zero");
02368         return k_rx_err_invalid_arg;
02369     }
02370
02371     if (length > k_riic_max_transfer_length) {
02372         rx_log_error(s_tag, "Write length exceeds maximum");
02373         return k_rx_err_invalid_arg;
02374     }
02375
02376     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02377     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02378         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02379         return k_rx_err_invalid_state;
02380     }
02381
02382     /* Get RIIC base */
02383     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02384
02385     /* Wait for bus ready */
02386     rx_err_t err = internal_wait_bus_ready(riic);
02387     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02388
02389     /* Pre-clear any status flags left over from a previous transaction (e.g.
02390     * NACKF, STOP, AL) that normal bus activity does not clear. Per RX72N
02391     * HW manual section 42.2.7, these are sticky software-clear flags that
02392     * IICRST wipes but normal transactions do not. Without this pre-clear a
02393     * stale NACKF survives into the next write and internal_write_byte
02394     * returns k_rx_err_nack on a byte that was actually ACKed; a stale AL
02395     * silently clears MST on the next START. */
02396     riic->icr2 = k_riic_register_clear;
02397
02398     /* Set controller transmit mode */
02399     riic->icr2 = k_riic_icr2_mst | k_riic_icr2_trs;
02400
02401     /* Send start condition */
02402     err = internal_send_start(riic);
02403     RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02404
02405     /* Send device address (write) */
02406     err = internal_write_byte(riic, (device_addr.value << k_riic_addr_shift) | k_riic_addr_write_bit);
02407     if (err != k_rx_ok) {
02408         rx_err_t stop_err = internal_send_stop(riic);
02409         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02410         return err;
02411     }
02412
02413     /* Send data bytes */
02414     for (uint16_t i = 0; i < length; i++) {
02415         err = internal_write_byte(riic, data[i]);
02416         if (err != k_rx_ok) {
02417             rx_err_t stop_err = internal_send_stop(riic);

```

```

02418     (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02419     return err;
02420 }
02421 }
02422
02423 /* Send stop condition */
02424 err = internal_send_stop(riic);
02425 RX_RETURN_ON_ERROR(err, s_tag, "Stop condition failed");
02426
02427 return k_rx_ok;
02428 }
02429
02430 /**
02431  * @brief Read data from an I2C peripheral device
02432  *
02433  * @details
02434  * Performs a complete I2C read transaction: START, address+R, receive data
02435  * bytes (ACK all except last), NACK last byte, STOP. This function is suitable
02436  * for reading device status, sensor data, or sequential memory reads where
02437  * the device auto-increments its internal pointer.
02438  *
02439  * ## Transaction Sequence
02440  *
02441  * ```
02442  * Controller                                Peripheral
02443  * |----- START ----->|
02444  * |--- Address + R (1) --->|
02445  * |<--- ACK -----<|
02446  * |<--- data[0] -----<|
02447  * |--- ACK ----->|
02448  * |<--- data[N-2] -----<|
02449  * |--- ACK ----->|
02450  * |<--- data[N-1] -----<|
02451  * |--- NACK ----->| <- Last byte
02452  * |--- STOP ----->|
02453  * ```
02454  *
02455  *
02456  * ## When to Use riic_read() vs riic_write_read()
02457  *
02458  * | Function | Use When |
02459  * |-----|-----|
02460  * | riic_read() | Device auto-increments address, or reading status |
02461  * | riic_write_read() | Need to specify register address first |
02462  *
02463  * ## Common Use Cases
02464  *
02465  * - Read device status register (if device supports direct read)
02466  * - Continue reading after previous write set address pointer
02467  * - Read from devices with auto-increment address mode
02468  *
02469  *
02470  *
02471  * @pre riic_init() called for this channel
02472  * @pre data != nullptr
02473  * @pre Buffer size >= length
02474  * @pre length >= 1 && length <= 256
02475  * @pre device_addr.value <= 127
02476  * @post On success: data contains received bytes, bus released
02477  * @post On failure: Bus released via STOP
02478  *
02479  * @note This is a blocking function - returns only after transfer complete.
02480  * @note NACK is automatically sent before reading last byte.
02481  *
02482  * @warning Most peripherals require riic_write_read() to specify which
02483  *         register to read from. Use this function only when the device
02484  *         supports direct reads or address auto-increment.
02485  *
02486  * @par Example: Read Temperature Sensor (Direct Read)
02487  * @code
02488  * // Read 2 bytes from temperature sensor at address 0x48
02489  * // (sensor auto-reports temperature on read)
02490  * uint8_t temp_data[2];
02491  * riic_channel_t channel = {.value = 0};
02492  * i2c_device_addr_t addr = {.value = 0x48};
02493  *
02494  * rx_err_t err = riic_read(channel, addr, temp_data, 2);
02495  * if (err == k_rx_ok) {
02496  *     int16_t raw = (temp_data[0] << 8) | temp_data[1];
02497  *     float celsius = raw * 0.0625f;
02498  * }
02499  * @endcode
02500  *
02501  * @see riic_init() Initialize channel before use
02502  * @see riic_write() Write-only transaction
02503  * @see riic_write_read() Combined write-then-read (register read)
02504  */

```



```

02505 * @callgraph
02506 */
02507 rx_err_t riic_read(const riic_channel_t channel,
02508                  const i2c_device_addr_t device_addr,
02509                  uint8_t* data,
02510                  const uint16_t length)
02511 {
02512     RX_CHECK_NULL_PTR(data, s_tag, "Data pointer is nullptr");
02513     RX_CHECK_RANGE_TAG(channel.value,
02514                        k_riic_channel_0,
02515                        (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02516                        k_rx_err_invalid_arg,
02517                        s_tag);
02518
02519     if (device_addr.value > k_riic_addr_max_7bit) {
02520         rx_log_error(s_tag, "Invalid device address");
02521         return k_rx_err_invalid_arg;
02522     }
02523
02524     if (length == k_riic_length_zero) {
02525         rx_log_error(s_tag, "Read length cannot be zero");
02526         return k_rx_err_invalid_arg;
02527     }
02528
02529     if (length > k_riic_max_transfer_length) {
02530         rx_log_error(s_tag, "Read length exceeds maximum");
02531         return k_rx_err_invalid_arg;
02532     }
02533
02534     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02535     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02536         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02537         return k_rx_err_invalid_state;
02538     }
02539
02540     /* Get RIIC base */
02541     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02542
02543     /* Wait for bus ready */
02544     rx_err_t err = internal_wait_bus_ready(riic);
02545     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02546
02547     /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02548     riic->icr2 = k_riic_register_clear;
02549
02550     /* Set controller receive mode */
02551     riic->iccr2 = k_riic_iccr2_mst;
02552
02553     /* Send start condition */
02554     err = internal_send_start(riic);
02555     RX_RETURN_ON_ERROR(err, s_tag, "Start condition failed");
02556
02557     /* Send device address (read) -- use the receive-mode helper, NOT
02558     * internal_write_byte: TEND/TDRE-reassertion don't fire after addr[R in
02559     * controller-receive mode; RDRF does. */
02560     err = internal_write_address_for_read(
02561         riic,
02562         (uint8_t)((device_addr.value « k_riic_addr_shift) | k_riic_addr_read_bit));
02563     if (err != k_rx_ok) {
02564         rx_err_t stop_err = internal_send_stop(riic);
02565         (void)stop_err; /* Preserve original error, stop is best-effort cleanup */
02566         return err;
02567     }
02568
02569     /* 1-byte read: pre-set ACKBT=1 before the dummy read so the sole byte's
02570     * 9th-clock ACK slot carries NACK (HUM Fig 42.10). */
02571     if (length == 1U) {
02572         riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02573     }
02574
02575     (void)riic->icdrr; /* dummy read kicks reception */
02576
02577     /* Mirror of the read loop in internal_riic_read_phase -- see that
02578     * function for the why-ACKBT-on-N-2 rationale. */
02579     for (uint16_t i = 0; i < length; i++) {
02580         uint32_t timeout = k_riic_timeout_us;
02581         while (!(riic->icr2 & k_riic_icr2_rdrf) && timeout > k_riic_timeout_zero) {
02582             timeout--;
02583         }
02584         if (timeout == k_riic_timeout_zero) {
02585             rx_log_error_val(s_tag, "Read byte RDRF timeout i=", (uint8_t)i);
02586             rx_err_t stop_err = internal_send_stop(riic);
02587             (void)stop_err;
02588             return k_rx_err_timeout;
02589         }
02590
02591         if (length >= 2U && i == (uint16_t)(length - 2U)) {

```



```

02592     riic->icmr3 |= k_riic_icmr3_ackbt_mask;
02593 }
02594
02595 if (i == (uint16_t)(length - k_riic_last_index_offset)) {
02596     riic->icsr2 &= (uint8_t) ~(uint8_t) k_riic_icsr2_stop;
02597     riic->iccr2 |= k_riic_iccr2_sp;
02598 }
02599
02600 data[i] = riic->icdrr;
02601 }
02602
02603 uint32_t stop_timeout = k_riic_timeout_us;
02604 while (!(riic->icsr2 & k_riic_icsr2_stop) && stop_timeout > k_riic_timeout_zero) {
02605     stop_timeout--;
02606 }
02607 if (stop_timeout == k_riic_timeout_zero) {
02608     rx_log_error(s_tag, "STOP after final ICDRR timeout");
02609     return k_rx_err_timeout;
02610 }
02611 riic->icsr2 &= (uint8_t) ~(uint8_t) (k_riic_icsr2_stop | k_riic_icsr2_nackf);
02612
02613 return k_rx_ok;
02614 }
02615
02616 /**
02617  * @brief Combined write-then-read I2C transaction (register read)
02618  *
02619  * @details
02620  * Performs a complete I2C write-read transaction using a repeated START
02621  * condition. This is the most common pattern for reading registers from
02622  * I2C peripherals: write the register address, then read the data.
02623  *
02624  * ## Transaction Sequence (Register Read)
02625  *
02626  * ```
02627  * Controller                                Peripheral
02628  * |----- START ----->|
02629  * |----- Address + W (0) ----->|
02630  * |<--- ACK -----<|
02631  * |----- Register Address ----->| <- write_data
02632  * |<--- ACK -----<|
02633  * |----- REPEATED START ----->| <- No STOP!
02634  * |----- Address + R (1) ----->|
02635  * |<--- ACK -----<|
02636  * |<--- Register Data -----<| <- read_data
02637  * |----- ACK/NACK ----->|
02638  * |----- STOP ----->|
02639  * ```
02640  *
02641  * ## Why Repeated START?
02642  *
02643  * The repeated START (Sr) is critical for atomic register access:
02644  *
02645  * | Approach | Bus Release | Risk |
02646  * |-----|-----|-----|
02647  * | STOP + START | Yes | Another controller could interrupt |
02648  * | Repeated START | No | Transaction is atomic |
02649  *
02650  * Many I2C peripherals require repeated START and may not work correctly
02651  * with separate write-then-read transactions.
02652  *
02653  * ## Common Use Cases
02654  *
02655  * | Peripheral Type | write_data | read_length | Purpose |
02656  * |-----|-----|-----|-----|
02657  * | Temperature | [reg_addr] | 2 | Read temp register |
02658  * | EEPROM | [addr_hi, addr_lo] | N | Read N bytes |
02659  * | Accelerometer | [reg_addr] | 6 | Read XYZ axes |
02660  * | RTC | [seconds_reg] | 7 | Read time |
02661  *
02662  * @pre riic_init() called for this channel
02663  * @pre write_data != nullptr
02664  * @pre read_data != nullptr
02665  * @pre write_length >= 1 && write_length <= 256
02666  * @pre read_length >= 1 && read_length <= 256
02667  * @pre device_addr.value <= 127
02668  * @post On success: read_data contains register contents, bus released
02669  * @post On failure: Bus released via STOP
02670  *
02671  * @note This is a blocking function - returns only after transfer complete.
02672  * @note Uses repeated START (no bus release between write and read phases).
02673  * @note NACK is automatically sent before reading last byte.

```

```

02679 *
02680 * @par Example: Read WHO_AM_I Register from IMU
02681 * @code
02682 * // Read WHO_AM_I register (0x0F) from MPU6050 at address 0x68
02683 * riic_channel_t channel = {.value = 0};
02684 * i2c_device_addr_t imu_addr = {.value = 0x68};
02685 *
02686 * uint8_t reg_addr = 0x0F; // WHO_AM_I register
02687 * uint8_t who_am_i;
02688 *
02689 * rx_err_t err = riic_write_read(channel, imu_addr,
02690 *                               &reg_addr, 1, // Write: register address
02691 *                               &who_am_i, 1); // Read: 1 byte
02692 *
02693 * if (err == k_rx_ok && who_am_i == 0x68) {
02694 *     // IMU detected and responding correctly
02695 * }
02696 * @endcode
02697 *
02698 * @par Example: Read 6 Bytes of Accelerometer Data
02699 * @code
02700 * // Read ACCEL_XOUT_H through ACCEL_ZOUT_L (6 bytes starting at 0x3B)
02701 * uint8_t reg_addr = 0x3B;
02702 * uint8_t accel_data[6];
02703 *
02704 * rx_err_t err = riic_write_read(channel, imu_addr,
02705 *                               &reg_addr, 1,
02706 *                               accel_data, 6);
02707 *
02708 * if (err == k_rx_ok) {
02709 *     int16_t accel_x = (accel_data[0] « 8) | accel_data[1];
02710 *     int16_t accel_y = (accel_data[2] « 8) | accel_data[3];
02711 *     int16_t accel_z = (accel_data[4] « 8) | accel_data[5];
02712 * }
02713 * @endcode
02714 *
02715 * @see riic_init() Initialize channel before use
02716 * @see riic_write() Write-only transaction
02717 * @see riic_read() Read-only transaction (no address write)
02718 *
02719 * @callgraph
02720 */
02721 rx_err_t riic_write_read(const riic_channel_t channel,
02722                          const i2c_device_addr_t device_addr,
02723                          const uint8_t* write_data,
02724                          const uint16_t write_length,
02725                          uint8_t* read_data,
02726                          const uint16_t read_length)
02727 {
02728     RX_CHECK_NULL_PTR(write_data, s_tag, "Write data pointer is nullptr");
02729     RX_CHECK_NULL_PTR(read_data, s_tag, "Read data pointer is nullptr");
02730     RX_CHECK_RANGE_TAG(channel.value,
02731                        k_riic_channel_0,
02732                        (uint8_t)(k_riic_max_channels - k_riic_last_index_offset),
02733                        k_rx_err_invalid_arg,
02734                        s_tag);
02735
02736     if (device_addr.value > k_riic_addr_max_7bit) {
02737         rx_log_error(s_tag, "Invalid device address");
02738         return k_rx_err_invalid_arg;
02739     }
02740
02741     if (write_length == k_riic_length_zero || read_length == k_riic_length_zero) {
02742         rx_log_error(s_tag, "Read/write length cannot be zero");
02743         return k_rx_err_invalid_arg;
02744     }
02745
02746     if (write_length > k_riic_max_transfer_length || read_length > k_riic_max_transfer_length) {
02747         rx_log_error(s_tag, "Read/write length exceeds maximum");
02748         return k_rx_err_invalid_arg;
02749     }
02750
02751     /* Validate channel mode (range already checked by RX_CHECK_RANGE_TAG above) */
02752     if (s_riic_channel_mode[channel.value] != k_riic_mode_controller) {
02753         rx_log_error(s_tag, "RIIC channel not initialized in controller mode");
02754         return k_rx_err_invalid_state;
02755     }
02756
02757     /* Get RIIC base */
02758     volatile rx_riic_regs_t* riic = internal_get_riic_base(channel.value);
02759
02760     /* Wait for bus ready */
02761     rx_err_t err = internal_wait_bus_ready(riic);
02762     RX_RETURN_ON_ERROR(err, s_tag, "Bus not ready");
02763
02764     /* Pre-clear stale flags from any previous cleanup. See riic_write(). */
02765     riic->icsr2 = k_riic_register_clear;

```

```

02766
02767 /* Perform write phase */
02768 err = internal_riic_write_phase(riic, device_addr, write_data, write_length);
02769 if (err != k_rx_ok) {
02770     return err;
02771 }
02772
02773 /* Perform read phase. On success the final byte's NACK + STOP is
02774  * issued atomically inside internal_read_byte(), so no separate
02775  * internal_send_stop() call is needed here. */
02776 err = internal_riic_read_phase(riic, device_addr, read_data, read_length);
02777 if (err != k_rx_ok) {
02778     return err;
02779 }
02780
02781 return k_rx_ok;
02782 }
02783
02784 /*
02785  * RIIC Peripheral (Device) Mode Functions
02786  *
02787  */
02788
02789 /**
02790  * @enum riic_peripheral_reg_constants_t
02791  * @brief Register bit-field constants for peripheral mode configuration
02792  *
02793  * @details
02794  * Bit field values used when configuring RIIC for peripheral (device) mode.
02795  * In peripheral mode the RX72N responds to an external I2C controller (RPi5).
02796  * All values fit in uint8_t since they are 8-bit register constants.
02797  *
02798  * @see riic_init_peripheral()
02799  * @since Version 1.0.0
02800  */
02801 typedef enum : uint8_t {
02802     k_riic_icmr1_peripheral_7bit = 0x00, /**< CKS=0, MS=0 (peripheral mode), 7-bit */
02803     k_riic_icser_sar0e           = 0x01, /**< ICSE bit 0: enable SAR0 address matching */
02804     k_riic_sarl_addr_shift       = 1,    /**< 7-bit addr stored in SARL[7:1] */
02805     k_riic_saru0_7bit            = 0x00, /**< SARU0 = 0 for 7-bit addressing */
02806 } riic_peripheral_reg_constants_t;
02807
02808 /**
02809  * @enum riic_peripheral_size_constants_t
02810  * @brief Size and timeout constants for peripheral mode operations
02811  *
02812  * @details
02813  * Timeout values for peripheral mode read/write.
02814  * k_riic_peripheral_transfer_limit is declared in hardware.h (public API) so
02815  * callers can statically size their buffers; only the timeout lives here.
02816  *
02817  * @see riic_peripheral_read()
02818  * @see riic_peripheral_write()
02819  * @since Version 1.0.0
02820  */
02821 typedef enum : uint16_t {
02822     k_riic_periph_timeout_us = 1000, /**< 1ms timeout for TDRE poll in write */
02823 } riic_peripheral_size_constants_t;
02824
02825 /**
02826  * @brief Initialize RIIC channel in peripheral (device) mode
02827  *
02828  * @details
02829  * Configures the RIIC peripheral to act as an I2C peripheral device, responding
02830  * to a fixed 7-bit I2C address issued by a controller (RPi5). Sets up the
02831  * peripheral address register (SARL0/SARU0), enables address match detection (ICSE.SAR0E),
02832  * and leaves the module in peripheral mode (MS=0 in ICMR1).
02833  *
02834  * ## Initialization Sequence
02835  *
02836  * 1. Validate channel number and device address
02837  * 2. Enable RIIC module clock (clear MSTPCRB bit)
02838  * 3. Reset RIIC internal state (ICRST)
02839  * 4. Configure peripheral address in SARL0/SARU0
02840  * 5. Enable SAR0 address match detection (ICSE.SAR0E)
02841  * 6. Configure peripheral mode (MS=0), 7-bit addressing
02842  * 7. Enable I2C bus interface (ICE)
02843  * 8. Mark channel as initialized
02844  *
02845  *
02846  *
02847  * @pre channel.value must be in range [0, k_riic_max_channels - 1]
02848  * @pre device_addr.value must be <= k_riic_addr_max_7bit (0x7F)
02849  * @post RIIC peripheral responds to device_addr on the I2C bus
02850  * @post s_riic_channel_mode[channel.value] == k_riic_mode_peripheral

```

```

02851 *
02852 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
02853 * @note Not thread-safe during initialization
02854 *
02855 * @see riic_peripheral_read() Read data written by I2C controller
02856 * @see riic_peripheral_write() Provide data to I2C controller read
02857 *
02858 * @since Version 1.0.0
02859 */
02860 rx_err_t riic_init_peripheral(const riic_channel_t channel, const i2c_device_addr_t device_addr)
02861 {
02862     if (channel.value >= k_riic_max_channels) {
02863         return k_rx_err_invalid_arg;
02864     }
02865
02866     if (device_addr.value > k_riic_addr_max_7bit) {
02867         rx_log_error(s_tag, "Invalid peripheral device address");
02868         return k_rx_err_invalid_arg;
02869     }
02870
02871     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02872     if (riic == nullptr) {
02873         return k_rx_err_invalid_arg;
02874     }
02875
02876     /* Enable RIIC module (clear module stop bit) */
02877     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
02878
02879     if (channel.value == k_riic_channel_0) {
02880         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic0;
02881     } else if (channel.value == k_riic_channel_1) {
02882         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic1;
02883     } else {
02884         system_regs()->mstpcrb &= ~(uint32_t)k_riic_mstpb_bit_value « k_riic_mstpb_riic2;
02885     }
02886
02887     *prcr_reg() = k_rx_prcr_lock;
02888
02889     /* Reset RIIC */
02890     riic->iccr1 = k_riic_iccr1_icrst;
02891     riic->iccr1 = k_riic_register_clear;
02892
02893     /* Set peripheral (device) address in SARL0/SARU0
02894      * SARL0[7:1] = device_addr.value (7-bit address)
02895      * SARU0 = 0 for 7-bit mode */
02896     riic->sarl0 = (uint8_t)(device_addr.value « k_riic_sarl_addr_shift);
02897     riic->saru0 = k_riic_saru0_7bit;
02898
02899     /* Enable SAR0 address match detection */
02900     riic->icser = k_riic_icser_sar0e;
02901
02902     /* Configure for peripheral mode (MS=0), 7-bit addressing, CKS=0 */
02903     riic->icmr1 = k_riic_icmr1_peripheral_7bit;
02904     riic->icmr2 = k_riic_icmr2_default;
02905     riic->icmr3 = k_riic_icmr2_default;
02906
02907     /* Enable I2C bus interface */
02908     riic->iccr1 = k_riic_iccr1_ice;
02909
02910     /* Mark channel as initialized in peripheral mode */
02911     s_riic_channel_mode[channel.value] = k_riic_mode_peripheral;
02912
02913     rx_log_debug(s_tag, "RIIC peripheral mode initialized");
02914
02915     return k_rx_ok;
02916 }
02917
02918 /**
02919 * @brief Read bytes written by the I2C controller to this peripheral
02920 *
02921 * @details
02922 * Polls ICSR2.RDRF (Receive Data Register Full) to detect incoming bytes from
02923 * the I2C controller (RPI5). Reads up to max_length bytes into the caller-supplied
02924 * buffer. Returns immediately with bytes_read == 0 if no data is currently available
02925 * (non-blocking). Each byte read clears the RDRF flag automatically on the RX72N.
02926 *
02927 * ## Read Algorithm
02928 *
02929 * 1. Validate pointers, channel, and max_length
02930 * 2. Clear *bytes_read to 0
02931 * 3. Poll ICSR2.RDRF; if set, read ICDRR into data[*bytes_read], increment count
02932 * 4. Repeat until RDRF clear or max_length reached
02933 * 5. Return k_rx_ok with actual count in *bytes_read
02934 *
02935 *
02936 *
02937 * @pre channel must be initialized via riic_init_peripheral()

```

```

02938 * @pre data != nullptr with at least max_length bytes of capacity
02939 * @pre max_length >= 1 && max_length <= k_riic_peripheral_transfer_limit
02940 * @post *bytes_read contains the number of bytes placed in data (0 to max_length)
02941 * @post data[0..(*bytes_read)-1] contain valid received bytes on k_rx_ok
02942 *
02943 * @note Non-blocking: returns immediately with available data (bytes_read may be 0)
02944 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
02945 * @note Not thread-safe, caller must provide external synchronization
02946 *
02947 * @see riic_init_peripheral() Initialize channel first
02948 * @see riic_peripheral_write() Provide read data to controller
02949 *
02950 * @since Version 1.0.0
02951 */
02952 rx_err_t riic_peripheral_read(const riic_channel_t channel,
02953                               uint8_t* data,
02954                               const uint16_t max_length,
02955                               uint16_t* bytes_read)
02956 {
02957     if (data == nullptr || bytes_read == nullptr) {
02958         return k_rx_err_null_ptr;
02959     }
02960
02961     if (channel.value >= k_riic_max_channels) {
02962         return k_rx_err_invalid_arg;
02963     }
02964
02965     if (max_length == 0) {
02966         return k_rx_err_invalid_arg;
02967     }
02968
02969     if (max_length > k_riic_peripheral_transfer_limit) {
02970         rx_log_error(s_tag, "Peripheral read max_length exceeds limit");
02971         return k_rx_err_invalid_arg;
02972     }
02973
02974     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
02975         return k_rx_err_invalid_state;
02976     }
02977
02978     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
02979     if (riic == nullptr) {
02980         return k_rx_err_invalid_arg;
02981     }
02982
02983     *bytes_read = 0;
02984
02985     /* Poll ICSR2.RDRF (bit 1) for each available byte - non-blocking */
02986     while (*bytes_read < max_length) {
02987         if (!(riic->icsr2 & k_riic_icsr2_rdrf)) {
02988             break; /* No more data available right now */
02989         }
02990         data[*bytes_read] = riic->icdrr;
02991         (*bytes_read)++;
02992     }
02993
02994     return k_rx_ok;
02995 }
02996
02997 /**
02998 * @brief Write bytes for the I2C controller to read from this peripheral
02999 *
03000 * @details
03001 * Loads data bytes into the RIIC transmit data register (ICDRT) one at a time.
03002 * For each byte, polls ICSR2.TDRE (Transmit Data Register Empty) until the
03003 * register is available or a 1 ms timeout expires, then writes the byte.
03004 * The controller clock-shifts bytes out automatically after each write.
03005 *
03006 * ## Write Algorithm
03007 *
03008 * 1. Validate pointer, channel, and length
03009 * 2. For each byte in data[0..length-1]:
03010 *     a. Poll ICSR2.TDRE with k_riic_periph_timeout_us countdown
03011 *     b. If timeout, return k_rx_err_timeout
03012 *     c. Write data[i] to ICDRT
03013 * 3. Return k_rx_ok
03014 *
03015 *
03016 *
03017 * @pre channel must be initialized via riic_init_peripheral()
03018 * @pre data != nullptr with at least length valid bytes
03019 * @pre length >= 1
03020 * @post On k_rx_ok: all length bytes have been written to ICDRT
03021 * @post On k_rx_err_timeout: partial write may have occurred (i bytes written)
03022 *
03023 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
03024 * @note Not thread-safe, caller must provide external synchronization

```

```

03025 *
03026 * @see riic_init_peripheral() Initialize channel first
03027 * @see riic_peripheral_read() Receive data from controller
03028 *
03029 * @since Version 1.0.0
03030 */
03031 rx_err_t
03032 riic_peripheral_write(const riic_channel_t channel, const uint8_t* data, const uint16_t length)
03033 {
03034     if (data == nullptr) {
03035         return k_rx_err_null_ptr;
03036     }
03037
03038     if (channel.value >= k_riic_max_channels) {
03039         return k_rx_err_invalid_arg;
03040     }
03041
03042     if (length == 0 || length > k_riic_peripheral_transfer_limit) {
03043         rx_log_error(
03044             s_tag,
03045             "riic_peripheral_write: length out of range (1..k_riic_peripheral_transfer_limit)");
03046         return k_rx_err_invalid_arg;
03047     }
03048
03049     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03050         return k_rx_err_invalid_state;
03051     }
03052
03053     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03054     if (riic == nullptr) {
03055         return k_rx_err_invalid_arg;
03056     }
03057
03058     for (uint16_t i = 0; i < length; i++) {
03059         /* Wait for ICSR2.TDRE (transmit data register empty) */
03060         uint32_t timeout = k_riic_periph_timeout_us;
03061         while (!(riic->icsr2 & k_riic_icsr2_tdre) && timeout > k_riic_timeout_zero) {
03062             timeout--;
03063         }
03064
03065         if (timeout == k_riic_timeout_zero) {
03066             return k_rx_err_timeout;
03067         }
03068
03069         riic->icdrt = data[i];
03070     }
03071
03072     rx_log_debug(s_tag, "RIIC peripheral write complete");
03073
03074     return k_rx_ok;
03075 }
03076
03077 /**
03078 * @brief Deinitialize an RIIC channel that was initialized in peripheral mode
03079 *
03080 * @details
03081 * Disables the peripheral address match (ICSER.SAR0E), clears the SARL0/SARU0
03082 * address registers, resets the RIIC module, and marks the channel as
03083 * uninitialized. After this call, riic_init() or riic_init_peripheral() may
03084 * be called again on the same channel.
03085 *
03086 *
03087 *
03088 * @pre channel.value must be in range [0, k_riic_max_channels - 1]
03089 * @pre Channel must have been initialized via riic_init_peripheral()
03090 * @post s_riic_channel_mode[channel.value] == k_riic_mode_uninitialized on success
03091 * @post RIIC peripheral address match disabled; hardware in reset state
03092 *
03093 * @note Only available on RX72N target (guarded by __RX__ preprocessor)
03094 * @note Not thread-safe; provide external synchronization if needed
03095 *
03096 * @see riic_init_peripheral() Complementary initialization function
03097 *
03098 * @since Version 1.0.0
03099 */
03100 rx_err_t riic_deinit_peripheral(const riic_channel_t channel)
03101 {
03102     if (channel.value >= k_riic_max_channels) {
03103         return k_rx_err_invalid_arg;
03104     }
03105
03106     if (s_riic_channel_mode[channel.value] != k_riic_mode_peripheral) {
03107         return k_rx_err_invalid_state;
03108     }
03109
03110     volatile rx_riic_regs_t* const riic = internal_get_riic_base(channel.value);
03111     if (riic == nullptr) {

```

```

03112     return k_rx_err_invalid_arg;
03113 }
03114
03115 /* Disable peripheral address match (clear SAR0E in IC SER) */
03116 riic->icser = (uint8_t)(riic->icser & ~(uint8_t)k_riic_icser_sar0e);
03117
03118 /* Clear peripheral address registers */
03119 riic->sarl0 = k_riic_register_clear;
03120 riic->saru0 = k_riic_register_clear;
03121
03122 /* Reset the RIIC module */
03123 riic->iccr1 = k_riic_iccr1_iicrst;
03124 riic->iccr1 = k_riic_register_clear;
03125
03126 /* Mark channel as uninitialized */
03127 s_riic_channel_mode[channel.value] = k_riic_mode_uninitialized;
03128
03129 rx_log_debug(s_tag, "RIIC peripheral mode deinitialized");
03130
03131 return k_rx_ok;
03132 }
03133
03134 #endif /* __RX__ */

```

8.107 libs/rx/hal/src/rspi.c File Reference

RSPI (Renesas Serial Peripheral Interface) Driver for RX72N.

```

#include <stdint.h>
#include <string.h>
#include "hardware.h"
#include "rx72n_clock.h"
#include "rx72n_regs.h"
#include "rx_gpio_constants.h"
#include "rx_port_constants.h"
#include "rx_port_utils.h"
#include "rx_register_protection.h"

```

Include dependency graph for rspi.c:



Data Structures

- struct [rspi_cs_config_t](#)
Controller mode CS pin configuration.

Enumerations

- enum [rspi_constants_t](#) : uint32_t {
[k_rspi_max_channels](#) = 3 , [k_rspi_timeout_us](#) = 10000 , [k_rspi_timeout_zero](#) = 0 ,
[k_rspi_len_zero](#) = 0 ,
[k_rspi_zero_u32](#) = 0 , [k_rspi_spbr_offset](#) = 1 , [k_rspi_loop_start](#) = 0 }
RSPI channel count, timeout, and general constants.
- enum [rspi_cs_defaults_t](#) : uint8_t { [k_rspi_cs_default_port](#) = 0 , [k_rspi_cs_default_pin](#) = 0 ,
[k_rspi_spbr_init](#) = 0 }
RSPI CS default constants (uninitialized state).
- enum [rspi_mstpcrb_bits_t](#) : uint8_t { [k_rspi_mstpb_rspi0](#) = 17 , [k_rspi_mstpb_rspi1](#) = 16 }
RSPI0/RSPI1 module stop bit positions in MSTPCRB (HW Manual p.408).

- enum `rspi_mstpcrc_bits_t` : `uint8_t` { `k_rspi_mstpc_rcspi2` = 22 }
RSPI2 module stop bit position in MSTPCRC (HW Manual p.410).
- enum `rspi_controller_constants_t` : `uint32_t` {
`k_rspi_min_freq_hz` = 100000 , `k_rspi_max_freq_hz` = 10000000 , `k_rspi_spbr_max` = 255 ,
`k_rspi_spbr_divisor` = 2 ,
`k_rspi_cs_setup_delay` = 10 , `k_rspi_cs_hold_delay` = 10 , `k_rspi_max_delay_cycles` = 10000
}
RSPI controller mode constants.
- enum `rspi_mode_limits_t` : `uint8_t` { `k_rspi_mode_min` = 0 , `k_rspi_mode_max` = 3 }
SPI mode limits.
- enum `rspi_spcmd_bits_t` : `uint16_t` {
`k_rspi_spcmd_cpha_mask` = 1U , `k_rspi_spcmd_cpol_mask` = 2U , `k_rspi_spcmd_cpol_pos`
= 1 , `k_rspi_spcmd_cpha_pos` = 0 ,
`k_rspi_spcmd_spl_shift` = 8 , `k_rspi_spcmd_8bit` = 7U , `k_rspi_spcmd_16bit` = 15U }
SPCMD register bit positions and masks.
- enum `rspi_spdcr_local_t` : `uint8_t` { `k_rspi_spdcr_splw_pos` = 5 , `k_rspi_spdcr_byte_mode`
= 0U }
SPDCR register bit positions (local).
- enum `rspi_register_defaults_t` : `uint8_t` { `k_rspi_sppcr_no_loopback` = 0U , `k_rspi_spcr_disabled`
= 0U }
RSPI register default values (reset/disabled state).
- enum `rspi_bit_constants_t` : `uint32_t` { `k_rspi_bit_set` = 1UL }
RSPI bit manipulation constants for register field operations.
- enum `rspi_transfer_constants_t` : `uint16_t` { `k_rspi_transfer_len_max` = 65535U ,
`k_rspi_spcmd_init` = 0U }
RSPI transfer length and command register constants.

Functions

- static volatile `rx_rspi_regs_t` * `internal_get_rspi_base` (const `rspi_channel_t` channel)
Get RSPI register base address from channel number.
- static void `internal_set_mstpcrb_for_channel` (const `rspi_channel_t` channel, const bool enable)
Set module stop bit for an RSPI channel.
- static `rx_err_t` `internal_rspi_validate_init` (const `rspi_channel_t` channel, const `rspi_config_t` *const config, volatile `rx_rspi_regs_t` **rspi_out)
Initialize RSPI channel in peripheral mode.
- static `uint16_t` `internal_rspi_build_spcmd` (volatile `rx_rspi_regs_t` *const rspi, const `rspi_config_t` *const config)
Build SPCMD0 value and program SPDCR for the requested data width.
- static void `internal_rspi_release_module_stop` (const `rspi_channel_t` channel)
Release the MSTPCRB module-stop bit for the requested channel.
- `rx_err_t` `rspi_init_peripheral` (const `rspi_channel_t` channel, const `rspi_config_t` *config)
Initialize RSPI channel in peripheral mode for RPi5 communication.
- static `rx_err_t` `internal_wait_tx_ready` (volatile `rx_rspi_regs_t` *rspi)
Wait for transmit buffer to become empty (SPTEF flag).
- static `rx_err_t` `internal_wait_rx_ready` (volatile `rx_rspi_regs_t` *rspi)
Wait for receive buffer to become full (SPRF flag).
- `rx_err_t` `rspi_peripheral_transfer` (const `rspi_channel_t` channel, const `uint8_t` *tx_data, `uint8_t` *rx_data, const `uint16_t` length)
Perform full-duplex SPI transfer in peripheral mode.
- `rx_err_t` `rspi_peripheral_read_available` (const `rspi_channel_t` channel, bool *available)
Check if data is available in the RSPI receive buffer.

- rx_err_t [rspi_peripheral_write_ready](#) (const [rspi_channel_t](#) channel, bool *ready)
Check if transmit buffer is ready for data.
- rx_err_t [rspi_deinit](#) (const [rspi_channel_t](#) channel)
Deinitialize and disable RSPI peripheral.
- static void [internal_timing_delay](#) (const uint16_t cycles)
Cycle-accurate timing delay using inline NOP instructions.
- static void [internal_configure_spcmd](#) (uint16_t *spcmd, const uint8_t spi_mode)
Configure SPCMD register for SPI mode (CPOL/CPHA).
- static rx_err_t [internal_calculate_spbr](#) (const uint32_t freq_hz, uint8_t *spbr)
Calculate SPBR value for desired SPI clock frequency.
- static rx_err_t [internal_configure_cs_gpio](#) (const rx_port_pin_t pin_config)
Configure GPIO pin for chip select output.
- static rx_err_t [internal_validate_controller_args](#) (const [rspi_channel_t](#) channel, const [rspi_controller_config_t](#) *config)
Validate controller initialization arguments.
- static rx_err_t [internal_prepare_controller](#) (const [rspi_channel_t](#) channel, const [rspi_controller_config_t](#) *config, volatile [rx_rspi_regs_t](#) **out_rspi, uint8_t *out_spbr)
Prepare controller hardware resources.
- static rx_err_t [internal_configure_registers](#) (const [rspi_channel_t](#) channel, volatile [rx_rspi_regs_t](#) *rspi, uint8_t spbr, uint16_t spcmd, const [rspi_controller_config_t](#) *config)
Configure RSPI hardware registers for controller mode.
- rx_err_t [rspi_init_controller](#) (const [rspi_channel_t](#) channel, const [rspi_controller_config_t](#) *config)
Initialize RSPI channel in controller mode.
- rx_err_t [rspi_controller_set_cs](#) (const [rspi_channel_t](#) channel, const bool active)
Set chip select line state for RSPI controller channel.
- static rx_err_t [internal_controller_assert_cs_with_setup](#) (const [rspi_channel_t](#) channel)
Assert CS with setup delay.
- static rx_err_t [internal_controller_deassert_cs_with_hold](#) (const [rspi_channel_t](#) channel)
Deassert CS with hold delay.
- static rx_err_t [internal_controller_do_16bit_transfer](#) (volatile [rx_rspi_regs_t](#) *rspi, uint16_t tx_data, uint16_t *rx_data)
Perform core 16-bit SPI transfer operation.
- rx_err_t [rspi_controller_transfer_16bit](#) (const [rspi_channel_t](#) channel, const uint16_t tx_data, uint16_t *const rx_data)
Perform 16-bit SPI transfer in controller mode with CS handling.
- rx_err_t [rspi_controller_deinit](#) (const [rspi_channel_t](#) channel)
Deinitialize RSPI controller mode.

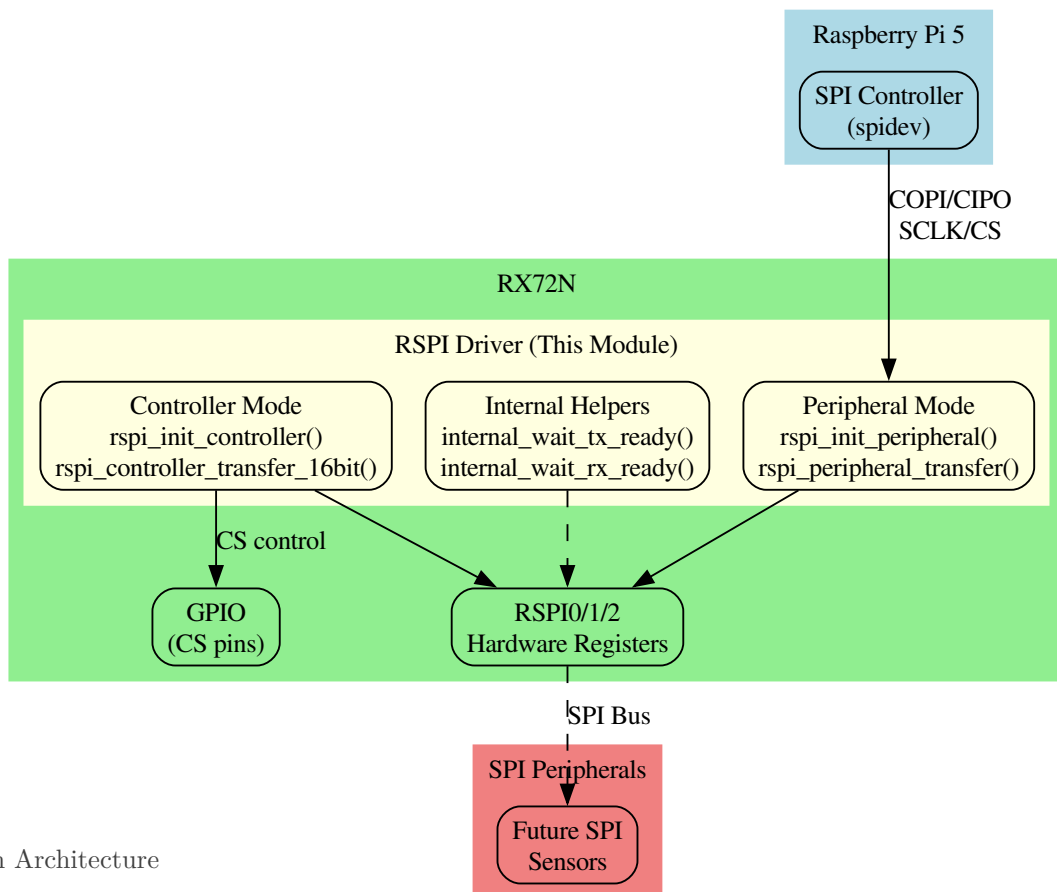
Variables

- static const char *const [s_tag](#) = "RSPI"
- static bool [s_rspi_channel_initialized](#) [[k_rspi_max_channels](#)] = {false, false, false}
- static bool [s_rspi_controller_initialized](#) [[k_rspi_max_channels](#)] = {false, false, false}
- static [rspi_cs_config_t](#) [s_rspi_cs_config](#) [[k_rspi_max_channels](#)]

8.107.1 Detailed Description

RSPI (Renesas Serial Peripheral Interface) Driver for RX72N.

Provides complete SPI driver implementation supporting both peripheral mode (RX72N as SPI peripheral with RPi5 as controller) and controller mode (RX72N as SPI controller for sensor communication). Implements full-duplex communication with configurable clock frequency, SPI modes 0-3, and automatic chip select management.



System Architecture

Supported Features

Feature	Peripheral Mode	Controller Mode
Full-duplex	↔ [OK]↔	↔ [OK]↔
8-bit data	↔ [OK]↔	-
16-bit data	↔ [OK]↔	↔ [OK]↔

Feature	Peripheral Mode	Controller Mode
SPI modes 0-3	↔ [OK]↔	↔ [OK]↔
Hardware CS	-	GPIO-based
Clock config	External	100kHz - 10MHz

RSPI Channels

Channel	Use Case	Status Tracking
RSPI0	RPi5 communication	s_rspi_channel_initialized↔ [0]
RSPI1	Not used in current design	s_rspi_controller_initialized↔ [1]
RSPI2	Not used	Available

Performance Characteristics

Metric	Value	Condition
Max clock (controller)	10 MHz	PCLKB=60MHz, SPBR=2
Min clock (controller)	100 kHz	PCLKB=60MHz, SPBR=255
Transfer timeout	10 ms	Per byte
CS setup/hold delay	~300 ns	10 NOP cycles @ 240MHz
16-bit transfer time	~2-50 us	Depends on clock

Memory Usage

Component	Size	Description
Code (.text)	~4 KB	All functions
Constants (.rodata)	~100 B	Static constants
Static data (.bss)	~20 B	Init flags, CS config
Stack (per call)	~50 B	Local variables

Hardware Register Access

Uses [rx72n_rspi_regs.h](#) register definitions with inline accessors:

- [rspi0\(\)](#), [rspi1\(\)](#), [rspi2\(\)](#) for channel base addresses
- MSTPCRB module stop control for power management
- PRCR register protection for system register access

Module Dependencies

- [hardware.h](#): Hardware definitions and inline accessors
- [rx72n_regs.h](#): Register structure definitions
- [rx_gpio_constants.h](#): GPIO pin definitions
- [rx_port_constants.h](#): Port constants
- [rx_port_utils.h](#): Port/pin extraction utilities
- [rx_register_protection.h](#): PRCR unlock/lock

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] All loops bounded (timeout counters, transfer length max)
- Rule 3: [OK] No dynamic memory allocation (static buffers only)
- Rule 4: [OK] Functions under 60 lines (refactored into helpers)
- Rule 5: [OK] Minimum 2 assertions per function (RX_ASSERT, RX_CHECK_NULL_PTR)
- Rule 6: [OK] Variables declared at smallest scope
- Rule 7: [OK] All return values checked
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [OK] No function pointers (direct hardware access)
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S (SRP): Driver handles only RSPI peripheral operations
- O (OCP): Mode/frequency configurable without code changes
- L (LSP): Peripheral/controller modes have consistent error semantics
- I (ISP): Separate init/transfer/deinit for each mode
- D (DIP): Depends on hardware abstractions ([rx72n_regs.h](#))

Thread Safety

- NOT thread-safe: Caller must provide external synchronization
- Static state (`s_rspi_*`) shared across calls
- Concurrent access to same channel requires mutex protection
- Different channels can be accessed independently

See also

[rx72n_rspi_regs.h](#) RSPI register definitions

[rx_spi_comm.h](#) Higher-level SPI communication layer

[docs/sections/03_hardware_pinout.tex](#) SPI pin assignments

Author

Locked, Inc.

Date

2026-01-01

Copyright

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Since

Version 1.0.0

Definition in file [rspi.c](#).

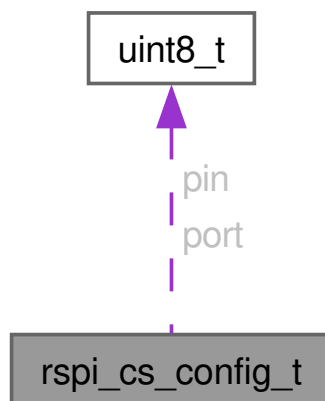
8.107.2 Data Structure Documentation

8.107.2.1 struct rspi`cs`config`_t

Controller mode CS pin configuration.

Definition at line [358](#) of file [rspi.c](#).

Collaboration diagram for rspi_cs_config_t:



Data Fields

<code>uint8_t</code>	<code>pin</code>	GPIO pin for chip select
<code>uint8_t</code>	<code>port</code>	GPIO port for chip select

8.107.3 Enumeration Type Documentation

8.107.3.1 rspi_bit_constants_t

```
enum rspi\_bit\_constants\_t : uint32_t
```

RSPI bit manipulation constants for register field operations.

Provides named constants for single-bit set operations used to construct register masks via left-shift. Using a named constant instead of a literal 1 improves readability and grep-ability of register manipulation code. All register mask construction in this module uses `k_rspi_bit_set` as the base value to ensure consistent, self-documenting bit manipulation.

Invariant

`k_rspi_bit_set` must always equal 1

```
// Construct a mask for MSTPCRB module stop bit
uint32_t mask = (k_rspi_bit_set « k_rspi_mstpb_rspi0);
system_regs()->mstpcrb &= ~mask;
```

See also

[internal_set_mstpcrb_for_channel\(\)](#) Uses for MSTPCRB mask construction

[internal_configure_spcmd\(\)](#) Uses for CPOL/CPHA bit setting

Since

Version 1.0.0

Enumerator

<code>k_rspi_bit_set</code>	Single bit set value (1) for shift-based register mask construction
-----------------------------	---

Definition at line 310 of file [rspi.c](#).

```
00310      : uint32_t {
00311  k_rspi_bit_set = 1UL, /**< Single bit set value (1) for shift-based register mask construction */
00312 } rspi_bit_constants_t;
```

8.107.3.2 rspi_constants_t

```
enum rspi\_constants\_t : uint32_t
```

RSPI channel count, timeout, and general constants.

Core configuration constants for RSPI driver. Timeout values are calibrated for worst-case SPI peripheral response times at minimum clock frequency.

Timeout Calculation

At 100 kHz minimum clock, one byte takes ~80 us. 10 ms timeout provides ~125 byte margin before declaring failure, sufficient for any single transfer operation.

Since

Version 1.0.0

Enumerator

k_rspi_max_channels	Number of RSPI channels: RSPi0, RSPi1, RSPi2. Value: 3
k_rspi_timeout_us	Timeout for TX/RX wait operations in loop iterations. Approximately 10 ms at typical loop overhead. Value: 10000 iterations (~10 ms)
k_rspi_timeout_zero	Sentinel value indicating timeout has expired. Value: 0
k_rspi_len_zero	Sentinel for zero-length transfer (invalid). Value: 0
k_rspi_zero_u32	Zero constant for uint32_t comparisons. Value: 0
k_rspi_spbr_offset	Offset for SPBR calculation formula. $SPBR = (PCLKB / (2 * freq)) - 1$ Value: 1
k_rspi_loop_start	Starting index for bounded loops (NASA Rule 2). Value: 0

Definition at line 176 of file [rspi.c](#).

```

00176         : uint32_t {
00177     k_rspi_max_channels = 3, /**< Number of RSPI channels: RSPi0, RSPi1, RSPi2.
00178                               @par Value: 3 */
00179     k_rspi_timeout_us = 10000, /**< Timeout for TX/RX wait operations in loop iterations.
00180                               Approximately 10 ms at typical loop overhead.
00181                               @par Value: 10000 iterations (~10 ms) */
00182     k_rspi_timeout_zero = 0, /**< Sentinel value indicating timeout has expired.
00183                               @par Value: 0 */
00184     k_rspi_len_zero = 0, /**< Sentinel for zero-length transfer (invalid).
00185                               @par Value: 0 */
00186     k_rspi_zero_u32 = 0, /**< Zero constant for uint32_t comparisons.
00187                               @par Value: 0 */
00188     k_rspi_spbr_offset = 1, /**< Offset for SPBR calculation formula.
00189                                $SPBR = (PCLKB / (2 * freq)) - 1$ 
00190                               @par Value: 1 */
00191     k_rspi_loop_start = 0, /**< Starting index for bounded loops (NASA Rule 2).
00192                               @par Value: 0 */

```

```

00193 k_rspi_spbr_offset = 1, /**< Offset for SPBR calculation formula.
00194          SPBR = (PCLKB / (2 * freq)) - 1
00195          @par Value: 1 */
00196
00197 k_rspi_loop_start = 0, /**< Starting index for bounded loops (NASA Rule 2).
00198          @par Value: 0 */
00199 } rspi_constants_t;

```

8.107.3.3 `rspi_controller_constants_t`

```
enum rspi\_controller\_constants\_t : uint32_t
```

RSPI controller mode constants.

PCLKB rate comes from [rx72n_clock.h](#) (`k_pclkb_hz`) – do not duplicate the literal here so the divisor math tracks any future clock-tree change.

Enumerator

<code>k_rspi_min_freq_hz</code>	Minimum SPI clock (100 kHz)
<code>k_rspi_max_freq_hz</code>	Maximum SPI clock (10 MHz)
<code>k_rspi_spbr_max</code>	Maximum SPBR value
<code>k_rspi_spbr_divisor</code>	SPBR divisor factor
<code>k_rspi_cs_setup_delay</code>	CS setup delay iterations (~300ns)
<code>k_rspi_cs_hold_delay</code>	CS hold delay iterations (~300ns)
<code>k_rspi_max_delay_cycles</code>	Sanity limit for timing delays

Definition at line 226 of file [rspi.c](#).

```

00226 : uint32_t {
00227 k_rspi_min_freq_hz = 100000, /**< Minimum SPI clock (100 kHz) */
00228 k_rspi_max_freq_hz = 10000000, /**< Maximum SPI clock (10 MHz) */
00229 k_rspi_spbr_max = 255, /**< Maximum SPBR value */
00230 k_rspi_spbr_divisor = 2, /**< SPBR divisor factor */
00231 k_rspi_cs_setup_delay = 10, /**< CS setup delay iterations (~300ns) */
00232 k_rspi_cs_hold_delay = 10, /**< CS hold delay iterations (~300ns) */
00233 k_rspi_max_delay_cycles = 10000, /**< Sanity limit for timing delays */
00234 } rspi_controller_constants_t;

```

8.107.3.4 `rspi_cs_defaults_t`

```
enum rspi\_cs\_defaults\_t : uint8_t
```

RSPI CS default constants (uninitialized state).

Enumerator

<code>k_rspi_cs_default_port</code>	Default CS port (unconfigured)
<code>k_rspi_cs_default_pin</code>	Default CS pin (unconfigured)
<code>k_rspi_spbr_init</code>	Initial SPBR value

Definition at line 202 of file [rspi.c](#).

```

00202 : uint8_t {
00203 k_rspi_cs_default_port = 0, /**< Default CS port (unconfigured) */
00204 k_rspi_cs_default_pin = 0, /**< Default CS pin (unconfigured) */
00205 k_rspi_spbr_init = 0, /**< Initial SPBR value */
00206 } rspi_cs_defaults_t;

```


8.107.3.5 rspi`mode`limits`t

```
enum rspi\_mode\_limits\_t : uint8_t
```

SPI mode limits.

Enumerator

k_rspi_mode_min	Minimum valid SPI mode (0)
k_rspi_mode_max	Maximum valid SPI mode (0-3)

Definition at line 237 of file [rspi.c](#).

```
00237      : uint8_t {
00238  k_rspi_mode_min = 0, /**< Minimum valid SPI mode (0) */
00239  k_rspi_mode_max = 3, /**< Maximum valid SPI mode (0-3) */
00240 } rspi_mode_limits_t;
```

8.107.3.6 rspi`mstpcrb`bits`t

```
enum rspi\_mstpcrb\_bits\_t : uint8_t
```

RSPI0/RSPI1 module stop bit positions in MSTPCRB (HW Manual p.408).

Enumerator

k_rspi_mstpb_rspi0	RSPI0 module stop bit in MSTPCRB
k_rspi_mstpb_rspi1	RSPI1 module stop bit in MSTPCRB

Definition at line 211 of file [rspi.c](#).

```
00211      : uint8_t {
00212  k_rspi_mstpb_rspi0 = 17, /**< RSPI0 module stop bit in MSTPCRB */
00213  k_rspi_mstpb_rspi1 = 16, /**< RSPI1 module stop bit in MSTPCRB */
00214 } rspi_mstpcrb_bits_t;
```

8.107.3.7 rspi`mstpcrc`bits`t

```
enum rspi\_mstpcrc\_bits\_t : uint8_t
```

RSPI2 module stop bit position in MSTPCRC (HW Manual p.410).

Enumerator

k_rspi_mstpc_rspi2	RSPI2 module stop bit in MSTPCRC
--------------------	----------------------------------

Definition at line 217 of file [rspi.c](#).

```
00217      : uint8_t {
00218  k_rspi_mstpc_rspi2 = 22, /**< RSPI2 module stop bit in MSTPCRC */
00219 } rspi_mstpcrc_bits_t;
```

8.107.3.8 `rspi_register_defaults_t`

```
enum rspi\_register\_defaults\_t : uint8_t
```

RSPI register default values (reset/disabled state).

Values written to RSPI registers during initialization to establish a known baseline configuration. These match the hardware reset defaults documented in the RX72N Hardware Manual Section 38.2. Writing these values before configuring mode-specific settings ensures a deterministic starting state regardless of prior peripheral activity.

Invariant

All values must match the hardware manual reset state

```
// Reset RSPI registers to defaults before reconfiguration
rspi->sppcr = k_rspi_sppcr_no_loopback;
rspi->sprcr = k_rspi_sprcr_disabled;
```

See also

[rspi_init_peripheral\(\)](#) Uses these during peripheral mode init

[internal_configure_registers\(\)](#) Uses these during controller mode init

Since

Version 1.0.0

Enumerator

<code>k_rspi_sppcr_no_loopback</code>	SPPCR: no loopback mode (normal operation)
<code>k_rspi_sprcr_disabled</code>	SPCR: SPI disabled (module in reset state)

Definition at line 282 of file [rspi.c](#).

```
00282     : uint8_t {
00283   k_rspi_sppcr_no_loopback = 0U, /**< SPPCR: no loopback mode (normal operation) */
00284   k_rspi_sprcr_disabled    = 0U, /**< SPCR: SPI disabled (module in reset state) */
00285 } rspi_register_defaults_t;
```

8.107.3.9 `rspi_spcmd_bits_t`

```
enum rspi\_spcmd\_bits\_t : uint16_t
```

SPCMD register bit positions and masks.

Enumerator

<code>k_rspi_spcmd_cpha_mask</code>	CPHA bit mask (bit 0)
<code>k_rspi_spcmd_cpol_mask</code>	CPOL bit mask (bit 1)
<code>k_rspi_spcmd_cpol_pos</code>	CPOL bit position
<code>k_rspi_spcmd_cpha_pos</code>	CPHA bit position

k_rspi_spcmd_spl_shift	SPL (data length) bit shift
k_rspi_spcmd_8bit	8-bit data length value
k_rspi_spcmd_16bit	16-bit data length value

Definition at line 243 of file [rspi.c](#).

```
00243      : uint16_t {
00244  k_rspi_spcmd_cpha_mask = 1U, /**< CPHA bit mask (bit 0) */
00245  k_rspi_spcmd_cpol_mask = 2U, /**< CPOL bit mask (bit 1) */
00246  k_rspi_spcmd_cpol_pos  = 1, /**< CPOL bit position */
00247  k_rspi_spcmd_cpha_pos  = 0, /**< CPHA bit position */
00248  k_rspi_spcmd_spl_shift = 8, /**< SPL (data length) bit shift */
00249  k_rspi_spcmd_8bit      = 7U, /**< 8-bit data length value */
00250  k_rspi_spcmd_16bit     = 15U, /**< 16-bit data length value */
00251 } rspi_spcmd_bits_t;
```

8.107.3.10 rspi`spdcr`local`t

```
enum rspi\_spdcr\_local\_t : uint8_t
```

SPDCR register bit positions (local).

Enumerator

k_rspi_spdcr_splw_pos	SPLW bit position (longword/32-bit access)
k_rspi_spdcr_byte_mode	Byte access mode

Definition at line 254 of file [rspi.c](#).

```
00254      : uint8_t {
00255  k_rspi_spdcr_splw_pos = 5, /**< SPLW bit position (longword/32-bit access) */
00256  k_rspi_spdcr_byte_mode = 0U, /**< Byte access mode */
00257 } rspi_spdcr_local_t;
```

8.107.3.11 rspi`transfer`constants`t

```
enum rspi\_transfer\_constants\_t : uint16_t
```

RSPI transfer length and command register constants.

Defines limits and initialization values for the RSPI transfer engine. The maximum transfer length is constrained by the 16-bit DMA counter. SPCMD initial value sets a clean baseline before configuring bit fields for CPOL, CPHA, and data length via bitwise OR operations.

Invariant

k_rspi_transfer_len_max must equal 65535 (uint16_t maximum, 16-bit DMA counter limit)

```
// Validate transfer length before starting peripheral transfer
if (length == k_rspi_len_zero || length > k_rspi_transfer_len_max) {
    return k_rx_err_invalid_arg;
}
// Initialize SPCMD before configuring mode-specific bits
uint16_t spcmd = k_rspi_spcmd_init;
```

See also

[rspi_peripheral_transfer\(\)](#) Uses [k_rspi_transfer_len_max](#) for length validation
[rspi_init_peripheral\(\)](#) Uses [k_rspi_spcmd_init](#) for register setup

Since

Version 1.0.0

Enumerator

k_rspi_transfer_len_max	Maximum transfer length (16-bit counter limit)
k_rspi_spcmd_init	SPCMD initial value (all fields at reset state)

Definition at line 339 of file [rspi.c](#).

```
00339         : uint16_t {
00340   k_rspi_transfer_len_max = 65535U, /**< Maximum transfer length (16-bit counter limit) */
00341   k_rspi_spcmd_init      = 0U,    /**< SPCMD initial value (all fields at reset state) */
00342 } rspi_transfer_constants_t;
```

8.107.4 Function Documentation

8.107.4.1 internal_calculate_spbr()

```
rx_err_t internal_calculate_spbr (
    const uint32_t freq_hz,
    uint8_t * spbr) [static]
```

Calculate SPBR value for desired SPI clock frequency.

Computes the RSPI Bit Rate Register (SPBR) value required to achieve the desired SPI clock frequency. Uses the formula: $SPBR = (PCLKB / (2 * freq_hz)) - 1$ with BRDV=0 (no additional divisor). The result is validated against the hardware-supported range [0, 255].

Precondition

`spbr` must be a valid non-null pointer
`freq_hz` must be in range [100000, 10000000] Hz

Postcondition

`*spbr` contains the calculated SPBR value if `k_rx_ok` returned
`*spbr` is unchanged on error

Note

Not thread-safe. Caller must provide synchronization.

See also

[rspi_init_controller\(\)](#) Calls this to configure clock frequency

[k_pclkb_hz](#) ([rx72n_clock.h](#)) PCLKB clock frequency used in calculation

Since

Version 1.0.0

Definition at line 1043 of file [rspi.c](#).

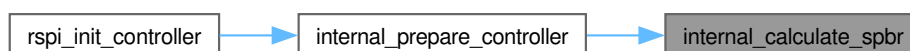
```

01044 {
01045     RX_CHECK_NULL_PTR(spbr, s_tag, "SPBR output pointer is nullptr");
01046
01047     if (freq_hz < k_rspi_min_freq_hz || freq_hz > k_rspi_max_freq_hz) {
01048         rx_log_error(s_tag, "SPI frequency out of range (100kHz-10MHz)");
01049         return k_rx_err_invalid_arg;
01050     }
01051
01052     /* SPBR formula: freq = PCLKB / (2 * (SPBR + 1) * 2^BRDV)
01053      * With BRDV=0: freq = PCLKB / (2 * (SPBR + 1))
01054      * Solving for SPBR: SPBR = (PCLKB / (2 * freq)) - 1 */
01055     const uint32_t divisor = k_rspi_spbr_divisor * freq_hz;
01056     const uint32_t spbr_val = (k_pclkb_hz / divisor) - k_rspi_spbr_offset;
01057
01058     /* Reject frequencies that are too low (would require SPBR > 255) */
01059     if (spbr_val > k_rspi_spbr_max) {
01060         rx_log_error(s_tag, "Requested SPI frequency too low; minimum achievable is ~117 kHz");
01061         return k_rx_err_invalid_arg;
01062     }
01063
01064     *spbr = (uint8_t)spbr_val;
01065
01066     return k_rx_ok;
01067 }
```

References [k_pclkb_hz](#), [k_rspi_max_freq_hz](#), [k_rspi_spbr_divisor](#), [k_rspi_spbr_max](#), [k_rspi_spbr_offset](#), and [s_tag](#).

Referenced by [internal_prepare_controller\(\)](#).

Here is the caller graph for this function:



8.107.4.2 internal`configure`cs`gpio()

```

rx_err_t internal_configure_cs_gpio (
    const rx_port_pin_t pin_config) [static]
```

Configure GPIO pin for chip select output.

Configures a GPIO pin as a digital output for use as an active-low SPI chip select signal. Extracts port and pin numbers from the type-safe `rx_port_pin_t` encoding, sets the pin direction to output via PDR, and drives the pin high (CS inactive) via PODR.

Precondition

`pin_config` must encode a valid port (0-J) and pin (0-7)

Port hardware must be available and not module-stopped

Postcondition

GPIO PDR bit set (output direction)
 GPIO PODR bit set (CS inactive / high)

Note

Not thread-safe. Caller must provide synchronization for port access.

See also

[rspi_init_controller\(\)](#) Calls this during controller initialization
[rx_port_from_pin\(\)](#) Extracts port number from `rx_port_pin_t`
[rx_pin_from_pin\(\)](#) Extracts pin number from `rx_port_pin_t`

Since

Version 1.0.0

Definition at line 1093 of file [rspi.c](#).

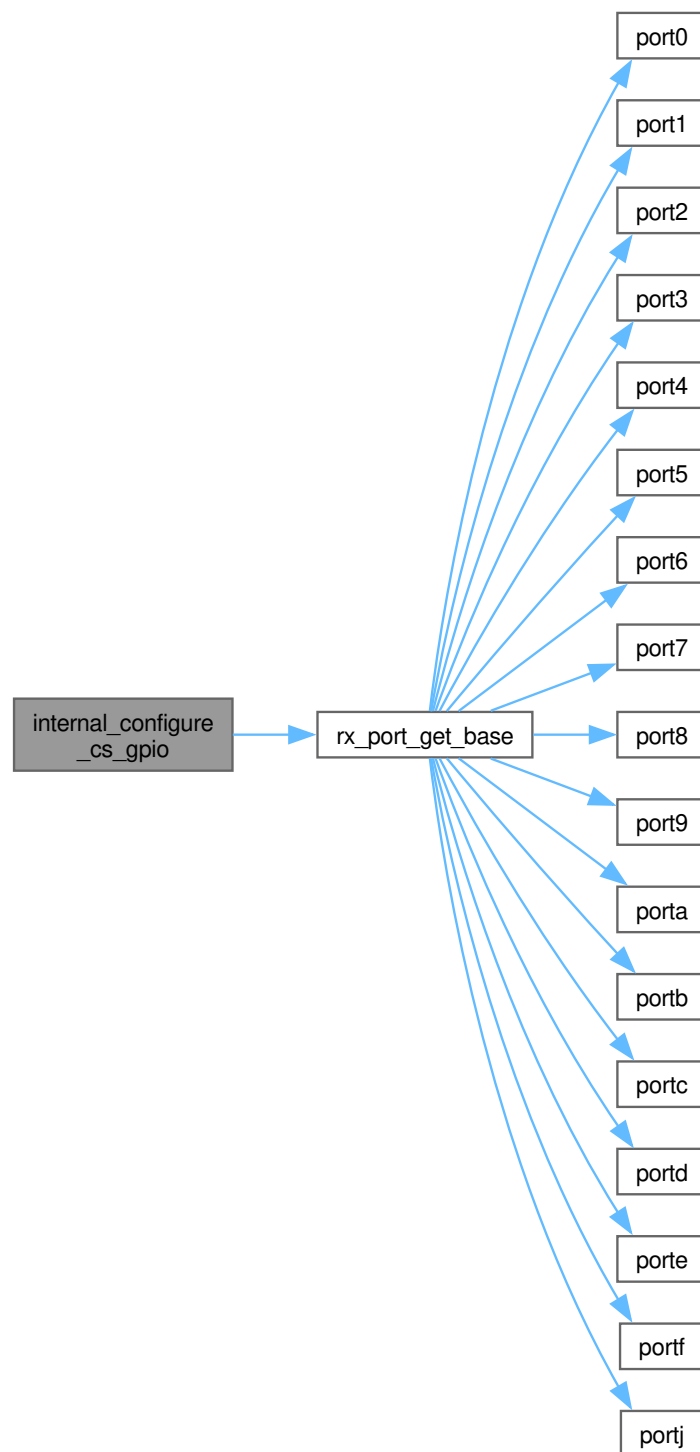
```

01094 {
01095     const uint8_t port = rx_port_from_pin(pin_config);
01096     const uint8_t pin = rx_pin_from_pin(pin_config);
01097
01098     volatile rx_port_regs_t* const port_regs = rx_port_get_base(port);
01099
01100     if (port_regs == nullptr) {
01101         rx_log_error(s_tag, "Invalid port number for CS pin");
01102         return k_rx_err_invalid_arg;
01103     }
01104
01105     /* Validate pin (lower bound check omitted - pin is uint8_t, k_rx_pin_min == 0,
01106      * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
01107     if (pin > k_rx_pin_max) {
01108         rx_log_error(s_tag, "Invalid pin number for CS pin (valid range 0-7)");
01109         return k_rx_err_invalid_arg;
01110     }
01111
01112     /* Configure as output */
01113     port_regs->pdr |= ((uint32_t)k_rspi_bit_set « pin);
01114
01115     /* Set high (CS inactive - active low) */
01116     port_regs->podr |= ((uint32_t)k_rspi_bit_set « pin);
01117
01118     return k_rx_ok;
01119 }
```

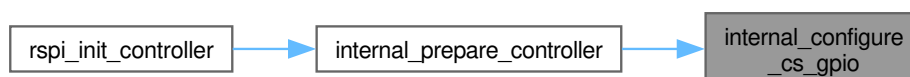
References [k_rspi_bit_set](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::podr](#), [rx_port_get_base\(\)](#), and [s_tag](#).

Referenced by [internal_prepare_controller\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.3 internal_configure_registers()

```

rx_err_t internal_configure_registers (
    const rspi\_channel\_t channel,

```

```
volatile rx_rspi_regs_t * rspi,
uint8_t spbr,
uint16_t spcmd,
const rspi_controller_config_t * config)  [static]
```

Configure RSPI hardware registers for controller mode.

Performs the complete RSPI hardware register configuration sequence for controller mode. Enables the module clock, disables SPI for safe register modification, programs bit rate, command, data control and pin control registers, then enables SPI in controller mode. Also stores the CS pin configuration and marks the channel as initialized. The register write order follows the RX72N Hardware Manual Section 38.3.6 initialization procedure.

Precondition

rspi must be a valid non-null pointer from [internal_get_rspi_base\(\)](#)
 spbr must be pre-calculated by [internal_calculate_spbr\(\)](#)
 spcmd must be pre-configured with CPOL/CPHA and data length bits

Postcondition

RSPI module clock enabled, registers configured, SPI enabled
 s_rspi_cs_config[channel] stores CS port and pin
 s_rspi_controller_initialized[channel] set to true

Note

Not thread-safe. Caller must provide synchronization.

Warning

PRCR register protection is temporarily unlocked during this call.

See also

[internal_prepare_controller\(\)](#) Provides rspi, spbr, and validates channel
[rspi_init_controller\(\)](#) Public API that orchestrates the full init flow

Since

Version 1.0.0

Definition at line 1253 of file [rspi.c](#).

```
01258 {
01259     /*
01260     * RSPI Controller Mode Register Write Sequence
01261     * =====
01262     * The register write order follows the RX72N Hardware Manual Section 38.3.6
01263     * (RSPI Initialization Procedure). The sequence is:
01264     *
01265     * 1. Enable module clock (MSTPCRB) - module must be powered before register access
01266     * 2. Disable SPI (SPCR.SPE=0) - registers must not be modified while SPI is active
01267     * 3. Configure SPBR - set bit rate divider while SPI is disabled
01268     * 4. Configure SPDCR and SPCMD0 - set data access width and mode while disabled
01269     * 5. Configure SPPCR - set pin control (no loopback) BEFORE enabling SPI
01270     * 6. Enable SPI (SPCR.SPE=1, MSTR=1) - must be the LAST register write to avoid
```



```

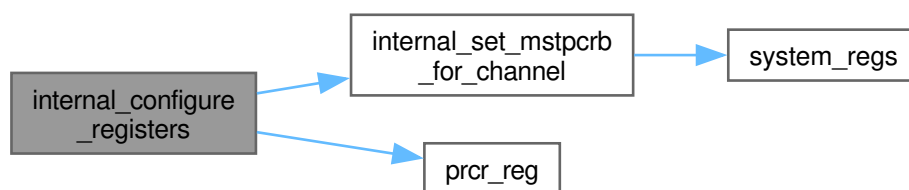
01271  *   partial configuration or clock glitches on the SPI bus
01272  */
01273
01274  /* Step 1: Enable RSPI module clock (clear module stop bit in MSTPCRB) */
01275  *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
01276  internal_set_mstpcrb_for_channel(channel, true);
01277  *prcr_reg() = k_rx_prcr_lock;
01278
01279  /* Step 2: Disable SPI before modifying configuration registers */
01280  rspi->spcr = k_rspi_spcr_disabled;
01281
01282  /* Step 3: Configure bit rate divider (freq = PCLKB / (2 * (SPBR + 1))) */
01283  rspi->spbr = spbr;
01284
01285  /* Step 4: Configure data access width (longword/32-bit mode) and command register */
01286  rspi->spdcr =
01287      (uint8_t)(k_rspi_bit_set « k_rspi_spdcr_splw_pos); /* Longword (32-bit) access mode */
01288  rspi->spcmd0 = spcmd;
01289
01290  /* Step 5: Configure pin control (no loopback) - MUST be written before SPCR.SPE
01291  * per HW manual to avoid glitches on COPI/CIPO lines during enable */
01292  rspi->sppcr = k_rspi_sppcr_no_loopback;
01293
01294  /* Step 6: Enable SPI in controller mode (MSTR=1, SPE=1) - MUST be the last register
01295  * write so all configuration is stable before the controller drives the bus */
01296  rspi->spcr = k_rspi_spcr_spe | k_rspi_spcr_mstr;
01297
01298  /* Store CS pin configuration */
01299  s_rspi_cs_config[channel].port = rx_port_from_pin(config->cs);
01300  s_rspi_cs_config[channel].pin = rx_pin_from_pin(config->cs);
01301
01302  /* Mark channel as initialized in controller mode */
01303  s_rspi_controller_initialized[channel] = true;
01304
01305  rx_log_debug(s_tag, "RSPI controller mode initialized");
01306
01307  return k_rx_ok;
01308 }

```

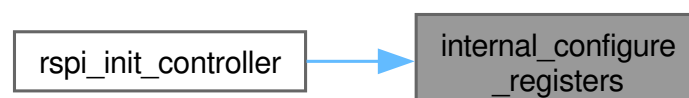
References [rspi_controller_config_t::cs](#), [internal_set_mstpcrb_for_channel\(\)](#), [k_rspi_bit_set](#), [k_rspi_spcr_disabled](#), [k_rspi_spcr_mstr](#), [k_rspi_spcr_spe](#), [k_rspi_spdcr_splw_pos](#), [k_rspi_sppcr_no_loopback](#), [prcr_reg\(\)](#), [s_rspi_controller_initialized](#), [s_rspi_cs_config](#), [s_tag](#), [rx_rspi_regs_t::spbr](#), [rx_rspi_regs_t::spcmd0](#), [rx_rspi_regs_t::spcr](#), [rx_rspi_regs_t::spdcr](#), and [rx_rspi_regs_t::sppcr](#).

Referenced by [rspi_init_controller\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.4 internal'configure'spcmd()

```
void internal_configure_spcmd (
    uint16_t * spcmd,
    const uint8_t spi_mode) [static]
```

Configure SPCMD register for SPI mode (CPOL/CPHA).

Sets CPOL and CPHA bits in an SPCMD register value based on SPI mode (0-3). The SPI mode encodes clock polarity (CPOL) and clock phase (CPHA) per the standard SPI specification. Bits are set via bitwise OR so pre-existing fields in the SPCMD value (such as data length) are preserved.

Mode	CPOL	CPHA	Clock Idle	Data Sampled
0	0	0	Low	Rising edge
1	0	1	Low	Falling edge
2	1	0	High	Falling edge
3	1	1	High	Rising edge

Precondition

spcmd must be a valid non-null pointer
 spi_mode must be in range [0, 3] (only bits 0-1 used)

Postcondition

spcmd CPOL/CPHA bits set according to spi_mode
 Other spcmd bits remain unchanged

Note

Not thread-safe. Caller must provide synchronization.

See also

[rspi_init_controller\(\)](#) Calls this during controller init
[rspi_init_peripheral\(\)](#) Performs equivalent inline configuration

Since

Version 1.0.0

Definition at line 1003 of file [rspi.c](#).

```
01004 {
01005     /* Rule 5: Pre-condition validation */
01006     RX_ASSERT(spcmd != nullptr, "spcmd pointer is nullptr");
01007     RX_ASSERT((spi_mode & ~(k_rspi_spcmd_cpol_mask | k_rspi_spcmd_cpha_mask)) == 0,
01008         "spi_mode contains invalid bits");
01009
01010     /* Configure SPI mode (CPOL and CPHA) */
01011     if (spi_mode & k_rspi_spcmd_cpol_mask) {
01012         *spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpol_pos); /* CPOL = 1 */
01013     }
01014     if (spi_mode & k_rspi_spcmd_cpha_mask) {
```

```

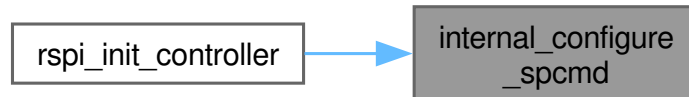
01015  *spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpha_pos); /* CPHA = 1 */
01016  }
01017  }

```

References [k_rspi_bit_set](#), [k_rspi_spcmd_cpha_mask](#), [k_rspi_spcmd_cpha_pos](#), [k_rspi_spcmd_cpol_mask](#), and [k_rspi_spcmd_cpol_pos](#).

Referenced by [rspi_init_controller\(\)](#).

Here is the caller graph for this function:



8.107.4.5 internal_controller_assert_cs_with_setup()

```

rx_err_t internal_controller_assert_cs_with_setup (
    const rspi\_channel\_t channel) [static]

```

Assert CS with setup delay.

Asserts the chip select line (drives low for active-low CS) and then applies a cycle-accurate setup time delay using NOP instructions. The setup delay ensures the SPI peripheral has sufficient time to recognize CS assertion before clock transitions begin. Delay is calibrated for ~300ns at 240 MHz core clock.

Precondition

Channel must be initialized via [rspi_init_controller\(\)](#)
[s_rspi_controller_initialized\[channel\]](#) must be true

Postcondition

CS GPIO pin driven low (active)
 Minimum setup delay of ~300ns has elapsed

Note

Not thread-safe. Caller must provide synchronization.

See also

[internal_controller_deassert_cs_with_hold\(\)](#) Companion deassert function
[internal_timing_delay\(\)](#) Provides cycle-accurate delay
[k_rspi_cs_setup_delay](#) Setup delay iteration count

Since

Version 1.0.0

CS setup time delay (~300ns). Inline assembly NOP is used to produce precise, cycle-accurate timing that C-level delays cannot guarantee. SPI peripherals typically require minimum CS setup time before SCLK transitions. Loop count `k_rspi_cs_setup_delay` is calibrated for RX72N at 240 MHz (4.17ns per cycle).

Definition at line 1437 of file [rspi.c](#).

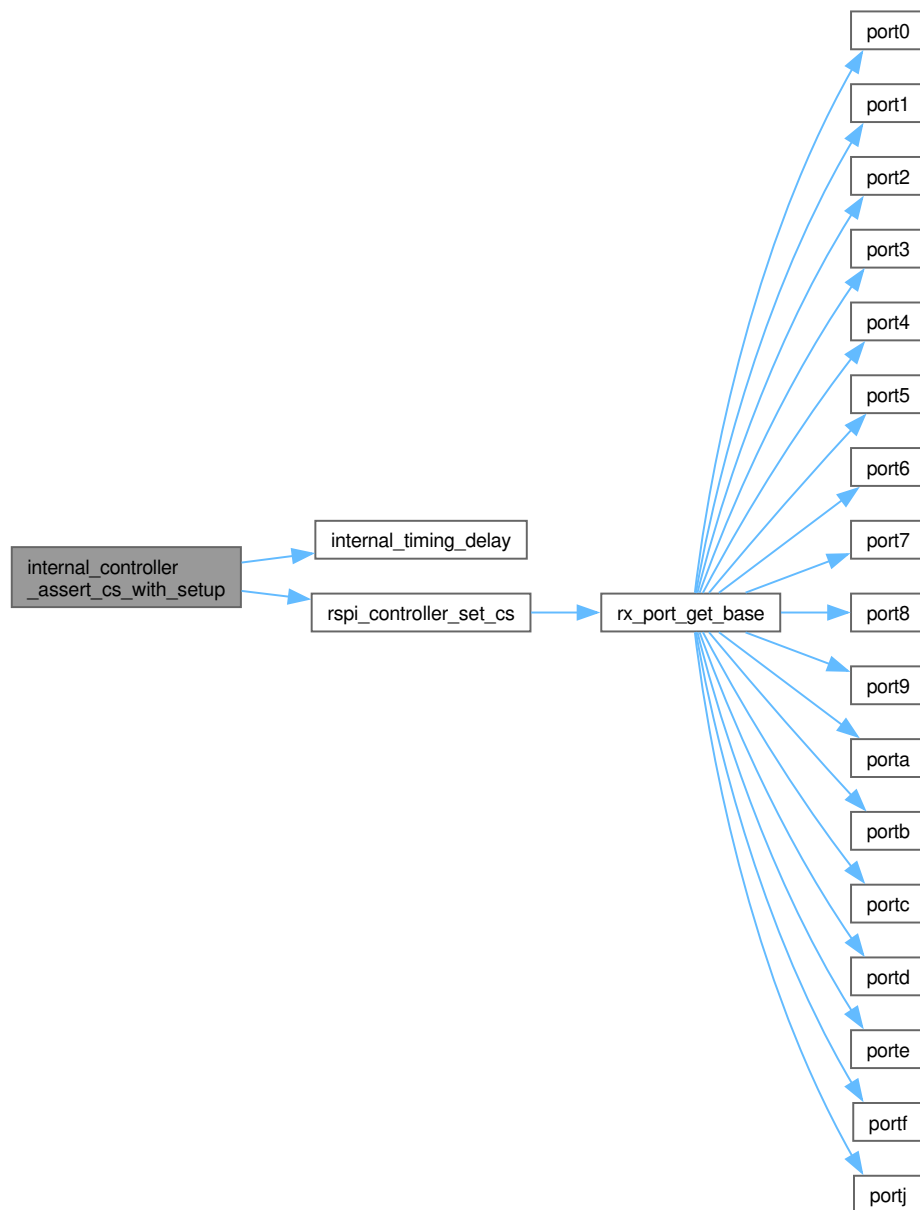
```

01438 {
01439     /* Rule 5: Pre-condition validation */
01440     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
01441              (channel == k_rspi_channel_2),
01442              "Invalid RSPI channel");
01443     RX_ASSERT(s_rspi_controller_initialized[channel], "RSPI controller channel not initialized");
01444
01445     /* Assert CS (active low) */
01446     const rx_err_t err = rspi_controller_set_cs(channel, true);
01447     if (err != k_rx_ok) {
01448         return err;
01449     }
01450
01451     /**
01452      * CS setup time delay (~300ns).
01453      * Inline assembly NOP is used to produce precise, cycle-accurate timing that
01454      * C-level delays cannot guarantee. SPI peripherals typically require minimum CS
01455      * setup time before SCLK transitions. Loop count k_rspi_cs_setup_delay is
01456      * calibrated for RX72N at 240 MHz (4.17ns per cycle).
01457      */
01458     internal_timing_delay(k_rspi_cs_setup_delay);
01459
01460     return k_rx_ok;
01461 }
```

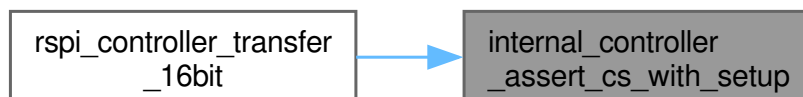
References [internal_timing_delay\(\)](#), [k_rspi_channel_0](#), [k_rspi_channel_1](#), [k_rspi_channel_2](#), [k_rspi_cs_setup_delay](#), [rspi_controller_set_cs\(\)](#), and [s_rspi_controller_initialized](#).

Referenced by [rspi_controller_transfer_16bit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.6 internal`controller`deassert`cs`with`hold()

```
rx_err_t internal_controller_deassert_cs_with_hold (
    const rspi\_channel\_t channel) [static]
```

Deassert CS with hold delay.

Applies a cycle-accurate hold time delay using NOP instructions, then deasserts the chip select line (drives high for active-low CS). The hold delay ensures the SPI peripheral has sufficient time to latch the final data bit before CS is released. Delay is calibrated for ~300ns at 240 MHz core clock.

Precondition

Channel must be initialized via [rspi_init_controller\(\)](#)
[s_rspi_controller_initialized\[channel\]](#) must be true

Postcondition

Minimum hold delay of $\sim 300\text{ns}$ has elapsed
 CS GPIO pin driven high (inactive)

Note

Not thread-safe. Caller must provide synchronization.

See also

[internal_controller_assert_cs_with_setup\(\)](#) Companion assert function
[internal_timing_delay\(\)](#) Provides cycle-accurate delay
[k_rspi_cs_hold_delay](#) Hold delay iteration count

Since

Version 1.0.0

CS hold time delay ($\sim 300\text{ns}$). Inline assembly NOP is used to produce precise, cycle-accurate timing that C-level delays cannot guarantee. SPI peripherals typically require minimum CS hold time after final SCLK transition before deassertion. Loop count [k_rspi_cs_hold_delay](#) is calibrated for RX72N at 240 MHz (4.17ns per cycle).

Definition at line 1488 of file [rspi.c](#).

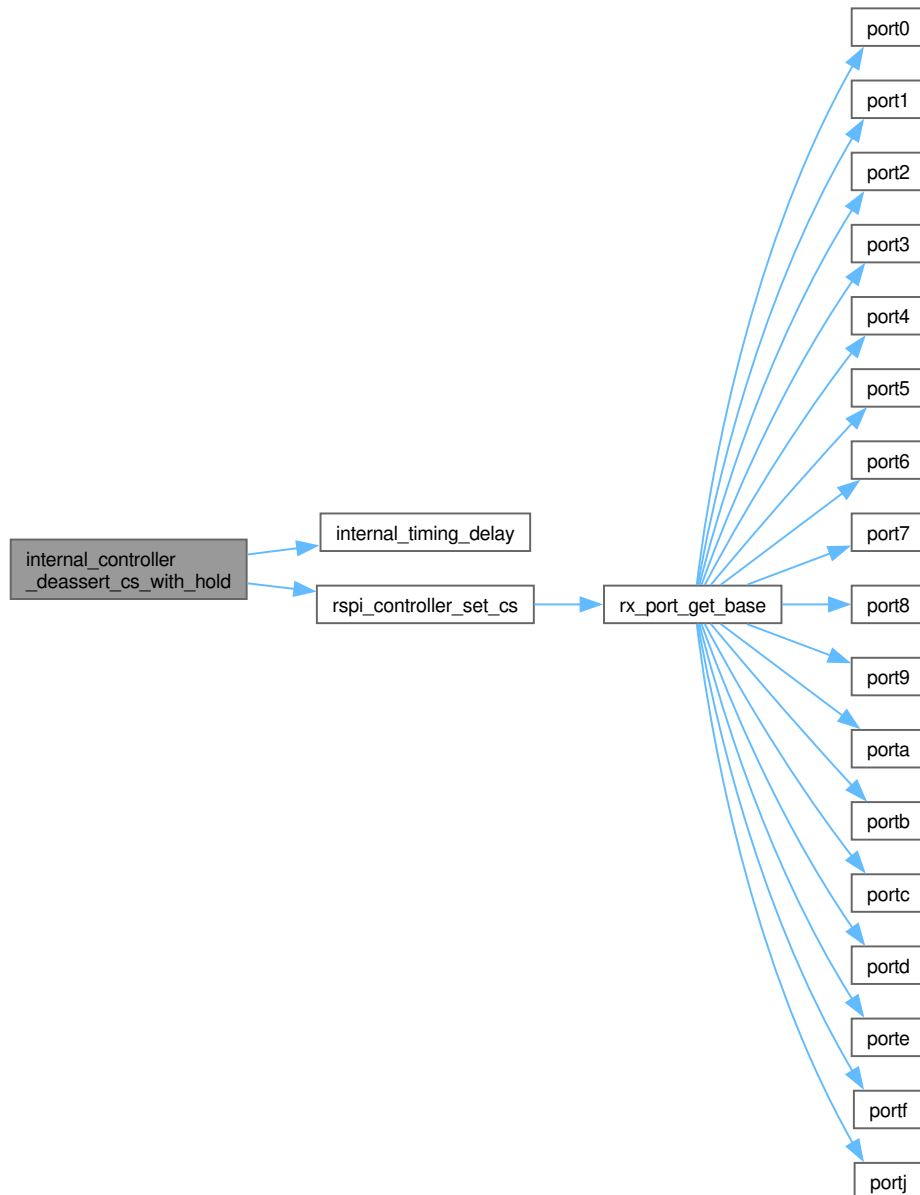
```

01489 {
01490     /* Rule 5: Pre-condition validation */
01491     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
01492             (channel == k_rspi_channel_2),
01493             "Invalid RSPI channel");
01494     RX_ASSERT(s_rspi_controller_initialized[channel], "RSPI controller channel not initialized");
01495
01496     /**
01497      * CS hold time delay (~300ns).
01498      * Inline assembly NOP is used to produce precise, cycle-accurate timing that
01499      * C-level delays cannot guarantee. SPI peripherals typically require minimum CS
01500      * hold time after final SCLK transition before deassertion. Loop count
01501      * k_rspi_cs_hold_delay is calibrated for RX72N at 240 MHz (4.17ns per cycle).
01502      */
01503     internal_timing_delay(k_rspi_cs_hold_delay);
01504
01505     /* Deassert CS (inactive high) */
01506     return rspi_controller_set_cs(channel, false);
01507 }
```

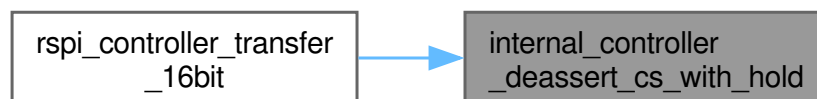
References [internal_timing_delay\(\)](#), [k_rspi_channel_0](#), [k_rspi_channel_1](#), [k_rspi_channel_2](#), [k_rspi_cs_hold_delay](#), [rspi_controller_set_cs\(\)](#), and [s_rspi_controller_initialized](#).

Referenced by [rspi_controller_transfer_16bit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.7 internal`controller`do`16bit`transfer()

```

rx_err_t internal_controller_do_16bit_transfer (
    volatile rx_rspi_regs_t * rspi,
    uint16_t tx_data,
    uint16_t * rx_data) [static]

```

Perform core 16-bit SPI transfer operation.

Executes a single full-duplex 16-bit SPI transfer without chip select management. Waits for the transmit buffer to become empty, writes 16-bit data to SPDR, waits for the receive buffer to fill, reads the 16-bit response from SPDR, and clears SPRF and OVRF status flags. This is the inner transfer primitive used by `rspi_controller_transfer_16bit()` after CS assertion and before CS deassertion.

Precondition

rspi must be a valid non-null pointer to initialized RSPI registers
 SPI must be enabled (SPCR.SPE=1) before calling

Postcondition

*rx_data contains the 16-bit value received during the transfer
 SPSR flags SPRF and OVRF are cleared

Note

Not thread-safe. Caller must provide synchronization.

See also

[rspi_controller_transfer_16bit\(\)](#) Public API that wraps this with CS handling
[internal_wait_tx_ready\(\)](#) TX buffer polling with timeout
[internal_wait_rx_ready\(\)](#) RX buffer polling with timeout

Since

Version 1.0.0

Definition at line 1535 of file [rspi.c](#).

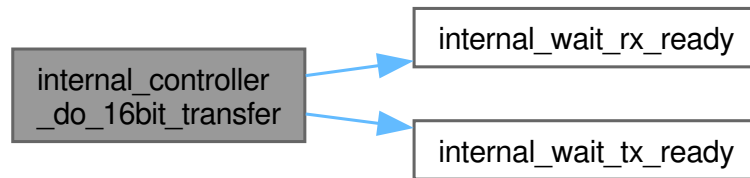
```

01538 {
01539     /* Rule 5: Pre-condition validation */
01540     RX_CHECK_NULL_PTR(rspi, s_tag, "RSPI register pointer is nullptr");
01541     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
01542
01543     /* Wait for transmit buffer empty */
01544     rx_err_t err = internal_wait_tx_ready(rspi);
01545     if (err != k_rx_ok) {
01546         rx_log_error(s_tag, "SPI controller transmit timeout");
01547         return err;
01548     }
01549
01550     /* Write 16-bit transmit data */
01551     rspi->spdr = tx_data;
01552
01553     /* Wait for receive buffer full */
01554     err = internal_wait_rx_ready(rspi);
01555     if (err != k_rx_ok) {
01556         rx_log_error(s_tag, "SPI controller receive timeout");
01557         return err;
01558     }
01559
01560     /* Read 16-bit receive data */
01561     *rx_data = (uint16_t)rspi->spdr;
01562
01563     /* Clear status flags */
01564     rspi->spsr &= (uint8_t) ~(uint8_t)(k_rspi_spsr_sprf | k_rspi_spsr_ovrf);
01565
01566     return k_rx_ok;
01567 }
```

References [internal_wait_rx_ready\(\)](#), [internal_wait_tx_ready\(\)](#), [k_rspi_spsr_ovrf](#), [k_rspi_spsr_sprf](#), [s_tag](#), [rx_rspi_regs_t::spdr](#), and [rx_rspi_regs_t::spsr](#).

Referenced by [rspi_controller_transfer_16bit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.8 internal`get`rspi`base()

```
volatile rx_rspi_regs_t * internal_get_rspi_base (
    const rspi_channel_t channel)  [static]
```

Get RSPI register base address from channel number.

Maps a channel index (0-2) to the corresponding RSPI peripheral register block via the inline accessor functions `rspi0()`/`rspi1()`/`rspi2()`. Returns nullptr for out-of-range channel numbers to allow callers to detect invalid input without risking a wild pointer dereference.

Precondition

- channel must be in range [0, k_rspi_max_channels)
- RSPI hardware must be present (RX72N 144-pin package)

Postcondition

- Returned pointer (if non-null) is aligned and points to valid HW registers
- No side effects on hardware state

Note

- Not thread-safe. Caller must provide synchronization if needed.

See also

- `rspi0()` Channel 0 register accessor
- `rspi1()` Channel 1 register accessor
- `rspi2()` Channel 2 register accessor

Since

Version 1.0.0

Definition at line 399 of file [rspi.c](#).

```

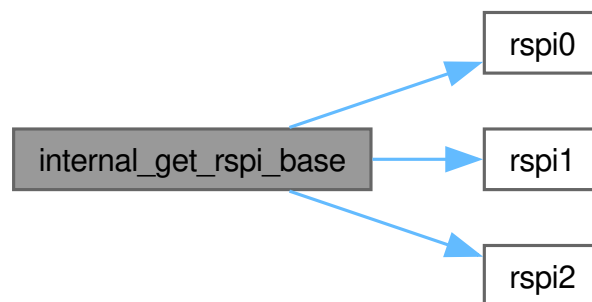
00400 {
00401     switch (channel) {
00402         case k_rspi_channel_0: {
00403             return rspi0();
00404         }
00405         case k_rspi_channel_1: {
00406             return rspi1();
00407         }
00408         case k_rspi_channel_2: {
00409             return rspi2();
00410         }
00411         default: {
00412             return nullptr;
00413         }
00414     }
00415 }

```

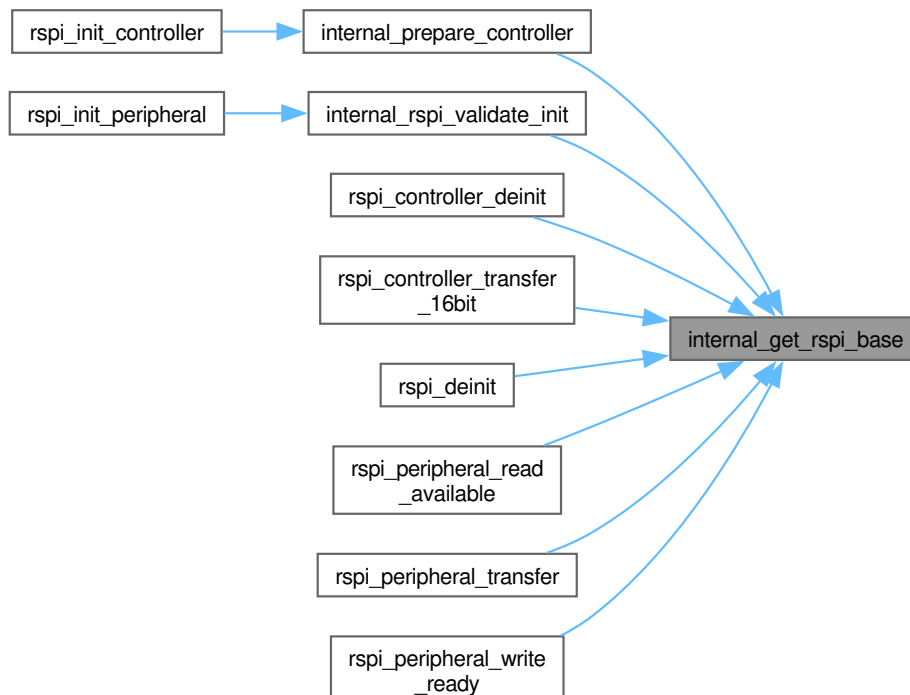
References [k_rspi_channel_0](#), [k_rspi_channel_1](#), [k_rspi_channel_2](#), [rspi0\(\)](#), [rspi1\(\)](#), and [rspi2\(\)](#).

Referenced by [internal_prepare_controller\(\)](#), [internal_rspi_validate_init\(\)](#), [rspi_controller_deinit\(\)](#), [rspi_controller_transfer_16bit\(\)](#), [rspi_deinit\(\)](#), [rspi_peripheral_read_available\(\)](#), [rspi_peripheral_transfer\(\)](#), and [rspi_peripheral_write_ready\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.9 `internal_prepare_controller()`

```
rx_err_t internal_prepare_controller (
    const rspi\_channel\_t channel,
    const rspi\_controller\_config\_t * config,
    volatile rx\_rspi\_regs\_t ** out_rspi,
    uint8_t * out_spbr) [static]
```

Prepare controller hardware resources.

Calculates the SPBR register value for the requested clock frequency, validates the RSPI register base address for the given channel, and configures the CS GPIO pin as an active-low output. Operations are ordered so that a failure in SPBR calculation avoids unnecessary GPIO configuration, and RSPI base validation occurs before GPIO setup to avoid leaving GPIO configured if the channel is invalid.

Precondition

config must be validated by [internal_validate_controller_args\(\)](#) first
channel must be in range [0, k_rspi_max_channels)

Postcondition

*out_rspi points to valid RSPI register base on success
*out_spbr contains calculated SPBR value on success
CS GPIO pin configured as output, driven high (inactive) on success

Note

Not thread-safe. Caller must provide synchronization.

See also

[internal_validate_controller_args\(\)](#) Must be called before this function
[internal_configure_registers\(\)](#) Called after this to program hardware
[internal_calculate_spbr\(\)](#) Called internally for frequency calculation

Since

Version 1.0.0

Definition at line 1197 of file [rspi.c](#).

```
01201 {
01202     /* Calculate SPBR for requested frequency */
01203     rx_err_t err = internal_calculate_spbr(config->freq_hz, out_spbr);
01204     if (err != k_rx_ok) {
01205         return err;
01206     }
01207
01208     /* Get RSPI base - validate before configuring GPIO to avoid leaving GPIO
01209      * configured if RSPI base lookup fails */
01210     *out_rspi = internal_get_rspi_base(channel);
01211     if (*out_rspi == nullptr) {
01212         return k_rx_err_invalid_arg;
01213     }
01214
01215     /* Configure CS GPIO pin (after RSPI base validation) */
01216     err = internal_configure_cs_gpio(config->cs);
```

```

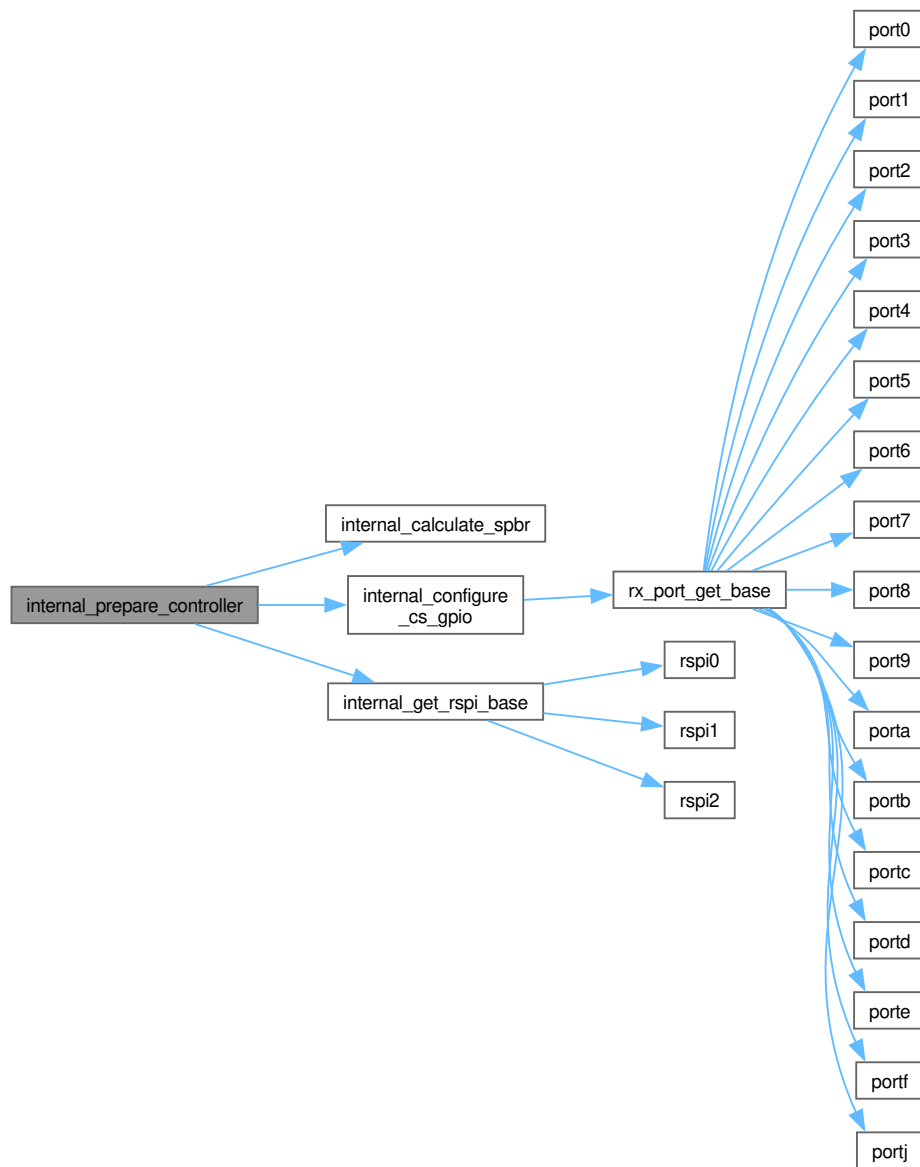
01217 if (err != k_rx_ok) {
01218     return err;
01219 }
01220
01221 return k_rx_ok;
01222 }

```

References [rspi_controller_config_t::cs](#), [rspi_controller_config_t::freq_hz](#), [internal_calculate_spbr\(\)](#), [internal_configure_cs_gpio\(\)](#), and [internal_get_rspi_base\(\)](#).

Referenced by [rspi_init_controller\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.10 internal`rspi`build`spcmd()

```
uint16_t internal_rspi_build_spcmd (
    volatile rx_rspi_regs_t *const rspi,
    const rspi_config_t *const config) [static]
```

Build SPCMD0 value and program SPDCR for the requested data width.

Encodes CPOL/CPHA bits from spi_mode and selects 8-bit or 16-bit data length. SPDCR is set to longword access for 16-bit transfers, byte access for 8-bit, matching the SPCMD0 SPL field.

Parameters

in	rspi	Peripheral base (non-null)
in	config	Validated init config

Returns

uint16_t Encoded SPCMD0 value

Precondition

SPCR.SPE=0 (caller has already disabled SPI)

Postcondition

SPDCR programmed; SPCMD0 not yet written

Since

Version 1.0.0

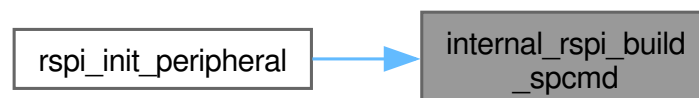
Definition at line 556 of file [rspi.c](#).

```
00558 {
00559     uint16_t spcmd = k_rspi_spcmd_init;
00560
00561     if (config->spi_mode & k_rspi_spcmd_cpol_mask) {
00562         spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpol_pos); /* CPOL = 1 */
00563     }
00564     if (config->spi_mode & k_rspi_spcmd_cpha_mask) {
00565         spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpha_pos); /* CPHA = 1 */
00566     }
00567
00568     if (config->use_16bit) {
00569         spcmd |= (k_rspi_spcmd_16bit « k_rspi_spcmd_spl_shift);
00570         rspi->spdcr = (uint8_t)(k_rspi_bit_set « k_rspi_spdcr_splw_pos); /* longword */
00571     } else {
00572         spcmd |= (k_rspi_spcmd_8bit « k_rspi_spcmd_spl_shift);
00573         rspi->spdcr = k_rspi_spdcr_byte_mode;
00574     }
00575
00576     return spcmd;
00577 }
```

References [k_rspi_bit_set](#), [k_rspi_spcmd_16bit](#), [k_rspi_spcmd_8bit](#), [k_rspi_spcmd_cpha_mask](#), [k_rspi_spcmd_cpha_pos](#), [k_rspi_spcmd_cpol_mask](#), [k_rspi_spcmd_cpol_pos](#), [k_rspi_spcmd_init](#), [k_rspi_spcmd_spl_shift](#), [k_rspi_spdcr_byte_mode](#), [k_rspi_spdcr_splw_pos](#), [rx_rspi_regs_t::spdcr](#), [rspi_config_t::spi_mode](#), and [rspi_config_t::use_16bit](#).

Referenced by [rspi_init_peripheral\(\)](#).

Here is the caller graph for this function:



8.107.4.11 `internal_rspi_release_module_stop()`

```
void internal_rspi_release_module_stop (
    const rspi\_channel\_t channel)  [static]
```

Release the MSTPCRB module-stop bit for the requested channel.

Wraps the protected-register PRCR unlock/lock around the MSTPCRB write that clears the channel's clock gate.

Parameters

in	channel	Validated channel index
----	---------	-------------------------

Precondition

`channel < k_rspi_max_channels`

Postcondition

Channel module clock running

Since

Version 1.0.0

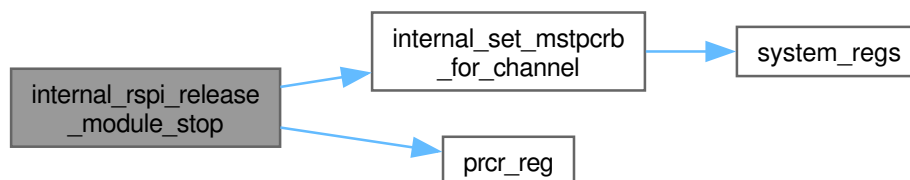
Definition at line 594 of file [rspi.c](#).

```
00595 {
00596   *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00597   internal_set_mstpcrb_for_channel(channel, true);
00598   *prcr_reg() = k_rx_prcr_lock;
00599 }
```

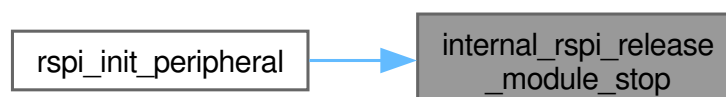
References [internal_set_mstpcrb_for_channel\(\)](#), and [prcr_reg\(\)](#).

Referenced by [rspi_init_peripheral\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.12 internal`rspi`validate`init()

```
rx_err_t internal_rspi_validate_init (
    const rspi_channel_t channel,
    const rspi_config_t *const config,
    volatile rx_rspi_regs_t ** rspi_out)  [static]
```

Initialize RSPI channel in peripheral mode.

Configures the specified RSPI channel as a SPI peripheral to communicate with the Raspberry Pi 5 (acting as SPI Controller). Enables the peripheral in the configured SPI mode (0-3) with optional 8-bit or 16-bit data frames. Uses register protection unlock/lock for module stop control.

Precondition

config pointer must be non-NULL
channel must be in range [0, k_rspi_max_channels)
config->spi_mode must be in range [0, 3]

Postcondition

If successful:

- RSPI module is enabled (module stop bit cleared via register protection)
- SPI mode (CPOL/CPHA) is configured per config->spi_mode
- Data frame length (8-bit or 16-bit) is set per config->use_16bit
- SPI is enabled in peripheral mode (SPCR.MSTR = 0)
- s_rspi_channel_initialized[channel] is set to true
- Peripheral is ready to receive transfers from SPI controller

Validate rspi_init_peripheral inputs and resolve peripheral base

Parameters

in	channel	Channel index
in	config	Caller's config (must be non-null)
out	rspi_out	On success, set to peripheral base

Returns

rx_err_t Error code

Return values

k_rx_ok	All checks pass
k_rx_err_null_ptr	config is nullptr
k_rx_err_invalid_arg	Bounds violation or unknown channel

Precondition

`rspi_out` non-null

Since

Version 1.0.0

Definition at line 515 of file [rspi.c](#).

```

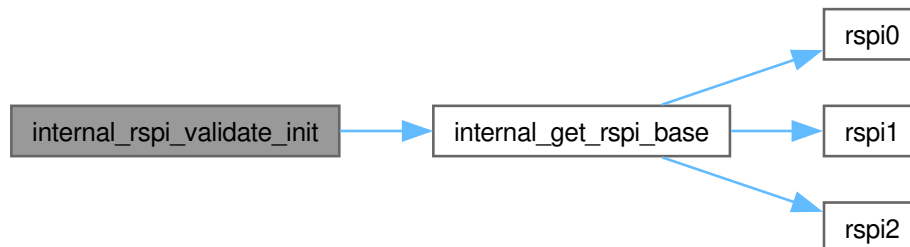
00518 {
00519     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00520
00521     if ((uint8_t)channel >= k_rspi_max_channels) {
00522         rx_log_error(s_tag, "Invalid RSPI channel");
00523         return k_rx_err_invalid_arg;
00524     }
00525     /* Lower bound check omitted: spi_mode is uint8_t and k_rspi_mode_min == 0
00526      * so spi_mode >= k_rspi_mode_min is always true (-Wtype-limits). */
00527     if ((uint8_t)config->spi_mode > k_rspi_mode_max) {
00528         rx_log_error(s_tag, "Invalid SPI mode (must be 0-3)");
00529         return k_rx_err_invalid_arg;
00530     }
00531     *rspi_out = internal_get_rspi_base(channel);
00532     if (*rspi_out == nullptr) {
00533         return k_rx_err_invalid_arg;
00534     }
00535     return k_rx_ok;
00536 }

```

References [internal_get_rspi_base\(\)](#), [k_rspi_max_channels](#), [k_rspi_mode_max](#), [s_tag](#), and [rspi_config_t::spi_mode](#).

Referenced by [rspi_init_peripheral\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.13 `internal_set_mstpcrb_for_channel()`

```
void internal_set_mstpcrb_for_channel (
    const rspi\_channel\_t channel,
    const bool enable)    [static]
```

Set module stop bit for an RSPI channel.

Controls the Module Stop Control Register bits that gate the clock to each RSPI channel. RSPI0 and RSPI1 are in MSTPCRB (bits 17 and 16); RSPI2 is in MSTPCRC (bit 22). Clearing a bit enables the module (starts the clock); setting it stops the module (reduces power consumption).

Precondition

[system_regs\(\)](#) must return a valid pointer
channel must be one of `k_rspi_channel_0`, `k_rspi_channel_1`, `k_rspi_channel_2`

Postcondition

MSTPCRB (channels 0/1) or MSTPCRC (channel 2) updated appropriately
Module clock enabled or disabled for the specified channel

Note

Not thread-safe. Caller must hold MSTPCR protection if applicable.

Warning

Disabling a module while it is actively transferring data will cause undefined hardware behavior.

See also

[internal_get_rspi_base\(\)](#) Maps channel to register base

Since

Version 1.0.0

Definition at line 440 of file [rspi.c](#).

```
00441 {
00442     RX_ASSERT(system_regs() != nullptr, "system_regs is nullptr");
00443     static_assert(k_rspi_bit_set != 0, "RSPI bit constant must be non-zero");
00444     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
00445         (channel == k_rspi_channel_2),
00446         "Invalid RSPI channel");
00447
00448     /* RSPI2 stop bit is in MSTPCRC (bit 22), not MSTPCRB (HW Manual p.410) */
00449     if (channel == k_rspi_channel_2) {
00450         const uint32_t mask = (k_rspi_bit_set « k_rspi_mstpc_rspi2);
00451         if (enable) {
00452             system_regs()->mstpcrc &= ~mask;
00453         } else {
00454             system_regs()->mstpcrc |= mask;
00455         }
00456         return;
00457     }
00458
00459     uint32_t mask = k_rspi_zero_u32;
```

```

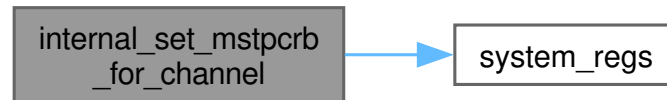
00460 if (channel == k_rspi_channel_0) {
00461     mask = (k_rspi_bit_set « k_rspi_mstpb_rspi0);
00462 } else if (channel == k_rspi_channel_1) {
00463     mask = (k_rspi_bit_set « k_rspi_mstpb_rspi1);
00464 }
00465 if (enable) {
00466     system_regs()->mstpcrb &= ~mask;
00467 } else {
00468     system_regs()->mstpcrb |= mask;
00469 }
00470 }

```

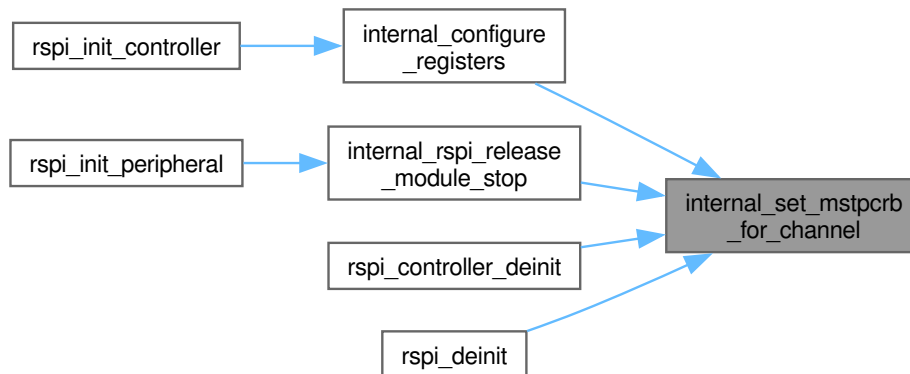
References [k_rspi_bit_set](#), [k_rspi_channel_0](#), [k_rspi_channel_1](#), [k_rspi_channel_2](#), [k_rspi_mstpb_rspi0](#), [k_rspi_mstpb_rspi1](#), [k_rspi_mstpc_rspi2](#), [k_rspi_zero_u32](#), [rx_system_regs_t::mstpcrb](#), [rx_system_regs_t::mstpcrb](#) and [system_regs\(\)](#).

Referenced by [internal_configure_registers\(\)](#), [internal_rspi_release_module_stop\(\)](#), [rspi_controller_deinit\(\)](#), and [rspi_deinit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.14 internal_timing_delay()

```

void internal_timing_delay (
    const uint16_t cycles) [static]

```

Cycle-accurate timing delay using inline NOP instructions.

Produces precise delays using inline assembly NOP operations for SPI timing requirements (CS setup/hold times). Each NOP cycle is 4.17 ns at 240 MHz core clock.

Precondition

- cycles must be non-zero
- cycles must not exceed `k_rspi_max_delay_cycles`

Postcondition

Minimum delay of (cycles * 4.17 ns) has elapsed
 No hardware state modified

Note

Timing is cycle-accurate on RX72N @ 240 MHz (4.17 ns per cycle)
 Using inline ASM ensures compiler does not optimize away the loop

Warning

Only accurate when compiled with known optimization level (-O2)

See also

[internal_controller_assert_cs_with_setup\(\)](#) Uses for CS setup time
[internal_controller_deassert_cs_with_hold\(\)](#) Uses for CS hold time

Since

Version 1.0.0

Definition at line 954 of file [rspi.c](#).

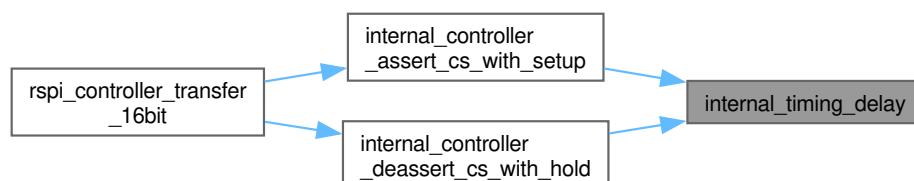
```

00955 {
00956  /* Rule 5: Pre-condition validation */
00957  RX_ASSERT(cycles != 0, "Timing delay cycles cannot be zero");
00958  RX_ASSERT(cycles <= k_rspi_max_delay_cycles, "Timing delay cycles exceeds sanity limit");
00959  /*
00960  * NASA Rule 2: Loop bounded by k_rspi_max_delay_cycles (compile-time constant)
00961  * Early break when delay >= cycles to execute only the required iterations.
00962  */
00963  volatile uint32_t delay;
00964  for (delay = k_rspi_loop_start; delay < k_rspi_max_delay_cycles; delay++) {
00965    if (delay >= cycles) {
00966      break;
00967    }
00968  }
00969  __asm__ volatile("nop");
00970 }
00971 }
```

References [k_rspi_loop_start](#), and [k_rspi_max_delay_cycles](#).

Referenced by [internal_controller_assert_cs_with_setup\(\)](#), and [internal_controller_deassert_cs_with_hold\(\)](#).

Here is the caller graph for this function:



8.107.4.15 `internal_validate_controller_args()`

```
rx_err_t internal_validate_controller_args (
    const rspi\_channel\_t channel,
    const rspi\_controller\_config\_t * config) [static]
```

Validate controller initialization arguments.

Performs comprehensive input validation for [rspi_init_controller\(\)](#) before any hardware modification occurs. Checks null pointer, channel range, duplicate initialization, and SPI mode range. Centralizes validation logic to keep the public API function concise.

Precondition

config may be nullptr (will be detected and rejected)
 channel may be out of range (will be detected and rejected)

Postcondition

No hardware state modified (validation only)
[s_rspi_controller_initialized\[\]](#) not modified

Note

Not thread-safe. Caller must provide synchronization.

See also

[rspi_init_controller\(\)](#) Calls this as first step of initialization
[internal_prepare_controller\(\)](#) Called after this validation passes

Since

Version 1.0.0

Definition at line 1144 of file [rspi.c](#).

```
01146 {
01147     RX_CHECK_NULL_PTR(config, s_tag, "Controller config pointer is nullptr");
01148
01149     /* Validate channel */
01150     if ((uint8_t)channel >= k_rspi_max_channels) {
01151         rx_log_error(s_tag, "Invalid RSPI channel");
01152         return k_rx_err_invalid_arg;
01153     }
01154
01155     /* Check if channel is already initialized */
01156     if (s_rspi_controller_initialized[channel]) {
01157         rx_log_error(s_tag, "RSPI channel already initialized");
01158         return k_rx_err_invalid_state;
01159     }
01160
01161     /* Validate mode (0-3) */
01162     if ((uint8_t)config->spi_mode > k_rspi_mode_max) {
01163         rx_log_error(s_tag, "Invalid SPI mode (must be 0-3)");
01164         return k_rx_err_invalid_arg;
01165     }
01166
01167     return k_rx_ok;
01168 }
```

References [k_rspi_max_channels](#), [k_rspi_mode_max](#), [s_rspi_controller_initialized](#), [s_tag](#), and [rspi_controller_config_t::spi_mode](#).

Referenced by [rspi_init_controller\(\)](#).

Here is the caller graph for this function:



8.107.4.16 `internal_wait_rx_ready()`

```
rx_err_t internal_wait_rx_ready (
    volatile rx\_rspi\_regs\_t * rspi) [static]
```

Wait for receive buffer to become full (SPRF flag).

Polls the SPSR.SPRF (SPI Receive Buffer Full Flag) in a busy-wait loop with a configurable timeout. When SPRF is set, received data is available in the SPDR register and can be read. The timeout prevents infinite loops if the SPI peripheral is stuck or no clock is being generated.

Precondition

- rspi must be a valid non-null pointer to initialized RSPI registers
- A transfer must be in progress or complete for SPRF to set

Postcondition

- No hardware state modified (read-only polling of SPSR)
- Timeout counter decremented to reflect remaining iterations

Note

- Not thread-safe. Caller must provide synchronization.

Warning

- Busy-wait loop; do not call from time-critical ISR context.

See also

- [internal_wait_tx_ready\(\)](#) Companion function for transmit buffer
- [k_rspi_timeout_us](#) Timeout constant

Since

- Version 1.0.0

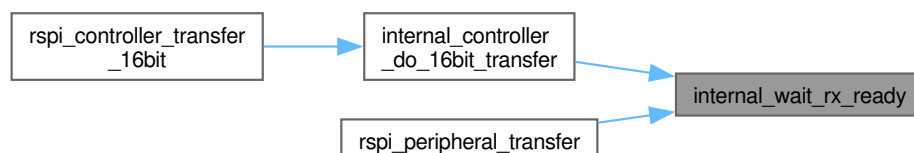
Definition at line 704 of file [rspi.c](#).

```
00705 {
00706  /* Pre-condition 1: RSPI register pointer must be valid */
00707  RX_ASSERT(rspi != nullptr, "RSPI register pointer is nullptr");
00708
00709  /* Pre-condition 2: Timeout constant must be non-zero (compile-time check) */
00710  static_assert(k_rspi_timeout_us > k_rspi_timeout_zero, "Timeout constant must be non-zero");
00711  uint32_t timeout = k_rspi_timeout_us;
00712
00713  while (!(rspi->spsr & k_rspi_spsr_sprf) && timeout > k_rspi_timeout_zero) {
00714      timeout--;
00715  }
00716
00717  if (timeout == k_rspi_timeout_zero) {
00718      return k_rx_err_timeout;
00719  }
00720
00721  return k_rx_ok;
00722 }
```

References [k_rspi_spsr_sprf](#), [k_rspi_timeout_us](#), [k_rspi_timeout_zero](#), and [rx_rspi_regs_t::spsr](#).

Referenced by [internal_controller_do_16bit_transfer\(\)](#), and [rspi_peripheral_transfer\(\)](#).

Here is the caller graph for this function:



8.107.4.17 `internal_wait_tx_ready()`

```
rx_err_t internal_wait_tx_ready (
    volatile rx_rspi_regs_t * rspi) [static]
```

Wait for transmit buffer to become empty (SPTEF flag).

Polls the SPSR.SPTEF (SPI Transmit Buffer Empty Flag) in a busy-wait loop with a configurable timeout. When SPTEF is set, the transmit buffer is empty and ready to accept new data via SPDR. The timeout prevents infinite loops if the SPI peripheral is stuck.

Precondition

`rspi` must be a valid non-null pointer to initialized RSPI registers
 RSPI module clock must be enabled (MSTPCRB bit cleared)

Postcondition

No hardware state modified (read-only polling of SPSR)
 Timeout counter decremented to reflect remaining iterations

Note

Not thread-safe. Caller must provide synchronization.

Warning

Busy-wait loop; do not call from time-critical ISR context.

See also

[internal_wait_rx_ready\(\)](#) Companion function for receive buffer
[k_rspi_timeout_us](#) Timeout constant

Since

Version 1.0.0

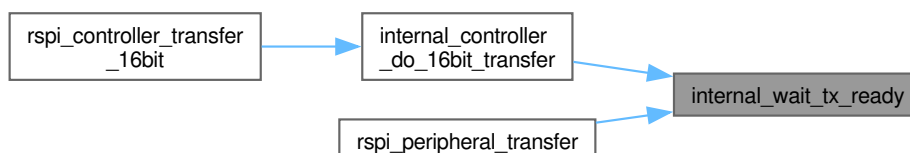
Definition at line 660 of file [rspi.c](#).

```
00661 {
00662  /* Pre-condition 1: RSPI register pointer must be valid */
00663  RX_ASSERT(rspi != nullptr, "RSPI register pointer is nullptr");
00664
00665  /* Pre-condition 2: Timeout constant must be non-zero (compile-time check) */
00666  static_assert(k_rspi_timeout_us > k_rspi_timeout_zero, "Timeout constant must be non-zero");
00667  uint32_t timeout = k_rspi_timeout_us;
00668
00669  while (!(rspi->spsr & k_rspi_spsr_sptef) && timeout > k_rspi_timeout_zero) {
00670      timeout--;
00671  }
00672
00673  if (timeout == k_rspi_timeout_zero) {
00674      return k_rx_err_timeout;
00675  }
00676
00677  return k_rx_ok;
00678 }
```

References [k_rspi_spsr_sptef](#), [k_rspi_timeout_us](#), [k_rspi_timeout_zero](#), and [rx_rspi_regs_t::spsr](#).

Referenced by [internal_controller_do_16bit_transfer\(\)](#), and [rspi_peripheral_transfer\(\)](#).

Here is the caller graph for this function:



8.107.4.18 `rspi_controller_deinit()`

```
rx_err_t rspi_controller_deinit (
    const rspi\_channel\_t channel)  [nodiscard]
```

Deinitialize RSPI controller mode.

Deinitialize and disable an RSPI controller-mode channel.

Disables the RSPI controller, deasserts chip select, clears configuration, and stops the module to reduce power consumption.

Precondition

Should be called after [rspi_init_controller\(\)](#) to properly clean up

Postcondition

If successful:

- CS is deasserted via `rspi_controller_set_cs(channel, false)`
- RSPI is disabled (`rspi->spr = k_rspi_spr_disabled`)
- Module clock is stopped via [internal_set_mstpcrb_for_channel\(\)](#)
- `s_rspi_cs_config[channel]` is cleared (`port=0`, `pin=0`)
- `s_rspi_controller_initialized[channel] = false`
- Register protection is unlocked/locked via `*prcr_reg()`

Note

Thread-safe only if caller ensures no concurrent access to the same channel

Definition at line 1656 of file [rspi.c](#).

```
01657 {
01658     /* Validate channel */
01659     if ((uint8_t)channel >= k_rspi_max_channels) {
01660         rx_log_error(s_tag, "Invalid RSPI channel");
01661         return k_rx_err_invalid_arg;
01662     }
01663
01664     /* Get RSPI base */
01665     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01666     if (rspi == nullptr) {
01667         return k_rx_err_invalid_arg;
01668     }
01669
01670     /*
01671     * RSPI Controller Deinitialization Sequence
01672     * =====
01673     * The deinit order ensures safe shutdown without bus glitches:
01674     * 1. Deassert CS (drive high) - release peripheral before disabling SPI
01675     * 2. Disable SPI (SPCR.SPE=0) - stop controller clock generation
01676     * 3. Disable module clock (MSTPCRB) - gate clock only after SPI is idle
01677     * 4. Clear software state - CS config and init flag after hardware is off
01678     */
01679
01680     /* Step 1: Deassert CS to release any connected peripheral before shutdown */
01681     if (s_rspi_controller_initialized[channel]) {
01682         (void)rspi_controller_set_cs(channel, false);
01683     }
01684
01685     /* Step 2: Disable SPI to stop controller clock generation */
01686     rspi->spr = k_rspi_spr_disabled;
01687
01688     /* Step 3: Disable RSPI module clock (set module stop bit) - must be done
01689     * after SPI is disabled to avoid undefined behavior from clock gating
```

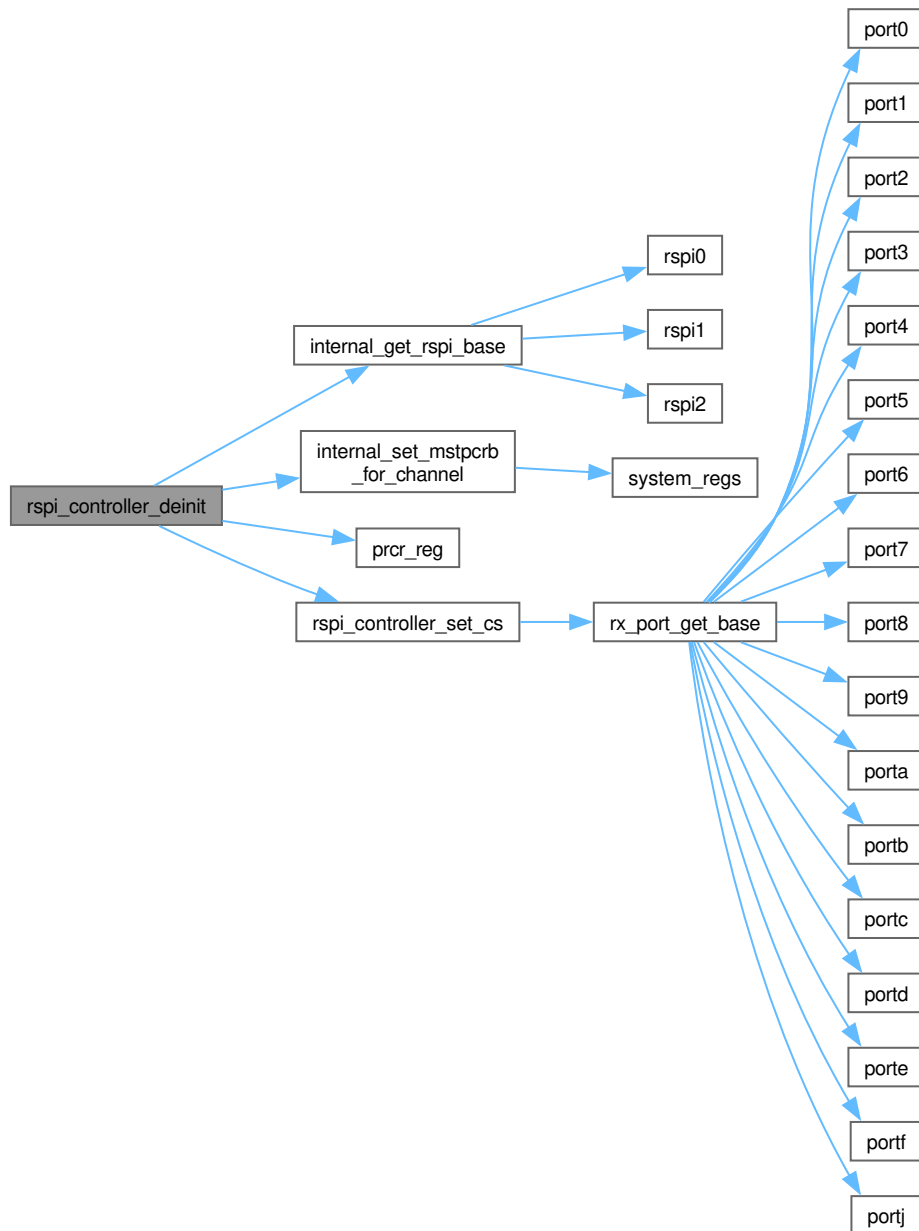
```

01690  * during an active transfer */
01691  *prcr_reg() = k_rx_pcr_unlock_prc1_prc3;
01692  internal_set_mstpcrb_for_channel(channel, false);
01693  *prcr_reg() = k_rx_pcr_lock;
01694
01695  /* Step 4: Clear software state after hardware is safely off */
01696  s_rspi_cs_config[channel].port = k_rspi_cs_default_port;
01697  s_rspi_cs_config[channel].pin = k_rspi_cs_default_pin;
01698
01699  s_rspi_controller_initialized[channel] = false;
01700
01701  rx_log_debug(s_tag, "RSPi controller deinitialized");
01702
01703  return k_rx_ok;
01704 }

```

References [internal_get_rspi_base\(\)](#), [internal_set_mstpcrb_for_channel\(\)](#), [k_rspi_cs_default_pin](#), [k_rspi_cs_default_port](#), [k_rspi_max_channels](#), [k_rspi_spcr_disabled](#), [prcr_reg\(\)](#), [rspi_controller_set_cs\(\)](#), [s_rspi_controller_initialized](#), [s_rspi_cs_config](#), [s_tag](#), and [rx_rspi_regs_t::spcr](#).

Here is the call graph for this function:



8.107.4.19 `rspi_controller_set_cs()`

```
rx_err_t rspi_controller_set_cs (
    const rspi\_channel\_t channel,
    const bool active) \[nodiscard\]
```

Set chip select line state for RSPI controller channel.

Set chip select line state for an RSPI controller-mode channel.

Controls the GPIO chip select pin for the specified RSPI channel operating in controller mode. CS is active-low (asserted low, deasserted high).

Precondition

```
Channel must be initialized via rspi\_init\_controller\(\)
channel < k_rspi_max_channels
s_rspi_controller_initialized[channel] == true
```

Postcondition

```
GPIO port_regs->podr is modified to assert/deassert CS pin
```

Note

```
Reads s_rspi_cs_config[channel] for port and pin configuration
```

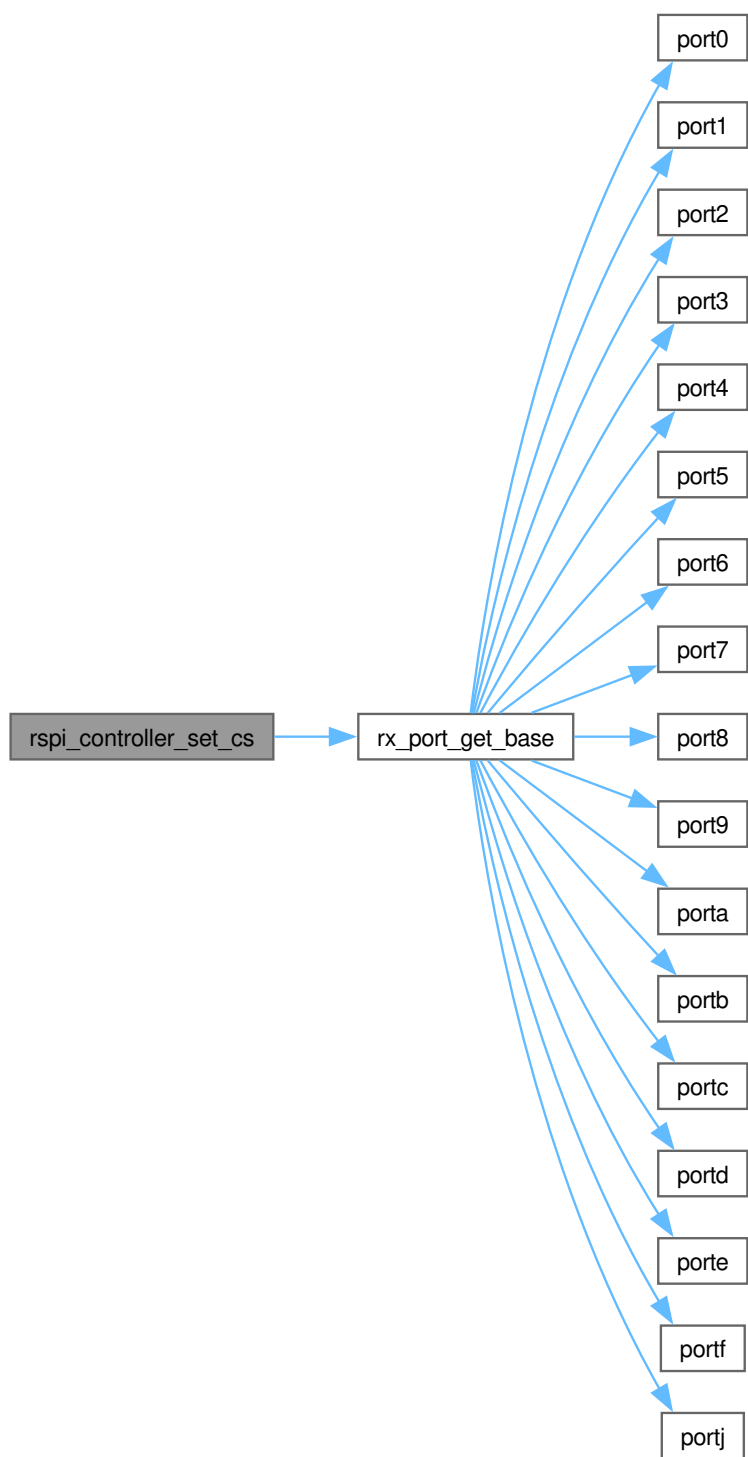
Definition at line 1382 of file [rspi.c](#).

```
01383 {
01384     /* Validate channel */
01385     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01386         rx_log_error(s_tag, "RSPI controller channel not initialized");
01387         return k_rx_err_invalid_state;
01388     }
01389
01390     const uint8_t port = s_rspi_cs_config[channel].port;
01391     const uint8_t pin = s_rspi_cs_config[channel].pin;
01392     volatile rx_port_regs_t* const port_regs = rx_port_get_base(port);
01393
01394     if (port_regs == nullptr) {
01395         return k_rx_err_invalid_arg;
01396     }
01397
01398     /* Compute pin mask once (identical for both assert and deassert) */
01399     const uint8_t pin_mask = (uint8_t)((uint32_t)k_rspi_bit_set « pin);
01400
01401     if (active) {
01402         /* CS active (low) */
01403         port_regs->podr &= (uint8_t)~pin_mask;
01404     } else {
01405         /* CS inactive (high) */
01406         port_regs->podr |= pin_mask;
01407     }
01408
01409     return k_rx_ok;
01410 }
```

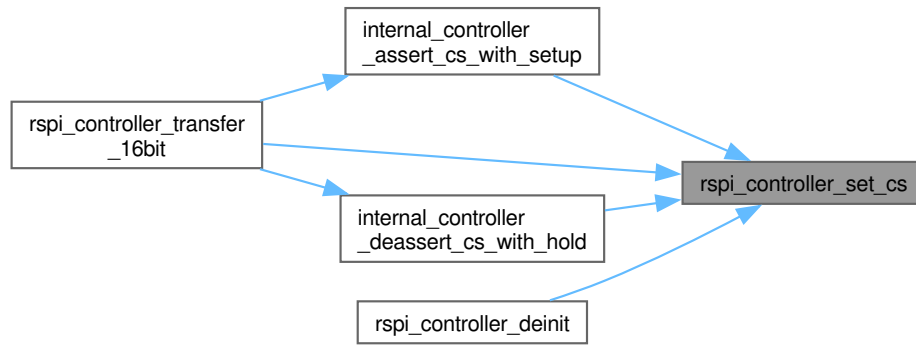
References [k_rspi_bit_set](#), [k_rspi_max_channels](#), [rx_port_regs_t::podr](#), [rx_port_get_base\(\)](#), [s_rspi_controller_initialized](#), [s_rspi_cs_config](#), and [s_tag](#).

Referenced by [internal_controller_assert_cs_with_setup\(\)](#), [internal_controller_deassert_cs_with_hold\(\)](#), [rspi_controller_deinit\(\)](#), and [rspi_controller_transfer_16bit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.107.4.20 rspi_controller_transfer_16bit()

```

rx_err_t rspi_controller_transfer_16bit (
    const rspi_channel_t channel,
    const uint16_t tx_data,
    uint16_t *const rx_data) [nodiscard]

```

Perform 16-bit SPI transfer in controller mode with CS handling.

Perform a 16-bit full-duplex SPI transfer in controller mode with automatic CS.

Executes a full-duplex 16-bit SPI transfer with automatic chip select assertion/deassertion and timing delays.

Sequence:

1. Assert CS (active low) with setup delay (~300ns)
2. Wait for TX buffer ready, then transmit 16-bit data
3. Wait for RX buffer full, then read 16-bit response
4. Hold CS for minimum hold time (~300ns)
5. Deassert CS (inactive high)
6. Clear status flags

Precondition

Channel must be initialized via [rspi_init_controller\(\)](#)

rx_data must be a valid non-nullptr

Postcondition

If successful:

- rx_data is filled with received 16-bit value
- Status flags (SPRF, OVRF) are cleared
- CS is deasserted (high)

On error:

- CS is deasserted if possible
- rx_data contents are undefined

Note

CS setup delay: k_rspi_cs_setup_delay NOP loops (~300ns)

CS hold delay: k_rspi_cs_hold_delay NOP loops (~300ns)

Uses inline assembly NOP for precise timing control

Definition at line 1600 of file [rspi.c](#).

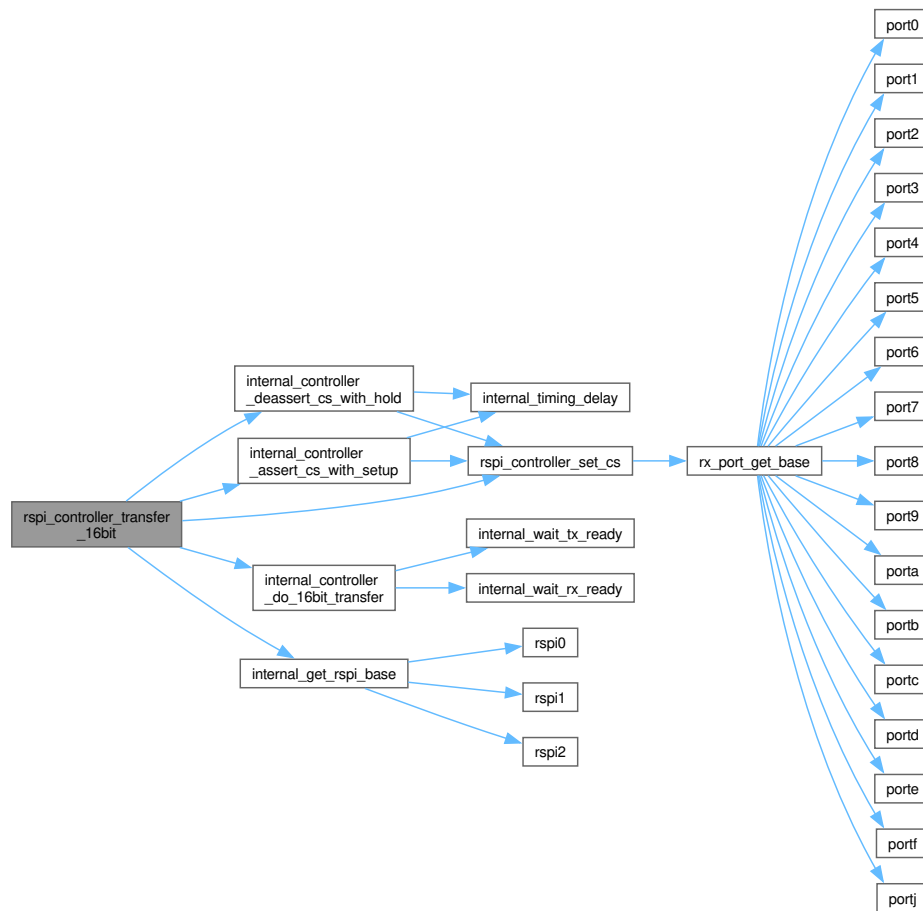
```

01603 {
01604     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
01605
01606     /* Validate channel */
01607     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01608         rx_log_error(s_tag, "RSPi controller channel not initialized");
01609         return k_rx_err_invalid_state;
01610     }
01611
01612     /* Get RSPi base */
01613     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01614     if (rspi == nullptr) {
01615         return k_rx_err_invalid_arg;
01616     }
01617
01618     /* Assert CS with setup delay */
01619     rx_err_t err = internal_controller_assert_cs_with_setup(channel);
01620     if (err != k_rx_ok) {
01621         return err;
01622     }
01623
01624     /* Perform 16-bit TX/RX transfer */
01625     err = internal_controller_do_16bit_transfer(rspi, tx_data, rx_data);
01626     if (err != k_rx_ok) {
01627         /* Deassert CS on error (ignore CS deassert errors) */
01628         (void)rspi_controller_set_cs(channel, false);
01629         return err;
01630     }
01631
01632     /* Deassert CS with hold delay */
01633     return internal_controller_deassert_cs_with_hold(channel);
01634 }

```

References [internal_controller_assert_cs_with_setup\(\)](#), [internal_controller_deassert_cs_with_hold\(\)](#), [internal_controller_do_16bit_transfer\(\)](#), [internal_get_rspi_base\(\)](#), [k_rspi_max_channels](#), [rspi_controller_set_cs\(\)](#), [s_rspi_controller_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.107.4.21 rspi_deinit()

```
rx_err_t rspi_deinit (
    const rspi\_channel\_t channel) [nodiscard]
```

Deinitialize and disable RSPI peripheral.

Deinitialize and disable an RSPI peripheral-mode channel.

Disables the specified RSPI channel, clears the initialization flag, and optionally disables the module (via module stop bit) to reduce power consumption.

Precondition

Channel should have been initialized via [rspi_init_peripheral\(\)](#) previously

Postcondition

If successful:

- RSPI peripheral is disabled (SPCR.SPE = 0)
- `s_rspi_channel_initialized[channel]` is set to false
- Channel is no longer available for SPI operations

Definition at line 884 of file [rspi.c](#).

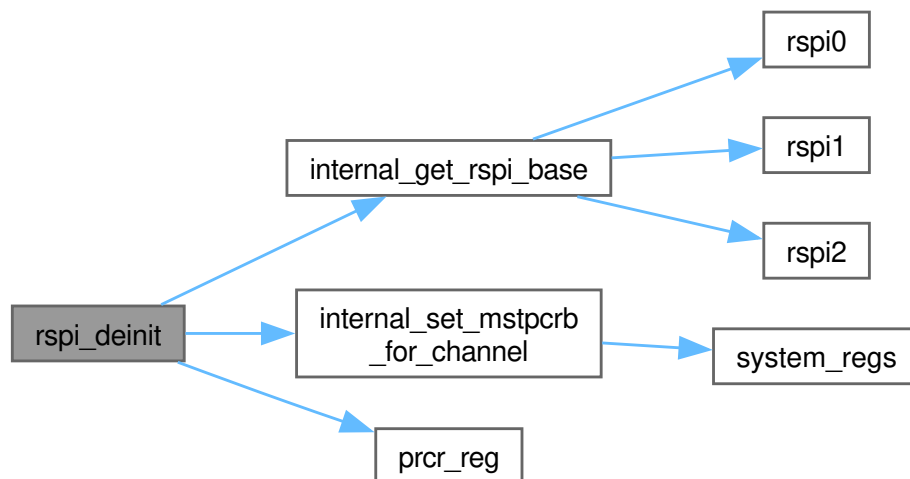
```

00885 {
00886     /* Validate channel */
00887     if ((uint8_t)channel >= k_rspi_max_channels) {
00888         rx_log_error(s_tag, "Invalid RSPI channel");
00889         return k_rx_err_invalid_arg;
00890     }
00891
00892     /* Get RSPI base */
00893     volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00894     if (rspi == nullptr) {
00895         return k_rx_err_invalid_arg;
00896     }
00897
00898     /*
00899     * RSPI Deinitialization Register Write Sequence
00900     * =====
00901     * The deinit order is the reverse of initialization:
00902     * 1. Disable SPI (SPCR.SPE=0) - stop all bus activity before removing clock
00903     * 2. Disable module clock (MSTPCRB) - safe to gate clock only after SPI is idle
00904     * 3. Clear software state - mark channel uninitialized after hardware is off
00905     */
00906
00907     /* Step 1: Disable SPI to stop all bus activity */
00908     rspi->spcr = k_rspi_spcr_disabled;
00909
00910     /* Step 2: Disable RSPI module clock (set module stop bit) - must be done
00911     * after SPI is disabled to avoid undefined behavior from clock gating
00912     * during an active transfer */
00913     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00914     internal_set_mstpcrb_for_channel(channel, false);
00915
00916     *prcr_reg() = k_rx_prcr_lock;
00917
00918     /* Step 3: Mark channel as uninitialized after hardware is safely off */
00919     s_rspi_channel_initialized[channel] = false;
00920
00921     rx_log_debug(s_tag, "RSPI deinitialized");
00922
00923     return k_rx_ok;
00924 }

```

References [internal_get_rspi_base\(\)](#), [internal_set_mstpcrb_for_channel\(\)](#), [k_rspi_max_channels](#), [k_rspi_spcr_disabled](#), [prcr_reg\(\)](#), [s_rspi_channel_initialized](#), [s_tag](#), and [rx_rspi_regs_t::spcr](#).

Here is the call graph for this function:



8.107.4.22 rspi_init_controller()

```

rx_err_t rspi_init_controller (
    const rspi_channel_t channel,
    const rspi_controller_config_t * config) [nodiscard]

```

Initialize RSPI channel in controller mode.

Initialize RSPI channel in controller mode for external device communication.

Configures the specified RSPI channel as a SPI controller to communicate with SPI peripheral devices. Enables the module, configures clock frequency, SPI mode, and chip select GPIO pin.

Precondition

config must be non-NULL
channel < k_rspi_max_channels
channel must not be already initialized
CS GPIO port and pin must be valid and available

Postcondition

If successful:

- RSPI module is enabled via [internal_set_mstpcrb_for_channel\(\)](#)
- *prcr_reg() is unlocked/locked for register protection
- RSPI registers configured: spbr, spcmd0, spcr, spdcr, sppcr
- CS GPIO configured as output, driven high (inactive)
- s_rspi_cs_config[channel] stores CS port/pin
- s_rspi_controller_initialized[channel] = true

On error:

- Hardware state is unchanged
- Channel remains uninitialized

Note

Bit rate calculation: $\text{freq} = \text{PCLKB} / (2 * (\text{SPBR} + 1))$

Always configures 16-bit data frames with word access mode

CS is active-low (asserted=0, deasserted=1)

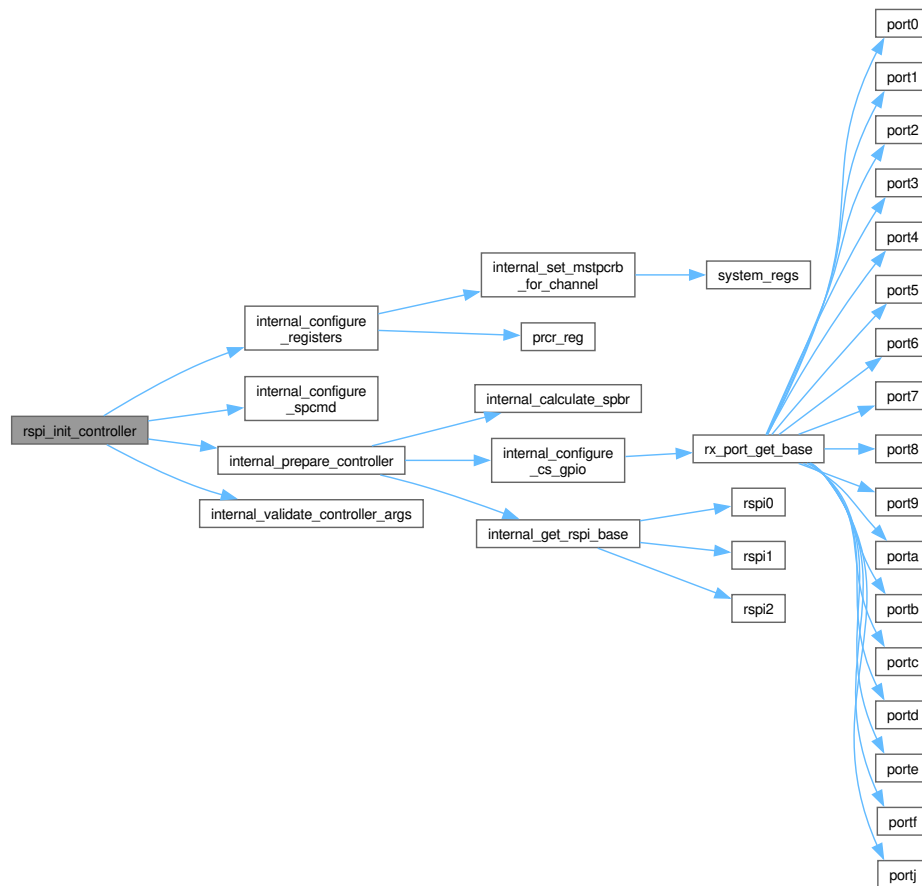
Definition at line 1339 of file [rspi.c](#).

```

01340 {
01341     /* Validate all arguments and preconditions */
01342     rx_err_t err = internal_validate_controller_args(channel, config);
01343     if (err != k_rx_ok) {
01344         return err;
01345     }
01346
01347     /* Prepare hardware resources (SPBR, RSPI base, CS GPIO) */
01348     volatile rx_rspi_regs_t* rspi = nullptr;
01349     uint8_t spbr = k_rspi_spbr_init;
01350     err = internal_prepare_controller(channel, config, &rspi, &spbr);
01351     if (err != k_rx_ok) {
01352         return err;
01353     }
01354
01355     /* Configure SPI mode (CPOL and CPHA) */
01356     uint16_t spcmd = k_rspi_spcmd_init;
01357     internal_configure_spcmd(&spcmd, config->spi_mode);
01358
01359     /* Configure 16-bit data length */
01360     spcmd |= (k_rspi_spcmd_16bit « k_rspi_spcmd_spl_shift);
01361
01362     /* Configure and enable RSPI hardware registers */
01363     return internal_configure_registers(channel, rspi, spbr, spcmd, config);
01364 }
```

References [internal_configure_registers\(\)](#), [internal_configure_spcmd\(\)](#), [internal_prepare_controller\(\)](#), [internal_validate_controller_args\(\)](#), [k_rspi_spbr_init](#), [k_rspi_spcmd_16bit](#), [k_rspi_spcmd_init](#), [k_rspi_spcmd_spl_shift](#), and [rspi_controller_config_t::spi_mode](#).

Here is the call graph for this function:



8.107.4.23 rspi_init_peripheral()

```

rx_err_t rspi_init_peripheral (
    rspi_channel_t channel,
    const rspi_config_t * config) [nodiscard]

```

Initialize RSPI channel in peripheral mode for RPi5 communication.

Configures the specified RSPI channel as an SPI peripheral device. In peripheral mode the RX72N receives the clock signal from an external SPI controller (Raspberry Pi 5). Performs full hardware initialization including module stop release, pin function selection (COPI/CIPO/CLK/CS via MPC), SPI mode register configuration, and frame size setup.

8.107.4.24 Algorithm

1. Validate channel (0-2) and config pointer
2. Release RSPI module from module stop (MSTPCRB)
3. Disable RSPI (SPCR.SPE = 0)
4. Configure pin functions via MPC (COPI, CIPO, CLK, CS)
5. Set SPI mode (CPOL/CPHA in SPCMD0)
6. Set frame length (8-bit or 16-bit in SPCMD0)

7. Configure as peripheral (SPCR.MSTR = 0)
8. Enable RSPI (SPCR.SPE = 1)

Parameters

in	channel	RSPI channel number (0-2)
in	config	Pointer to peripheral mode configuration • Must not be nullptr • config->spi_mode must be valid (0-3)

Returns

rx_err_t Error code indicating success or failure

Return values

k_rx_ok	Channel successfully initialized in peripheral mode
k_rx_err_invalid_arg	Channel > 2 or invalid SPI mode
k_rx_err_null_ptr	config pointer is nullptr
k_rx_err_invalid_state	Channel already initialized (call rspi_deinit() to reinitialize)

Precondition

Channel must not be already initialized (call [rspi_deinit\(\)](#) first to reinitialize)
 System clocks must be configured (PCLKB running at expected frequency)

Postcondition

RSPI channel configured as SPI peripheral, ready for transfers
 COPI/CIPO/CLK/CS pins configured for SPI peripheral function

Note

Thread Safety: NOT thread-safe. Initialize each channel from a single thread during system startup.

Example: RPi5 Communication Setup

```
rspi_config_t rpi_cfg = {
    .spi_mode = k_rspi_mode_0,
    .use_16bit = false
};
rx_err_t err = rspi_init_peripheral(k_rspi_channel_0, &rpi_cfg);
if (err != k_rx_ok) {
    rx_log_error_val("SPI", "RSPI0 peripheral init failed", err);
}
```

See also

[rspi_peripheral_transfer\(\)](#) Full-duplex SPI transfer

[rspi_deinit\(\)](#) Deinitialize RSPI channel

[rspi_config_t](#) Configuration structure

Since

Version 1.0.0

Definition at line 601 of file [rspi.c](#).

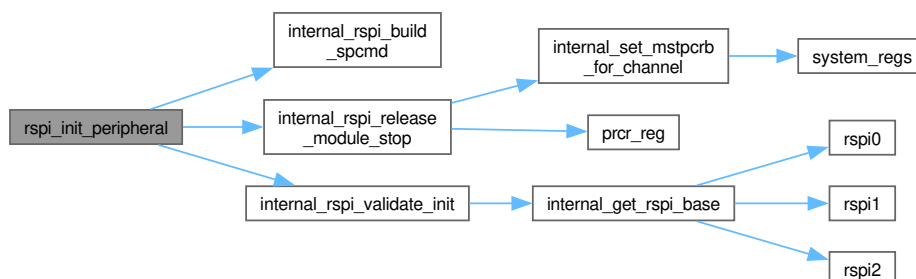
```

00602 {
00603     volatile rx_rspi_regs_t* rspi = nullptr;
00604     const rx_err_t check = internal_rspi_validate_init(channel, config, &rspi);
00605     if (check != k_rx_ok) {
00606         return check;
00607     }
00608
00609     /*
00610     * RSPI Peripheral Mode Register Write Sequence (RX72N HW Manual 38.3.6):
00611     * 1. Enable module clock (MSTPCRB)
00612     * 2. Disable SPI (SPCR.SPE=0)
00613     * 3. Configure SPCMD0 (CPOL/CPHA/data length)
00614     * 4. Configure SPDCR (byte vs longword access)
00615     * 5. Configure SPPCR (no loopback) BEFORE enabling SPI
00616     * 6. Enable SPI (SPCR.SPE=1) -- LAST write so partial config is never active
00617     */
00618     internal_rspi_release_module_stop(channel);
00619     rspi->spcr = k_rspi_spcr_disabled;
00620
00621     rspi->spcmd0 = internal_rspi_build_spcmd(rspi, config);
00622
00623     /* SPPCR MUST be written before SPCR.SPE per HW manual to avoid glitches
00624     * on COPI/CIPO lines during enable */
00625     rspi->sppcr = k_rspi_sppcr_no_loopback;
00626
00627     /* SPCR.SPE=1 MUST be the last register write so all configuration is
00628     * stable before the peripheral becomes active. MSTR=0 -> peripheral mode. */
00629     rspi->spcr = k_rspi_spcr_spe;
00630
00631     s_rspi_channel_initialized[channel] = true;
00632     rx_log_debug(s_tag, "RSPI peripheral mode initialized");
00633     return k_rx_ok;
00634 }

```

References [internal_rspi_build_spcmd\(\)](#), [internal_rspi_release_module_stop\(\)](#), [internal_rspi_validate_init\(\)](#), [k_rspi_spcr_disabled](#), [k_rspi_spcr_spe](#), [k_rspi_sppcr_no_loopback](#), [s_rspi_channel_initialized](#), [s_tag](#), [rx_rspi_regs_t::spcmd0](#), [rx_rspi_regs_t::spcr](#), and [rx_rspi_regs_t::sppcr](#).

Here is the call graph for this function:



8.107.4.25 `rspi_peripheral_read_available()`

```
rx_err_t rspi_peripheral_read_available (
    const rspi\_channel\_t channel,
    bool * available)    [nodiscard]
```

Check if data is available in the RSPI receive buffer.

Check if receive data is available in the RSPI receive buffer.

Polls the RSPI status register to determine if the receive buffer contains data ready for reading. Returns the status without blocking.

Precondition

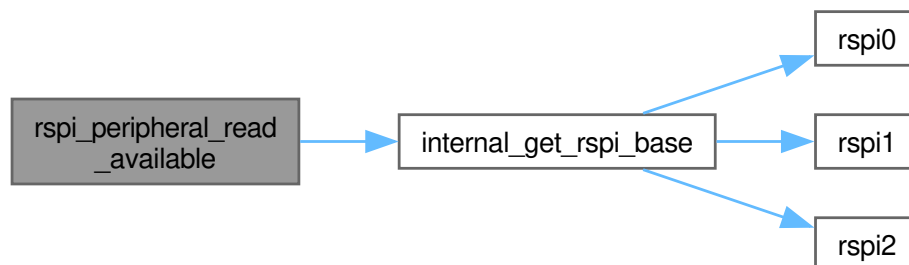
Channel must be initialized via [rspi_init_peripheral\(\)](#) before calling this function
available must be a valid non-nullptr

Definition at line 812 of file [rspi.c](#).

```
00813 {
00814     RX_CHECK_NULL_PTR(available, s_tag, "Available pointer is nullptr");
00815
00816     /* Validate channel */
00817     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00818         rx_log_error(s_tag, "RSPI channel not initialized");
00819         return k_rx_err_invalid_state;
00820     }
00821
00822     /* Get RSPI base */
00823     const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00824     if (rspi == nullptr) {
00825         *available = false;
00826         return k_rx_err_invalid_arg;
00827     }
00828
00829     /* Check if receive buffer has data */
00830     *available = (rspi->spsr & k_rspi_spsr_sprf) != 0;
00831
00832     return k_rx_ok;
00833 }
```

References [internal_get_rspi_base\(\)](#), [k_rspi_max_channels](#), [k_rspi_spsr_sprf](#), [s_rspi_channel_initialized](#), [s_tag](#), and [rx_rspi_regs_t::spsr](#).

Here is the call graph for this function:



8.107.4.26 `rspi_peripheral_transfer()`

```

rx_err_t rspi_peripheral_transfer (
    const rspi_channel_t channel,
    const uint8_t * tx_data,
    uint8_t * rx_data,
    const uint16_t length) [nodiscard]

```

Perform full-duplex SPI transfer in peripheral mode.

Full-duplex SPI transfer in peripheral mode.

Executes a full-duplex (simultaneous transmit and receive) SPI transfer on the specified RSPI channel operating in peripheral mode. Transfers data byte-by-byte with timeout protection on both transmit buffer ready and receive buffer full operations.

Precondition

- channel must be initialized via `rspi_init_peripheral()` before calling
- tx_data pointer must be non-NULL
- rx_data pointer must be non-NULL
- length must be in range [1, 65535]

Postcondition

- If successful, rx_data buffer is filled with received bytes corresponding to transmitted data at each byte offset

Definition at line 742 of file `rspi.c`.

```

00746 {
00747     RX_CHECK_NULL_PTR(tx_data, s_tag, "TX data pointer is nullptr");
00748     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
00749
00750     if (length == k_rspi_len_zero) {
00751         rx_log_error(s_tag, "Invalid transfer length");
00752         return k_rx_err_invalid_arg;
00753     }
00754
00755     /* Validate channel */
00756     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00757         rx_log_error(s_tag, "RSPI channel not initialized");
00758         return k_rx_err_invalid_state;
00759     }
00760
00761     /* Get RSPI base */
00762     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
00763     if (rspi == nullptr) {
00764         rx_log_error(s_tag, "Failed to get RSPI base address");
00765         return k_rx_err_invalid_arg;
00766     }
00767
00768     /* NASA Rule 2: Statically bounded loop */
00769     for (uint16_t i = k_rspi_loop_start; i < k_rspi_transfer_len_max; i++) {
00770         if (i >= length) {
00771             break;
00772         }
00773
00774         /* Wait for transmit buffer empty */
00775         rx_err_t err = internal_wait_tx_ready(rspi);
00776         if (err != k_rx_ok) {
00777             rx_log_error(s_tag, "SPI transmit timeout");
00778             return err;
00779         }
00780
00781         /* Write transmit data */
00782         rspi->spdr = tx_data[i];
00783
00784         /* Wait for receive buffer full */

```

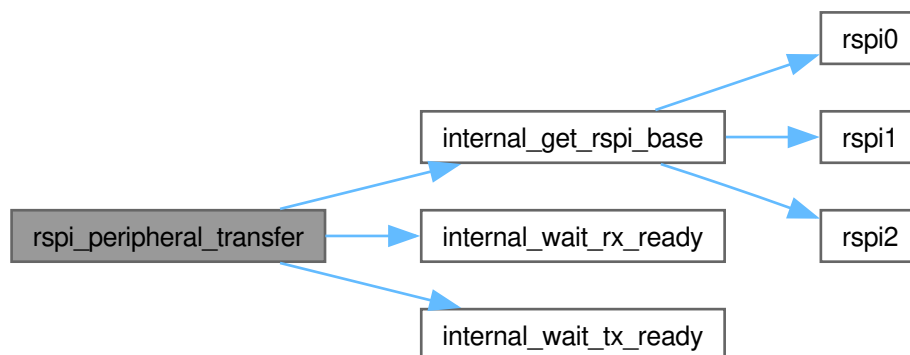
```

00785     err = internal_wait_rx_ready(rspi);
00786     if (err != k_rx_ok) {
00787         rx_log_error(s_tag, "SPI receive timeout");
00788         return err;
00789     }
00790
00791     /* Read receive data */
00792     rx_data[i] = (uint8_t)rspi->spdr;
00793
00794     /* Clear status flags */
00795     rspi->spsr &= (uint8_t) ~(uint8_t)(k_rspsr_sprf | k_rspsr_ovrf);
00796 }
00797
00798 return k_rx_ok;
00799 }

```

References [internal_get_rsps_base\(\)](#), [internal_wait_rx_ready\(\)](#), [internal_wait_tx_ready\(\)](#), [k_rsps_len_zero](#), [k_rsps_loop_start](#), [k_rsps_max_channels](#), [k_rsps_spsr_ovrf](#), [k_rsps_spsr_sprf](#), [k_rsps_transfer_len_max](#), [s_rsps_channel_initialized](#), [s_tag](#), [rx_rsps_regs_t::spdr](#), and [rx_rsps_regs_t::spsr](#).

Here is the call graph for this function:



8.107.4.27 rsps_peripheral_write_ready()

```

rx_err_t rsps_peripheral_write_ready (
    const rsps_channel_t channel,
    bool * ready) [nodiscard]

```

Check if transmit buffer is ready for data.

Check if the RSPI transmit buffer is ready for new data.

Polls the RSPI status register to determine if the transmit buffer is empty and ready to accept new data. Returns the status without blocking.

Precondition

Channel must be initialized via [rsps_init_peripheral\(\)](#)
 ready must be a valid non-nullptr

Definition at line 846 of file [rsps.c](#).

```

00847 {
00848     RX_CHECK_NULL_PTR(ready, s_tag, "Ready pointer is nullptr");
00849
00850     /* Validate channel */
00851     if ((uint8_t)channel >= k_rsps_max_channels || !s_rsps_channel_initialized[channel]) {
00852         rx_log_error(s_tag, "RSPI channel not initialized");
00853         return k_rx_err_invalid_state;

```

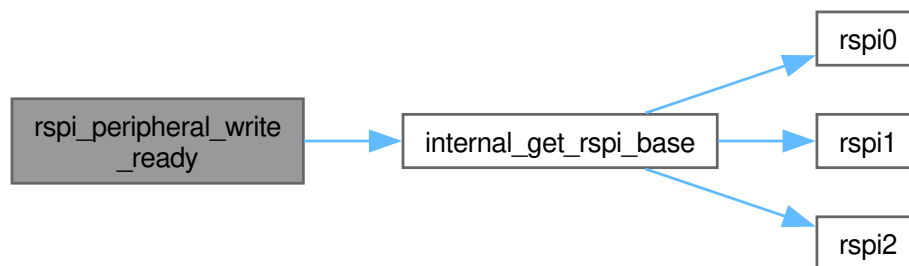
```

00854 }
00855
00856 /* Get RSPI base */
00857 const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00858 if (rspi == nullptr) {
00859     *ready = false;
00860     return k_rx_err_invalid_arg;
00861 }
00862
00863 /* Check if transmit buffer is empty */
00864 *ready = (rspi->spsr & k_rspi_spsr_sptef) != 0;
00865
00866 return k_rx_ok;
00867 }

```

References [internal_get_rspi_base\(\)](#), [k_rspi_max_channels](#), [k_rspi_spsr_sptef](#), [s_rspi_channel_initialized](#), [s_tag](#), and [rx_rspi_regs_t::spsr](#).

Here is the call graph for this function:



8.107.5 Variable Documentation

8.107.5.1 s'rspi'channel'initialized

```
bool s_rspi_channel_initialized[k_rspi_max_channels] = {false, false, false} [static]
```

Definition at line 352 of file [rspi.c](#).

```
00352 {false, false, false};
```

Referenced by [rspi_deinit\(\)](#), [rspi_init_peripheral\(\)](#), [rspi_peripheral_read_available\(\)](#), [rspi_peripheral_transfer\(\)](#), and [rspi_peripheral_write_ready\(\)](#).

8.107.5.2 s'rspi'controller'initialized

```
bool s_rspi_controller_initialized[k_rspi_max_channels] = {false, false, false} [static]
```

Definition at line 355 of file [rspi.c](#).

```
00355 {false, false, false};
```

Referenced by [internal_configure_registers\(\)](#), [internal_controller_assert_cs_with_setup\(\)](#), [internal_controller_deassert](#), [internal_validate_controller_args\(\)](#), [rspi_controller_deinit\(\)](#), [rspi_controller_set_cs\(\)](#), and [rspi_controller_transfer_10](#).

8.107.5.3 s'rspi'cs'config

```
rspi_cs_config_t s_rspi_cs_config[k_rspi_max_channels] [static]
```

Initial value:

```
= {
{k_rspi_cs_default_port, k_rspi_cs_default_pin},
{k_rspi_cs_default_port, k_rspi_cs_default_pin},
{k_rspi_cs_default_port, k_rspi_cs_default_pin},
}
```

Definition at line 364 of file [rspi.c](#).

```
00364 {
00365 {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00366 {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00367 {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00368 };
```

Referenced by [internal_configure_registers\(\)](#), [rspi_controller_deinit\(\)](#), and [rspi_controller_set_cs\(\)](#).

8.107.5.4 s'tag

```
const char* const s_tag = "RSPI" [static]
```

Definition at line 349 of file [rspi.c](#).

8.108 rspi.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rspi.c
00003  * @brief RSPi (Renesas Serial Peripheral Interface) Driver for RX72N
00004  *
00005  * @details
00006  * Provides complete SPI driver implementation supporting both peripheral mode
00007  * (RX72N as SPI peripheral with RPi5 as controller) and controller mode (RX72N
00008  * as SPI controller for sensor communication). Implements full-duplex
00009  * communication with configurable clock frequency, SPI modes 0-3, and automatic
00010  * chip select management.
00011  *
00012  * @par System Architecture
00013  * @dot
00014  * digraph rspi_architecture {
00015  *   rankdir=TB;
00016  *   node [shape=box, style=rounded];
00017  *
00018  *   subgraph cluster_rpi5 {
00019  *     label="Raspberry Pi 5";
00020  *     style=filled;
00021  *     color=lightblue;
00022  *     rpi5_spi [label="SPI Controller\n(spidev)"];
00023  *   }
00024  *
00025  *   subgraph cluster_rx72n {
00026  *     label="RX72N";
00027  *     style=filled;
00028  *     color=lightgreen;
00029  *
00030  *     subgraph cluster_driver {
00031  *       label="RSPI Driver (This Module)";
00032  *       style=filled;
00033  *       color=lightyellow;
00034  *       peripheral [label="Peripheral Mode\nrspi_init_peripheral()\nrspi_peripheral_transfer()"];
00035  *       controller [label="Controller Mode\nrspi_init_controller()\nrspi_controller_transfer_16bit()"];
00036  *       helpers [label="Internal Helpers\ninternal_wait_tx_ready()\ninternal_wait_rx_ready()"];
00037  *     }
00038  *
00039  *     rspi_hw [label="RSPI0/1/2\nHardware Registers"];
00040  *     gpio [label="GPIO\n(CS pins)"];
00041  *   }
00042  * }
```

```

00042 *
00043 * subgraph cluster_peripherals {
00044 *     label="SPI Peripherals";
00045 *     style=filled;
00046 *     color=lightcoral;
00047 *     sensors [label="Future SPI\nSensors"];
00048 * }
00049 *
00050 * rpi5_spi -> peripheral [label="COPI/CIPO\nSCLK/CS"];
00051 * peripheral -> rspi_hw;
00052 * controller -> rspi_hw;
00053 * controller -> gpio [label="CS control"];
00054 * helpers -> rspi_hw [style=dashed];
00055 * rspi_hw -> sensors [label="SPI Bus", style=dashed];
00056 * }
00057 * @enddot
00058 *
00059 * @par Supported Features
00060 * | Feature | Peripheral Mode | Controller Mode |
00061 * |-----|-----|-----|
00062 * | Full-duplex | [OK] | [OK] |
00063 * | 8-bit data | [OK] | - |
00064 * | 16-bit data | [OK] | [OK] |
00065 * | SPI modes 0-3 | [OK] | [OK] |
00066 * | Hardware CS | - | GPIO-based |
00067 * | Clock config | External | 100kHz - 10MHz |
00068 *
00069 * @par RSPI Channels
00070 * | Channel | Use Case | Status Tracking |
00071 * |-----|-----|-----|
00072 * | RSPi0 | RPi5 communication | s_rspi_channel_initialized[0] |
00073 * | RSPi1 | Not used in current design | s_rspi_controller_initialized[1] |
00074 * | RSPi2 | Not used | Available |
00075 *
00076 * @par Performance Characteristics
00077 * | Metric | Value | Condition |
00078 * |-----|-----|-----|
00079 * | Max clock (controller) | 10 MHz | PCLKB=60MHz, SPBR=2 |
00080 * | Min clock (controller) | 100 kHz | PCLKB=60MHz, SPBR=255 |
00081 * | Transfer timeout | 10 ms | Per byte |
00082 * | CS setup/hold delay | ~300 ns | 10 NOP cycles @ 240MHz |
00083 * | 16-bit transfer time | ~2-50 us | Depends on clock |
00084 *
00085 * @par Memory Usage
00086 * | Component | Size | Description |
00087 * |-----|-----|-----|
00088 * | Code (.text) | ~4 KB | All functions |
00089 * | Constants (.rodata) | ~100 B | Static constants |
00090 * | Static data (.bss) | ~20 B | Init flags, CS config |
00091 * | Stack (per call) | ~50 B | Local variables |
00092 *
00093 * @par Hardware Register Access
00094 * - Uses rx72n_rspi_regs.h register definitions with inline accessors:
00095 * - rspi0(), rspi1(), rspi2() for channel base addresses
00096 * - MSTPCRB module stop control for power management
00097 * - PRCR register protection for system register access
00098 *
00099 * @par Module Dependencies
00100 * - hardware.h: Hardware definitions and inline accessors
00101 * - rx72n_regs.h: Register structure definitions
00102 * - rx_gpio_constants.h: GPIO pin definitions
00103 * - rx_port_constants.h: Port constants
00104 * - rx_port_utils.h: Port/pin extraction utilities
00105 * - rx_register_protection.h: PRCR unlock/lock
00106 *
00107 * @par NASA Power of 10 Compliance
00108 * - Rule 1: [OK] No goto, setjmp, or recursion
00109 * - Rule 2: [OK] All loops bounded (timeout counters, transfer length max)
00110 * - Rule 3: [OK] No dynamic memory allocation (static buffers only)
00111 * - Rule 4: [OK] Functions under 60 lines (refactored into helpers)
00112 * - Rule 5: [OK] Minimum 2 assertions per function (RX_ASSERT, RX_CHECK_NULL_PTR)
00113 * - Rule 6: [OK] Variables declared at smallest scope
00114 * - Rule 7: [OK] All return values checked
00115 * - Rule 8: [OK] C23 typed enums for all constants
00116 * - Rule 9: [OK] No function pointers (direct hardware access)
00117 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00118 *
00119 * @par SOLID Principles
00120 * - **S (SRP):** Driver handles only RSPI peripheral operations
00121 * - **O (OCP):** Mode/frequency configurable without code changes
00122 * - **L (LSP):** Peripheral/controller modes have consistent error semantics
00123 * - **I (ISP):** Separate init/transfer/deinit for each mode
00124 * - **D (DIP):** Depends on hardware abstractions (rx72n_regs.h)
00125 *
00126 * @par Thread Safety
00127 * - **NOT thread-safe:** Caller must provide external synchronization
00128 * - Static state (s_rspi_*) shared across calls

```



```

00129 * - Concurrent access to same channel requires mutex protection
00130 * - Different channels can be accessed independently
00131 *
00132 * @see rx72n_rspi_regs.h RSPI register definitions
00133 * @see rx_spi_comm.h Higher-level SPI communication layer
00134 * @see docs/sections/03_hardware_pinout.tex SPI pin assignments
00135 *
00136 * @author Locked, Inc.
00137 * @date 2026-01-01
00138 * @copyright Copyright (c) 2026 Locked Inc.
00139 * SPDX-License-Identifier: MIT
00140 * @since Version 1.0.0
00141 */
00142
00143 #ifdef __RX__
00144
00145 #include <stdint.h>
00146 #include <string.h>
00147
00148 #include "hardware.h"
00149 #include "rx72n_clock.h"
00150 #include "rx72n_regs.h"
00151 #include "rx_gpio_constants.h"
00152 #include "rx_port_constants.h"
00153 #include "rx_port_utils.h"
00154 #include "rx_register_protection.h"
00155
00156 /*
=====
00157 * Constants
00158 *
=====
00159 */
00160
00161 /**
00162 * @enum rspi_constants_t
00163 * @brief RSPI channel count, timeout, and general constants
00164 *
00165 * @details
00166 * Core configuration constants for RSPI driver. Timeout values are calibrated
00167 * for worst-case SPI peripheral response times at minimum clock frequency.
00168 *
00169 * @par Timeout Calculation
00170 * At 100 kHz minimum clock, one byte takes ~80 us. 10 ms timeout provides
00171 * ~125 byte margin before declaring failure, sufficient for any single
00172 * transfer operation.
00173 *
00174 * @since Version 1.0.0
00175 */
00176 typedef enum : uint32_t {
00177     k_rspi_max_channels = 3, /**< Number of RSPI channels: RSPI0, RSPI1, RSPI2.
00178                               @par Value: 3 */
00179
00180     k_rspi_timeout_us = 10000, /**< Timeout for TX/RX wait operations in loop iterations.
00181                               Approximately 10 ms at typical loop overhead.
00182                               @par Value: 10000 iterations (~10 ms) */
00183
00184     k_rspi_timeout_zero = 0, /**< Sentinel value indicating timeout has expired.
00185                               @par Value: 0 */
00186
00187     k_rspi_len_zero = 0, /**< Sentinel for zero-length transfer (invalid).
00188                               @par Value: 0 */
00189
00190     k_rspi_zero_u32 = 0, /**< Zero constant for uint32_t comparisons.
00191                               @par Value: 0 */
00192
00193     k_rspi_spbr_offset = 1, /**< Offset for SPBR calculation formula.
00194                               SPBR = (PCLKB / (2 * freq)) - 1
00195                               @par Value: 1 */
00196
00197     k_rspi_loop_start = 0, /**< Starting index for bounded loops (NASA Rule 2).
00198                               @par Value: 0 */
00199 } rspi_constants_t;
00200
00201 /** @brief RSPI CS default constants (uninitialized state) */
00202 typedef enum : uint8_t {
00203     k_rspi_cs_default_port = 0, /**< Default CS port (unconfigured) */
00204     k_rspi_cs_default_pin = 0, /**< Default CS pin (unconfigured) */
00205     k_rspi_spbr_init = 0, /**< Initial SPBR value */
00206 } rspi_cs_defaults_t;
00207
00208 /** RSPI channel constants (k_rspi_channel_0/1/2) provided by hardware.h via rspi_channel_t */
00209
00210 /** @brief RSPI0/RSPI1 module stop bit positions in MSTPCRB (HW Manual p.408) */
00211 typedef enum : uint8_t {
00212     k_rspi_mstpb_rspi0 = 17, /**< RSPI0 module stop bit in MSTPCRB */
00213     k_rspi_mstpb_rspi1 = 16, /**< RSPI1 module stop bit in MSTPCRB */

```

```

00214 } rspi_mstpcrb_bits_t;
00215
00216 /** @brief RSPI2 module stop bit position in MSTPCRC (HW Manual p.410) */
00217 typedef enum : uint8_t {
00218     k_rspi_mstpcr_rsip2 = 22, /**< RSPI2 module stop bit in MSTPCRC */
00219 } rspi_mstpcrb_bits_t;
00220
00221 /** @brief RSPI controller mode constants
00222  *
00223  * PCLKB rate comes from rx72n_clock.h (k_pclkb_hz) -- do not duplicate the
00224  * literal here so the divisor math tracks any future clock-tree change.
00225  */
00226 typedef enum : uint32_t {
00227     k_rspi_min_freq_hz = 100000, /**< Minimum SPI clock (100 kHz) */
00228     k_rspi_max_freq_hz = 10000000, /**< Maximum SPI clock (10 MHz) */
00229     k_rspi_spbr_max = 255, /**< Maximum SPBR value */
00230     k_rspi_spbr_divisor = 2, /**< SPBR divisor factor */
00231     k_rspi_cs_setup_delay = 10, /**< CS setup delay iterations (~300ns) */
00232     k_rspi_cs_hold_delay = 10, /**< CS hold delay iterations (~300ns) */
00233     k_rspi_max_delay_cycles = 10000, /**< Sanity limit for timing delays */
00234 } rspi_controller_constants_t;
00235
00236 /** @brief SPI mode limits */
00237 typedef enum : uint8_t {
00238     k_rspi_mode_min = 0, /**< Minimum valid SPI mode (0) */
00239     k_rspi_mode_max = 3, /**< Maximum valid SPI mode (0-3) */
00240 } rspi_mode_limits_t;
00241
00242 /** @brief SPCMD register bit positions and masks */
00243 typedef enum : uint16_t {
00244     k_rspi_spcmd_cpha_mask = 1U, /**< CPHA bit mask (bit 0) */
00245     k_rspi_spcmd_cpol_mask = 2U, /**< CPOL bit mask (bit 1) */
00246     k_rspi_spcmd_cpol_pos = 1, /**< CPOL bit position */
00247     k_rspi_spcmd_cpha_pos = 0, /**< CPHA bit position */
00248     k_rspi_spcmd_spl_shift = 8, /**< SPL (data length) bit shift */
00249     k_rspi_spcmd_8bit = 7U, /**< 8-bit data length value */
00250     k_rspi_spcmd_16bit = 15U, /**< 16-bit data length value */
00251 } rspi_spcmd_bits_t;
00252
00253 /** @brief SPDCCR register bit positions (local) */
00254 typedef enum : uint8_t {
00255     k_rspi_spdcr_splw_pos = 5, /**< SPLW bit position (longword/32-bit access) */
00256     k_rspi_spdcr_byte_mode = 0U, /**< Byte access mode */
00257 } rspi_spdcr_local_t;
00258
00259 /**
00260  * @enum rspi_register_defaults_t
00261  * @brief RSPI register default values (reset/disabled state)
00262  *
00263  * @details
00264  * Values written to RSPI registers during initialization to establish
00265  * a known baseline configuration. These match the hardware reset defaults
00266  * documented in the RX72N Hardware Manual Section 38.2. Writing these values
00267  * before configuring mode-specific settings ensures a deterministic starting
00268  * state regardless of prior peripheral activity.
00269  *
00270  * @invariant All values must match the hardware manual reset state
00271  *
00272  * @code
00273  * // Reset RSPI registers to defaults before reconfiguration
00274  * rspi->sppcr = k_rspi_sppcr_no_loopback;
00275  * rspi->sprcr = k_rspi_sprcr_disabled;
00276  * @endcode
00277  *
00278  * @see rspi_init_peripheral() Uses these during peripheral mode init
00279  * @see rspi_internal_configure_registers() Uses these during controller mode init
00280  * @since Version 1.0.0
00281  */
00282 typedef enum : uint8_t {
00283     k_rspi_sppcr_no_loopback = 0U, /**< SPPCR: no loopback mode (normal operation) */
00284     k_rspi_sprcr_disabled = 0U, /**< SPCR: SPI disabled (module in reset state) */
00285 } rspi_register_defaults_t;
00286
00287 /**
00288  * @enum rspi_bit_constants_t
00289  * @brief RSPI bit manipulation constants for register field operations
00290  *
00291  * @details
00292  * Provides named constants for single-bit set operations used to construct
00293  * register masks via left-shift. Using a named constant instead of a literal
00294  * 1 improves readability and grep-ability of register manipulation code.
00295  * All register mask construction in this module uses k_rspi_bit_set as the
00296  * base value to ensure consistent, self-documenting bit manipulation.
00297  *
00298  * @invariant k_rspi_bit_set must always equal 1
00299  *
00300  * @code

```

```

00301 * // Construct a mask for MSTPCRB module stop bit
00302 * uint32_t mask = (k_rspi_bit_set « k_rspi_mstpb_rspi0);
00303 * system_regs()->mstpcrb &= ~mask;
00304 * @endcode
00305 *
00306 * @see internal_set_mstpcrb_for_channel() Uses for MSTPCRB mask construction
00307 * @see internal_configure_spcmd() Uses for CPOL/CPHA bit setting
00308 * @since Version 1.0.0
00309 */
00310 typedef enum : uint32_t {
00311     k_rspi_bit_set = 1UL, /**< Single bit set value (1) for shift-based register mask construction */
00312 } rspi_bit_constants_t;
00313
00314 /**
00315 * @enum rspi_transfer_constants_t
00316 * @brief RSPI transfer length and command register constants
00317 *
00318 * @details
00319 * Defines limits and initialization values for the RSPI transfer engine.
00320 * The maximum transfer length is constrained by the 16-bit DMA counter.
00321 * SPCMD initial value sets a clean baseline before configuring bit fields
00322 * for CPOL, CPHA, and data length via bitwise OR operations.
00323 *
00324 * @invariant k_rspi_transfer_len_max must equal 65535 (uint16_t maximum, 16-bit DMA counter limit)
00325 *
00326 * @code
00327 * // Validate transfer length before starting peripheral transfer
00328 * if (length == k_rspi_len_zero || length > k_rspi_transfer_len_max) {
00329 *     return k_rx_err_invalid_arg;
00330 * }
00331 * // Initialize SPCMD before configuring mode-specific bits
00332 * uint16_t spcmd = k_rspi_spcmd_init;
00333 * @endcode
00334 *
00335 * @see rspi_peripheral_transfer() Uses k_rspi_transfer_len_max for length validation
00336 * @see rspi_init_peripheral() Uses k_rspi_spcmd_init for register setup
00337 * @since Version 1.0.0
00338 */
00339 typedef enum : uint16_t {
00340     k_rspi_transfer_len_max = 65535U, /**< Maximum transfer length (16-bit counter limit) */
00341     k_rspi_spcmd_init = 0U, /**< SPCMD initial value (all fields at reset state) */
00342 } rspi_transfer_constants_t;
00343
00344 /*
=====
00345 * Static Variables
00346 *
=====
00347 */
00348
00349 static const char* const s_tag = "RSPI";
00350
00351 /* Track initialized RSPI channels (peripheral mode) */
00352 static bool s_rspi_channel_initialized[k_rspi_max_channels] = {false, false, false};
00353
00354 /* Track controller mode RSPI channels */
00355 static bool s_rspi_controller_initialized[k_rspi_max_channels] = {false, false, false};
00356
00357 /** @brief Controller mode CS pin configuration */
00358 typedef struct {
00359     uint8_t port; /**< GPIO port for chip select */
00360     uint8_t pin; /**< GPIO pin for chip select */
00361 } rspi_cs_config_t;
00362
00363 /* Track controller mode CS pins */
00364 static rspi_cs_config_t s_rspi_cs_config[k_rspi_max_channels] = {
00365     {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00366     {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00367     {k_rspi_cs_default_port, k_rspi_cs_default_pin},
00368 };
00369
00370 /*
=====
00371 * Internal Helper Functions
00372 *
=====
00373 */
00374
00375 /**
00376 * @brief Get RSPI register base address from channel number
00377 *
00378 * @details
00379 * Maps a channel index (0-2) to the corresponding RSPI peripheral register
00380 * block via the inline accessor functions rspi0()/rspi1()/rspi2(). Returns
00381 * nullptr for out-of-range channel numbers to allow callers to detect invalid
00382 * input without risking a wild pointer dereference.
00383 */

```

```

00384 *
00385 *
00386 * @pre channel must be in range [0, k_rspi_max_channels)
00387 * @pre RSPI hardware must be present (RX72N 144-pin package)
00388 * @post Returned pointer (if non-null) is aligned and points to valid HW registers
00389 * @post No side effects on hardware state
00390 *
00391 * @note Not thread-safe. Caller must provide synchronization if needed.
00392 *
00393 * @see rspi0() Channel 0 register accessor
00394 * @see rspi1() Channel 1 register accessor
00395 * @see rspi2() Channel 2 register accessor
00396 *
00397 * @since Version 1.0.0
00398 */
00399 static volatile rx_rspi_regs_t* internal_get_rspi_base(const rspi_channel_t channel)
00400 {
00401     switch (channel) {
00402         case k_rspi_channel_0: {
00403             return rspi0();
00404         }
00405         case k_rspi_channel_1: {
00406             return rspi1();
00407         }
00408         case k_rspi_channel_2: {
00409             return rspi2();
00410         }
00411         default: {
00412             return nullptr;
00413         }
00414     }
00415 }
00416
00417 /**
00418 * @brief Set module stop bit for an RSPI channel
00419 *
00420 * @details
00421 * Controls the Module Stop Control Register bits that gate the clock to each
00422 * RSPI channel. RSPI0 and RSPI1 are in MSTPCRB (bits 17 and 16); RSPI2 is in
00423 * MSTPCRC (bit 22). Clearing a bit enables the module (starts the clock);
00424 * setting it stops the module (reduces power consumption).
00425 *
00426 *
00427 * @pre system_regs() must return a valid pointer
00428 * @pre channel must be one of k_rspi_channel_0, k_rspi_channel_1, k_rspi_channel_2
00429 * @post MSTPCRB (channels 0/1) or MSTPCRC (channel 2) updated appropriately
00430 * @post Module clock enabled or disabled for the specified channel
00431 *
00432 * @note Not thread-safe. Caller must hold MSTPCR protection if applicable.
00433 * @warning Disabling a module while it is actively transferring data will
00434 *         cause undefined hardware behavior.
00435 *
00436 * @see internal_get_rspi_base() Maps channel to register base
00437 *
00438 * @since Version 1.0.0
00439 */
00440 static void internal_set_mstpcrb_for_channel(const rspi_channel_t channel, const bool enable)
00441 {
00442     RX_ASSERT(system_regs() != nullptr, "system_regs is nullptr");
00443     static_assert(k_rspi_bit_set != 0, "RSPI bit constant must be non-zero");
00444     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
00445              (channel == k_rspi_channel_2),
00446              "Invalid RSPI channel");
00447
00448     /* RSPI2 stop bit is in MSTPCRC (bit 22), not MSTPCRB (HW Manual p.410) */
00449     if (channel == k_rspi_channel_2) {
00450         const uint32_t mask = (k_rspi_bit_set « k_rspi_mstpc_rspi2);
00451         if (enable) {
00452             system_regs()->mstpcrc &= ~mask;
00453         } else {
00454             system_regs()->mstpcrc |= mask;
00455         }
00456         return;
00457     }
00458
00459     uint32_t mask = k_rspi_zero_u32;
00460     if (channel == k_rspi_channel_0) {
00461         mask = (k_rspi_bit_set « k_rspi_mstpb_rspi0);
00462     } else if (channel == k_rspi_channel_1) {
00463         mask = (k_rspi_bit_set « k_rspi_mstpb_rspi1);
00464     }
00465     if (enable) {
00466         system_regs()->mstpcrb &= ~mask;
00467     } else {
00468         system_regs()->mstpcrb |= mask;
00469     }
00470 }

```

```

00471
00472 /*
=====
00473 * Public API Implementation
00474 *
=====
00475 */
00476
00477 /**
00478 * @brief Initialize RSPI channel in peripheral mode
00479 *
00480 * Configures the specified RSPI channel as a SPI peripheral to communicate
00481 * with the Raspberry Pi 5 (acting as SPI Controller). Enables the peripheral
00482 * in the configured SPI mode (0-3) with optional 8-bit or 16-bit data frames.
00483 * Uses register protection unlock/lock for module stop control.
00484 *
00485 *
00486 *
00487 * @pre config pointer must be non-NULL
00488 * @pre channel must be in range [0, k_rspi_max_channels)
00489 * @pre config->spi_mode must be in range [0, 3]
00490 *
00491 * @post If successful:
00492 *   - RSPI module is enabled (module stop bit cleared via register protection)
00493 *   - SPI mode (CPOL/CPHA) is configured per config->spi_mode
00494 *   - Data frame length (8-bit or 16-bit) is set per config->use_16bit
00495 *   - SPI is enabled in peripheral mode (SPCR.MSTR = 0)
00496 *   - s_rspi_channel_initialized[channel] is set to true
00497 *   - Peripheral is ready to receive transfers from SPI controller
00498 */
00499 /**
00500 * @brief Validate rspi_init_peripheral inputs and resolve peripheral base
00501 *
00502 * @param[in] channel Channel index
00503 * @param[in] config Caller's config (must be non-null)
00504 * @param[out] rspi_out On success, set to peripheral base
00505 *
00506 * @return rx_err_t Error code
00507 * @retval k_rx_ok All checks pass
00508 * @retval k_rx_err_null_ptr config is nullptr
00509 * @retval k_rx_err_invalid_arg Bounds violation or unknown channel
00510 *
00511 * @pre rspi_out non-null
00512 *
00513 * @since Version 1.0.0
00514 */
00515 static rx_err_t internal_rspi_validate_init(const rspi_channel_t channel,
00516                                             const rspi_config_t* const config,
00517                                             volatile rx_rspi_regs_t** rspi_out)
00518 {
00519     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00520
00521     if ((uint8_t)channel >= k_rspi_max_channels) {
00522         rx_log_error(s_tag, "Invalid RSPI channel");
00523         return k_rx_err_invalid_arg;
00524     }
00525     /* Lower bound check omitted: spi_mode is uint8_t and k_rspi_mode_min == 0
00526      * so spi_mode >= k_rspi_mode_min is always true (-Wtype-limits). */
00527     if ((uint8_t)config->spi_mode > k_rspi_mode_max) {
00528         rx_log_error(s_tag, "Invalid SPI mode (must be 0-3)");
00529         return k_rx_err_invalid_arg;
00530     }
00531     *rspi_out = internal_get_rspi_base(channel);
00532     if (*rspi_out == nullptr) {
00533         return k_rx_err_invalid_arg;
00534     }
00535     return k_rx_ok;
00536 }
00537
00538 /**
00539 * @brief Build SPCMD0 value and program SPDCR for the requested data width
00540 *
00541 * @details
00542 * Encodes CPOL/CPHA bits from spi_mode and selects 8-bit or 16-bit data
00543 * length. SPDCR is set to longword access for 16-bit transfers, byte
00544 * access for 8-bit, matching the SPCMD0 SPL field.
00545 *
00546 * @param[in] rspi Peripheral base (non-null)
00547 * @param[in] config Validated init config
00548 *
00549 * @return uint16_t Encoded SPCMD0 value
00550 *
00551 * @pre SPCR.SPE=0 (caller has already disabled SPI)
00552 * @post SPDCR programmed; SPCMD0 not yet written
00553 *
00554 * @since Version 1.0.0
00555 */

```

```

00556 static uint16_t internal_rspi_build_spcmd(volatile rx_rspi_regs_t* const rspi,
00557                                           const rspi_config_t* const config)
00558 {
00559     uint16_t spcmd = k_rspi_spcmd_init;
00560
00561     if (config->spi_mode & k_rspi_spcmd_cpol_mask) {
00562         spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpol_pos); /* CPOL = 1 */
00563     }
00564     if (config->spi_mode & k_rspi_spcmd_cpha_mask) {
00565         spcmd |= (uint16_t)(k_rspi_bit_set « k_rspi_spcmd_cpha_pos); /* CPHA = 1 */
00566     }
00567
00568     if (config->use_16bit) {
00569         spcmd |= (k_rspi_spcmd_16bit « k_rspi_spcmd_spl_shift);
00570         rspi->spdcr = (uint8_t)(k_rspi_bit_set « k_rspi_spdcr_splw_pos); /* longword */
00571     } else {
00572         spcmd |= (k_rspi_spcmd_8bit « k_rspi_spcmd_spl_shift);
00573         rspi->spdcr = k_rspi_spdcr_byte_mode;
00574     }
00575
00576     return spcmd;
00577 }
00578
00579 /**
00580  * @brief Release the MSTPCRB module-stop bit for the requested channel
00581  *
00582  * @details
00583  * Wraps the protected-register PRCR unlock/lock around the MSTPCRB
00584  * write that clears the channel's clock gate.
00585  *
00586  * @param[in] channel Validated channel index
00587  *
00588  *
00589  * @pre channel < k_rspi_max_channels
00590  * @post Channel module clock running
00591  *
00592  * @since Version 1.0.0
00593  */
00594 static void internal_rspi_release_module_stop(const rspi_channel_t channel)
00595 {
00596     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00597     internal_set_mstpcrb_for_channel(channel, true);
00598     *prcr_reg() = k_rx_prcr_lock;
00599 }
00600
00601 rx_err_t rspi_init_peripheral(const rspi_channel_t channel, const rspi_config_t* config)
00602 {
00603     volatile rx_rspi_regs_t* rspi = nullptr;
00604     const rx_err_t check = internal_rspi_validate_init(channel, config, &rspi);
00605     if (check != k_rx_ok) {
00606         return check;
00607     }
00608
00609     /*
00610      * RSPI Peripheral Mode Register Write Sequence (RX72N HW Manual 38.3.6):
00611      * 1. Enable module clock (MSTPCRB)
00612      * 2. Disable SPI (SPCR.SPE=0)
00613      * 3. Configure SPCMD0 (CPOL/CPHA/data length)
00614      * 4. Configure SPDCE (byte vs longword access)
00615      * 5. Configure SPPCR (no loopback) BEFORE enabling SPI
00616      * 6. Enable SPI (SPCR.SPE=1) -- LAST write so partial config is never active
00617      */
00618     internal_rspi_release_module_stop(channel);
00619     rspi->spcr = k_rspi_spcr_disabled;
00620
00621     rspi->spcmd0 = internal_rspi_build_spcmd(rspi, config);
00622
00623     /* SPPCR MUST be written before SPCR.SPE per HW manual to avoid glitches
00624      * on COPI/CIPO lines during enable */
00625     rspi->sppcr = k_rspi_sppcr_no_loopback;
00626
00627     /* SPCR.SPE=1 MUST be the last register write so all configuration is
00628      * stable before the peripheral becomes active. MSTR=0 -> peripheral mode. */
00629     rspi->spcr = k_rspi_spcr_spe;
00630
00631     s_rspi_channel_initialized[channel] = true;
00632     rx_log_debug(s_tag, "RSPI peripheral mode initialized");
00633     return k_rx_ok;
00634 }
00635
00636 /**
00637  * @brief Wait for transmit buffer to become empty (SPTEF flag)
00638  *
00639  * @details
00640  * Polls the SPSR.SPTEF (SPI Transmit Buffer Empty Flag) in a busy-wait loop
00641  * with a configurable timeout. When SPTEF is set, the transmit buffer is empty
00642  * and ready to accept new data via SPDR. The timeout prevents infinite loops

```

```

00643 * if the SPI peripheral is stuck.
00644 *
00645 *
00646 *
00647 * @pre rspi must be a valid non-null pointer to initialized RSPI registers
00648 * @pre RSPI module clock must be enabled (MSTPCRB bit cleared)
00649 * @post No hardware state modified (read-only polling of SPSR)
00650 * @post Timeout counter decremented to reflect remaining iterations
00651 *
00652 * @note Not thread-safe. Caller must provide synchronization.
00653 * @warning Busy-wait loop; do not call from time-critical ISR context.
00654 *
00655 * @see internal_wait_rx_ready() Companion function for receive buffer
00656 * @see k_rspi_timeout_us Timeout constant
00657 *
00658 * @since Version 1.0.0
00659 */
00660 static rx_err_t internal_wait_tx_ready(volatile rx_rspi_regs_t* rspi)
00661 {
00662     /* Pre-condition 1: RSPI register pointer must be valid */
00663     RX_ASSERT(rspi != nullptr, "RSPI register pointer is nullptr");
00664
00665     /* Pre-condition 2: Timeout constant must be non-zero (compile-time check) */
00666     static_assert(k_rspi_timeout_us > k_rspi_timeout_zero, "Timeout constant must be non-zero");
00667     uint32_t timeout = k_rspi_timeout_us;
00668
00669     while (!(rspi->spsr & k_rspi_spsr_sptef) && timeout > k_rspi_timeout_zero) {
00670         timeout--;
00671     }
00672
00673     if (timeout == k_rspi_timeout_zero) {
00674         return k_rx_err_timeout;
00675     }
00676
00677     return k_rx_ok;
00678 }
00679
00680 /**
00681 * @brief Wait for receive buffer to become full (SPRF flag)
00682 *
00683 * @details
00684 * Polls the SPSR.SPRF (SPI Receive Buffer Full Flag) in a busy-wait loop
00685 * with a configurable timeout. When SPRF is set, received data is available
00686 * in the SPDR register and can be read. The timeout prevents infinite loops
00687 * if the SPI peripheral is stuck or no clock is being generated.
00688 *
00689 *
00690 *
00691 * @pre rspi must be a valid non-null pointer to initialized RSPI registers
00692 * @pre A transfer must be in progress or complete for SPRF to set
00693 * @post No hardware state modified (read-only polling of SPSR)
00694 * @post Timeout counter decremented to reflect remaining iterations
00695 *
00696 * @note Not thread-safe. Caller must provide synchronization.
00697 * @warning Busy-wait loop; do not call from time-critical ISR context.
00698 *
00699 * @see internal_wait_tx_ready() Companion function for transmit buffer
00700 * @see k_rspi_timeout_us Timeout constant
00701 *
00702 * @since Version 1.0.0
00703 */
00704 static rx_err_t internal_wait_rx_ready(volatile rx_rspi_regs_t* rspi)
00705 {
00706     /* Pre-condition 1: RSPI register pointer must be valid */
00707     RX_ASSERT(rspi != nullptr, "RSPI register pointer is nullptr");
00708
00709     /* Pre-condition 2: Timeout constant must be non-zero (compile-time check) */
00710     static_assert(k_rspi_timeout_us > k_rspi_timeout_zero, "Timeout constant must be non-zero");
00711     uint32_t timeout = k_rspi_timeout_us;
00712
00713     while (!(rspi->spsr & k_rspi_spsr_sprf) && timeout > k_rspi_timeout_zero) {
00714         timeout--;
00715     }
00716
00717     if (timeout == k_rspi_timeout_zero) {
00718         return k_rx_err_timeout;
00719     }
00720
00721     return k_rx_ok;
00722 }
00723
00724 /**
00725 * @brief Perform full-duplex SPI transfer in peripheral mode
00726 *
00727 * Executes a full-duplex (simultaneous transmit and receive) SPI transfer
00728 * on the specified RSPI channel operating in peripheral mode.
00729 * Transfers data byte-by-byte with timeout protection on both transmit

```



```

00730 * buffer ready and receive buffer full operations.
00731 *
00732 *
00733 *
00734 * @pre channel must be initialized via rspi_init_peripheral() before calling
00735 * @pre tx_data pointer must be non-NULL
00736 * @pre rx_data pointer must be non-NULL
00737 * @pre length must be in range [1, 65535]
00738 *
00739 * @post If successful, rx_data buffer is filled with received bytes
00740 *        corresponding to transmitted data at each byte offset
00741 */
00742 rx_err_t rspi_peripheral_transfer(const rspi_channel_t channel,
00743                                  const uint8_t* tx_data,
00744                                  uint8_t* rx_data,
00745                                  const uint16_t length)
00746 {
00747     RX_CHECK_NULL_PTR(tx_data, s_tag, "TX data pointer is nullptr");
00748     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
00749
00750     if (length == k_rspi_len_zero) {
00751         rx_log_error(s_tag, "Invalid transfer length");
00752         return k_rx_err_invalid_arg;
00753     }
00754
00755     /* Validate channel */
00756     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00757         rx_log_error(s_tag, "RSPI channel not initialized");
00758         return k_rx_err_invalid_state;
00759     }
00760
00761     /* Get RSPI base */
00762     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
00763     if (rspi == nullptr) {
00764         rx_log_error(s_tag, "Failed to get RSPI base address");
00765         return k_rx_err_invalid_arg;
00766     }
00767
00768     /* NASA Rule 2: Statically bounded loop */
00769     for (uint16_t i = k_rspi_loop_start; i < k_rspi_transfer_len_max; i++) {
00770         if (i >= length) {
00771             break;
00772         }
00773
00774         /* Wait for transmit buffer empty */
00775         rx_err_t err = internal_wait_tx_ready(rspi);
00776         if (err != k_rx_ok) {
00777             rx_log_error(s_tag, "SPI transmit timeout");
00778             return err;
00779         }
00780
00781         /* Write transmit data */
00782         rspi->spdr = tx_data[i];
00783
00784         /* Wait for receive buffer full */
00785         err = internal_wait_rx_ready(rspi);
00786         if (err != k_rx_ok) {
00787             rx_log_error(s_tag, "SPI receive timeout");
00788             return err;
00789         }
00790
00791         /* Read receive data */
00792         rx_data[i] = (uint8_t)rspi->spdr;
00793
00794         /* Clear status flags */
00795         rspi->spsr &= (uint8_t) ~(uint8_t)(k_rspi_spsr_sprf | k_rspi_spsr_ovrf);
00796     }
00797
00798     return k_rx_ok;
00799 }
00800
00801 /**
00802 * @brief Check if data is available in the RSPI receive buffer
00803 *
00804 * Polls the RSPI status register to determine if the receive buffer contains
00805 * data ready for reading. Returns the status without blocking.
00806 *
00807 *
00808 *
00809 * @pre Channel must be initialized via rspi_init_peripheral() before calling this function
00810 * @pre available must be a valid non-nullptr
00811 */
00812 rx_err_t rspi_peripheral_read_available(const rspi_channel_t channel, bool* available)
00813 {
00814     RX_CHECK_NULL_PTR(available, s_tag, "Available pointer is nullptr");
00815
00816     /* Validate channel */

```



```

00817 if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00818     rx_log_error(s_tag, "RSPi channel not initialized");
00819     return k_rx_err_invalid_state;
00820 }
00821
00822 /* Get RSPi base */
00823 const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00824 if (rspi == nullptr) {
00825     *available = false;
00826     return k_rx_err_invalid_arg;
00827 }
00828
00829 /* Check if receive buffer has data */
00830 *available = (rspi->spsr & k_rspi_spsr_sprf) != 0;
00831
00832 return k_rx_ok;
00833 }
00834
00835 /**
00836  * @brief Check if transmit buffer is ready for data
00837  *
00838  * Polls the RSPi status register to determine if the transmit buffer is empty
00839  * and ready to accept new data. Returns the status without blocking.
00840  *
00841  *
00842  *
00843  * @pre Channel must be initialized via rspi_init_peripheral()
00844  * @pre ready must be a valid non-nullptr
00845  */
00846 rx_err_t rspi_peripheral_write_ready(const rspi_channel_t channel, bool* ready)
00847 {
00848     RX_CHECK_NULL_PTR(ready, s_tag, "Ready pointer is nullptr");
00849
00850     /* Validate channel */
00851     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_channel_initialized[channel]) {
00852         rx_log_error(s_tag, "RSPi channel not initialized");
00853         return k_rx_err_invalid_state;
00854     }
00855
00856     /* Get RSPi base */
00857     const volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00858     if (rspi == nullptr) {
00859         *ready = false;
00860         return k_rx_err_invalid_arg;
00861     }
00862
00863     /* Check if transmit buffer is empty */
00864     *ready = (rspi->spsr & k_rspi_spsr_sptef) != 0;
00865
00866     return k_rx_ok;
00867 }
00868
00869 /**
00870  * @brief Deinitialize and disable RSPi peripheral
00871  *
00872  * Disables the specified RSPi channel, clears the initialization flag, and
00873  * optionally disables the module (via module stop bit) to reduce power consumption.
00874  *
00875  *
00876  *
00877  * @pre Channel should have been initialized via rspi_init_peripheral() previously
00878  *
00879  * @post If successful:
00880  *      - RSPi peripheral is disabled (SPCR.SPE = 0)
00881  *      - s_rspi_channel_initialized[channel] is set to false
00882  *      - Channel is no longer available for SPI operations
00883  */
00884 rx_err_t rspi_deinit(const rspi_channel_t channel)
00885 {
00886     /* Validate channel */
00887     if ((uint8_t)channel >= k_rspi_max_channels) {
00888         rx_log_error(s_tag, "Invalid RSPi channel");
00889         return k_rx_err_invalid_arg;
00890     }
00891
00892     /* Get RSPi base */
00893     volatile rx_rspi_regs_t* rspi = internal_get_rspi_base(channel);
00894     if (rspi == nullptr) {
00895         return k_rx_err_invalid_arg;
00896     }
00897
00898     /*
00899      * RSPi Deinitialization Register Write Sequence
00900      * =====
00901      * The deinit order is the reverse of initialization:
00902      * 1. Disable SPI (SPCR.SPE=0) - stop all bus activity before removing clock
00903      * 2. Disable module clock (MSTPCRB) - safe to gate clock only after SPI is idle
00904     */

```

```

00904 * 3. Clear software state - mark channel uninitialized after hardware is off
00905 */
00906
00907 /* Step 1: Disable SPI to stop all bus activity */
00908 rspi->spcr = k_rspi_spcr_disabled;
00909
00910 /* Step 2: Disable RSPI module clock (set module stop bit) - must be done
00911 * after SPI is disabled to avoid undefined behavior from clock gating
00912 * during an active transfer */
00913 *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00914 internal_set_mstpcrb_for_channel(channel, false);
00915
00916 *prcr_reg() = k_rx_prcr_lock;
00917
00918 /* Step 3: Mark channel as uninitialized after hardware is safely off */
00919 s_rspi_channel_initialized[channel] = false;
00920
00921 rx_log_debug(s_tag, "RSPI deinitialized");
00922
00923 return k_rx_ok;
00924 }
00925
00926 /*
=====
00927 * RSPI Controller Mode Functions
00928 *
=====
00929 */
00930
00931 /**
00932 * @brief Cycle-accurate timing delay using inline NOP instructions
00933 *
00934 * @details
00935 * Produces precise delays using inline assembly NOP operations for SPI
00936 * timing requirements (CS setup/hold times). Each NOP cycle is 4.17 ns
00937 * at 240 MHz core clock.
00938 *
00939 *
00940 * @pre cycles must be non-zero
00941 * @pre cycles must not exceed k_rspi_max_delay_cycles
00942 * @post Minimum delay of (cycles * 4.17 ns) has elapsed
00943 * @post No hardware state modified
00944 *
00945 * @note Timing is cycle-accurate on RX72N @ 240 MHz (4.17 ns per cycle)
00946 * @note Using inline ASM ensures compiler does not optimize away the loop
00947 * @warning Only accurate when compiled with known optimization level (-O2)
00948 *
00949 * @see internal_controller_assert_cs_with_setup() Uses for CS setup time
00950 * @see internal_controller_deassert_cs_with_hold() Uses for CS hold time
00951 *
00952 * @since Version 1.0.0
00953 */
00954 static void internal_timing_delay(const uint16_t cycles)
00955 {
00956     /* Rule 5: Pre-condition validation */
00957     RX_ASSERT(cycles != 0, "Timing delay cycles cannot be zero");
00958     RX_ASSERT(cycles <= k_rspi_max_delay_cycles, "Timing delay cycles exceeds sanity limit");
00959
00960     /*
00961     * NASA Rule 2: Loop bounded by k_rspi_max_delay_cycles (compile-time constant)
00962     * Early break when delay >= cycles to execute only the required iterations.
00963     */
00964     volatile uint32_t delay;
00965     for (delay = k_rspi_loop_start; delay < k_rspi_max_delay_cycles; delay++) {
00966         if (delay >= cycles) {
00967             break;
00968         }
00969         __asm__ volatile("nop");
00970     }
00971 }
00972
00973 /**
00974 * @brief Configure SPCMD register for SPI mode (CPOL/CPHA)
00975 *
00976 * @details
00977 * Sets CPOL and CPHA bits in an SPCMD register value based on SPI mode (0-3).
00978 * The SPI mode encodes clock polarity (CPOL) and clock phase (CPHA) per the
00979 * standard SPI specification. Bits are set via bitwise OR so pre-existing
00980 * fields in the SPCMD value (such as data length) are preserved.
00981 *
00982 * | Mode | CPOL | CPHA | Clock Idle | Data Sampled |
00983 * |-----|-----|-----|-----|-----|
00984 * | 0 | 0 | 0 | Low | Rising edge |
00985 * | 1 | 0 | 1 | Low | Falling edge |
00986 * | 2 | 1 | 0 | High | Falling edge |
00987 * | 3 | 1 | 1 | High | Rising edge |
00988 *

```

```

00989 *
00990 *
00991 * @pre spcmd must be a valid non-null pointer
00992 * @pre spi_mode must be in range [0, 3] (only bits 0-1 used)
00993 * @post spcmd CPOL/CPHA bits set according to spi_mode
00994 * @post Other spcmd bits remain unchanged
00995 *
00996 * @note Not thread-safe. Caller must provide synchronization.
00997 *
00998 * @see rspi_init_controller() Calls this during controller init
00999 * @see rspi_init_peripheral() Performs equivalent inline configuration
01000 *
01001 * @since Version 1.0.0
01002 */
01003 static void internal_configure_spcmd(uint16_t* spcmd, const uint8_t spi_mode)
01004 {
01005     /* Rule 5: Pre-condition validation */
01006     RX_ASSERT(spcmd != nullptr, "spcmd pointer is nullptr");
01007     RX_ASSERT((spi_mode & ~(k_rspi_spcmd_cpol_mask | k_rspi_spcmd_cpha_mask)) == 0,
01008         "spi_mode contains invalid bits");
01009
01010     /* Configure SPI mode (CPOL and CPHA) */
01011     if (spi_mode & k_rspi_spcmd_cpol_mask) {
01012         *spcmd |= (uint16_t)(k_rspi_bit_set << k_rspi_spcmd_cpol_pos); /* CPOL = 1 */
01013     }
01014     if (spi_mode & k_rspi_spcmd_cpha_mask) {
01015         *spcmd |= (uint16_t)(k_rspi_bit_set << k_rspi_spcmd_cpha_pos); /* CPHA = 1 */
01016     }
01017 }
01018
01019 /**
01020 * @brief Calculate SPBR value for desired SPI clock frequency
01021 *
01022 * @details
01023 * Computes the RSPI Bit Rate Register (SPBR) value required to achieve
01024 * the desired SPI clock frequency. Uses the formula:
01025 * SPBR = (PCLKB / (2 * freq_hz)) - 1
01026 * with BRDV=0 (no additional divisor). The result is validated against
01027 * the hardware-supported range [0, 255].
01028 *
01029 *
01030 *
01031 * @pre spbr must be a valid non-null pointer
01032 * @pre freq_hz must be in range [100000, 10000000] Hz
01033 * @post *spbr contains the calculated SPBR value if k_rx_ok returned
01034 * @post *spbr is unchanged on error
01035 *
01036 * @note Not thread-safe. Caller must provide synchronization.
01037 *
01038 * @see rspi_init_controller() Calls this to configure clock frequency
01039 * @see k_pclkb_hz (rx72n_clock.h) PCLKB clock frequency used in calculation
01040 *
01041 * @since Version 1.0.0
01042 */
01043 static rx_err_t internal_calculate_spbr(const uint32_t freq_hz, uint8_t* spbr)
01044 {
01045     RX_CHECK_NULL_PTR(spbr, s_tag, "SPBR output pointer is nullptr");
01046
01047     if (freq_hz < k_rspi_min_freq_hz || freq_hz > k_rspi_max_freq_hz) {
01048         rx_log_error(s_tag, "SPI frequency out of range (100kHz-10MHz)");
01049         return k_rx_err_invalid_arg;
01050     }
01051
01052     /* SPBR formula: freq = PCLKB / (2 * (SPBR + 1) * 2^BRDV)
01053      * With BRDV=0: freq = PCLKB / (2 * (SPBR + 1))
01054      * Solving for SPBR: SPBR = (PCLKB / (2 * freq)) - 1 */
01055     const uint32_t divisor = k_rspi_spbr_divisor * freq_hz;
01056     const uint32_t spbr_val = (k_pclkb_hz / divisor) - k_rspi_spbr_offset;
01057
01058     /* Reject frequencies that are too low (would require SPBR > 255) */
01059     if (spbr_val > k_rspi_spbr_max) {
01060         rx_log_error(s_tag, "Requested SPI frequency too low; minimum achievable is ~117 kHz");
01061         return k_rx_err_invalid_arg;
01062     }
01063
01064     *spbr = (uint8_t)spbr_val;
01065
01066     return k_rx_ok;
01067 }
01068
01069 /**
01070 * @brief Configure GPIO pin for chip select output
01071 *
01072 * @details
01073 * Configures a GPIO pin as a digital output for use as an active-low SPI
01074 * chip select signal. Extracts port and pin numbers from the type-safe
01075 * rx_port_pin_t encoding, sets the pin direction to output via PDR, and

```

```

01076 * drives the pin high (CS inactive) via PODR.
01077 *
01078 *
01079 *
01080 * @pre pin_config must encode a valid port (0-J) and pin (0-7)
01081 * @pre Port hardware must be available and not module-stopped
01082 * @post GPIO PDR bit set (output direction)
01083 * @post GPIO PODR bit set (CS inactive / high)
01084 *
01085 * @note Not thread-safe. Caller must provide synchronization for port access.
01086 *
01087 * @see rspi_init_controller() Calls this during controller initialization
01088 * @see rx_port_from_pin() Extracts port number from rx_port_pin_t
01089 * @see rx_pin_from_pin() Extracts pin number from rx_port_pin_t
01090 *
01091 * @since Version 1.0.0
01092 */
01093 static rx_err_t internal_configure_cs_gpio(const rx_port_pin_t pin_config)
01094 {
01095     const uint8_t port = rx_port_from_pin(pin_config);
01096     const uint8_t pin = rx_pin_from_pin(pin_config);
01097
01098     volatile rx_port_regs_t* const port_regs = rx_port_get_base(port);
01099
01100     if (port_regs == nullptr) {
01101         rx_log_error(s_tag, "Invalid port number for CS pin");
01102         return k_rx_err_invalid_arg;
01103     }
01104
01105     /* Validate pin (lower bound check omitted - pin is uint8_t, k_rx_pin_min == 0,
01106      * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
01107     if (pin > k_rx_pin_max) {
01108         rx_log_error(s_tag, "Invalid pin number for CS pin (valid range 0-7)");
01109         return k_rx_err_invalid_arg;
01110     }
01111
01112     /* Configure as output */
01113     port_regs->pdr |= ((uint32_t)k_rspi_bit_set « pin);
01114
01115     /* Set high (CS inactive - active low) */
01116     port_regs->podr |= ((uint32_t)k_rspi_bit_set « pin);
01117
01118     return k_rx_ok;
01119 }
01120
01121 /**
01122  * @brief Validate controller initialization arguments
01123  *
01124  * @details
01125  * Performs comprehensive input validation for rspi_init_controller() before
01126  * any hardware modification occurs. Checks null pointer, channel range,
01127  * duplicate initialization, and SPI mode range. Centralizes validation
01128  * logic to keep the public API function concise.
01129  *
01130  *
01131  *
01132  * @pre config may be nullptr (will be detected and rejected)
01133  * @pre channel may be out of range (will be detected and rejected)
01134  * @post No hardware state modified (validation only)
01135  * @post s_rspi_controller_initialized[] not modified
01136  *
01137  * @note Not thread-safe. Caller must provide synchronization.
01138  *
01139  * @see rspi_init_controller() Calls this as first step of initialization
01140  * @see internal_prepare_controller() Called after this validation passes
01141  *
01142  * @since Version 1.0.0
01143  */
01144 static rx_err_t internal_validate_controller_args(const rspi_channel_t channel,
01145                                                  const rspi_controller_config_t* config)
01146 {
01147     RX_CHECK_NULL_PTR(config, s_tag, "Controller config pointer is nullptr");
01148
01149     /* Validate channel */
01150     if ((uint8_t)channel >= k_rspi_max_channels) {
01151         rx_log_error(s_tag, "Invalid RSPI channel");
01152         return k_rx_err_invalid_arg;
01153     }
01154
01155     /* Check if channel is already initialized */
01156     if (s_rspi_controller_initialized[channel]) {
01157         rx_log_error(s_tag, "RSPI channel already initialized");
01158         return k_rx_err_invalid_state;
01159     }
01160
01161     /* Validate mode (0-3) */
01162     if ((uint8_t)config->spi_mode > k_rspi_mode_max) {

```

```

01163     rx_log_error(s_tag, "Invalid SPI mode (must be 0-3)");
01164     return k_rx_err_invalid_arg;
01165 }
01166
01167 return k_rx_ok;
01168 }
01169
01170 /**
01171  * @brief Prepare controller hardware resources
01172  *
01173  * @details
01174  * Calculates the SPBR register value for the requested clock frequency,
01175  * validates the RSPI register base address for the given channel, and
01176  * configures the CS GPIO pin as an active-low output. Operations are
01177  * ordered so that a failure in SPBR calculation avoids unnecessary GPIO
01178  * configuration, and RSPI base validation occurs before GPIO setup to
01179  * avoid leaving GPIO configured if the channel is invalid.
01180  *
01181  *
01182  *
01183  * @pre config must be validated by internal_validate_controller_args() first
01184  * @pre channel must be in range [0, k_rspi_max_channels)
01185  * @post *out_rspi points to valid RSPI register base on success
01186  * @post *out_spbr contains calculated SPBR value on success
01187  * @post CS GPIO pin configured as output, driven high (inactive) on success
01188  *
01189  * @note Not thread-safe. Caller must provide synchronization.
01190  *
01191  * @see internal_validate_controller_args() Must be called before this function
01192  * @see internal_configure_registers() Called after this to program hardware
01193  * @see internal_calculate_spbr() Called internally for frequency calculation
01194  *
01195  * @since Version 1.0.0
01196  */
01197 static rx_err_t internal_prepare_controller(const rspi_channel_t channel,
01198                                             const rspi_controller_config_t* config,
01199                                             volatile rx_rspi_regs_t** out_rspi,
01200                                             uint8_t* out_spbr)
01201 {
01202     /* Calculate SPBR for requested frequency */
01203     rx_err_t err = internal_calculate_spbr(config->freq_hz, out_spbr);
01204     if (err != k_rx_ok) {
01205         return err;
01206     }
01207
01208     /* Get RSPI base - validate before configuring GPIO to avoid leaving GPIO
01209      * configured if RSPI base lookup fails */
01210     *out_rspi = internal_get_rspi_base(channel);
01211     if (*out_rspi == nullptr) {
01212         return k_rx_err_invalid_arg;
01213     }
01214
01215     /* Configure CS GPIO pin (after RSPI base validation) */
01216     err = internal_configure_cs_gpio(config->cs);
01217     if (err != k_rx_ok) {
01218         return err;
01219     }
01220
01221     return k_rx_ok;
01222 }
01223
01224 /**
01225  * @brief Configure RSPI hardware registers for controller mode
01226  *
01227  * @details
01228  * Performs the complete RSPI hardware register configuration sequence for
01229  * controller mode. Enables the module clock, disables SPI for safe register
01230  * modification, programs bit rate, command, data control and pin control
01231  * registers, then enables SPI in controller mode. Also stores the CS pin
01232  * configuration and marks the channel as initialized. The register write
01233  * order follows the RX72N Hardware Manual Section 38.3.6 initialization
01234  * procedure.
01235  *
01236  *
01237  *
01238  * @pre rspi must be a valid non-null pointer from internal_get_rspi_base()
01239  * @pre spbr must be pre-calculated by internal_calculate_spbr()
01240  * @pre spcmd must be pre-configured with CPOL/CPHA and data length bits
01241  * @post RSPI module clock enabled, registers configured, SPI enabled
01242  * @post s_rspi_cs_config[channel] stores CS port and pin
01243  * @post s_rspi_controller_initialized[channel] set to true
01244  *
01245  * @note Not thread-safe. Caller must provide synchronization.
01246  * @warning PRCR register protection is temporarily unlocked during this call.
01247  *
01248  * @see internal_prepare_controller() Provides rspi, spbr, and validates channel
01249  * @see rspi_init_controller() Public API that orchestrates the full init flow

```

```

01250 *
01251 * @since Version 1.0.0
01252 */
01253 static rx_err_t internal_configure_registers(const rspi_channel_t channel,
01254                                             volatile rx_rspi_regs_t* rspi,
01255                                             uint8_t spbr,
01256                                             uint16_t spcmd,
01257                                             const rspi_controller_config_t* config)
01258 {
01259     /*
01260     * RSPI Controller Mode Register Write Sequence
01261     * =====
01262     * The register write order follows the RX72N Hardware Manual Section 38.3.6
01263     * (RSPI Initialization Procedure). The sequence is:
01264     *
01265     * 1. Enable module clock (MSTPCRB) - module must be powered before register access
01266     * 2. Disable SPI (SPCR.SPE=0) - registers must not be modified while SPI is active
01267     * 3. Configure SPBR - set bit rate divider while SPI is disabled
01268     * 4. Configure SPDCR and SPCMD0 - set data access width and mode while disabled
01269     * 5. Configure SPPCR - set pin control (no loopback) BEFORE enabling SPI
01270     * 6. Enable SPI (SPCR.SPE=1, MSTR=1) - must be the LAST register write to avoid
01271     *    partial configuration or clock glitches on the SPI bus
01272     */
01273
01274     /* Step 1: Enable RSPI module clock (clear module stop bit in MSTPCRB) */
01275     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
01276     internal_set_mstpcrb_for_channel(channel, true);
01277     *prcr_reg() = k_rx_prcr_lock;
01278
01279     /* Step 2: Disable SPI before modifying configuration registers */
01280     rspi->spcr = k_rspi_spcr_disabled;
01281
01282     /* Step 3: Configure bit rate divider (freq = PCLKB / (2 * (SPBR + 1))) */
01283     rspi->spbr = spbr;
01284
01285     /* Step 4: Configure data access width (longword/32-bit mode) and command register */
01286     rspi->spdcr =
01287         (uint8_t)(k_rspi_bit_set « k_rspi_spdcr_splw_pos); /* Longword (32-bit) access mode */
01288     rspi->spcmd0 = spcmd;
01289
01290     /* Step 5: Configure pin control (no loopback) - MUST be written before SPCR.SPE
01291     * per HW manual to avoid glitches on COPI/CIPO lines during enable */
01292     rspi->sppcr = k_rspi_sppcr_no_loopback;
01293
01294     /* Step 6: Enable SPI in controller mode (MSTR=1, SPE=1) - MUST be the last register
01295     * write so all configuration is stable before the controller drives the bus */
01296     rspi->spcr = k_rspi_spcr_spe | k_rspi_spcr_mstr;
01297
01298     /* Store CS pin configuration */
01299     s_rspi_cs_config[channel].port = rx_port_from_pin(config->cs);
01300     s_rspi_cs_config[channel].pin = rx_pin_from_pin(config->cs);
01301
01302     /* Mark channel as initialized in controller mode */
01303     s_rspi_controller_initialized[channel] = true;
01304
01305     rx_log_debug(s_tag, "RSPI controller mode initialized");
01306
01307     return k_rx_ok;
01308 }
01309
01310 /**
01311 * @brief Initialize RSPI channel in controller mode
01312 *
01313 * Configures the specified RSPI channel as a SPI controller to communicate
01314 * with SPI peripheral devices. Enables the module, configures clock frequency,
01315 * SPI mode, and chip select GPIO pin.
01316 *
01317 *
01318 *
01319 * @pre config must be non-NULL
01320 * @pre channel < k_rspi_max_channels
01321 * @pre channel must not be already initialized
01322 * @pre CS GPIO port and pin must be valid and available
01323 *
01324 * @post If successful:
01325 *   - RSPI module is enabled via internal_set_mstpcrb_for_channel()
01326 *   - *prcr_reg() is unlocked/locked for register protection
01327 *   - RSPI registers configured: spbr, spcmd0, spcr, spdcr, sppcr
01328 *   - CS GPIO configured as output, driven high (inactive)
01329 *   - s_rspi_cs_config[channel] stores CS port/pin
01330 *   - s_rspi_controller_initialized[channel] = true
01331 * @post On error:
01332 *   - Hardware state is unchanged
01333 *   - Channel remains uninitialized
01334 *
01335 * @note Bit rate calculation: freq = PCLKB / (2 * (SPBR + 1))
01336 * @note Always configures 16-bit data frames with word access mode

```

```

01337 * @note CS is active-low (asserted=0, deasserted=1)
01338 */
01339 rx_err_t rspi_init_controller(const rspi_channel_t channel, const rspi_controller_config_t* config)
01340 {
01341     /* Validate all arguments and preconditions */
01342     rx_err_t err = internal_validate_controller_args(channel, config);
01343     if (err != k_rx_ok) {
01344         return err;
01345     }
01346
01347     /* Prepare hardware resources (SPBR, RSPI base, CS GPIO) */
01348     volatile rx_rspi_regs_t* rspi = nullptr;
01349     uint8_t spbr = k_rspi_spbr_init;
01350     err = internal_prepare_controller(channel, config, &rspi, &spbr);
01351     if (err != k_rx_ok) {
01352         return err;
01353     }
01354
01355     /* Configure SPI mode (CPOL and CPHA) */
01356     uint16_t spcmd = k_rspi_spcmd_init;
01357     internal_configure_spcmd(&spcmd, config->spi_mode);
01358
01359     /* Configure 16-bit data length */
01360     spcmd |= (k_rspi_spcmd_16bit « k_rspi_spcmd_spl_shift);
01361
01362     /* Configure and enable RSPI hardware registers */
01363     return internal_configure_registers(channel, rspi, spbr, spcmd, config);
01364 }
01365
01366 /**
01367  * @brief Set chip select line state for RSPI controller channel
01368  *
01369  * Controls the GPIO chip select pin for the specified RSPI channel operating
01370  * in controller mode. CS is active-low (asserted low, deasserted high).
01371  *
01372  *
01373  *
01374  * @pre Channel must be initialized via rspi_init_controller()
01375  * @pre channel < k_rspi_max_channels
01376  * @pre s_rspi_controller_initialized[channel] == true
01377  *
01378  * @post GPIO port_regs->podr is modified to assert/deassert CS pin
01379  *
01380  * @note Reads s_rspi_cs_config[channel] for port and pin configuration
01381  */
01382 rx_err_t rspi_controller_set_cs(const rspi_channel_t channel, const bool active)
01383 {
01384     /* Validate channel */
01385     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01386         rx_log_error(s_tag, "RSPI controller channel not initialized");
01387         return k_rx_err_invalid_state;
01388     }
01389
01390     const uint8_t port = s_rspi_cs_config[channel].port;
01391     const uint8_t pin = s_rspi_cs_config[channel].pin;
01392     volatile rx_port_regs_t* const port_regs = rx_port_get_base(port);
01393
01394     if (port_regs == nullptr) {
01395         return k_rx_err_invalid_arg;
01396     }
01397
01398     /* Compute pin mask once (identical for both assert and deassert) */
01399     const uint8_t pin_mask = (uint8_t)((uint32_t)k_rspi_bit_set « pin);
01400
01401     if (active) {
01402         /* CS active (low) */
01403         port_regs->podr &= (uint8_t)~pin_mask;
01404     } else {
01405         /* CS inactive (high) */
01406         port_regs->podr |= pin_mask;
01407     }
01408
01409     return k_rx_ok;
01410 }
01411
01412 /**
01413  * @brief Assert CS with setup delay
01414  *
01415  * @details
01416  * Asserts the chip select line (drives low for active-low CS) and then
01417  * applies a cycle-accurate setup time delay using NOP instructions. The
01418  * setup delay ensures the SPI peripheral has sufficient time to recognize
01419  * CS assertion before clock transitions begin. Delay is calibrated for
01420  * ~300ns at 240 MHz core clock.
01421  *
01422  *
01423  */

```



```

01424 * @pre Channel must be initialized via rspi_init_controller()
01425 * @pre s_rspi_controller_initialized[channel] must be true
01426 * @post CS GPIO pin driven low (active)
01427 * @post Minimum setup delay of ~300ns has elapsed
01428 *
01429 * @note Not thread-safe. Caller must provide synchronization.
01430 *
01431 * @see internal_controller_deassert_cs_with_hold() Companion deassert function
01432 * @see internal_timing_delay() Provides cycle-accurate delay
01433 * @see k_rspi_cs_setup_delay Setup delay iteration count
01434 *
01435 * @since Version 1.0.0
01436 */
01437 static rx_err_t internal_controller_assert_cs_with_setup(const rspi_channel_t channel)
01438 {
01439     /* Rule 5: Pre-condition validation */
01440     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
01441              (channel == k_rspi_channel_2),
01442              "Invalid RSPI channel");
01443     RX_ASSERT(s_rspi_controller_initialized[channel], "RSPI controller channel not initialized");
01444
01445     /* Assert CS (active low) */
01446     const rx_err_t err = rspi_controller_set_cs(channel, true);
01447     if (err != k_rx_ok) {
01448         return err;
01449     }
01450
01451     /**
01452      * CS setup time delay (~300ns).
01453      * Inline assembly NOP is used to produce precise, cycle-accurate timing that
01454      * C-level delays cannot guarantee. SPI peripherals typically require minimum CS
01455      * setup time before SCLK transitions. Loop count k_rspi_cs_setup_delay is
01456      * calibrated for RX72N at 240 MHz (4.17ns per cycle).
01457      */
01458     internal_timing_delay(k_rspi_cs_setup_delay);
01459
01460     return k_rx_ok;
01461 }
01462
01463 /**
01464 * @brief Deassert CS with hold delay
01465 *
01466 * @details
01467 * Applies a cycle-accurate hold time delay using NOP instructions, then
01468 * deasserts the chip select line (drives high for active-low CS). The hold
01469 * delay ensures the SPI peripheral has sufficient time to latch the final
01470 * data bit before CS is released. Delay is calibrated for ~300ns at 240 MHz
01471 * core clock.
01472 *
01473 *
01474 *
01475 * @pre Channel must be initialized via rspi_init_controller()
01476 * @pre s_rspi_controller_initialized[channel] must be true
01477 * @post Minimum hold delay of ~300ns has elapsed
01478 * @post CS GPIO pin driven high (inactive)
01479 *
01480 * @note Not thread-safe. Caller must provide synchronization.
01481 *
01482 * @see internal_controller_assert_cs_with_setup() Companion assert function
01483 * @see internal_timing_delay() Provides cycle-accurate delay
01484 * @see k_rspi_cs_hold_delay Hold delay iteration count
01485 *
01486 * @since Version 1.0.0
01487 */
01488 static rx_err_t internal_controller_deassert_cs_with_hold(const rspi_channel_t channel)
01489 {
01490     /* Rule 5: Pre-condition validation */
01491     RX_ASSERT((channel == k_rspi_channel_0) || (channel == k_rspi_channel_1) ||
01492              (channel == k_rspi_channel_2),
01493              "Invalid RSPI channel");
01494     RX_ASSERT(s_rspi_controller_initialized[channel], "RSPI controller channel not initialized");
01495
01496     /**
01497      * CS hold time delay (~300ns).
01498      * Inline assembly NOP is used to produce precise, cycle-accurate timing that
01499      * C-level delays cannot guarantee. SPI peripherals typically require minimum CS
01500      * hold time after final SCLK transition before deassertion. Loop count
01501      * k_rspi_cs_hold_delay is calibrated for RX72N at 240 MHz (4.17ns per cycle).
01502      */
01503     internal_timing_delay(k_rspi_cs_hold_delay);
01504
01505     /* Deassert CS (inactive high) */
01506     return rspi_controller_set_cs(channel, false);
01507 }
01508
01509 /**
01510 * @brief Perform core 16-bit SPI transfer operation

```



```

01511 *
01512 * @details
01513 * Executes a single full-duplex 16-bit SPI transfer without chip select
01514 * management. Waits for the transmit buffer to become empty, writes 16-bit
01515 * data to SPDR, waits for the receive buffer to fill, reads the 16-bit
01516 * response from SPDR, and clears SPRF and OVRF status flags. This is the
01517 * inner transfer primitive used by rspi_controller_transfer_16bit() after
01518 * CS assertion and before CS deassertion.
01519 *
01520 *
01521 *
01522 * @pre rspi must be a valid non-null pointer to initialized RSPI registers
01523 * @pre SPI must be enabled (SPCR.SPE=1) before calling
01524 * @post *rx_data contains the 16-bit value received during the transfer
01525 * @post SPSR flags SPRF and OVRF are cleared
01526 *
01527 * @note Not thread-safe. Caller must provide synchronization.
01528 *
01529 * @see rspi_controller_transfer_16bit() Public API that wraps this with CS handling
01530 * @see internal_wait_tx_ready() TX buffer polling with timeout
01531 * @see internal_wait_rx_ready() RX buffer polling with timeout
01532 *
01533 * @since Version 1.0.0
01534 */
01535 static rx_err_t internal_controller_do_16bit_transfer(volatile rx_rspi_regs_t* rspi,
01536                                                     uint16_t tx_data,
01537                                                     uint16_t* rx_data)
01538 {
01539     /* Rule 5: Pre-condition validation */
01540     RX_CHECK_NULL_PTR(rspi, s_tag, "RSPI register pointer is nullptr");
01541     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
01542
01543     /* Wait for transmit buffer empty */
01544     rx_err_t err = internal_wait_tx_ready(rspi);
01545     if (err != k_rx_ok) {
01546         rx_log_error(s_tag, "SPI controller transmit timeout");
01547         return err;
01548     }
01549
01550     /* Write 16-bit transmit data */
01551     rspi->spdr = tx_data;
01552
01553     /* Wait for receive buffer full */
01554     err = internal_wait_rx_ready(rspi);
01555     if (err != k_rx_ok) {
01556         rx_log_error(s_tag, "SPI controller receive timeout");
01557         return err;
01558     }
01559
01560     /* Read 16-bit receive data */
01561     *rx_data = (uint16_t)rspi->spdr;
01562
01563     /* Clear status flags */
01564     rspi->spsr &= ~(uint8_t)(k_rspi_spsr_sprf | k_rspi_spsr_ovrf);
01565
01566     return k_rx_ok;
01567 }
01568
01569 /**
01570 * @brief Perform 16-bit SPI transfer in controller mode with CS handling
01571 *
01572 * Executes a full-duplex 16-bit SPI transfer with automatic chip select
01573 * assertion/deassertion and timing delays.
01574 *
01575 * Sequence:
01576 * 1. Assert CS (active low) with setup delay (~300ns)
01577 * 2. Wait for TX buffer ready, then transmit 16-bit data
01578 * 3. Wait for RX buffer full, then read 16-bit response
01579 * 4. Hold CS for minimum hold time (~300ns)
01580 * 5. Deassert CS (inactive high)
01581 * 6. Clear status flags
01582 *
01583 *
01584 *
01585 * @pre Channel must be initialized via rspi_init_controller()
01586 * @pre rx_data must be a valid non-nullptr
01587 *
01588 * @post If successful:
01589 *   - rx_data is filled with received 16-bit value
01590 *   - Status flags (SPRF, OVRF) are cleared
01591 *   - CS is deasserted (high)
01592 * @post On error:
01593 *   - CS is deasserted if possible
01594 *   - rx_data contents are undefined
01595 *
01596 * @note CS setup delay: k_rspi_cs_setup_delay NOP loops (~300ns)
01597 * @note CS hold delay: k_rspi_cs_hold_delay NOP loops (~300ns)

```

```

01598 * @note Uses inline assembly NOP for precise timing control
01599 */
01600 rx_err_t rspi_controller_transfer_16bit(const rspi_channel_t channel,
01601                                     const uint16_t tx_data,
01602                                     uint16_t* const rx_data)
01603 {
01604     RX_CHECK_NULL_PTR(rx_data, s_tag, "RX data pointer is nullptr");
01605
01606     /* Validate channel */
01607     if ((uint8_t)channel >= k_rspi_max_channels || !s_rspi_controller_initialized[channel]) {
01608         rx_log_error(s_tag, "RSPi controller channel not initialized");
01609         return k_rx_err_invalid_state;
01610     }
01611
01612     /* Get RSPi base */
01613     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01614     if (rspi == nullptr) {
01615         return k_rx_err_invalid_arg;
01616     }
01617
01618     /* Assert CS with setup delay */
01619     rx_err_t err = internal_controller_assert_cs_with_setup(channel);
01620     if (err != k_rx_ok) {
01621         return err;
01622     }
01623
01624     /* Perform 16-bit TX/RX transfer */
01625     err = internal_controller_do_16bit_transfer(rspi, tx_data, rx_data);
01626     if (err != k_rx_ok) {
01627         /* Deassert CS on error (ignore CS deassert errors) */
01628         (void)rspi_controller_set_cs(channel, false);
01629         return err;
01630     }
01631
01632     /* Deassert CS with hold delay */
01633     return internal_controller_deassert_cs_with_hold(channel);
01634 }
01635
01636 /**
01637  * @brief Deinitialize RSPi controller mode
01638  *
01639  * Disables the RSPi controller, deasserts chip select, clears configuration,
01640  * and stops the module to reduce power consumption.
01641  *
01642  *
01643  *
01644  * @pre Should be called after rspi_init_controller() to properly clean up
01645  *
01646  * @post If successful:
01647  *   - CS is deasserted via rspi_controller_set_cs(channel, false)
01648  *   - RSPi is disabled (rspi->spcr = k_rspi_spcr_disabled)
01649  *   - Module clock is stopped via internal_set_mstpcrb_for_channel()
01650  *   - s_rspi_cs_config[channel] is cleared (port=0, pin=0)
01651  *   - s_rspi_controller_initialized[channel] = false
01652  *   - Register protection is unlocked/locked via *prcr_reg()
01653  *
01654  * @note Thread-safe only if caller ensures no concurrent access to the same channel
01655  */
01656 rx_err_t rspi_controller_deinit(const rspi_channel_t channel)
01657 {
01658     /* Validate channel */
01659     if ((uint8_t)channel >= k_rspi_max_channels) {
01660         rx_log_error(s_tag, "Invalid RSPi channel");
01661         return k_rx_err_invalid_arg;
01662     }
01663
01664     /* Get RSPi base */
01665     volatile rx_rspi_regs_t* const rspi = internal_get_rspi_base(channel);
01666     if (rspi == nullptr) {
01667         return k_rx_err_invalid_arg;
01668     }
01669
01670     /*
01671      * RSPi Controller Deinitialization Sequence
01672      * =====
01673      * The deinit order ensures safe shutdown without bus glitches:
01674      * 1. Deassert CS (drive high) - release peripheral before disabling SPI
01675      * 2. Disable SPI (SPCR.SPE=0) - stop controller clock generation
01676      * 3. Disable module clock (MSTPCRB) - gate clock only after SPI is idle
01677      * 4. Clear software state - CS config and init flag after hardware is off
01678      */
01679
01680     /* Step 1: Deassert CS to release any connected peripheral before shutdown */
01681     if (s_rspi_controller_initialized[channel]) {
01682         (void)rspi_controller_set_cs(channel, false);
01683     }
01684 }

```

```

01685  /* Step 2: Disable SPI to stop controller clock generation */
01686  rspi->spr = k_rspi_spr_disabled;
01687
01688  /* Step 3: Disable RSPI module clock (set module stop bit) - must be done
01689   * after SPI is disabled to avoid undefined behavior from clock gating
01690   * during an active transfer */
01691  *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
01692  internal_set_mstpcrb_for_channel(channel, false);
01693  *prcr_reg() = k_rx_prcr_lock;
01694
01695  /* Step 4: Clear software state after hardware is safely off */
01696  s_rspi_cs_config[channel].port = k_rspi_cs_default_port;
01697  s_rspi_cs_config[channel].pin = k_rspi_cs_default_pin;
01698
01699  s_rspi_controller_initialized[channel] = false;
01700
01701  rx_log_debug(s_tag, "RSPI controller deinitialized");
01702
01703  return k_rx_ok;
01704 }
01705
01706 #endif /* __RX__ */

```

8.109 libs/rx`hal/src/rx`cac.c File Reference

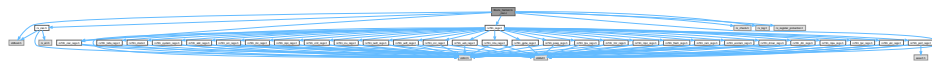
Clock Frequency Accuracy Measurement Circuit (CAC) Driver Implementation.

```

#include "rx_cac.h"
#include <stdbool.h>
#include <stddef.h>
#include "rx72n_cac_regs.h"
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_register_protection.h"

```

Include dependency graph for rx_cac.c:



Functions

- static void [internal_module_start](#) (void)
Release the CAC module from module-stop (MSTPCRC.MSTPC19 = 0).
- static void [internal_module_stop](#) (void)
Return the CAC module to module-stop (MSTPCRC.MSTPC19 = 1).
- static void [internal_apply_config](#) (const [rx_cac_config_t](#) *config)
Program CACR1, CACR2, CAULVR, CALLVR, and CAICR from a config struct.
- rx_err_t [rx_cac_init](#) (const [rx_cac_config_t](#) *config)
Initialize the CAC module with the supplied configuration.
- rx_err_t [rx_cac_start](#) (void)
Start the CAC measurement (implementation).
- rx_err_t [rx_cac_stop](#) (void)
Pause the CAC measurement (implementation).
- bool [rx_cac_check](#) (uint32_t *out_count)
Check the CAC frequency-error flag and snapshot count (implementation).
- rx_err_t [rx_cac_deinit](#) (void)
Fully deinitialize the CAC peripheral (implementation).

Variables

- static const char *const `s_tag` = "CAC"
Log tag for the CAC driver module.
- static bool `s_initialized` = false
Tracks whether `rx_cac_init()` has installed a valid configuration.
- static const uint8_t `s_fmcs_table` []
CACR1.FMCS bit pattern indexed by `rx_cac_clock_t`.
- static const uint8_t `s_rscs_table` []
CACR2.RSCS bit pattern indexed by `rx_cac_clock_t` (reference subset).
- static const uint8_t `s_tcscs_table` []
CACR1.TCSS bit pattern indexed by `rx_cac_target_div_t`.
- static const uint8_t `s_rcds_table` []
CACR2.RCDS bit pattern indexed by `rx_cac_ref_div_t`.

8.109.1 Detailed Description

Clock Frequency Accuracy Measurement Circuit (CAC) Driver Implementation.

Driver for the RX72N Clock Frequency Accuracy Measurement Circuit (CAC) per Renesas RX72N Group User's Manual (R01UH0824EJ0111) Chapter 11 "Clock Frequency Accuracy Measurement Circuit" (pp.↵ 392-397 of the Ch11 register map section referenced below).

The CAC compares a measurement-target clock against a reference clock, counts target-clock edges between reference-clock edges, and compares the accumulated count against CAULVR / CALLVR window bounds. A frequency deviation sets CASTR.FERRF; when CAICR.FERRIE is enabled the failure is routed through GROUPL0 (vector 110, bit 26) to the CPU.

Bring-up Sequence (RX72N HW Manual Ch11.3)

1. Unlock PRCR.PRC1, clear MSTPCRC.MSTPC19, re-lock PRCR.
2. Ensure CACR0.CFME = 0 while programming CACR1, CACR2, CAULVR, CALLVR, and CAICR. Writes to CAULVR / CALLVR require CFME = 0.
3. Clear any stale CASTR flags via CAICR write-1-clear bits.
4. Caller invokes `rx_cac_start()` to set CACR0.CFME = 1 and begin measuring.

NASA Power of 10 Compliance

- Rule 1: [OK] No goto / setjmp / recursion
- Rule 2: [OK] No loops
- Rule 3: [OK] Zero dynamic memory allocation
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] >= 2 validation checks per function
- Rule 6: [OK] File-scope statics, narrowest-scope locals
- Rule 7: [OK] All return values checked via RX_RETURN_ON_ERROR / explicit cast
- Rule 8: [OK] C23 typed enums for constants
- Rule 9: [OK] Single-level pointers
- Rule 10: [OK] -Wall -Wextra -Werror clean

See also

[rx_cac.h](#) Public API
[rx72n_cac_regs.h](#) Register definitions
RX72N HW Manual R01UH0824EJ0111 Chapter 11

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_cac.c](#).

8.109.2 Function Documentation

8.109.2.1 `internal`apply`config()`

```
void internal__apply__config (  
    const rx\_cac\_config\_t * config)    [static]
```

Program CACR1, CACR2, CAULVR, CALLVR, and CAICR from a config struct.

Precondition

config != nullptr and all enum fields validated by [rx_cac_init\(\)](#)
CACR0.CFME == 0 (writes to CAULVR/CALLVR require measurement halted)

Postcondition

Registers hold the requested configuration and CASTR flags are cleared

Since

Version 1.0.0

Definition at line 194 of file [rx_cac.c](#).

```

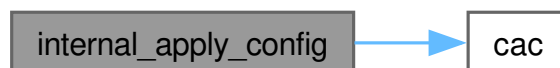
00195 {
00196     volatile rx_cac_regs_t* regs = cac();
00197     uint8_t      cacr1_val =
00198         (uint8_t)(s_fmcs_table[(uint8_t)config->measured_clock] |
00199             s_tcsc_table[(uint8_t)config->target_div] | k_cac_cacr1_edges_rising);
00200     uint8_t cacr2_val =
00201         (uint8_t)(s_rscs_table[(uint8_t)config->reference_clock] |
00202             s_rcds_table[(uint8_t)config->reference_div] | k_cac_cacr2_rps_internal);
00203     uint8_t caicr_val = k_cac_caicr_clear_all_flags;
00204     if (config->enable_ferrie) {
00205         caicr_val |= k_cac_caicr_ferrie_mask;
00206     }
00207     if (config->enable_mendie) {
00208         caicr_val |= k_cac_caicr_mendie_mask;
00209     }
00210     if (config->enable_ovfie) {
00211         caicr_val |= k_cac_caicr_ovfie_mask;
00212     }
00213     regs->cacr1 = cacr1_val;
00214     regs->cacr2 = cacr2_val;
00215     regs->caulvr = config->caulvr;
00216     regs->callvr = config->callvr;
00217     regs->caicr = caicr_val;
00218 }

```

References [cac\(\)](#), [rx_cac_regs_t::cacr1](#), [rx_cac_regs_t::cacr2](#), [rx_cac_regs_t::caicr](#), [rx_cac_config_t::callvr](#), [rx_cac_regs_t::callvr](#), [rx_cac_config_t::caulvr](#), [rx_cac_regs_t::caulvr](#), [rx_cac_config_t::enable_ferrie](#), [rx_cac_config_t::enable_mendie](#), [rx_cac_config_t::enable_ovfie](#), [k_cac_cacr1_edges_rising](#), [k_cac_cacr2_rps_internal](#), [k_cac_caicr_clear_all_flags](#), [k_cac_caicr_ferrie_mask](#), [k_cac_caicr_mendie_mask](#), [k_cac_caicr_ovfie_mask](#), [rx_cac_config_t::measured_clock](#), [rx_cac_config_t::reference_clock](#), [rx_cac_config_t::reference_div](#), [s_fmcs_table](#), [s_rcds_table](#), [s_rscs_table](#), [s_tcsc_table](#), and [rx_cac_config_t::target_div](#).

Referenced by [rx_cac_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.109.2.2 internal module start()

```

void internal_module_start (
    void ) [static]

```

Release the CAC module from module-stop (MSTPCRC.MSTPC19 = 0).

Unlocks PRC1, clears the MSTPC19 bit, re-locks PRCR.

Precondition

PRCR is accessible (not held by another critical section)

Postcondition

MSTPCRC.MSTPC19 = 0, PRCR re-locked

Since

Version 1.0.0

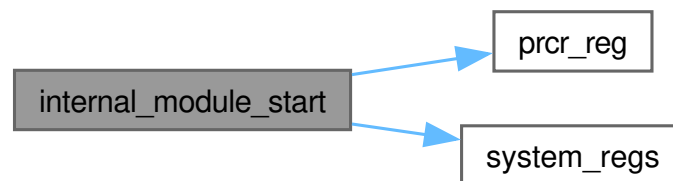
Definition at line 166 of file [rx_cac.c](#).

```
00167 {
00168  *prcr_reg() = k_rx_prcr_unlock_prc1;
00169  system_regs()->mstpcrc &= ~((uint32_t)1U « k_cac_mstpc_bit);
00170  *prcr_reg() = k_rx_prcr_lock;
00171 }
```

References [k_cac_mstpc_bit](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_cac_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.109.2.3 internal`module`stop()

```
void internal_module_stop (
    void ) [static]
```

Return the CAC module to module-stop (MSTPCRC.MSTPC19 = 1).

Unlocks PRC1, sets the MSTPC19 bit, re-locks PRCR.

Precondition

Measurement has been halted (CACR0.CFME = 0)

Postcondition

MSTPCRC.MSTPC19 = 1, PRCR re-locked

Since

Version 1.0.0

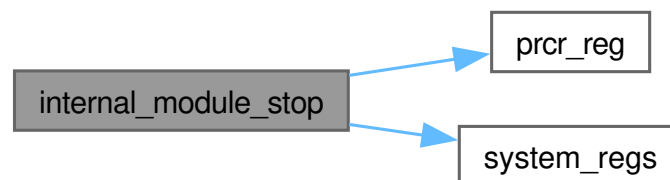
Definition at line 180 of file [rx_cac.c](#).

```
00181 {
00182  *prcr_reg() = k_rx_prcr_unlock_prc1;
00183  system_regs()->mstpcrc |= ((uint32_t)1U « k_cac_mstpc_bit);
00184  *prcr_reg() = k_rx_prcr_lock;
00185 }
```

References [k_cac_mstpc_bit](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_cac_deinit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.109.2.4 rx_cac_check()

```
bool rx_cac_check (
    uint32_t * out_count)
```

Check the CAC frequency-error flag and snapshot count (implementation).

Sample the CAC status and counter buffer, clearing acknowledged flags.

Reads CASTR (status), latches the current CACNTBR (counter buffer) value if `out_count` is non-NULL, then issues a write-1-clear to the IRQ status flags in CAICR (preserving the IRQ enables). Returns whether the FERRF (frequency-error) flag was set since the last check.

Note

Safe from any context including ISR (single register read + write, no blocking).

Definition at line 314 of file [rx_cac.c](#).

```

00315 {
00316     if (!s_initialized) {
00317         return false;
00318     }
00319     volatile rx_cac_regs_t* regs = cac();
00320     uint8_t status = regs->castr;
00321     if (out_count != nullptr) {
00322         *out_count = (uint32_t)regs->cacntbr;
00323     }
00324     /* Preserve enabled IRQ settings while issuing write-1-clear acknowledgements. */
00325     uint8_t ack =
00326         (uint8_t)(regs->caicr & (uint8_t)(k_cac_caicr_ferrie_mask | k_cac_caicr_mendie_mask |
00327             k_cac_caicr_ovfie_mask));
00328     ack |= k_cac_caicr_clear_all_flags;
00329     regs->caicr = ack;
00330     return (bool)((status & k_cac_castr_ferrf_mask) != 0U);
00331 }

```

References [cac\(\)](#), [rx_cac_regs_t::cacntbr](#), [rx_cac_regs_t::caicr](#), [rx_cac_regs_t::castr](#), [k_cac_caicr_clear_all_flags](#), [k_cac_caicr_ferrie_mask](#), [k_cac_caicr_mendie_mask](#), [k_cac_caicr_ovfie_mask](#), [k_cac_castr_ferrf_mask](#), and [s_initialized](#).

Here is the call graph for this function:



8.109.2.5 rx`cac`deinit()

```

rx_err_t rx_cac_deinit (
    void ) [nodiscard]

```

Fully deinitialize the CAC peripheral (implementation).

Tear down the CAC module and put it back in module-stop.

Stops the measurement (CFME=0), clears CACR1 / CACR2 (resetting the reference clock and tolerance), clears all IRQ flags via CAICR, then gates the CAC module clock through [internal_module_stop\(\)](#). Marks the driver as uninitialized.

Note

Not thread-safe. Serialize all `rx_cac_*` operations.

Definition at line 344 of file [rx_cac.c](#).

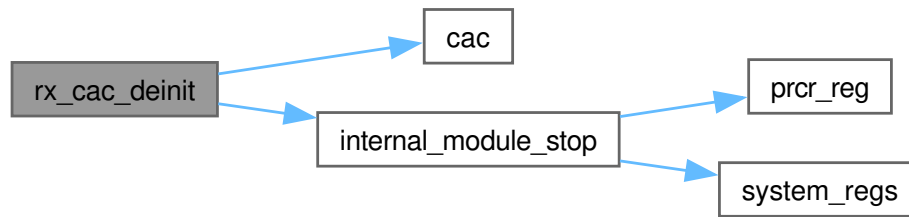
```

00345 {
00346     if (!s_initialized) {
00347         rx_log_error(s_tag, "rx_cac_deinit: not initialized");
00348         return k_rx_err_not_initialized;
00349     }
00350     volatile rx_cac_regs_t* regs = cac();
00351     regs->cacr0 = k_cac_cacr0_cfme_stop;
00352     regs->cacr1 = 0U;
00353     regs->cacr2 = 0U;
00354     regs->caicr = k_cac_caicr_clear_all_flags;
00355     internal_module_stop();
00356     s_initialized = false;
00357     return k_rx_ok;
00358 }

```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [rx_cac_regs_t::cacr1](#), [rx_cac_regs_t::cacr2](#), [rx_cac_regs_t::caicr](#), [internal_module_stop\(\)](#), [k_cac_cacr0_cfme_stop](#), [k_cac_caicr_clear_all_flags](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.109.2.6 rx_cac_init()

```

rx_err_t rx_cac_init (
    const rx_cac_config_t * config)  [nodiscard]
  
```

Initialize the CAC module with the supplied configuration.

Performs the full CAC bring-up sequence mandated by the RX72N HW Manual Ch10.3 "Operation":

1. Unlock PRCR.PRC1 and clear MSTPCRC.MSTPC19, then re-lock PRCR.
2. With CACR0.CFME = 0, program CACR1 (FMCS, TCSS, EDGES, CAIE), CACR2 (RSCS, RCDS, RPS, DFS), CAULVR, CALLVR, and CAICR enables.
3. Clear any stale CASTR flags via CAICR write-1-clear.
4. Leave CFME = 0 (caller must invoke `rx_cac_start()` to begin measuring).

Parameters

in	config	Pointer to configuration struct (must not be nullptr)
----	--------	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Initialization succeeded, CAC ready
k_rx_err_null_ptr	config is nullptr
k_rx_err_invalid_arg	Invalid clock/divider, or callvr >= caulvr
k_rx_err_invalid_state	CAC already initialized (call deinit first)

Precondition

```

config != nullptr
config->callvr < config->caulvr
config->reference_clock is in [k_cac_clock_main .. k_cac_clock_iwdtclk]
config->measured_clock is in [k_cac_clock_main .. k_cac_clock_clkout25]
config->target_div is in [k_cac_target_div_1 .. k_cac_target_div_32]
config->reference_div is in [k_cac_ref_div_32 .. k_cac_ref_div_8192]
  
```

Postcondition

CAC module is powered on (MSTPCRC.MSTPC19 = 0)
CACR0.CFME = 0 (measurement not yet running)
All configured limits and interrupt enables are loaded

Note

Not thread-safe. Call only during single-threaded initialization.

Since

Version 1.0.0

Definition at line 225 of file [rx_cac.c](#).

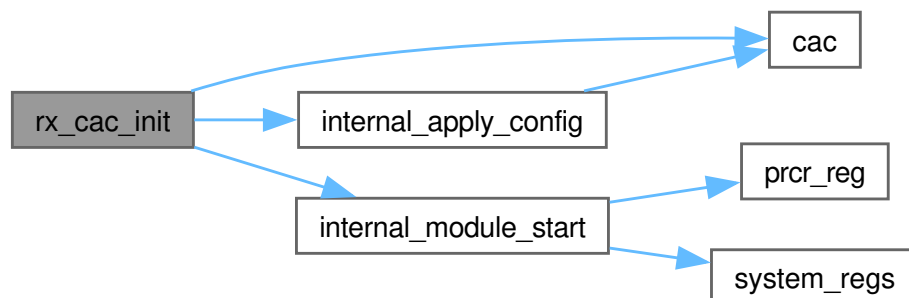
```

00226 {
00227     RX_CHECK_NULL_PTR(config, s_tag, "rx_cac_init: config is nullptr");
00228     if (s_initialized) {
00229         rx_log_error(s_tag, "rx_cac_init: already initialized");
00230         return k_rx_err_invalid_state;
00231     }
00232     if (config->callvr >= config->caulvr) {
00233         rx_log_error(s_tag, "rx_cac_init: callvr must be < caulvr");
00234         return k_rx_err_invalid_arg;
00235     }
00236     if ((uint8_t)config->reference_clock > k_cac_clock_iwdtclk) {
00237         rx_log_error(s_tag, "rx_cac_init: reference_clock out of range");
00238         return k_rx_err_invalid_arg;
00239     }
00240     if ((uint8_t)config->measured_clock > k_cac_clock_clkout25) {
00241         rx_log_error(s_tag, "rx_cac_init: measured_clock out of range");
00242         return k_rx_err_invalid_arg;
00243     }
00244     if ((uint8_t)config->target_div > k_cac_target_div_32) {
00245         rx_log_error(s_tag, "rx_cac_init: target_div out of range");
00246         return k_rx_err_invalid_arg;
00247     }
00248     if ((uint8_t)config->reference_div > k_cac_ref_div_8192) {
00249         rx_log_error(s_tag, "rx_cac_init: reference_div out of range");
00250         return k_rx_err_invalid_arg;
00251     }
00252     internal_module_start();
00253     volatile rx_cac_regs_t* regs = cac();
00254     regs->cacr0 = k_cac_cacr0_cfme_stop;
00255     internal_apply_config(config);
00256     s_initialized = true;
00257     return k_rx_ok;
00258 }

```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [rx_cac_config_t::callvr](#), [rx_cac_config_t::caulvr](#), [internal_apply_config\(\)](#), [internal_module_start\(\)](#), [k_cac_cacr0_cfme_stop](#), [k_cac_clock_clkout25](#), [k_cac_clock_iwdtclk](#), [k_cac_ref_div_8192](#), [k_cac_target_div_32](#), [rx_cac_config_t::measured_clock](#), [rx_cac_config_t::reference_clock](#), [rx_cac_config_t::reference_div](#), [s_initialized](#), [s_tag](#), and [rx_cac_config_t::target_div](#).

Here is the call graph for this function:



8.109.2.7 rx_cac_start()

```
rx_err_t rx_cac_start (
    void ) [nodiscard]
```

Start the CAC measurement (implementation).

Begin CAC measurement (CACR0.CFME = 1).

Sets CACR0.CFME=1 to enable comparison of the measurement clock against the reference clock configured by [rx_cac_init\(\)](#). Subsequent calls to [rx_cac_check\(\)](#) will report whether the measurement is within the configured tolerance window and return the latched count.

Note

Not thread-safe. Serialize all rx_cac_* operations.

Definition at line 271 of file [rx_cac.c](#).

```
00272 {
00273     if (!s_initialized) {
00274         rx_log_error(s_tag, "rx_cac_start: not initialized");
00275         return k_rx_err_not_initialized;
00276     }
00277     cac()->cacr0 = k_cac_cacr0_cfme_start;
00278     return k_rx_ok;
00279 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [k_cac_cacr0_cfme_start](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.109.2.8 rx_cac_stop()

```
rx_err_t rx_cac_stop (
    void ) [nodiscard]
```

Pause the CAC measurement (implementation).

Stop CAC measurement (CACR0.CFME = 0).

Clears CACR0.CFME, halting the measurement counter. The measurement configuration (reference clock, tolerance, IRQ enables) is preserved so a subsequent [rx_cac_start\(\)](#) resumes from a clean state. Useful when entering low-power modes or when switching reference clocks.

Note

Not thread-safe. Serialize all rx_cac_* operations.

Definition at line 292 of file [rx_cac.c](#).

```
00293 {
00294     if (!s_initialized) {
00295         rx_log_error(s_tag, "rx_cac_stop: not initialized");
00296         return k_rx_err_not_initialized;
00297     }
00298     cac()->cacr0 = k_cac_cacr0_cfme_stop;
00299     return k_rx_ok;
00300 }
```

References [cac\(\)](#), [rx_cac_regs_t::cacr0](#), [k_cac_cacr0_cfme_stop](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.109.3 Variable Documentation

8.109.3.1 s`fmcs`table

const uint8_t s_fmcs_table[] [static]

Initial value:

```

    = {
[k_cac_clock_main]   = k_cac_cacr1_fmcs_main,
[k_cac_clock_sub]    = k_cac_cacr1_fmcs_sub,
[k_cac_clock_hoco]   = k_cac_cacr1_fmcs_hoco,
[k_cac_clock_loco]   = k_cac_cacr1_fmcs_loco,
[k_cac_clock_pclkb]  = k_cac_cacr1_fmcs_pclkb,
[k_cac_clock_iwdtclk] = k_cac_cacr1_fmcs_iwdtclk,
[k_cac_clock_uclk]   = k_cac_cacr1_fmcs_uclk,
[k_cac_clock_clkout25] = k_cac_cacr1_fmcs_clkout25,
    }

```

CACR1.FMCS bit pattern indexed by [rx_cac_clock_t](#).

RX72N HW Manual p.393 (FMCS field of CACR1).

Since

Version 1.0.0

Definition at line 101 of file [rx_cac.c](#).

```

00101     {
00102 [k_cac_clock_main]   = k_cac_cacr1_fmcs_main,
00103 [k_cac_clock_sub]    = k_cac_cacr1_fmcs_sub,
00104 [k_cac_clock_hoco]   = k_cac_cacr1_fmcs_hoco,
00105 [k_cac_clock_loco]   = k_cac_cacr1_fmcs_loco,
00106 [k_cac_clock_pclkb]  = k_cac_cacr1_fmcs_pclkb,
00107 [k_cac_clock_iwdtclk] = k_cac_cacr1_fmcs_iwdtclk,
00108 [k_cac_clock_uclk]   = k_cac_cacr1_fmcs_uclk,
00109 [k_cac_clock_clkout25] = k_cac_cacr1_fmcs_clkout25,
00110 };

```

Referenced by [internal_apply_config\(\)](#).

8.109.3.2 s`initialized

bool s_initialized = false [static]

Tracks whether [rx_cac_init\(\)](#) has installed a valid configuration.

Cleared by [rx_cac_deinit\(\)](#) to permit a subsequent init call.

Since

Version 1.0.0

Definition at line 83 of file [rx_cac.c](#).

Referenced by [rx_cac_check\(\)](#), [rx_cac_deinit\(\)](#), [rx_cac_init\(\)](#), [rx_cac_start\(\)](#), [rx_cac_stop\(\)](#), [rx_eccram_init\(\)](#), [rx_eccram_register_error_isr\(\)](#), [rx_host_irq_assert\(\)](#), [rx_host_irq_deassert\(\)](#), [rx_host_irq_deinit\(\)](#), [rx_host_irq_init\(\)](#), [rx_lpc_enter_deep_software_standby\(\)](#), [rx_lpc_enter_sleep\(\)](#), [rx_lpc_enter_software_standby\(\)](#), [rx_lpc_get_wake_flags\(\)](#), [rx_lpc_init\(\)](#), and [rx_lpc_set_operating_power\(\)](#).

8.109.3.3 s'rcds'table

```
const uint8_t s_rcds_table[] [static]
```

Initial value:

```
= {
[k_cac_ref_div_32] = k_cac_cacr2_rcds_div_32,
[k_cac_ref_div_128] = k_cac_cacr2_rcds_div_128,
[k_cac_ref_div_1024] = k_cac_cacr2_rcds_div_1024,
[k_cac_ref_div_8192] = k_cac_cacr2_rcds_div_8192,
}
```

CACR2.RCDS bit pattern indexed by [rx_cac_ref_div_t](#).

RX72N HW Manual p.394 (RCDS field of CACR2).

Since

Version 1.0.0

Definition at line 147 of file [rx_cac.c](#).

```
00147 {
00148 [k_cac_ref_div_32] = k_cac_cacr2_rcds_div_32,
00149 [k_cac_ref_div_128] = k_cac_cacr2_rcds_div_128,
00150 [k_cac_ref_div_1024] = k_cac_cacr2_rcds_div_1024,
00151 [k_cac_ref_div_8192] = k_cac_cacr2_rcds_div_8192,
00152 };
```

Referenced by [internal_apply_config\(\)](#).

8.109.3.4 s'rscl'table

```
const uint8_t s_rscs_table[] [static]
```

Initial value:

```
= {
[k_cac_clock_main] = k_cac_cacr2_rscs_main,
[k_cac_clock_sub] = k_cac_cacr2_rscs_sub,
[k_cac_clock_hoco] = k_cac_cacr2_rscs_hoco,
[k_cac_clock_loco] = k_cac_cacr2_rscs_loco,
[k_cac_clock_pclkb] = k_cac_cacr2_rscs_pclkb,
[k_cac_clock_iwdtclk] = k_cac_cacr2_rscs_iwdtclk,
}
```

CACR2.RSCS bit pattern indexed by [rx_cac_clock_t](#) (reference subset).

RX72N HW Manual p.394 (RSCS field of CACR2). Only Main..IWDTCCLK (indices 0..5) are valid references; [rx_cac_init\(\)](#) rejects higher values.

Since

Version 1.0.0

Definition at line 119 of file [rx_cac.c](#).

```
00119 {
00120 [k_cac_clock_main] = k_cac_cacr2_rscs_main,
00121 [k_cac_clock_sub] = k_cac_cacr2_rscs_sub,
00122 [k_cac_clock_hoco] = k_cac_cacr2_rscs_hoco,
00123 [k_cac_clock_loco] = k_cac_cacr2_rscs_loco,
00124 [k_cac_clock_pclkb] = k_cac_cacr2_rscs_pclkb,
00125 [k_cac_clock_iwdtclk] = k_cac_cacr2_rscs_iwdtclk,
00126 };
```

Referenced by [internal_apply_config\(\)](#).

8.109.3.5 s`tag

```
const char* const s_tag = "CAC" [static]
```

Log tag for the CAC driver module.

String literal used by rx_log_* calls from this translation unit.

Since

Version 1.0.0

Definition at line 75 of file rx_cac.c.

8.109.3.6 s`tcss`table

```
const uint8_t s_tcss_table[] [static]
```

Initial value:

```
    = {  
[k_cac_target_div_1] = k_cac_cacr1_tcss_div_1,  
[k_cac_target_div_4] = k_cac_cacr1_tcss_div_4,  
[k_cac_target_div_8] = k_cac_cacr1_tcss_div_8,  
[k_cac_target_div_32] = k_cac_cacr1_tcss_div_32,  
}
```

CACR1.TCSS bit pattern indexed by rx_cac_target_div_t.

RX72N HW Manual p.393 (TCSS field of CACR1).

Since

Version 1.0.0

Definition at line 134 of file rx_cac.c.

```
00134 {  
00135 [k_cac_target_div_1] = k_cac_cacr1_tcss_div_1,  
00136 [k_cac_target_div_4] = k_cac_cacr1_tcss_div_4,  
00137 [k_cac_target_div_8] = k_cac_cacr1_tcss_div_8,  
00138 [k_cac_target_div_32] = k_cac_cacr1_tcss_div_32,  
00139 };
```

Referenced by internal_apply_config().

8.110 rx_cac.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_cac.c
00003  * @brief Clock Frequency Accuracy Measurement Circuit (CAC) Driver Implementation
00004  *
00005  * @details
00006  * Driver for the RX72N Clock Frequency Accuracy Measurement Circuit (CAC) per
00007  * Renesas RX72N Group User's Manual (R01UH0824EJ0111) Chapter 11 "Clock
00008  * Frequency Accuracy Measurement Circuit" (pp.392-397 of the Ch11 register map
00009  * section referenced below).
00010  *
00011  * The CAC compares a measurement-target clock against a reference clock, counts
00012  * target-clock edges between reference-clock edges, and compares the accumulated
00013  * count against CAULVR / CALLVR window bounds. A frequency deviation sets
00014  * CASTR.FERRF; when CAICR.FERRIE is enabled the failure is routed through
00015  * GROUPBL0 (vector 110, bit 26) to the CPU.
00016  *
00017  * @par Bring-up Sequence (RX72N HW Manual Ch11.3)
00018  * 1. Unlock PRCR.PRC1, clear MSTPCRC.MSTPC19, re-lock PRCR.
00019  * 2. Ensure CACR0.CFME = 0 while programming CACR1, CACR2, CAULVR, CALLVR,
00020  *    and CAICR. Writes to CAULVR / CALLVR require CFME = 0.
00021  * 3. Clear any stale CASTR flags via CAICR write-1-clear bits.
00022  * 4. Caller invokes rx_cac_start() to set CACR0.CFME = 1 and begin measuring.
00023  *
00024  * @par NASA Power of 10 Compliance
00025  * - Rule 1: [OK] No goto / setjmp / recursion
00026  * - Rule 2: [OK] No loops
00027  * - Rule 3: [OK] Zero dynamic memory allocation
00028  * - Rule 4: [OK] All functions < 60 lines
00029  * - Rule 5: [OK] >= 2 validation checks per function
00030  * - Rule 6: [OK] File-scope statics, narrowest-scope locals
00031  * - Rule 7: [OK] All return values checked via RX_RETURN_ON_ERROR / explicit cast
00032  * - Rule 8: [OK] C23 typed enums for constants
00033  * - Rule 9: [OK] Single-level pointers
00034  * - Rule 10: [OK] -Wall -Wextra -Werror clean
00035  *
00036  * @see rx_cac.h Public API
00037  * @see rx72n_cac_regs.h Register definitions
00038  * @see RX72N HW Manual R01UH0824EJ0111 Chapter 11
00039  *
00040  * @author Locked, Inc.
00041  * @date 2026-04-21
00042  * @copyright Copyright (c) 2026 Locked Inc.
00043  * SPDX-License-Identifier: MIT
00044  * @since Version 1.0.0
00045  */
00046
00047 #include "rx_cac.h"
00048
00049 #include <stdbool.h>
00050 #include <stddef.h>
00051
00052 #ifdef UNIT_TEST
00053 #include "mock_rx72n_regs.h"
00054 #include "mock_rx_cac.h"
00055 #else
00056 #include "rx72n_cac_regs.h"
00057 #include "rx72n_regs.h"
00058 #endif
00059
00060 #include "rx_check.h"
00061 #include "rx_log.h"
00062 #include "rx_register_protection.h"
00063
00064 /**
=====
00065  * Internal State
00066  *
=====
00067  */
00068
00069 /**
00070  * @var s_tag
00071  * @brief Log tag for the CAC driver module
00072  * @details String literal used by rx_log_* calls from this translation unit.
00073  * @since Version 1.0.0
00074  */
00075 static const char* const s_tag = "CAC";
00076
00077 /**
00078  * @var s_initialized
00079  * @brief Tracks whether rx_cac_init() has installed a valid configuration
00080  * @details Cleared by rx_cac_deinit() to permit a subsequent init call.

```



```

00081 * @since Version 1.0.0
00082 */
00083 static bool s_initialized = false;
00084
00085 /*
=====
00086 * Internal Translation Tables
00087
00088 * Lookup tables replace per-case switch statements so that the translators are
00089 * branch-free. rx_cac_init() validates each enum argument against the tables'
00090 * defined indices, so the caller's index is always in range by the time a
00091 * translator is invoked.
00092 *
=====
00093 */
00094
00095 /**
00096 * @var s_fmcs_table
00097 * @brief CACR1.FMCS bit pattern indexed by rx_cac_clock_t
00098 * @details RX72N HW Manual p.393 (FMCS field of CACR1).
00099 * @since Version 1.0.0
00100 */
00101 static const uint8_t s_fmcs_table[] = {
00102     [k_cac_clock_main]    = k_cac_cacr1_fmcs_main,
00103     [k_cac_clock_sub]     = k_cac_cacr1_fmcs_sub,
00104     [k_cac_clock_hoco]    = k_cac_cacr1_fmcs_hoco,
00105     [k_cac_clock_loco]    = k_cac_cacr1_fmcs_loco,
00106     [k_cac_clock_pclkb]   = k_cac_cacr1_fmcs_pclkb,
00107     [k_cac_clock_iwdtclk] = k_cac_cacr1_fmcs_iwdtclk,
00108     [k_cac_clock_uclk]    = k_cac_cacr1_fmcs_uclk,
00109     [k_cac_clock_clkout25] = k_cac_cacr1_fmcs_clkout25,
00110 };
00111
00112 /**
00113 * @var s_rscs_table
00114 * @brief CACR2.RSCS bit pattern indexed by rx_cac_clock_t (reference subset)
00115 * @details RX72N HW Manual p.394 (RSCS field of CACR2). Only Main..IWDTCLK
00116 * (indices 0..5) are valid references; rx_cac_init() rejects higher values.
00117 * @since Version 1.0.0
00118 */
00119 static const uint8_t s_rscs_table[] = {
00120     [k_cac_clock_main]    = k_cac_cacr2_rscs_main,
00121     [k_cac_clock_sub]     = k_cac_cacr2_rscs_sub,
00122     [k_cac_clock_hoco]    = k_cac_cacr2_rscs_hoco,
00123     [k_cac_clock_loco]    = k_cac_cacr2_rscs_loco,
00124     [k_cac_clock_pclkb]   = k_cac_cacr2_rscs_pclkb,
00125     [k_cac_clock_iwdtclk] = k_cac_cacr2_rscs_iwdtclk,
00126 };
00127
00128 /**
00129 * @var s_tcscs_table
00130 * @brief CACR1.TCSS bit pattern indexed by rx_cac_target_div_t
00131 * @details RX72N HW Manual p.393 (TCSS field of CACR1).
00132 * @since Version 1.0.0
00133 */
00134 static const uint8_t s_tcscs_table[] = {
00135     [k_cac_target_div_1]  = k_cac_cacr1_tcscs_div_1,
00136     [k_cac_target_div_4]  = k_cac_cacr1_tcscs_div_4,
00137     [k_cac_target_div_8]  = k_cac_cacr1_tcscs_div_8,
00138     [k_cac_target_div_32] = k_cac_cacr1_tcscs_div_32,
00139 };
00140
00141 /**
00142 * @var s_rcds_table
00143 * @brief CACR2.RCDS bit pattern indexed by rx_cac_ref_div_t
00144 * @details RX72N HW Manual p.394 (RCDS field of CACR2).
00145 * @since Version 1.0.0
00146 */
00147 static const uint8_t s_rcds_table[] = {
00148     [k_cac_ref_div_32]    = k_cac_cacr2_rcds_div_32,
00149     [k_cac_ref_div_128]   = k_cac_cacr2_rcds_div_128,
00150     [k_cac_ref_div_1024]  = k_cac_cacr2_rcds_div_1024,
00151     [k_cac_ref_div_8192]  = k_cac_cacr2_rcds_div_8192,
00152 };
00153
00154 /*
=====
00155 * Internal Helpers
00156 *
=====
00157 */
00158
00159 /**
00160 * @brief Release the CAC module from module-stop (MSTPCRC.MSTPC19 = 0)
00161 * @details Unlocks PRC1, clears the MSTPC19 bit, re-locks PRCR.
00162 * @pre PRCR is accessible (not held by another critical section)
00163 * @post MSTPCRC.MSTPC19 = 0, PRCR re-locked

```

```

00164 * @since Version 1.0.0
00165 */
00166 static void internal_module_start(void)
00167 {
00168     *prcr_reg() = k_rx_prcr_unlock_prc1;
00169     system_regs()->mstpcrc |= ~(uint32_t)1U « k_cac_mstpc_bit);
00170     *prcr_reg() = k_rx_prcr_lock;
00171 }
00172
00173 /**
00174 * @brief Return the CAC module to module-stop (MSTPCRC.MSTPC19 = 1)
00175 * @details Unlocks PRC1, sets the MSTPC19 bit, re-locks PRCR.
00176 * @pre Measurement has been halted (CACR0.CFME = 0)
00177 * @post MSTPCRC.MSTPC19 = 1, PRCR re-locked
00178 * @since Version 1.0.0
00179 */
00180 static void internal_module_stop(void)
00181 {
00182     *prcr_reg() = k_rx_prcr_unlock_prc1;
00183     system_regs()->mstpcrc |= ((uint32_t)1U « k_cac_mstpc_bit);
00184     *prcr_reg() = k_rx_prcr_lock;
00185 }
00186
00187 /**
00188 * @brief Program CACR1, CACR2, CAULVR, CALLVR, and CAICR from a config struct
00189 * @pre config != nullptr and all enum fields validated by rx_cac_init()
00190 * @pre CACR0.CFME == 0 (writes to CAULVR/CALLVR require measurement halted)
00191 * @post Registers hold the requested configuration and CASTR flags are cleared
00192 * @since Version 1.0.0
00193 */
00194 static void internal_apply_config(const rx_cac_config_t* config)
00195 {
00196     volatile rx_cac_regs_t* regs = cac();
00197     uint8_t cacr1_val =
00198         (uint8_t)(s_fmcs_table[(uint8_t)config->measured_clock] |
00199             s_tcsc_table[(uint8_t)config->target_div] | k_cac_cacr1_edges_rising);
00200     uint8_t cacr2_val =
00201         (uint8_t)(s_rscs_table[(uint8_t)config->reference_clock] |
00202             s_rcds_table[(uint8_t)config->reference_div] | k_cac_cacr2_rps_internal);
00203     uint8_t caicr_val = k_cac_caicr_clear_all_flags;
00204     if (config->enable_ferrie) {
00205         caicr_val |= k_cac_caicr_ferrie_mask;
00206     }
00207     if (config->enable_mendie) {
00208         caicr_val |= k_cac_caicr_mendie_mask;
00209     }
00210     if (config->enable_ovfie) {
00211         caicr_val |= k_cac_caicr_ovfie_mask;
00212     }
00213     regs->cacr1 = cacr1_val;
00214     regs->cacr2 = cacr2_val;
00215     regs->caulvr = config->caulvr;
00216     regs->callvr = config->callvr;
00217     regs->caicr = caicr_val;
00218 }
00219
00220 /*
=====
00221 * Public API
00222 *
=====
00223 */
00224
00225 rx_err_t rx_cac_init(const rx_cac_config_t* config)
00226 {
00227     RX_CHECK_NULL_PTR(config, s_tag, "rx_cac_init: config is nullptr");
00228     if (s_initialized) {
00229         rx_log_error(s_tag, "rx_cac_init: already initialized");
00230         return k_rx_err_invalid_state;
00231     }
00232     if (config->callvr >= config->caulvr) {
00233         rx_log_error(s_tag, "rx_cac_init: callvr must be < caulvr");
00234         return k_rx_err_invalid_arg;
00235     }
00236     if ((uint8_t)config->reference_clock > k_cac_clock_iwdtclk) {
00237         rx_log_error(s_tag, "rx_cac_init: reference_clock out of range");
00238         return k_rx_err_invalid_arg;
00239     }
00240     if ((uint8_t)config->measured_clock > k_cac_clock_clkout25) {
00241         rx_log_error(s_tag, "rx_cac_init: measured_clock out of range");
00242         return k_rx_err_invalid_arg;
00243     }
00244     if ((uint8_t)config->target_div > k_cac_target_div_32) {
00245         rx_log_error(s_tag, "rx_cac_init: target_div out of range");
00246         return k_rx_err_invalid_arg;
00247     }
00248     if ((uint8_t)config->reference_div > k_cac_ref_div_8192) {

```

```

00249     rx_log_error(s_tag, "rx_cac_init: reference_div out of range");
00250     return k_rx_err_invalid_arg;
00251 }
00252 internal_module_start();
00253 volatile rx_cac_regs_t* regs = cac();
00254 regs->caclr0 = k_cac_caclr0_cfme_stop;
00255 internal_apply_config(config);
00256 s_initialized = true;
00257 return k_rx_ok;
00258 }
00259
00260 /**
00261  * @brief Start the CAC measurement (implementation)
00262  *
00263  * @details
00264  * Sets CACR0.CFME=1 to enable comparison of the measurement clock against
00265  * the reference clock configured by rx_cac_init(). Subsequent calls to
00266  * rx_cac_check() will report whether the measurement is within the
00267  * configured tolerance window and return the latched count.
00268  *
00269  * @note Not thread-safe. Serialize all rx_cac_* operations.
00270  */
00271 rx_err_t rx_cac_start(void)
00272 {
00273     if (!s_initialized) {
00274         rx_log_error(s_tag, "rx_cac_start: not initialized");
00275         return k_rx_err_not_initialized;
00276     }
00277     cac()->caclr0 = k_cac_caclr0_cfme_start;
00278     return k_rx_ok;
00279 }
00280
00281 /**
00282  * @brief Pause the CAC measurement (implementation)
00283  *
00284  * @details
00285  * Clears CACR0.CFME, halting the measurement counter. The measurement
00286  * configuration (reference clock, tolerance, IRQ enables) is preserved
00287  * so a subsequent rx_cac_start() resumes from a clean state. Useful
00288  * when entering low-power modes or when switching reference clocks.
00289  *
00290  * @note Not thread-safe. Serialize all rx_cac_* operations.
00291  */
00292 rx_err_t rx_cac_stop(void)
00293 {
00294     if (!s_initialized) {
00295         rx_log_error(s_tag, "rx_cac_stop: not initialized");
00296         return k_rx_err_not_initialized;
00297     }
00298     cac()->caclr0 = k_cac_caclr0_cfme_stop;
00299     return k_rx_ok;
00300 }
00301
00302 /**
00303  * @brief Check the CAC frequency-error flag and snapshot count (implementation)
00304  *
00305  * @details
00306  * Reads CASTR (status), latches the current CACNTBR (counter buffer)
00307  * value if out_count is non-NULL, then issues a write-1-clear to the IRQ
00308  * status flags in CAICR (preserving the IRQ enables). Returns whether
00309  * the FERRF (frequency-error) flag was set since the last check.
00310  *
00311  * @note Safe from any context including ISR (single register read +
00312  *       write, no blocking).
00313  */
00314 bool rx_cac_check(uint32_t* out_count)
00315 {
00316     if (!s_initialized) {
00317         return false;
00318     }
00319     volatile rx_cac_regs_t* regs = cac();
00320     uint8_t status = regs->castr;
00321     if (out_count != nullptr) {
00322         *out_count = (uint32_t)regs->cacntbr;
00323     }
00324     /* Preserve enabled IRQ settings while issuing write-1-clear acknowledgements. */
00325     uint8_t ack =
00326         (uint8_t)(regs->caicr & (uint8_t)(k_cac_caicr_ferrie_mask | k_cac_caicr_mendie_mask |
00327                                         k_cac_caicr_ovfie_mask));
00328     ack |= k_cac_caicr_clear_all_flags;
00329     regs->caicr = ack;
00330     return (bool)((status & k_cac_castr_ferrf_mask) != 0U);
00331 }
00332
00333 /**
00334  * @brief Fully deinitialize the CAC peripheral (implementation)
00335  *

```

```

00336 * @details
00337 * Stops the measurement (CFME=0), clears CACR1 / CACR2 (resetting the
00338 * reference clock and tolerance), clears all IRQ flags via CAICR, then
00339 * gates the CAC module clock through internal_module_stop(). Marks the
00340 * driver as uninitialized.
00341 *
00342 * @note Not thread-safe. Serialize all rx_cac_* operations.
00343 */
00344 rx_err_t rx_cac_deinit(void)
00345 {
00346     if (!s_initialized) {
00347         rx_log_error(s_tag, "rx_cac_deinit: not initialized");
00348         return k_rx_err_not_initialized;
00349     }
00350     volatile rx_cac_regs_t* regs = cac();
00351     regs->cacr0 = k_cac_cacr0_cfme_stop;
00352     regs->cacr1 = 0U;
00353     regs->cacr2 = 0U;
00354     regs->caicr = k_cac_caicr_clear_all_flags;
00355     internal_module_stop();
00356     s_initialized = false;
00357     return k_rx_ok;
00358 }
00359
00360 #ifdef UNIT_TEST
00361 /**
00362 * @brief Test-only hook that clears the driver's initialization flag.
00363 * @details Allows unit tests to return the module to its power-on state without
00364 * touching the mock register bank.
00365 * @post s_initialized == false
00366 * @since Version 1.0.0
00367 */
00368 void rx_cac_test_reset(void)
00369 {
00370     s_initialized = false;
00371 }
00372 #endif

```

8.111 libs/rx/hal/src/rx_cmt.c File Reference

CMT (Compare Match Timer) Driver Implementation - Periodic Interrupt Generation.

```

#include "rx_cmt.h"
#include <stddef.h>
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_register_protection.h"

```

Include dependency graph for rx_cmt.c:



Data Structures

- struct [rx_cmt_interrupt_config_t](#)
CMT interrupt routing and priority configuration.

Enumerations

- enum [cmt_constants_t](#) : `uint8_t` {
`k_cmt_max_channels = 4` , `k_cmt_divider_start = 0` , `k_cmt_div_8 = 0` , `k_cmt_div_32 = 1`
, `k_cmt_div_128 = 2` , `k_cmt_div_512 = 3` , `k_cmt_num_dividers = 4` }
CMT channel and divider configuration constants.

- enum `cmt_divider_values_t` : `uint16_t` { `k_cmt_divider_val_8` = 8 , `k_cmt_divider_val_32` = 32 , `k_cmt_divider_val_128` = 128 , `k_cmt_divider_val_512` = 512 }
CMT divider values (actual divisor values).
- enum `cmt_module_stop_bits_t` : `uint8_t` { `k_cmt_mstpa_unit0` = 15 , `k_cmt_mstpa_unit1` = 14 }
CMT module stop bit positions in MSTPCRA (RX72N Manual Section 11.2.2, p.406).
- enum `cmt_cmcr_bits_t` : `uint8_t` { `k_cmt_cmcr_cks_shift` = 0 , `k_cmt_cmcr_cmie_pos` = 6 , `k_cmt_cmcr_rsvd7_pos` = 7 }
CMCR register bit positions (RX72N Manual Ch31, p.1583).
- enum `cmt_period_constants_t` : `uint16_t` { `k_cmt_period_min` = 0 , `k_cmt_period_max` = 0xFFFF , `k_cmt_period_adj` = 1 }
Period calculation constants.
- enum `cmt_cmstr_bits_t` : `uint8_t` { `k_cmt_cmstr_str0` = 0 , `k_cmt_cmstr_str1` = 1 }
CMSTR register bit positions.
- enum `cmt_bit_constants_t` : `uint8_t` { `k_cmt_bit_mask_lsb` = 1 , `k_cmt_value_zero` = 0 }
CMT bit manipulation constants.

Functions

- void `cmt1_isr` (void)
CMT1 (CMT channel 1) compare-match interrupt service routine.
- void `cmt2_isr` (void)
CMT2 (CMT channel 2) compare-match interrupt service routine.
- void `cmt3_isr` (void)
CMT3 (CMT channel 3) compare-match interrupt service routine.
- static volatile `rx_cmt_channel_regs_t` * `internal_get_cmt_base` (const `rx_cmt_channel_t` channel)
Map a CMT channel enum to its hardware register base pointer.
- static `rx_err_t` `internal_calculate_cmt_params` (const `uint32_t` frequency_hz, `uint8_t` *divider, `uint16_t` *cmcor)
Calculate the optimal CMT clock divider and compare-match value for a target frequency.
- static void `internal_enable_cmt_module_clock` (void)
Enable the CMT module clock by clearing the MSTPCRB module-stop bit.
- static void `internal_configure_cmt_timer_registers` (volatile `rx_cmt_channel_regs_t` *cmt, const `uint8_t` divider, const `uint16_t` cmcor)
Write clock divider, interrupt-enable, compare-match, and counter registers.
- static `rx_err_t` `internal_configure_cmt_interrupt` (const `rx_cmt_interrupt_config_t` config)
Configure the ICU interrupt vector priority and enable for a CMT channel.
- static `rx_err_t` `internal_validate_cmt_init_params` (const `rx_cmt_channel_t` channel, const `rx_cmt_config_t` *config)
Validate channel and configuration pointer before CMT initialisation.
- static void `internal_save_cmt_callback` (const `rx_cmt_channel_t` channel, const `rx_cmt_config_t` *config)
Persist the user callback, user-data pointer, and channel initialized flag.
- `rx_err_t` `rx_cmt_init` (const `rx_cmt_channel_t` channel, const `rx_cmt_config_t` *config)
Initialize a CMT channel for periodic interrupt generation.
- `rx_err_t` `rx_cmt_start` (const `rx_cmt_channel_t` channel)
Start CMT counter for a previously initialized channel.
- `rx_err_t` `rx_cmt_stop` (const `rx_cmt_channel_t` channel)
Stop CMT counter without deinitializing the channel.
- `rx_err_t` `rx_cmt_get_count` (const `rx_cmt_channel_t` channel, `uint16_t` *count)
Read current CMT counter (CMCNT) value.
- `rx_err_t` `rx_cmt_deinit` (const `rx_cmt_channel_t` channel)
Deinitialize CMT channel and release all resources.

Variables

- static const char *const `s_tag` = "CMT"
- static bool `s_cmt_initialized` [`k_cmt_max_channels`] = {false}
Per-channel initialization status flags.
- static `rx_cmt_callback_t` `s_cmt_callback` [`k_cmt_max_channels`] = {nullptr}
Per-channel interrupt callback function pointers.
- static void * `s_cmt_user_data` [`k_cmt_max_channels`] = {nullptr}
Per-channel user data for callbacks.

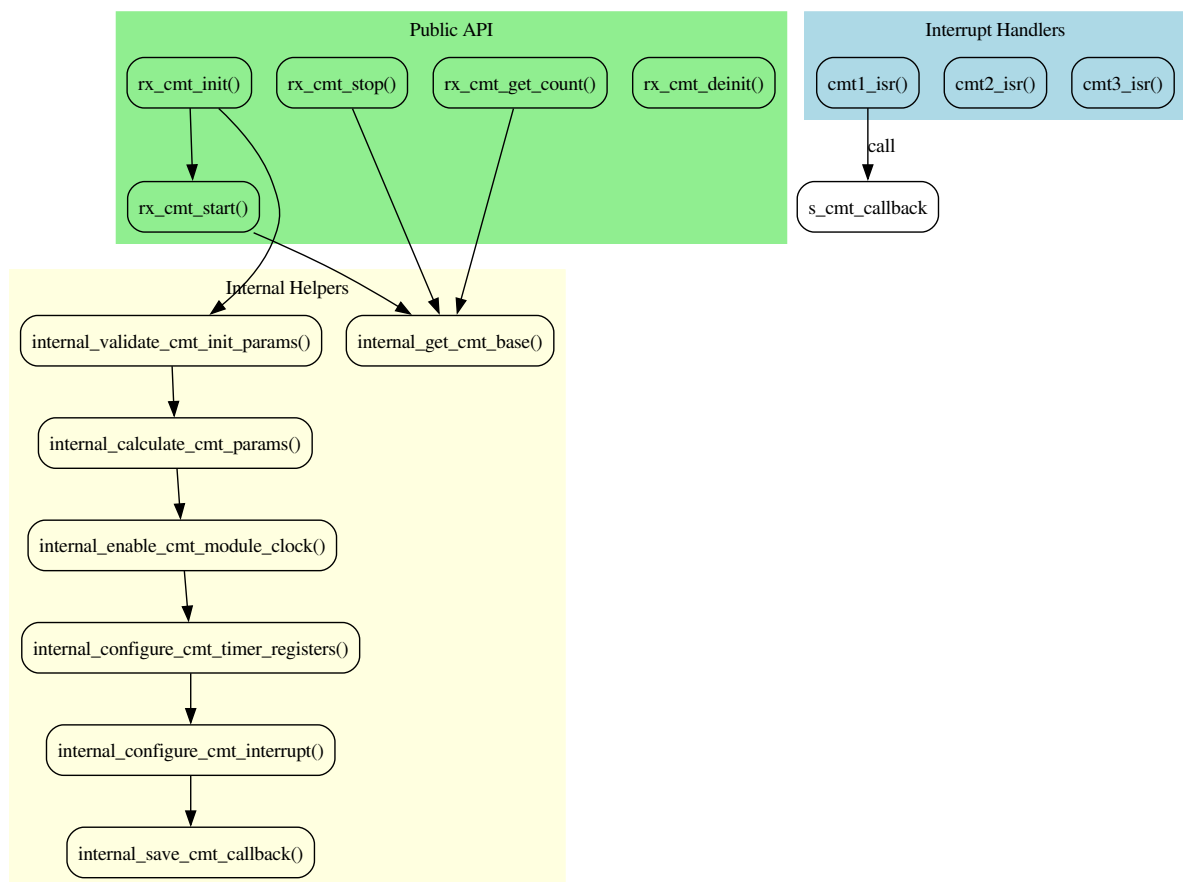
8.111.1 Detailed Description

CMT (Compare Match Timer) Driver Implementation - Periodic Interrupt Generation.

8.111.2 Overview

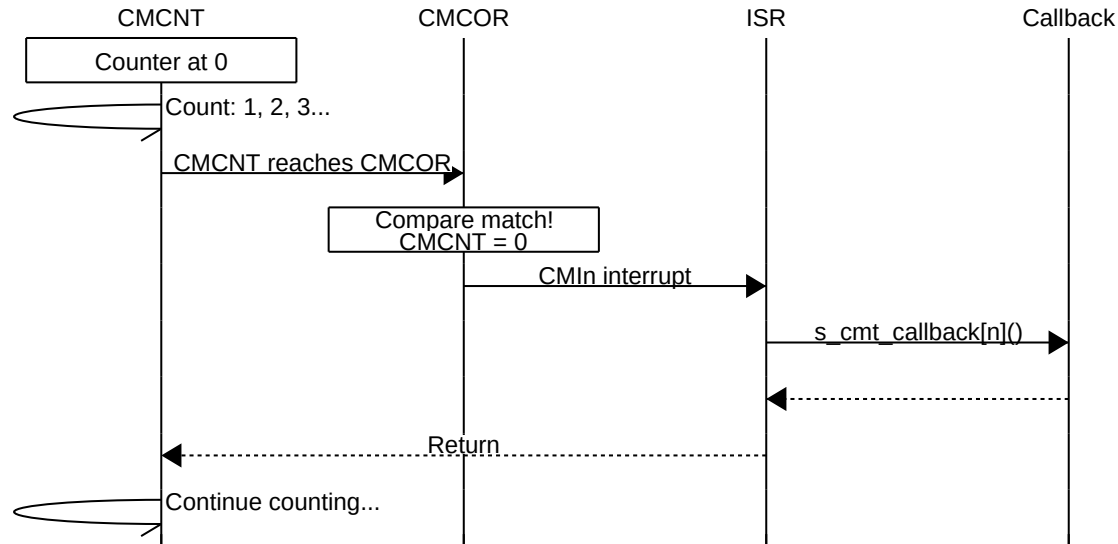
Implements the CMT driver for periodic interrupt generation on the RX72N. This file contains all internal helpers, interrupt handlers, and the public API implementation for CMT channels 0-3.

8.111.2.1 Implementation Architecture



8.111.2.2 CMT Timer Operation

The CMT operates as a 16-bit up-counter with compare match:



8.111.2.3 Clock Divider Selection Algorithm

The driver tries each divider in order ($/8$, $/32$, $/128$, $/512$) and selects the first one that produces a period fitting in 16 bits:

```

period = (PCLKB / divider) / frequency
if (0 < period <= 65535): use this divider
  
```

Divider	PCLKB/div	Min Period	Max Period
$/\leftarrow 8$	7.5 MHz	1 (7.5 MHz)	65535 (115 Hz)
$/32$	1.875 MHz	1 (1.875 MHz)	65535 (29 Hz)
$/128$	468.75 kHz	1 (469 kHz)	65535 (7.2 Hz)
$/512$	117.2 kHz	1 (117 kHz)	65535 (1.8 Hz)

8.111.2.4 Memory Map (Static State)

Symbol	Size	Description
<code>s_cmt_$\leftarrow$initialized[]</code>	4 bytes	Per-channel init flags
<code>s_cmt_$\leftarrow$callback[]</code>	16 bytes	Callback function pointers
<code>s_cmt_user_$\leftarrow$data[]</code>	16 bytes	User data pointers

Total static allocation: 36 bytes

8.111.2.5 Register Access Pattern

All register access uses inline accessor functions for type safety:

- [cmt0\(\)](#), [cmt1\(\)](#), [cmt2\(\)](#), [cmt3\(\)](#) - Channel registers
- [cmt_ctrl\(\)](#) - Control registers (CMSTR0, CMSTR1)
- [icu\(\)](#) - Interrupt controller

8.111.2.6 NASA Power of 10 Compliance

Rule	Implementation
Rule 1	No goto/recursion, single return paths
Rule 2	Bounded loop in internal_calculate_cmt_params()
Rule 3	Static arrays only, no heap
Rule 4	All functions under 60 lines
Rule 5	Parameter validation in all public functions
Rule 6	File-scope static with s_ prefix
Rule 7	All return values checked
Rule 8	C23 typed enums for all constants
Rule 9	Single-level pointers only
Rule 10	Clean with -Wall -Wextra -Werror

See also

[rx_cmt.h](#) Public API declarations

[rx72n_cmt_regs.h](#) CMT register definitions

[rx_register_protection.h](#) PRCR unlock for module stop

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_cmt.c](#).

8.111.3 Data Structure Documentation

8.111.3.1 struct rx`cmt`interrupt`config`

CMT interrupt routing and priority configuration.

Bundles the CMT channel identifier with the desired ICU interrupt priority level. Passed by value to [internal_configure_cmt_interrupt\(\)](#) so that the caller does not need to keep the structure alive after the call returns.

The channel field selects which ICU interrupt vector to program (CMI0-CMI3), and the priority field is written directly into the corresponding IPR[] register.

Invariant

- channel must be a valid [rx_cmt_channel_t](#) value (0-3)
- priority must be in [k_ipr_level_min, k_ipr_level_max]

Example:

```
rx_cmt_interrupt_config_t irq_cfg = {  
    .channel = k_cmt_channel_1,  
    .priority = 5,  
};  
rx_err_t err = internal_configure_cmt_interrupt(irq_cfg);
```

See also

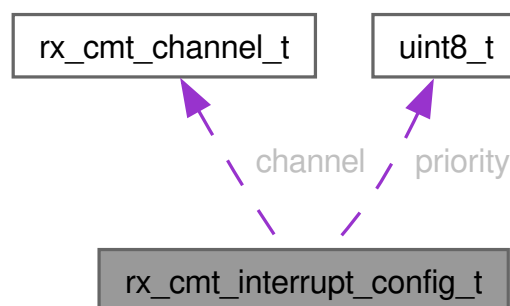
- [internal_configure_cmt_interrupt\(\)](#) Sole consumer of this struct
- [rx_cmt_init\(\)](#) Constructs this struct inline before calling the above

Since

Version 1.0.0

Definition at line 245 of file [rx_cmt.c](#).

Collaboration diagram for rx_cmt_interrupt_config_t:



Data Fields

rx_cmt_channel_t	channel	CMT channel identifier (k_cmt_channel_0 ... k_cmt_channel_3)
uint8_t	priority	ICU interrupt priority level (k_ipr_level_min ... k_ipr_level_max)

8.111.4 Enumeration Type Documentation

8.111.4.1 cmt_bit_constants_t

enum [cmt_bit_constants_t](#) : uint8_t

CMT bit manipulation constants.

Enumerator

k_cmt_bit_mask_lsb	Single bit set for shifts
k_cmt_value_zero	Zero value for comparisons

Definition at line 271 of file [rx_cmt.c](#).

```
00271      : uint8_t {
00272  k_cmt_bit_mask_lsb = 1, /**< Single bit set for shifts */
00273  k_cmt_value_zero   = 0, /**< Zero value for comparisons */
00274 } cmt_bit_constants_t;
```

8.111.4.2 cmt_cmcr_bits_t

enum [cmt_cmcr_bits_t](#) : uint8_t

CMCR register bit positions (RX72N Manual Ch31, p.1583).

Enumerator

k_cmt_cmcr_cks_shift	CKS (clock select) bit shift
k_cmt_cmcr_cmie_pos	CMIE (interrupt enable) bit position
k_cmt_cmcr_rsvd7_pos	Reserved bit 7: write value should be 1

Definition at line 251 of file [rx_cmt.c](#).

```
00251      : uint8_t {
00252  k_cmt_cmcr_cks_shift = 0, /**< CKS (clock select) bit shift */
00253  k_cmt_cmcr_cmie_pos  = 6, /**< CMIE (interrupt enable) bit position */
00254  k_cmt_cmcr_rsvd7_pos = 7, /**< Reserved bit 7: write value should be 1 */
00255 } cmt_cmcr_bits_t;
```

8.111.4.3 cmt`cmstr`bits``t`

```
enum cmt\_cmstr\_bits\_t : uint8_t
```

CMSTR register bit positions.

Enumerator

<code>k_cmt_cmstr_str0</code>	CMT0/CMT2 start bit
<code>k_cmt_cmstr_str1</code>	CMT1/CMT3 start bit

Definition at line [265](#) of file [rx_cmt.c](#).

```
00265     : uint8_t {
00266     k_cmt_cmstr_str0 = 0, /**< CMT0/CMT2 start bit */
00267     k_cmt_cmstr_str1 = 1, /**< CMT1/CMT3 start bit */
00268 } cmt_cmstr_bits_t;
```

8.111.4.4 cmt`constants``t`

```
enum cmt\_constants\_t : uint8_t
```

CMT channel and divider configuration constants.

Defines the channel count, divider settings, and loop bounds for CMT configuration. These values correspond to RX72N hardware capabilities.

8.111.4.5 Divider Settings (CMCR.CKS field)

CKS Value	Divider	Counter Clock @60MHz
0b00	$\frac{1}{8}$	7.5 MHz
0b01	$\frac{1}{32}$	1.875 MHz
0b10	$\frac{1}{128}$	468.75 kHz
0b11	$\frac{1}{512}$	117.19 kHz

Note

`k_cmt_divider_start` and `k_cmt_num_dividers` define loop bounds for NASA Power of 10 Rule 2 compliance.

Since

Version 1.0.0

Enumerator

<code>k_cmt_max_channels</code>	Total CMT channels: CMT0, CMT1, CMT2, CMT3
---------------------------------	--

k_cmt_divider_start	Loop start index for divider iteration (Rule 2)
k_cmt_div_8	CKS=0b00: PCLKB/8 divider (7.5 MHz @ 60 MHz PCLKB)
k_cmt_div_32	CKS=0b01: PCLKB/32 divider (1.875 MHz @ 60 MHz PCLKB)
k_cmt_div_128	CKS=0b10: PCLKB/128 divider (468.75 kHz @ 60 MHz PCLKB)
k_cmt_div_512	CKS=0b11: PCLKB/512 divider (117.19 kHz @ 60 MHz PCLKB)
k_cmt_num_dividers	Loop bound for divider iteration (Rule 2 compliance)

Definition at line 185 of file [rx_cmt.c](#).

```

00185      : uint8_t {
00186  k_cmt_max_channels = 4, /**< Total CMT channels: CMT0, CMT1, CMT2, CMT3 */
00187  k_cmt_divider_start = 0, /**< Loop start index for divider iteration (Rule 2) */
00188  k_cmt_div_8 = 0, /**< CKS=0b00: PCLKB/8 divider (7.5 MHz @ 60 MHz PCLKB) */
00189  k_cmt_div_32 = 1, /**< CKS=0b01: PCLKB/32 divider (1.875 MHz @ 60 MHz PCLKB) */
00190  k_cmt_div_128 = 2, /**< CKS=0b10: PCLKB/128 divider (468.75 kHz @ 60 MHz PCLKB) */
00191  k_cmt_div_512 = 3, /**< CKS=0b11: PCLKB/512 divider (117.19 kHz @ 60 MHz PCLKB) */
00192  k_cmt_num_dividers = 4, /**< Loop bound for divider iteration (Rule 2 compliance) */
00193 } cmt_constants_t;

```

8.111.4.6 cmt`divider`values`t

```
enum cmt\_divider\_values\_t : uint16_t
```

CMT divider values (actual divisor values).

Enumerator

k_cmt_divider_val_8	Divide by 8
k_cmt_divider_val_32	Divide by 32
k_cmt_divider_val_128	Divide by 128
k_cmt_divider_val_512	Divide by 512

Definition at line 196 of file [rx_cmt.c](#).

```

00196      : uint16_t {
00197  k_cmt_divider_val_8 = 8, /**< Divide by 8 */
00198  k_cmt_divider_val_32 = 32, /**< Divide by 32 */
00199  k_cmt_divider_val_128 = 128, /**< Divide by 128 */
00200  k_cmt_divider_val_512 = 512, /**< Divide by 512 */
00201 } cmt_divider_values_t;

```

8.111.4.7 cmt`module`stop`bits`t

```
enum cmt\_module\_stop\_bits\_t : uint8_t
```

CMT module stop bit positions in MSTPCRA (RX72N Manual Section 11.2.2, p.406).

CMT is controlled by MSTPCRA, NOT MSTPCRB.

- MSTPA15: CMT unit 0 (CMT0, CMT1) module stop
- MSTPA14: CMT unit 1 (CMT2, CMT3) module stop

Enumerator

k_cmt_mstpa_unit0	MSTPCRA bit 15: CMT unit 0 (CMT0, CMT1) module stop
k_cmt_mstpa_unit1	MSTPCRA bit 14: CMT unit 1 (CMT2, CMT3) module stop

Definition at line 210 of file [rx_cmt.c](#).

```
00210      : uint8_t {
00211  k_cmt_mstpa_unit0 = 15, /**< MSTPCRA bit 15: CMT unit 0 (CMT0, CMT1) module stop */
00212  k_cmt_mstpa_unit1 = 14, /**< MSTPCRA bit 14: CMT unit 1 (CMT2, CMT3) module stop */
00213 } cmt_module_stop_bits_t;
```

8.111.4.8 cmt'period'constants't

enum [cmt_period_constants_t](#) : uint16_t

Period calculation constants.

Enumerator

k_cmt_period_min	Minimum valid period
k_cmt_period_max	Maximum valid period (16-bit)
k_cmt_period_adj	Period adjustment (period - 1)

Definition at line 258 of file [rx_cmt.c](#).

```
00258      : uint16_t {
00259  k_cmt_period_min = 0, /**< Minimum valid period */
00260  k_cmt_period_max = 0xFFFF, /**< Maximum valid period (16-bit) */
00261  k_cmt_period_adj = 1, /**< Period adjustment (period - 1) */
00262 } cmt_period_constants_t;
```

8.111.5 Function Documentation

8.111.5.1 cmt1_isr()

```
void cmt1_isr (
    void )
```

CMT1 (CMT channel 1) compare-match interrupt service routine.

Invoked by the RX72N hardware whenever the CMT1 counter (CMCNT) reaches the value stored in CMCOR. The ISR checks whether a callback has been registered in `s_cmt_callback[k_cmt_channel_1]` and, if so, calls it with the associated user data pointer from `s_cmt_user_data[k_cmt_channel_1]`.

The compare-match interrupt flag is automatically cleared by the RX72N hardware upon entry; no explicit IR register write is needed.

Note

Executes in interrupt context; all code in this ISR and any registered callback must be ISR-safe (no blocking, no long operations)

Not thread-safe; the callback executes with the CMT1 interrupt masked

Warning

Do not call any ThreadX API that is not interrupt-safe from the registered callback (e.g., `tx_thread_sleep()` is forbidden; use ISR-safe variants such as `tx_semaphore_put_from_isr()`)

See also

[rx_cmt_init\(\)](#) Registers the callback invoked by this ISR
[rx_cmt_deinit\(\)](#) Clears the callback and disables this interrupt
[s_cmt_callback](#) Per-channel callback function pointer array

Since

Version 1.0.0

Definition at line 745 of file [rx_cmt.c](#).

```
00746 {
00747     /* Call user callback if registered */
00748     if (s_cmt_callback[k_cmt_channel_1] != nullptr) {
00749         s_cmt_callback[k_cmt_channel_1](s_cmt_user_data[k_cmt_channel_1]);
00750     }
00751 }
00752 /* Clear interrupt flag (read-modify-write not needed, auto-cleared) */
00753 }
```

References [k_cmt_channel_1](#), [s_cmt_callback](#), and [s_cmt_user_data](#).

8.111.5.2 cmt2_isr()

```
void cmt2_isr (
    void )
```

CMT2 (CMT channel 2) compare-match interrupt service routine.

Invoked by the RX72N hardware whenever the CMT2 counter (CMCNT) reaches the value stored in CMCOR. The ISR checks whether a callback has been registered in `s_cmt_callback[k_cmt_channel_2]` and, if so, calls it with the associated user data pointer from `s_cmt_user_data[k_cmt_channel_2]`.

The compare-match interrupt flag is automatically cleared by the RX72N hardware upon entry; no explicit IR register write is needed.

Note

Executes in interrupt context; all code in this ISR and any registered callback must be ISR-safe (no blocking, no long operations)

Not thread-safe; the callback executes with the CMT2 interrupt masked

Warning

Do not call any ThreadX API that is not interrupt-safe from the registered callback (e.g., `tx_thread_sleep()` is forbidden; use ISR-safe variants such as `tx_semaphore_put_from_isr()`)

See also

[rx_cmt_init\(\)](#) Registers the callback invoked by this ISR
[rx_cmt_deinit\(\)](#) Clears the callback and disables this interrupt
[s_cmt_callback](#) Per-channel callback function pointer array

Since

Version 1.0.0

Definition at line 781 of file [rx_cmt.c](#).

```
00782 {
00783     if (s_cmt_callback[k_cmt_channel_2] != nullptr) {
00784         s_cmt_callback[k_cmt_channel_2](s_cmt_user_data[k_cmt_channel_2]);
00785     }
00786 }
```

References [k_cmt_channel_2](#), [s_cmt_callback](#), and [s_cmt_user_data](#).

8.111.5.3 cmt3_isr()

```
void cmt3_isr (
    void )
```

CMT3 (CMT channel 3) compare-match interrupt service routine.

Invoked by the RX72N hardware whenever the CMT3 counter (CMCNT) reaches the value stored in CMCOR. The ISR checks whether a callback has been registered in [s_cmt_callback\[k_cmt_channel_3\]](#) and, if so, calls it with the associated user data pointer from [s_cmt_user_data\[k_cmt_channel_3\]](#).

The compare-match interrupt flag is automatically cleared by the RX72N hardware upon entry; no explicit IR register write is needed.

Note

Executes in interrupt context; all code in this ISR and any registered callback must be ISR-safe (no blocking, no long operations)

Not thread-safe; the callback executes with the CMT3 interrupt masked

Warning

Do not call any ThreadX API that is not interrupt-safe from the registered callback (e.g., [tx_thread_sleep\(\)](#) is forbidden; use ISR-safe variants such as [tx_semaphore_put_from_isr\(\)](#))

See also

[rx_cmt_init\(\)](#) Registers the callback invoked by this ISR
[rx_cmt_deinit\(\)](#) Clears the callback and disables this interrupt
[s_cmt_callback](#) Per-channel callback function pointer array

Since

Version 1.0.0

Definition at line 814 of file [rx_cmt.c](#).

```
00815 {
00816     if (s_cmt_callback[k_cmt_channel_3] != nullptr) {
00817         s_cmt_callback[k_cmt_channel_3](s_cmt_user_data[k_cmt_channel_3]);
00818     }
00819 }
```

References [k_cmt_channel_3](#), [s_cmt_callback](#), and [s_cmt_user_data](#).

8.111.5.4 internal'calculate'cmt'params()

```
rx_err_t internal_calculate_cmt_params (
    const uint32_t frequency_hz,
    uint8_t * divider,
    uint16_t * cmcor) [static]
```

Calculate the optimal CMT clock divider and compare-match value for a target frequency.

Iterates the four available PCLKB clock dividers (/8, /32, /128, /512) in ascending order and selects the first divider whose resulting 16-bit period value satisfies:

```
period = (PCLKB / divider) / frequency_hz
0 < period <= 65535
```

The chosen divider index is returned in *divider (corresponding to the CKS field of CMCR: 0=/8, 1=/32, 2=/128, 3=/512). The compare-match register value is returned in *cmcor; the caller must subtract the adjustment constant k_cmt_period_adj when writing to the hardware register.

Algorithm steps:

1. Validate output pointers are non-null
2. Reject frequency_hz == 0 or frequency_hz > PCLKB/8 (maximum achievable)
3. Loop i from k_cmt_divider_start to k_cmt_num_dividers - 1
4. Compute period_calc = (PCLKB / dividers[i]) / frequency_hz
5. If 0 < period_calc <= 65535, store i and period_calc, return k_rx_ok
6. If no divider fits, return k_rx_err_invalid_arg

Precondition

divider must point to a valid uint8_t storage location
cmcor must point to a valid uint16_t storage location

Postcondition

On k_rx_ok: *divider holds the CKS value (0-3) for CMCR
On k_rx_ok: *cmcor holds the period count; caller subtracts 1 before writing CMCR

Note

Not thread-safe; pure computation with no shared state modification
Uses a bounded loop (k_cmt_num_dividers iterations) – NASA Rule 2 compliant

Thread Safety:

Pure computation with no global or static state; safe to call from any context.

Performance:

Execution time: ~5 cycles (loop unrolled at -O2 for 4 divider steps)

Example:

```
uint8_t divider = 0;
uint16_t cmcor = 0;
rx_err_t err = internal_calculate_cmt_params(1000U, &divider, &cmcor);
if (err == k_rx_ok) {
    // divider and cmcor are ready for use in CMCR/CMCOR registers
}
```

See also

[internal_configure_cmt_timer_registers\(\)](#) Consumes the output of this function

[cmt_constants_t](#) Divider index constants ([k_cmt_div_8](#) ... [k_cmt_div_512](#))

[cmt_divider_values_t](#) Actual divisor values (8, 32, 128, 512)

Since

Version 1.0.0

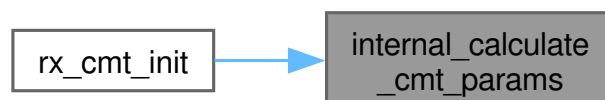
Definition at line 461 of file [rx_cmt.c](#).

```
00462 {
00463     static const uint16_t dividers[] = {k_cmt_divider_val_8,
00464                                         k_cmt_divider_val_32,
00465                                         k_cmt_divider_val_128,
00466                                         k_cmt_divider_val_512};
00467     const uint32_t pclk = k_pclk_hz;
00468     const uint32_t max_frequency = pclk / k_cmt_divider_val_8;
00469
00470     if (divider == nullptr || cmcor == nullptr) {
00471         return k_rx_err_null_ptr;
00472     }
00473
00474     if (frequency_hz <= (uint32_t)k_cmt_value_zero || frequency_hz > max_frequency) {
00475         return k_rx_err_invalid_arg;
00476     }
00477
00478     /* Try each divider to find one that fits in 16-bit */
00479     for (uint8_t i = k_cmt_divider_start; i < k_cmt_num_dividers; i++) {
00480         const uint32_t period_calc = (pclk / dividers[i]) / frequency_hz;
00481
00482         /* Check if period fits in 16-bit and is reasonable */
00483         if (period_calc > k_cmt_period_min && period_calc <= k_cmt_period_max) {
00484             *divider = i;
00485             *cmcor = (uint16_t)period_calc;
00486             return k_rx_ok;
00487         }
00488     }
00489
00490     rx_log_error(s_tag, "Error occurred");
00491     return k_rx_err_invalid_arg;
00492 }
```

References [k_cmt_divider_start](#), [k_cmt_divider_val_128](#), [k_cmt_divider_val_32](#), [k_cmt_divider_val_512](#), [k_cmt_divider_val_8](#), [k_cmt_num_dividers](#), [k_cmt_period_max](#), [k_cmt_period_min](#), [k_cmt_value_zero](#), [k_pclk_hz](#), and [s_tag](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the caller graph for this function:



8.111.5.5 internal_configure_cmt_interrupt()

```
rx_err_t internal_configure_cmt_interrupt (
    const rx_cmt_interrupt_config_t config) [static]
```

Configure the ICU interrupt vector priority and enable for a CMT channel.

Programs the Interrupt Controller Unit (ICU) to route and enable the compare-match interrupt (CMI_n) for the specified CMT channel. The three ICU registers modified are:

1. IPR[vector] - Sets the interrupt priority level for the channel's CMI_n vector to config.priority.
2. IER[index] - Sets the enable bit for the channel's interrupt vector within the IER register array. The bit index and byte index are derived from the vector number.
3. IR[vector] - Clears any pending interrupt flag before enabling, so a stale flag does not fire immediately after enable.

The vector number is computed as `k_vect_cmt0_cmi0 + config.channel`, which yields CMI₀ for channel 0, CMI₁ for channel 1, and so on.

Precondition

config.channel must be in the range [0, k_cmt_max_channels)
 config.priority must be in [k_ipr_level_min, k_ipr_level_max]

Postcondition

ICU IPR register for the channel's vector set to config.priority
 ICU IER bit for the channel's vector enabled
 ICU IR flag for the channel's vector cleared

Note

Not thread-safe; concurrent ICU modification from another context is unsafe
 Clears IR before enabling to prevent a spurious first interrupt

Thread Safety:

Not thread-safe. Must be called during initialization with interrupts disabled or before the RTOS scheduler starts.

Performance:

Execution time: ~4 register accesses; negligible overhead

Example:

```
rx_cmt_interrupt_config_t irq_cfg = {
    .channel = k_cmt_channel_1,
    .priority = 5,
};
rx_err_t err = internal_configure_cmt_interrupt(irq_cfg);
if (err != k_rx_ok) {
    return err;
}
```

See also

[rx_cmt_interrupt_config_t](#) Configuration structure passed to this function

[rx_cmt_deinit\(\)](#) Clears the IER bit to disable the interrupt

[rx_cmt_init\(\)](#) Constructs config and calls this function

Since

Version 1.0.0

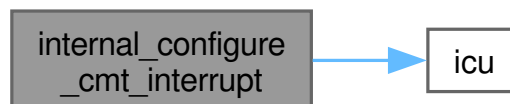
Definition at line 681 of file [rx_cmt.c](#).

```
00682 {
00683     /* Note: Lower bound check omitted - config.channel is uint8_t, k_cmt_channel_0 == 0,
00684      * so config.channel >= k_cmt_channel_0 is always true (-Wtype-limits) */
00685     if ((uint8_t)config.channel >= (uint8_t)k_cmt_max_channels) {
00686         return k_rx_err_invalid_arg;
00687     }
00688     if (config.priority < k_ipr_level_min || config.priority > k_ipr_level_max) {
00689         return k_rx_err_invalid_arg;
00690     }
00691 }
00692 /* CMI0/CMI1 have fixed ICU vectors (28, 29). CMI2/CMI3 are SELECTB
00693  * sources on RX72N and need SLIBRn programming, which is not yet
00694  * implemented in this driver. Reject interrupt-mode init on ch2/ch3
00695  * rather than enabling the wrong vectors (30/31 are CMWI0/CMWI1 for
00696  * CMTW timers). Per RX72N HW manual R01UH0824EJ0111 Section 15.3
00697  * (Interrupt Vector Table) and Section 15.4 (SELECTB sources). */
00698 if (config.channel == k_cmt_channel_2 || config.channel == k_cmt_channel_3) {
00699     return k_rx_err_not_supported;
00700 }
00701
00702 const uint8_t vector = k_vect_cmt0_cmi0 + config.channel;
00703 const uint8_t ier_index = vector / k_icu_ier_bits_per_reg;
00704 const uint8_t ier_bit = vector % k_icu_ier_bits_per_reg;
00705
00706 icu()->ipr[vector] = config.priority;
00707 const uint8_t ier_mask = (uint8_t)(k_cmt_bit_mask_lsb « ier_bit);
00708 icu()->ier[ier_index] |= ier_mask;
00709 icu()->ir[vector] = k_cmt_value_zero;
00710
00711 return k_rx_ok;
00712 }
```

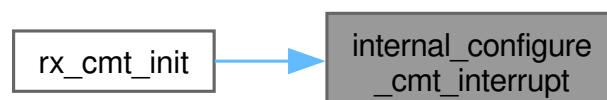
References [rx_cmt_interrupt_config_t::channel](#), [icu\(\)](#), [rx_icu_regs_t::ier](#), [rx_icu_regs_t::ipr](#), [rx_icu_regs_t::ir](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_2](#), [k_cmt_channel_3](#), [k_cmt_max_channels](#), [k_cmt_value_zero](#), [k_icu_ier_bits_per_reg](#), [k_ipr_level_max](#), [k_ipr_level_min](#), [k_vect_cmt0_cmi0](#), and [rx_cmt_interrupt_config_t::priority](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.111.5.6 internal_configure_cmt_timer_registers()

```
void internal_configure_cmt_timer_registers (  
    volatile rx_cmt_channel_regs_t * cmt,  
    const uint8_t divider,  
    const uint16_t cmcor)    [static]
```

Write clock divider, interrupt-enable, compare-match, and counter registers.

Programs the three CMT hardware registers for a single channel in the following order:

1. CMCOR - Sets the CKS (clock-select) field from divider and enables the compare-match interrupt (CMIE bit).
2. CMCOR - Loads the compare-match value after subtracting the adjustment constant `k_cmt_period_adj` (-1) so the hardware fires at the correct period.
3. CMCNT - Clears the counter to 0 so the first interrupt occurs exactly one full period after the timer is started.

The timer must be stopped (CMSTR bit cleared) before calling this function to avoid race conditions while writing to CMCOR and CMCNT.

Precondition

`cmt` must not be nullptr (enforced by `RX_ASSERT`)

The CMT timer must be stopped before calling this function

Postcondition

`cmt->cmcr` = (divider << CKS_SHIFT) | CMIE_BIT | RSVD7_BIT (bit7 must be written 1)

`cmt->cmcor` = `cmcor` - `k_cmt_period_adj`

`cmt->cmcnt` = 0

Note

Not thread-safe; caller must ensure the timer is stopped

`RX_ASSERT` fires in debug builds if `cmt` is nullptr

Thread Safety:

Not thread-safe. Must be called with the timer stopped and no concurrent ISR activity on this channel.

Performance:

Execution time: ~3 register writes; negligible overhead

Example:

```
uint8_t divider = 0;
uint16_t cmcor = 0;
(void)internal_calculate_cmt_params(1000U, &divider, &cmcor);
volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(k_cmt_channel_1);
internal_configure_cmt_timer_registers(cmt, divider, cmcor);
```

See also

[internal_calculate_cmt_params\(\)](#) Produces the divider and cmcor arguments
[rx_cmt_stop\(\)](#) Must be called before this function
[cmt_cmcr_bits_t](#) CKS shift and CMIE bit position constants

Since

Version 1.0.0

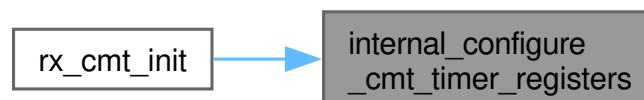
Definition at line 602 of file [rx_cmt.c](#).

```
00605 {
00606   RX_ASSERT(cmt != nullptr, "CMT registers are NULL");
00607
00608   /* Configure timer control register
00609    * - Set clock divider (bits 1-0)
00610    * - Enable compare match interrupt (bit 6)
00611    * - Set reserved bit 7 to 1 (write value should be 1 per RX72N manual p.1583)
00612    */
00613   cmt->cmcr =
00614     (divider « k_cmt_cmcr_cks_shift) | /* CKS: Clock Select */
00615     (k_cmt_bit_mask_lsb « k_cmt_cmcr_cmie_pos) | /* CMIE: Compare Match Interrupt Enable */
00616     (k_cmt_bit_mask_lsb « k_cmt_cmcr_rsvd7_pos); /* Reserved bit 7: write value should be 1 */
00617
00618   /* Set compare match value (period - 1) */
00619   cmt->cmcor = cmcor - k_cmt_period_adj;
00620
00621   /* Clear counter */
00622   cmt->cmcnt = k_cmt_value_zero;
00623 }
```

References [rx_cmt_channel_regs_t::cmcnt](#), [rx_cmt_channel_regs_t::cmcor](#), [rx_cmt_channel_regs_t::cmcr](#), [k_cmt_bit_mask_lsb](#), [k_cmt_cmcr_cks_shift](#), [k_cmt_cmcr_cmie_pos](#), [k_cmt_cmcr_rsvd7_pos](#), [k_cmt_period_adj](#), and [k_cmt_value_zero](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the caller graph for this function:



8.111.5.7 internal`enable`cmt`module`clock()

```
void internal_enable_cmt_module_clock (
    void ) [static]
```

Enable the CMT module clock by clearing the MSTPCRB module-stop bit.

Removes the CMT0-CMT3 peripheral from module-stop state so that the module clock is supplied and its registers become accessible. The sequence is:

1. Unlock the write-protect register (PRCR) for PRC1 and PRC3 bits
2. Clear bit `k_cmt_mstpa_unit0` (bit 15) in `MSTPCRA` to enable CMT unit 0 (CMT0/CMT1)
3. Clear bit `k_cmt_mstpa_unit1` (bit 14) in `MSTPCRA` to enable CMT unit 1 (CMT2/CMT3)
4. Re-lock the PRCR to prevent accidental modification

This function must be called before any CMT register access; reading or writing CMT registers while the module clock is stopped produces undefined hardware behaviour on the RX72N.

The function is idempotent: calling it when the module is already enabled simply clears an already-clear bit, which is harmless.

Precondition

The system clock (PCLKB) must be running before calling this function
No other task or ISR must be concurrently modifying `MSTPCRB`

Postcondition

`MSTPCRA` bits 15 and 14 are cleared; CMT module clocks for all 4 channels are active
PRCR is locked upon return (prevents accidental `MSTPCRB` changes)

Note

Not thread-safe; PRCR unlock/lock is a non-atomic read-modify-write
Called once per system lifetime; subsequent calls are benign

Thread Safety:

Not thread-safe. Call during system initialization before RTOS scheduler starts, or with interrupts disabled.

Performance:

Execution time: ~4 register writes; negligible overhead

Example:

```
// Called internally by rx_cmt_init(); not intended for direct use
internal_enable_cmt_module_clock();
// CMT registers are now accessible
```

See also

`rx_register_protection.h` PRCR lock/unlock constants
`rx_cmt_init()` Sole caller; invoked before register programming

Since

Version 1.0.0

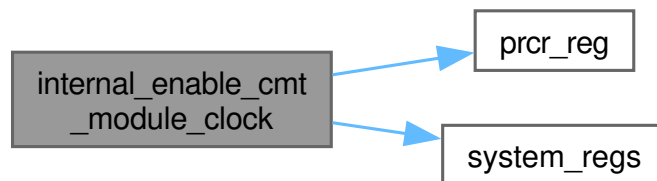
Definition at line 542 of file [rx_cmt.c](#).

```
00543 {
00544     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00545     /* CMT is in MSTPCRA (not MSTPCRB): bit 15 = unit 0 (CMT0/1), bit 14 = unit 1 (CMT2/3) */
00546     system_regs()->mstpcra &= ~((uint32_t)k_cmt_bit_mask_lsb « k_cmt_mstpa_unit0);
00547     system_regs()->mstpcra &= ~((uint32_t)k_cmt_bit_mask_lsb « k_cmt_mstpa_unit1);
00548     *prcr_reg() = k_rx_prcr_lock;
00549 }
```

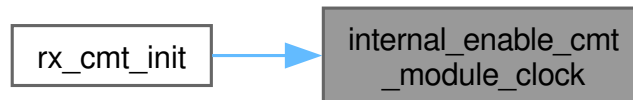
References [k_cmt_bit_mask_lsb](#), [k_cmt_mstpa_unit0](#), [k_cmt_mstpa_unit1](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.111.5.8 internal`get`cmt`base()

```
volatile rx_cmt_channel_regs_t * internal_get_cmt_base (
    const rx_cmt_channel_t channel) [static]
```

Map a CMT channel enum to its hardware register base pointer.

Uses a switch statement to convert a logical channel identifier into a pointer to the corresponding volatile [rx_cmt_channel_regs_t](#) register block. The four CMT channels (CMT0-CMT3) each have a distinct base address in the RX72N memory map; the inline accessor functions [cmt0\(\)](#)-[cmt3\(\)](#) encode those addresses in a type-safe manner.

The default case returns nullptr, which every caller is expected to check immediately after the call (NASA Rule 7 compliance).

Precondition

channel is a member of [rx_cmt_channel_t](#) (0-3)

Hardware register addresses must be accessible (clocks enabled before use)

Postcondition

Return value, if non-null, points to the live hardware register block
 Caller must validate the return value before dereferencing (Rule 7)

Note

Not thread-safe; the hardware registers are globally shared
 This function never writes to hardware; it only computes an address

Thread Safety:

Read-only address computation; no shared mutable state is accessed.

Performance:

Execution time: ~1 cycle (single branch/address load with -O2)

Example:

```
volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(k_cmt_channel_2);
if (cmt == nullptr) {
    return k_rx_err_invalid_arg;
}
cmt->cmcnt = 0;
```

See also

[cmt0\(\)](#) Inline accessor for CMT0 base address
[cmt1\(\)](#) Inline accessor for CMT1 base address
[cmt2\(\)](#) Inline accessor for CMT2 base address
[cmt3\(\)](#) Inline accessor for CMT3 base address

Since

Version 1.0.0

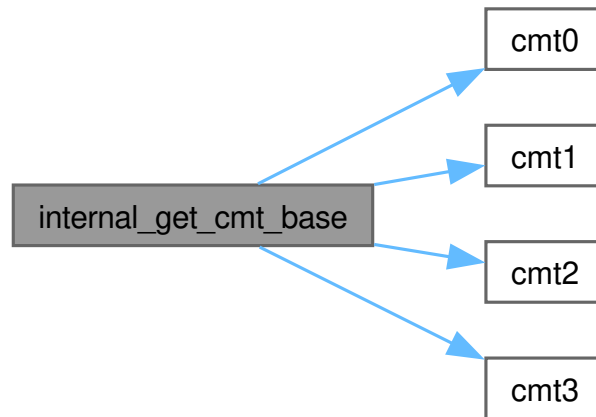
Definition at line 385 of file [rx_cmt.c](#).

```
00386 {
00387     switch (channel) {
00388         case k_cmt_channel_0:
00389             return cmt0();
00390         case k_cmt_channel_1:
00391             return cmt1();
00392         case k_cmt_channel_2:
00393             return cmt2();
00394         case k_cmt_channel_3:
00395             return cmt3();
00396         default:
00397             return nullptr;
00398     }
00399 }
```

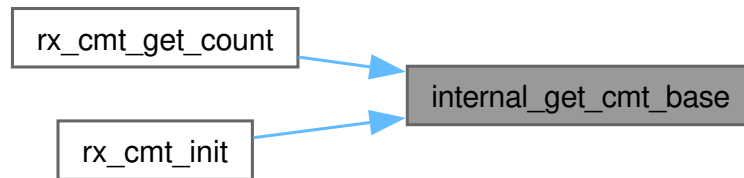
References [cmt0\(\)](#), [cmt1\(\)](#), [cmt2\(\)](#), [cmt3\(\)](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_channel_2](#), and [k_cmt_channel_3](#).

Referenced by [rx_cmt_get_count\(\)](#), and [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.111.5.9 internal_save_cmt_callback()

```

void internal_save_cmt_callback (
    const rx_cmt_channel_t channel,
    const rx_cmt_config_t * config) [static]
  
```

Persist the user callback, user-data pointer, and channel initialized flag.

Copies the callback function pointer and user-data pointer from the caller's configuration structure into the module-level static arrays, then marks the channel as initialized. After this call the ISR for the channel will invoke the registered callback on every compare-match event.

This function is the final step in the `rx_cmt_init()` sequence. It must only be called after all hardware registers have been programmed and the interrupt has been configured in the ICU, so that the ISR never encounters a partially initialised channel state.

Precondition

channel must be valid and in range [1, k_cmt_max_channels) (validated by caller)
 config must be non-null (validated by `internal_validate_cmt_init_params`)

Postcondition

```

s_cmt_callback[channel] = config->callback
s_cmt_user_data[channel] = config->user_data
s_cmt_initialized[channel] = true
  
```

Note

Not thread-safe; must be the last step of `rx_cmt_init()` with interrupts either disabled or not yet fired for this channel

After this call the channel is considered "live"; the ISR may fire at any point

Thread Safety:

Not thread-safe. The three static array writes are not atomic. Call only during initialization with the channel interrupt not yet active.

Performance:

Execution time: ~3 stores; negligible overhead

Example:

```
// Final step inside rx_cmt_init() after all hardware setup is complete
internal_save_cmt_callback(channel, config);
// s_cmt_initialized[channel] is now true; ISR will call config->callback
```

See also

[rx_cmt_init\(\)](#) Sole caller; calls this as the last initialisation step

[rx_cmt_deinit\(\)](#) Clears the state written by this function

[s_cmt_callback](#) Destination array for config->callback

[s_cmt_user_data](#) Destination array for config->user_data

[s_cmt_initialized](#) Destination array for the initialized flag

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Definition at line 955 of file [rx_cmt.c](#).

```
00957 {
00958   s_cmt_callback[channel] = config->callback;
00959   s_cmt_user_data[channel] = config->user_data;
00960   s_cmt_initialized[channel] = true;
00961 }
```

References [rx_cmt_config_t::callback](#), [s_cmt_callback](#), [s_cmt_initialized](#), [s_cmt_user_data](#), and [rx_cmt_config_t::user_data](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the caller graph for this function:



8.111.5.10 internal_validate_cmt_init_params()

```
rx_err_t internal_validate_cmt_init_params (
    const rx_cmt_channel_t channel,
    const rx_cmt_config_t * config) [static]
```

Validate channel and configuration pointer before CMT initialisation.

Performs all parameter checking required before `rx_cmt_init()` proceeds to hardware configuration. Validation is performed in this strict order:

1. Verify config pointer is non-null (`k_rx_err_null_ptr`)
2. Verify channel is within `[0, k_cmt_max_channels)` (`k_rx_err_invalid_arg`)
3. Reject channel 0, which is reserved for the ThreadX system-tick timer (`k_rx_err_conflict`)
4. Verify interrupt priority is within `[k_ipr_level_min, k_ipr_level_max]` (`k_rx_err_invalid_arg`)

Centralising these checks in a dedicated helper keeps the main `rx_cmt_init()` function concise and ensures that every check is logged before returning.

Precondition

config must be a pointer to a caller-owned `rx_cmt_config_t` (may be nullptr)
channel must be a member of `rx_cmt_channel_t`

Postcondition

On `k_rx_ok`: channel and `*config` are safe to use for hardware programming
On error: an error message has been emitted via `rx_log_error()`

Note

Not thread-safe; pure validation with no shared state modification

The lower-bound check `channel >= k_cmt_channel_0` is intentionally omitted to avoid -Wtype-limits: channel is `uint8_t` and `k_cmt_channel_0 == 0`, making the comparison always true

Thread Safety:

Pure validation; no shared mutable state is modified. Safe to call from any context, but the result may be stale if state changes after the call.

Performance:

Execution time: ~ 4 comparisons; negligible overhead

Example:

```
rx_err_t err = internal_validate_cmt_init_params(k_cmt_channel_1, &config);
if (err != k_rx_ok) {
    return err;
}
```

See also

[rx_cmt_init\(\)](#) Sole caller; uses this to guard hardware programming

[rx_cmt_config_t](#) Configuration structure whose fields are validated here

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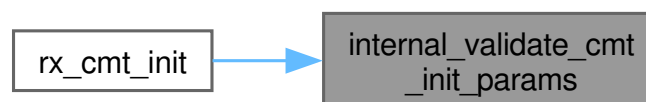
Definition at line 876 of file [rx_cmt.c](#).

```
00878 {
00879     if (config == nullptr) {
00880         rx_log_error(s_tag, "config pointer is nullptr");
00881         return k_rx_err_null_ptr;
00882     }
00883
00884     /* Note: Lower bound check omitted - channel is uint8_t, k_cmt_channel_0 == 0,
00885      * so channel >= k_cmt_channel_0 is always true (-Wtype-limits) */
00886     if ((uint8_t)channel >= (uint8_t)k_cmt_max_channels) {
00887         rx_log_error(s_tag, "Invalid CMT channel");
00888         return k_rx_err_invalid_arg;
00889     }
00890
00891     /* CMT0 is reserved for ThreadX system tick */
00892     if (channel == k_cmt_channel_0) {
00893         rx_log_error(s_tag, "CMT0 is reserved for ThreadX");
00894         return k_rx_err_conflict;
00895     }
00896
00897     if (config->priority < k_ipr_level_min || config->priority > k_ipr_level_max) {
00898         rx_log_error(s_tag, "Invalid interrupt priority");
00899         return k_rx_err_invalid_arg;
00900     }
00901
00902     return k_rx_ok;
00903 }
```

References [k_cmt_channel_0](#), [k_cmt_max_channels](#), [k_ipr_level_max](#), [k_ipr_level_min](#), [rx_cmt_config_t::priority](#), and [s_tag](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the caller graph for this function:



8.111.5.11 rx_cmt_deinit()

```
rx_err_t rx_cmt_deinit (
    const rx_cmt_channel_t channel) [nodiscard]
```

Deinitialize CMT channel and release all resources.

Deinitialize CMT channel.

Stops the CMT counter, disables the ICU interrupt for the channel, clears the stored callback and user data, and marks the channel as uninitialized. After this call the channel may be re-initialized with [rx_cmt_init\(\)](#).

Algorithm steps:

1. Validate channel range
2. Stop timer via [rx_cmt_stop\(\)](#) and propagate any error
3. Compute ICU vector, IER index, and IER bit for the channel
4. Clear the IER bit to disable the interrupt
5. Clear callback, user_data, and initialized flag

Precondition

channel must be in range [0, 3]

Caller should ensure no other task is using the channel callback

Postcondition

Counter halted (CMSTR bit cleared)

ICU interrupt disabled for channel

s_cmt_initialized[channel] set to false

s_cmt_callback[channel] and s_cmt_user_data[channel] set to nullptr

Note

Safe to call on an uninitialized channel ([rx_cmt_stop](#) tolerates it)

Module clock is NOT re-asserted - hardware remains clocked for re-use

Thread Safety:

Not thread-safe. Ensure no ISR or other thread is using the channel.

Example:

```
rx_err_t err = rx_cmt_deinit(k_cmt_channel_1);
if (err != k_rx_ok) {
    rx_log_error("APP", "CMT deinit failed");
}
```

See also

[rx_cmt_init\(\)](#) Re-initialize after deinit

[rx_cmt_stop\(\)](#) Stop without full deinit

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Definition at line 1390 of file [rx_cmt.c](#).

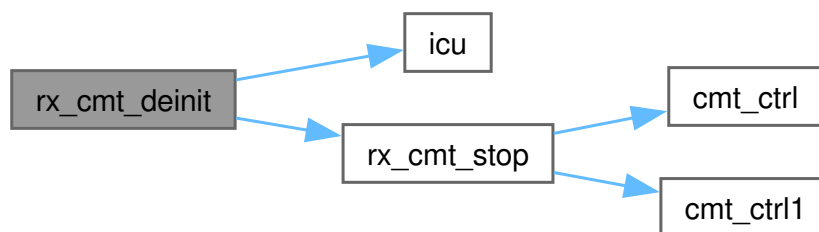
```

01391 {
01392     if ((uint8_t)channel >= k_cmt_max_channels) {
01393         return k_rx_err_invalid_arg;
01394     }
01395
01396     /* Stop timer and propagate any errors */
01397     rx_err_t err = rx_cmt_stop(channel);
01398     if (err != k_rx_ok) {
01399         return err;
01400     }
01401
01402     /* Disable interrupt only for channels with fixed ICU vectors (CMI0/CMI1).
01403      * CMI2/CMI3 are SELECTB sources; interrupt wiring for those is not
01404      * implemented (see internal_configure_cmt_interrupt() for details). */
01405     if (channel == k_cmt_channel_0 || channel == k_cmt_channel_1) {
01406         const uint8_t vector = k_vect_cmt0_cmi0 + channel;
01407         const uint8_t ier_index = vector / k_icu_ier_bits_per_reg;
01408         const uint8_t ier_bit = vector % k_icu_ier_bits_per_reg;
01409         const uint8_t ier_mask = (uint8_t)(k_cmt_bit_mask_lsb « ier_bit);
01410         icu()->ier[ier_index] &= (uint8_t)~ier_mask;
01411     }
01412
01413     /* Clear callback */
01414     s_cmt_callback[channel] = nullptr;
01415     s_cmt_user_data[channel] = nullptr;
01416     s_cmt_initialized[channel] = false;
01417
01418     rx_log_info(s_tag, "CMT deinitialized");
01419
01420     return k_rx_ok;
01421 }

```

References [icu\(\)](#), [rx_icu_regs_t::ier](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_max_channels](#), [k_icu_ier_bits_per_reg](#), [k_vect_cmt0_cmi0](#), [rx_cmt_stop\(\)](#), [s_cmt_callback](#), [s_cmt_initialized](#), [s_cmt_user_data](#), and [s_tag](#).

Here is the call graph for this function:



8.111.5.12 rx_cmt_get_count()

```

rx_err_t rx_cmt_get_count (
    const rx_cmt_channel_t channel,
    uint16_t * count) [nodiscard]

```

Read current CMT counter (CMCNT) value.

Get CMT counter value.

Returns the instantaneous value of the 16-bit CMCNT register for the specified channel. The counter increments at PCLKB divided by the configured divider (/8, /32, /128, or /512) and resets to 0 on compare match. Useful for measuring elapsed time within a period.

Precondition

Channel must be initialized via [rx_cmt_init\(\)](#)
 count must point to valid uint16_t storage

Postcondition

*count set to CMCNT register value at time of read
 Counter continues running (read is non-destructive)

Note

Value may change between consecutive reads if counter is running
 For precise time measurements disable interrupts around consecutive reads

Thread Safety:

Read-only; safe to call from multiple contexts, but value may be stale.

Example:

```
uint16_t ticks = 0;
rx_err_t err = rx_cmt_get_count(k_cmt_channel_1, &ticks);
if (err == k_rx_ok) {
    // ticks = current counter position within the period
}
```

See also

[rx_cmt_init\(\)](#) Initialize channel and set frequency
[rx_cmt_start\(\)](#) Start the counter

Since

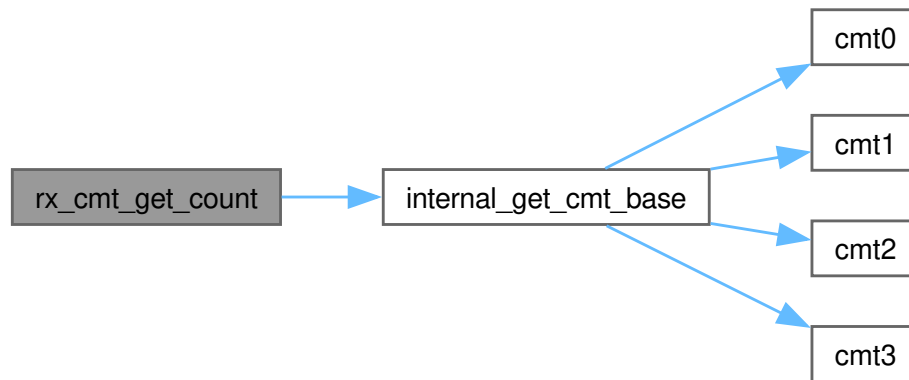
Version 1.0.0

Definition at line 1326 of file [rx_cmt.c](#).

```
01327 {
01328     const volatile rx_cmt_channel_regs_t* cmt = nullptr;
01329
01330     RX_CHECK_NULL_PTR(count, s_tag, "count pointer is nullptr");
01331
01332     if ((uint8_t)channel >= k_cmt_max_channels || !s_cmt_initialized[channel]) {
01333         return k_rx_err_invalid_state;
01334     }
01335
01336     cmt = internal_get_cmt_base(channel);
01337     if (cmt == nullptr) {
01338         return k_rx_err_invalid_arg;
01339     }
01340
01341     *count = cmt->cmcnt;
01342     return k_rx_ok;
01343 }
```

References [rx_cmt_channel_regs_t::cmcnt](#), [internal_get_cmt_base\(\)](#), [k_cmt_max_channels](#), [s_cmt_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.111.5.13 rx_cmt_init()

```

rx_err_t rx_cmt_init (
    const rx_cmt_channel_t channel,
    const rx_cmt_config_t * config) [nodiscard]

```

Initialize a CMT channel for periodic interrupt generation.

Initialize CMT channel for periodic interrupts.

Configures a CMT (Compare Match Timer) channel to fire a user callback at the requested frequency. Validates parameters, selects the optimal clock divider, enables the module clock, stops the timer for safe configuration, programs timer registers, configures the ICU interrupt routing, saves the callback, and starts the timer.

Algorithm steps:

1. Validate channel and config parameters via [internal_validate_cmt_init_params\(\)](#)
2. Obtain CMT channel register base via [internal_get_cmt_base\(\)](#)
3. Calculate optimal divider and CMCOR value via [internal_calculate_cmt_params\(\)](#)
4. Enable CMT module clock (clear MSTPCRB bit) via [internal_enable_cmt_module_clock\(\)](#)
5. Stop the timer to allow safe register writes via [rx_cmt_stop\(\)](#)
6. Program CMCR (divider + CMIE) and CMCOR via [internal_configure_cmt_timer_registers\(\)](#)
7. Configure ICU interrupt vector priority and enable via [internal_configure_cmt_interrupt\(\)](#)
8. Save callback pointer and mark channel initialized via [internal_save_cmt_callback\(\)](#)
9. Start the timer via [rx_cmt_start\(\)](#)

Clock divider selection: The driver iterates dividers /8, /32, /128, /512 and picks the first that produces a period fitting in 16 bits:

```

period = (PCLKB / divider) / frequency_hz
if (0 < period <= 65535): use this divider

```


Precondition

config must point to a valid `rx_cmt_config_t` structure
config->frequency_hz must be > 0 and $\leq \text{PCLKB} / 8$ (7.5 MHz at 60 MHz PCLKB)
channel must not be `k_cmt_channel_0`
System clocks must be initialized ($\text{PCLKB} = 60 \text{ MHz}$)

Postcondition

Timer running at requested frequency
config->callback registered and called from ISR at each compare match
`s_cmt_initialized[channel]` set to true
ICU interrupt for the channel enabled at specified priority

Note

Not thread-safe - call once per channel during system initialization
Callback executes in interrupt context; keep it short and non-blocking

Warning

CMT0 is reserved for ThreadX; do not use `k_cmt_channel_0`

Thread Safety:

Not thread-safe. Call once per channel during initialization before starting the RTOS scheduler or with interrupts disabled.

Performance:

- Execution time: $\sim 5 \text{ us}$ (mostly register writes)
- Stack usage: 48 bytes

Example:

```
static void my_timer_cb(void* user_data)
{
    // Called at 1 kHz from interrupt context
    (void)user_data;
    toggle_led();
}

const rx_cmt_config_t cfg = {
    .frequency_hz = 1000, // 1 kHz
    .priority     = 5,
    .callback     = my_timer_cb,
    .user_data    = nullptr,
};

rx_err_t err = rx_cmt_init(k_cmt_channel_1, &cfg);
if (err != k_rx_ok) {
    rx_log_error("APP", "CMT init failed");
}
```

See also

[rx_cmt_deinit\(\)](#) Stop and release channel
[rx_cmt_start\(\)](#) Start a previously stopped channel
[rx_cmt_stop\(\)](#) Stop the timer without deinitializing
[rx_cmt_config_t](#) Configuration structure definition

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Definition at line 1051 of file [rx_cmt.c](#).

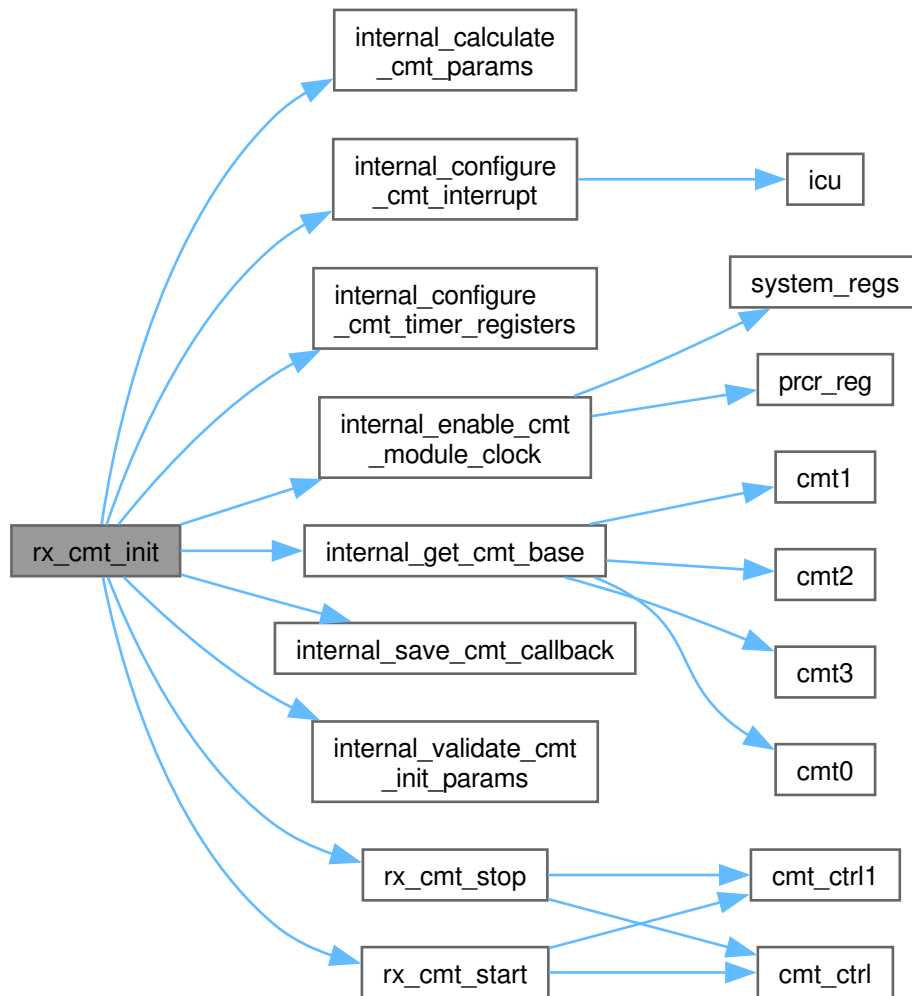
```

01052 {
01053     /* Validate parameters */
01054     rx_err_t err = internal_validate_cmt_init_params(channel, config);
01055     if (err != k_rx_ok) {
01056         return err;
01057     }
01058
01059     volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(channel);
01060     if (cmt == nullptr) {
01061         return k_rx_err_invalid_arg;
01062     }
01063
01064     /* Calculate divider and compare value */
01065     uint8_t divider;
01066     uint16_t cmcor;
01067     err = internal_calculate_cmt_params(config->frequency_hz, &divider, &cmcor);
01068     if (err != k_rx_ok) {
01069         return err;
01070     }
01071
01072     /* Enable CMT module (clear module stop bit) */
01073     internal_enable_cmt_module_clock();
01074
01075     /* Stop timer before configuration */
01076     err = rx_cmt_stop(channel);
01077     if (err != k_rx_ok) {
01078         return err;
01079     }
01080
01081     /* Configure timer registers */
01082     internal_configure_cmt_timer_registers(cmt, divider, cmcor);
01083
01084     /* Configure interrupt */
01085     err = internal_configure_cmt_interrupt(
01086         (rx_cmt_interrupt_config_t){.channel = channel, .priority = config->priority});
01087     if (err != k_rx_ok) {
01088         return err;
01089     }
01090
01091     /* Save callback and mark initialized */
01092     internal_save_cmt_callback(channel, config);
01093
01094     /* Start timer */
01095     err = rx_cmt_start(channel);
01096     if (err != k_rx_ok) {
01097         return err;
01098     }
01099
01100     rx_log_info(s_tag, "CMT initialized");
01101
01102     return k_rx_ok;
01103 }

```

References [rx_cmt_config_t::frequency_hz](#), [internal_calculate_cmt_params\(\)](#), [internal_configure_cmt_interrupt\(\)](#), [internal_configure_cmt_timer_registers\(\)](#), [internal_enable_cmt_module_clock\(\)](#), [internal_get_cmt_base\(\)](#), [internal_save_cmt_callback\(\)](#), [internal_validate_cmt_init_params\(\)](#), [rx_cmt_config_t::priority](#), [rx_cmt_start\(\)](#), [rx_cmt_stop\(\)](#), and [s_tag](#).

Here is the call graph for this function:



8.111.5.14 rx'cmt'start()

```
rx_err_t rx_cmt_start (
    const rx_cmt_channel_t channel) [nodiscard]
```

Start CMT counter for a previously initialized channel.

Start CMT timer.

Sets the appropriate bit in the CMSTR0 or CMSTR1 register to start the CMT counter running. After setting the bit the register is read back to verify the hardware accepted the write. Channels 0/1 share CMSTR0; channels 2/3 share CMSTR1. Channels 0 and 1 use STR0/STR1 bits respectively, as do channels 2 and 3.

Precondition

- channel must be initialized via [rx_cmt_init\(\)](#)
- channel must be in range [0, 3]

Postcondition

CMSTR bit for channel is set
Counter begins incrementing from its current value

Note

Usually called automatically by `rx_cmt_init()`; call explicitly only after `rx_cmt_stop()`
Performs a readback check to verify the hardware write succeeded

Thread Safety:

Not thread-safe; do not call concurrently with stop/deinit on the same channel.

Example:

```
// Pause then resume timer
rx_cmt_stop(k_cmt_channel_1);
// ... do critical work ...
rx_cmt_start(k_cmt_channel_1);
```

See also

[rx_cmt_stop\(\)](#) Stop the counter
[rx_cmt_init\(\)](#) Initialize channel (calls start internally)

Since

Version 1.0.0

Definition at line 1142 of file `rx_cmt.c`.

```
01143 {
01144     if ((uint8_t)channel >= k_cmt_max_channels) {
01145         return k_rx_err_invalid_arg;
01146     }
01147
01148     if (!s_cmt_initialized[channel]) {
01149         return k_rx_err_invalid_state;
01150     }
01151
01152     /* Set corresponding bit in CMSTR register */
01153     switch (channel) {
01154         case k_cmt_channel_0: {
01155             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01156             cmt_ctrl()->cmstr0 |= bit_mask;
01157             const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01158             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01159                 return k_rx_err_hw_error;
01160             }
01161             break;
01162         }
01163         case k_cmt_channel_1: {
01164             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01165             cmt_ctrl1()->cmstr0 |= bit_mask;
01166             const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01167             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01168                 return k_rx_err_hw_error;
01169             }
01170             break;
01171         }
01172         case k_cmt_channel_2: {
01173             const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01174             cmt_ctrl1()->cmstr0 |= bit_mask;
01175             const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01176             if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
```

```

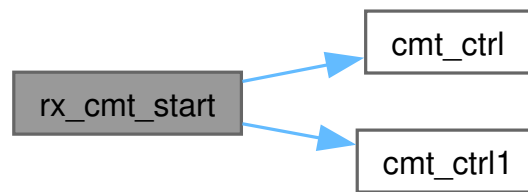
01177     return k_rx_err_hw_error;
01178 }
01179 break;
01180 }
01181 case k_cmt_channel_3: {
01182     const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01183     cmt_ctrl1()->cmstr0 |= bit_mask;
01184     const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01185     if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01186         return k_rx_err_hw_error;
01187     }
01188     break;
01189 }
01190 default:
01191     return k_rx_err_invalid_arg;
01192 }
01193
01194 return k_rx_ok;
01195 }

```

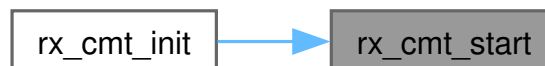
References [rx_cmt_control_regs_t::cmstr0](#), [cmt_ctrl\(\)](#), [cmt_ctrl1\(\)](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_channel_2](#), [k_cmt_channel_3](#), [k_cmt_cmstr_str0](#), [k_cmt_cmstr_str1](#), [k_cmt_max_channels](#), [k_cmt_value_zero](#), and [s_cmt_initialized](#).

Referenced by [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.111.5.15 rx`cmt`stop()

```

rx_err_t rx_cmt_stop (
    const rx_cmt_channel_t channel) [nodiscard]

```

Stop CMT counter without deinitializing the channel.

Stop CMT timer.

Clears the appropriate bit in CMSTR0 or CMSTR1 to halt the CMT counter. After clearing the bit the register is read back to verify the hardware accepted the write. The channel remains initialized; call [rx_cmt_start\(\)](#) to resume counting.

Note that [rx_cmt_stop\(\)](#) does NOT require the channel to be initialized (`s_cmt_initialized` check is omitted intentionally so that [rx_cmt_init\(\)](#) can call it safely before marking the channel as initialized).

Precondition

channel must be in range [0, 3]

Interrupts may still fire if one is already pending when stop is called

Postcondition

CMSTR bit for channel is cleared

Counter stops incrementing (current CMCNT value preserved)

Note

Does not clear the initialized flag - channel can be restarted

Called by `rx_cmt_init()` before programming registers (safe on uninitialized channel)

Thread Safety:

Not thread-safe; do not call concurrently with start/init/deinit on the same channel.

Example:

```
// Temporarily suspend timer
rx_cmt_stop(k_cmt_channel_2);
reconfigure_something();
rx_cmt_start(k_cmt_channel_2);
```

See also

[rx_cmt_start\(\)](#) Resume counting

[rx_cmt_deinit\(\)](#) Stop and release channel resources

Since

Version 1.0.0

Definition at line 1237 of file [rx_cmt.c](#).

```
01238 {
01239     if ((uint8_t)channel >= k_cmt_max_channels) {
01240         return k_rx_err_invalid_arg;
01241     }
01242
01243     /* Clear corresponding bit in CMSTR register */
01244     switch (channel) {
01245     case k_cmt_channel_0: {
01246         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01247         cmt_ctrl()->cmstr0 &= (uint16_t)~bit_mask;
01248         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01249         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01250             return k_rx_err_hw_error;
01251         }
01252         break;
01253     }
01254     case k_cmt_channel_1: {
01255         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01256         cmt_ctrl()->cmstr0 &= (uint16_t)~bit_mask;
01257         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01258         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01259             return k_rx_err_hw_error;
01260         }
01261         break;
01262     }
01263 }
```

```

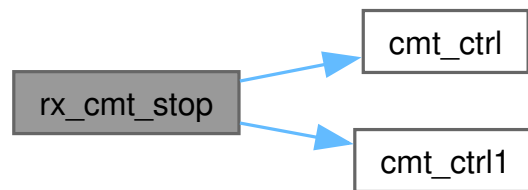
01263     case k_cmt_channel_2: {
01264         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01265         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01266         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01267         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01268             return k_rx_err_hw_error;
01269         }
01270         break;
01271     }
01272     case k_cmt_channel_3: {
01273         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01274         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01275         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01276         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01277             return k_rx_err_hw_error;
01278         }
01279         break;
01280     }
01281     default:
01282         return k_rx_err_invalid_arg;
01283 }
01284
01285 return k_rx_ok;
01286 }

```

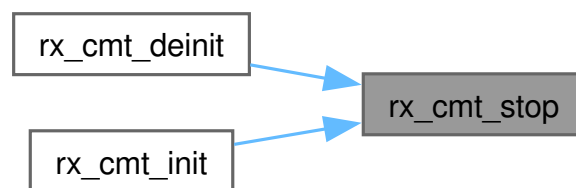
References [rx_cmt_control_regs_t::cmstr0](#), [cmt_ctrl\(\)](#), [cmt_ctrl1\(\)](#), [k_cmt_bit_mask_lsb](#), [k_cmt_channel_0](#), [k_cmt_channel_1](#), [k_cmt_channel_2](#), [k_cmt_channel_3](#), [k_cmt_cmstr_str0](#), [k_cmt_cmstr_str1](#), [k_cmt_max_channels](#), and [k_cmt_value_zero](#).

Referenced by [rx_cmt_deinit\(\)](#), and [rx_cmt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.111.6 Variable Documentation

8.111.6.1 s`cmt`callback

[rx_cmt_callback_t](#) `s_cmt_callback[k_cmt_max_channels]` = {nullptr} [static]

Per-channel interrupt callback function pointers.

Stores user-provided callback functions for each CMT channel. Called from the corresponding interrupt handler (`cmt1_isr`, `cmt2_isr`, `cmt3_isr`).

Note

NULL callbacks are valid - interrupt fires but no action taken.

Warning

Callback executes in interrupt context. Keep it fast!

See also

[rx_cmt_callback_t](#) Callback function type

Since

Version 1.0.0

Definition at line 318 of file [rx_cmt.c](#).

```
00318 {nullptr};
```

Referenced by [cmt1_isr\(\)](#), [cmt2_isr\(\)](#), [cmt3_isr\(\)](#), [internal_save_cmt_callback\(\)](#), and [rx_cmt_deinit\(\)](#).

8.111.6.2 s'cmt'initialized

```
bool s_cmt_initialized[k\_cmt\_max\_channels] = {false} [static]
```

Per-channel initialization status flags.

Tracks whether each CMT channel has been initialized. Set to true after successful [rx_cmt_init\(\)](#), cleared on [rx_cmt_deinit\(\)](#).

Index	Channel	Initial Value
0	CMT0	false (ThreadX manages)
1	CMT1	false
2	CMT2	false
3	CMT3	false

Note

Used to prevent operations on uninitialized channels.

Warning

Do not modify directly; managed by init/deinit functions.

Since

Version 1.0.0

Definition at line 301 of file [rx_cmt.c](#).

```
00301 {false};
```

Referenced by [internal_save_cmt_callback\(\)](#), [rx_cmt_deinit\(\)](#), [rx_cmt_get_count\(\)](#), and [rx_cmt_start\(\)](#).

8.111.6.3 s_cmt_user_data

```
void* s_cmt_user_data[k_cmt_max_channels] = {nullptr} [static]
```

Per-channel user data for callbacks.

Stores user-provided context data passed to callbacks. Allows callbacks to access application-specific state without global variables.

Note

May be NULL if callback doesn't need user data.

Since

Version 1.0.0

Definition at line 332 of file [rx_cmt.c](#).

```
00332 {nullptr};
```

Referenced by [cmt1_isr\(\)](#), [cmt2_isr\(\)](#), [cmt3_isr\(\)](#), [internal_save_cmt_callback\(\)](#), and [rx_cmt_deinit\(\)](#).

8.111.6.4 s_tag

```
const char* const s_tag = "CMT" [static]
```

Definition at line 156 of file [rx_cmt.c](#).

8.112 rx_cmt.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_cmt.c
00003  * @brief CMT (Compare Match Timer) Driver Implementation - Periodic Interrupt Generation
00004  *
00005  * @details
00006  * # Overview
00007  *
00008  * Implements the CMT driver for periodic interrupt generation on the RX72N.
00009  * This file contains all internal helpers, interrupt handlers, and the public
00010  * API implementation for CMT channels 0-3.
00011  *
00012  * ## Implementation Architecture
00013  *
00014  * @dot
00015  * digraph implementation {
00016  *   rankdir=TB;
00017  *   node [shape=box, style=rounded];
00018  *
00019  *   subgraph cluster_public {
00020  *     label="Public API";
00021  *     style=filled;
00022  *     color=lightgreen;
00023  *     init [label="rx_cmt_init()"];
00024  *     start [label="rx_cmt_start()"];
00025  *     stop [label="rx_cmt_stop()"];
00026  *     count [label="rx_cmt_get_count()"];
00027  *     deinit [label="rx_cmt_deinit()"];
00028  *   }
00029  *
00030  *   subgraph cluster_internal {
00031  *     label="Internal Helpers";
00032  *     style=filled;
```

```

00033 *   color=lightyellow;
00034 *   get_base [label="internal_get_cmt_base()"];
00035 *   calc [label="internal_calculate_cmt_params()"];
00036 *   enable [label="internal_enable_cmt_module_clock()"];
00037 *   configure [label="internal_configure_cmt_timer_registers()"];
00038 *   irq [label="internal_configure_cmt_interrupt()"];
00039 *   validate [label="internal_validate_cmt_init_params()"];
00040 *   save [label="internal_save_cmt_callback()"];
00041 * }
00042 *
00043 * subgraph cluster_isr {
00044 *   label="Interrupt Handlers";
00045 *   style=filled;
00046 *   color=lightblue;
00047 *   cmt1_isr [label="cmt1_isr()"];
00048 *   cmt2_isr [label="cmt2_isr()"];
00049 *   cmt3_isr [label="cmt3_isr()"];
00050 * }
00051 *
00052 * init -> validate -> calc -> enable -> configure -> irq -> save;
00053 * init -> start;
00054 * start -> get_base;
00055 * stop -> get_base;
00056 * count -> get_base;
00057 * cmt1_isr -> s_cmt_callback [label="call"];
00058 * }
00059 * @enddot
00060 *
00061 * ## CMT Timer Operation
00062 *
00063 * The CMT operates as a 16-bit up-counter with compare match:
00064 *
00065 * @msc
00066 *   width=600;
00067 *   CMCNT, CMCOR, ISR, Callback;
00068 *
00069 *   CMCNT box CMCNT [label="Counter at 0"];
00070 *   CMCNT -> CMCNT [label="Count: 1, 2, 3..."];
00071 *   CMCNT => CMCOR [label="CMCNT reaches CMCOR"];
00072 *   CMCOR box CMCOR [label="Compare match!\nCMCNT = 0"];
00073 *   CMCOR => ISR [label="CMIn interrupt"];
00074 *   ISR => Callback [label="s_cmt_callback[n]()"];
00075 *   Callback » ISR;
00076 *   ISR » CMCNT [label="Return"];
00077 *   CMCNT -> CMCNT [label="Continue counting..."];
00078 * @endmsc
00079 *
00080 * ## Clock Divider Selection Algorithm
00081 *
00082 * The driver tries each divider in order (/8, /32, /128, /512) and selects
00083 * the first one that produces a period fitting in 16 bits:
00084 *
00085 * ```
00086 * period = (PCLKB / divider) / frequency
00087 * if (0 < period <= 65535): use this divider
00088 * ```
00089 *
00090 * | Divider | PCLKB/div | Min Period | Max Period |
00091 * |-----|-----|-----|-----|
00092 * | /8 | 7.5 MHz | 1 (7.5 MHz) | 65535 (115 Hz) |
00093 * | /32 | 1.875 MHz | 1 (1.875 MHz) | 65535 (29 Hz) |
00094 * | /128 | 468.75 kHz | 1 (469 kHz) | 65535 (7.2 Hz) |
00095 * | /512 | 117.2 kHz | 1 (117 kHz) | 65535 (1.8 Hz) |
00096 *
00097 * ## Memory Map (Static State)
00098 *
00099 * | Symbol | Size | Description |
00100 * |-----|-----|-----|
00101 * | s_cmt_initialized[] | 4 bytes | Per-channel init flags |
00102 * | s_cmt_callback[] | 16 bytes | Callback function pointers |
00103 * | s_cmt_user_data[] | 16 bytes | User data pointers |
00104 *
00105 * ***Total static allocation:** 36 bytes
00106 *
00107 * ## Register Access Pattern
00108 *
00109 * All register access uses inline accessor functions for type safety:
00110 * - `cmt0()`, `cmt1()`, `cmt2()`, `cmt3()` - Channel registers
00111 * - `cmt_ctrl()` - Control registers (CMSTR0, CMSTR1)
00112 * - `icu()` - Interrupt controller
00113 *
00114 * ## NASA Power of 10 Compliance
00115 *
00116 * | Rule | Implementation |
00117 * |-----|-----|
00118 * | Rule 1 | No goto/recursion, single return paths |
00119 * | Rule 2 | Bounded loop in internal_calculate_cmt_params() |

```

```

00120 * | Rule 3 | Static arrays only, no heap |
00121 * | Rule 4 | All functions under 60 lines |
00122 * | Rule 5 | Parameter validation in all public functions |
00123 * | Rule 6 | File-scope static with s_ prefix |
00124 * | Rule 7 | All return values checked |
00125 * | Rule 8 | C23 typed enums for all constants |
00126 * | Rule 9 | Single-level pointers only |
00127 * | Rule 10 | Clean with -Wall -Wextra -Werror |
00128 *
00129 * @see rx_cmt.h Public API declarations
00130 * @see rx72n_cmt_regs.h CMT register definitions
00131 * @see rx_register_protection.h PRCR unlock for module stop
00132 *
00133 * @author Locked, Inc.
00134 * @date 2026-01-27
00135 * @copyright Copyright (c) 2026 Locked Inc.
00136 * SPDX-License-Identifier: MIT
00137 * @since Version 1.0.0
00138 */
00139
00140 #ifdef __RX__
00141
00142 #include "rx_cmt.h"
00143
00144 #include <stddef.h>
00145
00146 #include "rx72n_regs.h"
00147 #include "rx_check.h"
00148 #include "rx_log.h"
00149 #include "rx_register_protection.h"
00150
00151 /* Interrupt handler forward declarations (required for -Wmissing-declarations) */
00152 void cmt1_isr(void);
00153 void cmt2_isr(void);
00154 void cmt3_isr(void);
00155
00156 static const char* const s_tag = "CMT";
00157
00158 /*
=====
00159 * Constants
00160 *
=====
00161 */
00162
00163 /**
00164 * @enum cmt_constants_t
00165 * @brief CMT channel and divider configuration constants
00166 *
00167 * @details
00168 * Defines the channel count, divider settings, and loop bounds for CMT
00169 * configuration. These values correspond to RX72N hardware capabilities.
00170 *
00171 * ## Divider Settings (CMCR.CKS field)
00172 *
00173 * | CKS Value | Divider | Counter Clock @60MHz |
00174 * |-----|-----|-----|
00175 * | 0b00 | /8 | 7.5 MHz |
00176 * | 0b01 | /32 | 1.875 MHz |
00177 * | 0b10 | /128 | 468.75 kHz |
00178 * | 0b11 | /512 | 117.19 kHz |
00179 *
00180 * @note k_cmt_divider_start and k_cmt_num_dividers define loop bounds
00181 * for NASA Power of 10 Rule 2 compliance.
00182 *
00183 * @since Version 1.0.0
00184 */
00185 typedef enum : uint8_t {
00186     k_cmt_max_channels = 4, /**< Total CMT channels: CMT0, CMT1, CMT2, CMT3 */
00187     k_cmt_divider_start = 0, /**< Loop start index for divider iteration (Rule 2) */
00188     k_cmt_div_8 = 0, /**< CKS=0b00: PCLKB/8 divider (7.5 MHz @ 60 MHz PCLKB) */
00189     k_cmt_div_32 = 1, /**< CKS=0b01: PCLKB/32 divider (1.875 MHz @ 60 MHz PCLKB) */
00190     k_cmt_div_128 = 2, /**< CKS=0b10: PCLKB/128 divider (468.75 kHz @ 60 MHz PCLKB) */
00191     k_cmt_div_512 = 3, /**< CKS=0b11: PCLKB/512 divider (117.19 kHz @ 60 MHz PCLKB) */
00192     k_cmt_num_dividers = 4, /**< Loop bound for divider iteration (Rule 2 compliance) */
00193 } cmt_constants_t;
00194
00195 /** @brief CMT divider values (actual divisor values) */
00196 typedef enum : uint16_t {
00197     k_cmt_divider_val_8 = 8, /**< Divide by 8 */
00198     k_cmt_divider_val_32 = 32, /**< Divide by 32 */
00199     k_cmt_divider_val_128 = 128, /**< Divide by 128 */
00200     k_cmt_divider_val_512 = 512, /**< Divide by 512 */
00201 } cmt_divider_values_t;
00202
00203 /**
00204 * @brief CMT module stop bit positions in MSTPCRA (RX72N Manual Section 11.2.2, p.406)

```

```

00205 * @details
00206 * CMT is controlled by MSTPCRA, NOT MSTPCRB.
00207 * - MSTPA15: CMT unit 0 (CMT0, CMT1) module stop
00208 * - MSTPA14: CMT unit 1 (CMT2, CMT3) module stop
00209 */
00210 typedef enum : uint8_t {
00211     k_cmt_mstpa_unit0 = 15, /**< MSTPCRA bit 15: CMT unit 0 (CMT0, CMT1) module stop */
00212     k_cmt_mstpa_unit1 = 14, /**< MSTPCRA bit 14: CMT unit 1 (CMT2, CMT3) module stop */
00213 } cmt_module_stop_bits_t;
00214
00215 /**
00216 * @struct rx_cmt_interrupt_config_t
00217 * @brief CMT interrupt routing and priority configuration
00218 *
00219 * @details
00220 * Bundles the CMT channel identifier with the desired ICU interrupt priority
00221 * level. Passed by value to internal_configure_cmt_interrupt() so that the
00222 * caller does not need to keep the structure alive after the call returns.
00223 *
00224 * The channel field selects which ICU interrupt vector to program (CMI0-CMI3),
00225 * and the priority field is written directly into the corresponding IPR[]
00226 * register.
00227 *
00228 * @invariant channel must be a valid rx_cmt_channel_t value (0-3)
00229 * @invariant priority must be in [k_ipr_level_min, k_ipr_level_max]
00230 *
00231 * @par Example:
00232 * @code{.c}
00233 * rx_cmt_interrupt_config_t irq_cfg = {
00234 *     .channel = k_cmt_channel_1,
00235 *     .priority = 5,
00236 * };
00237 * rx_err_t err = internal_configure_cmt_interrupt(irq_cfg);
00238 * @endcode
00239 *
00240 * @see internal_configure_cmt_interrupt() Sole consumer of this struct
00241 * @see rx_cmt_init() Constructs this struct inline before calling the above
00242 *
00243 * @since Version 1.0.0
00244 */
00245 typedef struct {
00246     rx_cmt_channel_t channel; /**< CMT channel identifier (k_cmt_channel_0 ... k_cmt_channel_3) */
00247     uint8_t priority; /**< ICU interrupt priority level (k_ipr_level_min ... k_ipr_level_max) */
00248 } rx_cmt_interrupt_config_t;
00249
00250 /** @brief CMCR register bit positions (RX72N Manual Ch31, p.1583) */
00251 typedef enum : uint8_t {
00252     k_cmt_cmcr_cks_shift = 0, /**< CKS (clock select) bit shift */
00253     k_cmt_cmcr_cmie_pos = 6, /**< CMIE (interrupt enable) bit position */
00254     k_cmt_cmcr_rsvd7_pos = 7, /**< Reserved bit 7: write value should be 1 */
00255 } cmt_cmcr_bits_t;
00256
00257 /** @brief Period calculation constants */
00258 typedef enum : uint16_t {
00259     k_cmt_period_min = 0, /**< Minimum valid period */
00260     k_cmt_period_max = 0xFFFF, /**< Maximum valid period (16-bit) */
00261     k_cmt_period_adj = 1, /**< Period adjustment (period - 1) */
00262 } cmt_period_constants_t;
00263
00264 /** @brief CMSTR register bit positions */
00265 typedef enum : uint8_t {
00266     k_cmt_cmstr_str0 = 0, /**< CMT0/CMT2 start bit */
00267     k_cmt_cmstr_str1 = 1, /**< CMT1/CMT3 start bit */
00268 } cmt_cmstr_bits_t;
00269
00270 /** @brief CMT bit manipulation constants */
00271 typedef enum : uint8_t {
00272     k_cmt_bit_mask_lsb = 1, /**< Single bit set for shifts */
00273     k_cmt_value_zero = 0, /**< Zero value for comparisons */
00274 } cmt_bit_constants_t;
00275
00276 /*
=====
00277 * Static Variables
00278 *
=====
00279 */
00280
00281 /**
00282 * @var s_cmt_initialized
00283 * @brief Per-channel initialization status flags
00284 *
00285 * @details
00286 * Tracks whether each CMT channel has been initialized. Set to true after
00287 * successful rx_cmt_init(), cleared on rx_cmt_deinit().
00288 *
00289 * | Index | Channel | Initial Value |

```

```

00290 * |-----|-----|-----|
00291 * | 0 | CMT0 | false (ThreadX manages) |
00292 * | 1 | CMT1 | false |
00293 * | 2 | CMT2 | false |
00294 * | 3 | CMT3 | false |
00295 *
00296 * @note Used to prevent operations on uninitialized channels.
00297 * @warning Do not modify directly; managed by init/deinit functions.
00298 *
00299 * @since Version 1.0.0
00300 */
00301 static bool s_cmt_initialized[k_cmt_max_channels] = {false};
00302
00303 /**
00304 * @var s_cmt_callback
00305 * @brief Per-channel interrupt callback function pointers
00306 *
00307 * @details
00308 * Stores user-provided callback functions for each CMT channel. Called from
00309 * the corresponding interrupt handler (cmt1_isr, cmt2_isr, cmt3_isr).
00310 *
00311 * @note NULL callbacks are valid - interrupt fires but no action taken.
00312 * @warning Callback executes in interrupt context. Keep it fast!
00313 *
00314 * @see rx_cmt_callback_t Callback function type
00315 *
00316 * @since Version 1.0.0
00317 */
00318 static rx_cmt_callback_t s_cmt_callback[k_cmt_max_channels] = {nullptr};
00319
00320 /**
00321 * @var s_cmt_user_data
00322 * @brief Per-channel user data for callbacks
00323 *
00324 * @details
00325 * Stores user-provided context data passed to callbacks. Allows callbacks
00326 * to access application-specific state without global variables.
00327 *
00328 * @note May be NULL if callback doesn't need user data.
00329 *
00330 * @since Version 1.0.0
00331 */
00332 static void* s_cmt_user_data[k_cmt_max_channels] = {nullptr};
00333
00334 /*
=====
00335 * Internal Helper Functions
00336 *
=====
00337 */
00338
00339 /**
00340 * @brief Map a CMT channel enum to its hardware register base pointer
00341 *
00342 * @details
00343 * Uses a switch statement to convert a logical channel identifier into a
00344 * pointer to the corresponding volatile rx_cmt_channel_regs_t register block.
00345 * The four CMT channels (CMT0-CMT3) each have a distinct base address in the
00346 * RX72N memory map; the inline accessor functions cmt0()-cmt3() encode those
00347 * addresses in a type-safe manner.
00348 *
00349 * The default case returns nullptr, which every caller is expected to check
00350 * immediately after the call (NASA Rule 7 compliance).
00351 *
00352 *
00353 *
00354 * @pre channel is a member of rx_cmt_channel_t (0-3)
00355 * @pre Hardware register addresses must be accessible (clocks enabled before use)
00356 *
00357 * @post Return value, if non-null, points to the live hardware register block
00358 * @post Caller must validate the return value before dereferencing (Rule 7)
00359 *
00360 * @note Not thread-safe; the hardware registers are globally shared
00361 * @note This function never writes to hardware; it only computes an address
00362 *
00363 * @par Thread Safety:
00364 * Read-only address computation; no shared mutable state is accessed.
00365 *
00366 * @par Performance:
00367 * Execution time: ~1 cycle (single branch/address load with -O2)
00368 *
00369 * @par Example:
00370 * @code{.c}
00371 * volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(k_cmt_channel_2);
00372 * if (cmt == nullptr) {
00373 *     return k_rx_err_invalid_arg;
00374 * }

```

```

00375 * cmt->cmcnt = 0;
00376 * @endcode
00377 *
00378 * @see cmt0() Inline accessor for CMT0 base address
00379 * @see cmt1() Inline accessor for CMT1 base address
00380 * @see cmt2() Inline accessor for CMT2 base address
00381 * @see cmt3() Inline accessor for CMT3 base address
00382 *
00383 * @since Version 1.0.0
00384 */
00385 static volatile rx_cmt_channel_regs_t* internal_get_cmt_base(const rx_cmt_channel_t channel)
00386 {
00387     switch (channel) {
00388         case k_cmt_channel_0:
00389             return cmt0();
00390         case k_cmt_channel_1:
00391             return cmt1();
00392         case k_cmt_channel_2:
00393             return cmt2();
00394         case k_cmt_channel_3:
00395             return cmt3();
00396         default:
00397             return nullptr;
00398     }
00399 }
00400
00401 /**
00402  * @brief Calculate the optimal CMT clock divider and compare-match value for a target frequency
00403  *
00404  * @details
00405  * Iterates the four available PCLKB clock dividers (/8, /32, /128, /512) in
00406  * ascending order and selects the first divider whose resulting 16-bit period
00407  * value satisfies:
00408  *
00409  * ```
00410  * period = (PCLKB / divider) / frequency_hz
00411  * 0 < period <= 65535
00412  * ```
00413  *
00414  * The chosen divider index is returned in *divider (corresponding to the CKS
00415  * field of CMCR: 0=/8, 1=/32, 2=/128, 3=/512). The compare-match register
00416  * value is returned in *cmcor; the caller must subtract the adjustment constant
00417  * k_cmt_period_adj when writing to the hardware register.
00418  *
00419  * Algorithm steps:
00420  * 1. Validate output pointers are non-null
00421  * 2. Reject frequency_hz == 0 or frequency_hz > PCLKB/8 (maximum achievable)
00422  * 3. Loop i from k_cmt_divider_start to k_cmt_num_dividers - 1
00423  * 4. Compute period_calc = (PCLKB / dividers[i]) / frequency_hz
00424  * 5. If 0 < period_calc <= 65535, store i and period_calc, return k_rx_ok
00425  * 6. If no divider fits, return k_rx_err_invalid_arg
00426  *
00427  *
00428  *
00429  * @pre divider must point to a valid uint8_t storage location
00430  * @pre cmcor must point to a valid uint16_t storage location
00431  *
00432  * @post On k_rx_ok: *divider holds the CKS value (0-3) for CMCR
00433  * @post On k_rx_ok: *cmcor holds the period count; caller subtracts 1 before writing CMCR
00434  *
00435  * @note Not thread-safe; pure computation with no shared state modification
00436  * @note Uses a bounded loop (k_cmt_num_dividers iterations) -- NASA Rule 2 compliant
00437  *
00438  * @par Thread Safety:
00439  * Pure computation with no global or static state; safe to call from any context.
00440  *
00441  * @par Performance:
00442  * Execution time: ~5 cycles (loop unrolled at -O2 for 4 divider steps)
00443  *
00444  * @par Example:
00445  * @code{.c}
00446  * uint8_t divider = 0;
00447  * uint16_t cmcor = 0;
00448  * rx_err_t err = internal_calculate_cmt_params(1000U, &divider, &cmcor);
00449  * if (err == k_rx_ok) {
00450  *     // divider and cmcor are ready for use in CMCR/CMCOR registers
00451  * }
00452  * @endcode
00453  *
00454  * @see internal_configure_cmt_timer_registers() Consumes the output of this function
00455  * @see cmt_constants_t Divider index constants (k_cmt_div_8 ... k_cmt_div_512)
00456  * @see cmt_divider_values_t Actual divisor values (8, 32, 128, 512)
00457  *
00458  * @since Version 1.0.0
00459  */
00460 static rx_err_t
00461 internal_calculate_cmt_params(const uint32_t frequency_hz, uint8_t* divider, uint16_t* cmcor)

```

```

00462 {
00463     static const uint16_t dividers[] = {k_cmt_divider_val_8,
00464                                         k_cmt_divider_val_32,
00465                                         k_cmt_divider_val_128,
00466                                         k_cmt_divider_val_512};
00467     const uint32_t      pclkb      = k_pclkb_hz;
00468     const uint32_t      max_frequency = pclkb / k_cmt_divider_val_8;
00469
00470     if (divider == nullptr || cmcor == nullptr) {
00471         return k_rx_err_null_ptr;
00472     }
00473
00474     if (frequency_hz <= (uint32_t)k_cmt_value_zero || frequency_hz > max_frequency) {
00475         return k_rx_err_invalid_arg;
00476     }
00477
00478     /* Try each divider to find one that fits in 16-bit */
00479     for (uint8_t i = k_cmt_divider_start; i < k_cmt_num_dividers; i++) {
00480         const uint32_t period_calc = (pclkb / dividers[i]) / frequency_hz;
00481
00482         /* Check if period fits in 16-bit and is reasonable */
00483         if (period_calc > k_cmt_period_min && period_calc <= k_cmt_period_max) {
00484             *divider = i;
00485             *cmcor   = (uint16_t)period_calc;
00486             return k_rx_ok;
00487         }
00488     }
00489
00490     rx_log_error(s_tag, "Error occurred");
00491     return k_rx_err_invalid_arg;
00492 }
00493
00494 /**
00495  * @brief Enable the CMT module clock by clearing the MSTPCRB module-stop bit
00496  *
00497  * @details
00498  * Removes the CMT0-CMT3 peripheral from module-stop state so that the module
00499  * clock is supplied and its registers become accessible. The sequence is:
00500  *
00501  * 1. Unlock the write-protect register (PRCR) for PRC1 and PRC3 bits
00502  * 2. Clear bit k_cmt_mstpa_unit0 (bit 15) in MSTPCRA to enable CMT unit 0 (CMT0/CMT1)
00503  * 3. Clear bit k_cmt_mstpa_unit1 (bit 14) in MSTPCRA to enable CMT unit 1 (CMT2/CMT3)
00504  * 4. Re-lock the PRCR to prevent accidental modification
00505  *
00506  * This function must be called before any CMT register access; reading or
00507  * writing CMT registers while the module clock is stopped produces undefined
00508  * hardware behaviour on the RX72N.
00509  *
00510  * The function is idempotent: calling it when the module is already enabled
00511  * simply clears an already-clear bit, which is harmless.
00512  *
00513  *
00514  * @pre The system clock (PCLKB) must be running before calling this function
00515  * @pre No other task or ISR must be concurrently modifying MSTPCRB
00516  *
00517  * @post MSTPCRA bits 15 and 14 are cleared; CMT module clocks for all 4 channels are active
00518  * @post PRCR is locked upon return (prevents accidental MSTPCRB changes)
00519  *
00520  * @note Not thread-safe; PRCR unlock/lock is a non-atomic read-modify-write
00521  * @note Called once per system lifetime; subsequent calls are benign
00522  *
00523  * @par Thread Safety:
00524  * Not thread-safe. Call during system initialization before RTOS scheduler
00525  * starts, or with interrupts disabled.
00526  *
00527  * @par Performance:
00528  * Execution time: ~4 register writes; negligible overhead
00529  *
00530  * @par Example:
00531  * @code{.c}
00532  * // Called internally by rx_cmt_init(); not intended for direct use
00533  * internal_enable_cmt_module_clock();
00534  * // CMT registers are now accessible
00535  * @endcode
00536  *
00537  * @see rx_register_protection.h PRCR lock/unlock constants
00538  * @see rx_cmt_init() Sole caller; invoked before register programming
00539  *
00540  * @since Version 1.0.0
00541  */
00542 static void internal_enable_cmt_module_clock(void)
00543 {
00544     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00545     /* CMT is in MSTPCRA (not MSTPCRB): bit 15 = unit 0 (CMT0/1), bit 14 = unit 1 (CMT2/3) */
00546     system_regs()->mstpcra &= ~(uint32_t)k_cmt_bit_mask_lsb « k_cmt_mstpa_unit0;
00547     system_regs()->mstpcra &= ~(uint32_t)k_cmt_bit_mask_lsb « k_cmt_mstpa_unit1;
00548     *prcr_reg() = k_rx_prcr_lock;

```

```

00549 }
00550
00551 /**
00552  * @brief Write clock divider, interrupt-enable, compare-match, and counter registers
00553  *
00554  * @details
00555  * Programs the three CMT hardware registers for a single channel in the
00556  * following order:
00557  *
00558  * 1. **CMCR** - Sets the CKS (clock-select) field from divider and enables the
00559  *   compare-match interrupt (CMIE bit).
00560  * 2. **CMCOR** - Loads the compare-match value after subtracting the adjustment
00561  *   constant k_cmt_period_adj (-1) so the hardware fires at the correct period.
00562  * 3. **CMCNT** - Clears the counter to 0 so the first interrupt occurs exactly
00563  *   one full period after the timer is started.
00564  *
00565  * The timer must be stopped (CMSTR bit cleared) before calling this function
00566  * to avoid race conditions while writing to CMCOR and CMCNT.
00567  *
00568  *
00569  *
00570  * @pre cmt must not be nullptr (enforced by RX_ASSERT)
00571  * @pre The CMT timer must be stopped before calling this function
00572  *
00573  * @post cmt->cmcr = (divider « CKS_SHIFT) | CMIE_BIT | RSVD7_BIT (bit7 must be written 1)
00574  * @post cmt->cmcor = cmcor - k_cmt_period_adj
00575  * @post cmt->cmcnt = 0
00576  *
00577  * @note Not thread-safe; caller must ensure the timer is stopped
00578  * @note RX_ASSERT fires in debug builds if cmt is nullptr
00579  *
00580  * @par Thread Safety:
00581  * Not thread-safe. Must be called with the timer stopped and no concurrent ISR
00582  * activity on this channel.
00583  *
00584  * @par Performance:
00585  * Execution time: ~3 register writes; negligible overhead
00586  *
00587  * @par Example:
00588  * @code{.c}
00589  * uint8_t divider = 0;
00590  * uint16_t cmcor = 0;
00591  * (void)internal_calculate_cmt_params(1000U, &divider, &cmcor);
00592  * volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(k_cmt_channel_1);
00593  * internal_configure_cmt_timer_registers(cmt, divider, cmcor);
00594  * @endcode
00595  *
00596  * @see internal_calculate_cmt_params() Produces the divider and cmcor arguments
00597  * @see rx_cmt_stop() Must be called before this function
00598  * @see cmt_cmcr_bits_t CKS shift and CMIE bit position constants
00599  *
00600  * @since Version 1.0.0
00601  */
00602 static void internal_configure_cmt_timer_registers(volatile rx_cmt_channel_regs_t* cmt,
00603                                                  const uint8_t divider,
00604                                                  const uint16_t cmcor)
00605 {
00606     RX_ASSERT(cmt != nullptr, "CMT registers are NULL");
00607
00608     /* Configure timer control register
00609     * - Set clock divider (bits 1-0)
00610     * - Enable compare match interrupt (bit 6)
00611     * - Set reserved bit 7 to 1 (write value should be 1 per RX72N manual p.1583)
00612     */
00613     cmt->cmcr =
00614         (divider « k_cmt_cmcr_cks_shift) | /* CKS: Clock Select */
00615         (k_cmt_bit_mask_lsb « k_cmt_cmcr_cmie_pos) | /* CMIE: Compare Match Interrupt Enable */
00616         (k_cmt_bit_mask_lsb « k_cmt_cmcr_rsvd7_pos); /* Reserved bit 7: write value should be 1 */
00617
00618     /* Set compare match value (period - 1) */
00619     cmt->cmcor = cmcor - k_cmt_period_adj;
00620
00621     /* Clear counter */
00622     cmt->cmcnt = k_cmt_value_zero;
00623 }
00624
00625 /**
00626  * @brief Configure the ICU interrupt vector priority and enable for a CMT channel
00627  *
00628  * @details
00629  * Programs the Interrupt Controller Unit (ICU) to route and enable the
00630  * compare-match interrupt (CMIn) for the specified CMT channel. The three
00631  * ICU registers modified are:
00632  *
00633  * 1. **IPR[vector]** - Sets the interrupt priority level for the channel's
00634  *   CMIn vector to config.priority.
00635  * 2. **IER[index]** - Sets the enable bit for the channel's interrupt vector

```



```

00636 *   within the IER register array. The bit index and byte index are derived
00637 *   from the vector number.
00638 * 3. **IR[vector]** - Clears any pending interrupt flag before enabling, so
00639 *   a stale flag does not fire immediately after enable.
00640 *
00641 * The vector number is computed as k_vect_cmt0_cmi0 + config.channel, which
00642 * yields CMI0 for channel 0, CMI1 for channel 1, and so on.
00643 *
00644 *
00645 *
00646 * @pre config.channel must be in the range [0, k_cmt_max_channels)
00647 * @pre config.priority must be in [k_ipr_level_min, k_ipr_level_max]
00648 *
00649 * @post ICU IPR register for the channel's vector set to config.priority
00650 * @post ICU IER bit for the channel's vector enabled
00651 * @post ICU IR flag for the channel's vector cleared
00652 *
00653 * @note Not thread-safe; concurrent ICU modification from another context is unsafe
00654 * @note Clears IR before enabling to prevent a spurious first interrupt
00655 *
00656 * @par Thread Safety:
00657 * Not thread-safe. Must be called during initialization with interrupts disabled
00658 * or before the RTOS scheduler starts.
00659 *
00660 * @par Performance:
00661 * Execution time: ~4 register accesses; negligible overhead
00662 *
00663 * @par Example:
00664 * @code{.c}
00665 * rx_cmt_interrupt_config_t irq_cfg = {
00666 *   .channel = k_cmt_channel_1,
00667 *   .priority = 5,
00668 * };
00669 * rx_err_t err = internal_configure_cmt_interrupt(irq_cfg);
00670 * if (err != k_rx_ok) {
00671 *   return err;
00672 * }
00673 * @endcode
00674 *
00675 * @see rx_cmt_interrupt_config_t Configuration structure passed to this function
00676 * @see rx_cmt_deinit() Clears the IER bit to disable the interrupt
00677 * @see rx_cmt_init() Constructs config and calls this function
00678 *
00679 * @since Version 1.0.0
00680 */
00681 static rx_err_t internal_configure_cmt_interrupt(const rx_cmt_interrupt_config_t config)
00682 {
00683     /* Note: Lower bound check omitted - config.channel is uint8_t, k_cmt_channel_0 == 0,
00684      * so config.channel >= k_cmt_channel_0 is always true (-Wtype-limits) */
00685     if ((uint8_t)config.channel >= (uint8_t)k_cmt_max_channels) {
00686         return k_rx_err_invalid_arg;
00687     }
00688     if (config.priority < k_ipr_level_min || config.priority > k_ipr_level_max) {
00689         return k_rx_err_invalid_arg;
00690     }
00691 }
00692 /* CMI0/CMI1 have fixed ICU vectors (28, 29). CMI2/CMI3 are SELECTB
00693 * sources on RX72N and need SLIBRn programming, which is not yet
00694 * implemented in this driver. Reject interrupt-mode init on ch2/ch3
00695 * rather than enabling the wrong vectors (30/31 are CMWI0/CMWI1 for
00696 * CMTW timers). Per RX72N HW manual R01UH0824EJ0111 Section 15.3
00697 * (Interrupt Vector Table) and Section 15.4 (SELECTB sources). */
00698 if (config.channel == k_cmt_channel_2 || config.channel == k_cmt_channel_3) {
00699     return k_rx_err_not_supported;
00700 }
00701
00702 const uint8_t vector = k_vect_cmt0_cmi0 + config.channel;
00703 const uint8_t ier_index = vector / k_icu_ier_bits_per_reg;
00704 const uint8_t ier_bit = vector % k_icu_ier_bits_per_reg;
00705
00706 icu()->ipr[vector] = config.priority;
00707 const uint8_t ier_mask = (uint8_t)(k_cmt_bit_mask_lsb « ier_bit);
00708 icu()->ier[ier_index] |= ier_mask;
00709 icu()->ir[vector] = k_cmt_value_zero;
00710
00711 return k_rx_ok;
00712 }
00713
00714 /*
=====
00715 * Interrupt Handlers
00716 *
=====
00717 */
00718 /**
00719 */
00720 * @brief CMT1 (CMT channel 1) compare-match interrupt service routine

```

```

00721 *
00722 * @details
00723 * Invoked by the RX72N hardware whenever the CMT1 counter (CMCNT) reaches the
00724 * value stored in CMCOR. The ISR checks whether a callback has been registered
00725 * in s_cmt_callback[k_cmt_channel_1] and, if so, calls it with the associated
00726 * user data pointer from s_cmt_user_data[k_cmt_channel_1].
00727 *
00728 * The compare-match interrupt flag is automatically cleared by the RX72N
00729 * hardware upon entry; no explicit IR register write is needed.
00730 *
00731 * @note Executes in interrupt context; all code in this ISR and any registered
00732 *       callback must be ISR-safe (no blocking, no long operations)
00733 * @note Not thread-safe; the callback executes with the CMT1 interrupt masked
00734 *
00735 * @warning Do not call any ThreadX API that is not interrupt-safe from the
00736 *          registered callback (e.g., tx_thread_sleep() is forbidden; use
00737 *          ISR-safe variants such as tx_semaphore_put_from_isr())
00738 *
00739 * @see rx_cmt_init()      Registers the callback invoked by this ISR
00740 * @see rx_cmt_deinit()   Clears the callback and disables this interrupt
00741 * @see s_cmt_callback    Per-channel callback function pointer array
00742 *
00743 * @since Version 1.0.0
00744 */
00745 void cmt1_isr(void)
00746 {
00747     /* Call user callback if registered */
00748     if (s_cmt_callback[k_cmt_channel_1] != nullptr) {
00749         s_cmt_callback[k_cmt_channel_1](s_cmt_user_data[k_cmt_channel_1]);
00750     }
00751 }
00752 /* Clear interrupt flag (read-modify-write not needed, auto-cleared) */
00753 }
00754
00755 /**
00756 * @brief CMT2 (CMT channel 2) compare-match interrupt service routine
00757 *
00758 * @details
00759 * Invoked by the RX72N hardware whenever the CMT2 counter (CMCNT) reaches the
00760 * value stored in CMCOR. The ISR checks whether a callback has been registered
00761 * in s_cmt_callback[k_cmt_channel_2] and, if so, calls it with the associated
00762 * user data pointer from s_cmt_user_data[k_cmt_channel_2].
00763 *
00764 * The compare-match interrupt flag is automatically cleared by the RX72N
00765 * hardware upon entry; no explicit IR register write is needed.
00766 *
00767 * @note Executes in interrupt context; all code in this ISR and any registered
00768 *       callback must be ISR-safe (no blocking, no long operations)
00769 * @note Not thread-safe; the callback executes with the CMT2 interrupt masked
00770 *
00771 * @warning Do not call any ThreadX API that is not interrupt-safe from the
00772 *          registered callback (e.g., tx_thread_sleep() is forbidden; use
00773 *          ISR-safe variants such as tx_semaphore_put_from_isr())
00774 *
00775 * @see rx_cmt_init()      Registers the callback invoked by this ISR
00776 * @see rx_cmt_deinit()   Clears the callback and disables this interrupt
00777 * @see s_cmt_callback    Per-channel callback function pointer array
00778 *
00779 * @since Version 1.0.0
00780 */
00781 void cmt2_isr(void)
00782 {
00783     if (s_cmt_callback[k_cmt_channel_2] != nullptr) {
00784         s_cmt_callback[k_cmt_channel_2](s_cmt_user_data[k_cmt_channel_2]);
00785     }
00786 }
00787
00788 /**
00789 * @brief CMT3 (CMT channel 3) compare-match interrupt service routine
00790 *
00791 * @details
00792 * Invoked by the RX72N hardware whenever the CMT3 counter (CMCNT) reaches the
00793 * value stored in CMCOR. The ISR checks whether a callback has been registered
00794 * in s_cmt_callback[k_cmt_channel_3] and, if so, calls it with the associated
00795 * user data pointer from s_cmt_user_data[k_cmt_channel_3].
00796 *
00797 * The compare-match interrupt flag is automatically cleared by the RX72N
00798 * hardware upon entry; no explicit IR register write is needed.
00799 *
00800 * @note Executes in interrupt context; all code in this ISR and any registered
00801 *       callback must be ISR-safe (no blocking, no long operations)
00802 * @note Not thread-safe; the callback executes with the CMT3 interrupt masked
00803 *
00804 * @warning Do not call any ThreadX API that is not interrupt-safe from the
00805 *          registered callback (e.g., tx_thread_sleep() is forbidden; use
00806 *          ISR-safe variants such as tx_semaphore_put_from_isr())
00807 *

```

```

00808 * @see rx_cmt_init()      Registers the callback invoked by this ISR
00809 * @see rx_cmt_deinit()    Clears the callback and disables this interrupt
00810 * @see s_cmt_callback     Per-channel callback function pointer array
00811 *
00812 * @since Version 1.0.0
00813 */
00814 void cmt3_isr(void)
00815 {
00816     if (s_cmt_callback[k_cmt_channel_3] != nullptr) {
00817         s_cmt_callback[k_cmt_channel_3](s_cmt_user_data[k_cmt_channel_3]);
00818     }
00819 }
00820
00821 /*
=====
00822 * Additional Internal Helpers for rx_cmt_init
00823 *
=====
00824 */
00825
00826 /**
00827 * @brief Validate channel and configuration pointer before CMT initialisation
00828 *
00829 * @details
00830 * Performs all parameter checking required before rx_cmt_init() proceeds to
00831 * hardware configuration. Validation is performed in this strict order:
00832 *
00833 * 1. Verify config pointer is non-null (k_rx_err_null_ptr)
00834 * 2. Verify channel is within [0, k_cmt_max_channels) (k_rx_err_invalid_arg)
00835 * 3. Reject channel 0, which is reserved for the ThreadX system-tick timer
00836 *    (k_rx_err_conflict)
00837 * 4. Verify interrupt priority is within [k_ipr_level_min, k_ipr_level_max]
00838 *    (k_rx_err_invalid_arg)
00839 *
00840 * Centralising these checks in a dedicated helper keeps the main rx_cmt_init()
00841 * function concise and ensures that every check is logged before returning.
00842 *
00843 *
00844 *
00845 * @pre config must be a pointer to a caller-owned rx_cmt_config_t (may be nullptr)
00846 * @pre channel must be a member of rx_cmt_channel_t
00847 *
00848 * @post On k_rx_ok: channel and *config are safe to use for hardware programming
00849 * @post On error: an error message has been emitted via rx_log_error()
00850 *
00851 * @note Not thread-safe; pure validation with no shared state modification
00852 * @note The lower-bound check `channel >= k_cmt_channel_0` is intentionally
00853 *        omitted to avoid -Wtype-limits: channel is uint8_t and k_cmt_channel_0 == 0,
00854 *        making the comparison always true
00855 *
00856 * @par Thread Safety:
00857 * Pure validation; no shared mutable state is modified. Safe to call from any
00858 * context, but the result may be stale if state changes after the call.
00859 *
00860 * @par Performance:
00861 * Execution time: ~4 comparisons; negligible overhead
00862 *
00863 * @par Example:
00864 * @code{.c}
00865 * rx_err_t err = internal_validate_cmt_init_params(k_cmt_channel_1, &config);
00866 * if (err != k_rx_ok) {
00867 *     return err;
00868 * }
00869 * @endcode
00870 *
00871 * @see rx_cmt_init()      Sole caller; uses this to guard hardware programming
00872 * @see rx_cmt_config_t    Configuration structure whose fields are validated here
00873 *
00874 * @since Version 1.0.0
00875 */
00876 static rx_err_t internal_validate_cmt_init_params(const rx_cmt_channel_t channel,
00877                                                  const rx_cmt_config_t* config)
00878 {
00879     if (config == nullptr) {
00880         rx_log_error(s_tag, "config pointer is nullptr");
00881         return k_rx_err_null_ptr;
00882     }
00883
00884     /* Note: Lower bound check omitted - channel is uint8_t, k_cmt_channel_0 == 0,
00885      * so channel >= k_cmt_channel_0 is always true (-Wtype-limits) */
00886     if ((uint8_t)channel >= (uint8_t)k_cmt_max_channels) {
00887         rx_log_error(s_tag, "Invalid CMT channel");
00888         return k_rx_err_invalid_arg;
00889     }
00890
00891     /* CMT0 is reserved for ThreadX system tick */
00892     if (channel == k_cmt_channel_0) {

```

```

00893     rx_log_error(s_tag, "CMT0 is reserved for ThreadX");
00894     return k_rx_err_conflict;
00895 }
00896
00897 if (config->priority < k_ipr_level_min || config->priority > k_ipr_level_max) {
00898     rx_log_error(s_tag, "Invalid interrupt priority");
00899     return k_rx_err_invalid_arg;
00900 }
00901
00902 return k_rx_ok;
00903 }
00904
00905 /**
00906  * @brief Persist the user callback, user-data pointer, and channel initialized flag
00907  *
00908  * @details
00909  * Copies the callback function pointer and user-data pointer from the caller's
00910  * configuration structure into the module-level static arrays, then marks the
00911  * channel as initialized. After this call the ISR for the channel will invoke
00912  * the registered callback on every compare-match event.
00913  *
00914  * This function is the final step in the rx_cmt_init() sequence. It must only
00915  * be called after all hardware registers have been programmed and the interrupt
00916  * has been configured in the ICU, so that the ISR never encounters a partially
00917  * initialised channel state.
00918  *
00919  *
00920  *
00921  * @pre channel must be valid and in range [1, k_cmt_max_channels) (validated by caller)
00922  * @pre config must be non-null (validated by internal_validate_cmt_init_params)
00923  *
00924  * @post s_cmt_callback[channel] = config->callback
00925  * @post s_cmt_user_data[channel] = config->user_data
00926  * @post s_cmt_initialized[channel] = true
00927  *
00928  * @note Not thread-safe; must be the last step of rx_cmt_init() with interrupts
00929  *       either disabled or not yet fired for this channel
00930  * @note After this call the channel is considered "live"; the ISR may fire at
00931  *       any point
00932  *
00933  * @par Thread Safety:
00934  * Not thread-safe. The three static array writes are not atomic. Call only
00935  * during initialization with the channel interrupt not yet active.
00936  *
00937  * @par Performance:
00938  * Execution time: ~3 stores; negligible overhead
00939  *
00940  * @par Example:
00941  * @code{.c}
00942  * // Final step inside rx_cmt_init() after all hardware setup is complete
00943  * internal_save_cmt_callback(channel, config);
00944  * // s_cmt_initialized[channel] is now true; ISR will call config->callback
00945  * @endcode
00946  *
00947  * @see rx_cmt_init()      Sole caller; calls this as the last initialisation step
00948  * @see rx_cmt_deinit()   Clears the state written by this function
00949  * @see s_cmt_callback    Destination array for config->callback
00950  * @see s_cmt_user_data   Destination array for config->user_data
00951  * @see s_cmt_initialized Destination array for the initialized flag
00952  *
00953  * @since Version 1.0.0
00954  */
00955 static void internal_save_cmt_callback(const rx_cmt_channel_t channel,
00956                                       const rx_cmt_config_t* config)
00957 {
00958     s_cmt_callback[channel] = config->callback;
00959     s_cmt_user_data[channel] = config->user_data;
00960     s_cmt_initialized[channel] = true;
00961 }
00962
00963 /**
=====
00964  * Public API Implementation
00965  *
=====
00966  */
00967
00968 /**
00969  * @brief Initialize a CMT channel for periodic interrupt generation
00970  *
00971  * @details
00972  * Configures a CMT (Compare Match Timer) channel to fire a user callback at the
00973  * requested frequency. Validates parameters, selects the optimal clock divider,
00974  * enables the module clock, stops the timer for safe configuration, programs
00975  * timer registers, configures the ICU interrupt routing, saves the callback,
00976  * and starts the timer.
00977  *

```

```

00978 **Algorithm steps:**
00979 * 1. Validate channel and config parameters via internal_validate_cmt_init_params()
00980 * 2. Obtain CMT channel register base via internal_get_cmt_base()
00981 * 3. Calculate optimal divider and CMCOR value via internal_calculate_cmt_params()
00982 * 4. Enable CMT module clock (clear MSTPCRB bit) via internal_enable_cmt_module_clock()
00983 * 5. Stop the timer to allow safe register writes via rx_cmt_stop()
00984 * 6. Program CMCR (divider + CMIE) and CMCOR via internal_configure_cmt_timer_registers()
00985 * 7. Configure ICU interrupt vector priority and enable via internal_configure_cmt_interrupt()
00986 * 8. Save callback pointer and mark channel initialized via internal_save_cmt_callback()
00987 * 9. Start the timer via rx_cmt_start()
00988
00989 **Clock divider selection:**
00990 * The driver iterates dividers /8, /32, /128, /512 and picks the first that
00991 * produces a period fitting in 16 bits:
00992 * @code
00993 * period = (PCLKB / divider) / frequency_hz
00994 * if (0 < period <= 65535): use this divider
00995 * @endcode
00996 *
00997 *
00998 *
00999 *
01000 * @pre config must point to a valid rx_cmt_config_t structure
01001 * @pre config->frequency_hz must be > 0 and <= PCLKB / 8 (7.5 MHz at 60 MHz PCLKB)
01002 * @pre channel must not be k_cmt_channel_0
01003 * @pre System clocks must be initialized (PCLKB = 60 MHz)
01004 *
01005 * @post Timer running at requested frequency
01006 * @post config->callback registered and called from ISR at each compare match
01007 * @post s_cmt_initialized[channel] set to true
01008 * @post ICU interrupt for the channel enabled at specified priority
01009 *
01010 * @note Not thread-safe - call once per channel during system initialization
01011 * @note Callback executes in interrupt context; keep it short and non-blocking
01012 * @warning CMT0 is reserved for ThreadX; do not use k_cmt_channel_0
01013 *
01014 * @par Thread Safety:
01015 * Not thread-safe. Call once per channel during initialization before starting
01016 * the RTOS scheduler or with interrupts disabled.
01017 *
01018 * @par Performance:
01019 * - Execution time: ~5 us (mostly register writes)
01020 * - Stack usage: 48 bytes
01021 *
01022 * @par Example:
01023 * @code{.c}
01024 * static void my_timer_cb(void* user_data)
01025 * {
01026 * // Called at 1 kHz from interrupt context
01027 * (void)user_data;
01028 * toggle_led();
01029 * }
01030 *
01031 * const rx_cmt_config_t cfg = {
01032 * .frequency_hz = 1000, // 1 kHz
01033 * .priority = 5,
01034 * .callback = my_timer_cb,
01035 * .user_data = nullptr,
01036 * };
01037 *
01038 * rx_err_t err = rx_cmt_init(k_cmt_channel_1, &cfg);
01039 * if (err != k_rx_ok) {
01040 * rx_log_error("APP", "CMT init failed");
01041 * }
01042 * @endcode
01043 *
01044 * @see rx_cmt_deinit() Stop and release channel
01045 * @see rx_cmt_start() Start a previously stopped channel
01046 * @see rx_cmt_stop() Stop the timer without deinitializing
01047 * @see rx_cmt_config_t Configuration structure definition
01048 *
01049 * @since Version 1.0.0
01050 */
01051 rx_err_t rx_cmt_init(const rx_cmt_channel_t channel, const rx_cmt_config_t* config)
01052 {
01053 /* Validate parameters */
01054 rx_err_t err = internal_validate_cmt_init_params(channel, config);
01055 if (err != k_rx_ok) {
01056 return err;
01057 }
01058
01059 volatile rx_cmt_channel_regs_t* cmt = internal_get_cmt_base(channel);
01060 if (cmt == nullptr) {
01061 return k_rx_err_invalid_arg;
01062 }
01063
01064 /* Calculate divider and compare value */

```

```

01065 uint8_t divider;
01066 uint16_t cmcor;
01067 err = internal_calculate_cmt_params(config->frequency_hz, &divider, &cmcor);
01068 if (err != k_rx_ok) {
01069     return err;
01070 }
01071
01072 /* Enable CMT module (clear module stop bit) */
01073 internal_enable_cmt_module_clock();
01074
01075 /* Stop timer before configuration */
01076 err = rx_cmt_stop(channel);
01077 if (err != k_rx_ok) {
01078     return err;
01079 }
01080
01081 /* Configure timer registers */
01082 internal_configure_cmt_timer_registers(cmt, divider, cmcor);
01083
01084 /* Configure interrupt */
01085 err = internal_configure_cmt_interrupt(
01086     (rx_cmt_interrupt_config_t){.channel = channel, .priority = config->priority});
01087 if (err != k_rx_ok) {
01088     return err;
01089 }
01090
01091 /* Save callback and mark initialized */
01092 internal_save_cmt_callback(channel, config);
01093
01094 /* Start timer */
01095 err = rx_cmt_start(channel);
01096 if (err != k_rx_ok) {
01097     return err;
01098 }
01099
01100 rx_log_info(s_tag, "CMT initialized");
01101
01102 return k_rx_ok;
01103 }
01104
01105 /**
01106  * @brief Start CMT counter for a previously initialized channel
01107  *
01108  * @details
01109  * Sets the appropriate bit in the CMSTR0 or CMSTR1 register to start the
01110  * CMT counter running. After setting the bit the register is read back to
01111  * verify the hardware accepted the write. Channels 0/1 share CMSTR0;
01112  * channels 2/3 share CMSTR1. Channels 0 and 1 use STR0/STR1 bits respectively,
01113  * as do channels 2 and 3.
01114  *
01115  *
01116  *
01117  * @pre channel must be initialized via rx_cmt_init()
01118  * @pre channel must be in range [0, 3]
01119  *
01120  * @post CMSTR bit for channel is set
01121  * @post Counter begins incrementing from its current value
01122  *
01123  * @note Usually called automatically by rx_cmt_init(); call explicitly only after rx_cmt_stop()
01124  * @note Performs a readback check to verify the hardware write succeeded
01125  *
01126  * @par Thread Safety:
01127  * Not thread-safe; do not call concurrently with stop/deinit on the same channel.
01128  *
01129  * @par Example:
01130  * @code{.c}
01131  * // Pause then resume timer
01132  * rx_cmt_stop(k_cmt_channel_1);
01133  * // ... do critical work ...
01134  * rx_cmt_start(k_cmt_channel_1);
01135  * @endcode
01136  *
01137  * @see rx_cmt_stop() Stop the counter
01138  * @see rx_cmt_init() Initialize channel (calls start internally)
01139  *
01140  * @since Version 1.0.0
01141  */
01142 rx_err_t rx_cmt_start(const rx_cmt_channel_t channel)
01143 {
01144     if ((uint8_t)channel >= k_cmt_max_channels) {
01145         return k_rx_err_invalid_arg;
01146     }
01147
01148     if (!s_cmt_initialized[channel]) {
01149         return k_rx_err_invalid_state;
01150     }
01151 }

```

```

01152 /* Set corresponding bit in CMSTR register */
01153 switch (channel) {
01154     case k_cmt_channel_0: {
01155         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01156         cmt_ctrl()->cmstr0 |= bit_mask;
01157         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01158         if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01159             return k_rx_err_hw_error;
01160         }
01161         break;
01162     }
01163     case k_cmt_channel_1: {
01164         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01165         cmt_ctrl()->cmstr0 |= bit_mask;
01166         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01167         if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01168             return k_rx_err_hw_error;
01169         }
01170         break;
01171     }
01172     case k_cmt_channel_2: {
01173         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01174         cmt_ctrl()->cmstr0 |= bit_mask;
01175         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01176         if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01177             return k_rx_err_hw_error;
01178         }
01179         break;
01180     }
01181     case k_cmt_channel_3: {
01182         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01183         cmt_ctrl()->cmstr0 |= bit_mask;
01184         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01185         if ((cmstr_value & bit_mask) == k_cmt_value_zero) {
01186             return k_rx_err_hw_error;
01187         }
01188         break;
01189     }
01190     default:
01191         return k_rx_err_invalid_arg;
01192 }
01193
01194 return k_rx_ok;
01195 }
01196
01197 /**
01198  * @brief Stop CMT counter without deinitializing the channel
01199  *
01200  * @details
01201  * Clears the appropriate bit in CMSTR0 or CMSTR1 to halt the CMT counter.
01202  * After clearing the bit the register is read back to verify the hardware
01203  * accepted the write. The channel remains initialized; call rx_cmt_start()
01204  * to resume counting.
01205  *
01206  * Note that rx_cmt_stop() does NOT require the channel to be initialized
01207  * (s_cmt_initialized check is omitted intentionally so that rx_cmt_init()
01208  * can call it safely before marking the channel as initialized).
01209  *
01210  *
01211  * @pre channel must be in range [0, 3]
01212  * @pre Interrupts may still fire if one is already pending when stop is called
01213  *
01214  * @post CMSTR bit for channel is cleared
01215  * @post Counter stops incrementing (current CMCNT value preserved)
01216  *
01217  * @note Does not clear the initialized flag - channel can be restarted
01218  * @note Called by rx_cmt_init() before programming registers (safe on uninitialized channel)
01219  *
01220  * @par Thread Safety:
01221  * Not thread-safe; do not call concurrently with start/init/deinit on the same channel.
01222  *
01223  * @par Example:
01224  * @code{.c}
01225  * // Temporarily suspend timer
01226  * rx_cmt_stop(k_cmt_channel_2);
01227  * reconfigure_something();
01228  * rx_cmt_start(k_cmt_channel_2);
01229  * @endcode
01230  *
01231  * @see rx_cmt_start() Resume counting
01232  * @see rx_cmt_deinit() Stop and release channel resources
01233  *
01234  * @since Version 1.0.0
01235  */
01236
01237 rx_err_t rx_cmt_stop(const rx_cmt_channel_t channel)
01238 {

```



```

01239 if ((uint8_t)channel >= k_cmt_max_channels) {
01240     return k_rx_err_invalid_arg;
01241 }
01242
01243 /* Clear corresponding bit in CMSTR register */
01244 switch (channel) {
01245     case k_cmt_channel_0: {
01246         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01247         cmt_ctrl()->cmstr0 &= (uint16_t)~bit_mask;
01248         const uint16_t cmstr_value = cmt_ctrl()->cmstr0;
01249         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01250             return k_rx_err_hw_error;
01251         }
01252         break;
01253     }
01254     case k_cmt_channel_1: {
01255         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01256         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01257         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01258         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01259             return k_rx_err_hw_error;
01260         }
01261         break;
01262     }
01263     case k_cmt_channel_2: {
01264         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str0);
01265         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01266         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01267         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01268             return k_rx_err_hw_error;
01269         }
01270         break;
01271     }
01272     case k_cmt_channel_3: {
01273         const uint16_t bit_mask = (uint16_t)(k_cmt_bit_mask_lsb « k_cmt_cmstr_str1);
01274         cmt_ctrl1()->cmstr0 &= (uint16_t)~bit_mask;
01275         const uint16_t cmstr_value = cmt_ctrl1()->cmstr0;
01276         if ((cmstr_value & bit_mask) != k_cmt_value_zero) {
01277             return k_rx_err_hw_error;
01278         }
01279         break;
01280     }
01281     default:
01282         return k_rx_err_invalid_arg;
01283 }
01284
01285 return k_rx_ok;
01286 }
01287
01288 /**
01289  * @brief Read current CMT counter (CMCNT) value
01290  *
01291  * @details
01292  * Returns the instantaneous value of the 16-bit CMCNT register for the
01293  * specified channel. The counter increments at PCLKB divided by the
01294  * configured divider (/8, /32, /128, or /512) and resets to 0 on compare
01295  * match. Useful for measuring elapsed time within a period.
01296  *
01297  *
01298  *
01299  *
01300  * @pre Channel must be initialized via rx_cmt_init()
01301  * @pre count must point to valid uint16_t storage
01302  *
01303  * @post *count set to CMCNT register value at time of read
01304  * @post Counter continues running (read is non-destructive)
01305  *
01306  * @note Value may change between consecutive reads if counter is running
01307  * @note For precise time measurements disable interrupts around consecutive reads
01308  *
01309  * @par Thread Safety:
01310  * Read-only; safe to call from multiple contexts, but value may be stale.
01311  *
01312  * @par Example:
01313  * @code{.c}
01314  * uint16_t ticks = 0;
01315  * rx_err_t err = rx_cmt_get_count(k_cmt_channel_1, &ticks);
01316  * if (err == k_rx_ok) {
01317  *     // ticks = current counter position within the period
01318  * }
01319  * @endcode
01320  *
01321  * @see rx_cmt_init() Initialize channel and set frequency
01322  * @see rx_cmt_start() Start the counter
01323  *
01324  * @since Version 1.0.0
01325  */

```



```

01326 rx_err_t rx_cmt_get_count(const rx_cmt_channel_t channel, uint16_t* count)
01327 {
01328     const volatile rx_cmt_channel_regs_t* cmt = nullptr;
01329
01330     RX_CHECK_NULL_PTR(count, s_tag, "count pointer is nullptr");
01331
01332     if ((uint8_t)channel >= k_cmt_max_channels || !s_cmt_initialized[channel]) {
01333         return k_rx_err_invalid_state;
01334     }
01335
01336     cmt = internal_get_cmt_base(channel);
01337     if (cmt == nullptr) {
01338         return k_rx_err_invalid_arg;
01339     }
01340
01341     *count = cmt->cmcnt;
01342     return k_rx_ok;
01343 }
01344
01345 /**
01346  * @brief Deinitialize CMT channel and release all resources
01347  *
01348  * @details
01349  * Stops the CMT counter, disables the ICU interrupt for the channel,
01350  * clears the stored callback and user data, and marks the channel as
01351  * uninitialized. After this call the channel may be re-initialized with
01352  * rx_cmt_init().
01353  *
01354  * **Algorithm steps:**
01355  * 1. Validate channel range
01356  * 2. Stop timer via rx_cmt_stop() and propagate any error
01357  * 3. Compute ICU vector, IER index, and IER bit for the channel
01358  * 4. Clear the IER bit to disable the interrupt
01359  * 5. Clear callback, user_data, and initialized flag
01360  *
01361  *
01362  * @pre channel must be in range [0, 3]
01363  * @pre Caller should ensure no other task is using the channel callback
01364  *
01365  * @post Counter halted (CMSTR bit cleared)
01366  * @post ICU interrupt disabled for channel
01367  * @post s_cmt_initialized[channel] set to false
01368  * @post s_cmt_callback[channel] and s_cmt_user_data[channel] set to nullptr
01369  *
01370  * @note Safe to call on an uninitialized channel (rx_cmt_stop tolerates it)
01371  * @note Module clock is NOT re-asserted - hardware remains clocked for re-use
01372  *
01373  * @par Thread Safety:
01374  * Not thread-safe. Ensure no ISR or other thread is using the channel.
01375  *
01376  * @par Example:
01377  * @code{.c}
01378  * rx_err_t err = rx_cmt_deinit(k_cmt_channel_1);
01379  * if (err != k_rx_ok) {
01380  *     rx_log_error("APP", "CMT deinit failed");
01381  * }
01382  * @endcode
01383  *
01384  * @see rx_cmt_init() Re-initialize after deinit
01385  * @see rx_cmt_stop() Stop without full deinit
01386  *
01387  * @since Version 1.0.0
01388  */
01389
01390 rx_err_t rx_cmt_deinit(const rx_cmt_channel_t channel)
01391 {
01392     if ((uint8_t)channel >= k_cmt_max_channels) {
01393         return k_rx_err_invalid_arg;
01394     }
01395
01396     /* Stop timer and propagate any errors */
01397     rx_err_t err = rx_cmt_stop(channel);
01398     if (err != k_rx_ok) {
01399         return err;
01400     }
01401
01402     /* Disable interrupt only for channels with fixed ICU vectors (CMI0/CMI1).
01403      * CMI2/CMI3 are SELECTB sources; interrupt wiring for those is not
01404      * implemented (see internal_configure_cmt_interrupt() for details). */
01405     if (channel == k_cmt_channel_0 || channel == k_cmt_channel_1) {
01406         const uint8_t vector = k_vect_cmt0_cmi0 + channel;
01407         const uint8_t ier_index = vector / k_icu_ier_bits_per_reg;
01408         const uint8_t ier_bit = vector % k_icu_ier_bits_per_reg;
01409         const uint8_t ier_mask = (uint8_t)(k_cmt_bit_mask_lsb « ier_bit);
01410         icu()->ier[ier_index] &= (uint8_t)~ier_mask;
01411     }
01412 }

```

```

01413  /* Clear callback */
01414  s_cmt_callback[channel] = nullptr;
01415  s_cmt_user_data[channel] = nullptr;
01416  s_cmt_initialized[channel] = false;
01417
01418  rx_log_info(s_tag, "CMT deinitialized");
01419
01420  return k_rx_ok;
01421 }
01422
01423 #endif /* __RX__ */

```

8.113 libs/rx_hal/src/rx_doc.c File Reference

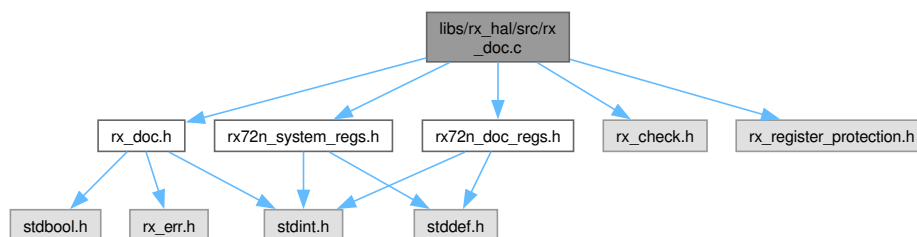
RX72N DOC (Data Operation Circuit) Driver Implementation.

```

#include "rx_doc.h"
#include "rx72n_doc_regs.h"
#include "rx72n_system_regs.h"
#include "rx_check.h"
#include "rx_register_protection.h"

```

Include dependency graph for rx_doc.c:



Enumerations

- enum `doc_init_state_t` : `uint8_t` { `k_doc_not_initialized` = 0 , `k_doc_initialized` = 1 }
DOC driver initialization state tracking.
- enum `doc_bit_widths_t` : `uint32_t` { `k_doc_mstpb_bit_one` = 1U }
Bit-width constants for shifting the module-stop bit into MSTPCRB.

Functions

- static `rx_err_t` `internal_docr_for_mode` (`rx_doc_mode_t` mode, `uint8_t` *docr_out)
Translate an `rx_doc_mode_t` into the corresponding DOCR bit pattern.
- static void `internal_enable_doc_clock` (void)
Enable the DOC module clock by clearing MSTPCRB.MSTPB6 under PRCR.PRC1.
- static void `internal_disable_doc_clock` (void)
Disable the DOC module clock by setting MSTPCRB.MSTPB6 under PRCR.PRC1.
- static void `internal_clear_dopcf` (void)
Clear the DOPCF flag while preserving the configured operating mode.
- static bool `internal_read_dopcf` (void)
Read and return the current DOPCF (operation completion flag).
- `rx_err_t` `rx_doc_init` (`rx_doc_mode_t` mode)
Initialize DOC in the selected operating mode.
- `rx_err_t` `rx_doc_set_reference` (`uint16_t` reference)

- Set the DOC compare reference (DODIR) register (implementation).
- `rx_err_t rx_doc_compare` (uint16_t value, bool *matched)
Compare value against the previously set DOC reference (implementation).
- `rx_err_t rx_doc_add` (uint16_t addend, uint16_t *sum, bool *overflow)
Perform a hardware-accelerated 16-bit add (implementation).
- `rx_err_t rx_doc_subtract` (uint16_t subtrahend, uint16_t *difference, bool *borrow)
Perform a hardware-accelerated 16-bit subtract (implementation).
- `rx_err_t rx_doc_deinit` (void)
Deinitialize the DOC peripheral and gate its clock (implementation).

Variables

- static const char *const `s_tag` = "DOC"
Log tag for this driver.
- static `doc_init_state_t s_doc_init_state` = `k_doc_not_initialized`
Tracks whether the DOC driver has been initialized.
- static `rx_doc_mode_t s_doc_mode` = `k_rx_doc_mode_compare`
Current DOC operating mode (valid only when `s_doc_init_state` == initialized).

8.113.1 Detailed Description

RX72N DOC (Data Operation Circuit) Driver Implementation.

Driver for the Renesas RX72N Data Operation Circuit – a small fixed-function 16-bit compute accelerator supporting four operating modes:

<code>rx`doc`mode` t</code>	OMS	DCSEL	Flag (DOPCF) Triggered On
<code>k_rx_doc_mode_compare</code>	00	1	DODIR == DODSR
<code>k_rx_doc_mode_compare_neq</code>	00	0	DODIR != DODSR
<code>k_rx_doc_mode_add</code>	01	x	Carry past 0xFFFF
<code>k_rx_doc_mode_subtract</code>	10	x	Borrow below 0x0000

8.113.1.1 Initialization Sequence

1. Validate the requested mode against `rx_doc_mode_t`.
2. Unlock PRCR (PRC1) and clear MSTPCRB.MSTPB6 to enable the DOC module clock.
3. Re-lock PRCR.
4. Write DOCR with the mode + DCSEL + DOPCFCL (clears any stale DOPCF).
5. Record the mode and flag the driver as initialized.

8.113.1.2 DOPCF Clearing

DOPCF (DOCR bit 5) is read-only; the peripheral clears it only when 1 is written to DOPCFCL (bit 6). Every operation writes DODSR to trigger the computation, reads DOPCF, then writes DOPCFCL | (current mode bits) back to DOCR to leave the mode untouched while clearing the flag.

8.113.1.3 Mode Switching

Re-calling `rx_doc_init()` with a different mode is the supported path to switch; mode mismatches between the configured operation and the API call return `k_rx_err_invalid_state`.

Manual Reference

RX72N Group User's Manual: Hardware (R01UH0824EJ0111), Chapter 59 "Data Operation Circuit", pages 2969-2978.

NASA Power of 10 Compliance

- Rule 1: No goto / setjmp / recursion
- Rule 2: No unbounded loops
- Rule 3: No dynamic allocation
- Rule 4: Every function ≤ 60 lines
- Rule 5: Minimum 2 validation checks per function
- Rule 6: File-scope state is static and private
- Rule 7: All returns checked or cast to void
- Rule 8: C23 typed enums for all constants
- Rule 9: Single-level pointers
- Rule 10: Compiles clean with -Wall -Wextra -Werror

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file `rx_doc.c`.

8.113.2 Enumeration Type Documentation

8.113.2.1 doc`bit`widths`t

```
enum doc\_bit\_widths\_t : uint32_t
```

Bit-width constants for shifting the module-stop bit into MSTPCRB.

Since

Version 1.0.0

Enumerator

<code>k_doc_mstpb_bit_one</code>	Single-bit value used for the MSTPCRB shift
----------------------------------	---

Definition at line 93 of file [rx_doc.c](#).

```
00093      : uint32_t {
00094  k_doc_mstpb_bit_one = 1U, /**< Single-bit value used for the MSTPCRB shift */
00095 } doc_bit_widths_t;
```

8.113.2.2 doc`init`state`t

```
enum doc\_init\_state\_t : uint8_t
```

DOC driver initialization state tracking.

Since

Version 1.0.0

Enumerator

<code>k_doc_not_initialized</code>	Driver not initialized; API calls will fail
<code>k_doc_initialized</code>	Driver initialized; operations permitted

Definition at line 83 of file [rx_doc.c](#).

```
00083      : uint8_t {
00084  k_doc_not_initialized = 0, /**< Driver not initialized; API calls will fail */
00085  k_doc_initialized    = 1, /**< Driver initialized; operations permitted */
00086 } doc_init_state_t;
```

8.113.3 Function Documentation

8.113.3.1 internal`clear`dopcf()

```
void internal_clear_dopcf (
    void ) [static]
```

Clear the DOPCF flag while preserving the configured operating mode.

Writes DOCR = (mode bits) | DOPCFCL. DOPCFCL is write-1-to-clear and leaves the mode fields (OMS, DCSEL) unchanged. `s_doc_mode` is guaranteed to be a valid enumerator whenever `s_doc_init_state == k_doc_initialized`, so the [internal_docr_for_mode\(\)](#) call cannot fail here.

Precondition

Driver must be initialized (`s_doc_init_state == k_doc_initialized`)

Postcondition

DOPCF is cleared

OMS / DCSEL / DOPCIE unchanged from their configured values

Since

Version 1.0.0

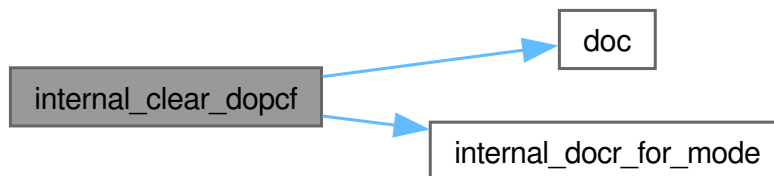
Definition at line 229 of file [rx_doc.c](#).

```
00230 {
00231     uint8_t mode_bits = 0U;
00232     (void)internal_docr_for_mode(s_doc_mode, &mode_bits);
00233     doc()->docr = (uint8_t)(mode_bits | k_doc_dopcfcl_clear);
00234 }
```

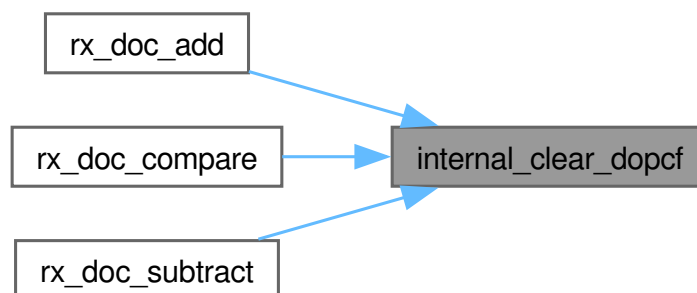
References [doc\(\)](#), [rx_doc_regs_t::docr](#), [internal_docr_for_mode\(\)](#), [k_doc_dopcfcl_clear](#), and [s_doc_mode](#).

Referenced by [rx_doc_add\(\)](#), [rx_doc_compare\(\)](#), and [rx_doc_subtract\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.113.3.2 `internal_disable_doc_clock()`

```
void internal_disable_doc_clock (  
    void ) [static]
```

Disable the DOC module clock by setting MSTPCRB.MSTPB6 under PRCR.PRC1.

Precondition

[prcr_reg\(\)](#) is writable

Postcondition

MSTPCRB.MSTPB6 == 1 (DOC module clock stopped)
PRCR is re-locked

Note

See [internal_enable_doc_clock\(\)](#) for the rationale on the absence of a [system_regs\(\)](#) null check.

Since

Version 1.0.0

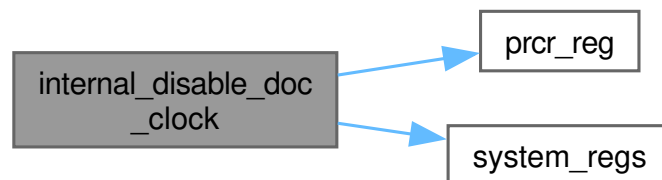
Definition at line 208 of file [rx_doc.c](#).

```
00209 {  
00210     *prcr_reg() = k_rx_prcr_unlock_prc1;  
00211     system_regs()->mstpcrb |= (k_doc_mstpb_bit_one « k_doc_mstpb_bit);  
00212     *prcr_reg() = k_rx_prcr_lock;  
00213 }
```

References [k_doc_mstpb_bit](#), [k_doc_mstpb_bit_one](#), [rx_system_regs_t::mstpcrb](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_doc_deinit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.113.3.3 `internal_docr_for_mode()`

```
rx_err_t internal_docr_for_mode (
    rx_doc_mode_t mode,
    uint8_t * docr_out) [static]
```

Translate an `rx_doc_mode_t` into the corresponding DOCR bit pattern.

Validates mode and, on success, writes the DOCR bit pattern (OMS + DCSEL

- DOPCIE disabled) for the requested mode into `docr_out`. DOPCFCL is left clear; the caller OR's it in when a flag clear is needed.

Combining the validation and the encoding in a single step eliminates the dead "unknown mode" branch that would otherwise exist in a pure encoder.

Precondition

`docr_out != NULL`

Postcondition

On success, `*docr_out` has `DOPCFCL == 0`

Manual Reference

RX72N HW Manual Section 59.2 "DOCR" (page 2970): OMS[1:0], DCSEL, DOPCIE layout.

Note

Pure function – no side effects, safe to call anywhere.

Since

Version 1.0.0

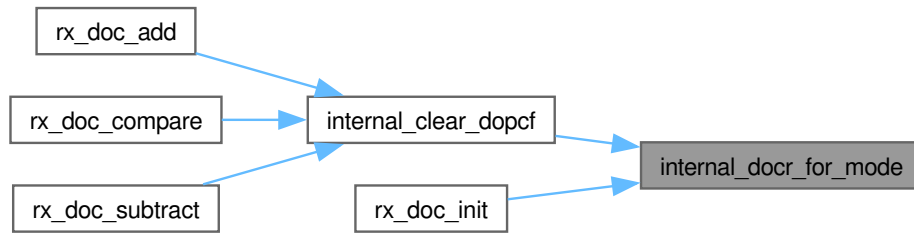
Definition at line 151 of file `rx_doc.c`.

```
00152 {
00153     switch (mode) {
00154         case k_rx_doc_mode_compare:
00155             *docr_out = (uint8_t)(k_doc_oms_compare | k_doc_dtsel_match | k_doc_dopcie_disabled);
00156             return k_rx_ok;
00157         case k_rx_doc_mode_compare_neq:
00158             *docr_out = (uint8_t)(k_doc_oms_compare | k_doc_dtsel_mismatch | k_doc_dopcie_disabled);
00159             return k_rx_ok;
00160         case k_rx_doc_mode_add:
00161             *docr_out = (uint8_t)(k_doc_oms_add | k_doc_dopcie_disabled);
00162             return k_rx_ok;
00163         case k_rx_doc_mode_subtract:
00164             *docr_out = (uint8_t)(k_doc_oms_subtract | k_doc_dopcie_disabled);
00165             return k_rx_ok;
00166         default:
00167             return k_rx_err_invalid_arg;
00168     }
00169 }
```

References `k_doc_dtsel_match`, `k_doc_dtsel_mismatch`, `k_doc_dopcie_disabled`, `k_doc_oms_add`, `k_doc_oms_compare`, `k_doc_oms_subtract`, `k_rx_doc_mode_add`, `k_rx_doc_mode_compare`, `k_rx_doc_mode_compare_neq`, and `k_rx_doc_mode_subtract`.

Referenced by `internal_clear_dopcf()`, and `rx_doc_init()`.

Here is the caller graph for this function:



8.113.3.4 internal`enable`doc`clock()

```
void internal_enable_doc_clock (
    void ) [static]
```

Enable the DOC module clock by clearing MSTPCRB.MSTPB6 under PRCR.PRC1.

Sequence:

1. Unlock PRCR (PRC1) via `k_rx_prcr_unlock_prc1`.
2. Clear bit `k_doc_mstpb_bit` (bit 6) in MSTPCRB.
3. Re-lock PRCR via `k_rx_prcr_lock`.

Precondition

`prcr_reg()` is writable

Postcondition

MSTPCRB.MSTPB6 == 0 (DOC module clock running)
 PRCR is re-locked (`k_rx_prcr_lock`)

Note

`system_regs()` is an inline accessor for a fixed memory-mapped address (`k_rx_system_regs_base`), so it is guaranteed non-NULL and no runtime check is performed (would be dead code per NASA Power of 10).

Not thread-safe; call during single-threaded init.

Since

Version 1.0.0

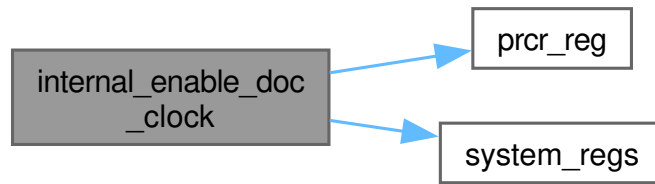
Definition at line 190 of file `rx_doc.c`.

```
00191 {
00192  *prcr_reg() = k_rx_prcr_unlock_prc1;
00193  system_regs()->mstpcrb &= ~(k_doc_mstpb_bit_one « k_doc_mstpb_bit);
00194  *prcr_reg() = k_rx_prcr_lock;
00195 }
```

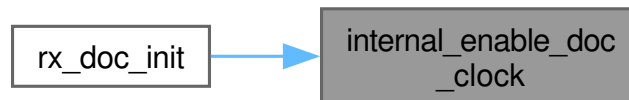
References `k_doc_mstpb_bit`, `k_doc_mstpb_bit_one`, `rx_system_regs_t::mstpcrb`, `prcr_reg()`, and `system_regs()`.

Referenced by `rx_doc_init()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.113.3.5 internal_read_dopcf()

```
bool internal_read_dopcf (
    void ) [static]
```

Read and return the current DOPCF (operation completion flag).

Since

Version 1.0.0

Definition at line 240 of file [rx_doc.c](#).

```
00241 {
00242     return (bool)((doc()->docr & k_doc_docr_dopcf_mask) != 0U);
00243 }
```

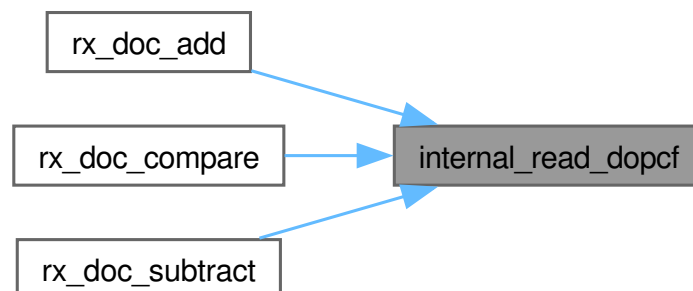
References [doc\(\)](#), [rx_doc_regs_t::docr](#), and [k_doc_docr_dopcf_mask](#).

Referenced by [rx_doc_add\(\)](#), [rx_doc_compare\(\)](#), and [rx_doc_subtract\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.113.3.6 rx`doc`add()

```

rx_err_t rx_doc_add (
    uint16_t addend,
    uint16_t * sum,
    bool * overflow)

```

Perform a hardware-accelerated 16-bit add (implementation).

Execute a 16-bit add using the DOC.

Writes the addend into DODSR; the DOC then computes $\text{sum} = \text{DODIR} + \text{addend}$ with overflow flagged in DOPCF. Reads the result back through DODSR (post-operation register) and reports the overflow flag, then clears DOPCF before return.

Note

Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 333 of file `rx_doc.c`.

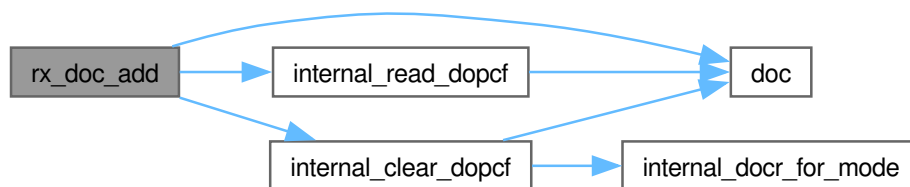
```

00334 {
00335     RX_CHECK_NULL_PTR(sum, s_tag, "rx_doc_add: sum is NULL");
00336     RX_CHECK_NULL_PTR(overflow, s_tag, "rx_doc_add: overflow is NULL");
00337
00338     if (s_doc_init_state != k_doc_initialized) {
00339         rx_log_error(s_tag, "rx_doc_add: driver not initialized");
00340         return k_rx_err_not_initialized;
00341     }
00342     if (s_doc_mode != k_rx_doc_mode_add) {
00343         rx_log_error(s_tag, "rx_doc_add: driver not in add mode");
00344         return k_rx_err_invalid_state;
00345     }
00346
00347     doc()->dodsr = addend;
00348     *sum        = doc()->dodsr;
00349     *overflow    = internal_read_dopcf();
00350     internal_clear_dopcf();
00351     return k_rx_ok;
00352 }

```

References `doc()`, `rx_doc_regs_t::dodsr`, `internal_clear_dopcf()`, `internal_read_dopcf()`, `k_doc_initialized`, `k_rx_doc_mode_add`, `s_doc_init_state`, `s_doc_mode`, and `s_tag`.

Here is the call graph for this function:



8.113.3.7 rx'doc'compare()

```
rx_err_t rx_doc_compare (
    uint16_t value,
    bool * matched)
```

Compare value against the previously set DOC reference (implementation).

Execute a compare operation against the loaded reference.

Writes value into DODSR, latches the comparator flag (DOPCF), and reports whether the configured compare condition is satisfied. In `k_rx_doc_mode_compare` the flag is set on equality; in `k_rx_doc_mode_compare_neq` the flag is set on inequality. The comparator flag is cleared internally before return so successive compares are independent.

Note

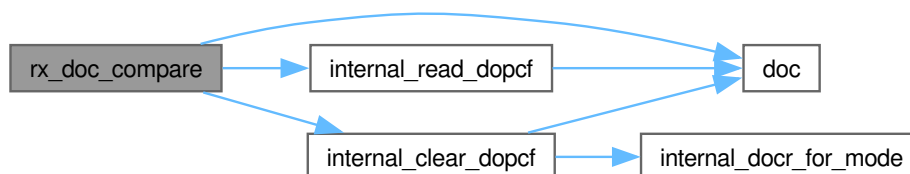
Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 303 of file `rx_doc.c`.

```
00304 {
00305     RX_CHECK_NULL_PTR(matched, s_tag, "rx_doc_compare: matched is NULL");
00306
00307     if (s_doc_init_state != k_doc_initialized) {
00308         rx_log_error(s_tag, "rx_doc_compare: driver not initialized");
00309         return k_rx_err_not_initialized;
00310     }
00311     if ((s_doc_mode != k_rx_doc_mode_compare) && (s_doc_mode != k_rx_doc_mode_compare_neq)) {
00312         rx_log_error(s_tag, "rx_doc_compare: driver not in a compare mode");
00313         return k_rx_err_invalid_state;
00314     }
00315
00316     doc()->dodsr = value;
00317     *matched = internal_read_dopcf();
00318     internal_clear_dopcf();
00319     return k_rx_ok;
00320 }
```

References `doc()`, `rx_doc_regs_t::dodsr`, `internal_clear_dopcf()`, `internal_read_dopcf()`, `k_doc_initialized`, `k_rx_doc_mode_compare`, `k_rx_doc_mode_compare_neq`, `s_doc_init_state`, `s_doc_mode`, and `s_tag`.

Here is the call graph for this function:



8.113.3.8 rx'doc'deinit()

```
rx_err_t rx_doc_deinit (
    void )
```

Deinitialize the DOC peripheral and gate its clock (implementation).

Deinitialize DOC and gate its clock.

Idempotent: returns `k_rx_ok` immediately if the driver was not initialized. Otherwise clears `DOC->DOCR` (mode register), disables the DOC module clock through `internal_disable_doc_clock()`, and marks the driver as uninitialized.

Note

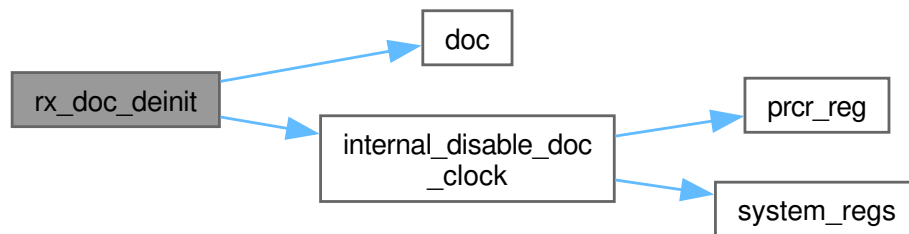
Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 397 of file [rx_doc.c](#).

```
00398 {
00399     if (s_doc_init_state != k_doc_initialized) {
00400         return k_rx_ok;
00401     }
00402
00403     doc()->docr = 0U;
00404     internal_disable_doc_clock();
00405
00406     s_doc_init_state = k_doc_not_initialized;
00407     return k_rx_ok;
00408 }
```

References [doc\(\)](#), [rx_doc_regs_t::docr](#), [internal_disable_doc_clock\(\)](#), [k_doc_initialized](#), [k_doc_not_initialized](#), and [s_doc_init_state](#).

Here is the call graph for this function:



8.113.3.9 rx`doc`init()

```
rx_err_t rx_doc_init (
    rx_doc_mode_t mode)
```

Initialize DOC in the selected operating mode.

Powers on the DOC (clears MSTPCRB.MSTPB6 under PRCR.PRC1 unlock), then configures DOCR with the requested operation mode, clears any stale DOPCF, and marks the driver initialized.

Subsequent operations expect the driver to remain in this mode until [rx_doc_init\(\)](#) is called again with a different mode (re-init is allowed and is the supported way to switch modes).

Parameters

in	mode	Operating mode to configure
----	------	-----------------------------

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success
<code>k_rx_err_invalid_arg</code>	mode is not a valid <code>rx_doc_mode_t</code> value

Precondition

PRCR accessor (`prcr_reg()`) is addressable
MSTPCRB is writable once PRCR.PRC1 is unlocked

Postcondition

DOC module stop bit cleared (`MSTPCRB.MSTPB6 = 0`)
DOCR configured for the requested mode, `DOPCIE = 0`, `DOPCF` cleared
Driver is marked initialized

Note

Not thread-safe. Caller must serialize DOC access.

See also

[rx_doc_deinit\(\)](#)

Since

Version 1.0.0

Definition at line 250 of file [rx_doc.c](#).

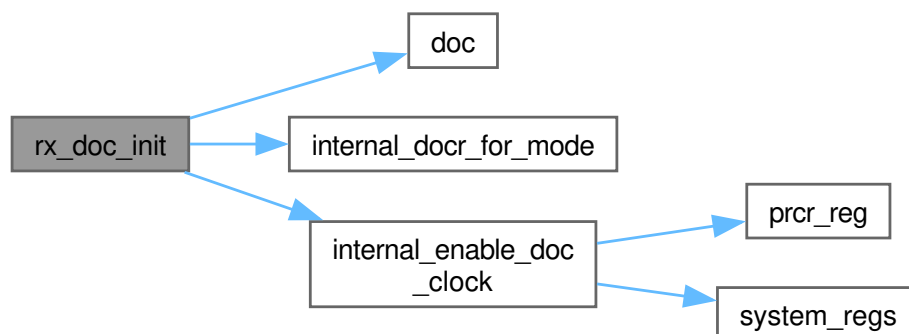
```

00251 {
00252     uint8_t mode_bits = 0U;
00253     const rx_err_t err = internal_docr_for_mode(mode, &mode_bits);
00254     if (err != k_rx_ok) {
00255         rx_log_error(s_tag, "rx_doc_init: invalid mode");
00256         return err;
00257     }
00258
00259     internal_enable_doc_clock();
00260     doc->docr = (uint8_t)(mode_bits | k_doc_dopcfcl_clear);
00261
00262     s_doc_mode = mode;
00263     s_doc_init_state = k_doc_initialized;
00264     return k_rx_ok;
00265 }

```

References [doc\(\)](#), [rx_doc_regs_t::docr](#), [internal_docr_for_mode\(\)](#), [internal_enable_doc_clock\(\)](#), [k_doc_dopcfcl_clear](#), [k_doc_initialized](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.113.3.10 rx`doc`set`reference()

```
rx_err_t rx_doc_set_reference (
    uint16_t reference)
```

Set the DOC compare reference (DODIR) register (implementation).

Load the DOC reference value (DODIR) used for the next operation.

Writes the supplied 16-bit reference value into the Data Operation Circuit Input register (DODIR) that subsequent [rx_doc_compare\(\)](#) / [rx_doc_add\(\)](#) / [rx_doc_subtract\(\)](#) calls operate against. Validates driver init state but does not validate or constrain the value range (any 16-bit value is meaningful in compare/add/subtract modes).

Note

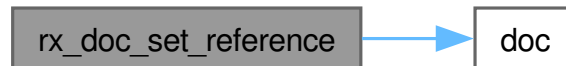
Not thread-safe. The DOC peripheral is global; serialize access at the application layer.

Definition at line 280 of file [rx_doc.c](#).

```
00281 {
00282     RX_VALIDATE_INIT(s_doc_init_state == k_doc_initialized,
00283                     s_tag,
00284                     "rx_doc_set_reference: driver not initialized");
00285
00286     doc()->dodir = reference;
00287     return k_rx_ok;
00288 }
```

References [doc\(\)](#), [rx_doc_regs_t::dodir](#), [k_doc_initialized](#), [s_doc_init_state](#), and [s_tag](#).

Here is the call graph for this function:



8.113.3.11 rx`doc`subtract()

```
rx_err_t rx_doc_subtract (
    uint16_t subtrahend,
    uint16_t * difference,
    bool * borrow)
```

Perform a hardware-accelerated 16-bit subtract (implementation).

Execute a 16-bit subtract using the DOC.

Writes subtrahend into DODSR; the DOC computes difference = DODIR - subtrahend with borrow flagged in DOPCF. Reads the result back through DODSR and reports the borrow flag, then clears DOPCF before return.

Note

Not thread-safe. Serialize access to the global DOC peripheral.

Definition at line 365 of file [rx_doc.c](#).

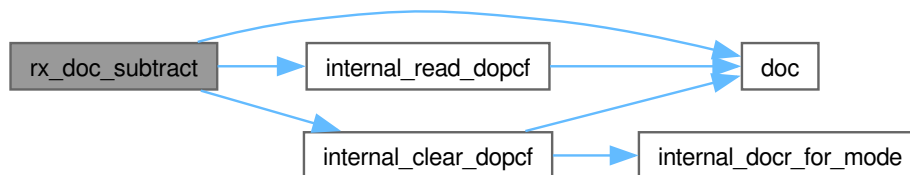
```

00366 {
00367     RX_CHECK_NULL_PTR(difference, s_tag, "rx_doc_subtract: difference is NULL");
00368     RX_CHECK_NULL_PTR(borrow, s_tag, "rx_doc_subtract: borrow is NULL");
00369
00370     if (s_doc_init_state != k_doc_initialized) {
00371         rx_log_error(s_tag, "rx_doc_subtract: driver not initialized");
00372         return k_rx_err_not_initialized;
00373     }
00374     if (s_doc_mode != k_rx_doc_mode_subtract) {
00375         rx_log_error(s_tag, "rx_doc_subtract: driver not in subtract mode");
00376         return k_rx_err_invalid_state;
00377     }
00378
00379     doc()->dodsr = subtrahend;
00380     *difference = doc()->dodsr;
00381     *borrow     = internal_read_dopcf();
00382     internal_clear_dopcf();
00383     return k_rx_ok;
00384 }

```

References [doc\(\)](#), [rx_doc_regs_t::dodsr](#), [internal_clear_dopcf\(\)](#), [internal_read_dopcf\(\)](#), [k_doc_initialized](#), [k_rx_doc_mode_subtract](#), [s_doc_init_state](#), [s_doc_mode](#), and [s_tag](#).

Here is the call graph for this function:



8.113.4 Variable Documentation

8.113.4.1 s'doc'init'state

[doc_init_state_t](#) s_doc_init_state = [k_doc_not_initialized](#) [static]

Tracks whether the DOC driver has been initialized.

Since

Version 1.0.0

Definition at line 114 of file [rx_doc.c](#).

Referenced by [rx_doc_add\(\)](#), [rx_doc_compare\(\)](#), [rx_doc_deinit\(\)](#), [rx_doc_init\(\)](#), [rx_doc_set_reference\(\)](#), and [rx_doc_subtract\(\)](#).

8.113.4.2 s`doc`mode

```
rx_doc_mode_t s_doc_mode = k_rx_doc_mode_compare [static]
```

Current DOC operating mode (valid only when s_doc_init_state == initialized).

Since

Version 1.0.0

Definition at line 121 of file rx_doc.c.

Referenced by [internal_clear_dopcf\(\)](#), [rx_doc_add\(\)](#), [rx_doc_compare\(\)](#), [rx_doc_init\(\)](#), and [rx_doc_subtract\(\)](#).

8.113.4.3 s`tag

```
const char* const s_tag = "DOC" [static]
```

Log tag for this driver.

Since

Version 1.0.0

Definition at line 107 of file rx_doc.c.

8.114 rx`doc.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_doc.c
00003  * @brief RX72N DOC (Data Operation Circuit) Driver Implementation
00004  *
00005  * @details
00006  * Driver for the Renesas RX72N Data Operation Circuit -- a small fixed-function
00007  * 16-bit compute accelerator supporting four operating modes:
00008  *
00009  * | rx_doc_mode_t | OMS | DCSEL | Flag (DOPCF) Triggered On |
00010  * |-----|-----|-----|
00011  * | k_rx_doc_mode_compare | 00 | 1 | DODIR == DODSR |
00012  * | k_rx_doc_mode_compare_neq | 00 | 0 | DODIR != DODSR |
00013  * | k_rx_doc_mode_add | 01 | x | Carry past 0xFFFF |
00014  * | k_rx_doc_mode_subtract | 10 | x | Borrow below 0x0000 |
00015  *
00016  * ## Initialization Sequence
00017  *
00018  * 1. Validate the requested mode against rx_doc_mode_t.
00019  * 2. Unlock PRCR (PRC1) and clear MSTPCRB.MSTPB6 to enable the DOC module clock.
00020  * 3. Re-lock PRCR.
00021  * 4. Write DOCR with the mode + DCSEL + DOPCFCL (clears any stale DOPCF).
00022  * 5. Record the mode and flag the driver as initialized.
00023  *
00024  * ## DOPCF Clearing
00025  *
00026  * DOPCF (DOCR bit 5) is read-only; the peripheral clears it only when 1 is
00027  * written to DOPCFCL (bit 6). Every operation writes DODSR to trigger the
00028  * computation, reads DOPCF, then writes DOPCFCL | (current mode bits) back to
00029  * DOCR to leave the mode untouched while clearing the flag.
00030  *
00031  * ## Mode Switching
00032  *
00033  * Re-calling rx_doc_init() with a different mode is the supported path to
```

```

00034 * switch; mode mismatches between the configured operation and the API call
00035 * return k_rx_err_invalid_state.
00036 *
00037 * @par Manual Reference
00038 * RX72N Group User's Manual: Hardware (R01UH0824EJ0111),
00039 * Chapter 59 "Data Operation Circuit", pages 2969-2978.
00040 *
00041 * @par NASA Power of 10 Compliance
00042 * - Rule 1: No goto / setjmp / recursion
00043 * - Rule 2: No unbounded loops
00044 * - Rule 3: No dynamic allocation
00045 * - Rule 4: Every function <= 60 lines
00046 * - Rule 5: Minimum 2 validation checks per function
00047 * - Rule 6: File-scope state is static and private
00048 * - Rule 7: All returns checked or cast to void
00049 * - Rule 8: C23 typed enums for all constants
00050 * - Rule 9: Single-level pointers
00051 * - Rule 10: Compiles clean with -Wall -Wextra -Werror
00052 *
00053 * @author Locked, Inc.
00054 * @date 2026-04-21
00055 * @copyright Copyright (c) 2026 Locked Inc.
00056 * SPDX-License-Identifier: MIT
00057 * @since Version 1.0.0
00058 */
00059
00060 #include "rx_doc.h"
00061
00062 #ifdef UNIT_TEST
00063 #include "mock_rx72n_system_regs.h"
00064 #include "mock_rx_doc.h"
00065 #else
00066 #include "rx72n_doc_regs.h"
00067 #include "rx72n_system_regs.h"
00068 #endif
00069
00070 #include "rx_check.h"
00071 #include "rx_register_protection.h"
00072
00073 /*
=====
00074 * Internal Constants
00075 *
=====
00076 */
00077
00078 /**
00079 * @enum doc_init_state_t
00080 * @brief DOC driver initialization state tracking
00081 * @since Version 1.0.0
00082 */
00083 typedef enum : uint8_t {
00084     k_doc_not_initialized = 0, /**< Driver not initialized; API calls will fail */
00085     k_doc_initialized     = 1, /**< Driver initialized; operations permitted */
00086 } doc_init_state_t;
00087
00088 /**
00089 * @enum doc_bit_widths_t
00090 * @brief Bit-width constants for shifting the module-stop bit into MSTPCRB
00091 * @since Version 1.0.0
00092 */
00093 typedef enum : uint32_t {
00094     k_doc_mstpb_bit_one = 1U, /**< Single-bit value used for the MSTPCRB shift */
00095 } doc_bit_widths_t;
00096
00097 /*
=====
00098 * Module State
00099 *
=====
00100 */
00101
00102 /**
00103 * @var s_tag
00104 * @brief Log tag for this driver
00105 * @since Version 1.0.0
00106 */
00107 static const char* const s_tag = "DOC";
00108
00109 /**
00110 * @var s_doc_init_state
00111 * @brief Tracks whether the DOC driver has been initialized
00112 * @since Version 1.0.0
00113 */
00114 static doc_init_state_t s_doc_init_state = k_doc_not_initialized;
00115
00116 /**

```

```

00117 * @var s_doc_mode
00118 * @brief Current DOC operating mode (valid only when s_doc_init_state == initialized)
00119 * @since Version 1.0.0
00120 */
00121 static rx_doc_mode_t s_doc_mode = k_rx_doc_mode_compare;
00122
00123 /*
=====
00124 * Internal Helpers
00125 *
=====
00126 */
00127
00128 /**
00129 * @brief Translate an rx_doc_mode_t into the corresponding DOCR bit pattern
00130 *
00131 * @details
00132 * Validates @p mode and, on success, writes the DOCR bit pattern (OMS + DCSEL
00133 * + DOPCIE disabled) for the requested mode into @p docr_out. DOPCFCL is left
00134 * clear; the caller OR's it in when a flag clear is needed.
00135 *
00136 * Combining the validation and the encoding in a single step eliminates the
00137 * dead "unknown mode" branch that would otherwise exist in a pure encoder.
00138 *
00139 *
00140 *
00141 * @pre docr_out != NULL
00142 * @post On success, @p *docr_out has DOPCFCL == 0
00143 *
00144 * @par Manual Reference
00145 * RX72N HW Manual Section 59.2 "DOCR" (page 2970): OMS[1:0], DCSEL, DOPCIE
00146 * layout.
00147 *
00148 * @note Pure function -- no side effects, safe to call anywhere.
00149 * @since Version 1.0.0
00150 */
00151 static rx_err_t internal_docr_for_mode(rx_doc_mode_t mode, uint8_t* docr_out)
00152 {
00153     switch (mode) {
00154         case k_rx_doc_mode_compare:
00155             *docr_out = (uint8_t)(k_doc_oms_compare | k_doc_dcsel_match | k_doc_dopcie_disabled);
00156             return k_rx_ok;
00157         case k_rx_doc_mode_compare_neg:
00158             *docr_out = (uint8_t)(k_doc_oms_compare | k_doc_dcsel_mismatch | k_doc_dopcie_disabled);
00159             return k_rx_ok;
00160         case k_rx_doc_mode_add:
00161             *docr_out = (uint8_t)(k_doc_oms_add | k_doc_dopcie_disabled);
00162             return k_rx_ok;
00163         case k_rx_doc_mode_subtract:
00164             *docr_out = (uint8_t)(k_doc_oms_subtract | k_doc_dopcie_disabled);
00165             return k_rx_ok;
00166         default:
00167             return k_rx_err_invalid_arg;
00168     }
00169 }
00170
00171 /**
00172 * @brief Enable the DOC module clock by clearing MSTPCRB.MSTPB6 under PRCR.PRC1
00173 *
00174 * @details
00175 * Sequence:
00176 * 1. Unlock PRCR (PRC1) via k_rx_prcr_unlock_prc1.
00177 * 2. Clear bit k_doc_mstpb_bit (bit 6) in MSTPCRB.
00178 * 3. Re-lock PRCR via k_rx_prcr_lock.
00179 *
00180 * @pre prcr_reg() is writable
00181 * @post MSTPCRB.MSTPB6 == 0 (DOC module clock running)
00182 * @post PRCR is re-locked (k_rx_prcr_lock)
00183 *
00184 * @note system_regs() is an inline accessor for a fixed memory-mapped address
00185 *       (k_rx_system_regs_base), so it is guaranteed non-NULL and no runtime
00186 *       check is performed (would be dead code per NASA Power of 10).
00187 * @note Not thread-safe; call during single-threaded init.
00188 * @since Version 1.0.0
00189 */
00190 static void internal_enable_doc_clock(void)
00191 {
00192     *prcr_reg() = k_rx_prcr_unlock_prc1;
00193     system_regs()->mstpcrb &= ~(k_doc_mstpb_bit_one < k_doc_mstpb_bit);
00194     *prcr_reg() = k_rx_prcr_lock;
00195 }
00196
00197 /**
00198 * @brief Disable the DOC module clock by setting MSTPCRB.MSTPB6 under PRCR.PRC1
00199 *
00200 * @pre prcr_reg() is writable
00201 * @post MSTPCRB.MSTPB6 == 1 (DOC module clock stopped)

```

```

00202 * @post PRCR is re-locked
00203 *
00204 * @note See internal_enable_doc_clock() for the rationale on the absence of a
00205 *       system_regs() null check.
00206 * @since Version 1.0.0
00207 */
00208 static void internal_disable_doc_clock(void)
00209 {
00210     *prcr_reg() = k_rx_prcr_unlock_prc1;
00211     system_regs()->mstpcrb |= (k_doc_mstpb_bit_one « k_doc_mstpb_bit);
00212     *prcr_reg() = k_rx_prcr_lock;
00213 }
00214
00215 /**
00216 * @brief Clear the DOPCF flag while preserving the configured operating mode
00217 *
00218 * @details
00219 * Writes DOCR = (mode bits) | DOPCFCL. DOPCFCL is write-1-to-clear and leaves
00220 * the mode fields (OMS, DCSEL) unchanged. s_doc_mode is guaranteed to be a
00221 * valid enumerator whenever s_doc_init_state == k_doc_initialized, so the
00222 * internal_docr_for_mode() call cannot fail here.
00223 *
00224 * @pre Driver must be initialized (s_doc_init_state == k_doc_initialized)
00225 * @post DOPCF is cleared
00226 * @post OMS / DCSEL / DOPCIE unchanged from their configured values
00227 * @since Version 1.0.0
00228 */
00229 static void internal_clear_dopcf(void)
00230 {
00231     uint8_t mode_bits = 0U;
00232     (void)internal_docr_for_mode(s_doc_mode, &mode_bits);
00233     doc()->docr = (uint8_t)(mode_bits | k_doc_dopcfcl_clear);
00234 }
00235
00236 /**
00237 * @brief Read and return the current DOPCF (operation completion flag)
00238 * @since Version 1.0.0
00239 */
00240 static bool internal_read_dopcf(void)
00241 {
00242     return (bool)((doc()->docr & k_doc_docr_dopcf_mask) != 0U);
00243 }
00244
00245 /*
=====
00246 * Public API
00247 *
=====
00248 */
00249
00250 rx_err_t rx_doc_init(rx_doc_mode_t mode)
00251 {
00252     uint8_t mode_bits = 0U;
00253     const rx_err_t err = internal_docr_for_mode(mode, &mode_bits);
00254     if (err != k_rx_ok) {
00255         rx_log_error(s_tag, "rx_doc_init: invalid mode");
00256         return err;
00257     }
00258
00259     internal_enable_doc_clock();
00260     doc()->docr = (uint8_t)(mode_bits | k_doc_dopcfcl_clear);
00261
00262     s_doc_mode = mode;
00263     s_doc_init_state = k_doc_initialized;
00264     return k_rx_ok;
00265 }
00266
00267 /**
00268 * @brief Set the DOC compare reference (DODIR) register (implementation)
00269 *
00270 * @details
00271 * Writes the supplied 16-bit reference value into the Data Operation
00272 * Circuit Input register (DODIR) that subsequent rx_doc_compare() /
00273 * rx_doc_add() / rx_doc_subtract() calls operate against. Validates
00274 * driver init state but does not validate or constrain the value range
00275 * (any 16-bit value is meaningful in compare/add/subtract modes).
00276 *
00277 * @note Not thread-safe. The DOC peripheral is global; serialize access
00278 *       at the application layer.
00279 */
00280 rx_err_t rx_doc_set_reference(uint16_t reference)
00281 {
00282     RX_VALIDATE_INIT(s_doc_init_state == k_doc_initialized,
00283                     s_tag,
00284                     "rx_doc_set_reference: driver not initialized");
00285
00286     doc()->dodir = reference;

```

```

00287     return k_rx_ok;
00288 }
00289
00290 /**
00291  * @brief Compare value against the previously set DOC reference (implementation)
00292  *
00293  * @details
00294  * Writes value into DODSR, latches the comparator flag (DOPCF), and
00295  * reports whether the configured compare condition is satisfied. In
00296  * k_rx_doc_mode_compare the flag is set on equality; in
00297  * k_rx_doc_mode_compare_neq the flag is set on inequality. The
00298  * comparator flag is cleared internally before return so successive
00299  * compares are independent.
00300  *
00301  * @note Not thread-safe. Serialize access to the global DOC peripheral.
00302  */
00303 rx_err_t rx_doc_compare(uint16_t value, bool* matched)
00304 {
00305     RX_CHECK_NULL_PTR(matched, s_tag, "rx_doc_compare: matched is NULL");
00306
00307     if (s_doc_init_state != k_doc_initialized) {
00308         rx_log_error(s_tag, "rx_doc_compare: driver not initialized");
00309         return k_rx_err_not_initialized;
00310     }
00311     if ((s_doc_mode != k_rx_doc_mode_compare) && (s_doc_mode != k_rx_doc_mode_compare_neq)) {
00312         rx_log_error(s_tag, "rx_doc_compare: driver not in a compare mode");
00313         return k_rx_err_invalid_state;
00314     }
00315
00316     doc()->dodsr = value;
00317     *matched = internal_read_dopcf();
00318     internal_clear_dopcf();
00319     return k_rx_ok;
00320 }
00321
00322 /**
00323  * @brief Perform a hardware-accelerated 16-bit add (implementation)
00324  *
00325  * @details
00326  * Writes the addend into DODSR; the DOC then computes
00327  * sum = DODIR + addend with overflow flagged in DOPCF. Reads the result
00328  * back through DODSR (post-operation register) and reports the overflow
00329  * flag, then clears DOPCF before return.
00330  *
00331  * @note Not thread-safe. Serialize access to the global DOC peripheral.
00332  */
00333 rx_err_t rx_doc_add(uint16_t addend, uint16_t* sum, bool* overflow)
00334 {
00335     RX_CHECK_NULL_PTR(sum, s_tag, "rx_doc_add: sum is NULL");
00336     RX_CHECK_NULL_PTR(overflow, s_tag, "rx_doc_add: overflow is NULL");
00337
00338     if (s_doc_init_state != k_doc_initialized) {
00339         rx_log_error(s_tag, "rx_doc_add: driver not initialized");
00340         return k_rx_err_not_initialized;
00341     }
00342     if (s_doc_mode != k_rx_doc_mode_add) {
00343         rx_log_error(s_tag, "rx_doc_add: driver not in add mode");
00344         return k_rx_err_invalid_state;
00345     }
00346
00347     doc()->dodsr = addend;
00348     *sum = doc()->dodsr;
00349     *overflow = internal_read_dopcf();
00350     internal_clear_dopcf();
00351     return k_rx_ok;
00352 }
00353
00354 /**
00355  * @brief Perform a hardware-accelerated 16-bit subtract (implementation)
00356  *
00357  * @details
00358  * Writes subtrahend into DODSR; the DOC computes
00359  * difference = DODIR - subtrahend with borrow flagged in DOPCF. Reads
00360  * the result back through DODSR and reports the borrow flag, then clears
00361  * DOPCF before return.
00362  *
00363  * @note Not thread-safe. Serialize access to the global DOC peripheral.
00364  */
00365 rx_err_t rx_doc_subtract(uint16_t subtrahend, uint16_t* difference, bool* borrow)
00366 {
00367     RX_CHECK_NULL_PTR(difference, s_tag, "rx_doc_subtract: difference is NULL");
00368     RX_CHECK_NULL_PTR(borrow, s_tag, "rx_doc_subtract: borrow is NULL");
00369
00370     if (s_doc_init_state != k_doc_initialized) {
00371         rx_log_error(s_tag, "rx_doc_subtract: driver not initialized");
00372         return k_rx_err_not_initialized;
00373     }

```

```

00374 if (s_doc_mode != k_rx_doc_mode_subtract) {
00375     rx_log_error(s_tag, "rx_doc_subtract: driver not in subtract mode");
00376     return k_rx_err_invalid_state;
00377 }
00378
00379 doc()->dodsr = subtrahend;
00380 *difference = doc()->dodsr;
00381 *borrow     = internal_read_dopcf();
00382 internal_clear_dopcf();
00383 return k_rx_ok;
00384 }
00385
00386 /**
00387  * @brief Deinitialize the DOC peripheral and gate its clock (implementation)
00388  *
00389  * @details
00390  * Idempotent: returns k_rx_ok immediately if the driver was not
00391  * initialized. Otherwise clears DOC->DOCR (mode register), disables the
00392  * DOC module clock through internal_disable_doc_clock(), and marks the
00393  * driver as uninitialized.
00394  *
00395  * @note Not thread-safe. Serialize access to the global DOC peripheral.
00396  */
00397 rx_err_t rx_doc_deinit(void)
00398 {
00399     if (s_doc_init_state != k_doc_initialized) {
00400         return k_rx_ok;
00401     }
00402
00403     doc()->docr = 0U;
00404     internal_disable_doc_clock();
00405
00406     s_doc_init_state = k_doc_not_initialized;
00407     return k_rx_ok;
00408 }
00409
00410 /**
00411  * Test Support
00412  */
00413 */
00414
00415 #ifndef UNIT_TEST
00416 /**
00417  * @brief Reset the DOC driver module state for unit testing
00418  *
00419  * @details
00420  * Available only in UNIT_TEST builds. Clears the initialized flag and resets
00421  * the tracked mode so individual tests can start from a known state without
00422  * relying on static initialization order.
00423  *
00424  * @post s_doc_init_state == k_doc_not_initialized
00425  * @post s_doc_mode == k_rx_doc_mode_compare
00426  * @since Version 1.0.0
00427  */
00428 void rx_doc_test_reset(void)
00429 {
00430     s_doc_init_state = k_doc_not_initialized;
00431     s_doc_mode       = k_rx_doc_mode_compare;
00432 }
00433 #endif

```

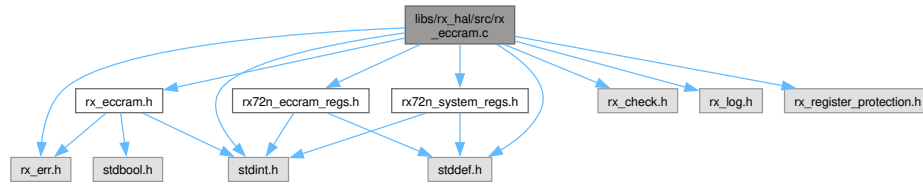
8.115 libs/rx/hal/src/rx_eccram.c File Reference

ECC-Protected RAM (ECCRAM) Driver Implementation.

```

#include "rx_eccram.h"
#include <stddef.h>
#include <stdint.h>
#include "rx72n_eccram_regs.h"
#include "rx72n_system_regs.h"
#include "rx_check.h"
#include "rx_err.h"
#include "rx_log.h"
#include "rx_register_protection.h"
Include dependency graph for rx_eccram.c:

```



Enumerations

- enum `rx_eccram_fill_t` : `uint32_t { k_rx_eccram_fill_pattern = 0x00000000U }`
Zero-initialisation fill pattern and word size.
- enum `rx_eccram_word_size_t` : `uint32_t { k_rx_eccram_word_bytes = 4 }`
Width of a single ECC-protected word in bytes.
- enum `rx_eccram_word_count_t` : `uint32_t { k_rx_eccram_word_count = (k_rx_eccram_region_size_bytes / k_rx_eccram_word_bytes) }`
Number of 32-bit words in the ECCRAM region.

Functions

- static `uint8_t internal_mode_to_reg(rx_eccram_mode_t mode)`
Translate public mode enum into ECCRAMMODE register value.
- static void `internal_enable_eccram_module_clock(void)`
Clear MSTPCRC.MSTPC6 so ECCRAM control block is clocked.
- static void `internal_write_eccram_mode(uint8_t mode_reg_value)`
Write ECCRAMMODE under ECCRAMPRCR unlock.
- static void `internal_enable_1bit_latch(void)`
Enable the 1-bit status-update latch.
- static void `internal_zero_fill_region(void)`
Zero-initialise the ECCRAM region with 32-bit stores.
- static void `internal_ram_error_isr_body(void)`
Shared RAM error handler core.
- void `rx_eccram_ram_error_isr(void)`
Vector-18 ISR entry (RAMERR) – ECCRAM 1-bit/2-bit error.
- `rx_err_t rx_eccram_init(rx_eccram_mode_t mode)`
Initialise ECCRAM in the requested operating mode.
- `rx_err_t rx_eccram_get_error_status(rx_eccram_status_t *out)`
Read a snapshot of ECCRAM error status and captured addresses.
- `rx_err_t rx_eccram_clear_errors(void)`
Clear 1-bit and 2-bit error status flags.
- `rx_err_t rx_eccram_register_error_isr(rx_eccram_on_1bit_fn_t on_1bit, rx_eccram_on_2bit_fn_t on_2bit, void *ctx)`
Register user callbacks for ECC error ISRs.
- `uintptr_t rx_eccram_region_start(void)`
Return the inclusive start address of the ECCRAM data region.
- `uintptr_t rx_eccram_region_end(void)`
Return the inclusive end address of the ECCRAM data region.

Variables

- static const char *const [s_tag](#) = "ECCRAM"
Module log tag for rx_log_* macros.
- static bool [s_initialized](#) = false
Driver initialised flag.
- static [rx_eccram_on_1bit_fn_t s_on_1bit](#) = NULL
Registered 1-bit error callback (may be nullptr).
- static [rx_eccram_on_2bit_fn_t s_on_2bit](#) = NULL
Registered 2-bit error callback (may be nullptr).
- static void * [s_isr_ctx](#) = NULL
Opaque context passed to registered callbacks.
- static volatile uintptr_t [s_last_2bit_addr](#) = 0
Latched address of most recent 2-bit error (captured in ISR).
- static volatile uintptr_t [s_last_1bit_addr](#) = 0
Latched address of most recent 1-bit error (captured in ISR).

8.115.1 Detailed Description

ECC-Protected RAM (ECCRAM) Driver Implementation.

Implements the [rx_eccram.h](#) API: module-stop clear, mode configuration under PRCR unlock, region zero-initialisation with 32-bit stores, error-status readout, flag clearing, ISR trampoline, and region introspection.

The RX72N has a single RAMERR vector (vector 18, RAM group) that fires for both 1-bit and 2-bit ECCRAM errors. The ISR trampoline inspects ECCRAM1STS and ECCRAM2STS to decide which registered handler (if any) to invoke, captures the failing address(es), and clears the latched status flags.

Register Protection Sequencing

Two independent protection registers are involved:

- System PRCR (0x000803FE) protects MSTPCRC (Manual Ch11). Unlock with `k_rx_prcr_unlock_prc1`, write MSTPCRC, re-lock with `k_rx_prcr_lock`.
- ECCRAMPRCR (0x000812C4) protects ECCRAMMODE and ECCRAM1STSEN (Manual section 60.2.13). Unlock with `k_rx_eccramprcr_unlock` (0xF1), write the guarded register, re-lock with `k_rx_eccramprcr_lock` (0xF0).

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_eccram.c](#).

8.115.2 Enumeration Type Documentation

8.115.2.1 rx`eccram`fill`_t

```
enum rx_eccram_fill_t : uint32_t
```

Zero-initialisation fill pattern and word size.

The ECC syndrome is computed per 32-bit word; filling with zero is the cheapest pattern that produces a well-defined, check-valid state.

Since

Version 1.0.0

Enumerator

k_rx_eccram_fill_pattern	Fill pattern used by the zero-initialisation loop.
--------------------------	--

Definition at line 73 of file [rx_eccram.c](#).

```
00073      : uint32_t {
00074  /** @brief Fill pattern used by the zero-initialisation loop */
00075  k_rx_eccram_fill_pattern = 0x00000000U,
00076 } rx_eccram_fill_t;
```

8.115.2.2 rx`eccram`word`count`_t

```
enum rx_eccram_word_count_t : uint32_t
```

Number of 32-bit words in the ECCRAM region.

Since

Version 1.0.0

Enumerator

k_rx_eccram_word_count	32 KB / 4 bytes per word = 8192 words
------------------------	---------------------------------------

Definition at line 99 of file [rx_eccram.c](#).

```
00099      : uint32_t {
00100  /** @brief 32 KB / 4 bytes per word = 8192 words */
00101  k_rx_eccram_word_count = (k_rx_eccram_region_size_bytes / k_rx_eccram_word_bytes),
00102 } rx_eccram_word_count_t;
```

8.115.2.3 rx_eccram_word_size_t

```
enum rx_eccram_word_size_t : uint32_t
```

Width of a single ECC-protected word in bytes.

Declared as a typed enum so it can bound the Rule-2 static loop upper in `rx_eccram_init()` without introducing a magic number.

Since

Version 1.0.0

Enumerator

k_rx_eccram_word_bytes	4 bytes per ECC word (uint32_t)
------------------------	---------------------------------

Definition at line 88 of file `rx_eccram.c`.

```
00088      : uint32_t {
00089      /** @brief 4 bytes per ECC word (uint32_t) */
00090      k_rx_eccram_word_bytes = 4,
00091  } rx_eccram_word_size_t;
```

8.115.3 Function Documentation

8.115.3.1 internal_enable_1bit_latch()

```
void internal_enable_1bit_latch (
    void ) [static]
```

Enable the 1-bit status-update latch.

Sets ECCRAM1STSEN=1 so that 1-bit errors are latched into ECCRAM1STS and ECCRAM1ECAD. Skipped for k_eccram_mode_disabled. Guarded by ECCRAMPRCR (shared with ECCRAMMODE).

Precondition

ECCRAM module clocked.

Postcondition

ECCRAM1STSEN.ECC1STSEN == 1.
ECCRAMPRCR is re-locked on return.

Since

Version 1.0.0

Definition at line 210 of file [rx_eccram.c](#).

```
00211 {
00212     volatile rx_eccram_regs_t* regs = eccram_regs();
00213     regs->eccramprcr = k_rx_eccramprcr_unlock;
00214     regs->eccram1tsen = k_rx_eccram1tsen_enable;
00215     regs->eccramprcr = k_rx_eccramprcr_lock;
00216 }
```

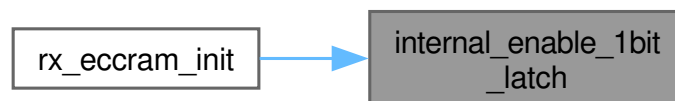
References [rx_eccram_regs_t::eccram1tsen](#), [eccram_regs\(\)](#), [rx_eccram_regs_t::eccramprcr](#), [k_rx_eccram1tsen_enable](#), [k_rx_eccramprcr_lock](#), and [k_rx_eccramprcr_unlock](#).

Referenced by [rx_eccram_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.115.3.2 internal_enable_eccram_module_clock()

```
void internal_enable_eccram_module_clock (
    void ) [static]
```

Clear MSTPCRC.MSTPC6 so ECCRAM control block is clocked.

Uses the standard STAR unlock/write/lock sequence on the system PRCR.

Precondition

System clock configured.

Caller holds exclusive access to MSTPCRC (single-threaded init).

Postcondition

MSTPCRC.MSTPC6 == 0.

System PRCR is re-locked on return.

See also

RX72N HW Manual Chapter 60, section 60.4.1 (MSTPCRC), page 2994

Since

Version 1.0.0

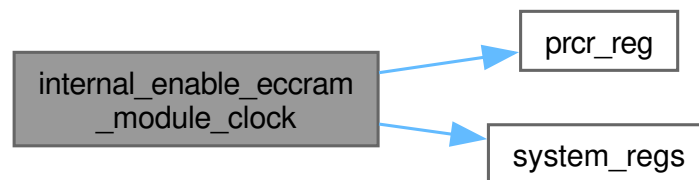
Definition at line 169 of file [rx_eccram.c](#).

```
00170 {
00171  *prcr_reg() = k_rx_prcr_unlock_prc1;
00172  system_regs()->mstpcrc &= ~(k_rx_eccram_mstpcrc_bit_mask);
00173  *prcr_reg() = k_rx_prcr_lock;
00174 }
```

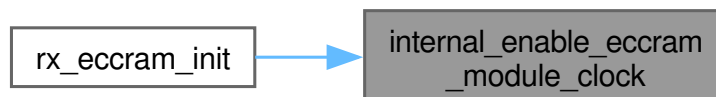
References [k_rx_eccram_mstpcrc_bit_mask](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_eccram_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.115.3.3 `internal_mode_to_reg()`

```
uint8_t internal_mode_to_reg (
    rx\_eccram\_mode\_t mode) [static]
```

Translate public mode enum into ECCRAMMODE register value.

Precondition

`mode` has been range-checked by the caller.

Postcondition

Return value is one of the `k_rx_eccrammode_*` constants.

Note

All error-visible modes share the same hardware encoding (RAMMOD=11b). The distinction between `correct_only`, `correct_and_detect`, and `detect_only` is enforced by which ISRs the driver registers (not by the hardware bits).

Since

Version 1.0.0

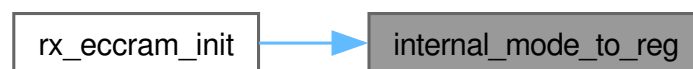
Definition at line 147 of file [rx_eccram.c](#).

```
00148 {
00149     if (mode == k_eccram_mode_disabled) {
00150         return k_rx_eccrammode_ecc_disabled;
00151     }
00152     return k_rx_eccrammode_ecc_with_check;
00153 }
```

References [k_eccram_mode_disabled](#), [k_rx_eccrammode_ecc_disabled](#), and [k_rx_eccrammode_ecc_with_check](#).

Referenced by [rx_eccram_init\(\)](#).

Here is the caller graph for this function:

8.115.3.4 `internal`ram`error`isr`body()`

```
void internal_ram_error_isr_body (
    void ) [static]
```

Shared RAM error handler core.

Called from the `__attribute__((interrupt))` wrapper below. Reads both status flags, captures the corresponding address registers, dispatches to registered user callbacks, and clears the flags so the next error can be detected.

The 2-bit branch captures ECCRAM2ECAD into `s_last_2bit_addr` before the flag-clearing write, matching the prompt requirement that the failing address be saved first.

Precondition

Called from IRQ context on vector 18 (RAMERR).

Postcondition

ECCRAM1STS and ECCRAM2STS flags cleared.

`s_last_1bit_addr` / `s_last_2bit_addr` updated on match.

Since

Version 1.0.0

Definition at line 310 of file [rx_eccram.c](#).

```

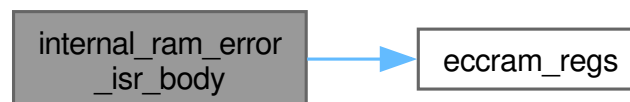
00311 {
00312     volatile rx_eccram_regs_t* regs = eccram_regs();
00313
00314     const uint8_t status2 = regs->eccram2sts;
00315     if ((status2 & k_rx_eccram2sts_ecc2err_mask) != 0U) {
00316         const uintptr_t addr2 = (uintptr_t)regs->eccram2ecad;
00317         s_last_2bit_addr = addr2;
00318         regs->eccram2sts = k_rx_eccram2sts_clear;
00319         if (s_on_2bit != NULL) {
00320             s_on_2bit(addr2, s_isr_ctx);
00321         }
00322     }
00323
00324     const uint8_t status1 = regs->eccram1sts;
00325     if ((status1 & k_rx_eccram1sts_ecc1err_mask) != 0U) {
00326         const uintptr_t addr1 = (uintptr_t)regs->eccram1ecad;
00327         s_last_1bit_addr = addr1;
00328         regs->eccram1sts = k_rx_eccram1sts_clear;
00329         if (s_on_1bit != NULL) {
00330             s_on_1bit(addr1, s_isr_ctx);
00331         }
00332     }
00333 }

```

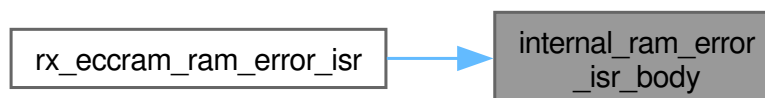
References [rx_eccram_regs_t::eccram1ecad](#), [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2ecad](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [k_rx_eccram1sts_clear](#), [k_rx_eccram1sts_ecc1err_mask](#), [k_rx_eccram2sts_clear](#), [k_rx_eccram2sts_ecc2err_mask](#), [s_isr_ctx](#), [s_last_1bit_addr](#), [s_last_2bit_addr](#), [s_on_1bit](#), and [s_on_2bit](#).

Referenced by [rx_eccram_ram_error_isr\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.115.3.5 internal_write_eccram_mode()

```

void internal_write_eccram_mode (
    uint8_t mode_reg_value) [static]

```

Write ECCRAMMODE under ECCRAMPCR unlock.

Precondition

ECCRAM module clocked (MSTPCRC.MSTPC6 == 0).
mode_reg_value is a valid [k_rx_eccrammode_*](#) constant.

Postcondition

ECCRAMMODE reflects mode_reg_value.

ECCRAMPRCR is re-locked on return.

See also

RX72N HW Manual Chapter 60, section 60.2.13, page 2985

Since

Version 1.0.0

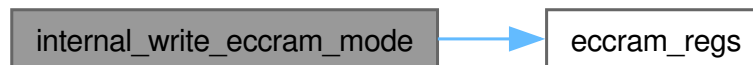
Definition at line 188 of file [rx_eccram.c](#).

```
00189 {
00190     volatile rx_eccram_regs_t* regs = eccram_regs();
00191     regs->eccramprcr = k_rx_eccramprcr_unlock;
00192     regs->eccrammode = mode_reg_value;
00193     regs->eccramprcr = k_rx_eccramprcr_lock;
00194 }
```

References [eccram_regs\(\)](#), [rx_eccram_regs_t::eccrammode](#), [rx_eccram_regs_t::eccramprcr](#), [k_rx_eccramprcr_lock](#), and [k_rx_eccramprcr_unlock](#).

Referenced by [rx_eccram_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.115.3.6 internal_zero_fill_region()

```
void internal_zero_fill_region (
    void ) [static]
```

Zero-initialise the ECCRAM region with 32-bit stores.

Iterates over every 32-bit word in ECCRAM and writes zero so that every ECC syndrome matches. The loop upper bound is a compile-time constant ([k_rx_eccram_word_count](#)), which satisfies NASA Rule 2.

Precondition

ECCRAMMODE == [k_rx_eccrammode_ecc_no_check](#) (so writes generate syndromes but reads do not trigger spurious errors).

Postcondition

Every 32-bit word of ECCRAM holds `k_rx_eccram_fill_pattern`.

Note

Runs in ~65 k cycles on RX72N at 240 MHz (~270 us).

Since

Version 1.0.0

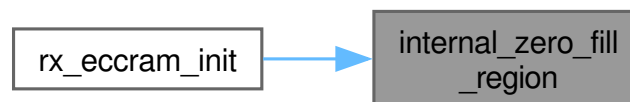
Definition at line 272 of file [rx_eccram.c](#).

```
00273 {
00274 #ifdef UNIT_TEST
00275     for (uint32_t i = 0; i < k_rx_eccram_word_count; i++) {
00276         g_mock_eccram_region[i] = k_rx_eccram_fill_pattern;
00277     }
00278 #else
00279     volatile uint32_t* dst = (volatile uint32_t*)k_rx_eccram_region_base_addr;
00280     for (uint32_t i = 0; i < k_rx_eccram_word_count; i++) {
00281         dst[i] = k_rx_eccram_fill_pattern;
00282     }
00283 #endif
00284 }
```

References [k_rx_eccram_fill_pattern](#), [k_rx_eccram_region_base_addr](#), and [k_rx_eccram_word_count](#).

Referenced by [rx_eccram_init\(\)](#).

Here is the caller graph for this function:



8.115.3.7 rx_eccram_clear_errors()

```
rx_err_t rx_eccram_clear_errors (
    void ) [nodiscard]
```

Clear 1-bit and 2-bit error status flags.

Writes `k_rx_eccram1sts_clear` and `k_rx_eccram2sts_clear` to acknowledge any outstanding error flags. Does not touch the captured addresses: those remain latched until the next error.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Flags cleared.
----------------------	----------------

Precondition

`rx_eccram_init()` has been called.

ECCRAMPRCR protection does not guard ECCRAM1STS/ECCRAM2STS (verified in Manual section 60.2.13 – both registers are directly writable without PRCR unlock).

Postcondition

`ECCRAM1STS.ECC1ERR == 0` and `ECCRAM2STS.ECC2ERR == 0`.

Note

Thread Safety: Call from ISR or task context; writes are atomic.

Since

Version 1.0.0

Definition at line 433 of file `rx_eccram.c`.

```
00434 {
00435     volatile rx_eccram_regs_t* regs = eccram_regs();
00436     regs->eccram1sts          = k_rx_eccram1sts_clear;
00437     regs->eccram2sts          = k_rx_eccram2sts_clear;
00438     return k_rx_ok;
00439 }
```

References `rx_eccram_regs_t::eccram1sts`, `rx_eccram_regs_t::eccram2sts`, `eccram_regs()`, `k_rx_eccram1sts_clear`, and `k_rx_eccram2sts_clear`.

Here is the call graph for this function:



8.115.3.8 rx`eccram`get`error`status()

```
rx_err_t rx_eccram_get_error_status (
    rx_eccram_status_t * out) [nodiscard]
```

Read a snapshot of ECCRAM error status and captured addresses.

Reads ECCRAM1STS, ECCRAM2STS, ECCRAM1ECAD, and ECCRAM2ECAD into a plain struct. No registers are modified – call `rx_eccram_clear_errors()` to acknowledge faults.

Parameters

out	out	Pointer to status struct populated on success.
-----	-----	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Status read successfully.
<code>k_rx_err_null_ptr</code>	out is nullptr.

Precondition

[rx_eccram_init\(\)](#) has been called.
out is a valid writable pointer.

Postcondition

out fields reflect the register state at call time.
No hardware state is modified by this call.

Note

Thread Safety: Safe for concurrent readers (pure register reads).

Since

Version 1.0.0

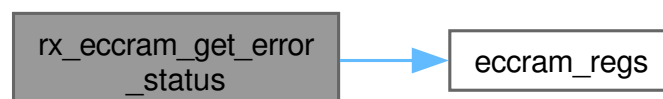
Definition at line 416 of file [rx_eccram.c](#).

```

00417 {
00418     RX_CHECK_NULL_PTR(out, s_tag, "out is nullptr");
00419
00420     volatile rx_eccram_regs_t* regs = eccram_regs();
00421
00422     const uint8_t s1 = regs->eccram1sts;
00423     const uint8_t s2 = regs->eccram2sts;
00424
00425     out->one_bit_error = (bool)((s1 & k_rx_eccram1sts_ecc1err_mask) != 0U);
00426     out->two_bit_error = (bool)((s2 & k_rx_eccram2sts_ecc2err_mask) != 0U);
00427     out->one_bit_addr = (uintptr_t)regs->eccram1ecad;
00428     out->two_bit_addr = (uintptr_t)regs->eccram2ecad;
00429
00430     return k_rx_ok;
00431 }
```

References [rx_eccram_regs_t::eccram1ecad](#), [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2ecad](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [k_rx_eccram1sts_ecc1err_mask](#), [k_rx_eccram2sts_ecc2err_mask](#), [rx_eccram_status_t::one_bit_addr](#), [rx_eccram_status_t::one_bit_error](#), [s_tag](#), [rx_eccram_status_t::two_bit_addr](#), and [rx_eccram_status_t::two_bit_error](#).

Here is the call graph for this function:



8.115.3.9 rx`eccram`init()

```
rx_err_t rx_eccram_init (
    rx_eccram_mode_t mode) [nodiscard]
```

Initialise ECCRAM in the requested operating mode.

Performs the full initialisation sequence required by the RX72N HW manual:

1. Clear MSTPCRC.MSTPC6 under system PRCR unlock so that the ECCRAM control block is clocked (Manual section 60.4.1, page 2994).
2. Unlock ECCRAMPRCR and set ECCRAMMODE = k_rx_eccrammode_ecc_no_check so ECC bits are generated but nothing is checked.
3. Zero all 32 KB of ECCRAM using aligned 32-bit stores (Manual section 60.3.3: uninitialised ECC syndromes would falsely flag).
4. Switch ECCRAMMODE to the mode that backs the requested mode argument and re-lock ECCRAMPRCR.
5. For any mode other than k_eccram_mode_disabled, enable the 1-bit status-update latch (ECCRAM1STSEN).

Parameters

in	mode	Desired operating mode (see rx_eccram_mode_t)
----	------	---

Returns

rx_err_t Error code

Return values

k_rx_ok	Initialisation successful.
k_rx_err_invalid_arg	mode is outside rx_eccram_mode_t range.
k_rx_err_hw_init_failed	Hardware mode read-back did not match.

Precondition

System clock is configured before this call.

Interrupts are globally disabled OR this function is called before any task can generate ECCRAM traffic.

Postcondition

MSTPCRC.MSTPC6 == 0 (ECCRAM control block clocked).

All 32 KB of ECCRAM are zeroed with matching ECC syndromes.

ECCRAMMODE reflects mode.

Note

Thread Safety: Call once from `hardware_init()`. Not reentrant.

Warning

This function must run before any data is placed in the `.eccram` linker section – it wipes the region.

NASA Power of 10 Compliance:

- Rule 2: Zero-fill loop bounded by `k_rx_eccram_region_size_bytes / 4`.
- Rule 5: 2 preconditions, 3 postconditions.
- Rule 7: Every internal helper's return value is checked.

See also

[rx_eccram_register_error_isr\(\)](#)
[rx_eccram_get_error_status\(\)](#)

Since

Version 1.0.0

Definition at line 368 of file `rx_eccram.c`.

```

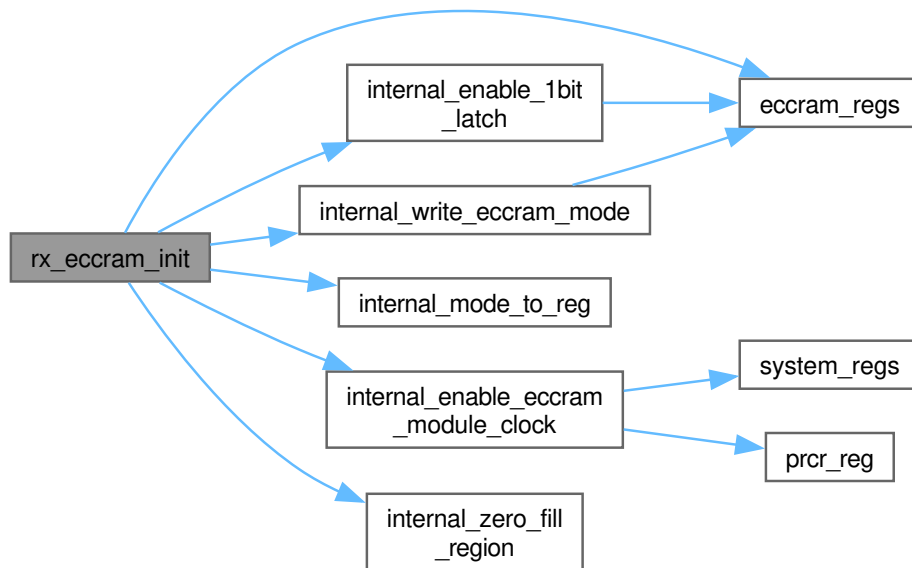
00369 {
00370     if (mode > k_eccram_mode_detect_only) {
00371         rx_log_error(s_tag, "invalid mode");
00372         return k_rx_err_invalid_arg;
00373     }
00374
00375     s_initialized = false;
00376     s_on_1bit     = NULL;
00377     s_on_2bit     = NULL;
00378     s_isr_ctx     = NULL;
00379     s_last_1bit_addr = 0;
00380     s_last_2bit_addr = 0;
00381
00382     internal_enable_eccram_module_clock();
00383
00384     internal_write_eccram_mode(k_rx_eccrammode_ecc_no_check);
00385
00386     internal_zero_fill_region();
00387
00388     const uint8_t final_mode_reg = internal_mode_to_reg(mode);
00389     internal_write_eccram_mode(final_mode_reg);
00390
00391     volatile rx_eccram_regs_t* regs = eccram_regs();
00392     const uint8_t actual = regs->eccrammode & k_rx_eccrammode_rammod_mask;
00393     /* HW-only failure: a stuck ECCRAMMODE bit can only happen on real silicon
00394      * (e.g. ECCRAMPRCR did not unlock, or the control block lost clock). The
00395      * host-side mock memory always reflects writes verbatim, so the true branch
00396      * is unreachable in unit tests. The behavior is still documented in the
00397      * header (@retval k_rx_err_hw_init_failed) and validated on hardware via
00398      * post-bringup smoke tests. */
00399     if (actual != (final_mode_reg & k_rx_eccrammode_rammod_mask)) { /* GCOVR_EXCL_BR_LINE */
00400         rx_log_error(s_tag, "ECCRAMMODE read-back mismatch"); /* GCOVR_EXCL_LINE */
00401         return k_rx_err_hw_init_failed; /* GCOVR_EXCL_LINE */
00402     }
00403
00404     regs->eccram1sts = k_rx_eccram1sts_clear;
00405     regs->eccram2sts = k_rx_eccram2sts_clear;
00406
00407     if (mode != k_eccram_mode_disabled) {
00408         internal_enable_1bit_latch();
00409     }
00410
00411     s_initialized = true;
00412     rx_log_info(s_tag, "ECCRAM initialised");

```

```
00413  return k_rx_ok;
00414 }
```

References [rx_eccram_regs_t::eccram1sts](#), [rx_eccram_regs_t::eccram2sts](#), [eccram_regs\(\)](#), [rx_eccram_regs_t::eccrammode](#), [internal_enable_1bit_latch\(\)](#), [internal_enable_eccram_module_clock\(\)](#), [internal_mode_to_reg\(\)](#), [internal_write_eccram_mode\(\)](#), [internal_zero_fill_region\(\)](#), [k_eccram_mode_detect_only](#), [k_eccram_mode_disabled](#), [k_rx_eccram1sts_clear](#), [k_rx_eccram2sts_clear](#), [k_rx_eccrammode_ecc_no_check](#), [k_rx_eccrammode_rammod_ma](#), [s_initialized](#), [s_isr_ctx](#), [s_last_1bit_addr](#), [s_last_2bit_addr](#), [s_on_1bit](#), [s_on_2bit](#), and [s_tag](#).

Here is the call graph for this function:



8.115.3.10 rx`eccram`ram`error`isr()

```
void rx_eccram_ram_error_isr (
    void )
```

Vector-18 ISR entry (RAMERR) – ECCRAM 1-bit/2-bit error.

The GNURX toolchain's vector table dispatches to this symbol; the interrupt attribute generates the correct prologue/epilogue.

Precondition

Vector 18 is enabled and wired to this symbol by the boot code.

Postcondition

[internal_ram_error_isr_body\(\)](#) has run to completion.

See also

[internal_ram_error_isr_body\(\)](#)

RX72N HW Manual Chapter 15 (ICU), RAMERR vector entry

Since

Version 1.0.0

Definition at line 351 of file [rx_eccram.c](#).

```
00352 {
00353     internal_ram_error_isr_body();
00354 }
```

References [internal_ram_error_isr_body\(\)](#).

Here is the call graph for this function:



8.115.3.11 rx_eccram_region_end()

```
uintptr_t rx_eccram_region_end (
    void )
```

Return the inclusive end address of the ECCRAM data region.

Returns

Physical address 0x00FFFFFF as a `uintptr_t`.

Precondition

None.

Postcondition

Return value equals `k_rx_eccram_region_end_addr`.

Return value - [rx_eccram_region_start\(\)](#) + 1 equals 32 KB.

Note

Thread Safety: Returns a compile-time constant; always safe.

See also

[rx_eccram_region_start\(\)](#)

RX72N HW Manual Chapter 60, Table 60.1, page 2977

Since

Version 1.0.0

Definition at line 465 of file [rx_eccram.c](#).

```
00466 {
00467     return k_rx_eccram_region_end_addr;
00468 }
```

References [k_rx_eccram_region_end_addr](#).

8.115.3.12 rx_eccram_region_start()

```
uintptr_t rx_eccram_region_start (  
    void )
```

Return the inclusive start address of the ECCRAM data region.

Exposes the hardware-defined base of the ECC-protected region so that linker scripts and runtime allocators can place objects inside it without hardcoding the literal address.

Returns

Physical address 0x00FF8000 as a `uintptr_t`.

Precondition

None.

Postcondition

Return value equals `k_rx_eccram_region_base_addr`.

Return value is word-aligned.

Note

Thread Safety: Returns a compile-time constant; always safe.

See also

[rx_eccram_region_end\(\)](#)

RX72N HW Manual Chapter 60, Table 60.1, page 2977

Since

Version 1.0.0

Definition at line 460 of file [rx_eccram.c](#).

```
00461 {  
00462     return k_rx_eccram_region_base_addr;  
00463 }
```

References [k_rx_eccram_region_base_addr](#).

8.115.3.13 `rx_eccram_register_error_isr()`

```
rx_err_t rx_eccram_register_error_isr (
    rx_eccram_on_1bit_fn_t on_1bit,
    rx_eccram_on_2bit_fn_t on_2bit,
    void * ctx) [nodiscard]
```

Register user callbacks for ECC error ISRs.

Installs 1-bit and/or 2-bit error callbacks that the driver's ISR trampoline will invoke on the respective RAM-error vectors. Passing nullptr for either callback leaves the previously installed handler in place, allowing callers to register them independently.

Dependency-Injection Rationale (NASA Rule 9 deviation)

Function pointers are permitted here because they are the exact pattern STAR uses to allow test mocks and application-level policy to substitute in without modifying the driver. See CLAUDE.md "SOLID Principles for C / Dependency Inversion".

Parameters

in	on_1bit	Handler invoked from ISR on 1-bit error (may be nullptr).
in	on_2bit	Handler invoked from ISR on 2-bit error (may be nullptr).
in	ctx	Opaque pointer passed through to the handlers.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Handlers installed.
<code>k_rx_err_invalid_state</code>	rx_eccram_init() has not yet been called.

Precondition

[rx_eccram_init\(\)](#) has been called.

Handlers, if non-null, are safe to call from interrupt context.

Postcondition

The driver's ISR trampolines dispatch to the supplied handlers.

ctx is retained until this function is called again.

Note

Thread Safety: Install once during init before interrupts are enabled, or guard calls at the task level.

Since

Version 1.0.0

Definition at line 441 of file [rx_eccram.c](#).

```
00444 {
00445     if (!s_initialized) {
00446         rx_log_error(s_tag, "not initialised");
00447         return k_rx_err_invalid_state;
00448     }
00449
00450     if (on_1bit != NULL) {
00451         s_on_1bit = on_1bit;
00452     }
00453     if (on_2bit != NULL) {
00454         s_on_2bit = on_2bit;
00455     }
00456     s_isr_ctx = ctx;
00457     return k_rx_ok;
00458 }
```

References [s_initialized](#), [s_isr_ctx](#), [s_on_1bit](#), [s_on_2bit](#), and [s_tag](#).

8.115.4 Variable Documentation

8.115.4.1 s`initialized

```
bool s_initialized = false    [static]
```

Driver initialised flag.

Definition at line 110 of file [rx_eccram.c](#).

8.115.4.2 s`isr`ctx

```
void* s_isr_ctx = NULL    [static]
```

Opaque context passed to registered callbacks.

Definition at line 119 of file [rx_eccram.c](#).

Referenced by [internal_ram_error_isr_body\(\)](#), [rx_eccram_init\(\)](#), and [rx_eccram_register_error_isr\(\)](#).

8.115.4.3 s`last`1bit`addr

```
volatile uintptr_t s_last_1bit_addr = 0    [static]
```

Latched address of most recent 1-bit error (captured in ISR).

Definition at line 125 of file [rx_eccram.c](#).

Referenced by [internal_ram_error_isr_body\(\)](#), and [rx_eccram_init\(\)](#).

8.115.4.4 s'last'2bit'addr

```
volatile uintptr_t s_last_2bit_addr = 0 [static]
```

Latched address of most recent 2-bit error (captured in ISR).

Definition at line 122 of file [rx_eccram.c](#).

Referenced by [internal_ram_error_isr_body\(\)](#), and [rx_eccram_init\(\)](#).

8.115.4.5 s'on'1bit

```
rx\_eccram\_on\_1bit\_fn\_t s_on_1bit = NULL [static]
```

Registered 1-bit error callback (may be nullptr).

Definition at line 113 of file [rx_eccram.c](#).

Referenced by [internal_ram_error_isr_body\(\)](#), [rx_eccram_init\(\)](#), and [rx_eccram_register_error_isr\(\)](#).

8.115.4.6 s'on'2bit

```
rx\_eccram\_on\_2bit\_fn\_t s_on_2bit = NULL [static]
```

Registered 2-bit error callback (may be nullptr).

Definition at line 116 of file [rx_eccram.c](#).

Referenced by [internal_ram_error_isr_body\(\)](#), [rx_eccram_init\(\)](#), and [rx_eccram_register_error_isr\(\)](#).

8.115.4.7 s'tag

```
const char* const s_tag = "ECCRAM" [static]
```

Module log tag for [rx_log_*](#) macros.

Definition at line 61 of file [rx_eccram.c](#).

8.116 rx_eccram.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_eccram.c
00003  * @brief ECC-Protected RAM (ECCRAM) Driver Implementation
00004  *
00005  * @details
00006  * Implements the rx_eccram.h API: module-stop clear, mode configuration
00007  * under PRCR unlock, region zero-initialisation with 32-bit stores,
00008  * error-status readout, flag clearing, ISR trampoline, and region
00009  * introspection.
00010  *
00011  * The RX72N has a single RAMERR vector (vector 18, RAM group) that
00012  * fires for both 1-bit and 2-bit ECCRAM errors. The ISR trampoline
00013  * inspects ECCRAM1STS and ECCRAM2STS to decide which registered
00014  * handler (if any) to invoke, captures the failing address(es), and
00015  * clears the latched status flags.
00016  *
00017  * @par Register Protection Sequencing
00018  * Two independent protection registers are involved:
00019  * - System PRCR (0x000803FE) protects MSTPCRC (Manual Ch11). Unlock
00020  *   with k_rx_prcr_unlock_prc1, write MSTPCRC, re-lock with
00021  *   k_rx_prcr_lock.
00022  * - ECCRAMPRCR (0x000812C4) protects ECCRAMMODE and ECCRAM1STSEN
00023  *   (Manual section 60.2.13). Unlock with k_rx_eccramprcr_unlock
00024  *   (0xF1), write the guarded register, re-lock with
00025  *   k_rx_eccramprcr_lock (0xF0).
00026  *
00027  * @author Locked, Inc.
00028  * @date 2026-04-21
00029  * @copyright Copyright (c) 2026 Locked Inc.
00030  * SPDX-License-Identifier: MIT
00031  * @since Version 1.0.0
00032  */
00033
00034 #include "rx_eccram.h"
00035
00036 #include <stddef.h>
00037 #include <stdint.h>
00038
00039 #ifdef UNIT_TEST
00040 #include "mock_rx72n_eccram_regs.h"
00041 #include "mock_rx72n_system_regs.h"
00042 #else
00043 #include "rx72n_eccram_regs.h"
00044 #include "rx72n_system_regs.h"
00045 #endif
00046
00047 #include "rx_check.h"
00048 #include "rx_err.h"
00049 #include "rx_log.h"
00050 #include "rx_register_protection.h"
00051
00052 /**
00053  * Constants
00054  */
00055
00056
00057 /**
00058  * @var s_tag
00059  * @brief Module log tag for rx_log_* macros
00060  */
00061 static const char* const s_tag = "ECCRAM";
00062
00063 /**
00064  * @enum rx_eccram_fill_t
00065  * @brief Zero-initialisation fill pattern and word size
00066  *
00067  * @details
00068  * The ECC syndrome is computed per 32-bit word; filling with zero is
00069  * the cheapest pattern that produces a well-defined, check-valid state.
00070  *
00071  * @since Version 1.0.0
00072  */
00073 typedef enum : uint32_t {
00074     /** @brief Fill pattern used by the zero-initialisation loop */
00075     k_rx_eccram_fill_pattern = 0x00000000U,
00076 } rx_eccram_fill_t;
00077
00078 /**
00079  * @enum rx_eccram_word_size_t
00080  * @brief Width of a single ECC-protected word in bytes

```

```

00081 *
00082 * @details
00083 * Declared as a typed enum so it can bound the Rule-2 static loop
00084 * upper in rx_eccram_init() without introducing a magic number.
00085 *
00086 * @since Version 1.0.0
00087 */
00088 typedef enum : uint32_t {
00089     /** @brief 4 bytes per ECC word (uint32_t) */
00090     k_rx_eccram_word_bytes = 4,
00091 } rx_eccram_word_size_t;
00092
00093 /**
00094  * @enum rx_eccram_word_count_t
00095  * @brief Number of 32-bit words in the ECCRAM region
00096  *
00097  * @since Version 1.0.0
00098  */
00099 typedef enum : uint32_t {
00100     /** @brief 32 KB / 4 bytes per word = 8192 words */
00101     k_rx_eccram_word_count = (k_rx_eccram_region_size_bytes / k_rx_eccram_word_bytes),
00102 } rx_eccram_word_count_t;
00103
00104 /*
=====
00105 * Static State
00106 *
=====
00107 */
00108
00109 /** @brief Driver initialised flag */
00110 static bool s_initialized = false;
00111
00112 /** @brief Registered 1-bit error callback (may be nullptr) */
00113 static rx_eccram_on_1bit_fn_t s_on_1bit = NULL;
00114
00115 /** @brief Registered 2-bit error callback (may be nullptr) */
00116 static rx_eccram_on_2bit_fn_t s_on_2bit = NULL;
00117
00118 /** @brief Opaque context passed to registered callbacks */
00119 static void* s_isr_ctx = NULL;
00120
00121 /** @brief Latched address of most recent 2-bit error (captured in ISR) */
00122 static volatile uintptr_t s_last_2bit_addr = 0;
00123
00124 /** @brief Latched address of most recent 1-bit error (captured in ISR) */
00125 static volatile uintptr_t s_last_1bit_addr = 0;
00126
00127 /*
=====
00128 * Internal Helpers
00129 *
=====
00130 */
00131
00132 /**
00133  * @brief Translate public mode enum into ECCRAMMODE register value
00134  *
00135  *
00136  *
00137  * @pre @p mode has been range-checked by the caller.
00138  * @post Return value is one of the k_rx_eccrammode_* constants.
00139  *
00140  * @note All error-visible modes share the same hardware encoding
00141  *       (RAMMOD=11b). The distinction between correct_only,
00142  *       correct_and_detect, and detect_only is enforced by which ISRs
00143  *       the driver registers (not by the hardware bits).
00144  *
00145  * @since Version 1.0.0
00146  */
00147 static uint8_t internal_mode_to_reg(rx_eccram_mode_t mode)
00148 {
00149     if (mode == k_eccram_mode_disabled) {
00150         return k_rx_eccrammode_ecc_disabled;
00151     }
00152     return k_rx_eccrammode_ecc_with_check;
00153 }
00154
00155 /**
00156  * @brief Clear MSTPCRC.MSTPC6 so ECCRAM control block is clocked
00157  *
00158  * @details
00159  * Uses the standard STAR unlock/write/lock sequence on the system PRCR.
00160  *
00161  * @pre System clock configured.
00162  * @pre Caller holds exclusive access to MSTPCRC (single-threaded init).
00163  * @post MSTPCRC.MSTPC6 == 0.

```

```

00164 * @post System PRCR is re-locked on return.
00165 *
00166 * @see RX72N HW Manual Chapter 60, section 60.4.1 (MSTPCRC), page 2994
00167 * @since Version 1.0.0
00168 */
00169 static void internal_enable_eccram_module_clock(void)
00170 {
00171     *prcr_reg() = k_rx_prcr_unlock_prc1;
00172     system_regs()->mstpcrc &= ~(k_rx_eccram_mstpcrc_bit_mask);
00173     *prcr_reg() = k_rx_prcr_lock;
00174 }
00175
00176 /**
00177 * @brief Write ECCRAMMODE under ECCRAMPRCR unlock
00178 *
00179 *
00180 * @pre ECCRAM module clocked (MSTPCRC.MSTPC6 == 0).
00181 * @pre @p mode_reg_value is a valid k_rx_eccrammode_* constant.
00182 * @post ECCRAMMODE reflects @p mode_reg_value.
00183 * @post ECCRAMPRCR is re-locked on return.
00184 *
00185 * @see RX72N HW Manual Chapter 60, section 60.2.13, page 2985
00186 * @since Version 1.0.0
00187 */
00188 static void internal_write_eccram_mode(uint8_t mode_reg_value)
00189 {
00190     volatile rx_eccram_regs_t* regs = eccram_regs();
00191     regs->eccramprcr = k_rx_eccramprcr_unlock;
00192     regs->eccrammode = mode_reg_value;
00193     regs->eccramprcr = k_rx_eccramprcr_lock;
00194 }
00195
00196 /**
00197 * @brief Enable the 1-bit status-update latch
00198 *
00199 * @details
00200 * Sets ECCRAM1STSEN=1 so that 1-bit errors are latched into ECCRAM1STS
00201 * and ECCRAM1ECAD. Skipped for k_eccram_mode_disabled. Guarded by
00202 * ECCRAMPRCR (shared with ECCRAMMODE).
00203 *
00204 * @pre ECCRAM module clocked.
00205 * @post ECCRAM1STSEN.ECC1STSEN == 1.
00206 * @post ECCRAMPRCR is re-locked on return.
00207 *
00208 * @since Version 1.0.0
00209 */
00210 static void internal_enable_1bit_latch(void)
00211 {
00212     volatile rx_eccram_regs_t* regs = eccram_regs();
00213     regs->eccramprcr = k_rx_eccramprcr_unlock;
00214     regs->eccram1stsen = k_rx_eccram1stsen_enable;
00215     regs->eccramprcr = k_rx_eccramprcr_lock;
00216 }
00217
00218 /**
00219 * @brief Zero-initialise the ECCRAM region with 32-bit stores
00220 *
00221 * @details
00222 * Iterates over every 32-bit word in ECCRAM and writes zero so that
00223 * every ECC syndrome matches. The loop upper bound is a compile-time
00224 * constant (k_rx_eccram_word_count), which satisfies NASA Rule 2.
00225 *
00226 * @pre ECCRAMMODE == k_rx_eccrammode_ecc_no_check (so writes generate
00227 * syndromes but reads do not trigger spurious errors).
00228 * @post Every 32-bit word of ECCRAM holds k_rx_eccram_fill_pattern.
00229 *
00230 * @note Runs in ~65 k cycles on RX72N at 240 MHz (~270 us).
00231 * @since Version 1.0.0
00232 */
00233 #ifdef UNIT_TEST
00234 /**
00235 * @brief Host-side stand-in for the ECCRAM region used by tests
00236 *
00237 * @details
00238 * The production zero-fill loop targets the hardware address
00239 * k_rx_eccram_region_base_addr (0x00FF8000). That address is not mapped on
00240 * the host under UNIT_TEST, so writes are redirected to this ordinary
00241 * static buffer, which has exactly the same element count as the real
00242 * ECCRAM region. Tests observe the effect of rx_eccram_init() by reading
00243 * this array.
00244 */
00245 uint32_t g_mock_eccram_region[k_rx_eccram_word_count];
00246
00247 /**
00248 * @brief Reset all driver static state (UNIT_TEST-only helper)
00249 *
00250 * @details

```

```

00251 * The driver holds static module state (s_initialized, callbacks, last
00252 * failing addresses). That state persists across individual Unity tests
00253 * because the module is linked once per test binary. This helper returns
00254 * the driver to its pre-rx_eccram_init() condition so each test starts
00255 * from a clean slate. Not exposed outside UNIT_TEST builds.
00256 *
00257 * @post s_initialized == false
00258 * @post All callback pointers == NULL
00259 * @post All latched failing addresses == 0
00260 */
00261 void rx_eccram_test_reset_state(void)
00262 {
00263     s_initialized = false;
00264     s_on_1bit     = NULL;
00265     s_on_2bit     = NULL;
00266     s_isr_ctx     = NULL;
00267     s_last_1bit_addr = 0;
00268     s_last_2bit_addr = 0;
00269 }
00270 #endif
00271
00272 static void internal_zero_fill_region(void)
00273 {
00274     #ifdef UNIT_TEST
00275     for (uint32_t i = 0; i < k_rx_eccram_word_count; i++) {
00276         g_mock_eccram_region[i] = k_rx_eccram_fill_pattern;
00277     }
00278     #else
00279     volatile uint32_t* dst = (volatile uint32_t*)k_rx_eccram_region_base_addr;
00280     for (uint32_t i = 0; i < k_rx_eccram_word_count; i++) {
00281         dst[i] = k_rx_eccram_fill_pattern;
00282     }
00283     #endif
00284 }
00285
00286 /*
=====
00287 * ISR Trampoline -- Vector 18 (RAMERR)
00288 *
=====
00289 */
00290
00291 /**
00292 * @brief Shared RAM error handler core
00293 *
00294 * @details
00295 * Called from the `__attribute__((interrupt))` wrapper below. Reads
00296 * both status flags, captures the corresponding address registers,
00297 * dispatches to registered user callbacks, and clears the flags so
00298 * the next error can be detected.
00299 *
00300 * The 2-bit branch captures ECCRAM2ECAD into s_last_2bit_addr *before*
00301 * the flag-clearing write, matching the prompt requirement that the
00302 * failing address be saved first.
00303 *
00304 * @pre Called from IRQ context on vector 18 (RAMERR).
00305 * @post ECCRAM1STS and ECCRAM2STS flags cleared.
00306 * @post s_last_1bit_addr / s_last_2bit_addr updated on match.
00307 *
00308 * @since Version 1.0.0
00309 */
00310 static void internal_ram_error_isr_body(void)
00311 {
00312     volatile rx_eccram_regs_t* regs = eccram_regs();
00313
00314     const uint8_t status2 = regs->eccram2sts;
00315     if ((status2 & k_rx_eccram2sts_ecc2err_mask) != 0U) {
00316         const uintptr_t addr2 = (uintptr_t)regs->eccram2ecad;
00317         s_last_2bit_addr = addr2;
00318         regs->eccram2sts = k_rx_eccram2sts_clear;
00319         if (s_on_2bit != NULL) {
00320             s_on_2bit(addr2, s_isr_ctx);
00321         }
00322     }
00323
00324     const uint8_t status1 = regs->eccram1sts;
00325     if ((status1 & k_rx_eccram1sts_ecc1err_mask) != 0U) {
00326         const uintptr_t addr1 = (uintptr_t)regs->eccram1ecad;
00327         s_last_1bit_addr = addr1;
00328         regs->eccram1sts = k_rx_eccram1sts_clear;
00329         if (s_on_1bit != NULL) {
00330             s_on_1bit(addr1, s_isr_ctx);
00331         }
00332     }
00333 }
00334
00335 /**

```

```

00336 * @brief Vector-18 ISR entry (RAMERR) -- ECCRAM 1-bit/2-bit error
00337 *
00338 * @details
00339 * The GNURX toolchain's vector table dispatches to this symbol; the
00340 * `interrupt' attribute generates the correct prologue/epilogue.
00341 *
00342 * @pre Vector 18 is enabled and wired to this symbol by the boot code.
00343 * @post internal_ram_error_isr_body() has run to completion.
00344 *
00345 * @see internal_ram_error_isr_body()
00346 * @see RX72N HW Manual Chapter 15 (ICU), RAMERR vector entry
00347 * @since Version 1.0.0
00348 */
00349 #ifndef UNIT_TEST
00350 void __attribute__((interrupt)) rx_eccram_ram_error_isr(void);
00351 void __attribute__((interrupt)) rx_eccram_ram_error_isr(void)
00352 {
00353     internal_ram_error_isr_body();
00354 }
00355 #else
00356 void rx_eccram_ram_error_isr(void);
00357 void rx_eccram_ram_error_isr(void)
00358 {
00359     internal_ram_error_isr_body();
00360 }
00361 #endif
00362
00363 /*
=====
00364 * Public API
00365 *
=====
00366 */
00367
00368 rx_err_t rx_eccram_init(rx_eccram_mode_t mode)
00369 {
00370     if (mode > k_eccram_mode_detect_only) {
00371         rx_log_error(s_tag, "invalid mode");
00372         return k_rx_err_invalid_arg;
00373     }
00374
00375     s_initialized = false;
00376     s_on_1bit = NULL;
00377     s_on_2bit = NULL;
00378     s_isr_ctx = NULL;
00379     s_last_1bit_addr = 0;
00380     s_last_2bit_addr = 0;
00381
00382     internal_enable_eccram_module_clock();
00383
00384     internal_write_eccram_mode(k_rx_eccrammode_ecc_no_check);
00385
00386     internal_zero_fill_region();
00387
00388     const uint8_t final_mode_reg = internal_mode_to_reg(mode);
00389     internal_write_eccram_mode(final_mode_reg);
00390
00391     volatile rx_eccram_regs_t* regs = eccram_regs();
00392     const uint8_t actual = regs->eccrammode & k_rx_eccrammode_rammod_mask;
00393     /* HW-only failure: a stuck ECCRAMMODE bit can only happen on real silicon
00394      * (e.g. ECCRAMPCR did not unlock, or the control block lost clock). The
00395      * host-side mock memory always reflects writes verbatim, so the true branch
00396      * is unreachable in unit tests. The behavior is still documented in the
00397      * header (@retval k_rx_err_hw_init_failed) and validated on hardware via
00398      * post-bringup smoke tests. */
00399     if (actual != (final_mode_reg & k_rx_eccrammode_rammod_mask)) { /* GCOVR_EXCL_BR_LINE */
00400         rx_log_error(s_tag, "ECCRAMMODE read-back mismatch"); /* GCOVR_EXCL_LINE */
00401         return k_rx_err_hw_init_failed; /* GCOVR_EXCL_LINE */
00402     }
00403
00404     regs->eccram1sts = k_rx_eccram1sts_clear;
00405     regs->eccram2sts = k_rx_eccram2sts_clear;
00406
00407     if (mode != k_eccram_mode_disabled) {
00408         internal_enable_1bit_latch();
00409     }
00410
00411     s_initialized = true;
00412     rx_log_info(s_tag, "ECCRAM initialised");
00413     return k_rx_ok;
00414 }
00415
00416 rx_err_t rx_eccram_get_error_status(rx_eccram_status_t* out)
00417 {
00418     RX_CHECK_NULL_PTR(out, s_tag, "out is nullptr");
00419
00420     volatile rx_eccram_regs_t* regs = eccram_regs();

```

```

00421
00422  const uint8_t s1 = regs->eccram1sts;
00423  const uint8_t s2 = regs->eccram2sts;
00424
00425  out->one_bit_error = (bool)((s1 & k_rx_eccram1sts_ecc1err_mask) != 0U);
00426  out->two_bit_error = (bool)((s2 & k_rx_eccram2sts_ecc2err_mask) != 0U);
00427  out->one_bit_addr = (uintptr_t)regs->eccram1ecad;
00428  out->two_bit_addr = (uintptr_t)regs->eccram2ecad;
00429
00430  return k_rx_ok;
00431 }
00432
00433 rx_err_t rx_eccram_clear_errors(void)
00434 {
00435  volatile rx_eccram_regs_t* regs = eccram_regs();
00436  regs->eccram1sts = k_rx_eccram1sts_clear;
00437  regs->eccram2sts = k_rx_eccram2sts_clear;
00438  return k_rx_ok;
00439 }
00440
00441 rx_err_t rx_eccram_register_error_isr(rx_eccram_on_1bit_fn_t on_1bit,
00442                                     rx_eccram_on_2bit_fn_t on_2bit,
00443                                     void* ctx)
00444 {
00445  if (!s_initialized) {
00446    rx_log_error(s_tag, "not initialised");
00447    return k_rx_err_invalid_state;
00448  }
00449
00450  if (on_1bit != NULL) {
00451    s_on_1bit = on_1bit;
00452  }
00453  if (on_2bit != NULL) {
00454    s_on_2bit = on_2bit;
00455  }
00456  s_isr_ctx = ctx;
00457  return k_rx_ok;
00458 }
00459
00460 uintptr_t rx_eccram_region_start(void)
00461 {
00462  return k_rx_eccram_region_base_addr;
00463 }
00464
00465 uintptr_t rx_eccram_region_end(void)
00466 {
00467  return k_rx_eccram_region_end_addr;
00468 }

```

8.117 libs/rx/hal/src/rx_gptw.c File Reference

GPTW PWM Driver Implementation for Motor Control.

```

#include "rx_gptw.h"
#include <stddef.h>
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_port_utils.h"
#include "rx_register_protection.h"
Include dependency graph for rx_gptw.c:

```



Enumerations

- enum `gptw_constants_t` : `uint8_t` { `k_gptw_max_channels` = 4 , `k_gptw_outputs_per_channel` = 2 }
- GPTW general constants.

- enum `gptw_period_constants_t` : `uint16_t` {
`k_gptw_period_divisor_saw` = 1 , `k_gptw_period_divisor_tri` = 2 , `k_gptw_period_min` = 10
, `k_gptw_period_zero` = 0 ,
`k_gptw_deadtime_disabled` = 0 }
- enum `gptw_duty_constants_t` : `uint8_t` { `k_gptw_duty_min` = 0 , `k_gptw_duty_max` = 100 ,
`k_gptw_duty_divisor` = 100 }
Duty cycle calculation constants.
- enum `mpc_pwpr_constants_t` : `uint8_t` { `k_mpc_pwpr_b0wi_clear` = 0x00 , `k_mpc_pwpr_pfswe_set`
= 0x40 , `k_mpc_pwpr_lock` = 0x80 }
- enum `mpc_pfs_offset_constants_t` : `uint16_t` { `k_mpc_pfs_base_offset` = 0x40 , `k_mpc_port_e_index`
= 0x0E , `k_mpc_pfs_port_stride` = 8 }
- enum `gptw_bit_constants_t` : `uint8_t` { `k_gptw_bit_set` = 1 }
- enum `gptw_channel_mask_t` : `uint8_t` { `k_gptw_all_channels_mask` = 0x0F }
- enum `gptw_direction_constants_t` : `uint32_t` { `k_gptw_gtuddtyc_ud_down` = 0 }
- enum `porte_gptw_pin_t` : `uint8_t` {
`k_porte_gptw0_a` = 5 , `k_porte_gptw0_b` = 2 , `k_porte_gptw1_a` = 4 , `k_porte_gptw1_b` =
1 ,
`k_porte_gptw2_a` = 3 , `k_porte_gptw2_b` = 0 , `k_porte_gptw3_a` = 7 , `k_porte_gptw3_b` =
6 }
- enum `pfs_block_layout_t` : `uintptr_t` { `k_pfs_block_base_addr` = 0x0008C140U , `k_pfs_port_stride`
= 8U }
- enum `gptw_phase_divisor_t` : `uint8_t` { `k_phase_divisor_none` = 1 , `k_phase_divisor_quarter`
= 4 , `k_phase_divisor_half` = 2 , `k_phase_divisor_three_quarter` = 4 }

Functions

- static volatile `rx_gptw_channel_regs_t` * `internal_get_gptw_base` (const `rx_gptw_channel_t`
channel)
Get GPTW channel register base address.
- static bool `internal_is_triangle_mode` (const `rx_gptw_wave_mode_t` mode)
Check if wave mode is triangle (center-aligned).
- static `uint32_t` `internal_get_gtcr_mode` (const `rx_gptw_wave_mode_t` mode)
Get GTCR mode bits for wave mode.
- static `rx_err_t` `internal_calculate_period` (const `uint32_t` frequency_hz, const `rx_gptw_wave_mode_t`
wave_mode, `uint32_t` *period)
Calculate period register value from frequency.
- static volatile `uint8_t` * `internal_pfs_addr` (`uint8_t` port_idx, `uint8_t` bit)
- static `rx_err_t` `internal_configure_mpc` (const `rx_gptw_config_t` *config)
- static void `internal_configure_port_pins` (const `rx_gptw_config_t` *config)
Configure Port E pins as peripheral outputs.
- static void `internal_enable_gptw_module_clock` (void)
Enable GPTW module clock.
- static `rx_err_t` `internal_configure_gptw_hardware` (volatile `rx_gptw_channel_regs_t` *gptw,
const `rx_gptw_config_t` *config, const `uint32_t` period)

- Configure GPTW hardware registers.
- static rx_err_t [internal_prepare_gptw_pwm_init](#) (const [rx_gptw_channel_t](#) channel, const [rx_gptw_config_t](#) *config, volatile [rx_gptw_channel_regs_t](#) **gptw_out, uint32_t *period_count, rx_err_t *out)
 - Validate channel and compute period for PWM initialization.
- rx_err_t [rx_gptw_init_pwm](#) (const [rx_gptw_channel_t](#) channel, const [rx_gptw_config_t](#) *config)
 - Initialize a single GPTW channel for PWM operation.
- rx_err_t [rx_gptw_set_duty](#) (const [rx_gptw_channel_id_t](#) channel, const [rx_gptw_output_id_t](#) output, const float duty_percent)
 - Set PWM duty cycle using floating-point percentage.
- rx_err_t [rx_gptw_set_duty_raw](#) (const [rx_gptw_channel_t](#) channel, [rx_gptw_output_t](#) output, uint32_t duty_count)
 - Set PWM duty cycle using a raw 32-bit timer count.
- rx_err_t [rx_gptw_get_duty](#) (const [rx_gptw_channel_t](#) channel, const [rx_gptw_output_t](#) output, float *duty_percent)
 - Read the current PWM duty cycle as a floating-point percentage.
- rx_err_t [rx_gptw_get_period](#) (const [rx_gptw_channel_t](#) channel, uint32_t *period_count)
 - Get PWM period count for GPTW channel.
- rx_err_t [rx_gptw_enable_output](#) (const [rx_gptw_channel_t](#) channel, const [rx_gptw_output_t](#) output, bool enable)
 - Enable or disable GPTW output pin.
- rx_err_t [rx_gptw_start](#) (const [rx_gptw_channel_t](#) channel)
 - Start GPTW counter (begin PWM generation).
- rx_err_t [rx_gptw_stop](#) (const [rx_gptw_channel_t](#) channel)
 - Stop GPTW counter (halt PWM generation).
- static void [internal_calculate_phase_offset](#) (const [rx_gptw_channel_t](#) channel, const uint32_t period, const bool is_triangle, uint32_t *init_count, bool *init_dir_up)
 - Calculate initial counter value and direction for phase staggering.
- static rx_err_t [internal_configure_channel_staggered](#) (const [rx_gptw_channel_t](#) channel, const [rx_gptw_config_t](#) *config, const uint32_t period, const bool is_triangle)
 - Configure a single GPTW channel for staggered PWM operation.
- rx_err_t [rx_gptw_init_all_staggered](#) (const [rx_gptw_config_t](#) *configs[k_rx_gptw_channel_count], rx_err_t *out)
 - Initialize all 4 GPTW channels with 90-degree phase staggering.
- rx_err_t [rx_gptw_deinit](#) (const [rx_gptw_channel_t](#) channel)
 - Deinitialize a GPTW channel.

Variables

- static const char *const [s_tag](#) = "GPTW"
- static const uint32_t [s_gptw_period_max](#) = 0xFFFFFFFFFUL
 - Period calculation constants.
- static const uint64_t [s_gptw_ns_per_second](#) = 1000000000ULL
- static bool [s_gptw_initialized](#) [k_gptw_max_channels] = {false}
 - Track initialized channels.
- static uint32_t [s_gptw_period](#) [k_gptw_max_channels] = {}
 - Period values for each channel.

8.117.1 Detailed Description

GPTW PWM Driver Implementation for Motor Control.

Production-grade implementation of the General PWM Timer (GPTW) driver for RX72N motor control applications. This file contains all internal helper functions and public API implementations for GPTW peripheral management.

8.117.1.1 Module Architecture

The implementation is structured in three layers:

1. Constants & Configuration: Typed enums for all magic numbers
2. Internal Helpers: Register access, calculations, hardware configuration
3. Public API: User-facing functions documented in [rx_gptw.h](#)

8.117.1.2 Supported PWM Modes

1. Sawtooth-wave PWM (edge-aligned):

- Counter: 0 -> GTPR -> 0 (reset at top)
- Period = PCLKA / frequency
- For 20kHz: period = 120MHz / 20kHz = 6000 counts (~12.5-bit resolution)
- Single update point at counter reset
- Best for: Simple PWM, LED dimming

2. Triangle-wave PWM (center-aligned):

- Counter: 0 -> GTPR -> 0 (up/down counting)
- Period = PCLKA / (2 x frequency)
- For 20kHz: period = 120MHz / 40kHz = 3000 counts (~11.5-bit resolution)
- Update at crest and/or trough (mode dependent)
- Lower EMI due to symmetric switching edges
- Reduced current ripple in motor windings
- Best for: Motor control with H-bridge

8.117.1.3 Implementation Notes

Register Access Pattern: All GPTW register modifications follow this sequence:

1. Unlock write protection (GTWP = 0xA500_0000)
2. Modify registers
3. Lock write protection (GTWP = 0x0000_0001)

Phase Staggering Implementation: For 4-motor systems, channels are initialized with counter offsets:

- Ch0: GTCNT = 0 (0deg reference)
- Ch1: GTCNT = period/4 (90deg offset)
- Ch2: GTCNT = period/2 (180deg offset)
- Ch3: GTCNT = 3xperiod/4 (270deg offset)

8.117.1.4 Memory Map

GPTW Register Base Addresses:

Module	Base Address	Size	Description
GPTW Common	0x000C2000	0x100	Shared start/stop registers
GPTW0	0x000C2100	0x80	Channel 0 registers
GPTW1	0x000C2180	0x80	Channel 1 registers
GPTW2	0x000C2200	0x80	Channel 2 registers
GPTW3	0x000C2280	0x80	Channel 3 registers

Module Clock:

- MSTPCRA.MSTPA7 controls GPTW module clock on RX72N
- Must be cleared (= 0) before register access
- Protected by PRCR register lock (PRC1, key 0xA502)

Clock Assumption:

This driver assumes PCKA = 96 MHz, which matches the STAR firmware clock_init: HOCO 16 MHz x12 PLL = 192 MHz, SCKCR=0x21C21211 giving PCKA = PLL/2 = 96 MHz. At 20 kHz PWM this yields GTPR = 96M/20k - 1 = 4799. If the board ever moves to a different clock plan the HAL must be updated (k_pclka_hz in [rx72n_clock.h](#)).

8.117.1.5 Static Variables

Variable	Type	Purpose
s_gptw_ initialized[]	bool [4]	Track which channels are initialized
s_gptw_ period[]	uint32_t [4]	Cache period values for duty calculation
s_tag	const char*	Log tag "GPTW"

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	 [OK]	No goto/setjmp/recursion in any function
2. Fixed loop bounds	 [OK]	Loops bounded by k_gptw_max_channels (4)
3. No dynamic allocation	 [OK]	All variables static or stack, zero malloc
4. Small functions	 [OK]	All functions < 60 lines
5. Assertions	 [OK]	RX_CHECK_NULL_PTR, switch validation, range checks
6. Narrow scope	 [OK]	Internal functions use static, minimal globals
7. Check return values	 [OK]	All rx_err_t returns propagated or handled
8. Limited preprocessor	 [OK]	C23 typed enums, macros only for RX_CHECK
9. Pointer restrictions	 [OK]	Single-level pointers, no function pointers
10. Compiler warnings	 [OK]	Builds clean with -Wall -Wextra -Werror

SOLID Principles:

Single Responsibility: This file handles ONLY GPTW hardware abstraction. Motor control logic, PID calculations, and H-bridge configuration are in separate modules.

Open/Closed: New waveform modes can be added by extending the [rx_gptw_wave_mode_t](#) enum and [internal_get_gtcr_mode\(\)](#) without modifying existing code paths.

See also

[rx_gptw.h](#) Public API documentation

[rx72n_gptw_regs.h](#) Register structure definitions

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[docs/sections/03_hardware_pinout.tex](#) Pin assignments

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Date

2026-01-27

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Since

Version 1.0.0

Definition in file [rx_gptw.c](#).

8.117.2 Enumeration Type Documentation

8.117.2.1 gptw_bit_constants_t

enum [gptw_bit_constants_t](#) : uint8_t

Single-bit value for shift operations.

Enumerator

<code>k_gptw_bit_set</code>	Value for single bit in shift operations
-----------------------------	--

Definition at line 178 of file [rx_gptw.c](#).

```
00178      : uint8_t {
00179  k_gptw_bit_set = 1, /**< Value for single bit in shift operations */
00180 } gptw_bit_constants_t;
```

8.117.2.2 gptw`channel`mask`t

```
enum gptw\_channel\_mask\_t : uint8_t
```

GPTW channel mask for start/stop operations.

Enumerator

<code>k_gptw_all_channels_mask</code>	Bitmask for all 4 GPTW channels (Ch0-3)
---------------------------------------	---

Definition at line 183 of file [rx_gptw.c](#).

```
00183      : uint8_t {
00184  k_gptw_all_channels_mask = 0x0F, /**< Bitmask for all 4 GPTW channels (Ch0-3) */
00185 } gptw_channel_mask_t;
```

8.117.2.3 gptw`constants`t

```
enum gptw\_constants\_t : uint8_t
```

GPTW general constants.

Enumerator

<code>k_gptw_max_channels</code>	GPTW0-GPTW3
<code>k_gptw_outputs_per_channel</code>	GTIOCA and GTIOCB

Definition at line 139 of file [rx_gptw.c](#).

```
00139      : uint8_t {
00140  k_gptw_max_channels      = 4, /**< GPTW0-GPTW3 */
00141  k_gptw_outputs_per_channel = 2, /**< GTIOCA and GTIOCB */
00142 } gptw_constants_t;
```

8.117.2.4 gptw`direction`constants`t

```
enum gptw\_direction\_constants\_t : uint32_t
```

GPTW count direction values for GTUDDTYC register.

Enumerator

<code>k_gptw_gtuddtyc_ud_down</code>	Count direction down (UD bit = 0)
--------------------------------------	-----------------------------------

Definition at line 188 of file [rx_gptw.c](#).

```
00188      : uint32_t {
00189  k_gptw_gtuddtyc_ud_down = 0, /**< Count direction down (UD bit = 0) */
00190 } gptw_direction_constants_t;
```

8.117.2.5 gptw`duty`constants`

enum [gptw_duty_constants_t](#) : uint8_t

Duty cycle calculation constants.

Enumerator

k_gptw_duty_min	Minimum duty cycle (0%)
k_gptw_duty_max	Maximum duty cycle (100%)
k_gptw_duty_divisor	Divisor for percentage conversion

Definition at line 157 of file [rx_gptw.c](#).

```
00157      : uint8_t {
00158  k_gptw_duty_min    = 0, /**< Minimum duty cycle (0%) */
00159  k_gptw_duty_max    = 100, /**< Maximum duty cycle (100%) */
00160  k_gptw_duty_divisor = 100, /**< Divisor for percentage conversion */
00161 } gptw_duty_constants_t;
```

8.117.2.6 gptw`period`constants`

enum [gptw_period_constants_t](#) : uint16_t

Enumerator

k_gptw_period_divisor_saw	Sawtooth: period = PCLKA / freq
k_gptw_period_divisor_tri	Triangle: period = PCLKA / (2 * freq)
k_gptw_period_min	Minimum valid period
k_gptw_period_zero	Zero period value
k_gptw_deadtime_disabled	No dead time

Definition at line 148 of file [rx_gptw.c](#).

```
00148      : uint16_t {
00149  k_gptw_period_divisor_saw = 1, /**< Sawtooth: period = PCLKA / freq */
00150  k_gptw_period_divisor_tri = 2, /**< Triangle: period = PCLKA / (2 * freq) */
00151  k_gptw_period_min        = 10, /**< Minimum valid period */
00152  k_gptw_period_zero       = 0, /**< Zero period value */
00153  k_gptw_deadtime_disabled = 0, /**< No dead time */
00154 } gptw_period_constants_t;
```

8.117.2.7 gptw`phase`divisor`

enum [gptw_phase_divisor_t](#) : uint8_t

Phase divisors for staggered PWM initialization.

Divisor constants used to calculate the initial GTCNT counter value for each GPTW channel during staggered initialization. Each channel is offset by 90 degrees from the previous one to spread switching edges across the PWM period.

For a given period P the offsets are:

- Ch0: 0 (0/4 of P)
- Ch1: P / 4 (quarter divisor)
- Ch2: P / 2 (half divisor)
- Ch3: 3 * P / 4 (three-quarter: multiply by 3, then divide by 4)

Invariant

All divisor values are ≥ 1 , preventing division-by-zero in [internal_calculate_phase_offset\(\)](#)

```
// Example: derive 90-degree offset for a period of 3000 counts
uint32_t offset_ch1 = 3000U / k_phase_divisor_quarter; // == 750
uint32_t offset_ch3 = (3000U * 3U) / k_phase_divisor_three_quarter; // == 2250
```

See also

[internal_calculate_phase_offset\(\)](#) Uses these divisors for GTCNT seeding
[rx_gptw_init_all_staggered\(\)](#) Top-level staggered initialization function

Enumerator

k_phase_divisor_none	0 deg: no offset (safe default, never causes div-by-zero)
k_phase_divisor_quarter	90 deg = period/4
k_phase_divisor_half	180 deg = period/2
k_phase_divisor_three_quarter	270 deg = 3*period/4 (multiplied by 3 then divided)

Definition at line 1459 of file [rx_gptw.c](#).

```
01459         : uint8_t {
01460   k_phase_divisor_none    = 1, /**< 0 deg: no offset (safe default, never causes div-by-zero) */
01461   k_phase_divisor_quarter = 4, /**< 90 deg = period/4 */
01462   k_phase_divisor_half    = 2, /**< 180 deg = period/2 */
01463   k_phase_divisor_three_quarter = 4, /**< 270 deg = 3*period/4 (multiplied by 3 then divided) */
01464 } gptw_phase_divisor_t;
```

8.117.2.8 mpc`pfs`offset`constants`t

```
enum mpc\_pfs\_offset\_constants\_t : uint16_t
```

MPC PFS register offset constants.

Enumerator

k_mpc_pfs_base_offset	Base offset for PFS registers
k_mpc_port_e_index	Port E index
k_mpc_pfs_port_stride	Bytes between port PFS groups

Definition at line 171 of file [rx_gptw.c](#).

```
00171         : uint16_t {
00172   k_mpc_pfs_base_offset = 0x40, /**< Base offset for PFS registers */
00173   k_mpc_port_e_index    = 0x0E, /**< Port E index */
00174   k_mpc_pfs_port_stride = 8,    /**< Bytes between port PFS groups */
00175 } mpc_pfs_offset_constants_t;
```

8.117.2.9 mpc`pwpr`constants`t

```
enum mpc\_pwpr\_constants\_t : uint8_t
```

MPC configuration constants.

Enumerator

k_mpc_pwpr_b0wi_clear	Clear B0WI to enable PFSWE write
k_mpc_pwpr_pfswe_set	Set PFSWE to enable PFS write
k_mpc_pwpr_lock	Set B0WI to lock PFS

Definition at line 164 of file `rx_gptw.c`.

```
00164      : uint8_t {
00165  k_mpc_pwpr_b0wi_clear = 0x00, /**< Clear B0WI to enable PFSWE write */
00166  k_mpc_pwpr_pfswe_set  = 0x40, /**< Set PFSWE to enable PFS write */
00167  k_mpc_pwpr_lock       = 0x80, /**< Set B0WI to lock PFS */
00168 } mpc_pwpr_constants_t;
```

8.117.2.10 pfs`block`layout`t

```
enum pfs_block_layout_t : uintptr_t
```

Configure MPC for GPTW output pins.

Configures the Multi-Function Pin Controller (MPC) to route GPTW timer outputs to the appropriate Port E pins. Each GPTW channel uses two pins for complementary PWM outputs (GTIOCA and GTIOCB).

8.117.2.11 MPC Configuration Sequence

1. Clear PWPR.B0WI to enable PFSWE bit access
2. Set PWPR.PFSWE to enable PFS register writes
3. Write PSEL value (0x1E) to select GPTW function for each pin
4. Set PWPR.B0WI to lock PFS registers

8.117.2.12 Pin Assignments

Channel	Output A (GTIOCA)	Output B (GTIOCB)	PSEL
GPTW0	PE5	PE2	0x1E
GPTW1	PE4	PE1	0x1E
GPTW2	PE3	PE0	0x1E
GPTW3	PE7	PE6	0x1E

Precondition

MPC module accessible (no separate clock enable required)

Pins not in use by other peripherals

Postcondition

Port E PFS registers configured for GPTW function
MPC write protection re-enabled

Invariant

MPC write protection always re-locked on exit (success or failure)

Note

Modifies global MPC registers, not thread-safe
Must be called before enabling port pin peripheral mode

Warning

Overwrites any previous pin function assignment
MPC must be unlocked before calling; function handles locking

Performance:

- Execution time: ~20 cycles (register writes)
- Memory: 0 bytes heap, ~8 bytes stack

See also

[internal_configure_port_pins\(\)](#) Configure PMR/PDR after MPC
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Since

Version 1.0.0

Enumerator

k_pfs_block_base_addr	
k_pfs_port_stride	

Definition at line 551 of file [rx_gptw.c](#).

```
00551     : uintptr_t {  
00552   k_pfs_block_base_addr = 0x0008C140U,  
00553   k_pfs_port_stride     = 8U,  
00554 } pfs_block_layout_t;
```

8.117.2.13 porte_gptw_pin_t

enum [porte_gptw_pin_t](#) : uint8_t

Port E pin indices for GPTW motor outputs.

Enumerator

k_porte_gptw0_a	PE5/GTIOC0A - Motor 0 phase
k_porte_gptw0_b	PE2/GTIOC0B - Motor 0 enable
k_porte_gptw1_a	PE4/GTIOC1A - Motor 1 phase
k_porte_gptw1_b	PE1/GTIOC1B - Motor 1 enable
k_porte_gptw2_a	PE3/GTIOC2A - Motor 2 phase
k_porte_gptw2_b	PE0/GTIOC2B - Motor 2 enable
k_porte_gptw3_a	PE7/GTIOC3A - Motor 3 phase
k_porte_gptw3_b	PE6/GTIOC3B - Motor 3 enable

Definition at line 193 of file [rx_gptw.c](#).

```

00193     : uint8_t {
00194 k_porte_gptw0_a = 5, /**< PE5/GTIOC0A - Motor 0 phase */
00195 k_porte_gptw0_b = 2, /**< PE2/GTIOC0B - Motor 0 enable */
00196 k_porte_gptw1_a = 4, /**< PE4/GTIOC1A - Motor 1 phase */
00197 k_porte_gptw1_b = 1, /**< PE1/GTIOC1B - Motor 1 enable */
00198 k_porte_gptw2_a = 3, /**< PE3/GTIOC2A - Motor 2 phase */
00199 k_porte_gptw2_b = 0, /**< PE0/GTIOC2B - Motor 2 enable */
00200 k_porte_gptw3_a = 7, /**< PE7/GTIOC3A - Motor 3 phase */
00201 k_porte_gptw3_b = 6, /**< PE6/GTIOC3B - Motor 3 enable */
00202 } porte_gptw_pin_t;

```

8.117.3 Function Documentation

8.117.3.1 internal_calculate_period()

```

rx_err_t internal_calculate_period (
    const uint32_t frequency_hz,
    const rx\_gptw\_wave\_mode\_t wave_mode,
    uint32_t * period) [static]

```

Calculate period register value from frequency.

Converts a desired PWM frequency to the GTPR (period register) count value. The calculation differs based on waveform mode:

8.117.3.2 Mathematical Formulas

Sawtooth (edge-aligned) mode:

$$\text{period} = \frac{f_{PCLKA}}{f_{PWM}}$$

Triangle (center-aligned) mode:

$$\text{period} = \frac{f_{PCLKA}}{2 \times f_{PWM}}$$

Where:

- f_{PCLKA} = 120 MHz (peripheral clock A)
- f_{PWM} = desired PWM frequency

8.117.3.3 Algorithm Steps

1. Validate period output pointer (RX_CHECK_NULL_PTR)
2. Validate wave_mode via exhaustive switch statement
3. Determine divisor based on mode (1 for sawtooth, 2 for triangle)
4. Check frequency_hz is non-zero (prevent division by zero)
5. Calculate period using 64-bit arithmetic to prevent overflow
6. Validate period is within hardware limits [10, 0xFFFFFFFF]
7. Store result and return success

8.117.3.4 Example Calculations

Frequency	Mode	Period Count	Resolution
20 kHz	Sawtooth	6000	0.0167%
20 kHz	Triangle	3000	0.0333%
100 Hz	Sawtooth	1,200,000	0.000083%
1 MHz	Sawtooth	120	0.833%

Precondition

PCLKA configured to 120 MHz (system clock setup complete)
 period pointer must point to valid uint32_t storage

Postcondition

On success, *period contains valid count [10, 0xFFFFFFFF]
 On failure, *period is unchanged

Invariant

period $\geq$ k_gptw_period_min (10) on success
 Calculation uses 64-bit math to prevent overflow

Note

Thread-safe: No shared state modified
 Uses RX_CHECK_NULL_PTR macro for pointer validation

Warning

Very low frequencies (< 1 Hz) may overflow 32-bit period
 Very high frequencies (> 12 MHz) produce periods < 10 (rejected)

Performance:

- Execution time: ~ 50 cycles (64-bit division)
- Memory: 0 bytes heap, ~ 16 bytes stack

NASA Power of 10 Compliance:

- Rule 5: [OK] 2 preconditions (NULL check, wave_mode validation)
- Rule 7: [OK] Returns error code, no silent failures

See also

[k_pclka_hz](#) PCLKA clock frequency constant
[internal_is_triangle_mode\(\)](#) Determines divisor selection

Since

Version 1.0.0

Definition at line 448 of file [rx_gptw.c](#).

```

00451 {
00452  /* Pre-condition: validate pointer (NASA Power of 10 Rule 5) */
00453  RX_CHECK_NULL_PTR(period, s_tag, "period pointer is nullptr");
00454
00455  /* Pre-condition: validate wave_mode using exhaustive switch (catches invalid/negative values) */
00456  switch (wave_mode) {
00457    case k_gptw_wave_saw_pwm:
00458    case k_gptw_wave_tri_pwm1:
00459    case k_gptw_wave_tri_pwm2:
00460    case k_gptw_wave_tri_pwm3:
00461      break;
00462    default:
00463      rx_log_error(s_tag, "Invalid wave mode");
00464      return k_rx_err_invalid_arg;
00465  }
00466
00467  /* Period calculation depends on waveform mode:
00468  * - Sawtooth: Period = PCLKA / frequency
00469  * - Triangle: Period = PCLKA / (2 * frequency), because counter goes up then down
00470  * PCLKA = 96 MHz on STAR (HOCO x12 PLL / 2); see rx72n_clock.h k_pclka_hz.
00471  */
00472  const uint32_t pclka = k_pclka_hz;
00473  const uint32_t divisor =
00474    internal_is_triangle_mode(wave_mode) ? k_gptw_period_divisor_tri : k_gptw_period_divisor_saw;

```

```

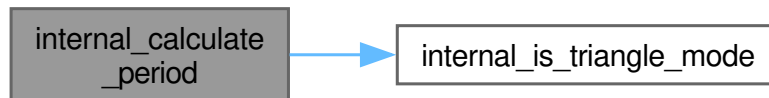
00475
00476 if (frequency_hz == k_gptw_period_zero) {
00477     rx_log_error(s_tag, "Frequency is zero");
00478     return k_rx_err_invalid_arg;
00479 }
00480
00481 /* Use 64-bit math to prevent overflow in divisor * frequency_hz */
00482 const uint64_t denominator = (uint64_t)divisor * (uint64_t)frequency_hz;
00483 const uint32_t period_calc = (uint32_t)(pclka / denominator);
00484
00485 /* Check if period fits in valid range */
00486 if (period_calc > s_gptw_period_max) {
00487     rx_log_error(s_tag, "Frequency too low");
00488     return k_rx_err_invalid_arg;
00489 }
00490
00491 if (period_calc < k_gptw_period_min) {
00492     rx_log_error(s_tag, "Frequency too high");
00493     return k_rx_err_invalid_arg;
00494 }
00495
00496 *period = period_calc;
00497 return k_rx_ok;
00498 }

```

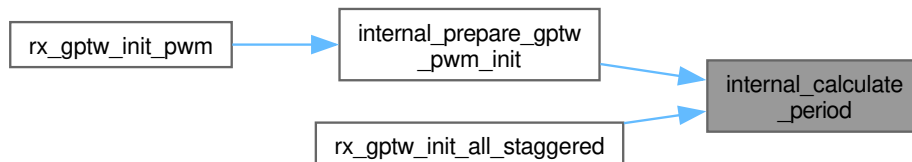
References [internal_is_triangle_mode\(\)](#), [k_gptw_period_divisor_saw](#), [k_gptw_period_divisor_tri](#), [k_gptw_period_min](#), [k_gptw_period_zero](#), [k_gptw_wave_saw_pwm](#), [k_gptw_wave_tri_pwm1](#), [k_gptw_wave_tri_pwm2](#), [k_gptw_wave_tri_pwm3](#), [k_pclka_hz](#), [s_gptw_period_max](#), and [s_tag](#).

Referenced by [internal_prepare_gptw_pwm_init\(\)](#), and [rx_gptw_init_all_staggered\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.5 internal_calculate_phase_offset()

```

void internal_calculate_phase_offset (
    const rx_gptw_channel_t channel,
    const uint32_t period,
    const bool is_triangle,
    uint32_t * init_count,
    bool * init_dir_up) [static]

```

Calculate initial counter value and direction for phase staggering.

Computes the initial counter value (GTCNT) and counting direction for a GPTW channel to achieve the desired phase offset. Phase staggering spreads PWM edges across the period to reduce peak current draw and EMI.

8.117.3.6 Phase Offset Table (Sawtooth Mode)

Channel	Phase	Counter Init	Direction
Ch0	0deg	0	Up (n/a)
Ch1	90deg	$\text{period} / \frac{1}{4}$	Up (n/a)
Ch2	180deg	$\text{period} / \frac{1}{2}$	Up (n/a)
Ch3	270deg	$3 \times \text{period} / \frac{1}{4}$	Up (n/a)

8.117.3.7 Phase Offset Table (Triangle Mode)

Channel	Phase	Counter Init	Direction
Ch0	0deg	0	Up
Ch1	90deg	$\text{period} / \frac{1}{2}$	Up
Ch2	180deg	period	Down
Ch3	270deg	$\text{period} / \frac{1}{2}$	Down

Precondition

Output pointers must be non-NULL

period should be > 0 for valid calculations

Postcondition

*init_count contains phase offset value

*init_dir_up contains initial direction (triangle mode)

Safe defaults (0, true) on any validation failure

Note

Returns void - validation failures produce safe defaults silently

For sawtooth mode, direction is ignored (always up)

NASA Power of 10 Compliance:

- Rule 5: [OK] nullptr checks, period validation
- Rule 2: [OK] No loops (direct calculation)

See also

[internal_configure_channel_staggered\(\)](#) Calls this function
[gptw_phase_divisor_t](#) Divisor constants

Definition at line 1510 of file [rx_gptw.c](#).

```

01515 {
01516 /* Pre-condition: validate output pointers (NASA Power of 10 Rule 5) */
01517 if (init_count == nullptr || init_dir_up == nullptr) {
01518 /* Cannot proceed without valid output pointers - caller error */
01519 return;
01520 }
01521
01522 /* Pre-condition: validate period is non-zero to avoid division by zero */
01523 if (period == 0) {
01524 *init_count = 0;
01525 *init_dir_up = true;
01526 return;
01527 }
01528
01529 /* Pre-condition: validate channel is in valid range */
01530 if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01531 *init_count = 0;
01532 *init_dir_up = true;
01533 return;
01534 }
01535
01536 *init_dir_up = true; /* Default to counting up */
01537
01538 if (lis_triangle) {
01539 /* Edge-Aligned (Sawtooth) Staggering:
01540 * Ch0 (0 deg): 0
01541 * Ch1 (90 deg): Period / 4
01542 * Ch2 (180 deg): Period / 2
01543 * Ch3 (270 deg): 3 * Period / 4
01544 */
01545 switch (channel) {
01546 case k_gptw_channel_0:
01547 *init_count = 0;
01548 break;
01549 case k_gptw_channel_1:
01550 *init_count = period / k_phase_divisor_quarter;
01551 break;
01552 case k_gptw_channel_2:
01553 *init_count = period / k_phase_divisor_half;
01554 break;
01555 case k_gptw_channel_3:
01556 *init_count = (period * 3) / k_phase_divisor_three_quarter;
01557 break;
01558 default:
01559 *init_count = 0;
01560 break;
01561 }
01562 } else {
01563 /* Center-Aligned (Triangle) Staggering:
01564 * Ch0 (0 deg): 0, Up
01565 * Ch1 (90 deg): Period/2, Up
01566 * Ch2 (180 deg): Period, Down (Wait: Period is Top)
01567 * Note: At Period, it turns around.
01568 * For 180 shift, we want the "center" of the OFF pulse to correspond
01569 * to the "center" of Ch0 ON pulse?
01570 * Standard 3-phase shift logic applied to 4-phase:
01571 * 0 deg: 0 (Valley) -> Up
01572 * 90 deg: Period/2 -> Up
01573 * 180 deg: Period (Peak) -> Down (naturally acts as peak)
01574 * 270 deg: Period/2 -> Down
01575 */
01576 switch (channel) {
01577 case k_gptw_channel_0:
01578 *init_count = 0;
01579 *init_dir_up = true;
01580 break;
01581 case k_gptw_channel_1:
01582 *init_count = period / k_phase_divisor_half;
01583 *init_dir_up = true;
01584 break;
01585 case k_gptw_channel_2:
01586 *init_count = period;
01587 *init_dir_up = false; /* Start at top, go down */
01588 break;
01589 case k_gptw_channel_3:
01590 *init_count = period / k_phase_divisor_half;
01591 *init_dir_up = false; /* Start at mid, go down */
01592 break;

```

```

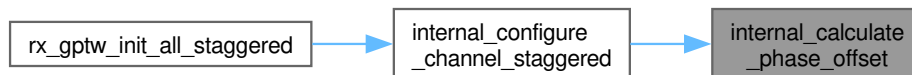
01593     default:
01594         *init_count = 0;
01595         break;
01596     }
01597 }
01598 }

```

References [k_gptw_channel_0](#), [k_gptw_channel_1](#), [k_gptw_channel_2](#), [k_gptw_channel_3](#), [k_gptw_max_channels](#), [k_phase_divisor_half](#), [k_phase_divisor_quarter](#), and [k_phase_divisor_three_quarter](#).

Referenced by [internal_configure_channel_staggered\(\)](#).

Here is the caller graph for this function:



8.117.3.8 `internal_configure_channel_staggered()`

```

rx_err_t internal_configure_channel_staggered (
    const rx_gptw_channel_t channel,
    const rx_gptw_config_t * config,
    const uint32_t period,
    const bool is_triangle) [static]

```

Configure a single GPTW channel for staggered PWM operation.

Performs complete hardware configuration for one GPTW channel including base timer setup, MPC pin function selection, port configuration, and phase staggering initialization.

8.117.3.9 Configuration Sequence

1. Validate channel range [0-3]
2. Get channel register base address
3. Configure base hardware (GTCR, GTIOR, GTPR, etc.)
4. Configure MPC for Port E alternate function
5. Configure port pins (PMR, PDR)
6. Calculate phase offset for this channel
7. Set initial counter value (GTCNT)
8. For triangle mode: seed initial count direction
9. Update initialization tracking state

8.117.3.10 Phase Staggering

Counter is initialized to a phase-offset value:

- Ch0: 0deg (reference, GTCNT=0)
- Ch1: 90deg (GTCNT=period/4)
- Ch2: 180deg (GTCNT=period/2)
- Ch3: 270deg (GTCNT=3xperiod/4)

Precondition

Module clock enabled
 Timer stopped
 config pointer valid (caller validates)

Postcondition

Channel fully configured with phase offset
`s_gptw_initialized[channel] = true`
`s_gptw_period[channel] = period`
 Timer NOT started (caller starts all channels together)

Note

Not thread-safe: modifies hardware registers and shared state arrays (`s_gptw_initialized`, `s_gptw_↵_period`); caller must ensure exclusive access
 Timer is NOT started by this function
 Called from `rx_gptw_init_all_staggered()` in a loop

NASA Power of 10 Compliance:

- Rule 5: [OK] Channel range validation
- Rule 7: [OK] All internal function returns checked

See also

[rx_gptw_init_all_staggered\(\)](#) Parent function
[internal_calculate_phase_offset\(\)](#) Phase calculation

Definition at line 1651 of file `rx_gptw.c`.

```
01655 {
01656     /* Pre-condition: validate channel range (NASA Power of 10 Rule 5) */
01657     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01658         return k_rx_err_invalid_arg;
01659     }
01660
01661     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01662
01663     /* Pre-condition: validate gptw pointer */
01664     if (gptw == nullptr) {
01665         return k_rx_err_invalid_arg;
```

```

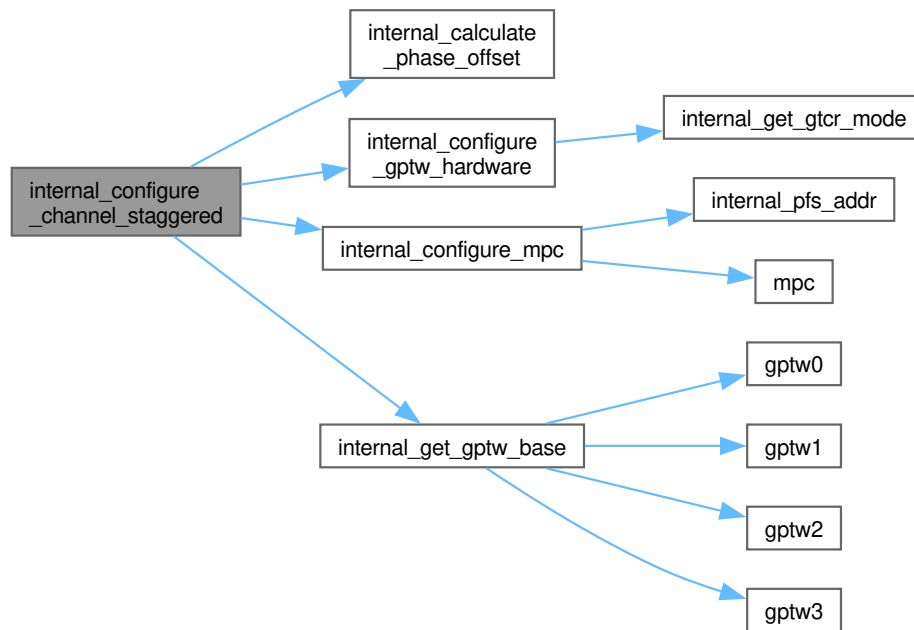
01666 }
01667
01668 /* Base hardware configuration */
01669 rx_err_t err = internal_configure_gptw_hardware(gptw, config, period);
01670 if (err != k_rx_ok) {
01671     return err;
01672 }
01673
01674 /* Configure MPC PFS for the GTIOC pads. PMR (peripheral-mode enable)
01675  * is set after the channels are atomically started in the caller. */
01676 err = internal_configure_mpc(config);
01677 if (err != k_rx_ok) {
01678     return err;
01679 }
01680
01681 /* Apply Phase Staggering */
01682 uint32_t init_count = 0;
01683 bool dir_up = true;
01684
01685 internal_calculate_phase_offset(channel, period, is_triangle, &init_count, &dir_up);
01686
01687 /* Modify counter value (unlocked by configure function, but we need to unlock again
01688  * since internal_configure_gptw_hardware locks it at the end)
01689  */
01690 gptw->gtwp = k_gptw_gtwp_unlock;
01691 gptw->gtcnt = init_count;
01692
01693 /* Handle Direction Seeding for Triangle Mode */
01694 if (is_triangle) {
01695     /* Use GTUDDTYC to force initial direction
01696      * UD (bit 0): 0=Down, 1=Up
01697      * UDF (bit 1): 1=Force
01698      * Sequence: Set Force -> Clear Force (latch direction)
01699      * Note: Manual verification confirmed UD=bit0, UDF=bit1.
01700      */
01701     uint32_t ud_val = dir_up ? k_gptw_gtuddtyc_ud : k_gptw_gtuddtyc_ud_down;
01702
01703     /* Force direction */
01704     gptw->gtuddtyc = k_gptw_gtuddtyc_udf | ud_val;
01705
01706     /* Clear force bit to allow normal counting operation */
01707     gptw->gtuddtyc = ud_val;
01708 }
01709
01710 gptw->gtwp = k_gptw_gtwp_lock;
01711
01712 /* Update internal state */
01713 s_gptw_period[channel] = period;
01714 s_gptw_initialized[channel] = true;
01715
01716 return k_rx_ok;
01717 }

```

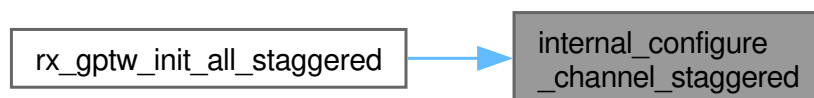
References [rx_gptw_channel_regs_t::gtcnt](#), [rx_gptw_channel_regs_t::gtuddtyc](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_calculate_phase_offset\(\)](#), [internal_configure_gptw_hardware\(\)](#), [internal_configure_mpc\(\)](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtuddtyc_ud](#), [k_gptw_gtuddtyc_ud_down](#), [k_gptw_gtuddtyc_udf](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), [k_gptw_max_channels](#), [s_gptw_initialized](#), and [s_gptw_period](#).

Referenced by [rx_gptw_init_all_staggered\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.11 internal`configure`gptw`hardware()

```

rx_err_t internal_configure_gptw_hardware (
    volatile rx_gptw_channel_regs_t * gptw,
    const rx_gptw_config_t * config,
    const uint32_t period) [static]

```

Configure GPTW hardware registers.

Performs the core hardware configuration of a GPTW channel. This function writes all necessary registers to set up PWM operation including control mode, I/O configuration, period, duty cycle initialization, buffer mode, and deadline.

8.117.3.12 Register Configuration Sequence

1. Unlock write protection: GTWP = 0xA500_0000
2. Configure control register (GTCR):
 - Set clock source to PCLKA/1 (120 MHz)
 - Set waveform mode (sawtooth or triangle)
3. Configure I/O control (GTIOR):
 - Output A: Initial low, toggle on compare match, enable
 - Output B: Initial low, toggle on compare match, enable

4. Set period register (GTPR): PWM cycle count
5. Initialize duty cycles (GTCCRA, GTCCRB): Both to 0
6. Clear counter (GTCNT): Start at 0
7. Enable buffer operation (GTBER): Glitch-free updates
8. Configure deadtime (GTDVU, GTDVD, GTDTCR): If requested
9. Lock write protection: GTWP = 0x0000_0001

8.117.3.13 Deadtime Calculation

$$\text{deadtime_counts} = \frac{\text{deadtime_ns} \times f_{PCLKA}}{10^9}$$

For 1000 ns at 96 MHz: deadtime_counts = 96

Precondition

gptw points to valid, unlocked GPTW registers
 config contains validated parameters
 period calculated and validated by [internal_calculate_period\(\)](#)

Postcondition

GPTW channel configured for PWM operation
 Duty cycles initialized to 0% (motor stopped)
 Buffer mode enabled for glitch-free updates
 Write protection re-enabled

Invariant

Write protection always re-locked on exit
 Duty cycles always start at 0 (safe state)

Note

Not thread-safe: Direct hardware register access
 Timer is NOT started by this function

Warning

gptw pointer must be valid - no runtime bounds checking
 Caller responsible for stopping timer before calling

Performance:

- Execution time: ~100 cycles (multiple register writes)
- Memory: 0 bytes heap, ~16 bytes stack

NASA Power of 10 Compliance:

- Rule 5: [OK] 3 preconditions (NULL checks, period range)

See also

[internal_calculate_period\(\)](#) Period value calculation

[rx72n_gptw_regs.h](#) Register structure definition

Since

Version 1.0.0

Definition at line 756 of file [rx_gptw.c](#).

```

00759 {
00760     /* Pre-condition: validate pointers (NASA Power of 10 Rule 5) */
00761     RX_CHECK_NULL_PTR(gptw, s_tag, "gptw pointer is nullptr");
00762     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00763
00764     /* Pre-condition: validate period range */
00765     if ((period < k_gptw_period_min) || (period > s_gptw_period_max)) {
00766         rx_log_error(s_tag, "Period out of range");
00767         return k_rx_err_invalid_arg;
00768     }
00769
00770     /* Unlock write protection for this channel */
00771     gptw->gtpw = k_gptw_gtpw_unlock;
00772
00773     /* Configure control register
00774     * - PCLKA/1 (96 MHz on STAR)
00775     * - Waveform mode from config (sawtooth or triangle)
00776     * NOTE: CST (bit 0) left clear; caller (rx_gptw_start) sets it.
00777     */
00778     const uint32_t gtrc_mode = internal_get_gtrc_mode(config->wave_mode);
00779     gptw->gtrc = k_gptw_gtrc_tpcs_1 | gtrc_mode;
00780
00781     /* Force UP-count direction. Reset value of GTUDDTYC is 0 (UD=0 = DOWN).
00782     * With GTIOR = 0x09 (compare-match-up event drives output LOW), a
00783     * down-counter never raises the compare-match-up event and the output
00784     * latches HIGH at end-of-cycle. UD=1 + UDF=1 forces up-count immediately. */
00785     gptw->gtuddtyc = k_gptw_gtuddtyc_ud | k_gptw_gtuddtyc_udf;
00786
00787     /* Configure I/O control register
00788     * - Initial low, high at end-of-cycle, low at compare match (sawtooth up)
00789     * - Enable both outputs
00790     */
00791     gptw->gtior =
00792         k_gptw_gtior_oa_init_low | k_gptw_gtior_oae | k_gptw_gtior_ob_init_low | k_gptw_gtior_obe;
00793
00794     /* Set period (GTPR = PWM cycle) */
00795     gptw->gtpw = period;
00796
00797     /* Set initial duty cycle to 0% for both outputs */
00798     gptw->gtccra = k_gptw_period_zero;
00799     gptw->gtccrb = k_gptw_period_zero;
00800
00801     /* Clear counter */
00802     gptw->gtcnt = k_gptw_period_zero;
00803
00804     /* Disable all buffer operation on GTCCRA/GTCCRB/GTPR.
00805     * Rationale: with single-buffer enabled (CCRA=0x01«16, CCRB=0x01«18)
00806     * the hardware copies GTCCRC->GTCCRA / GTCCRE->GTCCRB on each period
00807     * boundary. GTCCRC/E default to 0, so any duty written via
00808     * rx_gptw_set_duty_raw() gets clobbered to 0 on the next overflow --
00809     * outputs stick at initial state. With GTBER=0, direct writes to
00810     * GTCCRA/GTCCRB take effect immediately.
00811     * If double-buffered duty updates are ever needed, a separate opt-in
00812     * API should enable buffering deliberately. */
00813     gptw->gtber = 0U;
00814
00815     /* Configure dead time if requested */
00816     if (config->deadtime_ns > k_gptw_deadtime_disabled) {
00817         /* Calculate dead time count: deadtime_ns * (PCLKA / 1e9) */
00818         const uint32_t deadtime_count =
00819             (uint32_t)((uint64_t)config->deadtime_ns * k_pclka_hz / s_gptw_ns_per_second);
00820         gptw->gtdvu = deadtime_count;
00821         gptw->gtdvd = deadtime_count;
00822     }
00823 }
```

```

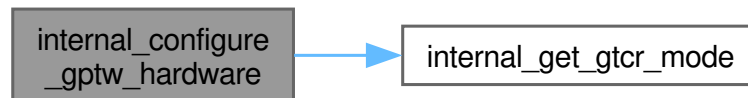
00822     gptw->gtdtcr = k_gptw_gtdtcr_tde; /* Enable dead time */
00823 }
00824
00825 /* Lock write protection */
00826 gptw->gtwp = k_gptw_gtwp_lock;
00827
00828 return k_rx_ok;
00829 }

```

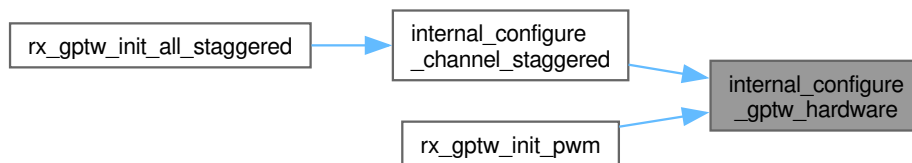
References [rx_gptw_config_t::deadtime_ns](#), [rx_gptw_channel_regs_t::gtber](#), [rx_gptw_channel_regs_t::gtccra](#), [rx_gptw_channel_regs_t::gtccrb](#), [rx_gptw_channel_regs_t::gtcnt](#), [rx_gptw_channel_regs_t::gtcr](#), [rx_gptw_channel_regs_t::gtdtcr](#), [rx_gptw_channel_regs_t::gtdvd](#), [rx_gptw_channel_regs_t::gtdvu](#), [rx_gptw_channel_regs_t::gtior](#), [rx_gptw_channel_regs_t::gtpr](#), [rx_gptw_channel_regs_t::gtuddtyc](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gtcr_mode\(\)](#), [k_gptw_deadtime_disabled](#), [k_gptw_gtcr_tpcs_1](#), [k_gptw_gtdtcr_tde](#), [k_gptw_gtior_oa_init_low](#), [k_gptw_gtior_oae](#), [k_gptw_gtior_ob_init_low](#), [k_gptw_gtior_obe](#), [k_gptw_gtuddtyc_ud](#), [k_gptw_gtuddtyc_udf](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), [k_gptw_period_min](#), [k_gptw_period_zero](#), [k_pclka_hz](#), [s_gptw_ns_per_second](#), [s_gptw_period_max](#), [s_tag](#), and [rx_gptw_config_t::wave_mode](#).

Referenced by [internal_configure_channel_staggered\(\)](#), and [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.14 internal_configure_mpc()

```

rx_err_t internal_configure_mpc (
    const rx_gptw_config_t * config) [static]

```

Definition at line 562 of file [rx_gptw.c](#).

```

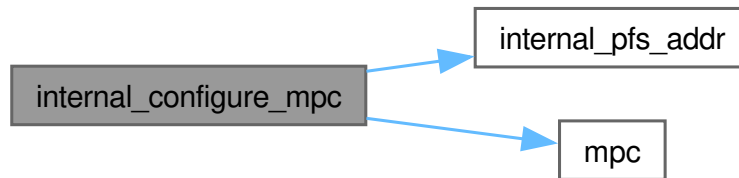
00563 {
00564     /* Unlock MPC: clear B0WI then set PFSWE so PFS becomes writable. */
00565     mpc()->pwpr = k_mpc_pwpr_b0wi_clear;
00566     mpc()->pwpr = k_mpc_pwpr_pfswe_set;
00567
00568     /* Caller supplies pin coordinates -- the lib does not assume a port.
00569     * PSEL value for GTIOC is k_pfs_psel_gptw on every port that has GTIOC
00570     * available (per RX72N HW manual Tables 23.4, 23.6, 23.10, 23.14, 23.16). */
00571     *internal_pfs_addr(config->port_a_idx, config->bit_a) = k_pfs_psel_gptw;
00572     *internal_pfs_addr(config->port_b_idx, config->bit_b) = k_pfs_psel_gptw;
00573
00574     mpc()->pwpr = k_mpc_pwpr_lock;
00575     return k_rx_ok;
00576 }

```

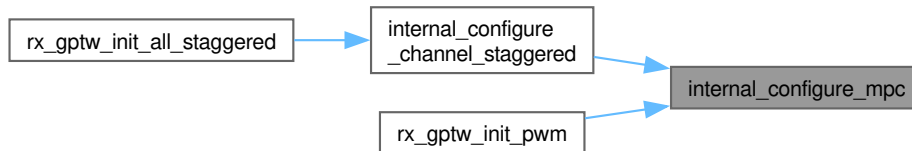
References [rx_gptw_config_t::bit_a](#), [rx_gptw_config_t::bit_b](#), [internal_pfs_addr\(\)](#), [k_mpc_pwpr_b0wi_clear](#), [k_mpc_pwpr_lock](#), [k_mpc_pwpr_pfswe_set](#), [k_pfs_psel_gptw](#), [mpc\(\)](#), [rx_gptw_config_t::port_a_idx](#), [rx_gptw_config_t::port_b_idx](#), and [rx_mpc_regs_t::pwpr](#).

Referenced by [internal_configure_channel_staggered\(\)](#), and [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.15 internal`configure`port`pins()

```
void internal_configure_port_pins (
    const rx\_gptw\_config\_t * config) [static]
```

Configure Port E pins as peripheral outputs.

Configures the Port E GPIO pins for GPTW peripheral output mode. Sets both the Peripheral Mode Register (PMR) and Port Direction Register (PDR) to enable GPTW timer outputs.

8.117.3.16 Register Configuration

For each pin used by the specified channel:

- PMR bit = 1: Enable peripheral mode (not GPIO)
- PDR bit = 1: Set pin direction to output

8.117.3.17 Pin Mapping

Channel	Output A Pin	Output B Pin
GPTW0	PE5 (bit 5)	PE2 (bit 2)
GPTW1	PE4 (bit 4)	PE1 (bit 1)
GPTW2	PE3 (bit 3)	PE0 (bit 0)
GPTW3	PE7 (bit 7)	PE6 (bit 6)

Precondition

MPC configured for GPTW function via [internal_configure_mpc\(\)](#)
 config->port_a_idx / config->port_b_idx are valid for [rx_port_get_base\(\)](#)

Postcondition

PMR bits set for both channel pins

PDR bits set for both channel pins

Note

No return value - silent no-op on invalid channel

Modifies Port E registers, not thread-safe

Uses OR-equals to preserve other pin configurations

Performance:

- Execution time: ~10 cycles (register read-modify-write)
- Memory: 0 bytes heap, 0 bytes stack

See also

[internal_configure_mpc\(\)](#) Must be called before this function

[porte\(\)](#) Port E register accessor

Since

Version 1.0.0

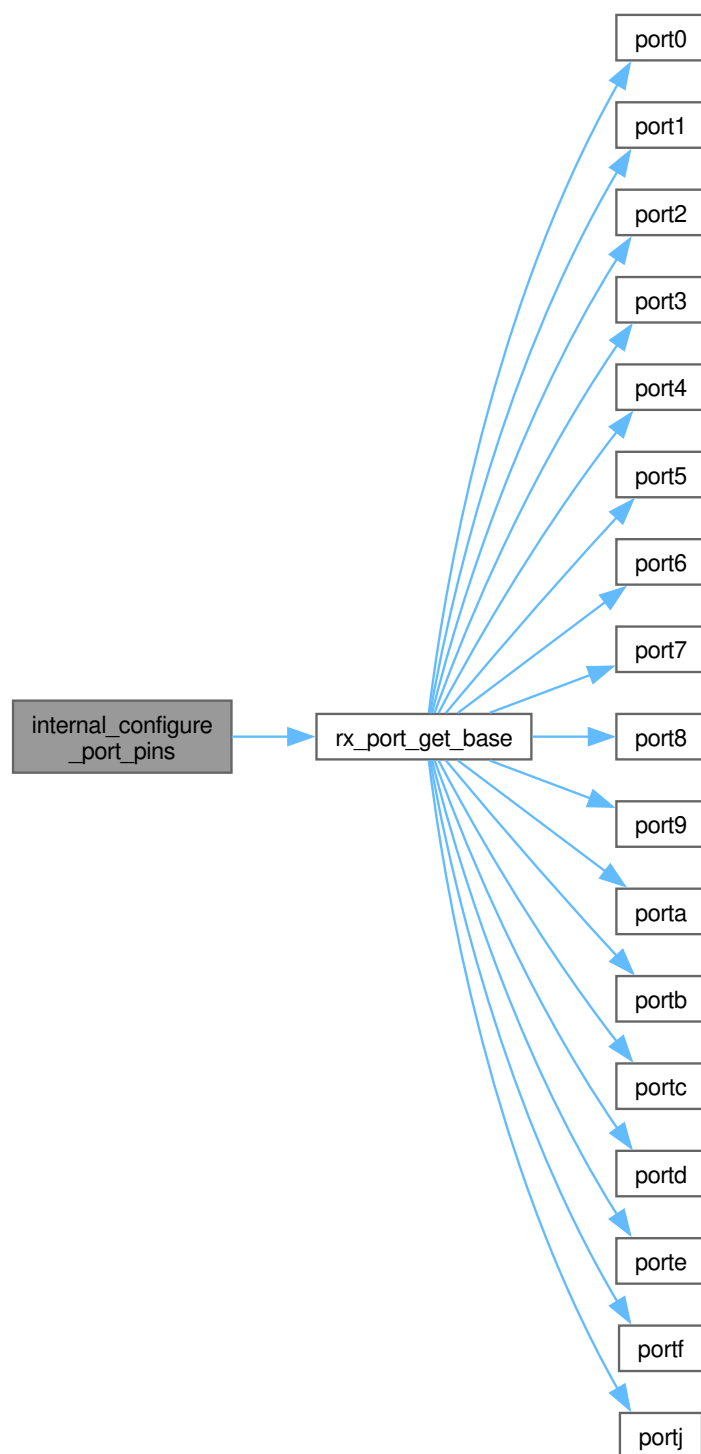
Definition at line 621 of file [rx_gptw.c](#).

```
00622 {
00623     /* Caller supplies pin coordinates. Use rx_port_get_base() so this works
00624      * for any port the board happens to route GTIOC outputs to. */
00625     volatile rx_port_regs_t* port_a = rx_port_get_base(config->port_a_idx);
00626     volatile rx_port_regs_t* port_b = rx_port_get_base(config->port_b_idx);
00627     if ((port_a == nullptr) || (port_b == nullptr)) {
00628         return;
00629     }
00630
00631     port_a->pdr |= (uint8_t)(1U « config->bit_a);
00632     port_b->pdr |= (uint8_t)(1U « config->bit_b);
00633     port_a->pmr |= (uint8_t)(1U « config->bit_a);
00634     port_b->pmr |= (uint8_t)(1U « config->bit_b);
00635 }
```

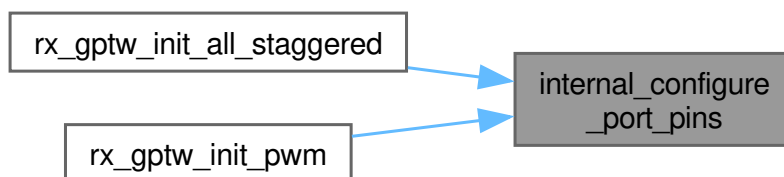
References [rx_gptw_config_t::bit_a](#), [rx_gptw_config_t::bit_b](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_gptw_config_t::port_a_idx](#), [rx_gptw_config_t::port_b_idx](#), and [rx_port_get_base\(\)](#).

Referenced by [rx_gptw_init_all_staggered\(\)](#), and [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.18 `internal_enable_gptw_module_clock()`

```
void internal_enable_gptw_module_clock (
    void ) [static]
```

Enable GPTW module clock.

Enables the GPTW peripheral by clearing its module stop bit in MSTPCRA. On RX72N, GPTW is controlled by MSTPCRA bit 7 (MSTPA7), per the RX72N Hardware Manual Chapter 11 and Renesas iodef.h. When set, the module clock is stopped and registers are inaccessible.

8.117.3.19 Register Access Sequence

1. Unlock PRCR: Write 0xA502 (PRC1) to enable writing to MSTPCR group
2. Clear MSTPA7: AND-mask to clear bit 7 in MSTPCRA (0x00080010)
3. Lock PRCR: Write 0xA500 to re-enable protection

8.117.3.20 Module Stop Control

Bit	Value	Effect
MSTPCRA.MSTPA7 = 1	Module stopped	Registers inaccessible, low power
MSTPCRA.MSTPA7 = 0	Module running	Registers accessible, clock active

Precondition

System clock configured

Postcondition

GPTW module clock enabled
 GPTW registers accessible
 PRCR protection re-enabled

Invariant

PRCR always re-locked after modification

Note

Not thread-safe: Modifies system-wide registers
 Idempotent: Safe to call multiple times

Warning

Must be called before any GPTW register access
 Modifies system protection register (PRCR)

Performance:

- Execution time: ~15 cycles (3 register writes)
- Memory: 0 bytes heap, 0 bytes stack

See also

[k_mstpa_gptw](#) GPTW module stop bit position (MSTPA7)

[k_rx_prcr_unlock_prc1](#) PRCR unlock value for MSTPCR group

RX72N Hardware Manual Chapter 11 - Low Power Consumption

Since

Version 1.0.0

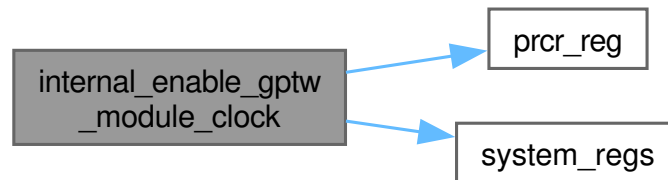
Definition at line 683 of file [rx_gptw.c](#).

```
00684 {
00685     /* RX72N: GPTW = MSTPCRA bit 7 (MSTPA7), NOT MSTPCRC bit 6 as earlier HAL
00686      * code claimed. Use PRC1 unlock (0xA502) which covers MSTPCR group. */
00687     *prcr_reg() = k_rx_prcr_unlock_prc1;
00688     system_regs()->mstpcra &= ~(uint32_t)k_gptw_bit_set « k_mstpa_gptw);
00689     *prcr_reg() = k_rx_prcr_lock;
00690 }
```

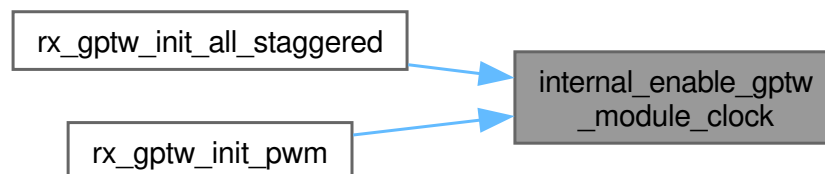
References [k_gptw_bit_set](#), [k_mstpa_gptw](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_gptw_init_all_staggered\(\)](#), and [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.21 internal`get`gptw`base()

```
volatile rx_gptw_channel_regs_t * internal_get_gptw_base (
    const rx_gptw_channel_t channel) [static]
```

Get GPTW channel register base address.

Maps a GPTW channel enumeration value to its corresponding hardware register base address. Each GPTW channel (0-3) has a dedicated register block at a fixed memory offset.

8.117.3.22 Memory Map

Channel	Base Address	Offset from GPTW0
k_gptw_channel_0	0x000C2100	0x000
k_gptw_channel_1	0x000C2180	0x080
k_gptw_channel_2	0x000C2200	0x100
k_gptw_channel_3	0x000C2280	0x180

Precondition

Module clock must be enabled before accessing returned pointer

Postcondition

Returned pointer is valid for hardware register access

No hardware state modified by this function

Note

Thread-safe: Returns constant address, no state modification

Caller must check for nullptr before using returned pointer

Performance:

- Execution time: ~10 cycles (switch statement)
- Memory: 0 bytes heap, 0 bytes stack allocation

See also

[gptw0\(\)](#), [gptw1\(\)](#), [gptw2\(\)](#), [gptw3\(\)](#) Underlying accessor functions

[rx72n_gptw_regs.h](#) Register structure definition

Since

Version 1.0.0

Definition at line 255 of file [rx_gptw.c](#).

```

00256 {
00257     switch (channel) {
00258         case k_gptw_channel_0:
00259             return gptw0();
00260         case k_gptw_channel_1:
00261             return gptw1();
00262         case k_gptw_channel_2:
00263             return gptw2();
00264         case k_gptw_channel_3:
00265             return gptw3();
00266         default:

```

```

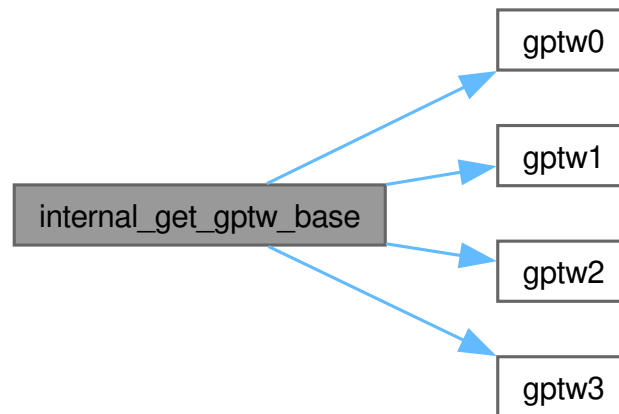
00267     return nullptr;
00268 }
00269 }

```

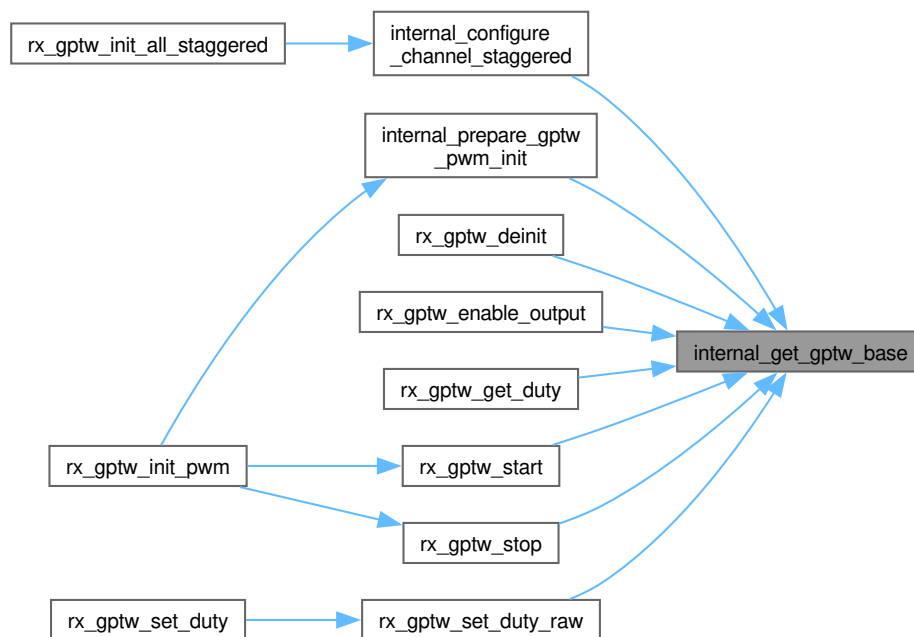
References [gptw0\(\)](#), [gptw1\(\)](#), [gptw2\(\)](#), [gptw3\(\)](#), [k_gptw_channel_0](#), [k_gptw_channel_1](#), [k_gptw_channel_2](#), and [k_gptw_channel_3](#).

Referenced by [internal_configure_channel_staggered\(\)](#), [internal_prepare_gptw_pwm_init\(\)](#), [rx_gptw_deinit\(\)](#), [rx_gptw_enable_output\(\)](#), [rx_gptw_get_duty\(\)](#), [rx_gptw_set_duty_raw\(\)](#), [rx_gptw_start\(\)](#), and [rx_gptw_stop\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.23 internal`get`gtcr`mode()

```

uint32_t internal_get_gtcr_mode (
    const rx_gptw_wave_mode_t mode) [static]

```

Get GTCR mode bits for wave mode.

Converts a waveform mode enumeration to the corresponding GTCR.MD register bit field value. The MD field (bits 18:16) in GTCR controls the timer's counting mode and PWM generation behavior.

8.117.3.24 GTCR.MD Field Mapping

Input Mode	GTCR.MD Value	Binary	Description
k_gptw_wave_saw_pwm	k_gptw_gtcr_md_saw_pwm	0b000	Sawtooth PWM
k_gptw_wave_tri_pwm1	k_gptw_gtcr_md_tri_pwm1	0b100	Triangle PWM 1
k_gptw_wave_tri_pwm2	k_gptw_gtcr_md_tri_pwm2	0b101	Triangle PWM 2
k_gptw_wave_tri_pwm3	k_gptw_gtcr_md_tri_pwm3	0b110	Triangle PWM 3

Precondition

mode should be validated before calling (caller's responsibility)
 GTCR write protection must be unlocked before writing returned value

Postcondition

No state modification (pure function)
 Returned value is ready for GTCR register write

Invariant

Invalid modes always return sawtooth as safe fallback

Note

Assert triggers on invalid mode in debug builds
 Returns safe default (sawtooth) on invalid mode in release builds
 Thread-safe: Pure function with no shared state

Warning

RX_ASSERT fires if mode is not a valid enum value

Performance:

- Execution time: ~10 cycles (switch statement)
- Memory: 0 bytes heap, 0 bytes stack

NASA Power of 10 Compliance:

- Rule 5: [OK] RX_ASSERT validates unexpected conditions

See also

[rx_gptw_wave_mode_t](#) Input enum definition

RX72N Hardware Manual Section 26.2.12 GTCR register description

Since

Version 1.0.0

Definition at line 358 of file [rx_gptw.c](#).

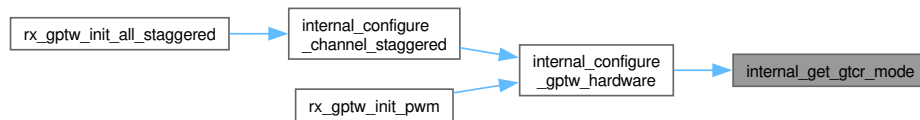
```

00359 {
00360     switch (mode) {
00361         case k_gptw_wave_tri_pwm1:
00362             return k_gptw_gtcr_md_tri_pwm1;
00363         case k_gptw_wave_tri_pwm2:
00364             return k_gptw_gtcr_md_tri_pwm2;
00365         case k_gptw_wave_tri_pwm3:
00366             return k_gptw_gtcr_md_tri_pwm3;
00367         case k_gptw_wave_saw_pwm:
00368             return k_gptw_gtcr_md_saw_pwm;
00369         default:
00370             /* Should never reach here if wave_mode was validated upstream (NASA Rule 5) */
00371             RX_ASSERT(false, "Unexpected wave mode in internal_get_gtcr_mode");
00372             return k_gptw_gtcr_md_saw_pwm;
00373     }
00374 }
```

References [k_gptw_gtcr_md_saw_pwm](#), [k_gptw_gtcr_md_tri_pwm1](#), [k_gptw_gtcr_md_tri_pwm2](#), [k_gptw_gtcr_md_tri_pwm3](#), [k_gptw_wave_saw_pwm](#), [k_gptw_wave_tri_pwm1](#), [k_gptw_wave_tri_pwm2](#), and [k_gptw_wave_tri_pwm3](#).

Referenced by [internal_configure_gptw_hardware\(\)](#).

Here is the caller graph for this function:



8.117.3.25 `internal_is_triangle_mode()`

```

bool internal_is_triangle_mode (
    const rx\_gptw\_wave\_mode\_t mode)    [inline], [static]
```

Check if wave mode is triangle (center-aligned).

Determines whether the specified waveform mode uses triangle wave (center-aligned) PWM generation. Triangle modes count up to period, then down to zero, resulting in symmetric PWM pulses centered within each period.

8.117.3.26 Mode Classification

Mode	Type	Returns
<code>k_gptw_wave_saw_pwm</code>	Sawtooth (edge-aligned)	false
<code>k_gptw_wave_tri_pwm1</code>	Triangle (center-aligned)	true
<code>k_gptw_wave_tri_pwm2</code>	Triangle (center-aligned)	true
<code>k_gptw_wave_tri_pwm3</code>	Triangle (center-aligned)	true

Precondition

mode should be a valid [rx_gptw_wave_mode_t](#) value

Postcondition

No state modification

Note

Pure function: Same input always produces same output

Thread-safe: No shared state access

Inlined for performance in period calculation loop

Performance:

- Execution time: ~ 3 cycles (two comparisons)
- Inlined at call site, no function call overhead

See also

[rx_gptw_wave_mode_t](#) Waveform mode definitions

[internal_calculate_period\(\)](#) Primary user of this function

Since

Version 1.0.0

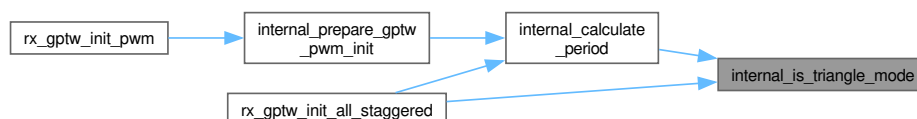
Definition at line 307 of file [rx_gptw.c](#).

```
00308 {
00309     return (mode == k_gptw_wave_tri_pwm1) || (mode == k_gptw_wave_tri_pwm2) ||
00310           (mode == k_gptw_wave_tri_pwm3);
00311 }
```

References [k_gptw_wave_tri_pwm1](#), [k_gptw_wave_tri_pwm2](#), and [k_gptw_wave_tri_pwm3](#).

Referenced by [internal_calculate_period\(\)](#), and [rx_gptw_init_all_staggered\(\)](#).

Here is the caller graph for this function:



8.117.3.27 `internal`pfs`addr()`

```
volatile uint8_t * internal_pfs_addr (
    uint8_t port_idx,
    uint8_t bit)    [inline], [static]
```

Definition at line 556 of file `rx_gptw.c`.

```
00557 {
00558     return (volatile uint8_t*)(k_pfs_block_base_addr + ((uint32_t)port_idx * k_pfs_port_stride) +
00559             bit);
00560 }
```

References `k_pfs_block_base_addr`, and `k_pfs_port_stride`.

Referenced by `internal_configure_mpc()`.

Here is the caller graph for this function:

8.117.3.28 `internal`prepare`gptw`pwm`init()`

```
rx_err_t internal_prepare_gptw_pwm_init (
    const rx_gptw_channel_t channel,
    const rx_gptw_config_t * config,
    volatile rx_gptw_channel_regs_t ** gptw_out,
    uint32_t * period_out)    [static]
```

Validate channel and compute period for PWM initialization.

Shared pre-initialization helper used by `rx_gptw_init_pwm()`. Performs two actions in a single function to reduce code duplication:

1. Validate channel range and resolve register base pointer
2. Calculate the GTPR period value from frequency and wave mode

Precondition

`*gptw_out` and `period_out` must be non-NULL (caller's responsibility)
`config` must be non-NULL (caller's responsibility)

Postcondition

`*gptw_out` points to valid GPTW channel registers on success
`*period_out` contains a value in `[k_gptw_period_min, s_gptw_period_max]` on success

Note

Pure helper: does not modify hardware registers or module state

Not thread-safe: relies on config content stability during call

See also

[internal_calculate_period\(\)](#) Period calculation details

[rx_gptw_init_pwm\(\)](#) Primary caller

Since

Version 1.0.0

Definition at line 856 of file [rx_gptw.c](#).

```

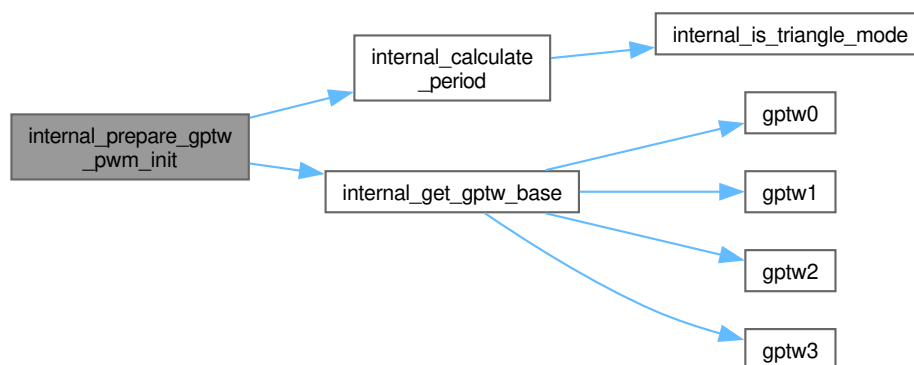
00860 {
00861     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
00862         rx_log_error(s_tag, "Invalid GPTW channel");
00863         return k_rx_err_invalid_arg;
00864     }
00865
00866     *gptw_out = internal_get_gptw_base(channel);
00867     if (*gptw_out == nullptr) {
00868         return k_rx_err_invalid_arg;
00869     }
00870
00871     rx_err_t err = internal_calculate_period(config->frequency_hz, config->wave_mode, period_out);
00872     if (err != k_rx_ok) {
00873         return err;
00874     }
00875
00876     return k_rx_ok;
00877 }

```

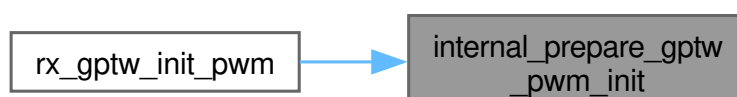
References [rx_gptw_config_t::frequency_hz](#), [internal_calculate_period\(\)](#), [internal_get_gptw_base\(\)](#), [k_gptw_max_channels](#), [s_tag](#), and [rx_gptw_config_t::wave_mode](#).

Referenced by [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.29 rx`gptw`deinit()

```
rx_err_t rx_gptw_deinit (  
    const rx_gptw_channel_t channel)  [nodiscard]
```

Deinitialize a GPTW channel.

Deinitialize a GPTW channel and release all resources.

Stops the timer, disables both PWM outputs (A and B), and resets the counter to zero. Call only when motor control is no longer active.

Precondition

Motor using this channel must be stopped before calling
channel must be in range [k_gptw_channel_0, k_gptw_channel_3]

Postcondition

Timer counter stopped (GTCR.CST = 0)
Both outputs disabled (GTIOR.OAE = 0, GTIOR.OBE = 0)
Counter register zeroed (GTCNT = k_gptw_period_zero)

Invariant

Write protection re-locked (GTWP) on all exit paths

Note

Not thread-safe: modifies hardware registers directly
Does not clear s_gptw_initialized[] or s_gptw_period[]

Warning

Ensure motor is at rest before calling to prevent abrupt stop

```
rx_err_t err = rx_gptw_deinit(k_gptw_channel_0);
```

See also

[rx_gptw_init_all_staggered\(\)](#) Re-initialize after deinit
[rx_gptw_stop\(\)](#) Stop without full teardown

Since

Version 1.0.0

Definition at line 1846 of file [rx_gptw.c](#).

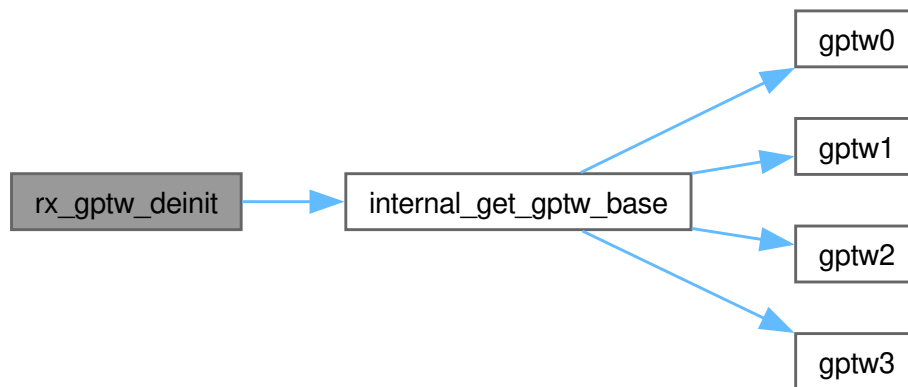
```

01847 {
01848     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01849         return k_rx_err_invalid_arg;
01850     }
01851
01852     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01853     if (gptw == nullptr) {
01854         return k_rx_err_invalid_arg;
01855     }
01856
01857     /* Unlock write protection */
01858     gptw->gtwp = k_gptw_gtwp_unlock;
01859
01860     /* Stop counter */
01861     gptw->gtcr &= ~k_gptw_gtcr_cst;
01862
01863     /* Disable both outputs (OAE and OBE) */
01864     gptw->gtior &= ~(k_gptw_gtior_oae | k_gptw_gtior_obe);
01865
01866     /* Reset counter to zero */
01867     gptw->gtcnt = k_gptw_period_zero;
01868
01869     /* Re-lock write protection */
01870     gptw->gtwp = k_gptw_gtwp_lock;
01871
01872     return k_rx_ok;
01873 }

```

References [rx_gptw_channel_regs_t::gtcnt](#), [rx_gptw_channel_regs_t::gtcr](#), [rx_gptw_channel_regs_t::gtior](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtcr_cst](#), [k_gptw_gtior_oae](#), [k_gptw_gtior_obe](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), [k_gptw_max_channels](#), and [k_gptw_period_zero](#).

Here is the call graph for this function:



8.117.3.30 rx_gptw_enable_output()

```

rx_err_t rx_gptw_enable_output (
    const rx_gptw_channel_t channel,
    const rx_gptw_output_t output,
    bool enable) [nodiscard]

```

Enable or disable GPTW output pin.

Enable or disable a PWM output pin.

Implementation of [rx_gptw_enable_output\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Controls the Output Enable bits (OAE/OBE) in the GTIOR register. When disabled, the corresponding output pin is held in its initial state (typically LOW).

8.117.3.31 Register Modification

Output	Bit Modified	Effect
Output A	GTIOR.OAE	Enable/disable GTIOCnA pin
Output B	GTIOR.OBE	Enable/disable GTIOCnB pin

Precondition

Channel initialized via rx_gptw_init_*()

Postcondition

GTIOR.OAE or GTIOR.OBE modified

Write protection re-enabled

Note

Not thread-safe: Modifies hardware registers

Timer continues running even when output disabled

Warning

Disable outputs before motor emergency stop to ensure safe state

See also

[rx_gptw.h](#) Complete API documentation

Definition at line 1286 of file rx_gptw.c.

```

01287 {
01288     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01289         return k_rx_err_invalid_state;
01290     }
01291
01292     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01293     if (gptw == nullptr) {
01294         return k_rx_err_invalid_arg;
01295     }
01296
01297     /* Unlock for modification */
01298     gptw->gtwp = k_gptw_gtwp_unlock;
01299
01300     /* Modify GTIOR to enable/disable output */
01301     switch (output) {
01302         case k_gptw_output_a:
01303             if (enable) {
01304                 gptw->gtior |= k_gptw_gtior_oae;
01305             } else {
01306                 gptw->gtior &= ~k_gptw_gtior_oae;
01307             }
01308             break;
01309         case k_gptw_output_b:
01310             if (enable) {
01311                 gptw->gtior |= k_gptw_gtior_obe;
01312             } else {
01313                 gptw->gtior &= ~k_gptw_gtior_obe;
01314             }
01315             break;
01316         default:
01317             gptw->gtwp = k_gptw_gtwp_lock;
01318             return k_rx_err_invalid_arg;

```

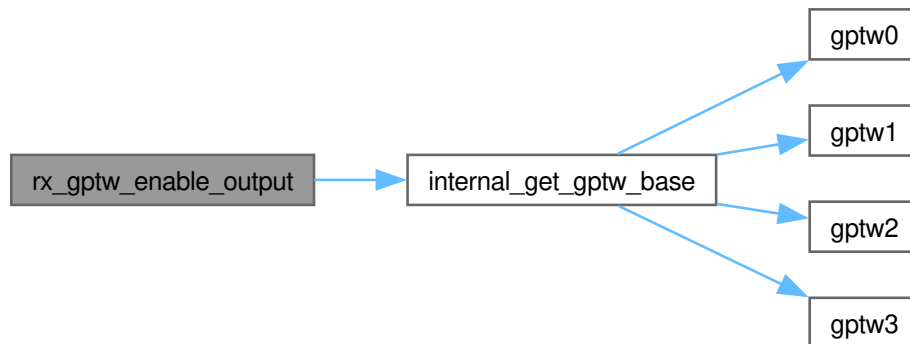
```

01319 }
01320
01321 gptw->gtpw = k_gptw_gtpw_lock;
01322
01323 return k_rx_ok;
01324 }

```

References [rx_gptw_channel_regs_t::gtior](#), [rx_gptw_channel_regs_t::gtpw](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtior_oae](#), [k_gptw_gtior_obe](#), [k_gptw_gtpw_lock](#), [k_gptw_gtpw_unlock](#), [k_gptw_max_channels](#), [k_gptw_output_a](#), [k_gptw_output_b](#), and [s_gptw_initialized](#).

Here is the call graph for this function:



8.117.3.32 rx_gptw_get_duty()

```

rx_err_t rx_gptw_get_duty (
    const rx_gptw_channel_t channel,
    const rx_gptw_output_t output,
    float * duty_percent) [nodiscard]

```

Read the current PWM duty cycle as a floating-point percentage.

Get current duty cycle as a floating-point percentage.

Reads the current compare register value (GTCCRA or GTCCRB) from hardware and converts it to a percentage using the cached period from `s_gptw_period[]`.

Algorithm:

$$\text{duty_percent} = \frac{\text{duty_count} \times 100}{\text{period}}$$

Precondition

- Channel must be initialized via `rx_gptw_init_*`
- `duty_percent` must point to valid float storage

Postcondition

- `*duty_percent` contains duty cycle in `[0.0, 100.0]` on success
- Hardware registers unchanged

Note

Returns actual register value; may differ from the last set value if buffer transfer has not yet occurred at period boundary

Not thread-safe: concurrent `rx_gptw_set_duty` calls may cause torn reads

```
float duty;
rx_err_t err = rx_gptw_get_duty(k_gptw_channel_0, k_gptw_output_a, &duty);
```

See also

[rx_gptw_set_duty\(\)](#) Set duty using floating-point percentage

[rx_gptw_get_period\(\)](#) Retrieve period count

Since

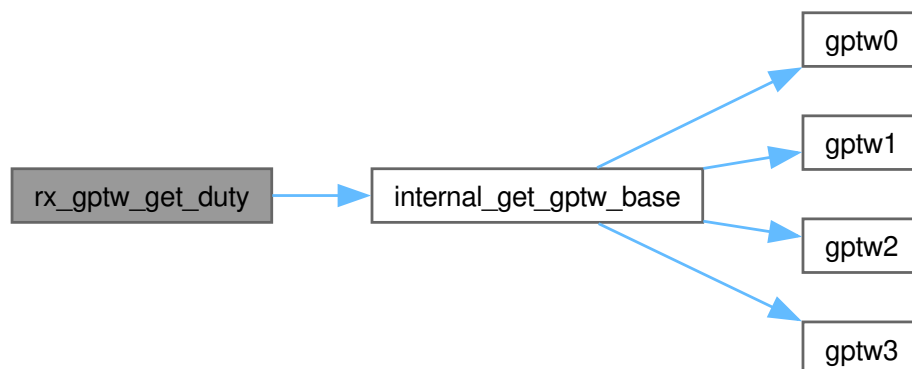
Version 1.0.0

Definition at line 1188 of file [rx_gptw.c](#).

```
01191 {
01192     RX_CHECK_NULL_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
01193
01194     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01195         return k_rx_err_invalid_state;
01196     }
01197
01198     const volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01199     if (gptw == nullptr) {
01200         return k_rx_err_invalid_arg;
01201     }
01202
01203     const uint32_t period = s_gptw_period[channel];
01204     uint32_t duty_count = k_gptw_period_zero;
01205
01206     switch (output) {
01207     case k_gptw_output_a:
01208         duty_count = gptw->gtccra;
01209         break;
01210     case k_gptw_output_b:
01211         duty_count = gptw->gtccrb;
01212         break;
01213     default:
01214         return k_rx_err_invalid_arg;
01215     }
01216
01217     *duty_percent = ((float)duty_count * (float)k_gptw_duty_max) / (float)period;
01218
01219     return k_rx_ok;
01220 }
```

References [rx_gptw_channel_regs_t::gtccra](#), [rx_gptw_channel_regs_t::gtccrb](#), [internal_get_gptw_base\(\)](#), [k_gptw_duty_max](#), [k_gptw_max_channels](#), [k_gptw_output_a](#), [k_gptw_output_b](#), [k_gptw_period_zero](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

Here is the call graph for this function:



8.117.3.33 rx_gptw_get_period()

```
rx_err_t rx_gptw_get_period (
    const rx_gptw_channel_t channel,
    uint32_t * period_count)  [nodiscard]
```

Get PWM period count for GPTW channel.

Get the 32-bit PWM period count for a GPTW channel.

Implementation of [rx_gptw_get_period\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Returns the cached period value from `s_gptw_period[]`, which matches the GTPR register value set during initialization.

Precondition

Channel initialized via `rx_gptw_init_*`()
`period_count` points to valid `uint32_t` storage

Postcondition

`*period_count` contains period value on success

Note

Thread-safe: Reads from cached value, not hardware
 Useful for calculating raw duty counts from percentages

See also

[rx_gptw.h](#) Complete API documentation

Definition at line 1243 of file [rx_gptw.c](#).

```
01244 {
01245     RX_CHECK_NULL_PTR(period_count, s_tag, "period_count pointer is nullptr");
01246
01247     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01248         return k_rx_err_invalid_state;
01249     }
01250
01251     *period_count = s_gptw_period[channel];
01252     return k_rx_ok;
01253 }
```

References [k_gptw_max_channels](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

8.117.3.34 rx_gptw_init_all_staggered()

```
rx_err_t rx_gptw_init_all_staggered (
    const rx_gptw_config_t * configs[k_rx_gptw_channel_count])  [nodiscard]
```

Initialize all 4 GPTW channels with 90-degree phase staggering.

Implementation of [rx_gptw_init_all_staggered\(\)](#) - see [rx_gptw.h](#) for complete documentation with diagrams, examples, and NASA compliance notes.

8.117.3.35 Implementation Overview

1. Validate inputs: NULL check, frequency > 0, valid wave_mode
2. Enable module clock: Clear MSTPCRA.MSTPA7
3. Stop all channels: Atomic stop via GTSTPA register
4. Calculate period: Common period for all channels
5. Configure each channel: Loop calls [internal_configure_channel_staggered\(\)](#)
6. Start all channels: Atomic start via GTSTRA register

8.117.3.36 Phase Staggering Benefits

- Reduces peak current by spreading PWM edges across period
- Lowers EMI from simultaneous switching transients
- Allows smaller power supply capacitors

See also

[rx_gptw.h](#) Complete API documentation with diagrams and examples

Definition at line 1747 of file [rx_gptw.c](#).

```

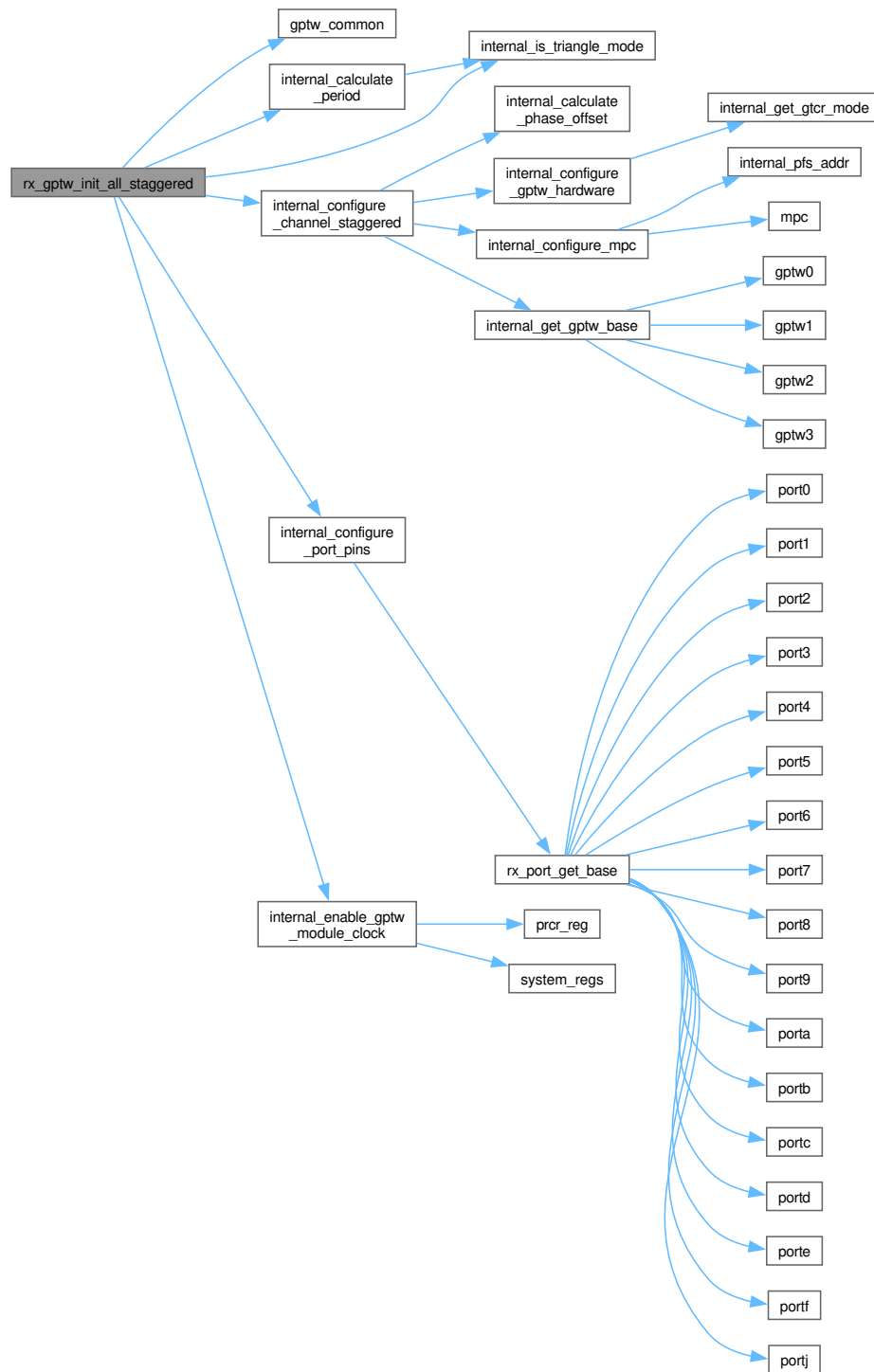
01748 {
01749     RX_CHECK_NULL_PTR(configs, s_tag, "configs array pointer is nullptr");
01750     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01751         RX_CHECK_NULL_PTR(configs[i], s_tag, "configs[i] pointer is nullptr");
01752     }
01753
01754     /* All channels must share frequency / wave_mode (otherwise the staggered
01755      * phasing math doesn't work). Validate once on configs[0]. */
01756     if (configs[0]->frequency_hz == 0) {
01757         rx_log_error(s_tag, "frequency_hz is zero");
01758         return k_rx_err_invalid_arg;
01759     }
01760     switch (configs[0]->wave_mode) {
01761         case k_gptw_wave_saw_pwm:
01762         case k_gptw_wave_tri_pwm1:
01763         case k_gptw_wave_tri_pwm2:
01764         case k_gptw_wave_tri_pwm3:
01765             break;
01766         default:
01767             rx_log_error(s_tag, "Invalid wave_mode");
01768             return k_rx_err_invalid_arg;
01769     }
01770
01771     rx_log_info(s_tag, "Initializing all GPTW channels (staggered)");
01772
01773     internal_enable_gptw_module_clock();
01774
01775     if (gptw_common() != nullptr) {
01776         gptw_common()->gtstpa = k_gptw_all_channels_mask; /* Stop Ch0-3 */
01777     }
01778
01779     uint32_t period;
01780     rx_err_t err =
01781         internal_calculate_period(configs[0]->frequency_hz, configs[0]->wave_mode, &period);
01782     if (err != k_rx_ok) {
01783         return err;
01784     }
01785
01786     const bool is_triangle = internal_is_triangle_mode(configs[0]->wave_mode);
01787     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01788         err =
01789             internal_configure_channel_staggered((rx_gptw_channel_t)i, configs[i], period, is_triangle);
01790         if (err != k_rx_ok) {
01791             return err;
01792         }
01793     }

```

```
01793     }
01794 }
01795
01796 /* Atomic start of all 4 channels. */
01797 if (gptw_common() == nullptr) {
01798     return k_rx_err_hw_error;
01799 }
01800 rx_log_info(s_tag, "Global start of GPTW channels");
01801 gptw_common()->gtstra = k_gptw_all_channels_mask;
01802
01803 /* Now that valid PWM is running on every channel, route the outputs
01804  * to their pads (PMR=1). Doing this after start avoids exposing the
01805  * downstream H-bridges to the GTIOR reset/initial state during the
01806  * brief window between mux and counter start. */
01807 for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01808     internal_configure_port_pins(configs[i]);
01809 }
01810
01811 return k_rx_ok;
01812 }
```

References [gptw_common\(\)](#), [rx_gptw_common_regs_t::gtstpa](#), [rx_gptw_common_regs_t::gtstra](#), [internal_calculate_period\(\)](#), [internal_configure_channel_staggered\(\)](#), [internal_configure_port_pins\(\)](#), [internal_enable_gptw_module_clock\(\)](#), [internal_is_triangle_mode\(\)](#), [k_gptw_all_channels_mask](#), [k_gptw_max_channels](#), [k_gptw_wave_saw_pwm](#), [k_gptw_wave_tri_pwm1](#), [k_gptw_wave_tri_pwm2](#), [k_gptw_wave_tri_pwm3](#), [k_rx_gptw_channel_count](#), and [s_tag](#).

Here is the call graph for this function:



8.117.3.37 rx`gptw`init`pwm()

```

rx_err_t rx_gptw_init_pwm (
    const rx_gptw_channel_t channel,
    const rx_gptw_config_t * config) [nodiscard]

```

Initialize a single GPTW channel for PWM operation.

Initialize a single GPTW channel for PWM output.

Configures one GPTW channel for PWM generation. This function performs complete hardware setup including module clock enable, MPC configuration, port pin setup, timer configuration, and automatic timer start. For 4-motor systems, prefer [rx_gptw_init_all_staggered\(\)](#) instead.

8.117.3.38 Initialization Sequence

1. Validate config pointer (RX_CHECK_NULL_PTR)
2. Validate channel range [0-3]
3. Get channel register base address
4. Calculate period from frequency and wave mode
5. Enable GPTW module clock (MSTPCRA.MSTPA7)
6. Stop timer if already running
7. Configure GPTW hardware registers (GTCR, GTIOR, GTPR, etc.)
8. Configure MPC for Port E alternate function
9. Configure port pins (PMR, PDR)
10. Store period value and set initialized flag
11. Start timer

Precondition

System clock configured (PCLKA = 120 MHz)
Port E pins not used by other peripherals

Postcondition

GPTW channel configured and running
Duty cycle initialized to 0%
MPC and port pins configured

Invariant

s_gptw_initialized[channel] == true on success
s_gptw_period[channel] contains valid period on success

Note

Not thread-safe: Call once during system initialization
Timer starts automatically - use [rx_gptw_stop\(\)](#) to halt if needed

Warning

Calling while channel is running will cause glitches

Attention

For 4-motor systems, use [rx_gptw_init_all_staggered\(\)](#) instead

Performance:

- Execution time: ~50 us at 240 MHz
- Memory: 0 bytes heap, ~32 bytes stack

Example (Single Channel for LED PWM):

```
rx_gptw_config_t led_config = {
    .frequency_hz = 1000,
    .deadtime_ns = 0,
    .wave_mode = k_gptw_wave_saw_pwm,
    .enable_complementary = false,
    .invert_polarity = false
};

rx_err_t err = rx_gptw_init_pwm(k_gptw_channel_0, &led_config);
if (err == k_rx_ok) {
    rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                    rx_gptw_output_id(k_gptw_output_a), 50.0f);
}
```

See also

[rx_gptw_init_all_staggered\(\)](#) Preferred for 4-motor systems
[rx_gptw_deinit\(\)](#) Cleanup function
[rx_gptw_config_t](#) Configuration structure

Since

Version 1.0.0

NASA Power of 10 Compliance:

- Rule 5: [OK] nullptr check, channel range check
- Rule 7: [OK] All internal function returns checked

Definition at line 958 of file [rx_gptw.c](#).

```
00959 {
00960     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00961     volatile rx_gptw_channel_regs_t* gptw = nullptr;
00962     uint32_t period = 0U;
00963     rx_err_t err = internal_prepare_gptw_pwm_init(channel, config, &gptw, &period);
00964     if (err != k_rx_ok) {
00965         return err;
00966     }
00967 }
00968
00969 rx_log_info(s_tag, "Initializing GPTW");
00970
00971 /* Enable GPTW module clock */
00972 internal_enable_gptw_module_clock();
00973
00974 /* Stop timer before configuration */
00975 err = rx_gptw_stop(channel);
00976 if (err != k_rx_ok) {
00977     rx_log_error(s_tag, "Failed to stop GPTW");
00978     return err;
00979 }
00980
00981 /* Configure GPTW hardware registers */
00982 err = internal_configure_gptw_hardware(gptw, config, period);
00983 if (err != k_rx_ok) {
00984     rx_log_error(s_tag, "Failed to configure GPTW hardware");
00985     return err;
00986 }
00987
00988 /* Write the PFS (peripheral function select) for the GTIOC pads. Pin
```

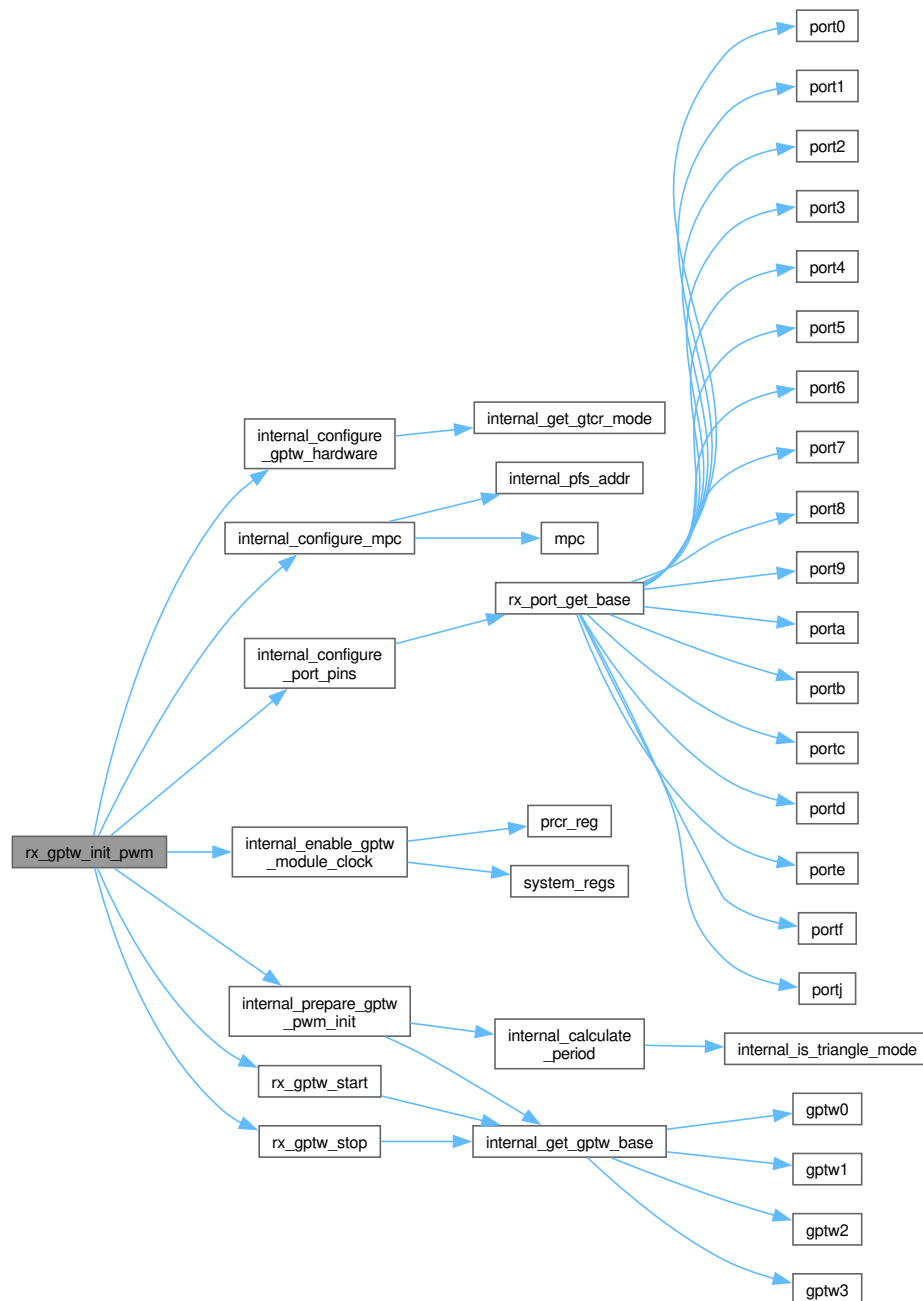
```

00989  * is still in GPIO mode at this point -- PMR is set later. */
00990  err = internal_configure_mpc(config);
00991  if (err != k_rx_ok) {
00992      rx_log_error(s_tag, "Failed to configure MPC");
00993      return err;
00994  }
00995
00996  /* Save period for duty cycle calculations */
00997  s_gptw_period[channel] = period;
00998  s_gptw_initialized[channel] = true;
00999
01000  /* Start timer FIRST -- the GTIOC output must be producing valid PWM
01001  * before we route it to the pad. Otherwise the brief window between
01002  * PMR=1 and counter start exposes downstream hardware (e.g. DRV8263H
01003  * H-bridge inputs) to whatever GTIOR's reset/initial state happens to
01004  * be, which has been observed to latch driver faults. */
01005  err = rx_gptw_start(channel);
01006  if (err != k_rx_ok) {
01007      rx_log_error(s_tag, "Failed to start GPTW");
01008      return err;
01009  }
01010
01011  /* Now route the running PWM to the pad: PDR=1 (output) and PMR=1
01012  * (peripheral mode). The pin transitions from GPIO straight to a
01013  * fully-running, valid PWM signal -- no glitch window. */
01014  internal_configure_port_pins(config);
01015
01016  rx_log_info(s_tag, "GPTW initialized successfully");
01017
01018  return k_rx_ok;
01019 }

```

References [internal_configure_gptw_hardware\(\)](#), [internal_configure_mpc\(\)](#), [internal_configure_port_pins\(\)](#), [internal_enable_gptw_module_clock\(\)](#), [internal_prepare_gptw_pwm_init\(\)](#), [rx_gptw_start\(\)](#), [rx_gptw_stop\(\)](#), [s_gptw_initialized](#), [s_gptw_period](#), and [s_tag](#).

Here is the call graph for this function:



8.117.3.39 rx'gptw'set'duty()

```

rx_err_t rx_gptw_set_duty (
    const rx_gptw_channel_id_t channel,
    const rx_gptw_output_id_t output,
    const float duty_percent) [nodiscard]

```

Set PWM duty cycle using floating-point percentage.

Set PWM duty cycle (floating-point percentage).

Converts floating-point duty percentage [0.0, 100.0] to raw count value and delegates to [rx_gptw_set_duty_raw\(\)](#) for hardware register update. The compare register (GTCCR or GTCCRB) is written and the change takes effect on the next PWM period boundary (glitch-free via buffer mode).

Algorithm:

$$\text{duty_count} = \left\lfloor \frac{\text{duty_percent} \times \text{period}}{100} \right\rfloor$$

Precondition

Channel must be initialized via `rx_gptw_init_*`()
 duty_percent must be in `[0.0, 100.0]`

Postcondition

Compare register written; change takes effect next period boundary
 Motor output drive updated on next PWM cycle

Note

Not thread-safe: concurrent writes to same output cause undefined behavior

```
rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
                rx_gptw_output_id(k_gptw_output_a), 75.0f);
```

See also

[rx_gptw_set_duty_raw\(\)](#) Set duty using raw count value
[rx_gptw_get_duty\(\)](#) Read current duty cycle

Since

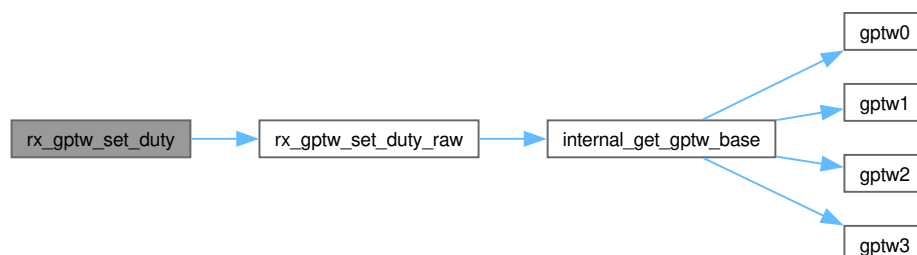
Version 1.0.0

Definition at line 1055 of file `rx_gptw.c`.

```
01058 {
01059     const rx_gptw_channel_t channel_value = channel.value;
01060     const rx_gptw_output_t output_value = output.value;
01061
01062     if (((int32_t)channel_value >= (int32_t)k_gptw_max_channels) ||
01063         !s_gptw_initialized[channel_value]) {
01064         return k_rx_err_invalid_state;
01065     }
01066
01067     if (duty_percent < (float)k_gptw_duty_min || duty_percent > (float)k_gptw_duty_max) {
01068         rx_log_error(s_tag, "Invalid duty cycle");
01069         return k_rx_err_invalid_arg;
01070     }
01071
01072     /* Convert percentage to count value */
01073     const uint32_t period = s_gptw_period[channel_value];
01074     const float period_f = (float)period;
01075     const float duty_count_f = (duty_percent * period_f) / (float)k_gptw_duty_divisor;
01076     const uint32_t duty_count = (uint32_t)duty_count_f;
01077
01078     return rx_gptw_set_duty_raw(channel_value, output_value, duty_count);
01079 }
```

References [k_gptw_duty_divisor](#), [k_gptw_duty_max](#), [k_gptw_duty_min](#), [k_gptw_max_channels](#), [rx_gptw_set_duty_raw\(\)](#), [s_gptw_initialized](#), [s_gptw_period](#), [s_tag](#), [rx_gptw_channel_id_t::value](#), and [rx_gptw_output_id_t::value](#).

Here is the call graph for this function:



8.117.3.40 rx`gptw`set`duty`raw()

```
rx_err_t rx_gptw_set_duty_raw (
    const rx_gptw_channel_t channel,
    rx_gptw_output_t output,
    uint32_t duty_count) [nodiscard]
```

Set PWM duty cycle using a raw 32-bit timer count.

Set PWM duty cycle using raw counter value.

Directly writes duty_count to the compare register (GTCCRA or GTCCRB). Buffer mode (enabled during init via GTBER) ensures the update takes effect on the next PWM period boundary without glitches. Values exceeding the period are clamped to period (100% duty).

Register Access:

- Output A: Writes to GTCCRA
- Output B: Writes to GTCCRB

Precondition

Channel must be initialized via rx_gptw_init_*

duty_count should be in [0, period] for meaningful duty; higher values clamped

Postcondition

GTCCRA or GTCCRB written with clamped duty_count

Change takes effect at next PWM period boundary

Note

Not thread-safe: concurrent writes to the same register cause undefined behavior

Values exceeding period are silently clamped to period

```
uint32_t period;
rx_gptw_get_period(k_gptw_channel_0, &period);
rx_gptw_set_duty_raw(k_gptw_channel_0, k_gptw_output_a, period / 2U); // 50%
```

See also

[rx_gptw_set_duty\(\)](#) Set duty using floating-point percentage

[rx_gptw_get_period\(\)](#) Retrieve period count for duty calculation

Since

Version 1.0.0

Definition at line 1117 of file `rx_gptw.c`.

```

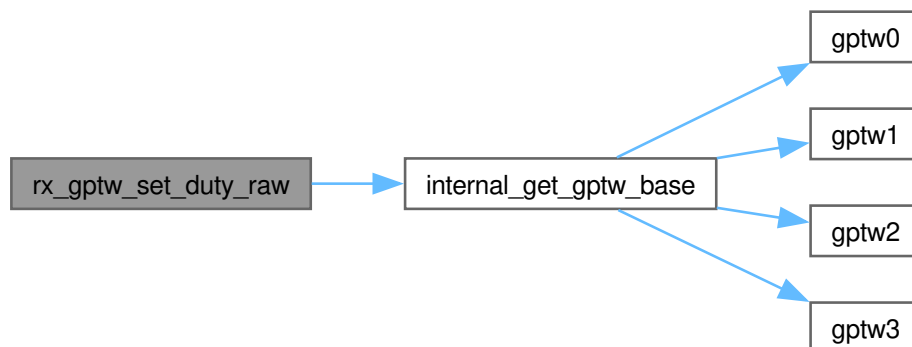
01118 {
01119     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01120         return k_rx_err_invalid_state;
01121     }
01122
01123     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01124     if (gptw == nullptr) {
01125         return k_rx_err_invalid_arg;
01126     }
01127
01128     /* Clamp to period */
01129     const uint32_t period = s_gptw_period[channel];
01130     if (duty_count > period) {
01131         duty_count = period;
01132     }
01133
01134     /* GTCCRA/GTCCRB are protected by GTWP.WP on RX72N. Unlock, write, re-lock.
01135      * With GTBER=0 (default after our fixed init), the written value takes
01136      * effect immediately rather than being double-buffered to the next period. */
01137     gptw->gtwp = k_gptw_gtwp_unlock;
01138     switch (output) {
01139     case k_gptw_output_a:
01140         gptw->gtccra = duty_count;
01141         break;
01142     case k_gptw_output_b:
01143         gptw->gtccrb = duty_count;
01144         break;
01145     default:
01146         gptw->gtwp = k_gptw_gtwp_lock;
01147         return k_rx_err_invalid_arg;
01148     }
01149     gptw->gtwp = k_gptw_gtwp_lock;
01150
01151     return k_rx_ok;
01152 }

```

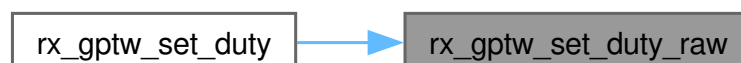
References `rx_gptw_channel_regs_t::gtccra`, `rx_gptw_channel_regs_t::gtccrb`, `rx_gptw_channel_regs_t::gtwp`, `internal_get_gptw_base()`, `k_gptw_gtwp_lock`, `k_gptw_gtwp_unlock`, `k_gptw_max_channels`, `k_gptw_output_a`, `k_gptw_output_b`, `s_gptw_initialized`, and `s_gptw_period`.

Referenced by `rx_gptw_set_duty()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.41 rx`gptw`start()

```
rx_err_t rx_gptw_start (
    const rx_gptw_channel_t channel) [nodiscard]
```

Start GPTW counter (begin PWM generation).

Start the GPTW timer counter for PWM generation.

Implementation of [rx_gptw_start\(\)](#) - see [rx_gptw.h](#) for complete documentation.

Sets the Count Start (CST) bit in GTCR to begin counter operation. The timer will count according to the configured waveform mode (sawtooth or triangle) and generate PWM output on the enabled pins.

8.117.3.42 Register Access

1. Unlock write protection (GTWP)
2. Set GTCR.CST bit
3. Lock write protection (GTWP)

Precondition

Channel configured via [rx_gptw_init_*](#)() (recommended)

Postcondition

Timer counting actively
PWM output generated on enabled pins

Note

Does NOT require channel to be "initialized" (no flag check)
Timer automatically started during [rx_gptw_init_*](#)()
Idempotent: Safe to call when already running

See also

[rx_gptw_stop\(\)](#) Stop the timer
[rx_gptw.h](#) Complete API documentation

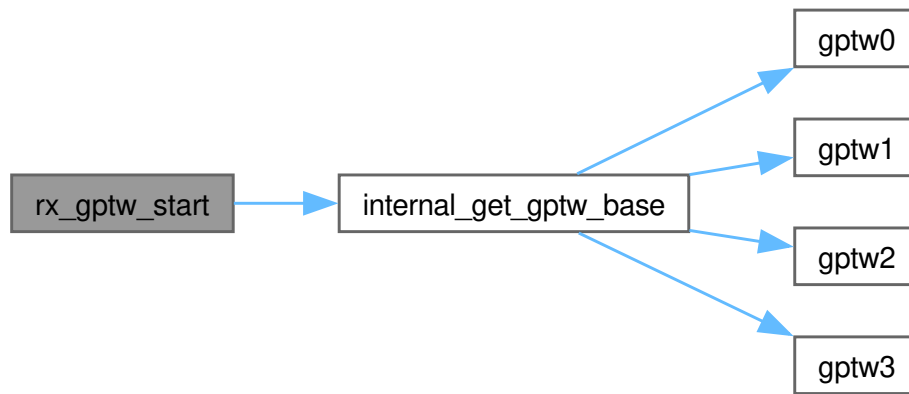
Definition at line 1356 of file [rx_gptw.c](#).

```
01357 {
01358     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01359         return k_rx_err_invalid_arg;
01360     }
01361
01362     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01363     if (gptw == nullptr) {
01364         return k_rx_err_invalid_arg;
01365     }
01366
01367     /* Unlock and set CST bit in GTCR */
01368     gptw->gtwp = k_gptw_gtwp_unlock;
01369     gptw->gtcr |= k_gptw_gtcr_cst;
01370     gptw->gtwp = k_gptw_gtwp_lock;
01371
01372     return k_rx_ok;
01373 }
```

References [rx_gptw_channel_regs_t::gtcr](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtcr_cst](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), and [k_gptw_max_channels](#).

Referenced by [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.3.43 rx_gptw_stop()

```
rx_err_t rx_gptw_stop (
    const rx_gptw_channel_t channel) [nodiscard]
```

Stop GPTW counter (halt PWM generation).

Stop the GPTW timer counter.

Implementation of `rx_gptw_stop()` - see [rx_gptw.h](#) for complete documentation.

Clears the Count Start (CST) bit in GTCR to halt counter operation. The timer counter value (GTCNT) is preserved and PWM outputs hold their current state.

8.117.3.44 Register Access

1. Unlock write protection (GTWP)
2. Clear GTCR.CST bit
3. Lock write protection (GTWP)

Precondition

None (can be called on uninitialized channel)

Postcondition

Timer stopped (GTCR.CST = 0)
 Counter value preserved (GTCNT unchanged)
 Output pins hold last state

Note

Does NOT require channel to be "initialized" (no flag check)

Idempotent: Safe to call when already stopped

For safety shutdown, also disable outputs via [rx_gptw_enable_output\(\)](#)

Warning

Outputs may remain HIGH when stopped - disable outputs for safe state

See also

[rx_gptw_start\(\)](#) Resume the timer

[rx_gptw_enable_output\(\)](#) Disable outputs for safe state

[rx_gptw.h](#) Complete API documentation

Definition at line 1408 of file [rx_gptw.c](#).

```

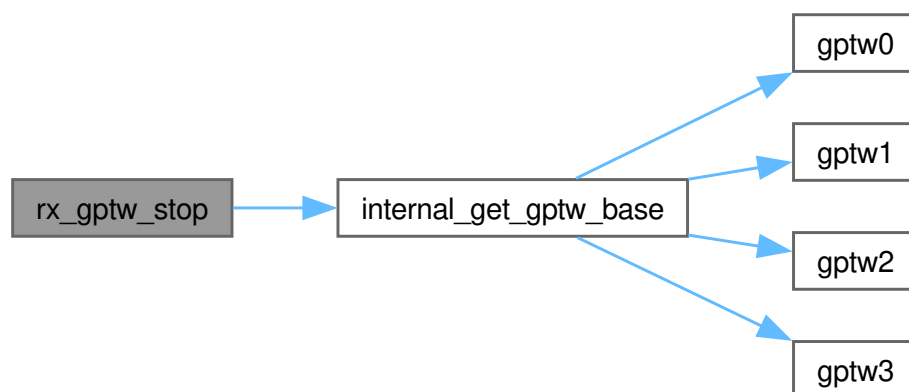
01409 {
01410     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01411         return k_rx_err_invalid_arg;
01412     }
01413
01414     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01415     if (gptw == nullptr) {
01416         return k_rx_err_invalid_arg;
01417     }
01418
01419     /* Unlock and clear CST bit in GTCR */
01420     gptw->gtwp = k_gptw_gtwp_unlock;
01421     gptw->gtcr &= ~k_gptw_gtcr_cst;
01422     gptw->gtwp = k_gptw_gtwp_lock;
01423
01424     return k_rx_ok;
01425 }

```

References [rx_gptw_channel_regs_t::gtcr](#), [rx_gptw_channel_regs_t::gtwp](#), [internal_get_gptw_base\(\)](#), [k_gptw_gtcr_cst](#), [k_gptw_gtwp_lock](#), [k_gptw_gtwp_unlock](#), and [k_gptw_max_channels](#).

Referenced by [rx_gptw_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.117.4 Variable Documentation

8.117.4.1 s'gptw'initialized

```
bool s_gptw_initialized[k_gptw_max_channels] = {false} [static]
```

Track initialized channels.

Definition at line 210 of file [rx_gptw.c](#).

```
00210 {false};
```

Referenced by [internal_configure_channel_staggered\(\)](#), [rx_gptw_enable_output\(\)](#), [rx_gptw_get_duty\(\)](#), [rx_gptw_get_period\(\)](#), [rx_gptw_init_pwm\(\)](#), [rx_gptw_set_duty\(\)](#), and [rx_gptw_set_duty_raw\(\)](#).

8.117.4.2 s'gptw'ns'per'second

```
const uint64_t s_gptw_ns_per_second = 1000000000ULL [static]
```

Nanoseconds per second

Definition at line 146 of file [rx_gptw.c](#).

Referenced by [internal_configure_gptw_hardware\(\)](#).

8.117.4.3 s'gptw'period

```
uint32_t s_gptw_period[k_gptw_max_channels] = {} [static]
```

Period values for each channel.

Definition at line 212 of file [rx_gptw.c](#).

```
00212 {};
```

Referenced by [internal_configure_channel_staggered\(\)](#), [rx_gptw_get_duty\(\)](#), [rx_gptw_get_period\(\)](#), [rx_gptw_init_pwm\(\)](#), [rx_gptw_set_duty\(\)](#), and [rx_gptw_set_duty_raw\(\)](#).

8.117.4.4 s'gptw'period'max

```
const uint32_t s_gptw_period_max = 0xFFFFFFFFFUL [static]
```

Period calculation constants.

Maximum valid period (32-bit)

Definition at line 145 of file [rx_gptw.c](#).

Referenced by [internal_calculate_period\(\)](#), and [internal_configure_gptw_hardware\(\)](#).

8.117.4.5 s'tag

```
const char* const s_tag = "GPTW" [static]
```

Definition at line 131 of file [rx_gptw.c](#).

8.118 rx_gptw.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_gptw.c
00003  * @brief GPTW PWM Driver Implementation for Motor Control
00004  *
00005  * @details
00006  * **Production-grade implementation** of the General PWM Timer (GPTW) driver for
00007  * RX72N motor control applications. This file contains all internal helper
00008  * functions and public API implementations for GPTW peripheral management.
00009  *
00010  * ## Module Architecture
00011  *
00012  * The implementation is structured in three layers:
00013  * 1. **Constants & Configuration**: Typed enums for all magic numbers
00014  * 2. **Internal Helpers**: Register access, calculations, hardware configuration
00015  * 3. **Public API**: User-facing functions documented in rx_gptw.h
00016  *
00017  * ## Supported PWM Modes
00018  *
00019  * **1. Sawtooth-wave PWM (edge-aligned)**:
00020  * - Counter: 0 -> GTPR -> 0 (reset at top)
00021  * - Period = PCLKA / frequency
00022  * - For 20kHz: period = 120MHz / 20kHz = 6000 counts (~12.5-bit resolution)
00023  * - Single update point at counter reset
00024  * - Best for: Simple PWM, LED dimming
00025  *
00026  * **2. Triangle-wave PWM (center-aligned)**:
00027  * - Counter: 0 -> GTPR -> 0 (up/down counting)
00028  * - Period = PCLKA / (2 x frequency)
00029  * - For 20kHz: period = 120MHz / 40kHz = 3000 counts (~11.5-bit resolution)
00030  * - Update at crest and/or trough (mode dependent)
00031  * - Lower EMI due to symmetric switching edges
00032  * - Reduced current ripple in motor windings
00033  * - Best for: Motor control with H-bridge
00034  *
00035  * ## Implementation Notes
00036  *
00037  * **Register Access Pattern**:
00038  * All GPTW register modifications follow this sequence:
00039  * 1. Unlock write protection (GTWP = 0xA500_0000)
00040  * 2. Modify registers
00041  * 3. Lock write protection (GTWP = 0x0000_0001)
00042  *
00043  * **Phase Staggering Implementation**:
00044  * For 4-motor systems, channels are initialized with counter offsets:
00045  * - Ch0: GTCNT = 0 (0deg reference)
00046  * - Ch1: GTCNT = period/4 (90deg offset)
00047  * - Ch2: GTCNT = period/2 (180deg offset)
00048  * - Ch3: GTCNT = 3xperiod/4 (270deg offset)
00049  *
00050  * ## Memory Map
00051  *
00052  * @par GPTW Register Base Addresses:
00053  *
00054  * | Module | Base Address | Size | Description |
00055  * |-----|-----|-----|-----|
00056  * | GPTW Common | 0x000C2000 | 0x100 | Shared start/stop registers |
00057  * | GPTW0 | 0x000C2100 | 0x80 | Channel 0 registers |
00058  * | GPTW1 | 0x000C2180 | 0x80 | Channel 1 registers |
00059  * | GPTW2 | 0x000C2200 | 0x80 | Channel 2 registers |
00060  * | GPTW3 | 0x000C2280 | 0x80 | Channel 3 registers |
00061  *
00062  * @par Module Clock:
00063  * - MSTPCRA.MSTPA7 controls GPTW module clock on RX72N
00064  * - Must be cleared (= 0) before register access
00065  * - Protected by PRCR register lock (PRC1, key 0xA502)
00066  *
00067  * @par Clock Assumption:
00068  * This driver assumes PCKA = 96 MHz, which matches the STAR firmware
00069  * clock_init: HOCO 16 MHz x12 PLL = 192 MHz, SCKCR=0x21C21211 giving
00070  * PCKA = PLL/2 = 96 MHz. At 20 kHz PWM this yields GTPR = 96M/20k - 1 = 4799.
00071  * If the board ever moves to a different clock plan the HAL must be
00072  * updated (k_pclka_hz in rx72n_clock.h).
00073  *
00074  * ## Static Variables
00075  *
00076  * | Variable | Type | Purpose |
00077  * |-----|-----|-----|
00078  * | s_gptw_initialized[] | bool[4] | Track which channels are initialized |
00079  * | s_gptw_period[] | uint32_t[4] | Cache period values for duty calculation |
00080  * | s_tag | const char* | Log tag "GPTW" |
00081  *
00082  * @par NASA Power of 10 Compliance:

```

```

00083 *
00084 * | Rule | Status | Implementation |
00085 * |-----|-----|-----|
00086 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion in any function |
00087 * | 2. Fixed loop bounds | [OK] | Loops bounded by k_gptw_max_channels (4) |
00088 * | 3. No dynamic allocation | [OK] | All variables static or stack, zero malloc |
00089 * | 4. Small functions | [OK] | All functions < 60 lines |
00090 * | 5. Assertions | [OK] | RX_CHECK_NULL_PTR, switch validation, range checks |
00091 * | 6. Narrow scope | [OK] | Internal functions use static, minimal globals |
00092 * | 7. Check return values | [OK] | All rx_err_t returns propagated or handled |
00093 * | 8. Limited preprocessor | [OK] | C23 typed enums, macros only for RX_CHECK |
00094 * | 9. Pointer restrictions | [OK] | Single-level pointers, no function pointers |
00095 * | 10. Compiler warnings | [OK] | Builds clean with -Wall -Wextra -Werror |
00096
00097 * @par SOLID Principles:
00098 *
00099 * **Single Responsibility**: This file handles ONLY GPTW hardware abstraction.
00100 * Motor control logic, PID calculations, and H-bridge configuration are in
00101 * separate modules.
00102 *
00103 * **Open/Closed**: New waveform modes can be added by extending the
00104 * rx_gptw_wave_mode_t enum and internal_get_gtcr_mode() without modifying
00105 * existing code paths.
00106 *
00107 * @see rx_gptw.h Public API documentation
00108 * @see rx72n_gptw_regs.h Register structure definitions
00109 * @see RX72N Hardware Manual Chapter 26 - General PWM Timer (GPTW)
00110 * @see docs/sections/03_hardware_pinout.tex Pin assignments
00111 *
00112 * @author Locked, Inc.
00113 * @date 2026-01-27
00114 * @copyright Copyright (c) 2026 Locked Inc.
00115 * SPDX-License-Identifier: MIT
00116 * @since Version 1.0.0
00117 */
00118
00119 #ifdef __RX__
00120
00121 #include "rx_gptw.h"
00122
00123 #include <stddef.h>
00124
00125 #include "rx72n_regs.h"
00126 #include "rx_check.h"
00127 #include "rx_log.h"
00128 #include "rx_port_utils.h"
00129 #include "rx_register_protection.h"
00130
00131 static const char* const s_tag = "GPTW";
00132
00133 /*
00134  * =====
00135  * Constants
00136  * =====
00137  */
00138 /** @brief GPTW general constants */
00139 typedef enum : uint8_t {
00140     k_gptw_max_channels = 4, /**< GPTW0-GPTW3 */
00141     k_gptw_outputs_per_channel = 2, /**< GTIOCA and GTIOCB */
00142 } gptw_constants_t;
00143
00144 /** @brief Period calculation constants */
00145 static const uint32_t s_gptw_period_max = 0xFFFFFFFFUL; /**< Maximum valid period (32-bit) */
00146 static const uint64_t s_gptw_ns_per_second = 1000000000ULL; /**< Nanoseconds per second */
00147
00148 typedef enum : uint16_t {
00149     k_gptw_period_divisor_saw = 1, /**< Sawtooth: period = PCLKA / freq */
00150     k_gptw_period_divisor_tri = 2, /**< Triangle: period = PCLKA / (2 * freq) */
00151     k_gptw_period_min = 10, /**< Minimum valid period */
00152     k_gptw_period_zero = 0, /**< Zero period value */
00153     k_gptw_deadtime_disabled = 0, /**< No dead time */
00154 } gptw_period_constants_t;
00155
00156 /** @brief Duty cycle calculation constants */
00157 typedef enum : uint8_t {
00158     k_gptw_duty_min = 0, /**< Minimum duty cycle (0%) */
00159     k_gptw_duty_max = 100, /**< Maximum duty cycle (100%) */
00160     k_gptw_duty_divisor = 100, /**< Divisor for percentage conversion */
00161 } gptw_duty_constants_t;
00162
00163 /** @brief MPC configuration constants */
00164 typedef enum : uint8_t {
00165     k_mpc_pwpr_b0wi_clear = 0x00, /**< Clear B0WI to enable PFSWE write */
00166     k_mpc_pwpr_pfswe_set = 0x40, /**< Set PFSWE to enable PFS write */
00167     k_mpc_pwpr_lock = 0x80, /**< Set B0WI to lock PFS */

```

```

00168 } mpc_pwpr_constants_t;
00169
00170 /** @brief MPC PFS register offset constants */
00171 typedef enum : uint16_t {
00172     k_mpc_pfs_base_offset = 0x40, /**< Base offset for PFS registers */
00173     k_mpc_port_e_index = 0x0E, /**< Port E index */
00174     k_mpc_pfs_port_stride = 8, /**< Bytes between port PFS groups */
00175 } mpc_pfs_offset_constants_t;
00176
00177 /** @brief Single-bit value for shift operations */
00178 typedef enum : uint8_t {
00179     k_gptw_bit_set = 1, /**< Value for single bit in shift operations */
00180 } gptw_bit_constants_t;
00181
00182 /** @brief GPTW channel mask for start/stop operations */
00183 typedef enum : uint8_t {
00184     k_gptw_all_channels_mask = 0x0F, /**< Bitmask for all 4 GPTW channels (Ch0-3) */
00185 } gptw_channel_mask_t;
00186
00187 /** @brief GPTW count direction values for GTUDDTYC register */
00188 typedef enum : uint32_t {
00189     k_gptw_gtuddtyc_ud_down = 0, /**< Count direction down (UD bit = 0) */
00190 } gptw_direction_constants_t;
00191
00192 /** @brief Port E pin indices for GPTW motor outputs */
00193 typedef enum : uint8_t {
00194     k_porte_gptw0_a = 5, /**< PE5/GTIOC0A - Motor 0 phase */
00195     k_porte_gptw0_b = 2, /**< PE2/GTIOC0B - Motor 0 enable */
00196     k_porte_gptw1_a = 4, /**< PE4/GTIOC1A - Motor 1 phase */
00197     k_porte_gptw1_b = 1, /**< PE1/GTIOC1B - Motor 1 enable */
00198     k_porte_gptw2_a = 3, /**< PE3/GTIOC2A - Motor 2 phase */
00199     k_porte_gptw2_b = 0, /**< PE0/GTIOC2B - Motor 2 enable */
00200     k_porte_gptw3_a = 7, /**< PE7/GTIOC3A - Motor 3 phase */
00201     k_porte_gptw3_b = 6, /**< PE6/GTIOC3B - Motor 3 enable */
00202 } porte_gptw_pin_t;
00203
00204 /*
=====
00205 * Static Variables
00206 *
=====
00207 */
00208
00209 /** @brief Track initialized channels */
00210 static bool s_gptw_initialized[k_gptw_max_channels] = {false};
00211 /** @brief Period values for each channel */
00212 static uint32_t s_gptw_period[k_gptw_max_channels] = {};
00213
00214 /*
=====
00215 * Internal Helper Functions
00216 *
=====
00217 */
00218
00219 /**
00220  * @brief Get GPTW channel register base address
00221  *
00222  * @details
00223  * Maps a GPTW channel enumeration value to its corresponding hardware register
00224  * base address. Each GPTW channel (0-3) has a dedicated register block at a
00225  * fixed memory offset.
00226  *
00227  * ## Memory Map
00228  *
00229  * | Channel | Base Address | Offset from GPTW0 |
00230  * |-----|-----|-----|
00231  * | k_gptw_channel_0 | 0x000C2100 | 0x000 |
00232  * | k_gptw_channel_1 | 0x000C2180 | 0x080 |
00233  * | k_gptw_channel_2 | 0x000C2200 | 0x100 |
00234  * | k_gptw_channel_3 | 0x000C2280 | 0x180 |
00235  *
00236  *
00237  *
00238  * @pre Module clock must be enabled before accessing returned pointer
00239  *
00240  * @post Returned pointer is valid for hardware register access
00241  * @post No hardware state modified by this function
00242  *
00243  * @note Thread-safe: Returns constant address, no state modification
00244  * @note Caller must check for nullptr before using returned pointer
00245  *
00246  * @par Performance:
00247  * - Execution time: ~10 cycles (switch statement)
00248  * - Memory: 0 bytes heap, 0 bytes stack allocation
00249  *
00250  * @see gptw0(), gptw1(), gptw2(), gptw3() Underlying accessor functions

```

```

00251 * @see rx72n_gptw_regs.h Register structure definition
00252 *
00253 * @since Version 1.0.0
00254 */
00255 static volatile rx_gptw_channel_regs_t* internal_get_gptw_base(const rx_gptw_channel_t channel)
00256 {
00257     switch (channel) {
00258         case k_gptw_channel_0:
00259             return gptw0();
00260         case k_gptw_channel_1:
00261             return gptw1();
00262         case k_gptw_channel_2:
00263             return gptw2();
00264         case k_gptw_channel_3:
00265             return gptw3();
00266         default:
00267             return nullptr;
00268     }
00269 }
00270
00271 /**
00272 * @brief Check if wave mode is triangle (center-aligned)
00273 *
00274 * @details
00275 * Determines whether the specified waveform mode uses triangle wave (center-aligned)
00276 * PWM generation. Triangle modes count up to period, then down to zero, resulting
00277 * in symmetric PWM pulses centered within each period.
00278 *
00279 * ## Mode Classification
00280 *
00281 * | Mode | Type | Returns |
00282 * |-----|-----|-----|
00283 * | k_gptw_wave_saw_pwm | Sawtooth (edge-aligned) | false |
00284 * | k_gptw_wave_tri_pwm1 | Triangle (center-aligned) | true |
00285 * | k_gptw_wave_tri_pwm2 | Triangle (center-aligned) | true |
00286 * | k_gptw_wave_tri_pwm3 | Triangle (center-aligned) | true |
00287 *
00288 *
00289 *
00290 * @pre mode should be a valid rx_gptw_wave_mode_t value
00291 *
00292 * @post No state modification
00293 *
00294 * @note Pure function: Same input always produces same output
00295 * @note Thread-safe: No shared state access
00296 * @note Inlined for performance in period calculation loop
00297 *
00298 * @par Performance:
00299 * - Execution time: ~3 cycles (two comparisons)
00300 * - Inlined at call site, no function call overhead
00301 *
00302 * @see rx_gptw_wave_mode_t Waveform mode definitions
00303 * @see internal_calculate_period() Primary user of this function
00304 *
00305 * @since Version 1.0.0
00306 */
00307 static inline bool internal_is_triangle_mode(const rx_gptw_wave_mode_t mode)
00308 {
00309     return (mode == k_gptw_wave_tri_pwm1) || (mode == k_gptw_wave_tri_pwm2) ||
00310         (mode == k_gptw_wave_tri_pwm3);
00311 }
00312
00313 /**
00314 * @brief Get GTCR mode bits for wave mode
00315 *
00316 * @details
00317 * Converts a waveform mode enumeration to the corresponding GTCR.MD register
00318 * bit field value. The MD field (bits 18:16) in GTCR controls the timer's
00319 * counting mode and PWM generation behavior.
00320 *
00321 * ## GTCR.MD Field Mapping
00322 *
00323 * | Input Mode | GTCR.MD Value | Binary | Description |
00324 * |-----|-----|-----|-----|
00325 * | k_gptw_wave_saw_pwm | k_gptw_gtcr_md_saw_pwm | 0b000 | Sawtooth PWM |
00326 * | k_gptw_wave_tri_pwm1 | k_gptw_gtcr_md_tri_pwm1 | 0b100 | Triangle PWM 1 |
00327 * | k_gptw_wave_tri_pwm2 | k_gptw_gtcr_md_tri_pwm2 | 0b101 | Triangle PWM 2 |
00328 * | k_gptw_wave_tri_pwm3 | k_gptw_gtcr_md_tri_pwm3 | 0b110 | Triangle PWM 3 |
00329 *
00330 *
00331 *
00332 * @pre mode should be validated before calling (caller's responsibility)
00333 * @pre GTCR write protection must be unlocked before writing returned value
00334 *
00335 * @post No state modification (pure function)
00336 * @post Returned value is ready for GTCR register write
00337 *

```

```

00338 * @invariant Invalid modes always return sawtooth as safe fallback
00339 *
00340 * @note Assert triggers on invalid mode in debug builds
00341 * @note Returns safe default (sawtooth) on invalid mode in release builds
00342 * @note Thread-safe: Pure function with no shared state
00343 *
00344 * @warning RX_ASSERT fires if mode is not a valid enum value
00345 *
00346 * @par Performance:
00347 * - Execution time: ~10 cycles (switch statement)
00348 * - Memory: 0 bytes heap, 0 bytes stack
00349 *
00350 * @par NASA Power of 10 Compliance:
00351 * - Rule 5: [OK] RX_ASSERT validates unexpected conditions
00352 *
00353 * @see rx_gptw_wave_mode_t Input enum definition
00354 * @see RX72N Hardware Manual Section 26.2.12 GTCR register description
00355 *
00356 * @since Version 1.0.0
00357 */
00358 static uint32_t internal_get_gtc_r_mode(const rx_gptw_wave_mode_t mode)
00359 {
00360     switch (mode) {
00361         case k_gptw_wave_tri_pwm1:
00362             return k_gptw_gtc_r_md_tri_pwm1;
00363         case k_gptw_wave_tri_pwm2:
00364             return k_gptw_gtc_r_md_tri_pwm2;
00365         case k_gptw_wave_tri_pwm3:
00366             return k_gptw_gtc_r_md_tri_pwm3;
00367         case k_gptw_wave_saw_pwm:
00368             return k_gptw_gtc_r_md_saw_pwm;
00369         default:
00370             /* Should never reach here if wave_mode was validated upstream (NASA Rule 5) */
00371             RX_ASSERT(false, "Unexpected wave mode in internal_get_gtc_r_mode");
00372             return k_gptw_gtc_r_md_saw_pwm;
00373     }
00374 }
00375
00376 /**
00377 * @brief Calculate period register value from frequency
00378 *
00379 * @details
00380 * Converts a desired PWM frequency to the GTPR (period register) count value.
00381 * The calculation differs based on waveform mode:
00382 *
00383 * ## Mathematical Formulas
00384 *
00385 * **Sawtooth (edge-aligned) mode:**
00386 * @f[
00387 * \text{period} = \frac{f_{\text{PCLKA}}}{f_{\text{PWM}}}
00388 * @f]
00389 *
00390 * **Triangle (center-aligned) mode:**
00391 * @f[
00392 * \text{period} = \frac{f_{\text{PCLKA}}}{2 \times f_{\text{PWM}}}
00393 * @f]
00394 *
00395 * Where:
00396 * - @f$ f_{\text{PCLKA}} $ @f$ = 120 MHz (peripheral clock A)
00397 * - @f$ f_{\text{PWM}} $ @f$ = desired PWM frequency
00398 *
00399 * ## Algorithm Steps
00400 *
00401 * 1. Validate period output pointer (RX_CHECK_NULL_PTR)
00402 * 2. Validate wave_mode via exhaustive switch statement
00403 * 3. Determine divisor based on mode (1 for sawtooth, 2 for triangle)
00404 * 4. Check frequency_hz is non-zero (prevent division by zero)
00405 * 5. Calculate period using 64-bit arithmetic to prevent overflow
00406 * 6. Validate period is within hardware limits [10, 0xFFFFFFFF]
00407 * 7. Store result and return success
00408 *
00409 * ## Example Calculations
00410 *
00411 * | Frequency | Mode | Period Count | Resolution |
00412 * |-----|-----|-----|-----|
00413 * | 20 kHz | Sawtooth | 6000 | 0.0167% |
00414 * | 20 kHz | Triangle | 3000 | 0.0333% |
00415 * | 100 Hz | Sawtooth | 1,200,000 | 0.000083% |
00416 * | 1 MHz | Sawtooth | 120 | 0.833% |
00417 *
00418 *
00419 *
00420 * @pre PCLKA configured to 120 MHz (system clock setup complete)
00421 * @pre period pointer must point to valid uint32_t storage
00422 *
00423 * @post On success, *period contains valid count [10, 0xFFFFFFFF]
00424 * @post On failure, *period is unchanged

```

```

00425 *
00426 * @invariant period >= k_gptw_period_min (10) on success
00427 * @invariant Calculation uses 64-bit math to prevent overflow
00428 *
00429 * @note Thread-safe: No shared state modified
00430 * @note Uses RX_CHECK_NULL_PTR macro for pointer validation
00431 *
00432 * @warning Very low frequencies (< 1 Hz) may overflow 32-bit period
00433 * @warning Very high frequencies (> 12 MHz) produce periods < 10 (rejected)
00434 *
00435 * @par Performance:
00436 * - Execution time: ~50 cycles (64-bit division)
00437 * - Memory: 0 bytes heap, ~16 bytes stack
00438 *
00439 * @par NASA Power of 10 Compliance:
00440 * - Rule 5: [OK] 2 preconditions (NULL check, wave_mode validation)
00441 * - Rule 7: [OK] Returns error code, no silent failures
00442 *
00443 * @see k_pclka_hz PCLKA clock frequency constant
00444 * @see internal_is_triangle_mode() Determines divisor selection
00445 *
00446 * @since Version 1.0.0
00447 */
00448 static rx_err_t internal_calculate_period(const uint32_t frequency_hz,
00449                                         const rx_gptw_wave_mode_t wave_mode,
00450                                         uint32_t* period)
00451 {
00452     /* Pre-condition: validate pointer (NASA Power of 10 Rule 5) */
00453     RX_CHECK_NULL_PTR(period, s_tag, "period pointer is nullptr");
00454
00455     /* Pre-condition: validate wave_mode using exhaustive switch (catches invalid/negative values) */
00456     switch (wave_mode) {
00457         case k_gptw_wave_saw_pwm:
00458         case k_gptw_wave_tri_pwm1:
00459         case k_gptw_wave_tri_pwm2:
00460         case k_gptw_wave_tri_pwm3:
00461             break;
00462         default:
00463             rx_log_error(s_tag, "Invalid wave mode");
00464             return k_rx_err_invalid_arg;
00465     }
00466
00467     /* Period calculation depends on waveform mode:
00468     * - Sawtooth: Period = PCLKA / frequency
00469     * - Triangle: Period = PCLKA / (2 * frequency), because counter goes up then down
00470     * PCLKA = 96 MHz on STAR (HOCO x12 PLL / 2); see rx72n_clock.h k_pclka_hz.
00471     */
00472     const uint32_t pclka = k_pclka_hz;
00473     const uint32_t divisor =
00474         internal_is_triangle_mode(wave_mode) ? k_gptw_period_divisor_tri : k_gptw_period_divisor_saw;
00475
00476     if (frequency_hz == k_gptw_period_zero) {
00477         rx_log_error(s_tag, "Frequency is zero");
00478         return k_rx_err_invalid_arg;
00479     }
00480
00481     /* Use 64-bit math to prevent overflow in divisor * frequency_hz */
00482     const uint64_t denominator = (uint64_t)divisor * (uint64_t)frequency_hz;
00483     const uint32_t period_calc = (uint32_t)(pclka / denominator);
00484
00485     /* Check if period fits in valid range */
00486     if (period_calc > s_gptw_period_max) {
00487         rx_log_error(s_tag, "Frequency too low");
00488         return k_rx_err_invalid_arg;
00489     }
00490
00491     if (period_calc < k_gptw_period_min) {
00492         rx_log_error(s_tag, "Frequency too high");
00493         return k_rx_err_invalid_arg;
00494     }
00495
00496     *period = period_calc;
00497     return k_rx_ok;
00498 }
00499 /**
00500 * @brief Configure MPC for GPTW output pins
00501 *
00502 * @details
00503 * Configures the Multi-Function Pin Controller (MPC) to route GPTW timer outputs
00504 * to the appropriate Port E pins. Each GPTW channel uses two pins for
00505 * complementary PWM outputs (GTIOCA and GTIOCB).
00506 *
00507 * ## MPC Configuration Sequence
00508 *
00509 * 1. Clear PWPR.BOWI to enable PFSWE bit access
00510 * 2. Set PWPR.PFSWE to enable PFS register writes

```

```

00512 * 3. Write PSEL value (0x1E) to select GPTW function for each pin
00513 * 4. Set PWPR.B0WI to lock PFS registers
00514 *
00515 * ## Pin Assignments
00516 *
00517 * | Channel | Output A (GTIOCnA) | Output B (GTIOCnB) | PSEL |
00518 * |-----|-----|-----|-----|
00519 * | GPTW0 | PE5 | PE2 | 0x1E |
00520 * | GPTW1 | PE4 | PE1 | 0x1E |
00521 * | GPTW2 | PE3 | PE0 | 0x1E |
00522 * | GPTW3 | PE7 | PE6 | 0x1E |
00523 *
00524 *
00525 *
00526 * @pre MPC module accessible (no separate clock enable required)
00527 * @pre Pins not in use by other peripherals
00528 *
00529 * @post Port E PFS registers configured for GPTW function
00530 * @post MPC write protection re-enabled
00531 *
00532 * @invariant MPC write protection always re-locked on exit (success or failure)
00533 *
00534 * @note Modifies global MPC registers, not thread-safe
00535 * @note Must be called before enabling port pin peripheral mode
00536 *
00537 * @warning Overwrites any previous pin function assignment
00538 * @warning MPC must be unlocked before calling; function handles locking
00539 *
00540 * @par Performance:
00541 * - Execution time: ~20 cycles (register writes)
00542 * - Memory: 0 bytes heap, ~8 bytes stack
00543 *
00544 * @see internal_configure_port_pins() Configure PMR/PDR after MPC
00545 * @see RX72N Hardware Manual Chapter 23 - Multi-Function Pin Controller
00546 *
00547 * @since Version 1.0.0
00548 */
00549 /* PFS register block starts at 0x0008C140; each port uses 8 bytes
00550 * (one PFS per pin, pins 0..7). */
00551 typedef enum : uintptr_t {
00552     k_pfs_block_base_addr = 0x0008C140U,
00553     k_pfs_port_stride      = 8U,
00554 } pfs_block_layout_t;
00555
00556 static inline volatile uint8_t* internal_pfs_addr(uint8_t port_idx, uint8_t bit)
00557 {
00558     return (volatile uint8_t*)(k_pfs_block_base_addr + ((uint32_t)port_idx * k_pfs_port_stride) +
00559                                bit);
00560 }
00561
00562 static rx_err_t internal_configure_mpc(const rx_gptw_config_t* config)
00563 {
00564     /* Unlock MPC: clear B0WI then set PFSWE so PFS becomes writable. */
00565     mpc()->pwpr = k_mpc_pwpr_b0wi_clear;
00566     mpc()->pwpr = k_mpc_pwpr_pfswe_set;
00567
00568     /* Caller supplies pin coordinates -- the lib does not assume a port.
00569     * PSEL value for GTIOC is k_pfs_psel_gptw on every port that has GTIOC
00570     * available (per RX72N HW manual Tables 23.4, 23.6, 23.10, 23.14, 23.16). */
00571     internal_pfs_addr(config->port_a_idx, config->bit_a) = k_pfs_psel_gptw;
00572     internal_pfs_addr(config->port_b_idx, config->bit_b) = k_pfs_psel_gptw;
00573
00574     mpc()->pwpr = k_mpc_pwpr_lock;
00575     return k_rx_ok;
00576 }
00577
00578 /**
00579 * @brief Configure Port E pins as peripheral outputs
00580 *
00581 * @details
00582 * Configures the Port E GPIO pins for GPTW peripheral output mode. Sets both the
00583 * Peripheral Mode Register (PMR) and Port Direction Register (PDR) to enable
00584 * GPTW timer outputs.
00585 *
00586 * ## Register Configuration
00587 *
00588 * For each pin used by the specified channel:
00589 * - **PMR bit = 1**: Enable peripheral mode (not GPIO)
00590 * - **PDR bit = 1**: Set pin direction to output
00591 *
00592 * ## Pin Mapping
00593 *
00594 * | Channel | Output A Pin | Output B Pin |
00595 * |-----|-----|-----|
00596 * | GPTW0 | PE5 (bit 5) | PE2 (bit 2) |
00597 * | GPTW1 | PE4 (bit 4) | PE1 (bit 1) |
00598 * | GPTW2 | PE3 (bit 3) | PE0 (bit 0) |

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00599 * | GPTW3 | PE7 (bit 7) | PE6 (bit 6) |
00600 *
00601 *
00602 * @pre MPC configured for GPTW function via internal_configure_mpc()
00603 * @pre config->port_a_idx / config->port_b_idx are valid for rx_port_get_base()
00604 *
00605 * @post PMR bits set for both channel pins
00606 * @post PDR bits set for both channel pins
00607 *
00608 * @note No return value - silent no-op on invalid channel
00609 * @note Modifies Port E registers, not thread-safe
00610 * @note Uses OR-equals to preserve other pin configurations
00611 *
00612 * @par Performance:
00613 * - Execution time: ~10 cycles (register read-modify-write)
00614 * - Memory: 0 bytes heap, 0 bytes stack
00615 *
00616 * @see internal_configure_mpc() Must be called before this function
00617 * @see porte() Port E register accessor
00618 *
00619 * @since Version 1.0.0
00620 */
00621 static void internal_configure_port_pins(const rx_gptw_config_t* config)
00622 {
00623     /* Caller supplies pin coordinates. Use rx_port_get_base() so this works
00624     * for any port the board happens to route GTIOC outputs to. */
00625     volatile rx_port_regs_t* port_a = rx_port_get_base(config->port_a_idx);
00626     volatile rx_port_regs_t* port_b = rx_port_get_base(config->port_b_idx);
00627     if ((port_a == nullptr) || (port_b == nullptr)) {
00628         return;
00629     }
00630
00631     port_a->pdr |= (uint8_t)(1U « config->bit_a);
00632     port_b->pdr |= (uint8_t)(1U « config->bit_b);
00633     port_a->pmr |= (uint8_t)(1U « config->bit_a);
00634     port_b->pmr |= (uint8_t)(1U « config->bit_b);
00635 }
00636
00637 /**
00638 * @brief Enable GPTW module clock
00639 *
00640 * @details
00641 * Enables the GPTW peripheral by clearing its module stop bit in MSTPCRA.
00642 * On RX72N, GPTW is controlled by MSTPCRA bit 7 (MSTPA7), per the
00643 * RX72N Hardware Manual Chapter 11 and Renesas iodef.h. When set, the
00644 * module clock is stopped and registers are inaccessible.
00645 *
00646 * ## Register Access Sequence
00647 *
00648 * 1. **Unlock PRCR**: Write 0xA502 (PRC1) to enable writing to MSTPCR group
00649 * 2. **Clear MSTPA7**: AND-mask to clear bit 7 in MSTPCRA (0x00080010)
00650 * 3. **Lock PRCR**: Write 0xA500 to re-enable protection
00651 *
00652 * ## Module Stop Control
00653 *
00654 * | Bit | Value | Effect |
00655 * |-----|-----|-----|
00656 * | MSTPCRA.MSTPA7 = 1 | Module stopped | Registers inaccessible, low power |
00657 * | MSTPCRA.MSTPA7 = 0 | Module running | Registers accessible, clock active |
00658 *
00659 * @pre System clock configured
00660 *
00661 * @post GPTW module clock enabled
00662 * @post GPTW registers accessible
00663 * @post PRCR protection re-enabled
00664 *
00665 * @invariant PRCR always re-locked after modification
00666 *
00667 * @note Not thread-safe: Modifies system-wide registers
00668 * @note Idempotent: Safe to call multiple times
00669 *
00670 * @warning Must be called before any GPTW register access
00671 * @warning Modifies system protection register (PRCR)
00672 *
00673 * @par Performance:
00674 * - Execution time: ~15 cycles (3 register writes)
00675 * - Memory: 0 bytes heap, 0 bytes stack
00676 *
00677 * @see k_mstpa_gptw GPTW module stop bit position (MSTPA7)
00678 * @see k_rx_prcr_unlock_prc1 PRCR unlock value for MSTPCR group
00679 * @see RX72N Hardware Manual Chapter 11 - Low Power Consumption
00680 *
00681 * @since Version 1.0.0
00682 */
00683 static void internal_enable_gptw_module_clock(void)
00684 {
00685     /* RX72N: GPTW = MSTPCRA bit 7 (MSTPA7), NOT MSTPCRC bit 6 as earlier HAL

```



```

00686  * code claimed. Use PRC1 unlock (0xA502) which covers MSTPCR group. */
00687  *prcr_reg() = k_rx_prcr_unlock_prc1;
00688  system_regs()->mstpcra &= ~(uint32_t)k_gptw_bit_set « k_mstpa_gptw);
00689  *prcr_reg() = k_rx_prcr_lock;
00690  }
00691
00692  /**
00693  * @brief Configure GPTW hardware registers
00694  *
00695  * @details
00696  * Performs the core hardware configuration of a GPTW channel. This function
00697  * writes all necessary registers to set up PWM operation including control mode,
00698  * I/O configuration, period, duty cycle initialization, buffer mode, and deadline.
00699  *
00700  * ## Register Configuration Sequence
00701  *
00702  * 1. **Unlock write protection**: GTWP = 0xA500_0000
00703  * 2. **Configure control register (GTCR)**:
00704  *   - Set clock source to PCLKA/1 (120 MHz)
00705  *   - Set waveform mode (sawtooth or triangle)
00706  * 3. **Configure I/O control (GTIOR)**:
00707  *   - Output A: Initial low, toggle on compare match, enable
00708  *   - Output B: Initial low, toggle on compare match, enable
00709  * 4. **Set period register (GTPR)**: PWM cycle count
00710  * 5. **Initialize duty cycles (GTCCRA, GTCCRB)**: Both to 0
00711  * 6. **Clear counter (GTCNT)**: Start at 0
00712  * 7. **Enable buffer operation (GTBER)**: Glitch-free updates
00713  * 8. **Configure deadline (GTDVU, GTDVD, GTDTCR)**: If requested
00714  * 9. **Lock write protection**: GTWP = 0x0000_0001
00715  *
00716  * ## Deadline Calculation
00717  *
00718  * @f[
00719  * \text{deadline\_counts} = \frac{\text{deadline\_ns}}{f_{PCLKA}} \{10^9\}
00720  * @f]
00721  *
00722  * For 1000 ns at 96 MHz: deadline_counts = 96
00723  *
00724  *
00725  *
00726  * @pre gptw points to valid, unlocked GPTW registers
00727  * @pre config contains validated parameters
00728  * @pre period calculated and validated by internal_calculate_period()
00729  *
00730  * @post GPTW channel configured for PWM operation
00731  * @post Duty cycles initialized to 0% (motor stopped)
00732  * @post Buffer mode enabled for glitch-free updates
00733  * @post Write protection re-enabled
00734  *
00735  * @invariant Write protection always re-locked on exit
00736  * @invariant Duty cycles always start at 0 (safe state)
00737  *
00738  * @note Not thread-safe: Direct hardware register access
00739  * @note Timer is NOT started by this function
00740  *
00741  * @warning gptw pointer must be valid - no runtime bounds checking
00742  * @warning Caller responsible for stopping timer before calling
00743  *
00744  * @par Performance:
00745  *   - Execution time: ~100 cycles (multiple register writes)
00746  *   - Memory: 0 bytes heap, ~16 bytes stack
00747  *
00748  * @par NASA Power of 10 Compliance:
00749  *   - Rule 5: [OK] 3 preconditions (NULL checks, period range)
00750  *
00751  * @see internal_calculate_period() Period value calculation
00752  * @see rx72n_gptw_regs.h Register structure definition
00753  *
00754  * @since Version 1.0.0
00755  */
00756  static rx_err_t internal_configure_gptw_hardware(volatile rx_gptw_channel_regs_t* gptw,
00757                                                  const rx_gptw_config_t* config,
00758                                                  const uint32_t period)
00759  {
00760  /* Pre-condition: validate pointers (NASA Power of 10 Rule 5) */
00761  RX_CHECK_NULL_PTR(gptw, s_tag, "gptw pointer is nullptr");
00762  RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00763
00764  /* Pre-condition: validate period range */
00765  if ((period < k_gptw_period_min) || (period > s_gptw_period_max)) {
00766      rx_log_error(s_tag, "Period out of range");
00767      return k_rx_err_invalid_arg;
00768  }
00769
00770  /* Unlock write protection for this channel */
00771  gptw->gtwp = k_gptw_gtwp_unlock;
00772

```

```

00773 /* Configure control register
00774 * - PCLKA/1 (96 MHz on STAR)
00775 * - Waveform mode from config (sawtooth or triangle)
00776 * NOTE: CST (bit 0) left clear; caller (rx_gptw_start) sets it.
00777 */
00778 const uint32_t gtrc_mode = internal_get_gtrc_mode(config->wave_mode);
00779 gptw->gtrc = k_gptw_gtrc_tpcs_1 | gtrc_mode;
00780
00781 /* Force UP-count direction. Reset value of GTUDDTYC is 0 (UD=0 = DOWN).
00782 * With GTIOR = 0x09 (compare-match-up event drives output LOW), a
00783 * down-counter never raises the compare-match-up event and the output
00784 * latches HIGH at end-of-cycle. UD=1 + UDF=1 forces up-count immediately. */
00785 gptw->gtuddtyc = k_gptw_gtuddtyc_ud | k_gptw_gtuddtyc_udf;
00786
00787 /* Configure I/O control register
00788 * - Initial low, high at end-of-cycle, low at compare match (sawtooth up)
00789 * - Enable both outputs
00790 */
00791 gptw->gtior =
00792     k_gptw_gtior_oa_init_low | k_gptw_gtior_oae | k_gptw_gtior_ob_init_low | k_gptw_gtior_obe;
00793
00794 /* Set period (GTPR = PWM cycle) */
00795 gptw->gtpr = period;
00796
00797 /* Set initial duty cycle to 0% for both outputs */
00798 gptw->gtccra = k_gptw_period_zero;
00799 gptw->gtccrb = k_gptw_period_zero;
00800
00801 /* Clear counter */
00802 gptw->gtcnt = k_gptw_period_zero;
00803
00804 /* Disable all buffer operation on GTCCRA/GTCCRB/GTPR.
00805 * Rationale: with single-buffer enabled (CCRA=0x01«16, CCRB=0x01«18)
00806 * the hardware copies GTCCRC->GTCCRA / GTCCRE->GTCCRB on each period
00807 * boundary. GTCCRC/E default to 0, so any duty written via
00808 * rx_gptw_set_duty_raw() gets clobbered to 0 on the next overflow --
00809 * outputs stick at initial state. With GTBER=0, direct writes to
00810 * GTCCRA/GTCCRB take effect immediately.
00811 * If double-buffered duty updates are ever needed, a separate opt-in
00812 * API should enable buffering deliberately. */
00813 gptw->gtber = 0U;
00814
00815 /* Configure dead time if requested */
00816 if (config->deadtime_ns > k_gptw_deadtime_disabled) {
00817     /* Calculate dead time count: deadtime_ns * (PCLKA / 1e9) */
00818     const uint32_t deadtime_count =
00819         (uint32_t)((uint64_t)config->deadtime_ns * k_pclka_hz / s_gptw_ns_per_second);
00820     gptw->gtdvu = deadtime_count;
00821     gptw->gtdvd = deadtime_count;
00822     gptw->gtdtcr = k_gptw_gtdtcr_tde; /* Enable dead time */
00823 }
00824
00825 /* Lock write protection */
00826 gptw->gtwp = k_gptw_gtwp_lock;
00827
00828 return k_rx_ok;
00829 }
00830
00831 /**
00832 * @brief Validate channel and compute period for PWM initialization
00833 *
00834 * @details
00835 * Shared pre-initialization helper used by rx_gptw_init_pwm(). Performs
00836 * two actions in a single function to reduce code duplication:
00837 * 1. Validate channel range and resolve register base pointer
00838 * 2. Calculate the GTPR period value from frequency and wave mode
00839 *
00840 *
00841 *
00842 * @pre gptw_out and period_out must be non-NULL (caller's responsibility)
00843 * @pre config must be non-NULL (caller's responsibility)
00844 *
00845 * @post *gptw_out points to valid GPTW channel registers on success
00846 * @post *period_out contains a value in [k_gptw_period_min, s_gptw_period_max] on success
00847 *
00848 * @note Pure helper: does not modify hardware registers or module state
00849 * @note Not thread-safe: relies on config content stability during call
00850 *
00851 * @see internal_calculate_period() Period calculation details
00852 * @see rx_gptw_init_pwm() Primary caller
00853 *
00854 * @since Version 1.0.0
00855 */
00856 static rx_err_t internal_prepare_gptw_pwm_init(const rx_gptw_channel_t channel,
00857     const rx_gptw_config_t* config,
00858     volatile rx_gptw_channel_regs_t** gptw_out,
00859     uint32_t* period_out)

```

```

00860 {
00861     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
00862         rx_log_error(s_tag, "Invalid GPTW channel");
00863         return k_rx_err_invalid_arg;
00864     }
00865
00866     *gptw_out = internal_get_gptw_base(channel);
00867     if (*gptw_out == nullptr) {
00868         return k_rx_err_invalid_arg;
00869     }
00870
00871     rx_err_t err = internal_calculate_period(config->frequency_hz, config->wave_mode, period_out);
00872     if (err != k_rx_ok) {
00873         return err;
00874     }
00875
00876     return k_rx_ok;
00877 }
00878
00879 /*
=====
00880 * Public API Implementation
00881 *
=====
00882 */
00883
00884 /**
00885  * @brief Initialize a single GPTW channel for PWM operation
00886  *
00887  * @details
00888  * Configures one GPTW channel for PWM generation. This function performs complete
00889  * hardware setup including module clock enable, MPC configuration, port pin setup,
00890  * timer configuration, and automatic timer start. For 4-motor systems, prefer
00891  * rx_gptw_init_all_staggered() instead.
00892  *
00893  * ## Initialization Sequence
00894  *
00895  * 1. Validate config pointer (RX_CHECK_NULL_PTR)
00896  * 2. Validate channel range [0-3]
00897  * 3. Get channel register base address
00898  * 4. Calculate period from frequency and wave mode
00899  * 5. Enable GPTW module clock (MSTPCRA.MSTPA7)
00900  * 6. Stop timer if already running
00901  * 7. Configure GPTW hardware registers (GTCR, GTIOR, GTPR, etc.)
00902  * 8. Configure MPC for Port E alternate function
00903  * 9. Configure port pins (PMR, PDR)
00904  * 10. Store period value and set initialized flag
00905  * 11. Start timer
00906  *
00907  *
00908  *
00909  * @pre System clock configured (PCLKA = 120 MHz)
00910  * @pre Port E pins not used by other peripherals
00911  *
00912  * @post GPTW channel configured and running
00913  * @post Duty cycle initialized to 0%
00914  * @post MPC and port pins configured
00915  *
00916  * @invariant s_gptw_initialized[channel] == true on success
00917  * @invariant s_gptw_period[channel] contains valid period on success
00918  *
00919  * @note Not thread-safe: Call once during system initialization
00920  * @note Timer starts automatically - use rx_gptw_stop() to halt if needed
00921  *
00922  * @warning Calling while channel is running will cause glitches
00923  * @attention For 4-motor systems, use rx_gptw_init_all_staggered() instead
00924  *
00925  * @par Performance:
00926  * - Execution time: ~50 us at 240 MHz
00927  * - Memory: 0 bytes heap, ~32 bytes stack
00928  *
00929  * @par Example (Single Channel for LED PWM):
00930  * @code{.c}
00931  * rx_gptw_config_t led_config = {
00932  *     .frequency_hz = 1000,
00933  *     .deadtime_ns = 0,
00934  *     .wave_mode = k_gptw_wave_saw_pwm,
00935  *     .enable_complementary = false,
00936  *     .invert_polarity = false
00937  * };
00938  *
00939  * rx_err_t err = rx_gptw_init_pwm(k_gptw_channel_0, &led_config);
00940  * if (err == k_rx_ok) {
00941  *     rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
00942  *                     rx_gptw_output_id(k_gptw_output_a), 50.0f);
00943  * }
00944  * @endcode

```

```

00945 *
00946 * @see rx_gptw_init_all_staggered() Preferred for 4-motor systems
00947 * @see rx_gptw_deinit() Cleanup function
00948 * @see rx_gptw_config_t Configuration structure
00949 *
00950 * @since Version 1.0.0
00951 *
00952 * @par NASA Power of 10 Compliance:
00953 * - Rule 5: [OK] nullptr check, channel range check
00954 * - Rule 7: [OK] All internal function returns checked
00955 *
00956 * @callgraph
00957 */
00958 rx_err_t rx_gptw_init_pwm(const rx_gptw_channel_t channel, const rx_gptw_config_t* config)
00959 {
00960     RX_CHECK_NULL_PTR(config, s_tag, "config pointer is nullptr");
00961
00962     volatile rx_gptw_channel_regs_t* gptw = nullptr;
00963     uint32_t period = 0U;
00964     rx_err_t err = internal_prepare_gptw_pwm_init(channel, config, &gptw, &period);
00965     if (err != k_rx_ok) {
00966         return err;
00967     }
00968
00969     rx_log_info(s_tag, "Initializing GPTW");
00970
00971     /* Enable GPTW module clock */
00972     internal_enable_gptw_module_clock();
00973
00974     /* Stop timer before configuration */
00975     err = rx_gptw_stop(channel);
00976     if (err != k_rx_ok) {
00977         rx_log_error(s_tag, "Failed to stop GPTW");
00978         return err;
00979     }
00980
00981     /* Configure GPTW hardware registers */
00982     err = internal_configure_gptw_hardware(gptw, config, period);
00983     if (err != k_rx_ok) {
00984         rx_log_error(s_tag, "Failed to configure GPTW hardware");
00985         return err;
00986     }
00987
00988     /* Write the PFS (peripheral function select) for the GTIOC pads. Pin
00989      * is still in GPIO mode at this point -- PMR is set later. */
00990     err = internal_configure_mpc(config);
00991     if (err != k_rx_ok) {
00992         rx_log_error(s_tag, "Failed to configure MPC");
00993         return err;
00994     }
00995
00996     /* Save period for duty cycle calculations */
00997     s_gptw_period[channel] = period;
00998     s_gptw_initialized[channel] = true;
00999
01000     /* Start timer FIRST -- the GTIOC output must be producing valid PWM
01001      * before we route it to the pad. Otherwise the brief window between
01002      * PMR=1 and counter start exposes downstream hardware (e.g. DRV8263H
01003      * H-bridge inputs) to whatever GTIOR's reset/initial state happens to
01004      * be, which has been observed to latch driver faults. */
01005     err = rx_gptw_start(channel);
01006     if (err != k_rx_ok) {
01007         rx_log_error(s_tag, "Failed to start GPTW");
01008         return err;
01009     }
01010
01011     /* Now route the running PWM to the pad: PDR=1 (output) and PMR=1
01012      * (peripheral mode). The pin transitions from GPIO straight to a
01013      * fully-running, valid PWM signal -- no glitch window. */
01014     internal_configure_port_pins(config);
01015
01016     rx_log_info(s_tag, "GPTW initialized successfully");
01017
01018     return k_rx_ok;
01019 }
01020 /**
01021 * @brief Set PWM duty cycle using floating-point percentage
01022 *
01023 * @details
01024 * Converts floating-point duty percentage [0.0, 100.0] to raw count value and
01025 * delegates to rx_gptw_set_duty_raw() for hardware register update. The compare
01026 * register (GTCCRA or GTCCRB) is written and the change takes effect on the next
01027 * PWM period boundary (glitch-free via buffer mode).
01028 *
01029 * @par Algorithm:
01030 * @f[

```

```

01032 * \text{duty\_count} = \left\lfloor \frac{\text{duty\_percent}}{100} \times \text{period} \right\rfloor
01033 * @f
01034 *
01035 *
01036 *
01037 * @pre Channel must be initialized via rx_gptw_init_*( )
01038 * @pre duty_percent must be in [0.0, 100.0]
01039 *
01040 * @post Compare register written; change takes effect next period boundary
01041 * @post Motor output drive updated on next PWM cycle
01042 *
01043 * @note Not thread-safe: concurrent writes to same output cause undefined behavior
01044 *
01045 * @code{.c}
01046 * rx_gptw_set_duty(rx_gptw_channel_id(k_gptw_channel_0),
01047 *                  rx_gptw_output_id(k_gptw_output_a), 75.0f);
01048 * @endcode
01049 *
01050 * @see rx_gptw_set_duty_raw() Set duty using raw count value
01051 * @see rx_gptw_get_duty() Read current duty cycle
01052 *
01053 * @since Version 1.0.0
01054 */
01055 rx_err_t rx_gptw_set_duty(const rx_gptw_channel_id_t channel,
01056                          const rx_gptw_output_id_t output,
01057                          const float duty_percent)
01058 {
01059     const rx_gptw_channel_t channel_value = channel.value;
01060     const rx_gptw_output_t output_value = output.value;
01061
01062     if (((int32_t)channel_value >= (int32_t)k_gptw_max_channels) ||
01063         !s_gptw_initialized[channel_value]) {
01064         return k_rx_err_invalid_state;
01065     }
01066
01067     if (duty_percent < (float)k_gptw_duty_min || duty_percent > (float)k_gptw_duty_max) {
01068         rx_log_error(s_tag, "Invalid duty cycle");
01069         return k_rx_err_invalid_arg;
01070     }
01071
01072     /* Convert percentage to count value */
01073     const uint32_t period = s_gptw_period[channel_value];
01074     const float period_f = (float)period;
01075     const float duty_count_f = (duty_percent * period_f) / (float)k_gptw_duty_divisor;
01076     const uint32_t duty_count = (uint32_t)duty_count_f;
01077
01078     return rx_gptw_set_duty_raw(channel_value, output_value, duty_count);
01079 }
01080
01081 /**
01082 * @brief Set PWM duty cycle using a raw 32-bit timer count
01083 *
01084 * @details
01085 * Directly writes duty_count to the compare register (GTCCRA or GTCCRB).
01086 * Buffer mode (enabled during init via GTBER) ensures the update takes effect
01087 * on the next PWM period boundary without glitches. Values exceeding the period
01088 * are clamped to period (100% duty).
01089 *
01090 * @par Register Access:
01091 * - Output A: Writes to GTCCRA
01092 * - Output B: Writes to GTCCRB
01093 *
01094 *
01095 *
01096 * @pre Channel must be initialized via rx_gptw_init_*( )
01097 * @pre duty_count should be in [0, period] for meaningful duty; higher values clamped
01098 *
01099 * @post GTCCRA or GTCCRB written with clamped duty_count
01100 * @post Change takes effect at next PWM period boundary
01101 *
01102 * @note Not thread-safe: concurrent writes to the same register cause undefined behavior
01103 * @note Values exceeding period are silently clamped to period
01104 *
01105 * @code{.c}
01106 * uint32_t period;
01107 * rx_gptw_get_period(k_gptw_channel_0, &period);
01108 * rx_gptw_set_duty_raw(k_gptw_channel_0, k_gptw_output_a, period / 2U); // 50%
01109 * @endcode
01110 *
01111 * @see rx_gptw_set_duty() Set duty using floating-point percentage
01112 * @see rx_gptw_get_period() Retrieve period count for duty calculation
01113 *
01114 * @since Version 1.0.0
01115 */
01116 rx_err_t
01117 rx_gptw_set_duty_raw(const rx_gptw_channel_t channel, rx_gptw_output_t output, uint32_t duty_count)
01118 {

```

```

01119 if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01120     return k_rx_err_invalid_state;
01121 }
01122
01123 volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01124 if (gptw == nullptr) {
01125     return k_rx_err_invalid_arg;
01126 }
01127
01128 /* Clamp to period */
01129 const uint32_t period = s_gptw_period[channel];
01130 if (duty_count > period) {
01131     duty_count = period;
01132 }
01133
01134 /* GTCCRA/GTCCRB are protected by GTWP.WP on RX72N. Unlock, write, re-lock.
01135  * With GTBER=0 (default after our fixed init), the written value takes
01136  * effect immediately rather than being double-buffered to the next period. */
01137 gptw->gtpw = k_gptw_gtpw_unlock;
01138 switch (output) {
01139     case k_gptw_output_a:
01140         gptw->gtccra = duty_count;
01141         break;
01142     case k_gptw_output_b:
01143         gptw->gtccrb = duty_count;
01144         break;
01145     default:
01146         gptw->gtpw = k_gptw_gtpw_lock;
01147         return k_rx_err_invalid_arg;
01148 }
01149 gptw->gtpw = k_gptw_gtpw_lock;
01150
01151 return k_rx_ok;
01152 }
01153
01154 /**
01155  * @brief Read the current PWM duty cycle as a floating-point percentage
01156  *
01157  * @details
01158  * Reads the current compare register value (GTCCRA or GTCCRB) from hardware
01159  * and converts it to a percentage using the cached period from s_gptw_period[].
01160  *
01161  * @par Algorithm:
01162  * @f[
01163  * \text{duty\_percent} = \frac{\text{duty\_count}}{\text{period}} \times 100
01164  * @f]
01165  *
01166  *
01167  *
01168  * @pre Channel must be initialized via rx_gptw_init_*( )
01169  * @pre duty_percent must point to valid float storage
01170  *
01171  * @post *duty_percent contains duty cycle in [0.0, 100.0] on success
01172  * @post Hardware registers unchanged
01173  *
01174  * @note Returns actual register value; may differ from the last set value if
01175  *       buffer transfer has not yet occurred at period boundary
01176  * @note Not thread-safe: concurrent rx_gptw_set_duty calls may cause torn reads
01177  *
01178  * @code{.c}
01179  * float duty;
01180  * rx_err_t err = rx_gptw_get_duty(k_gptw_channel_0, k_gptw_output_a, &duty);
01181  * @endcode
01182  *
01183  * @see rx_gptw_set_duty() Set duty using floating-point percentage
01184  * @see rx_gptw_get_period() Retrieve period count
01185  *
01186  * @since Version 1.0.0
01187  */
01188 rx_err_t rx_gptw_get_duty(const rx_gptw_channel_t channel,
01189                          const rx_gptw_output_t output,
01190                          float* duty_percent)
01191 {
01192     RX_CHECK_NULL_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
01193
01194     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01195         return k_rx_err_invalid_state;
01196     }
01197
01198     const volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01199     if (gptw == nullptr) {
01200         return k_rx_err_invalid_arg;
01201     }
01202
01203     const uint32_t period = s_gptw_period[channel];
01204     uint32_t duty_count = k_gptw_period_zero;
01205

```

```

01206 switch (output) {
01207     case k_gptw_output_a:
01208         duty_count = gptw->gtccra;
01209         break;
01210     case k_gptw_output_b:
01211         duty_count = gptw->gtccrb;
01212         break;
01213     default:
01214         return k_rx_err_invalid_arg;
01215 }
01216
01217 *duty_percent = ((float)duty_count * (float)k_gptw_duty_max) / (float)period;
01218
01219 return k_rx_ok;
01220 }
01221
01222 /**
01223  * @brief Get PWM period count for GPTW channel
01224  *
01225  * @details
01226  * Implementation of rx_gptw_get_period() - see rx_gptw.h for complete documentation.
01227  *
01228  * Returns the cached period value from s_gptw_period[], which matches the
01229  * GTPR register value set during initialization.
01230  *
01231  *
01232  *
01233  * @pre Channel initialized via rx_gptw_init_*( )
01234  * @pre period_count points to valid uint32_t storage
01235  *
01236  * @post *period_count contains period value on success
01237  *
01238  * @note Thread-safe: Reads from cached value, not hardware
01239  * @note Useful for calculating raw duty counts from percentages
01240  *
01241  * @see rx_gptw.h Complete API documentation
01242  */
01243 rx_err_t rx_gptw_get_period(const rx_gptw_channel_t channel, uint32_t* period_count)
01244 {
01245     RX_CHECK_NULL_PTR(period_count, s_tag, "period_count pointer is nullptr");
01246
01247     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01248         return k_rx_err_invalid_state;
01249     }
01250
01251     *period_count = s_gptw_period[channel];
01252     return k_rx_ok;
01253 }
01254
01255 /**
01256  * @brief Enable or disable GPTW output pin
01257  *
01258  * @details
01259  * Implementation of rx_gptw_enable_output() - see rx_gptw.h for complete documentation.
01260  *
01261  * Controls the Output Enable bits (OAE/OBE) in the GTIOR register. When disabled,
01262  * the corresponding output pin is held in its initial state (typically LOW).
01263  *
01264  * ## Register Modification
01265  *
01266  * | Output | Bit Modified | Effect |
01267  * |-----|-----|-----|
01268  * | Output A | GTIOR.OAE | Enable/disable GTIOCnA pin |
01269  * | Output B | GTIOR.OBE | Enable/disable GTIOCnB pin |
01270  *
01271  *
01272  *
01273  * @pre Channel initialized via rx_gptw_init_*( )
01274  *
01275  * @post GTIOR.OAE or GTIOR.OBE modified
01276  * @post Write protection re-enabled
01277  *
01278  * @note Not thread-safe: Modifies hardware registers
01279  * @note Timer continues running even when output disabled
01280  *
01281  * @warning Disable outputs before motor emergency stop to ensure safe state
01282  *
01283  * @see rx_gptw.h Complete API documentation
01284  */
01285 rx_err_t
01286 rx_gptw_enable_output(const rx_gptw_channel_t channel, const rx_gptw_output_t output, bool enable)
01287 {
01288     if (((int32_t)channel >= (int32_t)k_gptw_max_channels) || !s_gptw_initialized[channel]) {
01289         return k_rx_err_invalid_state;
01290     }
01291
01292     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);

```



```

01293 if (gptw == nullptr) {
01294     return k_rx_err_invalid_arg;
01295 }
01296
01297 /* Unlock for modification */
01298 gptw->gtwp = k_gptw_gtwp_unlock;
01299
01300 /* Modify GTIOR to enable/disable output */
01301 switch (output) {
01302     case k_gptw_output_a:
01303         if (enable) {
01304             gptw->gtior |= k_gptw_gtior_oae;
01305         } else {
01306             gptw->gtior &= ~k_gptw_gtior_oae;
01307         }
01308         break;
01309     case k_gptw_output_b:
01310         if (enable) {
01311             gptw->gtior |= k_gptw_gtior_obe;
01312         } else {
01313             gptw->gtior &= ~k_gptw_gtior_obe;
01314         }
01315         break;
01316     default:
01317         gptw->gtwp = k_gptw_gtwp_lock;
01318         return k_rx_err_invalid_arg;
01319 }
01320
01321 gptw->gtwp = k_gptw_gtwp_lock;
01322
01323 return k_rx_ok;
01324 }
01325
01326 /**
01327  * @brief Start GPTW counter (begin PWM generation)
01328  *
01329  * @details
01330  * Implementation of rx_gptw_start() - see rx_gptw.h for complete documentation.
01331  *
01332  * Sets the Count Start (CST) bit in GTCR to begin counter operation. The timer
01333  * will count according to the configured waveform mode (sawtooth or triangle)
01334  * and generate PWM output on the enabled pins.
01335  *
01336  * ## Register Access
01337  *
01338  * 1. Unlock write protection (GTWP)
01339  * 2. Set GTCR.CST bit
01340  * 3. Lock write protection (GTWP)
01341  *
01342  *
01343  *
01344  * @pre Channel configured via rx_gptw_init_*() (recommended)
01345  *
01346  * @post Timer counting actively
01347  * @post PWM output generated on enabled pins
01348  *
01349  * @note Does NOT require channel to be "initialized" (no flag check)
01350  * @note Timer automatically started during rx_gptw_init_*()
01351  * @note Idempotent: Safe to call when already running
01352  *
01353  * @see rx_gptw_stop() Stop the timer
01354  * @see rx_gptw.h Complete API documentation
01355  */
01356 rx_err_t rx_gptw_start(const rx_gptw_channel_t channel)
01357 {
01358     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01359         return k_rx_err_invalid_arg;
01360     }
01361
01362     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01363     if (gptw == nullptr) {
01364         return k_rx_err_invalid_arg;
01365     }
01366
01367     /* Unlock and set CST bit in GTCR */
01368     gptw->gtwp = k_gptw_gtwp_unlock;
01369     gptw->gtcr |= k_gptw_gtcr_cst;
01370     gptw->gtwp = k_gptw_gtwp_lock;
01371
01372     return k_rx_ok;
01373 }
01374
01375 /**
01376  * @brief Stop GPTW counter (halt PWM generation)
01377  *
01378  * @details
01379  * Implementation of rx_gptw_stop() - see rx_gptw.h for complete documentation.

```



```

01380 *
01381 * Clears the Count Start (CST) bit in GTCR to halt counter operation. The timer
01382 * counter value (GTCNT) is preserved and PWM outputs hold their current state.
01383 *
01384 * ## Register Access
01385 *
01386 * 1. Unlock write protection (GTWP)
01387 * 2. Clear GTCR.CST bit
01388 * 3. Lock write protection (GTWP)
01389 *
01390 *
01391 *
01392 * @pre None (can be called on uninitialized channel)
01393 *
01394 * @post Timer stopped (GTCR.CST = 0)
01395 * @post Counter value preserved (GTCNT unchanged)
01396 * @post Output pins hold last state
01397 *
01398 * @note Does NOT require channel to be "initialized" (no flag check)
01399 * @note Idempotent: Safe to call when already stopped
01400 * @note For safety shutdown, also disable outputs via rx_gptw_enable_output()
01401 *
01402 * @warning Outputs may remain HIGH when stopped - disable outputs for safe state
01403 *
01404 * @see rx_gptw_start() Resume the timer
01405 * @see rx_gptw_enable_output() Disable outputs for safe state
01406 * @see rx_gptw.h Complete API documentation
01407 */
01408 rx_err_t rx_gptw_stop(const rx_gptw_channel_t channel)
01409 {
01410     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01411         return k_rx_err_invalid_arg;
01412     }
01413
01414     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01415     if (gptw == nullptr) {
01416         return k_rx_err_invalid_arg;
01417     }
01418
01419     /* Unlock and clear CST bit in GTCR */
01420     gptw->gtwp = k_gptw_gtwp_unlock;
01421     gptw->gtcr &= ~k_gptw_gtcr_cst;
01422     gptw->gtwp = k_gptw_gtwp_lock;
01423
01424     return k_rx_ok;
01425 }
01426
01427 /*
=====
01428 * Phase Staggering Implementation
01429 *
=====
01430 */
01431
01432 /**
01433 * @enum gptw_phase_divisor_t
01434 * @brief Phase divisors for staggered PWM initialization
01435 *
01436 * @details
01437 * Divisor constants used to calculate the initial GTCNT counter value for each
01438 * GPTW channel during staggered initialization. Each channel is offset by
01439 * 90 degrees from the previous one to spread switching edges across the PWM period.
01440 *
01441 * For a given period P the offsets are:
01442 * - Ch0: 0 (0/4 of P)
01443 * - Ch1: P / 4 (quarter divisor)
01444 * - Ch2: P / 2 (half divisor)
01445 * - Ch3: 3 * P / 4 (three-quarter: multiply by 3, then divide by 4)
01446 *
01447 * @invariant All divisor values are >= 1, preventing division-by-zero in
01448 * internal_calculate_phase_offset()
01449 *
01450 * @code{.c}
01451 * // Example: derive 90-degree offset for a period of 3000 counts
01452 * uint32_t offset_ch1 = 3000U / k_phase_divisor_quarter; // == 750
01453 * uint32_t offset_ch3 = (3000U * 3U) / k_phase_divisor_three_quarter; // == 2250
01454 * @endcode
01455 *
01456 * @see internal_calculate_phase_offset() Uses these divisors for GTCNT seeding
01457 * @see rx_gptw_init_all_staggered() Top-level staggered initialization function
01458 */
01459 typedef enum : uint8_t {
01460     k_phase_divisor_none = 1, /**< 0 deg: no offset (safe default, never causes div-by-zero) */
01461     k_phase_divisor_quarter = 4, /**< 90 deg = period/4 */
01462     k_phase_divisor_half = 2, /**< 180 deg = period/2 */
01463     k_phase_divisor_three_quarter = 4, /**< 270 deg = 3*period/4 (multiplied by 3 then divided) */
01464 } gptw_phase_divisor_t;

```

```

01465
01466 /**
01467  * @brief Calculate initial counter value and direction for phase staggering
01468  *
01469  * @details
01470  * Computes the initial counter value (GTCNT) and counting direction for a GPTW
01471  * channel to achieve the desired phase offset. Phase staggering spreads PWM edges
01472  * across the period to reduce peak current draw and EMI.
01473  *
01474  * ## Phase Offset Table (Sawtooth Mode)
01475  *
01476  * | Channel | Phase | Counter Init | Direction |
01477  * |-----|-----|-----|-----|
01478  * | Ch0 | 0deg | 0 | Up (n/a) |
01479  * | Ch1 | 90deg | period/4 | Up (n/a) |
01480  * | Ch2 | 180deg | period/2 | Up (n/a) |
01481  * | Ch3 | 270deg | 3xperiod/4 | Up (n/a) |
01482  *
01483  * ## Phase Offset Table (Triangle Mode)
01484  *
01485  * | Channel | Phase | Counter Init | Direction |
01486  * |-----|-----|-----|-----|
01487  * | Ch0 | 0deg | 0 | Up |
01488  * | Ch1 | 90deg | period/2 | Up |
01489  * | Ch2 | 180deg | period | Down |
01490  * | Ch3 | 270deg | period/2 | Down |
01491  *
01492  *
01493  * @pre Output pointers must be non-NULL
01494  * @pre period should be > 0 for valid calculations
01495  *
01496  * @post *init_count contains phase offset value
01497  * @post *init_dir_up contains initial direction (triangle mode)
01498  * @post Safe defaults (0, true) on any validation failure
01499  *
01500  * @note Returns void - validation failures produce safe defaults silently
01501  * @note For sawtooth mode, direction is ignored (always up)
01502  *
01503  * @par NASA Power of 10 Compliance:
01504  * - Rule 5: [OK] nullptr checks, period validation
01505  * - Rule 2: [OK] No loops (direct calculation)
01506  *
01507  * @see internal_configure_channel_staggered() Calls this function
01508  * @see gptw_phase_divisor_t Divisor constants
01509  */
01510 static void internal_calculate_phase_offset(const rx_gptw_channel_t channel,
01511                                           const uint32_t period,
01512                                           const bool is_triangle,
01513                                           uint32_t* init_count,
01514                                           bool* init_dir_up)
01515 {
01516     /* Pre-condition: validate output pointers (NASA Power of 10 Rule 5) */
01517     if (init_count == nullptr || init_dir_up == nullptr) {
01518         /* Cannot proceed without valid output pointers - caller error */
01519         return;
01520     }
01521
01522     /* Pre-condition: validate period is non-zero to avoid division by zero */
01523     if (period == 0) {
01524         *init_count = 0;
01525         *init_dir_up = true;
01526         return;
01527     }
01528
01529     /* Pre-condition: validate channel is in valid range */
01530     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01531         *init_count = 0;
01532         *init_dir_up = true;
01533         return;
01534     }
01535
01536     *init_dir_up = true; /* Default to counting up */
01537
01538     if (is_triangle) {
01539         /* Edge-Aligned (Sawtooth) Staggering:
01540          * Ch0 (0 deg): 0
01541          * Ch1 (90 deg): Period / 4
01542          * Ch2 (180 deg): Period / 2
01543          * Ch3 (270 deg): 3 * Period / 4
01544          */
01545         switch (channel) {
01546             case k_gptw_channel_0:
01547                 *init_count = 0;
01548                 break;
01549             case k_gptw_channel_1:
01550                 *init_count = period / k_phase_divisor_quarter;
01551                 break;

```

```

01552     case k_gptw_channel_2:
01553         *init_count = period / k_phase_divisor_half;
01554         break;
01555     case k_gptw_channel_3:
01556         *init_count = (period * 3) / k_phase_divisor_three_quarter;
01557         break;
01558     default:
01559         *init_count = 0;
01560         break;
01561 }
01562 } else {
01563     /* Center-Aligned (Triangle) Staggering:
01564     * Ch0 (0 deg): 0, Up
01565     * Ch1 (90 deg): Period/2, Up
01566     * Ch2 (180 deg): Period, Down (Wait: Period is Top)
01567     *      Note: At Period, it turns around.
01568     *      For 180 shift, we want the "center" of the OFF pulse to correspond
01569     *      to the "center" of Ch0 ON pulse?
01570     *      Standard 3-phase shift logic applied to 4-phase:
01571     *      0 deg: 0 (Valley) -> Up
01572     *      90 deg: Period/2 -> Up
01573     *      180 deg: Period (Peak) -> Down (naturally acts as peak)
01574     *      270 deg: Period/2 -> Down
01575     */
01576     switch (channel) {
01577     case k_gptw_channel_0:
01578         *init_count = 0;
01579         *init_dir_up = true;
01580         break;
01581     case k_gptw_channel_1:
01582         *init_count = period / k_phase_divisor_half;
01583         *init_dir_up = true;
01584         break;
01585     case k_gptw_channel_2:
01586         *init_count = period;
01587         *init_dir_up = false; /* Start at top, go down */
01588         break;
01589     case k_gptw_channel_3:
01590         *init_count = period / k_phase_divisor_half;
01591         *init_dir_up = false; /* Start at mid, go down */
01592         break;
01593     default:
01594         *init_count = 0;
01595         break;
01596     }
01597 }
01598 }
01599
01600 /**
01601  * @brief Configure a single GPTW channel for staggered PWM operation
01602  *
01603  * @details
01604  * Performs complete hardware configuration for one GPTW channel including
01605  * base timer setup, MPC pin function selection, port configuration, and
01606  * phase staggering initialization.
01607  *
01608  * ## Configuration Sequence
01609  *
01610  * 1. Validate channel range [0-3]
01611  * 2. Get channel register base address
01612  * 3. Configure base hardware (GTCR, GTIOR, GTPR, etc.)
01613  * 4. Configure MPC for Port E alternate function
01614  * 5. Configure port pins (PMR, PDR)
01615  * 6. Calculate phase offset for this channel
01616  * 7. Set initial counter value (GTCNT)
01617  * 8. For triangle mode: seed initial count direction
01618  * 9. Update initialization tracking state
01619  *
01620  * ## Phase Staggering
01621  *
01622  * Counter is initialized to a phase-offset value:
01623  * - Ch0: 0deg (reference, GTCNT=0)
01624  * - Ch1: 90deg (GTCNT=period/4)
01625  * - Ch2: 180deg (GTCNT=period/2)
01626  * - Ch3: 270deg (GTCNT=3xperiod/4)
01627  *
01628  *
01629  *
01630  * @pre Module clock enabled
01631  * @pre Timer stopped
01632  * @pre config pointer valid (caller validates)
01633  *
01634  * @post Channel fully configured with phase offset
01635  * @post s_gptw_initialized[channel] = true
01636  * @post s_gptw_period[channel] = period
01637  * @post Timer NOT started (caller starts all channels together)
01638  *

```

```

01639 * @note Not thread-safe: modifies hardware registers and shared state arrays
01640 * (s_gptw_initialized, s_gptw_period); caller must ensure exclusive access
01641 * @note Timer is NOT started by this function
01642 * @note Called from rx_gptw_init_all_staggered() in a loop
01643 *
01644 * @par NASA Power of 10 Compliance:
01645 * - Rule 5: [OK] Channel range validation
01646 * - Rule 7: [OK] All internal function returns checked
01647 *
01648 * @see rx_gptw_init_all_staggered() Parent function
01649 * @see internal_calculate_phase_offset() Phase calculation
01650 */
01651 static rx_err_t internal_configure_channel_staggered(const rx_gptw_channel_t channel,
01652                                                     const rx_gptw_config_t* config,
01653                                                     const uint32_t period,
01654                                                     const bool is_triangle)
01655 {
01656     /* Pre-condition: validate channel range (NASA Power of 10 Rule 5) */
01657     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01658         return k_rx_err_invalid_arg;
01659     }
01660
01661     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01662
01663     /* Pre-condition: validate gptw pointer */
01664     if (gptw == nullptr) {
01665         return k_rx_err_invalid_arg;
01666     }
01667
01668     /* Base hardware configuration */
01669     rx_err_t err = internal_configure_gptw_hardware(gptw, config, period);
01670     if (err != k_rx_ok) {
01671         return err;
01672     }
01673
01674     /* Configure MPC PFS for the GTIOC pads. PMR (peripheral-mode enable)
01675      * is set after the channels are atomically started in the caller. */
01676     err = internal_configure_mpc(config);
01677     if (err != k_rx_ok) {
01678         return err;
01679     }
01680
01681     /* Apply Phase Staggering */
01682     uint32_t init_count = 0;
01683     bool dir_up = true;
01684
01685     internal_calculate_phase_offset(channel, period, is_triangle, &init_count, &dir_up);
01686
01687     /* Modify counter value (unlocked by configure function, but we need to unlock again
01688      * since internal_configure_gptw_hardware locks it at the end)
01689      */
01690     gptw->gtwp = k_gptw_gtwp_unlock;
01691     gptw->gtcnt = init_count;
01692
01693     /* Handle Direction Seeding for Triangle Mode */
01694     if (is_triangle) {
01695         /* Use GTUDDTYC to force initial direction
01696          * UD (bit 0): 0=Down, 1=Up
01697          * UDF (bit 1): 1=Force
01698          * Sequence: Set Force -> Clear Force (latch direction)
01699          * Note: Manual verification confirmed UD=bit0, UDF=bit1.
01700          */
01701         uint32_t ud_val = dir_up ? k_gptw_gtuddtyc_ud : k_gptw_gtuddtyc_ud_down;
01702
01703         /* Force direction */
01704         gptw->gtuddtyc = k_gptw_gtuddtyc_udf | ud_val;
01705
01706         /* Clear force bit to allow normal counting operation */
01707         gptw->gtuddtyc = ud_val;
01708     }
01709
01710     gptw->gtwp = k_gptw_gtwp_lock;
01711
01712     /* Update internal state */
01713     s_gptw_period[channel] = period;
01714     s_gptw_initialized[channel] = true;
01715
01716     return k_rx_ok;
01717 }
01718
01719 /**
01720 * @brief Initialize all 4 GPTW channels with 90-degree phase staggering
01721 *
01722 * @details
01723 * Implementation of rx_gptw_init_all_staggered() - see rx_gptw.h for complete
01724 * documentation with diagrams, examples, and NASA compliance notes.
01725 */

```

```

01726 * ## Implementation Overview
01727 *
01728 * 1. **Validate inputs**: NULL check, frequency > 0, valid wave_mode
01729 * 2. **Enable module clock**: Clear MSTPCRA.MSTPA7
01730 * 3. **Stop all channels**: Atomic stop via GTSTPA register
01731 * 4. **Calculate period**: Common period for all channels
01732 * 5. **Configure each channel**: Loop calls internal_configure_channel_staggered()
01733 * 6. **Start all channels**: Atomic start via GTSTRA register
01734 *
01735 * ## Phase Staggering Benefits
01736 *
01737 * - Reduces peak current by spreading PWM edges across period
01738 * - Lowers EMI from simultaneous switching transients
01739 * - Allows smaller power supply capacitors
01740 *
01741 *
01742 *
01743 * @see rx_gptw.h Complete API documentation with diagrams and examples
01744 *
01745 * @callgraph
01746 */
01747 rx_err_t rx_gptw_init_all_staggered(const rx_gptw_config_t* configs[k_rx_gptw_channel_count])
01748 {
01749     RX_CHECK_NULL_PTR(configs, s_tag, "configs array pointer is nullptr");
01750     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01751         RX_CHECK_NULL_PTR(configs[i], s_tag, "configs[i] pointer is nullptr");
01752     }
01753
01754     /* All channels must share frequency / wave_mode (otherwise the staggered
01755      * phasing math doesn't work). Validate once on configs[0]. */
01756     if (configs[0]->frequency_hz == 0) {
01757         rx_log_error(s_tag, "frequency_hz is zero");
01758         return k_rx_err_invalid_arg;
01759     }
01760     switch (configs[0]->wave_mode) {
01761         case k_gptw_wave_saw_pwm:
01762         case k_gptw_wave_tri_pwm1:
01763         case k_gptw_wave_tri_pwm2:
01764         case k_gptw_wave_tri_pwm3:
01765             break;
01766         default:
01767             rx_log_error(s_tag, "Invalid wave_mode");
01768             return k_rx_err_invalid_arg;
01769     }
01770
01771     rx_log_info(s_tag, "Initializing all GPTW channels (staggered)");
01772
01773     internal_enable_gptw_module_clock();
01774
01775     if (gptw_common() != nullptr) {
01776         gptw_common()->gtstpa = k_gptw_all_channels_mask; /* Stop Ch0-3 */
01777     }
01778
01779     uint32_t period;
01780     rx_err_t err =
01781         internal_calculate_period(configs[0]->frequency_hz, configs[0]->wave_mode, &period);
01782     if (err != k_rx_ok) {
01783         return err;
01784     }
01785
01786     const bool is_triangle = internal_is_triangle_mode(configs[0]->wave_mode);
01787
01788     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01789         err =
01790             internal_configure_channel_staggered((rx_gptw_channel_t)i, configs[i], period, is_triangle);
01791         if (err != k_rx_ok) {
01792             return err;
01793         }
01794     }
01795
01796     /* Atomic start of all 4 channels. */
01797     if (gptw_common() == nullptr) {
01798         return k_rx_err_hw_error;
01799     }
01800     rx_log_info(s_tag, "Global start of GPTW channels");
01801     gptw_common()->gtstra = k_gptw_all_channels_mask;
01802
01803     /* Now that valid PWM is running on every channel, route the outputs
01804      * to their pads (PMR=1). Doing this after start avoids exposing the
01805      * downstream H-bridges to the GTIOR reset/initial state during the
01806      * brief window between mux and counter start. */
01807     for (uint8_t i = 0; i < k_gptw_max_channels; i++) {
01808         internal_configure_port_pins(configs[i]);
01809     }
01810
01811     return k_rx_ok;
01812 }

```

```

01813
01814 /**
01815  * @brief Deinitialize a GPTW channel
01816  *
01817  * @details
01818  * Stops the timer, disables both PWM outputs (A and B), and resets
01819  * the counter to zero. Call only when motor control is no longer active.
01820  *
01821  *
01822  *
01823  * @pre Motor using this channel must be stopped before calling
01824  * @pre channel must be in range [k_gptw_channel_0, k_gptw_channel_3]
01825  *
01826  * @post Timer counter stopped (GTCR.CST = 0)
01827  * @post Both outputs disabled (GTIOR.OAE = 0, GTIOR.OBE = 0)
01828  * @post Counter register zeroed (GTCNT = k_gptw_period_zero)
01829  *
01830  * @invariant Write protection re-locked (GTWP) on all exit paths
01831  *
01832  * @note Not thread-safe: modifies hardware registers directly
01833  * @note Does not clear s_gptw_initialized[] or s_gptw_period[]
01834  *
01835  * @warning Ensure motor is at rest before calling to prevent abrupt stop
01836  *
01837  * @code{.c}
01838  * rx_err_t err = rx_gptw_deinit(k_gptw_channel_0);
01839  * @endcode
01840  *
01841  * @see rx_gptw_init_all_staggered() Re-initialize after deinit
01842  * @see rx_gptw_stop() Stop without full teardown
01843  *
01844  * @since Version 1.0.0
01845  */
01846 rx_err_t rx_gptw_deinit(const rx_gptw_channel_t channel)
01847 {
01848     if ((int32_t)channel >= (int32_t)k_gptw_max_channels) {
01849         return k_rx_err_invalid_arg;
01850     }
01851
01852     volatile rx_gptw_channel_regs_t* gptw = internal_get_gptw_base(channel);
01853     if (gptw == nullptr) {
01854         return k_rx_err_invalid_arg;
01855     }
01856
01857     /* Unlock write protection */
01858     gptw->gtwp = k_gptw_gtwp_unlock;
01859
01860     /* Stop counter */
01861     gptw->gtcr &= ~k_gptw_gtcr_cst;
01862
01863     /* Disable both outputs (OAE and OBE) */
01864     gptw->gtior &= ~(k_gptw_gtior_oae | k_gptw_gtior_obe);
01865
01866     /* Reset counter to zero */
01867     gptw->gtcnt = k_gptw_period_zero;
01868
01869     /* Re-lock write protection */
01870     gptw->gtwp = k_gptw_gtwp_lock;
01871
01872     return k_rx_ok;
01873 }
01874
01875 #endif /* __RX__ */

```

8.119 libs/rx/hal/src/rx_host_irq.c File Reference

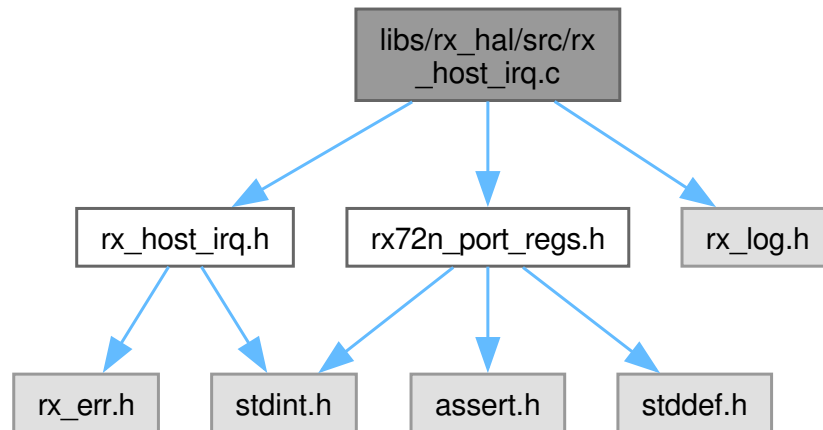
HOST_IRQ GPIO Output Driver Implementation.

```
#include "rx_host_irq.h"
```

```
#include "rx72n_port_regs.h"
```

```
#include "rx_log.h"
```

Include dependency graph for rx_host_irq.c:



Enumerations

- enum `host_irq_gpio_constants_t` : `uint8_t { k_host_irq_pin_bit = 7 , k_host_irq_pin_mask = 0x80 , k_host_irq_active = 0 , k_host_irq_idle = 1 }`
GPIO configuration constants for HOST_IRQ on P67.

Functions

- `rx_err_t rx_host_irq_init` (void)
Initialize HOST_IRQ pin as GPIO output (deasserted / HIGH).
- `rx_err_t rx_host_irq_deinit` (void)
Deinitialize the HOST_IRQ output (implementation).
- `rx_err_t rx_host_irq_assert` (void)
Assert the HOST_IRQ line (implementation).
- `rx_err_t rx_host_irq_deassert` (void)
De-assert the HOST_IRQ line (implementation).
- `rx_err_t rx_host_irq_is_asserted` (bool *asserted)
Check if HOST_IRQ is currently asserted.

Variables

- static const char *const `s_tag` = "HOST_IRQ"
- static bool `s_initialized` = false
Initialization flag.

8.119.1 Detailed Description

HOST_IRQ GPIO Output Driver Implementation.

Configures P67 (PORT6 pin 7) as a GPIO output for signaling the RPi5 host controller that the RX72N has data ready for SPI transfer. The pin is active-low: assert LOW = data ready, deassert HIGH = idle.

8.119.1.1 GPIO Configuration

```
rx_host_irq_init()
1. PMR6.B7 = 0    (GPIO mode)
2. PODR6.B7 = 1   (initial HIGH = idle)
3. PDR6.B7 = 1    (output direction)

rx_host_irq_assert()
PODR6.B7 = 0      (drive LOW = data ready)

rx_host_irq_deassert()
PODR6.B7 = 1      (drive HIGH = idle)
```

NASA Power of 10 Compliance:

- Rule 1: No goto, setjmp, recursion
- Rule 3: No dynamic memory (static variables only)
- Rule 4: All functions < 60 lines
- Rule 5: 2+ validation checks per function
- Rule 7: All return values checked
- Rule 8: C23 typed enums, no magic numbers
- Rule 10: -Wall -Wextra -Werror clean

See also

[rx_host_irq.h](#) Public API
[hardware_config.h](#) Pin assignments

Author

Locked, Inc.

Date

2026-02-10

Copyright

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Since

Version 1.0.0

Definition in file [rx_host_irq.c](#).

8.119.2 Enumeration Type Documentation

8.119.2.1 host`irq`gpio`constants`

enum [host_irq_gpio_constants_t](#) : uint8_t

GPIO configuration constants for HOST_IRQ on P67.

Enumerator

k_host_irq_pin_bit	Pin 7 within PORT6
k_host_irq_pin_mask	Bit mask for pin 7 (1 << 7)
k_host_irq_active	Active level: LOW
k_host_irq_idle	Idle level: HIGH

Definition at line 57 of file [rx_host_irq.c](#).

```
00057      : uint8_t {
00058  k_host_irq_pin_bit = 7,  /**< Pin 7 within PORT6 */
00059  k_host_irq_pin_mask = 0x80, /**< Bit mask for pin 7 (1 « 7) */
00060  k_host_irq_active = 0,  /**< Active level: LOW */
00061  k_host_irq_idle = 1,    /**< Idle level: HIGH */
00062 } host_irq_gpio_constants_t;
```

8.119.3 Function Documentation

8.119.3.1 rx`host`irq`assert()

```
rx_err_t rx_host_irq_assert (
    void ) [nodiscard]
```

Assert the HOST_IRQ line (implementation).

Assert HOST_IRQ (drive LOW to signal data ready).

Drives the P67 output LOW to signal the host SoC that the RX72N has data ready or wants attention. HOST_IRQ is an active-low, level-triggered line; the host will continue to see "asserted" until [rx_host_irq_deassert\(\)](#) drives it back HIGH. The function is a single register read-modify-write on PORT6.PODR; no interrupts are touched.

Note

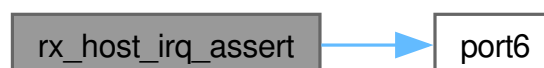
Safe to call from task context. Not protected against concurrent calls; serialize at the application layer.

Definition at line 148 of file [rx_host_irq.c](#).

```
00149 {
00150     if (!s_initialized) {
00151         return k_rx_err_invalid_state;
00152     }
00153
00154     volatile rx_port_regs_t* p6 = port6();
00155     p6->podr &= (uint8_t)~k_host_irq_pin_mask;
00156
00157     return k_rx_ok;
00158 }
```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::podr](#), [port6\(\)](#), and [s_initialized](#).

Here is the call graph for this function:



8.119.3.2 rx_host_irq_deassert()

```
rx_err_t rx_host_irq_deassert (
    void ) [nodiscard]
```

De-assert the HOST_IRQ line (implementation).

Deassert HOST_IRQ (drive HIGH, idle state).

Drives the P67 output HIGH to release the active-low HOST_IRQ line back to its idle state. Single register read-modify-write on PORT6.PODR; no interrupts touched.

Note

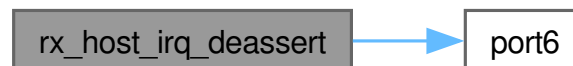
Safe to call from task context. Not protected against concurrent calls; serialize at the application layer.

Definition at line 171 of file [rx_host_irq.c](#).

```
00172 {
00173     if (!s_initialized) {
00174         return k_rx_err_invalid_state;
00175     }
00176
00177     volatile rx_port_regs_t* p6 = port6();
00178     p6->podr |= k_host_irq_pin_mask;
00179
00180     return k_rx_ok;
00181 }
```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::podr](#), [port6\(\)](#), and [s_initialized](#).

Here is the call graph for this function:



8.119.3.3 rx_host_irq_deinit()

```
rx_err_t rx_host_irq_deinit (
    void ) [nodiscard]
```

Deinitialize the HOST_IRQ output (implementation).

Deinitialize HOST_IRQ pin (revert to input).

Symmetric counterpart to [rx_host_irq_init\(\)](#). Drives P67 high one last time so the line is left in its idle (de-asserted) state, then changes the pin direction back to input, gating off any spurious downstream effects. Clears the `s_initialized` flag so subsequent assert/deassert calls return `k_rx_err_invalid_state`.

Note

Not thread-safe. Serialize all `rx_host_irq_*` operations at the application layer.

Definition at line 114 of file `rx_host_irq.c`.

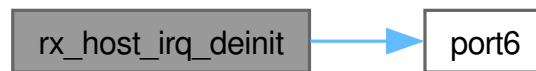
```

00115 {
00116     if (!s_initialized) {
00117         rx_log_error(s_tag, "Not initialized");
00118         return k_rx_err_invalid_state;
00119     }
00120
00121     volatile rx_port_regs_t* p6 = port6();
00122
00123     /* Deassert before reverting to input */
00124     p6->podr |= k_host_irq_pin_mask;
00125
00126     /* Revert to input (safe default) */
00127     p6->pdr &= (uint8_t)~k_host_irq_pin_mask;
00128
00129     s_initialized = false;
00130
00131     rx_log_info(s_tag, "HOST_IRQ P67 deinitialized (input)");
00132     return k_rx_ok;
00133 }

```

References `k_host_irq_pin_mask`, `rx_port_regs_t::pdr`, `rx_port_regs_t::podr`, `port6()`, `s_initialized`, and `s_tag`.

Here is the call graph for this function:



8.119.3.4 rx`host`irq`init()

```

rx_err_t rx_host_irq_init (
    void ) [nodiscard]

```

Initialize HOST_IRQ pin as GPIO output (deasserted / HIGH).

Configures P67 as a GPIO output in CMOS push-pull mode. Sets the initial output level to HIGH (deasserted / idle). The MPC function select is set to GPIO (not IRQ input).

Initialization steps:

1. Clear PMR bit 7 (GPIO mode, not peripheral)
2. Set PODR bit 7 HIGH (deasserted state)
3. Set PDR bit 7 (output direction)

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success, P67 configured as output
<code>k_rx_err_invalid_state</code>	Already initialized

Precondition

P67 not in use by another peripheral (RIIC2 etc.)

Postcondition

P67 configured as GPIO output, driven HIGH

Note

Not thread-safe. Call during system init only.

See also

[rx_host_irq_deinit\(\)](#) Disable the output

Since

Version 1.0.0

Definition at line 77 of file [rx_host_irq.c](#).

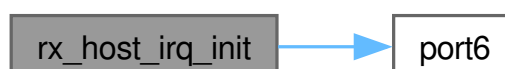
```

00078 {
00079     if (s_initialized) {
00080         rx_log_error(s_tag, "Already initialized");
00081         return k_rx_err_invalid_state;
00082     }
00083
00084     volatile rx_port_regs_t* p6 = port6();
00085
00086     /* Step 1: GPIO mode (clear PMR bit 7) */
00087     p6->pmr &= (uint8_t)~k_host_irq_pin_mask;
00088
00089     /* Step 2: Set output HIGH (deasserted / idle) before enabling output */
00090     p6->podr |= k_host_irq_pin_mask;
00091
00092     /* Step 3: Set direction to output */
00093     p6->pdr |= k_host_irq_pin_mask;
00094
00095     s_initialized = true;
00096
00097     rx_log_info(s_tag, "HOST_IRQ P67 initialized (output, idle HIGH)");
00098     return k_rx_ok;
00099 }

```

References [k_host_irq_pin_mask](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_port_regs_t::podr](#), [port6\(\)](#), [s_initialized](#), and [s_tag](#).

Here is the call graph for this function:



8.119.3.5 rx`host`irq`is`asserted()

```
rx_err_t rx_host_irq_is_asserted (
    bool * asserted) [nodiscard]
```

Check if HOST_IRQ is currently asserted.

Returns the current state of the HOST_IRQ output.

Parameters

out	asserted	Pointer to store state (true = asserted / LOW)
-----	----------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	asserted is nullptr

Precondition

[rx_host_irq_init\(\)](#) called

Postcondition

asserted set to current output state

Note

Thread-safe (reads volatile register)

Since

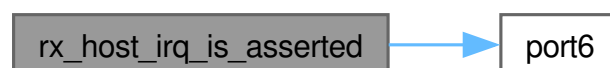
Version 1.0.0

Definition at line 183 of file [rx_host_irq.c](#).

```
00184 {
00185     if (asserted == nullptr) {
00186         return k_rx_err_null_ptr;
00187     }
00188
00189     volatile rx_port_regs_t* p6 = port6();
00190
00191     /* Active-low: pin LOW means asserted */
00192     *asserted = ((p6->podr & k_host_irq_pin_mask) == k_host_irq_active);
00193
00194     return k_rx_ok;
00195 }
```

References [k_host_irq_active](#), [k_host_irq_pin_mask](#), [rx_port_regs_t::podr](#), and [port6\(\)](#).

Here is the call graph for this function:



8.119.4 Variable Documentation

8.119.4.1 s_initialized

bool s_initialized = false [static]

Initialization flag.

Definition at line 70 of file [rx_host_irq.c](#).

8.119.4.2 s_tag

const char* const s_tag = "HOST_IRQ" [static]

Definition at line 49 of file [rx_host_irq.c](#).

8.120 rx_host_irq.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_host_irq.c
00003  * @brief HOST_IRQ GPIO Output Driver Implementation
00004  *
00005  * @details
00006  * Configures P67 (PORT6 pin 7) as a GPIO output for signaling the RPi5
00007  * host controller that the RX72N has data ready for SPI transfer. The pin
00008  * is active-low: assert LOW = data ready, deassert HIGH = idle.
00009  *
00010  * ## GPIO Configuration
00011  * ```
00012  * rx_host_irq_init()
00013  *   1. PMR6.B7 = 0 (GPIO mode)
00014  *   2. PODR6.B7 = 1 (initial HIGH = idle)
00015  *   3. PDR6.B7 = 1 (output direction)
00016  *
00017  * rx_host_irq_assert()
00018  *   PODR6.B7 = 0 (drive LOW = data ready)
00019  *
00020  * rx_host_irq_deassert()
00021  *   PODR6.B7 = 1 (drive HIGH = idle)
00022  * ```
00023  *
00024  * @par NASA Power of 10 Compliance:
00025  * - Rule 1: No goto, setjmp, recursion
00026  * - Rule 3: No dynamic memory (static variables only)
00027  * - Rule 4: All functions < 60 lines
00028  * - Rule 5: 2+ validation checks per function
00029  * - Rule 7: All return values checked
00030  * - Rule 8: C23 typed enums, no magic numbers
00031  * - Rule 10: -Wall -Wextra -Werror clean
00032  *
00033  * @see rx_host_irq.h Public API
00034  * @see hardware_config.h Pin assignments
00035  *
00036  * @author Locked, Inc.
00037  * @date 2026-02-10
00038  * @copyright Copyright (c) 2026 Locked Inc.
00039  * SPDX-License-Identifier: MIT
00040  * @since Version 1.0.0
00041  */
00042
00043 #include "rx_host_irq.h"
00044
00045 #include "rx72n_port_regs.h"
00046 #include "rx_log.h"
00047
00048 static const char* const s_tag = "HOST_IRQ";
00049
00050
```

```

00051 /*
=====
00052  * Constants
00053  *
=====
00054  */
00055
00056 /** @brief GPIO configuration constants for HOST_IRQ on P67 */
00057 typedef enum : uint8_t {
00058     k_host_irq_pin_bit = 7, /**< Pin 7 within PORT6 */
00059     k_host_irq_pin_mask = 0x80, /**< Bit mask for pin 7 (1 « 7) */
00060     k_host_irq_active = 0, /**< Active level: LOW */
00061     k_host_irq_idle = 1, /**< Idle level: HIGH */
00062 } host_irq_gpio_constants_t;
00063
00064 /*
=====
00065  * Static Variables
00066  *
=====
00067  */
00068
00069 /** @brief Initialization flag */
00070 static bool s_initialized = false;
00071
00072 /*
=====
00073  * Public API Implementation
00074  *
=====
00075  */
00076
00077 rx_err_t rx_host_irq_init(void)
00078 {
00079     if (s_initialized) {
00080         rx_log_error(s_tag, "Already initialized");
00081         return k_rx_err_invalid_state;
00082     }
00083
00084     volatile rx_port_regs_t* p6 = port6();
00085
00086     /* Step 1: GPIO mode (clear PMR bit 7) */
00087     p6->pmr &= (uint8_t)~k_host_irq_pin_mask;
00088
00089     /* Step 2: Set output HIGH (deasserted / idle) before enabling output */
00090     p6->podr |= k_host_irq_pin_mask;
00091
00092     /* Step 3: Set direction to output */
00093     p6->pdr |= k_host_irq_pin_mask;
00094
00095     s_initialized = true;
00096
00097     rx_log_info(s_tag, "HOST_IRQ P67 initialized (output, idle HIGH)");
00098     return k_rx_ok;
00099 }
00100
00101 /**
00102  * @brief Deinitialize the HOST_IRQ output (implementation)
00103  *
00104  * @details
00105  * Symmetric counterpart to rx_host_irq_init(). Drives P67 high one last
00106  * time so the line is left in its idle (de-asserted) state, then changes
00107  * the pin direction back to input, gating off any spurious downstream
00108  * effects. Clears the s_initialized flag so subsequent assert/deassert
00109  * calls return k_rx_err_invalid_state.
00110  *
00111  * @note Not thread-safe. Serialize all rx_host_irq_* operations at the
00112  * application layer.
00113  */
00114 rx_err_t rx_host_irq_deinit(void)
00115 {
00116     if (!s_initialized) {
00117         rx_log_error(s_tag, "Not initialized");
00118         return k_rx_err_invalid_state;
00119     }
00120
00121     volatile rx_port_regs_t* p6 = port6();
00122
00123     /* Deassert before reverting to input */
00124     p6->podr |= k_host_irq_pin_mask;
00125
00126     /* Revert to input (safe default) */
00127     p6->pdr &= (uint8_t)~k_host_irq_pin_mask;
00128
00129     s_initialized = false;
00130
00131     rx_log_info(s_tag, "HOST_IRQ P67 deinitialized (input)");

```

```

00132     return k_rx_ok;
00133 }
00134
00135 /**
00136  * @brief Assert the HOST_IRQ line (implementation)
00137  *
00138  * @details
00139  * Drives the P67 output LOW to signal the host SoC that the RX72N has
00140  * data ready or wants attention. HOST_IRQ is an active-low,
00141  * level-triggered line; the host will continue to see "asserted" until
00142  * rx_host_irq_deassert() drives it back HIGH. The function is a single
00143  * register read-modify-write on PORT6.PODR; no interrupts are touched.
00144  *
00145  * @note Safe to call from task context. Not protected against concurrent
00146  * calls; serialize at the application layer.
00147  */
00148 rx_err_t rx_host_irq_assert(void)
00149 {
00150     if (!s_initialized) {
00151         return k_rx_err_invalid_state;
00152     }
00153     volatile rx_port_regs_t* p6 = port6();
00154     p6->podr &= (uint8_t)~k_host_irq_pin_mask;
00155     return k_rx_ok;
00156 }
00157
00158 /**
00159  * @brief De-assert the HOST_IRQ line (implementation)
00160  *
00161  * @details
00162  * Drives the P67 output HIGH to release the active-low HOST_IRQ line back
00163  * to its idle state. Single register read-modify-write on PORT6.PODR;
00164  * no interrupts touched.
00165  *
00166  * @note Safe to call from task context. Not protected against concurrent
00167  * calls; serialize at the application layer.
00168  */
00169 rx_err_t rx_host_irq_deassert(void)
00170 {
00171     if (!s_initialized) {
00172         return k_rx_err_invalid_state;
00173     }
00174     volatile rx_port_regs_t* p6 = port6();
00175     p6->podr |= k_host_irq_pin_mask;
00176     return k_rx_ok;
00177 }
00178
00179 rx_err_t rx_host_irq_is_asserted(bool* asserted)
00180 {
00181     if (asserted == nullptr) {
00182         return k_rx_err_null_ptr;
00183     }
00184     volatile rx_port_regs_t* p6 = port6();
00185     /* Active-low: pin LOW means asserted */
00186     *asserted = ((p6->podr & k_host_irq_pin_mask) == k_host_irq_active);
00187     return k_rx_ok;
00188 }
00189
00190
00191
00192
00193
00194
00195

```

8.121 libs/rx72n/src/rx_irq_filter.c File Reference

IRQ Digital Filter Driver Implementation for RX72N.

```
#include "rx_irq_filter.h"
```

```
#include "rx72n_regs.h"
```

Include dependency graph for rx_irq_filter.c:



Enumerations

- enum [rx_irq_filter_constants_t](#) : uint8_t { [k_irq_filter_irqs_per_reg](#) = 8 , [k_irq_filter_clock_bits](#) = 2 , [k_irq_filter_clock_mask](#) = 0x03 }

IRQ filter register organization and bit field constants.

Functions

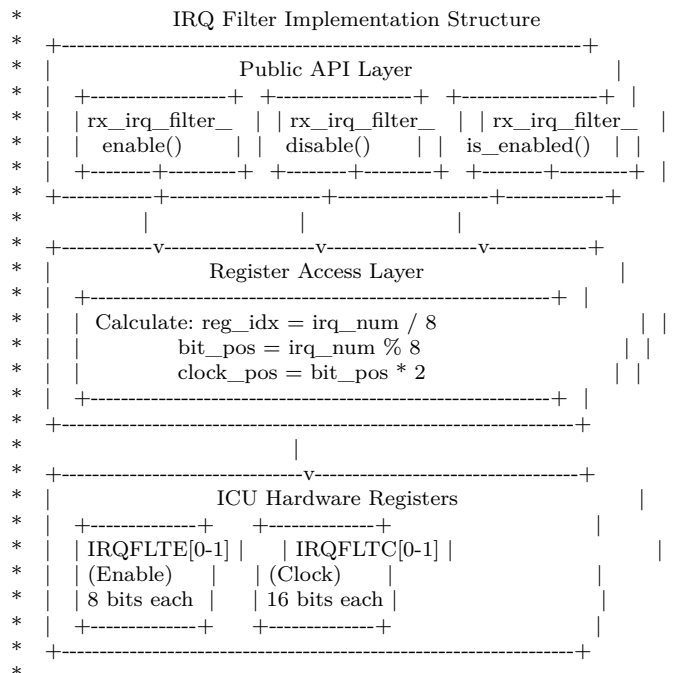
- rx_err_t [rx_irq_filter_enable](#) (uint8_t irq_num, [rx_irq_filter_clk_t](#) filter_clk)
Enable hardware digital filter on an IRQ pin.
- rx_err_t [rx_irq_filter_disable](#) (uint8_t irq_num)
Disable hardware digital filter on an IRQ pin.
- rx_err_t [rx_irq_filter_is_enabled](#) (uint8_t irq_num, bool *enabled)
Check if digital filter is enabled on an IRQ pin.
- rx_err_t [rx_irq_filter_get_clock](#) (uint8_t irq_num, [rx_irq_filter_clk_t](#) *filter_clk)
Get current filter clock divisor setting for an IRQ.

8.121.1 Detailed Description

IRQ Digital Filter Driver Implementation for RX72N.

Implements hardware digital filtering on IRQ pins using the ICU (Interrupt Controller Unit) peripheral. The filter provides noise rejection by requiring 3 consecutive samples at the same level before recognizing a valid transition.

Implementation Architecture



Register Organization

```

* IRQFLTE[0] (Filter Enable for IRQ0-7):
* +-----+-----+-----+-----+-----+-----+
* | IRQ7| IRQ6| IRQ5| IRQ4| IRQ3| IRQ2| IRQ1| IRQ0|
* | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
* +-----+-----+-----+-----+-----+-----+
*
* IRQFLTC[0] (Filter Clock for IRQ0-7):
* +-----+-----+-----+-----+-----+-----+
* | IRQ7 | IRQ6 | IRQ5 | IRQ4 | IRQ3 | IRQ2 | IRQ1 | IRQ0 |
* |b15:b14 |b13:b12 |b11:b10 | b9:b8 | b7:b6 | b5:b4 | b3:b2 | b1:b0 |
* +-----+-----+-----+-----+-----+-----+
* Clock values: 00=PCLK/1, 01=PCLK/8, 10=PCLK/32, 11=PCLK/64
*

```

Performance Characteristics

Function	Execution Time	Stack Usage
enable()	~200 ns	8 bytes
disable()	~100 ns	4 bytes
is_enabled()	~80 ns	4 bytes
get_clock()	~100 ns	8 bytes

Memory Usage

Category	Size	Notes
Code (.text)	~200 bytes	All functions
Constants (.rodata)	0 bytes	None
Static data	0 bytes	No static variables

Thread Safety

Not thread-safe. Register read-modify-write operations are not atomic. If filter configuration must change at runtime, use mutex protection.

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, recursion
- Rule 2: [OK] No loops (direct register operations)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 30 lines
- Rule 5: [OK] Input validation on all functions
- Rule 7: [OK] All return values checked
- Rule 8: [OK] C23 typed enums for constants
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

See also

[rx_irq_filter.h](#) Public API documentation

[rx72n_icu_regs.h](#) ICU register definitions

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_irq_filter.c](#).

8.121.2 Enumeration Type Documentation

8.121.2.1 irq`filter`constants`

enum [irq_filter_constants_t](#) : uint8_t

IRQ filter register organization and bit field constants.

Defines constants for navigating the IRQFLTE and IRQFLTC register arrays. The ICU organizes filter controls into two registers each:

- Index 0: IRQ0-7 (lower 8 IRQs)
- Index 1: IRQ8-15 (upper 8 IRQs)

Register Index Calculation

```
* For IRQ number N:  
* - Register index = N / 8 (0 for IRQ0-7, 1 for IRQ8-15)  
* - Bit position = N % 8 (0-7 within the register)  
* - Clock position = bit_position * 2 (2 bits per IRQ in IRQFLTC)  
*
```

Since

Version 1.0.0

Enumerator

k_irq_filter_irqs_per_reg	Number of IRQs controlled by each register pair. IRQFLTE[0]/↵ IRQFLTC[0] handle IRQ0-7, [1] handles IRQ8-15
k_irq_filter_clock_bits	Number of bits per clock divisor setting in IRQFLTC. Each IRQ uses 2 bits for 4 possible divisor values (0-3)
k_irq_filter_clock_mask	Bit mask for extracting/setting clock divisor value. 2 bits = 0b11 = 0x03

Definition at line 126 of file [rx_irq_filter.c](#).

```

00126      : uint8_t {
00127  /**
00128   * @brief Number of IRQs controlled by each register pair
00129   * @details IRQFLTE[0]/IRQFLTC[0] handle IRQ0-7, [1] handles IRQ8-15
00130   */
00131   k_irq_filter_irqs_per_reg = 8,
00132
00133  /**
00134   * @brief Number of bits per clock divisor setting in IRQFLTC
00135   * @details Each IRQ uses 2 bits for 4 possible divisor values (0-3)
00136   */
00137   k_irq_filter_clock_bits = 2,
00138
00139  /**
00140   * @brief Bit mask for extracting/setting clock divisor value
00141   * @details 2 bits = 0b11 = 0x03
00142   */
00143   k_irq_filter_clock_mask = 0x03,
00144 } irq_filter_constants_t;

```

8.121.3 Function Documentation

8.121.3.1 rx_irq_filter_disable()

```

rx_err_t rx_irq_filter_disable (
    uint8_t irq_num) [nodiscard]

```

Disable hardware digital filter on an IRQ pin.

Clears the filter enable bit in IRQFLTE for the specified IRQ. After disabling, raw unfiltered input passes to interrupt detector.

Note

Clock divisor setting preserved for potential re-enable

See also

[rx_irq_filter.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 236 of file [rx_irq_filter.c](#).

```
00237 {
00238     if (irq_num > k_rx_irq_max) {
00239         return k_rx_err_invalid_arg;
00240     }
00241
00242     #ifdef __RX__
00243     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00244     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00245
00246     /* Clear filter enable bit */
00247     icu()->irqflte[reg_idx] &= (uint8_t) ~(1 « bit_pos);
00248
00249     #endif
00250
00251     return k_rx_ok;
00252 }
```

References [icu\(\)](#), [rx_icu_regs_t::irqflte](#), [k_irq_filter_irqs_per_reg](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.121.3.2 rx`irq`filter`enable()

```
rx_err_t rx_irq_filter_enable (
    uint8_t irq_num,
    rx\_irq\_filter\_clk\_t filter_clk) [nodiscard]
```

Enable hardware digital filter on an IRQ pin.

Configures the ICU filter enable (IRQFLTE) and filter clock (IRQFLTC) registers for the specified IRQ. Sets clock divisor first, then enables the filter to avoid sampling at incorrect rate during transition.

Register Modification Sequence

1. IRQFLTC: Clear and set 2-bit clock field for this IRQ
2. IRQFLTE: Set enable bit for this IRQ

Note

On host (non-RX) builds, validates parameters but skips register access

See also

[rx_irq_filter.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 170 of file [rx_irq_filter.c](#).

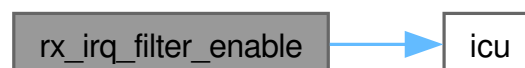
```

00171 {
00172     if (irq_num > k_rx_irq_max) {
00173         return k_rx_err_invalid_arg;
00174     }
00175
00176     if (filter_clk > k_irq_filter_pclk_64) {
00177         return k_rx_err_invalid_arg;
00178     }
00179
00180     #ifdef __RX__
00181     /*
00182      * Determine which register to use:
00183      * - IRQ0-7:  IRQFLTE[0], IRQFLTC[0]
00184      * - IRQ8-15: IRQFLTE[1], IRQFLTC[1]
00185      */
00186     uint8_t reg_idx  = irq_num / k_irq_filter_irqs_per_reg;
00187     uint8_t bit_pos  = irq_num % k_irq_filter_irqs_per_reg;
00188     uint8_t clock_pos = bit_pos * k_irq_filter_clock_bits;
00189
00190     /*
00191      * Set filter clock divisor first (before enabling)
00192      *
00193      * IRQFLTC[n] has 2 bits per IRQ:
00194      * - Bits [1:0]:  IRQ0/8 clock select
00195      * - Bits [3:2]:  IRQ1/9 clock select
00196      * - Bits [5:4]:  IRQ2/10 clock select
00197      * - etc.
00198      */
00199     uint16_t clock_mask = (uint16_t)(k_irq_filter_clock_mask « clock_pos);
00200     uint16_t clock_value = (uint16_t)(filter_clk « clock_pos);
00201
00202     icu()->irqftc[reg_idx] = (icu()->irqftc[reg_idx] & ~clock_mask) | clock_value;
00203
00204     /*
00205      * Enable filter for this IRQ
00206      *
00207      * IRQFLTE[n] has 1 bit per IRQ:
00208      * - Bit 0: IRQ0/8 filter enable
00209      * - Bit 1: IRQ1/9 filter enable
00210      * - etc.
00211      */
00212     icu()->irqfte[reg_idx] |= (uint8_t)(1 « bit_pos);
00213
00214     #else
00215     /* Host-side stub for unit testing */
00216     (void)filter_clk;
00217     #endif
00218
00219     return k_rx_ok;
00220 }

```

References [icu\(\)](#), [rx_icu_regs_t::irqftc](#), [rx_icu_regs_t::irqfte](#), [k_irq_filter_clock_bits](#), [k_irq_filter_clock_mask](#), [k_irq_filter_irqs_per_reg](#), [k_irq_filter_pclk_64](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.121.3.3 rx_irq_filter_get_clock()

```

rx_err_t rx_irq_filter_get_clock (
    uint8_t irq_num,
    rx_irq_filter_clk_t * filter_clk) [nodiscard]

```

Get current filter clock divisor setting for an IRQ.

Reads the 2-bit clock divisor field from IRQFLTC register for the specified IRQ. Returns the divisor setting even if filter is disabled.

Note

On host builds, always returns `k_irq_filter_pclk_1`

See also

[rx_irq_filter.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 305 of file `rx_irq_filter.c`.

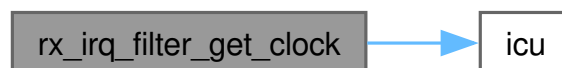
```

00306 {
00307     if (irq_num > k_rx_irq_max) {
00308         return k_rx_err_invalid_arg;
00309     }
00310
00311     if (filter_clk == nullptr) {
00312         return k_rx_err_invalid_arg;
00313     }
00314
00315     #ifdef __RX__
00316     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00317     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00318     uint8_t clock_pos = bit_pos * k_irq_filter_clock_bits;
00319
00320     uint16_t clock_value = (icu()->irqfltc[reg_idx] » clock_pos) & k_irq_filter_clock_mask;
00321     *filter_clk = (rx_irq_filter_clk_t)clock_value;
00322
00323     #else
00324     *filter_clk = k_irq_filter_pclk_1;
00325     #endif
00326
00327     return k_rx_ok;
00328 }

```

References `icu()`, `rx_icu_regs_t::irqfltc`, `k_irq_filter_clock_bits`, `k_irq_filter_clock_mask`, `k_irq_filter_irqs_per_reg`, `k_irq_filter_pclk_1`, and `k_rx_irq_max`.

Here is the call graph for this function:



8.121.3.4 rx`irq`filter`is`enabled()

```

rx_err_t rx_irq_filter_is_enabled (
    uint8_t irq_num,
    bool * enabled) [nodiscard]

```

Check if digital filter is enabled on an IRQ pin.

Reads the IRQFLTE register bit for the specified IRQ to determine current filter enable state.

Note

On host builds, always returns false (filter disabled)

See also

[rx_irq_filter.h](#) Full API documentation

Since

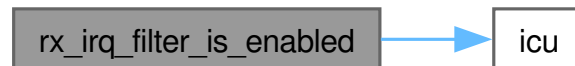
Version 1.0.0

Definition at line 268 of file [rx_irq_filter.c](#).

```
00269 {
00270     if (irq_num > k_rx_irq_max) {
00271         return k_rx_err_invalid_arg;
00272     }
00273
00274     if (enabled == nullptr) {
00275         return k_rx_err_invalid_arg;
00276     }
00277
00278     #ifdef __RX__
00279     uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00280     uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00281
00282     *enabled = (icu()->irqfte[reg_idx] & (1 « bit_pos)) != 0;
00283
00284     #else
00285     *enabled = false;
00286     #endif
00287
00288     return k_rx_ok;
00289 }
```

References [icu\(\)](#), [rx_icu_regs_t::irqfte](#), [k_irq_filter_irqs_per_reg](#), and [k_rx_irq_max](#).

Here is the call graph for this function:



8.122 rx_irq_filter.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_irq_filter.c
00003  * @brief IRQ Digital Filter Driver Implementation for RX72N
00004  *
00005  * @details
00006  * Implements hardware digital filtering on IRQ pins using the ICU (Interrupt
00007  * Controller Unit) peripheral. The filter provides noise rejection by requiring
00008  * 3 consecutive samples at the same level before recognizing a valid transition.
00009  *
00010  * @par Implementation Architecture
00011  * @verbatim
00012  *
00013  *   +-----+
00014  *   |               Public API Layer               |
00015  *   | +-----+ +-----+ +-----+ +-----+ |
00016  *   | | rx_irq_filter_ | | rx_irq_filter_ | | rx_irq_filter_ | |
00017  *   | | enable()      | | disable()      | | is_enabled()    | |
00018  *   | +-----+ +-----+ +-----+ +-----+ |
00019  *   +-----+
00020  *
00021  *   +-----+
00022  *   |               Register Access Layer           |
00023  *   | +-----+ +-----+ +-----+ +-----+ |
00024  *   | | Calculate: reg_idx = irq_num / 8 | | | | |
00025  *   | | bit_pos = irq_num % 8 | | | | |
00026  *   | | clock_pos = bit_pos * 2 | | | | |
00027  *   | +-----+ +-----+ +-----+ +-----+ |
00028  *   +-----+
```


Generated by Doxygen

```

00114 * - Index 1: IRQ8-15 (upper 8 IRQs)
00115 *
00116 * @par Register Index Calculation
00117 * @verbatim
00118 *   For IRQ number N:
00119 *   - Register index = N / 8 (0 for IRQ0-7, 1 for IRQ8-15)
00120 *   - Bit position = N % 8 (0-7 within the register)
00121 *   - Clock position = bit_position * 2 (2 bits per IRQ in IRQFLTC)
00122 * @endverbatim
00123 *
00124 * @since Version 1.0.0
00125 */
00126 typedef enum : uint8_t {
00127 /**
00128  * @brief Number of IRQs controlled by each register pair
00129  * @details IRQFLTE[0]/IRQFLTC[0] handle IRQ0-7, [1] handles IRQ8-15
00130  */
00131 k_irq_filter_irqs_per_reg = 8,
00132
00133 /**
00134  * @brief Number of bits per clock divisor setting in IRQFLTC
00135  * @details Each IRQ uses 2 bits for 4 possible divisor values (0-3)
00136  */
00137 k_irq_filter_clock_bits = 2,
00138
00139 /**
00140  * @brief Bit mask for extracting/setting clock divisor value
00141  * @details 2 bits = 0b11 = 0x03
00142  */
00143 k_irq_filter_clock_mask = 0x03,
00144 } irq_filter_constants_t;
00145
00146 /*
=====
00147 * Public API Implementation
00148 *
=====
00149 */
00150
00151 /**
00152  * @brief Enable hardware digital filter on an IRQ pin
00153  *
00154  * @details
00155  * Configures the ICU filter enable (IRQFLTE) and filter clock (IRQFLTC)
00156  * registers for the specified IRQ. Sets clock divisor first, then enables
00157  * the filter to avoid sampling at incorrect rate during transition.
00158  *
00159  * @par Register Modification Sequence
00160  * 1. IRQFLTC: Clear and set 2-bit clock field for this IRQ
00161  * 2. IRQFLTE: Set enable bit for this IRQ
00162  *
00163  *
00164  *
00165  * @note On host (non-RX) builds, validates parameters but skips register access
00166  *
00167  * @see rx_irq_filter.h Full API documentation
00168  * @since Version 1.0.0
00169  */
00170 rx_err_t rx_irq_filter_enable(uint8_t irq_num, rx_irq_filter_clk_t filter_clk)
00171 {
00172     if (irq_num > k_rx_irq_max) {
00173         return k_rx_err_invalid_arg;
00174     }
00175
00176     if (filter_clk > k_irq_filter_pclk_64) {
00177         return k_rx_err_invalid_arg;
00178     }
00179
00180 #ifdef __RX__
00181 /**
00182  * Determine which register to use:
00183  * - IRQ0-7: IRQFLTE[0], IRQFLTC[0]
00184  * - IRQ8-15: IRQFLTE[1], IRQFLTC[1]
00185  */
00186 uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00187 uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00188 uint8_t clock_pos = bit_pos * k_irq_filter_clock_bits;
00189
00190 /**
00191  * Set filter clock divisor first (before enabling)
00192  *
00193  * IRQFLTC[n] has 2 bits per IRQ:
00194  * - Bits [1:0]: IRQ0/8 clock select
00195  * - Bits [3:2]: IRQ1/9 clock select
00196  * - Bits [5:4]: IRQ2/10 clock select
00197  * - etc.
00198  */

```

```

00199 uint16_t clock_mask = (uint16_t)(k_irq_filter_clock_mask « clock_pos);
00200 uint16_t clock_value = (uint16_t)(filter_clk « clock_pos);
00201
00202 icu()->irqfite[reg_idx] = (icu()->irqfite[reg_idx] & ~clock_mask) | clock_value;
00203
00204 /*
00205  * Enable filter for this IRQ
00206  *
00207  * IRQFLTE[n] has 1 bit per IRQ:
00208  * - Bit 0: IRQ0/8 filter enable
00209  * - Bit 1: IRQ1/9 filter enable
00210  * - etc.
00211  */
00212 icu()->irqfite[reg_idx] |= (uint8_t)(1 « bit_pos);
00213
00214 #else
00215 /* Host-side stub for unit testing */
00216 (void)filter_clk;
00217 #endif
00218
00219 return k_rx_ok;
00220 }
00221
00222 /**
00223  * @brief Disable hardware digital filter on an IRQ pin
00224  *
00225  * @details
00226  * Clears the filter enable bit in IRQFLTE for the specified IRQ.
00227  * After disabling, raw unfiltered input passes to interrupt detector.
00228  *
00229  *
00230  *
00231  * @note Clock divisor setting preserved for potential re-enable
00232  *
00233  * @see rx_irq_filter.h Full API documentation
00234  * @since Version 1.0.0
00235  */
00236 rx_err_t rx_irq_filter_disable(uint8_t irq_num)
00237 {
00238     if (irq_num > k_rx_irq_max) {
00239         return k_rx_err_invalid_arg;
00240     }
00241
00242     #ifdef __RX__
00243         uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00244         uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00245
00246         /* Clear filter enable bit */
00247         icu()->irqfite[reg_idx] &= (uint8_t) ~(1 « bit_pos);
00248     #endif
00249
00250     return k_rx_ok;
00251 }
00252
00253
00254 /**
00255  * @brief Check if digital filter is enabled on an IRQ pin
00256  *
00257  * @details
00258  * Reads the IRQFLTE register bit for the specified IRQ to determine
00259  * current filter enable state.
00260  *
00261  *
00262  *
00263  * @note On host builds, always returns false (filter disabled)
00264  *
00265  * @see rx_irq_filter.h Full API documentation
00266  * @since Version 1.0.0
00267  */
00268 rx_err_t rx_irq_filter_is_enabled(uint8_t irq_num, bool* enabled)
00269 {
00270     if (irq_num > k_rx_irq_max) {
00271         return k_rx_err_invalid_arg;
00272     }
00273
00274     if (enabled == nullptr) {
00275         return k_rx_err_invalid_arg;
00276     }
00277
00278     #ifdef __RX__
00279         uint8_t reg_idx = irq_num / k_irq_filter_irqs_per_reg;
00280         uint8_t bit_pos = irq_num % k_irq_filter_irqs_per_reg;
00281
00282         *enabled = (icu()->irqfite[reg_idx] & (1 « bit_pos)) != 0;
00283     #else
00284         *enabled = false;
00285     #endif

```


Enumerations

- enum `iwdt_timeout_limits_t` : `uint32_t` { `k_iwdt_timeout_min_ms` = 1 , `k_iwdt_timeout_max_ms` = 16384 }
IWDT timeout validation limits.
- enum `iwdt_init_state_t` : `uint8_t` { `k_iwdt_not_initialized` = 0 , `k_iwdt_initialized` = 1 }
IWDT initialization state tracking.
- enum `iwdt_buffer_size_constants_t` : `uint8_t` { `k_task_name_max_len` = 16 , `k_task_name_cmp_len` = 15 , `k_log_msg_buffer_size` = 64 }
Task monitoring string buffer size constants.
- enum `task_registration_state_t` : `uint8_t` { `k_task_not_registered` = 0 , `k_task_registered` = 1 }
Task registration state tracking.
- enum `task_search_constants_t` : `int32_t` { `k_task_not_found` = -1 , `k_task_idx_min` = 0 }
Constants for task search operations.

Functions

- static `rx_err_t` `internal_find_timeout_config` (`const uint32_t` timeout_ms, `const iwdt_timeout_entry_t` **entry)
Find best timeout configuration for requested timeout.
- static `rx_err_t` `internal_configure_iwdt_control_register` (`const iwdt_timeout_entry_t` *config)
Configure IWDT control register (IWDT`r) from timeout table entry.
- static `rx_err_t` `internal_configure_iwdt_registers` (`void`)
Configure IWDT reset and sleep-count behavior registers.
- `rx_err_t` `rx_hal_iwdt_init` (`const uint32_t` timeout_ms)
Initialize the Independent Watchdog Timer.
- `void` `rx_hal_iwdt_feed` (`void`)
Feed the watchdog timer to prevent reset.
- `bool` `rx_hal_iwdt_was_reset` (`void`)
Check if system was reset by watchdog.
- `rx_iwdt_reset_cause_t` `rx_hal_iwdt_get_reset_cause` (`void`)
Get detailed watchdog reset cause.
- `void` `rx_hal_iwdt_clear_status` (`void`)
Clear watchdog status flags.
- static `int32_t` `internal_find_task` (`const char` *task_name)
Find task index by name in the registered tasks array.
- `rx_err_t` `rx_hal_iwdt_register_task` (`const char` *task_name, `uint32_t` timeout_ms)
Register a task for watchdog monitoring.
- `void` `rx_hal_iwdt_task_heartbeat` (`const char` *task_name)
Record a task heartbeat to indicate liveness.
- `rx_err_t` `rx_hal_iwdt_check_tasks` (`void`)
Check all monitored tasks for timeout violations.
- `const char` * `rx_hal_iwdt_get_failed_task` (`void`)
Get name of last failed task for post-mortem analysis.

Variables

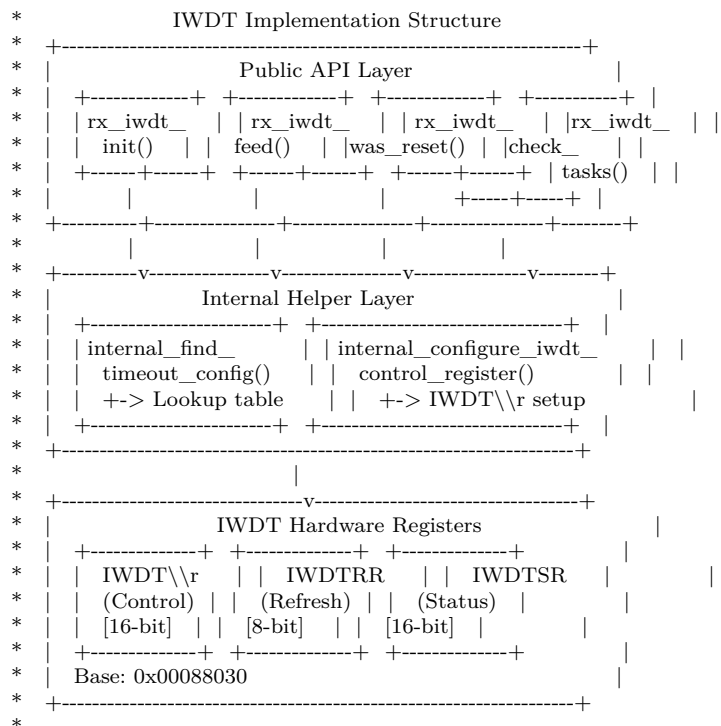
- static const `iwdt_timeout_entry_t s_timeout_table []`
IWDT timeout configuration lookup table.
- static const `uint32_t s_iwdt_timeout_table_size`
Number of entries in the timeout lookup table.
- static const char *const `s_tag = "IWDT"`
Log tag for IWDT module.
- static `iwdt_init_state_t s_iwdt_initialized = k_iwdt_not_initialized`
Flag indicating IWDT has been initialized.
- static `task_monitor_state_t s_task_monitor = {}`
Module state for task monitoring (static allocation).

8.123.1 Detailed Description

Independent Watchdog Timer (IWDT) Driver Implementation for RX72N.

Implements hardware and software watchdog functionality for safety-critical system monitoring. The IWDT provides protection against software hangs, deadlocks, and infinite loops through automatic system reset.

Implementation Architecture



Register Layout (IWDT\`r - 0x00088030)

```

* Bits [15:14] RPSS - Window Start Position
* +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+
* | - |RPSS1|RPSS0| - |RP- |RP- | - | - |CKS3|CKS2|CKS1|CKS0| - | - |TOPS1|TOPS0|
* | b15 | b14 | b13 | b12 |ES1 |ES0 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
* +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+ +-----+
* TOPS[1:0]: Timeout Period Select (00=1024, 01=4096, 10=8192, 11=16384 cycles)
* CKS[3:0]: Clock Division Ratio (0000=/1, 0010=/16, 0100=/32, 1111=/128)
* RPES[1:0]: Window End Position (00=75%, 01=50%, 10=25%, 11=0%)
* RPSS[1:0]: Window Start Position (00=100%, 01=75%, 10=50%, 11=25%)
*

```

IWDT Refresh Sequence (IWDTRR)

```

* Refresh Sequence (must not be interrupted):
* +-----+
* | Step 1: Write 0x00 to IWDTRR          |
* | Step 2: Write 0xFF to IWDTRR          |
* |           v                           |
* | Counter reloads to initial value      |
* +-----+
* WARNING: Incomplete sequence = Refresh Error -> System Reset
*

```

Timeout Calculation

```

* Timeout = (TOPS_cycles x CKS_divisor) / IWDTCCLK
*
* Where:
* - TOPS_cycles = 1024, 4096, 8192, or 16384 (based on TOPS bits)
* - CKS_divisor = 1, 16, 32, 64, or 128 (based on CKS bits)
* - IWDTCCLK = 120 kHz (dedicated oscillator)
*
* Example: TOPS=16384, CKS=16
* Timeout = (16384 x 16) / 120000 = 2.18 seconds
*

```

Performance Characteristics

Function	Execution Time	Stack Usage	Notes
rx_hal_iwdt_init()	~50 us	24 bytes	One-time setup
rx_hal_iwdt_feed()	~2 us	8 bytes	Critical path
rx_hal_iwdt_was_reset()	~5 us	8 bytes	Register read
rx_hal_iwdt_check_tasks()	~10 us x N	32 bytes	N = task count
rx_hal_iwdt_register_task()	~15 us	16 bytes	String copy
internal_find_timeout_config()	~3 us	8 bytes	Table lookup

Memory Usage

Category	Size	Notes
Code (.text)	~1.5 KB	All functions
Constants (.rodata)	~48 bytes	Timeout table (6 entries)
Static data (.bss)	~520 bytes	Task monitor state
Stack (max)	32 bytes	deepest call chain

Thread Safety

- [rx_hal_iwdt_feed\(\)](#): Safe from ISRs (uses atomic disable/enable)
- Task functions: Not thread-safe (caller must synchronize)
- Init/config: Call only from single thread during startup

Hardware Configuration

- Window mode: Disabled (refresh allowed anytime)
- Reset action: Full chip reset (not NMI)
- Sleep behavior: Continues counting during WAIT/STOP
- Clock source: Dedicated 120 kHz oscillator (independent of PCLK)

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, or recursion
- Rule 2: [OK] All loops bounded by s_iwdt_timeout_table_size or k_iwdt_max_tasks
- Rule 3: [OK] Zero dynamic memory allocation (all static)
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] Minimum 2 validation checks per function
- Rule 6: [OK] File-scope statics (s_), local variables at narrowest scope
- Rule 7: [OK] All return values checked or explicitly cast to (void)
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [OK] Single-level pointers, function pointers for DIP
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Module has single responsibility (watchdog management)
- O: Configurable via timeout table without code modification
- L: Consistent rx_err_t returns across all functions
- I: Separate APIs for hardware vs task monitoring
- D: Hardware access via [iwdt\(\)](#) accessor, mockable for testing

See also

[rx_hal_iwdt.h](#) Public API documentation and usage examples

[rx72n_iwdt_regs.h](#) IWDT register definitions

RX72N Hardware Manual Chapter 35 - Independent Watchdog Timer

Author

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Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_iwdt.c](#).

8.123.2 Data Structure Documentation

8.123.2.1 struct iwdt`timeout`entry`t

Timeout configuration lookup table entry.

Maps a human-readable timeout value (milliseconds) to the hardware register values (TOPS and CKS) needed to achieve approximately that timeout period.

Memory Layout (8 bytes)

```
* Offset 0: timeout_ms [4 bytes, uint32_t]
* Offset 4: tops      [2 bytes, uint16_t]
* Offset 6: cks       [2 bytes, uint16_t]
* Total: 8 bytes (no padding needed)
*
```

Timeout Calculation Formula

```
* Actual_Timeout = (TOPS_cycles x CKS_divisor) / IWDTCCLK
*
* Where IWDTCCLK = 120,000 Hz
*
```

Invariant

```
timeout_ms >= k_iwdt_timeout_min_ms
timeout_ms <= k_iwdt_timeout_max_ms
tops is valid TOPS register value
cks is valid CKS register value
```

Example:

```
// Look up a configuration entry and read its fields
const iwdt_timeout_entry_t* cfg = nullptr;
rx_err_t err = internal_find_timeout_config(1000, &cfg);
if (err == k_rx_ok) {
    // cfg->tops and cfg->cks are ready to combine into IWDTCCLK
    // cfg->timeout_ms reports the actual (>= requested) timeout
}
```

See also

[s_timeout_table](#) Table of pre-computed entries

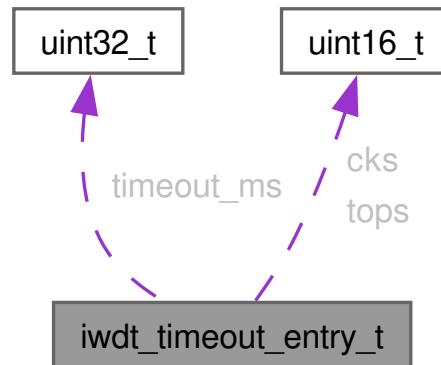
[internal_find_timeout_config\(\)](#) Lookup function that returns this type

Since

Version 1.0.0

Definition at line 299 of file [rx_iwdt.c](#).

Collaboration diagram for iwdt_timeout_entry_t:



Data Fields

uint16_t	cks	CKS bits [7:4] - Clock Division Select (/1, /16, /32, /64, /128)
uint32_t	timeout_ms	Target timeout in milliseconds (approximate)
uint16_t	tops	TOPS bits [1:0] - Timeout Period Select (1024-16384 cycles)

8.123.3 Enumeration Type Documentation

8.123.3.1 iwdt`buffer`size`constants`

```
enum iwdt\_buffer\_size\_constants\_t : uint8_t
```

Task monitoring string buffer size constants.

Defines buffer sizes for task names and log messages. Sized for embedded system constraints while allowing descriptive task names.

Size Rationale

- * `k_task_name_max_len` (16):
- * - Fits typical task names: "MotorCtrl", "SpiComm", "MainLoop"
- * - Keeps `task_monitor_t` struct compact (~32 bytes per task)
- * - Allows up to 8 tasks x 32 bytes = 256 bytes array
- *
- * `k_log_msg_buffer_size` (64):
- * - Fits formatted message: "Task deadlock: %s (timeout 16384 ms)"
- * - Stack-allocated (not static) to allow reentrant logging
- *

Invariant

```
k_task_name_cmp_len == k_task_name_max_len - 1
k_log_msg_buffer_size >= 48 (minimum for deadlock message)
```

Since

Version 1.0.0

Enumerator

k_task_name_max_len	Maximum task name length including null terminator
k_task_name_cmp_len	Length for strcmp comparison (excludes null)
k_log_msg_buffer_size	Log message buffer size for snprintf formatting

Definition at line 249 of file [rx_iwdt.c](#).

```
00249      : uint8_t {
00250  k_task_name_max_len  = 16, /**< Maximum task name length including null terminator */
00251  k_task_name_cmp_len  = 15, /**< Length for strcmp comparison (excludes null) */
00252  k_log_msg_buffer_size = 64, /**< Log message buffer size for snprintf formatting */
00253 } iwdt_buffer_size_constants_t;
```

8.123.3.2 iwdt`init`state`

```
enum iwdt\_init\_state\_t : uint8_t
```

IWDT initialization state tracking.

Tracks whether the IWDT has been initialized. Used to prevent:

- Double initialization (hardware watchdog cannot be reconfigured)
- API calls before initialization
- Feed operations on uninitialized watchdog

State Transitions

```
* [Power On] -> k_iwdt_not_initialized
* |
* | rx_hal_iwdt_init() succeeds
* v
* k_iwdt_initialized
* |
* | (No return path - IWDT cannot be stopped)
* v
* [System Reset] -> k_iwdt_not_initialized
*
```

Warning

Once initialized, IWDT cannot be disabled (hardware feature)

Since

Version 1.0.0

Enumerator

k_iwdt_not_initialized	IWDT not initialized - API calls will fail
k_iwdt_initialized	IWDT initialized - feed required periodically

Definition at line 219 of file [rx_iwdt.c](#).

```
00219      : uint8_t {
00220  k_iwdt_not_initialized = 0, /**< IWDT not initialized - API calls will fail */
00221  k_iwdt_initialized    = 1, /**< IWDT initialized - feed required periodically */
00222 } iwdt_init_state_t;
```

8.123.3.3 iwdt timeout limits

```
enum iwdt_timeout_limits_t : uint32_t
```

IWDT timeout validation limits.

Defines the valid range of timeout values for the IWDT. These limits are derived from hardware capabilities and practical system requirements.

Limit Rationale

```
* Minimum (1 ms):
* - Hardware can achieve ~8.5 ms minimum (1024/120kHz)
* - Using 1 ms as software minimum allows validation before hardware config
* - Actual hardware minimum will be selected from timeout table
*
* Maximum (16384 ms):
* - Hardware maximum = 16384 x 128 / 120000 ~ 17.5 seconds
* - Conservative limit (16.4 s) ensures reasonable failure detection
* - Longer timeouts make watchdog less effective
*
```

Invariant

```
k_iwdt_timeout_min_ms < k_iwdt_timeout_max_ms
```

Values are compile-time constants

Since

Version 1.0.0

Enumerator

k_iwdt_timeout_min_ms	Minimum valid timeout in milliseconds. Software validation limit (1 ms). Actual hardware minimum is ~128 ms with TOPS=16384, CKS=1. Used for early parameter validation.
k_iwdt_timeout_max_ms	Maximum valid timeout in milliseconds. Hardware maximum is ~17.5 seconds with TOPS=16384, CKS=128. Using 16384 ms as conservative limit for practical watchdog operation.

Definition at line 176 of file [rx_iwdt.c](#).

```

00176         : uint32_t {
00177     /**
00178     * @brief Minimum valid timeout in milliseconds
00179     * @details Software validation limit (1 ms). Actual hardware minimum
00180     * is ~128 ms with TOPS=16384, CKS=1. Used for early parameter validation.
00181     */
00182     k_iwdt_timeout_min_ms = 1,
00183
00184     /**
00185     * @brief Maximum valid timeout in milliseconds
00186     * @details Hardware maximum is ~17.5 seconds with TOPS=16384, CKS=128.
00187     * Using 16384 ms as conservative limit for practical watchdog operation.
00188     */
00189     k_iwdt_timeout_max_ms = 16384,
00190 } iwdt_timeout_limits_t;

```

8.123.4 Function Documentation

8.123.4.1 `internal_configure_iwdt_control_register()`

```

rx_err_t internal_configure_iwdt_control_register (
    const iwdt\_timeout\_entry\_t * config)    [static]

```

Configure IWDT control register (IWDT\`r) from timeout table entry.

Builds and writes the IWDT\`r register value from a timeout configuration entry. Combines TOPS, CKS, RPES, and RPSS fields into a single 16-bit value.

Register Construction

```

*   IWDT\`r = TOPS | CKS | RPES | RPSS
*
*   Where:
*   - TOPS[1:0]: Timeout Period Select (from config)
*   - CKS[7:4]: Clock Division Ratio (from config)
*   - RPES[9:8]: Window End = 0% (disabled)
*   - RPSS[13:12]: Window Start = 100% (full window)
*

```

Register Write Sequence

```

*   1. Validate config pointer (NULL check)
*   2. Validate hardware pointer (iwdt() check)
*   3. Build IWDT\`r value from fields
*   4. Write to IWDT\`r register
*

```

Precondition

config must point to valid [iwdt_timeout_entry_t](#)
 IWDT not yet started (first refresh not performed)
[iwdt\(\)](#) returns valid hardware register pointer

Postcondition

IWDT\`r configured with specified timeout and window settings

Note

Not thread-safe: must be called from single-threaded initialization context
 Window mode disabled (RPES=0%, RPSS=100%)
 Register can only be configured before first refresh

Warning

IWDT\`r becomes read-only after first refresh sequence

See also

[internal_find_timeout_config\(\)](#) Source of config parameter

Since

Version 1.0.0

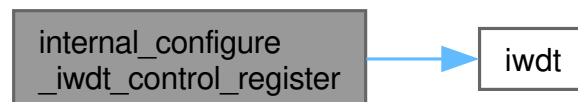
Definition at line 546 of file [rx_iwdt.c](#).

```
00547 {
00548     if (config == nullptr) {
00549         return k_rx_err_invalid_arg;
00550     }
00551
00552     if (iwdt() == nullptr) {
00553         return k_rx_err_hw_unmapped;
00554     }
00555
00556     uint16_t iwdtcr = 0;
00557     iwdtcr |= config->tops; /* Timeout period */
00558     iwdtcr |= config->cks; /* Clock divisor */
00559     iwdtcr |= k_iwdt_rpes_0; /* Window end at 0% (disabled) */
00560     iwdtcr |= k_iwdt_rpss_100; /* Window start at 100% (full) */
00561
00562     iwdt()->iwdtcr = iwdtcr;
00563     return k_rx_ok;
00564 }
```

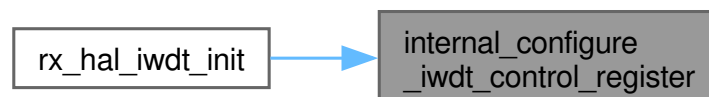
References [iwdt_timeout_entry_t::cks](#), [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtcr](#), [k_iwdt_rpes_0](#), [k_iwdt_rpss_100](#), and [iwdt_timeout_entry_t::tops](#).

Referenced by [rx_hal_iwdt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.123.4.2 internal`configure`iwdt`registers()

```
rx_err_t internal_configure_iwdt_registers (
    void ) [static]
```

Configure IWDTR reset and sleep-count behavior registers.

Configures IWDTR\`r (Reset Control) and IWDTCSTPR (Count Stop Control) registers for safety-critical operation:

- Full chip reset on timeout (not NMI)
- Continue counting during sleep modes

Register Configuration

```
* IWDTR\`r (Reset Control Register):
* +-----+
* | Bit 7 (RSTIRQS) = 1: Reset on underflow/error      |
* |              0: NMI on underflow/error (not used) |
* +-----+
* Using reset (1) for safety - NMI could be masked/ignored
*
* IWDTCSTPR (Count Stop Control Register):
* +-----+
* | Bit 7 (SLCSTP) = 0: Continue counting in sleep mode |
* |              1: Stop counting in sleep (not used) |
* +-----+
* Continue counting for safety - sleeping should not bypass watchdog
*
```

Precondition

IWDTR not yet initialized (s_iwdt_initialized == k_iwdt_not_initialized)
[iwdt\(\)](#) returns valid hardware register pointer
 Called after [internal_configure_iwdt_control_register\(\)](#)

Postcondition

IWDTR\`r configured for reset action
 IWDTCSTPR configured to continue counting in sleep

Note

Not thread-safe: must be called from single-threaded initialization context
 These registers can only be written before first refresh

Warning

Changing RSTIRQS to NMI may compromise system safety

Since

Version 1.0.0

Definition at line 606 of file [rx_iwdt.c](#).

```
00607 {
00608     if (iwdt() == nullptr) {
00609         return k_rx_err_hw_unmapped;
00610     }
00611     if (s_iwdt_initialized == k_iwdt_initialized) {
00612         return k_rx_err_invalid_state;
00613     }
00614 }
00615
00616 iwdt()->iwdtcr = k_iwdt_rstirqs_reset;
00617 iwdt()->iwdtcstpr = k_iwdt_slcstp_continue;
00618 return k_rx_ok;
00619 }
```

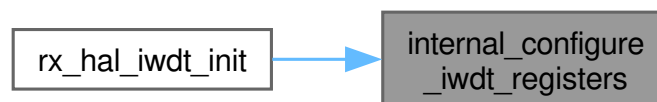
References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtcstpr](#), [rx_iwdt_regs_t::iwdtcr](#), [k_iwdt_initialized](#), [k_iwdt_rstirqs_reset](#), [k_iwdt_slcstp_continue](#), and [s_iwdt_initialized](#).

Referenced by [rx_hal_iwdt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.123.4.3 internal_find_timeout_config()

```
rx_err_t internal_find_timeout_config (
    const uint32_t timeout_ms,
    const iwdt_timeout_entry_t ** entry) [static]
```

Find best timeout configuration for requested timeout.

Performs linear search through `s_timeout_table` to find the first entry whose `timeout_ms` is $\geq$ the requested value. This provides the smallest timeout that meets or exceeds the requirement.

Algorithm

- * 1. Validate table is not empty
- * 2. Validate output pointer is not nullptr
- * 3. Linear scan through sorted table:
 - * - If `entry.timeout_ms` $\geq$ requested: return entry pointer
- * 4. If no entry found: requested timeout exceeds maximum
- *

Complexity

- Time: $O(n)$ where $n = s_iwdt_timeout_table_size$ (typically 6)
- Space: $O(1)$ - no additional memory allocated

Precondition

timeout_ms $\geq$ k_iwdt_timeout_min_ms
 entry $\neq$ nullptr
 s_timeout_table populated (compile-time guarantee)

Postcondition

*entry points to valid table entry with timeout_ms $\geq$ requested
 *entry is const - returned entry must not be modified

Note

Returns pointer to static data - valid for program lifetime
 Selected timeout may be longer than requested (next available)

Example:

```
const iwdt_timeout_entry_t* config = nullptr;
rx_err_t err = internal_find_timeout_config(1000, &config);
if (err == k_rx_ok) {
    // config->tops and config->cks ready for IWDT\
    // config->timeout_ms = actual timeout ( $\geq$  1000)
}
```

See also

[s_timeout_table](#) Source data for lookup
[iwdt_timeout_entry_t](#) Structure returned

Since

Version 1.0.0

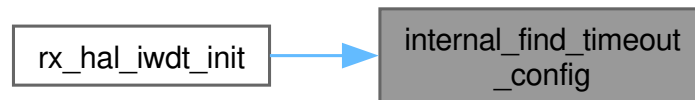
Definition at line 473 of file [rx_iwdt.c](#).

```
00475 {
00476     if (s_iwdt_timeout_table_size == 0U) {
00477         return k_rx_err_invalid_arg;
00478     }
00479
00480     if (entry == nullptr) {
00481         return k_rx_err_invalid_arg;
00482     }
00483
00484     /* Find first entry with timeout  $\geq$  requested */
00485     for (uint32_t i = 0; i < s_iwdt_timeout_table_size; i++) {
00486         if (s_timeout_table[i].timeout_ms  $\geq$  timeout_ms) {
00487             *entry = &s_timeout_table[i];
00488             return k_rx_ok;
00489         }
00490     }
00491
00492     /* No suitable configuration found */
00493     return k_rx_err_invalid_arg;
00494 }
```

References [s_iwdt_timeout_table_size](#), and [s_timeout_table](#).

Referenced by [rx_hal_iwdt_init\(\)](#).

Here is the caller graph for this function:



8.123.4.4 rx_hal_iwdt_clear_status()

```
void rx_hal_iwdt_clear_status (
    void )
```

Clear watchdog status flags.

Clears the UNDFE (underflow) and REFEE (refresh error) flags in the IWDT Status Register. These flags are "write-0-to-clear" type - writing 0 to a flag clears it, writing 1 has no effect.

Clear Sequence

- * 1. Read IWDTSR (get current status)
- * 2. AND with ~(UNDFE | REFEE) to clear both flags
- * 3. Write back to IWDTSR
- *
- * Note: Read-modify-write is safe because flags are write-0-to-clear
- *

Precondition

Status has been read via [rx_hal_iwdt_was_reset\(\)](#) or [rx_hal_iwdt_get_reset_cause\(\)](#)

Postcondition

UNDFE flag cleared
REFEE flag cleared

Note

On host builds, function is a no-op
Call after reading status at startup to prepare for next reset

Example:

```
// Startup diagnostic sequence
if (rx_hal_iwdt_was_reset()) {
    rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
    log_watchdog_reset(cause);
}
// Clear flags so next reset can be detected
rx_hal_iwdt_clear_status();
```

See also

[rx_hal_iwdt_was_reset\(\)](#) Check reset status
[rx_hal_iwdt_get_reset_cause\(\)](#) Get specific cause

Since

Version 1.0.0

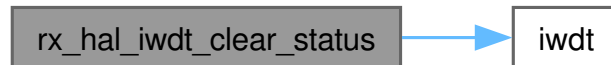
Definition at line 1037 of file rx_iwdt.c.

```

01038 {
01039 #ifdef __RX__
01040 /*
01041  * Clear status flags by writing 0 to the flag bits
01042  *
01043  * Bits 14-15 are write-0-to-clear
01044  */
01045 iwdt()->iwdtsr &= (uint16_t) ~(uint16_t)(k_iwdt_sr_undff | k_iwdt_sr_refef);
01046
01047 #else
01048 /* Host-side stub - no operation */
01049 #endif
01050 }
```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undff](#).

Here is the call graph for this function:



8.123.4.5 rx`hal`iwdt`feed()

```

void rx_hal_iwdt_feed (
    void )
```

Feed the watchdog timer to prevent reset.

Feed (refresh) the watchdog timer.

Performs the IWDT refresh sequence by writing the magic values 0x00 then 0xFF to the IWDTRR register. This resets the down-counter to its initial value, preventing timeout and system reset.

Refresh Sequence (Atomic Operation)

```

* +-----+
* | 1. Disable interrupts (save PSW)          |
* | 2. Write 0x00 to IWDTRR                   |
* | 3. Write 0xFF to IWDTRR                   |
* | 4. Restore interrupt state                 |
* +-----+
* \\rITICAL: Steps 2 and 3 must not be interrupted!
*
```

Timing Requirements

- * - Execution time: ~2 us including interrupt disable/enable
- * - Must be called faster than configured timeout period
- * - Typical call rate: Every 100-500 ms depending on timeout
- *

Precondition

IWDT must be initialized via [rx_hal_iwdt_init\(\)](#)
 Called at rate faster than configured timeout

Postcondition

IWDT counter reset to initial value
 Interrupt state restored to previous value

Note

Safe to call from ISRs (uses atomic disable/restore)
 On host builds, function is a no-op

Warning

Incomplete refresh sequence causes refresh error -> system reset
 Missing calls to this function cause timeout -> system reset

Example:

```
// Main supervision loop
while (1) {
    // Check all tasks for deadlock
    rx_err_t err = rx_hal_iwdt_check_tasks();
    if (err == k_rx_ok) {
        // All tasks healthy, feed hardware watchdog
        rx_hal_iwdt_feed();
    }
    // If task timeout detected, DON'T feed - let watchdog reset
    tx_thread_sleep(100); // 100 ms check interval
}
```

See also

[rx_hal_iwdt_init\(\)](#) Start the watchdog
[rx_hal_iwdt_check_tasks\(\)](#) Verify task health before feeding

Since

Version 1.0.0

Definition at line 821 of file [rx_iwdt.c](#).

```
00822 {
00823 #ifdef __RX__
00824 /*
00825  * IWDT Refresh Sequence (Section 35.3.3)
00826  *
00827  * Write 0x00 followed by 0xFF to IWDTRR.
00828  * This sequence resets the down-counter to its initial value.
00829  *
00830  * CRITICAL: This sequence must not be interrupted.
00831  * An incomplete sequence will trigger a refresh error.
00832  *
00833  * Uses inline assembly to save/restore PSW (Program Status Word) because
00834  * there is no C library function to manipulate interrupt state on RX architecture.
00835  */
00836
00837 /* Disable interrupts during refresh for atomicity */
```

```

00838  uint32_t psw;
00839
00840  __asm__ volatile("mvfc psw, %0" : "=r"(psw));
00841  __asm__ volatile("clrpsw i");
00842
00843  /* Perform refresh sequence */
00844  iwdt()->iwdtrr = k_iwdt_refresh_start; /* Write 0x00 */
00845  iwdt()->iwdtrr = k_iwdt_refresh_end;   /* Write 0xFF */
00846
00847  /* Restore interrupt state */
00848  __asm__ volatile("mvtc %0, psw" : : "r"(psw));
00849
00850  #else
00851  /* Host-side stub - no operation */
00852  #endif
00853  }

```

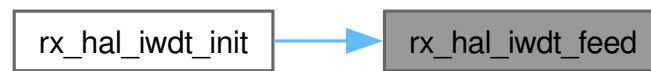
References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtrr](#), [k_iwdt_refresh_end](#), and [k_iwdt_refresh_start](#).

Referenced by [rx_hal_iwdt_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.123.4.6 rx_hal_iwdt_get_reset_cause()

```

rx_iwdt_reset_cause_t rx_hal_iwdt_get_reset_cause (
    void )

```

Get detailed watchdog reset cause.

Get detailed reset cause from IWDT.

Reads the IWDT Status Register to determine the specific cause of the last watchdog reset. Refresh errors take priority over underflow since they indicate a software bug in the refresh sequence.

Reset Cause Priority

- * 1. Refresh Error (REFEF=1): Invalid refresh sequence
 - * - Incomplete 0x00/0xFF write
 - * - Write occurred outside window (if window mode enabled)
- * 2. Underflow (UNDF=1): Counter reached zero
 - * - rx_hal_iwdt_feed() not called within timeout period
 - * - Software hang, deadlock, or infinite loop
- * 3. None: No watchdog reset occurred
 - * - Power-on reset, external reset, or other cause

Precondition

Can be called before [rx_hal_iwdt_init\(\)](#)

Status flags preserved across reset

Postcondition

Status flags NOT cleared (call [rx_hal_iwdt_clear_status\(\)](#) to clear)

Note

On host builds, always returns `k_iwdt_reset_none`

Refresh error has priority over underflow in return value

Example:

```
rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
switch (cause) {
    case k_iwdt_reset_refresh_error:
        uart_debug_puts("ERROR: Watchdog refresh sequence error!\r\n");
        break;
    case k_iwdt_reset_underflow:
        uart_debug_puts("WARN: Watchdog timeout - software hang detected\r\n");
        break;
    case k_iwdt_reset_none:
    default:
        uart_debug_puts("Normal startup\r\n");
        break;
}
rx_hal_iwdt_clear_status(); // Clear flags after reading
```

See also

[rx_hal_iwdt_was_reset\(\)](#) Quick check for any watchdog reset

[rx_hal_iwdt_clear_status\(\)](#) Clear status flags

Since

Version 1.0.0

Definition at line 976 of file [rx_iwdt.c](#).

```
00977 {
00978 #ifdef __RX__
00979     const uint16_t status = iwdt()->iwdtsr;
00980
00981     if (status & k_iwdt_sr_refef) {
00982         return k_iwdt_reset_refresh_error;
00983     }
00984
00985     if (status & k_iwdt_sr_undff) {
00986         return k_iwdt_reset_underflow;
00987     }
00988
00989     return k_iwdt_reset_none;
00990 #else
00991     return k_iwdt_reset_none;
00992 #endif
00993 }
00994 }
```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_reset_none](#), [k_iwdt_reset_refresh_error](#), [k_iwdt_reset_underflow](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undff](#).

Here is the call graph for this function:



8.123.4.7 rx`hal`iwdt`init()

```
rx_err_t rx_hal_iwdt_init (
    const uint32_t timeout_ms) [nodiscard]
```

Initialize the Independent Watchdog Timer.

Configures and starts the hardware IWDT with the specified timeout period. Once started, the watchdog cannot be stopped - periodic calls to [rx_hal_iwdt_feed\(\)](#) are required to prevent system reset.

Initialization Sequence

- * 1. Validate not already initialized
- * 2. Validate timeout_ms within range
- * 3. Look up timeout configuration in table
- * 4. Configure IWDTR\`r (timeout period, clock divisor, window)
- * 5. Configure IWDTR\`r (reset action - not NMI)
- * 6. Configure IWDTCSTPR (continue counting in sleep)
- * 7. Perform first refresh (starts the watchdog)
- * 8. Set initialized flag
- *

Point of No Return

- * After step 7 (first refresh), the IWDT is running and CANNOT be stopped.
- * Any failure after this point leaves system in watchdog-protected state.
- *

Precondition

IWDT must not be already initialized

timeout_ms in range [k_iwdt_timeout_min_ms, k_iwdt_timeout_max_ms]

System startup, single-threaded context

Postcondition

IWDT running with selected timeout period

s_iwdt_initialized set to k_iwdt_initialized

Periodic [rx_hal_iwdt_feed\(\)](#) calls now required

Note

On host builds (non-RX), skips hardware access but sets initialized flag

Actual timeout >= requested (selected from lookup table)

Warning

Once initialized, IWDT cannot be disabled (requires hard reset)

Must call [rx_hal_iwdt_feed\(\)](#) faster than timeout period

Example:

```
// Initialize with 2 second timeout
rx_err_t err = rx_hal_iwdt_init(2000);
if (err != k_rx_ok) {
    // Handle error - system cannot start safely
    while (1) { // halt }
}

// Now must feed periodically
while (1) {
    // ... main loop ...
    rx_hal_iwdt_feed(); // Must occur every < 2 seconds
    tx_thread_sleep(100);
}
```

See also

[rx_hal_iwdt_feed\(\)](#) Reset watchdog counter
[rx_hal_iwdt.h](#) Public API and configuration details

Since

Version 1.0.0

Definition at line 686 of file [rx_iwdt.c](#).

```
00687 {
00688 #ifdef __RX__
00689 if (s_iwdt_initialized == k_iwdt_initialized) {
00690     return k_rx_err_invalid_state;
00691 }
00692
00693 if (timeout_ms < k_iwdt_timeout_min_ms) {
00694     return k_rx_err_invalid_arg;
00695 }
00696
00697 if (timeout_ms > k_iwdt_timeout_max_ms) {
00698     return k_rx_err_invalid_arg;
00699 }
00700
00701 /* Find best timeout configuration */
00702 const iwdt_timeout_entry_t* config = nullptr;
00703 rx_err_t err = internal_find_timeout_config(timeout_ms, &config);
00704 if (err != k_rx_ok) {
00705     return err;
00706 }
00707
00708 /*
00709  * Configure IWDT Control Register (IWDTCR)
00710  *
00711  * Bits [1:0] TOPS - Timeout Period Select
00712  * Bits [7:4] CKS - Clock Division Ratio Select
00713  * Bits [9:8] RPES - Window End Position (0x03 = 0%, window disabled)
00714  * Bits [13:12] RPSS - Window Start Position (0x00 = 100%, full window)
00715  */
00716 err = internal_configure_iwdt_control_register(config);
00717 if (err != k_rx_ok) {
00718     return err;
00719 }
00720
00721 /*
00722  * Configure Reset Control Register (IWDTRCR)
00723  *
00724  * Bit 7 RSTIRQS - Reset Interrupt Request Select
00725  * 0 = Non-maskable interrupt request (NMI)
00726  * 1 = Reset
00727  *
00728  * We use reset (1) for safety - NMI could be masked or ignored.
00729  */
00730 err = internal_configure_iwdt_registers();
00731 if (err != k_rx_ok) {
00732     return err;
00733 }
00734
00735 /*
00736  * Configure Count Stop Control Register (IWDTCSTPR)
00737  */
```



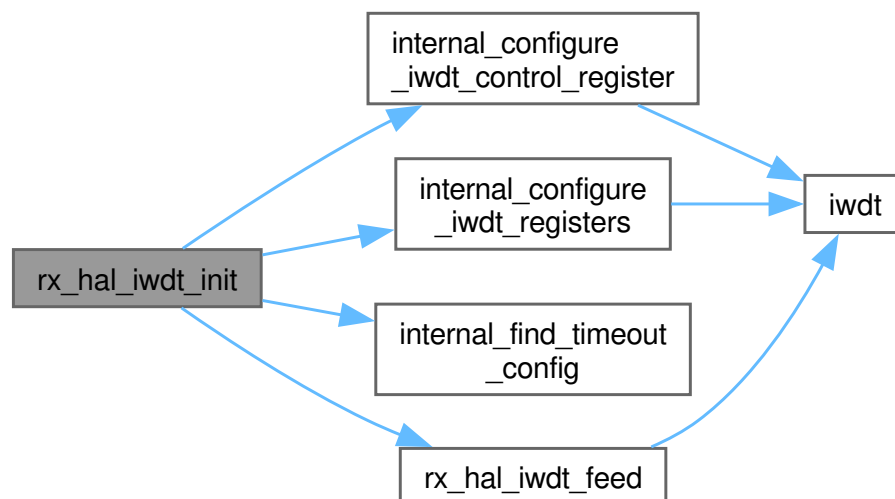
```

00738  * Bit 7 SLCSTP - Sleep Mode Count Stop Control
00739  * 0 = Count continues during sleep
00740  * 1 = Count stops during sleep
00741  *
00742  * We continue counting during sleep for safety.
00743  */
00744 /*
00745  * Start the IWDT by performing first refresh
00746  *
00747  * The IWDT starts counting after the first refresh sequence.
00748  * After this, the watchdog CANNOT be stopped.
00749  */
00750 rx_hal_iwdt_feed();
00751
00752 s_iwdt_initialized = k_iwdt_initialized;
00753 return k_rx_ok;
00754
00755 #else
00756 /* Host-side stub for unit testing */
00757 (void)timeout_ms;
00758 s_iwdt_initialized = k_iwdt_initialized;
00759 return k_rx_ok;
00760 #endif
00761 }

```

References [internal_configure_iwdt_control_register\(\)](#), [internal_configure_iwdt_registers\(\)](#), [internal_find_timeout_config\(\)](#), [k_iwdt_initialized](#), [k_iwdt_timeout_max_ms](#), [k_iwdt_timeout_min_ms](#), [rx_hal_iwdt_feed\(\)](#), and [s_iwdt_initialized](#).

Here is the call graph for this function:



8.123.4.8 rx_hal_iwdt_was_reset()

```

bool rx_hal_iwdt_was_reset (
    void ) [nodiscard]

```

Check if system was reset by watchdog.

Check if last reset was caused by watchdog timeout.

Reads the IWDT Status Register (IWDTSR) to determine if the previous system reset was caused by watchdog timeout (underflow) or refresh error. This function should be called early in startup for diagnostics.

Status Register Flags

```
* IWDTSR Bit 14 (UNDFE) - Underflow Flag:
* - 1: Counter reached 0 (timeout occurred)
* - 0: No underflow
*
* IWDTSR Bit 15 (REFEF) - Refresh Error Flag:
* - 1: Invalid refresh sequence detected
* - 0: No refresh error
*
```

Precondition

Can be called before [rx_hal_iwdt_init\(\)](#)
 Status flags preserved across reset (hardware feature)

Postcondition

Status flags NOT cleared (call [rx_hal_iwdt_clear_status\(\)](#) to clear)

Note

On host builds, always returns false
 Call [rx_hal_iwdt_get_reset_cause\(\)](#) for specific cause

Example:

```
// Check at startup
if (rx_hal_iwdt_was_reset()) {
    uart_debug_puts("[WARN] Recovered from watchdog reset!\r\n");
    const char* failed = rx_hal_iwdt_get_failed_task();
    if (failed != nullptr) {
        uart_debug_printf("Failed task: %s\r\n", failed);
    }
}
```

See also

[rx_hal_iwdt_get_reset_cause\(\)](#) Get specific reset cause
[rx_hal_iwdt_clear_status\(\)](#) Clear status flags after reading
[rx_hal_iwdt_get_failed_task\(\)](#) Get name of task that caused timeout

Since

Version 1.0.0

Definition at line 901 of file [rx_iwdt.c](#).

```
00902 {
00903 #ifdef __RX__
00904 /*
00905  * Check IWDTSR Status Register for underflow or refresh error
00906  *
00907  * Bit 14 UNDFE - Underflow Flag
00908  * 1 = Counter underflow occurred
00909  *
00910  * Bit 15 REFEF - Refresh Error Flag
00911  * 1 = Refresh error occurred
00912  */
00913 const uint16_t status = iwdt()->iwdtshr;
00914
```

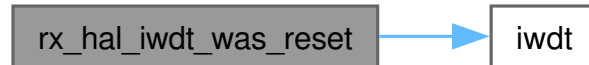
```

00915  return ((status & k_iwdt_sr_undff) != 0) || ((status & k_iwdt_sr_refef) != 0);
00916
00917  #else
00918  return false;
00919  #endif
00920 }

```

References [iwdt\(\)](#), [rx_iwdt_regs_t::iwdtsr](#), [k_iwdt_sr_refef](#), and [k_iwdt_sr_undff](#).

Here is the call graph for this function:



8.123.5 Variable Documentation

8.123.5.1 s`iwdt`initialized

```
iwdt\_init\_state\_t s_iwdt_initialized = k\_iwdt\_not\_initialized [static]
```

Flag indicating IWDT has been initialized.

Tracks initialization state to prevent:

- Multiple calls to [rx_hal_iwdt_init\(\)](#) (hardware cannot be reconfigured)
- API calls before init (undefined behavior prevention)
- Feed operations on uninitialized watchdog

Thread Safety

- Set only once during initialization (single-threaded startup)
- Read by all API functions (safe - single write, multiple reads)
- Never reset (IWDT cannot be stopped once started)

Invariant

Value is either [k_iwdt_not_initialized](#) or [k_iwdt_initialized](#)

Once set to [k_iwdt_initialized](#), never changes until reset

Warning

Do not modify directly - use [rx_hal_iwdt_init\(\)](#) only

Since

Version 1.0.0

Definition at line [418](#) of file [rx_iwdt.c](#).

Referenced by [internal_configure_iwdt_registers\(\)](#), [rx_hal_iwdt_check_tasks\(\)](#), [rx_hal_iwdt_get_failed_task\(\)](#), and [rx_hal_iwdt_init\(\)](#).

8.123.5.2 `s_iwdt_timeout_table_size`

```
const uint32_t s_iwdt_timeout_table_size [static]
```

Initial value:

```
sizeof(s_timeout_table) / sizeof(s_timeout_table[0])
```

Number of entries in the timeout lookup table.

Compile-time constant for table bounds checking. Used in [internal_find_timeout_config\(\)](#) for loop bounds (NASA Power of 10 Rule 2).

Invariant

```
s_iwdt_timeout_table_size == 6 (current implementation)
s_iwdt_timeout_table_size > 0
```

Since

Version 1.0.0

Definition at line 367 of file [rx_iwdt.c](#).

Referenced by [internal_find_timeout_config\(\)](#).

8.123.5.3 `s_tag`

```
const char* const s_tag = "IWDT" [static]
```

Log tag for IWDT module.

Identifies log messages from this module in system logs. Uses the tag "IWDT" (uppercase) to match S\rEAMING_SNAKE_CASE convention for log tags.

Log Message Format

```
* [IWDT] INFO: Registered task for monitoring
* [IWDT] ERROR: Task deadlock: MotorCtrl (timeout 500 ms)
*
```

Note

Pointer to string literal (stored in .rodata section)

Warning

Do not modify (const pointer to const data)

Since

Version 1.0.0

Definition at line 394 of file [rx_iwdt.c](#).

8.123.5.4 s`timeout`table

```
const iwdt_timeout_entry_t s_timeout_table[] [static]
```

Initial value:

```
= {
    { .timeout_ms = 128, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_1 },
    { .timeout_ms = 512, .tops = k_iwdt_tops_4096, .cks = k_iwdt_cks_div_16 },
    { .timeout_ms = 1000, .tops = k_iwdt_tops_8192, .cks = k_iwdt_cks_div_16 },
    { .timeout_ms = 2048, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_16 },
    { .timeout_ms = 8192, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_64 },
    { .timeout_ms = 16384, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_128 },
}
```

IWDT timeout configuration lookup table.

Static lookup table mapping requested timeout values to optimal IWDT register configurations. Entries are sorted by increasing timeout for linear search efficiency. The first entry with timeout_ms >= requested value is selected.

Table Organization

* Index	Request	TOPS	CKS	Actual Timeout	Use Case
* 0	128ms	16384	/1	~136ms	Fast control loops
* 1	512ms	4096	/16	~546ms	Normal operations
* 2	1000ms	8192	/16	~1.09s	Default timeout
* 3	2048ms	16384	/16	~2.18s	Moderate operations
* 4	8192ms	16384	/64	~8.74s	Slow processes
* 5	16384ms	16384	/128	~17.48s	Long operations
*					

Calculation Examples

```
* Entry 0: (16384 x 1) / 120000 = 0.1365s = 136.5ms
* Entry 2: (8192 x 16) / 120000 = 1.092s = 1092ms
* Entry 5: (16384 x 128) / 120000 = 17.48s = 17476ms
*
```

Note

Values are slightly longer than requested to provide safety margin

Table is const and placed in .rodata section (48 bytes total)

Warning

Do not modify table at runtime (const data)

See also

[internal_find_timeout_config\(\)](#) Table lookup function

[iwdt_timeout_entry_t](#) Entry structure definition

Since

Version 1.0.0

Definition at line 343 of file [rx_iwdt.c](#).

```
00343 {
00344     /* timeout_ms, TOPS (cycles), CKS (divisor) */
00345     /* Actual timeout = (cycles * divisor) / 120kHz */
00346     { .timeout_ms = 128, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_1 }, /* ~136ms actual */
00347     { .timeout_ms = 512, .tops = k_iwdt_tops_4096, .cks = k_iwdt_cks_div_16 }, /* ~546ms actual */
00348     { .timeout_ms = 1000, .tops = k_iwdt_tops_8192, .cks = k_iwdt_cks_div_16 }, /* ~1.09s actual */
00349     { .timeout_ms = 2048, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_16 }, /* ~2.18s actual */
00350     { .timeout_ms = 8192, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_64 }, /* ~8.74s actual */
00351     { .timeout_ms = 16384, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_128 }, /* ~17.48s actual */
00352 };
```

Referenced by [internal_find_timeout_config\(\)](#).

8.124 rx_iwdt.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_iwdt.c
00003  * @brief Independent Watchdog Timer (IWDT) Driver Implementation for RX72N
00004  *
00005  * @details
00006  * Implements hardware and software watchdog functionality for safety-critical
00007  * system monitoring. The IWDT provides protection against software hangs,
00008  * deadlocks, and infinite loops through automatic system reset.
00009  *
00010  * @par Implementation Architecture
00011  * @verbatim
00012  *
00013  *      +-----+
00014  *      |                                     |
00015  *      |      Public API Layer      |
00016  *      | +-----+ +-----+ +-----+ +-----+ |
00017  *      | | rx_iwdt_ | | rx_iwdt_ | | rx_iwdt_ | | rx_iwdt_ | |
00018  *      | | init()  | | feed()   | | was_reset() | | check_   | |
00019  *      | | +-----+ +-----+ +-----+ +-----+ |
00020  *      | +-----+ +-----+ +-----+ +-----+ |
00021  *      | |                                     | |
00022  *      | +-----+ +-----+ +-----+ +-----+ |
00023  *      |      Internal Helper Layer      |
00024  *      | +-----+ +-----+ +-----+ +-----+ |
00025  *      | | internal_find_ | | internal_configure_iwdt_ | | |
00026  *      | | timeout_config() | | control_register() | | |
00027  *      | | +-> Lookup table | | +-> IWDT\\r setup | | |
00028  *      | +-----+ +-----+ +-----+ +-----+ |
00029  *      | +-----+ +-----+ +-----+ +-----+ |
00030  *      | |                                     | |
00031  *      | +-----+ +-----+ +-----+ +-----+ |
00032  *      |      IWDT Hardware Registers      |
00033  *      | +-----+ +-----+ +-----+ +-----+ |
00034  *      | | IWDT\\r | | IWDTRR | | IWDTSR | | |
00035  *      | | (Control) | | (Refresh) | | (Status) | | |
00036  *      | | [16-bit] | | [8-bit] | | [16-bit] | | |
00037  *      | +-----+ +-----+ +-----+ +-----+ |
00038  *      | Base: 0x00088030 |
00039  *      +-----+
00040  * @endverbatim
00041  *
00042  * @par Register Layout (IWDT\\r - 0x00088030)
00043  * @verbatim
00044  *      Bits [15:14] RPSS - Window Start Position
00045  *      +-----+
00046  *      | - |RPSS1|RPSS0| - |RP- |RP- | - | - |CKS3 |CKS2 |CKS1 |CKS0 | - | - |TOPS1|TOPS0|
00047  *      | b15 | b14 | b13 | b12 | ES1 | ES0 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0 |
00048  *      +-----+
00049  *      TOPS[1:0]: Timeout Period Select (00=1024, 01=4096, 10=8192, 11=16384 cycles)
00050  *      CKS[3:0]: Clock Division Ratio (0000=/1, 0010=/16, 0100=/32, 1111=/128)
00051  *      RPES[1:0]: Window End Position (00=75%, 01=50%, 10=25%, 11=0%)
00052  *      RPSS[1:0]: Window Start Position (00=100%, 01=75%, 10=50%, 11=25%)
00053  * @endverbatim
00054  *
00055  * @par IWDT Refresh Sequence (IWDTRR)
00056  * @verbatim
00057  *      Refresh Sequence (must not be interrupted):
00058  *      +-----+
00059  *      | Step 1: Write 0x00 to IWDTRR |
00060  *      | Step 2: Write 0xFF to IWDTRR |
00061  *      | v |
00062  *      | Counter reloads to initial value |
00063  *      +-----+
00064  *      WARNING: Incomplete sequence = Refresh Error -> System Reset
00065  * @endverbatim
00066  *
00067  * @par Timeout Calculation
00068  * @verbatim
00069  *      Timeout = (TOPS_cycles x CKS_divisor) / IWDTCLK
00070  *
00071  *      Where:
00072  *      - TOPS_cycles = 1024, 4096, 8192, or 16384 (based on TOPS bits)
00073  *      - CKS_divisor = 1, 16, 32, 64, or 128 (based on CKS bits)
00074  *      - IWDTCLK = 120 kHz (dedicated oscillator)
00075  *
00076  *      Example: TOPS=16384, CKS=16
00077  *      Timeout = (16384 x 16) / 120000 = 2.18 seconds
00078  * @endverbatim
00079  *
00080  * @par Performance Characteristics
00081  * | Function | Execution Time | Stack Usage | Notes |
00082  * |-----|-----|-----|-----|

```

```

00083 * | rx_hal_iwtdt_init() | ~50 us | 24 bytes | One-time setup |
00084 * | rx_hal_iwtdt_feed() | ~2 us | 8 bytes | Critical path |
00085 * | rx_hal_iwtdt_was_reset() | ~5 us | 8 bytes | Register read |
00086 * | rx_hal_iwtdt_check_tasks() | ~10 us x N | 32 bytes | N = task count |
00087 * | rx_hal_iwtdt_register_task() | ~15 us | 16 bytes | String copy |
00088 * | internal_find_timeout_config() | ~3 us | 8 bytes | Table lookup |
00089 *
00090 * @par Memory Usage
00091 * | Category | Size | Notes |
00092 * |-----|-----|-----|
00093 * | Code (.text) | ~1.5 KB | All functions |
00094 * | Constants (.rodata) | ~48 bytes | Timeout table (6 entries) |
00095 * | Static data (.bss) | ~520 bytes | Task monitor state |
00096 * | Stack (max) | 32 bytes | deepest call chain |
00097 *
00098 * @par Thread Safety
00099 * - rx_hal_iwtdt_feed(): Safe from ISRs (uses atomic disable/enable)
00100 * - Task functions: Not thread-safe (caller must synchronize)
00101 * - Init/config: Call only from single thread during startup
00102 *
00103 * @par Hardware Configuration
00104 * - **Window mode**: Disabled (refresh allowed anytime)
00105 * - **Reset action**: Full chip reset (not NMI)
00106 * - **Sleep behavior**: Continues counting during WAIT/STOP
00107 * - **Clock source**: Dedicated 120 kHz oscillator (independent of PCLK)
00108 *
00109 * @par NASA Power of 10 Compliance
00110 * - **Rule 1**: [OK] No goto, setjmp, or recursion
00111 * - **Rule 2**: [OK] All loops bounded by s_iwtdt_timeout_table_size or k_iwtdt_max_tasks
00112 * - **Rule 3**: [OK] Zero dynamic memory allocation (all static)
00113 * - **Rule 4**: [OK] All functions < 60 lines
00114 * - **Rule 5**: [OK] Minimum 2 validation checks per function
00115 * - **Rule 6**: [OK] File-scope statics (s_), local variables at narrowest scope
00116 * - **Rule 7**: [OK] All return values checked or explicitly cast to (void)
00117 * - **Rule 8**: [OK] C23 typed enums for all constants
00118 * - **Rule 9**: [OK] Single-level pointers, function pointers for DIP
00119 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00120 *
00121 * @par SOLID Principles
00122 * - **S**: Module has single responsibility (watchdog management)
00123 * - **O**: Configurable via timeout table without code modification
00124 * - **I**: Consistent rx_err_t returns across all functions
00125 * - **I**: Separate APIs for hardware vs task monitoring
00126 * - **D**: Hardware access via iwtdt() accessor, mockable for testing
00127 *
00128 * @see rx_hal_iwtdt.h Public API documentation and usage examples
00129 * @see rx72n_iwtdt_regs.h IWDT register definitions
00130 * @see RX72N Hardware Manual Chapter 35 - Independent Watchdog Timer
00131 *
00132 * @author Locked, Inc.
00133 * @date 2026-01-29
00134 * @copyright Copyright (c) 2026 Locked Inc.
00135 * SPDX-License-Identifier: MIT
00136 * @since Version 1.0.0
00137 */
00138
00139 #include "rx_hal_iwtdt.h"
00140
00141 #ifdef __RX__
00142 #include "rx72n_regs.h"
00143 #endif
00144
00145 /*
=====
00146 * Internal Constants
00147 *
=====
*/
00148
00149 /**
00150 * @enum iwtdt_timeout_limits_t
00151 * @brief IWDT timeout validation limits
00152 *
00153 * @details
00154 * Defines the valid range of timeout values for the IWDT. These limits are
00155 * derived from hardware capabilities and practical system requirements.
00156 *
00157 * @par Limit Rationale
00158 * @verbatim
00159 * Minimum (1 ms):
00160 * - Hardware can achieve ~8.5 ms minimum (1024/120kHz)
00161 * - Using 1 ms as software minimum allows validation before hardware config
00162 * - Actual hardware minimum will be selected from timeout table
00163 *
00164 * Maximum (16384 ms):
00165 * - Hardware maximum = 16384 x 128 / 120000 ~ 17.5 seconds
00166 * - Conservative limit (16.4 s) ensures reasonable failure detection
00167 */

```

```

00168 * - Longer timeouts make watchdog less effective
00169 * @endverbatim
00170 *
00171 * @invariant k_iwdt_timeout_min_ms < k_iwdt_timeout_max_ms
00172 * @invariant Values are compile-time constants
00173 *
00174 * @since Version 1.0.0
00175 */
00176 typedef enum : uint32_t {
00177 /**
00178  * @brief Minimum valid timeout in milliseconds
00179  * @details Software validation limit (1 ms). Actual hardware minimum
00180  * is ~128 ms with TOPS=16384, CKS=1. Used for early parameter validation.
00181  */
00182 k_iwdt_timeout_min_ms = 1,
00183
00184 /**
00185  * @brief Maximum valid timeout in milliseconds
00186  * @details Hardware maximum is ~17.5 seconds with TOPS=16384, CKS=128.
00187  * Using 16384 ms as conservative limit for practical watchdog operation.
00188  */
00189 k_iwdt_timeout_max_ms = 16384,
00190 } iwdt_timeout_limits_t;
00191
00192 /**
00193 * @enum iwdt_init_state_t
00194 * @brief IWDT initialization state tracking
00195 *
00196 * @details
00197 * Tracks whether the IWDT has been initialized. Used to prevent:
00198 * - Double initialization (hardware watchdog cannot be reconfigured)
00199 * - API calls before initialization
00200 * - Feed operations on uninitialized watchdog
00201 *
00202 * @par State Transitions
00203 * @verbatim
00204 * [Power On] -> k_iwdt_not_initialized
00205 * |
00206 * | rx_hal_iwdt_init() succeeds
00207 * v
00208 * k_iwdt_initialized
00209 * |
00210 * | (No return path - IWDT cannot be stopped)
00211 * v
00212 * [System Reset] -> k_iwdt_not_initialized
00213 * @endverbatim
00214 *
00215 * @warning Once initialized, IWDT cannot be disabled (hardware feature)
00216 *
00217 * @since Version 1.0.0
00218 */
00219 typedef enum : uint8_t {
00220 k_iwdt_not_initialized = 0, /**< IWDT not initialized - API calls will fail */
00221 k_iwdt_initialized = 1, /**< IWDT initialized - feed required periodically */
00222 } iwdt_init_state_t;
00223
00224 /**
00225 * @enum iwdt_buffer_size_constants_t
00226 * @brief Task monitoring string buffer size constants
00227 *
00228 * @details
00229 * Defines buffer sizes for task names and log messages. Sized for embedded
00230 * system constraints while allowing descriptive task names.
00231 *
00232 * @par Size Rationale
00233 * @verbatim
00234 * k_task_name_max_len (16):
00235 * - Fits typical task names: "MotorCtrl", "SpiComm", "MainLoop"
00236 * - Keeps task_monitor_t struct compact (~32 bytes per task)
00237 * - Allows up to 8 tasks x 32 bytes = 256 bytes array
00238 *
00239 * k_log_msg_buffer_size (64):
00240 * - Fits formatted message: "Task deadlock: %s (timeout 16384 ms)"
00241 * - Stack-allocated (not static) to allow reentrant logging
00242 * @endverbatim
00243 *
00244 * @invariant k_task_name_cmp_len == k_task_name_max_len - 1
00245 * @invariant k_log_msg_buffer_size >= 48 (minimum for deadlock message)
00246 *
00247 * @since Version 1.0.0
00248 */
00249 typedef enum : uint8_t {
00250 k_task_name_max_len = 16, /**< Maximum task name length including null terminator */
00251 k_task_name_cmp_len = 15, /**< Length for strncmp comparison (excludes null) */
00252 k_log_msg_buffer_size = 64, /**< Log message buffer size for sprintf formatting */
00253 } iwdt_buffer_size_constants_t;
00254

```



```

00255 /**
00256  * @struct iwtdt_timeout_entry_t
00257  * @brief Timeout configuration lookup table entry
00258  *
00259  * @details
00260  * Maps a human-readable timeout value (milliseconds) to the hardware register
00261  * values (TOPS and CKS) needed to achieve approximately that timeout period.
00262  *
00263  * @par Memory Layout (8 bytes)
00264  * @verbatim
00265  *   Offset 0: timeout_ms [4 bytes, uint32_t]
00266  *   Offset 4: tops      [2 bytes, uint16_t]
00267  *   Offset 6: cks       [2 bytes, uint16_t]
00268  *   Total: 8 bytes (no padding needed)
00269  * @endverbatim
00270  *
00271  * @par Timeout Calculation Formula
00272  * @verbatim
00273  *   Actual_Timeout = (TOPS_cycles x CKS_divisor) / IWDTCLK
00274  *
00275  *   Where IWDTCLK = 120,000 Hz
00276  * @endverbatim
00277  *
00278  * @invariant timeout_ms >= k_iwtdt_timeout_min_ms
00279  * @invariant timeout_ms <= k_iwtdt_timeout_max_ms
00280  * @invariant tops is valid TOPS register value
00281  * @invariant cks is valid CKS register value
00282  *
00283  * @par Example:
00284  * @code
00285  * // Look up a configuration entry and read its fields
00286  * const iwtdt_timeout_entry_t* cfg = nullptr;
00287  * rx_err_t err = internal_find_timeout_config(1000, &cfg);
00288  * if (err == k_rx_ok) {
00289  *     // cfg->tops and cfg->cks are ready to combine into IWDT\\r
00290  *     // cfg->timeout_ms reports the actual (>= requested) timeout
00291  * }
00292  * @endcode
00293  *
00294  * @see s_timeout_table Table of pre-computed entries
00295  * @see internal_find_timeout_config() Lookup function that returns this type
00296  *
00297  * @since Version 1.0.0
00298  */
00299 typedef struct {
00300     uint32_t timeout_ms; /**< Target timeout in milliseconds (approximate) */
00301     uint16_t tops;       /**< TOPS bits [1:0] - Timeout Period Select (1024-16384 cycles) */
00302     uint16_t cks;        /**< CKS bits [7:4] - Clock Division Select (/1, /16, /32, /64, /128) */
00303 } iwtdt_timeout_entry_t;
00304
00305 /**
00306  * @var s_timeout_table
00307  * @brief IWDT timeout configuration lookup table
00308  *
00309  * @details
00310  * Static lookup table mapping requested timeout values to optimal IWDT register
00311  * configurations. Entries are sorted by increasing timeout for linear search
00312  * efficiency. The first entry with timeout_ms >= requested value is selected.
00313  *
00314  * @par Table Organization
00315  * @verbatim
00316  *   Index  Request  TOPS   CKS   Actual Timeout  Use Case
00317  *   -----
00318  *   0      128ms    16384  /1    ~136ms         Fast control loops
00319  *   1      512ms    4096   /16   ~546ms         Normal operations
00320  *   2      1000ms   8192   /16   ~1.09s         Default timeout
00321  *   3      2048ms   16384  /16   ~2.18s         Moderate operations
00322  *   4      8192ms   16384  /64   ~8.74s         Slow processes
00323  *   5      16384ms  16384  /128  ~17.48s        Long operations
00324  * @endverbatim
00325  *
00326  * @par Calculation Examples
00327  * @verbatim
00328  *   Entry 0: (16384 x 1) / 120000 = 0.1365s = 136.5ms
00329  *   Entry 2: (8192 x 16) / 120000 = 1.092s = 1092ms
00330  *   Entry 5: (16384 x 128) / 120000 = 17.48s = 17476ms
00331  * @endverbatim
00332  *
00333  * @note Values are slightly longer than requested to provide safety margin
00334  * @note Table is const and placed in .rodata section (48 bytes total)
00335  *
00336  * @warning Do not modify table at runtime (const data)
00337  *
00338  * @see internal_find_timeout_config() Table lookup function
00339  * @see iwtdt_timeout_entry_t Entry structure definition
00340  *
00341  * @since Version 1.0.0

```

```

00342 */
00343 static const iwdt_timeout_entry_t s_timeout_table[] = {
00344     /* timeout_ms, TOPS (cycles),      CKS (divisor) */
00345     /* Actual timeout = (cycles * divisor) / 120kHz */
00346     {.timeout_ms = 128, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_1}, /* ~136ms actual */
00347     {.timeout_ms = 512, .tops = k_iwdt_tops_4096, .cks = k_iwdt_cks_div_16}, /* ~546ms actual */
00348     {.timeout_ms = 1000, .tops = k_iwdt_tops_8192, .cks = k_iwdt_cks_div_16}, /* ~1.09s actual */
00349     {.timeout_ms = 2048, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_16}, /* ~2.18s actual */
00350     {.timeout_ms = 8192, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_64}, /* ~8.74s actual */
00351     {.timeout_ms = 16384, .tops = k_iwdt_tops_16384, .cks = k_iwdt_cks_div_128}, /* ~17.48s actual */
00352 };
00353
00354 /**
00355  * @var s_iwdt_timeout_table_size
00356  * @brief Number of entries in the timeout lookup table
00357  *
00358  * @details
00359  * Compile-time constant for table bounds checking. Used in
00360  * internal_find_timeout_config() for loop bounds (NASA Power of 10 Rule 2).
00361  *
00362  * @invariant s_iwdt_timeout_table_size == 6 (current implementation)
00363  * @invariant s_iwdt_timeout_table_size > 0
00364  *
00365  * @since Version 1.0.0
00366  */
00367 static const uint32_t s_iwdt_timeout_table_size =
00368     sizeof(s_timeout_table) / sizeof(s_timeout_table[0]);
00369
00370 /*
=====
00371  * Module State
00372  *
=====
00373  */
00374
00375 /**
00376  * @var s_tag
00377  * @brief Log tag for IWDT module
00378  *
00379  * @details
00380  * Identifies log messages from this module in system logs. Uses the tag "IWDT"
00381  * (uppercase) to match S\\rEAMING_SNAKE_CASE convention for log tags.
00382  *
00383  * @par Log Message Format
00384  * @verbatim
00385  * [IWDT] INFO: Registered task for monitoring
00386  * [IWDT] ERROR: Task deadlock: MotorCtrl (timeout 500 ms)
00387  * @endverbatim
00388  *
00389  * @note Pointer to string literal (stored in .rodata section)
00390  * @warning Do not modify (const pointer to const data)
00391  *
00392  * @since Version 1.0.0
00393  */
00394 static const char* const s_tag = "IWDT";
00395
00396 /**
00397  * @var s_iwdt_initialized
00398  * @brief Flag indicating IWDT has been initialized
00399  *
00400  * @details
00401  * Tracks initialization state to prevent:
00402  * - Multiple calls to rx_hal_iwdt_init() (hardware cannot be reconfigured)
00403  * - API calls before init (undefined behavior prevention)
00404  * - Feed operations on uninitialized watchdog
00405  *
00406  * @par Thread Safety
00407  * - Set only once during initialization (single-threaded startup)
00408  * - Read by all API functions (safe - single write, multiple reads)
00409  * - Never reset (IWDT cannot be stopped once started)
00410  *
00411  * @invariant Value is either k_iwdt_not_initialized or k_iwdt_initialized
00412  * @invariant Once set to k_iwdt_initialized, never changes until reset
00413  *
00414  * @warning Do not modify directly - use rx_hal_iwdt_init() only
00415  *
00416  * @since Version 1.0.0
00417  */
00418 static iwdt_init_state_t s_iwdt_initialized = k_iwdt_not_initialized;
00419
00420 /*
=====
00421  * Internal Helpers
00422  *
=====
00423  */
00424

```

```

00425 /**
00426  * @brief Find best timeout configuration for requested timeout
00427  *
00428  * @details
00429  * Performs linear search through s_timeout_table to find the first entry
00430  * whose timeout_ms is >= the requested value. This provides the smallest
00431  * timeout that meets or exceeds the requirement.
00432  *
00433  * @par Algorithm
00434  * @verbatim
00435  * 1. Validate table is not empty
00436  * 2. Validate output pointer is not nullptr
00437  * 3. Linear scan through sorted table:
00438  *    - If entry.timeout_ms >= requested: return entry pointer
00439  * 4. If no entry found: requested timeout exceeds maximum
00440  * @endverbatim
00441  *
00442  * @par Complexity
00443  * - Time: O(n) where n = s_iwtdt_timeout_table_size (typically 6)
00444  * - Space: O(1) - no additional memory allocated
00445  *
00446  *
00447  *
00448  * @pre timeout_ms >= k_iwtdt_timeout_min_ms
00449  * @pre entry != nullptr
00450  * @pre s_timeout_table populated (compile-time guarantee)
00451  *
00452  * @post *entry points to valid table entry with timeout_ms >= requested
00453  * @post *entry is const - returned entry must not be modified
00454  *
00455  * @note Returns pointer to static data - valid for program lifetime
00456  * @note Selected timeout may be longer than requested (next available)
00457  *
00458  * @par Example:
00459  * @code
00460  * const iwtdt_timeout_entry_t* config = nullptr;
00461  * rx_err_t err = internal_find_timeout_config(1000, &config);
00462  * if (err == k_rx_ok) {
00463  *     // config->tops and config->cks ready for IWDT\\r
00464  *     // config->timeout_ms = actual timeout (>= 1000)
00465  * }
00466  * @endcode
00467  *
00468  * @see s_timeout_table Source data for lookup
00469  * @see iwtdt_timeout_entry_t Structure returned
00470  *
00471  * @since Version 1.0.0
00472  */
00473 static rx_err_t internal_find_timeout_config(const uint32_t timeout_ms,
00474                                             const iwtdt_timeout_entry_t** entry)
00475 {
00476     if (s_iwtdt_timeout_table_size == 0U) {
00477         return k_rx_err_invalid_arg;
00478     }
00479
00480     if (entry == nullptr) {
00481         return k_rx_err_invalid_arg;
00482     }
00483
00484     /* Find first entry with timeout >= requested */
00485     for (uint32_t i = 0; i < s_iwtdt_timeout_table_size; i++) {
00486         if (s_timeout_table[i].timeout_ms >= timeout_ms) {
00487             *entry = &s_timeout_table[i];
00488             return k_rx_ok;
00489         }
00490     }
00491
00492     /* No suitable configuration found */
00493     return k_rx_err_invalid_arg;
00494 }
00495
00496 /**
=====
00497  * Public API Implementation
00498  *
=====
00499  */
00500
00501 #ifdef __RX__
00502 /**
00503  * @brief Configure IWDT control register (IWDT\\r) from timeout table entry
00504  *
00505  * @details
00506  * Builds and writes the IWDT\\r register value from a timeout configuration
00507  * entry. Combines TOPS, CKS, RPES, and RPSS fields into a single 16-bit value.
00508  *
00509  * @par Register Construction

```

```

00510 * @verbatim
00511 *   IWDT\|r = TOPS | CKS | RPES | RPSS
00512 *
00513 *   Where:
00514 *   - TOPS[1:0]: Timeout Period Select (from config)
00515 *   - CKS[7:4]: Clock Division Ratio (from config)
00516 *   - RPES[9:8]: Window End = 0% (disabled)
00517 *   - RPSS[13:12]: Window Start = 100% (full window)
00518 * @endverbatim
00519 *
00520 * @par Register Write Sequence
00521 * @verbatim
00522 *   1. Validate config pointer (NULL check)
00523 *   2. Validate hardware pointer (iwdt() check)
00524 *   3. Build IWDT\|r value from fields
00525 *   4. Write to IWDT\|r register
00526 * @endverbatim
00527 *
00528 *
00529 *
00530 * @pre config must point to valid iwdt_timeout_entry_t
00531 * @pre IWDT not yet started (first refresh not performed)
00532 * @pre iwdt() returns valid hardware register pointer
00533 *
00534 * @post IWDT\|r configured with specified timeout and window settings
00535 *
00536 * @note Not thread-safe: must be called from single-threaded initialization context
00537 * @note Window mode disabled (RPES=0%, RPSS=100%)
00538 * @note Register can only be configured before first refresh
00539 *
00540 * @warning IWDT\|r becomes read-only after first refresh sequence
00541 *
00542 * @see internal_find_timeout_config() Source of config parameter
00543 *
00544 * @since Version 1.0.0
00545 */
00546 static rx_err_t internal_configure_iwdt_control_register(const iwdt_timeout_entry_t* config)
00547 {
00548     if (config == nullptr) {
00549         return k_rx_err_invalid_arg;
00550     }
00551
00552     if (iwdt() == nullptr) {
00553         return k_rx_err_hw_unmapped;
00554     }
00555
00556     uint16_t iwdtcr = 0;
00557     iwdtcr |= config->tops; /* Timeout period */
00558     iwdtcr |= config->cks; /* Clock divisor */
00559     iwdtcr |= k_iwdt_rpes_0; /* Window end at 0% (disabled) */
00560     iwdtcr |= k_iwdt_rpss_100; /* Window start at 100% (full) */
00561
00562     iwdt()->iwdtcr = iwdtcr;
00563     return k_rx_ok;
00564 }
00565
00566 /**
00567 * @brief Configure IWDT reset and sleep-count behavior registers
00568 *
00569 * @details
00570 * Configures IWDTR\|r (Reset Control) and IWDTCSTPR (Count Stop Control)
00571 * registers for safety-critical operation:
00572 * - Full chip reset on timeout (not NMI)
00573 * - Continue counting during sleep modes
00574 *
00575 * @par Register Configuration
00576 * @verbatim
00577 *   IWDTR\|r (Reset Control Register):
00578 *   +-----+
00579 *   | Bit 7 (RSTIRQS) = 1: Reset on underflow/error |
00580 *   | 0: NMI on underflow/error (not used) |
00581 *   +-----+
00582 *   Using reset (1) for safety - NMI could be masked/ignored
00583 *
00584 *   IWDTCSTPR (Count Stop Control Register):
00585 *   +-----+
00586 *   | Bit 7 (SLCSTP) = 0: Continue counting in sleep mode |
00587 *   | 1: Stop counting in sleep (not used) |
00588 *   +-----+
00589 *   Continue counting for safety - sleeping should not bypass watchdog
00590 * @endverbatim
00591 *
00592 *
00593 * @pre IWDT not yet initialized (s_iwdt_initialized == k_iwdt_not_initialized)
00594 * @pre iwdt() returns valid hardware register pointer
00595 * @pre Called after internal_configure_iwdt_control_register()
00596 *

```

```

00597 * @post IWDTR\\r configured for reset action
00598 * @post IWDTCTPR configured to continue counting in sleep
00599 *
00600 * @note Not thread-safe: must be called from single-threaded initialization context
00601 * @note These registers can only be written before first refresh
00602 * @warning Changing RSTIRQS to NMI may compromise system safety
00603 *
00604 * @since Version 1.0.0
00605 */
00606 static rx_err_t internal_configure_iwtd_registers(void)
00607 {
00608     if (iwtd() == nullptr) {
00609         return k_rx_err_hw_unmapped;
00610     }
00611
00612     if (s_iwtd_initialized == k_iwtd_initialized) {
00613         return k_rx_err_invalid_state;
00614     }
00615
00616     iwtd()->iwtdtrcr = k_iwtd_rstirqs_reset;
00617     iwtd()->iwtdctpr = k_iwtd_slcstp_continue;
00618     return k_rx_ok;
00619 }
00620 #endif
00621 /**
00622  * @brief Initialize the Independent Watchdog Timer
00623  *
00624  * @details
00625  * Configures and starts the hardware IWDT with the specified timeout period.
00626  * Once started, the watchdog cannot be stopped - periodic calls to rx_hal_iwtd_feed()
00627  * are required to prevent system reset.
00628  *
00629  * @par Initialization Sequence
00630  * @verbatim
00631  * 1. Validate not already initialized
00632  * 2. Validate timeout_ms within range
00633  * 3. Look up timeout configuration in table
00634  * 4. Configure IWDTR\\r (timeout period, clock divisor, window)
00635  * 5. Configure IWDTR\\r (reset action - not NMI)
00636  * 6. Configure IWDTCTPR (continue counting in sleep)
00637  * 7. Perform first refresh (starts the watchdog)
00638  * 8. Set initialized flag
00639  * @endverbatim
00640  *
00641  * @par Point of No Return
00642  * @verbatim
00643  * After step 7 (first refresh), the IWDT is running and CANNOT be stopped.
00644  * Any failure after this point leaves system in watchdog-protected state.
00645  * @endverbatim
00646  *
00647  *
00648  *
00649  * @pre IWDT must not be already initialized
00650  * @pre timeout_ms in range [k_iwtd_timeout_min_ms, k_iwtd_timeout_max_ms]
00651  * @pre System startup, single-threaded context
00652  *
00653  * @post IWDT running with selected timeout period
00654  * @post s_iwtd_initialized set to k_iwtd_initialized
00655  * @post Periodic rx_hal_iwtd_feed() calls now required
00656  *
00657  * @note On host builds (non-RX), skips hardware access but sets initialized flag
00658  * @note Actual timeout >= requested (selected from lookup table)
00659  *
00660  * @warning Once initialized, IWDT cannot be disabled (requires hard reset)
00661  * @warning Must call rx_hal_iwtd_feed() faster than timeout period
00662  *
00663  * @par Example:
00664  * @code
00665  * // Initialize with 2 second timeout
00666  * rx_err_t err = rx_hal_iwtd_init(2000);
00667  * if (err != k_rx_ok) {
00668  *     // Handle error - system cannot start safely
00669  *     while (1) { // halt }
00670  * }
00671  *
00672  * // Now must feed periodically
00673  * while (1) {
00674  *     // ... main loop ...
00675  *     rx_hal_iwtd_feed(); // Must occur every < 2 seconds
00676  *     tx_thread_sleep(100);
00677  * }
00678  * @endcode
00679  *
00680  * @see rx_hal_iwtd_feed() Reset watchdog counter
00681  * @see rx_hal_iwtd.h Public API and configuration details
00682  *
00683  *

```

```

00684 * @since Version 1.0.0
00685 */
00686 rx_err_t rx_hal_iwdt_init(const uint32_t timeout_ms)
00687 {
00688     #ifdef RX
00689         if (s_iwdt_initialized == k_iwdt_initialized) {
00690             return k_rx_err_invalid_state;
00691         }
00692     #endif
00693     if (timeout_ms < k_iwdt_timeout_min_ms) {
00694         return k_rx_err_invalid_arg;
00695     }
00696     if (timeout_ms > k_iwdt_timeout_max_ms) {
00697         return k_rx_err_invalid_arg;
00698     }
00699 }
00700
00701 /* Find best timeout configuration */
00702 const iwdt_timeout_entry_t* config = nullptr;
00703 rx_err_t err = internal_find_timeout_config(timeout_ms, &config);
00704 if (err != k_rx_ok) {
00705     return err;
00706 }
00707
00708 /*
00709  * Configure IWDT Control Register (IWDTCR)
00710  *
00711  * Bits [1:0] TOPS - Timeout Period Select
00712  * Bits [7:4] CKS - Clock Division Ratio Select
00713  * Bits [9:8] RPES - Window End Position (0x03 = 0%, window disabled)
00714  * Bits [13:12] RPSS - Window Start Position (0x00 = 100%, full window)
00715  */
00716 err = internal_configure_iwdt_control_register(config);
00717 if (err != k_rx_ok) {
00718     return err;
00719 }
00720
00721 /*
00722  * Configure Reset Control Register (IWDTRCR)
00723  *
00724  * Bit 7 RSTIRQS - Reset Interrupt Request Select
00725  * 0 = Non-maskable interrupt request (NMI)
00726  * 1 = Reset
00727  *
00728  * We use reset (1) for safety - NMI could be masked or ignored.
00729  */
00730 err = internal_configure_iwdt_registers();
00731 if (err != k_rx_ok) {
00732     return err;
00733 }
00734
00735 /*
00736  * Configure Count Stop Control Register (IWDTCSTPR)
00737  *
00738  * Bit 7 SLCSTP - Sleep Mode Count Stop Control
00739  * 0 = Count continues during sleep
00740  * 1 = Count stops during sleep
00741  *
00742  * We continue counting during sleep for safety.
00743  */
00744 /*
00745  * Start the IWDT by performing first refresh
00746  *
00747  * The IWDT starts counting after the first refresh sequence.
00748  * After this, the watchdog CANNOT be stopped.
00749  */
00750 rx_hal_iwdt_feed();
00751
00752 s_iwdt_initialized = k_iwdt_initialized;
00753 return k_rx_ok;
00754
00755 #else
00756 /* Host-side stub for unit testing */
00757 (void)timeout_ms;
00758 s_iwdt_initialized = k_iwdt_initialized;
00759 return k_rx_ok;
00760 #endif
00761 }
00762
00763 /**
00764  * @brief Feed the watchdog timer to prevent reset
00765  *
00766  * @details
00767  * Performs the IWDT refresh sequence by writing the magic values 0x00 then 0xFF
00768  * to the IWDTRR register. This resets the down-counter to its initial value,
00769  * preventing timeout and system reset.
00770  */

```

```

00771 * @par Refresh Sequence (Atomic Operation)
00772 * @verbatim
00773 * +-----+
00774 * | 1. Disable interrupts (save PSW) |
00775 * | 2. Write 0x00 to IWDTRR          |
00776 * | 3. Write 0xFF to IWDTRR          |
00777 * | 4. Restore interrupt state       |
00778 * +-----+
00779 * \\CRITICAL: Steps 2 and 3 must not be interrupted!
00780 * @endverbatim
00781
00782 * @par Timing Requirements
00783 * @verbatim
00784 * - Execution time: ~2 us including interrupt disable/enable
00785 * - Must be called faster than configured timeout period
00786 * - Typical call rate: Every 100-500 ms depending on timeout
00787 * @endverbatim
00788
00789 * @pre IWDT must be initialized via rx_hal_iwdt_init()
00790 * @pre Called at rate faster than configured timeout
00791
00792 * @post IWDT counter reset to initial value
00793 * @post Interrupt state restored to previous value
00794
00795 * @note Safe to call from ISRs (uses atomic disable/restore)
00796 * @note On host builds, function is a no-op
00797
00798 * @warning Incomplete refresh sequence causes refresh error -> system reset
00799 * @warning Missing calls to this function cause timeout -> system reset
00800
00801 * @par Example:
00802 * @code
00803 * // Main supervision loop
00804 * while (1) {
00805 *     // Check all tasks for deadlock
00806 *     rx_err_t err = rx_hal_iwdt_check_tasks();
00807 *     if (err == k_rx_ok) {
00808 *         // All tasks healthy, feed hardware watchdog
00809 *         rx_hal_iwdt_feed();
00810 *     }
00811 *     // If task timeout detected, DON'T feed - let watchdog reset
00812 *     tx_thread_sleep(100); // 100 ms check interval
00813 * }
00814 * @endcode
00815
00816 * @see rx_hal_iwdt_init() Start the watchdog
00817 * @see rx_hal_iwdt_check_tasks() Verify task health before feeding
00818
00819 * @since Version 1.0.0
00820 */
00821 void rx_hal_iwdt_feed(void)
00822 {
00823 #ifdef __RX__
00824 /*
00825  * IWDT Refresh Sequence (Section 35.3.3)
00826  *
00827  * Write 0x00 followed by 0xFF to IWDTRR.
00828  * This sequence resets the down-counter to its initial value.
00829  *
00830  * CRITICAL: This sequence must not be interrupted.
00831  * An incomplete sequence will trigger a refresh error.
00832  *
00833  * Uses inline assembly to save/restore PSW (Program Status Word) because
00834  * there is no C library function to manipulate interrupt state on RX architecture.
00835  */
00836 /* Disable interrupts during refresh for atomicity */
00837 uint32_t psw;
00838
00839 __asm__ volatile("mvfc psw, %0" : "=r"(psw));
00840 __asm__ volatile("clrpsw i");
00841
00842 /* Perform refresh sequence */
00843 iwdt()->iwtdrr = k_iwdt_refresh_start; /* Write 0x00 */
00844 iwdt()->iwtdrr = k_iwdt_refresh_end;   /* Write 0xFF */
00845
00846 /* Restore interrupt state */
00847 __asm__ volatile("mvtc %0, psw" : "=r"(psw));
00848
00849 #else
00850 /* Host-side stub - no operation */
00851 #endif
00852 }
00853
00854 /**
00855 * @brief Check if system was reset by watchdog
00856 */
00857

```

```

00858 * @details
00859 * Reads the IWDTSR Status Register (IWDTSR) to determine if the previous
00860 * system reset was caused by watchdog timeout (underflow) or refresh error.
00861 * This function should be called early in startup for diagnostics.
00862 *
00863 * @par Status Register Flags
00864 * @verbatim
00865 *   IWDTSR Bit 14 (UNDFF) - Underflow Flag:
00866 *   - 1: Counter reached 0 (timeout occurred)
00867 *   - 0: No underflow
00868 *
00869 *   IWDTSR Bit 15 (REFEF) - Refresh Error Flag:
00870 *   - 1: Invalid refresh sequence detected
00871 *   - 0: No refresh error
00872 * @endverbatim
00873 *
00874 *
00875 * @pre Can be called before rx_hal_iwdt_init()
00876 * @pre Status flags preserved across reset (hardware feature)
00877 *
00878 * @post Status flags NOT cleared (call rx_hal_iwdt_clear_status() to clear)
00879 *
00880 * @note On host builds, always returns false
00881 * @note Call rx_hal_iwdt_get_reset_cause() for specific cause
00882 *
00883 * @par Example:
00884 * @code
00885 * // Check at startup
00886 * if (rx_hal_iwdt_was_reset()) {
00887 *     uart_debug_puts("[WARN] Recovered from watchdog reset!\r\n");
00888 *     const char* failed = rx_hal_iwdt_get_failed_task();
00889 *     if (failed != nullptr) {
00890 *         uart_debug_printf("Failed task: %s\r\n", failed);
00891 *     }
00892 * }
00893 * @endcode
00894 *
00895 * @see rx_hal_iwdt_get_reset_cause() Get specific reset cause
00896 * @see rx_hal_iwdt_clear_status() Clear status flags after reading
00897 * @see rx_hal_iwdt_get_failed_task() Get name of task that caused timeout
00898 *
00899 * @since Version 1.0.0
00900 */
00901 bool rx_hal_iwdt_was_reset(void)
00902 {
00903 #ifdef __RX__
00904 /*
00905  * Check IWDTSR Status Register for underflow or refresh error
00906  *
00907  * Bit 14 UNDFF - Underflow Flag
00908  * 1 = Counter underflow occurred
00909  *
00910  * Bit 15 REFEF - Refresh Error Flag
00911  * 1 = Refresh error occurred
00912  */
00913 const uint16_t status = iwdt()->iwdtsr;
00914
00915 return ((status & k_iwdt_sr_undff) != 0) || ((status & k_iwdt_sr_refef) != 0);
00916 #else
00917 return false;
00918 #endif
00919 }
00920 /**
00921 * @brief Get detailed watchdog reset cause
00922 *
00923 * @details
00924 * Reads the IWDTSR Status Register to determine the specific cause of the
00925 * last watchdog reset. Refresh errors take priority over underflow since
00926 * they indicate a software bug in the refresh sequence.
00927 *
00928 * @par Reset Cause Priority
00929 * @verbatim
00930 *   1. Refresh Error (REFEF=1): Invalid refresh sequence
00931 *   - Incomplete 0x00/0xFF write
00932 *   - Write occurred outside window (if window mode enabled)
00933 *
00934 *   2. Underflow (UNDFF=1): Counter reached zero
00935 *   - rx_hal_iwdt_feed() not called within timeout period
00936 *   - Software hang, deadlock, or infinite loop
00937 *
00938 *   3. None: No watchdog reset occurred
00939 *   - Power-on reset, external reset, or other cause
00940 * @endverbatim
00941 *
00942 * @endverbatim
00943 *
00944 *

```



```

00945 * @pre Can be called before rx_hal_iwdt_init()
00946 * @pre Status flags preserved across reset
00947 *
00948 * @post Status flags NOT cleared (call rx_hal_iwdt_clear_status() to clear)
00949 *
00950 * @note On host builds, always returns k_iwdt_reset_none
00951 * @note Refresh error has priority over underflow in return value
00952 *
00953 * @par Example:
00954 * @code
00955 * rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
00956 * switch (cause) {
00957 *     case k_iwdt_reset_refresh_error:
00958 *         uart_debug_puts("ERROR: Watchdog refresh sequence error!\r\n");
00959 *         break;
00960 *     case k_iwdt_reset_underflow:
00961 *         uart_debug_puts("WARN: Watchdog timeout - software hang detected\r\n");
00962 *         break;
00963 *     case k_iwdt_reset_none:
00964 *     default:
00965 *         uart_debug_puts("Normal startup\r\n");
00966 *         break;
00967 * }
00968 * rx_hal_iwdt_clear_status(); // Clear flags after reading
00969 * @endcode
00970 *
00971 * @see rx_hal_iwdt_was_reset() Quick check for any watchdog reset
00972 * @see rx_hal_iwdt_clear_status() Clear status flags
00973 *
00974 * @since Version 1.0.0
00975 */
00976 rx_iwdt_reset_cause_t rx_hal_iwdt_get_reset_cause(void)
00977 {
00978 #ifdef __RX__
00979     const uint16_t status = iwdt()->iwdtsr;
00980
00981     if (status & k_iwdt_sr_refef) {
00982         return k_iwdt_reset_refresh_error;
00983     }
00984
00985     if (status & k_iwdt_sr_undff) {
00986         return k_iwdt_reset_underflow;
00987     }
00988
00989     return k_iwdt_reset_none;
00990 #else
00991     return k_iwdt_reset_none;
00992 #endif
00993 }
00994 /**
00995 * @brief Clear watchdog status flags
00996 *
00997 * @details
00998 * Clears the UNDF (underflow) and REFEF (refresh error) flags in the
00999 * IWDTSR Register. These flags are "write-0-to-clear" type - writing
01000 * 0 to a flag clears it, writing 1 has no effect.
01001 *
01002 * @par Clear Sequence
01003 * @verbatim
01004 * 1. Read IWDTSR (get current status)
01005 * 2. AND with ~(UNDF | REFEF) to clear both flags
01006 * 3. Write back to IWDTSR
01007 *
01008 * Note: Read-modify-write is safe because flags are write-0-to-clear
01009 * @endverbatim
01010 *
01011 * @pre Status has been read via rx_hal_iwdt_was_reset() or rx_hal_iwdt_get_reset_cause()
01012 *
01013 * @post UNDF flag cleared
01014 * @post REFEF flag cleared
01015 *
01016 * @note On host builds, function is a no-op
01017 * @note Call after reading status at startup to prepare for next reset
01018 *
01019 * @par Example:
01020 * @code
01021 * // Startup diagnostic sequence
01022 * if (rx_hal_iwdt_was_reset()) {
01023 *     rx_iwdt_reset_cause_t cause = rx_hal_iwdt_get_reset_cause();
01024 *     log_watchdog_reset(cause);
01025 * }
01026 * // Clear flags so next reset can be detected
01027 * rx_hal_iwdt_clear_status();
01028 * @endcode
01029 *
01030 */
01031

```



```

01117 * @since Version 1.0.0
01118 */
01119 typedef enum : uint8_t {
01120     k_task_not_registered = 0, /**< Slot available for new task registration */
01121     k_task_registered     = 1, /**< Slot in use by registered task */
01122 } task_registration_state_t;
01123
01124 /**
01125 * @enum task_search_constants_t
01126 * @brief Constants for task search operations
01127 *
01128 * @details
01129 * Defines sentinel values and bounds for task array operations.
01130 * Using signed int32_t allows -1 sentinel for "not found" result.
01131 *
01132 * @since Version 1.0.0
01133 */
01134 typedef enum : int32_t {
01135     k_task_not_found = -1, /**< Return value when task name not in array */
01136     k_task_idx_min   = 0, /**< Minimum valid task index (array start) */
01137 } task_search_constants_t;
01138
01139 /**
01140 * @struct task_monitor_t
01141 * @brief Task monitoring state for a single registered task
01142 *
01143 * @details
01144 * Stores all information needed to track one task's heartbeat status.
01145 * Each registered task gets one slot in the s_task_monitor.tasks[] array.
01146 *
01147 * @par Memory Layout (~32 bytes per task)
01148 * @verbatim
01149 *   Offset 0: task_name[16]    [16 bytes] - Null-terminated name
01150 *   Offset 16: timeout_ms      [4 bytes]  - Heartbeat timeout
01151 *   Offset 20: last_heartbeat_ms [4 bytes] - Last heartbeat timestamp
01152 *   Offset 24: registered      [1 byte]   - Registration state
01153 *   Offset 25: padding          [7 bytes]  - Alignment padding
01154 *   Total: ~32 bytes (depends on alignment)
01155 * @endverbatim
01156 *
01157 * @par Timeout Check Algorithm
01158 * @verbatim
01159 *   elapsed = current_time - last_heartbeat_ms
01160 *   if (elapsed > timeout_ms) -> Task has timed out
01161 * @endverbatim
01162 *
01163 * @invariant task_name is null-terminated
01164 * @invariant timeout_ms >= k_iwtdt_timeout_min_ms
01165 * @invariant If registered == k_task_registered, task_name[0] != '\0'
01166 *
01167 * @since Version 1.0.0
01168 */
01169 typedef struct {
01170     char
01171     task_name[k_task_name_max_len]; /**< Task identifier (null-terminated, truncated to 15 chars) */
01172     uint32_t timeout_ms; /**< Maximum time between heartbeats in milliseconds */
01173     uint32_t last_heartbeat_ms; /**< Timestamp (tx_time_get) of last heartbeat */
01174     task_registration_state_t registered; /**< Whether this slot is in use */
01175 } task_monitor_t;
01176
01177 /**
01178 * @struct task_monitor_state_t
01179 * @brief Complete task monitoring module state
01180 *
01181 * @details
01182 * Contains all state for the task monitoring subsystem. Statically allocated
01183 * to avoid dynamic memory (NASA Power of 10 Rule 3). One instance exists
01184 * as s_task_monitor.
01185 *
01186 * @par Memory Layout (~520 bytes total)
01187 * @verbatim
01188 *   Offset 0: tasks[8]          [-256 bytes] - Task array (8 x 32 bytes)
01189 *   Offset 256: task_count      [1 byte]   - Number of registered tasks
01190 *   Offset 257: failed_task[16] [16 bytes] - Failed task name buffer
01191 *   Offset 273: padding         [7 bytes]  - Alignment
01192 *   Total: ~280 bytes (actual layout depends on alignment)
01193 * @endverbatim
01194 *
01195 * @invariant task_count <= k_iwtdt_max_tasks
01196 * @invariant failed_task is null-terminated (or all zeros if no failure)
01197 *
01198 * @since Version 1.0.0
01199 */
01200 typedef struct {
01201     task_monitor_t tasks[k_iwtdt_max_tasks]; /**< Array of task monitoring slots */
01202     uint8_t task_count; /**< Number of registered tasks (0-8) */
01203     char failed_task[k_task_name_max_len]; /**< Name of most recent failed task */

```

```

01204 } task_monitor_state_t;
01205
01206 /**
01207  * @var s_task_monitor
01208  * @brief Module state for task monitoring (static allocation)
01209  *
01210  * @details
01211  * Single instance of task monitoring state. Zero-initialized at startup,
01212  * ensuring all tasks start unregistered and no failed task recorded.
01213  *
01214  * @par Initial State
01215  * @verbatim
01216  * - task_count = 0
01217  * - All tasks[].registered = k_task_not_registered
01218  * - failed_task[0] = '\0'
01219  * @endverbatim
01220  *
01221  * @par Thread Safety
01222  * - Registration: Call from single thread during initialization
01223  * - Heartbeats: Not thread-safe, caller must synchronize
01224  * - Check: Not thread-safe, typically called from single monitoring task
01225  *
01226  * @warning Do not access directly - use API functions
01227  *
01228  * @since Version 1.0.0
01229  */
01230 static task_monitor_state_t s_task_monitor = {};
01231
01232 /**
01233  * @brief Find task index by name in the registered tasks array
01234  *
01235  * @details
01236  * Performs linear search through s_task_monitor.tasks[] to find a task
01237  * matching the given name. Uses strncmp for safe string comparison.
01238  *
01239  * @par Algorithm
01240  * @verbatim
01241  * 1. Validate task_name is not nullptr
01242  * 2. Validate task_name is not empty string
01243  * 3. Linear scan through registered tasks (0 to task_count-1):
01244  *    - Compare names using strncmp(task_name, k_task_name_cmp_len)
01245  *    - If match found: return index
01246  * 4. If no match: return k_task_not_found (-1)
01247  * @endverbatim
01248  *
01249  * @par Complexity
01250  * - Time: O(n x m) where n = task_count, m = k_task_name_cmp_len
01251  * - Space: O(1) - no additional memory
01252  *
01253  *
01254  * @pre task_name != nullptr
01255  * @pre task_name[0] != '\0'
01256  *
01257  * @post Return value < s_task_monitor.task_count (if found)
01258  * @post Return value == k_task_not_found (if not found or invalid input)
01259  *
01260  * @note Uses k_task_name_cmp_len (15) for comparison, not full buffer
01261  * @note Empty string returns k_task_not_found (invalid task name)
01262  *
01263  * @see rx_hal_iwdt_task_heartbeat() Primary caller
01264  * @see rx_hal_iwdt_register_task() Uses for duplicate detection
01265  *
01266  * @since Version 1.0.0
01267  */
01268
01269 static int32_t internal_find_task(const char* task_name)
01270 {
01271     if (task_name == nullptr) {
01272         return k_task_not_found;
01273     }
01274
01275     if (task_name[0] == '\0') {
01276         return k_task_not_found; /* Empty task name invalid */
01277     }
01278
01279     for (uint32_t i = (uint32_t)k_task_idx_min; i < s_task_monitor.task_count; i++) {
01280         if (strncmp(s_task_monitor.tasks[i].task_name, task_name, k_task_name_cmp_len) == 0) {
01281             return (int32_t)i;
01282         }
01283     }
01284
01285     return k_task_not_found;
01286 }
01287
01288 /**
01289  * @brief Register a task for watchdog monitoring
01290  *

```

```

01291 * @details
01292 * Adds a task to the monitoring array with specified timeout period. The task
01293 * must subsequently call rx_hal_iwdt_task_heartbeat() within each timeout period
01294 * to indicate liveness. Failed tasks are detected by rx_hal_iwdt_check_tasks().
01295 *
01296 * @par Registration Process
01297 * @verbatim
01298 * 1. Validate task_name (not nullptr, not empty)
01299 * 2. Validate timeout_ms (within allowed range)
01300 * 3. Check capacity (task_count < k_iwdt_max_tasks)
01301 * 4. Check for duplicate registration (same name)
01302 * 5. Copy task name to slot (truncate if > 15 chars)
01303 * 6. Initialize timeout and heartbeat timestamp
01304 * 7. Mark slot as registered
01305 * 8. Increment task_count
01306 * @endverbatim
01307 *
01308 *
01309 *
01310 * @pre IWDT module initialized (but not strictly required for registration)
01311 * @pre task_name points to valid null-terminated string
01312 * @pre timeout_ms appropriate for task's execution frequency
01313 *
01314 * @post Task added to s_task_monitor.tasks[]
01315 * @post s_task_monitor.task_count incremented
01316 * @post Initial heartbeat timestamp set to current time
01317 *
01318 * @invariant s_task_monitor.task_count <= k_iwdt_max_tasks at all times
01319 *
01320 * @note Task name truncated to 15 characters if longer
01321 * @note First heartbeat auto-recorded at registration time
01322 *
01323 * @warning Not thread-safe - call during initialization only
01324 *
01325 * @par Example:
01326 * @code
01327 * // Register motor control task (must heartbeat every 500ms)
01328 * rx_err_t err = rx_hal_iwdt_register_task("MotorCtrl", 500);
01329 * if (err != k_rx_ok) {
01330 *     // Handle registration failure
01331 *     return err;
01332 * }
01333 *
01334 * // In motor task loop
01335 * while (1) {
01336 *     motor_control_update();
01337 *     rx_hal_iwdt_task_heartbeat("MotorCtrl");
01338 *     tx_thread_sleep(10); // 100Hz loop
01339 * }
01340 * @endcode
01341 *
01342 * @see rx_hal_iwdt_task_heartbeat() Send heartbeat from registered task
01343 * @see rx_hal_iwdt_check_tasks() Check all tasks for timeout
01344 *
01345 * @since Version 1.0.0
01346 */
01347 rx_err_t rx_hal_iwdt_register_task(const char* task_name, uint32_t timeout_ms)
01348 {
01349     /* Validate parameters */
01350     if (task_name == nullptr) {
01351         return k_rx_err_invalid_arg;
01352     }
01353
01354     if (task_name[0] == '\0') {
01355         return k_rx_err_invalid_arg;
01356     }
01357
01358     if (timeout_ms < k_iwdt_timeout_min_ms || timeout_ms > k_iwdt_timeout_max_ms) {
01359         return k_rx_err_invalid_arg;
01360     }
01361
01362     if (s_task_monitor.task_count >= k_iwdt_max_tasks) {
01363         return k_rx_err_no_mem;
01364     }
01365
01366     /* Check for duplicate registration */
01367     if (internal_find_task(task_name) != k_task_not_found) {
01368         return k_rx_err_exists;
01369     }
01370
01371     /* Register new task */
01372     task_monitor_t* task = &s_task_monitor.tasks[s_task_monitor.task_count];
01373
01374     strncpy(task->task_name, task_name, k_task_name_cmp_len);
01375     task->task_name[k_task_name_cmp_len] = '\0'; /* Ensure null termination */
01376
01377     task->timeout_ms = timeout_ms;

```

```

01378 task->last_heartbeat_ms = tx_time_get();
01379 task->registered      = k_task_registered;
01380
01381 s_task_monitor.task_count++;
01382
01383 rx_log_info(s_tag, "Registered task for monitoring");
01384
01385 return k_rx_ok;
01386 }
01387
01388 /**
01389  * @brief Record a task heartbeat to indicate liveness
01390  *
01391  * @details
01392  * Updates the last_heartbeat_ms timestamp for the specified task. Must be
01393  * called by each monitored task at a rate faster than its configured timeout
01394  * to avoid being flagged as failed by rx_hal_iwdt_check_tasks().
01395  *
01396  * @par Heartbeat Algorithm
01397  * @verbatim
01398  * 1. Validate task_name (not nullptr)
01399  * 2. Find task in registered array (internal_find_task)
01400  * 3. If found: Update last_heartbeat_ms = tx_time_get()
01401  * 4. If not found: Log error (graceful degradation)
01402  * @endverbatim
01403  *
01404  * @par Timing Requirements
01405  * @verbatim
01406  * - Must be called faster than task's timeout_ms
01407  * - Typical pattern: Call at end of task's main loop iteration
01408  * - Example: 500ms timeout -> call at least every 400ms
01409  * @endverbatim
01410  *
01411  *
01412  * @pre task_name must be non-NULL
01413  * @pre Task must be registered via rx_hal_iwdt_register_task()
01414  * @pre Called faster than task's configured timeout
01415  *
01416  * @post Task's last_heartbeat_ms updated to current time
01417  * @post No state change if task not found (logs error only)
01418  *
01419  * @note Void return - errors logged but not returned
01420  * @note Unregistered task heartbeats are logged and ignored
01421  * @note Uses tx_time_get() for timestamp (ThreadX tick count)
01422  *
01423  * @warning Missing heartbeats cause rx_hal_iwdt_check_tasks() to detect timeout
01424  *
01425  * @par Example:
01426  * @code
01427  * static void communication_task(ULONG input) {
01428  *     while (1) {
01429  *         // Process incoming messages
01430  *         process_messages();
01431  *
01432  *         // Indicate task is alive
01433  *         rx_hal_iwdt_task_heartbeat("CommTask");
01434  *
01435  *         // Wait for next iteration
01436  *         tx_thread_sleep(50); // 50ms loop
01437  *     }
01438  * }
01439  * @endcode
01440  *
01441  * @see rx_hal_iwdt_register_task() Register task before heartbeating
01442  * @see rx_hal_iwdt_check_tasks() Verifies heartbeats are received
01443  *
01444  * @since Version 1.0.0
01445  */
01446 void rx_hal_iwdt_task_heartbeat(const char* task_name)
01447 {
01448     if (task_name == nullptr) {
01449         rx_log_error(s_tag, "Heartbeat called with nullptr task_name");
01450         return;
01451     }
01452
01453     int32_t idx = internal_find_task(task_name);
01454
01455     if (idx == k_task_not_found) {
01456         /* Task not registered - log warning but don't fail */
01457         rx_log_error(s_tag, "Heartbeat from unregistered task");
01458         return;
01459     }
01460
01461     /* Update heartbeat timestamp */
01462     s_task_monitor.tasks[idx].last_heartbeat_ms = tx_time_get();
01463 }
01464

```

```

01465 /**
01466  * @brief Check all monitored tasks for timeout violations
01467  *
01468  * @details
01469  * Iterates through all registered tasks and compares elapsed time since
01470  * last heartbeat against each task's configured timeout. If any task has
01471  * exceeded its timeout, logs the failure and records the task name.
01472  *
01473  * @par Check Algorithm
01474  * @verbatim
01475  * 1. Validate IWDT is initialized
01476  * 2. Validate task_count is within bounds (detect corruption)
01477  * 3. Get current time from ThreadX
01478  * 4. For each registered task:
01479  *    a. Calculate elapsed = current_time - last_heartbeat_ms
01480  *    b. If elapsed > timeout_ms:
01481  *        - Log error with task name and timeout
01482  *        - Record failed task name
01483  *        - Set result = k_rx_err_timeout
01484  * 5. Return result (k_rx_ok if all pass, k_rx_err_timeout if any fail)
01485  * @endverbatim
01486  *
01487  * @par Typical Usage Pattern
01488  * @verbatim
01489  * Monitoring Task Loop:
01490  * +-----+
01491  * | 1. rx_hal_iwdt_check_tasks() |
01492  * | 2. If all tasks OK: rx_hal_iwdt_feed() |
01493  * | 3. If timeout: DON'T feed (let WDT reset) |
01494  * | 4. tx_thread_sleep(check_interval) |
01495  * | 5. Repeat |
01496  * +-----+
01497  * @endverbatim
01498  *
01499  *
01500  * @pre IWDT must be initialized via rx_hal_iwdt_init()
01501  * @pre Called periodically from monitoring task
01502  *
01503  * @post If timeout detected, failed task name recorded in s_task_monitor.failed_task
01504  * @post Error logged for each timed-out task
01505  * @post Continues checking all tasks even if one fails (logs all failures)
01506  *
01507  * @note Does NOT automatically feed hardware watchdog
01508  * @note Checks all tasks even after finding failure (comprehensive logging)
01509  * @note Uses tx_time_get() which returns uint32_t tick count
01510  *
01511  * @warning Do NOT feed watchdog if this function returns k_rx_err_timeout
01512  *
01513  * @par Example:
01514  * @code
01515  * static void watchdog_monitor_task(ULONG input) {
01516  *     while (1) {
01517  *         rx_err_t result = rx_hal_iwdt_check_tasks();
01518  *
01519  *         if (result == k_rx_ok) {
01520  *             // All tasks healthy - feed hardware watchdog
01521  *             rx_hal_iwdt_feed();
01522  *         } else if (result == k_rx_err_timeout) {
01523  *             // Task deadlock detected!
01524  *             // DON'T feed - let hardware watchdog reset system
01525  *             const char* failed = rx_hal_iwdt_get_failed_task();
01526  *             uart_debug_printf("DEADLOCK: %s\\r\\n", failed);
01527  *         }
01528  *
01529  *         tx_thread_sleep(100); // Check every 100ms
01530  *     }
01531  * }
01532  * @endcode
01533  *
01534  * @see rx_hal_iwdt_register_task() Register tasks before checking
01535  * @see rx_hal_iwdt_task_heartbeat() Tasks report liveness
01536  * @see rx_hal_iwdt_get_failed_task() Get name of failed task
01537  * @see rx_hal_iwdt_feed() Feed hardware watchdog if all tasks OK
01538  *
01539  * @since Version 1.0.0
01540  */
01541 rx_err_t rx_hal_iwdt_check_tasks(void)
01542 {
01543     uint32_t current_time_ms;
01544     rx_err_t result;
01545
01546     const task_monitor_t* task = nullptr;
01547     uint32_t elapsed_ms;
01548     char log_msg[k_log_msg_buffer_size];
01549     int32_t snprintf_result;
01550
01551     if (s_iwdt_initialized != k_iwdt_initialized) {

```

```

01552     return k_rx_err_invalid_state;
01553 }
01554
01555 if (s_task_monitor.task_count > k_iwdt_max_tasks) {
01556     return k_rx_err_invalid_state; /* Corrupted state */
01557 }
01558
01559 current_time_ms = tx_time_get();
01560 result          = k_rx_ok;
01561
01562 for (uint32_t i = (uint32_t)k_task_idx_min; i < s_task_monitor.task_count; i++) {
01563     task          = &s_task_monitor.tasks[i];
01564     elapsed_ms    = current_time_ms - task->last_heartbeat_ms;
01565
01566     if (elapsed_ms > task->timeout_ms) {
01567         /* Task has exceeded heartbeat timeout - deadlock detected */
01568         snprintf_result = snprintf(log_msg,
01569                                   sizeof(log_msg),
01570                                   "Task deadlock: %s (timeout %" PRIu32 " ms)",
01571                                   task->task_name,
01572                                   task->timeout_ms);
01573
01574         if (snprintf_result > 0 && (uint32_t)snprintf_result < sizeof(log_msg)) {
01575             rx_log_error(s_tag, log_msg);
01576         }
01577
01578         /* Record failed task name for post-mortem */
01579         strncpy(s_task_monitor.failed_task, task->task_name, k_task_name_cmp_len);
01580         s_task_monitor.failed_task[k_task_name_cmp_len] = '\0';
01581
01582         result = k_rx_err_timeout;
01583         /* Continue checking other tasks to log all failures */
01584     }
01585 }
01586
01587 return result;
01588 }
01589
01590 /**
01591  * @brief Get name of last failed task for post-mortem analysis
01592  *
01593  * @details
01594  * Returns the name of the task that most recently exceeded its heartbeat
01595  * timeout. This information is recorded by rx_hal_iwdt_check_tasks() when it
01596  * detects a timeout. Useful for diagnosing which task caused a watchdog reset.
01597  *
01598  * @par Use Cases
01599  * @verbatim
01600  * 1. Post-reset diagnostics:
01601  *   - After watchdog reset, call rx_hal_iwdt_get_failed_task()
01602  *   - Log failed task name for debugging
01603  *   - Helps identify deadlock/hang location
01604  *
01605  * 2. Pre-reset logging:
01606  *   - When rx_hal_iwdt_check_tasks() returns k_rx_err_timeout
01607  *   - Get failed task name and log before reset occurs
01608  * @endverbatim
01609  *
01610  *
01611  * @pre IWDT must be initialized via rx_hal_iwdt_init()
01612  * @pre rx_hal_iwdt_check_tasks() has been called and detected a timeout
01613  *
01614  * @post Returns pointer to s_task_monitor.failed_task (do not modify)
01615  * @post Returned pointer valid until next timeout detection
01616  *
01617  * @note Returns pointer to internal static buffer - do not free
01618  * @note Buffer contents may change on next rx_hal_iwdt_check_tasks() failure
01619  * @note Returns NULL if failed_task[0] == '\0' (no failure recorded)
01620  *
01621  * @warning Do not modify returned buffer (const pointer)
01622  * @warning Buffer may be overwritten by next timeout detection
01623  *
01624  * @par Example:
01625  * @code
01626  * // After detecting timeout
01627  * rx_err_t result = rx_hal_iwdt_check_tasks();
01628  * if (result == k_rx_err_timeout) {
01629  *     const char* failed = rx_hal_iwdt_get_failed_task();
01630  *     if (failed != nullptr) {
01631  *         uart_debug_printf("FATAL: Task '%s' deadlocked\\r\\n", failed);
01632  *     }
01633  *     // Let watchdog reset the system...
01634  * }
01635  *
01636  * // After system restart
01637  * if (rx_hal_iwdt_was_reset()) {
01638  *     const char* failed = rx_hal_iwdt_get_failed_task();

```



```

01639 * // Note: May be NULL if buffer not preserved across reset
01640 * }
01641 * @endcode
01642 *
01643 * @see rx_hal_iwdt_check_tasks() Records failed task name
01644 * @see rx_hal_iwdt_was_reset() Check if watchdog caused reset
01645 *
01646 * @since Version 1.0.0
01647 */
01648 const char* rx_hal_iwdt_get_failed_task(void)
01649 {
01650     if (s_iwdt_initialized != k_iwdt_initialized) {
01651         return nullptr;
01652     }
01653     if (s_task_monitor.failed_task[0] != '\0') {
01654         return s_task_monitor.failed_task;
01655     }
01656     return nullptr;
01657 }
01658
01659 }
01660
01661 /** @} */ /* End of iwdt_task_monitoring group */

```

8.125 libs/rx_hal/src/rx_lpc.c File Reference

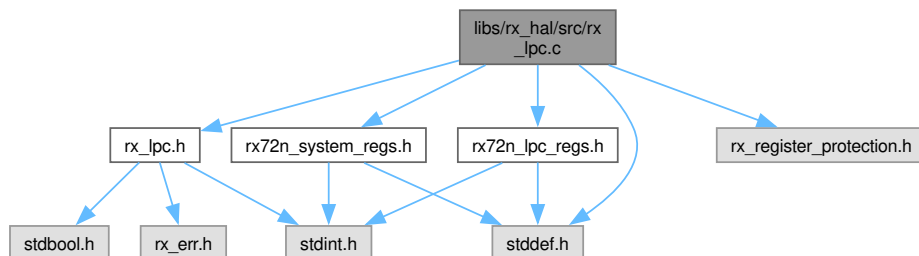
Low Power Consumption (LPC) HAL Driver Implementation for RX72N.

```

#include "rx_lpc.h"
#include <stddef.h>
#include "rx72n_lpc_regs.h"
#include "rx72n_system_regs.h"
#include "rx_register_protection.h"

```

Include dependency graph for rx_lpc.c:



Enumerations

- enum `rx_lpc_init_state_t` : `uint8_t` { `k_lpc_not_initialized` = 0 , `k_lpc_initialized` = 1 }
Driver initialization tracking.
- enum `rx_lpc_sbycr_bits_t` : `uint16_t` { `k_sbycr_sby` = 0x8000U }
SBYCR bit definitions (RX72N Manual Section 11.2.1, page 405).
- enum `rx_lpc_poll_limits_t` : `uint16_t` { `k_opcmstf_poll_max` = 1024U }
Bounded-retry limits for register polls.

Functions

- static rx_err_t [internal_translate_opcc_mode](#) (rx_lpc_opcc_mode_t mode, uint8_t *opcm_↔ bits)
Translate OPCCR mode enum to OPCM[2:0] field value.
- static uint8_t [internal_assemble_dpsbycr](#) (rx_lpc_deep_power_t deep_power, bool keep_io)
Program DPSIER0..DPSIER3 from the 32-bit wake mask.
- rx_err_t [rx_lpc_init](#) (void)
Initialize the Low Power Consumption driver.
- rx_err_t [rx_lpc_set_operating_power](#) (rx_lpc_opcc_mode_t mode)
Switch the MCU to a different operating-power-control mode (implementation).
- rx_err_t [rx_lpc_enter_sleep](#) (void)
Enter Sleep low-power mode (implementation).
- rx_err_t [rx_lpc_enter_software_standby](#) (void)
Enter Software Standby mode (implementation).
- rx_err_t [rx_lpc_enter_deep_software_standby](#) (uint32_t wake_mask, rx_lpc_deep_power_t deep_power, bool keep_io)
Enter Deep Software Standby mode.
- bool [rx_lpc_was_deep_standby_wake](#) (void)
Report whether the most recent reset was a Deep Standby wake.
- rx_err_t [rx_lpc_get_wake_flags](#) (uint32_t *flags)
Read the latched Deep Software Standby wake-source bitfield (implementation).

Variables

- static rx_lpc_init_state_t [s_initialized](#) = [k_lpc_not_initialized](#)
Driver initialization flag.
- static rx_lpc_last_mode_t [s_last_mode](#) = [k_lpc_mode_none](#)
Last low-power mode the driver attempted to enter.
- static bool [s_was_deep_standby_wake](#) = false
Latched "last reset was deep-standby wake" flag (set by rx_lpc_init).
- static uint32_t [s_latched_wake_flags](#) = 0U
Latched wake flags read from DPSIFR0..DPSIFR3 at init time.

8.125.1 Detailed Description

Low Power Consumption (LPC) HAL Driver Implementation for RX72N.

Implements the Sleep, Software Standby, and Deep Software Standby mode entry paths plus the OPCCR (Operating Power Control) selector described in the RX72N Hardware Manual Chapter 11.

Register Write Protocol

All writes to SBYCR, OPCCR, RSTCKCR, and DPSBYCR require PRC1 to be unlocked via PRCR (key 0xA5, PRC1=1). The driver unlocks, writes, and re-locks PRCR around every such access. PRCR writes are guarded by `#ifdef __RX__` so host-side unit tests can exercise the driver without hardware-mapped register access.

Low-Power Entry Instruction

All three low-power modes are entered by executing the WAIT instruction on RX. SBYCR.SSBY and DPSBYCR.DPSBY are set beforehand to select which flavour of low power the WAIT enters:

```
* SSBY=0, DPSBY=0 -> Sleep (WAIT)
* SSBY=1, DPSBY=0 -> Software Standby (WAIT)
* SSBY=1, DPSBY=1 -> Deep Software Standby (WAIT)
*
```

NASA Power of 10 Compliance

- Rule 1: [OK] No goto/setjmp/recursion
- Rule 2: [OK] OPCMTSF poll bounded by k_opcmtsf_poll_max retries
- Rule 3: [OK] Zero dynamic allocation (file-scope statics only)
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] >= 2 pre/post conditions per function
- Rule 6: [OK] s_*-prefixed file-scope statics, locals at narrowest scope
- Rule 7: [OK] All return values checked or explicitly cast to (void)
- Rule 8: [OK] C23 typed enums, preprocessor only for RX/UNIT_TEST guards
- Rule 9: [OK] Single-level pointers
- Rule 10: [OK] Compiles with -Wall -Wextra -Werror

See also

[rx_lpc.h](#) Public API

[rx72n_lpc_regs.h](#) Register layout and bit definitions

RX72N Hardware Manual Chapter 11 - Low Power Consumption

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_lpc.c](#).

8.125.2 Enumeration Type Documentation

8.125.2.1 rx_lpc_init_state_t

enum [rx_lpc_init_state_t](#) : uint8_t

Driver initialization tracking.

Since

Version 1.0.0

Enumerator

k_lpc_not_initialized	rx_lpc_init() has not been called
k_lpc_initialized	rx_lpc_init() completed successfully

Definition at line 73 of file [rx_lpc.c](#).

```
00073      : uint8_t {
00074  k_lpc_not_initialized = 0, /**< rx_lpc_init() has not been called */
00075  k_lpc_initialized    = 1, /**< rx_lpc_init() completed successfully */
00076 } rx_lpc_init_state_t;
```

8.125.2.2 rx_lpc_poll_limits_t

enum [rx_lpc_poll_limits_t](#) : uint16_t

Bounded-retry limits for register polls.

OPCCR.OPCMTSF typically clears within a few hundred cycles. 1024 is a generous ceiling that still keeps the poll bounded per NASA Rule 2.

Since

Version 1.0.0

Enumerator

k_opcmtsf_poll_max	Max OPCMTSF poll iterations
--------------------	-----------------------------

Definition at line 102 of file [rx_lpc.c](#).

```
00102      : uint16_t {
00103  k_opcmtsf_poll_max = 1024U, /**< Max OPCMTSF poll iterations */
00104 } rx_lpc_poll_limits_t;
```

8.125.2.3 rx`lpc`sbycr`bits`

```
enum rx_lpc_sbycr_bits_t : uint16_t
```

SBYCR bit definitions (RX72N Manual Section 11.2.1, page 405).

Only the two bits manipulated by this driver are declared here. Other bits in SBYCR (OPE) are left at their reset values.

Since

Version 1.0.0

Enumerator

k_sbycr_ssbby	Bit 15: Software Standby Select (0=sleep, 1=standby)
---------------	--

Definition at line 88 of file rx_lpc.c.

```
00088      : uint16_t {
00089      k_sbycr_ssbby = 0x8000U, /**< Bit 15: Software Standby Select (0=sleep, 1=standby) */
00090 } rx_lpc_sbycr_bits_t;
```

8.125.3 Function Documentation

8.125.3.1 internal`assemble`dpsbycr()

```
uint8_t internal_assemble_dpsbycr (
    rx_lpc_deep_power_t deep_power,
    bool keep_io) [static]
```

Program DPSIER0..DPSIER3 from the 32-bit wake mask.

Each byte of the mask maps directly to one DPSIER register as described in rx_lpc.h. The NMI bit (DPSIER2.DNMIE) is documented as write-once by the hardware; this driver re-asserts it on every call and the hardware will ignore the redundant writes after the first.

Precondition

dps_regs() returns a valid (hardware or host-mocked) pointer

Postcondition

DPSIER0..DPSIER3 reflect the enabled wake sources

Since

Version 1.0.0

Assemble DPSBYCR value for a deep-standby entry

Internal helper. The only caller (rx_lpc_enter_deep_software_standby) validates deep_power against the three defined rx_lpc_deep_power_t enumerators before calling, so the switch does not need a default branch (NASA Rule 5: validate at boundaries, not inside private helpers). The DPSBY bit is always set because this value is only used on the deep-standby entry path.

Precondition

deep_power is one of k_lpc_deep_ram_usb_on / k_lpc_deep_ram_usb_off / k_lpc_deep_lvd_off (enforced by caller)

Postcondition

Return value contains at most the bits DEEPCUT[1:0], IOKEEP, DPSBY

Since

Version 1.0.0

Definition at line 215 of file rx_lpc.c.

```

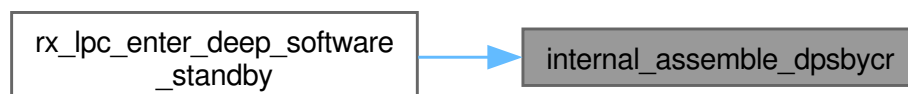
00216 {
00217 /* LUT keyed by rx_lpc_deep_power_t enumerator value (0..2). Using a LUT
00218  * rather than a switch eliminates the synthetic "unreachable default"
00219  * branch that gcov tracks: once deep_power is pre-validated by the
00220  * caller, a switch still produces a default fall-through branch that
00221  * no public API path can execute. The LUT has exactly one branch-free
00222  * path per input and keeps the function trivially verifiable. */
00223 static const uint8_t s_deepcut_bits[] = {
00224     [k_lpc_deep_ram_usb_on] = k_dpsbycr_deepcut_ram_usb_on,
00225     [k_lpc_deep_ram_usb_off] = k_dpsbycr_deepcut_ram_usb_off,
00226     [k_lpc_deep_lvd_off] = k_dpsbycr_deepcut_lvd_off,
00227 };
00228
00229 uint8_t value = s_deepcut_bits[deep_power];
00230 if (keep_io) {
00231     value |= k_dpsbycr_iokeep;
00232 }
00233 value |= k_dpsbycr_dpsby;
00234 return value;
00235 }

```

References [k_dpsbycr_deepcut_lvd_off](#), [k_dpsbycr_deepcut_ram_usb_off](#), [k_dpsbycr_deepcut_ram_usb_on](#), [k_dpsbycr_dpsby](#), [k_dpsbycr_iokeep](#), [k_lpc_deep_lvd_off](#), [k_lpc_deep_ram_usb_off](#), and [k_lpc_deep_ram_usb_on](#).

Referenced by [rx_lpc_enter_deep_software_standby\(\)](#).

Here is the caller graph for this function:



8.125.3.2 internal_translate_opcc_mode()

```

rx_err_t internal_translate_opcc_mode (
    rx_lpc_opcc_mode_t mode,
    uint8_t * opcm_bits) [static]

```

Translate OPCCR mode enum to OPCMR[2:0] field value.

Internal helper. The only caller ([rx_lpc_set_operating_power](#)) always passes the address of a stack-local `uint8_t`, so a NULL `opcm_bits` cannot occur from any public API path. No defensive NULL check is therefore required (NASA Rule 5: validate at boundaries, not inside private helpers).

Precondition

`opcm_bits != NULL` (enforced by callers)

Postcondition

`opcm_bits` contains one of `k_opccr_opcm_` constants on success

Since

Version 1.0.0

Definition at line 153 of file `rx_lpc.c`.

```
00154 {
00155     switch (mode) {
00156         case k_lpc_opcc_high_speed:
00157             *opcm_bits = k_opccr_opcm_highspeed;
00158             return k_rx_ok;
00159         case k_lpc_opcc_low_speed_1:
00160             *opcm_bits = k_opccr_opcm_lowspeed1;
00161             return k_rx_ok;
00162         case k_lpc_opcc_low_speed_2:
00163             *opcm_bits = k_opccr_opcm_lowspeed2;
00164             return k_rx_ok;
00165         default:
00166             return k_rx_err_invalid_arg;
00167     }
00168 }
```

References `k_lpc_opcc_high_speed`, `k_lpc_opcc_low_speed_1`, `k_lpc_opcc_low_speed_2`, `k_opccr_opcm_highspeed`, `k_opccr_opcm_lowspeed1`, and `k_opccr_opcm_lowspeed2`.

Referenced by `rx_lpc_set_operating_power()`.

Here is the caller graph for this function:



8.125.3.3 rx'lpc'enter'deep'software'standby()

```
rx_err_t rx_lpc_enter_deep_software_standby (
    uint32_t wake_mask,
    rx_lpc_deep_power_t deep_power,
    bool keep_io) [nodiscard]
```

Enter Deep Software Standby mode.

Configures `DPSIER0..DPSIER3` from `wake_mask`, sets `DPSBYCR.DEEPCUT`, optionally sets `DPSBYCR.IOKEEP`, sets `SBYCR.SSBY=1` and `DPSBYCR.DPSBY=1` under `PRC1` protection, then executes `WAIT`.

On deep-standby wake the MCU restarts from the reset vector. This function therefore does not return in that case on real hardware; the return path only exists for host-side unit tests. After reset, `rx_lpc_was_deep_standby_wake()` returns true and `rx_lpc_get_wake_flags()` reports which source caused the wake.

Parameters

in	wake_mask	Bitwise OR of rx_lpc_wake_flags_t values; must not be 0
in	deep_power	Deep-cut power configuration (DEEPCUT[1:0])
in	keep_io	If true, set DPSBYCR.IOKEEP=1 so I/O state is retained across the wake-up until the application clears it

Returns

`rx_err_t` Error code (returned only on host builds / on error)

Return values

<code>k_rx_ok</code>	Configuration accepted (host build)
<code>k_rx_err_invalid_arg</code>	<code>wake_mask == 0</code> or contains undefined bits; <code>deep_power</code> not one of rx_lpc_deep_power_t
<code>k_rx_err_not_initialized</code>	rx_lpc_init() has not been called

Precondition

[rx_lpc_init\(\)](#) has been called

`wake_mask != 0` and `(wake_mask & ~k_lpc_wake_all_mask) == 0`

Corresponding pins / peripherals configured by the caller

Postcondition

DPSIER0..DPSIER3 reflect the enabled wake sources

DPSBYCR.DEEPCUT/IOKEEP/DPSBY configured per arguments

`s_last_mode == k_lpc_mode_deep_software_standby` (test-visible)

Warning

This function does not return on real hardware. Any cleanup required after deep standby must be performed at the reset vector / application startup path.

Since

Version 1.0.0

Definition at line 403 of file [rx_lpc.c](#).

```

00404 {
00405     if (s_initialized != k_lpc_initialized) {
00406         return k_rx_err_not_initialized;
00407     }
00408     if (wake_mask == 0U) {
00409         return k_rx_err_invalid_arg;
00410     }
00411     if ((wake_mask & ~k_lpc_wake_all_mask) != 0U) {
00412         return k_rx_err_invalid_arg;
00413     }
00414     if ((deep_power != k_lpc_deep_ram_usb_on) && (deep_power != k_lpc_deep_ram_usb_off) &&
00415         (deep_power != k_lpc_deep_lvd_off)) {

```



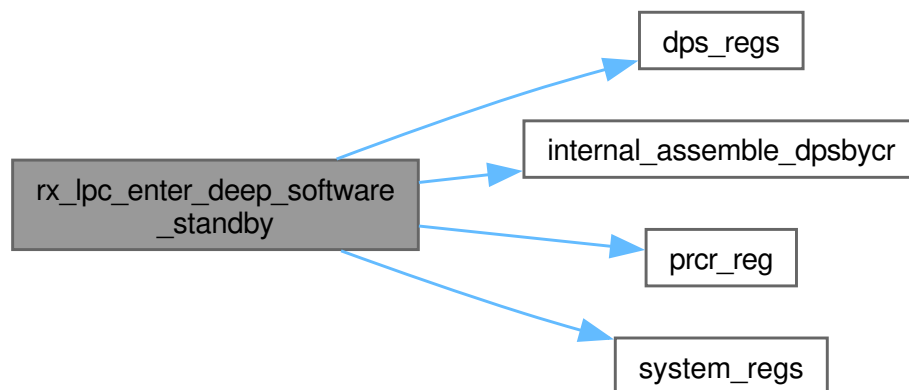
```

00416     return k_rx_err_invalid_arg;
00417 }
00418
00419 const uint8_t dpsbycr_value = internal_assemble_dpsbycr(deep_power, keep_io);
00420
00421 #ifdef __RX__
00422 volatile uint16_t* prcr = prcr_reg();
00423 volatile rx_system_regs_t* sys = system_regs();
00424 volatile rx_dps_regs_t* dps = dps_regs();
00425
00426 *prcr = k_rx_prcr_unlock_prc1;
00427
00428 /* 1. Enable wake sources */
00429 dps->dpsier0 = (uint8_t)((wake_mask » 0U) & 0xFFU);
00430 dps->dpsier1 = (uint8_t)((wake_mask » 8U) & 0xFFU);
00431 dps->dpsier2 = (uint8_t)((wake_mask » 16U) & 0xFFU);
00432 dps->dpsier3 = (uint8_t)((wake_mask » 24U) & 0xFFU);
00433
00434 /* 2. Clear any stale wake flags */
00435 dps->dpsifr0 = 0U;
00436 dps->dpsifr1 = 0U;
00437 dps->dpsifr2 = 0U;
00438 dps->dpsifr3 = 0U;
00439
00440 /* 3. Program DPSBYCR (DEEPCUT, IOKEEP, DPSBY=1) */
00441 dps->dpsbycr = dpsbycr_value;
00442
00443 /* 4. Set SBYCR.SSBY so WAIT enters standby-family */
00444 sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssbby);
00445
00446 *prcr = k_rx_prcr_lock;
00447 #else
00448 (void)dpsbycr_value;
00449 internal_program_wake_enables(wake_mask);
00450 #endif
00451
00452 s_last_mode = k_lpc_mode_deep_software_standby;
00453
00454 #ifdef __RX__
00455 __asm__ volatile("wait");
00456 /* On real hardware, execution never reaches here - wake restarts at reset */
00457 #endif
00458
00459 return k_rx_ok;
00460 }

```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsbycr](#), [rx_dps_regs_t::dpsier0](#), [rx_dps_regs_t::dpsier1](#), [rx_dps_regs_t::dpsier2](#), [rx_dps_regs_t::dpsier3](#), [rx_dps_regs_t::dpsifr0](#), [rx_dps_regs_t::dpsifr1](#), [rx_dps_regs_t::dpsifr2](#), [rx_dps_regs_t::dpsifr3](#), [internal_assemble_dpsbycr\(\)](#), [k_lpc_deep_lvd_off](#), [k_lpc_deep_ram_usb_off](#), [k_lpc_deep_ram_usb_on](#), [k_lpc_initialized](#), [k_lpc_mode_deep_software_standby](#), [k_lpc_wake_all_mask](#), [k_sbycr_ssbby](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.125.3.4 rx_lpc_enter_sleep()

```
rx_err_t rx_lpc_enter_sleep (
    void ) [nodiscard]
```

Enter Sleep low-power mode (implementation).

Enter Sleep mode (CPU stop, peripherals run).

Configures SBYCR for Sleep mode (SSBY=0) under PRCR.PRC1 protection, then issues a WAIT instruction to halt the CPU. Peripheral clocks continue running, so any enabled interrupt resumes execution at the instruction following WAIT. Records the last entered mode for post-mortem diagnostics. On the unit-test target the WAIT is skipped.

Note

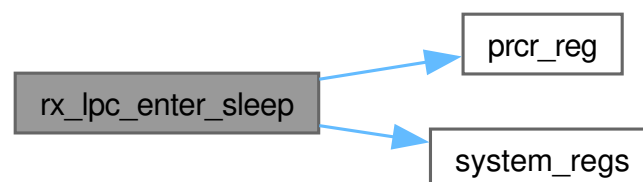
Not thread-safe. Serialize all rx_lpc_* operations.

Definition at line 338 of file [rx_lpc.c](#).

```
00339 {
00340     if (s_initialized != k_lpc_initialized) {
00341         return k_rx_err_not_initialized;
00342     }
00343
00344     #ifdef __RX__
00345     volatile uint16_t* prcr = prcr_reg();
00346     volatile rx_system_regs_t* sys = system_regs();
00347
00348     *prcr = k_rx_prcr_unlock_prc1;
00349     sys->sbycr = (uint16_t)(sys->sbycr & (uint16_t)~k_sbycr_ssb);
00350     *prcr = k_rx_prcr_lock;
00351     #endif
00352
00353     s_last_mode = k_lpc_mode_sleep;
00354
00355     #ifdef __RX__
00356     __asm__ volatile("wait");
00357     #endif
00358
00359     return k_rx_ok;
00360 }
```

References [k_lpc_initialized](#), [k_lpc_mode_sleep](#), [k_sbycr_ssb](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.125.3.5 rx`lpc`enter`software`standby()

```
rx_err_t rx_lpc_enter_software_standby (
    void ) [nodiscard]
```

Enter Software Standby mode (implementation).

Enter Software Standby mode.

Configures SBYCR.SSBY=1 and DPSBYCR.DPSBY=0 under PRCR.PRC1 so that the upcoming WAIT instruction enters Software Standby (not Sleep, not Deep Standby). Most peripheral clocks are gated off; only configured wake sources (level-sensitive IRQ pins, RTC, watchdog) can resume execution. On the unit-test target the WAIT is skipped.

Note

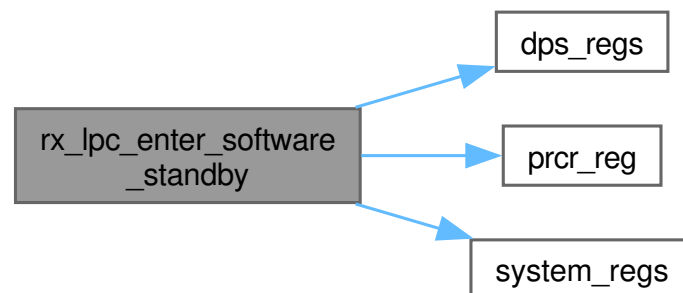
Not thread-safe. Serialize all rx_lpc_* operations.

Definition at line 374 of file rx_lpc.c.

```
00375 {
00376     if (s_initialized != k_lpc_initialized) {
00377         return k_rx_err_not_initialized;
00378     }
00379
00380     #ifdef __RX__
00381     volatile uint16_t* prcr = prcr_reg();
00382     volatile rx_system_regs_t* sys = system_regs();
00383     volatile rx_dps_regs_t* dps = dps_regs();
00384
00385     *prcr = k_rx_prcr_unlock_prc1;
00386     /* Clear DPSBY so WAIT enters software standby, not deep standby */
00387     dps->dpsbycr = (uint8_t)(dps->dpsbycr & (uint8_t)~k_dpsbycr_dpsby);
00388     /* Set SSBY so WAIT enters standby, not sleep */
00389     sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssb);
00390     *prcr = k_rx_prcr_lock;
00391     #endif
00392
00393     s_last_mode = k_lpc_mode_software_standby;
00394
00395     #ifdef __RX__
00396     __asm__ volatile("wait");
00397     #endif
00398
00399     return k_rx_ok;
00400 }
```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsbycr](#), [k_dpsbycr_dpsby](#), [k_lpc_initialized](#), [k_lpc_mode_software_standby](#), [k_sbycr_ssb](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [rx_system_regs_t::sbycr](#), and [system_regs\(\)](#).

Here is the call graph for this function:



8.125.3.6 rx_lpc_get_wake_flags()

```
rx_err_t rx_lpc_get_wake_flags (
    uint32_t * flags)  [nodiscard]
```

Read the latched Deep Software Standby wake-source bitfield (implementation).

Read the deep-standby wake flags (DPSIFR0..DPSIFR3).

Copies the cached `s_latched_wake_flags` into the caller-supplied destination. The bitfield has one bit per wake source; the layout is defined by `rx_lpc_wake_t`. Useful after wake to determine which event resumed execution.

Note

Not safe from ISR (acquires no lock but reads driver state that could be torn during init).

Definition at line 503 of file `rx_lpc.c`.

```
00504 {
00505     if (flags == NULL) {
00506         return k_rx_err_invalid_arg;
00507     }
00508     if (s_initialized != k_lpc_initialized) {
00509         return k_rx_err_not_initialized;
00510     }
00511
00512     *flags = s_latched_wake_flags;
00513     return k_rx_ok;
00514 }
```

References [k_lpc_initialized](#), [s_initialized](#), and [s_latched_wake_flags](#).

8.125.3.7 rx_lpc_init()

```
rx_err_t rx_lpc_init (
    void )  [nodiscard]
```

Initialize the Low Power Consumption driver.

Clears the driver's internal state, latches whether the most recent reset was a deep-standby wake-up (so application code can read it back via [rx_lpc_was_deep_standby_wake\(\)](#)), and pre-clears any pending deep-standby wake flags (DPSIFR0..DPSIFR3) that may have been set by the preceding deep-standby exit.

This function does not change the operating power mode; applications should call [rx_lpc_set_operating_power\(\)](#) separately if they want to switch away from the default high-speed mode.

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Driver initialized successfully
----------------------	---------------------------------

Precondition

May be called exactly once per reset
System clocks must already be configured

Postcondition

Driver state tracking is reset
Deep-standby wake flags are cleared

Note

Safe to call from single-threaded startup only.

Since

Version 1.0.0

Definition at line 242 of file [rx_lpc.c](#).

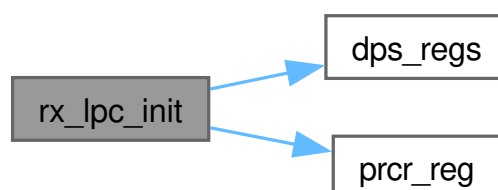
```

00243 {
00244     /* Latch whatever the boot path told us about the previous reset */
00245     #ifdef UNIT_TEST
00246         s_was_deep_standby_wake = s_inject_was_wake;
00247         s_latched_wake_flags    = s_inject_pending_flags;
00248         s_inject_was_wake       = false;
00249         s_inject_pending_flags  = 0U;
00250     #else
00251         s_was_deep_standby_wake = false;
00252         s_latched_wake_flags    = 0U;
00253     #endif
00254
00255     #ifdef __RX__
00256         /* Read DPSIFR0..DPSIFR3, OR them into the latched flags, then clear them */
00257         volatile rx_dps_regs_t* dps = dps_regs();
00258         uint32_t flags = 0U;
00259         flags |= ((uint32_t)dps->dpsifr0) « 0U;
00260         flags |= ((uint32_t)dps->dpsifr1) « 8U;
00261         flags |= ((uint32_t)dps->dpsifr2) « 16U;
00262         flags |= ((uint32_t)dps->dpsifr3) « 24U;
00263         s_latched_wake_flags |= flags;
00264
00265         /* Write 0 to each flag register to clear any pending wake bits */
00266         volatile uint16_t* prcr = prcr_reg();
00267         *prcr                    = k_rx_prcr_unlock_prc1;
00268         dps->dpsifr0              = 0U;
00269         dps->dpsifr1              = 0U;
00270         dps->dpsifr2              = 0U;
00271         dps->dpsifr3              = 0U;
00272         *prcr                    = k_rx_prcr_lock;
00273     #endif
00274
00275     s_last_mode = k_lpc_mode_none;
00276     s_initialized = k_lpc_initialized;
00277     return k_rx_ok;
00278 }

```

References [dps_regs\(\)](#), [rx_dps_regs_t::dpsifr0](#), [rx_dps_regs_t::dpsifr1](#), [rx_dps_regs_t::dpsifr2](#), [rx_dps_regs_t::dpsifr3](#), [k_lpc_initialized](#), [k_lpc_mode_none](#), [prcr_reg\(\)](#), [s_initialized](#), [s_last_mode](#), [s_latched_wake_flags](#), and [s_was_deep_standby_wake](#).

Here is the call graph for this function:



8.125.3.8 rx_lpc_set_operating_power()

```
rx_err_t rx_lpc_set_operating_power (
    rx_lpc_opcc_mode_t mode) [nodiscard]
```

Switch the MCU to a different operating-power-control mode (implementation).

Select the operating power control (OPCCR) mode.

Programs the OPCCR register with the OPCM bits derived from the requested mode (high-speed / middle-speed / low-power, etc.). Performs the standard PRCR.PRC1 unlock-write-lock dance, then bounded-polls the OPCMTSF transition-status flag for completion (max k_opcmtsf_poll_max iterations). On the unit-test target the register write is skipped.

Note

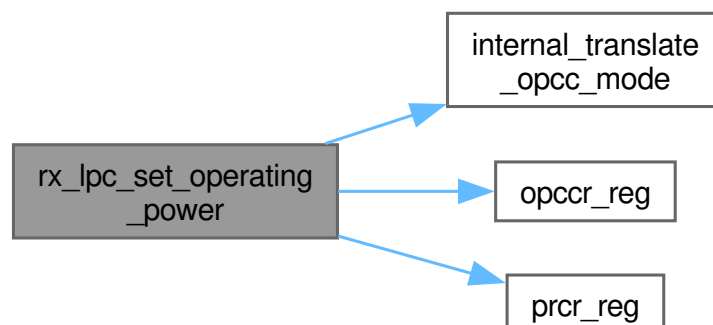
Not thread-safe. Serialize all rx_lpc_* operations.

Definition at line 293 of file rx_lpc.c.

```
00294 {
00295     if (s_initialized != k_lpc_initialized) {
00296         return k_rx_err_not_initialized;
00297     }
00298
00299     uint8_t opcm_bits = 0U;
00300     rx_err_t err = internal_translate_opcc_mode(mode, &opcm_bits);
00301     if (err != k_rx_ok) {
00302         return err;
00303     }
00304
00305     #ifdef __RX__
00306     volatile uint16_t* prcr = prcr_reg();
00307     volatile uint8_t* ocr = opccr_reg();
00308
00309     *prcr = k_rx_prcr_unlock_prc1;
00310     *ocr = opcm_bits;
00311     *prcr = k_rx_prcr_lock;
00312
00313     /* Bounded poll for OPCMTSF=0 (Manual 11.2.6, page 414) */
00314     for (uint16_t i = 0U; i < k_opcmtsf_poll_max; i++) {
00315         if ((*ocr & k_opccr_opcmtsf) == 0U) {
00316             return k_rx_ok;
00317         }
00318     }
00319     return k_rx_err_hw_timeout;
00320 #else
00321     (void)opcm_bits;
00322     return k_rx_ok;
00323 #endif
00324 }
```

References [internal_translate_opcc_mode\(\)](#), [k_lpc_initialized](#), [k_opccr_opcmtsf](#), [k_opcmtsf_poll_max](#), [opccr_reg\(\)](#), [prcr_reg\(\)](#), and [s_initialized](#).

Here is the call graph for this function:



8.125.3.9 rx`lpc`was`deep`standby`wake()

```
bool rx_lpc_was_deep_standby_wake (
    void ) [nodiscard]
```

Report whether the most recent reset was a Deep Standby wake.

Query whether the last reset was caused by a deep-standby wake.

Returns the cached `s_was_deep_standby_wake` flag latched at boot time. Allows application code to distinguish between a power-on reset and a resume from Deep Software Standby and take different paths (e.g., skip full re-initialization when waking).

Returns

true if the most recent reset was a Deep Software Standby wake; false otherwise (or before [rx_lpc_init\(\)](#) ran).

Precondition

None (function is safe to call at any time after init or before).

Postcondition

No state mutated.

Return value is stable across multiple calls in the same boot session.

Note

Safe from any context including ISR; pure read of static storage.

See also

[rx_lpc_get_wake_flags\(\)](#) Detailed wake-source bitfield.

Since

Version 1.0.0

Definition at line 486 of file [rx_lpc.c](#).

```
00487 {
00488     return s_was_deep_standby_wake;
00489 }
```

References [s_was_deep_standby_wake](#).

8.125.4 Variable Documentation

8.125.4.1 s`initialized

```
rx_lpc_init_state_t s_initialized = k_lpc_not_initialized [static]
```

Driver initialization flag.

Definition at line 112 of file [rx_lpc.c](#).

8.125.4.2 s'last'mode

```
rx_lpc_last_mode_t s_last_mode = k_lpc_mode_none [static]
```

Last low-power mode the driver attempted to enter.

Definition at line 115 of file [rx_lpc.c](#).

Referenced by [rx_lpc_enter_deep_software_standby\(\)](#), [rx_lpc_enter_sleep\(\)](#), [rx_lpc_enter_software_standby\(\)](#), and [rx_lpc_init\(\)](#).

8.125.4.3 s'latched'wake'flags

```
uint32_t s_latched_wake_flags = 0U [static]
```

Latched wake flags read from DPSIFR0..DPSIFR3 at init time.

Definition at line 121 of file [rx_lpc.c](#).

Referenced by [rx_lpc_get_wake_flags\(\)](#), and [rx_lpc_init\(\)](#).

8.125.4.4 s'was'deep'standby'wake

```
bool s_was_deep_standby_wake = false [static]
```

Latched "last reset was deep-standby wake" flag (set by [rx_lpc_init\(\)](#)).

Definition at line 118 of file [rx_lpc.c](#).

Referenced by [rx_lpc_init\(\)](#), and [rx_lpc_was_deep_standby_wake\(\)](#).

8.126 rx'lpc.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_lpc.c
00003  * @brief Low Power Consumption (LPC) HAL Driver Implementation for RX72N
00004  *
00005  * @details
00006  * Implements the Sleep, Software Standby, and Deep Software Standby mode
00007  * entry paths plus the OPCCR (Operating Power Control) selector described
00008  * in the RX72N Hardware Manual Chapter 11.
00009  *
00010  * @par Register Write Protocol
00011  * All writes to SBYCR, OPCCR, RSTCKCR, and DPSBYCR require PRC1 to be
00012  * unlocked via PRCR (key 0xA5, PRC1=1). The driver unlocks, writes, and
00013  * re-locks PRCR around every such access. PRCR writes are guarded by
00014  * `#ifdef __RX__` so host-side unit tests can exercise the driver without
00015  * hardware-mapped register access.
00016  *
00017  * @par Low-Power Entry Instruction
00018  * All three low-power modes are entered by executing the WAIT instruction
00019  * on RX. SBYCR.SSBY and DPSBYCR.DPSBY are set beforehand to select which
00020  * flavour of low power the WAIT enters:
00021  *
00022  * @verbatim
00023  * SSBY=0, DPSBY=0 -> Sleep (WAIT)
00024  * SSBY=1, DPSBY=0 -> Software Standby (WAIT)
00025  * SSBY=1, DPSBY=1 -> Deep Software Standby (WAIT)
00026  * @endverbatim
00027  */
```



```

00028 * @par NASA Power of 10 Compliance
00029 * - Rule 1: [OK] No goto/setjmp/recursion
00030 * - Rule 2: [OK] OPCMTSF poll bounded by k_opcmsf_poll_max retries
00031 * - Rule 3: [OK] Zero dynamic allocation (file-scope statics only)
00032 * - Rule 4: [OK] All functions < 60 lines
00033 * - Rule 5: [OK] >= 2 pre/post conditions per function
00034 * - Rule 6: [OK] s_*-prefixed file-scope statics, locals at narrowest scope
00035 * - Rule 7: [OK] All return values checked or explicitly cast to (void)
00036 * - Rule 8: [OK] C23 typed enums, preprocessor only for __RX__/UNIT_TEST guards
00037 * - Rule 9: [OK] Single-level pointers
00038 * - Rule 10: [OK] Compiles with -Wall -Wextra -Werror
00039 *
00040 * @see rx_lpc.h      Public API
00041 * @see rx72n_lpc_regs.h Register layout and bit definitions
00042 * @see RX72N Hardware Manual Chapter 11 - Low Power Consumption
00043 *
00044 * @author Locked, Inc.
00045 * @date 2026-04-21
00046 * @copyright Copyright (c) 2026 Locked Inc.
00047 * SPDX-License-Identifier: MIT
00048 * @since Version 1.0.0
00049 */
00050
00051 #include "rx_lpc.h"
00052
00053 #include <stddef.h>
00054
00055 #include "rx72n_lpc_regs.h"
00056
00057 #ifdef __RX__
00058 #include "rx72n_system_regs.h"
00059 #include "rx_register_protection.h"
00060 #endif
00061
00062 /*
=====
00063 * Internal Constants
00064 *
=====
00065 */
00066
00067 /**
00068 * @enum rx_lpc_init_state_t
00069 * @brief Driver initialization tracking
00070 *
00071 * @since Version 1.0.0
00072 */
00073 typedef enum : uint8_t {
00074     k_lpc_not_initialized = 0, /**< rx_lpc_init() has not been called */
00075     k_lpc_initialized = 1, /**< rx_lpc_init() completed successfully */
00076 } rx_lpc_init_state_t;
00077
00078 /**
00079 * @enum rx_lpc_sbycr_bits_t
00080 * @brief SBYCR bit definitions (RX72N Manual Section 11.2.1, page 405)
00081 *
00082 * @details
00083 * Only the two bits manipulated by this driver are declared here. Other
00084 * bits in SBYCR (OPE) are left at their reset values.
00085 *
00086 * @since Version 1.0.0
00087 */
00088 typedef enum : uint16_t {
00089     k_sbycr_ssby = 0x8000U, /**< Bit 15: Software Standby Select (0=sleep, 1=standby) */
00090 } rx_lpc_sbycr_bits_t;
00091
00092 /**
00093 * @enum rx_lpc_poll_limits_t
00094 * @brief Bounded-retry limits for register polls
00095 *
00096 * @details
00097 * OPCCR.OPCMTSF typically clears within a few hundred cycles. 1024 is a
00098 * generous ceiling that still keeps the poll bounded per NASA Rule 2.
00099 *
00100 * @since Version 1.0.0
00101 */
00102 typedef enum : uint16_t {
00103     k_opcmsf_poll_max = 1024U, /**< Max OPCMTSF poll iterations */
00104 } rx_lpc_poll_limits_t;
00105
00106 /*
=====
00107 * Module State
00108 *
=====
00109 */
00110

```

```

00111 /** @brief Driver initialization flag */
00112 static rx_lpc_init_state_t s_initialized = k_lpc_not_initialized;
00113
00114 /** @brief Last low-power mode the driver attempted to enter */
00115 static rx_lpc_last_mode_t s_last_mode = k_lpc_mode_none;
00116
00117 /** @brief Latched "last reset was deep-standby wake" flag (set by rx_lpc_init) */
00118 static bool s_was_deep_standby_wake = false;
00119
00120 /** @brief Latched wake flags read from DPSIFR0..DPSIFR3 at init time */
00121 static uint32_t s_latched_wake_flags = 0U;
00122
00123 #ifdef UNIT_TEST
00124 /** @brief Test-injected value for the next init: simulated "was deep-standby wake" */
00125 static bool s_inject_was_wake = false;
00126
00127 /** @brief Test-injected value for the next init: simulated pending DPSIFR bitmask */
00128 static uint32_t s_inject_pending_flags = 0U;
00129 #endif
00130
00131 /*
=====
00132 * Internal Helpers
00133 *
=====
00134 */
00135
00136 /**
00137  * @brief Translate OPCCR mode enum to OPCM[2:0] field value
00138  *
00139  * @details
00140  * Internal helper. The only caller (rx_lpc_set_operating_power) always
00141  * passes the address of a stack-local uint8_t, so a NULL @p opcm_bits
00142  * cannot occur from any public API path. No defensive NULL check is
00143  * therefore required (NASA Rule 5: validate at boundaries, not inside
00144  * private helpers).
00145  *
00146  *
00147  *
00148  * @pre opcm_bits != NULL (enforced by callers)
00149  * @post *opcm_bits contains one of k_opccr_opcm_* constants on success
00150  *
00151  * @since Version 1.0.0
00152  */
00153 static rx_err_t internal_translate_opcc_mode(rx_lpc_opcc_mode_t mode, uint8_t* opcm_bits)
00154 {
00155     switch (mode) {
00156     case k_lpc_opcc_high_speed:
00157         *opcm_bits = k_opccr_opcm_highspeed;
00158         return k_rx_ok;
00159     case k_lpc_opcc_low_speed_1:
00160         *opcm_bits = k_opccr_opcm_lowspeed1;
00161         return k_rx_ok;
00162     case k_lpc_opcc_low_speed_2:
00163         *opcm_bits = k_opccr_opcm_lowspeed2;
00164         return k_rx_ok;
00165     default:
00166         return k_rx_err_invalid_arg;
00167     }
00168 }
00169
00170 /**
00171  * @brief Program DPSIER0..DPSIER3 from the 32-bit wake mask
00172  *
00173  * @details
00174  * Each byte of the mask maps directly to one DPSIER register as described
00175  * in rx_lpc.h. The NMI bit (DPSIER2.DNMIE) is documented as write-once by
00176  * the hardware; this driver re-asserts it on every call and the hardware
00177  * will ignore the redundant writes after the first.
00178  *
00179  *
00180  * @pre dps_regs() returns a valid (hardware or host-mocked) pointer
00181  * @post DPSIER0..DPSIER3 reflect the enabled wake sources
00182  *
00183  * @since Version 1.0.0
00184  */
00185 /* Only used by the host-build (#else) branch of rx_lpc_enter_deep_software_standby;
00186  * on RX target the same writes are inlined directly under #ifdef __RX__. Guard
00187  * the helper definition so the cross-compile build doesn't trip
00188  * -Werror=unused-function. */
00189 #ifndef __RX__
00190 static void internal_program_wake_enables(uint32_t wake_mask)
00191 {
00192     (void)wake_mask;
00193 }
00194 #endif
00195

```

```

00196 /**
00197  * @brief Assemble DPSBYCR value for a deep-standby entry
00198  *
00199  * @details
00200  * Internal helper. The only caller (rx_lpc_enter_deep_software_standby)
00201  * validates @p deep_power against the three defined rx_lpc_deep_power_t
00202  * enumerators before calling, so the switch does not need a default
00203  * branch (NASA Rule 5: validate at boundaries, not inside private
00204  * helpers). The DPSBY bit is always set because this value is only used
00205  * on the deep-standby entry path.
00206  *
00207  *
00208  *
00209  * @pre deep_power is one of k_lpc_deep_ram_usb_on / k_lpc_deep_ram_usb_off
00210  *      / k_lpc_deep_lvd_off (enforced by caller)
00211  * @post Return value contains at most the bits DEEPCUT[1:0], IOKEEP, DPSBY
00212  *
00213  * @since Version 1.0.0
00214  */
00215 static uint8_t internal_assemble_dpsbycr(rx_lpc_deep_power_t deep_power, bool keep_io)
00216 {
00217     /* LUT keyed by rx_lpc_deep_power_t enumerator value (0..2). Using a LUT
00218      * rather than a switch eliminates the synthetic "unreachable default"
00219      * branch that gcov tracks: once deep_power is pre-validated by the
00220      * caller, a switch still produces a default fall-through branch that
00221      * no public API path can execute. The LUT has exactly one branch-free
00222      * path per input and keeps the function trivially verifiable. */
00223     static const uint8_t s_deepcut_bits[] = {
00224         [k_lpc_deep_ram_usb_on] = k_dpsbycr_deepcut_ram_usb_on,
00225         [k_lpc_deep_ram_usb_off] = k_dpsbycr_deepcut_ram_usb_off,
00226         [k_lpc_deep_lvd_off] = k_dpsbycr_deepcut_lvd_off,
00227     };
00228
00229     uint8_t value = s_deepcut_bits[deep_power];
00230     if (keep_io) {
00231         value |= k_dpsbycr_iokeep;
00232     }
00233     value |= k_dpsbycr_dpsby;
00234     return value;
00235 }
00236
00237 /**
=====
00238  * Public API
00239  *
=====
00240  */
00241
00242 rx_err_t rx_lpc_init(void)
00243 {
00244     /* Latch whatever the boot path told us about the previous reset */
00245     #ifdef UNIT_TEST
00246         s_was_deep_standby_wake = s_inject_was_wake;
00247         s_latched_wake_flags = s_inject_pending_flags;
00248         s_inject_was_wake = false;
00249         s_inject_pending_flags = 0U;
00250     #else
00251         s_was_deep_standby_wake = false;
00252         s_latched_wake_flags = 0U;
00253     #endif
00254
00255     #ifdef __RX__
00256         /* Read DPSIFR0..DPSIFR3, OR them into the latched flags, then clear them */
00257         volatile rx_dps_regs_t* dps = dps_regs();
00258         uint32_t flags = 0U;
00259         flags |= ((uint32_t)dps->dpsifr0) « 0U;
00260         flags |= ((uint32_t)dps->dpsifr1) « 8U;
00261         flags |= ((uint32_t)dps->dpsifr2) « 16U;
00262         flags |= ((uint32_t)dps->dpsifr3) « 24U;
00263         s_latched_wake_flags |= flags;
00264
00265         /* Write 0 to each flag register to clear any pending wake bits */
00266         volatile uint16_t* prcr = prcr_reg();
00267         *prcr = k_rx_prcr_unlock_prc1;
00268         dps->dpsifr0 = 0U;
00269         dps->dpsifr1 = 0U;
00270         dps->dpsifr2 = 0U;
00271         dps->dpsifr3 = 0U;
00272         *prcr = k_rx_prcr_lock;
00273     #endif
00274
00275     s_last_mode = k_lpc_mode_none;
00276     s_initialized = k_lpc_initialized;
00277     return k_rx_ok;
00278 }
00279
00280 /**

```

```

00281 * @brief Switch the MCU to a different operating-power-control mode (implementation)
00282 *
00283 * @details
00284 * Programs the OPCCR register with the OPCM bits derived from the
00285 * requested mode (high-speed / middle-speed / low-power, etc.). Performs
00286 * the standard PRCR.PRC1 unlock-write-lock dance, then bounded-polls the
00287 * OPCMTSF transition-status flag for completion (max
00288 * k_opcmtsf_poll_max iterations). On the unit-test target the register
00289 * write is skipped.
00290 *
00291 * @note Not thread-safe. Serialize all rx_lpc_* operations.
00292 */
00293 rx_err_t rx_lpc_set_operating_power(rx_lpc_opcc_mode_t mode)
00294 {
00295     if (s_initialized != k_lpc_initialized) {
00296         return k_rx_err_not_initialized;
00297     }
00298
00299     uint8_t opcm_bits = 0U;
00300     rx_err_t err = internal_translate_opcc_mode(mode, &opcm_bits);
00301     if (err != k_rx_ok) {
00302         return err;
00303     }
00304
00305 #ifdef __RX__
00306     volatile uint16_t* prcr = prcr_reg();
00307     volatile uint8_t* ocr = opccr_reg();
00308
00309     *prcr = k_rx_prcr_unlock_prc1;
00310     *ocr = opcm_bits;
00311     *prcr = k_rx_prcr_lock;
00312
00313     /* Bounded poll for OPCMTSF=0 (Manual 11.2.6, page 414) */
00314     for (uint16_t i = 0U; i < k_opcmtsf_poll_max; i++) {
00315         if ((*ocr & k_opccr_opcmtsf) == 0U) {
00316             return k_rx_ok;
00317         }
00318     }
00319     return k_rx_err_hw_timeout;
00320 #else
00321     (void)opcm_bits;
00322     return k_rx_ok;
00323 #endif
00324 }
00325
00326 /**
00327 * @brief Enter Sleep low-power mode (implementation)
00328 *
00329 * @details
00330 * Configures SBYCR for Sleep mode (SSBY=0) under PRCR.PRC1 protection,
00331 * then issues a WAIT instruction to halt the CPU. Peripheral clocks
00332 * continue running, so any enabled interrupt resumes execution at the
00333 * instruction following WAIT. Records the last entered mode for
00334 * post-mortem diagnostics. On the unit-test target the WAIT is skipped.
00335 *
00336 * @note Not thread-safe. Serialize all rx_lpc_* operations.
00337 */
00338 rx_err_t rx_lpc_enter_sleep(void)
00339 {
00340     if (s_initialized != k_lpc_initialized) {
00341         return k_rx_err_not_initialized;
00342     }
00343
00344 #ifdef __RX__
00345     volatile uint16_t* prcr = prcr_reg();
00346     volatile rx_system_regs_t* sys = system_regs();
00347
00348     *prcr = k_rx_prcr_unlock_prc1;
00349     sys->sbycr = (uint16_t)(sys->sbycr & (uint16_t)~k_sbycr_ssbby);
00350     *prcr = k_rx_prcr_lock;
00351 #endif
00352
00353     s_last_mode = k_lpc_mode_sleep;
00354
00355 #ifdef __RX__
00356     __asm__ volatile("wait");
00357 #endif
00358     return k_rx_ok;
00359 }
00360
00361 /**
00362 * @brief Enter Software Standby mode (implementation)
00363 *
00364 * @details
00365 * Configures SBYCR.SSBY=1 and DPSBYCR.DPSBY=0 under PRCR.PRC1 so that the
00366 * upcoming WAIT instruction enters Software Standby (not Sleep, not Deep

```

```

00368 * Standby). Most peripheral clocks are gated off; only configured wake
00369 * sources (level-sensitive IRQ pins, RTC, watchdog) can resume execution.
00370 * On the unit-test target the WAIT is skipped.
00371 *
00372 * @note Not thread-safe. Serialize all rx_lpc_* operations.
00373 */
00374 rx_err_t rx_lpc_enter_software_standby(void)
00375 {
00376     if (s_initialized != k_lpc_initialized) {
00377         return k_rx_err_not_initialized;
00378     }
00379
00380 #ifdef __RX__
00381     volatile uint16_t* prcr = prcr_reg();
00382     volatile rx_system_regs_t* sys = system_regs();
00383     volatile rx_dps_regs_t* dps = dps_regs();
00384
00385     *prcr = k_rx_prcr_unlock_prc1;
00386     /* Clear DPSBY so WAIT enters software standby, not deep standby */
00387     dps->dpsbycr = (uint8_t)(dps->dpsbycr & (uint8_t)~k_dpsbycr_dpsby);
00388     /* Set SSBY so WAIT enters standby, not sleep */
00389     sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssb);
00390     *prcr = k_rx_prcr_lock;
00391 #endif
00392
00393     s_last_mode = k_lpc_mode_software_standby;
00394
00395 #ifdef __RX__
00396     __asm__ volatile("wait");
00397 #endif
00398
00399     return k_rx_ok;
00400 }
00401
00402 rx_err_t
00403 rx_lpc_enter_deep_software_standby(uint32_t wake_mask, rx_lpc_deep_power_t deep_power, bool keep_io)
00404 {
00405     if (s_initialized != k_lpc_initialized) {
00406         return k_rx_err_not_initialized;
00407     }
00408     if (wake_mask == 0U) {
00409         return k_rx_err_invalid_arg;
00410     }
00411     if ((wake_mask & ~k_lpc_wake_all_mask) != 0U) {
00412         return k_rx_err_invalid_arg;
00413     }
00414     if ((deep_power != k_lpc_deep_ram_usb_on) && (deep_power != k_lpc_deep_ram_usb_off) &&
00415         (deep_power != k_lpc_deep_lvd_off)) {
00416         return k_rx_err_invalid_arg;
00417     }
00418
00419     const uint8_t dpsbycr_value = internal_assemble_dpsbycr(deep_power, keep_io);
00420
00421 #ifdef __RX__
00422     volatile uint16_t* prcr = prcr_reg();
00423     volatile rx_system_regs_t* sys = system_regs();
00424     volatile rx_dps_regs_t* dps = dps_regs();
00425
00426     *prcr = k_rx_prcr_unlock_prc1;
00427
00428     /* 1. Enable wake sources */
00429     dps->dpsier0 = (uint8_t)((wake_mask » 0U) & 0xFFU);
00430     dps->dpsier1 = (uint8_t)((wake_mask » 8U) & 0xFFU);
00431     dps->dpsier2 = (uint8_t)((wake_mask » 16U) & 0xFFU);
00432     dps->dpsier3 = (uint8_t)((wake_mask » 24U) & 0xFFU);
00433
00434     /* 2. Clear any stale wake flags */
00435     dps->dpsifr0 = 0U;
00436     dps->dpsifr1 = 0U;
00437     dps->dpsifr2 = 0U;
00438     dps->dpsifr3 = 0U;
00439
00440     /* 3. Program DPSBYCR (DEEPCUT, IOKEEP, DPSBY=1) */
00441     dps->dpsbycr = dpsbycr_value;
00442
00443     /* 4. Set SBYCR.SSBY so WAIT enters standby-family */
00444     sys->sbycr = (uint16_t)(sys->sbycr | k_sbycr_ssb);
00445
00446     *prcr = k_rx_prcr_lock;
00447 #else
00448     (void)dpsbycr_value;
00449     internal_program_wake_enables(wake_mask);
00450 #endif
00451
00452     s_last_mode = k_lpc_mode_deep_software_standby;
00453
00454 #ifdef __RX__

```

```

00455     __asm__ volatile("wait");
00456     /* On real hardware, execution never reaches here - wake restarts at reset */
00457 #endif
00458
00459     return k_rx_ok;
00460 }
00461
00462 /**
00463  * @brief Report whether the most recent reset was a Deep Standby wake
00464  *
00465  * @details
00466  * Returns the cached s_was_deep_standby_wake flag latched at boot time.
00467  * Allows application code to distinguish between a power-on reset and a
00468  * resume from Deep Software Standby and take different paths
00469  * (e.g., skip full re-initialization when waking).
00470  *
00471  * @return true if the most recent reset was a Deep Software Standby wake;
00472  *         false otherwise (or before rx_lpc_init() ran).
00473  *
00474  * @pre None (function is safe to call at any time after init or before).
00475  *
00476  * @post No state mutated.
00477  * @post Return value is stable across multiple calls in the same boot
00478  *       session.
00479  *
00480  * @note Safe from any context including ISR; pure read of static storage.
00481  *
00482  * @see rx_lpc_get_wake_flags() Detailed wake-source bitfield.
00483  *
00484  * @since Version 1.0.0
00485  */
00486 bool rx_lpc_was_deep_standby_wake(void)
00487 {
00488     return s_was_deep_standby_wake;
00489 }
00490
00491 /**
00492  * @brief Read the latched Deep Software Standby wake-source bitfield (implementation)
00493  *
00494  * @details
00495  * Copies the cached s_latched_wake_flags into the caller-supplied
00496  * destination. The bitfield has one bit per wake source; the layout is
00497  * defined by rx_lpc_wake_t. Useful after wake to determine which event
00498  * resumed execution.
00499  *
00500  * @note Not safe from ISR (acquires no lock but reads driver state that
00501  *       could be torn during init).
00502  */
00503 rx_err_t rx_lpc_get_wake_flags(uint32_t* flags)
00504 {
00505     if (flags == NULL) {
00506         return k_rx_err_invalid_arg;
00507     }
00508     if (s_initialized != k_lpc_initialized) {
00509         return k_rx_err_not_initialized;
00510     }
00511
00512     *flags = s_latched_wake_flags;
00513     return k_rx_ok;
00514 }
00515
00516 /*
00517  * Test-Only Helpers
00518  */
00519
00520 #ifdef UNIT_TEST
00521 void rx_lpc_test_reset(void)
00522 {
00523     s_initialized = k_lpc_not_initialized;
00524     s_last_mode = k_lpc_mode_none;
00525     s_was_deep_standby_wake = false;
00526     s_latched_wake_flags = 0U;
00527     s_inject_was_wake = false;
00528     s_inject_pending_flags = 0U;
00529 }
00530
00531 rx_lpc_last_mode_t rx_lpc_test_get_last_mode(void)
00532 {
00533     return s_last_mode;
00534 }
00535
00536 void rx_lpc_test_set_deep_standby_wake(bool was_wake)
00537 {
00538     s_inject_was_wake = was_wake;
00539 }

```

```

00540 }
00541
00542 void rx_lpc_test_set_pending_wake_flags(uint32_t flags)
00543 {
00544     s_inject_pending_flags = flags;
00545 }
00546 #endif /* UNIT_TEST */

```

8.127 libs/rx`hal/src/rx`mpc.c File Reference

Multi-Function Pin Controller (MPC) Driver Implementation.

```

#include "rx_mpc.h"
#include <stddef.h>
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_port_constants.h"
#include "rx_port_utils.h"

```

Include dependency graph for rx_mpc.c:



Enumerations

- enum `mpc_pin_constants_t` : `uint8_t` { `k_mpc_min_pin` = 0 , `k_mpc_max_pin` = `k_rx_pin_max` }
- Pin number validation constants for MPC driver.
- enum `mpc_pfs_offset_t` : `uint8_t` { `k_mpc_pfs_base_offset` }
- PFS register base offset from MPC structure start.
- enum `mpc_port_number_t` : `uint8_t` {
`k_mpc_port_start` = `k_rx_port_0` , `k_mpc_port_0` = `k_rx_port_0` , `k_mpc_port_1` = `k_rx_port_1` , `k_mpc_port_2` = `k_rx_port_2` ,
`k_mpc_port_3` = `k_rx_port_3` , `k_mpc_port_4` = `k_rx_port_4` , `k_mpc_port_5` = `k_rx_port_5` , `k_mpc_port_6` = `k_rx_port_6` ,
`k_mpc_port_7` = `k_rx_port_7` , `k_mpc_port_8` = `k_rx_port_8` , `k_mpc_port_9` = `k_rx_port_9` , `k_mpc_port_a` = `k_rx_port_a` ,
`k_mpc_port_b` = `k_rx_port_b` , `k_mpc_port_c` = `k_rx_port_c` , `k_mpc_port_d` = `k_rx_port_d` , `k_mpc_port_e` = `k_rx_port_e` ,
`k_mpc_port_f` = `k_rx_port_f` , `k_mpc_port_g` = `k_rx_port_g` , `k_mpc_port_j` = `k_rx_port_j` }
- Port number aliases for MPC driver validation.
- enum `mpc_port_offset_t` : `uint16_t` {
`k_mpc_port0_offset` = 0 , `k_mpc_port1_offset` = 8 , `k_mpc_port2_offset` = 16 ,
`k_mpc_port3_offset` = 24 ,
`k_mpc_port4_offset` = 32 , `k_mpc_port5_offset` = 40 , `k_mpc_port6_offset` = 48 ,
`k_mpc_port7_offset` = 56 ,
`k_mpc_port8_offset` = 64 , `k_mpc_port9_offset` = 72 , `k_mpc_porta_offset` = 80 ,
`k_mpc_portb_offset` = 88 ,
`k_mpc_portc_offset` = 96 , `k_mpc_portd_offset` = 104 , `k_mpc_porte_offset` = 112 ,
`k_mpc_portf_offset` = 120 ,
`k_mpc_portg_offset` = 128 , `k_mpc_portj_offset` = 144 }
- PFS register byte offsets for each port.
- enum `mpc_psel_constants_t` : `uint8_t` { `k_mpc_psel_max` = 31 }

- PSEL field validation constants.
- enum `mpc_pfs_gpio_t` : `uint8_t` { `k_mpc_pfs_gpio` = 0 }
PFS register value for GPIO mode.
- enum `mpc_pwpr_t` : `uint8_t` { `k_mpc_pwpr_clear` = 0 }
PWPR register control values.
- enum `mpc_port_offset_sentinel_t` : `uint16_t` { `k_mpc_port_offset_invalid` = 0xFFFFU }
Calculate PFS register address for a given port and pin.

Functions

- static `uint16_t` `internal_mpc_port_offset` (`const uint8_t` port)
Resolve a port enum to its PFS-register offset.
- static volatile `uint8_t` * `internal_get_pfs_register` (`const uint8_t` port, `uint8_t` pin)
- static void `internal_mpc_unlock` (void)
Unlock MPC PFS register write protection.
- static void `internal_mpc_lock` (void)
Lock MPC PFS register write protection.
- static `rx_err_t` `internal_write_pfs` (`const uint8_t` port, `uint8_t` pin, `uint8_t` value)
Write value to PFS register with automatic unlock/lock.
- `rx_err_t` `rx_mpc_set_gpio` (`const rx_port_pin_t` pin)
Configure pin for GPIO (General Purpose I/O) function.
- `rx_err_t` `rx_mpc_set_peripheral` (`const rx_mpc_peripheral_config_t` *config)
Configure pin for peripheral function with specified PSEL code.
- `rx_err_t` `rx_mpc_set_mtu_pwm` (`const rx_port_pin_t` pin)
Configure pin for MTU3a timer PWM output (MTIOC).
- `rx_err_t` `rx_mpc_set_mtu_encoder` (`const rx_port_pin_t` pin)
Configure pin for MTU3a encoder/phase counter input (MTCLK).
- `rx_err_t` `rx_mpc_set_tpu_encoder` (`const rx_port_pin_t` pin)
Configure pin for TPU encoder/phase counter input (TCLK).
- `rx_err_t` `rx_mpc_set_adc` (`const rx_port_pin_t` pin)
Configure pin for ADC analog input.
- `rx_err_t` `rx_mpc_set_sci` (`const rx_port_pin_t` pin)
Configure pin for SCI (Serial Communication Interface) UART function.
- `rx_err_t` `rx_mpc_set_riic` (`const rx_port_pin_t` pin)
Configure pin for RIIC (I2C) bus function.
- `rx_err_t` `rx_mpc_set_rspi` (`const rx_port_pin_t` pin)
Configure pin for RSPI (SPI) bus function.
- `rx_err_t` `rx_mpc_set_gptw` (`const rx_port_pin_t` pin)
Configure pin for GPTW complementary PWM output.
- `rx_err_t` `rx_mpc_set_usb_vbus` (`const rx_port_pin_t` pin)
Configure pin for USB VBUS detection.
- `rx_err_t` `rx_mpc_set_irq` (`const rx_port_pin_t` pin)
Configure pin for IRQ (external interrupt) function.

Variables

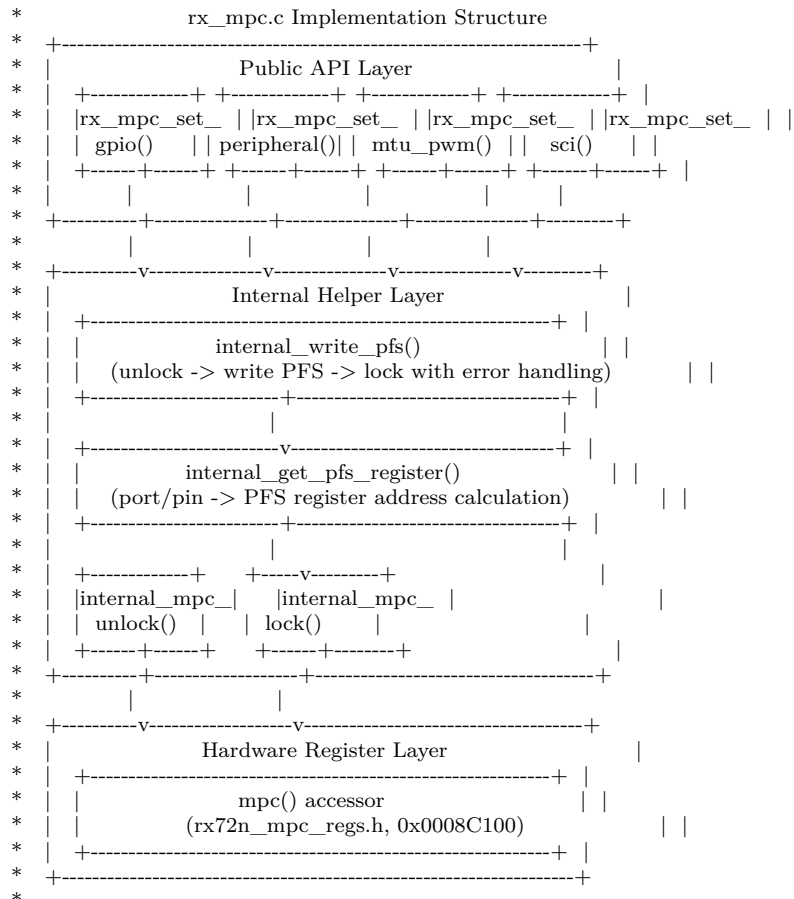
- static const char *const `s_tag` = "MPC"
Logging tag for MPC module.

8.127.1 Detailed Description

Multi-Function Pin Controller (MPC) Driver Implementation.

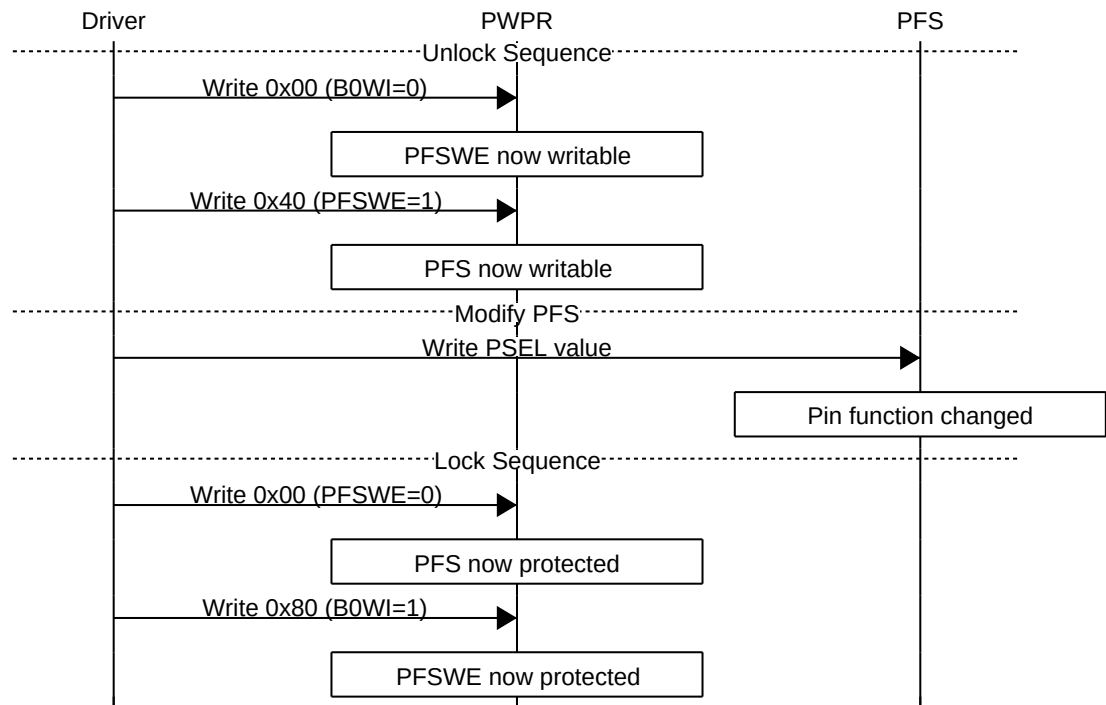
Implementation of the MPC driver for RX72N pin multiplexer configuration. This module provides the core functionality for connecting GPIO pins to peripheral functions through the PFS (Pin Function Select) registers.

Implementation Architecture



PWPR Write Protection Sequence

The PFS registers are protected by a two-stage lock. The unlock/lock sequence must be followed exactly:



PFS Register Address Calculation

PFS registers are laid out contiguously in memory with 8 bytes per port:

```

* PFS Base = MPC Base + 0x40 = 0x0008C140
*
* Address = PFS_Base + (port_offset) + pin_number
*
* Port Offsets (8 bytes each):
* Port 0: +0   Port 1: +8   Port 2: +16  Port 3: +24
* Port 4: +32  Port 5: +40  Port 6: +48  Port 7: +56
* Port 8: +64  Port 9: +72  Port A: +80  Port B: +88
* Port C: +96  Port D: +104 Port E: +112 Port J: +144
*
* Example: PB7 (Port B, Pin 7)
* Address = 0x0008C140 + 88 + 7 = 0x0008C19F
*

```

Performance Characteristics

Function	Typical Time	Stack Usage	Notes
rx_mpc_set_gpio()	1.2 us	24 bytes	Direct PFS write
rx_mpc_set_peripheral()	1.5 us	32 bytes	With validation
rx_mpc_set_mtu_pwm()	1.6 us	40 bytes	Calls set_peripheral
Bulk config (10 pins)	15 us	-	Sequential calls

Memory Usage

Category	Size	Description
Code (.text)	~800 bytes	All functions
Constants (.rodata)	4 bytes	Tag string "MPC"
Static data (.data)	0 bytes	None
Stack (max)	40 bytes	Deepest call chain

Thread Safety Analysis

NOT THREAD-SAFE. Critical section analysis:

- [internal_mpc_unlock\(\)](#) -> [internal_write_pfs\(\)](#) -> [internal_mpc_lock\(\)](#)
- This sequence is non-atomic (3 register writes)
- Interrupt or thread switch during unlock leaves PFS unprotected
- Race condition: Thread A unlocks, Thread B unlocks, Thread A writes, Thread B writes (corrupted configuration possible)

Recommended Usage Pattern

```
// Option 1: Configure all pins during single-threaded init
void hardware_init(void) // Called before RTOS starts
{
    rx_mpc_set_peripheral(&pwm_config);
    rx_mpc_set_sci(k_port_b_pin_7);
    // ... all MPC configuration here
}

// Option 2: Use mutex if runtime reconfiguration needed
void configure_pin_safe(const rx_mpc_peripheral_config_t* config)
{
    tx_mutex_get(&mpc_mutex, TX_WAIT_FOREVER);
    rx_mpc_set_peripheral(config);
    tx_mutex_put(&mpc_mutex);
}
```

Error Handling Strategy

- All public functions validate inputs before hardware access
- Errors logged via `rx_log_error()` with descriptive messages
- No partial configuration: if validation fails, no registers modified
- Lock always restored even on error path (no dangling unlock)

NASA Power of 10 Compliance

- Rule 1: [OK] No goto, setjmp, longjmp, recursion
- Rule 2: [OK] Fixed loop bounds (switch/case only, no iteration)
- Rule 3: [OK] No dynamic memory allocation
- Rule 4: [OK] All functions < 60 lines
- Rule 5: [OK] Assertions via `RX_CHECK_*` macros (2+ per function)
- Rule 6: [OK] Variables at smallest scope (function-local)
- Rule 7: [OK] All return values checked and propagated
- Rule 8: [OK] C23 typed enums for all constants
- Rule 9: [WARN] No function pointers (direct calls only)
- Rule 10: [OK] Compiles with `-Wall -Wextra -Werror`

SOLID Principles

- Single Responsibility: Each internal function has one purpose
- Open/Closed: New peripherals added via `set_peripheral()`
- Dependency Inversion: Uses [rx72n_mpc_regs.h](#) abstraction

See also

[rx_mpc.h](#) Public API documentation

[rx72n_mpc_regs.h](#) Hardware register definitions

`docs/sections/03_hardware_pinout.tex` Pin assignment documentation

Author

Locked, Inc.

Date

2026-01-29

Copyright

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Since

Version 1.0.0

Definition in file [rx_mpc.c](#).

8.127.2 Enumeration Type Documentation

8.127.2.1 `mpc_pfs_gpio_t`

enum [mpc_pfs_gpio_t](#) : `uint8_t`

PFS register value for GPIO mode.

GPIO mode requires all bits in PFS register to be 0:

- `PSEL[4:0] = 0` (no peripheral function)
- `ISEL = 0` (no interrupt input)
- `ASEL = 0` (digital mode, not analog)

See also

[rx_mpc_set_gpio\(\)](#) Uses this value

Since

Version 1.0.0

Enumerator

k_mpc_pfs_gpio	GPIO mode: all zeros (PSEL=0, ISEL=0, ASEL=0)
----------------	---

Definition at line 357 of file rx_mpc.c.

```
00357      : uint8_t {
00358  k_mpc_pfs_gpio = 0, /**< GPIO mode: all zeros (PSEL=0, ISEL=0, ASEL=0) */
00359 } mpc_pfs_gpio_t;
```

8.127.2.2 mpc`pfs`offset`

```
enum mpc_pfs_offset_t : uint8_t
```

PFS register base offset from MPC structure start.

The PFS registers start at offset 0x40 from MPC base (0x0008C100), which is address 0x0008C140. However, when accessing via the rx_mpc_regs_t struct, we need to account for the structure layout.

This offset is used with pointer arithmetic to calculate PFS register addresses from the mpc() accessor.

Calculation

$$\text{PFS_addr} = (\text{uint8_t}*)\text{mpc}() + \text{k_mpc_pfs_base_offset} + \text{port_offset} + \text{pin}$$

Since

Version 1.0.0

Enumerator

k_mpc_pfs_base_offset	Byte offset from mpc() to P00PFS register (0x40: P00PFS=0x0008C140, base=0x0008C100)
-----------------------	--

Definition at line 232 of file rx_mpc.c.

```
00232      : uint8_t {
00233  k_mpc_pfs_base_offset =
00234    64, /**< Byte offset from mpc() to P00PFS register (0x40: P00PFS=0x0008C140, base=0x0008C100) */
00235 } mpc_pfs_offset_t;
```

8.127.2.3 mpc`pin`constants`

```
enum mpc_pin_constants_t : uint8_t
```

Pin number validation constants for MPC driver.

Defines the valid range for pin numbers within a port. RX72N GPIO ports support up to 8 pins per port (pins 0-7), though not all pins are available on all ports or packages.

See also

[rx_port_constants.h](#) Source of `k_rx_pin_max` definition

Since

Version 1.0.0

Enumerator

<code>k_mpc_min_pin</code>	Minimum valid pin number (always 0)
<code>k_mpc_max_pin</code>	Maximum valid pin number (7 for 8 pins per port)

Definition at line 210 of file [rx_mpc.c](#).

```
00210     : uint8_t {
00211   k_mpc_min_pin = 0,           /**< Minimum valid pin number (always 0) */
00212   k_mpc_max_pin = k_rx_pin_max, /**< Maximum valid pin number (7 for 8 pins per port) */
00213 } mpc_pin_constants_t;
```

8.127.2.4 mpc_port_number_t

```
enum mpc\_port\_number\_t : uint8_t
```

Port number aliases for MPC driver validation.

Maps the generic `rx_port_*` constants to MPC-specific names for use in the switch statement in [internal_get_pfs_register\(\)](#). This provides compile-time verification that all ports are handled.

144-pin LFQFP Package Availability

Port	Available	Notes
0-5	Partial	Some pins available
6-9	Partial	Some pins available (P67, P86/P87 on 144-pin)
A-E	Full	All 8 pins per port
F	Partial	PF0-PF2, PF5 available on 144-pin
G-H	None	Not on 144-pin package
J	Partial	PJ3, PJ5 available on 144-pin

See also

[rx_port_constants.h](#) Source definitions

Since

Version 1.0.0

Enumerator

k_mpc_port_start	First valid port number for range check
k_mpc_port_0	Port 0 (GPIO P00-P07)
k_mpc_port_1	Port 1 (GPIO P10-P17)
k_mpc_port_2	Port 2 (GPIO P20-P27, encoder inputs)
k_mpc_port_3	Port 3 (GPIO P30-P34, P35=NMI)
k_mpc_port_4	Port 4 (GPIO P40-P47, ADC inputs)
k_mpc_port_5	Port 5 (GPIO P50-P57)
k_mpc_port_6	Port 6 (GPIO P60-P67)
k_mpc_port_7	Port 7 (GPIO P71-P77, P70 N/A)
k_mpc_port_8	Port 8 (GPIO P80-P87)
k_mpc_port_9	Port 9 (GPIO P90-P97)
k_mpc_port_a	Port A (GPIO PA0-PA7, SPI pins)
k_mpc_port_b	Port B (GPIO PB0-PB7, UART pins)
k_mpc_port_c	Port C (GPIO PC0-PC7, encoder inputs)
k_mpc_port_d	Port D (GPIO PD0-PD7)
k_mpc_port_e	Port E (GPIO PE0-PE7, PWM outputs)
k_mpc_port_f	Port F (partial on 144-pin: PF0-PF2, PF5)
k_mpc_port_g	Port G (not on 144-pin package)
k_mpc_port_j	Port J (partial: PJ3, PJ5 on 144-pin)

Definition at line 259 of file rx_mpc.c.

```

00259      : uint8_t {
00260  k_mpc_port_start = k_rx_port_0, /**< First valid port number for range check */
00261  k_mpc_port_0    = k_rx_port_0, /**< Port 0 (GPIO P00-P07) */
00262  k_mpc_port_1    = k_rx_port_1, /**< Port 1 (GPIO P10-P17) */
00263  k_mpc_port_2    = k_rx_port_2, /**< Port 2 (GPIO P20-P27, encoder inputs) */
00264  k_mpc_port_3    = k_rx_port_3, /**< Port 3 (GPIO P30-P34, P35=NMI) */
00265  k_mpc_port_4    = k_rx_port_4, /**< Port 4 (GPIO P40-P47, ADC inputs) */
00266  k_mpc_port_5    = k_rx_port_5, /**< Port 5 (GPIO P50-P57) */
00267  k_mpc_port_6    = k_rx_port_6, /**< Port 6 (GPIO P60-P67) */

```

```

00268 k_mpc_port_7    = k_rx_port_7, /**< Port 7 (GPIO P71-P77, P70 N/A) */
00269 k_mpc_port_8    = k_rx_port_8, /**< Port 8 (GPIO P80-P87) */
00270 k_mpc_port_9    = k_rx_port_9, /**< Port 9 (GPIO P90-P97) */
00271 k_mpc_port_a    = k_rx_port_a, /**< Port A (GPIO PA0-PA7, SPI pins) */
00272 k_mpc_port_b    = k_rx_port_b, /**< Port B (GPIO PB0-PB7, UART pins) */
00273 k_mpc_port_c    = k_rx_port_c, /**< Port C (GPIO PC0-PC7, encoder inputs) */
00274 k_mpc_port_d    = k_rx_port_d, /**< Port D (GPIO PD0-PD7) */
00275 k_mpc_port_e    = k_rx_port_e, /**< Port E (GPIO PE0-PE7, PWM outputs) */
00276 k_mpc_port_f    = k_rx_port_f, /**< Port F (partial on 144-pin: PF0-PF2, PF5) */
00277 k_mpc_port_g    = k_rx_port_g, /**< Port G (not on 144-pin package) */
00278 k_mpc_port_j    = k_rx_port_j, /**< Port J (partial: PJ3, PJ5 on 144-pin) */
00279 } mpc_port_number_t;

```

8.127.2.5 mpc_port_offset_sentinel_t

```
enum mpc_port_offset_sentinel_t : uint16_t
```

Calculate PFS register address for a given port and pin.

Computes the memory address of the Pin Function Select (PFS) register for the specified port and pin combination. Uses a lookup table (switch) to get the port offset, then adds the pin number.

Algorithm Steps

1. Validate pin number is in range [0, 7]
2. Get base address of PFS registers from `mpc()` accessor
3. Look up port offset in switch statement
4. If port invalid, log error and return `nullptr`
5. Return `base + offset + pin`

Address Calculation

```

* PFS_addr = (mpc() + k_mpc_pfs_base_offset) + port_offset + pin
*
* Example: PB7 (Port B, Pin 7)
* = (0x0008C100 + 64) + 88 + 7
* = 0x0008C140 + 88 + 7
* = 0x0008C19F (PB7PFS)
*

```

Precondition

PCLKB clock must be running for MPC register access

Caller has verified channel is in valid range for the target package

Postcondition

No registers modified (read-only address calculation)

Note

Internal function - not exported in header

Port G is explicitly rejected (returns `NULL`) since it is absent on the 144-pin LFQFP package

Port H has no named constant in `rx_port_constants.h`; a raw value in that range falls to the default case and returns `NULL`

Warning

Returned pointer is volatile - must dereference with care
 Caller must still unlock PWPR before writing to the register

Performance

- Execution time: ~200 ns @ 240 MHz (switch lookup + pointer math)
- Stack usage: 8 bytes (local variables)

See also

[mpc\(\)](#) MPC register accessor
[mpc_port_offset_t](#) Port offset constants

Since

Version 1.0.0

Sentinel value indicating a port that is not pin-muxable

Used in the `s_port_pfs_offsets` table to flag ports that are unavailable on the 144-pin LFQFP package (port G, port H). Looking up such a port returns `nullptr` from [internal_get_pfs_register\(\)](#).

Since

Version 1.0.0

Enumerator

<code>k_mpc_port_offset_invalid</code>	Port not available on this package.
--	-------------------------------------

Definition at line 440 of file [rx_mpc.c](#).

```
00440      : uint16_t {
00441  k_mpc_port_offset_invalid = 0xFFFFU, /**< @brief Port not available on this package */
00442 } mpc_port_offset_sentinel_t;
```

8.127.2.6 mpc`port`offset`t

enum [mpc_port_offset_t](#) : `uint16_t`

PFS register byte offsets for each port.

PFS registers are laid out contiguously in memory with 8 bytes allocated per port (one byte per pin), even if not all pins are physically present. These offsets are added to the PFS base address to reach each port's registers.

Memory Layout

```

*   PFS Base (0x0008C140):
*   +0x00: P00PFS - P07PFS (Port 0)
*   +0x08: P10PFS - P17PFS (Port 1)
*   +0x10: P20PFS - P27PFS (Port 2)
*   ...
*   +0x50: PA0PFS - PA7PFS (Port A)
*   +0x58: PB0PFS - PB7PFS (Port B)
*   ...
*   +0x90: PJ0PFS - PJ7PFS (Port J)
*

```

Note

Each port occupies 8 bytes regardless of actual pin count

See also

[internal_get_pfs_register\(\)](#) Uses these offsets

Since

Version 1.0.0

Enumerator

k_mpc_port0_offset	Port 0 offset: 0x00 (P00-P07)
k_mpc_port1_offset	Port 1 offset: 0x08 (P10-P17)
k_mpc_port2_offset	Port 2 offset: 0x10 (P20-P27)
k_mpc_port3_offset	Port 3 offset: 0x18 (P30-P34)
k_mpc_port4_offset	Port 4 offset: 0x20 (P40-P47, ADC)
k_mpc_port5_offset	Port 5 offset: 0x28 (P50-P57)
k_mpc_port6_offset	Port 6 offset: 0x30 (P60-P67)
k_mpc_port7_offset	Port 7 offset: 0x38 (P71-P77)
k_mpc_port8_offset	Port 8 offset: 0x40 (P80-P87)
k_mpc_port9_offset	Port 9 offset: 0x48 (P90-P97)
k_mpc_porta_offset	Port A offset: 0x50 (PA0-PA7, SPI)
k_mpc_portb_offset	Port B offset: 0x58 (PB0-PB7, UART)
k_mpc_portc_offset	Port C offset: 0x60 (PC0-PC7, MTU)
k_mpc_portd_offset	Port D offset: 0x68 (PD0-PD7)
k_mpc_porte_offset	Port E offset: 0x70 (PE0-PE7, PWM)
k_mpc_portf_offset	Port F offset: 0x78 (partial on 144-pin)
k_mpc_portg_offset	Port G offset: 0x80 (not on 144-pin)
k_mpc_portj_offset	Port J offset: 0x90 (PJ3, PJ5 on 144-pin)

Definition at line 308 of file [rx_mpc.c](#).

```

00308     : uint16_t {
00309 k_mpc_port0_offset = 0,  /**< Port 0 offset: 0x00 (P00-P07) */
00310 k_mpc_port1_offset = 8,  /**< Port 1 offset: 0x08 (P10-P17) */
00311 k_mpc_port2_offset = 16, /**< Port 2 offset: 0x10 (P20-P27) */

```

```

00312 k_mpc_port3_offset = 24, /**< Port 3 offset: 0x18 (P30-P34) */
00313 k_mpc_port4_offset = 32, /**< Port 4 offset: 0x20 (P40-P47, ADC) */
00314 k_mpc_port5_offset = 40, /**< Port 5 offset: 0x28 (P50-P57) */
00315 k_mpc_port6_offset = 48, /**< Port 6 offset: 0x30 (P60-P67) */
00316 k_mpc_port7_offset = 56, /**< Port 7 offset: 0x38 (P71-P77) */
00317 k_mpc_port8_offset = 64, /**< Port 8 offset: 0x40 (P80-P87) */
00318 k_mpc_port9_offset = 72, /**< Port 9 offset: 0x48 (P90-P97) */
00319 k_mpc_porta_offset = 80, /**< Port A offset: 0x50 (PA0-PA7, SPI) */
00320 k_mpc_portb_offset = 88, /**< Port B offset: 0x58 (PB0-PB7, UART) */
00321 k_mpc_portc_offset = 96, /**< Port C offset: 0x60 (PC0-PC7, MTU) */
00322 k_mpc_portd_offset = 104, /**< Port D offset: 0x68 (PD0-PD7) */
00323 k_mpc_porte_offset = 112, /**< Port E offset: 0x70 (PE0-PE7, PWM) */
00324 k_mpc_portf_offset = 120, /**< Port F offset: 0x78 (partial on 144-pin) */
00325 k_mpc_portg_offset = 128, /**< Port G offset: 0x80 (not on 144-pin) */
00326 k_mpc_portj_offset = 144, /**< Port J offset: 0x90 (PJ3, PJ5 on 144-pin) */
00327 } mpc_port_offset_t;

```

8.127.2.7 mpc`psel`constants`t

```
enum mpc_psel_constants_t : uint8_t
```

PSEL field validation constants.

The PSEL field in PFS registers is 5 bits wide, allowing values 0-31. This constant defines the maximum valid value for input validation.

See also

[rx_mpc_set_peripheral\(\)](#) Validates PSEL against this maximum

Since

Version 1.0.0

Enumerator

k_mpc_psel_max	Maximum PSEL value (5 bits = 0x1F)
----------------	------------------------------------

Definition at line 340 of file [rx_mpc.c](#).

```

00340         : uint8_t {
00341 k_mpc_psel_max = 31, /**< Maximum PSEL value (5 bits = 0x1F) */
00342 } mpc_psel_constants_t;

```

8.127.2.8 mpc`pwpr`t

```
enum mpc_pwpr_t : uint8_t
```

PWPR register control values.

Values used to control the PWPR (Write Protect Register) during the unlock/lock sequence. Clearing PWPR is the first step in both sequences.

See also

[internal_mpc_unlock\(\)](#) Uses [k_mpc_pwpr_clear](#) then [k_mpc_pwpr_pfswe](#)

[internal_mpc_lock\(\)](#) Uses [k_mpc_pwpr_clear](#) then [k_mpc_pwpr_b0wi](#)

Since

Version 1.0.0

Enumerator

k_mpc_pwpr_clear	Clear all PWPR bits (first step of unlock/lock)
------------------	---

Definition at line 373 of file [rx_mpc.c](#).

```
00373      : uint8_t {
00374  k_mpc_pwpr_clear = 0, /**< Clear all PWPR bits (first step of unlock/lock) */
00375 } mpc_pwpr_t;
```

8.127.3 Function Documentation

8.127.3.1 internal_get_pfs_register()

```
volatile uint8_t * internal_get_pfs_register (
    const uint8_t port,
    uint8_t pin) [static]
```

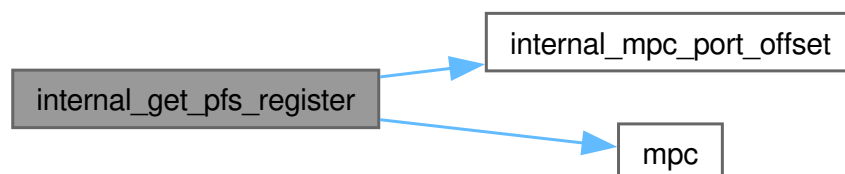
Definition at line 506 of file [rx_mpc.c](#).

```
00507 {
00508     if (pin > k_mpc_max_pin) {
00509         return nullptr;
00510     }
00511
00512     const uint16_t port_offset = internal_mpc_port_offset(port);
00513     if (port_offset == (uint16_t)k_mpc_port_offset_invalid) {
00514         rx_log_error(s_tag,
00515             (port == k_mpc_port_g) ? "Port not available on this package" : "Invalid port");
00516         return nullptr;
00517     }
00518
00519     /* PFS registers start at MPC base + 0x40 (offset for P00PFS). Each port
00520      * has 8 pins laid out P00-P07, P10-P17, P20-P27, etc. */
00521     volatile uint8_t* pfs_base = (volatile uint8_t*)mpc() + k_mpc_pfs_base_offset;
00522     return pfs_base + port_offset + pin;
00523 }
```

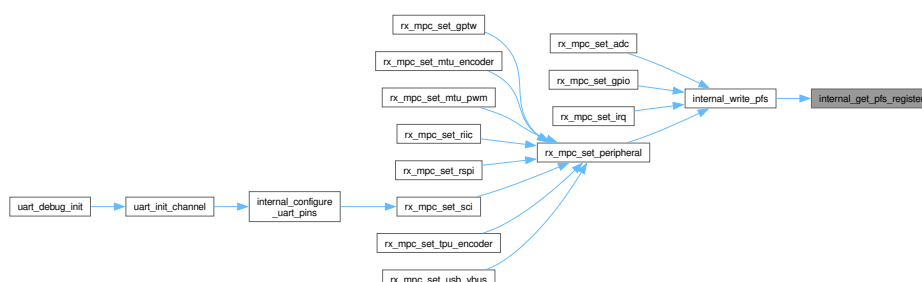
References [internal_mpc_port_offset\(\)](#), [k_mpc_max_pin](#), [k_mpc_pfs_base_offset](#), [k_mpc_port_g](#), [k_mpc_port_offset_invalid](#), [mpc\(\)](#), and [s_tag](#).

Referenced by [internal_write_pfs\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.2 internal`mpc`lock()

```
void internal__mpc_lock (  
    void ) [static]
```

Lock MPC PFS register write protection.

Executes the two-step lock sequence to protect PFS registers after modification. This restores write protection and prevents accidental changes to pin functions.

Lock Sequence (per RX72N Hardware Manual Ch20)

1. Write 0x00 to PWPR (clears PFSWE bit, protects PFS registers)
2. Write 0x80 to PWPR (sets B0WI bit, prevents PFSWE modification)

Timing

- Two 8-bit register writes
- No wait states required between writes
- Total time: ~100 ns @ 240 MHz

Precondition

[internal__mpc_unlock\(\)](#) was called previously
PFS modifications are complete

Postcondition

PWPR.PFSWE = 0 (PFS registers protected)
PWPR.B0WI = 1 (PFSWE bit protected)

Note

Internal function - not exported in header

Example:

```
// Typical usage: unlock, modify PFS, then lock immediately  
internal__mpc_unlock();  
*pfs_reg = k_psel_rspi_clk; // Write peripheral select value  
internal__mpc_lock();       // Restore write protection
```

See also

[internal__mpc_unlock\(\)](#) Must be called before PFS writes
[k_mpc_pwpr_b0wi](#) B0WI enable constant (0x80)

Since

Version 1.0.0

Definition at line 614 of file [rx_mpc.c](#).

```
00615 {
00616     /* Lock sequence:
00617      * 1. Clear PFSWE to protect PFS
00618      * 2. Set B0WI to protect PFSWE
00619      */
00620     mpc()->pwpr = k_mpc_pwpr_clear; /* Clear PFSWE */
00621     mpc()->pwpr = k_mpc_pwpr_b0wi; /* Set B0WI */
00622 }
```

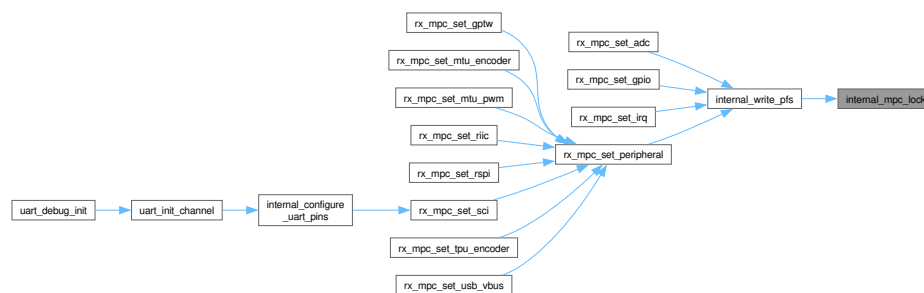
References [k_mpc_pwpr_b0wi](#), [k_mpc_pwpr_clear](#), [mpc\(\)](#), and [rx_mpc_regs_t::pwpr](#).

Referenced by [internal_write_pfs\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.3 internal_mpc_port_offset()

```
uint16_t internal_mpc_port_offset (
    const uint8_t port) [static]
```

Resolve a port enum to its PFS-register offset.

Lookup table indexed by `k_mpc_port_*` values. Returns `k_mpc_port_offset_invalid` for ports unavailable on this package. Replaces the prior 60+ line switch statement that this function used.

Parameters

in	port	Port enum value (<code>k_mpc_port_0 .. k_mpc_port_j</code>)
----	------	---

Returns

uint16_t PFS port offset, or `k_mpc_port_offset_invalid` if absent

Precondition

None

Postcondition

Return value is either a valid offset or the sentinel

Note

Pure function, no side effects.

Since

Version 1.0.0

Definition at line 463 of file [rx_mpc.c](#).

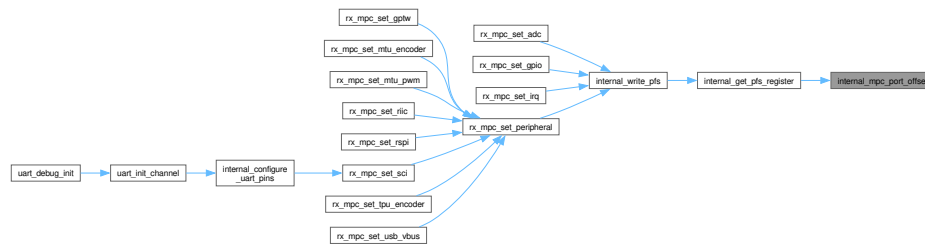
```

00464 {
00465     switch (port) {
00466         case k_mpc_port_0:
00467             return k_mpc_port0_offset;
00468         case k_mpc_port_1:
00469             return k_mpc_port1_offset;
00470         case k_mpc_port_2:
00471             return k_mpc_port2_offset;
00472         case k_mpc_port_3:
00473             return k_mpc_port3_offset;
00474         case k_mpc_port_4:
00475             return k_mpc_port4_offset;
00476         case k_mpc_port_5:
00477             return k_mpc_port5_offset;
00478         case k_mpc_port_6:
00479             return k_mpc_port6_offset;
00480         case k_mpc_port_7:
00481             return k_mpc_port7_offset;
00482         case k_mpc_port_8:
00483             return k_mpc_port8_offset;
00484         case k_mpc_port_9:
00485             return k_mpc_port9_offset;
00486         case k_mpc_port_a:
00487             return k_mpc_porta_offset;
00488         case k_mpc_port_b:
00489             return k_mpc_portb_offset;
00490         case k_mpc_port_c:
00491             return k_mpc_portc_offset;
00492         case k_mpc_port_d:
00493             return k_mpc_portd_offset;
00494         case k_mpc_port_e:
00495             return k_mpc_porte_offset;
00496         case k_mpc_port_f:
00497             return k_mpc_portf_offset;
00498         case k_mpc_port_j:
00499             return k_mpc_portj_offset;
00500         case k_mpc_port_g: /* not on 144-pin LFQFP */
00501             default:
00502                 return k_mpc_port_offset_invalid;
00503     }
00504 }
```

References [k_mpc_port0_offset](#), [k_mpc_port1_offset](#), [k_mpc_port2_offset](#), [k_mpc_port3_offset](#), [k_mpc_port4_offset](#), [k_mpc_port5_offset](#), [k_mpc_port6_offset](#), [k_mpc_port7_offset](#), [k_mpc_port8_offset](#), [k_mpc_port9_offset](#), [k_mpc_port_0](#), [k_mpc_port_1](#), [k_mpc_port_2](#), [k_mpc_port_3](#), [k_mpc_port_4](#), [k_mpc_port_5](#), [k_mpc_port_6](#), [k_mpc_port_7](#), [k_mpc_port_8](#), [k_mpc_port_9](#), [k_mpc_port_a](#), [k_mpc_port_b](#), [k_mpc_port_c](#), [k_mpc_port_d](#), [k_mpc_port_e](#), [k_mpc_port_f](#), [k_mpc_port_g](#), [k_mpc_port_j](#), [k_mpc_port_offset_invalid](#), [k_mpc_porta_offset](#), [k_mpc_portb_offset](#), [k_mpc_portc_offset](#), [k_mpc_portd_offset](#), [k_mpc_porte_offset](#), [k_mpc_portf_offset](#), and [k_mpc_portj_offset](#).

Referenced by [internal_get_pfs_register\(\)](#).

Here is the caller graph for this function:



8.127.3.4 internal'mpc'unlock()

```
void internal_mpc_unlock (
    void ) [static]
```

Unlock MPC PFS register write protection.

Executes the two-step unlock sequence required before writing to any PFS register. The PWPR (Write Protect Register) guards against accidental modification of pin functions.

Unlock Sequence (per RX72N Hardware Manual Ch20)

1. Write 0x00 to PWPR (clears B0WI bit, allows PFSWE modification)
2. Write 0x40 to PWPR (sets PFSWE bit, enables PFS register writes)

Timing

- Two 8-bit register writes
- No wait states required between writes
- Total time: ~100 ns @ 240 MHz

Precondition

MPC module must be active (not in module stop)

Postcondition

PWPR.B0WI = 0 (PFSWE modifiable)

PWPR.PFSWE = 1 (PFS registers writable)

Note

Must call [internal_mpc_lock\(\)](#) after PFS modification!

Internal function - not exported in header

Warning

Not thread-safe - window between unlock and lock is vulnerable

Failing to lock after modification leaves PFS unprotected

Example:

```
// Typical usage: unlock, modify PFS, then lock immediately
internal_mpc_unlock();
*pfs_reg = k_psel_sci_tx; // Write peripheral select value
internal_mpc_lock();
```

See also

[internal_mpc_lock\(\)](#) Must be called after PFS writes
[k_mpc_pwpr_pfswe](#) PFSWE enable constant (0x40)

Since

Version 1.0.0

Definition at line 566 of file [rx_mpc.c](#).

```
00567 {
00568     /* Unlock sequence:
00569      * 1. Clear B0WI to allow PFSWE write
00570      * 2. Set PFSWE to allow PFS write
00571      */
00572     mpc()->pwpr = k_mpc_pwpr_clear; /* Clear B0WI */
00573     mpc()->pwpr = k_mpc_pwpr_pfswe; /* Set PFSWE */
00574 }
```

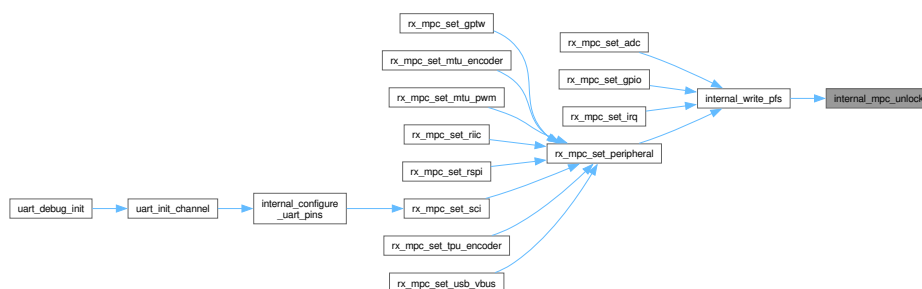
References [k_mpc_pwpr_clear](#), [k_mpc_pwpr_pfswe](#), [mpc\(\)](#), and [rx_mpc_regs_t::pwpr](#).

Referenced by [internal_write_pfs\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.5 `internal_write_pfs()`

```

rx_err_t internal_write_pfs (
    const uint8_t port,
    uint8_t pin,
    uint8_t value)  [static]

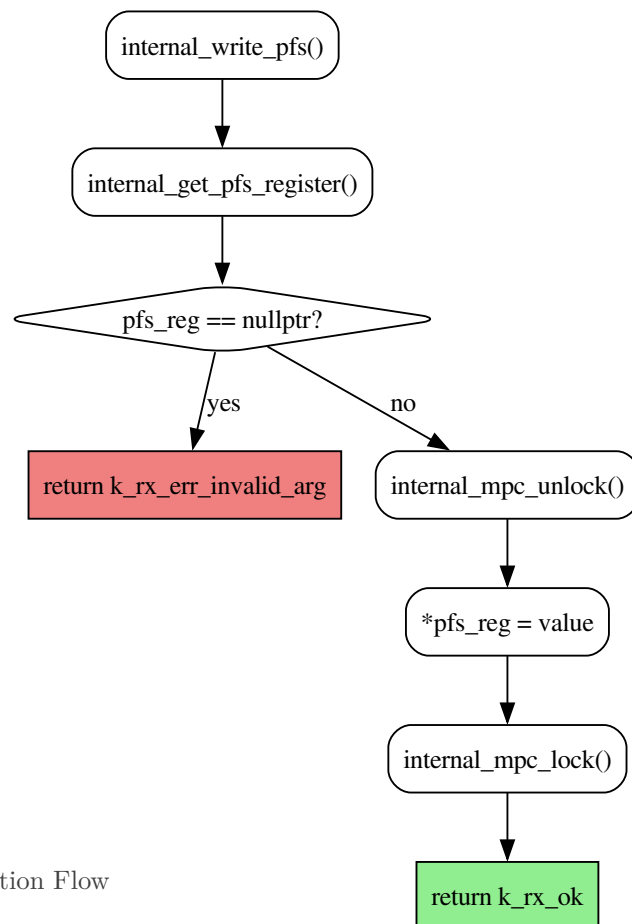
```

Write value to PFS register with automatic unlock/lock.

Complete PFS modification sequence: calculates register address, unlocks protection, writes the value, and re-locks protection. This function handles the entire write-protected register access pattern.

Algorithm Steps

1. Call [internal_get_pfs_register\(\)](#) to get register address
2. If address is nullptr, return error (no unlock performed)
3. Call [internal_mpc_unlock\(\)](#) to enable PFS writes
4. Write value to PFS register
5. Call [internal_mpc_lock\(\)](#) to protect PFS registers
6. Return success



Execution Flow

Precondition

- PCLKB clock must be running
- Value should be a valid PFS configuration

Postcondition

- PFS register contains specified value
- PWPR is locked (B0WI=1)

Note

- Internal function - not exported in header
- Lock is always executed even on error path (defensive coding)

Warning

- Not thread-safe - the unlock/write/lock sequence is non-atomic

Performance

- Execution time: ~500 ns @ 240 MHz
- Stack usage: 16 bytes

See also

- [internal_get_pfs_register\(\)](#) Address calculation
- [internal_mpc_unlock\(\)](#) Protection unlock
- [internal_mpc_lock\(\)](#) Protection lock

Since

- Version 1.0.0

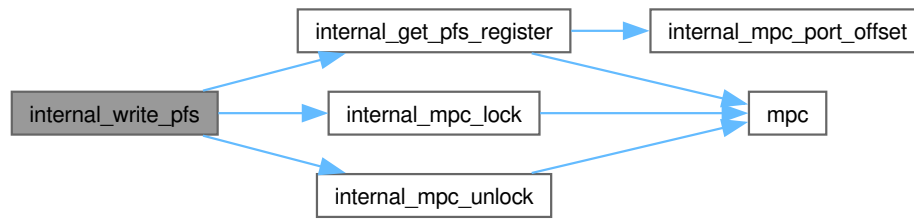
Definition at line 688 of file [rx_mpc.c](#).

```
00689 {  
00690     volatile uint8_t* pfs_reg = internal_get_pfs_register(port, pin);  
00691     if (pfs_reg == nullptr) {  
00692         return k_rx_err_invalid_arg;  
00693     }  
00694  
00695     /* Unlock, write, lock */  
00696     internal_mpc_unlock();  
00697     *pfs_reg = value;  
00698     internal_mpc_lock();  
00699  
00700     return k_rx_ok;  
00701 }
```

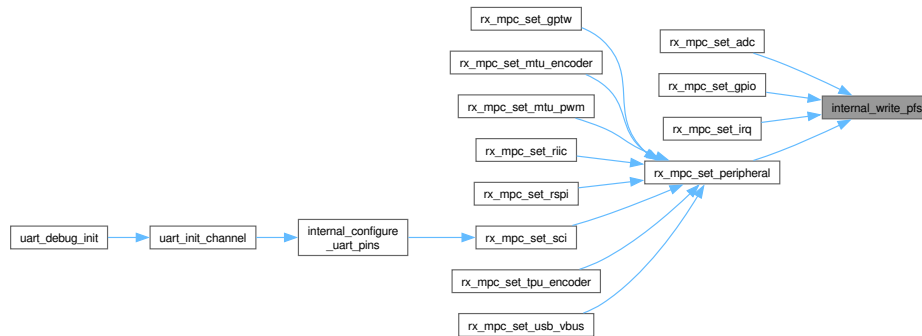
References [internal_get_pfs_register\(\)](#), [internal_mpc_lock\(\)](#), and [internal_mpc_unlock\(\)](#).

Referenced by [rx_mpc_set_adc\(\)](#), [rx_mpc_set_gpio\(\)](#), [rx_mpc_set_irq\(\)](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.6 rx_mpc_set_adc()

```

rx_err_t rx_mpc_set_adc (
    const rx_port_pin_t pin) [nodiscard]
  
```

Configure pin for ADC analog input.

Configures a pin for analog-to-digital conversion by setting the ASEL bit in the PFS register. This disables the digital input buffer to prevent noise during analog measurement.

Precondition

- PCLKB clock must be running, MPC not in module stop
- pin must encode a valid port/pin combination with analog capability

Postcondition

- PFS register contains k_pfs_asel (0x80) for the specified pin
- PWPR locked after operation

Note

- Thread safety: Not thread-safe
- Sets ASEL bit (0x80) rather than PSEL field
- Also configure S12ADC for actual conversion

```

// Configure P40 (AN000) for ADC analog input
rx_err_t err = rx_mpc_set_adc(RX_PORT_PIN(4, 0));
if (err != k_rx_ok) {
    // handle error
}
  
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

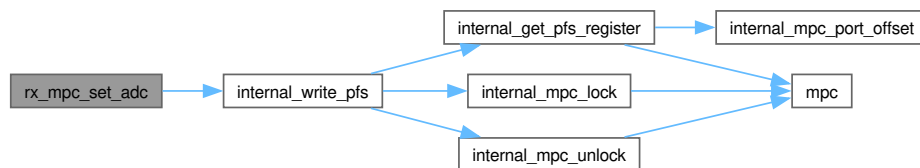
Definition at line 1009 of file [rx_mpc.c](#).

```

01010 {
01011     /* Extract port and pin number from type-safe enum */
01012     const uint8_t port  = rx_port_from_pin(pin);
01013     const uint8_t pin_num = rx_pin_from_pin(pin);
01014
01015     /* Validate the decoded port and pin are in valid range */
01016     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
01017         rx_log_error(s_tag, "Invalid port or pin number");
01018         return k_rx_err_invalid_arg;
01019     }
01020
01021     /* ADC mode: Set ASEL = 1 to disable digital I/O
01022      * Common ADC pins: P40-P47 (AN000-AN007)
01023      */
01024     return internal_write_pfs(port, pin_num, k_pfs_asel);
01025 }
```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_pfs_asel](#), and [s_tag](#).

Here is the call graph for this function:



8.127.3.7 rx`mpc`set`gpio()

```

rx_err_t rx_mpc_set_gpio (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for GPIO (General Purpose I/O) function.

Sets the specified pin to GPIO mode by writing 0x00 to its PFS register. This disconnects the pin from all peripheral functions and allows it to be controlled by the PORT registers (PDR, PODR, PIDR).

Implementation Details

1. Extract port and pin number from encoded `rx_port_pin_t`
2. Validate port is in range `[0, J]`
3. Validate pin is in range `[0, 7]`
4. Write 0x00 to PFS register (via `internal_write_pfs`)

Precondition

PCLKB clock running, MPC not in module stop
pin must be a valid `rx_port_pin_t` constant

Postcondition

PFS register = 0x00 (GPIO mode)

PWPR locked after operation

Note

Thread safety: Not thread-safe

Also set PORT PMR bit to 0 for complete GPIO configuration

```
// Configure P30 (port 3, pin 0) for GPIO use
rx_err_t err = rx_mpc_set_gpio(RX_PORT_PIN(3, 0));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

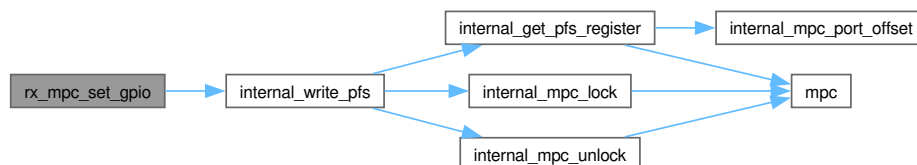
Version 1.0.0

Definition at line 744 of file [rx_mpc.c](#).

```
00745 {
00746     /* Extract port and pin number from rx_port_pin_t */
00747     /* Pre-condition: port and pin must be in valid range */
00748     const uint8_t port = rx_port_from_pin(pin);
00749     const uint8_t pin_num = rx_pin_from_pin(pin);
00750
00751     RX_CHECK_RANGE_TAG(port, k_mpc_port_start, k_mpc_port_j, k_rx_err_invalid_arg, s_tag);
00752     RX_CHECK_RANGE_TAG(pin_num, k_mpc_min_pin, k_mpc_max_pin, k_rx_err_invalid_arg, s_tag);
00753
00754     /* GPIO mode: PSEL = 0, ISEL = 0, ASEL = 0 */
00755     return internal_write_pfs(port, pin_num, k_mpc_pfs_gpio);
00756 }
```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_min_pin](#), [k_mpc_pfs_gpio](#), [k_mpc_port_j](#), [k_mpc_port_start](#), and [s_tag](#).

Here is the call graph for this function:



8.127.3.8 rx_mpc_set_gptw()

```
rx_err_t rx_mpc_set_gptw (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for GPTW complementary PWM output.

Configure pin for GPTW (General PWM Timer) complementary output.

Convenience wrapper that configures a pin for GPTW operation using PSEL = k_psel_gptw (0x1E). GPTW provides complementary PWM output with dead-time insertion for motor control.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with GPTW multiplexing

Postcondition

PFS register contains `k_psel_gptw` (0x1E) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe
Used for 4 GPTW channels (0-3) with 8 total pins (IN2 + IN1 per motor)
Phase staggering configured separately via `rx_gptw` driver

```
// Configure motor 2 GPTW IN2 and IN1 pins (Port E)
rx_err_t err = rx_mpc_set_gptw(RX_PORT_PIN(0xE, 3)); // PE3 GTIOC2A (IN2)
if (err == k_rx_ok) {
    err = rx_mpc_set_gptw(RX_PORT_PIN(8, 6)); // P8.6 GTIOC2B (IN1)
}
```

See also

[rx_mpc.h](#) Full API documentation with usage examples
[rx_gptw_init_all_staggered\(\)](#) GPTW channel configuration

Since

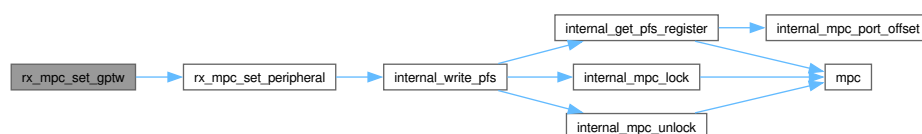
Version 1.0.0

Definition at line 1234 of file `rx_mpc.c`.

```
01235 {
01236     /* GPTW pins use PSEL = 0x1E
01237      * Supports 4 GPTW channels (0-3) for motor IN2/IN1 control */
01238     const rx_mpc_peripheral_config_t config = {
01239         .pin = pin,
01240         .psel = k_psel_gptw /* 0x1E - defined in rx_pin_psel_t enum */
01241     };
01242
01243     return rx_mpc_set_peripheral(&config);
01244 }
```

References [k_psel_gptw](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.3.9 rx_mpc_set_irq()

```
rx_err_t rx_mpc_set_irq (
    const rx_port_pin_t pin)  [nodiscard]
```

Configure pin for IRQ (external interrupt) function.

Sets pin function to IRQ mode by writing the ISEL bit (bit 6, value 0x40) directly to the PFS register via [internal_write_pfs\(\)](#). This bypasses [rx_mpc_set_peripheral\(\)](#) because ISEL=0x40 exceeds the 5-bit PSEL field maximum (k_mpc_psel_max = 31).

Mirrors the approach used by [rx_mpc_set_adc\(\)](#) for ASEL (bit 7): both functions write their respective bit directly rather than going through the PSEL validation path.

Algorithm Steps:

1. Extract port number and pin number from encoded pin parameter
2. Validate port in range [0, J]
3. Validate pin in range [0, 7]
4. Write ISEL bit (0x40) to PFS register via [internal_write_pfs\(\)](#)

Precondition

Pin must have IRQ multiplexing capability (P00-P07 for IRQ8-15)
PCLKB clock must be running

Postcondition

Pin ISEL bit (0x40) set in PFS register
PSEL field remains 0 (no peripheral function conflict)

Note

Thread safety: Not thread-safe
Typical pins: P00-P07 for IRQ8-IRQ15 on RX72N
Used by HC-SR04 ultrasonic sensors for echo pulse measurement

```
// Configure P00 for IRQ8 (HC-SR04 echo input)
rx_err_t err = rx_mpc_set_irq(RX_PORT_PIN(0, 0));
if (err != k_rx_ok) {
    // handle error
}
// Then configure ICU for rising/falling edge detection
```

See also

[rx_mpc_set_adc\(\)](#) Same pattern: writes ASEL bit (0x80) directly
[rx_mpc.h](#) Full API documentation with usage examples
[rx_hcsr04_icu_configure\(\)](#) Configure ICU after calling this function

Since

Version 1.0.0

Definition at line 1336 of file rx_mpc.c.

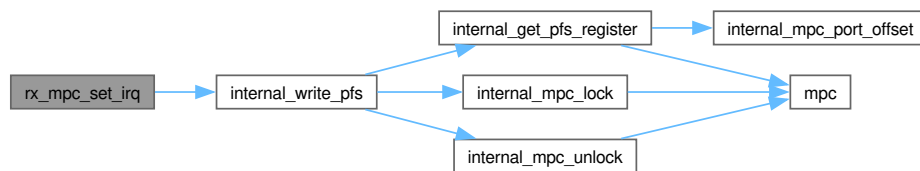
```

01337 {
01338 /* Extract port and pin number from type-safe enum */
01339 const uint8_t port = rx_port_from_pin(pin);
01340 const uint8_t pin_num = rx_pin_from_pin(pin);
01341
01342 /* Validate the decoded port and pin are in valid range */
01343 if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
01344     rx_log_error(s_tag, "Invalid port or pin number for IRQ function");
01345     return k_rx_err_invalid_arg;
01346 }
01347
01348 /* IRQ mode: Set ISEL = 1 (bit 6) to enable IRQ input function.
01349 * This is NOT a PSEL value; k_pfs_isel is 0x40 which exceeds the 5-bit
01350 * PSEL field. Write directly like rx_mpc_set_adc() does for ASEL.
01351 * Typical IRQ pins: P00-P07 (IRQ8-IRQ15) for HC-SR04 echo measurement. */
01352 return internal_write_pfs(port, pin_num, k_pfs_isel);
01353 }

```

References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_pfs_isel](#), and [s_tag](#).

Here is the call graph for this function:



8.127.3.10 rx`mpc`set`mtu`encoder()

```

rx_err_t rx_mpc_set_mtu_encoder (
    const rx_port_pin_t pin) [nodiscard]

```

Configure pin for MTU3a encoder/phase counter input (MTCLK).

Convenience wrapper that configures a pin for MTU external clock / phase counting mode using PSEL = k_psel_mtu_phase (0x02). Used for quadrature encoders on pins that expose MTCLKA/B/C/D.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination supported by MTCLK input

Postcondition

PFS register contains k_psel_mtu_phase (0x02) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe

Configure both Phase A and Phase B pins for proper operation

```
// Configure P24 and P25 for MTU phase counting (MTCLKA/B)
rx_err_t err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 4));
if (err == k_rx_ok) {
    err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 5));
}
```

See also

[rx_mpc.h](#) Full API documentation

[rx_mtu_encoder.h](#) Encoder driver using MTU phase counting

Since

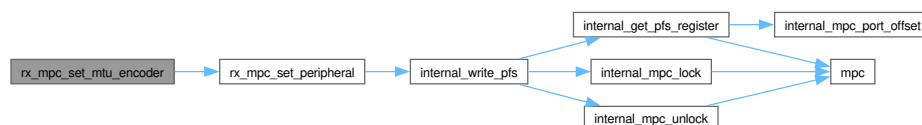
Version 1.0.0

Definition at line 906 of file [rx_mpc.c](#).

```
00907 {
00908     /* MTU external clock / phase counting pins use PSEL = 0x02 on every
00909     * RX72N MTCLK candidate pin. See RX72N Group Hardware Manual
00910     * (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)",
00911     * Tables 23.5 / 23.6 / 23.17 / 23.19 -- every row that selects MTCLKA,
00912     * MTCLKB, MTCLKC, or MTCLKD on a 144-pin LFQFP pin is PSEL = 000010b.
00913     * The old value 0x03 was wrong -- it caused MTU TCNT to stay at 0x0000
00914     * because no MTCLK source was actually connected through the PFS mux.
00915     *
00916     * Common STAR pins:
00917     * P24/P25 (Motor 0 front-right MTCLKA/B)
00918     * PA1/PC5 (Motor 1 front-left MTCLKC/D)
00919     */
00920     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_phase};
00921
00922     return rx_mpc_set_peripheral(&config);
00923 }
```

References [k_psel_mtu_phase](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.3.11 rx_mpc_set_mtu_pwm()

```
rx_err_t rx_mpc_set_mtu_pwm (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for MTU3a timer PWM output (MTIOC).

Convenience wrapper that configures a pin for MTU I/O compare output using PSEL = [k_psel_mtu_ioc](#) (0x01). Commonly used for motor PWM.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination supported by MTU MTIOC

Postcondition

PFS register contains k_psel_mtu_ioc (0x01) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe

STAR project uses GPTW for motor PWM (see [rx_mpc_set_gptw\(\)](#), PSEL = k_psel_gptw (0x1E)), not MTU

```
// Configure P14 for MTU3 PWM output (MTIOCA)
rx_err_t err = rx_mpc_set_mtu_pwm(RX_PORT_PIN(1, 4));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

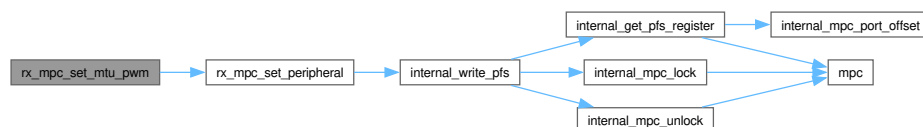
Version 1.0.0

Definition at line 863 of file [rx_mpc.c](#).

```
00864 {
00865     /* MTU PWM (MTIOC) pins typically use PSEL = 0x01
00866     * Common pins:
00867     * - P14-P17: MTU3 MTIOCA/B/C/D
00868     * - P24-P27: MTU4 MTIOCA/B/C/D
00869     */
00870     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_ioc};
00871
00872     return rx_mpc_set_peripheral(&config);
00873 }
```

References [k_psel_mtu_ioc](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.3.12 rx_mpc_set_peripheral()

```
rx_err_t rx_mpc_set_peripheral (
    const rx_mpc_peripheral_config_t * config)  [nodiscard]
```

Configure pin for peripheral function with specified PSEL code.

Core function for peripheral pin configuration. Writes the specified PSEL value to the pin's PFS register, connecting it to a peripheral.

Implementation Details

1. Validate config pointer is not nullptr
2. Validate PSEL value is in range [0, 31]
3. Extract port and pin from config->pin
4. Validate extracted values
5. Write PSEL to PFS register (via internal_write_pfs)

Precondition

config must be non-NULL with valid fields
PCLKB clock running

Postcondition

PFS register contains config->psel
PWPR locked after operation

Note

Thread safety: Not thread-safe
Also set PORT PMR bit to 1 for peripheral mode

```
// Configure PA0 for RSPI clock (PSEL = 0x0D)
const rx_mpc_peripheral_config_t cfg = {
    .pin = RX_PORT_PIN(0xA, 0),
    .psel = 0x0D,
};
rx_err_t err = rx_mpc_set_peripheral(&cfg);
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 798 of file [rx_mpc.c](#).

```

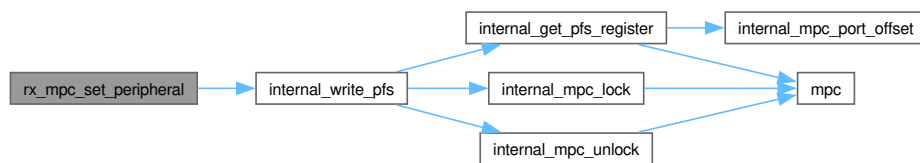
00799 {
00800     if (config == nullptr) {
00801         rx_log_error(s_tag, "Invalid configuration pointer");
00802         return k_rx_err_invalid_arg;
00803     }
00804
00805     if (config->psel > k_mpc_psel_max) {
00806         rx_log_error(s_tag, "Invalid PSEL value exceeds maximum");
00807         rx_log_error_val(s_tag, "psel=", config->psel);
00808         rx_log_error_val(s_tag, "k_mpc_psel_max=", (uint8_t)k_mpc_psel_max);
00809         return k_rx_err_invalid_arg;
00810     }
00811
00812     /* Extract port and pin number from rx_port_pin_t */
00813     const uint8_t port = rx_port_from_pin(config->pin);
00814     const uint8_t pin_num = rx_pin_from_pin(config->pin);
00815
00816     /* Validate the decoded port and pin are in valid range */
00817     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
00818         rx_log_error(s_tag, "Invalid port or pin number");
00819         rx_log_error_val(s_tag, "port=", port);
00820         rx_log_error_val(s_tag, "pin=", pin_num);
00821         return k_rx_err_invalid_arg;
00822     }
00823
00824     /* Peripheral mode: PSEL = specified, ISEL = 0, ASEL = 0.
00825     *
00826     * NOTE: PORT.PMR bit is intentionally NOT set here. Setting PMR=1 before
00827     * the peripheral itself (e.g. GPTW timer) is fully configured exposes
00828     * the pin to undefined/glitchy output that can latch a fault in
00829     * downstream silicon (e.g. DRV8263H motor driver). Caller is responsible
00830     * for setting PMR=1 AFTER the peripheral is fully initialized. */
00831     return internal_write_pfs(port, pin_num, config->psel);
00832 }

```

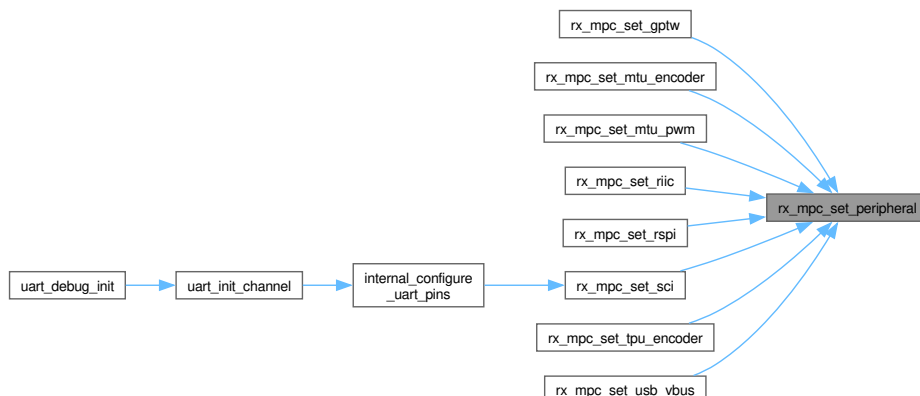
References [internal_write_pfs\(\)](#), [k_mpc_max_pin](#), [k_mpc_port_j](#), [k_mpc_psel_max](#), [rx_mpc_peripheral_config_t::rx_mpc_peripheral_config_t::psel](#), and [s_tag](#).

Referenced by [rx_mpc_set_gptw\(\)](#), [rx_mpc_set_mtu_encoder\(\)](#), [rx_mpc_set_mtu_pwm\(\)](#), [rx_mpc_set_riic\(\)](#), [rx_mpc_set_rspi\(\)](#), [rx_mpc_set_sci\(\)](#), [rx_mpc_set_tpu_encoder\(\)](#), and [rx_mpc_set_usb_vbus\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.13 rx_mpc_set_riic()

```
rx_err_t rx_mpc_set_riic (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for RIIC (I2C) bus function.

Convenience wrapper that configures a pin for RIIC operation using PSEL = k_psel_riic_scl (0x0F). Works for both SCL and SDA pins.

Unlike [rx_mpc_set_gptw\(\)](#), this function also sets PORT.PMR=1 for the pin before returning – RIIC cannot drive SDA/SCL while PMR=0 and the immediate-vs-deferred PMR trade-off differs from the motor-PWM case:

Peripheral	When to raise PMR	Reason
GPTW/PWM	AFTER timer is running	Push-pull drive: an un-
	(rx_gptw_init_pwm handles	initialized output could
	this after rx_gptw_start)	momentarily glitch
		DRV8263H H-bridge inputs
		to Vm/2 during the
		GTIOR reset state window.
RIIC/I2C	IMMEDIATELY after PFS	Open-drain + external
	(here, inside this wrapper)	pull-ups: idle bus is
		both lines high (3.3V);
		routing an un-initialized
		RIIC just leaves the
		pull-ups floating the
		lines high. No hazard.

Without PMR=1 on the RIIC pins, rx72n peripheral bits still toggle internally but the pad stays disconnected from RIIC; every Start condition times out with k_rx_err_timeout (0x108) at the first [riic_write\(\)](#)/[riic_read\(\)](#) call. Discovered 2026-04-22: production imu_task saw tight "[ERROR] [RIIC] Start condition timeout" loops while imu_test/main.c (which sets PMR=1 at line 227 before calling its own [riic_init\(\)](#)) talked to BNO055 + BMP280 cleanly on the same board.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with RIIC multiplexing

Postcondition

PFS register contains k_psel_riic_scl (0x0F) for the specified pin
PORT.PMR bit for the pin is set (pin routed to RIIC peripheral)
PWPR locked after operation

Note

Thread safety: Not thread-safe

Configure both SCL and SDA pins; external pull-ups required

```
// Configure P12 (SCL) and P13 (SDA) for RIIC0
rx_err_t err = rx_mpc_set_riic(RX_PORT_PIN(1, 2));
if (err == k_rx_ok) {
    err = rx_mpc_set_riic(RX_PORT_PIN(1, 3));
}
```

See also

[rx_mpc.h](#) Full API documentation

[rx_mpc_set_gptw](#) Contrast: deferred-PMR case

star-rx72n-firmware/imu_test/main.c::riic1_init Bench reference

Since

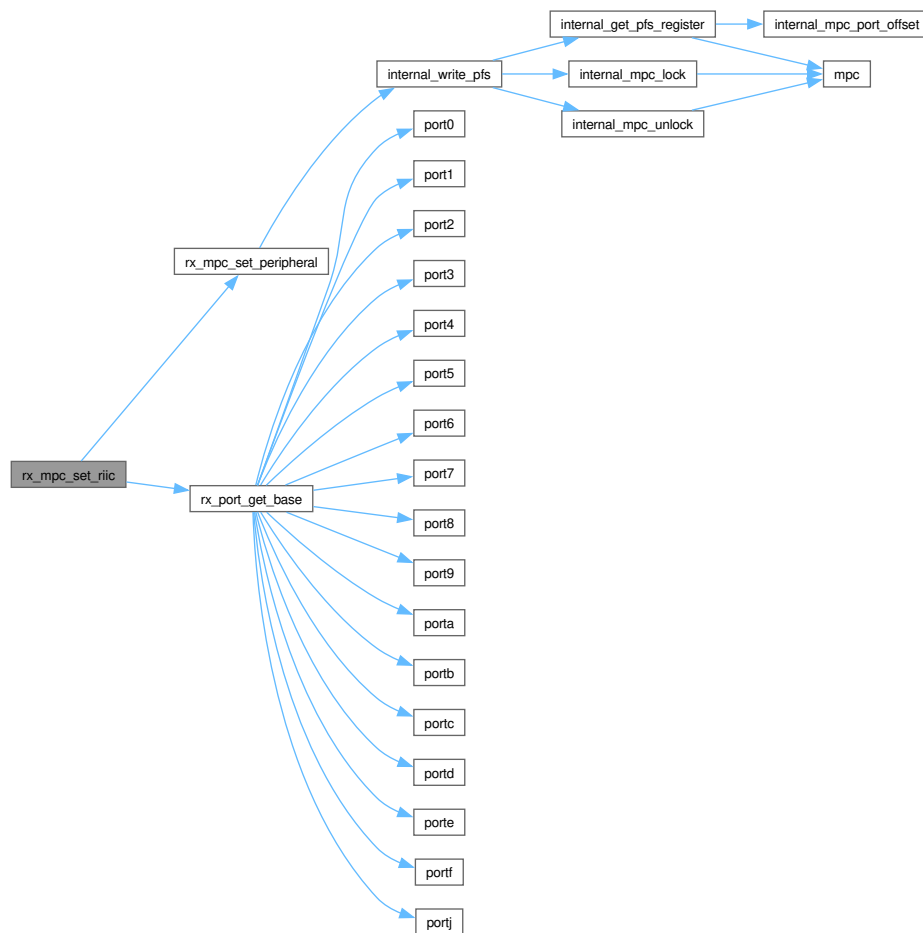
Version 1.0.0

Definition at line 1126 of file [rx_mpc.c](#).

```
01127 {
01128     /* RIIC pins use PSEL = 0x0F
01129     * SCL and SDA use the same PSEL code
01130     */
01131     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_riic_scl};
01132
01133     const rx_err_t err = rx_mpc_set_peripheral(&config);
01134     if (err != k_rx_ok) {
01135         return err;
01136     }
01137
01138     /* Set PORT.PMR=1 so SCL/SDA actually route to the RIIC peripheral.
01139     *
01140     * Unlike GPTW (where PMR must be deferred until after the timer is
01141     * running to avoid exposing DRV8263H inputs to GTIOR reset state),
01142     * I2C pins are open-drain with external pull-ups: driving them high
01143     * through the peripheral mux is the idle-bus state and perfectly safe.
01144     *
01145     * Without this bit set, P20/P21 remain GPIO (PMR=0) and RIIC1 can
01146     * never toggle the lines -- every Start condition times out with
01147     * k_rx_err_timeout (0x108). The bench-proven imu_test/main.c sets
01148     * PMR identically at rx72n-firmware/imu_test/main.c:227. */
01149     const uint8_t port_idx = rx_port_from_pin(pin);
01150     const uint8_t bit = rx_pin_from_pin(pin);
01151     volatile rx_port_regs_t* const port = rx_port_get_base(port_idx);
01152     if (port == nullptr) {
01153         return k_rx_err_invalid_arg;
01154     }
01155     port->pmr |= (uint8_t)(1U « bit);
01156     return k_rx_ok;
01157 }
```

References [k_psel_riic_scl](#), [rx_port_regs_t::pmr](#), [rx_mpc_set_peripheral\(\)](#), and [rx_port_get_base\(\)](#).

Here is the call graph for this function:



8.127.3.14 rx_mpc_set_rspi()

```
rx_err_t rx_mpc_set_rspi (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for RSPI (SPI) bus function.

Convenience wrapper that configures a pin for RSPI operation using PSEL = k_psel_rspi_clk (0x0D). Works for all SPI signals (CLK, COPI, CIPO, SSL).

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with RSPI multiplexing

Postcondition

PFS register contains k_psel_rspi_clk (0x0D) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe

Configure all 4 SPI pins (CLK, COPI, CIPO, CS)

Uses OSHWA-approved inclusive terminology

```
// Configure PA0-PA4 for RSPI0 (CLK, COPI, CIPO, SSL0, SSL1)
rx_err_t err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 0)); // CLK
if (err == k_rx_ok) {
    err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 1)); // COPI
}
```

See also

[rx_mpc.h](#) Full API documentation

[rx_spi_comm.h](#) SPI communication driver

Since

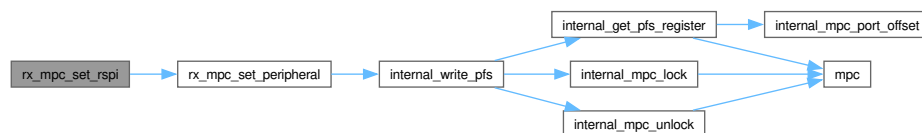
Version 1.0.0

Definition at line 1191 of file [rx_mpc.c](#).

```
01192 {
01193     /* RSPI pins use PSEL = 0x0D
01194      * All RSPI signals (CLK, COPI, CIPO, SSL) use same PSEL
01195      */
01196     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_rspi_clk};
01197
01198     return rx_mpc_set_peripheral(&config);
01199 }
```

References [k_psel_rspi_clk](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.3.15 rx`mpc`set`sci()

```
rx_err_t rx_mpc_set_sci (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for SCI (Serial Communication Interface) UART function.

Convenience wrapper that configures a pin for SCI UART operation using `PSEL = k_psel_sci_tx (0x0A)`. Works for both TXD and RXD pins.

Precondition

PCLKB clock must be running, MPC not in module stop

pin must encode a valid port/pin combination with SCI multiplexing

Postcondition

PFS register contains `k_psel_sci_tx` (0x0A) for the specified pin
 PWPR locked after operation

Note

Thread safety: Not thread-safe
 Configure both TXD and RXD pins for bidirectional UART

```
// Configure PB7 for SCI9 TXD
rx_err_t err = rx_mpc_set_sci(RX_PORT_PIN(0xB, 7));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation

Since

Version 1.0.0

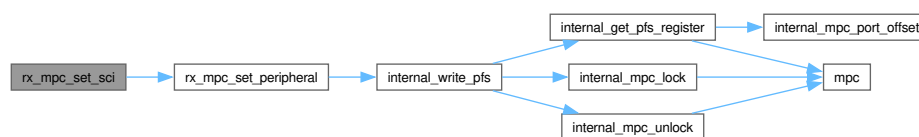
Definition at line 1056 of file `rx_mpc.c`.

```
01057 {
01058     /* SCI pins use PSEL = 0x0A
01059      * TX and RX use the same PSEL code
01060      */
01061     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_sci_tx};
01062
01063     return rx_mpc_set_peripheral(&config);
01064 }
```

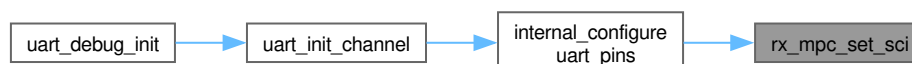
References [k_psel_sci_tx](#), and [rx_mpc_set_peripheral\(\)](#).

Referenced by [internal_configure_uart_pins\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.127.3.16 rx`mpc`set`tpu`encoder()

```
rx_err_t rx_mpc_set_tpu_encoder (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for TPU encoder/phase counter input (TCLK).

Convenience wrapper that configures a pin for TPU external clock input (TCLKA/B/C/D). Picks the correct PSEL value based on which port the pin is on, because RX72N uses two different alternate-function slots for the SAME TCLK signal depending on candidate:

- Port C pins (PC0/PC1/PC2/PC3) use PSEL = 0x03 (k_psel_tpu_clk_portc)
- Port 1 / Port A / Port B pins use PSEL = 0x04 (k_psel_tpu_clk_alt)

Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)". TCLK rows in Table 23.19 (Port C) use PSEL = 000011b, while TCLK rows in Tables 23.5 (P1x), 23.17 (PAx), and 23.18 (PBx) use PSEL = 000100b. Before this split, every TPU pin was configured with PSEL=0x03, which silently misconfigured the non-Port-C halves of STAR's rear-wheel encoders (PA3 for Motor 2 Phase B, PB3 for Motor 3 Phase B) so those edges were never counted.

Precondition

PCLKB clock must be running, MPC not in module stop

pin must encode a valid port/pin combination supported by TCLK input

Postcondition

PFS register contains the correct TPU-TCLK PSEL for that pin's port

PWPR locked after operation

Note

Thread safety: Not thread-safe

Configure both Phase A and Phase B pins for proper operation

```
// Configure PC2 (TCLKA) and PA3 (TCLKB) for Motor 2 rear-left encoder
rx_err_t err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xC, 2)); // -> 0x03
if (err == k_rx_ok) {
    err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xA, 3));      // -> 0x04
}
```

See also

[rx_mpc.h](#) Full API documentation

[rx_encoder_tpu.h](#) TPU encoder driver using phase counting

Since

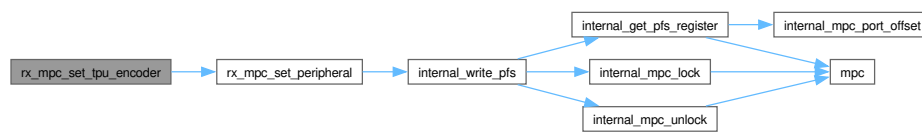
Version 1.0.0

Definition at line 967 of file [rx_mpc.c](#).

```
00968 {
00969     /* Pick the right PSEL for the TCLK function on this pin's port. */
00970     const uint8_t port = rx_port_from_pin(pin);
00971     const uint8_t psel =
00972         (port == k_mpc_port_c) ? (uint8_t)k_psel_tpu_clk_portc : (uint8_t)k_psel_tpu_clk_alt;
00973
00974     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = psel};
00975     return rx_mpc_set_peripheral(&config);
00976 }
```

References [k_mpc_port_c](#), [k_psel_tpu_clk_alt](#), [k_psel_tpu_clk_portc](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.3.17 rx_mpc_set_usb_vbus()

```
rx_err_t rx_mpc_set_usb_vbus (
    const rx_port_pin_t pin) [nodiscard]
```

Configure pin for USB VBUS detection.

Configure pin for USB VBUS detection function.

Convenience wrapper that configures a pin for USB VBUS detection using PSEL = 0x11. Required for USB CDC enumeration and power monitoring.

Precondition

PCLKB clock must be running, MPC not in module stop
pin must encode a valid port/pin combination with USB VBUS multiplexing

Postcondition

PFS register contains k_psel_usb_vbus (0x11) for the specified pin
PWPR locked after operation

Note

Thread safety: Not thread-safe
Active-high VBUS detect (pin = 1 when 5V present)
Also requires USB PHY and controller configuration

```
// Configure P16 for USB0_VBUS detection
rx_err_t err = rx_mpc_set_usb_vbus(RX_PORT_PIN(1, 6));
if (err != k_rx_ok) {
    // handle error
}
```

See also

[rx_mpc.h](#) Full API documentation with usage examples
[rx_usb.h](#) USB controller driver

Since

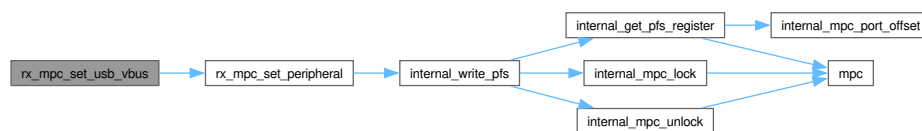
Version 1.0.0

Definition at line 1278 of file [rx_mpc.c](#).

```
01279 {
01280     /* USB VBUS detect uses PSEL = 0x11
01281      * Typical pin: P1.6 for USB0_VBUS on RX72N */
01282     const rx_mpc_peripheral_config_t config = {
01283         .pin = pin,
01284         .psel = k_psel_usb_vbus /* 0x11 - defined in rx_pin_psel_t enum */
01285     };
01286     return rx_mpc_set_peripheral(&config);
01287 }
01288 }
```

References [k_psel_usb_vbus](#), and [rx_mpc_set_peripheral\(\)](#).

Here is the call graph for this function:



8.127.4 Variable Documentation

8.127.4.1 s`tag

```
const char* const s_tag = "MPC" [static]
```

Logging tag for MPC module.

Static constant string used as the tag parameter for all `rx_log_*`() calls within this module. Provides consistent identification of MPC-related log messages in debug output.

Value: "MPC"

Memory: 4 bytes in .rodata section

Note

Statically allocated, never modified

Since

Version 1.0.0

Definition at line 191 of file [rx_mpc.c](#).

8.128 rx_mpc.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_mpc.c
00003  * @brief Multi-Function Pin Controller (MPC) Driver Implementation
00004  *
00005  * @details
00006  * Implementation of the MPC driver for RX72N pin multiplexer configuration.
00007  * This module provides the core functionality for connecting GPIO pins to
00008  * peripheral functions through the PFS (Pin Function Select) registers.
00009  *
00010  * @par Implementation Architecture
00011  * @verbatim
00012  * rx_mpc.c Implementation Structure
00013  * +-----+
00014  * | Public API Layer |
00015  * | +-----+ +-----+ +-----+ +-----+ |
00016  * | rx_mpc_set_ | rx_mpc_set_ | rx_mpc_set_ | rx_mpc_set_ | |
00017  * | gpio() | peripheral() | mtu_pwm() | sci() | |
00018  * | +-----+ +-----+ +-----+ +-----+ |
00019  * | | | | | | | | |
00020  * +-----+
00021  * | | | | |
00022  * | +-----+ +-----+ +-----+ +-----+ |
00023  * | Internal Helper Layer |
00024  * | +-----+ +-----+ |
00025  * | | internal_write_pfs() | |
00026  * | | (unlock -> write PFS -> lock with error handling) | |
00027  * | | +-----+ +-----+ |
00028  * | | | | |
00029  * | | +-----+ +-----+ |
00030  * | | internal_get_pfs_register() | |
00031  * | | (port/pin -> PFS register address calculation) | |
00032  * | | +-----+ +-----+ |
00033  * | | | | |
00034  * | | +-----+ +-----+ |
00035  * | | internal_mpc_ | internal_mpc_ |
00036  * | | unlock() | lock() |
00037  * | | +-----+ +-----+ |
00038  * | | +-----+ +-----+ |
00039  * | | | | |
00040  * | +-----+ +-----+ |
00041  * | Hardware Register Layer |
00042  * | +-----+ +-----+ |
00043  * | | mpc() accessor | |
00044  * | | (rx72n_mpc_regs.h, 0x0008C100) | |
00045  * | | +-----+ +-----+ |
00046  * | | +-----+ +-----+ |
00047  * @endverbatim
00048  *
00049  * @par PWPR Write Protection Sequence
00050  * The PFS registers are protected by a two-stage lock. The unlock/lock
00051  * sequence must be followed exactly:
00052  * @msc
00053  * Driver, PWPR, PFS;
00054  *
00055  * --- [label="Unlock Sequence"];
00056  * Driver => PWPR [label="Write 0x00 (B0WI=0)"];
00057  * PWPR box PWPR [label="PFSWE now writable"];
00058  * Driver => PWPR [label="Write 0x40 (PFSWE=1)"];
00059  * PWPR box PWPR [label="PFS now writable"];
00060  *
00061  * --- [label="Modify PFS"];
00062  * Driver => PFS [label="Write PSEL value"];
00063  * PFS box PFS [label="Pin function changed"];
00064  *
00065  * --- [label="Lock Sequence"];
00066  * Driver => PWPR [label="Write 0x00 (PFSWE=0)"];
00067  * PWPR box PWPR [label="PFS now protected"];
00068  * Driver => PWPR [label="Write 0x80 (B0WI=1)"];
00069  * PWPR box PWPR [label="PFSWE now protected"];
00070  * @endmsc
00071  *
00072  * @par PFS Register Address Calculation
00073  * PFS registers are laid out contiguously in memory with 8 bytes per port:
00074  * @verbatim
00075  * PFS Base = MPC Base + 0x40 = 0x0008C140
00076  *
00077  * Address = PFS_Base + (port_offset) + pin_number
00078  *
00079  * Port Offsets (8 bytes each):
00080  * Port 0: +0 Port 1: +8 Port 2: +16 Port 3: +24
00081  * Port 4: +32 Port 5: +40 Port 6: +48 Port 7: +56
00082  * Port 8: +64 Port 9: +72 Port A: +80 Port B: +88

```

```

00083 *   Port C: +96   Port D: +104   Port E: +112   Port J: +144
00084 *
00085 *   Example: PB7 (Port B, Pin 7)
00086 *   Address = 0x0008C140 + 88 + 7 = 0x0008C19F
00087 * @endverbatim
00088 *
00089 * @par Performance Characteristics
00090 * | Function | Typical Time | Stack Usage | Notes |
00091 * |-----|-----|-----|-----|
00092 * | rx_mpc_set_gpio() | 1.2 us | 24 bytes | Direct PFS write |
00093 * | rx_mpc_set_peripheral() | 1.5 us | 32 bytes | With validation |
00094 * | rx_mpc_set_mtu_pwm() | 1.6 us | 40 bytes | Calls set_peripheral |
00095 * | Bulk config (10 pins) | 15 us | - | Sequential calls |
00096 *
00097 * @par Memory Usage
00098 * | Category | Size | Description |
00099 * |-----|-----|-----|
00100 * | Code (.text) | ~800 bytes | All functions |
00101 * | Constants (.rodata) | 4 bytes | Tag string "MPC" |
00102 * | Static data (.data) | 0 bytes | None |
00103 * | Stack (max) | 40 bytes | Deepest call chain |
00104 *
00105 * @par Thread Safety Analysis
00106 * **NOT THREAD-SAFE.** Critical section analysis:
00107 * - internal_mpc_unlock() -> internal_write_pfs() -> internal_mpc_lock()
00108 * - This sequence is non-atomic (3 register writes)
00109 * - Interrupt or thread switch during unlock leaves PFS unprotected
00110 * - Race condition: Thread A unlocks, Thread B unlocks, Thread A writes,
00111 *   Thread B writes (corrupted configuration possible)
00112 *
00113 * @par Recommended Usage Pattern
00114 * @code{.c}
00115 * // Option 1: Configure all pins during single-threaded init
00116 * void hardware_init(void) // Called before RTOS starts
00117 * {
00118 *     rx_mpc_set_peripheral(&pwm_config);
00119 *     rx_mpc_set_sci(k_port_b_pin_7);
00120 *     // ... all MPC configuration here
00121 * }
00122 *
00123 * // Option 2: Use mutex if runtime reconfiguration needed
00124 * void configure_pin_safe(const rx_mpc_peripheral_config_t* config)
00125 * {
00126 *     tx_mutex_get(&mpc_mutex, TX_WAIT_FOREVER);
00127 *     rx_mpc_set_peripheral(config);
00128 *     tx_mutex_put(&mpc_mutex);
00129 * }
00130 * @endcode
00131 *
00132 * @par Error Handling Strategy
00133 * - All public functions validate inputs before hardware access
00134 * - Errors logged via rx_log_error() with descriptive messages
00135 * - No partial configuration: if validation fails, no registers modified
00136 * - Lock always restored even on error path (no dangling unlock)
00137 *
00138 * @par NASA Power of 10 Compliance
00139 * - **Rule 1**: [OK] No goto, setjmp, longjmp, recursion
00140 * - **Rule 2**: [OK] Fixed loop bounds (switch/case only, no iteration)
00141 * - **Rule 3**: [OK] No dynamic memory allocation
00142 * - **Rule 4**: [OK] All functions < 60 lines
00143 * - **Rule 5**: [OK] Assertions via RX_CHECK_* macros (2+ per function)
00144 * - **Rule 6**: [OK] Variables at smallest scope (function-local)
00145 * - **Rule 7**: [OK] All return values checked and propagated
00146 * - **Rule 8**: [OK] C23 typed enums for all constants
00147 * - **Rule 9**: [WARN] No function pointers (direct calls only)
00148 * - **Rule 10**: [OK] Compiles with -Wall -Wextra -Werror
00149 *
00150 * @par SOLID Principles
00151 * - **Single Responsibility**: Each internal function has one purpose
00152 * - **Open/Closed**: New peripherals added via set_peripheral()
00153 * - **Dependency Inversion**: Uses rx72n_mpc_regs.h abstraction
00154 *
00155 * @see rx_mpc.h Public API documentation
00156 * @see rx72n_mpc_regs.h Hardware register definitions
00157 * @see docs/sections/03_hardware_pinout.tex Pin assignment documentation
00158 *
00159 * @author Locked, Inc.
00160 * @date 2026-01-29
00161 * @copyright Copyright (c) 2026 Locked Inc.
00162 * SPDX-License-Identifier: MIT
00163 * @since Version 1.0.0
00164 */
00165
00166 #include "rx_mpc.h"
00167
00168 #include <stddef.h>
00169

```

```

00170 #include "rx72n_regs.h"
00171 #include "rx_check.h"
00172 #include "rx_log.h"
00173 #include "rx_port_constants.h"
00174 #include "rx_port_utils.h"
00175
00176 /**
00177  * @var s_tag
00178  * @brief Logging tag for MPC module
00179  *
00180  * @details
00181  * Static constant string used as the tag parameter for all rx_log_*() calls
00182  * within this module. Provides consistent identification of MPC-related
00183  * log messages in debug output.
00184  *
00185  * @par Value: "MPC"
00186  * @par Memory: 4 bytes in .rodata section
00187  *
00188  * @note Statically allocated, never modified
00189  * @since Version 1.0.0
00190  */
00191 static const char* const s_tag = "MPC";
00192
00193 /*
=====
00194  * Constants
00195  *
=====
00196  */
00197
00198 /**
00199  * @enum mpc_pin_constants_t
00200  * @brief Pin number validation constants for MPC driver
00201  *
00202  * @details
00203  * Defines the valid range for pin numbers within a port. RX72N GPIO ports
00204  * support up to 8 pins per port (pins 0-7), though not all pins are
00205  * available on all ports or packages.
00206  *
00207  * @see rx_port_constants.h Source of k_rx_pin_max definition
00208  * @since Version 1.0.0
00209  */
00210 typedef enum : uint8_t {
00211     k_mpc_min_pin = 0, /**< Minimum valid pin number (always 0) */
00212     k_mpc_max_pin = k_rx_pin_max, /**< Maximum valid pin number (7 for 8 pins per port) */
00213 } mpc_pin_constants_t;
00214
00215 /**
00216  * @enum mpc_pfs_offset_t
00217  * @brief PFS register base offset from MPC structure start
00218  *
00219  * @details
00220  * The PFS registers start at offset 0x40 from MPC base (0x0008C100), which
00221  * is address 0x0008C140. However, when accessing via the rx_mpc_regs_t struct,
00222  * we need to account for the structure layout.
00223  *
00224  * This offset is used with pointer arithmetic to calculate PFS register
00225  * addresses from the mpc() accessor.
00226  *
00227  * @par Calculation
00228  * PFS_addr = (uint8_t*)mpc() + k_mpc_pfs_base_offset + port_offset + pin
00229  *
00230  * @since Version 1.0.0
00231  */
00232 typedef enum : uint8_t {
00233     k_mpc_pfs_base_offset =
00234         64, /**< Byte offset from mpc() to P00PFS register (0x40: P00PFS=0x0008C140, base=0x0008C100) */
00235 } mpc_pfs_offset_t;
00236
00237 /**
00238  * @enum mpc_port_number_t
00239  * @brief Port number aliases for MPC driver validation
00240  *
00241  * @details
00242  * Maps the generic rx_port_* constants to MPC-specific names for use in
00243  * the switch statement in internal_get_pfs_register(). This provides
00244  * compile-time verification that all ports are handled.
00245  *
00246  * @par 144-pin LFQFP Package Availability
00247  * | Port | Available | Notes |
00248  * |-----|-----|-----|
00249  * | 0-5 | Partial | Some pins available |
00250  * | 6-9 | Partial | Some pins available (P67, P86/P87 on 144-pin) |
00251  * | A-E | Full | All 8 pins per port |
00252  * | F | Partial | PF0-PF2, PF5 available on 144-pin |
00253  * | G-H | None | Not on 144-pin package |
00254  * | J | Partial | PJ3, PJ5 available on 144-pin |

```



```

00255 *
00256 * @see rx_port_constants.h Source definitions
00257 * @since Version 1.0.0
00258 */
00259 typedef enum : uint8_t {
00260     k_mpc_port_start = k_rx_port_0, /**< First valid port number for range check */
00261     k_mpc_port_0 = k_rx_port_0, /**< Port 0 (GPIO P00-P07) */
00262     k_mpc_port_1 = k_rx_port_1, /**< Port 1 (GPIO P10-P17) */
00263     k_mpc_port_2 = k_rx_port_2, /**< Port 2 (GPIO P20-P27, encoder inputs) */
00264     k_mpc_port_3 = k_rx_port_3, /**< Port 3 (GPIO P30-P34, P35=NMI) */
00265     k_mpc_port_4 = k_rx_port_4, /**< Port 4 (GPIO P40-P47, ADC inputs) */
00266     k_mpc_port_5 = k_rx_port_5, /**< Port 5 (GPIO P50-P57) */
00267     k_mpc_port_6 = k_rx_port_6, /**< Port 6 (GPIO P60-P67) */
00268     k_mpc_port_7 = k_rx_port_7, /**< Port 7 (GPIO P71-P77, P70 N/A) */
00269     k_mpc_port_8 = k_rx_port_8, /**< Port 8 (GPIO P80-P87) */
00270     k_mpc_port_9 = k_rx_port_9, /**< Port 9 (GPIO P90-P97) */
00271     k_mpc_port_a = k_rx_port_a, /**< Port A (GPIO PA0-PA7, SPI pins) */
00272     k_mpc_port_b = k_rx_port_b, /**< Port B (GPIO PB0-PB7, UART pins) */
00273     k_mpc_port_c = k_rx_port_c, /**< Port C (GPIO PC0-PC7, encoder inputs) */
00274     k_mpc_port_d = k_rx_port_d, /**< Port D (GPIO PD0-PD7) */
00275     k_mpc_port_e = k_rx_port_e, /**< Port E (GPIO PE0-PE7, PWM outputs) */
00276     k_mpc_port_f = k_rx_port_f, /**< Port F (partial on 144-pin: PF0-PF2, PF5) */
00277     k_mpc_port_g = k_rx_port_g, /**< Port G (not on 144-pin package) */
00278     k_mpc_port_j = k_rx_port_j, /**< Port J (partial: PJ3, PJ5 on 144-pin) */
00279 } mpc_port_number_t;
00280
00281 /**
00282 * @enum mpc_port_offset_t
00283 * @brief PFS register byte offsets for each port
00284 *
00285 * @details
00286 * PFS registers are laid out contiguously in memory with 8 bytes allocated
00287 * per port (one byte per pin), even if not all pins are physically present.
00288 * These offsets are added to the PFS base address to reach each port's
00289 * registers.
00290 *
00291 * @par Memory Layout
00292 * @verbatim
00293 *   PFS Base (0x0008C140):
00294 *   +0x00: P00PFS - P07PFS (Port 0)
00295 *   +0x08: P10PFS - P17PFS (Port 1)
00296 *   +0x10: P20PFS - P27PFS (Port 2)
00297 *   ...
00298 *   +0x50: PA0PFS - PA7PFS (Port A)
00299 *   +0x58: PB0PFS - PB7PFS (Port B)
00300 *   ...
00301 *   +0x90: PJ0PFS - PJ7PFS (Port J)
00302 * @endverbatim
00303 *
00304 * @note Each port occupies 8 bytes regardless of actual pin count
00305 * @see internal_get_pfs_register() Uses these offsets
00306 * @since Version 1.0.0
00307 */
00308 typedef enum : uint16_t {
00309     k_mpc_port0_offset = 0, /**< Port 0 offset: 0x00 (P00-P07) */
00310     k_mpc_port1_offset = 8, /**< Port 1 offset: 0x08 (P10-P17) */
00311     k_mpc_port2_offset = 16, /**< Port 2 offset: 0x10 (P20-P27) */
00312     k_mpc_port3_offset = 24, /**< Port 3 offset: 0x18 (P30-P34) */
00313     k_mpc_port4_offset = 32, /**< Port 4 offset: 0x20 (P40-P47, ADC) */
00314     k_mpc_port5_offset = 40, /**< Port 5 offset: 0x28 (P50-P57) */
00315     k_mpc_port6_offset = 48, /**< Port 6 offset: 0x30 (P60-P67) */
00316     k_mpc_port7_offset = 56, /**< Port 7 offset: 0x38 (P71-P77) */
00317     k_mpc_port8_offset = 64, /**< Port 8 offset: 0x40 (P80-P87) */
00318     k_mpc_port9_offset = 72, /**< Port 9 offset: 0x48 (P90-P97) */
00319     k_mpc_porta_offset = 80, /**< Port A offset: 0x50 (PA0-PA7, SPI) */
00320     k_mpc_portb_offset = 88, /**< Port B offset: 0x58 (PB0-PB7, UART) */
00321     k_mpc_portc_offset = 96, /**< Port C offset: 0x60 (PC0-PC7, MTU) */
00322     k_mpc_portd_offset = 104, /**< Port D offset: 0x68 (PD0-PD7) */
00323     k_mpc_porte_offset = 112, /**< Port E offset: 0x70 (PE0-PE7, PWM) */
00324     k_mpc_portf_offset = 120, /**< Port F offset: 0x78 (partial on 144-pin) */
00325     k_mpc_portg_offset = 128, /**< Port G offset: 0x80 (not on 144-pin) */
00326     k_mpc_portj_offset = 144, /**< Port J offset: 0x90 (PJ3, PJ5 on 144-pin) */
00327 } mpc_port_offset_t;
00328
00329 /**
00330 * @enum mpc_psel_constants_t
00331 * @brief PSEL field validation constants
00332 *
00333 * @details
00334 * The PSEL field in PFS registers is 5 bits wide, allowing values 0-31.
00335 * This constant defines the maximum valid value for input validation.
00336 *
00337 * @see rx_mpc_set_peripheral() Validates PSEL against this maximum
00338 * @since Version 1.0.0
00339 */
00340 typedef enum : uint8_t {
00341     k_mpc_psel_max = 31, /**< Maximum PSEL value (5 bits = 0x1F) */

```

```

00342 } mpc_psel_constants_t;
00343
00344 /**
00345  * @enum mpc_pfs_gpio_t
00346  * @brief PFS register value for GPIO mode
00347  *
00348  * @details
00349  * GPIO mode requires all bits in PFS register to be 0:
00350  * - PSEL[4:0] = 0 (no peripheral function)
00351  * - ISEL = 0 (no interrupt input)
00352  * - ASEL = 0 (digital mode, not analog)
00353  *
00354  * @see rx_mpc_set_gpio() Uses this value
00355  * @since Version 1.0.0
00356  */
00357 typedef enum : uint8_t {
00358     k_mpc_pfs_gpio = 0, /**< GPIO mode: all zeros (PSEL=0, ISEL=0, ASEL=0) */
00359 } mpc_pfs_gpio_t;
00360
00361 /**
00362  * @enum mpc_pwpr_t
00363  * @brief PWPR register control values
00364  *
00365  * @details
00366  * Values used to control the PWPR (Write Protect Register) during the
00367  * unlock/lock sequence. Clearing PWPR is the first step in both sequences.
00368  *
00369  * @see internal_mpc_unlock() Uses k_mpc_pwpr_clear then k_mpc_pwpr_pfswe
00370  * @see internal_mpc_lock() Uses k_mpc_pwpr_clear then k_mpc_pwpr_b0wi
00371  * @since Version 1.0.0
00372  */
00373 typedef enum : uint8_t {
00374     k_mpc_pwpr_clear = 0, /**< Clear all PWPR bits (first step of unlock/lock) */
00375 } mpc_pwpr_t;
00376
00377 /*
=====
00378  * Internal Helper Functions
00379  *
=====
00380  */
00381
00382 /**
00383  * @brief Calculate PFS register address for a given port and pin
00384  *
00385  * @details
00386  * Computes the memory address of the Pin Function Select (PFS) register
00387  * for the specified port and pin combination. Uses a lookup table (switch)
00388  * to get the port offset, then adds the pin number.
00389  *
00390  * @par Algorithm Steps
00391  * 1. Validate pin number is in range [0, 7]
00392  * 2. Get base address of PFS registers from mpc() accessor
00393  * 3. Look up port offset in switch statement
00394  * 4. If port invalid, log error and return nullptr
00395  * 5. Return base + offset + pin
00396  *
00397  * @par Address Calculation
00398  * @verbatim
00399  *   PFS_addr = (mpc() + k_mpc_pfs_base_offset) + port_offset + pin
00400  *
00401  *   Example: PB7 (Port B, Pin 7)
00402  *   = (0x0008C100 + 64) + 88 + 7
00403  *   = 0x0008C140 + 88 + 7
00404  *   = 0x0008C19F (PB7PFS)
00405  * @endverbatim
00406  *
00407  *
00408  *
00409  * @pre PCLKB clock must be running for MPC register access
00410  * @pre Caller has verified channel is in valid range for the target package
00411  *
00412  * @post No registers modified (read-only address calculation)
00413  *
00414  * @note Internal function - not exported in header
00415  * @note Port G is explicitly rejected (returns NULL) since it is absent on the 144-pin LFQFP package
00416  * @note Port H has no named constant in rx_port_constants.h; a raw value in that range falls to the default case and
    returns NULL
00417  *
00418  * @warning Returned pointer is volatile - must dereference with care
00419  * @warning Caller must still unlock PWPR before writing to the register
00420  *
00421  * @par Performance
00422  * - Execution time: ~200 ns @ 240 MHz (switch lookup + pointer math)
00423  * - Stack usage: 8 bytes (local variables)
00424  *
00425  * @see mpc() MPC register accessor

```

```

00426 * @see mpc_port_offset_t Port offset constants
00427 *
00428 * @since Version 1.0.0
00429 */
00430 /**
00431 * @brief Sentinel value indicating a port that is not pin-muxable
00432 *
00433 * @details
00434 * Used in the s_port_pfs_offsets table to flag ports that are unavailable
00435 * on the 144-pin LFQFP package (port G, port H). Looking up such a port
00436 * returns nullptr from internal_get_pfs_register().
00437 *
00438 * @since Version 1.0.0
00439 */
00440 typedef enum : uint16_t {
00441     k_mpc_port_offset_invalid = 0xFFFFU, /**< @brief Port not available on this package */
00442 } mpc_port_offset_sentinel_t;
00443
00444 /**
00445 * @brief Resolve a port enum to its PFS-register offset
00446 *
00447 * @details
00448 * Lookup table indexed by k_mpc_port_* values. Returns
00449 * k_mpc_port_offset_invalid for ports unavailable on this package.
00450 * Replaces the prior 60+ line switch statement that this function used.
00451 *
00452 * @param[in] port Port enum value (k_mpc_port_0 .. k_mpc_port_j)
00453 *
00454 * @return uint16_t PFS port offset, or k_mpc_port_offset_invalid if absent
00455 *
00456 * @pre None
00457 * @post Return value is either a valid offset or the sentinel
00458 *
00459 * @note Pure function, no side effects.
00460 *
00461 * @since Version 1.0.0
00462 */
00463 static uint16_t internal_mpc_port_offset(const uint8_t port)
00464 {
00465     switch (port) {
00466         case k_mpc_port_0:
00467             return k_mpc_port0_offset;
00468         case k_mpc_port_1:
00469             return k_mpc_port1_offset;
00470         case k_mpc_port_2:
00471             return k_mpc_port2_offset;
00472         case k_mpc_port_3:
00473             return k_mpc_port3_offset;
00474         case k_mpc_port_4:
00475             return k_mpc_port4_offset;
00476         case k_mpc_port_5:
00477             return k_mpc_port5_offset;
00478         case k_mpc_port_6:
00479             return k_mpc_port6_offset;
00480         case k_mpc_port_7:
00481             return k_mpc_port7_offset;
00482         case k_mpc_port_8:
00483             return k_mpc_port8_offset;
00484         case k_mpc_port_9:
00485             return k_mpc_port9_offset;
00486         case k_mpc_port_a:
00487             return k_mpc_porta_offset;
00488         case k_mpc_port_b:
00489             return k_mpc_portb_offset;
00490         case k_mpc_port_c:
00491             return k_mpc_portc_offset;
00492         case k_mpc_port_d:
00493             return k_mpc_portd_offset;
00494         case k_mpc_port_e:
00495             return k_mpc_porte_offset;
00496         case k_mpc_port_f:
00497             return k_mpc_portf_offset;
00498         case k_mpc_port_j:
00499             return k_mpc_portj_offset;
00500         case k_mpc_port_g: /* not on 144-pin LFQFP */
00501             default:
00502                 return k_mpc_port_offset_invalid;
00503     }
00504 }
00505
00506 static volatile uint8_t* internal_get_pfs_register(const uint8_t port, uint8_t pin)
00507 {
00508     if (pin > k_mpc_max_pin) {
00509         return nullptr;
00510     }
00511
00512     const uint16_t port_offset = internal_mpc_port_offset(port);

```

```

00513 if (port_offset == (uint16_t)k_mpc_port_offset_invalid) {
00514     rx_log_error(s_tag,
00515         (port == k_mpc_port_g) ? "Port not available on this package" : "Invalid port");
00516     return nullptr;
00517 }
00518
00519 /* PFS registers start at MPC base + 0x40 (offset for P00PFS). Each port
00520  * has 8 pins laid out P00-P07, P10-P17, P20-P27, etc. */
00521 volatile uint8_t* pfs_base = (volatile uint8_t*)mpc() + k_mpc_pfs_base_offset;
00522 return pfs_base + port_offset + pin;
00523 }
00524
00525 /**
00526  * @brief Unlock MPC PFS register write protection
00527  *
00528  * @details
00529  * Executes the two-step unlock sequence required before writing to any
00530  * PFS register. The PWPR (Write Protect Register) guards against accidental
00531  * modification of pin functions.
00532  *
00533  * @par Unlock Sequence (per RX72N Hardware Manual Ch20)
00534  * 1. Write 0x00 to PWPR (clears B0WI bit, allows PFSWE modification)
00535  * 2. Write 0x40 to PWPR (sets PFSWE bit, enables PFS register writes)
00536  *
00537  * @par Timing
00538  * - Two 8-bit register writes
00539  * - No wait states required between writes
00540  * - Total time: ~100 ns @ 240 MHz
00541  *
00542  * @pre MPC module must be active (not in module stop)
00543  *
00544  * @post PWPR.B0WI = 0 (PFSWE modifiable)
00545  * @post PWPR.PFSWE = 1 (PFS registers writable)
00546  *
00547  * @note Must call internal_mpc_lock() after PFS modification!
00548  * @note Internal function - not exported in header
00549  *
00550  * @warning Not thread-safe - window between unlock and lock is vulnerable
00551  * @warning Failing to lock after modification leaves PFS unprotected
00552  *
00553  * @par Example:
00554  * @code
00555  * // Typical usage: unlock, modify PFS, then lock immediately
00556  * internal_mpc_unlock();
00557  * *pfs_reg = k_psel_sci_tx; // Write peripheral select value
00558  * internal_mpc_lock();
00559  * @endcode
00560  *
00561  * @see internal_mpc_lock() Must be called after PFS writes
00562  * @see k_mpc_pwpr_pfswe PFSWE enable constant (0x40)
00563  *
00564  * @since Version 1.0.0
00565  */
00566 static void internal_mpc_unlock(void)
00567 {
00568     /* Unlock sequence:
00569     * 1. Clear B0WI to allow PFSWE write
00570     * 2. Set PFSWE to allow PFS write
00571     */
00572     mpc()->pwpr = k_mpc_pwpr_clear; /* Clear B0WI */
00573     mpc()->pwpr = k_mpc_pwpr_pfswe; /* Set PFSWE */
00574 }
00575
00576 /**
00577  * @brief Lock MPC PFS register write protection
00578  *
00579  * @details
00580  * Executes the two-step lock sequence to protect PFS registers after
00581  * modification. This restores write protection and prevents accidental
00582  * changes to pin functions.
00583  *
00584  * @par Lock Sequence (per RX72N Hardware Manual Ch20)
00585  * 1. Write 0x00 to PWPR (clears PFSWE bit, protects PFS registers)
00586  * 2. Write 0x80 to PWPR (sets B0WI bit, prevents PFSWE modification)
00587  *
00588  * @par Timing
00589  * - Two 8-bit register writes
00590  * - No wait states required between writes
00591  * - Total time: ~100 ns @ 240 MHz
00592  *
00593  * @pre internal_mpc_unlock() was called previously
00594  * @pre PFS modifications are complete
00595  *
00596  * @post PWPR.PFSWE = 0 (PFS registers protected)
00597  * @post PWPR.B0WI = 1 (PFSWE bit protected)
00598  *
00599  * @note Internal function - not exported in header

```

```

00600 *
00601 * @par Example:
00602 * @code
00603 * // Typical usage: unlock, modify PFS, then lock immediately
00604 * internal_mpc_unlock();
00605 * *pfs_reg = k_psel_rspi_clk; // Write peripheral select value
00606 * internal_mpc_lock(); // Restore write protection
00607 * @endcode
00608 *
00609 * @see internal_mpc_unlock() Must be called before PFS writes
00610 * @see k_mpc_pwpr_b0wi B0WI enable constant (0x80)
00611 *
00612 * @since Version 1.0.0
00613 */
00614 static void internal_mpc_lock(void)
00615 {
00616     /* Lock sequence:
00617     * 1. Clear PFSWE to protect PFS
00618     * 2. Set B0WI to protect PFSWE
00619     */
00620     mpc()->pwpr = k_mpc_pwpr_clear; /* Clear PFSWE */
00621     mpc()->pwpr = k_mpc_pwpr_b0wi; /* Set B0WI */
00622 }
00623
00624 /**
00625 * @brief Write value to PFS register with automatic unlock/lock
00626 *
00627 * @details
00628 * Complete PFS modification sequence: calculates register address, unlocks
00629 * protection, writes the value, and re-locks protection. This function
00630 * handles the entire write-protected register access pattern.
00631 *
00632 * @par Algorithm Steps
00633 * 1. Call internal_get_pfs_register() to get register address
00634 * 2. If address is nullptr, return error (no unlock performed)
00635 * 3. Call internal_mpc_unlock() to enable PFS writes
00636 * 4. Write value to PFS register
00637 * 5. Call internal_mpc_lock() to protect PFS registers
00638 * 6. Return success
00639 *
00640 * @par Execution Flow
00641 * @dot
00642 * digraph write_pfs {
00643 *     rankdir=TB;
00644 *     node [shape=box, style=rounded];
00645 *
00646 *     start [label="internal_write_pfs()"];
00647 *     get_reg [label="internal_get_pfs_register()"];
00648 *     check [label="pfs_reg == nullptr?", shape=diamond];
00649 *     error [label="return k_rx_err_invalid_arg", fillcolor=lightcoral, style=filled];
00650 *     unlock [label="internal_mpc_unlock()"];
00651 *     write [label="*pfs_reg = value"];
00652 *     lock [label="internal_mpc_lock()"];
00653 *     success [label="return k_rx_ok", fillcolor=lightgreen, style=filled];
00654 *
00655 *     start -> get_reg;
00656 *     get_reg -> check;
00657 *     check -> error [label="yes"];
00658 *     check -> unlock [label="no"];
00659 *     unlock -> write;
00660 *     write -> lock;
00661 *     lock -> success;
00662 * }
00663 * @enddot
00664 *
00665 *
00666 *
00667 * @pre PCLKB clock must be running
00668 * @pre Value should be a valid PFS configuration
00669 *
00670 * @post PFS register contains specified value
00671 * @post PWPR is locked (B0WI=1)
00672 *
00673 * @note Internal function - not exported in header
00674 * @note Lock is always executed even on error path (defensive coding)
00675 *
00676 * @warning Not thread-safe - the unlock/write/lock sequence is non-atomic
00677 *
00678 * @par Performance
00679 * - Execution time: ~500 ns @ 240 MHz
00680 * - Stack usage: 16 bytes
00681 *
00682 * @see internal_get_pfs_register() Address calculation
00683 * @see internal_mpc_unlock() Protection unlock
00684 * @see internal_mpc_lock() Protection lock
00685 *
00686 * @since Version 1.0.0

```

```

00687 */
00688 static rx_err_t internal_write_pfs(const uint8_t port, uint8_t pin, uint8_t value)
00689 {
00690     volatile uint8_t* pfs_reg = internal_get_pfs_register(port, pin);
00691     if (pfs_reg == nullptr) {
00692         return k_rx_err_invalid_arg;
00693     }
00694
00695     /* Unlock, write, lock */
00696     internal_mpc_unlock();
00697     *pfs_reg = value;
00698     internal_mpc_lock();
00699
00700     return k_rx_ok;
00701 }
00702
00703 /*
00704 * Public API Implementation
00705 *
00706 */
00707
00708 /**
00709 * @brief Configure pin for GPIO (General Purpose I/O) function
00710 *
00711 * @details
00712 * Sets the specified pin to GPIO mode by writing 0x00 to its PFS register.
00713 * This disconnects the pin from all peripheral functions and allows it to
00714 * be controlled by the PORT registers (PDR, PODR, PIDR).
00715 *
00716 * @par Implementation Details
00717 * 1. Extract port and pin number from encoded rx_port_pin_t
00718 * 2. Validate port is in range [0, J]
00719 * 3. Validate pin is in range [0, 7]
00720 * 4. Write 0x00 to PFS register (via internal_write_pfs)
00721 *
00722 *
00723 *
00724 * @pre PCLKB clock running, MPC not in module stop
00725 * @pre pin must be a valid rx_port_pin_t constant
00726 *
00727 * @post PFS register = 0x00 (GPIO mode)
00728 * @post PWPR locked after operation
00729 *
00730 * @note Thread safety: Not thread-safe
00731 * @note Also set PORT PMR bit to 0 for complete GPIO configuration
00732 *
00733 * @code
00734 * // Configure P30 (port 3, pin 0) for GPIO use
00735 * rx_err_t err = rx_mpc_set_gpio(RX_PORT_PIN(3, 0));
00736 * if (err != k_rx_ok) {
00737 *     // handle error
00738 * }
00739 * @endcode
00740 *
00741 * @see rx_mpc.h Full API documentation
00742 * @since Version 1.0.0
00743 */
00744 rx_err_t rx_mpc_set_gpio(const rx_port_pin_t pin)
00745 {
00746     /* Extract port and pin number from rx_port_pin_t */
00747     /* Pre-condition: port and pin must be in valid range */
00748     const uint8_t port = rx_port_from_pin(pin);
00749     const uint8_t pin_num = rx_pin_from_pin(pin);
00750
00751     RX_CHECK_RANGE_TAG(port, k_mpc_port_start, k_mpc_port_j, k_rx_err_invalid_arg, s_tag);
00752     RX_CHECK_RANGE_TAG(pin_num, k_mpc_min_pin, k_mpc_max_pin, k_rx_err_invalid_arg, s_tag);
00753
00754     /* GPIO mode: PSEL = 0, ISEL = 0, ASEL = 0 */
00755     return internal_write_pfs(port, pin_num, k_mpc_pfs_gpio);
00756 }
00757
00758 /**
00759 * @brief Configure pin for peripheral function with specified PSEL code
00760 *
00761 * @details
00762 * Core function for peripheral pin configuration. Writes the specified
00763 * PSEL value to the pin's PFS register, connecting it to a peripheral.
00764 *
00765 * @par Implementation Details
00766 * 1. Validate config pointer is not nullptr
00767 * 2. Validate PSEL value is in range [0, 31]
00768 * 3. Extract port and pin from config->pin
00769 * 4. Validate extracted values
00770 * 5. Write PSEL to PFS register (via internal_write_pfs)
00771 *

```

```

00772 *
00773 *
00774 * @pre config must be non-NULL with valid fields
00775 * @pre PCLKB clock running
00776 *
00777 * @post PFS register contains config->psel
00778 * @post PWPR locked after operation
00779 *
00780 * @note Thread safety: Not thread-safe
00781 * @note Also set PORT PMR bit to 1 for peripheral mode
00782 *
00783 * @code
00784 * // Configure PA0 for RSPI clock (PSEL = 0x0D)
00785 * const rx_mpc_peripheral_config_t cfg = {
00786 *     .pin = RX_PORT_PIN(0xA, 0),
00787 *     .psel = 0x0D,
00788 * };
00789 * rx_err_t err = rx_mpc_set_peripheral(&cfg);
00790 * if (err != k_rx_ok) {
00791 *     // handle error
00792 * }
00793 * @endcode
00794 *
00795 * @see rx_mpc.h Full API documentation
00796 * @since Version 1.0.0
00797 */
00798 rx_err_t rx_mpc_set_peripheral(const rx_mpc_peripheral_config_t* config)
00799 {
00800     if (config == nullptr) {
00801         rx_log_error(s_tag, "Invalid configuration pointer");
00802         return k_rx_err_invalid_arg;
00803     }
00804
00805     if (config->psel > k_mpc_psel_max) {
00806         rx_log_error(s_tag, "Invalid PSEL value exceeds maximum");
00807         rx_log_error_val(s_tag, "psel=", config->psel);
00808         rx_log_error_val(s_tag, "k_mpc_psel_max=", (uint8_t)k_mpc_psel_max);
00809         return k_rx_err_invalid_arg;
00810     }
00811
00812     /* Extract port and pin number from rx_port_pin_t */
00813     const uint8_t port = rx_port_from_pin(config->pin);
00814     const uint8_t pin_num = rx_pin_from_pin(config->pin);
00815
00816     /* Validate the decoded port and pin are in valid range */
00817     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
00818         rx_log_error(s_tag, "Invalid port or pin number");
00819         rx_log_error_val(s_tag, "port=", port);
00820         rx_log_error_val(s_tag, "pin=", pin_num);
00821         return k_rx_err_invalid_arg;
00822     }
00823
00824     /* Peripheral mode: PSEL = specified, ISEL = 0, ASEL = 0.
00825     *
00826     * NOTE: PORT.PMR bit is intentionally NOT set here. Setting PMR=1 before
00827     * the peripheral itself (e.g. GPTW timer) is fully configured exposes
00828     * the pin to undefined/glitchy output that can latch a fault in
00829     * downstream silicon (e.g. DRV8263H motor driver). Caller is responsible
00830     * for setting PMR=1 AFTER the peripheral is fully initialized. */
00831     return internal_write_pfs(port, pin_num, config->psel);
00832 }
00833
00834 /**
00835 * @brief Configure pin for MTU3a timer PWM output (MTIOC)
00836 *
00837 * @details
00838 * Convenience wrapper that configures a pin for MTU I/O compare output
00839 * using PSEL = k_psel_mtu_ioc (0x01). Commonly used for motor PWM.
00840 *
00841 *
00842 *
00843 * @pre PCLKB clock must be running, MPC not in module stop
00844 * @pre pin must encode a valid port/pin combination supported by MTU MTIOC
00845 *
00846 * @post PFS register contains k_psel_mtu_ioc (0x01) for the specified pin
00847 * @post PWPR locked after operation
00848 *
00849 * @note Thread safety: Not thread-safe
00850 * @note STAR project uses GPTW for motor PWM (see rx_mpc_set_gptw(), PSEL = k_psel_gptw (0x1E)), not MTU
00851 *
00852 * @code
00853 * // Configure P14 for MTU3 PWM output (MTIOCA)
00854 * rx_err_t err = rx_mpc_set_mtu_pwm(RX_PORT_PIN(1, 4));
00855 * if (err != k_rx_ok) {
00856 *     // handle error
00857 * }
00858 * @endcode

```



```

00859 *
00860 * @see rx_mpc.h Full API documentation
00861 * @since Version 1.0.0
00862 */
00863 rx_err_t rx_mpc_set_mtu_pwm(const rx_port_pin_t pin)
00864 {
00865     /* MTU PWM (MTIOC) pins typically use PSEL = 0x01
00866     * Common pins:
00867     * - P14-P17: MTU3 MTIOCA/B/C/D
00868     * - P24-P27: MTU4 MTIOCA/B/C/D
00869     */
00870     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_ioc};
00871
00872     return rx_mpc_set_peripheral(&config);
00873 }
00874
00875 /**
00876 * @brief Configure pin for MTU3a encoder/phase counter input (MTCLK)
00877 *
00878 * @details
00879 * Convenience wrapper that configures a pin for MTU external clock / phase
00880 * counting mode using PSEL = k_psel_mtu_phase (0x02). Used for quadrature
00881 * encoders on pins that expose MTCLKA/B/C/D.
00882 *
00883 *
00884 *
00885 * @pre PCLKB clock must be running, MPC not in module stop
00886 * @pre pin must encode a valid port/pin combination supported by MTCLK input
00887 *
00888 * @post PFS register contains k_psel_mtu_phase (0x02) for the specified pin
00889 * @post PWPR locked after operation
00890 *
00891 * @note Thread safety: Not thread-safe
00892 * @note Configure both Phase A and Phase B pins for proper operation
00893 *
00894 * @code
00895 * // Configure P24 and P25 for MTU phase counting (MTCLKA/B)
00896 * rx_err_t err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 4));
00897 * if (err == k_rx_ok) {
00898 *     err = rx_mpc_set_mtu_encoder(RX_PORT_PIN(2, 5));
00899 * }
00900 * @endcode
00901 *
00902 * @see rx_mpc.h Full API documentation
00903 * @see rx_mtu_encoder.h Encoder driver using MTU phase counting
00904 * @since Version 1.0.0
00905 */
00906 rx_err_t rx_mpc_set_mtu_encoder(const rx_port_pin_t pin)
00907 {
00908     /* MTU external clock / phase counting pins use PSEL = 0x02 on every
00909     * RX72N MTCLK candidate pin. See RX72N Group Hardware Manual
00910     * (R01UH0824EJ0111), chapter 23 "Multi-Function Pin Controller (MPC)",
00911     * Tables 23.5 / 23.6 / 23.17 / 23.19 -- every row that selects MTCLKA,
00912     * MTCLKB, MTCLKC, or MTCLKD on a 144-pin LFQFP pin is PSEL = 000010b.
00913     * The old value 0x03 was wrong -- it caused MTU TCNT to stay at 0x0000
00914     * because no MTCLK source was actually connected through the PFS mux.
00915     *
00916     * Common STAR pins:
00917     * P24/P25 (Motor 0 front-right MTCLKA/B)
00918     * PA1/PC5 (Motor 1 front-left MTCLKC/D)
00919     */
00920     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_mtu_phase};
00921
00922     return rx_mpc_set_peripheral(&config);
00923 }
00924
00925 /**
00926 * @brief Configure pin for TPU encoder/phase counter input (TCLK)
00927 *
00928 * @details
00929 * Convenience wrapper that configures a pin for TPU external clock input
00930 * (TCLKA/B/C/D). Picks the correct PSEL value based on which port the
00931 * pin is on, because RX72N uses two different alternate-function slots
00932 * for the SAME TCLK signal depending on candidate:
00933 * - Port C pins (PC0/PC1/PC2/PC3) use PSEL = 0x03 (k_psel_tpu_clk_portc)
00934 * - Port 1 / Port A / Port B pins use PSEL = 0x04 (k_psel_tpu_clk_alt)
00935 *
00936 * Reference: RX72N Group Hardware Manual (R01UH0824EJ0111), chapter 23
00937 * "Multi-Function Pin Controller (MPC)". TCLK rows in Table 23.19 (Port C)
00938 * use PSEL = 000011b, while TCLK rows in Tables 23.5 (P1x), 23.17 (PAx),
00939 * and 23.18 (PBx) use PSEL = 000100b. Before this split, every TPU pin was
00940 * configured with PSEL=0x03, which silently misconfigured the non-Port-C
00941 * halves of STAR's rear-wheel encoders (PA3 for Motor 2 Phase B, PB3 for
00942 * Motor 3 Phase B) so those edges were never counted.
00943 *
00944 *
00945 */

```



```

00946 * @pre PCLKB clock must be running, MPC not in module stop
00947 * @pre pin must encode a valid port/pin combination supported by TCLK input
00948 *
00949 * @post PFS register contains the correct TPU-TCLK PSEL for that pin's port
00950 * @post PWPR locked after operation
00951 *
00952 * @note Thread safety: Not thread-safe
00953 * @note Configure both Phase A and Phase B pins for proper operation
00954 *
00955 * @code
00956 * // Configure PC2 (TCLKA) and PA3 (TCLKB) for Motor 2 rear-left encoder
00957 * rx_err_t err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xC, 2)); // -> 0x03
00958 * if (err == k_rx_ok) {
00959 *     err = rx_mpc_set_tpu_encoder(RX_PORT_PIN(0xA, 3)); // -> 0x04
00960 * }
00961 * @endcode
00962 *
00963 * @see rx_mpc.h Full API documentation
00964 * @see rx_encoder_tpu.h TPU encoder driver using phase counting
00965 * @since Version 1.0.0
00966 */
00967 rx_err_t rx_mpc_set_tpu_encoder(const rx_port_pin_t pin)
00968 {
00969     /* Pick the right PSEL for the TCLK function on this pin's port. */
00970     const uint8_t port = rx_port_from_pin(pin);
00971     const uint8_t psel =
00972         (port == k_mpc_port_c) ? (uint8_t)k_psel_tpu_clk_portc : (uint8_t)k_psel_tpu_clk_alt;
00973
00974     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = psel};
00975     return rx_mpc_set_peripheral(&config);
00976 }
00977
00978 /**
00979 * @brief Configure pin for ADC analog input
00980 *
00981 * @details
00982 * Configures a pin for analog-to-digital conversion by setting the ASEL
00983 * bit in the PFS register. This disables the digital input buffer to
00984 * prevent noise during analog measurement.
00985 *
00986 *
00987 *
00988 * @pre PCLKB clock must be running, MPC not in module stop
00989 * @pre pin must encode a valid port/pin combination with analog capability
00990 *
00991 * @post PFS register contains k_pfs_asel (0x80) for the specified pin
00992 * @post PWPR locked after operation
00993 *
00994 * @note Thread safety: Not thread-safe
00995 * @note Sets ASEL bit (0x80) rather than PSEL field
00996 * @note Also configure S12ADC for actual conversion
00997 *
00998 * @code
00999 * // Configure P40 (AN000) for ADC analog input
01000 * rx_err_t err = rx_mpc_set_adc(RX_PORT_PIN(4, 0));
01001 * if (err != k_rx_ok) {
01002 *     // handle error
01003 * }
01004 * @endcode
01005 *
01006 * @see rx_mpc.h Full API documentation
01007 * @since Version 1.0.0
01008 */
01009 rx_err_t rx_mpc_set_adc(const rx_port_pin_t pin)
01010 {
01011     /* Extract port and pin number from type-safe enum */
01012     const uint8_t port = rx_port_from_pin(pin);
01013     const uint8_t pin_num = rx_pin_from_pin(pin);
01014
01015     /* Validate the decoded port and pin are in valid range */
01016     if (port > k_mpc_port_j || pin_num > k_mpc_max_pin) {
01017         rx_log_error(s_tag, "Invalid port or pin number");
01018         return k_rx_err_invalid_arg;
01019     }
01020
01021     /* ADC mode: Set ASEL = 1 to disable digital I/O
01022     * Common ADC pins: P40-P47 (AN000-AN007)
01023     */
01024     return internal_write_pfs(port, pin_num, k_pfs_asel);
01025 }
01026
01027 /**
01028 * @brief Configure pin for SCI (Serial Communication Interface) UART function
01029 *
01030 * @details
01031 * Convenience wrapper that configures a pin for SCI UART operation using
01032 * PSEL = k_psel_sci_tx (0x0A). Works for both TXD and RXD pins.

```

```

01033 *
01034 *
01035 *
01036 * @pre PCLKB clock must be running, MPC not in module stop
01037 * @pre pin must encode a valid port/pin combination with SCI multiplexing
01038 *
01039 * @post PFS register contains k_psel_sci_tx (0x0A) for the specified pin
01040 * @post PWPR locked after operation
01041 *
01042 * @note Thread safety: Not thread-safe
01043 * @note Configure both TXD and RXD pins for bidirectional UART
01044 *
01045 * @code
01046 * // Configure PB7 for SCI9 TXD
01047 * rx_err_t err = rx_mpc_set_sci(RX_PORT_PIN(0xB, 7));
01048 * if (err != k_rx_ok) {
01049 *     // handle error
01050 * }
01051 * @endcode
01052 *
01053 * @see rx_mpc.h Full API documentation
01054 * @since Version 1.0.0
01055 */
01056 rx_err_t rx_mpc_set_sci(const rx_port_pin_t pin)
01057 {
01058     /* SCI pins use PSEL = 0x0A
01059     * TX and RX use the same PSEL code
01060     */
01061     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_sci_tx};
01062
01063     return rx_mpc_set_peripheral(&config);
01064 }
01065
01066 /**
01067 * @brief Configure pin for RIIC (I2C) bus function
01068 *
01069 * @details
01070 * Convenience wrapper that configures a pin for RIIC operation using
01071 * PSEL = k_psel_riic_scl (0x0F). Works for both SCL and SDA pins.
01072 *
01073 * Unlike rx_mpc_set_gptw(), this function also sets PORT.PMR=1 for the
01074 * pin before returning -- RIIC cannot drive SDA/SCL while PMR=0 and the
01075 * immediate-vs-deferred PMR trade-off differs from the motor-PWM case:
01076 *
01077 * | Peripheral | When to raise PMR | Reason |
01078 * |-----|-----|-----|
01079 * | GPTW/PWM | AFTER timer is running | Push-pull drive: an un-
01080 * | | (rx_gptw_init_pwm handles | initialized output could
01081 * | | this after rx_gptw_start) | momentarily glitch
01082 * | | | DRV8263H H-bridge inputs
01083 * | | | to Vm/2 during the
01084 * | | | GTIOR reset state window.
01085 * | RIIC/I2C | IMMEDIATELY after PFS | Open-drain + external
01086 * | | (here, inside this wrapper) | pull-ups: idle bus is
01087 * | | | both lines high (3.3V);
01088 * | | | routing an un-initialized
01089 * | | | RIIC just leaves the
01090 * | | | pull-ups floating the
01091 * | | | lines high. No hazard.
01092 *
01093 * Without PMR=1 on the RIIC pins, rx72n peripheral bits still toggle
01094 * internally but the pad stays disconnected from RIIC; every Start
01095 * condition times out with k_rx_err_timeout (0x108) at the first
01096 * riic_write()/riic_read() call. Discovered 2026-04-22: production
01097 * imu_task saw tight "[ERROR] [RIIC] Start condition timeout" loops while
01098 * imu_test/main.c (which sets PMR=1 at line 227 before calling its own
01099 * riic_init()) talked to BNO055 + BMP280 cleanly on the same board.
01100 *
01101 *
01102 *
01103 * @pre PCLKB clock must be running, MPC not in module stop
01104 * @pre pin must encode a valid port/pin combination with RIIC multiplexing
01105 *
01106 * @post PFS register contains k_psel_riic_scl (0x0F) for the specified pin
01107 * @post PORT.PMR bit for the pin is set (pin routed to RIIC peripheral)
01108 * @post PWPR locked after operation
01109 *
01110 * @note Thread safety: Not thread-safe
01111 * @note Configure both SCL and SDA pins; external pull-ups required
01112 *
01113 * @code
01114 * // Configure P12 (SCL) and P13 (SDA) for RIIC0
01115 * rx_err_t err = rx_mpc_set_riic(RX_PORT_PIN(1, 2));
01116 * if (err == k_rx_ok) {
01117 *     err = rx_mpc_set_riic(RX_PORT_PIN(1, 3));
01118 * }
01119 * @endcode

```

```

01120 *
01121 * @see rx_mpc.h Full API documentation
01122 * @see rx_mpc_set_gptw Contrast: deferred-PMR case
01123 * @see star-rx72n-firmware/imu_test/main.c::riic1_init Bench reference
01124 * @since Version 1.0.0
01125 */
01126 rx_err_t rx_mpc_set_riic(const rx_port_pin_t pin)
01127 {
01128     /* RIIC pins use PSEL = 0x0F
01129     * SCL and SDA use the same PSEL code
01130     */
01131     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_riic_scl};
01132
01133     const rx_err_t err = rx_mpc_set_peripheral(&config);
01134     if (err != k_rx_ok) {
01135         return err;
01136     }
01137
01138     /* Set PORT.PMR=1 so SCL/SDA actually route to the RIIC peripheral.
01139     *
01140     * Unlike GPTW (where PMR must be deferred until after the timer is
01141     * running to avoid exposing DRV8263H inputs to GTIOR reset state),
01142     * I2C pins are open-drain with external pull-ups: driving them high
01143     * through the peripheral mux is the idle-bus state and perfectly safe.
01144     *
01145     * Without this bit set, P20/P21 remain GPIO (PMR=0) and RIIC1 can
01146     * never toggle the lines -- every Start condition times out with
01147     * k_rx_err_timeout (0x108). The bench-proven imu_test/main.c sets
01148     * PMR identically at rx72n-firmware/imu_test/main.c:227. */
01149     const uint8_t port_idx = rx_port_from_pin(pin);
01150     const uint8_t bit = rx_pin_from_pin(pin);
01151     volatile rx_port_regs_t* const port = rx_port_get_base(port_idx);
01152     if (port == nullptr) {
01153         return k_rx_err_invalid_arg;
01154     }
01155     port->pmr |= (uint8_t)(1U << bit);
01156     return k_rx_ok;
01157 }
01158
01159 /**
01160 * @brief Configure pin for RSPI (SPI) bus function
01161 *
01162 * @details
01163 * Convenience wrapper that configures a pin for RSPI operation using
01164 * PSEL = k_psel_rspi_clk (0x0D). Works for all SPI signals (CLK, COPI,
01165 * CIPO, SSL).
01166 *
01167 *
01168 *
01169 * @pre PCLKB clock must be running, MPC not in module stop
01170 * @pre pin must encode a valid port/pin combination with RSPI multiplexing
01171 *
01172 * @post PFS register contains k_psel_rspi_clk (0x0D) for the specified pin
01173 * @post PWPR locked after operation
01174 *
01175 * @note Thread safety: Not thread-safe
01176 * @note Configure all 4 SPI pins (CLK, COPI, CIPO, CS)
01177 * @note Uses OSHWA-approved inclusive terminology
01178 *
01179 * @code
01180 * // Configure PA0-PA4 for RSPI0 (CLK, COPI, CIPO, SSL0, SSL1)
01181 * rx_err_t err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 0)); // CLK
01182 * if (err == k_rx_ok) {
01183 *     err = rx_mpc_set_rspi(RX_PORT_PIN(0xA, 1)); // COPI
01184 * }
01185 * @endcode
01186 *
01187 * @see rx_mpc.h Full API documentation
01188 * @see rx_spi_comm.h SPI communication driver
01189 * @since Version 1.0.0
01190 */
01191 rx_err_t rx_mpc_set_rspi(const rx_port_pin_t pin)
01192 {
01193     /* RSPI pins use PSEL = 0x0D
01194     * All RSPI signals (CLK, COPI, CIPO, SSL) use same PSEL
01195     */
01196     const rx_mpc_peripheral_config_t config = {.pin = pin, .psel = k_psel_rspi_clk};
01197
01198     return rx_mpc_set_peripheral(&config);
01199 }
01200
01201 /**
01202 * @brief Configure pin for GPTW complementary PWM output
01203 *
01204 * @details
01205 * Convenience wrapper that configures a pin for GPTW operation using
01206 * PSEL = k_psel_gptw (0x1E). GPTW provides complementary PWM output with dead-time

```

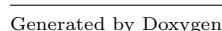
```

01207 * insertion for motor control.
01208 *
01209 *
01210 *
01211 * @pre PCLKB clock must be running, MPC not in module stop
01212 * @pre pin must encode a valid port/pin combination with GPTW multiplexing
01213 *
01214 * @post PFS register contains k_psel_gptw (0x1E) for the specified pin
01215 * @post PWPR locked after operation
01216 *
01217 * @note Thread safety: Not thread-safe
01218 * @note Used for 4 GPTW channels (0-3) with 8 total pins (IN2 + IN1 per motor)
01219 * @note Phase staggering configured separately via rx_gptw driver
01220 *
01221 * @code
01222 * // Configure motor 2 GPTW IN2 and IN1 pins (Port E)
01223 * rx_err_t err = rx_mpc_set_gptw(RX_PORT_PIN(0xE, 3)); // PE3 GTIOC2A (IN2)
01224 * if (err == k_rx_ok) {
01225 *     err = rx_mpc_set_gptw(RX_PORT_PIN(8, 6)); // P8.6 GTIOC2B (IN1)
01226 * }
01227 * @endcode
01228 *
01229 * @see rx_mpc.h Full API documentation with usage examples
01230 * @see rx_gptw_init_all_staggered() GPTW channel configuration
01231 *
01232 * @since Version 1.0.0
01233 */
01234 rx_err_t rx_mpc_set_gptw(const rx_port_pin_t pin)
01235 {
01236     /* GPTW pins use PSEL = 0x1E
01237     * Supports 4 GPTW channels (0-3) for motor IN2/IN1 control */
01238     const rx_mpc_peripheral_config_t config = {
01239         .pin = pin,
01240         .psel = k_psel_gptw /* 0x1E - defined in rx_pin_psel_t enum */
01241     };
01242
01243     return rx_mpc_set_peripheral(&config);
01244 }
01245
01246 /**
01247 * @brief Configure pin for USB VBUS detection
01248 *
01249 * @details
01250 * Convenience wrapper that configures a pin for USB VBUS detection using
01251 * PSEL = 0x11. Required for USB CDC enumeration and power monitoring.
01252 *
01253 *
01254 *
01255 * @pre PCLKB clock must be running, MPC not in module stop
01256 * @pre pin must encode a valid port/pin combination with USB VBUS multiplexing
01257 *
01258 * @post PFS register contains k_psel_usb_vbus (0x11) for the specified pin
01259 * @post PWPR locked after operation
01260 *
01261 * @note Thread safety: Not thread-safe
01262 * @note Active-high VBUS detect (pin = 1 when 5V present)
01263 * @note Also requires USB PHY and controller configuration
01264 *
01265 * @code
01266 * // Configure P16 for USB0_VBUS detection
01267 * rx_err_t err = rx_mpc_set_usb_vbus(RX_PORT_PIN(1, 6));
01268 * if (err != k_rx_ok) {
01269 *     // handle error
01270 * }
01271 * @endcode
01272 *
01273 * @see rx_mpc.h Full API documentation with usage examples
01274 * @see rx_usb.h USB controller driver
01275 *
01276 * @since Version 1.0.0
01277 */
01278 rx_err_t rx_mpc_set_usb_vbus(const rx_port_pin_t pin)
01279 {
01280     /* USB VBUS detect uses PSEL = 0x11
01281     * Typical pin: P1.6 for USB0_VBUS on RX72N */
01282     const rx_mpc_peripheral_config_t config = {
01283         .pin = pin,
01284         .psel = k_psel_usb_vbus /* 0x11 - defined in rx_pin_psel_t enum */
01285     };
01286
01287     return rx_mpc_set_peripheral(&config);
01288 }
01289
01290 /**
01291 * @brief Configure pin for IRQ (external interrupt) function
01292 *
01293 * @details

```

8.129 `libs/rx-hal/src/rx-mtu.c` File Reference

```
#include "rx_mtu.h"
#include <stddef.h>
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_register_protection.h"
Include dependency graph for rx_mtu.c:
```



Enumerations

- enum `mtu_constants_t` : `uint8_t` { `k_mtu_max_channels` = 8 }
MTU general constants.
- enum `mtu_module_stop_bits_t` : `uint8_t` { `k_mtu_mstpa_mtu0_4` = 9 }
MTU module stop bit position in MSTPCRA.
- enum `mtu_bit_constants_t` : `uint32_t` { `k_mtu_bit_one` = 1UL }
Bit manipulation constants.
- enum `mtu_period_constants_t` : `uint16_t` { `k_mtu_period_divisor` = 2 , `k_mtu_period_max` = 0xFFFF , `k_mtu_period_min` = 10 , `k_mtu_period_zero` = 0 }
Period calculation constants.
- enum `mtu_tior_shift_t` : `uint8_t` { `k_mtu_tior_low_shift` = 0 , `k_mtu_tior_high_shift` = 4 }
TIOR register shift positions.
- enum `mtu_tior_mask_t` : `uint8_t` { `k_mtu_tior_keep_high_clear_low` = 0xF0 , `k_mtu_tior_keep_low_clear_high` = 0x0F }
TIOR register clear-masks (value AND'd against TIOR to zero a nibble).
- enum `mtu_tior_disabled_t` : `uint8_t` { `k_mtu_tior_disabled` = 0x00 }
TIOR output disabled value.
- enum `mtu_duty_constants_t` : `uint8_t` { `k_mtu_duty_min` = 0 , `k_mtu_duty_max` = 100 , `k_mtu_duty_divisor` = 100 }
Duty cycle calculation constants.

Functions

- static volatile void * `internal_get_mtu_base` (const `rx_mtu_channel_t` channel)
Get MTU channel register base address.
- static bool `internal_is_valid_channel` (const `rx_mtu_channel_t` channel)
Check if MTU channel is valid.
- static `rx_err_t` `internal_clear_tstr_bit` (volatile `rx_mtu_tstr_regs_t` *tstr, const `uint8_t` mask)
Clear a counter-start bit in a TSTR register and verify it cleared.
- static `rx_err_t` `internal_calculate_period` (const `uint32_t` frequency_hz, `uint16_t` *period)
Calculate period register value from frequency.
- static volatile `uint16_t` * `internal_get_tgr_register` (volatile `rx_mtu_channel_regs_t` *mtu, const `rx_mtu_output_t` output)
Get TGR register pointer for output channel.
- static `rx_err_t` `internal_set_duty_raw_mtu` (volatile `rx_mtu_channel_regs_t` *mtu, const `rx_mtu_output_t` output, const `uint16_t` duty_count)
Write a raw duty count to the TGR register for the specified output.
- `rx_err_t` `rx_mtu_init_pwm` (const `rx_mtu_channel_t` channel, const `rx_mtu_config_t` *config)
Initialize MTU channel for PWM operation.
- `rx_err_t` `rx_mtu_set_duty` (const `rx_mtu_channel_t` channel, `rx_mtu_output_t` output, const float duty_percent)
Set PWM duty cycle as a floating-point percentage.
- `rx_err_t` `rx_mtu_set_duty_raw` (const `rx_mtu_channel_t` channel, `rx_mtu_output_t` output, `uint16_t` duty_count)
Set PWM duty cycle using a raw 16-bit timer count.
- `rx_err_t` `rx_mtu_get_duty` (const `rx_mtu_channel_t` channel, const `rx_mtu_output_t` output, float *duty_percent)
Read the current PWM duty cycle as a floating-point percentage.
- `rx_err_t` `rx_mtu_get_period` (const `rx_mtu_channel_t` channel, `uint16_t` *period_count)
Get PWM period count for MTU channel.

- rx_err_t [rx_mtu_enable_output](#) (const [rx_mtu_channel_t](#) channel, [rx_mtu_output_t](#) output, const bool enable)
Enable or disable a single MTU PWM output pin.
- rx_err_t [rx_mtu_start](#) (const [rx_mtu_channel_t](#) channel)
Start MTU counter (begin PWM generation).
- rx_err_t [rx_mtu_stop](#) (const [rx_mtu_channel_t](#) channel)
Stop MTU counter (halt PWM generation).
- rx_err_t [rx_mtu_deinit](#) (const [rx_mtu_channel_t](#) channel)
Deinitialize an MTU PWM channel.

Variables

- static const char *const [s_tag](#) = "MTU"
- static bool [s_mtu_initialized](#) [[k_mtu_max_channels](#)] = {false}
Track which MTU channels have been initialized.
- static uint16_t [s_mtu_period](#) [[k_mtu_max_channels](#)] = {}
Cached PWM period register values for each MTU channel.

8.129.1 Detailed Description

MTU PWM Driver Implementation for RX72N.

Production-grade implementation of the Multi-Function Timer Unit (MTU) peripheral driver for PWM generation on RX72N. The MTU provides 16-bit timer resolution with multiple outputs per channel.

8.129.1.1 Module Architecture

The MTU peripheral is organized as:

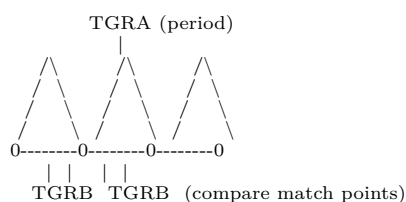
- MTU0-MTU4: First group, controlled by MSTPCRA.MSTPA9 and TSTRA
- MTU6-MTU7: Second group, controlled by MSTPCRA.MSTPA9 and TSTRB
- Note: MTU5 exists but is excluded from general PWM use

8.129.1.2 PWM Mode 1 (Triangle Wave - Center-Aligned)

This driver configures MTU in PWM Mode 1 for center-aligned PWM:

- Counter counts up from 0 to TGRA (period), then down to 0
- Compare matches on TGRB/TGRC/TGRD control duty cycle
- Symmetric PWM pulses reduce harmonic content
- Lower EMI compared to edge-aligned PWM

PWM Timing Diagram:



8.129.1.3 Frequency and Resolution

Calculation:

$$\text{Period} = \frac{f_{PCLKA}}{2 \times f_{PWM}}$$

Example: 20kHz PWM with PCLKA=120MHz:

Parameter	Value	Notes
Period	3000 counts	120MHz / (2 x 20kHz)
Resolution	~11.5 bits	$\log_2(3000) \sim 11.5$
Duty range	0-3000	0-100% in integer counts
Step size	0.033%	100% / 3000

8.129.1.4 Register Base Addresses

Channel	Base Address	TSTR Register
MTU0	0x000C1300	TSTRA (bit 0)
MTU1	0x000C1380	TSTRA (bit 1)
MTU2	0x000C1400	TSTRA (bit 2)
MTU3	0x000C1200	TSTRA (bit 6)
MTU4	0x000C1201	TSTRA (bit 7)
MTU6	0x000C1A00	TSTRB (bit 6)
MTU7	0x000C1A01	TSTRB (bit 7)

8.129.1.5 Static Variables

Variable	Type	Purpose
s_mtu_initialized[]	bool↔ [8]	Track which channels are initialized
s_mtu_period[]	uint16_t↔ [8]	Cache period values for duty calculation
s_tag	const char*	Log tag "MTU"

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	↔ [OK]↔	No goto/setjmp/recursion
2. Fixed loop bounds	↔ [OK]↔	No unbounded loops
3. No dynamic allocation	↔ [OK]↔	All static, zero malloc
4. Small functions	↔ [OK]↔	All functions < 60 lines
5. Assertions	↔ [OK]↔	RX_VALIDATE_PTR, RX_VALIDATE_INIT
6. Narrow scope	↔ [OK]↔	Static variables, internal functions
7. Check return values	↔ [OK]↔	All rx_err_t returns checked
8. Limited preprocessor	↔ [OK]↔	C23 typed enums for constants
9. Pointer restrictions	↔ [OK]↔	Single-level pointers
10. Compiler warnings	↔ [OK]↔	Clean with -Wall -Wextra -Werror

SOLID Principles:

Single Responsibility: This file handles ONLY MTU PWM generation. Motor control logic and encoder reading are in separate modules.

See also

[rx_mtu.h](#) Public API documentation

[rx72n_mtu_regs.h](#) Register structure definitions

RX72N Hardware Manual Chapter 24 - Multi-Function Timer Unit

Author

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Date

2026-01-27

Copyright

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Since

Version 1.0.0

Definition in file [rx_mtu.c](#).

8.129.2 Enumeration Type Documentation

8.129.2.1 mtu'bit'constants't

```
enum mtu\_bit\_constants\_t : uint32_t
```

Bit manipulation constants.

Enumerator

<code>k_mtu_bit_one</code>	Single bit value for shifts
----------------------------	-----------------------------

Definition at line [135](#) of file [rx_mtu.c](#).

```
00135      : uint32_t {
00136  k_mtu_bit_one = 1UL, /**< Single bit value for shifts */
00137 } mtu_bit_constants_t;
```

8.129.2.2 mtu'constants't

```
enum mtu\_constants\_t : uint8_t
```

MTU general constants.

Enumerator

<code>k_mtu_max_channels</code>	MTU0-MTU4, MTU6-MTU7 (sparse indexing, max index is 7)
---------------------------------	--

Definition at line [125](#) of file [rx_mtu.c](#).

```
00125      : uint8_t {
00126  k_mtu_max_channels = 8, /**< MTU0-MTU4, MTU6-MTU7 (sparse indexing, max index is 7) */
00127 } mtu_constants_t;
```

8.129.2.3 mtu`duty`constants`t

enum [mtu_duty_constants_t](#) : uint8_t

Duty cycle calculation constants.

Enumerator

k_mtu_duty_min	Minimum duty cycle (0%)
k_mtu_duty_max	Maximum duty cycle (100%)
k_mtu_duty_divisor	Divisor for percentage conversion

Definition at line 171 of file [rx_mtu.c](#).

```
00171      : uint8_t {
00172  k_mtu_duty_min    = 0,  /**< Minimum duty cycle (0%) */
00173  k_mtu_duty_max    = 100, /**< Maximum duty cycle (100%) */
00174  k_mtu_duty_divisor = 100, /**< Divisor for percentage conversion */
00175 } mtu_duty_constants_t;
```

8.129.2.4 mtu`module`stop`bits`t

enum [mtu_module_stop_bits_t](#) : uint8_t

MTU module stop bit position in MSTPCRA.

Enumerator

k_mtu_mstpa_mtu0_4	MTU module stop bit (all channels) – MSTPCRA.MSTPA9
--------------------	---

Definition at line 130 of file [rx_mtu.c](#).

```
00130      : uint8_t {
00131  k_mtu_mstpa_mtu0_4 = 9, /**< MTU module stop bit (all channels) -- MSTPCRA.MSTPA9 */
00132 } mtu_module_stop_bits_t;
```

8.129.2.5 mtu`period`constants`t

enum [mtu_period_constants_t](#) : uint16_t

Period calculation constants.

Enumerator

k_mtu_period_divisor	Triangle wave period divisor
k_mtu_period_max	Maximum valid period (16-bit)
k_mtu_period_min	Minimum valid period
k_mtu_period_zero	Zero period value

Definition at line 140 of file [rx_mtu.c](#).

```
00140      : uint16_t {
00141  k_mtu_period_divisor = 2,  /**< Triangle wave period divisor */
00142  k_mtu_period_max    = 0xFFFF, /**< Maximum valid period (16-bit) */
00143  k_mtu_period_min    = 10,  /**< Minimum valid period */
00144  k_mtu_period_zero   = 0,  /**< Zero period value */
00145 } mtu_period_constants_t;
```

8.129.2.6 mtu_tior_disabled_t

```
enum mtu_tior_disabled_t : uint8_t
```

TIOR output disabled value.

Enumerator

k_mtu_tior_disabled	Output disabled
---------------------	-----------------

Definition at line 166 of file [rx_mtu.c](#).

```
00166      : uint8_t {
00167  k_mtu_tior_disabled = 0x00, /**< Output disabled */
00168 } mtu_tior_disabled_t;
```

8.129.2.7 mtu_tior_mask_t

```
enum mtu_tior_mask_t : uint8_t
```

TIOR register clear-masks (value AND'd against TIOR to zero a nibble).

The low nibble (bits 3:0) controls MTIOCA/MTIOCC and the high nibble (bits 7:4) controls MTIOCB/↵ MTIOCD. To zero the low nibble without disturbing the high nibble you AND with 0xF0; to zero the high nibble you AND with 0x0F. Names therefore describe which nibble is kept.

Enumerator

k_mtu_tior_keep_high_clear_low	AND-mask: keep high nibble, clear low (MTIOCA/C)
k_mtu_tior_keep_low_clear_high	AND-mask: keep low nibble, clear high (MTIOCB/D)

Definition at line 160 of file [rx_mtu.c](#).

```
00160      : uint8_t {
00161  k_mtu_tior_keep_high_clear_low = 0xF0, /**< AND-mask: keep high nibble, clear low (MTIOCA/C) */
00162  k_mtu_tior_keep_low_clear_high = 0x0F, /**< AND-mask: keep low nibble, clear high (MTIOCB/D) */
00163 } mtu_tior_mask_t;
```

8.129.2.8 mtu_tior_shift_t

```
enum mtu_tior_shift_t : uint8_t
```

TIOR register shift positions.

Enumerator

k_mtu_tior_low_shift	Low nibble shift (MTIOCA/MTIOCC)
k_mtu_tior_high_shift	High nibble shift (MTIOCB/MTIOCD)

Definition at line 148 of file [rx_mtu.c](#).

```
00148      : uint8_t {
00149  k_mtu_tior_low_shift = 0, /**< Low nibble shift (MTIOCA/MTIOCC) */
00150  k_mtu_tior_high_shift = 4, /**< High nibble shift (MTIOCB/MTIOCD) */
00151 } mtu_tior_shift_t;
```

8.129.3 Function Documentation

8.129.3.1 internal`calculate`period()

```
rx_err_t internal_calculate_period (
    const uint32_t frequency_hz,
    uint16_t * period) [static]
```

Calculate period register value from frequency.

Converts desired PWM frequency to the TGRA (period) register value for PWM Mode 1 (triangle wave, center-aligned).

8.129.3.2 Calculation

$$\text{period} = \frac{f_{PCLKA}}{2 \times f_{PWM}}$$

Where:

- $f_{PCLKA} = 120 \text{ MHz}$
- Divisor of 2 accounts for up/down counting in triangle mode

8.129.3.3 Valid Frequency Range

Frequency	Period	Status
< 915 Hz	> 65535	Rejected (overflow)
915 Hz - 6 MHz	10-65535	Valid
> 6 MHz	< 10	Rejected (too small)

Precondition

PCLKA configured to 120 MHz
period points to valid uint16_t storage

Postcondition

*period contains value in range [10, 65535]

Note

16-bit resolution limits frequency range compared to 32-bit GPTW

See also

[k_pclka_hz](#) PCLKA clock frequency
[k_mtu_period_max](#) Maximum 16-bit period value

Since

Version 1.0.0

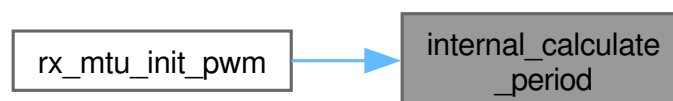
Definition at line 376 of file [rx_mtu.c](#).

```
00377 {
00378     if (period == nullptr) {
00379         return k_rx_err_null_ptr;
00380     }
00381
00382     if (frequency_hz == k_mtu_period_zero) {
00383         return k_rx_err_invalid_arg;
00384     }
00385
00386     /* For PWM mode 1 (triangle wave):
00387      * Period = PCLKA / (2 * frequency)
00388      * PCLKA = 120 MHz
00389      */
00390     const uint32_t pclka = k_pclka_hz;
00391     const uint32_t period_calc = pclka / (k_mtu_period_divisor * frequency_hz);
00392
00393     /* Check if period fits in 16-bit register */
00394     if (period_calc > k_mtu_period_max) {
00395         rx_log_error(s_tag, "Frequency too low");
00396         return k_rx_err_invalid_arg;
00397     }
00398
00399     if (period_calc < k_mtu_period_min) {
00400         rx_log_error(s_tag, "Frequency too high");
00401         return k_rx_err_invalid_arg;
00402     }
00403
00404     *period = (uint16_t)period_calc;
00405     return k_rx_ok;
00406 }
```

References [k_mtu_period_divisor](#), [k_mtu_period_max](#), [k_mtu_period_min](#), [k_mtu_period_zero](#), [k_pclka_hz](#), and [s_tag](#).

Referenced by [rx_mtu_init_pwm\(\)](#).

Here is the caller graph for this function:



8.129.3.4 internal_clear_tstr_bit()

```
rx_err_t internal_clear_tstr_bit (
    volatile rx_mtu_tstr_regs_t * tstr,
    const uint8_t mask) [static]
```

Clear a counter-start bit in a TSTR register and verify it cleared.

Performs a read-modify-write on the TSTR register to clear the specified bit mask, then reads back the register to confirm the bit was cleared. Returns a hardware error if the bit is still set after the write, which can indicate a peripheral fault.

Precondition

tstr points to a valid TSTRA or TSTRB hardware register
 mask is a valid CSTn bit mask from [mtu_tstr_bits_t](#)

Postcondition

Specified CSTn bit is cleared (counter stopped) on success
 tstr register unchanged on error

Note

Not thread-safe: caller must ensure exclusive access to TSTR register
 Readback verification adds one extra register read cycle

See also

[rx_mtu_stop\(\)](#) Primary caller of this function
[k_mtu_tstr_cst0](#) Counter-start bit constants

Since

Version 1.0.0

Definition at line 323 of file [rx_mtu.c](#).

```
00324 {
00325     if (tstr == nullptr) {
00326         return k_rx_err_invalid_arg;
00327     }
00328     tstr->tstr &= (uint8_t)~mask;
00329     if ((tstr->tstr & mask) != 0) {
00330         return k_rx_err_hw_error;
00331     }
00332 }
00333
00334 return k_rx_ok;
00335 }
```

References [rx_mtu_tstr_regs_t::tstr](#).

Referenced by [rx_mtu_stop\(\)](#).

Here is the caller graph for this function:



8.129.3.5 internal`get`mtu`base()

```
volatile void * internal__get__mtu__base (
    const rx\_mtu\_channel\_t channel) [static]
```

Get MTU channel register base address.

Maps an MTU channel enumeration to its hardware register base address. The MTU peripheral has sparse channel numbering (0-4, 6-7 with no 5).

8.129.3.6 Memory Map

Channel	Base Address	Description
MTU0	0x000C1290	16-bit timer with 4 outputs
MTU1	0x000C1290	16-bit timer with 2 outputs
MTU2	0x000C1292	16-bit timer with 2 outputs
MTU3	(nullptr)	sparse layout, cannot use standard struct API
MTU4	(nullptr)	sparse layout, cannot use standard struct API
MTU6	(nullptr)	interleaved layout, use mtu6() with rx_mtu3_channel_regs_t
MTU7	(nullptr)	interleaved layout, use mtu7() with direct register offsets

MTU3, MTU4, MTU6, and MTU7 all have an interleaved register layout that is incompatible with [rx_mtu_channel_regs_t](#). Use [mtu3\(\)](#)/[mtu6\(\)](#) with [rx_mtu3_channel_regs_t](#) directly; this function returns nullptr for all four channels to prevent callers from applying the wrong struct offset mapping.

Precondition

Module clock must be enabled before accessing returned pointer

Postcondition

Returned pointer valid for hardware access

No hardware state modified

Note

Thread-safe: Returns constant address

Returns void* to accommodate different register structure types

See also

[mtu0\(\)](#), [mtu1\(\)](#), etc. Underlying accessor functions

[rx72n_mtu_regs.h](#) Register structure definitions

Since

Version 1.0.0

Definition at line 257 of file [rx_mtu.c](#).

```

00258 {
00259     switch (channel) {
00260         case k_mtu_channel_0:
00261             return (volatile void*)mtu0();
00262         case k_mtu_channel_1:
00263             return (volatile void*)mtu1();
00264         case k_mtu_channel_2:
00265             return (volatile void*)mtu2();
00266         case k_mtu_channel_3: /* fall through -- sparse layout, incompatible with standard API */
00267         case k_mtu_channel_4: /* MTU3/4 use rx_mtu3_channel_regs_t; call mtu3()/mtu4() directly */
00268             return nullptr;
00269         case k_mtu_channel_6: /* fall through -- interleaved layout, use mtu6() with rx_mtu3_channel_regs_t */
00270         case k_mtu_channel_7: /* interleaved layout, use mtu7() with direct address offsets */
00271             return nullptr;
00272         default:

```



```

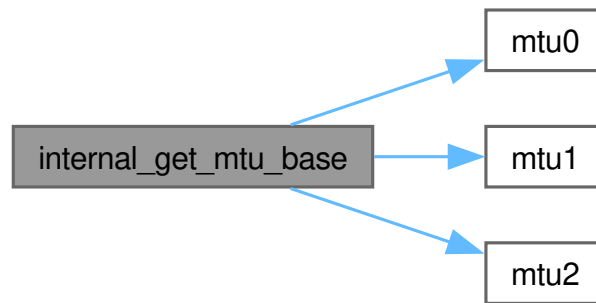
00273     return nullptr;
00274 }
00275 }

```

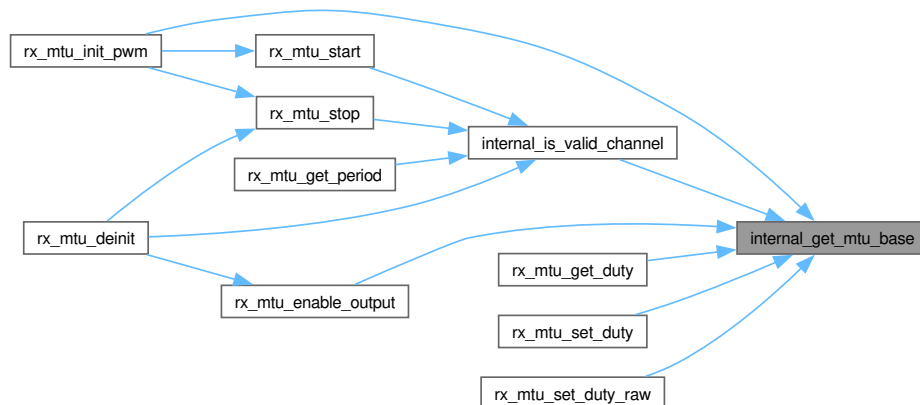
References [k_mtu_channel_0](#), [k_mtu_channel_1](#), [k_mtu_channel_2](#), [k_mtu_channel_3](#), [k_mtu_channel_4](#), [k_mtu_channel_6](#), [k_mtu_channel_7](#), [mtu0\(\)](#), [mtu1\(\)](#), and [mtu2\(\)](#).

Referenced by [internal_is_valid_channel\(\)](#), [rx_mtu_enable_output\(\)](#), [rx_mtu_get_duty\(\)](#), [rx_mtu_init_pwm\(\)](#), [rx_mtu_set_duty\(\)](#), and [rx_mtu_set_duty_raw\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.3.7 internal`get`tgr`register()

```

volatile uint16_t * internal_get_tgr_register (
    volatile rx_mtu_channel_regs_t * mtu,
    const rx_mtu_output_t output) [static]

```

Get TGR register pointer for output channel.

Definition at line 413 of file [rx_mtu.c](#).

```

00415 {
00416     if (mtu == nullptr) {
00417         return nullptr;
00418     }
00419     switch (output) {
00420     case k_mtu_output_a:
00421         return &mtu->tgra;
00422     case k_mtu_output_b:

```

```

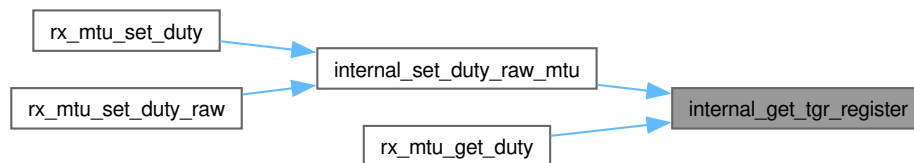
00424     return &mtu->tgrb;
00425 case k_mtu_output_c:
00426     return &mtu->tgrc;
00427 case k_mtu_output_d:
00428     return &mtu->tgrd;
00429 default:
00430     return nullptr;
00431 }
00432 }

```

References [k_mtu_output_a](#), [k_mtu_output_b](#), [k_mtu_output_c](#), [k_mtu_output_d](#), [rx_mtu_channel_regs_t::tgra](#), [rx_mtu_channel_regs_t::tgrb](#), [rx_mtu_channel_regs_t::tgrc](#), and [rx_mtu_channel_regs_t::tgrd](#).

Referenced by [internal_set_duty_raw_mtu\(\)](#), and [rx_mtu_get_duty\(\)](#).

Here is the caller graph for this function:



8.129.3.8 internal_is_valid_channel()

```

bool internal_is_valid_channel (
    const rx_mtu_channel_t channel) [static]

```

Check if MTU channel is valid.

Validates an MTU channel by checking if it has a valid register base address. MTU channels 0-4 and 6-7 are valid; channel 5 does not exist.

Note

Thread-safe: Pure function

Delegates to [internal_get_mtu_base\(\)](#) for actual validation

See also

[internal_get_mtu_base\(\)](#) Underlying validation logic

Since

Version 1.0.0

Definition at line 293 of file [rx_mtu.c](#).

```

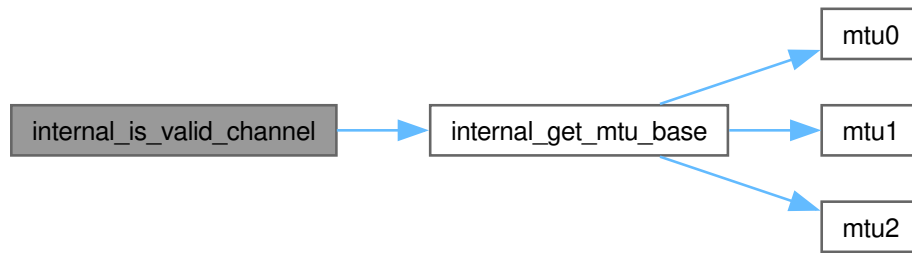
00294 {
00295     return internal_get_mtu_base(channel) != nullptr;
00296 }

```

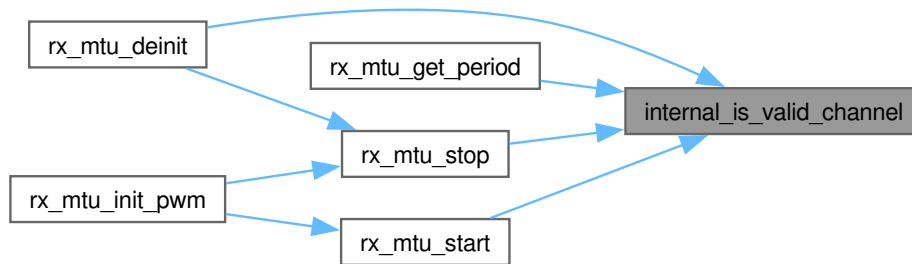
References [internal_get_mtu_base\(\)](#).

Referenced by [rx_mtu_deinit\(\)](#), [rx_mtu_get_period\(\)](#), [rx_mtu_start\(\)](#), and [rx_mtu_stop\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.3.9 internal`set`duty`raw`mtu()

```

rx_err_t internal_set_duty_raw_mtu (
    volatile rx_mtu_channel_regs_t * mtu,
    const rx_mtu_output_t output,
    const uint16_t duty_count) [static]

```

Write a raw duty count to the TGR register for the specified output.

Resolves the TGR register pointer for the requested output via `internal_get_tgr_register()` and writes `duty_count` to it. The write is buffered by hardware and takes effect at the next PWM period boundary, producing a glitch-free duty cycle update.

Algorithm:

1. Validate mtu pointer is non-NULL
2. Resolve TGR register pointer for output via `internal_get_tgr_register()`
3. Validate resolved TGR pointer is non-NULL
4. Write `duty_count` to TGR register

Precondition

mtu points to a valid, initialized MTU channel register block
 duty_count has been validated or clamped by the caller to [0, period]

Postcondition

TGR register written with `duty_count`
 Change takes effect at next PWM period boundary (buffered by hardware)

Note

Not thread-safe: caller must ensure exclusive access to the TGR register

Duty count is not bounds-checked; caller must clamp before calling

```
volatile rx_mtu_channel_regs_t* mtu =
    (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(k_mtu_channel_0);
rx_err_t err = internal_set_duty_raw_mtu(mtu, k_mtu_output_b, 1500U);
```

See also

[internal_get_tgr_register\(\)](#) Resolves TGR pointer from output selector

[rx_mtu_set_duty_raw\(\)](#) Public API caller that clamps duty_count first

Since

Version 1.0.0

Definition at line 471 of file [rx_mtu.c](#).

```
00474 {
00475     if (mtu == nullptr) {
00476         return k_rx_err_invalid_arg;
00477     }
00478
00479     volatile uint16_t* tgr = internal_get_tgr_register(mtu, output);
00480     if (tgr == nullptr) {
00481         return k_rx_err_invalid_arg;
00482     }
00483
00484     /* Update duty cycle (buffered, takes effect on next period) */
00485     *tgr = duty_count;
00486     return k_rx_ok;
00487 }
```

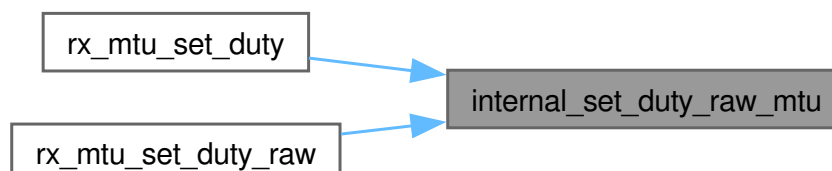
References [internal_get_tgr_register\(\)](#).

Referenced by [rx_mtu_set_duty\(\)](#), and [rx_mtu_set_duty_raw\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.3.10 rx`mtu`deinit()

```
rx_err_t rx_mtu_deinit (
    const rx_mtu_channel_t channel) [nodiscard]
```

Deinitialize an MTU PWM channel.

Deinitialize MTU channel.

Performs an orderly shutdown of the specified MTU channel:

1. Stop the timer counter ([rx_mtu_stop\(\)](#))
2. Disable all four outputs (A, B, C, D) via [rx_mtu_enable_output\(\)](#)
3. Clear the cached period to 0
4. Clear the initialized flag

After this call the channel may be safely re-initialized with different parameters via [rx_mtu_init_pwm\(\)](#).

Precondition

Motor or load driven by this channel must be stopped externally before call
channel must be a valid MTU channel (0-4, 6-7)

Postcondition

Timer stopped and outputs disabled
`s_mtu_initialized[channel] == false`
`s_mtu_period[channel] == k_mtu_period_zero`

Note

Not thread-safe: do not call while another task is accessing this channel
Idempotent with respect to re-initialization (can call init again after)

Warning

Ensure motor is at rest before calling to avoid abrupt stop

```
rx_err_t err = rx_mtu_deinit(k_mtu_channel_0);
```

See also

[rx_mtu_init_pwm\(\)](#) Re-initialize after deinit
[rx_mtu_stop\(\)](#) Stop timer without full teardown

Since

Version 1.0.0

Definition at line 1087 of file [rx_mtu.c](#).

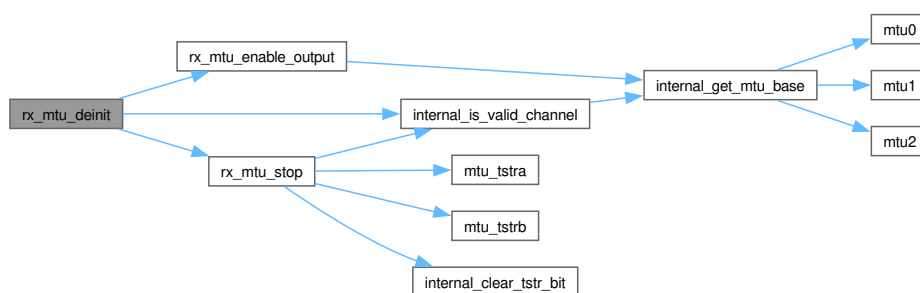
```

01088 {
01089     if (!internal_is_valid_channel(channel)) {
01090         return k_rx_err_invalid_arg;
01091     }
01092
01093     /* Stop timer */
01094     rx_err_t err = rx_mtu_stop(channel);
01095     if (err != k_rx_ok) {
01096         return err;
01097     }
01098
01099     /* Disable all outputs */
01100     err = rx_mtu_enable_output(channel, k_mtu_output_a, false);
01101     if (err != k_rx_ok) {
01102         return err;
01103     }
01104     err = rx_mtu_enable_output(channel, k_mtu_output_b, false);
01105     if (err != k_rx_ok) {
01106         return err;
01107     }
01108     err = rx_mtu_enable_output(channel, k_mtu_output_c, false);
01109     if (err != k_rx_ok) {
01110         return err;
01111     }
01112     err = rx_mtu_enable_output(channel, k_mtu_output_d, false);
01113     if (err != k_rx_ok) {
01114         return err;
01115     }
01116
01117     /* Mark as uninitialized */
01118     s_mtu_initialized[channel] = false;
01119     s_mtu_period[channel] = k_mtu_period_zero;
01120
01121     rx_log_info(s_tag, "MTU channel deinitialized");
01122
01123     return k_rx_ok;
01124 }

```

References [internal_is_valid_channel\(\)](#), [k_mtu_output_a](#), [k_mtu_output_b](#), [k_mtu_output_c](#), [k_mtu_output_d](#), [k_mtu_period_zero](#), [rx_mtu_enable_output\(\)](#), [rx_mtu_stop\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.129.3.11 rx_mtu_enable_output()

```

rx_err_t rx_mtu_enable_output (
    const rx_mtu_channel_t channel,
    rx_mtu_output_t output,
    const bool enable) [nodiscard]

```

Enable or disable a single MTU PWM output pin.

Enable or disable PWM output.

Controls whether the MTIOCA/B/C/D pin actively drives the PWM waveform or is held in its initial state (off). Modifies the TIORH or TIORL register nibble corresponding to the requested output without disturbing other outputs.

Output	Register	Nibble
A	TIORH	Low (bits 3:0)
B	TIORH	High (bits 7:4)
C	TIORL	Low (bits 3:0)
D	TIORL	High (bits 7:4)

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
output must be a valid [rx_mtu_output_t](#) enum value

Postcondition

TIORH or TIORL nibble updated; timer continues running
Other outputs on the same channel are unchanged

Note

Not thread-safe: caller must prevent concurrent register access
Timer keeps running; only output pin drive changes

Warning

Outputs may remain HIGH when disabled, depending on last compare state

```
rx_mtu_enable_output(k_mtu_channel_0, k_mtu_output_a, false); // disable
```

See also

[rx_mtu_init_pwm\(\)](#) Initialize channel before calling
[rx_mtu_stop\(\)](#) Halt the entire timer

Since

Version 1.0.0

Definition at line 883 of file [rx_mtu.c](#).

```
00884 {
00885     volatile rx_mtu_channel_regs_t* const mtu =
00886         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00887     if (mtu == nullptr) {
00888         return k_rx_err_invalid_arg;
00889     }
00890
00891     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00892
00893     /* Enable/disable output by modifying TIOR registers */
```

```

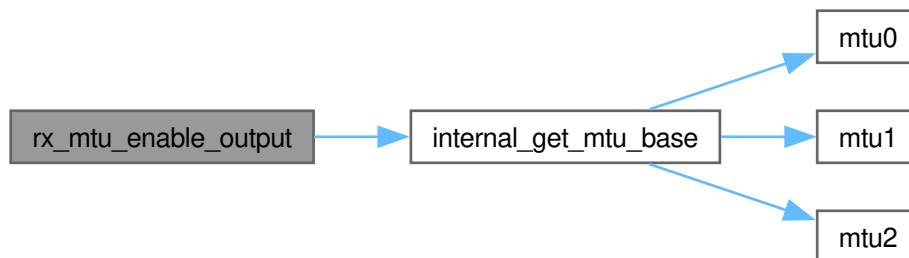
00894  const uint8_t tior_value = enable ? k_mtu_tior_init_low : k_mtu_tior_disabled;
00895
00896  switch (output) {
00897      case k_mtu_output_a:
00898          mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_high_clear_low) |
00899                               (tior_value « k_mtu_tior_low_shift));
00900          break;
00901      case k_mtu_output_b:
00902          mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_low_clear_high) |
00903                               (tior_value « k_mtu_tior_high_shift));
00904          break;
00905      case k_mtu_output_c:
00906          mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_high_clear_low) |
00907                               (tior_value « k_mtu_tior_low_shift));
00908          break;
00909      case k_mtu_output_d:
00910          mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_low_clear_high) |
00911                               (tior_value « k_mtu_tior_high_shift));
00912          break;
00913      default:
00914          return k_rx_err_invalid_arg;
00915  }
00916
00917  return k_rx_ok;
00918 }

```

References [internal_get_mtu_base\(\)](#), [k_mtu_output_a](#), [k_mtu_output_b](#), [k_mtu_output_c](#), [k_mtu_output_d](#), [k_mtu_tior_disabled](#), [k_mtu_tior_high_shift](#), [k_mtu_tior_init_low](#), [k_mtu_tior_keep_high_clear_low](#), [k_mtu_tior_keep_low_clear_high](#), [k_mtu_tior_low_shift](#), [s_mtu_initialized](#), [s_tag](#), [rx_mtu_channel_regs_t::tiorh](#), and [rx_mtu_channel_regs_t::tiorl](#).

Referenced by [rx_mtu_deinit\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.3.12 rx_mtu_get_duty()

```

rx_err_t rx_mtu_get_duty (
    const rx_mtu_channel_t channel,
    const rx_mtu_output_t output,
    float * duty_percent) [nodiscard]

```

Read the current PWM duty cycle as a floating-point percentage.

Get current duty cycle.

Reads the TGR register for the specified output and converts the raw count to a percentage using the cached period value.

$$\text{duty_percent} = \frac{\text{duty_count} \times 100}{\text{period}}$$

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
duty_percent must point to valid float storage

Postcondition

*duty_percent contains duty cycle in [0.0, 100.0] on success
Hardware TGR register unchanged

Note

Not thread-safe: concurrent set_duty calls may cause torn reads

```
float duty;
rx_err_t err = rx_mtu_get_duty(k_mtu_channel_0, k_mtu_output_b, &duty);
```

See also

[rx_mtu_set_duty\(\)](#) Set duty using percentage
[rx_mtu_get_period\(\)](#) Retrieve period count

Since

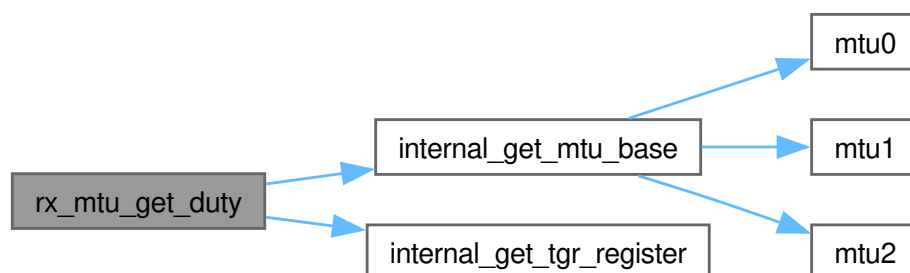
Version 1.0.0

Definition at line 775 of file rx_mtu.c.

```
00776 {
00777   RX_VALIDATE_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
00778
00779   volatile rx_mtu_channel_regs_t* const mtu =
00780     (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00781   if (mtu == nullptr) {
00782     return k_rx_err_invalid_arg;
00783   }
00784
00785   RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00786
00787   const volatile uint16_t* const tgr = internal_get_tgr_register(mtu, output);
00788   if (tgr == nullptr) {
00789     return k_rx_err_invalid_arg;
00790   }
00791
00792   const uint16_t period = s_mtu_period[channel];
00793   const uint16_t duty_count = *tgr;
00794
00795   *duty_percent = (float)(duty_count * (float)k_mtu_duty_max) / period;
00796
00797   return k_rx_ok;
00798 }
```

References [internal_get_mtu_base\(\)](#), [internal_get_tgr_register\(\)](#), [k_mtu_duty_max](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.129.3.13 rx_mtu_get_period()

```
rx_err_t rx_mtu_get_period (
    const rx_mtu_channel_t channel,
    uint16_t * period_count) [nodiscard]
```

Get PWM period count for MTU channel.

Get PWM period count.

Returns the cached period value (TGRA) from initialization. Useful for calculating raw duty counts from percentages without floating-point.

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
 period_count must point to valid uint16_t storage

Postcondition

*period_count contains the TGRA register value on success
 Hardware registers unchanged

Note

Thread-safe: reads from static cache, no hardware access

```
uint16_t period;
rx_err_t err = rx_mtu_get_period(k_mtu_channel_0, &period);
if (err == k_rx_ok) {
    uint16_t half_duty = period / 2U; // 50%
    rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, half_duty);
}
```

See also

[rx_mtu.h](#) Complete API documentation

[rx_mtu_set_duty_raw\(\)](#) Use period count to set duty

Since

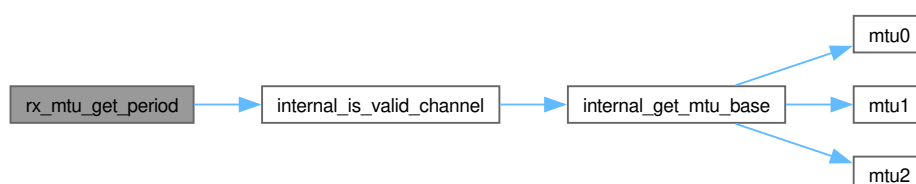
Version 1.0.0

Definition at line 831 of file [rx_mtu.c](#).

```
00832 {
00833   RX_VALIDATE_PTR(period_count, s_tag, "period_count pointer is nullptr");
00834
00835   if (!internal_is_valid_channel(channel)) {
00836     return k_rx_err_invalid_arg;
00837   }
00838
00839   RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00840
00841   *period_count = s_mtu_period[channel];
00842   return k_rx_ok;
00843 }
```

References [internal_is_valid_channel\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.129.3.14 rx`mtu`init`pwm()

```
rx_err_t rx_mtu_init_pwm (
    const rx_mtu_channel_t channel,
    const rx_mtu_config_t * config) [nodiscard]
```

Initialize MTU channel for PWM operation.

Initialize MTU channel for PWM output.

Configures an MTU channel for PWM Mode 1 (triangle wave, center-aligned) operation. Performs complete hardware setup including module clock enable, timer configuration, and automatic start.

8.129.3.15 Initialization Sequence

1. Validate config pointer
2. Get channel register base address
3. Calculate period from frequency
4. Enable MTU module clock (clear MSTPA bit)
5. Stop timer if running
6. Configure TCR (clock source, clear condition)
7. Configure TMDR (PWM Mode 1)
8. Configure TIORH/TIORL (output behavior)
9. Set TGRA (period), clear TGRB/TGRC/TGRD (duty = 0)
10. Clear TCNT counter
11. Store period and set initialized flag
12. Start timer

8.129.3.16 PWM Mode 1 Configuration

Register	Value	Effect
TCR	TPSC=1, CCLR=TGRA	PCLKA/1, clear on TGRA match
TMDR	MD=PWM1	Triangle wave mode
TIORH/ L	Init low	Outputs toggle on compare match

Precondition

System clock configured (PCLKA = 120 MHz)
 config pointer valid with valid frequency

Postcondition

MTU channel configured and timer running
 All duty cycles initialized to 0%
 s_mtu_initialized[channel] = true

Note

Timer starts automatically
 Call once during system initialization

Warning

Not thread-safe: Configure before starting motor tasks

Example:

```
rx_mtu_config_t config = {
    .frequency_hz = 20000 // 20 kHz
};
rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
```

See also

[rx_mtu.h](#) Complete API documentation
[rx_mtu_deinit\(\)](#) Cleanup function

Since

Version 1.0.0

Definition at line 554 of file [rx_mtu.c](#).

```
00555 {
00556     RX_VALIDATE_PTR(config, s_tag, "config pointer is nullptr");
00557
00558     volatile rx_mtu_channel_regs_t* const mtu =
00559         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00560     if (mtu == nullptr) {
00561         return k_rx_err_invalid_arg;
00562     }
00563
00564     /* Calculate period from frequency */
00565     uint16_t period;
00566     rx_err_t err = internal_calculate_period(config->frequency_hz, &period);
00567     if (err != k_rx_ok) {
00568         return err;
00569     }
00570
00571     rx_log_info(s_tag, "Initializing MTU");
00572
00573     /* Enable MTU module (clear module stop bit) */
00574     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00575
00576     system_regs()->mstpcra &= ~(k_mtu_bit_one « k_mtu_mstpa_mtu0_4);
00577
00578     *prcr_reg() = k_rx_prcr_lock;
00579
00580     /* Stop timer before configuration */
00581     err = rx_mtu_stop(channel);
00582     if (err != k_rx_ok) {
00583         return err;
00584     }
00585
00586     /* Configure timer control register
```

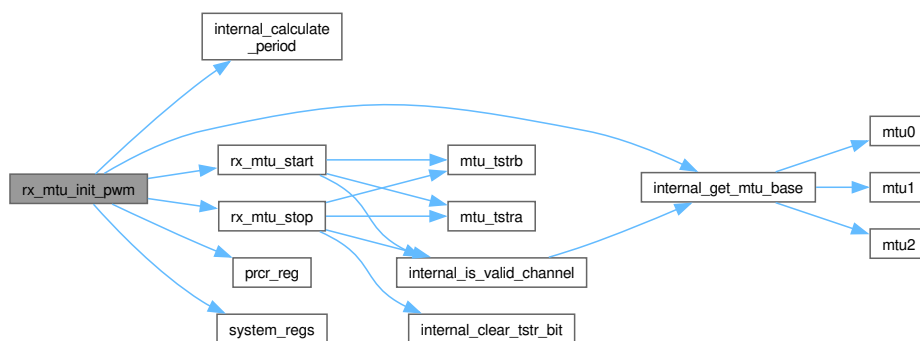
```

00587  * - PCLKA/1 (120 MHz)
00588  * - Clear counter on TGRA compare match
00589  */
00590  mtu->tcr = k_mtu_tcr_tpsec_1 | k_mtu_tcr_cclr_tgra;
00591
00592  /* Configure PWM mode 1 (triangle wave, center-aligned) */
00593  mtu->tmdr = k_mtu_tmdr_md_pwm1;
00594
00595  /* Configure I/O control registers for PWM outputs
00596  * TIORH: Controls MTIOCA (output A) and MTIOCB (output B)
00597  * TIORL: Controls MTIOCC (output C) and MTIOCD (output D)
00598  *
00599  * For PWM mode 1:
00600  * - Initial low, high on up-count compare, low on down-count compare
00601  */
00602  mtu->tiorh = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCA */
00603              (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCB */
00604  mtu->tiorl = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCC */
00605              (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCD */
00606
00607  /* Set period (TGRA = top of triangle wave) */
00608  mtu->tgra = period;
00609
00610  /* Set initial duty cycle to 0% for all outputs */
00611  mtu->tgrb = k_mtu_period_zero; /* MTIOCB duty */
00612  mtu->tgrc = k_mtu_period_zero; /* MTIOCC duty */
00613  mtu->tgrd = k_mtu_period_zero; /* MTIOCD duty */
00614
00615  /* Clear counter */
00616  mtu->tcnt = k_mtu_period_zero;
00617
00618  /* Save period for duty cycle calculations */
00619  s_mtu_period[channel] = period;
00620  s_mtu_initialized[channel] = true;
00621
00622  /* Start timer */
00623  err = rx_mtu_start(channel);
00624  if (err != k_rx_ok) {
00625      return err;
00626  }
00627
00628  rx_log_info(s_tag, "MTU initialized successfully");
00629
00630  return k_rx_ok;
00631 }

```

References [rx_mtu_config_t::frequency_hz](#), [internal_calculate_period\(\)](#), [internal_get_mtu_base\(\)](#), [k_mtu_bit_one](#), [k_mtu_mstpa_mtu0_4](#), [k_mtu_period_zero](#), [k_mtu_tcr_cclr_tgra](#), [k_mtu_tcr_tpsec_1](#), [k_mtu_tior_high_shift](#), [k_mtu_tior_init_low](#), [k_mtu_tior_low_shift](#), [k_mtu_tmdr_md_pwm1](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), [rx_mtu_start\(\)](#), [rx_mtu_stop\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), [s_tag](#), [system_regs\(\)](#), [rx_mtu_channel_regs_t::tcnt](#), [rx_mtu_channel_regs_t::tcr](#), [rx_mtu_channel_regs_t::tgra](#), [rx_mtu_channel_regs_t::tgrb](#), [rx_mtu_channel_regs_t::tgrc](#), [rx_mtu_channel_regs_t::tgrd](#), [rx_mtu_channel_regs_t::tiorh](#), [rx_mtu_channel_regs_t::tiorl](#), and [rx_mtu_channel_regs_t::tmdr](#).

Here is the call graph for this function:



8.129.3.17 rx_mtu_set_duty()

```
rx_err_t rx_mtu_set_duty (  
    const rx_mtu_channel_t channel,  
    rx_mtu_output_t output,  
    const float duty_percent) [nodiscard]
```

Set PWM duty cycle as a floating-point percentage.

Set PWM duty cycle.

Converts a percentage value to raw timer counts using the cached period and writes it to the appropriate TGR register. The update is buffered and takes effect at the next PWM period boundary (glitch-free).

Algorithm:

1. Validate channel and initialization state
2. Validate duty_percent in range [0.0, 100.0]
3. Retrieve cached period from s_mtu_period[channel]
4. Compute duty_count = (duty_percent * period) / 100
5. Write duty_count to TGR register via [internal_set_duty_raw_mtu\(\)](#)

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)
duty_percent must be in [0.0, 100.0]

Postcondition

TGR register updated with new duty count
Change takes effect at next PWM period boundary

Note

Not thread-safe: caller must prevent concurrent access

```
rx_err_t err = rx_mtu_set_duty(k_mtu_channel_0, k_mtu_output_b, 50.0f);
```

See also

[rx_mtu_set_duty_raw\(\)](#) Set duty using raw count value
[rx_mtu_get_duty\(\)](#) Read current duty cycle

Since

Version 1.0.0

Definition at line 668 of file [rx_mtu.c](#).

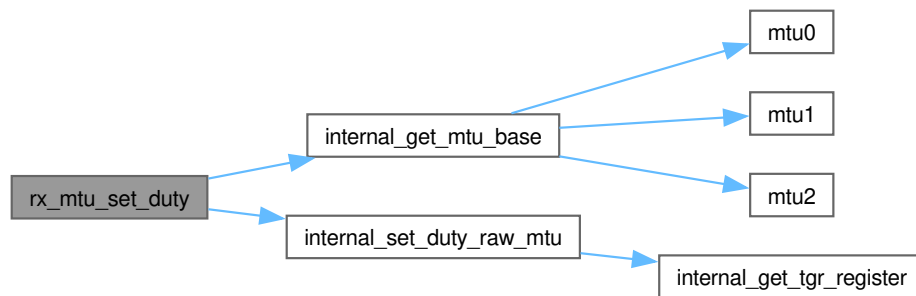
```

00669 {
00670     volatile rx_mtu_channel_regs_t* const mtu =
00671         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00672     if (mtu == nullptr) {
00673         return k_rx_err_invalid_arg;
00674     }
00675     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00676     if (duty_percent < (float)k_mtu_duty_min || duty_percent > (float)k_mtu_duty_max) {
00677         rx_log_error(s_tag, "Invalid duty cycle");
00678         return k_rx_err_invalid_arg;
00679     }
00680     /* Convert percentage to count value */
00681     const uint16_t period = s_mtu_period[channel];
00682     const uint16_t duty_count = (uint16_t)((duty_percent * period) / (float)k_mtu_duty_divisor);
00683     return internal_set_duty_raw_mtu(mtu, output, duty_count);
00684 }

```

References [internal_get_mtu_base\(\)](#), [internal_set_duty_raw_mtu\(\)](#), [k_mtu_duty_divisor](#), [k_mtu_duty_max](#), [k_mtu_duty_min](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.129.3.18 rx`mtu`set`duty`raw()

```

rx_err_t rx_mtu_set_duty_raw (
    const rx_mtu_channel_t channel,
    rx_mtu_output_t output,
    uint16_t duty_count) [nodiscard]

```

Set PWM duty cycle using a raw 16-bit timer count.

Set PWM duty cycle (raw count value).

Directly sets the compare register (TGR) for the specified output to duty_count timer ticks. Values larger than the channel period are clamped to the period value (100% duty). The update is buffered.

Use this function when the caller has already converted a duty percentage to counts (e.g. via [rx_mtu_get_period\(\)](#)) to avoid redundant floating-point multiplication.

Precondition

Channel must be initialized via [rx_mtu_init_pwm\(\)](#)

duty_count should be in [0, period]; values above period are clamped

Postcondition

TGR register written with (possibly clamped) duty_count

Change takes effect at next PWM period boundary

Note

Not thread-safe: caller must prevent concurrent access

```
uint16_t period;
rx_mtu_get_period(k_mtu_channel_0, &period);
rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, period / 2U); // 50%
```

See also

[rx_mtu_set_duty\(\)](#) Set duty using floating-point percentage

[rx_mtu_get_period\(\)](#) Retrieve period for count calculation

Since

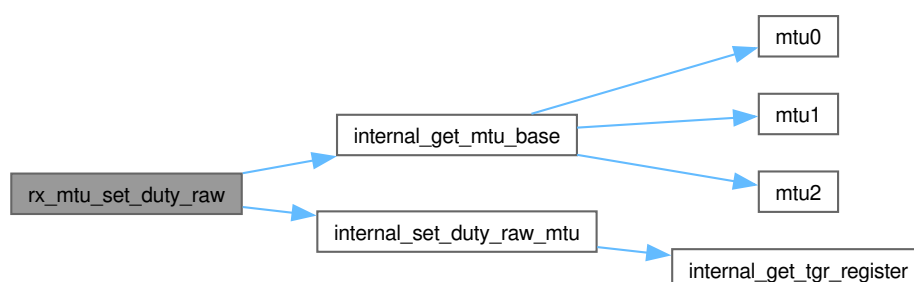
Version 1.0.0

Definition at line 724 of file [rx_mtu.c](#).

```
00725 {
00726     volatile rx_mtu_channel_regs_t* const mtu =
00727         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00728     if (mtu == nullptr) {
00729         return k_rx_err_invalid_arg;
00730     }
00731
00732     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00733
00734     /* Clamp to period */
00735     const uint16_t period = s_mtu_period[channel];
00736     if (duty_count > period) {
00737         duty_count = period;
00738     }
00739
00740     return internal_set_duty_raw_mtu(mtu, output, duty_count);
00741 }
```

References [internal_get_mtu_base\(\)](#), [internal_set_duty_raw_mtu\(\)](#), [s_mtu_initialized](#), [s_mtu_period](#), and [s_tag](#).

Here is the call graph for this function:



8.129.3.19 rx`mtu`start()

```
rx_err_t rx_mtu_start (  
    const rx_mtu_channel_t channel) [nodiscard]
```

Start MTU counter (begin PWM generation).

Start MTU timer.

Sets the Count Start (CSTn) bit in the appropriate TSTR register to begin counter operation. Channels 0-4 use TSTRA, channels 6-7 use TSTRB.

8.129.3.20 TSTR Bit Mapping

Channel	Register	Bit
MTU0	TSTRA	CST0
MTU1	TSTRA	CST1
MTU2	TSTRA	CST2
MTU3	TSTRA	CST3
MTU4	TSTRA	CST4
MTU6	TSTRB	CST6
MTU7	TSTRB	CST7

Precondition

Channel initialized via [rx_mtu_init_pwm\(\)](#)

Postcondition

Timer counting, PWM output active

Note

Timer automatically started during init

Idempotent: Safe to call when already running

See also

[rx_mtu_stop\(\)](#) Stop the timer

Since

Version 1.0.0

Definition at line 952 of file [rx_mtu.c](#).

```

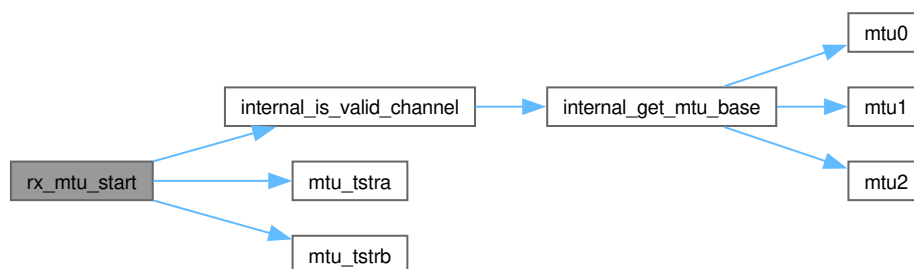
00953 {
00954     if (internal_is_valid_channel(channel)) {
00955         return k_rx_err_invalid_arg;
00956     }
00957     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00959     /* Set corresponding bit in TSTR register */
00961     switch (channel) {
00962     case k_mtu_channel_0:
00963         mtu_tstra()->tstr |= k_mtu_tstr_cst0;
00964         break;
00965     case k_mtu_channel_1:
00966         mtu_tstra()->tstr |= k_mtu_tstr_cst1;
00967         break;
00968     case k_mtu_channel_2:
00969         mtu_tstra()->tstr |= k_mtu_tstr_cst2;
00970         break;
00971     case k_mtu_channel_3:
00972         mtu_tstra()->tstr |= k_mtu_tstr_cst3;
00973         break;
00974     case k_mtu_channel_4:
00975         mtu_tstra()->tstr |= k_mtu_tstr_cst4;
00976         break;
00977     case k_mtu_channel_6:
00978         mtu_tstrb()->tstr |= k_mtu_tstr_cst6;
00979         break;
00980     case k_mtu_channel_7:
00981         mtu_tstrb()->tstr |= k_mtu_tstr_cst7;
00982         break;
00983     default:
00984         return k_rx_err_invalid_arg;
00985     }
00986     return k_rx_ok;
00987 }
00988

```

References [internal_is_valid_channel\(\)](#), [k_mtu_channel_0](#), [k_mtu_channel_1](#), [k_mtu_channel_2](#), [k_mtu_channel_3](#), [k_mtu_channel_4](#), [k_mtu_channel_6](#), [k_mtu_channel_7](#), [k_mtu_tstr_cst0](#), [k_mtu_tstr_cst1](#), [k_mtu_tstr_cst2](#), [k_mtu_tstr_cst3](#), [k_mtu_tstr_cst4](#), [k_mtu_tstr_cst6](#), [k_mtu_tstr_cst7](#), [mtu_tstra\(\)](#), [mtu_tstrb\(\)](#), [s_mtu_initialized](#), [s_tag](#), and [rx_mtu_tstr_regs_t::tstr](#).

Referenced by [rx_mtu_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.3.21 rx`mtu`stop()

```
rx_err_t rx_mtu_stop (
    const rx_mtu_channel_t channel) [nodiscard]
```

Stop MTU counter (halt PWM generation).

Stop MTU timer.

Clears the Count Start (CSTn) bit in the appropriate TSTR register to halt counter operation. Counter value and output state are preserved.

Precondition

None (can be called on uninitialized channel)

Postcondition

Timer stopped, counter preserved

Output pins hold current state

Note

Does NOT check initialization flag (safe for use during init)

Idempotent: Safe to call when already stopped

Warning

Outputs may remain HIGH - disable outputs for safe state

See also

[rx_mtu_start\(\)](#) Resume the timer

[rx_mtu_enable_output\(\)](#) Disable outputs

Since

Version 1.0.0

Definition at line 1014 of file [rx_mtu.c](#).

```
01015 {
01016     if (!internal_is_valid_channel(channel)) {
01017         return k_rx_err_invalid_arg;
01018     }
01019
01020     /* Clear corresponding bit in TSTR register */
01021     rx_err_t err;
01022     switch (channel) {
01023     case k_mtu_channel_0:
01024         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst0);
01025         break;
01026     case k_mtu_channel_1:
01027         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst1);
01028         break;
01029     case k_mtu_channel_2:
01030         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst2);
01031         break;
```

```

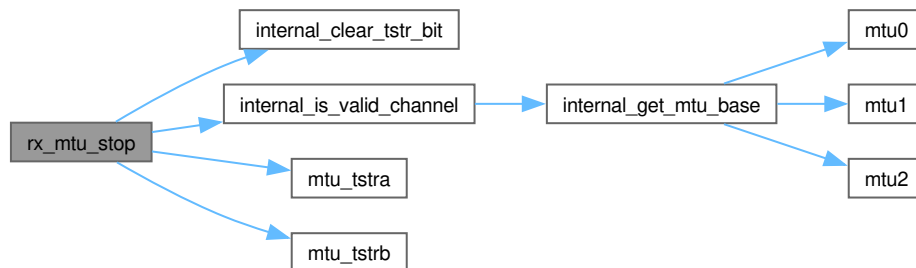
01032     case k_mtu_channel_3:
01033         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst3);
01034         break;
01035     case k_mtu_channel_4:
01036         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst4);
01037         break;
01038     case k_mtu_channel_6:
01039         err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst6);
01040         break;
01041     case k_mtu_channel_7:
01042         err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst7);
01043         break;
01044     default:
01045         return k_rx_err_invalid_arg;
01046 }
01047
01048 return err;
01049 }

```

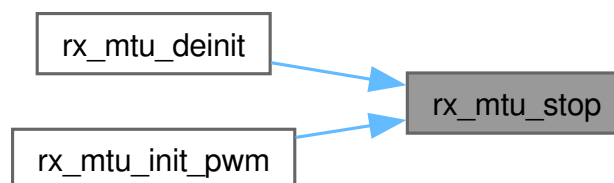
References [internal_clear_tstr_bit\(\)](#), [internal_is_valid_channel\(\)](#), [k_mtu_channel_0](#), [k_mtu_channel_1](#), [k_mtu_channel_2](#), [k_mtu_channel_3](#), [k_mtu_channel_4](#), [k_mtu_channel_6](#), [k_mtu_channel_7](#), [k_mtu_tstr_cst0](#), [k_mtu_tstr_cst1](#), [k_mtu_tstr_cst2](#), [k_mtu_tstr_cst3](#), [k_mtu_tstr_cst4](#), [k_mtu_tstr_cst6](#), [k_mtu_tstr_cst7](#), [mtu_tstra\(\)](#), and [mtu_tstrb\(\)](#).

Referenced by [rx_mtu_deinit\(\)](#), and [rx_mtu_init_pwm\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.129.4 Variable Documentation

8.129.4.1 s'mtu'initialized

```
bool s_mtu_initialized[k_mtu_max_channels] = {false} [static]
```

Track which MTU channels have been initialized.

Boolean array indexed by MTU channel number (0-7). Elements are set to true by [rx_mtu_init_pwm\(\)](#) and cleared to false by [rx_mtu_deinit\(\)](#). Guards all public API functions that require prior initialization.

Note

Indexed directly by [rx_mtu_channel_t](#) value (sparse: indices 0-4, 6-7)

Warning

Do not modify directly - use the public API functions

Since

Version 1.0.0

Definition at line 195 of file [rx_mtu.c](#).

```
00195 {false};
```

Referenced by [rx_mtu_deinit\(\)](#), [rx_mtu_enable_output\(\)](#), [rx_mtu_get_duty\(\)](#), [rx_mtu_get_period\(\)](#), [rx_mtu_init_pwm\(\)](#), [rx_mtu_set_duty\(\)](#), [rx_mtu_set_duty_raw\(\)](#), and [rx_mtu_start\(\)](#).

8.129.4.2 s`mtu`period

```
uint16_t s_mtu_period[k_mtu_max_channels] = {} [static]
```

Cached PWM period register values for each MTU channel.

Stores the TGRA (period) register value written during [rx_mtu_init_pwm\(\)](#). Used by [rx_mtu_set_duty\(\)](#) to convert duty percentage to raw counts without re-reading the hardware register, and by [rx_mtu_get_period\(\)](#) to report the current period.

Note

Indexed directly by [rx_mtu_channel_t](#) value (sparse: indices 0-4, 6-7)

Warning

Do not modify directly - value must match hardware TGRA register

Since

Version 1.0.0

Definition at line 211 of file [rx_mtu.c](#).

```
00211 {};
```

Referenced by [rx_mtu_deinit\(\)](#), [rx_mtu_get_duty\(\)](#), [rx_mtu_get_period\(\)](#), [rx_mtu_init_pwm\(\)](#), [rx_mtu_set_duty\(\)](#), and [rx_mtu_set_duty_raw\(\)](#).

8.129.4.3 s`tag

```
const char* const s_tag = "MTU" [static]
```

Definition at line 117 of file [rx_mtu.c](#).

8.130 rx_mtu.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_mtu.c
00003  * @brief MTU PWM Driver Implementation for RX72N
00004  *
00005  * @details
00006  * **Production-grade implementation** of the Multi-Function Timer Unit (MTU)
00007  * peripheral driver for PWM generation on RX72N. The MTU provides 16-bit
00008  * timer resolution with multiple outputs per channel.
00009  *
00010  * ## Module Architecture
00011  *
00012  * The MTU peripheral is organized as:
00013  * - **MTU0-MTU4**: First group, controlled by MSTPCRA.MSTPA9 and TSTRA
00014  * - **MTU6-MTU7**: Second group, controlled by MSTPCRA.MSTPA9 and TSTRB
00015  * - **Note**: MTU5 exists but is excluded from general PWM use
00016  *
00017  * ## PWM Mode 1 (Triangle Wave - Center-Aligned)
00018  *
00019  * This driver configures MTU in PWM Mode 1 for center-aligned PWM:
00020  * - Counter counts up from 0 to TGRA (period), then down to 0
00021  * - Compare matches on TGRB/TGRC/TGRD control duty cycle
00022  * - Symmetric PWM pulses reduce harmonic content
00023  * - Lower EMI compared to edge-aligned PWM
00024  *
00025  * @par PWM Timing Diagram:
00026  *
00027  * ```
00028  *          TGRA (period)
00029  *          |
00030  *      /\  /\  /\
00031  *     /\  /\  /\
00032  *    /\  /\  /\
00033  *   /\  /\  /\
00034  *  /\  /\  /\
00035  * /\  /\  /\
00036  * 0-----0-----0-----0
00037  *   | | | | | | | |
00038  *   TGRB TGRB (compare match points)
00039  * ```
00040  *
00041  * ## Frequency and Resolution
00042  *
00043  * @par Calculation:
00044  * @f
00045  * \text{Period} = \frac{f_{PCLKA}}{2 \times f_{PWM}}
00046  * @f
00047  *
00048  * @par Example: 20kHz PWM with PCLKA=120MHz:
00049  *
00050  * | Parameter | Value | Notes |
00051  * |-----|-----|-----|
00052  * | Period | 3000 counts | 120MHz / (2 x 20kHz) |
00053  * | Resolution | ~11.5 bits | log2(3000) ~ 11.5 |
00054  * | Duty range | 0-3000 | 0-100% in integer counts |
00055  * | Step size | 0.033% | 100% / 3000 |
00056  *
00057  * ## Register Base Addresses
00058  *
00059  * | Channel | Base Address | TSTR Register |
00060  * |-----|-----|-----|
00061  * | MTU0 | 0x000C1300 | TSTRA (bit 0) |
00062  * | MTU1 | 0x000C1380 | TSTRA (bit 1) |
00063  * | MTU2 | 0x000C1400 | TSTRA (bit 2) |
00064  * | MTU3 | 0x000C1200 | TSTRA (bit 6) |
00065  * | MTU4 | 0x000C1201 | TSTRA (bit 7) |
00066  * | MTU6 | 0x000C1A00 | TSTRB (bit 6) |
00067  * | MTU7 | 0x000C1A01 | TSTRB (bit 7) |
00068  *
00069  * ## Static Variables
00070  *
00071  * | Variable | Type | Purpose |
00072  * |-----|-----|-----|
00073  * | s_mtu_initialized[] | bool[8] | Track which channels are initialized |
00074  * | s_mtu_period[] | uint16_t[8] | Cache period values for duty calculation |
00075  * | s_tag | const char* | Log tag "MTU" |
00076  *
00077  * @par NASA Power of 10 Compliance:
00078  *
00079  * | Rule | Status | Implementation |
00080  * |-----|-----|-----|
00081  * | 1. Simple control flow | [OK] | No goto/setjmp/recursion |
00082  * | 2. Fixed loop bounds | [OK] | No unbounded loops |
00083  * | 3. No dynamic allocation | [OK] | All static, zero malloc |
00084  * | 4. Small functions | [OK] | All functions < 60 lines |

```

```

00083 * | 5. Assertions | [OK] | RX_VALIDATE_PTR, RX_VALIDATE_INIT |
00084 * | 6. Narrow scope | [OK] | Static variables, internal functions |
00085 * | 7. Check return values | [OK] | All rx_err_t returns checked |
00086 * | 8. Limited preprocessor | [OK] | C23 typed enums for constants |
00087 * | 9. Pointer restrictions | [OK] | Single-level pointers |
00088 * | 10. Compiler warnings | [OK] | Clean with -Wall -Wextra -Werror |
00089 *
00090 * @par SOLID Principles:
00091 *
00092 * **Single Responsibility**: This file handles ONLY MTU PWM generation.
00093 * Motor control logic and encoder reading are in separate modules.
00094 *
00095 * @see rx_mtu.h Public API documentation
00096 * @see rx72n_mtu_regs.h Register structure definitions
00097 * @see RX72N Hardware Manual Chapter 24 - Multi-Function Timer Unit
00098 *
00099 * @author Locked, Inc.
00100 * @date 2026-01-27
00101 * @copyright Copyright (c) 2026 Locked Inc.
00102 * SPDX-License-Identifier: MIT
00103 * @since Version 1.0.0
00104 */
00105
00106 #ifdef __RX__
00107
00108 #include "rx_mtu.h"
00109
00110 #include <stddef.h>
00111
00112 #include "rx72n_regs.h"
00113 #include "rx_check.h"
00114 #include "rx_log.h"
00115 #include "rx_register_protection.h"
00116
00117 static const char* const s_tag = "MTU";
00118
00119 /*
=====
00120 * Constants
00121 *
=====
00122 */
00123
00124 /** @brief MTU general constants */
00125 typedef enum : uint8_t {
00126     k_mtu_max_channels = 8, /**< MTU0-MTU4, MTU6-MTU7 (sparse indexing, max index is 7) */
00127 } mtu_constants_t;
00128
00129 /** @brief MTU module stop bit position in MSTPCRA */
00130 typedef enum : uint8_t {
00131     k_mtu_mstpa_mtu0_4 = 9, /**< MTU module stop bit (all channels) -- MSTPCRA.MSTPA9 */
00132 } mtu_module_stop_bits_t;
00133
00134 /** @brief Bit manipulation constants */
00135 typedef enum : uint32_t {
00136     k_mtu_bit_one = 1UL, /**< Single bit value for shifts */
00137 } mtu_bit_constants_t;
00138
00139 /** @brief Period calculation constants */
00140 typedef enum : uint16_t {
00141     k_mtu_period_divisor = 2, /**< Triangle wave period divisor */
00142     k_mtu_period_max = 0xFFFF, /**< Maximum valid period (16-bit) */
00143     k_mtu_period_min = 10, /**< Minimum valid period */
00144     k_mtu_period_zero = 0, /**< Zero period value */
00145 } mtu_period_constants_t;
00146
00147 /** @brief TIOR register shift positions */
00148 typedef enum : uint8_t {
00149     k_mtu_tior_low_shift = 0, /**< Low nibble shift (MTIOCA/MTIOCC) */
00150     k_mtu_tior_high_shift = 4, /**< High nibble shift (MTIOCB/MTIOCD) */
00151 } mtu_tior_shift_t;
00152
00153 /** @brief TIOR register clear-masks (value AND'd against TIOR to zero a nibble)
00154 *
00155 * The low nibble (bits 3:0) controls MTIOCA/MTIOCC and the high nibble
00156 * (bits 7:4) controls MTIOCB/MTIOCD. To zero the low nibble without
00157 * disturbing the high nibble you AND with 0xF0; to zero the high nibble
00158 * you AND with 0x0F. Names therefore describe which nibble is *kept*.
00159 */
00160 typedef enum : uint8_t {
00161     k_mtu_tior_keep_high_clear_low = 0xF0, /**< AND-mask: keep high nibble, clear low (MTIOCA/C) */
00162     k_mtu_tior_keep_low_clear_high = 0x0F, /**< AND-mask: keep low nibble, clear high (MTIOCB/D) */
00163 } mtu_tior_mask_t;
00164
00165 /** @brief TIOR output disabled value */
00166 typedef enum : uint8_t {
00167     k_mtu_tior_disabled = 0x00, /**< Output disabled */

```

```

00168 } mtu_tior_disabled_t;
00169
00170 /** @brief Duty cycle calculation constants */
00171 typedef enum : uint8_t {
00172     k_mtu_duty_min    = 0, /**< Minimum duty cycle (0%) */
00173     k_mtu_duty_max    = 100, /**< Maximum duty cycle (100%) */
00174     k_mtu_duty_divisor = 100, /**< Divisor for percentage conversion */
00175 } mtu_duty_constants_t;
00176
00177 /*
=====
00178  * Static Variables
00179  *
=====
00180  */
00181
00182 /**
00183  * @var s_mtu_initialized
00184  * @brief Track which MTU channels have been initialized
00185  *
00186  * @details
00187  * Boolean array indexed by MTU channel number (0-7). Elements are set to
00188  * true by rx_mtu_init_pwm() and cleared to false by rx_mtu_deinit(). Guards
00189  * all public API functions that require prior initialization.
00190  *
00191  * @note Indexed directly by rx_mtu_channel_t value (sparse: indices 0-4, 6-7)
00192  * @warning Do not modify directly - use the public API functions
00193  * @since Version 1.0.0
00194  */
00195 static bool s_mtu_initialized[k_mtu_max_channels] = {false};
00196
00197 /**
00198  * @var s_mtu_period
00199  * @brief Cached PWM period register values for each MTU channel
00200  *
00201  * @details
00202  * Stores the TGRA (period) register value written during rx_mtu_init_pwm().
00203  * Used by rx_mtu_set_duty() to convert duty percentage to raw counts without
00204  * re-reading the hardware register, and by rx_mtu_get_period() to report
00205  * the current period.
00206  *
00207  * @note Indexed directly by rx_mtu_channel_t value (sparse: indices 0-4, 6-7)
00208  * @warning Do not modify directly - value must match hardware TGRA register
00209  * @since Version 1.0.0
00210  */
00211 static uint16_t s_mtu_period[k_mtu_max_channels] = {};
00212
00213 /*
=====
00214  * Internal Helper Functions
00215  *
=====
00216  */
00217
00218 /**
00219  * @brief Get MTU channel register base address
00220  *
00221  * @details
00222  * Maps an MTU channel enumeration to its hardware register base address.
00223  * The MTU peripheral has sparse channel numbering (0-4, 6-7 with no 5).
00224  *
00225  * ## Memory Map
00226  *
00227  * | Channel | Base Address | Description |
00228  * |-----|-----|-----|
00229  * | MTU0 | 0x000C1290 | 16-bit timer with 4 outputs |
00230  * | MTU1 | 0x000C1290 | 16-bit timer with 2 outputs |
00231  * | MTU2 | 0x000C1292 | 16-bit timer with 2 outputs |
00232  * | MTU3 | (nullptr) | sparse layout, cannot use standard struct API |
00233  * | MTU4 | (nullptr) | sparse layout, cannot use standard struct API |
00234  * | MTU6 | (nullptr) | interleaved layout, use mtu6() with rx_mtu3_channel_regs_t |
00235  * | MTU7 | (nullptr) | interleaved layout, use mtu7() with direct register offsets |
00236  *
00237  * MTU3, MTU4, MTU6, and MTU7 all have an interleaved register layout that is
00238  * incompatible with rx_mtu_channel_regs_t. Use mtu3()/mtu6() with
00239  * rx_mtu3_channel_regs_t directly; this function returns nullptr for all four
00240  * channels to prevent callers from applying the wrong struct offset mapping.
00241  *
00242  *
00243  *
00244  * @pre Module clock must be enabled before accessing returned pointer
00245  *
00246  * @post Returned pointer valid for hardware access
00247  * @post No hardware state modified
00248  *
00249  * @note Thread-safe: Returns constant address
00250  * @note Returns void* to accommodate different register structure types

```



```

00251 *
00252 * @see mtu0(), mtu1(), etc. Underlying accessor functions
00253 * @see rx72n_mtu_regs.h Register structure definitions
00254 *
00255 * @since Version 1.0.0
00256 */
00257 static volatile void* internal_get_mtu_base(const rx_mtu_channel_t channel)
00258 {
00259     switch (channel) {
00260         case k_mtu_channel_0:
00261             return (volatile void*)mtu0();
00262         case k_mtu_channel_1:
00263             return (volatile void*)mtu1();
00264         case k_mtu_channel_2:
00265             return (volatile void*)mtu2();
00266         case k_mtu_channel_3: /* fall through -- sparse layout, incompatible with standard API */
00267         case k_mtu_channel_4: /* MTU3/4 use rx_mtu3_channel_regs_t; call mtu3()/mtu4() directly */
00268             return nullptr;
00269         case k_mtu_channel_6: /* fall through -- interleaved layout, use mtu6() with rx_mtu3_channel_regs_t */
00270         case k_mtu_channel_7: /* interleaved layout, use mtu7() with direct address offsets */
00271             return nullptr;
00272         default:
00273             return nullptr;
00274     }
00275 }
00276
00277 /**
00278 * @brief Check if MTU channel is valid
00279 *
00280 * @details
00281 * Validates an MTU channel by checking if it has a valid register base address.
00282 * MTU channels 0-4 and 6-7 are valid; channel 5 does not exist.
00283 *
00284 *
00285 *
00286 * @note Thread-safe: Pure function
00287 * @note Delegates to internal_get_mtu_base() for actual validation
00288 *
00289 * @see internal_get_mtu_base() Underlying validation logic
00290 *
00291 * @since Version 1.0.0
00292 */
00293 static bool internal_is_valid_channel(const rx_mtu_channel_t channel)
00294 {
00295     return internal_get_mtu_base(channel) != nullptr;
00296 }
00297
00298 /**
00299 * @brief Clear a counter-start bit in a TSTR register and verify it cleared
00300 *
00301 * @details
00302 * Performs a read-modify-write on the TSTR register to clear the specified
00303 * bit mask, then reads back the register to confirm the bit was cleared.
00304 * Returns a hardware error if the bit is still set after the write, which
00305 * can indicate a peripheral fault.
00306 *
00307 *
00308 *
00309 * @pre tstr points to a valid TSTRA or TSTRB hardware register
00310 * @pre mask is a valid CSTn bit mask from mtu_tstr_bits_t
00311 *
00312 * @post Specified CSTn bit is cleared (counter stopped) on success
00313 * @post tstr register unchanged on error
00314 *
00315 * @note Not thread-safe: caller must ensure exclusive access to TSTR register
00316 * @note Readback verification adds one extra register read cycle
00317 *
00318 * @see rx_mtu_stop() Primary caller of this function
00319 * @see k_mtu_tstr_cst0 Counter-start bit constants
00320 *
00321 * @since Version 1.0.0
00322 */
00323 static rx_err_t internal_clear_tstr_bit(volatile rx_mtu_tstr_regs_t* tstr, const uint8_t mask)
00324 {
00325     if (tstr == nullptr) {
00326         return k_rx_err_invalid_arg;
00327     }
00328
00329     tstr->tstr &= (uint8_t)~mask;
00330     if ((tstr->tstr & mask) != 0) {
00331         return k_rx_err_hw_error;
00332     }
00333
00334     return k_rx_ok;
00335 }
00336
00337 /**

```

```

00338 * @brief Calculate period register value from frequency
00339 *
00340 * @details
00341 * Converts desired PWM frequency to the TGRA (period) register value for
00342 * PWM Mode 1 (triangle wave, center-aligned).
00343 *
00344 * ## Calculation
00345 *
00346 * @f[
00347 * \text{period} = \frac{f_{\text{PCLKA}}}{2 \times f_{\text{PWM}}}
00348 * @f]
00349 *
00350 * Where:
00351 * - @f$ f_{\text{PCLKA}} @f$ = 120 MHz
00352 * - Divisor of 2 accounts for up/down counting in triangle mode
00353 *
00354 * ## Valid Frequency Range
00355 *
00356 * | Frequency | Period | Status |
00357 * |-----|-----|-----|
00358 * | < 915 Hz | > 65535 | Rejected (overflow) |
00359 * | 915 Hz - 6 MHz | 10-65535 | Valid |
00360 * | > 6 MHz | < 10 | Rejected (too small) |
00361 *
00362 *
00363 *
00364 * @pre PCLKA configured to 120 MHz
00365 * @pre period points to valid uint16_t storage
00366 *
00367 * @post *period contains value in range [10, 65535]
00368 *
00369 * @note 16-bit resolution limits frequency range compared to 32-bit GPTW
00370 *
00371 * @see k_pclka_hz PCLKA clock frequency
00372 * @see k_mtu_period_max Maximum 16-bit period value
00373 *
00374 * @since Version 1.0.0
00375 */
00376 static rx_err_t internal_calculate_period(const uint32_t frequency_hz, uint16_t* period)
00377 {
00378     if (period == nullptr) {
00379         return k_rx_err_null_ptr;
00380     }
00381
00382     if (frequency_hz == k_mtu_period_zero) {
00383         return k_rx_err_invalid_arg;
00384     }
00385
00386     /* For PWM mode 1 (triangle wave):
00387      * Period = PCLKA / (2 * frequency)
00388      * PCLKA = 120 MHz
00389      */
00390     const uint32_t pclka = k_pclka_hz;
00391     const uint32_t period_calc = pclka / (k_mtu_period_divisor * frequency_hz);
00392
00393     /* Check if period fits in 16-bit register */
00394     if (period_calc > k_mtu_period_max) {
00395         rx_log_error(s_tag, "Frequency too low");
00396         return k_rx_err_invalid_arg;
00397     }
00398
00399     if (period_calc < k_mtu_period_min) {
00400         rx_log_error(s_tag, "Frequency too high");
00401         return k_rx_err_invalid_arg;
00402     }
00403
00404     *period = (uint16_t)period_calc;
00405     return k_rx_ok;
00406 }
00407
00408 /**
00409 * @brief Get TGR register pointer for output channel
00410 *
00411 *
00412 */
00413 static volatile uint16_t* internal_get_tgr_register(volatile rx_mtu_channel_regs_t* mtu,
00414                                                     const rx_mtu_output_t output)
00415 {
00416     if (mtu == nullptr) {
00417         return nullptr;
00418     }
00419
00420     switch (output) {
00421         case k_mtu_output_a:
00422             return &mtu->tgra;
00423         case k_mtu_output_b:
00424             return &mtu->tgrb;
00425     }

```

```

00425     case k_mtu_output_c:
00426         return &mtu->tgrc;
00427     case k_mtu_output_d:
00428         return &mtu->tgrd;
00429     default:
00430         return nullptr;
00431 }
00432 }
00433
00434 /**
00435  * @brief Write a raw duty count to the TGR register for the specified output
00436  *
00437  * @details
00438  * Resolves the TGR register pointer for the requested output via
00439  * internal_get_tgr_register() and writes duty_count to it. The write is
00440  * buffered by hardware and takes effect at the next PWM period boundary,
00441  * producing a glitch-free duty cycle update.
00442  *
00443  * Algorithm:
00444  * 1. Validate mtu pointer is non-NULL
00445  * 2. Resolve TGR register pointer for output via internal_get_tgr_register()
00446  * 3. Validate resolved TGR pointer is non-NULL
00447  * 4. Write duty_count to TGR register
00448  *
00449  *
00450  *
00451  * @pre mtu points to a valid, initialized MTU channel register block
00452  * @pre duty_count has been validated or clamped by the caller to [0, period]
00453  *
00454  * @post TGR register written with duty_count
00455  * @post Change takes effect at next PWM period boundary (buffered by hardware)
00456  *
00457  * @note Not thread-safe: caller must ensure exclusive access to the TGR register
00458  * @note Duty count is not bounds-checked; caller must clamp before calling
00459  *
00460  * @code{.c}
00461  * volatile rx_mtu_channel_regs_t* mtu =
00462  *     (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(k_mtu_channel_0);
00463  * rx_err_t err = internal_set_duty_raw_mtu(mtu, k_mtu_output_b, 1500U);
00464  * @endcode
00465  *
00466  * @see internal_get_tgr_register() Resolves TGR pointer from output selector
00467  * @see rx_mtu_set_duty_raw() Public API caller that clamps duty_count first
00468  *
00469  * @since Version 1.0.0
00470  */
00471 static rx_err_t internal_set_duty_raw_mtu(volatile rx_mtu_channel_regs_t* mtu,
00472                                           const rx_mtu_output_t      output,
00473                                           const uint16_t             duty_count)
00474 {
00475     if (mtu == nullptr) {
00476         return k_rx_err_invalid_arg;
00477     }
00478
00479     volatile uint16_t* tgr = internal_get_tgr_register(mtu, output);
00480     if (tgr == nullptr) {
00481         return k_rx_err_invalid_arg;
00482     }
00483
00484     /* Update duty cycle (buffered, takes effect on next period) */
00485     *tgr = duty_count;
00486     return k_rx_ok;
00487 }
00488
00489 /**
00490  * Public API Implementation
00491  *
00492  *
00493  *
00494  */
00495 /**
00496  * @brief Initialize MTU channel for PWM operation
00497  *
00498  * @details
00499  * Configures an MTU channel for PWM Mode 1 (triangle wave, center-aligned)
00500  * operation. Performs complete hardware setup including module clock enable,
00501  * timer configuration, and automatic start.
00502  *
00503  * ## Initialization Sequence
00504  * 1. Validate config pointer
00505  * 2. Get channel register base address
00506  * 3. Calculate period from frequency
00507  * 4. Enable MTU module clock (clear MSTPA bit)
00508  * 5. Stop timer if running
00509  * 6. Configure TCR (clock source, clear condition)

```

```

00510 * 7. Configure TMDR (PWM Mode 1)
00511 * 8. Configure TIORH/TIOLR (output behavior)
00512 * 9. Set TGRA (period), clear TGRB/TGRC/TGRD (duty = 0)
00513 * 10. Clear TCNT counter
00514 * 11. Store period and set initialized flag
00515 * 12. Start timer
00516 *
00517 * ## PWM Mode 1 Configuration
00518 *
00519 * | Register | Value | Effect |
00520 * |-----|-----|-----|
00521 * | TCR | TPSC=1, CCLR=TGRA | PCLKA/1, clear on TGRA match |
00522 * | TMDR | MD=PWM1 | Triangle wave mode |
00523 * | TIORH/L | Init low | Outputs toggle on compare match |
00524 *
00525 *
00526 *
00527 * @pre System clock configured (PCLKA = 120 MHz)
00528 * @pre config pointer valid with valid frequency
00529 *
00530 * @post MTU channel configured and timer running
00531 * @post All duty cycles initialized to 0%
00532 * @post s_mtu_initialized[channel] = true
00533 *
00534 * @note Timer starts automatically
00535 * @note Call once during system initialization
00536 *
00537 * @warning Not thread-safe: Configure before starting motor tasks
00538 *
00539 * @par Example:
00540 * @code{.c}
00541 * rx_mtu_config_t config = {
00542 *     .frequency_hz = 20000 // 20 kHz
00543 * };
00544 * rx_err_t err = rx_mtu_init_pwm(k_mtu_channel_0, &config);
00545 * @endcode
00546 *
00547 * @see rx_mtu.h Complete API documentation
00548 * @see rx_mtu_deinit() Cleanup function
00549 *
00550 * @since Version 1.0.0
00551 *
00552 * @callgraph
00553 */
00554 rx_err_t rx_mtu_init_pwm(const rx_mtu_channel_t channel, const rx_mtu_config_t* config)
00555 {
00556     RX_VALIDATE_PTR(config, s_tag, "config pointer is nullptr");
00557
00558     volatile rx_mtu_channel_regs_t* const mtu =
00559         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00560     if (mtu == nullptr) {
00561         return k_rx_err_invalid_arg;
00562     }
00563
00564     /* Calculate period from frequency */
00565     uint16_t period;
00566     rx_err_t err = internal_calculate_period(config->frequency_hz, &period);
00567     if (err != k_rx_ok) {
00568         return err;
00569     }
00570
00571     rx_log_info(s_tag, "Initializing MTU");
00572
00573     /* Enable MTU module (clear module stop bit) */
00574     *prcr_reg() = k_rx_prcr_unlock_prc1_prc3;
00575
00576     system_regs()->mstpcra &= ~(k_mtu_bit_one « k_mtu_mstpa_mtu0_4);
00577
00578     *prcr_reg() = k_rx_prcr_lock;
00579
00580     /* Stop timer before configuration */
00581     err = rx_mtu_stop(channel);
00582     if (err != k_rx_ok) {
00583         return err;
00584     }
00585
00586     /* Configure timer control register
00587     * - PCLKA/1 (120 MHz)
00588     * - Clear counter on TGRA compare match
00589     */
00590     mtu->tcr = k_mtu_tcr_tpdc_1 | k_mtu_tcr_cclr_tgra;
00591
00592     /* Configure PWM mode 1 (triangle wave, center-aligned) */
00593     mtu->tmdr = k_mtu_tmdr_md_pwm1;
00594
00595     /* Configure I/O control registers for PWM outputs
00596     * TIORH: Controls MTIOCA (output A) and MTIOCB (output B)

```

```

00597 * TIORL: Controls MTIOCC (output C) and MTIOCD (output D)
00598 *
00599 * For PWM mode 1:
00600 * - Initial low, high on up-count compare, low on down-count compare
00601 */
00602 mtu->tiorh = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCA */
00603 (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCB */
00604 mtu->tiorl = (k_mtu_tior_init_low « k_mtu_tior_low_shift) | /* MTIOCC */
00605 (k_mtu_tior_init_low « k_mtu_tior_high_shift); /* MTIOCD */
00606
00607 /* Set period (TGRA = top of triangle wave) */
00608 mtu->tgra = period;
00609
00610 /* Set initial duty cycle to 0% for all outputs */
00611 mtu->tgrb = k_mtu_period_zero; /* MTIOCB duty */
00612 mtu->tgrc = k_mtu_period_zero; /* MTIOCC duty */
00613 mtu->tgrd = k_mtu_period_zero; /* MTIOCD duty */
00614
00615 /* Clear counter */
00616 mtu->tcnt = k_mtu_period_zero;
00617
00618 /* Save period for duty cycle calculations */
00619 s_mtu_period[channel] = period;
00620 s_mtu_initialized[channel] = true;
00621
00622 /* Start timer */
00623 err = rx_mtu_start(channel);
00624 if (err != k_rx_ok) {
00625     return err;
00626 }
00627
00628 rx_log_info(s_tag, "MTU initialized successfully");
00629
00630 return k_rx_ok;
00631 }
00632
00633 /**
00634 * @brief Set PWM duty cycle as a floating-point percentage
00635 *
00636 * @details
00637 * Converts a percentage value to raw timer counts using the cached period and
00638 * writes it to the appropriate TGR register. The update is buffered and takes
00639 * effect at the next PWM period boundary (glitch-free).
00640 *
00641 * Algorithm:
00642 * 1. Validate channel and initialization state
00643 * 2. Validate duty_percent in range [0.0, 100.0]
00644 * 3. Retrieve cached period from s_mtu_period[channel]
00645 * 4. Compute duty_count = (duty_percent * period) / 100
00646 * 5. Write duty_count to TGR register via internal_set_duty_raw_mtu()
00647 *
00648 *
00649 *
00650 * @pre Channel must be initialized via rx_mtu_init_pwm()
00651 * @pre duty_percent must be in [0.0, 100.0]
00652 *
00653 * @post TGR register updated with new duty count
00654 * @post Change takes effect at next PWM period boundary
00655 *
00656 * @note Not thread-safe: caller must prevent concurrent access
00657 *
00658 * @code{.c}
00659 * rx_err_t err = rx_mtu_set_duty(k_mtu_channel_0, k_mtu_output_b, 50.0f);
00660 * @endcode
00661 *
00662 * @see rx_mtu_set_duty_raw() Set duty using raw count value
00663 * @see rx_mtu_get_duty() Read current duty cycle
00664 *
00665 * @since Version 1.0.0
00666 */
00667 rx_err_t
00668 rx_mtu_set_duty(const rx_mtu_channel_t channel, rx_mtu_output_t output, const float duty_percent)
00669 {
00670     volatile rx_mtu_channel_regs_t* const mtu =
00671         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00672     if (mtu == nullptr) {
00673         return k_rx_err_invalid_arg;
00674     }
00675
00676     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00677
00678     if (duty_percent < (float)k_mtu_duty_min || duty_percent > (float)k_mtu_duty_max) {
00679         rx_log_error(s_tag, "Invalid duty cycle");
00680         return k_rx_err_invalid_arg;
00681     }
00682
00683     /* Convert percentage to count value */

```

```

00684  const uint16_t period    = s_mtu_period[channel];
00685  const uint16_t duty_count = (uint16_t)((duty_percent * period) / (float)k_mtu_duty_divisor);
00686
00687  return internal_set_duty_raw_mtu(mtu, output, duty_count);
00688 }
00689
00690 /**
00691  * @brief Set PWM duty cycle using a raw 16-bit timer count
00692  *
00693  * @details
00694  * Directly sets the compare register (TGR) for the specified output to
00695  * duty_count timer ticks. Values larger than the channel period are clamped
00696  * to the period value (100% duty). The update is buffered.
00697
00698  * Use this function when the caller has already converted a duty percentage
00699  * to counts (e.g. via rx_mtu_get_period()) to avoid redundant floating-point
00700  * multiplication.
00701  *
00702  *
00703  *
00704  * @pre Channel must be initialized via rx_mtu_init_pwm()
00705  * @pre duty_count should be in [0, period]; values above period are clamped
00706  *
00707  * @post TGR register written with (possibly clamped) duty_count
00708  * @post Change takes effect at next PWM period boundary
00709  *
00710  * @note Not thread-safe: caller must prevent concurrent access
00711  *
00712  * @code{.c}
00713  * uint16_t period;
00714  * rx_mtu_get_period(k_mtu_channel_0, &period);
00715  * rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, period / 2U); // 50%
00716  * @endcode
00717  *
00718  * @see rx_mtu_set_duty() Set duty using floating-point percentage
00719  * @see rx_mtu_get_period() Retrieve period for count calculation
00720  *
00721  * @since Version 1.0.0
00722  */
00723 rx_err_t
00724 rx_mtu_set_duty_raw(const rx_mtu_channel_t channel, rx_mtu_output_t output, uint16_t duty_count)
00725 {
00726     volatile rx_mtu_channel_regs_t* const mtu =
00727         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00728     if (mtu == nullptr) {
00729         return k_rx_err_invalid_arg;
00730     }
00731
00732     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00733
00734     /* Clamp to period */
00735     const uint16_t period = s_mtu_period[channel];
00736     if (duty_count > period) {
00737         duty_count = period;
00738     }
00739
00740     return internal_set_duty_raw_mtu(mtu, output, duty_count);
00741 }
00742
00743 /**
00744  * @brief Read the current PWM duty cycle as a floating-point percentage
00745  *
00746  * @details
00747  * Reads the TGR register for the specified output and converts the raw count
00748  * to a percentage using the cached period value.
00749  *
00750  * @f[
00751  * \text{duty\_percent} = \frac{\text{duty\_count}}{\text{period}} \times 100
00752  * @f]
00753  *
00754  *
00755  *
00756  * @pre Channel must be initialized via rx_mtu_init_pwm()
00757  * @pre duty_percent must point to valid float storage
00758  *
00759  * @post *duty_percent contains duty cycle in [0.0, 100.0] on success
00760  * @post Hardware TGR register unchanged
00761  *
00762  * @note Not thread-safe: concurrent set_duty calls may cause torn reads
00763  *
00764  * @code{.c}
00765  * float duty;
00766  * rx_err_t err = rx_mtu_get_duty(k_mtu_channel_0, k_mtu_output_b, &duty);
00767  * @endcode
00768  *
00769  * @see rx_mtu_set_duty() Set duty using percentage
00770  * @see rx_mtu_get_period() Retrieve period count

```

```

00771 *
00772 * @since Version 1.0.0
00773 */
00774 rx_err_t
00775 rx_mtu_get_duty(const rx_mtu_channel_t channel, const rx_mtu_output_t output, float* duty_percent)
00776 {
00777     RX_VALIDATE_PTR(duty_percent, s_tag, "duty_percent pointer is nullptr");
00778
00779     volatile rx_mtu_channel_regs_t* const mtu =
00780         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00781     if (mtu == nullptr) {
00782         return k_rx_err_invalid_arg;
00783     }
00784
00785     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00786
00787     const volatile uint16_t* const tgr = internal_get_tgr_register(mtu, output);
00788     if (tgr == nullptr) {
00789         return k_rx_err_invalid_arg;
00790     }
00791
00792     const uint16_t period = s_mtu_period[channel];
00793     const uint16_t duty_count = *tgr;
00794
00795     *duty_percent = (float)(duty_count * (float)k_mtu_duty_max) / period;
00796
00797     return k_rx_ok;
00798 }
00799
00800 /**
00801 * @brief Get PWM period count for MTU channel
00802 *
00803 * @details
00804 * Returns the cached period value (TGRA) from initialization. Useful for
00805 * calculating raw duty counts from percentages without floating-point.
00806 *
00807 *
00808 *
00809 * @pre Channel must be initialized via rx_mtu_init_pwm()
00810 * @pre period_count must point to valid uint16_t storage
00811 *
00812 * @post *period_count contains the TGRA register value on success
00813 * @post Hardware registers unchanged
00814 *
00815 * @note Thread-safe: reads from static cache, no hardware access
00816 *
00817 * @code{.c}
00818 * uint16_t period;
00819 * rx_err_t err = rx_mtu_get_period(k_mtu_channel_0, &period);
00820 * if (err == k_rx_ok) {
00821 *     uint16_t half_duty = period / 2U; // 50%
00822 *     rx_mtu_set_duty_raw(k_mtu_channel_0, k_mtu_output_b, half_duty);
00823 * }
00824 * @endcode
00825 *
00826 * @see rx_mtu.h Complete API documentation
00827 * @see rx_mtu_set_duty_raw() Use period count to set duty
00828 *
00829 * @since Version 1.0.0
00830 */
00831 rx_err_t rx_mtu_get_period(const rx_mtu_channel_t channel, uint16_t* period_count)
00832 {
00833     RX_VALIDATE_PTR(period_count, s_tag, "period_count pointer is nullptr");
00834
00835     if (!internal_is_valid_channel(channel)) {
00836         return k_rx_err_invalid_arg;
00837     }
00838
00839     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00840
00841     *period_count = s_mtu_period[channel];
00842     return k_rx_ok;
00843 }
00844
00845 /**
00846 * @brief Enable or disable a single MTU PWM output pin
00847 *
00848 * @details
00849 * Controls whether the MTIOCA/B/C/D pin actively drives the PWM waveform or
00850 * is held in its initial state (off). Modifies the TIORH or TIORL register
00851 * nibble corresponding to the requested output without disturbing other outputs.
00852 *
00853 * | Output | Register | Nibble |
00854 * |-----|-----|-----|
00855 * | A      | TIORH    | Low (bits 3:0) |
00856 * | B      | TIORH    | High (bits 7:4) |
00857 * | C      | TIORL    | Low (bits 3:0) |

```

```

00858 * | D      | TIORL   | High (bits 7:4) |
00859 *
00860 *
00861 *
00862 * @pre Channel must be initialized via rx_mtu_init_pwm()
00863 * @pre output must be a valid rx_mtu_output_t enum value
00864 *
00865 * @post TIORH or TIORL nibble updated; timer continues running
00866 * @post Other outputs on the same channel are unchanged
00867 *
00868 * @note Not thread-safe: caller must prevent concurrent register access
00869 * @note Timer keeps running; only output pin drive changes
00870 *
00871 * @warning Outputs may remain HIGH when disabled, depending on last compare state
00872 *
00873 * @code{.c}
00874 * rx_mtu_enable_output(k_mtu_channel_0, k_mtu_output_a, false); // disable
00875 * @endcode
00876 *
00877 * @see rx_mtu_init_pwm() Initialize channel before calling
00878 * @see rx_mtu_stop() Halt the entire timer
00879 *
00880 * @since Version 1.0.0
00881 */
00882 rx_err_t
00883 rx_mtu_enable_output(const rx_mtu_channel_t channel, rx_mtu_output_t output, const bool enable)
00884 {
00885     volatile rx_mtu_channel_regs_t* const mtu =
00886         (volatile rx_mtu_channel_regs_t*)internal_get_mtu_base(channel);
00887     if (mtu == nullptr) {
00888         return k_rx_err_invalid_arg;
00889     }
00890
00891     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00892
00893     /* Enable/disable output by modifying TIOR registers */
00894     const uint8_t tior_value = enable ? k_mtu_tior_init_low : k_mtu_tior_disabled;
00895
00896     switch (output) {
00897     case k_mtu_output_a:
00898         mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_high_clear_low) |
00899                               (tior_value « k_mtu_tior_low_shift));
00900         break;
00901     case k_mtu_output_b:
00902         mtu->tiorh = (uint8_t)((mtu->tiorh & k_mtu_tior_keep_low_clear_high) |
00903                               (tior_value « k_mtu_tior_high_shift));
00904         break;
00905     case k_mtu_output_c:
00906         mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_high_clear_low) |
00907                               (tior_value « k_mtu_tior_low_shift));
00908         break;
00909     case k_mtu_output_d:
00910         mtu->tiorl = (uint8_t)((mtu->tiorl & k_mtu_tior_keep_low_clear_high) |
00911                               (tior_value « k_mtu_tior_high_shift));
00912         break;
00913     default:
00914         return k_rx_err_invalid_arg;
00915     }
00916
00917     return k_rx_ok;
00918 }
00919
00920 /**
00921 * @brief Start MTU counter (begin PWM generation)
00922 *
00923 * @details
00924 * Sets the Count Start (CSTn) bit in the appropriate TSTR register to
00925 * begin counter operation. Channels 0-4 use TSTRA, channels 6-7 use TSTRB.
00926 *
00927 * ## TSTR Bit Mapping
00928 *
00929 * | Channel | Register | Bit |
00930 * |-----|-----|-----|
00931 * | MTU0 | TSTRA | CST0 |
00932 * | MTU1 | TSTRA | CST1 |
00933 * | MTU2 | TSTRA | CST2 |
00934 * | MTU3 | TSTRA | CST3 |
00935 * | MTU4 | TSTRA | CST4 |
00936 * | MTU6 | TSTRB | CST6 |
00937 * | MTU7 | TSTRB | CST7 |
00938 *
00939 *
00940 *
00941 * @pre Channel initialized via rx_mtu_init_pwm()
00942 *
00943 * @post Timer counting, PWM output active
00944 *

```



```

00945 * @note Timer automatically started during init
00946 * @note Idempotent: Safe to call when already running
00947 *
00948 * @see rx_mtu_stop() Stop the timer
00949 *
00950 * @since Version 1.0.0
00951 */
00952 rx_err_t rx_mtu_start(const rx_mtu_channel_t channel)
00953 {
00954     if (!internal_is_valid_channel(channel)) {
00955         return k_rx_err_invalid_arg;
00956     }
00957     RX_VALIDATE_INIT(s_mtu_initialized[channel], s_tag, "MTU channel not initialized");
00958
00959     /* Set corresponding bit in TSTR register */
00960     switch (channel) {
00961     case k_mtu_channel_0:
00962         mtu_tstra()->tstr |= k_mtu_tstr_cst0;
00963         break;
00964     case k_mtu_channel_1:
00965         mtu_tstra()->tstr |= k_mtu_tstr_cst1;
00966         break;
00967     case k_mtu_channel_2:
00968         mtu_tstra()->tstr |= k_mtu_tstr_cst2;
00969         break;
00970     case k_mtu_channel_3:
00971         mtu_tstra()->tstr |= k_mtu_tstr_cst3;
00972         break;
00973     case k_mtu_channel_4:
00974         mtu_tstra()->tstr |= k_mtu_tstr_cst4;
00975         break;
00976     case k_mtu_channel_6:
00977         mtu_tstrb()->tstr |= k_mtu_tstr_cst6;
00978         break;
00979     case k_mtu_channel_7:
00980         mtu_tstrb()->tstr |= k_mtu_tstr_cst7;
00981         break;
00982     default:
00983         return k_rx_err_invalid_arg;
00984     }
00985 }
00986
00987 return k_rx_ok;
00988 }
00989
00990 /**
00991 * @brief Stop MTU counter (halt PWM generation)
00992 *
00993 * @details
00994 * Clears the Count Start (CSTn) bit in the appropriate TSTR register to
00995 * halt counter operation. Counter value and output state are preserved.
00996 *
00997 *
00998 *
00999 * @pre None (can be called on uninitialized channel)
01000 *
01001 * @post Timer stopped, counter preserved
01002 * @post Output pins hold current state
01003 *
01004 * @note Does NOT check initialization flag (safe for use during init)
01005 * @note Idempotent: Safe to call when already stopped
01006 *
01007 * @warning Outputs may remain HIGH - disable outputs for safe state
01008 *
01009 * @see rx_mtu_start() Resume the timer
01010 * @see rx_mtu_enable_output() Disable outputs
01011 *
01012 * @since Version 1.0.0
01013 */
01014 rx_err_t rx_mtu_stop(const rx_mtu_channel_t channel)
01015 {
01016     if (!internal_is_valid_channel(channel)) {
01017         return k_rx_err_invalid_arg;
01018     }
01019
01020     /* Clear corresponding bit in TSTR register */
01021     rx_err_t err;
01022     switch (channel) {
01023     case k_mtu_channel_0:
01024         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst0);
01025         break;
01026     case k_mtu_channel_1:
01027         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst1);
01028         break;
01029     case k_mtu_channel_2:
01030         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst2);
01031         break;

```

```

01032     case k_mtu_channel_3:
01033         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst3);
01034         break;
01035     case k_mtu_channel_4:
01036         err = internal_clear_tstr_bit(mtu_tstra(), k_mtu_tstr_cst4);
01037         break;
01038     case k_mtu_channel_6:
01039         err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst6);
01040         break;
01041     case k_mtu_channel_7:
01042         err = internal_clear_tstr_bit(mtu_tstrb(), k_mtu_tstr_cst7);
01043         break;
01044     default:
01045         return k_rx_err_invalid_arg;
01046     }
01047
01048     return err;
01049 }
01050
01051 /**
01052  * @brief Deinitialize an MTU PWM channel
01053  *
01054  * @details
01055  * Performs an orderly shutdown of the specified MTU channel:
01056  * 1. Stop the timer counter (rx_mtu_stop())
01057  * 2. Disable all four outputs (A, B, C, D) via rx_mtu_enable_output()
01058  * 3. Clear the cached period to 0
01059  * 4. Clear the initialized flag
01060  *
01061  * After this call the channel may be safely re-initialized with different
01062  * parameters via rx_mtu_init_pwm().
01063  *
01064  *
01065  *
01066  * @pre Motor or load driven by this channel must be stopped externally before call
01067  * @pre channel must be a valid MTU channel (0-4, 6-7)
01068  *
01069  * @post Timer stopped and outputs disabled
01070  * @post s_mtu_initialized[channel] == false
01071  * @post s_mtu_period[channel] == k_mtu_period_zero
01072  *
01073  * @note Not thread-safe: do not call while another task is accessing this channel
01074  * @note Idempotent with respect to re-initialization (can call init again after)
01075  *
01076  * @warning Ensure motor is at rest before calling to avoid abrupt stop
01077  *
01078  * @code{.c}
01079  * rx_err_t err = rx_mtu_deinit(k_mtu_channel_0);
01080  * @endcode
01081  *
01082  * @see rx_mtu_init_pwm() Re-initialize after deinit
01083  * @see rx_mtu_stop() Stop timer without full teardown
01084  *
01085  * @since Version 1.0.0
01086  */
01087 rx_err_t rx_mtu_deinit(const rx_mtu_channel_t channel)
01088 {
01089     if (internal_is_valid_channel(channel)) {
01090         return k_rx_err_invalid_arg;
01091     }
01092
01093     /* Stop timer */
01094     rx_err_t err = rx_mtu_stop(channel);
01095     if (err != k_rx_ok) {
01096         return err;
01097     }
01098
01099     /* Disable all outputs */
01100     err = rx_mtu_enable_output(channel, k_mtu_output_a, false);
01101     if (err != k_rx_ok) {
01102         return err;
01103     }
01104     err = rx_mtu_enable_output(channel, k_mtu_output_b, false);
01105     if (err != k_rx_ok) {
01106         return err;
01107     }
01108     err = rx_mtu_enable_output(channel, k_mtu_output_c, false);
01109     if (err != k_rx_ok) {
01110         return err;
01111     }
01112     err = rx_mtu_enable_output(channel, k_mtu_output_d, false);
01113     if (err != k_rx_ok) {
01114         return err;
01115     }
01116
01117     /* Mark as uninitialized */
01118     s_mtu_initialized[channel] = false;

```

```

01119 s_mtu_period[channel]    = k_mtu_period_zero;
01120
01121 rx_log_info(s_tag, "MTU channel deinitialized");
01122
01123 return k_rx_ok;
01124 }
01125
01126 #endif /* __RX__ */

```

8.131 libs/rx_hal/src/rx_poeg.c File Reference

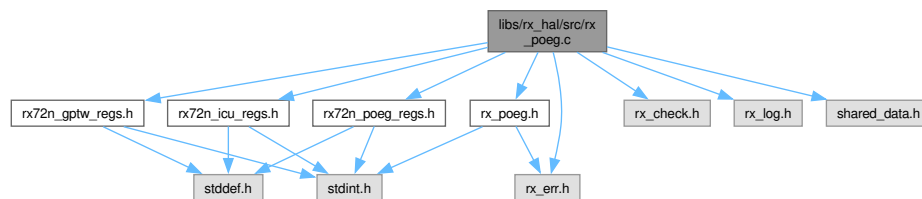
POEG Motor Fault Protection Driver Implementation.

```

#include "rx_poeg.h"
#include "rx72n_gptw_regs.h"
#include "rx72n_icu_regs.h"
#include "rx72n_poeg_regs.h"
#include "rx_check.h"
#include "rx_err.h"
#include "rx_log.h"
#include "shared_data.h"

```

Include dependency graph for rx_poeg.c:



Enumerations

- enum `poeg_icu_constants_t` : `uint8_t` { `k_icu_ir_clear` = 0 , `k_ier_bit_enable_base` = 1 , `k_ier_bits_per_reg` = 8 }
ICU configuration constants for POEG interrupts.
- enum `poeg_init_config_t` : `uint32_t` { `k_poeg_init_value` }
POEG initialization configuration values.
- enum `poeg_gtintad_values_t` : `uint32_t` { `k_gtintad_motor_0` = (`k_gptw_gtintad_grp_a` | `k_gptw_gtintad_grpdte` | `k_gptw_gtintad_grpabl`) , `k_gtintad_motor_1` = (`k_gptw_gtintad_grp_b` | `k_gptw_gtintad_grpdte` | `k_gptw_gtintad_grpabl`) , `k_gtintad_motor_2` = (`k_gptw_gtintad_grp_c` | `k_gptw_gtintad_grpdte` | `k_gptw_gtintad_grpabl`) , `k_gtintad_motor_3` = (`k_gptw_gtintad_grp_d` | `k_gptw_gtintad_grpdte` | `k_gptw_gtintad_grpabl`) }
Pre-computed GTINTAD values for each GPTW-POEG link.

Functions

- static volatile `uint32_t` * `poeg_grpbl2_reg` (void)
Accessors for the GROUPBL2 status/enable registers.
- static volatile `uint32_t` * `poeg_genbl2_reg` (void)
- static void `internal_enable_poeg_groupbl2_irq` (`uint8_t` priority)
Configure ICU for the shared POEG GROUPBL2 vector.
- static void `internal_configure_gtintad` (`uint8_t` motor_index)
Configure GPTW GTINTAD for POEG group linkage.

- void [poeg_groupbl2_isr](#) (void)
GROUPBL2 ISR - dispatches POEGGA/B/C/D fault interrupts.
- rx_err_t [rx_poeg_init](#) (void)
Initialize POEG fault protection for all 4 motor channels.
- rx_err_t [rx_poeg_get_fault_status](#) (uint8_t motor_index, bool *fault_active)
Read latched POEG fault status (implementation).
- rx_err_t [rx_poeg_clear_fault](#) (uint8_t motor_index)
Acknowledge and clear all latched POEG fault flags (implementation).
- rx_err_t [rx_poeg_software_stop](#) (uint8_t motor_index)
Trigger a Software Stop (implementation).
- rx_err_t [rx_poeg_clear_software_stop](#) (uint8_t motor_index)
Release a Software Stop (implementation).

Variables

- static const char *const [s_tag](#) = "POEG"
Module log tag.
- static volatile [rx_poegg_regs_t](#) *const [s_poeg_groups](#) [[k_poeg_motor_count](#)]
POEG group register accessor lookup table (indexed by motor).
- static volatile [rx_gptw_channel_regs_t](#) *const [s_gptw_channels](#) [[k_poeg_motor_count](#)]
GPTW channel register accessor lookup table (indexed by motor).
- static const uint32_t [s_gtintad_values](#) [[k_poeg_motor_count](#)]
GTINTAD values lookup table (indexed by motor).
- static const uint8_t [s_poeg_grpbl2_bits](#) [[k_poeg_motor_count](#)]
GRPBL2 sub-source bit for each POEG group (indexed by motor).

8.131.1 Detailed Description

POEG Motor Fault Protection Driver Implementation.

Implements hardware-level emergency PWM output disable for the STAR motor controller. Each of the 4 DRV8263H motor drivers has its nFAULT output connected to a GTETRG pin, which feeds a POEG group. When a fault is detected, POEG immediately disables GPTW PWM output (Hi-Z).

POEGGA/B/C/D are sub-sources of the shared GROUPBL2 vector (107) on RX72N (HW manual R01↔UH0824EJ0111 Section 15.3, Table 15.4). A single GROUPBL2 ISR reads GRPBL2, dispatches to each faulted group, and:

1. Clears the ICU GROUPBL2 IR flag
2. Logs which POEG groups asserted
3. Triggers e-stop via shared_data

Implementation Notes

- POEG PIDE (pin detection enable) is write-once after reset
- nFAULT is active-low, so INV bit is set to detect low-level
- Noise filter enabled with PCLKB/8 sampling (~400ns at 60MHz)
- GTINTAD links each GPTW channel to its corresponding POEG group

Author

Locked, Inc.

Date

2026-02-10

Copyright

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Since

Version 1.0.0

Definition in file [rx_poeg.c](#).

8.131.2 Enumeration Type Documentation

8.131.2.1 poeg`gtintad`values`t

enum [poeg_gtintad_values_t](#) : uint32_t

Pre-computed GTINTAD values for each GPTW-POEG link.

Each GPTW channel's GTINTAD register is configured to:

- Select the corresponding POEG group (A/B/C/D)
- Enable dead-time error detection (GRPDTE)
- Enable simultaneous low output detection (GRPABL)

Since

Version 1.0.0

Enumerator

k_gtintad_motor_0	
k_gtintad_motor_1	
k_gtintad_motor_2	
k_gtintad_motor_3	

Definition at line 90 of file [rx_poeg.c](#).

```
00090      : uint32_t {
00091  k_gtintad_motor_0 = (k_gptw_gtintad_grp_a | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00092  k_gtintad_motor_1 = (k_gptw_gtintad_grp_b | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00093  k_gtintad_motor_2 = (k_gptw_gtintad_grp_c | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00094  k_gtintad_motor_3 = (k_gptw_gtintad_grp_d | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00095 } poeg_gtintad_values_t;
```

8.131.2.2 poeg`icu`constants``t`

```
enum poeg_icu_constants_t : uint8_t
```

ICU configuration constants for POEG interrupts.

Since

Version 1.0.0

Enumerator

k_icu_ir_clear	Write 0 to clear IR flag
k_ier_bit_enable_base	Base for IER bit shift
k_ier_bits_per_reg	8 bits per IER register

Definition at line 54 of file [rx_poeg.c](#).

```
00054      : uint8_t {
00055  k_icu_ir_clear      = 0, /**< Write 0 to clear IR flag */
00056  k_ier_bit_enable_base = 1, /**< Base for IER bit shift */
00057  k_ier_bits_per_reg   = 8, /**< 8 bits per IER register */
00058 } poeg_icu_constants_t;
```

8.131.2.3 poeg`init`config``t`

```
enum poeg_init_config_t : uint32_t
```

POEG initialization configuration values.

Combined register value for each POEG group initialization:

- PIDE = 1 (enable pin detection)
- INV = 1 (invert for active-low nFAULT)
- NFEN = 1 (enable noise filter)
- NFCS = 01 (PCLKB/8, ~400ns sampling)

Since

Version 1.0.0

Enumerator

k_poeg_init_value	
-------------------	--

Definition at line 73 of file [rx_poeg.c](#).

```
00073      : uint32_t {
00074  k_poeg_init_value =
00075  (k_poeg_pide_enable | k_poeg_inv_invert | k_poeg_nfen_enable | k_poeg_nfcs_pclkb_div8),
00076 } poeg_init_config_t;
```

8.131.3 Function Documentation

8.131.3.1 `internal`configure`gtintad()`

```
void internal__configure__gtintad (
    uint8_t motor_index) [static]
```

Configure GPTW GTINTAD for POEG group linkage.

Precondition

GPTW channel initialized and outputs not active during GRP write

Postcondition

GTINTAD configured with POEG group and detection enables

Since

Version 1.0.0

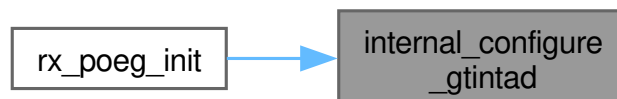
Definition at line 192 of file `rx_poeg.c`.

```
00193 {
00194     volatile rx_gptw_channel_regs_t* gptw = s_gptw_channels[motor_index];
00195
00196     /* Unlock write protection */
00197     gptw->gtwp = k_gptw_gtwp_unlock;
00198
00199     /* Set GTINTAD: POEG group + dead-time error + simultaneous low detection */
00200     gptw->gtintad = s_gtintad_values[motor_index];
00201
00202     /* Lock write protection */
00203     gptw->gtwp = k_gptw_gtwp_lock;
00204 }
```

References `rx_gptw_channel_regs_t::gtintad`, `rx_gptw_channel_regs_t::gtwp`, `k_gptw_gtwp_lock`, `k_gptw_gtwp_unlock`, `s_gptw_channels`, and `s_gtintad_values`.

Referenced by `rx_poeg_init()`.

Here is the caller graph for this function:



8.131.3.2 internal_enable_poeg_groupbl2_irq()

```
void internal_enable_poeg_groupbl2_irq (
    uint8_t priority) [static]
```

Configure ICU for the shared POEG GROUPBL2 vector.

Precondition

ICU registers accessible

Postcondition

GROUPBL2 vector (107) enabled at specified priority with all four POEGG sub-sources unmasked in GENBL2.

Since

Version 1.0.0

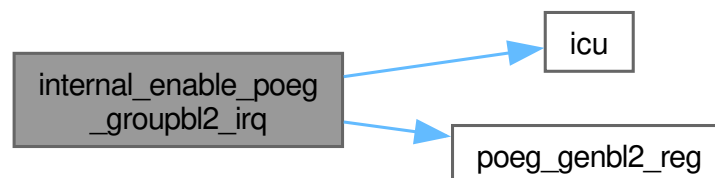
Definition at line 160 of file [rx_poeg.c](#).

```
00161 {
00162     volatile rx_icu_regs_t* icu_regs = icu();
00163     const uint16_t vector = k_poeg_irq_groupbl2_vector;
00164
00165     /* Enable all four POEGG sub-sources in GENBL2 */
00166     *poeg_genbl2_reg() |=
00167         (uint32_t)((1UL « k_poeg_grpbl2_bit_poeggai) | (1UL « k_poeg_grpbl2_bit_poeggbi) |
00168             (1UL « k_poeg_grpbl2_bit_poeggci) | (1UL « k_poeg_grpbl2_bit_poeggdi));
00169
00170     /* Clear any pending interrupt */
00171     icu_regs->ir[vector] = k_icu_ir_clear;
00172
00173     /* Set interrupt priority */
00174     icu_regs->ipr[vector] = priority;
00175
00176     /* Enable in IER register */
00177     const uint8_t ier_idx = (uint8_t)(vector / k_ier_bits_per_reg);
00178     const uint8_t ier_bit = (uint8_t)(vector % k_ier_bits_per_reg);
00179     const uint8_t ier_mask = (uint8_t)(k_ier_bit_enable_base « ier_bit);
00180     icu_regs->ier[ier_idx] |= ier_mask;
00181 }
```

References [icu\(\)](#), [rx_icu_regs_t::ier](#), [rx_icu_regs_t::ipr](#), [rx_icu_regs_t::ir](#), [k_icu_ir_clear](#), [k_ier_bit_enable_base](#), [k_ier_bits_per_reg](#), [k_poeg_grpbl2_bit_poeggai](#), [k_poeg_grpbl2_bit_poeggbi](#), [k_poeg_grpbl2_bit_poeggci](#), [k_poeg_grpbl2_bit_poeggdi](#), [k_poeg_irq_groupbl2_vector](#), and [poeg_genbl2_reg\(\)](#).

Referenced by [rx_poeg_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.131.3.3 poeg`genbl2`reg()

```
volatile uint32_t * poeg_genbl2_reg (  
    void ) [inline], [static]
```

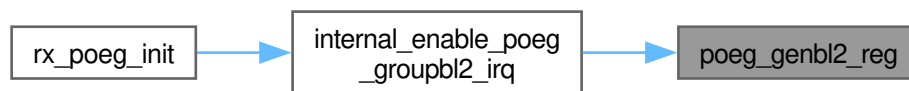
Definition at line 140 of file [rx_poeg.c](#).

```
00141 {  
00142     return (volatile uint32_t*)k_poeg_genbl2_addr;  
00143 }
```

References [k_poeg_genbl2_addr](#).

Referenced by [internal_enable_poeg_groupbl2_irq\(\)](#).

Here is the caller graph for this function:



8.131.3.4 poeg`groupbl2`isr()

```
void poeg_groupbl2_isr (  
    void )
```

GROUPBL2 ISR - dispatches POEGGA/B/C/D fault interrupts.

Handles the shared GROUPBL2 vector (107). Reads GRPBL2 to identify which POEG sub-sources asserted, logs each, and triggers a single e-stop.

Precondition

POEG groups initialized via [rx_poeg_init\(\)](#)

Postcondition

For every POEGG bit set in GRPBL2, the fault is logged
Global e-stop triggered once if any POEG source fired

Note

Thread Safety: ISR context, runs at priority 14

Since

Version 1.0.0

Definition at line 226 of file [rx_poeg.c](#).

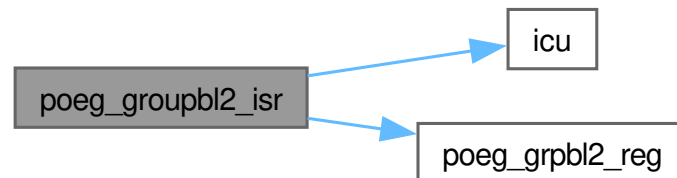
```

00227 {
00228     const uint32_t grpbl2 = *poeg_grpbl2_reg();
00229     bool faulted = false;
00230
00231     for (uint8_t i = 0; i < k_poeg_motor_count; i++) {
00232         if ((grpbl2 & (1UL « s_poeg_grpbl2_bits[i])) == 0U) {
00233             continue;
00234         }
00235
00236         volatile rx_poegg_regs_t* poeg = s_poeg_groups[i];
00237         const uint32_t status = poeg->poeggn;
00238         if ((status & k_poeg_pidf_detected) != 0U) {
00239             rx_log_error_val(s_tag, "nFAULT motor", (uint32_t)i);
00240         }
00241         if ((status & k_poeg_iocf_detected) != 0U) {
00242             rx_log_error_val(s_tag, "GPTW output err motor", (uint32_t)i);
00243         }
00244         faulted = true;
00245     }
00246
00247     /* Clear the shared GROUPBL2 request flag */
00248     icu()->ir[k_poeg_irq_groupbl2_vector] = k_icu_ir_clear;
00249
00250     if (faulted) {
00251         shared_data_trigger_estop_isr_safe(k_estop_reason_driver_fault);
00252     }
00253 }

```

References [icu\(\)](#), [rx_icu_regs_t::ir](#), [k_icu_ir_clear](#), [k_poeg_iocf_detected](#), [k_poeg_irq_groupbl2_vector](#), [k_poeg_motor_count](#), [k_poeg_pidf_detected](#), [poeg_grpbl2_reg\(\)](#), [rx_poegg_regs_t::poeggn](#), [s_poeg_groups](#), [s_poeg_grpbl2_bits](#), and [s_tag](#).

Here is the call graph for this function:



8.131.3.5 poeg_grpbl2_reg()

```

volatile uint32_t * poeg_grpbl2_reg (
    void ) [inline], [static]

```

Accessors for the GROUPBL2 status/enable registers.

Definition at line 135 of file [rx_poeg.c](#).

```

00136 {
00137     return (volatile uint32_t*)k_poeg_grpbl2_addr;
00138 }

```

References [k_poeg_grpbl2_addr](#).

Referenced by [poeg_groupbl2_isr\(\)](#).

Here is the caller graph for this function:



8.131.3.6 rx`poeg`clear`fault()

```
rx_err_t rx_poeg_clear_fault (
    uint8_t motor_index)
```

Acknowledge and clear all latched POEG fault flags (implementation).

Clear POEG fault and re-enable motor output.

Reads POEGGn, masks off the PIDF / IOCF / OSTPF latched-fault bits, and writes the result back. Refuses to clear (returns k_rx_err_busy) while the underlying nFAULT pin is still active (POEGGn.ST==0 indicates nFAULT still asserted under INV=1). After a successful clear, also clears any driver-fault E-Stop in shared_data when applicable, so the system can resume motor control if the underlying physical fault has been cleared.

Note

Not thread-safe with concurrent access to shared_data E-Stop fields. Typically called from the safety supervisor task.

Definition at line 327 of file rx_poeg.c.

```
00328 {
00329     if (motor_index >= k_poeg_motor_count) {
00330         return k_rx_err_invalid_arg;
00331     }
00332
00333     volatile rx_poegg_regs_t* poeg = s_poeg_groups[motor_index];
00334
00335     /* Check if fault source is still active (ST bit shows pin state) */
00336     /* With INV=1, ST=0 means nFAULT is still asserted (fault active) */
00337     if ((poeg->poeggn & k_poeg_st_high) == 0) {
00338         rx_log_warn_val(s_tag, "Cannot clear: nFAULT still active motor", (uint32_t)motor_index);
00339         return k_rx_err_busy;
00340     }
00341
00342     /* Clear PIDF and IOCF flags by writing 0 to them */
00343     uint32_t reg_val = poeg->poeggn;
00344     reg_val &= ~(k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected);
00345     poeg->poeggn = reg_val;
00346
00347     /* Clear e-stop if this was the cause */
00348     if (shared_data_get_estop_reason() == k_estop_reason_driver_fault) {
00349         (void)shared_data_clear_estop();
00350     }
00351
00352     rx_log_info_val(s_tag, "Fault cleared motor", (uint32_t)motor_index);
00353     return k_rx_ok;
00354 }
```

References [k_poeg_iocf_detected](#), [k_poeg_motor_count](#), [k_poeg_ostpf_detected](#), [k_poeg_pidf_detected](#), [k_poeg_st_high](#), [rx_poegg_regs_t::poeggn](#), [s_poeg_groups](#), and [s_tag](#).

8.131.3.7 rx`poeg`clear`software`stop()

```
rx_err_t rx_poeg_clear_software_stop (
    uint8_t motor_index)
```

Release a Software Stop (implementation).

Release software emergency stop on a specific motor.

Reads POEGGn, clears the SSF bit, and writes the result back. This re-enables the GPTW outputs of the targeted motor (subject to any hardware-latched fault flags still being cleared first).

Note

Safe from any context including ISR; single read-modify-write.

Definition at line 388 of file [rx_poeg.c](#).

```
00389 {
00390     if (motor_index >= k_poeg_motor_count) {
00391         return k_rx_err_invalid_arg;
00392     }
00393
00394     uint32_t reg_val = s_poeg_groups[motor_index]->poeggn;
00395     reg_val &= ~k_poeg_ssf_stop;
00396     s_poeg_groups[motor_index]->poeggn = reg_val;
00397     return k_rx_ok;
00398 }
```

References [k_poeg_motor_count](#), [k_poeg_ssf_stop](#), and [s_poeg_groups](#).

8.131.3.8 rx_poeg_get_fault_status()

```
rx_err_t rx_poeg_get_fault_status (
    uint8_t motor_index,
    bool * fault_active)
```

Read latched POEG fault status (implementation).

Check if a motor has an active POEG fault.

Reads POEGGn for the specified motor's POEG group and checks the OR of the three fault-source bits: PIDF (port input fault), IOCF (output comparator fault), and OSTPF (over-stop signal fault). Reports whether any of them are currently latched.

Note

Safe from any context including ISR; single register read.

Definition at line 297 of file [rx_poeg.c](#).

```
00298 {
00299     RX_CHECK_NULL_PTR(fault_active, s_tag, "fault_active is nullptr");
00300
00301     if (motor_index >= k_poeg_motor_count) {
00302         return k_rx_err_invalid_arg;
00303     }
00304
00305     const uint32_t status = s_poeg_groups[motor_index]->poeggn;
00306     *fault_active =
00307         ((status & (k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected)) != 0);
00308
00309     return k_rx_ok;
00310 }
```

References [k_poeg_iocf_detected](#), [k_poeg_motor_count](#), [k_poeg_ostpf_detected](#), [k_poeg_pidf_detected](#), [s_poeg_groups](#), and [s_tag](#).

8.131.3.9 rx`poeg`init()

```
rx_err_t rx_poeg_init (
    void )
```

Initialize POEG fault protection for all 4 motor channels.

Configures POEG Groups A-D for hardware motor fault protection:

1. Enables external pin detection (PIDE) on each group
2. Enables noise filter (NFEN) with PCLKB/8 sampling ($\sim 400\text{ns}$)
3. Inverts input (INV) since DRV8263H nFAULT is active-low
4. Links each GPTW channel to its POEG group via GTINTAD
5. Configures ICU interrupts for vectors 188-191 at priority 14

After initialization, a DRV8263H fault will:

- Immediately disable GPTW PWM output (hardware, $\sim \text{nanoseconds}$)
- Fire POEG ISR which sets fault flags in shared_data

Returns

rx_err_t Error code

Return values

k_rx_ok	All 4 POEG groups initialized successfully
k_rx_err_hw_init_failed	POEG register write verification failed

Precondition

GPTW module enabled (MSTPCRA.MSTPA7 = 0)

GTETRGR pins configured via MPC: PDR=0 (input), PSEL=0x1E (GTETRGR function select)

shared_data initialized

Postcondition

POEG Groups A-D enabled with pin detection + noise filter

GPTW0-3 GTINTAD linked to POEG Groups A-D respectively

ICU vectors 188-191 enabled at priority 14

Any existing nFAULT assertion will trigger immediately

Note

Thread Safety: Call once during hardware_init, before motor tasks start

Warning

PIDE enable bit is write-once after reset – cannot be disabled

NASA Power of 10 Compliance:

- Rule 1: No goto, setjmp, recursion
- Rule 2: Loop bounded by k_poeg_motor_count (4)
- Rule 3: No dynamic allocation
- Rule 5: 3 preconditions, 4 postconditions
- Rule 7: Return value checked
- Rule 8: All constants as C23 typed enums

See also

[rx_poeg_clear_fault\(\)](#) Clear fault after resolution

[rx_poeg_get_fault_status\(\)](#) Query fault state

Since

Version 1.0.0

Definition at line 260 of file [rx_poeg.c](#).

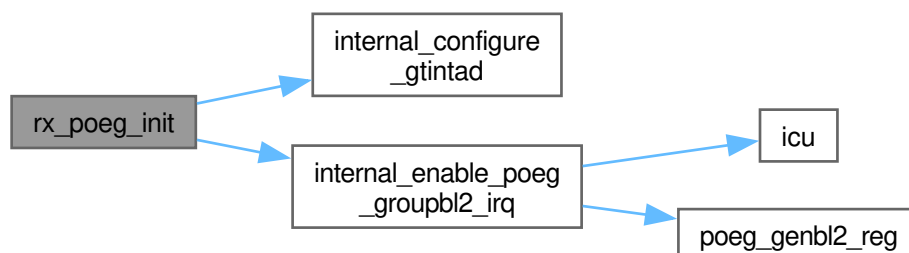
```

00261 {
00262   rx_log_info(s_tag, "Initializing POEG fault protection");
00263
00264   for (uint8_t i = 0; i < k_poeg_motor_count; i++) {
00265     /* Configure POEG group: enable pin detection, invert, noise filter */
00266     volatile rx_poegg_regs_t* poeg = s_poeg_groups[i];
00267     poeg->poeggn                = k_poeg_init_value;
00268
00269     /* Verify write-once enable bits were accepted */
00270     if ((poeg->poeggn & k_poeg_pide_enable) == 0) {
00271       rx_log_error_val(s_tag, "POEG PIDE write failed motor", (uint32_t)i);
00272       return k_rx_err_hw_init_failed;
00273     }
00274
00275     /* Link GPTW channel to POEG group via GTINTAD */
00276     internal_configure_gtintad(i);
00277   }
00278
00279   /* Enable shared GROUPBL2 ICU vector (one dispatch for all POEG groups) */
00280   internal_enable_poeg_groupbl2_irq(k_poeg_isr_priority);
00281
00282   rx_log_info(s_tag, "POEG fault protection active (4 groups on GROUPBL2, priority 14)");
00283   return k_rx_ok;
00284 }

```

References [internal_configure_gtintad\(\)](#), [internal_enable_poeg_groupbl2_irq\(\)](#), [k_poeg_init_value](#), [k_poeg_isr_priority](#), [k_poeg_motor_count](#), [k_poeg_pide_enable](#), [rx_poegg_regs_t::poeggn](#), [s_poeg_groups](#), and [s_tag](#).

Here is the call graph for this function:



8.131.3.10 rx`poeg`software`stop()

```
rx_err_t rx_poeg_software_stop (
    uint8_t motor_index)
```

Trigger a Software Stop (implementation).

Trigger software emergency stop on a specific motor.

Sets POEGn.SSF=1, force-disabling the GPTW outputs of the targeted motor as if a hardware fault had occurred. Does not touch the latched fault flags; once a Software Stop is set, the user must call [rx_poeg_clear_software_stop\(\)](#) to release.

Note

Safe from any context including ISR; single register write. Used by the safety supervisor for software-initiated E-Stop.

Definition at line 368 of file [rx_poeg.c](#).

```
00369 {
00370     if (motor_index >= k_poeg_motor_count) {
00371         return k_rx_err_invalid_arg;
00372     }
00373
00374     s_poeg_groups[motor_index]->poeggn |= k_poeg_ssf_stop;
00375     return k_rx_ok;
00376 }
```

References [k_poeg_motor_count](#), [k_poeg_ssf_stop](#), and [s_poeg_groups](#).

8.131.4 Variable Documentation

8.131.4.1 s`gptw`channels

```
volatile rx_gptw_channel_regs_t* const s_gptw_channels[k_poeg_motor_count] [static]
```

Initial value:

```
= {
    (volatile rx_gptw_channel_regs_t*)k_gptw0_base_addr,
    (volatile rx_gptw_channel_regs_t*)k_gptw1_base_addr,
    (volatile rx_gptw_channel_regs_t*)k_gptw2_base_addr,
    (volatile rx_gptw_channel_regs_t*)k_gptw3_base_addr,
}
```

GPTW channel register accessor lookup table (indexed by motor).

Definition at line 111 of file [rx_poeg.c](#).

```
00111 {
00112     (volatile rx_gptw_channel_regs_t*)k_gptw0_base_addr,
00113     (volatile rx_gptw_channel_regs_t*)k_gptw1_base_addr,
00114     (volatile rx_gptw_channel_regs_t*)k_gptw2_base_addr,
00115     (volatile rx_gptw_channel_regs_t*)k_gptw3_base_addr,
00116 };
```

Referenced by [internal_configure_gtintad\(\)](#).

8.131.4.2 s'gtintad'values

const uint32_t s_gtintad_values[k_poeg_motor_count] [static]

Initial value:

```

                                = {
    k_gtintad_motor_0,
    k_gtintad_motor_1,
    k_gtintad_motor_2,
    k_gtintad_motor_3,
}

```

GTINTAD values lookup table (indexed by motor).

Definition at line 119 of file [rx_poeg.c](#).

```

00119                                     {
00120   k_gtintad_motor_0,
00121   k_gtintad_motor_1,
00122   k_gtintad_motor_2,
00123   k_gtintad_motor_3,
00124 };

```

Referenced by [internal_configure_gtintad\(\)](#).

8.131.4.3 s'poeg'groups

volatile rx_poegg_regs_t* const s_poeg_groups[k_poeg_motor_count] [static]

Initial value:

```

                                = {
    (volatile rx_poegg_regs_t*)k_poegga_base_addr,
    (volatile rx_poegg_regs_t*)k_poeggb_base_addr,
    (volatile rx_poegg_regs_t*)k_poeggc_base_addr,
    (volatile rx_poegg_regs_t*)k_poeggd_base_addr,
}

```

POEG group register accessor lookup table (indexed by motor).

Definition at line 103 of file [rx_poeg.c](#).

```

00103                                     {
00104   (volatile rx_poegg_regs_t*)k_poegga_base_addr,
00105   (volatile rx_poegg_regs_t*)k_poeggb_base_addr,
00106   (volatile rx_poegg_regs_t*)k_poeggc_base_addr,
00107   (volatile rx_poegg_regs_t*)k_poeggd_base_addr,
00108 };

```

Referenced by [poeg_groupbl2_isr\(\)](#), [rx_poeg_clear_fault\(\)](#), [rx_poeg_clear_software_stop\(\)](#), [rx_poeg_get_fault_status\(\)](#), [rx_poeg_init\(\)](#), and [rx_poeg_software_stop\(\)](#).

8.131.4.4 s'poeg'grpbl2'bits

const uint8_t s_poeg_grpbl2_bits[k_poeg_motor_count] [static]

Initial value:

```

                                = {
    k_poeg_grpbl2_bit_poeggai,
    k_poeg_grpbl2_bit_poeggbi,
    k_poeg_grpbl2_bit_poeggci,
    k_poeg_grpbl2_bit_poeggdi,
}

```

GRPBL2 sub-source bit for each POEG group (indexed by motor).

Definition at line 127 of file [rx_poeg.c](#).

```

00127                                     {
00128   k_poeg_grpbl2_bit_poeggai,
00129   k_poeg_grpbl2_bit_poeggbi,
00130   k_poeg_grpbl2_bit_poeggci,
00131   k_poeg_grpbl2_bit_poeggdi,
00132 };

```

Referenced by [poeg_groupbl2_isr\(\)](#).

8.131.4.5 s_tag

```
const char* const s_tag = "POEG" [static]
```

Module log tag.

Definition at line 47 of file rx_poeg.c.

8.132 rx_poeg.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_poeg.c
00003  * @brief POEG Motor Fault Protection Driver Implementation
00004  *
00005  * @details
00006  * Implements hardware-level emergency PWM output disable for the STAR
00007  * motor controller. Each of the 4 DRV8263H motor drivers has its nFAULT
00008  * output connected to a GTETRG pin, which feeds a POEG group. When a
00009  * fault is detected, POEG immediately disables GPTW PWM output (Hi-Z).
00010  *
00011  * POEGGA/B/C/D are sub-sources of the shared GROUPBL2 vector (107) on
00012  * RX72N (HW manual R01UH0824EJ0111 Section 15.3, Table 15.4). A single
00013  * GROUPBL2 ISR reads GRPBL2, dispatches to each faulted group, and:
00014  * 1. Clears the ICU GROUPBL2 IR flag
00015  * 2. Logs which POEG groups asserted
00016  * 3. Triggers e-stop via shared_data
00017  *
00018  * @par Implementation Notes
00019  * - POEG PIDE (pin detection enable) is write-once after reset
00020  * - nFAULT is active-low, so INV bit is set to detect low-level
00021  * - Noise filter enabled with PCLKB/8 sampling (~400ns at 60MHz)
00022  * - GTINTAD links each GPTW channel to its corresponding POEG group
00023  *
00024  * @author Locked, Inc.
00025  * @date 2026-02-10
00026  * @copyright Copyright (c) 2026 Locked Inc.
00027  * SPDX-License-Identifier: MIT
00028  * @since Version 1.0.0
00029  */
00030
00031 #include "rx_poeg.h"
00032
00033 #include "rx72n_gptw_regs.h"
00034 #include "rx72n_icu_regs.h"
00035 #include "rx72n_poeg_regs.h"
00036 #include "rx_check.h"
00037 #include "rx_err.h"
00038 #include "rx_log.h"
00039 #include "shared_data.h"
00040
00041 /**
00042  * Constants
00043  *
00044  */
00045
00046 /** @brief Module log tag */
00047 static const char* const s_tag = "POEG";
00048
00049 /**
00050  * @enum poeg_icu_constants_t
00051  * @brief ICU configuration constants for POEG interrupts
00052  * @since Version 1.0.0
00053  */
00054 typedef enum : uint8_t {
00055     k_icu_ir_clear = 0, /**< Write 0 to clear IR flag */
00056     k_ier_bit_enable_base = 1, /**< Base for IER bit shift */
00057     k_ier_bits_per_reg = 8, /**< 8 bits per IER register */
00058 } poeg_icu_constants_t;
00059
00060 /**
00061  * @enum poeg_init_config_t
00062  * @brief POEG initialization configuration values
00063  *
00064  * @details
```

```

00065 * Combined register value for each POEG group initialization:
00066 * - PIDE = 1 (enable pin detection)
00067 * - INV = 1 (invert for active-low nFAULT)
00068 * - NFEN = 1 (enable noise filter)
00069 * - NFCS = 01 (PCLKB/8, ~400ns sampling)
00070 *
00071 * @since Version 1.0.0
00072 */
00073 typedef enum : uint32_t {
00074     k_poeg_init_value =
00075         (k_poeg_pide_enable | k_poeg_inv_invert | k_poeg_nfen_enable | k_poeg_nfcs_pclkb_div8),
00076 } poeg_init_config_t;
00077 /**
00078 * @enum poeg_gtintad_values_t
00079 * @brief Pre-computed GTINTAD values for each GPTW-POEG link
00080 *
00081 * @details
00082 * Each GPTW channel's GTINTAD register is configured to:
00083 * - Select the corresponding POEG group (A/B/C/D)
00084 * - Enable dead-time error detection (GRPDTE)
00085 * - Enable simultaneous low output detection (GRPABL)
00086 *
00087 * @since Version 1.0.0
00088 */
00089 typedef enum : uint32_t {
00090     k_gtintad_motor_0 = (k_gptw_gtintad_grp_a | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00091     k_gtintad_motor_1 = (k_gptw_gtintad_grp_b | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00092     k_gtintad_motor_2 = (k_gptw_gtintad_grp_c | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00093     k_gtintad_motor_3 = (k_gptw_gtintad_grp_d | k_gptw_gtintad_grpdte | k_gptw_gtintad_grpabl),
00094 } poeg_gtintad_values_t;
00095
00096
00097 /*
=====
00098 * Static Variables
00099 *
=====
00100 */
00101
00102 /** @brief POEG group register accessor lookup table (indexed by motor) */
00103 static volatile rx_poegg_regs_t* const s_poeg_groups[k_poeg_motor_count] = {
00104     (volatile rx_poegg_regs_t*)k_poegga_base_addr,
00105     (volatile rx_poegg_regs_t*)k_poeggb_base_addr,
00106     (volatile rx_poegg_regs_t*)k_poeggc_base_addr,
00107     (volatile rx_poegg_regs_t*)k_poeggd_base_addr,
00108 };
00109
00110 /** @brief GPTW channel register accessor lookup table (indexed by motor) */
00111 static volatile rx_gptw_channel_regs_t* const s_gptw_channels[k_poeg_motor_count] = {
00112     (volatile rx_gptw_channel_regs_t*)k_gptw0_base_addr,
00113     (volatile rx_gptw_channel_regs_t*)k_gptw1_base_addr,
00114     (volatile rx_gptw_channel_regs_t*)k_gptw2_base_addr,
00115     (volatile rx_gptw_channel_regs_t*)k_gptw3_base_addr,
00116 };
00117
00118 /** @brief GTINTAD values lookup table (indexed by motor) */
00119 static const uint32_t s_gtintad_values[k_poeg_motor_count] = {
00120     k_gtintad_motor_0,
00121     k_gtintad_motor_1,
00122     k_gtintad_motor_2,
00123     k_gtintad_motor_3,
00124 };
00125
00126 /** @brief GRPBL2 sub-source bit for each POEG group (indexed by motor) */
00127 static const uint8_t s_poeg_grpbl2_bits[k_poeg_motor_count] = {
00128     k_poeg_grpbl2_bit_poeggai,
00129     k_poeg_grpbl2_bit_poeggbi,
00130     k_poeg_grpbl2_bit_poeggci,
00131     k_poeg_grpbl2_bit_poeggdi,
00132 };
00133
00134 /** @brief Accessors for the GROUPBL2 status/enable registers */
00135 static inline volatile uint32_t* poeg_grpbl2_reg(void)
00136 {
00137     return (volatile uint32_t*)k_poeg_grpbl2_addr;
00138 }
00139
00140 static inline volatile uint32_t* poeg_genbl2_reg(void)
00141 {
00142     return (volatile uint32_t*)k_poeg_genbl2_addr;
00143 }
00144
00145 /*
=====
00146 * Internal Helper Functions
00147 *
=====

```

```

00148 */
00149
00150 /**
00151  * @brief Configure ICU for the shared POEG GROUPBL2 vector
00152  *
00153  *
00154  * @pre ICU registers accessible
00155  * @post GROUPBL2 vector (107) enabled at specified priority with all four
00156  *       POEGG sub-sources unmasked in GENBL2.
00157  *
00158  * @since Version 1.0.0
00159  */
00160 static void internal_enable_poeg_groupbl2_irq(uint8_t priority)
00161 {
00162     volatile rx_icu_regs_t* icu_regs = icu();
00163     const uint16_t vector = k_poeg_irq_groupbl2_vector;
00164
00165     /* Enable all four POEGG sub-sources in GENBL2 */
00166     *poeg_genbl2_reg() |=
00167         (uint32_t)((1UL « k_poeg_grpbl2_bit_poeggai) | (1UL « k_poeg_grpbl2_bit_poeggbi) |
00168             (1UL « k_poeg_grpbl2_bit_poeggci) | (1UL « k_poeg_grpbl2_bit_poeggd));
00169
00170     /* Clear any pending interrupt */
00171     icu_regs->ir[vector] = k_icu_ir_clear;
00172
00173     /* Set interrupt priority */
00174     icu_regs->ipr[vector] = priority;
00175
00176     /* Enable in IER register */
00177     const uint8_t ier_idx = (uint8_t)(vector / k_ier_bits_per_reg);
00178     const uint8_t ier_bit = (uint8_t)(vector % k_ier_bits_per_reg);
00179     const uint8_t ier_mask = (uint8_t)(k_ier_bit_enable_base « ier_bit);
00180     icu_regs->ier[ier_idx] |= ier_mask;
00181 }
00182
00183 /**
00184  * @brief Configure GPTW GTINTAD for POEG group linkage
00185  *
00186  *
00187  * @pre GPTW channel initialized and outputs not active during GRP write
00188  * @post GTINTAD configured with POEG group and detection enables
00189  *
00190  * @since Version 1.0.0
00191  */
00192 static void internal_configure_gtintad(uint8_t motor_index)
00193 {
00194     volatile rx_gptw_channel_regs_t* gptw = s_gptw_channels[motor_index];
00195
00196     /* Unlock write protection */
00197     gptw->gtwp = k_gptw_gtwp_unlock;
00198
00199     /* Set GTINTAD: POEG group + dead-time error + simultaneous low detection */
00200     gptw->gtintad = s_gtintad_values[motor_index];
00201
00202     /* Lock write protection */
00203     gptw->gtwp = k_gptw_gtwp_lock;
00204 }
00205
00206 /*
=====
00207  * ISR Handler - Single GROUPBL2 dispatcher for POEGGA/B/C/D
00208  *
=====
00209  */
00210
00211 /**
00212  * @brief GROUPBL2 ISR - dispatches POEGGA/B/C/D fault interrupts
00213  *
00214  * @details
00215  * Handles the shared GROUPBL2 vector (107). Reads GRPBL2 to identify which
00216  * POEG sub-sources asserted, logs each, and triggers a single e-stop.
00217  *
00218  * @pre POEG groups initialized via rx_poeg_init()
00219  * @post For every POEGG bit set in GRPBL2, the fault is logged
00220  * @post Global e-stop triggered once if any POEG source fired
00221  *
00222  * @note Thread Safety: ISR context, runs at priority 14
00223  * @since Version 1.0.0
00224  */
00225 void __attribute__((interrupt)) poeg_groupbl2_isr(void);
00226 void __attribute__((interrupt)) poeg_groupbl2_isr(void)
00227 {
00228     const uint32_t grpbl2 = *poeg_grpbl2_reg();
00229     bool faulted = false;
00230
00231     for (uint8_t i = 0; i < k_poeg_motor_count; i++) {
00232         if ((grpbl2 & (1UL « s_poeg_grpbl2_bits[i])) == 0U) {

```

```

00233     continue;
00234 }
00235
00236 volatile rx_poegg_regs_t* poeg = s_poeg_groups[i];
00237 const uint32_t status = poeg->poeggn;
00238 if ((status & k_poeg_pidf_detected) != 0U) {
00239     rx_log_error_val(s_tag, "nFAULT motor", (uint32_t)i);
00240 }
00241 if ((status & k_poeg_iocf_detected) != 0U) {
00242     rx_log_error_val(s_tag, "GPTW output err motor", (uint32_t)i);
00243 }
00244 faulted = true;
00245 }
00246
00247 /* Clear the shared GROUPBL2 request flag */
00248 icu()->ir[k_poeg_irq_groupbl2_vector] = k_icu_ir_clear;
00249
00250 if (faulted) {
00251     shared_data_trigger_estop_isr_safe(k_estop_reason_driver_fault);
00252 }
00253 }
00254
00255 /*
=====
00256 * Public API
00257 *
=====
00258 */
00259 rx_err_t rx_poeg_init(void)
00260 {
00261     rx_log_info(s_tag, "Initializing POEG fault protection");
00262
00263     for (uint8_t i = 0; i < k_poeg_motor_count; i++) {
00264         /* Configure POEG group: enable pin detection, invert, noise filter */
00265         volatile rx_poegg_regs_t* poeg = s_poeg_groups[i];
00266         poeg->poeggn = k_poeg_init_value;
00267
00268         /* Verify write-once enable bits were accepted */
00269         if ((poeg->poeggn & k_poeg_pide_enable) == 0) {
00270             rx_log_error_val(s_tag, "POEG PIDE write failed motor", (uint32_t)i);
00271             return k_rx_err_hw_init_failed;
00272         }
00273     }
00274
00275     /* Link GPTW channel to POEG group via GTINTAD */
00276     internal_configure_gtintad(i);
00277 }
00278
00279 /* Enable shared GROUPBL2 ICU vector (one dispatch for all POEG groups) */
00280 internal_enable_poeg_groupbl2_irq(k_poeg_isr_priority);
00281
00282 rx_log_info(s_tag, "POEG fault protection active (4 groups on GROUPBL2, priority 14)");
00283 return k_rx_ok;
00284 }
00285
00286 /**
00287  * @brief Read latched POEG fault status (implementation)
00288  *
00289  * @details
00290  * Reads POEGGn for the specified motor's POEG group and checks the OR of
00291  * the three fault-source bits: PIDF (port input fault), IOCF (output
00292  * comparator fault), and OSTPF (over-stop signal fault). Reports whether
00293  * any of them are currently latched.
00294  *
00295  * @note Safe from any context including ISR; single register read.
00296  */
00297 rx_err_t rx_poeg_get_fault_status(uint8_t motor_index, bool* fault_active)
00298 {
00299     RX_CHECK_NULL_PTR(fault_active, s_tag, "fault_active is nullptr");
00300
00301     if (motor_index >= k_poeg_motor_count) {
00302         return k_rx_err_invalid_arg;
00303     }
00304
00305     const uint32_t status = s_poeg_groups[motor_index]->poeggn;
00306     *fault_active =
00307         ((status & (k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected)) != 0);
00308
00309     return k_rx_ok;
00310 }
00311
00312 /**
00313  * @brief Acknowledge and clear all latched POEG fault flags (implementation)
00314  *
00315  * @details
00316  * Reads POEGGn, masks off the PIDF / IOCF / OSTPF latched-fault bits, and
00317  * writes the result back. Refuses to clear (returns k_rx_err_busy) while

```

```

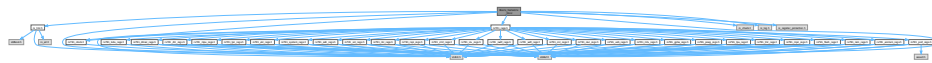
00318 * the underlying nFAULT pin is still active (POEGn.ST==0 indicates
00319 * nFAULT still asserted under INV=1). After a successful clear, also
00320 * clears any driver-fault E-Stop in shared_data when applicable, so the
00321 * system can resume motor control if the underlying physical fault has
00322 * been cleared.
00323 *
00324 * @note Not thread-safe with concurrent access to shared_data E-Stop
00325 * fields. Typically called from the safety supervisor task.
00326 */
00327 rx_err_t rx_poeg_clear_fault(uint8_t motor_index)
00328 {
00329     if (motor_index >= k_poeg_motor_count) {
00330         return k_rx_err_invalid_arg;
00331     }
00332
00333     volatile rx_poegg_regs_t* poeg = s_poeg_groups[motor_index];
00334
00335     /* Check if fault source is still active (ST bit shows pin state) */
00336     /* With INV=1, ST=0 means nFAULT is still asserted (fault active) */
00337     if ((poeg->poeggn & k_poeg_st_high) == 0) {
00338         rx_log_warn_val(s_tag, "Cannot clear: nFAULT still active motor", (uint32_t)motor_index);
00339         return k_rx_err_busy;
00340     }
00341
00342     /* Clear PIDF and IOCF flags by writing 0 to them */
00343     uint32_t reg_val = poeg->poeggn;
00344     reg_val &= ~(k_poeg_pidf_detected | k_poeg_iocf_detected | k_poeg_ostpf_detected);
00345     poeg->poeggn = reg_val;
00346
00347     /* Clear e-stop if this was the cause */
00348     if (shared_data_get_estop_reason() == k_estop_reason_driver_fault) {
00349         (void)shared_data_clear_estop();
00350     }
00351
00352     rx_log_info_val(s_tag, "Fault cleared motor", (uint32_t)motor_index);
00353     return k_rx_ok;
00354 }
00355
00356 /**
00357 * @brief Trigger a Software Stop (implementation)
00358 *
00359 * @details
00360 * Sets POEGn.SSF=1, force-disabling the GPTW outputs of the targeted
00361 * motor as if a hardware fault had occurred. Does not touch the latched
00362 * fault flags; once a Software Stop is set, the user must call
00363 * rx_poeg_clear_software_stop() to release.
00364 *
00365 * @note Safe from any context including ISR; single register write.
00366 * Used by the safety supervisor for software-initiated E-Stop.
00367 */
00368 rx_err_t rx_poeg_software_stop(uint8_t motor_index)
00369 {
00370     if (motor_index >= k_poeg_motor_count) {
00371         return k_rx_err_invalid_arg;
00372     }
00373
00374     s_poeg_groups[motor_index]->poeggn |= k_poeg_ssf_stop;
00375     return k_rx_ok;
00376 }
00377
00378 /**
00379 * @brief Release a Software Stop (implementation)
00380 *
00381 * @details
00382 * Reads POEGn, clears the SSF bit, and writes the result back. This
00383 * re-enables the GPTW outputs of the targeted motor (subject to any
00384 * hardware-latched fault flags still being cleared first).
00385 *
00386 * @note Safe from any context including ISR; single read-modify-write.
00387 */
00388 rx_err_t rx_poeg_clear_software_stop(uint8_t motor_index)
00389 {
00390     if (motor_index >= k_poeg_motor_count) {
00391         return k_rx_err_invalid_arg;
00392     }
00393
00394     uint32_t reg_val = s_poeg_groups[motor_index]->poeggn;
00395     reg_val &= ~k_poeg_ssf_stop;
00396     s_poeg_groups[motor_index]->poeggn = reg_val;
00397     return k_rx_ok;
00398 }

```

8.133 libs/rx'hal/src/rx'tmr.c File Reference

TMR HAL Driver Implementation (8-bit + 16-bit cascade).

```
#include "rx_tmr.h"
#include <stddef.h>
#include <stdint.h>
#include "rx72n_regs.h"
#include "rx72n_clock.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_register_protection.h"
Include dependency graph for rx_tmr.c:
```



Enumerations

- enum `tmr_period_limits_t` : `uint32_t` { `k_tmr_us_per_sec` = 1000000U , `k_tmr_max_tcora` = 0xFFU , `k_tmr_max_tcora_16bit` = 0xFFFFU }
Compile-time period / divisor limits.
- enum `tmr_channel_count_t` : `uint8_t` { `k_tmr_channel_count_internal` = 4 }
Array sizing constant for per-channel state.
- enum `tmr_pclk_divisor_t` : `uint32_t` {
 `k_tmr_pclk_div_1` = 1U , `k_tmr_pclk_div_2` = 2U , `k_tmr_pclk_div_8` = 8U ,
 `k_tmr_pclk_div_32` = 32U ,
 `k_tmr_pclk_div_64` = 64U , `k_tmr_pclk_div_1024` = 1024U , `k_tmr_pclk_div_8192` = 8192U
 , `k_tmr_pclk_div_none` = 0U }
Integer prescaler ratios for internal clock sources.
- enum `tmr_byte_shift_t` : `uint8_t` { `k_tmr_byte_shift` = 8U , `k_tmr_byte_mask` = 0xFFU }
Shift / mask constants for packing cascaded 16-bit counts.

Functions

- `RX_STATIC_TESTABLE volatile rx_tmr_channel_regs_t * internal_tmr_get_regs(rx_tmr_channel_t channel)`
Get register pointer for a TMR channel.
- static `uint32_t internal_tmr_unit_mstp_bit(rx_tmr_channel_t channel)`
Return the MSTPCRA bit mask for a TMR channel's unit.
- static void `internal_tmr_enable_module_clock(rx_tmr_channel_t channel)`
Enable the module clock for the channel's TMR unit.
- static `rx_err_t internal_tmr_clock_bits(rx_tmr_clock_source_t source, uint8_t *bits)`
Map a clock source to TCCR CSS|CKS bits.
- static `uint32_t internal_tmr_clock_divider(rx_tmr_clock_source_t source)`
Return the integer divider for an internal clock source.
- static `uint8_t internal_tmr_compose_tcr(const rx_tmr_config_t *config)`
Compose TCR value from config.
- static `uint8_t internal_tmr_compose_tcsr(const rx_tmr_config_t *config)`
Compose TCSR value from config (OSA/OSB fields).
- static `rx_err_t internal_tmr_validate_config(const rx_tmr_config_t *config)`

- Validate enum fields of a configuration.
- static void `internal_tmr_program_channel` (`rx_tmr_channel_t` channel, const `rx_tmr_config_t` *config)
 - Write register fields for a single channel during init.
- static void `internal_tmr_init_cascade_pair` (const `rx_tmr_config_t` *config)
 - Program the paired odd channel in a 16-bit cascade configuration.
- `rx_err_t rx_tmr_init` (const `rx_tmr_config_t` *config)
 - Initialise a TMR channel (or cascaded pair).
- `rx_err_t rx_tmr_start` (`rx_tmr_channel_t` channel)
 - Start counting on a TMR channel.
- `rx_err_t rx_tmr_stop` (`rx_tmr_channel_t` channel)
 - Stop a TMR channel counter.
- `rx_err_t rx_tmr_read` (`rx_tmr_channel_t` channel, `uint16_t` *value)
 - Read the current TMR counter value.
- `rx_err_t rx_tmr_set_period_us` (`rx_tmr_channel_t` channel, `uint32_t` period_us)
 - Set the compare-match period in microseconds.
- `rx_err_t rx_tmr_register_compare_match_isr` (`rx_tmr_channel_t` channel, `rx_tmr_isr_callback_t` callback)
 - Register an interrupt callback for compare-match / overflow events.
- `rx_err_t rx_tmr_deinit` (`rx_tmr_channel_t` channel)
 - Deinitialise a TMR channel.
- void `rx_tmr_isr_dispatch` (`rx_tmr_channel_t` channel)
 - Internal dispatcher invoked by the SELECTB vector shim.

Variables

- static const char *const `s_tag` = "TMR"
 - Logging tag for the TMR module.
- static bool `s_tmr_initialized` [`k_tmr_channel_count_internal`] = {false, false, false, false}
 - Per-channel init flags indexed by `rx_tmr_channel_t`.
- static `rx_tmr_mode_t` `s_tmr_mode` [`k_tmr_channel_count_internal`]
 - Per-channel operating mode (independent vs cascade).
- static `uint8_t` `s_tmr_clock_bits` [`k_tmr_channel_count_internal`] = {0U, 0U, 0U, 0U}
 - Cached TCCR clock bits (CSS|CKS) applied at `rx_tmr_start()`.
- static `rx_tmr_clock_source_t` `s_tmr_clock_source` [`k_tmr_channel_count_internal`]
 - Cached clock-source enum per channel (needed by `set_period_us`).
- static `rx_tmr_isr_callback_t` `s_tmr_callback` [`k_tmr_channel_count_internal`]
 - Per-channel compare-match / overflow callback pointers.

8.133.1 Detailed Description

TMR HAL Driver Implementation (8-bit + 16-bit cascade).

Implementation of the TMR hardware abstraction layer for the RX72N 8-bit Timer peripheral. Provides per-channel init/start/stop, counter readout, microsecond period programming, and SELECTB ISR-callback registration.

8.133.1.1 Initialisation Sequence

```
* rx_tmr_init(&config)
* |
* +-- 1. Validate config pointer (NASA Rule 5)
* +-- 2. Validate channel / clock / clear / outputs
* +-- 3. In cascade mode, require even channel
* +-- 4. Check not already initialised
* +-- 5. Enable unit module clock (MSTPA5 or MSTPA4, PRCR-gated)
* +-- 6. Stop counter (TCCR = 0)
* +-- 7. Program TCR (interrupt enables + CCLR)
* +-- 8. Program TCSR (output select A/B)
* +-- 9. Preload TCCR clock bits into internal state (start writes them)
* +-- 10. Clear TCNT to 0
* +-- 11. Mark channel initialised
* +-- 12. In cascade mode, also configure paired odd channel
*
```

8.133.1.2 Channel Register Access

Because even and odd TMR channels share a 16-byte block at different parity byte offsets, the driver uses [rx72n_tmr_regs.h](#) inline accessors ([tmr0\(\)..tmr3\(\)](#)) that each return a volatile pointer landing on the correct byte. Each struct field then accesses only the target channel's own byte.

NASA Power of 10 Compliance

- Rule 1: No goto / recursion / setjmp
- Rule 2: All loops bounded
- Rule 3: No dynamic memory
- Rule 4: Functions kept short (decomposed helpers)
- Rule 5: ≥ 2 validation checks per public function
- Rule 6: Variables at smallest scope
- Rule 7: All return values checked / explicitly cast (void)
- Rule 8: C23 typed enums for constants
- Rule 9: Function pointer permitted for ISR DIP
- Rule 10: -Wall -Wextra -Werror clean

See also

[rx_tmr.h](#) Public API

[rx72n_tmr_regs.h](#) Register structures and bit definitions

Author

Locked, Inc.

Date

2026-04-21

Copyright

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Since

Version 1.0.0

Definition in file [rx_tmr.c](#).

8.133.2 Enumeration Type Documentation

8.133.2.1 tmr`byte`shift`

enum [tmr_byte_shift_t](#) : uint8_t

Shift / mask constants for packing cascaded 16-bit counts.

Since

Version 1.0.0

Enumerator

k_tmr_byte_shift	Shift to move high byte to bits [15:8]
k_tmr_byte_mask	Mask for isolating one byte

Definition at line [138](#) of file [rx_tmr.c](#).

```
00138         : uint8_t {
00139   k_tmr_byte_shift = 8U,    /**< Shift to move high byte to bits [15:8] */
00140   k_tmr_byte_mask  = 0xFFU, /**< Mask for isolating one byte */
00141 } tmr_byte_shift_t;
```

8.133.2.2 tmr`channel`count`

enum [tmr_channel_count_t](#) : uint8_t

Array sizing constant for per-channel state.

Since

Version 1.0.0

Enumerator

k_tmr_channel_count_internal	TMR0..TMR3
------------------------------	------------

Definition at line [108](#) of file [rx_tmr.c](#).

```
00108         : uint8_t {
00109   k_tmr_channel_count_internal = 4, /**< TMR0..TMR3 */
00110 } tmr_channel_count_t;
```

8.133.2.3 tmr_pclk_divisor_t

```
enum tmr_pclk_divisor_t : uint32_t
```

Integer prescaler ratios for internal clock sources.

The RX72N TMR peripheral divides PCLKB by one of these ratios before feeding the counter. Each value mirrors a `k_tmr_clock_pclk_div_*` enum value and is used by `internal_tmr_clock_divider()` to compute the effective tick rate. Named constants avoid magic-number literals.

Since

Version 1.0.0

Enumerator

<code>k_tmr_pclk_div_1</code>	Prescaler /1
<code>k_tmr_pclk_div_2</code>	Prescaler /2
<code>k_tmr_pclk_div_8</code>	Prescaler /8
<code>k_tmr_pclk_div_32</code>	Prescaler /32
<code>k_tmr_pclk_div_64</code>	Prescaler /64
<code>k_tmr_pclk_div_1024</code>	Prescaler /1024
<code>k_tmr_pclk_div_8192</code>	Prescaler /8192
<code>k_tmr_pclk_div_none</code>	Sentinel: external clock (no PCLK divider)

Definition at line 122 of file `rx_tmr.c`.

```
00122     : uint32_t {
00123 k_tmr_pclk_div_1  = 1U,    /**< Prescaler /1 */
00124 k_tmr_pclk_div_2  = 2U,    /**< Prescaler /2 */
00125 k_tmr_pclk_div_8  = 8U,    /**< Prescaler /8 */
00126 k_tmr_pclk_div_32 = 32U,    /**< Prescaler /32 */
00127 k_tmr_pclk_div_64 = 64U,    /**< Prescaler /64 */
00128 k_tmr_pclk_div_1024 = 1024U, /**< Prescaler /1024 */
00129 k_tmr_pclk_div_8192 = 8192U, /**< Prescaler /8192 */
00130 k_tmr_pclk_div_none = 0U,    /**< Sentinel: external clock (no PCLK divider) */
00131 } tmr_pclk_divisor_t;
```

8.133.2.4 tmr_period_limits_t

```
enum tmr_period_limits_t : uint32_t
```

Compile-time period / divisor limits.

Since

Version 1.0.0

Enumerator

<code>k_tmr_us_per_sec</code>	Microseconds per second
-------------------------------	-------------------------

<code>k_tmr_max_tcora</code>	8-bit TCORA maximum value
<code>k_tmr_max_tcora_16bit</code>	16-bit cascade TCORA maximum value

Definition at line 97 of file [rx_tmr.c](#).

```
00097      : uint32_t {
00098  k_tmr_us_per_sec      = 1000000U, /**< Microseconds per second */
00099  k_tmr_max_tcora       = 0xFFU,   /**< 8-bit TCORA maximum value */
00100  k_tmr_max_tcora_16bit = 0xFFFFU, /**< 16-bit cascade TCORA maximum value */
00101 } tmr_period_limits_t;
```

8.133.3 Function Documentation

8.133.3.1 `internal_tmr_clock_bits()`

```
rx_err_t internal_tmr_clock_bits (
    rx_tmr_clock_source_t source,
    uint8_t * bits) [static]
```

Map a clock source to TCCR CSS|CKS bits.

Since

Version 1.0.0

Definition at line 267 of file [rx_tmr.c](#).

```
00268 {
00269  /* GCOVR_EXCL_BR_START(validate_config blocks out-of-range clock_source) */
00270  switch (source) {
00271  case k_tmr_clock_pclk_div_1:
00272      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div1);
00273      return k_rx_ok;
00274  case k_tmr_clock_pclk_div_2:
00275      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div2);
00276      return k_rx_ok;
00277  case k_tmr_clock_pclk_div_8:
00278      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div8);
00279      return k_rx_ok;
00280  case k_tmr_clock_pclk_div_32:
00281      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div32);
00282      return k_rx_ok;
00283  case k_tmr_clock_pclk_div_64:
00284      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div64);
00285      return k_rx_ok;
00286  case k_tmr_clock_pclk_div_1024:
00287      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div1024);
00288      return k_rx_ok;
00289  case k_tmr_clock_pclk_div_8192:
00290      *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div8192);
00291      return k_rx_ok;
00292  case k_tmr_clock_external_rising:
00293      *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_rising);
00294      return k_rx_ok;
00295  case k_tmr_clock_external_falling:
00296      *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_falling);
00297      return k_rx_ok;
00298  case k_tmr_clock_external_both:
00299      *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_both);
00300      return k_rx_ok;
00301  default: /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range clock_source */
00302      return k_rx_err_invalid_arg; /* GCOVR_EXCL_LINE */
00303  }
00304  /* GCOVR_EXCL_BR_STOP */
00305 }
```

References [k_tmr_clock_external_both](#), [k_tmr_clock_external_falling](#), [k_tmr_clock_external_rising](#), [k_tmr_clock_pclk_div_1](#), [k_tmr_clock_pclk_div_1024](#), [k_tmr_clock_pclk_div_2](#), [k_tmr_clock_pclk_div_32](#), [k_tmr_clock_pclk_div_64](#), [k_tmr_clock_pclk_div_8](#), [k_tmr_clock_pclk_div_8192](#), [k_tmr_tccr_cks_ext_both](#),

[k_tmr_tccr_cks_ext_falling](#), [k_tmr_tccr_cks_ext_rising](#), [k_tmr_tccr_cks_pclk_div1](#), [k_tmr_tccr_cks_pclk_div10](#), [k_tmr_tccr_cks_pclk_div2](#), [k_tmr_tccr_cks_pclk_div32](#), [k_tmr_tccr_cks_pclk_div64](#), [k_tmr_tccr_cks_pclk_div8](#), [k_tmr_tccr_cks_pclk_div8192](#), [k_tmr_tccr_css_external](#), and [k_tmr_tccr_css_internal](#).

Referenced by [rx_tmr_init\(\)](#).

Here is the caller graph for this function:



8.133.3.2 internal_tmr_clock_divider()

```
uint32_t internal_tmr_clock_divider (
    rx_tmr_clock_source_t source) [static]
```

Return the integer divider for an internal clock source.

Since

Version 1.0.0

Definition at line 311 of file [rx_tmr.c](#).

```
00312 {
00313     switch (source) {
00314         case k_tmr_clock_pclk_div_1:
00315             return k_tmr_pclk_div_1;
00316         case k_tmr_clock_pclk_div_2:
00317             return k_tmr_pclk_div_2;
00318         case k_tmr_clock_pclk_div_8:
00319             return k_tmr_pclk_div_8;
00320         case k_tmr_clock_pclk_div_32:
00321             return k_tmr_pclk_div_32;
00322         case k_tmr_clock_pclk_div_64:
00323             return k_tmr_pclk_div_64;
00324         case k_tmr_clock_pclk_div_1024:
00325             return k_tmr_pclk_div_1024;
00326         case k_tmr_clock_pclk_div_8192:
00327             return k_tmr_pclk_div_8192;
00328         default:
00329             return k_tmr_pclk_div_none; /* External or invalid */
00330     }
00331 }
```

References [k_tmr_clock_pclk_div_1](#), [k_tmr_clock_pclk_div_1024](#), [k_tmr_clock_pclk_div_2](#), [k_tmr_clock_pclk_div_32](#), [k_tmr_clock_pclk_div_64](#), [k_tmr_clock_pclk_div_8](#), [k_tmr_clock_pclk_div_8192](#), [k_tmr_pclk_div_1](#), [k_tmr_pclk_div_1024](#), [k_tmr_pclk_div_2](#), [k_tmr_pclk_div_32](#), [k_tmr_pclk_div_64](#), [k_tmr_pclk_div_8](#), [k_tmr_pclk_div_8192](#), and [k_tmr_pclk_div_none](#).

Referenced by [rx_tmr_set_period_us\(\)](#).

Here is the caller graph for this function:



8.133.3.3 internal`tmr`compose`tcr()

```
uint8_t internal_tmr_compose_tcr (
    const rx_tmr_config_t * config) [static]
```

Compose TCR value from config.

Since

Version 1.0.0

Definition at line 337 of file [rx_tmr.c](#).

```
00338 {
00339     uint8_t tcr = 0U;
00340
00341     if ((config->irq_mask & k_tmr_irq_cmp_match_a) != 0U) {
00342         tcr |= k_tmr_tcr_cmiea;
00343     }
00344     if ((config->irq_mask & k_tmr_irq_cmp_match_b) != 0U) {
00345         tcr |= k_tmr_tcr_cmieb;
00346     }
00347     if ((config->irq_mask & k_tmr_irq_overflow) != 0U) {
00348         tcr |= k_tmr_tcr_ovie;
00349     }
00350
00351     /* GCOVR_EXCL_BR_START(validate_config blocks out-of-range counter_clear) */
00352     switch (config->counter_clear) {
00353     case k_tmr_clear_disabled:
00354         tcr |= k_tmr_tcr_cclr_disabled;
00355         break;
00356     case k_tmr_clear_cmp_match_a:
00357         tcr |= k_tmr_tcr_cclr_cmp_match_a;
00358         break;
00359     case k_tmr_clear_cmp_match_b:
00360         tcr |= k_tmr_tcr_cclr_cmp_match_b;
00361         break;
00362     case k_tmr_clear_external_reset:
00363         tcr |= k_tmr_tcr_cclr_external_sig;
00364         break;
00365     default: /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range counter_clear */
00366         break; /* GCOVR_EXCL_LINE */
00367     }
00368     /* GCOVR_EXCL_BR_STOP */
00369     return tcr;
00370 }
```

References [rx_tmr_config_t::counter_clear](#), [rx_tmr_config_t::irq_mask](#), [k_tmr_clear_cmp_match_a](#), [k_tmr_clear_cmp_match_b](#), [k_tmr_clear_disabled](#), [k_tmr_clear_external_reset](#), [k_tmr_irq_cmp_match_a](#), [k_tmr_irq_cmp_match_b](#), [k_tmr_irq_overflow](#), [k_tmr_tcr_cclr_cmp_match_a](#), [k_tmr_tcr_cclr_cmp_match_b](#), [k_tmr_tcr_cclr_disabled](#), [k_tmr_tcr_cclr_external_sig](#), [k_tmr_tcr_cmiea](#), [k_tmr_tcr_cmieb](#), and [k_tmr_tcr_ovie](#).

Referenced by [internal_tmr_program_channel\(\)](#).

Here is the caller graph for this function:



8.133.3.4 internal'tmr'compose'tcsr()

```
uint8_t internal_tmr_compose_tcsr (
    const rx_tmr_config_t * config)  [static]
```

Compose TCSR value from config (OSA/OSB fields).

Since

Version 1.0.0

Definition at line 376 of file [rx_tmr.c](#).

```
00377 {
00378     const uint8_t osa = (uint8_t)((((uint8_t)config->output_a) « k_tmr_tcsr_osa_shift);
00379     const uint8_t osb = (uint8_t)((((uint8_t)config->output_b) « k_tmr_tcsr_osb_shift);
00380     return (uint8_t)(osa | osb);
00381 }
```

References [k_tmr_tcsr_osa_shift](#), [k_tmr_tcsr_osb_shift](#), [rx_tmr_config_t::output_a](#), and [rx_tmr_config_t::output_b](#).

Referenced by [internal_tmr_program_channel\(\)](#).

Here is the caller graph for this function:



8.133.3.5 internal'tmr'enable'module'clock()

```
void internal_tmr_enable_module_clock (
    rx_tmr_channel_t channel)  [static]
```

Enable the module clock for the channel's TMR unit.

Precondition

PRCR register accessible

Postcondition

MSTPCRA bit cleared for the unit

Since

Version 1.0.0

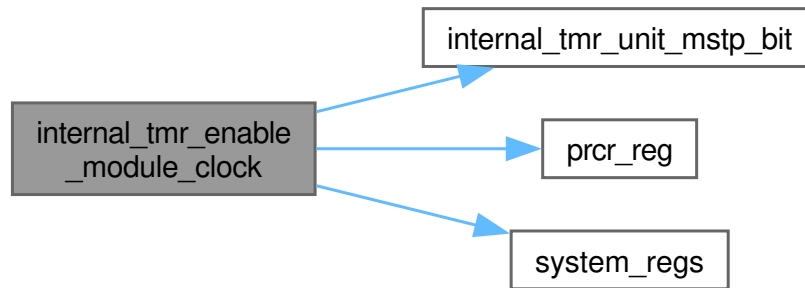
Definition at line 254 of file [rx_tmr.c](#).

```
00255 {
00256     const uint32_t mstp_bit = internal_tmr_unit_mstp_bit(channel);
00257
00258     *prcr_reg() = k_rx_prcr_unlock_prc1;
00259     system_regs()->mstpcra &= ~mstp_bit;
00260     *prcr_reg() = k_rx_prcr_lock;
00261 }
```

References [internal_tmr_unit_mstp_bit\(\)](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_tmr_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.133.3.6 internal`tmr`get`regs()

```
RX_STATIC_TESTABLE volatile rx\_tmr\_channel\_regs\_t * internal_tmr_get_regs (
    rx\_tmr\_channel\_t channel)
```

Get register pointer for a TMR channel.

Since

Version 1.0.0

Definition at line 215 of file [rx_tmr.c](#).

```

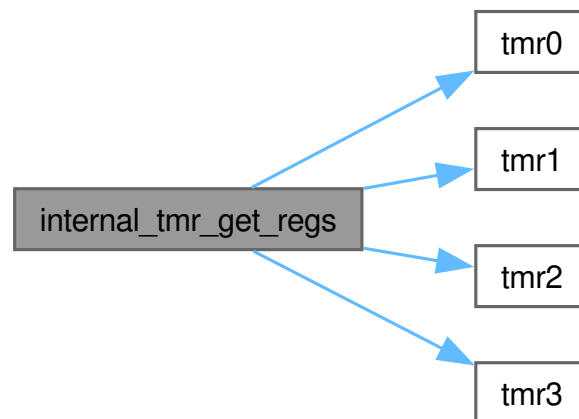
00216 {
00217     /* GCOVR_EXCL_BR_START(validate_config restricts channel to 0..3) */
00218     switch (channel) {
00219         case k_tmr_channel_0:
00220             return tmr0();
00221         case k_tmr_channel_1:
00222             return tmr1();
00223         case k_tmr_channel_2:
00224             return tmr2();
00225         case k_tmr_channel_3:
00226             return tmr3();
00227         default:
00228             /* GCOVR_EXCL_LINE */
00229             RX_ASSERT(false, "TMR channel out of range"); /* GCOVR_EXCL_LINE */
00230             return NULL; /* GCOVR_EXCL_LINE */
00231     }
00232     /* GCOVR_EXCL_BR_STOP */

```

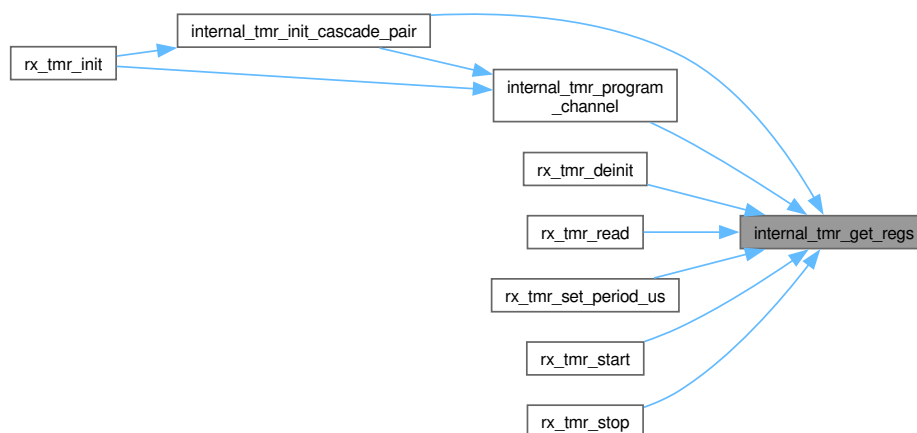
References [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_2](#), [k_tmr_channel_3](#), [tmr0\(\)](#), [tmr1\(\)](#), [tmr2\(\)](#), and [tmr3\(\)](#).

Referenced by [internal_tmr_init_cascade_pair\(\)](#), [internal_tmr_program_channel\(\)](#), [rx_tmr_deinit\(\)](#), [rx_tmr_read\(\)](#), [rx_tmr_set_period_us\(\)](#), [rx_tmr_start\(\)](#), and [rx_tmr_stop\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.133.3.7 `internal_tmr_init_cascade_pair()`

```
void internal_tmr_init_cascade_pair (
    const rx_tmr_config_t * config) [static]
```

Program the paired odd channel in a 16-bit cascade configuration.

Extracted from `rx_tmr_init()` to keep the caller under the function-size threshold. The even channel counts on its selected clock source; the odd channel counts on the even channel's TCORA compare match via `CSS = cascade`.

Since

Version 1.0.0

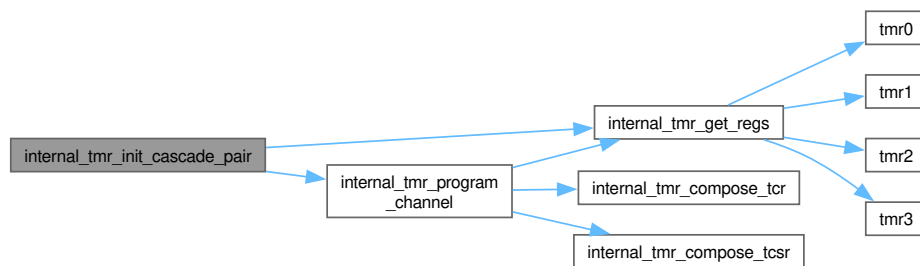
Definition at line 446 of file `rx_tmr.c`.

```
00447 {
00448     const rx_tmr_channel_t odd =
00449         (config->channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00450
00451     rx_tmr_config_t odd_cfg = *config;
00452     odd_cfg.channel = odd;
00453     internal_tmr_program_channel(odd, &odd_cfg);
00454
00455     volatile rx_tmr_channel_regs_t* odd_regs = internal_tmr_get_regs(odd);
00456     odd_regs->tccr = k_tmr_tccr_css_cascade;
00457
00458     s_tmr_mode[(uint8_t)odd] = k_tmr_mode_16bit_cascade;
00459     s_tmr_initialized[(uint8_t)odd] = true;
00460     s_tmr_clock_bits[(uint8_t)odd] = k_tmr_tccr_css_cascade;
00461     s_tmr_clock_source[(uint8_t)odd] = config->clock_source;
00462 }
```

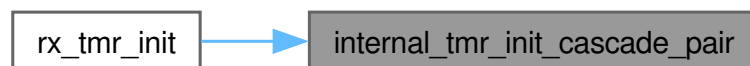
References `rx_tmr_config_t::channel`, `rx_tmr_config_t::clock_source`, `internal_tmr_get_regs()`, `internal_tmr_program_channel()`, `k_tmr_channel_0`, `k_tmr_channel_1`, `k_tmr_channel_3`, `k_tmr_mode_16bit_cascade`, `k_tmr_tccr_css_cascade`, `s_tmr_clock_bits`, `s_tmr_clock_source`, `s_tmr_initialized`, `s_tmr_mode`, and `rx_tmr_channel_regs_t::tccr`.

Referenced by `rx_tmr_init()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.133.3.8 `internal_tmr_program_channel()`

```
void internal_tmr_program_channel (
    rx_tmr_channel_t channel,
    const rx_tmr_config_t * config) [static]
```

Write register fields for a single channel during init.

Since

Version 1.0.0

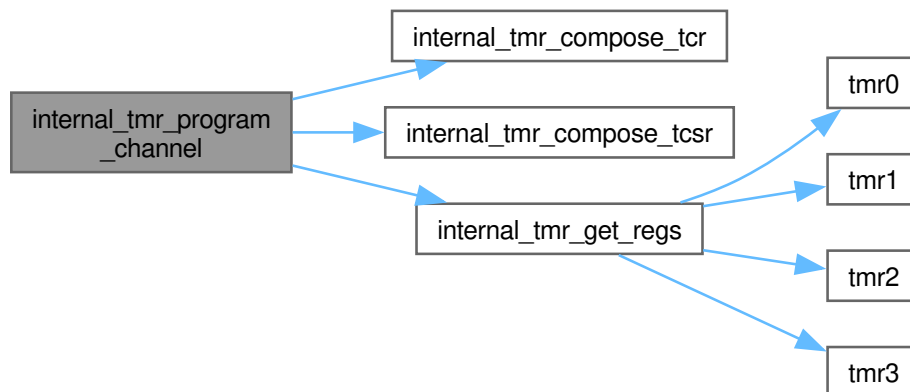
Definition at line 420 of file `rx_tmr.c`.

```
00421 {
00422     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00423
00424     regs->tccr = 0U; /* Stop counter before programming */
00425     regs->tcr = internal_tmr_compose_tcr(config);
00426     regs->tcsr = internal_tmr_compose_tcsr(config);
00427     regs->tcora = k_tmr_max_tcora;
00428     regs->tcorb = k_tmr_max_tcora;
00429     regs->tcnt = 0U;
00430 }
```

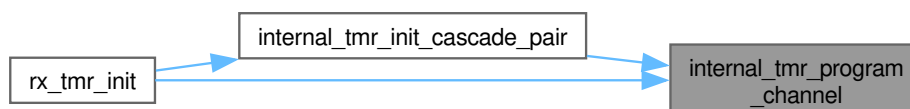
References `internal_tmr_compose_tcr()`, `internal_tmr_compose_tcsr()`, `internal_tmr_get_regs()`, `k_tmr_max_tcora`, `rx_tmr_channel_regs_t::tccr`, `rx_tmr_channel_regs_t::tcnt`, `rx_tmr_channel_regs_t::tcora`, `rx_tmr_channel_regs_t::tcorb`, `rx_tmr_channel_regs_t::tcr`, and `rx_tmr_channel_regs_t::tcsr`.

Referenced by `internal_tmr_init_cascade_pair()`, and `rx_tmr_init()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.133.3.9 `internal`tmr`unit`mstp`bit()`

```
uint32_t internal_tmr_unit_mstp_bit (
    rx_tmr_channel_t channel) [static]
```

Return the MSTPCRA bit mask for a TMR channel's unit.

Since

Version 1.0.0

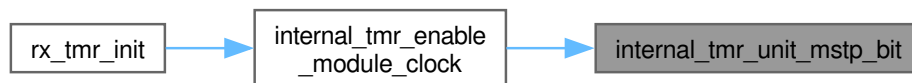
Definition at line 238 of file `rx_tmr.c`.

```
00239 {
00240     /* Public callers pre-validate channel in 0..3 via internal_tmr_validate_config, so
00241      * anything not in unit 0 (TMR0/1) must belong to unit 1 (TMR2/3). */
00242     if ((channel == k_tmr_channel_0) || (channel == k_tmr_channel_1)) {
00243         return k_tmr_mstpcra_mstp5;
00244     }
00245     return k_tmr_mstpcra_mstp4;
00246 }
```

References `k_tmr_channel_0`, `k_tmr_channel_1`, `k_tmr_mstpcra_mstp4`, and `k_tmr_mstpcra_mstp5`.

Referenced by `internal_tmr_enable_module_clock()`.

Here is the caller graph for this function:

8.133.3.10 `internal`tmr`validate`config()`

```
rx_err_t internal_tmr_validate_config (
    const rx_tmr_config_t * config) [static]
```

Validate enum fields of a configuration.

Since

Version 1.0.0

Definition at line 387 of file `rx_tmr.c`.

```
00388 {
00389     if ((uint8_t)config->channel >= k_tmr_channel_count_internal) {
00390         return k_rx_err_invalid_arg;
00391     }
00392     if ((config->mode != k_tmr_mode_8bit_independent) && (config->mode != k_tmr_mode_16bit_cascade)) {
00393         return k_rx_err_invalid_arg;
00394     }
00395     if ((uint8_t)config->counter_clear > k_tmr_clear_external_reset) {
00396         return k_rx_err_invalid_arg;
00397     }
00398     if ((uint8_t)config->output_a > k_tmr_output_toggle) {
00399         return k_rx_err_invalid_arg;
00400     }
00401     if ((uint8_t)config->output_b > k_tmr_output_toggle) {
00402         return k_rx_err_invalid_arg;
00403     }
00404     if ((uint8_t)config->clock_source > k_tmr_clock_external_both) {
00405         return k_rx_err_invalid_arg;
00406     }
00407 }
```

```

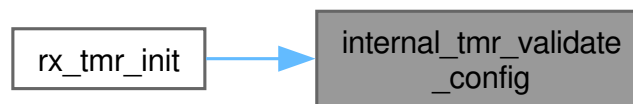
00406 }
00407 /* Cascade mode requires an even channel */
00408 if (config->mode == k_tmr_mode_16bit_cascade) {
00409     if ((config->channel != k_tmr_channel_0) && (config->channel != k_tmr_channel_2)) {
00410         return k_rx_err_invalid_arg;
00411     }
00412 }
00413 return k_rx_ok;
00414 }

```

References [rx_tmr_config_t::channel](#), [rx_tmr_config_t::clock_source](#), [rx_tmr_config_t::counter_clear](#), [k_tmr_channel_0](#), [k_tmr_channel_2](#), [k_tmr_channel_count_internal](#), [k_tmr_clear_external_reset](#), [k_tmr_clock_external_both](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_mode_8bit_independent](#), [k_tmr_output_toggle](#), [rx_tmr_config_t::mode](#), [rx_tmr_config_t::output_a](#), and [rx_tmr_config_t::output_b](#).

Referenced by [rx_tmr_init\(\)](#).

Here is the caller graph for this function:



8.133.3.11 rx_tmr_deinit()

```

rx_err_t rx_tmr_deinit (
    rx_tmr_channel_t channel) [nodiscard]

```

Deinitialise a TMR channel.

Stops the counter, clears TCR/TCSR/TCCR, zeroes TCNT, removes any installed ISR callback, and marks the channel as uninitialised. The module stop bit is NOT asserted because the paired channel in the same unit may still be active.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Channel released
k_rx_err_invalid_arg	Channel out of range

Precondition

(none - safe to call on uninitialised channel)

Postcondition

Channel registers cleared
Internal init flag cleared

Note

Thread Safety: Not thread-safe.

Since

Version 1.0.0

Definition at line 625 of file [rx_tmr.c](#).

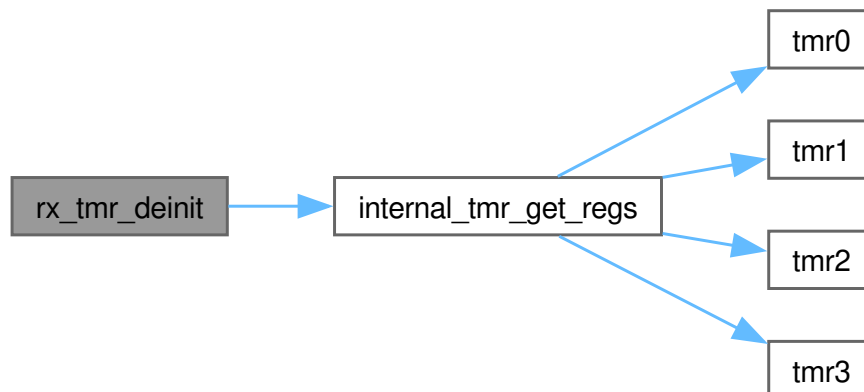
```

00626 {
00627     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00628         return k_rx_err_invalid_arg;
00629     }
00630
00631     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00632     regs->tccr = 0U;
00633     regs->tcr = 0U;
00634     regs->tcsr = 0U;
00635     regs->tcora = 0U;
00636     regs->tcorb = 0U;
00637     regs->tcnt = 0U;
00638
00639     const uint8_t idx = (uint8_t)channel;
00640     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00641         const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00642         volatile rx_tmr_channel_regs_t* odd_regs = internal_tmr_get_regs(odd);
00643         odd_regs->tccr = 0U;
00644         odd_regs->tcr = 0U;
00645         odd_regs->tcsr = 0U;
00646         odd_regs->tcora = 0U;
00647         odd_regs->tcorb = 0U;
00648         odd_regs->tcnt = 0U;
00649         s_tmr_initialized[(uint8_t)odd] = false;
00650         s_tmr_mode[(uint8_t)odd] = k_tmr_mode_8bit_independent;
00651         s_tmr_clock_bits[(uint8_t)odd] = 0U;
00652         s_tmr_callback[(uint8_t)odd] = NULL;
00653     }
00654
00655     s_tmr_initialized[idx] = false;
00656     s_tmr_mode[idx] = k_tmr_mode_8bit_independent;
00657     s_tmr_clock_bits[idx] = 0U;
00658     s_tmr_callback[idx] = NULL;
00659     return k_rx_ok;
00660 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_mode_8bit_independent](#), [s_tmr_callback](#), [s_tmr_clock_bits](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [rx_tmr_channel_regs_t::tcnt](#), [rx_tmr_channel_regs_t::tcora](#), [rx_tmr_channel_regs_t::tcorb](#), [rx_tmr_channel_regs_t::tcr](#), and [rx_tmr_channel_regs_t::tcsr](#).

Here is the call graph for this function:



8.133.3.12 rx_tmr_init()

```
rx_err_t rx_tmr_init (
    const rx_tmr_config_t * config) [nodiscard]
```

Initialise a TMR channel (or cascaded pair).

Enables the module stop clock for the channel's unit (MSTPA5 for unit 0, MSTPA4 for unit 1), stops the counter, writes TCR/TCSR/TCCR/TCORA/ TCORB according to the supplied configuration, and clears TCNT to zero. In cascade mode, both channels of the unit are configured in one call.

Parameters

in	config	Pointer to configuration (non-NULL)
----	--------	-------------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	config is NULL
k_rx_err_invalid_arg	Invalid channel, clock, clear, output, or a cascade request on an odd channel
k_rx_err_invalid_state	Channel already initialised

Precondition

System clocks configured (PCLKB running)

In cascade mode, config->channel is k_tmr_channel_0 or k_tmr_channel_2 (even channel)

Postcondition

Channel counter stopped at 0

Module clock enabled (MSTPCRA bit cleared)

Internal init flag set

Note

Not thread-safe; call from main-task context at startup.

See also

[rx_tmr_start\(\)](#) Begin counting

[rx_tmr_deinit\(\)](#) Release the channel

Since

Version 1.0.0

Definition at line 464 of file [rx_tmr.c](#).

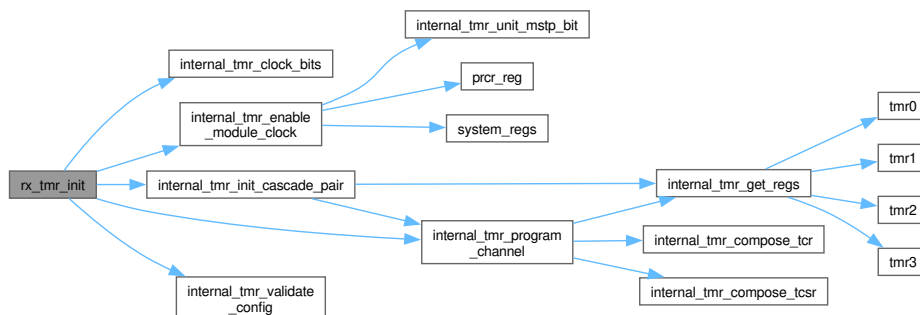
```

00465 {
00466     /* Pre-condition 1: non-null config */
00467     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is NULL");
00468
00469     /* Pre-condition 2: enum ranges */
00470     rx_err_t err = internal_tmr_validate_config(config);
00471     if (err != k_rx_ok) {
00472         rx_log_error(s_tag, "Invalid TMR config");
00473         return err;
00474     }
00475
00476     const uint8_t idx = (uint8_t)config->channel;
00477
00478     /* Pre-condition 3: not already initialised */
00479     if (s_tmr_initialized[idx]) {
00480         rx_log_error(s_tag, "Channel already initialised");
00481         return k_rx_err_invalid_state;
00482     }
00483
00484     /* Compose clock bits (validated via enum range above) */
00485     uint8_t clock_bits = 0U;
00486     err = internal_tmr_clock_bits(config->clock_source, &clock_bits);
00487     if (err != k_rx_ok) { /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range clock_source */
00488         return err; /* GCOVR_EXCL_LINE */
00489     } /* GCOVR_EXCL_BR_LINE */
00490
00491     /* 1. Enable unit module clock */
00492     internal_tmr_enable_module_clock(config->channel);
00493
00494     /* 2. Program the primary channel */
00495     internal_tmr_program_channel(config->channel, config);
00496
00497     /* 3. For cascade mode, program the paired odd channel */
00498     if (config->mode == k_tmr_mode_16bit_cascade) {
00499         internal_tmr_init_cascade_pair(config);
00500     }
00501
00502     /* 4. Cache state for the primary channel */
00503     s_tmr_mode[idx] = config->mode;
00504     s_tmr_clock_bits[idx] = clock_bits;
00505     s_tmr_clock_source[idx] = config->clock_source;
00506     s_tmr_initialized[idx] = true;
00507
00508     rx_log_info_val(s_tag, "TMR channel initialised", (uint32_t)config->channel);
00509     return k_rx_ok;
00510 }

```

References [rx_tmr_config_t::channel](#), [rx_tmr_config_t::clock_source](#), [internal_tmr_clock_bits\(\)](#), [internal_tmr_enable_module_clock\(\)](#), [internal_tmr_init_cascade_pair\(\)](#), [internal_tmr_program_channel\(\)](#), [internal_tmr_validate_config\(\)](#), [k_tmr_mode_16bit_cascade](#), [rx_tmr_config_t::mode](#), [s_tag](#), [s_tmr_clock_bits](#), [s_tmr_clock_source](#), [s_tmr_initialized](#), and [s_tmr_mode](#).

Here is the call graph for this function:



8.133.3.13 rx_tmr_isr_dispatch()

```
void rx_tmr_isr_dispatch (
    rx_tmr_channel_t channel)
```

Internal dispatcher invoked by the SELECTB vector shim.

Called from the low-level vector when a TMR CMIA/CMIB/OVI interrupt fires. Clears the source flag in the TMR (compare-match sets no persistent status bit, so the hardware does it automatically; this function exists so the test harness can synthesise ISR invocations) and invokes the registered user callback.

Since

Version 1.0.0

Definition at line 679 of file [rx_tmr.c](#).

```
00680 {
00681     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00682         return;
00683     }
00684     rx_tmr_isr_callback_t cb = s_tmr_callback[(uint8_t)channel];
00685     if (cb != NULL) {
00686         cb(channel);
00687     }
00688 }
```

References [k_tmr_channel_count_internal](#), and [s_tmr_callback](#).

8.133.3.14 rx_tmr_read()

```
rx_err_t rx_tmr_read (
    rx_tmr_channel_t channel,
    uint16_t * value) [nodiscard]
```

Read the current TMR counter value.

For 8-bit mode, *value holds the 8-bit TCNT zero-extended to uint16_t. For 16-bit cascade mode, *value holds the full 16-bit value (even channel = upper byte, odd channel = lower byte).

Parameters

in	channel	TMR channel (cascade: pass the even channel)
out	value	Pointer receiving counter value (non-NULL)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success
k_rx_err_null_ptr	value is NULL
k_rx_err_invalid_arg	Channel out of range

k_rx_err_invalid_state	Channel not initialised
------------------------	-------------------------

Precondition

Channel initialised

Postcondition

*value contains counter at time of read

Note

Thread Safety: Safe - single volatile read.

Since

Version 1.0.0

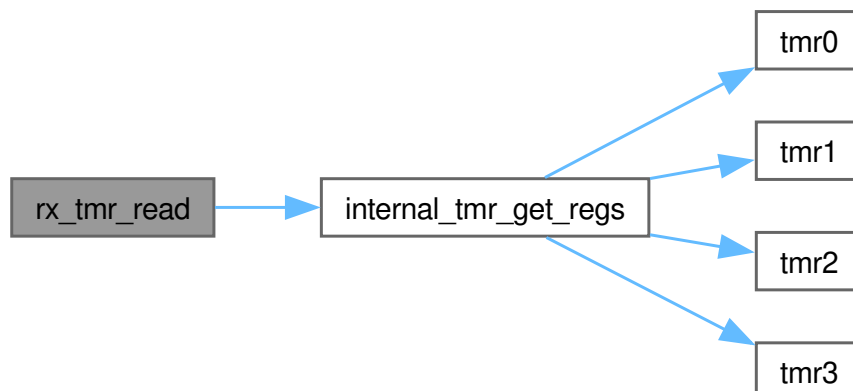
Definition at line 551 of file [rx_tmr.c](#).

```

00552 {
00553   RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is NULL");
00554   if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00555     return k_rx_err_invalid_arg;
00556   }
00557   const uint8_t idx = (uint8_t)channel;
00558   RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00559   if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00560     const uint8_t upper = internal_tmr_get_regs(channel)->tcnt;
00561     const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00562     const uint8_t lower = internal_tmr_get_regs(odd)->tcnt;
00563     *value = (uint16_t)(((uint16_t)upper « (uint16_t)k_tmr_byte_shift) | (uint16_t)lower);
00564   } else {
00565     *value = (uint16_t)internal_tmr_get_regs(channel)->tcnt;
00566   }
00567   return k_rx_ok;
00568 }
00569 }
```

References [internal_tmr_get_regs\(\)](#), [k_tmr_byte_shift](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [s_tag](#), [s_tmr_initialized](#), [s_tmr_mode](#), and [rx_tmr_channel_regs_t::tcnt](#).

Here is the call graph for this function:



8.133.3.15 `rx_tmr_register_compare_match_isr()`

```
rx_err_t rx_tmr_register_compare_match_isr (
    rx_tmr_channel_t channel,
    rx_tmr_isr_callback_t callback) [nodiscard]
```

Register an interrupt callback for compare-match / overflow events.

Installs a user callback invoked from the ICU SELECTB dispatcher when any of the channel's enabled TMR interrupts (CMIA, CMIB, or OVI) fire. The driver's ISR shim clears the source flag before calling back. Pass NULL to deregister.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
in	callback	User callback, or NULL to deregister

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success
<code>k_rx_err_invalid_arg</code>	Channel out of range

Precondition

Channel initialised with desired `irq_mask`

Postcondition

callback is installed (or cleared) atomically

Note

Thread Safety: Safe - function-pointer write is atomic on RX.

Since

Version 1.0.0

Definition at line 616 of file [rx_tmr.c](#).

```
00617 {
00618     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00619         return k_rx_err_invalid_arg;
00620     }
00621     s_tmr_callback[(uint8_t)channel] = callback;
00622     return k_rx_ok;
00623 }
```

References [k_tmr_channel_count_internal](#), and [s_tmr_callback](#).

8.133.3.16 rx`tmr`set`period`us()

```
rx_err_t rx_tmr_set_period_us (
    rx_tmr_channel_t channel,
    uint32_t period_us) [nodiscard]
```

Set the compare-match period in microseconds.

Recomputes and writes TCORA so that a TCORA compare match fires every period_us microseconds using the channel's configured clock source. TCORB is untouched. The caller must ensure the period fits into the channel counter width (8-bit single channel max = ~35 ms at PCLK/8192; 16-bit cascade max ~ = 8.9 s at PCLK/8192).

The underlying formula is $tcora = \text{round}(\text{period_us} * \text{PCLK_Hz} / (\text{divider} * 1e6)) - 1$, where divider is 1/2/8/32/64/1024/8192 according to the configured clock_source. External clock sources are not supported.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
in	period_us	Desired period in microseconds (1..max for mode)

Returns

rx_err_t Error code

Return values

k_rx_ok	Success, TCORA updated
k_rx_err_invalid_arg	Channel out of range, period=0, period too large for the selected divider, or external clock source
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised with an internal-clock clock_source

Postcondition

TCORA holds the rounded tick count - 1

Note

Thread Safety: Safe - single register write.

Since

Version 1.0.0

Definition at line 571 of file [rx_tmr.c](#).

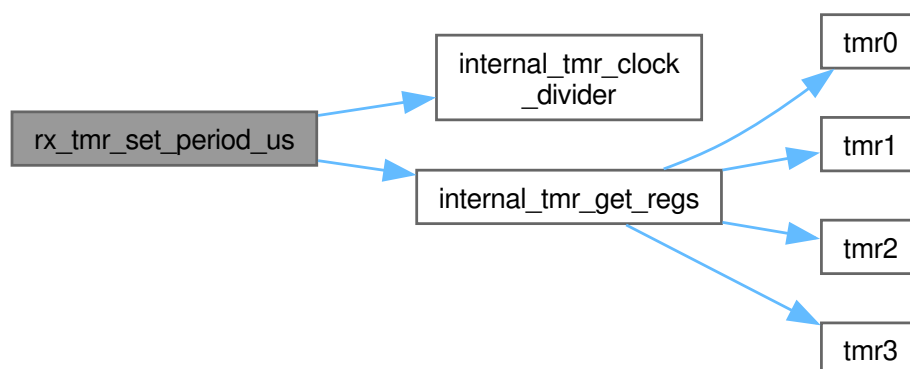
```

00572 {
00573     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00574         return k_rx_err_invalid_arg;
00575     }
00576     const uint8_t idx = (uint8_t)channel;
00577     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00578
00579     if (period_us == 0U) {
00580         return k_rx_err_invalid_arg;
00581     }
00582
00583     const uint32_t divider = internal_tmr_clock_divider(s_tmr_clock_source[idx]);
00584     if (divider == 0U) {
00585         /* External clock selected - cannot compute period */
00586         return k_rx_err_invalid_arg;
00587     }
00588
00589     /* tick_hz = PCLKB / divider; ticks = period_us * tick_hz / 1e6
00590      * compare = ticks - 1 (TCORA fires when TCNT reaches compare value) */
00591     const uint64_t ticks =
00592         ((uint64_t)period_us * (uint64_t)k_pclkb_hz) / ((uint64_t)divider * (uint64_t)k_tmr_us_per_sec);
00593     if (ticks == 0U) {
00594         return k_rx_err_invalid_arg;
00595     }
00596     const uint64_t compare = ticks - 1U;
00597
00598     const uint32_t max_compare =
00599         (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) ? k_tmr_max_tcora_16bit : k_tmr_max_tcora;
00600     if (compare > (uint64_t)max_compare) {
00601         rx_log_error_val(s_tag, "Period too large for divider", period_us);
00602         return k_rx_err_invalid_arg;
00603     }
00604
00605     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00606         const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00607         internal_tmr_get_regs(channel)->tcora =
00608             (uint8_t)((compare » (uint64_t)k_tmr_byte_shift) & (uint64_t)k_tmr_byte_mask);
00609         internal_tmr_get_regs(odd)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00610     } else {
00611         internal_tmr_get_regs(channel)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00612     }
00613     return k_rx_ok;
00614 }

```

References [internal_tmr_clock_divider\(\)](#), [internal_tmr_get_regs\(\)](#), [k_pclkb_hz](#), [k_tmr_byte_mask](#), [k_tmr_byte_shift](#), [k_tmr_channel_0](#), [k_tmr_channel_1](#), [k_tmr_channel_3](#), [k_tmr_channel_count_internal](#), [k_tmr_max_tcora](#), [k_tmr_max_tcora_16bit](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_us_per_sec](#), [s_tag](#), [s_tmr_clock_source](#), [s_tmr_initialized](#), [s_tmr_mode](#), and [rx_tmr_channel_regs_t::tcora](#).

Here is the call graph for this function:



8.133.3.17 rx`tmr`start()

```
rx_err_t rx_tmr_start (
    rx_tmr_channel_t channel) [nodiscard]
```

Start counting on a TMR channel.

Writes the configured clock-source bits into TCCR so the counter begins advancing on each selected clock edge. Channel must have been initialised via `rx_tmr_init()`.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Counter started
k_rx_err_invalid_arg	Channel out of range
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised via `rx_tmr_init()`

Postcondition

TCCR programmed with clock source (counter running)

Note

Thread Safety: Safe - single register write.

Since

Version 1.0.0

Definition at line 512 of file `rx_tmr.c`.

```
00513 {
00514     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00515         return k_rx_err_invalid_arg;
00516     }
00517     const uint8_t idx = (uint8_t)channel;
00518     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00519
00520     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00521     regs->tccr = s_tmr_clock_bits[idx];
00522
00523     /* In cascade mode the paired odd channel also needs its cascade bits
00524      * kept after a stop/start cycle. Cascade is only valid on even channels
00525      * (enforced by validate_config), so the ternary below is exhaustive. */
```

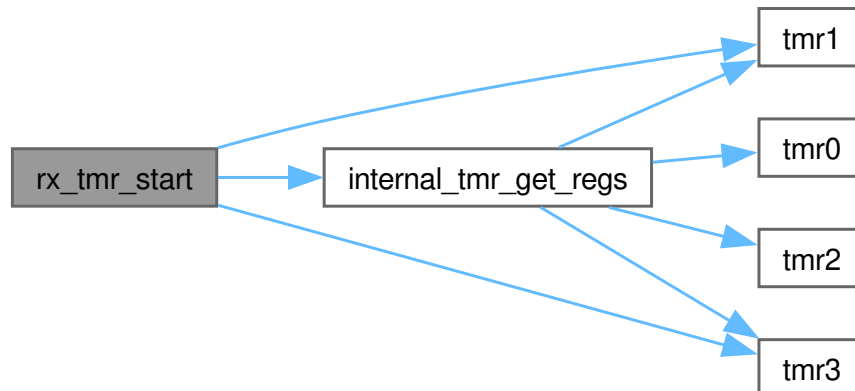
```

00526 if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00527     volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00528     odd_regs->tccr = k_tmr_tccr_css_cascade;
00529 }
00530 return k_rx_ok;
00531 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [k_tmr_tccr_css_cascade](#), [s_tag](#), [s_tmr_clock_bits](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [tmr1\(\)](#), and [tmr3\(\)](#).

Here is the call graph for this function:



8.133.3.18 rx_tmr_stop()

```

rx_err_t rx_tmr_stop (
    rx_tmr_channel_t channel) [nodiscard]

```

Stop a TMR channel counter.

Writes 0 into TCCR CKS/CSS to disable the clock input. The TCNT value is preserved.

Parameters

in	channel	TMR channel (cascade: pass the even channel)
----	---------	--

Returns

rx_err_t Error code

Return values

k_rx_ok	Counter stopped
k_rx_err_invalid_arg	Channel out of range
k_rx_err_invalid_state	Channel not initialised

Precondition

Channel initialised

Postcondition

Counter no longer advances

TCNT preserved

Note

Thread Safety: Safe - single register write.

Since

Version 1.0.0

Definition at line 533 of file [rx_tmr.c](#).

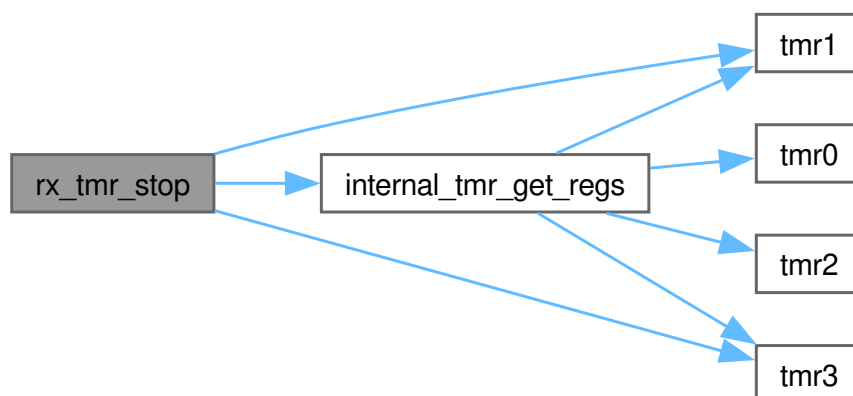
```

00534 {
00535     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00536         return k_rx_err_invalid_arg;
00537     }
00538     const uint8_t idx = (uint8_t)channel;
00539     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00540
00541     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00542     regs->tccr = 0U;
00543
00544     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00545         volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00546         odd_regs->tccr = 0U;
00547     }
00548     return k_rx_ok;
00549 }

```

References [internal_tmr_get_regs\(\)](#), [k_tmr_channel_0](#), [k_tmr_channel_count_internal](#), [k_tmr_mode_16bit_cascade](#), [s_tag](#), [s_tmr_initialized](#), [s_tmr_mode](#), [rx_tmr_channel_regs_t::tccr](#), [tmr1\(\)](#), and [tmr3\(\)](#).

Here is the call graph for this function:



8.133.4 Variable Documentation

8.133.4.1 s`tag

```
const char* const s_tag = "TMR" [static]
```

Logging tag for the TMR module.

4 bytes in .rodata.

Since

Version 1.0.0

Definition at line 86 of file [rx_tmr.c](#).

8.133.4.2 s`tmr`callback

```
rx_tmr_isr_callback_t s_tmr_callback[k_tmr_channel_count_internal] [static]
```

Initial value:

```

                                = {
    NULL,
    NULL,
    NULL,
    NULL,
}

```

Per-channel compare-match / overflow callback pointers.

Since

Version 1.0.0

Definition at line 199 of file [rx_tmr.c](#).

```

00199
00200 NULL,
00201 NULL,
00202 NULL,
00203 NULL,
00204 };
                                {

```

Referenced by [rx_tmr_deinit\(\)](#), [rx_tmr_isr_dispatch\(\)](#), and [rx_tmr_register_compare_match_isr\(\)](#).

8.133.4.3 s`tmr`clock`bits

```
uint8_t s_tmr_clock_bits[k_tmr_channel_count_internal] = {0U, 0U, 0U, 0U} [static]
```

Cached TCCR clock bits (CSS|CKS) applied at [rx_tmr_start\(\)](#).

[rx_tmr_init\(\)](#) computes the TCCR clock-field value but does NOT write it immediately because that would start the counter. [rx_tmr_start\(\)](#) OR's these bits into TCCR when requested. [rx_tmr_stop\(\)](#) clears the clock field so the counter halts.

Since

Version 1.0.0

Definition at line 180 of file [rx_tmr.c](#).

```
00180 {0U, 0U, 0U, 0U};
```

Referenced by [internal_tmr_init_cascade_pair\(\)](#), [rx_tmr_deinit\(\)](#), [rx_tmr_init\(\)](#), and [rx_tmr_start\(\)](#).

8.133.4.4 s'tmr'clock'source

[rx_tmr_clock_source_t](#) [s_tmr_clock_source\[k_tmr_channel_count_internal\]](#) [static]

Initial value:

```

                                = {
    k_tmr_clock_pclk_div_1,
    k_tmr_clock_pclk_div_1,
    k_tmr_clock_pclk_div_1,
    k_tmr_clock_pclk_div_1,
}

```

Cached clock-source enum per channel (needed by [set_period_us](#)).

Since

Version 1.0.0

Definition at line [187](#) of file [rx_tmr.c](#).

```

00187
00188 k_tmr_clock_pclk_div_1,
00189 k_tmr_clock_pclk_div_1,
00190 k_tmr_clock_pclk_div_1,
00191 k_tmr_clock_pclk_div_1,
00192 };

```

Referenced by [internal_tmr_init_cascade_pair\(\)](#), [rx_tmr_init\(\)](#), and [rx_tmr_set_period_us\(\)](#).

8.133.4.5 s'tmr'initialized

[bool](#) [s_tmr_initialized\[k_tmr_channel_count_internal\]](#) = {false, false, false, false} [static]

Per-channel init flags indexed by [rx_tmr_channel_t](#).

Warning

Do not modify directly - use [rx_tmr_init\(\)](#) / [rx_tmr_deinit\(\)](#)

Since

Version 1.0.0

Definition at line [154](#) of file [rx_tmr.c](#).

```

00154 {false, false, false, false};

```

Referenced by [internal_tmr_init_cascade_pair\(\)](#), [rx_tmr_deinit\(\)](#), [rx_tmr_init\(\)](#), [rx_tmr_read\(\)](#), [rx_tmr_set_period_us\(\)](#), [rx_tmr_start\(\)](#), and [rx_tmr_stop\(\)](#).

8.133.4.6 s'tmr'mode

`rx_tmr_mode_t s_tmr_mode[k_tmr_channel_count_internal]` [static]

Initial value:

```
= {
    k_tmr_mode_8bit_independent,
    k_tmr_mode_8bit_independent,
    k_tmr_mode_8bit_independent,
    k_tmr_mode_8bit_independent,
}
```

Per-channel operating mode (independent vs cascade).

Since

Version 1.0.0

Definition at line 161 of file `rx_tmr.c`.

```
00161                                     {
00162     k_tmr_mode_8bit_independent,
00163     k_tmr_mode_8bit_independent,
00164     k_tmr_mode_8bit_independent,
00165     k_tmr_mode_8bit_independent,
00166 };
```

Referenced by `internal_tmr_init_cascade_pair()`, `rx_tmr_deinit()`, `rx_tmr_init()`, `rx_tmr_read()`, `rx_tmr_set_period_us()`, `rx_tmr_start()`, and `rx_tmr_stop()`.

8.134 rx'tmr.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_tmr.c
00003  * @brief TMR HAL Driver Implementation (8-bit + 16-bit cascade)
00004  *
00005  * @details
00006  * Implementation of the TMR hardware abstraction layer for the RX72N
00007  * 8-bit Timer peripheral. Provides per-channel init/start/stop, counter
00008  * readout, microsecond period programming, and SELECTB ISR-callback
00009  * registration.
00010  *
00011  * ## Initialisation Sequence
00012  *
00013  * @verbatim
00014  *   rx_tmr_init(&config)
00015  *   |
00016  *   +-- 1. Validate config pointer (NASA Rule 5)
00017  *   +-- 2. Validate channel / clock / clear / outputs
00018  *   +-- 3. In cascade mode, require even channel
00019  *   +-- 4. Check not already initialised
00020  *   +-- 5. Enable unit module clock (MSTPA5 or MSTPA4, PRCR-gated)
00021  *   +-- 6. Stop counter (TCCR = 0)
00022  *   +-- 7. Program TCR (interrupt enables + CCLR)
00023  *   +-- 8. Program TCSR (output select A/B)
00024  *   +-- 9. Preload TCCR clock bits into internal state (start writes them)
00025  *   +-- 10. Clear TCNT to 0
00026  *   +-- 11. Mark channel initialised
00027  *   +-- 12. In cascade mode, also configure paired odd channel
00028  * @endverbatim
00029  *
00030  * ## Channel Register Access
00031  *
00032  * Because even and odd TMR channels share a 16-byte block at different
00033  * parity byte offsets, the driver uses rx72n_tmr_regs.h inline accessors
00034  * (tmr0().tmr3()) that each return a volatile pointer landing on the
00035  * correct byte. Each struct field then accesses only the target channel's
00036  * own byte.
00037  *
00038  * @par NASA Power of 10 Compliance
00039  * - Rule 1: No goto / recursion / setjmp
```

```

00040 * - Rule 2: All loops bounded
00041 * - Rule 3: No dynamic memory
00042 * - Rule 4: Functions kept short (decomposed helpers)
00043 * - Rule 5: >= 2 validation checks per public function
00044 * - Rule 6: Variables at smallest scope
00045 * - Rule 7: All return values checked / explicitly cast (void)
00046 * - Rule 8: C23 typed enums for constants
00047 * - Rule 9: Function pointer permitted for ISR DIP
00048 * - Rule 10: -Wall -Wextra -Werror clean
00049 *
00050 * @see rx_tmr.h Public API
00051 * @see rx72n_tmr_regs.h Register structures and bit definitions
00052 *
00053 * @author Locked, Inc.
00054 * @date 2026-04-21
00055 * @copyright Copyright (c) 2026 Locked Inc.
00056 * SPDX-License-Identifier: MIT
00057 * @since Version 1.0.0
00058 */
00059
00060 #include "rx_tmr.h"
00061
00062 #include <stddef.h>
00063 #include <stdint.h>
00064
00065 #ifdef UNIT_TEST
00066 #include "mock_rx72n_regs.h"
00067 #else
00068 #include "rx72n_regs.h"
00069 #endif
00070 #include "rx72n_clock.h"
00071 #include "rx_check.h"
00072 #include "rx_log.h"
00073 #include "rx_register_protection.h"
00074
00075 /*
=====
00076 * Module-private Constants
00077 *
=====
00078 */
00079
00080 /**
00081 * @var s_tag
00082 * @brief Logging tag for the TMR module
00083 * @details 4 bytes in .rodata.
00084 * @since Version 1.0.0
00085 */
00086 static const char* const s_tag = "TMR";
00087
00088 /* Peripheral clock rate for internal-clock period calculations comes from
00089 * the authoritative rx72n_clock.h definition (k_pclkb_hz). Avoid duplicating
00090 * the literal so the value tracks any future clock-tree change. */
00091
00092 /**
00093 * @enum tmr_period_limits_t
00094 * @brief Compile-time period / divisor limits
00095 * @since Version 1.0.0
00096 */
00097 typedef enum : uint32_t {
00098     k_tmr_us_per_sec = 1000000U, /**< Microseconds per second */
00099     k_tmr_max_tcora = 0xFFU, /**< 8-bit TCORA maximum value */
00100     k_tmr_max_tcora_16bit = 0xFFFFU, /**< 16-bit cascade TCORA maximum value */
00101 } tmr_period_limits_t;
00102
00103 /**
00104 * @enum tmr_channel_count_t
00105 * @brief Array sizing constant for per-channel state
00106 * @since Version 1.0.0
00107 */
00108 typedef enum : uint8_t {
00109     k_tmr_channel_count_internal = 4, /**< TMR0..TMR3 */
00110 } tmr_channel_count_t;
00111
00112 /**
00113 * @enum tmr_pclk_divisor_t
00114 * @brief Integer prescaler ratios for internal clock sources
00115 * @details
00116 * The RX72N TMR peripheral divides PCLKB by one of these ratios before
00117 * feeding the counter. Each value mirrors a `k_tmr_clock_pclk_div_*`
00118 * enum value and is used by internal_tmr_clock_divider() to compute the
00119 * effective tick rate. Named constants avoid magic-number literals.
00120 * @since Version 1.0.0
00121 */
00122 typedef enum : uint32_t {
00123     k_tmr_pclk_div_1 = 1U, /**< Prescaler /1 */
00124     k_tmr_pclk_div_2 = 2U, /**< Prescaler /2 */

```

```

00125 k_tmr_pclk_div_8   = 8U,    /**< Prescaler /8 */
00126 k_tmr_pclk_div_32  = 32U,    /**< Prescaler /32 */
00127 k_tmr_pclk_div_64  = 64U,    /**< Prescaler /64 */
00128 k_tmr_pclk_div_1024 = 1024U, /**< Prescaler /1024 */
00129 k_tmr_pclk_div_8192 = 8192U, /**< Prescaler /8192 */
00130 k_tmr_pclk_div_none = 0U,    /**< Sentinel: external clock (no PCLK divider) */
00131 } tmr_pclk_divisor_t;
00132
00133 /**
00134 * @enum tmr_byte_shift_t
00135 * @brief Shift / mask constants for packing cascaded 16-bit counts
00136 * @since Version 1.0.0
00137 */
00138 typedef enum : uint8_t {
00139     k_tmr_byte_shift = 8U,    /**< Shift to move high byte to bits [15:8] */
00140     k_tmr_byte_mask  = 0xFFU, /**< Mask for isolating one byte */
00141 } tmr_byte_shift_t;
00142
00143 /*
=====
00144 * Module-private State
00145 *
=====
00146 */
00147
00148 /**
00149 * @var s_tmr_initialized
00150 * @brief Per-channel init flags indexed by rx_tmr_channel_t
00151 * @warning Do not modify directly - use rx_tmr_init() / rx_tmr_deinit()
00152 * @since Version 1.0.0
00153 */
00154 static bool s_tmr_initialized[k_tmr_channel_count_internal] = {false, false, false, false};
00155
00156 /**
00157 * @var s_tmr_mode
00158 * @brief Per-channel operating mode (independent vs cascade)
00159 * @since Version 1.0.0
00160 */
00161 static rx_tmr_mode_t s_tmr_mode[k_tmr_channel_count_internal] = {
00162     k_tmr_mode_8bit_independent,
00163     k_tmr_mode_8bit_independent,
00164     k_tmr_mode_8bit_independent,
00165     k_tmr_mode_8bit_independent,
00166 };
00167
00168 /**
00169 * @var s_tmr_clock_bits
00170 * @brief Cached TCCR clock bits (CSS|CKS) applied at rx_tmr_start()
00171 *
00172 * @details
00173 * rx_tmr_init() computes the TCCR clock-field value but does NOT write it
00174 * immediately because that would start the counter. rx_tmr_start() OR's
00175 * these bits into TCCR when requested. rx_tmr_stop() clears the clock
00176 * field so the counter halts.
00177 *
00178 * @since Version 1.0.0
00179 */
00180 static uint8_t s_tmr_clock_bits[k_tmr_channel_count_internal] = {0U, 0U, 0U, 0U};
00181
00182 /**
00183 * @var s_tmr_clock_source
00184 * @brief Cached clock-source enum per channel (needed by set_period_us)
00185 * @since Version 1.0.0
00186 */
00187 static rx_tmr_clock_source_t s_tmr_clock_source[k_tmr_channel_count_internal] = {
00188     k_tmr_clock_pclk_div_1,
00189     k_tmr_clock_pclk_div_1,
00190     k_tmr_clock_pclk_div_1,
00191     k_tmr_clock_pclk_div_1,
00192 };
00193
00194 /**
00195 * @var s_tmr_callback
00196 * @brief Per-channel compare-match / overflow callback pointers
00197 * @since Version 1.0.0
00198 */
00199 static rx_tmr_isr_callback_t s_tmr_callback[k_tmr_channel_count_internal] = {
00200     NULL,
00201     NULL,
00202     NULL,
00203     NULL,
00204 };
00205
00206 /*
=====
00207 * Internal Helpers
00208 *
=====

```

```

=====
00209 */
00210
00211 /**
00212  * @brief Get register pointer for a TMR channel
00213  * @since Version 1.0.0
00214  */
00215 RX_STATIC_TESTABLE volatile rx_tmr_channel_regs_t* internal_tmr_get_regs(rx_tmr_channel_t channel)
00216 {
00217     /* GCOVR_EXCL_BR_START(validate_config restricts channel to 0..3) */
00218     switch (channel) {
00219         case k_tmr_channel_0:
00220             return tmr0();
00221         case k_tmr_channel_1:
00222             return tmr1();
00223         case k_tmr_channel_2:
00224             return tmr2();
00225         case k_tmr_channel_3:
00226             return tmr3();
00227         default:
00228             RX_ASSERT(false, "TMR channel out of range"); /* GCOVR_EXCL_LINE */
00229             return NULL; /* GCOVR_EXCL_LINE */
00230     }
00231     /* GCOVR_EXCL_BR_STOP */
00232 }
00233
00234 /**
00235  * @brief Return the MSTPCRA bit mask for a TMR channel's unit
00236  * @since Version 1.0.0
00237  */
00238 static uint32_t internal_tmr_unit_mstp_bit(rx_tmr_channel_t channel)
00239 {
00240     /* Public callers pre-validate channel in 0..3 via internal_tmr_validate_config, so
00241      * anything not in unit 0 (TMR0/1) must belong to unit 1 (TMR2/3). */
00242     if ((channel == k_tmr_channel_0) || (channel == k_tmr_channel_1)) {
00243         return k_tmr_mstpcra_mstpa5;
00244     }
00245     return k_tmr_mstpcra_mstpa4;
00246 }
00247
00248 /**
00249  * @brief Enable the module clock for the channel's TMR unit
00250  * @pre PRCR register accessible
00251  * @post MSTPCRA bit cleared for the unit
00252  * @since Version 1.0.0
00253  */
00254 static void internal_tmr_enable_module_clock(rx_tmr_channel_t channel)
00255 {
00256     const uint32_t mstp_bit = internal_tmr_unit_mstp_bit(channel);
00257
00258     *prcr_reg() = k_rx_prcr_unlock_prc1;
00259     system_regs()->mstpcra &= ~mstp_bit;
00260     *prcr_reg() = k_rx_prcr_lock;
00261 }
00262
00263 /**
00264  * @brief Map a clock source to TCCR CSS|CKS bits
00265  * @since Version 1.0.0
00266  */
00267 static rx_err_t internal_tmr_clock_bits(rx_tmr_clock_source_t source, uint8_t* bits)
00268 {
00269     /* GCOVR_EXCL_BR_START(validate_config blocks out-of-range clock_source) */
00270     switch (source) {
00271         case k_tmr_clock_pclk_div_1:
00272             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div1);
00273             return k_rx_ok;
00274         case k_tmr_clock_pclk_div_2:
00275             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div2);
00276             return k_rx_ok;
00277         case k_tmr_clock_pclk_div_8:
00278             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div8);
00279             return k_rx_ok;
00280         case k_tmr_clock_pclk_div_32:
00281             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div32);
00282             return k_rx_ok;
00283         case k_tmr_clock_pclk_div_64:
00284             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div64);
00285             return k_rx_ok;
00286         case k_tmr_clock_pclk_div_1024:
00287             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div1024);
00288             return k_rx_ok;
00289         case k_tmr_clock_pclk_div_8192:
00290             *bits = (uint8_t)(k_tmr_tccr_css_internal | k_tmr_tccr_cks_pclk_div8192);
00291             return k_rx_ok;
00292         case k_tmr_clock_external_rising:
00293             *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_rising);
00294             return k_rx_ok;

```

```

00295     case k_tmr_clock_external_falling:
00296         *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_falling);
00297         return k_rx_ok;
00298     case k_tmr_clock_external_both:
00299         *bits = (uint8_t)(k_tmr_tccr_css_external | k_tmr_tccr_cks_ext_both);
00300         return k_rx_ok;
00301     default: /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range clock_source */
00302         return k_rx_err_invalid_arg; /* GCOVR_EXCL_LINE */
00303 }
00304 /* GCOVR_EXCL_BR_STOP */
00305 }
00306
00307 /**
00308  * @brief Return the integer divider for an internal clock source
00309  * @since Version 1.0.0
00310  */
00311 static uint32_t internal_tmr_clock_divider(rx_tmr_clock_source_t source)
00312 {
00313     switch (source) {
00314         case k_tmr_clock_pclk_div_1:
00315             return k_tmr_pclk_div_1;
00316         case k_tmr_clock_pclk_div_2:
00317             return k_tmr_pclk_div_2;
00318         case k_tmr_clock_pclk_div_8:
00319             return k_tmr_pclk_div_8;
00320         case k_tmr_clock_pclk_div_32:
00321             return k_tmr_pclk_div_32;
00322         case k_tmr_clock_pclk_div_64:
00323             return k_tmr_pclk_div_64;
00324         case k_tmr_clock_pclk_div_1024:
00325             return k_tmr_pclk_div_1024;
00326         case k_tmr_clock_pclk_div_8192:
00327             return k_tmr_pclk_div_8192;
00328         default:
00329             return k_tmr_pclk_div_none; /* External or invalid */
00330     }
00331 }
00332
00333 /**
00334  * @brief Compose TCR value from config
00335  * @since Version 1.0.0
00336  */
00337 static uint8_t internal_tmr_compose_tcr(const rx_tmr_config_t* config)
00338 {
00339     uint8_t tcr = 0U;
00340
00341     if ((config->irq_mask & k_tmr_irq_cmp_match_a) != 0U) {
00342         tcr |= k_tmr_tcr_cmiea;
00343     }
00344     if ((config->irq_mask & k_tmr_irq_cmp_match_b) != 0U) {
00345         tcr |= k_tmr_tcr_cmieb;
00346     }
00347     if ((config->irq_mask & k_tmr_irq_overflow) != 0U) {
00348         tcr |= k_tmr_tcr_ovie;
00349     }
00350
00351     /* GCOVR_EXCL_BR_START(validate_config blocks out-of-range counter_clear) */
00352     switch (config->counter_clear) {
00353         case k_tmr_clear_disabled:
00354             tcr |= k_tmr_tcr_cclr_disabled;
00355             break;
00356         case k_tmr_clear_cmp_match_a:
00357             tcr |= k_tmr_tcr_cclr_cmp_match_a;
00358             break;
00359         case k_tmr_clear_cmp_match_b:
00360             tcr |= k_tmr_tcr_cclr_cmp_match_b;
00361             break;
00362         case k_tmr_clear_external_reset:
00363             tcr |= k_tmr_tcr_cclr_external_sig;
00364             break;
00365         default: /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range counter_clear */
00366             break; /* GCOVR_EXCL_LINE */
00367     }
00368     /* GCOVR_EXCL_BR_STOP */
00369     return tcr;
00370 }
00371
00372 /**
00373  * @brief Compose TCSR value from config (OSA/OSB fields)
00374  * @since Version 1.0.0
00375  */
00376 static uint8_t internal_tmr_compose_tcsr(const rx_tmr_config_t* config)
00377 {
00378     const uint8_t osa = (uint8_t)((((uint8_t)config->output_a) << k_tmr_tcsr_osa_shift);
00379     const uint8_t osb = (uint8_t)((((uint8_t)config->output_b) << k_tmr_tcsr_osb_shift);
00380     return (uint8_t)(osa | osb);
00381 }

```

```

00382
00383 /**
00384  * @brief Validate enum fields of a configuration
00385  * @since Version 1.0.0
00386  */
00387 static rx_err_t internal_tmr_validate_config(const rx_tmr_config_t* config)
00388 {
00389     if ((uint8_t)config->channel >= k_tmr_channel_count_internal) {
00390         return k_rx_err_invalid_arg;
00391     }
00392     if ((config->mode != k_tmr_mode_8bit_independent) && (config->mode != k_tmr_mode_16bit_cascade)) {
00393         return k_rx_err_invalid_arg;
00394     }
00395     if ((uint8_t)config->counter_clear > k_tmr_clear_external_reset) {
00396         return k_rx_err_invalid_arg;
00397     }
00398     if ((uint8_t)config->output_a > k_tmr_output_toggle) {
00399         return k_rx_err_invalid_arg;
00400     }
00401     if ((uint8_t)config->output_b > k_tmr_output_toggle) {
00402         return k_rx_err_invalid_arg;
00403     }
00404     if ((uint8_t)config->clock_source > k_tmr_clock_external_both) {
00405         return k_rx_err_invalid_arg;
00406     }
00407     /* Cascade mode requires an even channel */
00408     if (config->mode == k_tmr_mode_16bit_cascade) {
00409         if ((config->channel != k_tmr_channel_0) && (config->channel != k_tmr_channel_2)) {
00410             return k_rx_err_invalid_arg;
00411         }
00412     }
00413     return k_rx_ok;
00414 }
00415
00416 /**
00417  * @brief Write register fields for a single channel during init
00418  * @since Version 1.0.0
00419  */
00420 static void internal_tmr_program_channel(rx_tmr_channel_t channel, const rx_tmr_config_t* config)
00421 {
00422     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00423
00424     regs->tccr = 0U; /* Stop counter before programming */
00425     regs->tcr = internal_tmr_compose_tcr(config);
00426     regs->tcsr = internal_tmr_compose_tcsr(config);
00427     regs->tcora = k_tmr_max_tcora;
00428     regs->tcorb = k_tmr_max_tcora;
00429     regs->tcnt = 0U;
00430 }
00431
00432 /*
=====
00433  * Public API
00434  *
=====
00435  */
00436
00437 /**
00438  * @brief Program the paired odd channel in a 16-bit cascade configuration
00439  * @details
00440  * Extracted from rx_tmr_init() to keep the caller under the function-size
00441  * threshold. The even channel counts on its selected clock source; the
00442  * odd channel counts on the even channel's TCORA compare match via
00443  * CSS = cascade.
00444  * @since Version 1.0.0
00445  */
00446 static void internal_tmr_init_cascade_pair(const rx_tmr_config_t* config)
00447 {
00448     const rx_tmr_channel_t odd =
00449         (config->channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00450
00451     rx_tmr_config_t odd_cfg = *config;
00452     odd_cfg.channel = odd;
00453     internal_tmr_program_channel(odd, &odd_cfg);
00454
00455     volatile rx_tmr_channel_regs_t* odd_regs = internal_tmr_get_regs(odd);
00456     odd_regs->tccr = k_tmr_tccr_css_cascade;
00457
00458     s_tmr_mode[(uint8_t)odd] = k_tmr_mode_16bit_cascade;
00459     s_tmr_initialized[(uint8_t)odd] = true;
00460     s_tmr_clock_bits[(uint8_t)odd] = k_tmr_tccr_css_cascade;
00461     s_tmr_clock_source[(uint8_t)odd] = config->clock_source;
00462 }
00463
00464 rx_err_t rx_tmr_init(const rx_tmr_config_t* config)
00465 {
00466     /* Pre-condition 1: non-null config */

```

```

00467 RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is NULL");
00468
00469 /* Pre-condition 2: enum ranges */
00470 rx_err_t err = internal_tmr_validate_config(config);
00471 if (err != k_rx_ok) {
00472     rx_log_error(s_tag, "Invalid TMR config");
00473     return err;
00474 }
00475
00476 const uint8_t idx = (uint8_t)config->channel;
00477
00478 /* Pre-condition 3: not already initialised */
00479 if (s_tmr_initialized[idx]) {
00480     rx_log_error(s_tag, "Channel already initialised");
00481     return k_rx_err_invalid_state;
00482 }
00483
00484 /* Compose clock bits (validated via enum range above) */
00485 uint8_t clock_bits = 0U;
00486 err = internal_tmr_clock_bits(config->clock_source, &clock_bits);
00487 if (err != k_rx_ok) { /* GCOVR_EXCL_LINE -- validate_config blocks out-of-range clock_source */
00488     return err; /* GCOVR_EXCL_LINE */
00489 } /* GCOVR_EXCL_BR_LINE */
00490
00491 /* 1. Enable unit module clock */
00492 internal_tmr_enable_module_clock(config->channel);
00493
00494 /* 2. Program the primary channel */
00495 internal_tmr_program_channel(config->channel, config);
00496
00497 /* 3. For cascade mode, program the paired odd channel */
00498 if (config->mode == k_tmr_mode_16bit_cascade) {
00499     internal_tmr_init_cascade_pair(config);
00500 }
00501
00502 /* 4. Cache state for the primary channel */
00503 s_tmr_mode[idx] = config->mode;
00504 s_tmr_clock_bits[idx] = clock_bits;
00505 s_tmr_clock_source[idx] = config->clock_source;
00506 s_tmr_initialized[idx] = true;
00507
00508 rx_log_info_val(s_tag, "TMR channel initialised", (uint32_t)config->channel);
00509 return k_rx_ok;
00510 }
00511
00512 rx_err_t rx_tmr_start(rx_tmr_channel_t channel)
00513 {
00514     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00515         return k_rx_err_invalid_arg;
00516     }
00517     const uint8_t idx = (uint8_t)channel;
00518     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00519
00520     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00521     regs->tccr = s_tmr_clock_bits[idx];
00522
00523     /* In cascade mode the paired odd channel also needs its cascade bits
00524      * kept after a stop/start cycle. Cascade is only valid on even channels
00525      * (enforced by validate_config), so the ternary below is exhaustive. */
00526     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00527         volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00528         odd_regs->tccr = k_tmr_tccr_css_cascade;
00529     }
00530     return k_rx_ok;
00531 }
00532
00533 rx_err_t rx_tmr_stop(rx_tmr_channel_t channel)
00534 {
00535     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00536         return k_rx_err_invalid_arg;
00537     }
00538     const uint8_t idx = (uint8_t)channel;
00539     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00540
00541     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00542     regs->tccr = 0U;
00543
00544     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00545         volatile rx_tmr_channel_regs_t* odd_regs = (channel == k_tmr_channel_0) ? tmr1() : tmr3();
00546         odd_regs->tccr = 0U;
00547     }
00548     return k_rx_ok;
00549 }
00550
00551 rx_err_t rx_tmr_read(rx_tmr_channel_t channel, uint16_t* value)
00552 {
00553     RX_CHECK_NULL_PTR(value, s_tag, "Value pointer is NULL");

```



```

00554 if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00555     return k_rx_err_invalid_arg;
00556 }
00557 const uint8_t idx = (uint8_t)channel;
00558 RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00559
00560 if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00561     const uint8_t upper = internal_tmr_get_regs(channel)->tcnt;
00562     const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00563     const uint8_t lower = internal_tmr_get_regs(odd)->tcnt;
00564     *value = (uint16_t)((uint16_t)upper « (uint16_t)k_tmr_byte_shift) | (uint16_t)lower;
00565 } else {
00566     *value = (uint16_t)internal_tmr_get_regs(channel)->tcnt;
00567 }
00568 return k_rx_ok;
00569 }
00570
00571 rx_err_t rx_tmr_set_period_us(rx_tmr_channel_t channel, uint32_t period_us)
00572 {
00573     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00574         return k_rx_err_invalid_arg;
00575     }
00576     const uint8_t idx = (uint8_t)channel;
00577     RX_VALIDATE_INIT(s_tmr_initialized[idx], s_tag, "Channel not initialised");
00578
00579     if (period_us == 0U) {
00580         return k_rx_err_invalid_arg;
00581     }
00582
00583     const uint32_t divider = internal_tmr_clock_divider(s_tmr_clock_source[idx]);
00584     if (divider == 0U) {
00585         /* External clock selected - cannot compute period */
00586         return k_rx_err_invalid_arg;
00587     }
00588
00589     /* tick_hz = PCLKB / divider; ticks = period_us * tick_hz / 1e6
00590     * compare = ticks - 1 (TCORA fires when TCNT reaches compare value) */
00591     const uint64_t ticks =
00592         ((uint64_t)period_us * (uint64_t)k_pclkb_hz) / ((uint64_t)divider * (uint64_t)k_tmr_us_per_sec);
00593     if (ticks == 0U) {
00594         return k_rx_err_invalid_arg;
00595     }
00596     const uint64_t compare = ticks - 1U;
00597
00598     const uint32_t max_compare =
00599         (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) ? k_tmr_max_tcora_16bit : k_tmr_max_tcora;
00600     if (compare > (uint64_t)max_compare) {
00601         rx_log_error_val(s_tag, "Period too large for divider", period_us);
00602         return k_rx_err_invalid_arg;
00603     }
00604
00605     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {
00606         const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00607         internal_tmr_get_regs(channel)->tcora =
00608             (uint8_t)((compare » (uint64_t)k_tmr_byte_shift) & (uint64_t)k_tmr_byte_mask);
00609         internal_tmr_get_regs(odd)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00610     } else {
00611         internal_tmr_get_regs(channel)->tcora = (uint8_t)(compare & (uint64_t)k_tmr_byte_mask);
00612     }
00613     return k_rx_ok;
00614 }
00615
00616 rx_err_t rx_tmr_register_compare_match_isr(rx_tmr_channel_t channel, rx_tmr_isr_callback_t callback)
00617 {
00618     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00619         return k_rx_err_invalid_arg;
00620     }
00621     s_tmr_callback[(uint8_t)channel] = callback;
00622     return k_rx_ok;
00623 }
00624
00625 rx_err_t rx_tmr_deinit(rx_tmr_channel_t channel)
00626 {
00627     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00628         return k_rx_err_invalid_arg;
00629     }
00630
00631     volatile rx_tmr_channel_regs_t* regs = internal_tmr_get_regs(channel);
00632     regs->tccr = 0U;
00633     regs->tcr = 0U;
00634     regs->tcsr = 0U;
00635     regs->tcora = 0U;
00636     regs->tcorb = 0U;
00637     regs->tcnt = 0U;
00638
00639     const uint8_t idx = (uint8_t)channel;
00640     if (s_tmr_mode[idx] == k_tmr_mode_16bit_cascade) {

```

```

00641     const rx_tmr_channel_t odd = (channel == k_tmr_channel_0) ? k_tmr_channel_1 : k_tmr_channel_3;
00642     volatile rx_tmr_channel_regs_t* odd_regs = internal_tmr_get_regs(odd);
00643     odd_regs->tccr      = 0U;
00644     odd_regs->tcr       = 0U;
00645     odd_regs->tcsr      = 0U;
00646     odd_regs->tcora     = 0U;
00647     odd_regs->tcorb     = 0U;
00648     odd_regs->tcnt      = 0U;
00649     s_tmr_initialized[(uint8_t)odd] = false;
00650     s_tmr_mode[(uint8_t)odd]       = k_tmr_mode_8bit_independent;
00651     s_tmr_clock_bits[(uint8_t)odd] = 0U;
00652     s_tmr_callback[(uint8_t)odd]   = NULL;
00653 }
00654
00655 s_tmr_initialized[idx] = false;
00656 s_tmr_mode[idx]       = k_tmr_mode_8bit_independent;
00657 s_tmr_clock_bits[idx] = 0U;
00658 s_tmr_callback[idx]   = NULL;
00659 return k_rx_ok;
00660 }
00661
00662 /*
=====
00663 * ISR Dispatch Hooks (SELECTB)
00664 *
=====
00665 */
00666
00667 /**
00668 * @brief Internal dispatcher invoked by the SELECTB vector shim
00669 *
00670 * @details
00671 * Called from the low-level vector when a TMR CMIA/CMIB/OVI interrupt
00672 * fires. Clears the source flag in the TMR (compare-match sets no
00673 * persistent status bit, so the hardware does it automatically; this
00674 * function exists so the test harness can synthesise ISR invocations)
00675 * and invokes the registered user callback.
00676 *
00677 * @since Version 1.0.0
00678 */
00679 void rx_tmr_isr_dispatch(rx_tmr_channel_t channel)
00680 {
00681     if ((uint8_t)channel >= k_tmr_channel_count_internal) {
00682         return;
00683     }
00684     rx_tmr_isr_callback_t cb = s_tmr_callback[(uint8_t)channel];
00685     if (cb != NULL) {
00686         cb(channel);
00687     }
00688 }

```

8.135 libs/rx/hal/src/rx_tpu.c File Reference

TPU HAL Driver Implementation for Phase Counting Mode.

```

#include "rx_tpu.h"
#include <stddef.h>
#include "rx72n_regs.h"
#include "rx_check.h"
#include "rx_log.h"
#include "rx_register_protection.h"
Include dependency graph for rx_tpu.c:

```



Enumerations

- enum [tpu_phase_channel_count_t](#) : uint8_t { [k_tpu_phase_channel_count](#) = 4 }
- Number of phase-counting-capable TPU channels.

- enum `tpu_channel_index_t` : `uint8_t` { `k_tpu_idx_ch1` = 0 , `k_tpu_idx_ch2` = 1 , `k_tpu_idx_ch4` = 2 , `k_tpu_idx_ch5` = 3 }
Array index mapping for phase-counting channels.
- enum `tpu_tmdr_reset_t` : `uint8_t` { `k_tpu_tmdr_reset` = 0x00 }
TMDR register reset value (normal mode).
- enum `tpu_tcr_phase_count_t` : `uint8_t` { `k_tpu_tcr_phase_count` = 0x00 }
TCR register value for phase counting mode.
- enum `tpu_tier_disable_t` : `uint8_t` { `k_tpu_tier_all_disabled` = 0x40 }
TIER register value with all interrupts disabled.
- enum `tpu_tcmt_zero_t` : `uint16_t` { `k_tpu_tcmt_zero` = 0x0000 }
TCNT register zero value.

Functions

- static `rx_err_t` `internal_channel_to_index` (const `rx_tpu_channel_t` channel, `uint8_t` *index)
Map TPU channel number to array index.
- `RX_STATIC_TESTABLE` volatile `rx_tpu_regs_t` * `internal_get_regs` (const `rx_tpu_channel_t` channel)
Get register pointer for a phase-counting TPU channel.
- `RX_STATIC_TESTABLE` `uint8_t` `internal_get_cst_bit` (const `rx_tpu_channel_t` channel)
Get TSTR CST bit mask for a TPU channel.
- static `uint8_t` `internal_get_tmdr_mode` (const `rx_tpu_phase_mode_t` mode)
Map `rx_tpu_phase_mode_t` to TMDR.MD register value.
- static void `internal_enable_tpu_module_clock` (void)
Enable TPU module clock by clearing MSTPCRA.MSTPA13.
- `rx_err_t` `rx_tpu_init_phase_count` (const `rx_tpu_config_t` *config)
Initialize TPU channel for phase counting mode.
- `rx_err_t` `rx_tpu_start` (const `rx_tpu_channel_t` channel)
Start the TPU channel counter (begin phase counting).
- `rx_err_t` `rx_tpu_stop` (const `rx_tpu_channel_t` channel)
Stop the TPU channel counter (halt phase counting).
- `rx_err_t` `rx_tpu_read_count` (const `rx_tpu_channel_t` channel, `uint16_t` *count)
Read the current 16-bit encoder counter value.
- `rx_err_t` `rx_tpu_read_direction` (const `rx_tpu_channel_t` channel, bool *counting_up)
Read the current encoder counting direction.
- `rx_err_t` `rx_tpu_reset_count` (const `rx_tpu_channel_t` channel)
Reset the TPU channel encoder counter to zero.
- `rx_err_t` `rx_tpu_deinit` (const `rx_tpu_channel_t` channel)
Deinitialize a TPU phase-counting channel.

Variables

- static const char *const `s_tag` = "TPU"
Logging tag for TPU module.
- static bool `s_tpu_initialized` [`k_tpu_phase_channel_count`] = {false, false, false, false}
Track which phase-counting channels are initialized.

8.135.1 Detailed Description

TPU HAL Driver Implementation for Phase Counting Mode.

Implementation of the TPU hardware abstraction layer for RX72N quadrature encoder phase counting. This driver configures TPU channels 1, 2, 4, and 5 for hardware-based encoder position tracking with 4x resolution.

8.135.1.1 Module Architecture

The implementation is structured in three layers:

1. Constants & State: Typed enums for all magic numbers, static tracking
2. Internal Helpers: Channel validation, register access, bit mapping
3. Public API: User-facing functions documented in [rx_tpu.h](#)

8.135.1.2 Initialization Sequence

```
* rx_tpu_init_phase_count(&config)
* |
* +-- 1. Validate config (NULL, channel, phase_mode)
* +-- 2. Check not already initialized
* +-- 3. Enable TPU module clock (MSTPCRA.MSTPA13 = 0)
* +-- 4. Stop channel counter (TSTR.CSTn = 0)
* +-- 5. Set phase counting mode (TMDR.MD = mode)
* +-- 6. Set TCR to free-running (CCLR = disabled)
* +-- 7. Clear TCNT to zero
* +-- 8. Disable interrupts (TIER = 0x40; bit 6 reserved/must-write-1)
* +-- 9. Mark channel as initialized
*
```

8.135.1.3 Static Variables

Variable	Type	Purpose
s_tpu_initialized[]	bool[4]	Track which channels are initialized
s_tag	const char*	Log tag "TPU"

8.135.1.4 Channel-to-Index Mapping

Phase counting channels (1, 2, 4, 5) are mapped to array indices 0-3:

Channel	Index	Use in STAR	Clock Pins
TPU1	0	Back-right encoder	TCLKA + TCLKB
TPU2	1	Back-left encoder	TCLKC + TCLKD
TPU4	2	Available	TCLKC + TCLKD

Channel	Index	Use in STAR	Clock Pins
TPU5	3	Available	TCLKA + TCLKB

Performance Characteristics

Function	Time	Stack	Notes
rx_tpu_init_phase_count()	~20 us	32 B	One-time setup per channel
rx_tpu_start()	~0.5 us	8 B	Single register write
rx_tpu_stop()	~0.5 us	8 B	Single register write
rx_tpu_read_count()	~0.5 us	8 B	Single register read
rx_tpu_read_direction()	~0.5 us	8 B	Single register read
rx_tpu_reset_count()	~0.5 us	8 B	Single register write
rx_tpu_deinit()	~1 us	16 B	Stop + reset + cleanup

Memory Usage

Category	Size	Description
Code (.text)	~600 bytes	All functions
Constants	4 bytes	Tag string "TPU"
Static data	4 bytes	Initialized flags array
Stack (max)	32 bytes	Deepest call chain

NASA Power of 10 Compliance

- Rule 1: No goto, setjmp, or recursion
- Rule 2: All loops bounded (no loops in this module)
- Rule 3: No dynamic memory allocation
- Rule 4: All functions < 60 lines
- Rule 5: Minimum 2 validation checks per function
- Rule 6: Variables at smallest scope
- Rule 7: All return values checked
- Rule 8: C23 typed enums for all constants
- Rule 9: Single-level pointers only
- Rule 10: Compiles with -Wall -Wextra -Werror

SOLID Principles

- S: Only TPU hardware access (no encoder logic, no motor control)
- O: Extensible via [rx_tpu_config_t](#) without modifying existing code
- L: Consistent [rx_err_t](#) return semantics across all functions
- I: Focused API: init, start, stop, read, reset, deinit
- D: Depends on register abstractions ([rx72n_tpu_regs.h](#)), not raw addresses

See also

[rx_tpu.h](#) Public API documentation
[rx72n_tpu_regs.h](#) TPU register definitions
[rx_mtu_encoder.h](#) MTU encoder interface (front wheels)

Author

Locked, Inc.

Date

2026-02-10

Copyright

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Since

Version 1.0.0

Definition in file [rx_tpu.c](#).

8.135.2 Enumeration Type Documentation

8.135.2.1 tpu_channel_index_t

enum [tpu_channel_index_t](#) : uint8_t

Array index mapping for phase-counting channels.

Maps the non-contiguous channel numbers (1, 2, 4, 5) to contiguous array indices (0-3) for the `s_tpu_initialized[]` tracking array.

Since

Version 1.0.0

Enumerator

<code>k_tpu_idx_ch1</code>	TPU1 -> index 0 (back-right encoder)
<code>k_tpu_idx_ch2</code>	TPU2 -> index 1 (back-left encoder)
<code>k_tpu_idx_ch4</code>	TPU4 -> index 2 (available)
<code>k_tpu_idx_ch5</code>	TPU5 -> index 3 (available)

Definition at line 157 of file [rx_tpu.c](#).

```
00157     : uint8_t {
00158 k_tpu_idx_ch1 = 0, /**< TPU1 -> index 0 (back-right encoder) */
00159 k_tpu_idx_ch2 = 1, /**< TPU2 -> index 1 (back-left encoder) */
00160 k_tpu_idx_ch4 = 2, /**< TPU4 -> index 2 (available) */
00161 k_tpu_idx_ch5 = 3, /**< TPU5 -> index 3 (available) */
00162 } tpu_channel_index_t;
```

8.135.2.2 tpu`phase`channel`count`t

enum [tpu_phase_channel_count_t](#) : uint8_t

Number of phase-counting-capable TPU channels.

TPU1, TPU2, TPU4, and TPU5 support phase counting mode. TPU0 and TPU3 do NOT support phase counting.

Since

Version 1.0.0

Enumerator

k_tpu_phase_channel_count	Channels 1, 2, 4, 5
---------------------------	---------------------

Definition at line [143](#) of file [rx_tpu.c](#).

```
00143      : uint8_t {
00144  k_tpu_phase_channel_count = 4, /**< Channels 1, 2, 4, 5 */
00145 } tpu_phase_channel_count_t;
```

8.135.2.3 tpu`tcnt`zero`t

enum [tpu_tcnt_zero_t](#) : uint16_t

TCNT register zero value.

Since

Version 1.0.0

Enumerator

k_tpu_tcnt_zero	Counter cleared to zero
-----------------	-------------------------

Definition at line [209](#) of file [rx_tpu.c](#).

```
00209      : uint16_t {
00210  k_tpu_tcnt_zero = 0x0000, /**< Counter cleared to zero */
00211 } tpu_tcnt_zero_t;
```

8.135.2.4 tpu`tcr`phase`count`t

enum [tpu_tcr_phase_count_t](#) : uint8_t

TCR register value for phase counting mode.

In phase counting mode, the TPSC and CKEG fields in TCR are ignored because the external clock pins provide the counting signal. CCLR is set to disabled (free-running counter).

Since

Version 1.0.0

Enumerator

k_tpu_tcr_phase_count	Free-running: TPSC=0, CKEG=0, CCLR=0
-----------------------	--------------------------------------

Definition at line 184 of file [rx_tpu.c](#).

```
00184      : uint8_t {
00185 k_tpu_tcr_phase_count = 0x00, /**< Free-running: TPSC=0, CKEG=0, CCLR=0 */
00186 } tpu_tcr_phase_count_t;
```

8.135.2.5 tpu`tier`disable`

```
enum tpu_tier_disable_t : uint8_t
```

TIER register value with all interrupts disabled.

Bit 6 of TIER is reserved and must always be written as 1 (manual p.1461: "This bit is read as 1. The write value should be 1."). The reset value of TIER is 0x40 (bit 6 set). Writing 0x00 violates the specification; 0x40 disables all interrupt sources while preserving the reserved bit.

Since

Version 1.0.0

Enumerator

k_tpu_tier_all_disabled	All interrupts disabled; bit 6 reserved, must be 1
-------------------------	--

Definition at line 200 of file [rx_tpu.c](#).

```
00200      : uint8_t {
00201 k_tpu_tier_all_disabled = 0x40, /**< All interrupts disabled; bit 6 reserved, must be 1 */
00202 } tpu_tier_disable_t;
```

8.135.2.6 tpu`tmdr`reset`

```
enum tpu_tmdr_reset_t : uint8_t
```

TMDR register reset value (normal mode).

Since

Version 1.0.0

Enumerator

k_tpu_tmdr_reset	Normal operation mode (reset default)
------------------	---------------------------------------

Definition at line 169 of file [rx_tpu.c](#).

```
00169      : uint8_t {
00170 k_tpu_tmdr_reset = 0x00, /**< Normal operation mode (reset default) */
00171 } tpu_tmdr_reset_t;
```


8.135.3 Function Documentation

8.135.3.1 `internal_channel_to_index()`

```
rx_err_t internal_channel_to_index (
    const rx_tpu_channel_t channel,
    uint8_t * index)  [static]
```

Map TPU channel number to array index.

Converts the non-contiguous channel enumeration (1, 2, 4, 5) to a contiguous array index (0-3) for use with `s_tpu_initialized[]`.

Precondition

- index must be non-NULL (caller responsibility)
- channel must be a valid `rx_tpu_channel_t` enumeration value

Postcondition

- *index contains valid array index on success
- *index is unchanged on `k_rx_err_invalid_arg`

Note

Thread Safety: Safe - pure function with no shared state

Since

Version 1.0.0

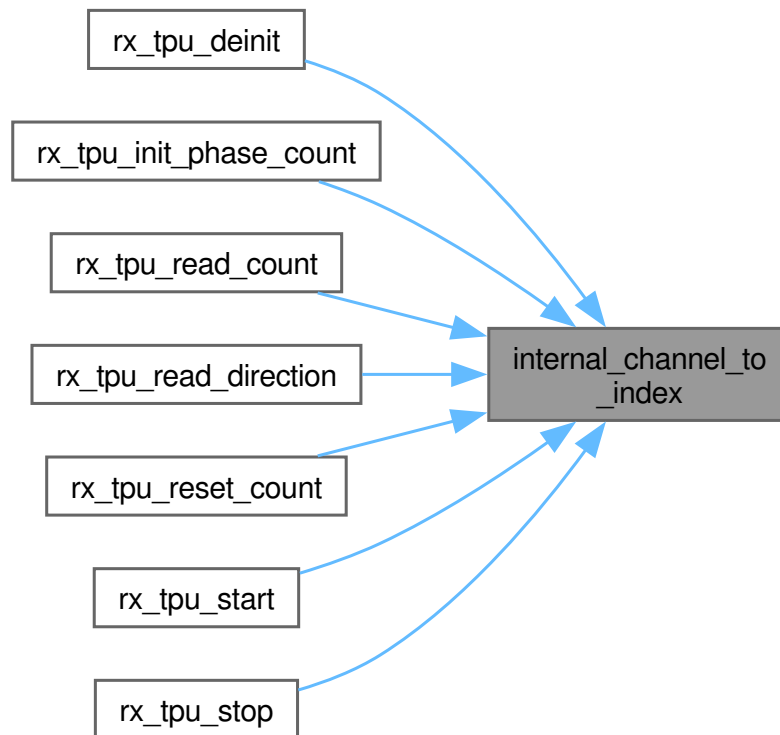
Definition at line 254 of file `rx_tpu.c`.

```
00255 {
00256     switch (channel) {
00257         case k_tpu_channel_1:
00258             *index = k_tpu_idx_ch1;
00259             return k_rx_ok;
00260         case k_tpu_channel_2:
00261             *index = k_tpu_idx_ch2;
00262             return k_rx_ok;
00263         case k_tpu_channel_4:
00264             *index = k_tpu_idx_ch4;
00265             return k_rx_ok;
00266         case k_tpu_channel_5:
00267             *index = k_tpu_idx_ch5;
00268             return k_rx_ok;
00269         default:
00270             rx_log_error_val(s_tag, "Invalid TPU channel", (uint32_t)channel);
00271             return k_rx_err_invalid_arg;
00272     }
00273 }
```

References `k_tpu_channel_1`, `k_tpu_channel_2`, `k_tpu_channel_4`, `k_tpu_channel_5`, `k_tpu_idx_ch1`, `k_tpu_idx_ch2`, `k_tpu_idx_ch4`, `k_tpu_idx_ch5`, and `s_tag`.

Referenced by `rx_tpu_deinit()`, `rx_tpu_init_phase_count()`, `rx_tpu_read_count()`, `rx_tpu_read_direction()`, `rx_tpu_reset_count()`, `rx_tpu_start()`, and `rx_tpu_stop()`.

Here is the caller graph for this function:



8.135.3.2 internal_enable_tpu_module_clock()

```
void internal_enable_tpu_module_clock (
    void ) [static]
```

Enable TPU module clock by clearing MSTPCRA.MSTPA13.

Enables the TPU peripheral by clearing its module stop bit in MSTPCRA. Uses the PRCR register protection unlock/lock sequence. Idempotent - safe to call multiple times.

Precondition

System clock configured

PRCR register accessible (system not in protected mode that blocks PRC1)

Postcondition

TPU module clock enabled (MSTPCRA.MSTPA13 = 0)

PRCR protection re-enabled

Note

Thread Safety: Not thread-safe (modifies system-wide registers)

Since

Version 1.0.0

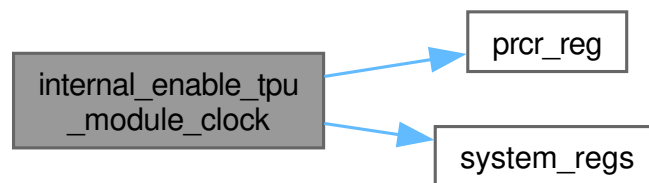
Definition at line 403 of file [rx_tpu.c](#).

```
00404 {
00405     *prcr_reg() = k_rx_prcr_unlock_prc1;
00406     system_regs()->mstpcra &= ~k_tpu_mstpcra_mstp13;
00407     *prcr_reg() = k_rx_prcr_lock;
00408 }
```

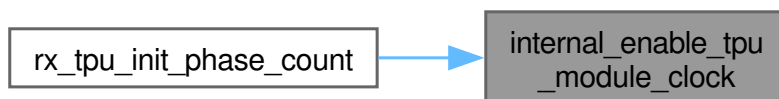
References [k_tpu_mstpcra_mstp13](#), [rx_system_regs_t::mstpcra](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [rx_tpu_init_phase_count\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.135.3.3 internal`get`cst`bit()

```
RX_STATIC_TESTABLE uint8_t internal_get_cst_bit (
    const rx\_tpu\_channel\_t channel)
```

Get TSTR CST bit mask for a TPU channel.

Returns the counter start bit mask in TSTR for the specified channel. This bit is set to start counting and cleared to stop counting. Each phase-counting-capable channel has a dedicated bit position in the shared TSTR register: CST1, CST2, CST4, and CST5.

Precondition

channel must be one of `k_tpu_channel_1`, `_2`, `_4`, or `_5`

Caller validates that a non-zero return indicates a valid channel

Postcondition

No hardware registers modified

Returned value is safe to OR/AND into TSTR register

Note

Thread Safety: Safe - pure function

Since

Version 1.0.0

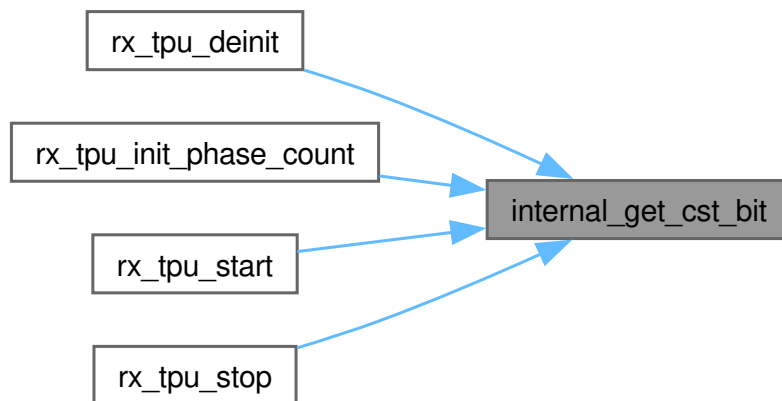
Definition at line 332 of file [rx_tpu.c](#).

```
00333 {
00334     switch (channel) {
00335         case k_tpu_channel_1:
00336             return k_tpu_tstr_cst1;
00337         case k_tpu_channel_2:
00338             return k_tpu_tstr_cst2;
00339         case k_tpu_channel_4:
00340             return k_tpu_tstr_cst4;
00341         case k_tpu_channel_5:
00342             return k_tpu_tstr_cst5;
00343         default:
00344             RX_ASSERT(false, "Post-condition: channel must be valid (1,2,4,5) - caller validates");
00345             return 0;
00346     }
00347 }
```

References [k_tpu_channel_1](#), [k_tpu_channel_2](#), [k_tpu_channel_4](#), [k_tpu_channel_5](#), [k_tpu_tstr_cst1](#), [k_tpu_tstr_cst2](#), [k_tpu_tstr_cst4](#), and [k_tpu_tstr_cst5](#).

Referenced by [rx_tpu_deinit\(\)](#), [rx_tpu_init_phase_count\(\)](#), [rx_tpu_start\(\)](#), and [rx_tpu_stop\(\)](#).

Here is the caller graph for this function:



8.135.3.4 `internal_get_regs()`

```
RX_STATIC_TESTABLE volatile rx\_tpu\_regs\_t * internal_get_regs (
    const rx\_tpu\_channel\_t channel)
```

Get register pointer for a phase-counting TPU channel.

Maps a TPU channel identifier to the corresponding hardware register base address pointer. Only returns pointers for phase-counting-capable channels (1, 2, 4, 5).

Precondition

TPU module enabled (MSTPCRA.MSTPA13 = 0)
 channel must be a valid [rx_tpu_channel_t](#) enumeration value

Postcondition

No registers modified
 Returned pointer (if non-NULL) points to valid hardware register block

Note

Thread Safety: Safe - returns constant hardware address

See also

[tpu1\(\)](#), [tpu2\(\)](#), [tpu4\(\)](#), [tpu5\(\)](#) Underlying accessor functions

Since

Version 1.0.0

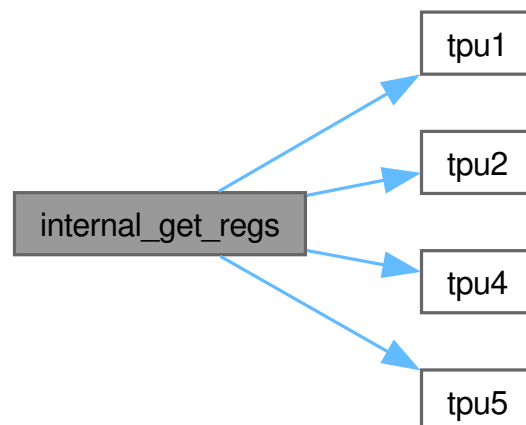
Definition at line 295 of file [rx_tpu.c](#).

```
00296 {
00297     switch (channel) {
00298     case k_tpu_channel_1:
00299         return tpu1();
00300     case k_tpu_channel_2:
00301         return tpu2();
00302     case k_tpu_channel_4:
00303         return tpu4();
00304     case k_tpu_channel_5:
00305         return tpu5();
00306     default:
00307         RX_ASSERT(false, "Post-condition: channel must be valid (1,2,4,5) - caller validates");
00308         return nullptr;
00309     }
00310 }
```

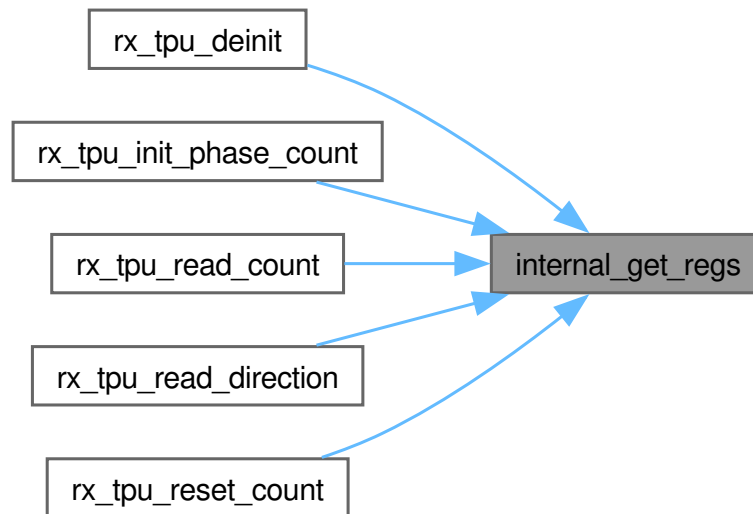
References [k_tpu_channel_1](#), [k_tpu_channel_2](#), [k_tpu_channel_4](#), [k_tpu_channel_5](#), [tpu1\(\)](#), [tpu2\(\)](#), [tpu4\(\)](#), and [tpu5\(\)](#).

Referenced by [rx_tpu_deinit\(\)](#), [rx_tpu_init_phase_count\(\)](#), [rx_tpu_read_count\(\)](#), [rx_tpu_read_direction\(\)](#), and [rx_tpu_reset_count\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.135.3.5 internal_get_tmdr_mode()

```
uint8_t internal_get_tmdr_mode (
    const rx\_tpu\_phase\_mode\_t mode) [static]
```

Map [rx_tpu_phase_mode_t](#) to TMDR.MD register value.

Converts the phase mode enumeration (1-4) to the corresponding TMDR register value for phase counting mode selection. The caller must check whether the return value equals `k_tpu_tmdr_md_normal` (0x00) to detect an invalid mode argument before writing to hardware.

Precondition

mode must be a valid [rx_tpu_phase_mode_t](#) enumeration value

Caller must check return != `k_tpu_tmdr_md_normal` before hardware write

Postcondition

No hardware registers modified

Returned value is safe to write directly to TMDR.MD field

Note

Thread Safety: Safe - pure function

See also

[rx_tpu_tmdr_md_t](#) TMDR mode register values

Since

Version 1.0.0

Definition at line 370 of file [rx_tpu.c](#).

```
00371 {
00372     switch (mode) {
00373         case k_tpu_phase_mode_1:
00374             return k_tpu_tmdr_md_phase_count_1;
00375         case k_tpu_phase_mode_2:
00376             return k_tpu_tmdr_md_phase_count_2;
00377         case k_tpu_phase_mode_3:
00378             return k_tpu_tmdr_md_phase_count_3;
00379         case k_tpu_phase_mode_4:
00380             return k_tpu_tmdr_md_phase_count_4;
00381         default:
00382             return k_tpu_tmdr_md_normal;
00383     }
00384 }
```

References [k_tpu_phase_mode_1](#), [k_tpu_phase_mode_2](#), [k_tpu_phase_mode_3](#), [k_tpu_phase_mode_4](#), [k_tpu_tmdr_md_normal](#), [k_tpu_tmdr_md_phase_count_1](#), [k_tpu_tmdr_md_phase_count_2](#), [k_tpu_tmdr_md_phase_count_3](#), and [k_tpu_tmdr_md_phase_count_4](#).

Referenced by [rx_tpu_init_phase_count\(\)](#).

Here is the caller graph for this function:



8.135.3.6 rx`tpu`deinit()

```
rx_err_t rx_tpu_deinit (
    const rx\_tpu\_channel\_t channel) [nodiscard]
```

Deinitialize a TPU phase-counting channel.

Deinitialize TPU channel.

Performs an orderly shutdown of the specified TPU channel:

1. Validate channel and get array index
2. Stop counter (clear CSTn in TSTR)
3. Reset TMDR to normal operation mode (0x00)
4. Disable all interrupts (TIER = 0x40; bit 6 reserved/must-write-1)
5. Clear counter (TCNT = 0)
6. Mark channel as uninitialized in `s_tpu_initialized[]`

This function does NOT disable the TPU module clock (MSTPCRA.MSTPA13), as other channels may still be in use.

Precondition

channel must be a valid phase-counting TPU channel (1, 2, 4, or 5)
 Motor driven by this encoder must be stopped before calling

Postcondition

Counter stopped and zeroed
 TMDR reset to normal mode
 s_tpu_initialized[idx] == false

Note

Not thread-safe: do not access channel concurrently during deinit
 Channel may be partially initialized; deinit succeeds regardless

```
rx_err_t err = rx_tpu_deinit(k_tpu_channel_1);
```

See also

[rx_tpu_init_phase_count\(\)](#) Re-initialize after deinit
[rx_tpu_stop\(\)](#) Stop counter without full teardown

Since

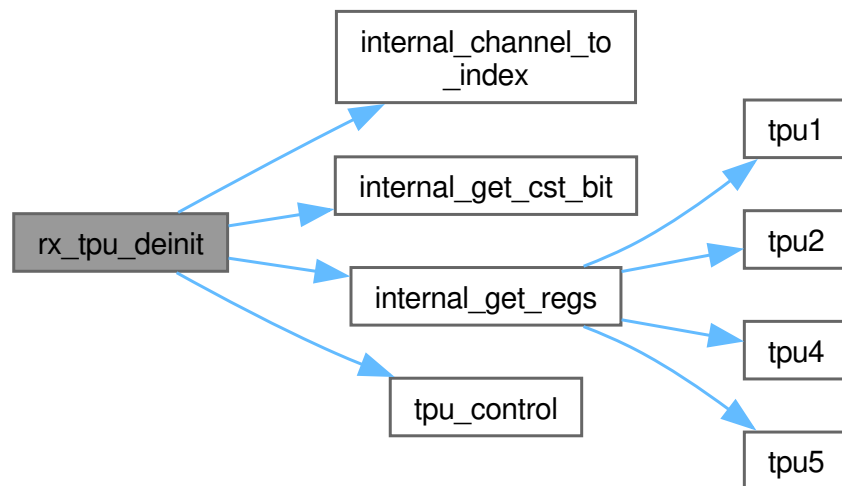
Version 1.0.0

Definition at line 777 of file [rx_tpu.c](#).

```
00778 {
00779     /* Pre-condition 1: validate channel */
00780     uint8_t idx = 0U;
00781     const rx_err_t err = internal_channel_to_index(channel, &idx);
00782     if (err != k_rx_ok) {
00783         return err;
00784     }
00785
00786     /* Get register pointer (no init check - allow deinit of partially init'd channel) */
00787     volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00788
00789     /* 1. Stop counter */
00790     const uint8_t cst_bit = internal_get_cst_bit(channel);
00791     tpu_control()->tstr &= (uint8_t)~cst_bit;
00792
00793     /* 2. Reset mode to normal operation */
00794     regs->tmdr = k_tpu_tmdr_reset;
00795
00796     /* 3. Disable all interrupts */
00797     regs->tier = k_tpu_tier_all_disabled;
00798
00799     /* 4. Clear counter */
00800     regs->tcnt = k_tpu_tcnt_zero;
00801
00802     /* 5. Mark channel as uninitialized */
00803     s_tpu_initialized[idx] = false;
00804
00805     rx_log_info_val(s_tag, "TPU channel deinitialized", (uint32_t)channel);
00806
00807     return k_rx_ok;
00808 }
```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tcnt_zero](#), [k_tpu_tier_all_disabled](#), [k_tpu_tmdr_reset](#), [s_tag](#), [s_tpu_initialized](#), [rx_tpu_regs_t::tcnt](#), [rx_tpu_regs_t::tier](#), [rx_tpu_regs_t::tmdr](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.135.3.7 rx`tpu`init`phase`count()

```
rx_err_t rx_tpu_init_phase_count (
    const rx_tpu_config_t * config) [nodiscard]
```

Initialize TPU channel for phase counting mode.

Implementation of [rx_tpu_init_phase_count\(\)](#) - see [rx_tpu.h](#) for complete API documentation.

Register Configuration Sequence

1. Unlock PRCR, clear MSTPCRA.MSTPA13, lock PRCR
2. Clear CSTn in TSTR (stop counter)
3. Write phase counting mode to TMDR.MD
4. Write 0x00 to TCR (free-running, external clock)
5. Write 0x00 to TIER (no interrupts)
6. Write 0x0000 to TCNT (clear counter)

See also

[rx_tpu.h](#) Full API documentation

Since

Version 1.0.0

Definition at line 433 of file [rx_tpu.c](#).

```

00434 {
00435     /* Pre-condition 1: validate config pointer (NASA Rule 5) */
00436     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
00437
00438     /* Pre-condition 2: validate channel is phase-counting capable */
00439     uint8_t idx = 0U;
00440     const rx_err_t err = internal_channel_to_index(config->channel, &idx);
00441     if (err != k_rx_ok) {
00442         return err;
00443     }
00444 }
```

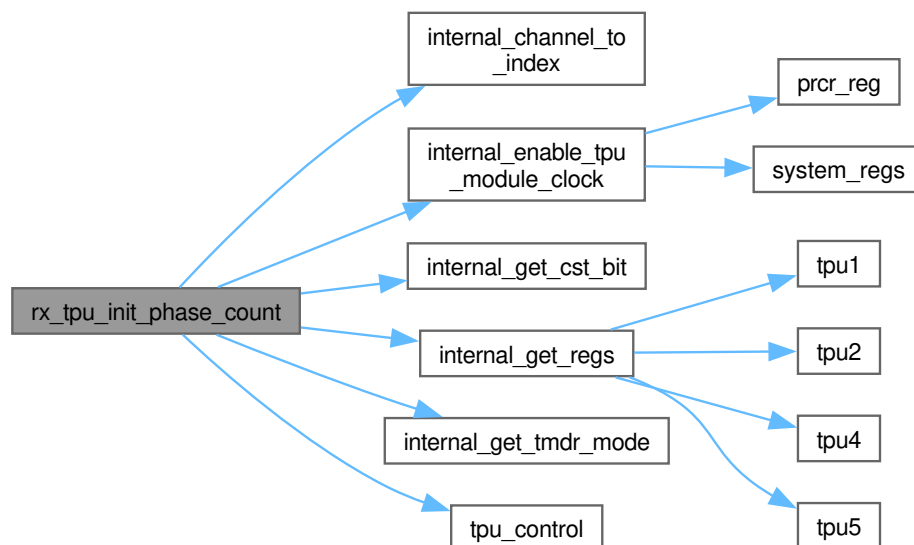
```

00445  /* Pre-condition 3: validate phase mode range */
00446  const uint8_t tmdr_mode = internal_get_tmdr_mode(config->phase_mode);
00447  if (tmdr_mode == k_tpu_tmdr_md_normal) {
00448      rx_log_error_val(s_tag, "Invalid phase mode", (uint32_t)config->phase_mode);
00449      return k_rx_err_invalid_arg;
00450  }
00451
00452  /* Pre-condition 4: check not already initialized */
00453  if (s_tpu_initialized[idx]) {
00454      rx_log_error(s_tag, "Channel already initialized");
00455      return k_rx_err_invalid_state;
00456  }
00457
00458  /* 1. Enable TPU module clock */
00459  internal_enable_tpu_module_clock();
00460
00461  /* 2. Get register pointer */
00462  volatile rx_tpu_regs_t* regs = internal_get_regs(config->channel);
00463
00464  /* 3. Stop counter */
00465  const uint8_t cst_bit = internal_get_cst_bit(config->channel);
00466  tpu_control()->tstr &= (uint8_t)~cst_bit;
00467
00468  /* 4. Configure TCR - free-running (TPSC/CKEG ignored in phase count mode) */
00469  regs->tcr = k_tpu_tcr_phase_count;
00470
00471  /* 5. Set phase counting mode in TMDR */
00472  regs->tmdr = tmdr_mode;
00473
00474  /* 6. Disable all interrupts */
00475  regs->tier = k_tpu_tier_all_disabled;
00476
00477  /* 7. Clear counter */
00478  regs->tcnt = k_tpu_tcnt_zero;
00479
00480  /* 8. Mark channel as initialized */
00481  s_tpu_initialized[idx] = true;
00482
00483  rx_log_info(s_tag, "TPU phase count initialized");
00484  rx_log_info_val(s_tag, " channel", (uint32_t)config->channel);
00485  rx_log_info_val(s_tag, " mode", (uint32_t)config->phase_mode);
00486
00487  return k_rx_ok;
00488 }

```

References [rx_tpu_config_t::channel](#), [internal_channel_to_index\(\)](#), [internal_enable_tpu_module_clock\(\)](#), [internal_get_cst_bit\(\)](#), [internal_get_regs\(\)](#), [internal_get_tmdr_mode\(\)](#), [k_tpu_tcnt_zero](#), [k_tpu_tcr_phase_count](#), [k_tpu_tier_all_disabled](#), [k_tpu_tmdr_md_normal](#), [rx_tpu_config_t::phase_mode](#), [s_tag](#), [s_tpu_initialized](#), [rx_tpu_regs_t::tcnt](#), [rx_tpu_regs_t::tcr](#), [rx_tpu_regs_t::tier](#), [rx_tpu_regs_t::tmdr](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.135.3.8 rx`tpu`read`count()

```
rx_err_t rx_tpu_read_count (
    const rx_tpu_channel_t channel,
    uint16_t * count) [nodiscard]
```

Read the current 16-bit encoder counter value.

Read TPU channel 16-bit counter value.

Performs a single volatile read of the TCNT register for the specified channel. In 4x phase counting mode, the counter increments or decrements on every edge of both A and B encoder channels.

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)
count must point to valid uint16_t storage

Postcondition

*count contains the TCNT register value at the time of the read
Hardware registers unchanged

Note

Not thread-safe: counter may wrap between multiple reads
Wraps naturally at 0 and 65535 (16-bit overflow)

```
uint16_t count;
rx_err_t err = rx_tpu_read_count(k_tpu_channel_1, &count);
```

See also

[rx_tpu_reset_count\(\)](#) Clear counter to zero
[rx_tpu_read_direction\(\)](#) Read counting direction

Since

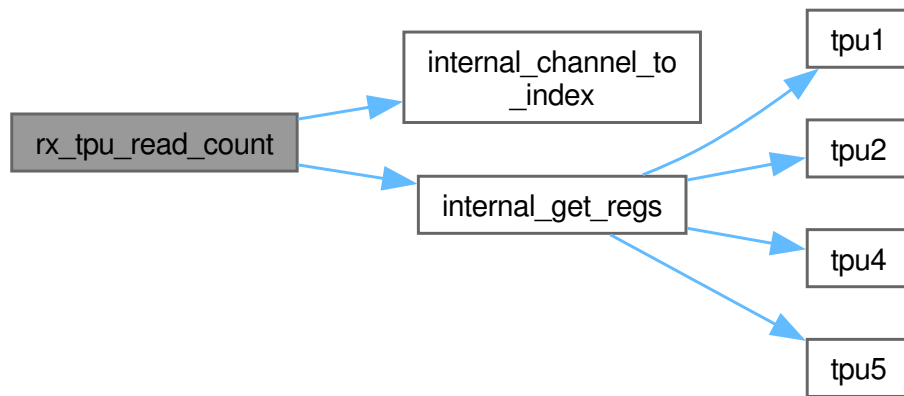
Version 1.0.0

Definition at line 613 of file [rx_tpu.c](#).

```
00614 {
00615     /* Pre-condition 1: validate output pointer */
00616     RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
00617
00618     /* Pre-condition 2: validate channel */
00619     uint8_t idx = 0U;
00620     const rx_err_t err = internal_channel_to_index(channel, &idx);
00621     if (err != k_rx_ok) {
00622         return err;
00623     }
00624
00625     /* Pre-condition 3: check initialized */
00626     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00627
00628     /* Read TCNT register */
00629     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00630
00631     *count = regs->tcnt;
00632
00633     return k_rx_ok;
00634 }
```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tcnt](#).

Here is the call graph for this function:



8.135.3.9 rx_tpu_read_direction()

```

rx_err_t rx_tpu_read_direction (
    const rx_tpu_channel_t channel,
    bool * counting_up) [nodiscard]

```

Read the current encoder counting direction.

Read TPU channel counting direction.

Reads the Timer Counter Flag Direction (TCFD) bit in the TSR register. TCFD reflects the last count event direction:

- TCFD = 1: Counter incremented on the last edge (counting up / forward)
- TCFD = 0: Counter decremented on the last edge (counting down / reverse)

Precondition

Channel must be initialized via `rx_tpu_init_phase_count()`
`counting_up` must point to valid bool storage

Postcondition

*`counting_up` reflects TSR.TCFD at the time of the read
 Hardware registers unchanged

Note

Not thread-safe: value may change between reads if encoder is moving
 Direction is unreliable if counter is stationary (no recent edge)

```

bool forward;
rx_err_t err = rx_tpu_read_direction(k_tpu_channel_1, &forward);

```

See also

`rx_tpu_read_count()` Read counter position
`k_tpu_tsr_tcf` TSR direction bit mask

Since

Version 1.0.0

Definition at line 666 of file rx_tpu.c.

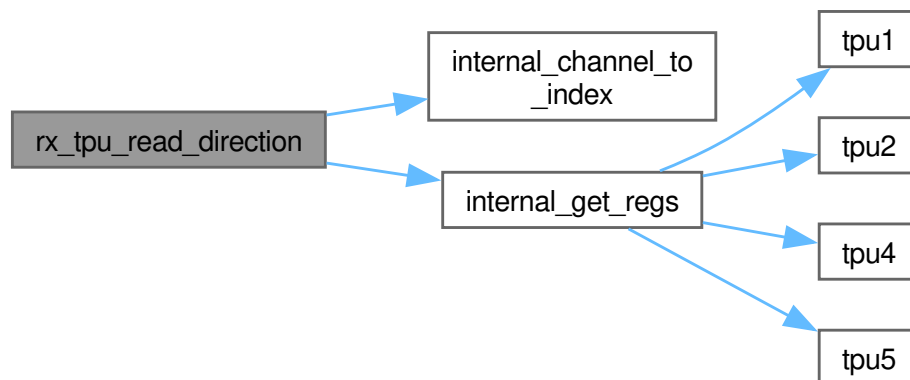
```

00667 {
00668     /* Pre-condition 1: validate output pointer */
00669     RX_CHECK_NULL_PTR(counting_up, s_tag, "Direction pointer is nullptr");
00670
00671     /* Pre-condition 2: validate channel */
00672     uint8_t idx = 0U;
00673     const rx_err_t err = internal_channel_to_index(channel, &idx);
00674     if (err != k_rx_ok) {
00675         return err;
00676     }
00677
00678     /* Pre-condition 3: check initialized */
00679     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00680
00681     /* Read TSR.TCFD bit */
00682     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00683
00684     *counting_up = (bool)((regs->tsr & k_tpu_tsr_tcfid) != 0U);
00685
00686     return k_rx_ok;
00687 }

```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tsr_tcfid](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tsr](#).

Here is the call graph for this function:



8.135.3.10 rx`tpu`reset`count()

```

rx_err_t rx_tpu_reset_count (
    const rx_tpu_channel_t channel) [nodiscard]

```

Reset the TPU channel encoder counter to zero.

Reset TPU channel counter to zero.

Writes `k_tpu_tcmt_zero` (0x0000) to the TCNT register, resetting the encoder position counter. The counter continues running from the new zero baseline.

Typical use: zero encoder position at a known reference point (home).

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)

Channel should be stopped ([rx_tpu_stop\(\)](#)) before resetting for atomic 0

Postcondition

TCNT register written to 0x0000

Counter resumes from zero if running

Note

Not thread-safe: write races with live counter updates from encoder

May cause a single spurious velocity estimate if called while running

```
rx_tpu_stop(k_tpu_channel_1);
rx_tpu_reset_count(k_tpu_channel_1);
rx_tpu_start(k_tpu_channel_1);
```

See also

[rx_tpu_stop\(\)](#) Stop counter before resetting for deterministic result

[rx_tpu_read_count\(\)](#) Verify counter after reset

Since

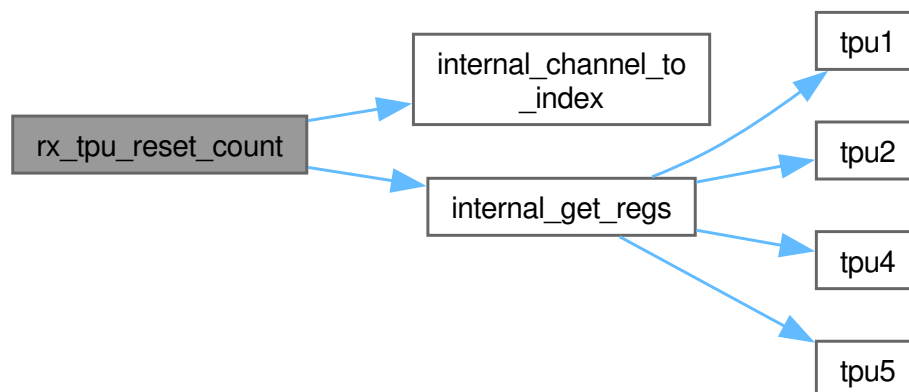
Version 1.0.0

Definition at line 721 of file [rx_tpu.c](#).

```
00722 {
00723  /* Pre-condition 1: validate channel */
00724  uint8_t idx = 0U;
00725  const rx_err_t err = internal_channel_to_index(channel, &idx);
00726  if (err != k_rx_ok) {
00727    return err;
00728  }
00729
00730  /* Pre-condition 2: check initialized */
00731  RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00732
00733  /* Write 0 to TCNT */
00734  volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00735
00736  regs->tcnt = k_tpu_tcnt_zero;
00737
00738  return k_rx_ok;
00739 }
```

References [internal_channel_to_index\(\)](#), [internal_get_regs\(\)](#), [k_tpu_tcnt_zero](#), [s_tag](#), [s_tpu_initialized](#), and [rx_tpu_regs_t::tcnt](#).

Here is the call graph for this function:



8.135.3.11 rx`tpu`start()

```
rx_err_t rx_tpu_start (
    const rx_tpu_channel_t channel) [nodiscard]
```

Start the TPU channel counter (begin phase counting).

Start TPU channel counter.

Sets the Count Start (CSTn) bit in the TSTR register to begin counter operation. Once started, the TCNT register increments or decrements on each external encoder clock edge according to the configured phase mode.

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)

External encoder clock pins must be connected and valid

Postcondition

TSTR.CSTn bit set, counter begins counting encoder pulses

TCNT updates in real time from encoder edges

Note

Not thread-safe: modifies shared TSTR register

Idempotent: safe to call when already running

```
rx_err_t err = rx_tpu_start(k_tpu_channel_1);
```

See also

[rx_tpu_stop\(\)](#) Stop the counter

[rx_tpu_read_count\(\)](#) Read current counter value

Since

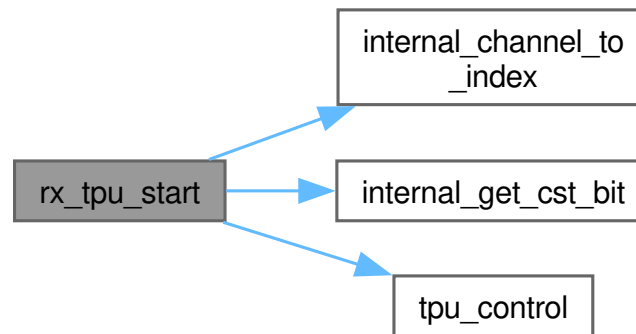
Version 1.0.0

Definition at line 518 of file [rx_tpu.c](#).

```
00519 {
00520     /* Pre-condition 1: validate channel */
00521     uint8_t idx = 0U;
00522     rx_err_t const err = internal_channel_to_index(channel, &idx);
00523     if (err != k_rx_ok) {
00524         return err;
00525     }
00526
00527     /* Pre-condition 2: check initialized */
00528     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00529
00530     /* Set CSTn bit in TSTR to start counter */
00531     const uint8_t cst_bit = internal_get_cst_bit(channel);
00532     tpu_control()->tstr |= cst_bit;
00533
00534     return k_rx_ok;
00535 }
```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [s_tag](#), [s_tpu_initialized](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.135.3.12 rx_tpu_stop()

```
rx_err_t rx_tpu_stop (
    const rx_tpu_channel_t channel) [nodiscard]
```

Stop the TPU channel counter (halt phase counting).

Stop TPU channel counter.

Clears the Count Start (CSTn) bit in the TSTR register to halt counter operation. The TCNT value is preserved and encoder edges no longer update it. The channel remains initialized and can be restarted with [rx_tpu_start\(\)](#).

Precondition

Channel must be initialized via [rx_tpu_init_phase_count\(\)](#)

TSTR register must be accessible (module clock enabled)

Postcondition

TSTR.CSTn bit cleared, counter halted

TCNT value preserved at last count

Note

Not thread-safe: modifies shared TSTR register

Idempotent: safe to call when already stopped

```
rx_err_t err = rx_tpu_stop(k_tpu_channel_1);
```

See also

[rx_tpu_start\(\)](#) Resume counter operation

[rx_tpu_reset_count\(\)](#) Clear counter to zero

Since

Version 1.0.0

Definition at line 565 of file [rx_tpu.c](#).

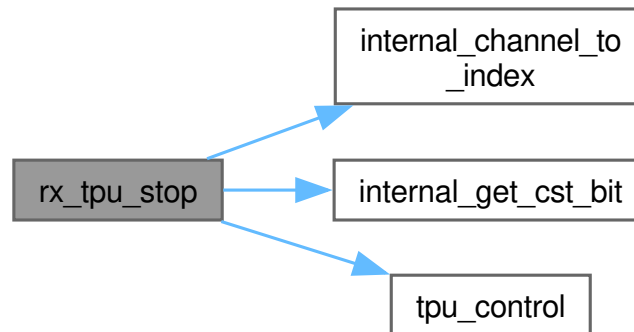
```

00566 {
00567     /* Pre-condition 1: validate channel */
00568     uint8_t idx = 0U;
00569     rx_err_t const err = internal_channel_to_index(channel, &idx);
00570     if (err != k_rx_ok) {
00571         return err;
00572     }
00573
00574     /* Pre-condition 2: check initialized */
00575     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00576
00577     /* Clear CSTn bit in TSTR to stop counter */
00578     const uint8_t cst_bit = internal_get_cst_bit(channel);
00579     tpu_control()->tstr &= (uint8_t)~cst_bit;
00580
00581     return k_rx_ok;
00582 }

```

References [internal_channel_to_index\(\)](#), [internal_get_cst_bit\(\)](#), [s_tag](#), [s_tpu_initialized](#), [tpu_control\(\)](#), and [rx_tpu_control_regs_t::tstr](#).

Here is the call graph for this function:



8.135.4 Variable Documentation

8.135.4.1 s`tag

```
const char* const s_tag = "TPU" [static]
```

Logging tag for TPU module.

Static constant string used as the tag parameter for all `rx_log_*`() calls within this module.

Value: "TPU"

Memory: 4 bytes in .rodata section

Note

Statically allocated, never modified

Since

Version 1.0.0

Definition at line 131 of file [rx_tpu.c](#).

8.135.4.2 s'tpu'initialized

```
bool s_tpu_initialized[k_tpu_phase_channel_count] = {false, false, false, false} [static]
```

Track which phase-counting channels are initialized.

Boolean array indexed by `tpu_channel_index_t` values (0-3). Set to true by `rx_tpu_init_phase_count()`, false by `rx_tpu_deinit()`.

Warning

Do not modify directly - use public API functions

Since

Version 1.0.0

Definition at line 229 of file `rx_tpu.c`.

```
00229 {false, false, false, false};
```

Referenced by `rx_tpu_deinit()`, `rx_tpu_init_phase_count()`, `rx_tpu_read_count()`, `rx_tpu_read_direction()`, `rx_tpu_reset_count()`, `rx_tpu_start()`, and `rx_tpu_stop()`.

8.136 rx'tpu.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_tpu.c
00003  * @brief TPU HAL Driver Implementation for Phase Counting Mode
00004  *
00005  * @details
00006  * Implementation of the TPU hardware abstraction layer for RX72N quadrature
00007  * encoder phase counting. This driver configures TPU channels 1, 2, 4, and 5
00008  * for hardware-based encoder position tracking with 4x resolution.
00009  *
00010  * ### Module Architecture
00011  *
00012  * The implementation is structured in three layers:
00013  * 1. **Constants & State**: Typed enums for all magic numbers, static tracking
00014  * 2. **Internal Helpers**: Channel validation, register access, bit mapping
00015  * 3. **Public API**: User-facing functions documented in rx_tpu.h
00016  *
00017  * ## Initialization Sequence
00018  *
00019  * @verbatim
00020  rx_tpu_init_phase_count(&config)
00021  * |
00022  * +-- 1. Validate config (NULL, channel, phase_mode)
00023  * +-- 2. Check not already initialized
00024  * +-- 3. Enable TPU module clock (MSTPCRA.MSTPA13 = 0)
00025  * +-- 4. Stop channel counter (TSTR.CSTn = 0)
00026  * +-- 5. Set phase counting mode (TMDR.MD = mode)
00027  * +-- 6. Set TCR to free-running (OCLR = disabled)
00028  * +-- 7. Clear TCNT to zero
00029  * +-- 8. Disable interrupts (TIER = 0x40; bit 6 reserved/must-write-1)
00030  * +-- 9. Mark channel as initialized
00031  * @endverbatim
00032  *
00033  * ## Static Variables
00034  *
00035  * | Variable | Type | Purpose |
00036  * |-----|-----|-----|
00037  * | s_tpu_initialized[] | bool[4] | Track which channels are initialized |
00038  * | s_tag | const char* | Log tag "TPU" |
00039  *
00040  * ## Channel-to-Index Mapping
00041  *
```

```

00042 * Phase counting channels (1, 2, 4, 5) are mapped to array indices 0-3:
00043 * | Channel | Index | Use in STAR          | Clock Pins          |
00044 * |-----|-----|-----|-----|
00045 * | TPU1    | 0    | Back-right encoder   | TCLKA + TCLKB      |
00046 * | TPU2    | 1    | Back-left encoder    | TCLKC + TCLKD      |
00047 * | TPU4    | 2    | Available            | TCLKC + TCLKD      |
00048 * | TPU5    | 3    | Available            | TCLKA + TCLKB      |
00049 *
00050 * @par Performance Characteristics
00051 * | Function          | Time      | Stack | Notes          |
00052 * |-----|-----|-----|-----|
00053 * | rx_tpu_init_phase_count() | ~20 us | 32 B | One-time setup per channel |
00054 * | rx_tpu_start()          | ~0.5 us | 8 B  | Single register write      |
00055 * | rx_tpu_stop()           | ~0.5 us | 8 B  | Single register write      |
00056 * | rx_tpu_read_count()     | ~0.5 us | 8 B  | Single register read       |
00057 * | rx_tpu_read_direction() | ~0.5 us | 8 B  | Single register read       |
00058 * | rx_tpu_reset_count()    | ~0.5 us | 8 B  | Single register write      |
00059 * | rx_tpu_deinit()        | ~1 us  | 16 B | Stop + reset + cleanup     |
00060 *
00061 * @par Memory Usage
00062 * | Category          | Size      | Description          |
00063 * |-----|-----|-----|
00064 * | Code (.text)      | ~600 bytes | All functions        |
00065 * | Constants          | 4 bytes   | Tag string "TPU"     |
00066 * | Static data        | 4 bytes   | Initialized flags array |
00067 * | Stack (max)        | 32 bytes  | Deepest call chain   |
00068 *
00069 * @par NASA Power of 10 Compliance
00070 * - Rule 1: No goto, setjmp, or recursion
00071 * - Rule 2: All loops bounded (no loops in this module)
00072 * - Rule 3: No dynamic memory allocation
00073 * - Rule 4: All functions < 60 lines
00074 * - Rule 5: Minimum 2 validation checks per function
00075 * - Rule 6: Variables at smallest scope
00076 * - Rule 7: All return values checked
00077 * - Rule 8: C23 typed enums for all constants
00078 * - Rule 9: Single-level pointers only
00079 * - Rule 10: Compiles with -Wall -Wextra -Werror
00080 *
00081 * @par SOLID Principles
00082 * - **S**: Only TPU hardware access (no encoder logic, no motor control)
00083 * - **O**: Extensible via rx_tpu_config_t without modifying existing code
00084 * - **I**: Consistent rx_err_t return semantics across all functions
00085 * - **I**: Focused API: init, start, stop, read, reset, deinit
00086 * - **D**: Depends on register abstractions (rx72n_tpu_regs.h), not raw addresses
00087 *
00088 * @see rx_tpu.h Public API documentation
00089 * @see rx72n_tpu_regs.h TPU register definitions
00090 * @see rx_mtu_encoder.h MTU encoder interface (front wheels)
00091 *
00092 * @author Locked, Inc.
00093 * @date 2026-02-10
00094 * @copyright Copyright (c) 2026 Locked Inc.
00095 * SPDX-License-Identifier: MIT
00096 * @since Version 1.0.0
00097 */
00098
00099 #include "rx_tpu.h"
00100
00101 #include <stddef.h>
00102
00103 #ifdef UNIT_TEST
00104 #include "mock_rx72n_regs.h"
00105 #else
00106 #include "rx72n_regs.h"
00107 #endif
00108 #include "rx_check.h"
00109 #include "rx_log.h"
00110 #include "rx_register_protection.h"
00111
00112 /*
=====
00113 * Constants
00114 *
=====
00115 */
00116
00117 /**
00118 * @var s_tag
00119 * @brief Logging tag for TPU module
00120 *
00121 * @details
00122 * Static constant string used as the tag parameter for all rx_log_*() calls
00123 * within this module.
00124 *
00125 * @par Value: "TPU"
00126 * @par Memory: 4 bytes in .rodata section

```

```

00127 *
00128 * @note Statically allocated, never modified
00129 * @since Version 1.0.0
00130 */
00131 static const char* const s_tag = "TPU";
00132
00133 /**
00134 * @enum tpu_phase_channel_count_t
00135 * @brief Number of phase-counting-capable TPU channels
00136 *
00137 * @details
00138 * TPU1, TPU2, TPU4, and TPU5 support phase counting mode.
00139 * TPU0 and TPU3 do NOT support phase counting.
00140 *
00141 * @since Version 1.0.0
00142 */
00143 typedef enum : uint8_t {
00144     k_tpu_phase_channel_count = 4, /**< Channels 1, 2, 4, 5 */
00145 } tpu_phase_channel_count_t;
00146
00147 /**
00148 * @enum tpu_channel_index_t
00149 * @brief Array index mapping for phase-counting channels
00150 *
00151 * @details
00152 * Maps the non-contiguous channel numbers (1, 2, 4, 5) to contiguous
00153 * array indices (0-3) for the s_tpu_initialized[] tracking array.
00154 *
00155 * @since Version 1.0.0
00156 */
00157 typedef enum : uint8_t {
00158     k_tpu_idx_ch1 = 0, /**< TPU1 -> index 0 (back-right encoder) */
00159     k_tpu_idx_ch2 = 1, /**< TPU2 -> index 1 (back-left encoder) */
00160     k_tpu_idx_ch4 = 2, /**< TPU4 -> index 2 (available) */
00161     k_tpu_idx_ch5 = 3, /**< TPU5 -> index 3 (available) */
00162 } tpu_channel_index_t;
00163
00164 /**
00165 * @enum tpu_tmdr_reset_t
00166 * @brief TMDR register reset value (normal mode)
00167 * @since Version 1.0.0
00168 */
00169 typedef enum : uint8_t {
00170     k_tpu_tmdr_reset = 0x00, /**< Normal operation mode (reset default) */
00171 } tpu_tmdr_reset_t;
00172
00173 /**
00174 * @enum tpu_tcr_phase_count_t
00175 * @brief TCR register value for phase counting mode
00176 *
00177 * @details
00178 * In phase counting mode, the TPSC and CKEG fields in TCR are ignored
00179 * because the external clock pins provide the counting signal. CCLR is
00180 * set to disabled (free-running counter).
00181 *
00182 * @since Version 1.0.0
00183 */
00184 typedef enum : uint8_t {
00185     k_tpu_tcr_phase_count = 0x00, /**< Free-running: TPSC=0, CKEG=0, CCLR=0 */
00186 } tpu_tcr_phase_count_t;
00187
00188 /**
00189 * @enum tpu_tier_disable_t
00190 * @brief TIER register value with all interrupts disabled
00191 *
00192 * @details
00193 * Bit 6 of TIER is reserved and must always be written as 1 (manual p.1461:
00194 * "This bit is read as 1. The write value should be 1."). The reset value of
00195 * TIER is 0x40 (bit 6 set). Writing 0x00 violates the specification; 0x40
00196 * disables all interrupt sources while preserving the reserved bit.
00197 *
00198 * @since Version 1.0.0
00199 */
00200 typedef enum : uint8_t {
00201     k_tpu_tier_all_disabled = 0x40, /**< All interrupts disabled; bit 6 reserved, must be 1 */
00202 } tpu_tier_disable_t;
00203
00204 /**
00205 * @enum tpu_tcmt_zero_t
00206 * @brief TCNT register zero value
00207 * @since Version 1.0.0
00208 */
00209 typedef enum : uint16_t {
00210     k_tpu_tcmt_zero = 0x0000, /**< Counter cleared to zero */
00211 } tpu_tcmt_zero_t;
00212
00213 */

```

```

=====
00214 * Static Variables
00215 *
=====
00216 */
00217
00218 /**
00219 * @var s_tpu_initialized
00220 * @brief Track which phase-counting channels are initialized
00221 *
00222 * @details
00223 * Boolean array indexed by tpu_channel_index_t values (0-3).
00224 * Set to true by rx_tpu_init_phase_count(), false by rx_tpu_deinit().
00225 *
00226 * @warning Do not modify directly - use public API functions
00227 * @since Version 1.0.0
00228 */
00229 static bool s_tpu_initialized[k_tpu_phase_channel_count] = {false, false, false, false};
00230
00231 /*
=====
00232 * Internal Helper Functions
00233 *
=====
00234 */
00235
00236 /**
00237 * @brief Map TPU channel number to array index
00238 *
00239 * @details
00240 * Converts the non-contiguous channel enumeration (1, 2, 4, 5) to a
00241 * contiguous array index (0-3) for use with s_tpu_initialized[].
00242 *
00243 *
00244 *
00245 * @pre index must be non-NULL (caller responsibility)
00246 * @pre channel must be a valid rx_tpu_channel_t enumeration value
00247 * @post *index contains valid array index on success
00248 * @post *index is unchanged on k_rx_err_invalid_arg
00249 *
00250 * @note Thread Safety: Safe - pure function with no shared state
00251 *
00252 * @since Version 1.0.0
00253 */
00254 static rx_err_t internal_channel_to_index(const rx_tpu_channel_t channel, uint8_t* index)
00255 {
00256     switch (channel) {
00257         case k_tpu_channel_1:
00258             *index = k_tpu_idx_ch1;
00259             return k_rx_ok;
00260         case k_tpu_channel_2:
00261             *index = k_tpu_idx_ch2;
00262             return k_rx_ok;
00263         case k_tpu_channel_4:
00264             *index = k_tpu_idx_ch4;
00265             return k_rx_ok;
00266         case k_tpu_channel_5:
00267             *index = k_tpu_idx_ch5;
00268             return k_rx_ok;
00269         default:
00270             rx_log_error_val(s_tag, "Invalid TPU channel", (uint32_t)channel);
00271             return k_rx_err_invalid_arg;
00272     }
00273 }
00274
00275 /**
00276 * @brief Get register pointer for a phase-counting TPU channel
00277 *
00278 * @details
00279 * Maps a TPU channel identifier to the corresponding hardware register
00280 * base address pointer. Only returns pointers for phase-counting-capable
00281 * channels (1, 2, 4, 5).
00282 *
00283 *
00284 *
00285 * @pre TPU module enabled (MSTPCRA.MSTPA13 = 0)
00286 * @pre channel must be a valid rx_tpu_channel_t enumeration value
00287 * @post No registers modified
00288 * @post Returned pointer (if non-NULL) points to valid hardware register block
00289 *
00290 * @note Thread Safety: Safe - returns constant hardware address
00291 *
00292 * @see tpu1(), tpu2(), tpu4(), tpu5() Underlying accessor functions
00293 * @since Version 1.0.0
00294 */
00295 RX_STATIC_TESTABLE volatile rx_tpu_regs_t* internal_get_regs(const rx_tpu_channel_t channel)
00296 {

```

```

00297 switch (channel) {
00298     case k_tpu_channel_1:
00299         return tpu1();
00300     case k_tpu_channel_2:
00301         return tpu2();
00302     case k_tpu_channel_4:
00303         return tpu4();
00304     case k_tpu_channel_5:
00305         return tpu5();
00306     default:
00307         RX_ASSERT(false, "Post-condition: channel must be valid (1,2,4,5) - caller validates");
00308         return nullptr;
00309 }
00310 }
00311
00312 /**
00313  * @brief Get TSTR CST bit mask for a TPU channel
00314  *
00315  * @details
00316  * Returns the counter start bit mask in TSTR for the specified channel.
00317  * This bit is set to start counting and cleared to stop counting.
00318  * Each phase-counting-capable channel has a dedicated bit position in the
00319  * shared TSTR register: CST1, CST2, CST4, and CST5.
00320  *
00321  *
00322  *
00323  * @pre channel must be one of k_tpu_channel_1, _2, _4, or _5
00324  * @pre Caller validates that a non-zero return indicates a valid channel
00325  * @post No hardware registers modified
00326  * @post Returned value is safe to OR/AND into TSTR register
00327  *
00328  * @note Thread Safety: Safe - pure function
00329  *
00330  * @since Version 1.0.0
00331  */
00332 RX_STATIC_TESTABLE uint8_t internal_get_cst_bit(const rx_tpu_channel_t channel)
00333 {
00334     switch (channel) {
00335         case k_tpu_channel_1:
00336             return k_tpu_tstr_cst1;
00337         case k_tpu_channel_2:
00338             return k_tpu_tstr_cst2;
00339         case k_tpu_channel_4:
00340             return k_tpu_tstr_cst4;
00341         case k_tpu_channel_5:
00342             return k_tpu_tstr_cst5;
00343         default:
00344             RX_ASSERT(false, "Post-condition: channel must be valid (1,2,4,5) - caller validates");
00345             return 0;
00346     }
00347 }
00348
00349 /**
00350  * @brief Map rx_tpu_phase_mode_t to TMDR.MD register value
00351  *
00352  * @details
00353  * Converts the phase mode enumeration (1-4) to the corresponding TMDR
00354  * register value for phase counting mode selection. The caller must check
00355  * whether the return value equals k_tpu_tmdr_md_normal (0x00) to detect
00356  * an invalid mode argument before writing to hardware.
00357  *
00358  *
00359  *
00360  * @pre mode must be a valid rx_tpu_phase_mode_t enumeration value
00361  * @pre Caller must check return != k_tpu_tmdr_md_normal before hardware write
00362  * @post No hardware registers modified
00363  * @post Returned value is safe to write directly to TMDR.MD field
00364  *
00365  * @note Thread Safety: Safe - pure function
00366  *
00367  * @see rx_tpu_tmdr_md_t TMDR mode register values
00368  * @since Version 1.0.0
00369  */
00370 static uint8_t internal_get_tmdr_mode(const rx_tpu_phase_mode_t mode)
00371 {
00372     switch (mode) {
00373         case k_tpu_phase_mode_1:
00374             return k_tpu_tmdr_md_phase_count_1;
00375         case k_tpu_phase_mode_2:
00376             return k_tpu_tmdr_md_phase_count_2;
00377         case k_tpu_phase_mode_3:
00378             return k_tpu_tmdr_md_phase_count_3;
00379         case k_tpu_phase_mode_4:
00380             return k_tpu_tmdr_md_phase_count_4;
00381         default:
00382             return k_tpu_tmdr_md_normal;
00383     }

```

```

00384 }
00385
00386 /**
00387  * @brief Enable TPU module clock by clearing MSTPCRA.MSTPA13
00388  *
00389  * @details
00390  * Enables the TPU peripheral by clearing its module stop bit in MSTPCRA.
00391  * Uses the PRCR register protection unlock/lock sequence. Idempotent -
00392  * safe to call multiple times.
00393  *
00394  * @pre System clock configured
00395  * @pre PRCR register accessible (system not in protected mode that blocks PRC1)
00396  * @post TPU module clock enabled (MSTPCRA.MSTPA13 = 0)
00397  * @post PRCR protection re-enabled
00398  *
00399  * @note Thread Safety: Not thread-safe (modifies system-wide registers)
00400  *
00401  * @since Version 1.0.0
00402  */
00403 static void internal_enable_tpu_module_clock(void)
00404 {
00405     *prcr_reg() = k_rx_prcr_unlock_prc1;
00406     system_regs()->mstpcra &= ~k_tpu_mstpcra_mstpa13;
00407     *prcr_reg() = k_rx_prcr_lock;
00408 }
00409
00410 /*
=====
00411  * Public API Implementation
00412  *
=====
00413  */
00414
00415 /**
00416  * @brief Initialize TPU channel for phase counting mode
00417  *
00418  * @details
00419  * Implementation of rx_tpu_init_phase_count() - see rx_tpu.h for complete
00420  * API documentation.
00421  *
00422  * @par Register Configuration Sequence
00423  * 1. Unlock PRCR, clear MSTPCRA.MSTPA13, lock PRCR
00424  * 2. Clear CSTn in TSTR (stop counter)
00425  * 3. Write phase counting mode to TMDR.MD
00426  * 4. Write 0x00 to TCR (free-running, external clock)
00427  * 5. Write 0x00 to TIER (no interrupts)
00428  * 6. Write 0x0000 to TCNT (clear counter)
00429  *
00430  * @see rx_tpu.h Full API documentation
00431  * @since Version 1.0.0
00432  */
00433 rx_err_t rx_tpu_init_phase_count(const rx_tpu_config_t* config)
00434 {
00435     /* Pre-condition 1: validate config pointer (NASA Rule 5) */
00436     RX_CHECK_NULL_PTR(config, s_tag, "Config pointer is nullptr");
00437
00438     /* Pre-condition 2: validate channel is phase-counting capable */
00439     uint8_t idx = 0U;
00440     const rx_err_t err = internal_channel_to_index(config->channel, &idx);
00441     if (err != k_rx_ok) {
00442         return err;
00443     }
00444
00445     /* Pre-condition 3: validate phase mode range */
00446     const uint8_t tmdr_mode = internal_get_tmdr_mode(config->phase_mode);
00447     if (tmdr_mode == k_tpu_tmdr_md_normal) {
00448         rx_log_error_val(s_tag, "Invalid phase mode", (uint32_t)config->phase_mode);
00449         return k_rx_err_invalid_arg;
00450     }
00451
00452     /* Pre-condition 4: check not already initialized */
00453     if (s_tpu_initialized[idx]) {
00454         rx_log_error(s_tag, "Channel already initialized");
00455         return k_rx_err_invalid_state;
00456     }
00457
00458     /* 1. Enable TPU module clock */
00459     internal_enable_tpu_module_clock();
00460
00461     /* 2. Get register pointer */
00462     volatile rx_tpu_regs_t* regs = internal_get_regs(config->channel);
00463
00464     /* 3. Stop counter */
00465     const uint8_t cst_bit = internal_get_cst_bit(config->channel);
00466     tpu_control()->tstr &= (uint8_t)-cst_bit;
00467
00468     /* 4. Configure TCR - free-running (TPSC/KEG ignored in phase count mode) */

```

```

00469 regs->tcr = k_tpu_tcr_phase_count;
00470
00471 /* 5. Set phase counting mode in TMDR */
00472 regs->tmdr = tmdr_mode;
00473
00474 /* 6. Disable all interrupts */
00475 regs->tier = k_tpu_tier_all_disabled;
00476
00477 /* 7. Clear counter */
00478 regs->tcnt = k_tpu_tcnt_zero;
00479
00480 /* 8. Mark channel as initialized */
00481 s_tpu_initialized[idx] = true;
00482
00483 rx_log_info(s_tag, "TPU phase count initialized");
00484 rx_log_info_val(s_tag, " channel", (uint32_t)config->channel);
00485 rx_log_info_val(s_tag, " mode", (uint32_t)config->phase_mode);
00486
00487 return k_rx_ok;
00488 }
00489
00490 /**
00491  * @brief Start the TPU channel counter (begin phase counting)
00492  *
00493  * @details
00494  * Sets the Count Start (CSTn) bit in the TSTR register to begin counter
00495  * operation. Once started, the TCNT register increments or decrements on
00496  * each external encoder clock edge according to the configured phase mode.
00497  *
00498  *
00499  *
00500  * @pre Channel must be initialized via rx_tpu_init_phase_count()
00501  * @pre External encoder clock pins must be connected and valid
00502  *
00503  * @post TSTR.CSTn bit set, counter begins counting encoder pulses
00504  * @post TCNT updates in real time from encoder edges
00505  *
00506  * @note Not thread-safe: modifies shared TSTR register
00507  * @note Idempotent: safe to call when already running
00508  *
00509  * @code{.c}
00510  * rx_err_t err = rx_tpu_start(k_tpu_channel_1);
00511  * @endcode
00512  *
00513  * @see rx_tpu_stop() Stop the counter
00514  * @see rx_tpu_read_count() Read current counter value
00515  *
00516  * @since Version 1.0.0
00517  */
00518 rx_err_t rx_tpu_start(const rx_tpu_channel_t channel)
00519 {
00520     /* Pre-condition 1: validate channel */
00521     uint8_t idx = 0U;
00522     rx_err_t const err = internal_channel_to_index(channel, &idx);
00523     if (err != k_rx_ok) {
00524         return err;
00525     }
00526
00527     /* Pre-condition 2: check initialized */
00528     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00529
00530     /* Set CSTn bit in TSTR to start counter */
00531     const uint8_t cst_bit = internal_get_cst_bit(channel);
00532     tpu_control()->tstr |= cst_bit;
00533
00534     return k_rx_ok;
00535 }
00536
00537 /**
00538  * @brief Stop the TPU channel counter (halt phase counting)
00539  *
00540  * @details
00541  * Clears the Count Start (CSTn) bit in the TSTR register to halt counter
00542  * operation. The TCNT value is preserved and encoder edges no longer update
00543  * it. The channel remains initialized and can be restarted with rx_tpu_start().
00544  *
00545  *
00546  *
00547  * @pre Channel must be initialized via rx_tpu_init_phase_count()
00548  * @pre TSTR register must be accessible (module clock enabled)
00549  *
00550  * @post TSTR.CSTn bit cleared, counter halted
00551  * @post TCNT value preserved at last count
00552  *
00553  * @note Not thread-safe: modifies shared TSTR register
00554  * @note Idempotent: safe to call when already stopped
00555  */

```



```

00556 * @code{.c}
00557 * rx_err_t err = rx_tpu_stop(k_tpu_channel_1);
00558 * @endcode
00559 *
00560 * @see rx_tpu_start() Resume counter operation
00561 * @see rx_tpu_reset_count() Clear counter to zero
00562 *
00563 * @since Version 1.0.0
00564 */
00565 rx_err_t rx_tpu_stop(const rx_tpu_channel_t channel)
00566 {
00567     /* Pre-condition 1: validate channel */
00568     uint8_t idx = 0U;
00569     rx_err_t const err = internal_channel_to_index(channel, &idx);
00570     if (err != k_rx_ok) {
00571         return err;
00572     }
00573
00574     /* Pre-condition 2: check initialized */
00575     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00576
00577     /* Clear CSTn bit in TSTR to stop counter */
00578     const uint8_t cst_bit = internal_get_cst_bit(channel);
00579     tpu_control()->tstr &= (uint8_t)~cst_bit;
00580
00581     return k_rx_ok;
00582 }
00583
00584 /**
00585 * @brief Read the current 16-bit encoder counter value
00586 *
00587 * @details
00588 * Performs a single volatile read of the TCNT register for the specified
00589 * channel. In 4x phase counting mode, the counter increments or decrements
00590 * on every edge of both A and B encoder channels.
00591 *
00592 *
00593 *
00594 * @pre Channel must be initialized via rx_tpu_init_phase_count()
00595 * @pre count must point to valid uint16_t storage
00596 *
00597 * @post *count contains the TCNT register value at the time of the read
00598 * @post Hardware registers unchanged
00599 *
00600 * @note Not thread-safe: counter may wrap between multiple reads
00601 * @note Wraps naturally at 0 and 65535 (16-bit overflow)
00602 *
00603 * @code{.c}
00604 * uint16_t count;
00605 * rx_err_t err = rx_tpu_read_count(k_tpu_channel_1, &count);
00606 * @endcode
00607 *
00608 * @see rx_tpu_reset_count() Clear counter to zero
00609 * @see rx_tpu_read_direction() Read counting direction
00610 *
00611 * @since Version 1.0.0
00612 */
00613 rx_err_t rx_tpu_read_count(const rx_tpu_channel_t channel, uint16_t* count)
00614 {
00615     /* Pre-condition 1: validate output pointer */
00616     RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
00617
00618     /* Pre-condition 2: validate channel */
00619     uint8_t idx = 0U;
00620     rx_err_t const err = internal_channel_to_index(channel, &idx);
00621     if (err != k_rx_ok) {
00622         return err;
00623     }
00624
00625     /* Pre-condition 3: check initialized */
00626     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00627
00628     /* Read TCNT register */
00629     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00630
00631     *count = regs->tcnt;
00632
00633     return k_rx_ok;
00634 }
00635
00636 /**
00637 * @brief Read the current encoder counting direction
00638 *
00639 * @details
00640 * Reads the Timer Counter Flag Direction (TCFD) bit in the TSR register.
00641 * TCFD reflects the last count event direction:
00642 * - TCFD = 1: Counter incremented on the last edge (counting up / forward)

```

```

00643 * - TCFD = 0: Counter decremented on the last edge (counting down / reverse)
00644 *
00645 *
00646 *
00647 * @pre Channel must be initialized via rx_tpu_init_phase_count()
00648 * @pre counting_up must point to valid bool storage
00649 *
00650 * @post *counting_up reflects TSR.TCFD at the time of the read
00651 * @post Hardware registers unchanged
00652 *
00653 * @note Not thread-safe: value may change between reads if encoder is moving
00654 * @note Direction is unreliable if counter is stationary (no recent edge)
00655 *
00656 * @code{.c}
00657 * bool forward;
00658 * rx_err_t err = rx_tpu_read_direction(k_tpu_channel_1, &forward);
00659 * @endcode
00660 *
00661 * @see rx_tpu_read_count() Read counter position
00662 * @see k_tpu_tsr_tcfid TSR direction bit mask
00663 *
00664 * @since Version 1.0.0
00665 */
00666 rx_err_t rx_tpu_read_direction(const rx_tpu_channel_t channel, bool* counting_up)
00667 {
00668     /* Pre-condition 1: validate output pointer */
00669     RX_CHECK_NULL_PTR(counting_up, s_tag, "Direction pointer is nullptr");
00670
00671     /* Pre-condition 2: validate channel */
00672     uint8_t idx = 0U;
00673     const rx_err_t err = internal_channel_to_index(channel, &idx);
00674     if (err != k_rx_ok) {
00675         return err;
00676     }
00677
00678     /* Pre-condition 3: check initialized */
00679     RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00680
00681     /* Read TSR.TCFD bit */
00682     const volatile rx_tpu_regs_t* regs = internal_get_regs(channel);
00683
00684     *counting_up = (bool)((regs->tsr & k_tpu_tsr_tcfid) != 0U);
00685
00686     return k_rx_ok;
00687 }
00688
00689 /**
00690 * @brief Reset the TPU channel encoder counter to zero
00691 *
00692 * @details
00693 * Writes k_tpu_tcmt_zero (0x0000) to the TCNT register, resetting the
00694 * encoder position counter. The counter continues running from the new
00695 * zero baseline.
00696 *
00697 * Typical use: zero encoder position at a known reference point (home).
00698 *
00699 *
00700 *
00701 * @pre Channel must be initialized via rx_tpu_init_phase_count()
00702 * @pre Channel should be stopped (rx_tpu_stop()) before resetting for atomic 0
00703 *
00704 * @post TCNT register written to 0x0000
00705 * @post Counter resumes from zero if running
00706 *
00707 * @note Not thread-safe: write races with live counter updates from encoder
00708 * @note May cause a single spurious velocity estimate if called while running
00709 *
00710 * @code{.c}
00711 * rx_tpu_stop(k_tpu_channel_1);
00712 * rx_tpu_reset_count(k_tpu_channel_1);
00713 * rx_tpu_start(k_tpu_channel_1);
00714 * @endcode
00715 *
00716 * @see rx_tpu_stop() Stop counter before resetting for deterministic result
00717 * @see rx_tpu_read_count() Verify counter after reset
00718 *
00719 * @since Version 1.0.0
00720 */
00721 rx_err_t rx_tpu_reset_count(const rx_tpu_channel_t channel)
00722 {
00723     /* Pre-condition 1: validate channel */
00724     uint8_t idx = 0U;
00725     const rx_err_t err = internal_channel_to_index(channel, &idx);
00726     if (err != k_rx_ok) {
00727         return err;
00728     }
00729 }

```

```

00730 /* Pre-condition 2: check initialized */
00731 RX_VALIDATE_INIT(s_tpu_initialized[idx], s_tag, "Channel not initialized");
00732
00733 /* Write 0 to TCNT */
00734 volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00735
00736 regs->tcnt = k_tpu_tcnt_zero;
00737
00738 return k_rx_ok;
00739 }
00740
00741 /**
00742 * @brief Deinitialize a TPU phase-counting channel
00743 *
00744 * @details
00745 * Performs an orderly shutdown of the specified TPU channel:
00746 * 1. Validate channel and get array index
00747 * 2. Stop counter (clear CSTn in TSTR)
00748 * 3. Reset TMDR to normal operation mode (0x00)
00749 * 4. Disable all interrupts (TIER = 0x40; bit 6 reserved/must-write-1)
00750 * 5. Clear counter (TCNT = 0)
00751 * 6. Mark channel as uninitialized in s_tpu_initialized[]
00752 *
00753 * This function does NOT disable the TPU module clock (MSTPCRA.MSTPA13),
00754 * as other channels may still be in use.
00755 *
00756 *
00757 *
00758 * @pre channel must be a valid phase-counting TPU channel (1, 2, 4, or 5)
00759 * @pre Motor driven by this encoder must be stopped before calling
00760 *
00761 * @post Counter stopped and zeroed
00762 * @post TMDR reset to normal mode
00763 * @post s_tpu_initialized[idx] == false
00764 *
00765 * @note Not thread-safe: do not access channel concurrently during deinit
00766 * @note Channel may be partially initialized; deinit succeeds regardless
00767 *
00768 * @code{.c}
00769 rx_err_t err = rx_tpu_deinit(k_tpu_channel_1);
00770 @endcode
00771
00772 * @see rx_tpu_init_phase_count() Re-initialize after deinit
00773 * @see rx_tpu_stop() Stop counter without full teardown
00774 *
00775 * @since Version 1.0.0
00776 */
00777 rx_err_t rx_tpu_deinit(const rx_tpu_channel_t channel)
00778 {
00779 /* Pre-condition 1: validate channel */
00780 uint8_t idx = 0U;
00781 const rx_err_t err = internal_channel_to_index(channel, &idx);
00782 if (err != k_rx_ok) {
00783 return err;
00784 }
00785
00786 /* Get register pointer (no init check - allow deinit of partially init'd channel) */
00787 volatile rx_tpu_regs_t* const regs = internal_get_regs(channel);
00788
00789 /* 1. Stop counter */
00790 const uint8_t cst_bit = internal_get_cst_bit(channel);
00791 tpu_control()->tstr &= (uint8_t)~cst_bit;
00792
00793 /* 2. Reset mode to normal operation */
00794 regs->tmdr = k_tpu_tmdr_reset;
00795
00796 /* 3. Disable all interrupts */
00797 regs->tier = k_tpu_tier_all_disabled;
00798
00799 /* 4. Clear counter */
00800 regs->tcnt = k_tpu_tcnt_zero;
00801
00802 /* 5. Mark channel as uninitialized */
00803 s_tpu_initialized[idx] = false;
00804
00805 rx_log_info_val(s_tag, "TPU channel deinitialized", (uint32_t)channel);
00806
00807 return k_rx_ok;
00808 }

```

8.137 libs/rx_hal/src/timer.c File Reference

ThreadX system tick timer implementation using RX72N CMT0.

```
#include <stdint.h>
#include "hardware.h"
#include "rx72n_regs.h"
#include "rx_check.h"
Include dependency graph for timer.c:
```



Enumerations

- enum [cmt0_config_t](#) : uint16_t { [k_cmt0_compare_match](#) = 18749 , [k_cmt0_irq_priority](#) = 3 , [k_cmt0_ipr_index](#) = 4 }
- CMT0 timer configuration constants for 100 Hz ThreadX system tick.
- enum [cmt0_cmcr_values_t](#) : uint16_t { [k_cmt0_cmcr_config](#) = 0x00C1 }
- CMT0 Control Register (CMCR) configuration values.
- enum [cmt0_cmstr_bits_t](#) : uint8_t { [k_cmt0_cmstr_start_bit](#) = 0x01 }
- CMT Start Register (CMSTR0) bit masks.
- enum [cmt0_ier_constants_t](#) : uint8_t { [k_ier_bit_enable_base](#) = 1 }
- ICU Interrupt Enable Register (IER) calculation constants.
- enum [cmt0_counter_values_t](#) : uint16_t { [k_cmt0_counter_init](#) = 0 }
- CMT0 Counter (CMCNT) register values.
- enum [icu_ir_values_t](#) : uint8_t { [k_icu_ir_clear](#) = 0 }
- ICU Interrupt Request (IR) flag values.

Functions

- void [cmt0_isr](#) (void)
- CMT0 compare match interrupt service routine (ISR).
- void [_tx_timer_interrupt](#) (void)
- rx_err_t [timer_init](#) (void)
- Initialize CMT0 timer for ThreadX system tick at 100 Hz.
- rx_err_t [timer_stop](#) (void)
- Stop CMT0 system tick timer.
- rx_err_t [timer_get_count](#) (uint16_t *count)
- Read current CMT0 counter value.

Variables

- static const char *const [s_tag](#) = "TIMER"
- Log tag for this module.
- const uint32_t [rx_cmt0_vector_anchor](#)
- const void *const [g_cmt0_vector_keep](#) = (const void*)&[rx_cmt0_vector_anchor](#)

8.137.1 Detailed Description

ThreadX system tick timer implementation using RX72N CMT0.

This file implements the system tick timer for the ThreadX RTOS using the RX72N's CMT0 (Compare Match Timer 0) peripheral. The timer generates precise periodic interrupts at 100 Hz (10 ms period) to drive the ThreadX scheduler, time-slice preemption, and system time tracking.

8.137.1.1 Purpose and Design Rationale

The system tick is the heartbeat of the RTOS - all time-based operations (delays, timeouts, scheduling) depend on this timer. Key design decisions:

- CMT0 peripheral: Simplest timer on RX72N, dedicated to system tick
- 100 Hz tick rate: ThreadX standard (TX_TIMER_TICKS_PER_SECOND = 100)
- PCLKB/32 divider: Achieves exact 100.00 Hz with 60 MHz PCLKB
- Priority 3/15: Higher than app tasks, lower than critical motor control
- Zero drift: Hardware compare match ensures no timing accumulation error

8.137.1.2 System Architecture Context

```

Hardware:
  PCLKB (60 MHz)
    v
  CMT0 (/32) -> 1.875 MHz
    v
  Compare: 18749 -> 100.00 Hz interrupt
    v
Software:
  cmt0_isr() (this file)
    v
  _tx_timer_interrupt() (ThreadX kernel)
    v
  tx_timer_expiration_process() (ThreadX kernel)
    v
  Application thread scheduling, tx_thread_sleep(), tx_event_flags_get()

```

All ThreadX timing operations depend on this timer:

- Thread sleep/delay (tx_thread_sleep)
- Event flag timeouts (tx_event_flags_get)
- Mutex timeouts (tx_mutex_get)
- Message queue timeouts (tx_queue_receive)
- Time-slice round-robin scheduling

8.137.1.3 Hardware Requirements

Requirement	Specification
MCU	Renesas RX72N
Peripheral	CMT0 (Compare Match Timer 0)
Clock Source	PCLKB = 60 MHz
Interrupt Vector	VECT_CMT0_CMI0 (28)
ICU Priority	3 (configurable 0-15)
CPU Frequency	240 MHz (for timing accuracy)

8.137.1.4 Performance Characteristics

- Tick period: 10.000 ms (100.00 Hz)
- Tick accuracy: +/-0 ticks (hardware compare match, no drift)
- ISR execution time: ~2-5 us @ 240 MHz (measured with logic analyzer)
- ISR overhead: 0.02% - 0.05% CPU utilization
- Jitter: <100 ns (hardware interrupt latency only)
- **Memory usage**: 0 bytes RAM (all static/code)

8.137.1.5 Timer Calculation Details

Target: 100 Hz tick (10 ms period)

Clock tree:

$$f_{CMT} = \frac{f_{PCLKB}}{Divider} = \frac{60 \text{ MHz}}{32} = 1.875 \text{ MHz}$$

Compare match value:

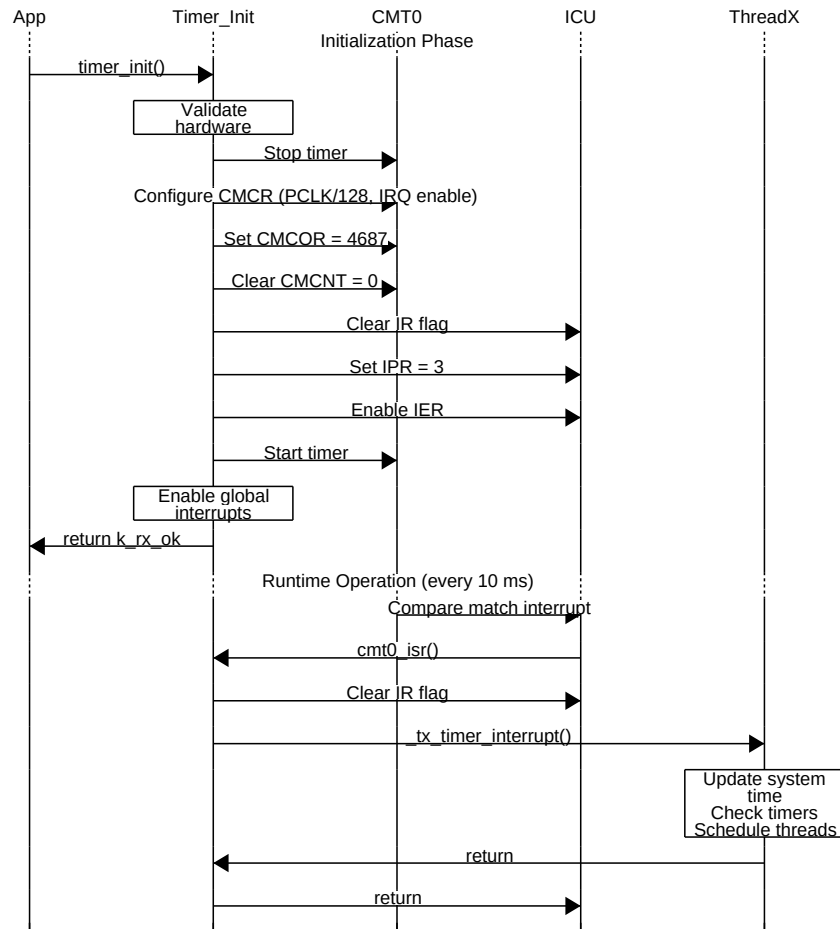
$$CMCOR = \frac{f_{CMT}}{f_{tick}} - 1 = \frac{1875000 \text{ Hz}}{100 \text{ Hz}} - 1 = 18749$$

Actual tick frequency:

$$f_{actual} = \frac{1875000}{18749 + 1} = 100.0000 \text{ Hz}$$

Timing accuracy: Exact 10.000 ms (zero error due to integer division)

8.137.1.6 Initialization Sequence



8.137.1.7 Usage Example

```

#include "timer.h"

// Initialize system tick before starting ThreadX
rx_err_t err = timer_init();
if (err != k_rx_ok) {
    rx_log_error("MAIN", "Failed to initialize system tick");
    return err;
}

// Timer now runs at 100 Hz, driving ThreadX scheduler
// Application code uses ThreadX timing functions:
tx_thread_sleep(100); // Sleep for 1 second (100 ticks)

// Stop timer when shutting down
timer_stop();
  
```

Module Dependencies

- [rx72n_regs.h](#): CMT0 and ICU register definitions
- [hardware.h](#): System clock configuration
- [tx_api.h](#): ThreadX kernel API
- [rx_check.h](#): Validation macros

NASA Power of 10 Compliance

Rule	Compliance	Details
1	↔ [OK]↔	No goto, setjmp, recursion - simple sequential code
2	↔ [OK]↔	No loops (register configuration only)
3	↔ [OK]↔	No dynamic allocation (all static configuration)
4	↔ [OK]↔	All functions under 60 lines
5	↔ [OK]↔	Minimum 2 pre/post conditions per function
6	↔ [OK]↔	Minimal scope (locals declared at use point)
7	↔ [OK]↔	All return values validated
8	↔ [OK]↔	C23 typed enums for ALL constants
9	↔ [OK]↔	Single level of dereferencing for registers
10	↔ [OK]↔	Compiles with -Wall -Wextra -Werror

SOLID Principles

Single Responsibility: Timer module has one job - provide system tick for ThreadX. Does not handle application timers, delays, or scheduling (those are ThreadX responsibilities).

Open/Closed: Tick rate configurable via `k_cmt0_compare_match` constant. Can extend to different rates without modifying core logic.

Liskov Substitution: CMT0 can be replaced with any other timer (CMT1, TPU, TMR) by implementing same initialization interface.

Interface Segregation: Minimal API (`init`, `stop`, `get_count`) - callers only depend on what they need. ThreadX only depends on ISR.

Dependency Inversion: ThreadX depends on timer abstraction (`_tx_timer_interrupt`), not on CMT0 hardware details. Easy to port to different MCU.

See also

docs/sections/08_threadx_integration.tex ThreadX system tick configuration
RX72N User's Manual Chapter 23: Compare Match Timer (CMT)
tx_initialize_low_level.c ThreadX port configuration

Author

Locked, Inc.

Date

2026-01-27

Copyright

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Definition in file [timer.c](#).

8.137.2 Enumeration Type Documentation

8.137.2.1 cmt0_cmcr_values_t

```
enum cmt0\_cmcr\_values\_t : uint16_t
```

CMT0 Control Register (CMCR) configuration values.

Bit field values for CMT0.CMCR register configuration. CMCR controls clock source, divider, and interrupt enable.

Bit breakdown for 0x00C1:

- Bit 7: Reserved; write value should be 1 (RX72N Manual Ch31, p.1583)
- Bit 6 (CMIE) = 1: Compare match interrupt enabled
- Bits 1-0 (CKS) = 01: Clock source = PCLKB/32
- All other bits = 0: Reserved/default

See also

RX72N User's Manual Chapter 31, Section 31.2.3 - CMCR register description

Enumerator

k_cmt0_cmcr_config	CMCR register configuration for 100 Hz tick with interrupts. Configures CMT0 for PCLKB/32 clock source with compare match interrupts. Value: 0x00C1 (binary: 0000_0000_1100_0001) Bit 7: Reserved, write value should be 1 (per RX72N manual p.1583) Bit 6 (CMIE): 1 = Interrupt enabled Bits 1-0 (CKS): 01 = PCLKB/32 divider
--------------------	---

Definition at line 322 of file [timer.c](#).

```

00322         : uint16_t {
00323     /**
00324     * @brief CMCR register configuration for 100 Hz tick with interrupts
00325     * @details
00326     * Configures CMT0 for PCLKB/32 clock source with compare match interrupts.
00327     * @par Value: 0x00C1 (binary: 0000_0000_1100_0001)
00328     * @par Bit 7: Reserved, write value should be 1 (per RX72N manual p.1583)
00329     * @par Bit 6 (CMIE): 1 = Interrupt enabled
00330     * @par Bits 1-0 (CKS): 01 = PCLKB/32 divider
00331     */
00332     k_cmt0_cmcr_config = 0x00C1,
00333 } cmt0_cmcr_values_t;

```

8.137.2.2 cmt0_cmstr_bits_t

```
enum cmt0\_cmstr\_bits\_t : uint8_t
```

CMT Start Register (CMSTR0) bit masks.

Bit positions for CMT0.CMSTR0 register, which controls timer start/stop. CMSTR0 bit 0 starts/stops CMT0 counter.

See also

RX72N User's Manual Section 23.2.2 - CMSTR0 register

Enumerator

k_cmt0_cmstr_start_bit	CMT0 start/stop control bit in CMSTR0 register. Writing 1 starts CMT0 counter. Writing 0 stops counter. Value: 0x01 (bit 0) Usage: cmstr0 = k_cmt0_cmstr_start_bit (start) Usage: cmstr0 &= ~k_cmt0_cmstr_start_bit (stop)
------------------------	---

Definition at line 345 of file [timer.c](#).

```

00345         : uint8_t {
00346     /**
00347     * @brief CMT0 start/stop control bit in CMSTR0 register
00348     * @details
00349     * Writing 1 starts CMT0 counter. Writing 0 stops counter.
00350     * @par Value: 0x01 (bit 0)
00351     * @par Usage: cmstr0 |= k_cmt0_cmstr_start_bit (start)
00352     * @par Usage: cmstr0 &= ~k_cmt0_cmstr_start_bit (stop)
00353     */
00354     k_cmt0_cmstr_start_bit = 0x01,
00355 } cmt0_cmstr_bits_t;

```

8.137.2.3 cmt0`config`

```
enum cmt0\_config\_t : uint16_t
```

CMT0 timer configuration constants for 100 Hz ThreadX system tick.

Configuration values for CMT0 hardware timer to achieve precise 100.00 Hz periodic interrupts for ThreadX RTOS scheduling. Values are derived from clock tree analysis and interrupt priority allocation.

Clock calculation derivation:

$$CMCOR = \frac{f_{PCLKB}}{Divider \times f_{target}} - 1 = \frac{60,000,000}{32 \times 100} - 1 = 18749$$

Priority allocation rationale:

- Priority 3/15 balances responsiveness with critical motor interrupts
- Higher than application tasks (default priority 0)
- Lower than motor control (priority 5) and fault handlers (priority 7)

Value Constraints:

Constant	Min	Max	Rationale
k_cmt0_compare_match	99	65535	Min: 10 kHz max tick, Max: uint16_t
k_cmt0_irq_priority	0	15	0=lowest, 15=highest (NMI)

Usage Example:

```
// Set compare match register
cmt0()->cmcor = k_cmt0_compare_match;

// Set interrupt priority in ICU
icu()->ipr[k_vect_cmt0_cmi0] = k_cmt0_irq_priority;
```

See also

[timer_init\(\)](#) Initialization function using these constants
 RX72N User's Manual Section 23.2.3 for CMCOR calculation

Enumerator

k_cmt0_compare_match	Compare match value for 100.00 Hz tick rate. CMT0 CMCOR register value. When CMCNT reaches this value, interrupt fires and counter resets to 0. Calculated for exact 100 Hz with zero drift. Value: 18749 counts Derivation: $(60 \text{ MHz} / 32 / 100 \text{ Hz}) - 1 = 18749$ Accuracy: 100.0000 Hz (integer division, zero error)
k_cmt0_irq_priority	CMT0 interrupt priority level in ICU. Interrupt priority for system tick. Priority 3 ensures timely scheduling without starving critical motor control interrupts. Value: 3 (out of 0-15 range) Rationale: Balance between responsiveness and critical motor control Context: App tasks = 0, System tick = 3, Motor = 5, Faults = 7

k_cmt0_ipr_index	ICU IPR register index for CMT0_CMI0 interrupt on RX72N. RX72N's ICU IPR registers are SPARSELY mapped – IPR[] array index is NOT the vector number. Per the RX72N HW manual section 14 (ICU), vector 28 (CMT0_CMI0) shares IPR[4] (0x00087304) with other group-4 peripherals. Writing to ipr[28] (0x0008731C) is silently dropped, so the interrupt priority would remain at 0 (= disabled) and the CMT0 compare-match ISR would never be accepted by the CPU even though CMT0 itself was counting. This was the root cause of ThreadX ticks never firing in the production firmware's original timer_init(); tasks' tx_thread_sleep calls hung forever because _tx_timer_system_clock was frozen. Verified on hardware 2026-04-21: writing to IPR[4] made the ISR counter advance (0 -> 604+ in ~6 s). Writing to IPR[28] produced zero ISR entries. Same workaround applied in motors_rtos/cmt0.c.
------------------	--

Definition at line 261 of file [timer.c](#).

```

00261         : uint16_t {
00262     /**
00263     * @brief Compare match value for 100.00 Hz tick rate
00264     * @details
00265     * CMT0 CMCOR register value. When CMCNT reaches this value, interrupt fires
00266     * and counter resets to 0. Calculated for exact 100 Hz with zero drift.
00267     * @par Value: 18749 counts
00268     * @par Derivation: (60 MHz / 32 / 100 Hz) - 1 = 18749
00269     * @par Accuracy: 100.0000 Hz (integer division, zero error)
00270     */
00271     k_cmt0_compare_match = 18749,
00272
00273     /**
00274     * @brief CMT0 interrupt priority level in ICU
00275     * @details
00276     * Interrupt priority for system tick. Priority 3 ensures timely scheduling
00277     * without starving critical motor control interrupts.
00278     * @par Value: 3 (out of 0-15 range)
00279     * @par Rationale: Balance between responsiveness and critical motor control
00280     * @par Context: App tasks = 0, System tick = 3, Motor = 5, Faults = 7
00281     */
00282     k_cmt0_irq_priority = 3,
00283
00284     /**
00285     * @brief ICU IPR register index for CMT0_CMI0 interrupt on RX72N
00286     * @details
00287     * RX72N's ICU IPR registers are SPARSELY mapped -- IPR[] array index
00288     * is NOT the vector number. Per the RX72N HW manual section 14
00289     * (ICU), vector 28 (CMT0_CMI0) shares IPR[4] (0x00087304) with other
00290     * group-4 peripherals. Writing to ipr[28] (0x0008731C) is silently
00291     * dropped, so the interrupt priority would remain at 0 (= disabled)
00292     * and the CMT0 compare-match ISR would never be accepted by the CPU
00293     * even though CMT0 itself was counting.
00294     *
00295     * This was the root cause of ThreadX ticks never firing in the
00296     * production firmware's original timer_init(); tasks' tx_thread_sleep
00297     * calls hung forever because _tx_timer_system_clock was frozen.
00298     *
00299     * Verified on hardware 2026-04-21: writing to IPR[4] made the ISR
00300     * counter advance (0 -> 604+ in ~6 s). Writing to IPR[28] produced
00301     * zero ISR entries. Same workaround applied in motors_rtos/cmt0.c.
00302     */
00303     k_cmt0_ipr_index = 4,
00304 } cmt0_config_t;

```

8.137.2.4 cmt0`counter`values`t

```
enum cmt0_counter_values_t : uint16_t
```

CMT0 Counter (CMCMT) register values.

Standard values for CMT0.CMCNT counter register. Counter increments every clock cycle and resets to 0 on compare match.

Enumerator

k_cmt0_counter_init	Counter initialization value (reset state). Written to CMCNT to reset counter to known state before starting timer. Value: 0 Usage: <code>cmt0()->cmcnt = k_cmt0_counter_init</code>
---------------------	--

Definition at line 384 of file [timer.c](#).

```

00384         : uint16_t {
00385     /**
00386      * @brief Counter initialization value (reset state)
00387      * @details
00388      * Written to CMCNT to reset counter to known state before starting timer.
00389      * @par Value: 0
00390      * @par Usage: cmt0()->cmcnt = k_cmt0_counter_init
00391      */
00392     k_cmt0_counter_init = 0,
00393 } cmt0_counter_values_t;

```

8.137.2.5 cmt0`ier`constants`

enum [cmt0_ier_constants_t](#) : uint8_t

ICU Interrupt Enable Register (IER) calculation constants.

Constants for calculating IER bit position. IER registers are byte arrays where each bit enables a specific interrupt vector.

Enumerator

k_ier_bit_enable_base	Base value for IER bit enable shift calculation. Used in expression: $(1 \ll (\text{vector} \% 8))$ to set correct bit in IER byte. Value: 1 Usage: <code>ier[vector/8] = (k_ier_bit_enable_base << (vector % 8))</code>
-----------------------	---

Definition at line 365 of file [timer.c](#).

```

00365         : uint8_t {
00366     /**
00367      * @brief Base value for IER bit enable shift calculation
00368      * @details
00369      * Used in expression:  $(1 \ll (\text{vector} \% 8))$  to set correct bit in IER byte.
00370      * @par Value: 1
00371      * @par Usage: ier[vector/8] |= (k_ier_bit_enable_base << (vector % 8))
00372      */
00373     k_ier_bit_enable_base = 1,
00374 } cmt0_ier_constants_t;

```

8.137.2.6 icu`ir`values`t

enum `icu_ir_values_t` : `uint8_t`

ICU Interrupt Request (IR) flag values.

Values for clearing ICU interrupt request flags. IR flags must be cleared in ISR before processing interrupt to prevent spurious re-triggering.

See also

RX72N User's Manual Chapter 13 - ICU (Interrupt Controller Unit)

Enumerator

<code>k_icu_ir_clear</code>	Value to clear ICU IR interrupt request flag. Writing 0 to IR flag clears pending interrupt. Must be done at start of ISR to acknowledge hardware and prevent immediate re-entry. Value: 0 Usage: <code>icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear</code> Warning Must clear IR flag BEFORE processing interrupt handler logic
-----------------------------	--

Definition at line 413 of file [timer.c](#).

```

00413      : uint8_t {
00414  /**
00415   * @brief Value to clear ICU IR interrupt request flag
00416   * @details
00417   * Writing 0 to IR flag clears pending interrupt. Must be done at start
00418   * of ISR to acknowledge hardware and prevent immediate re-entry.
00419   * @par Value: 0
00420   * @par Usage: icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear
00421   * @warning Must clear IR flag BEFORE processing interrupt handler logic
00422   */
00423   k_icu_ir_clear = 0,
00424 } icu_ir_values_t;
```

8.137.3 Function Documentation

8.137.3.1 tx`timer`interrupt()

```
void tx_timer_interrupt (
    void ) [extern]
```

Referenced by [cmt0_isr\(\)](#).

Here is the caller graph for this function:



8.137.3.2 cmt0_isr()

```
void cmt0_isr (  
    void )
```

CMT0 compare match interrupt service routine (ISR).

Hardware interrupt handler called by ICU when CMT0 counter matches CMCOR value (every 10.000 ms). This ISR clears the interrupt flag and delegates to ThreadX kernel to handle timer processing and thread scheduling.

Execution context: Interrupt context (cannot block, minimal stack)

Call frequency: 100 Hz (every 10 ms)

Algorithm steps:

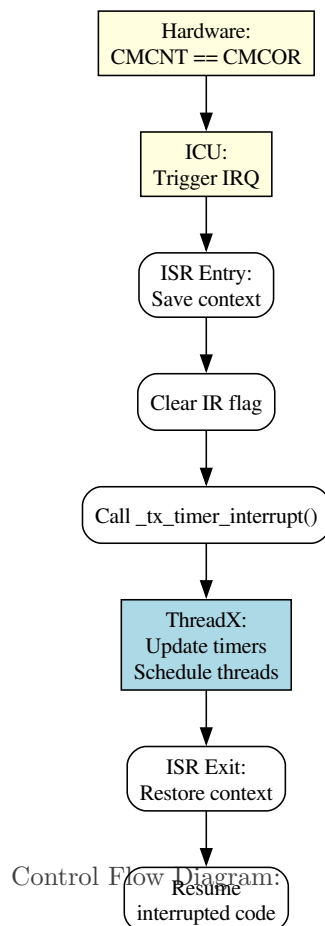
1. Hardware triggers interrupt when CMCNT == CMCOR
2. CPU vectors to this ISR via vector table entry VECT_CMT0_CMI0
3. Clear ICU IR flag to acknowledge interrupt
4. Call ThreadX [__tx_timer_interrupt\(\)](#) to process timers and scheduling
5. Return from interrupt (hardware restores context)

ThreadX processing (inside [__tx_timer_interrupt\(\)](#)):

- Increment system time counter
- Check timer expiration list
- Wake sleeping threads whose timeout expired
- Perform time-slice round-robin if enabled
- Trigger context switch if higher-priority thread ready

Performance:

- Execution time: 2-5 us @ 240 MHz (measured)
- CPU overhead: 0.02% - 0.05% (5 us x 100 Hz / 1 second)
- Stack usage: ~64 bytes (interrupt frame + [__tx_timer_interrupt\(\)](#))
- Jitter: <100 ns (hardware interrupt latency)



Precondition

CMT0 must be initialized and running via [timer_init\(\)](#)
 ICU interrupt must be enabled (IER bit set)
 Global interrupts must be enabled (PSW I-flag set)

Postcondition

ICU IR flag cleared (prevents spurious re-trigger)
 ThreadX system time incremented by 1 tick
 Expired timers processed, blocked threads potentially woken
 Context switch may occur if higher-priority thread ready

Note

Thread Safety: Runs in interrupt context. Must not call blocking ThreadX APIs ([tx_mutex_get](#), [tx_thread_sleep](#)). Only non-blocking operations allowed.

Re-entrancy: Not reentrant (interrupt priority protects). CMT0 interrupt cannot nest on itself. Higher-priority interrupts (motor faults) can preempt this ISR.

Critical Timing: This ISR must complete quickly (<10 us target) to maintain system responsiveness. Excessive ISR time causes scheduler latency and motor control jitter.

Warning

****No blocking operations**:** Never call `tx_thread_sleep`, `tx_mutex_get`, `tx_queue_receive`, or any blocking API from ISR. These will cause system crash.

Attention

****IR flag must be cleared first**:** Clearing IR flag before processing prevents race condition where new interrupt triggers while still processing previous one.

Example - System Timeline:

```
// Timeline (measured with oscilloscope):
t=0.000ms: Application thread running
t=10.000ms: CMT0 compare match -> Hardware interrupt
t=10.001ms: ISR entry (CPU context save: ~500ns)
t=10.002ms: Clear IR flag (register write: ~100ns)
t=10.003ms: Call _tx_timer_interrupt() (~2us processing)
t=10.005ms: ISR exit (CPU context restore: ~500ns)
t=10.006ms: Resume application thread (or switch to higher priority)
```

See also

[_tx_timer_interrupt\(\)](#) ThreadX kernel timer processing function
[timer_init\(\)](#) CMT0 initialization that enables this ISR
 RX72N User's Manual Section 23.3 - CMT Interrupt Operation
 ThreadX User Guide Section 3.4 - Interrupt Handling

Since

Version 1.0.0

NASA Power of 10 Compliance

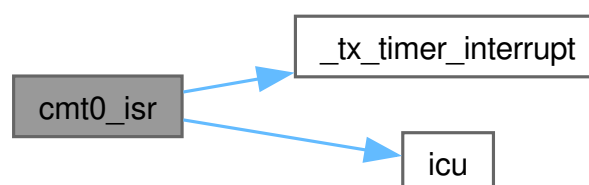
- Rule 1 [OK] No goto, setjmp, recursion
- Rule 2 [OK] No loops
- Rule 3 [OK] No dynamic allocation
- Rule 4 [OK] Function is 7 lines (under 60 line limit)
- Rule 5 [OK] 3 preconditions, 4 postconditions
- Rule 8 [OK] Uses C23 typed enum (`k_icu_ir_clear`)
- Rule 9 [OK] Single level of dereferencing

Definition at line 540 of file [timer.c](#).

```
00541 {
00542  /* Clear ICU interrupt request flag */
00543  icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear;
00544
00545  /* Call ThreadX timer interrupt handler */
00546  _tx_timer_interrupt();
00547 }
```

References [_tx_timer_interrupt\(\)](#), [icu\(\)](#), [rx_icu_regs_t::ir](#), [k_icu_ir_clear](#), and [k_vect_cmt0_cmi0](#).

Here is the call graph for this function:



8.137.3.3 timer`get`count()

```
rx_err_t timer_get_count (
    uint16_t * count)  [nodiscard]
```

Read current CMT0 counter value.

Get current CMT0 counter value.

Reads the current value of CMT0.CMCNT register, which increments from 0 to CMCOR (4687) at 468.75 kHz. This provides sub-millisecond timing resolution for diagnostics, performance measurement, and jitter analysis.

Algorithm steps:

1. Validate count pointer is not nullptr
2. Validate CMT0 hardware accessibility
3. Read CMCNT register (atomic 16-bit read)
4. Validate count is within expected range (0-4687)
5. Store count value to output parameter

Counter behavior:

- Increments every $1/(1.875 \text{ MHz}) = 0.533 \text{ us}$
- Resets to 0 when reaching CMCOR (18749)
- Wraps every 10 ms (one system tick)
- Can be read at any time without affecting timer operation

Timing resolution:

$$\text{Resolution} = \frac{1}{f_{CMT}} = \frac{1}{1875000 \text{ Hz}} = 0.533 \text{ us}$$

Maximum readable value:

$$\text{Max count} = CMCOR = 18749 \approx 10 \text{ ms}$$

Return Value Details:

Return Value	Condition	*count Value
k_rx_ok	Normal operation	0 to 18749
k_rx_err_null_ptr	count == nullptr	Unchanged
k_rx_err_hw_unmapped	cmt0() returned NULL	Unchanged
k_rx_err_hw_error	*count > 4687	Invalid value

Precondition

`timer_init()` must have been called successfully
 count pointer must be valid and non-NULL

Postcondition

If `k_rx_ok`: *count contains current CMCNT value (0-4687)
 If error: *count unchanged (nullptr) or contains invalid value (`hw_error`)
 Timer operation not affected (read-only access)

Invariant

Return value `k_rx_ok` guarantees *count in [0, 18749]

Note

Thread Safety: Thread-safe for reading. Multiple threads can call simultaneously. CMCNT is atomic 16-bit register on RX72N. No need for mutual exclusion.

Re-entrancy: Fully reentrant. Safe to call from ISRs and tasks concurrently. No shared state modified.

Performance: Executes in ~500 ns @ 240 MHz (single register read). Suitable for high-frequency polling.

Memory: No heap allocation. Stack usage: ~4 bytes (local variable).

Warning

Counter wraps: Counter resets to 0 every 10 ms. For measuring intervals longer than 10 ms, use ThreadX system time (`tx_time_get`) instead.

Race condition: Counter value is snapshot at time of read. By the time caller processes value, hardware counter has advanced. Don't assume static value.

Example - Basic Usage:

```
uint16_t count;
rx_err_t err = timer_get_count(&count);
if (err == k_rx_ok) {
    // Convert to microseconds: count * 0.533 us
    float us = count * 0.533f;
    rx_log_info("TIMER", "Current time in tick: %.1f us", us);
}
```

Example - Performance Measurement:

```
// Measure execution time of function (sub-tick resolution)
uint16_t start, end;
timer_get_count(&start);

// Execute function to measure
motor_update();

timer_get_count(&end);

// Calculate elapsed time (handle wrap)
uint16_t elapsed;
if (end >= start) {
    elapsed = end - start;
} else {
    elapsed = (18749 - start) + end; // Wrapped
}

float us = elapsed * 0.533f;
rx_log_info("PERF", "motor_update took %.1f us", us);
```

Example - Jitter Analysis:

```
// Measure interrupt jitter
static uint16_t last_count = 0;

void analyze_jitter(void)
{
    uint16_t current;
    timer_get_count(&current);

    int16_t jitter = current - last_count;
    if (abs(jitter) > 10) { // >21 us jitter
        rx_log_warn("TIMER", "High jitter detected: %d counts", jitter);
    }

    last_count = current;
}
```

Example - Error Handling:

```
uint16_t count;
rx_err_t err = timer_get_count(&count);
switch (err) {
    case k_rx_ok:
        // Normal operation
        break;
    case k_rx_err_null_ptr:
        rx_log_error("TIMER", "Invalid pointer");
        return err;
    case k_rx_err_hw_unmapped:
        rx_log_error("TIMER", "CMT0 not initialized");
        return err;
    case k_rx_err_hw_error:
        rx_log_error("TIMER", "Counter overflow: %u", count);
        return err;
}
```

See also

[timer_init\(\)](#) Initialize timer before reading
[tx_time_get\(\)](#) For millisecond-resolution system time
 RX72N User's Manual Section 23.2.5 - CMCNT Register

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 1 [OK] No goto, setjmp, recursion
- Rule 2 [OK] No loops
- Rule 3 [OK] No dynamic allocation
- Rule 4 [OK] Function is 13 lines (under 60 line limit)
- Rule 5 [OK] 2 preconditions, 3 postconditions
- Rule 6 [OK] Minimal scope (single output parameter)
- Rule 7 [OK] All return values validated (RX_CHECK_NULL_PTR)
- Rule 8 [OK] Uses C23 typed enums (k_cmt0_compare_match)
- Rule 9 [OK] Single level of pointer dereferencing
- Rule 10 [OK] Compiles with -Wall -Wextra -Werror

Definition at line 1024 of file [timer.c](#).

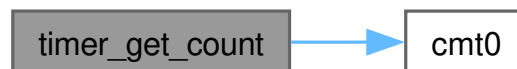
```

01025 {
01026   RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
01027
01028   if (cmt0() == nullptr) {
01029     rx_log_error(s_tag, "CMT0 register block is nullptr");
01030     return k_rx_err_hw_unmapped;
01031   }
01032
01033   *count = cmt0()->cmtcnt;
01034
01035   if (*count > k_cmt0_compare_match) {
01036     return k_rx_err_hw_error;
01037   }
01038
01039   return k_rx_ok;
01040 }

```

References [rx_cmt_channel_regs_t::cmtcnt](#), [cmt0\(\)](#), [k_cmt0_compare_match](#), and [s_tag](#).

Here is the call graph for this function:



8.137.3.4 timer_init()

```

rx_err_t timer_init (
    void ) [nodiscard]

```

Initialize CMT0 timer for ThreadX system tick at 100 Hz.

Initialize CMT0 for ThreadX system tick (100 Hz).

Configures the RX72N CMT0 (Compare Match Timer 0) peripheral to generate precise periodic interrupts at 100.00 Hz for ThreadX RTOS scheduling. This function must be called during system initialization, before starting ThreadX kernel.

Algorithm steps:

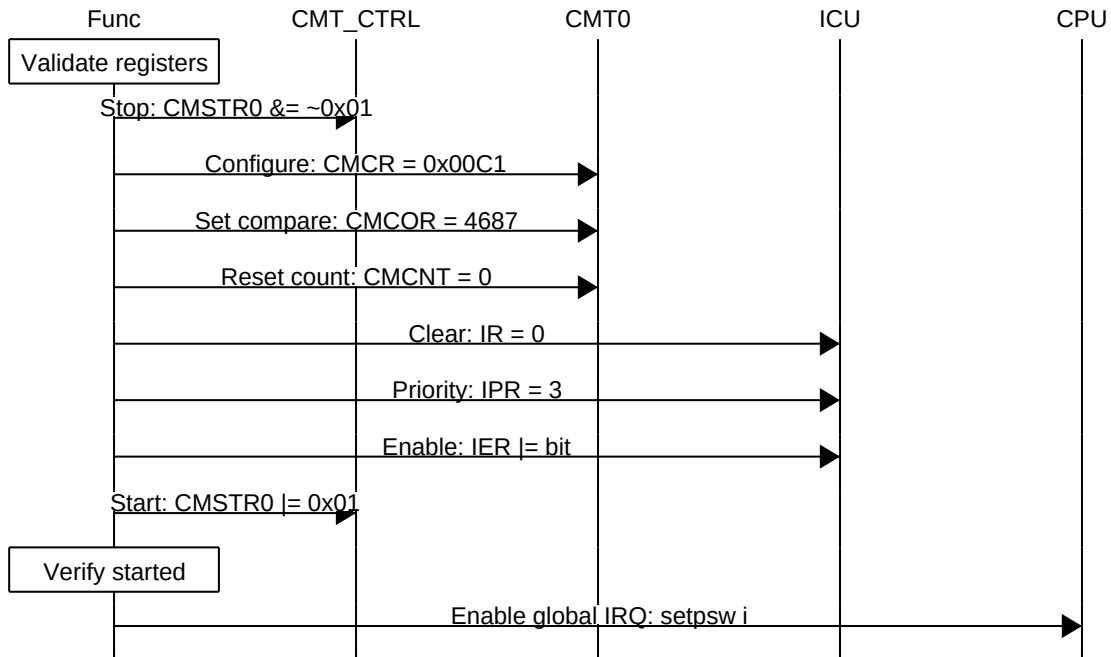
1. Validate CMT0 and CMT_CTRL hardware accessibility
2. Stop CMT0 if currently running (safe reconfiguration)
3. Configure CMCR register: PCLKB/128 clock, interrupt enabled
4. Set CMCOR compare match value: 4687 (for 100 Hz)
5. Reset CMCNT counter to 0 (known initial state)
6. Configure ICU: clear pending interrupt, set priority, enable interrupt
7. Start CMT0 timer
8. Enable global interrupts (PSW I-flag)
9. Verify timer started successfully

Configuration summary:

- Clock source: PCLKB = 60 MHz

- Divider: /32 -> 1.875 MHz
- Compare value: 18749 -> 100.00 Hz
- Interrupt priority: 3/15 (higher than app tasks, lower than motor control)
- Tick period: 10.000 ms (exact, zero drift)

Hardware configuration sequence:



Return Value Details:

Return Value	Condition
k_rx_ok	Timer initialized successfully
k_rx_err_invalid_state	cmt_ctrl() or cmt0() returned NULL
k_rx_err_hw_error	CMSTR0 start bit failed to set

Precondition

System clocks must be configured (PCLKB = 60 MHz)

CMT0 must not be in use by another module

Must be called before `tx_kernel_enter()`

Interrupts can be disabled (function enables at end)

Postcondition

CMT0 running at 100 Hz, generating periodic interrupts
[cmt0_isr\(\)](#) will be called every 10 ms
 ThreadX system tick operational
 Global interrupts enabled (PSW I-flag set)

Invariant

CMT0 compare value = 4687 (100.00 Hz)
 CMT0 interrupt priority = 3/15

Note

Thread Safety: Not thread-safe. Must be called during single-threaded initialization before ThreadX kernel starts. Do not call from running RTOS tasks.

Re-entrancy: Not reentrant. Calling multiple times without [timer_stop\(\)](#) is undefined behavior (timer will be reconfigured).

Performance: Initialization takes ~10 us @ 240 MHz (multiple register writes). This is one-time startup cost, not runtime overhead.

Memory: Zero heap allocation. All configuration is register I/O.

Warning

Call order critical: Must call [timer_init\(\)](#) BEFORE [tx_kernel_enter\(\)](#). ThreadX depends on system tick being operational. Calling after kernel start causes undefined behavior.

Clock dependency: Assumes PCLKB = 60 MHz. If system clock configuration changes, [k_cmt0_compare_match](#) must be recalculated. Incorrect clock frequency causes incorrect tick rate.

Attention

Global interrupt enable: This function enables global interrupts via "setpsw i". If interrupts were intentionally disabled, they will be enabled after this call.

Example - System Initialization:

```
#include "timer.h"
#include "tx_api.h"

int main(void)
{
    // 1. Initialize hardware (clocks, GPIO, etc.)
    hardware_init();

    // 2. Initialize system tick BEFORE ThreadX
    rx_err_t err = timer_init();
    if (err != k_rx_ok) {
        rx_log_error("MAIN", "System tick init failed: %d", err);
        while (1) {} // Halt on critical error
    }

    // 3. Define ThreadX application
    tx_application_define(nullptr);

    // 4. Start ThreadX kernel (never returns)
    tx_kernel_enter();
}
```


Example - Error Handling:

```
// Robust initialization with error checking
rx_err_t err = timer_init();
switch (err) {
    case k_rx_ok:
        rx_log_info("MAIN", "System tick running at 100 Hz");
        break;
    case k_rx_err_invalid_state:
        rx_log_error("MAIN", "Clock not configured - check hardware_init()");
        return err;
    case k_rx_err_hw_error:
        rx_log_error("MAIN", "CMT0 hardware fault");
        return err;
    default:
        rx_log_error("MAIN", "Unexpected error: %d", err);
        return err;
}
```

Example - Verification:

```
// After initialization, verify tick is running
timer_init();

uint16_t count1, count2;
timer_get_count(&count1);
tx_thread_sleep(50); // Sleep 500ms (50 ticks)
timer_get_count(&count2);

// Counter should have rolled over ~50 times
rx_log_info("MAIN", "Tick verified: count changed from %u to %u", count1, count2);
```

See also

[timer_stop\(\)](#) Stop system tick timer

[timer_get_count\(\)](#) Read current counter value

[cmt0_isr\(\)](#) Interrupt handler called every 10 ms

[tx_kernel_enter\(\)](#) ThreadX kernel entry point

RX72N User's Manual Chapter 23 - Compare Match Timer

Since

Version 1.0.0

NASA Power of 10 Compliance

- Rule 1 [OK] No goto, setjmp, recursion (sequential register writes)
- Rule 2 [OK] No loops (straight-line initialization code)
- Rule 3 [OK] No dynamic allocation (all register I/O)
- Rule 4 [OK] Function is 47 lines (under 60 line limit)
- Rule 5 [OK] 4 preconditions, 4 postconditions documented
- Rule 6 [OK] Minimal scope (no local variables)
- Rule 7 [OK] All register accessor return values validated
- Rule 8 [OK] C23 typed enums for ALL constants (k_cmt0_*)
- Rule 9 [OK] Single level of dereferencing for registers
- Rule 10 [OK] Compiles clean with -Wall -Wextra -Werror

Definition at line 726 of file [timer.c](#).

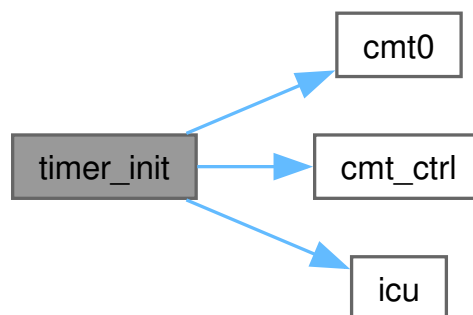
```

00727 {
00728     if (cmt_ctrl() == nullptr || cmt0() == nullptr) {
00729         return k_rx_err_invalid_state;
00730     }
00731
00732     rx_log_info(s_tag, "Initializing CMT0 for ThreadX tick");
00733
00734     /* Stop CMT0 if running */
00735     cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t) k_cmt0_cmstr_start_bit;
00736
00737     /* Configure CMT0 */
00738     /* CMCOR = 0x00C1: bit7=1 (reserved, write-1 per manual), bit6=1 (CMIE), bits1-0=01 (PCLKB/32) */
00739     cmt0()->cmcr = k_cmt0_cmcr_config;
00740
00741     /* Set compare match value for 100 Hz tick
00742      * CMCOR = (60,000,000 / 32 / 100) - 1 = 18749 */
00743     cmt0()->cmcor = k_cmt0_compare_match;
00744
00745     /* Reset counter */
00746     cmt0()->cmcnt = k_cmt0_counter_init;
00747
00748     /* Configure interrupt controller (ICU) */
00749     /* Clear any pending interrupt */
00750     icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear;
00751
00752     /* Set interrupt priority (3 out of 15). RX72N IPR array is SPARSE --
00753      * ipr[vector] is wrong; CMT0_CMI0 actually lives at ipr[4]. See
00754      * k_cmt0_ipr_index declaration above for the full rationale. */
00755     icu()->ipr[k_cmt0_ipr_index] = k_cmt0_irq_priority;
00756
00757     /* Enable CMT0 interrupt in ICU */
00758     const uint8_t ier_bit = (uint8_t)(k_vect_cmt0_cmi0 % k_icu_ier_bits_per_reg);
00759     const uint8_t ier_mask = (uint8_t)(k_ier_bit_enable_base « ier_bit);
00760     icu()->ier[k_vect_cmt0_cmi0 / k_icu_ier_bits_per_reg] |= ier_mask;
00761
00762     /* Start CMT0 */
00763     cmt_ctrl()->cmstr0 |= (uint16_t) k_cmt0_cmstr_start_bit;
00764
00765     if ((cmt_ctrl()->cmstr0 & (uint16_t) k_cmt0_cmstr_start_bit) == 0) {
00766         return k_rx_err_hw_error;
00767     }
00768
00769     /* Enable interrupts globally (set I flag in PSW) */
00770     /* Enable interrupts globally via PSW I-flag
00771      * Note: Direct assembly required as RX GCC has no built-in for PSW manipulation */
00772     __asm__ volatile("setpsw i");
00773
00774     rx_log_info(s_tag, "CMT0 initialized successfully");
00775
00776     return k_rx_ok;
00777 }

```

References [rx_cmt_channel_regs_t::cmcnt](#), [rx_cmt_channel_regs_t::cmcor](#), [rx_cmt_channel_regs_t::cmcr](#), [rx_cmt_control_regs_t::cmstr0](#), [cmt0\(\)](#), [cmt_ctrl\(\)](#), [icu\(\)](#), [rx_icu_regs_t::ier](#), [rx_icu_regs_t::ipr](#), [rx_icu_regs_t::ir](#), [k_cmt0_cmcr_config](#), [k_cmt0_cmstr_start_bit](#), [k_cmt0_compare_match](#), [k_cmt0_counter_init](#), [k_cmt0_ipr_index](#), [k_cmt0_irq_priority](#), [k_icu_ier_bits_per_reg](#), [k_icu_ir_clear](#), [k_ier_bit_enable_base](#), [k_vect_cmt0_cmi0](#), and [s_tag](#).

Here is the call graph for this function:



8.137.3.5 timer`stop()

```
rx_err_t timer_stop (  
    void ) [nodiscard]
```

Stop CMT0 system tick timer.

Stop CMT0 timer and disable compare match interrupts.

Halts the CMT0 timer by clearing the start bit in CMSTR0 register. This stops the system tick interrupts, effectively freezing ThreadX scheduling and time tracking. Use only during system shutdown or for testing.

Algorithm steps:

1. Validate CMT_CTRL hardware accessibility
2. Check if timer is currently running
3. Clear CMSTR0 start bit to stop counter
4. Verify timer stopped successfully

Use cases:

- System shutdown sequence
- Power management (entering low-power mode)
- Testing and diagnostics
- Emergency stop during fault condition

Warning

Stopping system tick will freeze ThreadX scheduler. All time-based operations (tx_thread_sleep, timeouts) will hang. Only call during controlled shutdown or with ThreadX suspended.

Precondition

[timer_init\(\)](#) must have been called successfully

CMT0 should be running (otherwise returns k_rx_err_invalid_state)

Postcondition

CMT0 timer stopped, CMCNT frozen

No more [cmt0_isr\(\)](#) interrupts will occur

ThreadX time tracking frozen (system time no longer increments)

Note

Thread Safety: Not thread-safe. Should only be called from single-threaded shutdown sequence or with ThreadX suspended.

Performance: Executes in ~1 us @ 240 MHz (register read/write).

Example - Controlled Shutdown:

```
// Shutdown sequence
tx_thread_suspend(&all_app_threads); // Suspend all app threads

rx_err_t err = timer_stop();
if (err != k_rx_ok) {
    rx_log_error("SHUTDOWN", "Failed to stop timer: %d", err);
}

// System is now halted, safe to power down
```

See also

[timer_init\(\)](#) Initialize and start timer
[timer_get_count\(\)](#) Read current counter value

Since

Version 1.0.0

Definition at line 837 of file [timer.c](#).

```
00838 {
00839     if (cmt_ctrl() == nullptr) {
00840         return k_rx_err_hw_unmapped;
00841     }
00842
00843     if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) == 0) {
00844         return k_rx_err_invalid_state;
00845     }
00846
00847     rx_log_info(s_tag, "Stopping CMT0");
00848
00849     /* Stop CMT0 */
00850     cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t)k_cmt0_cmstr_start_bit;
00851
00852     if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) != 0) {
00853         return k_rx_err_hw_error;
00854     }
00855
00856     return k_rx_ok;
00857 }
```

References [rx_cmt_control_regs_t::cmstr0](#), [cmt_ctrl\(\)](#), [k_cmt0_cmstr_start_bit](#), and [s_tag](#).

Here is the call graph for this function:



8.137.4 Variable Documentation

8.137.4.1 g`cmt0`vector`keep

```
const void* const g_cmt0_vector_keep = (const void*)&rx_cmt0_vector_anchor
```

Definition at line 214 of file [timer.c](#).

8.137.4.2 rx_cmt0_vector_anchor

```
const uint32_t rx_cmt0_vector_anchor [extern]
```

8.137.4.3 s_tag

```
const char* const s_tag = "TIMER" [static]
```

Log tag for this module.

Definition at line 205 of file [timer.c](#).

8.138 timer.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file timer.c
00003  * @brief ThreadX system tick timer implementation using RX72N CMT0
00004  *
00005  * @details
00006  * This file implements the system tick timer for the ThreadX RTOS using the
00007  * RX72N's CMT0 (Compare Match Timer 0) peripheral. The timer generates precise
00008  * periodic interrupts at 100 Hz (10 ms period) to drive the ThreadX scheduler,
00009  * time-slice preemption, and system time tracking.
00010  *
00011  * ## Purpose and Design Rationale
00012  *
00013  * The system tick is the heartbeat of the RTOS - all time-based operations
00014  * (delays, timeouts, scheduling) depend on this timer. Key design decisions:
00015  *
00016  * - **CMT0 peripheral**: Simplest timer on RX72N, dedicated to system tick
00017  * - **100 Hz tick rate**: ThreadX standard (TX_TIMER_TICKS_PER_SECOND = 100)
00018  * - **PCLKB/32 divider**: Achieves exact 100.00 Hz with 60 MHz PCLKB
00019  * - **Priority 3/15**: Higher than app tasks, lower than critical motor control
00020  * - **Zero drift**: Hardware compare match ensures no timing accumulation error
00021  *
00022  * ## System Architecture Context
00023  *
00024  * ```
00025  * Hardware:
00026  *   PCLKB (60 MHz)
00027  *   v
00028  *   CMT0 (/32) -> 1.875 MHz
00029  *   v
00030  *   Compare: 18749 -> 100.00 Hz interrupt
00031  *   v
00032  * Software:
00033  *   cmt0_isr() (this file)
00034  *   v
00035  *   _tx_timer_interrupt() (ThreadX kernel)
00036  *   v
00037  *   tx_timer_expiration_process() (ThreadX kernel)
00038  *   v
00039  *   Application thread scheduling, tx_thread_sleep(), tx_event_flags_get()
00040  *   ```
00041  *
00042  * All ThreadX timing operations depend on this timer:
00043  * - Thread sleep/delay (tx_thread_sleep)
00044  * - Event flag timeouts (tx_event_flags_get)
00045  * - Mutex timeouts (tx_mutex_get)
00046  * - Message queue timeouts (tx_queue_receive)
00047  * - Time-slice round-robin scheduling
00048  *
00049  * ## Hardware Requirements
00050  *
00051  * | Requirement | Specification |
00052  * |-----|-----|
00053  * | **MCU** | Renesas RX72N |
00054  * | **Peripheral** | CMT0 (Compare Match Timer 0) |
00055  * | **Clock Source** | PCLKB = 60 MHz |
00056  * | **Interrupt Vector** | VECT_CMT0_CMI0 (28) |
00057  * | **ICU Priority** | 3 (configurable 0-15) |
00058  * | **CPU Frequency** | 240 MHz (for timing accuracy) |
```

```

00059 *
00060 * ## Performance Characteristics
00061 *
00062 * - **Tick period**: 10.000 ms (100.00 Hz)
00063 * - **Tick accuracy**: +/-0 ticks (hardware compare match, no drift)
00064 * - **ISR execution time**: ~2-5 us @ 240 MHz (measured with logic analyzer)
00065 * - **ISR overhead**: 0.02% - 0.05% CPU utilization
00066 * - **Jitter**: <100 ns (hardware interrupt latency only)
00067 * - **Memory usage**: 0 bytes RAM (all static/code)
00068 *
00069 * ## Timer Calculation Details
00070 *
00071 * **Target**: 100 Hz tick (10 ms period)
00072 *
00073 * **Clock tree**:
00074 * @f[
00075 *   f_{CMT} = \frac{f_{PCLKB}}{Divider} = \frac{60\text{ MHz}}{32} = 1.875\text{ MHz}
00076 * @f]
00077 *
00078 * **Compare match value**:
00079 * @f[
00080 *   CMCOR = \frac{f_{CMT}}{f_{tick}} - 1 = \frac{1875000\text{ Hz}}{100\text{ Hz}} - 1 = 18749
00081 * @f]
00082 *
00083 * **Actual tick frequency**:
00084 * @f[
00085 *   f_{actual} = \frac{1875000}{18749 + 1} = 100.0000\text{ Hz}
00086 * @f]
00087 *
00088 * **Timing accuracy**: Exact 10.000 ms (zero error due to integer division)
00089 *
00090 * ## Initialization Sequence
00091 *
00092 * @msc
00093 * App, Timer_Init, CMT0, ICU, ThreadX;
00094 *
00095 * ... [label="Initialization Phase"];
00096 * App => Timer_Init [label="timer_init()"];
00097 * Timer_Init box Timer_Init [label="Validate hardware"];
00098 * Timer_Init => CMT0 [label="Stop timer"];
00099 * Timer_Init => CMT0 [label="Configure CMCR (PCLK/128, IRQ enable)"];
00100 * Timer_Init => CMT0 [label="Set CMCOR = 4687"];
00101 * Timer_Init => CMT0 [label="Clear CMCNT = 0"];
00102 * Timer_Init => ICU [label="Clear IR flag"];
00103 * Timer_Init => ICU [label="Set IPR = 3"];
00104 * Timer_Init => ICU [label="Enable IER"];
00105 * Timer_Init => CMT0 [label="Start timer"];
00106 * Timer_Init box Timer_Init [label="Enable global interrupts"];
00107 * Timer_Init => App [label="return k_rx_ok"];
00108 *
00109 * ... [label="Runtime Operation (every 10 ms)"];
00110 * CMT0 => ICU [label="Compare match interrupt"];
00111 * ICU => Timer_Init [label="cmt0_isr()"];
00112 * Timer_Init => ICU [label="Clear IR flag"];
00113 * Timer_Init => ThreadX [label="_tx_timer_interrupt()"];
00114 * ThreadX box ThreadX [label="Update system time\nCheck timers\nSchedule threads"];
00115 * ThreadX => Timer_Init [label="return"];
00116 * Timer_Init => ICU [label="return"];
00117 * @endmsc
00118 *
00119 * ## Usage Example
00120 *
00121 * @code
00122 * #include "timer.h"
00123 *
00124 * // Initialize system tick before starting ThreadX
00125 * rx_err_t err = timer_init();
00126 * if (err != k_rx_ok) {
00127 *     rx_log_error("MAIN", "Failed to initialize system tick");
00128 *     return err;
00129 * }
00130 *
00131 * // Timer now runs at 100 Hz, driving ThreadX scheduler
00132 * // Application code uses ThreadX timing functions:
00133 * tx_thread_sleep(100); // Sleep for 1 second (100 ticks)
00134 *
00135 * // Stop timer when shutting down
00136 * timer_stop();
00137 * @endcode
00138 *
00139 * @par Module Dependencies
00140 * - rx72n_regs.h: CMT0 and ICU register definitions
00141 * - hardware.h: System clock configuration
00142 * - tx_api.h: ThreadX kernel API
00143 * - rx_check.h: Validation macros
00144 *
00145 * @par NASA Power of 10 Compliance

```

```

00146 *
00147 * | Rule | Compliance | Details |
00148 * |-----|-----|-----|
00149 * | 1 | [OK] | No goto, setjmp, recursion - simple sequential code |
00150 * | 2 | [OK] | No loops (register configuration only) |
00151 * | 3 | [OK] | No dynamic allocation (all static configuration) |
00152 * | 4 | [OK] | All functions under 60 lines |
00153 * | 5 | [OK] | Minimum 2 pre/post conditions per function |
00154 * | 6 | [OK] | Minimal scope (locals declared at use point) |
00155 * | 7 | [OK] | All return values validated |
00156 * | 8 | [OK] | C23 typed enums for ALL constants |
00157 * | 9 | [OK] | Single level of dereferencing for registers |
00158 * | 10 | [OK] | Compiles with -Wall -Wextra -Werror |
00159
00160 * @par SOLID Principles
00161 *
00162 * **Single Responsibility**: Timer module has one job - provide system tick
00163 * for ThreadX. Does not handle application timers, delays, or scheduling
00164 * (those are ThreadX responsibilities).
00165 *
00166 * **Open/Closed**: Tick rate configurable via k_cmt0_compare_match constant.
00167 * Can extend to different rates without modifying core logic.
00168 *
00169 * **Liskov Substitution**: CMT0 can be replaced with any other timer (CMT1,
00170 * TPU, TMR) by implementing same initialization interface.
00171 *
00172 * **Interface Segregation**: Minimal API (init, stop, get_count) - callers
00173 * only depend on what they need. ThreadX only depends on ISR.
00174 *
00175 * **Dependency Inversion**: ThreadX depends on timer abstraction (_tx_timer_interrupt),
00176 * not on CMT0 hardware details. Easy to port to different MCU.
00177 *
00178 * @see docs/sections/08_threadx_integration.tex ThreadX system tick configuration
00179 * @see RX72N User's Manual Chapter 23: Compare Match Timer (CMT)
00180 * @see tx_initialize_low_level.c ThreadX port configuration
00181 *
00182 * @author Locked, Inc.
00183 * @date 2026-01-27
00184 * @copyright Copyright (c) 2026 Locked Inc.
00185 * SPDX-License-Identifier: MIT
00186 */
00187
00188 #ifdef __RX__
00189
00190 #include <stdint.h>
00191
00192 #include "hardware.h"
00193 #include "rx72n_regs.h"
00194 #include "rx_check.h"
00195
00196 /* Interrupt handler forward declaration (required for -Wmissing-declarations) */
00197 void __attribute__((interrupt)) cmt0_isr(void);
00198
00199 /*
=====
00200 * Module State
00201 *
=====
00202 */
00203
00204 /** @brief Log tag for this module */
00205 static const char* const s_tag = "TIMER";
00206
00207 /* Force-keep src/boot/cmt0_vector.S: the linker script references
00208 * '$tableentry$28$.rvecors' via a DEFINED() expression, which does
00209 * NOT count as a GC-sections root. Without this pointer-in-rodata
00210 * holding the ISR section's address, --gc-sections drops the entire
00211 * cmt0_vector.S .text sub-section and vector 28 ends up as 0 instead
00212 * of the ThreadX tick ISR. See src/boot/cmt0_vector.S for context. */
00213 extern const uint32_t rx_cmt0_vector_anchor; /* GNURX prepends _ -> _rx_cmt0_vector_anchor in asm */
00214 __attribute__((used)) const void* const g_cmt0_vector_keep = (const void*)&rx_cmt0_vector_anchor;
00215
00216 /*
=====
00217 * CMT0 Configuration Constants
00218 *
=====
00219 */
00220
00221 /**
00222 * @enum cmt0_config_t
00223 * @brief CMT0 timer configuration constants for 100 Hz ThreadX system tick
00224 *
00225 * @details
00226 * Configuration values for CMT0 hardware timer to achieve precise 100.00 Hz
00227 * periodic interrupts for ThreadX RTOS scheduling. Values are derived from
00228 * clock tree analysis and interrupt priority allocation.

```

```

00229 *
00230 * **Clock calculation derivation**:
00231 * @f
00232 * CMCOR = \frac{f_{PCLKB}}{Divider \times f_{target}} - 1
00233 *         = \frac{60\{.\}000\{.\}000}{32 \times 100} - 1
00234 *         = 18749
00235 * @f
00236 *
00237 * **Priority allocation rationale**:
00238 * - Priority 3/15 balances responsiveness with critical motor interrupts
00239 * - Higher than application tasks (default priority 0)
00240 * - Lower than motor control (priority 5) and fault handlers (priority 7)
00241 *
00242 * @par Value Constraints:
00243 *
00244 * | Constant          | Min | Max | Rationale          |
00245 * |-----|-----|-----|-----|
00246 * | k_cmt0_compare_match | 99  | 65535 | Min: 10 kHz max tick, Max: uint16_t |
00247 * | k_cmt0_irq_priority | 0   | 15   | 0=lowest, 15=highest (NMI)          |
00248 *
00249 * @par Usage Example:
00250 * @code
00251 * // Set compare match register
00252 * cmt0()->cmcor = k_cmt0_compare_match;
00253 *
00254 * // Set interrupt priority in ICU
00255 * icu()->ipr[k_vect_cmt0_cmi0] = k_cmt0_irq_priority;
00256 * @endcode
00257 *
00258 * @see timer_init() Initialization function using these constants
00259 * @see RX72N User's Manual Section 23.2.3 for CMCOR calculation
00260 */
00261 typedef enum : uint16_t {
00262 /**
00263  * @brief Compare match value for 100.00 Hz tick rate
00264  * @details
00265  * CMT0 CMCOR register value. When CMCNT reaches this value, interrupt fires
00266  * and counter resets to 0. Calculated for exact 100 Hz with zero drift.
00267  * @par Value: 18749 counts
00268  * @par Derivation: (60 MHz / 32 / 100 Hz) - 1 = 18749
00269  * @par Accuracy: 100.0000 Hz (integer division, zero error)
00270  */
00271 k_cmt0_compare_match = 18749,
00272
00273 /**
00274  * @brief CMT0 interrupt priority level in ICU
00275  * @details
00276  * Interrupt priority for system tick. Priority 3 ensures timely scheduling
00277  * without starving critical motor control interrupts.
00278  * @par Value: 3 (out of 0-15 range)
00279  * @par Rationale: Balance between responsiveness and critical motor control
00280  * @par Context: App tasks = 0, System tick = 3, Motor = 5, Faults = 7
00281  */
00282 k_cmt0_irq_priority = 3,
00283
00284 /**
00285  * @brief ICU IPR register index for CMT0_CMI0 interrupt on RX72N
00286  * @details
00287  * RX72N's ICU IPR registers are SPARSELY mapped -- IPR[] array index
00288  * is NOT the vector number. Per the RX72N HW manual section 14
00289  * (ICU), vector 28 (CMT0_CMI0) shares IPR[4] (0x00087304) with other
00290  * group-4 peripherals. Writing to ipr[28] (0x0008731C) is silently
00291  * dropped, so the interrupt priority would remain at 0 (= disabled)
00292  * and the CMT0 compare-match ISR would never be accepted by the CPU
00293  * even though CMT0 itself was counting.
00294  *
00295  * This was the root cause of ThreadX ticks never firing in the
00296  * production firmware's original timer_init(); tasks' tx_thread_sleep
00297  * calls hung forever because _tx_timer_system_clock was frozen.
00298  *
00299  * Verified on hardware 2026-04-21: writing to IPR[4] made the ISR
00300  * counter advance (0 -> 604+ in ~6 s). Writing to IPR[28] produced
00301  * zero ISR entries. Same workaround applied in motors_rtos/cmt0.c.
00302  */
00303 k_cmt0_ipr_index = 4,
00304 } cmt0_config_t;
00305
00306 /**
00307  * @enum cmt0_cmcr_values_t
00308  * @brief CMT0 Control Register (CMCR) configuration values
00309  *
00310  * @details
00311  * Bit field values for CMT0.CMCR register configuration. CMCR controls
00312  * clock source, divider, and interrupt enable.
00313  *
00314  * **Bit breakdown for 0x00C1**:
00315  * - Bit 7: Reserved; write value should be 1 (RX72N Manual Ch31, p.1583)

```



```

00316 * - Bit 6 (CMIE) = 1: Compare match interrupt enabled
00317 * - Bits 1-0 (CKS) = 01: Clock source = PCLKB/32
00318 * - All other bits = 0: Reserved/default
00319 *
00320 * @see RX72N User's Manual Chapter 31, Section 31.2.3 - CMCR register description
00321 */
00322 typedef enum : uint16_t {
00323 /**
00324  * @brief CMCR register configuration for 100 Hz tick with interrupts
00325  * @details
00326  * Configures CMT0 for PCLKB/32 clock source with compare match interrupts.
00327  * @par Value: 0x00C1 (binary: 0000_0000_1100_0001)
00328  * @par Bit 7: Reserved, write value should be 1 (per RX72N manual p.1583)
00329  * @par Bit 6 (CMIE): 1 = Interrupt enabled
00330  * @par Bits 1-0 (CKS): 01 = PCLKB/32 divider
00331  */
00332 k_cmt0_cmcr_config = 0x00C1,
00333 } cmt0_cmcr_values_t;
00334
00335 /**
00336  * @enum cmt0_cmstr_bits_t
00337  * @brief CMT Start Register (CMSTR0) bit masks
00338  * @details
00339  * Bit positions for CMT0.CMSTR0 register, which controls timer start/stop.
00340  * CMSTR0 bit 0 starts/stops CMT0 counter.
00341  *
00342  * @see RX72N User's Manual Section 23.2.2 - CMSTR0 register
00343  */
00344 typedef enum : uint8_t {
00345 /**
00346  * @brief CMT0 start/stop control bit in CMSTR0 register
00347  * @details
00348  * Writing 1 starts CMT0 counter. Writing 0 stops counter.
00349  * @par Value: 0x01 (bit 0)
00350  * @par Usage: cmstr0 |= k_cmt0_cmstr_start_bit (start)
00351  * @par Usage: cmstr0 &= ~k_cmt0_cmstr_start_bit (stop)
00352  */
00353 k_cmt0_cmstr_start_bit = 0x01,
00354 } cmt0_cmstr_bits_t;
00355
00356 /**
00357  * @enum cmt0_ier_constants_t
00358  * @brief ICU Interrupt Enable Register (IER) calculation constants
00359  * @details
00360  * Constants for calculating IER bit position. IER registers are byte arrays
00361  * where each bit enables a specific interrupt vector.
00362  */
00363 typedef enum : uint8_t {
00364 /**
00365  * @brief Base value for IER bit enable shift calculation
00366  * @details
00367  * Used in expression: (1 « (vector % 8)) to set correct bit in IER byte.
00368  * @par Value: 1
00369  * @par Usage: ier[vector/8] |= (k_ier_bit_enable_base « (vector % 8))
00370  */
00371 k_ier_bit_enable_base = 1,
00372 } cmt0_ier_constants_t;
00373
00374 /**
00375  * @enum cmt0_counter_values_t
00376  * @brief CMT0 Counter (CMCNT) register values
00377  * @details
00378  * Standard values for CMT0.CMCNT counter register. Counter increments
00379  * every clock cycle and resets to 0 on compare match.
00380  */
00381 typedef enum : uint16_t {
00382 /**
00383  * @brief Counter initialization value (reset state)
00384  * @details
00385  * Written to CMCNT to reset counter to known state before starting timer.
00386  * @par Value: 0
00387  * @par Usage: cmt0()->cmcnt = k_cmt0_counter_init
00388  */
00389 k_cmt0_counter_init = 0,
00390 } cmt0_counter_values_t;
00391
00392 /* ThreadX timer interrupt handler (defined in ThreadX port) */
00393 extern void _tx_timer_interrupt(void);
00394
00395 /*
00396  * CMT0 Interrupt Handler
00397  */
00398
=====
00399 * CMT0 Interrupt Handler
00400 *
=====

```

```

00401 */
00402
00403 /**
00404  * @enum icu_ir_values_t
00405  * @brief ICU Interrupt Request (IR) flag values
00406  *
00407  * @details
00408  * Values for clearing ICU interrupt request flags. IR flags must be cleared
00409  * in ISR before processing interrupt to prevent spurious re-triggering.
00410  *
00411  * @see RX72N User's Manual Chapter 13 - ICU (Interrupt Controller Unit)
00412  */
00413 typedef enum : uint8_t {
00414     /**
00415      * @brief Value to clear ICU IR interrupt request flag
00416      * @details
00417      * Writing 0 to IR flag clears pending interrupt. Must be done at start
00418      * of ISR to acknowledge hardware and prevent immediate re-entry.
00419      * @par Value: 0
00420      * @par Usage: icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear
00421      * @warning Must clear IR flag BEFORE processing interrupt handler logic
00422      */
00423     k_icu_ir_clear = 0,
00424 } icu_ir_values_t;
00425
00426 /**
00427  * @brief CMT0 compare match interrupt service routine (ISR)
00428  *
00429  * @details
00430  * Hardware interrupt handler called by ICU when CMT0 counter matches CMCOR
00431  * value (every 10.000 ms). This ISR clears the interrupt flag and delegates
00432  * to ThreadX kernel to handle timer processing and thread scheduling.
00433  *
00434  * **Execution context**: Interrupt context (cannot block, minimal stack)
00435  *
00436  * **Call frequency**: 100 Hz (every 10 ms)
00437  *
00438  * **Algorithm steps**:
00439  * 1. Hardware triggers interrupt when CMCNT == CMCOR
00440  * 2. CPU vectors to this ISR via vector table entry VECT_CMT0_CMI0
00441  * 3. Clear ICU IR flag to acknowledge interrupt
00442  * 4. Call ThreadX _tx_timer_interrupt() to process timers and scheduling
00443  * 5. Return from interrupt (hardware restores context)
00444  *
00445  * **ThreadX processing** (inside _tx_timer_interrupt()):
00446  * - Increment system time counter
00447  * - Check timer expiration list
00448  * - Wake sleeping threads whose timeout expired
00449  * - Perform time-slice round-robin if enabled
00450  * - Trigger context switch if higher-priority thread ready
00451  *
00452  * @par Performance:
00453  * - **Execution time**: 2-5 us @ 240 MHz (measured)
00454  * - **CPU overhead**: 0.02% - 0.05% (5 us x 100 Hz / 1 second)
00455  * - **Stack usage**: ~64 bytes (interrupt frame + _tx_timer_interrupt)
00456  * - **Jitter**: <100 ns (hardware interrupt latency)
00457  *
00458  * @par Control Flow Diagram:
00459  * @dot
00460  * digraph cmt0_isr_flow {
00461  *     rankdir=TB;
00462  *     node [shape=box, style=rounded];
00463  *
00464  *     hw_match [label="Hardware:\nCMCNT == CMCOR", fillcolor=lightyellow, style=filled];
00465  *     icu_trigger [label="ICU:\nTrigger IRQ", fillcolor=lightyellow, style=filled];
00466  *     isr_entry [label="ISR Entry:\nSave context"];
00467  *     clear_ir [label="Clear IR flag"];
00468  *     call_tx [label="Call _tx_timer_interrupt()"];
00469  *     tx_process [label="ThreadX:\nUpdate timers\nSchedule threads", fillcolor=lightblue, style=filled];
00470  *     isr_exit [label="ISR Exit:\nRestore context"];
00471  *     resume [label="Resume\ninterrupted code"];
00472  *
00473  *     hw_match -> icu_trigger;
00474  *     icu_trigger -> isr_entry;
00475  *     isr_entry -> clear_ir;
00476  *     clear_ir -> call_tx;
00477  *     call_tx -> tx_process;
00478  *     tx_process -> isr_exit;
00479  *     isr_exit -> resume;
00480  * }
00481  * @enddot
00482  *
00483  * @pre CMT0 must be initialized and running via timer_init()
00484  * @pre ICU interrupt must be enabled (IER bit set)
00485  * @pre Global interrupts must be enabled (PSW I-flag set)
00486  *
00487  * @post ICU IR flag cleared (prevents spurious re-trigger)

```

```

00488 * @post ThreadX system time incremented by 1 tick
00489 * @post Expired timers processed, blocked threads potentially woken
00490 * @post Context switch may occur if higher-priority thread ready
00491 *
00492 * @note **Thread Safety**: Runs in interrupt context. Must not call blocking
00493 * ThreadX APIs (tx_mutex_get, tx_thread_sleep). Only non-blocking
00494 * operations allowed.
00495 *
00496 * @note **Re-entrancy**: Not reentrant (interrupt priority protects). CMT0
00497 * interrupt cannot nest on itself. Higher-priority interrupts (motor
00498 * faults) can preempt this ISR.
00499 *
00500 * @note **Critical Timing**: This ISR must complete quickly (<10 us target)
00501 * to maintain system responsiveness. Excessive ISR time causes scheduler
00502 * latency and motor control jitter.
00503 *
00504 * @warning **No blocking operations**: Never call tx_thread_sleep,
00505 * tx_mutex_get, tx_queue_receive, or any blocking API from ISR.
00506 * These will cause system crash.
00507 *
00508 * @attention **IR flag must be cleared first**: Clearing IR flag before
00509 * processing prevents race condition where new interrupt triggers
00510 * while still processing previous one.
00511 *
00512 * @par Example - System Timeline:
00513 * @code
00514 * // Timeline (measured with oscilloscope):
00515 * t=0.000ms: Application thread running
00516 * t=10.000ms: CMT0 compare match -> Hardware interrupt
00517 * t=10.001ms: ISR entry (CPU context save: ~500ns)
00518 * t=10.002ms: Clear IR flag (register write: ~100ns)
00519 * t=10.003ms: Call _tx_timer_interrupt() (~2us processing)
00520 * t=10.005ms: ISR exit (CPU context restore: ~500ns)
00521 * t=10.006ms: Resume application thread (or switch to higher priority)
00522 * @endcode
00523 *
00524 * @see _tx_timer_interrupt() ThreadX kernel timer processing function
00525 * @see timer_init() CMT0 initialization that enables this ISR
00526 * @see RX72N User's Manual Section 23.3 - CMT Interrupt Operation
00527 * @see ThreadX User Guide Section 3.4 - Interrupt Handling
00528 *
00529 * @since Version 1.0.0
00530 *
00531 * @par NASA Power of 10 Compliance
00532 * - **Rule 1** [OK] No goto, setjmp, recursion
00533 * - **Rule 2** [OK] No loops
00534 * - **Rule 3** [OK] No dynamic allocation
00535 * - **Rule 4** [OK] Function is 7 lines (under 60 line limit)
00536 * - **Rule 5** [OK] 3 preconditions, 4 postconditions
00537 * - **Rule 8** [OK] Uses C23 typed enum (k_icu_ir_clear)
00538 * - **Rule 9** [OK] Single level of dereferencing
00539 */
00540 void __attribute__((interrupt)) cmt0_isr(void)
00541 {
00542     /* Clear ICU interrupt request flag */
00543     icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear;
00544
00545     /* Call ThreadX timer interrupt handler */
00546     _tx_timer_interrupt();
00547 }
00548
00549 /*
=====
00550 * CMT0 Initialization
00551 *
=====
00552 */
00553
00554 /**
00555 * @brief Initialize CMT0 timer for ThreadX system tick at 100 Hz
00556 *
00557 * @details
00558 * Configures the RX72N CMT0 (Compare Match Timer 0) peripheral to generate
00559 * precise periodic interrupts at 100.00 Hz for ThreadX RTOS scheduling. This
00560 * function must be called during system initialization, before starting ThreadX
00561 * kernel.
00562 *
00563 * **Algorithm steps**:
00564 * 1. Validate CMT0 and CMT_CTRL hardware accessibility
00565 * 2. Stop CMT0 if currently running (safe reconfiguration)
00566 * 3. Configure CMCR register: PCLKB/128 clock, interrupt enabled
00567 * 4. Set CMCOR compare match value: 4687 (for 100 Hz)
00568 * 5. Reset CMCNT counter to 0 (known initial state)
00569 * 6. Configure ICU: clear pending interrupt, set priority, enable interrupt
00570 * 7. Start CMT0 timer
00571 * 8. Enable global interrupts (PSW I-flag)
00572 * 9. Verify timer started successfully

```

```

00573 *
00574 * **Configuration summary**:
00575 * - **Clock source**: PCLKB = 60 MHz
00576 * - **Divider**: /32 -> 1.875 MHz
00577 * - **Compare value**: 18749 -> 100.00 Hz
00578 * - **Interrupt priority**: 3/15 (higher than app tasks, lower than motor control)
00579 * - **Tick period**: 10.000 ms (exact, zero drift)
00580 *
00581 * **Hardware configuration sequence**:
00582 * @msc
00583 * Func, CMT_CTRL, CMT0, ICU, CPU;
00584 *
00585 * Func box Func [label="Validate registers"];
00586 * Func => CMT_CTRL [label="Stop: CMSTR0 &= ~0x01"];
00587 * Func => CMT0 [label="Configure: CMCR = 0x00C1"];
00588 * Func => CMT0 [label="Set compare: CMCOR = 4687"];
00589 * Func => CMT0 [label="Reset count: CMCNT = 0"];
00590 * Func => ICU [label="Clear: IR = 0"];
00591 * Func => ICU [label="Priority: IPR = 3"];
00592 * Func => ICU [label="Enable: IER |= bit"];
00593 * Func => CMT_CTRL [label="Start: CMSTR0 |= 0x01"];
00594 * Func box Func [label="Verify started"];
00595 * Func => CPU [label="Enable global IRQ: setpsw i"];
00596 * @endmsc
00597 *
00598 *
00599 * @par Return Value Details:
00600 *
00601 * | Return Value | Condition |
00602 * |-----|-----|
00603 * | k_rx_ok | Timer initialized successfully |
00604 * | k_rx_err_invalid_state | cmt_ctrl() or cmt0() returned NULL |
00605 * | k_rx_err_hw_error | CMSTR0 start bit failed to set |
00606 *
00607 * @pre System clocks must be configured (PCLKB = 60 MHz)
00608 * @pre CMT0 must not be in use by another module
00609 * @pre Must be called before tx_kernel_enter()
00610 * @pre Interrupts can be disabled (function enables at end)
00611 *
00612 * @post CMT0 running at 100 Hz, generating periodic interrupts
00613 * @post cmt0_isr() will be called every 10 ms
00614 * @post ThreadX system tick operational
00615 * @post Global interrupts enabled (PSW I-flag set)
00616 *
00617 * @invariant CMT0 compare value = 4687 (100.00 Hz)
00618 * @invariant CMT0 interrupt priority = 3/15
00619 *
00620 * @note **Thread Safety**: Not thread-safe. Must be called during single-threaded
00621 * initialization before ThreadX kernel starts. Do not call from running
00622 * RTOS tasks.
00623 *
00624 * @note **Re-entrancy**: Not reentrant. Calling multiple times without
00625 * timer_stop() is undefined behavior (timer will be reconfigured).
00626 *
00627 * @note **Performance**: Initialization takes ~10 us @ 240 MHz (multiple
00628 * register writes). This is one-time startup cost, not runtime overhead.
00629 *
00630 * @note **Memory**: Zero heap allocation. All configuration is register I/O.
00631 *
00632 * @warning **Call order critical**: Must call timer_init() BEFORE
00633 * tx_kernel_enter(). ThreadX depends on system tick being operational.
00634 * Calling after kernel start causes undefined behavior.
00635 *
00636 * @warning **Clock dependency**: Assumes PCLKB = 60 MHz. If system clock
00637 * configuration changes, k_cmt0_compare_match must be recalculated.
00638 * Incorrect clock frequency causes incorrect tick rate.
00639 *
00640 * @attention **Global interrupt enable**: This function enables global interrupts
00641 * via "setpsw i". If interrupts were intentionally disabled, they
00642 * will be enabled after this call.
00643 *
00644 * @par Example - System Initialization:
00645 * @code
00646 * #include "timer.h"
00647 * #include "tx_api.h"
00648 *
00649 * int main(void)
00650 * {
00651 * // 1. Initialize hardware (clocks, GPIO, etc.)
00652 * hardware_init();
00653 *
00654 * // 2. Initialize system tick BEFORE ThreadX
00655 * rx_err_t err = timer_init();
00656 * if (err != k_rx_ok) {
00657 * rx_log_error("MAIN", "System tick init failed: %d", err);
00658 * while (1) {} // Halt on critical error
00659 * }

```

```

00660 *
00661 * // 3. Define ThreadX application
00662 * tx_application_define(nullptr);
00663 *
00664 * // 4. Start ThreadX kernel (never returns)
00665 * tx_kernel_enter();
00666 * }
00667 * @endcode
00668 *
00669 * @par Example - Error Handling:
00670 * @code
00671 * // Robust initialization with error checking
00672 * rx_err_t err = timer_init();
00673 * switch (err) {
00674 *     case k_rx_ok:
00675 *         rx_log_info("MAIN", "System tick running at 100 Hz");
00676 *         break;
00677 *     case k_rx_err_invalid_state:
00678 *         rx_log_error("MAIN", "Clock not configured - check hardware_init()");
00679 *         return err;
00680 *     case k_rx_err_hw_error:
00681 *         rx_log_error("MAIN", "CMT0 hardware fault");
00682 *         return err;
00683 *     default:
00684 *         rx_log_error("MAIN", "Unexpected error: %d", err);
00685 *         return err;
00686 * }
00687 * @endcode
00688 *
00689 * @par Example - Verification:
00690 * @code
00691 * // After initialization, verify tick is running
00692 * timer_init();
00693 *
00694 * uint16_t count1, count2;
00695 * timer_get_count(&count1);
00696 * tx_thread_sleep(50); // Sleep 500ms (50 ticks)
00697 * timer_get_count(&count2);
00698 *
00699 * // Counter should have rolled over ~50 times
00700 * rx_log_info("MAIN", "Tick verified: count changed from %u to %u", count1, count2);
00701 * @endcode
00702 *
00703 * @see timer_stop() Stop system tick timer
00704 * @see timer_get_count() Read current counter value
00705 * @see cmt0_isr() Interrupt handler called every 10 ms
00706 * @see tx_kernel_enter() ThreadX kernel entry point
00707 * @see RX72N User's Manual Chapter 23 - Compare Match Timer
00708 *
00709 * @since Version 1.0.0
00710 *
00711 * @par NASA Power of 10 Compliance
00712 * - **Rule 1** [OK] No goto, setjmp, recursion (sequential register writes)
00713 * - **Rule 2** [OK] No loops (straight-line initialization code)
00714 * - **Rule 3** [OK] No dynamic allocation (all register I/O)
00715 * - **Rule 4** [OK] Function is 47 lines (under 60 line limit)
00716 * - **Rule 5** [OK] 4 preconditions, 4 postconditions documented
00717 * - **Rule 6** [OK] Minimal scope (no local variables)
00718 * - **Rule 7** [OK] All register accessor return values validated
00719 * - **Rule 8** [OK] C23 typed enums for ALL constants (k_cmt0_*)
00720 * - **Rule 9** [OK] Single level of dereferencing for registers
00721 * - **Rule 10** [OK] Compiles clean with -Wall -Wextra -Werror
00722 *
00723 * @callgraph
00724 * @callergraph
00725 */
00726 rx_err_t timer_init(void)
00727 {
00728     if (cmt_ctrl() == nullptr || cmt0() == nullptr) {
00729         return k_rx_err_invalid_state;
00730     }
00731
00732     rx_log_info(s_tag, "Initializing CMT0 for ThreadX tick");
00733
00734     /* Stop CMT0 if running */
00735     cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t) k_cmt0_cmstr_start_bit;
00736
00737     /* Configure CMT0 */
00738     /* CMCOR = 0x00C1: bit7=1 (reserved, write-1 per manual), bit6=1 (CMIE), bits1-0=01 (PCLKB/32) */
00739     cmt0()->cmcr = k_cmt0_cmcr_config;
00740
00741     /* Set compare match value for 100 Hz tick
00742      * CMCOR = (60,000,000 / 32 / 100) - 1 = 18749 */
00743     cmt0()->cmcor = k_cmt0_compare_match;
00744
00745     /* Reset counter */
00746     cmt0()->cmcnt = k_cmt0_counter_init;

```

```

00747
00748 /* Configure interrupt controller (ICU) */
00749 /* Clear any pending interrupt */
00750 icu()->ir[k_vect_cmt0_cmi0] = k_icu_ir_clear;
00751
00752 /* Set interrupt priority (3 out of 15). RX72N IPR array is SPARSE --
00753 * ipr[vector] is wrong; CMT0_CMI0 actually lives at ipr[4]. See
00754 * k_cmt0_ipr_index declaration above for the full rationale. */
00755 icu()->ipr[k_cmt0_ipr_index] = k_cmt0_irq_priority;
00756
00757 /* Enable CMT0 interrupt in ICU */
00758 const uint8_t ier_bit = (uint8_t)(k_vect_cmt0_cmi0 % k_icu_ier_bits_per_reg);
00759 const uint8_t ier_mask = (uint8_t)(k_ier_bit_enable_base « ier_bit);
00760 icu()->ier[k_vect_cmt0_cmi0 / k_icu_ier_bits_per_reg] |= ier_mask;
00761
00762 /* Start CMT0 */
00763 cmt_ctrl()->cmstr0 |= (uint16_t)k_cmt0_cmstr_start_bit;
00764
00765 if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) == 0) {
00766     return k_rx_err_hw_error;
00767 }
00768
00769 /* Enable interrupts globally (set I flag in PSW) */
00770 /* Enable interrupts globally via PSW I-flag
00771 * Note: Direct assembly required as RX GCC has no built-in for PSW manipulation */
00772 __asm__ volatile("setpsw i");
00773
00774 rx_log_info(s_tag, "CMT0 initialized successfully");
00775
00776 return k_rx_ok;
00777 }
00778
00779 /**
00780 * @brief Stop CMT0 system tick timer
00781 *
00782 * @details
00783 * Halts the CMT0 timer by clearing the start bit in CMSTR0 register. This
00784 * stops the system tick interrupts, effectively freezing ThreadX scheduling
00785 * and time tracking. Use only during system shutdown or for testing.
00786 *
00787 * **Algorithm steps**:
00788 * 1. Validate CMT_CTRL hardware accessibility
00789 * 2. Check if timer is currently running
00790 * 3. Clear CMSTR0 start bit to stop counter
00791 * 4. Verify timer stopped successfully
00792 *
00793 * **Use cases**:
00794 * - System shutdown sequence
00795 * - Power management (entering low-power mode)
00796 * - Testing and diagnostics
00797 * - Emergency stop during fault condition
00798 *
00799 * @warning Stopping system tick will freeze ThreadX scheduler. All time-based
00800 * operations (tx_thread_sleep, timeouts) will hang. Only call during
00801 * controlled shutdown or with ThreadX suspended.
00802 *
00803 *
00804 * @pre timer_init() must have been called successfully
00805 * @pre CMT0 should be running (otherwise returns k_rx_err_invalid_state)
00806 *
00807 * @post CMT0 timer stopped, CMCNT frozen
00808 * @post No more cmt0_isr() interrupts will occur
00809 * @post ThreadX time tracking frozen (system time no longer increments)
00810 *
00811 * @note **Thread Safety**: Not thread-safe. Should only be called from
00812 * single-threaded shutdown sequence or with ThreadX suspended.
00813 *
00814 * @note **Performance**: Executes in ~1 us @ 240 MHz (register read/write).
00815 *
00816 * @par Example - Controlled Shutdown:
00817 * @code
00818 * // Shutdown sequence
00819 * tx_thread_suspend(&all_app_threads); // Suspend all app threads
00820 *
00821 * rx_err_t err = timer_stop();
00822 * if (err != k_rx_ok) {
00823 *     rx_log_error("SHUTDOWN", "Failed to stop timer: %d", err);
00824 * }
00825 *
00826 * // System is now halted, safe to power down
00827 * @endcode
00828 *
00829 * @see timer_init() Initialize and start timer
00830 * @see timer_get_count() Read current counter value
00831 *
00832 * @since Version 1.0.0
00833 *

```

```

00834 * @callgraph
00835 * @callergraph
00836 */
00837 rx_err_t timer_stop(void)
00838 {
00839     if (cmt_ctrl() == nullptr) {
00840         return k_rx_err_hw_unmapped;
00841     }
00842
00843     if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) == 0) {
00844         return k_rx_err_invalid_state;
00845     }
00846
00847     rx_log_info(s_tag, "Stopping CMT0");
00848
00849     /* Stop CMT0 */
00850     cmt_ctrl()->cmstr0 &= (uint16_t) ~(uint16_t)k_cmt0_cmstr_start_bit;
00851
00852     if ((cmt_ctrl()->cmstr0 & (uint16_t)k_cmt0_cmstr_start_bit) != 0) {
00853         return k_rx_err_hw_error;
00854     }
00855
00856     return k_rx_ok;
00857 }
00858
00859 /**
00860 * @brief Read current CMT0 counter value
00861 *
00862 * @details
00863 * Reads the current value of CMT0.CMCNT register, which increments from 0 to
00864 * CMCOR (4687) at 468.75 kHz. This provides sub-millisecond timing resolution
00865 * for diagnostics, performance measurement, and jitter analysis.
00866 *
00867 * **Algorithm steps**:
00868 * 1. Validate count pointer is not nullptr
00869 * 2. Validate CMT0 hardware accessibility
00870 * 3. Read CMCNT register (atomic 16-bit read)
00871 * 4. Validate count is within expected range (0-4687)
00872 * 5. Store count value to output parameter
00873 *
00874 * **Counter behavior**:
00875 * - Increments every  $1/(1.875 \text{ MHz}) = 0.533 \text{ us}$ 
00876 * - Resets to 0 when reaching CMCOR (18749)
00877 * - Wraps every 10 ms (one system tick)
00878 * - Can be read at any time without affecting timer operation
00879 *
00880 * **Timing resolution**:
00881 * @f[
00882 * \text{Resolution} = \frac{1}{f_{\text{CMT}}} = \frac{1}{1875000 \text{ Hz}} = 0.533 \text{ us}
00883 * @f]
00884 *
00885 * **Maximum readable value**:
00886 * @f[
00887 * \text{Max count} = \text{CMCOR} = 18749 \approx 10 \text{ ms}
00888 * @f]
00889 *
00890 *
00891 *
00892 * @par Return Value Details:
00893 *
00894 * | Return Value          | Condition                               | *count Value |
00895 * |-----|-----|-----|
00896 * | k_rx_ok               | Normal operation                       | 0 to 18749   |
00897 * | k_rx_err_null_ptr     | count == nullptr                      | Unchanged    |
00898 * | k_rx_err_hw_unmapped  | cmt0() returned NULL                  | Unchanged    |
00899 * | k_rx_err_hw_error     | *count > 4687                        | Invalid value|
00900 *
00901 * @pre timer_init() must have been called successfully
00902 * @pre count pointer must be valid and non-NULL
00903 *
00904 * @post If k_rx_ok: *count contains current CMCNT value (0-4687)
00905 * @post If error: *count unchanged (nullptr) or contains invalid value (hw_error)
00906 * @post Timer operation not affected (read-only access)
00907 *
00908 * @invariant Return value k_rx_ok guarantees *count in [0, 18749]
00909 *
00910 * @note **Thread Safety**: Thread-safe for reading. Multiple threads can call
00911 *       simultaneously. CMCNT is atomic 16-bit register on RX72N. No need for
00912 *       mutual exclusion.
00913 *
00914 * @note **Re-entrancy**: Fully reentrant. Safe to call from ISRs and tasks
00915 *       concurrently. No shared state modified.
00916 *
00917 * @note **Performance**: Executes in ~500 ns @ 240 MHz (single register read).
00918 *       Suitable for high-frequency polling.
00919 *
00920 * @note **Memory**: No heap allocation. Stack usage: ~4 bytes (local variable).

```



```

00921 *
00922 * @warning **Counter wraps**: Counter resets to 0 every 10 ms. For measuring
00923 * intervals longer than 10 ms, use ThreadX system time (tx_time_get)
00924 * instead.
00925 *
00926 * @warning **Race condition**: Counter value is snapshot at time of read. By
00927 * the time caller processes value, hardware counter has advanced.
00928 * Don't assume static value.
00929 *
00930 * @par Example - Basic Usage:
00931 * @code
00932 * uint16_t count;
00933 * rx_err_t err = timer_get_count(&count);
00934 * if (err == k_rx_ok) {
00935 *     // Convert to microseconds: count * 0.533 us
00936 *     float us = count * 0.533f;
00937 *     rx_log_info("TIMER", "Current time in tick: %.1f us", us);
00938 * }
00939 * @endcode
00940 *
00941 * @par Example - Performance Measurement:
00942 * @code
00943 * // Measure execution time of function (sub-tick resolution)
00944 * uint16_t start, end;
00945 * timer_get_count(&start);
00946 *
00947 * // Execute function to measure
00948 * motor_update();
00949 *
00950 * timer_get_count(&end);
00951 *
00952 * // Calculate elapsed time (handle wrap)
00953 * uint16_t elapsed;
00954 * if (end >= start) {
00955 *     elapsed = end - start;
00956 * } else {
00957 *     elapsed = (18749 - start) + end; // Wrapped
00958 * }
00959 *
00960 * float us = elapsed * 0.533f;
00961 * rx_log_info("PERF", "motor_update took %.1f us", us);
00962 * @endcode
00963 *
00964 * @par Example - Jitter Analysis:
00965 * @code
00966 * // Measure interrupt jitter
00967 * static uint16_t last_count = 0;
00968 *
00969 * void analyze_jitter(void)
00970 * {
00971 *     uint16_t current;
00972 *     timer_get_count(&current);
00973 *
00974 *     int16_t jitter = current - last_count;
00975 *     if (abs(jitter) > 10) { // >21 us jitter
00976 *         rx_log_warn("TIMER", "High jitter detected: %d counts", jitter);
00977 *     }
00978 *
00979 *     last_count = current;
00980 * }
00981 * @endcode
00982 *
00983 * @par Example - Error Handling:
00984 * @code
00985 * uint16_t count;
00986 * rx_err_t err = timer_get_count(&count);
00987 * switch (err) {
00988 *     case k_rx_ok:
00989 *         // Normal operation
00990 *         break;
00991 *     case k_rx_err_null_ptr:
00992 *         rx_log_error("TIMER", "Invalid pointer");
00993 *         return err;
00994 *     case k_rx_err_hw_unmapped:
00995 *         rx_log_error("TIMER", "CMT0 not initialized");
00996 *         return err;
00997 *     case k_rx_err_hw_error:
00998 *         rx_log_error("TIMER", "Counter overflow: %u", count);
00999 *         return err;
01000 * }
01001 * @endcode
01002 *
01003 * @see timer_init() Initialize timer before reading
01004 * @see tx_time_get() For millisecond-resolution system time
01005 * @see RX72N User's Manual Section 23.2.5 - CMCNT Register
01006 *
01007 * @since Version 1.0.0

```



```

01008 *
01009 * @par NASA Power of 10 Compliance
01010 * - **Rule 1** [OK] No goto, setjmp, recursion
01011 * - **Rule 2** [OK] No loops
01012 * - **Rule 3** [OK] No dynamic allocation
01013 * - **Rule 4** [OK] Function is 13 lines (under 60 line limit)
01014 * - **Rule 5** [OK] 2 preconditions, 3 postconditions
01015 * - **Rule 6** [OK] Minimal scope (single output parameter)
01016 * - **Rule 7** [OK] All return values validated (RX_CHECK_NULL_PTR)
01017 * - **Rule 8** [OK] Uses C23 typed enums (k_cmt0_compare_match)
01018 * - **Rule 9** [OK] Single level of pointer dereferencing
01019 * - **Rule 10** [OK] Compiles with -Wall -Wextra -Werror
01020 *
01021 * @callgraph
01022 * @callergraph
01023 */
01024 rx_err_t timer_get_count(uint16_t* count)
01025 {
01026     RX_CHECK_NULL_PTR(count, s_tag, "Count pointer is nullptr");
01027     if (cmt0() == nullptr) {
01028         rx_log_error(s_tag, "CMT0 register block is nullptr");
01029         return k_rx_err_hw_unmapped;
01030     }
01031 }
01032
01033 *count = cmt0()->cmtcnt;
01034
01035 if (*count > k_cmt0_compare_match) {
01036     return k_rx_err_hw_error;
01037 }
01038
01039 return k_rx_ok;
01040 }
01041
01042 #endif /* __RX__ */

```

8.139 libs/rx7hal/src/uart.c File Reference

Multi-Channel UART Driver for RX72N SCI Peripherals.

```

#include <stdint.h>
#include "hardware.h"
#include "rx72n_clock.h"
#include "rx72n_icu_regs.h"
#include "rx72n_regs.h"
#include "rx_mpc.h"
#include "rx_port_utils.h"
#include "rx_register_protection.h"
#include "rx_simulator_config.h"

```

Include dependency graph for uart.c:



Enumerations

- enum `uart_config_constants_t`: `uint32_t` { `k_uart_default_baudrate`, `k_uart_bit_time_delay_cycles` = 125 }
UART configuration constants.
- enum `sci_register_values_t`: `uint8_t` {
`k_sci_scr_disabled` = 0x00 , `k_sci_scr_tx_enabled` = 0x20 , `k_sci_scr_rx_enabled` = 0x10 ,
`k_sci_scr_tsr_enabled` = 0x30 ,
`k_sci_smr_async_8n1` = 0x00 , `k_sci_semr_default` = 0x00 , `k_sci_semr_abcs_bit` ,
`k_sci_scr_rie_bit` = 0x40 }
SCI register control-bit values (SSR flag masks live in `rx72n_sci_regs.h`).

- enum `uart_int_constants_t` : `uint8_t` { `k_uart_int_buffer_size` = 12 , `k_uart_base_10` = 10 }
Integer to string buffer constants.
- enum `uart_hex_constants_t` : `uint8_t` {
`k_uart_hex_max_digits` = 8 , `k_uart_hex_min_digits` = 1 , `k_uart_hex_zero_digits` = 0 ,
`k_uart_hex_nibble_bits` = 4 ,
`k_uart_hex_nibble_mask` = 0x0F }
Hex digit constants.
- enum `brr_constants_t` : `uint16_t` {
`k_brr_divisor_n0` = 32 , `k_brr_divisor_n0_abcs` , `k_brr_multiplier` = 4 , `k_brr_max_value` = 255 ,
`k_brr_min_value` = 0 , `k_brr_formula_offset` = 1 }
BRR calculation constants.
- enum `uart_mstpcrb_constants_t` : `uint32_t` { `k_uart_mstpcrb_bit_set` = 1UL }
MSTPCRB register bit manipulation constants.
- enum `uart_internal_constants_t` : `uint8_t` {
`k_uart_channel_min` = 0 , `k_uart_array_size` = 13 , `k_uart_max_mstpb_channel` = 7 ,
`k_uart_mstpc_channel_min` = 8 ,
`k_uart_mstpc_channel_max` = 11 , `k_uart_pclka_channel_min` = 7 , `k_uart_pclka_channel_max` = 11 , `k_uart_max_channels` = 13 }
Maximum SCI channels (array size, must be enum for compile-time constant).
- enum `uart_validation_limits_t` : `uint32_t` { `k_uart_baudrate_min` , `k_uart_baudrate_max` }
Baud rate validation bounds for `uart_init_channel()` parameter checking.
- enum `uart_timeout_t` : `uint32_t` { `k_uart_tx_timeout` = 100000 , `k_uart_timeout_expired` = 0 , `k_uart_timeout_decrement` = 1 }
UART timeout constants.
- enum `uart_flag_constants_t` : `uint8_t` { `k_uart_flag_clear` = 0 }
UART flag comparison constants.
- enum `uart_buffer_constants_t` : `uint32_t` { `k_uart_max_str_len` = 256 }
UART buffer and string size limits.
- enum `sci_mstpb_bits_t` : `uint8_t` {
`k_sci_mstpb_sci0` = 31 , `k_sci_mstpb_sci1` = 30 , `k_sci_mstpb_sci2` = 29 , `k_sci_mstpb_sci3` = 28 ,
`k_sci_mstpb_sci4` = 27 , `k_sci_mstpb_sci5` = 26 , `k_sci_mstpb_sci6` = 25 , `k_sci_mstpb_sci7` = 24 ,
`k_sci_mstpb_sci12` = 4 }
SCI module stop bit positions in MSTP\rB.
- enum `sci_mstpc_bits_t` : `uint8_t` { `k_sci_mstpc_sci8` = 27 , `k_sci_mstpc_sci9` = 26 ,
`k_sci_mstpc_sci10` = 25 , `k_sci_mstpc_sci11` = 24 }
SCI module stop bit positions in MSTP\rC.
- enum `uart_debug_pins_t` : `uint16_t` { `k_uart_debug_tx_gpio` = `k_rx_pb_7` , `k_uart_debug_rx_gpio` = `k_rx_pb_6` }
Debug UART pins (SCI9 on RX72N).
- enum `uart_gpio_constants_t` : `uint8_t` { `k_uart_gpio_bit_set` = 1 }
GPIO register bit manipulation constants for UART pin configuration.
- enum `sci9_rx_ring_consts_t` : `uint16_t` { `k_sci9_rx_ring_size` , `k_sci9_rx_ring_mask` = `k_sci9_rx_ring_size` - 1U , `k_sci9_rx_ring_priority` = 3U }

Functions

- static `uint8_t` `internal_calculate_brr` (const `uint32_t` baudrate, const `uint32_t` pclk_hz, const bool abcs)
Calculate the 8-bit BRR register value for a target baud rate.
- static void `internal_clear_errors` (volatile `rx_sci_regs_t` *sci)

- Clear receive error flags in the SCI Serial Status Register (SSR).
- static int8_t [internal_get_mstpb_bit](#) (const uint8_t channel)
 - Return the MSTP\B bit position for a channel, or -1 for MSTP\RC channels.
- static rx_err_t [internal_enable_sci_clock](#) (const uint8_t channel)
 - Enable the peripheral clock for an SCI channel.
- static rx_err_t [internal_configure_uart_pins](#) (const rx_port_pin_t tx_gpio, rx_port_pin_t rx_gpio)
 - Configure the GPIO and MPC pin-mux settings for a UART TX/RX pin pair.
- static rx_err_t [internal_uart_validate_init](#) (const [uart_channel_config_t](#) *const config, volatile [rx_sci_regs_t](#) **const sci_out)
 - Initialize UART channel with specified configuration.
- static uint32_t [internal_uart_select_pclk](#) (const [uart_channel_t](#) channel)
 - Resolve the peripheral clock frequency for a given SCI channel.
- static void [internal_uart_program_baud](#) (volatile [rx_sci_regs_t](#) *const sci, const [uart_channel_t](#) channel, const uint32_t baudrate)
 - Program SMR/SEMR/BRR for the requested channel and baud.
- static void [internal_uart_enable_sci9_rx_isr](#) (volatile [rx_sci_regs_t](#) *const sci)
 - Switch SCI9 to interrupt-driven RX mode.
- rx_err_t [uart_init_channel](#) (const [uart_channel_config_t](#) *config)
 - Initialize SCI channel for UART communication.
- rx_err_t [uart_deinit_channel](#) (const [uart_channel_t](#) channel)
 - Deinitialize UART channel and disable TX/RX.
- rx_err_t [uart_putc_channel](#) (const [uart_channel_t](#) channel, const char data)
 - Transmit single character on UART channel.
- rx_err_t [uart_puts_channel](#) (const [uart_channel_t](#) channel, const char *str)
 - Transmit null-terminated string on UART channel with newline conversion.
- rx_err_t [uart_write_channel](#) (const [uart_channel_t](#) channel, const uint8_t *data, uint16_t length)
 - Write binary data to UART channel without newline conversion.
- rx_err_t [uart_getc_channel](#) (const [uart_channel_t](#) channel, char *data)
 - Receive single character from UART channel (non-blocking).
- rx_err_t [uart_read_channel](#) (const [uart_channel_t](#) channel, uint8_t *data, uint16_t length, uint16_t *bytes_read)
 - Read available bytes from UART channel (non-blocking).
- rx_err_t [uart_rx_available](#) (const [uart_channel_t](#) channel, bool *available)
 - Check if receive data is available on UART channel.
- rx_err_t [uart_clear_rx_errors](#) ([uart_channel_t](#) channel)
 - Clear sticky receive error flags (ORER/FER/PER) on a UART channel.
- void [INT_SCI9_RXI9](#) (void)
 - SCI9 receive-data-full interrupt (vector 102).
- rx_err_t [uart_debug_init](#) (void)
 - Initialize default debug UART channel (SCI9 at 115200 baud).
- void [uart_debug_putc](#) (const char data)
 - Transmit single character on debug UART (SCI9).
- void [uart_debug_puts](#) (const char *str)
 - Transmit null-terminated string on debug UART with newline conversion.
- void [uart_debug_putint](#) (const int32_t value)
 - Transmit signed 32-bit integer as decimal string on debug UART.
- void [uart_debug_puthex](#) (const uint32_t value, uint8_t digits)
 - Transmit 32-bit unsigned integer as hexadecimal string on debug UART.

Variables

- static bool `s_channel_initialized` [`k_uart_array_size`] = {false}
Per-channel initialization state.
- static volatile uint8_t `s_sci9_rx_ring` [`k_sci9_rx_ring_size`]
- static volatile uint16_t `s_sci9_rx_head` = 0U
- static volatile uint16_t `s_sci9_rx_tail` = 0U

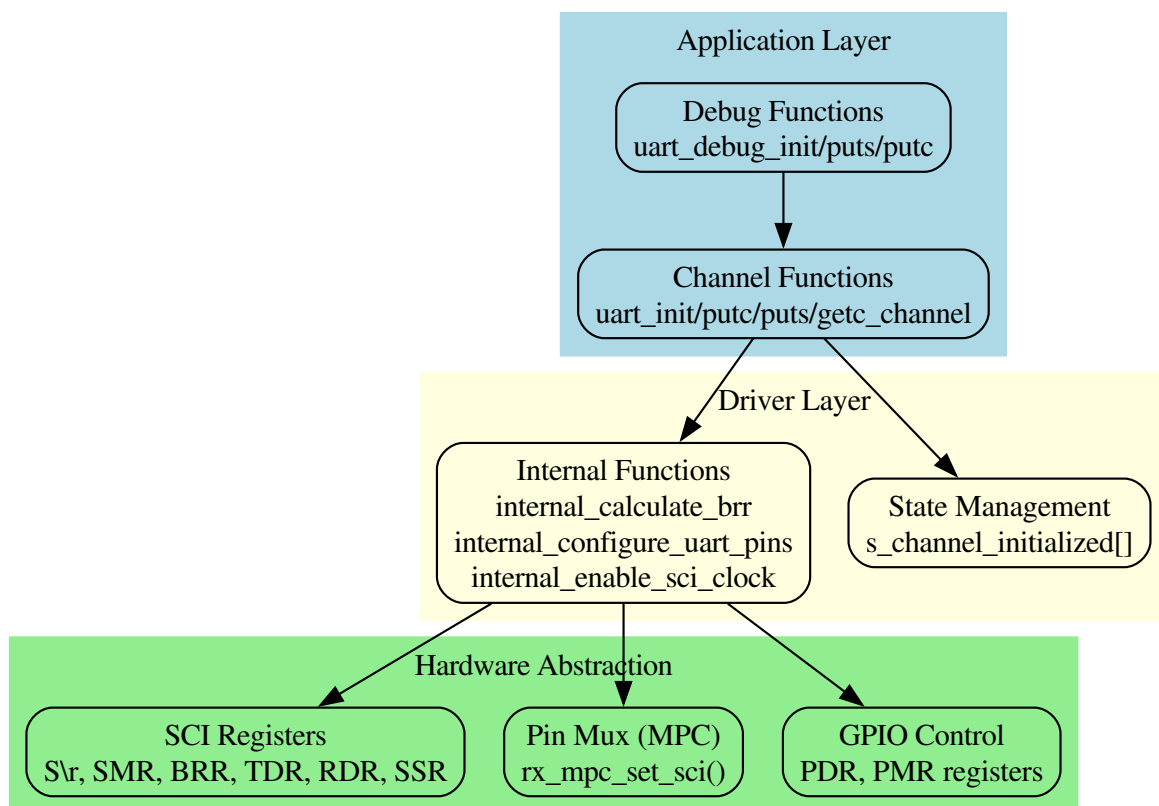
8.139.1 Detailed Description

Multi-Channel UART Driver for RX72N SCI Peripherals.

8.139.1.1 Overview

Production-quality UART driver for RX72N Serial Communication Interface (SCI) peripherals. Provides synchronous TX and RX functionality across 13 SCI channels (SCI0-SCI12) with configurable baud rates, automatic newline conversion, and comprehensive error handling.

8.139.1.2 Architecture



8.139.1.3 Hardware Details

Hardware Requirements:

Component	Requirement	Notes
MCU	Renesas RX72N	SCI peripheral required
Channels	SCI0-SCI12	13 channels total
Clock	PCLKB (SCI0-6,12) / PCLKA (SCI7-11)	Per-channel BRR clock
TX Pin	Port pin configured as output	Requires MPC configuration
RX Pin	Port pin configured as input	Requires MPC configuration
USB Bridge	CY7C65213 (default)	For SCI9 debug channel

Default Debug Configuration (SCI9):

Parameter	Value	Notes
Channel	SCI9	Connected to USB-UART bridge
TX Pin	PB7 (TXD9)	Output, peripheral mode
RX Pin	PB6 (RXD9)	Input, peripheral mode
Baud Rate	115200 bps	Standard debug speed
Data Bits	8	Fixed
Parity	None	Fixed
Stop Bits	1	Fixed

8.139.1.4 Baud Rate Calculation

The SCI baud rate formula for n=0 (CKS=00, PCLK/1):

$$\text{BRR} = \frac{\text{PCLK}}{32 \times B} - 1$$

f_PCLK depends on the channel group (manual section 5 I/O register table):

- SCI0-SCI6, SCI12 (SCIj/SCIh): PCLKB (k_pclkb_hz, 60 MHz)
- SCI7-SCI11 (SCIi, extended 0x000D0xxx): PCLKA (k_pclka_hz, 120 MHz)

Note: [rx72n_clock.h](#) is authoritative for PCLK frequencies; trust the header over any older bench note. PCLKB = 60 MHz on this platform.

Supported Baud Rates (at PCLKB = 60 MHz, for SCI0-SCI6/SCI12):

Baud Rate	BRR	Actual Rate	Error
9600	194	9615	+0.16%
19200	96	19231	+0.16%
38400	47	38462	+0.16%
57600	31	58594	+1.73%
115200	15	117188	+1.73%
230400	7	234375	+1.73%
460800	3	468750	+1.73%
921600	1	937500	+1.73%

8.139.1.5 Performance Characteristics

Operation	Execution Time	Stack Usage	Notes
uart_init_channel	~50 us	48 bytes	Includes MPC configuration
uart_putc_channel	~90 us @ 115200	16 bytes	Includes wait for TDRE
uart_puts_channel	~90 us/char	24 bytes	With LF->\r\n conversion
uart_getc_channel	~5 us	16 bytes	Non-blocking if no data
uart_debug_init	~50 us	64 bytes	Wrapper for SCI9

Memory Footprint:

- Code size: ~2 KB (all functions)
- Static data: 13 bytes (s_channel_initialized array)
- Stack usage: < 64 bytes (deepest call)
- Heap usage: 0 bytes (zero dynamic allocation)

8.139.1.6 Thread Safety

Function	Thread Safe?	Notes
uart_init_channel	[FAIL] No	Call once per channel during init
uart_deinit_channel	[FAIL] No	Call once during shutdown
uart_putc_channel	[WARN] Partial	Safe if different channels
uart_puts_channel	[WARN] Partial	Safe if different channels
uart_getc_channel	[WARN] Partial	Safe if different channels
uart_debug↵_*	[WARN] Partial	All use same channel (SCI9)

Note: Multiple threads can safely use different UART channels simultaneously. For shared channel access, use external mutex protection.

Module Dependencies:

This module depends on:

- [hardware.h](#) - Hardware configuration and channel constants
- [rx72n_clock.h](#) - PCLKB frequency definition
- [rx72n_regs.h](#) - SCI register definitions
- [rx_mpc.h](#) - Pin mux configuration
- [rx_port_utils.h](#) - GPIO port utilities
- [rx_register_protection.h](#) - PR\r register protection

This module is used by:

- [rx_log.c](#) - Logging subsystem output
- [main.c](#) - Debug output during initialization
- [comm_task.c](#) - Communication task debug messages

NASA Power of 10 Compliance:

Rule	Status	Implementation
1. Simple control flow	↔ [OK]↔	No goto/setjmp/recursion
2. Fixed loop bounds	↔ [OK]↔	All loops use k_uart_max_str_len limit
3. No dynamic allocation	↔ [OK]↔	Zero malloc/free, static buffers only
4. Small functions	↔ [OK]↔	All functions < 60 lines
5. Assertions (>=2/func)	↔ [OK]↔	Parameter validation + state checks
6. Narrow scope	↔ [OK]↔	Static file-scope state, local variables
7. Check return values	↔ [OK]↔	All rx_err_t returns propagated

Rule	Status	Implementation
8. Limited preprocessor	↔ [OK]↔	C23 typed enums only
9. Pointer restrictions	↔ [OK]↔	Single-level pointers, no arithmetic
10. Compiler warnings	↔ [OK]↔	-Wall -Wextra -Werror clean

SOLID Principles:

Single Responsibility (S):

- This module handles ONLY UART communication
- Separate from higher-level logging (rx_log.c)
- Separate from protocol parsing (rx_frame.c)

Open/Closed (O):

- New baud rates supported without code changes
- New channels added via configuration structure
- Pin assignments configurable per-channel

Liskov Substitution (L):

- All channel functions accept any valid channel
- Consistent rx_err_t return semantics
- Debug functions are simple wrappers

Interface Segregation (I):

- Separate init/deinit from TX/RX operations
- Debug functions provide simplified interface
- Channel functions provide full control

Dependency Inversion (D):

- Uses abstract rx_port_pin_t for pin specification
- Uses rx_mpc module for pin configuration
- Uses rx_err_t for standardized error handling

See also

[hardware.h](#) Hardware configuration and constants
[rx72n_sci_regs.h](#) SCI register definitions
[rx_mpc.h](#) Pin mux configuration API
[rx_log.h](#) Higher-level logging that uses this driver

Author

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Date

2026-01-28

Copyright

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Since

Version 1.0.0

Definition in file [uart.c](#).

8.139.2 Enumeration Type Documentation

8.139.2.1 brr`constants` t

enum [brr_constants_t](#) : uint16_t

BRR calculation constants.

Enumerator

k_brr_divisor_n0	Divisor for n=0, ABCS=0 (default 16-cycle base): $64 * 2^{(2n-1)} = 32$
k_brr_divisor_n0_abcs	Divisor for n=0, ABCS=1 (8-cycle base): halves the divisor, HUM 34.2.12 ↵
k_brr_multiplier	Multiplier per CKS increment (2^2)
k_brr_max_value	Maximum BRR register value
k_brr_min_value	Minimum BRR register value
k_brr_formula_offset	BRR formula subtract-1 offset: $BRR = (PCLK / (div * B)) - 1$

Definition at line 271 of file [uart.c](#).

```
00271     : uint16_t {
00272 k_brr_divisor_n0 = 32, /**< Divisor for n=0, ABCS=0 (default 16-cycle base): 64 * 2^(2n-1) = 32 */
00273 k_brr_divisor_n0_abcs =
00274     16, /**< Divisor for n=0, ABCS=1 (8-cycle base): halves the divisor, HUM 34.2.12 */
00275 k_brr_multiplier     = 4, /**< Multiplier per CKS increment (2^2) */
00276 k_brr_max_value      = 255, /**< Maximum BRR register value */
00277 k_brr_min_value      = 0, /**< Minimum BRR register value */
00278 k_brr_formula_offset = 1, /**< BRR formula subtract-1 offset: BRR = (PCLK/(div*B)) - 1 */
00279 } brr_constants_t;
```

8.139.2.2 sci9'rx'ring'consts't

```
enum sci9\_rx\_ring\_consts\_t : uint16_t
```

Enumerator

k_sci9_rx_ring_size	Power-of-2 mask-friendly size; covers 4.4 ms of 115200-baud traffic
k_sci9_rx_ring_mask	
k_sci9_rxi9_priority	IPR[102]: lower than motor ctrl, higher than CMT tick

Definition at line [462](#) of file [uart.c](#).

```
00462      : uint16_t {
00463  k_sci9_rx_ring_size =
00464      512U, /**< Power-of-2 mask-friendly size; covers 4.4 ms of 115200-baud traffic */
00465  k_sci9_rx_ring_mask = k_sci9_rx_ring_size - 1U,
00466  k_sci9_rxi9_priority = 3U, /**< IPR[102]: lower than motor ctrl, higher than CMT tick */
00467 } sci9_rx_ring_consts_t;
```

8.139.2.3 sci'mstpb'bits't

```
enum sci\_mstpb\_bits\_t : uint8_t
```

SCI module stop bit positions in MSTP\rB.

SCI0-SCI7 occupy contiguous bits 31-24 (MSTPB31 down to MSTPB24). SCI12 (SCIh module) is a non-contiguous outlier at MSTPB4. SCI8-SCI11 live in MSTP\rC, not here – see [sci_mstpc_bits_t](#).

Manual: section 11.2.3 MSTP\rB, page 408.

Enumerator

k_sci_mstpb_sci0	SCI0 module stop bit in MSTPCRB (MSTPB31)
k_sci_mstpb_sci1	SCI1 module stop bit in MSTPCRB (MSTPB30)
k_sci_mstpb_sci2	SCI2 module stop bit in MSTPCRB (MSTPB29)
k_sci_mstpb_sci3	SCI3 module stop bit in MSTPCRB (MSTPB28)
k_sci_mstpb_sci4	SCI4 module stop bit in MSTPCRB (MSTPB27)
k_sci_mstpb_sci5	SCI5 module stop bit in MSTPCRB (MSTPB26)
k_sci_mstpb_sci6	SCI6 module stop bit in MSTPCRB (MSTPB25)
k_sci_mstpb_sci7	SCI7 module stop bit in MSTPCRB (MSTPB24)
k_sci_mstpb_sci12	SCI12 (SCIh) module stop bit in MSTPCRB (MSTPB4)

Definition at line [373](#) of file [uart.c](#).

```
00373      : uint8_t {
00374  k_sci_mstpb_sci0 = 31, /**< SCI0 module stop bit in MSTPCRB (MSTPB31) */
00375  k_sci_mstpb_sci1 = 30, /**< SCI1 module stop bit in MSTPCRB (MSTPB30) */
00376  k_sci_mstpb_sci2 = 29, /**< SCI2 module stop bit in MSTPCRB (MSTPB29) */
00377  k_sci_mstpb_sci3 = 28, /**< SCI3 module stop bit in MSTPCRB (MSTPB28) */
00378  k_sci_mstpb_sci4 = 27, /**< SCI4 module stop bit in MSTPCRB (MSTPB27) */
00379  k_sci_mstpb_sci5 = 26, /**< SCI5 module stop bit in MSTPCRB (MSTPB26) */
00380  k_sci_mstpb_sci6 = 25, /**< SCI6 module stop bit in MSTPCRB (MSTPB25) */
00381  k_sci_mstpb_sci7 = 24, /**< SCI7 module stop bit in MSTPCRB (MSTPB24) */
00382  k_sci_mstpb_sci12 = 4, /**< SCI12 (SCIh) module stop bit in MSTPCRB (MSTPB4) */
00383 } sci_mstpb_bits_t;
```

8.139.2.4 sci_mstpc_bits_t

```
enum sci_mstpc_bits_t : uint8_t
```

SCI module stop bit positions in MSTP\RC.

SCI8-SCI11 (SCIi module, extended peripheral region 0x000D0000) are controlled by MSTP\RC, NOT MSTP\RB. Clearing the corresponding bit releases the channel from the module-stop state.

Manual: section 11.2.4 MSTP\RC, page 410.

Enumerator

k_sci_mstpc_sci8	SCI8 module stop bit in MSTPCRC (MSTPC27)
k_sci_mstpc_sci9	SCI9 module stop bit in MSTPCRC (MSTPC26)
k_sci_mstpc_sci10	SCI10 module stop bit in MSTPCRC (MSTPC25)
k_sci_mstpc_sci11	SCI11 module stop bit in MSTPCRC (MSTPC24)

Definition at line 394 of file [uart.c](#).

```
00394     : uint8_t {
00395     k_sci_mstpc_sci8 = 27, /**< SCI8 module stop bit in MSTPCRC (MSTPC27) */
00396     k_sci_mstpc_sci9 = 26, /**< SCI9 module stop bit in MSTPCRC (MSTPC26) */
00397     k_sci_mstpc_sci10 = 25, /**< SCI10 module stop bit in MSTPCRC (MSTPC25) */
00398     k_sci_mstpc_sci11 = 24, /**< SCI11 module stop bit in MSTPCRC (MSTPC24) */
00399 } sci_mstpc_bits_t;
```

8.139.2.5 sci_register_values_t

```
enum sci_register_values_t : uint8_t
```

SCI register control-bit values (SSR flag masks live in [rx72n_sci_regs.h](#)).

Enumerator

k_sci_scr_disabled	SCR: All functions disabled
k_sci_scr_tx_enabled	SCR: Transmit enabled (TE=1)
k_sci_scr_rx_enabled	SCR: Receive enabled (RE=1)
k_sci_scr_txrx_enabled	SCR: TX + RX enabled (TE=1, RE=1)
k_sci_smr_async_8n1	SMR: Async mode, 8 data bits, no parity, 1 stop bit, PCLK/1
k_sci_semr_default	SEMR: Default extended mode (ABCS=0, 16-cycle base)
k_sci_semr_abcs_bit	SEMR.ABCS: 1=8-cycle base clock (halves BRR divisor, HUM 34.2.12)
k_sci_scr_rie_bit	SCR.RIE: receive interrupt enable (HUM 34.2.5)

Definition at line 243 of file [uart.c](#).

```
00243     : uint8_t {
00244     k_sci_scr_disabled = 0x00, /**< SCR: All functions disabled */
00245     k_sci_scr_tx_enabled = 0x20, /**< SCR: Transmit enabled (TE=1) */
00246     k_sci_scr_rx_enabled = 0x10, /**< SCR: Receive enabled (RE=1) */
00247     k_sci_scr_txrx_enabled = 0x30, /**< SCR: TX + RX enabled (TE=1, RE=1) */
00248     k_sci_smr_async_8n1 = 0x00, /**< SMR: Async mode, 8 data bits, no parity, 1 stop bit, PCLK/1 */
00249     k_sci_semr_default = 0x00, /**< SEMR: Default extended mode (ABCS=0, 16-cycle base) */
00250     k_sci_semr_abcs_bit =
00251     0x10, /**< SEMR.ABCS: 1=8-cycle base clock (halves BRR divisor, HUM 34.2.12) */
00252     k_sci_scr_rie_bit = 0x40, /**< SCR.RIE: receive interrupt enable (HUM 34.2.5) */
00253 } sci_register_values_t;
```

8.139.2.6 `uart`buffer`constants`t`

```
enum uart\_buffer\_constants\_t : uint32_t
```

UART buffer and string size limits.

Enumerator

<code>k_uart_max_str_len</code>	Maximum string length for <code>uart_puts_channel</code>
---------------------------------	--

Definition at line 360 of file [uart.c](#).

```
00360      : uint32_t {
00361  k_uart_max_str_len = 256, /**< Maximum string length for uart_puts_channel */
00362 } uart_buffer_constants_t;
```

8.139.2.7 `uart`config`constants`t`

```
enum uart\_config\_constants\_t : uint32_t
```

UART configuration constants.

Enumerator

<code>k_uart_default_baudrate</code>	Default baud rate: 921600 bps. Max standard rate that hits <2%% error on SCI9 with PCLKA=120 MHz. Formula (ABCS=0/CKS=00): $BRR = PCLKA / (32 * B) - 1$. $120e6 / (32 * 921600) - 1 = 4.07 \rightarrow BRR = 3$ Actual rate = $120e6 / (32 * (3 + 1)) = 937500$ bps Error vs 921600 = $(937500 - 921600) / 921600 = +1.72\%$ (well within ~3%% UART tolerance) Well within the CY7C65213 USB-UART bridge's 3 Mbps ceiling.
<code>k_uart_bit_time_delay_cycles</code>	~1.09us at 921600 bps, ~260 cycles @240 MHz

Definition at line 225 of file [uart.c](#).

```
00225      : uint32_t {
00226  /**
00227   * @brief Default baud rate: 921600 bps
00228   *
00229   * Max standard rate that hits <2%% error on SCI9 with PCLKA=120 MHz.
00230   * Formula (ABCS=0/CKS=00):  $BRR = PCLKA / (32 * B) - 1$ .
00231   *  $120e6 / (32 * 921600) - 1 = 4.07 \rightarrow BRR = 3$ 
00232   * Actual rate =  $120e6 / (32 * (3 + 1)) = 937500$  bps
00233   * Error vs 921600 =  $(937500 - 921600) / 921600 = +1.72\%$  (well within ~3%% UART tolerance)
00234   *
00235   * Well within the CY7C65213 USB-UART bridge's 3 Mbps ceiling.
00236   */
00237  k_uart_default_baudrate =
00238    115200, /** matches motors_rtos/HIL-tests; 921600 at PCLKB=60 MHz has 1.7% BRR rounding error */
00239  k_uart_bit_time_delay_cycles = 125, /**< ~1.09us at 921600 bps, ~260 cycles @240 MHz */
00240 } uart_config_constants_t;
```

8.139.2.8 `uart`debug`pins`t`

```
enum uart\_debug\_pins\_t : uint16_t
```

Debug UART pins (SCI9 on RX72N).

Enumerator

k_uart_debug_tx_gpio	PB7 = TXD9
k_uart_debug_rx_gpio	PB6 = RXD9

Definition at line 402 of file [uart.c](#).

```
00402      : uint16_t {
00403  k_uart_debug_tx_gpio = k_rx_pb_7, /**< PB7 = TXD9 */
00404  k_uart_debug_rx_gpio = k_rx_pb_6, /**< PB6 = RXD9 */
00405 } uart_debug_pins_t;
```

8.139.2.9 uart`flag`constants`t

```
enum uart\_flag\_constants\_t : uint8_t
```

UART flag comparison constants.

Enumerator

k_uart_flag_clear	Flag bit is not set (comparison result)
-------------------	---

Definition at line 355 of file [uart.c](#).

```
00355      : uint8_t {
00356  k_uart_flag_clear = 0, /**< Flag bit is not set (comparison result) */
00357 } uart_flag_constants_t;
```

8.139.2.10 uart`gpio`constants`t

```
enum uart\_gpio\_constants\_t : uint8_t
```

GPIO register bit manipulation constants for UART pin configuration.

Provides the single-bit seed value used when constructing per-pin bitmasks for the PDR (Port Direction Register) and PMR (Port Mode Register) during UART TX/RX pin setup. A bit mask for a specific pin is formed by shifting this value left by the pin index obtained from `rx_pin_from_pin()`.

Invariant

k_uart_gpio_bit_set must equal 1 so that left-shifting by a pin index produces an isolated single-bit mask.

Example:

```
// Build the TX pin mask and set the direction bit
const uint8_t tx_pin      = rx_pin_from_pin(tx_gpio);
const uint8_t tx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « tx_pin);
tx_port_base->pdr |= tx_pin_mask; // Set output direction
tx_port_base->pmr |= tx_pin_mask; // Switch to peripheral mode
```

See also

[internal_configure_uart_pins\(\)](#) Only consumer of this constant
[uart_init_channel\(\)](#) Top-level function that triggers pin configuration

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Enumerator

k_uart_gpio_bit_set	Seed value for constructing a single-pin bitmask via left-shift
---------------------	---

Definition at line 434 of file [uart.c](#).

```
00434         : uint8_t {
00435   k_uart_gpio_bit_set = 1, /**< Seed value for constructing a single-pin bitmask via left-shift */
00436 } uart_gpio_constants_t;
```

8.139.2.11 `uart`hex`constants``

```
enum uart\_hex\_constants\_t : uint8_t
```

Hex digit constants.

Enumerator

k_uart_hex_max_digits	Maximum hex digits to print (32-bit value)
k_uart_hex_min_digits	Minimum hex digits to print
k_uart_hex_zero_digits	Zero digits value
k_uart_hex_nibble_bits	Bits per hex nibble
k_uart_hex_nibble_mask	Mask for hex nibble

Definition at line 262 of file [uart.c](#).

```
00262         : uint8_t {
00263   k_uart_hex_max_digits = 8, /**< Maximum hex digits to print (32-bit value) */
00264   k_uart_hex_min_digits = 1, /**< Minimum hex digits to print */
00265   k_uart_hex_zero_digits = 0, /**< Zero digits value */
00266   k_uart_hex_nibble_bits = 4, /**< Bits per hex nibble */
00267   k_uart_hex_nibble_mask = 0x0F, /**< Mask for hex nibble */
00268 } uart_hex_constants_t;
```

8.139.2.12 `uart`int`constants``

```
enum uart\_int\_constants\_t : uint8_t
```

Integer to string buffer constants.

Enumerator

k_uart_int_buffer_size	Buffer size for int32 to string (enough for -2147483648)
k_uart_base_10	Base 10 for decimal conversion

Definition at line 256 of file [uart.c](#).

```
00256         : uint8_t {
00257   k_uart_int_buffer_size = 12, /**< Buffer size for int32 to string (enough for -2147483648) */
00258   k_uart_base_10 = 10, /**< Base 10 for decimal conversion */
00259 } uart_int_constants_t;
```

8.139.2.13 uart`internal`constants``t`

```
enum uart\_internal\_constants\_t : uint8_t
```

Maximum SCI channels (array size, must be enum for compile-time constant).

Enumerator

<code>k_uart_channel_min</code>	Minimum UART channel (SCI0)
<code>k_uart_array_size</code>	Array size for <code>s_channel_initialized</code>
<code>k_uart_max_mstpb_channel</code>	Last contiguous MSTPCRB channel (SCI0-SCI7, bits 31-24)
<code>k_uart_mstpc_channel_min</code>	First MSTPCRC channel (SCI8 at MSTPCRC bit 27)
<code>k_uart_mstpc_channel_max</code>	Last MSTPCRC channel (SCI11 at MSTPCRC bit 24)
<code>k_uart_pclka_channel_min</code>	First channel using PCLKA for BRR (SCIi: SCI7-SCI11)
<code>k_uart_pclka_channel_max</code>	Last channel using PCLKA for BRR (SCIi: SCI7-SCI11)
<code>k_uart_max_channels</code>	Maximum valid channel value (SCI channels 0-12)

Definition at line 287 of file [uart.c](#).

```
00287     : uint8_t {
00288   k_uart_channel_min    = 0, /**< Minimum UART channel (SCI0) */
00289   k_uart_array_size     = 13, /**< Array size for s_channel_initialized */
00290   k_uart_max_mstpb_channel = 7, /**< Last contiguous MSTPCRB channel (SCI0-SCI7, bits 31-24) */
00291   k_uart_mstpc_channel_min = 8, /**< First MSTPCRC channel (SCI8 at MSTPCRC bit 27) */
00292   k_uart_mstpc_channel_max = 11, /**< Last MSTPCRC channel (SCI11 at MSTPCRC bit 24) */
00293   k_uart_pclka_channel_min = 7, /**< First channel using PCLKA for BRR (SCIi: SCI7-SCI11) */
00294   k_uart_pclka_channel_max = 11, /**< Last channel using PCLKA for BRR (SCIi: SCI7-SCI11) */
00295   k_uart_max_channels    = 13, /**< Maximum valid channel value (SCI channels 0-12) */
00296 } uart_internal_constants_t;
```

8.139.2.14 uart`mstpcrb`constants``t`

```
enum uart\_mstpcrb\_constants\_t : uint32_t
```

MSTPCRB register bit manipulation constants.

Enumerator

<code>k_uart_mstpcrb_bit_set</code>	Single-bit mask for MSTPCRB bit-clear operations
-------------------------------------	--

Definition at line 282 of file [uart.c](#).

```
00282     : uint32_t {
00283   k_uart_mstpcrb_bit_set = 1UL, /**< Single-bit mask for MSTPCRB bit-clear operations */
00284 } uart_mstpcrb_constants_t;
```

8.139.2.15 uart`timeout``t`

```
enum uart\_timeout\_t : uint32_t
```

UART timeout constants.

Enumerator

k_uart_tx_timeout	Transmit buffer wait timeout (prevents infinite loop)
k_uart_timeout_expired	Timeout counter expired value
k_uart_timeout_decrement	Timeout counter decrement value

Definition at line 348 of file [uart.c](#).

```
00348      : uint32_t {
00349  k_uart_tx_timeout      = 100000, /**< Transmit buffer wait timeout (prevents infinite loop) */
00350  k_uart_timeout_expired = 0,      /**< Timeout counter expired value */
00351  k_uart_timeout_decrement = 1,    /**< Timeout counter decrement value */
00352 } uart_timeout_t;
```

8.139.2.16 `uart_validation_limits_t`

```
enum uart\_validation\_limits\_t : uint32_t
```

Baud rate validation bounds for [uart_init_channel\(\)](#) parameter checking.

Defines the closed interval [`k_uart_baudrate_min`, `k_uart_baudrate_max`] that every requested baud rate must satisfy before the driver attempts to program the BRR register.

The maximum is derived directly from the BRR formula for `n=0` (`CKS=00`):

$$\text{BRR} = \frac{\text{PCLKB}}{32 \times B} - 1 \geq 0 \implies B \leq \frac{\text{PCLKB}}{32}$$

A BRR of 0 corresponds to the fastest achievable baud rate at the current PCLKB frequency, so requesting anything higher would underflow the register.

The minimum is 1 bps, which prevents a divide-by-zero in the BRR formula. In practice, any baud rate below ~9600 is impractical on a 60 MHz PCLKB, but the lower bound is kept permissive to avoid false negatives during testing.

Invariant

`k_uart_baudrate_min` must be > 0 to prevent division by zero in [internal_calculate_brr\(\)](#).

`k_uart_baudrate_max` must equal `k_pclkb_hz / k_brr_divisor_n0` so that the computed BRR is always ≥ 0 (i.e., no register underflow).

Example:

```
// Validation performed inside uart_init_channel()
if ((config->baudrate < k_uart_baudrate_min) ||
    (config->baudrate > k_uart_baudrate_max)) {
    return k_rx_err_invalid_arg;
}
```

See also

[uart_init_channel\(\)](#) Uses these bounds to validate the baudrate field

[internal_calculate_brr\(\)](#) Computes the actual BRR register value

[brr_constants_t](#) BRR formula divisor constants

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Enumerator

k_uart_baudrate_min	Minimum valid baud rate (bps); must be > 0 to avoid divide-by-zero in BRR formula
k_uart_baudrate_max	Maximum valid baud rate (bps); BRR = 0 at this rate, higher values would underflow

Definition at line 339 of file [uart.c](#).

```
00339     : uint32_t {
00340   k_uart_baudrate_min =
00341     1, /**< Minimum valid baud rate (bps); must be > 0 to avoid divide-by-zero in BRR formula */
00342   k_uart_baudrate_max =
00343     (k_pclkb_hz /
00344      k_brr_divisor_n0), /**< Maximum valid baud rate (bps); BRR = 0 at this rate, higher values would underflow */
00345 } uart_validation_limits_t;
```

8.139.3 Function Documentation

8.139.3.1 INT`SCI9`RXI9()

```
void INT_SCI9_RXI9 (
    void )
```

SCI9 receive-data-full interrupt (vector 102).

Fires when a byte lands in SCI9.RDR (RDRF rising edge). Reads the byte, clears RDRF by writing-0 after a read of SSR per HUM 34.2.8, and pushes the byte into s_sci9_rx_ring. ORER (overrun error) is checked and cleared so a dropped byte doesn't wedge the peripheral; the drop counter is incremented so telemetry can surface it.

ISR execution time: ~200 ns at 240 MHz (8 register ops + ring push). Ring capacity: k_sci9_rx_↵ ring_size = 512 bytes = 4.4 ms of 115200 baud which comfortably covers the ~10 ms comm_task poll period worst case.

Registered at relocatable-vector-table slot 102 via the \$tableentry\$102\$.rvecrors symbol below. Without that symbol the linker script (src/boot/linker_script_rvecrors.inc) falls through to \$tableentry\$default\$.rvecrors (0xFFFFFFFF), and the interrupt is effectively disabled even with IER↵ [12].bit6 set.

attribute((interrupt)) makes GCC generate the RTE / full register save+restore for the ISR body; the \$tableentry alias just places the function at the right vector slot.

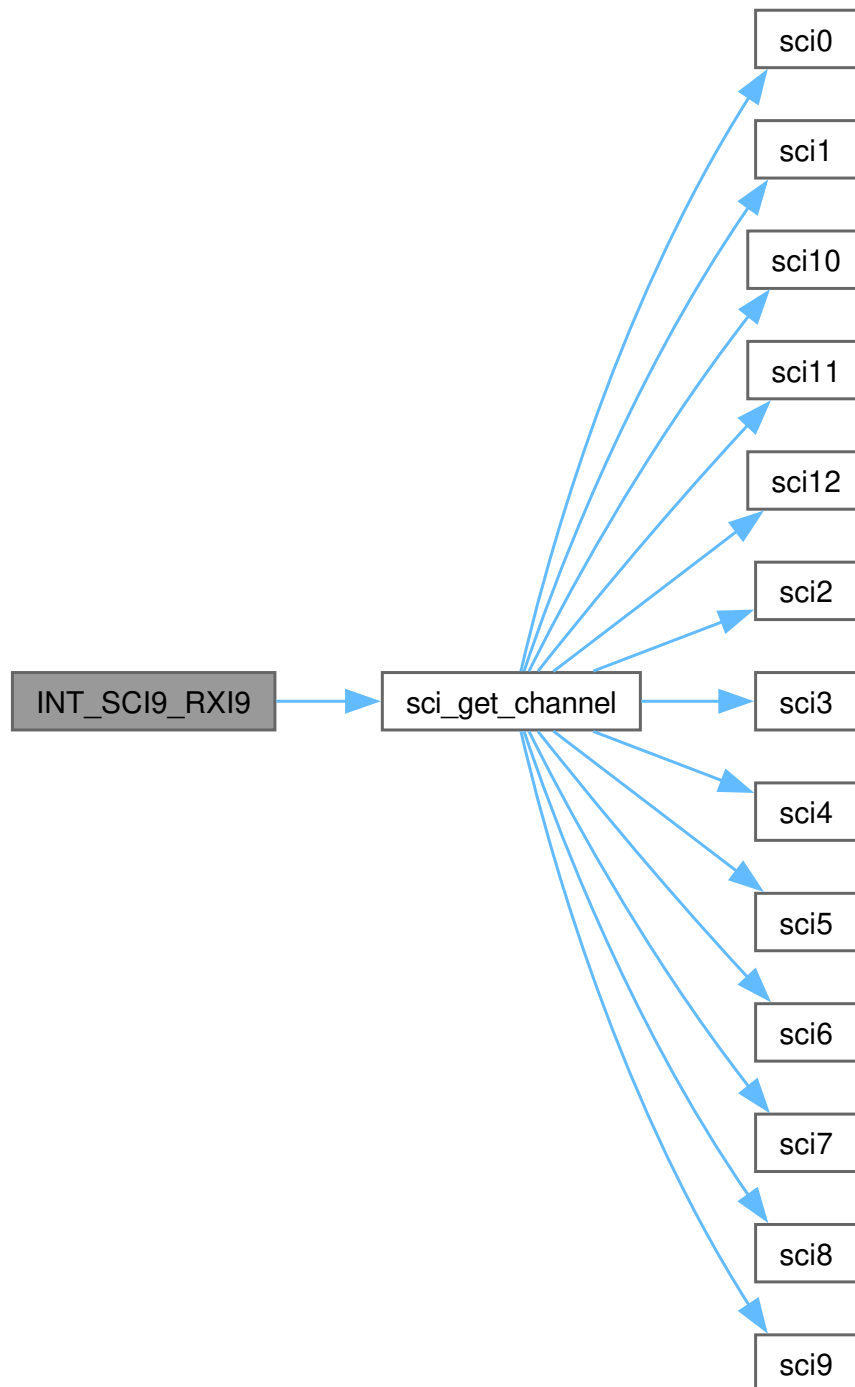
Definition at line 1874 of file [uart.c](#).

```
01875 {
01876 #ifdef __RX__
01877 volatile rx_sci_regs_t* sci = sci_get_channel(k_uart_channel_9);
01878 if (sci == nullptr) {
01879     return;
01880 }
01881
01882 /* Clear error flags first so the RDR read is not masked by a sticky
01883  * ORER/FER/PER -- HUM 34.2.7 requires these to be cleared before the
01884  * peripheral will accept further bytes. */
01885 if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01886     const uint8_t ssr_err = sci->ssr;
01887     sci->ssr = (uint8_t)(ssr_err & ~k_sci_ssr_error_mask);
01888 }
01889 }
```

```
01890  const uint8_t byte = (uint8_t)sci->rdr;
01891
01892  /* Clear RDRF: read SSR, write it back with RDRF=0. */
01893  const uint8_t ssr = sci->ssr;
01894  sci->ssr          = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01895
01896  const uint16_t next_head = (uint16_t)((s_sci9_rx_head + 1U) & k_sci9_rx_ring_mask);
01897  if (next_head != s_sci9_rx_tail) {
01898      s_sci9_rx_ring[s_sci9_rx_head] = byte;
01899      s_sci9_rx_head                  = next_head;
01900  }
01901  /* Ring full: byte is dropped. comm_task polls every 10 ms and can
01902   * drain ~1150 bytes in that window, vs 115 bytes arriving at 115200
01903   * baud -- so the 512-byte ring has >4x headroom. If comm_task stalls,
01904   * the next well-formed frame the gateway sends will fail CRC and
01905   * trigger a retransmit, so no silent data loss. */
01906  #endif
01907 }
```

References [k_sci9_rx_ring_mask](#), [k_sci_ssr_error_mask](#), [k_sci_ssr_rdrf_flag](#), [k_uart_channel_9](#), [k_uart_flag_clear](#), [rx_sci_regs_t::rdr](#), [s_sci9_rx_head](#), [s_sci9_rx_ring](#), [s_sci9_rx_tail](#), [sci_get_channel\(\)](#), and [rx_sci_regs_t::ssr](#).

Here is the call graph for this function:



8.139.3.2 internal`calculate`brr()

```
uint8_t internal_calculate_brr (  
    const uint32_t baudrate,  
    const uint32_t pclk_hz,  
    const bool abcs) [static]
```

Calculate the 8-bit BRR register value for a target baud rate.

Computes the value to be written to the SCI Bit Rate Register (BRR) so that the SCI peripheral generates the requested baud rate from PCLKB. The general RX72N SCI baud rate formula for asynchronous mode is:

$$\text{BRR} = \frac{\text{PCLKB}}{64 \times 2^{2n-1} \times B} - 1$$

This driver always uses $n=0$ (CKS bits = 0b00, clock source = PCLK/1), which simplifies to:

$$\text{BRR} = \frac{\text{PCLK}}{32 \times B} - 1$$

The caller supplies the correct PCLK frequency for the target channel:

- SCI0-SCI6, SCI12 (SCIj/SCIh, standard region): use `k_pclkb_hz`
- SCI7-SCI11 (SCIi, extended region 0x00D0xxx): use `k_pclka_hz`

This distinction matters when $\text{PCLKA} \neq \text{PCLKB}$. [rx72n_clock.h](#) is the authoritative source for PCLK frequencies (`k_pclka_hz`, `k_pclkb_hz`, etc.); trust the header over any older bench note.

Algorithm steps:

1. Guard against `baudrate == 0` (return `k_brr_max_value` as a safe sentinel).
2. Guard against `baudrate > pclk_hz/32` (return `k_brr_min_value`).
3. Compute `divisor_result = pclk_hz / (32 * B)` using integer arithmetic.
4. Guard against `divisor_result <= 1` to prevent underflow from the -1 offset.
5. Compute `BRR = divisor_result - 1`.
6. Clamp the result to `k_brr_max_value` (255) if the integer exceeds the 8-bit range (indicates a baud rate too slow for this clock).
7. Return the clamped 8-bit result.

Precondition

baudrate validated by caller; `pclk_hz` must be non-zero
`k_brr_divisor_n0` must be non-zero compile-time constant

Postcondition

Return value is always in `[0, 255]`
 Caller must write the returned value to `sci->brr` before enabling TX/RX

Note

Stateless pure function; safe to call from any context including ISR
 Always uses $n=0$ (CKS=00); support for $n=1..3$ is not implemented

Example:

```
// SCI9 uses PCLKA (SCIi extended region), ABCS=1 for baud accuracy
sci->brr = internal_calculate_brr(115200U, k_pclka_hz, true);
```

See also

[uart_init_channel\(\)](#) Caller that selects pclk_hz and writes sci->brr
[brr_constants_t](#) Constants used in the BRR formula

Since

Version 1.0.0

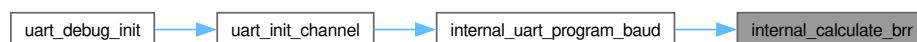
Definition at line 538 of file [uart.c](#).

```
00539 {
00540     /* Pre-condition: reject zero baudrate (division by zero) */
00541     if (baudrate == 0) {
00542         return k_brr_max_value;
00543     }
00544
00545     /* CKS=0, so the divisor is either 32 (ABCS=0, default 16-cycle base clock)
00546      * or 16 (ABCS=1, 8-cycle base clock). HUM 34.2.12 -- the 8-cycle base
00547      * doubles BRR resolution, which cuts 115200/120MHz baud error from
00548      * +1.72% (unusable, causes FER on every byte) to +0.16%. */
00549     const uint32_t divisor = abcs ? (uint32_t)k_brr_divisor_n0_abcs : (uint32_t)k_brr_divisor_n0;
00550
00551     /* Pre-condition: reject baudrate above maximum for this clock */
00552     if (baudrate > (pclk_hz / divisor)) {
00553         return k_brr_min_value;
00554     }
00555
00556     /* For n=0 (CKS=00): BRR = (PCLK / (divisor * B)) - 1 */
00557     const uint32_t divisor_result = pclk_hz / (divisor * baudrate);
00558
00559     /* Guard against underflow: if divisor_result is 0, subtraction would wrap */
00560     if (divisor_result <= k_brr_formula_offset) {
00561         return k_brr_min_value;
00562     }
00563
00564     const uint32_t brr_value = divisor_result - k_brr_formula_offset;
00565
00566     /* Clamp to maximum BRR register value */
00567     if (brr_value > k_brr_max_value) {
00568         return k_brr_max_value;
00569     }
00570
00571     const uint8_t result = (uint8_t)brr_value;
00572
00573     return result;
00574 }
```

References [k_brr_divisor_n0](#), [k_brr_divisor_n0_abcs](#), [k_brr_formula_offset](#), [k_brr_max_value](#), and [k_brr_min_value](#).

Referenced by [internal_uart_program_baud\(\)](#).

Here is the caller graph for this function:



8.139.3.3 internal_clear_errors()

```
void internal_clear_errors (
    volatile rx_sci_regs_t * sci) [static]
```

Clear receive error flags in the SCI Serial Status Register (SSR).

Reads the SSR register and writes it back with the three receive error flag bits masked out, which clears any pending ORER (Overrun Error), FER (Framing Error), and PER (Parity Error) conditions on the SCI channel.

The RX72N SCI peripheral requires a read-modify-write sequence to clear error flags: the hardware only allows clearing a flag by writing 0 to it while keeping the rest of the register unchanged. Writing 1 to an already-set flag has no effect (the write is ignored).

Algorithm steps:

1. Guard: return immediately if sci is nullptr (NASA Power of 10 Rule 5).
2. Read the current value of sci->ssr into a local volatile variable.
3. Cast-to-void the read result to suppress the unused-variable warning while still ensuring the volatile read is not optimised away.
4. Write sci->ssr = (ssr & ~k_sci_ssr_error_mask) to clear bits [5:3] (ORER, FER, PER) while preserving all other SSR bits.

Error flag bit positions in SSR:

Bit	Flag	Description
5	ORER	Overflow Error - new data arrived before previous data was read
4	FER	Framing Error - no valid stop bit detected
3	PER	Parity Error - parity mismatch (unused in 8N1 mode)

Precondition

sci must point to a valid, hardware-mapped `rx_sci_regs_t` register block

The SCI module clock must be enabled (MSTP\RB bit cleared) before any SSR access; undefined behaviour otherwise

Postcondition

ORER, FER, and PER bits in sci->ssr are cleared (written to 0)

All other SSR bits (TDRE, RDRF, TEND, MPB, MPBT) are preserved

Note

This function is intentionally void-returning; the caller (`uart_getc_channel`) checks for errors before calling and does not need a return code

The intermediate volatile read prevents the compiler from merging the read and write into a single store, which would miss the read requirement

Thread Safety:

Not thread-safe. SSR is a read-modify-write target; concurrent access from two threads on the same channel can corrupt flag state. External mutex protection is required if multiple threads share a channel.

Performance:

- Execution time: ~3 cycles (volatile read + mask + volatile write)
- Stack usage: 4 bytes (one volatile `uint8_t` local)

Example:

```
volatile rx_sci_regs_t* sci = sci_get_channel(k_uart_channel_9);
if (sci != nullptr) {
    // Clear any stale error flags before reading receive data
    internal_clear_errors(sci);
    const char data = (char)sci->rdr;
}
```

See also

[uart_getc_channel\(\)](#) Caller that invokes this before reading RDR
[sci_register_values_t k_sci_ssr_error_mask](#) bitmask definition
[uart_rx_available\(\)](#) Non-destructive receive availability check

Since

Version 1.0.0

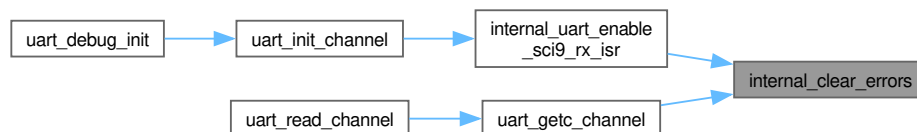
Definition at line 642 of file [uart.c](#).

```
00643 {
00644     /* Pre-condition: Validate pointer (NASA Power of 10 Rule 5) */
00645     if (sci == nullptr) {
00646         return;
00647     }
00648     volatile uint8_t ssr = 0;
00649     /* Read SSR then clear error flags */
00650     ssr = sci->ssr;
00651     (void)ssr; /* Suppress unused variable warning */
00652     sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_error_mask);
00653 }
00654
00655 }
```

References [k_sci_ssr_error_mask](#), and [rx_sci_regs_t::ssr](#).

Referenced by [internal_uart_enable_sci9_rx_isr\(\)](#), and [uart_getc_channel\(\)](#).

Here is the caller graph for this function:



8.139.3.4 internal`configure`uart`pins()

```
rx_err_t internal_configure_uart_pins (
    const rx_port_pin_t tx_gpio,
    rx_port_pin_t rx_gpio) [static]
```

Configure the GPIO and MPC pin-mux settings for a UART TX/RX pin pair.

Prepares the two GPIO pins required for SCI UART operation by programming the Multi-Function Pin Controller (MPC) and the Port Direction/Mode registers in the correct order. The RX72N hardware requires that:

- MPC is configured before switching a pin to peripheral mode (PMR)
- TX pin is driven as an output; RX pin is configured as an input
- Both pins are switched to peripheral mode (PMR bit = 1) last

Algorithm steps:

1. Extract the port number and pin number from each `rx_port_pin_t` using `rx_port_from_pin()` and `rx_pin_from_pin()`.
2. Validate that both pin numbers are $\leq k_rx_pin_max$ (7); return `k_rx_err_invalid_arg` if out of range. (Lower-bound check is omitted because pin numbers are `uint8_t` and `k_rx_pin_min == 0`, which would trigger -Wtype-limits.)
3. Obtain volatile port register base pointers via `rx_port_get_base()`; return `k_rx_err_invalid_arg` if either is `nullptr` (invalid port number).
4. Call `rx_mpc_set_sci(tx_gpio)` to set the TX pin's PFS register to the SCI TXD function; propagate any error immediately.
5. Call `rx_mpc_set_sci(rx_gpio)` to set the RX pin's PFS register to the SCI RXD function; propagate any error immediately.
6. Build per-pin bitmasks (`k_uart_gpio_bit_set << pin_number`).
7. Set PDR bit for TX pin (output direction) and PMR bit for TX pin (peripheral mode).
8. Clear PDR bit for RX pin (input direction) and set PMR bit for RX pin (peripheral mode).
9. Return `k_rx_ok`.

MPC write protection: `rx_mpc_set_sci()` internally handles the PFSWE unlock/lock sequence, so this function does not need to touch the PFSWE bit directly.

Pin direction and mode summary:

Pin	PDR (direction)	PMR (mode)
TX	1 (output)	1 (peripheral)
RX	0 (input)	1 (peripheral)

Precondition

`tx_gpio` and `rx_gpio` must correspond to physically valid RX72N port pins

The SCI module clock must already be enabled (MSTP\RB bit cleared) so that the SCI TXD/RXD MPC function codes are accepted

Postcondition

TX pin: PDR bit set (output), PMR bit set (peripheral function)

RX pin: PDR bit cleared (input), PMR bit set (peripheral function)

Both pins' PFS registers configured for SCI TX/RX function via MPC

Note

Pin validation only checks upper bound (`>k_rx_pin_max`); lower bound (0) is omitted to avoid the `-Wtype-limits` compiler warning triggered by comparing an unsigned type to 0

Warning

Passing the same pin for both TX and RX produces undefined hardware behaviour; no duplicate-pin check is performed

Thread Safety:

Not thread-safe. PDR/PMR are read-modify-write targets shared with all GPIO operations on the same port. Call only during single-threaded initialization.

Performance:

- Execution time: ~10 us (dominated by two `rx_mpc_set_sci()` calls)
- Stack usage: 32 bytes (four `uint8_t` locals + two pointer locals)

Example:

```
// Configure SCI9 default debug pins (PB7=TX, PB6=RX)
rx_err_t err = internal_configure_uart_pins(
    (rx_port_pin_t)k_uart_debug_tx_gpio,
    (rx_port_pin_t)k_uart_debug_rx_gpio);
if (err != k_rx_ok) {
    return err; // Invalid pin specification or MPC error
}
```

See also

`uart_init_channel()` Caller that passes validated `tx_gpio/rx_gpio`
`rx_mpc_set_sci()` MPC pin-function assignment (with PFSWE unlock)
`rx_port_get_base()` Port register base address lookup
`uart_gpio_constants_t k_uart_gpio_bit_set` used to build pin bitmasks

Since

Version 1.0.0

Definition at line 836 of file `uart.c`.

```
00837 {
00838     /* Extract port and pin numbers for hardware register access */
00839     const uint8_t tx_port = rx_port_from_pin(tx_gpio);
00840     const uint8_t tx_pin  = rx_pin_from_pin(tx_gpio);
00841     const uint8_t rx_port = rx_port_from_pin(rx_gpio);
00842     const uint8_t rx_pin  = rx_pin_from_pin(rx_gpio);
00843
00844     /* Validate pin numbers (lower bound checks omitted - pins are uint8_t, k_rx_pin_min == 0,
00845      * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
00846     if (tx_pin > k_rx_pin_max || rx_pin > k_rx_pin_max) {
00847         return k_rx_err_invalid_arg;
00848     }
00849
00850     /* Get port bases */
00851     volatile rx_port_regs_t* tx_port_base = rx_port_get_base(tx_port);
00852     volatile rx_port_regs_t* rx_port_base = rx_port_get_base(rx_port);
00853 }
```

```

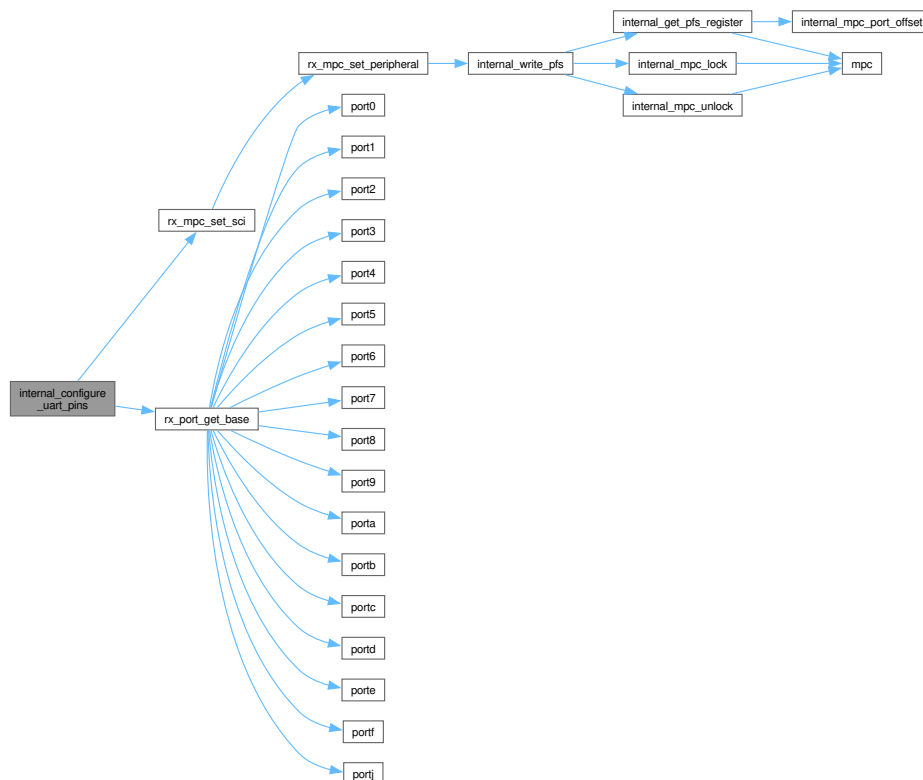
00854 if (tx_port_base == nullptr || rx_port_base == nullptr) {
00855     return k_rx_err_invalid_arg;
00856 }
00857
00858 /* Configure MPC for SCI function */
00859 rx_err_t err = rx_mpc_set_sci(tx_gpio);
00860 if (err != k_rx_ok) {
00861     return err;
00862 }
00863
00864 err = rx_mpc_set_sci(rx_gpio);
00865 if (err != k_rx_ok) {
00866     return err;
00867 }
00868
00869 /* Configure TX pin: output direction, peripheral mode */
00870 const uint8_t tx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « tx_pin);
00871 const uint8_t rx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « rx_pin);
00872
00873 tx_port_base->pdr |= tx_pin_mask; /* Output */
00874 tx_port_base->pmr |= tx_pin_mask; /* Peripheral mode */
00875
00876 /* Configure RX pin: input direction, peripheral mode */
00877 rx_port_base->pdr &= (uint8_t)~rx_pin_mask; /* Input */
00878 rx_port_base->pmr |= rx_pin_mask; /* Peripheral mode */
00879
00880 return k_rx_ok;
00881 }

```

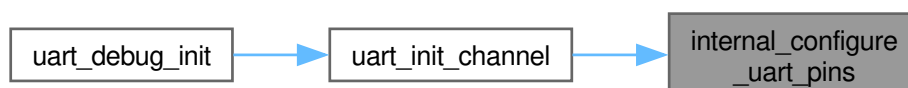
References [k_uart_gpio_bit_set](#), [rx_port_regs_t::pdr](#), [rx_port_regs_t::pmr](#), [rx_mpc_set_sci\(\)](#), and [rx_port_get_base\(\)](#).

Referenced by [uart_init_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.5 internal`enable`sci`clock()

```
rx_err_t internal_enable_sci_clock (
    const uint8_t channel) [static]
```

Enable the peripheral clock for an SCI channel.

Clears the module-stop bit for the given SCI channel so its clock runs. Different channel groups use different control registers (manual 11.2.3/11.2.4):

Channels	Register	Bits
SCI0-SCI7	MSTP\`rB	31-24
SCI8-SCI11	MSTP\`rC	27-24
SCI12	MSTP\`rB	4

Both MSTP\`rB and MSTP\`rC are PR\`r-protected; this function performs the required unlock/modify/↔lock sequence for whichever register applies.

Precondition

channel validated by [uart_init_channel\(\)](#) to be in [0, 12]
PR\`r not already unlocked by another context

Postcondition

Module-stop bit cleared; SCI clock gate open
PR\`r returned to locked state

Warning

Not thread-safe; unlock/modify/lock must be atomic

Since

Version 1.0.0

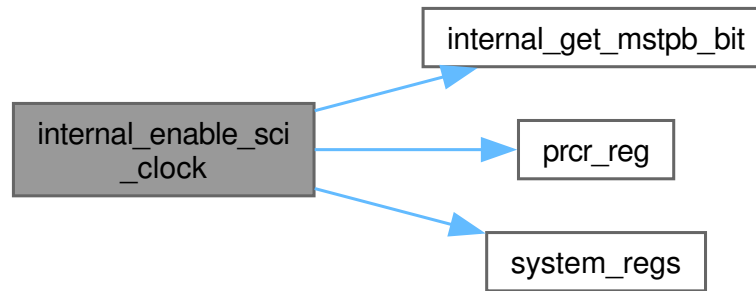
Definition at line 730 of file [uart.c](#).

```
00731 {
00732     const int8_t mstpb_bit = internal_get_mstpb_bit(channel);
00733
00734     *prcr_reg() = k_rx_prcr_unlock_all;
00735
00736     if (mstpb_bit >= 0) {
00737         /* SCI0-SCI7 and SCI12: clear bit in MSTPCRB */
00738         system_regs()->mstpcrb &= ~(k_uart_mstpcrb_bit_set « (uint8_t)mstpb_bit);
00739     } else if (channel >= k_uart_mstpc_channel_min && channel <= k_uart_mstpc_channel_max) {
00740         /* SCI8-SCI11: clear bit in MSTPCRC (bits 27-24) */
00741         const uint8_t mstpc_bit = (uint8_t)(k_sci_mstpc_sci8 - (channel - k_uart_mstpc_channel_min));
00742         system_regs()->mstpcrc &= ~(k_uart_mstpcrb_bit_set « mstpc_bit);
00743     } else {
00744         *prcr_reg() = k_rx_prcr_lock;
00745         return k_rx_err_invalid_arg;
00746     }
00747
00748     *prcr_reg() = k_rx_prcr_lock;
00749     return k_rx_ok;
00750 }
```

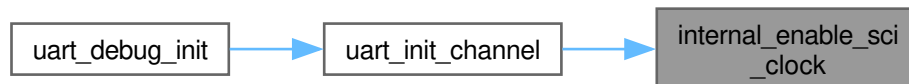
References [internal_get_mstpb_bit\(\)](#), [k_sci_mstpc_sci8](#), [k_uart_mstpc_channel_max](#), [k_uart_mstpc_channel_min](#), [k_uart_mstpcrb_bit_set](#), [rx_system_regs_t::mstpcrb](#), [rx_system_regs_t::mstpcrc](#), [prcr_reg\(\)](#), and [system_regs\(\)](#).

Referenced by [uart_init_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.6 internal_get_mstpb_bit()

```
int8_t internal_get_mstpb_bit (
    const uint8_t channel) [static]
```

Return the MSTP\RB bit position for a channel, or -1 for MSTP\RC channels.

Per manual section 11.2.3/11.2.4:

- SCI0-SCI7: MSTP\RB bits 31-24 (contiguous, descending)
- SCI8-SCI11: MSTP\RC bits 27-24 – returns -1, caller uses MSTP\RC
- SCI12 (SCIh): MSTP\RB bit 4 (non-contiguous outlier)

Bit position table:

Channel	Register	Bit	Enum constant
SCI0	MSTP\RB	31	k_sci_mstpb_sci0
SCI1	MSTP\RB	30	k_sci_mstpb_sci1
...	MSTP\RB	...	...
SCI7	MSTP\RB	24	k_sci_mstpb_sci7
SCI8	MSTP\RC	27	(returns -1)
SCI9	MSTP\RC	26	(returns -1)
SCI10	MSTP\RC	25	(returns -1)
SCI11	MSTP\RC	24	(returns -1)
SCI12	MSTP\RB	4	k_sci_mstpb_sci12

Precondition

`k_sci_mstpb_sci0 == 31` for the descending formula to be correct

Postcondition

If return ≥ 0 , it is a valid MSTP\`rB bit index

Note

int8_t so -1 fits without consuming a valid bit position

Since

Version 1.0.0

Definition at line 688 of file [uart.c](#).

```
00689 {
00690     if (channel <= k_uart_max_mstpb_channel) {
00691         /* SCI0-SCI7: MSTPCRB bits 31-24, descending */
00692         return (int8_t)(k_sci_mstpb_sci0 - channel);
00693     }
00694     if (channel > k_uart_mstpc_channel_max) {
00695         /* SCI12 (SCIh): MSTPCRB bit 4 -- only remaining valid channel above 11 */
00696         return (int8_t)k_sci_mstpb_sci12;
00697     }
00698     /* SCI8-SCI11: live in MSTPCRC, not MSTPCRB */
00699     return -1;
00700 }
```

References [k_sci_mstpb_sci0](#), [k_sci_mstpb_sci12](#), [k_uart_max_mstpb_channel](#), and [k_uart_mstpc_channel_max](#).

Referenced by [internal_enable_sci_clock\(\)](#).

Here is the caller graph for this function:



8.139.3.7 internal`uart`enable`sci9`rx`isr()

```
void internal_uart_enable_sci9_rx_isr (
    volatile rx_sci_regs_t *const sci) [static]
```

Switch SCI9 to interrupt-driven RX mode.

Resets the ring-buffer head/tail, drains any pending RDR + clears boot-time error flags, programs the RXI9 vector priority/IER/IR, then sets SCR.RIE=1 so the RXI9 ISR begins firing on RDRF. HUM vector 102 = SCI9_RXI9, IER index 12 ((102>>3)=12), IER bit 6 ((102&7)=6). See comm_task RX overrun note at the top of this file.

Parameters

in	sci	SCI9 peripheral base (must be non-null)
----	-----	---

Precondition

Caller has set SCR=TX|RE before this call

Postcondition

SCR.RIE asserted, RXI9 vector enabled, ring drained

Since

Version 1.0.0

Definition at line 1109 of file [uart.c](#).

```

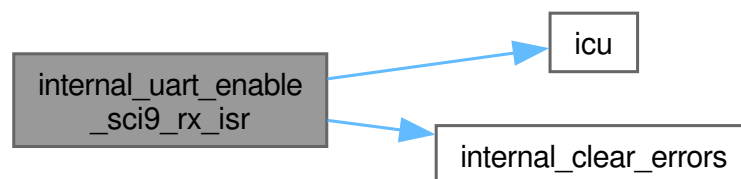
01110 {
01111     s_sci9_rx_head = 0U;
01112     s_sci9_rx_tail = 0U;
01113
01114     /* Drain any pending RDR byte and clear error flags BEFORE enabling the
01115      * RXI interrupt, so the first real byte doesn't find an ORER set from
01116      * boot-time noise (which would gate further interrupts). */
01117     if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01118         internal_clear_errors(sci);
01119     }
01120     if ((sci->ssr & k_sci_ssr_rdrf_flag) != k_uart_flag_clear) {
01121         (void)sci->rdr;
01122         const uint8_t ssr = sci->ssr;
01123         sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01124     }
01125
01126     const uint16_t vec_rxi9 = (uint16_t)k_vect_sci9_rxi9;
01127     icu()->ipr[vec_rxi9] = k_sci9_rxi9_priority;
01128     icu()->ir[vec_rxi9] = 0U;
01129     icu()->ier[vec_rxi9 / k_icu_ier_bits_per_reg] |=
01130         (uint8_t)(1U « (vec_rxi9 % k_icu_ier_bits_per_reg));
01131
01132     sci->scr = (uint8_t)(k_sci_scr_txxr_enabled | k_sci_scr_rie_bit);
01133 }

```

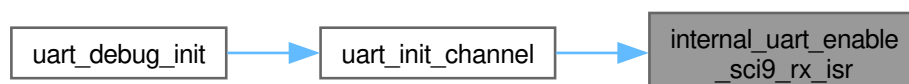
References [icu\(\)](#), [rx_icu_regs_t::ier](#), [internal_clear_errors\(\)](#), [rx_icu_regs_t::ipr](#), [rx_icu_regs_t::ir](#), [k_icu_ier_bits_per_reg](#), [k_sci9_rxi9_priority](#), [k_sci_scr_rie_bit](#), [k_sci_scr_txxr_enabled](#), [k_sci_ssr_error_mask](#), [k_sci_ssr_rdrf_flag](#), [k_uart_flag_clear](#), [k_vect_sci9_rxi9](#), [rx_sci_regs_t::rdr](#), [s_sci9_rx_head](#), [s_sci9_rx_tail](#), [rx_sci_regs_t::scr](#), and [rx_sci_regs_t::ssr](#).

Referenced by [uart_init_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.8 internal`uart`program`baud()

```
void internal_uart_program_baud (
    volatile rx_sci_regs_t *const sci,
    const uart_channel_t channel,
    const uint32_t baudrate) [static]
```

Program SMR/SEMR/BRR for the requested channel and baud.

Sets SMR=async-8N1, picks SEMR.ABCS=1 for SCI9 (lowers baud error from +1.72% to +0.16% at 115200 on PCLKA=120 MHz, see HUM 34.2.12), computes BRR via internal_calculate_brr, then waits k_uart_bit_time_ - delay_cycles for the new rate to settle. SEMR MUST be written before BRR because BRR is interpreted against whatever base clock SEMR selects.

Parameters

in	sci	SCI peripheral base (must be non-null)
in	channel	Validated SCI channel index
in	baudrate	Validated requested baud

Precondition

SCR=disabled before this call

Postcondition

SMR/SEMR/BRR programmed

Since

Version 1.0.0

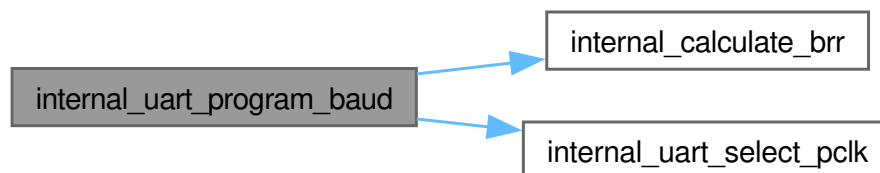
Definition at line 1076 of file [uart.c](#).

```
01079 {
01080     sci->smr      = k_sci_smr_async_8n1;
01081     const bool use_abcs = (channel == k_uart_channel_9);
01082     sci->semr     = use_abcs ? (uint8_t)k_sci_semr_abcs_bit : (uint8_t)k_sci_semr_default;
01083     sci->brr      = internal_calculate_brr(baudrate, internal_uart_select_pclk(channel), use_abcs);
01084
01085     /* NOTE: Busy-wait required - may run before ThreadX initialization */
01086     for (volatile uint32_t i = 0; i < k_uart_bit_time_delay_cycles; i++) {
01087         __asm__ volatile("nop");
01088     }
01089 }
```

References [rx_sci_regs_t::brr](#), [internal_calculate_brr\(\)](#), [internal_uart_select_pclk\(\)](#), [k_sci_semr_abcs_bit](#), [k_sci_semr_default](#), [k_sci_smr_async_8n1](#), [k_uart_bit_time_delay_cycles](#), [k_uart_channel_9](#), [rx_sci_regs_t::semr](#), and [rx_sci_regs_t::smr](#).

Referenced by [uart_init_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.9 internal_uart_select_pclk()

```
uint32_t internal_uart_select_pclk (
    const uart\_channel\_t channel) [static]
```

Resolve the peripheral clock frequency for a given SCI channel.

SCIj (SCI0-SCI6) and SCIl (SCI12) are clocked by PCLKB. SCIn (SCI7-SCI11, extended region 0x000↵D0xxx) are clocked by PCLKA. Using the wrong clock produces a wrong BRR and a wrong baud rate.

Parameters

in	channel	Validated SCI channel index
----	---------	-----------------------------

Returns

uint32_t Peripheral clock in Hz

Precondition

channel < k_uart_max_channels

Since

Version 1.0.0

Definition at line 1047 of file [uart.c](#).

```
01048 {
01049     return (((uint8_t)channel >= k_uart_pclka_channel_min) &&
01050         ((uint8_t)channel <= k_uart_pclka_channel_max))
01051         ? k_pclka_hz
01052         : k_pclkb_hz;
01053 }
```

References [k_pclka_hz](#), [k_pclkb_hz](#), [k_uart_pclka_channel_max](#), and [k_uart_pclka_channel_min](#).

Referenced by [internal_uart_program_baud\(\)](#).

Here is the caller graph for this function:



8.139.3.10 internal_uart_validate_init()

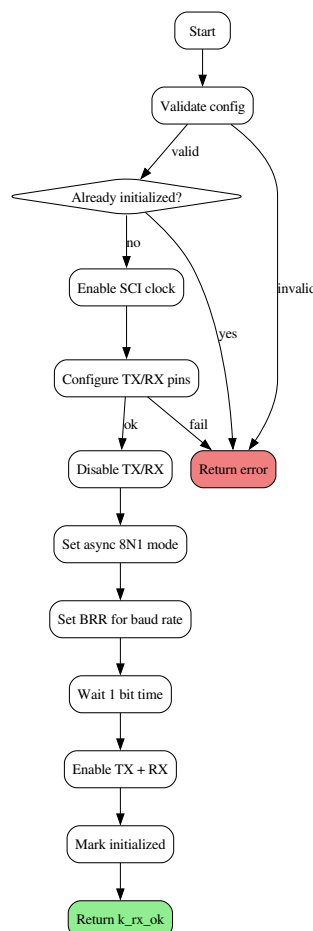
```
rx_err_t internal_uart_validate_init (
    const uart_channel_config_t *const config,
    volatile rx_sci_regs_t **const sci_out) [static]
```

Initialize UART channel with specified configuration.

Configures and enables an SCI peripheral for asynchronous UART operation. Handles module clock enable, pin mux configuration, and SCI register setup.

Algorithm steps:

1. Validate configuration parameters (NULL check, channel range, baud rate)
2. Get SCI register base address for channel
3. Check if channel is already initialized (prevent double-init)
4. Enable SCI module clock (clear MSTPB bit)
5. Configure TX/RX pins via MPC and GPIO registers
6. Disable TX/RX during configuration ($S\backslash r = 0$)
7. Set serial mode: async, 8N1, PCLK/1 (SMR)
8. Calculate and set baud rate divisor (BRR)
9. Wait for 1 bit time (synchronization)
10. Enable TX and RX ($S\backslash r = 0x30$)
11. Mark channel as initialized



Precondition

config must point to valid [uart_channel_config_t](#) structure
config->channel must be in range [0, 12]
config->baudrate must be in valid range for PCLKB
Channel must not be already initialized

Postcondition

Channel configured for 8N1 async operation
TX and RX enabled and ready for use
s_channel_initialized[channel] set to true
TX/RX pins configured for peripheral function

Note

Not thread-safe - call once during initialization
Includes ~1ms delay for baud rate synchronization

Warning

Does not validate pin assignments against hardware

Thread Safety:

Not thread-safe. Call once per channel during system initialization.

Performance:

- Execution time: ~50 us (includes MPC and delay)
- Stack usage: 48 bytes

Example:

```
const uart_channel_config_t config = {
    .channel = k_uart_channel_9,
    .baudrate = 115200,
    .tx_gpio = k_rx_pb_7,
    .rx_gpio = k_rx_pb_6,
};

rx_err_t err = uart_init_channel(&config);
if (err != k_rx_ok) {
    // Handle error
}
```

See also

[uart_deinit_channel\(\)](#) Deinitialize channel
[uart_putc_channel\(\)](#) Transmit single character
[uart_channel_config_t](#) Configuration structure

Since

Version 1.0.0

Validate `uart_init_channel` arguments and resolve SCI base

Checks for null config, channel range, baud range, channel-already- initialized state. Resolves config->channel to its SCI peripheral base address.

Parameters

in	config	Caller's channel config (must be non-null)
out	sci_out	On success, set to the SCI peripheral base

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	All checks pass, *sci_out non-null
<code>k_rx_err_null_ptr</code>	config is nullptr
<code>k_rx_err_invalid_arg</code>	Bounds violation
<code>k_rx_err_invalid_state</code>	Channel already initialized

Precondition

sci_out non-null

Since

Version 1.0.0

Definition at line 1009 of file `uart.c`.

```

01011 {
01012     if (config == nullptr) {
01013         return k_rx_err_null_ptr;
01014     }
01015     if ((uint8_t)config->channel >= k_uart_max_channels) {
01016         return k_rx_err_invalid_arg;
01017     }
01018     if ((config->baudrate < k_uart_baudrate_min) || (config->baudrate > k_uart_baudrate_max)) {
01019         return k_rx_err_invalid_arg;
01020     }
01021     *sci_out = sci_get_channel(config->channel);
01022     if (*sci_out == nullptr) {
01023         return k_rx_err_invalid_arg;
01024     }
01025     if (s_channel_initialized[config->channel]) {
01026         return k_rx_err_invalid_state;
01027     }
01028     return k_rx_ok;
01029 }
```

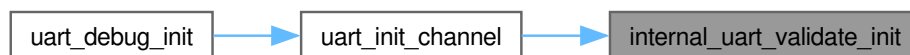
References `uart_channel_config_t::baudrate`, `uart_channel_config_t::channel`, `k_uart_baudrate_max`, `k_uart_baudrate_min`, `k_uart_max_channels`, `s_channel_initialized`, and `sci_get_channel()`.

Referenced by `uart_init_channel()`.

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.11 uart_clear_rx_errors()

```
rx_err_t uart_clear_rx_errors (
    uart_channel_t channel) [nodiscard]
```

Clear sticky receive error flags (ORER/FER/PER) on a UART channel.

Reads SSR and writes back with the receive error bits masked to 0. Per HUM 34.2.7, those flags must be cleared before the receiver will latch further bytes. The RXI ISR clears them on every byte; this helper is the periodic-fallback path when an error latches without RDRF firing (e.g. line glitch on an idle bus).

Parameters

in	channel	SCI channel (0-12).
----	---------	---------------------

Returns

k_rx_ok on success, k_rx_err_invalid_arg if channel is invalid.

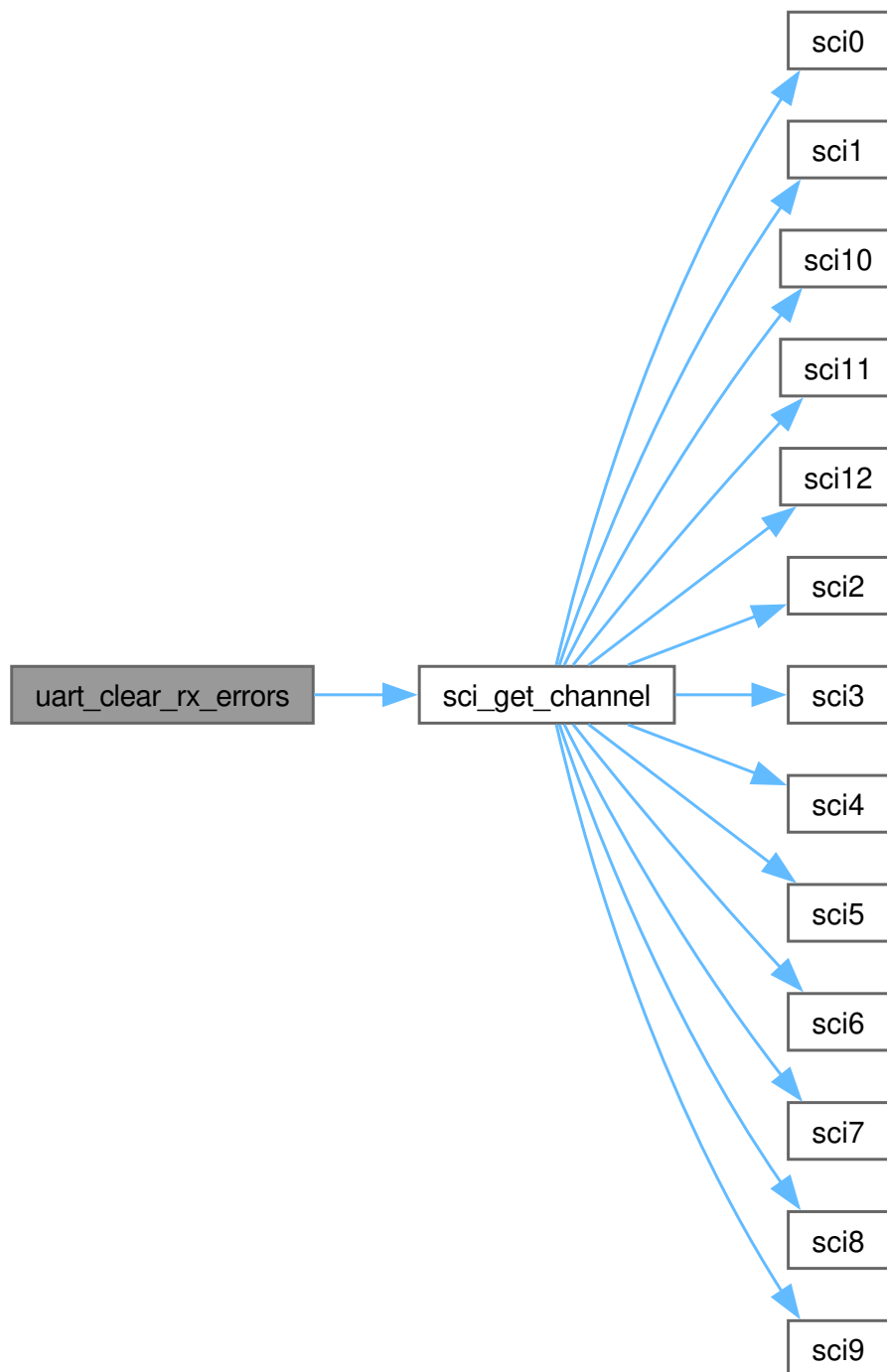
Definition at line 1834 of file [uart.c](#).

```

01835 {
01836     if ((uint8_t)channel >= k_uart_max_channels) {
01837         return k_rx_err_invalid_arg;
01838     }
01839     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01840     if (sci == nullptr) {
01841         return k_rx_err_invalid_arg;
01842     }
01843     const uint8_t ssr = sci->ssr;
01844     if ((ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01845         sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_error_mask);
01846     }
01847     return k_rx_ok;
01848 }
```

References [k_sci_ssr_error_mask](#), [k_uart_flag_clear](#), [k_uart_max_channels](#), [sci_get_channel\(\)](#), and [rx_sci_regs_t::ssr](#).

Here is the call graph for this function:



8.139.3.12 `uart_debug_init()`

```
rx_err_t uart_debug_init (  
    void ) [nodiscard]
```

Initialize default debug UART channel (SCI9 at 115200 baud).

Initialize debug UART (SCI9, 115200 bps, 8N1).

Convenience function to initialize SCI9 with default debug settings:

- Channel: SCI9
- Baud rate: 115200 bps
- TX pin: PB7 (TXD9)
- RX pin: PB6 (RXD9)
- Format: 8N1 (8 data bits, no parity, 1 stop bit)

SCI9 is connected to the CY7C65213 USB-UART bridge on the STAR PCB, providing a virtual COM port when connected via USB.

Algorithm steps:

1. Create `uart_channel_config_t` with default values
2. Call `uart_init_channel()` with the configuration
3. Return result

Precondition

System clocks must be initialized (PCLKB = 60 MHz)

SCI9 must not be already initialized

Postcondition

SCI9 configured and ready for TX/RX at 115200 baud

`uart_debug_puts()`, `uart_debug_putc()` etc. are functional

Note

Call once during early system initialization

Call before any other debug output functions

Warning

Must be called before ThreadX starts if debug output is needed during init

Thread Safety:

Not thread-safe. Call once during startup before ThreadX.

Performance:

- Execution time: ~50 us
- Stack usage: 64 bytes

Example:

```
void system_init(void)
{
    // Initialize clocks first
    rx_clock_init();

    // Initialize debug UART for early output
    rx_err_t err = uart_debug_init();
    if (err != k_rx_ok) {
        // Cannot print error - no UART!
        // Could blink LED or set error flag
        return;
    }

    // Now we can print debug messages
    uart_debug_puts("[INFO] System starting...\n");
}
```

See also

[uart_init_channel\(\)](#) Generic channel initialization

[uart_debug_puts\(\)](#) Debug string output

[uart_debug_putc\(\)](#) Debug character output

Since

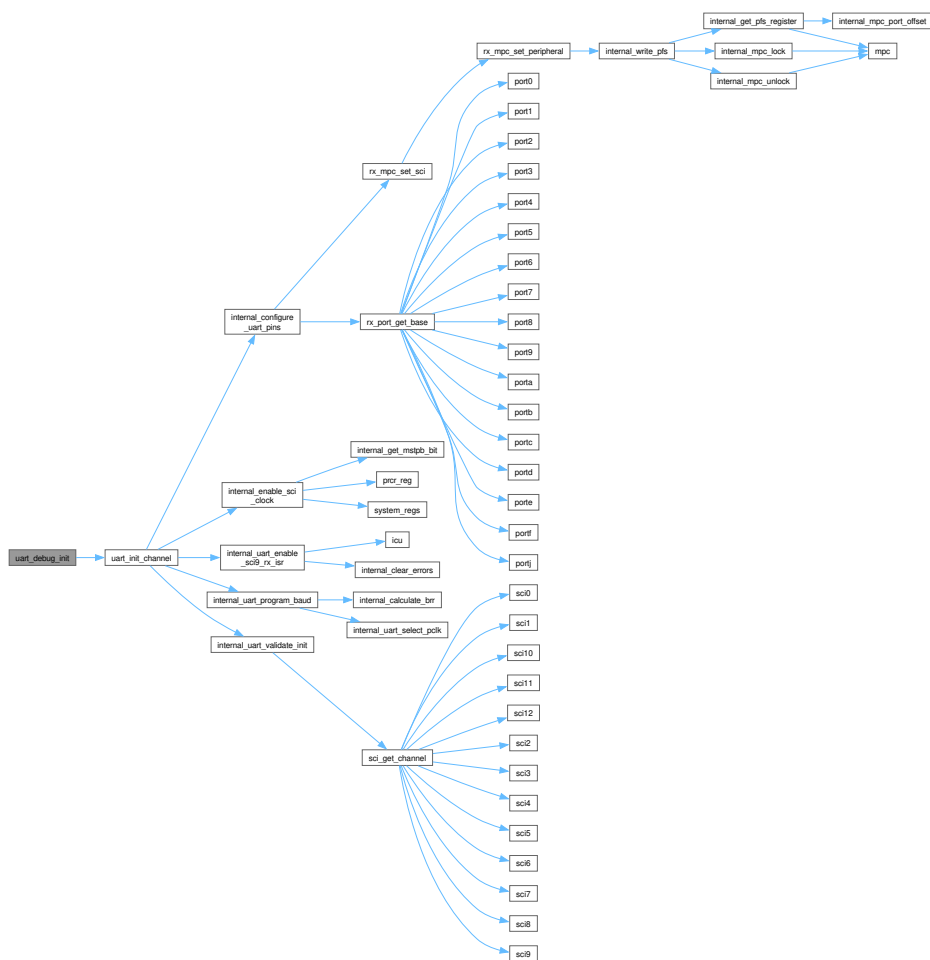
Version 1.0.0

Definition at line 1977 of file [uart.c](#).

```
01978 {
01979     const uart_channel_config_t config = {
01980         .channel = (uart_channel_t)k_uart_debug_channel,
01981         .baudrate = k_uart_default_baudrate,
01982         .tx_gpio = (rx_port_pin_t)k_uart_debug_tx_gpio,
01983         .rx_gpio = (rx_port_pin_t)k_uart_debug_rx_gpio,
01984     };
01985     return uart_init_channel(&config);
01986 }
```

References [k_uart_debug_channel](#), [k_uart_debug_rx_gpio](#), [k_uart_debug_tx_gpio](#), [k_uart_default_baudrate](#), and [uart_init_channel\(\)](#).

Here is the call graph for this function:



```
void uart_debug_putc (
    const char data)
```

Transmit a single character on debug UART (SCI9).

Algorithm steps:

1. Call `uart_putc_channel(k_uart_debug_channel, data)`
2. Cast return value to `(void)` - errors discarded

uart_debug_init() must have been called successfully
SCI9 must be initialized and TX enabled

Postcondition

Character written to SCI9 TDR and transmission started
Any errors are silently discarded

Note

Error return from [uart_putc_channel\(\)](#) is intentionally discarded
For error-checked output use [uart_putc_channel\(\)](#) directly

Warning

Only available when RX_IS_SIMULATOR is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us @ 115200 baud
- Stack usage: 16 bytes

Example:

```
uart_debug_putc('A');    // Send 'A'  
uart_debug_putc('\r');  // Carriage return  
uart_debug_putc('\n');  // Line feed
```

See also

[uart_debug_puts\(\)](#) Transmit string with newline conversion
[uart_putc_channel\(\)](#) Error-checked single character transmit

Since

Version 1.0.0

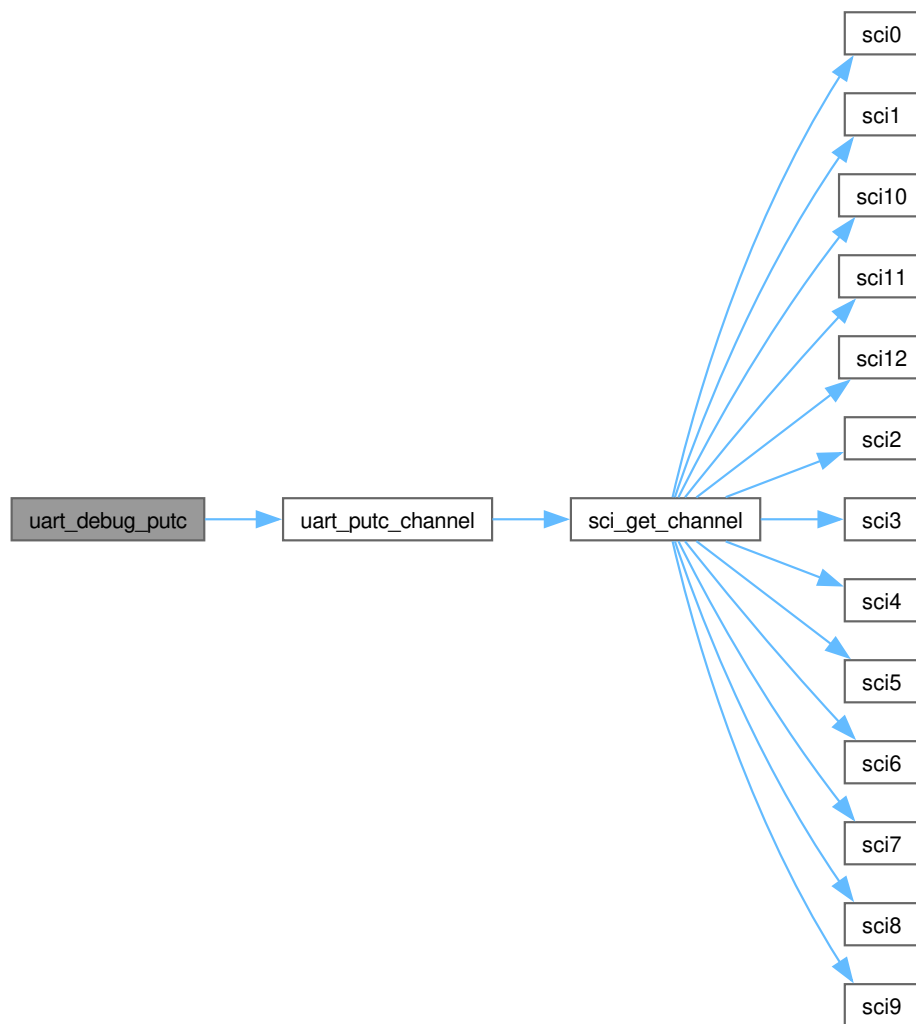
Definition at line 2040 of file [uart.c](#).

```
02041 {  
02042  /* Ignore errors for debug output (used in early init before error handling) */  
02043  (void)uart_putc_channel((uart_channel_t)k_uart_debug_channel, data);  
02044 }
```

References [k_uart_debug_channel](#), and [uart_putc_channel\(\)](#).

Referenced by [uart_debug_puthex\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.14 uart`debug`puthex()

```
void uart__debug__puthex (
    const uint32_t value,
    uint8_t digits)
```

Transmit 32-bit unsigned integer as hexadecimal string on debug UART.

Print a hexadecimal value on debug UART (SCI9).

Transmits "0x" prefix followed by the specified number of uppercase hex digits of value on SCI9. Digits are always printed most-significant first. The digits parameter is clamped to [k_uart_hex_min_digits, k_uart_hex_max_digits] (1-8) before use.

Algorithm steps:

1. Send "0x" prefix via `uart_debug_puts()`
2. Clamp digits to [1, 8]
3. Iterate nibbles from MSN to LSN (statically bounded by `k_uart_hex_max_digits`)
4. For each nibble index < digits, extract nibble and print uppercase hex char

Precondition

`uart_debug_init()` must have been called successfully
`uart_debug_puts()` and `uart_debug_putc()` must be functional

Postcondition

"0x" followed by digits uppercase hex characters transmitted on SCI9
digits clamped to [1, 8] if out of range

Note

Uses static lookup table `s_hex[]` for digit-to-character conversion
Always prefixes output with "0x"

Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes

Example:

```
uart_debug_puts("Addr: ");  
uart_debug_puthex(0xDEADBEEF, 8); // Sends "0xDEADBEEF"  
uart_debug_puthex(0x42, 2);      // Sends "0x42"  
uart_debug_putc('\n');
```

See also

`uart_debug_putint()` Print signed decimal value
`uart_debug_puts()` Underlying string transmit

Since

Version 1.0.0

Definition at line 2230 of file [uart.c](#).

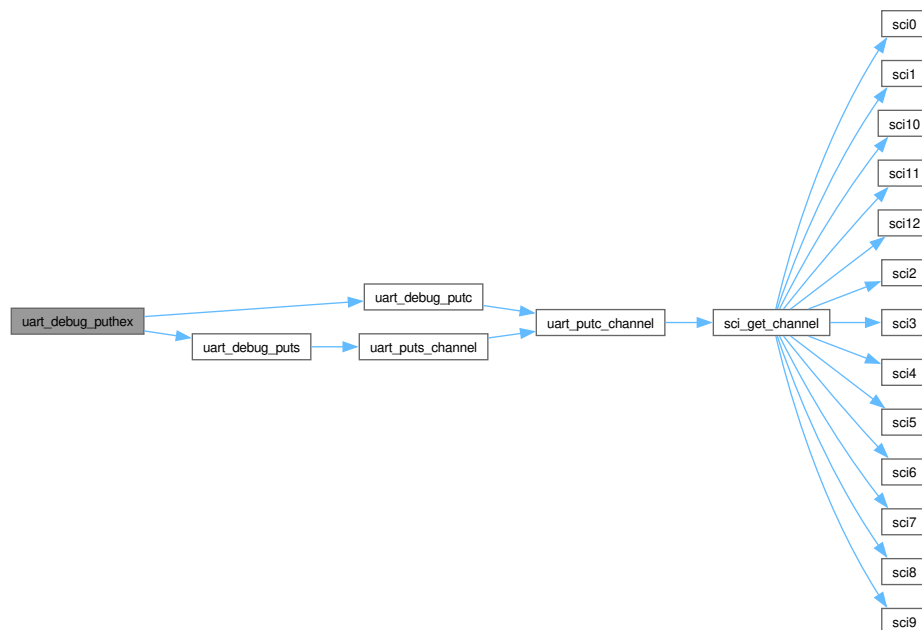
```

02231 {
02232     static const char s_hex[] = "0123456789ABCDEF";
02233
02234     uart_debug_puts("0x");
02235
02236     /* Clamp digits to valid range */
02237     if (digits > k_uart_hex_max_digits) {
02238         digits = k_uart_hex_max_digits;
02239     }
02240     if (digits == k_uart_hex_zero_digits) {
02241         digits = k_uart_hex_min_digits;
02242     }
02243
02244     /* Print hex digits from most significant (statically bounded, unsigned counter) */
02245     for (uint8_t i = k_uart_hex_zero_digits; i < k_uart_hex_max_digits; ++i) {
02246         const uint8_t digit_idx = (k_uart_hex_max_digits - k_uart_hex_min_digits) - i;
02247         if (digit_idx < digits) {
02248             const uint8_t nibble =
02249                 (uint8_t)((value » (digit_idx * k_uart_hex_nibble_bits)) & k_uart_hex_nibble_mask);
02250             uart_debug_putc(s_hex[nibble]);
02251         }
02252     }
02253 }

```

References [k_uart_hex_max_digits](#), [k_uart_hex_min_digits](#), [k_uart_hex_nibble_bits](#), [k_uart_hex_nibble_mask](#), [k_uart_hex_zero_digits](#), [uart_debug_putc\(\)](#), and [uart_debug_puts\(\)](#).

Here is the call graph for this function:



8.139.3.15 uart'debug'putint()

```

void uart_debug_putint (
    const int32_t value)

```

Transmit signed 32-bit integer as decimal string on debug UART.

Print a signed integer on debug UART (SCI9).

Converts a signed 32-bit integer to a decimal ASCII string and transmits it on SCI9. Handles INT32_↔MIN correctly by using int64_t for the negation. Uses a fixed-size stack buffer (k_uart_int_buffer_size = 12) built in reverse then passed to `uart_debug_puts()`.

Algorithm steps:

1. Determine sign; compute abs_value as uint32_t
2. Null-terminate the buffer end
3. Build digits right-to-left (statically bounded by k_uart_int_buffer_size)
4. Prepend '-' if negative
5. Call `uart_debug_puts()` with the resulting substring pointer

Precondition

`uart_debug_init()` must have been called successfully
`uart_debug_puts()` must be functional

Postcondition

Decimal representation of value transmitted on SCI9
Leading zeros suppressed; negative values prefixed with '-'

Note

No return value - output errors are silently discarded
Buffer is stack-allocated; safe for re-entrant calls on different tasks

Warning

Only available when RX_IS_SIMULATOR is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us per digit @ 115200 baud + digit-loop overhead
- Stack usage: 32 bytes (buffer + locals)

Example:

```
uart_debug_puts("Value: ");  
uart_debug_putint(42); // Sends "42"  
uart_debug_putint(-2147483648); // Sends "-2147483648"  
uart_debug_putc('\n');
```

See also

[uart_debug_puthex\(\)](#) Print value in hexadecimal
[uart_debug_puts\(\)](#) Underlying string transmit

Since

Version 1.0.0

Definition at line 2150 of file [uart.c](#).

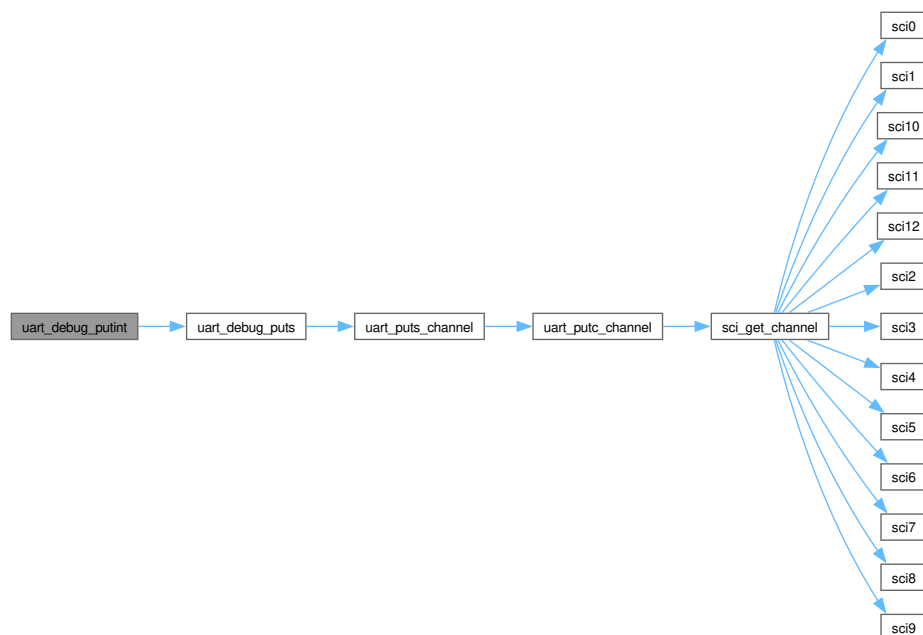
```

02151 {
02152     char buffer[k_uart_int_buffer_size]; /* Enough for -2147483648 */
02153     char* p = buffer + sizeof(buffer) - 1;
02154
02155     /* Handle negative numbers */
02156     const bool is_negative = (value < 0);
02157     uint32_t abs_value = is_negative ? (uint32_t)(-(int64_t)value) : (uint32_t)value;
02158
02159     /* Null terminate */
02160     *p = '\0';
02161
02162     /* Convert to string (reverse order, statically bounded) */
02163     for (uint8_t i = 0; i < k_uart_int_buffer_size - 1; ++i) {
02164         *(--p) = (char)('0' + (abs_value % k_uart_base_10));
02165         abs_value /= k_uart_base_10;
02166         if (abs_value == 0) {
02167             break; /* All digits processed */
02168         }
02169         if (p <= buffer) {
02170             break; /* Buffer limit reached (should not happen with correct sizing) */
02171         }
02172     }
02173
02174     /* Add minus sign if negative */
02175     if (is_negative && p > buffer) {
02176         *(--p) = '-';
02177     }
02178
02179     /* Transmit the string */
02180     uart_debug_puts(p);
02181 }

```

References [k_uart_base_10](#), [k_uart_int_buffer_size](#), and [uart_debug_puts\(\)](#).

Here is the call graph for this function:



8.139.3.16 `uart_debug_puts()`

```
void uart_debug_puts (  
    const char * str)
```

Transmit null-terminated string on debug UART with newline conversion.

Transmit a null-terminated string on debug UART (SCI9).

Transmits a null-terminated string to SCI9 with automatic LF-to-`\r+LF` conversion for terminal compatibility. Enforces a maximum length of `k_uart_max_str_len` (256) characters per NASA Power of 10 Rule 2. Silently returns on NULL pointer to allow safe use in early init.

Algorithm steps:

1. Return immediately if `str` is `nullptr` (defensive, no error return)
2. For each character up to `k_uart_max_str_len`:
 - a. Stop at null terminator
 - b. If LF, send `\r` first
 - c. Send character via `uart_debug_putc()`

Precondition

`uart_debug_init()` must have been called successfully
`str` should point to a null-terminated string in valid memory

Postcondition

All characters up to null terminator transmitted to SCI9
Each LF replaced by `\r\n` in the transmitted output

Note

No return value - errors from `uart_debug_putc()` are silently discarded
String truncated at `k_uart_max_str_len` (256) characters

Warning

Only available when `RX_IS_SIMULATOR` is 0 (hardware builds)

Thread Safety:

Not safe for concurrent access on SCI9 without external mutex.

Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes

Example:

```
uart_debug_puts("Hello, World!\n"); // Sends "Hello, World!\\r\\n"
uart_debug_puts("[INFO] Boot complete\n");
```

See also

[uart_debug_putc\(\)](#) Transmit single character
[uart_debug_putint\(\)](#) Transmit decimal integer
[uart_puts_channel\(\)](#) Error-checked string transmit

Since

Version 1.0.0

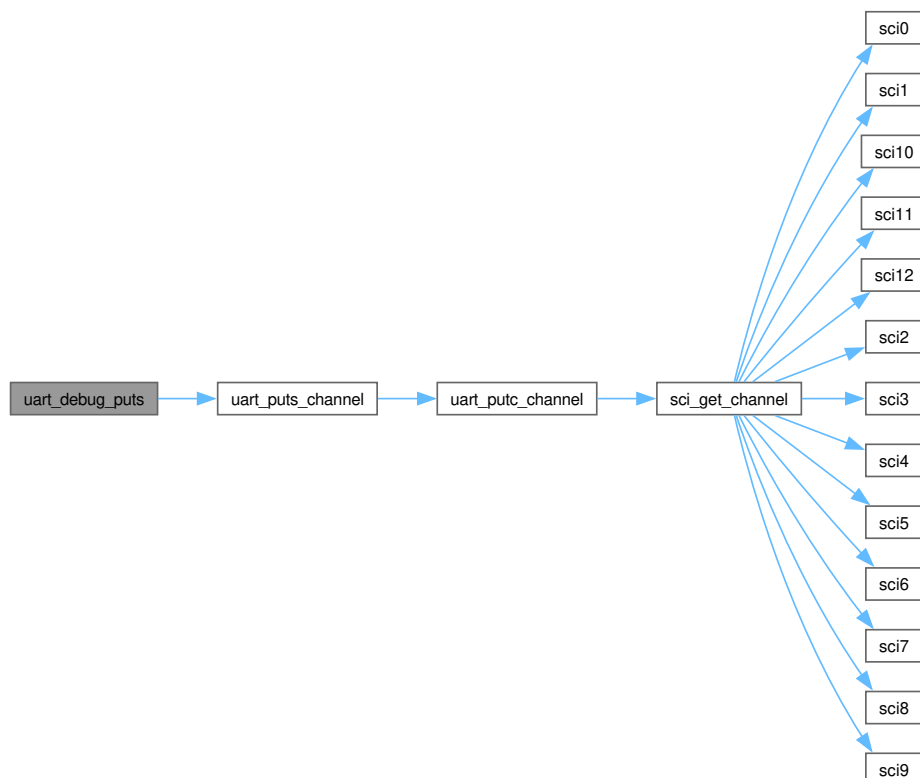
Definition at line 2092 of file [uart.c](#).

```
02093 {
02094     if (str == nullptr) {
02095         return;
02096     }
02097
02098     /* Delegate to uart_puts_channel which already handles LF->CRLF conversion,
02099      * bounded iteration (k_uart_max_str_len), and per-character error checking. */
02100     (void)uart_puts_channel((uart_channel_t)k_uart_debug_channel, str);
02101 }
```

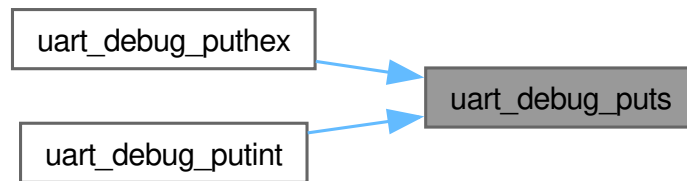
References [k_uart_debug_channel](#), and [uart_puts_channel\(\)](#).

Referenced by [uart_debug_puthex\(\)](#), and [uart_debug_putint\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.17 `uart_deinit_channel()`

```
rx_err_t uart_deinit_channel (
    const uart\_channel\_t channel) [nodiscard]
```

Deinitialize UART channel and disable TX/RX.

Deinitialize SCI channel.

Disables transmit and receive on the specified SCI channel and marks it as uninitialized. The module clock is left running (safe to re-initialize without re-enabling). After this call the channel may be re-initialized with [uart_init_channel\(\)](#).

Algorithm steps:

1. Validate channel number (0-12)
2. Get SCI register base address
3. Write `S\r = 0` to disable TX and RX
4. Clear `s_channel_initialized[channel]`

Precondition

channel must be in range [0, 12]

Channel should be initialized before deinitializing (safe to call on uninit channel)

Postcondition

TX and RX disabled (`S\r = 0`)

`s_channel_initialized[channel]` set to false

Note

Not thread-safe - call during shutdown, not during normal operation

Module stop clock is NOT re-asserted (peripheral clock left enabled)

Thread Safety:

Not thread-safe. Do not call while another thread is using the channel.

Performance:

- Execution time: ~ 1 us
- Stack usage: 16 bytes

Example:

```
rx_err_t err = uart_deinit_channel(k_uart_channel_9);
if (err != k_rx_ok) {
    // Handle error (invalid channel)
}
// Channel is now disabled and can be re-initialized
```

See also

[uart_init_channel\(\)](#) Initialize channel
[uart_debug_init\(\)](#) Initialize debug channel (SCI9)

Since

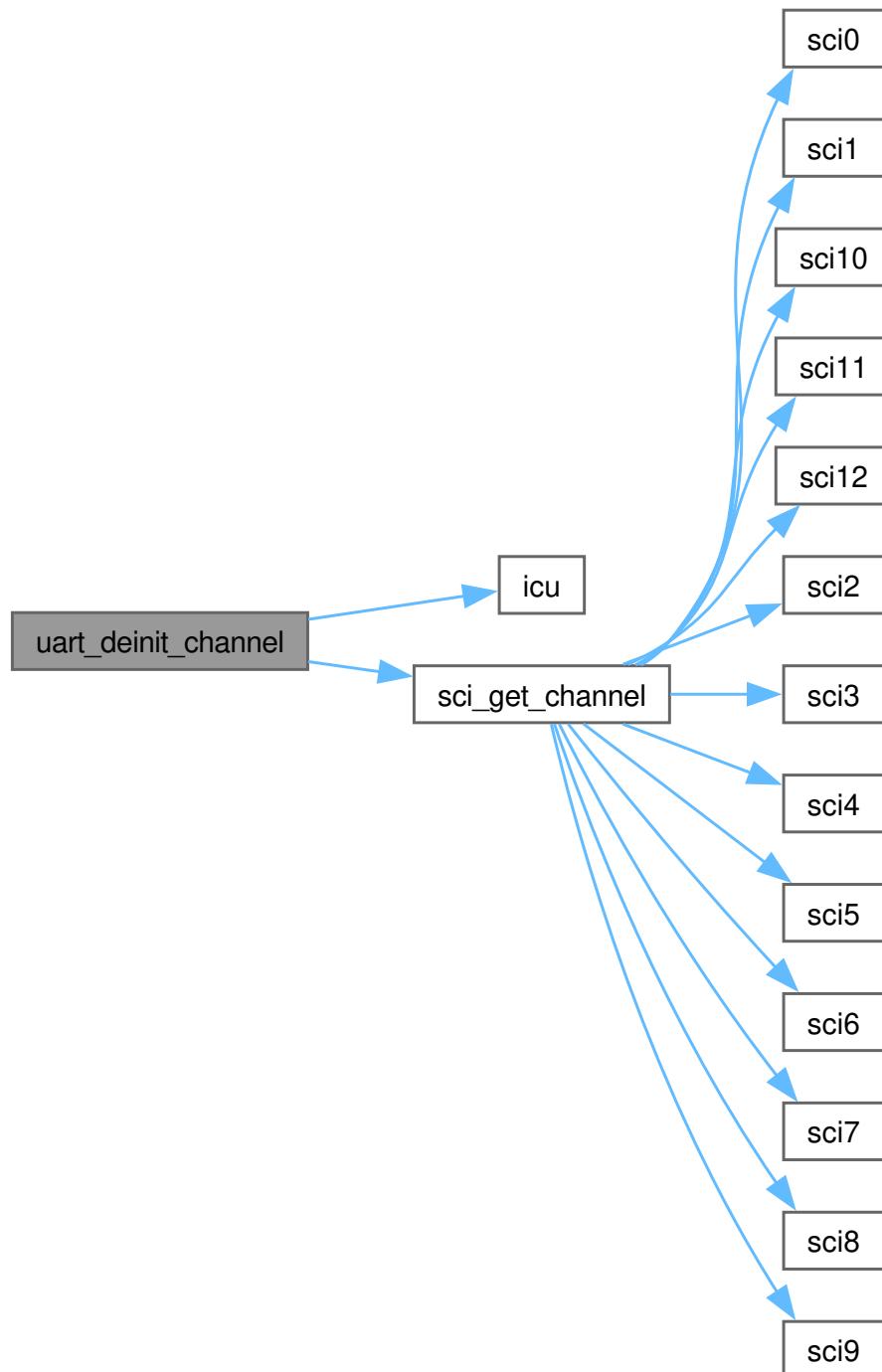
Version 1.0.0

Definition at line 1215 of file [uart.c](#).

```
01216 {
01217     /* Validate channel */
01218     if ((uint8_t)channel >= k_uart_max_channels) {
01219         return k_rx_err_invalid_arg;
01220     }
01221
01222     /* Get SCI register base */
01223     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01224     if (sci == nullptr) {
01225         return k_rx_err_invalid_arg;
01226     }
01227
01228     /* Disable TX/RX (this also drops SCR.RIE for SCI9). */
01229     sci->scr = k_sci_scr_disabled;
01230
01231     /* For SCI9, mask the RXI9 interrupt in the ICU and flush the ring so a
01232      * subsequent re-init starts clean. */
01233     if (channel == k_uart_channel_9) {
01234         const uint16_t vec_rxi9 = (uint16_t)k_vect_sci9_rxi9;
01235         icu()->ier[vec_rxi9 / k_icu_ier_bits_per_reg] &=
01236             (uint8_t)~(1U « (vec_rxi9 % k_icu_ier_bits_per_reg));
01237         icu()->ir[vec_rxi9] = 0U;
01238         s_sci9_rx_head = 0U;
01239         s_sci9_rx_tail = 0U;
01240     }
01241
01242     /* Mark channel as not initialized */
01243     s_channel_initialized[channel] = false;
01244
01245     return k_rx_ok;
01246 }
```

References [icu\(\)](#), [rx_icu_regs_t::ier](#), [rx_icu_regs_t::ir](#), [k_icu_ier_bits_per_reg](#), [k_sci_scr_disabled](#), [k_uart_channel_9](#), [k_uart_max_channels](#), [k_vect_sci9_rxi9](#), [s_channel_initialized](#), [s_sci9_rx_head](#), [s_sci9_rx_tail](#), [sci_get_channel\(\)](#), and [rx_sci_regs_t::scr](#).

Here is the call graph for this function:



8.139.3.18 `uart_getc_channel()`

```

rx_err_t uart_getc_channel (
    const uart\_channel\_t channel,
    char * data) [nodiscard]
  
```

Receive single character from UART channel (non-blocking).

Receive a single character from specified channel (non-blocking).

Attempts to read a single byte from the specified SCI channel's receive data register (RDR). This is a non-blocking operation - returns immediately with `k_rx_err_empty` if no data is available.

Algorithm steps:

1. Validate data pointer and channel number
2. Check channel initialization state
3. Get SCI register base address
4. Clear any error flags (ORER, FER, PER)
5. Check RDRF flag (Receive Data Register Full)
 - If not set, return k_rx_err_empty immediately
6. Read data byte from RDR register
7. Clear RDRF flag (read SSR, write back with RDRF=0)

Error flag handling:

- ORER (Overrun Error): Previous data overwritten before read
- FER (Framing Error): Stop bit not detected
- PER (Parity Error): Parity mismatch (not used in 8N1 mode)

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
data must point to valid memory for single char

Postcondition

On success, *data contains received byte
RDRF flag cleared (ready for next byte)
Error flags cleared if any were set

Note

Non-blocking - returns immediately if no data
Clears error flags automatically before read
Single byte buffer - call frequently to avoid overrun

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~5 us (no wait)
- Stack usage: 16 bytes

Example (Polling Loop):

```
char c;
rx_err_t err;

// Poll for incoming data
do {
    err = uart_getc_channel(k_uart_channel_9, &c);
    if (err == k_rx_ok) {
        process_character(c);
    }
} while (err == k_rx_ok); // Continue while data available
```

Example (Timeout Loop):

```
char c;
uint32_t timeout = 100000;

while (timeout > 0) {
    rx_err_t err = uart_getc_channel(k_uart_channel_9, &c);
    if (err == k_rx_ok) {
        // Got data
        break;
    }
    timeout--;
}

if (timeout == 0) {
    // No data received within timeout
}
```

See also

[uart_read_channel\(\)](#) Read multiple bytes
[uart_rx_available\(\)](#) Check if data is available

Since

Version 1.0.0

Definition at line 1606 of file [uart.c](#).

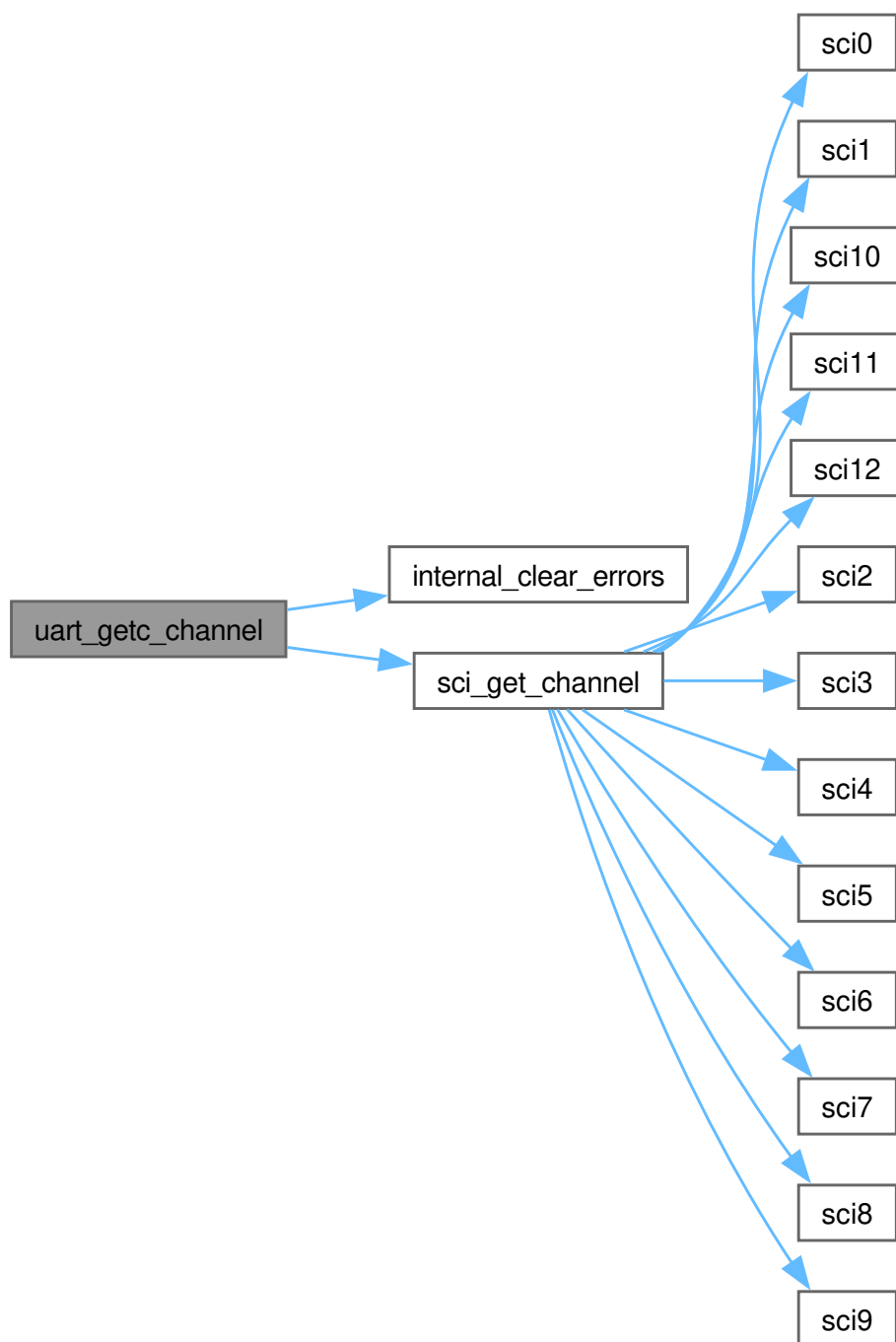
```
01607 {
01608     /* Validate parameters */
01609     if (data == nullptr) {
01610         return k_rx_err_null_ptr;
01611     }
01612
01613     if ((uint8_t)channel >= k_uart_max_channels) {
01614         return k_rx_err_invalid_arg;
01615     }
01616
01617     if (!s_channel_initialized[channel]) {
01618         return k_rx_err_invalid_state;
01619     }
01620
01621     /* SCI9 reads pull from the RXI9-fed ring buffer so polled callers cannot
01622      * lose bytes that arrive between polls. Head/tail word accesses are atomic
01623      * on RX72N (aligned 16-bit loads/stores are single-cycle), so lock-free
01624      * SPSC is safe between the ISR producer and the task consumer. */
01625     if (channel == k_uart_channel_9) {
01626         if (s_sci9_rx_tail == s_sci9_rx_head) {
01627             return k_rx_err_empty;
01628         }
01629         *data = (char)s_sci9_rx_ring[s_sci9_rx_tail];
01630         s_sci9_rx_tail = (uint16_t)((s_sci9_rx_tail + 1U) & k_sci9_rx_ring_mask);
01631         return k_rx_ok;
01632     }
01633
01634     /* Get SCI register base */
01635     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01636     if (sci == nullptr) {
01637         return k_rx_err_invalid_arg;
01638     }
```

```
01639
01640 /* Clear any error flags first */
01641 if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01642     internal_clear_errors(sci);
01643 }
01644
01645 /* Check if receive data is available (RDRF flag) */
01646 if ((sci->ssr & k_sci_ssr_rdrf_flag) == k_uart_flag_clear) {
01647     return k_rx_err_empty;
01648 }
01649
01650 /* Read received data */
01651 *data = (char)sci->rdr;
01652
01653 /* Clear RDRF flag by reading SSR then writing 0 */
01654 /* Some RX MCUs require explicit clear after reading RDR */
01655 const uint8_t ssr = sci->ssr;
01656 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01657
01658 return k_rx_ok;
01659 }
```

References [internal_clear_errors\(\)](#), [k_sci9_rx_ring_mask](#), [k_sci_ssr_error_mask](#), [k_sci_ssr_rdrf_flag](#), [k_uart_channel_9](#), [k_uart_flag_clear](#), [k_uart_max_channels](#), [rx_sci_regs_t::rdr](#), [s_channel_initialized](#), [s_sci9_rx_head](#), [s_sci9_rx_ring](#), [s_sci9_rx_tail](#), [sci_get_channel\(\)](#), and [rx_sci_regs_t::ssr](#).

Referenced by [uart_read_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.19 uart_init_channel()

```
rx_err_t uart_init_channel (
    const uart_channel_config_t * config) [nodiscard]
```

Initialize SCI channel for UART communication.

Performs full hardware initialization including:

- Module stop control (MSTPCRB)
- Pin function configuration (MPC)
- GPIO direction setup (PDR/PMR)
- SCI register configuration

Parameters

in	config	UART channel configuration (must not be NULL)
----	--------	---

Returns

k_rx_ok on success, k_rx_err_invalid_arg if config is nullptr or contains invalid values, k_rx_err_invalid_state if channel already initialized

Definition at line 1135 of file [uart.c](#).

```

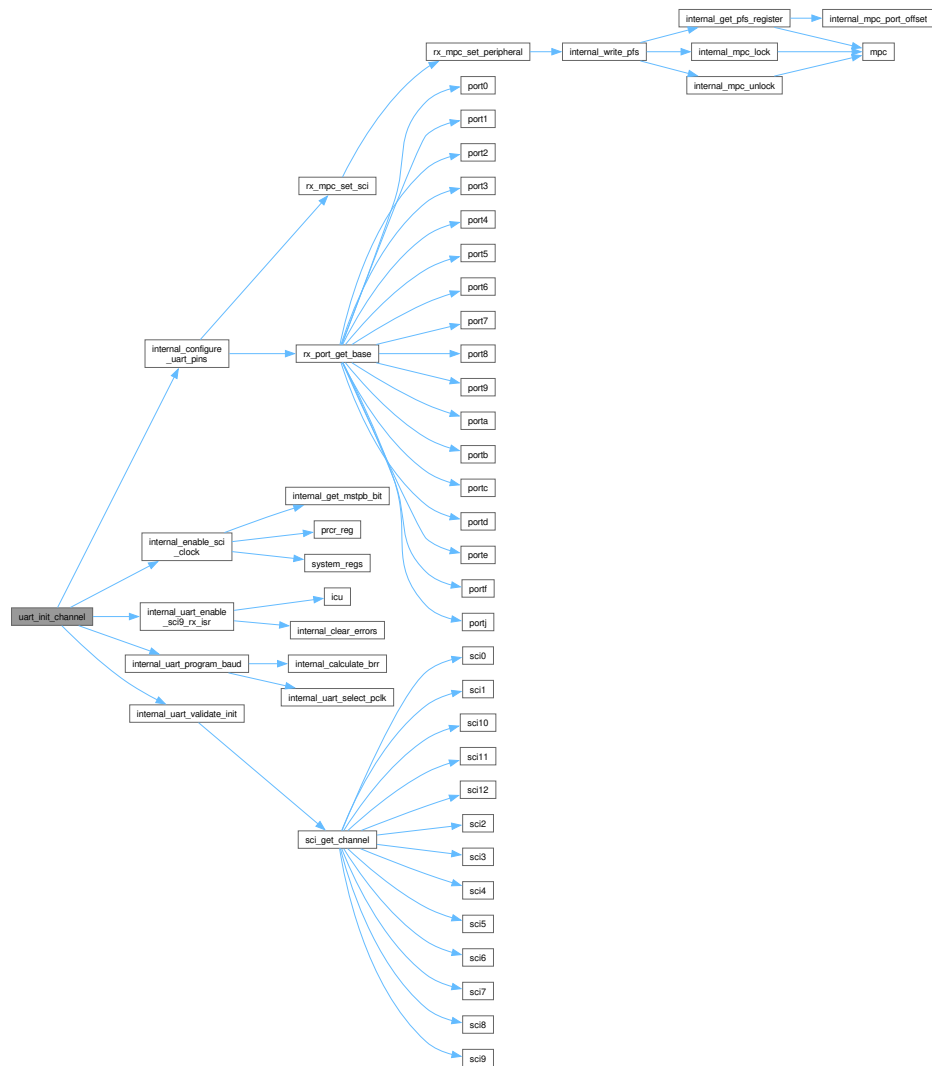
01136 {
01137     volatile rx_sci_regs_t* sci = nullptr;
01138     const rx_err_t check = internal_uart_validate_init(config, &sci);
01139     if (check != k_rx_ok) {
01140         return check;
01141     }
01142
01143     rx_err_t err = internal_enable_sci_clock(config->channel);
01144     if (err != k_rx_ok) {
01145         return err;
01146     }
01147     err = internal_configure_uart_pins(config->tx_gpio, config->rx_gpio);
01148     if (err != k_rx_ok) {
01149         return err;
01150     }
01151
01152     /* Disable TX/RX while reprogramming registers */
01153     sci->scr = k_sci_scr_disabled;
01154
01155     internal_uart_program_baud(sci, config->channel, config->baudrate);
01156
01157     /* Configure serial control: Enable TX and RX */
01158     sci->scr = k_sci_scr_txrx_enabled;
01159
01160     if (config->channel == k_uart_channel_9) {
01161         internal_uart_enable_sci9_rx_isr(sci);
01162     }
01163
01164     s_channel_initialized[config->channel] = true;
01165     return k_rx_ok;
01166 }

```

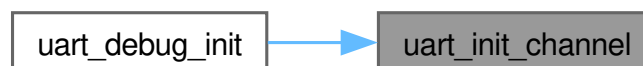
References [uart_channel_config_t::baudrate](#), [uart_channel_config_t::channel](#), [internal_configure_uart_pins\(\)](#), [internal_enable_sci_clock\(\)](#), [internal_uart_enable_sci9_rx_isr\(\)](#), [internal_uart_program_baud\(\)](#), [internal_uart_validate_init\(\)](#), [k_sci_scr_disabled](#), [k_sci_scr_txrx_enabled](#), [k_uart_channel_9](#), [uart_channel_config_t::rx_gpio](#), [s_channel_initialized](#), [rx_sci_regs_t::scr](#), and [uart_channel_config_t::tx_gpio](#).

Referenced by [uart_debug_init\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.20 uart_putc_channel()

```

rx_err_t uart_putc_channel (
    const uart_channel_t channel,
    const char data) [nodiscard]

```

Transmit single character on UART channel.

Transmit a single character on specified channel.

Transmits a single byte on the specified SCI channel. Waits for the transmit data register to become empty (TDRE flag) with timeout protection, then writes the data to TDR. Uses polling mode (no interrupts).

Algorithm steps:

1. Validate channel number (0-12)
2. Check channel is initialized
3. Get SCI register base address
4. Wait for TDRE flag with timeout (`k_uart_tx_timeout` iterations)
5. Write data byte to TDR register
6. Clear TDRE flag (read SSR, write back with TDRE=0)

Character transmission timing at 115200 baud:

- Start bit: 8.68 us
- 8 data bits: 69.44 us
- Stop bit: 8.68 us
- Total: ~86.8 us per character

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
UART TX must be enabled (happens during init)

Postcondition

Character written to TDR and transmission started
TDRE flag cleared (TDR occupied)

Note

Blocking call - waits for TDRE flag
Timeout prevents infinite wait if hardware fails

Warning

May block for up to ~100ms if TX line is held low

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~90 us @ 115200 baud (one character time)
- Stack usage: 16 bytes

Example:

```
// Send single character
rx_err_t err = uart_putc_channel(k_uart_channel_9, 'A');
if (err != k_rx_ok) {
    // Handle error
}

// Send carriage return + line feed
uart_putc_channel(k_uart_channel_9, '\\r');
uart_putc_channel(k_uart_channel_9, '\\n');
```

See also

[uart_puts_channel\(\)](#) Transmit string with newline conversion
[uart_write_channel\(\)](#) Transmit binary data

Since

Version 1.0.0

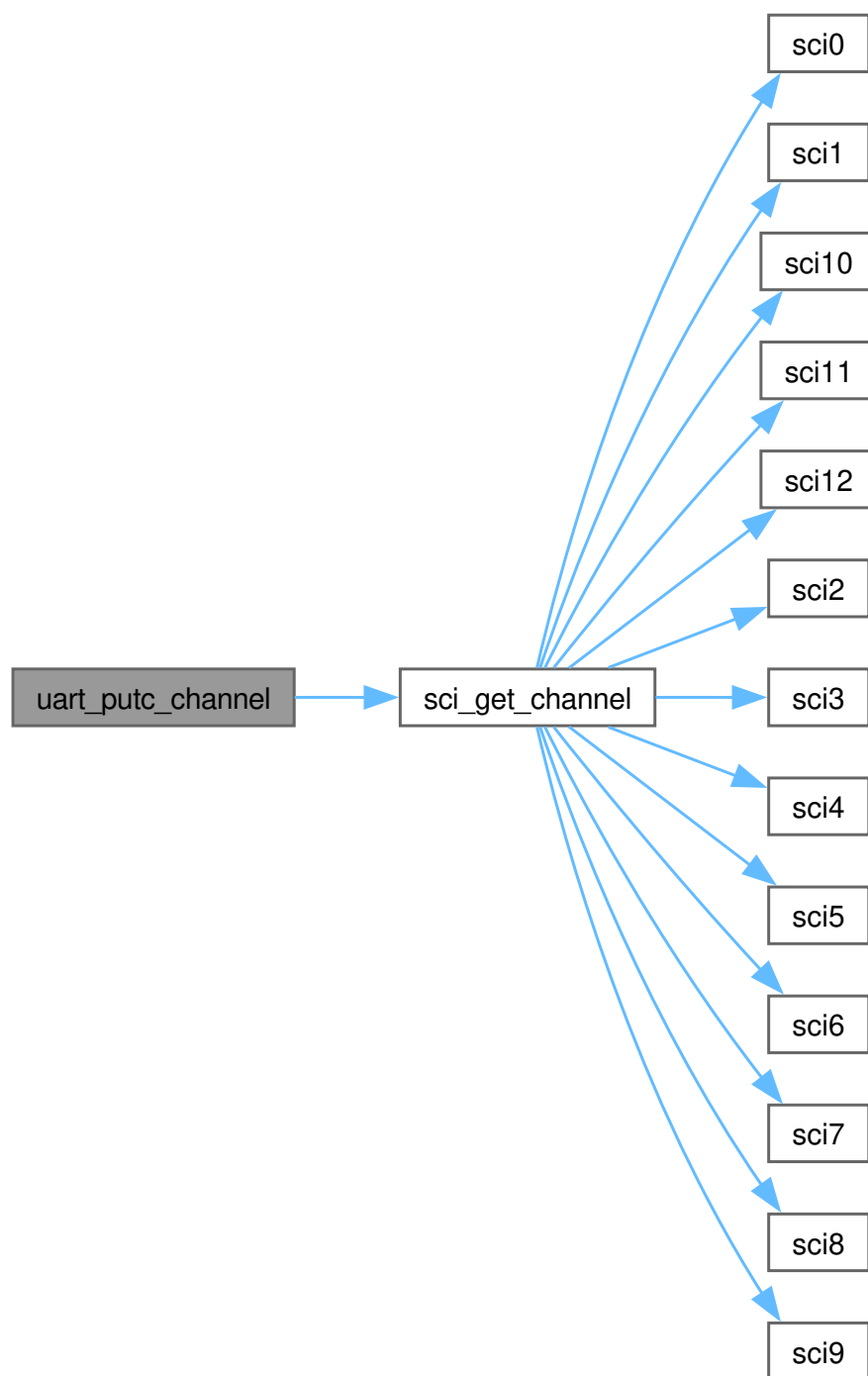
Definition at line 1308 of file [uart.c](#).

```
01309 {
01310     /* Validate channel */
01311     if ((uint8_t)channel >= k_uart_max_channels) {
01312         return k_rx_err_invalid_arg;
01313     }
01314
01315     /* Check initialization */
01316     if (!s_channel_initialized[channel]) {
01317         return k_rx_err_invalid_state;
01318     }
01319
01320     /* Get SCI register base */
01321     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01322     if (sci == nullptr) {
01323         return k_rx_err_invalid_arg;
01324     }
01325
01326     /* Wait for transmit buffer to be empty (TDRE flag) with timeout */
01327     uint32_t timeout = k_uart_tx_timeout;
01328     while ((sci->ssr & k_sci_ssr_tdre_flag) == k_uart_flag_clear &&
01329         timeout > k_uart_timeout_expired) {
01330         timeout -= k_uart_timeout_decrement;
01331     }
01332
01333     /* Check if timeout occurred */
01334     if (timeout == k_uart_timeout_expired) {
01335         return k_rx_err_timeout;
01336     }
01337
01338     /* Write data to transmit register */
01339     sci->tdr = (uint8_t)data;
01340
01341     /* Clear TDRE flag by reading SSR then writing 0 */
01342     const uint8_t ssr = sci->ssr;
01343     sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_tdre_flag);
01344
01345     return k_rx_ok;
01346 }
```

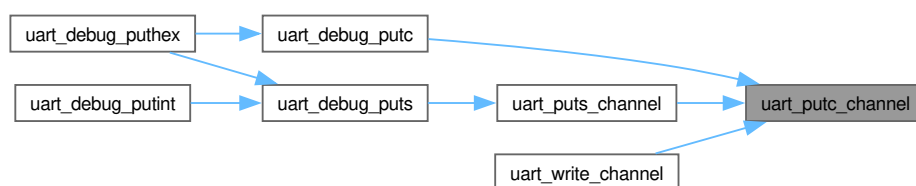
References [k_sci_ssr_tdre_flag](#), [k_uart_flag_clear](#), [k_uart_max_channels](#), [k_uart_timeout_decrement](#), [k_uart_timeout_expired](#), [k_uart_tx_timeout](#), [s_channel_initialized](#), [sci_get_channel\(\)](#), [rx_sci_regs_t::ssr](#), and [rx_sci_regs_t::tdr](#).

Referenced by [uart_debug_putc\(\)](#), [uart_puts_channel\(\)](#), and [uart_write_channel\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.21 `uart_puts_channel()`

```
rx_err_t uart_puts_channel (
    const uart\_channel\_t channel,
    const char * str)  [nodiscard]
```

Transmit null-terminated string on UART channel with newline conversion.

Transmit a null-terminated string on specified channel.

Transmits a null-terminated string with automatic LF to \r+LF conversion for terminal compatibility. Enforces maximum string length (`k_uart_max_str_len = 256`) to comply with NASA Power of 10 Rule 2 (bounded loops).

Algorithm steps:

1. Validate string pointer (NULL check)
2. Validate channel number and initialization state
3. For each character up to `k_uart_max_str_len`: a. Check for null terminator (success exit) b. If ' ', send '\r' first (\r+LF conversion) c. Transmit character via [uart_putc_channel\(\)](#) d. Propagate any errors immediately
4. If limit reached without terminator, return `k_rx_err_invalid_size`

Newline conversion:

- Input ' ' (LF) -> Output '\r\n' (\r+LF)
- Required for Windows terminals (e.g., PuTTY, TeraTerm)
- Ensures cursor returns to column 0 before line feed

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
 str must point to null-terminated string in valid memory

Postcondition

All characters transmitted including converted newlines
 On error, partial string may have been transmitted

Note

Blocking call - waits for each character to transmit
 String length limited to 256 characters for safety

Warning

Long strings may take significant time (256 chars ~ 22ms @ 115200)

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~90 us per character @ 115200 baud
- Stack usage: 24 bytes
- Example: 100-char string ~ 9 ms

Example:

```
// Simple string
uart_puts_channel(k_uart_channel_9, "Hello, World!\n");
// Output: "Hello, World!\r\n"

// Multiple lines
uart_puts_channel(k_uart_channel_9, "Line 1\nLine 2\nLine 3\n");
// Output: "Line 1\r\nLine 2\r\nLine 3\r\n"

// Error handling
rx_err_t err = uart_puts_channel(k_uart_channel_9, msg);
if (err != k_rx_ok) {
    // Partial transmission may have occurred
}
```

See also

[uart_putc_channel\(\)](#) Transmit single character
[uart_write_channel\(\)](#) Transmit binary data (no conversion)
[uart_debug_puts\(\)](#) Simplified debug wrapper

Since

Version 1.0.0

Definition at line 1415 of file [uart.c](#).

```
01416 {
01417     /* Validate parameters */
01418     if (str == nullptr) {
01419         return k_rx_err_null_ptr;
01420     }
01421
01422     if ((uint8_t)channel >= k_uart_max_channels) {
01423         return k_rx_err_invalid_arg;
01424     }
01425
01426     if (!s_channel_initialized[channel]) {
01427         return k_rx_err_invalid_state;
01428     }
01429
01430     /* Transmit string with \n to \r\n conversion (statically bounded) */
01431     for (uint32_t i = 0; i < k_uart_max_str_len; ++i) {
01432         if (str[i] == '\0') {
01433             return k_rx_ok; /* Terminator found, success */
01434         }
01435         if (str[i] == '\n') {
01436             const rx_err_t err = uart_putc_channel(channel, '\r');
01437             if (err != k_rx_ok) {
01438                 return err;
01439             }
01440         }
01441         const rx_err_t err = uart_putc_channel(channel, str[i]);
01442         if (err != k_rx_ok) {
01443             return err;
01444         }
01445     }
01446 }
```

```

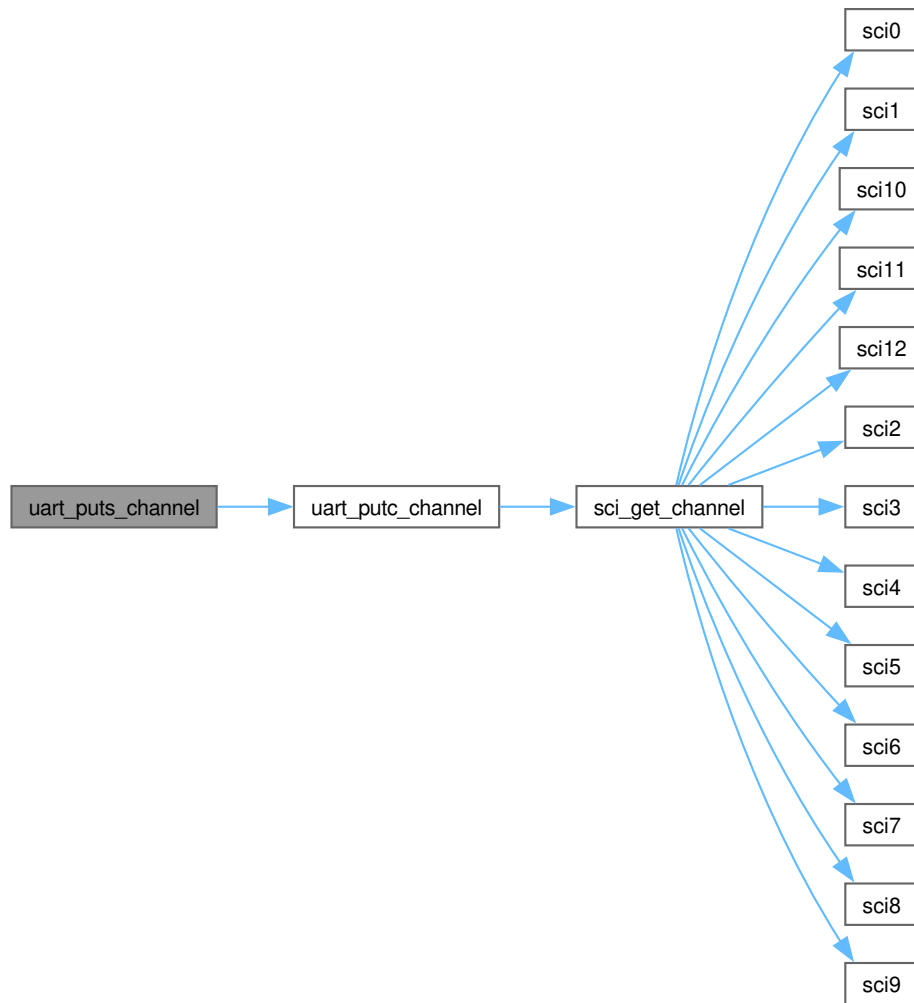
01447  /* Reached limit without finding terminator */
01448  return k_rx_err_invalid_size;
01449  }

```

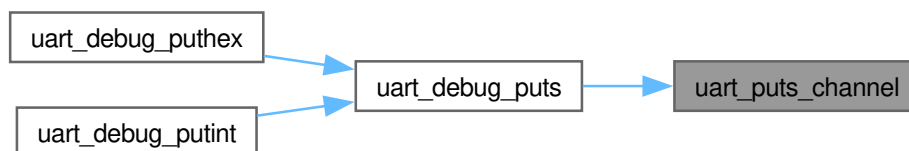
References [k_uart_max_channels](#), [k_uart_max_str_len](#), [s_channel_initialized](#), and [uart_putc_channel\(\)](#).

Referenced by [uart_debug_puts\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



8.139.3.22 `uart_read_channel()`

```
rx_err_t uart_read_channel (  
    const uart\_channel\_t channel,  
    uint8_t * data,  
    uint16_t length,  
    uint16_t * bytes_read) [nodiscard]
```

Read available bytes from UART channel (non-blocking).

Read available data from specified channel (non-blocking).

Reads up to length bytes from the specified SCI channel using [uart_getc_channel\(\)](#). Stops immediately when no more data is available (RDRE flag not set) rather than waiting. Actual bytes received is reported via bytes_read.

Algorithm steps:

1. Validate data and bytes_read pointers (NULL check)
2. Validate channel number and initialization state
3. Set *bytes_read = 0
4. For each slot in [0, length): a. Call [uart_getc_channel\(\)](#); if k_rx_err_empty, break (done) b. On other error, propagate immediately c. Store byte and increment *bytes_read
5. Return k_rx_ok (even if zero bytes were read)

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
data must point to at least length bytes of writable memory

Postcondition

*bytes_read set to actual number of bytes received
data[0..*bytes_read-1] contain the received bytes

Note

Non-blocking - returns immediately with whatever data is available
k_rx_ok with *bytes_read == 0 means no data was available

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~5 us per byte received + ~5 us when no data
- Stack usage: 24 bytes

Example:

```
uint8_t  buf[64];
uint16_t count = 0;
rx_err_t err  = uart_read_channel(k_uart_channel_9, buf, sizeof(buf), &count);
if (err == k_rx_ok && count > 0) {
    process_bytes(buf, count);
}
```

See also

[uart_getc_channel\(\)](#) Read single character
[uart_rx_available\(\)](#) Check if data is available
[uart_write_channel\(\)](#) Write binary data

Since

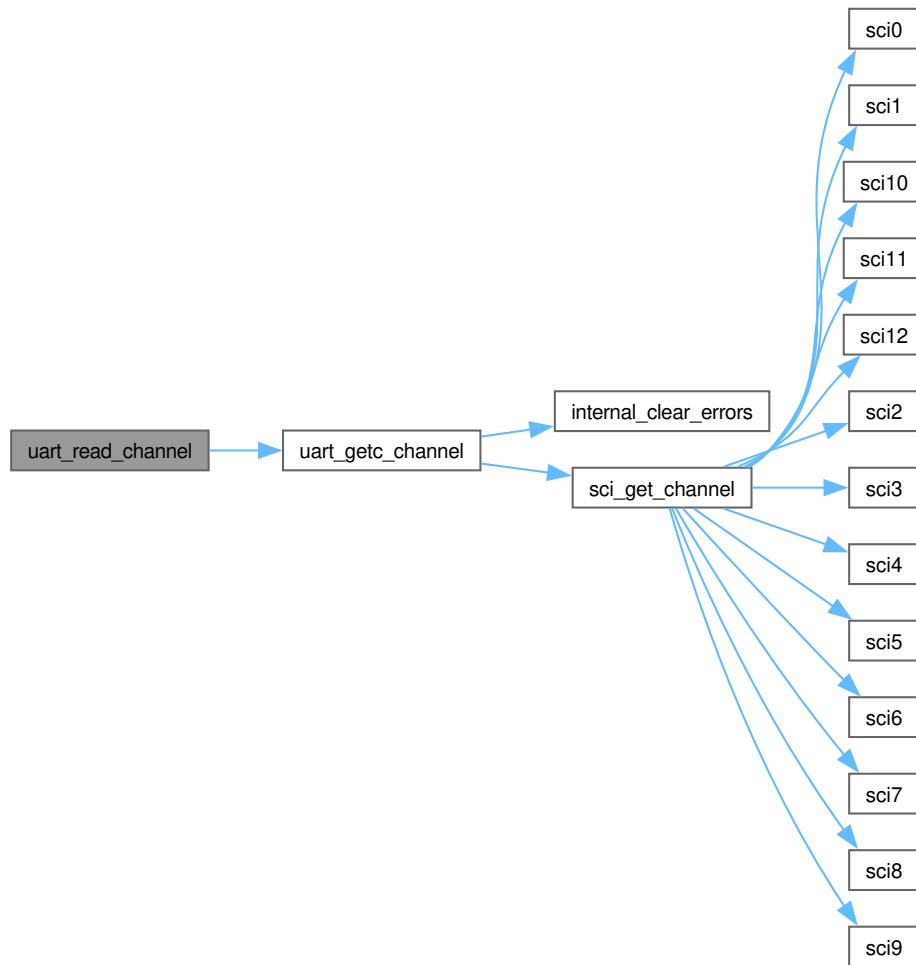
Version 1.0.0

Definition at line 1717 of file [uart.c](#).

```
01721 {
01722     /* Validate parameters */
01723     if (data == nullptr || bytes_read == nullptr) {
01724         return k_rx_err_null_ptr;
01725     }
01726
01727     if ((uint8_t)channel >= k_uart_max_channels) {
01728         return k_rx_err_invalid_arg;
01729     }
01730
01731     if (!s_channel_initialized[channel]) {
01732         return k_rx_err_invalid_state;
01733     }
01734
01735     /* Read available bytes */
01736     *bytes_read = 0;
01737     for (uint16_t i = 0; i < length; i++) {
01738         char c;
01739         const rx_err_t err = uart_getc_channel(channel, &c);
01740         if (err == k_rx_err_empty) {
01741             /* No more data available */
01742             break;
01743         }
01744         if (err != k_rx_ok) {
01745             return err;
01746         }
01747         data[i] = (uint8_t)c;
01748         (*bytes_read)++;
01749     }
01750
01751     return k_rx_ok;
01752 }
```

References [k_uart_max_channels](#), [s_channel_initialized](#), and [uart_getc_channel\(\)](#).

Here is the call graph for this function:



8.139.3.23 uart'rx'available()

```

rx_err_t uart_rx_available (
    const uart\_channel\_t channel,
    bool * available) [nodiscard]

```

Check if receive data is available on UART channel.

Check if receive data is available on channel.

Checks the RDRF (Receive Data Register Full) flag in the SSR register of the specified SCI channel. Provides a non-destructive peek at receive availability without consuming any data from the buffer.

Algorithm steps:

1. Validate available pointer (NULL check)
2. Validate channel number and initialization state
3. Get SCI register base address
4. Read SSR and test RDRF bit; store boolean result in *available

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
available must point to valid bool storage

Postcondition

*available reflects current RDRF state
No data is consumed from the receive buffer

Note

Non-destructive - does not read RDR or clear any flags
RDRF can be set again immediately after reading if UART is receiving

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~2 us
- Stack usage: 16 bytes

Example:

```
bool ready = false;
if (uart_rx_available(k_uart_channel_9, &ready) == k_rx_ok && ready) {
    char c;
    uart_getc_channel(k_uart_channel_9, &c);
}
```

See also

[uart_getc_channel\(\)](#) Read single character
[uart_read_channel\(\)](#) Read multiple bytes

Since

Version 1.0.0

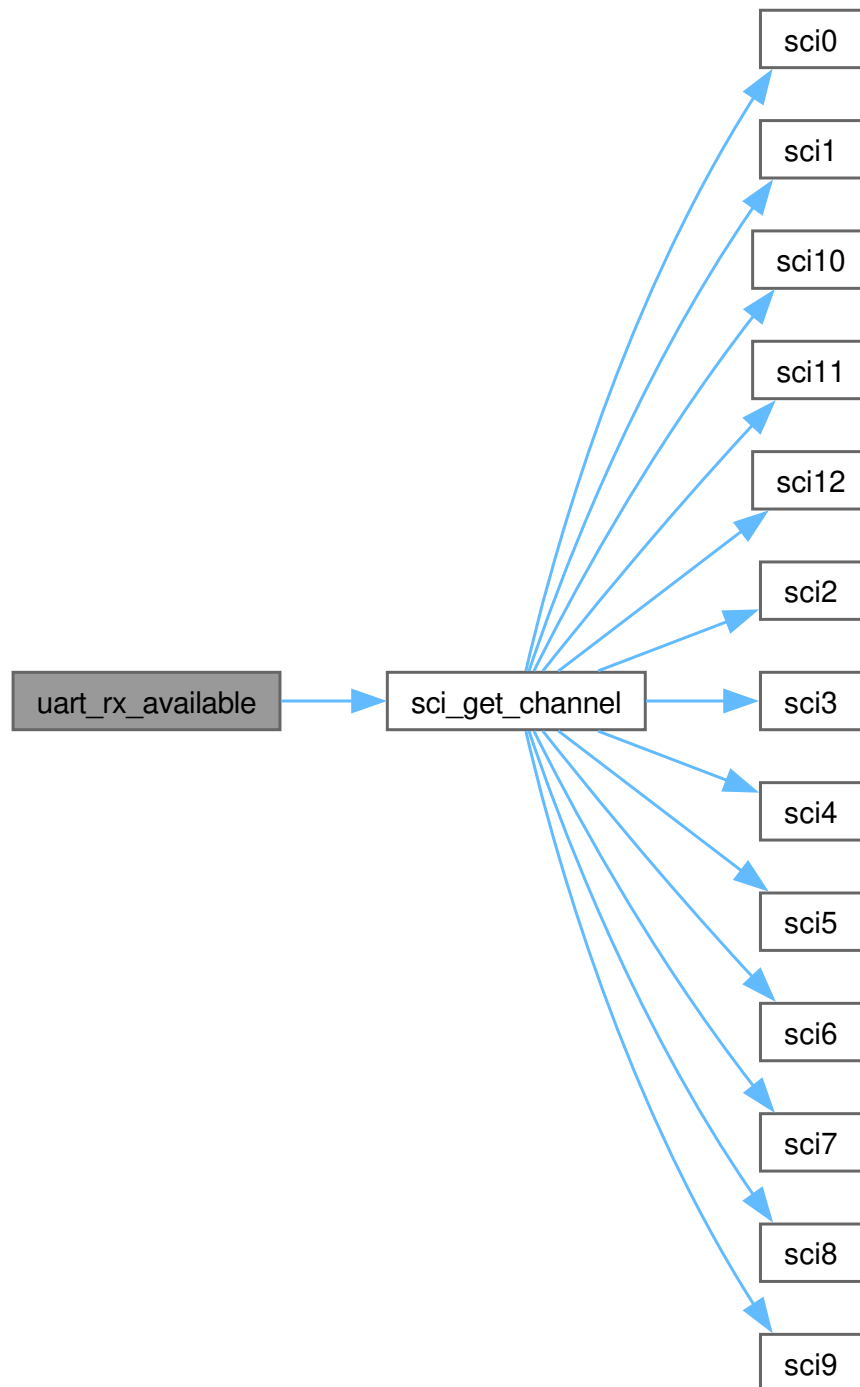
Definition at line 1801 of file [uart.c](#).

```
01802 {
01803     /* Validate parameters */
01804     if (available == nullptr) {
01805         return k_rx_err_null_ptr;
01806     }
01807
01808     if ((uint8_t)channel >= k_uart_max_channels) {
01809         return k_rx_err_invalid_arg;
01810     }
01811
01812     if (!s_channel_initialized[channel]) {
01813         return k_rx_err_invalid_state;
01814     }
01815
01816     /* SCI9 uses the RXI9 ring buffer; non-empty = bytes available. */
```

```
01817 if (channel == k_uart_channel_9) {
01818     *available = (s_sci9_rx_head != s_sci9_rx_tail);
01819     return k_rx_ok;
01820 }
01821
01822 /* Get SCI register base */
01823 const volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01824 if (sci == nullptr) {
01825     return k_rx_err_invalid_arg;
01826 }
01827
01828 /* Check RDRF flag */
01829 *available = ((sci->ssr & k_sci_ssr_rdrf_flag) != k_uart_flag_clear);
01830
01831 return k_rx_ok;
01832 }
```

References [k_sci_ssr_rdrf_flag](#), [k_uart_channel_9](#), [k_uart_flag_clear](#), [k_uart_max_channels](#), [s_channel_initialized](#), [s_sci9_rx_head](#), [s_sci9_rx_tail](#), [sci_get_channel\(\)](#), and [rx_sci_regs_t::ssr](#).

Here is the call graph for this function:



8.139.3.24 `uart_write_channel()`

```
rx_err_t uart_write_channel (  
    const uart\_channel\_t channel,  
    const uint8_t * data,  
    uint16_t length) [nodiscard]
```

Write binary data to UART channel without newline conversion.

Write buffer to specified channel.

Transmits a block of raw bytes on the specified SCI channel using [uart_putc_channel\(\)](#) for each byte. No LF-to-`\r`+LF conversion is performed, making this function suitable for binary protocol data.

Algorithm steps:

1. Validate data pointer (NULL check)
2. Validate channel number and initialization state
3. For each byte in `[0, length)`: call [uart_putc_channel\(\)](#) and propagate errors

Precondition

Channel must be initialized via [uart_init_channel\(\)](#)
data must point to at least length bytes of valid memory

Postcondition

All length bytes transmitted in order
On error, partial data may have been transmitted

Note

Blocking call - waits for each byte to be accepted by TDR
No newline conversion - use [uart_puts_channel\(\)](#) for text output

Thread Safety:

Thread-safe for different channels. Not safe for same channel without mutex.

Performance:

- Execution time: ~90 us per byte @ 115200 baud
- Stack usage: 24 bytes

Example:

```
const uint8_t frame[] = {0xAA, 0x55, 0x01, 0x02, 0x03};
rx_err_t err = uart_write_channel(k_uart_channel_9, frame, sizeof(frame));
if (err != k_rx_ok) {
    // Handle partial write
}
```

See also

[uart_puts_channel\(\)](#) Transmit text string with newline conversion
[uart_read_channel\(\)](#) Read binary data

Since

Version 1.0.0

Definition at line 1498 of file [uart.c](#).

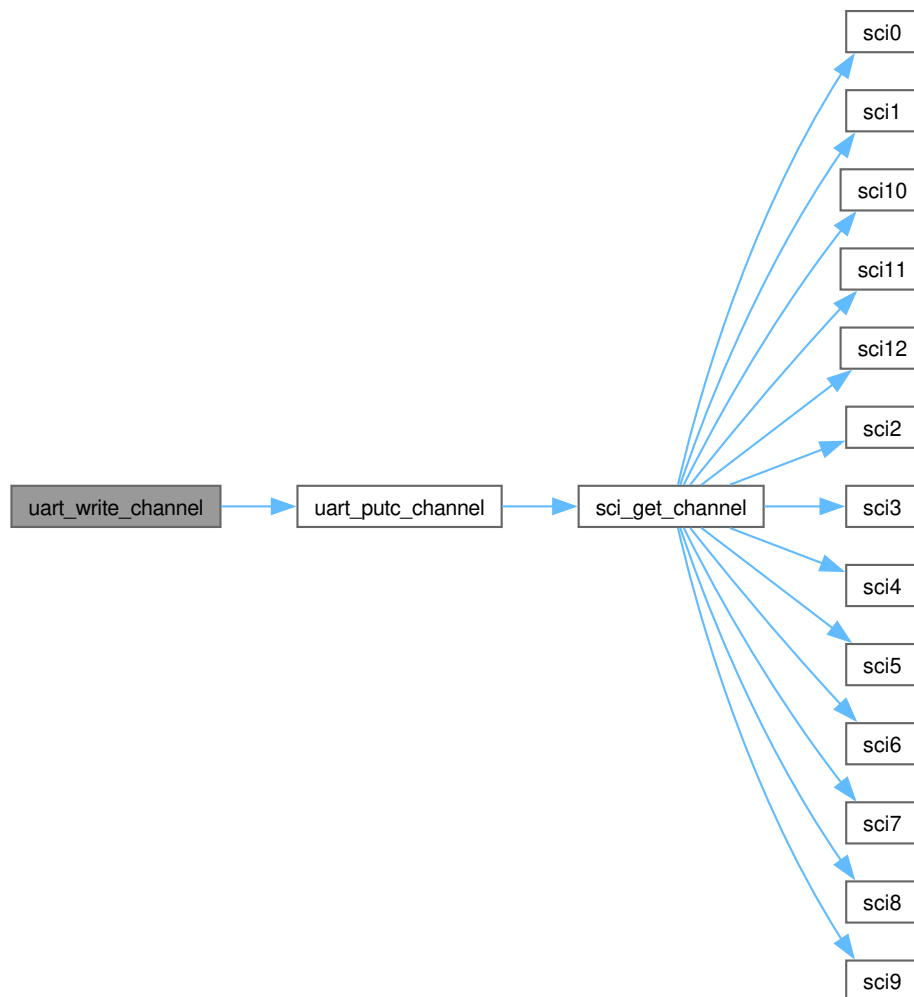
```

01499 {
01500     /* Validate parameters */
01501     if (data == nullptr) {
01502         return k_rx_err_null_ptr;
01503     }
01504
01505     if ((uint8_t)channel >= k_uart_max_channels) {
01506         return k_rx_err_invalid_arg;
01507     }
01508
01509     if (!s_channel_initialized[channel]) {
01510         return k_rx_err_invalid_state;
01511     }
01512
01513     /* Write each byte */
01514     for (uint16_t i = 0; i < length; i++) {
01515         const rx_err_t err = uart_putc_channel(channel, (char)data[i]);
01516         if (err != k_rx_ok) {
01517             return err;
01518         }
01519     }
01520
01521     return k_rx_ok;
01522 }

```

References [k_uart_max_channels](#), [s_channel_initialized](#), and [uart_putc_channel\(\)](#).

Here is the call graph for this function:



8.139.4 Variable Documentation

8.139.4.1 s'channel'initialized

```
bool s_channel_initialized[k_uart_array_size] = {false} [static]
```

Per-channel initialization state.

Definition at line 444 of file [uart.c](#).

```
00444 {false};
```

Referenced by [internal_uart_validate_init\(\)](#), [uart_deinit_channel\(\)](#), [uart_getc_channel\(\)](#), [uart_init_channel\(\)](#), [uart_putc_channel\(\)](#), [uart_puts_channel\(\)](#), [uart_read_channel\(\)](#), [uart_rx_available\(\)](#), and [uart_write_channel\(\)](#).

8.139.4.2 s'sci9'rx'head

```
volatile uint16_t s_sci9_rx_head = 0U [static]
```

ISR writes here, task reads

Definition at line 470 of file [uart.c](#).

Referenced by [INT_SCI9_RXI9\(\)](#), [internal_uart_enable_sci9_rx_isr\(\)](#), [uart_deinit_channel\(\)](#), [uart_getc_channel\(\)](#), and [uart_rx_available\(\)](#).

8.139.4.3 s'sci9'rx'ring

```
volatile uint8_t s_sci9_rx_ring[k_sci9_rx_ring_size] [static]
```

Definition at line 469 of file [uart.c](#).

Referenced by [INT_SCI9_RXI9\(\)](#), and [uart_getc_channel\(\)](#).

8.139.4.4 s'sci9'rx'tail

```
volatile uint16_t s_sci9_rx_tail = 0U [static]
```

Task reads here, ISR never touches

Definition at line 471 of file [uart.c](#).

Referenced by [INT_SCI9_RXI9\(\)](#), [internal_uart_enable_sci9_rx_isr\(\)](#), [uart_deinit_channel\(\)](#), [uart_getc_channel\(\)](#), and [uart_rx_available\(\)](#).

8.140 uart.c

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file uart.c
00003  * @brief Multi-Channel UART Driver for RX72N SCI Peripherals
00004  *
00005  * @details
00006  * ### Overview
00007  *
00008  * Production-quality UART driver for RX72N Serial Communication Interface (SCI)
00009  * peripherals. Provides synchronous TX and RX functionality across 13 SCI channels
00010  * (SCI0-SCI12) with configurable baud rates, automatic newline conversion, and
00011  * comprehensive error handling.
00012  *
00013  * ### Architecture
00014  *
00015  * @dot
00016  * digraph uart_architecture {
00017  *   rankdir=TB;
00018  *   node [shape=box, style=rounded];
00019  *
00020  *   subgraph cluster_app {
00021  *     label="Application Layer";
00022  *     style=filled;
00023  *     color=lightblue;
00024  *     debug_api [label="Debug Functions\nuart_debug_init/puts/putc"];
00025  *     channel_api [label="Channel Functions\nuart_init/putc/puts/getc_channel"];
00026  *   }
00027  *
00028  *   subgraph cluster_driver {
00029  *     label="Driver Layer";
00030  *     style=filled;
00031  *     color=lightyellow;
00032  *     internal [label="Internal
Functions\ninternal_calculate_brr\ninternal_configure_uart_pins\ninternal_enable_sci_clock"];
00033  *     state [label="State Management\ns_channel_initialized"];
00034  *   }
00035  *
00036  *   subgraph cluster_hal {
00037  *     label="Hardware Abstraction";
00038  *     style=filled;
00039  *     color=lightgreen;
00040  *     sci_regs [label="SCI Registers\nS\\r, SMR, BRR, TDR, RDR, SSR"];
00041  *     mpc [label="Pin Mux (MPC)\nrx_mpc_set_sci"];
00042  *     gpio [label="GPIO Control\nPDR, PMR registers"];
00043  *   }
00044  *
00045  *   debug_api -> channel_api;
00046  *   channel_api -> internal;
00047  *   channel_api -> state;
00048  *   internal -> sci_regs;
00049  *   internal -> mpc;
00050  *   internal -> gpio;
00051  * }
00052  * @enddot
00053  *
00054  * ### Hardware Details
00055  *
00056  * @par Hardware Requirements:
00057  *
00058  * | Component | Requirement | Notes |
00059  * |-----|-----|-----|
00060  * | MCU | Renesas RX72N | SCI peripheral required |
00061  * | Channels | SCI0-SCI12 | 13 channels total |
00062  * | Clock | PCLKB (SCI0-6,12) / PCLKA (SCI7-11) | Per-channel BRR clock |
00063  * | TX Pin | Port pin configured as output | Requires MPC configuration |
00064  * | RX Pin | Port pin configured as input | Requires MPC configuration |
00065  * | USB Bridge | CY7C65213 (default) | For SCI9 debug channel |
00066  *
00067  * @par Default Debug Configuration (SCI9):
00068  *
00069  * | Parameter | Value | Notes |
00070  * |-----|-----|-----|
00071  * | Channel | SCI9 | Connected to USB-UART bridge |
00072  * | TX Pin | PB7 (TXD9) | Output, peripheral mode |
00073  * | RX Pin | PB6 (RXD9) | Input, peripheral mode |
00074  * | Baud Rate | 115200 bps | Standard debug speed |
00075  * | Data Bits | 8 | Fixed |
00076  * | Parity | None | Fixed |
00077  * | Stop Bits | 1 | Fixed |
00078  *
00079  * ### Baud Rate Calculation
00080  *
00081  * The SCI baud rate formula for n=0 (CKS=00, PCLK/1):

```

```

00082 *
00083 * @f
00084 * \text{BRR} = \frac{\text{PCLK}}{32 \times B} - 1
00085 * @f
00086 *
00087 * f_PCLK depends on the channel group (manual section 5 I/O register table):
00088 * - SCI0-SCI6, SCI12 (SCIj/SCIh): PCLKB (k_pclkb_hz, 60 MHz)
00089 * - SCI7-SCI11 (SCIi, extended 0x00D0xxx): PCLKA (k_pclka_hz, 120 MHz)
00090 *
00091 * Note: rx72n_clock.h is authoritative for PCLK frequencies; trust the header
00092 * over any older bench note. PCLKB = 60 MHz on this platform.
00093 *
00094 * @par Supported Baud Rates (at PCLKB = 60 MHz, for SCI0-SCI6/SCI12):
00095 *
00096 * | Baud Rate | BRR | Actual Rate | Error |
00097 * |-----|-----|-----|-----|
00098 * | 9600 | 194 | 9615 | +0.16% |
00099 * | 19200 | 96 | 19231 | +0.16% |
00100 * | 38400 | 47 | 38462 | +0.16% |
00101 * | 57600 | 31 | 58594 | +1.73% |
00102 * | 115200 | 15 | 117188 | +1.73% |
00103 * | 230400 | 7 | 234375 | +1.73% |
00104 * | 460800 | 3 | 468750 | +1.73% |
00105 * | 921600 | 1 | 937500 | +1.73% |
00106 *
00107 * ## Performance Characteristics
00108 *
00109 * | Operation | Execution Time | Stack Usage | Notes |
00110 * |-----|-----|-----|-----|
00111 * | uart_init_channel | ~50 us | 48 bytes | Includes MPC configuration |
00112 * | uart_putc_channel | ~90 us @ 115200 | 16 bytes | Includes wait for TDRE |
00113 * | uart_puts_channel | ~90 us/char | 24 bytes | With LF->\r\n conversion |
00114 * | uart_getc_channel | ~5 us | 16 bytes | Non-blocking if no data |
00115 * | uart_debug_init | ~50 us | 64 bytes | Wrapper for SCI9 |
00116 *
00117 * @par Memory Footprint:
00118 * - Code size: ~2 KB (all functions)
00119 * - Static data: 13 bytes (s_channel_initialized array)
00120 * - Stack usage: < 64 bytes (deepest call)
00121 * - Heap usage: 0 bytes (zero dynamic allocation)
00122 *
00123 * ## Thread Safety
00124 *
00125 * | Function | Thread Safe? | Notes |
00126 * |-----|-----|-----|
00127 * | uart_init_channel | [FAIL] No | Call once per channel during init |
00128 * | uart_deinit_channel | [FAIL] No | Call once during shutdown |
00129 * | uart_putc_channel | [WARN] Partial | Safe if different channels |
00130 * | uart_puts_channel | [WARN] Partial | Safe if different channels |
00131 * | uart_getc_channel | [WARN] Partial | Safe if different channels |
00132 * | uart_debug_* | [WARN] Partial | All use same channel (SCI9) |
00133 *
00134 * **Note:** Multiple threads can safely use different UART channels simultaneously.
00135 * For shared channel access, use external mutex protection.
00136 *
00137 * @par Module Dependencies:
00138 *
00139 * **This module depends on:**
00140 * - `hardware.h` - Hardware configuration and channel constants
00141 * - `rx72n_clock.h` - PCLKB frequency definition
00142 * - `rx72n_regs.h` - SCI register definitions
00143 * - `rx_mpc.h` - Pin mux configuration
00144 * - `rx_port_utils.h` - GPIO port utilities
00145 * - `rx_register_protection.h` - PR register protection
00146 *
00147 * **This module is used by:**
00148 * - `rx_log.c` - Logging subsystem output
00149 * - `main.c` - Debug output during initialization
00150 * - `comm_task.c` - Communication task debug messages
00151 *
00152 * @par NASA Power of 10 Compliance:
00153 *
00154 * | Rule | Status | Implementation |
00155 * |-----|-----|-----|
00156 * | 1. Simple control flow | [OK] | No goto/setjmp/recursion |
00157 * | 2. Fixed loop bounds | [OK] | All loops use k_uart_max_str_len limit |
00158 * | 3. No dynamic allocation | [OK] | Zero malloc/free, static buffers only |
00159 * | 4. Small functions | [OK] | All functions < 60 lines |
00160 * | 5. Assertions (>=2/func) | [OK] | Parameter validation + state checks |
00161 * | 6. Narrow scope | [OK] | Static file-scope state, local variables |
00162 * | 7. Check return values | [OK] | All rx_err_t returns propagated |
00163 * | 8. Limited preprocessor | [OK] | C23 typed enums only |
00164 * | 9. Pointer restrictions | [OK] | Single-level pointers, no arithmetic |
00165 * | 10. Compiler warnings | [OK] | -Wall -Wextra -Werror clean |
00166 *
00167 * @par SOLID Principles:
00168 *

```

```

00169 **Single Responsibility (S):**
00170 * - This module handles ONLY UART communication
00171 * - Separate from higher-level logging (rx_log.c)
00172 * - Separate from protocol parsing (rx_frame.c)
00173 *
00174 **Open/Closed (O):**
00175 * - New baud rates supported without code changes
00176 * - New channels added via configuration structure
00177 * - Pin assignments configurable per-channel
00178 *
00179 **Liskov Substitution (L):**
00180 * - All channel functions accept any valid channel
00181 * - Consistent rx_err_t return semantics
00182 * - Debug functions are simple wrappers
00183 *
00184 **Interface Segregation (I):**
00185 * - Separate init/deinit from TX/RX operations
00186 * - Debug functions provide simplified interface
00187 * - Channel functions provide full control
00188 *
00189 **Dependency Inversion (D):**
00190 * - Uses abstract rx_port_pin_t for pin specification
00191 * - Uses rx_mpc module for pin configuration
00192 * - Uses rx_err_t for standardized error handling
00193 *
00194 * @see hardware.h Hardware configuration and constants
00195 * @see rx72n_sci_regs.h SCI register definitions
00196 * @see rx_mpc.h Pin mux configuration API
00197 * @see rx_log.h Higher-level logging that uses this driver
00198 *
00199 * @author Locked, Inc.
00200 * @date 2026-01-28
00201 * @copyright Copyright (c) 2026 Locked Inc.
00202 * SPDX-License-Identifier: MIT
00203 * @since Version 1.0.0
00204 */
00205
00206 #ifdef __RX__
00207
00208 #include <stdint.h>
00209
00210 #include "hardware.h"
00211 #include "rx72n_clock.h"
00212 #include "rx72n_icu_regs.h" /* icu() for SCI9_RXI9 interrupt enable */
00213 #include "rx72n_regs.h"
00214 #include "rx_mpc.h"
00215 #include "rx_port_utils.h"
00216 #include "rx_register_protection.h"
00217 #include "rx_simulator_config.h" /* For RX_IS_SIMULATOR conditional compilation */
00218
00219 /*
=====
00220 * Private Definitions
00221 *
=====
00222 */
00223
00224 /** @brief UART configuration constants */
00225 typedef enum : uint32_t {
00226 /**
00227 * @brief Default baud rate: 921600 bps
00228 *
00229 * Max standard rate that hits <2%% error on SCI9 with PCLKA=120 MHz.
00230 * Formula (ABCS=0/CKS=00): BRR = PCLKA / (32 * B) - 1.
00231 * 120e6 / (32 * 921600) - 1 = 4.07 -> BRR = 3
00232 * Actual rate = 120e6 / (32 * (3 + 1)) = 937500 bps
00233 * Error vs 921600 = (937500 - 921600) / 921600 = +1.72%% (well within ~3%% UART tolerance)
00234 *
00235 * Well within the CY7C65213 USB-UART bridge's 3 Mbps ceiling.
00236 */
00237 k_uart_default_baudrate =
00238 115200, /* matches motors_rtos/HIL-tests; 921600 at PCLKB=60 MHz has 1.7% BRR rounding error */
00239 k_uart_bit_time_delay_cycles = 125, /**< ~1.09us at 921600 bps, ~260 cycles @240 MHz */
00240 } uart_config_constants_t;
00241
00242 /** @brief SCI register control-bit values (SSR flag masks live in rx72n_sci_regs.h) */
00243 typedef enum : uint8_t {
00244 k_sci_scr_disabled = 0x00, /**< SCR: All functions disabled */
00245 k_sci_scr_tx_enabled = 0x20, /**< SCR: Transmit enabled (TE=1) */
00246 k_sci_scr_rx_enabled = 0x10, /**< SCR: Receive enabled (RE=1) */
00247 k_sci_scr_txrx_enabled = 0x30, /**< SCR: TX + RX enabled (TE=1, RE=1) */
00248 k_sci_smr_async_8n1 = 0x00, /**< SMR: Async mode, 8 data bits, no parity, 1 stop bit, PCLK/1 */
00249 k_sci_semr_default = 0x00, /**< SEMR: Default extended mode (ABCS=0, 16-cycle base) */
00250 k_sci_semr_abcs_bit =
00251 0x10, /**< SEMR.ABCS: 1=8-cycle base clock (halves BRR divisor, HUM 34.2.12) */
00252 k_sci_scr_rie_bit = 0x40, /**< SCR.RIE: receive interrupt enable (HUM 34.2.5) */
00253 } sci_register_values_t;

```

```

00254
00255 /** @brief Integer to string buffer constants */
00256 typedef enum : uint8_t {
00257     k_uart_int_buffer_size = 12, /**< Buffer size for int32 to string (enough for -2147483648) */
00258     k_uart_base_10 = 10, /**< Base 10 for decimal conversion */
00259 } uart_int_constants_t;
00260
00261 /** @brief Hex digit constants */
00262 typedef enum : uint8_t {
00263     k_uart_hex_max_digits = 8, /**< Maximum hex digits to print (32-bit value) */
00264     k_uart_hex_min_digits = 1, /**< Minimum hex digits to print */
00265     k_uart_hex_zero_digits = 0, /**< Zero digits value */
00266     k_uart_hex_nibble_bits = 4, /**< Bits per hex nibble */
00267     k_uart_hex_nibble_mask = 0x0F, /**< Mask for hex nibble */
00268 } uart_hex_constants_t;
00269
00270 /** @brief BRR calculation constants */
00271 typedef enum : uint16_t {
00272     k_brr_divisor_n0 = 32, /**< Divisor for n=0, ABCS=0 (default 16-cycle base):  $64 * 2^{(2n-1)} = 32$  */
00273     k_brr_divisor_n0_abcs =
00274         16, /**< Divisor for n=0, ABCS=1 (8-cycle base): halves the divisor, HUM 34.2.12 */
00275     k_brr_multiplier = 4, /**< Multiplier per CKS increment ( $2^2$ ) */
00276     k_brr_max_value = 255, /**< Maximum BRR register value */
00277     k_brr_min_value = 0, /**< Minimum BRR register value */
00278     k_brr_formula_offset = 1, /**< BRR formula subtract-1 offset:  $BRR = (PCLK/(div*B)) - 1$  */
00279 } brr_constants_t;
00280
00281 /** @brief MSTPCRB register bit manipulation constants */
00282 typedef enum : uint32_t {
00283     k_uart_mstpcrb_bit_set = 1UL, /**< Single-bit mask for MSTPCRB bit-clear operations */
00284 } uart_mstpcrb_constants_t;
00285
00286 /** @brief Maximum SCI channels (array size, must be enum for compile-time constant) */
00287 typedef enum : uint8_t {
00288     k_uart_channel_min = 0, /**< Minimum UART channel (SCI0) */
00289     k_uart_array_size = 13, /**< Array size for s_channel_initialized */
00290     k_uart_max_mstpb_channel = 7, /**< Last contiguous MSTPCRB channel (SCI0-SCI7, bits 31-24) */
00291     k_uart_mstpc_channel_min = 8, /**< First MSTPCRC channel (SCI8 at MSTPCRC bit 27) */
00292     k_uart_mstpc_channel_max = 11, /**< Last MSTPCRC channel (SCI11 at MSTPCRC bit 24) */
00293     k_uart_pclka_channel_min = 7, /**< First channel using PCLKA for BRR (SCIi: SCI7-SCI11) */
00294     k_uart_pclka_channel_max = 11, /**< Last channel using PCLKA for BRR (SCIi: SCI7-SCI11) */
00295     k_uart_max_channels = 13, /**< Maximum valid channel value (SCI channels 0-12) */
00296 } uart_internal_constants_t;
00297
00298 /**
00299  * @enum uart_validation_limits_t
00300  * @brief Baud rate validation bounds for uart_init_channel() parameter checking
00301  *
00302  * @details
00303  * Defines the closed interval [k_uart_baudrate_min, k_uart_baudrate_max] that
00304  * every requested baud rate must satisfy before the driver attempts to program
00305  * the BRR register.
00306  *
00307  * The maximum is derived directly from the BRR formula for n=0 (CKS=00):
00308  * @f
00309  * \text{BRR} = \frac{\text{PCLKB}}{32 \times B} - 1 \geq 0
00310  * \implies B \leq \frac{\text{PCLKB}}{32}
00311  * @f
00312  * A BRR of 0 corresponds to the fastest achievable baud rate at the current
00313  * PCLKB frequency, so requesting anything higher would underflow the register.
00314  *
00315  * The minimum is 1 bps, which prevents a divide-by-zero in the BRR formula.
00316  * In practice, any baud rate below ~9600 is impractical on a 60 MHz PCLKB, but
00317  * the lower bound is kept permissive to avoid false negatives during testing.
00318  *
00319  * @invariant k_uart_baudrate_min must be > 0 to prevent division by zero in
00320  * internal_calculate_brr().
00321  * @invariant k_uart_baudrate_max must equal k_pclkb_hz / k_brr_divisor_n0 so
00322  * that the computed BRR is always >= 0 (i.e., no register underflow).
00323  *
00324  * @par Example:
00325  * @code{.c}
00326  * // Validation performed inside uart_init_channel()
00327  * if ((config->baudrate < k_uart_baudrate_min) ||
00328      (config->baudrate > k_uart_baudrate_max)) {
00329     return k_rx_err_invalid_arg;
00330 }
00331 * @endcode
00332 *
00333 * @see uart_init_channel() Uses these bounds to validate the baudrate field
00334 * @see internal_calculate_brr() Computes the actual BRR register value
00335 * @see brr_constants_t BRR formula divisor constants
00336 *
00337 * @since Version 1.0.0
00338 */
00339 typedef enum : uint32_t {
00340     k_uart_baudrate_min =

```

```

00341     1, /**< Minimum valid baud rate (bps); must be > 0 to avoid divide-by-zero in BRR formula */
00342     k_uart_baudrate_max =
00343     (k_pclkb_hz /
00344     k_brr_divisor_n0), /**< Maximum valid baud rate (bps); BRR = 0 at this rate, higher values would underflow */
00345 } uart_validation_limits_t;
00346
00347 /** @brief UART timeout constants */
00348 typedef enum : uint32_t {
00349     k_uart_tx_timeout = 100000, /**< Transmit buffer wait timeout (prevents infinite loop) */
00350     k_uart_timeout_expired = 0, /**< Timeout counter expired value */
00351     k_uart_timeout_decrement = 1, /**< Timeout counter decrement value */
00352 } uart_timeout_t;
00353
00354 /** @brief UART flag comparison constants */
00355 typedef enum : uint8_t {
00356     k_uart_flag_clear = 0, /**< Flag bit is not set (comparison result) */
00357 } uart_flag_constants_t;
00358
00359 /** @brief UART buffer and string size limits */
00360 typedef enum : uint32_t {
00361     k_uart_max_str_len = 256, /**< Maximum string length for uart_puts_channel */
00362 } uart_buffer_constants_t;
00363
00364 /**
00365  * @brief SCI module stop bit positions in MSTP\\rB
00366  *
00367  * SCI0-SCI7 occupy contiguous bits 31-24 (MSTPB31 down to MSTPB24).
00368  * SCI12 (SCIh module) is a non-contiguous outlier at MSTPB4.
00369  * SCI8-SCI11 live in MSTP\\rC, not here -- see sci_mstpc_bits_t.
00370  *
00371  * Manual: section 11.2.3 MSTP\\rB, page 408.
00372  */
00373 typedef enum : uint8_t {
00374     k_sci_mstpb_sci0 = 31, /**< SCI0 module stop bit in MSTPCRB (MSTPB31) */
00375     k_sci_mstpb_sci1 = 30, /**< SCI1 module stop bit in MSTPCRB (MSTPB30) */
00376     k_sci_mstpb_sci2 = 29, /**< SCI2 module stop bit in MSTPCRB (MSTPB29) */
00377     k_sci_mstpb_sci3 = 28, /**< SCI3 module stop bit in MSTPCRB (MSTPB28) */
00378     k_sci_mstpb_sci4 = 27, /**< SCI4 module stop bit in MSTPCRB (MSTPB27) */
00379     k_sci_mstpb_sci5 = 26, /**< SCI5 module stop bit in MSTPCRB (MSTPB26) */
00380     k_sci_mstpb_sci6 = 25, /**< SCI6 module stop bit in MSTPCRB (MSTPB25) */
00381     k_sci_mstpb_sci7 = 24, /**< SCI7 module stop bit in MSTPCRB (MSTPB24) */
00382     k_sci_mstpb_sci12 = 4, /**< SCI12 (SCIh) module stop bit in MSTPCRB (MSTPB4) */
00383 } sci_mstpb_bits_t;
00384
00385 /**
00386  * @brief SCI module stop bit positions in MSTP\\rC
00387  *
00388  * SCI8-SCI11 (SCIi module, extended peripheral region 0x000D0000) are
00389  * controlled by MSTP\\rC, NOT MSTP\\rB. Clearing the corresponding bit
00390  * releases the channel from the module-stop state.
00391  *
00392  * Manual: section 11.2.4 MSTP\\rC, page 410.
00393  */
00394 typedef enum : uint8_t {
00395     k_sci_mstpc_sci8 = 27, /**< SCI8 module stop bit in MSTPCRC (MSTPC27) */
00396     k_sci_mstpc_sci9 = 26, /**< SCI9 module stop bit in MSTPCRC (MSTPC26) */
00397     k_sci_mstpc_sci10 = 25, /**< SCI10 module stop bit in MSTPCRC (MSTPC25) */
00398     k_sci_mstpc_sci11 = 24, /**< SCI11 module stop bit in MSTPCRC (MSTPC24) */
00399 } sci_mstpc_bits_t;
00400
00401 /** @brief Debug UART pins (SCI9 on RX72N) */
00402 typedef enum : uint16_t {
00403     k_uart_debug_tx_gpio = k_rx_pb_7, /**< PB7 = TXD9 */
00404     k_uart_debug_rx_gpio = k_rx_pb_6, /**< PB6 = RXD9 */
00405 } uart_debug_pins_t;
00406
00407 /**
00408  * @enum uart_gpio_constants_t
00409  * @brief GPIO register bit manipulation constants for UART pin configuration
00410  *
00411  * @details
00412  * Provides the single-bit seed value used when constructing per-pin bitmasks
00413  * for the PDR (Port Direction Register) and PMR (Port Mode Register) during
00414  * UART TX/RX pin setup. A bit mask for a specific pin is formed by shifting
00415  * this value left by the pin index obtained from rx_pin_from_pin().
00416  *
00417  * @invariant k_uart_gpio_bit_set must equal 1 so that left-shifting by a pin
00418  * index produces an isolated single-bit mask.
00419  *
00420  * @par Example:
00421  * @code{.c}
00422  * // Build the TX pin mask and set the direction bit
00423  * const uint8_t tx_pin = rx_pin_from_pin(tx_gpio);
00424  * const uint8_t tx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « tx_pin);
00425  * tx_port_base->pdr |= tx_pin_mask; // Set output direction
00426  * tx_port_base->pmr |= tx_pin_mask; // Switch to peripheral mode
00427  * @endcode

```

```

00428 *
00429 * @see internal_configure_uart_pins() Only consumer of this constant
00430 * @see uart_init_channel() Top-level function that triggers pin configuration
00431 *
00432 * @since Version 1.0.0
00433 */
00434 typedef enum : uint8_t {
00435     k_uart_gpio_bit_set = 1, /**< Seed value for constructing a single-pin bitmask via left-shift */
00436 } uart_gpio_constants_t;
00437
00438 /*
=====
00439 * Private State
00440 *
=====
00441 */
00442
00443 /** @brief Per-channel initialization state */
00444 static bool s_channel_initialized[k_uart_array_size] = {false};
00445
00446 /*
=====
00447 * Interrupt-driven RX ring buffer for SCI9
00448 *
=====
00449 *
00450 * SCI9 on the RX72N has no hardware RX FIFO. Polling RDRF in a 10 ms task
00451 * loop (rx_comm_manager's 100 Hz cadence) catches roughly 1 byte per poll
00452 * while the gateway can send ~115 bytes / 10 ms at 115200 baud. Any byte
00453 * arriving while RDRF is still set from the previous byte triggers an
00454 * overrun (SSR.ORER) and is lost.
00455 *
00456 * Fix: enable the SCI9_RXI9 interrupt (vector 102) and copy every byte
00457 * into a lock-free single-producer / single-consumer ring buffer as it
00458 * arrives. uart_getc_channel (and uart_rx_available) for SCI9 then reads
00459 * from the ring instead of polling RDR directly. HUM section 34.2.8
00460 * describes the RXI request behavior.
00461 */
00462 typedef enum : uint16_t {
00463     k_sci9_rx_ring_size =
00464         512U, /**< Power-of-2 mask-friendly size; covers 4.4 ms of 115200-baud traffic */
00465     k_sci9_rx_ring_mask = k_sci9_rx_ring_size - 1U,
00466     k_sci9_rx_i9_priority = 3U, /**< IPR[102]: lower than motor ctrl, higher than CMT tick */
00467 } sci9_rx_ring_consts_t;
00468
00469 static volatile uint8_t s_sci9_rx_ring[k_sci9_rx_ring_size];
00470 static volatile uint16_t s_sci9_rx_head = 0U; /**< ISR writes here, task reads */
00471 static volatile uint16_t s_sci9_rx_tail = 0U; /**< Task reads here, ISR never touches */
00472
00473 /*
=====
00474 * Private Functions
00475 *
=====
00476 */
00477
00478 /**
00479 * @brief Calculate the 8-bit BRR register value for a target baud rate
00480 *
00481 * @details
00482 * Computes the value to be written to the SCI Bit Rate Register (BRR) so that
00483 * the SCI peripheral generates the requested baud rate from PCLKB. The
00484 * general RX72N SCI baud rate formula for asynchronous mode is:
00485 *
00486 * @f[
00487 * \text{BRR} = \frac{\text{PCLKB}}{64 \times 2^{2n-1} \times B} - 1
00488 * @f]
00489 *
00490 * This driver always uses n=0 (CKS bits = 0b00, clock source = PCLK/1), which
00491 * simplifies to:
00492 *
00493 * @f[
00494 * \text{BRR} = \frac{\text{PCLK}}{32 \times B} - 1
00495 * @f]
00496 *
00497 * The caller supplies the correct PCLK frequency for the target channel:
00498 * - SCI0-SCI6, SCI12 (SCIj/SCIh, standard region): use k_pclkb_hz
00499 * - SCI7-SCI11 (SCIi, extended region 0x000D0xxx): use k_pclka_hz
00500 *
00501 * This distinction matters when PCLKA != PCLKB. rx72n_clock.h is the
00502 * authoritative source for PCLK frequencies (k_pclka_hz, k_pclkb_hz, etc.);
00503 * trust the header over any older bench note.
00504 *
00505 * **Algorithm steps:**
00506 * 1. Guard against baudrate == 0 (return k_brr_max_value as a safe sentinel).
00507 * 2. Guard against baudrate > pclk_hz/32 (return k_brr_min_value).
00508 * 3. Compute divisor_result = pclk_hz / (32 * B) using integer arithmetic.

```



```

00509 * 4. Guard against divisor_result <= 1 to prevent underflow from the -1 offset.
00510 * 5. Compute BRR = divisor_result - 1.
00511 * 6. Clamp the result to k_brr_max_value (255) if the integer exceeds the
00512 * 8-bit range (indicates a baud rate too slow for this clock).
00513 * 7. Return the clamped 8-bit result.
00514 *
00515 *
00516 *
00517 * @pre baudrate validated by caller; pclk_hz must be non-zero
00518 * @pre k_brr_divisor_n0 must be non-zero compile-time constant
00519 *
00520 * @post Return value is always in [0, 255]
00521 * @post Caller must write the returned value to sci->brr before enabling TX/RX
00522 *
00523 * @note Stateless pure function; safe to call from any context including ISR
00524 * @note Always uses n=0 (CKS=00); support for n=1..3 is not implemented
00525 *
00526 * @par Example:
00527 * @code{.c}
00528 * // SCI9 uses PCLKA (SCIi extended region), ABCS=1 for baud accuracy
00529 * sci->brr = internal_calculate_brr(115200U, k_pclka_hz, true);
00530 * @endcode
00531 *
00532 * @see uart_init_channel() Caller that selects pclk_hz and writes sci->brr
00533 * @see brr_constants_t Constants used in the BRR formula
00534 *
00535 * @since Version 1.0.0
00536 */
00537 static uint8_t
00538 internal_calculate_brr(const uint32_t baudrate, const uint32_t pclk_hz, const bool abcs)
00539 {
00540     /* Pre-condition: reject zero baudrate (division by zero) */
00541     if (baudrate == 0) {
00542         return k_brr_max_value;
00543     }
00544
00545     /* CKS=0, so the divisor is either 32 (ABCS=0, default 16-cycle base clock)
00546      * or 16 (ABCS=1, 8-cycle base clock). HUM 34.2.12 -- the 8-cycle base
00547      * doubles BRR resolution, which cuts 115200/120MHz baud error from
00548      * +1.72% (unusable, causes FER on every byte) to +0.16%. */
00549     const uint32_t divisor = abcs ? (uint32_t)k_brr_divisor_n0_abcs : (uint32_t)k_brr_divisor_n0;
00550
00551     /* Pre-condition: reject baudrate above maximum for this clock */
00552     if (baudrate > (pclk_hz / divisor)) {
00553         return k_brr_min_value;
00554     }
00555
00556     /* For n=0 (CKS=00): BRR = (PCLK / (divisor * B)) - 1 */
00557     const uint32_t divisor_result = pclk_hz / (divisor * baudrate);
00558
00559     /* Guard against underflow: if divisor_result is 0, subtraction would wrap */
00560     if (divisor_result <= k_brr_formula_offset) {
00561         return k_brr_min_value;
00562     }
00563
00564     const uint32_t brr_value = divisor_result - k_brr_formula_offset;
00565
00566     /* Clamp to maximum BRR register value */
00567     if (brr_value > k_brr_max_value) {
00568         return k_brr_max_value;
00569     }
00570
00571     const uint8_t result = (uint8_t)brr_value;
00572
00573     return result;
00574 }
00575
00576 /**
00577 * @brief Clear receive error flags in the SCI Serial Status Register (SSR)
00578 *
00579 * @details
00580 * Reads the SSR register and writes it back with the three receive error flag
00581 * bits masked out, which clears any pending ORER (Overrun Error), FER
00582 * (Framing Error), and PER (Parity Error) conditions on the SCI channel.
00583 *
00584 * The RX72N SCI peripheral requires a read-modify-write sequence to clear
00585 * error flags: the hardware only allows clearing a flag by writing 0 to it
00586 * while keeping the rest of the register unchanged. Writing 1 to an already-
00587 * set flag has no effect (the write is ignored).
00588 *
00589 * **Algorithm steps:**
00590 * 1. Guard: return immediately if sci is nullptr (NASA Power of 10 Rule 5).
00591 * 2. Read the current value of sci->ssr into a local volatile variable.
00592 * 3. Cast-to-void the read result to suppress the unused-variable warning while
00593 *    still ensuring the volatile read is not optimised away.
00594 * 4. Write sci->ssr = (ssr & ~k_sci_ssr_error_mask) to clear bits [5:3]
00595 *    (ORER, FER, PER) while preserving all other SSR bits.

```



```

00596 *
00597 * **Error flag bit positions in SSR:**
00598 * | Bit | Flag | Description |
00599 * |-----|-----|
00600 * | 5 | ORER | Overrun Error - new data arrived before previous data was read |
00601 * | 4 | FER | Framing Error - no valid stop bit detected |
00602 * | 3 | PER | Parity Error - parity mismatch (unused in 8N1 mode) |
00603 *
00604 *
00605 * @pre sci must point to a valid, hardware-mapped rx_sci_regs_t register block
00606 * @pre The SCI module clock must be enabled (MSTP\|rB bit cleared) before
00607 * any SSR access; undefined behaviour otherwise
00608 *
00609 * @post ORER, FER, and PER bits in sci->ssr are cleared (written to 0)
00610 * @post All other SSR bits (TDRE, RDRF, TEND, MPB, MPBT) are preserved
00611 *
00612 * @note This function is intentionally void-returning; the caller (uart_getc_channel)
00613 * checks for errors before calling and does not need a return code
00614 * @note The intermediate volatile read prevents the compiler from merging the
00615 * read and write into a single store, which would miss the read requirement
00616 *
00617 * @par Thread Safety:
00618 * Not thread-safe. SSR is a read-modify-write target; concurrent access from
00619 * two threads on the same channel can corrupt flag state. External mutex
00620 * protection is required if multiple threads share a channel.
00621 *
00622 * @par Performance:
00623 * - Execution time: ~3 cycles (volatile read + mask + volatile write)
00624 * - Stack usage: 4 bytes (one volatile uint8_t local)
00625 *
00626 * @par Example:
00627 * @code{.c}
00628 * volatile rx_sci_regs_t* sci = sci_get_channel(k_uart_channel_9);
00629 * if (sci != nullptr) {
00630 * // Clear any stale error flags before reading receive data
00631 * internal_clear_errors(sci);
00632 * const char data = (char)sci->rdr;
00633 * }
00634 * @endcode
00635 *
00636 * @see uart_getc_channel() Caller that invokes this before reading RDR
00637 * @see sci_register_values_t k_sci_ssr_error_mask bitmask definition
00638 * @see uart_rx_available() Non-destructive receive availability check
00639 *
00640 * @since Version 1.0.0
00641 */
00642 static void internal_clear_errors(volatile rx_sci_regs_t* sci)
00643 {
00644 /* Pre-condition: Validate pointer (NASA Power of 10 Rule 5) */
00645 if (sci == nullptr) {
00646 return;
00647 }
00648
00649 volatile uint8_t ssr = 0;
00650
00651 /* Read SSR then clear error flags */
00652 ssr = sci->ssr;
00653 (void)ssr; /* Suppress unused variable warning */
00654 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_error_mask);
00655 }
00656
00657 /**
00658 * @brief Return the MSTP\|rB bit position for a channel, or -1 for MSTP\|rC channels
00659 *
00660 * @details
00661 * Per manual section 11.2.3/11.2.4:
00662 * - SCI0-SCI7: MSTP\|rB bits 31-24 (contiguous, descending)
00663 * - SCI8-SCI11: MSTP\|rC bits 27-24 -- returns -1, caller uses MSTP\|rC
00664 * - SCI12 (SCIh): MSTP\|rB bit 4 (non-contiguous outlier)
00665 *
00666 * **Bit position table:**
00667 * | Channel | Register | Bit | Enum constant |
00668 * |-----|-----|-----|
00669 * | SCI0 | MSTP\|rB | 31 | k_sci_mstpb_sci0 |
00670 * | SCI1 | MSTP\|rB | 30 | k_sci_mstpb_sci1 |
00671 * | ... | MSTP\|rB | ... | ... |
00672 * | SCI7 | MSTP\|rB | 24 | k_sci_mstpb_sci7 |
00673 * | SCI8 | MSTP\|rC | 27 | (returns -1) |
00674 * | SCI9 | MSTP\|rC | 26 | (returns -1) |
00675 * | SCI10 | MSTP\|rC | 25 | (returns -1) |
00676 * | SCI11 | MSTP\|rC | 24 | (returns -1) |
00677 * | SCI12 | MSTP\|rB | 4 | k_sci_mstpb_sci12 |
00678 *
00679 *
00680 *
00681 * @pre k_sci_mstpb_sci0 == 31 for the descending formula to be correct
00682 * @post If return >= 0, it is a valid MSTP\|rB bit index

```

```

00683 *
00684 * @note int8_t so -1 fits without consuming a valid bit position
00685 *
00686 * @since Version 1.0.0
00687 */
00688 static int8_t internal_get_mstpb_bit(const uint8_t channel)
00689 {
00690     if (channel <= k_uart_max_mstpb_channel) {
00691         /* SCI0-SCI7: MSTPCRB bits 31-24, descending */
00692         return (int8_t)(k_sci_mstpb_sci0 - channel);
00693     }
00694     if (channel > k_uart_mstpc_channel_max) {
00695         /* SCI12 (SCIh): MSTPCRB bit 4 -- only remaining valid channel above 11 */
00696         return (int8_t)k_sci_mstpb_sci12;
00697     }
00698     /* SCI8-SCI11: live in MSTPCRC, not MSTPCRB */
00699     return -1;
00700 }
00701
00702 /**
00703 * @brief Enable the peripheral clock for an SCI channel
00704 *
00705 * @details
00706 * Clears the module-stop bit for the given SCI channel so its clock runs.
00707 * Different channel groups use different control registers (manual 11.2.3/11.2.4):
00708 *
00709 * | Channels | Register | Bits |
00710 * |-----|-----|-----|
00711 * | SCI0-SCI7 | MSTP\\rB | 31-24 |
00712 * | SCI8-SCI11 | MSTP\\rC | 27-24 |
00713 * | SCI12 | MSTP\\rB | 4 |
00714 *
00715 * Both MSTP\\rB and MSTP\\rC are PR\\r-protected; this function performs the
00716 * required unlock/modify/lock sequence for whichever register applies.
00717 *
00718 *
00719 *
00720 * @pre channel validated by uart_init_channel() to be in [0, 12]
00721 * @pre PR\\r not already unlocked by another context
00722 *
00723 * @post Module-stop bit cleared; SCI clock gate open
00724 * @post PR\\r returned to locked state
00725 *
00726 * @warning Not thread-safe; unlock/modify/lock must be atomic
00727 *
00728 * @since Version 1.0.0
00729 */
00730 static rx_err_t internal_enable_sci_clock(const uint8_t channel)
00731 {
00732     const int8_t mstpb_bit = internal_get_mstpb_bit(channel);
00733
00734     *prcr_reg() = k_rx_prcr_unlock_all;
00735
00736     if (mstpb_bit >= 0) {
00737         /* SCI0-SCI7 and SCI12: clear bit in MSTPCRB */
00738         system_regs()->mstpcrb &= ~(k_uart_mstpcrb_bit_set « (uint8_t)mstpb_bit);
00739     } else if (channel >= k_uart_mstpc_channel_min && channel <= k_uart_mstpc_channel_max) {
00740         /* SCI8-SCI11: clear bit in MSTPCRC (bits 27-24) */
00741         const uint8_t mstpc_bit = (uint8_t)(k_sci_mstpc_sci8 - (channel - k_uart_mstpc_channel_min));
00742         system_regs()->mstpcrc &= ~(k_uart_mstpcrb_bit_set « mstpc_bit);
00743     } else {
00744         *prcr_reg() = k_rx_prcr_lock;
00745         return k_rx_err_invalid_arg;
00746     }
00747
00748     *prcr_reg() = k_rx_prcr_lock;
00749     return k_rx_ok;
00750 }
00751
00752 /**
00753 * @brief Configure the GPIO and MPC pin-mux settings for a UART TX/RX pin pair
00754 *
00755 * @details
00756 * Prepares the two GPIO pins required for SCI UART operation by programming
00757 * the Multi-Function Pin Controller (MPC) and the Port Direction/Mode registers
00758 * in the correct order. The RX72N hardware requires that:
00759 * - MPC is configured before switching a pin to peripheral mode (PMR)
00760 * - TX pin is driven as an output; RX pin is configured as an input
00761 * - Both pins are switched to peripheral mode (PMR bit = 1) last
00762 *
00763 * **Algorithm steps:**
00764 * 1. Extract the port number and pin number from each rx_port_pin_t using
00765 *    rx_port_from_pin() and rx_pin_from_pin().
00766 * 2. Validate that both pin numbers are <= k_rx_pin_max (7); return
00767 *    k_rx_err_invalid_arg if out of range. (Lower-bound check is omitted
00768 *    because pin numbers are uint8_t and k_rx_pin_min == 0, which would
00769 *    trigger -Wtype-limits.)

```

```

00770 * 3. Obtain volatile port register base pointers via rx_port_get_base(); return
00771 * k_rx_err_invalid_arg if either is nullptr (invalid port number).
00772 * 4. Call rx_mpc_set_sci(tx_gpio) to set the TX pin's PFS register to the SCI
00773 * TXD function; propagate any error immediately.
00774 * 5. Call rx_mpc_set_sci(rx_gpio) to set the RX pin's PFS register to the SCI
00775 * RXD function; propagate any error immediately.
00776 * 6. Build per-pin bitmasks (k_uart_gpio_bit_set « pin_number).
00777 * 7. Set PDR bit for TX pin (output direction) and PMR bit for TX pin
00778 * (peripheral mode).
00779 * 8. Clear PDR bit for RX pin (input direction) and set PMR bit for RX pin
00780 * (peripheral mode).
00781 * 9. Return k_rx_ok.
00782 *
00783 * **MPC write protection:**
00784 * rx_mpc_set_sci() internally handles the PFSWE unlock/lock sequence, so this
00785 * function does not need to touch the PFSWE bit directly.
00786 *
00787 * **Pin direction and mode summary:**
00788 * | Pin | PDR (direction) | PMR (mode) |
00789 * |-----|-----|
00790 * | TX | 1 (output) | 1 (peripheral) |
00791 * | RX | 0 (input) | 1 (peripheral) |
00792 *
00793 *
00794 *
00795 *
00796 * @pre tx_gpio and rx_gpio must correspond to physically valid RX72N port pins
00797 * @pre The SCI module clock must already be enabled (MSTP\\rB bit cleared) so
00798 * that the SCI TXD/RXD MPC function codes are accepted
00799 *
00800 * @post TX pin: PDR bit set (output), PMR bit set (peripheral function)
00801 * @post RX pin: PDR bit cleared (input), PMR bit set (peripheral function)
00802 * @post Both pins' PFS registers configured for SCI TX/RX function via MPC
00803 *
00804 * @note Pin validation only checks upper bound (>k_rx_pin_max); lower bound
00805 * (0) is omitted to avoid the -Wtype-limits compiler warning triggered by
00806 * comparing an unsigned type to 0
00807 * @warning Passing the same pin for both TX and RX produces undefined hardware
00808 * behaviour; no duplicate-pin check is performed
00809 *
00810 * @par Thread Safety:
00811 * Not thread-safe. PDR/PMR are read-modify-write targets shared with all GPIO
00812 * operations on the same port. Call only during single-threaded initialization.
00813 *
00814 * @par Performance:
00815 * - Execution time: ~10 us (dominated by two rx_mpc_set_sci() calls)
00816 * - Stack usage: 32 bytes (four uint8_t locals + two pointer locals)
00817 *
00818 * @par Example:
00819 * @code{.c}
00820 * // Configure SCI9 default debug pins (PB7=TX, PB6=RX)
00821 * rx_err_t err = internal_configure_uart_pins(
00822 * (rx_port_pin_t)k_uart_debug_tx_gpio,
00823 * (rx_port_pin_t)k_uart_debug_rx_gpio);
00824 * if (err != k_rx_ok) {
00825 * return err; // Invalid pin specification or MPC error
00826 * }
00827 * @endcode
00828 *
00829 * @see uart_init_channel() Caller that passes validated tx_gpio/rx_gpio
00830 * @see rx_mpc_set_sci() MPC pin-function assignment (with PFSWE unlock)
00831 * @see rx_port_get_base() Port register base address lookup
00832 * @see uart_gpio_constants_t k_uart_gpio_bit_set used to build pin bitmasks
00833 *
00834 * @since Version 1.0.0
00835 */
00836 static rx_err_t internal_configure_uart_pins(const rx_port_pin_t tx_gpio, rx_port_pin_t rx_gpio)
00837 {
00838 /* Extract port and pin numbers for hardware register access */
00839 const uint8_t tx_port = rx_port_from_pin(tx_gpio);
00840 const uint8_t tx_pin = rx_pin_from_pin(tx_gpio);
00841 const uint8_t rx_port = rx_port_from_pin(rx_gpio);
00842 const uint8_t rx_pin = rx_pin_from_pin(rx_gpio);
00843
00844 /* Validate pin numbers (lower bound checks omitted - pins are uint8_t, k_rx_pin_min == 0,
00845 * so pin >= k_rx_pin_min is always true, avoiding -Wtype-limits warning) */
00846 if (tx_pin > k_rx_pin_max || rx_pin > k_rx_pin_max) {
00847 return k_rx_err_invalid_arg;
00848 }
00849
00850 /* Get port bases */
00851 volatile rx_port_regs_t* tx_port_base = rx_port_get_base(tx_port);
00852 volatile rx_port_regs_t* rx_port_base = rx_port_get_base(rx_port);
00853
00854 if (tx_port_base == nullptr || rx_port_base == nullptr) {
00855 return k_rx_err_invalid_arg;
00856 }

```

```

00857
00858 /* Configure MPC for SCI function */
00859 rx_err_t err = rx_mpc_set_sci(tx_gpio);
00860 if (err != k_rx_ok) {
00861     return err;
00862 }
00863
00864 err = rx_mpc_set_sci(rx_gpio);
00865 if (err != k_rx_ok) {
00866     return err;
00867 }
00868
00869 /* Configure TX pin: output direction, peripheral mode */
00870 const uint8_t tx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « tx_pin);
00871 const uint8_t rx_pin_mask = (uint8_t)(k_uart_gpio_bit_set « rx_pin);
00872
00873 tx_port_base->pdr |= tx_pin_mask; /* Output */
00874 tx_port_base->pmr |= tx_pin_mask; /* Peripheral mode */
00875
00876 /* Configure RX pin: input direction, peripheral mode */
00877 rx_port_base->pdr &= (uint8_t)~rx_pin_mask; /* Input */
00878 rx_port_base->pmr |= rx_pin_mask; /* Peripheral mode */
00879
00880 return k_rx_ok;
00881 }
00882
00883 /*
=====
00884 * Multi-Channel UART Functions
00885 *
=====
00886 */
00887
00888 /**
00889 * @brief Initialize UART channel with specified configuration
00890 *
00891 * @details
00892 * Configures and enables an SCI peripheral for asynchronous UART operation.
00893 * Handles module clock enable, pin mux configuration, and SCI register setup.
00894 *
00895 * **Algorithm steps:**
00896 * 1. Validate configuration parameters (NULL check, channel range, baud rate)
00897 * 2. Get SCI register base address for channel
00898 * 3. Check if channel is already initialized (prevent double-init)
00899 * 4. Enable SCI module clock (clear MSTPB bit)
00900 * 5. Configure TX/RX pins via MPC and GPIO registers
00901 * 6. Disable TX/RX during configuration (S\\r = 0)
00902 * 7. Set serial mode: async, 8N1, PCLK/1 (SMR)
00903 * 8. Calculate and set baud rate divisor (BRR)
00904 * 9. Wait for 1 bit time (synchronization)
00905 * 10. Enable TX and RX (S\\r = 0x30)
00906 * 11. Mark channel as initialized
00907 *
00908 * @dot
00909 * digraph uart_init {
00910 *     rankdir=TB;
00911 *     node [shape=box, style=rounded];
00912 *
00913 *     start [label="Start"];
00914 *     validate [label="Validate config"];
00915 *     check_init [label="Already initialized?", shape=diamond];
00916 *     enable_clk [label="Enable SCI clock"];
00917 *     config_pins [label="Configure TX/RX pins"];
00918 *     disable_txrx [label="Disable TX/RX"];
00919 *     set_mode [label="Set async 8N1 mode"];
00920 *     set_baud [label="Set BRR for baud rate"];
00921 *     wait [label="Wait 1 bit time"];
00922 *     enable_txrx [label="Enable TX + RX"];
00923 *     mark_init [label="Mark initialized"];
00924 *     success [label="Return k_rx_ok", fillcolor=lightgreen, style="rounded, filled"];
00925 *     error [label="Return error", fillcolor=lightcoral, style="rounded, filled"];
00926 *
00927 *     start -> validate;
00928 *     validate -> error [label="invalid"];
00929 *     validate -> check_init [label="valid"];
00930 *     check_init -> error [label="yes"];
00931 *     check_init -> enable_clk [label="no"];
00932 *     enable_clk -> config_pins;
00933 *     config_pins -> error [label="fail"];
00934 *     config_pins -> disable_txrx [label="ok"];
00935 *     disable_txrx -> set_mode;
00936 *     set_mode -> set_baud;
00937 *     set_baud -> wait;
00938 *     wait -> enable_txrx;
00939 *     enable_txrx -> mark_init;
00940 *     mark_init -> success;
00941 * }

```

```

00942 * @enddot
00943 *
00944 *
00945 *
00946 * @pre config must point to valid uart_channel_config_t structure
00947 * @pre config->channel must be in range [0, 12]
00948 * @pre config->baudrate must be in valid range for PCLKB
00949 * @pre Channel must not be already initialized
00950 *
00951 * @post Channel configured for 8N1 async operation
00952 * @post TX and RX enabled and ready for use
00953 * @post s_channel_initialized[channel] set to true
00954 * @post TX/RX pins configured for peripheral function
00955 *
00956 * @note Not thread-safe - call once during initialization
00957 * @note Includes ~1ms delay for baud rate synchronization
00958 * @warning Does not validate pin assignments against hardware
00959 *
00960 * @par Thread Safety:
00961 * Not thread-safe. Call once per channel during system initialization.
00962 *
00963 * @par Performance:
00964 * - Execution time: ~50 us (includes MPC and delay)
00965 * - Stack usage: 48 bytes
00966 *
00967 * @par Example:
00968 * @code{.c}
00969 * const uart_channel_config_t config = {
00970 *     .channel = k_uart_channel_9,
00971 *     .baudrate = 115200,
00972 *     .tx_gpio = k_rx_pb_7,
00973 *     .rx_gpio = k_rx_pb_6,
00974 * };
00975 *
00976 * rx_err_t err = uart_init_channel(&config);
00977 * if (err != k_rx_ok) {
00978 *     // Handle error
00979 * }
00980 * @endcode
00981 *
00982 * @see uart_deinit_channel() Deinitialize channel
00983 * @see uart_putc_channel() Transmit single character
00984 * @see uart_channel_config_t Configuration structure
00985 *
00986 * @since Version 1.0.0
00987 */
00988 /**
00989 * @brief Validate uart_init_channel arguments and resolve SCI base
00990 *
00991 * @details
00992 * Checks for null config, channel range, baud range, channel-already-
00993 * initialized state. Resolves config->channel to its SCI peripheral
00994 * base address.
00995 *
00996 * @param[in] config Caller's channel config (must be non-null)
00997 * @param[out] sci_out On success, set to the SCI peripheral base
00998 *
00999 * @return rx_err_t Error code
01000 * @retval k_rx_ok All checks pass, *sci_out non-null
01001 * @retval k_rx_err_null_ptr config is nullptr
01002 * @retval k_rx_err_invalid_arg Bounds violation
01003 * @retval k_rx_err_invalid_state Channel already initialized
01004 *
01005 * @pre sci_out non-null
01006 *
01007 * @since Version 1.0.0
01008 */
01009 static rx_err_t internal_uart_validate_init(const uart_channel_config_t* const config,
01010                                           volatile rx_sci_regs_t** const sci_out)
01011 {
01012     if (config == nullptr) {
01013         return k_rx_err_null_ptr;
01014     }
01015     if ((uint8_t)config->channel >= k_uart_max_channels) {
01016         return k_rx_err_invalid_arg;
01017     }
01018     if ((config->baudrate < k_uart_baudrate_min) || (config->baudrate > k_uart_baudrate_max)) {
01019         return k_rx_err_invalid_arg;
01020     }
01021     *sci_out = sci_get_channel(config->channel);
01022     if (*sci_out == nullptr) {
01023         return k_rx_err_invalid_arg;
01024     }
01025     if (s_channel_initialized[config->channel]) {
01026         return k_rx_err_invalid_state;
01027     }
01028     return k_rx_ok;

```

```

01029 }
01030
01031 /**
01032  * @brief Resolve the peripheral clock frequency for a given SCI channel
01033  *
01034  * @details
01035  * SCILj (SCI0-SCI6) and SCILh (SCI12) are clocked by PCLKB. SCILi
01036  * (SCI7-SCI11, extended region 0x00D0xxx) are clocked by PCLKA. Using
01037  * the wrong clock produces a wrong BRR and a wrong baud rate.
01038  *
01039  * @param[in] channel Validated SCI channel index
01040  *
01041  * @return uint32_t Peripheral clock in Hz
01042  *
01043  * @pre channel < k_uart_max_channels
01044  *
01045  * @since Version 1.0.0
01046  */
01047 static uint32_t internal_uart_select_pclk(const uart_channel_t channel)
01048 {
01049     return (((uint8_t)channel >= k_uart_pclka_channel_min) &&
01050            ((uint8_t)channel <= k_uart_pclka_channel_max))
01051            ? k_pclka_hz
01052            : k_pclkb_hz;
01053 }
01054
01055 /**
01056  * @brief Program SMR/SEMR/BRR for the requested channel and baud
01057  *
01058  * @details
01059  * Sets SMR=async-8N1, picks SEMR.ABCS=1 for SCI9 (lowers baud error
01060  * from +1.72% to +0.16% at 115200 on PCLKA=120 MHz, see HUM 34.2.12),
01061  * computes BRR via internal_calculate_brr, then waits k_uart_bit_time_
01062  * delay_cycles for the new rate to settle. SEMR MUST be written before
01063  * BRR because BRR is interpreted against whatever base clock SEMR
01064  * selects.
01065  *
01066  * @param[in] sci SCI peripheral base (must be non-null)
01067  * @param[in] channel Validated SCI channel index
01068  * @param[in] baudrate Validated requested baud
01069  *
01070  *
01071  * @pre SCR=disabled before this call
01072  * @post SMR/SEMR/BRR programmed
01073  *
01074  * @since Version 1.0.0
01075  */
01076 static void internal_uart_program_baud(volatile rx_sci_regs_t* const sci,
01077                                       const uart_channel_t channel,
01078                                       const uint32_t baudrate)
01079 {
01080     sci->smr = k_sci_smr_async_8n1;
01081     const bool use_abcs = (channel == k_uart_channel_9);
01082     sci->semr = use_abcs ? (uint8_t)k_sci_semr_abcs_bit : (uint8_t)k_sci_semr_default;
01083     sci->brr = internal_calculate_brr(baudrate, internal_uart_select_pclk(channel), use_abcs);
01084
01085     /* NOTE: Busy-wait required - may run before ThreadX initialization */
01086     for (volatile uint32_t i = 0; i < k_uart_bit_time_delay_cycles; i++) {
01087         __asm__ volatile("nop");
01088     }
01089 }
01090
01091 /**
01092  * @brief Switch SCI9 to interrupt-driven RX mode
01093  *
01094  * @details
01095  * Resets the ring-buffer head/tail, drains any pending RDR + clears
01096  * boot-time error flags, programs the RXI9 vector priority/IER/IR, then
01097  * sets SCR.RIE=1 so the RXI9 ISR begins firing on RDRF. HUM vector 102
01098  * = SCI9_RXI9, IER index 12 ((102>3)=12), IER bit 6 ((102&7)=6). See
01099  * comm_task RX overrun note at the top of this file.
01100  *
01101  * @param[in] sci SCI9 peripheral base (must be non-null)
01102  *
01103  *
01104  * @pre Caller has set SCR=TX|RE before this call
01105  * @post SCR.RIE asserted, RXI9 vector enabled, ring drained
01106  *
01107  * @since Version 1.0.0
01108  */
01109 static void internal_uart_enable_sci9_rx_isr(volatile rx_sci_regs_t* const sci)
01110 {
01111     s_sci9_rx_head = 0U;
01112     s_sci9_rx_tail = 0U;
01113
01114     /* Drain any pending RDR byte and clear error flags BEFORE enabling the
01115      * RXI interrupt, so the first real byte doesn't find an ORER set from

```

```

01116  * boot-time noise (which would gate further interrupts). */
01117  if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01118      internal_clear_errors(sci);
01119  }
01120  if ((sci->ssr & k_sci_ssr_rdrf_flag) != k_uart_flag_clear) {
01121      (void)sci->rdr;
01122      const uint8_t ssr = sci->ssr;
01123      sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01124  }
01125
01126  const uint16_t vec_rxi9 = (uint16_t)k_vect_sci9_rxi9;
01127  icu()->ipr[vec_rxi9] = k_sci9_rxi9_priority;
01128  icu()->ir[vec_rxi9] = 0U;
01129  icu()->ier[vec_rxi9 / k_icu_ier_bits_per_reg] |=
01130      (uint8_t)(1U « (vec_rxi9 % k_icu_ier_bits_per_reg));
01131
01132  sci->scr = (uint8_t)(k_sci_scr_txrx_enabled | k_sci_scr_rie_bit);
01133  }
01134
01135  rx_err_t uart_init_channel(const uart_channel_config_t* config)
01136  {
01137      volatile rx_sci_regs_t* sci = nullptr;
01138      const rx_err_t check = internal_uart_validate_init(config, &sci);
01139      if (check != k_rx_ok) {
01140          return check;
01141      }
01142
01143      rx_err_t err = internal_enable_sci_clock(config->channel);
01144      if (err != k_rx_ok) {
01145          return err;
01146      }
01147      err = internal_configure_uart_pins(config->tx_gpio, config->rx_gpio);
01148      if (err != k_rx_ok) {
01149          return err;
01150      }
01151
01152      /* Disable TX/RX while reprogramming registers */
01153      sci->scr = k_sci_scr_disabled;
01154
01155      internal_uart_program_baud(sci, config->channel, config->baudrate);
01156
01157      /* Configure serial control: Enable TX and RX */
01158      sci->scr = k_sci_scr_txrx_enabled;
01159
01160      if (config->channel == k_uart_channel_9) {
01161          internal_uart_enable_sci9_rx_isr(sci);
01162      }
01163
01164      s_channel_initialized[config->channel] = true;
01165      return k_rx_ok;
01166  }
01167
01168  /**
01169   * @brief Deinitialize UART channel and disable TX/RX
01170   *
01171   * @details
01172   * Disables transmit and receive on the specified SCI channel and marks it
01173   * as uninitialized. The module clock is left running (safe to re-initialize
01174   * without re-enabling). After this call the channel may be re-initialized
01175   * with uart_init_channel().
01176   *
01177   * **Algorithm steps:**
01178   * 1. Validate channel number (0-12)
01179   * 2. Get SCI register base address
01180   * 3. Write S\\r = 0 to disable TX and RX
01181   * 4. Clear s_channel_initialized[channel]
01182   *
01183   *
01184   *
01185   * @pre channel must be in range [0, 12]
01186   * @pre Channel should be initialized before deinitializing (safe to call on uninit channel)
01187   *
01188   * @post TX and RX disabled (S\\r = 0)
01189   * @post s_channel_initialized[channel] set to false
01190   *
01191   * @note Not thread-safe - call during shutdown, not during normal operation
01192   * @note Module stop clock is NOT re-asserted (peripheral clock left enabled)
01193   *
01194   * @par Thread Safety:
01195   * Not thread-safe. Do not call while another thread is using the channel.
01196   *
01197   * @par Performance:
01198   * - Execution time: ~1 us
01199   * - Stack usage: 16 bytes
01200   *
01201   * @par Example:
01202   * @code{.c}

```



```

01203 * rx_err_t err = uart_deinit_channel(k_uart_channel_9);
01204 * if (err != k_rx_ok) {
01205 *     // Handle error (invalid channel)
01206 * }
01207 * // Channel is now disabled and can be re-initialized
01208 * @endcode
01209 *
01210 * @see uart_init_channel() Initialize channel
01211 * @see uart_debug_init() Initialize debug channel (SCI9)
01212 *
01213 * @since Version 1.0.0
01214 */
01215 rx_err_t uart_deinit_channel(const uart_channel_t channel)
01216 {
01217     /* Validate channel */
01218     if ((uint8_t)channel >= k_uart_max_channels) {
01219         return k_rx_err_invalid_arg;
01220     }
01221
01222     /* Get SCI register base */
01223     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01224     if (sci == nullptr) {
01225         return k_rx_err_invalid_arg;
01226     }
01227
01228     /* Disable TX/RX (this also drops SCR.RIE for SCI9). */
01229     sci->scr = k_sci_scr_disabled;
01230
01231     /* For SCI9, mask the RXI9 interrupt in the ICU and flush the ring so a
01232     * subsequent re-init starts clean. */
01233     if (channel == k_uart_channel_9) {
01234         const uint16_t vec_rx19 = (uint16_t)k_vect_sci9_rx19;
01235         icu()->ier[vec_rx19 / k_icu_ier_bits_per_reg] &=
01236             (uint8_t) ~(1U « (vec_rx19 % k_icu_ier_bits_per_reg));
01237         icu()->ir[vec_rx19] = 0U;
01238         s_sci9_rx_head = 0U;
01239         s_sci9_rx_tail = 0U;
01240     }
01241
01242     /* Mark channel as not initialized */
01243     s_channel_initialized[channel] = false;
01244
01245     return k_rx_ok;
01246 }
01247
01248 /**
01249 * @brief Transmit single character on UART channel
01250 *
01251 * @details
01252 * Transmits a single byte on the specified SCI channel. Waits for the transmit
01253 * data register to become empty (TDRE flag) with timeout protection, then writes
01254 * the data to TDR. Uses polling mode (no interrupts).
01255 *
01256 * **Algorithm steps:**
01257 * 1. Validate channel number (0-12)
01258 * 2. Check channel is initialized
01259 * 3. Get SCI register base address
01260 * 4. Wait for TDRE flag with timeout (k_uart_tx_timeout iterations)
01261 * 5. Write data byte to TDR register
01262 * 6. Clear TDRE flag (read SSR, write back with TDRE=0)
01263 *
01264 * **Character transmission timing at 115200 baud:**
01265 * - Start bit: 8.68 us
01266 * - 8 data bits: 69.44 us
01267 * - Stop bit: 8.68 us
01268 * - **Total**: ~86.8 us per character
01269 *
01270 *
01271 *
01272 *
01273 * @pre Channel must be initialized via uart_init_channel()
01274 * @pre UART TX must be enabled (happens during init)
01275 *
01276 * @post Character written to TDR and transmission started
01277 * @post TDRE flag cleared (TDR occupied)
01278 *
01279 * @note Blocking call - waits for TDRE flag
01280 * @note Timeout prevents infinite wait if hardware fails
01281 * @warning May block for up to ~100ms if TX line is held low
01282 *
01283 * @par Thread Safety:
01284 * Thread-safe for different channels. Not safe for same channel without mutex.
01285 *
01286 * @par Performance:
01287 * - Execution time: ~90 us @ 115200 baud (one character time)
01288 * - Stack usage: 16 bytes
01289 *

```



```

01290 * @par Example:
01291 * @code{.c}
01292 * // Send single character
01293 * rx_err_t err = uart_putc_channel(k_uart_channel_9, 'A');
01294 * if (err != k_rx_ok) {
01295 *     // Handle error
01296 * }
01297 *
01298 * // Send carriage return + line feed
01299 * uart_putc_channel(k_uart_channel_9, '\\r');
01300 * uart_putc_channel(k_uart_channel_9, '\\n');
01301 * @endcode
01302 *
01303 * @see uart_puts_channel() Transmit string with newline conversion
01304 * @see uart_write_channel() Transmit binary data
01305 *
01306 * @since Version 1.0.0
01307 */
01308 rx_err_t uart_putc_channel(const uart_channel_t channel, const char data)
01309 {
01310     /* Validate channel */
01311     if ((uint8_t)channel >= k_uart_max_channels) {
01312         return k_rx_err_invalid_arg;
01313     }
01314
01315     /* Check initialization */
01316     if (!s_channel_initialized[channel]) {
01317         return k_rx_err_invalid_state;
01318     }
01319
01320     /* Get SCI register base */
01321     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01322     if (sci == nullptr) {
01323         return k_rx_err_invalid_arg;
01324     }
01325
01326     /* Wait for transmit buffer to be empty (TDRE flag) with timeout */
01327     uint32_t timeout = k_uart_tx_timeout;
01328     while ((sci->ssr & k_sci_ssr_tdre_flag) == k_uart_flag_clear &&
01329         timeout > k_uart_timeout_expired) {
01330         timeout -= k_uart_timeout_decrement;
01331     }
01332
01333     /* Check if timeout occurred */
01334     if (timeout == k_uart_timeout_expired) {
01335         return k_rx_err_timeout;
01336     }
01337
01338     /* Write data to transmit register */
01339     sci->tdr = (uint8_t)data;
01340
01341     /* Clear TDRE flag by reading SSR then writing 0 */
01342     const uint8_t ssr = sci->ssr;
01343     sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_tdre_flag);
01344
01345     return k_rx_ok;
01346 }
01347 /**
01348 * @brief Transmit null-terminated string on UART channel with newline conversion
01349 *
01350 * @details
01351 * Transmits a null-terminated string with automatic LF to \\r+LF conversion for
01352 * terminal compatibility. Enforces maximum string length (k_uart_max_str_len = 256)
01353 * to comply with NASA Power of 10 Rule 2 (bounded loops).
01354 *
01355 * **Algorithm steps:**
01356 * 1. Validate string pointer (NULL check)
01357 * 2. Validate channel number and initialization state
01358 * 3. For each character up to k_uart_max_str_len:
01359 *     a. Check for null terminator (success exit)
01360 *     b. If '\\n', send '\\r' first (\\r+LF conversion)
01361 *     c. Transmit character via uart_putc_channel()
01362 *     d. Propagate any errors immediately
01363 * 4. If limit reached without terminator, return k_rx_err_invalid_size
01364 *
01365 * **Newline conversion:**
01366 * - Input '\\n' (LF) -> Output '\\r\\n' (\\r+LF)
01367 * - Required for Windows terminals (e.g., PuTTY, TeraTerm)
01368 * - Ensures cursor returns to column 0 before line feed
01369 *
01370 *
01371 *
01372 *
01373 *
01374 * @pre Channel must be initialized via uart_init_channel()
01375 * @pre str must point to null-terminated string in valid memory
01376 *

```

```

01377 * @post All characters transmitted including converted newlines
01378 * @post On error, partial string may have been transmitted
01379 *
01380 * @note Blocking call - waits for each character to transmit
01381 * @note String length limited to 256 characters for safety
01382 * @warning Long strings may take significant time (256 chars ~ 22ms @ 115200)
01383 *
01384 * @par Thread Safety:
01385 * Thread-safe for different channels. Not safe for same channel without mutex.
01386 *
01387 * @par Performance:
01388 * - Execution time: ~90 us per character @ 115200 baud
01389 * - Stack usage: 24 bytes
01390 * - Example: 100-char string ~ 9 ms
01391 *
01392 * @par Example:
01393 * @code{.c}
01394 * // Simple string
01395 * uart_puts_channel(k_uart_channel_9, "Hello, World!\n");
01396 * // Output: "Hello, World!\r\n"
01397 *
01398 * // Multiple lines
01399 * uart_puts_channel(k_uart_channel_9, "Line 1\nLine 2\nLine 3\n");
01400 * // Output: "Line 1\r\nLine 2\r\nLine 3\r\n"
01401 *
01402 * // Error handling
01403 * rx_err_t err = uart_puts_channel(k_uart_channel_9, msg);
01404 * if (err != k_rx_ok) {
01405 *     // Partial transmission may have occurred
01406 * }
01407 * @endcode
01408 *
01409 * @see uart_putc_channel() Transmit single character
01410 * @see uart_write_channel() Transmit binary data (no conversion)
01411 * @see uart_debug_puts() Simplified debug wrapper
01412 *
01413 * @since Version 1.0.0
01414 */
01415 rx_err_t uart_puts_channel(const uart_channel_t channel, const char* str)
01416 {
01417     /* Validate parameters */
01418     if (str == nullptr) {
01419         return k_rx_err_null_ptr;
01420     }
01421
01422     if ((uint8_t)channel >= k_uart_max_channels) {
01423         return k_rx_err_invalid_arg;
01424     }
01425
01426     if (!s_channel_initialized[channel]) {
01427         return k_rx_err_invalid_state;
01428     }
01429
01430     /* Transmit string with \n to \r\n conversion (statically bounded) */
01431     for (uint32_t i = 0; i < k_uart_max_str_len; ++i) {
01432         if (str[i] == '\0') {
01433             return k_rx_ok; /* Terminator found, success */
01434         }
01435         if (str[i] == '\n') {
01436             const rx_err_t err = uart_putc_channel(channel, '\r');
01437             if (err != k_rx_ok) {
01438                 return err;
01439             }
01440         }
01441         const rx_err_t err = uart_putc_channel(channel, str[i]);
01442         if (err != k_rx_ok) {
01443             return err;
01444         }
01445     }
01446
01447     /* Reached limit without finding terminator */
01448     return k_rx_err_invalid_size;
01449 }
01450
01451 /**
01452 * @brief Write binary data to UART channel without newline conversion
01453 *
01454 * @details
01455 * Transmits a block of raw bytes on the specified SCI channel using
01456 * uart_putc_channel() for each byte. No LF-to-\r+LF conversion is performed,
01457 * making this function suitable for binary protocol data.
01458 *
01459 * **Algorithm steps:**
01460 * 1. Validate data pointer (NULL check)
01461 * 2. Validate channel number and initialization state
01462 * 3. For each byte in [0, length): call uart_putc_channel() and propagate errors
01463 */

```

```

01464 *
01465 *
01466 *
01467 *
01468 * @pre Channel must be initialized via uart_init_channel()
01469 * @pre data must point to at least `length` bytes of valid memory
01470 *
01471 * @post All `length` bytes transmitted in order
01472 * @post On error, partial data may have been transmitted
01473 *
01474 * @note Blocking call - waits for each byte to be accepted by TDR
01475 * @note No newline conversion - use uart_puts_channel() for text output
01476 *
01477 * @par Thread Safety:
01478 * Thread-safe for different channels. Not safe for same channel without mutex.
01479 *
01480 * @par Performance:
01481 * - Execution time: ~90 us per byte @ 115200 baud
01482 * - Stack usage: 24 bytes
01483 *
01484 * @par Example:
01485 * @code{.c}
01486 * const uint8_t frame[] = {0xAA, 0x55, 0x01, 0x02, 0x03};
01487 * rx_err_t err = uart_write_channel(k_uart_channel_9, frame, sizeof(frame));
01488 * if (err != k_rx_ok) {
01489 *     // Handle partial write
01490 * }
01491 * @endcode
01492 *
01493 * @see uart_puts_channel() Transmit text string with newline conversion
01494 * @see uart_read_channel() Read binary data
01495 *
01496 * @since Version 1.0.0
01497 */
01498 rx_err_t uart_write_channel(const uart_channel_t channel, const uint8_t* data, uint16_t length)
01499 {
01500     /* Validate parameters */
01501     if (data == nullptr) {
01502         return k_rx_err_null_ptr;
01503     }
01504
01505     if ((uint8_t)channel >= k_uart_max_channels) {
01506         return k_rx_err_invalid_arg;
01507     }
01508
01509     if (!s_channel_initialized[channel]) {
01510         return k_rx_err_invalid_state;
01511     }
01512
01513     /* Write each byte */
01514     for (uint16_t i = 0; i < length; i++) {
01515         const rx_err_t err = uart_putc_channel(channel, (char)data[i]);
01516         if (err != k_rx_ok) {
01517             return err;
01518         }
01519     }
01520
01521     return k_rx_ok;
01522 }
01523
01524 /**
01525 * @brief Receive single character from UART channel (non-blocking)
01526 *
01527 * @details
01528 * Attempts to read a single byte from the specified SCI channel's receive data
01529 * register (RDR). This is a non-blocking operation - returns immediately
01530 * with k_rx_err_empty if no data is available.
01531 *
01532 * **Algorithm steps:**
01533 * 1. Validate data pointer and channel number
01534 * 2. Check channel initialization state
01535 * 3. Get SCI register base address
01536 * 4. Clear any error flags (ORER, FER, PER)
01537 * 5. Check RDRF flag (Receive Data Register Full)
01538 *    - If not set, return k_rx_err_empty immediately
01539 * 6. Read data byte from RDR register
01540 * 7. Clear RDRF flag (read SSR, write back with RDRF=0)
01541 *
01542 * **Error flag handling:**
01543 * - ORER (Overrun Error): Previous data overwritten before read
01544 * - FER (Framing Error): Stop bit not detected
01545 * - PER (Parity Error): Parity mismatch (not used in 8N1 mode)
01546 *
01547 *
01548 *
01549 *
01550 * @pre Channel must be initialized via uart_init_channel()

```

```

01551 * @pre data must point to valid memory for single char
01552 *
01553 * @post On success, *data contains received byte
01554 * @post RDRF flag cleared (ready for next byte)
01555 * @post Error flags cleared if any were set
01556 *
01557 * @note Non-blocking - returns immediately if no data
01558 * @note Clears error flags automatically before read
01559 * @note Single byte buffer - call frequently to avoid overrun
01560 *
01561 * @par Thread Safety:
01562 * Thread-safe for different channels. Not safe for same channel without mutex.
01563 *
01564 * @par Performance:
01565 * - Execution time: ~5 us (no wait)
01566 * - Stack usage: 16 bytes
01567 *
01568 * @par Example (Polling Loop):
01569 * @code{.c}
01570 * char c;
01571 * rx_err_t err;
01572 *
01573 * // Poll for incoming data
01574 * do {
01575 *   err = uart_getc_channel(k_uart_channel_9, &c);
01576 *   if (err == k_rx_ok) {
01577 *     process_character(c);
01578 *   }
01579 * } while (err == k_rx_ok); // Continue while data available
01580 * @endcode
01581 *
01582 * @par Example (Timeout Loop):
01583 * @code{.c}
01584 * char c;
01585 * uint32_t timeout = 100000;
01586 *
01587 * while (timeout > 0) {
01588 *   rx_err_t err = uart_getc_channel(k_uart_channel_9, &c);
01589 *   if (err == k_rx_ok) {
01590 *     // Got data
01591 *     break;
01592 *   }
01593 *   timeout--;
01594 * }
01595 *
01596 * if (timeout == 0) {
01597 *   // No data received within timeout
01598 * }
01599 * @endcode
01600 *
01601 * @see uart_read_channel() Read multiple bytes
01602 * @see uart_rx_available() Check if data is available
01603 *
01604 * @since Version 1.0.0
01605 */
01606 rx_err_t uart_getc_channel(const uart_channel_t channel, char* data)
01607 {
01608     /* Validate parameters */
01609     if (data == nullptr) {
01610         return k_rx_err_null_ptr;
01611     }
01612
01613     if ((uint8_t)channel >= k_uart_max_channels) {
01614         return k_rx_err_invalid_arg;
01615     }
01616
01617     if (!s_channel_initialized[channel]) {
01618         return k_rx_err_invalid_state;
01619     }
01620
01621     /* SCI9 reads pull from the RXI9-fed ring buffer so polled callers cannot
01622     * lose bytes that arrive between polls. Head/tail word accesses are atomic
01623     * on RX72N (aligned 16-bit loads/stores are single-cycle), so lock-free
01624     * SPSC is safe between the ISR producer and the task consumer. */
01625     if (channel == k_uart_channel_9) {
01626         if (s_sci9_rx_tail == s_sci9_rx_head) {
01627             return k_rx_err_empty;
01628         }
01629         *data = (char)s_sci9_rx_ring[s_sci9_rx_tail];
01630         s_sci9_rx_tail = (uint16_t)((s_sci9_rx_tail + 1U) & k_sci9_rx_ring_mask);
01631         return k_rx_ok;
01632     }
01633
01634     /* Get SCI register base */
01635     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01636     if (sci == nullptr) {
01637         return k_rx_err_invalid_arg;

```

```

01638 }
01639
01640 /* Clear any error flags first */
01641 if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01642     internal_clear_errors(sci);
01643 }
01644
01645 /* Check if receive data is available (RDRF flag) */
01646 if ((sci->ssr & k_sci_ssr_rdrf_flag) == k_uart_flag_clear) {
01647     return k_rx_err_empty;
01648 }
01649
01650 /* Read received data */
01651 *data = (char)sci->rdr;
01652
01653 /* Clear RDRF flag by reading SSR then writing 0 */
01654 /* Some RX MCUs require explicit clear after reading RDR */
01655 const uint8_t ssr = sci->ssr;
01656 sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01657
01658 return k_rx_ok;
01659 }
01660
01661 /**
01662  * @brief Read available bytes from UART channel (non-blocking)
01663  *
01664  * @details
01665  * Reads up to 'length' bytes from the specified SCI channel using
01666  * uart_getc_channel(). Stops immediately when no more data is available
01667  * (RDRF flag not set) rather than waiting. Actual bytes received is
01668  * reported via 'bytes_read'.
01669  *
01670  * **Algorithm steps:**
01671  * 1. Validate data and bytes_read pointers (NULL check)
01672  * 2. Validate channel number and initialization state
01673  * 3. Set *bytes_read = 0
01674  * 4. For each slot in [0, length):
01675  *     a. Call uart_getc_channel(); if k_rx_err_empty, break (done)
01676  *     b. On other error, propagate immediately
01677  *     c. Store byte and increment *bytes_read
01678  * 5. Return k_rx_ok (even if zero bytes were read)
01679  *
01680  *
01681  *
01682  *
01683  *
01684  *
01685  * @pre Channel must be initialized via uart_init_channel()
01686  * @pre data must point to at least 'length' bytes of writable memory
01687  *
01688  * @post *bytes_read set to actual number of bytes received
01689  * @post data[0..*bytes_read-1] contain the received bytes
01690  *
01691  * @note Non-blocking - returns immediately with whatever data is available
01692  * @note k_rx_ok with *bytes_read == 0 means no data was available
01693  *
01694  * @par Thread Safety:
01695  * Thread-safe for different channels. Not safe for same channel without mutex.
01696  *
01697  * @par Performance:
01698  * - Execution time: ~5 us per byte received + ~5 us when no data
01699  * - Stack usage: 24 bytes
01700  *
01701  * @par Example:
01702  * @code{.c}
01703  * uint8_t buf[64];
01704  * uint16_t count = 0;
01705  * rx_err_t err = uart_read_channel(k_uart_channel_9, buf, sizeof(buf), &count);
01706  * if (err == k_rx_ok && count > 0) {
01707  *     process_bytes(buf, count);
01708  * }
01709  * @endcode
01710  *
01711  * @see uart_getc_channel() Read single character
01712  * @see uart_rx_available() Check if data is available
01713  * @see uart_write_channel() Write binary data
01714  *
01715  * @since Version 1.0.0
01716  */
01717 rx_err_t uart_read_channel(const uart_channel_t channel,
01718                          uint8_t* data,
01719                          uint16_t length,
01720                          uint16_t* bytes_read)
01721 {
01722     /* Validate parameters */
01723     if (data == nullptr || bytes_read == nullptr) {
01724         return k_rx_err_null_ptr;

```

```

01725 }
01726
01727 if ((uint8_t)channel >= k_uart_max_channels) {
01728     return k_rx_err_invalid_arg;
01729 }
01730
01731 if (!s_channel_initialized[channel]) {
01732     return k_rx_err_invalid_state;
01733 }
01734
01735 /* Read available bytes */
01736 *bytes_read = 0;
01737 for (uint16_t i = 0; i < length; i++) {
01738     char c;
01739     const rx_err_t err = uart_getc_channel(channel, &c);
01740     if (err == k_rx_err_empty) {
01741         /* No more data available */
01742         break;
01743     }
01744     if (err != k_rx_ok) {
01745         return err;
01746     }
01747     data[i] = (uint8_t)c;
01748     (*bytes_read)++;
01749 }
01750
01751 return k_rx_ok;
01752 }
01753
01754 /**
01755  * @brief Check if receive data is available on UART channel
01756  *
01757  * @details
01758  * Checks the RDRF (Receive Data Register Full) flag in the SSR register of
01759  * the specified SCI channel. Provides a non-destructive peek at receive
01760  * availability without consuming any data from the buffer.
01761  *
01762  * **Algorithm steps:**
01763  * 1. Validate available pointer (NULL check)
01764  * 2. Validate channel number and initialization state
01765  * 3. Get SCI register base address
01766  * 4. Read SSR and test RDRF bit; store boolean result in *available
01767  *
01768  *
01769  *
01770  *
01771  * @pre Channel must be initialized via uart_init_channel()
01772  * @pre available must point to valid bool storage
01773  *
01774  * @post *available reflects current RDRF state
01775  * @post No data is consumed from the receive buffer
01776  *
01777  * @note Non-destructive - does not read RDR or clear any flags
01778  * @note RDRF can be set again immediately after reading if UART is receiving
01779  *
01780  * @par Thread Safety:
01781  * Thread-safe for different channels. Not safe for same channel without mutex.
01782  *
01783  * @par Performance:
01784  * - Execution time: ~2 us
01785  * - Stack usage: 16 bytes
01786  *
01787  * @par Example:
01788  * @code{.c}
01789  * bool ready = false;
01790  * if (uart_rx_available(k_uart_channel_9, &ready) == k_rx_ok && ready) {
01791  *     char c;
01792  *     uart_getc_channel(k_uart_channel_9, &c);
01793  * }
01794  * @endcode
01795  *
01796  * @see uart_getc_channel() Read single character
01797  * @see uart_read_channel() Read multiple bytes
01798  *
01799  * @since Version 1.0.0
01800  */
01801 rx_err_t uart_rx_available(const uart_channel_t channel, bool* available)
01802 {
01803     /* Validate parameters */
01804     if (available == nullptr) {
01805         return k_rx_err_null_ptr;
01806     }
01807
01808     if ((uint8_t)channel >= k_uart_max_channels) {
01809         return k_rx_err_invalid_arg;
01810     }
01811

```

```

01812 if (!s_channel_initialized[channel]) {
01813     return k_rx_err_invalid_state;
01814 }
01815
01816 /* SCI9 uses the RXI9 ring buffer; non-empty = bytes available. */
01817 if (channel == k_uart_channel_9) {
01818     *available = (s_sci9_rx_head != s_sci9_rx_tail);
01819     return k_rx_ok;
01820 }
01821
01822 /* Get SCI register base */
01823 const volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01824 if (sci == nullptr) {
01825     return k_rx_err_invalid_arg;
01826 }
01827
01828 /* Check RDRF flag */
01829 *available = ((sci->ssr & k_sci_ssr_rdrf_flag) != k_uart_flag_clear);
01830
01831 return k_rx_ok;
01832 }
01833
01834 rx_err_t uart_clear_rx_errors(uart_channel_t channel)
01835 {
01836     if ((uint8_t)channel >= k_uart_max_channels) {
01837         return k_rx_err_invalid_arg;
01838     }
01839     volatile rx_sci_regs_t* sci = sci_get_channel(channel);
01840     if (sci == nullptr) {
01841         return k_rx_err_invalid_arg;
01842     }
01843     const uint8_t ssr = sci->ssr;
01844     if ((ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01845         sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_error_mask);
01846     }
01847     return k_rx_ok;
01848 }
01849
01850 /**
01851  * @brief SCI9 receive-data-full interrupt (vector 102)
01852  *
01853  * @details
01854  * Fires when a byte lands in SCI9.RDR (RDRF rising edge). Reads the byte,
01855  * clears RDRF by writing-0 after a read of SSR per HUM 34.2.8, and pushes
01856  * the byte into s_sci9_rx_ring. ORER (overflow error) is checked and cleared
01857  * so a dropped byte doesn't wedge the peripheral; the drop counter is
01858  * incremented so telemetry can surface it.
01859  *
01860  * ISR execution time: ~200 ns at 240 MHz (8 register ops + ring push).
01861  * Ring capacity: k_sci9_rx_ring_size = 512 bytes = 4.4 ms of 115200 baud
01862  * which comfortably covers the ~10 ms comm_task poll period worst case.
01863  *
01864  * Registered at relocatable-vector-table slot 102 via the
01865  * `Stableentry$102$.rVectors` symbol below. Without that symbol the
01866  * linker script (src/boot/linker_script_rVectors.inc) falls through to
01867  * $Stableentry$default$.rVectors (0xFFFFFFFF), and the interrupt is
01868  * effectively disabled even with IER[12].bit6 set.
01869  *
01870  * __attribute__((interrupt)) makes GCC generate the RTE / full register
01871  * save+restore for the ISR body; the $Stableentry alias just places the
01872  * function at the right vector slot.
01873  */
01874 __attribute__((interrupt, used)) void INT_SCI9_RXI9(void)
01875 {
01876     #ifdef __RX__
01877     volatile rx_sci_regs_t* sci = sci_get_channel(k_uart_channel_9);
01878     if (sci == nullptr) {
01879         return;
01880     }
01881
01882     /* Clear error flags first so the RDR read is not masked by a sticky
01883     * ORER/FER/PER -- HUM 34.2.7 requires these to be cleared before the
01884     * peripheral will accept further bytes. */
01885     if ((sci->ssr & k_sci_ssr_error_mask) != k_uart_flag_clear) {
01886         const uint8_t ssr_err = sci->ssr;
01887         sci->ssr = (uint8_t)(ssr_err & ~k_sci_ssr_error_mask);
01888     }
01889
01890     const uint8_t byte = (uint8_t)sci->rdr;
01891
01892     /* Clear RDRF: read SSR, write it back with RDRF=0. */
01893     const uint8_t ssr = sci->ssr;
01894     sci->ssr = (uint8_t)(ssr & ~k_sci_ssr_rdrf_flag);
01895
01896     const uint16_t next_head = (uint16_t)((s_sci9_rx_head + 1U) & k_sci9_rx_ring_mask);
01897     if (next_head != s_sci9_rx_tail) {
01898         s_sci9_rx_ring[s_sci9_rx_head] = byte;

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01899     s_sci9_rx_head          = next_head;
01900 }
01901 /* Ring full: byte is dropped. comm_task polls every 10 ms and can
01902 * drain ~1150 bytes in that window, vs 115 bytes arriving at 115200
01903 * baud -- so the 512-byte ring has >4x headroom. If comm_task stalls,
01904 * the next well-formed frame the gateway sends will fail CRC and
01905 * trigger a retransmit, so no silent data loss. */
01906 #endif
01907 }
01908
01909 /*
=====
01910 * UART Debug Functions - Convenience Wrappers for SCI9
01911 *
=====
01912 */
01913
01914 /**
01915 * @brief Initialize default debug UART channel (SCI9 at 115200 baud)
01916 *
01917 * @details
01918 * Convenience function to initialize SCI9 with default debug settings:
01919 * - **Channel**: SCI9
01920 * - **Baud rate**: 115200 bps
01921 * - **TX pin**: PB7 (TXD9)
01922 * - **RX pin**: PB6 (RXD9)
01923 * - **Format**: 8N1 (8 data bits, no parity, 1 stop bit)
01924 *
01925 * SCI9 is connected to the CY7C65213 USB-UART bridge on the STAR PCB,
01926 * providing a virtual COM port when connected via USB.
01927 *
01928 * **Algorithm steps:**
01929 * 1. Create uart_channel_config_t with default values
01930 * 2. Call uart_init_channel() with the configuration
01931 * 3. Return result
01932 *
01933 *
01934 * @pre System clocks must be initialized (PCLKB = 60 MHz)
01935 * @pre SCI9 must not be already initialized
01936 *
01937 * @post SCI9 configured and ready for TX/RX at 115200 baud
01938 * @post uart_debug_puts(), uart_debug_putc() etc. are functional
01939 *
01940 * @note Call once during early system initialization
01941 * @note Call before any other debug output functions
01942 * @warning Must be called before ThreadX starts if debug output is needed during init
01943 *
01944 * @par Thread Safety:
01945 * Not thread-safe. Call once during startup before ThreadX.
01946 *
01947 * @par Performance:
01948 * - Execution time: ~50 us
01949 * - Stack usage: 64 bytes
01950 *
01951 * @par Example:
01952 * @code{.c}
01953 * void system_init(void)
01954 * {
01955 *     // Initialize clocks first
01956 *     rx_clock_init();
01957 *
01958 *     // Initialize debug UART for early output
01959 *     rx_err_t err = uart_debug_init();
01960 *     if (err != k_rx_ok) {
01961 *         // Cannot print error - no UART!
01962 *         // Could blink LED or set error flag
01963 *         return;
01964 *     }
01965 *
01966 *     // Now we can print debug messages
01967 *     uart_debug_puts("[INFO] System starting...\n");
01968 * }
01969 * @endcode
01970 *
01971 * @see uart_init_channel() Generic channel initialization
01972 * @see uart_debug_puts() Debug string output
01973 * @see uart_debug_putc() Debug character output
01974 *
01975 * @since Version 1.0.0
01976 */
01977 rx_err_t uart_debug_init(void)
01978 {
01979     const uart_channel_config_t config = {
01980         .channel = (uart_channel_t)k_uart_debug_channel,
01981         .baudrate = k_uart_default_baudrate,
01982         .tx_gpio = (rx_port_pin_t)k_uart_debug_tx_gpio,
01983         .rx_gpio = (rx_port_pin_t)k_uart_debug_rx_gpio,

```



```

01984 };
01985 return uart_init_channel(&config);
01986 }
01987
01988 #if !RX_IS_SIMULATOR
01989 /*
=====
01990 * Hardware Mode: Debug UART Functions
01991 *
=====
01992 * These functions use the SCI9 UART hardware for debug output.
01993 * In simulator mode, these are replaced by inline implementations in rx_log.h
01994 * that redirect to stdout.
01995 *
=====
01996 */
01997
01998 /**
01999 * @brief Transmit single character on debug UART (SCI9)
02000 *
02001 * @details
02002 * Convenience wrapper around uart_putc_channel() for the fixed debug channel
02003 * (SCI9). Errors are silently ignored to allow use in early initialization
02004 * contexts where error propagation is not yet possible.
02005 *
02006 * **Algorithm steps:**
02007 * 1. Call uart_putc_channel(k_uart_debug_channel, data)
02008 * 2. Cast return value to (void) - errors discarded
02009 *
02010 *
02011 * @pre uart_debug_init() must have been called successfully
02012 * @pre SCI9 must be initialized and TX enabled
02013 *
02014 * @post Character written to SCI9 TDR and transmission started
02015 * @post Any errors are silently discarded
02016 *
02017 * @note Error return from uart_putc_channel() is intentionally discarded
02018 * @note For error-checked output use uart_putc_channel() directly
02019 * @warning Only available when RX_IS_SIMULATOR is 0 (hardware builds)
02020 *
02021 * @par Thread Safety:
02022 * Not safe for concurrent access on SCI9 without external mutex.
02023 *
02024 * @par Performance:
02025 * - Execution time: ~90 us @ 115200 baud
02026 * - Stack usage: 16 bytes
02027 *
02028 * @par Example:
02029 * @code{.c}
02030 * uart_debug_putc('A'); // Send 'A'
02031 * uart_debug_putc('\r'); // Carriage return
02032 * uart_debug_putc('\n'); // Line feed
02033 * @endcode
02034 *
02035 * @see uart_debug_puts() Transmit string with newline conversion
02036 * @see uart_putc_channel() Error-checked single character transmit
02037 *
02038 * @since Version 1.0.0
02039 */
02040 void uart_debug_putc(const char data)
02041 {
02042     /* Ignore errors for debug output (used in early init before error handling) */
02043     (void)uart_putc_channel((uart_channel_t)k_uart_debug_channel, data);
02044 }
02045
02046 /**
02047 * @brief Transmit null-terminated string on debug UART with newline conversion
02048 *
02049 * @details
02050 * Transmits a null-terminated string to SCI9 with automatic LF-to-\\r+LF
02051 * conversion for terminal compatibility. Enforces a maximum length of
02052 * k_uart_max_str_len (256) characters per NASA Power of 10 Rule 2.
02053 * Silently returns on NULL pointer to allow safe use in early init.
02054 *
02055 * **Algorithm steps:**
02056 * 1. Return immediately if str is nullptr (defensive, no error return)
02057 * 2. For each character up to k_uart_max_str_len:
02058 *     a. Stop at null terminator
02059 *     b. If LF, send \\r first
02060 *     c. Send character via uart_debug_putc()
02061 *
02062 *
02063 * @pre uart_debug_init() must have been called successfully
02064 * @pre str should point to a null-terminated string in valid memory
02065 *
02066 * @post All characters up to null terminator transmitted to SCI9
02067 * @post Each LF replaced by \\r\\n in the transmitted output

```

```

02068 *
02069 * @note No return value - errors from uart_debug_putc() are silently discarded
02070 * @note String truncated at k_uart_max_str_len (256) characters
02071 * @warning Only available when RX_IS_SIMULATOR is 0 (hardware builds)
02072 *
02073 * @par Thread Safety:
02074 * Not safe for concurrent access on SCI9 without external mutex.
02075 *
02076 * @par Performance:
02077 * - Execution time: ~90 us per character @ 115200 baud
02078 * - Stack usage: 24 bytes
02079 *
02080 * @par Example:
02081 * @code{.c}
02082 * uart_debug_puts("Hello, World!\n"); // Sends "Hello, World!\r\n"
02083 * uart_debug_puts("[INFO] Boot complete\n");
02084 * @endcode
02085 *
02086 * @see uart_debug_putc() Transmit single character
02087 * @see uart_debug_putint() Transmit decimal integer
02088 * @see uart_puts_channel() Error-checked string transmit
02089 *
02090 * @since Version 1.0.0
02091 */
02092 void uart_debug_puts(const char* str)
02093 {
02094     if (str == nullptr) {
02095         return;
02096     }
02097
02098     /* Delegate to uart_puts_channel which already handles LF->CRLF conversion,
02099      * bounded iteration (k_uart_max_str_len), and per-character error checking. */
02100     (void)uart_puts_channel((uart_channel_t)k_uart_debug_channel, str);
02101 }
02102
02103 /**
02104 * @brief Transmit signed 32-bit integer as decimal string on debug UART
02105 *
02106 * @details
02107 * Converts a signed 32-bit integer to a decimal ASCII string and transmits
02108 * it on SCI9. Handles INT32_MIN correctly by using int64_t for the negation.
02109 * Uses a fixed-size stack buffer (k_uart_int_buffer_size = 12) built in
02110 * reverse then passed to uart_debug_puts().
02111 *
02112 * **Algorithm steps:**
02113 * 1. Determine sign; compute abs_value as uint32_t
02114 * 2. Null-terminate the buffer end
02115 * 3. Build digits right-to-left (statically bounded by k_uart_int_buffer_size)
02116 * 4. Prepend '-' if negative
02117 * 5. Call uart_debug_puts() with the resulting substring pointer
02118 *
02119 *
02120 * @pre uart_debug_init() must have been called successfully
02121 * @pre uart_debug_puts() must be functional
02122 *
02123 * @post Decimal representation of value transmitted on SCI9
02124 * @post Leading zeros suppressed; negative values prefixed with '-'
02125 *
02126 * @note No return value - output errors are silently discarded
02127 * @note Buffer is stack-allocated; safe for re-entrant calls on different tasks
02128 * @warning Only available when RX_IS_SIMULATOR is 0 (hardware builds)
02129 *
02130 * @par Thread Safety:
02131 * Not safe for concurrent access on SCI9 without external mutex.
02132 *
02133 * @par Performance:
02134 * - Execution time: ~90 us per digit @ 115200 baud + digit-loop overhead
02135 * - Stack usage: 32 bytes (buffer + locals)
02136 *
02137 * @par Example:
02138 * @code{.c}
02139 * uart_debug_puts("Value: ");
02140 * uart_debug_putint(42); // Sends "42"
02141 * uart_debug_putint(-2147483648); // Sends "-2147483648"
02142 * uart_debug_putc('\n');
02143 * @endcode
02144 *
02145 * @see uart_debug_puthex() Print value in hexadecimal
02146 * @see uart_debug_puts() Underlying string transmit
02147 *
02148 * @since Version 1.0.0
02149 */
02150 void uart_debug_putint(const int32_t value)
02151 {
02152     char buffer[k_uart_int_buffer_size]; /* Enough for -2147483648 */
02153     char* p = buffer + sizeof(buffer) - 1;
02154 }

```

```

02155 /* Handle negative numbers */
02156 const bool is_negative = (value < 0);
02157 uint32_t abs_value = is_negative ? (uint32_t)(-(int64_t)value) : (uint32_t)value;
02158
02159 /* Null terminate */
02160 *p = '\0';
02161
02162 /* Convert to string (reverse order, statically bounded) */
02163 for (uint8_t i = 0; i < k_uart_int_buffer_size - 1; ++i) {
02164     *(&p) = (char)('0' + (abs_value % k_uart_base_10));
02165     abs_value /= k_uart_base_10;
02166     if (abs_value == 0) {
02167         break; /* All digits processed */
02168     }
02169     if (p <= buffer) {
02170         break; /* Buffer limit reached (should not happen with correct sizing) */
02171     }
02172 }
02173
02174 /* Add minus sign if negative */
02175 if (is_negative && p > buffer) {
02176     *(&p) = '-';
02177 }
02178
02179 /* Transmit the string */
02180 uart_debug_puts(p);
02181 }
02182
02183 /**
02184  * @brief Transmit 32-bit unsigned integer as hexadecimal string on debug UART
02185  *
02186  * @details
02187  * Transmits "0x" prefix followed by the specified number of uppercase hex
02188  * digits of `value` on SCI9. Digits are always printed most-significant first.
02189  * The `digits` parameter is clamped to [k_uart_hex_min_digits, k_uart_hex_max_digits]
02190  * (1-8) before use.
02191  *
02192  * **Algorithm steps:**
02193  * 1. Send "0x" prefix via uart_debug_puts()
02194  * 2. Clamp digits to [1, 8]
02195  * 3. Iterate nibbles from MSN to LSN (statically bounded by k_uart_hex_max_digits)
02196  * 4. For each nibble index < digits, extract nibble and print uppercase hex char
02197  *
02198  *
02199  *
02200  * @pre uart_debug_init() must have been called successfully
02201  * @pre uart_debug_puts() and uart_debug_putc() must be functional
02202  *
02203  * @post "0x" followed by `digits` uppercase hex characters transmitted on SCI9
02204  * @post digits clamped to [1, 8] if out of range
02205  *
02206  * @note Uses static lookup table s_hex[] for digit-to-character conversion
02207  * @note Always prefixes output with "0x"
02208  * @warning Only available when RX_IS_SIMULATOR is 0 (hardware builds)
02209  *
02210  * @par Thread Safety:
02211  * Not safe for concurrent access on SCI9 without external mutex.
02212  *
02213  * @par Performance:
02214  * - Execution time: ~90 us per character @ 115200 baud
02215  * - Stack usage: 24 bytes
02216  *
02217  * @par Example:
02218  * @code{.c}
02219  * uart_debug_puts("Addr: ");
02220  * uart_debug_puthex(0xDEADBEEF, 8); // Sends "0xDEADBEEF"
02221  * uart_debug_puthex(0x42, 2);      // Sends "0x42"
02222  * uart_debug_putc('\n');
02223  * @endcode
02224  *
02225  * @see uart_debug_putint() Print signed decimal value
02226  * @see uart_debug_puts() Underlying string transmit
02227  *
02228  * @since Version 1.0.0
02229  */
02230 void uart_debug_puthex(const uint32_t value, uint8_t digits)
02231 {
02232     static const char s_hex[] = "0123456789ABCDEF";
02233
02234     uart_debug_puts("0x");
02235
02236     /* Clamp digits to valid range */
02237     if (digits > k_uart_hex_max_digits) {
02238         digits = k_uart_hex_max_digits;
02239     }
02240     if (digits == k_uart_hex_zero_digits) {
02241         digits = k_uart_hex_min_digits;

```

```
02242 }
02243
02244 /* Print hex digits from most significant (statically bounded, unsigned counter) */
02245 for (uint8_t i = k_uart_hex_zero_digits; i < k_uart_hex_max_digits; ++i) {
02246     const uint8_t digit_idx = (k_uart_hex_max_digits - k_uart_hex_min_digits) - i;
02247     if (digit_idx < digits) {
02248         const uint8_t nibble =
02249             (uint8_t)((value » (digit_idx * k_uart_hex_nibble_bits)) & k_uart_hex_nibble_mask);
02250         uart_debug_putc(s_hex[nibble]);
02251     }
02252 }
02253 }
02254
02255 #endif /* !RX_IS_SIMULATOR */
02256
02257 #endif /* __RX__ */
```